Modern Quantum Field Theory

From Foundations to the Frontier

Preface

Why Another QFT Textbook?

Quantum Field Theory stands as one of humanity's greatest intellectual achievements. It provides a unified framework for understanding the fundamental forces of nature, has yielded predictions verified to extraordinary precision, and continues to generate deep connections between physics and mathematics. Yet despite its importance, learning QFT remains a notoriously challenging endeavor.

The existing textbooks on QFT are, by and large, excellent. Peskin and Schroeder's comprehensive treatment has trained generations of physicists. Weinberg's magisterial three volumes offer unparalleled depth and rigor. Schwartz provides a modern perspective with clarity and precision. Srednicki's streamlined organization makes the path to loop calculations concrete. Zee's intuitive approach brings joy to the subject, and Coleman's legendary lectures offer a masterclass in physical reasoning.

So why write another textbook? This book aims to fill several gaps in the existing literature:

(i) **Conceptual motivation before formalism.** Many texts plunge into the mathematical machinery of field quantization without adequately explaining *why* quantum field theory is necessary. We begin with Chapter 1, "Why Quantum Field Theory?," which demonstrates that QFT is not merely one possible approach to relativistic quantum mechanics—it is the *only* consistent approach.

(ii) **Complete, self-contained derivations.** Rather than leaving crucial steps as exercises or referring the reader to external sources for technical arguments, every theorem, loop integral, trace identity, and renormalization-group coefficient that the development relies upon is derived in full within the text or its six appendices. Worked examples and explicit computations carry every calculation from start to finish on the page.

(iii) **A complete path to the frontier.** Most textbooks stop at the Standard Model or one-loop QCD. This book continues at full technical depth

through supersymmetry, conformal field theory and the conformal bootstrap, holography and quantum information, modern on-shell scattering amplitude methods, and quantum field theory in curved spacetime.

(iv) **The modern perspective throughout.** Effective field theory and the renormalization group are not afterthoughts but organizing principles. Symmetry guides every development, and three recurring theories—real scalar $\lambda\phi^4$ theory, quantum electrodynamics, and non-Abelian Yang–Mills theory—serve as unifying threads from classical field theory through the research frontier.

Prerequisites

This book is designed for first-year graduate students in physics, though advanced undergraduates with strong preparation will also find it accessible. The required background includes:

- **Quantum mechanics:** Familiarity with Dirac notation, angular momentum, time-dependent perturbation theory, and elementary scattering theory.
- **Classical mechanics:** Lagrangian and Hamiltonian formulations, canonical transformations.
- **Electrodynamics:** Maxwell's equations in covariant form, potentials, and gauge transformations.
- **Special relativity:** Four-vector notation, Lorentz transformations, and relativistic kinematics.
- **Mathematics:** Linear algebra, multivariable calculus, Fourier analysis, and complex contour integration (reviewed and extended in Part I). Representation theory, functional methods, and differential-geometric tools are developed from scratch as needed.

How to Use This Book

The book is organized into seventeen parts (Chapters 1–65) and six technical appendices (Appendices A–F), progressing systematically from foundations to the research frontier:

Parts I–III: Foundations (Chapters 1–13). Why quantum fields are forced by relativity and quantum mechanics; representation theory of the Lorentz

and Poincaré groups, functional methods, and analytic tools; classical Lagrangian field theory, Noether's theorem, and the gauge principle; canonical and path-integral quantization of free scalar, Dirac, and electromagnetic fields.

Parts IV–VI: Interacting Fields, QED, and Renormalization (Chapters 14–24). The S-matrix, LSZ reduction, Wick's theorem, Feynman rules, and cross sections; tree-level and one-loop quantum electrodynamics, infrared divergences, and precision tests ($g-2$, the Lamb shift); ultraviolet divergences, perturbative renormalization, the Wilsonian renormalization group, and the Callan–Symanzik equation.

Parts VII–IX: Non-Abelian Gauge Theory and the Standard Model (Chapters Yang–Mills theory, Faddeev–Popov and BRST quantization, asymptotic freedom, and perturbative QCD; spontaneous symmetry breaking, Goldstone's theorem, the Higgs mechanism, electroweak unification, and CKM/PMNS flavor physics; the complete Standard Model, precision electroweak and Higgs tests, and physics beyond the Standard Model.

Parts X–XIII: Anomalies, EFT, Non-Perturbative Physics, and Supersymmetr Chiral and gauge anomalies, anomaly matching, and Wess–Zumino–Witten terms; the effective field theory paradigm, chiral perturbation theory, HQET, and SMEFT; solitons, monopoles, instantons, the θ-vacuum, and lattice gauge theory; supersymmetry algebras, supersymmetric gauge theories, Seiberg duality, and Seiberg–Witten theory.

Parts XIV–XVII: The Research Frontier (Chapters 47–65). Conformal field theory in general dimensions and in two dimensions, the numerical and analytic conformal bootstrap; AdS/CFT correspondence, holographic correlators, entanglement entropy, Ryu–Takayanagi and quantum extremal surfaces, and the black hole information problem; spinor-helicity methods, BCFW recursion, generalized unitarity, color-kinematics duality and the double copy, and positive geometries; quantum fields in curved spacetime, the Unruh effect, and Hawking radiation.

A standard two-semester graduate course can cover Parts I–IX (supplemented by Appendices A–D), with selected chapters from Parts X–XII. Parts XIII–XVII can be used for advanced graduate courses or independent study leading directly into current research literature.

Notation and Conventions

We use the "mostly minus" metric signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units with $\hbar = c = 1$, and Hermitian Lie algebra generators satisfying

$\left[T^a, T^b\right] = if^{abc}T^c$ with $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ in the fundamental representation. Complete convention tables, Fourier and propagator normalizations, group-theory invariants, gamma-matrix identities, master loop integrals, functional determinants, and Standard Model reference data are collected in Appendices A–F.

Acknowledgments

This book could not have been written without the profound influence of the authors whose works we build upon. We are especially indebted to the pedagogical insights of Sidney Coleman, whose lectures set a standard that all physics textbooks should aspire to match, and to the classic texts of Peskin and Schroeder, Weinberg, Schwartz, Srednicki, and Zee.

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Chapter 1

Why Quantum Field Theory?

"The career of a young theoretical physicist consists of treating the harmonic oscillator in ever-increasing levels of abstraction."
—Sidney Coleman

This book is about what happens when two of the deepest theories of twentieth-century physics are forced to share a single description of the world. The reader arrives already fluent in both. Nonrelativistic quantum mechanics supplies a Hilbert space of states, observables as Hermitian operators, the canonical commutator $[\hat{x}, \hat{p}] = i$, and unitary time evolution generated by a Hamiltonian. Special relativity supplies a spacetime in which no influence outruns light and in which the laws take the same form for every inertial observer. Each theory is, in its own arena, ruthlessly well tested. The thesis of this chapter is that one cannot simply bolt them together: the most natural attempt — a relativistic version of the single-particle Schrödinger equation — fails, and it fails not through a technical clumsiness to be patched but through a structural impossibility. What replaces it is a quantum theory not of particles but of *fields*, and particles emerge from it as a derived notion. Establishing why this replacement is forced, and not merely one option among many, is the whole business of Chapter 1.

The argument is constructive rather than axiomatic. We will write down the obvious relativistic wave equations, the Klein–Gordon equation and the Dirac equation, and watch each one break: negative energies with no stable ground state, probability densities that go negative, and amplitudes that leak outside the light cone. Each failure, examined closely, points the same direction. The energy uncertainty of a particle confined to its own Compton wavelength is enough to make a new particle–antiparticle pair, so particle number cannot be conserved; and the apparent acausality of relativistic propagation is repaired only when a particle going one way is allowed to interfere with an antiparticle going the other. Both repairs demand a framework with a variable number of particles and a local, causal structure built in from the start. That framework is quantum field theory,

and the principle that organizes it — microscopic causality, the commuting of observables at spacelike separation — is important enough that it earns its own section (Section 1.5), one that is referred back to from the amplitudes chapters at the very end of the book.

We fix conventions now, in force for the whole chapter and matching the book-wide canon collected in Appendix A. The metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, so that an on-shell particle satisfies $p^2 = m^2$; we work in natural units $\hbar = c = 1$, restoring factors only when we put numbers to an estimate and saying so when we do; and repeated indices are summed, with $p \cdot x = p^0 x^0 - \mathbf{p}\cdot\mathbf{x}$. Where this chapter previews a later construction we point forward honestly: the scalar field is quantized in Chapter 9, the Dirac field in Chapter 11, and the electromagnetic field in Chapter 13, and nothing asserted here as a preview is left unproved there. The three running examples that thread the book — the real scalar $\lambda\phi^4$ theory, quantum electrodynamics, and Yang–Mills theory — make their first appearance at the end of this chapter as destinations on a map, not yet as machinery.

1.1 Relativity versus the single particle

The Schrödinger equation is not Lorentz invariant, and the reason is plain on the face of it. Quantizing the nonrelativistic dispersion relation $E = \mathbf{p}^2/2m$ by the substitutions $E \to i\partial_t$ and $\mathbf{p} \to -i\nabla$ gives

$$i\,\frac{\partial\psi}{\partial t} = -\frac{1}{2m}\nabla^2\psi, \tag{1.1}$$

which is first order in time but second order in space. Under a Lorentz boost time and space are mixed, so an equation that treats ∂_t and ∇ so asymmetrically cannot retain its form in every frame. This is not a deficiency of Schrödinger's equation — it is an exact and superbly accurate description of slow electrons — but it tells us that any relativistic wave equation must put time and space derivatives on the same footing. The natural place to start is therefore not with the dynamics but with the relativistic dispersion relation it must encode,

$$E^2 = \mathbf{p}^2 + m^2, \tag{1.2}$$

which is just the statement $p^2 = m^2$ in mostly-minus signature, written out with $p^0 = E$. Equation (1.2) is the seed from which every candidate relativistic wave equation grows, and it already contains the germ of the trouble: solving it for the energy gives $E = \pm\sqrt{\mathbf{p}^2 + m^2}$, two branches, one of them negative. A nonrelativistic theory never has to confront the negative branch because $E = \mathbf{p}^2/2m \geq 0$ automatically. A relativistic theory cannot avoid it, because the square in (1.2) is exactly what Lorentz invariance requires. Holding this observation in mind, we now build the simplest equation consistent with (1.2) and follow its consequences without flinching.

1.2 The Klein–Gordon equation

Applying the operator substitutions $E \to i\partial_t$, $\mathbf{p} \to -i\nabla$ directly to the squared relation (1.2) gives a wave equation for a single complex function $\phi(x)$,

$$\boxed{\left(\partial_t^2 - \nabla^2 + m^2\right)\phi = 0} \qquad \Longleftrightarrow \qquad (\Box + m^2)\phi = 0, \tag{1.3}$$

the *Klein–Gordon equation*, where $\Box \equiv \partial_\mu \partial^\mu = \partial_t^2 - \nabla^2$ is the d'Alembertian. Because $\Box$ is a Lorentz scalar and m^2 a constant, (1.3) has manifestly the same form in every inertial frame: a solution in one frame is a solution in all. In this narrow sense Klein–Gordon succeeds where Schrödinger could not. As the equation of motion of a *classical* relativistic field it is entirely sound, and it reappears in exactly that role in Chapter 6 as the Euler–Lagrange equation of the free scalar Lagrangian. The difficulties we are about to expose are difficulties of *interpretation*: they arise only when one insists, as the analogy with Schrödinger's equation invites, on reading ϕ as the probability amplitude of a single particle.

1.2.1 The conserved current, and why its density is not a probability

A probability interpretation needs a conserved, positive density. In the non-relativistic theory the Schrödinger equation supplies one: $\rho = |\psi|^2 \geq 0$ obeys $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$. We ask whether Klein–Gordon supplies an analogue, and we construct it honestly rather than quoting the answer. Multiply (1.3) by ϕ^* on the left, multiply the complex conjugate equation $(\Box + m^2)\phi^* = 0$ by ϕ, and subtract:

$$\phi^* \Box \phi - \phi \, \Box \phi^* = 0. \tag{1.4}$$

The mass terms cancel exactly because m^2 is real, which is the first hint that the surviving object is a genuine conservation law rather than an accident. The left-hand side is a total divergence. To see this, note

$$\partial_\mu(\phi^* \partial^\mu \phi - \phi \, \partial^\mu \phi^*) = \partial_\mu \phi^* \partial^\mu \phi + \phi^* \Box \phi - \partial_\mu \phi \, \partial^\mu \phi^* - \phi \, \Box \phi^* = \phi^* \Box \phi - \phi \, \Box \phi^*, \tag{1.5}$$

where the two gradient-cross terms cancel. Comparing (1.4) with (1.5), the four-current

$$j^\mu = i \left(\phi^* \partial^\mu \phi - \phi \, \partial^\mu \phi^*\right) \tag{1.6}$$

satisfies $\partial_\mu j^\mu = 0$; the factor of i is inserted so that j^μ is real, since under conjugation the bracket changes sign. Writing out the time component with $\partial^0 = \partial_t$ gives the candidate density

$$\rho \equiv j^0 = i \left(\phi^* \partial_t \phi - \phi \, \partial_t \phi^*\right), \tag{1.7}$$

and $\partial_t \rho + \nabla \cdot \mathbf{j} = 0$ with $\mathbf{j} = -i(\phi^* \nabla\phi - \phi \nabla\phi^*)$. So a conserved density exists. The fatal question is its sign.

Here the second-order character of (1.3) takes its revenge. Specifying a solution requires both ϕ and $\partial_t \phi$ as independent initial data, and (1.7) depends on both; nothing constrains their relative phase, so ρ may take either sign. The cleanest demonstration is to evaluate ρ on the plane-wave solutions of (1.3). Substituting $\phi = e^{-ip\cdot x} = e^{-iEt + i\mathbf{p}\cdot\mathbf{x}}$ into Klein–Gordon reproduces $E^2 = \mathbf{p}^2 + m^2$, so both signs $E = \pm E_{\mathbf{p}}$ with $E_{\mathbf{p}} \equiv +\sqrt{\mathbf{p}^2 + m^2}$ are admissible. For such a mode $\partial_t \phi = -iE\phi$, and (1.7) gives

$$\rho = i(\phi^*(-iE\phi) - \phi(iE\phi^*)) = 2E\,|\phi|^2. \tag{1.8}$$

The density is positive on the positive-energy branch and *negative* on the negative-energy branch, in lockstep with the sign of E. A negative probability is meaningless, and one cannot escape by discarding the negative-energy modes, both because that projection is not Lorentz invariant — a boost mixes the branches — and because, as (1.8) makes plain, the bad density is welded to the negative energies that (1.2) forced upon us in the first place. The two diseases, negative energy and negative probability, are one disease.

The density was never a probability

Equation (1.8) carries the resolution as well as the problem, though we cannot read it yet. The quantity $\rho = 2E|\phi|^2$ flips sign with E exactly as an electric *charge* density would for a particle versus its antiparticle. The Klein–Gordon current (1.6) is not a probability current at all; it is a *charge* current, and the field ϕ describes both a particle and an oppositely charged antiparticle on the same footing. This is invisible from inside single-particle quantum mechanics, where there is one particle and its norm must be a probability. It becomes the natural reading only once ϕ is an operator and j^μ a conserved Noether charge of a many-particle theory — the quantized complex scalar of Chapter 9.

1.2.2 Negative energy and the absence of a ground state

Even setting probability aside, the negative-energy branch is independently catastrophic for any would-be quantum theory. Quantum mechanics rests on the existence of a state of lowest energy, the ground state, both because the spectrum must be bounded below for the theory to be stable and because the whole edifice of thermodynamics and of perturbation theory is built on it. The dispersion relation (1.2) admits energies $E = -\sqrt{\mathbf{p}^2 + m^2}$ unbounded below as $|\mathbf{p}| \to \infty$. A single particle coupled to anything that can carry away energy — a photon, say — would cascade down this bottomless ladder forever, radiating without limit. No such cascade is observed; ordinary matter is stable. A consistent relativistic

quantum theory must therefore find a way to live with the negative-energy solutions of (1.2) that it cannot discard, while nonetheless possessing a stable lowest-energy state. Reconciling these two demands is the single technical knot that the rest of the chapter unties.

1.3 Dirac's equation and the sea

Dirac approached the negative-probability problem from a sharp diagnosis: the density (1.7) is dangerous because it involves $\partial_t\phi$, and it involves $\partial_t\phi$ because Klein–Gordon is second order in time. A first-order equation would give a density built from ψ alone, as in the nonrelativistic theory. The price of first order in time is, by Lorentz symmetry, first order in space as well; and an equation linear in all four derivatives that nonetheless squares to the Klein–Gordon operator cannot be satisfied by a single function. Demanding

$$(i\gamma^\mu\partial_\mu - m)(i\gamma^\nu\partial_\nu + m) = -(\Box + m^2) \tag{1.9}$$

forces $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}$, the Clifford algebra, whose lowest-dimensional realization in four spacetime dimensions is by 4×4 matrices. The wavefunction must accordingly carry four components, and Dirac's equation reads

$$\boxed{(i\gamma^\mu\partial_\mu - m)\psi = 0} \tag{1.10}$$

with ψ a four-component spinor. The full representation theory behind the γ^μ and the spinor ψ is developed in Chapter 11; here we need only the algebra $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$ and the Dirac adjoint $\bar\psi = \psi^\dagger\gamma^0$.

1.3.1 A positive density at last

The promise of the first-order construction was a positive, conserved density, and we verify it in full rather than assert it. Begin with (1.10), written $i\gamma^\mu\partial_\mu\psi = m\psi$. Take the Hermitian conjugate, $-i\,\partial_\mu\psi^\dagger\gamma^{\mu\dagger} = m\psi^\dagger$, and use the identity $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ (a consequence of the Clifford algebra together with $(\gamma^0)^2 = 1$, established in Chapter 11) after multiplying on the right by γ^0:

$$-i\,\partial_\mu\psi^\dagger\gamma^0\gamma^\mu = m\,\psi^\dagger\gamma^0, \qquad \text{i.e.} \qquad -i\,\partial_\mu\bar\psi\,\gamma^\mu = m\,\bar\psi. \tag{1.11}$$

Now contract the original equation with $\bar\psi$ on the left and the adjoint equation (1.11) with ψ on the right, and add:

$$i\,\bar\psi\gamma^\mu\partial_\mu\psi + i\,(\partial_\mu\bar\psi)\gamma^\mu\psi = m\,\bar\psi\psi - m\,\bar\psi\psi = 0 \implies \partial_\mu\Big(\bar\psi\gamma^\mu\psi\Big) = 0. \tag{1.12}$$

The Dirac current $j^\mu = \bar\psi\gamma^\mu\psi$ is conserved, and its time component is

$$j^0 = \bar\psi\gamma^0\psi = \psi^\dagger\gamma^0\gamma^0\psi = \psi^\dagger\psi = \sum_{a=1}^{4}|\psi_a|^2 \geq 0, \tag{1.13}$$

using $(\gamma^0)^2 = \eta^{00} = 1$. The density is manifestly non-negative. Dirac's equation has accomplished exactly what it set out to do: it has cured the negative-probability disease of Klein–Gordon.

1.3.2 But the negative energies remain

What it has *not* cured is the negative energies, because those came from the dispersion relation (1.2), which Dirac's equation reproduces component by component — that was the design requirement. Plane-wave solutions of (1.10) again come with $E = \pm E_{\mathbf{p}}$, and the negative-energy spinors cannot be thrown away: they are needed for the four-component spinor basis to be complete, so that an arbitrary initial ψ can be expanded at all. A positive density over a spectrum with no bottom is no improvement on the stability problem of Section 1.2; an electron in a positive-energy state could still radiate its way down into the negative-energy continuum.

Dirac's escape was as bold as the equation. Electrons are fermions, obeying the Pauli exclusion principle, so suppose that what we call the vacuum is in fact the state in which *every* negative-energy level is already occupied — an infinite filled "sea." A positive-energy electron then cannot fall into a negative-energy state because all of them are taken. The sea is by fiat the ground state. Excitations above it are of two kinds. Adding energy can promote a sea electron to a positive-energy state, leaving behind a *hole*: the absence of a negative-energy, negatively charged electron, which presents to the world as the presence of a positive-energy, positively charged particle of the *same mass*. The hole is an *antiparticle*.

From the proton to the positron

Dirac at first hoped the hole might be the proton, the only positively charged particle then known, reluctant to predict a new one. The identification was untenable: the hole must have the electron's mass, not the proton's two-thousand-fold larger one. By 1931 Dirac accepted the conclusion and predicted a positive electron [151]. Carl Anderson found it in cosmic-ray tracks in 1932. The episode is the first instance of a pattern that recurs throughout this book: a consistency requirement of relativistic quantum theory — here, what to do with the unavoidable negative-energy states — predicts new physics, and the new physics is there.

1.3.3 Why the sea is not the final word

The hole picture is a triumph, but it is scaffolding, not foundation, and it is worth being precise about why. First, it is irredeemably tied to the exclusion principle and so to fermions. Bosons — the quanta of the Klein–Gordon field,

for instance — have no exclusion principle; one cannot fill their negative-energy states to keep particles out, since any number of bosons may occupy each state. Yet charged spin-zero particles and their antiparticles plainly exist (charged pions, for one), and the negative-energy and negative-density problems of Section 1.2 are every bit as real for them. A construction that works only for fermions cannot be the general resolution. Second, the sea is an infinite reservoir of negative charge and energy whose properties must be quietly subtracted away to recover sensible answers; the bookkeeping is awkward and frame-dependent in just the way a fundamental description should not be. Third, and most tellingly, the very language has shifted under our feet: the moment we admit holes we have abandoned the single-particle theory we set out to build. A hole is a second excitation; pair-creation of an electron and a hole changes the particle count. The Dirac sea is a many-body system wearing single-particle clothing.

The lesson of the sea

The success of the hole picture and the awkwardness of its mechanism point past themselves to a cleaner statement. What relativistic quantum mechanics is really telling us is that *particle number is not a fixed label of a state*. Antiparticles must exist; particles and antiparticles can be made and unmade together; the natural arena is one in which the number of quanta is itself dynamical. The Dirac sea is one — fermion-only, frame-dependent — realization of this truth. Quantum field theory will be its frame-independent, species-blind realization, in which particle and antiparticle, boson and fermion, all appear symmetrically as excitations of a field and the vacuum needs no infinite reservoir at all.

1.4 Pair creation: particle number is not conserved

We have reached the central physical fact by two routes — negative-energy stability and the limits of the sea — and we now reach it a third way, by an estimate so elementary it can be made on the back of an envelope and so robust that no escape from it exists. The claim is that the attempt to localize a relativistic quantum particle within its own Compton wavelength supplies enough energy to create new particles, so that "one particle, sharply located" is not a state the theory can hold.

The estimate combines two principles the reader already trusts. From quantum mechanics, confining a particle to a spatial region of size Δx costs a momentum spread $\Delta p \gtrsim 1/\Delta x$ (this is $\hbar/\Delta x$ with $\hbar$ restored), and for a relativistic particle a momentum of order Δp carries an energy of order $\Delta E \sim \Delta p \sim 1/\Delta x$. From relativity, energy of order $2m$ (that is, $2mc^2$) deposited in a small region is enough to materialize a particle–antiparticle pair, since each member has rest

energy m. Setting the available energy equal to the pair threshold,

$$\Delta E \sim \frac{1}{\Delta x} \gtrsim 2m \qquad \Longleftrightarrow \qquad \Delta x \lesssim \frac{1}{2m}, \tag{1.14}$$

so as soon as one tries to pin a particle down to within (a fraction of) its reduced Compton wavelength $\lambda_C \equiv 1/m = \hbar/mc$, the localization energy suffices to make a new pair. The would-be single particle has spawned company. Probing shorter distances does not sharpen the picture of one particle; it guarantees many.

Worked Example 1.1 (title=Compton wavelengths and pair-creation thresholds)**.** We put numbers to (1.14) for three masses, which also fixes the distance and energy scales at which a single-particle description must be abandoned. The one constant needed is $\hbar c = 197.3\,\text{MeV fm}$, so the reduced Compton wavelength is $\lambda_C = \hbar c/(mc^2)$ and the pair threshold is $2mc^2$.

Electron ($mc^2 = 0.511\,\text{MeV}$):

$$\lambda_C = \frac{197.3\,\text{MeV fm}}{0.511\,\text{MeV}} \approx 386\,\text{fm}, \qquad 2mc^2 \approx 1.02\,\text{MeV}. \tag{1.15}$$

This is the scale of atomic and low-energy collision physics; it is exactly where positrons and pair production first intrude on the single-electron description, as in the 1.02 MeV threshold for e^+e^- production by a photon.

Proton ($mc^2 = 938\,\text{MeV}$):

$$\lambda_C = \frac{197.3}{938}\,\text{fm} \approx 0.21\,\text{fm}, \qquad 2mc^2 \approx 1.88\,\text{GeV}. \tag{1.16}$$

The proton's Compton wavelength is a fifth of a femtometre, smaller than the proton itself ($\sim 0.84\,\text{fm}$), a first sign that the proton is not elementary.

A light nucleus (carbon-12, $mc^2 \approx 11.2\,\text{GeV}$):

$$\lambda_C \approx \frac{197.3}{11200}\,\text{fm} \approx 1.8 \times 10^{-2}\,\text{fm}, \qquad 2mc^2 \approx 22\,\text{GeV}. \tag{1.17}$$

The heavier the object, the shorter the distance over which single-particle quantum mechanics survives and the higher the energy needed to break it — which is why nonrelativistic quantum mechanics is so accurate for nuclei and atoms in their ground states, and why it fails precisely in the relativistic, high-energy collisions where new particles are made.

The estimate is a threshold, not a switch

Equation (1.14) should not be read as "below the Compton wavelength, pairs; above it, none." Virtual pair creation contributes at *all* scales — it is responsible, for example, for the vacuum polarization that makes the electric charge run with energy (Chapter 13 and Part VI). What the

estimate establishes is the sharper, structural point: there is *no* regime in which a relativistic quantum theory can consistently hold particle number fixed, because the very act of localization that a single-particle wavefunction presupposes injects pair-creating energy. Particle number is not a good quantum number, full stop.

1.5 Locality, causality, and microcausality

A relativistic theory must respect causality: no measurement, no disturbance, no signal may propagate faster than light, because in another frame a superluminal influence runs backward in time and effect precedes cause. We now show that relativistic single-particle quantum mechanics violates this requirement, that the violation is repaired only by the existence of antiparticles, and that the repair is codified in a condition — microcausality — which a field theory must satisfy and which, remarkably, turns out to dictate the connection between spin and statistics. This section is the conceptual hinge of the chapter, and it is referred back to from the synthesis of scattering theory at the end of the book, where causality is reinstated as one of the small handful of inputs from which amplitudes can be reconstructed.

1.5.1 Amplitudes leak outside the light cone

Consider the amplitude for a particle prepared at the spacetime origin to be found at a point $\mathbf{x}$ a time t later. In nonrelativistic quantum mechanics this is the Schrödinger propagator

$$U(\mathbf{x},t) = \left\langle \mathbf{x}\middle| e^{-i\hat{H}t}\, 0\right\rangle = \int \frac{d^3p}{(2\pi)^3}\, e^{i\mathbf{p}\cdot\mathbf{x}}\, e^{-i\mathbf{p}^2 t/2m} = \left(\frac{m}{2\pi i t}\right)^{3/2} e^{\,im\mathbf{x}^2/2t}, \quad (1.18)$$

the last step being a Gaussian integral. This amplitude is nonzero for *every* $\mathbf{x}$ at any $t > 0$: a particle localized at the origin has instantly a nonzero amplitude to be found arbitrarily far away. Strictly this already violates relativity, but harmlessly, because the nonrelativistic theory makes no claim to relativistic causality and the offending amplitude is a pure phase of fixed modulus that carries no usable signal. The question is what becomes of (1.18) when we do the same computation relativistically, replacing the energy $\mathbf{p}^2/2m$ by the relativistic $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$:

$$U(\mathbf{x},t) = \int \frac{d^3p}{(2\pi)^3}\, e^{i\mathbf{p}\cdot\mathbf{x}}\, e^{-iE_{\mathbf{p}}t}. \quad (1.19)$$

This integral is evaluated in full in Chapter 9, where the propagator is studied as the central object of the free scalar theory; the result is that for a spacelike-separated final point, $\mathbf{x}^2 > t^2$, the amplitude does not vanish but falls off as $U \sim$

$e^{-m\sqrt{\mathbf{x}^2-t^2}}$. The amplitude to propagate outside the light cone is exponentially small — suppressed once the spacelike separation exceeds a Compton wavelength, the same scale as everywhere in this chapter — but it is not zero. Taken at face value as a single-particle propagation amplitude, (1.19) says a particle can be found beyond the reach of light. Causality appears to be broken.

1.5.2 How antiparticles restore causality

The resolution is the one foreshadowed by the negative-energy density of (1.8) and by the holes of Section 1.3, and it is worth stating precisely even though its proof must wait for the operator formalism of Chapter 9. Causality, in a quantum theory, is not the demand that out-of-cone amplitudes vanish individually; it is the demand that no *measurement* at one point can affect the statistics of a *measurement* at a spacelike-separated point. The amplitude (1.19) for a particle to travel from a point A to a spacelike-separated point B is, by itself, nonzero — but so is the amplitude for an *antiparticle* to travel from B to A. To a Lorentz observer the two processes are not even distinct: a boost can reverse the temporal order of two spacelike-separated events, turning "particle from A to B" into "antiparticle from B to A," since reversing the direction of travel of a particle of charge $+q$ in time is indistinguishable from a charge $-q$ moving the other way. Any frame-independent, causal quantity must therefore include both. When one computes the genuinely observable object — the commutator of two field operators at A and B — the particle and antiparticle contributions *cancel exactly* outside the light cone. The leftover noncausal tail of (1.19) is precisely the piece that the antiparticle process subtracts.

This cancellation is impossible to arrange in a single-particle theory, which has particles but no antiparticles to do the cancelling, and it requires the two species to have equal masses and opposite charges and to be created and destroyed in concert — exactly the content of antiparticle symmetry. It works only in a theory where both are present from the start, on equal footing, with no infinite sea: a quantum field theory. The full computation, including the exact form of the cancellation and the role of the time-ordering that defines the Feynman propagator, is carried out in Chapter 9; what matters here is the logic, that *causality forces antiparticles*, the converse of the lesson of Section 1.4 that pair creation forces a many-particle theory.

1.5.3 Microcausality and the seed of spin-statistics

We can now state the principle that the field-theoretic framework will be built to satisfy. Observables are represented by local operators $\mathcal{O}(x)$, built from fields at the point x. Two measurements at spacelike separation must not influence each other, which in quantum mechanics means their operators must be simultaneously diagonalizable — they must commute. This is the principle of *microcausality*, or

local commutativity:

$$\boxed{[\mathcal{O}_1(x), \mathcal{O}_2(y)] = 0 \qquad \text{whenever} \qquad (x-y)^2 < 0.} \tag{1.20}$$

Spacelike-separated observables commute; the commutator is the carrier of mutual influence, and outside the light cone there must be none. Equation (1.20) is not a theorem to be proved within single-particle quantum mechanics — it cannot be, as we have seen — but a structural requirement *imposed* on a relativistic quantum theory, and it is one of the load-bearing axioms of the subject.

Its consequences run deeper than causality alone. When the free fields are quantized in Part III, the requirement (1.20) that observables built from them commute at spacelike separation can be met for a spin-zero field only if the field is expanded with *commuting* creation and annihilation operators, and for a spin-one-half field only if it is expanded with *anticommuting* ones. Demanding causality thus forces integer-spin particles to be bosons and half-integer-spin particles to be fermions: this is the spin-statistics connection, and Chapter 9 exhibits it explicitly for the scalar while Chapter 11 does so for the Dirac field. The Pauli exclusion principle, postulated by Dirac as an external input to build his sea, is in the field-theoretic framework a *derived* consequence of relativistic causality. That a single principle — no influence outside the light cone — should both repair the causality paradox and explain why electrons exclude one another is among the deepest results in physics, and it is a result that has no home in single-particle quantum mechanics. It lives only in quantum field theory.

1.6 Fields as the fundamental objects

Every thread of the argument now converges on the same prescription. Particle number is not conserved (Section 1.4); antiparticles must exist and interfere with particles (Section 1.5); the negative-energy states cannot be discarded yet must not destabilize the vacuum (Sections 1.2 and 1.3). A framework meeting all of these at once cannot take "the position of the particle" as its basic variable, because there may be no definite number of particles to have positions. It must instead take as fundamental an object defined everywhere in spacetime, able to support any number of excitations, with a stable lowest-energy state. That object is a *quantum field*: an operator $\hat{\phi}(x)$, or $\hat{\psi}(x)$, attached to each spacetime point, acting on a single Hilbert space that accommodates states of zero, one, two, or arbitrarily many quanta.

1.6.1 Particles as quanta

The reader has, in fact, already met a quantum field and quantized it, perhaps without the name. The electromagnetic field is a classical field $\mathbf{E}(\mathbf{x}, t)$, $\mathbf{B}(\mathbf{x}, t)$

filling all of space, with energy

$$H_{\text{EM}} = \frac{1}{2}\int d^3x\,(\mathbf{E}^2 + \mathbf{B}^2). \tag{1.21}$$

Resolved into normal modes, this is a collection of independent harmonic oscillators, one for each wavevector and polarization. Quantizing each oscillator by the ladder method of elementary quantum mechanics gives a mode of frequency ω the energy spectrum $E = \omega\,(n + \frac{1}{2})$ with $n = 0, 1, 2, \ldots$, and we name the integer n the *number of photons* in that mode. The photon is not a primitive entity; it is a *quantized excitation of the electromagnetic field*, and the photon number is exactly the kind of dynamical, non-conserved quantity that Section 1.4 demanded. This construction — the quantization of the electromagnetic field and the emergence of the photon, together with the gauge subtleties unique to a massless spin-one field — is carried out in Chapter 13.

The same idea, applied to every species, is the ontology of the subject. Each kind of particle is the quantum of a field. Electrons and positrons are the excitations of the Dirac field $\psi(x)$ of Chapter 11; the spinless quanta of a real scalar field are the simplest case of all, quantized first in Chapter 9; quarks and gluons are excitations of quark and gluon fields. The field is primary; particles are what its quanta are called. In this language the vacuum is simply the state with no quanta in any mode — a unique, stable, lowest-energy state, with no infinite sea to subtract — and a multiparticle state is built by exciting modes above it. The negative-energy solutions that wrecked the single-particle reading are reinterpreted without contradiction: the negative-frequency part of the field operator multiplies an operator that *creates an antiparticle*, rather than describing a particle of negative energy. Every energy in the spectrum is positive; the ground state exists; antiparticles are automatic; particle number floats. The entire list of demands is met at one stroke.

Why "second quantization" is a misleading name

The passage from quantum mechanics to field theory is often called *second quantization*, and the term, though entrenched, misleads. In first quantization one promotes the classical phase-space variables of a particle to operators, $x, p \to \hat{x}, \hat{p}$ with $[\hat{x}, \hat{p}] = i$. In "second" quantization one promotes the wavefunction itself — already a quantum object — to an operator-valued field $\hat{\psi}(\mathbf{x})$ obeying $\left[\hat{\psi}(\mathbf{x}), \hat{\psi}^\dagger(\mathbf{y})\right] = \delta^{(3)}(\mathbf{x}-\mathbf{y})$ (anticommutators for fermions). But nothing is quantized twice. What has happened is a change of fundamental variable: the field, not the particle coordinate, is the thing whose modes are quantized. The single-particle Schrödinger wavefunction reappears only as the matrix element $\langle 0|\hat{\psi}(\mathbf{x})|1 \text{ particle}\rangle$, a derived quantity in a theory whose Hilbert space — the *Fock space*, the direct sum of the n-particle spaces for all $n \geq 0$ — contains states of every

particle number at once.

1.6.2 The three theories that will recur

Having argued for fields in general, we name the three specific field theories that will serve as the book's running examples, so the reader can recognize them as old friends when each returns, one layer deeper, throughout the chapters ahead. We do not develop them here; each is properly introduced as the logical development requires.

The simplest is a single real scalar field with a quartic self-interaction, the $\lambda\phi^4$ *theory*, whose classical Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \tag{1.22}$$

is the universal pedagogical workhorse: simple enough to compute everything explicitly, rich enough to exhibit nearly every phenomenon in the book. We will quantize its free part in Chapter 9, compute its first scattering amplitude, renormalize it, follow its coupling under the renormalization group, and finally meet it at a nontrivial fixed point that is a conformal field theory — the handoff from the textbook core to the frontier. The second is *quantum electrodynamics*, the theory of the electron field ψ coupled to the photon field A_μ, whose interaction we will derive in Chapter 13 and Part V from nothing but the principle of local U(1) gauge invariance; it carries us to the most precisely verified prediction in all of science, the electron's anomalous magnetic moment. The third is *Yang–Mills theory*, the non-abelian generalization in which the gauge group is SU(N) rather than U(1); it underlies both the strong force (SU(3)) and, in broken form, the weak force, and its peculiar property of growing weak at short distances — asymptotic freedom — is one of the high points of Part VII. These three theories, scalar, abelian gauge, and non-abelian gauge, are the spine along which the entire development is organized.

1.7 A map of the book

With the necessity of fields established, we close by laying out the route from here to the frontier, so the reader can see the destination from the trailhead. The logical order is cumulative: each part supplies tools the next one cannot do without.

We begin with *mathematical foundations* (Part I): the representation theory of the Lorentz and Poincaré groups, which classifies what a relativistic particle can be — the scalars, spinors, and vectors that became our fields above — together with the functional and analytic machinery the rest of the book quotes.

Classical field theory (Part II) then treats fields as classical dynamical systems: the action principle, Noether's theorem linking symmetries to conservation laws, and the gauge principle that derives electromagnetism from a symmetry. With the classical theory in hand we *quantize the free fields* (Part III), making rigorous the heuristic "modes are oscillators" of Section 1.6 for the scalar, Dirac, and electromagnetic fields, and proving there the microcausality and spin-statistics results promised in Section 1.5.

Interactions enter next (Parts IV–VI). The S-matrix and the Dyson series turn the interacting Hamiltonian into a perturbative expansion; Feynman diagrams organize that expansion; and the cross-section formulae connect it to what detectors measure. Applied to electrodynamics (Part V) this yields real numbers for real processes. The infinities that appear along the way are not a defect but a signpost, and *renormalization and the renormalization group* (Part VI) extract from them the modern understanding that every quantum field theory is an effective description valid up to some scale, with physics organized by how couplings flow as that scale changes. From there the structure of the Standard Model unfolds: *non-abelian gauge theories* (Part VII), *spontaneous symmetry breaking* and the Higgs mechanism (Part VIII), the *Standard Model* itself (Part IX), and the *anomalies* (Part X) that constrain which theories are even consistent. Beyond perturbation theory lie *effective field theories* (Part XI), genuinely *non-perturbative methods* — instantons, solitons, the lattice (Part XII) — and *supersymmetry* (Part XIII), where exact results become possible.

The final third of the book is the frontier, developed at the same depth as the core rather than surveyed. *Conformal field theory and the bootstrap* (Part XIV) study quantum field theories at their scale-invariant fixed points — where $\lambda\phi^4$ theory terminates, at the Wilson–Fisher fixed point and the three-dimensional Ising model — and reconstruct them from symmetry and consistency alone. *Holography* (Part XV) develops the correspondence by which a quantum field theory without gravity is exactly equivalent to a gravitational theory in one higher dimension, knitting together quantum information and spacetime geometry. *Modern scattering amplitudes* (Part XVI) returns to the gauge theories of QED and Yang–Mills theory and rebuilds their amplitudes from analyticity, unitarity, and the very causality of Section 1.5, revealing structures — the spinor-helicity formalism, on-shell recursion, the double copy relating gauge theory to gravity — invisible in the Feynman-diagram expansion that first produced them. A short final part treats *quantum field theory in curved spacetime*, where the vacuum itself becomes observer-dependent and black holes are found to radiate.

Some of these frontier results are settled mathematics and some are the live edge of current research; we will be explicit, and will date our claims, about which is which. The reader who reaches the end will have followed a single chain of reasoning, with no gaps and no external lookups, from the failure we diagnosed in this chapter — that quantum mechanics and special relativity cannot share a

single particle — all the way to the structures at the boundary of what is known. The first link in that chain is the one we have just forged: relativity and the quantum, taken together, are a theory of fields.

Further reading

The motivation for quantum field theory from the failure of single-particle relativistic quantum mechanics is given with particular care in Peskin and Schroeder [396] ch. 1–2 and Schwartz [432] ch. 1–2; the localization/pair-creation argument of Section 1.4 follows the spirit of both. Weinberg [500] ch. 1 tells the same story as intellectual history, from the Dirac equation through the discovery that fields, not particles, are fundamental, and is the definitive statement of the view that quantum field theory is the inevitable consequence of relativity, the quantum, and cluster decomposition. Zee [532] ch. I.1–I.2 gives the quickest physical route to the same conclusion. The original papers repay reading: Dirac's equation [150] and his prediction of the antielectron [151], and the Feynman–Stückelberg reinterpretation of antiparticles as negative-energy solutions running backward in time [191]. For the structural results previewed in Section 1.5 — microcausality and the spin-statistics connection at the axiomatic level — see Streater and Wightman [458]; the algebraic perspective on locality is developed in Haag [246]. Coleman [129] is unmatched as a sustained pedagogical companion to the whole of the textbook core.

Part I

Mathematical Foundations

Chapter 2

Groups and Representations

Chapter 1 argued that relativity and quantum mechanics cannot coexist as separate theories: insist on both, and the number of particles ceases to be conserved, localized one-particle wavefunctions lose their meaning, and we are forced to describe nature by quantum *fields* with infinitely many degrees of freedom. The question this raises is organizational. With so many degrees of freedom, what disciplines them? What tells us which fields can exist, how they combine, and which interactions are allowed? The answer, running through every chapter that follows, is *symmetry*. And the lesson that makes symmetry the north star of this book is Wigner's: an elementary particle is not a little ball with properties attached, but *is* an irreducible unitary representation of the symmetry group of spacetime. Mass and spin are not data we measure and tabulate; they are the two labels (the Casimir invariants) that classify those representations. Before we can say in Chapter 3 what a relativistic particle can possibly be, or in Chapters 8 and 25 how the gauge principle conjures the forces between particles out of a demand for local symmetry, we must first learn what a representation is and how to find them all. That is the work of this chapter.

We build the language from the group axioms upward, from first principles, without assuming prior acquaintance with representation theory. What *is* assumed is linear algebra—eigendecompositions, inner products, unitary and Hermitian operators—and the angular-momentum ladder of nonrelativistic quantum mechanics at the Sakurai level. The reader has met J^2, J_z, and the raising and lowering operators $J_\pm$, and has used the spin-j multiplets of dimension $2j+1$. Here those familiar objects are recast as what they honestly are: the complete representation theory of the Lie algebra $\mathfrak{su}(2)$. Nothing about the ladder is taken on faith as physics folklore; it is rederived as a theorem about how an abstract three-element bracket can be realized on a finite-dimensional space. This recasting is the single most important transferable skill in the chapter, because the method that solves $\mathfrak{su}(2)$ is the method that, suitably generalized, solves every compact simple Lie algebra.

The chapter is one road, not a list of topics. A symmetry acts on the quantum state space, and consistency with the probability interpretation forces that action to be *unitary*; a representation is exactly such a linear action, and the first task is to tame it. A general (reducible) action is a stack of indivisible pieces, so we isolate the atoms—the irreducible representations—and prove the structural facts that make the decomposition unique and computable: Maschke's complete-reducibility theorem, Schur's lemma, and the orthogonality of characters. We then pass from finite groups to *continuous* ones, where a single new idea does almost all the work: a continuous symmetry is captured near the identity by its linearization, the Lie algebra, and the full group element is recovered by exponentiation. With that bridge in hand we solve $\mathfrak{su}(2)$ completely—the worked spine of the chapter—and find the spin-j irreducibles falling out as the only possibilities. Combining two systems then forces the tensor product and its decomposition into irreducibles, the Clebsch–Gordan problem, which is where addition of angular momenta comes from. Finally the $\mathfrak{su}(2)$ method is industrialized: Cartan and Weyl replace the single ladder by a ladder for each *root*, and the highest-weight construction classifies the irreducibles of any compact simple algebra. We carry this far enough to solve $\mathfrak{su}(3)$ in full.

The payoff is that this foundations chapter is where two of the book's three benchmark theories receive their alphabet. The complete $\mathfrak{su}(2)$ representation theory built here is the template every later spin and isospin computation silently reuses; in particular the explicit spin-$\frac{1}{2}$ matrices seed QED, where the electron enters as a two-component object that Chapter 11 quantizes as a Dirac field. And $\mathfrak{su}(3)$, with its Hermitian generators T^a, structure constants f^{abc}, and quadratic Casimir, is the exact language of color that Yang–Mills theory will speak: the non-abelian gauge theory of Chapter 25 is built on precisely these data, and its color factors C_A and C_F are the Casimirs defined here. The hand-off to Chapter 3 is sharp: the Lie-algebra/exponential machinery and the full $\mathfrak{su}(2)$ representation theory are exactly what is needed to recognize the Lorentz algebra as $\mathfrak{so}(1,3) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ and to read off the scalar, Weyl, Dirac, and vector fields as its representations.

Throughout we use the standing conventions of Chapter A. Generators of a Lie algebra are taken *Hermitian*, so that group elements $\exp(-i\theta_a T^a)$ are unitary, with the bracket and fundamental normalization

$$\left[T^a, T^b\right] = i f^{abc} T^c, \qquad \operatorname{tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab} \quad \text{(fundamental representation)}. \tag{2.1}$$

The structural definitions live here; the catalogued numerical data—Casimirs and Dynkin indices for general $\mathrm{SU}(N)$, the $\mathrm{SU}(N)$ identities, and the explicit Gell-Mann and Pauli matrices—are arbitrated by Chapter B. With conventions fixed, we begin where all of group theory begins: with the idea of a symmetry and the axioms that pin it down.

2.1 Symmetry, Groups, and Why We Represent Them

The chapter opened with a claim that sounds more like a slogan than a theorem: an elementary particle *is* an irreducible unitary representation of the spacetime symmetry group. A slogan is only as good as the words in it, and three of these words have so far been used without being defined. What is a *representation*? When is one *unitary*? What does it mean for a representation to be *irreducible* — the property that, Wigner tells us, distinguishes an elementary particle from a composite one? This section answers the first two questions and sets up the third. We begin where the physics begins, with the notion of a symmetry, extract from it the abstract object a symmetry generates — a group — and then confront the fact that in quantum mechanics a group never acts on states directly but always through linear operators. That action is the representation. By the end of the section we will have a working vocabulary (group, homomorphism, representation, faithfulness, equivalence, unitarity, direct sum), a stocked zoo of examples to test every later statement against, and one genuine theorem: over a finite group, the word "unitary" costs nothing, because every representation can be made unitary by a change of basis. That theorem, due to Weyl, is the finite-group shadow of a fact that will organize the whole book.

Throughout we use the conventions fixed in Chapter A; the structural data (dimensions, Casimir eigenvalues, Dynkin indices) for the continuous groups named below are collected in Chapter B and will be derived where the development needs them.

2.1.1 What a symmetry is, and why it must act linearly

Physically, a symmetry is a transformation of a system that leaves its dynamics unchanged: rotate the apparatus, or translate it, or reflect it, and the same experiment gives statistically the same results. The transformations that qualify share a structure so rigid that it is worth isolating before we name it. First, doing nothing is a symmetry. Second, any symmetry can be undone — if a rotation by θ preserves the dynamics, so does the rotation by $-\theta$ that reverses it. Third, two symmetries performed in succession give a third: rotate, then rotate again, and the net operation is still a rotation that preserves the dynamics. Fourth, when we compose three of them the order of bracketing is irrelevant, because composition of transformations of a set is automatically associative. These four facts — an identity, inverses, closure under composition, and associativity — are not a coincidence of rotations. They are forced on *any* collection of invertible transformations closed under composition, and they are exactly the axioms of a group. The group is the abstract residue of the symmetry: it remembers how the transformations combine while forgetting what they act on.

But in quantum mechanics a symmetry cannot remain abstract, because the system it acts on is a definite mathematical object — a complex Hilbert space $\mathcal{H}$

whose rays (one-dimensional subspaces) are the physical states. A symmetry must send states to states in a way that preserves everything physically measurable, and the only physically measurable thing built from two states $|\psi\rangle, |\phi\rangle$ is the transition probability $|\langle\psi|\phi\rangle|^2 / (\langle\psi|\psi\rangle \langle\phi|\phi\rangle)$. So a quantum symmetry is, at minimum, a map of rays to rays that preserves all such probabilities. The content of Wigner's theorem — which we state here for orientation and prove in its proper home, the spacetime setting of Chapter 3 — is that any such map is induced by an operator U on $\mathcal{H}$ that is either *unitary* (linear, with $\langle U\psi|U\phi\rangle = \langle\psi|\phi\rangle$) or *antiunitary* (antilinear, with $\langle U\psi|U\phi\rangle = \langle\phi|\psi\rangle$), and that the operator is unique up to an overall phase. Antiunitary operators are needed only for symmetries that reverse the direction of time; every continuous symmetry, and every symmetry we build up from the identity, is realized unitarily.

Physical Insight

The lesson to carry forward is that the abstract group is never the real object of study in quantum mechanics. A group of symmetries is given to us, but what acts on the states is a family of unitary operators $U(g)$, one for each group element g, composing the way the group does: $U(g)U(h) = U(gh)$. That family — a homomorphism from the group into the unitary operators on $\mathcal{H}$ — is a *unitary representation*. The phase ambiguity in Wigner's theorem means the composition law need only hold up to a phase, $U(g)U(h) = e^{i\xi(g,h)}U(gh)$, giving a *projective* representation; we note this now and return to it for the rotation and Lorentz groups in Chapter 3, where the phases turn out to encode half-integer spin. Until then we work with honest (non-projective) representations, where $U(g)U(h) = U(gh)$ exactly.

This is why representation theory, and not abstract group theory, is the language of this book: the physics lives in how the symmetry group acts by linear operators on a vector space, and different actions of the same group are genuinely different physics. An electron and a photon are acted on by the same rotation group, but through inequivalent representations, and that inequivalence is the difference between spin $\frac{1}{2}$ and spin 1. We now make each italicized word precise.

2.1.2 Groups

Definition 2.1 (Group)**.** A *group* is a set G equipped with a binary operation $G \times G \to G$, written $(g, h) \mapsto gh$ and called the *product*, satisfying:

(G1) *Associativity:* $(gh)k = g(hk)$ for all $g, h, k \in G$.

(G2) *Identity:* there is an element $e \in G$ with $eg = ge = g$ for all $g \in G$.

(G3) *Inverses:* for each $g \in G$ there is an element $g^{-1} \in G$ with $g\,g^{-1} = g^{-1}g = e$.

The group is *abelian* (or commutative) if in addition $gh = hg$ for all g, h. The number of elements $|G|$ is the *order* of G; it may be finite or infinite. A *subgroup* is a subset $H \subseteq G$ that contains e and is closed under products and inverses, so that H is itself a group under the inherited operation.

The identity is unique, for if e and e' both satisfy (G2) then $e = ee' = e'$. Inverses are unique, for if h and h' both invert g then $h = h(gh') = (hg)h' = h'$. These one-line arguments are the reason we may speak of *the* identity and *the* inverse; we use them silently from now on. Two more facts of constant use: $(gh)^{-1} = h^{-1}g^{-1}$ (check: $(gh)(h^{-1}g^{-1}) = g(hh^{-1})g^{-1} = e$), and the *cancellation law* $gx = gy \Rightarrow x = y$, obtained by multiplying on the left by g^{-1}. Cancellation has an immediate consequence we will exploit repeatedly: for fixed g, the map $x \mapsto gx$ is a bijection of G onto itself, since it is injective by cancellation and surjective because $g^{-1}y$ maps to y. "Left translation permutes the group" is the engine behind both the regular representation and the averaging trick below.

The point of the axioms is the examples they govern, so we stock a zoo. We split it into the finite groups, where counting arguments rule, and the continuous (Lie) groups, where the physics of spacetime and gauge symmetry lives.

Finite groups. The simplest infinite family is the *cyclic group*.

Example 2.2 (Cyclic group $\mathbb{Z}_n$)**.** Fix an integer $n \geq 1$ and let $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$ with the product given by addition modulo n. Associativity is inherited from integer addition; the identity is 0; the inverse of k is $n - k$ (and of 0 is 0). The group is abelian and of order n. A more revealing model realizes the *same* group as the n-th roots of unity $\{\, \omega^k : k = 0, \ldots, n-1 \,\}$ with $\omega = e^{2\pi i/n}$, under multiplication of complex numbers: $\omega^j \omega^k = \omega^{j+k} = \omega^{(j+k) \bmod n}$ because $\omega^n = 1$. Geometrically these are the rotations of a regular n-gon about its center by multiples of $2\pi/n$. The cyclic group is generated by a single element (the rotation ω, or the integer 1): every element is a power of it. We meet $\mathbb{Z}_2$ again as the parity $\phi \to -\phi$ of ϕ^4 theory.

Allowing reflections as well as rotations of the n-gon enlarges $\mathbb{Z}_n$ to a non-abelian group, our first.

Example 2.3 (Dihedral group D_n)**.** The *dihedral group* D_n is the symmetry group of a regular n-gon, comprising the n rotations of $\mathbb{Z}_n$ and n reflections, so $|D_n| = 2n$. Write r for the rotation by $2\pi/n$ and s for any one reflection. Then $r^n = e$, $s^2 = e$, and the two generators fail to commute: reflecting, rotating, and reflecting back is the same as rotating the other way,

$$srs^{-1} = r^{-1}. \tag{2.2}$$

These relations determine the group completely — every element is uniquely r^k or $s\,r^k$ with $0 \le k < n$ — and Eq. (2.2) shows D_n is nonabelian for $n \ge 3$. A presentation like this, listing generators and the relations they satisfy, is often the most economical way to specify a finite group.

Example 2.4 (Symmetric group S_n)**.** The *symmetric group* S_n is the set of all bijections (permutations) of $\{1, 2, \dots, n\}$, with composition of functions as the product. The identity is the permutation that fixes everything; inverses exist because bijections are invertible; associativity is automatic for composition of maps. Its order is $|S_n| = n!$, and it is nonabelian for $n \ge 3$. The symmetric group is not an exotic special case but the universal one: *every* finite group of order n is a subgroup of S_n, because (by the cancellation law) left translation embeds G into the permutations of its own underlying set. This embedding is Cayley's theorem, and it is the prototype of the regular representation we build in Section 2.1.3. S_n also governs the quantum statistics of identical particles: the permutation symmetry of an n-body wavefunction is an action of S_n, and bosons and fermions are the two one-dimensional representations we will meet by name below.

Continuous groups. The groups that organize field theory are not finite but *continuous*: their elements are labeled by real parameters varying smoothly, and the group product and inversion are smooth functions of those parameters. Such a group is a *Lie group*; we treat the continuous parameters and the calculus they support systematically in Section 2.3, and here only name the cast and their defining conditions as matrix groups. A *matrix group* is a subgroup of the invertible $n \times n$ complex matrices $\mathrm{GL}(n, \mathbb{C})$ under matrix multiplication; the identity is the unit matrix $\mathbf{1}$ and inverses are matrix inverses, so the group axioms hold automatically and the only question is which matrices we keep.

Example 2.5 (The unitary and orthogonal families)**.** The following matrix groups recur throughout the book.

- $\mathrm{U}(1) = \{\, e^{i\theta} : \theta \in \mathbb{R} \,\}$, the unit circle of phases under multiplication, with $e^{i\theta}e^{i\theta'} = e^{i(\theta+\theta')}$. It is abelian, and is the gauge group of electromagnetism (QED): the phase rotations of a complex field. Topologically it is a circle.

- $\mathrm{SU}(N) = \{\, U \in \mathrm{GL}(N, \mathbb{C}) : U^\dagger U = \mathbf{1},\ \det U = 1 \,\}$, the special unitary group. The condition $U^\dagger U = \mathbf{1}$ preserves the Hermitian inner product $\langle u|v\rangle = u^\dagger v$ on $\mathbb{C}^N$; the condition $\det U = 1$ removes an overall phase. $\mathrm{SU}(2)$ is the group of spin (QED's electron) and the worked spine of this chapter; $\mathrm{SU}(3)$ is the color group of Yang–Mills theory and, separately, the approximate flavor symmetry of the light quarks. We solve $\mathrm{SU}(2)$ completely and reach $\mathrm{SU}(3)$ by the same method.

- $\mathrm{SO}(N) = \{\, R \in \mathrm{GL}(N,\mathbb{R}) : R^{\mathsf{T}}R = \mathbf{1},\ \det R = 1 \,\}$, the special orthogonal group, the rotations of real N-dimensional Euclidean space: $R^{\mathsf{T}}R = \mathbf{1}$ preserves the Euclidean length $|\mathbf{x}|^2 = \mathbf{x}^{\mathsf{T}}\mathbf{x}$, and $\det R = 1$ keeps orientation (excluding reflections). $\mathrm{SO}(3)$ is the rotation group of ordinary space.

The group that ultimately matters for relativistic physics is the *Lorentz group* $\mathrm{SO}(1,3)^{+}$, the analogue of $\mathrm{SO}(N)$ for the indefinite Minkowski metric of Chapter A. Its representations are the entire subject of Chapter 3; we name it here so the reader knows what the machinery of this chapter is being built to attack. A first hint of the trouble it will cause: $\mathrm{U}(1)$, $\mathrm{SU}(N)$ and $\mathrm{SO}(N)$ are *compact* (their parameter ranges are bounded and closed — the phases live on a circle, the unitary and orthogonal matrices in a bounded region), whereas the Lorentz group is *noncompact* because boosts run off to infinite rapidity. Compactness is exactly the property that will let us unitarize representations, first for finite groups in this section and then, by an integral in place of a sum, for compact Lie groups. Its failure for the Lorentz group is why a relativistic particle has no finite-dimensional unitary representation, a fact Chapter 3 must confront head-on.

To compare groups, and to recognize when two apparently different groups are "the same," we need the structure-preserving maps between them.

Definition 2.6 (Homomorphism, isomorphism)**.** A *homomorphism* of groups is a map $\varphi : G \to H$ that respects the products,

$$\varphi(g_1 g_2) = \varphi(g_1)\,\varphi(g_2) \qquad \text{for all } g_1, g_2 \in G. \tag{2.3}$$

It automatically respects the identity and inverses: $\varphi(e_G) = \varphi(e_G e_G) = \varphi(e_G)^2$ forces $\varphi(e_G) = e_H$ after cancellation, and then $\varphi(g)\varphi(g^{-1}) = \varphi(e_G) = e_H$ gives $\varphi(g^{-1}) = \varphi(g)^{-1}$. The *kernel* $\ker\varphi = \{\, g \in G : \varphi(g) = e_H \,\}$ and the *image* $\varphi(G) \subseteq H$ are subgroups. The map φ is injective if and only if $\ker\varphi = \{e_G\}$, since $\varphi(g_1) = \varphi(g_2) \Leftrightarrow \varphi(g_1 g_2^{-1}) = e_H$. An *isomorphism* is a bijective homomorphism; when one exists we write $G \cong H$ and treat the two groups as the same abstract object presented differently — exactly the relation between the two models of $\mathbb{Z}_n$ in Theorem 2.2.

Homomorphisms are the lens through which one group "sees" another. The representations of the next subsection are nothing but the homomorphisms into the most concrete target groups of all: the invertible linear operators on a vector space.

2.1.3 Representations

Definition 2.7 (Representation)**.** Let G be a group and V a finite-dimensional complex vector space. A *representation* of G on V is a homomorphism

$$\rho : G \longrightarrow \mathrm{GL}(V), \qquad \rho(g_1 g_2) = \rho(g_1)\rho(g_2), \tag{2.4}$$

where $\mathrm{GL}(V)$ is the group of invertible linear operators on V. The *dimension* of the representation is $\dim V$. Choosing a basis of V turns each $\rho(g)$ into an invertible matrix, and Eq. (2.4) becomes a statement about matrix products; we move freely between the basis-free operator picture and the matrix picture, calling the latter a *matrix representation*. The space V is the *representation space*, and its vectors carry the physical states (or fields) on which the symmetry acts.

A representation is precisely the realization of an abstract symmetry as honest linear operators, the object Wigner's theorem hands us in the quantum setting. The defining property Eq. (2.4) guarantees consistency: composing symmetries and then acting is the same as acting one after another, and $\rho(e) = \mathbf{1}_V$ while $\rho(g^{-1}) = \rho(g)^{-1}$, so the operators reproduce the group's bookkeeping faithfully. Three representations exist for *every* group and anchor all later examples.

Example 2.8 (Trivial, defining, and regular representations)**.** *(i) The trivial representation* sends every g to the identity on a one-dimensional space, $\rho(g) = 1$ on $V = \mathbb{C}$. It is the mathematical home of any quantity invariant under the symmetry (a scalar, a singlet). For S_n it is the symmetric (bosonic) one-dimensional representation; the *sign representation* $\rho(\pi) = \mathrm{sgn}(\pi) \in \{\pm 1\}$, which records the parity of a permutation, is the antisymmetric (fermionic) one.

(ii) The defining (fundamental) representation of a matrix group is the tautological one: $\rho(g) = g$ acting on the column vectors $\mathbb{C}^N$ (or $\mathbb{R}^N$) it is built from. For $\mathrm{SU}(N)$ this is the N-dimensional representation that quarks transform in; for $\mathrm{SO}(3)$ it is the action on ordinary vectors.

(iii) The regular representation is the universal construction that exists for any finite group G. Take $V = \mathbb{C}[G]$, the $|G|$-dimensional complex vector space with one basis vector $|h\rangle$ for each group element $h \in G$, and let G act by permuting the basis through left multiplication,

$$\rho_{\mathrm{reg}}(g)\ |h\rangle = |gh\rangle\,. \tag{2.5}$$

This is a representation: $\rho_{\mathrm{reg}}(g_1)\rho_{\mathrm{reg}}(g_2)\,|h\rangle = \rho_{\mathrm{reg}}(g_1)\,|g_2 h\rangle = |g_1 g_2 h\rangle = \rho_{\mathrm{reg}}(g_1 g_2)\,|h\rangle$, and each operator is invertible because left translation is a bijection (the cancellation law again). In a basis, $\rho_{\mathrm{reg}}(g)$ is a permutation matrix — exactly one 1 in each row and column. The regular representation packages the entire multiplication table of G into linear algebra, and we will see in Section 2.2.6 and the character section that it contains every irreducible representation of G, each appearing a number of times equal to its own dimension. It is the finite-group analogue of the statement that the space of functions on a symmetry group decomposes into all the field multiplets the group allows.

When does a representation lose information about the group? Exactly when its homomorphism collapses distinct elements.

Definition 2.9 (Faithful representation)**.** A representation $\rho : G \to \mathrm{GL}(V)$ is *faithful* if it is injective, equivalently if $\ker \rho = \{e\}$: distinct group elements act by distinct operators. A faithful representation is a perfect copy of G inside $\mathrm{GL}(V)$, so that the abstract group can be recovered entirely from its action.

The trivial representation is maximally unfaithful (its kernel is all of G); the regular representation is always faithful, because $\rho_{\mathrm{reg}}(g)\,|e\rangle = |g\rangle$ already distinguishes every g. Cayley's theorem, mentioned above, is the observation that this faithful permutation action embeds G into $S_{|G|}$. Faithfulness matters physically: the sign representation of S_n is unfaithful for $n \geq 3$ (it cannot tell a 3-cycle from the identity, both having $\mathrm{sgn} = +1$), so a quantity transforming in it sees only the parity of a permutation, not the permutation itself.

Two representations can be the same in all that matters yet look different because they are written in different bases. We must declare such pairs equivalent or the classification program is hopeless.

Definition 2.10 (Equivalence)**.** Two representations $\rho : G \to \mathrm{GL}(V)$ and $\rho' : G \to \mathrm{GL}(V')$ are *equivalent* (written $\rho \sim \rho'$) if there is an invertible linear map $S : V \to V'$ intertwining them,

$$\rho'(g) = S\,\rho(g)\,S^{-1} \qquad \text{for all } g \in G. \tag{2.6}$$

When $V = V'$ this is a change of basis: ρ and ρ' are the same operators expressed in coordinates related by S. Equivalence is an equivalence relation (reflexive with $S = \mathbf{1}$, symmetric with S^{-1}, transitive by composing intertwiners), and the real objects of representation theory are the equivalence classes, not the individual matrices.

Of all the bases one might choose, one class is singled out by physics. In quantum mechanics the representation space carries an inner product — it is (a piece of) a Hilbert space — and probabilities are inner products. A symmetry must preserve them.

Definition 2.11 (Unitary representation)**.** Let V carry a Hermitian inner product $\langle\,\cdot\,,\cdot\,\rangle$. A representation $\rho : G \to \mathrm{GL}(V)$ is *unitary* if every operator $\rho(g)$ is unitary,

$$\langle \rho(g)u,\, \rho(g)v\rangle = \langle u, v\rangle \qquad \text{for all } g \in G,\ u, v \in V, \tag{2.7}$$

equivalently $\rho(g)^{\dagger}\rho(g) = \mathbf{1}$ for every g, with the adjoint taken in $\langle\,\cdot\,,\cdot\,\rangle$. Unitarity is precisely the condition that the representation conserves quantum-mechanical probability: $\langle\psi,\psi\rangle$ is left invariant, so a normalized state stays normalized under the symmetry.

Unitarity is the property Wigner's theorem promises for quantum symmetries, and it is the reason the chapter's slogan reads "unitary representation" rather than just "representation." Finally, representations can be combined. The simplest combination stacks two systems that do not interact under the symmetry.

Definition 2.12 (Direct sum)**.** Given representations $\rho_1 : G \to \mathrm{GL}(V_1)$ and $\rho_2 : G \to \mathrm{GL}(V_2)$, their *direct sum* is the representation $\rho_1 \oplus \rho_2$ on $V_1 \oplus V_2$ acting blockwise,

$$(\rho_1 \oplus \rho_2)(g) = \begin{pmatrix} \rho_1(g) & 0 \\ 0 & \rho_2(g) \end{pmatrix}, \tag{2.8}$$

a $(\dim V_1 + \dim V_2)$-dimensional representation in which the two blocks never mix: V_1 and V_2 are each carried to themselves by every $\rho(g)$. A subspace with this last property — invariant under all of $\rho(G)$ — is called a *subrepresentation*, and a representation is a direct sum exactly when its space splits into two complementary subrepresentations. If a representation can be brought, by a single change of basis as in Eq. (2.6), to the block-diagonal form Eq. (2.8) with both blocks nonzero, it is *reducible*; if no nontrivial invariant subspace exists, it is *irreducible*.

The direct sum is the bookkeeping of "and": a spin-$\frac{1}{2}$ particle together with a spin-1 particle transforms in the direct sum of the two rotation representations. The deep question — which the rest of the chapter exists to answer — is the reverse one. Given a representation, can we always split it into a direct sum of irreducibles, the indivisible atoms out of which all others are assembled? If so, classifying the atoms classifies everything. Reducibility is a basis-dependent statement (Eq. (2.8) asks for a particular block form), and that is precisely where unitarity earns its keep: in a unitary representation, the orthogonal complement of an invariant subspace is again invariant, so every reducible unitary representation *does* split. We prove that complete reducibility, and the lemma of Schur that organizes the atoms, in the next section. The present section secures the prerequisite: that for a finite group, unitarity may always be arranged.

2.1.4 The Weyl unitarian trick

The decomposition program just sketched needs unitary representations to run. Yet Theorem 2.7 demands no inner-product compatibility at all — a generic representation by invertible matrices is not unitary. The remarkable fact, and the one substantial theorem of this section, is that for a finite group this never obstructs us: *every* finite-dimensional complex representation of a finite group is equivalent to a unitary one. The construction is averaging, the most characteristic move in the subject. An inner product handed to us need not be invariant under $\rho(G)$, but we can force invariance by averaging it over the group — and because the group is finite, the average is a finite sum that manifestly exists.

Theorem 2.13 (Unitarizability of finite-group representations; Weyl)**.** *Let G be a finite group and $\rho : G \to \mathrm{GL}(V)$ a finite-dimensional complex representation. Then ρ is equivalent to a unitary representation. Equivalently, there exists a Hermitian inner product on V with respect to which every $\rho(g)$ is unitary.*

Proof. Start with any Hermitian inner product $\langle \cdot, \cdot \rangle_0$ on V; one always exists — pick a basis and declare it orthonormal, so that $\langle u, v \rangle_0 = u^\dagger v$ in that basis. This inner product need not satisfy Eq. (2.7). Define a new pairing by averaging it over the group:

$$\langle u, v \rangle \;\equiv\; \frac{1}{|G|} \sum_{g \in G} \langle \rho(g)u,\, \rho(g)v \rangle_0. \tag{2.9}$$

Step 1: Eq. (2.9) is a Hermitian inner product. Each summand is linear in v and conjugate-linear in u (because $\langle \cdot, \cdot \rangle_0$ is, and $\rho(g)$ is linear), so the sum is sesquilinear. Conjugate symmetry, $\langle v, u \rangle = \overline{\langle u, v \rangle}$, follows termwise from that of $\langle \cdot, \cdot \rangle_0$. For positivity, set $u = v$:

$$\langle u, u \rangle = \frac{1}{|G|} \sum_{g \in G} \langle \rho(g)u,\, \rho(g)u \rangle_0 = \frac{1}{|G|} \sum_{g \in G} \|\rho(g)u\|_0^2 \;\geq\; 0, \tag{2.10}$$

a sum of nonnegative terms. It vanishes only if every term vanishes, in particular the $g = e$ term $\|u\|_0^2$, forcing $u = 0$. (Equivalently, each $\rho(g)$ is invertible, so $\rho(g)u = 0 \Leftrightarrow u = 0$.) Hence Eq. (2.9) is positive-definite and is a bona fide Hermitian inner product.

Step 2: the new inner product is G-invariant. Fix $h \in G$ and compute, using that ρ is a homomorphism so $\rho(g)\rho(h) = \rho(gh)$,

$$\langle \rho(h)u,\, \rho(h)v \rangle = \frac{1}{|G|} \sum_{g \in G} \langle \rho(g)\rho(h)u,\, \rho(g)\rho(h)v \rangle_0 = \frac{1}{|G|} \sum_{g \in G} \langle \rho(gh)u,\, \rho(gh)v \rangle_0. \tag{2.11}$$

As g runs once over G, the product gh also runs once over G — this is the bijection $g \mapsto gh$ established by the cancellation law. Reindexing the sum by $g' = gh$ therefore leaves it unchanged:

$$\langle \rho(h)u,\, \rho(h)v \rangle = \frac{1}{|G|} \sum_{g' \in G} \langle \rho(g')u,\, \rho(g')v \rangle_0 = \langle u, v \rangle. \tag{2.12}$$

Thus every $\rho(h)$ preserves $\langle \cdot, \cdot \rangle$: with respect to the averaged inner product, ρ is already unitary. This proves the second ("equivalently") form of the theorem. The averaging has built the invariance that the original inner product lacked.

Step 3: realize the invariance as a similarity to the standard unitary group. It remains to exhibit the intertwiner S of Eq. (2.6) that conjugates ρ into operators unitary for the *standard* inner product $\langle u, v \rangle_0 = u^\dagger v$ — this is what "equivalent to a unitary representation" means in matrix terms. Both inner products are written through a matrix: from Eq. (2.9),

$$\langle u, v \rangle = \frac{1}{|G|} \sum_{g \in G} \big(\rho(g)u\big)^\dagger \big(\rho(g)v\big) = u^\dagger\, H\, v, \qquad H \equiv \frac{1}{|G|} \sum_{g \in G} \rho(g)^\dagger \rho(g). \tag{2.13}$$

The matrix H is Hermitian, $H^\dagger = H$, since each $\rho(g)^\dagger \rho(g)$ is, and it is positive-definite because $u^\dagger H u = \langle u, u \rangle > 0$ for $u \neq 0$ by Step 1. The invariance Eq. (2.12),

written in matrix form $\langle \rho(h)u, \rho(h)v\rangle = (\rho(h)u)^\dagger H(\rho(h)v) = u^\dagger(\rho(h)^\dagger H\rho(h))v$ and equated to $u^\dagger Hv$ for all u, v, is the operator identity

$$\rho(h)^\dagger\, H\, \rho(h) = H \qquad \text{for all } h \in G. \tag{2.14}$$

Because H is positive-definite Hermitian it has a unique positive-definite Hermitian square root $S \equiv H^{1/2}$: diagonalize $H = UDU^\dagger$ with U unitary and D diagonal with strictly positive entries, and set $S = UD^{1/2}U^\dagger$, so that $S = S^\dagger$, S is invertible, and $S^2 = H$. Now define the equivalent representation $\rho'(g) \equiv S\,\rho(g)\,S^{-1}$. It is a representation (conjugation by a fixed S preserves products), and it is unitary for the standard inner product:

$$\begin{aligned}\rho'(g)^\dagger\rho'(g) &= \big(S\rho(g)S^{-1}\big)^\dagger\big(S\rho(g)S^{-1}\big) = S^{-1}\rho(g)^\dagger S \;\cdot\; S\rho(g)S^{-1} \\ &= S^{-1}\,\rho(g)^\dagger S^2\rho(g)\,S^{-1} = S^{-1}\,\rho(g)^\dagger H\rho(g)\,S^{-1} = S^{-1}HS^{-1} = S^{-1}S^2S^{-1} = \mathbf{1},\end{aligned} \tag{2.15}$$

where we used $S^\dagger = S$, $(S^{-1})^\dagger = S^{-1}$, then $S^2 = H$, then the invariance Eq. (2.14), and finally $S^2 = H$ once more. Hence ρ' is a unitary representation equivalent to ρ, which is the first form of the theorem. □

Physical Insight

Every line of the proof used finiteness in exactly one place: the average $\frac{1}{|G|}\sum_{g\in G}$ must converge. For a *compact* Lie group the finite sum is replaced by an integral against the (finite, normalized) invariant Haar measure, $\int_G dg\,(\cdots)$, and the same three steps go through verbatim — so compact Lie groups (in particular U(1), SU(N), SO(N)) also have only unitary representations, up to equivalence. This is the deep reason compactness was flagged in Theorem 2.5. The construction breaks for the noncompact Lorentz group precisely because its invariant measure has infinite total volume: there is no normalized average, no invariant inner product on a finite-dimensional space, and indeed no finite-dimensional unitary representation at all — the structural fact that forces the relativistic particle into an infinite-dimensional Hilbert space in Chapter 3.

The trick is best absorbed on a concrete representation that starts out non-unitary.

Worked Example 2.14 (Unitarizing a representation of $\mathbb{Z}_2$)**.** Let $G = \mathbb{Z}_2 = \{e, r\}$ with $r^2 = e$, and define a two-dimensional representation by

$$\rho(e) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \qquad \rho(r) = \begin{pmatrix} 1 & 1 \\ 0 & -1 \end{pmatrix}. \tag{2.16}$$

This is a representation: $\rho(r)^2 = \left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1-1 \\ 0 & 1 \end{smallmatrix}\right) = \mathbf{1} = \rho(e)$, as required by $r^2 = e$. It is *not* unitary, since $\rho(r)^\dagger\rho(r) = \left(\begin{smallmatrix} 1 & 0 \\ 1 & -1 \end{smallmatrix}\right)\left(\begin{smallmatrix} 1 & 1 \\ 0 & -1 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right) \neq \mathbf{1}$. We

unitarize it by the recipe of Theorem 2.13. The averaged metric matrix Eq. (2.13) is

$$H = \tfrac{1}{2}\Big(\rho(e)^\dagger\rho(e) + \rho(r)^\dagger\rho(r)\Big) = \tfrac{1}{2}\left[\begin{pmatrix}1&0\\0&1\end{pmatrix} + \begin{pmatrix}1&1\\1&2\end{pmatrix}\right] = \begin{pmatrix}1&\frac{1}{2}\\\frac{1}{2}&\frac{3}{2}\end{pmatrix}, \tag{2.17}$$

which is Hermitian and positive-definite (trace $\frac{5}{2} > 0$, determinant $\frac{3}{2} - \frac{1}{4} = \frac{5}{4} > 0$). We check the invariance Eq. (2.14) directly. First

$$H\rho(r) = \begin{pmatrix}1&\frac{1}{2}\\\frac{1}{2}&\frac{3}{2}\end{pmatrix}\begin{pmatrix}1&1\\0&-1\end{pmatrix} = \begin{pmatrix}1&1-\frac{1}{2}\\\frac{1}{2}&\frac{1}{2}-\frac{3}{2}\end{pmatrix} = \begin{pmatrix}1&\frac{1}{2}\\\frac{1}{2}&-1\end{pmatrix}, \tag{2.18}$$

and then

$$\rho(r)^\dagger(H\rho(r)) = \begin{pmatrix}1&0\\1&-1\end{pmatrix}\begin{pmatrix}1&\frac{1}{2}\\\frac{1}{2}&-1\end{pmatrix} = \begin{pmatrix}1&\frac{1}{2}\\1-\frac{1}{2}&\frac{1}{2}+1\end{pmatrix} = \begin{pmatrix}1&\frac{1}{2}\\\frac{1}{2}&\frac{3}{2}\end{pmatrix} = H, \tag{2.19}$$

confirming $\rho(r)^\dagger H\rho(r) = H$. So ρ is genuinely unitary with respect to the averaged inner product $\langle u, v\rangle = u^\dagger H v$, even though it was not unitary for the standard one. Taking the positive square root $S = H^{1/2}$ and forming $\rho'(g) = S\rho(g)S^{-1}$ produces, by Eq. (2.15), an equivalent representation built from ordinary unitary matrices — the same abstract representation, now written in an orthonormal basis adapted to the symmetry. The averaging has not changed the representation; it has changed the geometry of the space to one the symmetry respects.

A second example shows the trick's payoff: once unitarity is free, a representation visibly falls apart into atoms, which is the program of the next section.

Worked Example 2.15 (The regular representation of $\mathbb{Z}_n$ as a sum of atoms)**.** Take $G = \mathbb{Z}_n$ and its regular representation Eq. (2.5) on $V = \mathbb{C}[\mathbb{Z}_n]$, with basis $|0\rangle, |1\rangle, \ldots, |n-1\rangle$ and the generator acting as $\rho_{\text{reg}}(1)\,|j\rangle = |(j+1) \bmod n\rangle$ — a cyclic shift, hence the $n \times n$ permutation matrix with 1's on the (cyclic) subdiagonal. This is already unitary (permutation matrices are), and being abelian, $\mathbb{Z}_n$ can have its single generator diagonalized. The eigenvectors of a cyclic shift are the discrete Fourier modes

$$|\chi_k\rangle = \frac{1}{\sqrt{n}}\sum_{j=0}^{n-1}\omega^{-jk}\,|j\rangle\,, \qquad \omega = e^{2\pi i/n}, \quad k = 0, 1, \ldots, n-1, \tag{2.20}$$

for which $\rho_{\text{reg}}(1)\,|\chi_k\rangle = \frac{1}{\sqrt{n}}\sum_j \omega^{-jk}\,|j+1\rangle = \omega^k\,\frac{1}{\sqrt{n}}\sum_{j'}\omega^{-j'k}\,|j'\rangle = \omega^k\,|\chi_k\rangle$, after relabeling $j' = j+1$ and using $\omega^{-(j'-1)k} = \omega^k\omega^{-j'k}$. Thus the n-dimensional

regular representation decomposes, in the orthonormal Fourier basis, into a direct sum of n one-dimensional representations,

$$\rho_{\text{reg}} \cong \bigoplus_{k=0}^{n-1} \chi_k, \qquad \chi_k(m) = \omega^{mk} = e^{2\pi i m k/n}, \tag{2.21}$$

one for each character χ_k of $\mathbb{Z}_n$. Each χ_k is irreducible — a one-dimensional representation can have no nontrivial invariant subspace — so these are the atoms, and every one of them appears exactly once in the regular representation. This is the promised statement (Theorem 2.8) that the regular representation contains every irreducible, each with multiplicity equal to its dimension; for an abelian group all irreducibles are one-dimensional, so "each appears dim times" reads "each appears once," and the count $\sum_k (\dim \chi_k)^2 = n \cdot 1^2 = n = |G|$ already foreshadows the general dimension formula that the character theory of the next section will prove.

We have now fixed the language in which the rest of the chapter — and much of the book — is written: a symmetry acts on quantum states through a representation; physically that representation is unitary, and over a finite group (and, by averaging against an invariant measure, over a compact one) unitarity may always be arranged; representations combine by direct sum, and the same notion run backward asks whether a given representation splits. Theorem 2.15 has shown a representation visibly breaking into one-dimensional atoms, and Theorem 2.14 has shown unitarity manufactured where none was given. What remains is the heart of the matter, deferred to the next section: the proof that every finite-dimensional unitary representation splits completely into irreducible pieces, the lemma of Schur that pins down how those pieces can map to one another, and the character calculus that counts and identifies them — the bookkeeping that turns "a particle is an irreducible unitary representation" from a slogan into an algorithm.

2.2 Irreducibility, Schur's Lemma, and Characters

The previous section established the object we actually study: not the abstract group G, but a *representation*, a homomorphism $\rho\colon G \to \mathrm{GL}(V)$ realizing the symmetry by linear operators on a vector space, and we proved that for a finite group every representation may be taken unitary (Theorem 2.13). That left a question dangling. A representation can be a clumsy package: the operators $\rho(g)$ may leave some subspace of V untouched, so that the representation secretly contains a smaller one, padded out by directions the group never mixes into it. The natural impulse—the impulse behind essentially every use of symmetry in physics, from the angular-momentum multiplets of the hydrogen atom to the multiplets of the quark model—is to break a representation into its smallest

indivisible pieces and study those. This section makes "indivisible" precise (irreducibility), proves that for finite groups every representation is built cleanly out of indivisible pieces (Maschke's theorem), proves the two rigidity statements that make the indivisible pieces behave like atoms (Schur's lemma), and then introduces the bookkeeping device—the *character*—that lets us recognize which atoms appear in a given representation and with what multiplicity, by nothing more strenuous than computing a few traces. The whole apparatus is the template the rest of the chapter executes: for $\mathfrak{su}(2)$, for $\mathfrak{su}(3)$, and, ported to the continuous setting, for every symmetry algebra of the Standard Model.

We work over the complex numbers $\mathbb{C}$ throughout. This is not a convenience: the second half of Schur's lemma, and with it the entire clean theory of characters, depends on $\mathbb{C}$ being algebraically closed, and physics lives on complex Hilbert spaces in any case. Unless stated otherwise G is a finite group of order $|G|$ and every vector space is finite-dimensional over $\mathbb{C}$.

2.2.1 Invariant subspaces and irreducibility

Definition 2.16 (Invariant subspace)**.** Let (ρ, V) be a representation of G. A linear subspace $W \subseteq V$ is *invariant* (or a *subrepresentation*) if it is mapped into itself by every group element:

$$\rho(g)\, W \subseteq W \qquad \text{for all } g \in G. \tag{2.22}$$

Because each $\rho(g)$ is invertible and W is finite-dimensional, $\rho(g)W \subseteq W$ forces $\rho(g)W = W$; the restriction $\rho|_W(g) \equiv \rho(g)|_W$ is then itself a representation of G on W.

Every representation has two invariant subspaces for free: the zero subspace $\{0\}$ and the whole space V. These are the *trivial* invariant subspaces. A representation is interesting—indivisible—precisely when it has no others.

Definition 2.17 (Irreducible and completely reducible)**.** A representation (ρ, V) with $V \neq \{0\}$ is *irreducible* if its only invariant subspaces are $\{0\}$ and V. It is *reducible* otherwise. It is *completely reducible* (or *semisimple*) if V decomposes as a direct sum of invariant subspaces each of which is irreducible,

$$V = W_1 \oplus W_2 \oplus \cdots \oplus W_r, \qquad \text{each } W_k \text{ invariant and irreducible.} \tag{2.23}$$

The distinction between merely reducible and completely reducible is the crux. Reducibility says V contains a proper invariant subspace W. Complete reducibility demands more: that W have an invariant *complement* W', a second invariant subspace with $V = W \oplus W'$, so that the representation genuinely splits as the direct sum $\rho|_W \oplus \rho|_{W'}$ rather than merely containing W as a sub-block.

In a basis adapted to a single invariant subspace W, the matrices $\rho(g)$ are block upper-triangular,

$$\rho(g) \;=\; \begin{pmatrix} \rho|_W(g) & B(g) \\ 0 & C(g) \end{pmatrix}, \tag{2.24}$$

the lower-left block vanishing exactly because W is invariant. Reducibility only guarantees this triangular form. Complete reducibility is the statement that one can choose the complement so that the off-diagonal block $B(g)$ also vanishes for *all* g simultaneously, leaving the matrices block-*diagonal*. There is no reason in pure linear algebra to expect such a complement to exist—a single nilpotent operator already provides a counterexample to "every invariant subspace has an invariant complement." That it always exists for representations of a finite group is the content of the next theorem, and it is what makes the finite theory so clean.

2.2.2 Maschke's theorem

Theorem 2.18 (Maschke)**.** *Every representation of a finite group G on a complex vector space is completely reducible. Equivalently: every invariant subspace $W \subseteq V$ admits an invariant complement W', an invariant subspace with $V = W \oplus W'$.*

The two formulations are equivalent by an obvious induction. Given the complement statement, decompose any V: if V is irreducible we are done; if not, it has a proper nontrivial invariant W, which by hypothesis has an invariant complement W', and $\dim W, \dim W' < \dim V$, so by induction on dimension each splits into irreducibles, and their union is a decomposition Eq. (2.23) of V. So it suffices to prove the complement statement.

The mechanism is *averaging over the group*—the single most important trick of finite-group representation theory, and the discrete ancestor of the integration over a compact Lie group that will replace it in later sections. The idea: any gadget that is *not* G-invariant can be forced to become G-invariant by averaging its G-translates, and for a finite group the average is a genuine finite sum divided by $|G|$.

Proof. Let $W \subseteq V$ be invariant. Choose, by ordinary linear algebra, *any* linear complement U to W, so $V = W \oplus U$ as vector spaces (this ignores G entirely; U need not be invariant). Let $p\colon V \to V$ be the corresponding projection onto W along U: it satisfies $p(V) = W$, $p|_W = \mathrm{id}_W$, and $p^2 = p$. This p is generally not an intertwiner; it does not commute with the group action. We repair that by averaging. Define

$$P \;\equiv\; \frac{1}{|G|} \sum_{g \in G} \rho(g)\, p\, \rho(g)^{-1}. \tag{2.25}$$

We claim P is a G-equivariant projection onto W. There are three things to check.

(i) P maps V into W. For any $v \in V$ and any g, we have $\rho(g)^{-1}v \in V$, then $p\rho(g)^{-1}v \in W$, and then $\rho(g)$ maps W into W by invariance, so $\rho(g)p\rho(g)^{-1}v \in W$. A sum of vectors in W lies in W, so $Pv \in W$.

(ii) P acts as the identity on W. Let $w \in W$. Then $\rho(g)^{-1}w \in W$ (invariance again), so p acts on it as the identity: $p\,\rho(g)^{-1}w = \rho(g)^{-1}w$. Hence each term is $\rho(g)\rho(g)^{-1}w = w$, and

$$Pw = \frac{1}{|G|}\sum_{g\in G} w = w. \tag{2.26}$$

Combining (i) and (ii): $P(V) = W$, and $P^2 = P$ because P first lands in W and then fixes W pointwise. So P is a projection onto W.

(iii) P commutes with the representation. This is where the average earns its keep. For any $h \in G$,

$$\begin{aligned}\rho(h)\,P\,\rho(h)^{-1} &= \frac{1}{|G|}\sum_{g\in G}\rho(h)\rho(g)\,p\,\rho(g)^{-1}\rho(h)^{-1}\\ &= \frac{1}{|G|}\sum_{g\in G}\rho(hg)\,p\,\rho(hg)^{-1} \;=\; \frac{1}{|G|}\sum_{g'\in G}\rho(g')\,p\,\rho(g')^{-1} \;=\; P,\end{aligned} \tag{2.27}$$

where we used the homomorphism property $\rho(h)\rho(g) = \rho(hg)$ and then the substitution $g' = hg$, which is a bijection of G onto itself (left multiplication by a fixed h permutes the group), leaving the sum unchanged. Thus $\rho(h)P = P\rho(h)$ for all h.

Now set $W' \equiv \ker P$. Since P is a projection, $V = \operatorname{im} P \oplus \ker P = W \oplus W'$. And W' is invariant: if $Pv = 0$ then $P\rho(h)v = \rho(h)Pv = 0$ by Eq. (2.27), so $\rho(h)v \in \ker P = W'$. Thus W' is the desired invariant complement. □

Remark 2.19 (Two faces of the same averaging). The construction in Theorem 2.13 of an invariant inner product, and the construction Eq. (2.25) of an invariant projector, are the same averaging idea wearing different clothes. If one starts instead from a G-invariant Hermitian inner product $\langle\cdot,\cdot\rangle$ (built by averaging an arbitrary one, exactly as in the previous section) then the orthogonal complement $W^{\perp}$ of an invariant W is automatically invariant: for $w \in W$, $u \in W^{\perp}$, $g \in G$, invariance of the inner product gives $\langle w, \rho(g)u\rangle = \langle \rho(g)^{-1}w, u\rangle = 0$ because $\rho(g)^{-1}w \in W$. So $\rho(g)u \in W^{\perp}$, and $W^{\perp}$ is the invariant complement. This is Maschke's theorem proved through unitarity; we have given the projector proof in full because the projector Eq. (2.25) reappears, lightly disguised, in the orthogonality relations below.

Maschke's theorem reduces the study of all finite-group representations to the study of the irreducible ones: every representation is a direct sum of irreducibles,

so to "know all representations" it suffices to (a) list the irreducibles and (b) say how a given representation decomposes into them. Step (a) is a finite problem, as the sum-of-squares theorem below will show; step (b) is solved by characters. Before either, we need the rigidity statement that makes the decomposition essentially unique.

2.2.3 Schur's lemma

The organizing notion is the *intertwiner*: a linear map that respects two group actions.

Definition 2.20 (Intertwiner)**.** Let (ρ, V) and (σ, W) be representations of G. A linear map $\phi\colon V \to W$ is an *intertwiner* (or *G-equivariant map*, or morphism of representations) if it commutes with the group actions,

$$\phi\,\rho(g) = \sigma(g)\,\phi \qquad \text{for all } g \in G. \tag{2.28}$$

An invertible intertwiner is an *equivalence* (isomorphism) of representations; ρ and σ are *equivalent*, written $\rho \cong \sigma$, when one exists. Recall (Theorem 2.7) that equivalent representations are "the same representation in different coordinates," related by $\sigma(g) = S\rho(g)S^{-1}$ with $S = \phi$.

Theorem 2.21 (Schur's lemma)**.** *Let (ρ, V) and (σ, W) be irreducible representations of G over $\mathbb{C}$, and let $\phi\colon V \to W$ be an intertwiner.*

(I) *If ρ and σ are inequivalent, then $\phi = 0$. More generally, any intertwiner between irreducibles is either zero or an isomorphism.*

(II) *If $(\rho, V) = (\sigma, W)$ is one and the same irreducible representation, then ϕ is a scalar multiple of the identity: $\phi = \lambda\,\mathrm{id}_V$ for some $\lambda \in \mathbb{C}$.*

Proof. Part (I). The proof rests on a single observation: the kernel and image of an intertwiner are invariant subspaces. Consider $\ker\phi \subseteq V$. If $v \in \ker\phi$, then for any g,

$$\phi(\rho(g)v) = \sigma(g)\,\phi(v) = \sigma(g)\,0 = 0, \tag{2.29}$$

using the intertwining relation Eq. (2.28), so $\rho(g)v \in \ker\phi$. Thus $\ker\phi$ is invariant under ρ. Since (ρ, V) is irreducible, $\ker\phi$ is either $\{0\}$ or V.

Now consider $\mathrm{im}\,\phi \subseteq W$. If $w = \phi(v) \in \mathrm{im}\,\phi$, then for any g,

$$\sigma(g)\,w = \sigma(g)\,\phi(v) = \phi(\rho(g)v) \in \mathrm{im}\,\phi, \tag{2.30}$$

again by Eq. (2.28), so $\mathrm{im}\,\phi$ is invariant under σ. Since (σ, W) is irreducible, $\mathrm{im}\,\phi$ is either $\{0\}$ or W.

There are now two cases. Either $\phi = 0$ (then $\ker\phi = V$ and $\mathrm{im}\,\phi = \{0\}$, consistent), or $\phi \neq 0$. In the latter case $\ker\phi \neq V$, forcing $\ker\phi = \{0\}$, so ϕ is

injective; and $\operatorname{im}\phi \neq \{0\}$, forcing $\operatorname{im}\phi = W$, so ϕ is surjective. Hence a nonzero intertwiner between irreducibles is a bijection, i.e. an isomorphism, and then Eq. (2.28) reads $\sigma(g) = \phi\,\rho(g)\,\phi^{-1}$, exhibiting $\rho \cong \sigma$. Contrapositively, if $\rho \ncong \sigma$ then no nonzero intertwiner can exist, so $\phi = 0$. This proves (I).

Part (II). Here $\phi\colon V \to V$ is an intertwiner of an irreducible representation with itself, and we use that $\mathbb{C}$ is algebraically closed. The linear operator ϕ on the finite-dimensional complex space V has a characteristic polynomial, which (since $\mathbb{C}$ is algebraically closed) has a root $\lambda \in \mathbb{C}$; that root is an eigenvalue of ϕ, so $\phi - \lambda\,\mathrm{id}_V$ has nontrivial kernel. But $\phi - \lambda\,\mathrm{id}_V$ is itself an intertwiner: it is a linear combination of the intertwiner ϕ and the identity (the identity intertwines any representation with itself, trivially), and intertwiners form a vector space closed under linear combination, since

$$(\phi - \lambda\,\mathrm{id})\rho(g) = \phi\rho(g) - \lambda\rho(g) = \rho(g)\phi - \rho(g)\lambda\,\mathrm{id} = \rho(g)(\phi - \lambda\,\mathrm{id}). \tag{2.31}$$

By Part (I) applied to the intertwiner $\phi - \lambda\,\mathrm{id}\colon V \to V$ (same irreducible at both ends), it is either zero or an isomorphism. It is *not* an isomorphism, because it has nontrivial kernel (the λ-eigenspace). Therefore it is zero: $\phi = \lambda\,\mathrm{id}_V$. □

Algebraic closure is essential to Part (II)

Part (II) genuinely uses $\mathbb{C}$. Over $\mathbb{R}$ it can fail: the rotation representation of the cyclic group on $\mathbb{R}^2$ is irreducible, yet the rotation by 90° is an intertwiner that is manifestly *not* a real scalar multiple of the identity (it has no real eigenvalue). The complexification of that representation splits into two one-dimensional pieces, restoring Schur; this is exactly why the honest classification of fields by the Lorentz group in Chapter 3 is carried out over $\mathbb{C}$.

Schur's lemma is the engine of everything that follows. Two immediate consequences set the tone.

Corollary 2.22 (Irreducibles of an abelian group are one-dimensional)**.** *If G is abelian, every irreducible complex representation is one-dimensional.*

Proof. Let (ρ, V) be irreducible and fix any $h \in G$. For every $g \in G$, commutativity of G gives $\rho(h)\rho(g) = \rho(hg) = \rho(gh) = \rho(g)\rho(h)$, so $\rho(h)$ commutes with every $\rho(g)$; that is, $\rho(h)$ is an intertwiner of (ρ, V) with itself. By Schur (II), $\rho(h) = \lambda_h\,\mathrm{id}_V$ for some scalar $\lambda_h \in \mathbb{C}$. But then *every* subspace of V is invariant (a scalar multiplies any subspace into itself), so irreducibility forces $\dim V = 1$. □

This corollary is the abstract reason the irreducible representations of the translation/rotation phases U(1) are the single Fourier modes $e^{in\theta}$, and it is the seed of the weight-lattice picture: an abelian subalgebra (the Cartan subalgebra)

can be diagonalized simultaneously in any representation, and its eigenvalues are the weights that organize $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ in later sections.

Corollary 2.23 (Uniqueness of the intertwiner up to scale)**.** *If (ρ, V) and (σ, W) are equivalent irreducibles, the intertwiner between them is unique up to an overall complex scalar. In particular the decomposition of a representation into irreducibles is unique up to isomorphism and reordering.*

Proof. If $\phi_1, \phi_2 \colon V \to W$ are two nonzero intertwiners, both are isomorphisms by Schur (I), so $\phi_2^{-1}\phi_1 \colon V \to V$ is an intertwiner of ρ with itself, hence $\phi_2^{-1}\phi_1 = \lambda\,\mathrm{id}$ by Schur (II), i.e. $\phi_1 = \lambda\phi_2$. The uniqueness of the decomposition—that the list of irreducible constituents W_k in Eq. (2.23), with multiplicities, does not depend on how the decomposition was performed—follows because the multiplicity of a given irreducible U in V equals $\dim \mathrm{Hom}_G(U, V)$ (the dimension of the space of intertwiners $U \to V$), and Schur makes that dimension well defined: each irreducible summand isomorphic to U contributes exactly 1 by (I)+(II), each summand inequivalent to U contributes 0 by (I). We will recover this multiplicity far more cheaply as a character inner product in Theorem 2.28. □

2.2.4 Characters

We can now decompose anything into irreducibles in principle, but to do it in practice we need a computable fingerprint that identifies an irreducible and reads off multiplicities. That fingerprint is the trace of the representation matrices.

Definition 2.24 (Character)**.** The *character* of a representation (ρ, V) is the function $\chi_\rho \colon G \to \mathbb{C}$,

$$\chi_\rho(g) \;\equiv\; \mathrm{tr}\,\rho(g). \tag{2.32}$$

The trace discards a vast amount of information—it forgets everything about $\rho(g)$ except the sum of its eigenvalues—yet, remarkably, it retains *exactly* enough to identify a representation up to equivalence. The reason it is the right invariant is a short list of properties, each following from elementary trace identities.

Proposition 2.25 (Elementary properties of characters)**.** *Let (ρ, V), (σ, W) be representations of G with $d_\rho = \dim V$. Then:*

1. *$\chi_\rho(e) = d_\rho$ (the character at the identity is the dimension).*
2. *χ_ρ is a* class function*: it is constant on conjugacy classes, $\chi_\rho(hgh^{-1}) = \chi_\rho(g)$ for all g, h.*
3. *Equivalent representations have equal characters: $\rho \cong \sigma \Rightarrow \chi_\rho = \chi_\sigma$.*
4. *The character of a direct sum is the sum of characters: $\chi_{\rho\oplus\sigma} = \chi_\rho + \chi_\sigma$.*
5. *If ρ is unitary, $\chi_\rho(g^{-1}) = \overline{\chi_\rho(g)}$.*

Proof. (1) $\rho(e) = \mathrm{id}_V$, so $\operatorname{tr}\rho(e) = \dim V = d_\rho$.

(2) Using $\rho(hgh^{-1}) = \rho(h)\rho(g)\rho(h)^{-1}$ and the cyclicity of the trace, $\operatorname{tr}\big(\rho(h)\rho(g)\rho(h)^{-1}\big) = \operatorname{tr}\big(\rho(h)^{-1}\rho(h)\rho(g)\big) = \operatorname{tr}\rho(g)$. Conjugate group elements have equal character; the character is determined by its values on the conjugacy classes.

(3) If $\sigma(g) = S\rho(g)S^{-1}$ then $\operatorname{tr}\sigma(g) = \operatorname{tr}\big(S\rho(g)S^{-1}\big) = \operatorname{tr}\rho(g)$ by cyclicity.

(4) In a basis adapted to $V \oplus W$, $(\rho \oplus \sigma)(g)$ is block-diagonal with blocks $\rho(g)$ and $\sigma(g)$, so its trace is $\operatorname{tr}\rho(g) + \operatorname{tr}\sigma(g)$.

(5) A unitary $\rho(g)$ has $\rho(g)^{-1} = \rho(g)^\dagger$, and the eigenvalues of a unitary matrix are phases $e^{i\theta_k}$ with $\left|e^{i\theta_k}\right| = 1$, so the eigenvalues of $\rho(g)^{-1}$ are $e^{-i\theta_k} = \overline{e^{i\theta_k}}$. Hence $\chi_\rho(g^{-1}) = \operatorname{tr}\rho(g)^{-1} = \sum_k e^{-i\theta_k} = \overline{\sum_k e^{i\theta_k}} = \overline{\chi_\rho(g)}$. Since every finite-group representation may be taken unitary (Theorem 2.13), and characters are invariant under equivalence by (3), property (5) holds for the character of *any* finite-group representation. □

Property (2) tells us where characters live: in the space of *class functions*,

$$\mathcal{C}(G) \;\equiv\; \{\, f : G \to \mathbb{C} \;:\; f(hgh^{-1}) = f(g)\ \forall g, h \,\}, \tag{2.33}$$

a complex vector space whose dimension is the number of conjugacy classes of G (a class function is freely specified by one value per class). We equip $\mathcal{C}(G)$ with the Hermitian inner product

$$\langle f_1, f_2 \rangle \;\equiv\; \frac{1}{|G|} \sum_{g \in G} \overline{f_1(g)}\, f_2(g). \tag{2.34}$$

The central theorem of the subject is that the irreducible characters form an *orthonormal* set in this inner product. Everything practical—identifying irreducibles, extracting multiplicities, counting irreducibles—falls out of that one fact.

2.2.5 Orthogonality of characters

We prove orthogonality at the finer level of matrix entries first, because that form is what Schur's lemma directly delivers, and then trace down to characters. The tool is again averaging Eq. (2.25), now applied to an arbitrary linear map between two irreducibles.

Theorem 2.26 (Schur orthogonality for matrix coefficients)**.** *Let (ρ, V) and (σ, W) be irreducible* unitary *representations of G, of dimensions d_ρ and d_σ, with matrix entries $\rho_{ij}(g)$ and $\sigma_{kl}(g)$ in fixed orthonormal bases. Then:*

1. *if $\rho \not\cong \sigma$,*

$$\frac{1}{|G|} \sum_{g \in G} \overline{\rho_{ij}(g)}\, \sigma_{kl}(g) = 0 \qquad \text{for all } i, j, k, l; \tag{2.35}$$

2. if $\rho = \sigma$ (the same irreducible),

$$\frac{1}{|G|}\sum_{g\in G}\overline{\rho_{ij}(g)}\,\rho_{kl}(g) = \frac{1}{d_\rho}\,\delta_{ik}\,\delta_{jl}. \tag{2.36}$$

Proof. Let $A\colon W \to V$ be *any* linear map, and form its G-average

$$A^0 \;\equiv\; \frac{1}{|G|}\sum_{g\in G}\rho(g)\,A\,\sigma(g)^{-1}. \tag{2.37}$$

Exactly as in Eq. (2.27), A^0 is an intertwiner: for any h,

$$\rho(h)A^0\sigma(h)^{-1} = \frac{1}{|G|}\sum_{g}\rho(hg)A\sigma(hg)^{-1} = A^0, \tag{2.38}$$

so $\rho(h)A^0 = A^0\sigma(h)$ for all h. Schur's lemma now constrains A^0 completely.

Case $\rho \not\cong \sigma$. By Schur (I), $A^0 = 0$ for every choice of A.

Case $\rho = \sigma$ (so $V = W$, $d_\rho = d_\sigma \equiv d$). By Schur (II), $A^0 = \lambda(A)\,\mathrm{id}_V$ for some scalar $\lambda(A)$ depending linearly on A. Taking the trace of Eq. (2.37) and using cyclicity, $\operatorname{tr} A^0 = \frac{1}{|G|}\sum_g \operatorname{tr}(\rho(g)A\rho(g)^{-1}) = \frac{1}{|G|}\sum_g \operatorname{tr} A = \operatorname{tr} A$, while $\operatorname{tr}(\lambda\,\mathrm{id}_V) = \lambda d$; hence

$$\lambda(A) = \frac{\operatorname{tr} A}{d}, \qquad \text{i.e.} \qquad A^0 = \frac{\operatorname{tr} A}{d}\,\mathrm{id}_V. \tag{2.39}$$

To extract Eqs. (2.35) and (2.36) we choose A to be a single matrix unit: let $A = E_{jk}$, the linear map whose only nonzero entry is a 1 in row j, column k, so $(E_{jk})_{ab} = \delta_{aj}\delta_{bk}$. Then in components, using unitarity $\sigma(g)^{-1} = \sigma(g)^\dagger$ so that $\big(\sigma(g)^{-1}\big)_{lk'} = \overline{\sigma_{k'l}(g)}$,

$$\begin{aligned}(A^0)_{il} &= \frac{1}{|G|}\sum_g\sum_{a,b}\rho_{ia}(g)\,(E_{jk})_{ab}\,\big(\sigma(g)^{-1}\big)_{bl}\\ &= \frac{1}{|G|}\sum_g \rho_{ij}(g)\,\big(\sigma(g)^{-1}\big)_{kl} = \frac{1}{|G|}\sum_g \rho_{ij}(g)\,\overline{\sigma_{lk}(g)}.\end{aligned} \tag{2.40}$$

For inequivalent ρ, σ the left side vanishes, giving $\frac{1}{|G|}\sum_g \rho_{ij}(g)\overline{\sigma_{lk}(g)} = 0$; relabelling indices ($l \leftrightarrow k$, complex-conjugating the whole equation, which is allowed since the right side is 0) yields Eq. (2.35). For $\rho = \sigma$, Eq. (2.39) gives $(A^0)_{il} = \frac{\operatorname{tr} E_{jk}}{d}\delta_{il} = \frac{\delta_{jk}}{d}\delta_{il}$, so

$$\frac{1}{|G|}\sum_g \rho_{ij}(g)\,\overline{\rho_{lk}(g)} = \frac{1}{d}\,\delta_{il}\,\delta_{jk}. \tag{2.41}$$

Conjugating both sides and relabelling indices to match Eq. (2.36) (rename $i \to j$, $j \to i$, $l \to k$, $k \to l$ consistently) gives the stated form. The numbers $\delta_{ik}\delta_{jl}$ and $\delta_{il}\delta_{jk}$ on the two sides agree after the relabelling; both record that the average is nonzero only when the "row pattern" and the "column pattern" of the two matrix coefficients match. □

The character orthogonality relation is now a one-line contraction.

Theorem 2.27 (First orthogonality relation for characters)**.** *Let χ_i and χ_j be the characters of two irreducible representations of G. Then, in the inner product Eq. (2.34),*

$$\langle \chi_i, \chi_j \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_i(g)}\, \chi_j(g) = \delta_{ij}, \tag{2.42}$$

where $\delta_{ij} = 1$ if the two representations are equivalent and 0 otherwise. The irreducible characters of G form an orthonormal system in $\mathcal{C}(G)$.

Proof. The character is the trace, $\chi_i(g) = \sum_a \rho^{(i)}_{aa}(g)$, so

$$\langle \chi_i, \chi_j \rangle = \frac{1}{|G|} \sum_g \sum_a \overline{\rho^{(i)}_{aa}(g)} \sum_b \rho^{(j)}_{bb}(g) = \sum_{a,b} \frac{1}{|G|} \sum_g \overline{\rho^{(i)}_{aa}(g)}\, \rho^{(j)}_{bb}(g). \tag{2.43}$$

If the two irreducibles are inequivalent, every inner sum vanishes by Eq. (2.35), so $\langle \chi_i, \chi_j \rangle = 0$. If they are equivalent we may take them equal (characters are equivalence invariants), and Eq. (2.36) with $i \to a$, $j \to a$, $k \to b$, $l \to b$ gives $\frac{1}{|G|} \sum_g \overline{\rho_{aa}(g)} \rho_{bb}(g) = \frac{1}{d} \delta_{ab} \delta_{ab} = \frac{1}{d} \delta_{ab}$, whence

$$\langle \chi_i, \chi_i \rangle = \sum_{a,b} \frac{1}{d} \delta_{ab} = \frac{1}{d} \sum_{a=1}^{d} 1 = \frac{d}{d} = 1. \tag{2.44}$$

Thus $\langle \chi_i, \chi_j \rangle = \delta_{ij}$. □

This single relation does all the bookkeeping the chapter promised.

Corollary 2.28 (Multiplicities and the irreducibility test)**.** *Let (ρ, V) be any representation of G, decomposed by Maschke as $V \cong \bigoplus_i m_i V_i$ where the V_i are the distinct irreducibles and $m_i \geq 0$ is the multiplicity of V_i. Then:*

1. *$\chi_\rho = \sum_i m_i \chi_i$, and the multiplicity is recovered by an inner product,*

$$m_i = \langle \chi_i, \chi_\rho \rangle = \frac{1}{|G|} \sum_{g \in G} \overline{\chi_i(g)}\, \chi_\rho(g). \tag{2.45}$$

2. *$\langle \chi_\rho, \chi_\rho \rangle = \sum_i m_i^2$. In particular ρ is irreducible* if and only if *$\langle \chi_\rho, \chi_\rho \rangle = 1$.*

3. *Two representations are equivalent if and only if they have the same character.*

Proof. (1) Additivity of the character over direct sums (Theorem 2.25(4)) gives $\chi_\rho = \sum_i m_i \chi_i$; taking $\langle \chi_i, \cdot \rangle$ and using orthonormality Eq. (2.42), $\langle \chi_i, \chi_\rho \rangle = \sum_j m_j \langle \chi_i, \chi_j \rangle = \sum_j m_j \delta_{ij} = m_i$.

(2) $\langle \chi_\rho, \chi_\rho \rangle = \sum_{i,j} m_i m_j \langle \chi_i, \chi_j \rangle = \sum_{i,j} m_i m_j \delta_{ij} = \sum_i m_i^2$. A sum of squares of non-negative integers equals 1 precisely when one $m_i = 1$ and the rest vanish, i.e. precisely when ρ is itself irreducible.

(3) If $\rho \cong \sigma$ then $\chi_\rho = \chi_\sigma$ by Theorem 2.25(3). Conversely if $\chi_\rho = \chi_\sigma$ then by Eq. (2.45) each irreducible occurs in both with the same multiplicity $m_i = \langle \chi_i, \chi_\rho \rangle = \langle \chi_i, \chi_\sigma \rangle$, so the two decompositions into irreducibles coincide and $\rho \cong \sigma$. □

Part (3) is the punchline that justifies calling the character a fingerprint: over $\mathbb{C}$, the trace function determines a finite-group representation completely. The infinite data of the matrices $\rho(g)$ is compressed, with no loss for classification purposes, into one number per conjugacy class.

2.2.6 The regular representation and the sum-of-squares theorem

How many irreducibles does a finite group have, and how big are they? Both questions are answered by decomposing one particularly rich representation in which every irreducible is guaranteed to appear: the regular representation, built from the group acting on itself.

Definition 2.29 (Regular representation)**.** Let $\mathbb{C}[G]$ be the $|G|$-dimensional complex vector space with basis $\{e_g\}_{g \in G}$ labelled by the group elements. The (left) *regular representation* ρ_{reg} acts by permuting the basis according to left multiplication,

$$\rho_{\mathrm{reg}}(h)\, e_g \;=\; e_{hg}. \tag{2.46}$$

This is a genuine representation: $\rho_{\mathrm{reg}}(h_1)\rho_{\mathrm{reg}}(h_2)e_g = \rho_{\mathrm{reg}}(h_1)e_{h_2 g} = e_{h_1 h_2 g} = \rho_{\mathrm{reg}}(h_1 h_2)e_g$. Its character is immediate.

Proposition 2.30 (Character of the regular representation)**.**

$$\chi_{\mathrm{reg}}(g) = \begin{cases} |G|\,, & g = e, \\ 0, & g \neq e. \end{cases} \tag{2.47}$$

Proof. $\chi_{\mathrm{reg}}(g) = \operatorname{tr} \rho_{\mathrm{reg}}(g)$ counts the diagonal entries of the permutation matrix of g, i.e. the number of basis vectors $e_{g'}$ left fixed: those with $gg' = g'$, equivalently $g = e$. For $g = e$ all $|G|$ basis vectors are fixed, giving $\operatorname{tr} = |G|$; for $g \neq e$ none are fixed (left multiplication by $g \neq e$ has no fixed point because $gg' = g' \Rightarrow g = e$), giving $\operatorname{tr} = 0$. □

Theorem 2.31 (Decomposition of the regular representation; sum of squares)**.** *Let $V_1, \dots, V_r$ be the distinct irreducible representations of the finite group G, of dimensions $d_1, \dots, d_r$ (so $d_i = \chi_i(e)$). Then the regular representation contains each irreducible with multiplicity equal to its dimension,*

$$\mathbb{C}[G] \;\cong\; \bigoplus_{i=1}^{r} d_i\, V_i, \tag{2.48}$$

and consequently the order of the group is the sum of the squares of the dimensions of its inequivalent irreducibles,

$$\boxed{\,|G| = \sum_{i=1}^{r} d_i^2\,.\,} \tag{2.49}$$

Proof. By Theorem 2.28(1) the multiplicity of V_i in $\mathbb{C}[G]$ is

$$m_i = \langle \chi_i, \chi_{\mathrm{reg}} \rangle = \frac{1}{|G|} \sum_{g} \overline{\chi_i(g)}\, \chi_{\mathrm{reg}}(g) = \frac{1}{|G|}\, \overline{\chi_i(e)}\; |G| = \overline{\chi_i(e)} = d_i, \tag{2.50}$$

where only the $g = e$ term of Eq. (2.47) survives, and $\chi_i(e) = d_i$ is real (a dimension). This is Eq. (2.48). Taking the dimension of both sides, $\dim \mathbb{C}[G] = |G|$ on the left and $\sum_i m_i \dim V_i = \sum_i d_i \cdot d_i = \sum_i d_i^2$ on the right, gives Eq. (2.49). □

The identity Eq. (2.49) is a powerful finiteness statement: it bounds both the number of irreducibles and their sizes by the group order, and frequently pins down the dimensions outright. For a group of order 6, for instance, the only ways to write 6 as a sum of positive squares are $6 = 1+1+1+1+1+1$ and $6 = 1+1+4$; the abelian case forces the first (all irreducibles one-dimensional, by Theorem 2.22), and a nonabelian group of order 6 is therefore forced into the second, with irreducible dimensions $1, 1, 2$. We will see this play out for S_3 in a moment.

One question remains for the counting: how many irreducibles r are there? The orthonormality Eq. (2.42) already shows $r \le \dim \mathcal{C}(G) = \#\{\text{conjugacy classes}\}$ (orthonormal vectors are linearly independent). The reverse inequality—that the irreducible characters *span* the class functions, so the count is exact—is the natural completion and is worth proving in full, because the chapter's strategy ("expand everything in irreducibles") is only as good as our ability to know we have found them all.

Theorem 2.32 (Completeness; number of irreducibles equals number of classes)**.** *The irreducible characters $\chi_1, \dots, \chi_r$ form an orthonormal* basis *of the space $\mathcal{C}(G)$ of class functions. Hence the number r of inequivalent irreducible representations of G equals the number of conjugacy classes of G.*

Proof. By Theorem 2.27 the χ_i are orthonormal, hence linearly independent; we must show they span $\mathcal{C}(G)$. It suffices to show that the only class function orthogonal to every χ_i is the zero function, for then the orthogonal complement of $\operatorname{span}\{\chi_i\}$ in $\mathcal{C}(G)$ is trivial.

Let $f \in \mathcal{C}(G)$ satisfy $\langle \chi_i, f\rangle = 0$ for all i. For each irreducible (ρ_i, V_i) form the operator

$$T_i \;\equiv\; \sum_{g\in G} \overline{f(g)}\,\rho_i(g). \tag{2.51}$$

Because f is a class function, T_i is an intertwiner of ρ_i with itself: for any h,

$$\rho_i(h)\,T_i\,\rho_i(h)^{-1} = \sum_{g} \overline{f(g)}\,\rho_i(hgh^{-1}) = \sum_{g'} \overline{f(h^{-1}g'h)}\,\rho_i(g') = \sum_{g'} \overline{f(g')}\,\rho_i(g') = T_i, \tag{2.52}$$

where we substituted $g' = hgh^{-1}$ (a bijection of G) and used $f(h^{-1}g'h) = f(g')$. Hence $\rho_i(h)T_i = T_i\rho_i(h)$, and Schur (II) gives $T_i = \mu_i \,\mathrm{id}_{V_i}$. Taking the trace, $\mu_i d_i = \operatorname{tr} T_i = \sum_g \overline{f(g)}\,\chi_i(g) = |G|\,\langle f, \chi_i\rangle = |G|\,\overline{\langle \chi_i, f\rangle} = 0$, so $\mu_i = 0$ and therefore $T_i = 0$ for every irreducible ρ_i.

Now $T_i = 0$ for all irreducibles means $\sum_g \overline{f(g)}\,\rho(g) = 0$ in every representation ρ, because every representation is a direct sum of irreducibles (Maschke) and the operator is block-diagonal across the sum. Apply this to the regular representation and act on the basis vector e_e:

$$0 = \sum_{g} \overline{f(g)}\,\rho_{\mathrm{reg}}(g)\,e_e = \sum_{g} \overline{f(g)}\,e_g. \tag{2.53}$$

The e_g are linearly independent, so every coefficient vanishes: $\overline{f(g)} = 0$ for all g, i.e. $f = 0$. Thus no nonzero class function is orthogonal to all irreducible characters, the χ_i span $\mathcal{C}(G)$, and $r = \dim \mathcal{C}(G) = \#\{\text{conjugacy classes}\}$. □

We now have a complete, finite algorithm for the representation theory of any finite group: list the conjugacy classes (their number is r); find r irreducible characters, orthonormal under Eq. (2.34), with dimensions satisfying $\sum d_i^2 = |G|$; assemble them into the *character table*, an $r \times r$ array of class-versus-irreducible values; and read off the decomposition of any representation by the inner products Eq. (2.45). The worked example below carries this out start to finish for the smallest nonabelian group.

2.2.7 Worked example: the character table of S_3

Worked Example 2.33 (Complete representation theory of S_3)**.** We determine, with no input beyond the theorems just proved, every irreducible representation of the symmetric group S_3—equivalently the symmetry group D_3 of the equilateral triangle (Fig. 2.1)—its character table, and the decomposition of two natural representations. This is the smallest nonabelian example and the prototype for the $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ multiplet analyses of later sections.

Step 1: conjugacy classes. S_3 consists of the 6 permutations of $\{1,2,3\}$. In a symmetric group, two permutations are conjugate iff they have the same cycle type, so the conjugacy classes are organized by cycle structure:

- the identity $\{e\}$, one element;
- the transpositions $\{(12),(13),(23)\}$, three elements (call a representative τ);
- the 3-cycles $\{(123),(132)\}$, two elements (call a representative c).

There are 3 conjugacy classes, with sizes $1+3+2=6=|S_3|$. By Theorem 2.32 there are exactly 3 inequivalent irreducible representations.

Step 2: dimensions. By the sum-of-squares theorem Eq. (2.49), $d_1^2+d_2^2+d_3^2=6$. The only solution in positive integers (up to ordering) is $1^2+1^2+2^2=6$. So S_3 has two one-dimensional irreducibles and one two-dimensional irreducible. We exhibit all three.

Step 3: the two one-dimensional irreducibles. A one-dimensional representation is a homomorphism $S_3\to\mathbb{C}^\times$; its image is abelian, so it factors through the abelianization, and concretely it must send each transposition to a root of unity squaring to 1 (since $\tau^2=e$). The two choices are:

- the *trivial* representation, $\rho_{\mathrm{triv}}(g)=1$ for all g, with character $\chi_{\mathrm{triv}}\equiv 1$;
- the *sign* representation, $\rho_{\mathrm{sgn}}(g)=\mathrm{sgn}(g)$, equal to $+1$ on even permutations (the identity and the 3-cycles) and -1 on the odd permutations (the transpositions), with character $\chi_{\mathrm{sgn}}(e)=1$, $\chi_{\mathrm{sgn}}(\tau)=-1$, $\chi_{\mathrm{sgn}}(c)=1$.

Both are manifestly homomorphisms and one-dimensional, hence irreducible.

Step 4: the two-dimensional irreducible. Realize $S_3=D_3$ geometrically as the rigid symmetries of an equilateral triangle centered at the origin (Fig. 2.1): a 3-cycle is a rotation by $2\pi/3$, a transposition is a reflection across an axis through one vertex. In an orthonormal frame of the plane, the rotation c by 120° and a reflection τ are

$$\rho_{\mathrm{std}}(c)=\begin{pmatrix}-\frac{1}{2} & -\frac{\sqrt{3}}{2}\\ \frac{\sqrt{3}}{2} & -\frac{1}{2}\end{pmatrix},\qquad \rho_{\mathrm{std}}(\tau)=\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}. \tag{2.54}$$

These two matrices generate a two-dimensional representation (every group element is a product of rotations and reflections). Its character is read off from the traces: $\chi_{\mathrm{std}}(e)=\mathrm{tr}\,\mathbf{1}_2=2$, $\chi_{\mathrm{std}}(\tau)=1+(-1)=0$, and $\chi_{\mathrm{std}}(c)=-\frac{1}{2}-\frac{1}{2}=-1$.

We verify irreducibility by the test of Theorem 2.28(2), weighting each class by its size in the inner product Eq. (2.34):

$$\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle = \frac{1}{6}\Big[\underbrace{1 \cdot 2^2}_{e} + \underbrace{3 \cdot 0^2}_{\tau} + \underbrace{2 \cdot (-1)^2}_{c}\Big] = \frac{1}{6}(4 + 0 + 2) = 1, \tag{2.55}$$

so ρ_{std} is irreducible. It is the third irreducible, the "standard" representation.

Step 5: the character table. Collecting the three characters, with the class sizes recorded in the header:

	$\{e\}$	$\{\tau\}$	$\{c\}$
class size	1	3	2
χ_{triv}	1	1	1
χ_{sgn}	1	-1	1
χ_{std}	2	0	-1

Step 6: orthonormality check. The rows must be orthonormal under Eq. (2.34) (characters are real here, so conjugation is invisible). The diagonal entries are $\langle \chi_{\text{triv}}, \chi_{\text{triv}} \rangle = \frac{1}{6}(1 + 3 + 2) = 1$, $\langle \chi_{\text{sgn}}, \chi_{\text{sgn}} \rangle = \frac{1}{6}(1 + 3 + 2) = 1$, and $\langle \chi_{\text{std}}, \chi_{\text{std}} \rangle = 1$ from Step 4. The off-diagonal entries are

$$\begin{aligned}
\langle \chi_{\text{triv}}, \chi_{\text{sgn}} \rangle &= \tfrac{1}{6}\big[1 \cdot 1 \cdot 1 + 3 \cdot 1 \cdot (-1) + 2 \cdot 1 \cdot 1\big] = \tfrac{1}{6}(1 - 3 + 2) = 0, \\
\langle \chi_{\text{triv}}, \chi_{\text{std}} \rangle &= \tfrac{1}{6}\big[1 \cdot 1 \cdot 2 + 3 \cdot 1 \cdot 0 + 2 \cdot 1 \cdot (-1)\big] = \tfrac{1}{6}(2 + 0 - 2) = 0, \\
\langle \chi_{\text{sgn}}, \chi_{\text{std}} \rangle &= \tfrac{1}{6}\big[1 \cdot 1 \cdot 2 + 3 \cdot (-1) \cdot 0 + 2 \cdot 1 \cdot (-1)\big] = \tfrac{1}{6}(2 + 0 - 2) = 0.
\end{aligned} \tag{2.56}$$

All three vanish: the irreducible characters are orthonormal, as Theorem 2.27 demands. The dimension sum checks too: $1^2 + 1^2 + 2^2 = 6 = |S_3|$, in accord with Eq. (2.49).

Step 7: decomposing two representations. First, the *permutation representation* ρ_{perm} on $\mathbb{C}^3$, where S_3 permutes the standard basis vectors $\hat{e}_1, \hat{e}_2, \hat{e}_3$. Its character counts fixed basis vectors: $\chi_{\text{perm}}(g) = \#\{\text{points fixed by } g\}$, giving $\chi_{\text{perm}}(e) = 3$, $\chi_{\text{perm}}(\tau) = 1$ (a transposition fixes one point), $\chi_{\text{perm}}(c) = 0$ (a 3-cycle fixes none). The multiplicities follow from Eq. (2.45):

$$\begin{aligned}
m_{\text{triv}} &= \langle \chi_{\text{triv}}, \chi_{\text{perm}} \rangle = \tfrac{1}{6}(1 \cdot 3 + 3 \cdot 1 + 2 \cdot 0) = \tfrac{1}{6}(6) = 1, \\
m_{\text{sgn}} &= \langle \chi_{\text{sgn}}, \chi_{\text{perm}} \rangle = \tfrac{1}{6}(1 \cdot 3 + 3 \cdot (-1) \cdot 1 + 2 \cdot 1 \cdot 0) = \tfrac{1}{6}(0) = 0, \\
m_{\text{std}} &= \langle \chi_{\text{std}}, \chi_{\text{perm}} \rangle = \tfrac{1}{6}(1 \cdot 2 \cdot 3 + 3 \cdot 0 \cdot 1 + 2 \cdot (-1) \cdot 0) = \tfrac{1}{6}(6) = 1.
\end{aligned} \tag{2.57}$$

Hence $\rho_{\text{perm}} \cong \rho_{\text{triv}} \oplus \rho_{\text{std}}$, the familiar split of $\mathbb{C}^3$ into the diagonal line $\mathbb{C}(\hat{e}_1 + \hat{e}_2 + \hat{e}_3)$ (on which S_3 acts trivially) and its orthogonal complement, the plane

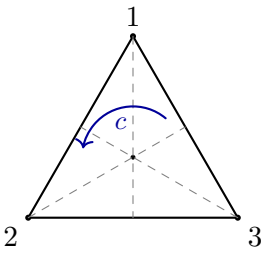


Figure 2.1: The equilateral triangle whose rigid symmetry group is $D_3 \cong S_3$. The three dashed lines are the reflection axes (each fixing one vertex and swapping the other two: the transpositions τ); the blue arc is the 120° rotation c, a 3-cycle. The three reflections form one conjugacy class and the two nontrivial rotations another, matching the class structure used in Theorem 2.33.

$\{x_1 + x_2 + x_3 = 0\}$ (on which it acts as the standard ρ_{std}). The arithmetic confirms what geometry suggested.

Second, the *regular representation* on $\mathbb{C}[S_3]$ (dim = 6). Theorem 2.31 predicts $\mathbb{C}[S_3] \cong d_{\text{triv}}V_{\text{triv}} \oplus d_{\text{sgn}}V_{\text{sgn}} \oplus d_{\text{std}}V_{\text{std}} = 1 \cdot V_{\text{triv}} \oplus 1 \cdot V_{\text{sgn}} \oplus 2 \cdot V_{\text{std}}$, of total dimension $1 \cdot 1 + 1 \cdot 1 + 2 \cdot 2 = 6$. As a check via characters, $\chi_{\text{reg}} = (6, 0, 0)$ on the three classes (Eq. (2.47)), and $\langle \chi_{\text{std}}, \chi_{\text{reg}} \rangle = \frac{1}{6}(2 \cdot 6) = 2 = d_{\text{std}}$, exactly the predicted multiplicity. Every irreducible appears in the regular representation precisely as many times as its dimension, and the books balance.

Why characters are the physicist's tool

The lesson the S_3 table makes concrete is the one that recurs for every symmetry in this book. To know how a symmetry G organizes a system, one does not need the full matrices of the action—one needs only the character, a single number per conjugacy class, and the orthonormal character table of G. Then "which multiplets appear, and how many of each" is the trivial linear-algebra problem of expanding a vector (χ_ρ) in an orthonormal basis (the χ_i), Eq. (2.45). When G is the rotation group, this is precisely how a reducible collection of angular-momentum states is sorted into spin multiplets; when G is a flavor or color group, it is how a tensor product of quarks is sorted into hadron multiplets. The discrete machinery of this section is the rehearsal; the continuous version, with sums over G replaced by invariant integrals, is the performance.

2.2.8 From finite to continuous

We have built the complete decomposition program for finite groups: Maschke guarantees that representations break into irreducible atoms, Schur's lemma

makes those atoms rigid, and characters—orthonormal class functions—let us name and count them with traces. The conceptual content is entirely portable. But the engine that drove every proof was the finite average $\frac{1}{|G|}\sum_{g\in G}$, and the symmetry groups of physics—rotations, Lorentz transformations, gauge groups—are continuous, with infinitely many elements and no finite sum to take. Two routes reopen the program in that setting. For *compact* continuous groups the average survives as an invariant integral $\int_G d\mu(g)$ over the group, and Maschke, Schur, and character orthogonality carry over almost verbatim (this is the path $\mathfrak{su}(2)$ will follow). More powerfully, a continuous group is determined near the identity by its *linearization*—the tangent space at the identity, with its commutator bracket—which is a finite-dimensional algebraic object even when the group is infinite. The next section makes that linearization precise, introduces the Lie algebra and the exponential map that climbs back to the group, and so converts the decomposition philosophy established here into the highest-weight machinery that will solve $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ completely.

2.3 Lie Groups, Lie Algebras, and the Exponential Bridge

The program assembled in the previous two sections — list the irreducible representations of a group, then expand every representation as a direct sum of those atoms (Section 2.2.1) — was built for finite groups, where the words "list" and "sum" are literal: there are finitely many irreducibles, characters are finite tables, and orthogonality relations are finite sums over group elements. The groups that organize physics are not finite. The rotations of space, the boosts of relativity, the phase rotations of electromagnetism, and the color rotations of the strong interaction all form *continua*: their elements are labelled by real parameters varying smoothly, and a sum over group elements becomes an integral over a manifold. We cannot tabulate such a group element by element. What saves the program is that a continuous group is almost entirely determined by its behaviour in an infinitesimal neighbourhood of the identity. Differentiating the group at the identity collapses the curved, infinite object into a flat, finite one — a vector space equipped with a single bilinear operation, the *Lie algebra* — and the exponential map climbs back from this linear shadow to the group itself. This section builds that bridge in both directions. We define a matrix Lie group and its Lie algebra, show that the algebra is closed under the commutator bracket, extract the structure constants f^{abc} in the Hermitian-generator convention that is canonical for the rest of this book, prove the Jacobi identity and use it to represent the algebra on itself through the adjoint representation, and finally read off from the Baker–Campbell–Hausdorff formula exactly how much of the group's multiplication law the bracket alone remembers.

Throughout, $\mathbf{1}$ denotes the $n\times n$ identity matrix and all matrices are complex

$n \times n$ unless stated otherwise; the matrix groups of interest sit inside the general linear group $\mathrm{GL}(n, \mathbb{C})$ of invertible such matrices.

2.3.1 Matrix Lie groups and the tangent space at the identity

The continuous groups of physics are without exception groups of matrices, and we lose nothing essential by restricting attention to those. The only structural input we need beyond the group axioms of Section 2.1.2 is that the group is a *closed* subset of $\mathrm{GL}(n, \mathbb{C})$: a limit of group elements that happens to remain invertible is again a group element.

Definition 2.34 (Matrix Lie group)**.** A *matrix Lie group* G is a subgroup of $\mathrm{GL}(n, \mathbb{C})$ that is closed as a subset: if $A_k \in G$ is a sequence of group elements and $A_k \to A$ with A invertible, then $A \in G$. The smallest n for which G arises this way fixes the *defining* (or *fundamental*) representation, the matrices being the representation of G on $\mathbb{C}^n$ already in the sense of Theorem 2.7.

The examples that carry the book are all defined by an algebraic constraint imposed on a matrix, and closedness is automatic because the constraint is preserved under limits. The unitary group $\mathrm{U}(n) = \{U : U^\dagger U = \mathbf{1}\}$, its unimodular subgroup $\mathrm{SU}(n) = \{U \in \mathrm{U}(n) : \det U = 1\}$, the orthogonal group $\mathrm{O}(n) = \{O \in \mathrm{GL}(n, \mathbb{R}) : O^{\mathsf{T}} O = \mathbf{1}\}$ and its identity component $\mathrm{SO}(n)$, and the Lorentz group $\mathrm{O}(1,3)$ of Theorem 3.1 are each cut out of $\mathrm{GL}(n)$ by polynomial equations in the matrix entries, hence closed. The defining relation in each case is the single equation we will now differentiate.

The strategic idea is geometric. A matrix Lie group is a smooth surface — a manifold — sitting inside the linear space of all $n \times n$ matrices, and it passes through the identity $\mathbf{1}$. Near $\mathbf{1}$ the surface is well approximated by its tangent plane at that point, and that tangent plane is a genuine linear space: we can add its vectors and scale them, even though we cannot add or scale group elements themselves (the sum of two unitary matrices is almost never unitary). The tangent plane is the linearization we are after.

Definition 2.35 (Tangent vector at the identity)**.** A matrix X is *tangent to G at the identity* if there is a smooth curve $\gamma : (-\delta, \delta) \to G$ in the group with $\gamma(0) = \mathbf{1}$ and $\gamma'(0) = X$. The set of all such X is the tangent space $T_{\mathbf{1}}G$.

To see what $T_{\mathbf{1}}G$ is concretely, differentiate the defining relation along such a curve. Take $G = \mathrm{SU}(n)$, the case that fixes our generator conventions. A curve $U(t) \in \mathrm{SU}(n)$ with $U(0) = \mathbf{1}$ obeys $U(t)^\dagger U(t) = \mathbf{1}$ and $\det U(t) = 1$ identically in t. Differentiating the first relation at $t = 0$ with $U(0) = \mathbf{1}$,

$$\dot{U}(0)^\dagger U(0) + U(0)^\dagger \dot{U}(0) = 0 \quad \Longrightarrow \quad X^\dagger + X = 0, \tag{2.58}$$

so the tangent vector $X = \dot{U}(0)$ is *anti-Hermitian*. Differentiating $\det U(t) = 1$ requires Jacobi's formula $\frac{d}{dt} \det M(t) = \det M(t)\ \mathrm{tr}\big(M(t)^{-1}\dot{M}(t)\big)$, which at $t = 0$ gives

$$\mathrm{tr}\, X = 0 : \tag{2.59}$$

the tangent vector is *traceless*. Thus the tangent space to $\mathrm{SU}(n)$ at the identity is the real vector space of traceless anti-Hermitian matrices. It is a *real* vector space — closed under real linear combinations, since a real combination of traceless anti-Hermitian matrices is again traceless and anti-Hermitian, but not closed under multiplication by i, which would turn anti-Hermitian into Hermitian. We record its dimension for $n = 2$, which we will need below: a traceless anti-Hermitian 2×2 matrix has the form i times a traceless Hermitian one, and the traceless Hermitian 2×2 matrices are spanned over $\mathbb{R}$ by the three Pauli matrices, so $\dim_{\mathbb{R}} T_{\mathbf{1}}\mathrm{SU}(2) = 3$.

This tangent space is the Lie algebra, but to make the definition independent of the choice of curve — and to provide the return route from algebra to group — we need the matrix exponential.

2.3.2 The matrix exponential and one-parameter subgroups

The exponential of a matrix is defined exactly as the scalar exponential, by its power series, and the same estimate that proves convergence of the scalar series proves convergence here.

Definition 2.36 (Matrix exponential)**.** For any $n \times n$ complex matrix X,

$$\exp(X) \;=\; \sum_{k=0}^{\infty} \frac{X^k}{k!} \;=\; \mathbf{1} + X + \tfrac{1}{2}X^2 + \tfrac{1}{6}X^3 + \cdots . \tag{2.60}$$

The series converges absolutely in any submultiplicative matrix norm, since $\sum_k \|X\|^k / k! = e^{\|X\|} < \infty$ bounds the partial sums; the limit is therefore a well-defined matrix, entire in X.

We collect the properties used below. Each is the matrix analogue of a familiar scalar fact, and each is proved by manipulating the absolutely convergent series.

Lemma 2.37 (Properties of the matrix exponential)**.** *Let X, Y be $n \times n$ matrices and A invertible. Then*

1. $\dfrac{d}{dt}\exp(tX) = X\exp(tX) = \exp(tX)\,X$ *and* $\exp(0) = \mathbf{1}$*;*
2. $\exp\big((s+t)X\big) = \exp(sX)\exp(tX)$*, so* $\exp(X)$ *is always invertible with inverse* $\exp(-X)$*;*

3. *if* $XY = YX$ *then* $\exp(X+Y) = \exp(X)\exp(Y)$ *(commutativity is required);*

4. $A\exp(X)A^{-1} = \exp(AXA^{-1})$ *and* $\exp(X)^\dagger = \exp(X^\dagger)$*;*

5. $\det\exp(X) = \exp(\operatorname{tr} X)$.

Proof. For (1), differentiate Eq. (2.60) term by term (legitimate by absolute and uniform convergence on compact t-intervals): $\frac{d}{dt}\sum_k t^k X^k/k! = \sum_{k\geq 1} t^{k-1}X^k/(k-1)! = X\sum_{j\geq 0} t^j X^j/j!$, and X may be pulled out on either side since it commutes with its own powers. For (2), both $t\mapsto \exp((s+t)X)$ and $t\mapsto \exp(sX)\exp(tX)$ satisfy the same first-order linear matrix differential equation $F' = XF$ (using (1)) with the same value $\exp(sX)$ at $t=0$; by uniqueness of solutions of linear ODEs they coincide, and setting $t=-s$ gives $\exp(sX)\exp(-sX) = \mathbf{1}$. Statement (3): when $XY=YX$ the binomial theorem holds for $(X+Y)^k$, and the Cauchy product of the absolutely convergent series for $\exp X$ and $\exp Y$ reproduces $\sum_k (X+Y)^k/k!$ exactly as in the scalar proof. Statement (4) follows from $AX^kA^{-1} = (AXA^{-1})^k$ applied term by term, and from $(X^k)^\dagger = (X^\dagger)^k$. For (5), let $f(t) = \det\exp(tX)$; by Jacobi's formula and (1), $f'(t) = f(t)\,\operatorname{tr}\big(\exp(-tX)\,X\exp(tX)\big) = f(t)\operatorname{tr}X$, using cyclicity of the trace, so $f' = (\operatorname{tr}X)f$ with $f(0)=1$, whence $f(t) = e^{t\operatorname{tr}X}$; set $t=1$. □

Property (2) says that $t\mapsto\exp(tX)$ is a group homomorphism from the additive group $(\mathbb{R},+)$ into $\mathrm{GL}(n,\mathbb{C})$. Such homomorphisms are the smooth threads through the identity, and they are *all* of exponential form.

Proposition 2.38 (One-parameter subgroups)**.** *Let* $\gamma:\mathbb{R}\to\mathrm{GL}(n,\mathbb{C})$ *be a continuous homomorphism, i.e.* $\gamma(s+t)=\gamma(s)\gamma(t)$ *and* $\gamma(0)=\mathbf{1}$*. Then* γ *is smooth and* $\gamma(t)=\exp(tX)$ *with* $X=\gamma'(0)$*. Conversely, every* X *gives such a homomorphism through Eq. (2.60).*

Proof. The converse is Theorem 2.37(2). For the forward direction, continuity of a homomorphism of $(\mathbb{R},+)$ into a matrix group upgrades to differentiability: choose τ small enough that $C=\int_0^\tau \gamma(u)\,du$ is invertible (it tends to $\tau\mathbf{1}$ as $\tau\to 0$), then $\gamma(t)C = \int_0^\tau \gamma(t+u)\,du = \int_t^{t+\tau}\gamma(v)\,dv$ is differentiable in t as an integral of a continuous function, so $\gamma(t) = \big(\int_t^{t+\tau}\gamma\big)C^{-1}$ is differentiable. Now differentiate the homomorphism property $\gamma(s+t)=\gamma(s)\gamma(t)$ with respect to s at $s=0$: $\gamma'(t) = \gamma'(0)\gamma(t) = X\gamma(t)$. This is the linear ODE whose unique solution with $\gamma(0)=\mathbf{1}$ is $\gamma(t)=\exp(tX)$ by Theorem 2.37(1). □

This is the bridge. A tangent direction X at the identity generates a whole one-parameter subgroup $\exp(tX)$ that lies on the curved group surface and is tangent to X at $t=0$; conversely every smooth one-dimensional "flow" through the identity is of this form. Figure 2.2 pictures the construction.

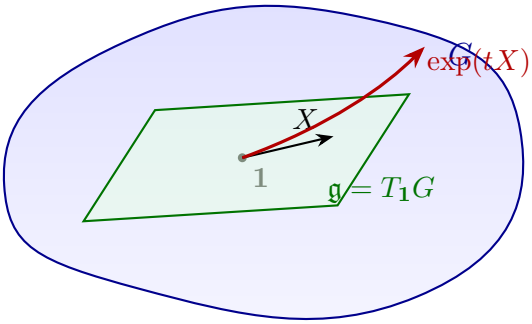


Figure 2.2: The exponential bridge. A matrix Lie group G is a curved surface through the identity $\mathbf{1}$; its tangent space there is the Lie algebra $\mathfrak{g} = T_{\mathbf{1}}G$, a flat vector space on which addition and scaling make sense. A tangent direction $X \in \mathfrak{g}$ generates the one-parameter subgroup $t \mapsto \exp(tX)$, a curve lying *on* G and tangent to X at the identity. The bracket on $\mathfrak{g}$, extracted in Section 2.3.3, records how two such flows fail to commute.

We can now define the Lie algebra so that it manifestly is a real vector space and so that the exponential genuinely maps it back into the group.

Definition 2.39 (Lie algebra of a matrix Lie group)**.** The *Lie algebra* $\mathfrak{g} = \mathrm{Lie}(G)$ of a matrix Lie group G is the set of matrices X such that $\exp(tX) \in G$ for *all* real t.

This set coincides with the tangent space $T_{\mathbf{1}}G$ of Theorem 2.35: if $\exp(tX) \in G$ for all t then $t \mapsto \exp(tX)$ is a curve in G through $\mathbf{1}$ with derivative X, so $X \in T_{\mathbf{1}}G$; conversely any tangent vector arises, by Theorem 2.38, from a one-parameter subgroup inside G. The definition through exp has the advantage that the return statement is built in: for every $X \in \mathfrak{g}$, setting $t = 1$ gives $\exp(X) \in G$. The exponential map genuinely climbs from the algebra to the group.

The earlier computation Eq. (2.58)–Eq. (2.59) can now be read forward as well as backward. If X is traceless and anti-Hermitian, then $\exp(X)$ is unitary and unimodular: $\exp(X)^\dagger \exp(X) = \exp(X^\dagger)\exp(X) = \exp(-X)\exp(X) = \mathbf{1}$ by Theorem 2.37(2),(4) (the two exponents commute), and $\det \exp(X) = \exp(\operatorname{tr} X) = \exp(0) = 1$ by Theorem 2.37(5). Hence exp carries the traceless anti-Hermitian matrices into $\mathrm{SU}(n)$, and we have identified

$$\mathfrak{su}(n) \;=\; \mathrm{Lie}\big(\mathrm{SU}(n)\big) \;=\; \{\, X \in M_n(\mathbb{C}) : X^\dagger = -X,\ \operatorname{tr} X = 0 \,\}, \qquad \dim_{\mathbb{R}} \mathfrak{su}(n) = n^2 - 1. \tag{2.61}$$

The dimension count is the number of independent real parameters in a traceless anti-Hermitian matrix: a Hermitian $n \times n$ matrix has n^2 real parameters, tracelessness removes one, and multiplication by i converts to anti-Hermitian.

2.3.3 The Lie algebra: closure under the bracket

So far $\mathfrak{g}$ is only a vector space. Its defining *algebraic* structure — the thing that remembers the non-commutativity of the group — emerges when we ask how two one-parameter subgroups fail to commute. Consider the group commutator

$$C(s,t) \;=\; \exp(sX)\,\exp(tY)\,\exp(-sX)\,\exp(-tY), \tag{2.62}$$

which equals $\mathbf{1}$ exactly when the two flows commute. Expanding each factor to second order using Eq. (2.60),

$$\begin{aligned}\exp(sX)\exp(tY) &= \Big(\mathbf{1} + sX + \tfrac{1}{2}s^2X^2\Big)\Big(\mathbf{1} + tY + \tfrac{1}{2}t^2Y^2\Big) + \cdots \\ &= \mathbf{1} + sX + tY + st\,XY + \tfrac{1}{2}s^2X^2 + \tfrac{1}{2}t^2Y^2 + \cdots,\end{aligned} \tag{2.63}$$

and similarly $\exp(-sX)\exp(-tY) = \mathbf{1} - sX - tY + st\,XY + \tfrac{1}{2}s^2X^2 + \tfrac{1}{2}t^2Y^2 + \cdots$. Multiplying the two and keeping terms through second order in (s,t), the linear terms cancel, the pure s^2 and t^2 terms cancel, and the surviving cross term is

$$C(s,t) \;=\; \mathbf{1} + st\,(XY - YX) + \cdots \;=\; \mathbf{1} + st\;[X,Y] + O\big((s,t)^3\big), \tag{2.64}$$

where the *commutator* (or *Lie bracket*) is

$$[X,Y] \;\equiv\; XY - YX. \tag{2.65}$$

The leading deviation of the group commutator from the identity is precisely st times the bracket. This is the sense in which the bracket measures the group's non-commutativity to leading order, and it is the reason the bracket — not the matrix product — is the intrinsic operation of the algebra.

The decisive fact is that the bracket stays inside the algebra.

Theorem 2.40 (Closure of the Lie algebra under the bracket)**.** *If $X, Y \in \mathfrak{g} = \mathrm{Lie}(G)$, then $[X,Y] \in \mathfrak{g}$. Together with its real vector-space structure this makes $\mathfrak{g}$ a* Lie algebra*: a real vector space equipped with a bilinear, antisymmetric bracket that, as we prove in Section 2.3.5, satisfies the Jacobi identity.*

Proof. We first confirm that $\mathfrak{g}$ is a real vector space. Closure under real scaling is immediate: if $\exp(tX) \in G$ for all t then $\exp\big(t(cX)\big) = \exp\big((tc)X\big) \in G$ for all t and any real c. Closure under addition uses the *Lie product formula*

$$\exp\big(t(X+Y)\big) \;=\; \lim_{m\to\infty}\Big(\exp\tfrac{tX}{m}\,\exp\tfrac{tY}{m}\Big)^m. \tag{2.66}$$

To prove Eq. (2.66), expand the single bracketed factor: $\exp(\frac{tX}{m})\exp(\frac{tY}{m}) = \mathbf{1} + \frac{t}{m}(X+Y) + O(1/m^2)$. Writing this as $\exp\big(\frac{t}{m}(X+Y) + O(1/m^2)\big)$ — valid since the logarithm series inverts the exponential near $\mathbf{1}$ — and raising to the m-th power multiplies the exponent by m, giving $\exp\big(t(X+Y) + O(1/m)\big) \to \exp\big(t(X+Y)\big)$

as $m \to \infty$. Now each factor $\exp(\frac{tX}{m})\exp(\frac{tY}{m})$ lies in G because $X, Y \in \mathfrak{g}$, so does its m-th power, and since G is closed (Theorem 2.34) the limit $\exp(t(X+Y))$ lies in G for every t. Hence $X + Y \in \mathfrak{g}$, and $\mathfrak{g}$ is a real vector space. Being a finite-dimensional subspace of $M_n(\mathbb{C})$, it is in particular topologically closed.

For the bracket, fix $X, Y \in \mathfrak{g}$ and consider, for each real s, the *conjugated* matrix

$$Y(s) \;\equiv\; \exp(sX)\, Y\, \exp(-sX). \tag{2.67}$$

We claim $Y(s) \in \mathfrak{g}$ for every s. Indeed, by Theorem 2.37(4),

$$\exp\big(t\,Y(s)\big) = \exp\big(t\,\exp(sX)\,Y\,\exp(-sX)\big) = \exp(sX)\,\exp(tY)\,\exp(-sX), \tag{2.68}$$

which is a product of three elements of G, hence in G, for all t; so $Y(s) \in \mathfrak{g}$ by Theorem 2.39. Thus $s \mapsto Y(s)$ is a differentiable curve lying entirely in the closed subspace $\mathfrak{g}$. The derivative of such a curve is a limit of difference quotients $(Y(s) - Y(0))/s$, each of which lies in the subspace $\mathfrak{g}$ (a subspace is closed under subtraction and scaling), and since $\mathfrak{g}$ is closed the limit lies in $\mathfrak{g}$ as well. Computing the derivative at $s = 0$ with Theorem 2.37(1),

$$\left.\frac{d}{ds}\right|_{s=0} Y(s) = \left.\frac{d}{ds}\right|_{s=0} \Big(\exp(sX)\,Y\,\exp(-sX)\Big) = XY - YX = [X, Y]\,. \tag{2.69}$$

Therefore $[X, Y] \in \mathfrak{g}$. Bilinearity and antisymmetry $[X, Y] = -\,[Y, X]$ of the bracket are immediate from its definition Eq. (2.65). □

Theorem 2.40 is what makes the linearization useful. The infinitely many group elements have been replaced by a finite-dimensional vector space, and the entire group multiplication law, to leading nontrivial order, has been compressed into one bilinear operation on that space. The next task is to write that operation in coordinates.

2.3.4 Structure constants and the Hermitian-generator convention

Choose a basis of the Lie algebra. Because $\mathfrak{g}$ is closed under the bracket, the bracket of two basis vectors is again a linear combination of basis vectors, and the coefficients are numbers intrinsic to the group. These numbers are the *structure constants*; they encode everything the Lie algebra knows.

Two conventions for the basis are in use, and they differ by a factor of i. The mathematician's convention takes the basis vectors to be elements of $\mathfrak{g}$ itself — for $\mathfrak{su}(n)$, traceless *anti*-Hermitian matrices — so that exp of a real combination lands in the group. The physicist's convention, which is the canon of this book (Chapter A, Chapter B), strips out the i and works with *Hermitian* generators T^a, writing a group element near the identity as $\exp(-i\alpha^a T^a)$ with

real parameters α^a. The two are related by $X = -i\alpha^a T^a$: the algebra element X is anti-Hermitian, while the generator $T^a = i\,\partial X/\partial\alpha^a$ is Hermitian. We adopt the physicist's convention once and for all.

Notation Watch

Hermitian generators T^a (with $T^{a\dagger} = T^a$) are *not* themselves elements of the real Lie algebra $\mathfrak{g}$; the algebra elements are the anti-Hermitian combinations iT^a. The Hermitian T^a are a convenience that makes the generators observables (they exponentiate to unitaries via $\exp(-i\alpha^a T^a)$) and keeps the structure constants real. Whenever this book writes "the generators T^a," the Hermitian convention with the normalization Eq. (2.72) below is meant, in agreement with Chapter B and Yang–Mills theory of Chapter 25. The mathematician's $\left[iT^a, iT^b\right] = -\left[T^a, T^b\right]$ accounts for every apparent sign discrepancy with the math literature.

In the Hermitian convention the closure Theorem 2.40 reads as follows. The algebra elements iT^a span $\mathfrak{g}$ over $\mathbb{R}$, and their bracket lies in $\mathfrak{g}$: $\left[iT^a, iT^b\right] = -\left[T^a, T^b\right] \in \mathfrak{g} = \mathrm{span}_{\mathbb{R}}\{iT^c\}$. Hence $\left[T^a, T^b\right]$ is i times a real combination of the T^c, which we write as

$$\boxed{\left[T^a, T^b\right] \;=\; i\, f^{abc}\, T^c} \tag{2.70}$$

with real coefficients f^{abc}, the *structure constants*, summed over c. That the f^{abc} are real is the content of the previous sentence; it can also be checked directly by taking the Hermitian conjugate of Eq. (2.70): the left side gives $\left[T^b, T^a\right] = -\left[T^a, T^b\right]$, while the right side gives $-i\,(f^{abc})^* T^c$, and consistency forces $(f^{abc})^* = f^{abc}$. Antisymmetry in the first two indices,

$$f^{abc} = -f^{bac}, \tag{2.71}$$

is immediate from $\left[T^a, T^b\right] = -\left[T^b, T^a\right]$.

The structure constants enjoy a stronger symmetry once we fix the normalization of the generators. The trace of a product of two Hermitian generators defines a symmetric, positive-definite bilinear form on the algebra; for a simple Lie algebra this form is unique up to scale, and we fix the scale in the defining representation by the canonical normalization

$$\boxed{\mathrm{tr}(T^a T^b) \;=\; \tfrac{1}{2}\,\delta^{ab}} \tag{2.72}$$

(the Dynkin index $T_F = \frac{1}{2}$ of Chapter B). With this normalization the structure constants become *totally antisymmetric* in all three indices.

Proposition 2.41 (Total antisymmetry of f^{abc})**.** *In a basis of Hermitian generators obeying Eq. (2.72), the structure constants defined by Eq. (2.70) satisfy*

$$f^{abc} \;=\; -2i\,\mathrm{tr}\Big(\,\big[T^a,T^b\big]\,T^c\Big), \tag{2.73}$$

and f^{abc} is totally antisymmetric: it changes sign under the interchange of any two indices and is invariant under cyclic permutations.

Proof. Multiply Eq. (2.70) on the right by T^d and take the trace, using Eq. (2.72):

$$\mathrm{tr}\Big(\,\big[T^a,T^b\big]\,T^d\Big) = i\,f^{abc}\,\mathrm{tr}(T^cT^d) = i\,f^{abc}\,\tfrac{1}{2}\delta^{cd} = \tfrac{i}{2}\,f^{abd}. \tag{2.74}$$

Solving for f^{abd} and renaming $d \to c$ gives Eq. (2.73). From this expression antisymmetry under $a \leftrightarrow b$ is Eq. (2.71) again (the commutator flips sign). For cyclic invariance, expand the commutator and use cyclicity of the trace, $\mathrm{tr}(ABC) = \mathrm{tr}(CAB) = \mathrm{tr}(BCA)$:

$$\begin{aligned} f^{abc} &= -2i\,\mathrm{tr}(T^aT^bT^c - T^bT^aT^c),\\ f^{bca} &= -2i\,\mathrm{tr}(T^bT^cT^a - T^cT^bT^a) = -2i\,\mathrm{tr}(T^aT^bT^c - T^bT^aT^c) = f^{abc}, \end{aligned} \tag{2.75}$$

where in the second line we cycled $\mathrm{tr}(T^bT^cT^a) = \mathrm{tr}(T^aT^bT^c)$ and $\mathrm{tr}(T^cT^bT^a) = \mathrm{tr}(T^aT^cT^b) = \mathrm{tr}(T^bT^aT^c)$. Thus f^{abc} is invariant under cyclic permutations and antisymmetric under the transposition $a \leftrightarrow b$. Since the cyclic permutations together with one transposition generate the full permutation group S_3 on three letters, and every transposition is conjugate to $a \leftrightarrow b$ under cyclic permutations, f^{abc} is antisymmetric under *every* transposition, i.e. totally antisymmetric. □

The structure constants are thus a totally antisymmetric real array, the same array that will appear as the self-coupling of the gauge field in the Yang–Mills Lagrangian of Chapter 25; the conventions Eq. (2.70) and Eq. (2.72) fixed here are exactly the ones that thread uses.

2.3.5 The Jacobi identity and the adjoint representation

The bracket is not an arbitrary antisymmetric product. Because it is built from the associative matrix product, it obeys one further constraint, the Jacobi identity, which is the precise statement that the algebra can represent itself.

Theorem 2.42 (Jacobi identity)**.** *For any X, Y, Z in a Lie algebra of matrices,*

$$\big[[X,Y],Z\big] + \big[[Y,Z],X\big] + \big[[Z,X],Y\big] \;=\; 0. \tag{2.76}$$

Equivalently, in terms of structure constants,

$$f^{abe}f^{ecd} + f^{bce}f^{ead} + f^{cae}f^{ebd} \;=\; 0. \tag{2.77}$$

Proof. Expand each double commutator using $[A,B]=AB-BA$, which is the only place associativity of matrix multiplication enters:

$$\begin{aligned}[[X,Y],Z] &= XYZ - YXZ - ZXY + ZYX,\\ [[Y,Z],X] &= YZX - ZYX - XYZ + XZY,\\ [[Z,X],Y] &= ZXY - XZY - YZX + YXZ.\end{aligned} \tag{2.78}$$

Adding the three lines, every one of the twelve triple products cancels against an identical product of opposite sign: XYZ in the first line cancels $-XYZ$ in the second, $-YXZ$ in the first cancels $+YXZ$ in the third, $-ZXY$ in the first cancels $+ZXY$ in the third, $+ZYX$ in the first cancels $-ZYX$ in the second, and the remaining four terms $YZX, XZY, -YZX, -XZY$ cancel pairwise. The sum is zero, which is Eq. (2.76). To obtain Eq. (2.77), set $X=T^a$, $Y=T^b$, $Z=T^c$ and use Eq. (2.70) twice. The first term is $\left[\left[T^a,T^b\right],T^c\right] = \left[if^{abe}T^e,T^c\right] = if^{abe}\,if^{ecd}T^d = -f^{abe}f^{ecd}\,T^d$; the other two follow by cyclic relabelling. Summing and dropping the overall factor $-T^d$ (the T^d are linearly independent) gives Eq. (2.77). □

The Jacobi identity acquires its meaning when rewritten as a statement about representations. Define, for each algebra element X, the linear operator ad_X that acts on the algebra itself by bracketing:

$$\mathrm{ad}_X : \mathfrak{g}\to\mathfrak{g}, \qquad \mathrm{ad}_X(Y) \;=\; [X,Y]\,. \tag{2.79}$$

By Theorem 2.40 the result lies in $\mathfrak{g}$, so ad_X is a genuine linear map of the vector space $\mathfrak{g}$ to itself, an element of the matrix algebra $\mathfrak{gl}(\mathfrak{g})$ of operators on $\mathfrak{g}$. The assignment $X\mapsto\mathrm{ad}_X$ is itself a representation of the Lie algebra.

Definition 2.43 (Adjoint representation of the algebra)**.** The *adjoint representation* of $\mathfrak{g}$ is the linear map $\mathrm{ad}:\mathfrak{g}\to\mathfrak{gl}(\mathfrak{g})$, $X\mapsto\mathrm{ad}_X$, of Eq. (2.79). Its dimension equals $\dim\mathfrak{g}$; for $\mathfrak{su}(n)$ it is the (n^2-1)-dimensional representation.

Proposition 2.44 (ad is a representation; this is the Jacobi identity)**.** *The adjoint map respects the bracket,*

$$\mathrm{ad}_{[X,Y]} \;=\; [\mathrm{ad}_X,\mathrm{ad}_Y] \quad \textit{(as operators on } \mathfrak{g}\textit{)}, \tag{2.80}$$

so ad *is a representation of* $\mathfrak{g}$ *in the sense of Theorem 2.7. The identity Eq. (2.80) is logically equivalent to the Jacobi identity Eq. (2.76).*

Proof. Apply both sides of Eq. (2.80) to an arbitrary $Z\in\mathfrak{g}$. The left side is $\mathrm{ad}_{[X,Y]}(Z)=[[X,Y],Z]$. The right side is

$$[\mathrm{ad}_X,\mathrm{ad}_Y](Z) = \mathrm{ad}_X\,\mathrm{ad}_Y(Z) - \mathrm{ad}_Y\,\mathrm{ad}_X(Z) = [X,[Y,Z]] - [Y,[X,Z]]\,. \tag{2.81}$$

Equality of the two for all Z reads $[[X,Y],Z] = [X,[Y,Z]] - [Y,[X,Z]]$. Rearranging and using antisymmetry, $[[X,Y],Z] - [X,[Y,Z]] + [Y,[X,Z]] = [[X,Y],Z] + [[Y,Z],X] + [[Z,X],Y] = 0$, which is exactly Eq. (2.76). Hence Eq. (2.80) holds if and only if the Jacobi identity holds. □

In the Hermitian-generator basis the adjoint representation has explicit matrices built from the structure constants. Acting on the basis, $\mathrm{ad}_{T^a}(T^b) = \left[T^a, T^b\right] = i f^{abc} T^c$, so reading off the coefficient of T^c and defining the matrix elements with the standard physics normalization,

$$\left(T^a_{\mathrm{adj}}\right)_{bc} \;=\; -\,i\, f^{abc}. \tag{2.82}$$

These are Hermitian: $(T^a_{\mathrm{adj}})^*_{cb} = (-if^{acb})^* = +if^{acb} = -if^{abc} = (T^a_{\mathrm{adj}})_{bc}$, using reality and total antisymmetry of f, and they satisfy the same algebra $\left[T^a_{\mathrm{adj}}, T^b_{\mathrm{adj}}\right] = i f^{abc} T^c_{\mathrm{adj}}$, which is once more Eq. (2.77). Thus the structure constants *are* the adjoint generators: a Lie algebra always represents itself, in dimension equal to its own. We will meet Eq. (2.82) again in the very next worked example, where the adjoint of $\mathfrak{su}(2)$ turns out to be the rotation generators of $\mathfrak{so}(3)$.

2.3.6 The exponentiated adjoint and the action of the group on its algebra

The adjoint representation of the *algebra* integrates to a representation of the *group*. A group element $g \in G$ acts on the algebra by conjugation,

$$\mathrm{Ad}_g : \mathfrak{g} \to \mathfrak{g}, \qquad \mathrm{Ad}_g(X) = g\, X\, g^{-1}, \tag{2.83}$$

which lands in $\mathfrak{g}$ because $\exp(t\, gXg^{-1}) = g \exp(tX) g^{-1} \in G$ for all t by Theorem 2.37(4), so $gXg^{-1} \in \mathfrak{g}$ by Theorem 2.39. The map $\mathrm{Ad} : G \to \mathrm{GL}(\mathfrak{g})$, $g \mapsto \mathrm{Ad}_g$, is a representation of the group on its own algebra, the *adjoint representation of the group*. Its relation to the algebra adjoint ad is the cleanest illustration of the exponential bridge.

Proposition 2.45 ($\mathrm{Ad} \circ \exp = \exp \circ \mathrm{ad}$)**.** *For all $X, Y \in \mathfrak{g}$,*

$$\mathrm{Ad}_{\exp(tX)}\, Y \;=\; \exp(t\ \mathrm{ad}_X)\, Y \;=\; Y + t\,[X,Y] + \tfrac{t^2}{2}\,[X,[X,Y]] + \cdots . \tag{2.84}$$

In words: conjugating by the group element $\exp(tX)$ acts on the algebra as the exponential of the operator ad_X. The group adjoint is the exponentiated algebra adjoint.

Proof. Let $F(t) = \mathrm{Ad}_{\exp(tX)}\, Y = \exp(tX)\, Y \exp(-tX)$. Differentiating with Theorem 2.37(1),

$$F'(t) = X \exp(tX)\, Y \exp(-tX) - \exp(tX)\, Y \exp(-tX)\, X = [X, F(t)] = \mathrm{ad}_X\ F(t), \tag{2.85}$$

since X commutes with $\exp(\pm tX)$ and may be moved through. This is a linear ODE for F in the (finite-dimensional) space $\mathfrak{g}$, with constant operator ad_X and initial value $F(0) = Y$; its unique solution is $F(t) = \exp(t\,\operatorname{ad}_X)Y$ by the same argument as Theorem 2.37(1). Expanding the operator exponential gives the explicit nested-commutator series. □

Theorem 2.45 is the first instance of a general principle: *differentiating a representation of the group at the identity yields a representation of the algebra, and exponentiating a representation of the algebra recovers (the identity component of) the representation of the group.* The differential of the group adjoint Ad is the algebra adjoint ad. This correspondence between group representations and algebra representations, of which $(\mathrm{Ad}, \mathrm{ad})$ is the example carried by the structure constants themselves, is what allows the entire classification problem to be moved down to the algebra — the strategy executed in full for $\mathfrak{su}(2)$ in the next section.

2.3.7 The Baker–Campbell–Hausdorff bridge

We have seen that the bracket controls the group commutator to leading order (Eq. (2.64)). The Baker–Campbell–Hausdorff (BCH) formula makes the sharper statement that the bracket controls the entire product of two exponentials: there is a single algebra element Z, built from X and Y using only brackets, with $\exp(X)\exp(Y) = \exp(Z)$. We derive its leading nontrivial term, which is all that the rest of the book needs and which already carries the qualitative message.

Theorem 2.46 (Leading BCH term)**.** *For matrices X, Y small enough that the logarithm series converges,*

$$\exp(X)\,\exp(Y) \;=\; \exp\Big(X + Y + \tfrac{1}{2}\,[X,Y] + \tfrac{1}{12}\,[X,[X,Y]] - \tfrac{1}{12}\,[Y,[X,Y]] + \cdots\Big), \tag{2.86}$$

where every term beyond $X+Y$ is a nested commutator. In particular, to second order,

$$\exp(X)\exp(Y) \;=\; \exp\Big(X + Y + \tfrac{1}{2}\,[X,Y] + O(\textit{order } 3)\Big). \tag{2.87}$$

Proof. We compute $Z = \log\big(\exp(X)\exp(Y)\big)$ to second order in the joint degree of X and Y. Multiply the two exponential series and keep terms of total degree ≤ 2:

$$\begin{aligned}\exp(X)\exp(Y) &= \Big(\mathbf{1} + X + \tfrac{1}{2}X^2\Big)\Big(\mathbf{1} + Y + \tfrac{1}{2}Y^2\Big) + \cdots \\ &= \mathbf{1} + (X+Y) + \Big(\tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2\Big) + \cdots \;\equiv\; \mathbf{1} + W. \end{aligned} \tag{2.88}$$

Here $W = (X+Y) + \big(\tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2\big) + \cdots$ collects everything beyond the identity, with a first-degree part $W_1 = X + Y$ and a second-degree part $W_2 = \tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2$. Now apply the logarithm series $\log(\mathbf{1} + W) = W -$

$\frac{1}{2}W^2 + \cdots$, keeping degree ≤ 2. Only W_1 contributes to W^2 at this order, so $W^2 = W_1^2 + \cdots = (X+Y)^2 + \cdots = X^2 + XY + YX + Y^2 + \cdots$. Therefore

$$\begin{aligned} Z &= W - \tfrac{1}{2}W^2 + \cdots = (X+Y) + \left(\tfrac{1}{2}X^2 + XY + \tfrac{1}{2}Y^2\right) - \tfrac{1}{2}(X^2 + XY + YX + Y^2) + \cdots \\ &= (X+Y) + \tfrac{1}{2}(XY - YX) + \cdots = X + Y + \tfrac{1}{2}[X,Y] + \cdots . \end{aligned} \tag{2.89}$$

The pure X^2 and Y^2 terms cancel between the two contributions, leaving precisely $\frac{1}{2}[X,Y]$ at second order. This establishes Eq. (2.87); the third-order terms in Eq. (2.86) are obtained by carrying the same computation one degree further, and one finds they assemble into the displayed nested commutators. The crucial structural fact — that *every* correction to $X+Y$ is a commutator, so that Z lies in the Lie algebra whenever X, Y do — is the theorem of Baker, Campbell, and Hausdorff in its full form; the closure under brackets proved in Theorem 2.40 is exactly what guarantees each such commutator stays in $\mathfrak{g}$. □

Physical Insight

Equation (2.87) is the quantitative meaning of "the algebra remembers the group to leading order." If the group were abelian, $[X,Y] = 0$ and $\exp(X)\exp(Y) = \exp(X+Y)$ exactly: exponentials would simply add their exponents, as for ordinary numbers. The first obstruction to this is the bracket, appearing at second order with the universal coefficient $\frac{1}{2}$. Every higher correction is built from brackets as well, so a group with a trivial bracket (an abelian Lie algebra) has, near the identity, the additive structure of a vector space, while all non-commutativity — and hence all of the interesting physics of non-abelian gauge theory in Chapter 25 — is generated by iterated brackets of the generators. The Lie algebra is therefore not a poor approximation to the group but a faithful one: the connected group is reconstructed from its algebra by Eq. (2.86), and the entire chapter's strategy of classifying representations at the level of the algebra is justified.

2.3.8 Worked example: $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ exponentiated

We close by carrying out the exponential map explicitly for the two algebras that dominate the rest of the chapter and the book. The computation produces closed forms we will reuse without comment when we solve $\mathfrak{su}(2)$ completely in Section 2.4, and it exhibits in miniature the relationship between a group and its double cover.

Worked Example 2.47 (Exponentiating the $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ generators)**.**

The $\mathfrak{su}(2)$ generators. By Eq. (2.61), $\mathfrak{su}(2)$ consists of traceless anti-Hermitian 2×2 matrices, the real span of $\{iT^a\}$ with Hermitian generators

$$T^a=\tfrac{1}{2}\sigma^a,\qquad \sigma^1=\begin{pmatrix}0&1\\1&0\end{pmatrix},\ \sigma^2=\begin{pmatrix}0&-i\\i&0\end{pmatrix},\ \sigma^3=\begin{pmatrix}1&0\\0&-1\end{pmatrix}. \tag{2.90}$$

The Pauli matrices obey $\sigma^a\sigma^b=\delta^{ab}\mathbf{1}+i\epsilon^{abc}\sigma^c$, from which both defining data of the generators follow at once. The normalization is $\operatorname{tr}(T^aT^b)=\frac{1}{4}\operatorname{tr}(\sigma^a\sigma^b)=\frac{1}{4}\cdot 2\delta^{ab}=\frac{1}{2}\delta^{ab}$, confirming Eq. (2.72). The bracket is

$$\left[T^a,T^b\right]=\tfrac{1}{4}\left[\sigma^a,\sigma^b\right]=\tfrac{1}{4}(2i\epsilon^{abc}\sigma^c)=i\,\epsilon^{abc}\,T^c, \tag{2.91}$$

so the structure constants of $\mathfrak{su}(2)$ are the totally antisymmetric Levi-Civita symbol, $f^{abc}=\epsilon^{abc}$, in accordance with Theorem 2.41.

Exponentiating $\mathfrak{su}(2)$. Take a unit vector $\hat n\in\mathbb{R}^3$ and an angle $\theta\in\mathbb{R}$, and form the group element $U=\exp(-i\theta\,\hat n\cdot\mathbf{T})=\exp(-i\tfrac{\theta}{2}\,\hat n\cdot\boldsymbol{\sigma})$, where $\hat n\cdot\boldsymbol{\sigma}=n^a\sigma^a$. The key simplification is that $\hat n\cdot\boldsymbol{\sigma}$ squares to the identity:

$$(\hat n\cdot\boldsymbol{\sigma})^2=n^an^b\,\sigma^a\sigma^b=n^an^b(\delta^{ab}\mathbf{1}+i\epsilon^{abc}\sigma^c)=|\hat n|^2\ \mathbf{1}=\mathbf{1}, \tag{2.92}$$

the antisymmetric piece vanishing because n^an^b is symmetric. Splitting the exponential series Eq. (2.60) into even and odd powers and using $(\hat n\cdot\boldsymbol{\sigma})^{2k}=\mathbf{1}$, $(\hat n\cdot\boldsymbol{\sigma})^{2k+1}=\hat n\cdot\boldsymbol{\sigma}$,

$$\begin{aligned}U=\exp\Big(-i\tfrac{\theta}{2}\,\hat n\cdot\boldsymbol{\sigma}\Big)&=\sum_{k=0}^{\infty}\frac{1}{(2k)!}\Big(-i\tfrac{\theta}{2}\Big)^{2k}\mathbf{1}+\sum_{k=0}^{\infty}\frac{1}{(2k+1)!}\Big(-i\tfrac{\theta}{2}\Big)^{2k+1}\hat n\cdot\boldsymbol{\sigma}\\&=\cos\frac{\theta}{2}\,\mathbf{1}-i\sin\frac{\theta}{2}\,\hat n\cdot\boldsymbol{\sigma}.\end{aligned} \tag{2.93}$$

This is the general element of SU(2) in axis–angle form. It is manifestly unitary, $U^\dagger U=(\cos\frac{\theta}{2}\,\mathbf{1}+i\sin\frac{\theta}{2}\,\hat n\cdot\boldsymbol{\sigma})(\cos\frac{\theta}{2}\,\mathbf{1}-i\sin\frac{\theta}{2}\,\hat n\cdot\boldsymbol{\sigma})=\cos^2\frac{\theta}{2}\,\mathbf{1}+\sin^2\frac{\theta}{2}\,(\hat n\cdot\boldsymbol{\sigma})^2=\mathbf{1}$, and unimodular, $\det U=\cos^2\frac{\theta}{2}+\sin^2\frac{\theta}{2}=1$ (the determinant of $a\mathbf{1}+i\,\hat n\cdot\boldsymbol{\sigma}\,b$ is a^2+b^2 for the Pauli matrices), so indeed $U\in\mathrm{SU}(2)$ as the bridge guarantees.

The $\mathfrak{so}(3)$ generators as the adjoint of $\mathfrak{su}(2)$. By Eq. (2.82) the adjoint representation of $\mathfrak{su}(2)$ has generators $(J^a)_{bc}=-if^{abc}=-i\epsilon^{abc}$, three Hermitian 3×3 matrices,

$$J^1=\begin{pmatrix}0&0&0\\0&0&-i\\0&i&0\end{pmatrix},\quad J^2=\begin{pmatrix}0&0&i\\0&0&0\\-i&0&0\end{pmatrix},\quad J^3=\begin{pmatrix}0&-i&0\\i&0&0\\0&0&0\end{pmatrix}, \tag{2.94}$$

satisfying the *same* algebra $\left[J^a,J^b\right]=i\epsilon^{abc}J^c$. These are exactly the generators of rotations of three-dimensional space, the Lie algebra $\mathfrak{so}(3)$ of Theorem 3.7: $\mathfrak{so}(3)=\mathfrak{su}(2)$ as Lie algebras, the adjoint of $\mathfrak{su}(2)$ being the defining representation

of SO(3). Acting on a vector $\mathbf{v} \in \mathbb{R}^3$ the generator realizes the cross product, $(\hat{n}\cdot\mathbf{J})\,\mathbf{v} = i\,\hat{n}\times\mathbf{v}$, since $\big((\hat{n}\cdot\mathbf{J})\mathbf{v}\big)_b = -i\epsilon^{abc}n^a v^c = i\,(\hat{n}\times\mathbf{v})_b$.

Exponentiating $\mathfrak{so}(3)$. Form $R = \exp(-i\theta\,\hat{n}\cdot\mathbf{J})$. Now the relevant identity is cubic rather than quadratic. Writing $B = \hat{n}\cdot\mathbf{J}$, the relation $(\hat{n}\cdot\mathbf{J})\mathbf{v} = i\,\hat{n}\times\mathbf{v}$ iterated gives $B^3\mathbf{v} = i\,\hat{n}\times\big(i\,\hat{n}\times(i\,\hat{n}\times\mathbf{v})\big) = -i\,\hat{n}\times\big(\hat{n}(\hat{n}\cdot\mathbf{v})-\mathbf{v}\big) = i\,\hat{n}\times\mathbf{v} = B\mathbf{v}$, so

$$B^3 = B, \qquad B = \hat{n}\cdot\mathbf{J}. \tag{2.95}$$

Splitting the series with $B^{2k+1} = B$ and $B^{2k} = B^2$ for $k \geq 1$,

$$\begin{aligned} R = \exp(-i\theta B) &= \mathbf{1} + B\sum_{k=0}^{\infty}\frac{(-i\theta)^{2k+1}}{(2k+1)!} + B^2\sum_{k=1}^{\infty}\frac{(-i\theta)^{2k}}{(2k)!} \\ &= \mathbf{1} - i\sin\theta\,B + (\cos\theta - 1)\,B^2, \end{aligned} \tag{2.96}$$

where $\sum_{k\geq 0}(-i\theta)^{2k+1}/(2k+1)! = \sinh(-i\theta) = -i\sin\theta$ and $\sum_{k\geq 1}(-i\theta)^{2k}/(2k)! = \cosh(-i\theta) - 1 = \cos\theta - 1$. Substituting $(B)_{bc} = -i\epsilon^{abc}n^a$ and $(B^2)_{bc} = \delta_{bc} - n_b n_c$ (which follows from $B^2\mathbf{v} = i\hat{n}\times(i\hat{n}\times\mathbf{v}) = -\big(\hat{n}(\hat{n}\cdot\mathbf{v}) - \mathbf{v}\big) = \mathbf{v} - \hat{n}(\hat{n}\cdot\mathbf{v})$),

$$R_{bc} = \cos\theta\,\delta_{bc} + (1-\cos\theta)\,n_b n_c - \sin\theta\,\epsilon_{bca}\,n_a, \tag{2.97}$$

which is the Rodrigues rotation formula: R is the rotation by angle θ about the axis $\hat{n}$. (The cross-term sign comes from the $-i\sin\theta\,B$ term: $-i\sin\theta\,(-i\epsilon^{abc}n^a) = -\sin\theta\,\epsilon^{abc}n^a = -\sin\theta\,\epsilon_{bca}n_a$, using the cyclic identity $\epsilon^{abc} = \epsilon_{bca}$, matching Eq. (2.97).) Thus exp carries $\mathfrak{so}(3)$ onto the rotation group SO(3), with θ the literal rotation angle.

The factor of two. Comparing Eq. (2.93) and Eq. (2.97), the same axis–angle data $(\hat{n},\theta)$ produces an SU(2) element through the *half*-angle $\theta/2$ but an SO(3) rotation through the full angle θ. The consequence is decisive. A full rotation $\theta = 2\pi$ returns the spatial rotation to the identity, $R(\hat{n},2\pi) = \mathbf{1}$ from Eq. (2.97), but sends the SU(2) element to

$$U(\hat{n},2\pi) = \cos\pi\,\mathbf{1} - i\sin\pi\,\hat{n}\cdot\boldsymbol{\sigma} = -\mathbf{1} \neq \mathbf{1}, \tag{2.98}$$

and only at $\theta = 4\pi$ does U return to $\mathbf{1}$. Both $U = \pm\mathbf{1}$ map to the single rotation $R = \mathbf{1}$, so the assignment $U \mapsto R$ is two-to-one. The map $\mathrm{SU}(2) \to \mathrm{SO}(3)$ defined by Ad — conjugation of the Pauli vector $\hat{n}\cdot\boldsymbol{\sigma}$ by U realizes the rotation R via Theorem 2.45 — is the two-to-one covering homomorphism with kernel $\{\pm\mathbf{1}\}$. This is the algebraic origin of the topological double cover $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$ recorded in Theorem 3.10, and the reason half-integer spin exists: representations of $\mathfrak{su}(2)$ that are single-valued on SU(2) but double-valued on SO(3) are physically admissible because the symmetry group acting on quantum states is the cover SU(2), not SO(3) itself. We take up that fact, and build every irreducible representation of $\mathfrak{su}(2)$ from these generators, in the next section.

With the Lie algebra, its bracket and structure constants, the Jacobi identity, the adjoint representations ad and Ad, and the exponential map all in hand — and with $\mathfrak{su}(2)$ and $\mathfrak{so}(3)$ exponentiated in closed form — we have the complete machinery for turning a continuous symmetry into a finite linear-algebra problem. The next section spends that machinery on its most important single application: solving $\mathfrak{su}(2)$ completely by the highest-weight and ladder construction, reading off the dimensions and Casimir spectrum of its irreducible representations, and settling the SU(2)-over-SO(3) relationship just glimpsed in Eq. (2.98) into the explanation of half-integer spin.

2.4 The Worked Spine: $\mathfrak{su}(2)$, Its Representations, and SU(2) over SO(3)

The previous section gave us the general machinery: a matrix Lie group is captured near the identity by its Lie algebra (Theorem 2.39), the bracket and its structure constants (Eq. (2.70)) encode the group multiplication to leading nontrivial order, and the exponential map (Eq. (2.60)) climbs back from the algebra to the group. That machinery is abstract until it is run, in full, on a concrete algebra; and there is exactly one algebra on which every later representation-theoretic move in this book is first rehearsed. It is $\mathfrak{su}(2)$: the smallest nonabelian compact simple Lie algebra, the rank-one case from which the entire theory of roots and weights is the patient generalization. We now solve it completely and by hand. Every irreducible representation will be built from scratch by the highest-weight/ladder construction; its dimension and Casimir eigenvalue will be derived rather than quoted; the low-spin matrices will be written out explicitly; and the relationship between the group SU(2) and the rotation group SO(3)—the relationship that forces the existence of half-integer spin—will be settled. The reader has met the angular-momentum ladder in nonrelativistic quantum mechanics; here we recast it as honest representation theory, deriving as theorems what there was assumed as the spectrum of $\mathbf{J}^2$ and J_3.

2.4.1 The algebra, the Casimir, and the ladder operators

Definition 2.48 (The Lie algebra $\mathfrak{su}(2)$)**.** The real Lie algebra $\mathfrak{su}(2)$ is the three-dimensional algebra spanned by Hermitian generators J^1, J^2, J^3 with brackets

$$\left[J^i, J^j\right] = i\,\epsilon^{ijk}\,J^k, \qquad i,j,k \in \{1,2,3\}, \tag{2.99}$$

ϵ^{ijk} the totally antisymmetric symbol with $\epsilon^{123} = +1$, and a sum over the repeated index k understood.

The structure constants of $\mathfrak{su}(2)$ in this basis are therefore $f^{ijk} = \epsilon^{ijk}$, fully antisymmetric, in keeping with the general normalization $\left[T^a, T^b\right] = i f^{abc} T^c$ fixed in the chapter introduction and tabulated in Chapter B. We choose the generators Hermitian so that the group elements $\exp(-i\theta_a J^a)$ are unitary; this is the convention under which physical symmetry operators act on Hilbert space. A *representation* of $\mathfrak{su}(2)$ on a finite-dimensional complex vector space V is, as in Theorem 2.39, a set of three operators on V—we use the same letters J^i, the context fixing whether the abstract element or its image is meant—satisfying Eq. (2.99). To classify the representations is to find every way these brackets can be realized by operators on a finite-dimensional space, up to the change of basis that identifies equivalent representations.

The first move is to build the operator that organizes the whole problem. Although no single generator commutes with the others—that is the content of Eq. (2.99)—a particular quadratic combination does.

Definition 2.49 (Casimir operator)**.** The *quadratic Casimir* of $\mathfrak{su}(2)$ is

$$\mathbf{J}^2 \;\equiv\; (J^1)^2 + (J^2)^2 + (J^3)^2. \tag{2.100}$$

Proposition 2.50 (The Casimir is central)**.** $\left[\mathbf{J}^2, J^i\right] = 0$ *for every* i.

Proof. It suffices to check $i = 3$; the other two follow by relabelling. Expand the commutator term by term, using the Leibniz identity $[AB, C] = A\,[B, C] + [A, C]\,B$ and the algebra Eq. (2.99) in the form $\left[J^1, J^3\right] = -iJ^2$, $\left[J^2, J^3\right] = +iJ^1$, $\left[J^3, J^3\right] = 0$:

$$\left[(J^1)^2, J^3\right] = J^1\left[J^1, J^3\right] + \left[J^1, J^3\right]J^1 = J^1(-iJ^2) + (-iJ^2)J^1 = -i(J^1J^2 + J^2J^1), \tag{2.101}$$

$$\left[(J^2)^2, J^3\right] = J^2\left[J^2, J^3\right] + \left[J^2, J^3\right]J^2 = J^2(iJ^1) + (iJ^1)J^2 = +i(J^2J^1 + J^1J^2), \tag{2.102}$$

$$\left[(J^3)^2, J^3\right] = 0. \tag{2.103}$$

The first two lines are negatives of one another, so their sum vanishes, and adding the third leaves $\left[\mathbf{J}^2, J^3\right] = 0$. □

The centrality of $\mathbf{J}^2$ is what makes it the lever of the construction. On an *irreducible* representation, an operator commuting with every generator commutes with the whole image of the algebra; by Schur's lemma (Theorem 2.21) it must then be a scalar multiple of the identity. Thus on each irreducible V the Casimir takes a single fixed value, $\mathbf{J}^2 = \lambda\,\mathbf{1}$, and labelling the irreducibles by λ will turn out to label them completely.

To resolve the representation we diagonalize as much as we can simultaneously. The operators $\mathbf{J}^2$ and J^3 commute (Theorem 2.50), and J^3 is Hermitian, so we

may choose a basis of common eigenvectors. The remaining two generators are repackaged not as Hermitian operators but as a raising/lowering pair, exactly as the harmonic oscillator's x and p are repackaged into $a^\dagger$ and a.

Definition 2.51 (Ladder operators)**.**

$$J_\pm \equiv J^1 \pm i\, J^2. \tag{2.104}$$

Because J^1, J^2 are Hermitian, $J_\pm^\dagger = J^1 \mp i J^2 = J_\mp$: the raising and lowering operators are mutual adjoints. Their algebra with J^3 and with each other is the engine of everything that follows, so we compute it in full.

Lemma 2.52 (Ladder algebra)**.**

$$\left[J^3, J_\pm\right] = \pm\, J_\pm, \tag{2.105}$$

$$[J_+, J_-] = 2\, J^3, \tag{2.106}$$

$$\mathbf{J}^2 = J^3(J^3 \pm 1) + J_\mp J_\pm. \tag{2.107}$$

Proof. For Eq. (2.105), use Eq. (2.99):

$$\left[J^3, J_\pm\right] = \left[J^3, J^1\right] \pm i \left[J^3, J^2\right] = iJ^2 \pm i(-iJ^1) = iJ^2 \pm J^1 = \pm(J^1 \pm iJ^2) = \pm J_\pm. \tag{2.108}$$

For Eq. (2.106),

$$[J_+, J_-] = \left[J^1 + iJ^2, J^1 - iJ^2\right] = -i\left[J^1, J^2\right] + i\left[J^2, J^1\right] = -2i\left[J^1, J^2\right] = -2i\,(iJ^3) = 2J^3. \tag{2.109}$$

For Eq. (2.107), compute the product $J_\mp J_\pm$ directly:

$$J_\mp J_\pm = (J^1 \mp iJ^2)(J^1 \pm iJ^2) = (J^1)^2 + (J^2)^2 \pm i\left[J^1, J^2\right] = (J^1)^2 + (J^2)^2 \mp J^3, \tag{2.110}$$

where the cross terms combined into the commutator and $\left[J^1, J^2\right] = iJ^3$ was used. Adding and subtracting $(J^3)^2$,

$$J_\mp J_\pm = \mathbf{J}^2 - (J^3)^2 \mp J^3 = \mathbf{J}^2 - J^3(J^3 \pm 1), \tag{2.111}$$

which rearranges to Eq. (2.107). □

Equation (2.105) is the decisive relation: it says that $J_\pm$ shifts the eigenvalue of J^3 by exactly ± 1. If $|m\rangle$ is an eigenvector of J^3 with eigenvalue m, then

$$J^3\left(J_\pm |m\rangle\right) = \left(J_\pm J^3 + \left[J^3, J_\pm\right]\right)|m\rangle = (m \pm 1)\, J_\pm |m\rangle\,, \tag{2.112}$$

so $J_\pm |m\rangle$ is either zero or an eigenvector with eigenvalue $m \pm 1$. The eigenvalue of J^3 is traditionally called the *weight* m; Eq. (2.112) is the statement that J_+ raises and J_- lowers the weight in unit steps. This is the abstract content of the angular-momentum ladder, here proved from the bracket alone rather than assumed.

2.4.2 The highest-weight construction

We can now state and prove the central classification theorem of the chapter. Everything later—the Lorentz representations of Chapter 3, the isospin and color multiplets, the Clebsch–Gordan series of the next section—is an application or a generalization of it.

Theorem 2.53 (Irreducible representations of $\mathfrak{su}(2)$)**.** *For each value $j \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \dots\}$ there is, up to equivalence, exactly one irreducible representation $D^{(j)}$ of $\mathfrak{su}(2)$, the* spin-j *representation, and these exhaust the finite-dimensional irreducibles. It has the following properties.*

1. *Its dimension is $2j+1$.*
2. *The Casimir acts as the scalar $\mathbf{J}^2 = j(j+1)\,\mathbf{1}$.*
3. *There is an orthonormal basis $\{|j,m\rangle : m = -j, -j+1, \dots, j-1, j\}$ of simultaneous eigenvectors,*

$$\mathbf{J}^2\,|j,m\rangle = j(j+1)\,|j,m\rangle\,, \qquad J^3\,|j,m\rangle = m\;|j,m\rangle\,, \tag{2.113}$$

   with the weights m spaced by unit steps and symmetric about zero.
4. *The ladder operators act by*

$$J_\pm\,|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}\;\;|j,m\pm 1\rangle\,. \tag{2.114}$$

Proof. Let V be a finite-dimensional irreducible representation. By Theorem 2.50 and Schur's lemma (Theorem 2.21), $\mathbf{J}^2 = \lambda\,\mathbf{1}$ for a single number λ. Because J^3 is Hermitian on the finite-dimensional space V, it is diagonalizable with real eigenvalues; let the largest one be j (it exists, the spectrum being a finite set of reals), and let $|v\rangle$ be a corresponding unit eigenvector, $J^3\,|v\rangle = j\,|v\rangle$.

Step 1: the top of the ladder. Apply J_+. By Eq. (2.112), $J_+\,|v\rangle$ has weight $j+1$. But j was the largest weight, so no vector of weight $j+1$ exists; hence

$$J_+\,|v\rangle = 0. \tag{2.115}$$

We call $|v\rangle$ a *highest-weight vector*. Now evaluate the Casimir on it using the upper sign of Eq. (2.107):

$$\lambda\,|v\rangle = \mathbf{J}^2\,|v\rangle = \big(J^3(J^3+1) + J_-J_+\big)\,|v\rangle = \big(j(j+1) + 0\big)\,|v\rangle\,, \tag{2.116}$$

the last term vanishing by Eq. (2.115). Therefore

$$\lambda = j(j+1), \tag{2.117}$$

which fixes the Casimir eigenvalue in terms of the top weight and proves property (2) once we identify j as the spin label.

Step 2: descending the ladder. Define a chain of vectors by repeated lowering,

$$|v_0\rangle \equiv |v\rangle\,, \qquad |v_{n+1}\rangle \equiv J_-\,|v_n\rangle \quad (n = 0, 1, 2, \dots). \tag{2.118}$$

By Eq. (2.112), $|v_n\rangle$ (when nonzero) has weight $j - n$; distinct weights mean distinct eigenvectors of J^3, so the nonzero $|v_n\rangle$ are linearly independent. Since V is finite-dimensional, the chain must terminate: there is a smallest integer $N \geq 0$ with $|v_N\rangle \neq 0$ but $J_-\,|v_N\rangle = |v_{N+1}\rangle = 0$. The bottom vector $|v_N\rangle$ has weight $m_{\min} = j - N$ and is annihilated by J_-. Evaluate the Casimir on it, now with the lower sign of Eq. (2.107):

$$\lambda\,|v_N\rangle = \big(J^3(J^3 - 1) + J_+J_-\big)\,|v_N\rangle = m_{\min}(m_{\min} - 1)\,|v_N\rangle\,, \tag{2.119}$$

so $\lambda = m_{\min}(m_{\min} - 1)$. Combined with Eq. (2.117) this requires

$$m_{\min}(m_{\min} - 1) = j(j + 1). \tag{2.120}$$

This quadratic in $m_{\min}$ has the two roots $m_{\min} = -j$ and $m_{\min} = j + 1$. The second is excluded because $m_{\min} \leq j$, so

$$m_{\min} = -j. \tag{2.121}$$

Step 3: quantization of j and the dimension. The bottom weight is $j - N = -j$, hence $N = 2j$. But N is a nonnegative integer by construction, so

$$2j \in \{0, 1, 2, 3, \dots\}, \qquad \text{i.e.} \qquad j \in \left\{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \dots\right\}. \tag{2.122}$$

This is the entire reason spin is quantized in half-integer units, and it has come from nothing but the algebra Eq. (2.99) and the finiteness of the representation: the ladder must close on itself, and it can only do so after an integer number of unit steps that carry $+j$ to $-j$, a span of $2j$. The weights are then $m \in \{-j, -j+1, \dots, j\}$, a set of $N + 1 = 2j + 1$ values, and the corresponding vectors $|v_0\rangle, \dots, |v_N\rangle$ span a $(2j + 1)$-dimensional subspace $W \subseteq V$.

Step 4: W is the whole space, and the matrix elements. The subspace W is invariant under J^3 (it is spanned by J^3-eigenvectors) and under J_- (which maps each $|v_n\rangle$ to the next, or to zero). It is also invariant under J_+: applying J_+ to $|v_n\rangle$ gives a weight-$(j - n + 1)$ vector, which lies in W because W contains the unique-up-to-scale weight space at each weight. To see the uniqueness and pin the coefficients at once, normalize. Set $|j, j\rangle \equiv |v\rangle$ and define $|j, m\rangle$ as the unit-normalized lowering chain. The norm of $J_\pm\,|j, m\rangle$ follows from Eq. (2.107) and $J_\pm^\dagger = J_\mp$:

$$\big\|J_\pm\,|j, m\rangle\big\|^2 = \langle j, m|J_\mp J_\pm|j, m\rangle = \Big\langle j, m\Big|\mathbf{J}^2 - J^3(J^3 \pm 1)\Big|j, m\Big\rangle = j(j{+}1) - m(m{\pm}1). \tag{2.123}$$

The right-hand side is automatically ≥ 0 on the allowed weights, vanishing exactly at the ends $m = \pm j$ (consistent with $J_+ |j, j\rangle = 0$ and $J_- |j, -j\rangle = 0$), and choosing the phases so the square roots are positive gives Eq. (2.114), property (4). Because J_+ and J_- never leave W and J^3 acts diagonally on it, W is a nonzero invariant subspace; irreducibility of V forces $W = V$, so $\dim V = 2j + 1$, property (1), and the basis $\{|j, m\rangle\}$ is the one in property (3). Since Eqs. (2.113) and (2.114) determine the action of all generators completely in terms of j alone, any two irreducibles with the same top weight j are equivalent, and Eq. (2.122) shows every half-integer j is realized; this is uniqueness and exhaustiveness. □

Physical Insight

The construction never invoked Hilbert space, wavefunctions, or rotations—only three operators and one bracket. That is the lesson worth carrying forward: an irreducible representation is rigid. Once the highest weight is fixed, the dimension, the spectrum, the Casimir, and every matrix element are forced. The ladder is not a calculational trick borrowed from quantum mechanics; it is the structure of the algebra made visible. When we attack $\mathfrak{su}(3)$ and the higher-rank algebras later, the only thing that changes is that the single ladder becomes a lattice of ladders—one for each positive root—but the logic, "go up until you must stop, then descend and demand closure," is exactly the one proved here.

A picture helps fix the combinatorics of the weights. Each irreducible is a single string of equally spaced points on the m-axis, symmetric about the origin, with $J_\pm$ hopping along it; the half-integer strings sit on the half-integer lattice and the integer strings on the integer lattice.

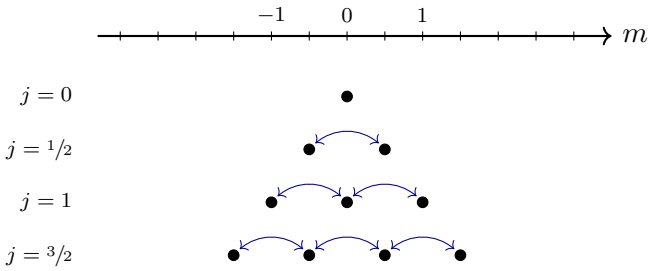


Figure 2.3: Weight diagrams of the first few $\mathfrak{su}(2)$ irreducibles. Each row is a single irreducible $D^{(j)}$; the dots are the weights $m \in \{-j, \ldots, j\}$, the arrows are the unit hops of $J_\pm$ from Eq. (2.114). Integer-j strings live on the integer lattice (and descend to genuine SO(3) representations), half-integer strings on the half-integer lattice (genuinely SU(2)). The string for $D^{(j)}$ has $2j + 1$ dots.

2.4.3 The fundamental and adjoint matrices

The theorem is a recipe; running it on the two smallest nontrivial spins produces the matrices that the rest of the book uses without further comment.

Spin $\frac{1}{2}$: the fundamental. For $j = \frac{1}{2}$ the space is two-dimensional, with weights $m = +\frac{1}{2}, -\frac{1}{2}$. In the ordered basis $\left(\left|\frac{1}{2}, \frac{1}{2}\right\rangle, \left|\frac{1}{2}, -\frac{1}{2}\right\rangle\right)$, Eq. (2.113) gives $J^3 = \frac{1}{2}\,\mathrm{diag}(1, -1)$, and Eq. (2.114) gives the single nonzero matrix element of each ladder operator: $J_+ \left|\frac{1}{2}, -\frac{1}{2}\right\rangle = \sqrt{\frac{3}{4} - (-\frac{1}{2})(\frac{1}{2})}\, \left|\frac{1}{2}, \frac{1}{2}\right\rangle = \left|\frac{1}{2}, \frac{1}{2}\right\rangle$ and $J_- \left|\frac{1}{2}, \frac{1}{2}\right\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$. Hence

$$J^3 = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad J_+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \tag{2.124}$$

Recombining $J^1 = \frac{1}{2}(J_+ + J_-)$ and $J^2 = \frac{1}{2i}(J_+ - J_-)$ reproduces the Pauli matrices:

$$J^i = \tfrac{1}{2}\,\sigma^i, \qquad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \; \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \; \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{2.125}$$

The Pauli matrices satisfy $\sigma^i\sigma^j = \delta^{ij}\mathbf{1} + i\epsilon^{ijk}\sigma^k$, from which $\left[\frac{1}{2}\sigma^i, \frac{1}{2}\sigma^j\right] = \frac{1}{4}\left[\sigma^i, \sigma^j\right] = \frac{1}{4}(2i\epsilon^{ijk}\sigma^k) = i\epsilon^{ijk}(\frac{1}{2}\sigma^k)$, confirming Eq. (2.99). This spin-$\frac{1}{2}$ representation is the *fundamental* (or *defining*) representation of $\mathfrak{su}(2)$; the generators carry the standard normalization

$$\mathrm{tr}(J^i J^j) = \tfrac{1}{4}\,\mathrm{tr}(\sigma^i\sigma^j) = \tfrac{1}{4}\cdot 2\delta^{ij} = \tfrac{1}{2}\,\delta^{ij}, \tag{2.126}$$

matching the convention $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ of the chapter introduction and Chapter B. This two-component object is the algebraic skeleton of the electron's spin; it is the representation QED will quantize when the electron enters as a Dirac field in Chapter 11, and it is one of the two $\mathfrak{su}(2)$ factors that Chapter 3 stacks to build the Lorentz group.

Spin 1: the adjoint. For $j = 1$ the space is three-dimensional with weights $m = 1, 0, -1$. The Casimir value is $j(j+1) = 2$. The ladder coefficients from Eq. (2.114) are $\sqrt{2 - m(m+1)}$: for $m = -1$ and $m = 0$ this is $\sqrt{2}$, and zero at the top. In the ordered basis $(|1,1\rangle, |1,0\rangle, |1,-1\rangle)$,

$$J^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}, \quad J_+ = \begin{pmatrix} 0 & \sqrt{2} & 0 \\ 0 & 0 & \sqrt{2} \\ 0 & 0 & 0 \end{pmatrix}, \quad J_- = \begin{pmatrix} 0 & 0 & 0 \\ \sqrt{2} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{pmatrix}. \tag{2.127}$$

The name "adjoint" is earned because $\dim D^{(1)} = 3$ equals the dimension of $\mathfrak{su}(2)$ itself: the spin-1 representation is the action of the algebra on itself by the

bracket, $\mathrm{ad}_X(Y) = [X, Y]$, the adjoint representation of Section 2.3. We verify this identification, and the consistency of the whole apparatus, in the worked example.

Worked Example 2.54 (The spin-1 representation, by hand)**.** We build the spin-1 generators in two bases, check the $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ brackets, and exponentiate to recover an ordinary rotation, closing the loop with the covering map of the next subsection.

(a) Cartesian (adjoint) form. The adjoint representation acts on the three-dimensional algebra; choosing the basis $\{J^1, J^2, J^3\}$ itself as the carrier space, the matrix of ad_{J^a} has entries

$$(\mathcal{J}^a)_{bc} = \text{coefficient of } J^b \text{ in } [J^a, J^c] = -i\,\epsilon^{abc}, \tag{2.128}$$

the last equality because $[J^a, J^c] = i\epsilon^{acd}J^d = i\epsilon^{acb}J^b = -i\epsilon^{abc}J^b$. Writing the three matrices out,

$$\mathcal{J}^1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \quad \mathcal{J}^2 = \begin{pmatrix} 0 & 0 & i \\ 0 & 0 & 0 \\ -i & 0 & 0 \end{pmatrix}, \quad \mathcal{J}^3 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{2.129}$$

These are Hermitian, and one checks the bracket directly. Multiplying the matrices Eq. (2.129) entry by entry,

$$\mathcal{J}^1\mathcal{J}^2 = \begin{pmatrix} 0 & 0 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \qquad \mathcal{J}^2\mathcal{J}^1 = \begin{pmatrix} 0 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \tag{2.130}$$

so that

$$\left[\mathcal{J}^1, \mathcal{J}^2\right] = \mathcal{J}^1\mathcal{J}^2 - \mathcal{J}^2\mathcal{J}^1 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = i\,\mathcal{J}^3, \tag{2.131}$$

the last equality because $i\,\mathcal{J}^3 = i\begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. Cyclic relabelling gives $[\mathcal{J}^2, \mathcal{J}^3] = i\mathcal{J}^1$ and $[\mathcal{J}^3, \mathcal{J}^1] = i\mathcal{J}^2$, so Eq. (2.129) satisfies Eq. (2.99). The same conclusion follows in index form without writing out matrices: with $(\mathcal{J}^a)_{ce} = -i\epsilon^{ace}$,

$$\left[\mathcal{J}^a, \mathcal{J}^b\right]_{ce} = -\sum_d (\epsilon^{acd}\epsilon^{bde} - \epsilon^{bcd}\epsilon^{ade}) = \delta^{ac}\delta^{be} - \delta^{ae}\delta^{bc} = i\,\epsilon^{abf}\,(\mathcal{J}^f)_{ce}, \tag{2.132}$$

where the middle equality used $\epsilon^{acd}\epsilon^{bde} = \epsilon^{dac}\epsilon^{deb} = \delta^{ae}\delta^{cb} - \delta^{ab}\delta^{ce}$ (and the same with $a \leftrightarrow b$), and the last used $i\epsilon^{abf}(\mathcal{J}^f)_{ce} = \epsilon^{abf}\epsilon^{fce} = \delta^{ac}\delta^{be} - \delta^{ae}\delta^{bc}$. This is the very Levi-Civita identity that will produce the $\mathfrak{so}(3)$ algebra in Chapter 3. The Casimir is likewise computed in index form,

$$(\boldsymbol{\mathcal{J}}^2)_{ce} = \sum_a (\mathcal{J}^a\mathcal{J}^a)_{ce} = -\sum_{a,d} \epsilon^{acd}\epsilon^{ade} = -\sum_d (\delta^{cd}\delta^{de} - 3\,\delta^{ce}) = -(\delta^{ce} - 3\,\delta^{ce}) = 2\,\delta^{ce}, \tag{2.133}$$

so $\mathcal{J}^2 = 2\,\mathbf{1}_3$, confirming $j(j+1) = 2$ for $j = 1$.

(b) Equivalence to the spherical form. The matrices Eq. (2.127) and Eq. (2.129) are the same representation in different bases: $\mathcal{J}^3$ is brought to the diagonal form $\mathrm{diag}(1, 0, -1)$ by the unitary change of basis to the eigenvectors $|1, \pm 1\rangle = \mp\frac{1}{\sqrt{2}}(\hat{e}_1 \pm i\hat{e}_2)$, $|1, 0\rangle = \hat{e}_3$ (the spherical basis). Both therefore have Casimir 2 and weights $\{1, 0, -1\}$, and by the uniqueness in Theorem 2.53 they must be unitarily equivalent, as the explicit basis change confirms.

(c) Exponentiation. A rotation by angle θ about the third axis is $R_3(\theta) = \exp(-i\theta\,\mathcal{J}^3)$. Since $\mathcal{J}^3$ restricted to the 1–2 block is $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right) = \sigma^2$ with $(\sigma^2)^2 = \mathbf{1}_2$, the series splits into even and odd powers,

$$\exp(-i\theta\,\mathcal{J}^3) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2.134}$$

the ordinary 3×3 rotation of a vector about the z-axis. This is the "section's exp computation" the spin-1 representation must reproduce: the adjoint of $\mathfrak{su}(2)$ exponentiates to the rotation group acting on three-vectors. Where the spin-$\frac{1}{2}$ matrix $\exp(-i\theta\sigma^3/2) = \mathrm{diag}(e^{-i\theta/2}, e^{+i\theta/2})$ needs $\theta = 4\pi$ to return to the identity, the spin-1 matrix Eq. (2.134) returns at $\theta = 2\pi$. The factor of two between these two periods is the entire subject of the next subsection.

2.4.4 SU(2) over SO(3): the double cover and half-integer spin

We have classified representations of the *algebra* $\mathfrak{su}(2)$. The physics, however, asks for representations of a *group*—of the actual rotations a particle can undergo—and there are two candidate groups sharing the algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$. One is SO(3), the rotations of three-dimensional space, 3×3 real orthogonal matrices of determinant one. The other is SU(2), the 2×2 complex unitary matrices of determinant one, in which the spin-$\frac{1}{2}$ matrices Eq. (2.125) naturally live. They are not the same group, and the precise way they differ is what determines which of our representations describe genuine particles and which the famous sign change under a 2π rotation.

Theorem 2.55 (The covering homomorphism)**.** *There is a surjective group homomorphism*

$$\varphi : \mathrm{SU}(2) \longrightarrow \mathrm{SO}(3) \tag{2.135}$$

whose kernel is $\{+\mathbf{1}, -\mathbf{1}\}$*. It is two-to-one:* $\varphi(U) = \varphi(-U)$ *for all* U*, and these are the only coincidences. Consequently* $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm\mathbf{1}\}$*, and* SU(2) *is the (simply connected) double cover of* SO(3)*.*

Proof. The carrier of the action. The key is to find a copy of $\mathbb{R}^3$ on which SU(2) acts. Take the real vector space of traceless Hermitian 2×2 matrices. It is

three-dimensional, with the Pauli matrices Eq. (2.125) as a basis, so every such matrix is

$$X = \mathbf{x}\cdot\boldsymbol{\sigma} = x_1\sigma^1 + x_2\sigma^2 + x_3\sigma^3 = \begin{pmatrix} x_3 & x_1 - ix_2 \\ x_1 + ix_2 & -x_3 \end{pmatrix}, \qquad \mathbf{x}\in\mathbb{R}^3. \tag{2.136}$$

The determinant encodes the Euclidean length:

$$\det X = -x_3^2 - (x_1 - ix_2)(x_1 + ix_2) = -(x_1^2 + x_2^2 + x_3^2) = -|\mathbf{x}|^2. \tag{2.137}$$

The action. For $U\in\mathrm{SU}(2)$ define $X' = UXU^\dagger$. The map $X\mapsto X'$ is linear in X, and it stays inside the same space: X' is Hermitian because $(UXU^\dagger)^\dagger = UX^\dagger U^\dagger = UXU^\dagger$, and traceless because $\mathrm{tr}(UXU^\dagger) = \mathrm{tr}(U^\dagger UX) = \mathrm{tr}\,X = 0$ by cyclicity. So $X' = \mathbf{x}'\cdot\boldsymbol{\sigma}$ for some $\mathbf{x}'\in\mathbb{R}^3$, and since the map is linear in $\mathbf{x}$ there is a real 3×3 matrix $R(U)$ with $\mathbf{x}' = R(U)\,\mathbf{x}$.

It is a rotation. The determinant is preserved:

$$\det X' = \det(UXU^\dagger) = \det U\,\det X\,\det U^\dagger = \det X, \tag{2.138}$$

using $\det U = 1$ and $\det U^\dagger = \overline{\det U} = 1$. By Eq. (2.137) this says $|\mathbf{x}'|^2 = |\mathbf{x}|^2$ for all $\mathbf{x}$, so $R(U)$ is a linear isometry of $\mathbb{R}^3$: $R(U)\in\mathrm{O}(3)$, i.e. $R(U)^{\mathsf{T}}R(U) = \mathbf{1}$. To rule out the reflections (determinant -1) we use continuity. The group $\mathrm{SU}(2)$ is connected—indeed, writing $U = a\,\mathbf{1} + i\mathbf{b}\cdot\boldsymbol{\sigma}$ with $a^2 + |\mathbf{b}|^2 = 1$ identifies it with the unit three-sphere S^3, which is connected (and simply connected). The map $U\mapsto\det R(U)$ is continuous, takes values in the discrete set $\{+1,-1\}$, and equals $+1$ at $U = \mathbf{1}$ (where $R = \mathbf{1}$); a continuous function from a connected space into $\{\pm1\}$ is constant, so $\det R(U) = +1$ for all U. Hence $R(U)\in\mathrm{SO}(3)$, and we set $\varphi(U)\equiv R(U)$.

It is a homomorphism. For $U_1, U_2\in\mathrm{SU}(2)$, the action composes as $X\mapsto U_1U_2\,X\,(U_1U_2)^\dagger = U_1(U_2XU_2^\dagger)U_1^\dagger$, which is the action of U_2 followed by that of U_1. Translating to the vectors, $R(U_1U_2) = R(U_1)R(U_2)$, i.e. $\varphi(U_1U_2) = \varphi(U_1)\varphi(U_2)$.

The kernel. Suppose $\varphi(U) = \mathbf{1}$, i.e. $UXU^\dagger = X$ for every traceless Hermitian X, equivalently $U\sigma^i = \sigma^i U$ for $i = 1,2,3$. An operator commuting with all three Pauli matrices commutes with the entire algebra they generate, which acts irreducibly on $\mathbb{C}^2$ (the spin-$\frac{1}{2}$ representation is irreducible, having no proper invariant subspace); by Schur's lemma (Theorem 2.21) $U = c\,\mathbf{1}$ for a scalar c. The constraints $U\in\mathrm{SU}(2)$ then force $\det(c\,\mathbf{1}) = c^2 = 1$, so $c = \pm1$. Both $+\mathbf{1}$ and $-\mathbf{1}$ lie in $\mathrm{SU}(2)$ and both clearly act trivially (the minus signs cancel in $UXU^\dagger$), so

$$\ker\varphi = \{+\mathbf{1},\,-\mathbf{1}\}. \tag{2.139}$$

Surjectivity and the two-to-one count. We exhibit a preimage of every rotation. Consider

$$U(\theta,\hat{\mathbf{n}}) = \exp\!\Big(-\tfrac{i}{2}\,\theta\,\hat{\mathbf{n}}\cdot\boldsymbol{\sigma}\Big) = \cos\tfrac{\theta}{2}\,\mathbf{1} - i\sin\tfrac{\theta}{2}\,\hat{\mathbf{n}}\cdot\boldsymbol{\sigma}, \tag{2.140}$$

the second equality from $(\hat{\mathbf{n}}\cdot\boldsymbol{\sigma})^2 = \mathbf{1}$ summing the even/odd series as in the worked example. Take $\hat{\mathbf{n}} = \hat{e}_3$, so $U = \mathrm{diag}(e^{-i\theta/2}, e^{+i\theta/2})$, and compute its action on the basis. Using $\sigma^1, \sigma^2, \sigma^3$ and $U\sigma U^\dagger$:

$$\begin{aligned} U\sigma^3 U^\dagger &= \sigma^3, \\ U\sigma^1 U^\dagger &= \cos\theta\,\sigma^1 + \sin\theta\,\sigma^2, \\ U\sigma^2 U^\dagger &= -\sin\theta\,\sigma^1 + \cos\theta\,\sigma^2, \end{aligned} \tag{2.141}$$

the middle two obtained from $e^{-i\theta\sigma^3/2}\sigma^1 e^{+i\theta\sigma^3/2}$ by expanding and using $\sigma^3\sigma^1 = i\sigma^2$, $\sigma^3\sigma^2 = -i\sigma^1$. Reading off the coordinates, $R(U)$ is precisely the rotation by angle θ about the third axis,

$$\varphi(U(\theta,\hat{e}_3)) = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}, \tag{2.142}$$

identical to the spin-1 exponential Eq. (2.134). Every rotation is a rotation about *some* axis through *some* angle, and by rotating the axis $\hat{\mathbf{n}}$ over the sphere and θ over $[0, 2\pi)$ the elements Eq. (2.140) cover all of SO(3); hence φ is surjective. Finally, given any $R \in \mathrm{SO}(3)$, the fiber $\varphi^{-1}(R)$ is a coset of the kernel Eq. (2.139), so it contains exactly two elements $\{U, -U\}$. The map is two-to-one and $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm\mathbf{1}\}$. □

The two-to-one structure has a sharp consequence visible already in Eq. (2.140): a rotation by $\theta = 2\pi$ about any axis is the identity of SO(3), but its SU(2) preimage is

$$U(2\pi, \hat{\mathbf{n}}) = \cos\pi\,\mathbf{1} - i\sin\pi\,\hat{\mathbf{n}}\cdot\boldsymbol{\sigma} = -\mathbf{1} \neq +\mathbf{1}. \tag{2.143}$$

Only at $\theta = 4\pi$ does one return to $+\mathbf{1}$. Topologically, $\mathrm{SU}(2) = S^3$ is simply connected while $\mathrm{SO}(3) = S^3/\{\pm\mathbf{1}\} = \mathbb{RP}^3$ has fundamental group $\mathbb{Z}_2$: the loop in SO(3) traced by a full 2π rotation cannot be contracted, but traversing it *twice* can. This is the algebraic and topological content of the "belt trick" familiar from demonstrations of spin.

Theorem 2.56 (Which spins are genuine SO(3) representations)**.** *A spin-j representation $D^{(j)}$ of $\mathfrak{su}(2)$ descends to a single-valued representation of* SO(3) *if and only if j is an integer. For half-integer j, $D^{(j)}$ is a genuine (single-valued) representation of* SU(2) *but only a double-valued (projective) representation of* SO(3)*.*

Proof. A representation of the algebra exponentiates to a single-valued representation of SO(3) exactly when it is trivial on the kernel Eq. (2.139) of φ—i.e. when $D^{(j)}(-\mathbf{1}) = +\mathbf{1}$—because SO(3) is SU(2) modulo that kernel, so a function on SO(3) is the same as a function on SU(2) constant on the cosets $\{U, -U\}$. From Eq. (2.143), the group element $-\mathbf{1} \in \mathrm{SU}(2)$ is the 2π rotation about (say)

the third axis, $-\mathbf{1} = U(2\pi, \hat{e}_3) = \exp(-2\pi i J^3)$ in any representation. On the basis $|j, m\rangle$ this acts diagonally,

$$D^{(j)}(-\mathbf{1})\,|j, m\rangle = e^{-2\pi i\, m}\,|j, m\rangle = e^{-2\pi i\, j}\;|j, m\rangle\,, \tag{2.144}$$

where the second equality holds because the weights m all differ from j by integers, so $e^{-2\pi i m} = e^{-2\pi i j}$ for every m in the multiplet (this is also why the answer is a single scalar, as Schur demands). Now $e^{-2\pi i j} = +1$ if $j \in \mathbb{Z}$ and $e^{-2\pi i j} = -1$ if $j \in \mathbb{Z} + \frac{1}{2}$. Thus for integer j, $D^{(j)}(-\mathbf{1}) = +\mathbf{1}$ and the representation is single-valued on SO(3); for half-integer j, $D^{(j)}(-\mathbf{1}) = -\mathbf{1}$, so the two preimages U and $-U$ of a given rotation are represented by matrices differing by an overall sign—the representation assigns *two* matrices to each rotation and is double-valued (a projective representation with the $\mathbb{Z}_2$ phase) on SO(3), while remaining perfectly single-valued on SU(2). □

Physical Insight

This theorem is the reason the electron is described by SU(2) and not SO(3). A half-integer spin state is multiplied by -1 under a 2π rotation, a transformation that does nothing to the spatial coordinates. Such a sign is invisible in any single measurement—$|\psi|^2$ is unchanged—but it is real and observable in interference, and it is mandatory once one accepts that the rotation group acting on a quantum system need only be represented projectively (states being rays, not vectors). Nature uses the full SU(2): the spin-$\frac{1}{2}$ fundamental Eq. (2.125) is a legitimate building block, and the price of admission is that SU(2), not SO(3), is the true rotation group of quantum mechanics. The integer-spin representations lose nothing by this—they are single-valued either way—but the half-integer ones exist *only* because we passed to the cover.

Caution

"Double-valued representation" is shorthand, not a contradiction. A representation, by definition, is single-valued; the honest statement is that the half-integer $D^{(j)}$ are bona fide single-valued representations of SU(2) that fail to factor through φ, hence define only *projective* representations of SO(3). The clean resolution—always take the symmetry group to be the simply connected cover and demand ordinary (linear) representations of it—is the policy this book follows everywhere, and it is exactly the policy that Chapter 3 will apply to the Lorentz group, whose simply connected cover $\mathrm{SL}(2)\mathbb{C}$ plays the role SU(2) plays here.

With $\mathfrak{su}(2)$ now solved end to end—every irreducible constructed, dimensioned, and matrixed, and the group-level subtlety of half-integer spin settled—we

hold a complete catalogue $\{D^{(j)}\}_{j\in\frac{1}{2}\mathbb{Z}_{\geq 0}}$ and, more valuable than the catalogue, the highest-weight method that produced it. Two roads lead onward. The first asks what happens when two such systems are combined: the tensor product $D^{(j_1)}\otimes D^{(j_2)}$ is reducible, and decomposing it into irreducibles is the Clebsch–Gordan problem taken up in Section 2.5, where the same ladder reappears as the bookkeeping of the combined weights. The second asks how the method generalizes when the algebra has rank greater than one: the single ladder becomes a system of ladders indexed by roots, and the integer $2j$ becomes a set of integers, the Dynkin labels. Both roads start from the rank-one spine built here.

2.5 Tensor Products, Clebsch–Gordan, and Invariants

The previous section solved $\mathfrak{su}(2)$ completely: every irreducible representation is a spin-j multiplet of dimension $2j+1$, built by the highest-weight construction and labelled by the eigenvalue $j(j+1)$ of the Casimir (Theorem 2.53). That catalogue answers the question *"how can one system carry the symmetry?"* The physics, however, almost never presents one system. Two electrons each carry spin $\frac{1}{2}$; a quark carries colour while an antiquark carries the conjugate colour; a field and its derivative carry Lorentz indices that must be tied together into a scalar Lagrangian. In every such case we must combine two carriers of the symmetry into one composite carrier and ask how the composite decomposes into the irreducible atoms we have just classified. That operation is the *tensor product* of representations, and its decomposition into irreducibles is the *Clebsch–Gordan problem*. This section sets up the tensor product in general, solves the Clebsch–Gordan problem for $\mathfrak{su}(2)$ in full, and then isolates the single most important piece of the answer for field theory: the *singlet*, the invariant atom whose existence is exactly the statement that a particular combination of fields is symmetry-invariant. Every gauge-invariant interaction term in this book is, at bottom, a singlet picked out of a tensor product.

2.5.1 The tensor product of two representations

Let a group G act on two finite-dimensional complex vector spaces V_1 and V_2 through representations $\rho_1\colon G\to \mathrm{GL}(V_1)$ and $\rho_2\colon G\to\mathrm{GL}(V_2)$. The state space of the composite system is the tensor product $V_1\otimes V_2$, the vector space spanned by formal products $v_1\otimes v_2$ with $v_1\in V_1$, $v_2\in V_2$, bilinear in its two slots. If $\{e_i\}_{i=1}^{n_1}$ is a basis of V_1 and $\{f_a\}_{a=1}^{n_2}$ a basis of V_2, then $\{e_i\otimes f_a\}$ is a basis of $V_1\otimes V_2$, which therefore has dimension n_1n_2. The physical content is the familiar rule of quantum mechanics: the joint states of two independent systems are spanned by products of their individual states.

Definition 2.57 (Tensor product representation)**.** The *tensor product representation* $\rho_1 \otimes \rho_2$ of G on $V_1 \otimes V_2$ is defined on product vectors by

$$(\rho_1 \otimes \rho_2)(g)\,(v_1 \otimes v_2) \;=\; (\rho_1(g)\,v_1) \otimes (\rho_2(g)\,v_2), \tag{2.145}$$

and extended by linearity to all of $V_1 \otimes V_2$. The *same* group element g acts on both factors simultaneously.

That Eq. (2.145) is a representation is immediate from the homomorphism property of ρ_1 and ρ_2: for $g, h \in G$,

$$\begin{aligned}(\rho_1 \otimes \rho_2)(g)\,(\rho_1 \otimes \rho_2)(h)\,(v_1 \otimes v_2) &= (\rho_1(g)\rho_1(h)v_1) \otimes (\rho_2(g)\rho_2(h)v_2) \\ &= (\rho_1(gh)v_1) \otimes (\rho_2(gh)v_2) = (\rho_1 \otimes \rho_2)(gh)\,(v_1 \otimes v_2).\end{aligned} \tag{2.146}$$

In a product basis the matrix of $(\rho_1 \otimes \rho_2)(g)$ is the Kronecker product of the two factor matrices, $\big[(\rho_1 \otimes \rho_2)(g)\big]_{ia,jb} = \rho_1(g)_{ij}\,\rho_2(g)_{ab}$.

For a continuous symmetry the more useful object is the action of the Lie algebra, obtained by differentiating Eq. (2.145) along a one-parameter subgroup. Recall from the previous sections that a generator X of the algebra acts on a representation V through $\rho(e^{tX}) = \exp(t\,\rho_*(X))$, with $\rho_*(X) = \frac{d}{dt}\rho(e^{tX})\big|_{t=0}$. Apply this to the product. Writing $\rho_{1*}(X) \equiv X^{(1)}$ and $\rho_{2*}(X) \equiv X^{(2)}$ for the generators on the two factors, and differentiating the product $g(t) = e^{tX}$ acting through Eq. (2.145),

$$\frac{d}{dt}\Big[(e^{tX^{(1)}}v_1) \otimes (e^{tX^{(2)}}v_2)\Big]_{t=0} = (X^{(1)}v_1) \otimes v_2 \;+\; v_1 \otimes (X^{(2)}v_2), \tag{2.147}$$

by the Leibniz rule for the derivative of a product. Hence the generator on the tensor product is the *co-product*

$$(\rho_1 \otimes \rho_2)_*(X) \;=\; X^{(1)} \otimes \mathbf{1}_2 \;+\; \mathbf{1}_1 \otimes X^{(2)}, \tag{2.148}$$

where $\mathbf{1}_1, \mathbf{1}_2$ are the identities on V_1, V_2. This is the rule "angular momenta add": for $\mathfrak{su}(2)$, the total generator is $J^a = J^a_{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes J^a_{(2)}$, the familiar $\mathbf{J} = \mathbf{J}_{(1)} + \mathbf{J}_{(2)}$ of the addition of angular momentum.

That Eq. (2.148) again satisfies the Lie bracket is a short check that exhibits why the cross terms cancel. Suppressing the identity factors and writing $X^{(1)}$ for $X^{(1)} \otimes \mathbf{1}$ (which commutes with every $Y^{(2)} = \mathbf{1} \otimes Y^{(2)}$ because they act on different factors),

$$\begin{aligned}\Big[X^{(1)} + X^{(2)}, Y^{(1)} + Y^{(2)}\Big] &= \Big[X^{(1)}, Y^{(1)}\Big] + \Big[X^{(2)}, Y^{(2)}\Big] + \Big[X^{(1)}, Y^{(2)}\Big] + \Big[X^{(2)}, Y^{(1)}\Big] \\ &= [X, Y]^{(1)} + [X, Y]^{(2)}\,,\end{aligned} \tag{2.149}$$

since the two mixed commutators vanish, the operators acting on independent factors. The co-product therefore defines a genuine representation of the algebra, with the same structure constants.

Physical Insight

The co-product is the algebraic heart of "conservation of total charge." Because $X^{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes X^{(2)}$ generates the symmetry of the *whole* system, it is the additive total quantum number—total momentum, total spin, total isospin, total colour. Two particles that individually carry charges q_1 and q_2 carry total charge $q_1 + q_2$ for exactly the reason Eq. (2.148) is a sum. The Clebsch–Gordan problem below is the question of which total charges the composite can have.

The new feature, absent for a single system, is that a tensor product of two *irreducible* representations is generally *reducible*. The Kronecker product matrices in Eq. (2.145) do not in general act irreducibly on $V_1 \otimes V_2$; there are invariant subspaces. By complete reducibility for compact groups (Theorem 2.18) the product splits into a direct sum of irreducibles,

$$V_1 \otimes V_2 \;\cong\; \bigoplus_{\lambda} m_\lambda \, V_\lambda, \tag{2.150}$$

where V_λ runs over the irreducibles of G and the non-negative integer m_λ is the multiplicity with which V_λ appears. Computing the multiplicities m_λ *is* the Clebsch–Gordan decomposition. The change of basis that realizes Eq. (2.150) concretely—from the product basis $\{e_i \otimes f_a\}$ to a basis adapted to the irreducible blocks—is recorded in the Clebsch–Gordan coefficients of Section 2.5.3.

2.5.2 Weights add: the $\mathfrak{su}(2)$ Clebsch–Gordan series

For $\mathfrak{su}(2)$ the multiplicities in Eq. (2.150) are determined by a single elementary fact, which is the engine of all of representation theory beyond rank one: *under the tensor product, weights add.* The eigenvalue of J^3 (the weight) on a product state is the sum of the two individual eigenvalues, because J^3 acts through the co-product Eq. (2.148):

$$J^3 \left(|j_1\, m_1\rangle \otimes |j_2\, m_2\rangle \right) = (m_1 + m_2) \left(|j_1\, m_1\rangle \otimes |j_2\, m_2\rangle \right). \tag{2.151}$$

The product basis is therefore already a weight basis, with the product state $|j_1 m_1\rangle \otimes |j_2 m_2\rangle$ carrying weight $m = m_1 + m_2$. We can read off the decomposition by counting how many product states sit at each weight and matching that profile against the known weight content of the irreducibles—each spin-j multiplet contributes exactly one state at each weight $m \in \{-j, -j+1, \ldots, j\}$.

Theorem 2.58 ($\mathfrak{su}(2)$ Clebsch–Gordan series)**.** *For non-negative half-integers j_1, j_2, the tensor product of the spin-j_1 and spin-j_2 irreducibles of $\mathfrak{su}(2)$ decomposes, with unit multiplicity, as*

$$j_1 \otimes j_2 \;=\; \bigoplus_{j=|j_1-j_2|}^{j_1+j_2} j \;=\; (j_1+j_2) \oplus (j_1+j_2-1) \oplus \cdots \oplus |j_1-j_2|, \tag{2.152}$$

the spin j running in integer steps from $|j_1 - j_2|$ to $j_1 + j_2$. Each value of j in this range occurs exactly once; no value outside it occurs.

Proof. Assume without loss of generality $j_1 \geq j_2$. Let $N(m)$ be the number of product basis states $|j_1 m_1\rangle \otimes |j_2 m_2\rangle$ of total weight $m = m_1 + m_2$. Because m_1 ranges over $\{-j_1, \ldots, j_1\}$ and m_2 over $\{-j_2, \ldots, j_2\}$, the count $N(m)$ is the number of ways to write $m = m_1 + m_2$ with both constraints satisfied. The weight content is symmetric, $N(m) = N(-m)$, so it suffices to track $m \geq 0$. The maximum weight is $m = j_1 + j_2$, attained only by $m_1 = j_1, m_2 = j_2$, so $N(j_1 + j_2) = 1$. As m decreases by one, one more choice of (m_1, m_2) becomes available at each step—m_2 can be lowered while m_1 is raised to compensate, until m_2 hits its floor $-j_2$. Concretely, for m in the upper range,

$$N(m) = j_1 + j_2 - m + 1, \qquad j_1 - j_2 \leq m \leq j_1 + j_2, \tag{2.153}$$

which counts the admissible $m_2 \in \{m - j_1, \ldots, j_2\}$, i.e. $j_2 - (m - j_1) + 1$ values. Once m drops below $j_1 - j_2$, the constraint from j_2 saturates and the count plateaus at its maximum,

$$N(m) = 2j_2 + 1, \qquad -(j_1 - j_2) \leq m \leq j_1 - j_2, \tag{2.154}$$

the full width of the smaller multiplet. For $m > j_1 + j_2$ there are no states, $N(m) = 0$. The profile $N(m)$ is thus a symmetric trapezoid: it climbs linearly from 1 at the edge to $2j_2 + 1$ on the plateau, holds, and descends.

Now decompose this profile into irreducibles. Let n_j be the number of spin-j multiplets in the product. Since a spin-j multiplet contributes exactly one state to every weight m with $|m| \leq j$ and none with $|m| > j$, the count at weight $m \geq 0$ is the number of multiplets large enough to reach it,

$$N(m) = \sum_{j \geq m} n_j. \tag{2.155}$$

Inverting this telescoping sum gives the multiplicities directly,

$$n_j = N(j) - N(j+1). \tag{2.156}$$

On the descending edge $j_1 - j_2 \leq j \leq j_1 + j_2$ we use Eq. (2.153):

$$n_j = [(j_1 + j_2 - j + 1)] - [(j_1 + j_2 - (j+1) + 1)] = 1. \tag{2.157}$$

On the plateau $j < j_1 - j_2$ both $N(j)$ and $N(j+1)$ equal $2j_2 + 1$, so $n_j = 0$; and for $j > j_1 + j_2$ both counts vanish, so $n_j = 0$. Hence $n_j = 1$ precisely for $|j_1 - j_2| \leq j \leq j_1 + j_2$ and zero otherwise, which is Eq. (2.152). □

The decomposition must conserve dimension, and the check is a clean test of the arithmetic. The product space has dimension $(2j_1+1)(2j_2+1)$; the right-hand side of Eq. (2.152) has dimension

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1). \tag{2.158}$$

This is an arithmetic series with $(j_1+j_2)-|j_1-j_2|+1 = 2j_2+1$ terms (taking $j_1 \geq j_2$), first term $2|j_1-j_2|+1 = 2(j_1-j_2)+1$ and last term $2(j_1+j_2)+1$. The sum of an arithmetic series is the number of terms times the average of first and last,

$$\sum_{j=|j_1-j_2|}^{j_1+j_2} (2j+1) = (2j_2+1)\cdot\frac{[2(j_1-j_2)+1]+[2(j_1+j_2)+1]}{2} = (2j_2+1)(2j_1+1), \tag{2.159}$$

matching $\dim(V_{j_1}\otimes V_{j_2})$ exactly. Dimension is conserved.

The same series, read through characters

Theorem 2.58 can also be obtained from the character orthogonality of the earlier section (Theorem 2.27), and the two viewpoints illuminate each other. The character of the spin-j representation evaluated on a rotation by angle θ is the weight-generating polynomial

$$\chi_j(\theta) = \sum_{m=-j}^{j} e^{im\theta} = \frac{\sin((2j+1)\theta/2)}{\sin(\theta/2)}. \tag{2.160}$$

Because the character of a tensor product is the product of characters ($\mathrm{tr}(A\otimes B) = \mathrm{tr}\,A\ \mathrm{tr}\,B$ applied to Eq. (2.145)), the Clebsch–Gordan series is the statement that the product of two such Dirichlet kernels is the sum $\chi_{j_1}\chi_{j_2} = \sum_j \chi_j$ over the range Eq. (2.152). Multiplying the finite geometric sums, $\chi_{j_1}\chi_{j_2} = \sum_{m_1,m_2} e^{i(m_1+m_2)\theta}$, and regrouping the exponentials by total weight reproduces exactly the trapezoidal profile $N(m)$ of the proof—weight counting and character multiplication are the same computation in two notations.

2.5.3 Clebsch–Gordan coefficients

Theorem 2.58 tells us *which* total spins appear; the Clebsch–Gordan coefficients tell us *how* the coupled states are built from the product states. They are the entries of the unitary change of basis between the two natural bases of $V_{j_1}\otimes V_{j_2}$: the *product (uncoupled) basis* $\{|j_1m_1\rangle\otimes|j_2m_2\rangle\}$, in which $J^3_{(1)}$ and

$J^3_{(2)}$ are separately diagonal, and the *coupled basis* $\{|(j_1 j_2)\, j\, m\rangle\}$, in which the total Casimir $\mathbf{J}^2$ and J^3 are diagonal.

Definition 2.59 (Clebsch–Gordan coefficients)**.** The *Clebsch–Gordan coefficients* are the overlaps

$$\langle j_1\, m_1;\, j_2\, m_2 \,|\, j\, m\rangle \;\equiv\; \big(\langle j_1 m_1| \otimes \langle j_2 m_2|\big)\; |(j_1 j_2)\, j\, m\rangle\,, \tag{2.161}$$

defined so that the coupled states expand in the product basis as

$$|(j_1 j_2)\, j\, m\rangle \;=\; \sum_{m_1, m_2} \langle j_1\, m_1;\, j_2\, m_2 \,|\, j\, m\rangle \;\; |j_1 m_1\rangle \otimes |j_2 m_2\rangle\,. \tag{2.162}$$

Three structural facts fix these numbers almost completely. First, the *selection rule*: because weights add (Eq. (2.151)), the overlap vanishes unless $m = m_1 + m_2$. Second, the triangle condition $|j_1 - j_2| \le j \le j_1 + j_2$ of Theorem 2.58, outside which $|(j_1 j_2) j m\rangle$ does not exist. Third, *unitarity*: both bases are orthonormal, so the coefficients form a real orthogonal matrix once a phase convention is fixed, and they satisfy the completeness and orthogonality relations

$$\sum_{m_1, m_2} \langle j_1 m_1; j_2 m_2 | j\, m\rangle \, \langle j_1 m_1; j_2 m_2 | j'\, m'\rangle = \delta_{j j'}\, \delta_{m m'}, \tag{2.163}$$

$$\sum_{j, m} \langle j_1 m_1; j_2 m_2 | j\, m\rangle \, \langle j_1 m_1'; j_2 m_2' | j\, m\rangle = \delta_{m_1 m_1'}\, \delta_{m_2 m_2'}. \tag{2.164}$$

The remaining freedom is the phase of each coupled multiplet, fixed once and for all by the *Condon–Shortley convention*: the highest-weight overlap $\langle j_1\, j_1;\, j_2\, (j - j_1) \,|\, j\, j\rangle$ is taken real and positive. With that convention all Clebsch–Gordan coefficients are real, and the matrix is genuinely orthogonal.

The constructive route to the coefficients is the ladder algorithm, the direct extension of the highest-weight method of the previous section to the product space. It runs as follows. The unique product state of maximal weight, $|j_1 j_1\rangle \otimes |j_2 j_2\rangle$, must be the top of the largest multiplet, so $|(j_1 j_2)\,(j_1{+}j_2)\,(j_1{+}j_2)\rangle = |j_1 j_1\rangle \otimes |j_2 j_2\rangle$. Applying the total lowering operator $J^- = J^-_{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes J^-_{(2)}$ repeatedly generates the whole spin-$(j_1{+}j_2)$ tower, using the action established in the previous section (Eq. (2.114)),

$$J^{\mp}\, |j\, m\rangle = \sqrt{(j \pm m)(j \mp m + 1)}\;\; |j\, m \mp 1\rangle\,. \tag{2.165}$$

At weight $m = j_1 + j_2 - 1$ the product space is two-dimensional; one combination is the descendant just computed, and the orthogonal combination (with Condon–Shortley sign) is the highest-weight state of the spin-$(j_1{+}j_2{-}1)$ multiplet. Lowering that generates its tower; proceeding down through each weight, peeling off one new highest-weight state at each level, produces every coupled state and hence every coefficient. The next subsection's worked example carries this algorithm out in complete detail for the smallest nontrivial case.

2.5.4 Invariant tensors and the singlet

Among the irreducibles, one is singled out for field theory: the trivial, one-dimensional representation $j = 0$, on which every group element acts as the identity. A vector in a copy of this representation inside some tensor product is called a *singlet*, and it is precisely an *invariant*: it is annihilated by every generator and therefore unchanged by every group transformation. The hunt for singlets inside tensor products is, almost verbatim, the hunt for symmetry-invariant combinations of fields.

Definition 2.60 (Singlet and invariant tensor)**.** A vector $|s\rangle \in V_1 \otimes \cdots \otimes V_k$ is a *singlet* (an invariant) if the total generator annihilates it, $(\rho_1 \otimes \cdots \otimes \rho_k)_*(X)\ |s\rangle = 0$ for every X in the algebra; equivalently $(\rho_1 \otimes \cdots \otimes \rho_k)(g)\,|s\rangle = |s\rangle$ for all $g \in G$. The component array of such a state, with one index for each factor, is an *invariant tensor*: a collection of numbers $T_{ia\ldots}$ unchanged when each index is rotated by the corresponding representation matrix.

The first question is when a singlet exists at all. For a product of two irreducibles the answer is immediate from the Clebsch–Gordan series and Schur's lemma.

Theorem 2.61 (The singlet in $j_1 \otimes j_2$)**.** *The tensor product $j_1 \otimes j_2$ of two $\mathfrak{su}(2)$ irreducibles contains a singlet if and only if $j_1 = j_2$, and then exactly one. Equivalently, an invariant bilinear pairing between V_{j_1} and V_{j_2} exists if and only if the two spins are equal, and is unique up to scale.*

Proof. By Theorem 2.58 the trivial representation $j = 0$ appears in $j_1 \otimes j_2$ iff 0 lies in the range $|j_1 - j_2| \le 0 \le j_1 + j_2$, i.e. iff $|j_1 - j_2| = 0$, i.e. $j_1 = j_2$; and the series has unit multiplicity, so it appears exactly once. The equivalence with the existence of an invariant pairing is Schur's lemma (Theorem 2.21): an invariant bilinear form on $V_{j_1} \otimes V_{j_2}$ is an intertwiner $V_{j_1} \to V_{j_2}^*$, and $V_{j_2}^*$ carries the representation conjugate to j_2, which for $\mathfrak{su}(2)$ is again spin j_2 (Theorem 2.63 below). A nonzero intertwiner between irreducibles exists iff they are isomorphic, i.e. $j_1 = j_2$, and is then unique up to a scalar. □

The fundamental case $j = \frac{1}{2}$ produces the invariant tensor that recurs throughout the book. The singlet in $\frac{1}{2} \otimes \frac{1}{2}$ is the antisymmetric combination (computed explicitly in Theorem 2.64 below), and the array of its coefficients is the two-index Levi–Civita symbol

$$\epsilon_{ab} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}_{ab}, \qquad \epsilon_{12} = -\epsilon_{21} = 1, \tag{2.166}$$

with $a, b \in \{1, 2\}$ labelling the two components of the doublet. That ϵ is an invariant tensor of SU(2) is the statement that contracting two doublets with it,

$\epsilon_{ab}\,\psi^a\chi^b = \psi^1\chi^2 - \psi^2\chi^1$, yields an SU(2)-invariant. The proof is one line at the group level: for $U \in \mathrm{SU}(2)$,

$$\epsilon_{ab}\,U^a{}_c\,U^b{}_d = (\det U)\,\epsilon_{cd} = \epsilon_{cd}, \tag{2.167}$$

because contracting the antisymmetric symbol with two columns of a matrix returns its determinant, and $\det U = 1$ for $U \in \mathrm{SU}(2)$. Infinitesimally, writing $U = \mathbf{1} + i\omega_a T^a + O(\omega^2)$ with $T^a = \sigma^a/2$, the invariance Eq. (2.167) becomes the generator condition

$$(T^a)^{\top}\epsilon + \epsilon\,T^a = 0 \quad\Longleftrightarrow\quad \epsilon\,\sigma^a\,\epsilon^{-1} = -(\sigma^a)^{\top}, \tag{2.168}$$

the familiar identity that $i\sigma^2 = \epsilon$ conjugates each Pauli matrix into minus its transpose, which one verifies directly on $\sigma^1, \sigma^2, \sigma^3$. This single tensor is the seed of SU(2) invariants: in the electroweak sector the Yukawa coupling of the Higgs doublet to a lepton doublet is written $\epsilon_{ab}H^aL^b$, and the antisymmetric contraction is exactly what makes the term gauge invariant.

Singlets are the gauge-invariant Lagrangian terms

This is the structural payoff of the section, and it underlies every interaction in QED and Yang–Mills theory. A term in a Lagrangian is admissible only if it is invariant under the symmetry group—a scalar, hence a singlet in the tensor product of the representations carried by the fields it contains. Building an invariant Lagrangian is therefore the combinatorial problem of finding all singlets in products of the field representations. For a Dirac mass term the relevant product is (fundamental) $\otimes$ (antifundamental), whose singlet is the contraction $\bar\psi\psi$; for the non-abelian field strength it is the singlet in the adjoint $\otimes$ adjoint product, realized by the Killing form δ^{ab} and giving $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$. The invariant tensors δ^{ab}, ϵ_{ab}, f^{abc}, and d^{abc} (Chapter B) are nothing but the realizations of singlets in the products that recur, and writing down a gauge theory amounts to contracting indices with them until everything is a singlet.

It is often useful to refine the decomposition by permutation symmetry. When the two factors are copies of the *same* representation, $V \otimes V$, the exchange of the two factors is itself a symmetry that commutes with the group action, so the product splits into its symmetric and antisymmetric parts,

$$V \otimes V \;=\; \mathrm{Sym}^2 V \;\oplus\; \Lambda^2 V, \tag{2.169}$$

of dimensions $\frac{1}{2}n(n+1)$ and $\frac{1}{2}n(n-1)$ for $n = \dim V$, and each summand is itself a subrepresentation. For $\mathfrak{su}(2)$ the spins in $j \otimes j$ alternate in exchange parity, the top spin $2j$ being symmetric and the parities alternating down the series; the singlet ($j' = 0$) therefore sits in the symmetric part when $2j$ is

even (integer j) and in the antisymmetric part when $2j$ is odd (half-integer j). For the doublet $j = \frac{1}{2}$ this reproduces what we found above: the singlet is *antisymmetric*, realized by ϵ_{ab}, while the triplet is the symmetric part. This symmetric/antisymmetric distinction of the invariant pairing is exactly the data that classifies a representation as real or pseudoreal, to which we turn next.

2.5.5 Conjugate representations and the real/pseudoreal/complex trichotomy

The proof of Theorem 2.61 invoked the conjugate representation, the last structural ingredient and the one that will distinguish a quark from an antiquark in the next section. Given a representation ρ, complex-conjugating the matrices produces another set of matrices acting on the conjugate space.

Definition 2.62 (Conjugate representation)**.** The *conjugate* (or complex-conjugate) representation $\bar{\rho}$ of ρ acts by $\bar{\rho}(g) = \rho(g)^*$, with Lie-algebra generators

$$\bar{T}^a = -\,(T^a)^*. \tag{2.170}$$

A representation carrying $\bar{\rho}$ on a copy $\bar{V}$ of the underlying space is denoted $\bar{V}$, and its weights are minus those of V.

The sign in Eq. (2.170) is forced by the convention $\rho(e^{-i\omega_a T^a}) = e^{-i\omega_a T^a}$ in the representation: conjugating $e^{-i\omega_a T^a}$ gives $e^{i\omega_a (T^a)^*} = e^{-i\omega_a (-(T^a)^*)}$, so the conjugate generator is $-(T^a)^*$. That $\bar{T}^a$ obeys the same algebra follows by conjugating the bracket: with real structure constants f^{abc},

$$\left[\bar{T}^a, \bar{T}^b\right] = \left[(T^a)^*, (T^b)^*\right] = \left(\left[T^a, T^b\right]\right)^* = \left(i f^{abc} T^c\right)^* = -i f^{abc} (T^c)^* = i f^{abc}\, \bar{T}^c, \tag{2.171}$$

so $\bar{\rho}$ is a bona fide representation with the same structure constants. Its J^3 weights are negated because $\bar{J}^3 = -(J^3)^* = -J^3$ (the Cartan generator is real and diagonal), which flips every weight; this is precisely why antiparticles carry opposite additive charges.

The decisive question is whether $\bar{\rho}$ is a *new* representation or merely ρ in disguise. There are exactly three possibilities, distinguished by whether, and how, an invariant bilinear form exists on V.

Theorem 2.63 (Real/pseudoreal/complex trichotomy)**.** *Let ρ be an irreducible representation of a compact group on a complex space V. Exactly one of the following holds.*

1. ***Complex:*** *ρ is not isomorphic to $\bar{\rho}$. There is no invariant bilinear form on V.*

2. ***Real (orthogonal):*** *$\rho \cong \bar{\rho}$ through an intertwiner that corresponds to a* symmetric *invariant bilinear form on V. The matrices can be brought to real form in a suitable basis.*

3. ***Pseudoreal (symplectic):*** *$\rho \cong \bar{\rho}$ through an intertwiner that corresponds to an* antisymmetric *invariant bilinear form on V. The matrices cannot be made real, but ρ and $\bar{\rho}$ are nonetheless isomorphic.*

Proof. A representation is isomorphic to its conjugate iff there is an invertible intertwiner C with $C\,\rho(g)\,C^{-1} = \rho(g)^* = \bar{\rho}(g)$ for all g, i.e. $C\,T^a = \bar{T}^a\,C = -(T^a)^*C$ at the algebra level. Such a C is the same data as a nondegenerate invariant bilinear form $B(u,v) = u^\top C\,v$ on V: invariance $B(\rho(g)u, \rho(g)v) = B(u,v)$ is exactly $\rho(g)^\top C\,\rho(g) = C$, which on differentiating reproduces the intertwining condition. If no such C exists, ρ is complex (case 1).

Suppose C exists, so V carries an invariant nondegenerate bilinear form B. Its transpose $B^\top$ is also an invariant bilinear form. By Schur's lemma (Theorem 2.21) the space of invariant bilinear forms on an irreducible V is at most one-dimensional—an invariant form is an intertwiner $V \to V^*$, unique up to scale when nonzero. Hence $B^\top = c\,B$ for some scalar c. Transposing once more, $B = (B^\top)^\top = c\,B^\top = c^2 B$, so $c^2 = 1$ and $c = \pm 1$. Thus B is either symmetric ($c = +1$, case 2) or antisymmetric ($c = -1$, case 3), and the two cases are mutually exclusive because the form is unique up to scale. A symmetric nondegenerate invariant form is, after a basis change, the standard inner product preserved by an orthogonal group, and the matrices are real in that basis; an antisymmetric one is a symplectic form, and no real basis exists because reality would force a symmetric invariant inner product. This exhausts the possibilities. □

The discriminator is computed by a single class function, the *Frobenius–Schur indicator*, which extracts c without finding C. Averaging the character over the group at squared arguments,

$$\nu(\rho) = \frac{1}{|G|}\sum_{g\in G} \chi_\rho(g^2) \quad (\text{sum} \to \text{Haar integral for a Lie group}), \tag{2.172}$$

returns $\nu = +1$ for real, $\nu = -1$ for pseudoreal, and $\nu = 0$ for complex representations; the value is exactly the scalar c of the proof when a form exists, and zero when none does.

For $\mathfrak{su}(2)$ the verdict is uniform and important: *every $\mathfrak{su}(2)$ irreducible is self-conjugate*, and the conjugating intertwiner is the rotation by π about the 2-axis,

$$C = e^{i\pi J^2}, \tag{2.173}$$

which sends $J^3 \to -J^3$ and so maps the spin-j weight ladder onto the reversed ladder of $\bar{\rho}$. Concretely on the doublet, $J^2 = \sigma^2/2$ gives $C = e^{i\pi\sigma^2/2} = i\sigma^2 = \epsilon$,

the very invariant tensor of Eq. (2.166); the intertwining relation Eq. (2.168), $\epsilon\,\sigma^a\epsilon^{-1} = -(\sigma^a)^*$ (using that the Hermitian σ^a satisfy $(\sigma^a)^\top = (\sigma^a)^*$), is exactly $CT^aC^{-1} = -(T^a)^* = \bar{T}^a$. Since ϵ is *antisymmetric*, the doublet is *pseudoreal* by Theorem 2.63: the spin-$\frac{1}{2}$ representation is equivalent to its conjugate, yet cannot be written with real matrices. More generally the invariant form on spin j is symmetric for integer j and antisymmetric for half-integer j—the very alternation found in Eq. (2.169)—so the Frobenius–Schur indicator is

$$\nu(j) = (-1)^{2j} = \begin{cases} +1, & j \text{ integer (real)}, \\ -1, & j \text{ half-integer (pseudoreal)}. \end{cases} \tag{2.174}$$

Caution

That spin $\frac{1}{2}$ is pseudoreal, not real, is a genuine subtlety with physical teeth. "Self-conjugate" is weaker than "real": the doublet equals its conjugate ($\mathbf{2} \cong \bar{\mathbf{2}}$ for SU(2), via ϵ), but no choice of basis makes the Pauli matrices real, because the invariant pairing is antisymmetric rather than symmetric. The contrast with SU(3) in the next section is exactly here: the fundamental $\mathbf{3}$ of SU(3) is genuinely *complex*, $\mathbf{3} \ncong \bar{\mathbf{3}}$, which is why a quark and an antiquark are distinct objects, whereas an SU(2) doublet and its conjugate are not.

2.5.6 Worked example: $\frac{1}{2} \otimes \frac{1}{2} = 0 \oplus 1$

We now carry out the Clebsch–Gordan construction in full for the smallest nontrivial product, the coupling of two spin-$\frac{1}{2}$ systems. This is the addition of two electron spins; the result organizes the two-electron spin states into the spin-singlet and spin-triplet that govern, for instance, the exchange splitting in atoms and the structure of the deuteron.

Worked Example 2.64 (Two spin-$\frac{1}{2}$: singlet and triplet)**.** Take $j_1 = j_2 = \frac{1}{2}$, writing $|\uparrow\rangle \equiv \left|\frac{1}{2}, +\frac{1}{2}\right\rangle$ and $|\downarrow\rangle \equiv \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$ for each factor and abbreviating product states as $|\uparrow\downarrow\rangle \equiv |\uparrow\rangle \otimes |\downarrow\rangle$. The series Eq. (2.152) predicts $\frac{1}{2} \otimes \frac{1}{2} = 1 \oplus 0$, dimensions $4 = 3 + 1$. We construct both multiplets explicitly with the ladder algorithm.

Top of the triplet. The unique product state of maximal weight $m = 1$ is the top of the spin-1 multiplet,

$$|1, +1\rangle = |\uparrow\uparrow\rangle\,. \tag{2.175}$$

Lowering once. Apply $J^- = J^-_{(1)} \otimes \mathbf{1} + \mathbf{1} \otimes J^-_{(2)}$. On the left, Eq. (2.165) with $j = 1, m = 1$ gives $J^-\,|1,1\rangle = \sqrt{(1+1)(1-1+1)}\;|1,0\rangle = \sqrt{2}\;|1,0\rangle$. On the right, each single-spin lowering gives $J^-\,|\uparrow\rangle = \sqrt{(\frac{1}{2}+\frac{1}{2})(\frac{1}{2}-\frac{1}{2}+1)}\;|\downarrow\rangle = |\downarrow\rangle$,

so $J^-|\uparrow\uparrow\rangle = |\downarrow\uparrow\rangle + |\uparrow\downarrow\rangle$. Equating,

$$|1,0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle + |\downarrow\uparrow\rangle\big). \tag{2.176}$$

Lowering again. With $j=1, m=0$, $J^-|1,0\rangle = \sqrt{(1)(2)}\,|1,-1\rangle = \sqrt{2}\,|1,-1\rangle$, while applying J^- to the right-hand combination gives $\frac{1}{\sqrt{2}}(|\downarrow\downarrow\rangle + |\downarrow\downarrow\rangle) = \sqrt{2}\,|\downarrow\downarrow\rangle$. Hence

$$|1,-1\rangle = |\downarrow\downarrow\rangle\,, \tag{2.177}$$

completing the symmetric spin-1 triplet.

The singlet. The weight-0 subspace is two-dimensional, spanned by $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. One direction is the triplet state $|1,0\rangle$ just found; the orthogonal direction, normalized and signed by the Condon–Shortley convention $\langle \frac{1}{2},+\frac{1}{2};\frac{1}{2},-\frac{1}{2}\,|\,0,0\rangle > 0$, is the spin-0 singlet,

$$|0,0\rangle = \frac{1}{\sqrt{2}}\big(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\big). \tag{2.178}$$

One checks it is a singlet directly: $J^3|0,0\rangle = 0$ since both terms have weight 0; $J^-|0,0\rangle = \frac{1}{\sqrt{2}}(|\downarrow\downarrow\rangle - |\downarrow\downarrow\rangle) = 0$; and $J^+|0,0\rangle = \frac{1}{\sqrt{2}}(|\uparrow\uparrow\rangle - |\uparrow\uparrow\rangle) = 0$. Annihilated by all three generators, it is invariant.

The coefficients and the ϵ tensor. Reading off the expansion coefficients gives the Clebsch–Gordan table

$$\langle \tfrac{1}{2}m_1;\tfrac{1}{2}m_2|1\,m\rangle: \quad \langle\uparrow\uparrow|1,1\rangle = 1,\ \langle\uparrow\downarrow|1,0\rangle = \langle\downarrow\uparrow|1,0\rangle = \tfrac{1}{\sqrt{2}},\ \langle\downarrow\downarrow|1,-1\rangle = 1, \tag{2.179}$$

$$\langle \tfrac{1}{2}m_1;\tfrac{1}{2}m_2|0\,0\rangle: \quad \langle\uparrow\downarrow|0,0\rangle = \tfrac{1}{\sqrt{2}}, \quad \langle\downarrow\uparrow|0,0\rangle = -\tfrac{1}{\sqrt{2}}. \tag{2.180}$$

The singlet is the antisymmetric combination, and writing $1\equiv\uparrow$, $2\equiv\downarrow$ for the doublet index its coefficient array is, up to the normalization $1/\sqrt{2}$, exactly the invariant tensor of Eq. (2.166),

$$|0,0\rangle = \frac{1}{\sqrt{2}}\,\epsilon_{ab}\,|a\rangle\otimes|b\rangle\,, \qquad \epsilon_{12} = -\epsilon_{21} = 1. \tag{2.181}$$

The triplet is the symmetric part of $\mathbf{2}\otimes\mathbf{2}$ and the singlet the antisymmetric part, in accord with Eq. (2.169) and the pseudoreality Eq. (2.174) of the doublet. This single computation—two spins giving one singlet and one triplet, with the singlet realized by ϵ—is the prototype reused for every later addition of angular momentum, isospin, or colour, and the spin-$\frac{1}{2}$ doublet here is the same electron spin that enters QED.

The weight-bookkeeping behind the whole construction is worth seeing at a glance. Figure 2.4 plots the four product states of $\frac{1}{2}\otimes\frac{1}{2}$ on the weight axis and regroups them into the triplet and the singlet: the trapezoidal weight profile

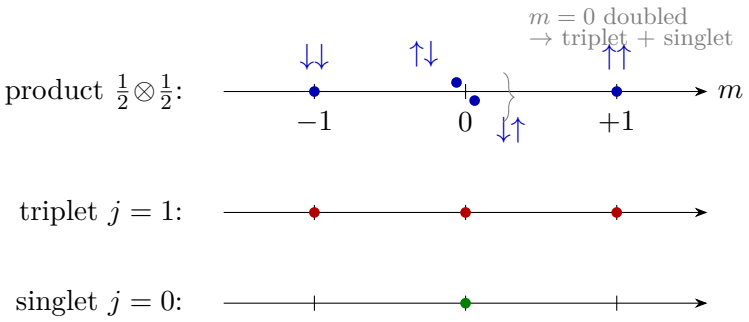


Figure 2.4: Weight addition for $\frac{1}{2}\otimes\frac{1}{2}$. The four product states (top) carry weights $m = m_1 + m_2 \in \{+1, 0, 0, -1\}$, the central weight appearing twice. Regrouping into irreducibles splits the doubled $m = 0$ level into the symmetric triplet ($j = 1$, weights $\{+1, 0, -1\}$) and the antisymmetric singlet ($j = 0$, weight $\{0\}$). The trapezoidal weight profile $N(m) = (1, 2, 1)$ is the $j_1 = j_2 = \frac{1}{2}$ case of the count in the proof of Theorem 2.58.

$N(m) = (1, 2, 1)$ at $m = (1, 0, -1)$ of the proof is laid out directly, with the doubled central weight $m = 0$ splitting into one triplet state and one singlet.

With the tensor product solved for $\mathfrak{su}(2)$ and the singlet identified as the carrier of invariance, the two ideas that propagate forward are now in hand. The first is the *weight-addition viewpoint*: a tensor product is decomposed by adding the weights of its factors and re-sorting the resulting multiset into the triangular weight patterns of irreducibles—a procedure that, recast as addition of weight *vectors* in the next section, decomposes products for any semisimple algebra through its weight diagrams. The second is the *singlet/invariant-tensor* construction, the algebraic statement of gauge invariance, which generalizes verbatim once the higher-rank weights and the conjugate representations are in place. Both are delivered to the Cartan–Weyl machinery of the following section, where the maximal commuting set of generators replaces the single J^3 and the entire structure is bootstrapped from the copies of $\mathfrak{su}(2)$ attached to each root.

2.6 Cartan–Weyl: Cartan Subalgebra, Roots, and the Highest-Weight Theorem

The solution of $\mathfrak{su}(2)$ in Section 2.4 was not a lucky accident of a small algebra; it was a method. We picked one generator, J_3, that could be diagonalized; we labelled states by its eigenvalue m; we found two more generators, $J_\pm = J_1 \pm iJ_2$, that did not commute with J_3 but instead *shifted* its eigenvalue by a fixed amount, $[J_3, J_\pm] = \pm J_\pm$; and we let those raising and lowering operators sweep out an entire irreducible representation from a single highest-weight state. The tensor-product bookkeeping of Section 2.5 added one more ingredient: weights

add when representations are multiplied, so the eigenvalue label is an additive quantum number living in a vector space. Cartan and Weyl observed that nothing in this procedure used the smallness of $\mathfrak{su}(2)$. For *any* semisimple Lie algebra one may diagonalize a maximal commuting set of generators, label states by the tuple of eigenvalues, and find that the remaining generators shift those eigenvalues by fixed vectors. The miracle that makes the general theory finite and computable is that each of those shift vectors comes with its own copy of $\mathfrak{su}(2)$ — so the entire structure of every simple Lie algebra is bootstrapped from the one example we have already solved completely. This section builds that machinery in full and ends by handing the rank-two instance to $\mathfrak{su}(3)$.

2.6.1 The Cartan subalgebra

We work with a compact semisimple Lie algebra $\mathfrak{g}$, the relevant case for the gauge groups of physics; $\mathfrak{su}(N)$ is the running example and the only one we will need concretely. In the physics convention fixed in the chapter introduction (Eq. (2.70)) the generators T^a are Hermitian, $\left[T^a, T^b\right] = if^{abc}T^c$, with the fundamental normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$. The trace form

$$\langle X, Y\rangle \;\equiv\; \mathrm{tr}(XY), \qquad X, Y \in \mathfrak{g}, \tag{2.182}$$

evaluated in the fundamental representation, is real, symmetric, and *positive definite* on the real span of the Hermitian generators, since $\langle X, X\rangle = \mathrm{tr}(X^2) = \mathrm{tr}(X^\dagger X) \geq 0$ with equality only for $X = 0$. It is moreover *invariant*, $\langle [Z, X]\,, Y\rangle + \langle X, [Z, Y]\rangle = 0$, by cyclicity of the trace. This nondegenerate invariant form is all we will need; it plays the role that the Killing form plays in the abstract theory, and for a compact algebra it is manifestly nondegenerate without invoking Cartan's criterion.

Definition 2.65 (Cartan subalgebra and rank)**.** A *Cartan subalgebra* (CSA) $\mathfrak{h} \subset \mathfrak{g}$ is a maximal abelian subalgebra consisting of elements that are simultaneously diagonalizable in every representation. Its dimension $r = \dim \mathfrak{h}$ is the *rank* of $\mathfrak{g}$.

For $\mathfrak{su}(2)$ the Cartan subalgebra is one-dimensional, spanned by J_3 alone: there is no second generator commuting with it, so $\mathfrak{su}(2)$ has rank one. For $\mathfrak{su}(N)$ one may take the $N-1$ diagonal traceless Hermitian matrices, so $\mathfrak{su}(N)$ has rank $N-1$; we will see this explicitly for $\mathfrak{su}(3)$ (rank two) in Section 2.7. That a maximal commuting set of Hermitian generators can be *simultaneously* diagonalized is exactly the spectral theorem for commuting Hermitian operators, so for compact $\mathfrak{g}$ the CSA exists and the diagonalizability clause in Theorem 2.65 is automatic. Choose once and for all a basis $H_1, \ldots, H_r$ of $\mathfrak{h}$, orthonormal in the trace form, $\langle H_i, H_j\rangle = \frac{1}{2}\delta_{ij}$.

2.6.2 Weights of a representation

Let ρ be a representation of $\mathfrak{g}$ on a vector space V. The operators $\rho(H_1), \dots, \rho(H_r)$ are commuting Hermitian operators, hence simultaneously diagonalizable: V decomposes into common eigenspaces.

Definition 2.66 (Weight)**.** A nonzero vector $|\mu\rangle \in V$ with

$$\rho(H_i)\,|\mu\rangle = \mu_i\,|\mu\rangle\,, \qquad i = 1, \dots, r, \tag{2.183}$$

is a *weight vector*; the real r-tuple $\mu = (\mu_1, \dots, \mu_r)$, viewed as a linear functional $\mu(H) = \sum_i \mu_i\, c_i$ on $H = \sum_i c_i H_i \in \mathfrak{h}$, is its *weight*. The number of independent weight vectors sharing a weight μ is the *multiplicity* of μ.

Weights live in the dual space $\mathfrak{h}^*$; the eigenvalues are real because the $\rho(H_i)$ are Hermitian. For $\mathfrak{su}(2)$ a weight is a single number, the eigenvalue m of J_3, and Theorem 2.53 is precisely the statement that the weights of the spin-j irreducible are $m = -j, -j+1, \dots, j$, each with multiplicity one. The content of Section 2.5 that weights add on tensor products is now transparent: if $|\mu\rangle \in V_1$ and $|\nu\rangle \in V_2$ then, because H_i acts on $V_1 \otimes V_2$ as $\rho_1(H_i) \otimes \mathbf{1} + \mathbf{1} \otimes \rho_2(H_i)$,

$$(\rho_1 \otimes \rho_2)(H_i)\,(|\mu\rangle \otimes |\nu\rangle) = (\mu_i + \nu_i)\;|\mu\rangle \otimes |\nu\rangle\,, \tag{2.184}$$

so the weights of a tensor product are the (multiset) sums of the weights of the factors. This additivity is the engine that drives every weight-diagram computation in higher rank.

2.6.3 The root-space decomposition of the algebra

The deepest application of weights is to the algebra acting on *itself*. The adjoint representation $\mathrm{ad}_X(Y) = [X, Y]$ makes $\mathfrak{g}$ a representation of itself (Theorem 2.39); its weights carry the entire structure of $\mathfrak{g}$. Restricting to the Cartan subalgebra, the operators $\mathrm{ad}_{H_1}, \dots, \mathrm{ad}_{H_r}$ commute — because $\mathrm{ad}_{[H_i,H_j]} = \big[\mathrm{ad}_{H_i}, \mathrm{ad}_{H_j}\big]$ and the H_i commute among themselves — and they are Hermitian with respect to the trace form, since $\langle \mathrm{ad}_H\, X, Y\rangle = -\langle X, \mathrm{ad}_H\, Y\rangle$ for H in the (real, Hermitian) compact form means ad_H is antisymmetric, hence $i\,\mathrm{ad}_H$ is symmetric and ad_H is diagonalizable with imaginary eigenvalues over $\mathbb{R}$, i.e. real eigenvalues once we pass to the complexified algebra. We therefore complexify: let $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes \mathbb{C}$. This is the same step that turned J_1, J_2 into $J_\pm = J_1 \pm i J_2$ — the ladder operators were never Hermitian; they lived in the complexification. In $\mathfrak{g}_{\mathbb{C}}$ the commuting operators ad_{H_i} are simultaneously diagonalizable.

Theorem 2.67 (Root-space decomposition)**.** *The complexified algebra decomposes as a direct sum of common eigenspaces of* $\mathrm{ad}_{\mathfrak{h}}$*,*

$$\mathfrak{g}_{\mathbb{C}} \;=\; \mathfrak{h}_{\mathbb{C}} \;\oplus\; \bigoplus_{\alpha \in \Phi} \mathfrak{g}_\alpha, \qquad \mathfrak{g}_\alpha = \big\{\, E \in \mathfrak{g}_{\mathbb{C}} : [H, E] = \alpha(H)\, E \;\; \forall H \in \mathfrak{h} \,\big\}, \tag{2.185}$$

where the roots α *run over a finite set* $\Phi \subset \mathfrak{h}^* \setminus \{0\}$ *of nonzero weights of the adjoint representation. The zero-eigenvalue space is exactly* $\mathfrak{h}_{\mathbb{C}}$*, and each root space* $\mathfrak{g}_\alpha$ *is one-dimensional.*

The phrase "roots are the weights of the adjoint representation" is the whole idea: a root α is just the weight of a generator $E_\alpha \in \mathfrak{g}_\alpha$ under the Cartan action,

$$[H_i, E_\alpha] = \alpha_i \, E_\alpha, \qquad i = 1, \ldots, r, \tag{2.186}$$

the direct analogue of $[J_3, J_\pm] = \pm J_\pm$ with $\alpha = \pm 1$. We prove the theorem in stages.

Diagonal action and the zero space. Simultaneous diagonalizability of $\mathrm{ad}_\mathfrak{h}$ was established above. Let E be a common eigenvector: $\mathrm{ad}_{H_i} E = c_i E$ for scalars c_i. Define $\alpha \in \mathfrak{h}^*$ by $\alpha(H_i) = c_i$ and extend linearly; then $[H, E] = \alpha(H)E$ for all $H \in \mathfrak{h}$, which is Eq. (2.186). The eigenvectors with $\alpha = 0$ are precisely the elements commuting with all of $\mathfrak{h}$, i.e. the centralizer of $\mathfrak{h}$ in $\mathfrak{g}_{\mathbb{C}}$. We claim this centralizer is $\mathfrak{h}_{\mathbb{C}}$ itself. Suppose X commutes with all of $\mathfrak{h}$ but $X \notin \mathfrak{h}_{\mathbb{C}}$; writing $X = H_0 + \sum_\alpha x_\alpha E_\alpha$ in the eigenbasis, $[H, X] = \sum_\alpha \alpha(H) x_\alpha E_\alpha$ must vanish for all H, forcing $x_\alpha = 0$ whenever $\alpha \neq 0$, so $X = H_0 \in \mathfrak{h}_{\mathbb{C}}$. Hence the centralizer is $\mathfrak{h}_{\mathbb{C}}$: the Cartan subalgebra is *self-centralizing*, which is the precise content of its maximality as an abelian subalgebra. This proves the zero space is exactly $\mathfrak{h}_{\mathbb{C}}$.

Conjugate roots and the reality of $E_\alpha^\dagger$. Taking the Hermitian conjugate of Eq. (2.186) and using $H_i^\dagger = H_i$ gives $\left[H_i, E_\alpha^\dagger\right] = -\alpha_i E_\alpha^\dagger$, so $E_\alpha^\dagger \in \mathfrak{g}_{-\alpha}$. Therefore $\alpha \in \Phi \Rightarrow -\alpha \in \Phi$, and we may always normalize the root vectors so that

$$E_{-\alpha} = E_\alpha^\dagger, \tag{2.187}$$

exactly as $J_- = J_+^\dagger$ in $\mathfrak{su}(2)$. The proof that each $\mathfrak{g}_\alpha$ is one-dimensional uses the $\mathfrak{su}(2)$ attached to α and is given in Section 2.6.4 once that structure is in place.

2.6.4 The form on weights and the $\mathfrak{su}(2)$ attached to each root

Because the trace form is nondegenerate on $\mathfrak{h}$, it sets up an isomorphism between $\mathfrak{h}$ and its dual $\mathfrak{h}^*$: to each linear functional $\lambda \in \mathfrak{h}^*$ corresponds a unique element $t_\lambda \in \mathfrak{h}_{\mathbb{C}}$ with

$$\lambda(H) = \langle t_\lambda, H \rangle \quad \text{for all } H \in \mathfrak{h}. \tag{2.188}$$

This lets us transport the form to weights and roots: define the (symmetric, real-valued on real weights) inner product

$$(\lambda, \mu) \;\equiv\; \langle t_\lambda, t_\mu \rangle = \lambda(t_\mu) = \mu(t_\lambda). \tag{2.189}$$

Concretely, in the orthonormal Cartan basis with $\langle H_i, H_j\rangle = \frac{1}{2}\delta_{ij}$ one has $t_\lambda = 2\sum_i \lambda(H_i)H_i$ and $(\lambda,\mu) = 2\sum_i \lambda(H_i)\mu(H_i)$; the constants will not matter because every formula below is invariant under rescaling the form.

Now fix a root α and pick generators $E_\alpha \in \mathfrak{g}_\alpha$, $E_{-\alpha} = E_\alpha^\dagger \in \mathfrak{g}_{-\alpha}$. Their commutator lies in $\mathfrak{h}_{\mathbb{C}}$, because for any $H \in \mathfrak{h}$,

$$[H,[E_\alpha,E_{-\alpha}]] = [[H,E_\alpha],E_{-\alpha}]+[E_\alpha,[H,E_{-\alpha}]] = \big(\alpha(H)-\alpha(H)\big)\,[E_\alpha,E_{-\alpha}] = 0, \tag{2.190}$$

so $[E_\alpha, E_{-\alpha}]$ commutes with all of $\mathfrak{h}$ and therefore, by self-centralizing, lies in $\mathfrak{h}_{\mathbb{C}}$. A short computation identifies it: for any $H \in \mathfrak{h}$, invariance of the trace form gives

$$\langle [E_\alpha,E_{-\alpha}],H\rangle = \langle E_\alpha,[E_{-\alpha},H]\rangle = \alpha(H)\,\langle E_\alpha,E_{-\alpha}\rangle, \tag{2.191}$$

using $[H,E_{-\alpha}] = -\alpha(H)E_{-\alpha}$. Comparing with Eq. (2.188), and normalizing the root vectors so that $\langle E_\alpha, E_{-\alpha}\rangle = 1$ (possible because the form pairs $\mathfrak{g}_\alpha$ nondegenerately with $\mathfrak{g}_{-\alpha}$),

$$[E_\alpha,E_{-\alpha}] = t_\alpha \in \mathfrak{h}_{\mathbb{C}}. \tag{2.192}$$

Finally $[t_\alpha, E_{\pm\alpha}] = \pm\alpha(t_\alpha)E_{\pm\alpha} = \pm(\alpha,\alpha)E_{\pm\alpha}$ from Eqs. (2.186) and (2.189). The three operators do not yet have the canonical $\mathfrak{su}(2)$ normalization, but rescaling fixes that. Define the *coroot*

$$\alpha^\vee \;\equiv\; \frac{2\,t_\alpha}{(\alpha,\alpha)} \in \mathfrak{h}_{\mathbb{C}}, \qquad e_\alpha \equiv E_\alpha, \quad f_\alpha \equiv \frac{2}{(\alpha,\alpha)}E_{-\alpha}, \quad h_\alpha \equiv \alpha^\vee. \tag{2.193}$$

Then, directly from Eqs. (2.186) and (2.192),

$$[h_\alpha,e_\alpha] = 2e_\alpha, \qquad [h_\alpha,f_\alpha] = -2f_\alpha, \qquad [e_\alpha,f_\alpha] = h_\alpha. \tag{2.194}$$

Proposition 2.68 (An $\mathfrak{su}(2)$ for every root)**.** *For each root α, the triple $\{e_\alpha, f_\alpha, h_\alpha\}$ spans a subalgebra $\mathfrak{su}(2)_\alpha \subset \mathfrak{g}_{\mathbb{C}}$ isomorphic to (the complexification of) $\mathfrak{su}(2)$, with h_α playing the role of $2J_3$, e_α of J_+, and f_α of J_-.*

The factor of 2 in $h_\alpha = 2J_3$ is the entire reason the theory is governed by *integers*: in the basis Eq. (2.194) the eigenvalues of h_α are $2m$, twice the magnetic quantum number, hence even or odd integers but always integers. This is the conventional normalization in which the structural numbers of Lie theory come out integral.

Root spaces are one-dimensional. We can now finish Theorem 2.67. Consider the subspace

$$V_\alpha = \mathbb{C}\,f_\alpha \;\oplus\; \mathbb{C}\,h_\alpha \;\oplus\; \bigoplus_{k\geq 1}\mathfrak{g}_{k\alpha}, \tag{2.195}$$

which is invariant under $\mathfrak{su}(2)_\alpha$ (acting by ad): e_α raises the h_α-eigenvalue by 2, f_α lowers it by 2, and h_α is diagonal. The h_α-eigenvalue on $\mathfrak{g}_{k\alpha}$ is $k\alpha(h_\alpha) = 2k$,

on h_α it is 0, and on f_α it is -2. As an $\mathfrak{su}(2)_\alpha$-representation V_α decomposes into irreducibles by Theorem 2.53; we read off constraints from its weight content. The weight -2 appears once (on f_α), and the weight 0 appears once (on h_α): every other zero-weight vector of $\mathfrak{g}_\mathbb{C}$ lies in $\mathfrak{h}_\mathbb{C}$ but is annihilated by both e_α and f_α — indeed for $H \in \mathfrak{h}$ with $\alpha(H) = 0$ one has $[e_\alpha, H] = -\alpha(H)e_\alpha = 0$ — so such vectors form $\mathfrak{su}(2)_\alpha$-singlets and do not belong to V_α. An $\mathfrak{su}(2)$ multiplet of spin j contains the weight 0 iff j is an integer, and then exactly once; it contains the weight -2 (i.e. $2m = -2$, $m = -1$) iff $j \geq 1$. Since V_α contains weight -2 exactly once and weight 0 exactly once, V_α must be a single irreducible of spin $j = 1$ — the adjoint of $\mathfrak{su}(2)_\alpha$, with weights $\{-2, 0, +2\}$. Consequently the only positive multiple of α that is a root is α itself ($\mathfrak{g}_{k\alpha} = 0$ for $k \geq 2$), and $\dim \mathfrak{g}_\alpha = 1$ (the weight $+2$ space). In particular 2α is never a root. This completes Theorem 2.67.

2.6.5 The root string and the master formula

The same $\mathfrak{su}(2)_\alpha$ now constrains how a root or weight may sit relative to α. This is the central quantization argument, generalizing the ladder of $\mathfrak{su}(2)$.

Theorem 2.69 (Root/weight strings)**.** *Let μ be any weight of any finite-dimensional representation (in particular μ may be a root, taking the representation to be the adjoint), and let α be a root. The set of weights of the form $\mu + j\alpha$, $j \in \mathbb{Z}$, that actually occur forms an unbroken string*

$$\mu - p\alpha,\ \mu - (p-1)\alpha,\ \ldots,\ \mu,\ \ldots,\ \mu + q\alpha, \qquad p, q \in \mathbb{Z}_{\geq 0}, \tag{2.196}$$

with no gaps, and the endpoints are tied to the geometry by the master formula

$$\boxed{\langle \mu, \alpha^\vee \rangle \equiv \frac{2(\mu,\alpha)}{(\alpha,\alpha)} = p - q \in \mathbb{Z}.} \tag{2.197}$$

Proof. Apply the operators $e_\alpha, f_\alpha, h_\alpha$ of $\mathfrak{su}(2)_\alpha$ to the weight vectors lying along the line $\mu + j\alpha$. The h_α-eigenvalue on a vector of weight $\mu + j\alpha$ is

$$(\mu + j\alpha)(h_\alpha) = \frac{2(\mu + j\alpha, \alpha)}{(\alpha,\alpha)} = \frac{2(\mu,\alpha)}{(\alpha,\alpha)} + 2j = \langle \mu, \alpha^\vee \rangle + 2j, \tag{2.198}$$

using Eq. (2.189) and $(\alpha,\alpha) = (\alpha,\alpha)$. Thus successive vectors along the string differ by 2 in h_α-eigenvalue, exactly the spacing of an $\mathfrak{su}(2)$ ladder. The span of the weight vectors on this line is an $\mathfrak{su}(2)_\alpha$-submodule (raising/lowering moves $j \to j \pm 1$ and stays on the line), so by Theorem 2.53 its h_α-weights run consecutively and symmetrically about 0: if the top occurs at $j = q$ and the bottom at $j = -p$, the corresponding h_α-eigenvalues $\langle \mu, \alpha^\vee \rangle + 2q$ and $\langle \mu, \alpha^\vee \rangle - 2p$ are $+2J$ and $-2J$ for the same J, hence sum to zero:

$$\big(\langle \mu, \alpha^\vee \rangle + 2q\big) + \big(\langle \mu, \alpha^\vee \rangle - 2p\big) = 0 \implies \langle \mu, \alpha^\vee \rangle = p - q. \tag{2.199}$$

The string is unbroken because an $\mathfrak{su}(2)$ irreducible has no missing internal weights (Theorem 2.53). The right-hand side is an integer, so $\langle \mu, \alpha^\vee \rangle \in \mathbb{Z}$. □

The integers $\langle \mu, \alpha^\vee \rangle$ are the *Cartan integers*. The master formula is the single most powerful constraint in the theory: it says that twice the projection of any weight onto any root, measured in units of that root's length-squared, is an integer determined by how far the weight can be lowered minus how far it can be raised along α. Applied with μ a root, it becomes a self-consistency condition on the root system itself, which we now exploit.

2.6.6 Root geometry: the allowed angles

Take two roots α, β that are not proportional. The master formula Eq. (2.197), applied once with $(\mu, \alpha) = (\beta, \alpha)$ and once with the roles swapped, gives two integers,

$$n_{\beta\alpha} \equiv \langle \beta, \alpha^\vee \rangle = \frac{2(\beta,\alpha)}{(\alpha,\alpha)} \in \mathbb{Z}, \qquad n_{\alpha\beta} \equiv \langle \alpha, \beta^\vee \rangle = \frac{2(\alpha,\beta)}{(\beta,\beta)} \in \mathbb{Z}. \tag{2.200}$$

Because the trace form is positive definite on the real span of the roots, the Cauchy–Schwarz angle θ between α and β is well defined through $(\alpha, \beta) = |\alpha|\,|\beta| \cos\theta$, where $|\alpha|^2 = (\alpha, \alpha)$. Multiplying the two integers,

$$n_{\alpha\beta}\, n_{\beta\alpha} = \frac{4(\alpha,\beta)^2}{(\alpha,\alpha)(\beta,\beta)} = 4\cos^2\theta. \tag{2.201}$$

Proposition 2.70 (Quantized root angles)**.** *For non-proportional roots α, β, the product $n_{\alpha\beta} n_{\beta\alpha} = 4\cos^2\theta$ is an integer in $\{0, 1, 2, 3\}$. Consequently the angle between two roots can only be $90°, 60°, 120°, 45°, 135°, 30°$, or $150°$, and whenever the roots are not orthogonal their squared-length ratio is fixed,*

$$\frac{|\beta|^2}{|\alpha|^2} = \frac{n_{\beta\alpha}}{n_{\alpha\beta}}. \tag{2.202}$$

Proof. The product is a product of two integers, hence an integer; it equals $4\cos^2\theta$, which lies in $[0, 4]$. The value 4 requires $\cos^2\theta = 1$, i.e. $\alpha \parallel \beta$, excluded. So $n_{\alpha\beta} n_{\beta\alpha} \in \{0, 1, 2, 3\}$. Solving $4\cos^2\theta \in \{0, 1, 2, 3\}$ gives $\cos^2\theta \in \{0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}\}$, hence $|\cos\theta| \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}, \frac{\sqrt{3}}{2}\}$ and the listed angles. The length ratio follows by dividing the two definitions in Eq. (2.200), valid whenever both integers are nonzero. □

Figure 2.5 shows the four root systems this constraint allows in rank two; they are exactly the rank-two simple algebras plus the trivial product, and $\mathfrak{su}(3)$ is the case $\theta = 120°$ with equal lengths. The enumeration of these four is worked in full in Theorem 2.77.

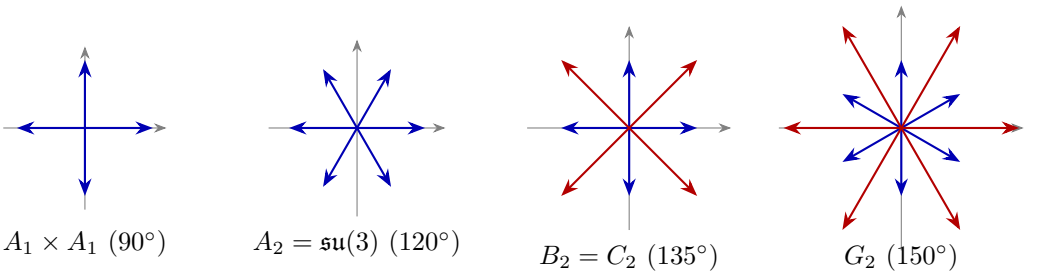


Figure 2.5: The four root systems allowed in rank two by Theorem 2.70, one for each value $4\cos^2\theta \in \{0, 1, 2, 3\}$ of the angle between a pair of simple roots. Short roots in blue, long roots in red. A_2 is the root system of $\mathfrak{su}(3)$, taken up in Section 2.7.

2.6.7 Simple roots and the Cartan matrix

The root system Φ contains every $\pm\alpha$, so it has a built-in redundancy; we extract a minimal generating set. Choose a generic linear functional (equivalently a hyperplane through the origin missing every root) and call a root *positive* if it lies on the chosen side; write $\alpha \succ 0$. Each root is then positive or negative, and $\Phi = \Phi^+ \sqcup (-\Phi^+)$.

Definition 2.71 (Simple roots)**.** A positive root is *simple* if it cannot be written as a sum of two positive roots. The set of simple roots is denoted $\Delta = \{\alpha_1, \dots, \alpha_r\}$.

The number of simple roots equals the rank r, and the simple roots are a basis of $\mathfrak{h}^*$: every positive root is a *non-negative integer* combination of simple roots (lower a positive root by simple roots until it is itself simple, the "height" decreasing at each step), and the simple roots are linearly independent because they all lie strictly on one side of the chosen hyperplane and pairwise satisfy the inequality of the next lemma, which forbids linear relations with positive coefficients. The key geometric fact is that simple roots meet only at obtuse or right angles.

Lemma 2.72 (Simple roots are non-acute)**.** *For distinct simple roots $\alpha_i \neq \alpha_j$, $(\alpha_i, \alpha_j) \leq 0$, hence the Cartan integer $\langle \alpha_i, \alpha_j^\vee \rangle \leq 0$.*

Proof. We first show $\alpha_i - \alpha_j$ is *not* a root. If it were a positive root $\gamma = \alpha_i - \alpha_j \succ 0$, then $\alpha_i = \gamma + \alpha_j$ would be a sum of two positive roots, contradicting the simplicity of α_i; if $\alpha_i - \alpha_j$ were a negative root, then $\alpha_j - \alpha_i \succ 0$ and $\alpha_j = (\alpha_j - \alpha_i) + \alpha_i$ would contradict the simplicity of α_j. So $\alpha_i - \alpha_j \notin \Phi$. Now consider the α_i-string through α_j from Theorem 2.69: since $\alpha_j - \alpha_i$ is not a root, the string cannot extend in the $-\alpha_i$ direction, so $p = 0$ and the master formula Eq. (2.197) gives $\langle \alpha_j, \alpha_i^\vee \rangle = p - q = -q \leq 0$. As (α_i, α_j) and $\langle \alpha_j, \alpha_i^\vee \rangle$ have the same sign (they differ by the positive factor $2/(\alpha_i, \alpha_i)$), $(\alpha_i, \alpha_j) \leq 0$. □

Definition 2.73 (Cartan matrix)**.** The *Cartan matrix* of $\mathfrak{g}$ is the $r \times r$ integer matrix

$$A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)}. \tag{2.203}$$

By the results above the Cartan matrix has a rigid shape: $A_{ii} = 2$ on the diagonal (each root makes angle 0 with itself, $\langle \alpha_i, \alpha_i^\vee \rangle = 2$); off-diagonal entries $A_{ij} \leq 0$ are non-positive integers (Theorem 2.72); and $A_{ij} = 0 \Leftrightarrow A_{ji} = 0$, since by Eq. (2.201) their product is $4\cos^2\theta_{ij}$ and one vanishes iff the angle is 90°. Moreover $A_{ij}A_{ji} = 4\cos^2\theta_{ij} \in \{0, 1, 2, 3\}$. The matrix A encodes the entire algebra: from it one reconstructs all roots (by repeatedly adding simple roots subject to the master formula), the full set of commutation relations (the Chevalley–Serre presentation), and every representation. The Cartan matrix need not be symmetric — the asymmetry records the length ratios of Eq. (2.202) — but it is symmetrizable, DA symmetric for the positive diagonal $D = \mathrm{diag}((\alpha_i, \alpha_i))$.

2.6.8 Dynkin diagrams

The Cartan matrix is conveniently drawn. To each simple root assign a node; join nodes i and j by $A_{ij}A_{ji} \in \{0, 1, 2, 3\}$ lines (no line for orthogonal roots), and when the roots have different lengths put an arrowhead on the multiple line pointing from the long root to the short root. This is the *Dynkin diagram*. It contains exactly the same information as the Cartan matrix — the number of lines fixes the angle and the length ratio, the arrow disambiguates which root is long — but it is read at a glance.

2.6.9 The highest-weight theorem

We can now state the higher-rank generalization of the $\mathfrak{su}(2)$ classification of irreducibles by their top spin. Choose the simple roots $\Delta = \{\alpha_i\}$ as above. A weight μ is *dominant integral* if every Cartan integer against a simple coroot is a non-negative integer,

$$\langle \mu, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}, \qquad i = 1, \ldots, r. \tag{2.204}$$

Within a representation, call a weight vector $|\Lambda\rangle$ a *highest-weight vector* if it is annihilated by every raising operator of every simple root, $e_{\alpha_i}|\Lambda\rangle = 0$ for all i — the direct generalization of $J_+|j, j\rangle = 0$.

Theorem 2.74 (Highest weight)**.** *Every finite-dimensional irreducible representation of $\mathfrak{g}$ contains a unique (up to scale) highest-weight vector, whose weight Λ is dominant integral. Conversely, for every dominant integral weight Λ there is exactly one finite-dimensional irreducible representation with that highest weight,*

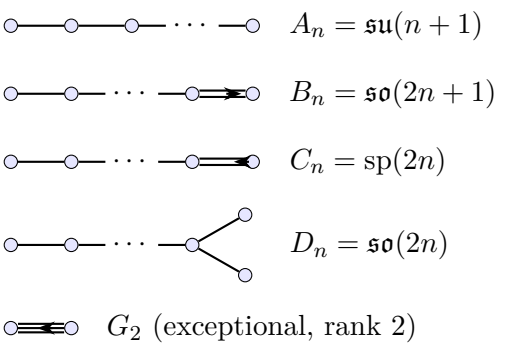


Figure 2.6: The connected Dynkin diagrams of the simple Lie algebras: the four infinite families A_n, B_n, C_n, D_n and the five exceptionals G_2, F_4, E_6, E_7, E_8. An arrow on a multiple bond points from the long root to the short root. The number of bonds between two nodes is $A_{ij}A_{ji} = 4\cos^2\theta_{ij}$.

and the assignment

$$\{\textit{finite-dimensional irreducibles of } \mathfrak{g}\} \longleftrightarrow \{\textit{dominant integral weights } \Lambda\} \tag{2.205}$$

is a bijection. The whole representation is generated from $|\Lambda\rangle$ *by repeated application of the lowering operators* f_{α_i}*, and every other weight has the form* $\Lambda - \sum_i k_i\alpha_i$ *with* $k_i \in \mathbb{Z}_{\geq 0}$.

Proof of the structural part. Order the weights by height $\mathrm{ht}(\mu) = \sum_i c_i$ where $\mu = \sum_i c_i\alpha_i$; in a finite-dimensional representation there are finitely many weights, so a maximal one Λ exists. A weight vector $|\Lambda\rangle$ of maximal height is annihilated by every e_{α_i}, since $e_{\alpha_i}|\Lambda\rangle$ would have weight $\Lambda + \alpha_i$ of greater height; thus $|\Lambda\rangle$ is a highest-weight vector. For each simple root, apply the $\mathfrak{su}(2)_{\alpha_i}$ analysis (Theorem 2.69) to the α_i-string through Λ: the string cannot rise ($q = 0$ since $\Lambda + \alpha_i$ is not a weight), so $\langle \Lambda, \alpha_i^\vee\rangle = p - q = p \geq 0$ and integral — hence Λ is dominant integral. Repeated lowering by the f_{α_i} generates an invariant subspace containing $|\Lambda\rangle$; by irreducibility it is the whole space, so every weight

is Λ minus a non-negative integer combination of simple roots, and Λ is the unique maximal weight (any other highest-weight vector would generate the same module from a weight $\leq \Lambda$ and $\geq \Lambda$, forcing equality). The remaining existence-and-uniqueness statement — that every dominant integral Λ is realized by a finite-dimensional irreducible, and only one — is proved by constructing the module as the irreducible quotient of the Verma module generated by a formal highest-weight vector and showing finite-dimensionality follows from dominance; we use this construction but do not reproduce it here, deferring the full argument to the references in Section 2.6 Further reading. The $\mathfrak{su}(2)$ case of the whole theorem is exactly Theorem 2.53, proved in full earlier. □

Caution

The honest scope of Theorem 2.74 is this: the structural half — that an irreducible *has* a unique dominant highest weight, is generated by lowering from it, and is determined by it — is proved above using only the $\mathfrak{su}(2)$ atom and finite-dimensionality. The *existence* half — that every dominant weight is actually attained by a finite-dimensional irreducible — requires the Verma-module construction and the Weyl dimension/character machinery, which we state and use but whose full proof we send to the literature. Every concrete representation we build in this book (the $\mathfrak{su}(3)$ multiplets of Section 2.7, the Lorentz fields of Chapter 3) is constructed explicitly by lowering, so existence is never assumed where we actually compute.

The dominant weights have a canonical basis. Define the *fundamental weights* $\omega_1, \ldots, \omega_r$ as the dual basis to the simple coroots,

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}. \tag{2.206}$$

Then every dominant integral weight is a non-negative integer combination

$$\Lambda = \sum_{i=1}^{r} a_i \, \omega_i, \qquad a_i = \langle \Lambda, \alpha_i^\vee \rangle \in \mathbb{Z}_{\geq 0}, \tag{2.207}$$

and the integers $(a_1, \ldots, a_r)$ are the *Dynkin labels* of the representation — the higher-rank replacement for the single label $2j$ of $\mathfrak{su}(2)$. The *fundamental representations* are those with a single $a_i = 1$; from them, by taking tensor products and projecting onto highest-weight components (Section 2.5), every irreducible is built. For $\mathfrak{su}(2)$, $r = 1$, the lone fundamental weight is $\omega_1 =$ the weight of the spin-$\frac{1}{2}$ doublet, and $\Lambda = 2j\,\omega_1$ recovers the familiar labelling.

2.6.10 The classification of simple Lie algebras

We have reduced the structure of a simple Lie algebra to a single combinatorial object — its Cartan matrix, equivalently its Dynkin diagram — subject to the

rigid constraints of Theorems 2.70 and 2.72: each pair of nodes carries $0, 1, 2$, or 3 bonds, and the matrix must be positive definite (the trace form is positive definite, so its restriction to the simple roots, of which the symmetrized Cartan matrix is the Gram matrix up to scale, must be too). The classification problem is therefore purely Euclidean: *which connected diagrams of unit vectors at the allowed angles have a positive-definite Gram matrix?* The answer is one of the cleanest results in mathematics.

Theorem 2.75 (Cartan–Killing classification)**.** *Every finite-dimensional simple complex Lie algebra is isomorphic to exactly one of:*

- *the four infinite families* A_n $(n \geq 1)$, B_n $(n \geq 2)$, C_n $(n \geq 3)$, D_n $(n \geq 4)$, *realized as*

$$A_n = \mathfrak{su}(n+1), \quad B_n = \mathfrak{so}(2n+1), \quad C_n = \mathrm{sp}(2n), \quad D_n = \mathfrak{so}(2n); \tag{2.208}$$

- *the five exceptional algebras* G_2, F_4, E_6, E_7, E_8, *of ranks* $2, 4, 6, 7, 8$.

Their Dynkin diagrams are precisely the connected diagrams of Fig. 2.6; equivalently, these are exactly the connected positive-definite diagrams allowed by the bond rules.

The honest scope: the *tools* for this theorem are all in hand — the bond rules from the master formula, the positive-definiteness of the trace form, and the reconstruction of the algebra from its Cartan matrix. The remaining step is a finite (if intricate) Euclidean enumeration: rule out diagrams with cycles, with too-high vertex degree, or with too many/too-long multiple bonds, by exhibiting a vector combination of zero or negative norm. We state the result and use it; the complete enumeration is in the references of the Further reading. The classical families are the algebras of the matrix groups physics already cares about: $A_{N-1} = \mathfrak{su}(N)$ is the gauge group of every Yang–Mills theory in Yang–Mills theory, with $A_1 = \mathfrak{su}(2)$ the rank-one atom of this whole development and $A_2 = \mathfrak{su}(3)$ the case taken up next. The exceptional algebras are not idle curiosities; E_6, E_7, E_8 are candidate grand-unified and string gauge groups returned to in Chapter 35, and the affine extensions of these same diagrams organize two-dimensional conformal field theory in Chapter 48.

2.6.11 Worked examples

Worked Example 2.76 (The A_2 root system and the Cartan matrix of $\mathfrak{su}(3)$)**.** We construct, from Theorem 2.70 alone, the rank-two root system with two simple roots at 120°, and read off its data. This is the root system of $\mathfrak{su}(3)$, and the construction is the template Section 2.7 fills in with explicit Gell-Mann matrices.

Take two simple roots α_1, α_2 of equal length, normalized to $(\alpha_i, \alpha_i) = 2$, meeting at $\theta = 120°$. Then $(\alpha_1, \alpha_2) = |\alpha_1||\alpha_2| \cos 120° = 2 \cdot (-\frac{1}{2}) = -1$, so the Cartan integers are

$$A_{12} = \frac{2(\alpha_1, \alpha_2)}{(\alpha_2, \alpha_2)} = \frac{2(-1)}{2} = -1, \qquad A_{21} = \frac{2(\alpha_2, \alpha_1)}{(\alpha_1, \alpha_1)} = -1, \tag{2.209}$$

giving the symmetric Cartan matrix

$$A = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \qquad \det A = 3 > 0, \tag{2.210}$$

which is positive definite, confirming A_2 is a genuine simple algebra. The product $A_{12}A_{21} = 1 = 4\cos^2 120°$ means the Dynkin diagram is two nodes joined by a single line, with no arrow (equal lengths): $\circ\!-\!\circ$.

Now generate the positive roots by adding simple roots subject to the master formula Eq. (2.197). Start with α_1 and ask whether $\alpha_1 + \alpha_2$ is a root, using the α_2-string through α_1. Since $\alpha_1 - \alpha_2$ is not a root (both simple, Theorem 2.72), the string has $p = 0$, so $q = -\langle \alpha_1, \alpha_2^\vee \rangle = -A_{12} = 1$: the string is $\alpha_1,\ \alpha_1 + \alpha_2$, and $\alpha_1 + \alpha_2$ *is* a root. Repeating in the α_1 direction through $\alpha_1 + \alpha_2$: the α_1-string downward already includes α_2 (one step down) so $p = 1$; and $\langle \alpha_1 + \alpha_2, \alpha_1^\vee \rangle = A_{11} + A_{21} = 2 - 1 = 1 = p - q$ gives $q = 0$, so $\alpha_1 + (\alpha_1 + \alpha_2) = 2\alpha_1 + \alpha_2$ is *not* a root. Likewise $\alpha_1 + 2\alpha_2$ is not a root. The positive roots are therefore

$$\Phi^+ = \{\alpha_1,\ \alpha_2,\ \alpha_1 + \alpha_2\}, \tag{2.211}$$

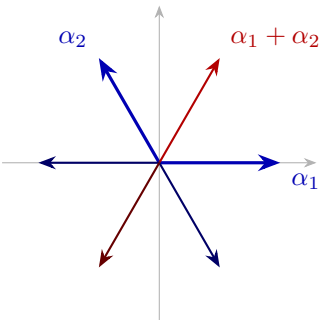
and with their negatives the full system has 6 roots, the hexagon of Fig. 2.7. Adding the $r = 2$ Cartan generators gives $\dim \mathfrak{su}(3) = 6 + 2 = 8$, the eight Gell-Mann generators, exactly as expected.

The fundamental weights solve Eq. (2.206), $\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}$. Writing $\omega_i = \sum_k c_{ik}\alpha_k$ and using $\langle \alpha_k, \alpha_j^\vee \rangle = A_{kj}$ gives $\sum_k c_{ik}A_{kj} = \delta_{ij}$, i.e. $C = A^{-1}$ with

$$A^{-1} = \frac{1}{3}\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \qquad \omega_1 = \tfrac{1}{3}(2\alpha_1 + \alpha_2), \quad \omega_2 = \tfrac{1}{3}(\alpha_1 + 2\alpha_2). \tag{2.212}$$

The fundamental representation with Dynkin labels $(1, 0)$ has highest weight ω_1 — this is the $\mathbf{3}$ of $\mathfrak{su}(3)$ — and $(0, 1)$ has highest weight ω_2 — the conjugate $\bar{\mathbf{3}}$. The adjoint, with highest root $\alpha_1 + \alpha_2$, has Dynkin labels $(1, 1)$ and dimension 8. All of this is the abstract skeleton; Section 2.7 clothes it in matrices and weight diagrams.

Worked Example 2.77 (All rank-two root systems)**.** We enumerate every rank-two simple root system directly from Theorem 2.70, confirming Fig. 2.5 and illustrating the classification in the smallest nontrivial case. Two simple roots



A_2 root system of $\mathfrak{su}(3)$

Figure 2.7: The six roots of $\mathfrak{su}(3)$ (A_2). The simple roots α_1, α_2 (heavy blue) sit at 120°; their sum $\alpha_1 + \alpha_2$ (red) is the highest root, and the remaining three are the negatives. Together with the two Cartan generators they give $\dim \mathfrak{su}(3) = 8$.

α_1, α_2 are characterized by the single Cartan-integer product $A_{12}A_{21} = 4\cos^2\theta \in \{0, 1, 2, 3\}$ (the value 4 being excluded as parallel). We treat each value.

Product 0 *($\theta = 90°$).* The simple roots are orthogonal, $A_{12} = A_{21} = 0$. The algebra is not simple: its Dynkin diagram is two disconnected nodes, $\mathfrak{su}(2) \oplus \mathfrak{su}(2) = A_1 \times A_1$. The root system is the four roots $\pm\alpha_1, \pm\alpha_2$ on perpendicular axes.

Product 1 *($\theta = 120°$).* Necessarily $A_{12} = A_{21} = -1$, so equal lengths, by Eq. (2.202). This is $A_2 = \mathfrak{su}(3)$ of Theorem 2.76, six roots forming a hexagon. Diagram $\circ\!-\!\circ$.

Product 2 *($\theta = 135°$).* Then $\{A_{12}, A_{21}\} = \{-1, -2\}$, so by Eq. (2.202) the roots have squared-length ratio 2: one short, one long. Generating roots as in Theorem 2.76, the α-strings now produce $\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2$ (with α_1 short), giving four positive roots and eight roots in all. This is $B_2 = \mathfrak{so}(5)$, equal as a root system to $C_2 = \mathrm{sp}(4)$ (the two differ only by which root is called long, i.e. the direction of the arrow). Diagram $\circ\Rightarrow\circ$.

Product 3 *($\theta = 150°$).* Then $\{A_{12}, A_{21}\} = \{-1, -3\}$, squared-length ratio 3. The strings now generate six positive roots, $\alpha_1, \alpha_2, \alpha_1 + \alpha_2, 2\alpha_1 + \alpha_2, 3\alpha_1 + \alpha_2, 3\alpha_1 + 2\alpha_2$ (with α_1 short), twelve roots total — the star-of-David configuration of G_2, the smallest exceptional algebra, $\dim G_2 = 12 + 2 = 14$. Diagram $\circ\Rrightarrow\circ$.

Thus the only rank-two simple root systems are A_2, $B_2 (= C_2)$, and G_2, with $A_1 \times A_1$ the lone (semisimple, non-simple) extra. The positive-definiteness check is automatic here since every 2×2 Cartan matrix $\left(\begin{smallmatrix} 2 & -a \\ -b & 2 \end{smallmatrix}\right)$ with $ab \leq 3$ has $\det = 4 - ab > 0$; rank three is where positive-definiteness first starts excluding diagrams, which is the seed of the full classification.

The machinery is now complete: a Cartan subalgebra to diagonalize, weights to label states, a root system carrying one $\mathfrak{su}(2)$ per root, the master formula

quantizing all geometry, simple roots and a Cartan matrix to package it, and the highest-weight theorem to enumerate representations. In the next section we stop working in the abstract and turn the crank on the first genuinely higher-rank case, $\mathfrak{su}(3)$ — building its weight diagrams, the fundamental **3** and $\bar{\mathbf{3}}$, the adjoint **8**, and the tensor-product decompositions that are the working language of QCD color and flavor.

2.7 su(3): The Physicist's Workhorse

The Cartan–Weyl construction of Section 2.6 is a machine that takes any semisimple Lie algebra apart into a maximal commuting set — the Cartan subalgebra — and a collection of ladder operators, each labelled by a root and each generating its own copy of $\mathfrak{su}(2)$. For $\mathfrak{su}(2)$ itself the machine is almost idle: the Cartan subalgebra is one-dimensional, there is a single pair of roots, and the whole of Section 2.4 is the result. The first time the construction has anything genuinely new to say is at rank two, and the first rank-two algebra a physicist meets is $\mathfrak{su}(3)$. It is not an academic example. $\mathfrak{su}(3)$ is the algebra of the color SU(3) gauge symmetry that binds quarks into hadrons, and, in a separate and approximate guise, the algebra of the flavor SU(3) that organized the particle zoo into the multiplets of the Eightfold Way before quarks were known to exist. The generators T^a, the structure constants f^{abc}, the triplet **3** and its distinct conjugate $\bar{\mathbf{3}}$, the octet **8**, and the invariants C_F, C_A, T_F that this section assembles are the literal alphabet of the Yang–Mills and QCD chapters in Part VII. We build them once, here, by running the Cartan–Weyl machine to completion on $\mathfrak{su}(3)$, so that when Yang–Mills theory begins in Chapter 25 the color bookkeeping is already in hand.

2.7.1 The algebra, the Cartan subalgebra, and the rank

The compact real algebra $\mathfrak{su}(3)$ is the space of traceless anti-Hermitian 3×3 matrices, with the matrix commutator as bracket. As always in this book we work with Hermitian generators T^a, related to the anti-Hermitian elements by a factor of i; the physical normalization is $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ and the bracket is $\left[T^a, T^b\right] = i f^{abc} T^c$ (Chapter A). A Hermitian traceless 3×3 matrix has $3^2 - 1 = 8$ real parameters, so the algebra is eight-dimensional and carries eight generators. The standard basis is $T^a = \frac{1}{2}\lambda^a$, where the λ^a are the Gell-Mann matrices written out in Eq. (B.59) and shown there to satisfy $\mathrm{tr}(\lambda^a \lambda^b) = 2\delta^{ab}$ (Theorem B.17); the full set of structure constants f^{abc} and the symmetric symbols d^{abc} are computed by direct trace in Theorem B.18. We import that explicit matrix data rather than re-derive it, and concentrate here on what the Cartan–Weyl analysis makes of it.

The first question the machine asks is: how large a set of the T^a can be simultaneously diagonalized? Two of the Gell-Mann matrices are already diagonal,

$$H_1 \;=\; T^3 \;=\; \tfrac{1}{2}\,\mathrm{diag}(1,-1,0), \qquad H_2 \;=\; T^8 \;=\; \tfrac{1}{2\sqrt{3}}\,\mathrm{diag}(1,1,-2), \quad (2.213)$$

and they commute, $[H_1, H_2] = 0$, because diagonal matrices always do. No third generator commutes with both: a matrix commuting with H_1 and H_2, both of which have distinct diagonal entries in the generic combination, must itself be diagonal and traceless, and the diagonal traceless matrices form exactly the two-dimensional span of H_1, H_2. The Cartan subalgebra is therefore $\mathfrak{h} = \mathrm{span}\{H_1, H_2\}$, and the *rank* of $\mathfrak{su}(3)$ is two. This is the structural fact that makes $\mathfrak{su}(3)$ the first interesting case: with two commuting charges, states are labelled by a point in a two-dimensional plane, and the entire representation theory becomes genuinely geometric — weights are lattice points, roots are vectors, and the diagrams one draws are honest pictures rather than the points-on-a-line of $\mathfrak{su}(2)$.

2.7.2 The root system: the $\mathfrak{su}(3)$ hexagon

The remaining six generators are organized by the adjoint action of the Cartan subalgebra. Following Section 2.6, we look for ladder operators E_α — eigenvectors of ad_{H_1} and ad_{H_2} simultaneously — and the pair of eigenvalues $\alpha = (\alpha_1, \alpha_2)$ defined by

$$[H_i, E_\alpha] \;=\; \alpha_i\, E_\alpha, \qquad i = 1, 2, \quad (2.214)$$

is the root carried by E_α. The six off-diagonal directions of $\mathfrak{su}(3)$ combine into three conjugate pairs of ladder operators, each built from two Gell-Mann matrices supported on one off-diagonal block:

$$I_\pm = \tfrac{1}{2}(\lambda^1 \pm i\lambda^2), \qquad V_\pm = \tfrac{1}{2}(\lambda^4 \pm i\lambda^5), \qquad U_\pm = \tfrac{1}{2}(\lambda^6 \pm i\lambda^7). \quad (2.215)$$

In matrix-unit notation $I_+ = E_{12}$, $V_+ = E_{13}$, $U_+ = E_{23}$, where E_{ij} is the matrix with a single 1 in row i, column j; the lowering operators are the transposes. The eigenvalues (2.214) are now a one-line computation. For $I_+ = E_{12}$, using $[\mathrm{diag}(d_1, d_2, d_3), E_{ij}] = (d_i - d_j)E_{ij}$ together with the diagonals in (2.213),

$$[H_1, E_{12}] = (\tfrac{1}{2} - (-\tfrac{1}{2}))E_{12} = 1 \cdot E_{12}, \qquad [H_2, E_{12}] = (\tfrac{1}{2\sqrt{3}} - \tfrac{1}{2\sqrt{3}})E_{12} = 0, \quad (2.216)$$

so I_+ carries the root $\alpha_1 = (1, 0)$. The same arithmetic on E_{13} and E_{23} gives

$$V_+ = E_{13} : \; \Big(\tfrac{1}{2},\, \tfrac{1}{2\sqrt{3}} + \tfrac{1}{\sqrt{3}}\Big) = \Big(\tfrac{1}{2},\, \tfrac{\sqrt{3}}{2}\Big), \qquad U_+ = E_{23} : \; \Big(-\tfrac{1}{2},\, \tfrac{1}{2\sqrt{3}} + \tfrac{1}{\sqrt{3}}\Big) = \Big(-\tfrac{1}{2},\, \tfrac{\sqrt{3}}{2}\Big), \quad (2.217)$$

and the three lowering operators carry the opposite roots. The six roots of $\mathfrak{su}(3)$ are therefore

$$\pm(1, 0), \qquad \pm\Big(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\Big), \qquad \pm\Big(\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\Big), \quad (2.218)$$

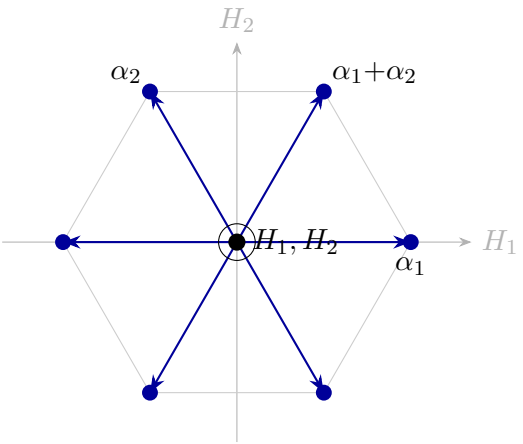


Figure 2.8: The root system of $\mathfrak{su}(3)$: six roots of equal length at the vertices of a regular hexagon in the orthonormal Cartan plane (H_1, H_2), together with the two-dimensional zero-root space (the Cartan subalgebra) at the centre. A choice of positive half-plane singles out the simple roots $\alpha_1 = (1, 0)$ and $\alpha_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$; their sum $\alpha_1 + \alpha_2$ is the third positive root, the highest root, and the highest weight of the adjoint representation $\mathbf{8}$.

and they are the great geometric punchline of rank two: each has length one in the (H_1, H_2) plane, and consecutive roots are separated by 60°. They sit at the vertices of a regular hexagon (Fig. 2.8). Every root α comes paired with $-\alpha$ and generates an $\mathfrak{su}(2)$: the trio (I_+, I_-, H_1) closes on the isospin $\mathfrak{su}(2)$, $(V_+, V_-, \cdot)$ and $(U_+, U_-, \cdot)$ on the V-spin and U-spin $\mathfrak{su}(2)$s. That every root reproduces the algebra we solved completely in Section 2.4 is the content of the Cartan–Weyl bootstrap, made concrete.

Notation Watch

The physics literature on the quark model uses coordinates (t_3, y) — isospin projection and hypercharge — with $t_3 = T^3$ and $y = \frac{1}{\sqrt{3}}\lambda^8$, so that $y = \frac{2}{\sqrt{3}}H_2$. In those coordinates the fundamental weights are $(\frac{1}{2}, \frac{1}{3}), (-\frac{1}{2}, \frac{1}{3}), (0, -\frac{2}{3})$, the familiar hypercharges of (u, d, s). The map from (H_1, H_2) to (t_3, y) stretches the vertical axis and shears the hexagon: lengths and angles are no longer the Killing-form lengths and angles, so the "hexagon" looks elongated. We draw all diagrams in the orthonormal (H_1, H_2) basis, where the geometry is the true root geometry, and convert to (t_3, y) only when comparing to hadron charge assignments. The same (t_3, y) plot appears in Fig. B.1.

Simple roots, the Cartan matrix, and the Dynkin diagram. To organize the hexagon we pick a positive half-plane — say, roots with $\alpha_1 > 0$, or $\alpha_1 = 0$ and $\alpha_2 > 0$ — giving the positive roots $\alpha_1 = (1, 0)$, $\alpha_2 = (-\frac{1}{2}, \frac{\sqrt{3}}{2})$, and $\alpha_1 + \alpha_2 = (\frac{1}{2}, \frac{\sqrt{3}}{2})$. A positive root is *simple* if it is not the sum of two positive roots; here α_1 and α_2 are simple and $\alpha_1 + \alpha_2$ is not. With the inner products $(\alpha_1, \alpha_1) = (\alpha_2, \alpha_2) = 1$ and $(\alpha_1, \alpha_2) = -\frac{1}{2}$, the Cartan matrix of Section 2.6 is

$$A_{ij} = \frac{2(\alpha_i, \alpha_j)}{(\alpha_j, \alpha_j)} = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}. \tag{2.219}$$

The off-diagonal entries $A_{12}A_{21} = 1$ mean the two simple roots meet at 120° and have equal length, which is the Dynkin diagram A_2: two nodes joined by a single line. This is the rank-two entry in the $A_n = \mathfrak{su}(n+1)$ family of the classification stated in Section 2.6.

Dual to the simple roots are the two *fundamental weights* ω_1, ω_2, defined by $2(\omega_i, \alpha_j)/(\alpha_j, \alpha_j) = \delta_{ij}$. Solving the linear system with (2.219) gives

$$\omega_1 = \left(\tfrac{1}{2}, \tfrac{1}{2\sqrt{3}}\right), \qquad \omega_2 = \left(0, \tfrac{1}{\sqrt{3}}\right), \tag{2.220}$$

and the highest-weight theorem (Theorem 2.74) tells us that every finite-dimensional irreducible representation of $\mathfrak{su}(3)$ is labelled by a pair of non-negative integers, its *Dynkin labels* (p, q), through the dominant highest weight $\Lambda = p\,\omega_1 + q\,\omega_2$. The whole representation theory of $\mathfrak{su}(3)$ is now indexed by the quadrant $p, q \geq 0$. We meet the four smallest members below.

2.7.3 The fundamental 3, the conjugate $\bar{\mathbf{3}}$, and why they differ

The defining representation acts on $\mathbb{C}^3$; its three basis vectors e_1, e_2, e_3 are simultaneous eigenvectors of H_1, H_2, and reading the diagonals of (2.213) gives their weights,

$$\mu_1 = \left(\tfrac{1}{2}, \tfrac{1}{2\sqrt{3}}\right), \quad \mu_2 = \left(-\tfrac{1}{2}, \tfrac{1}{2\sqrt{3}}\right), \quad \mu_3 = \left(0, -\tfrac{1}{\sqrt{3}}\right). \tag{2.221}$$

They sum to zero (the generators are traceless) and form an equilateral triangle. The highest of them is $\mu_1 = \omega_1$, so this is the irrep with Dynkin labels $(1, 0)$: the *fundamental* $\mathbf{3}$. A ladder operator moves a state by its root: I_+ sends $\mu_2 \to \mu_1$ (difference $\mu_1 - \mu_2 = (1, 0) = \alpha_1$), exactly as a raising operator should, and the three weights are connected into the triangle by the three positive roots.

The conjugate representation $\bar{\mathbf{3}}$ is built from generators $\bar{T}^a = -(T^a)^* = -(T^a)^\mathsf{T}$ (Hermiticity makes the conjugate and the negative-transpose coincide). These satisfy the same algebra, $\left[\bar{T}^a, \bar{T}^b\right] = i f^{abc} \bar{T}^c$, because complex-conjugating $\left[T^a, T^b\right] = i f^{abc} T^c$ flips the sign of the explicit i and the extra minus signs cancel.

Since the Cartan generators are real and diagonal, $\bar{H}_i = -H_i$, and the weights of $\bar{\mathbf{3}}$ are the negatives of (2.221),

$$-\mu_1 = \left(-\tfrac{1}{2}, -\tfrac{1}{2\sqrt{3}}\right), \quad -\mu_2 = \left(\tfrac{1}{2}, -\tfrac{1}{2\sqrt{3}}\right), \quad -\mu_3 = \left(0, \tfrac{1}{\sqrt{3}}\right), \tag{2.222}$$

with highest weight $-\mu_3 = \omega_2$, Dynkin labels $(0, 1)$.

Theorem 2.78 (The fundamental of $\mathfrak{su}(3)$ is complex)**.** *The representations* $\mathbf{3}$ *and* $\bar{\mathbf{3}}$ *are inequivalent. Equivalently,* $\mathbf{3}$ *is a* complex *representation: there is no change of basis taking the generators* T^a *to their conjugates* $-(T^a)^*$.

Proof. Two finite-dimensional representations of a semisimple Lie algebra are equivalent if and only if they have the same multiset of weights (a corollary of the highest-weight theorem, Theorem 2.74: the weight system is a complete invariant). The weight set of $\mathbf{3}$ is $\{\mu_1, \mu_2, \mu_3\}$, an equilateral triangle with its apex μ_3 pointing toward negative H_2; the weight set of $\bar{\mathbf{3}}$ is $\{-\mu_1, -\mu_2, -\mu_3\}$, the same triangle reflected through the origin, apex pointing toward positive H_2. A multiset of three points equals its own reflection through the origin only if the points are centrally symmetric, i.e. closed under $\mu \mapsto -\mu$; but $\mu_1 = (\frac{1}{2}, \frac{1}{2\sqrt{3}})$ and $-\mu_1$ are both required to lie in $\{\mu_1, \mu_2, \mu_3\}$, and $-\mu_1$ is none of the three. The weight sets differ, so the representations are inequivalent. □

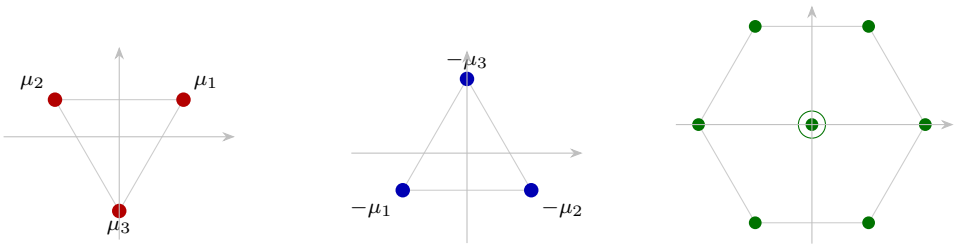
This is the first place the rank-one intuition fails. Every representation of $\mathfrak{su}(2)$ is real or pseudoreal — spin-j and its conjugate carry the same weights $\{-j, \dots, j\}$, a set symmetric about zero — so in $\mathfrak{su}(2)$ one never has to distinguish a representation from its conjugate. In $\mathfrak{su}(3)$ one must: a quark $q \sim \mathbf{3}$ and an antiquark $\bar{q} \sim \bar{\mathbf{3}}$ are genuinely different objects, and the physical distinction between matter and antimatter in QCD color is exactly the statement of Theorem 2.78.

2.7.4 The adjoint 8

The algebra acts on itself by the bracket, and this eight-dimensional adjoint representation is the octet $\mathbf{8}$. Its weights are read off the Cartan–Weyl decomposition directly: each ladder operator E_α is a weight vector of weight equal to its root α (that is precisely what (2.214) says, with ad_{H_i} the adjoint action), and each Cartan generator H_i has weight zero, since $[H_i, H_j] = 0$. The weight system of $\mathbf{8}$ is therefore

$$\underbrace{\{\pm(1,0),\ \pm(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}),\ \pm(\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2})\}}_{\text{six roots}} \cup \underbrace{\{0,\ 0\}}_{H_1,\, H_2}, \tag{2.223}$$

the six hexagon vertices plus a doubly occupied centre (Fig. 2.9). Its highest weight is the highest root $\alpha_1 + \alpha_2 = \omega_1 + \omega_2$, Dynkin labels $(1, 1)$; for any simple



(a) $\mathbf{3} = (1,0)$ (b) $\bar{\mathbf{3}} = (0,1)$ (c) $\mathbf{8} = (1,1)$

Figure 2.9: Weight diagrams in the Cartan plane (H_1, H_2). The fundamental $\mathbf{3}$ and its conjugate $\bar{\mathbf{3}}$ are equilateral triangles of opposite orientation — the geometric image of Theorem 2.78, that the two are inequivalent. The adjoint $\mathbf{8}$ is the root hexagon with a doubly occupied centre (the circled point), the double occupancy being the rank, 2.

algebra the adjoint highest weight is the highest root, and the zero weight appears with multiplicity equal to the rank, which is why the centre of the octet diagram is doubly occupied. Physically $\mathbf{8}$ is the color content of the gluons: eight gauge bosons, one per generator, transforming in the adjoint — the fact that the gluon octet *is* the adjoint is the statement that gauge fields live in the Lie algebra, which Yang–Mills theory takes as its starting point in Chapter 25.

2.7.5 Tensor products by weight addition

The physically central products are decomposed by the rule established in Section 2.5: the weights of $R_1 \otimes R_2$ are the pairwise sums of the weights of the factors, and one peels off irreducibles by repeatedly identifying the highest remaining weight, attaching its full irrep, and subtracting that irrep's weight system. We carry this out for the two products on which the quark model and the color structure of QCD rest.

Worked Example 2.79 ($\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}$: mesons and the color singlet)**.** We combine a fundamental (weights μ_i, Eq. (2.221)) with a conjugate (weights $-\mu_j$, Eq. (2.222)). The $3 \times 3 = 9$ product weights are the differences $\mu_i - \mu_j$. Split them by whether the indices agree.

Diagonal terms $i = j$. Each gives $\mu_i - \mu_i = 0$; there are three of them, so the zero weight occurs with multiplicity three.

Off-diagonal terms $i \neq j$. The six differences are

$$\mu_1 - \mu_2 = (1,0), \quad \mu_1 - \mu_3 = (\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}), \quad \mu_2 - \mu_3 = (-\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}), \tag{2.224}$$

together with their negatives $\mu_2 - \mu_1, \mu_3 - \mu_1, \mu_3 - \mu_2$. Comparing with (2.218), these are exactly the six roots of $\mathfrak{su}(3)$.

Assembling irreps. The full product weight system is {six roots} ∪ {0, 0, 0}. The highest weight present is the highest root $\alpha_1 + \alpha_2 = \mu_1 - \mu_3$, with Dynkin labels (1, 1): this is the adjoint **8**, whose weight system (2.223) is the six roots plus a *double* zero. Subtracting it leaves $3 - 2 = 1$ copy of the zero weight and nothing else — a single state at the origin, the trivial representation **1**. Therefore

$$\mathbf{3} \otimes \bar{\mathbf{3}} = \mathbf{8} \oplus \mathbf{1}, \tag{2.225}$$

and the dimensions check, $3 \times 3 = 9 = 8 + 1$. The singlet is the invariant: writing a fundamental index up, $q^i \sim \mathbf{3}$, and a conjugate index down, $\bar{q}_j \sim \bar{\mathbf{3}}$, the unique invariant bilinear is the trace $\bar{q}_i q^i = \delta^i_j\, \bar{q}_i q^j$, which is annihilated by every generator because δ^i_j is the invariant tensor of SU(3). It is the only piece of $q\bar{q}$ carrying no color, and the statement that physical hadrons are color singlets is the statement that only the **1** in (2.225) can appear as an asymptotic state. In flavor SU(3) the same decomposition is the meson nonet: a quark–antiquark pair makes an octet of mesons (the pions, kaons, and η) plus a flavor-singlet η'.

Physical Insight

Theorem 2.79 is the algebraic skeleton of two superficially different physical statements. In *color* SU(3) it says a quark and an antiquark can neutralize their color into the invariant **1**; confinement is the dynamical demand that asymptotic states sit in this singlet. In *flavor* SU(3) the very same $\mathbf{8} \oplus \mathbf{1}$ is the observed pattern of light mesons. One decomposition, two readings — which is exactly why the same algebra earned the name "the physicist's workhorse" twice over.

The second product is the two-quark combinatorics. Adding two fundamentals, the nine weights $\mu_i + \mu_j$ split into the six symmetric combinations $\{2\mu_1, 2\mu_2, 2\mu_3, \mu_1 + \mu_2, \mu_1 + \mu_3, \mu_2 + \mu_3\}$ and the three antisymmetric ones $\{\mu_1 + \mu_2, \mu_1 + \mu_3, \mu_2 + \mu_3\}$. The highest symmetric weight is $2\mu_1 = 2\omega_1$, Dynkin labels (2, 0): the sextet **6**. For the antisymmetric part, use $\mu_1 + \mu_2 + \mu_3 = 0$ to rewrite $\mu_1 + \mu_2 = -\mu_3$, $\mu_1 + \mu_3 = -\mu_2$, $\mu_2 + \mu_3 = -\mu_1$; these are precisely the weights (2.222) of $\bar{\mathbf{3}}$. Hence

$$\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}, \tag{2.226}$$

with $3 \times 3 = 9 = 6 + 3$. The antisymmetric pair of quarks transforms as an antiquark in color — the group-theoretic root of the diquark and of the way three quarks build a baryon, since $\mathbf{3}\otimes\mathbf{3}\otimes\mathbf{3} \supset \bar{\mathbf{3}}\otimes\mathbf{3} \supset \mathbf{1}$ delivers the color-singlet baryon. (Iterating the same weight-addition on (2.226) gives $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$, the baryon decuplet, octets, and singlet of the Eightfold Way.)

2.7.6 Dimensions and the Young-tableau shortcut

Reading dimensions off weight diagrams works but does not scale. For $\mathfrak{su}(3)$ the highest-weight theorem is sharpened by the Weyl dimension formula into a

closed expression in the Dynkin labels (p, q),

$$\dim(p,q) = \tfrac{1}{2}\,(p+1)(q+1)(p+q+2), \tag{2.227}$$

which reproduces every dimension met above: $(1,0)$ and $(0,1)$ give 3; $(1,1)$ gives $\tfrac{1}{2}\cdot 2\cdot 2\cdot 4 = 8$; $(2,0)$ gives $\tfrac{1}{2}\cdot 3\cdot 1\cdot 4 = 6$; and $(3,0)$ gives $\tfrac{1}{2}\cdot 4\cdot 1\cdot 5 = 10$, the decuplet. Conjugation acts by $(p,q)\mapsto(q,p)$, which leaves (2.227) invariant — a representation and its conjugate always share a dimension, and $\mathbf{3}, \bar{\mathbf{3}}$ are the smallest pair where the two are nonetheless distinct.

For SU(N) generally the efficient bookkeeping is the *Young tableau*. A representation is encoded by a left-justified diagram of boxes whose row lengths weakly decrease, with at most $N-1$ rows (a full column of N boxes is a singlet and may be deleted). The fundamental $\mathbf{N}$ is a single box $\square$; the conjugate $\bar{\mathbf{N}}$ is a column of $N-1$ boxes (its weights complement the fundamental's to fill a column). Tensor products are computed by the Littlewood–Richardson rule for gluing tableaux, and the symmetric/antisymmetric split of two boxes,

$$\square \otimes \square = \underbrace{(\text{two boxes in a row})}_{\text{symmetric}} \;\oplus\; \underbrace{(\text{two boxes in a column})}_{\text{antisymmetric}}, \tag{2.228}$$

is exactly (2.226) for $N=3$: the row is the $\mathbf{6}$, the column (two of three boxes) is the $\bar{\mathbf{3}}$. The dimension of a tableau in SU(N) is the ratio

$$\dim_{\mathrm{SU}(N)}(\text{tableau}) = \frac{\prod_{\text{cells}}(N + \text{col} - \text{row})}{\prod_{\text{cells}}(\text{hook length})}, \tag{2.229}$$

where for a cell in row r, column c the numerator factor is $N + c - r$ and the hook length counts that cell plus the cells directly below and directly to its right. A single box gives $N/1 = N$; the two-box column gives, for $N=3$, numerator $(3)(3-1) = 6$ and hooks $(2)(1)$, so dim $= 3 = \bar{\mathbf{3}}$; the two-box row gives numerator $(3)(3+1) = 12$ and hooks $(2)(1)$, so dim $= 6 = \mathbf{6}$ — reproducing (2.226) dimensionally without ever drawing a weight. We use the tableau only as a shortcut here; the catalogued SU(N) tableau dimensions and product rules are collected in Chapter B.

2.7.7 The quadratic Casimir and the Dynkin index

Two invariants attach to every representation R and are the numbers that gauge theory actually consumes. The *Dynkin index* $T(R)$ is fixed by the trace normalization (Theorem B.2),

$$\mathrm{tr}(T_R^a\, T_R^b) = T(R)\,\delta^{ab}, \tag{2.230}$$

and the *quadratic Casimir* $C_2(R)$ by the value of the (basis-independent) operator $\sum_a T_R^a T_R^a$, which commutes with all generators and so, by Schur's lemma on an

irreducible R, is a multiple of the identity (Theorem B.4),

$$\sum_{a=1}^{8} T_R^a T_R^a = C_2(R)\,\mathbf{1}_R. \tag{2.231}$$

Taking the trace of (2.231) and using (2.230) on the left ties the two together for any R:

$$\dim(R)\,C_2(R) = \dim(G)\,T(R), \tag{2.232}$$

the master identity Theorem B.5, with $\dim(G) = 8$ for $\mathfrak{su}(3)$. We now compute both invariants for the fundamental and the adjoint and verify (2.232) in each case.

Worked Example 2.80 (C_2 and T for **3** and **8**)**.** *Fundamental.* The index is the normalization itself: with $T^a = \frac{1}{2}\lambda^a$ and $\mathrm{tr}(\lambda^a\lambda^b) = 2\delta^{ab}$ (Eq. (B.60)), one has $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, so

$$T(\mathbf{3}) = T_F = \tfrac{1}{2}. \tag{2.233}$$

For the Casimir, trace (2.231) over the three-dimensional space. The left side is $\sum_a \mathrm{tr}(T^aT^a) = 8\cdot\frac{1}{2} = 4$; the right side is $C_2(\mathbf{3})\ \mathrm{tr}\,\mathbf{1}_3 = 3\,C_2(\mathbf{3})$. Hence

$$C_2(\mathbf{3}) = C_F = \frac{4}{3}. \tag{2.234}$$

This is the $N = 3$ value of the general SU(N) result $C_F = (N^2-1)/2N$ (Eq. (B.70)): $(9-1)/6 = 4/3$. The master identity (2.232) reads $3\cdot\frac{4}{3} = 8\cdot\frac{1}{2}$, i.e. $4 = 4$. ✓

Adjoint. Here the generators are the structure constants, $(T^a_{\mathrm{adj}})_{bc} = -if^{abc}$, so the index and Casimir are both expressed through the contraction $f^{acd}f^{bcd}$. By Theorem B.28 this contraction is $f^{acd}f^{bcd} = C_A\,\delta^{ab}$ with $C_A = N = 3$ for $\mathfrak{su}(3)$. The Casimir of the adjoint follows from (2.231) by writing out the matrix product, summing the internal index e and the generator label a:

$$\begin{aligned}\left(\textstyle\sum_a T^a_{\mathrm{adj}}T^a_{\mathrm{adj}}\right)_{bc} &= \sum_{a,e}(T^a_{\mathrm{adj}})_{be}(T^a_{\mathrm{adj}})_{ec} = \sum_{a,e}(-if^{abe})(-if^{aec}) = -\sum_{a,e} f^{abe}f^{aec}\\ &= \sum_{a,e} f^{abe}f^{ace} = \sum_{a,e} f^{bae}f^{cae} = C_A\,\delta_{bc}.\end{aligned} \tag{2.235}$$

The fourth equality uses $f^{aec} = -f^{ace}$ to cancel the leading minus sign, and the fifth uses the antisymmetry $f^{abe} = -f^{bae}$, $f^{ace} = -f^{cae}$ (whose two sign flips cancel); the resulting sum $\sum_{a,e} f^{bae}f^{cae}$ is exactly the adjoint contraction $f^{acd}f^{bcd} = C_A\delta^{ab}$ of Theorem B.28, with the free pair (b,c) and summed pair (a,e). Hence $C_2(\mathbf{8}) = C_A = 3$. The index is read from the same contraction through (2.230):

$$\mathrm{tr}(T^a_{\mathrm{adj}}T^b_{\mathrm{adj}}) = \sum_{c,d}(T^a_{\mathrm{adj}})_{cd}(T^b_{\mathrm{adj}})_{dc} = \sum_{c,d}(-if^{acd})(-if^{bdc}) = -\sum_{c,d} f^{acd}f^{bdc} = \sum_{c,d} f^{acd}f^{bcd} = C \tag{2.236}$$

using $f^{bdc} = -f^{bcd}$ in the penultimate step, so $T(\mathbf{8}) = C_A = 3$. The adjoint is the case where index and Casimir coincide, which the master identity makes inevitable: since $\dim(\mathbf{8}) = \dim(G) = 8$, (2.232) collapses to $C_2(\mathbf{8}) = T(\mathbf{8})$, and indeed $8 \cdot 3 = 8 \cdot 3$. ✓

We record the $\mathfrak{su}(3)$ data the gauge-theory chapters will quote:

$$T_F = \tfrac{1}{2}, \qquad C_F = \tfrac{4}{3}, \qquad C_A = 3, \tag{2.237}$$

the $N = 3$ specialization of the SU(N) family $T_F = \frac{1}{2}$, $C_F = (N^2-1)/2N$, $C_A = N$ catalogued, with full proofs, in Chapter B. The numbers are not decorative. In the one-loop Yang–Mills β-function the combination $\frac{11}{3}C_A - \frac{4}{3}T_F n_f$ decides the sign of the running coupling; with $C_A = 3$ and $T_F = \frac{1}{2}$ it becomes $11 - \frac{2}{3}n_f$, positive for the six quark flavors of QCD, which is asymptotic freedom. The Casimir $C_F = \frac{4}{3}$ sets the strength with which a quark radiates gluons. Chapter 27 computes that β-function; the inputs are exactly (2.237).

With $\mathfrak{su}(3)$ the Cartan–Weyl programme of Chapter 2 has paid off into a complete, computable example of the higher-rank theory, and it hands forward the two pieces of machinery the rest of the book runs on. To Chapter 3 it hands the confidence that the root/weight method survives beyond rank one, the very technique by which the Lorentz algebra is unwound into $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$. To Yang–Mills theory — the classical non-abelian Lagrangian of Chapter 25, its quantization in Chapter 26, and the asymptotic-freedom computation of Chapter 27 — it hands the literal alphabet: the Hermitian generators T^a and structure constants f^{abc}, the inequivalent color representations $\mathbf{3}$, $\bar{\mathbf{3}}$, and $\mathbf{8}$, and the invariants $C_F = \frac{4}{3}$, $C_A = 3$, $T_F = \frac{1}{2}$ of (2.237), so that the gauge-theory part of this book reads as growing out of these foundations rather than starting afresh.

Further reading

The physicist's route through this material, emphasizing $\mathfrak{su}(2)$, $\mathfrak{su}(3)$, roots, and weights with an eye to particle physics, is Georgi [218]; Zee [532] (the group-theory chapters) covers the same ground with the same motivations and many worked examples. For the representation-theoretic perspective developed rigorously—Schur, characters, the highest-weight theorem, and the full classification of simple Lie algebras whose proof we only scoped—see the mathematics texts Fulton and Harris [206] and Hall [248], the latter especially careful about the exponential map and the Lie group/Lie algebra correspondence used in Section 2.3. Nakahara [355] places Lie groups and algebras inside the broader differential-geometric setting that later chapters on gauge theory and topology will need. Weinberg [500] (chapter 2) gives the physical companion to Chapter 3: how these representations organize the states of a relativistic quantum theory, the form in which group theory will be used for the rest of the book.

Chapter 3

The Lorentz and Poincaré Groups

Chapter 2 ended on a promise. It built the language of groups and their representations, solved $\mathfrak{su}(2)$ completely, and argued—following Wigner—that an elementary particle is not an object with properties bolted on but *is* an irreducible unitary representation of the symmetry group of spacetime, its mass and spin the two invariant labels of that representation. This chapter redeems the promise for the group that relativity actually hands us. It asks one question and answers it exhaustively: *what can a relativistic particle be?* The premise is the principle of relativity, that the laws of physics look the same to every inertial observer; the transformations relating those observers form the Lorentz group, enlarged by spacetime translations to the Poincaré group, and a relativistic field or particle is forced to carry one of its representations. Determining those representations—finite-dimensional for fields, infinite-dimensional and unitary for particles—is the whole of the chapter.

We import freely from the foundations laid next door. The matrix-Lie-group machinery, the exponential map (Eq. (2.60)), and the linearization that trades a connected group for its Lie algebra are the tools of Section 2.3; the entire engine is the complete representation theory of $\mathfrak{su}(2)$, its spin-j irreducibles of dimension $2j+1$ (Theorem 2.53), Schur's lemma (Theorem 2.21), and the Clebsch–Gordan series (Section 2.5). Even the topological input we need—that a 2π rotation is a noncontractible loop, so that SU(2) double covers SO(3)—is the $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$ established there. Throughout we hold the global conventions fixed in Chapter A and mirrored in the chapter introduction: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, and Hermitian generators normalized so that group elements are $\exp(-\frac{i}{2}\omega_{\mu\nu}M^{\mu\nu})$. The two-component spinor conventions and the chiral gamma-matrix basis introduced here are the ones collected in Chapter C.

The argument runs as a single road. We begin with the Lorentz group itself (Section 3.1), reading off its global shape—four disconnected components, non-compactness, and the $\mathbb{Z}_2$ fundamental group that will permit spinors—and

isolating the proper orthochronous component $\mathrm{SO}(1,3)^+$ as the piece whose representations we seek, with parity and time reversal standing outside it as discrete symmetries. Linearizing $\mathrm{SO}(1,3)^+$ produces the Lorentz Lie algebra and its six generators (Section 3.2); a single complex change of basis then splits the complexified algebra into two commuting copies of $\mathfrak{su}(2)$, and the spin classification of Chapter 2 delivers at once the (j_-, j_+) catalogue of fields—the scalar, the two Weyl spinors, the four-vector, the field strength (Section 3.3). We make those spinor labels concrete by building the double cover $\mathrm{SL}(2,\mathbb{C})$ and its Weyl, Dirac, and Majorana representations, and we show that parity is what forces the electron to be a Dirac, not a Weyl, field (Section 3.4).

These finite-dimensional representations describe fields, never states: being non-unitary, they cannot act on a Hilbert space of probabilities. The remedy is to restore the rest of the symmetry. Adjoining translations builds the Poincaré group and its algebra (Section 3.5), whose energy–momentum generators have continuous spectrum and therefore force the unitary representations to be infinite-dimensional. Two Casimir operators—the mass-squared P^2 and the Pauli–Lubański square W^2—survive to label them (Section 3.6), and Wigner's method of induced representations reduces the entire classification to the representation theory of a single small subgroup, the little group. Executing that program orbit by orbit (Section 3.7) yields the chapter's thesis: a relativistic particle is a unitary irreducible representation of the Poincaré group, one for each $(m > 0, s)$ and each massless helicity h—and the massless case, with its non-compact little group, forces gauge redundancy upon any covariant description of the photon. The cargo is then handed forward to the threads: the scalar of ϕ^4 theory, the electron of QED, and the photon and gluon of QED and Yang–Mills theory each leave this chapter knowing exactly what they are, ready to be quantized in Part III.

3.1 The Lorentz group: spacetime symmetry and its components

The chapter's guiding question — *what can a relativistic particle be?* — becomes a precise mathematical problem the moment we accept its premise: the laws of physics must look the same to all inertial observers, and the transformations relating those observers form a group. If a particle, or a field, is to be a physical object at all, the way it changes under a change of observer cannot be arbitrary; it must furnish a representation of that group, in the exact sense built abstractly in Chapter 2. The entire chapter is the determination of those representations. Before we can classify the representations of a group, however, we must know the group itself — not merely its multiplication law but its global shape: how many disconnected pieces it has, which piece is the one continuously connected to "no transformation at all," whether that piece is compact, and what its loops

look like. Each of these topological facts will turn out to control a piece of the physics — the existence of parity and time reversal as separate operations, the impossibility of finite-dimensional *unitary* representations, the existence of half-integer spin. This first section establishes those facts. We define the group, isolate the component whose representations we will spend the chapter classifying, and read off the topology that drives everything later.

We work throughout in natural units $\hbar = c = 1$ and with the mostly-minus Minkowski metric

$$\eta_{\mu\nu} = \eta^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \tag{3.1}$$

the particle-physics convention fixed in Chapter A and used in every chapter of this book where a sign matters. Greek indices $\mu, \nu, \ldots$ run over $0, 1, 2, 3$; Latin indices $i, j, \ldots$ run over the spatial values $1, 2, 3$; the Einstein summation convention is in force, and indices are raised and lowered with η. A point of spacetime is $x^\mu = (x^0, \mathbf{x}) = (t, \mathbf{x})$.

3.1.1 Minkowski space and the defining relation

Special relativity asserts that the geometry of spacetime is encoded not in a notion of separate spatial distance and elapsed time — both of which are observer-dependent — but in a single invariant, the spacetime interval between two events. For events separated by Δx^μ this interval is the quadratic form

$$\Delta s^2 \;=\; \eta_{\mu\nu}\,\Delta x^\mu \Delta x^\nu \;=\; (\Delta x^0)^2 - |\Delta \mathbf{x}|^2 \,. \tag{3.2}$$

Its sign classifies the separation as timelike ($\Delta s^2 > 0$), spacelike ($\Delta s^2 < 0$), or null ($\Delta s^2 = 0$), and these classifications are physical: they decide whether one event can causally influence another. The principle of relativity, in its kinematic form, is the statement that the admissible changes of inertial frame are exactly the coordinate transformations that leave Eq. (3.2) invariant. We restrict attention here to the *linear, homogeneous* such transformations — those fixing the origin; adjoining the spacetime translations to obtain the full Poincaré group is the business of a later section. A linear change of coordinates is a matrix Λ acting by

$$x^\mu \;\longmapsto\; x'^\mu = \Lambda^\mu{}_\nu \, x^\nu . \tag{3.3}$$

Definition 3.1 (Lorentz transformation and the Lorentz group)**.** A *Lorentz transformation* is a real 4×4 matrix $\Lambda = (\Lambda^\mu{}_\nu)$ that leaves the Minkowski interval Eq. (3.2) invariant, i.e. such that $\eta_{\mu\nu}\, x'^\mu x'^\nu = \eta_{\rho\sigma}\, x^\rho x^\sigma$ for every $x^\mu \in \mathbb{R}^4$. The set of all such matrices is the *Lorentz group* $\mathrm{O}(1,3)$.

The defining property translates into a single algebraic relation. Substituting Eq. (3.3) into the invariance condition gives $\eta_{\mu\nu}\Lambda^\mu{}_\rho \Lambda^\nu{}_\sigma \, x^\rho x^\sigma = \eta_{\rho\sigma}\, x^\rho x^\sigma$ for all

x. Because the quadratic form on each side is symmetric in ρ, σ and must agree for arbitrary x^ρ, the coefficient tensors must be equal:

$$\eta_{\mu\nu}\,\Lambda^{\mu}{}_{\rho}\,\Lambda^{\nu}{}_{\sigma} \;=\; \eta_{\rho\sigma}. \tag{3.4}$$

In matrix notation, writing Λ^{T} for the transpose $(\Lambda^{\mathsf{T}})_{\rho}{}^{\mu} = \Lambda^{\mu}{}_{\rho}$, this is the compact statement

$$\boxed{\Lambda^{\mathsf{T}}\,\eta\,\Lambda \;=\; \eta}\,. \tag{3.5}$$

Equation (3.5) is the defining relation of the Lorentz group, the exact analogue of the relation $R^{\mathsf{T}}\mathbf{1}\,R = \mathbf{1}$ defining the orthogonal group $\mathrm{O}(3)$, with the Euclidean metric $\mathbf{1}$ replaced by the Lorentzian η. Everything in this section is squeezed out of it.

That $\mathrm{O}(1,3)$ is a group is immediate from Eq. (3.5) and the general criterion for matrix groups in Chapter 2. The identity $\mathbf{1}$ satisfies it trivially. If Λ_1, Λ_2 both satisfy it, so does the product, since $(\Lambda_1\Lambda_2)^{\mathsf{T}}\eta(\Lambda_1\Lambda_2) = \Lambda_2^{\mathsf{T}}(\Lambda_1^{\mathsf{T}}\eta\,\Lambda_1)\Lambda_2 = \Lambda_2^{\mathsf{T}}\eta\,\Lambda_2 = \eta$. Taking determinants in Eq. (3.5) (anticipating the next subsection) shows $\det\Lambda \neq 0$, so Λ is invertible; and multiplying $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$ on the left by $(\Lambda^{\mathsf{T}})^{-1}$ and on the right by Λ^{-1} gives $\eta = (\Lambda^{-1})^{\mathsf{T}}\eta\,\Lambda^{-1}$, so $\Lambda^{-1} \in \mathrm{O}(1,3)$ as well. Moreover $\mathrm{O}(1,3)$ is closed as a subset of the 16-dimensional space of real 4×4 matrices — it is the common zero set of the polynomial equations Eq. (3.4) — and the group operations are polynomial in the entries, so $\mathrm{O}(1,3)$ is a matrix Lie group in the sense of Chapter 2. What we do not yet know, and what the rest of the section is about, is its *global* structure.

3.1.2 Two invariant signs: the determinant and the time component

The defining relation forces two numbers built from Λ to take only the values ± 1 and to be "locally constant" across the group. These two discrete invariants are what break $\mathrm{O}(1,3)$ into pieces.

Proposition 3.2 (The two discrete invariants)**.** *Every* $\Lambda \in \mathrm{O}(1,3)$ *satisfies*

$$\det\Lambda = \pm 1, \tag{3.6}$$

$$(\Lambda^0{}_0)^2 = 1 + \sum_{i=1}^{3}(\Lambda^i{}_0)^2 \;\geq\; 1, \qquad \textit{so}\ \ \left|\Lambda^0{}_0\right| \geq 1. \tag{3.7}$$

Consequently $\det\Lambda \in \{+1,-1\}$ *and* $\Lambda^0{}_0 \in [1,\infty)\cup(-\infty,-1]$*; in particular* $\Lambda^0{}_0$ *never takes a value in the open interval* $(-1,1)$*.*

Proof. For Eq. (3.6), take the determinant of both sides of Eq. (3.5). Using $\det(ABC) = \det A\det B\det C$, $\det(\Lambda^{\mathsf{T}}) = \det\Lambda$, and $\det\eta = -1$,

$$(\det\Lambda)^2\,\det\eta = \det\eta \;\Longrightarrow\; (\det\Lambda)^2(-1) = -1 \;\Longrightarrow\; (\det\Lambda)^2 = 1, \tag{3.8}$$

hence $\det\Lambda = \pm 1$. In particular $\det\Lambda \neq 0$, justifying the invertibility claimed above.

For Eq. (3.7), evaluate the component form Eq. (3.4) at $\rho = \sigma = 0$. The right side is $\eta_{00} = +1$, and the left side is

$$\eta_{\mu\nu}\,\Lambda^{\mu}{}_{0}\,\Lambda^{\nu}{}_{0} = (\Lambda^{0}{}_{0})^{2} - \sum_{i=1}^{3}(\Lambda^{i}{}_{0})^{2}, \tag{3.9}$$

where the relative minus signs come from the spatial entries of η. Setting the two equal and rearranging gives exactly Eq. (3.7). The sum of squares on the right is ≥ 0, so $(\Lambda^{0}{}_{0})^{2} \geq 1$ and $|\Lambda^{0}{}_{0}| \geq 1$. □

The geometric content of Eq. (3.7) is worth naming. The first column $\Lambda^{\mu}{}_{0}$ of Λ is the image of the unit time vector $e_0 = (1, 0, 0, 0)$, i.e. the velocity-four-vector of the transformed time axis. The relation says this image is a *unit timelike* vector ($\eta_{\mu\nu}\Lambda^{\mu}{}_{0}\Lambda^{\nu}{}_{0} = 1 > 0$); a Lorentz transformation cannot tilt the time axis so far that it becomes spacelike. The two sheets $\Lambda^{0}{}_{0} \geq 1$ and $\Lambda^{0}{}_{0} \leq -1$ correspond to the transformed time axis pointing into the future or into the past light cone respectively. This is the kinematic root of the arrow of time being a discrete, not continuous, attribute of a Lorentz transformation.

We now record the names attached to the two signs.

Definition 3.3 (Proper/improper, orthochronous/non-orthochronous)**.** A Lorentz transformation Λ is called *proper* if $\det\Lambda = +1$ and *improper* if $\det\Lambda = -1$; it is called *orthochronous* if $\Lambda^{0}{}_{0} \geq 1$ and *non-orthochronous* if $\Lambda^{0}{}_{0} \leq -1$. The four sign combinations partition $\mathrm{O}(1,3)$ into four disjoint subsets,

$$L^{\uparrow}_{+}, \quad L^{\uparrow}_{-}, \quad L^{\downarrow}_{+}, \quad L^{\downarrow}_{-}, \tag{3.10}$$

where the subscript records $\mathrm{sgn}(\det\Lambda)$ and the arrow records $\mathrm{sgn}(\Lambda^{0}{}_{0})$ (up for orthochronous, down for non-orthochronous). The *proper orthochronous* subset is denoted $L^{\uparrow}_{+}$, also written $\mathrm{SO}(1,3)^{+}$.

These two signs are not merely labels assigned to a static set; their decisive feature is that the orthochronous sign is *multiplicative*, exactly as the determinant is. This is the technical heart of the section.

Lemma 3.4 (Conservation of the time sign under composition)**.** *For any* $\Lambda, \Lambda' \in \mathrm{O}(1,3)$*,*

$$\mathrm{sgn}\left((\Lambda\Lambda')^{0}{}_{0}\right) = \mathrm{sgn}(\Lambda^{0}{}_{0})\ \mathrm{sgn}(\Lambda'^{0}{}_{0}). \tag{3.11}$$

Moreover $(\Lambda^{-1})^{0}{}_{0} = \Lambda^{0}{}_{0}$*, so the inverse has the same time sign. Together with the multiplicativity of the determinant, this makes the two signs a homomorphism* $\mathrm{O}(1,3) \to \{\pm 1\} \times \{\pm 1\}$.

Proof. Write the matrix product in components:

$$(\Lambda\Lambda')^0{}_0 = \Lambda^0{}_\mu\,\Lambda'^\mu{}_0 = \Lambda^0{}_0\,\Lambda'^0{}_0 + \sum_{i=1}^{3}\Lambda^0{}_i\,\Lambda'^i{}_0. \tag{3.12}$$

The claim is that the cross term $\sum_i \Lambda^0{}_i\Lambda'^i{}_0$ is never large enough to overturn the sign of the leading term $\Lambda^0{}_0\Lambda'^0{}_0$. Introduce the two spatial vectors $\mathbf{a} = (\Lambda^0{}_1, \Lambda^0{}_2, \Lambda^0{}_3)$ and $\mathbf{b} = (\Lambda'^1{}_0, \Lambda'^2{}_0, \Lambda'^3{}_0)$, so that the cross term is the Euclidean dot product $\mathbf{a}\cdot\mathbf{b}$.

We need the Euclidean norms of $\mathbf{a}$ and $\mathbf{b}$. For $\mathbf{b}$, Eq. (3.7) applied to Λ' gives directly $|\mathbf{b}|^2 = \sum_i(\Lambda'^i{}_0)^2 = (\Lambda'^0{}_0)^2 - 1$. For $\mathbf{a}$ we need the analogue of Eq. (3.7) for the top *row* of Λ. This comes from the companion relation $\Lambda\,\eta\,\Lambda^{\mathsf T} = \eta$, which follows from Eq. (3.5): since $\Lambda^{\mathsf T}\eta\Lambda = \eta$ and $\eta^2 = \mathbf{1}$, we have $\Lambda^{-1} = \eta\,\Lambda^{\mathsf T}\eta$, and inserting this into $\Lambda\Lambda^{-1} = \mathbf{1}$ yields $\Lambda\,\eta\,\Lambda^{\mathsf T}\eta = \mathbf{1}$, i.e. $\Lambda\,\eta\,\Lambda^{\mathsf T} = \eta$. Its $(0,0)$ component reads $(\Lambda^0{}_0)^2 - \sum_i(\Lambda^0{}_i)^2 = 1$, so $|\mathbf{a}|^2 = (\Lambda^0{}_0)^2 - 1$.

By the Cauchy–Schwarz inequality $|\mathbf{a}\cdot\mathbf{b}| \le |\mathbf{a}|\;|\mathbf{b}| = \sqrt{\big((\Lambda^0{}_0)^2-1\big)\big((\Lambda'^0{}_0)^2-1\big)}$. Abbreviating $u = |\Lambda^0{}_0| \ge 1$ and $v = \left|\Lambda'^0{}_0\right| \ge 1$, the radicand is $(u^2-1)(v^2-1) = u^2v^2 - (u^2+v^2-1)$. Since $u^2, v^2 \ge 1$ we have $u^2+v^2-1 \ge 1 > 0$, hence $(u^2-1)(v^2-1) < u^2v^2$ and therefore

$$|\mathbf{a}\cdot\mathbf{b}| \;<\; uv \;=\; \left|\Lambda^0{}_0\,\Lambda'^0{}_0\right|. \tag{3.13}$$

The leading term in Eq. (3.12) therefore strictly dominates the cross term in magnitude, so the sum has the same sign as $\Lambda^0{}_0\Lambda'^0{}_0$, which is Eq. (3.11).

For the inverse, recall $\Lambda^{-1} = \eta\,\Lambda^{\mathsf T}\eta$. Because η is diagonal, the $(0,0)$ entry of the right side is $\eta_{00}(\Lambda^{\mathsf T})^0{}_0\eta_{00} = (+1)\,\Lambda^0{}_0\,(+1) = \Lambda^0{}_0$, so $(\Lambda^{-1})^0{}_0 = \Lambda^0{}_0$ as claimed. The determinant is multiplicative on any group of matrices and equal on Λ and Λ^{-1} up to the value ±1 here, so the pair $\big(\operatorname{sgn}\det\Lambda,\ \operatorname{sgn}\Lambda^0{}_0\big)$ is a group homomorphism into $\{\pm1\}^2$. □

Theorem 3.4 has an immediate structural payoff. The two signs being a homomorphism into $\{\pm1\}^2$ means the preimage of the identity element $(+1,+1)$ — namely $L_+^\uparrow = \mathrm{SO}(1,3)^+$, the proper orthochronous set — is a *subgroup*, indeed a normal subgroup, and the other three sets are its cosets.

Proposition 3.5 ($\mathrm{SO}(1,3)^+$ is a normal subgroup of index four)**.** *$L_+^\uparrow = \mathrm{SO}(1,3)^+$ is a normal subgroup of $\mathrm{O}(1,3)$, and the four sets Eq. (3.10) are its four cosets. The quotient $\mathrm{O}(1,3)/\mathrm{SO}(1,3)^+$ is the Klein four-group $\mathbb{Z}_2\times\mathbb{Z}_2$.*

Proof. $\mathrm{SO}(1,3)^+$ is the kernel of the homomorphism $\Phi : \mathrm{O}(1,3) \to \{\pm1\}\times\{\pm1\}$, $\Phi(\Lambda) = (\det\Lambda, \operatorname{sgn}\Lambda^0{}_0)$, of Theorem 3.4, hence a normal subgroup. By the first isomorphism theorem (Chapter 2) the cosets are the fibres of Φ, which are precisely the four sign-sectors, and $\mathrm{O}(1,3)/\mathrm{SO}(1,3)^+ \cong \operatorname{im}\Phi$. That the image is all of $\{\pm1\}^2$ — so the quotient is $\mathbb{Z}_2\times\mathbb{Z}_2$ and there are genuinely four nonempty sectors — is witnessed by the four representatives in Eq. (3.14) below. □

3.1.3 The discrete operations and the four components

The cosets are reached from $\mathrm{SO}(1,3)^+$ by two distinguished discrete transformations, which flip the two signs independently. *Parity* (space inversion) and *time reversal* are the matrices

$$P = \mathrm{diag}(+1,-1,-1,-1) = \eta, \qquad T = \mathrm{diag}(-1,+1,+1,+1) = -\eta, \tag{3.14}$$

together with their product $PT = -\mathbf{1} = \mathrm{diag}(-1,-1,-1,-1)$. Each manifestly satisfies Eq. (3.5) (for $P = \eta$, $P^{\mathsf{T}}\eta P = \eta\,\eta\,\eta = \eta$ since $\eta^2 = \mathbf{1}$; likewise for $T = -\eta$ and $PT = -\mathbf{1}$), so all three are genuine Lorentz transformations. Their invariants are read straight off the diagonals:

	det	$\Lambda^0{}_0$	sector
$\mathbf{1}$	$+1$	$+1$	$L^{\uparrow}_{+} = \mathrm{SO}(1,3)^+$
P	-1	$+1$	$L^{\uparrow}_{-}$
T	-1	-1	$L^{\downarrow}_{-}$
PT	$+1$	-1	$L^{\downarrow}_{+}$

(3.15)

Thus $\{\mathbf{1}, P, T, PT\}$ is a set of coset representatives, one in each sector, and every Lorentz transformation factors uniquely as one of these times an element of $\mathrm{SO}(1,3)^+$. The names matter physically: P and T are the discrete spacetime operations whose action on every field of the book — scalar, spinor, vector — is inherited from Eq. (3.14), and whose possible *violation* by the weak interaction is one of the central facts of the Standard Model. Here they enter purely as the topological connectors between the components, but we fix the conventions Eq. (3.14) once and for all, as flagged in the chapter program; all three benchmark theories inherit them.

We can now state precisely how the four sectors sit inside the group as a topological space. The key point is that the two sign invariants are not only locally constant algebraically; they are *continuous* functions on $\mathrm{O}(1,3)$, and a continuous function valued in $\{\pm 1\}$ cannot change without passing through forbidden values.

Theorem 3.6 (Four disconnected components)**.** *The Lorentz group* $\mathrm{O}(1,3)$ *has exactly four connected components, the four sets Eq. (3.10). The identity component is* $\mathrm{SO}(1,3)^+$*, and the other three are obtained from it by left multiplication by* P*,* T*, and* PT *respectively.*

Proof. The four sectors are mutually separated. The map $\Lambda \mapsto \det\Lambda$ is a polynomial in the entries of Λ, hence continuous on $\mathrm{O}(1,3)$, and takes only the values ± 1 by Eq. (3.6). The map $\Lambda \mapsto \Lambda^0{}_0$ is the projection onto a single matrix entry, also continuous, and by Eq. (3.7) it never takes a value in the open interval $(-1,1)$; therefore $\Lambda \mapsto \mathrm{sgn}\,\Lambda^0{}_0$ is continuous on $\mathrm{O}(1,3)$ (its only possible discontinuity would be at a zero crossing, which is excluded). A continuous map into the

discrete two-point set $\{\pm 1\}$ is constant on each connected component. Hence both signs are constant on each component, so each component lies entirely within one of the four sectors; equivalently, no continuous path in $\mathrm{O}(1,3)$ can join two different sectors. The four sectors are therefore unions of components, pairwise disjoint and open-and-closed in $\mathrm{O}(1,3)$.

Each sector is itself connected. It suffices to prove this for $\mathrm{SO}(1,3)^+$, since left multiplication by the fixed invertible matrix P, T, or PT is a homeomorphism of $\mathrm{O}(1,3)$ carrying $\mathrm{SO}(1,3)^+$ onto the other three sectors and preserving connectedness; this is established as Theorem 3.7 below. Granting it, each sector is connected, so the four sectors are exactly the four connected components, and the one containing $\mathbf{1}$ is $\mathrm{SO}(1,3)^+$. □

It remains to prove that $\mathrm{SO}(1,3)^+$ is connected. We do this with an argument that will also be the prototype for the Wigner classification later in the chapter: a group that acts transitively on a connected space, with connected stabilizer, is itself connected.

Proposition 3.7 ($\mathrm{SO}(1,3)^+$ is path-connected)**.** *$\mathrm{SO}(1,3)^+$ is path-connected; in particular it is the connected component of the identity in $\mathrm{O}(1,3)$.*

Proof. Consider the action of $\mathrm{SO}(1,3)^+$ on the *forward unit hyperboloid*

$$H^+ = \{\, p^\mu \in \mathbb{R}^4 : \eta_{\mu\nu} p^\mu p^\nu = 1,\ p^0 > 0 \,\}, \tag{3.16}$$

the upper sheet of the two-sheeted hyperboloid of unit future-pointing timelike vectors (H^+ is the set of four-velocities, and after rescaling the mass shell of a particle of unit mass). As a graph $p^0 = \sqrt{1+|\mathbf{p}|^2}$ over $\mathbf{p} \in \mathbb{R}^3$, H^+ is homeomorphic to $\mathbb{R}^3$, hence connected.

The action is transitive. Take any $p \in H^+$. We exhibit a proper orthochronous Λ with $\Lambda\, e_0 = p$, where $e_0 = (1,0,0,0)$. If $\mathbf{p} = 0$ then $p = e_0$ and $\Lambda = \mathbf{1}$ works. Otherwise write $|\mathbf{p}| = \sinh\zeta$ with $\zeta > 0$ (so $p^0 = \cosh\zeta$, consistent with $p \in H^+$), let $\hat{\mathbf{n}} = \mathbf{p}/|\mathbf{p}|$ be the unit spatial direction, and define the boost of rapidity ζ along $\hat{\mathbf{n}}$,

$$B(\zeta\hat{\mathbf{n}})^0{}_0 = \cosh\zeta, \quad B^0{}_i = B^i{}_0 = \hat{n}_i \sinh\zeta, \quad B^i{}_j = \delta_{ij} + (\cosh\zeta - 1)\hat{n}_i\hat{n}_j. \tag{3.17}$$

A direct check (carried out for the special case $\hat{\mathbf{n}} = \hat{\mathbf{x}}$ in Theorem 3.8, and reducing to it after a spatial rotation) shows $B(\zeta\hat{\mathbf{n}}) \in \mathrm{SO}(1,3)^+$ and $B(\zeta\hat{\mathbf{n}})\, e_0 = (\cosh\zeta, \sinh\zeta\, \hat{\mathbf{n}}) = p$. Hence the action is transitive.

The stabilizer of e_0 is connected. If $\Lambda \in \mathrm{SO}(1,3)^+$ fixes e_0, then its first column is $\Lambda^\mu{}_0 = \delta^\mu_0$. The defining relation $\Lambda\eta\Lambda^{\mathsf{T}} = \eta$ at $(0,i)$ then forces $\Lambda^0{}_i = 0$ as well, so Λ is block diagonal, $\Lambda = \mathrm{diag}(1, R)$ with R a real 3×3 matrix. The remaining content of the defining relation is $R^{\mathsf{T}}R = \mathbf{1}$, i.e. $R \in \mathrm{O}(3)$, and $\det\Lambda = \det R = +1$ selects $R \in \mathrm{SO}(3)$. The stabilizer is therefore a copy of the

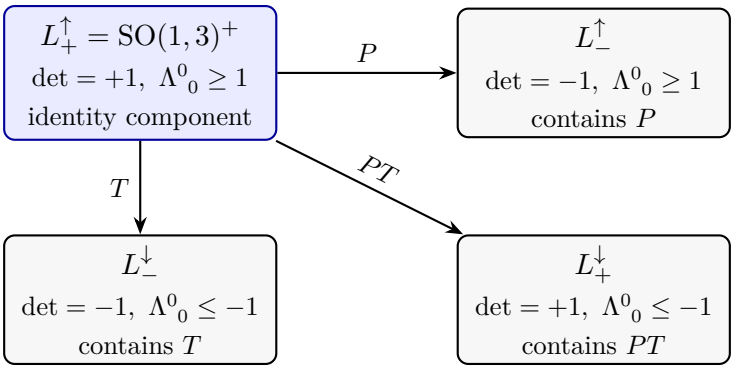


Figure 3.1: The four connected components of the Lorentz group $\mathrm{O}(1,3)$, labelled by the two invariant signs $\det\Lambda = \pm 1$ and $\operatorname{sgn}\Lambda^0{}_0 = \pm 1$. Only the proper orthochronous component $\mathrm{SO}(1,3)^+$ contains the identity and is a subgroup; the discrete operations $P = \mathrm{diag}(+,-,-,-)$, $T = \mathrm{diag}(-,+,+,+)$, and $PT = -\mathbf{1}$ map it onto the other three. The representation theory of this chapter is the representation theory of $\mathrm{SO}(1,3)^+$; P and T are imposed afterwards as discrete symmetries.

rotation group $\mathrm{SO}(3)$, which is path-connected (Chapter 2: every rotation is exp of an antisymmetric generator, and the segment $s \mapsto \exp(s\,\omega)$, $s \in [0,1]$, joins it to the identity).

Connectedness of the total group. The orbit map $\pi : \mathrm{SO}(1,3)^+ \to H^+$, $\pi(\Lambda) = \Lambda e_0$, identifies H^+ with the homogeneous space $\mathrm{SO}(1,3)^+/\mathrm{SO}(3)$ and, for a closed subgroup of a Lie group, is a locally trivial fibre bundle (Chapter 2); in particular it has the path-lifting property. Now let $\Lambda \in \mathrm{SO}(1,3)^+$ be arbitrary. Since H^+ is path-connected, choose a path γ in H^+ from $\pi(\Lambda) = \Lambda e_0$ to $\pi(\mathbf{1}) = e_0$; lift it through π to a path in $\mathrm{SO}(1,3)^+$ starting at Λ and ending at some h with $\pi(h) = e_0$, i.e. $h \in \mathrm{SO}(3)$. Since $\mathrm{SO}(3)$ is path-connected, join h to $\mathbf{1}$ by a path inside $\mathrm{SO}(3) \subset \mathrm{SO}(1,3)^+$. Concatenating, Λ is joined to $\mathbf{1}$ by a path in $\mathrm{SO}(1,3)^+$. As Λ was arbitrary, $\mathrm{SO}(1,3)^+$ is path-connected. □

Theorem 3.7 completes the proof of Theorem 3.6: $\mathrm{O}(1,3)$ has exactly four connected components, the proper orthochronous one $\mathrm{SO}(1,3)^+$ is the identity component, and P, T, PT are the discrete operations that connect it to the other three. The geometry is summarized in Fig. 3.1.

There is a strategic conclusion to draw before we go on. Because P and T are disconnected from the identity, no continuous (let alone infinitesimal) symmetry argument can produce them; they are extra, optional structure. The representations we can build by exponentiating infinitesimal generators — the program of the next sections — are representations of the *identity component* $\mathrm{SO}(1,3)^+$ alone. Whether a given field or particle also carries a representation of P or T is a separate question, answered theory by theory (the neutrino's

answer differs from the electron's). This is why we single out $\mathrm{SO}(1,3)^+$ as "the connected piece whose representations we seek": it is the largest part of the symmetry group reachable from the identity, and hence the part forced on us by relativity together with the requirement of a continuous symmetry.

3.1.4 A worked transformation: boosts, rotations, and rapidity

Before extracting the topology of $\mathrm{SO}(1,3)^+$ we make the abstract definition concrete by exhibiting its two basic kinds of element and verifying the defining relation by hand. This also pins down the parametrization that will make the non-compactness transparent.

Worked Example 3.8 (A boost and a rotation, checked against $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$)**.**

The boost. The standard boost along the x^1-axis with rapidity $\zeta \in \mathbb{R}$ is

$$B_1(\zeta) = \begin{pmatrix} \cosh\zeta & \sinh\zeta & 0 & 0 \\ \sinh\zeta & \cosh\zeta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.18}$$

The rapidity is related to the relative velocity $\beta = v/c$ and Lorentz factor $\gamma = (1-\beta^2)^{-1/2}$ by $\cosh\zeta = \gamma$, $\sinh\zeta = \gamma\beta$, $\tanh\zeta = \beta$; the familiar form of the boost is recovered by this substitution. We verify $B_1^{\mathsf{T}}\eta B_1 = \eta$. Only the upper-left 2×2 block is nontrivial, so it suffices to check that block. With $\eta_2 = \mathrm{diag}(+1,-1)$ and $b = \left(\begin{smallmatrix} \cosh\zeta & \sinh\zeta \\ \sinh\zeta & \cosh\zeta \end{smallmatrix}\right)$ (which is symmetric, $b^{\mathsf{T}} = b$),

$$b^{\mathsf{T}}\eta_2 b = b\,\eta_2\,b = \begin{pmatrix} \cosh\zeta & \sinh\zeta \\ \sinh\zeta & \cosh\zeta \end{pmatrix}\begin{pmatrix} \cosh\zeta & \sinh\zeta \\ -\sinh\zeta & -\cosh\zeta \end{pmatrix} = \begin{pmatrix} \cosh^2\zeta - \sinh^2\zeta & 0 \\ 0 & \sinh^2\zeta - \cosh^2\zeta \end{pmatrix} \tag{3.19}$$

using $\cosh^2\zeta - \sinh^2\zeta = 1$. Thus $B_1(\zeta) \in \mathrm{O}(1,3)$. Its invariants are $\det B_1 = \cosh^2\zeta - \sinh^2\zeta = 1$ and ${B_1}^0{}_0 = \cosh\zeta \geq 1$, so $B_1(\zeta) \in \mathrm{SO}(1,3)^+$ for every ζ, as the topology demands. Acting on the time axis, $B_1(\zeta)\,e_0 = (\cosh\zeta, \sinh\zeta, 0, 0)$, a point of the hyperboloid H^+ of Eq. (3.16), confirming the construction used in Theorem 3.7.

Rapidity adds. Composing two boosts along the same axis,

$$B_1(\zeta_1)B_1(\zeta_2)^0{}_0 = \cosh\zeta_1\cosh\zeta_2 + \sinh\zeta_1\sinh\zeta_2 = \cosh(\zeta_1+\zeta_2), \tag{3.20}$$

and similarly for the other entries, so $B_1(\zeta_1)B_1(\zeta_2) = B_1(\zeta_1+\zeta_2)$. *Collinear boosts add their rapidities*; the map $\zeta \mapsto B_1(\zeta)$ is a group isomorphism from $(\mathbb{R},+)$ onto the one-parameter subgroup of x^1-boosts. The nonlinear velocity-addition law $\beta = (\beta_1+\beta_2)/(1+\beta_1\beta_2)$ is just $\tanh(\zeta_1+\zeta_2)$ rewritten, and $\operatorname{sgn} B^0{}_0 = +1$ is preserved under the addition, in concrete agreement with Theorem 3.4.

The rotation. A rotation by angle θ about the x^3-axis is

$$R_3(\theta) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3.21}$$

which leaves x^0 untouched and rotates (x^1, x^2). Here ${R_3}^0{}_0 = 1$ and $\det R_3 = +1$, so again $R_3(\theta) \in \mathrm{SO}(1,3)^+$, and the defining relation reduces to the Euclidean orthogonality of the spatial block, $\left(\begin{smallmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{smallmatrix}\right)$, which is manifest. Unlike the boost, the rotation is *periodic*: $R_3(\theta)$ has det and ${\Lambda^0}_0$ bounded, all entries lying in $[-1, 1]$, and $R_3(\theta + 2\pi) = R_3(\theta)$. The contrast between the unbounded $\cosh\zeta$ of the boost and the bounded $\cos\theta$ of the rotation is precisely the contrast between the non-compact and compact directions of the group, to which we now turn.

3.1.5 Non-compactness

The boost of Theorem 3.8 already contains the decisive topological fact that separates the Lorentz group from the rotation group and forces the entire representation-theoretic story that follows.

Proposition 3.9 ($\mathrm{SO}(1,3)^+$ is non-compact)**.** *The group* $\mathrm{SO}(1,3)^+$ *is not compact.*

Proof. View $\mathrm{SO}(1,3)^+$ as a subset of the space $M_4(\mathbb{R}) \cong \mathbb{R}^{16}$ of real 4×4 matrices with its usual topology. By the Heine–Borel theorem a subset of a finite-dimensional Euclidean space is compact if and only if it is closed and bounded. The one-parameter family of boosts $\{B_1(\zeta) : \zeta \in \mathbb{R}\} \subset \mathrm{SO}(1,3)^+$ from Eq. (3.18) has matrix entries $\cosh\zeta$ and $\sinh\zeta$, which tend to $+\infty$ as $\zeta \to \infty$. Hence the operator norm of $B_1(\zeta)$ is unbounded over the group, so $\mathrm{SO}(1,3)^+$ is an unbounded subset of $\mathbb{R}^{16}$ and therefore not compact. □

The physical reading is that the boost parameter — the rapidity ζ — ranges over the entire real line $\mathbb{R}$, with no largest value, reflecting the fact that one can boost to a frame of arbitrarily large relative velocity (approaching but never reaching c, since $\beta = \tanh\zeta \to 1$). The rotations, by contrast, form a compact subgroup $\mathrm{SO}(3)$ with the angle running over a circle. The Lorentz group is thus a product, in a sense made precise by the polar decomposition below, of a compact rotational factor and a non-compact boost factor.

This single fact will reverberate through the rest of the chapter, so it is worth saying now why it matters, even though the payoff comes later.

Physical Insight: why non-compactness forbids finite-dimensional unitary representations

A continuous unitary representation of a non-compact group on a *finite*-dimensional Hilbert space cannot be faithful and nontrivial on the non-compact directions. The reason is structural: the boost generators, being non-compact, must be represented by operators whose one-parameter groups $e^{i\zeta K}$ are unbounded unless K is represented trivially, whereas a unitary operator on a finite-dimensional space has all eigenvalues of modulus one and is therefore bounded. Concretely, the finite-dimensional representations we construct in Section 3.2 and the sections following it will represent boosts by Hermitian, *non*-unitary matrices like $e^{\zeta K}$ with real exponent — exactly right for transforming *field components*, which need not have a Lorentz-invariant inner product. The unitary representations, which describe *particle states* and must preserve probabilities, are forced to be infinite-dimensional; building them is the content of Wigner's classification later in the chapter. The split between "fields" (finite-dimensional, non-unitary) and "states" (infinite-dimensional, unitary) traces directly to the non-compactness proved here.

3.1.6 The fundamental group: a first look at the double cover

The last topological invariant we need is the fundamental group $\pi_1(\mathrm{SO}(1,3)^+)$, which counts the inequivalent classes of loops based at the identity. Its value is the single most consequential topological fact about the Lorentz group, because a nontrivial π_1 is exactly what permits *double-valued* representations — spinors — and hence half-integer spin.

Theorem 3.10 (Fundamental group of the Lorentz group)**.** $\mathrm{SO}(1,3)^+$ *is connected but not simply connected; its fundamental group is*

$$\pi_1(\mathrm{SO}(1,3)^+) = \mathbb{Z}_2. \tag{3.22}$$

We give here the topological reduction that makes the result transparent and computes it modulo the single input $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$ established in Chapter 2; the *constructive* proof, which exhibits the connected double cover explicitly as the group $\mathrm{SL}(2,\mathbb{C})$ and is what we actually use to build spinors, is given in Section 3.4.

Proof via deformation retraction onto $\mathrm{SO}(3)$. The mechanism is the polar (Cartan) decomposition of the Lorentz group: every $\Lambda \in \mathrm{SO}(1,3)^+$ factors *uniquely* as a pure boost times a rotation,

$$\Lambda = B\,R, \qquad B = e^{\boldsymbol{\zeta}\cdot\mathbf{K}} \text{ (symmetric, positive)}, \quad R \in \mathrm{SO}(3) \text{ (block-diagonal)}, \tag{3.23}$$

where $\boldsymbol{\zeta} \in \mathbb{R}^3$ is the boost rapidity-vector and $\mathbf{K}$ are the boost generators constructed in Section 3.2. The decomposition is the Lorentzian analogue of writing an invertible matrix as positive-symmetric times orthogonal; existence and uniqueness follow from the polar decomposition of the restriction of Λ to its action on H^+, with the boost factor recovered as the unique positive square root. The factorization gives a homeomorphism

$$\mathrm{SO}(1,3)^+ \;\cong\; \mathbb{R}^3 \times \mathrm{SO}(3) \tag{3.24}$$

as topological spaces (*not* as groups): the boost factor is parametrized by $\boldsymbol{\zeta} \in \mathbb{R}^3$, which is contractible, and the rotation factor by SO(3). Contracting the Euclidean $\mathbb{R}^3$ to a point — explicitly, the deformation $r_s(\boldsymbol{\zeta}, R) = (s\,\boldsymbol{\zeta}, R)$ for s running from 1 to 0 — is a deformation retraction of $\mathrm{SO}(1,3)^+$ onto its maximal compact subgroup SO(3). A deformation retraction preserves the fundamental group, so

$$\pi_1(\mathrm{SO}(1,3)^+) \;\cong\; \pi_1(\mathrm{SO}(3)) \;=\; \mathbb{Z}_2, \tag{3.25}$$

the last equality being the result, proved in Chapter 2, that a 2π rotation is a noncontractible loop in SO(3) while a 4π rotation is contractible (the "belt trick" / plate trick). This establishes Eq. (3.22). □

The factor of $\mathbb{R}^3$ in Eq. (3.24) confirms once more the non-compactness of Theorem 3.9: $\mathrm{SO}(1,3)^+$ retracts onto a compact core SO(3) but is itself the product of that core with three unbounded boost directions. And the surviving $\mathbb{Z}_2$ in π_1 is the seed of everything spinorial. A group with $\pi_1 = \mathbb{Z}_2$ possesses a unique connected double cover, and the genuine, single-valued representations of that double cover include extra "half-integer" ones that descend only to *projective* (double-valued) representations of $\mathrm{SO}(1,3)^+$ itself. Those are the spinor representations; the electron of QED is one of them. We will identify the double cover concretely as $\mathrm{SL}(2,\mathbb{C})$, and use it as the actual engine of the representation theory, in Section 3.4.

We have now isolated the object whose representations are the relativistic fields and particles: the proper orthochronous Lorentz group $\mathrm{SO}(1,3)^+$, the identity component of O(1, 3), a non-compact, doubly-connected matrix Lie group, with the discrete operations P and T standing outside it as separate symmetries to be imposed by hand. Connectedness is what now lets us trade the group for its infinitesimal version: every element of $\mathrm{SO}(1,3)^+$ is reached from the identity along a continuous path, so the group is generated by the tangent vectors to such paths at the identity. The next section takes that derivative — linearizing the defining relation $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$ at $\Lambda = \mathbf{1}$ to extract the six generators $M^{\mu\nu}$ and their commutation relations, the Lorentz Lie algebra from which the entire classification of fields will be assembled.

3.2 The Lorentz Lie algebra

The previous section singled out the proper orthochronous Lorentz group $\mathrm{SO}(1,3)^+$ as the connected component of the identity: the largest piece of the full transformation group that can be reached from the identity by a continuous path, and therefore the piece on which the methods of Chapter 2 can act. A connected matrix Lie group is, however, an awkward object to classify directly. Its representation theory is much easier to attack through its *tangent space at the identity*, the Lie algebra, because a representation of a connected, simply-connected group is in one-to-one correspondence with a representation of its algebra, and even when the group is only connected (as $\mathrm{SO}(1,3)^+$ is, being doubly connected rather than simply connected) the algebra captures every local fact and very nearly every global one. The strategy of this section is the standard linearization of Section 2.3: sit at the identity element $\Lambda = \mathbf{1}$, move an infinitesimal amount, and read off the constraints that the defining relation imposes on the allowed directions of motion. Those directions are the generators; their commutators are the structure constants; and the structure constants are the entire content we will spend the next section decoding.

We work throughout in the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units fixed in the chapter introduction. The starting point is the defining relation of the Lorentz group, which we recall from the previous section: a real 4×4 matrix Λ with components $\Lambda^{\mu}{}_{\nu}$ is a Lorentz transformation iff it preserves the Minkowski form, $\eta_{\rho\sigma}\,\Lambda^{\rho}{}_{\mu}\,\Lambda^{\sigma}{}_{\nu} = \eta_{\mu\nu}$, which in matrix form is the relation $\Lambda^{\mathsf{T}}\eta\,\Lambda = \eta$ of Eq. (3.5). Everything in this section is the systematic exploitation of Eq. (3.5) in a neighborhood of $\Lambda = \mathbf{1}$.

3.2.1 Infinitesimal transformations and the antisymmetry constraint

A curve through the identity in $\mathrm{SO}(1,3)^+$ is a one-parameter family $\Lambda(t)$ with $\Lambda(0) = \mathbf{1}$. Its tangent at $t = 0$ is the object we want, so we write the most general infinitesimal transformation as

$$\Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} + \omega^{\mu}{}_{\nu} + \mathcal{O}(\omega^2), \tag{3.26}$$

where the entries of $\omega^{\mu}{}_{\nu}$ are first-order small and otherwise arbitrary; they parametrize the directions one may move away from the identity. Not every ω is allowed, however: Λ must still satisfy Eq. (3.5). Substituting Eq. (3.26) into the defining relation and keeping only terms through first order,

$$\begin{aligned} \eta_{\mu\nu} &= \eta_{\rho\sigma}\,(\delta^{\rho}{}_{\mu} + \omega^{\rho}{}_{\mu})\,(\delta^{\sigma}{}_{\nu} + \omega^{\sigma}{}_{\nu}) \\ &= \eta_{\rho\sigma}\,\delta^{\rho}{}_{\mu}\,\delta^{\sigma}{}_{\nu} + \eta_{\rho\sigma}\,\omega^{\rho}{}_{\mu}\,\delta^{\sigma}{}_{\nu} + \eta_{\rho\sigma}\,\delta^{\rho}{}_{\mu}\,\omega^{\sigma}{}_{\nu} + \mathcal{O}(\omega^2) \\ &= \eta_{\mu\nu} + \eta_{\sigma\nu}\,\omega^{\sigma}{}_{\mu} + \eta_{\rho\mu}\,\omega^{\rho}{}_{\nu} + \mathcal{O}(\omega^2). \end{aligned} \tag{3.27}$$

The zeroth-order term reproduces $\eta_{\mu\nu}$, so the first-order terms must cancel among themselves. It is convenient to lower the free index on ω by defining $\omega_{\mu\nu} \equiv \eta_{\mu\rho}\,\omega^{\rho}{}_{\nu}$; the surviving condition in Eq. (3.27) is then $\omega_{\nu\mu} + \omega_{\mu\nu} = 0$, that is,

$$\omega_{\mu\nu} = -\,\omega_{\nu\mu}. \tag{3.28}$$

The infinitesimal Lorentz parameters form an antisymmetric rank-two tensor. An antisymmetric 4×4 matrix has $\frac{1}{2} \cdot 4 \cdot 3 = 6$ independent entries, so the Lorentz group is six-dimensional: three of the parameters will turn out to generate rotations and three to generate boosts. This is the linear-algebraic origin of the fact, anticipated in the previous section, that $\mathrm{SO}(1,3)^{+}$ is a six-parameter group.

3.2.2 The six generators $M^{\mu\nu}$

To extract a basis of generators we exhibit the six independent components of $\omega_{\mu\nu}$ explicitly. Because the index pair $(\mu\nu)$ ranges over an antisymmetric tensor, we may write $\omega^{\mu}{}_{\nu}$ as a linear combination of six fixed basis matrices, one for each independent component, with $\omega_{\mu\nu}$ supplying the coefficients. The conventional way to package this—and the one that makes the generators carry the same index structure as the parameters—is to write the finite group element as an exponential, in the manner of the exponential map Eq. (2.60), and to define the generators as the coefficients in the expansion. We set

$$\Lambda = \exp\!\Big(-\tfrac{i}{2}\,\omega_{\rho\sigma}\,M^{\rho\sigma}\Big), \qquad \Lambda^{\mu}{}_{\nu} = \delta^{\mu}{}_{\nu} - \tfrac{i}{2}\,\omega_{\rho\sigma}\,(M^{\rho\sigma})^{\mu}{}_{\nu} + \mathcal{O}(\omega^{2}), \tag{3.29}$$

where the second expression is the first-order expansion. Here $M^{\rho\sigma}$ is a set of six 4×4 matrices labelled by the antisymmetric pair $\rho\sigma$ (so $M^{\rho\sigma} = -M^{\sigma\rho}$, matching the antisymmetry of $\omega_{\rho\sigma}$), and $(M^{\rho\sigma})^{\mu}{}_{\nu}$ denotes the (μ,ν) entry of the matrix $M^{\rho\sigma}$. The factor of $-\frac{i}{2}$ is pure convention: the i makes the generators Hermitian in the unitary representations we eventually want (so that Λ there is unitary), and the $\frac{1}{2}$ compensates the double-counting that comes from summing the antisymmetric pair $\rho\sigma$ over all ordered values rather than over $\rho < \sigma$.

Comparing the two expansions Eq. (3.26) and Eq. (3.29) gives

$$\omega^{\mu}{}_{\nu} = -\tfrac{i}{2}\,\omega_{\rho\sigma}\,(M^{\rho\sigma})^{\mu}{}_{\nu}, \tag{3.30}$$

and this identity, valid for arbitrary antisymmetric $\omega_{\rho\sigma}$, determines the generator matrices. To solve it we re-express the left-hand side in terms of $\omega_{\rho\sigma}$ as well. Raising one index, $\omega^{\mu}{}_{\nu} = \eta^{\mu\alpha}\omega_{\alpha\nu}$, and writing $\omega_{\alpha\nu}$ through its antisymmetry as the manifestly antisymmetric combination

$$\omega_{\alpha\nu} = \tfrac{1}{2}\,\omega_{\rho\sigma}\big(\delta^{\rho}{}_{\alpha}\,\delta^{\sigma}{}_{\nu} - \delta^{\sigma}{}_{\alpha}\,\delta^{\rho}{}_{\nu}\big), \tag{3.31}$$

we obtain

$$\omega^{\mu}{}_{\nu} = \eta^{\mu\alpha}\,\tfrac{1}{2}\,\omega_{\rho\sigma}\big(\delta^{\rho}{}_{\alpha}\,\delta^{\sigma}{}_{\nu} - \delta^{\sigma}{}_{\alpha}\,\delta^{\rho}{}_{\nu}\big) = \tfrac{1}{2}\,\omega_{\rho\sigma}\big(\eta^{\mu\rho}\,\delta^{\sigma}{}_{\nu} - \eta^{\mu\sigma}\,\delta^{\rho}{}_{\nu}\big). \tag{3.32}$$

Setting Eq. (3.32) equal to the right-hand side of Eq. (3.30) and using that $\omega_{\rho\sigma}$ is an arbitrary antisymmetric tensor—so its coefficient, already antisymmetric in $\rho \leftrightarrow \sigma$ on both sides, may be equated—gives $-\frac{i}{2}(M^{\rho\sigma})^{\mu}{}_{\nu} = \frac{1}{2}(\eta^{\mu\rho}\delta^{\sigma}{}_{\nu} - \eta^{\mu\sigma}\delta^{\rho}{}_{\nu})$. Solving for the generator and using the symmetry $\eta^{\mu\rho} = \eta^{\rho\mu}$,

$$\boxed{(M^{\rho\sigma})^{\mu}{}_{\nu} = i(\eta^{\rho\mu}\,\delta^{\sigma}{}_{\nu} - \eta^{\sigma\mu}\,\delta^{\rho}{}_{\nu}).} \tag{3.33}$$

This is the explicit defining representation of the six Lorentz generators. We record its salient features. The matrix $M^{\rho\sigma}$ is antisymmetric in its label pair, $(M^{\rho\sigma})^{\mu}{}_{\nu} = -(M^{\sigma\rho})^{\mu}{}_{\nu}$, as required. Each generator has exactly two nonzero entries: $M^{\rho\sigma}$ acts in the ρ–σ plane and annihilates the two orthogonal directions. And the position of the metric controls a crucial sign difference between the two types of plane—a 0–i plane (one timelike direction) versus an i–j plane (two spacelike directions)—which is exactly the sign that will distinguish boosts from rotations below.

Definition 3.11 (Lorentz generators)**.** The six *Lorentz generators* in the defining (vector) representation are the antisymmetric set of matrices $M^{\rho\sigma}$, $\rho, \sigma \in \{0, 1, 2, 3\}$, with components given by Eq. (3.33). A finite proper orthochronous Lorentz transformation is recovered as $\Lambda = \exp\!\big(-\frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma}\big)$ for a real antisymmetric parameter tensor $\omega_{\rho\sigma}$.

3.2.3 The covariant commutation relations

The Lie algebra is the set of commutators of the generators. Because Eq. (3.33) gives the generators as explicit matrices, we can compute the commutator directly by matrix multiplication—no abstraction required—and the result, being expressed entirely through η and M, will then hold in *every* representation, not just the defining one. This is the central structural fact of the chapter.

Proposition 3.12 (The Lorentz algebra $\mathfrak{so}(1,3)$)**.** *The Lorentz generators satisfy*

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - \eta^{\nu\sigma}M^{\mu\rho} + \eta^{\mu\sigma}M^{\nu\rho}). \tag{3.34}$$

Proof. We compute the matrix product $(M^{\mu\nu}M^{\rho\sigma})^{\alpha}{}_{\beta}$ by inserting an intermediate index γ and using Eq. (3.33) twice:

$$\begin{aligned}(M^{\mu\nu}M^{\rho\sigma})^{\alpha}{}_{\beta} &= (M^{\mu\nu})^{\alpha}{}_{\gamma}\,(M^{\rho\sigma})^{\gamma}{}_{\beta} \\ &= i(\eta^{\mu\alpha}\delta^{\nu}{}_{\gamma} - \eta^{\nu\alpha}\delta^{\mu}{}_{\gamma})\; i(\eta^{\rho\gamma}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\gamma}\delta^{\rho}{}_{\beta}) \\ &= -(\eta^{\mu\alpha}\delta^{\nu}{}_{\gamma} - \eta^{\nu\alpha}\delta^{\mu}{}_{\gamma})(\eta^{\rho\gamma}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\gamma}\delta^{\rho}{}_{\beta}).\end{aligned} \tag{3.35}$$

The sum over γ collapses each Kronecker delta against a metric, e.g. $\delta^{\nu}{}_{\gamma}\eta^{\rho\gamma} = \eta^{\rho\nu}$, giving the four terms

$$(M^{\mu\nu}M^{\rho\sigma})^{\alpha}{}_{\beta} = -\Big[\eta^{\mu\alpha}\eta^{\rho\nu}\delta^{\sigma}{}_{\beta} - \eta^{\mu\alpha}\eta^{\sigma\nu}\delta^{\rho}{}_{\beta} - \eta^{\nu\alpha}\eta^{\rho\mu}\delta^{\sigma}{}_{\beta} + \eta^{\nu\alpha}\eta^{\sigma\mu}\delta^{\rho}{}_{\beta}\Big]. \tag{3.36}$$

The opposite ordering $(M^{\rho\sigma}M^{\mu\nu})^{\alpha}{}_{\beta}$ follows from Eq. (3.36) by the simultaneous swap $(\mu\nu) \leftrightarrow (\rho\sigma)$:

$$(M^{\rho\sigma}M^{\mu\nu})^{\alpha}{}_{\beta} = -\Big[\eta^{\rho\alpha}\eta^{\mu\sigma}\delta^{\nu}{}_{\beta} - \eta^{\rho\alpha}\eta^{\nu\sigma}\delta^{\mu}{}_{\beta} - \eta^{\sigma\alpha}\eta^{\mu\rho}\delta^{\nu}{}_{\beta} + \eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta}\Big]. \tag{3.37}$$

Subtracting Eq. (3.37) from Eq. (3.36) and using the symmetry of the metric throughout, the commutator is the eight-term expression

$$\begin{aligned}[M^{\mu\nu}, M^{\rho\sigma}]^{\alpha}{}_{\beta} = -\Big[\ &\eta^{\mu\alpha}\eta^{\nu\rho}\delta^{\sigma}{}_{\beta} - \eta^{\mu\alpha}\eta^{\nu\sigma}\delta^{\rho}{}_{\beta} - \eta^{\nu\alpha}\eta^{\mu\rho}\delta^{\sigma}{}_{\beta} + \eta^{\nu\alpha}\eta^{\mu\sigma}\delta^{\rho}{}_{\beta} \\ &- \eta^{\rho\alpha}\eta^{\mu\sigma}\delta^{\nu}{}_{\beta} + \eta^{\rho\alpha}\eta^{\nu\sigma}\delta^{\mu}{}_{\beta} + \eta^{\sigma\alpha}\eta^{\mu\rho}\delta^{\nu}{}_{\beta} - \eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta}\Big].\end{aligned} \tag{3.38}$$

It remains to recognize the right-hand side as a combination of single generators. From Eq. (3.33), the matrix entry of a generator times the imaginary unit is $i\,(M^{ab})^{\alpha}{}_{\beta} = -(\eta^{a\alpha}\delta^{b}{}_{\beta} - \eta^{b\alpha}\delta^{a}{}_{\beta})$. Expanding the claimed right-hand side of Eq. (3.34) in this way,

$$\begin{aligned}i(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - \eta^{\nu\sigma}M^{\mu\rho} + \eta^{\mu\sigma}M^{\nu\rho})^{\alpha}{}_{\beta} = -\Big[\ &\eta^{\nu\rho}(\eta^{\mu\alpha}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\alpha}\delta^{\mu}{}_{\beta}) \\ &- \eta^{\mu\rho}(\eta^{\nu\alpha}\delta^{\sigma}{}_{\beta} - \eta^{\sigma\alpha}\delta^{\nu}{}_{\beta}) \\ &- \eta^{\nu\sigma}(\eta^{\mu\alpha}\delta^{\rho}{}_{\beta} - \eta^{\rho\alpha}\delta^{\mu}{}_{\beta}) \\ &+ \eta^{\mu\sigma}(\eta^{\nu\alpha}\delta^{\rho}{}_{\beta} - \eta^{\rho\alpha}\delta^{\nu}{}_{\beta})\Big].\end{aligned} \tag{3.39}$$

Multiplying out the eight products in Eq. (3.39) and comparing with Eq. (3.38) term by term, each of the eight terms matches exactly: for instance the term $-\eta^{\mu\alpha}\eta^{\nu\rho}\delta^{\sigma}{}_{\beta}$ appears in both, the term $+\eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta}$ in Eq. (3.39) matches $-(-\eta^{\sigma\alpha}\eta^{\nu\rho}\delta^{\mu}{}_{\beta})$ in Eq. (3.38), and so on through all eight. The two expressions are identical, which establishes Eq. (3.34). □

A few remarks fix the meaning of Eq. (3.34). First, the right-hand side is again a Lorentz generator, so the commutator closes: the $M^{\mu\nu}$ span a six-dimensional Lie algebra, denoted $\mathfrak{so}(1,3)$, with the metric components $\eta^{\mu\nu}$ playing the role of (part of) the structure constants in the sense of Chapter 2. Second, every appearance of η on the right is what carries the indefinite signature of spacetime into the algebra; had we worked with the Euclidean metric δ we would have obtained the compact rotation algebra $\mathfrak{so}(4)$ instead, and the entire representation theory would have been different. Third—and this is the feature we will pursue—Eq. (3.34) is manifestly covariant but it mixes the timelike index 0 and the spacelike indices $1, 2, 3$ on an equal footing, which hides the physical content. The next step is to split it.

3.2.4 Rotations, boosts, and the J–K algebra

The covariant algebra Eq. (3.34) treats all six generators symmetrically, but physically they are of two kinds: the three that rotate space among itself, and the three that mix time with a spatial direction. We make this split explicit. The three generators with both indices spatial, M^{jk} with $j,k \in \{1,2,3\}$, are antisymmetric in j,k and so encode three independent matrices; we package them, exactly as orbital angular momentum is packaged in nonrelativistic quantum mechanics, by contracting with the Levi-Civita symbol. The three with a timelike index, M^{0i}, are the remaining three.

Definition 3.13 (Rotation and boost generators)**.** The *rotation generators* J_i and *boost generators* K_i, $i \in \{1,2,3\}$, are

$$J_i \;\equiv\; \tfrac{1}{2}\,\epsilon_{ijk}\,M^{jk}, \qquad K_i \;\equiv\; M^{0i}. \tag{3.40}$$

The definition of J_i is invertible: contracting with ϵ and using $\epsilon_{ijk}\epsilon_{iab} = \delta_{ja}\delta_{kb} - \delta_{jb}\delta_{ka}$ gives $\epsilon_{ijk}J_i = \frac{1}{2}\epsilon_{ijk}\epsilon_{iab}M^{ab} = \frac{1}{2}(M^{jk} - M^{kj}) = M^{jk}$, so

$$M^{jk} = \epsilon_{ijk}\,J_i, \qquad M^{0i} = K_i. \tag{3.41}$$

Throughout this reduction we use that in the mostly-minus convention the spatial metric components are $\eta^{ij} = -\delta^{ij}$, while $\eta^{00} = +1$ and $\eta^{0i} = 0$. The single relative minus sign between η^{00} and η^{ij} is the whole story; we now watch it propagate through the three families of commutators.

Proposition 3.14 (The J–K algebra)**.** *The rotation and boost generators obey*

$$[J_i, J_j] = i\,\epsilon_{ijk}\,J_k, \tag{3.42}$$
$$[J_i, K_j] = i\,\epsilon_{ijk}\,K_k, \tag{3.43}$$
$$[K_i, K_j] = -\,i\,\epsilon_{ijk}\,J_k. \tag{3.44}$$

Proof. We evaluate the three cases in turn from the covariant relation Eq. (3.34), substituting $\eta^{ij} = -\delta^{ij}$ wherever both indices are spatial and dropping any term containing a single $\eta^{0i} = 0$.

(i) Rotation–rotation. Using Eq. (3.40),

$$[J_i, J_j] = \tfrac{1}{4}\,\epsilon_{iab}\,\epsilon_{jcd}\,\Big[M^{ab}, M^{cd}\Big]. \tag{3.45}$$

With all four indices a,b,c,d spatial, every metric in Eq. (3.34) is of the form $\eta^{ab} = -\delta^{ab}$, so

$$\Big[M^{ab}, M^{cd}\Big] = i(-\delta^{bc}M^{ad} + \delta^{ac}M^{bd} + \delta^{bd}M^{ac} - \delta^{ad}M^{bc}). \tag{3.46}$$

The four terms contribute equally after the contraction with $\epsilon_{iab}\epsilon_{jcd}$, because M^{ab} and the ϵ's are each antisymmetric: relabelling dummy indices turns each term into the first. Taking four times the first term,

$$[J_i, J_j] = \tfrac{1}{4} \cdot 4 \cdot (-i)\, \epsilon_{iab}\, \epsilon_{jcd}\, \delta^{bc} M^{ad} = -\, i\, \epsilon_{iab}\, \epsilon_{jbd}\, M^{ad}. \tag{3.47}$$

Now $\epsilon_{iab}\epsilon_{jbd} = -\epsilon_{iab}\epsilon_{jdb} = -(\delta_{ij}\delta_{ad} - \delta_{id}\delta_{aj})$, so

$$[J_i, J_j] = i\,(\delta_{ij}\delta_{ad} - \delta_{id}\delta_{aj}) M^{ad} = i\,\delta_{ij}\, M^{aa} - i\, M^{ji}. \tag{3.48}$$

The first term vanishes because $M^{aa} = 0$ by antisymmetry; in the second, $-iM^{ji} = +iM^{ij} = i\,\epsilon_{ijk} J_k$ by Eq. (3.41). Hence $[J_i, J_j] = i\,\epsilon_{ijk} J_k$, which is Eq. (3.42). This is precisely the angular-momentum algebra $\mathfrak{su}(2)$ of Chapter 2: the rotation subgroup of the Lorentz group is the rotation group of space, and the J_i are its generators.

(ii) Rotation–boost. Here

$$[J_i, K_j] = \tfrac{1}{2}\, \epsilon_{iab} \left[M^{ab}, M^{0j} \right], \tag{3.49}$$

with a, b spatial and one timelike index on the second factor. From Eq. (3.34) with $(\mu\nu\rho\sigma) = (a\, b\, 0\, j)$,

$$\left[M^{ab}, M^{0j} \right] = i(\eta^{b0} M^{aj} - \eta^{a0} M^{bj} - \eta^{bj} M^{a0} + \eta^{aj} M^{b0}). \tag{3.50}$$

The first two terms vanish since $\eta^{a0} = \eta^{b0} = 0$. In the remaining two, $\eta^{bj} = -\delta^{bj}$, $\eta^{aj} = -\delta^{aj}$, and $M^{a0} = -M^{0a} = -K_a$, so

$$\left[M^{ab}, M^{0j} \right] = i(\delta^{bj} M^{a0} - \delta^{aj} M^{b0}) = i(-\delta^{bj} K_a + \delta^{aj} K_b). \tag{3.51}$$

Therefore

$$[J_i, K_j] = \tfrac{i}{2}\, \epsilon_{iab}(-\delta^{bj} K_a + \delta^{aj} K_b) = \tfrac{i}{2}(-\epsilon_{iaj} K_a + \epsilon_{ijb} K_b) = \tfrac{i}{2}(\epsilon_{ija} K_a + \epsilon_{ijb} K_b) = i\,\epsilon_{ijk} K_k, \tag{3.52}$$

where we used $-\epsilon_{iaj} = \epsilon_{ija}$. This is Eq. (3.43): under rotations, the three boost generators transform as the components of an ordinary spatial vector—as they must, since $\mathbf{K}$ points along the boost direction.

(iii) Boost–boost. Finally

$$[K_i, K_j] = \left[M^{0i}, M^{0j} \right]. \tag{3.53}$$

Applying Eq. (3.34) with $(\mu\nu\rho\sigma) = (0\, i\, 0\, j)$,

$$\left[M^{0i}, M^{0j} \right] = i(\eta^{i0} M^{0j} - \eta^{00} M^{ij} - \eta^{ij} M^{00} + \eta^{0j} M^{i0}). \tag{3.54}$$

The terms with η^{i0} and η^{0j} vanish, and $M^{00} = 0$. Only the second term survives, and here is the decisive point: it carries the timelike metric component $\eta^{00} = +1$, not a spatial $-\delta$. Thus

$$[K_i, K_j] = -\,i\,\eta^{00}\,M^{ij} = -\,i\,M^{ij} = -\,i\,\epsilon_{ijk}\,J_k, \tag{3.55}$$

using Eq. (3.41) in the last step. This is Eq. (3.44). □

The sign in Eq. (3.44) deserves to be stated plainly, because it is the hinge on which the rest of the chapter turns. Trace it back through the proof: it came from $\eta^{00} = +1$ in the boost–boost commutator, whereas the corresponding sign in the rotation–rotation commutator came from $\eta^{ij} = -\delta^{ij}$. The lone relative minus sign of the mostly-minus metric, sitting between the time and space components, has reappeared as the lone minus sign in $[K_i, K_j] = -i\epsilon_{ijk}J_k$. Had spacetime been Euclidean, $\mathfrak{so}(4)$, this sign would have been $+i$ and boosts would close into a second, independent rotation algebra; the Lorentzian signature instead makes the commutator of two boosts produce a rotation *with the wrong sign*.

Physical Insight

The minus sign is not a bookkeeping curiosity; it is the geometric statement that boosts do not form a closed subgroup the way rotations do. Two successive rotations about different axes compose into a single rotation (the J's close: Eq. (3.42)). Two successive boosts in different directions do *not* compose into a boost—their composition contains a rotation, the Wigner rotation responsible for Thomas precession—and the algebraic fingerprint of this fact is exactly the appearance of J on the right of Eq. (3.44). Geometrically, the rotations sweep out a compact sphere of directions while the boosts run off to infinite rapidity along non-compact hyperbolas; the next subsection makes this compact-versus-non-compact contrast fully explicit at the level of finite transformations.

3.2.5 Finite rotations and boosts

To connect the generators back to the transformations they generate, we feed the split Eq. (3.40) into the exponential map of Theorem 3.11. Restrict the antisymmetric parameter tensor first to its purely spatial part, writing $\omega_{jk} = \epsilon_{jkl}\,\theta_l$ with $\boldsymbol{\theta} = (\theta_1, \theta_2, \theta_3)$ and setting the mixed components ω_{0i} to zero. Then

$$-\tfrac{i}{2}\,\omega_{\rho\sigma}M^{\rho\sigma} = -\tfrac{i}{2}\,\omega_{jk}M^{jk} = -\tfrac{i}{2}\,\epsilon_{jkl}\theta_l\,M^{jk} = -\,i\,\theta_l\big(\tfrac{1}{2}\epsilon_{ljk}M^{jk}\big) = -\,i\,\boldsymbol{\theta}\cdot\mathbf{J}, \tag{3.56}$$

so that finite rotations are

$$R(\boldsymbol{\theta}) = \exp\!\big(-\,i\,\boldsymbol{\theta}\cdot\mathbf{J}\big). \tag{3.57}$$

Restricting instead to the mixed components, write $\omega_{0i} = \eta_i$ with $\boldsymbol{\eta} = (\eta_1, \eta_2, \eta_3)$ the *rapidity*, and set $\omega_{jk} = 0$. Using the antisymmetry $\omega_{i0} = -\eta_i$ together with $M^{i0} = -M^{0i}$,

$$-\tfrac{i}{2}\,\omega_{\rho\sigma}M^{\rho\sigma} = -\tfrac{i}{2}(\omega_{0i}M^{0i}+\omega_{i0}M^{i0}) = -\tfrac{i}{2}(\eta_i M^{0i}+(-\eta_i)(-M^{0i})) = -\,i\,\eta_i M^{0i} = -\,i\,\boldsymbol{\eta}\cdot\mathbf{K}, \tag{3.58}$$

so that finite boosts are

$$B(\boldsymbol{\eta}) = \exp\!\big(-\,i\,\boldsymbol{\eta}\cdot\mathbf{K}\big). \tag{3.59}$$

Every element of $\mathrm{SO}(1,3)^+$ can be written, in a neighborhood of the identity and in fact globally by the polar decomposition, as a product of a rotation Eq. (3.57) and a boost Eq. (3.59).

Notation Watch

The symbol η does double duty in this section: $\eta_{\mu\nu}$ (always with two spacetime indices) is the Minkowski metric, while $\boldsymbol{\eta}$ (boldface, three-component) is the rapidity vector parametrizing a boost in Eq. (3.59). The two never share an index structure, so no ambiguity arises, but the coincidence of letters is standard and worth flagging once. The rapidity is additive for collinear boosts—unlike the velocity—which is precisely why it, and not v, is the natural exponential parameter. The single-boost worked example of Section 3.1 wrote the rapidity as a scalar ζ (in $B_1(\zeta)$); it is the same parameter, $\boldsymbol{\eta} = \eta\,\hat{\mathbf{n}}$ being its vector form along the boost axis $\hat{\mathbf{n}}$. Likewise, the boost factor of the polar decomposition Eq. (3.23) was written there as $e^{\boldsymbol{\eta}\cdot\mathbf{K}}$ with a *real, symmetric* exponent and no explicit i; that is consistent with the present $B(\boldsymbol{\eta}) = \exp(-i\boldsymbol{\eta}\cdot\mathbf{K})$ because, in the Hermitian-generator convention Eq. (3.33) adopted here, the matrices $-i\mathbf{K} = -iM^{0i}$ are exactly real and symmetric (see Theorem 3.15), so $\exp(-i\boldsymbol{\eta}\cdot\mathbf{K}) = \exp(\boldsymbol{\eta}\cdot(-i\mathbf{K}))$ displays the boost as the exponential of a real symmetric generator. We use the i-convention Eq. (3.33) uniformly, since it is the one in which $\mathbf{J}$ is Hermitian and the $\mathfrak{su}(2)\oplus\mathfrak{su}(2)$ split of the next section is cleanest.

The two finite transformations Eqs. (3.57) and (3.59) look symmetric, but the metric makes them behave very differently. The following worked example computes both explicitly in the defining representation and exposes the contrast that drives the next section.

Worked Example 3.15 (A boost and a rotation, exponentiated)**.** We construct K_1 and J_3 as explicit 4×4 matrices from Eq. (3.33) and exponentiate them.

The boost generator. With $\rho\sigma = 01$, Eq. (3.33) gives $(K_1)^{\mu}{}_{\nu} = (M^{01})^{\mu}{}_{\nu} = i(\eta^{0\mu}\delta^{1}{}_{\nu} - \eta^{1\mu}\delta^{0}{}_{\nu})$. Only $\eta^{00} = 1$ and $\eta^{11} = -1$ are nonzero among the relevant components, so the only nonvanishing entries are $(K_1)^{0}{}_{1} = i\,\eta^{00} = i$ and $(K_1)^{1}{}_{0} =$

$-i\,\eta^{11} = i$. Ordering the rows and columns as $(0,1,2,3)$,

$$K_1 = \begin{pmatrix} 0 & i & 0 & 0 \\ i & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \tag{3.60}$$

To exponentiate, form $-i\,\eta_1 K_1$, which restricted to the 0–1 block is $\eta_1\,S$ with $S = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Since $S^2 = \mathbf{1}_2$, the exponential series splits into even and odd powers,

$$\exp(\eta_1 S) = \mathbf{1}_2 \cosh\eta_1 + S \sinh\eta_1 = \begin{pmatrix} \cosh\eta_1 & \sinh\eta_1 \\ \sinh\eta_1 & \cosh\eta_1 \end{pmatrix}, \tag{3.61}$$

so the finite boost Eq. (3.59) along the x^1-axis is

$$B(\eta_1 \hat{\mathbf{e}}_1) = \exp(-i\eta_1 K_1) = \begin{pmatrix} \cosh\eta_1 & \sinh\eta_1 & 0 & 0 \\ \sinh\eta_1 & \cosh\eta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \tag{3.62}$$

This is the familiar Lorentz boost $x^{0\prime} = x^0 \cosh\eta_1 + x^1 \sinh\eta_1$, $x^{1\prime} = x^0 \sinh\eta_1 + x^1 \cosh\eta_1$, with the rapidity related to the velocity by $\tanh\eta_1 = v$ (so $\cosh\eta_1 = \gamma$). The matrix entries grow without bound as $\eta_1 \to \infty$: the boost orbit is a non-compact hyperbola, $(\cosh\eta_1)^2 - (\sinh\eta_1)^2 = 1$.

The rotation generator. With $\rho\sigma = 12$, $(J_3)^\mu{}_\nu = (M^{12})^\mu{}_\nu = i(\eta^{1\mu}\delta^2{}_\nu - \eta^{2\mu}\delta^1{}_\nu)$, whose only nonzero entries are $(J_3)^1{}_2 = i\,\eta^{11} = -i$ and $(J_3)^2{}_1 = -i\,\eta^{22} = i$. In the 1–2 block this is $\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)$, the Pauli matrix σ_2, which satisfies $\sigma_2^2 = \mathbf{1}_2$. The same even/odd split now yields trigonometric, not hyperbolic, functions,

$$R(\theta_3 \hat{\mathbf{e}}_3) = \exp(-i\theta_3 J_3) = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta_3 & -\sin\theta_3 & 0 \\ 0 & \sin\theta_3 & \cos\theta_3 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \tag{3.63}$$

the ordinary rotation by angle θ_3 in the x^1–x^2 plane, periodic in θ_3 with period 2π: the rotation orbit is a compact circle.

The lesson. Inspect the generators as matrices. The rotation generator J_3 is Hermitian, $J_3^\dagger = J_3$, so $R = \exp(-i\theta_3 J_3)$ is unitary. The boost generator K_1 is *anti*-Hermitian, $K_1^\dagger = -K_1$, so $B = \exp(-i\eta_1 K_1)$ is *not* unitary—indeed it is symmetric and real, with eigenvalues $e^{\pm\eta_1}$ running off to infinity. In this finite-dimensional defining representation the boosts cannot be represented unitarily. This is no accident of the choice of representation: it is forced by the algebra. The sign in $[K_i, K_j] = -i\epsilon_{ijk}J_k$ means that one cannot make all six generators Hermitian at once in any finite-dimensional representation, because doing so would require the boosts to generate a compact subgroup, contradicting the non-compact hyperbolic orbits we have just exhibited.

Caution

The generators $M^{\mu\nu}$ are not Hermitian matrices in the naive sense in the defining representation, and one should not expect them to be: only the three rotation generators are Hermitian, while the three boost generators are anti-Hermitian. The defining four-vector representation is therefore a perfectly good *finite-dimensional* representation of the Lorentz algebra, but it is *not* a unitary one. This is the generic situation for finite-dimensional representations of a non-compact group, and it is exactly why fields (which carry finite-dimensional representations) and states (which must carry unitary ones) are governed by different parts of the representation theory—a division the chapter introduction flagged and that the Poincaré analysis later in the chapter resolves.

We have linearized $\mathrm{SO}(1,3)^+$ completely. The six generators $M^{\mu\nu}$ obey the covariant algebra Eq. (3.34); split into rotations and boosts they obey the J–K algebra Eqs. (3.42) to (3.44); and the finite transformations are the exponentials Eqs. (3.57) and (3.59). The one feature that does not fit the comfortable pattern of Chapter 2 is the minus sign in $[K_i, K_j] = -i\epsilon_{ijk}J_k$, which—as Theorem 3.15 showed concretely—blocks any finite-dimensional unitary representation and prevents the J's and K's from organizing into two ordinary, independent angular-momentum algebras. The next section turns this apparent obstruction into the central tool of the chapter: by complexifying the algebra and forming the combinations $\mathbf{J} \pm i\mathbf{K}$, the troublesome sign is absorbed, the boosts and rotations decouple into two commuting copies of $\mathfrak{su}(2)$, and the entire finite-dimensional representation theory of the Lorentz group falls out of the spin classification already in hand.

3.3 $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ and the (j_-, j_+) classification of fields

The previous section linearised $\mathrm{SO}(1,3)^+$ and split its six generators into the three rotations J_i and the three boosts K_i, ending with the algebra

$$[J_i, J_j] = i\,\epsilon_{ijk}J_k, \qquad [J_i, K_j] = i\,\epsilon_{ijk}K_k, \qquad [K_i, K_j] = -\,i\,\epsilon_{ijk}J_k, \tag{3.64}$$

established in Eqs. (3.42) to (3.44) of Section 3.2. (Throughout this chapter the metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and $\hbar = c = 1$; ϵ_{ijk} is the flat three-dimensional Levi-Civita symbol with $\epsilon_{123} = +1$.) The first two relations are reassuring: the J_i close among themselves into the familiar rotation algebra $\mathfrak{su}(2)$, and the K_i transform as a vector under rotations, exactly as a boost direction should. The third relation is the source of every difficulty and every richness in the representation theory that follows. Its minus sign says that the commutator of two boosts is a *rotation*, and it is the algebraic fingerprint of the

noncompactness of $\mathrm{SO}(1,3)^+$: boosting in two different planes and undoing the boosts does not return you to the identity but leaves a residual rotation, while the rapidity that parametrises a boost runs over all of $\mathbb{R}$ rather than a compact circle.

That same minus sign blocks the most naive hope. One might wish that J_i and K_i each closed into its own copy of $\mathfrak{su}(2)$, so that the entire representation theory would reduce twice over to the $\mathfrak{su}(2)$ theory already built in Chapter 2. They do not: $[K_i, K_j]$ produces a J, not a K, so the boosts alone do not close. The remedy is to allow ourselves complex coefficients. We complexify the algebra, and inside the complexification we will discover that two genuinely decoupled $\mathfrak{su}(2)$'s have been hiding all along. This is the central structural fact of the chapter, and from it the entire catalogue of relativistic fields — scalar, the two Weyl spinors, the four-vector, the field strength — drops out mechanically.

3.3.1 Complexifying the algebra: the two hidden $\mathfrak{su}(2)$'s

The obstruction in Eq. (3.64) is that J and K are coupled by real structure constants in a way that cannot be diagonalised over $\mathbb{R}$. Over $\mathbb{C}$ it can. Consider the complex linear combinations that follow.

Definition 3.16 (Self-dual and anti-self-dual generators)**.** On the complexification of the Lorentz algebra — the complex span of $\{J_i, K_i\}$ with complex coefficients — define

$$N_i^{\pm} \;=\; \tfrac{1}{2}(J_i \pm i\,K_i), \qquad i = 1,2,3. \tag{3.65}$$

We also write $\vec{A} \equiv \vec{N}^{+} = \tfrac{1}{2}(\vec{J} + i\vec{K})$ and $\vec{B} \equiv \vec{N}^{-} = \tfrac{1}{2}(\vec{J} - i\vec{K})$ for the two triples; these are the names carried into the spinor section. The relations invert to

$$J_i = N_i^{+} + N_i^{-}, \qquad K_i = -\,i\,(N_i^{+} - N_i^{-}). \tag{3.66}$$

These six combinations are not elements of the real Lorentz algebra $\mathfrak{so}(1,3)$ — they involve i times a real generator — but they are perfectly good elements of its complexification $\mathfrak{so}(1,3)_{\mathbb{C}}$, and that is all we need to organise the representation theory. We compute their brackets directly from Eq. (3.64). The computation is short and worth doing in full because the cancellations are the whole point.

For the bracket of two N^{+}'s, expand bilinearly,

$$\begin{aligned}\left[N_i^{+}, N_j^{+}\right] &= \tfrac{1}{4}\left[J_i + iK_i,\; J_j + iK_j\right] \\ &= \tfrac{1}{4}\Big([J_i, J_j] + i\,[J_i, K_j] + i\,[K_i, J_j] + i^2\,[K_i, K_j]\Big).\end{aligned} \tag{3.67}$$

Insert Eq. (3.64), using $[K_i, J_j] = -\,[J_j, K_i] = -i\,\epsilon_{jik}K_k = +\,i\,\epsilon_{ijk}K_k$ for the third term and $-\,[K_i, K_j] = +\,i\,\epsilon_{ijk}J_k$ for the fourth:

$$\left[N_i^{+}, N_j^{+}\right] = \tfrac{1}{4}\Big(i\epsilon_{ijk}J_k + i\,(i\epsilon_{ijk}K_k) + i\,(i\epsilon_{ijk}K_k) + i\epsilon_{ijk}J_k\Big)$$

$$= \tfrac{1}{4}\Big(2i\,\epsilon_{ijk}J_k - 2\,\epsilon_{ijk}K_k\Big) = \tfrac{1}{4}\,(2i\,\epsilon_{ijk})\,(J_k + iK_k) = i\,\epsilon_{ijk}\,N_k^+. \quad (3.68)$$

The identical manipulation with $i \to -i$ throughout gives the N^- bracket; carrying it out,

$$\begin{aligned}\Big[N_i^-, N_j^-\Big] &= \tfrac{1}{4}\Big([J_i, J_j] - i\,[J_i, K_j] - i\,[K_i, J_j] + i^2\,[K_i, K_j]\Big) \\ &= \tfrac{1}{4}\Big(i\epsilon_{ijk}J_k + \epsilon_{ijk}K_k + \epsilon_{ijk}K_k + i\epsilon_{ijk}J_k\Big) = \tfrac{1}{4}(2i\epsilon_{ijk})\,(J_k - iK_k) = i\,\epsilon_{ijk}\,N_k^-.\end{aligned} \quad (3.69)$$

Finally the cross bracket, where the cancellation is the decisive one:

$$\begin{aligned}\Big[N_i^+, N_j^-\Big] &= \tfrac{1}{4}\,[J_i + iK_i,\; J_j - iK_j\,] \\ &= \tfrac{1}{4}\Big([J_i, J_j] - i\,[J_i, K_j] + i\,[K_i, J_j] + [K_i, K_j]\Big) \\ &= \tfrac{1}{4}\Big(i\epsilon_{ijk}J_k - i(i\epsilon_{ijk}K_k) + i(i\epsilon_{ijk}K_k) - i\epsilon_{ijk}J_k\Big) = 0.\end{aligned} \quad (3.70)$$

The J-terms cancel against each other and the K-terms cancel against each other. We have proved the following.

Theorem 3.17 (Decoupling of the complexified Lorentz algebra)**.** *The generators $N_i^\pm$ of Theorem 3.16 satisfy*

$$\Big[N_i^+, N_j^+\Big] = i\,\epsilon_{ijk}N_k^+, \qquad \Big[N_i^-, N_j^-\Big] = i\,\epsilon_{ijk}N_k^-, \qquad \Big[N_i^+, N_j^-\Big] = 0. \quad (3.71)$$

Hence the complexified Lorentz algebra is a direct sum of two commuting copies of the complexified $\mathfrak{su}(2)$,

$$\mathfrak{so}(1,3)_{\mathbb{C}} \;\cong\; \mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}, \quad (3.72)$$

the first copy spanned by $\{N_i^+\} = \{\vec{A}\}$ and the second by $\{N_i^-\} = \{\vec{B}\}$.

Proof. The brackets Eq. (3.71) were computed above. The six elements $N_i^\pm$ are related to the six real generators J_i, K_i by the invertible complex-linear change of basis Eq. (3.65), so they span the same complex vector space $\mathfrak{so}(1,3)_{\mathbb{C}}$. The first two relations say each triple closes into the $\mathfrak{su}(2)$ bracket; the third says the two triples commute. A Lie algebra that is the vector-space direct sum of two subalgebras which commute elementwise is, by definition, their Lie-algebra direct sum. Each summand has the structure constants of $\mathfrak{su}(2)$ over $\mathbb{C}$, i.e. is $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2,\mathbb{C})$. □

Two remarks fix the meaning of this isomorphism, because it is easy to over-read.

Notation Watch

The decomposition Eq. (3.72) holds for the *complexified* algebra. The real Lorentz algebra $\mathfrak{so}(1,3)$ is *not* a direct sum of two real $\mathfrak{su}(2)$'s: the $N_i^{\pm}$ are not real generators, and there is no real change of basis achieving the split. (Indeed $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ is the real algebra of the compact group SO(4), which is a *different* real form of the same complexification — the precise sense in which "SO(1,3) is the analytic continuation of SO(4)."") What survives over $\mathbb{R}$ is a reality structure: complex conjugation of Eq. (3.65) swaps $N_i^+ \leftrightarrow N_i^-$, equivalently $\vec{A} \leftrightarrow \vec{B}$, because it flips the sign of the i multiplying the real K_i while leaving the real J_i alone. This swap of the two factors is the algebraic seed of the relation between the two Weyl spinors under complex conjugation and parity, taken up in Section 3.4. The $\pm$ labels are attached to $\frac{1}{2}(J \pm iK)$ once and for all by Theorem 3.16; some texts use the opposite sign, so when comparing formulas one must check which combination is being called "+."

Physical Insight

Physically, $\vec{A} = \vec{N}^+$ and $\vec{B} = \vec{N}^-$ measure two independent "angular momenta" built from rotations and boosts together. Because they commute and each generates an $\mathfrak{su}(2)$, a relativistic field carries *two* half-integer charges, not one. The single spin of nonrelativistic quantum mechanics is the diagonal combination $\vec{J} = \vec{A} + \vec{B}$ that survives when boosts are forgotten; relativity refines it into a pair. The two charges are what we will come to call left- and right-handed (chirality), and a theory that treats them asymmetrically is a chiral theory — the structural origin of parity violation in the weak interaction lives in this very decoupling.

3.3.2 The (j_-, j_+) classification

Theorem 3.17 reduces the representation theory of the Lorentz algebra to a problem we have already solved. We import two facts from Chapter 2. First, the finite-dimensional irreducible representations of $\mathfrak{su}(2)$ are labelled by a single nonnegative half-integer, the spin $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, with the spin-$j$ representation D^j acting on a space of dimension $2j+1$ on which the Casimir $\mathbf{J}^2 = \sum_i J_i J_i$ takes the value $j(j+1)$ (Theorem 2.53). This classification is unchanged when we pass to the complexification $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$: a complex-linear representation of $\mathfrak{sl}(2,\mathbb{C})$ on a finite-dimensional complex space is the same data as a representation of the real $\mathfrak{su}(2)$, because $\mathfrak{su}(2)$ spans $\mathfrak{sl}(2,\mathbb{C})$ over $\mathbb{C}$ and a Lie-algebra homomorphism out of it extends uniquely complex-linearly. Second, we need the rule by which representations of a direct sum are built from those of the summands, which rests on Schur's lemma (Theorem 2.21).

Lemma 3.18 (Irreps of a direct sum are external tensor products)**.** *Let $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$ be a direct sum of Lie algebras and let (ρ, V) be a finite-dimensional irreducible complex representation. Then there exist irreducible representations (ρ_1, V_1) of $\mathfrak{g}_1$ and (ρ_2, V_2) of $\mathfrak{g}_2$ such that $V \cong V_1 \otimes V_2$ with*

$$\rho(X_1 \oplus X_2) = \rho_1(X_1) \otimes \mathbf{1}_{V_2} + \mathbf{1}_{V_1} \otimes \rho_2(X_2), \qquad X_i \in \mathfrak{g}_i. \tag{3.73}$$

Conversely every such external tensor product of irreducibles is an irreducible representation of $\mathfrak{g}$, and these exhaust the irreducibles up to isomorphism.

Proof. Because $[\mathfrak{g}_1, \mathfrak{g}_2] = 0$ inside $\mathfrak{g}$, the operators $\rho(X_1)$ commute with all $\rho(X_2)$. Restrict the representation to $\mathfrak{g}_1$; as a $\mathfrak{g}_1$-module V decomposes into isotypic components, each a sum of copies of a fixed irreducible $\mathfrak{g}_1$-module. Fix one irreducible $\mathfrak{g}_1$-module V_1 that actually occurs, and form the multiplicity space $M = \mathrm{Hom}_{\mathfrak{g}_1}(V_1, V)$, the space of $\mathfrak{g}_1$-intertwiners from V_1 into V. The evaluation map

$$V_1 \otimes M \longrightarrow V, \qquad v \otimes f \mapsto f(v), \tag{3.74}$$

is a $\mathfrak{g}_1$-module map onto the V_1-isotypic component of V; it is injective because the f's are linearly independent intertwiners, hence an isomorphism onto that component. Now each $\rho(X_2)$ commutes with $\mathfrak{g}_1$, so it preserves the $\mathfrak{g}_1$-isotypic decomposition and maps the V_1-component into itself. By Schur's lemma (Theorem 2.21) a $\mathfrak{g}_1$-endomorphism of the isotypic component $V_1 \otimes M$ acts as $\mathbf{1}_{V_1} \otimes \sigma(X_2)$ for a unique operator $\sigma(X_2)$ on M, since it must act as a scalar (in fact the identity, being a module map) on the irreducible factor V_1 and arbitrarily on M. The assignment $X_2 \mapsto \sigma(X_2)$ is a representation of $\mathfrak{g}_2$ on M. The subspace $V_1 \otimes M$ is thus invariant under all of $\mathfrak{g} = \mathfrak{g}_1 \oplus \mathfrak{g}_2$; since V is irreducible and $V_1 \otimes M$ is nonzero and invariant, $V = V_1 \otimes M$. Finally, irreducibility of V forces M to be an irreducible $\mathfrak{g}_2$-module V_2: any proper invariant $W \subsetneq M$ would give the proper invariant subspace $V_1 \otimes W$ of V. The converse — that $V_1 \otimes V_2$ is irreducible for irreducible factors, and that these exhaust the list — follows by reversing the argument: an invariant subspace of $V_1 \otimes V_2$, restricted to either tensor factor, would yield a proper invariant subspace there. □

Combining the lemma with the $\mathfrak{su}(2)$ classification gives the result that organises the entire chapter.

Theorem 3.19 (The (j_-, j_+) classification of Lorentz representations)**.** *The finite-dimensional irreducible representations of the (complexified) Lorentz algebra are in one-to-one correspondence with ordered pairs*

$$(j_-, j_+), \qquad j_-, j_+ \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \dots\}. \tag{3.75}$$

The representation $D^{(j_-, j_+)}$ acts on $V_{j_-} \otimes V_{j_+}$, where the first factor (generators $\vec{A} = \vec{N}^+$) carries the spin-j_- irreducible and the second factor (generators $\vec{B} = \vec{N}^-$) carries the spin-j_+ irreducible. Its dimension is

$$\dim D^{(j_-, j_+)} = (2j_- + 1)(2j_+ + 1). \tag{3.76}$$

Proof. By Theorem 3.17 the complexified Lorentz algebra is $\mathfrak{su}(2)_{\mathbb{C}} \oplus \mathfrak{su}(2)_{\mathbb{C}}$, with the first summand spanned by $\vec{A} = \vec{N}^+$ and the second by $\vec{B} = \vec{N}^-$. By Theorem 3.18 its finite-dimensional irreducibles are the external tensor products $D^{j_-} \boxtimes D^{j_+}$ of irreducibles of the two factors, and by Theorem 2.53 each factor's irreducibles are the spin representations labelled by a half-integer. The dimension is the product $(2j_- + 1)(2j_+ + 1)$. A finite-dimensional representation of the real algebra $\mathfrak{so}(1,3)$ on a complex vector space is the same as a complex-linear representation of its complexification, so the same list classifies the (complex) representations of the real Lorentz algebra. □

The convention deserves one sentence of explanation, since the two factors are on an equal footing and the choice of which to call "first" is a convention that propagates through the rest of the chapter. We assign j_- to the $\vec{A} = \vec{N}^+ = \frac{1}{2}(\vec{J} + i\vec{K})$ factor. With this choice, the smallest nontrivial representation in which only $\vec{A}$ acts is $(\frac{1}{2}, 0)$, and Section 3.4 will identify it with the *left-handed* Weyl spinor of the standard physics convention — the spinor whose boost generator is $\vec{K} = -i\vec{\sigma}/2$. The opposite assignment merely relabels every representation by the reflection $(j_-, j_+) \mapsto (j_+, j_-)$ and is just as consistent; we fix ours to match the spinor labelling used universally below.

The single most useful corollary tells us how an ordinary spin — the quantity measured in the rest frame, generated by the rotation subgroup — sits inside (j_-, j_+). The rotation generator is the *diagonal* sum $\vec{J} = \vec{A} + \vec{B} = \vec{N}^+ + \vec{N}^-$, so as a representation of the rotation $\mathfrak{su}(2)$ alone, $D^{(j_-,j_+)}$ is the tensor product of a spin-j_- and a spin-j_+ representation. Its decomposition is exactly the Clebsch–Gordan series (Section 2.5):

$$D^{(j_-,j_+)}\Big|_{\text{rotations}} \cong \bigoplus_{j=|j_- - j_+|}^{j_- + j_+} D^j, \tag{3.77}$$

so the rotational spins present in the field are $j = |j_- - j_+|, |j_- - j_+| + 1, \dots, j_- + j_+$, each once. Counting dimensions, $\sum_{j=|j_- - j_+|}^{j_- + j_+} (2j + 1) = (2j_- + 1)(2j_+ + 1)$, consistent with Eq. (3.76). This is the bridge between the abstract pair (j_-, j_+) and the physical content, "what spins does this field describe."

3.3.3 The catalogue of low-spin fields

We now read off the smallest representations. They are not a random list: they are precisely the fields out of which every Lagrangian in this book is built, and naming them here fixes the objects that the rest of Part I and the benchmark theories will quantise.

$(0,0)$ — **the scalar.** The trivial representation, $j_- = j_+ = 0$, of dimension 1. Both $\vec{A}$ and $\vec{B}$ act as zero, so $J_i = K_i = 0$: the single component is unchanged by

every Lorentz transformation. A field carrying this representation is a Lorentz scalar $\phi(x)$, transforming as $\phi'(x') = \phi(x)$. This is the representation of ϕ^4 theory: the real scalar field of $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$, whose classical theory is built in Chapter 6 and which is quantised in Chapter 9. The rest-frame spin from Eq. (3.77) is $j = 0$, as it must be for a scalar particle.

$(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$ — the two Weyl spinors. Each has dimension $(2\cdot\frac{1}{2}+1)(1) = 2$, and by Eq. (3.77) each carries rotational spin $j = \frac{1}{2}$: these are two-component spin-$\frac{1}{2}$ objects. In $(\frac{1}{2}, 0)$ the first factor $\vec{A} = \vec{N}^+$ acts as the spin-$\frac{1}{2}$ generator $\vec{\sigma}/2$ and the second $\vec{B} = \vec{N}^-$ acts trivially; in $(0, \frac{1}{2})$ it is the reverse. Reading off the physical generators from Eq. (3.66),

$$(\tfrac{1}{2}, 0): \quad \vec{J} = \tfrac{1}{2}\vec{\sigma}, \quad \vec{K} = -\tfrac{i}{2}\vec{\sigma}; \qquad\qquad (0, \tfrac{1}{2}): \quad \vec{J} = \tfrac{1}{2}\vec{\sigma}, \quad \vec{K} = +\tfrac{i}{2}\vec{\sigma}, \tag{3.78}$$

since for $(\frac{1}{2}, 0)$ we have $\vec{A} = \vec{\sigma}/2, \vec{B} = 0$, hence $\vec{J} = \vec{A} = \vec{\sigma}/2$ and $\vec{K} = -i(\vec{A} - \vec{B}) = -i\vec{\sigma}/2$, and the signs flip for $(0, \frac{1}{2})$. The two representations are inequivalent — no Lorentz transformation turns one into the other, because they differ in the sign of the boost generator — but they are exchanged by the reality structure (complex conjugation) and, as we will see, by parity. We call $(\frac{1}{2}, 0)$ the *left-handed* Weyl spinor and $(0, \frac{1}{2})$ the *right-handed* Weyl spinor; this assignment, and the concrete realisation through $\mathrm{SL}(2, \mathbb{C})$ with undotted and dotted two-component indices, is the business of Section 3.4. For now the content is purely algebraic: there are exactly two distinct two-component spinor representations because there are two factors in Eq. (3.72), and the boost generators act on them with opposite signs.

$(\frac{1}{2}, \frac{1}{2})$ — the four-vector. Dimension $2 \cdot 2 = 4$. By Eq. (3.77) the rotational content is $j = 0 \oplus 1$: a scalar (one component) plus a three-vector (three components), which is exactly how the four components of a Lorentz vector $V^\mu = (V^0, \mathbf{V})$ look to a nonrelativistic observer — a time component that is a rotational scalar and a spatial part that is a rotational vector. We prove in Theorem 3.20 below that $(\frac{1}{2}, \frac{1}{2})$ is genuinely the vector representation V^μ, by constructing the explicit intertwiner between a four-vector index μ and a pair of spinor indices. This representation underlies the gauge potential A_μ of QED and Yang–Mills theory: A_μ transforms as a four-vector (up to the gauge ambiguity dissected in Chapters 8 and 25), i.e. it lives in $(\frac{1}{2}, \frac{1}{2})$.

$(1, 0) \oplus (0, 1)$ — the field strength. Each of $(1, 0)$ and $(0, 1)$ has dimension 3; their sum has dimension 6, the number of independent components of an antisymmetric rank-two tensor $F_{\mu\nu} = -F_{\nu\mu}$ in four dimensions. By Eq. (3.77) the spin content of $(1, 0)$ alone is $j = 1$, and likewise for $(0, 1)$. We show in Theorem 3.21 that the antisymmetric tensor representation decomposes precisely

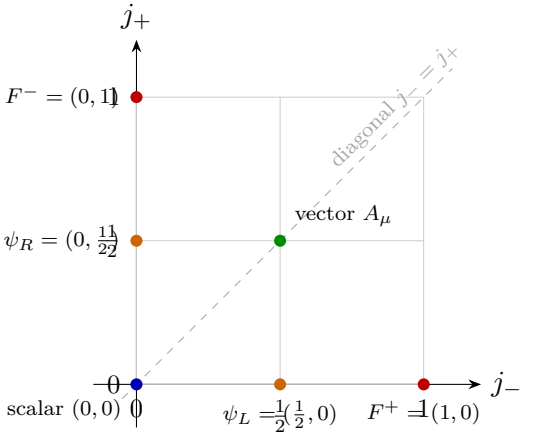


Figure 3.2: The lattice of finite-dimensional irreducible Lorentz representations (j_-, j_+), with j_- (the spin of $\vec{A} = \vec{N}^+$) horizontal and j_+ (the spin of $\vec{B} = \vec{N}^-$) vertical. The diagonal $j_- = j_+$ carries the tensor (bosonic) fields with integer rest-frame spin: $(0,0)$ the scalar, $(\frac{1}{2}, \frac{1}{2})$ the four-vector A_μ. Off-diagonal points carry half-integer spin (Weyl spinors $\psi_L = (\frac{1}{2}, 0)$, $\psi_R = (0, \frac{1}{2})$) or the self-dual/anti-self-dual field strengths $F^\pm = (1,0), (0,1)$. Complex conjugation, and (Section 3.4) parity, reflect the lattice across the diagonal, $(j_-, j_+) \mapsto (j_+, j_-)$.

as $(1,0) \oplus (0,1)$, the two pieces being the self-dual and anti-self-dual parts of $F_{\mu\nu}$, equivalently the combinations $\mathbf{E} \pm i\mathbf{B}$ of the electric and magnetic fields. This is the representation of the electromagnetic and Yang–Mills field strengths.

The pattern visible in Fig. 3.2 is worth naming. Representations on the diagonal $j_- = j_+$ contain integer spins only and describe *tensor* (bosonic) fields; representations with $j_- + j_+$ half-odd-integer contain half-integer spins and describe *spinor* (fermionic) fields. The reflection across the diagonal, $(j_-, j_+) \mapsto (j_+, j_-)$, is the complex-conjugation/parity operation flagged in the notation watch above; a representation fixed by it ($j_- = j_+$, or a reducible sum like $(1,0) \oplus (0,1)$) admits real, parity-even field content, while a single off-diagonal representation does not — the deep reason a single Weyl field is intrinsically chiral.

Worked Example 3.20 (The $(\frac{1}{2}, \frac{1}{2})$ representation is the four-vector)**.** We show explicitly that $(\frac{1}{2}, \frac{1}{2})$ is the four-vector representation by constructing an invertible intertwiner between the four-vector index μ and the pair of two-component spinor indices carried by $(\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$, and verifying that it commutes with the Lorentz action.

Setup. A vector in $(\frac{1}{2}, \frac{1}{2}) = (\frac{1}{2}, 0) \otimes (0, \frac{1}{2})$ carries one index $a \in \{1,2\}$ transforming under the first factor $(\vec{A})$ and one index $\dot{b} \in \{1,2\}$ transforming under the second $(\vec{B})$; write its components $v_{a\dot{b}}$, a 2×2 array. (The dot distinguishes the second-factor index from the first; the full index calculus is set

up in Section 3.4.) Introduce the four Hermitian 2×2 matrices

$$\sigma^\mu_{a\dot{b}} = (\mathbf{1}, \sigma^1, \sigma^2, \sigma^3), \qquad \sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \ \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \ \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{3.79}$$

the unit matrix together with the three Pauli matrices, and define the map from four-vectors to 2×2 matrices

$$v^\mu \ \longmapsto \ \underline{v}_{a\dot{b}} \ := \ v_\mu \, \sigma^\mu_{a\dot{b}} \ = \ v^0 \, \mathbf{1} - \sum_{i=1}^{3} v^i \, \sigma^i, \tag{3.80}$$

where $v_\mu = \eta_{\mu\nu} v^\nu = (v^0, -\mathbf{v})$ in the mostly-minus metric.

It is a real-linear bijection onto Hermitian matrices. Writing out Eq. (3.80),

$$\underline{v} = \begin{pmatrix} v^0 - v^3 & -v^1 + iv^2 \\ -v^1 - iv^2 & v^0 + v^3 \end{pmatrix}. \tag{3.81}$$

Every 2×2 Hermitian matrix has exactly four real parameters (two real diagonal entries and one complex off-diagonal entry), and reading Eq. (3.81) off entry by entry recovers $v^0, \ldots, v^3$ uniquely. Hence Eq. (3.80) is a real-linear isomorphism from $\mathbb{R}^{1,3}$ onto the four-real-dimensional space of Hermitian 2×2 matrices.

The intertwining property, infinitesimally. An infinitesimal element of the first factor $(\frac{1}{2}, 0)$ acts on the undotted index by the matrix $\delta M = -\frac{i}{2}\vec{\theta} \cdot \vec{\sigma} - \frac{1}{2}\vec{\eta} \cdot \vec{\sigma}$, read off from the generators Eq. (3.78) ($\delta M = -i\vec{\theta} \cdot \vec{J} - i\vec{\eta} \cdot \vec{K}$ with $\vec{J} = \vec{\sigma}/2$, $\vec{K} = -i\vec{\sigma}/2$). The dotted index, transforming in the complex-conjugate factor, is acted on from the right by $\delta M^\dagger = +\frac{i}{2}\vec{\theta} \cdot \vec{\sigma} - \frac{1}{2}\vec{\eta} \cdot \vec{\sigma}$. The induced change of the matrix $\underline{v}$ is therefore

$$\delta \underline{v} = \delta M \, \underline{v} + \underline{v} \, \delta M^\dagger. \tag{3.82}$$

We evaluate the rotation and boost parts separately. For a rotation ($\vec{\eta} = 0$) the $\vec{\theta}$-terms combine into a commutator,

$$\begin{aligned} \delta \underline{v}\big|_{\text{rot}} &= -\tfrac{i}{2} \, \theta^a \, [\sigma^a, \underline{v}] = -\tfrac{i}{2} \, \theta^a \big(-v^k \left[\sigma^a, \sigma^k\right]\big) = -\tfrac{i}{2} \, \theta^a \big(-v^k \, 2i\epsilon^{akl} \sigma^l\big) \\ &= -\, \theta^a v^k \epsilon^{akl} \sigma^l = -(\vec{\theta} \times \mathbf{v})^l \sigma^l, \end{aligned} \tag{3.83}$$

using $\left[\sigma^a, \sigma^k\right] = 2i\epsilon^{akl}\sigma^l$ and $\epsilon^{akl}\theta^a v^k = (\vec{\theta} \times \mathbf{v})^l$. This is precisely $-\delta v^l \sigma^l$ for the ordinary rotation $\delta \mathbf{v} = \vec{\theta} \times \mathbf{v}$, $\delta v^0 = 0$ of a four-vector. For a boost ($\vec{\theta} = 0$) the $\vec{\eta}$-terms combine into an anticommutator,

$$\begin{aligned} \delta \underline{v}\big|_{\text{boost}} &= -\tfrac{1}{2} \, \eta^a \, \{\sigma^a, \underline{v}\} = -\tfrac{1}{2} \, \eta^a \big(2v^0 \sigma^a - v^k \left\{\sigma^a, \sigma^k\right\}\big) \\ &= -\tfrac{1}{2} \, \eta^a \big(2v^0 \sigma^a - 2v^a\big) = (\vec{\eta} \cdot \mathbf{v}) \, \mathbf{1} - v^0 \, \eta^a \sigma^a, \end{aligned} \tag{3.84}$$

using $\left\{\sigma^a, \sigma^k\right\} = 2\delta^{ak}\mathbf{1}$. This is exactly $\delta v^0 \, \mathbf{1} - \delta v^a \sigma^a$ for the four-vector boost $\delta v^0 = \vec{\eta} \cdot \mathbf{v}$, $\delta \mathbf{v} = v^0 \vec{\eta}$. In both cases Eq. (3.82) reproduces the four-vector

transformation of v^μ, so the map Eq. (3.80) intertwines the spinor-pair action $(\frac{1}{2},0)\otimes(0,\frac{1}{2})$ with the vector action. Being also a bijection, it is an isomorphism of representations: $(\frac{1}{2},\frac{1}{2}) \cong$ the four-vector.

The invariant is the determinant. As a bonus the construction exposes the Minkowski norm. Computing the determinant of Eq. (3.81),

$$\det \underline{v} = (v^0-v^3)(v^0+v^3)-(-v^1+iv^2)(-v^1-iv^2) = (v^0)^2-(v^3)^2-(v^1)^2-(v^2)^2 = v_\mu v^\mu. \tag{3.85}$$

The finite version of the transformation is $\underline{v} \mapsto A\,\underline{v}\,A^\dagger$ with $A \in \mathrm{SL}(2,\mathbb{C})$ the exponential of δM; it preserves Hermiticity and, since $\det A = 1$, the determinant, hence the norm $v\cdot v$. That the resulting map on v^μ is exactly the proper orthochronous Lorentz group, two-to-one, is the double-cover statement built in Theorem 3.23; here we needed only the infinitesimal intertwining, which already pins down the representation.

Worked Example 3.21 (The antisymmetric tensor is $(1,0)\oplus(0,1)$)**.** An antisymmetric rank-two tensor $F_{\mu\nu} = -F_{\nu\mu}$ has six independent components; we show it carries the reducible representation $(1,0)\oplus(0,1)$ and identify the two irreducible pieces.

Decomposition by symmetry of the two factors. An antisymmetric tensor is the antisymmetric square $\Lambda^2 V$ of the vector representation $V = (\frac{1}{2},\frac{1}{2})$. Using Theorem 3.18 the full tensor square is

$$V\otimes V = \left((\tfrac{1}{2},0)\otimes(0,\tfrac{1}{2})\right)^{\otimes 2} = (\tfrac{1}{2}\otimes\tfrac{1}{2})_{\vec{A}} \otimes (\tfrac{1}{2}\otimes\tfrac{1}{2})_{\vec{B}}, \tag{3.86}$$

where each parenthesis is a tensor product of two spin-$\frac{1}{2}$ representations of one $\mathfrak{su}(2)$ factor. By the $\mathfrak{su}(2)$ Clebsch–Gordan rule (Section 2.5), $\frac{1}{2}\otimes\frac{1}{2} = 1\oplus 0$ in each factor, with the spin-1 triplet *symmetric* and the spin-0 singlet *antisymmetric* under exchange of the two copies. The overall exchange symmetry of a term is the product of the symmetries in the two factors, so

$$V\otimes V = \underbrace{(1,1)}_{\text{sym}\otimes\text{sym}} \oplus \underbrace{(0,0)}_{\text{asym}\otimes\text{asym}} \oplus \underbrace{(1,0)}_{\text{sym}\otimes\text{asym}} \oplus \underbrace{(0,1)}_{\text{asym}\otimes\text{sym}}, \tag{3.87}$$

of dimensions $9+1+3+3 = 16$. The symmetric square $\mathrm{Sym}^2 V$ (dimension 10) is $(1,1)\oplus(0,0)$ — a traceless symmetric tensor plus a trace — while the antisymmetric square is

$$\Lambda^2 V = (1,0)\oplus(0,1), \qquad \dim = 3+3 = 6, \tag{3.88}$$

matching the six components of $F_{\mu\nu}$. This proves the decomposition.

Identifying the pieces as $\mathbf{E}\pm i\mathbf{B}$. Write the electric and magnetic fields as $E^i = F^{i0}$ and $B^i = -\frac{1}{2}\epsilon^{ijk}F_{jk}$, the standard repackaging of the six components

of $F_{\mu\nu}$. Under the Lorentz group both rotate as ordinary three-vectors, $\vec{J}$ acting as the spin-1 generator. Under an infinitesimal boost of rapidity $\vec{\eta}$ they mix as

$$\delta\mathbf{E} = \vec{\eta} \times \mathbf{B}, \qquad \delta\mathbf{B} = -\vec{\eta} \times \mathbf{E}, \tag{3.89}$$

which follows from $F_{\mu\nu}$ being a Lorentz two-tensor acted on by the boost matrix of Theorem 3.15. Forming the complex combinations $\mathbf{F}^{\pm} = \mathbf{E} \pm i\mathbf{B}$, the boost Eq. (3.89) becomes a single closed transformation,

$$\delta\mathbf{F}^{+} = \vec{\eta} \times \mathbf{B} - i\,\vec{\eta} \times \mathbf{E} = -\,i\,\vec{\eta} \times (\mathbf{E} + i\mathbf{B}) = -\,i\,\vec{\eta} \times \mathbf{F}^{+}, \tag{3.90}$$

and likewise $\delta\mathbf{F}^{-} = +i\,\vec{\eta} \times \mathbf{F}^{-}$: the boost acts on $\mathbf{F}^{\pm}$ as $\mp i$ times a rotation, i.e. $\vec{K} = \mp i\vec{J}$ on $\mathbf{F}^{\pm}$. Substituting into Eq. (3.65), on $\mathbf{F}^{+}$ we get $\vec{N}^{-} = \frac{1}{2}(\vec{J} - i\vec{K}) = \frac{1}{2}(\vec{J} - i(-i\vec{J})) = 0$ and $\vec{N}^{+} = \frac{1}{2}(\vec{J} + i\vec{K}) = \vec{J}$, a spin-1 representation of the first factor only: this is $(1, 0)$. On $\mathbf{F}^{-}$ the signs reverse, $\vec{N}^{+} = 0$, $\vec{N}^{-} = \vec{J}$, which is $(0, 1)$. Hence

$$\mathbf{F}^{+} = \mathbf{E} + i\mathbf{B} \;\in\; (1, 0), \qquad \mathbf{F}^{-} = \mathbf{E} - i\mathbf{B} \;\in\; (0, 1). \tag{3.91}$$

The Lorentz-invariant projectors are the Hodge dual. The splitting Eq. (3.91) can be written covariantly. Define the dual tensor

$$\widetilde{F}_{\mu\nu} = \tfrac{1}{2}\,\epsilon_{\mu\nu\rho\sigma} F^{\rho\sigma}, \tag{3.92}$$

with the totally antisymmetric symbol normalised by $\epsilon_{0123} = +1$ (the convention of Chapter A). In Lorentzian signature the dual squares to minus the identity on two-forms,

$$\widetilde{(\widetilde{F})}_{\mu\nu} = \tfrac{1}{4}\,\epsilon_{\mu\nu\rho\sigma}\epsilon^{\rho\sigma\alpha\beta} F_{\alpha\beta} = -\,F_{\mu\nu}, \tag{3.93}$$

where we used $\epsilon_{\mu\nu\rho\sigma}\epsilon^{\rho\sigma\alpha\beta} = -2(\delta^{\alpha}_{\mu}\delta^{\beta}_{\nu} - \delta^{\beta}_{\mu}\delta^{\alpha}_{\nu})$, the extra minus sign relative to Euclidean signature coming from $\det\eta = -1$. Because the dual map is built from the invariant tensors ϵ and η it commutes with Lorentz transformations, and because $\widetilde{(\,\cdot\,)}^{2} = -1$ its eigenvalues are $\pm i$. Its eigenprojections,

$$F^{\pm}_{\mu\nu} = \tfrac{1}{2}(F_{\mu\nu} \mp i\,\widetilde{F}_{\mu\nu}), \qquad \text{satisfying} \qquad \widetilde{F^{\pm}}_{\mu\nu} = \pm\,i\,F^{\pm}_{\mu\nu}, \tag{3.94}$$

are the Lorentz-invariant *self-dual* and *anti-self-dual* parts, each a three-complex-dimensional space, and a short component check identifies them with $\mathbf{F}^{\pm} = \mathbf{E} \pm i\mathbf{B}$ of Eq. (3.91). Reality of the physical $F_{\mu\nu} = F^{+}_{\mu\nu} + F^{-}_{\mu\nu}$ ties the two halves: complex conjugation sends $\mathbf{E} + i\mathbf{B} \mapsto \mathbf{E} - i\mathbf{B}$, i.e. $(1, 0) \leftrightarrow (0, 1)$, exactly the diagonal reflection of Fig. 3.2. A single real antisymmetric tensor field — the electromagnetic field strength of QED, the Yang–Mills field strength $F^{a}_{\mu\nu}$ of Yang–Mills theory — therefore carries the *reducible but parity-symmetric* representation $(1, 0) \oplus (0, 1)$.

3.3.4 Why none of these are unitary

There is a price for the elegance of the (j_-, j_+) classification, and stating it now prevents a serious confusion later. The finite-dimensional representations we have just catalogued describe *fields*; they do not describe *quantum states*. A quantum state lives in a Hilbert space, and a symmetry must act on it by unitary operators so that probabilities are preserved. The representations above are, with the single exception of the trivial $(0,0)$, never unitary. The obstruction is exactly the noncompactness traced to the minus sign in Eq. (3.64).

Theorem 3.22 (No nontrivial finite-dimensional unitary representation)**.** *Let $D^{(j_-,j_+)}$ be a finite-dimensional irreducible representation of the Lorentz algebra with $(j_-, j_+) \neq (0,0)$. There is no positive-definite Hermitian inner product on its representation space with respect to which every Lorentz transformation acts unitarily; equivalently, the boost generators cannot all be represented by Hermitian operators.*

Proof. On $V = V_{j_-} \otimes V_{j_+}$ equip each factor with the standard $\mathfrak{su}(2)$-invariant inner product, so that the angular-momentum operators $\vec{A} = \vec{N}^+$ on V_{j_-} and $\vec{B} = \vec{N}^-$ on V_{j_+} are Hermitian (this is the unitarity of the compact group's representations, Theorem 2.53). With respect to this product the rotation generator $J_i = N_i^+ + N_i^-$ is Hermitian, but the boost generator

$$K_i = -\,i\,(N_i^+ - N_i^-) \tag{3.95}$$

from Eq. (3.66) is *anti*-Hermitian, being $-i$ times the Hermitian operator $N_i^+ - N_i^-$. An anti-Hermitian operator is normal, hence diagonalisable, and its eigenvalues are purely imaginary.

Now suppose, for contradiction, that some positive-definite inner product $\langle \cdot, \cdot \rangle_1$ made the whole representation unitary. Then each boost generator K_i, which generates the one-parameter subgroup $e^{-i\eta K_i}$, would have to be Hermitian with respect to $\langle \cdot, \cdot \rangle_1$, hence would have *real* eigenvalues. But the eigenvalues of a fixed linear operator are the roots of its characteristic polynomial and do not depend on any choice of inner product. They would therefore have to be simultaneously purely imaginary and real, that is, all zero. A diagonalisable operator whose eigenvalues all vanish is the zero operator, so $K_i = 0$ for every i. Then Eq. (3.66) forces $N_i^+ = N_i^-$ for all i, and the vanishing cross-commutator $\left[N_i^+, N_j^-\right] = 0$ collapses to $\left[N_i^+, N_j^+\right] = 0$; comparing with $\left[N_i^+, N_j^+\right] = i\epsilon_{ijk} N_k^+$ forces $N_k^+ = 0$, whence $J_i = K_i = 0$ and the representation is trivial, $(j_-, j_+) = (0,0)$, contrary to hypothesis. □

Physical Insight

The result is not a technicality; it is the structural fact that splits the rest of the chapter in two. *Fields* transform in the finite-dimensional, non-unitary (j_-, j_+) representations — and there is no contradiction, because a classical or quantum field is not a probability amplitude and need not preserve a norm. *One-particle states*, by contrast, transform in genuinely unitary representations, which by the theorem must be infinite-dimensional; the Lorentz group acts unitarily only on function spaces, never on a finite tuple of indices. Constructing those infinite-dimensional unitary representations is the work of Wigner's classification, which we reach after adjoining translations to form the Poincaré group (Sections 3.5 and 3.7). The mass and spin of a particle are the labels of *that* classification; the pair (j_-, j_+) is the label of the field that creates it.

With the catalogue in hand, the abstract pieces now demand a concrete realisation. The two two-component representations $(\tfrac{1}{2}, 0)$ and $(0, \tfrac{1}{2})$ have so far been distinguished only by which factor acts and by the opposite sign of their boost generators in Eq. (3.78); we have not yet written down the matrices $A \in \mathrm{SL}(2,\mathbb{C})$ that implement them, nor explained why the reflection $(j_-, j_+) \mapsto (j_+, j_-)$ that exchanges them should be physically realised by *parity*. The next section builds $\mathrm{SL}(2,\mathbb{C})$ as the universal double cover of $\mathrm{SO}(1,3)^+$, gives the two Weyl spinors their explicit transformation laws with dotted and undotted indices, and shows how demanding parity invariance glues $(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})$ into the Dirac representation that carries the electron of QED.

3.4 Spinors: Weyl, Dirac, Majorana, and the SL(2,ℂ) double cover

The classification of Section 3.3 handed us, among the finite-dimensional irreducible representations labelled by a pair of half-integers (j_-, j_+), two objects that are equally simple and yet stubbornly abstract: the representations $(\tfrac{1}{2}, 0)$ and $(0, \tfrac{1}{2})$. They are the smallest faithful representations of the Lorentz algebra that are not the trivial scalar, and the decomposition $\mathfrak{so}(1,3)_{\mathbb{C}} \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$ told us they exist and what their dimensions are. But that argument worked entirely at the level of the complexified *algebra*; it never produced an honest matrix group acting on an honest two-dimensional space, and it left untouched two physical questions that the algebra alone cannot answer. First, what *group* do these representations represent — is there a Lie group whose representation theory contains $(\tfrac{1}{2}, 0)$ and $(0, \tfrac{1}{2})$, and how is it related to the rotation–boost group $\mathrm{SO}^+(1,3)$ we started from? Second, how does *parity*, which is a symmetry of the full Lorentz group $\mathrm{O}(1,3)$ but lies *outside* the identity component, act on

these labels? The answer to the first question is the group $\mathrm{SL}(2,\mathbb{C})$, the universal double cover of $\mathrm{SO}^+(1,3)$; the answer to the second is that parity exchanges the two labels, which is exactly why the electron — a parity-respecting spin-$\frac{1}{2}$ particle — cannot be a single Weyl spinor but must be the reducible Dirac pair $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$. This section makes both statements precise and proves them in full, realising the spinor representations concretely and, along the way, discharging the topological claim $\pi_1(\mathrm{SO}^+(1,3)) = \mathbb{Z}_2$ that we left open in Section 3.1.

The whole construction rests on a single, almost magical, change of viewpoint: a real four-vector can be repackaged as a 2×2 Hermitian matrix in such a way that its Minkowski length becomes a determinant. We start there.

3.4.1 The Pauli map: four-vectors as Hermitian matrices

Let σ^i $(i=1,2,3)$ be the Pauli matrices and write $\sigma^\mu = (\mathbf{1},\vec{\sigma}) = (\mathbf{1},\sigma^1,\sigma^2,\sigma^3)$, a four-tuple of 2×2 matrices indexed by a spacetime index μ. To a four-vector $x^\mu = (x^0,\vec{x})$ we associate the matrix

$$X \;\equiv\; x_\mu\,\sigma^\mu \;=\; x^0\,\mathbf{1} - \vec{x}\cdot\vec{\sigma} \;=\; \begin{pmatrix} x^0 - x^3 & -x^1 + ix^2 \\ -x^1 - ix^2 & x^0 + x^3 \end{pmatrix}, \tag{3.96}$$

where we used the mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$ so that $x_\mu = (x^0,-\vec{x})$. The map $x \mapsto X$ is a real-linear isomorphism from $\mathbb{R}^{1,3}$ onto the four-real-dimensional space $\mathcal{H}_2$ of Hermitian 2×2 matrices: it is linear and invertible (the inverse is $x^\mu = \frac{1}{2}\,\mathrm{tr}(\bar{\sigma}^\mu X)$, with $\bar{\sigma}^\mu$ defined in Eq. (3.106) below), and X is Hermitian precisely because the x^μ are real and the σ^μ are Hermitian.

The point of the repackaging is the determinant. Reading off the entries of Eq. (3.96),

$$\det X \;=\; (x^0{-}x^3)(x^0{+}x^3) - (-x^1{+}ix^2)(-x^1{-}ix^2) \;=\; (x^0)^2 - (x^3)^2 - (x^1)^2 - (x^2)^2 \;=\; x\cdot x. \tag{3.97}$$

The Minkowski square of x *is* the determinant of its matrix avatar. This is the hinge on which everything turns: any operation on matrices that preserves both Hermiticity and the determinant will, transported back through the isomorphism, act on $\mathbb{R}^{1,3}$ as a linear transformation preserving $x\cdot x$ — that is, as a Lorentz transformation.

There is exactly such an operation. For any matrix $A \in \mathrm{SL}(2,\mathbb{C})$ — the group of complex 2×2 matrices with $\det A = 1$ — consider

$$X \;\longmapsto\; X' \;=\; A\,X\,A^\dagger. \tag{3.98}$$

This sends Hermitian matrices to Hermitian matrices, since $(AXA^\dagger)^\dagger = AX^\dagger A^\dagger = AXA^\dagger$ when $X^\dagger = X$. And it preserves the determinant, because $\det(AXA^\dagger) = \det A\,\det X\,\overline{\det A} = |\det A|^2 \det X = \det X$, using $\det A = 1$. By Eq. (3.97), the four-vector x' defined by $X' = x'_\mu\sigma^\mu$ satisfies $x'\cdot x' = x\cdot x$. The transformation

$x \mapsto x'$ is real-linear (because Eq. (3.98) is, and the isomorphism is real-linear), so it is a genuine element of $\mathrm{O}(1,3)$. We have produced a map

$$\Lambda : \mathrm{SL}(2,\mathbb{C}) \longrightarrow \mathrm{O}(1,3), \qquad A\,X\,A^{\dagger} \;=\; \big(\Lambda(A)\,x\big)_{\mu}\,\sigma^{\mu}. \tag{3.99}$$

The next subsection establishes exactly which Lorentz transformations are reached and how many matrices A produce each one.

3.4.2 $\mathrm{SL}(2,\mathbb{C})$ double covers $\mathrm{SO}^{+}(1,3)$

This is the central structural theorem of the section, and the concrete spinorial counterpart of the abstract $\mathrm{SU}(2) \to \mathrm{SO}(3)$ double cover that Chapter 2 built as a template. We state it in full and prove every clause.

Theorem 3.23 (The spinor double cover)**.** *The map Λ of Eq. (3.99) is a surjective group homomorphism from $\mathrm{SL}(2,\mathbb{C})$ onto the proper orthochronous Lorentz group $\mathrm{SO}^{+}(1,3)$, with kernel $\{+\mathbf{1}, -\mathbf{1}\}$. It is therefore exactly two-to-one. Moreover $\mathrm{SL}(2,\mathbb{C})$ is connected and simply connected, so it is the universal cover of $\mathrm{SO}^{+}(1,3)$ and consequently*

$$\pi_1\big(\mathrm{SO}^{+}(1,3)\big) \;\cong\; \ker\Lambda \;\cong\; \mathbb{Z}_2. \tag{3.100}$$

Proof. Homomorphism. For $A, B \in \mathrm{SL}(2,\mathbb{C})$ we compute the action of the product directly from Eq. (3.99): $(AB)X(AB)^{\dagger} = A\,(BXB^{\dagger})\,A^{\dagger}$. The inner factor $BXB^{\dagger}$ is the matrix avatar of $\Lambda(B)x$, and applying $A(\cdot)A^{\dagger}$ to it produces the avatar of $\Lambda(A)(\Lambda(B)x)$. Hence $\Lambda(AB) = \Lambda(A)\Lambda(B)$.

Kernel. Suppose $\Lambda(A) = \mathbf{1}$, i.e. $AXA^{\dagger} = X$ for every Hermitian X. Choosing $X = \mathbf{1}$ gives $AA^{\dagger} = \mathbf{1}$, so A is unitary. Then $AXA^{\dagger} = AXA^{-1} = X$ for all Hermitian X; since every complex 2×2 matrix is a complex linear combination of Hermitian matrices (write $M = \frac{1}{2}(M + M^{\dagger}) + i\cdot\frac{1}{2i}(M - M^{\dagger})$, both summands Hermitian), A commutes with *every* 2×2 matrix. By Schur's lemma (Chapter 2) $A = c\,\mathbf{1}$ for some scalar c, and $\det A = 1$ forces $c^2 = 1$, so $c = \pm 1$. Thus $\ker\Lambda = \{\pm\mathbf{1}\}$, and because the kernel has two elements, Λ is exactly two-to-one onto its image.

Image lands in $\mathrm{SO}^{+}(1,3)$. The group $\mathrm{SL}(2,\mathbb{C})$ is connected: any A can be joined to $\mathbf{1}$ by a path inside $\mathrm{SL}(2,\mathbb{C})$, as the explicit polar decomposition below makes manifest. A continuous homomorphism carries a connected group into a connected subgroup, so $\Lambda(\mathrm{SL}(2,\mathbb{C}))$ lies in the connected component of the identity of $\mathrm{O}(1,3)$, which is precisely $\mathrm{SO}^{+}(1,3)$ (the transformations of determinant $+1$ that preserve the direction of time). In particular every $\Lambda(A)$ is proper and orthochronous; no parity or time reversal can arise from Eq. (3.98).

Surjectivity onto $\mathrm{SO}^{+}(1,3)$. The differential of Λ at the identity is an isomorphism of Lie algebras. Concretely, write $A = \mathbf{1} + \frac{1}{2}\,\zeta_j\,\sigma^{j} + \cdots$ with complex parameters $\zeta_j = -i\theta_j - \eta_j$; tracelessness of $\frac{1}{2}\zeta_j\sigma^{j}$ is automatic and is exactly

the linearised condition $\det A = 1$, so the six real parameters $(\vec{\theta}, \vec{\eta})$ parametrise the tangent space $\mathfrak{sl}(2) = \mathfrak{su}(2) \oplus i\,\mathfrak{su}(2)$ at $\mathbf{1}$. Substituting into Eq. (3.98) and keeping first order reproduces, as the explicit computation of Section 3.4.3 will show, an infinitesimal rotation by $\vec{\theta}$ and boost by rapidity $\vec{\eta}$ — the full six-dimensional Lorentz algebra of Section 3.2. Hence $d\Lambda$ is a linear isomorphism $\mathfrak{sl}(2)(\mathbb{C})_{\mathbb{R}} \xrightarrow{\sim} \mathfrak{so}(1,3)$ of six-dimensional real Lie algebras, so the image of Λ contains an open neighbourhood of the identity in $\mathrm{SO}^+(1,3)$. A connected Lie group is generated by any neighbourhood of its identity (every element is a finite product of elements near $\mathbf{1}$, since the union of all powers of a symmetric identity-neighbourhood is open, closed, and nonempty, hence everything), so Λ is onto $\mathrm{SO}^+(1,3)$.

Simple connectedness and π_1. Polar decomposition realises every $A \in \mathrm{SL}(2,\mathbb{C})$ uniquely as $A = PU$ with $P = (AA^\dagger)^{1/2}$ a positive-definite Hermitian matrix and $U = P^{-1}A$ unitary. Taking determinants, $\det P \det U = 1$ with $\det P > 0$ real and $|\det U| = 1$; the only solution is $\det P = 1$, $\det U = 1$, so $U \in \mathrm{SU}(2)$ and P is a positive Hermitian matrix of unit determinant. The set of such P is the image of the traceless Hermitian matrices under the exponential $P = \exp(\frac{1}{2}\vec{\eta}\cdot\vec{\sigma})$ and is homeomorphic to $\mathbb{R}^3$ (parametrised freely by $\vec{\eta}$), hence contractible; and $\mathrm{SU}(2) \cong S^3$ is simply connected (Chapter 2). Therefore, as a topological space,

$$\mathrm{SL}(2,\mathbb{C}) \;\cong\; \mathbb{R}^3 \times S^3, \tag{3.101}$$

which is connected and simply connected. A connected double cover of $\mathrm{SO}^+(1,3)$ by a simply connected group is its universal cover, and for a covering of connected groups the fibre over the identity — here $\ker\Lambda = \{\pm\mathbf{1}\}$ — is isomorphic to the fundamental group of the base. This proves Eq. (3.100), discharging the claim deferred from Section 3.1. □

Physical Insight

The two sheets are not a bookkeeping nuisance; they are the reason spinors exist. A rotation by 2π about a fixed axis is the *identity* in $\mathrm{SO}^+(1,3)$, but the path tracing it lifts to a path in $\mathrm{SU}(2) \subset \mathrm{SL}(2,\mathbb{C})$ from $+\mathbf{1}$ to $-\mathbf{1}$: a 2π rotation sends $A \to -A$, and only a 4π rotation returns to $+\mathbf{1}$. Any object that transforms under A rather than under $\Lambda(A)$ — a spinor — therefore picks up a sign under a full rotation. The non-trivial loop generating $\mathbb{Z}_2 = \pi_1(\mathrm{SO}^+(1,3))$ is exactly the 2π rotation, and the spinor's minus sign is the holonomy of the double cover around that loop. This single $\mathbb{Z}_2$ is what the spin–statistics connection of Chapter 11 will trade for the antisymmetry of fermionic wavefunctions.

3.4.3 Weyl spinors and their transformation laws

Theorem 3.23 tells us that $\mathrm{SL}(2,\mathbb{C})$ is the group whose representations we should be cataloguing, and it comes equipped with an obvious two-dimensional one: its own defining action on $\mathbb{C}^2$.

Definition 3.24 (Left- and right-handed Weyl spinors)**.** A *left-handed Weyl spinor* is a pair $\psi_L \in \mathbb{C}^2$ transforming in the defining (fundamental) representation of $\mathrm{SL}(2,\mathbb{C})$,

$$\psi_L \;\longrightarrow\; A\,\psi_L, \qquad A \in \mathrm{SL}(2,\mathbb{C}). \tag{3.102}$$

A *right-handed Weyl spinor* is $\psi_R \in \mathbb{C}^2$ transforming in the conjugate-inverse-transpose, or anti-fundamental, representation,

$$\psi_R \;\longrightarrow\; (A^\dagger)^{-1}\psi_R. \tag{3.103}$$

To see that these are the abstract $(\tfrac12,0)$ and $(0,\tfrac12)$ we read off their generators. Parametrise a general element near the identity, exactly as in the surjectivity argument, by a rotation angle $\vec\theta$ and a rapidity $\vec\eta$:

$$A(\vec\theta,\vec\eta) \;=\; \exp\!\Big(-\tfrac{i}{2}\,\vec\theta\cdot\vec\sigma - \tfrac12\,\vec\eta\cdot\vec\sigma\Big). \tag{3.104}$$

Then Eqs. (3.102) and (3.103) become, using $(A^\dagger)^{-1} = \exp(-\frac{i}{2}\vec\theta\cdot\vec\sigma + \frac12\vec\eta\cdot\vec\sigma)$ (the rapidity term is anti-Hermitian-conjugated and flips sign while the rotation term, being anti-Hermitian, does not),

$$\psi_L \to \exp\!\Big(\tfrac{(-i\vec\theta-\vec\eta)\cdot\vec\sigma}{2}\Big)\psi_L, \qquad \psi_R \to \exp\!\Big(\tfrac{(-i\vec\theta+\vec\eta)\cdot\vec\sigma}{2}\Big)\psi_R. \tag{3.105}$$

Both spinors carry the *same* rotation generator $\vec J = \vec\sigma/2$, the spin-$\frac12$ generator of $\mathfrak{su}(2)$ from Chapter 2; they differ only in the sign of the boost generator, $\vec K = \pm i\vec\sigma/2$. In terms of the commuting combinations $\vec A = \frac12(\vec J + i\vec K)$ and $\vec B = \frac12(\vec J - i\vec K)$ introduced in Section 3.3, the left spinor has $\vec A = \vec\sigma/2,\ \vec B = 0$ and the right spinor has $\vec A = 0,\ \vec B = \vec\sigma/2$. Thus ψ_L realises $(\tfrac12,0)$ and ψ_R realises $(0,\tfrac12)$, concretely and explicitly. This is the realisation promised in the section opening.

We should check that the parametrisation Eq. (3.104) is the one consistent with Eq. (3.98); verifying it on a boost both confirms the claim and shows where the second set of Pauli matrices, $\bar\sigma^\mu$, comes from.

Worked Example 3.25 (A boost from $\mathrm{SL}(2,\mathbb{C})$, and the appearance of $\bar\sigma^\mu$)**.** Take a pure boost along $\hat z$ with rapidity η, so $\vec\theta = 0$, $\vec\eta = \eta\,\hat z$, and from Eq. (3.104),

$$A = \exp(-\tfrac{\eta}{2}\sigma^3) = \mathrm{diag}(e^{-\eta/2},\, e^{+\eta/2}), \qquad A^\dagger = A.$$

With $X = x^0\mathbf{1} - \vec{x}\cdot\vec{\sigma}$ from Eq. (3.96), the congruence Eq. (3.98) acts only on the diagonal entries, which carry $x^0 \mp x^3$:

$$(AXA^\dagger)_{11} = e^{-\eta}(x^0 - x^3), \qquad (AXA^\dagger)_{22} = e^{+\eta}(x^0 + x^3),$$

while the off-diagonal entries (carrying x^1, x^2) are multiplied by $e^{-\eta/2}e^{+\eta/2} = 1$ and are unchanged. Writing $X' = x'^0\mathbf{1} - \vec{x}'\cdot\vec{\sigma}$ and matching, $x'^0 - x'^3 = e^{-\eta}(x^0 - x^3)$ and $x'^0 + x'^3 = e^{+\eta}(x^0 + x^3)$. Adding and subtracting,

$$x'^0 = x^0\cosh\eta + x^3\sinh\eta, \qquad x'^3 = x^0\sinh\eta + x^3\cosh\eta, \qquad x'^1 = x^1,\ x'^2 = x^2,$$

which is exactly a boost of rapidity η along $\hat{z}$. So Eq. (3.104) is the correct dictionary, and $\psi_L \to A\psi_L$ is genuinely a left Weyl spinor under this boost.

The same computation reveals the natural objects for forming Lorentz invariants and covariants. Define

$$\bar{\sigma}^\mu \;=\; (\mathbf{1}, -\vec{\sigma}) \;=\; (\mathbf{1}, -\sigma^1, -\sigma^2, -\sigma^3), \tag{3.106}$$

the spatial sign-flipped partner of $\sigma^\mu = (\mathbf{1}, \vec{\sigma})$. Consider the bilinear $j^\mu = \psi_L^\dagger\,\bar{\sigma}^\mu\,\psi_L$. Under Eq. (3.102) it becomes $\psi_L^\dagger A^\dagger\,\bar{\sigma}^\mu\,A\,\psi_L$, so it is a four-vector iff $A^\dagger\bar{\sigma}^\mu A = \Lambda^\mu{}_\nu\,\bar{\sigma}^\nu$. For the boost at hand, $A^\dagger\bar{\sigma}^0 A = A^2 = \mathrm{diag}(e^{-\eta}, e^{\eta}) = \cosh\eta\,\mathbf{1} - \sinh\eta\,\sigma^3 = \cosh\eta\,\bar{\sigma}^0 + \sinh\eta\,\bar{\sigma}^3$, since $\bar{\sigma}^3 = -\sigma^3$. The right-hand side is $\Lambda^0{}_0\,\bar{\sigma}^0 + \Lambda^0{}_3\,\bar{\sigma}^3$ with precisely the boost matrix entries found above. Hence $\psi_L^\dagger\bar{\sigma}^\mu\psi_L$ transforms as a contravariant four-vector. By the mirror argument $\psi_R^\dagger\,\sigma^\mu\,\psi_R$ is also a four-vector, with σ^μ and $\bar{\sigma}^\mu$ trading places. These two bilinears are the kinetic-term building blocks of the Weyl Lagrangians quantised in Chapter 11.

There is one more invariant we will need constantly. The defining and anti-fundamental representations are not independent: $\mathrm{SL}(2,\mathbb{C})$ leaves invariant the antisymmetric symbol

$$\varepsilon = i\sigma^2 = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \varepsilon^{-1} = -i\sigma^2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \tag{3.107}$$

which plays for spinor indices the role the metric $\eta_{\mu\nu}$ plays for vector indices. The key identity is that for *any* 2×2 matrix A,

$$A^{\mathsf{T}}\,\varepsilon\,A = (\det A)\,\varepsilon, \tag{3.108}$$

which one checks by direct multiplication (both sides are antisymmetric 2×2 matrices, so each is a multiple of ε, and comparing the $(1,2)$ entries gives the cofactor expansion $\det A$). For $A \in \mathrm{SL}(2,\mathbb{C})$ this reads $A^{\mathsf{T}}\varepsilon A = \varepsilon$, the precise statement that ε is an invariant bilinear form: the combination $\psi_L^{\mathsf{T}}\varepsilon\,\chi_L = \psi_L^1\chi_L^2 - \psi_L^2\chi_L^1$ of two left spinors is Lorentz invariant. Rearranging $A^\dagger\varepsilon A^* = \varepsilon$

(which is Eq. (3.108) conjugated, using $\varepsilon^* = \varepsilon$) — left-multiplying by $(A^\dagger)^{-1}$ and right-multiplying by ε^{-1} — gives the equally important

$$\varepsilon\, A^*\, \varepsilon^{-1} = (A^\dagger)^{-1}. \tag{3.109}$$

This says that $\varepsilon\, \psi_L^*$ transforms with $(A^\dagger)^{-1}$ — that is, as a *right*-handed Weyl spinor. The complex conjugate of a left spinor, dressed with ε, is a right spinor; this is the algebraic seed of both the Majorana condition (Section 3.4.6) and charge conjugation (Chapter 11).

3.4.4 The Dirac spinor and the Clifford algebra

A left and a right Weyl spinor can always be stacked into a single four-component object on which the Lorentz group acts reducibly. Doing so is not merely notational: it is forced by parity (next subsection), and it is the natural arena for the Clifford algebra of gamma matrices.

Definition 3.26 (Dirac spinor)**.** A *Dirac spinor* is the four-component column

$$\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \in \mathbb{C}^4, \tag{3.110}$$

carrying the reducible representation $(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})$ of $\mathrm{SL}(2,\mathbb{C})$, with each block transforming by Eq. (3.105).

The reason this stacking is so productive is that the two-component matrices σ^μ and $\bar\sigma^\mu$ assemble into a set of four 4×4 matrices satisfying a single clean relation. Define, in the *Weyl* (chiral) basis,

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{pmatrix}, \qquad \sigma^\mu = (\mathbf{1}, \vec\sigma), \quad \bar\sigma^\mu = (\mathbf{1}, -\vec\sigma). \tag{3.111}$$

These obey the Clifford (Dirac) algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\, \mathbf{1}_4, \tag{3.112}$$

the master relation of Chapter C (Appendix C), whose full set of trace and contraction identities we collect there and use here only without full proof. We can verify Eq. (3.112) directly from Eq. (3.111): the anticommutator is block-diagonal,

$$\{\gamma^\mu, \gamma^\nu\} = \begin{pmatrix} \sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu & 0 \\ 0 & \bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu \end{pmatrix},$$

and the two-component blocks satisfy $\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu = 2\eta^{\mu\nu}\mathbf{1}$. The latter is checked case by case: for $\mu = \nu = 0$ it is $2\mathbf{1} = 2\eta^{00}\mathbf{1}$; for $\mu = \nu = i$ it is

$2\sigma^i(-\sigma^i) = -2\mathbf{1} = 2\eta^{ii}\mathbf{1}$ (no sum); and for $\mu = 0, \nu = i$ it is $(-\sigma^i)+\sigma^i = 0 = 2\eta^{0i}$. The same holds for the lower block. The Weyl basis thus *solves* the Clifford algebra, and conversely any solution of Eq. (3.112) in 4×4 matrices is unitarily equivalent to it.

In this basis the chirality matrix is diagonal:

$$\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & +\mathbf{1} \end{pmatrix}, \qquad P_L = \tfrac{1-\gamma^5}{2} = \begin{pmatrix} \mathbf{1} & 0 \\ 0 & 0 \end{pmatrix}, \quad P_R = \tfrac{1+\gamma^5}{2} = \begin{pmatrix} 0 & 0 \\ 0 & \mathbf{1} \end{pmatrix}. \tag{3.113}$$

The projectors P_L, P_R extract the upper and lower blocks, so $P_L\psi = (\psi_L, 0)^{\mathsf{T}}$ and $P_R\psi = (0, \psi_R)^{\mathsf{T}}$ — chirality is literally the choice of Weyl component, which is why Eq. (3.110) writes ψ_L above ψ_R.

The Lorentz generators on a Dirac spinor are built from the gammas as

$$S^{\mu\nu} = \frac{i}{4}\,[\gamma^\mu, \gamma^\nu]\,, \tag{3.114}$$

and the Lorentz transformation of the spinor is $\psi \to \exp(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu})\psi$, with $\omega_{\mu\nu}$ the antisymmetric parameter encoding $(\vec{\theta}, \vec{\eta})$. Two facts must be established: that Eq. (3.114) *is* a representation of the Lorentz algebra, and that it is the *reducible* representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ and no other. The first is a computation we owe the reader in full; the second follows by inspection of the blocks.

Worked Example 3.27 ($S^{\mu\nu} = \frac{i}{4}\,[\gamma^\mu, \gamma^\nu]$ satisfies the Lorentz algebra)**.** We show that the matrices Eq. (3.114) reproduce the Lorentz commutation relations of Section 3.2,

$$[S^{\mu\nu}, S^{\rho\sigma}] = i\big(\eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} - \eta^{\nu\sigma}S^{\mu\rho} + \eta^{\mu\sigma}S^{\nu\rho}\big), \tag{3.115}$$

using only the Clifford relation Eq. (3.112). First rewrite $S^{\mu\nu}$ without the commutator. Since $[\gamma^\mu, \gamma^\nu] = 2\gamma^\mu\gamma^\nu - \{\gamma^\mu, \gamma^\nu\} = 2\gamma^\mu\gamma^\nu - 2\eta^{\mu\nu}$,

$$S^{\mu\nu} = \frac{i}{2}\big(\gamma^\mu\gamma^\nu - \eta^{\mu\nu}\big), \qquad \text{equivalently} \qquad \gamma^\mu\gamma^\nu = -2i\,S^{\mu\nu} + \eta^{\mu\nu}. \tag{3.116}$$

Step 1 (the gamma is a vector operator). We first compute $[S^{\mu\nu}, \gamma^\rho]$. Only the $\gamma^\mu\gamma^\nu$ piece of Eq. (3.116) contributes, so $[S^{\mu\nu}, \gamma^\rho] = \frac{i}{2}\,[\gamma^\mu\gamma^\nu, \gamma^\rho]$. Move γ^ρ through using $\gamma^\nu\gamma^\rho = 2\eta^{\nu\rho} - \gamma^\rho\gamma^\nu$ and $\gamma^\rho\gamma^\mu = 2\eta^{\rho\mu} - \gamma^\mu\gamma^\rho$:

$$\begin{aligned} \gamma^\mu\gamma^\nu\gamma^\rho &= \gamma^\mu\big(2\eta^{\nu\rho} - \gamma^\rho\gamma^\nu\big) = 2\eta^{\nu\rho}\gamma^\mu - \gamma^\mu\gamma^\rho\gamma^\nu, \\ \gamma^\rho\gamma^\mu\gamma^\nu &= \big(2\eta^{\rho\mu} - \gamma^\mu\gamma^\rho\big)\gamma^\nu = 2\eta^{\rho\mu}\gamma^\nu - \gamma^\mu\gamma^\rho\gamma^\nu, \end{aligned}$$

so that $[\gamma^\mu\gamma^\nu, \gamma^\rho] = 2\eta^{\nu\rho}\gamma^\mu - 2\eta^{\mu\rho}\gamma^\nu$, and therefore

$$[S^{\mu\nu}, \gamma^\rho] = i\big(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu\big). \tag{3.117}$$

This is precisely the statement that γ^ρ transforms as a four-vector under the action generated by $S^{\mu\nu}$.

Step 2 (close the algebra). Now use Eq. (3.116) to write $S^{\rho\sigma}$ in terms of $\gamma^\rho\gamma^\sigma$ and apply the Leibniz rule for the commutator together with Eq. (3.117):

$$\begin{aligned}[S^{\mu\nu}, S^{\rho\sigma}] &= \frac{i}{2}\,[S^{\mu\nu}, \gamma^\rho\gamma^\sigma] = \frac{i}{2}\Big([S^{\mu\nu}, \gamma^\rho]\,\gamma^\sigma + \gamma^\rho\,[S^{\mu\nu}, \gamma^\sigma]\Big) \\ &= \frac{i}{2}\Big(i(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu)\gamma^\sigma + \gamma^\rho\, i(\eta^{\nu\sigma}\gamma^\mu - \eta^{\mu\sigma}\gamma^\nu)\Big) \\ &= -\frac{1}{2}\Big(\eta^{\nu\rho}\gamma^\mu\gamma^\sigma - \eta^{\mu\rho}\gamma^\nu\gamma^\sigma + \eta^{\nu\sigma}\gamma^\rho\gamma^\mu - \eta^{\mu\sigma}\gamma^\rho\gamma^\nu\Big).\end{aligned}$$

Replace each product $\gamma^\alpha\gamma^\beta = -2iS^{\alpha\beta} + \eta^{\alpha\beta}$ from Eq. (3.116). The four pure-metric terms are $\eta^{\nu\rho}\eta^{\mu\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\nu\sigma}\eta^{\rho\mu} - \eta^{\mu\sigma}\eta^{\rho\nu}$, which cancel in pairs ($\eta^{\nu\rho}\eta^{\mu\sigma} = \eta^{\mu\sigma}\eta^{\rho\nu}$ and $\eta^{\mu\rho}\eta^{\nu\sigma} = \eta^{\nu\sigma}\eta^{\rho\mu}$). The S-terms give

$$\begin{aligned}[S^{\mu\nu}, S^{\rho\sigma}] &= -\frac{1}{2}(-2i)\Big(\eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} + \eta^{\nu\sigma}S^{\rho\mu} - \eta^{\mu\sigma}S^{\rho\nu}\Big) \\ &= i\Big(\eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} - \eta^{\nu\sigma}S^{\mu\rho} + \eta^{\mu\sigma}S^{\nu\rho}\Big),\end{aligned}$$

where in the last step we used the antisymmetry $S^{\rho\mu} = -S^{\mu\rho}$, $S^{\rho\nu} = -S^{\nu\rho}$. This is Eq. (3.115) verbatim. The Dirac matrices therefore furnish an honest representation of the Lorentz algebra of Section 3.2.

That this representation is $(\tfrac12, 0) \oplus (0, \tfrac12)$ is now visible by inspection, since Eq. (3.114) is block-diagonal. Using Eq. (3.111), the commutator $[\gamma^\mu, \gamma^\nu]$ has vanishing off-diagonal blocks, so $S^{\mu\nu} = \mathrm{diag}(S_L^{\mu\nu}, S_R^{\mu\nu})$ acts independently on ψ_L and ψ_R. Evaluating the blocks, the rotation generators $J^k = \frac12\epsilon^{kij}S^{ij}$ come out to $\vec\sigma/2$ on *both* blocks, while the boost generators $K^k = S^{0k}$ are $-\frac{i}{2}\vec\sigma$ on the upper (ψ_L) block and $+\frac{i}{2}\vec\sigma$ on the lower (ψ_R) block. For the upper block, for instance, $S_L^{ij} = \frac{i}{4}(\sigma^i\bar\sigma^j - \sigma^j\bar\sigma^i) = \frac{i}{4}(\sigma^i(-\sigma^j) - \sigma^j(-\sigma^i)) = -\frac{i}{4}\,[\sigma^i, \sigma^j] = \frac12\epsilon^{ijk}\sigma^k$, whence $J_L^k = \frac12\sigma^k$, and $S_L^{0k} = \frac{i}{4}(\sigma^0\bar\sigma^k - \sigma^k\bar\sigma^0) = \frac{i}{4}(-\sigma^k - \sigma^k) = -\frac{i}{2}\sigma^k$. Exponentiating, the upper block of $\exp(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu})$ is exactly the left-spinor law Eq. (3.105) and the lower block the right-spinor law. The Dirac representation is precisely the Weyl pair — nothing more, nothing less — which is the decomposition $(\tfrac12, 0) \oplus (0, \tfrac12)$ we set out to prove.

3.4.5 Parity and the necessity of the Dirac spinor

Everything so far lives inside the identity component $\mathrm{SO}^+(1,3)$, reached by $\mathrm{SL}(2,\mathbb{C})$. Parity $P : (x^0, \vec x) \mapsto (x^0, -\vec x)$ has $\det = -1$ and lies in a *disconnected* sheet of $\mathrm{O}(1,3)$, so it is not of the form $\Lambda(A)$ for any A. Its effect on representations is nonetheless completely fixed by its effect on the generators.

Proposition 3.28 (Parity exchanges the representation labels)**.** *Under parity the rotation generators are unchanged and the boost generators reverse sign,*

$$P: \quad \vec{J} \mapsto \vec{J}, \qquad \vec{K} \mapsto -\vec{K}. \tag{3.118}$$

Consequently the commuting $\mathfrak{su}(2)$ *generators of Section 3.3 are interchanged,* $\vec{A} \leftrightarrow \vec{B}$*, and parity maps the irreducible representation* (j_-, j_+) *to* (j_+, j_-)*. In particular it exchanges the two Weyl spinors,* $(\frac{1}{2}, 0) \leftrightarrow (0, \frac{1}{2})$*.*

Proof. Rotations are spatial and survive parity unchanged: $\vec{J}$ generates rotations of $\vec{x}$, and conjugating by P leaves a rotation a rotation, $P\, e^{-i\vec{\theta}\cdot\vec{J}} P^{-1} = e^{-i\vec{\theta}\cdot\vec{J}}$, so $\vec{J} \mapsto \vec{J}$. A boost, on the other hand, mixes x^0 with $\vec{x}$; conjugating a boost of rapidity $\vec{\eta}$ by P reverses the spatial direction and hence the boost, $P\, e^{-i\vec{\eta}\cdot\vec{K}} P^{-1} = e^{+i\vec{\eta}\cdot\vec{K}}$, giving $\vec{K} \mapsto -\vec{K}$. This is Eq. (3.118). (Equivalently and more compactly, parity acts on the generators $M^{\mu\nu}$ of Section 3.2 by $M^{\mu\nu} \mapsto P^{\mu}{}_{\rho} P^{\nu}{}_{\sigma} M^{\rho\sigma}$ with $P = \mathrm{diag}(+,-,-,-)$; the purely spatial $M^{ij} = \epsilon^{ijk} J_k$ pick up two sign flips and are unchanged, while the mixed $M^{0i} = K^i$ pick up one and flip sign.)

Now recall the decoupling combinations of Section 3.3, $\vec{A} = \frac{1}{2}(\vec{J} + i\vec{K})$ and $\vec{B} = \frac{1}{2}(\vec{J} - i\vec{K})$, the two commuting $\mathfrak{su}(2)$ factors labelled by spins (j_-, j_+). Applying Eq. (3.118),

$$\vec{A} = \tfrac{1}{2}(\vec{J} + i\vec{K}) \xmapsto{P} \tfrac{1}{2}(\vec{J} - i\vec{K}) = \vec{B}, \qquad \vec{B} \xmapsto{P} \vec{A}.$$

Parity swaps the two $\mathfrak{su}(2)$ factors. A representation in which the first factor carries spin j_- and the second j_+ is therefore carried, after parity, into the representation with the spins exchanged: $(j_-, j_+) \mapsto (j_+, j_-)$. Setting $(j_-, j_+) = (\frac{1}{2}, 0)$ gives $(0, \frac{1}{2})$, and vice versa. □

Figure 3.3 displays the action as a reflection of the (j_-, j_+) lattice across its diagonal. The physical consequence is immediate and decisive. A field that is to describe a spin-$\frac{1}{2}$ particle in a parity-respecting theory — and electromagnetism, the binding force of the electron, conserves parity — must transform into itself (up to a fixed unitary) under P. A single left Weyl spinor cannot: parity would map it to a right Weyl spinor, an object that does not live in the representation we started with. The smallest parity-invariant representation containing spin $\frac{1}{2}$ is therefore the *reducible* sum

$$(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2}), \tag{3.119}$$

on which parity acts by exchanging the two summands — concretely, by $\psi \mapsto \gamma^0 \psi$, since $\gamma^0 = \left(\begin{smallmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{smallmatrix}\right)$ in the Weyl basis swaps ψ_L and ψ_R. This is the Dirac spinor of Theorem 3.26. *Parity forces the Dirac representation.* The electron of QED is a Dirac field for exactly this reason; the neutrino, which participates only in the parity-violating weak interaction, can be (and in the minimal Standard Model is) a single Weyl spinor.

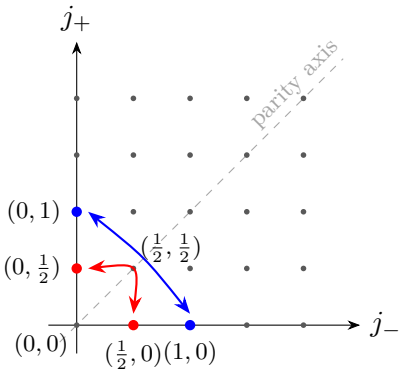


Figure 3.3: The lattice of finite-dimensional Lorentz representations (j_-, j_+). Parity reflects the lattice across the diagonal $j_- = j_+$, exchanging $(j_-, j_+) \leftrightarrow (j_+, j_-)$ (Theorem 3.28). The two Weyl spinors $(\frac{1}{2}, 0)$ (red) and $(0, \frac{1}{2})$ (red) are swapped, as are $(1, 0)$ and $(0, 1)$ (blue, the self-dual and anti-self-dual parts of the field strength). Only diagonal representations (j, j) — the scalar $(0, 0)$ and the vector $(\frac{1}{2}, \frac{1}{2})$ — are parity-invariant by themselves.

Physical Insight

The arithmetic of Theorem 3.28 also explains the rest of the field content of Section 3.3 at a glance. The scalar $(0, 0)$ and the four-vector $(\frac{1}{2}, \frac{1}{2})$ sit *on* the diagonal of Fig. 3.3 and are individually parity-invariant — consistent with a single scalar field and a single vector potential being perfectly good parity-even objects. The field strength $F_{\mu\nu}$, an antisymmetric tensor, sits at $(1, 0) \oplus (0, 1)$: its self-dual and anti-self-dual parts $(1, 0)$ and $(0, 1)$ are individually *off*-diagonal and are swapped by parity, which is the representation-theoretic statement that parity exchanges $\vec{E} \pm i\vec{B}$, i.e. sends $\vec{E} \to \vec{E}$, $\vec{B} \to -\vec{B}$. Parity is not an afterthought bolted onto the classification; it is a reflection symmetry of the whole lattice.

3.4.6 The Majorana condition

The Dirac spinor Eq. (3.110) carries four complex components, twice the count of a Weyl spinor. We saw in Eq. (3.109) that complex conjugation, dressed with ε, turns a left spinor into a right one. This opens the possibility of a *reality* constraint that ties the right block to the conjugate of the left block, halving the degrees of freedom while keeping a four-component, parity-friendly object.

Definition 3.29 (Majorana spinor)**.** A Dirac spinor satisfies the *Majorana condition* if its right-handed part is the ε-conjugate of its left-handed part,

$$\psi_R = \varepsilon\, \psi_L^* = i\sigma^2\, \psi_L^*. \tag{3.120}$$

A spinor obeying Eq. (3.120) is called a *Majorana spinor*; it is built from a single Weyl spinor ψ_L and carries two complex (four real) degrees of freedom, half those of a Dirac spinor.

The condition Eq. (3.120) is Lorentz covariant: this is exactly what Eq. (3.109) guarantees. Under a Lorentz transformation $\psi_L \to A\psi_L$, the left side of Eq. (3.120) transforms as $\psi_R \to (A^\dagger)^{-1}\psi_R$, while the right side transforms as $\varepsilon\psi_L^* \to \varepsilon A^*\psi_L^* = (\varepsilon A^*\varepsilon^{-1})\,\varepsilon\psi_L^* = (A^\dagger)^{-1}\varepsilon\psi_L^*$. Both sides transform identically, so a spinor satisfying the constraint in one frame satisfies it in all frames. The constraint is thus a Lorentz-invariant statement, and the Majorana spinor is a bona fide field, not a frame-dependent artifact.

Two remarks fix its meaning. First, the Majorana condition is the spinor analogue of the reality condition $\phi = \phi^*$ that distinguishes a real scalar from a complex one: it is the maximal reality constraint compatible with Lorentz invariance for a spin-$\frac{1}{2}$ field. Second, because $\varepsilon\psi_L^*$ is precisely the combination that will define the charge-conjugate field in Chapter 11, Eq. (3.120) states that a Majorana spinor equals its own charge conjugate; physically, the corresponding particle is its own antiparticle. Whether nature realises this for the neutrino is an open experimental question, pursued through the search for neutrinoless double beta decay; we flag it here only to mark that the algebraic possibility is forced on us by Eq. (3.109) and will be quantised in Chapter 11.

Notation Watch

The two-component sign conventions adopted here — $\sigma^\mu = (\mathbf{1}, \vec{\sigma})$, $\bar{\sigma}^\mu = (\mathbf{1}, -\vec{\sigma})$ of Eq. (3.106), the invariant $\varepsilon = i\sigma^2$ of Eq. (3.107), and the chiral gamma basis Eq. (3.111) with $\gamma^5 = \mathrm{diag}(-\mathbf{1}, +\mathbf{1})$ — are the ones used throughout this book and collected in Chapter C. They match the four-component spinor of Chapter 11 and the two-component (van der Waerden) notation revisited in the amplitudes chapters Chapter 58. References differ in the sign of γ^5 and in whether ψ_L sits above or below ψ_R in Eq. (3.110); we fix both choices once, here.

We now have the complete finite-dimensional dictionary: every field of the theory is a definite (j_-, j_+), the spin-$\frac{1}{2}$ entries are realised concretely as Weyl, Dirac, and Majorana spinors carried by the double cover $\mathrm{SL}(2,\mathbb{C})$, and parity organises them by reflection. What is still missing is the rest of the symmetry: these representations describe how fields respond to rotations and boosts *at a point*, but a relativistic theory must also be invariant under spacetime translations. Adjoining translations to the Lorentz group enlarges it to the Poincaré group, the actual symmetry algebra of relativistic quantum field theory, and turns the finite-dimensional (field) picture into the infinite-dimensional unitary (state) picture; that enlargement, and the semidirect-product structure $\mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3)$ it produces, is the business of Section 3.5.

3.5 The Poincaré group and algebra

The finite-dimensional classification just completed answers a sharp question: it lists every way a single multiplet of field components can transform linearly under the proper orthochronous Lorentz group $\mathrm{SO}^+(1,3)$, and it labels those ways by the pair (j_-, j_+) of half-integers. Scalars, Weyl spinors, the Dirac bispinor, the four-vector — the entire field content of the theories in this book sits somewhere in that list. Yet the list describes *fields*, the local carriers of the dynamics, and not the *states* of the quantum theory, the vectors in the physical Hilbert space whose inner products are probabilities. For states the relevant representations must be *unitary*, so that probability is conserved under symmetry transformations; and we have already seen that $\mathrm{SO}^+(1,3)$, being noncompact, has *no* nontrivial finite-dimensional unitary representation (Theorem 3.22). Something is missing.

What is missing is the rest of the symmetry group of Minkowski spacetime. The Lorentz group rotates and boosts about a fixed origin; but the laws of physics do not single out an origin of spacetime any more than they single out a rest frame. Spacetime is *homogeneous*: a translation $x \mapsto x + a$ is as good a symmetry as a rotation. Adjoining these translations to the Lorentz group produces the full isometry group of Minkowski space — the *Poincaré group* — and the generators of the translations are precisely the energy and momentum operators P^μ. Their spectrum, as we will see at the end of this section, is forced to be continuous, and that single fact is what makes the unitary representations infinite-dimensional: they are spaces of states labelled by momentum, not finite multiplets of field components. The Poincaré group, not the Lorentz group alone, is therefore the actual symmetry algebra of relativistic quantum field theory. This section builds it: the group, its semidirect-product structure, and its complete Lie algebra. (Throughout we keep the conventions fixed for this chapter: the metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and $\hbar = c = 1$.)

3.5.1 Spacetime translations and the inhomogeneous Lorentz group

A Lorentz transformation acts on a spacetime point by the linear map $x^\mu \mapsto \Lambda^\mu{}_\nu x^\nu$, with $\Lambda \in \mathrm{SO}^+(1,3)$ preserving the quadratic form $\eta_{\mu\nu}x^\mu x^\nu$ and fixing the origin $x = 0$. The most general transformation that preserves the *interval* between any two points — the genuine isometry of Minkowski space — need not fix the origin. It is the affine map

$$x^\mu \longmapsto x'^\mu = \Lambda^\mu{}_\nu\, x^\nu + a^\mu, \qquad \Lambda \in \mathrm{SO}^+(1,3), \quad a \in \mathbb{R}^{1,3}, \tag{3.121}$$

the composition of a Lorentz transformation with a rigid translation by the constant four-vector a^μ. That this preserves intervals is immediate: for two points x, y, $x' - y' = \Lambda(x - y)$, so $\eta_{\mu\nu}(x'-y')^\mu(x'-y')^\nu = \eta_{\mu\nu}(x-y)^\mu(x-y)^\nu$.

Conversely, any interval-preserving map of Minkowski space onto itself is of this form; the translation part is $a^\mu = x'^\mu(0)$, the image of the origin, and the linear part is then forced to be Lorentz. We package the data as an ordered pair.

Definition 3.30 (Poincaré group)**.** The *Poincaré group* (the inhomogeneous Lorentz group) is the set

$$\mathrm{ISO}^+(1,3) \;=\; \{\,(\Lambda, a) : \Lambda \in \mathrm{SO}^+(1,3),\; a \in \mathbb{R}^{1,3}\,\} \tag{3.122}$$

of pairs, each acting on Minkowski space by Eq. (3.121), with the group operation defined by composition of these actions.

The composition law is fixed by demanding that the action Eq. (3.121) be a genuine left action: applying (Λ_2, a_2) and then (Λ_1, a_1) must equal applying their product. Track a point through both:

$$\begin{aligned} x \;&\xmapsto{(\Lambda_2,a_2)}\; \Lambda_2 x + a_2 \;\xmapsto{(\Lambda_1,a_1)}\; \Lambda_1(\Lambda_2 x + a_2) + a_1 \\ &=\; (\Lambda_1\Lambda_2)\, x + (a_1 + \Lambda_1 a_2). \end{aligned} \tag{3.123}$$

The right-hand side is the action Eq. (3.121) of the single pair $(\Lambda_1\Lambda_2,\; a_1 + \Lambda_1 a_2)$. Hence the group law is

$$\boxed{(\Lambda_1, a_1)\,(\Lambda_2, a_2) \;=\; (\Lambda_1\Lambda_2,\; a_1 + \Lambda_1 a_2).} \tag{3.124}$$

The structure is not a direct product: the translation parts do not simply add, because the second translation a_2 is first rotated by Λ_1 before being combined with a_1. This entanglement of the two factors — the rotation acting on the translation — is the whole content of what follows, and it is exactly what will produce the nonzero commutator $[M^{\mu\nu}, P^\rho]$ at the level of the algebra.

From Eq. (3.124) the unit element is $(\mathbf{1}, 0)$, and the inverse is found by solving $(\Lambda, a)(\Lambda', a') = (\mathbf{1}, 0)$: the Lorentz part gives $\Lambda' = \Lambda^{-1}$, and the translation part $a + \Lambda a' = 0$ gives $a' = -\Lambda^{-1} a$. Thus

$$(\Lambda, a)^{-1} \;=\; (\Lambda^{-1},\, -\Lambda^{-1} a), \tag{3.125}$$

which one checks also satisfies $(\Lambda, a)^{-1}(\Lambda, a) = (\mathbf{1}, 0)$. Associativity follows from associativity of matrix multiplication together with the linearity of Λ acting on a; we verify it once, since the same algebra recurs below:

$$\begin{aligned} \big[(\Lambda_1, a_1)(\Lambda_2, a_2)\big](\Lambda_3, a_3) &= (\Lambda_1\Lambda_2,\; a_1 + \Lambda_1 a_2)(\Lambda_3, a_3) \\ &= \big(\Lambda_1\Lambda_2\Lambda_3,\; a_1 + \Lambda_1 a_2 + \Lambda_1\Lambda_2 a_3\big), \end{aligned} \tag{3.126}$$

while grouping the other way,

$$(\Lambda_1, a_1)\big[(\Lambda_2, a_2)(\Lambda_3, a_3)\big] = (\Lambda_1, a_1)(\Lambda_2\Lambda_3,\; a_2 + \Lambda_2 a_3)$$

$$= (\Lambda_1\Lambda_2\Lambda_3,\ a_1 + \Lambda_1(a_2 + \Lambda_2 a_3)), \tag{3.127}$$

and the two translation parts agree because $\Lambda_1(a_2+\Lambda_2 a_3) = \Lambda_1 a_2 + \Lambda_1\Lambda_2 a_3$. The Poincaré group is therefore a genuine group; because both factors are smooth manifolds and the operations are smooth, it is a ten-dimensional Lie group — six Lorentz parameters and four translation parameters.

3.5.2 The semidirect-product structure

The composition law Eq. (3.124) is the defining instance of a construction introduced abstractly in Chapter 2: the *semidirect product*. We recall it here in the concrete setting we need, because the structure dictates the form of the algebra.

Two subgroups sit inside the Poincaré group. The *translations*

$$T \;=\; \{\, (\mathbf{1}, a) : a \in \mathbb{R}^{1,3} \,\} \tag{3.128}$$

form a copy of the additive group $\mathbb{R}^{1,3}$, since by Eq. (3.124) $(\mathbf{1}, a)(\mathbf{1}, b) = (\mathbf{1}, a+b)$; this subgroup is abelian. The *Lorentz transformations*

$$L \;=\; \{\, (\Lambda, 0) : \Lambda \in \mathrm{SO}^+(1,3) \,\} \tag{3.129}$$

form a copy of $\mathrm{SO}^+(1,3)$, with $(\Lambda_1, 0)(\Lambda_2, 0) = (\Lambda_1\Lambda_2, 0)$. Every Poincaré element factorizes *uniquely* as a translation followed by a Lorentz transformation,

$$(\Lambda, a) \;=\; (\mathbf{1}, a)\,(\Lambda, 0), \tag{3.130}$$

which is immediate from Eq. (3.124) since the right-hand side is $(\mathbf{1}\cdot\Lambda,\ a + \mathbf{1}\cdot 0) = (\Lambda, a)$. So as a set the group is the product $T \times L \cong \mathbb{R}^{1,3} \times \mathrm{SO}^+(1,3)$; what is nontrivial is how the two factors fail to commute.

The decisive structural fact is that the translation subgroup is *normal*: it is stable under conjugation by the whole group. Conjugating a pure translation $(\mathbf{1}, b)$ by an arbitrary element (Λ, a),

$$\begin{aligned}(\Lambda, a)\,(\mathbf{1}, b)\,(\Lambda, a)^{-1} &= (\Lambda,\ a + \Lambda b)\,(\Lambda^{-1},\ -\Lambda^{-1}a) \\ &= (\Lambda\Lambda^{-1},\ a + \Lambda b + \Lambda(-\Lambda^{-1}a)) \\ &= (\mathbf{1},\ a + \Lambda b - a) \;=\; (\mathbf{1},\ \Lambda b),\end{aligned} \tag{3.131}$$

where we used Eq. (3.125) for the inverse and Eq. (3.124) twice. The result $(\mathbf{1}, \Lambda b)$ is again a pure translation, so $T \trianglelefteq \mathrm{ISO}^+(1,3)$ is normal. The Lorentz subgroup L is *not* normal — conjugating $(\Lambda', 0)$ by a translation produces a translation part — which is the asymmetry built into the semidirect, as opposed to direct, product.

Eq. (3.131) does more than establish normality: it tells us exactly *how* the Lorentz factor acts on the normal factor. Conjugation by $(\Lambda, 0)$ sends the translation b to Λb — the Lorentz group acts on the space of translations by its defining four-vector representation. This is the data of a semidirect product.

Definition 3.31 (Semidirect product)**.** Let N and H be groups and let $\phi : H \to \mathrm{Aut}(N)$ be a homomorphism from H into the automorphisms of N, written $\phi_h(n)$. The *(outer) semidirect product* $N \rtimes_\phi H$ is the set $N \times H$ with the product

$$(n_1, h_1)(n_2, h_2) \;=\; (n_1\, \phi_{h_1}(n_2),\; h_1 h_2). \tag{3.132}$$

In it $N \cong \{(n, e)\}$ is normal, $H \cong \{(e, h)\}$ is a subgroup, $N \cap H = \{e\}$, and NH is the whole group; conjugation realizes the action, $(e, h)(n, e)(e, h)^{-1} = (\phi_h(n), e)$.

Theorem 3.32 (Poincaré as a semidirect product)**.** *The Poincaré group is the semidirect product*

$$\mathrm{ISO}^+(1,3) \;=\; \mathbb{R}^{1,3} \rtimes \mathrm{SO}^+(1,3), \tag{3.133}$$

where $\mathrm{SO}^+(1,3)$ *acts on the translation group* $\mathbb{R}^{1,3}$ *through its defining representation,* $\phi_\Lambda(b) = \Lambda b$*.*

Proof. Take $N = \mathbb{R}^{1,3}$ (written additively, with neutral element 0), $H = \mathrm{SO}^+(1,3)$, and $\phi_\Lambda(b) = \Lambda b$. That ϕ is a homomorphism into $\mathrm{Aut}(\mathbb{R}^{1,3})$ is clear: Λ acts linearly, so it is an additive automorphism, and $\phi_{\Lambda_1\Lambda_2} = \phi_{\Lambda_1} \circ \phi_{\Lambda_2}$ because $(\Lambda_1\Lambda_2)b = \Lambda_1(\Lambda_2 b)$. With these choices the abstract product Eq. (3.132) reads $(a_1, \Lambda_1)(a_2, \Lambda_2) = (a_1 + \Lambda_1 a_2,\, \Lambda_1\Lambda_2)$, which is exactly Eq. (3.124) after writing the pair in the order (Λ, a). The identification of the normal factor with the translations and of the conjugation action with ϕ is the content of Eq. (3.131). Hence the two groups coincide. □

Physical Insight

A semidirect product is the precise algebraic statement of "a symmetry that moves you around space, sitting inside a larger symmetry that also rotates the directions you can move in." Translations form the normal, abelian backbone; the Lorentz group is a non-normal factor that *stirs* that backbone by rotating and boosting the translation vectors. The same pattern — an abelian normal subgroup acted on by a nonabelian factor — recurs throughout physics: the Euclidean group $\mathbb{R}^3 \rtimes \mathrm{SO}(3)$ of nonrelativistic space, and the gauge groups of spontaneously broken theories. The nonabelian commutator that this structure forces between "momentum" and "rotation" is below.

3.5.3 The Lie algebra: generators and the translation block

We now linearize. By the Lie-group/Lie-algebra correspondence of Chapter 2, the ten-dimensional Poincaré group has a ten-dimensional Lie algebra, spanned by the infinitesimal generators of its one-parameter subgroups. We are interested in unitary representations on the Hilbert space of states, so we write the generators

as the Hermitian operators that represent the group there, $(\Lambda, a) \mapsto U(\Lambda, a)$ with U unitary and $U(\Lambda_1, a_1)U(\Lambda_2, a_2) = U((\Lambda_1, a_1)(\Lambda_2, a_2))$.

For the Lorentz factor we keep the convention fixed in Eq. (3.29): a Lorentz transformation $\Lambda^\mu{}_\nu = \delta^\mu_\nu + \omega^\mu{}_\nu$ with antisymmetric parameter $\omega_{\mu\nu} = -\omega_{\nu\mu}$ is represented by

$$U(\Lambda, 0) \;=\; \exp\!\Big(-\tfrac{i}{2}\,\omega_{\mu\nu} M^{\mu\nu}\Big), \qquad M^{\mu\nu} = -M^{\nu\mu}, \tag{3.134}$$

with the six Hermitian generators $M^{\mu\nu}$ obeying the Lorentz algebra established in Section 3.2,

$$[M^{\mu\nu}, M^{\rho\sigma}] \;=\; i(\eta^{\nu\rho} M^{\mu\sigma} - \eta^{\mu\rho} M^{\nu\sigma} - \eta^{\nu\sigma} M^{\mu\rho} + \eta^{\mu\sigma} M^{\nu\rho}). \tag{3.135}$$

For the translations we introduce four Hermitian generators P^μ, the *energy–momentum operators*, through

$$U(\mathbf{1}, a) \;=\; \exp\!\big(i\, a^\mu P_\mu\big) \;=\; \exp\!\big(i\, a \cdot P\big), \qquad a \cdot P \equiv a^\mu P_\mu = \eta_{\mu\nu} a^\mu P^\nu. \tag{3.136}$$

The sign and the factor of i are the standard quantum-mechanical convention: $P^0 = H$ is the Hamiltonian, generating time translations $U(\mathbf{1}, a^0) = e^{iHa^0}$, and P^i the momentum, with eigenvalues that will turn out to be the physical four-momenta. We read off the four-vector index on P^μ as a label for now; that it genuinely *is* a four-vector is the theorem of Section 3.5.4.

The first commutator is the easiest and follows directly from the abelian multiplication of translations. Since $(\mathbf{1}, a)(\mathbf{1}, b) = (\mathbf{1}, a + b) = (\mathbf{1}, b)(\mathbf{1}, a)$, the representing operators commute, $U(\mathbf{1}, a)U(\mathbf{1}, b) = U(\mathbf{1}, a + b) = U(\mathbf{1}, b)U(\mathbf{1}, a)$. Expanding both sides to second order in the parameters, $(1 + ia \cdot P - \frac{1}{2}(a \cdot P)^2 + \dots)(1 + ib \cdot P + \dots)$, the terms linear in each parameter cancel against the reversed product, and the surviving condition is $a_\mu b_\nu\, P^\mu P^\nu = a_\mu b_\nu\, P^\nu P^\mu$ for all a, b, i.e. $P^\mu P^\nu = P^\nu P^\mu$. Hence

$$\boxed{[P^\mu, P^\nu] \;=\; 0.} \tag{3.137}$$

The four energy–momentum operators commute: energy and the three momenta can be simultaneously diagonalized, and their common eigenvalues will label states. This is the algebraic shadow of the homogeneity of spacetime — translating first in one direction and then another gives the same result either way.

3.5.4 P^μ is a four-vector: the $[M, P]$ commutator

The remaining nontrivial commutator, $[M^{\mu\nu}, P^\rho]$, is forced entirely by the semidirect structure Eq. (3.131): it encodes the fact that conjugating a translation by a Lorentz transformation rotates the translation vector. We derive it cleanly by representing that conjugation statement and then linearizing.

Lift Eq. (3.131) with $a = 0$ to the representation. Because U is a homomorphism, $U(\Lambda,0)U(\mathbf{1},b)U(\Lambda,0)^{-1} = U((\Lambda,0)(\mathbf{1},b)(\Lambda,0)^{-1}) = U(\mathbf{1},\Lambda b)$. Writing the translations through Eq. (3.136),

$$U(\Lambda)\, e^{\,i\, b^\mu P_\mu}\, U(\Lambda)^{-1} \;=\; e^{\,i\,(\Lambda b)^\mu P_\mu}, \qquad U(\Lambda) \equiv U(\Lambda,0). \tag{3.138}$$

The exponent on the left is $i\, b^\mu\, U(\Lambda) P_\mu U(\Lambda)^{-1}$, since conjugation distributes over the power series of the exponential; the exponent on the right is $i\,(\Lambda b)^\mu P_\mu = i\,\Lambda^\mu{}_\nu b^\nu P_\mu = i\, b^\nu(\Lambda^\mu{}_\nu P_\mu)$. These hold for every b, so matching the coefficient of b^ν gives the finite transformation law of the momentum operator,

$$U(\Lambda)\, P_\nu\, U(\Lambda)^{-1} \;=\; \Lambda^\mu{}_\nu\, P_\mu. \tag{3.139}$$

This is precisely the statement that P_μ transforms as a covariant four-vector under the Lorentz group: conjugation by $U(\Lambda)$ acts on the index of P exactly as Λ acts on the index of a coordinate. (Raising the index with η and using $\eta^{\rho\nu}\Lambda^\mu{}_\nu\eta_{\mu\sigma} = (\Lambda^{-1})^\rho{}_\sigma$, the same law reads $U(\Lambda)P^\rho U(\Lambda)^{-1} = (\Lambda^{-1})^\rho{}_\sigma P^\sigma$, the contravariant form.)

Now linearize. Put $\Lambda^\mu{}_\nu = \delta^\mu_\nu + \omega^\mu{}_\nu$ with ω infinitesimal, and expand both sides of Eq. (3.139) to first order in ω. On the left, with $U(\Lambda) = 1 - \frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma} + \dots$ from Eq. (3.134) and $U(\Lambda)^{-1} = 1 + \frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma} + \dots$,

$$U(\Lambda)\, P_\nu\, U(\Lambda)^{-1} \;=\; P_\nu - \tfrac{i}{2}\,\omega_{\rho\sigma}\,[M^{\rho\sigma}, P_\nu] + O(\omega^2). \tag{3.140}$$

On the right,

$$\Lambda^\mu{}_\nu P_\mu = (\delta^\mu_\nu + \omega^\mu{}_\nu)P_\mu = P_\nu + \omega^\mu{}_\nu P_\mu. \tag{3.141}$$

We rewrite the inhomogeneous term to expose the antisymmetric parameter $\omega_{\rho\sigma}$. Lowering with the metric, $\omega^\mu{}_\nu = \eta^{\mu\alpha}\omega_{\alpha\nu}$, so $\omega^\mu{}_\nu P_\mu = \eta^{\mu\alpha}\omega_{\alpha\nu}P_\mu = \omega_{\alpha\nu}P^\alpha = \omega_{\rho\nu}P^\rho$ after renaming $\alpha \to \rho$. To match the structure $\omega_{\rho\sigma}(\cdots)$ on the left, insert δ^σ_ν and antisymmetrize in $\rho\sigma$, which is legitimate because $\omega_{\rho\sigma}$ is antisymmetric:

$$\omega_{\rho\nu}P^\rho \;=\; \omega_{\rho\sigma}\,\delta^\sigma_\nu\,P^\rho \;=\; \tfrac{1}{2}\,\omega_{\rho\sigma}\big(\delta^\sigma_\nu P^\rho - \delta^\rho_\nu P^\sigma\big). \tag{3.142}$$

Equating Eq. (3.140) with Eq. (3.142) and cancelling P_ν,

$$-\tfrac{i}{2}\,\omega_{\rho\sigma}\,[M^{\rho\sigma}, P_\nu] \;=\; \tfrac{1}{2}\,\omega_{\rho\sigma}\big(\delta^\sigma_\nu P^\rho - \delta^\rho_\nu P^\sigma\big). \tag{3.143}$$

The parameter $\omega_{\rho\sigma}$ is an arbitrary antisymmetric array and both sides are already antisymmetric in $\rho\sigma$, so the coefficients must match: $-i\,[M^{\rho\sigma}, P_\nu] = \delta^\sigma_\nu P^\rho - \delta^\rho_\nu P^\sigma$. Dividing by $-i$ (so that $1/(-i) = i$),

$$[M^{\rho\sigma}, P_\nu] \;=\; i\big(\delta^\sigma_\nu P^\rho - \delta^\rho_\nu P^\sigma\big). \tag{3.144}$$

Finally raise the index $\nu \to \lambda$ with $\eta^{\lambda\nu}$, using $\eta^{\lambda\nu}\delta^\sigma_\nu = \eta^{\lambda\sigma}$ and likewise for ρ, and rename $(\rho,\sigma,\lambda) \to (\mu,\nu,\rho)$:

$$\boxed{\;[M^{\mu\nu}, P^\rho] \;=\; i\big(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu\big).\;} \tag{3.145}$$

This is the second nontrivial relation of the Poincaré algebra. It has the identical index structure to the action of $M^{\mu\nu}$ on the single index of any four-vector operator V^ρ — which is the invariant way of saying that P^ρ *is* a four-vector under the Lorentz generators, not merely a quartet of operators that happen to be labelled by a spacetime index.

It is worth confirming Eq. (3.145) against the nonrelativistic expectation in a concrete component, because a sign slip here propagates through the whole representation theory. Take $\mu\nu = 12$, so that $M^{12} = J^3$ is the generator of rotations about the third axis, and $\rho = 1$:

$$\left[J^3, P^1\right] = \left[M^{12}, P^1\right] = i(\eta^{21}P^1 - \eta^{11}P^2) = i(0 - (-1)P^2) = i\,P^2, \quad (3.146)$$

using $\eta^{11} = -1$. Lowering both spatial indices ($P^1 = -P_1$, $P^2 = -P_2$) this is $[J^3, P_1] = iP_2$, the familiar $[J_i, P_j] = i\epsilon_{ijk}P_k$ of elementary quantum mechanics. The relativistic relation Eq. (3.145) contains the nonrelativistic angular-momentum algebra as its rotation sub-block, and in addition tells us how boosts ($M^{0i} = K^i$) mix energy and momentum — for $\mu\nu = 0i$ and $\rho = 0$, $[K^i, P^0] = i(\eta^{i0}P^0 - \eta^{00}P^i) = -iP^i$, the statement that a boost generator does not commute with the Hamiltonian, exactly as it must since boosting changes the energy.

3.5.5 The full Poincaré algebra

Assembling Eqs. (3.135), (3.137) and (3.145) gives the complete Lie algebra of the Poincaré group, $\mathfrak{iso}(1,3)$, the symmetry algebra of every relativistic quantum field theory in this book:

$$[P^\mu, P^\nu] = 0, \quad (3.147)$$

$$[M^{\mu\nu}, P^\rho] = i(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu), \quad (3.148)$$

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - \eta^{\nu\sigma}M^{\mu\rho} + \eta^{\mu\sigma}M^{\nu\rho}). \quad (3.149)$$

Three structural features deserve to be named. First, the translations Eq. (3.147) form an abelian ideal: $[P, P] = 0$ and, by Eq. (3.148), $[M, P] \sim P$ closes back into the translations. An abelian normal subalgebra of this kind is exactly the infinitesimal counterpart of the normal subgroup T found in Eq. (3.131); it is the reason the Poincaré algebra is not semisimple, which in turn is why its representation theory (next section) cannot be read off the Cartan classification and instead requires Wigner's separate method. Second, Eq. (3.149) alone is the Lorentz subalgebra, closed in itself: setting all P^μ to zero recovers the homogeneous theory of Section 3.2. Third, the cross relation Eq. (3.148) is the only place the two factors talk to each other, and it says nothing more than "P is a four-vector": the entire nontriviality of gluing translations to Lorentz transformations is concentrated in that one line.

Notation Watch

The signs in Eqs. (3.148) and (3.149) are tied to the three conventions fixed above: the mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$, the Lorentz exponential $U(\Lambda,0) = \exp(-\frac{i}{2}\omega_{\mu\nu}M^{\mu\nu})$ of Eq. (3.134), and the translation exponential $U(\mathbf{1},a) = \exp(+i\,a^\mu P_\mu)$ of Eq. (3.136). References that adopt mostly-plus, or the opposite sign in either exponent, will display $[M,M]$ or $[M,P]$ with flipped signs; the physical content — the rotation sub-block $[J_i,P_j] = i\epsilon_{ijk}P_k$ and $[J_i,J_j] = i\epsilon_{ijk}J_k$ — is convention-independent and is what we checked componentwise above.

To see the algebra realized in a setting where every commutator can be computed by hand, and to make contact with the fields that will carry it, we work out the action of the generators on functions of spacetime.

Worked Example 3.33 (The generators as differential operators on a scalar field)**.** A scalar field φ transforms under a Poincaré element by being dragged along with the spacetime point, $(U(\Lambda,a)\varphi)(x) = \varphi((\Lambda,a)^{-1}x) = \varphi(\Lambda^{-1}(x-a))$, the value at the new point equalling the old value at the pre-image. We extract the generators by linearizing this action and verify that they satisfy Eqs. (3.147) and (3.148) directly.

Translations. For $\Lambda = \mathbf{1}$ and infinitesimal a, $\varphi(x-a) = \varphi(x) - a^\mu\partial_\mu\varphi(x) + O(a^2)$. Matching against $U(\mathbf{1},a)\varphi = (1 + i\,a^\mu P_\mu)\varphi + O(a^2)$ from Eq. (3.136) gives $i\,a^\mu P_\mu\varphi = -a^\mu\partial_\mu\varphi$, so

$$P_\mu = i\,\partial_\mu, \qquad \text{equivalently} \qquad P^\mu = i\,\partial^\mu. \tag{3.150}$$

On a plane wave $\varphi = e^{-ip\cdot x}$ this gives $P_\mu\, e^{-ip\cdot x} = i\partial_\mu e^{-ip\cdot x} = p_\mu\, e^{-ip\cdot x}$, so the eigenvalue of P_μ is the four-momentum p_μ and that of $P^0 = H$ is the energy p^0, confirming the physical identification. The commutator Eq. (3.147) is now immediate: $[P^\mu,P^\nu] = -\,[\partial^\mu,\partial^\nu] = 0$, since partial derivatives commute.

Lorentz transformations. For $a = 0$ and $\Lambda^{-1\mu}{}_\nu = \delta^\mu_\nu - \omega^\mu{}_\nu$, the pre-image is $(\Lambda^{-1}x)^\mu = x^\mu - \omega^\mu{}_\nu x^\nu$, so $\varphi(\Lambda^{-1}x) = \varphi(x) - \omega^\mu{}_\nu x^\nu\partial_\mu\varphi + O(\omega^2)$. Writing the inhomogeneous piece as in Eq. (3.142), $\omega^\mu{}_\nu x^\nu\partial_\mu = \omega_{\rho\sigma}x^\sigma\partial^\rho = \frac{1}{2}\omega_{\rho\sigma}(x^\sigma\partial^\rho - x^\rho\partial^\sigma)$, and matching against $U(\Lambda,0)\varphi = (1 - \frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma})\varphi$ gives

$$M^{\mu\nu} = i(x^\mu\partial^\nu - x^\nu\partial^\mu). \tag{3.151}$$

This is the orbital angular-momentum operator promoted to spacetime; for a field carrying internal indices there would be an additional spin piece, but the scalar has none.

Verification of $[M^{\mu\nu},P^\rho]$. Using Eqs. (3.150) and (3.151) and the elementary relation $[x^\mu,\partial^\rho] = -\eta^{\mu\rho}$ (from $\partial^\rho x^\mu = \eta^{\rho\alpha}\partial_\alpha x^\mu = \eta^{\rho\alpha}\delta^\mu_\alpha = \eta^{\rho\mu}$),

$$[M^{\mu\nu},P^\rho] = [\,i(x^\mu\partial^\nu - x^\nu\partial^\mu)\,,\,i\,\partial^\rho\,] = -\,[x^\mu\partial^\nu - x^\nu\partial^\mu,\partial^\rho]$$

$$= -\Big([x^\mu, \partial^\rho]\,\partial^\nu - [x^\nu, \partial^\rho]\,\partial^\mu\Big) = -(-\eta^{\mu\rho}\partial^\nu + \eta^{\nu\rho}\partial^\mu)$$
$$= \eta^{\mu\rho}\partial^\nu - \eta^{\nu\rho}\partial^\mu, \tag{3.152}$$

where the $x^\mu\,[\partial^\nu, \partial^\rho]$ terms vanished. Substituting $\partial^\mu = -iP^\mu$ from Eq. (3.150),

$$[M^{\mu\nu}, P^\rho] = \eta^{\mu\rho}(-iP^\nu) - \eta^{\nu\rho}(-iP^\mu) = i(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu), \tag{3.153}$$

in exact agreement with Eq. (3.148). A parallel computation reproduces Eq. (3.149); the differential realization Eqs. (3.150) and (3.151) thus furnishes an explicit, faithful representation of the entire Poincaré algebra on functions of spacetime, the representation that every classical field carries before its internal spin indices are switched on.

3.5.6 Unitary representations are infinite-dimensional

We close by drawing the structural conclusion that motivated adjoining translations in the first place, and that sets the agenda for the rest of the chapter. The physical states of a quantum theory live in a Hilbert space on which the symmetry acts by *unitary* operators; the generators P^μ and $M^{\mu\nu}$ are then Hermitian. We claim that any such representation, if it is nontrivial, is necessarily infinite-dimensional.

Theorem 3.34 (Nontrivial unitary Poincaré representations are infinite-dimensional)**.** *Let U be a unitary representation of* $\mathrm{ISO}^+(1,3)$ *on a Hilbert space $\mathcal{H}$ in which the energy–momentum operators P^μ are not all zero. Then* $\dim \mathcal{H} = \infty$.

Proof. Suppose, for contradiction, that $\dim \mathcal{H} = N < \infty$. The operators P^μ are Hermitian (because U is unitary and Eq. (3.136) holds) and mutually commuting by Eq. (3.147). On a finite-dimensional space a commuting family of Hermitian operators is simultaneously diagonalizable, so $\mathcal{H}$ has a basis of joint eigenvectors $|p, \ldots\rangle$ with $P^\mu\,|p, \ldots\rangle = p^\mu\,|p, \ldots\rangle$, and the set of eigenvalues $\mathrm{Spec} = \{p^\mu\} \subset \mathbb{R}^{1,3}$ is *finite*, of size at most N.

Now use Lorentz covariance. For any $\Lambda \in \mathrm{SO}^+(1,3)$ and any joint eigenvector $|p\rangle$, the finite transformation law Eq. (3.139) in its contravariant form gives, for the rotated state $|p'\rangle \equiv U(\Lambda)\,|p\rangle$,

$$P^\mu\,|p'\rangle = P^\mu U(\Lambda)\,|p\rangle = U(\Lambda)\,\big(U(\Lambda)^{-1}P^\mu U(\Lambda)\big)\,|p\rangle = U(\Lambda)\,\Lambda^\mu{}_\nu P^\nu\,|p\rangle = (\Lambda p)^\mu\;|p'\rangle\,. \tag{3.154}$$

So $|p'\rangle$ is again a joint eigenvector, with eigenvalue Λp. The spectrum Spec is therefore invariant under the action of the entire Lorentz group: if $p \in \mathrm{Spec}$ then $\Lambda p \in \mathrm{Spec}$ for every Λ. But the orbit of any nonzero four-vector p under $\mathrm{SO}^+(1,3)$ is an entire mass hyperboloid (when $p^2 = \eta_{\mu\nu}p^\mu p^\nu \neq 0$) or an entire light-cone sheet (when $p^2 = 0$, $p \neq 0$) — in every case an uncountably infinite set, since boosts move p continuously along the hyperboloid. A finite set cannot

contain an infinite one, so Spec can contain no nonzero vector; the only Lorentz-invariant finite subset of $\mathbb{R}^{1,3}$ is $\{0\}$. Hence $p^\mu = 0$ on all of $\mathcal{H}$, i.e. $P^\mu = 0$, contradicting the hypothesis. Therefore no nontrivial unitary representation is finite-dimensional. □

Physical Insight

This theorem is the precise sense in which *fields are not states*. The finite-dimensional representations classified earlier in this chapter — the scalar, the Weyl and Dirac spinors, the vector — are non-unitary, and they describe the local field operators: finite multiplets of components that mix under Lorentz transformations at a point. The states of the theory, by contrast, fill out infinite-dimensional unitary representations, organized by a *continuous* label — the momentum p^μ ranging over a Lorentz orbit — together with whatever discrete data distinguishes states of the same momentum. A one-particle state is not a point in a small matrix multiplet; it is a wavefunction on a mass shell. The tension between the two kinds of representation, finite-dimensional for fields and infinite-dimensional for states, is resolved by the field operators of later chapters, which are built precisely to intertwine them.

The proof has done more than establish a dimension count: it has shown that the *orbit* of p^μ under the Lorentz group — the mass hyperboloid or light-cone — is the natural arena, and that classifying the unitary representations means classifying these orbits together with the residual structure that survives at a fixed momentum. Two questions are thereby forced upon us. First, which Lorentz-invariant quantities distinguish the orbits and the representations built on them — that is, what are the *Casimir operators* of the Poincaré algebra Eqs. (3.147) to (3.149)? Second, what is the residual symmetry that fixes a chosen momentum p^μ — the *little group* — and how do its representations complete the classification? The next section constructs the two Casimirs, P^2 and the square of the Pauli–Lubański vector W^2, proves their invariance, and through them sets Wigner's method of induced representations in motion, reducing the entire problem to the pair (mass, little-group representation) — the data that, we will find, *is* the definition of a relativistic particle.

3.6 Casimir operators: mass and spin

The previous section closed the algebraic part of the story: adjoining translations to the Lorentz group produced the Poincaré algebra $\mathfrak{iso}(1,3)$, whose full set of commutators we recorded in Eqs. (3.147) to (3.149). That algebra is the symmetry algebra of every relativistic quantum theory in this book. What we have *not* yet done is the thing the whole chapter was built to do: classify the

unitary irreducible representations of this algebra, the representations that can act on a quantum-mechanical Hilbert space of states. The finite-dimensional representations classified in Sections 3.3 and 3.4 are non-unitary (the boosts are represented by non-Hermitian matrices, as Section 3.3.4 showed) and describe *fields*; the unitary representations describe *states*, and they are necessarily infinite-dimensional because the Lorentz group is non-compact. The Cartan machinery of Chapter 2 does not apply directly, because $\mathfrak{iso}(1,3)$ is not semisimple — the translations form an abelian ideal. The replacement, due to Wigner, rests on two operators that commute with everything in the algebra. By Theorem 2.21 such an operator acts as a single scalar on each irreducible representation, so its eigenvalue is a label that tells the representations apart. These are the *Casimir operators*, and this section constructs both of them, proves they are invariants, and then turns them into a complete classification scheme by reducing the problem to the representation theory of a single small subgroup — the little group. The orbit-by-orbit execution of that scheme is the business of Section 3.7; here we build the apparatus.

Throughout we use the conventions fixed earlier in the chapter: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, the Hermitian Lorentz generators $M^{\mu\nu} = -M^{\nu\mu}$ of Theorem 3.11 and the translation generators P^μ, and the totally antisymmetric Levi–Civita symbol normalized to $\epsilon_{0123} = +1$, so that with all indices raised $\epsilon^{0123} = -1$ (collected in Chapter A). We will lean repeatedly on the three relations

$$[P^\mu, P^\nu] = 0, \tag{3.155}$$

$$[M^{\mu\nu}, P^\rho] = i(\eta^{\nu\rho}P^\mu - \eta^{\mu\rho}P^\nu), \tag{3.156}$$

$$[M^{\mu\nu}, M^{\rho\sigma}] = i(\eta^{\nu\rho}M^{\mu\sigma} - \eta^{\mu\rho}M^{\nu\sigma} - \eta^{\nu\sigma}M^{\mu\rho} + \eta^{\mu\sigma}M^{\nu\rho}), \tag{3.157}$$

which are Eqs. (3.147) to (3.149) of Section 3.5.5, recalled here for convenience.

3.6.1 The first Casimir: P^2 and mass

The simplest invariant is built from the translations alone. Because the four momenta commute among themselves by Eq. (3.155), their Lorentz-invariant contraction is a natural candidate for an operator that commutes with the whole algebra.

Definition 3.35 (First Casimir)**.** The *first Casimir operator* of the Poincaré algebra is the mass-squared operator

$$C_1 \;\equiv\; P^2 \;=\; P^\mu P_\mu \;=\; \eta_{\mu\nu}\,P^\mu P^\nu. \tag{3.158}$$

Proposition 3.36 (P^2 is a Casimir)**.** *The operator P^2 commutes with every generator of the Poincaré algebra:* $\left[P^2, P^\lambda\right] = 0$ *and* $\left[P^2, M^{\rho\sigma}\right] = 0$.

Proof. The first identity is immediate from Eq. (3.155): since each P^μ commutes with P^λ,

$$\left[P^2, P^\lambda\right] = \left[P^\mu P_\mu, P^\lambda\right] = \left[P^\mu, P^\lambda\right] P_\mu + P^\mu \left[P_\mu, P^\lambda\right] = 0. \tag{3.159}$$

For the second, we use the derivation property of the commutator together with Eq. (3.156). Lowering the free index in Eq. (3.156) gives $[M^{\rho\sigma}, P_\mu] = i(\delta^\sigma{}_\mu P^\rho - \delta^\rho{}_\mu P^\sigma)$, so

$$\begin{aligned}\left[M^{\rho\sigma}, P^2\right] &= [M^{\rho\sigma}, P^\mu]\, P_\mu + P^\mu\, [M^{\rho\sigma}, P_\mu] \\ &= i(\eta^{\sigma\mu} P^\rho - \eta^{\rho\mu} P^\sigma) P_\mu + i\, P^\mu (\delta^\sigma{}_\mu P^\rho - \delta^\rho{}_\mu P^\sigma) \\ &= i(P^\rho P^\sigma - P^\sigma P^\rho) + i(P^\sigma P^\rho - P^\rho P^\sigma) = 0,\end{aligned} \tag{3.160}$$

where in the last line we contracted the metrics ($\eta^{\sigma\mu} P_\mu = P^\sigma$, etc.) and then used $[P^\rho, P^\sigma] = 0$ to see each parenthesis vanish separately. Hence P^2 commutes with all ten generators. □

By Theorem 2.21, an operator that commutes with every element of an irreducible representation is a multiple of the identity. On a given unitary irreducible representation, therefore, P^2 takes a single fixed value, which we write as

$$P^2 = m^2\, \mathbf{1}. \tag{3.161}$$

The real number m^2 is the first invariant label of the representation. For physical one-particle states it is non-negative and $m \geq 0$ is the *mass* of the particle; on shell $p^2 = m^2$ is exactly the relativistic mass-shell condition, recovered here not as an equation of motion but as the statement that mass is a Casimir eigenvalue. The representations with $m^2 < 0$ (tachyonic) and the degenerate case $m^2 = 0$ are not excluded by the algebra and will be handled, with the physical verdict on each, in Section 3.7.1.

Physical Insight

That mass is a Casimir eigenvalue, not a parameter we are free to vary within a single particle species, is the representation-theoretic content of the statement "a particle has a definite mass." A superposition of states with different m^2 is not a state in any single irreducible representation; it is a state in a reducible one, and decomposing it is exactly what diagonalizing the mass matrix accomplishes in theories with mixing. The mass shell is the orbit of the momentum operator, and we will see in Section 3.6.3 that the entire representation is built by sweeping a single internal multiplet over that orbit.

3.6.2 The Pauli–Lubanski vector and the second Casimir

One invariant is not enough. The momentum-squared distinguishes representations by mass, but it says nothing about the internal degrees of freedom — the spin — that a particle of fixed mass can carry. A second invariant must be built from the Lorentz generators $M^{\mu\nu}$. The naive contraction $M^{\mu\nu}M_{\mu\nu}$ is Lorentz invariant but does *not* commute with the translations, so it is not a Casimir of the full Poincaré algebra. The correct object, which entangles the spin generators with the momentum in precisely the way that survives translations, is the Pauli–Lubanski vector.

Definition 3.37 (Pauli–Lubanski vector)**.** The *Pauli–Lubanski (pseudo)vector* is

$$W^{\mu} \;\equiv\; -\tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}\,P_{\nu}M_{\rho\sigma}. \tag{3.162}$$

The ordering of P_ν and $M_{\rho\sigma}$ in Eq. (3.162) is immaterial. Their commutator is $[P_\nu, M_{\rho\sigma}] = -\,[M_{\rho\sigma}, P_\nu] = -i(\eta_{\nu\sigma}P_\rho - \eta_{\nu\rho}P_\sigma)$, and contracting it with $\epsilon^{\mu\nu\rho\sigma}$ gives zero, because ϵ is antisymmetric in $(\nu\sigma)$ and in $(\nu\rho)$ while the two terms are symmetric in exactly those pairs. Thus $\epsilon^{\mu\nu\rho\sigma}P_\nu M_{\rho\sigma} = \epsilon^{\mu\nu\rho\sigma}M_{\rho\sigma}P_\nu$, and W^μ is unambiguous. Lowering the free index and relabelling dummies, Eq. (3.162) is identical to the form $W_\mu = -\frac{1}{2}\epsilon_{\mu\nu\rho\sigma}M^{\nu\rho}P^\sigma$ used in Section 3.7.

The Pauli–Lubanski vector has three defining properties, which together make W^2 a Casimir.

Proposition 3.38 (Properties of W^μ)**.** *The Pauli–Lubanski vector satisfies*

$$\left[W^{\mu}, P^{\lambda}\right] = 0, \tag{3.163}$$

$$W^{\mu}P_{\mu} = 0. \tag{3.164}$$

Moreover W^μ is a Lorentz four-vector operator:

$$[M^{\rho\sigma}, W^{\mu}] = i(\eta^{\sigma\mu}W^{\rho} - \eta^{\rho\mu}W^{\sigma}). \tag{3.165}$$

Proof. (i) $\left[W^{\mu}, P^{\lambda}\right] = 0$. Since P_ν commutes with P^λ, only the $M_{\rho\sigma}$ factor contributes:

$$\left[W^{\mu}, P^{\lambda}\right] = -\tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}\,P_{\nu}\,\left[M_{\rho\sigma}, P^{\lambda}\right]. \tag{3.166}$$

Lowering both Lorentz indices in Eq. (3.156) gives $\left[M_{\rho\sigma}, P^{\lambda}\right] = i(\delta^{\lambda}{}_{\sigma}P_\rho - \delta^{\lambda}{}_{\rho}P_\sigma)$, whence

$$\begin{aligned}\left[W^{\mu}, P^{\lambda}\right] &= -\tfrac{i}{2}\,\epsilon^{\mu\nu\rho\sigma}\left(\delta^{\lambda}{}_{\sigma}P_{\nu}P_{\rho} - \delta^{\lambda}{}_{\rho}P_{\nu}P_{\sigma}\right)\\ &= -\tfrac{i}{2}\Big(\epsilon^{\mu\nu\rho\lambda}P_{\nu}P_{\rho} - \epsilon^{\mu\nu\lambda\sigma}P_{\nu}P_{\sigma}\Big) = 0.\end{aligned} \tag{3.167}$$

In the first term $\epsilon^{\mu\nu\rho\lambda}$ is antisymmetric in $(\nu\rho)$ while $P_\nu P_\rho$ is symmetric (the momenta commute), so the contraction vanishes; the second term vanishes by the same argument in $(\nu\sigma)$. This proves Eq. (3.163).

(ii) $W^\mu P_\mu = 0$. Write $W^\mu P_\mu = -\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu M_{\rho\sigma} P_\mu$ and move P_μ to the left past $M_{\rho\sigma}$:

$$M_{\rho\sigma} P_\mu = P_\mu M_{\rho\sigma} + [M_{\rho\sigma}, P_\mu], \qquad [M_{\rho\sigma}, P_\mu] = i(\eta_{\mu\sigma} P_\rho - \eta_{\mu\rho} P_\sigma). \tag{3.168}$$

The first piece gives $-\frac{1}{2}\epsilon^{\mu\nu\rho\sigma} P_\nu P_\mu M_{\rho\sigma} = 0$, because ϵ is antisymmetric in $(\mu\nu)$ and $P_\nu P_\mu$ is symmetric. The second piece gives

$$-\tfrac{i}{2}\,\epsilon^{\mu\nu\rho\sigma} P_\nu (\eta_{\mu\sigma} P_\rho - \eta_{\mu\rho} P_\sigma) = -\tfrac{i}{2}\Big(\epsilon^{\mu\nu\rho}{}_{\mu}\, P_\nu P_\rho - \epsilon^{\mu\nu}{}_{\mu}{}^{\sigma} P_\nu P_\sigma\Big) = 0, \tag{3.169}$$

since each surviving symbol carries the index μ in two slots (one raised, one lowered by the metric) and is therefore antisymmetric under the exchange of those slots while the metric contracting them is symmetric. Hence Eq. (3.164) holds: W is orthogonal to P, leaving it with at most three independent components on a momentum eigenstate.

(iii) W^μ *is a four-vector.* This is the statement Eq. (3.165). We prove it by a structural argument that we isolate as a lemma, because it recurs whenever a covariant operator is assembled by contracting tensor operators with invariant symbols. □

Lemma 3.39 (Invariant symbols and tensor operators)**.** *Call a collection of operators* $T^{\mu_1\cdots\mu_n}$ *a* tensor operator *if each index transforms in the vector representation under the Lorentz generators, i.e. if for every index slot*

$$\Big[M^{ab}, T^{\cdots c\cdots}\Big] = i(\eta^{bc}\, T^{\cdots a\cdots} - \eta^{ac}\, T^{\cdots b\cdots}), \tag{3.170}$$

the replacement acting on the displayed slot and summed over all slots. Then:

1. *the momentum* P^μ *is a tensor operator (Eq. (3.156)), and the Lorentz generator* $M^{\rho\sigma}$ *is a tensor operator (Eq. (3.157));*
2. *the product of two tensor operators is a tensor operator with the union of their indices;*
3. *a constant tensor* $C^{\mu_1\cdots\mu_n}$ *that is* Lorentz invariant*, meaning*

$$\sum_{k=1}^{n} \omega^{\mu_k}{}_{\alpha}\, C^{\mu_1\cdots\alpha\cdots\mu_k} = 0 \qquad \text{for every antisymmetric } \omega_{ab}, \tag{3.171}$$

contracts a tensor operator into a tensor operator on the uncontracted indices. The metric $\eta^{\mu\nu}$ *and the Levi–Civita symbol* $\epsilon^{\mu\nu\rho\sigma}$ *are both Lorentz invariant in this sense.*

Proof. Statement (1) for P is Eq. (3.156) read as Eq. (3.170) with one index. For $M^{\rho\sigma}$, applying the single-index rule Eq. (3.170) to each of the two indices of $M^{\rho\sigma}$ and using $M^{ab} = -M^{ba}$ reproduces exactly Eq. (3.157); this is the computation already carried out around Eq. (3.34), now reinterpreted as "M transforms as an antisymmetric two-tensor." Statement (2) is the Leibniz rule for the commutator: $\big[M^{ab}, TS\big] = \big[M^{ab}, T\big]\, S + T\, \big[M^{ab}, S\big]$, and each term on the right is the vector action on the indices of T and of S respectively, so the sum is the vector action on all indices of TS.

For statement (3), recall the defining multilinear property of the determinant: for any 4×4 matrix Λ,

$$\Lambda^{\mu}{}_{a}\Lambda^{\nu}{}_{b}\Lambda^{\rho}{}_{c}\Lambda^{\sigma}{}_{d}\,\epsilon^{abcd} = (\det\Lambda)\,\epsilon^{\mu\nu\rho\sigma}. \tag{3.172}$$

Setting $\Lambda^{\mu}{}_{a} = \delta^{\mu}{}_{a} + \omega^{\mu}{}_{a}$ and expanding both sides to first order in ω, with $\det(\mathbf{1}+\omega) = 1 + \omega^{\alpha}{}_{\alpha} + O(\omega^2)$, the linear terms give

$$\omega^{\mu}{}_{\alpha}\epsilon^{\alpha\nu\rho\sigma} + \omega^{\nu}{}_{\alpha}\epsilon^{\mu\alpha\rho\sigma} + \omega^{\rho}{}_{\alpha}\epsilon^{\mu\nu\alpha\sigma} + \omega^{\sigma}{}_{\alpha}\epsilon^{\mu\nu\rho\alpha} = \big(\omega^{\alpha}{}_{\alpha}\big)\,\epsilon^{\mu\nu\rho\sigma}. \tag{3.173}$$

For ω in the Lorentz algebra the trace vanishes: $\omega^{\alpha}{}_{\alpha} = \eta^{\alpha\beta}\omega_{\beta\alpha} = 0$, the contraction of the symmetric η with the antisymmetric ω. Hence the left side of Eq. (3.173) is zero, which is precisely Eq. (3.171) for $C = \epsilon$. The same differentiation of the defining relation $\Lambda^{\mu}{}_{a}\Lambda^{\nu}{}_{b}\,\eta^{ab} = \eta^{\mu\nu}$ gives $\omega^{\mu}{}_{\alpha}\eta^{\alpha\nu} + \omega^{\nu}{}_{\alpha}\eta^{\mu\alpha} = \omega^{\mu\nu} + \omega^{\nu\mu} = 0$, so η is invariant too.

Finally, given an invariant C and a tensor operator T, form the contraction $U^{\cdots} = C^{\cdots}{}_{\nu_1\cdots\nu_k}\,T^{\cdots\nu_1\cdots\nu_k}$. Taking the commutator with M^{ab}, only T responds, and by Eq. (3.170) it produces the vector action on each of its indices. The actions on the *contracted* indices can be moved onto C at the cost of a sign, by the invariance identity Eq. (3.171) rewritten with the algebra index ab in place of the generic ω; these telescope and cancel, leaving exactly the vector action on the *free* indices. Thus U is a tensor operator. □

Equation (3.165) now follows in one line. By Theorem 3.39, the product $P_\nu M_{\rho\sigma}$ is a tensor operator with three lower indices, and $\epsilon^{\mu\nu\rho\sigma}$ is an invariant symbol; contracting ν, ρ, σ leaves the single free index μ, so $W^{\mu} = -\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}P_\nu M_{\rho\sigma}$ is a vector operator, which is the content of Eq. (3.165). (The explicit index computation, expanding $\big[M^{ab}, P_\nu M_{\rho\sigma}\big]$ with Eqs. (3.156) and (3.157) and resumming, reproduces the same result; the lemma packages that algebra once and for all.) An independent check at the level of components is provided in Theorem 3.45 of the next section, where W^{μ} is evaluated in the rest frame and found to be the angular momentum.

With these properties in hand the second invariant is immediate.

Definition 3.40 (Second Casimir)**.** The *second Casimir operator* is the Lorentz square of the Pauli–Lubanski vector,

$$C_2 \;\equiv\; W^2 \;=\; W^{\mu}W_{\mu}. \tag{3.174}$$

Proposition 3.41 (W^2 is a Casimir)**.** $\left[W^2, P^\lambda\right] = 0$ *and* $\left[W^2, M^{\rho\sigma}\right] = 0$.

Proof. For the translations, Eq. (3.163) gives

$$\left[W^2, P^\lambda\right] = \left[W^\mu, P^\lambda\right] W_\mu + W^\mu \left[W_\mu, P^\lambda\right] = 0. \tag{3.175}$$

For the Lorentz generators, W^2 is a scalar contraction of the vector operator W^μ with the invariant metric, so it is invariant; explicitly, using Eq. (3.165) and its lowered form $[M^{\rho\sigma}, W_\mu] = i(\delta^\sigma{}_\mu W^\rho - \delta^\rho{}_\mu W^\sigma)$,

$$\begin{aligned}\left[M^{\rho\sigma}, W^2\right] &= \left[M^{\rho\sigma}, W^\mu\right] W_\mu + W^\mu \left[M^{\rho\sigma}, W_\mu\right] \\ &= i(\eta^{\sigma\mu} W^\rho - \eta^{\rho\mu} W^\sigma) W_\mu + i\, W^\mu (\delta^\sigma{}_\mu W^\rho - \delta^\rho{}_\mu W^\sigma) \\ &= i(W^\rho W^\sigma - W^\sigma W^\rho) + i(W^\sigma W^\rho - W^\rho W^\sigma) = 0. \end{aligned} \tag{3.176}$$

The two contributions cancel term by term, irrespective of whether the components of W commute among themselves; this is the algebraic signature of a Lorentz scalar. □

Again by Theorem 2.21, W^2 is a fixed scalar on each unitary irreducible representation. Its eigenvalue is the second invariant label; we will see in Section 3.7.2 that for $m > 0$ it equals $-m^2 s(s+1)$ with s the spin, and in Section 3.7.3 that for $m = 0$ it degenerates and helicity takes over. Before that, one further computation exposes *why* W^2 encodes spin: the components of W close, on a mass shell, into the angular-momentum algebra.

Worked Example 3.42 (The Pauli–Lubanski algebra and the emergence of spin)**.** We compute the commutator of two Pauli–Lubanski components and show it reduces, on a fixed momentum, to an $\mathfrak{su}(2)$. Start from Eq. (3.162) and use that P_α commutes with W^μ by Eq. (3.163), so only the $M_{\beta\gamma}$ factor responds:

$$[W^\mu, W^\nu] = -\tfrac{1}{2}\, \epsilon^{\nu\alpha\beta\gamma} P_\alpha \left[W^\mu, M_{\beta\gamma}\right]. \tag{3.177}$$

Because W^μ is a vector operator, Eq. (3.165) (lowered on the second pair) gives $[W^\mu, M_{\beta\gamma}] = -\left[M_{\beta\gamma}, W^\mu\right] = i(\delta^\mu{}_\beta W_\gamma - \delta^\mu{}_\gamma W_\beta)$. Substituting,

$$\begin{aligned}[W^\mu, W^\nu] &= -\tfrac{i}{2}\, \epsilon^{\nu\alpha\beta\gamma} P_\alpha (\delta^\mu{}_\beta W_\gamma - \delta^\mu{}_\gamma W_\beta) \\ &= -\tfrac{i}{2}\left(\epsilon^{\nu\alpha\mu\gamma} P_\alpha W_\gamma - \epsilon^{\nu\alpha\beta\mu} P_\alpha W_\beta\right) = -i\, \epsilon^{\nu\alpha\mu\gamma} P_\alpha W_\gamma, \end{aligned} \tag{3.178}$$

where the two terms combined after relabelling $\beta \to \gamma$ in the second and using $\epsilon^{\nu\alpha\gamma\mu} = -\epsilon^{\nu\alpha\mu\gamma}$. Renaming $\alpha \to \sigma$, $\gamma \to \rho$ and reordering the (commuting) W and P, the index string $\epsilon^{\nu\sigma\mu\rho}$ is brought to standard order by an odd permutation, $\epsilon^{\nu\sigma\mu\rho} = -\epsilon^{\mu\nu\rho\sigma}$, giving the covariant Pauli–Lubanski algebra

$$[W^\mu, W^\nu] = i\, \epsilon^{\mu\nu\rho\sigma} W_\rho P_\sigma. \tag{3.179}$$

Now specialize to a state of definite timelike momentum and pass to its rest frame, $P^\sigma \to (m, \mathbf{0})$, so that only $\sigma = 0$ contributes on the right:

$$\left[W^i, W^j\right] = i\, m\, \epsilon^{ijk0}\, W_k = i\, m\, \epsilon^{ijk}\, W^k, \tag{3.180}$$

using $\epsilon^{ijk0} = -\epsilon^{0ijk} = \epsilon^{ijk}_{(3)}$ (the three-dimensional symbol, since $\epsilon^{0123} = -1$) and $W_k = -W^k$. Defining the rest-frame spin $S^i \equiv W^i/m$, this reads $[S^i, S^j] = i\epsilon^{ijk}S^k$: *the spatial Pauli–Lubanski components, divided by the mass, satisfy exactly the angular-momentum algebra* $\mathfrak{su}(2)$ *of Chapter 2.* The Casimir $W^2 = -m^2\mathbf{S}^2$ is then $-m^2$ times the $\mathfrak{su}(2)$ Casimir, and on the spin-s irreducible representation $\mathbf{S}^2 = s(s+1)$. This is the algebraic reason a massive particle's internal states form a spin multiplet, and it identifies the relevant subgroup — the rotations that fix the rest momentum — as the object whose representations we must classify. The full extraction of $W^2 = -m^2 s(s+1)$ in components is carried out in Theorem 3.45.

3.6.3 The little group and Wigner's induced representations

We now have two invariant labels, (m^2, w^2), where w^2 is the eigenvalue of W^2. The remaining task is to construct the representation space itself and to see exactly what additional data, beyond these two numbers, specify it. The construction is Wigner's method of *induced representations*: build the entire Hilbert space by acting with the group on a single, carefully chosen multiplet of states sitting over one standard momentum. The structure that organizes this is a subgroup of the Lorentz group called the little group.

States and the momentum orbit. Because the four momenta commute, Eq. (3.155), we may diagonalize them simultaneously. A basis of the one-particle Hilbert space is

$$P^\mu\ |p, \sigma\rangle = p^\mu\ |p, \sigma\rangle\,, \tag{3.181}$$

where p^μ is the momentum eigenvalue and σ is a discrete label distinguishing states of the same momentum — the internal degeneracy we are trying to characterize. The first Casimir fixes $p^2 = m^2$ on the representation, so every accessible momentum lies on the mass shell. How a Lorentz transformation acts on these states follows from P^μ being a four-vector operator. The finite form of Eq. (3.156), established in Eq. (3.139), is $U(\Lambda)^{-1}P^\mu U(\Lambda) = \Lambda^\mu{}_\nu P^\nu$, equivalently $U(\Lambda)\, P^\mu\, U(\Lambda)^{-1} = (\Lambda^{-1})^\mu{}_\nu P^\nu$. Acting on a momentum eigenstate,

$$P^\mu \left(U(\Lambda)\,|p,\sigma\rangle\right) = U(\Lambda)\left(U(\Lambda)^{-1}P^\mu U(\Lambda)\right)|p,\sigma\rangle = U(\Lambda)\,\Lambda^\mu{}_\nu P^\nu\,|p,\sigma\rangle = (\Lambda p)^\mu\left(U(\Lambda)\,|p,\sigma\rangle\right). \tag{3.182}$$

Thus $U(\Lambda)\,|p,\sigma\rangle$ is a momentum eigenstate with eigenvalue Λp: *Lorentz transformations move momenta along the mass shell.* This intertwining property —

inherited unchanged by Section 3.7.1 — means that within a single irreducible representation the momenta sweep out exactly one Lorentz orbit, and $U(\Lambda)$ can change the internal label σ but cannot leave the orbit. The whole representation is therefore determined by (i) which orbit, and (ii) how the residual symmetry acts on σ at one point of the orbit.

Standard momentum and the little group. Choose, once and for all, a convenient reference point k^μ on the orbit — the *standard momentum* (for example $k = (m, \mathbf{0})$ on the massive shell, or $k = (\omega, 0, 0, \omega)$ on the light cone). Every other momentum p on the orbit is reached from k by some Lorentz transformation. The transformations that leave k alone form the key subgroup.

Definition 3.43 (Little group)**.** The *little group* of the standard momentum k is the stabilizer

$$G_k \;\equiv\; \{\, \Lambda \in \mathrm{SO}(1,3)^+ \;:\; \Lambda^\mu{}_\nu k^\nu = k^\mu \,\}, \tag{3.183}$$

the subgroup of proper orthochronous Lorentz transformations that fix k.

The little group is the precise carrier of spin. Its elements act on the standard momentum trivially, so they map the fibre of states $\{|k,\sigma\rangle\}$ over k to itself, and the way they shuffle the labels σ is a genuine finite-dimensional problem — a representation of G_k.

The induced construction. Pick for each momentum p on the orbit a specific "standard boost" $L(p)$, a Lorentz transformation with

$$L(p)^\mu{}_\nu\, k^\nu = p^\mu, \qquad L(k) = \mathbf{1}, \tag{3.184}$$

chosen smoothly in p (a choice of representatives for the cosets $\mathrm{SO}(1,3)^+/G_k$, which is the orbit). *Define* the states over a general momentum by boosting the fibre over k:

$$|p,\sigma\rangle \;\equiv\; U\big(L(p)\big)\;|k,\sigma\rangle\,. \tag{3.185}$$

We use the *covariant* normalization throughout, in which this factor-free definition is the right one and the resulting inner product is the Lorentz-invariant $\langle p',\sigma'|p,\sigma\rangle = (2\pi)^3 2E_{\mathbf{p}}\,\delta^3(\mathbf{p}-\mathbf{p}')\delta_{\sigma'\sigma}$ recorded below; the alternative non-covariant convention and the explicit energy factor it introduces are discussed honestly at the end of the subsection. The whole Hilbert space is thereby generated from the single fibre over k and a choice of boosts; nothing has been assumed about how $U(\Lambda)$ acts except on that one fibre.

Now compute the action of an arbitrary Lorentz transformation Λ on a general state. Using $U(\Lambda)U(L(p)) = U(\Lambda L(p))$ and inserting $\mathbf{1} = U(L(\Lambda p))U(L(\Lambda p))^{-1}$,

$$U(\Lambda)\,|p,\sigma\rangle = U\big(L(\Lambda p)\big)\,\underbrace{U\big(L(\Lambda p)^{-1}\,\Lambda\,L(p)\big)}_{W(\Lambda,p)}\,|k,\sigma\rangle\,. \tag{3.186}$$

The element $W(\Lambda, p) \equiv L(\Lambda p)^{-1} \Lambda\, L(p)$ is in the little group:

$$W(\Lambda, p)\, k = L(\Lambda p)^{-1}\, \Lambda\, L(p)\, k = L(\Lambda p)^{-1}\, \Lambda\, p = L(\Lambda p)^{-1} (\Lambda p) = k, \tag{3.187}$$

since by construction $L(\Lambda p)$ carries k to Λp. This $W(\Lambda, p) \in G_k$ is the *Wigner rotation*: the net little-group element produced by boosting up from k to p, transforming by Λ, and boosting back down from Λp to k. Because it fixes k, it acts only on the internal labels of the fibre,

$$U\big(W(\Lambda, p)\big)\, |k, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}\big(W(\Lambda, p)\big)\ |k, \sigma'\rangle\,, \tag{3.188}$$

where the matrices $D(W)$ form a representation of G_k. Substituting back into Eq. (3.186) and recognizing $U(L(\Lambda p))\, |k, \sigma'\rangle = |\Lambda p, \sigma'\rangle$ from Eq. (3.185),

$$\boxed{U(\Lambda)\, |p, \sigma\rangle = \sum_{\sigma'} D_{\sigma'\sigma}\big(W(\Lambda, p)\big)\ |\Lambda p, \sigma'\rangle\,.} \tag{3.189}$$

This is the master formula — the covariant-normalization form, carrying no extra energy factor, exactly as it is reused in Eq. (3.199). It says the full, infinite-dimensional Poincaré representation is completely determined by the finite-dimensional little-group representation D: the momentum index is carried along the orbit by the geometric action $p \mapsto \Lambda p$, and *all* the nontrivial internal dynamics is the Wigner-rotation matrix $D(W(\Lambda, p))$.

That the matrices D genuinely form a representation is forced by associativity. From the definition of W one checks the composition law (a cocycle identity)

$$W(\Lambda_1 \Lambda_2,\, p) = W(\Lambda_1,\, \Lambda_2 p)\, W(\Lambda_2,\, p), \tag{3.190}$$

obtained by substituting the definitions and cancelling the intermediate $L(\Lambda_2 p)^{-1} L(\Lambda_2 p) = \mathbf{1}$. Applying Eq. (3.189) twice and demanding consistency with $U(\Lambda_1)U(\Lambda_2) = U(\Lambda_1\Lambda_2)$ then requires $D(W(\Lambda_1, \Lambda_2 p))D(W(\Lambda_2, p)) = D(W(\Lambda_1\Lambda_2, p))$, which is exactly Eq. (3.190): D is a homomorphism on G_k.

Theorem 3.44 (Wigner's classification, induced form)**.** *Let U be a unitary representation of the Poincaré group on a one-particle Hilbert space, built by induction from the standard momentum k as in Eqs. (3.185) and (3.189). Then U is irreducible if and only if the little-group representation D on the fibre over k is irreducible. Consequently the unitary irreducible representations of the Poincaré group are classified by*

> (the choice of momentum orbit, equivalently the value of P^2 and the sign of p^0) together with a unitary irreducible representation D of the little group G_k.

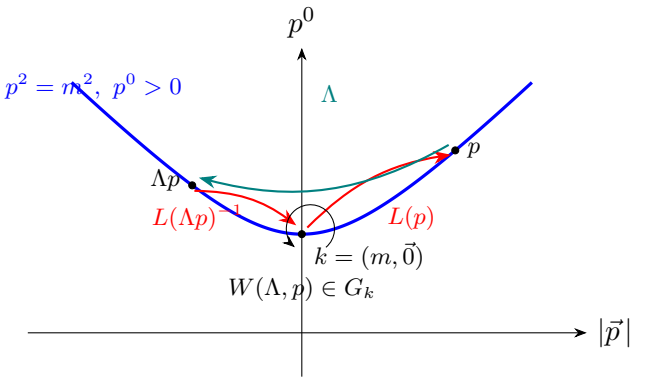


Figure 3.4: Wigner's induced construction on the massive shell. States over a general momentum p are defined by boosting the fibre over the standard momentum k with $L(p)$. A Lorentz transformation Λ moves $p \mapsto \Lambda p$ along the orbit; following the boost up from k, the transformation, and the boost back down, $W(\Lambda, p) = L(\Lambda p)^{-1} \Lambda L(p)$ returns to k and so lies in the little group G_k. All internal (spin) structure is the little-group matrix $D(W)$.

Proof sketch. The momentum operators act diagonally and transitively over the orbit, by Eq. (3.182): any two momenta are connected by some $U(\Lambda)$, so a nonzero Poincaré-invariant subspace must contain states of *every* momentum in the orbit, obtained by boosting. Restricted to the fibre over k, such a subspace is invariant under $U(G_k)$ — the little group maps the k-fibre to itself. Hence an invariant subspace of the whole representation is in one-to-one correspondence with a D-invariant subspace of the fibre: the representation is irreducible exactly when D is. The map (orbit, D) $\mapsto U$ is, conversely, surjective onto the irreducibles, because every irreducible U singles out an orbit (its momentum support) and an irreducible little-group action on the fibre. This is the content of Mackey's theory of induced representations for a semidirect product with abelian normal subgroup, of which $\mathbb{R}^{1,3} \rtimes \mathrm{SO}(1,3)^+$ is the case at hand. □

The classification has thus been reduced, completely, to a finite problem repeated once per orbit: find the little group, and list its unitary irreducible representations. The two Casimirs are the invariant fingerprints of the result — $P^2 = m^2$ names the orbit, and W^2 (on the massive orbit, $-m^2 s(s+1)$) names the little-group irrep — so the abstract pair (m^2, w^2) and the concrete pair (orbit, D) carry the same information.

Measure, normalization, and unitarity — stated honestly. The construction above is exact at the level of the group action, but a few analytic points must be named rather than glossed, because they are where the formal manipulation meets the demands of a genuine Hilbert space. The states

Eq. (3.181) are momentum eigenstates and hence not normalizable; they are continuum-normalized distributions, and the rigorous object is the Hilbert space $L^2(\mathcal{O}_m, d\tilde{p}) \otimes V_D$ of square-integrable, V_D-valued functions on the orbit $\mathcal{O}_m$, with V_D the little-group representation space. The natural measure on the mass shell is the Lorentz-invariant one,

$$d\tilde{p} \;=\; \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}, \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} = p^0. \tag{3.191}$$

Its invariance is the identity

$$\int \frac{d^3p}{2E_{\mathbf{p}}}\, f(p) = \int d^4p\; \delta(p^2 - m^2)\, \theta(p^0)\, f(p), \tag{3.192}$$

obtained by performing the p^0 integral against $\delta(p^2 - m^2) = \delta\big((p^0)^2 - \mathbf{p}^2 - m^2\big) = \frac{1}{2E_{\mathbf{p}}}\delta(p^0 - E_{\mathbf{p}})$ on the forward sheet; the right-hand side is manifestly Lorentz invariant (each factor is), so the left-hand side is too. In the covariant convention adopted at Eq. (3.185) the states then satisfy the invariant normalization

$$\langle p', \sigma' | p, \sigma \rangle = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p} - \mathbf{p}')\, \delta_{\sigma'\sigma}. \tag{3.193}$$

Two facts make the representation Eq. (3.189) unitary on this space. First, the measure Eq. (3.191) is invariant under $p \mapsto \Lambda p$, so the geometric part of the action preserves the L^2 inner product with the weight $2E_{\mathbf{p}}$ of Eq. (3.193) — this invariance is exactly why the covariant transformation law Eq. (3.189) carries *no* extra energy factor. Second, the internal part is the Wigner matrix $D(W)$, and *the whole construction is unitary if and only if D is a unitary representation of the little group.* This is the reason the classification demands *unitary* irreps of G_k, not arbitrary ones: only those give a unitary — hence physically admissible — Poincaré representation.

It is worth naming the alternative convention, since the discrepancy is a perennial source of sign-and-factor confusion. Rescaling to the non-covariant states $|p, \sigma\rangle_0 \equiv \big[(2\pi)^3 2E_{\mathbf{p}}\big]^{-1/2} |p, \sigma\rangle$ replaces Eq. (3.193) by the flat normalization $\langle p', \sigma' | p, \sigma \rangle_0 = \delta^3(\mathbf{p} - \mathbf{p}')\delta_{\sigma'\sigma}$, at the price of an explicit factor in the transformation law,

$$U(\Lambda)\, |p, \sigma\rangle_0 = \sqrt{\frac{E_{\Lambda\mathbf{p}}}{E_{\mathbf{p}}}} \sum_{\sigma'} D_{\sigma'\sigma}(W(\Lambda, p))\; |\Lambda p, \sigma'\rangle_0 \,, \tag{3.194}$$

the square root being precisely the Jacobian of $p \mapsto \Lambda p$ that the invariant measure absorbed above. The two conventions are unitarily equivalent and describe the same representation; we use the covariant one throughout, including in Section 3.7. Finally, the construction depends on the arbitrary choice of standard boost $L(p)$ (a section of $\mathrm{SO}(1,3)^+ \to \mathcal{O}_m$): a different section $L'(p) = L(p)\, h(p)$ with $h(p) \in G_k$ changes D by a p-dependent conjugation $D \mapsto D(h)^{-1} D\, D(h)$, which

is a unitary equivalence. The *equivalence class* of D — and therefore the physical content, the spin or helicity — is independent of this choice. We will make the standard choices explicit, orbit by orbit, in the next section.

This reduces the entire classification to a checklist: for each Lorentz orbit on momentum space, identify the little group G_k as a stabilizer, and read off its unitary irreducible representations. The forward timelike orbit will give the rotation group and its spins; the forward null orbit will give the Euclidean group of the plane and its helicities — and, as the degeneracy of that non-compact little group, the seed of gauge redundancy. Those orbits, together with the negative-energy, vacuum, and spacelike cases, are precisely what Section 3.7 now works out one by one, delivering the chapter's final answer to what a relativistic particle can be. The two invariants built here, $P^2 = m^2$ and W^2, are the labels that survive into dynamics: they fix the one-particle asymptotic states that enter the LSZ reduction of Chapter 14 for all three benchmark theories — the spin-0 scalar of ϕ^4 theory, the spin-$\frac{1}{2}$ electron of QED, and the massless helicity-1 photon and gluon of QED and Yang–Mills theory.

3.7 Wigner's classification: what a relativistic particle can be

The previous section reduced an enormous problem to a finite one. We constructed the two Poincaré Casimirs — the mass-squared operator P^2 and the Pauli–Lubanski square W^2 — proved that they commute with every generator, and set up Wigner's method of induced representations: a unitary irreducible representation of the Poincaré group is fixed by the value of P^2 on the orbit it lives on, together with an irreducible representation of the *little group*, the subgroup of the Lorentz group that fixes a chosen standard momentum on that orbit (Section 3.6). The classification of *all* one-particle states therefore amounts to one finite task repeated orbit by orbit: list the orbits of the Lorentz group on momentum space, identify the little group of each, and classify its unitary irreducible representations. That is the work of this section. When it is done we will have the complete catalogue of what a relativistic particle can be — the chapter's thesis — and, as a byproduct that will reshape everything that follows, we will see the massless case force the existence of gauge redundancy.

3.7.1 The orbits: where a momentum can live

A one-particle state is, before anything else, an eigenstate of the four-momentum operator P^μ with some eigenvalue p^μ. Acting with a Lorentz transformation $U(\Lambda)$ sends $|p, \sigma\rangle$ to a state of momentum Λp (this is the defining intertwining property of the unitary representation, Section 3.6). Hence within a single irreducible representation the accessible momenta form exactly one *Lorentz orbit*: a set

of the form $\{\Lambda p : \Lambda \in \mathrm{SO}(1,3)^{\uparrow}\}$. Because $p^2 = \eta_{\mu\nu}p^\mu p^\nu$ and the sign of p^0 are Lorentz invariants (the latter on account of the orthochronous restriction, Section 3.1.2), the orbits are precisely the level sets of these invariants. Writing $m^2 \equiv p^2$, the possibilities are:

(I) $m^2 > 0$, $p^0 > 0$: the *forward mass hyperboloid*. Standard momentum $k = (m, 0, 0, 0)$ — the rest frame.

(II) $m^2 > 0$, $p^0 < 0$: the backward mass hyperboloid. Standard momentum $k = (-m, 0, 0, 0)$.

(III) $m^2 = 0$, $p^0 > 0$: the *forward light cone* (minus its apex). Standard momentum $k = (\omega, 0, 0, \omega)$, any fixed $\omega > 0$.

(IV) $m^2 = 0$, $p^0 < 0$: the backward light cone. Standard momentum $k = (-\omega, 0, 0, \omega)$.

(V) $p^\mu = 0$: the single point at the origin. The vacuum.

(VI) $m^2 < 0$: the spacelike (single-sheeted) hyperboloid. Standard momentum $k = (0, 0, 0, \kappa)$, $\kappa > 0$. Tachyons.

Figure 3.5 shows the six families in the $(p^0, |\vec{p}\,|)$ half-plane. They exhaust momentum space: every p^μ has a definite $(p^2, \operatorname{sgn} p^0)$ and so lies in exactly one. The classification problem now splits into six independent sub-problems, one per orbit, and three of them we can dispose of immediately on physical grounds before doing any group theory.

The vacuum orbit (V) is trivial: the little group is the whole Lorentz group, and the physical vacuum is the unique Lorentz-invariant state, the one-dimensional trivial representation. It is a "particle" only in the degenerate sense of carrying no momentum and no spin.

The tachyonic orbit (VI) has little group $\mathrm{SO}(2,1)$, the stabilizer of a spacelike vector. This is a *non-compact* group; its nontrivial unitary representations are all infinite-dimensional, so a tachyon would carry an infinite tower of internal states. Worse, the orbit contains momenta with p^0 of either sign connected by orthochronous transformations only through the spacelike region, and a field built on it has spacelike-separated excitations that fail to commute — microcausality is lost. Tachyons therefore do not describe stable propagating particles. (When a "tachyonic mass" $m^2 < 0$ appears in a Lagrangian it signals that one is expanding about an unstable point of the potential; the cure is to find the true vacuum, the mechanism behind spontaneous symmetry breaking, taken up in Part VIII — it is never an honest one-particle state.) We set (VI) aside.

The backward orbits (II) and (IV) have $p^0 < 0$: they would be negative-energy states. In the quantum theory these are not discarded but *reinterpreted*: the

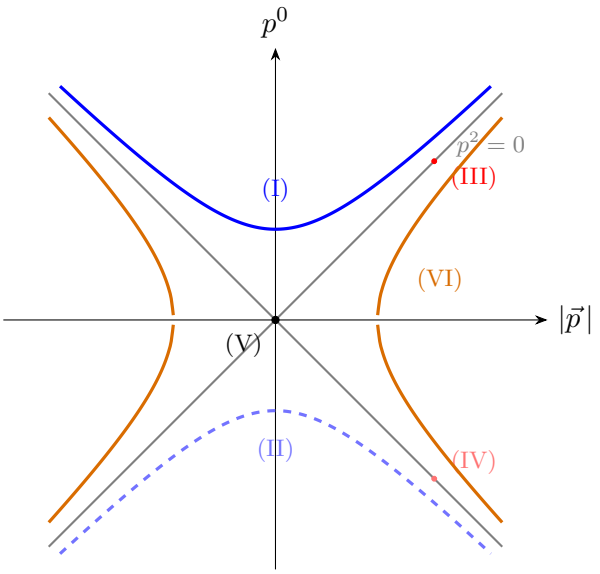


Figure 3.5: The Lorentz orbits in momentum space, drawn in the $(p^0, |\vec{p}\,|)$ half-plane (each point off the axis is a full SO(3) shell). The forward (I) and backward (II) mass hyperboloids carry the massive particles and antiparticles; the forward (III) and backward (IV) light cones carry the massless ones; the origin (V) is the vacuum; the spacelike hyperboloid (VI) would carry tachyons and is excluded. The light cone $p^2 = 0$ separates the timelike interior from the spacelike exterior.

construction of local fields will require, alongside each positive-energy particle, an equal-mass partner built from the backward orbit, the antiparticle, with the relabelling restoring positive energy. The little-group analysis of (II) and (IV) is identical to that of (I) and (III) — same stabilizer, same standard-momentum group — so the representation content is the same and we lose nothing by analyzing only the forward orbits here. The pairing of (I) with (II) and of (III) with (IV) is exactly the particle/antiparticle doubling that the spin–statistics and CPT discussion of Chapter 11 will make precise.

Two orbits remain to be worked in full: the massive forward hyperboloid (I) and the massless forward cone (III). They are genuinely different because their little groups are different — one compact, one not — and that single distinction is the origin of the deepest dichotomy in particle physics: massive particles carry *spin*, massless particles carry *helicity*, and only the massless case forces gauge redundancy.

3.7.2 The massive case: little group SU(2) and spin

Fix the standard momentum of orbit (I) at the rest frame value

$$k^\mu = (m, 0, 0, 0), \qquad m > 0. \tag{3.195}$$

The little group G_k is the set of $\Lambda \in \mathrm{SO}(1,3)^\uparrow$ with $\Lambda k = k$. A Lorentz transformation fixes the timelike vector $(m, 0, 0, 0)$ iff it leaves the time axis pointwise fixed, i.e. iff it acts only on the three spatial axes as a rotation:

$$G_k = \mathrm{SO}(3). \tag{3.196}$$

This is the heart of the massive case: *in the rest frame, the residual symmetry is exactly the rotation group*, the same SO(3) whose representation theory we already possess. The unitary representations of the little group are therefore the unitary representations of SO(3), or — because a quantum state is defined only up to phase and we have the option of representing the double cover — of its universal cover SU(2). Which one nature uses is not ours to choose a priori; both are admissible, and we have already classified them: the irreducible unitary representations of SU(2) are labelled by a spin

$$s \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, 2, \dots\}, \tag{3.197}$$

of dimension $2s+1$, spanned by the eigenstates $|s, s_3\rangle$ of J_3 with $s_3 \in \{-s, -s+1, \dots, s\}$ (Theorem 2.53). The integer-s representations are genuine representations of SO(3); the half-integer ones are faithful only on SU(2), the double cover whose existence we traced to the $\mathbb{Z}_2$ topology of the Lorentz group in Section 3.4.2. That a half-integer-spin state changes sign under a 2π rotation is not a paradox but the defining signature of the spinor representations built there.

Putting the pieces together through Wigner's induction, a massive one-particle representation is labelled by (m, s) and is spanned by states $|p, s_3\rangle$ where p ranges over the forward hyperboloid $p^2 = m^2,\ p^0 > 0$ and s_3 over the $2s+1$ values for fixed s. The boost $L(p)$ that carries k to p defines the states off the rest frame,

$$|p, s_3\rangle \equiv U(L(p))\ |k, s_3\rangle\,, \tag{3.198}$$

and a general Lorentz transformation acts by

$$U(\Lambda)\ |p, s_3\rangle = \sum_{s_3'} D^{(s)}_{s_3' s_3}(\mathcal{W}(\Lambda, p))\ |\Lambda p,\ s_3'\rangle\,, \qquad \mathcal{W}(\Lambda, p) \equiv L(\Lambda p)^{-1}\,\Lambda\,L(p) \in \mathrm{SU}(2), \tag{3.199}$$

where $D^{(s)}$ is the spin-s Wigner matrix and $\mathcal{W}$ is the *Wigner rotation*: the little-group element produced by going to p, transforming, and returning. The momentum index transforms by the orbit action; all the spin information is

carried by the finite-dimensional unitary matrix $D^{(s)}(\mathcal{W})$. The representation is manifestly unitary because $D^{(s)}$ is unitary, and it is irreducible because SU(2) acts irreducibly on each fixed-p fibre and the boosts move between fibres transitively. This is a complete, finite description of an infinite-dimensional Hilbert space: $2s+1$ numbers attached to each point of a three-dimensional hyperboloid.

It remains to connect this labelling to the invariant we built for exactly this purpose, the Pauli–Lubanski square, and confirm that "spin s" is the value of a Casimir and not merely a bookkeeping choice.

Worked Example 3.45 (W^2 **in the rest frame:** $W^2 = -m^2 s(s+1)$**).** Recall the Pauli–Lubanski vector from Section 3.6,

$$W_\mu = -\tfrac{1}{2}\,\varepsilon_{\mu\nu\rho\sigma}\,J^{\nu\rho}P^\sigma, \qquad \varepsilon_{0123} = +1, \tag{3.200}$$

with $J^{\nu\rho}$ the Lorentz generators and P^σ the translation generators. We evaluate $W^2 = W^\mu W_\mu$ directly on a rest-frame state, where $P^\sigma \to k^\sigma = m\,\delta^\sigma{}_0$ by Eq. (3.195). Substituting,

$$W_\mu = -\tfrac{1}{2}\,m\,\varepsilon_{\mu\nu\rho 0}\,J^{\nu\rho}. \tag{3.201}$$

The symbol $\varepsilon_{\mu\nu\rho 0}$ is nonzero only when μ,ν,ρ are the three *spatial* indices, so first $W_0 = 0$ (its index would have to be spatial). For a spatial index i,

$$W_i = -\tfrac{1}{2}\,m\,\varepsilon_{ijk0}\,J^{jk} = -\tfrac{1}{2}\,m\,(-\varepsilon_{ijk})\,J^{jk} = \tfrac{1}{2}\,m\,\varepsilon_{ijk}\,J^{jk}, \tag{3.202}$$

where $\varepsilon_{ijk0} = -\varepsilon_{0ijk} = -\varepsilon_{ijk}$ (one transposition moving the index 0 past three slots gives $(-1)^3$) and ε_{ijk} is the ordinary three-dimensional Levi-Civita symbol. Now $\frac{1}{2}\varepsilon_{ijk}J^{jk}$ is precisely the rotation generator J^i, the angular momentum, so

$$W_0 = 0, \qquad W_i = m\,J^i. \tag{3.203}$$

In the rest frame the Pauli–Lubanski vector *is* the angular momentum, rescaled by the mass — the precise sense in which W^μ generalizes spin covariantly. Contracting with the mostly-minus metric, and remembering $W^0 = W_0$, $W^i = -W_i$,

$$W^2 = W^\mu W_\mu = (W_0)^2 - \sum_{i=1}^{3}(W_i)^2 = 0 - m^2\sum_i (J^i)^2 = -m^2\,\mathbf{J}^2. \tag{3.204}$$

On the spin-s irreducible representation of the little group $\mathbf{J}^2 = s(s+1)$ (Theorem 2.53), so

$$\boxed{W^2 = -m^2\,s(s+1)\,.} \tag{3.205}$$

Because W^2 commutes with every Poincaré generator, this value is frame-independent: although we computed it in the rest frame, it holds on every state of the representation. The pair of Casimir eigenvalues $(P^2, W^2) = (m^2, -m^2 s(s+1))$ is the invariant fingerprint of the massive irrep, confirming that (m, s) are good labels and that "spin" is genuinely a Casimir of the Poincaré algebra, not an artefact of a frame.

The massive catalogue is now closed. For each $m > 0$ and each $s \in \{0, \frac{1}{2}, 1, \dots\}$ there is exactly one unitary irreducible representation, with $2s+1$ spin states. The scalar ($s = 0$) will be the quantum of ϕ^4 theory; the $s = \frac{1}{2}$ representation will be the electron of QED. Their field-theoretic homes are fixed in Section 3.7.5.

3.7.3 The massless case: little group ISO(2) and helicity

For a massless particle there is no rest frame: the orbit (III) is the light cone, and the standard momentum cannot be brought to $(m, 0, 0, 0)$. We choose instead a fixed null momentum along the z-axis,

$$k^\mu = (\omega, 0, 0, \omega), \qquad \omega > 0, \tag{3.206}$$

and ask which Lorentz transformations fix it. The answer is structurally richer than SO(3), and the difference is the whole story.

The little-group algebra. We work infinitesimally. Write the Lorentz generators as the rotations J_i and boosts K_i obeying the algebra established in Section 3.6,

$$[J_i, J_j] = i\varepsilon_{ijk} J_k, \qquad [J_i, K_j] = i\varepsilon_{ijk} K_k, \qquad [K_i, K_j] = -i\varepsilon_{ijk} J_k. \tag{3.207}$$

A generator X belongs to the little group iff its action annihilates k, i.e. iff the matrix $X^\mu{}_\nu$ satisfies $X^\mu{}_\nu k^\nu = 0$. The rotation about z obviously qualifies: J_3 fixes the z-axis and hence k. To find the rest, act with the elementary generators on $k = (\omega, 0, 0, \omega)$, using the explicit 4×4 matrices (boosts mix p^0 with a spatial component symmetrically, rotations mix two spatial components antisymmetrically). The boost K_1 sends $k \mapsto (0, \omega, 0, 0)$ (it raises a p^1 component from p^0), while the rotation J_2 sends $k \mapsto (0, \omega, 0, 0)$ as well (it rotates the z-component into x); their difference therefore annihilates k. The same in the y-direction relates K_2 and J_1. Define

$$A \equiv K_1 - J_2, \qquad B \equiv K_2 + J_1. \tag{3.208}$$

By construction $Ak = Bk = 0$: together with J_3 these are the three independent generators of the little group. Their algebra follows from Eq. (3.207) by direct computation. For $[J_3, A]$,

$$[J_3, A] = [J_3, K_1] - [J_3, J_2] = i\,\varepsilon_{312} K_2 - i\,\varepsilon_{321} J_1 = iK_2 + iJ_1 = iB, \tag{3.209}$$

using $\varepsilon_{312} = +1$, $\varepsilon_{321} = -1$. Likewise

$$[J_3, B] = [J_3, K_2] + [J_3, J_1] = i\,\varepsilon_{321} K_1 + i\,\varepsilon_{312} J_2 = -iK_1 + iJ_2 = -iA. \tag{3.210}$$

Finally

$$\begin{aligned}[A,B] &= [K_1 - J_2,\, K_2 + J_1] \\ &= [K_1,K_2] + [K_1,J_1] - [J_2,K_2] - [J_2,J_1] \\ &= (-i\,\varepsilon_{123}J_3) + 0 - 0 - (i\,\varepsilon_{213}J_3) \\ &= -iJ_3 + iJ_3 = 0,\end{aligned} \tag{3.211}$$

where the middle two brackets vanish because $[K_1, J_1] \propto \varepsilon_{11k} = 0$ and $[J_2, K_2] \propto \varepsilon_{22k} = 0$. Collecting,

$$[A,B] = 0, \qquad [J_3,A] = iB, \qquad [J_3,B] = -iA. \tag{3.212}$$

This is the Lie algebra $\mathfrak{iso}(2)$ of $\mathrm{ISO}(2) \cong E(2)$, the group of rigid motions of the Euclidean plane: A and B are the two commuting "translations," and J_3 is the "rotation" that mixes them like x and y. The massless little group is the symmetry group of a two-dimensional flat space — which is itself a miniature Poincaré group, with A, B playing the role of momenta and J_3 the role of angular momentum.

Representations and the continuous-spin trap. We now classify the unitary irreducible representations of ISO(2). The strategy mirrors Wigner's own method one dimension down. The translations commute, so we may diagonalize them simultaneously,

$$A\,|a,b\rangle = a\,|a,b\rangle\,, \qquad B\,|a,b\rangle = b\,|a,b\rangle\,. \tag{3.213}$$

The "rotation" J_3 acts on the eigenvalue pair (a, b) as a rotation of the plane — this is exactly content of $[J_3, A] = iB$, $[J_3, B] = -iA$ — so it preserves the invariant $A^2 + B^2$, the single Casimir of $\mathfrak{iso}(2)$. Set

$$A^2 + B^2 = \rho^2, \qquad \rho \geq 0. \tag{3.214}$$

Two cases arise, and the physical consequence of the distinction is enormous.

If $\rho > 0$, then (a, b) lives on a circle of radius ρ, and acting with finite J_3 rotations sweeps out the whole circle: the representation contains a continuum of states $|a, b\rangle$ parametrized by the angle on that circle, and a particle in it would carry a continuous internal degree of freedom together with an infinite tower of J_3 eigenvalues. These are the *continuous-spin* (or infinite-spin) representations. They are mathematically perfectly good unitary irreps of the Poincaré group, but they are not realized by any known particle, and there are strong reasons to expect they cannot be: ρ is a dimensionful continuous label, so such a particle would have a continuum of polarization states, giving, for instance, a divergent specific heat for any thermal gas of such particles and a thermodynamics incompatible

with observation. No local, finite-component field describes them. We therefore impose, as the physical selection rule that defines an ordinary massless particle,

$$\rho = 0 \quad \Longleftrightarrow \quad A\,|\psi\rangle = B\,|\psi\rangle = 0 \ \text{ on all physical states.} \tag{3.215}$$

(That $A^2 + B^2 = 0$ forces $A = B = 0$ follows because A, B are Hermitian: $\langle\psi|\, A^2 + B^2\,|\psi\rangle = \|A\psi\|^2 + \|B\psi\|^2 = 0$ makes each norm vanish.)

Caution

The vanishing of A and B is a *physical* restriction, not a theorem: continuous-spin representations exist and are unitary. What we have shown is that an ordinary massless particle — one with a finite number of polarization states and a local field description — is exactly the $\rho = 0$ case. The infinite-spin representations remain a live, if exotic, theoretical possibility; that they are absent from nature is an empirical input, dated and honest, not a consequence of Poincaré invariance alone.

Helicity. With $A = B = 0$ the little group collapses to the single remaining generator J_3. Its representations are one-dimensional — a massless particle of the ordinary type has *no internal multiplet at all*, only a single state per momentum — labelled by the eigenvalue

$$J_3\,|k\rangle = h\,|k\rangle\,, \tag{3.216}$$

the *helicity*. Because J_3 generates rotations about the direction of motion $\hat{k} = \hat{z}$, helicity is the projection of angular momentum onto the momentum, $h = \mathbf{J}\cdot\hat{p}$ — a Lorentz-invariant statement for a massless particle precisely because no boost can overtake it and flip $\hat{p}$. Evaluating the Casimirs on this representation, $P^2 = 0$ trivially and, since W^μ is built from $J^{\nu\rho}P^\sigma$ and reduces on the little group to a multiple of the momentum, $W^\mu = h\,P^\mu$; hence $W^2 = h^2P^2 = 0$. Both Casimirs vanish: the massless irreps are *not* distinguished by W^2 but by the single number h, which is why helicity, not spin, is the right label here.

Quantization of helicity. The allowed values of h are fixed by the same topology that gave us half-integer spin. A rotation by 2π about $\hat{z}$ is the identity in SO(3) but equals the nontrivial central element $-\mathbf{1}$ in its double cover; consistency of the representation under the closed loop of 2π rotations requires $e^{-2\pi i h} = \pm 1$, the sign being fixed once and for all by whether the state lies in a genuine SO(3) representation or in the double cover. This is exactly the $\mathbb{Z}_2 = \pi_1(\mathrm{SO}(1,3)^\uparrow)$ we identified in Sections 3.1.6 and 3.4.2: the universal cover $\mathrm{SL}(2,\mathbb{C})$ is a double cover, and its center acts on physical states by ± 1. Hence

$$h \in \left\{0,\ \pm\tfrac{1}{2},\ \pm 1,\ \pm\tfrac{3}{2},\ \ldots\right\}, \tag{3.217}$$

quantized in half-integers for precisely the reason spin is. Helicity quantization is not an independent postulate; it is the $\mathbb{Z}_2$ topology of the Lorentz group read off on the light cone.

The discrete operations and the pairing of $\pm h$. A given irreducible representation has a *single* helicity h; the positive- and negative-helicity states are, group-theoretically, different representations. Whether a particle carries both is decided by the discrete spacetime symmetries, which lie outside the connected component $\mathrm{SO}(1,3)^{\uparrow}$ (Section 3.1.3). Under parity P, the momentum reverses, $\vec{p} \to -\vec{p}$, while angular momentum, an axial vector, does not, $\mathbf{J} \to \mathbf{J}$; therefore helicity flips,

$$P: \quad h \longmapsto -h. \tag{3.218}$$

A parity-invariant massless theory must therefore contain *both* helicities $\pm h$: they are exchanged by P and so cannot be split apart without breaking it. The photon ($h = \pm 1$) and the graviton ($h = \pm 2$) accordingly come with two states each. The neutrinos of the Standard Model, by contrast, are described — in the idealization of exact masslessness — by single-helicity Weyl representations, and the theory is correspondingly *not* parity invariant: this is the chiral structure that Chapter 11's Weyl spinors realize at field level. Either way, the count is fixed: an ordinary massless particle is either one state (a single helicity, parity violated) or two (the pair $\pm h$, parity respected) — never the $2s+1$ of the massive case.

Physical Insight

The compactness of the little group is destiny. For $m > 0$ the little group $\mathrm{SU}(2)$ is compact, its irreps are finite-dimensional, and a massive particle of spin s carries the full $2s+1$-fold multiplet — a spin-1 massive vector has *three* polarizations. For $m = 0$ the little group $\mathrm{ISO}(2)$ is non-compact; demanding a finite number of states forces $A = B = 0$ and leaves only the abelian J_3, so a massless particle carries at most *two* helicity states regardless of how large $|h|$ is. The disappearing polarizations as $m \to 0$ — three becoming two for a vector — are not a smooth limit but a change in the little group itself, and the "lost" longitudinal state is exactly what gauge invariance will have to account for.

3.7.4 Helicity one cannot be a four-vector: the seed of gauge invariance

We have learned that a massless helicity-± 1 particle — the photon, the gluon — carries exactly two physical states. We will eventually want to describe it with a field that transforms simply under the Lorentz group, and the natural candidate

is a four-vector field $A^\mu(x)$, transforming in the $(\frac{1}{2},\frac{1}{2})$ representation built in Section 3.3.3. But A^μ has *four* components, and as an SO(3) representation $(\frac{1}{2},\frac{1}{2})$ decomposes as $\mathbf{4} = \mathbf{3} \oplus \mathbf{1}$, i.e. spin-1 plus spin-0. There is a mismatch between four field components and two physical helicity states, and the resolution of that mismatch is forced on us by the little group. It is the origin of gauge invariance.

To see it cleanly, track what the little group does to the polarization vectors directly.

Worked Example 3.46 (The little group shifts polarizations by k^μ: gauge redundancy is forced)**.** For the standard null momentum $k^\mu = (\omega, 0, 0, \omega)$ of Eq. (3.206), a one-photon state is created by a four-vector field contracted with a polarization vector $\varepsilon^\mu(k)$. The two physical (transverse) polarizations corresponding to helicity ± 1 are

$$\varepsilon^\mu_\pm(k) = \tfrac{1}{\sqrt{2}}\,(0,\ 1,\ \pm i,\ 0), \tag{3.219}$$

the same circular-polarization combinations familiar from classical electromagnetism, chosen so that a rotation by θ about $\hat{z}$ multiplies $\varepsilon_\pm$ by $e^{\mp i\theta}$, confirming helicity ± 1. Now apply a little-group "translation," the generator $A = K_1 - J_2$ of Eq. (3.208), to ε_+. Acting on a four-vector V, the boost contributes $K_1 : V^1 \to V^1 + \alpha V^0$, $V^0 \to V^0 + \alpha V^1$, and the rotation $J_2 : V^1 \to V^1 + \alpha V^3$, $V^3 \to V^3 - \alpha V^1$ at first order in the parameter α. On $\varepsilon_+ = \frac{1}{\sqrt{2}}(0,1,i,0)$, only the $V^1 = 1/\sqrt{2}$ component sources anything:

$$A\,\varepsilon_+ = K_1\,\varepsilon_+ - J_2\,\varepsilon_+ = \tfrac{1}{\sqrt{2}}(1,0,0,0) - \tfrac{1}{\sqrt{2}}(0,0,0,-1) = \tfrac{1}{\sqrt{2}}(1,0,0,1) = \frac{1}{\sqrt{2}\,\omega}\,k^\mu. \tag{3.220}$$

The action of the little group on the polarization vector is *not* homogeneous: it shifts ε^μ_+ by a multiple of the momentum k^μ itself. Exponentiating, a finite little-group element W acts as

$$\varepsilon^\mu_\pm(k) \ \longmapsto\ \varepsilon^\mu_\pm(k) + \beta(W)\,k^\mu, \tag{3.221}$$

with β a real parameter determined by W. The transverse polarizations do *not* transform among themselves as the components of a genuine four-vector would; the inhomogeneous k^μ term is unavoidable, a direct consequence of the non-compact translations A, B in $\mathfrak{iso}(2)$.

There are only two ways to live with Eq. (3.221). Either ε^μ is not a Lorentz four-vector — but then the field built from it is not Lorentz covariant, and manifest relativistic invariance is lost. Or we keep the covariant four-vector field A^μ and *declare the shift Eq. (3.221) to be a redundancy of description*: two polarization vectors differing by a multiple of k^μ describe the same physical state,

$$\varepsilon^\mu(k) \ \sim\ \varepsilon^\mu(k) + \beta\,k^\mu. \tag{3.222}$$

This is exactly a gauge transformation. In position space $\varepsilon^\mu(k)e^{-ik\cdot x}$ is the plane-wave content of $A^\mu(x)$, and adding $\beta k^\mu e^{-ik\cdot x}$ is adding $\partial^\mu\chi$ for $\chi \propto e^{-ik\cdot x}$: the equivalence Eq. (3.222) is the momentum-space form of

$$A^\mu(x) \;\sim\; A^\mu(x) + \partial^\mu\chi(x). \tag{3.223}$$

The bookkeeping now closes perfectly. The four components of A^μ are reduced by one through the equivalence Eq. (3.222) (the freedom to add k^μ) and by one more through the masslessness constraint that removes the would-be time-like/longitudinal mode ($k\cdot\varepsilon = 0$ together with the shift), leaving $4 - 2 = 2$ physical degrees of freedom — exactly the two helicities. Gauge invariance is not an aesthetic principle imposed from outside; it is the price of writing a massless helicity-1 particle, which intrinsically has two states, inside a Lorentz-covariant four-component field.

This is the chapter's most consequential conclusion, so it is worth stating plainly. A massive spin-1 particle has three states and is described, without redundancy, by a constrained four-vector (the Proca field, with $\partial_\mu A^\mu = 0$ eliminating the fourth component). A massless helicity-1 particle has only two states, and *no* unconstrained or simply-constrained four-vector field can carry exactly two Lorentz-covariant states: the little group forces the inhomogeneous transformation Eq. (3.221), and the only Lorentz-covariant way to absorb it is to gauge it away. Every gauge theory in this book — the electromagnetism of QED, the Yang–Mills theory of Yang–Mills theory — is, at bottom, a response to this single representation-theoretic fact. The full development is owed to later chapters: the classical gauge principle in Chapter 8, the quantization of the photon with its physical-state problem in Chapter 13, and the non-abelian generalization in Chapter 25. What Theorem 3.46 establishes is that those chapters are not introducing a new postulate but discharging a debt incurred here, on the light cone, by Wigner's classification.

3.7.5 The catalogue: fields to particles

We can now lay the two halves of the chapter side by side. Sections 3.3 and 3.4 classified the finite-dimensional — and therefore non-unitary — representations of the Lorentz algebra by a pair (j_-, j_+) of $\mathfrak{su}(2)$ spins; these are the *fields*, the local objects that carry Lorentz indices and out of which Lagrangians are built. The present section classified the infinite-dimensional unitary representations of the Poincaré group by (m, s) or by $m = 0$ and helicity h; these are the *states*, the actual particles in the Hilbert space. Table 3.1 is the dictionary between them, and it is the destination the whole chapter has been travelling toward.

Three features of the table deserve emphasis, because they encode the tension that drives the rest of the book. First, the scalar is the clean case: one field component, one particle state, no mismatch — which is why ϕ^4 is the workhorse

Field	(j_-, j_+)	**Particle (unitary irrep)**	**Theory / chapter**
Scalar ϕ	$(0,0)$	$m \geq 0$, spin/helicity 0	ϕ^4 theory (Chapter 9
Weyl ψ_L, ψ_R	$(\frac{1}{2},0)$, $(0,\frac{1}{2})$	$m = 0$, helicity $\mp\frac{1}{2}$	chiral fermions (Cha
Dirac Ψ	$(\frac{1}{2},0)\oplus(0,\frac{1}{2})$	$m > 0$, spin $\frac{1}{2}$ (2 states)	QED, electron (Chap
Vector A^μ	$(\frac{1}{2},\frac{1}{2})$	$m = 0$, helicity ±1 (2 states, *gauge*)	QED / Yang–Mills (

Table 3.1: The dictionary between field representations (finite-dimensional, non-unitary, Section 3.3) and particle representations (infinite-dimensional, unitary, this section). The left column transforms under the Lorentz algebra by (j_-, j_+); the right column is the one-particle Hilbert space content, labelled by mass and by spin (massive) or helicity (massless). The vector field carries gauge redundancy precisely because its two physical helicity states cannot fill its four covariant components, as shown in Theorem 3.46.

on which every technique is first tried. Second, the Dirac field bundles the two Weyl representations together precisely so that parity can act (it exchanges $(\frac{1}{2},0) \leftrightarrow (0,\frac{1}{2})$, the field-level image of the helicity flip $h \to -h$), and a massive Dirac particle's two states are the spin-$\frac{1}{2}$ doublet of the massive little group SU(2). Third, the vector is the case where field and particle counts genuinely disagree, and the entry "gauge" in its row is the residue of Theorem 3.46: a four-component covariant field for a two-state particle is necessarily redundant. The systematic study of *fields* as classical objects begins in Part II; their *quantization* into the particle states catalogued here begins in Part III.

This completes the chapter's central argument. We began by asking what a relativistic particle can be, and we have answered it exhaustively: a one-particle state is a unitary irreducible representation of the Poincaré group, and there is precisely one for every $(m > 0,\ s)$ with $s \in \{0, \frac{1}{2}, 1, \dots\}$, one for every massless helicity $h \in \{0, \pm\frac{1}{2}, \pm1, \dots\}$, and the vacuum — the rest excluded by causality or unobserved. The threads now collect their cargo from this catalogue and carry it forward into dynamics: the scalar of ϕ^4 theory is quantized in Chapter 9, the spin-$\frac{1}{2}$ electron of QED in Chapter 11, and the massless helicity-1 photon and gluon of QED and Yang–Mills theory in Chapters 13 and 25 — where the gauge redundancy seeded in Theorem 3.46, and first met classically in Chapter 8, is finally confronted head on.

Further reading

The definitive physical treatment of the unitary representations of the Poincaré group, and the source of the field-versus-particle organizing principle used through-

out this chapter, is Weinberg [500] (chapter 2); it is the natural companion to everything here. The classification itself is Wigner's, in the founding paper [509], with the finite-dimensional field representations and the relativistic wave equations laid out in Bargmann and Wigner [50]. The standard textbook accounts of the Lorentz algebra, the (j_-, j_+) classification, and the spinor formalism are Peskin and Schroeder [396], Schwartz [432], and Srednicki [454] (the last especially careful with the two-component dotted/undotted calculus); Zee [532] gives the same material with the group-theoretic motivation foregrounded. For the topology underlying the double cover and $\pi_1(\mathrm{SO}(1,3)^+) = \mathbb{Z}_2$—covering spaces, $\mathrm{SL}(2,\mathbb{C})$, and the spin groups—see Nakahara [355] and, for the representation-theoretic backbone, Fulton and Harris [206] and Hall [248]. Readers wanting the axiomatic setting in which Poincaré covariance, spin, and statistics are tied together rigorously should consult Streater and Wightman [458] and Haag [246].

Chapter 4

Functional Methods

Quantum field theory is, at bottom, a theory of *functions of functions*. The action that defines a theory assigns a number to each field configuration; the amplitude for one configuration to evolve into another is a sum over all configurations interpolating between them; the correlation functions that encode every observable are functional derivatives of a single generating object. To speak this language fluently one needs a calculus whose variables are not points in $\mathbb{R}^n$ but entire fields, an integration theory over the infinite-dimensional space of such fields, and a Legendre-transform technology that converts between a description in terms of sources and a description in terms of the fields they produce. This chapter builds that toolbox. It is the second of the two foundational machineries Part I lays down: Chapters 2 and 3 supplied the symmetry-and-representation language, and the present chapter supplies the variational and functional-integral language, leaving only the analytic machinery of Chapter 5 before the field theory of Part II can begin.

The chapter is deliberately staged from the rigorous to the formal, and it is honest at every step about which is which. We begin on solid ground in Section 4.1, where the functional derivative is defined precisely in the Gâteaux and Fréchet senses, its computational rules are established, the functional Taylor expansion is set up, and the Euler–Lagrange equation is derived as the exact stationarity condition of an action functional with its boundary terms tracked honestly rather than discarded. We then descend to finite dimensions in Section 4.2, where everything can be proved: the real and complex Gaussian integral formulas with their determinant prefactors and source terms, and Wick's theorem as a purely combinatorial statement about moments of a Gaussian measure, the two-point function being the inverse of the quadratic form. These finite-dimensional identities are the templates for every Gaussian manipulation in the rest of the book; we prove them once, in full, so that the field-theoretic versions later can be read as their formal continuum limits rather than as new postulates.

With the finite-dimensional theory secured, Section 4.3 confronts the cen-

tral object directly. The functional integral is constructed as a limit of finite-dimensional integrals over a lattice discretization; we exhibit exactly where that limit is a theorem (the Euclidean Gaussian case, where the measure genuinely exists) and exactly where it is a formal device (the oscillatory Minkowski case, whose convergence is a question deferred to the $i\epsilon$ and Wick-rotation analysis of Chapter 5). This section is the chapter's rigor summary: it states precisely which path-integral results in this book are theorems and which are controlled formal limits, so that no later chapter has to pretend the continuum measure is something it is not. The remaining sections then develop the formalism that makes the integral useful regardless of its measure-theoretic status.

Section 4.4 introduces the generating functional $Z[J]$, its logarithm $W[J] = \log Z[J]$, and the hierarchy of correlation functions they produce; the linked-cluster (cumulant) theorem, proved here in finite dimensions, identifies W as the generator of *connected* correlators. Section 4.5 performs the Legendre transform of W to obtain the effective action $\Gamma[\phi]$, establishing the convex-analysis facts it rests on (involution, the $\Gamma'' W'' = 1$ identity), showing that Γ generates the one-particle-irreducible vertices, and that Γ reduces to the classical action S at tree level. Finally Section 4.6 develops Laplace and steepest-descent asymptotics in finite dimensions with controlled error, organizes the resulting expansion as a series in $\hbar$ (the loop expansion), and identifies the functional determinant of the second variation as the one-loop Gaussian-fluctuation prefactor, so that the one-loop effective action is $\frac{i}{2}\log\det$ of the second variation.

Throughout, a zero-dimensional ϕ^4 "field theory" – a single ordinary integral with a quartic weight – serves as the carried model computation. It is not yet ϕ^4 theory proper, which is not defined as a Lagrangian field theory until Chapter 6; rather it is the testbed in which Z, W, Γ, the loop expansion, and the asymptotic (and Borel-summable) character of the perturbative series can all be exhibited *exactly*, with no functional-analytic difficulty, before any of them is asked to survive the continuum limit. The genuine field-theoretic versions – the free propagator, Wick's theorem for fields, and the field-theory generating functional – are assembled in Chapter 10, which imports this chapter's finite-dimensional results wholesale. We are stocking a toolbox; we name on the box exactly which later chapters each tool is for.

4.1 The Calculus of Functionals

Classical mechanics is dynamics on a finite-dimensional configuration space: a particle is a point $q \in \mathbb{R}^n$, its history is a path $t \mapsto q(t)$, and the action $S(q)$ is an ordinary — if multivariable — function whose stationary points are the trajectories. Field theory replaces the point q by an entire field configuration ϕ, a function on spacetime. The configuration space is now the space of all such functions, which is infinite-dimensional, and the action becomes a machine that

eats a whole field and returns a number. To do dynamics on this space — to write "$\delta S = 0$", to expand around a classical solution, to define a generating functional and differentiate it — we need a calculus on the space of field configurations: a notion of derivative for maps whose argument is a function. That calculus is the subject of this section. We build it carefully, because every later manipulation in this book, from the Euler–Lagrange equations to the loop expansion of the effective action, is an application of the rules established here.

The guiding principle is conservative: infinite-dimensional calculus should reduce to ordinary calculus the moment we discretise spacetime onto a finite lattice of points. Every definition below is engineered so that this reduction is transparent, and we will point it out as we go. The one genuinely new object is the *functional derivative*, whose "components" are labelled by a continuous index x rather than a discrete index i, and whose Kronecker delta is therefore a Dirac delta. We use the delta function at the naive level throughout this section; its rigorous incarnation as a tempered distribution, and the function spaces on which all of this is genuinely well defined, are the business of Chapter 5. Here we keep the analysis honest by stating the hypotheses each result needs and flagging where the naive manipulation hides a distributional subtlety.

4.1.1 Functionals

We must first say precisely what kind of object we are differentiating. Fix a domain $\Omega \subseteq \mathbb{R}^4$ — either a bounded region with smooth boundary $\partial\Omega$, or all of Minkowski space $\mathbb{R}^{1,3}$ — and a real vector space V of field configurations on it. For concreteness V may be taken as the smooth fields $\phi : \Omega \to \mathbb{R}$ that, together with all their derivatives, decay rapidly at infinity (the Schwartz space $\mathcal{S}(\mathbb{R}^4)$ when $\Omega = \mathbb{R}^4$) or vanish appropriately on $\partial\Omega$; the precise choice is dictated by the boundary conditions of the problem and is discussed where it matters. The only structure we need now is that V is a real topological vector space, so that "$\phi + \varepsilon\eta$" makes sense for $\phi, \eta \in V$ and $\varepsilon \in \mathbb{R}$, and so that we have a notion of one field being close to another.

Definition 4.1 (Functional)**.** A *functional* on V is a map $F : U \to \mathbb{R}$ (or $\to \mathbb{C}$) defined on an open subset $U \subseteq V$, assigning to each field configuration $\phi \in U$ a single number $F[\phi]$. We write the argument in square brackets, $F[\phi]$, to signal that the input is an entire function and not the value of ϕ at one point.

The square-bracket notation is not decorative: $F[\phi]$ depends on *all* the values $\{\phi(x) : x \in \Omega\}$ at once, in general nonlocally. Three families of examples recur constantly and are worth fixing now, because the entire computational apparatus of the section is calibrated against them.

Example 4.2 (The building-block functionals)**.**

1. **Local functionals.** Given a smooth function $g : \mathbb{R} \to \mathbb{R}$, set

$$F[\phi] = \int_\Omega \mathrm{d}^4x \; g(\phi(x)). \tag{4.1}$$

   More generally the integrand may depend on ϕ and finitely many of its derivatives, $F[\phi] = \int_\Omega \mathrm{d}^4x \; \mathcal{L}(\phi(x), \partial_\mu\phi(x))$; this is the *action functional*, the central object of the chapter.

2. **Bilocal functionals.** Given a kernel $K(x, y)$,

$$F[\phi] = \int_\Omega\int_\Omega \mathrm{d}^4x \, \mathrm{d}^4y \; \phi(x)\, K(x, y)\, \phi(y). \tag{4.2}$$

   Such quadratic forms are the integrands of every Gaussian integral in the next section, with K the inverse propagator.

3. **The evaluation (point) functional.** For a fixed $y \in \Omega$,

$$E_y[\phi] = \phi(y) = \int_\Omega \mathrm{d}^4x \; \delta^4(x - y)\, \phi(x). \tag{4.3}$$

   This trivial-looking functional, which simply reads off the field's value at one point, is what forces the Dirac delta into the formalism; the rewriting on the right, legitimate at the naive level, anticipates exactly that.

Physical Insight

The discretised picture makes the abstraction concrete. Replace Ω by a finite lattice of N points x_i with cell volume Δ, so a field configuration is the finite list of numbers $\phi_i \equiv \phi(x_i) \in \mathbb{R}^N$ and an integral becomes a sum, $\int_\Omega \mathrm{d}^4x \, (\cdots) \to \Delta \sum_i (\cdots)$. A functional $F[\phi]$ is then literally an ordinary function $F(\phi_1, \ldots, \phi_N)$ of N real variables. The local functional (4.1) becomes $\Delta \sum_i g(\phi_i)$, the bilocal (4.2) becomes $\Delta^2 \sum_{ij} \phi_i K_{ij} \phi_j$, and the evaluation functional (4.3) just picks out one component. Functional calculus is multivariable calculus in the limit $N \to \infty$, $\Delta \to 0$; we keep returning to this dictionary to sanity-check every formula.

4.1.2 The functional derivative

To differentiate a functional we mimic the directional derivative of multivariable calculus. There, $\partial_v f(q) = \frac{d}{d\varepsilon} f(q + \varepsilon v)\big|_0$ measures the rate of change of f as we move from q in the direction v. For a functional, the "direction" is itself a field $\eta \in V$, and we move along the straight line $\phi + \varepsilon\eta$ in configuration space.

Definition 4.3 (First variation / Gateaux derivative)**.** Let F be a functional and $\phi \in U$. The *first variation* of F at ϕ in the direction $\eta \in V$ is

$$\delta F[\phi;\eta] \;\equiv\; \frac{d}{d\varepsilon}\, F[\phi+\varepsilon\eta]\Big|_{\varepsilon=0} \;=\; \lim_{\varepsilon\to 0} \frac{F[\phi+\varepsilon\eta]-F[\phi]}{\varepsilon}, \tag{4.4}$$

whenever the limit exists. This is the *Gateaux derivative* of F at ϕ.

By construction $\delta F[\phi;\eta]$ is the ordinary derivative at the origin of the one-real-variable function $\varepsilon \mapsto F[\phi+\varepsilon\eta]$; all the familiar one-dimensional differentiation rules apply to it before we do anything field-theoretic. The map $\eta \mapsto \delta F[\phi;\eta]$ is in good cases linear and continuous — it then is a continuous linear functional on V, i.e. a distribution. When such a distribution can be represented by integration against a function (or distribution) $\frac{\delta F}{\delta\phi(x)}$, that representative is what we call the functional derivative.

Definition 4.4 (Functional derivative)**.** Suppose the first variation (4.4) exists for all $\eta \in V$, is linear and continuous in η, and can be written in the integral form

$$\delta F[\phi;\eta] \;=\; \int_\Omega \mathrm{d}^4x\, \frac{\delta F}{\delta\phi(x)}\, \eta(x) \tag{4.5}$$

for some locally integrable kernel (or distribution) $\dfrac{\delta F}{\delta\phi(x)}$. That kernel is the *functional derivative* of F with respect to ϕ at the point x. It depends on the base configuration ϕ and is itself, for each fixed ϕ, a function of x.

Two points of interpretation. First, $\dfrac{\delta F}{\delta\phi(x)}$ is the continuum analogue of the gradient component $\partial f/\partial\phi_i$: in the discretised picture, (4.5) reads $\delta F = \Delta \sum_i \frac{\delta F}{\delta\phi(x_i)}\,\eta_i$, and comparing with the chain rule $\delta F = \sum_i \frac{\partial F}{\partial\phi_i}\,\eta_i$ identifies

$$\frac{\delta F}{\delta\phi(x_i)} \;=\; \frac{1}{\Delta}\,\frac{\partial F}{\partial\phi_i}. \tag{4.6}$$

The factor $1/\Delta$ (units of inverse volume) is the seed of the Dirac delta that will appear in place of the Kronecker delta. Second, the representation (4.5) is a genuine restriction: not every continuous linear functional of η is an integral against an ordinary function (the evaluation functional already needs a delta function), but in the cases we meet — local and bilocal functionals, and the actions built from them — the kernel exists, possibly as a distribution, and is what we compute.

Uniqueness of the kernel. For the definition to be meaningful, the kernel representing a given first variation must be unique. It is, and the proof is the foundational lemma of the entire variational calculus.

Lemma 4.5 (Fundamental lemma of the calculus of variations)**.** *Let f be a continuous function on Ω. If*

$$\int_\Omega \mathrm{d}^4x\, f(x)\,\eta(x) = 0 \qquad \text{for every smooth } \eta \text{ with compact support in the interior of } \Omega, \tag{4.7}$$

then $f \equiv 0$ on Ω.

Proof. Suppose not. Then $f(x_0) \neq 0$ for some interior point x_0; without loss of generality $f(x_0) > 0$. By continuity there is a closed ball $\bar{B}_r(x_0)$ of radius $r > 0$ contained in the interior of Ω on which $f(x) \geq \tfrac{1}{2} f(x_0) > 0$. Choose a *bump function*: a smooth $\eta \geq 0$ supported in $B_r(x_0)$ with $\eta(x_0) > 0$, for instance the standard mollifier

$$\eta(x) = \begin{cases} \exp\!\Big(-\dfrac{r^2}{r^2 - |x - x_0|^2}\Big), & |x - x_0| < r, \\ 0, & |x - x_0| \geq r, \end{cases} \tag{4.8}$$

which is smooth (all derivatives vanish as $|x - x_0| \to r^-$), nonnegative, compactly supported, and strictly positive at x_0. With this choice the integrand $f\eta \geq 0$ everywhere and is strictly positive on a neighbourhood of x_0, so $\int_\Omega f\eta > 0$, contradicting (4.7). Hence no such x_0 exists and $f \equiv 0$. $\square$

Theorem 4.6 (Uniqueness of the functional derivative)**.** *If two continuous kernels $K_1(x)$ and $K_2(x)$ both represent the first variation of F at ϕ in the sense of (4.5) for all admissible η, then $K_1 = K_2$. The functional derivative, when it exists as a continuous function, is unique.*

Proof. Subtracting the two representations gives $\int_\Omega \mathrm{d}^4x\, [K_1(x) - K_2(x)]\,\eta(x) = 0$ for all test functions η. By Theorem 4.5 the continuous difference $K_1 - K_2$ vanishes identically. $\square$

When the kernel is only a distribution the same conclusion holds with "equal as distributions," which is the appropriate statement once Chapter 5 puts the delta function on a rigorous footing; the bump-function argument above is exactly the seed of the distributional uniqueness theorem proved there.

The delta-function kernel. The functional derivative of the field with respect to itself is the basic input to all subsequent computations. Apply the definition to the evaluation functional (4.3), $E_y[\phi] = \phi(y)$. Its first variation is immediate,

$$\delta E_y[\phi;\eta] = \frac{d}{d\varepsilon}\,\big(\phi(y) + \varepsilon\,\eta(y)\big)\Big|_{0} = \eta(y) = \int_\Omega \mathrm{d}^4x\; \delta^4(x - y)\,\eta(x), \tag{4.9}$$

where the last step is the defining property of the delta function. Comparing with (4.5) we read off the kernel,

$$\boxed{\frac{\delta\phi(y)}{\delta\phi(x)} = \delta^4(x-y).} \tag{4.10}$$

This is the continuum statement that the components of a field are independent coordinates: varying ϕ at x has no effect on its value at any other point y, and unit effect at $y = x$. It is the exact analogue of $\partial\phi_i/\partial\phi_j = \delta_{ij}$, with the Dirac delta replacing the Kronecker delta precisely as the correspondence (4.6) demands: $\delta_{ij}/\Delta \to \delta^4(x-y)$ as the lattice is refined. Every functional derivative we compute is, at bottom, repeated use of (4.10) together with the rules of the next subsection.

Gateaux versus Frechet. The first variation (4.4) is a *directional* derivative: it probes F one direction η at a time. The stronger notion is the *Frechet derivative*, a single bounded linear map $DF[\phi]$ approximating the increment uniformly in the direction,

$$F[\phi+\eta] - F[\phi] - DF[\phi]\,\eta = o(\|\eta\|) \quad \text{as } \|\eta\| \to 0, \tag{4.11}$$

with the remainder controlled uniformly over all η in a norm ball. The relation between the two is exactly as in finite dimensions:

- If F is Frechet differentiable at ϕ, then the Gateaux derivative exists and equals it, $\delta F[\phi;\eta] = DF[\phi]\,\eta$; the functional derivative (4.5) is then the integral kernel of the bounded operator $DF[\phi]$.
- The converse can fail: a functional may have a directional derivative in every direction without being Frechet differentiable (as already in finite dimensions). But if the Gateaux derivative $\delta F[\phi;\eta]$ exists in a neighbourhood of ϕ, is linear in η, and depends *continuously* on the base point ϕ, then F is Frechet differentiable and the two coincide.

For the functionals of physics — polynomials in the field and its derivatives, integrated against smooth kernels — this continuity holds, and we use "functional derivative" for the common value without further comment. We compute everything via the workable Gateaux definition (4.4) and rely on the equivalence to know the result is the honest derivative.

4.1.3 Computational rules

The first variation is an ordinary derivative in ε, so the rules of single-variable calculus transfer to functionals essentially for free. We prove each one from (4.4) rather than assert it, then read off its consequence for the kernel via (4.5).

Proposition 4.7 (Linearity, product rule, chain rule)**.** *Let F, G be functionals differentiable at ϕ, and a, b constants. Then*

$$\frac{\delta(aF + bG)}{\delta\phi(x)} = a\,\frac{\delta F}{\delta\phi(x)} + b\,\frac{\delta G}{\delta\phi(x)}, \tag{4.12}$$

$$\frac{\delta(FG)}{\delta\phi(x)} = \frac{\delta F}{\delta\phi(x)}\,G[\phi] + F[\phi]\,\frac{\delta G}{\delta\phi(x)}, \tag{4.13}$$

and, for an ordinary smooth function $h : \mathbb{R} \to \mathbb{R}$ composed with a real-valued functional, $\frac{\delta\, h(F)}{\delta\phi(x)} = h'(F[\phi])\,\frac{\delta F}{\delta\phi(x)}$. If $F[\phi] = G[\psi]$ where $\psi = \psi[\phi]$ is itself a field-valued functional, the chain rule reads

$$\frac{\delta F}{\delta\phi(x)} = \int_\Omega \mathrm{d}^4 y\; \frac{\delta G}{\delta\psi(y)}\,\frac{\delta\psi(y)}{\delta\phi(x)}. \tag{4.14}$$

Proof. Linearity is immediate: $\frac{d}{d\varepsilon}(aF + bG)[\phi + \varepsilon\eta]|_0 = a\,\delta F[\phi;\eta] + b\,\delta G[\phi;\eta]$, and matching to (4.5) gives (4.12).

Product rule. Define the one-variable functions $f(\varepsilon) = F[\phi + \varepsilon\eta]$ and $g(\varepsilon) = G[\phi + \varepsilon\eta]$, so $(FG)[\phi + \varepsilon\eta] = f(\varepsilon)g(\varepsilon)$. The ordinary product rule in ε gives

$$\delta(FG)[\phi;\eta] = \frac{d}{d\varepsilon}(fg)\Big|_0 = f'(0)g(0) + f(0)g'(0) = \delta F[\phi;\eta]\,G[\phi] + F[\phi]\,\delta G[\phi;\eta]. \tag{4.15}$$

Substituting the kernel representation (4.5) for each first variation,

$$\int_\Omega \mathrm{d}^4 x\,\frac{\delta(FG)}{\delta\phi(x)}\eta(x) = \int_\Omega \mathrm{d}^4 x\Big[\frac{\delta F}{\delta\phi(x)}G[\phi] + F[\phi]\frac{\delta G}{\delta\phi(x)}\Big]\eta(x), \tag{4.16}$$

and since this holds for every η, Theorem 4.5 forces the integrands to agree, which is (4.13). The composition rule with h is the same argument run with the single-variable chain rule: $\frac{d}{d\varepsilon}h(f(\varepsilon))|_0 = h'(f(0))f'(0) = h'(F[\phi])\,\delta F[\phi;\eta]$.

Chain rule for field-valued ψ. Now $F[\phi + \varepsilon\eta] = G[\psi[\phi + \varepsilon\eta]]$. Write $\psi_\varepsilon \equiv \psi[\phi + \varepsilon\eta]$ and differentiate at $\varepsilon = 0$. The inner variation is, by definition of the functional derivative of ψ,

$$\frac{d}{d\varepsilon}\psi_\varepsilon(y)\Big|_0 = \int_\Omega \mathrm{d}^4 x\,\frac{\delta\psi(y)}{\delta\phi(x)}\,\eta(x), \tag{4.17}$$

i.e. the curve $\varepsilon \mapsto \psi_\varepsilon$ in configuration space leaves $\psi[\phi]$ at $\varepsilon = 0$ with this "velocity" field. Applying the Gateaux derivative of G in that velocity direction,

$$\begin{aligned}\delta F[\phi;\eta] &= \frac{d}{d\varepsilon}G[\psi_\varepsilon]\Big|_0 = \int_\Omega \mathrm{d}^4 y\,\frac{\delta G}{\delta\psi(y)}\,\frac{d}{d\varepsilon}\psi_\varepsilon(y)\Big|_0 \\ &= \int_\Omega \mathrm{d}^4 y\,\frac{\delta G}{\delta\psi(y)}\int_\Omega \mathrm{d}^4 x\,\frac{\delta\psi(y)}{\delta\phi(x)}\,\eta(x) = \int_\Omega \mathrm{d}^4 x\,\Big[\int_\Omega \mathrm{d}^4 y\,\frac{\delta G}{\delta\psi(y)}\,\frac{\delta\psi(y)}{\delta\phi(x)}\Big]\eta(x).\end{aligned} \tag{4.18}$$

The bracketed quantity is, by Theorem 4.6, the functional derivative $\delta F/\delta\phi(x)$, which is (4.14). The intermediate spacetime point y is summed over precisely as the intermediate index is summed in the matrix chain rule $\partial F/\partial\phi_i = \sum_j (\partial G/\partial\psi_j)(\partial\psi_j/\partial\phi_i)$; the integral $\int \mathrm{d}^4y$ is the continuum $\sum_j$. □

These three rules, together with the delta-kernel (4.10), suffice to differentiate every functional we encounter. We now apply them to the building blocks of Theorem 4.2, since these results are used verbatim throughout the book.

Worked Example 4.8 (Functional derivatives of the building blocks)**.** We compute $\delta F/\delta\phi(z)$ for each functional of Theorem 4.2.

(a) Ultralocal density. For $F[\phi] = \int_\Omega \mathrm{d}^4x\, g(\phi(x))$, the first variation is

$$\delta F[\phi;\eta] = \frac{d}{d\varepsilon}\int_\Omega \mathrm{d}^4x\, g(\phi(x)+\varepsilon\eta(x))\Big|_0 = \int_\Omega \mathrm{d}^4x\, g'(\phi(x))\,\eta(x), \tag{4.19}$$

where we differentiated under the integral and used the ordinary chain rule on g. Comparison with (4.5) gives

$$\frac{\delta F}{\delta\phi(z)} = g'(\phi(z)). \tag{4.20}$$

The functional derivative of an ultralocal functional is itself ultralocal: it is just g' evaluated at the same point. As a check, take $g(\phi) = \tfrac12\phi^2$, so $F = \tfrac12\int\phi^2$ and $\delta F/\delta\phi(z) = \phi(z)$, the field itself.

(b) A density with one derivative. The kinetic term $F[\phi] = \tfrac12\int_\Omega \mathrm{d}^4x\,(\partial_\mu\phi)(\partial^\mu\phi)$ introduces the only genuinely new step, an integration by parts. Its first variation is

$$\begin{aligned}\delta F[\phi;\eta] &= \frac{d}{d\varepsilon}\tfrac12\int_\Omega \mathrm{d}^4x\,\partial_\mu(\phi+\varepsilon\eta)\,\partial^\mu(\phi+\varepsilon\eta)\Big|_0 = \int_\Omega \mathrm{d}^4x\,(\partial_\mu\phi)(\partial^\mu\eta)\\ &= \int_\Omega \mathrm{d}^4x\,\partial_\mu[(\partial^\mu\phi)\eta] - \int_\Omega \mathrm{d}^4x\,(\partial_\mu\partial^\mu\phi)\,\eta = -\int_\Omega \mathrm{d}^4x\,(\Box\phi)\,\eta + \oint_{\partial\Omega}\mathrm{d}S_\mu\,(\partial^\mu\phi)\,\eta,\end{aligned} \tag{4.21}$$

where $\Box \equiv \partial_\mu\partial^\mu = \partial_0^2 - \nabla^2$ is the d'Alembertian in the mostly-minus metric, and the boundary integral comes from the divergence theorem applied to $\partial_\mu[(\partial^\mu\phi)\eta]$. For test functions η supported in the interior (or fields decaying at infinity) the boundary term vanishes, and

$$\frac{\delta F}{\delta\phi(z)} = -\Box\,\phi(z). \tag{4.22}$$

The boundary term is not optional bookkeeping — it is the connection where boundary conditions enter the variational principle, and we track it explicitly when we turn to the action in Section 4.1.5.

(c) Bilocal form. For $F[\phi] = \int\!\int \mathrm{d}^4x\,\mathrm{d}^4y\,\phi(x)K(x,y)\phi(y)$, linearity of the first variation across the two factors gives, using (4.10) on each,

$$\begin{aligned}\delta F[\phi;\eta] &= \iint \mathrm{d}^4x\,\mathrm{d}^4y\,\big[\eta(x)K(x,y)\phi(y) + \phi(x)K(x,y)\eta(y)\big] \\ &= \int_\Omega \mathrm{d}^4z\,\eta(z)\int_\Omega \mathrm{d}^4y\,\big[K(z,y)+K(y,z)\big]\phi(y),\end{aligned} \tag{4.23}$$

where the second line relabels the integration variable in each term to z. Hence

$$\frac{\delta F}{\delta\phi(z)} = \int_\Omega \mathrm{d}^4y\,\big[K(z,y)+K(y,z)\big]\phi(y) \xrightarrow{\;K\text{ symmetric}\;} 2\int_\Omega \mathrm{d}^4y\,K(z,y)\phi(y). \tag{4.24}$$

This is the continuum gradient of a quadratic form, $\partial_i(\phi^{\mathsf{T}}K\phi) = (K+K^{\mathsf{T}})_{ij}\phi_j$, with the matrix index j promoted to the spacetime integral $\int \mathrm{d}^4y$. It is exactly the combination that, set to zero in the next section, yields the linear field equation $K\phi = 0$, and whose kernel is the quadratic operator of the Gaussian integral.

4.1.4 The functional Taylor expansion

Single derivatives let us locate stationary configurations; to expand a functional *around* one — the central operation of the entire perturbative and semiclassical apparatus — we need the full Taylor series. We obtain it, and the higher functional derivatives that are its coefficients, by reducing to the ordinary Taylor expansion in the single variable ε.

Higher functional derivatives are defined by iterating Theorem 4.4. The second functional derivative $\dfrac{\delta^2 F}{\delta\phi(x_1)\delta\phi(x_2)}$ is the functional derivative, with respect to $\phi(x_2)$, of the function $x_1 \mapsto \delta F/\delta\phi(x_1)$; in terms of first variations it is read off from the double variation

$$\frac{d}{d\varepsilon_1}\frac{d}{d\varepsilon_2}F[\phi+\varepsilon_1\eta_1+\varepsilon_2\eta_2]\Big|_{\varepsilon_1=\varepsilon_2=0} = \iint \mathrm{d}^4x_1\,\mathrm{d}^4x_2\,\frac{\delta^2 F}{\delta\phi(x_1)\,\delta\phi(x_2)}\,\eta_1(x_1)\,\eta_2(x_2), \tag{4.25}$$

and in general the n-th derivative is the symmetric kernel appearing when n independent variations are taken,

$$\frac{\mathrm{d}^n}{\mathrm{d}\varepsilon_1\cdots\mathrm{d}\varepsilon_n}F\Big[\phi+\sum_a \varepsilon_a\eta_a\Big]\Big|_{\varepsilon=0} = \int\cdots\int \prod_{a=1}^{n}\mathrm{d}^4x_a\,\frac{\delta^n F}{\delta\phi(x_1)\cdots\delta\phi(x_n)}\prod_{a=1}^{n}\eta_a(x_a). \tag{4.26}$$

Proposition 4.9 (Symmetry of the higher kernels)**.** *Whenever the mixed variations exist and are continuous, the n-th functional derivative is symmetric under any permutation of its arguments $x_1,\dots,x_n$:*

$$\frac{\delta^n F}{\delta\phi(x_{\sigma(1)})\cdots\delta\phi(x_{\sigma(n)})} = \frac{\delta^n F}{\delta\phi(x_1)\cdots\delta\phi(x_n)} \qquad \text{for every permutation } \sigma. \tag{4.27}$$

Proof. It suffices to prove symmetry under interchange of two adjacent arguments, since transpositions of neighbours generate the full permutation group. Consider the two-parameter real function $g(\varepsilon_i, \varepsilon_j) = F[\phi + \varepsilon_i \eta_i + \varepsilon_j \eta_j + \cdots]$, the dots denoting any other fixed variations held at their values. The left-hand sides of (4.25) computed in the two orders are $\partial_{\varepsilon_i}\partial_{\varepsilon_j} g$ and $\partial_{\varepsilon_j}\partial_{\varepsilon_i} g$. By the equality of mixed partial derivatives (Clairaut–Schwarz) for the ordinary smooth function g of two real variables, these are equal. Since they equal the kernel integrated against $\eta_i \otimes \eta_j$ in the two orders, and this holds for all test functions η_i, η_j, Theorem 4.5 applied in each variable forces the kernels themselves to be equal, which is (4.27) for the transposition (ij). □

This symmetry is the functional shadow of the symmetry of the Hessian matrix $\partial_i \partial_j F = \partial_j \partial_i F$, and it is the reason the Taylor coefficients below can be treated as symmetric multilinear kernels — equivalently, as symmetric tensors with continuous indices. We now assemble the series.

Theorem 4.10 (Functional Taylor expansion)**.** *Let F be a functional with continuous functional derivatives to all orders in a neighbourhood of ϕ. Then for η small enough that the line $\phi + \varepsilon\eta$, $\varepsilon \in [0, 1]$, stays in that neighbourhood,*

$$F[\phi + \eta] = \sum_{n=0}^{\infty} \frac{1}{n!} \int \cdots \int \prod_{a=1}^{n} \mathrm{d}^4 x_a \, \frac{\delta^n F}{\delta\phi(x_1) \cdots \delta\phi(x_n)}\bigg|_{\phi} \eta(x_1) \cdots \eta(x_n), \tag{4.28}$$

whenever the series converges, with the understanding that it is at least the asymptotic expansion of $F[\phi + \eta]$ in powers of η.

Proof. Fix ϕ and η and define the single-variable function $g(\varepsilon) \equiv F[\phi + \varepsilon\eta]$. Its ordinary Maclaurin expansion is

$$g(1) = \sum_{n=0}^{\infty} \frac{1}{n!} g^{(n)}(0), \qquad g^{(n)}(0) = \frac{\mathrm{d}^n}{\mathrm{d}\varepsilon^n} F[\phi + \varepsilon\eta]\Big|_0. \tag{4.29}$$

The n-th derivative is the special case of (4.26) with all directions equal, $\eta_1 = \cdots = \eta_n = \eta$ (set $\varepsilon_a = \varepsilon$ and use $\frac{d}{d\varepsilon} = \sum_a \partial_{\varepsilon_a}$ on the diagonal, which produces $n!$ identical cross terms only after the symmetric kernel collapses them — here they are automatically accounted because we take all η_a equal):

$$g^{(n)}(0) = \int \cdots \int \prod_{a=1}^{n} \mathrm{d}^4 x_a \, \frac{\delta^n F}{\delta\phi(x_1) \cdots \delta\phi(x_n)}\bigg|_{\phi} \eta(x_1) \cdots \eta(x_n). \tag{4.30}$$

Substituting into $g(1) = \sum_n g^{(n)}(0)/n!$ and recognising $g(1) = F[\phi + \eta]$ yields (4.28). Convergence is governed by the radius of convergence of the scalar series $\sum g^{(n)}(0)/n!$; for the polynomial functionals of field theory the series in η *terminates* (a degree-k polynomial functional has $\delta^n F = 0$ for $n > k$), so (4.28) is an exact finite sum and convergence is not at issue. For non-polynomial F it is the Taylor expansion of g about $\varepsilon = 0$ evaluated at $\varepsilon = 1$, valid with its usual remainder. □

The first three terms of (4.28) are the ones we use constantly, so we display them with names:

$$F[\phi+\eta] = \underbrace{F[\phi]}_{\text{value}} + \underbrace{\int \mathrm{d}^4x\, \frac{\delta F}{\delta\phi(x)}\,\eta(x)}_{\text{first variation } \delta F} + \underbrace{\frac{1}{2}\iint \mathrm{d}^4x\,\mathrm{d}^4y\,\eta(x)\,\frac{\delta^2 F}{\delta\phi(x)\delta\phi(y)}\,\eta(y)}_{\text{second variation } \frac{1}{2}\delta^2 F} + \cdots \tag{4.31}$$

The linear term vanishes precisely at a stationary configuration (the topic of the next subsection); there the leading nontrivial term is the second variation, the *quadratic form*

$$\delta^2 F[\phi;\eta] = \iint \mathrm{d}^4x\,\mathrm{d}^4y\;\eta(x)\,\frac{\delta^2 F}{\delta\phi(x)\delta\phi(y)}\bigg|_{\phi}\,\eta(y), \tag{4.32}$$

whose kernel $\dfrac{\delta^2 F}{\delta\phi(x)\delta\phi(y)}$ is the continuum Hessian. By Theorem 4.9 it is a symmetric kernel, hence a symmetric operator on the space of variations; this is the object whose determinant and inverse the Gaussian integration of the next section is built to compute. The whole point of expanding around a stationary configuration is that (4.32) becomes the leading term, and a Gaussian.

4.1.5 The Euler–Lagrange equation as a stationarity condition

We can now state and prove the central result of classical field dynamics in the language just built. An action functional assigns to each field history a number, and the physical histories are its stationary points: those configurations from which the action does not change to first order under any allowed variation. "Stationary" is "$\delta S/\delta\phi = 0$," and computing that functional derivative honestly — with its boundary term — yields the Euler–Lagrange equation.

Theorem 4.11 (Euler–Lagrange equation and the variational boundary term)**.** *Let*

$$S[\phi] = \int_\Omega \mathrm{d}^4x\;\mathcal{L}(\phi(x), \partial_\mu\phi(x)) \tag{4.33}$$

be a local action with $\mathcal{L}$ a smooth function of the field and its first derivatives. Then the first variation of S in the direction η is

$$\delta S[\phi;\eta] = \int_\Omega \mathrm{d}^4x\,\Big[\frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\Big)\Big]\,\eta + \oint_{\partial\Omega} \mathrm{d}S_\mu\;\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\,\eta, \tag{4.34}$$

so the functional derivative, valid wherever the boundary term is annihilated, is the Euler–Lagrange combination

$$\frac{\delta S}{\delta\phi(x)} = \frac{\partial\mathcal{L}}{\partial\phi} - \partial_\mu\Big(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\Big). \tag{4.35}$$

If S is stationary, $\delta S[\phi;\eta]=0$ for all admissible variations η, and the boundary term vanishes (by either of the conditions below), then ϕ satisfies the Euler–Lagrange equation

$$\boxed{\frac{\partial \mathcal{L}}{\partial \phi}-\partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)=0.} \tag{4.36}$$

Proof. Form $S[\phi+\varepsilon\eta]=\int_\Omega \mathrm{d}^4x\,\mathcal{L}(\phi+\varepsilon\eta,\partial_\mu\phi+\varepsilon\partial_\mu\eta)$ and differentiate at $\varepsilon=0$, using the ordinary chain rule on $\mathcal{L}$ as a function of its two slots ϕ and $\partial_\mu\phi$ (the latter varying because $\partial_\mu(\phi+\varepsilon\eta)=\partial_\mu\phi+\varepsilon\,\partial_\mu\eta$):

$$\delta S[\phi;\eta]=\int_\Omega \mathrm{d}^4x\,\Big[\frac{\partial \mathcal{L}}{\partial \phi}\,\eta+\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu\eta\Big]. \tag{4.37}$$

The second term carries the derivative on η; to expose the kernel we move that derivative off η by the Leibniz identity

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\partial_\mu\eta=\partial_\mu\Big[\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\,\eta\Big]-\partial_\mu\Big(\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\Big)\,\eta. \tag{4.38}$$

Substituting (4.38) into (4.37) and applying the four-dimensional divergence theorem to the total-derivative piece, $\int_\Omega \mathrm{d}^4x\,\partial_\mu V^\mu=\oint_{\partial\Omega}\mathrm{d}S_\mu\,V^\mu$ with $V^\mu=\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\eta$, gives exactly (4.34). The bracket multiplying η in the bulk integral is, by Theorem 4.4, the functional derivative (4.35), provided the boundary term does not contribute — which holds under either standard condition:

1. *Fixed (Dirichlet) data.* If the field's values are prescribed on $\partial\Omega$, then admissible variations are those that respect this prescription, $\eta|_{\partial\Omega}=0$, and the boundary integrand $\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}\eta$ vanishes pointwise on $\partial\Omega$. This is the case in the action principle "with fixed endpoints": in mechanics one holds $q(t_1),q(t_2)$ fixed; in field theory one holds the field fixed on the boundary of the spacetime region. It is also the case for $\Omega=\mathbb{R}^4$ with variations of compact support, or fields and variations decaying fast enough that the surface integral at infinity vanishes.

2. *Natural (Neumann-type) boundary conditions.* If the field is *not* fixed on $\partial\Omega$, stationarity for arbitrary η — including those nonzero on the boundary — forces *both* the bulk Euler–Lagrange equation (4.36) *and* the vanishing of the boundary integrand, $\frac{\partial \mathcal{L}}{\partial(\partial_\mu\phi)}n_\mu=0$ on $\partial\Omega$, with n_μ the outward normal. These are the natural boundary conditions; the variational principle generates them automatically rather than having them imposed by hand.

In either case the boundary term is annihilated, and stationarity reduces to $\int_\Omega \mathrm{d}^4x\,[\partial\mathcal{L}/\partial\phi-\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi))]\,\eta=0$ for all admissible η. By the fundamental lemma Theorem 4.5 the bracket vanishes identically, which is (4.36). □

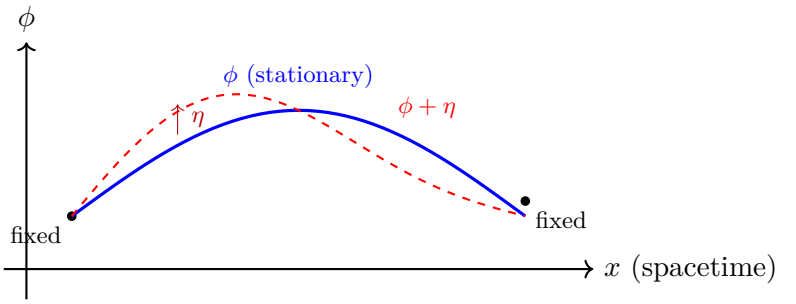


Figure 4.1: The variational setup behind Theorem 4.11, drawn with one spacetime direction. The stationary configuration ϕ (solid) and a nearby varied one $\phi + \eta$ (dashed) share the same data on the boundary, so the variation η vanishes there ($\eta|_{\partial\Omega} = 0$); this is condition 1 of the proof, which kills the surface term in (4.34). Stationarity demands $\delta S = 0$ for *every* such η, and the fundamental lemma then forces the Euler–Lagrange equation pointwise in the bulk.

Caution

The boundary term in (4.34) is the single most frequently dropped — and most frequently legitimately dropped — object in field theory, but it is never *automatically* zero. Whether it vanishes is a statement about boundary conditions, not about the action, and it is exactly the place where physics beyond the bulk equations of motion lives: it determines which boundary terms one is free to add to an action without changing the dynamics (they shift the surface integral, not the Euler–Lagrange equation), it underlies the Gibbons–Hawking term in gravity, and it governs the consistency of an action principle on a manifold with boundary. When this book "integrates by parts and drops the surface term," it is invoking one of the two conditions above; we state which one whenever the distinction could matter.

It remains to compute the second variation of a generic local action, since that quadratic operator is precisely what the next section will integrate. We do so as a worked computation, both to exhibit the second functional derivative of an action explicitly and to hand off the operator cleanly.

Worked Example 4.12 (First and second variation of a generic local action)**.** Take the canonical local action of a single scalar field,

$$S[\phi] = \int_\Omega \mathrm{d}^4x \left[\tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - V(\phi)\right], \tag{4.39}$$

with V a smooth potential. We compute its first and second variations from scratch.

First variation. With $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - V(\phi)$ we have $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$ and $\partial\mathcal{L}/\partial\phi = -V'(\phi)$. Substituting into the general result (4.34),

$$\delta S[\phi;\eta] = -\int_\Omega \mathrm{d}^4x\,\left[\Box\phi + V'(\phi)\right]\eta + \oint_{\partial\Omega}\mathrm{d}S_\mu\,(\partial^\mu\phi)\,\eta, \tag{4.40}$$

so that, dropping the surface term under the conditions of Theorem 4.11,

$$\frac{\delta S}{\delta\phi(x)} = -\left[\Box\phi(x) + V'(\phi(x))\right], \qquad \text{stationary} \iff \Box\phi + V'(\phi) = 0. \tag{4.41}$$

This is the classical equation of motion for the field (4.39); we have built the operator $\delta S/\delta\phi$ but deliberately do not specialise V here. The genuine $\lambda\phi^4$ stage — the choice $V = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$ giving the equation of motion $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$ of ϕ^4 theory — is owned by Chapter 6, which opens the classical dynamics chapter; here we only construct the $\delta/\delta\phi$ machinery it will be applied with.

Second variation. Differentiate once more. The cleanest route is to expand $S[\phi+\eta]$ directly to second order in η and read off the quadratic piece per (4.31). Inserting $\phi \to \phi + \eta$ into (4.39) and using $(\partial(\phi+\eta))^2 = (\partial\phi)^2 + 2(\partial_\mu\phi)(\partial^\mu\eta) + (\partial_\mu\eta)(\partial^\mu\eta)$ and the Taylor expansion $V(\phi+\eta) = V(\phi) + V'(\phi)\eta + \frac{1}{2}V''(\phi)\eta^2 + \cdots$, the part quadratic in η is

$$\tfrac{1}{2}\,\delta^2 S[\phi;\eta] = \int_\Omega \mathrm{d}^4x\,\Big[\tfrac{1}{2}(\partial_\mu\eta)(\partial^\mu\eta) - \tfrac{1}{2}V''(\phi)\,\eta^2\Big]. \tag{4.42}$$

Integrate the kinetic piece by parts exactly as in (4.21), discarding the surface term (the variation η already vanishes on $\partial\Omega$ by the same boundary condition that fixed the first variation):

$$\delta^2 S[\phi;\eta] = \int_\Omega \mathrm{d}^4x\,\eta(x)\,\big[-\Box - V''(\phi(x))\big]\,\eta(x). \tag{4.43}$$

Comparing with the definition (4.32) of the second variation as a bilinear form $\int\!\!\int \eta(x)\,\frac{\delta^2 S}{\delta\phi(x)\delta\phi(y)}\,\eta(y)$, we read off the second functional derivative as a kernel,

$$\left.\frac{\delta^2 S}{\delta\phi(x)\,\delta\phi(y)}\right|_\phi = \big[-\Box_x - V''(\phi(x))\big]\,\delta^4(x-y). \tag{4.44}$$

The delta function expresses that, at the level of two derivatives, the action of (4.39) is still local: differentiating twice produces a differential operator acting at a single point, not a genuinely bilocal smearing. The operator

$$\mathcal{K}_\phi \;\equiv\; -\Box - V''(\phi) \tag{4.45}$$

is the *second-variation operator* (or fluctuation operator) about the configuration ϕ. It is symmetric, as Theorem 4.9 guarantees, since $\delta^4(x-y)$ and $-\Box_x$ are each symmetric under $x \leftrightarrow y$ when paired between two copies of η. For the

free field $V = \frac{1}{2}m^2\phi^2$ it is the field-independent operator $\mathcal{K} = -(\Box + m^2)$, the inverse propagator whose Gaussian integral is computed next; for an interacting potential it acquires the background-dependent piece $-V''(\phi)$ evaluated on the configuration about which we expand.

The computation just completed is the bridge to the next section. We have shown that expanding any local action to second order about a configuration ϕ produces, as the leading nontrivial term, the quadratic form $\frac{1}{2}\int \eta\, \mathcal{K}_\phi\, \eta$ built from the symmetric second-variation operator $\mathcal{K}_\phi = -\Box - V''(\phi)$ of (4.45). When ϕ is a stationary configuration — a solution of the Euler–Lagrange equation (4.36), so that the first-order term vanishes — this Gaussian integrand is precisely what controls the leading fluctuations about the classical solution. The next section develops the one integral that can be done exactly, the multidimensional Gaussian $\int \mathrm{d}^n\eta\, \exp(-\frac{1}{2}\eta^{\mathsf{T}}\mathcal{K}\eta)$, and so learns to integrate exactly the quadratic form $\mathcal{K}_\phi$ we have just constructed — the operation on which every usable formula of quantum field theory is a formal limit.

4.2 Gaussian Integration in Finite Dimensions

The previous section closed by handing us a quadratic form. Expanding an action functional about a stationary configuration ϕ_{cl}, the functional Taylor expansion of Section 4.1 gave

$$S[\phi_{\mathrm{cl}} + \eta] = S[\phi_{\mathrm{cl}}] + \frac{1}{2}\iint d^dx\, d^dy\, \eta(x)\, \frac{\delta^2 S}{\delta\phi(x)\,\delta\phi(y)}\bigg|_{\phi_{\mathrm{cl}}} \eta(y) + \cdots , \tag{4.46}$$

with the linear term absent precisely because ϕ_{cl} solves the Euler–Lagrange equation. The object that controls the leading behaviour of the theory near a classical solution is therefore the *second variation*: a symmetric bilinear form on the space of fluctuations η. Every program that follows in this book—the path integral, perturbation theory, the loop expansion, the effective action—rests on giving meaning to a "sum over fluctuations" weighted by the exponential of such a form, $\exp(-\frac{1}{2}\,\eta\cdot K\eta)$ in Euclidean signature or $\exp(\frac{i}{2}\,\eta\cdot K\eta)$ in Lorentzian. That sum is an integral over infinitely many variables, and there is exactly one such integral that mathematics permits us to do in closed form: the Gaussian. It is no exaggeration that every usable formula in quantum field theory is, in the end, a Gaussian integral—either an exact one, or one treated by expanding a non-Gaussian weight around a Gaussian and integrating term by term.

For this reason we now stop and develop the Gaussian integral in finite dimensions to complete rigor, because the field-theoretic object of the next section will be *defined* as a limit of these. We work throughout with a real symmetric *positive-definite* matrix A, i.e. a quadratic form $x^{\mathsf{T}}Ax$ that is strictly positive for $x \neq 0$; this is the Euclidean case, where the integrals converge

absolutely as ordinary Riemann integrals and no regularization is needed. The oscillatory Lorentzian weight, where A is merely real symmetric and the integral converges only conditionally, is handled by the $i\epsilon$ prescription and analytic continuation developed in Chapter 5; we flag the distinction where it matters but do not need it here. We assume the linear algebra of symmetric positive-definite matrices—orthogonal diagonalization, the relation between the determinant and the product of eigenvalues, and that $A > 0$ has a unique positive inverse A^{-1}—as a prerequisite.

4.2.1 The one-dimensional Gaussian

Everything multidimensional is bootstrapped from a single one-dimensional integral, so we begin there and carry the constant precisely.

Lemma 4.13 (The basic Gaussian integral)**.** *For any real $a > 0$ and any real j,*

$$\int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2}ax^2+jx} \;=\; \sqrt{\frac{2\pi}{a}}\; e^{j^2/2a}. \tag{4.47}$$

Proof. Take first $j = 0$ and call the integral $I = \int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2}ax^2}$. The integrand is positive and decays faster than any power, so I is a finite positive number, and we may evaluate I^2 as a two-dimensional integral over the plane and pass to polar coordinates:

$$\begin{aligned} I^2 &= \int_{-\infty}^{\infty} dx \int_{-\infty}^{\infty} dy\; e^{-\frac{1}{2}a(x^2+y^2)} = \int_0^{2\pi} d\theta \int_0^{\infty} r\, dr\; e^{-\frac{1}{2}ar^2} \\ &= 2\pi \int_0^{\infty} r\, dr\; e^{-\frac{1}{2}ar^2} = 2\pi \left[-\frac{1}{a}\, e^{-\frac{1}{2}ar^2} \right]_0^{\infty} = \frac{2\pi}{a}. \end{aligned} \tag{4.48}$$

The substitution $u = \frac{1}{2}ar^2$, $du = a\, r\, dr$ makes the radial integral elementary, as shown. Since $I > 0$ we conclude $I = \sqrt{2\pi/a}$.

For general j complete the square in the exponent,

$$-\frac{1}{2}ax^2 + jx = -\frac{a}{2}\left(x - \frac{j}{a}\right)^2 + \frac{j^2}{2a}, \tag{4.49}$$

and translate the integration variable $x \to x + j/a$, which leaves the measure dx and the infinite limits unchanged. The integral factorizes into the constant $e^{j^2/2a}$ times the $j = 0$ integral $\sqrt{2\pi/a}$, giving (4.47). □

The exponent $j^2/2a$ is the first appearance of the structure that will dominate the whole section: the value of a sourced Gaussian is the exponential of a quadratic in the source, with the coefficient $1/a$ being the *inverse* of the quadratic form. We now lift this to n variables.

4.2.2 The multidimensional Gaussian and the determinant prefactor

Let A be a real symmetric $n\times n$ matrix with $A>0$, and write $x=(x_1,\dots,x_n)^{\mathsf T}\in\mathbb{R}^n$, $x^{\mathsf T}Ax=\sum_{i,j}x_iA_{ij}x_j$, and $d^nx=dx_1\cdots dx_n$.

Proposition 4.14 (Source-free n-dimensional Gaussian)**.** *For real symmetric $A>0$,*

$$\int_{\mathbb{R}^n} d^nx\, e^{-\frac{1}{2}x^{\mathsf T}Ax} \;=\; \frac{(2\pi)^{n/2}}{\sqrt{\det A}}\,. \tag{4.50}$$

Proof. Because A is real and symmetric, the spectral theorem provides an orthogonal matrix O ($O^{\mathsf T}O=\mathbf{1}$, hence $|\det O|=1$) that diagonalizes it:

$$A=O^{\mathsf T}D\,O,\qquad D=\operatorname{diag}(a_1,\dots,a_n), \tag{4.51}$$

and positive-definiteness means every eigenvalue $a_i>0$. Change variables to the principal axes by $y=Ox$, equivalently $x=O^{\mathsf T}y$. The Jacobian of a linear change of variables is the absolute value of its determinant, $d^nx=|\det O^{\mathsf T}|\,d^ny=d^ny$, and the domain $\mathbb{R}^n$ is preserved. In the new variables the quadratic form is diagonal,

$$x^{\mathsf T}Ax=(O^{\mathsf T}y)^{\mathsf T}\,O^{\mathsf T}DO\,(O^{\mathsf T}y)=y^{\mathsf T}(OO^{\mathsf T})D(OO^{\mathsf T})y=y^{\mathsf T}Dy=\sum_{i=1}^{n}a_i\,y_i^2, \tag{4.52}$$

using $OO^{\mathsf T}=\mathbf{1}$. The integral therefore factorizes into n independent one-dimensional integrals, each evaluated by Theorem 4.13 at $j=0$:

$$\int_{\mathbb{R}^n} d^nx\, e^{-\frac{1}{2}x^{\mathsf T}Ax}=\prod_{i=1}^{n}\int_{-\infty}^{\infty}dy_i\, e^{-\frac{1}{2}a_iy_i^2}=\prod_{i=1}^{n}\sqrt{\frac{2\pi}{a_i}}=\frac{(2\pi)^{n/2}}{\sqrt{\prod_i a_i}}. \tag{4.53}$$

Finally $\det A=\det(O^{\mathsf T}DO)=\det D=\prod_i a_i$ because $\det O^{\mathsf T}\det O=1$, so $\prod_i a_i=\det A$ and (4.50) follows. □

The result deserves a pause. The integral has produced a pure number times $(\det A)^{-1/2}$. In field theory A becomes a differential operator—typically the operator K appearing in the second variation (4.46)—and this prefactor becomes its *functional determinant*, the source of the entire one-loop contribution to the effective action. The single inverse square root of a determinant computed here is, quite literally, the seed of every one-loop calculation in the book.

4.2.3 The source term and completing the square

The structurally decisive formula adds a linear source $J\in\mathbb{R}^n$. It is the finite-dimensional ancestor of the generating functional $Z[J]$.

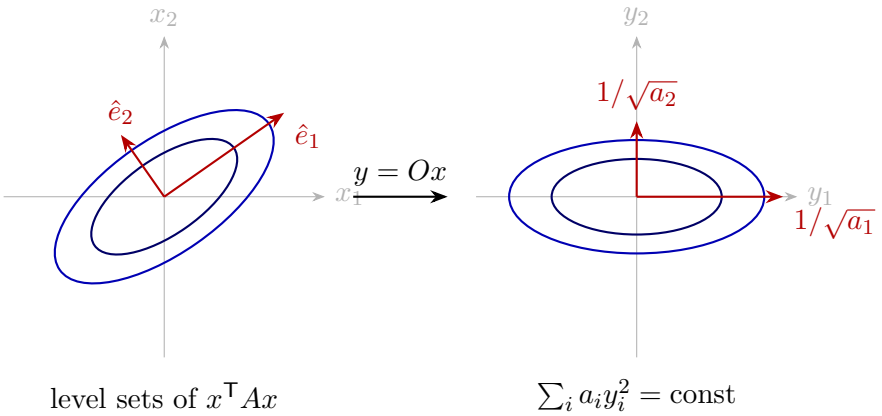


Figure 4.2: The orthogonal change of variables $y = Ox$ underlying Theorem 4.14. The level sets of $x^{\mathsf{T}}Ax$ are ellipsoids; rotating to the principal axes $\hat{e}_i$ (eigenvectors of A) makes the form a sum of squares with weights a_i (the eigenvalues), along which the Gaussian factorizes. The rotation preserves the volume element, $d^n x = d^n y$, so the only surviving geometric data is $\prod_i a_i = \det A$.

Theorem 4.15 (The sourced Gaussian integral)**.** *For real symmetric $A > 0$ and any $J \in \mathbb{R}^n$,*

$$Z(J) \;\equiv\; \int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2}x^{\mathsf{T}}Ax + J^{\mathsf{T}}x} \;=\; \frac{(2\pi)^{n/2}}{\sqrt{\det A}}\, \exp\Big(\tfrac{1}{2}\, J^{\mathsf{T}} A^{-1} J\Big). \tag{4.54}$$

Proof. Complete the square. Since A is invertible we may shift $x = y + A^{-1}J$, a translation that leaves $d^n x = d^n y$ and the domain unchanged. Using $A^{\mathsf{T}} = A$ and $(A^{-1})^{\mathsf{T}} = A^{-1}$, expand

$$\begin{aligned} -\frac{1}{2}x^{\mathsf{T}}Ax + J^{\mathsf{T}}x &= -\frac{1}{2}(y + A^{-1}J)^{\mathsf{T}}A(y + A^{-1}J) + J^{\mathsf{T}}(y + A^{-1}J) \\ &= -\frac{1}{2}\Big(y^{\mathsf{T}}Ay + y^{\mathsf{T}}J + J^{\mathsf{T}}y + J^{\mathsf{T}}A^{-1}J\Big) + J^{\mathsf{T}}y + J^{\mathsf{T}}A^{-1}J, \end{aligned} \tag{4.55}$$

where we used $A\,A^{-1}J = J$ and $(A^{-1}J)^{\mathsf{T}}A = J^{\mathsf{T}}A^{-1}A = J^{\mathsf{T}}$. Each of $y^{\mathsf{T}}J$ and $J^{\mathsf{T}}y$ is the same scalar, so $-\frac{1}{2}y^{\mathsf{T}}J - \frac{1}{2}J^{\mathsf{T}}y = -J^{\mathsf{T}}y$, which cancels the $+J^{\mathsf{T}}y$; and $-\frac{1}{2}J^{\mathsf{T}}A^{-1}J + J^{\mathsf{T}}A^{-1}J = +\frac{1}{2}J^{\mathsf{T}}A^{-1}J$. The exponent collapses to

$$-\frac{1}{2}x^{\mathsf{T}}Ax + J^{\mathsf{T}}x = -\frac{1}{2}y^{\mathsf{T}}Ay + \frac{1}{2}J^{\mathsf{T}}A^{-1}J. \tag{4.56}$$

The second term is x-independent and comes out of the integral; the first reproduces the source-free integral (4.50). Hence $Z(J) = e^{\frac{1}{2}J^{\mathsf{T}}A^{-1}J}\,(2\pi)^{n/2}(\det A)^{-1/2}$, which is (4.54). □

Two features of (4.54) will be used relentlessly. First, the J-dependence is entirely in the exponential of a quadratic form whose kernel is A^{-1}, *not* A: the object that appears in correlators is always the inverse of the operator in the action. Second, the J-independent prefactor $Z(0) = (2\pi)^{n/2}(\det A)^{-1/2}$ drops out of every normalized average, so it is convenient to factor it off once and for all.

Definition 4.16 (Gaussian average)**.** For a polynomially-bounded function $f : \mathbb{R}^n \to \mathbb{C}$ define its *Gaussian average* with respect to $A > 0$ by

$$\langle f \rangle \;\equiv\; \frac{\displaystyle\int_{\mathbb{R}^n} d^n x\; f(x)\, e^{-\frac{1}{2}x^{\mathsf{T}}Ax}}{\displaystyle\int_{\mathbb{R}^n} d^n x\; e^{-\frac{1}{2}x^{\mathsf{T}}Ax}}\,. \tag{4.57}$$

Dividing (4.54) by $Z(0)$ gives the *normalized generating function*

$$\frac{Z(J)}{Z(0)} \;=\; \langle e^{J^{\mathsf{T}}x} \rangle \;=\; \exp\!\Big(\tfrac{1}{2}\, J^{\mathsf{T}} A^{-1} J\Big), \tag{4.58}$$

the central identity from which all moments will be read off by differentiation.

4.2.4 The complex Gaussian

Charged fields, and later (by formal analogy) fermions, are integrated against a *Hermitian* quadratic form in complex variables. The result differs from the real case in a way worth isolating now: the determinant appears to the first inverse power, not the inverse square root.

Let $z = (z_1, \dots, z_n)^{\mathsf{T}} \in \mathbb{C}^n$ with conjugate $\bar z$ and $z^\dagger = \bar z^{\mathsf{T}}$, and let A be a Hermitian positive-definite matrix, $A^\dagger = A > 0$ (real positive eigenvalues). We integrate over the underlying real plane with the measure

$$d^{2n}z \;\equiv\; \prod_{i=1}^{n} \frac{d(\operatorname{Re} z_i)\, d(\operatorname{Im} z_i)}{\pi}, \tag{4.59}$$

the factor $1/\pi$ per complex variable chosen so that the normalization comes out clean.

Theorem 4.17 (The complex (Hermitian) Gaussian)**.** *For Hermitian $A > 0$ and any $w \in \mathbb{C}^n$,*

$$\int d^{2n}z\; e^{-z^\dagger A z + w^\dagger z + z^\dagger w} \;=\; \frac{1}{\det A}\; \exp(w^\dagger A^{-1} w). \tag{4.60}$$

In particular the source-free integral equals $(\det A)^{-1}$.

Proof. Take first $w = 0$. Diagonalize the Hermitian matrix by a unitary U ($U^\dagger U = \mathbf{1}$), $A = U^\dagger D U$ with $D = \mathrm{diag}(a_1, \dots, a_n)$ and each $a_i > 0$, and change variables $z = U^\dagger \zeta$. As a real-linear map on $\mathbb{R}^{2n}$, multiplication by the unitary $U^\dagger$ has Jacobian determinant $|\det U^\dagger|^2 = 1$ (each complex eigenvalue $e^{i\theta}$ of a unitary contributes a 2×2 rotation block of unit determinant), so the measure (4.59) is invariant, $d^{2n}z = d^{2n}\zeta$. The form becomes diagonal, $z^\dagger A z = \zeta^\dagger D \zeta = \sum_i a_i |\zeta_i|^2$, and the integral factorizes. Writing $\zeta_i = u_i + i v_i$, each factor is a real two-dimensional Gaussian,

$$\int \frac{du_i\, dv_i}{\pi}\, e^{-a_i(u_i^2+v_i^2)} = \frac{1}{\pi} \cdot \frac{\pi}{a_i} = \frac{1}{a_i}, \tag{4.61}$$

using (4.48) with $a \to 2a_i$ in each of the two real directions (or directly the polar evaluation). The product gives $\prod_i a_i^{-1} = (\det D)^{-1} = (\det A)^{-1}$, since $\det A = \det D$ for a unitary similarity. This proves the $w = 0$ case.

For general w, complete the square exactly as in the real case but now respecting Hermiticity. Shift $z = \zeta + A^{-1}w$, so that $z^\dagger = \zeta^\dagger + w^\dagger A^{-1}$ because A^{-1} is Hermitian. Then

$$\begin{aligned} -z^\dagger A z + w^\dagger z + z^\dagger w &= -(\zeta^\dagger + w^\dagger A^{-1}) A (\zeta + A^{-1}w) + w^\dagger(\zeta + A^{-1}w) + (\zeta^\dagger + w^\dagger A^{-1})w \\ &= -\zeta^\dagger A \zeta - \zeta^\dagger w - w^\dagger \zeta - w^\dagger A^{-1} w + w^\dagger \zeta + w^\dagger A^{-1} w + \zeta^\dagger w + w^\dagger A^{-1} w \\ &= -\zeta^\dagger A \zeta + w^\dagger A^{-1} w, \end{aligned} \tag{4.62}$$

where the cross terms $\zeta^\dagger w$ and $w^\dagger \zeta$ cancel pairwise. The constant $w^\dagger A^{-1} w$ exits the integral and the remaining ζ-integral is the $w = 0$ result, giving (4.60). □

Notation Watch

The contrast between $(\det A)^{-1/2}$ for one real field (Theorem 4.15) and $(\det A)^{-1}$ for one complex field (Theorem 4.17) is exactly the statement that a complex field is "two real fields," so its determinant prefactor is squared relative to a single real degree of freedom. The same Hermitian-form integral, but with the integration variables promoted to anticommuting Grassmann numbers, will instead yield $\det A$ in the numerator—the inverse of the bosonic answer. That sign in the exponent of the determinant is the bookkeeping origin of the closed fermion loop's minus sign; we build Grassmann/Berezin integration from scratch and prove the $+\det A$ rule in Chapter 12, and merely note the analogy here.

4.2.5 Moments and Wick's theorem

We now harvest the moments of the real Gaussian. The two-point function comes out first, and most cleanly, directly from the definition.

Proposition 4.18 (The two-point function is the inverse form)**.** *For the Gaussian average* (4.57) *with respect to* $A > 0$*,*

$$\langle x_i \rangle = 0, \qquad \langle x_i\, x_j \rangle = (A^{-1})_{ij}. \tag{4.63}$$

Proof. We use the only fact about the measure we shall ever need: it is a total derivative when differentiated, so boundary terms vanish. Concretely, for any polynomially-bounded $g(x)$ the Gaussian weight $\rho(x) = e^{-\frac{1}{2}x^{\mathsf{T}}Ax}$ decays faster than any polynomial grows, so

$$\int_{\mathbb{R}^n} d^n x\, \frac{\partial}{\partial x_\ell}\Big[\, g(x)\, \rho(x) \,\Big] = 0, \tag{4.64}$$

the integral of an exact derivative over all of $\mathbb{R}^n$. Since $\partial_{x_\ell}\rho = -(Ax)_\ell\, \rho = -\sum_p A_{\ell p} x_p\, \rho$, expanding (4.64) gives the *Schwinger–Dyson relation*

$$\Big\langle \frac{\partial g}{\partial x_\ell} \Big\rangle = \sum_p A_{\ell p}\, \langle x_p\, g \rangle. \tag{4.65}$$

Take $g = 1$: the left side vanishes, so $\sum_p A_{\ell p}\langle x_p\rangle = 0$ for every ℓ; as A is invertible, $\langle x_p\rangle = 0$. Take $g = x_j$: the left side is $\langle \delta_{\ell j}\rangle = \delta_{\ell j}$, so $\delta_{\ell j} = \sum_p A_{\ell p}\langle x_p x_j\rangle$, i.e. $A\,\langle x x_j\rangle = e_j$ componentwise. Multiplying by A^{-1} yields $\langle x_i x_j\rangle = (A^{-1})_{ij}$. □

This is the result the whole section was built to reach: *the two-point function is the inverse of the quadratic form in the action.* The matrix A^{-1} is the finite-dimensional propagator. With this in hand, all higher moments are fixed combinatorially.

Theorem 4.19 (Wick's theorem in finite dimensions)**.** *Let* $A > 0$ *be real symmetric and write* $G \equiv A^{-1}$ *for the propagator. Then all odd moments vanish, and the* $2k$*-point moment is the sum over perfect pairings of its indices:*

$$\langle x_{i_1} x_{i_2} \cdots x_{i_{2k}} \rangle \;=\; \sum_{\text{pairings } P}\; \prod_{(a,b)\in P} G_{i_a i_b} \;=\; \sum_{\text{pairings } P}\; \prod_{(a,b)\in P} (A^{-1})_{i_a i_b}, \tag{4.66}$$

where a pairing *(perfect matching) of* $\{1, 2, \ldots, 2k\}$ *is a partition into* k *unordered pairs, and the number of such pairings is* $(2k-1)!! = (2k)!/(2^k k!)$*.*

Proof. We give the generating-function proof, which makes the combinatorics explicit, and then note the equivalent inductive proof from (4.65).

Differentiate the normalized generating function (4.58). Each factor of x_i inside the average is produced by a derivative with respect to the source:

$$\frac{\partial}{\partial J_i}\, \langle e^{J^{\mathsf{T}}x} \rangle = \langle x_i\, e^{J^{\mathsf{T}}x} \rangle, \tag{4.67}$$

the interchange of derivative and integral being justified by the uniform Gaussian decay. Applying m such derivatives and then setting $J = 0$ removes the exponential and leaves the moment,

$$\langle x_{i_1} \cdots x_{i_m} \rangle = \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_m}} \exp\Big(\tfrac{1}{2}\, J^{\mathsf{T}} A^{-1} J\Big)\Big|_{J=0}. \tag{4.68}$$

Write $Q(J) = \frac{1}{2}\sum_{a,b} J_a G_{ab} J_b$ with $G = A^{-1}$ symmetric, and expand $e^{Q} = \sum_{r\geq 0} Q^r/r!$. Each power Q^r is a homogeneous polynomial of degree $2r$ in J. A polynomial term of degree D has a nonvanishing m-fold derivative at $J = 0$ only when $D = m$: if $D < m$ some derivative kills it, and if $D > m$ a factor of J survives and is set to zero. Hence:

Odd moments vanish. If m is odd, no power Q^r has degree m (all degrees $2r$ are even), so (4.68) is zero.

Even moments. Set $m = 2k$; only the term $r = k$ survives,

$$Q^k = \frac{1}{2^k}\Big(\sum_{a,b} J_a G_{ab} J_b\Big)^k = \frac{1}{2^k} \sum_{a_1,b_1,\ldots,a_k,b_k} \Big(\prod_{s=1}^{k} G_{a_s b_s}\Big)\, J_{a_1} J_{b_1} \cdots J_{a_k} J_{b_k}, \tag{4.69}$$

so that

$$\langle x_{i_1} \cdots x_{i_{2k}} \rangle = \frac{1}{2^k k!} \sum_{a_1,b_1,\ldots,a_k,b_k} \Big(\prod_{s=1}^{k} G_{a_s b_s}\Big) \frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_{2k}}} (J_{a_1} J_{b_1} \cdots J_{a_k} J_{b_k}). \tag{4.70}$$

The derivative of a product of $2k$ first-order monomials $J_{c_1} \cdots J_{c_{2k}}$ (here the c's are the list $a_1, b_1, \ldots, a_k, b_k$) by $2k$ distinct derivatives is

$$\frac{\partial}{\partial J_{i_1}} \cdots \frac{\partial}{\partial J_{i_{2k}}} (J_{c_1} \cdots J_{c_{2k}}) = \sum_{\sigma \in S_{2k}} \prod_{t=1}^{2k} \delta_{i_t,\, c_{\sigma(t)}}, \tag{4.71}$$

the sum over all $(2k)!$ bijections matching the derivative slots i_t to the factor slots $c_{\sigma(t)}$: each derivative removes exactly one J, and every assignment of which derivative removes which factor contributes one term.

Inserting (4.71) into (4.70) and performing the free sums over the dummy indices a_s, b_s, each permutation σ assigns the external indices $i_1, \ldots, i_{2k}$ to the $2k$ slots of the k propagator factors $G_{a_s b_s}$, producing a product of k factors of G, one for each pair of slots (a_s, b_s). Thus the result is a sum over the $(2k)!$ permutations of $\prod_s G_{(\text{index in slot } a_s)(\text{index in slot } b_s)}$. This overcounts the distinct *pairings* of $\{i_1, \ldots, i_{2k}\}$: a given pairing arises from

- the $k!$ orderings of which pair occupies which factor s, and
- the 2^k swaps $a_s \leftrightarrow b_s$ within each factor, all equal because G is symmetric, $G_{ab} = G_{ba}$,

giving $2^k k!$ permutations per pairing. These $2^k k!$ copies are exactly cancelled by the prefactor $1/(2^k k!)$ in (4.70). What remains is one term per distinct pairing, $\prod_{(a,b)\in P} G_{i_a i_b}$, summed over all pairings P, which is (4.66). The count of pairings is the number of permutations modulo this symmetry, $(2k)!/(2^k k!) = (2k-1)!!$. □

Remark 4.20 (The inductive form: Wick's recursion). The same theorem follows from the Schwinger–Dyson relation (4.65) by induction on the number of insertions, a route worth recording because it is exactly how Wick's theorem is organized for fields in Chapter 10. Put $g = x_{i_2}\cdots x_{i_m}$ in (4.65). Its derivative by x_ℓ hits each factor in turn, $\partial_{x_\ell} g = \sum_{r=2}^{m} \delta_{\ell,i_r}\, x_{i_2}\cdots \widehat{x_{i_r}} \cdots x_{i_m}$ (the hat denoting omission), so

$$\sum_{r=2}^{m} \delta_{\ell,i_r} \,\langle x_{i_2}\cdots \widehat{x_{i_r}} \cdots x_{i_m}\rangle = \sum_{p} A_{\ell p}\,\langle x_p\, x_{i_2}\cdots x_{i_m}\rangle. \tag{4.72}$$

Contracting with $(A^{-1})_{i_1\ell}$ and summing over ℓ turns the right side into $\langle x_{i_1} x_{i_2}\cdots x_{i_m}\rangle$ and gives the recursion

$$\langle x_{i_1} x_{i_2}\cdots x_{i_m}\rangle = \sum_{r=2}^{m} (A^{-1})_{i_1 i_r}\,\langle x_{i_2}\cdots \widehat{x_{i_r}} \cdots x_{i_m}\rangle. \tag{4.73}$$

In words: the first variable x_{i_1} must be contracted with one of the remaining $m-1$ variables, contributing a propagator $(A^{-1})_{i_1 i_r}$ times the average of what is left. Starting from $\langle 1\rangle = 1$ and $\langle x_i\rangle = 0$, this recursion manifestly generates the sum over all pairings: it pairs i_1 with every partner and recurses on the rest, and it forces every odd moment to zero (an odd recursion always bottoms out on a single unpaired $\langle x_i\rangle = 0$). For $m = 2$ it reproduces $\langle x_{i_1} x_{i_2}\rangle = (A^{-1})_{i_1 i_2}$ of Theorem 4.18.

It is worth writing the lowest nontrivial case in full, since its pattern is the one that recurs for fields. The four-point function has $(4-1)!! = 3$ pairings:

$$\langle x_i x_j x_k x_l\rangle = (A^{-1})_{ij}(A^{-1})_{kl} + (A^{-1})_{ik}(A^{-1})_{jl} + (A^{-1})_{il}(A^{-1})_{jk}. \tag{4.74}$$

Physical Insight

Wick's theorem says that a Gaussian random vector has *no genuine higher correlation*: every moment is built from the single two-point datum A^{-1} by the purely combinatorial rule of pairing. This is the precise sense in which the Gaussian is the "free" or "non-interacting" distribution—all structure resides in the propagator, and all the diagrams one will ever draw for a free field are just the perfect matchings counted here. Interactions, when we add them, are exactly the failure of (4.66): the deviation of the true

moments from their Gaussian (pairing) values.

4.2.6 The carried toy: the zero-dimensional ϕ^4 model

To see the whole apparatus at work on the model we will return to repeatedly, we set up the simplest interacting theory imaginable: a field theory in zero spacetime dimensions, where the "field" is a single real number x and the "functional integral" is one ordinary integral. This is a self-contained model computation, not the formal ϕ^4 theory of Chapter 6 onward; it is a rigorous laboratory in which every statement of the preceding subsections becomes an honest, convergent integral. The genuine free scalar propagator and Wick's theorem *for fields*—where A^{-1} becomes the Feynman propagator and the indices become spacetime points—are built in Chapter 10; here we exercise only the zero-dimensional shadow.

Worked Example 4.21 (The zero-dimensional ϕ^4 partition function)**.** Define, for a coupling $\lambda \geq 0$ and source $J \in \mathbb{R}$, the partition function

$$Z(\lambda, J) \;=\; \int_{-\infty}^{\infty} dx \;\exp\Big(-\tfrac{1}{2}x^2 - \tfrac{\lambda}{4!}x^4 + Jx\Big). \tag{4.75}$$

Convergence and status. For $\lambda > 0$ the quartic term $-\frac{\lambda}{4!}x^4$ dominates the exponent at large $|x|$ and the integrand decays like $e^{-\lambda x^4/4!}$, so the integral converges absolutely for *every* real J and defines a smooth function of J; at $\lambda = 0$ it reduces to the convergent Gaussian. (For $\lambda < 0$ the integral diverges—the would-be "vacuum" is unstable—which is the zero-dimensional face of the statement that the ϕ^4 perturbation series is asymptotic, not convergent; we set that question up in Section 4.3 and treat asymptotic series in Chapter 5.) Thus (4.75) is, in contrast to the field-theoretic integral of the next section, an ordinary integral about which everything is rigorous.

The free value. At $\lambda = 0$ this is the one-dimensional sourced Gaussian of Theorem 4.13 with $a = 1$, $j = J$:

$$Z(0, J) = \int_{-\infty}^{\infty} dx\; e^{-\frac{1}{2}x^2 + Jx} = \sqrt{2\pi}\; e^{J^2/2}, \qquad Z(0,0) = \sqrt{2\pi}. \tag{4.76}$$

The quadratic form is the 1×1 matrix $A = 1$, so the propagator is $A^{-1} = 1$. The normalized generating function is $Z(0, J)/Z(0, 0) = e^{J^2/2}$, the $n = 1$ instance of (4.58).

Free moments by Wick. With $G = A^{-1} = 1$, Theorem 4.19 gives the free moments as plain pairing counts,

$$\langle x^{2k}\rangle_0 = (2k-1)!! \cdot (A^{-1})^k = (2k-1)!!, \qquad \langle x^{2k+1}\rangle_0 = 0, \tag{4.77}$$

where $\langle\,\cdot\,\rangle_0$ is the average (4.57) at $\lambda = 0$. One can verify the first two directly from (4.76): differentiating $e^{J^2/2}$ twice and four times and setting $J = 0$ gives $\langle x^2\rangle_0 = 1$ and $\langle x^4\rangle_0 = 3$, matching $1!! = 1$ and $3!! = 3$.

Enumerating the contractions of $\langle x^4\rangle_0$. The quartic interaction will be expanded as a power series in λ, and each term is a free average of a power of x; the lowest is $\langle x^4\rangle_0$. By (4.74) with all indices equal and $A^{-1} = 1$, the three pairings of the four factors $x\,x\,x\,x$ are

$$\underbrace{(12)(34)}_{1} \;+\; \underbrace{(13)(24)}_{1} \;+\; \underbrace{(14)(23)}_{1} \;=\; 3, \tag{4.78}$$

each contributing $(A^{-1})^2 = 1$. The number $3 = 3!!$ is precisely the combinatorial weight that will reappear as the symmetry factor of the one-vertex ("figure-eight" tadpole) diagram of the ϕ^4 theory: writing the first correction to the free energy,

$$\frac{Z(\lambda,0)}{Z(0,0)} = \left\langle e^{-\frac{\lambda}{4!}x^4}\right\rangle_0 = 1 - \frac{\lambda}{4!}\,\langle x^4\rangle_0 + O(\lambda^2) = 1 - \frac{\lambda}{4!}\cdot 3 + O(\lambda^2) = 1 - \frac{\lambda}{8} + O(\lambda^2), \tag{4.79}$$

where the coefficient $3/4! = 1/8$ is the $4!$ of the vertex partly cancelled by the 3 ways of contracting—the zero-dimensional avatar of a Feynman symmetry factor. We stop the expansion here: it is the *setup* that this section owes. The status of the series (4.79) as λ-power series—convergent or merely asymptotic—and its diagrammatic organization belong to the following sections and to Chapter 15, where symmetry factors are done with full care.

We have now extracted, from one convergent integral and its sources, the complete finite-dimensional kit: the Gaussian value with its $(\det A)^{-1/2}$ prefactor (4.54), its complex/Hermitian counterpart (4.60), the normalized generating function (4.58), the identification of the two-point function with the inverse form A^{-1} (4.63), and Wick's pairing theorem (4.66). Every one of these statements was a theorem about an ordinary integral over $\mathbb{R}^n$. The next section takes the only remaining step that separates this from quantum field theory: it lets $n \to \infty$, replacing the matrix A by the differential operator of the second variation (4.46) and the sum over $\mathbb{R}^n$ by a sum over field configurations, and asks—carefully, and with the honesty the chapter's scope demands—which of these clean finite-dimensional theorems survive the limit as theorems and which survive only as disciplined formal definitions.

4.3 From Finite to Infinite: The Functional Integral and Its Status

The previous section closed by handing us a complete, fully rigorous kit of finite-dimensional theorems: the sourced Gaussian value with its $(\det A)^{-1/2}$

prefactor (4.54), the identification of the two-point function with the inverse form A^{-1} (4.63), and Wick's pairing theorem (4.66). Each was a theorem about an honest integral over $\mathbb{R}^n$. What separates that finite-dimensional calculus from quantum field theory is a single, enormous step: $n \to \infty$. A field configuration is not a finite list of numbers but a function on spacetime—an element of an infinite-dimensional space—and the "sum over configurations" weighted by e^{iS} or e^{-S_E} is an integral over that space. The purpose of this section is to take that step *explicitly*, by the only honest route: discretize spacetime so that the field becomes a finite vector and the integral becomes one of the finite-dimensional Gaussians we have already proved, and then remove the discretization. We will find that the algebra goes through cleanly and reproduces the formulas one writes in practice—but that the limit itself is a theorem in some cases and only a disciplined formal definition in others. The chapter's scope demands that we say, precisely and without bluffing, which is which. By the end we will have a field generating functional $Z[J]$, an explicit formula for the free propagator as the inverse of the quadratic operator, and an Assessment of Current Status of the standing of the symbol $\int \mathcal{D}\phi$.

4.3.1 Lattice discretization of a field on a finite box

The infinity in $\int \mathcal{D}\phi$ is of two distinct kinds, and it pays to separate them. The first is the *ultraviolet* infinity: a field has independent values at every point of a continuum, uncountably many degrees of freedom in any region. The second is the *infrared* infinity: spacetime is unbounded, so even after coarse-graining there are infinitely many regions. We tame both at once by working in a finite box with a finite mesh, leaving a genuinely finite-dimensional integral to which Section 4.2 applies verbatim.

We work in Euclidean signature throughout this construction—the case in which the weight e^{-S_E} is a decaying exponential rather than an oscillatory phase, so the finite-dimensional integrals converge absolutely and the limit has a chance of being a genuine measure. The passage from the physical Lorentzian theory to this Euclidean one is the Wick rotation $t \to -i\tau$ introduced in Chapter 5 and carried out for fields in Chapter 10; we return to the oscillatory Minkowski integral in Section 4.3.3 and treat it as what it is, a formal object made meaningful by analytic continuation from here. We carry the running example, the real scalar of ϕ^4 theory (Chapter 6), with Euclidean action

$$S_E[\phi] = \int d^d x \left[\tfrac{1}{2} (\partial_\mu \phi)(\partial_\mu \phi) + \tfrac{1}{2} m^2 \phi^2 + \tfrac{\lambda}{4!} \phi^4 \right], \tag{4.80}$$

where the index μ is now summed with the Euclidean (Kronecker) metric, all signs in the quadratic part are positive, and d is the Euclidean spacetime dimension ($d = 4$ for the physical theory, but we keep d general because the rigor verdict will depend on it).

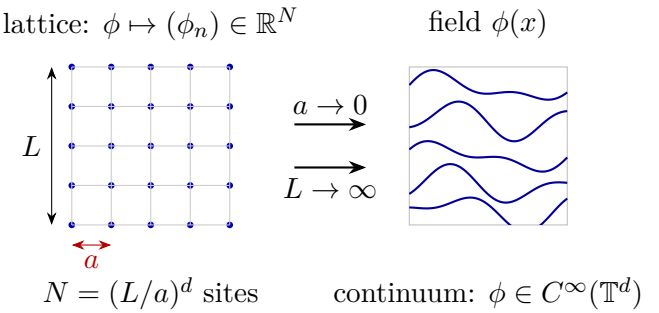


Figure 4.3: The discretize-then-continuum strategy of this section. A field on a box of side L is sampled on a mesh of spacing a, becoming a finite vector $(\phi_n) \in \mathbb{R}^N$ with $N = (L/a)^d$; the functional integral becomes the finite-dimensional Gaussian of Section 4.2. The continuum field theory is the limit $a \to 0$ (ultraviolet, infinitely many degrees of freedom per unit volume) and $L \to \infty$ (infrared, infinite volume). Whether this limit exists as a *theorem* or only as a *formal* prescription is the subject of Section 4.3.4.

The box and the mesh. Replace $\mathbb{R}^d$ by a cubic box of side L with periodic boundary conditions—topologically a torus $\mathbb{T}^d$, which removes boundary effects and makes Fourier analysis a finite sum. Introduce a lattice spacing a dividing L, so that the sites are

$$x_n = a\,n, \qquad n \in \Lambda \equiv \{0, 1, \dots, \mathcal{N}-1\}^d, \qquad \mathcal{N} = \frac{L}{a}, \tag{4.81}$$

with site indices read modulo $\mathcal{N}$ in each direction (periodicity). There are

$$N = \mathcal{N}^d = (L/a)^d \tag{4.82}$$

sites in all, a finite number. A field configuration is now specified by its values at the sites, i.e. by the finite vector

$$\phi \;\longmapsto\; (\phi_n)_{n\in\Lambda} \in \mathbb{R}^N, \qquad \phi_n \equiv \phi(x_n). \tag{4.83}$$

This is the substitution "field $\to$ finite vector" that the construction demands; everything else follows by translating each piece of S_E and of the measure into this finite data.

The action becomes a quadratic-plus-interaction form. Each ingredient of (4.80) has a canonical lattice counterpart. The spacetime integral becomes a Riemann sum weighted by the cell volume,

$$\int d^d x \;\longrightarrow\; a^d \sum_{n\in\Lambda} . \tag{4.84}$$

The gradient becomes a finite difference. Writing $\hat{\mu}$ for the unit lattice vector in direction μ, the forward difference

$$(\partial_\mu \phi)(x_n) \;\longrightarrow\; \frac{\phi_{n+\hat{\mu}} - \phi_n}{a} \tag{4.85}$$

reproduces the derivative to $O(a)$. The potential terms are local and simply sample the field at each site. Assembling these, the lattice action is

$$S_E^{\mathrm{lat}}[\phi] = a^d \sum_{n \in \Lambda} \left[\frac{1}{2} \sum_{\mu=1}^{d} \Big(\frac{\phi_{n+\hat{\mu}} - \phi_n}{a} \Big)^2 + \frac{1}{2} m^2 \phi_n^2 + \frac{\lambda}{4!} \phi_n^4 \right]. \tag{4.86}$$

The first two terms are quadratic in the vector (ϕ_n) and the last is the interaction; this is exactly the "quadratic + interaction form" structure of the finite-dimensional theory. To expose the quadratic form, expand the square and use periodicity to relabel the shifted sums (a discrete summation by parts: the cross term $\sum_n \phi_n \phi_{n+\hat{\mu}}$ is unchanged under $n \to n - \hat{\mu}$ because the sum runs over the whole torus). Then

$$\begin{aligned} a^d \sum_n \frac{1}{2} \sum_\mu \frac{(\phi_{n+\hat{\mu}} - \phi_n)^2}{a^2} &= \frac{a^d}{2a^2} \sum_n \sum_\mu \left(\phi_{n+\hat{\mu}}^2 + \phi_n^2 - 2\phi_n \phi_{n+\hat{\mu}} \right) \\ &= -\frac{a^d}{2a^2} \sum_n \sum_\mu \phi_n \left(\phi_{n+\hat{\mu}} + \phi_{n-\hat{\mu}} - 2\phi_n \right) \\ &= -\frac{a^d}{2} \sum_n \phi_n \, (\Delta_{\mathrm{lat}} \phi)_n, \end{aligned} \tag{4.87}$$

where the *lattice Laplacian* is

$$(\Delta_{\mathrm{lat}} \phi)_n \equiv \frac{1}{a^2} \sum_{\mu=1}^{d} \left(\phi_{n+\hat{\mu}} + \phi_{n-\hat{\mu}} - 2\phi_n \right). \tag{4.88}$$

Therefore the quadratic part of (4.86) is $\frac{1}{2} \sum_{n,n'} \phi_n K_{nn'} \phi_{n'}$ with the real symmetric $N \times N$ matrix

$$K_{nn'} = a^d \left[(-\Delta_{\mathrm{lat}} + m^2) \right]_{nn'}, \tag{4.89}$$

which is precisely the discretized quadratic operator $-\Delta + m^2$ of the second variation (4.46), dressed by the cell volume a^d. It is positive definite: for any nonzero ϕ, $\sum_{n,n'} \phi_n K_{nn'} \phi_{n'} = a^d \sum_n \left[\frac{1}{2} \sum_\mu \big(\frac{\phi_{n+\hat{\mu}} - \phi_n}{a} \big)^2 \cdot 2 + m^2 \phi_n^2 \right] \geq a^d m^2 \sum_n \phi_n^2 > 0$, so $K > 0$ and all of Section 4.2 applies.

The measure becomes a product of Lebesgue measures. The remaining ingredient is the "sum over configurations." On the lattice this is the ordinary Lebesgue integral over the value at each site,

$$\mathcal{D}\phi \;\longrightarrow\; \prod_{n \in \Lambda} d\phi_n, \tag{4.90}$$

a product of N copies of $d\phi$, i.e. Lebesgue measure on $\mathbb{R}^N$. The source term $\int d^dx\, J\phi$ becomes $a^d \sum_n J_n\phi_n$ with $J_n \equiv J(x_n)$. We have arrived at a completely well-defined finite-dimensional integral.

Definition 4.22 (The regularized (lattice) functional integral)**.** The *lattice-regularized generating functional* of the scalar theory on the box $\mathbb{T}_L^d$ with spacing a is the ordinary N-dimensional integral

$$Z_a^L[J] \;\equiv\; \int_{\mathbb{R}^N} \Big(\prod_{n\in\Lambda} d\phi_n\Big)\, \exp\Big(-S_E^{\text{lat}}[\phi] + a^d \sum_n J_n\phi_n\Big). \tag{4.91}$$

The (still purely formal) continuum functional integral is the symbol

$$Z[J] \;\equiv\; \int \mathcal{D}\phi\; e^{-S_E[\phi] + \int d^dx\, J\phi}, \tag{4.92}$$

whose precise meaning—when it has one—is the double limit $\lim_{a\to 0,\, L\to\infty} Z_a^L[J]$ (after the field- and coupling-renormalizations of Chapter 22 are inserted), and whose meaning otherwise is exactly (4.91) at fixed regulators.

Physical Insight

The lattice is not an approximation scheme bolted on after the fact; for the interacting continuum theory it is, as far as anyone can prove, the *definition*. The symbol $\int \mathcal{D}\phi$ has no content independent of some regulator that renders it a finite-dimensional integral. When we manipulate $\int \mathcal{D}\phi\, e^{-S}$ in later chapters as though it were an integral—shifting variables, completing the square, differentiating under the sign—each such move is legitimate because it is a move on (4.91), where it is an ordinary theorem of calculus, that survives the limit. The honest reading of every path integral in this book is "the $a \to 0$, $L \to \infty$ limit of this lattice expression, when it exists."

4.3.2 The free generating functional and the continuum propagator

Set $\lambda = 0$. Then $S_E^{\text{lat}} = \frac{1}{2}\phi^{\mathsf{T}} K \phi$ is purely quadratic and (4.91) is *exactly* the sourced Gaussian of Theorem 4.15 with matrix $A = K$ and source vector $\tilde{J}_n = a^d J_n$ (the cell volume is part of the source because the source term carries the measure factor a^d). The theorem applies with no modification and gives, at once,

$$Z_{a,0}^L[J] = \frac{(2\pi)^{N/2}}{\sqrt{\det K}}\, \exp\Big(\tfrac{1}{2}\, \tilde{J}^{\mathsf{T}} K^{-1} \tilde{J}\Big) = Z_{a,0}^L[0]\, \exp\Big(\tfrac{1}{2} \sum_{n,n'} a^d J_n\, (K^{-1})_{nn'}\, a^d J_{n'}\Big). \tag{4.93}$$

This is the central point of fidelity: *the discretized free generating functional is literally the finite-dimensional Gaussian result.* Nothing has been added or guessed; we have only renamed the variables of Theorem 4.15. By Theorem 4.18 the lattice two-point function is the inverse matrix,

$$\langle \phi_n \, \phi_{n'} \rangle = (K^{-1})_{nn'}, \tag{4.94}$$

and all higher correlators are its Wick pairings (4.66). Everything now reduces to one computation: the continuum limit of the matrix K^{-1}. We carry it in full, because the result—that the continuum propagator is the kernel of the inverse of the quadratic operator—is the single most-used fact in the rest of the book.

Diagonalizing by the discrete Fourier transform. The lattice operator $-\Delta_{\text{lat}} + m^2$ is translation invariant on the torus and is therefore diagonalized by plane waves. The momenta compatible with the period-L boundary condition are quantized, and the spacing a truncates them to a *Brillouin zone*:

$$p_\mu = \frac{2\pi}{L}\, k_\mu, \qquad k_\mu \in \Big\{ -\tfrac{\mathcal{N}}{2}, \dots, \tfrac{\mathcal{N}}{2} - 1 \Big\}, \qquad \text{so} \qquad p_\mu \in \Big[-\frac{\pi}{a}, \frac{\pi}{a} \Big), \tag{4.95}$$

exactly N momenta in all—as many as there are sites, as it must be. The normalized plane waves $u_n^{(p)} = N^{-1/2} e^{ip\cdot x_n}$ form an orthonormal basis of $\mathbb{C}^N$. Acting on one of them, the lattice Laplacian (4.88) returns the same wave times a number,

$$\begin{aligned} (-\Delta_{\text{lat}} u^{(p)})_n &= -\frac{1}{a^2} \sum_\mu \big(e^{ip_\mu a} + e^{-ip_\mu a} - 2 \big) u_n^{(p)} = \frac{1}{a^2} \sum_\mu \big(2 - 2\cos p_\mu a \big)\, u_n^{(p)} \\ &= \underbrace{\Big[\sum_{\mu=1}^{d} \frac{4}{a^2} \sin^2 \frac{p_\mu a}{2} \Big]}_{\equiv\ \hat{p}^{\,2}} \, u_n^{(p)}, \end{aligned} \tag{4.96}$$

using $2 - 2\cos\theta = 4\sin^2(\theta/2)$. The quantity $\hat{p}^{\,2}$ is the *lattice momentum squared*; it agrees with $p^2 = \sum_\mu p_\mu^2$ when $|p_\mu| a \ll 1$ and bends over to a finite maximum at the zone boundary. Thus the eigenvalues of $K = a^d(-\Delta_{\text{lat}} + m^2)$ are $a^d(\hat{p}^{\,2} + m^2)$, and the inverse matrix in position space is the spectral sum

$$(K^{-1})_{nn'} = \sum_p u_n^{(p)} \, \frac{1}{a^d(\hat{p}^{\,2} + m^2)} \, (u_{n'}^{(p)})^* = \frac{1}{Na^d} \sum_p \frac{e^{ip\cdot(x_n - x_{n'})}}{\hat{p}^{\,2} + m^2}. \tag{4.97}$$

Because $Na^d = (L/a)^d a^d = L^d$, the prefactor is simply L^{-d}, the inverse volume, and the lattice propagator is

$$G_a^L(x_n, x_{n'}) \equiv (K^{-1})_{nn'} = \frac{1}{L^d} \sum_p \frac{e^{ip\cdot(x_n - x_{n'})}}{\hat{p}^{\,2} + m^2}. \tag{4.98}$$

The two limits. Take infinite volume first, $L \to \infty$ at fixed a. The momenta (4.95) have spacing $\Delta p_\mu = 2\pi/L \to 0$ and fill the Brillouin zone densely; the Riemann-sum rule $\frac{1}{L^d}\sum_p \to \int_{\mathrm{BZ}} \frac{d^d p}{(2\pi)^d}$ (each momentum cell has volume $(2\pi/L)^d$) converts the sum to an integral,

$$G_a(x-y) = \int_{-\pi/a}^{\pi/a} \frac{d^d p}{(2\pi)^d}\, \frac{e^{ip\cdot(x-y)}}{\hat{p}^{\,2} + m^2}. \tag{4.99}$$

Now take the continuum, $a \to 0$. The integrand converges pointwise: $\hat{p}^{\,2} = \sum_\mu \frac{4}{a^2}\sin^2(p_\mu a/2) \to \sum_\mu p_\mu^2 = p^2$ as $a \to 0$ (Taylor: $\sin(p_\mu a/2) = p_\mu a/2 + O(a^3)$), and the integration domain opens up to all of $\mathbb{R}^d$. The result is the *Euclidean propagator*,

$$\boxed{G(x-y) = \int \frac{d^d p}{(2\pi)^d}\, \frac{e^{ip\cdot(x-y)}}{p^2 + m^2}} \tag{4.100}$$

which is exactly the kernel of the inverse operator $(-\Delta + m^2)^{-1}$: applying $-\Delta + m^2$ in x brings down $p^2 + m^2$ inside the integral, cancelling the denominator and leaving $\int \frac{d^d p}{(2\pi)^d} e^{ip(x-y)} = \delta^d(x-y)$, i.e.

$$(-\Delta_x + m^2)\, G(x-y) = \delta^d(x-y). \tag{4.101}$$

We have therefore *derived*, by an explicit lattice-to-continuum limit, the statement that in field theory the two-point function is the inverse of the quadratic operator in the action—the infinite-dimensional incarnation of Theorem 4.18, with the matrix index i replaced by the spacetime point x and the matrix inverse replaced by the Green's function. Feeding (4.100) back into (4.93), and using $a^d \sum_n \to \int d^d x$ to turn the source pairing into a double spacetime integral, gives the free continuum generating functional

$$\frac{Z[J]}{Z[0]} = \exp\Big(\tfrac{1}{2}\iint d^d x\, d^d y\; J(x)\, G(x-y)\, J(y)\Big), \qquad G = (-\Delta + m^2)^{-1}, \tag{4.102}$$

the field-theoretic analogue of the normalized Gaussian generating function (4.58). This is the object whose functional derivatives the next section differentiates to extract correlators.

The prefactor and the functional determinant. The source-independent prefactor of (4.93) is $Z^L_{a,0}[0] = (2\pi)^{N/2}(\det K)^{-1/2}$. As $a \to 0$ the number of sites $N \to \infty$ and $\det K = \prod_p a^d(\hat{p}^{\,2} + m^2)$ runs off to a meaningless value: this is the *functional determinant* $\det(-\Delta + m^2)$, and it is divergent. The continuum symbol $(\det(-\Delta + m^2))^{-1/2}$ is therefore the most formal part of the whole construction. Two things rescue it in practice. First, the prefactor is J-independent and cancels in every normalized correlator, which is why the next section divides it out at the start. Second, physical quantities depend only on *ratios* of such

determinants—e.g. the determinant in a background field divided by the free one—and these ratios are finite after renormalization; this is the content of the one-loop effective action computed in Section 4.6 and made systematic with zeta-function or heat-kernel regularization there. We flag the divergence now and discharge it there.

4.3.3 The Minkowski integral and the oscillatory obstruction

The physical theory lives in Lorentzian signature, where the weight is the phase e^{iS} rather than the decaying e^{-S_E}. Repeating the lattice construction with the mostly-minus action $S[\phi] = \int d^dx\,[\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4]$ gives, in the free case, the quadratic form $\frac{i}{2}\phi^{\mathsf{T}}K_M\phi$ with $K_M = a^d\,[-(\Box + m^2)]$ (an integration by parts turns $\frac{i}{2}\phi(-\Box - m^2)\phi$ into iS). Formally completing the square as in the one-dimensional oscillatory integral $\int dx\, e^{\frac{i}{2}ax^2+ijx} = \sqrt{2\pi i/a}\; e^{-ij^2/2a}$—the analytic continuation $a \to ia$ of Theorem 4.13, justified by the $i\epsilon$ device of Chapter 5—one obtains

$$\frac{Z[J]}{Z[0]} = \exp\Big(-\tfrac{1}{2}\iint d^dx\, d^dy\; J(x)\, D_F(x-y)\, J(y)\Big), \tag{4.103}$$

with the Feynman propagator

$$D_F(x-y) = \int \frac{d^dp}{(2\pi)^d}\,\frac{i\,e^{-ip\cdot(x-y)}}{p^2 - m^2 + i\epsilon}, \qquad (\Box_x + m^2)\, D_F(x-y) = -i\,\delta^d(x-y). \tag{4.104}$$

The $+i\epsilon$ is not optional decoration. The naive Lorentzian quadratic operator $\Box + m^2$ is *not invertible*: it annihilates every on-shell mode $p^2 = m^2$, so it has a kernel and "$(\Box + m^2)^{-1}$" is ill-defined until the singular denominator is displaced off the real axis. The $+i\epsilon$ is the unique displacement that selects the boundary condition making D_F the *time-ordered* two-point function—the propagator that emerges automatically because the (formal) Lorentzian path integral, like its rigorous Euclidean continuation, weights configurations in a way that orders operators in time. We state this here for the free field; the systematic proof that the path integral computes vacuum expectation values of time-ordered products, and the derivation of (4.104) by Wick rotation from (4.100) ($p_E^0 = -ip^0$, $p_E^2 = -p^2 + i\epsilon$), are carried out in Chapter 10 and used in Chapter 15. The deeper point for the present section is that the Minkowski integral is the place where rigor genuinely breaks: a bounded oscillatory "measure" $e^{iS}\mathcal{D}\phi$ has infinite total variation and is not a measure at all even in finite dimensions (the integral $\int e^{iax^2}dx$ converges only conditionally, as an improper Riemann integral). The honest object is the Euclidean one of Section 4.3.2; the Minkowski formulas are its analytic continuation.

4.3.4 The rigor summary: what is a theorem and what is formal

We can now state plainly, as precision requires, the standing of the symbol $\int \mathcal{D}\phi$. There are three distinct verdicts.

(i) The Euclidean, super-renormalizable, low-dimensional case is a theorem. In low enough Euclidean dimension the double limit of Theorem 4.22 exists and defines a genuine probability measure on a space of (distributional) field configurations, with all moments finite and Euclidean-invariant. This is the achievement of *constructive quantum field theory*: the $P(\phi)_2$ theories (polynomial self-interactions of a scalar in $d = 2$) and ϕ^4_3 (the quartic scalar in $d = 3$) were constructed rigorously, the lattice (or continuum cutoff) removed, and the limiting correlation functions shown to satisfy the axioms below. In these cases "$\int \mathcal{D}\phi\, e^{-S_E}$" is not a figure of speech: it is an integral against an honest measure, and the manipulations of this chapter are theorems. The constructive machinery itself lies beyond our scope; we record it without full proof and point to the lattice formulation of Chapter 43 and to the further reading for the constructions.

(ii) The Minkowski/oscillatory integral is formal. As just discussed, $e^{iS}\mathcal{D}\phi$ is not a measure; the Lorentzian functional integral acquires meaning only through Wick rotation to case (i) and analytic continuation back. We use it freely and productively, but as a generating device for a perturbation series and as the continuation of a Euclidean object, never as an integral in the measure-theoretic sense.

(iii) The $d = 4$ interacting case is, as of the mid-2020s, formal. The physically central theories—ϕ^4 in four dimensions in particular—are not known to exist as nontrivial continuum quantum field theories. The accumulated evidence has now become a theorem: ϕ^4_4 is *trivial*, meaning that any continuum limit of the lattice theory (4.91) with $a \to 0$ forces the renormalized coupling to zero, so the only survivor is the free theory. Triviality for $d > 4$ was established by Aizenman and Fröhlich [10, 202], and the marginal physical case $d = 4$ (together with the four-dimensional critical Ising model) was settled rigorously by Aizenman and Duminil-Copin in 2021 [11]. Four-dimensional Yang–Mills with a mass gap is the subject of one of the Clay Millennium Problems and is open. For these theories the path integral is a formal perturbative definition—an asymptotic series generated by the manipulations we are developing—together with the lattice at finite a as a non-perturbative regulator; it is not (yet) a constructed measure. This is the honest status, and it is the contract Chapter 10 inherits and that the renormalization program of Part V works within.

Behind verdicts (ii)–(iii) sits a single, elementary mathematical fact that explains why the difficulty is structural rather than a failure of cleverness: there

is no infinite-dimensional analogue of Lebesgue measure. The phrase $\mathcal{D}\phi$, taken literally as "$\prod_x d\phi(x)$," would be a translation-invariant measure on an infinite-dimensional space, and no such object exists. We prove this now, in complete elementary detail, so that the reader *sees* why $\int \mathcal{D}\phi$ cannot be a measure on its own.

Theorem 4.23 (No translation-invariant Lebesgue measure in infinite dimensions)**.** *Let H be an infinite-dimensional separable Hilbert space (e.g. the L^2 space of field configurations). There is no nonzero Borel measure μ on H that is both* translation invariant*—$\mu(v+E)=\mu(E)$ for every Borel set E and every $v\in H$—and* locally finite*, in the sense of assigning finite measure to every bounded set. Equivalently, any translation-invariant Borel measure on H that is finite on some ball is identically zero.*

Proof. Suppose μ is translation invariant and locally finite; we show $\mu\equiv 0$. Since H is infinite dimensional it contains an orthonormal sequence $\{e_k\}_{k=1}^{\infty}$. Any two distinct members are at distance

$$\|e_k-e_l\|^2=\|e_k\|^2+\|e_l\|^2-2\langle e_k,e_l\rangle=1+1-0=2,\qquad k\neq l, \tag{4.105}$$

so $\|e_k-e_l\|=\sqrt{2}$. Consider the open balls $B(e_k,r)$ of radius $r=\frac{1}{2}$ centred at these points. They are pairwise disjoint: if some z lay in both $B(e_k,r)$ and $B(e_l,r)$ then by the triangle inequality $\sqrt{2}=\|e_k-e_l\|\le\|e_k-z\|+\|z-e_l\|<r+r=1<\sqrt{2}$, a contradiction. Each $B(e_k,r)$ is a translate of the fixed ball $B(0,r)$ by the vector e_k, so by translation invariance they all have the *same* measure,

$$\mu\big(B(e_k,r)\big)=\mu\big(B(0,r)\big)\equiv c,\qquad \text{for every } k. \tag{4.106}$$

Moreover every $B(e_k,r)$ is contained in the single bounded ball $B(0,1+r)$, because $\|e_k\|=1$ implies any point within r of e_k is within $1+r$ of the origin. Countable additivity over the disjoint family then gives

$$\mu\big(B(0,1+r)\big)\;\ge\;\sum_{k=1}^{\infty}\mu\big(B(e_k,r)\big)\;=\;\sum_{k=1}^{\infty}c. \tag{4.107}$$

Local finiteness says the left-hand side is finite. A sum of infinitely many copies of a fixed nonnegative number c is finite only if $c=0$. Hence $\mu(B(0,\frac{1}{2}))=c=0$, and by translation invariance *every* ball of radius $\frac{1}{2}$ has measure zero. Finally, H is separable, so it has a countable dense subset $\{x_j\}$, and the balls $\{B(x_j,\frac{1}{2})\}_j$ cover H (every point is within $\frac{1}{2}$ of some x_j). Countable subadditivity gives

$$\mu(H)\le\sum_{j=1}^{\infty}\mu\big(B(x_j,\tfrac{1}{2})\big)=\sum_{j=1}^{\infty}0=0, \tag{4.108}$$

so $\mu(H)=0$ and μ is the zero measure. □

Caution

Theorem 4.23 is the precise reason $\int \mathcal{D}\phi$ is *not* an integral against a measure $\mathcal{D}\phi$. The flat, translation-invariant "$\prod_x d\phi(x)$" one is tempted to write down cannot exist on the infinite-dimensional configuration space: it would have to assign the same value to the infinitely many disjoint unit balls packed into a bounded region, and no nonzero locally finite measure can do that. What *does* exist on such spaces is the *Gaussian measure*—the rigorous object underlying (4.102)—which is not translation invariant: shifting a configuration changes its Gaussian weight, and that weight (the $e^{-\frac{1}{2}\phi K\phi}$ of the free action) is exactly what provides the missing decay. The interacting measure, when it exists (case (i)), is built by reweighting the Gaussian one by $e^{-S_{\text{int}}}$. So the meaningful decomposition is never "measure = flat × weight"; it is "measure = Gaussian reference measure, possibly reweighted." The symbol $\mathcal{D}\phi$ in isolation is shorthand for the lattice product (4.90) awaiting a limit, nothing more.

Notation Watch

Throughout the book the symbol $\int \mathcal{D}\phi\, e^{iS[\phi]}$ (or its Euclidean $\int \mathcal{D}\phi\, e^{-S_E[\phi]}$) is used as a complete unit. Read it as: "the $a \to 0$, $L \to \infty$ limit of the finite-dimensional integral (4.91), with renormalizations inserted, when that limit exists; and otherwise as that finite-dimensional integral at fixed regulators, generating an asymptotic perturbation series." No separate meaning is ever attached to $\mathcal{D}\phi$ by itself. This is why every later "derivation from the path integral" is legitimate: it is a manipulation valid at finite N that commutes with the limit.

4.3.5 The Euclidean route and Osterwalder–Schrader reconstruction

It is worth naming, without full proof, what a rigorous construction would have to accomplish, because it organizes the entire constructive program and clarifies the status of the perturbative manipulations that follow. The strategy of case (i)—and the only known route to a rigorous interacting QFT—is the *Euclidean route*: construct the Euclidean correlation functions (the moments of the limiting measure of Theorem 4.22), verify that they satisfy a short list of axioms, and then *reconstruct* the physical Lorentzian theory from them by analytic continuation.

Theorem 4.24 (Osterwalder–Schrader reconstruction)**.** *Suppose a family of Euclidean n-point functions $\{S_n(x_1, \dots, x_n)\}_{n\geq 0}$ satisfies (OS0) suitable temperedness/growth bounds, (OS1) Euclidean invariance under the rotations and*

translations of $\mathbb{R}^d$, (OS2) symmetry under permutations of the points, (OS3) reflection positivity*—a positivity condition under reflection in a hyperplane that is the Euclidean shadow of unitarity—and (OS4) a clustering property at large separation. Then the S_n are the analytic continuations to imaginary time of the Wightman correlation functions of a unique relativistic quantum field theory in Minkowski signature, satisfying the Wightman axioms; in particular reflection positivity (OS3) is what guarantees a Hilbert space with positive-definite inner product and a unitary time evolution.*

This is the precise sense in which the Euclidean lattice integral (4.91), if its continuum limit can be controlled, *would* make the path integral rigorous: the limit supplies the S_n, the lattice construction manifestly delivers (OS1)–(OS2) and, with a reflection-positive discretization, (OS3), and Theorem 4.24 converts them into an honest operator theory. We prove neither the reconstruction theorem nor any specific construction here—they are the province of constructive field theory and the lattice chapter, Chapter 43, and we cite them for further reading rather than reproving them—but the reader should carry away the logical shape: *Euclidean measure (rigorous in low d) $\to$ OS axioms $\to$ Lorentzian QFT*. The perturbative path integral of the rest of this Part is, in effect, an algorithm for computing the would-be S_n order by order in the coupling, valid whether or not the underlying measure has been constructed.

4.3.6 The carried toy: the one case where the path integral is an honest integral

The zero-dimensional ϕ^4 model of Theorem 4.21 is the unique member of the family that escapes every difficulty catalogued above, and that makes it the perfect place to see the rigorous-versus-formal contrast in sharp relief. In $d = 0$ there is a single spacetime point, the "field" is one real number, and the "functional integral" is one ordinary integral—there is no ultraviolet limit to take (a never appears), no infrared limit (L never appears), no infinite-dimensional space, and hence no obstruction from Theorem 4.23. The lattice count (4.82) reads $N = (L/a)^0 = 1$: the regulator is already absent.

Worked Example 4.25 (The $d = 0$ path integral converges, but its perturbation series does not)**.** We continue ϕ^4 theory from Theorem 4.21. The partition function (4.75) at zero source,

$$Z(\lambda) = \int_{-\infty}^{\infty} dx\, e^{-\frac{1}{2}x^2 - \frac{\lambda}{4!}x^4}, \tag{4.109}$$

is, for every $\lambda > 0$, an absolutely convergent integral defining a smooth (indeed real-analytic in $\lambda > 0$) function: the integrand is dominated by $e^{-x^2/2}$ for all $\lambda \geq 0$, so the integral exists and may be differentiated under the sign in λ any number of times. By the three-part verdict of Section 4.3.4 this is the whole

content of "the path integral exists rigorously" in $d = 0$: (4.109) *is* the measure $e^{-S}\,dx$ on $\mathbb{R}^1$, with no limit pending. None of cases (ii)–(iii) can arise.

Now contrast its *perturbation series*. Expanding the interaction and integrating term by term using the free moments (4.77), $\langle x^{4k}\rangle_0 = (4k-1)!!$,

$$\frac{Z(\lambda)}{Z(0)} = \Big\langle e^{-\frac{\lambda}{4!}x^4}\Big\rangle_0 = \sum_{k=0}^{\infty}\frac{1}{k!}\Big(-\frac{\lambda}{4!}\Big)^k\langle x^{4k}\rangle_0 = \sum_{k=0}^{\infty} a_k\,(-\lambda)^k, \qquad a_k = \frac{(4k-1)!!}{(4!)^k\,k!}. \tag{4.110}$$

This is the perturbative "Feynman expansion" of the toy: a_k is the sum of symmetry-weighted vacuum diagrams at order k. We show its radius of convergence is *zero*. Using $(4k+3)!! = (4k+3)(4k+1)(4k-1)!!$ for consecutive odd factors,

$$\frac{a_{k+1}}{a_k} = \frac{(4(k+1)-1)!!}{(4k-1)!!}\cdot\frac{(4!)^k\,k!}{(4!)^{k+1}\,(k+1)!} = \frac{(4k+3)(4k+1)}{4!\,(k+1)} \xrightarrow[k\to\infty]{} \infty, \tag{4.111}$$

since the numerator grows like $16k^2$ while the denominator grows like $24k$, so the ratio behaves as $\frac{2}{3}k \to \infty$. By the ratio test the series (4.110) converges only at $\lambda = 0$: its radius of convergence is 0. The terms eventually grow factorially, the universal hallmark of an *asymptotic* series.

The lesson is exactly the chapter's: the object $Z(\lambda)$ is rigorous and finite, yet the series of Feynman diagrams that "computes" it does not converge to it—it is asymptotic, faithful only as $\lambda \to 0^+$ term by term. The divergence is not a defect of the integral but of the expansion: interchanging the (convergent) x-integral with the (divergent) λ-sum is illegitimate, which one can see directly from the fact that at $\lambda < 0$ the integral (4.109) diverges outright, so no power series in λ with positive radius could represent a function analytic in a disc about $\lambda = 0$. This is the zero-dimensional shadow of the status verdicts: even where the path integral is a perfectly honest integral, its perturbative definition is only a formal (asymptotic) series. Its resummation—the recovery of $Z(\lambda)$ from the divergent coefficients a_k by Borel summation—is set up without full proof in Chapter 5, and the diagrammatic reading of the a_k as symmetry factors is completed in Chapter 15. The constructive analogues for $d \le 3$ and the lattice definition for general d are the further-reading endpoint of this thread and the business of Chapter 43.

We have now done what this section set out to do. The functional integral has been built honestly—discretize on a box, recognize the finite-dimensional Gaussian, take the continuum and infinite-volume limits—and the free generating functional (4.102) together with the propagator-as-inverse-operator (4.100) have been derived, not asserted. Its standing has been stated without evasion: a theorem in low-d Euclidean signature, a Wick-rotated formal object in Minkowski, and a perturbatively defined formal object (with the lattice as its non-perturbative

regulator) for the physical four-dimensional interacting theories, with the non-existence of a flat infinite-dimensional measure (Theorem 4.23) as the structural reason. This $Z[J]$—and the precise contract attached to it—is the object the rest of the chapter organizes; the next section takes it as given and asks how to extract from it the physically irreducible data, the connected correlators generated by $W[J] = \log Z[J]$.

4.4 Generating Functionals and Connected Correlators

The previous section handed us the generating functional $Z[J]$ and, with it, a careful accounting of which of its manipulations are theorems and which are disciplined formal limits (Section 4.3). A generating functional is a bookkeeping device: it packs an entire hierarchy of correlation functions into a single object whose functional derivatives, evaluated at vanishing source, peel the correlators off one at a time. But $Z[J]$ packs them in a redundant way. Its derivatives produce *all* correlators, including the disconnected ones that are mere products of lower correlators and carry no new dynamical information. The central discovery of this section is that the redundancy is removed in one stroke by taking a logarithm: $W[J] = \log Z[J]$ generates exactly the *connected* correlators, the irreducible data of the theory, with every disconnected piece automatically subtracted. This is the *linked-cluster* (cumulant) theorem, and we prove it in full in the rigorous finite-dimensional setting before transcribing it to fields. Along the way we identify the first derivative $\frac{\delta W[J]}{\delta J}$ as the source-dependent expectation value of the field — the *classical field* $\phi_c[J]$ — and the second derivative as the connected two-point function, i.e. the covariance. These two objects are precisely the input the Legendre transform of the next section consumes, so the present section both completes the statistical structure of $Z[J]$ and prepares its Legendre conjugate.

Throughout we work primarily in the Euclidean convention, where $Z[J] = \int \mathcal{D}\phi\, e^{-S_E[\phi]+\int J\phi}$ is a genuine (formal limit of) absolutely convergent integral and $W = \log Z$ is real; the Minkowski bookkeeping, which differs only by tracked factors of i, is recorded in Section 4.4.2. The finite-dimensional model that makes every statement a theorem is the one fixed in Section 4.2: a measure $d\mu(x) = e^{-S(x)}\, d^N x$ on $\mathbb{R}^N$ with S bounded below and growing fast enough that all moments exist.

4.4.1 Full correlators as derivatives of Z

We begin with the unnormalized generating functional of the finite-dimensional model,

$$Z[J] \;=\; \int_{\mathbb{R}^N} d^N x\; e^{-S(x)+J\cdot x}, \qquad J\cdot x = \sum_{i=1}^{N} J_i x_i, \tag{4.112}$$

which is nothing but the moment generating function of the measure $d\mu$. From the derivative dictionary established in Section 4.2 — that each $\partial/\partial J_i$ acting under the integral brings down a factor of x_i — repeated differentiation produces the moments,

$$\left.\frac{\partial^n Z[J]}{\partial J_{i_1}\cdots\partial J_{i_n}}\right|_{J=0} \;=\; \int d^N x\; x_{i_1}\cdots x_{i_n}\, e^{-S(x)} \;=\; Z[0]\,\langle x_{i_1}\cdots x_{i_n}\rangle. \tag{4.113}$$

Here the (normalized) expectation value of any function f is $\langle f\rangle \equiv Z[0]^{-1}\int f\, e^{-S}\, d^N x$. Dividing (4.113) by $Z[0]$ defines the *full* (or *complete*) n-point functions as the normalized derivatives of Z. The normalization matters: $Z[0] = \int e^{-S}$ is the partition function, and in the field theory it is the sum of all vacuum diagrams, which must be divided out for any correlator to be finite and physical. It is convenient to absorb it once and for all into a normalized functional.

Definition 4.26 (Full correlators)**.** Let $\widehat{Z}[J] \equiv Z[J]/Z[0]$, so $\widehat{Z}[0] = 1$. In the field theory, with continuous index x replacing the discrete i, the *full n-point correlation function* is

$$\begin{aligned} G^{(n)}(x_1,\ldots,x_n) \;&\equiv\; \langle\phi(x_1)\cdots\phi(x_n)\rangle \\ &=\; \frac{1}{Z[0]}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0} \;=\; \frac{\delta^n \widehat{Z}[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}. \end{aligned} \tag{4.114}$$

The functional Taylor expansion of $\widehat{Z}[J]$ in powers of the source therefore has the full correlators as its coefficients,

$$\widehat{Z}[J] \;=\; \sum_{n=0}^{\infty}\frac{1}{n!}\int d^d x_1\cdots d^d x_n\; G^{(n)}(x_1,\ldots,x_n)\, J(x_1)\cdots J(x_n). \tag{4.115}$$

In the Euclidean field theory the bracket $\langle\cdot\rangle$ is the path-integral average; on Minkowski continuation it becomes the vacuum expectation value of a *time-ordered* product, $\langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|0\rangle$, because the path integral builds correlators in time order automatically — a point established for the free theory in Section 4.3 and carried forward to Feynman rules in Chapter 15. The functional (4.115) is honest but wasteful: as we now show, most of the information in $G^{(n)}$ is already contained in the lower correlators.

4.4.2 The connected generating functional W

The waste is visible already in the free Gaussian theory. There Wick's theorem (Section 4.2) gives $G^{(4)}$ as a sum of three products of two-point functions; the four-point function contains *nothing* the propagator does not already contain. We want a generating functional whose derivatives strip away these factorized pieces and retain only the genuinely new, "connected" part at each order. The remarkable fact is that the operation that does this is the logarithm.

Definition 4.27 (Connected generating functional and connected correlators)**.** The *connected generating functional* is

$$W[J] \;\equiv\; \log Z[J] \qquad \text{(Euclidean)}, \tag{4.116}$$

and the *connected n-point functions* are its functional derivatives,

$$G_c^{(n)}(x_1,\dots,x_n) \;\equiv\; \left.\frac{\delta^n W[J]}{\delta J(x_1)\cdots\delta J(x_n)}\right|_{J=0}, \qquad n \geq 1. \tag{4.117}$$

Two remarks fix the conventions. First, W and $\widehat{W} \equiv \log \widehat{Z} = W - \log Z[0]$ differ only by the additive constant $\log Z[0]$, which is killed by any derivative with $n \geq 1$; the connected correlators (4.117) are therefore insensitive to the vacuum normalization, exactly as they should be. The constant $W[0] = \log Z[0]$ is itself meaningful — it is (minus) the free energy, the sum of connected vacuum diagrams — but it never contaminates the $n \geq 1$ correlators. Second, the *logarithm is the whole content*: the claim, proved in Section 4.4.3, is that the coefficients of $W[J]$ are precisely the connected parts of the coefficients of log-free $Z[J]$.

Notation Watch

Minkowski factors of i. In Minkowski signature the generating functional carries the phase e^{iS}, $Z[J] = \int \mathcal{D}\phi\, e^{i(S[\phi]+\int J\phi)}$, and the standard definition is

$$W[J] \;\equiv\; -\,i \log Z[J]. \tag{4.118}$$

Each $\delta/\delta J(x)$ now brings down $i\phi(x)$, so the full correlator carries $G^{(n)} = (-i)^n Z[0]^{-1}\delta^n Z/\delta J^n|_0$ and, after the same algebra as below, the connected correlator is $G_c^{(n)} = (-i)^{\,n-1}\,\delta^n W/\delta J(x_1)\cdots\delta J(x_n)\big|_0$. The single residual factor $(-i)^{n-1}$ is bookkeeping; the combinatorial statement of this section — which W generates which graphs — is identical in both signatures. We use $W = \log Z$ (Euclidean) for every proof and translate at the end.

Before proving the theorem we extract its two lowest consequences, which are the connections this section is responsible for. Differentiating $W = \log Z$ once,

$$\frac{\delta W[J]}{\delta J(x)} \;=\; \frac{1}{Z[J]}\,\frac{\delta Z[J]}{\delta J(x)} \;=\; \frac{\int \mathcal{D}\phi\;\phi(x)\,e^{-S+\int J\phi}}{\int \mathcal{D}\phi\; e^{-S+\int J\phi}} \;=\; \big\langle \phi(x) \big\rangle_J \;\equiv\; \phi_c(x;J). \tag{4.119}$$

The first derivative of W is the expectation value of the field *in the presence of the source* — the source-dependent mean, which we name the *classical field* $\phi_c[J]$. At $J = 0$ it is the genuine one-point function (the vacuum condensate), which vanishes in a $\mathbb{Z}_2$-symmetric vacuum but need not in general. Differentiating once

more,

$$\frac{\delta^2 W[J]}{\delta J(x)\,\delta J(y)} = \frac{\delta}{\delta J(y)}\left[\frac{1}{Z}\frac{\delta Z}{\delta J(x)}\right] = \frac{1}{Z}\frac{\delta^2 Z}{\delta J(x)\,\delta J(y)} - \frac{1}{Z^2}\frac{\delta Z}{\delta J(x)}\frac{\delta Z}{\delta J(y)}$$
$$= \big\langle \phi(x)\phi(y)\big\rangle_J - \big\langle \phi(x)\big\rangle_J \big\langle \phi(y)\big\rangle_J . \tag{4.120}$$

The second derivative of W is the *covariance* of the field — the connected two-point function at arbitrary source. At $J = 0$,

$$G_c^{(2)}(x,y) \;=\; \langle \phi(x)\phi(y)\rangle - \langle \phi(x)\rangle\langle \phi(y)\rangle, \tag{4.121}$$

which in a symmetric vacuum ($\langle\phi\rangle = 0$) is just the full propagator. This is the bridge to the free theory: at $\lambda = 0$ the bracket is Gaussian and (4.121) reduces to the Feynman propagator $D_F(x-y)$ of Chapters 9 and 10. The connected two-point function *is* the propagator, dressed by interactions; the logarithm has already subtracted the disconnected $\langle\phi\rangle\langle\phi\rangle$ piece in (4.120).

4.4.3 The linked-cluster theorem

We now prove, in the finite-dimensional setting where it is a theorem and not a formal manipulation, that the derivatives of $W = \log Z$ are exactly the connected parts of the derivatives of Z. The statement is purely combinatorial: it is the moment-cumulant relation of probability theory, and its diagrammatic shadow is the statement that "log of all diagrams = connected diagrams."

To state it cleanly, fix a finite index set and let $S = \{i_1, \dots, i_n\}$ label the external legs (repetitions allowed; the i_a need not be distinct). For a subset $B \subseteq S$ write $\partial_B \equiv \prod_{i\in B} \partial/\partial J_i$, and for a partition π of S into nonempty disjoint blocks B write $|\pi|$ for the number of blocks. Define the connected correlator over a block by $\langle x_B\rangle_c \equiv \partial_B W\big|_{J=0}$.

Theorem 4.28 (Linked-cluster / cumulant theorem)**.** *With $Z[J] = \int e^{-S(x)+J\cdot x} d^N x$ and $W[J] = \log Z[J]$, for every finite multiset S of indices,*

$$\partial_S Z \;=\; Z \sum_{\pi \vdash S} \prod_{B\in\pi} \partial_B W, \tag{4.122}$$

where the sum runs over all set partitions π of S. Consequently, dividing by $Z[0]$ and setting $J = 0$, the full correlators are sums over partitions of products of connected correlators,

$$\big\langle x_{i_1}\cdots x_{i_n}\big\rangle \;=\; \sum_{\pi\vdash\{i_1,\dots,i_n\}} \prod_{B\in\pi} \big\langle x_B\big\rangle_c . \tag{4.123}$$

Proof. We prove (4.122) by induction on $|S|$; (4.123) then follows by setting $J = 0$, dividing by $Z[0] = Z|_{J=0}$, and using $\partial_S Z|_0 / Z[0] = \langle x_S\rangle$ from (4.113) and $\partial_B W|_0 = \langle x_B\rangle_c$ from (4.117).

Base case. For $S = \{i\}$ there is a single partition, the one block $\{i\}$, and the definition $W = \log Z$ gives $\partial_i Z = (\partial_i \log Z)\, Z = Z\, \partial_i W$, which is (4.122).

Inductive step. Assume (4.122) for a set S, and adjoin a new index $j \notin S$. Apply ∂_j to both sides and use the product rule on the right, together with $\partial_j Z = (\partial_j W) Z$:

$$\begin{aligned}
\partial_j \partial_S Z &= \partial_j \left(Z \sum_{\pi \vdash S} \prod_{B \in \pi} \partial_B W \right) \\
&= (\partial_j Z) \sum_{\pi \vdash S} \prod_{B \in \pi} \partial_B W \;+\; Z \sum_{\pi \vdash S} \partial_j \left(\prod_{B \in \pi} \partial_B W \right) \\
&= Z\, (\partial_j W) \sum_{\pi \vdash S} \prod_{B \in \pi} \partial_B W \;+\; Z \sum_{\pi \vdash S} \sum_{B \in \pi} \left(\partial_{B \cup \{j\}} W \right) \prod_{B' \in \pi,\, B' \neq B} \partial_{B'} W .
\end{aligned} \tag{4.124}$$

Read the two terms as a sum over partitions of $S \cup \{j\}$. In the first term, the factor $\partial_j W$ adjoins the *singleton* block $\{j\}$ to a partition π of S; this generates exactly those partitions of $S \cup \{j\}$ in which j sits alone. In the second term, the operator ∂_j has merged j into one existing block B, turning $\partial_B W$ into $\partial_{B \cup \{j\}} W$; ranging over $\pi \vdash S$ and over the blocks $B \in \pi$ generates exactly those partitions of $S \cup \{j\}$ in which j joins a nonsingleton placement, i.e. shares its block with at least one element of S. Every partition π' of $S \cup \{j\}$ falls into precisely one of these two cases — either $\{j\} \in \pi'$ or not — and is produced exactly once. Hence

$$\partial_j \partial_S Z \;=\; Z \sum_{\pi' \vdash S \cup \{j\}} \prod_{B \in \pi'} \partial_B W, \tag{4.125}$$

which is (4.122) for $S \cup \{j\}$. This closes the induction. □

Theorem 4.28 is the rigorous core. It says that the full correlators are recovered from the connected ones by summing over all ways of partitioning the external legs into connected clusters — each cluster a connected correlator, the partition encoding how the legs group. Equivalently, inverting the relation (done explicitly in Section 4.4.4), the connected correlator $G_c^{(n)}$ is the full $G^{(n)}$ with all partitioned products of lower correlators subtracted off. The logarithm performs this subtraction to all orders at once.

Physical Insight

The content of Theorem 4.28 is the exponential formula: a structure built from labelled pieces, in which a configuration is an arbitrary disjoint collection of "connected" pieces, has a generating function that is the exponential of the generating function of the connected pieces alone. Here the configuration is a correlator, the connected pieces are the $\langle x_B \rangle_c$, and

$Z = e^W$ is precisely the exponential. The same identity appears as the cumulant expansion in statistics (where W is the cumulant generating function), as the cluster expansion in statistical mechanics (where W is the free energy and the connected pieces are clusters), and — in the next subsection — as the diagrammatic statement that the logarithm of the sum of all Feynman diagrams is the sum of the connected ones.

Diagrammatic restatement. In perturbation theory each correlator is a sum of Feynman diagrams with external legs at $x_1, \dots, x_n$, vertices from the interaction, and propagators from the free covariance (the rules are derived in Chapter 15; we anticipate only their gross structure). A general diagram need not be connected: it decomposes uniquely into connected components. Summing over all diagrams contributing to $Z[J]$ is the same as summing over all ways of choosing a multiset of connected diagrams and juxtaposing them. If $\mathcal{C}$ denotes the sum of all distinct connected diagrams (each weighted by its symmetry factor), then a configuration containing k copies of connected pieces is counted with a $1/k!$ for the interchange of identical copies, and summing k from 0 to ∞ assembles the exponential series,

$$\begin{aligned} Z[J] \;&=\; \sum_{k=0}^{\infty} \frac{1}{k!}\, \mathcal{C}^{\,k} \;=\; e^{\mathcal{C}}, \\ \Longrightarrow \qquad W[J] \;&=\; \log Z[J] \;=\; \mathcal{C} \;=\; (\text{sum of connected diagrams}). \end{aligned} \tag{4.126}$$

This is the same statement as (4.122), now read on graphs: a partition of the external legs into blocks is the decomposition of a diagram into its connected components. In particular the disconnected vacuum diagrams that plague $Z[0]$ exponentiate into $W[0] = \log Z[0]$ and cancel out of every $n \geq 1$ connected correlator — the diagrammatic face of the normalization remark after Theorem 4.27. The finer field-theoretic taxonomy (connected vs. *amputated* vs. *one-particle irreducible*) and the precise symmetry-factor accounting belong to Chapter 15; here we have established the first and most basic layer, connectedness, and proved it cleanly in the finite-dimensional model.

4.4.4 Low-order connected–full relations

We now make (4.123) explicit at low order and invert it, both to exhibit the combinatorics concretely and because the $n = 2$ and $n = 4$ relations are quoted constantly later. Write $\langle \cdots \rangle$ for full correlators and $\langle \cdots \rangle_c$ for connected ones, and label external points by their indices. The partitions of an n-element set number the Bell numbers $B_n = 1, 1, 2, 5, 15, \dots$, and each partition contributes one product term.

For $n = 1$ there is one partition, so moment and cumulant coincide:

$$\langle x_1 \rangle \;=\; \langle x_1 \rangle_c . \tag{4.127}$$

For $n = 2$ the two partitions $\{12\}$ and $\{1\}\{2\}$ give

$$\langle x_1 x_2 \rangle \;=\; \langle x_1 x_2 \rangle_c \;+\; \langle x_1 \rangle_c \langle x_2 \rangle_c , \tag{4.128}$$

which inverts immediately (using (4.127)) to the connected two-point function as the covariance,

$$\boxed{\langle x_1 x_2 \rangle_c \;=\; \langle x_1 x_2 \rangle - \langle x_1 \rangle \langle x_2 \rangle} \tag{4.129}$$

in agreement with the direct computation (4.121). For $n = 3$ the $B_3 = 5$ partitions ($\{123\}$; three of type $\{ij\}\{k\}$; and $\{1\}\{2\}\{3\}$) give

$$\langle x_1 x_2 x_3 \rangle = \langle x_1 x_2 x_3 \rangle_c + \big[\langle x_1 x_2 \rangle_c \langle x_3 \rangle_c + 2 \text{ perms} \big] + \langle x_1 \rangle_c \langle x_2 \rangle_c \langle x_3 \rangle_c , \tag{4.130}$$

the bracket denoting the sum over the three ways to choose the singleton.

The case $n = 4$ is the one that recurs throughout perturbation theory, so we carry it in full. The $B_4 = 15$ partitions of $\{1, 2, 3, 4\}$ fall into five types: one block of four (1 partition); a block of three and a singleton (4 partitions, one for each choice of singleton); two blocks of two (3 partitions); a block of two and two singletons (6 partitions); and four singletons (1 partition). Summing one product per partition,

$$\begin{aligned} \langle x_1 x_2 x_3 x_4 \rangle = {} & \langle x_1 x_2 x_3 x_4 \rangle_c \\ & + \big[\langle x_1 x_2 x_3 \rangle_c \langle x_4 \rangle_c + 3 \text{ perms} \big] \\ & + \big[\langle x_1 x_2 \rangle_c \langle x_3 x_4 \rangle_c + \langle x_1 x_3 \rangle_c \langle x_2 x_4 \rangle_c + \langle x_1 x_4 \rangle_c \langle x_2 x_3 \rangle_c \big] \\ & + \big[\langle x_1 x_2 \rangle_c \langle x_3 \rangle_c \langle x_4 \rangle_c + 5 \text{ perms} \big] + \langle x_1 \rangle_c \langle x_2 \rangle_c \langle x_3 \rangle_c \langle x_4 \rangle_c . \end{aligned} \tag{4.131}$$

The five term-types use $1 + 4 + 3 + 6 + 1 = 15$ partitions, as required. To obtain the connected four-point function we solve (4.131) for $\langle x_1 x_2 x_3 x_4 \rangle_c$. In the physically central case of a symmetric vacuum, all one-point functions vanish, $\langle x_i \rangle = \langle x_i \rangle_c = 0$, which kills every term containing a singleton cumulant; by (4.129) the surviving two-point cumulants equal the full two-point functions. What remains is the clean relation

$$\boxed{\langle x_1 x_2 x_3 x_4 \rangle_c = \langle x_1 x_2 x_3 x_4 \rangle - \langle x_1 x_2 \rangle \langle x_3 x_4 \rangle - \langle x_1 x_3 \rangle \langle x_2 x_4 \rangle - \langle x_1 x_4 \rangle \langle x_2 x_3 \rangle} \tag{4.132}$$

The three subtracted products are exactly the three disconnected pairings of the four legs — the Wick contractions that the free theory produces and that carry no interaction information. The connected four-point function is what is left after removing them, and at leading order in $\lambda\phi^4$ it is the single connected "X" graph joining all four legs at one vertex. Figure 4.4 draws the decomposition. For a generic (non-symmetric) vacuum one simply retains the singleton terms; the inversion is mechanical, and we have written it once at $n = 2$ and $n = 4$ to display the pattern.

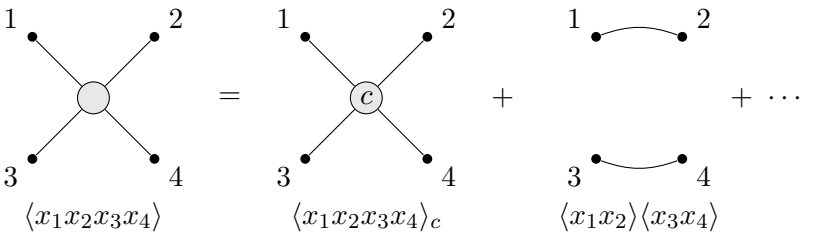


Figure 4.4: The full four-point function as connected part plus disconnected pairings, (4.132) (symmetric vacuum). The connected blob c is the new information at fourth order; the three disconnected pairings $\langle x_1x_2\rangle\langle x_3x_4\rangle$, $\langle x_1x_3\rangle\langle x_2x_4\rangle$, $\langle x_1x_4\rangle\langle x_2x_3\rangle$ are the Wick products already fixed by the two-point function. Taking $W = \log Z$ removes the latter automatically.

4.4.5 The classical field and the bridge to the Legendre transform

We return to the first derivative (4.119) and develop it into the object the next section needs. The classical field

$$\phi_c(x;J) \;=\; \frac{\delta W[J]}{\delta J(x)} \;=\; \langle\phi(x)\rangle_J \tag{4.133}$$

is the expectation value of $\phi(x)$ computed in the source-tilted measure $\propto e^{-S+\int J\phi}$. It is not the operator field and not, in general, a solution of the classical equation of motion; it is the quantum-averaged field profile that the source J sustains. The terminology "classical" is justified in the next section, where ϕ_c turns out to extremize the effective action $\Gamma[\phi_c]$ in exactly the way the true classical field extremizes S. The verification that $\phi_c = \langle\phi\rangle_J$ was already given in (4.119); the point to record is that ϕ_c is a functional of J, and that this functional is invertible precisely when the map is monotone — which is governed by the second derivative.

That second derivative we computed in (4.120): at arbitrary source,

$$\frac{\delta\phi_c(x;J)}{\delta J(y)} = \frac{\delta^2 W[J]}{\delta J(x)\,\delta J(y)} = \langle\phi(x)\phi(y)\rangle_J - \langle\phi(x)\rangle_J\langle\phi(y)\rangle_J \;\equiv\; G_c^{(2)}(x,y;J). \tag{4.134}$$

Two facts follow that are the entire input to the Legendre transform. First, the right-hand side is a covariance, hence (in the Euclidean theory) a positive semi-definite kernel: for any test function f,

$$\int f(x)\,G_c^{(2)}(x,y;J)\,f(y) = \big\langle\big(\textstyle\int f\,\phi - \langle\int f\,\phi\rangle_J\big)^2\big\rangle_J \;\geq\; 0. \tag{4.135}$$

Thus $W[J]$ is a convex functional of J — its Hessian $\delta^2 W/\delta J\,\delta J$ is positive semi-definite — which is exactly the property the Legendre transform requires for

its conjugate to be well defined and the transform to be an involution. Second, (4.134) identifies the Hessian of W as the Jacobian of the map $J \mapsto \phi_c$. When this Jacobian is invertible the map can be inverted to give $J = J[\phi_c]$, and its inverse kernel — the functional inverse of $G_c^{(2)}$ — will turn out to be the second derivative of the effective action. The relation between Γ'' and W'' as mutual inverses, $\int \Gamma'' W'' = \mathbf{1}$, is the technical heart of the next section; the present section has supplied its two ingredients, the conjugate variable $\phi_c = \delta W/\delta J$ and the positive-definite covariance $\delta^2 W/\delta J\,\delta J = G_c^{(2)}$.

Worked Example 4.29 (Connected correlators of the zero-dimensional ϕ^4 model)**.** We advance the carried zero-dimensional ϕ^4 toy of Section 4.2 — the $N = 1$ instance of (4.112) — by computing $W(J) = \log Z(J)$ to first order in the coupling and reading off its connected correlators, watching the disconnected pieces drop out. With $m^2 = 1$ for arithmetic cleanliness (so the free covariance, the propagator, equals 1) the model is

$$Z(J) \;=\; \int_{-\infty}^{\infty} dx\; \exp\Big(-\tfrac{1}{2}x^2 - \tfrac{\lambda}{4!}\,x^4 + Jx \Big), \qquad \lambda > 0. \tag{4.136}$$

This is an ordinary, absolutely convergent integral — a genuine 0-dimensional "field theory" — so every step is rigorous and the linked-cluster theorem (Theorem 4.28) applies verbatim with the single index dropped.

Source trick. Pull the quartic out of the integral by replacing $x \to \partial_J$ acting on the free generating function. With $Z_0(J) = \int dx\, e^{-x^2/2 + Jx} = \sqrt{2\pi}\, e^{J^2/2}$,

$$Z(J) = e^{-\frac{\lambda}{4!}\partial_J^4} Z_0(J) = \sqrt{2\pi}\, e^{J^2/2}\Big[1 - \tfrac{\lambda}{4!}\, e^{-J^2/2}\, \partial_J^4\, e^{J^2/2} + O(\lambda^2)\Big]. \tag{4.137}$$

We need $\partial_J^4 e^{J^2/2}$. Differentiating $f = e^{J^2/2}$ repeatedly, $\partial_J f = Jf$, $\partial_J^2 f = (1 + J^2)f$, $\partial_J^3 f = (3J + J^3)f$, and

$$\partial_J^4 e^{J^2/2} = (3 + 6J^2 + J^4)\, e^{J^2/2} \;\Longrightarrow\; e^{-J^2/2}\partial_J^4 e^{J^2/2} = J^4 + 6J^2 + 3. \tag{4.138}$$

Therefore, to first order in λ,

$$Z(J) = \sqrt{2\pi}\, e^{J^2/2}\Big[1 - \tfrac{\lambda}{24}(J^4 + 6J^2 + 3) + O(\lambda^2)\Big]. \tag{4.139}$$

Take the logarithm. Using $\log(1 - u) = -u + O(u^2)$ with $u = O(\lambda)$,

$$W(J) = \log Z(J) = \underbrace{\log\sqrt{2\pi}}_{\text{const}} + \tfrac{1}{2}J^2 - \tfrac{\lambda}{24}(J^4 + 6J^2 + 3) + O(\lambda^2). \tag{4.140}$$

The logarithm has linearized what was a ratio: there is no longer any denominator $Z(0) = \sqrt{2\pi}\,(1 - \lambda/8 + \cdots)$ to expand, because log converted the division by $Z(0)$ into the subtraction of the constant $W(0) = \log\sqrt{2\pi} - \lambda/8 + \cdots$. This is the cancellation of disconnected vacuum pieces in action.

Read off the connected correlators as derivatives of W at $J=0$, per (4.117). The model is $\mathbb{Z}_2$-symmetric, so odd derivatives vanish; the nonzero ones through $O(\lambda)$ are

$$\langle x^2\rangle_c = W''(0) = \frac{d^2}{dJ^2}\Big[\tfrac{1}{2}J^2 - \tfrac{\lambda}{24}\cdot 6J^2\Big]_{J=0} = 1 - \frac{\lambda}{2} + O(\lambda^2), \tag{4.141}$$

$$\langle x^4\rangle_c = W^{(4)}(0) = -\frac{\lambda}{24}\frac{d^4}{dJ^4}J^4\Big|_{J=0} = -\frac{\lambda}{24}\cdot 24 = -\lambda + O(\lambda^2). \tag{4.142}$$

The connected two-point function is the propagator 1 corrected at one loop by $-\lambda/2$; the connected four-point function is, at leading order, simply $-\lambda$ — the bare vertex (the 0-dimensional avatar of the $-i\lambda$ that ϕ^4 theory meets in Chapter 15). Nothing disconnected appears in W by construction.

Cross-check against the full correlators and the inversion (4.132). Compute the full moments directly from (4.136), normalizing by $Z(0)$ and using the Gaussian moments $\langle x^2\rangle_0 = 1$, $\langle x^4\rangle_0 = 3$, $\langle x^6\rangle_0 = 15$, $\langle x^8\rangle_0 = 105$. With $Z(0)/\sqrt{2\pi} = 1 - \frac{\lambda}{24}\cdot 3 = 1 - \lambda/8$,

$$\langle x^2\rangle = \frac{\langle x^2\rangle_0 - \frac{\lambda}{24}\langle x^6\rangle_0}{1-\lambda/8} = \frac{1-\frac{15\lambda}{24}}{1-\lambda/8} = \Big(1-\tfrac{15\lambda}{24}\Big)\Big(1+\tfrac{\lambda}{8}\Big) + O(\lambda^2) = 1 - \frac{\lambda}{2} + O(\lambda^2), \tag{4.143}$$

$$\langle x^4\rangle = \frac{\langle x^4\rangle_0 - \frac{\lambda}{24}\langle x^8\rangle_0}{1-\lambda/8} = \frac{3-\frac{105\lambda}{24}}{1-\lambda/8} = \Big(3-\tfrac{105\lambda}{24}\Big)\Big(1+\tfrac{\lambda}{8}\Big) + O(\lambda^2) = 3 - 4\lambda + O(\lambda^2). \tag{4.144}$$

The full two-point function agrees with the connected one, (4.141), because $\langle x\rangle = 0$. For the four-point function the inversion (4.132) with all points equal reads $\langle x^4\rangle_c = \langle x^4\rangle - 3\langle x^2\rangle^2$, and indeed

$$\langle x^4\rangle - 3\langle x^2\rangle^2 = (3-4\lambda) - 3\Big(1-\tfrac{\lambda}{2}\Big)^2 = (3-4\lambda) - 3(1-\lambda) + O(\lambda^2) = -\lambda + O(\lambda^2), \tag{4.145}$$

exactly $W^{(4)}(0)$ from (4.142). The arithmetic displays the combinatorics of Fig. 4.4 concretely: the full $\langle x^4\rangle = 3 - 4\lambda$ splits into the connected vertex $-\lambda$ and the three disconnected pairings $3\langle x^2\rangle^2 = 3 - 3\lambda$, the "3" being the three Wick contractions of the free theory and the -3λ their propagator corrections. The logarithm subtracts the whole $3-3\lambda$ in one operation, leaving the connected $-\lambda$. This is the linked-cluster theorem made completely explicit in the simplest model that has any interactions at all.

Handoff

We have organized $Z[J]$ into its irreducible content: $W[J] = \log Z[J]$ (or $-i\log Z$ in Minkowski) generates exactly the connected correlators, a fact proved as

the linked-cluster theorem (4.123) and made explicit through the $n = 2$ and $n = 4$ relations and the zero-dimensional ϕ^4 computation. The two derivatives we singled out — the classical field $\phi_c[J] = \delta W/\delta J$, the source-dependent expectation value, and the connected propagator $\delta^2 W/\delta J\,\delta J = G_c^{(2)}$, a positive-definite covariance making W convex — are precisely the data a Legendre transform consumes. The next section trades the independent variable J for its conjugate ϕ_c and builds the effective action $\Gamma[\phi_c]$, whose second derivative we will prove is the functional inverse of $G_c^{(2)}$.

4.5 The Legendre Transform and the Effective Action

The previous section handed us three objects and one promise. The objects are the connected generating functional $W[J] = \log Z[J]$ (Theorem 4.27), its first derivative the *classical field*

$$\phi_c(x;J) \;=\; \frac{\delta W[J]}{\delta J(x)} \;=\; \langle \phi(x) \rangle_J, \tag{4.146}$$

and its second derivative, the connected two-point function at arbitrary source,

$$\frac{\delta \phi_c(x;J)}{\delta J(y)} = \frac{\delta^2 W[J]}{\delta J(x)\,\delta J(y)} = G_c^{(2)}(x,y;J), \tag{4.147}$$

a positive semi-definite covariance (Eqs. (4.134) and (4.135)). The promise, recorded at the end of Section 4.4.5, is that the inverse of this covariance is the second derivative of a new functional — the *effective action* $\Gamma[\phi_c]$ — obtained from $W[J]$ by trading the source J for its conjugate variable ϕ_c. That trade is the Legendre transform. It is the same operation that, in classical mechanics, passes from the Lagrangian $L(q,\dot q)$ to the Hamiltonian $H(q,p)$ by trading $\dot q$ for $p = \partial L/\partial \dot q$, and in thermodynamics passes from the energy to the free energies by trading extensive for intensive variables. Here it trades the external source for the field it sustains, and the functional it produces is the central object of the chapter: $\Gamma[\phi_c]$ is the quantum-corrected action whose stationary points are the true field configurations of the interacting theory, whose Hessian is the full inverse propagator, and whose higher derivatives are the one-particle-irreducible vertices from which every connected amplitude is assembled as a tree. We build it in four steps: the Legendre transform reviewed rigorously in finite dimensions; the definition of Γ and the conjugate relation $\delta\Gamma/\delta\phi_c = J$; the reciprocity identity $\Gamma'' W'' = \mathbf{1}$ and its consequence that Γ generates the one-particle-irreducible (1PI) vertices; and the tree-level identity $\Gamma = S$ that names Γ "the effective action."

4.5.1 The Legendre transform of a convex function

We first establish the transform where it is a theorem of ordinary calculus: on $\mathbb{R}^N$, for the smooth strictly convex functions to which the finite-dimensional

W of Theorem 4.28 belongs. The functional statements we need follow by the dictionary "index $i \to$ spacetime point x, sum $\to$ integral" that has organized the whole chapter; we make the transcription explicit at the end. Throughout, a function $f\colon \mathbb{R}^N \to \mathbb{R}$ is *convex* if its Hessian $f''_{ij} = \partial^2 f/\partial J_i \partial J_j$ is positive semi-definite everywhere, and *strictly convex* if f'' is positive definite; the latter is the case for W, whose Hessian (4.147) is the covariance of a non-degenerate distribution and hence strictly positive (Eq. (4.135)).

Definition 4.30 (Legendre transform)**.** Let $f\colon \mathbb{R}^N \to \mathbb{R}$ be convex. Its *Legendre transform* (or *convex conjugate*) is the function $f^\star\colon \mathbb{R}^N \to \mathbb{R} \cup \{+\infty\}$

$$f^\star(\xi) \;\equiv\; \sup_{J \in \mathbb{R}^N} \Big(\xi \cdot J - f(J) \Big), \qquad \xi \cdot J = \sum_{i=1}^{N} \xi_i J_i. \tag{4.148}$$

The variable ξ is *conjugate* to J.

The geometric content of (4.148) is worth stating before the algebra, because it makes every subsequent property obvious. Fix the slope ξ and slide the hyperplane $J \mapsto \xi \cdot J - c$ of that slope downward (increasing c) until it last touches the graph of f from below; the value of c at that contact is $f^\star(\xi)$. The transform thus records, for each slope ξ, how far below the origin the supporting hyperplane of that slope sits (Fig. 4.5). Because f is convex its graph lies above all its tangents, so for a smooth f the supremum is attained where the tangent plane has slope ξ, i.e. where $\nabla f(J) = \xi$. This single picture contains the conjugate relations, Fenchel's inequality, and the involution; we now extract them.

Theorem 4.31 (Conjugate relations, Fenchel–Young, and involution)**.** *Let $f \in C^2(\mathbb{R}^N)$ be strictly convex with ∇f a bijection of $\mathbb{R}^N$ onto its image. Then:*

(i) (Fenchel–Young inequality.) *For all J, ξ,*

$$f(J) + f^\star(\xi) \;\geq\; \xi \cdot J, \tag{4.149}$$

with equality if and only if $\xi = \nabla f(J)$.

(ii) (Conjugate variables.) *The supremum in (4.148) is attained at the unique $J = J^\star(\xi)$ solving $\nabla f(J) = \xi$, and there*

$$f^\star(\xi) = \xi \cdot J^\star(\xi) - f\big(J^\star(\xi)\big), \qquad \nabla f^\star(\xi) = J^\star(\xi). \tag{4.150}$$

Thus $\xi = \nabla f(J) \iff J = \nabla f^\star(\xi)$: the gradient maps of f and $f^\star$ are mutual inverses.

(iii) (Convexity of the conjugate and involution.) *$f^\star$ is convex, and $f^{\star\star} = f$.*

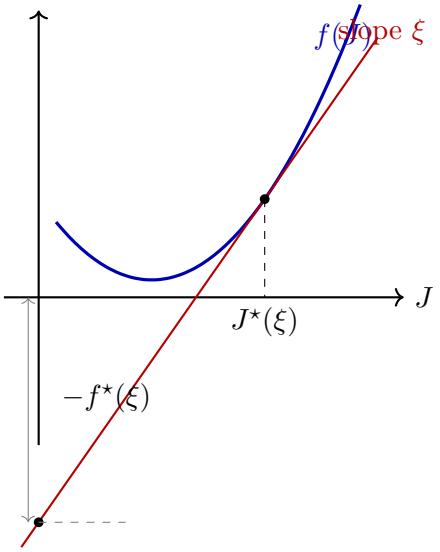


Figure 4.5: The Legendre transform geometrically. For a fixed slope ξ the supporting line $J \mapsto \xi J - c$ touches the convex graph of f from below at the point $J^\star(\xi)$ where $f' = \xi$; its intercept with the vertical axis is $-f^\star(\xi)$. As ξ ranges over all slopes the family of supporting lines reconstructs f from below, which is the geometric statement of the involution $f^{\star\star} = f$.

Proof. (i). Fix J, ξ. By definition $f^\star(\xi) \geq \xi \cdot J' - f(J')$ for every J'; taking $J' = J$ gives $f^\star(\xi) \geq \xi \cdot J - f(J)$, which is (4.149). Equality holds iff J itself realizes the supremum.

(ii). The objective $g_\xi(J) \equiv \xi \cdot J - f(J)$ has gradient $\nabla g_\xi = \xi - \nabla f(J)$ and Hessian $-f''(J) \prec 0$ (negative definite, since f is strictly convex). A function with everywhere negative-definite Hessian is strictly concave, so any critical point is the unique global maximum. The critical-point equation $\nabla g_\xi = 0$ reads $\nabla f(J) = \xi$, which by hypothesis has the unique solution $J^\star(\xi)$. Hence the supremum is attained there and equals $\xi \cdot J^\star(\xi) - f(J^\star(\xi))$, the first equation of (4.150). For the gradient of $f^\star$, differentiate this expression in ξ, treating $J^\star = J^\star(\xi)$ as a function of ξ and using the chain rule:

$$\frac{\partial f^\star}{\partial \xi_i} = J_i^\star + \sum_j \xi_j \frac{\partial J_j^\star}{\partial \xi_i} - \sum_j \frac{\partial f}{\partial J_j}\Big|_{J^\star} \frac{\partial J_j^\star}{\partial \xi_i}. \tag{4.151}$$

At $J^\star$ one has $\partial f / \partial J_j = \xi_j$, so the last two sums cancel term by term, leaving $\partial f^\star / \partial \xi_i = J_i^\star$, i.e. $\nabla f^\star(\xi) = J^\star(\xi)$. This is the second equation of (4.150). The cancellation in (4.151) is the engine of the whole construction; it is the reason a Legendre transform of a function of J is a clean function of ξ alone, with no leftover dependence on the stationary point $J^\star(\xi)$. It is the envelope theorem: the derivative of an optimized objective with respect to a parameter is the partial

derivative of the objective, the implicit variation of the optimizer contributing nothing because the objective is stationary there.

(iii). Convexity of $f^\star$ is immediate from (4.148): for each fixed J the map $\xi \mapsto \xi \cdot J - f(J)$ is affine (hence convex), and a pointwise supremum of convex functions is convex. For the involution, apply (4.150) to $f^\star$ in place of f. By (ii), $\nabla f^\star$ is the inverse map of ∇f; in particular it is a bijection, so $f^{\star\star}$ is defined and, by the conjugate relation read backwards,

$$f^{\star\star}(J) = \sup_{\xi} \big(J \cdot \xi - f^\star(\xi) \big) = J \cdot \xi^\star(J) - f^\star(\xi^\star(J)), \tag{4.152}$$

where $\xi^\star(J)$ is the maximizer, characterized by $\nabla f^\star(\xi) = J$. By (ii) this last equation says $J = J^\star(\xi)$, i.e. $\xi^\star(J) = \nabla f(J)$. Substituting the value of $f^\star$ from (4.150) at $\xi = \xi^\star(J)$,

$$f^{\star\star}(J) = J \cdot \xi^\star(J) - \big[\xi^\star(J) \cdot J - f(J) \big] = f(J), \tag{4.153}$$

the cross terms cancelling identically. Hence $f^{\star\star} = f$. □

The transform is therefore an involution on smooth strictly convex functions: applied twice it returns the original. This symmetry is what makes W and Γ equal partners rather than one being primary; either determines the other, and the relation is the same in both directions. The remaining structural fact is what the transform does to second derivatives. It is the identity the previous section promised, and it is the technical heart of everything that follows.

Proposition 4.32 (Reciprocity of Hessians)**.** *Under the hypotheses of Theorem 4.31, the Hessians of f and $f^\star$ are matrix inverses of one another at conjugate points. Writing $f''(J)_{ij} = \partial^2 f / \partial J_i \partial J_j$ and $f^{\star\prime\prime}(\xi)_{ij} = \partial^2 f^\star / \partial \xi_i \partial \xi_j$, with $\xi = \nabla f(J)$,*

$$\sum_k f''(J)_{ik}\, f^{\star\prime\prime}(\xi)_{kj} \;=\; \delta_{ij}, \qquad \text{i.e.} \qquad f^{\star\prime\prime}(\xi) = \big[f''(J) \big]^{-1}. \tag{4.154}$$

Proof. We have two mutually inverse maps from Theorem 4.31(ii): the gradient $\nabla f \colon J \mapsto \xi$ and the gradient $\nabla f^\star \colon \xi \mapsto J$. Their Jacobians are exactly the Hessians:

$$\frac{\partial \xi_i}{\partial J_k} = \frac{\partial}{\partial J_k} \frac{\partial f}{\partial J_i} = f''(J)_{ik}, \qquad \frac{\partial J_k}{\partial \xi_j} = \frac{\partial}{\partial \xi_j} \frac{\partial f^\star}{\partial \xi_k} = f^{\star\prime\prime}(\xi)_{kj}. \tag{4.155}$$

Because the maps are inverse, $\xi(J(\xi)) = \xi$, so by the chain rule the product of their Jacobians is the identity:

$$\delta_{ij} = \frac{\partial \xi_i}{\partial \xi_j} = \sum_k \frac{\partial \xi_i}{\partial J_k} \frac{\partial J_k}{\partial \xi_j} = \sum_k f''(J)_{ik}\, f^{\star\prime\prime}(\xi)_{kj}. \tag{4.156}$$

This is (4.154). Equivalently, it is the inverse-function theorem applied to ∇f: the Jacobian of the inverse map is the inverse of the Jacobian. □

Three remarks transcribe these results to the functional setting and prepare the application. First, the dictionary is literal: a functional $W[J]$ of a field $J(x)$ has a "gradient" $\delta W/\delta J(x)$ and a "Hessian" $\delta^2 W/\delta J(x)\delta J(y)$, the latter a two-point kernel; (4.154) becomes the statement that two kernels are inverse in the convolution sense, $\int dz\, A(x,z)B(z,y) = \delta(x-y)$. Second, the bijectivity hypothesis on ∇f is, functionally, the invertibility of the map $J \mapsto \phi_c$, which the previous section tied to the positive-definiteness of the covariance (4.147); we assume it holds on the domain of interest, i.e. that the source uniquely determines the classical field and vice versa. Third, the strict convexity of W established in Eq. (4.135) is exactly the hypothesis of Theorem 4.31: W is a legitimate input to the Legendre transform, its conjugate is well defined, convex, and involutive, and the reciprocity (4.154) holds. We may now define the effective action.

4.5.2 The effective action and the conjugate relation

Definition 4.33 (Effective action)**.** The *effective action* is the Legendre transform of the connected generating functional with respect to the source,

$$\Gamma[\phi_c] \;\equiv\; W^\star[\phi_c] \;=\; \sup_J \Big(\int d^d x\, J(x)\,\phi_c(x) - W[J] \Big) \;=\; \int d^d x\, J(x)\,\phi_c(x) - W[J]\Big|_{J=J[\phi_c]}, \tag{4.157}$$

where $J[\phi_c]$ is the source that produces the prescribed classical field, i.e. the solution of $\phi_c(x) = \delta W/\delta J(x)$ read as an equation for J.

The defining property is the conjugate relation, which Theorem 4.31(ii) delivers directly. Differentiating (4.157) with respect to $\phi_c(y)$ and using the envelope cancellation (4.151) verbatim — now with the continuous index — the implicit variation of $J[\phi_c]$ drops out and

$$\boxed{\frac{\delta \Gamma[\phi_c]}{\delta \phi_c(x)} \;=\; J(x)\,.} \tag{4.158}$$

Let us see the cancellation once in the functional notation, since it is the relation on which the entire interpretation rests. Writing $J = J[\phi_c]$ for the stationary source,

$$\begin{aligned} \frac{\delta \Gamma}{\delta \phi_c(x)} &= J(x) + \int d^d y\, \phi_c(y)\, \frac{\delta J(y)}{\delta \phi_c(x)} - \int d^d y\, \frac{\delta W}{\delta J(y)}\, \frac{\delta J(y)}{\delta \phi_c(x)} \\ &= J(x) + \int d^d y\, \big[\phi_c(y) - \phi_c(y)\big]\, \frac{\delta J(y)}{\delta \phi_c(x)} = J(x), \end{aligned} \tag{4.159}$$

where the second line used $\delta W/\delta J(y) = \phi_c(y)$ from (4.146). The relation (4.158) is the precise sense in which Γ is dual to W: where the first derivative of W returns the field sustained by a source, the first derivative of Γ returns the source

required to sustain a field. Setting the external source to zero, $J = 0$, the physical configurations of the theory are the stationary points of the effective action,

$$\left.\frac{\delta\Gamma[\phi_c]}{\delta\phi_c(x)}\right|_{J=0} = 0, \tag{4.160}$$

the quantum-mechanical replacement for the classical equation of motion $\delta S/\delta\phi = 0$. This is why ϕ_c earned the name "classical field" in Section 4.4.5: it extremizes Γ exactly as the genuine classical field extremizes S, with Γ playing the role of an action that has absorbed all quantum fluctuations. The solution of (4.160) is the true vacuum expectation value $\langle\phi\rangle$, equation of state and all radiative corrections included.

Notation Watch: Euclidean versus Minkowski sign of the effective action

We work in Euclidean signature ($W = \log Z$, Theorem 4.27), where W is genuinely convex (4.135) and the Legendre theory of Section 4.5.1 applies as stated; this is the convention of the previous section and of the effective-potential and Wilsonian-flow chapters that consume this object (Chapters 23 and 29). In Minkowski signature the generating functional carries e^{iS} and one sets $W = -i\log Z$ (Section 4.4.2); the standard field-theory effective action is then written with the opposite overall sign,

$$\Gamma[\phi_c] = W[J] - \int d^dx\, J\,\phi_c, \qquad \frac{\delta\Gamma}{\delta\phi_c(x)} = -J(x), \tag{4.161}$$

the flip being part of the same factor-of-i bookkeeping that turned $\log Z$ into $-i\log Z$. Every structural statement below — the reciprocity, the 1PI generation, the tree-level identity — holds in both conventions; only signs and factors of i differ, and we flag them where they matter. We use the Euclidean form (4.157) for all proofs.

We can now redeem the reciprocity promise. The effective action is $W^\star$, so Theorem 4.32 applies with $f = W$, $f^\star = \Gamma$, $\xi = \phi_c$. The finite-dimensional identity (4.154) becomes, in the functional setting, an identity of two-point kernels.

Theorem 4.34 (The Hessian of Γ is the inverse propagator)**.** *At conjugate points ($\phi_c = \delta W/\delta J$), the second functional derivative of Γ is the kernel inverse of the connected two-point function,*

$$\int d^dz\, \frac{\delta^2 W[J]}{\delta J(x)\,\delta J(z)}\,\frac{\delta^2\Gamma[\phi_c]}{\delta\phi_c(z)\,\delta\phi_c(y)} = \delta^{(d)}(x-y). \tag{4.162}$$

Equivalently, with $G_c^{(2)} = \delta^2 W/\delta J\,\delta J$ the full connected propagator and $\Gamma^{(2)} \equiv$

$\delta^2\Gamma/\delta\phi_c\,\delta\phi_c$,

$$\Gamma^{(2)} = \left(G_c^{(2)}\right)^{-1}. \tag{4.163}$$

Proof. Differentiate the conjugate relation (4.158), $\delta\Gamma/\delta\phi_c(z) = J(z)$, with respect to $\phi_c(y)$, and likewise $\phi_c = \delta W/\delta J$ with respect to J. The two maps $J \mapsto \phi_c$ and $\phi_c \mapsto J$ are functional inverses, so by the chain rule, exactly as in (4.156) but with the index sum replaced by an integral over the intermediate point z,

$$\delta^{(d)}(x-y) = \frac{\delta\phi_c(x)}{\delta\phi_c(y)} = \int d^dz\, \frac{\delta\phi_c(x)}{\delta J(z)}\,\frac{\delta J(z)}{\delta\phi_c(y)} = \int d^dz\, \frac{\delta^2 W}{\delta J(x)\delta J(z)}\,\frac{\delta^2\Gamma}{\delta\phi_c(z)\delta\phi_c(y)}, \tag{4.164}$$

using (4.146) for $\delta\phi_c/\delta J = W''$ and (4.158) for $\delta J/\delta\phi_c = \Gamma''$. This is (4.162); (4.163) is the same statement in operator notation. □

Theorem 4.34 is the first concrete payoff. In a symmetric vacuum the connected two-point function $G_c^{(2)}$ *is* the full propagator of the interacting theory (Eq. (4.121)); its inverse is therefore the full inverse propagator. The theorem says that this full inverse propagator is nothing but the second derivative of Γ. At the free level we can check it against objects already in hand. For the free scalar, W is Gaussian and $G_c^{(2)}$ reduces to the Euclidean propagator $(-\partial^2 + m^2)^{-1}$ (Chapters 9 and 10); its inverse kernel is the operator $-\partial^2 + m^2$, which is exactly $\delta^2 S/\delta\phi\,\delta\phi$ for the free action. Thus at the free level $\Gamma^{(2)} = S^{(2)}$ — the second derivative of Γ equals the second derivative of the classical action. This is the $n = 2$ instance of a general fact proved below, that $\Gamma = S$ at tree level. In Minkowski signature the same computation, carried with the factors of i of (4.161) and the Feynman propagator $i/(p^2 - m^2 + i\epsilon)$, gives $\Gamma^{(2)}(p) = p^2 - m^2$, again equal to the momentum-space inverse free propagator $S^{(2)}$; the interacting $\Gamma^{(2)}(p) = p^2 - m^2 - \Pi(p^2)$ then defines the self-energy Π, the object ϕ^4 theory computes at one loop in Chapters 21 and 22.

Physical Insight

The Legendre transform exchanges "cause" and "effect." The functional $W[J]$ answers "given a source, what field response do I get, and how do connected clusters of the field correlate?" Its conjugate $\Gamma[\phi_c]$ answers the inverse question: "to hold the field at a prescribed profile ϕ_c, what source must I apply?" — and at zero source this is the equation of state of the vacuum. Because the two questions are inverse, their second derivatives are inverse kernels: the response function (propagator) and the stiffness (inverse propagator) are reciprocal, just as compliance and elastic modulus are in a material. The interacting vacuum behaves like a polarizable medium, and Γ'' measures its stiffness against being displaced.

4.5.3 Γ generates the one-particle-irreducible vertices

The reciprocity identity is the $n = 2$ member of an infinite family. Differentiating it repeatedly expresses every connected correlator $W^{(n)} = \delta^n W/\delta J^n$ in terms of the derivatives of Γ,

$$\Gamma^{(n)}(x_1, \ldots, x_n) \;\equiv\; \left.\frac{\delta^n \Gamma[\phi_c]}{\delta\phi_c(x_1)\cdots\delta\phi_c(x_n)}\right|_{\phi_c=\bar\phi}, \tag{4.165}$$

evaluated at the stationary configuration $\bar\phi$ (the solution of (4.160); $\bar\phi = 0$ in a symmetric vacuum). These $\Gamma^{(n)}$ are the *proper vertices*. The central structural theorem of this section is that they are precisely the *one-particle-irreducible* (1PI) vertices — the amputated correlation functions that cannot be disconnected by cutting a single internal line — and that the full connected correlators are reconstructed from them by summing *tree* graphs. We prove this by carrying the differentiation to fourth order, where the pattern becomes unambiguous, working in one variable (the zero-dimensional reduction) for algebraic transparency and quoting the index-carrying multivariate form alongside; the multivariate proof is identical with sums over intermediate points reinstated.

Write $W' = \phi_c$, $\Gamma' = J$ (the conjugate relations), and abbreviate $W^{(n)}$, $\Gamma^{(n)}$ for the J- and ϕ_c-derivatives respectively. The reciprocity (4.163) reads

$$W''\,\Gamma'' = 1, \qquad \text{so} \qquad W'' = \frac{1}{\Gamma''}. \tag{4.166}$$

Treat Γ'' as a function of ϕ_c and recall $d\phi_c/dJ = W''$. Differentiate (4.166) with respect to J:

$$W''' = \frac{d}{dJ}\frac{1}{\Gamma''} = -\frac{1}{(\Gamma'')^2}\,\frac{d\Gamma''}{d\phi_c}\,\frac{d\phi_c}{dJ} = -\frac{\Gamma'''}{(\Gamma'')^2}\,W'' = -\frac{\Gamma'''}{(\Gamma'')^3} = -\,(W'')^3\,\Gamma'''. \tag{4.167}$$

The connected three-point function is minus the proper three-vertex Γ''' dressed with three full propagators W'', one on each external leg. In the multivariate (functional) form, with all factors understood as kernels contracted over the suppressed intermediate points,

$$W^{(3)}_{ijk} = -\sum_{abc} W_{ia} W_{jb} W_{kc}\,\Gamma^{(3)}_{abc}. \tag{4.168}$$

Differentiate (4.167) once more, again using $d/dJ = W''\,d/d\phi_c$ on the ϕ_c-dependent factors Γ''' and Γ'':

$$\begin{aligned} W'''' &= \frac{d}{dJ}\Big[-(W'')^3\Gamma'''\Big] = -3(W'')^2\,W'''\,\Gamma''' - (W'')^3\,\frac{d\Gamma'''}{d\phi_c}\,W'' \\ &= -3(W'')^2\big[-(W'')^3\Gamma'''\big]\Gamma''' - (W'')^4\,\Gamma'''' \end{aligned}$$

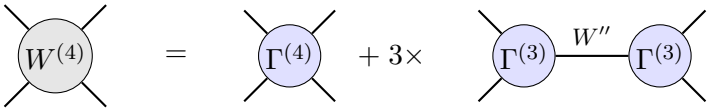


Figure 4.6: The tree expansion (4.169) of the connected four-point function. The proper vertices $\Gamma^{(n)}$ (1PI, light blobs) are joined by full propagators W'' into tree graphs; external legs also carry a W''. The single internal line in the second topology makes it one-particle-reducible, and the factor 3 is the sum over s, t, u channels. No closed loops appear: Γ generates the skeleton, and W is built from it by trees.

$$= -(W'')^4\,\Gamma'''' + 3\,(W'')^5\,(\Gamma''')^2. \tag{4.169}$$

This is the decomposition we were after. The connected four-point function is the sum of two topologically distinct contributions: a *single* proper four-vertex Γ'''' with four external propagators, and a *pair* of proper three-vertices Γ''' joined by one *internal* propagator (the fifth W'') and carrying four external propagators, the factor 3 counting the three ways of partitioning the four external legs into two pairs — the s, t, u channels. Figure 4.6 draws the two topologies. Crucially, the second graph is a *tree*: it has no closed loop, and it is one-particle-reducible (cutting the single internal W'' line disconnects it). All such reducibility in $W^{(4)}$ is exhibited explicitly, built from the irreducible pieces Γ'''. The proper vertex Γ'''' is, by contrast, what remains after every such reducible tree has been stripped: it is one-particle-irreducible by construction.

The pattern of (4.167)–(4.169) continues to all orders and is the content of the following theorem, whose inductive proof is the systematic version of the two differentiations just performed.

Theorem 4.35 (Γ generates the 1PI vertices)**.** *Each connected n-point function $W^{(n)}$ ($n \geq 3$) is the sum over all* tree *graphs whose vertices are the proper vertices $\Gamma^{(k)}$ ($3 \leq k \leq n$), whose internal and external lines are full propagators $W'' = (\Gamma'')^{-1}$, weighted by the symmetry factor of the tree and an overall sign $(-1)^{(\#\,\text{vertices})-1}$. Equivalently, the proper vertices $\Gamma^{(n)}$ are the amputated one-particle-irreducible n-point functions: those connected, amputated diagrams that remain connected after any single internal line is cut.*

Proof (induction on n). The base cases $n = 2, 3, 4$ are (4.163), (4.167), (4.169). For the inductive step, suppose $W^{(n)}$ is given by the stated tree sum. Differentiate with respect to J once more, using the operator identity

$$\frac{d}{dJ} = \sum_a \frac{\delta\phi_c(a)}{\delta J}\,\frac{\delta}{\delta\phi_c(a)} = \int W''\,\frac{\delta}{\delta\phi_c}, \tag{4.170}$$

which converts a J-derivative into a ϕ_c-derivative preceded by attachment of one full propagator W''. Acting on the tree sum for $W^{(n)}$, the new derivative either

(i) lands on one of the proper vertices $\Gamma^{(k)}$, raising it to $\Gamma^{(k+1)}$ and attaching the new external leg through the inserted W'', or (ii) lands on one of the propagator lines $W'' = (\Gamma'')^{-1}$, and since $\frac{d}{dJ}(\Gamma'')^{-1} = -(\Gamma'')^{-1}\Gamma'''(\Gamma'')^{-1}W''$ this splits the line into two propagators joined by a new proper three-vertex Γ''', again attaching the new leg through a W''. Case (i) grows an existing vertex; case (ii) inserts a new trivalent vertex onto an existing line. Both operations preserve the tree topology (no loop is ever created, since each step adds exactly one new external leg through one new propagator and either one or zero new vertices, keeping the relation $\#\text{lines} = \#\text{vertices} + \#\text{external} - 1$ that characterizes a tree), and together they generate *exactly* all trees with $n+1$ external legs from all trees with n legs, each with the correct symmetry factor and the sign incremented by one factor of -1 whenever case (ii) adds a vertex. This is the stated tree sum at order $n+1$. The irreducibility characterization follows: a graph contributing to $W^{(n)}$ that is one-particle-reducible is, by definition, cut by some internal line into two connected pieces; in the tree sum that internal line is one of the propagators W'' joining two sub-trees, so the reducible content is precisely the part of the sum with at least one internal line, leaving the single-vertex term $\Gamma^{(n)}$ (no internal line) as the irreducible remainder. □

Physical Insight

Theorem 4.35 is the organizing principle behind the entire diagrammatic expansion to come. It says the connected correlators contain no information beyond the proper vertices: W is reconstructed from Γ by drawing trees. All the genuinely new structure at each order — everything that is not just two simpler processes glued by a propagator — lives in the 1PI vertices $\Gamma^{(n)}$. This is why renormalization is organized around 1PI quantities (Chapters 21 and 22): one need only render the finitely many divergent *proper* vertices finite, and the reducible trees take care of themselves. It is also why Γ, not W, is the natural carrier of the quantum dynamics: it is the irreducible skeleton from which the flesh is hung.

4.5.4 $\Gamma = S$ at tree level and the loop expansion

We have called Γ the effective action and the quantum-corrected classical action; both names are justified by a single computation. Reinstate $\hbar$ as a bookkeeping parameter multiplying the action in the exponent — the loop-counting parameter of the next section — so that the Euclidean generating functional reads

$$Z[J] = \int \mathcal{D}\phi \, \exp\Big[-\tfrac{1}{\hbar}\big(S[\phi] - \textstyle\int J\phi\big)\Big], \qquad W[J] = \hbar \log Z[J]. \tag{4.171}$$

The normalization $W = \hbar \log Z$ keeps $\phi_c = \delta W/\delta J = \langle\phi\rangle_J$ finite as $\hbar \to 0$. In that limit the integral (4.171) is dominated by its saddle point, the configuration

ϕ_J extremizing the exponent,

$$\left.\frac{\delta S}{\delta \phi}\right|_{\phi_J} = J, \tag{4.172}$$

which is the classical equation of motion in the presence of the source. The leading-order Laplace evaluation (Section 4.2; the systematic asymptotic expansion is the business of Section 4.6) gives

$$W[J] = -\big(S[\phi_J] - \textstyle\int J\phi_J\big) + O(\hbar) = \sup_{\phi}\Big(\textstyle\int J\phi - S[\phi]\Big) + O(\hbar) = S^\star[J] + O(\hbar). \tag{4.173}$$

The supremum is over field configurations because, S being bounded below and convex near the relevant configurations, (4.172) locates its maximum; the result is precisely the Legendre transform $S^\star$ of the classical action. Now apply the effective action's own definition, $\Gamma = W^\star$, and the involution Theorem 4.31(iii):

$$\Gamma[\phi_c] = W^\star[\phi_c] = \big(S^\star\big)^\star[\phi_c] + O(\hbar) = S^{\star\star}[\phi_c] + O(\hbar) = S[\phi_c] + O(\hbar). \tag{4.174}$$

Theorem 4.36 (Tree-level identity)**.** *To leading order in the loop-counting parameter, the effective action equals the classical action,*

$$\Gamma[\phi_c] = S[\phi_c] + O(\hbar). \tag{4.175}$$

The quantum corrections begin at order $\hbar$ (one loop).

The proof is (4.173)–(4.174): W is the Legendre transform of S at leading order, and Γ is the Legendre transform of W, so by the involution Γ returns to S. The two Legendre transforms undo each other, and the tree-level effective action is simply the classical action. This is the cleanest possible statement of why Γ deserves to be called *the* effective action: it begins life, at $\hbar^0$, as the action S itself, and the loop expansion dresses it order by order in $\hbar$ with quantum corrections, $\Gamma = S + \hbar\,\Gamma_1 + \hbar^2\,\Gamma_2 + \cdots$. The $n = 2$ shadow of (4.175) is the free-level identity $\Gamma^{(2)} = S^{(2)}$ verified after Theorem 4.34; the full statement upgrades it to every vertex.

That the corrections are organized by powers of $\hbar$ as a *loop* expansion is the content of the next section, but the counting is already visible. A connected graph with V vertices and I internal lines carries, from (4.171), a factor $\hbar^{-1}$ per vertex and $\hbar$ per propagator, hence $\hbar^{I-V}$; with the topological identity $L = I - V + 1$ for the number of independent loops, this is $\hbar^{L-1}$. The tree graphs ($L = 0$) carry $\hbar^{-1}$ and assemble, through Theorem 4.35, into $\Gamma = S$ at $O(\hbar^0)$ after the overall normalization of (4.171); each additional loop costs one power of $\hbar$. Thus the loop expansion of Γ is literally its expansion in $\hbar$, with the classical action as the zeroth term.

Physical Insight

For spacetime-constant field configurations $\phi_c(x) = \varphi$, the effective action reduces to a single function, the *effective potential*: in Euclidean signature $\Gamma[\varphi] = \left(\int d^d x\right) V_{\text{eff}}(\varphi)$, the volume factor emerging because nothing depends on position. By Theorem 4.36 its leading term is the classical potential, $V_{\text{eff}}(\varphi) = V(\varphi) + O(\hbar)$, and the stationarity condition (4.160) becomes $V'_{\text{eff}}(\varphi) = 0$: the vacuum sits at the minimum of the quantum-corrected potential, not the classical one. This is the engine of spontaneous symmetry breaking, where the one-loop term can shift, tilt, or even create the minimum. We compute V_{eff} for the real scalar and read off its consequences in Chapter 29; the convexity of Γ guaranteed by Theorem 4.31(iii) is, in that context, the exact statement of the Maxwell construction that flattens V_{eff} between degenerate vacua.

4.5.5 The carried toy: the effective action of the zero-dimensional ϕ^4 model

We close by carrying ϕ^4 theory one step further, computing Γ for the zero-dimensional model and watching the 1PI structure assemble with no integrals to obscure it. We continue directly from Theorem 4.29, which gave $W(J) = \log Z(J)$ for the $N = 1$ model with $m^2 = 1$ to first order in the coupling.

Worked Example 4.37 (Effective action of the zero-dimensional ϕ^4 model)**.** Recall from (4.140) the connected generating functional of the model $Z(J) = \int dx\, \exp(-\frac{1}{2}x^2 - \frac{\lambda}{4!}x^4 + Jx)$, namely (dropping the additive constant, which no derivative sees)

$$W(J) = \tfrac{1}{2}J^2 - \frac{\lambda}{24}(J^4 + 6J^2) + O(\lambda^2). \tag{4.176}$$

The classical field is the first derivative,

$$\phi_c = W'(J) = J - \frac{\lambda}{6}J^3 - \frac{\lambda}{2}J + O(\lambda^2) = \Big(1 - \frac{\lambda}{2}\Big)J - \frac{\lambda}{6}J^3 + O(\lambda^2). \tag{4.177}$$

To build $\Gamma(\phi_c)$ we need the inverse map $J(\phi_c)$, which by the conjugate relation (4.158) equals $\Gamma'(\phi_c)$. Invert (4.177) perturbatively, writing $J = \phi_c + \lambda J_1 + O(\lambda^2)$ and matching:

$$\phi_c = \Big(1 - \tfrac{\lambda}{2}\Big)(\phi_c + \lambda J_1) - \tfrac{\lambda}{6}\phi_c^3 + O(\lambda^2) = \phi_c + \lambda\Big(J_1 - \tfrac{1}{2}\phi_c - \tfrac{1}{6}\phi_c^3\Big) + O(\lambda^2), \tag{4.178}$$

so the $O(\lambda)$ term must vanish, giving $J_1 = \frac{1}{2}\phi_c + \frac{1}{6}\phi_c^3$ and

$$\Gamma'(\phi_c) = J(\phi_c) = \phi_c + \lambda\Big(\tfrac{1}{2}\phi_c + \tfrac{1}{6}\phi_c^3\Big) + O(\lambda^2). \tag{4.179}$$

Integrate in ϕ_c (the constant of integration is irrelevant):

$$\Gamma(\phi_c) = \tfrac{1}{2}\phi_c^2 + \frac{\lambda}{24}\phi_c^4 + \frac{\lambda}{4}\phi_c^2 + O(\lambda^2). \tag{4.180}$$

The first two terms are exactly the classical action $S(\phi_c) = \frac{1}{2}\phi_c^2 + \frac{\lambda}{4!}\phi_c^4$ with $m^2 = 1$, confirming Theorem 4.36: $\Gamma = S$ plus a correction that starts at one loop. That correction is the term $\frac{\lambda}{4}\phi_c^2$, and it is precisely the one-loop functional determinant of the next section reduced to zero dimensions. Indeed the one-loop formula $\Gamma = S + \frac{1}{2}\log S''(\phi_c)$, which Section 4.6 derives in general, gives here

$$\tfrac{1}{2}\log S''(\phi_c) = \tfrac{1}{2}\log\left(1 + \tfrac{\lambda}{2}\phi_c^2\right) = \frac{\lambda}{4}\phi_c^2 + O(\lambda^2), \tag{4.181}$$

matching (4.180) term by term: the "functional determinant" is the single logarithm $\log S''$, and its $O(\lambda)$ piece is the toy's one-loop mass renormalization.

Reciprocity check. From (4.180), the proper two-point "vertex" is $\Gamma''(0) = 1 + \frac{\lambda}{2} + O(\lambda^2)$, while the connected two-point function computed in (4.141) is $W''(0) = \langle x^2\rangle_c = 1 - \frac{\lambda}{2} + O(\lambda^2)$. Their product is

$$\Gamma''(0)\,W''(0) = \left(1 + \tfrac{\lambda}{2}\right)\left(1 - \tfrac{\lambda}{2}\right) + O(\lambda^2) = 1 + O(\lambda^2), \tag{4.182}$$

verifying $\Gamma''W'' = 1$ (Eq. (4.163)) to the order computed.

1PI structure. The proper four-vertex is, from (4.180),

$$\Gamma^{(4)}(0) = \frac{d^4}{d\phi_c^4}\Big[\frac{\lambda}{24}\phi_c^4\Big] = \lambda + O(\lambda^2), \tag{4.183}$$

the classical coupling itself — the zero-dimensional 1PI vertex is the bare λ at tree level (the Euclidean avatar of the Minkowski Feynman-rule vertex $-i\lambda$ that ϕ^4 theory meets in Chapter 15). The model is $\mathbb{Z}_2$-symmetric, so $\Gamma^{(3)} = 0$ and the tree formula (4.169) collapses to its first term,

$$W^{(4)}(0) = -\Gamma^{(4)}(0)\,\big(W''(0)\big)^4 = -\lambda\cdot(1)^4 + O(\lambda^2) = -\lambda + O(\lambda^2), \tag{4.184}$$

which is exactly the connected four-point function $\langle x^4\rangle_c = -\lambda$ found independently in (4.142). The 1PI structure is thus completely explicit: the connected $W^{(4)}$ is the single proper vertex $\Gamma^{(4)} = \lambda$ with its four legs amputated — in zero dimensions "amputation" is just stripping four factors of the full propagator $W''(0) = 1 + O(\lambda)$ — and there is no reducible piece because $\Gamma^{(3)}$ vanishes. The same arithmetic in a theory with a cubic coupling would populate the second topology of Fig. 4.6.

This completes the toolbox's central construction. We have built $\Gamma[\phi_c]$ as the Legendre transform of $W[J]$, proved it convex and involutive with the source

as conjugate variable ($\delta\Gamma/\delta\phi_c = J$), shown its Hessian to be the full inverse propagator ($\Gamma'' W'' = \mathbf{1}$), identified its higher derivatives as the 1PI vertices from which all connected amplitudes assemble as trees, and established that it reduces to the classical action S at tree level, with quantum corrections organized in powers of $\hbar$. The tree-level identity $\Gamma = S$ already exposed, in (4.173), the object the final section needs: the leading-order evaluation of the functional integral by its saddle point. Section 4.6 makes that evaluation systematic — the Laplace expansion with controlled error, transcribed to the functional integral as the loop expansion, with the one-loop term emerging as the functional determinant $\frac{1}{2}\log\det S''$ whose zero-dimensional shadow we just met in (4.181). The effective action built here is the stationary-action functional that Chapter 23 flows under the Wilsonian renormalization group and that Chapter 29 evaluates as the effective potential governing spontaneous symmetry breaking.

4.6 Saddle Points, Functional Determinants, and the Loop Expansion

The previous section handed us the effective action $\Gamma[\phi_c]$, the Legendre transform of $W[J]$, together with the single most important fact about it: at lowest order it is the classical action, $\Gamma[\phi_c] = S[\phi_c]$, and its second variation $\delta^2\Gamma/\delta\phi_c^2$ is the inverse of the connected two-point function (Section 4.5). What we did not yet do is *compute* Γ beyond that leading statement. To do so we must finally confront the integral that defines $W[J]$ and learn to evaluate it. We have no hope of doing the functional integral exactly outside the Gaussian case of Section 4.2; what we can do is evaluate it *asymptotically*, dominated by the configurations that extremize the exponent. This is the method of steepest descent, and transcribing it to the functional integral is the single most productive idea in perturbative quantum field theory. It produces the loop expansion, exposes it as an expansion in powers of $\hbar$ about the classical limit, and identifies its first quantum correction as a functional determinant of the second-variation operator $\delta^2 S/\delta\phi\,\delta\phi$ already isolated in Section 4.1.

We build the method honestly: first in one finite-dimensional integral, with the error term controlled; then in n dimensions, where the Gaussian prefactor becomes a determinant of the Hessian; then we organize the corrections as connected vacuum graphs and read off the loop-counting rule. Only then do we restore $\hbar$, transcribe everything to the functional integral, and write down the one-loop effective action. We close by carrying the zero-dimensional ϕ^4 toy of ϕ^4 theory to its asymptotic loop series and proving that this series has zero radius of convergence — the first appearance of a theme (the divergence of perturbation theory) that Chapter 5 and Chapter 42 develop in earnest.

4.6.1 Laplace's method in one dimension

Consider the one-parameter family of integrals

$$I(\lambda) = \int_a^b g(x)\, e^{-\lambda f(x)}\, \mathrm{d}x, \qquad \lambda \to +\infty, \tag{4.185}$$

with $-\infty \le a < b \le +\infty$. The idea is elementary and physical: as λ grows, $e^{-\lambda f}$ becomes overwhelmingly concentrated near the point where f is smallest, so the integral is controlled by the value of g there and by the local curvature of f. The content of the method is to make "overwhelmingly" quantitative and to bound the error.

Theorem 4.38 (Laplace's method, one dimension)**.** *Let $f, g \in C^\infty$ on $[a,b]$. Suppose f attains its global minimum over $[a,b]$ at a single interior point $x_0 \in (a,b)$, that this minimum is nondegenerate, $f'(x_0) = 0$ and $f''(x_0) > 0$, and that the integral (4.185) converges absolutely for some $\lambda = \lambda_0$. Then as $\lambda \to +\infty$,*

$$I(\lambda) = e^{-\lambda f(x_0)} \sqrt{\frac{2\pi}{\lambda f''(x_0)}} \left[g(x_0) + \frac{c_1}{\lambda} + \frac{c_2}{\lambda^2} + \cdots \right], \tag{4.186}$$

an asymptotic expansion in $1/\lambda$ whose leading term carries a relative error $O(1/\lambda)$. The coefficients c_k are universal polynomials in the derivatives of f and g at x_0; c_1 is computed in Theorem 4.39.

Proof. Write $\phi(x) \equiv f(x) - f(x_0) \ge 0$, so that $\phi(x_0) = 0$ and $\phi(x) > 0$ for $x \neq x_0$; pull the constant $e^{-\lambda f(x_0)}$ out front. It remains to show

$$J(\lambda) \equiv \int_a^b g(x)\, e^{-\lambda \phi(x)}\, \mathrm{d}x = \sqrt{\frac{2\pi}{\lambda f''(x_0)}} \left[g(x_0) + O(1/\lambda) \right]. \tag{4.187}$$

Step 1: localization. Fix $\delta > 0$ small enough that $[x_0 - \delta, x_0 + \delta] \subset (a,b)$ and $f'' > 0$ throughout this interval (possible since f'' is continuous and $f''(x_0) > 0$). Split $J = J_{\mathrm{in}} + J_{\mathrm{out}}$ over $|x - x_0| < \delta$ and $|x - x_0| \ge \delta$. On the outer region ϕ is bounded below by a positive constant, $\eta \equiv \inf_{|x-x_0|\ge\delta} \phi(x) > 0$, because x_0 is the *unique* global minimum. For $\lambda > \lambda_0$ we estimate, using $\phi \ge \eta$ on the outer region and absolute convergence at λ_0,

$$|J_{\mathrm{out}}| \le \int_{|x-x_0|\ge\delta} |g|\, e^{-(\lambda-\lambda_0)\phi}\, e^{-\lambda_0 \phi}\, \mathrm{d}x \le e^{-(\lambda-\lambda_0)\eta} \int_a^b |g|\, e^{-\lambda_0 \phi}\, \mathrm{d}x = C\, e^{-(\lambda-\lambda_0)\eta}, \tag{4.188}$$

with $C < \infty$ independent of λ. The outer contribution is therefore exponentially small in λ — smaller than every power λ^{-k} — and contributes nothing to the asymptotic series.

Step 2: rescaling the core. On the inner interval Taylor's theorem with remainder gives

$$\phi(x) = \tfrac{1}{2} f''(x_0)(x-x_0)^2 + \rho(x), \qquad |\rho(x)| \le M\,|x-x_0|^3, \tag{4.189}$$

where $M = \frac{1}{6}\sup_{|x-x_0|\le\delta}|f'''| < \infty$. Substitute $x = x_0 + u/\sqrt{\lambda}$, so $\mathrm{d}x = \lambda^{-1/2}\mathrm{d}u$ and the inner region becomes $|u| \le \delta\sqrt{\lambda}$:

$$J_{\mathrm{in}} = \frac{1}{\sqrt{\lambda}} \int_{|u|\le\delta\sqrt{\lambda}} g\Big(x_0 + \tfrac{u}{\sqrt{\lambda}}\Big) \exp\Big(-\tfrac{1}{2} f''(x_0)\,u^2 - \lambda\,\rho(x_0 + \tfrac{u}{\sqrt{\lambda}})\Big)\,\mathrm{d}u. \tag{4.190}$$

The remainder in the exponent obeys $|\lambda\,\rho| \le M|u|^3/\sqrt{\lambda}$, so it vanishes pointwise as $\lambda\to\infty$; likewise $g(x_0+u/\sqrt{\lambda}) \to g(x_0)$. We need a λ-independent integrable majorant to pass the limit inside. Shrink δ if necessary so that $M\delta \le \frac{1}{4}f''(x_0)$. Then for $|u| \le \delta\sqrt{\lambda}$ we have $|u|^3/\sqrt{\lambda} = |u|^2\cdot(|u|/\sqrt{\lambda}) \le \delta\,u^2$, hence $|\lambda\rho| \le M\delta\,u^2 \le \frac{1}{4}f''(x_0)u^2$, and the integrand is bounded in modulus by

$$\Big(\sup_{[x_0-\delta,x_0+\delta]}|g|\Big)\, e^{-\frac{1}{2}f''(x_0)u^2 + \frac{1}{4}f''(x_0)u^2} = \Big(\sup|g|\Big)\, e^{-\frac{1}{4}f''(x_0)u^2}, \tag{4.191}$$

which is integrable over $\mathbb{R}$ and independent of λ. By dominated convergence,

$$\sqrt{\lambda}\,J_{\mathrm{in}} \xrightarrow{\lambda\to\infty} g(x_0)\int_{-\infty}^{\infty} e^{-\frac{1}{2}f''(x_0)u^2}\,\mathrm{d}u = g(x_0)\sqrt{\frac{2\pi}{f''(x_0)}}, \tag{4.192}$$

which is (4.187) at leading order. Combining with the exponentially small J_{out} proves the leading term of (4.186).

Step 3: the full series. To extract the subleading coefficients, do not take the limit but instead expand the two λ-dependent factors in (4.190) in powers of $\lambda^{-1/2}$:

$$g\Big(x_0+\tfrac{u}{\sqrt{\lambda}}\Big) = \sum_{m\ge 0}\frac{g^{(m)}(x_0)}{m!}\,\frac{u^m}{\lambda^{m/2}}, \qquad e^{-\lambda\rho} = \exp\Big(-\sum_{j\ge 3}\frac{f^{(j)}(x_0)}{j!}\,\frac{u^j}{\lambda^{(j-2)/2}}\Big). \tag{4.193}$$

Each factor is a power series in $\lambda^{-1/2}$ with coefficients that are polynomials in u. Multiplying out and integrating term by term against the Gaussian weight $e^{-\frac{1}{2}f''(x_0)u^2}$, every *odd* power of u integrates to zero by parity, so only integer powers of $1/\lambda$ survive — this is why (4.186) is a series in $1/\lambda$ rather than $1/\sqrt{\lambda}$. Extending each truncated u-integral from $|u| \le \delta\sqrt{\lambda}$ to all of $\mathbb{R}$ costs only an exponentially small error of the type (4.188), and the Gaussian moments $\int u^{2k}e^{-\frac{1}{2}f''u^2}\mathrm{d}u = (2k-1)!!\,(f'')^{-k}\sqrt{2\pi/f''}$ are finite. Truncating (4.193) after the terms that produce λ^{-N} leaves a remainder bounded, by the same dominated-convergence estimate applied to the first omitted monomial, by $O(\lambda^{-N-1})$. This is precisely the statement that (4.186) is a genuine asymptotic expansion. $\square$

The mechanism in Step 3 is worth naming, because it recurs verbatim in the field theory. The quadratic part of f supplies a Gaussian measure; the cubic and higher derivatives of f are *interaction vertices*; the factor g is an *insertion*; and the asymptotic series is generated by Wick-contracting the vertices and insertions against the Gaussian. The bookkeeping device — expand the exponential of the non-Gaussian part, contract with Gaussian moments — is the entire content of perturbation theory, stripped of spacetime.

Worked Example 4.39 (The first Laplace correction)**.** We compute c_1 in (4.186) for $g \equiv 1$, the case relevant to a partition function. Abbreviate $A \equiv f''(x_0)$, $B \equiv f'''(x_0)$, $C \equiv f''''(x_0)$. Only the contributions of order $1/\lambda$ are kept. From (4.193) with $g \equiv 1$, the exponent of the non-Gaussian factor is $-\lambda\rho = -\frac{B}{6}\frac{u^3}{\sqrt{\lambda}} - \frac{C}{24}\frac{u^4}{\lambda} + \cdots$. Expanding $e^{-\lambda\rho}$ and retaining all terms of order λ^{-1} in the integrand,

$$e^{-\lambda\rho} = 1 - \frac{C}{24}\frac{u^4}{\lambda} + \frac{1}{2}\Big(\frac{B}{6}\Big)^2\frac{u^6}{\lambda} + O(\lambda^{-3/2}), \tag{4.194}$$

where the u^6 term is the square of the cubic vertex (the cross term $u^3 \cdot u^4$ is odd and the linear u^3 term is odd, so both vanish on integration). Writing $\langle u^{2k}\rangle \equiv (2k-1)!!\,A^{-k}$ for the normalized Gaussian moments, the bracket in (4.186) equals $1 + \lambda^{-1}c_1 + \cdots$ with

$$c_1 = -\frac{C}{24}\langle u^4\rangle + \frac{B^2}{72}\langle u^6\rangle = -\frac{C}{24}\frac{3}{A^2} + \frac{B^2}{72}\frac{15}{A^3} = -\frac{f''''(x_0)}{8\,f''(x_0)^2} + \frac{5\,f'''(x_0)^2}{24\,f''(x_0)^3}. \tag{4.195}$$

The two terms are exactly the two connected vacuum diagrams at this order: the quartic "figure-eight" (one C vertex, two propagators A^{-1}) and the cubic "sunset" (two B vertices, three propagators). We will meet them again, now drawn as Feynman graphs, in Section 4.6.2.

4.6.2 Steepest descent in n dimensions and the determinant prefactor

The step that matters for field theory is the passage from one integration variable to many, because the functional integral is the limit of an n-dimensional integral as $n \to \infty$. Consider

$$I(\lambda) = \int_{\mathbb{R}^n} g(\mathbf{x})\, e^{-\lambda f(\mathbf{x})}\, \mathrm{d}^n x, \tag{4.196}$$

with f attaining a unique, nondegenerate global minimum at $\mathbf{x}_0$: the gradient vanishes, $\nabla f(\mathbf{x}_0) = 0$, and the Hessian $H_{ij} \equiv \partial_i\partial_j f(\mathbf{x}_0)$ is positive definite. This is the finite-dimensional ancestor of the saddle-point condition $\delta S/\delta\phi = 0$ and of the second-variation operator $\delta^2 S/\delta\phi\,\delta\phi$.

Theorem 4.40 (Laplace's method, n dimensions)**.** *Under the hypotheses above, with f, g smooth and the integral absolutely convergent for some λ_0, as $\lambda \to +\infty$*

$$I(\lambda) = e^{-\lambda f(\mathbf{x}_0)} \left(\frac{2\pi}{\lambda}\right)^{n/2} \frac{1}{\sqrt{\det H}} \left[g(\mathbf{x}_0) + O(1/\lambda)\right]. \tag{4.197}$$

Proof. The localization argument of Theorem 4.38 carries over unchanged: outside any ball $|\mathbf{x} - \mathbf{x}_0| \geq \delta$ the exponent exceeds its minimum by a fixed positive amount and the contribution is exponentially small. For the core, diagonalize the Hessian. Since H is real symmetric positive definite, there is an orthogonal O with $O^{\mathsf{T}} H O = \operatorname{diag}(\omega_1, \dots, \omega_n)$, all $\omega_i > 0$. Change variables $\mathbf{x} - \mathbf{x}_0 = O\,\mathbf{y}$ (Jacobian 1, since $\det O = \pm 1$), then rescale $y_i = u_i/\sqrt{\lambda}$. The quadratic part of the exponent becomes $\frac{1}{2}\sum_i \omega_i u_i^2$, the measure becomes $\lambda^{-n/2}\prod_i \mathrm{d}u_i$, and exactly as in one dimension the cubic and higher terms are suppressed by powers of $\lambda^{-1/2}$. Dominated convergence gives, at leading order,

$$I(\lambda) \sim e^{-\lambda f(\mathbf{x}_0)}\, \lambda^{-n/2}\, g(\mathbf{x}_0) \prod_{i=1}^{n} \int_{-\infty}^{\infty} e^{-\frac{1}{2}\omega_i u_i^2}\, \mathrm{d}u_i = e^{-\lambda f(\mathbf{x}_0)}\, \lambda^{-n/2}\, g(\mathbf{x}_0) \prod_{i=1}^{n} \sqrt{\frac{2\pi}{\omega_i}}. \tag{4.198}$$

Now $\prod_i \omega_i = \det H$, so $\prod_i \sqrt{2\pi/\omega_i} = (2\pi)^{n/2} (\det H)^{-1/2}$, which is (4.197). □

Equation (4.197) is the formula on which everything below rests. The exponential is the value of the integrand at the saddle; the power of λ counts the dimensions; and the Gaussian fluctuations around the saddle produce the prefactor $(\det H)^{-1/2}$, the determinant of the Hessian. When we let the integral become functional, H becomes the second-variation operator $\delta^2 S/\delta\phi\,\delta\phi$ and $(\det H)^{-1/2}$ becomes a *functional determinant*. The whole one-loop apparatus is contained, in embryo, in this prefactor.

Corrections as connected vacuum graphs. To organize the subleading terms, set $g \equiv 1$ and write the exponent as its quadratic part plus everything else,

$$f(\mathbf{x}) = f(\mathbf{x}_0) + \tfrac{1}{2}(\mathbf{x}-\mathbf{x}_0)^{\mathsf{T}} H (\mathbf{x}-\mathbf{x}_0) + V(\mathbf{x}-\mathbf{x}_0), \qquad V = \sum_{r \geq 3} \frac{1}{r!} f_{i_1 \cdots i_r}\, \xi^{i_1} \cdots \xi^{i_r}, \tag{4.199}$$

where $\xi \equiv \mathbf{x} - \mathbf{x}_0$ and $f_{i_1 \cdots i_r}$ are the higher derivatives of f at $\mathbf{x}_0$. Factoring out the saddle and the Gaussian, define the normalized Gaussian average

$$\langle\, \mathcal{O} \,\rangle \equiv \frac{\displaystyle\int \mathrm{d}^n\xi\, \mathcal{O}\, e^{-\frac{\lambda}{2}\xi^{\mathsf{T}} H \xi}}{\displaystyle\int \mathrm{d}^n\xi\, e^{-\frac{\lambda}{2}\xi^{\mathsf{T}} H \xi}}, \qquad \left\langle \xi^i \xi^j \right\rangle = \frac{1}{\lambda}\,(H^{-1})^{ij}. \tag{4.200}$$

The key fact, the finite-dimensional avatar of the propagator, is the second line: each contraction of two fluctuation variables carries one inverse power of λ. Then

$$I(\lambda) = e^{-\lambda f(\mathbf{x}_0)} \left(\tfrac{2\pi}{\lambda}\right)^{n/2} (\det H)^{-1/2} \left\langle e^{-\lambda V(\xi)} \right\rangle, \tag{4.201}$$

and $\langle e^{-\lambda V} \rangle$ is evaluated by expanding the exponential and Wick-contracting, exactly as in Theorem 4.39. By the linked-cluster theorem — which states that the logarithm of a sum of all Wick contractions equals the sum over *connected* contractions, proved for the generating functional in Section 4.4.3 — the logarithm collects only connected vacuum graphs:

$$\log \left\langle e^{-\lambda V} \right\rangle = \sum_{\Gamma \text{ connected vacuum}} \frac{1}{\sigma(\Gamma)} \, \lambda^{V(\Gamma) - I(\Gamma)} \, (\text{amplitude}), \tag{4.202}$$

with $V(\Gamma)$ the number of vertices, $I(\Gamma)$ the number of internal lines (contractions), and $\sigma(\Gamma)$ the symmetry factor. Each vertex of (4.199) carries one factor of λ (it comes from $-\lambda V$); each internal line carries λ^{-1} by (4.200); hence the power λ^{V-I}.

The loop-counting relation. The exponent $V - I$ is not arbitrary: it is fixed by the topology of the graph. For a connected graph the number of independent loops — the first Betti number L — obeys

$$\boxed{L = I - V + 1}. \tag{4.203}$$

Proof. Pick a spanning tree of the connected graph: a maximal subset of internal lines containing no loop and touching every vertex. A tree on V vertices has exactly $V - 1$ lines. Each of the remaining $I - (V - 1)$ internal lines, when added back, closes exactly one independent loop, and these are a basis for the loop space; hence $L = I - (V - 1) = I - V + 1$. □

There is a second reading of (4.203) that we will use constantly once momenta appear: each internal line carries an integration variable (a loop momentum), each vertex imposes one conservation constraint, and one such constraint is redundant (it fixes only the overall external balance), leaving $I - (V - 1) = L$ free integrations — one per loop. Either way, combining (4.203) with the power λ^{V-I} gives

$$\lambda^{V(\Gamma) - I(\Gamma)} = \lambda^{-(L(\Gamma) - 1)} = \lambda^{1 - L(\Gamma)}. \tag{4.204}$$

A connected vacuum graph with L loops is suppressed by $\lambda^{-(L-1)}$ relative to the Gaussian (one-loop) term. The leading Gaussian factor $(\det H)^{-1/2}$ is itself the $L = 1$ contribution; the figure-eight and sunset of Theorem 4.39 have $L = 2$ and contribute the $1/\lambda$ correction; and so on. The small parameter that orders the whole expansion is $1/\lambda$, and it counts loops. Figure 4.7 shows the two-loop graphs.

figure-eight: $V=1,\ I=2,\ L=2$ sunset: $V=2,\ I=3,\ L=2$

Figure 4.7: The two connected two-loop vacuum graphs that produce the first correction c_1/λ in Laplace's method, Theorem 4.39. Each obeys $L=I-V+1=2$, so each is suppressed by one power of $1/\lambda$ relative to the Gaussian determinant. These are the finite-dimensional ancestors of the two-loop vacuum diagrams of ϕ^4 theory.

4.6.3 Restoring $\hbar$: the saddle expansion is the loop expansion

We now transcribe the finite-dimensional method to the functional integral and identify the expansion parameter physically. The generating functional of Section 4.5, with $\hbar$ restored to the exponent, is

$$Z[J]=e^{iW[J]/\hbar}=\int \mathcal{D}\phi\ \exp\Big[\tfrac{i}{\hbar}\Big(S[\phi]+\textstyle\int J\phi\Big)\Big]. \tag{4.205}$$

The integral (4.205) is the formal $n\to\infty$ limit of (4.196) with the identification $\lambda\leftrightarrow 1/\hbar$ and a rotation of the contour that turns the oscillatory $e^{iS/\hbar}$ into the convergent steepest-descent $e^{-(\,\cdot\,)/\hbar}$; the rotation is the Wick rotation, defined and justified in Chapter 10, and at the level of asymptotics it does not affect the combinatorics. With that identification, $\hbar\to 0$ is exactly the $\lambda\to\infty$ limit, and *the steepest-descent expansion in $1/\lambda$ becomes an expansion in powers of $\hbar$.*

The saddle point is the stationary configuration of the exponent,

$$\frac{\delta}{\delta\phi}\Big(S[\phi]+\textstyle\int J\phi\Big)\Big|_{\phi=\phi_J}=0 \quad\Longleftrightarrow\quad \frac{\delta S}{\delta\phi(x)}\Big|_{\phi_J}+J(x)=0, \tag{4.206}$$

which is nothing but the classical equation of motion in the presence of the source J. The leading term of W is the action at this saddle, $W[J]=S[\phi_J]+\int J\phi_J+O(\hbar)$, reproducing the tree-level identification $\Gamma=S$ of Section 4.5. The classical solution *is* the $\hbar\to 0$ limit; quantum corrections are the fluctuations around it.

To count $\hbar$ in the corrections, write $\phi=\phi_J+\sqrt{\hbar}\,\eta$ and expand the exponent. The shift by the factor $\sqrt{\hbar}$ is chosen precisely so the quadratic fluctuation term is $\hbar$-independent. Using the saddle condition (4.206) to kill the linear term,

$$\begin{aligned}\frac{1}{\hbar}\Big(S[\phi]+\textstyle\int J\phi\Big)=\frac{1}{\hbar}\Big(S[\phi_J]+\int J\phi_J\Big)+\tfrac12\textstyle\int\eta\,\frac{\delta^2 S}{\delta\phi\,\delta\phi}\Big|_{\phi_J}\eta\\ +\sqrt{\hbar}\,\frac{1}{3!}\int S^{(3)}[\phi_J]\,\eta^3+\hbar\,\frac{1}{4!}\int S^{(4)}[\phi_J]\,\eta^4+\cdots,\end{aligned} \tag{4.207}$$

where $S^{(r)}[\phi_J] \equiv \delta^r S/\delta\phi^r|_{\phi_J}$ and the schematic $\int \eta^r$ denotes the r-fold spacetime integral against $S^{(r)}$. A vertex with r legs thus carries a factor $\hbar^{r/2-1}$: cubic vertices come with $\sqrt{\hbar}$, quartic with $\hbar$, and so on. The Gaussian contraction $\langle \eta\,\eta \rangle$ is now $\hbar$-independent (the $\hbar$ was scaled out by the $\sqrt{\hbar}$ in $\phi = \phi_J + \sqrt{\hbar}\eta$).

For a connected vacuum graph with V vertices and I internal lines, the total power of $\hbar$ from the vertices is $\sum_v (\frac{r_v}{2} - 1)$. Because every internal line joins two vertex-legs, $\sum_v r_v = 2I$, so the total is $I - V = L - 1$ by the loop relation (4.203). We have proved the central combinatorial fact:

Proposition 4.41 (Loop expansion = $\hbar$ expansion)**.** *In the saddle-point evaluation of (4.205), a connected vacuum graph with L loops contributes at order $\hbar^{L-1}$ to $\log Z$ and at order $\hbar^L$ to $W[J] = -i\hbar \log Z$. Equivalently, if one keeps $S/\hbar$ without rescaling the field, each propagator carries $\hbar$ and each vertex carries $\hbar^{-1}$, so a connected graph carries $\hbar^{I-V} = \hbar^{L-1}$; the overall factor $\hbar$ in $W = -i\hbar \log Z$ raises this to $\hbar^L$. The expansion of W (and of Γ) in powers of $\hbar$ is therefore an expansion in the number of loops.*

The two counting routes in Theorem 4.41 agree because they differ only by the field redefinition $\phi = \phi_J + \sqrt{\hbar}\,\eta$, whose Jacobian is a field-independent constant absorbed into the normalization of $\mathcal{D}\phi$. The physical reading is the content of the insight box at the end of this section: $\hbar$ is the size of quantum fluctuations, and counting loops is counting powers of $\hbar$, i.e. counting how far one has moved from the classical limit.

4.6.4 The one-loop effective action and the functional determinant

We can now extract the first quantum correction to the effective action. Keep only the terms through order $\hbar$ in Theorem 4.41. From (4.207), dropping the interaction vertices (which start at order $\hbar$ in $\log Z$ through two loops), the fluctuation integral is purely Gaussian:

$$Z[J] = \exp\Big[\tfrac{i}{\hbar}\big(S[\phi_J] + \textstyle\int J\phi_J\big)\Big] \int \mathcal{D}\eta\, \exp\Big[\tfrac{i}{2}\textstyle\int \eta\, \frac{\delta^2 S}{\delta\phi\,\delta\phi}\Big|_{\phi_J} \eta\Big]\,(1 + O(\hbar)). \tag{4.208}$$

The fluctuation integral is the infinite-dimensional Gaussian of Section 4.2. With the operator

$$\mathcal{O}_J(x,y) \equiv \frac{\delta^2 S}{\delta\phi(x)\,\delta\phi(y)}\Big|_{\phi=\phi_J}, \tag{4.209}$$

the Gaussian formula $\int \mathcal{D}\eta\, \exp(\frac{i}{2}\int \eta\, \mathcal{O}_J\, \eta) = \mathcal{N}\,(\det \mathcal{O}_J)^{-1/2}$, where $\mathcal{N}$ is the field-independent normalization, gives

$$W[J] = S[\phi_J] + \int J\phi_J + \frac{i\hbar}{2} \log\det \mathcal{O}_J + O(\hbar^2), \tag{4.210}$$

using $W = -i\hbar \log Z$ and $\log(\det \mathcal{O}_J)^{-1/2} = -\frac{1}{2}\log\det \mathcal{O}_J$, with $\mathcal{N}$ absorbed into the (physically irrelevant) additive normalization of W.

To convert this into a statement about Γ rather than W, we use the background-field reading of the Legendre transform. The cleanest route is to choose the source so that the classical field is a prescribed background $\bar\phi$, i.e. set $\phi_c = \bar\phi$ and shift $\phi = \bar\phi + \eta$ from the start. Then $\Gamma[\bar\phi]$ is the generating functional of one-particle-irreducible (1PI) vacuum graphs computed in the presence of $\bar\phi$ (Section 4.5): the linear-in-η term is removed by the very definition of the Legendre transform (it is cancelled against J), and disconnected and one-particle-reducible pieces drop out. At one loop there is a single 1PI vacuum graph — the closed loop of the fluctuation η in the quadratic potential $S^{(2)}[\bar\phi]$ — and it is exactly the Gaussian determinant. Hence

$$\boxed{\Gamma[\bar\phi] = S[\bar\phi] + \frac{i\hbar}{2}\log\det\left(\frac{\delta^2 S}{\delta\phi\,\delta\phi}\Big|_{\bar\phi}\right) + O(\hbar^2)} \tag{4.211}$$

With $\hbar = 1$ this is the formula promised by this section: $\Gamma = S + \frac{i}{2}\log\det S'' +$ higher loops. The tree term $S[\bar\phi]$ is the classical action; the one-loop term is the logarithm of the functional determinant of the second-variation operator; the two-loop and higher terms are the sums of connected 1PI vacuum graphs with $L \geq 2$, each carrying $\hbar^{L-1}$ more than the tree.

What the functional determinant means, and why it diverges. The object $\det \mathcal{O}_J$ is defined as the product of eigenvalues of the operator (4.209). If $\mathcal{O}_J$ has a discrete spectrum $\{\lambda_n\}$ (as it does, for example, on a compact Euclidean spacetime, or after putting the theory in a box), then

$$\log\det \mathcal{O}_J = \operatorname{Tr}\log \mathcal{O}_J = \sum_n \log\lambda_n. \tag{4.212}$$

This is a formal expression: in field theory the spectrum is unbounded above, $\lambda_n \to \infty$, so the sum $\sum_n \log\lambda_n$ diverges. The divergence is the ultraviolet divergence of the one-loop vacuum energy, seen here in its cleanest guise — it is the statement that summing the zero-point energies $\frac{1}{2}\hbar\omega$ of infinitely many fluctuation modes gives infinity. It is a genuine divergence of the unrenormalized expression, not a flaw in the derivation, and it is cured by the renormalization program of Part VI; in the regulated, renormalized theory (4.211) delivers finite, physical one-loop corrections such as the Coleman–Weinberg effective potential of Chapter 29.

To make (4.212) a well-defined finite object before renormalization one introduces a *regularized determinant*. Two standard definitions, equivalent where both apply, are

- the *zeta-function determinant*: from the spectral zeta function $\zeta_{\mathcal{O}}(s) \equiv \sum_n \lambda_n^{-s}$, analytically continued from the region $\mathrm{Re}\, s$ large where it converges, one defines $\log\det \mathcal{O} \equiv -\zeta'_{\mathcal{O}}(0)$, which is finite because $\zeta_{\mathcal{O}}(s)$ is analytic at $s = 0$ for the elliptic operators of interest;
- the *heat-kernel* (proper-time) representation: $\log\det \mathcal{O} = -\int_0^\infty \frac{\mathrm{d}t}{t}\ \mathrm{Tr}\ e^{-t\mathcal{O}}$, regulated by a short-time cutoff $t \geq \epsilon^2$ whose divergences as $\epsilon \to 0$ are local and absorbed into counterterms.

We *state* these here so that (4.211) is not left dangling on a divergent symbol; the constructions are carried out in full, including the analytic continuation of $\zeta_{\mathcal{O}}$ and the small-t heat-kernel (Seeley–DeWitt) expansion, in Chapter E. The one-loop effective potential of ϕ^4 and the running it encodes are computed there and in Chapters 23 and 29; the genuine ϕ^4 theory loop integrals begin at Chapter 14.

Physical Insight

The loop expansion is the semiclassical expansion. Restoring $\hbar$ in $e^{iS/\hbar}$ shows that $\hbar$ multiplies the fluctuations and $1/\hbar$ multiplies the classical action; the stationary-phase (saddle) point is the classical trajectory, and each additional loop is one more power of $\hbar$ — one step further from classical mechanics into the quantum. The effective action $\Gamma[\bar{\phi}]$ is the object that makes this precise: it is the quantity whose own stationarity, $\delta\Gamma/\delta\bar{\phi} = 0$ at $J = 0$, gives the *full quantum* equations of motion for the field expectation value, just as $\delta S/\delta\phi = 0$ gives the classical ones. In this sense Γ is the "quantum-corrected action": replace S by Γ and treat the resulting variational problem classically, and you have resummed all the quantum fluctuations into the equations of motion. The one-loop term $\frac{i}{2}\hbar \log\det S''$ is the leading such correction — the back-reaction of the zero-point fluctuations of every mode on the dynamics of the background.

4.6.5 The zero-dimensional ϕ^4 integral as an asymptotic series

We close by carrying ϕ^4 theory to the point this section makes possible. Strip spacetime down to a single point, so the field is one real number $\phi \equiv x$ and the functional integral is one ordinary integral. With the Euclidean ϕ^4 action, $S_E = \frac{1}{2}\phi^2 + \frac{\lambda}{4!}\phi^4$ (mass set to one), the partition function is

$$Z(\lambda) = \int_{-\infty}^{\infty} \mathrm{d}x\ \exp\Big(-\tfrac{1}{2}x^2 - \tfrac{\lambda}{4!}\,x^4\Big), \qquad \lambda \geq 0, \tag{4.213}$$

the interacting continuation of the Gaussian ($\lambda = 0$) integral of Section 4.2. It converges for every $\lambda \geq 0$ (and, as an analytic function, for all complex λ with $\mathrm{Re}\,\lambda > 0$), so $Z(\lambda)$ is a perfectly well-defined, finite number for each such λ. The question is what its perturbative (loop) expansion in λ looks like.

Worked Example 4.42 (The asymptotic loop series of the zero-dimensional toy)**.** *The coefficients.* Expand the interaction and integrate term by term, using the Gaussian moments $\int_{-\infty}^{\infty} x^{2k} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}\,(2k-1)!! = \sqrt{2\pi}\,(2k)!/(2^k k!)$:

$$Z(\lambda) = \sum_{n=0}^{\infty} \frac{1}{n!}\Big(-\frac{\lambda}{4!}\Big)^n \int_{-\infty}^{\infty} x^{4n} e^{-x^2/2}\,\mathrm{d}x = \sqrt{2\pi}\sum_{n=0}^{\infty} a_n\,\lambda^n, \qquad a_n = \frac{(-1)^n\,(4n)!}{n!\,(2n)!\,96^{\,n}}, \tag{4.214}$$

where $96 = 24\cdot 4$ collects the $4!$ from the vertex and the 2^{2n} from the double factorial. The first few coefficients are

$$a_0 = 1, \quad a_1 = -\tfrac{1}{8}, \quad a_2 = \tfrac{35}{384}, \quad a_3 = -\tfrac{385}{3072}, \quad \dots \tag{4.215}$$

The term of order λ^n is the sum of all vacuum diagrams built from n quartic vertices: by (4.203), n quartic vertices ($V = n$, hence $I = 2n$ lines) give $L = n+1$ loops, so successive powers of λ are successive loop orders. The coefficient $a_1 = -\frac{1}{8}$ is precisely the figure-eight of Fig. 4.7 with its symmetry factor; the loop expansion of this toy is literally the series (4.214).

The radius of convergence is zero. Form the ratio of consecutive coefficients exactly,

$$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(4n+1)(4n+2)(4n+3)(4n+4)}{(n+1)(2n+1)(2n+2)\cdot 96} \xrightarrow{\;n\to\infty\;} \frac{256\,n^4}{384\,n^3} = \frac{2}{3}\,n \;\to\; \infty. \tag{4.216}$$

For any fixed $\lambda \neq 0$ the terms $a_n\lambda^n$ eventually grow without bound, so the series diverges: its radius of convergence is 0. Sharpening (4.216) with Stirling's formula gives the large-order behavior

$$a_n \;\sim\; \frac{(-1)^n}{\pi\sqrt{2}}\left(\frac{2}{3}\right)^n \frac{(n-1)!}{1}, \qquad n\to\infty, \tag{4.217}$$

the hallmark factorial growth $a_n \sim n!\,(2/3)^n$ of a perturbative series. The series (4.214) is therefore *asymptotic* to $Z(\lambda)$ as $\lambda \to 0^+$ — each partial sum approximates Z with an error smaller than the first omitted term — but it does not converge to it.

There is a second, entirely physical, argument for the zero radius of convergence, due to Dyson, that we record because it generalizes verbatim to the real theory. Suppose the series converged in some disk $|\lambda| < r$. Then it would converge for small *negative* λ as well. But for any $\lambda < 0$ the integrand of (4.213) behaves like $e^{+|\lambda| x^4/4!}$ at large $|x|$ and the integral *diverges*: the would-be ground state is unstable because the potential is unbounded below. A function that fails to exist on one side of the origin cannot have a convergent Taylor series centered there. The divergence of perturbation theory is thus not a technical accident but a reflection of the nonanalyticity of the theory at zero coupling —

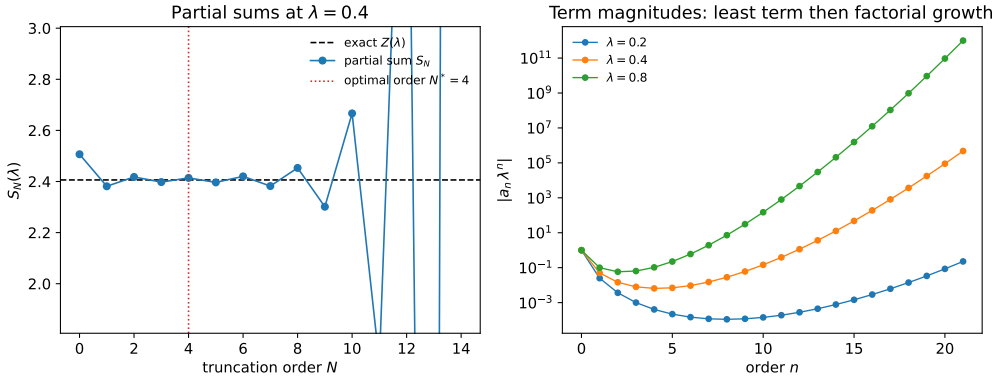


Figure 4.8: The asymptotic loop series of the zero-dimensional ϕ^4 integral (4.213). *Left:* partial sums $S_N(\lambda) = \sqrt{2\pi}\sum_{n\le N} a_n\lambda^n$ at $\lambda = 0.4$ converge toward the exact $Z(\lambda)$ (dashed) up to the optimal truncation N^*, then diverge. *Right:* the magnitudes $|a_n\lambda^n|$ shrink to a least term near N^* and thereafter grow factorially, the universal signature of an asymptotic series. The sign-alternating factorial growth (4.217) with ratio 2/3 places the leading Borel singularity at $t = -3/2$, on the negative real axis and so off the positive integration contour — exactly the configuration that makes this series Borel summable.

the same nonanalyticity that, in four dimensions, is organized by instantons and renormalons in Chapter 42.

Figure 4.8 shows both faces of the asymptotic series: the partial sums chase the exact value, achieve their best accuracy at an optimal truncation order $N^* \sim 1/(\text{const}\cdot\lambda)$, and then diverge as the factorially growing terms take over.

That the series is Borel summable — that one can recover the exact $Z(\lambda)$ from its divergent coefficients by forming the Borel transform $B(t) = \sum_n (a_n/n!)\, t^n$, which by (4.217) has finite radius 3/2, continuing it, and evaluating $Z(\lambda) = \int_0^\infty e^{-t} B(\lambda t)\, \mathrm{d}t$ — is a statement we can make here but not yet justify. The notions of asymptotic series and Borel summation are developed without full proof in Chapter 5, and the obstruction to Borel summability in genuine field theories (Borel-plane singularities from instantons and renormalons) is the subject of Chapter 42.

This completes the functional toolbox. We now hold the variational derivative and the second variation (Section 4.1), the exact Gaussian integral and its determinant (Section 4.2), the generating functionals $Z[J]$ and $W[J]$ and the effective action $\Gamma[\phi_c]$ (Section 4.5), and now the saddle-point/loop expansion with its one-loop functional determinant. Chapter 6 puts the variational calculus to work on classical field dynamics; Chapter 10 builds the scalar path integral and turns the loop expansion derived here into the working perturbation theory of quantum fields; and Chapters 23 and 29 take the effective action and its

potential to the renormalization group and to spontaneous symmetry breaking.

Further reading

The functional and path-integral methods of this chapter are developed, with the physicist's emphasis, in Zee [532], Peskin and Schroeder [396], Schwartz [432], and Srednicki [454]; Weinberg [500] gives the formalism with unusual care about what the generating functionals do and do not assume. Coleman [129] is the classic source for the effective action, the loop expansion, and the saddle-point organization of perturbation theory. For the convex-analysis foundation of the Legendre transform and the effective action as a true convex envelope, and for the asymptotic and Borel-summation theory of the loop series, see Zinn-Justin [534]. The honest status of the continuum measure – which path-integral statements are theorems and which are formal – is the subject of constructive field theory, surveyed in Glimm and Jaffe [225], with the axiomatic backdrop in Streater and Wightman [458].

Chapter 5

Analysis Tools: Distributions, Complex Analysis, Asymptotics

This is the last toolbox laid down before field theory begins. The previous chapters of Part I supplied two of the three languages the rest of the book speaks. Chapters 2 and 3 built the symmetry-and-representation language – finite and Lie groups, the Lorentz and Poincaré algebras, Wigner's classification of what a relativistic particle can be – and Chapter 4 built the functional-calculus machinery – variations, functional derivatives, Gaussian and generating functionals, the Legendre transform. What is still missing is the *analytic* language: the precise theory of the singular, the complex, and the divergent. That is what this chapter supplies, and with it Part I is complete and Part II can begin. We assume the reader's complex analysis through Cauchy's theorem and the residue theorem, together with elementary Fourier analysis and Gaussian integration. These are *reviewed for deployment, not rebuilt*: we restate exactly the residue and Jordan-lemma machinery we will lean on, in our notation, and then use it. Everything past that line – distribution theory, the $i\varepsilon$ prescription, the legality of Wick rotation, dispersion relations, and the asymptotic/Borel circle of ideas – is *built* here from the ground up.

The chapter exists to resolve two difficulties that are not optional decorations on quantum field theory but structural features of it, unavoidable the moment one writes down a propagator or a perturbative amplitude. The first is that the central objects of the theory – propagators, vertex functions, scattering amplitudes – are *not ordinary functions*. A free propagator has a pole exactly where energy and momentum sit on the mass shell, the place one most wants to evaluate it; the time-ordered two-point function is a boundary value of an analytic function, well defined only once a rule is given for going around that pole. That rule is the $i\varepsilon$ prescription, and an amplitude is meaningful only relative to it. The second difficulty is that the central computational engine of the theory – perturbation theory in a coupling – produces power series with

zero radius of convergence. The number of Feynman diagrams grows factorially with the order, and the resulting series diverges for every nonzero coupling. A perturbative "answer" is therefore not the sum of a convergent series; it is an asymptotic expansion, and extracting a number from it requires a theory of what such an expansion means. Both difficulties must be given precise meaning before a single field is quantized, or every later computation rests on undefined symbols.

These two threads – singular objects and divergent series – are in fact one argument, and the single word that unifies them is *analytic structure*. The chapter is organized so that each section hands the next link of one chain. Singular distributions – the delta function, the Cauchy principal value, the boundary values $1/(x \pm i\varepsilon)$ – are introduced first (Section 5.1) and shown by the Sokhotski–Plemelj formula to encode, in one identity, both a principal value and a delta function: the analytic content of an $i\varepsilon$. The Fourier transform on tempered distributions (Section 5.2) turns these objects into momentum space, where the $1/(x \pm i\varepsilon)$ structure becomes a *pole* in a complex energy variable. Poles are then handled by contour integration (Section 5.3): the retarded, advanced, and Feynman propagators are revealed to be one and the same integrand evaluated on three different contours, three prescriptions for going around the same two poles. The Feynman pole placement is exactly what makes one contour deformation legal, and that deformation is the *Wick rotation* (Section 5.4) connecting Minkowski and Euclidean signature – the prototype of analytic continuation as a computational tool. Continuation in the *external* invariants, rather than the loop variables, is the subject of Section 5.5: there analyticity in the upper half of a complex energy plane is shown to be, read physically, nothing but *causality*, and from causality one derives the Kramers–Kronig and subtracted dispersion relations and the sum rules they imply. Finally, the analytic functions these methods produce are approximated in practice by the divergent perturbative series of the second difficulty; Section 5.6 develops the asymptotic-series technology – optimal truncation, Watson's lemma, Laplace and steepest-descent – that controls them term by term, and Section 5.7 develops Borel summation, which reconstructs the function from the series when it can, and exposes the nonperturbative ambiguity when it cannot. Singular distribution $\to$ pole $\to$ contour $\to$ continuation $\to$ causality $\to$ dispersion relation $\to$ asymptotic series $\to$ Borel sum: one chain, seven sections.

A brief word on scope and rigor. The bulk of this chapter is proved in full and stands on its own: the construction of the distribution spaces and the universal differentiability of distributions, the Sokhotski–Plemelj formula, the Fourier transform on $\mathcal{S}'$ and the transform dictionary, the residue/Jordan-lemma evaluation of the propagator contours, the legality of the Wick rotation traced directly to the Feynman pole placement, the Kramers–Kronig relations, Watson's lemma, and the steepest-descent expansion are all established here with every step shown. A smaller set of results is given as precise statements with explicit forward pointers to where the book discharges them: the Källén–Lehmann

spectral representation, whose nonperturbative positivity underlies the analytic assumptions used here, is stated now and proved in Chapter 9; the optical theorem, whose imaginary part feeds the dispersion relations, is stated now and proved in Chapter 14; and the deepest questions – whether the full perturbative series of a realistic four-dimensional QFT is Borel summable, the renormalon obstruction in asymptotically free theories, and the resurgence/transseries program – are flagged as genuinely open and dated to the mid-2020s, not papered over. The reader should know which sentences are theorems and which are signposts.

A word on the threads, and a word on what the worked computations are doing. no earlier model is formally advanced in this chapter: the ϕ^4, QED, and Yang–Mills models are not defined until Part II, so there is no Lagrangian here to continue. What the chapter does instead is *plant the exact analytic objects* those threads will later need, as self-contained model calculations. The momentum-space scalar propagator $i/(p^2 - m^2 + i\varepsilon)$ assembled here is, verbatim, the internal line that ϕ^4 theory will quantize in Chapter 9 and feed into the path integral in Chapter 10; the Wick rotation made legal here is the one every loop integral in Chapter 19 performs before reaching the master integrals of Chapter D; the dispersion-relation and optical-theorem machinery is the analytic backbone of the unitarity methods in Chapter 60; and the Borel-summation and saddle-point analysis is exactly the language in which the semiclassical expansion of Chapter 10 and the instanton physics of Chapter 42 are later phrased. The conventions in force throughout match the book-wide canon collected in Appendix A (Chapter A): the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, the Fourier convention with $e^{-ip\cdot x}$ and a $(2\pi)^{-4}$ per momentum integral, and the Feynman $i\varepsilon$. We are stocking a toolbox, and we name on the box exactly which later chapters each tool is for.

5.1 Distributions and the $i\epsilon$ prescription

The chapter introduction made a promise that this section must now redeem. The central objects of quantum field theory — propagators, vertex functions, amplitudes — are not functions in the sense of the calculus. The free Feynman propagator in momentum space, $i/(p^2 - m^2 + i\epsilon)$, is singular on the entire mass shell $p^2 = m^2$; in position space it oscillates without decaying and cannot be evaluated pointwise on the light cone; and the small imaginary part $i\epsilon$, which the textbook reader is taught to "send to zero at the end," is doing genuine analytic work that no naive limit of functions can reproduce. We met this already at the naive level in Chapter 4, where the functional derivative produced a Dirac delta as its "Kronecker delta" and we agreed to postpone the question of what that object really is. The postponement ends here. The right framework is L. Schwartz's theory of *distributions*: one enlarges the class of admissible objects from functions to continuous linear functionals on a space of very well-behaved

test functions, recovering ordinary functions as a special case but admitting also δ, its derivatives, the principal value $\mathrm{PV}\,1/x$, and — the result this section is built toward — the boundary values $1/(x \pm i\epsilon)$. Once these live in a vector space on which differentiation is always defined and limits have a precise meaning, the statement "the propagator is a distribution" stops being folklore and becomes a theorem we can compute with. Throughout we use the Fourier and sign conventions fixed in Chapter A; the Fourier transform itself is built on the present foundation in Section 5.2.

5.1.1 Test functions: $\mathcal{D}$ and $\mathcal{S}$

The strategy is to define a singular object not by its values — it has none — but by how it *acts*, by the numbers it returns when integrated against a smooth, rapidly-decaying probe. Everything therefore rests on a good choice of probe. Two spaces of probes are standard, and we will need both.

Definition 5.1 (Test function spaces)**.** Let $\mathbb{R}^n$ be the underlying space.

1. The space of *compactly supported test functions* is
$$\mathcal{D}(\mathbb{R}^n) = C_c^\infty(\mathbb{R}^n) = \{\, \varphi \in C^\infty(\mathbb{R}^n) : \ \operatorname{supp}\varphi \text{ is compact} \,\}, \tag{5.1}$$
the infinitely differentiable functions that vanish outside some bounded set.

2. The *Schwartz space* of *rapidly decreasing functions* is
$$\mathcal{S}(\mathbb{R}^n) = \Big\{ \varphi \in C^\infty(\mathbb{R}^n) : \ \|\varphi\|_{\alpha,\beta} := \sup_{x\in\mathbb{R}^n} \big|x^\alpha\, \partial^\beta \varphi(x)\big| < \infty \ \text{ for all } \alpha, \beta \Big\}, \tag{5.2}$$
the smooth functions all of whose derivatives decay faster than any inverse power of $|x|$. The countable family of *seminorms* $\|\cdot\|_{\alpha,\beta}$ defines the topology of $\mathcal{S}$.

Here $\alpha = (\alpha_1, \dots, \alpha_n)$ is a multi-index, $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$, $\partial^\beta = \partial_1^{\beta_1} \cdots \partial_n^{\beta_n}$, and $|\alpha| = \sum_i \alpha_i$. The condition in Eq. (5.2) says that multiplying any derivative of φ by any polynomial still leaves a bounded function: φ and all its derivatives die off at infinity faster than $|x|^{-N}$ for every N. The Gaussian $e^{-|x|^2}$ is the prototype and lives in $\mathcal{S}$ but not in $\mathcal{D}$ (it never exactly vanishes). The standard bump
$$\varphi_0(x) = \begin{cases} \exp\!\big(-1/(1-|x|^2)\big), & |x| < 1, \\ 0, & |x| \geq 1, \end{cases} \tag{5.3}$$
lives in $\mathcal{D}$: it is smooth (all derivatives vanish as $|x| \uparrow 1$ because the essential singularity of $e^{-1/t}$ at $t = 0^+$ is flatter than any power) and is supported in the closed unit ball. Such bumps exist in abundance; their existence is what makes the

whole theory nonvacuous. Since compact support forces faster-than-polynomial decay trivially, we have the inclusion

$$\mathcal{D}(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset C^\infty(\mathbb{R}^n), \tag{5.4}$$

and the smaller the space of probes, the larger the space of objects that can be tested against them.

Physical Insight

Why *two* spaces? The compactly supported $\mathcal{D}$ is the more permissive probe, and its dual $\mathcal{D}'$ accordingly contains the most objects — including wildly growing ones like e^{x^2}, read as a functional. But Schwartz space is closed under the Fourier transform (a Gaussian maps to a Gaussian, and rapid decay in x is exactly smoothness in p and vice versa), whereas $\mathcal{D}$ is *not*: a compactly supported function has an entire, hence non-compactly-supported, Fourier transform. Since QFT lives and breathes in momentum space, the Fourier-friendly pair $\mathcal{S}$ and its dual $\mathcal{S}'$ are the ones we will marry to the Fourier transform in Section 5.2. We develop both here because $\mathcal{D}$ gives the cleanest definitions and $\mathcal{S}$ gives the objects we actually need.

5.1.2 Distributions and tempered distributions

A distribution is a rule that assigns a number to each test function, linearly and continuously. Linearity is obvious; continuity is what excludes pathologies and must be stated with care, because the test-function spaces are not normed but are topologized by families of seminorms.

Definition 5.2 (Distributions and tempered distributions)**.** A *distribution* is a continuous linear functional $T : \mathcal{D}(\mathbb{R}^n) \to \mathbb{C}$. The space of distributions is denoted $\mathcal{D}'(\mathbb{R}^n)$. A *tempered distribution* is a continuous linear functional $T : \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$; their space is $\mathcal{S}'(\mathbb{R}^n)$. We write the action of T on a test function as a pairing, $\langle T, \varphi\rangle \in \mathbb{C}$, linear in φ.

Concretely, continuity is the following sequential criterion, which is all we will ever use.

Proposition 5.3 (Continuity criteria)**.** *A linear functional T on $\mathcal{S}(\mathbb{R}^n)$ is a tempered distribution if and only if there exist a constant $C \geq 0$ and an integer $N \geq 0$ such that*

$$\big|\langle T, \varphi\rangle\big| \;\leq\; C \max_{|\alpha|,|\beta|\leq N} \|\varphi\|_{\alpha,\beta} \qquad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{5.5}$$

Equivalently, $\langle T, \varphi_k\rangle \to 0$ whenever $\|\varphi_k\|_{\alpha,\beta} \to 0$ for every α, β. For $T \in \mathcal{D}'$ the analogous statement holds with the seminorms taken over a fixed compact set containing the supports.

The estimate Eq. (5.5) is the working definition of "tempered": the functional is controlled by *finitely many* seminorms, so it grows at most polynomially. We verify this bound for every functional we introduce; doing so is what certifies membership in $\mathcal{S}'$.

The simplest distributions are the functions themselves. Any locally integrable function f that grows at most polynomially, $|f(x)| \le C(1+|x|)^M$, defines a *regular* tempered distribution T_f by integration,

$$\langle T_f, \varphi \rangle = \int_{\mathbb{R}^n} f(x)\,\varphi(x)\,\mathrm{d}^n x, \tag{5.6}$$

the integral converging absolutely because φ decays faster than f grows. The map $f \mapsto T_f$ is injective (if $\int f\varphi = 0$ for all φ then $f = 0$ almost everywhere), so we may and do identify f with T_f and write $\langle f, \varphi\rangle$ for the integral. This is the precise sense in which distributions *enlarge* the functions rather than replacing them: every reasonable function is a distribution, and the new objects are the ones — like δ — that are *not* of the form T_f for any function f.

We will also need a notion of limit. Because a distribution is known through its pairings, the natural convergence is pairing-by-pairing.

Definition 5.4 (Weak / distributional convergence)**.** A sequence (or net) T_k in $\mathcal{S}'$ *converges weakly* (equivalently, in $\mathcal{S}'$, or in the sense of distributions) to $T \in \mathcal{S}'$ if

$$\langle T_k, \varphi\rangle \;\longrightarrow\; \langle T, \varphi\rangle \qquad \text{for every } \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{5.7}$$

The same definition with $\mathcal{S}$ replaced by $\mathcal{D}$ gives convergence in $\mathcal{D}'$.

This is exactly the topology in which "$i\epsilon \to 0^+$" will be made precise: the family $1/(x+i\epsilon)$, indexed by $\epsilon > 0$, is a net of regular distributions, and its $\epsilon \downarrow 0$ limit is taken in the weak sense of Eq. (5.7), never pointwise.

5.1.3 The Dirac delta, multiplication, and the distributional derivative

Definition 5.5 (Dirac delta)**.** The *Dirac delta* at the origin is the tempered distribution $\delta \in \mathcal{S}'(\mathbb{R}^n)$ defined by evaluation,

$$\langle \delta, \varphi\rangle = \varphi(0), \tag{5.8}$$

and δ_a, the delta at a, by $\langle \delta_a, \varphi\rangle = \varphi(a)$.

That $\delta \in \mathcal{S}'$ is immediate from Theorem 5.3: $|\langle\delta,\varphi\rangle| = |\varphi(0)| \le \sup|\varphi| = \|\varphi\|_{0,0}$, the bound Eq. (5.5) with $C = 1$, $N = 0$. It is *not* a regular distribution: no locally integrable f satisfies $\int f\varphi = \varphi(0)$ for all φ, as one sees by taking bumps φ concentrated away from the origin (forcing $f = 0$ a.e.) and yet demanding

$\varphi(0) \neq 0$. The delta is the cleanest example of a genuinely new object, and the physicist's "function that is zero everywhere but infinite at the origin with unit integral" is just an evocative but incoherent paraphrase of Eq. (5.8).

Two operations on functions extend to distributions *by duality*: one defines the operation on T so that, on regular distributions, it agrees with the ordinary operation under the integral sign. This single principle generates the entire calculus of distributions.

Multiplication by a smooth function. If $T = T_f$ is regular and g is smooth with all derivatives polynomially bounded (so that $g\varphi \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$), then $\langle gT_f, \varphi\rangle = \int gf\varphi = \int f(g\varphi) = \langle T_f, g\varphi\rangle$. We *define* the product gT for any $T \in \mathcal{S}'$ by this last expression:

$$\langle gT, \varphi\rangle := \langle T, g\varphi\rangle. \tag{5.9}$$

It is a tempered distribution because $\varphi \mapsto g\varphi$ is continuous on $\mathcal{S}$. In particular $\langle g\,\delta, \varphi\rangle = \langle \delta, g\varphi\rangle = g(0)\varphi(0)$, i.e.

$$g(x)\,\delta(x) = g(0)\,\delta(x), \qquad \text{whence} \qquad x\,\delta(x) = 0. \tag{5.10}$$

Note the sharp limitation: *multiplication of two distributions is not defined in general.* There is no consistent meaning for $\delta(x)^2$, and the ultraviolet divergences of QFT are, at root, the failure of products of propagators (themselves distributions) to be distributions — a point we return to when renormalization is built (Chapter 21).

The distributional derivative. Let $f \in C^1$ define a regular distribution. For $\varphi \in \mathcal{D}$ (compact support, so no boundary terms) or $\varphi \in \mathcal{S}$ (rapid decay, so boundary terms vanish at infinity), integration by parts gives, with $\partial_j = \partial/\partial x_j$,

$$\langle \partial_j f, \varphi\rangle = \int (\partial_j f)\,\varphi\,\mathrm{d}^n x = -\int f\,(\partial_j \varphi)\,\mathrm{d}^n x = -\langle f, \partial_j \varphi\rangle. \tag{5.11}$$

The right-hand side makes sense for *any* distribution, since $\partial_j\varphi$ is again a test function. We therefore define the derivative of any $T \in \mathcal{S}'$ (or $\mathcal{D}'$) by

$$\langle \partial_j T, \varphi\rangle := -\langle T, \partial_j \varphi\rangle, \qquad \text{and more generally} \qquad \langle \partial^\alpha T, \varphi\rangle = (-1)^{|\alpha|}\langle T, \partial^\alpha \varphi\rangle. \tag{5.12}$$

This definition has a remarkable consequence, the single structural fact that makes distributions so powerful in field theory.

Theorem 5.6 (Distributions are infinitely differentiable)**.** *Every $T \in \mathcal{D}'(\mathbb{R}^n)$ has derivatives $\partial^\alpha T \in \mathcal{D}'(\mathbb{R}^n)$ of all orders, given by Eq. (5.12); the same holds for every $T \in \mathcal{S}'(\mathbb{R}^n)$ with derivatives in $\mathcal{S}'$. Moreover, mixed partials commute, $\partial_i\partial_j T = \partial_j\partial_i T$.*

Proof. Fix $T \in \mathcal{S}'$ and a multi-index α. The map $\varphi \mapsto (-1)^{|\alpha|}\langle T, \partial^\alpha \varphi\rangle$ is linear. It is continuous: differentiation $\varphi \mapsto \partial^\alpha \varphi$ is continuous on $\mathcal{S}$ because $\|\partial^\alpha \varphi\|_{\gamma,\beta} = \|\varphi\|_{\gamma,\beta+\alpha}$ is itself one of the defining seminorms, so if T obeys the bound Eq. (5.5) with data (C, N) then $\partial^\alpha T$ obeys it with $(C, N + |\alpha|)$. Hence $\partial^\alpha T \in \mathcal{S}'$. Commutativity of mixed partials is inherited from the test functions, where it is the equality of smooth mixed partials of φ: $\langle \partial_i \partial_j T, \varphi\rangle = \langle T, \partial_j \partial_i \varphi\rangle = \langle T, \partial_i \partial_j \varphi\rangle = \langle \partial_j \partial_i T, \varphi\rangle$. The argument for $\mathcal{D}'$ is identical with seminorms restricted to compact sets. □

The content is worth pausing on. An ordinary function need not be differentiable, or even continuous; but *as a distribution* it is differentiable infinitely often, the derivative simply being moved onto the always-smooth test function. The step function, the absolute value, the propagator's discontinuity across the light cone — all are differentiable in $\mathcal{D}'$. The price, and it is the right price, is that the derivative may be a more singular object than the original.

Worked Example 5.7 (The derivatives of the delta)**.** We establish that $\delta' \in \mathcal{S}'(\mathbb{R})$ acts by $\langle \delta', \varphi\rangle = -\varphi'(0)$ and check the definition against an explicit regularization, exactly the kind of regularize-then-remove manoeuvre that the $i\epsilon$ prescription will formalize.

From the definition. Applying Eq. (5.12) with $T = \delta$,

$$\langle \delta', \varphi\rangle = -\langle \delta, \varphi'\rangle = -\varphi'(0), \qquad \text{and} \qquad \langle \delta^{(k)}, \varphi\rangle = (-1)^k \varphi^{(k)}(0). \tag{5.13}$$

This is a tempered distribution: $|\langle \delta', \varphi\rangle| = |\varphi'(0)| \le \|\varphi\|_{0,1}$, the bound Eq. (5.5) with $N = 1$. So δ' has *order one*, one notch more singular than δ.

Justification by regularization. Represent δ as the weak limit of the smooth, normalized Lorentzians

$$\eta_\epsilon(x) = \frac{1}{\pi}\,\frac{\epsilon}{x^2 + \epsilon^2}, \qquad \int_{-\infty}^{\infty} \eta_\epsilon(x)\,\mathrm{d}x = 1, \tag{5.14}$$

which we prove converge to δ in Eq. (5.28) below. Each η_ϵ is a genuine smooth function, so its derivative is the ordinary one, and integration by parts (the boundary terms vanish because η_ϵ and φ decay) gives

$$\langle \eta_\epsilon', \varphi\rangle = \int_{-\infty}^{\infty} \eta_\epsilon'(x)\,\varphi(x)\,\mathrm{d}x = -\int_{-\infty}^{\infty} \eta_\epsilon(x)\,\varphi'(x)\,\mathrm{d}x \xrightarrow[\epsilon \to 0^+]{} -\varphi'(0). \tag{5.15}$$

The limit is $-\varphi'(0)$ because $\eta_\epsilon \to \delta$ weakly applied to the test function $\varphi' \in \mathcal{S}$. Thus $\eta_\epsilon' \to \delta'$ in $\mathcal{S}'$, and the abstract definition Eq. (5.13) agrees with the regularized object in the weak limit. This is the template for the whole section: a singular distribution is the weak limit of a one-parameter family of honest functions, and operations on it are operations on the family, taken under the limit.

5.1.4 The Cauchy principal value

The function $1/x$ is not locally integrable at the origin, so the naive prescription Eq. (5.6) fails: $\int \varphi(x)/x\,\mathrm{d}x$ diverges for a generic φ with $\varphi(0) \neq 0$. The cure is to excise a symmetric neighborhood of the singularity and let it shrink, so that the divergent contributions from $x > 0$ and $x < 0$ cancel before they can blow up.

Definition 5.8 (Cauchy principal value)**.** The *principal value* of $1/x$ is the functional

$$\left\langle \mathrm{PV}\tfrac{1}{x},\ \varphi \right\rangle = \mathrm{PV}\int_{-\infty}^{\infty} \frac{\varphi(x)}{x}\,\mathrm{d}x := \lim_{\delta\to 0^+} \int_{|x|\geq\delta} \frac{\varphi(x)}{x}\,\mathrm{d}x, \qquad \varphi \in \mathcal{S}(\mathbb{R}). \tag{5.16}$$

That the symmetric limit exists, and defines a tempered distribution, is not automatic and must be shown; the symmetric exclusion is doing real work.

Worked Example 5.9 ($\mathrm{PV}\,1/x$ is a well-defined tempered distribution)**.** We prove the limit Eq. (5.16) exists for every $\varphi \in \mathcal{S}(\mathbb{R})$ and obeys the continuity bound Eq. (5.5).

Existence. Split the excised integral at $|x| = 1$ and combine the contributions from $\pm x$ on the inner region. On $\delta \leq |x| \leq 1$, pairing x with $-x$,

$$\int_{\delta\leq|x|\leq 1} \frac{\varphi(x)}{x}\,\mathrm{d}x = \int_{\delta}^{1} \frac{\varphi(x) - \varphi(-x)}{x}\,\mathrm{d}x. \tag{5.17}$$

The integrand is the heart of the matter. By the mean value theorem (or by Taylor with remainder), $\varphi(x) - \varphi(-x) = 2x\,\varphi'(\xi_x)$ for some $\xi_x \in (-x, x)$, so

$$\frac{\varphi(x) - \varphi(-x)}{x} \xrightarrow[x\to 0^+]{} 2\varphi'(0), \qquad \left|\frac{\varphi(x) - \varphi(-x)}{x}\right| \leq 2 \sup_{|t|\leq 1} |\varphi'(t)| \leq 2\,\|\varphi\|_{0,1}\,. \tag{5.18}$$

The integrand is therefore bounded on $(0, 1]$ and extends continuously to $x = 0$, so $\int_\delta^1$ converges to $\int_0^1$ as $\delta \to 0^+$: the inner limit exists. The outer piece $\int_{|x|\geq 1} \varphi(x)/x\,\mathrm{d}x$ is independent of δ and converges absolutely, since $|\varphi(x)/x| \leq |\varphi(x)|$ there and φ is integrable. Hence the full limit Eq. (5.16) exists.

Temperedness. Bound the two pieces. From Eq. (5.18),

$$\left|\int_0^1 \frac{\varphi(x) - \varphi(-x)}{x}\,\mathrm{d}x\right| \leq 2\,\|\varphi\|_{0,1}\,. \tag{5.19}$$

For the tail, insert and remove a factor x^2 and use that $(1 + x^2)\varphi$ is bounded by a seminorm:

$$\left|\int_{|x|\geq 1} \frac{\varphi(x)}{x}\,\mathrm{d}x\right| \leq \int_{|x|\geq 1} \frac{|(1+x^2)\varphi(x)|}{|x|\,(1+x^2)}\,\mathrm{d}x \leq (\|\varphi\|_{0,0} + \|\varphi\|_{2,0}) \int_{|x|\geq 1} \frac{\mathrm{d}x}{|x|(1+x^2)}. \tag{5.20}$$

The remaining integral is a finite number ($2\int_1^\infty \mathrm{d}x/(x+x^3) = \ln 2$). Adding the two estimates,

$$\left|\langle \mathrm{PV}\tfrac{1}{x}, \varphi\rangle\right| \le 2\,\|\varphi\|_{0,1} + (\ln 2)(\|\varphi\|_{0,0} + \|\varphi\|_{2,0}) \le C \max_{|\alpha|,|\beta|\le 2} \|\varphi\|_{\alpha,\beta}\,, \tag{5.21}$$

which is the bound Eq. (5.5) with $N = 2$. Therefore $\mathrm{PV}\,1/x \in \mathcal{S}'(\mathbb{R})$, of order at most one (the only derivative of φ that entered is the first).

The principal value is exactly the odd part of $1/x$ tamed: the even singularity would survive the symmetric limit and diverge, but $1/x$ is odd, and the symmetric excision lets the antisymmetry cancel the pole. This is the structural fact that the Sokhotski–Plemelj formula now turns into an identity between boundary values.

5.1.5 The Sokhotski–Plemelj formula

We can now prove the first load-bearing result of the chapter. The objects $1/(x \pm i\epsilon)$ are, for each fixed $\epsilon > 0$, perfectly ordinary smooth bounded functions of the real variable x (the denominator never vanishes); each is a regular tempered distribution. The claim is that their $\epsilon \downarrow 0$ limits exist in $\mathcal{S}'$ and split cleanly into the principal value and a delta.

Theorem 5.10 (Sokhotski–Plemelj)**.** *As $\epsilon \to 0^+$, the following limits hold in $\mathcal{S}'(\mathbb{R})$:*

$$\frac{1}{x \pm i\epsilon} \xrightarrow[\epsilon\to 0^+]{} \mathrm{PV}\frac{1}{x} \;\mp\; i\pi\,\delta(x). \tag{5.22}$$

Explicitly, for every $\varphi \in \mathcal{S}(\mathbb{R})$,

$$\lim_{\epsilon\to 0^+} \int_{-\infty}^{\infty} \frac{\varphi(x)}{x \pm i\epsilon}\,\mathrm{d}x = \mathrm{PV}\int_{-\infty}^{\infty} \frac{\varphi(x)}{x}\,\mathrm{d}x \;\mp\; i\pi\,\varphi(0). \tag{5.23}$$

Proof. Take the upper sign throughout; the lower sign follows by complex conjugation, since $1/(x - i\epsilon) = \overline{1/(x + i\epsilon)}$ for real x. Rationalize the denominator by multiplying by the conjugate,

$$\frac{1}{x + i\epsilon} = \frac{x - i\epsilon}{(x + i\epsilon)(x - i\epsilon)} = \frac{x}{x^2 + \epsilon^2} \;-\; i\,\frac{\epsilon}{x^2 + \epsilon^2}, \tag{5.24}$$

a clean separation into a real part (odd in x) and an imaginary part (even in x). We treat the two parts as separate weak limits and add the results.

Imaginary part: a nascent delta. The imaginary part is $-i$ times $\epsilon/(x^2+\epsilon^2) = \pi\,\eta_\epsilon(x)$, the normalized Lorentzian of Eq. (5.14). First, its mass is fixed for every ϵ: with $x = \epsilon u$,

$$\int_{-\infty}^{\infty} \frac{\epsilon}{x^2 + \epsilon^2}\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{\mathrm{d}u}{1 + u^2} = \big[\arctan u\big]_{-\infty}^{\infty} = \pi. \tag{5.25}$$

Now pair against φ and subtract the expected answer, using Eq. (5.25) to write $\pi\varphi(0) = \int \frac{\epsilon}{x^2+\epsilon^2}\varphi(0)\,\mathrm{d}x$:

$$\int_{-\infty}^{\infty} \frac{\epsilon}{x^2+\epsilon^2}\,\varphi(x)\,\mathrm{d}x - \pi\,\varphi(0) = \int_{-\infty}^{\infty} \frac{\epsilon}{x^2+\epsilon^2}\,\big[\varphi(x)-\varphi(0)\big]\,\mathrm{d}x. \tag{5.26}$$

Substitute $x = \epsilon u$ again ($\mathrm{d}x = \epsilon\,\mathrm{d}u$, and $\epsilon\cdot\epsilon/(\epsilon^2u^2+\epsilon^2) = 1/(1+u^2)$):

$$\int_{-\infty}^{\infty} \frac{\epsilon}{x^2+\epsilon^2}\,\big[\varphi(x)-\varphi(0)\big]\,\mathrm{d}x = \int_{-\infty}^{\infty} \frac{\varphi(\epsilon u)-\varphi(0)}{1+u^2}\,\mathrm{d}u. \tag{5.27}$$

The integrand converges pointwise to 0 as $\epsilon \to 0^+$ (because φ is continuous, $\varphi(\epsilon u)\to\varphi(0)$ for each fixed u), and it is dominated by the fixed integrable function $2\,\|\varphi\|_{0,0}/(1+u^2)$ since $|\varphi(\epsilon u)-\varphi(0)| \le 2\sup|\varphi|$. By dominated convergence the integral tends to 0. Therefore

$$\int_{-\infty}^{\infty} \frac{\epsilon}{x^2+\epsilon^2}\,\varphi(x)\,\mathrm{d}x \xrightarrow[\epsilon\to0^+]{} \pi\,\varphi(0), \qquad \text{i.e.} \qquad \frac{\epsilon}{x^2+\epsilon^2} \xrightarrow{\mathcal{S}'} \pi\,\delta(x). \tag{5.28}$$

This also discharges the promise made in Theorem 5.7: $\eta_\epsilon = \frac{1}{\pi}\frac{\epsilon}{x^2+\epsilon^2} \to \delta$ weakly. The imaginary part of Eq. (5.24) thus contributes $-i\pi\,\varphi(0)$ in the limit.

Real part: the principal value. We show that, against any $\varphi \in \mathcal{S}$,

$$\int_{-\infty}^{\infty} \frac{x}{x^2+\epsilon^2}\,\varphi(x)\,\mathrm{d}x \xrightarrow[\epsilon\to0^+]{} \mathrm{PV}\int_{-\infty}^{\infty} \frac{\varphi(x)}{x}\,\mathrm{d}x. \tag{5.29}$$

Split the real integral at $|x| = 1$, exactly as in Theorem 5.9. On $|x| \le 1$, the function $x/(x^2+\epsilon^2)$ is odd, so the constant part of φ integrates to zero for every ϵ:

$$\int_{-1}^{1} \frac{x}{x^2+\epsilon^2}\,\varphi(0)\,\mathrm{d}x = 0. \tag{5.30}$$

Write $\varphi(x) = \varphi(0) + x\,\psi(x)$, where Hadamard's lemma gives the explicit smooth remainder

$$\psi(x) = \int_0^1 \varphi'(tx)\,\mathrm{d}t, \qquad \psi \in C^\infty, \quad \psi(0) = \varphi'(0). \tag{5.31}$$

Then, using Eq. (5.30),

$$\int_{-1}^{1} \frac{x}{x^2+\epsilon^2}\,\varphi(x)\,\mathrm{d}x = \int_{-1}^{1} \frac{x^2}{x^2+\epsilon^2}\,\psi(x)\,\mathrm{d}x. \tag{5.32}$$

As $\epsilon\to0^+$ the factor $x^2/(x^2+\epsilon^2)$ increases to 1 for $x \neq 0$ and is bounded by 1; since ψ is bounded on $[-1,1]$, dominated convergence gives $\int_{-1}^{1}\frac{x^2}{x^2+\epsilon^2}\psi\,\mathrm{d}x \to \int_{-1}^{1}\psi(x)\,\mathrm{d}x$. On the other side, the principal value over the same region is, by the identical decomposition (the $\varphi(0)/x$ term vanishes in the symmetric limit because it is odd),

$$\mathrm{PV}\int_{-1}^{1}\frac{\varphi(x)}{x}\,\mathrm{d}x = \lim_{\delta\to0^+}\int_{\delta\le|x|\le1}\frac{\varphi(0)+x\psi(x)}{x}\,\mathrm{d}x = \int_{-1}^{1}\psi(x)\,\mathrm{d}x. \tag{5.33}$$

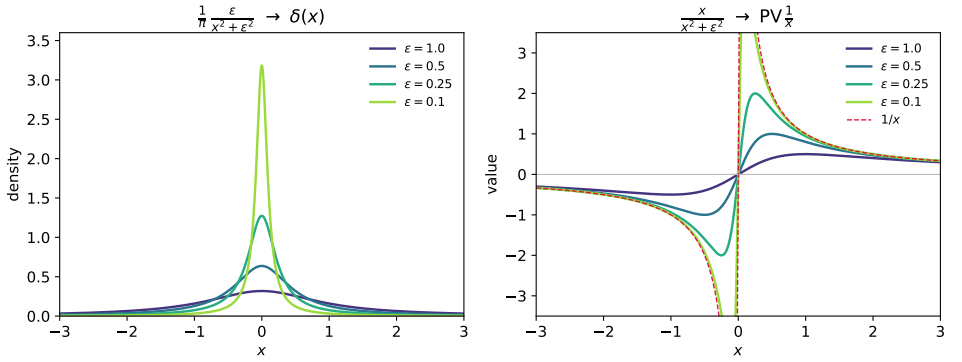


Figure 5.1: The Sokhotski–Plemelj decomposition $1/(x - i\epsilon) = x/(x^2 + \epsilon^2) + i\,\epsilon/(x^2 + \epsilon^2)$. Left: the imaginary part is the normalized Lorentzian $\frac{1}{\pi}\frac{\epsilon}{x^2+\epsilon^2}$, a nascent delta of fixed unit mass that concentrates at the origin as $\epsilon \to 0^+$ (Eq. (5.28)). Right: the real part $x/(x^2 + \epsilon^2)$ (odd in x) regularizes the pole of $1/x$ (dashed) at the origin and reproduces it elsewhere, converging weakly to $\mathrm{PV}\,1/x$ (Eq. (5.29)).

The inner contributions match. For the tail $|x| \geq 1$, the integrand $x\varphi(x)/(x^2+\epsilon^2)$ converges pointwise to $\varphi(x)/x$ and is dominated by the integrable $|\varphi(x)|/|x| \leq |\varphi(x)|$ uniformly in $\epsilon \leq 1$ (since $x^2 + \epsilon^2 \geq x^2$ there), so dominated convergence gives $\int_{|x|\geq 1} \to \int_{|x|\geq 1} \varphi(x)/x\,\mathrm{d}x$, which is the (absolutely convergent) tail of the principal value. Adding inner and tail establishes Eq. (5.29).

Assembly. Combining the real and imaginary limits in Eq. (5.24),

$$\int_{-\infty}^{\infty} \frac{\varphi(x)}{x + i\epsilon}\,\mathrm{d}x \xrightarrow[\epsilon\to 0^+]{} \mathrm{PV}\int_{-\infty}^{\infty} \frac{\varphi(x)}{x}\,\mathrm{d}x \;-\; i\pi\,\varphi(0), \tag{5.34}$$

which is Eq. (5.23) with the upper sign. Conjugation flips the sign of the imaginary part and fixes the (real) principal value, giving the lower sign and completing the proof. □

Figure 5.1 shows the two halves of Eq. (5.24) in action: the even Lorentzian narrowing into the delta, and the odd $x/(x^2 + \epsilon^2)$ flattening onto $1/x$ away from the origin while staying finite at it.

Physical Insight

The sign of the $i\epsilon$ chooses which way an infinitesimal contour skirts the pole, and the $\mp i\pi\delta$ records the half-residue picked up in passing. This is the seed of every contour prescription in the book. The same formula, read in energy, separates a real *dispersive* part (PV) from an imaginary *absorptive* part (δ); read on the propagator's mass shell it is the statement that the real part of $i/(p^2 - m^2 + i\epsilon)$ is a delta enforcing $p^2 = m^2$, i.e.

on-shell particle production. The Feynman prescription's $+i\epsilon$ is precisely the choice in Eq. (5.22); the link between this analytic choice and causality is developed when the propagator is constructed in Chapter 9, and its physical face — the optical theorem — in Chapter 14.

Caution

The decomposition Eq. (5.22) is an identity *between distributions*, valid only under an integral against a test function. Reading it pointwise — "$1/(x+i\epsilon)$ equals $1/x$ except at the origin where it is $-i\pi\delta$" — is meaningless: δ has no values, and $1/x$ is not locally integrable. Every manipulation of $i\epsilon$ factors in this book is, at bottom, an application of Theorem 5.10 inside a (momentum) integral, and is legitimate exactly to the extent that such an integral is present.

5.1.6 Order, support, and the structure of tempered distributions

We have repeatedly read off how many derivatives of φ a functional needs: δ needed none, δ' one, $\mathrm{PV}\,1/x$ one. This is a genuine invariant.

Definition 5.11 (Order and support)**.** A distribution T has *finite order* if the bound Eq. (5.5) (restricted to a fixed compact set, for $T \in \mathcal{D}'$) holds for some N; the least such N is the *order* $\operatorname{ord} T$. The *support* $\operatorname{supp} T$ is the complement of the largest open set U on which T vanishes, where T *vanishes on* U means $\langle T, \varphi\rangle = 0$ for every test function φ with $\operatorname{supp}\varphi \subset U$.

Thus $\operatorname{ord}\delta = 0$, $\operatorname{ord}\delta^{(k)} = k$, $\operatorname{ord}(\mathrm{PV}\,1/x) = 1$; and $\operatorname{supp}\delta = \{0\} = \operatorname{supp}\delta^{(k)}$, while $\operatorname{supp}(\mathrm{PV}\,1/x) = \mathbb{R}$. A regular distribution T_f has order 0 and support equal to the (essential) support of f. Distributions supported at a single point are completely classified — they are exactly the finite linear combinations of δ and its derivatives there — a fact we will invoke when local counterterms are characterized in Chapter 21.

The examples are not accidental in their tameness. Every tempered distribution, however singular it looks, is a finite number of derivatives of an ordinary, mildly-growing *function* — which is why distribution theory is exactly the right enlargement of the functions and no larger. We state this at the level the book uses it (the proof is a standard application of the Fourier characterization of $\mathcal{S}'$ developed in Section 5.2, and we do not need its details).

Theorem 5.12 (Structure of tempered distributions)**.** *Every $T \in \mathcal{S}'(\mathbb{R}^n)$ has finite order, and there exist a multi-index α and a continuous function $F : \mathbb{R}^n \to \mathbb{C}$*

of polynomial growth, $|F(x)| \le C(1+|x|)^M$*, such that*

$$T = \partial^\alpha F \qquad \textit{(distributional derivative).} \tag{5.35}$$

That is, the tempered distributions are precisely the finite-order distributional derivatives of polynomially bounded continuous functions.

The theorem dignifies every singular object we have met: $\delta = \theta'$ is one derivative of the bounded Heaviside step θ (a fact we verify directly once the step's distributional derivative is computed in Section 5.2), and $\mathrm{PV}\,1/x = (\ln|x|)'$ is one derivative of the locally integrable, polynomially-bounded $\ln|x|$ — as one checks by integration by parts against a test function, using oddness exactly as in Eq. (5.17). The propagators of QFT will be no worse: distributions of finite order, derivatives of honest functions, manipulated only inside integrals.

With the singular objects of daily QFT now precisely defined — δ, its derivatives, $\mathrm{PV}\,1/x$, and the boundary values $1/(x \pm i\epsilon)$ — and with weak convergence as the meaning of every limit, we have the vector space $\mathcal{S}'$ on which differentiation is unconditional and the $i\epsilon$ rule is a theorem rather than a slogan. The Sokhotski–Plemelj formula already hints that these objects are most transparent in momentum language: an $i\epsilon$ turned into a δ, a pole into an on-shell condition. To exploit that, we need to carry the Fourier transform from Schwartz functions onto all of $\mathcal{S}'$ by the same duality that built the derivative here — the task of Section 5.2.

5.2 Tempered distributions and the Fourier transform

The previous section gave singular objects a rigorous home: the delta, its derivatives, the principal value, and the boundary values $1/(x \pm i\epsilon)$ are all distributions, and the Sokhotski–Plemelj identity Theorem 5.10 showed that an $i\epsilon$ in a denominator is precisely a packaging of a principal value together with a delta. That identity already contains, in miniature, the entire analytic content of a propagator. What it does not yet contain is the *momentum-space* face of the same object. Every propagator the reader will meet—the scalar $i/(p^2 - m^2 + i\epsilon)$, the photon, the Dirac $i(\not{p} + m)/(p^2 - m^2 + i\epsilon)$—is written in momentum space, and every loop integral is an integral over momenta. The bridge between the position-space distributions of Section 5.1 (the previous section) and these momentum-space expressions is the Fourier transform. We therefore now do for the Fourier transform what Section 5.1 did for differentiation: we extend it from nice functions, where it is elementary, to distributions, where it is indispensable.

The extension is not automatic. The Fourier transform trades smoothness for decay and decay for smoothness, so a test space adapted to it must control *both* at once; the compactly supported test functions of the previous section will not

do, for a sharp reason we make precise below. The right space is Schwartz space, and the distributions it supports are the *tempered* distributions—exactly the class to which every propagator, vertex function, and correlation function in this book belongs. Once we have the transform on this class, the rules that physicists use daily ($\partial_\mu \leftrightarrow$ momentum, product$\leftrightarrow$ convolution, translation$\leftrightarrow$phase) become theorems, and the handful of transforms that underlie every Feynman computation ($\hat{\delta} = 1$, $\widehat{1} = (2\pi)^n \delta$, the self-dual Gaussian, the transform of the step function) become one-line consequences. We close by writing the Feynman propagator as the tempered distribution it is, reading its $i\epsilon$ through Sokhotski–Plemelj, and handing the resulting momentum integral to the contour calculus of the next section.

5.2.1 Why a new test space: Schwartz functions

The test functions of the previous section were the smooth, compactly supported functions $\mathcal{D} = C_c^\infty(\mathbb{R}^n)$. They are an excellent probe for *local* questions, but they are catastrophically ill-suited to the Fourier transform, and the reason is structural rather than technical.

Proposition 5.13 (Compact support is not Fourier-stable)**.** *If $\varphi \in C_c^\infty(\mathbb{R}^n)$ is not identically zero, then its Fourier transform $\hat{\varphi}$ extends to an entire analytic function on $\mathbb{C}^n$ and therefore cannot have compact support.*

Proof. Write $\hat{\varphi}(k) = \int_{\mathbb{R}^n} d^n x\, e^{ik\cdot x}\varphi(x)$ with φ supported in a ball of radius R. For complex $k = u + iv$ the integrand is $e^{iu\cdot x}e^{-v\cdot x}\varphi(x)$, bounded in modulus by $e^{R|v|}\,|\varphi(x)|$ on the compact support, so the integral converges for every $k \in \mathbb{C}^n$ and may be differentiated under the integral sign in each complex variable; $\hat{\varphi}$ is entire. An entire function that vanishes on an open set vanishes identically (it is determined by its Taylor series at any point), and an entire function of one variable with compact support would vanish on the complement of a compact set, hence everywhere; the same holds variable by variable. Thus $\hat{\varphi} \equiv 0$, contradicting the inversion formula $\varphi = \mathcal{F}^{-1}\hat{\varphi}$ unless $\varphi \equiv 0$. □

So the Fourier transform pushes us out of $\mathcal{D}$ immediately. The cure is to ask of test functions not compact support but *rapid decay*, symmetrically in position and momentum: a function should decay faster than any inverse polynomial, and so should all its derivatives. This is exactly the property that the derivative$\leftrightarrow$multiplication rule of the Fourier transform preserves, which is why the resulting space is the natural one.

This is exactly the Schwartz space introduced in the previous section, and we use it without redefinition. We recall (Theorem 5.1) that $\mathcal{S}(\mathbb{R}^n)$ consists of the $f \in C^\infty(\mathbb{R}^n)$ for which every seminorm

$$\|f\|_{\alpha,\beta} \;:=\; \sup_{x\in\mathbb{R}^n} \left| x^\alpha\, \partial^\beta f(x) \right| \tag{5.36}$$

is finite for all multi-indices $\alpha, \beta \in \mathbb{N}_0^n$ (here $x^\alpha = x_1^{\alpha_1} \cdots x_n^{\alpha_n}$ and $\partial^\beta = \partial_1^{\beta_1} \cdots \partial_n^{\beta_n}$): every derivative decays faster than every inverse power, so smoothness and decay are controlled on an equal footing. Convergence $f_j \to f$ in $\mathcal{S}$ means $\|f_j - f\|_{\alpha,\beta} \to 0$ for every α, β. The Gaussian $e^{-|x|^2/2}$ is the prototype, and the previous section recorded the strict inclusions $C_c^\infty(\mathbb{R}^n) \subset \mathcal{S}(\mathbb{R}^n) \subset C^\infty(\mathbb{R}^n)$ (Eq. (5.4)). The one new structural remark we need is its consequence for the dual spaces: because $\mathcal{D} \subset \mathcal{S}$, every continuous linear functional on $\mathcal{S}$ restricts to one on $\mathcal{D}$, so the tempered distributions form a *subclass* of the distributions of the previous section, $\mathcal{S}'(\mathbb{R}^n) \subset \mathcal{D}'(\mathbb{R}^n)$ — the price of demanding continuity against the larger probe space $\mathcal{S}$ is that the functional may grow no faster than a polynomial.

The objects we transform are therefore the tempered distributions $\mathcal{S}'(\mathbb{R}^n)$ already defined in Theorem 5.2: the continuous linear functionals $T\colon \mathcal{S}(\mathbb{R}^n) \to \mathbb{C}$, where continuity is the bound of Theorem 5.3, namely the existence of a constant C and an integer N with $|\langle T, \varphi\rangle| \le C \max_{|\alpha|,|\beta|\le N} \|\varphi\|_{\alpha,\beta}$ for all $\varphi \in \mathcal{S}$. We restate the membership criterion here only because the entire Fourier construction below is one long verification of it: every transform we build is shown to obey this bound.

The word "tempered" refers to temperate (at most polynomial) growth, and the examples make the point. Every polynomially bounded locally integrable function g—in particular every bounded function, every L^1 or L^2 function, the constant 1, and any polynomial—defines a tempered distribution by $\langle g, \varphi\rangle = \int g\varphi$, the integral converging because φ kills any polynomial growth. The delta δ, its derivatives $\partial^\beta \delta$, the principal value $\mathrm{PV}(1/x)$, and the boundary values $1/(x \pm i\epsilon)$ from Section 5.1 are all tempered (each is bounded by finitely many Schwartz seminorms, as the previous section's estimates show). What is *not* tempered is anything growing faster than every polynomial: e^x on $\mathbb{R}$ is not a tempered distribution, because pairing it against a Schwartz function need not converge and, more to the point, its "Fourier transform" would be a delta displaced into the complex plane, outside $\mathcal{S}'$. The restriction to temperate growth is exactly the price of a well-defined Fourier transform, and it is a price QFT is always willing to pay: fields, propagators, and amplitudes are tempered.

5.2.2 The Fourier transform on Schwartz space

We fix the convention once, in line with the book's standing choice (Appendix A, Chapter A); every (2π) and every sign below follows from it.

Notation Watch

Throughout the book the Fourier transform and its inverse are

$$\hat{f}(k) \equiv (\mathcal{F}f)(k) = \int_{\mathbb{R}^n} d^n x\, e^{ik\cdot x}\, f(x), \qquad f(x) = \int_{\mathbb{R}^n} \frac{d^n k}{(2\pi)^n}\, e^{-ik\cdot x}\, \hat{f}(k), \tag{5.37}$$

so that *all* factors of $(2\pi)^{-1}$ sit on the momentum-integral (inverse) side, and the forward transform carries the sign $e^{+ik\cdot x}$. This is the spacetime convention of Chapter A read with the *Euclidean* dot product $k\cdot x = \sum_j k_j x_j$; for the Minkowski spacetime transform one replaces $k\cdot x$ by the metric pairing $k\cdot x = k^0 x^0 - \mathbf{k}\cdot\mathbf{x}$, which is the same formula after a relabelling of signs (a linear change of variables of unit Jacobian, since $|\det \eta| = 1$) and changes none of the results of this subsection. We write n for the dimension in the abstract development and specialize to $n = 4$ for the propagator.

Because $f \in \mathcal{S} \subset L^1$, the integral in Eq. (5.37) converges absolutely and $\hat{f}$ is a bounded continuous function. The first task is to show that $\hat{f}$ is again Schwartz, and the engine is the pair of rules that turn calculus in x into algebra in k. They follow from a single integration by parts and a single differentiation under the integral sign.

The transform dictionary. For $f \in \mathcal{S}$ and any index j, integrate $\int e^{ik\cdot x}\partial_{x_j} f\, d^n x$ by parts (boundary terms vanish by rapid decay) and differentiate $\hat{f}$ under the integral; one finds

$$\widehat{\partial_{x_j} f}(k) = -\,i\,k_j\,\hat{f}(k), \tag{5.38}$$

$$\widehat{x_j f}\,(k) = -\,i\,\partial_{k_j}\hat{f}(k). \tag{5.39}$$

Differentiation in position becomes multiplication by momentum, and multiplication by position becomes differentiation in momentum. Iterating, $\widehat{\partial^\beta f} = (-ik)^\beta \hat{f}$ and $\widehat{x^\alpha f} = (-i\partial_k)^\alpha \hat{f}$, where $(-ik)^\beta = \prod_j (-ik_j)^{\beta_j}$. Two further rules round out the dictionary, each a one-line change of variables in Eq. (5.37): for the translate $(\tau_a f)(x) := f(x-a)$ and the modulated function $e^{ib\cdot x} f(x)$,

$$\widehat{\tau_a f}(k) = e^{\,ik\cdot a}\,\hat{f}(k), \qquad \widehat{e^{\,ib\cdot x} f}(k) = \hat{f}(k+b), \tag{5.40}$$

so *translation becomes a phase and a phase becomes a translation*; and for the rescaling $f(\lambda x)$ with $\lambda > 0$,

$$\widehat{f(\lambda\,\cdot)}(k) = \lambda^{-n}\,\hat{f}(k/\lambda), \tag{5.41}$$

the contraction in position dilating into momentum. Finally the two rules that the interacting theory lives on. Recall the convolution $(f*g)(x) = \int f(y)\, g(x-y)\, d^n y$;

for $f, g \in \mathcal{S}$ a direct application of Fubini's theorem to Eq. (5.37) gives the *convolution theorem* and its dual,

$$\widehat{f * g} = \hat{f}\,\hat{g}, \tag{5.42}$$

$$\widehat{f\,g} = \tfrac{1}{(2\pi)^n}\,\hat{f} * \hat{g}. \tag{5.43}$$

The first is immediate: $\widehat{f*g}(k) = \int d^n y\, f(y) e^{iky} \int d^n z\, g(z) e^{ikz} = \hat{f}(k)\hat{g}(k)$ after writing $z = x - y$. The second follows by inserting the inverse transform of f and using Eq. (5.40); the $(2\pi)^{-n}$ is the price of the convention's asymmetric placement of 2π's. *Convolution becomes pointwise product, and pointwise product becomes convolution*—the statement that interactions, which are pointwise products of fields, become momentum-space convolutions, i.e. the momentum-conserving integrals at vertices. On $\mathcal{S}$ both sides are unambiguous because the product and convolution of Schwartz functions are again Schwartz; on $\mathcal{S}'$ they hold only when one factor is regular enough to make the operation legal, a caveat we return to in Section 5.2.5.

Equipped with the dictionary, the central structural fact is now within reach.

Theorem 5.14 ($\mathcal{F}$ is a continuous bijection of $\mathcal{S}$)**.** *The Fourier transform Eq. (5.37) maps $\mathcal{S}(\mathbb{R}^n)$ into itself, continuously, and is a bijection whose inverse is the second formula in Eq. (5.37). Explicitly, $(\mathcal{F}^{-1}g)(x) = (2\pi)^{-n}\hat{g}(-x)$, so $\mathcal{F}^4 = (2\pi)^{2n}\,(\text{parity})^2 = (2\pi)^{2n}\,\mathrm{id}$ after the appropriate rescaling.*

Proof. Step 1: $\hat{f} \in \mathcal{S}$. For any multi-indices α, β, combine Eqs. (5.38) and (5.39):

$$k^\alpha\, \partial_k^\beta\, \hat{f}(k) \;=\; k^\alpha\, \widehat{(ix)^\beta f}(k) \;=\; (+i)^{|\beta|}(+i)^{|\alpha|}\, \widehat{\partial_x^\alpha (x^\beta f)}(k), \tag{5.44}$$

where in the last step we used Eq. (5.38) in reverse, turning the factor k^α into the transform of ∂_x^α of the argument. The function $\partial_x^\alpha(x^\beta f)$ is a finite sum of terms $x^{\beta'}\partial_x^{\alpha'} f$, each in $\mathcal{S} \subset L^1$; hence its Fourier transform is bounded, with

$$\left\|\hat{f}\right\|_{\alpha,\beta} = \sup_k \left|k^\alpha \partial_k^\beta \hat{f}(k)\right| \le \int_{\mathbb{R}^n} \left|\partial_x^\alpha (x^\beta f)\right| d^n x \le C_{\alpha,\beta} \sum_{\alpha' \le \alpha,\, \beta' \le \beta} \|f\|_{\beta',\alpha'}\ , \tag{5.45}$$

the last inequality because the L^1 norm of a Schwartz function is controlled by finitely many of its sup-seminorms (multiply and divide by $(1+|x|^2)^n$, whose reciprocal is integrable). Every Schwartz seminorm of $\hat{f}$ is finite, so $\hat{f} \in \mathcal{S}$; and Eq. (5.45) is exactly the estimate stating that $\mathcal{F}\colon \mathcal{S} \to \mathcal{S}$ is *continuous*.

Step 2: the multiplication formula. For $f, g \in \mathcal{S}$, Fubini's theorem applied to the absolutely convergent double integral gives the symmetric identity that lets one "move the hat":

$$\int_{\mathbb{R}^n} \hat{f}(x)\, g(x)\, d^n x = \iint e^{ix\cdot y} f(y) g(x)\, d^n y\, d^n x = \int_{\mathbb{R}^n} f(y)\, \hat{g}(y)\, d^n y. \tag{5.46}$$

This is Theorem 5.15 below; it is the seed of the duality definition in the next subsection.

Step 3: inversion. We must show $f = \mathcal{F}^{-1}\mathcal{F}f$, i.e. $f(x) = \int \frac{d^n k}{(2\pi)^n} e^{-ik\cdot x}\hat{f}(k)$. Regulate the k-integral with a Gaussian and write $\hat{f}$ out:

$$I_\epsilon(x) := \int \frac{d^n k}{(2\pi)^n}\, e^{-ik\cdot x}\, e^{-\epsilon|k|^2/2}\, \hat{f}(k) = \int_{\mathbb{R}^n} d^n y\, f(y) \underbrace{\int \frac{d^n k}{(2\pi)^n}\, e^{\,ik\cdot(y-x)-\epsilon|k|^2/2}}_{=\,G_\epsilon(y-x)}. \tag{5.47}$$

The inner k-integral is the Gaussian transform computed in Theorem 5.16 below: with the normalization there, $G_\epsilon(b) = (2\pi\epsilon)^{-n/2} e^{-|b|^2/2\epsilon}$, a Gaussian of total mass $\int G_\epsilon = 1$ and width $\sqrt{\epsilon}$. Thus $I_\epsilon = f * G_\epsilon$ is the convolution of f with an approximate identity, so $I_\epsilon(x) \to f(x)$ pointwise (indeed uniformly) as $\epsilon \to 0^+$, because f is continuous and G_ϵ concentrates at the origin. On the other hand, since $\hat{f} \in \mathcal{S} \subset L^1$, dominated convergence gives $I_\epsilon(x) \to \int \frac{d^n k}{(2\pi)^n} e^{-ik\cdot x}\hat{f}(k)$. Equating the two limits proves the inversion formula. Inversion makes $\mathcal{F}$ injective (it has a left inverse) and surjective (it has a right inverse), hence a bijection; the explicit form $\mathcal{F}^{-1}g = (2\pi)^{-n}\hat{g}(-\cdot)$ is read directly from Eq. (5.37). □

Lemma 5.15 (Multiplication formula / Parseval on $\mathcal{S}$)**.** *For all $f, g \in \mathcal{S}(\mathbb{R}^n)$, $\int \hat{f}\, g = \int f\, \hat{g}$, and consequently the Plancherel identity $\int |f|^2 = (2\pi)^{-n} \int \left|\hat{f}\right|^2$ holds.*

Proof. The first identity is Eq. (5.46). For Plancherel apply it with $g = \overline{\mathcal{F}^{-1}\bar{f}}$ and use inversion; the constant $(2\pi)^{-n}$ is again the convention's bookkeeping. We will not need Plancherel until the spectral discussions of later chapters, but record it here for completeness. □

The Gaussian computation that powers inversion is worth isolating, both because the result is used constantly and because the contour shift that appears in it is the first instance of the analytic-continuation move that organizes the whole chapter.

Worked Example 5.16 (The Gaussian is its own transform)**.** We evaluate $\hat{g}(k) = \int_{\mathbb{R}^n} d^n x\, e^{ik\cdot x}\, e^{-a|x|^2/2}$ for a constant a with $\operatorname{Re} a > 0$. The integrand factorizes over coordinates, so it suffices to do the one-dimensional integral

$$\mathcal{I}(\kappa) = \int_{-\infty}^{\infty} dx\, e^{\,i\kappa x - \frac{a}{2}x^2}. \tag{5.48}$$

Complete the square in the exponent: $i\kappa x - \frac{a}{2}x^2 = -\frac{a}{2}\left(x - \frac{i\kappa}{a}\right)^2 - \frac{\kappa^2}{2a}$. Hence

$$\mathcal{I}(\kappa) = e^{-\kappa^2/2a} \int_{-\infty}^{\infty} dx\, e^{-\frac{a}{2}\,(x - i\kappa/a)^2}. \tag{5.49}$$

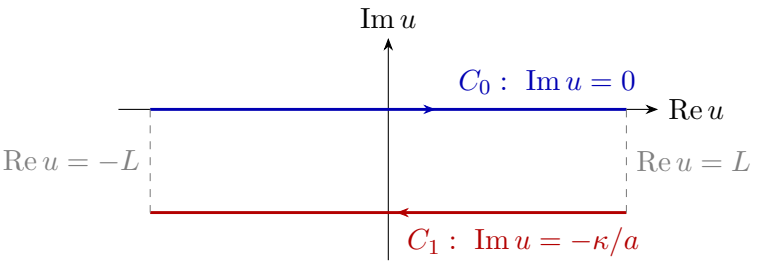


Figure 5.2: The contour shift in Theorem 5.16. The integrand $e^{-au^2/2}$ is entire; on the closed rectangle Cauchy's theorem holds, and the dashed vertical sides vanish as $L \to \infty$ because $\operatorname{Re} a > 0$. Hence the integral over the displaced line C_1 equals the integral over the real axis C_0. This is the prototype of every contour shift in the chapter; the oscillatory case a purely imaginary (the Minkowski path integral) is handled by the same move regularized by an $i\epsilon$, taken up in Section 5.3 (the next section).

The remaining integral runs over the horizontal line $\{x - i\kappa/a : x \in \mathbb{R}\}$, displaced from the real axis into the complex u-plane by the constant imaginary amount $-i\kappa/a$. We claim it equals the integral over the real axis. The function $u \mapsto e^{-au^2/2}$ is entire (no poles to cross), and on the rectangle with horizontal sides the real axis and the shifted line, closed by two vertical segments at $\operatorname{Re} u = \pm L$, Cauchy's theorem gives a vanishing contour integral. The two vertical contributions are bounded by $|\kappa/a| \cdot \sup e^{-a\operatorname{Re}(u^2)/2} \sim e^{-aL^2/2} \to 0$ as $L \to \infty$, because $\operatorname{Re} a > 0$ makes the Gaussian decay dominate the bounded imaginary excursion (Fig. 5.2). The shifted integral therefore equals the real one, $\int_{\mathbb{R}} e^{-au^2/2}\,du = \sqrt{2\pi/a}$ (the standard real Gaussian, analytically continued in a from the positive axis, with the principal branch of the square root). Collecting,

$$\mathcal{I}(\kappa) = \sqrt{\tfrac{2\pi}{a}}\, e^{-\kappa^2/2a}, \qquad \Longrightarrow \qquad \boxed{\hat{g}(k) = \left(\tfrac{2\pi}{a}\right)^{n/2} e^{-|k|^2/2a}} \tag{5.50}$$

In particular the choice $a = 1$ gives $\widehat{e^{-|x|^2/2}} = (2\pi)^{n/2} e^{-|k|^2/2}$: the Gaussian is its own Fourier transform up to the convention's normalization, the unique fixed point (up to scale) of $\mathcal{F}$. Setting $a = \epsilon$ and reading Eq. (5.50) as an inverse transform reproduces the nascent-delta kernel G_ϵ of Eq. (5.47): $\int \frac{dk}{(2\pi)} e^{ikb - \epsilon k^2/2} = (2\pi\epsilon)^{-1/2} e^{-b^2/2\epsilon}$ per coordinate.

5.2.3 Extension to tempered distributions by duality

We now have everything needed to define the Fourier transform of a distribution. The guiding principle is the multiplication formula Theorem 5.15: for an ordinary function f that is itself a tempered distribution, the identity $\int \hat{f}\,\varphi = \int f\,\hat{\varphi}$ says

that "pairing $\hat{f}$ against φ" equals "pairing f against $\hat{\varphi}$." The right-hand side makes sense for *any* tempered distribution in place of f, because $\hat{\varphi} \in \mathcal{S}$ whenever $\varphi \in \mathcal{S}$ (Theorem 5.14). We therefore promote the identity to a definition.

Definition 5.17 (Fourier transform of a tempered distribution)**.** For $T \in \mathcal{S}'(\mathbb{R}^n)$, its Fourier transform $\hat{T} = \mathcal{F}T$ is the tempered distribution defined by

$$\langle \hat{T}, \varphi \rangle := \langle T, \hat{\varphi} \rangle, \qquad \text{for all } \varphi \in \mathcal{S}(\mathbb{R}^n). \tag{5.51}$$

Theorem 5.18 (The transform is a bijection of $\mathcal{S}'$)**.** *Equation (5.51) defines $\hat{T}$ as an element of $\mathcal{S}'$; the map $\mathcal{F}\colon \mathcal{S}' \to \mathcal{S}'$ is a bijection with inverse $\langle \mathcal{F}^{-1}S, \varphi \rangle = \langle S, \mathcal{F}^{-1}\varphi \rangle$; and it agrees with the classical transform Eq. (5.37) on every $f \in \mathcal{S}$ (and more generally on every L^1 or L^2 function), so no notational clash arises.*

Proof. Well-defined and tempered. The right-hand side of Eq. (5.51) is finite because $\hat{\varphi} \in \mathcal{S}$ and $T \in \mathcal{S}'$. It is linear in φ. For continuity, if $\varphi_j \to \varphi$ in $\mathcal{S}$ then $\hat{\varphi}_j \to \hat{\varphi}$ in $\mathcal{S}$ by the continuity half of Theorem 5.14 (Eq. (5.45)), whence $\langle T, \hat{\varphi}_j \rangle \to \langle T, \hat{\varphi} \rangle$ by continuity of T; so $\langle \hat{T}, \varphi_j \rangle \to \langle \hat{T}, \varphi \rangle$ and $\hat{T} \in \mathcal{S}'$. *Bijection.* Define $\mathcal{F}^{-1}$ on $\mathcal{S}'$ by the displayed transpose formula. For $\varphi \in \mathcal{S}$, $\langle \mathcal{F}^{-1}\hat{T}, \varphi \rangle = \langle \hat{T}, \mathcal{F}^{-1}\varphi \rangle = \langle T, \mathcal{F}\mathcal{F}^{-1}\varphi \rangle = \langle T, \varphi \rangle$ using inversion on $\mathcal{S}$; likewise $\mathcal{F}\mathcal{F}^{-1} = \mathrm{id}$, so $\mathcal{F}$ is bijective on $\mathcal{S}'$ with the stated inverse. *Consistency.* If $T = f \in \mathcal{S}$ (or $f \in L^1$), then for every φ, $\langle \hat{T}, \varphi \rangle = \langle f, \hat{\varphi} \rangle = \int f \hat{\varphi} = \int \hat{f}\, \varphi = \langle \hat{f}, \varphi \rangle$, the middle equality being exactly Theorem 5.15; thus the distributional transform coincides with the function-theoretic one. □

The entire dictionary Eqs. (5.38) to (5.41) transfers verbatim to $\mathcal{S}'$, because each rule is the transpose of the corresponding rule on $\mathcal{S}$. For instance, the derivative rule on distributions reads

$$\begin{aligned} \langle \widehat{\partial_j T}, \varphi \rangle &= \langle \partial_j T, \hat{\varphi} \rangle = -\langle T, \partial_j \hat{\varphi} \rangle = -i \, \langle T, \widehat{x_j \varphi} \rangle \\ &= -i \, \langle \hat{T}, x_j \varphi \rangle = \langle -i k_j \, \hat{T}, \varphi \rangle \quad \Longrightarrow \quad \widehat{\partial_j T} = -i k_j \, \hat{T}, \end{aligned} \tag{5.52}$$

where we used the multiplication rule Eq. (5.39) on the test function in the form $\partial_j \hat{\varphi} = i \, \widehat{x_j \varphi}$, then the duality Eq. (5.51) to move the hat back onto T, and where k_j denotes the coordinate of the momentum space on which $\hat{T}$ lives. The distributional derivative $\partial_j T$ was defined in Section 5.1 (the previous section); the chain of pairings just pushes Eq. (5.38) through the duality. The convolution theorem Eq. (5.42) survives in the legal cases: if $\psi \in \mathcal{S}$ and $T \in \mathcal{S}'$, the convolution $T * \psi$ is a smooth, polynomially bounded function (hence tempered) and $\widehat{T * \psi} = \hat{\psi}\, \hat{T}$, the product now of a Schwartz function with a tempered distribution—a product that *is* defined precisely because one factor is smooth. We flag for emphasis, and develop in Section 5.2.5, that the unrestricted product of two tempered distributions is *not* defined; the convolution theorem is the safe face of a generally dangerous operation.

5.2.4 The catalogue every propagator uses

We can now compute, in two or three lines apiece, the transforms that every momentum-space computation in this book silently invokes. Each follows by feeding Eq. (5.51) a test function and using inversion.

The delta and the constant. The delta is the easiest: for any $\varphi \in \mathcal{S}$,

$$\langle \hat{\delta}, \varphi \rangle = \langle \delta, \hat{\varphi} \rangle = \hat{\varphi}(0) = \int_{\mathbb{R}^n} d^n x \, e^{i0\cdot x} \varphi(x) = \int \varphi = \langle 1, \varphi \rangle, \qquad \Longrightarrow \qquad \boxed{\hat{\delta} = 1} \tag{5.53}$$

The delta, the perfectly localized object, transforms to the constant 1, the perfectly delocalized one—the uncertainty principle in its starkest form. The reciprocal statement, the transform of the constant, is the single most-used Fourier identity in field theory, since it is the origin of every momentum-conserving $(2\pi)^n \delta^{(n)}$ at a vertex.

Worked Example 5.19 ($\widehat{1} = (2\pi)^n \delta$)**.** Apply Eq. (5.51) to $T = 1$:

$$\langle \hat{1}, \varphi \rangle = \langle 1, \hat{\varphi} \rangle = \int_{\mathbb{R}^n} d^n k \; \hat{\varphi}(k). \tag{5.54}$$

Now read the inverse transform Eq. (5.37) at $x = 0$: $\varphi(0) = \int \frac{d^n k}{(2\pi)^n} \hat{\varphi}(k)$, i.e. $\int d^n k \, \hat{\varphi}(k) = (2\pi)^n \varphi(0)$. Therefore $\langle \hat{1}, \varphi \rangle = (2\pi)^n \varphi(0) = (2\pi)^n \langle \delta, \varphi \rangle$ for every φ, which is the assertion

$$\boxed{\widehat{1} = (2\pi)^n \, \delta} \tag{5.55}$$

It is illuminating to see this as the width-to-zero limit of the Gaussian, the computation the spec of this chapter calls for and the one that makes the $(2\pi)^n$ tangible. The constant is the pointwise limit $1 = \lim_{a \to 0^+} e^{-a|x|^2/2}$, and this limit also holds in $\mathcal{S}'$ (dominated convergence in each pairing). Since $\mathcal{F}$ is continuous on $\mathcal{S}'$, we may transform inside the limit and use Eq. (5.50):

$$\widehat{1} = \lim_{a \to 0^+} \widehat{e^{-a|x|^2/2}} = \lim_{a \to 0^+} \left(\tfrac{2\pi}{a}\right)^{n/2} e^{-|k|^2/2a} = (2\pi)^n \lim_{a \to 0^+} \underbrace{\tfrac{1}{(2\pi a)^{n/2}} e^{-|k|^2/2a}}_{\text{unit-mass Gaussian}} = (2\pi)^n \delta(k). \tag{5.56}$$

The bracketed factor is a Gaussian of total integral 1 and width $\sqrt{a} \to 0$: a nascent delta. The leftover $(2\pi)^n$ is exactly the normalization factor of Eq. (5.55), now seen to be the volume factor the convention parks on the inverse transform. This is the identity that, in four spacetime dimensions, produces the $(2\pi)^4 \delta^{(4)}(\sum p)$ enforcing energy–momentum conservation at every interaction vertex (Chapters 9 and 13).

The Gaussian, again. The transform Eq. (5.50) is already a $\mathcal{S}'$ statement when read for $\operatorname{Re} a > 0$, and it is the workhorse of every Euclidean (Wick-rotated) propagator and every saddle-point evaluation later in the chapter. We record only that the purely imaginary case $a = -i\beta$ (the oscillatory Fresnel integral of the Minkowski path integral) is obtained from Eq. (5.50) by analytic continuation $a \to -i\beta + 0^+$, the limit existing as a tempered distribution; the $+0^+$ is the very $i\epsilon$ that the next two sections install as a contour prescription.

Step and sign: the appearance of PV and δ together. The last entry is the one that ties the Fourier transform back to Sokhotski–Plemelj and forward to the propagator. We work in one dimension and write the variable as ω in momentum space, anticipating its role as an energy.

Worked Example 5.20 (Transform of the Heaviside step and the sign)**.** Let $\theta(t)$ be the Heaviside step ($\theta = 1$ for $t > 0$, 0 for $t < 0$). It is bounded, hence tempered. Its forward transform $\int_0^\infty e^{i\omega t}\,dt$ does not converge as an ordinary integral, but it converges as a distribution, and the clean way to see the answer is to insert the convergence factor that defines θ as the limit of an integrable function: $\theta(t) = \lim_{\eta\to 0^+} \theta(t)e^{-\eta t}$ in $\mathcal{S}'$. For $\eta > 0$,

$$\widehat{\theta\, e^{-\eta t}}(\omega) = \int_0^\infty e^{\,i\omega t - \eta t}\,dt = \frac{1}{\eta - i\omega} = \frac{i}{\omega + i\eta}. \tag{5.57}$$

Taking $\eta \to 0^+$ inside the pairing and applying Sokhotski–Plemelj Theorem 5.10, $\dfrac{1}{\omega + i\eta} \to \mathrm{PV}\,\dfrac{1}{\omega} - i\pi\delta(\omega)$, we obtain

$$\boxed{\hat\theta(\omega) = i\Big(\mathrm{PV}\,\tfrac{1}{\omega} - i\pi\delta(\omega)\Big) = \pi\,\delta(\omega) + i\,\mathrm{PV}\,\frac{1}{\omega}} \tag{5.58}$$

The transform of the step splits into a delta and a principal value—the same two pieces that Sokhotski–Plemelj extracted from $1/(\omega + i\eta)$ in the previous section, now arising as the real (even) and imaginary (odd) parts of a causal object's spectrum. The sign function follows at once. Since $\operatorname{sgn}(t) = 2\theta(t) - 1$, linearity together with Eq. (5.55) (here $\widehat{1} = 2\pi\delta$ in $n = 1$) gives

$$\widehat{\operatorname{sgn}}(\omega) = 2\hat\theta(\omega) - \widehat{1} = 2\pi\delta(\omega) + 2i\,\mathrm{PV}\,\tfrac{1}{\omega} - 2\pi\delta(\omega) = 2i\,\mathrm{PV}\,\frac{1}{\omega}. \tag{5.59}$$

The delta pieces cancel: the sign function, being odd and unbounded only as a constant at infinity, has a purely imaginary, purely principal-value transform. Inverting Eq. (5.59) also records the companion fact $\widehat{\mathrm{PV}(1/t)} = i\pi\,\operatorname{sgn}(\omega)$, the transform of the principal value, which reappears in the dispersion relations of a later section.

Physical Insight

The decomposition $\hat{\theta} = \pi\delta + i\,\mathrm{PV}(1/\omega)$ is the spectral signature of *causality*. The step function $\theta(t)$ is what multiplies a field commutator to build a *retarded* (cause-precedes-effect) Green function. Its transform having a δ part (an on-shell, real, dissipative piece) and a PV part (an off-shell, reactive piece) tied together is, read at the level of response functions, the Kramers–Kronig relation between absorption and dispersion. The same PV/δ pairing, repackaged as $1/(\omega \pm i\eta)$, is what distinguishes the retarded, advanced, and Feynman propagators in the next section. Here is the chapter's through-line in one formula: a singular distribution has become a pole, and the side on which the pole sits will become a contour prescription.

5.2.5 Why products of distributions are dangerous

The Fourier dictionary turns pointwise products into convolutions (Eq. (5.43)). In the interacting theory the central objects are products of propagators *at the same spacetime point*, and these are products of distributions. It is essential to understand, before any field is quantized, that such products are in general *undefined*, because this single fact is the structural origin of the ultraviolet divergences that renormalization exists to treat.

The obstruction is already visible for the delta. There is no consistent way to assign a value to $\delta(x)^2$: approximating δ by nascent deltas δ_ϵ of width ϵ, the candidate $\delta_\epsilon(x)^2$ has integral $\sim 1/\epsilon \to \infty$, so no limit exists in $\mathcal{D}'$. More sharply, a theorem of L. Schwartz states that there is no associative, commutative bilinear product on $\mathcal{D}'(\mathbb{R})$ that restricts to the ordinary product on continuous functions and respects differentiation by the Leibniz rule: the distributions cannot be made into such an algebra. The product of two distributions is therefore a *partial* operation, legal only when the singularities of the two factors do not collide.

The precise criterion, which we state at the level we will need and not reprove here, is Hörmander's. To each distribution u one attaches its *wavefront set* $\mathrm{WF}(u) \subset (\mathbb{R}^n \times (\mathbb{R}^n \setminus \{0\}))$, a refinement of the singular support that records not only *where* u is singular but *in which momentum directions* that singularity points (the directions in which $\hat{u}$ fails to decay rapidly when u is localized near the point). For the delta on $\mathbb{R}$, $\mathrm{WF}(\delta) = \{(0,\xi) : \xi \neq 0\}$: singular at the origin, in *all* momentum directions.

Proposition 5.21 (Hörmander's product criterion)**.** *Let $u, v \in \mathcal{D}'(\mathbb{R}^n)$. If their wavefront sets satisfy the transversality condition*

$$(x,\xi) \in \mathrm{WF}(u) \implies (x,-\xi) \notin \mathrm{WF}(v) \qquad \text{for all } x,\ \xi \neq 0, \tag{5.60}$$

then the product uv is well-defined as a distribution, independent of the approximating sequences, and $\mathrm{WF}(uv)$ *is contained in the natural sum of the two wavefront sets. If the condition fails—if some singular point carries opposite momentum directions in u and in v, so that the convolution $\hat{u} * \hat{v}$ of Eq. (5.43) diverges—the product is not defined by any limiting procedure.*

The square $\delta \cdot \delta$ violates Eq. (5.60) at the origin, where $(0, \xi)$ and $(0, -\xi)$ both lie in $\mathrm{WF}(\delta)$: hence δ^2 is meaningless, in agreement with the divergent estimate above. By contrast $\theta \cdot \delta$ at points away from the origin is fine, and a smooth function times any distribution is always legal because a smooth factor has empty wavefront set.

Physical Insight

This is the mathematical reason quantum field theory needs renormalization, stated before we have written down a single loop. A propagator is a distribution whose wavefront set sits on the light cone; a Feynman diagram with a loop multiplies two or more propagators that share spacetime arguments, and at coincident points their wavefront sets point in opposite directions along the cone—exactly the collision Eq. (5.60) forbids. The product of propagators at a vertex is therefore an ill-defined distribution, and the divergent loop integral is the symptom. Renormalization is, from this vantage, the systematic procedure for *extending* these ill-defined products to genuine distributions by subtracting their coincident-point ambiguity—the Epstein–Glaser viewpoint. We take this up properly in Chapter 21, where the superficial degree of divergence is precisely a measure of how badly Eq. (5.60) fails.

5.2.6 The Feynman propagator as a tempered distribution

We have assembled enough to give the central object of the chapter its precise status. In four-dimensional Minkowski spacetime, with the metric pairing of Chapter A and $n = 4$, the *Feynman propagator* of a free scalar of mass m is the tempered distribution $D_F \in \mathcal{S}'(\mathbb{R}^{1,3})$ whose Fourier transform is the locally integrable function

Definition 5.22 (Momentum-space Feynman propagator)**.**

$$\widehat{D_F}(p) \;=\; \frac{i}{p^2 - m^2 + i\epsilon}, \qquad D_F(x) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i\, e^{-ip\cdot x}}{p^2 - m^2 + i\epsilon}, \tag{5.61}$$

with $p^2 = p_\mu p^\mu = (p^0)^2 - |\mathbf{p}|^2$ and $\epsilon \to 0^+$ understood after pairing against test functions.

Three points justify every word of this definition with the machinery just built. First, *it is a legitimate tempered distribution.* For fixed $\epsilon > 0$ the denominator $p^2 - m^2 + i\epsilon$ never vanishes (its imaginary part is ϵ), so $\widehat{D_F}$ is a smooth, polynomially bounded—indeed bounded by $1/\epsilon$—function of p, manifestly an element of $\mathcal{S}'$. By Theorem 5.18 its inverse transform D_F is then a well-defined tempered distribution on position space, and the $\epsilon \to 0^+$ limit exists in $\mathcal{S}'$ because the boundary value $1/(p^2 - m^2 + i\epsilon)$ does, exactly as in the previous section. The propagator is not a function—it is singular on the mass shell $p^2 = m^2$ and along the light cone in x—but it *is* a distribution, and the slogan "the propagator is a distribution" is now a theorem.

Second, *the $i\epsilon$ is read through Sokhotski–Plemelj.* Treating $p^2 - m^2$ as the real variable in Theorem 5.10,

$$\frac{i}{p^2 - m^2 + i\epsilon} \;=\; i\,\mathrm{PV}\,\frac{1}{p^2 - m^2} \;+\; \pi\,\delta(p^2 - m^2), \tag{5.62}$$

so the propagator carries a principal-value part (the off-shell, virtual propagation) and a $\delta(p^2 - m^2)$ part (the on-shell, real-particle content) welded together in exactly the proportion the step function displayed in Eq. (5.58). The same analytic anatomy—a pole plus a delta, a PV plus an on-shell piece—runs from $\theta(t)$ through $1/(x \pm i\epsilon)$ to the propagator: this is one object wearing three costumes.

Third, *this is the object the threads will need.* The momentum-space form Eq. (5.61) is precisely the scalar propagator of ϕ^4 theory, to be derived from the free path integral and the mode expansion in Chapter 9, and its photon analogue underlies QED in Chapter 13; no thread is formally advanced in this foundations chapter, but the analytic object every thread later manipulates has now been built and certified. What remains is purely computational and is the business of the next section: the position-space content of Eq. (5.61) is locked inside the energy integral $\int dp^0\, e^{-ip^0x^0}/(p^2 - m^2 + i\epsilon)$, whose integrand has two poles in the complex p^0 plane displaced off the real axis by the $i\epsilon$. To turn the distribution into a number one must choose how a contour passes those poles—and the retarded, advanced, and Feynman propagators are nothing but the three such choices. The contour calculus that makes those choices, and evaluates the integral, is where we go now.

5.3 Contour integration and the propagator prescriptions

The previous section handed us the Feynman propagator as a *distribution* in momentum space,

$$\widetilde{D}_F(p) = \frac{i}{p^2 - m^2 + i\varepsilon}, \tag{5.63}$$

the object Eq. (5.61), with the $i\varepsilon$ understood as the boundary-value prescription that makes the singular denominator a well-defined tempered distribution through the Sokhotski–Plemelj identity Eq. (5.22). That is a complete *definition*, but it is not yet a number. To extract a number—the propagator in position space, the amplitude for a particle to get from one event to another—we must perform the Fourier integral $\int d^4p\, e^{-ip\cdot x}\, \widetilde{D}_F(p)/(2\pi)^4$, and the heart of that integral is the contour integral over the energy variable p^0. The purpose of this section is to make the $i\varepsilon$ *operational*: to show that it is precisely an instruction about how a contour in the complex p^0-plane is to step around two poles, and that the celebrated retarded, advanced, and Feynman propagators are not three different functions handed down from on high but three choices of that single instruction, applied to one and the same integrand.

We proceed in three movements. First we recall—reviewing, not rebuilding, since complex analysis through the residue theorem is a prerequisite—exactly the residue calculus we will deploy, and we prove in full the one estimate that is genuinely a working tool rather than a memory: Jordan's lemma, which controls the arc that closes a Fourier contour at infinity. Second, we locate the poles of Eq. (5.63) in the complex p^0-plane and read off how the $i\varepsilon$ displaces them. Third, in a single self-contained worked computation, we evaluate the energy integral three times—retarded, advanced, Feynman—from the same integrand, and watch the contour decide which poles it encloses. The Feynman case will deliver exactly the time-ordered combination $\theta(t)e^{-iE_{\mathbf{p}}t} + \theta(-t)e^{+iE_{\mathbf{p}}t}$ that Chapter 9 will recognize as the vacuum expectation value $\langle 0|\mathcal{T}\phi\phi|0\rangle$, and that ϕ^4 theory (from Chapter 9) and QED (from Chapter 13) will quantize. No thread is formally advanced here—the interacting field theories are not defined until Part II—so the computation stands on its own as the model calculation that *plants* the analytic object those threads later need.

5.3.1 The residue calculus, reviewed for deployment

We assume the reader has met holomorphic functions, contour integrals, and the residue theorem; we restate them only to fix notation and to isolate the one fact we must prove. Throughout, $z = x + iy$ is a complex variable, a *contour* is a piecewise-C^1 curve, and "positively oriented" means counterclockwise. The residue of a meromorphic f at an isolated singularity z_0 is the coefficient a_{-1} in its Laurent expansion $f(z) = \sum_n a_n (z - z_0)^n$ about z_0, written $\operatorname{Res}_{z=z_0} f$. For a *simple* pole the only case we need below,

$$\operatorname{Res}_{z=z_0} f = \lim_{z\to z_0} (z-z_0)\, f(z), \qquad \text{and if } f = \frac{g}{h} \text{ with } h(z_0) = 0,\ h'(z_0) \neq 0,\ \operatorname{Res}_{z=z_0} \frac{g}{h} = \frac{g}{h'} \tag{5.64}$$

Proposition 5.23 (Cauchy's theorem and the residue theorem; reviewed, not proved)**.** *Let $D \subset \mathbb{C}$ be open and simply connected, and let f be holomorphic on*

D except at finitely many isolated singularities. Let γ be a positively oriented simple closed contour in D enclosing the singularities $z_1, \dots, z_n$ and no others. Then

$$\oint_\gamma f(z)\,dz = 2\pi i \sum_{k=1}^{n} \operatorname{Res}_{z=z_k} f. \tag{5.65}$$

In particular, if f is holomorphic everywhere inside γ (no singularities enclosed), the integral vanishes (Cauchy's integral theorem). A clockwise *(negatively oriented) closed contour gives instead $-2\pi i \sum_k \operatorname{Res}_{z=z_k} f$.*

These are standard and are reviewed here only to be used; the proofs belong to a complex-analysis course and are not reproduced. What is *not* merely a quoted fact, and what we therefore prove in full, is the lemma that licenses the entire method we are about to use: when we evaluate a Fourier-type integral $\int_{-\infty}^{\infty} e^{ikx} f(k)\,dk$ by closing the real axis into a large semicircle and applying Eq. (5.65), we must know that the contribution of the semicircular arc vanishes as its radius grows. For an integrand decaying as fast as $1/k^2$—which is the case for the propagator denominator $1/(k^2 - \dots)$—this is immediate, since the integrand falls as R^{-2} while the arc length grows only as R. But many integrals in this book carry only a single power, $f(k) \sim 1/k$ (the Fourier transforms of the sign and Heaviside distributions from §2 are exactly of this type, and so are individual terms in spectral representations), and there the cruder estimate fails: $R^{-1} \cdot R = O(1)$. The oscillatory factor e^{ikx} must be made to do the work. That is Jordan's lemma.

5.3.2 Jordan's lemma

Lemma 5.24 (Jordan's lemma)**.** *Let C_R^+ denote the semicircular arc $z = Re^{i\theta}$, $\theta \in [0, \pi]$, in the upper half-plane, and suppose f is continuous on C_R^+ for all large R with*

$$M_R \equiv \max_{0 \le \theta \le \pi} \left| f(Re^{i\theta}) \right| \longrightarrow 0 \qquad \text{as } R \to \infty. \tag{5.66}$$

Then for every real $a > 0$,

$$\lim_{R \to \infty} \int_{C_R^+} e^{iaz} f(z)\,dz = 0. \tag{5.67}$$

The mirror statement holds in the lower half-plane: if C_R^- is the arc $\theta \in [-\pi, 0]$ and $M_R \to 0$ there, then $\int_{C_R^-} e^{iaz} f(z)\,dz \to 0$ for every $a < 0$.

Proof. Parametrize C_R^+ by $z = Re^{i\theta}$, $dz = iRe^{i\theta}\,d\theta$, so that $|dz| = R\,d\theta$. The exponent is

$$iaz = iaR(\cos\theta + i\sin\theta) = iaR\cos\theta - aR\sin\theta, \tag{5.68}$$

whose modulus factor is $|e^{iaz}| = e^{-aR\sin\theta}$. Bounding the integrand by its modulus,

$$\left|\int_{C_R^+} e^{iaz} f(z)\,dz\right| \le \int_0^\pi e^{-aR\sin\theta}\,|f(Re^{i\theta})|\,R\,d\theta \le M_R\,R\int_0^\pi e^{-aR\sin\theta}\,d\theta. \quad (5.69)$$

The integrand $e^{-aR\sin\theta}$ is symmetric about $\theta=\pi/2$, since $\sin\theta=\sin(\pi-\theta)$, so

$$\int_0^\pi e^{-aR\sin\theta}\,d\theta = 2\int_0^{\pi/2} e^{-aR\sin\theta}\,d\theta. \quad (5.70)$$

Now we use the elementary *Jordan inequality*: on $[0,\pi/2]$ the sine lies above its chord,

$$\sin\theta \ge \frac{2\theta}{\pi}, \qquad 0\le\theta\le\frac{\pi}{2}. \quad (5.71)$$

This is geometrically the statement that the concave function $\sin\theta$, which agrees with the straight line $2\theta/\pi$ at the endpoints $\theta=0$ and $\theta=\pi/2$, lies above that line between them; analytically, $g(\theta)\equiv\sin\theta-2\theta/\pi$ satisfies $g(0)=g(\pi/2)=0$ and $g''(\theta)=-\sin\theta\le 0$, so g is concave and hence nonnegative on the interval. Because $a>0$ and $R>0$, the exponential is decreasing in its argument, so Eq. (5.71) gives the pointwise bound $e^{-aR\sin\theta}\le e^{-2aR\theta/\pi}$ on $[0,\pi/2]$. Therefore

$$2\int_0^{\pi/2} e^{-aR\sin\theta}\,d\theta \le 2\int_0^{\pi/2} e^{-2aR\theta/\pi}\,d\theta = 2\cdot\frac{\pi}{2aR}\Big(1-e^{-aR}\Big) = \frac{\pi}{aR}\Big(1-e^{-aR}\Big) \le \frac{\pi}{aR}. \quad (5.72)$$

Combining Eqs. (5.69), (5.70) and (5.72),

$$\left|\int_{C_R^+} e^{iaz} f(z)\,dz\right| \le M_R\,R\cdot\frac{\pi}{aR} = \frac{\pi}{a}\,M_R \xrightarrow[R\to\infty]{} 0, \quad (5.73)$$

since $M_R\to 0$ by hypothesis. The factors of R have cancelled exactly: this is the whole content of the lemma, that the linear growth of the arc length is defeated by the integrated exponential suppression even though f alone need only vanish. The lower-half-plane statement follows by the substitution $\theta\to-\theta$, which sends $\sin\theta\to-\sin\theta$ and therefore requires $a<0$ to keep the exponent $-|a|R|\sin\theta|$ negative. □

The hypothesis $a>0\Leftrightarrow$ close upward, $a<0\Leftrightarrow$ close downward is the crucial bookkeeping, and it is worth stating as the operational rule that we will invoke without further comment for the rest of the book.

Proposition 5.25 (Closure rule for Fourier-type integrals)**.** *Let f be meromorphic with finitely many poles, none on the real axis, and with $M_R\to 0$ on large*

semicircles in at least one half-plane. Then for real b,

$$\int_{-\infty}^{\infty} e^{ibk} f(k)\,dk = \begin{cases} 2\pi i \displaystyle\sum_{\text{poles in UHP}} \operatorname{Res} f, & b > 0 \quad (\textit{close in the upper half-plane}), \\ -2\pi i \displaystyle\sum_{\text{poles in LHP}} \operatorname{Res} f, & b < 0 \quad (\textit{close in the lower half-plane}). \end{cases} \tag{5.74}$$

Proof. For $b > 0$, close the real-axis integral with the arc C_R^+. The closed contour $[-R, R] \cup C_R^+$ is positively oriented and, for R larger than the modulus of every pole, encloses exactly the poles in the upper half-plane, so Eq. (5.65) gives $\int_{-R}^{R} + \int_{C_R^+} = 2\pi i \sum_{\text{UHP}} \operatorname{Res} f$. By Jordan's lemma Theorem 5.24 with $a = b > 0$ the arc term vanishes as $R \to \infty$, leaving the first line. For $b < 0$ close downward with C_R^-; this closed contour is *clockwise* (it runs $-R \to R$ along the real axis and then back through the lower arc), hence carries the sign $-2\pi i$, and Jordan's lemma with $a = b < 0$ kills the arc, giving the second line. □

The sign of the exponent's coefficient b dictates the closure: this is the single most-used fact in every propagator and amplitude computation in this book, and it is what turns the abstract $i\varepsilon$ into the concrete $\theta(\pm t)$ that physically encodes the direction of time. We now apply it.

5.3.3 Pole structure of the momentum-space propagator

Fix the spatial momentum $\mathbf{p}$ and regard the Feynman denominator in Eq. (5.63) as a function of the complex energy p^0. With the mostly-minus metric, $p^2 = (p^0)^2 - \mathbf{p}^2$, so writing

$$E_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + m^2} \; > 0 \tag{5.75}$$

for the on-shell energy, the denominator becomes

$$p^2 - m^2 + i\varepsilon = (p^0)^2 - (\mathbf{p}^2 + m^2) + i\varepsilon = (p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon. \tag{5.76}$$

This is a quadratic in p^0 with roots at $p^0 = \pm\sqrt{E_{\mathbf{p}}^2 - i\varepsilon}$. Since ε is a positive infinitesimal we may expand the square root to first order, $\sqrt{E_{\mathbf{p}}^2 - i\varepsilon} = E_{\mathbf{p}}\sqrt{1 - i\varepsilon/E_{\mathbf{p}}^2} = E_{\mathbf{p}} - i\varepsilon/(2E_{\mathbf{p}}) + O(\varepsilon^2)$. Absorbing the positive constant $1/(2E_{\mathbf{p}})$ into the redefinition of the infinitesimal (a standard and harmless abuse: only the *sign* of ε carries meaning, never its magnitude, and a positive multiple of a positive infinitesimal is again a positive infinitesimal), the two roots are

$$p_+^0 = +E_{\mathbf{p}} - i\varepsilon \qquad \text{and} \qquad p_-^0 = -E_{\mathbf{p}} + i\varepsilon, \tag{5.77}$$

and the denominator factorizes as

$$(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon = (p^0 - E_{\mathbf{p}} + i\varepsilon)(p^0 + E_{\mathbf{p}} - i\varepsilon). \tag{5.78}$$

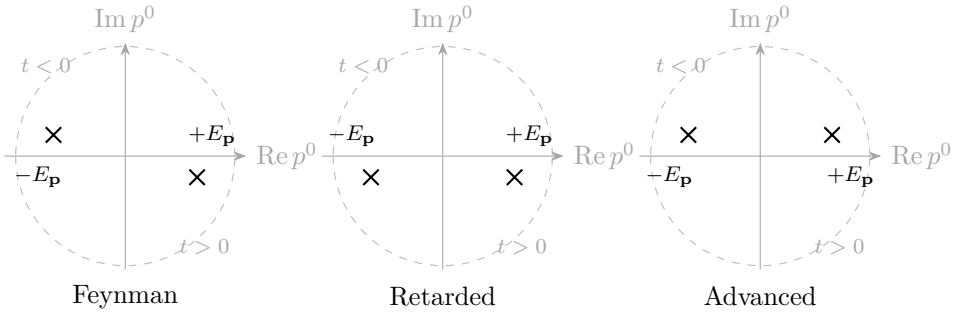


Figure 5.3: The three propagator prescriptions as pole placements in the complex p^0-plane, for fixed spatial momentum $\mathbf{p}$, together with the semicircular arcs that close the energy integral. The exponential $e^{-ip^0 t}$ forces closure *downward* for $t > 0$ and *upward* for $t < 0$ (Theorem 5.25). The Feynman prescription straddles the axis ($+E_{\mathbf{p}}$ below, $-E_{\mathbf{p}}$ above); the retarded keeps both poles below, so the upward $t < 0$ contour encloses nothing and the propagator vanishes for $t < 0$; the advanced keeps both above, vanishing for $t > 0$.

The reader can verify the factorization by multiplying out: the cross terms in ε cancel and the ε^2 term is dropped, leaving $(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon(p^0 + E_{\mathbf{p}}) - i\varepsilon(p^0 - E_{\mathbf{p}}) = (p^0)^2 - E_{\mathbf{p}}^2 + 2i\varepsilon E_{\mathbf{p}}$, which is $(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon$ after the same rescaling of the infinitesimal.

The content of Eq. (5.77) is the geometric statement around which the whole section turns: *the Feynman $i\varepsilon$ pushes the positive-energy pole $+E_{\mathbf{p}}$ infinitesimally below the real axis and the negative-energy pole $-E_{\mathbf{p}}$ infinitesimally above it.* The two poles straddle the real p^0-axis on opposite sides. This asymmetric placement is the algebraic fingerprint of time-ordering, as the worked computation will show. It is also exactly the placement that will permit the real p^0-contour to be rotated onto the imaginary axis without crossing a pole—the Wick rotation that Section 5.4 builds in the next section—because a contour swung from the positive real axis up to the positive imaginary axis sweeps through the first quadrant, which the Feynman prescription has kept clear of poles.

The retarded and advanced prescriptions correspond to placing *both* poles on the same side of the real axis. Concretely they replace $p^0 \to p^0 + i\varepsilon$ (retarded) or $p^0 \to p^0 - i\varepsilon$ (advanced) uniformly in the energy argument, i.e. they use the denominators

$$\text{retarded:}\quad (p^0 + i\varepsilon)^2 - E_{\mathbf{p}}^2, \qquad \text{advanced:}\quad (p^0 - i\varepsilon)^2 - E_{\mathbf{p}}^2. \tag{5.79}$$

For the retarded denominator the roots are $p^0 = -i\varepsilon \pm E_{\mathbf{p}}$, i.e. *both* poles sit just below the real axis at $\pm E_{\mathbf{p}} - i\varepsilon$; for the advanced denominator both sit just above, at $\pm E_{\mathbf{p}} + i\varepsilon$. Figure 5.3 collects the three pole layouts together with the contours that close them.

5.3.4 The three propagators from one integrand

We now compute. The position-space propagator is the four-dimensional Fourier transform of Eq. (5.63), in the book's fixed convention (Chapter A),

$$D_F(x) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\, \frac{i}{p^2 - m^2 + i\varepsilon} = \int \frac{d^3p}{(2\pi)^3}\, e^{i\mathbf{p}\cdot\mathbf{x}} \underbrace{\int_{-\infty}^{\infty} \frac{dp^0}{2\pi}\, \frac{i\, e^{-ip^0 t}}{(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon}}_{\equiv\, I_F(t)}, \tag{5.80}$$

where we used $p\cdot x = p^0 t - \mathbf{p}\cdot\mathbf{x}$ with $t \equiv x^0$, and the spatial part has been separated out cleanly. Every physical and analytic feature of the propagator lives in the inner one-dimensional energy integral $I_F(t)$; the spatial integral is a spectator that merely reattaches the $\mathbf{x}$-dependence. The worked example below evaluates I_F and its retarded and advanced cousins from the identical integrand, the only difference being the placement of the two poles fixed in Section 5.3.3.

Worked Example 5.26 (The three energy integrals)**.** We evaluate the energy integral

$$I(t) = \int_{-\infty}^{\infty} \frac{dp^0}{2\pi}\, \frac{i\, e^{-ip^0 t}}{(p^0)^2 - E_{\mathbf{p}}^2} \tag{5.81}$$

for each of the three pole prescriptions of Fig. 5.3, recovering the Feynman, retarded, and advanced propagators in turn.

Setting up the closure. The exponent is $-ip^0 t$, so in the notation of the closure rule Theorem 5.25 the coefficient of ip^0 is $b = -t$. Hence:

- for $t > 0$ we have $b < 0$ and must close *downward*, in the lower half-plane, picking up $-2\pi i$ times the residues there;
- for $t < 0$ we have $b > 0$ and must close *upward*, in the upper half-plane, picking up $+2\pi i$ times the residues there.

The integrand decays as $|p^0|^{-2}$ on the arcs, so the arc contribution vanishes by the elementary estimate (and a fortiori by Jordan's lemma Theorem 5.24); the closure is legitimate in every case below.

Two residues, computed once. Writing the denominator in factored form $(p^0 - E_{\mathbf{p}})(p^0 + E_{\mathbf{p}})$ (the $i\varepsilon$ only moves the poles; it does not change the residues, which we evaluate at the limiting locations $\pm E_{\mathbf{p}}$), the residues of the integrand $i\, e^{-ip^0 t}/[(p^0 - E_{\mathbf{p}})(p^0 + E_{\mathbf{p}})]$ at the two poles are, by Eq. (5.64),

$$\mathrm{Res}_{p^0 = +E_{\mathbf{p}}} = \frac{i\, e^{-iE_{\mathbf{p}}t}}{(E_{\mathbf{p}} + E_{\mathbf{p}})} = \frac{i\, e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}, \tag{5.82}$$

$$\mathrm{Res}_{p^0 = -E_{\mathbf{p}}} = \frac{i\, e^{+iE_{\mathbf{p}}t}}{(-E_{\mathbf{p}} - E_{\mathbf{p}})} = -\frac{i\, e^{+iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}. \tag{5.83}$$

These two numbers, assembled according to which poles each contour encloses, produce all three propagators.

Feynman ($+E_{\mathbf{p}}$ below, $-E_{\mathbf{p}}$ above). For $t > 0$ close downward; the lower half-plane contains only the pole at $p^0 = +E_{\mathbf{p}} - i\varepsilon$. The contour is clockwise, so

$$I_F(t>0) = \frac{1}{2\pi}\,(-2\pi i)\,\mathrm{Res}_{p^0=+E_{\mathbf{p}}} = \frac{1}{2\pi}\,(-2\pi i)\,\frac{i\,e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}} = \frac{e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}, \tag{5.84}$$

using $(-2\pi i)(i) = 2\pi$. For $t < 0$ close upward; the upper half-plane contains only the pole at $p^0 = -E_{\mathbf{p}} + i\varepsilon$. The contour is counterclockwise, so

$$I_F(t<0) = \frac{1}{2\pi}\,(+2\pi i)\,\mathrm{Res}_{p^0=-E_{\mathbf{p}}} = \frac{1}{2\pi}\,(2\pi i)\left(-\frac{i\,e^{+iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}\right) = \frac{e^{+iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}, \tag{5.85}$$

using $(2\pi i)(-i) = 2\pi$. Combining the two cases with Heaviside step functions,

$$\boxed{I_F(t) = \frac{1}{2E_{\mathbf{p}}}\Big[\theta(t)\,e^{-iE_{\mathbf{p}}t} + \theta(-t)\,e^{+iE_{\mathbf{p}}t}\Big].} \tag{5.86}$$

Retarded (both poles below). For $t > 0$ close downward; now *both* poles lie in the lower half-plane, so the clockwise contour encloses both:

$$\begin{aligned} I_R(t>0) &= \frac{1}{2\pi}\,(-2\pi i)\Big(\mathrm{Res}_{+E_{\mathbf{p}}} + \mathrm{Res}_{-E_{\mathbf{p}}}\Big) = (-i)\left(\frac{i\,e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}} - \frac{i\,e^{+iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}\right) \\ &= \frac{1}{2E_{\mathbf{p}}}\big(e^{-iE_{\mathbf{p}}t} - e^{+iE_{\mathbf{p}}t}\big) = -\frac{i\,\sin(E_{\mathbf{p}}t)}{E_{\mathbf{p}}}, \end{aligned} \tag{5.87}$$

where we used $e^{-i\phi} - e^{+i\phi} = -2i\sin\phi$. For $t < 0$ close upward; the upper half-plane now contains *no* poles, so by Cauchy's theorem $I_R(t<0) = 0$. Thus

$$\boxed{I_R(t) = -\theta(t)\,\frac{\sin(E_{\mathbf{p}}t)}{E_{\mathbf{p}}}\,i \;=\; -\,\frac{i\,\theta(t)}{E_{\mathbf{p}}}\,\sin(E_{\mathbf{p}}t).} \tag{5.88}$$

The retarded propagator vanishes identically for $t < 0$: it has *support* only in the forward light-cone in time, which is exactly the causal requirement that an effect cannot precede its source. The two poles sitting on the same (lower) side is what makes the $t < 0$ contour enclose nothing.

Advanced (both poles above). The computation mirrors the retarded one. For $t < 0$ close upward, enclosing both poles counterclockwise:

$$I_A(t<0) = \frac{1}{2\pi}\,(+2\pi i)\Big(\mathrm{Res}_{+E_{\mathbf{p}}} + \mathrm{Res}_{-E_{\mathbf{p}}}\Big) = (i)\left(\frac{i\,e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}} - \frac{i\,e^{+iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}\right)$$

$$= -\frac{1}{2E_{\mathbf{p}}}(e^{-iE_{\mathbf{p}}t} - e^{+iE_{\mathbf{p}}t}) = \frac{i\,\sin(E_{\mathbf{p}}t)}{E_{\mathbf{p}}}, \tag{5.89}$$

while for $t > 0$ the downward contour encloses no poles and $I_A(t > 0) = 0$. Hence

$$\boxed{I_A(t) = +\frac{i\,\theta(-t)}{E_{\mathbf{p}}}\,\sin(E_{\mathbf{p}}t),} \tag{5.90}$$

supported only in the backward time-cone $t < 0$, as the name advertises. Comparing Eqs. (5.88) and (5.90) one sees $I_A(t) = I_R(-t)$, the time-reversal of the retarded propagator, exactly as the mirror-image pole placements in Fig. 5.3 demand.

It is worth pausing on what the worked example demonstrates, because it is the conceptual payload of the section. One integrand—the manifestly meromorphic $i\,e^{-ip^0t}/[(p^0)^2 - E_{\mathbf{p}}^2]$—produced three physically distinct propagators, and the *only* input that distinguished them was an infinitesimal displacement of two poles off the real axis. The $i\varepsilon$ is not a regulator one bolts on and forgets; it is the entire physical content, dictating which residues a closed contour sees and therefore which combinations of $e^{\pm iE_{\mathbf{p}}t}$ appear and with what step-function support. Causality (the vanishing of D_R for $t < 0$), and the particle/antiparticle bookkeeping of the Feynman propagator, are read off the geometry of Fig. 5.3.

Physical Insight: the $i\varepsilon$ *is* the boundary condition

A Green's function of a differential operator is only determined once boundary conditions are imposed; for the wave operator $\Box + m^2$ the homogeneous solutions are precisely the on-shell modes $e^{\mp iE_{\mathbf{p}}t}$, so a Green's function is ambiguous up to their addition. In the frequency domain that ambiguity is exactly the freedom to slide poles across the real axis. The retarded prescription chooses "no signal before the source," the advanced "no signal after," and the Feynman prescription chooses the mixed boundary condition—positive frequencies forward in time, negative frequencies backward—that, as Chapter 9 will show, is what the vacuum of a relativistic quantum field actually realizes. There is one analytic object, the propagator-as-distribution; the three names are three boundary conditions, and the $i\varepsilon$ is how a boundary condition is written in the integrand.

5.3.5 The Feynman propagator and time ordering

The Feynman result Eq. (5.86) deserves to be examined on its own, because it is the object the rest of the book is built to compute. Reattaching the spatial

integral from Eq. (5.80),

$$D_F(x) = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\mathbf{p}}}\,e^{i\mathbf{p}\cdot\mathbf{x}}\Big[\theta(t)\,e^{-iE_{\mathbf{p}}t} + \theta(-t)\,e^{+iE_{\mathbf{p}}t}\Big]. \tag{5.91}$$

Read the bracket. For $t > 0$ the propagator carries the *positive-frequency* factor $e^{-iE_{\mathbf{p}}t}$: a disturbance created at the earlier time propagates forward with the energy of a particle of mass m. For $t < 0$ it carries $e^{+iE_{\mathbf{p}}t} = e^{-iE_{\mathbf{p}}|t|}$, again a positive-energy phase but running in the opposite time sense—the contribution one will later interpret as an antiparticle propagating forward, or equivalently a particle propagating backward. The combination is precisely the *time-ordered* structure

$$I_F(t) = \frac{1}{2E_{\mathbf{p}}}\Big[\theta(t)\,e^{-iE_{\mathbf{p}}t} + \theta(-t)\,e^{+iE_{\mathbf{p}}t}\Big], \tag{5.92}$$

where the step functions place the larger-time factor on the left in each term. In Chapter 9 this exact object will be identified, mode by mode, with the vacuum two-point function $\langle 0|\mathcal{T}\phi(x)\phi(0)|0\rangle$ of the free scalar field, the time-ordering symbol $\mathcal{T}$ being the field-theoretic origin of the very $\theta(\pm t)$ we have just produced by contour choice. We have, in other words, computed $\langle 0|\mathcal{T}\phi\phi|0\rangle$ before we have a field ϕ: the analytic structure determines the answer, and quantization in Part III will only supply its operator interpretation. This is why the present calculation is the model computation that "plants" the object the threads need rather than itself advancing a thread.

Historical Note: Stückelberg, Feynman, and the two-sided propagator

The recognition that a single causal-looking propagator could carry positive frequencies forward and negative frequencies backward in time—unifying particle and antiparticle into one analytic object—is due to Stückelberg in the late 1930s and to Feynman in his 1949 spacetime approach to quantum electrodynamics. The $i\varepsilon$ that selects this combination is sometimes still called the Feynman prescription for that reason. The retarded Green's function, by contrast, is the classical object familiar from electrodynamics, where causality is imposed by hand through the choice of boundary condition; the quantum theory makes the Feynman combination, not the retarded one, the natural propagator inside the path integral, for reasons that become transparent once the generating functional is built in Chapters 9 and 10.

5.3.6 Poles versus branch points: a preview of cuts

Every singularity we have met in this section is a *pole*: an isolated point where the function blows up like $(z - z_0)^{-n}$, around which the function is single-valued

and the residue theorem applies verbatim. The propagator $1/(p^2 - m^2 + i\varepsilon)$ is meromorphic in p^0, with the two simple poles of Eq. (5.77) and nothing else. This is the entire reason the contour technology of this section works so cleanly: closing the contour reduces an integral to a finite sum of residues.

The integrals produced by *loops*, which the book will meet from Chapter 19 onward, are not so simple. A one-loop diagram in ϕ^4 or QED produces, after the loop momentum is integrated, functions such as $\log(m^2 - p^2 - i\varepsilon)$ or $\sqrt{1 - 4m^2/s}$, whose singularities are *branch points*, not poles. We must distinguish the two carefully, since the cut structure they generate is the analytic shadow of physical thresholds and the engine of the dispersion relations of Section 5.5.

Definition 5.27 (Branch point and branch cut)**.** A point z_0 is a *branch point* of a function f if f is not single-valued in any punctured neighborhood of z_0: continuing f analytically once around a small loop encircling z_0 returns a *different* value. To work with a single-valued branch one removes a curve emanating from z_0—a *branch cut*—across which f jumps by the discontinuity $\operatorname{disc} f(z) \equiv f(z + i0) - f(z - i0)$, the difference of the boundary values from the two sides.

The contrast with a pole is sharp and worth stating explicitly. Encircle a simple pole of $g(z) = 1/(z - z_0)$ once and the function returns to its starting value; the integral around it is the nonzero number $2\pi i$, but the function is single-valued. Encircle the branch point of $h(z) = \log(z - z_0)$ once and the function does *not* return: it shifts by $2\pi i$, because log is genuinely multivalued. A pole contributes a residue; a branch cut contributes an integral of the discontinuity along the cut. The simplest field-theoretic example, which Chapter 19 computes in full, is the scalar one-loop "bubble," whose value is proportional to a function with a branch point at the two-particle threshold $s = 4m^2$ in the Mandelstam invariant $s = (p_1 + p_2)^2$; below threshold the function is real, above threshold it develops an imaginary part—the discontinuity across the cut—which the optical theorem ties to the physical production rate of the two-particle intermediate state. The same analytic continuation that we used here innocuously, to move a contour past poles, becomes in Section 5.5 the deep statement that the propagator and amplitudes are boundary values of functions analytic in their invariants, with poles at stable particles and cuts at multiparticle thresholds, the whole structure dictated by causality.

For the present chapter the lesson is one of bookkeeping discipline: *before* closing a contour one must know whether the singularities inside are poles, to be summed by residues, or branch points, around which the contour must instead be deformed onto the cut. In this section there were only poles, and the residue theorem sufficed; from Chapter 19 the cuts arrive, and the residue theorem must be supplemented by the discontinuity calculus previewed here.

The worked computation has established the operational meaning of the $i\varepsilon$

as a contour instruction, and Fig. 5.3 shows that the Feynman prescription is the one placement that leaves the first and third quadrants of the p^0-plane free of poles. That clearance is not an accident of bookkeeping; it is precisely what will permit the real p^0-contour to be rotated onto the imaginary axis without crossing a singularity—and a contour that may be moved freely so long as it crosses no pole is the seed of analytic continuation. The next section promotes this freedom to a principle and carries out the rotation in earnest, establishing the Wick rotation that links Minkowski and Euclidean signature and identifying the Feynman $i\varepsilon$ as exactly the ingredient that makes the rotation legal.

5.4 Analytic continuation and Wick rotation

The previous section ended with a freedom and a puzzle. The freedom is that the energy integral defining a propagator may be evaluated along any contour in the complex p^0 plane that begins and ends at the same places and does not cross a pole; deforming the contour changes nothing. The puzzle is that this freedom looks like a mere convenience for doing residue arithmetic, when in fact it is an instance of something far larger. Once an integral is recognized as the boundary value of an analytic function, the integral is no longer tied to the real axis at all: it can be slid, rotated, and re-expressed wherever the analyticity permits, and two expressions that look completely different—a Minkowski integral oscillating along the real p^0 axis and a Euclidean integral decaying along the imaginary axis—turn out to be the same number. This section promotes contour deformation to the principle of *analytic continuation*, and then spends that principle on the single most-used continuation in the book: the *Wick rotation* that links Lorentzian and Euclidean signature. We will prove that the Feynman $i\epsilon$ is exactly the prescription that makes the rotation legal, and we will introduce the Schwinger proper-time representation, which turns the indefinite, conditionally convergent Minkowski denominators into manifestly convergent Euclidean Gaussians.

5.4.1 The principle of analytic continuation

We assume, as a reviewed prerequisite, the basic structure theory of holomorphic functions: a function analytic on an open set has a convergent power series about every point of that set, and conversely. The one statement we need to isolate, because everything in this section rests on it, is the rigidity of analytic functions.

Theorem 5.28 (Identity theorem)**.** *Let f and g be analytic on a connected open set $U \subseteq \mathbb{C}$, and suppose $f = g$ on a subset $S \subseteq U$ that has a limit point in U (for instance, S contains a line segment, an open subdomain, or any sequence converging to an interior point). Then $f = g$ everywhere on U.*

Proof. Set $h = f - g$, analytic on U and vanishing on S. Let $z_\star \in U$ be a limit

point of S. Expanding h in its power series about $z_\star$, suppose h is not identically zero near $z_\star$; then there is a smallest n with $h^{(n)}(z_\star) \neq 0$, so $h(z) = (z - z_\star)^n\, u(z)$ with u analytic and $u(z_\star) \neq 0$. By continuity $u \neq 0$ in a punctured neighbourhood of $z_\star$, so h has no zeros there other than $z_\star$ itself—contradicting that $z_\star$ is a limit of zeros of h. Hence h vanishes to all orders at $z_\star$ and is identically zero on a neighbourhood. The set on which h vanishes to all orders is therefore open; it is also closed in U (an intersection of the closed sets $\{h^{(k)} = 0\}$); since U is connected and the set is nonempty, it is all of U. □

The identity theorem has one consequence we use relentlessly: *analytic continuation, when it exists, is unique.* If a function is defined and analytic on a domain D_1, and one wishes to extend it to a larger connected domain $D_2 \supset D_1$ as an analytic function, then *any* such extension must agree with any other on D_1, hence—by Theorem 5.28—everywhere on D_2. There is at most one analytic continuation. This turns "find a formula that agrees with the original where the original makes sense, and is analytic on a larger region" into a rigorous and unambiguous operation: the new formula *is* the old function, merely viewed where its first definition broke down.

In quantum field theory the objects we continue are almost never bare functions; they are integrals depending on an external parameter. The pattern is the following.

Proposition 5.29 (Continuation of a parameter integral)**.** *Suppose $F(z) = \int_\Omega \kappa(x, z)\, dx$ converges absolutely and uniformly on compact subsets of an open set $D_1 \subseteq \mathbb{C}$, with $z \mapsto \kappa(x, z)$ analytic for each x. Then F is analytic on D_1. If G is analytic on a larger connected $D_2 \supset D_1$ and $G = F$ on D_1, then G is the unique analytic continuation of F, and one is entitled to* use *G*'s formula as the value of the original integral *for all $z \in D_2$—even at points z where the original integral $\int_\Omega \kappa(x, z)\, dx$ no longer converges.*

Proof. Analyticity of F on D_1 is Morera's theorem combined with Fubini: for any closed contour γ in a compact subset of D_1, uniform convergence lets us exchange $\oint_\gamma dz$ with $\int_\Omega dx$, and $\oint_\gamma \kappa(x, z)\, dz = 0$ for each x by Cauchy's theorem, so $\oint_\gamma F = 0$ and F is analytic. Uniqueness of the continuation is Theorem 5.28. □

The clause that matters is the last one. The Euler integral $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt$ converges only for $\operatorname{Re} z > 0$, yet the gamma function is defined and analytic on all of $\mathbb{C}$ except the nonpositive integers; the recursion $\Gamma(z) = \Gamma(z+1)/z$ is the continuation, and Theorem 5.29 is the licence to read off $\Gamma(-\tfrac{1}{2})$ from it even though the defining integral diverges there. Dimensional regularization (Chapter D) is the same move performed on the spacetime dimension d: a loop integral convergent only for d in some strip is *defined*, for all other d, as its unique analytic continuation. And the Wick rotation we now build is this principle applied to the contour of the energy integral.

5.4.2 The Wick rotation

Minkowski-space calculations carry oscillatory weights—e^{iS} in the path integral, $e^{-ip\cdot x}$ in Fourier transforms, and propagator denominators $p^2 - m^2$ that change sign and vanish on the mass shell. None of these is absolutely convergent without the $i\epsilon$ regulator, and the oscillations make both analytic estimates and numerical work treacherous. The Wick rotation trades the Lorentzian signature for a Euclidean one in which every weight becomes a real, decaying Gaussian. We first fix the rotation as a change of variables; the next subsection proves it is legal.

Definition 5.30 (Wick rotation)**.** The *Wick rotation* continues the time coordinate and the energy into the complex plane by

$$x^0 = t \;\longrightarrow\; -i\,\tau, \qquad p^0 \;\longrightarrow\; i\,p_E^4, \tag{5.93}$$

with $\tau \in \mathbb{R}$ the *Euclidean time* and $p_E^4 \in \mathbb{R}$ the *Euclidean energy*. Spatial components are unchanged, $x^i = x_E^i$, $p^i = p_E^i$. The Euclidean four-vectors are $x_E = (\tau, \mathbf{x})$, $p_E = (p_E^4, \mathbf{p})$, contracted with the positive-definite metric $\delta_{\mu\nu} = \mathrm{diag}(+,+,+,+)$.

Throughout, p_E^2 denotes the Euclidean square $p_E^2 = (p_E^4)^2 + \mathbf{p}^2 \geq 0$, in contrast to the Minkowski $p^2 = (p^0)^2 - \mathbf{p}^2$.

Notation Watch

Two unrelated objects in this section are both spelled with an epsilon. The *regulator* $i\epsilon$ in the Feynman propagator is the infinitesimal positive quantity sent to 0^+ at the end; it sets the analytic structure. The dimensional-regularization parameter in $d = 4 - 2\varepsilon$, which appears only in the worked example below, is written ε and is a finite continuation variable. They are never the same symbol and never interact; keep them visually distinct.

The two substitutions in Eq. (5.93) are consistent with the Fourier kernel and with the metric. Under them the Minkowski square continues to minus the Euclidean square,

$$p^2 = (p^0)^2 - \mathbf{p}^2 \;\longrightarrow\; (ip_E^4)^2 - \mathbf{p}^2 = -\big[(p_E^4)^2 + \mathbf{p}^2\big] = -\,p_E^2, \tag{5.94}$$

and the phase of a plane wave becomes a real exponent, $p\cdot x = p^0 x^0 - \mathbf{p}\cdot\mathbf{x} \to (ip_E^4)(-i\tau) - \mathbf{p}\cdot\mathbf{x} = p_E^4\tau - \mathbf{p}\cdot\mathbf{x}$.

The deepest consequence is for the action, and it is what makes the rotation the backbone of the Euclidean path integral of Chapter 10. For the carried scalar theory with Lagrangian density $\mathcal{L} = \tfrac{1}{2}(\partial_t\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - V(\phi)$, the Minkowski action is $S = \int dt\, d^3x\, \mathcal{L}$. Substituting $t = -i\tau$, so that $\partial_t = i\,\partial_\tau$

and the temporal kinetic term flips sign, $(\partial_t\phi)^2 \to -(\partial_\tau\phi)^2$, while the measure becomes $dt\, d^3x = -i\, d\tau\, d^3x = -i\, d^4x_E$, one finds

$$\begin{aligned} iS &= i\int(-i\, d^4x_E)\left[-\tfrac{1}{2}(\partial_\tau\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - V(\phi)\right] \\ &= -\int d^4x_E\left[\tfrac{1}{2}(\partial_\tau\phi)^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + V(\phi)\right] \;\equiv\; -S_E, \end{aligned} \tag{5.95}$$

so that the oscillatory weight becomes a real, bounded, decaying one,

$$e^{iS} \;\longrightarrow\; e^{-S_E}, \qquad S_E = \int d^4x_E\Big[\tfrac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) + \tfrac{1}{2}m^2\phi^2 + V(\phi)\Big] \geq 0, \tag{5.96}$$

with all indices in S_E summed with the Euclidean metric. The Euclidean action is bounded below; e^{-S_E} is a genuine probability weight. This is exactly the S_E frozen for ϕ^4 theory, whose quartic version $V = \frac{\lambda}{4!}\phi^4$ gives the Euclidean action used in Chapter 10; the present rotation is the operation that chapter invokes by name.

Physical Insight

A Lorentzian path integral is an interference problem: e^{iS} assigns each history a unit-modulus phase, and observables emerge from delicate cancellation. A Euclidean path integral is a statistics problem: e^{-S_E} assigns each history a positive Boltzmann weight, and observables are expectation values in an ordinary probability measure. The Wick rotation says these are two analytic faces of one object. Everything hard about oscillatory integrals—conditional convergence, sign problems, the meaning of "$\int e^{iS}$"—is bought off by paying the price of working at imaginary time and continuing back at the end.

5.4.3 Why the rotation is legal: the $i\epsilon$ does the work

A change of variables in a one-dimensional integral is only as good as the contour deformation that justifies it. Replacing $\int dp^0$ along the real axis by $\int dp^4_E$ along the imaginary axis is the deformation $\theta : 0 \to \pi/2$ of the ray $p^0 = |p^0|e^{i\theta}$, swinging the contour counterclockwise through the first and third quadrants. This is legitimate precisely when (i) the integrand has no poles in the swept quadrants and (ii) the contributions from the connecting arcs at infinity vanish. The content of the legality theorem is that the *Feynman* prescription, and only it, places the propagator poles out of the way of this particular rotation. We import the pole positions established in the previous section.

The pole structure is read off from the Feynman denominator. Setting $p^2 - m^2 + i\epsilon = 0$ at fixed real spatial momentum $\mathbf{p}$ gives $(p^0)^2 = \mathbf{p}^2 + m^2 - i\epsilon =$

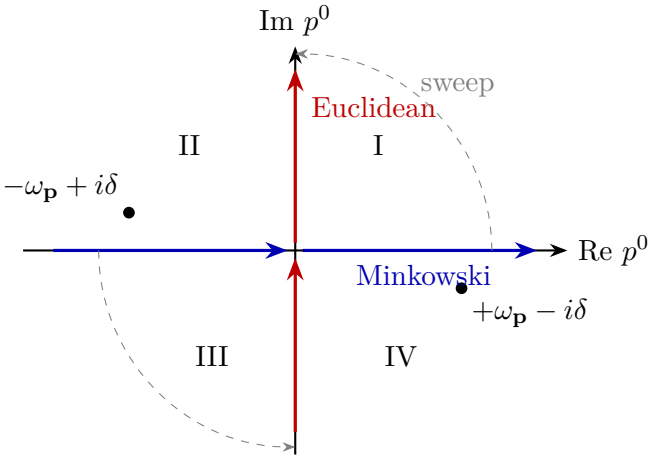


Figure 5.4: The complex p^0 plane for the Feynman propagator. The poles sit in quadrants II ($-\omega_{\mathbf{p}} + i\delta$) and IV ($+\omega_{\mathbf{p}} - i\delta$), with $\delta = \epsilon/2\omega_{\mathbf{p}} \to 0^+$. Wick rotation swings the integration contour counterclockwise from the real axis (Minkowski, blue) to the imaginary axis (Euclidean, red) through the pole-free quadrants I and III; the quarter-circle arcs at infinity (gray) contribute nothing, and no pole is crossed.

$\omega_{\mathbf{p}}^2 - i\epsilon$, with $\omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} > 0$, so to first order in ϵ,

$$p^0 = \pm\sqrt{\omega_{\mathbf{p}}^2 - i\epsilon} = \pm\Big(\omega_{\mathbf{p}} - \tfrac{i\epsilon}{2\omega_{\mathbf{p}}}\Big) + O(\epsilon^2) = \pm(\omega_{\mathbf{p}} - i\delta), \qquad \delta = \frac{\epsilon}{2\omega_{\mathbf{p}}} > 0. \tag{5.97}$$

The pole at $+\omega_{\mathbf{p}} - i\delta$ lies in the fourth quadrant (positive real part, negative imaginary part); the pole at $-\omega_{\mathbf{p}} + i\delta$ lies in the second quadrant. This is precisely the placement derived in Eq. (5.77), and it is what distinguishes the Feynman prescription from the retarded ($i\epsilon\,\mathrm{sgn}\,p^0$, both poles below the axis) and advanced (both above) choices. Quadrants I and III—the regions the rotation sweeps—are free of poles. Figure 5.4 shows the geometry.

Theorem 5.31 (Legality of the Wick rotation)**.** *Let $f(p^0)$ be meromorphic in p^0 at fixed real $\mathbf{p}$, with all poles confined to the open second and fourth quadrants (the Feynman placement (5.97)), and suppose there are constants $C > 0$ and $\eta > 0$ such that $|f(p^0)| \le C\,|p^0|^{-1-\eta}$ for all sufficiently large $|p^0|$ with $\arg p^0 \in [0, \frac{\pi}{2}] \cup [\pi, \frac{3\pi}{2}]$. Then the Minkowski and Euclidean energy integrals agree:*

$$\int_{-\infty}^{\infty} dp^0\, f(p^0) \;=\; i\int_{-\infty}^{\infty} dp_E^4\, f(i\,p_E^4). \tag{5.98}$$

Proof. Work first in the first quadrant. Consider the closed, positively oriented contour Γ_R consisting of: the segment C_1 along the real axis from 0 to R; the quarter-circle arc C_2, $p^0 = Re^{i\theta}$ with $\theta : 0 \to \frac{\pi}{2}$; and the segment C_3 down the

imaginary axis from iR to 0. Since all poles of f lie in quadrant II or IV, Γ_R encloses none, and Cauchy's theorem gives

$$\int_{C_1} f\,dp^0 + \int_{C_2} f\,dp^0 + \int_{C_3} f\,dp^0 = 0. \tag{5.99}$$

On the arc, the ML estimate together with the decay hypothesis bounds the length $\frac{\pi}{2}R$ times the maximum of $|f|$:

$$\left|\int_{C_2} f\,dp^0\right| \le \frac{\pi}{2}R \cdot CR^{-1-\eta} = \frac{\pi C}{2}\,R^{-\eta} \xrightarrow[R\to\infty]{} 0. \tag{5.100}$$

On C_3 parametrize $p^0 = ir$ with r running from R down to 0, so $dp^0 = i\,dr$ and $\int_{C_3} f\,dp^0 = \int_R^0 f(ir)\,i\,dr = -i\int_0^R f(ir)\,dr$. Inserting this and (5.100) into (5.99) and letting $R \to \infty$,

$$\int_0^\infty f(p^0)\,dp^0 = i\int_0^\infty f(ir)\,dr. \tag{5.101}$$

The third quadrant is treated identically with the closed contour built from the negative real axis ($-R \to 0$), the arc $\theta : \pi \to \frac{3\pi}{2}$, and the negative imaginary axis; the arc again vanishes and the same parametrization gives

$$\int_{-\infty}^0 f(p^0)\,dp^0 = i\int_{-\infty}^0 f(ir)\,dr. \tag{5.102}$$

Adding (5.101) and (5.102) and renaming the dummy variable $r = p_E^4$ yields (5.98). □

Two remarks fix the meaning. First, the decay hypothesis $|f| \lesssim |p^0|^{-1-\eta}$ holds for a single Feynman propagator ($f \sim (p^0)^{-2}$, so $\eta = 1$) and a fortiori for any product of propagators, which is the only case we ever rotate; when an integrand grows—a loop integral that is genuinely divergent—the arcs do not vanish in $d = 4$, and the rotation is justified instead in $d = 4 - 2\varepsilon$ dimensions, where the integral is made convergent first (Chapter D). Second, and this is the whole point of the section heading: had we used the retarded or advanced prescription, both poles would sit in the lower (or upper) half-plane, one of them *inside* a swept quadrant, and Eq. (5.99) would pick up a residue; the rotation would change the answer. The Feynman $i\epsilon$ is not one acceptable choice among several—it is the unique pole placement for which Lorentzian and Euclidean integrals are literally equal. This is the precise sense in which the $i\epsilon$ "makes the rotation legal."

Caution

The rotation $p^0 \to ip_E^4$ in momentum space and $t \to -i\tau$ in position space turn the same way only because energy and time are conjugate; do not rotate one clockwise and the other counterclockwise. A sign error here

flips e^{-S_E} to a divergent e^{+S_E} and is among the most common slips in the subject. The fixed convention is Eq. (5.93), matching Chapter 10.

5.4.4 The Schwinger proper-time representation

The Wick rotation makes denominators convergent, but it is paired with a second device that converts those denominators into Gaussians, at which point the Euclidean integrals of Chapter 19 and Chapter D can actually be done. That device is the Schwinger, or proper-time, representation: it writes a denominator as an integral of an exponential over an auxiliary "time" $s \geq 0$, exchanging $1/A$ for $\int ds\, e^{\pm isA}$. The $i\epsilon$ resurfaces here in a new guise—now as the regulator that guarantees convergence at the upper limit $s \to \infty$.

Proposition 5.32 (Schwinger representation)**.** *For real A and $\epsilon > 0$,*

$$\frac{1}{A + i\epsilon} = -\,i \int_0^\infty ds\; e^{\,is\,(A+i\epsilon)}, \tag{5.103}$$

the integral converging absolutely because the $i\epsilon$ supplies a damping factor $e^{-\epsilon s}$. Applied to the Feynman propagator denominator ($A = p^2 - m^2$) this is the compact and manifestly convergent identity

$$\frac{i}{p^2 - m^2 + i\epsilon} = \int_0^\infty ds\; e^{\,is\,(p^2 - m^2 + i\epsilon)}. \tag{5.104}$$

In Euclidean signature, where the denominator $A = p_E^2 + m^2 > 0$ is strictly positive, no regulator is needed and the representation is the elementary Laplace identity

$$\frac{1}{A} = \int_0^\infty ds\; e^{-sA}, \qquad A > 0, \tag{5.105}$$

together with its powered version, for $n > 0$,

$$\frac{1}{A^{\,n}} = \frac{1}{\Gamma(n)} \int_0^\infty ds\; s^{\,n-1} e^{-sA}. \tag{5.106}$$

Proof. For (5.103), the integrand is $e^{is(A+i\epsilon)} = e^{isA}\, e^{-\epsilon s}$, whose modulus is $e^{-\epsilon s}$; hence the integral converges absolutely for $\epsilon > 0$. Integrating directly,

$$\int_0^\infty ds\; e^{\,is(A+i\epsilon)} = \left[\frac{e^{\,is(A+i\epsilon)}}{i(A+i\epsilon)}\right]_0^\infty = \frac{0 - 1}{i(A+i\epsilon)} = \frac{-1}{i(A+i\epsilon)} = \frac{i}{A+i\epsilon}, \tag{5.107}$$

where the upper endpoint vanishes precisely because $|e^{is(A+i\epsilon)}| = e^{-\epsilon s} \to 0$, and we used $-1/i = i$. Multiplying (5.107) by $-i$ gives (5.103); and (5.104) is (5.107) evaluated at $A = p^2 - m^2$. For the Euclidean identity (5.105), $A > 0$ makes $\int_0^\infty e^{-sA}\, ds = [-e^{-sA}/A]_0^\infty = 1/A$ with no regulator. The powered form (5.106) follows by substituting $s \to s/A$ in the definition $\Gamma(n) = \int_0^\infty u^{n-1} e^{-u}\, du$, which gives $\Gamma(n) = A^n \int_0^\infty s^{n-1} e^{-sA}\, ds$. □

The structure to internalize is the parallel between the two regulators. In the Lorentzian form (5.104) the exponent $is(p^2 - m^2) - \epsilon s$ has a real part $-\epsilon s$ that comes *entirely* from the Feynman $i\epsilon$; without it the integral over s would be a pure phase $e^{is(p^2-m^2)}$ and would not converge at $s \to \infty$. After Wick rotation $p^2 \to -p_E^2$, the same exponent becomes $-s(p_E^2 + m^2)$, a real decaying Gaussian-in-the-making whose convergence is automatic. The $i\epsilon$ is the Lorentzian shadow of Euclidean positivity: it is exactly the infinitesimal needed to push the Minkowski integral onto the convergent side, and once one has rotated to Euclidean signature it has done its job and is dropped. This is why every practical loop calculation in the book rotates first and parametrizes second.

5.4.5 A worked Wick rotation: the scalar bubble integral

We now run the entire machine on the representative one-loop scalar integral that recurs throughout Chapter 19 and is tabulated in Chapter D. It is the simplest integral that is only conditionally defined in Minkowski space yet perfectly convergent (away from $d = 4$) in Euclidean space, so it shows the payoff cleanly.

Worked Example 5.33 (Wick rotation of $\int d^4p\,/(p^2 - m^2 + i\epsilon)^2$)**.** Consider

$$I = \int \frac{d^4p}{(2\pi)^4} \frac{1}{(p^2 - m^2 + i\epsilon)^2}. \tag{5.108}$$

The integrand is the square of a Feynman propagator denominator; by (5.97) it has *double* poles in p^0 at $\pm(\omega_{\mathbf{p}} - i\delta)$, both in quadrants II and IV. At fixed $\mathbf{p}$ it falls as $(p^0)^{-4}$, so the hypotheses of Theorem 5.31 hold with room to spare, and the p^0 contour may be rotated to the imaginary axis.

Step 1: rotate. Apply Eq. (5.93) to the energy integration inside the four-dimensional measure. With $p^0 = ip_E^4$ we have $dp^0 = i\,dp_E^4$, hence $d^4p = i\,d^4p_E$, while by (5.94) $p^2 - m^2 + i\epsilon \to -(p_E^2 + m^2)$, the regulator now superfluous because the denominator no longer vanishes. The squared denominator becomes $(p_E^2 + m^2)^2$, and so

$$I = \int \frac{i\,d^4p_E}{(2\pi)^4} \frac{1}{(p_E^2 + m^2)^2} = i \int \frac{d^4p_E}{(2\pi)^4} \frac{1}{(p_E^2 + m^2)^2}. \tag{5.109}$$

The conditionally convergent Minkowski oscillation has become a positive, rotationally symmetric Euclidean integrand. Only the overall factor of i records that we ever left Lorentzian signature.

Step 2: regulate the dimension. In $d = 4$ the Euclidean integral (5.109) diverges logarithmically at large p_E ($\int d^4p_E/p_E^4$). We therefore continue the dimension to $d = 4 - 2\varepsilon$, as licensed by Theorem 5.29 and detailed in Chapter D, and compute

$$I_E(d) \equiv \int \frac{d^dp_E}{(2\pi)^d} \frac{1}{(p_E^2 + m^2)^2}. \tag{5.110}$$

Step 3: Schwinger-parametrize and Gaussian-integrate. Apply the powered Euclidean representation (5.106) with $n = 2$ and $A = p_E^2 + m^2$, so that $\frac{1}{(p_E^2+m^2)^2} = \int_0^\infty ds\, s\, e^{-s(p_E^2+m^2)}$. Inserting this and exchanging the orders of integration (legitimate by absolute convergence, the momentum integral being Gaussian for each $s > 0$),

$$I_E(d) = \int_0^\infty ds\, s\, e^{-sm^2} \int \frac{d^d p_E}{(2\pi)^d}\, e^{-s\, p_E^2}. \tag{5.111}$$

The inner integral is the d-dimensional Gaussian transform established in Eq. (5.50); with all d components decoupling, $\int d^d p_E\, e^{-sp_E^2} = (\pi/s)^{d/2}$, so

$$\int \frac{d^d p_E}{(2\pi)^d}\, e^{-s\, p_E^2} = \frac{1}{(2\pi)^d}\Big(\frac{\pi}{s}\Big)^{d/2} = \frac{1}{(4\pi)^{d/2}}\, s^{-d/2}. \tag{5.112}$$

Substituting (5.112) into (5.111) collapses the problem to a single proper-time integral,

$$I_E(d) = \frac{1}{(4\pi)^{d/2}} \int_0^\infty ds\, s^{\,1-d/2}\, e^{-sm^2} = \frac{1}{(4\pi)^{d/2}}\, \Gamma\Big(2 - \tfrac{d}{2}\Big)\, (m^2)^{\frac{d}{2}-2}, \tag{5.113}$$

where the last equality is $\int_0^\infty s^{a-1} e^{-sm^2}\, ds = \Gamma(a)(m^2)^{-a}$ with $a = 2 - \frac{d}{2}$.

Step 4: restore the rotation factor and read off the divergence. Combining (5.109) with (5.113),

$$\boxed{I = \frac{i}{(4\pi)^{d/2}}\, \Gamma\Big(2 - \tfrac{d}{2}\Big)\, (m^2)^{\frac{d}{2}-2}} \xrightarrow{d=4-2\varepsilon} \frac{i}{(4\pi)^2}\Big[\frac{1}{\varepsilon} - \gamma_E + \ln\frac{4\pi}{m^2} + O(\varepsilon)\Big], \tag{5.114}$$

the expansion using $\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_E + O(\varepsilon)$ and $(m^2)^{-\varepsilon} = 1 - \varepsilon \ln m^2 + \cdots$ (with the usual $\mu^{2\varepsilon}$ restoring dimensions, supplied in Chapter D). The logarithmic divergence of the original $d = 4$ integral has become the simple pole $\Gamma(2 - d/2) = \Gamma(\varepsilon) \sim 1/\varepsilon$, which is exactly what the renormalization machinery of Chapter 19 subtracts.

Three threads tie together in Theorem 5.33, and it is worth naming them. The pole placement of Eq. (5.77) told us *which way* to rotate; the legality theorem Theorem 5.31 told us we were *allowed* to; the Schwinger identity Theorem 5.32 and the Gaussian of Eq. (5.50) did the *work*. The general one-loop master integral $\int d^d p_E/(p_E^2 + \Delta)^n$, of which (5.110) is the case $n = 2$, $\Delta = m^2$, is obtained by the identical steps and is recorded in Chapter D; every loop computation in QED (Chapter 19) and ϕ^4 theory proceeds by rotating to Euclidean signature, combining denominators, Schwinger- or Feynman-parametrizing, and reducing to this single Gaussian.

With the Wick rotation established, the correlators and amplitudes of the book are seen to be Euclidean analytic functions of their external invariants, continued

back to Lorentzian signature at the end. The location of the singularities that this continuation cannot avoid—the poles and branch cuts that survive on the real-momentum axis—is not a defect of the continuation but a physical datum: it encodes the spectrum and the thresholds of the theory. The next section reads that analytic structure physically, showing that analyticity in the external invariants is the imprint of causality, and turns it into the working dispersion relations that reconstruct an amplitude from its discontinuity.

5.5 Analyticity as physics: causality and dispersion relations

The previous section, Section 5.4, established analyticity as a *technical* fact: contour deformation promoted to a principle, the Wick rotation made legal by the Feynman $i\epsilon$, Euclidean correlation functions read as the boundary values of objects holomorphic in a domain of complexified momenta. In Euclidean signature that holomorphy is almost featureless — the Euclidean propagator $1/(p_E^2 + m^2)$ has no singularity for real p_E at all — and it is tempting to regard analyticity as nothing more than a convenience that lets one rotate integrals into a region where they converge. The purpose of this section is to overturn that impression. Read in Minkowski signature, the singularities of an amplitude are not blemishes to be rotated away; they *are* the physics. A pole on the real axis is a stable particle; a branch cut is the opening of a decay or scattering channel; the location of the nearest singularity sets the range of a force. And the very fact that the amplitude is the boundary value of an analytic function turns out to be the mathematical shadow of a single physical principle: *causality* — no effect before its cause.

This is the most economical idea in the analytic toolkit. We will prove that a response which respects causality, in the sharp sense that it vanishes before the stimulus that produced it, has a Fourier transform that is holomorphic in an entire half-plane of complex frequency. From holomorphy plus a mild bound at infinity, Cauchy's theorem then squeezes out a rigid constraint: the real and imaginary parts of the response are not independent functions but are determined by each other through a principal-value integral — the Kramers–Kronig relations. The same machinery, applied to a scattering amplitude regarded as a function of a Mandelstam invariant, becomes a *dispersion relation*: a formula that reconstructs the whole amplitude from the discontinuity across its cuts, with the discontinuity itself fixed — via the Sokhotski–Plemelj identity of Theorem 5.10 — to be twice the imaginary part, the *absorptive* part. We close by stating, at the level of precise definitions awaiting their proofs, the two results that this circle of ideas exists to serve: the Källén–Lehmann spectral representation of the two-point function and the optical theorem $2\,\mathrm{Im}\,\mathcal{M} \sim \sigma_{\mathrm{tot}}$. The worked computation that ends the section — reconstructing a model two-point function with a single threshold from

its discontinuity — is self-contained, but it is not idle: it builds, with all algebra shown, exactly the analytic object that QED's vacuum polarization $\Pi(q^2)$ will be in Chapter 19, and the dispersive logic that organizes the unitarity methods of Chapter 60.

5.5.1 Causality as a support condition

The cleanest setting in which to see analyticity emerge from causality is linear response. Suppose a system in equilibrium is poked by an external force $f(t)$ and responds with some observable $x(t)$. For small forces the response is linear and time-translation invariant, so the most general possibility is a convolution

$$x(t) = \int_{-\infty}^{\infty} G_R(t - t')\, f(t')\, \mathrm{d}t', \tag{5.115}$$

in which the *response function* (or retarded Green's function) G_R encodes everything about the system's dynamics. The kernel $G_R(\tau)$ is the response measured at a time τ after a sharp unit kick. Causality is the statement that the system cannot respond before it is kicked:

Definition 5.34 (Causal (retarded) response)**.** A response function G_R is *causal* if

$$G_R(\tau) = 0 \qquad \text{for all } \tau < 0. \tag{5.116}$$

Equivalently, $\operatorname{supp} G_R \subseteq [0, \infty)$.

This is a *support condition* — a statement about where the function is allowed to be nonzero — and it is the entire physical input. Everything that follows is its analytic consequence. We work with the Fourier transform in the frequency variable ω, adopting for this subsection the response-theory convention

$$\widehat{G}(\omega) = \int_{-\infty}^{\infty} G_R(\tau)\, e^{i\omega\tau}\, \mathrm{d}\tau \overset{(5.116)}{=} \int_{0}^{\infty} G_R(\tau)\, e^{i\omega\tau}\, \mathrm{d}\tau, \tag{5.117}$$

where the second equality is the only place the support condition enters. (The sign in the exponent is the standard one for retarded functions; it is the time-component of the book's spacetime convention of Chapter 3 restricted to one dimension, and it is what makes the half-plane below come out to be the *upper* one. We flag the clash with the spatial Fourier sign in Section 5.5.1.) The transform variable will now be allowed to become complex, $\omega = \operatorname{Re}\omega + i \operatorname{Im}\omega$.

Theorem 5.35 (Causality forces upper-half-plane analyticity)**.** *Let $G_R \in L^1(0, \infty)$ satisfy the support condition (5.116). Then the function $\widehat{G}(\omega)$ defined by (5.117) is holomorphic in the open upper-half plane*

$$\mathbb{C}_+ = \{\omega \in \mathbb{C} : \operatorname{Im}\omega > 0\}, \tag{5.118}$$

and is bounded there by $|\widehat{G}(\omega)| \le \|G_R\|_{L^1}$. *If moreover* G_R *is bounded with at most exponential growth* $|G_R(\tau)| \le Ce^{\gamma\tau}$, *the same conclusions hold in the shifted half-plane* $\operatorname{Im}\omega > \gamma$.

Proof. Write $\omega = \sigma + i\eta$ with $\eta = \operatorname{Im}\omega > 0$. Inside the integral (5.117) the exponential factorizes,

$$e^{i\omega\tau} = e^{i\sigma\tau}\, e^{-\eta\tau}, \tag{5.119}$$

and here the support condition does its work: because τ ranges only over $[0,\infty)$, the factor $e^{-\eta\tau}$ is a *decaying* exponential, never a growing one. The integrand is therefore dominated in modulus by an integrable function independent of the real part σ,

$$\bigl|G_R(\tau)\, e^{i\omega\tau}\bigr| = |G_R(\tau)|\, e^{-\eta\tau} \le |G_R(\tau)| \in L^1(0,\infty). \tag{5.120}$$

This already gives absolute convergence of (5.117) throughout $\mathbb{C}_+$ and the bound

$$|\widehat{G}(\omega)| \le \int_0^\infty |G_R(\tau)|\, e^{-\eta\tau}\, \mathrm{d}\tau \le \int_0^\infty |G_R(\tau)|\, \mathrm{d}\tau = \|G_R\|_{L^1}. \tag{5.121}$$

It remains to show holomorphy. We verify it by Morera's theorem, which sidesteps any need to differentiate under the integral sign. Let Δ be any closed triangle contained in $\mathbb{C}_+$; then $\eta \ge \eta_0 > 0$ on Δ for some η_0, and the bound (5.120) sharpens to the uniform, τ-integrable majorant $|G_R(\tau)|\, e^{-\eta_0\tau}$. The function $\omega \mapsto e^{i\omega\tau}$ is entire, so for each fixed τ its contour integral around $\partial\Delta$ vanishes by Cauchy's theorem. The uniform integrable bound licenses Fubini's theorem to exchange the two integrations,

$$\oint_{\partial\Delta} \widehat{G}(\omega)\, \mathrm{d}\omega = \oint_{\partial\Delta} \left(\int_0^\infty G_R(\tau)\, e^{i\omega\tau}\, \mathrm{d}\tau \right) \mathrm{d}\omega = \int_0^\infty G_R(\tau) \left(\oint_{\partial\Delta} e^{i\omega\tau}\, \mathrm{d}\omega \right) \mathrm{d}\tau = 0. \tag{5.122}$$

Since the integral of $\widehat{G}$ around every triangle in $\mathbb{C}_+$ vanishes and $\widehat{G}$ is continuous there (again by dominated convergence applied to (5.117)), Morera's theorem makes $\widehat{G}$ holomorphic on $\mathbb{C}_+$. The exponential-growth refinement is identical with $e^{-\eta\tau}$ replaced by $e^{-(\eta-\gamma)\tau}$, integrable precisely when $\eta > \gamma$. □

The logic is worth restating in one sentence, because the same sentence will recur in every later appearance of analyticity. *The support of the function in time controls the half-plane of analyticity in frequency:* support in $\tau > 0$ buys analyticity for $\operatorname{Im}\omega > 0$. The exponent $e^{i\omega\tau}$ converts the geometric fact "τ is positive" into the analytic fact "$e^{i\omega\tau}$ decays upward," and decay is exactly what convergence and holomorphy require.

Physical Insight

The half-plane is not arbitrary. A retarded function lives on $\tau > 0$ and is analytic *above*; an advanced function lives on $\tau < 0$ and is analytic *below*; the Feynman propagator, which is neither purely retarded nor advanced, is the boundary value of a function analytic in neither half-plane alone but built — via the $i\epsilon$ of Section 5.4 — to have its positive- and negative-frequency parts each propagate causally. The single phrase "analytic in the upper-half plane" is thus a precise, checkable encoding of "the future does not influence the past."

The converse of Theorem 5.35 is also true, and it is the deeper half of the equivalence; we state it without proof, as it is not needed downstream and its proof is a genuine excursion into Hardy-space theory.

Theorem 5.36 (Titchmarsh)**.** *Let $g \in L^2(\mathbb{R})$. The following are equivalent:* (i) *g is the boundary value of a function holomorphic and uniformly square-integrable on the upper-half plane (a function in the Hardy space $H^2(\mathbb{C}_+)$);* (ii) *the inverse Fourier transform of g is supported in $[0, \infty)$, i.e. g is causal;* (iii) *the real and imaginary parts of g on the real axis are Hilbert transforms of one another — the Kramers–Kronig relations of Theorem 5.37 below. This is the Paley–Wiener/Titchmarsh circle of theorems; Theorem 5.35 proves the implication* (ii)⇒(i) *that we actually use.*

Notation Watch

In this section the temporal transform carries $e^{+i\omega\tau}$ (Eq. (5.117)), so that causal functions are analytic in the *upper* ω-plane. The book's standing spacetime convention (Chapter 3) is $f(x) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot x} \tilde{f}(p)$ with $p \cdot x = p^0 t - \mathbf{p} \cdot \mathbf{x}$; the time-component there is $e^{-ip^0 t}$. The two agree once one identifies $\omega = p^0$ and remembers that the response kernel is integrated against the *source at earlier times*, which flips the sign relative to the field expansion. When we pass to amplitudes as functions of $s = p^2$ in Section 5.5.5, the relevant analyticity is in s and this sign bookkeeping no longer intrudes.

5.5.2 The Kramers–Kronig relations

Analyticity in a half-plane, together with a decay bound, is enough to make the real and imaginary parts of a response function rigidly interdependent. The mechanism is Cauchy's formula evaluated on the boundary of the half-plane, and the technical heart is the Sokhotski–Plemelj identity, which tells us precisely what happens when a Cauchy kernel is pushed onto the contour of integration.

Assume henceforth that $\widehat{G}$ is holomorphic in $\mathbb{C}_+$ (guaranteed by Theorem 5.35) and that it *decays* on large semicircles in the closed upper-half plane: $|\widehat{G}(\omega)| \to 0$ as $|\omega| \to \infty$ with $\operatorname{Im}\omega \geq 0$. (Physically $\widehat{G}$ is a susceptibility, and the system cannot respond to an infinitely fast drive, so $\widehat{G} \to 0$ at high frequency; we lift this assumption in Section 5.5.3.) Fix a real frequency ω and approach it from inside the half-plane, writing the interior point as $\omega + i\epsilon$ with $\epsilon \to 0^+$. Cauchy's integral formula on the contour Γ_R consisting of the real axis from $-R$ to R closed by the semicircle of radius R in $\mathbb{C}_+$ gives, since $\widehat{G}$ is holomorphic inside,

$$\widehat{G}(\omega + i\epsilon) = \frac{1}{2\pi i} \oint_{\Gamma_R} \frac{\widehat{G}(\omega')}{\omega' - \omega - i\epsilon}\, \mathrm{d}\omega'. \tag{5.123}$$

The semicircular arc contributes a term bounded by $\max_{\text{arc}} |\widehat{G}| \cdot \pi R/R = \pi \max_{\text{arc}} |\widehat{G}| \to 0$ as $R \to \infty$, by the decay assumption (this is Jordan's lemma in its elementary form, the $1/R$ from the denominator beating the πR arclength once $\widehat{G} \to 0$). Sending $R \to \infty$ leaves only the real-axis integral,

$$\widehat{G}(\omega + i\epsilon) = \frac{1}{2\pi i} \int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{\omega' - \omega - i\epsilon}\, \mathrm{d}\omega'. \tag{5.124}$$

Now take $\epsilon \to 0^+$. The boundary value of $\widehat{G}$ on the left is its physical real-frequency value $\widehat{G}(\omega)$. On the right the Cauchy kernel hits the real axis, and this is exactly the situation the Sokhotski–Plemelj identity (Theorem 5.10) governs:

$$\frac{1}{\omega' - \omega - i\epsilon} \xrightarrow{\epsilon \to 0^+} \operatorname{PV} \frac{1}{\omega' - \omega} + i\pi\, \delta(\omega' - \omega). \tag{5.125}$$

Inserting (5.125) into (5.124) and doing the delta-function integral,

$$\widehat{G}(\omega) = \frac{1}{2\pi i} \operatorname{PV} \int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{\omega' - \omega}\, \mathrm{d}\omega' + \frac{1}{2\pi i}\, i\pi\, \widehat{G}(\omega) = \frac{1}{2\pi i} \operatorname{PV} \int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{\omega' - \omega}\, \mathrm{d}\omega' + \tfrac{1}{2}\widehat{G}(\omega). \tag{5.126}$$

The term $\frac{1}{2}\widehat{G}(\omega)$ — the contribution of the boundary pole, geometrically the half of a residue picked up by an indentation that skirts the singularity (Fig. 5.5) — moves to the left, and multiplying through by 2 yields the master relation

$$\boxed{\widehat{G}(\omega) = \frac{1}{i\pi} \operatorname{PV} \int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{\omega' - \omega}\, \mathrm{d}\omega'.} \tag{5.127}$$

This says $\widehat{G}$ is, up to the factor $1/i$, its own Hilbert transform — a statement that has bite only because the real and imaginary parts can now be separated.

Write $\widehat{G} = \operatorname{Re}\widehat{G} + i \operatorname{Im}\widehat{G}$ on the real axis and take the real and imaginary parts of (5.127). Because $1/i = -i$, the real part of the left side equals the imaginary part of the integral and vice versa:

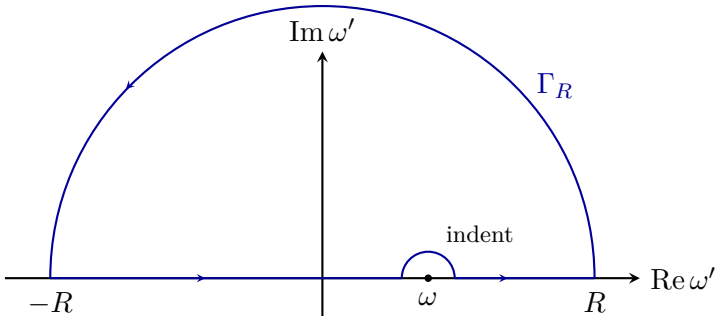


Figure 5.5: The contour behind the Kramers–Kronig relation (5.127). Because the causal response $\widehat{G}$ is holomorphic in the upper-half plane (Theorem 5.35) and decays on the large arc, the closed contour Γ_R encloses no singularity. Pushing the evaluation point onto the real axis produces the simple pole at $\omega' = \omega$; the small semicircular indentation skirting it contributes half a residue, the $\frac{1}{2}\widehat{G}(\omega)$ of (5.126), equivalently the $i\pi\delta$ of Sokhotski–Plemelj (5.125).

Theorem 5.37 (Kramers–Kronig relations)**.** *For a causal response function $\widehat{G}$ holomorphic in $\mathbb{C}_+$ and vanishing at infinity there,*

$$\operatorname{Re}\widehat{G}(\omega) = \frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{\operatorname{Im}\widehat{G}(\omega')}{\omega' - \omega}\,\mathrm{d}\omega', \tag{5.128}$$

$$\operatorname{Im}\widehat{G}(\omega) = -\frac{1}{\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{\operatorname{Re}\widehat{G}(\omega')}{\omega' - \omega}\,\mathrm{d}\omega'. \tag{5.129}$$

The real and imaginary parts are Hilbert transforms of each other: knowing one on the whole real axis determines the other completely.

The physical content is striking. The imaginary part of a susceptibility is the *dissipative* response — the part of the system's reaction that is out of phase with the drive and therefore absorbs energy from it. The real part is the *reactive* (dispersive) response, in phase with the drive. Theorem 5.37 says these cannot be chosen independently: a medium that absorbs at some frequencies is *obliged*, by causality alone, to be dispersive at others. In optics, with $\widehat{G}$ the complex refractive index, this is the statement that the frequency dependence of the refractive index (the cause of a prism's rainbow) is fixed by the absorption spectrum, and conversely.

For a *real* response kernel $G_R(\tau) \in \mathbb{R}$ — the physical case, since $x(t)$ and $f(t)$ in (5.115) are real — there is an extra symmetry that folds these integrals onto positive frequencies. Conjugating (5.117) and using G_R real,

$$\widehat{G}(\omega)^* = \int_0^\infty G_R(\tau)\, e^{-i\omega^*\tau}\,\mathrm{d}\tau = \widehat{G}(-\omega^*), \qquad \text{so on the real axis} \quad \operatorname{Re}\widehat{G}(-\omega) = \operatorname{Re}\widehat{G}(\omega), \quad \text{I} \tag{5.130}$$

The real part is even and the imaginary part odd. Splitting the integral (5.128) at the origin and substituting $\omega' \to -\omega'$ on the negative half,

$$\int_{-\infty}^{0} \frac{\operatorname{Im}\widehat{G}(\omega')}{\omega'-\omega}\,\mathrm{d}\omega' \overset{\omega'\to-\omega'}{=} \int_{0}^{\infty} \frac{-\operatorname{Im}\widehat{G}(\omega')}{-\omega'-\omega}\,\mathrm{d}\omega' = \int_{0}^{\infty} \frac{\operatorname{Im}\widehat{G}(\omega')}{\omega'+\omega}\,\mathrm{d}\omega', \tag{5.131}$$

so that the two halves combine over the common denominator $(\omega'-\omega)^{-1}+(\omega'+\omega)^{-1} = 2\omega'/(\omega'^2-\omega^2)$, giving the form in which the relations are usually quoted:

$$\operatorname{Re}\widehat{G}(\omega) = \frac{2}{\pi}\,\mathrm{PV}\int_{0}^{\infty} \frac{\omega'\,\operatorname{Im}\widehat{G}(\omega')}{\omega'^2-\omega^2}\,\mathrm{d}\omega', \tag{5.132}$$

$$\operatorname{Im}\widehat{G}(\omega) = -\frac{2\omega}{\pi}\,\mathrm{PV}\int_{0}^{\infty} \frac{\operatorname{Re}\widehat{G}(\omega')}{\omega'^2-\omega^2}\,\mathrm{d}\omega'. \tag{5.133}$$

These are the Kramers–Kronig relations as they appear in optics and many-body theory. We have proved them from nothing but the support condition (5.116) and a decay bound.

5.5.3 Subtractions: the price of slow decay

The derivation above leaned on $\widehat{G}\to 0$ at large $|\omega|$ so that the arc in (5.123) would drop. Many functions of interest — and essentially every amplitude in field theory — do not decay; they approach a constant or grow. The cure is standard and carries a name that will reappear, with the same arithmetic, at the centre of renormalization: *subtraction*.

Suppose $\widehat{G}$ is holomorphic in $\mathbb{C}_+$ but tends to a nonzero constant, or more generally grows slower than linearly, so that the arc contribution in (5.123) does not vanish. Choose a reference point ω_0 on the real axis where $\widehat{G}(\omega_0)$ is known (typically a low frequency, or threshold, where the response is measured or where it is real). The *once-subtracted* function

$$\widehat{G}_1(\omega) \equiv \frac{\widehat{G}(\omega)-\widehat{G}(\omega_0)}{\omega-\omega_0} \tag{5.134}$$

is still holomorphic in $\mathbb{C}_+$ — the apparent pole at $\omega=\omega_0$ is removable because the numerator vanishes there — and it decays one power faster than $\widehat{G}$, fast enough that the arc now drops. Applying the unsubtracted master relation (5.127) to $\widehat{G}_1$ and multiplying back through by $(\omega-\omega_0)$ gives the once-subtracted dispersion relation

$$\widehat{G}(\omega) = \widehat{G}(\omega_0) + \frac{\omega-\omega_0}{i\pi}\,\mathrm{PV}\int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{(\omega'-\omega)(\omega'-\omega_0)}\,\mathrm{d}\omega'. \tag{5.135}$$

The extra factor $(\omega'-\omega_0)^{-1}$ in the integrand improves its convergence at large ω' by one power; the price is the appearance of one undetermined constant,

$\widehat{G}(\omega_0)$, which the dispersion integral does *not* fix and which must be supplied from outside — a measurement, a symmetry, or a renormalization condition. If $\widehat{G}$ grows linearly, one subtraction is not enough and one subtracts twice, at the cost of two constants:

$$\widehat{G}(\omega) = \widehat{G}(\omega_0) + (\omega-\omega_0)\,\widehat{G}'(\omega_0) + \frac{(\omega-\omega_0)^2}{i\pi}\,\mathrm{PV}\!\int_{-\infty}^{\infty} \frac{\widehat{G}(\omega')}{(\omega'-\omega)(\omega'-\omega_0)^2}\,\mathrm{d}\omega'. \tag{5.136}$$

The pattern is clear: each power of high-energy growth that the function exhibits costs one subtraction, hence one undetermined constant, which is fixed by one piece of low-energy data. The number of subtractions needed is set by the behaviour at infinity; the constants are the data that the dispersion relation cannot generate from the discontinuity alone.

Physical Insight

The arithmetic "one power of growth = one subtraction = one undetermined constant fixed by data" is the same arithmetic that will organize renormalization. There, the superficial degree of divergence of a diagram (Chapter 21) counts the powers of large loop momentum, hence the number of subtractions needed to render the amplitude finite, hence the number of counterterms whose finite parts are fixed by renormalization conditions (Chapter 22). A dispersion relation and a renormalized amplitude are the same analytic function written two ways: as an integral over its discontinuity plus a polynomial of undetermined constants. We have met the bookkeeping of renormalization here, in its gentlest form, before any field is quantized.

5.5.4 The discontinuity across a cut is twice the imaginary part

We now pass from the frequency variable of linear response to the natural variable of relativistic amplitudes, a Mandelstam invariant we call s (for the two-point function below it is $s = p^2$, the squared momentum flowing through). An amplitude $\Pi(s)$ is, generically, a function holomorphic in the complex s-plane except for singularities forced on it by physical processes: a pole at $s = m^2$ where an intermediate state can go on shell as a single stable particle, and a *branch cut* running along the real axis from a threshold s_{th} (the squared invariant mass of the lightest multiparticle state into which the system can decay or scatter) to $+\infty$, where a continuum of intermediate states opens up. The cut is the analytic signature of those open channels.

The key structural fact is that the size of the jump across the cut is not free: it is fixed to be twice the imaginary part of the boundary value. The reason is a reflection symmetry inherited from reality.

Theorem 5.38 (Discontinuity equals twice the imaginary part)**.** *Let* Π *be holomorphic in the cut plane* $\mathbb{C}\setminus[s_{\rm th},\infty)$ *and real on the segment of the real axis below threshold,* $\Pi(s)\in\mathbb{R}$ *for* $s<s_{\rm th}$*. Then* Π *obeys the Schwarz reflection principle* $\Pi(s^*)=\Pi(s)^*$*, and the discontinuity across the cut is*

$$\operatorname{disc}\Pi(s)\equiv\lim_{\epsilon\to0^+}\big[\Pi(s+i\epsilon)-\Pi(s-i\epsilon)\big]=2i\,\operatorname{Im}\Pi(s+i\epsilon),\qquad s>s_{\rm th}. \tag{5.137}$$

Proof. The function $h(s)\equiv\Pi(s^*)^*$ is holomorphic wherever Π is (composition with the antiholomorphic conjugation done twice is holomorphic), and on the lower-threshold segment it agrees with Π because Π is real there: for $s<s_{\rm th}$ real, $h(s)=\Pi(s)^*=\Pi(s)$. Two functions holomorphic on the (connected) cut plane that agree on a segment agree everywhere there, by the identity theorem (Section 5.4). Hence $\Pi(s^*)^*=\Pi(s)$, i.e. $\Pi(s^*)=\Pi(s)^*$ throughout. Apply this with $s\to s+i\epsilon$, whose conjugate is $s-i\epsilon$:

$$\Pi(s-i\epsilon)=\Pi\big((s+i\epsilon)^*\big)=\Pi(s+i\epsilon)^*. \tag{5.138}$$

Therefore the jump is

$$\Pi(s+i\epsilon)-\Pi(s-i\epsilon)=\Pi(s+i\epsilon)-\Pi(s+i\epsilon)^*=2i\,\operatorname{Im}\Pi(s+i\epsilon), \tag{5.139}$$

which is (5.137). □

This little theorem is the hinge of the whole subject. It says that the *imaginary part* of an amplitude on the upper lip of its cut — a quantity that, by the optical theorem of Section 5.5.6, is a sum of physical probabilities and is therefore directly measurable — is exactly one half of the analytic discontinuity that a contour integral around the cut would pick up. The Sokhotski–Plemelj identity (Theorem 5.10) is the same statement at the level of the elementary kernel: writing a propagator denominator as $1/(s'-s\mp i\epsilon)$, the two sign choices straddle the cut, and their difference,

$$\frac{1}{s'-s-i\epsilon}-\frac{1}{s'-s+i\epsilon}=2\pi i\,\delta(s'-s), \tag{5.140}$$

is the discontinuity of $1/(s'-s)$ regarded as a function of s across its (point) singularity. Every cut in field theory is, microscopically, a coherent sum of such elementary $i\epsilon$ discontinuities, one for each intermediate particle put on shell.

Physical Insight

The physical reading of (5.137) is that *an amplitude acquires an imaginary part exactly when an intermediate state can go on shell.* Below threshold there is no real process the intermediate state can undergo, the propagators stay off the real-axis poles, and Π is real. Above threshold the intermediate

particles can be produced as real, on-shell quanta; the propagator denominators hit their poles, Sokhotski–Plemelj converts $1/(p^2 - m^2 + i\epsilon)$ into a $-i\pi\delta(p^2 - m^2)$ that enforces the on-shell condition, and the amplitude develops an absorptive (imaginary) part. "Absorptive" is literal: it is the part of forward scattering that removes flux from the beam into open channels.

5.5.5 Dispersion relations: reconstructing an amplitude from its cut

With Theorem 5.38 in hand, the dispersion relation for an amplitude writes itself. Suppose $\Pi(s)$ is holomorphic in the cut plane, real below threshold, and bounded at infinity well enough that the following arc drops. By Cauchy's formula, for s off the cut,

$$\Pi(s) = \frac{1}{2\pi i} \oint_C \frac{\Pi(s')}{s' - s}\, \mathrm{d}s', \tag{5.141}$$

where C is a small counterclockwise circle around s. Now *deform* C outward (Fig. 5.6). The integrand is holomorphic everywhere except at the cut $[s_{\mathrm{th}}, \infty)$ and on the arc at infinity, so the circle around s can be blown up until it becomes a large circle at radius R minus a contour that wraps tightly around the cut (the deformation sweeps across no singularity except the explicit pole at $s' = s$, which is what we are computing). If $\Pi(s') \to 0$ as $|s'| \to \infty$, the large circle drops, and what remains is the integral down the upper lip of the cut and back along the lower lip:

$$\Pi(s) = \frac{1}{2\pi i} \int_{s_{\mathrm{th}}}^{\infty} \frac{\Pi(s' + i\epsilon) - \Pi(s' - i\epsilon)}{s' - s}\, \mathrm{d}s' = \frac{1}{2\pi i} \int_{s_{\mathrm{th}}}^{\infty} \frac{\operatorname{disc}\Pi(s')}{s' - s}\, \mathrm{d}s'. \tag{5.142}$$

Inserting the discontinuity (5.137) converts this into an integral over the imaginary part alone:

$$\boxed{\Pi(s) = \frac{1}{\pi} \int_{s_{\mathrm{th}}}^{\infty} \frac{\operatorname{Im}\Pi(s')}{s' - s}\, \mathrm{d}s'.} \tag{5.143}$$

The amplitude everywhere in the cut plane is reconstructed from its imaginary part on the cut — from its absorptive part, hence (looking ahead) from total cross sections.

When Π does not vanish at infinity the large circle does not drop, and we subtract exactly as in Section 5.5.3, now most conveniently at $s = 0$. The once-subtracted dispersion relation reads

$$\Pi(s) = \Pi(0) + \frac{s}{\pi} \int_{s_{\mathrm{th}}}^{\infty} \frac{\operatorname{Im}\Pi(s')}{s'\,(s' - s)}\, \mathrm{d}s', \tag{5.144}$$

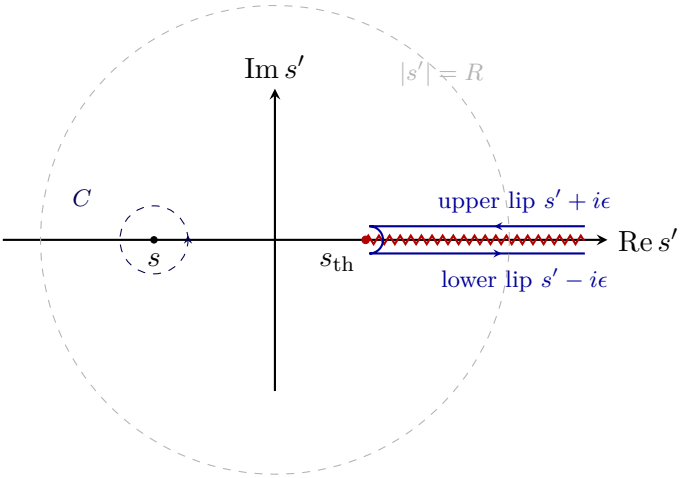


Figure 5.6: The contour deformation behind the dispersion relation (5.143). Cauchy's formula on the small circle C around s equals the same integrand wrapped around all other singularities. The only other singularity is the branch cut $[s_{\text{th}}, \infty)$ (zigzag); deforming C out to the large circle $|s'| = R$ and collapsing onto the cut leaves an integral down the upper lip and back along the lower lip, whose integrand is the discontinuity $\operatorname{disc}\Pi = 2i\,\operatorname{Im}\Pi$ of (5.137). If $\Pi \to 0$ on the large circle it drops; otherwise it must be tamed by subtractions (5.144).

the extra $1/s'$ in the integrand again buying one power of convergence at the cost of the single constant $\Pi(0)$. A twice-subtracted version,

$$\Pi(s) = \Pi(0) + s\,\Pi'(0) + \frac{s^2}{\pi}\int_{s_{\text{th}}}^{\infty} \frac{\operatorname{Im}\Pi(s')}{s'^2\,(s'-s)}\,\mathrm{d}s', \tag{5.145}$$

costs two constants, $\Pi(0)$ and $\Pi'(0)$. These are the workhorse formulas of Chapters 19 and 60.

5.5.6 Two results stated, awaiting their proofs

The dispersive machinery exists to serve two structural results of quantum field theory. Neither can be proved before fields are quantized, so we state them precisely here, with forward pointers to the chapters that discharge them, and note how each is an instance of what we have just built.

The Källén–Lehmann spectral representation. For an interacting scalar field ϕ with a single stable particle of mass m, the exact (Heisenberg-picture) two-point function has a Fourier transform of the form

$$\Delta(p^2) = \int_0^{\infty} \mathrm{d}\mu^2\,\rho(\mu^2)\,\frac{i}{p^2 - \mu^2 + i\epsilon}, \qquad \rho(\mu^2) \geq 0, \tag{5.146}$$

with a nonnegative *spectral density* ρ that consists of a delta function $Z\,\delta(\mu^2 - m^2)$ at the one-particle mass (the residue $0 \le Z \le 1$ being the field-strength renormalization) plus a continuum supported above the multiparticle threshold $\mu^2 \ge (2m)^2$. Comparing (5.146) with the dispersion relation (5.143) shows they are the *same statement*: the spectral density is the discontinuity of the propagator, $\rho(\mu^2) = \frac{1}{\pi}\operatorname{Im}\Delta(\mu^2)$, and the representation (5.146) is precisely the reconstruction of Δ from its cut. The nonnegativity $\rho \ge 0$ is the new physical input — it follows from inserting a complete set of states and is the spectral shadow of unitarity. The full construction, including the proof that $\rho \ge 0$ and the identification of Z, is carried out in Chapter 9 (free case) and completed for the interacting field alongside LSZ in Chapter 14.

The optical theorem. Unitarity of the S-matrix, $\mathbf{S}^\dagger\mathbf{S} = 1$ written as $\mathbf{S} = 1 + i\mathcal{T}$, forces $-i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger\mathcal{T}$, whose diagonal matrix element relates the imaginary part of a forward amplitude to a sum over all states the initial state can scatter into:

$$2\,\operatorname{Im}\mathcal{M}(i \to i) = \sum_X \int \mathrm{d}\Pi_X\, \left|\mathcal{M}(i \to X)\right|^2, \tag{5.147}$$

the sum running over all kinematically allowed final states X with their Lorentz-invariant phase space $\mathrm{d}\Pi_X$. For a two-particle initial state at centre-of-mass energy squared s, the right-hand side is proportional to the total cross section, giving the form usually quoted,

$$\operatorname{Im}\mathcal{M}(s, t=0) = 2\,E_{\mathrm{cm}}\,p_{\mathrm{cm}}\,\sigma_{\mathrm{tot}}(s) \xrightarrow{\;s\to\infty\;} s\,\sigma_{\mathrm{tot}}(s). \tag{5.148}$$

This is the promised physical meaning of the imaginary part that the dispersion integrals (5.143)–(5.145) integrate over: $\operatorname{Im}\Pi$ is not an abstract analytic datum but a sum of squared physical amplitudes, hence a total rate. A dispersion relation thus reconstructs the full amplitude — real and imaginary parts at all s — from cross-section data alone, up to the subtraction constants. The derivation of (5.147) from unitarity is given in Chapter 14, its diagrammatic incarnation (Cutkosky's cutting rules, which compute $\operatorname{disc}\Pi$ diagram by diagram by putting internal lines on shell exactly as in (5.140)) in Chapter 60.

5.5.7 Worked computation: a two-point function from its discontinuity

We now make the whole circle of ideas concrete on a model that is not a toy: it is, up to spin and charge factors supplied in Chapter 19, the vacuum polarization of QED. Consider the one-loop "bubble" two-point function built from two propagators of a particle of mass m,

$$\Pi(s) = \frac{g^2}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2 - m^2 + i\epsilon)\,((k+q)^2 - m^2 + i\epsilon)}, \qquad s = q^2, \tag{5.149}$$

with g a coupling of mass dimension making Π dimensionless. We do *not* evaluate the loop integral directly here — that is the business of Chapter 19, with the dimensional regularization of Chapter 5's companion machinery. Instead we exploit analyticity: we compute only the *discontinuity*, which is elementary, and then *reconstruct* the whole function by a dispersion relation. This is the strategy the worked example is built to teach, because it is how hard amplitudes are actually obtained when the direct integral is intractable.

Worked Example 5.39 (Reconstructing the bubble from its cut)**. Step 1: locate the cut.** The integrand of (5.149) develops a singularity in s when both internal lines can be simultaneously on shell with positive energy — when the external momentum q can create a real pair. The lightest such intermediate state is two quanta of mass m, with invariant mass squared $(2m)^2$. Hence the cut runs along

$$s \in [s_{\rm th}, \infty), \qquad s_{\rm th} = 4m^2, \tag{5.150}$$

and $\Pi(s)$ is real and analytic for $s < 4m^2$, in particular at $s = 0$.

Step 2: the discontinuity from on-shell phase space. By Theorem 5.38, the discontinuity is $2i\,\mathrm{Im}\,\Pi$, and by the optical-theorem logic (5.147) the imaginary part is the two-body phase space of the on-shell pair weighted by the (here constant) production amplitude. Put the two lines on shell with the Sokhotski–Plemelj replacement (5.140), $1/(k^2 - m^2 + i\epsilon) \to -i\pi\,\delta(k^2 - m^2)$ for each line, keeping positive energies. The resulting integral is the standard two-body Lorentz-invariant phase space,

$$\mathrm{Im}\,\Pi(s) = \frac{g^2}{2}\cdot\frac{1}{8\pi}\,\beta(s)\,\theta(s-4m^2) = \frac{g^2}{16\pi}\,\beta(s)\,\theta(s-4m^2), \qquad \beta(s) \equiv \sqrt{1-\frac{4m^2}{s}}, \tag{5.151}$$

where $\beta(s)$ is the velocity of each produced quantum in the centre-of-mass frame and the factor $1/8\pi$ is the two-body phase-space volume $\int \mathrm{d}\Pi_2 = \beta/8\pi$ (derived in Chapter 16; we quote only the kinematic number, which is all the example needs). The key qualitative facts are visible immediately: $\mathrm{Im}\,\Pi$ vanishes as $\beta \to 0$ at threshold ($s \to 4m^2$, the pair produced at rest, vanishing phase space) and approaches the constant $g^2/16\pi$ as $s \to \infty$ ($\beta \to 1$). Because $\mathrm{Im}\,\Pi$ tends to a nonzero constant, the unsubtracted integral (5.143) would diverge logarithmically: *one subtraction is mandatory.*

Step 3: the once-subtracted dispersion relation. Subtracting at $s = 0$ (allowed, since Π is analytic there), Eq. (5.144) gives

$$\Pi(s) - \Pi(0) = \frac{s}{\pi}\int_{4m^2}^{\infty} \frac{\mathrm{Im}\,\Pi(s')}{s'\,(s'-s)}\,\mathrm{d}s' = \frac{g^2}{16\pi^2}\,s\int_{4m^2}^{\infty}\frac{\beta(s')}{s'\,(s'-s)}\,\mathrm{d}s'. \tag{5.152}$$

We now evaluate the integral in closed form. The natural variable is β itself: set $v = \beta(s') = \sqrt{1-4m^2/s'}$, so $s' = 4m^2/(1-v^2)$ and $\mathrm{d}s' = 4m^2\cdot 2v\,(1-v^2)^{-2}\,\mathrm{d}v$,

with $v:0\to 1$ as $s':4m^2\to\infty$. A short reduction, using $s'-s=[4m^2-s(1-v^2)]/(1-v^2)$, collapses the integrand to a rational function of v:

$$\frac{\beta(s')\,\mathrm{d}s'}{s'\,(s'-s)}=\frac{v\cdot 4m^2\,\dfrac{2v}{(1-v^2)^2}\,\mathrm{d}v}{\dfrac{4m^2}{1-v^2}\cdot\dfrac{4m^2-s(1-v^2)}{1-v^2}}=\frac{2v^2}{4m^2-s(1-v^2)}\,\mathrm{d}v=\frac{2v^2}{(4m^2-s)+s\,v^2}\,\mathrm{d}v. \tag{5.153}$$

Thus

$$K(s)\equiv\int_{4m^2}^{\infty}\frac{\beta(s')}{s'(s'-s)}\,\mathrm{d}s'=\int_0^1\frac{2v^2}{(4m^2-s)+s\,v^2}\,\mathrm{d}v=\frac{2}{s}\int_0^1\left[1-\frac{4m^2-s}{(4m^2-s)+s\,v^2}\right]\mathrm{d}v, \tag{5.154}$$

where we split $sv^2=[(4m^2-s)+sv^2]-(4m^2-s)$ in the numerator. For $0<s<4m^2$ the constant $4m^2-s>0$, and writing $b\equiv\sqrt{4m^2/s-1}=\sqrt{(4m^2-s)/s}>0$ the remaining elementary integral is an arctangent,

$$\int_0^1\frac{\mathrm{d}v}{(4m^2-s)+s\,v^2}=\frac{1}{s}\int_0^1\frac{\mathrm{d}v}{b^2+v^2}=\frac{1}{s\,b}\arctan\frac{1}{b}, \tag{5.155}$$

so that, assembling (5.154) with $4m^2-s=s\,b^2$,

$$K(s)=\frac{2}{s}\Big[\,1-s\,b^2\cdot\frac{1}{s\,b}\arctan\frac{1}{b}\,\Big]=\frac{2}{s}\Big[\,1-b\arctan\frac{1}{b}\,\Big]. \tag{5.156}$$

Multiplying by the prefactor in (5.152), the once-subtracted dispersion relation delivers the two-point function below threshold in closed form,

$$\boxed{\Pi(s)-\Pi(0)=\frac{g^2}{8\pi^2}\left[1-\sqrt{\frac{4m^2}{s}-1}\,\arctan\frac{1}{\sqrt{\frac{4m^2}{s}-1}}\right],\qquad 0<s<4m^2.} \tag{5.157}$$

Step 4: continue across the cut and recover the input. The expression (5.157) is real and analytic for $0<s<4m^2$. Continue it to $s>4m^2$ by following $s\to s+i\epsilon$, the physical prescription. There $b=\sqrt{4m^2/s-1}$ becomes imaginary; with the $+i\epsilon$ branch it continues to $b\to -i\,\beta$ with $\beta=\sqrt{1-4m^2/s}\in(0,1)$. Using $\arctan(z)=\frac{1}{2i}\ln\frac{1+iz}{1-iz}$, one finds $b\arctan(1/b)\to\beta\,\mathrm{artanh}(1/\beta)=\frac{\beta}{2}\ln\frac{1+\beta}{1-\beta}-i\,\frac{\pi\beta}{2}$, so that

$$\Pi(s)-\Pi(0)=\frac{g^2}{8\pi^2}\left[1-\frac{\beta}{2}\ln\frac{1+\beta}{1-\beta}\right]+i\,\frac{g^2}{16\pi}\,\beta,\qquad s>4m^2. \tag{5.158}$$

The imaginary part of the reconstructed function is $\dfrac{g^2}{16\pi}\beta$ — *exactly the input* (5.151) we started from. The dispersion relation has reproduced its own discontinuity, the internal consistency check guaranteed by Theorem 5.38: the

analytic function built from a given imaginary part has that imaginary part. The real part, the genuinely new output, is the logarithm $-\frac{g^2}{16\pi^2}\beta \ln\frac{1+\beta}{1-\beta}$ that one would otherwise have to extract from the full loop integral. Figure 5.7 plots both parts: the imaginary part switches on at $s = 4m^2$ with the square-root threshold behaviour of β, while the real part is smooth and develops a logarithmic shoulder there.

Step 5: read off the lessons. Three features of (5.157)–(5.158) are permanent. First, the branch point sits exactly at the physical threshold $s = 4m^2$ for producing the two-particle intermediate state: *singularities are kinematics*. Second, the single subtraction constant $\Pi(0)$ is undetermined by the dispersion integral; it is fixed by a renormalization condition, and the choice $\Pi(0)$ as the value at zero momentum is precisely the on-shell-like scheme used for QED vacuum polarization in Chapters 19 and 20, where this same $\Pi(0)$ subtraction is what renders the photon self-energy finite and defines the renormalized charge. Third, the logarithm $\ln\frac{1+\beta}{1-\beta}$ — with its imaginary part above threshold and its real, smooth behaviour below — is the universal one-loop bubble function; the QED computation of Chapter 19 differs only by spin-dependent numerator factors and the replacement of the scalar pair by an e^+e^- pair, and yields the running of the fine-structure constant of Chapter 24 through this very logarithm.

We have closed the loop promised at the start: a support condition in time (causality) became analyticity in a half-plane; analyticity plus a growth bound became the Kramers–Kronig and dispersion relations; and a single elementary discontinuity, fixed by Sokhotski–Plemelj to be twice an imaginary part, was enough to reconstruct an entire nontrivial two-point function, with the only freedom an undetermined subtraction constant that anticipates renormalization. The lesson is that an analytic amplitude is *rigid*: it is completely fixed by the locations and strengths of its singularities. There is, however, a sting in the tail. Perturbation theory does not hand us this rigid analytic function directly; it hands us a power series in the coupling that *approximates* it — and, as the zero-dimensional computation of Section 4.6 already warned and Chapter 42 will confirm, that series has zero radius of convergence. To make sense of the approximation we must understand what it means for a divergent series to represent an analytic function, and how to extract numbers from it. That is the asymptotic-analysis toolkit — asymptotic series, Watson's lemma, and the method of steepest descent — to which the next section turns.

5.6 Asymptotic series, Watson's lemma, and steepest descent

The dispersion relations of the previous section (Section 5.5) reconstruct an analytic amplitude from its discontinuity, exhibiting it as a genuine, well-defined

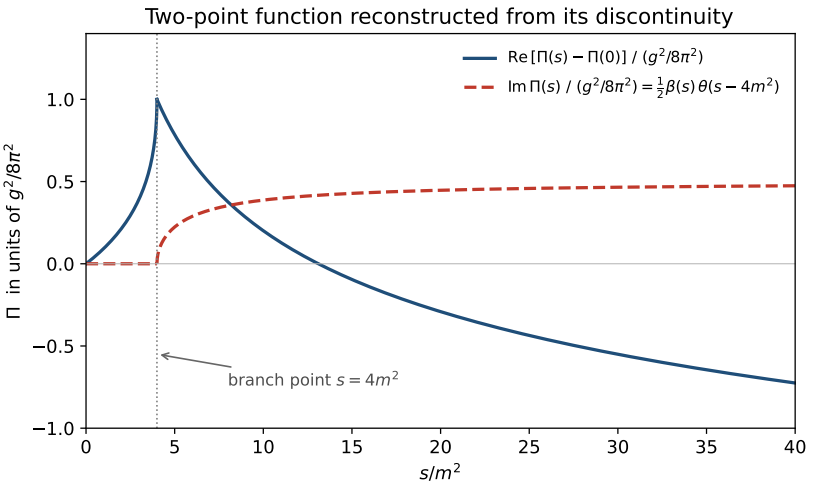


Figure 5.7: The model two-point function $\Pi(s)$ of Theorem 5.39, reconstructed from its discontinuity by the once-subtracted dispersion relation (5.152). The imaginary part (dashed) vanishes below the threshold $s = 4m^2$, switches on there as the phase-space factor $\beta(s) = \sqrt{1 - 4m^2/s}$, and saturates at high energy. The real part (solid), the new output of the dispersion integral, is everywhere smooth and carries the logarithmic shoulder (5.158) at threshold. This is, up to spin factors, the vacuum-polarization function $\Pi(q^2)$ of QED computed in Chapter 19. Both parts are in units of $g^2/8\pi^2$; the imaginary curve shows $\frac{1}{2}\beta$ in those units.

analytic function of the external invariants. Yet the only systematic way we will ever have to *compute* such an amplitude is perturbation theory: an expansion in a small coupling, generated diagram by diagram. The uncomfortable fact, foreshadowed already in the zero-dimensional toy of Section 4.6.5 and made unavoidable by Dyson's argument below, is that this expansion does not converge. For any fixed nonzero coupling the successive terms eventually grow without bound; the series has zero radius of convergence. A naive reading would conclude that perturbation theory is meaningless. The correct reading, which this section establishes, is that the perturbative series is *asymptotic*: it is a precisely defined object that approximates the true amplitude with an error controlled by its own terms, and from which physical numbers are extracted by stopping at the right place. To make this precise we need three things, developed in turn: the Poincaré definition of an asymptotic expansion and the phenomenon of optimal truncation; Watson's lemma, which reads off the asymptotics of a Laplace-type integral from the small-time behaviour of its integrand and exposes the factorial growth of the coefficients; and the method of steepest descent, which evaluates the oscillatory and exponential integrals — propagators, partition functions, semiclassical path integrals — that generate these series in the first place. The whole apparatus is the analytic engine of every saddle-point evaluation in the book, from the

loop expansion (Chapter 10) to instantons (Chapter 42); here we build it on self-contained model integrals that plant exactly the objects those later chapters consume.

5.6.1 Asymptotic (Poincaré) series

The first task is to say precisely what it means for a divergent series to "represent" a function. The definition is due to Poincaré, and its entire content is to reverse the order of two limits.

Definition 5.40 (Asymptotic expansion)**.** Let f be defined on a punctured neighbourhood of $g = 0$ (for definiteness, on an interval $0 < g < g_0$, or in a sector of the complex g-plane). A formal power series $\sum_{n=0}^{\infty} a_n g^n$ is an *asymptotic expansion* of f as $g \to 0^+$, written

$$f(g) \sim \sum_{n=0}^{\infty} a_n g^n \qquad (g \to 0^+), \tag{5.159}$$

if for every fixed integer $N \geq 0$ the remainder after $N+1$ terms is of higher order than the last term retained:

$$R_N(g) \equiv f(g) - \sum_{n=0}^{N} a_n g^n = o(g^N) \quad \text{as } g \to 0^+, \quad \text{equivalently} \quad R_N(g) = O(g^{N+1}). \tag{5.160}$$

The two estimates in (5.160) are equivalent whenever, as in every case we meet, the next term $a_{N+1} g^{N+1}$ is itself the leading part of R_N; we use the sharper $O(g^{N+1})$ form throughout. The crucial point is what Theorem 5.40 does *not* require: nothing whatsoever is said about the limit $N \to \infty$ at fixed g. A *convergent* series demands that, holding g fixed, the partial sums tend to $f(g)$ as $N \to \infty$. An *asymptotic* series demands that, holding N fixed, the error shrinks faster than g^N as $g \to 0$. These are independent conditions. A convergent power series with a positive radius of convergence is automatically asymptotic (its tail is $O(g^{N+1})$), but the converse fails spectacularly: a series can be asymptotic to a function and yet diverge for every $g \neq 0$. That is precisely the situation in quantum field theory, and the cleanest model exhibiting it is the following.

The canonical model: the Euler–Stieltjes integral. Consider the function defined for $g > 0$ by the convergent integral

$$E(g) \equiv \int_0^{\infty} \frac{e^{-t}}{1 + gt}\, \mathrm{d}t. \tag{5.161}$$

Because $0 < 1/(1+gt) \leq 1$ on the domain of integration, the integral is bounded by $\int_0^{\infty} e^{-t} \mathrm{d}t = 1$ and converges absolutely for every $g \geq 0$; E is a smooth,

finite, decreasing function with $E(0) = 1$. (It is, up to a substitution, the exponential integral; here we need only the elementary bounds.) Now generate its perturbative expansion in the naive way, by expanding the geometric factor under the integral sign,

$$\frac{1}{1+gt} = \sum_{n=0}^{N} (-gt)^n + \frac{(-gt)^{N+1}}{1+gt}, \tag{5.162}$$

an *exact* algebraic identity (the finite geometric sum plus its remainder), valid for all $t \geq 0$ and $g \geq 0$. Multiplying by e^{-t} and integrating term by term using the elementary moment $\int_0^\infty t^n e^{-t}\,\mathrm{d}t = n!$, we obtain

$$E(g) = \sum_{n=0}^{N} (-1)^n\, n!\, g^n + R_N(g), \qquad R_N(g) = (-1)^{N+1} g^{N+1} \int_0^\infty \frac{t^{N+1} e^{-t}}{1+gt}\,\mathrm{d}t. \tag{5.163}$$

The series is forced upon us: its coefficients are $a_n = (-1)^n\, n!$. They grow factorially, so by the ratio test the radius of convergence is

$$\lim_{n\to\infty} \left| \frac{a_n g^n}{a_{n+1} g^{n+1}} \right| = \lim_{n\to\infty} \frac{1}{(n+1)\,|g|} = 0, \tag{5.164}$$

i.e. the series $\sum (-1)^n n!\, g^n$ *diverges for every* $g \neq 0$. Nonetheless it is asymptotic to E. To see this we do not discard the remainder but bound it. Since $1 + gt \geq 1$ for $g, t \geq 0$,

$$|R_N(g)| = g^{N+1} \int_0^\infty \frac{t^{N+1} e^{-t}}{1+gt}\,\mathrm{d}t \leq g^{N+1} \int_0^\infty t^{N+1} e^{-t}\,\mathrm{d}t = (N+1)!\, g^{N+1}. \tag{5.165}$$

The error after summing through $a_N g^N$ is bounded by $(N+1)!\, g^{N+1}$ — exactly the magnitude of the *first omitted term*. Thus $R_N(g) = O(g^{N+1})$ for each fixed N, which is Theorem 5.40: the divergent series (5.163) is the asymptotic expansion of the perfectly good function (5.161). This single example contains, in miniature, the entire situation of perturbative QFT: a finite analytic observable, a factorially divergent series that nonetheless represents it, and a remainder bounded by its own first dropped term. The factorial $(N+1)!$ in (5.165) is the same factorial that Watson's lemma will trace, in Section 5.6.2, to the $n!$ moments of the Laplace measure.

The coefficients are unique; the function is not. Two facts about asymptotic series are decisive for everything that follows, and they pull in opposite directions.

Proposition 5.41 (Uniqueness of asymptotic coefficients)**.** *If f admits an asymptotic expansion (5.159), its coefficients are determined by f through the*

iterated limits

$$a_0 = \lim_{g\to 0^+} f(g), \qquad a_N = \lim_{g\to 0^+} \frac{1}{g^N}\Big(f(g) - \sum_{n=0}^{N-1} a_n g^n\Big). \tag{5.166}$$

In particular a given function has at most one asymptotic power series.

Proof. By (5.160) with N replaced by $N-1$, $f(g)-\sum_{n=0}^{N-1} a_n g^n = a_N g^N + o(g^N)$. Dividing by g^N and letting $g \to 0^+$ kills the $o(g^N)$ term and isolates a_N, which proves (5.166) by induction on N starting from $a_0 = \lim f$. Each a_N is thereby fixed by f alone. □

The reverse statement is the one that creates the central problem of the chapter. The map from functions to asymptotic series is *not injective*. Consider $h(g) = e^{-1/g}$ for $g > 0$ (and $h(0) = 0$). Every derivative of h at the origin vanishes: by induction $h^{(k)}(g) = P_k(1/g)\, e^{-1/g}$ for a polynomial P_k, and $g^{-m} e^{-1/g} \to 0$ as $g \to 0^+$ for every m (set $u = 1/g \to \infty$ and use $u^m e^{-u} \to 0$). Hence

$$e^{-1/g} \sim 0 + 0\cdot g + 0\cdot g^2 + \cdots \qquad (g \to 0^+)\,: \tag{5.167}$$

the function $e^{-1/g}$ has the *zero* asymptotic series, indistinguishable order by order from the constant function 0. Consequently, if $f \sim \sum a_n g^n$ then $f + c\, e^{-1/g}$ has the very same asymptotic series for any constant c. The coefficients $\{a_n\}$ remember f only up to an additive term that is smaller than every power of g — a contribution "beyond all orders" in perturbation theory. This exponentially small ambiguity is not a defect of the definition; it is a true feature of the objects. In QFT the missing piece $e^{-1/g}$ is no abstraction: with g a coupling it is the weight $e^{-S_{\text{inst}}/g}$ of an instanton (Chapter 42), a genuine non-perturbative contribution to which the perturbative series is, by construction, blind. The question of whether and how the function can nevertheless be recovered from its coefficients is exactly what Borel summation answers in Section 5.7.

Optimal truncation and the irreducible error. If the series diverges, one cannot sum it; one truncates it. The art is to truncate at the order that minimises the error. Take the Euler model, where by (5.165) the error after N terms is no larger than the first omitted term, $|R_N(g)| \lesssim (N+1)!\, g^{N+1}$, and (since the terms alternate in sign and the remainder integral is positive) is in fact bracketed between the first two omitted terms. The size of the term of order n is $b_n(g) \equiv n!\, g^n$. As n increases at fixed g the terms first *decrease*, then reach a minimum, then *increase* factorially. The turning point is where consecutive terms are equal in magnitude,

$$\frac{b_{n+1}}{b_n} = (n+1)\, g \approx 1 \quad \Longrightarrow \quad N^*(g) \approx \frac{1}{g}, \tag{5.168}$$

the *optimal truncation order*: sum up to the least term and stop. The accuracy achievable there is the size of that least term. Using Stirling's formula $N! \sim \sqrt{2\pi N}\,(N/e)^N$ (derived in Theorem 5.45) at $N = N^* = 1/g$,

$$b_{N^*}(g) = N^*!\, g^{N^*} \sim \sqrt{2\pi N^*}\left(\frac{N^*}{e}\right)^{N^*} g^{N^*} = \sqrt{\frac{2\pi}{g}}\left(\frac{1}{eg}\right)^{1/g} g^{1/g} = \sqrt{\frac{2\pi}{g}}\; e^{-1/g}. \tag{5.169}$$

The best the divergent series can do is to reproduce $E(g)$ with an error of order $e^{-1/g}$ — precisely the size of the invisible term (5.167). This is no coincidence: the irreducible error of optimal truncation and the ambiguity in reconstructing f from its coefficients are the same exponentially small object, seen from two sides. Optimal truncation is how perturbative QED achieves its astonishing accuracy: with $g \sim \alpha/\pi \approx 2 \times 10^{-3}$, the least term sits near $N^* \sim 1/g \sim 400$, far beyond any order ever computed, so in practice every accessible term still lies on the decreasing part of the curve and adding terms keeps improving the answer. The divergence is real but operationally invisible at the couplings of nature. Figure 4.8 already displayed this behaviour for the zero-dimensional ϕ^4 partition function; we record the quantitative law here.

Worked Example 5.42 (Optimal truncation of the Euler series)**.** We make (5.168)–(5.169) sharp for $E(g)$ at a definite coupling and exhibit the alternating bracket. Take $g = \frac{1}{5}$. The terms $b_n = (-1)^n n!\, 5^{-n}$ have magnitudes

$$|b_0| = 1,\ |b_1| = 0.2,\ |b_2| = 0.08,\ |b_3| = 0.048,\ |b_4| = 0.0384,\ |b_5| = 0.0384,\ |b_6| = 0.046, \ldots \tag{5.170}$$

The magnitudes decrease to a least term at $n = 4, 5$ (equal, since $b_{n+1}/b_n = (n+1)/5 = 1$ at $n = 4$, confirming $N^* = 1/g = 5$) and increase thereafter. Because the partial sums of an alternating series with the remainder of one sign bracket the true value, summing through $n = 4$ and through $n = 5$ traps $E(\frac{1}{5})$ between consecutive partial sums:

$$S_4 = 1 - 0.2 + 0.08 - 0.048 + 0.0384 = 0.8704, \qquad S_5 = S_4 - 0.0384 = 0.8320, \tag{5.171}$$

so $0.8320 < E(\frac{1}{5}) < 0.8704$, an uncertainty of 0.0384 — exactly the least term. The midpoint 0.8512 estimates $E(\frac{1}{5})$ with error below 0.02, and the predicted irreducible scale $\sqrt{2\pi/g}\, e^{-1/g} = \sqrt{10\pi}\, e^{-5} \approx 5.6 \times 0.0067 \approx 0.038$ matches the least term (5.169) to the accuracy of the estimate. No truncation of the divergent series can do better than this $O(e^{-1/g})$ window; to cross it one needs the non-perturbative information of Section 5.7.

5.6.2 Watson's lemma

Where do the factorially growing coefficients of a perturbative series come from? The generic source is a *Laplace integral* — an integral over a positive variable

with an exponential weight $e^{-t/g}$ (or e^{-xt} with $x = 1/g \to \infty$). Such integrals are ubiquitous: every Schwinger proper-time representation of a propagator (Eq. (5.104)) is of this form, and after Wick rotation so is every Euclidean partition function. Watson's lemma is the theorem that converts the *small-t* behaviour of the integrand directly into the *small-g* asymptotic expansion of the integral, and in doing so makes the origin of the factorials completely transparent.

The mechanism is a single elementary moment. With $x = 1/g$,

$$\int_0^\infty e^{-t/g}\, t^{\,n}\, \mathrm{d}t = \int_0^\infty e^{-xt}\, t^{\,n}\, \mathrm{d}t = \frac{n!}{x^{\,n+1}} = n!\; g^{\,n+1}, \tag{5.172}$$

the substitution $s = xt$ reducing it to $x^{-(n+1)} \int_0^\infty s^n e^{-s} \mathrm{d}s = x^{-(n+1)}\, \Gamma(n+1)$. More generally, for any exponent $\beta > -1$,

$$\int_0^\infty e^{-t/g}\, t^{\,\beta}\, \mathrm{d}t = \Gamma(\beta+1)\, g^{\,\beta+1}. \tag{5.173}$$

The factorial in (5.172) is the seed: a term $a_n t^n$ in the integrand contributes $a_n\, n!\, g^{n+1}$ to the integral, and summing such contributions produces a series whose coefficients carry an explicit $n!$. The factorial divergence of perturbation theory is, at bottom, the factorial in the moments of the exponential measure.

Theorem 5.43 (Watson's lemma)**.** *Let $\phi : [0,\infty) \to \mathbb{C}$ be measurable, and suppose:*

(i) ϕ has an asymptotic expansion at the origin in (possibly non-integer) powers,

$$\phi(t) \sim \sum_{n=0}^{\infty} a_n\, t^{\,\alpha+\beta n} \qquad (t \to 0^+), \qquad \alpha > -1,\ \beta > 0, \tag{5.174}$$

meaning that for each N, $\phi(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n} = O(t^{\alpha+\beta(N+1)})$ as $t \to 0^+$;

(ii) ϕ does not grow too fast at infinity: there are constants C, b with $|\phi(t)| \le C\, e^{bt}$ for all $t \ge T_0$.

Then the Laplace integral $F(g) \equiv \int_0^\infty e^{-t/g}\, \phi(t)\, \mathrm{d}t$ converges for $0 < g < 1/b$ and admits the asymptotic expansion

$$F(g) \sim \sum_{n=0}^{\infty} a_n\, \Gamma(\alpha+\beta n+1)\; g^{\,\alpha+\beta n+1} \qquad (g \to 0^+). \tag{5.175}$$

Proof. Convergence. For $0 < g < 1/b$ the weight $e^{-t/g}$ beats the growth bound (ii): on $[T_0,\infty)$, $|e^{-t/g}\phi(t)| \le C\, e^{-(1/g-b)t}$ is integrable, while on $[0,T_0]$ the integrand is bounded by hypothesis (i) (ϕ is $O(t^\alpha)$ with $\alpha > -1$ near 0, hence integrable). So $F(g)$ is finite.

Splitting. Fix N. By hypothesis (i) there are constants $\delta > 0$ and K_N such that, on the core interval $[0, \delta]$,

$$\Big|\phi(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n}\Big| \le K_N\, t^{\alpha+\beta(N+1)}, \qquad 0 \le t \le \delta. \tag{5.176}$$

Write $F(g) = \int_0^\delta + \int_\delta^\infty$. We treat the tail, the principal terms, and the core remainder separately.

Step 1: the tail is exponentially small. On $[\delta, \infty)$, splitting off the growth at T_0 and using (ii),

$$\Big|\int_\delta^\infty e^{-t/g}\phi(t)\,\mathrm{d}t\Big| \le e^{-\delta/2g}\int_\delta^\infty e^{-t/2g}|\phi(t)|\,\mathrm{d}t \le e^{-\delta/2g}\,C', \tag{5.177}$$

where we factored $e^{-t/g} = e^{-\delta/2g}\,e^{-(t/g-\delta/2g)}$ and, for g small enough that $1/2g > b$, bounded the remaining integral by a g-independent constant C' (the integral $\int_\delta^\infty e^{-t/2g}|\phi|\,\mathrm{d}t$ is largest at the smallest admissible g and is dominated by $\int_\delta^\infty e^{-(1/2g-b)t}C\,\mathrm{d}t$, uniformly bounded for g below a fixed threshold). The factor $e^{-\delta/2g}$ is smaller than every power g^M, so the tail contributes nothing to the asymptotic series.

Step 2: the principal terms. For each $n \le N$, the integral of $a_n t^{\alpha+\beta n}$ over $[0, \delta]$ differs from the integral over $[0, \infty)$ by a tail of the same exponentially small type as (5.177). Hence, by (5.173),

$$\int_0^\delta e^{-t/g}\,a_n t^{\alpha+\beta n}\,\mathrm{d}t = a_n\,\Gamma(\alpha+\beta n+1)\,g^{\alpha+\beta n+1} + O(e^{-\delta/2g}). \tag{5.178}$$

Summed over $n = 0, \dots, N$ this produces exactly the first $N+1$ terms of (5.175), up to exponentially small corrections.

Step 3: the core remainder is the right order. Using the bound (5.176) and again (5.173),

$$\Big|\int_0^\delta e^{-t/g}\Big(\phi(t) - \sum_{n=0}^{N} a_n t^{\alpha+\beta n}\Big)\mathrm{d}t\Big| \le K_N\int_0^\infty e^{-t/g}\,t^{\alpha+\beta(N+1)}\,\mathrm{d}t = K_N\,\Gamma\big(\alpha+\beta(N+1)+1\big)\,g^{\alpha+\beta(} \tag{5.179}$$

where extending the upper limit from δ to ∞ only enlarges the (positive) integral. This is $O(g^{\alpha+\beta(N+1)+1})$, i.e. higher order than the last term $n = N$ retained in (5.175).

Assembling the three steps, $F(g) - \sum_{n=0}^{N} a_n\Gamma(\alpha+\beta n+1)g^{\alpha+\beta n+1} = O(g^{\alpha+\beta(N+1)+1}) + O(e^{-\delta/2g})$, which is $o(g^{\alpha+\beta N+1})$. Since N was arbitrary, this is precisely Theorem 5.40 for the series (5.175). □

Watson's lemma is exactly the rigorous content of "expand the integrand and integrate term by term": the term-by-term integration is legitimate *as*

an asymptotic statement even when the resulting series diverges, because each truncation carries the controlled remainder (5.179). Applied to the Euler integral (5.161) it reproduces (5.163) instantly: there $\phi(t) = e^{-t}/(1+gt)$ is not quite of the pure form, but writing $E(g) = \int_0^\infty e^{-t}(1+gt)^{-1}\mathrm{d}t$ and expanding $(1+gt)^{-1} = \sum(-gt)^n$ is the same manoeuvre, with the moment $\int_0^\infty e^{-t}t^n\mathrm{d}t = n! = \Gamma(n+1)$ playing the role of (5.172). The general lesson is structural: *the more singular (the higher the powers in) the small-t behaviour of ϕ, the faster the coefficients $a_n\Gamma(\alpha+\beta n+1)$ grow*, because Γ grows factorially in its argument. Large orders of perturbation theory are governed by the integrand near the endpoint of integration — a principle that, in the path integral, becomes the statement that large orders are governed by the configurations nearest the saddle, which is the bridge to instantons.

5.6.3 Laplace's method and the method of steepest descent

Watson's lemma handles integrals already in Laplace form. The integrals that actually arise — amplitudes, partition functions, the path integral itself — are not: they carry an exponent $e^{\lambda f(z)}$ with f a nontrivial function of an integration variable that may range over the real line or a contour in the complex plane, and the large parameter λ (a large mass, an inverse coupling, $1/\hbar$) sits in the exponent. The method that evaluates them asymptotically is *Laplace's method* when f is real and *steepest descent* when f is complex; *stationary phase* is the purely oscillatory limit. The strategy is always the same: the integral is dominated by the neighbourhood of the point where the exponent is stationary, and the local quadratic behaviour there produces a Gaussian whose width and orientation supply the prefactor.

The real-variable case was established in full in Chapter 4 as the engine of the loop expansion: Theorem 4.38 proves that for f, g smooth with a unique nondegenerate interior minimum x_0 of f,

$$\int_a^b g(x)\, e^{-\lambda f(x)}\,\mathrm{d}x = e^{-\lambda f(x_0)}\sqrt{\frac{2\pi}{\lambda f''(x_0)}}\left[g(x_0) + \frac{c_1}{\lambda} + O(\lambda^{-2})\right] \qquad (\lambda \to +\infty), \tag{5.180}$$

with the subleading coefficient $c_1 = -\frac{f''''(x_0)}{8f''(x_0)^2} + \frac{5f'''(x_0)^2}{24f''(x_0)^3}$ computed in Theorem 4.39, and Theorem 4.40 extends this to n dimensions with the Gaussian fluctuation prefactor $(\det H)^{-1/2}$, H the Hessian of f at the saddle. We *import* these results rather than reprove them. What Chapter 4 deliberately deferred — the honest treatment of the contour rotation that turns the oscillatory e^{iS} into the convergent e^{-S_E} — is exactly the complex story we now build.

Why one must go into the complex plane. Consider the prototype

$$I(\lambda) = \int_C g(z)\, e^{\lambda f(z)}\,\mathrm{d}z, \tag{5.181}$$

with f, g holomorphic in a domain containing the contour C, and $\lambda \to +\infty$. Two features distinguish this from the real case. First, f is complex, so "minimise f" is meaningless; the natural object is the modulus $|e^{\lambda f}| = e^{\lambda \operatorname{Re} f}$, governed by the real part $u \equiv \operatorname{Re} f$. Second, u is harmonic ($\nabla^2 u = 0$, since f is holomorphic and the Cauchy–Riemann equations give $\partial_x^2 u = -\partial_y^2 u$), and a nonconstant harmonic function has *no* interior maxima or minima — only *saddles*, by the maximum principle. There is therefore no peak to sit on; the best one can do is find a saddle of u and route the contour through it along the direction in which u *descends fastest in both directions*, so that the saddle is a local maximum *along the contour*. This is the geometric content of the name.

Saddle, steepest-descent direction, constant phase. A saddle point is a zero of the derivative,

$$f'(z_0) = 0, \tag{5.182}$$

and we assume it nondegenerate, $f''(z_0) \neq 0$. Write the holomorphic increment to quadratic order,

$$f(z) = f(z_0) + \tfrac{1}{2} f''(z_0)\,(z - z_0)^2 + O\big((z - z_0)^3\big). \tag{5.183}$$

Set $f''(z_0) = |f''(z_0)|\, e^{i\theta}$ and parametrise the displacement by its direction, $z - z_0 = \rho\, e^{i\varphi}$ with $\rho \geq 0$. Then the quadratic term is

$$\tfrac{1}{2} f''(z_0)(z - z_0)^2 = \tfrac{1}{2} |f''(z_0)|\, \rho^2\, e^{i(\theta + 2\varphi)}. \tag{5.184}$$

Along the contour we want this term to be *real and negative*, for two reasons at once: real, so that the integrand does not oscillate near the saddle (the *constant-phase* or *stationary-phase* condition, $\operatorname{Im} f = \text{const}$, which is what makes the Gaussian approximation legitimate rather than a delicate cancellation of oscillations); and negative, so that $e^{\lambda f}$ *decays* away from the saddle (the *steepest-descent* condition). Both are secured by

$$\theta + 2\varphi = \pi \quad \Longrightarrow \quad \varphi = \frac{\pi - \theta}{2} \ \ (\text{or } \varphi + \pi), \tag{5.185}$$

which fixes the two opposite directions (the contour passes straight through) of steepest descent. That "constant phase" and "steepest descent" coincide is not luck but the Cauchy–Riemann structure: with $f = u + iv$, the curves $v = \text{const}$ and $u = \text{const}$ are orthogonal families, and the gradient of u (the direction of steepest change of u) is tangent to the curves $v = \text{const}$. So the path along which $|e^{\lambda f}|$ falls off fastest is automatically a path on which $\operatorname{Im} f$ is constant. Figure 5.8 draws this for the oscillatory prototype $f = iz^2$.

Reduction to a real Laplace integral. With the steepest-descent contour located, the evaluation is mechanical. Deform C to a contour C' that passes through z_0 in the direction (5.185); the deformation is legitimate by Cauchy's theorem provided the integrand is holomorphic in the swept region and the arcs at infinity give no contribution — precisely the contour-deformation discipline established in Section 5.3, applied here to a saddle rather than a pole. On C' parametrise $z = z_0 + s\,e^{i\varphi}$ with s real (arc length, signed), so $\mathrm{d}z = e^{i\varphi}\mathrm{d}s$ and, by construction (5.185),

$$f(z) = f(z_0) - \tfrac{1}{2}|f''(z_0)|\,s^2 + O(s^3), \qquad e^{\lambda f(z)} = e^{\lambda f(z_0)}\,e^{-\frac{\lambda}{2}|f''(z_0)|\,s^2}\big[1 + O(s^3)\big]. \tag{5.186}$$

The integral has become a real Laplace integral of exactly the type (5.180), with the exponent $-\frac{1}{2}|f''(z_0)|s^2$ playing the role of $-f(s)$ and an overall constant phase $e^{i\varphi}$ from the measure:

$$I(\lambda) = e^{i\varphi}\,e^{\lambda f(z_0)} \int_{-\infty}^{\infty} \big[g(z_0) + O(s)\big]\, e^{-\frac{\lambda}{2}|f''(z_0)|s^2}\,\mathrm{d}s\,\big[1 + O(\lambda^{-1})\big]. \tag{5.187}$$

The leading Gaussian integral is $\int_{-\infty}^{\infty} e^{-\frac{\lambda}{2}|f''|s^2}\mathrm{d}s = \sqrt{2\pi/(\lambda|f''(z_0)|)}$. Restoring $e^{i\varphi} = e^{i(\pi-\theta)/2}$ and noting $|f''(z_0)| = f''(z_0)e^{-i\theta}$, the direction phase and the modulus combine into a single clean branch,

$$e^{i\varphi}\sqrt{\frac{2\pi}{\lambda|f''(z_0)|}} = \sqrt{\frac{2\pi}{\lambda\,|f''(z_0)|}}\;e^{i(\pi-\theta)/2} = \sqrt{\frac{2\pi}{-\lambda\,f''(z_0)}}, \tag{5.188}$$

where the square root on the right is the one whose phase is $(\pi-\theta)/2$, i.e. the branch selected by the steepest-descent direction. We have proved:

Theorem 5.44 (Method of steepest descent, leading order)**.** *Let f, g be holomorphic near a contour C, let z_0 be a simple saddle, $f'(z_0) = 0$, $f''(z_0) \neq 0$, dominating* $\operatorname{Re} f$ *on C, and suppose C may be deformed to the steepest-descent contour through z_0 without crossing singularities and with negligible contribution from infinity. Then*

$$\boxed{I(\lambda) = \int_C g(z)\,e^{\lambda f(z)}\,\mathrm{d}z = g(z_0)\sqrt{\frac{2\pi}{-\lambda\,f''(z_0)}}\;e^{\lambda f(z_0)}\Big[1 + O(\lambda^{-1})\Big] \quad (\lambda \to +\infty),} \tag{5.189}$$

with the branch of the square root fixed by the steepest-descent direction (5.185). The full asymptotic series in $1/\lambda$ has the same form as the real case: writing $f(z) = f(z_0) + \frac{1}{2}f''(z_0)w^2$ locally and expanding g and the cubic-and-higher part of f in powers of w, term-by-term Gaussian integration gives

$$I(\lambda) = \sqrt{\frac{2\pi}{-\lambda f''(z_0)}}\;e^{\lambda f(z_0)}\Big[g(z_0) + \frac{1}{\lambda}\Big(-\frac{g''(z_0)}{2f''(z_0)} + \frac{g'(z_0)f'''(z_0)}{2f''(z_0)^2} + g(z_0)\Big(\frac{f''''(z_0)}{8f''(z_0)^2} - \frac{5f'''(z_0)^2}{24f''(z_0)^3}\Big)\Big) + \tag{5.190}$$

Proof. The leading order is (5.187)–(5.188). For the series, the localization and dominated-convergence estimates that justify replacing the deformed contour by the full real s-line, expanding, and integrating term by term are identical to those proved in Theorem 4.38 (Step 3 there); the only change is the complex prefactor $e^{i\varphi}$ and the use of the holomorphic Taylor expansion (5.183) in place of the real one. With $g \equiv \text{const}$ the bracket reduces to $1+\lambda^{-1}c_1$ with $c_1 = \frac{f''''}{8f''^2}\cdot(-1)+\frac{5f'''^2}{24f''^3}$ evaluated with the sign convention $e^{+\lambda f}$ here versus $e^{-\lambda f}$ in (5.180) (which flips $f \to -f$, hence the displayed signs); this is exactly the negative of Eq. (4.195), as the change $e^{-\lambda f} \to e^{+\lambda f}$ demands. The g' and g'' terms come from expanding the insertion $g(z_0 + e^{i\varphi}s)$ to second order and keeping the even moment. □

Caution

Steepest descent is only as good as the global contour analysis. Equation (5.189) is the contribution of *one* saddle; when the original contour deforms onto several saddles, one sums their contributions, and the dominant one is the saddle of largest $\operatorname{Re} f(z_0)$. When two saddles have equal $\operatorname{Re} f$, their contributions can interfere — this is the Stokes phenomenon, and the abrupt switching-on of a subdominant exponential as a parameter varies is a Stokes line. Honestly tracking which saddles contribute on which side of a Stokes line is the analytic heart of resurgence (Section 5.7, Chapter 42); the naive single-saddle formula is the first term of a structure that can be much richer.

Stationary phase and the Fresnel $e^{\pm i\pi/4}$. The purely oscillatory case — the one relevant to the Minkowski path integral e^{iS} before Wick rotation — is the special instance $\operatorname{Re} f \equiv 0$ on the original contour. Take $f(x) = i\psi(x)$ with ψ real on the real axis and a nondegenerate stationary point $\psi'(x_0) = 0$, $\psi''(x_0) \neq 0$, and consider

$$I(\lambda) = \int_{-\infty}^{\infty} g(x)\, e^{i\lambda\psi(x)}\, \mathrm{d}x. \tag{5.191}$$

Here $\operatorname{Re} f = 0$ everywhere on the real axis, so there is no decay at all: the integrand has constant modulus and converges only by oscillation. The saddle is at x_0, where $f''(x_0) = i\psi''(x_0)$, so $\theta = \arg(i\psi'') = \pm\pi/2$ according to the sign of $\psi''(x_0)$. The steepest-descent direction (5.185) is then $\varphi = (\pi \mp \pi/2)/2 = \pm\pi/4$: one must rotate the contour off the real axis by 45° into the complex plane to find the path on which the oscillation becomes a Gaussian decay (this is precisely the rotation drawn in Fig. 5.8). Applying Theorem 5.44 with $\lambda f'' = i\lambda\psi''$ and $-\lambda f'' = -i\lambda\psi'' = \lambda|\psi''|e^{\mp i\pi/2}$,

$$I(\lambda) = g(x_0)\sqrt{\frac{2\pi}{\lambda|\psi''(x_0)|}}\, e^{\pm i\pi/4}\, e^{i\lambda\psi(x_0)}\,[1 + O(\lambda^{-1})], \qquad \pm = \operatorname{sgn}\psi''(x_0). \tag{5.192}$$

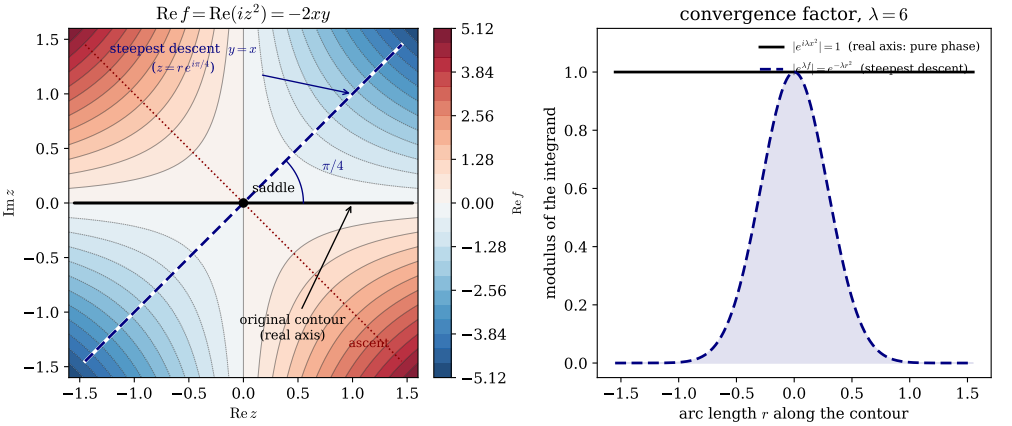


Figure 5.8: Steepest descent and stationary phase for the oscillatory prototype $I(\lambda) = \int g(z)\, e^{i\lambda z^2} \mathrm{d}z$, exponent $f(z) = iz^2$ with a saddle at the origin. *Left:* filled contours of $\mathrm{Re}\, f = \mathrm{Re}(iz^2) = -2xy$; the level sets $\mathrm{Im}\, f = \text{const}$ are the hyperbolas $xy = \text{const}$. The real axis (the original contour, black) is a constant-phase curve on which $\mathrm{Re}\, f \equiv 0$, so the integrand merely oscillates. Rotating by $\varphi = \pi/4$ onto the diagonal $y = x$ (navy dashed) reaches the steepest-descent ray $z = r\, e^{i\pi/4}$, where $iz^2 = -r^2$ is real and negative; the orthogonal diagonal $y = -x$ is the steepest-ascent direction. *Right:* along the real axis the modulus $|e^{i\lambda x^2}| = 1$ never decays, whereas along the rotated contour $|e^{\lambda f}| = e^{-\lambda r^2}$ is a genuine Gaussian. The 45° rotation is the geometric origin of the Fresnel phase $e^{i\pi/4}$ in (5.192).

The extra phase $e^{\pm i\pi/4} = \sqrt{\pm i}$ is the celebrated Fresnel factor; it is nothing but the branch (5.188) of the Gaussian, $\sqrt{2\pi/(-i\lambda\psi'')} = \sqrt{2\pi/(\lambda|\psi''|)}\, e^{\pm i\pi/4}$. In the semiclassical path integral the accumulation of these $\pi/4$ phases along a trajectory is the Maslov index; it is the seed, planted here in one dimension, of the phase conventions that Chapter 10 will need when the saddle is a classical field configuration. The leading factor $\lambda^{-1/2}$ per stationary point is, in d dimensions, $\lambda^{-d/2}(\det)^{-1/2}$ by the same diagonalization as Theorem 4.40.

5.6.4 Stirling's formula from the saddle point

The Gamma function provides the canonical worked output of the whole machinery: its asymptotics are pure Laplace's method, its integral representation is exactly of the form (5.181), and the answer — Stirling's series — is the formula we have already used twice above (in (5.169) and in the loop-counting of Chapter 4). We now derive it in full, leading term and first correction, directly from the saddle point.

Worked Example 5.45 (Stirling's series for Γ)**.** Start from the Euler integral,

valid for $\operatorname{Re} z > 0$,

$$\Gamma(z+1) = \int_0^\infty t^{\,z}\, e^{-t}\, \mathrm{d}t = \int_0^\infty \exp\!\big(z \ln t - t\big)\, \mathrm{d}t. \tag{5.193}$$

The exponent $z \ln t - t$ is *not* yet in Laplace form with a large parameter multiplying a fixed function, because z multiplies only part of it. The standard rescaling cures this: substitute $t = z\,\tau$, so $\mathrm{d}t = z\,\mathrm{d}\tau$, $t^z = z^z \tau^z$, $e^{-t} = e^{-z\tau}$, and

$$\Gamma(z+1) = z^{\,z+1} \int_0^\infty \exp\!\big(-z\,\varphi(\tau)\big)\, \mathrm{d}\tau, \qquad \varphi(\tau) \equiv \tau - \ln \tau. \tag{5.194}$$

Now z multiplies the entire fixed function φ, and we may take $z \to +\infty$ by Laplace's method. The exponent φ has a unique minimum on $(0,\infty)$:

$$\varphi'(\tau) = 1 - \frac{1}{\tau} = 0 \;\Rightarrow\; \tau_0 = 1, \qquad \varphi(1) = 1, \qquad \varphi''(\tau) = \frac{1}{\tau^2} \;\Rightarrow\; \varphi''(1) = 1 > 0, \tag{5.195}$$

so $\tau_0 = 1$ is a nondegenerate minimum, exactly the hypothesis of Theorem 4.38. The leading term follows immediately from (5.180) with $g \equiv 1$, $\lambda = z$, $f = \varphi$:

$$\Gamma(z{+}1) = z^{z+1}\, e^{-z\varphi(1)} \sqrt{\frac{2\pi}{z\,\varphi''(1)}}\,[1{+}O(1/z)] = z^{z+1}\, e^{-z} \sqrt{\frac{2\pi}{z}}\,[1{+}\cdots] = \sqrt{2\pi z}\,\left(\frac{z}{e}\right)^{\!z}[1{+}\cdots]. \tag{5.196}$$

This is Stirling's formula. For integer $z = n$ it reads $n! \sim \sqrt{2\pi n}\,(n/e)^n$; for $\Gamma(n) = (n-1)!$ the same calculation with $z \to z-1$ in the prefactor gives the equivalent form $\Gamma(z) \sim \sqrt{2\pi/z}\,(z/e)^z$ quoted in the section blueprint, the two differing only by the factor z that distinguishes $\Gamma(z+1) = z\,\Gamma(z)$.

The first correction. For the $1/z$ term we invoke the universal Laplace coefficient $c_1 = -\frac{\varphi''''(1)}{8\varphi''(1)^2} + \frac{5\varphi'''(1)^2}{24\varphi''(1)^3}$ of Eq. (4.195) with $f = \varphi$. The derivatives at $\tau_0 = 1$ are

$$\varphi''(1) = 1, \qquad \varphi'''(\tau) = -\frac{2}{\tau^3} \Rightarrow \varphi'''(1) = -2, \qquad \varphi''''(\tau) = \frac{6}{\tau^4} \Rightarrow \varphi''''(1) = 6, \tag{5.197}$$

so that

$$c_1 = -\frac{6}{8\cdot 1^2} + \frac{5\cdot(-2)^2}{24\cdot 1^3} = -\frac{3}{4} + \frac{20}{24} = -\frac{3}{4} + \frac{5}{6} = \frac{-9+10}{12} = \frac{1}{12}. \tag{5.198}$$

The expansion parameter is $1/\lambda = 1/z$, so the bracket in (5.196) is $1 + \frac{1}{12z} + O(z^{-2})$, giving the first two terms of Stirling's series:

$$\boxed{\Gamma(z+1) = \sqrt{2\pi z}\,\left(\frac{z}{e}\right)^{\!z}\left(1 + \frac{1}{12z} + \frac{1}{288 z^2} - \cdots\right)} \tag{5.199}$$

where we have appended the next coefficient $1/288$ (obtainable by carrying the expansion one order further, exactly as c_1 was obtained) for completeness. Two

checks confirm the result. Numerically, at $z = 10$: $\sqrt{20\pi}\,(10/e)^{10} = 3598695.6\ldots$, and multiplying by $(1 + \frac{1}{120} + \frac{1}{28800}) = 1.008368$ gives 3628810, against the exact $10! = 3628800$ — agreement to seven figures from two correction terms. Structurally, the series (5.199) is itself *asymptotic, not convergent*: its coefficients eventually grow factorially (the Stirling coefficients are essentially Bernoulli numbers, $B_{2k} \sim (2k)!$), so it is a textbook instance of the Poincaré series of Theorem 5.40 — to be summed only up to its least term, with an irreducible error of the $O(e^{-2\pi z})$ type. Stirling's formula is thus not merely an application of the method; it is a complete miniature of the entire section.

5.6.5 Dyson's argument: why the QFT series must be asymptotic

We have built the tools that generate and tame asymptotic series. It remains to state, at the level of principle, why the perturbation series of a genuine quantum field theory is one of them — why it cannot possibly converge. The argument is Dyson's, first made for quantum electrodynamics in 1952, and we already met its arithmetic in the zero-dimensional toy of Section 4.6.5. Its virtue is that it requires no computation of any diagram: it is a statement about the analytic structure of the theory as a function of the coupling.

The physical observables of QED (or of ϕ^4) are computed as power series in the coupling — in $\alpha = e^2/4\pi$ for QED, in λ for ϕ^4. Suppose, for contradiction, that such a series converged in some disk of positive radius about the origin in the complex coupling plane. A convergent power series defines an analytic function throughout its disk of convergence, in particular along the negative real axis: the theory at small *negative* coupling would then be the analytic continuation of the theory at small positive coupling, a well-defined physical world. But consider what negative coupling means.

- For ϕ^4 with $\lambda < 0$, the potential $\frac{\lambda}{4!}\phi^4$ is unbounded *below*: arbitrarily large field configurations have arbitrarily negative energy. The would-be vacuum is unstable; there is no ground state, and the partition function (the $\int e^{-S_E}$ of Eq. (4.75), generalised to the field theory) diverges. The theory simply does not exist for $\lambda < 0$.

- For QED with $e^2 < 0$, the Coulomb interaction reverses sign: like charges *attract* and unlike charges *repel*. The vacuum becomes catastrophically unstable to the spontaneous creation of widely separated regions of like charge, whose negative interaction energy can overcompensate the rest mass of the produced pairs. Again there is no stable ground state.

In both cases the theory fails to exist for any negative coupling, however small. But an analytic function cannot fail to exist on one side of a point where it is analytic: if the series converged in a disk, the theory would exist in that

whole disk, negative axis included. The only way to reconcile a perfectly good theory at $\lambda > 0$ (or $\alpha > 0$) with a non-existent one at $\lambda < 0$ is for the coupling origin to be a *singular point* — a point of non-analyticity — of every physical quantity. A function nonanalytic at the origin has *zero radius of convergence* for its Taylor series there. The perturbation series therefore cannot converge; it can at best be *asymptotic*, in exactly the Poincaré sense of Theorem 5.40.

This argument fixes more than the mere fact of divergence; it predicts its rate. The same instability that forbids convergence is, quantitatively, a configuration that tunnels out of the false vacuum, and the amplitude for that tunnelling is the exponentially small $e^{-A/\lambda}$ weight of an instanton. Reading the singularity at $\lambda = 0$ through this instanton lens (as Chapter 42 does in detail) predicts that the perturbative coefficients grow factorially,

$$a_n \;\sim\; C\,(-A)^{-n}\,n! \qquad (n \to \infty), \tag{5.200}$$

with A the instanton action — precisely the behaviour we computed directly for the zero-dimensional toy in Eq. (4.217) (there $A^{-1} = 2/3$), and precisely the behaviour that makes the series divergent and Watson's lemma the natural language for it. The constant A is the location of the nearest singularity in the Borel plane, the object Section 5.7 is about to define.

This leaves us in a sharp position. We possess, in the perturbative coefficients $\{a_n\}$, a complete and computable summary of a definite analytic observable; yet the series $\sum a_n g^n$ they form has zero radius of convergence and so cannot be summed in any ordinary sense, while the coefficients themselves — by Theorem 5.41 and the flat-function ambiguity (5.167) — determine the underlying function only up to an exponentially small remainder. The natural question is whether that remainder can be pinned down: can the original function be reconstructed from its divergent asymptotic coefficients, and if so, exactly when? The answer, and the obstruction to it posed by the very Borel-plane singularities at $g = -A$ that Dyson's argument has just located, is the Borel summation developed in Section 5.7.

5.7 Divergent series and Borel summation

The previous section left us with a sharp and uncomfortable question. We showed that the perturbative expansions of quantum field theory are at best *asymptotic*: by Dyson's argument (Section 5.6.5 and Eq. (5.200)) the radius of convergence is zero, and Watson's lemma (Theorem 5.43) exposed *why*—the coefficients grow like $n!$, so the would-be power series $\sum a_n g^n$ diverges for every nonzero coupling g. Optimal truncation salvaged a finite, exponentially accurate *number* from such a series, but it did not give us back a *function*. The asymptotic series knows the function only up to an irreducible error of order $e^{-\text{const}/g}$, and many distinct functions share the same asymptotic series (the standard witness is $e^{-1/g}$, whose

every Taylor coefficient at $g = 0^+$ vanishes). So: can a function be recovered *exactly* from its divergent asymptotic coefficients, and if so, under what conditions is the recovery unique?

The affirmative answer, when it exists, is *Borel summation*. The idea is disarmingly simple. The series diverges because its coefficients carry an extra $n!$; divide it out, sum the resulting (now convergent) series to a genuine function $B(t)$ in an auxiliary *Borel plane*, and then reinstate the $n!$ by a Laplace integral $\int_0^\infty e^{-t} t^n \, dt = n!$. The divergence has been laundered through an integral transform. The deep content—and the reason this section closes the analytic toolbox rather than merely decorating it—is that the entire question of *when* the procedure reconstructs the original function, and *when* it fails, is governed by the analytic structure of $B(t)$: by where its singularities sit in the Borel plane. Singularities off the integration ray cost nothing; singularities *on* it make the answer ambiguous by an exponentially small, purely imaginary amount. Those obstructing singularities are not pathologies of toy models—in real quantum field theory they are the fingerprints of instantons and of the running coupling (renormalons), and their organization is the subject of the modern *resurgence* program. We develop the machinery rigorously, prove the prototype reconstruction in full, exhibit the ambiguity by direct computation, and then state honestly what is and is not known about the field-theoretic case as of the mid-2020s.

5.7.1 The Borel transform and the Borel sum

We begin with the formal manipulation and then make it precise.

Definition 5.46 (Borel transform)**.** Let $\hat{f}(g) = \sum_{n=0}^{\infty} a_n g^n$ be a formal power series. Its *Borel transform* is the formal power series in the auxiliary variable t obtained by dividing each coefficient by $n!$,

$$B(t) \;:=\; \sum_{n=0}^{\infty} \frac{a_n}{n!}\, t^n. \tag{5.201}$$

If the original coefficients grow factorially, $|a_n| \le A\, \sigma^n \, n!$ for some constants A, σ—precisely the growth Watson's lemma attached to a QFT amplitude—then $|a_n/n!| \le A\sigma^n$, so Eq. (5.201) converges in the disk $|t| < 1/\sigma$ and defines an honest analytic function near the origin of the Borel plane. The $n!$ in the denominator is doing exactly the job we asked of it: it converts a series of zero radius of convergence into one of *finite* radius. The function $B(t)$ is the first genuine object the divergent series has produced.

To undo the division we use the elementary Laplace identity $\int_0^\infty e^{-t}\, t^n \, dt = \Gamma(n+1) = n!$. Substituting $B(gt) = \sum_n (a_n/n!)(gt)^n$ under the integral and integrating term by term reinstates each coefficient:

$$\int_0^\infty e^{-t}\, B(gt)\, dt \;=\; \sum_{n=0}^{\infty} \frac{a_n}{n!}\, g^n \int_0^\infty e^{-t}\, t^n \, dt \;=\; \sum_{n=0}^{\infty} a_n\, g^n \;=\; \hat{f}(g). \tag{5.202}$$

The interchange of sum and integral in Eq. (5.202) is legitimate *inside the disk of convergence* $|g| < 1/\sigma$ of $\hat{f}$, where the series converges anyway and Borel summation tells us nothing new. The entire utility of the construction is that the *integral* on the left may exist and define an analytic function even where the *series* on the right diverges, provided B continues analytically from its disk along the positive real axis and does not grow too fast there. This motivates the operative definition.

Definition 5.47 (Borel sum)**.** A formal power series $\hat{f}(g) = \sum_n a_n g^n$ is *Borel summable* (in the direction of the positive real axis) if (i) its Borel transform Eq. (5.201) has a nonzero radius of convergence, (ii) $B(t)$ admits analytic continuation to an open neighbourhood of the half-line $[0, \infty)$, and (iii) the continuation grows at most exponentially there, $|B(t)| \le K e^{|t|/R}$ for some $K, R > 0$. The *Borel sum* is then the function

$$\mathcal{B}\hat{f}(g) \;:=\; \int_0^\infty e^{-t}\, B(gt)\, dt \;=\; \frac{1}{g}\int_0^\infty e^{-s/g}\, B(s)\, ds, \tag{5.203}$$

the two forms related by the substitution $s = gt$. The integral converges absolutely for $0 < g < R$ and, more generally, for complex g with $\mathrm{Re}(1/g) > 1/R$.

The second form in Eq. (5.203) displays the structure most clearly: the Borel sum is a Laplace transform of B evaluated at $1/g$, and the convergence condition $\mathrm{Re}(1/g) > 1/R$ is exactly what tames the exponential growth $e^{|s|/R}$ against the kernel $e^{-s/g}$. We have purchased a function from a divergent series at the price of an analytic-continuation requirement on B. The rest of the section is, in essence, the study of when that requirement can be met.

Physical Insight

The Borel sum is an instruction to read the divergent series as data about a *different* function—B, living in the Borel plane—and then to perform a single, ordinary integral. All the difficulty of resummation has been relocated to a question of complex analysis: the analytic continuation and singularity structure of $B(t)$. This is the same move that organized the whole chapter: singular distributions became poles, poles became contour prescriptions, contour prescriptions became analytic continuations. Borel summation is that doctrine applied one last time, now to the perturbative series itself.

Directional and lateral Borel sums

The integration ray in Eq. (5.203) need not be the positive real axis. If B continues along the ray $\arg t = \theta$ with the same exponential bound, we may

define the *directional Borel sum*

$$\mathcal{B}_\theta \hat{f}(g) \;=\; \frac{1}{g}\int_0^{e^{i\theta}\infty} e^{-s/g}\, B(s)\, ds, \tag{5.204}$$

the contour running from the origin to infinity along $\arg s = \theta$. When B is analytic in a sector containing several such rays the answer is independent of θ by Cauchy's theorem, and the series is unambiguously summable. The interesting and physically generic case is when a singularity of B sits *on* the positive axis, $\theta = 0$. Then the two one-sided limits

$$\mathcal{B}_{0^\pm} \hat{f}(g) \;=\; \lim_{\theta \to 0^\pm} \mathcal{B}_\theta \hat{f}(g), \tag{5.205}$$

the *lateral* Borel sums obtained by skirting the singularity above ($\theta \to 0^+$) or below ($\theta \to 0^-$), need not agree. Their disagreement is the analytic statement of a nonperturbative ambiguity, and we compute it exactly in Theorem 5.51.

5.7.2 When the Borel sum reproduces the function: the Nevanlinna–Sokal theorem

Definition 5.47 produces *a* function from a divergent series, but it does not by itself guarantee that this function is *the* one we started from. We need a converse: a theorem stating that, under stated hypotheses on the original function f, its asymptotic series is Borel summable and its Borel sum equals f—and, crucially, that f is the *unique* function with that asymptotic series obeying the hypotheses. The sharpest such statement is due to Sokal, building on a 1919 theorem of F. Nevanlinna; it improves Watson's classical Borel-summability criterion by replacing a sector by a disk and is the standard working tool.

Theorem 5.48 (Nevanlinna–Sokal)**.** *Let $R > 0$ and let f be analytic in the open disk*

$$C_R \;=\; \Big\{ z \in \mathbb{C} \;:\; \operatorname{Re} \tfrac{1}{z} > \tfrac{1}{R} \Big\} \;=\; \Big\{ z : \; |z - \tfrac{R}{2}| < \tfrac{R}{2} \Big\}, \tag{5.206}$$

the open disk of radius $R/2$ centred at $R/2$, tangent to the imaginary axis at the origin. Suppose f obeys, uniformly in $z \in C_R$*, the asymptotic estimates*

$$\Big| f(z) - \sum_{n=0}^{N-1} a_n z^n \Big| \;\le\; A\, \sigma^N\, N!\, |z|^N \qquad \text{for all } N \ge 0, \tag{5.207}$$

with constants A, σ independent of N and z. Then:

1. *the Borel transform $B(t) = \sum_n a_n t^n/n!$ converges for $|t| < 1/\sigma$ and continues analytically to the strip $S_\sigma = \{t : \operatorname{dist}(t, [0,\infty)) < 1/\sigma\}$ about the positive real axis;*

2. *on S_σ the continuation obeys the exponential bound $|B(t)| \le K\, e^{|t|/R}$ for some $K > 0$;*

3. *f is recovered by its Borel sum: for every $z \in C_R$,*

$$f(z) \;=\; \frac{1}{z}\int_0^\infty e^{-t/z}\, B(t)\, dt, \tag{5.208}$$

the integral converging absolutely.

Conversely, these properties pin f down: any two functions analytic in C_R and satisfying Eq. (5.207) with the same *coefficients a_n coincide throughout C_R.*

The geometry of the disk Eq. (5.206) is the whole point and deserves a word, since it is where Sokal's version is stronger than Watson's. The equivalence of the two descriptions in Eq. (5.206) is immediate algebra: writing $z = x + iy$,

$$\operatorname{Re}\tfrac{1}{z} = \frac{x}{x^2+y^2} > \tfrac{1}{R} \iff x^2 + y^2 < Rx \iff \left(x - \tfrac{R}{2}\right)^2 + y^2 < \left(\tfrac{R}{2}\right)^2, \tag{5.209}$$

a disk tangent to the imaginary axis at the origin. The reason it must be a disk and not merely a sector $|\arg z| < \pi/2 + \delta$ is sharp: a sector of opening just greater than π is exactly the regime in which uniqueness can fail—there one can add a function like $e^{-c/z}$, which is analytic and asymptotically zero in the sector but does not vanish, and the asymptotic series no longer determines the function. Requiring analyticity and the remainder bound on the full disk C_R, which approaches the boundary tangentially rather than along straight rays, removes exactly that freedom and restores uniqueness. The improvement from Watson's theorem is precisely this passage from a sector to a tangent disk; it is what makes the criterion usable, because the disk condition is the one that QFT amplitudes plausibly satisfy.

Remark 5.49 (Why we state rather than reprove the hard direction). The forward implications (i)–(iii) are a direct computation: the bound Eq. (5.207) at order N controls a_n through Cauchy's estimates on C_R, which yields both the radius of convergence of B and the exponential bound, and Eq. (5.208) then follows by the same term-by-term Laplace identity as in Eq. (5.202) together with a remainder estimate. The uniqueness (converse) direction is the substantive part and rests on a Phragmén–Lindelöf argument in the Borel plane; we use it without full proof and record the precise reference [363, 449]. What matters for the book is the *shape* of the statement, which we will invoke verbatim: a factorially-bounded remainder, *uniform on a tangent disk*, is necessary and sufficient for the asymptotic series to reconstruct its function by Borel summation. The corresponding analyticity-in-g statement for honest field theories is exactly what is not known in general (Section 5.7.5).

We now make the entire circle of ideas concrete on the series that every later chapter's perturbative expansion resembles.

Worked Example 5.50 (Borel summation of the prototype factorial series)**.** Consider the divergent series

$$\hat{f}(g) \;=\; \sum_{n=0}^{\infty} (-1)^n\, n!\, g^{n+1} \;=\; g - 1!\, g^2 + 2!\, g^3 - 3!\, g^4 + \cdots, \tag{5.210}$$

the *Euler series*, the cleanest model of an alternating factorially divergent expansion. By the ratio test its radius of convergence is 0: for any $g \neq 0$ the terms eventually grow without bound. We Borel-sum it and show that (a) the procedure reproduces Eq. (5.210) coefficient by coefficient, and (b) it returns a definite finite function, the exponential integral.

Step 1: the Borel transform. It is convenient to strip the overall factor g and Borel-transform the bracket $\hat{E}(g) = \hat{f}(g)/g = \sum_{n\geq 0} (-1)^n n!\, g^n$, whose coefficients are $a_n = (-1)^n n!$. Dividing by $n!$ collapses the factorial entirely,

$$B(t) \;=\; \sum_{n=0}^{\infty} \frac{(-1)^n n!}{n!}\, t^n \;=\; \sum_{n=0}^{\infty} (-t)^n \;=\; \frac{1}{1+t}, \qquad |t| < 1. \tag{5.211}$$

The Borel transform is a single pole at $t = -1$. This location—on the *negative* real axis, off the integration ray—is the reason the Euler series is unambiguously Borel summable, as we now see.

Step 2: the Borel integral. The function Eq. (5.211) continues analytically to all of $\mathbb{C} \setminus \{-1\}$, in particular along the whole positive axis, where it is bounded by 1; the exponential-bound hypothesis of Theorem 5.47 holds trivially with $R = \infty$. Hence for every $g > 0$,

$$\mathcal{B}\hat{E}(g) \;=\; \int_0^{\infty} e^{-t}\, B(gt)\, dt \;=\; \int_0^{\infty} \frac{e^{-t}}{1+gt}\, dt, \tag{5.212}$$

a perfectly finite, positive number for each $g > 0$. The full prototype is $\mathcal{B}\hat{f}(g) = g\, \mathcal{B}\hat{E}(g) = \int_0^{\infty} g\, e^{-t}/(1+gt)\, dt$.

Step 3: identification with the exponential integral. Substitute $u = 1 + gt$, so $t = (u-1)/g$ and $dt = du/g$, with $u : 1 \to \infty$:

$$\int_0^{\infty} \frac{e^{-t}}{1+gt}\, dt = \frac{e^{1/g}}{g} \int_1^{\infty} \frac{e^{-u/g}}{u}\, du = \frac{e^{1/g}}{g} \int_{1/g}^{\infty} \frac{e^{-v}}{v}\, dv = \frac{e^{1/g}}{g}\, E_1\!\left(\tfrac{1}{g}\right), \tag{5.213}$$

where in the middle step $v = u/g$ and $E_1(x) = \int_x^{\infty} e^{-v} v^{-1}\, dv$ is the standard exponential integral. Therefore the Borel sum of the prototype series is the explicit function

$$\mathcal{B}\hat{f}(g) \;=\; g \cdot \frac{e^{1/g}}{g}\, E_1\!\left(\tfrac{1}{g}\right) \;=\; e^{1/g}\, E_1\!\left(\tfrac{1}{g}\right), \tag{5.214}$$

finite and analytic for all $g > 0$. The divergent series has been replaced by a genuine closed-form function.

Step 4: it really is the asymptotic series of Eq. (5.214). Expand the geometric kernel in Eq. (5.212) to finite order with exact remainder,

$$\frac{1}{1+gt} = \sum_{n=0}^{N-1} (-gt)^n + \frac{(-gt)^N}{1+gt}, \tag{5.215}$$

and integrate against e^{-t} using $\int_0^\infty e^{-t} t^n \, dt = n!$:

$$\mathcal{B}\hat{E}(g) = \sum_{n=0}^{N-1} (-1)^n n! \, g^n + R_N(g), \qquad R_N(g) = (-1)^N g^N \int_0^\infty \frac{e^{-t} t^N}{1+gt} \, dt. \tag{5.216}$$

For $g > 0$ and $t > 0$ the denominator satisfies $1 + gt > 1$, so $0 < 1/(1+gt) < 1$ and the remainder is bounded by

$$|R_N(g)| \;\le\; g^N \int_0^\infty e^{-t} t^N \, dt \;=\; N! \, g^N. \tag{5.217}$$

Two conclusions follow. First, the partial sums of Eq. (5.216) *are* the prototype series Eq. (5.210) (after restoring the factor g): Borel summation has reproduced every coefficient. Second, the bound Eq. (5.217) is exactly a Poincaré asymptotic estimate with the tell-tale $N!$ growth, so the explicit function Eq. (5.214) has Eq. (5.210) as its asymptotic series and no more—and, by the uniqueness clause of Theorem 5.48, it is the *only* such function. Finally, the error $N! \, g^N$ is minimized over N near $N_* \sim 1/g$, where Stirling gives $N_*! \, g^{N_*} \sim \sqrt{2\pi/g} \, e^{-1/g}$: optimal truncation (Eq. (5.168)) recovers the function only up to an error of order $e^{-1/g}$, while Borel summation recovers it exactly. The gap between the two is the nonperturbative information that the bare series cannot hold and the Borel integral can.

The Euler example is benign because its Borel singularity lies at $t = -1$, away from the integration ray. The defining feature of *realistic* perturbation theory is that the singularities land on the positive axis, and that is where the analysis becomes physics.

5.7.3 Singularities in the Borel plane and nonperturbative ambiguity

Suppose the coefficients do not alternate—the generic situation when the large order is dominated by a single sign, as for a series whose terms count configurations of fixed sign. The cleanest model is $a_n = n!$, giving the Borel transform

$$B(t) = \sum_{n \ge 0} \frac{n!}{n!} t^n = \sum_{n \ge 0} t^n = \frac{1}{1-t}, \tag{5.218}$$

with a pole now at $t = +1$, squarely *on* the integration ray. The Borel integral $\int_0^\infty e^{-t}/(1-gt)\,dt$ runs straight through the singularity at $t = 1/g$ and is not defined as it stands. Something must be specified to give it meaning, and whatever we specify will leave a residue of arbitrariness. This is the analytic origin of the nonperturbative ambiguity of perturbation theory, and the next worked example computes it exactly. Figure 5.9 contrasts the two situations.

Figure 5.9: Singularity structure in the Borel t-plane. Left: the Euler series of Theorem 5.50 has its only singularity at $t = -1$, off the integration ray $\theta = 0$; the Borel sum is unambiguous. Right: a same-sign factorial series has a singularity at $t_0 = 1/g_0 > 0$ on the ray. The two lateral sums $\mathcal{B}_{0\pm}$ skirt it above and below and differ by the exponentially small, purely imaginary amount computed in Theorem 5.51.

Worked Example 5.51 (A Borel-plane pole and its two-fold ambiguity)**.** Model a single nonperturbative singularity by the Borel transform

$$B(t) = \frac{t_0}{t_0 - t}, \qquad t_0 = \frac{1}{g_0} > 0, \tag{5.219}$$

a simple pole at $t = t_0$ on the positive axis (normalized so $B(0) = 1$). We compute the two lateral Borel sums of Theorem 5.47 and show their difference is purely imaginary and exponentially small in $1/g$.

Step 1: lateral contours. Using the directional form Eq. (5.204), the lateral sums skirt the pole above ($\theta \to 0^+$) and below ($\theta \to 0^-$):

$$\mathcal{B}_{0\pm}(g) = \frac{1}{g}\int_{\mathcal{C}_\pm} e^{-t/g}\,\frac{t_0}{t_0 - t}\,dt, \tag{5.220}$$

where $\mathcal{C}_+$ ($\mathcal{C}_-$) runs from 0 to $+\infty$ passing infinitesimally above (below) t_0. Both integrals exist because the kernel $e^{-t/g}$ enforces convergence at infinity for $g > 0$ and the contour avoids the pole.

Step 2: the difference is a residue. The two contours agree everywhere except in a neighbourhood of t_0. Subtracting, $\mathcal{B}_{0+} - \mathcal{B}_{0-}$ is $1/g$ times the integral over the

closed loop that goes out along $\mathcal{C}_+$ (above) and back along $\mathcal{C}_-$ (below)—a loop encircling t_0 once *clockwise*. By the residue theorem,

$$\mathcal{B}_{0^+}(g) - \mathcal{B}_{0^-}(g) = \frac{1}{g}\,(-2\pi i)\,\mathrm{Res}_{t=t_0}\Big[e^{-t/g}\,\frac{t_0}{t_0 - t}\Big]. \tag{5.221}$$

Near t_0 we have $t_0/(t_0-t) = -t_0/(t-t_0)$, a simple pole of residue $-t_0$; multiplying by the regular factor $e^{-t/g}$ evaluated at $t = t_0$ gives

$$\mathrm{Res}_{t=t_0}\Big[e^{-t/g}\frac{t_0}{t_0 - t}\Big] = -t_0\, e^{-t_0/g}. \tag{5.222}$$

Substituting Eq. (5.222) into Eq. (5.221),

$$\boxed{\mathcal{B}_{0^+}(g) - \mathcal{B}_{0^-}(g) = \frac{2\pi i\, t_0}{g}\; e^{-t_0/g} = \frac{2\pi i}{g_0\, g}\; e^{-1/(g_0 g)}.} \tag{5.223}$$

Step 3: read off the ambiguity. The two prescriptions share the same real part—the Cauchy principal value, well defined by the Sokhotski–Plemelj theorem proved in Theorem 5.10—and differ only by the imaginary term Eq. (5.223). Concretely, writing the common real part as $\mathrm{Re}\;\mathcal{B}_0(g) = \frac{1}{g}\,\mathrm{PV}\!\int_0^\infty e^{-t/g}\,\frac{t_0}{t_0 - t}\,dt$, the lateral sums are

$$\mathcal{B}_{0^\pm}(g) = \mathrm{Re}\;\mathcal{B}_0(g)\;\pm\;\frac{i\pi t_0}{g}\,e^{-t_0/g}, \tag{5.224}$$

the same split into principal value plus $\pm i\pi\,\delta$-contribution that Sokhotski–Plemelj attaches to $1/(t_0 - t \mp i0)$. The ambiguity in the resummed function is the imaginary piece

$$\mathrm{Im}\;\mathcal{B}_{0^\pm}(g) = \pm\frac{\pi t_0}{g}\,e^{-t_0/g} = \pm\frac{\pi}{g_0 g}\,e^{-1/(g_0 g)}, \tag{5.225}$$

smaller than every power of g as $g \to 0^+$ and therefore invisible to the perturbative series, which sees only the real Taylor coefficients. A singularity on the positive Borel axis converts the divergent series into a function defined only up to an exponentially small, purely imaginary nonperturbative ambiguity.

Three features of Theorem 5.51 carry directly into field theory. First, the obstruction is *local* in the Borel plane: it is governed entirely by the nearest singularity to the origin on the positive axis, whose position t_0 sets the size $e^{-t_0/g}$ of the ambiguity. Second, the ambiguity is purely *imaginary* for real coupling, even though the physical quantity (a cross section, an energy level) is real—a paradox resolved only when the ambiguity is cancelled against another imaginary contribution from elsewhere in the nonperturbative expansion. Third, the position t_0 of the singularity is not arbitrary: in honest theories it is fixed by a physical action or by the running of the coupling, which is what we turn to now.

5.7.4 Physical identification: instanton and renormalon singularities

Where do the Borel singularities of a real quantum field theory sit, and what puts them there? The large-order behaviour of perturbation theory—the very $n!$ that Watson's lemma exposed—is controlled by the singularity of $B(t)$ nearest the origin, through the elementary correspondence between a power-law branch point of B and factorial growth of its Taylor coefficients: if $B(t) \sim (1 - t/t_0)^{-\gamma}$ near t_0, then $a_n = n!\,[t^n]B(t) \sim n!\,t_0^{-n}\,n^{\gamma-1}/\Gamma(\gamma)$, so that

$$a_n \;\sim\; \frac{n!}{t_0^{n}}\, n^{\gamma-1} \qquad (n \to \infty). \tag{5.226}$$

Reading Eq. (5.226) backwards: the *rate* of the factorial divergence measured in perturbation theory *is* the location t_0 of the leading Borel singularity. Two distinct mechanisms generate such singularities, and they are physically very different.

Instanton singularities. The first source is semiclassical. In the path-integral evaluation of a perturbative coefficient (ϕ^4 theory and QED, once they are quantized in Part II onward), the $n!$ growth arises because the number of Feynman diagrams at order n grows factorially, and Lipatov's method [324] shows that this combinatorial growth is captured by a nontrivial saddle point of the action—an *instanton*, a finite-action solution of the Euclidean equations of motion. Such a saddle, of action S_{inst}, places a Borel singularity at

$$t_0 \;=\; S_{\mathrm{inst}} \tag{5.227}$$

in the conventions where the coupling sits as $e^{-S/g}$ in the Borel kernel, so that the associated nonperturbative ambiguity $e^{-t_0/g} = e^{-S_{\mathrm{inst}}/g}$ of Theorem 5.51 is exactly the weight of one instanton. The factorial divergence of perturbation theory and the existence of instantons are thus two faces of one analytic fact. We construct instantons explicitly, compute S_{inst} for Yang–Mills (Yang–Mills theory), and exhibit the θ-vacuum in Chapter 42; the present section establishes *why* they must appear as Borel singularities before they are constructed.

Renormalon singularities. The second source has no semiclassical avatar and was identified by 't Hooft [465]. Even in a theory or observable with no relevant instanton, factorial growth arises from a single chain of diagrams—a string of bubble insertions on a propagator—once one integrates the *running* coupling over loop momenta. A subintegral of the schematic form $\int dk^2\,\alpha(k^2)\,(k^2/Q^2)^p$ with $\alpha(k^2) \propto 1/\ln(k^2/\Lambda^2)$ produces, upon expansion in the coupling at the external scale, coefficients growing like $n!\,(b_0/p)^{-n}$, hence a Borel singularity at $t_0 \propto p/b_0$ set by the one-loop coefficient b_0 of the beta function rather than by

any action. These are the *renormalons*. The region of small loop momentum produces *infrared renormalons*, which sit on the positive Borel axis and therefore generate genuine ambiguities of the kind computed above; the region of large loop momentum produces *ultraviolet renormalons* on the negative axis, which do not. The crucial physical statement, which we can only flag here and develop with the operator product expansion and the running coupling later, is that the infrared-renormalon ambiguity is not a defect of the theory: it is exactly cancelled by the ambiguity in defining the nonperturbative power corrections (vacuum condensates) of the operator product expansion, so that the sum of perturbative and power-suppressed pieces is unambiguous. The renormalon and its cancellation belong to the renormalization and running-coupling development; we forward-reference the running coupling and Λ_{QCD} of Chapter 27 and the Callan–Symanzik machinery of Chapter 24, where b_0 is computed, and the modern review [60].

Caution

Borel summability is not a property a quantum field theory enjoys by default. The benign Euler series of Theorem 5.50 is Borel summable precisely because its singularity lies off the positive axis; the same-sign factorial series of Eq. (5.218), the infrared renormalon, and the instanton–anti-instanton sector all put singularities *on* the positive axis and are therefore Borel summable only after a lateral prescription is chosen, at the cost of the ambiguity Eq. (5.225). Stating that "the QFT perturbation series is Borel summable" without specifying the direction and without exhibiting the cancellation of the resulting ambiguity is, in general, a false statement of closure. What is rigorously established (Borel summability of specific super-renormalizable models and of quantum-mechanical examples) and what is conjectural (full four-dimensional gauge theory) must be kept apart; see Section 5.7.5.

5.7.5 Resurgence and transseries: the mid-2020s frontier

The picture that has emerged is that perturbation theory is one sector of a larger object. The imaginary ambiguity Eq. (5.225) of the lateral Borel sum cannot be the final word, because physical answers are real and unambiguous; the resolution is that the perturbative series is accompanied by *additional* sectors—one-instanton, two-instanton, and so on—each itself an asymptotic series with its own Borel ambiguity, and these ambiguities cancel among the sectors. The bookkeeping device that makes this precise is the *transseries*: instead of a single power series one writes

$$f(g) \;\sim\; \sum_{k\geq 0} e^{-k\,S/g}\, g^{-k\beta_k} \sum_{n\geq 0} a_n^{(k)}\, g^n, \tag{5.228}$$

a double expansion in powers of g *and* in the nonperturbative exponentials $e^{-S/g}$, with $k = 0$ the ordinary perturbative series. The content of *resurgence*, the theory founded by Écalle [169], is that these sectors are not independent: the late-order coefficients $a_n^{(0)}$ of the perturbative series *re-surge*, encoding through Eq. (5.226) the data of the very instanton sectors $k \geq 1$ whose ambiguities cancel theirs. The lateral Borel ambiguity of the $k = 0$ sector, $\pm i\pi$ times an $e^{-S/g}$-weighted residue, is cancelled by the explicitly imaginary ambiguity of the $k = 1$ sector, leaving a real, prescription-independent *median* resummation. Écalle's alien calculus and the associated bridge equations make the cancellation an exact and computable statement.

This is the correct modern framework for everything this chapter has been building toward, and it is also, honestly stated, an active research frontier rather than a closed theory. As of the mid-2020s, resurgence is established rigorously and used quantitatively in quantum mechanics (the double well and periodic potentials), matrix models, topological string theory, and a growing list of integrable and two-dimensional field theories; in those settings the transseries Eq. (5.228) is constructed and the cancellation of ambiguities is demonstrated. For full four-dimensional gauge theory the program is *conjectural and incomplete*: the relevant Borel singularities (instanton and renormalon) are identified, the structure Eq. (5.228) is expected, but a constructive proof that the trans-series reproduces the theory, with all ambiguities cancelling, does not exist. We will meet the first concrete ingredients—explicit instantons and their action—in Chapter 42, and the renormalon side in the renormalization and running-coupling chapters; the synthesis into a complete resurgent description of QCD remains open. The honest register here matches the one set at the start of the chapter: the analytic objects are exact and in hand, while their full assembly into a nonperturbative definition of the gauge theories is a problem of the current era [25, 159].

Historical Note

The taming of divergent series has a long arc. Euler summed $\sum(-1)^n n!\, g^{n+1}$ to the exponential integral by exactly the manipulation of Theorem 5.50 a century before it was justified; Borel's $n!$-dividing transform and its Laplace inverse date to the 1890s, and Hardy's *Divergent Series* [250] codified the classical theory. F. Nevanlinna's 1919 uniqueness theorem [363] lay essentially unused by physicists until Sokal [449] sharpened it to the tangent-disk form of Theorem 5.48 and made it the working criterion. Dyson's 1952 argument [168] that QED perturbation theory must diverge (Section 5.6.5), Lipatov's 1977 instanton calculus of large orders [324], and 't Hooft's 1979 identification of renormalons [465] turned the abstract theory into physics. Écalle's *Les fonctions résurgentes*

[169] supplied the language that, four decades on, organizes the subject.

With Borel summation in place, the analytic toolbox of Part I is complete. Every propagator the next part writes down is a boundary value fixed by the $i\epsilon$ prescription and the Sokhotski–Plemelj theorem of Section 5.1.5; every Euclidean computation is licensed by the Wick rotation; every dispersive and unitarity argument rests on the analyticity in external invariants of Section 5.5; and every perturbative series those chapters generate is now understood as an asymptotic expansion to be read, at best, through Borel resummation, with its instanton and renormalon singularities already named. We are ready to leave the foundations and begin classical field theory in Part II, where the threads ϕ^4, QED, and Yang–Mills are at last defined and the analytic machinery stocked here is put to work.

Further reading

The functional-analytic theory of distributions is developed in Reed and Simon [416] and, with the physicist's emphasis on generalized functions and their transforms, in Gel'fand and Shilov [214]; the definitive modern treatment, including the wavefront-set technology behind the products of distributions deferred to renormalization, is Hörmander [270]. The axiomatic role of tempered distributions and analyticity in quantum field theory proper is the subject of Streater and Wightman [458]. The analytic-S-matrix program – causality, analyticity in invariants, and dispersion relations as physical principles – is laid out in Eden et al. [170]. For the asymptotic and divergent-series machinery, Bender and Orszag [59] is the standard working reference for Watson's lemma and steepest descent, while the divergence of QFT perturbation theory traces to Dyson [168]; the large-order behaviour, Borel summation, and the renormalon story are collected in Le Guillou and Zinn-Justin [318] and developed in depth in Zinn-Justin [534]. The use of all of these tools in context appears throughout Weinberg [500], Peskin and Schroeder [396], and Schwartz [432].

Part II

Classical Field Theory

Chapter 6

Lagrangian and Hamiltonian Field Theory

Part I built the stage. In Chapter 3 we asked what a relativistic field is even *allowed* to be, and the answer came from representation theory: a field must transform in some finite-dimensional representation of the Lorentz group, and the classification of those representations (predicted Theorem 3.19, Section 3.3) handed us the entire kinematic vocabulary of the subject—scalars, Weyl and Dirac spinors, vectors—before a single equation of motion was written. In Chapter 4 we built the analytic machinery for objects that depend on a continuum of variables: the functional derivative, the action functional, the fundamental lemma of the calculus of variations, and the Legendre transform (predicted Section 4.1.2, Section 4.5.1). What Part I conspicuously did not supply was *motion*. We know the alphabet; we have not yet written a sentence. A scalar field at rest and a scalar field in violent oscillation are the same kinematic object; nothing said so far distinguishes them, because nothing so far is dynamical.

This chapter supplies the dynamics, and it does so from a single organizing principle. Just as the entire mechanics of a finite collection of particles flows from Hamilton's principle—trajectories are the ones that make an action stationary—so the entire classical dynamics of fields flows from *stationary action* applied to a system with a continuum of degrees of freedom, one (or a few) at every point of spacetime. From that one postulate we will extract the equations of motion, the conserved energy and momentum, the canonical bracket structure, and the obstructions (constraints) that appear whenever the action is built with redundancy. Everything in this chapter is purely classical. The fields here are c-number functions on spacetime (with a flagged exception for the anticommuting Dirac field, whose Grassmann nature we note but do not yet exploit); there are no operators, no Hilbert space, no $\hbar$ doing any work. This is deliberate. The cleanest route to understanding quantization, which occupies Part III and beyond, is to first hold in hand—completely, with no loose ends—the classical

theory that quantization acts upon.

In the arc set out in Chapter 1, quantum field theory is the reconciliation of special relativity with quantum mechanics: the framework in which a fixed, finite number of particles is no longer a consistent assumption and fields become the fundamental dynamical entities. That reconciliation has two halves. One half is relativistic and classical—a Lorentz-invariant dynamics for fields—and the other is quantum. This chapter completes the first half. We build relativistic classical field theory to a finish, so that when Part III promotes Poisson brackets to commutators there is a definite, fully understood structure being promoted, and the only genuinely new ingredient is the quantum one.

The chapter also introduces the book's standing cast. Three classical field theories recur from here to the research frontier, and all three make their first appearance in these pages with their Lagrangians, equations of motion, Hamiltonians, and stress tensors derived once and imported everywhere after. The real scalar field becomes ϕ^4 theory the instant we add the quartic self-interaction $-\frac{\lambda}{4!}\phi^4$, and we carry its full classical layer in this chapter—action, the equation of motion $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$, Hamiltonian, Poisson brackets, and stress tensor. The Dirac field, the protagonist of QED, appears with its first-order Lagrangian and the Dirac equation. The Maxwell field appears with $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, the inhomogeneous equations $\partial_\mu F^{\mu\nu} = J^\nu$, and the gauge redundancy that will dominate so much of the later book. These are not toy illustrations; they are the actual theories the rest of the book quantizes, renormalizes, and pushes to the frontier.

Conventions are those fixed for the whole book and collected in Chapter A: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, and the Fourier and index conventions stated there; the gamma-matrix algebra and the $\bar{\psi} = \psi^\dagger\gamma^0$ convention used for the Dirac field live in Chapter C. The forward map is equally definite. The chapter ends by extracting, from invariance of the action under spacetime translations, the canonical stress–energy tensor and the conserved four-momentum; Chapter 7 generalizes that single computation into the full Noether correspondence between continuous symmetries and conserved currents, and repairs the canonical tensor into the symmetric, gauge-invariant Belinfante form. The canonical momenta and equal-time Poisson brackets assembled here are, object for object, exactly what Chapter 9 promotes to canonical commutators when the scalar field is quantized. Read this way, the chapter is the hinge on which the book turns from kinematics to dynamics, and from classical to quantum.

6.1 From Particles to Fields: The Action Functional

Part I built a stage and left it empty. Chapter 3 classified what a relativistic field is *allowed* to be — the catalogue of Lorentz/Poincaré representations: scalar, Weyl and Dirac spinor, vector — and Chapter 4 assembled the variational

calculus, the notion of a functional and its derivative, that we will need to put those objects in motion. What neither chapter supplied is dynamics: a rule that selects, out of all conceivable histories of a field, the ones nature actually realizes. This chapter supplies that rule, and it does so with a single organizing principle, stationary action, applied to systems carrying a continuum of degrees of freedom. The present section builds the object that the principle acts on. We first show, by taking a concrete mechanical system — a lattice of masses joined by springs — to its continuum limit, that a field is nothing more exotic than a mechanical system with one degree of freedom at every point of space, and that the energetics of such a system organize themselves into a *Lagrangian density*. We then abstract from the example to the general definition, fix the three structural requirements that every Lagrangian density in this book obeys — locality, Lorentz invariance, and dependence on at most first derivatives — and justify each one rather than decree it. The section closes by putting the book's first protagonist on stage: the free real scalar field, whose Lagrangian is the seed of ϕ^4 theory.

Everything here is purely classical. The fields are c-number functions on spacetime; there is no Hilbert space, no operator, no $\hbar$. This is deliberate. The cleanest route to understanding quantization (Part III) is to hold the classical theory it quantizes completely in hand first, and the classical theory is interesting and structured in its own right.

6.1.1 A mechanical system with infinitely many degrees of freedom

The conceptual leap from particle mechanics to field theory is the leap from finitely many degrees of freedom to a continuum of them. The honest way to take that leap is to watch it happen in an example where every step is ordinary mechanics. We therefore start not with a field but with a long chain of identical masses connected by springs, write down its Lagrangian as taught in any course on classical mechanics, and ask what becomes of that Lagrangian when the masses are packed infinitely densely. A field, and the density whose integral is its action, will fall out.

Consider N point masses m arranged along a line at equilibrium spacing a, the n-th mass tethered to its neighbours by identical springs of stiffness κ. Let $q_n(t)$ be the longitudinal displacement of the n-th mass from its equilibrium position $x_n = na$ (Fig. 6.1). The kinetic energy is the usual sum $\sum_n \frac{1}{2} m \dot{q}_n^2$. The potential energy is stored in the springs; the spring between masses n and $n+1$ is stretched by $q_{n+1} - q_n$, so it holds energy $\frac{1}{2}\kappa\,(q_{n+1} - q_n)^2$. The Lagrangian of the chain is therefore

$$L \;=\; \sum_n \Big[\tfrac{1}{2}\, m\, \dot{q}_n^2 \;-\; \tfrac{1}{2}\, \kappa\, (q_{n+1} - q_n)^2 \Big]. \tag{6.1}$$

This is a system of N coupled oscillators: N degrees of freedom, one displacement

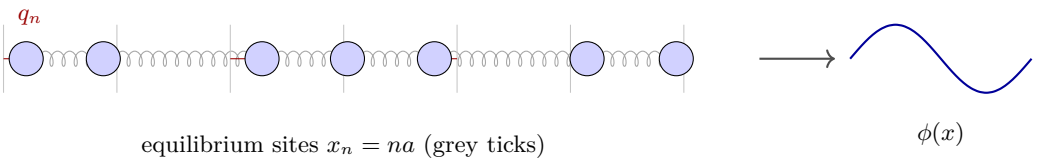


Figure 6.1: From a chain to a field. A line of masses m at equilibrium spacing a (grey ticks), each displaced by q_n (red) and coupled to its neighbours by springs of stiffness κ. As $a \to 0$ at fixed mass density $\mu = m/a$ and fixed elastic modulus $Y = \kappa a$, the discrete displacements $q_n(t)$ become a smooth field $\phi(x,t)$, the finite difference $q_{n+1} - q_n$ becomes $a\,\partial_x\phi$, and the sum over sites becomes an integral. The chain's N degrees of freedom become one degree of freedom at each point of space.

per site. Nothing about Eq. (6.1) is unfamiliar; it is the standard model problem whose normal modes are lattice phonons.

We now take the continuum limit. Hold fixed two macroscopic quantities as $a \to 0$ and $N \to \infty$ with $L_{\text{tot}} = Na$ fixed: the linear *mass density* $\mu \equiv m/a$ and the elastic *modulus* $Y \equiv \kappa a$ (the one-dimensional analogue of Young's modulus). Both are intensive: they describe the material, not the discretization. Introduce a smooth field $\phi(x,t)$ that interpolates the displacements, $\phi(x_n,t) = q_n(t)$. For neighbouring sites Taylor expansion gives

$$q_{n+1} - q_n \;=\; \phi(x_n + a, t) - \phi(x_n, t) \;=\; a\,\frac{\partial\phi}{\partial x}\Big|_{x_n} + O(a^2), \tag{6.2}$$

so the finite difference is $a\,\partial_x\phi$ to leading order. Rewrite Eq. (6.1) by inserting one factor of a per term and grouping it with the sum, which is what turns $\sum_n a\,(\cdots)$ into a Riemann sum for $\int dx\,(\cdots)$:

$$L \;=\; \sum_n a\left[\tfrac{1}{2}\,\frac{m}{a}\,\dot{q}_n^2 \;-\; \tfrac{1}{2}\,\kappa a\left(\frac{q_{n+1}-q_n}{a}\right)^2\right] \;=\; \sum_n a\left[\tfrac{1}{2}\,\mu\,\dot{\phi}^2 \;-\; \tfrac{1}{2}\,Y\,(\partial_x\phi)^2\right] + O(a), \tag{6.3}$$

where in the second equality we used $m/a = \mu$, $\kappa a = Y$, the interpolation $\dot{q}_n = \dot{\phi}(x_n)$, and Eq. (6.2). The error term collects the $O(a^2)$ remainder of the Taylor expansion, which carries an extra power of a relative to the kept terms and vanishes in the limit. Taking $a \to 0$, the Riemann sum becomes an integral and the Lagrangian becomes a *spatial integral of a density*:

$$L \;=\; \int dx\,\mathcal{L}, \qquad \mathcal{L}(x,t) \;=\; \tfrac{1}{2}\,\mu\,\dot{\phi}^2 \;-\; \tfrac{1}{2}\,Y\,(\partial_x\phi)^2. \tag{6.4}$$

This $\mathcal{L}$ is the *Lagrangian density*, and it is the central object of the rest of this chapter. Three of its features were forced on us by the derivation and are worth naming, because they survive the abstraction to come.

First, the appearance of a degree of freedom at every point. The label n that indexed the masses has become the continuous label x; the configuration of the system at one instant is no longer a list of N numbers but the entire function $x \mapsto \phi(x,t)$. A field is, precisely, a mechanical system whose degrees of freedom are indexed by the points of space — uncountably many of them, one real number per point. This is the sense in which field theory *is* mechanics with infinitely many degrees of freedom, and it is the reason the variational principle of Chapter 4 will transcribe to fields with only cosmetic change.

Second, the appearance of $(\partial_x\phi)^2$. The gradient term is not put in by hand; it is the continuum residue of the spring coupling $\frac{1}{2}\kappa(q_{n+1}-q_n)^2$ between neighbours. Because the potential energy of a spring depends only on the *difference* of adjacent displacements — a uniform displacement $q_n \to q_n + \text{const}$ stretches no spring and costs no energy — the leading continuum cost of a nonuniform configuration is quadratic in the first spatial derivative. The squared gradient is the field-theoretic statement that it costs energy to bend the field but not to slide it rigidly. This term is the origin of every $(\partial\phi)^2$ in the book.

Third, the kinetic term $\frac{1}{2}\mu\dot{\phi}^2$ is the obvious continuum sum of the site kinetic energies $\frac{1}{2}m\dot{q}_n^2$, with m/a collected into the density μ. Kinetic energy depends on the *time* derivative of the field, potential energy on its *space* derivatives; the Lagrangian density is their difference, exactly as $L = T - V$ in particle mechanics.

The relativistic idealization

The density Eq. (6.4) is not yet Lorentz invariant, and it should not be: a lattice of masses is a Galilean mechanical system with a preferred rest frame, the frame of the material. To expose its near-relativistic structure, factor out the density,

$$\mathcal{L} = \tfrac{1}{2}\,\mu\Big[\dot{\phi}^2 - v^2(\partial_x\phi)^2\Big], \qquad v^2 \equiv \frac{Y}{\mu} = \frac{\kappa a^2}{m}. \tag{6.5}$$

The combination v has dimensions of speed; a plane wave $\phi \sim e^{i(kx-\omega t)}$ inserted into the Euler–Lagrange equation of Eq. (6.5) (the equation of motion derived in general in the next section) obeys $\omega^2 = v^2k^2$, so v is the speed at which long-wavelength disturbances propagate down the chain — the speed of sound in this elastic medium. The density Eq. (6.5) is invariant under the boosts of an "acoustic" Lorentz group built with v in place of c.

A genuinely relativistic field is the idealization in which the propagation speed of the medium is the universal speed of light, $v = c$, which in natural units ($\hbar = c = 1$, Chapter A) means $v = 1$. Setting $v = 1$ and absorbing the constant μ by the canonical rescaling $\phi \to \phi/\sqrt{\mu}$ (a field redefinition that leaves the dynamics unchanged, since L is quadratic and homogeneous in ϕ) collapses Eq. (6.5) to

$$\mathcal{L} = \tfrac{1}{2}\,\dot{\phi}^2 - \tfrac{1}{2}\,(\partial_x\phi)^2. \tag{6.6}$$

In one space and one time dimension, with the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-)$ of Chapter A, the two terms are exactly the two pieces of a Lorentz scalar:

$$\partial_\mu\phi\,\partial^\mu\phi = (\partial_0\phi)^2 - (\partial_1\phi)^2 = \dot\phi^2 - (\partial_x\phi)^2, \quad \text{so} \quad \mathcal{L} = \tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi. \tag{6.7}$$

The same construction in three space dimensions — a cubic lattice of masses with springs along each axis — produces, by the identical Taylor-and-collect argument applied independently to each direction, the kinetic term $\frac{1}{2}\dot\phi^2$ and a gradient term $\frac{1}{2}\sum_{i=1}^{3}(\partial_i\phi)^2 = \frac{1}{2}(\nabla\phi)^2$, whose difference is again $\frac{1}{2}\partial_\mu\phi\,\partial^\mu\phi$ with $\mu = 0,1,2,3$. We have manufactured, from balls and springs, the kinetic part of the free scalar field theory.

Physical Insight

The squared gradient is the price of inhomogeneity, and the relativistic theory is the one in which that price is the same in every frame. The lattice "knows" a rest frame — the frame in which the masses sit still — and its sound speed v is generically far below c. Promoting $v \to c$ is not a cosmetic relabeling but a genuine physical postulate: it asserts that the field's excitations are causal at the universal speed, i.e. that the field is a relativistic, not a material, degree of freedom. This is why Lorentz invariance must be *imposed* as a requirement on field Lagrangians (Section 6.1.4) rather than discovered — the mechanical model that motivates the structure does not, on its own, supply it.

The mass term

The free scalar field carries one more term, the mass term, and the mechanical model produces it too once we add one ingredient. Suppose that in addition to the springs between neighbours, each mass is tethered to its own equilibrium point by an on-site spring of stiffness κ' — physically, a restoring force that resists *absolute* displacement, not just relative displacement. This adds $-\sum_n \frac{1}{2}\kappa' q_n^2$ to the potential energy. Collecting a factor of a as before, $\sum_n \frac{1}{2}\kappa' q_n^2 = \sum_n a\,\frac{1}{2}(\kappa'/a)\phi^2 \to \int dx\,\frac{1}{2}(\kappa'/a)\phi^2$, so after the same rescaling $\phi \to \phi/\sqrt{\mu}$ and $v = 1$ the density acquires a term $-\frac{1}{2}m^2\phi^2$ with

$$m^2 \equiv \frac{\kappa'}{\mu\, a}. \tag{6.8}$$

The dispersion relation now reads $\omega^2 = k^2 + m^2$: a plane wave of wavenumber k oscillates at frequency $\omega = \sqrt{k^2+m^2}$, so even the longest-wavelength mode $k \to 0$ has a nonzero frequency $\omega \to m$. The on-site spring opens a *gap* in the spectrum, and m is the size of that gap. This is the first appearance of the deepest single fact about the mass term: in a relativistic field theory the mass is

the energy of the field's lowest excitation, the gap below which no propagating mode exists, and equivalently the inverse of the length scale $1/m$ over which static disturbances of the field decay. We will meet $1/m$ again as the Compton wavelength, as the range of a force, and as the correlation length of the quantum theory; all are the same length, first seen here as the reciprocal of an on-site spring constant.

6.1.2 What a classical field is

The example has done its work. We now state, once and cleanly, what a classical field is in this book, importing the representation-theoretic content from Chapter 3 rather than rebuilding it.

Definition 6.1 (Classical field)**.** Let M be Minkowski spacetime with coordinates $x^\mu = (x^0, \mathbf{x})$ and metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ (Chapter A), and let (R, V) be a finite-dimensional representation of the Lorentz group $\mathrm{O}(1,3)$ (or of its Poincaré extension), as classified in Section 3.3. A *classical field of type* R is a smooth assignment

$$x \longmapsto \phi_a(x) \in V, \qquad a = 1, \dots, \dim V, \tag{6.9}$$

of an element of the representation space V to each spacetime point, transforming under a Lorentz transformation $x' = \Lambda x$ as

$$\phi'_a(x') \;=\; R(\Lambda)_a{}^b \, \phi_b(x). \tag{6.10}$$

The index a runs over the components of V; for a multi-field theory the collection $\{\phi_a\}$ may assemble several irreducible representations, and a then runs over all of them.

The cases that recur throughout the book are exactly those catalogued in Section 3.3.3, and we name them here only to fix the notation; their transformation laws and the proof that the list is complete belong to Chapter 3.

A *scalar field* $\phi(x)$ carries the trivial representation, $R(\Lambda) = 1$, so Eq. (6.10) reads $\phi'(x') = \phi(x)$: the value of the field is unchanged, merely re-evaluated at the transformed point. This is the simplest field and the protagonist of this chapter.

A *Dirac spinor field* $\psi_a(x)$, $a = 1, \dots, 4$, carries the representation $(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})$, with $R(\Lambda) = S(\Lambda)$ the spinor representation built from the gamma matrices of Chapter C; it describes the electron of QED.

A *vector field* $A^\mu(x)$ carries the $(\tfrac{1}{2}, \tfrac{1}{2})$ representation, transforming as $A'^\mu(x') = \Lambda^\mu{}_\nu A^\nu(x)$; it describes the photon of QED and, in its non-abelian generalization, the gluon of Yang–Mills theory.

These are introduced as the recurring cast of the whole book. Their classical Lagrangians, equations of motion, Hamiltonians, and stress tensors are derived once, in this chapter, and imported everywhere afterward. For the remainder of this section the scalar suffices.

6.1.3 The action functional

In particle mechanics the dynamics of $q(t)$ is encoded in the action $S = \int dt\, L(q, \dot{q}, t)$, a number assigned to each trajectory, made stationary by the physical path (Chapter 4). For a field the trajectory is replaced by a field configuration $\phi_a(x)$ over a region of spacetime, and the time integral by an integral over that region. We have just seen, in Eq. (6.4), that the natural field Lagrangian is itself a spatial integral of a density; combining the two integrals, the action of a field theory is a single integral over spacetime of a Lagrangian *density*.

Definition 6.2 (Lagrangian density and action)**.** Let $\{\phi_a\}$ be a collection of classical fields. A *Lagrangian density* is a function

$$\mathcal{L}(\phi_a(x),\, \partial_\mu \phi_a(x),\, x) \tag{6.11}$$

of the fields, their first spacetime derivatives, and (at most) the point x explicitly. The *action* of the field configuration over a spacetime region $\Omega \subseteq M$ is the functional

$$S[\phi_a] \;=\; \int_\Omega d^4x\; \mathcal{L}(\phi_a,\, \partial_\mu \phi_a,\, x), \qquad d^4x \equiv dx^0\, dx^1\, dx^2\, dx^3, \tag{6.12}$$

where the notion of a functional and its variation are those of Section 4.1.1. When Ω is unspecified it is understood to be all of M, with the fields assumed to fall off fast enough at infinity for the integral to converge.

The functional $S[\phi_a]$ assigns a real number to each field configuration; the dynamics, derived in the next section, is the statement that the physically realized configurations are those at which S is stationary against all variations vanishing on the boundary of Ω. Before we can vary it, however, we must pin down the form of $\mathcal{L}$. Theorem 6.2 already encodes two strong assumptions — that $\mathcal{L}$ depends on the fields at a single point and on no derivative higher than the first — which we have so far merely read off the balls-and-springs example. We now elevate them to principles and justify them.

6.1.4 The three structural requirements

Not every functional of ϕ_a is an admissible action. Three requirements constrain the Lagrangian density of every theory in this book. They are not arbitrary aesthetics; each is the formal expression of a physical demand, and we give the demand together with the constraint it imposes.

Locality

> *The Lagrangian density at x depends on the fields and their derivatives evaluated at the single point x, not on their values elsewhere.*

This is the assumption that produced the *density* in Eq. (6.4): each term in the chain's Lagrangian involved a single site or a pair of infinitesimally separated sites, and in the continuum that became a function of ϕ and $\partial_\mu\phi$ at one point. A nonlocal alternative — a Lagrangian containing, say, $\int d^4y\,\phi(x)\,K(x-y)\,\phi(y)$ with K of finite range — is mathematically conceivable, but it would let the field at x feel the field at a finite distance instantaneously across the term, in tension with the causal structure of relativity: influences should propagate through the field, point to neighbouring point, not jump across gaps. Locality is the demand that all interaction be mediated by the field itself, contact-like at each point, with finite-range effects arising only *dynamically*, by the field carrying influence from point to point at speeds bounded by c. Every fundamental Lagrangian we write is local in this sense; apparent nonlocality (as in an effective theory where a heavy field has been integrated out) is always the shadow of an underlying local description, expandable in local operators with more and more derivatives. We will return to this point when we discuss effective field theory; for now locality is a standing assumption, built into Theorem 6.2 by allowing only pointwise dependence.

Lorentz invariance

> *The action is a Lorentz scalar: S takes the same value in every inertial frame. Because the spacetime measure d^4x is itself invariant, this holds if and only if the Lagrangian density $\mathcal{L}$ is a Lorentz scalar.*

The physical content is the principle of relativity: the laws of motion, which are the stationarity conditions of S, must be the same in every inertial frame, and the cleanest way to guarantee this is to make the object they descend from, S, frame-independent. The requirement splits into a statement about the measure and a statement about the integrand, and the first half is a short computation worth doing once.

Proposition 6.3 (The spacetime measure is Lorentz invariant)**.** *Under a Lorentz transformation $x'^\mu = \Lambda^\mu{}_\nu x^\nu$ with $\Lambda \in \mathrm{O}(1,3)$, the four-volume element is unchanged: $d^4x' = d^4x$.*

Proof. A change of variables in a multiple integral rescales the volume element by the absolute value of the Jacobian determinant,

$$d^4x' = \left|\det\Big(\frac{\partial x'^\mu}{\partial x^\nu}\Big)\right| d^4x = |\det\Lambda|\; d^4x, \tag{6.13}$$

since $\partial x'^\mu/\partial x^\nu = \Lambda^\mu{}_\nu$ for a linear transformation. It remains to show $|\det\Lambda| = 1$. By definition a Lorentz transformation preserves the metric, $\Lambda^\mu{}_\rho\,\eta_{\mu\nu}\,\Lambda^\nu{}_\sigma = \eta_{\rho\sigma}$,

which in matrix form is $\Lambda^{\mathsf{T}}\eta\,\Lambda = \eta$. Taking determinants of both sides and using $\det(ABC) = \det A \det B \det C$ and $\det \Lambda^{\mathsf{T}} = \det \Lambda$,

$$(\det \Lambda)^2 \, \det \eta = \det \eta. \tag{6.14}$$

Since $\det \eta = -1 \neq 0$, it cancels, leaving $(\det \Lambda)^2 = 1$, hence $\det \Lambda = \pm 1$ and $|\det \Lambda| = 1$. Substituting into Eq. (6.13) gives $d^4x' = d^4x$. □

The two signs $\det \Lambda = \pm 1$ distinguish the proper Lorentz transformations ($\det \Lambda = +1$, continuously connected to the identity — rotations and boosts) from the improper ones ($\det \Lambda = -1$, involving a reflection such as parity or time reversal), a distinction analyzed in Chapter 3. For the invariance of the measure only the absolute value matters, and both branches give $|\det \Lambda| = 1$. Translations, the remaining Poincaré generators, shift $x^\mu \to x^\mu + a^\mu$ with unit Jacobian trivially, so d^4x is invariant under the full Poincaré group.

Proposition 6.4 (Invariance of the action)**.** *With d^4x Lorentz invariant by Theorem 6.3, the action $S = \int d^4x\, \mathcal{L}$ is Lorentz invariant if and only if the Lagrangian density transforms as a Lorentz scalar, $\mathcal{L}'(x') = \mathcal{L}(x)$.*

Proof. Under $x' = \Lambda x$ the transformed action is $S' = \int d^4x'\, \mathcal{L}'(x')$. Changing the integration variable back to x via Theorem 6.3, $d^4x' = d^4x$, gives $S' = \int d^4x\, \mathcal{L}'(x'(x))$. This equals $S = \int d^4x\, \mathcal{L}(x)$ for every region and every configuration precisely when the integrands agree, $\mathcal{L}'(x') = \mathcal{L}(x)$, which is the defining transformation law of a Lorentz scalar. Conversely, if $\mathcal{L}$ is a scalar the two integrals are equal term by term, so $S' = S$. □

In practice Theorem 6.4 is a constructive recipe: to build a Lorentz-invariant action one assembles $\mathcal{L}$ out of Lorentz scalars formed from the fields and their derivatives. For a scalar field the available scalars are ϕ, $\partial_\mu\phi\,\partial^\mu\phi$, ϕ^2, ϕ^4, and so on — any expression in which every Lorentz index is contracted. For a Dirac field the invariants are bilinears such as $\bar\psi\psi$ and $\bar\psi\, i\partial\!\!\!/\psi$ (built with the gamma matrices of Chapter C); for a vector field, contractions such as $F_{\mu\nu}F^{\mu\nu}$. The transformation law Eq. (6.10) guarantees that any fully contracted polynomial in the fields and their derivatives is a scalar, hence an admissible contribution to $\mathcal{L}$. This is how every Lagrangian in the book is written down: list the fields, list the scalars one can build from them up to the required dimension, and take a linear combination.

Dependence on at most first derivatives

> *The Lagrangian density depends on the fields and their* first *derivatives only — not on $\partial_\mu\partial_\nu\phi$ or anything higher.*

This is the requirement that the example handed us for free — the spring energy produced $(\partial_x\phi)^2$, never $(\partial_x^2\phi)^2$ — but unlike locality and Lorentz invariance it is not a matter of taste or even of relativity. It is forced by the demand that the theory have a stable ground state and a well-posed initial-value problem. A Lagrangian with first derivatives only yields second-order equations of motion (as we derive in the next section), whose solution is fixed by two pieces of initial data per degree of freedom, ϕ and $\dot\phi$ — exactly the Cauchy data of mechanics. A Lagrangian containing genuine higher derivatives yields higher-order equations needing more initial data, and the price of those extra data is catastrophic: the Hamiltonian is unbounded below. This is the content of Ostrogradsky's theorem, which we now state and demonstrate in its simplest nontrivial instance, because the instability is the honest reason for the first-derivative rule and deserves to be seen, not merely asserted.

Worked Example 6.5 (The Ostrogradsky instability)**.** We work in ordinary mechanics, one degree of freedom; the field case is the same statement at every point of space. Consider a Lagrangian that depends on the acceleration, $L(q,\dot q,\ddot q)$, and is *nondegenerate*, meaning $\partial^2 L/\partial\ddot q^2 \neq 0$ so that $\ddot q$ genuinely appears. The Euler–Lagrange equation generalizes to

$$\frac{\partial L}{\partial q} - \frac{d}{dt}\frac{\partial L}{\partial \dot q} + \frac{d^2}{dt^2}\frac{\partial L}{\partial \ddot q} = 0, \tag{6.15}$$

a *fourth*-order differential equation: its solution requires four initial data $(q,\dot q,\ddot q,\dddot q)$, twice the usual two. Ostrogradsky's prescription handles such a system canonically by treating q and $\dot q$ as two independent coordinates,

$$Q_1 \equiv q, \qquad Q_2 \equiv \dot q, \tag{6.16}$$

with two conjugate momenta dictated by the higher-derivative Legendre transform,

$$P_1 \equiv \frac{\partial L}{\partial \dot q} - \frac{d}{dt}\frac{\partial L}{\partial \ddot q}, \qquad P_2 \equiv \frac{\partial L}{\partial \ddot q}. \tag{6.17}$$

Nondegeneracy, $\partial^2 L/\partial\ddot q^2 \neq 0$, is exactly the condition that the second relation can be inverted to express the acceleration as a function $\ddot q = \mathcal{A}(Q_1,Q_2,P_2)$ of the phase-space variables. The Hamiltonian is the higher-derivative Legendre transform,

$$H = P_1\dot q + P_2\ddot q - L = P_1\,Q_2 + P_2\,\mathcal{A}(Q_1,Q_2,P_2) - L(Q_1,Q_2,\mathcal{A}). \tag{6.18}$$

The decisive feature is the very first term: H is *linear* in the momentum P_1, which appears nowhere else in Eq. (6.18) (the remaining terms depend only on Q_1,Q_2,P_2). A function linear in P_1 with a nonzero coefficient Q_2 takes arbitrarily large positive *and* arbitrarily large negative values as $P_1 \to \pm\infty$. The Hamiltonian is therefore unbounded below: there is no state of lowest energy.

To see that this is a genuine instability and not a coordinate artifact, take the explicit nondegenerate example

$$L = \tfrac{1}{2}\,\ddot{q}^{\,2} - \tfrac{1}{2}\,\omega^2 \dot{q}^{\,2}, \tag{6.19}$$

a free higher-derivative oscillator. Here $\partial^2 L/\partial \ddot{q}^2 = 1 \neq 0$, the momenta Eq. (6.17) are $P_1 = -\omega^2 \dot{q} - \dddot{q} = -\omega^2 Q_2 - \dddot{q}$ and $P_2 = \ddot{q}$, so $\mathcal{A} = P_2$, and Eq. (6.18) becomes

$$H = P_1 Q_2 + \tfrac{1}{2} P_2^2 + \tfrac{1}{2}\omega^2 Q_2^2. \tag{6.20}$$

Along the direction $P_1 \to -\infty$ at fixed positive Q_2, $H \to -\infty$ without bound. The equation of motion Eq. (6.15) reads $\ddddot{q} + \omega^2 \ddot{q} = 0$, with general solution $q(t) = c_1 + c_2 t + c_3 \cos\omega t + c_4 \sin\omega t$: alongside the ordinary oscillation at frequency ω there is a mode (the $c_2 t$ piece, and in interacting versions a second oscillator with the *wrong*-sign kinetic energy, a "ghost") that carries negative energy. Coupling such a system to any normal positive-energy degree of freedom lets the total energy run away — the negative-energy mode is excited without bound while the positive-energy mode is excited to compensate, conserving energy while both amplitudes diverge. This runaway is fatal classically and worse quantum-mechanically.

The lesson of Theorem 6.5 is the justification for Theorem 6.2: restricting $\mathcal{L}$ to first derivatives is the condition that the Ostrogradsky construction never runs, so that the Hamiltonian is bounded below and a stable vacuum can exist. We state the result at the level we will use it.

Proposition 6.6 (Ostrogradsky)**.** *A nondegenerate Lagrangian depending on time derivatives of order higher than the first has a canonical Hamiltonian that is linear in at least one canonical momentum, and is therefore unbounded below. Consequently the Lagrangian densities of stable, local field theories are taken to depend on the fields and their first derivatives only; higher-derivative terms, when they appear in effective descriptions, are treated perturbatively as small corrections rather than as defining a new propagating degree of freedom.*

The full statement and proof of Ostrogradsky's theorem for arbitrary order are a direct generalization of the worked computation above and add no new idea; the linearity in the lowest Ostrogradsky momentum is generic, and the parenthetical qualification about effective theories is the standard escape hatch, made precise when we organize derivative expansions in the renormalization chapters. For all the fundamental Lagrangians of this book — scalar, Dirac, Maxwell, Yang–Mills — the first-derivative rule holds exactly.

Caution

The first-derivative rule constrains the Lagrangian *density*, not the action. Adding a total derivative $\partial_\mu K^\mu$ to $\mathcal{L}$ changes S only by a boundary term and so contains no new dynamics, even though $\partial_\mu K^\mu$ may involve second derivatives of the fields when K^μ depends on $\partial\phi$; such terms are harmless precisely because they integrate to the boundary and drop out of the bulk equations of motion. The honest statement is that the *bulk* dynamics depends on $\mathcal{L}$ only up to total derivatives, and within that equivalence class one may always choose a representative with at most first derivatives. The careful treatment of these boundary terms is the technical heart of the next section.

6.1.5 The first protagonist: the free real scalar field

We now write down the Lagrangian that the rest of the book will quantize, perturb, and renormalize. Following the structural requirements, we want the simplest local, Lorentz-invariant Lagrangian for a single real scalar field $\phi(x)$, built from at most first derivatives. The Lorentz scalars available at lowest order are the kinetic invariant $\partial_\mu\phi\,\partial^\mu\phi$ and the field invariants $\phi^2, \phi^4, \ldots$; reflection symmetry $\phi \to -\phi$ (the $\mathbb{Z}_2$ symmetry that becomes the organizing symmetry of ϕ^4 theory in Chapter 7) forbids odd powers. Keeping the kinetic term and the quadratic potential — the two pieces the balls-and-springs limit handed us in Eq. (6.7) and Eq. (6.8) — gives the *free real scalar Lagrangian*:

$$\boxed{\mathcal{L} = \tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,m^2\phi^2} \tag{6.21}$$

in the mostly-minus metric, with $\partial_\mu\phi\,\partial^\mu\phi = \dot\phi^2 - (\nabla\phi)^2$. The relative minus sign between kinetic and mass terms is the $L = T - V$ of mechanics: $\frac{1}{2}\dot\phi^2$ is kinetic energy density, while $\frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2$ is potential energy density (gradient cost plus on-site cost), and the Lagrangian density is their difference. The coefficient $\frac{1}{2}$ is the conventional normalization for a real field, fixed by the requirement that the kinetic term be canonically normalized; it is the canonical normalization of ϕ^4 theory, and we adopt it without alteration here and everywhere the scalar appears.

Notation Watch

Equation (6.21) is the $\lambda \to 0$ (free) limit of the ϕ^4 Lagrangian. The full interacting Lagrangian adds the quartic self-interaction $-\frac{\lambda}{4!}\phi^4$, introduced in the next section once the variational machinery is in place; the present section deliberately stops at the free theory so that the equation of motion derived next is the Klein–Gordon equation in its purest form. Throughout

the book, the scalar field is denoted ϕ, the mass m, and the quartic coupling λ with its $1/4!$ normalization.

Dimensional analysis

Before leaving the scalar we fix its mass dimension, a number used constantly in later chapters to decide which interactions are renormalizable and which operators are relevant or irrelevant. The action is dimensionless in natural units, because it appears in the quantum theory only through $e^{iS/\hbar}$ and $\hbar = 1$ (Chapter A); writing $[\,\cdot\,]$ for the mass dimension (the power of mass, with length counted as mass^{-1} since $\hbar = c = 1$), this is the statement

$$[S] = 0. \tag{6.22}$$

In d spacetime dimensions the measure carries $[d^d x] = -d$ (it is a d-fold product of lengths), so from $S = \int d^d x\, \mathcal{L}$ and $[S] = 0$ the Lagrangian density must carry $[\mathcal{L}] = d$. In the physical case $d = 4$,

$$[\mathcal{L}] = 4. \tag{6.23}$$

Now read the dimension of ϕ off the kinetic term. Each derivative carries one power of mass, $[\partial_\mu] = 1$, so $[\partial_\mu \phi\, \partial^\mu \phi] = 2 + 2[\phi]$, and demanding this equal $[\mathcal{L}] = 4$ gives $2 + 2[\phi] = 4$, hence

$$[\phi] = 1 \qquad \text{(a scalar field has the dimension of mass, in } d = 4\text{)}. \tag{6.24}$$

The mass term is a consistency check: $[m^2\phi^2] = 2[m] + 2[\phi] = 2 + 2 = 4 = [\mathcal{L}]$, confirming that the parameter m in Eq. (6.21) indeed has $[m] = 1$, the dimension of a mass, as its name promised. The same accounting in general d gives $[\phi] = (d-2)/2$, a formula we will need when continuing to $d = 4 - 2\varepsilon$ in dimensional regularization; the quartic coupling then has $[\lambda] = 4 - d = 2\varepsilon$, the fact that makes ϕ^4 marginally renormalizable in four dimensions. We record these here only to note that the dimensional bookkeeping of the entire renormalization program starts from Eq. (6.24).

We have built the stage's first piece of machinery. A classical field is a mechanical system with one degree of freedom per point of space (Theorem 6.1); its dynamics is to be extracted from an action $S = \int d^4x\, \mathcal{L}$ (Theorem 6.2) whose density is local, a Lorentz scalar, and a function of the fields and their first derivatives alone — three requirements we have justified rather than imposed. The free real scalar Lagrangian Eq. (6.21) is the first concrete protagonist, written in the canonical normalization of ϕ^4 theory. What we do not yet have is its equation of motion. The action is a number assigned to each history; the physical history is the one that makes it stationary. Carrying out that variation

— handling the boundary term with the care the Theorem 6.5 caution demanded, and reading off the Euler–Lagrange field equations — is the work of the next section, where the free scalar's stationarity condition turns out to be the Klein–Gordon equation and the engine that generates every equation of motion in the book is built.

6.2 Stationary Action and the Euler–Lagrange Field Equations

The previous section built the object whose stationarity is to generate all of classical dynamics: the action functional

$$S[\phi] \;=\; \int_{\Omega} \mathrm{d}^4x \; \mathcal{L}(\phi_a(x),\, \partial_\mu \phi_a(x),\, x), \tag{6.25}$$

a number assigned to each field configuration ϕ_a on a spacetime region Ω, built from a local Lagrangian density $\mathcal{L}$ that depends on the fields and their first derivatives (Section 6.1). Here the index $a = 1, \dots, N$ runs over the components of whatever multiplet of fields the theory contains — a single real scalar has $N = 1$, the Dirac field will have several complex components, the electromagnetic field four — and the explicit x-dependence in $\mathcal{L}$ is retained for generality, though every fundamental Lagrangian in this book turns out to have none (translation invariance, the subject of Chapter 7). Mechanics taught us that the physical trajectory of a system is the one that makes its action stationary against small variations with fixed endpoints. We now carry that principle, intact, to the continuum: the physical field configuration is the one for which S is stationary against small variations of ϕ_a that vanish on the boundary $\partial\Omega$. This single demand, written out, is the engine of the entire chapter — indeed of the entire book. Every equation of motion that follows, for the scalar, the Dirac field, the Maxwell field, and every interacting theory built from them, is a special case of the covariant Euler–Lagrange equation derived below.

6.2.1 The variational problem and admissible variations

We must first say precisely what "small variation with fixed endpoints" means for a field. Fix a configuration ϕ_a and a second smooth function $\eta_a(x)$ on Ω, and form the one-parameter family

$$\phi_a(x;\, s) \;=\; \phi_a(x) \;+\; s\,\eta_a(x), \qquad s \in \mathbb{R}. \tag{6.26}$$

The function η_a encodes the direction of the variation and s its magnitude; the *variation* of the field is

$$\delta\phi_a(x) \;\equiv\; \frac{\partial}{\partial s}\,\phi_a(x;s)\Big|_{s=0} = \eta_a(x). \tag{6.27}$$

The boundary condition of mechanics — endpoints held fixed — becomes, in the field setting, the requirement that the variation vanish on the boundary of the integration region:

$$\delta\phi_a(x) \;=\; 0 \qquad \text{for all } x \in \partial\Omega. \tag{6.28}$$

A variation η_a satisfying Eq. (6.28) (and otherwise smooth) we call *admissible*. It is worth pausing on what $\partial\Omega$ is. The natural region Ω for a dynamical problem is a spacetime slab bounded by two equal-time hypersurfaces $t = t_1$ and $t = t_2$ together with a timelike "wall" at large spatial radius. On the two temporal faces, Eq. (6.28) is the direct analogue of holding the initial and final configurations fixed — exactly the endpoint condition of Hamilton's principle in mechanics. On the spatial wall, Eq. (6.28) is guaranteed in practice by the standing physical assumption that fields (and hence their variations) fall off to zero at spatial infinity, so that the system carries finite energy; we take this fall-off as part of the definition of an admissible configuration. The two faces play genuinely different roles — one is the boundary of a finite control interval, the other is a decay condition — but both contribute to $\partial\Omega$, and both will be needed when we drop the boundary term below.

Definition 6.7 (Stationary action)**.** A configuration ϕ_a is a *stationary point* of the action Eq. (6.25) if, for every admissible variation η_a,

$$\frac{\mathrm{d}}{\mathrm{d}s}\, S[\phi_a + s\,\eta_a]\Big|_{s=0} \;=\; 0. \tag{6.29}$$

We write this condition as $\delta S = 0$ and call Eq. (6.29) the *principle of stationary action*.

Physical Insight

The classical principle is one of *stationary*, not least, action. For the systems of this book the stationary configuration is rarely a minimum — the relativistic free particle maximizes its proper time along the physical worldline, so its action $S = -m\int d\tau$ is a local minimum, while field actions are typically saddle points. What is universal is that the first-order change vanishes: the physical configuration is one from which the action does not change to linear order in any allowed deformation. This is also exactly the feature that survives quantization. In the path integral of Chapter 10 every configuration contributes with weight e^{iS}, and the stationary configurations are those of stationary phase — the points where neighbouring histories interfere constructively rather than wash out. The classical equation of motion is thus the leading term of an $\hbar \to 0$ stationary-phase expansion, which is the precise sense in which classical field theory is the skeleton of the quantum theory.

6.2.2 Derivation of the covariant Euler–Lagrange equations

We now evaluate the stationarity condition Eq. (6.29) explicitly and read off the differential equation it imposes. The result is the central theorem of the chapter.

Theorem 6.8 (Covariant Euler–Lagrange field equations)**.** *Let $\mathcal{L}(\phi_a, \partial_\mu \phi_a, x)$ be a Lagrangian density that is differentiable in its arguments, and let S be the action Eq. (6.25) on a region Ω. A smooth configuration ϕ_a is a stationary point of S against all admissible variations (Theorem 6.7) if and only if it satisfies, at every interior point of Ω and for each component a,*

$$\boxed{\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0} \tag{6.30}$$

with the repeated index μ summed over the four spacetime directions.

Proof. Insert the family Eq. (6.26) into the action and differentiate under the integral sign with respect to the parameter s. Because Ω is fixed and $\mathcal{L}$ is differentiable, the s-derivative passes through the integral and acts on the integrand by the chain rule:

$$\frac{\mathrm{d}}{\mathrm{d}s} S[\phi + s\eta]\Big|_{s=0} = \int_\Omega \mathrm{d}^4 x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \frac{\partial \phi_a}{\partial s} + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \frac{\partial(\partial_\mu \phi_a)}{\partial s} \right]_{s=0}, \tag{6.31}$$

where a sum over both a and μ is understood. The first factor in each term is evaluated on the unperturbed configuration ϕ_a. For the derivative factors, Eq. (6.27) gives $\partial \phi_a / \partial s = \eta_a = \delta \phi_a$, and the corresponding statement for the field gradient follows because the spacetime derivative ∂_μ and the parameter derivative ∂_s act on independent variables and therefore commute:

$$\frac{\partial}{\partial s} (\partial_\mu \phi_a(x; s))\Big|_{s=0} = \partial_\mu \left(\frac{\partial \phi_a(x; s)}{\partial s}\Big|_{s=0} \right) = \partial_\mu (\delta \phi_a). \tag{6.32}$$

This identity — that the variation of the derivative is the derivative of the variation, $\delta(\partial_\mu \phi_a) = \partial_\mu(\delta \phi_a)$ — is the field-theoretic content of "δ commutes with ∂_μ" and holds simply because $\delta \phi_a(x)$ is the difference of two field configurations evaluated at the *same* point x. Substituting both results,

$$\delta S = \int_\Omega \mathrm{d}^4 x \left[\frac{\partial \mathcal{L}}{\partial \phi_a} \delta \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\mu (\delta \phi_a) \right]. \tag{6.33}$$

The two terms are not yet in a common form: the first multiplies $\delta \phi_a$, the second multiplies its gradient. To extract a single overall factor of $\delta \phi_a$ we integrate the second term by parts, using the Leibniz rule in the guise

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \partial_\mu (\delta \phi_a) = \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \delta \phi_a \right) - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \right) \delta \phi_a. \tag{6.34}$$

Inserting Eq. (6.34) into Eq. (6.33) and grouping, the variation splits into a "bulk" piece proportional to $\delta\phi_a$ and a total-divergence "boundary" piece:

$$\delta S \;=\; \underbrace{\int_\Omega \mathrm{d}^4x \,\left[\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right)\right]\delta\phi_a}_{\text{bulk}} \;+\; \underbrace{\int_\Omega \mathrm{d}^4x\;\partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\delta\phi_a\right)}_{\text{boundary}}. \tag{6.35}$$

The boundary piece is the integral of a four-divergence $\partial_\mu V^\mu$ with $V^\mu \equiv (\partial\mathcal{L}/\partial(\partial_\mu\phi_a))\,\delta\phi_a$. By the divergence theorem applied to the region Ω,

$$\int_\Omega \mathrm{d}^4x\;\partial_\mu V^\mu \;=\; \oint_{\partial\Omega} \mathrm{d}\Sigma_\mu\, V^\mu \;=\; \oint_{\partial\Omega} \mathrm{d}\Sigma_\mu\,\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\delta\phi_a, \tag{6.36}$$

where $\mathrm{d}\Sigma_\mu$ is the outward-directed surface element on $\partial\Omega$. This is the step where the boundary condition earns its keep. Because every admissible variation satisfies $\delta\phi_a = 0$ on all of $\partial\Omega$ by Eq. (6.28), the integrand of Eq. (6.36) vanishes identically on the boundary, and the entire boundary piece is zero. Hence the variation reduces to the bulk integral alone,

$$\delta S \;=\; \int_\Omega \mathrm{d}^4x\,\left[\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right)\right]\delta\phi_a. \tag{6.37}$$

Stationarity demands that this vanish for *every* admissible $\eta_a = \delta\phi_a$. The bracket multiplying $\delta\phi_a$ is a fixed continuous function of x (it is built from the configuration ϕ_a and its derivatives, which are given), and $\delta\phi_a$ ranges over all smooth functions vanishing on $\partial\Omega$. The fundamental lemma of the calculus of variations (Section 4.1.2) states precisely that if the integral of a continuous function against every such test function vanishes, then the function itself vanishes at every interior point. Applying it component by component — choosing variations supported near any chosen interior point and in any single component a — forces the bracket to vanish for each a:

$$\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right) = 0, \tag{6.38}$$

which is Eq. (6.30) up to an overall sign. Conversely, if Eq. (6.30) holds then the bulk integrand vanishes identically and $\delta S = 0$ for every admissible variation. The two conditions are therefore equivalent. □

A few structural remarks fix the meaning of Eq. (6.30) before we use it. There is one equation for each field component a, so a multiplet of N fields obeys N coupled equations. Each is manifestly Lorentz covariant: $\partial/\partial\phi_a$ is a scalar operation, while $\partial/\partial(\partial_\mu\phi_a)$ carries an upper index μ that is contracted with the divergence ∂_μ, so the whole left-hand side transforms as a scalar (for a scalar field) or, more generally, in the same representation as ϕ_a itself. Because $\mathcal{L}$

depends on first derivatives, the term $\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi_a))$ contains *second* derivatives of the fields: the Euler–Lagrange equations are second-order partial differential equations, exactly as the Newtonian intuition "$\ddot{q} =$ force" would predict. That second-order structure is why specifying a field and its time derivative on one hypersurface determines the evolution — the Cauchy data that the Hamiltonian formulation of Section 6.4 will repackage into canonical pairs.

6.2.3 The functional-derivative restatement

The content of Theorem 6.8 can be compressed into a single statement about the functional derivative of S. Recall from Section 4.1.2 that the functional derivative $\delta S/\delta\phi_a(x)$ is defined as the coefficient of $\delta\phi_a(x)$ in the first-order change of S, i.e. by the relation $\delta S = \int \mathrm{d}^4x\,(\delta S/\delta\phi_a(x))\,\delta\phi_a(x)$ holding for all admissible variations. Comparing with Eq. (6.37) reads off the functional derivative of the action directly:

$$\frac{\delta S}{\delta\phi_a(x)} \;=\; \frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right), \tag{6.39}$$

all quantities evaluated at the point x. The principle of stationary action is then the statement that the physical configuration is a zero of this functional derivative,

$$\frac{\delta S}{\delta\phi_a(x)} \;=\; 0 \qquad \text{for all } x, \tag{6.40}$$

which is identical in form to the finite-dimensional condition "$\partial S/\partial q_i = 0$ at a critical point," now with the discrete index i replaced by the continuous label (a, x). This is the most useful packaging of the equations of motion for later chapters: the generating functionals of Chapter 10 are differentiated by exactly this $\delta/\delta\phi_a(x)$, and the classical equation of motion will reappear as the saddle-point condition $\delta S/\delta\phi = -J$ in the presence of a source J.

6.2.4 Total-derivative ambiguity of the Lagrangian

The action does not determine the Lagrangian density uniquely. Two Lagrangians that differ by a total spacetime divergence,

$$\mathcal{L}' \;=\; \mathcal{L} \;+\; \partial_\mu K^\mu, \tag{6.41}$$

with $K^\mu = K^\mu(\phi_a, x)$ any differentiable function of the fields and position, yield *the same equations of motion*. The quickest argument is the one already assembled in the proof. The extra term contributes to the action a pure boundary integral,

$$\int_\Omega \mathrm{d}^4x\;\partial_\mu K^\mu \;=\; \oint_{\partial\Omega} \mathrm{d}\Sigma_\mu\, K^\mu, \tag{6.42}$$

which depends only on the values of the fields on $\partial\Omega$. But those boundary values are held fixed under admissible variations, so the boundary integral is a constant

along the family Eq. (6.26) and its variation vanishes: $\delta\int_\Omega \partial_\mu K^\mu = 0$. Therefore $\delta S' = \delta S$ for every admissible variation, and S' and S have identical stationary points. Adding a total derivative to the Lagrangian is invisible to the dynamics.

This freedom is not a mere curiosity; it is a structural feature we will exploit and must guard against in equal measure. On the constructive side, it lets us drop total derivatives freely when simplifying a Lagrangian, and it underlies the freedom to integrate by parts inside an action — a manipulation used constantly from Chapter 7 onward. On the cautionary side, it means that quantities computed *from* the Lagrangian rather than from the equations of motion need not be unique. The canonical stress–energy tensor derived at the end of this chapter (Section 6.6) inherits exactly this ambiguity — it is defined only up to an "improvement" term — and the proper treatment of boundary terms becomes essential the moment we consider variations that do *not* vanish on $\partial\Omega$, as in the derivation of Noether currents (Chapter 7) and in gauge theories where surface terms carry physical charge. The following worked example verifies the invariance Eq. (6.41) by direct computation, exhibiting in detail why the Euler–Lagrange operator annihilates any total divergence.

Worked Example 6.9 (A total divergence is annihilated by the Euler–Lagrange operator)**.** Define the *Euler–Lagrange operator* acting on any density $f(\phi_a, \partial_\mu\phi_a, x)$ by

$$E_a[f] \;\equiv\; \frac{\partial f}{\partial \phi_a} - \partial_\mu\!\left(\frac{\partial f}{\partial(\partial_\mu \phi_a)}\right), \tag{6.43}$$

so that the equation of motion Eq. (6.30) reads $E_a[\mathcal{L}] = 0$. We show that for $K^\nu = K^\nu(\phi_b, x)$ depending on the fields and position but not on field derivatives,

$$E_a\big[\partial_\nu K^\nu\big] \;=\; 0 \qquad \text{identically (for every configuration).} \tag{6.44}$$

This is stronger than the boundary argument above: it shows the cancellation happens algebraically in the field equations themselves, with no appeal to boundary conditions.

Setup. Write the total divergence out with the chain rule. Since K^ν depends on x both through the fields $\phi_b(x)$ and explicitly,

$$F \;\equiv\; \partial_\nu K^\nu \;=\; \frac{\partial K^\nu}{\partial \phi_b}\,\partial_\nu\phi_b \;+\; \left.\frac{\partial K^\nu}{\partial x^\nu}\right|_{\phi}, \tag{6.45}$$

where $|_\phi$ denotes the partial derivative at fixed fields (the explicit x-dependence) and sums over b and ν are understood. Note that F depends on the fields ϕ_b and on the first derivatives $\partial_\nu\phi_b$, so it is a legitimate argument of the operator Eq. (6.43).

First piece, $\partial F/\partial\phi_a$. Differentiating Eq. (6.45) with respect to ϕ_a at fixed field derivatives,

$$\frac{\partial F}{\partial \phi_a} \;=\; \frac{\partial^2 K^\nu}{\partial\phi_a\partial\phi_b}\,\partial_\nu\phi_b \;+\; \left.\frac{\partial^2 K^\nu}{\partial\phi_a\partial x^\nu}\right|_{\phi}. \tag{6.46}$$

Only the field-dependent coefficients are differentiated; the factor $\partial_\nu \phi_b$ is an independent variable and is untouched.

Second piece, $\partial F/\partial(\partial_\mu \phi_a)$. The only place a field derivative appears in Eq. (6.45) is the explicit factor $\partial_\nu \phi_b$ in the first term. Differentiating,

$$\frac{\partial F}{\partial(\partial_\mu \phi_a)} \;=\; \frac{\partial K^\nu}{\partial \phi_b}\,\frac{\partial(\partial_\nu \phi_b)}{\partial(\partial_\mu \phi_a)} \;=\; \frac{\partial K^\nu}{\partial \phi_b}\,\delta^\mu_\nu\,\delta_{ab} \;=\; \frac{\partial K^\mu}{\partial \phi_a}, \tag{6.47}$$

using that distinct components and distinct derivative directions are independent variables, $\partial(\partial_\nu \phi_b)/\partial(\partial_\mu \phi_a) = \delta^\mu_\nu \delta_{ab}$. Now take the spacetime divergence of Eq. (6.47). The function $\partial K^\mu/\partial \phi_a$ depends on x through the fields and explicitly, so the chain rule gives

$$\partial_\mu\!\left(\frac{\partial F}{\partial(\partial_\mu \phi_a)}\right) \;=\; \partial_\mu\!\left(\frac{\partial K^\mu}{\partial \phi_a}\right) \;=\; \frac{\partial^2 K^\mu}{\partial \phi_a \partial \phi_b}\,\partial_\mu \phi_b \;+\; \frac{\partial^2 K^\mu}{\partial \phi_a \partial x^\mu}\Big|_\phi . \tag{6.48}$$

Subtraction. Form $E_a[F] = \partial F/\partial \phi_a - \partial_\mu(\partial F/\partial(\partial_\mu \phi_a))$ by subtracting Eq. (6.48) from Eq. (6.46):

$$E_a[F] \;=\; \left(\frac{\partial^2 K^\nu}{\partial \phi_a \partial \phi_b}\,\partial_\nu \phi_b + \frac{\partial^2 K^\nu}{\partial \phi_a \partial x^\nu}\Big|_\phi\right) - \left(\frac{\partial^2 K^\mu}{\partial \phi_a \partial \phi_b}\,\partial_\mu \phi_b + \frac{\partial^2 K^\mu}{\partial \phi_a \partial x^\mu}\Big|_\phi\right). \tag{6.49}$$

The two summed indices μ and ν are dummies; relabelling $\nu \to \mu$ in the first parenthesis makes the two parentheses term-by-term identical, so they cancel exactly:

$$E_a\big[\partial_\nu K^\nu\big] \;=\; 0. \tag{6.50}$$

The cancellation hinges only on the equality of mixed partial derivatives of K^ν (Schwarz's theorem), and holds off-shell, for every field configuration. This confirms from the inside that shifting $\mathcal{L} \to \mathcal{L} + \partial_\mu K^\mu$ leaves $E_a[\mathcal{L}]$ — and hence the equation of motion — untouched, in complete agreement with the boundary argument of Eq. (6.42).

6.2.5 The free scalar: the Klein–Gordon equation

We now turn the machine on its first concrete target, the simplest relativistic field: a single real scalar ϕ. The unique Lorentz-invariant Lagrangian built from ϕ and its first derivatives, quadratic in the field (hence "free") and containing the canonical kinetic term, is

$$\mathcal{L}_0 \;=\; \tfrac{1}{2}\,(\partial_\mu \phi)(\partial^\mu \phi) \;-\; \tfrac{1}{2}\,m^2 \phi^2 \;=\; \tfrac{1}{2}\,\eta^{\rho\sigma}(\partial_\rho \phi)(\partial_\sigma \phi) - \tfrac{1}{2} m^2 \phi^2, \tag{6.51}$$

with m a real constant of mass dimension one (in natural units) that will turn out to be the mass of the associated particle. The mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ is in force, so the kinetic term $\tfrac{1}{2}(\partial_\mu \phi)(\partial^\mu \phi) = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}(\nabla \phi)^2$

has the "kinetic minus gradient" form of a mechanical energy density, the sign that makes the energy bounded below (Section 6.6). To apply Eq. (6.30) we need the two partial derivatives of $\mathcal{L}_0$.

The field-derivative is immediate, since only the mass term depends on ϕ undifferentiated:

$$\frac{\partial \mathcal{L}_0}{\partial \phi} \;=\; -m^2 \phi. \tag{6.52}$$

The gradient-derivative requires care with the two factors in the kinetic term, which must be treated as a function of the independent variables $\partial_\mu \phi$. Writing the kinetic term as $\frac{1}{2}\eta^{\rho\sigma}(\partial_\rho\phi)(\partial_\sigma\phi)$ and differentiating with the product rule,

$$\begin{aligned} \frac{\partial \mathcal{L}_0}{\partial(\partial_\mu\phi)} &= \tfrac{1}{2}\,\eta^{\rho\sigma}\left[\frac{\partial(\partial_\rho\phi)}{\partial(\partial_\mu\phi)}\,\partial_\sigma\phi + \partial_\rho\phi\,\frac{\partial(\partial_\sigma\phi)}{\partial(\partial_\mu\phi)}\right] = \tfrac{1}{2}\,\eta^{\rho\sigma}\left[\delta^\mu_\rho\,\partial_\sigma\phi + \partial_\rho\phi\,\delta^\mu_\sigma\right] \\ &= \tfrac{1}{2}\left(\eta^{\mu\sigma}\partial_\sigma\phi + \eta^{\rho\mu}\partial_\rho\phi\right) = \tfrac{1}{2}\left(\partial^\mu\phi + \partial^\mu\phi\right) = \partial^\mu\phi. \end{aligned} \tag{6.53}$$

The two terms are equal because the metric is symmetric, and they combine to cancel the $\frac{1}{2}$ — a small calculation worth doing once explicitly, because the same pattern (the kinetic normalization $\frac{1}{2}$ existing precisely to undo the factor of two from the two identical factors) recurs for every field in the book. Now assemble the Euler–Lagrange equation. Taking the divergence of Eq. (6.53) gives $\partial_\mu(\partial^\mu\phi) = \partial_\mu\partial^\mu\phi \equiv \Box\phi$, the d'Alembertian, and Eq. (6.30) reads

$$\partial_\mu\left(\frac{\partial \mathcal{L}_0}{\partial(\partial_\mu\phi)}\right) - \frac{\partial \mathcal{L}_0}{\partial\phi} \;=\; \Box\phi - (-m^2\phi) \;=\; \Box\phi + m^2\phi \;=\; 0, \tag{6.54}$$

that is,

$$\boxed{\left(\Box + m^2\right)\phi \;=\; 0} \tag{6.55}$$

the *Klein–Gordon equation*. With the mostly-minus convention, $\Box = \partial_\mu\partial^\mu = \partial_t^2 - \nabla^2$, so Eq. (6.55) is the wave equation with a mass term. Its plane-wave solutions $\phi \sim e^{-ip\cdot x}$ require $p_\mu p^\mu = p^2 = m^2$, the relativistic mass-shell condition — the first sign that the parameter m in Eq. (6.51) is a physical mass, and the seed of the particle interpretation built in Chapter 9. The Klein–Gordon equation is the free equation of motion that all of ϕ^4 theory perturbs around.

6.2.6 Scalar ϕ^4 theory: the interacting scalar and its equation of motion

We now introduce the first of the book's three carried protagonists. ϕ^4 *theory* is the self-interacting real scalar field, the universal pedagogical workhorse that we will follow from this classical Lagrangian through quantization, its first scattering amplitude, renormalization, and finally a nontrivial renormalization-group fixed point. Its standard Lagrangian, in the symbols adds a single quartic

self-interaction term to the free scalar Eq. (6.51):

$$\boxed{\mathcal{L} \;=\; \tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}\,m^2\phi^2 \;-\; \frac{\lambda}{4!}\,\phi^4\,, \qquad \phi \text{ real},\ \lambda > 0.} \tag{6.56}$$

The coupling λ is dimensionless in four spacetime dimensions (a fact whose significance for renormalizability is taken up in Chapter 21), and the conventional factor $1/4!$ is chosen so that the Feynman rule for the four-point vertex will come out as the clean $-i\lambda$ of Chapter 15; the same $4! = 24$ that looks arbitrary here is exactly the number of ways to attach four identical lines to a vertex, and it cancels neatly in the equation of motion, as we now see. The condition $\lambda > 0$ ensures the potential $V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$ is bounded below, so that the energy of Section 6.6 is bounded below and the theory has a stable vacuum. The quartic term is the minimal interaction consistent with the discrete symmetry $\phi \to -\phi$ (the $\mathbb{Z}_2$ symmetry whose consequences Chapter 7 draws out); a cubic term ϕ^3 would break it and render the potential unbounded below.

The following worked example derives the exact classical equation of motion for Eq. (6.56) by direct application of the Euler–Lagrange machine, completing the Chapter 6 classical foundation of ϕ^4 theory.

Worked Example 6.10 (The classical ϕ^4 equation of motion)**.** We apply Eq. (6.30) to the Lagrangian Eq. (6.56). The kinetic part is identical to the free scalar, so its derivatives are quoted from Eqs. (6.52) and (6.53); only the potential changes the field-derivative.

Gradient-derivative. The interaction $-\frac{\lambda}{4!}\phi^4$ contains no field derivatives, so it does not contribute to $\partial\mathcal{L}/\partial(\partial_\mu\phi)$, and the result is unchanged from the free case:

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} \;=\; \partial^\mu\phi, \qquad \partial_\mu\!\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\right) \;=\; \Box\phi. \tag{6.57}$$

Field-derivative. Differentiating Eq. (6.56) with respect to ϕ at fixed gradient,

$$\frac{\partial\mathcal{L}}{\partial\phi} \;=\; -m^2\phi \;-\; \frac{\lambda}{4!}\,\frac{\partial(\phi^4)}{\partial\phi} \;=\; -m^2\phi \;-\; \frac{\lambda}{4!}\,4\phi^3 \;=\; -m^2\phi \;-\; \frac{\lambda}{3!}\,\phi^3, \tag{6.58}$$

where the factor $4/4! = 4/24 = 1/6 = 1/3!$ is the promised cancellation: differentiating the quartic brings down the four, and $4!/4 = 3!$.

Assemble. The Euler–Lagrange equation $\partial_\mu(\partial\mathcal{L}/\partial(\partial_\mu\phi)) - \partial\mathcal{L}/\partial\phi = 0$ becomes

$$\Box\phi \;-\; \left(-m^2\phi - \frac{\lambda}{3!}\phi^3\right) \;=\; \Box\phi + m^2\phi + \frac{\lambda}{3!}\phi^3 \;=\; 0, \tag{6.59}$$

which rearranges to the carried equation of motion of ϕ^4 theory,

$$\boxed{(\Box + m^2)\,\phi \;=\; -\,\frac{\lambda}{3!}\,\phi^3\,.} \tag{6.60}$$

The structure is transparent and sets the template for every interacting equation of motion to come: the free Klein–Gordon operator $(\Box + m^2)$ on the left, and on the right a source term $-\frac{\lambda}{3!}\phi^3$ built from the field itself — the field acting as its own source through the self-interaction. The free equation Eq. (6.55) is recovered exactly in the limit $\lambda \to 0$. This nonlinearity is precisely what makes the theory interacting and nontrivial: it has no general closed-form solution, and the perturbative expansion in powers of λ that organizes its quantum treatment (Chapters 14 and 15) is, at its root, the iterative solution of Eq. (6.60) treating the right-hand side as a small correction.

Caution

The equations of motion are insensitive to total-derivative terms and to boundary data *only* because admissible variations vanish on $\partial\Omega$. The instant one considers variations that do not vanish there — which is exactly the situation in the derivation of Noether's theorem, where the variation is generated by a symmetry transformation and need not respect any boundary condition — the boundary term in Eq. (6.35) no longer drops and carries genuine physical information (it becomes the conserved current). Likewise, the total-derivative ambiguity Eq. (6.41) that is harmless for the equations of motion is *not* harmless for the canonical stress–energy tensor, which is read off from the Lagrangian rather than from the field equations. Keep the boundary term in view: it is dropped here only under a stated hypothesis, and later chapters lift that hypothesis deliberately.

With the Euler–Lagrange equations established as a general machine, the functional-derivative restatement in hand, the total-derivative ambiguity understood, and ϕ^4 theory advanced to its exact classical equation of motion $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$, we are ready to stress-test the machine on the two protagonists whose Lagrangians hide structural surprises. The next section assembles the rest of the cast — the Dirac field, whose action is first-order in derivatives, and the Maxwell field, whose action carries a gauge redundancy — deriving their equations of motion from the same principle and surfacing the two features that the Hamiltonian sections must then confront.

6.3 The Cast Assembled: Dirac and Maxwell Lagrangians

The previous section forged a single tool. From $\delta S = 0$ we extracted the covariant Euler–Lagrange equation for a field ϕ,

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} - \frac{\partial \mathcal{L}}{\partial \phi} = 0, \tag{6.61}$$

and tested it on the real scalar, recovering the Klein–Gordon equation and, with the quartic term, the classical equation of motion of ϕ^4 theory (Section 6.2). One protagonist is now in hand. This section assembles the rest of the cast. We turn the same crank (6.61) on the two remaining fields that recur throughout the book — the Dirac spinor and the Maxwell potential — and on the way meet a third, the complex scalar, whose global phase symmetry is the seed from which the entire gauge principle of Chapter 8 will grow.

The point of doing all three here, side by side, is not merely to collect equations of motion. It is to expose two *structural* features that the scalar did not display and that the Hamiltonian sections to come must confront head-on. The Dirac action turns out to be *first order* in derivatives: its Lagrangian is, up to a total derivative, already its own canonical one-form, so the Legendre transform that defines the Hamiltonian will be degenerate before we even begin. The Maxwell action turns out to be *gauge redundant*: distinct potentials describe the same physics, so the Euler–Lagrange operator has a kernel and cannot be inverted. Both surprises are visible at the purely classical level, and both are flagged here so that Section 6.4 and Chapter 13 know exactly what they inherit. Everything in this section is classical — c-number fields, and (flagged where it matters) anticommuting fields — with no operators and no Hilbert space.

6.3.1 The complex scalar and its global $U(1)$

Before the spinor, a gentle generalization of the scalar we already understand. Instead of one real field we take a single *complex* field $\phi(x)$, equivalently two real fields packaged as $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$. The natural Lorentz-invariant, $U(1)$-symmetric Lagrangian with the right (mass) dimension is

$$\mathcal{L} = (\partial_\mu \phi^*)(\partial^\mu \phi) - m^2\, \phi^* \phi, \tag{6.62}$$

with $\phi^*\phi = |\phi|^2 \geq 0$. The relative coefficients are fixed by the same logic as for the real scalar: the kinetic term must be real and have the canonical normalization that makes the field's quantum a particle of mass m, and the only Lorentz scalar quadratic in ϕ without derivatives is $|\phi|^2$.

Notation Watch

Within this chapter the symbol ϕ does double duty. In ϕ^4 theory (Section 6.2) it is a *real* field; here in Section 6.3.1 it is *complex*. The two never appear in the same computation, and the complex case is always signalled by the presence of ϕ^*. We never write ϕ^* for the real field.

To extract the equations of motion we must vary (6.62). The cleanest route is the one used for every complex or charged field in this book, so it is worth stating once with care. The Lagrangian (6.62) depends on the two real fields

ϕ_1, ϕ_2; stationarity of the action requires $\delta S = 0$ under independent variations of *both*. But the map

$$(\phi_1, \phi_2) \longleftrightarrow (\phi, \phi^*), \qquad \phi = \tfrac{1}{\sqrt{2}}(\phi_1 + i\phi_2), \quad \phi^* = \tfrac{1}{\sqrt{2}}(\phi_1 - i\phi_2), \tag{6.63}$$

is an invertible linear change of variables (with constant, nonsingular coefficients). Requiring $\delta S = 0$ for all independent $\delta\phi_1, \delta\phi_2$ is therefore exactly equivalent to requiring $\delta S = 0$ for all independent $\delta\phi, \delta\phi^*$. We may thus treat ϕ and ϕ^* as two independent fields and apply the Euler–Lagrange equation (6.61) to each separately. This is not a trick or an approximation; it is a legitimate, invertible relabelling of the real degrees of freedom, and the same observation will license treating ψ and $\bar{\psi}$ as independent for the Dirac field below.

Varying ϕ^* in (6.62): the Lagrangian depends on $\partial_\mu\phi^*$ through $(\partial_\mu\phi^*)(\partial^\mu\phi)$ and on ϕ^* through $-m^2\phi^*\phi$, so

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)} = \partial^\mu\phi, \qquad \frac{\partial\mathcal{L}}{\partial\phi^*} = -m^2\phi, \tag{6.64}$$

and (6.61) gives

$$\partial_\mu\partial^\mu\phi + m^2\phi = (\Box + m^2)\phi = 0. \tag{6.65}$$

Varying ϕ instead yields the complex conjugate equation $(\Box + m^2)\phi^* = 0$, as it must, since the action is real. The complex scalar is thus two Klein–Gordon fields of equal mass — exactly what the decomposition (6.63) promised, since ϕ_1 and ϕ_2 each satisfy $(\Box + m^2)\phi_i = 0$.

What distinguishes (6.62) from a pair of unrelated real scalars is a symmetry the real case did not possess. The Lagrangian is invariant under the rigid (spacetime-independent) phase rotation

$$\phi(x) \longrightarrow e^{i\alpha}\phi(x), \qquad \phi^*(x) \longrightarrow e^{-i\alpha}\phi^*(x), \qquad \alpha \in \mathbb{R} \text{ constant}, \tag{6.66}$$

because every term in (6.62) pairs one ϕ with one ϕ^*, so the phases cancel: $\phi^*\phi \to e^{-i\alpha}e^{i\alpha}\phi^*\phi = \phi^*\phi$, and likewise for the kinetic term, since α is constant and passes through ∂_μ. The transformations (6.66) form the group $U(1)$, and the invariance is a *global continuous internal symmetry*. By the machinery of the next chapter this symmetry implies a conserved Noether current $j^\mu = i(\phi^*\partial^\mu\phi - \phi\,\partial^\mu\phi^*)$ and a conserved charge; we state this here only to place the marker and defer the systematic derivation to Chapter 7.

Physical Insight

The invariance (6.66) is rigid: the *same* phase α is applied at every spacetime point. A natural and far-reaching question — what if we demand invariance under a phase $\alpha(x)$ that varies from point to point? — breaks (6.62) immediately, because $\partial_\mu(e^{i\alpha(x)}\phi) = e^{i\alpha(x)}(\partial_\mu + i\,\partial_\mu\alpha)\phi$

produces an unwanted $\partial_\mu \alpha$ term. Repairing the kinetic term so that local phase rotations *are* a symmetry forces the introduction of a new vector field that transforms to cancel $\partial_\mu \alpha$ — and that field is the photon. This is the gauge principle, and (6.66) is its seed. We do not pursue it now; it is the central business of Chapter 8.

6.3.2 The Dirac Lagrangian and equation of motion

We now meet the field whose quanta are spin-$\frac{1}{2}$ particles — the electron of QED. From Chapter 3 we import the Dirac spinor $\psi(x)$, a four-component object transforming in the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ representation of the Lorentz group (Section 3.3.3), and from Chapter C the gamma-matrix algebra

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}, \qquad \bar{\psi} \equiv \psi^\dagger \gamma^0, \tag{6.67}$$

together with the conjugation conventions fixed there (Theorem C.1). The Dirac adjoint $\bar{\psi}$, rather than $\psi^\dagger$, is the object that combines with ψ into Lorentz scalars, because γ^0 supplies precisely the factor that turns the non-unitary spinor representation into one under which $\bar{\psi}\,\Gamma\,\psi$ transforms correctly for each gamma-matrix structure Γ.

The unique Lorentz-invariant Lagrangian that is first order in derivatives, bilinear in the spinor, and yields a sensible (mass-m) wave equation is the *Dirac Lagrangian*

$$\mathcal{L} = \bar{\psi}\,(i\gamma^\mu \partial_\mu - m)\,\psi = \bar{\psi}\,(i\partial\!\!\!/ - m)\,\psi, \tag{6.68}$$

where $\partial\!\!\!/ \equiv \gamma^\mu \partial_\mu$. Each term is a Lorentz scalar: $\bar{\psi}\psi$ is the scalar bilinear, and $\bar{\psi}\gamma^\mu\psi$ is a four-vector contracted with the four-vector ∂_μ. The factor of i makes the kinetic term Hermitian up to a total derivative (we verify this below), and the relative sign and coefficient of the mass term are fixed by demanding that (6.68) reproduce the Dirac equation with the standard mass.

Caution

The classical Dirac field is *Grassmann-valued*: its components anticommute, $\psi_a(x)\psi_b(y) = -\psi_b(y)\psi_a(x)$. This is not optional decoration; it is what the spin–statistics connection (Chapter 11) will force on the quantum theory, and keeping it classically is what makes the classical limit consistent. In the variations below it changes intermediate *signs* but never the final equation of motion, because each term in (6.68) is bilinear: an even number of Grassmann fields, so reordering two of them costs two sign flips that cancel. We carry out the variation in the order that keeps the bookkeeping transparent and note the one place a sign must be tracked.

Worked Example 6.11 (The Dirac equation from the Dirac Lagrangian)**.** We vary the action $S = \int d^4x\, \bar\psi(i\partial\!\!\!/ - m)\psi$ treating ψ and $\bar\psi$ as independent fields, the justification being the same invertible relabelling (6.63) that licensed independent ϕ, ϕ^* variation: the real and imaginary parts of the components of ψ are the independent real fields, and $(\psi, \bar\psi)$ is a constant invertible linear repackaging of them.

Variation with respect to $\bar\psi$. The Lagrangian (6.68) contains $\bar\psi$ undifferentiated — every derivative falls on ψ. Hence

$$\frac{\partial \mathcal{L}}{\partial(\partial_\mu \bar\psi)} = 0, \qquad \frac{\partial \mathcal{L}}{\partial \bar\psi} = (i\partial\!\!\!/ - m)\psi, \tag{6.69}$$

and the Euler–Lagrange equation (6.61), applied to $\bar\psi$, reduces to its second term:

$$\boxed{(i\partial\!\!\!/ - m)\,\psi = 0} \qquad \Longleftrightarrow \qquad (i\gamma^\mu \partial_\mu - m)\psi = 0. \tag{6.70}$$

This is the Dirac equation. Because $\bar\psi$ appears without derivatives, no integration by parts is needed: the equation of motion drops out algebraically. That the *adjoint* field is the convenient variable to vary — it sits to the left, undifferentiated — is the first hint of the first-order structure we dissect below.

Variation with respect to ψ. Now the derivative is in play. Writing the variation explicitly,

$$\delta S = \int d^4x\; \bar\psi\,(i\gamma^\mu \partial_\mu - m)\,\delta\psi = \int d^4x\; \big[\, i\,\bar\psi\gamma^\mu\, \partial_\mu(\delta\psi) - m\,\bar\psi\,\delta\psi\,\big]. \tag{6.71}$$

Integrate the derivative term by parts, discarding the boundary term (the variation $\delta\psi$ vanishes on the boundary of the integration region, exactly as in Section 6.2):

$$\int d^4x\; i\,\bar\psi\gamma^\mu\,\partial_\mu(\delta\psi) = -\int d^4x\; i\,(\partial_\mu\bar\psi)\gamma^\mu\,\delta\psi \;+\; \underbrace{\oint d\Sigma_\mu\; i\,\bar\psi\gamma^\mu\,\delta\psi}_{=\,0}\,. \tag{6.72}$$

Collecting the coefficient of $\delta\psi$ (which is itself Grassmann-odd; the integration by parts and the factoring of $\delta\psi$ to the right are precisely where the even-bilinear remark in the Caution guarantees no stray sign survives) gives

$$\delta S = \int d^4x\; \big[\, -i\,(\partial_\mu\bar\psi)\gamma^\mu - m\,\bar\psi\,\big]\,\delta\psi = 0 \quad \text{for all } \delta\psi, \tag{6.73}$$

hence

$$\boxed{i\,(\partial_\mu\bar\psi)\gamma^\mu + m\,\bar\psi = 0} \qquad \Longleftrightarrow \qquad i\,\partial_\mu\bar\psi\,\gamma^\mu + m\,\bar\psi = 0, \tag{6.74}$$

the *Dirac adjoint equation*, often written $\bar\psi\,(i\overleftarrow{\partial\!\!\!/} + m) = 0$ with the arrow indicating that the derivative acts to the left.

The two equations (6.70) and (6.74) are not independent statements but Dirac conjugates of one another, and checking this consistency is a sharp test of the gamma-matrix conventions. Take the Hermitian conjugate of (6.70). Using $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$ (the defining relation that makes $\bar\psi$ the right adjoint, Chapter C),

$$[(i\gamma^\mu\partial_\mu - m)\psi]^\dagger = -i\,\partial_\mu\psi^\dagger\,(\gamma^\mu)^\dagger - m\,\psi^\dagger = -i\,\partial_\mu\psi^\dagger\,\gamma^0\gamma^\mu\gamma^0 - m\,\psi^\dagger = 0. \quad (6.75)$$

Multiply on the right by γ^0 and use $(\gamma^0)^2 = \mathbf{1}$ together with $\bar\psi = \psi^\dagger\gamma^0$:

$$-i\,\partial_\mu(\psi^\dagger\gamma^0)\,\gamma^\mu - m\,\psi^\dagger\gamma^0 = -i\,\partial_\mu\bar\psi\,\gamma^\mu - m\,\bar\psi = 0, \quad (6.76)$$

which is exactly -1 times (6.74). The two variations are consistent, and the reality (Hermiticity) of the action is confirmed: varying ψ and varying $\bar\psi$ give conjugate content, not new content.

The first structural surprise: the Dirac action is first order

The scalar Lagrangian was *quadratic* in derivatives — a term $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$ with two factors of velocity — and this is what made its Legendre transform routine: the canonical momentum $\pi = \partial\mathcal{L}/\partial\dot\phi = \dot\phi$ was a genuine, invertible function of the velocity. The Dirac Lagrangian (6.68) is *linear* in derivatives. Isolating the time derivative,

$$\mathcal{L} = i\,\bar\psi\gamma^0\partial_0\psi + i\,\bar\psi\gamma^i\partial_i\psi - m\,\bar\psi\psi = i\,\psi^\dagger\dot\psi + (\text{no further } \dot\psi,\ \text{no } \dot{\bar\psi}), \quad (6.77)$$

where we used $\bar\psi\gamma^0 = \psi^\dagger\gamma^0\gamma^0 = \psi^\dagger$. The entire dependence on the velocity $\dot\psi$ sits in the single term $i\,\psi^\dagger\dot\psi$, which is *linear* in $\dot\psi$, and there is no dependence on $\dot{\bar\psi}$ at all.

This is structurally the form $p\,\dot q$ — a canonical one-form (a symplectic potential) $\theta = p\,\dot q$ — rather than a kinetic energy $\frac{1}{2}p^2$. Reading off the momentum conjugate to ψ,

$$\pi_\psi = \frac{\partial\mathcal{L}}{\partial\dot\psi} = i\,\psi^\dagger, \quad (6.78)$$

we find that the momentum is *not* an independent quantity expressing the velocity — it is simply the field $\psi^\dagger$ itself, with no velocity appearing anywhere. The relation $\pi_\psi - i\psi^\dagger = 0$ is a *constraint* on phase space, not a definition that can be inverted to solve for $\dot\psi$. The Legendre transform is therefore degenerate from the outset, and the naive Hamiltonian $\mathcal{H} = \pi_\psi\dot\psi - \mathcal{L}$ has its velocity dependence cancel identically. We flag this now and resolve it properly in Section 6.4: the Dirac field's phase space must be handled as a *constrained* system, and the constraint (6.78) is the classical origin of the anticommutator structure that quantization (Chapter 11) will impose.

6.3.3 The Maxwell Lagrangian and equation of motion

The last protagonist is the electromagnetic field, carried as the photon of QED. From Chapter 3 the gauge potential $A_\mu(x)$ is a Lorentz four-vector field, transforming in the $(\frac{1}{2},\frac{1}{2})$ representation (Section 3.3.3). The gauge-invariant object built from it is the antisymmetric *field-strength tensor*

$$F_{\mu\nu} \equiv \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{6.79}$$

whose components reproduce the electric and magnetic fields ($F^{0i} = -E^i$, $F^{ij} = -\epsilon^{ijk}B^k$ in our conventions, Chapter A). The unique Lorentz-invariant, parity-even Lagrangian quadratic in F and yielding second-order linear field equations is the *Maxwell Lagrangian*

$$\mathcal{L} = -\tfrac{1}{4}\, F_{\mu\nu}F^{\mu\nu}. \tag{6.80}$$

The overall normalization $-\frac{1}{4}$ is the standard one that gives the canonically normalized kinetic term for the propagating photon polarizations and the conventional Maxwell equations; the sign makes the energy density positive (the time-derivative, "electric" part enters $\mathcal{L}$ with a $+\frac{1}{2}E^2$). We now derive its equation of motion, carrying every index contraction explicitly because the antisymmetrization at the heart of it is exactly the step where sign and factor errors breed.

Worked Example 6.12 (The inhomogeneous Maxwell equations from $\mathcal{L} = -\frac{1}{4}F^2$)**.** The dynamical field is A_μ, so the Euler–Lagrange equation (6.61) reads, with β the free index,

$$\partial_\alpha \frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} - \frac{\partial \mathcal{L}}{\partial A_\beta} = 0. \tag{6.81}$$

The Lagrangian (6.80) contains A_μ only through its derivatives, so $\partial\mathcal{L}/\partial A_\beta = 0$ and the whole equation lives in the first term. We must compute $\partial\mathcal{L}/\partial(\partial_\alpha A_\beta)$.

Step 1: differentiate the square. Write $\mathcal{L} = -\frac{1}{4}\, F_{\mu\nu}F^{\mu\nu}$ and differentiate, using the chain rule and the symmetry $F_{\mu\nu}F^{\mu\nu}$ being a sum over both indices:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -\tfrac{1}{4}\,\frac{\partial(F_{\mu\nu}F^{\mu\nu})}{\partial(\partial_\alpha A_\beta)} = -\tfrac{1}{4}\cdot 2\, F^{\mu\nu}\,\frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)} = -\tfrac{1}{2}\, F^{\mu\nu}\,\frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)}, \tag{6.82}$$

where the factor of 2 arises because both F's in the product depend on ∂A (equivalently, $\partial F^{\mu\nu}/\partial(\partial_\alpha A_\beta)$ and $\partial F_{\mu\nu}/\partial(\partial_\alpha A_\beta)$ contribute equally after raising/lowering with the metric).

Step 2: differentiate the field strength. From (6.79), $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and treating each $\partial_\alpha A_\beta$ as an independent variable,

$$\frac{\partial(\partial_\mu A_\nu)}{\partial(\partial_\alpha A_\beta)} = \delta_\mu{}^\alpha\,\delta_\nu{}^\beta \quad\Longrightarrow\quad \frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)} = \delta_\mu{}^\alpha\delta_\nu{}^\beta - \delta_\nu{}^\alpha\delta_\mu{}^\beta. \tag{6.83}$$

This is the antisymmetrization: differentiating F, which is itself an antisymmetrized derivative of A, returns an antisymmetrized pair of Kronecker deltas.

Step 3: contract. Insert (6.83) and contract the deltas against $F^{\mu\nu}$:

$$\frac{\partial \mathcal{L}}{\partial(\partial_\alpha A_\beta)} = -\tfrac{1}{2}\,F^{\mu\nu}(\delta_\mu{}^\alpha\delta_\nu{}^\beta - \delta_\nu{}^\alpha\delta_\mu{}^\beta) = -\tfrac{1}{2}(F^{\alpha\beta} - F^{\beta\alpha}) = -\tfrac{1}{2}(F^{\alpha\beta} + F^{\alpha\beta}) = -F^{\alpha\beta}, \tag{6.84}$$

where the last step used the antisymmetry $F^{\beta\alpha} = -F^{\alpha\beta}$. The two terms of the antisymmetrized delta combine to cancel the $\frac{1}{2}$ exactly — the factor and the antisymmetrization conspire to give the clean result $\partial\mathcal{L}/\partial(\partial_\alpha A_\beta) = -F^{\alpha\beta}$.

Step 4: assemble. With $\partial\mathcal{L}/\partial A_\beta = 0$, the Euler–Lagrange equation (6.81) becomes

$$\partial_\alpha(-F^{\alpha\beta}) = 0 \quad\Longrightarrow\quad \boxed{\partial_\mu F^{\mu\nu} = 0}. \tag{6.85}$$

These are the source-free inhomogeneous Maxwell equations. If we add a coupling to an external conserved current, $\mathcal{L} \to \mathcal{L} - A_\mu J^\mu$, the only change is $\partial\mathcal{L}/\partial A_\beta = -J^\beta$, and (6.81) gives

$$\partial_\alpha(-F^{\alpha\beta}) - (-J^\beta) = 0 \quad\Longrightarrow\quad \boxed{\partial_\mu F^{\mu\nu} = J^\nu}, \tag{6.86}$$

the inhomogeneous Maxwell equations with a source: $\nu = 0$ is Gauss's law $\nabla\cdot\mathbf{E} = \rho$, and $\nu = i$ is the Ampère–Maxwell law. Consistency demands $\partial_\nu J^\nu = 0$: contracting (6.86) with ∂_ν gives $\partial_\nu\partial_\mu F^{\mu\nu} = 0$ automatically, because $\partial_\nu\partial_\mu$ is symmetric while $F^{\mu\nu}$ is antisymmetric, so the current's conservation is not an extra assumption but a built-in requirement of the field equation.

The Bianchi identity: the homogeneous equations for free

The two equations (6.86) are only half of Maxwell's system. The other half — $\nabla\cdot\mathbf{B} = 0$ and Faraday's law $\nabla\times\mathbf{E} + \partial_t\mathbf{B} = 0$ — are *not* equations of motion at all. They follow identically, off-shell, from the mere fact that F is built as $F = dA$, i.e. from the definition (6.79). Form the totally antisymmetrized derivative

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu}. \tag{6.87}$$

Substitute $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ into each term:

$$\begin{aligned}
\partial_\lambda F_{\mu\nu} &= \partial_\lambda\partial_\mu A_\nu - \partial_\lambda\partial_\nu A_\mu,\\
\partial_\mu F_{\nu\lambda} &= \partial_\mu\partial_\nu A_\lambda - \partial_\mu\partial_\lambda A_\nu,\\
\partial_\nu F_{\lambda\mu} &= \partial_\nu\partial_\lambda A_\mu - \partial_\nu\partial_\mu A_\lambda.
\end{aligned} \tag{6.88}$$

Adding the three lines, the six terms cancel in pairs, each pair related by the commuting of partial derivatives $\partial_a\partial_b = \partial_b\partial_a$. Hence

$$\boxed{\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0}, \tag{6.89}$$

the *Bianchi identity*, holding for any A_μ whatsoever, on-shell or off. It is an identity, not a dynamical law: it encodes that $\mathbf{E}$ and $\mathbf{B}$ descend from a single potential. In differential-form language (6.89) is just $dF = d(dA) = 0$, the statement $d^2 = 0$. The split is conceptually clean: the homogeneous Maxwell equations are kinematics (they say $F = dA$), the inhomogeneous ones are dynamics (they come from $\delta S = 0$).

The second structural surprise: gauge redundancy and a degenerate operator

We turn to the feature that will reverberate through the entire book. The combination (6.79) is invariant under shifting the potential by the gradient of an arbitrary scalar function $\chi(x)$:

$$A_\mu(x) \longrightarrow A_\mu(x) + \partial_\mu \chi(x). \tag{6.90}$$

The verification is immediate and worth seeing in full:

$$F_{\mu\nu} \longrightarrow \partial_\mu(A_\nu + \partial_\nu\chi) - \partial_\nu(A_\mu + \partial_\mu\chi) = F_{\mu\nu} + (\partial_\mu\partial_\nu\chi - \partial_\nu\partial_\mu\chi) = F_{\mu\nu}, \tag{6.91}$$

the added piece vanishing because partial derivatives commute. Since $F_{\mu\nu}$ is unchanged, so is $\mathcal{L} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$: the transformation (6.90) is an *exact* symmetry of the Maxwell action, for *every* function $\chi(x)$. This is a redundancy, not a physical symmetry: it does not relate distinct physical states but rather identifies many potentials A_μ as descriptions of the *same* electromagnetic field. The physical content is $F_{\mu\nu}$; the potential carries a surplus of unphysical, longitudinal information.

This redundancy has a sharp algebraic fingerprint: the Euler–Lagrange operator is *non-invertible*. Expand the left-hand side of the source-free equation (6.85) in terms of the potential,

$$\partial_\mu F^{\mu\nu} = \partial_\mu(\partial^\mu A^\nu - \partial^\nu A^\mu) = \Box A^\nu - \partial^\nu(\partial_\mu A^\mu) = (\eta^{\nu\mu}\Box - \partial^\nu\partial^\mu)A_\mu \equiv M^{\nu\mu}A_\mu, \tag{6.92}$$

which defines the second-order differential operator $M^{\nu\mu} = \eta^{\nu\mu}\Box - \partial^\nu\partial^\mu$. In momentum space, with $A_\mu(x) \propto e^{-ik\cdot x}$ so that $\partial_\mu \to -ik_\mu$, this operator becomes the matrix

$$M^{\nu\mu}(k) = -(\eta^{\nu\mu}k^2 - k^\nu k^\mu). \tag{6.93}$$

Now test it on a pure-gauge configuration. A gauge transformation (6.90) with $\chi \propto e^{-ik\cdot x}$ shifts A_μ by $\partial_\mu\chi \propto -ik_\mu\chi$ — that is, by a vector *proportional to* k_μ (a "longitudinal" mode). Acting on k_μ,

$$M^{\nu\mu}(k)\,k_\mu = -(\eta^{\nu\mu}k^2 - k^\nu k^\mu)k_\mu = -(k^\nu k^2 - k^\nu k^2) = 0. \tag{6.94}$$

The operator *annihilates* every pure-gauge mode. Its kernel is non-empty, so $M^{\nu\mu}$ is not invertible: there is no matrix $G_{\nu\rho}$ with $M^{\nu\mu}G_{\mu\rho} = \delta^\nu{}_\rho$. The equation

(6.94) is, of course, the same statement as gauge invariance (6.90): the directions in field space along which the operator does nothing are exactly the directions a gauge transformation moves you, the directions along which the action is flat (Fig. 6.2).

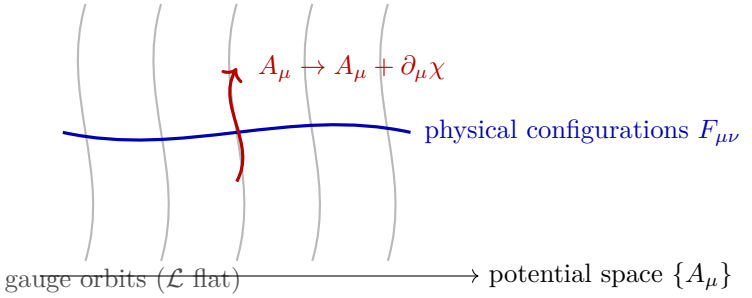


Figure 6.2: Gauge redundancy as flatness of the action. Each grey fibre is a gauge orbit, the set of potentials $A_\mu + \partial_\mu \chi$ related by (6.90); the Maxwell Lagrangian is constant along it, so the Euler–Lagrange operator $M^{\nu\mu}$ has no curvature in the fibre direction and annihilates it, equation (6.94). Only the transverse, physical data — the field strength $F_{\mu\nu}$ — carry dynamics. Inverting $M^{\nu\mu}$ (to define a propagator) requires first picking one representative per orbit: gauge fixing.

Physical Insight

Non-invertibility of $M^{\nu\mu}$ is not a technical nuisance to be patched; it is the classical seed of every gauge subtlety in this book. To define a photon propagator one must invert the kinetic operator, and one cannot invert an operator with a kernel. The cure — choosing one representative from each gauge orbit, i.e. *gauge fixing* — is forced on us already at the classical level and becomes unavoidable at quantization (Chapter 13, and in the non-abelian case the Faddeev–Popov construction of Chapter 26). Equivalently, in the Hamiltonian language of the next section, the same redundancy appears as a *constraint*: the momentum conjugate to A_0 vanishes identically, $\pi^0 = \partial\mathcal{L}/\partial\dot{A}_0 = -F^{00} = 0$, exactly as the Dirac momentum (6.78) was constrained. Two fields, two constraints, one moral: the naive Legendre transform fails whenever the Lagrangian is degenerate.

The cast is now complete. We have, derived from first principles and on equal footing, the real scalar (Section 6.2), the complex scalar with its global $U(1)$ (6.66), the Dirac field with its first-order action (6.77), and the Maxwell field with its gauge redundancy (6.90). These are the protagonists of the entire book, and their classical Lagrangians and equations of motion are fixed here once and for all. Two of them — the first-order Dirac action and the degenerate Maxwell action — arrive carrying constraints that the ordinary Legendre transform cannot

digest. It is to that transform, and to the constrained phase space these two surprises demand, that Section 6.4 now turns; the gauge principle that will weld the complex-scalar phase symmetry and the Maxwell field into a single structure waits for Chapter 8 and QED proper.

6.4 The Hamiltonian Formulation of Fields

The previous two sections gave us the dynamics of fields in their most economical form: a single scalar functional, the action, whose stationarity encodes the entire content of the equations of motion. From it we read off the Klein–Gordon equation for the scalar, the Dirac equation for the spinor, and Maxwell's equations for the vector, each as an Euler–Lagrange condition (Section 6.2). The Lagrangian picture is supremely well suited to relativity, because the action is a Lorentz scalar and the symmetry of a theory is visible at a glance. But it is not the picture in which one quantizes. Quantization—the business of Part III—needs a notion of *state at an instant*, of the data one specifies on a spatial slice and then propagates forward, and of the algebra obeyed by the basic observables on that slice. That is the Hamiltonian picture. It trades the manifest covariance of the action for a phase space: a configuration variable and its conjugate momentum at each point of space, evolving under a Hamiltonian, with a Poisson bracket that measures how observables fail to commute under canonical flow.

This section performs that trade for fields. We define the canonical momentum density by a Legendre transform in the time derivative, build the Hamiltonian, derive Hamilton's field equations from the variational principle and check that they reproduce the Euler–Lagrange equations on the sector where the transform is honest, and then install the equal-time Poisson-bracket algebra. The payoff is twofold. First, we obtain for the real scalar—and hence for ϕ^4 theory—the exact classical objects that Chapter 9 will promote to operators and canonical commutators: the Hamiltonian density and the bracket $\{\phi(\mathbf{x}), \pi(\mathbf{y})\}_{\mathrm{P}} = \delta^3(\mathbf{x}-\mathbf{y})$. Second, we discover that the construction is not automatic. The Legendre transform presupposes that the momenta can be solved for the velocities, and for two of our three protagonists—the first-order Dirac field and the gauge-degenerate Maxwell field—that presupposition fails. Isolating exactly where and how it fails is the work of Section 6.5; here we lay the regular-case machinery cleanly enough that the failure, when it comes, is unmistakable.

6.4.1 The field Legendre transform and the canonical momentum density

In point-particle mechanics one passes from $L(q, \dot{q})$ to $H(q, p)$ by defining the conjugate momentum $p = \partial L/\partial \dot{q}$ and Legendre-transforming in $\dot{q}$, replacing the velocity by the momentum as an independent coordinate (Section 4.5.1). A

classical field theory is a mechanical system with a continuous label: instead of finitely many coordinates $q_i(t)$ we have a field value $\phi_a(\mathbf{x}, t)$ at every point $\mathbf{x}$ of space, the spatial position playing the role of the index i. The Legendre transform is taken in the time derivative $\dot{\phi}_a \equiv \partial_0 \phi_a$ alone—time is singled out the moment we choose a frame in which to define "an instant"—while the spatial gradients $\partial_i \phi_a$ are spectators, carried along as part of the configuration.

Throughout, the dynamics derives from a local Lagrangian density

$$\mathcal{L} = \mathcal{L}(\phi_a,\, \partial_0 \phi_a,\, \partial_i \phi_a), \tag{6.95}$$

a function of the fields, their time derivatives, and their spatial gradients, with the index a running over all field components in play (for several real scalars, the four components of A^μ, the spinor components of ψ, and so on). We isolate the time derivative because the Hamiltonian formalism does.

Definition 6.13 (Canonical momentum density)**.** The *canonical momentum density conjugate to* ϕ_a is

$$\pi^a(x) \;\equiv\; \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_a)} \;=\; \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a}\,, \tag{6.96}$$

the partial derivative of the Lagrangian density with respect to the velocity $\dot{\phi}_a$ at the spacetime point x, taken at fixed ϕ_a and fixed spatial gradients $\partial_i \phi_a$.

The object $\pi^a(x)$ is a *density*: it carries the same mass dimension as $\mathcal{L}/\dot{\phi}_a$, and the quantity with the dimensions of a momentum is $\int \mathrm{d}^3x\, \pi^a$. The placement of the field index is a bookkeeping choice; we write π^a with an upper index to pair against ϕ_a and will lower or raise it freely when there is no ambiguity, since for the cases at hand the field-space metric is trivial.

For the Legendre transform to be invertible we must be able to read Eq. (6.96) backwards—to solve it for the velocities $\dot{\phi}_a$ as functions of $(\phi_b, \partial_i \phi_b, \pi^b)$. By the implicit function theorem this is possible, locally, precisely when the Jacobian of the map $\dot{\phi}_a \mapsto \pi^a$ is nonsingular.

Definition 6.14 (Regular and singular Lagrangians)**.** The *velocity Hessian* of $\mathcal{L}$ is the matrix in field-component space

$$W_{ab} \;\equiv\; \frac{\partial^2 \mathcal{L}}{\partial \dot{\phi}_a \partial \dot{\phi}_b}\,. \tag{6.97}$$

The Lagrangian is *regular* (or non-singular) at a configuration if $\det W_{ab} \neq 0$ there, and *singular* if $\det W_{ab} = 0$. On the regular sector the relation Eq. (6.96) can be inverted to give $\dot{\phi}_a = \dot{\phi}_a(\phi, \partial_i \phi, \pi)$.

Caution

Regularity is exactly the hypothesis that will fail for two of our three protagonists. The Dirac Lagrangian is *first order* in derivatives, so $\mathcal{L}$ is linear in $\dot{\psi}$ and $W_{ab} = 0$ identically; the Maxwell Lagrangian contains no $\dot{A}_0$ at all, so the momentum conjugate to A_0 vanishes identically and W_{ab} has a zero eigenvalue. In both cases Eq. (6.96) cannot be solved for all the velocities; it instead imposes *constraints* relating the π's and ϕ's. The whole of Section 6.5 is devoted to that situation. For the remainder of *this* section we assume a regular Lagrangian, which the real scalar—and ϕ^4 theory—satisfies.

With invertibility in hand we define the Hamiltonian by the field Legendre transform.

Definition 6.15 (Hamiltonian density and Hamiltonian)**.** For a regular Lagrangian the *Hamiltonian density* is

$$\mathcal{H} \;\equiv\; \pi^a\,\dot{\phi}_a - \mathcal{L}\,, \tag{6.98}$$

expressed as a function of $(\phi_a, \partial_i\phi_a, \pi^a)$ by using the inverse relation $\dot{\phi}_a = \dot{\phi}_a(\phi, \partial_i\phi, \pi)$ to eliminate every velocity. The *Hamiltonian* is its spatial integral over a constant-time slice,

$$H[\phi, \pi] \;=\; \int \mathrm{d}^3x\; \mathcal{H}(\phi_a(\mathbf{x}),\, \partial_i\phi_a(\mathbf{x}),\, \pi^a(\mathbf{x}))\,, \tag{6.99}$$

a functional of the field configuration and the momentum configuration on that slice.

Two points deserve emphasis before we differentiate anything. First, $\mathcal{H}$ is built to have *no explicit dependence on the velocities*: every $\dot{\phi}_a$ has been traded for a π^a. This is the entire purpose of the Legendre transform, and it is what makes the variation in the next subsection collapse so cleanly. Second, H lives on a Cauchy slice. The pair $(\phi_a(\mathbf{x}), \pi^a(\mathbf{x}))$ specified at one instant constitutes a point in the field-theoretic phase space, and the Hamiltonian is a number assigned to each such point. Whereas the action is a functional of an entire spacetime history, the Hamiltonian is a functional of instantaneous data—which is exactly the form one needs to speak of a state and its evolution.

6.4.2 Hamilton's field equations

We now show that the single first-order-in-time system of Hamilton's equations is equivalent, on the regular sector, to the second-order Euler–Lagrange equations. The derivation is a variation of H carried with all the algebra in view; the key step is the cancellation that the Legendre transform was designed to produce.

Recall (Section 4.1.2) that for a local functional $F[\phi,\pi]=\int \mathrm{d}^3x\, f(\phi_a,\partial_i\phi_a,\pi^a)$ the functional derivatives are the coefficients in

$$\delta F=\int \mathrm{d}^3x\left[\frac{\delta F}{\delta\phi_a}\,\delta\phi_a+\frac{\delta F}{\delta\pi^a}\,\delta\pi^a\right], \tag{6.100}$$

where, after any integration by parts that moves derivatives off the variations, the boundary terms are discarded under the standing assumption that variations have compact spatial support (or fall off fast enough that surface integrals at spatial infinity vanish). For a density depending on the field and its first gradient this gives the explicit rule

$$\frac{\delta F}{\delta\phi_a}=\frac{\partial f}{\partial\phi_a}-\partial_i\!\left(\frac{\partial f}{\partial(\partial_i\phi_a)}\right), \qquad \frac{\delta F}{\delta\pi^a}=\frac{\partial f}{\partial\pi^a}, \tag{6.101}$$

the second being simpler because $\mathcal{H}$ for our cases contains no spatial derivatives of π^a.

Varying the Hamiltonian. Take the differential of Eq. (6.98) treating $\mathcal{H}$ momentarily as a function of all of $\phi_a,\partial_i\phi_a,\pi^a$ *and* the velocities $\dot\phi_a$, before imposing the inverse relation. Using the definition $\mathcal{H}=\pi^a\dot\phi_a-\mathcal{L}$ and the chain rule,

$$\delta\mathcal{H}=\dot\phi_a\,\delta\pi^a+\pi^a\,\delta\dot\phi_a-\frac{\partial\mathcal{L}}{\partial\phi_a}\,\delta\phi_a-\frac{\partial\mathcal{L}}{\partial(\partial_i\phi_a)}\,\delta(\partial_i\phi_a)-\frac{\partial\mathcal{L}}{\partial\dot\phi_a}\,\delta\dot\phi_a\,. \tag{6.102}$$

The defining relation $\pi^a=\partial\mathcal{L}/\partial\dot\phi_a$ from Eq. (6.96) makes the second and last terms cancel identically:

$$\pi^a\,\delta\dot\phi_a-\frac{\partial\mathcal{L}}{\partial\dot\phi_a}\,\delta\dot\phi_a=0\,. \tag{6.103}$$

This is the heart of the Legendre transform: the velocity variation drops out, and $\mathcal{H}$ is genuinely a function of $(\phi_a,\partial_i\phi_a,\pi^a)$ with no residual $\dot\phi_a$ dependence. What survives is

$$\delta\mathcal{H}=\dot\phi_a\,\delta\pi^a-\frac{\partial\mathcal{L}}{\partial\phi_a}\,\delta\phi_a-\frac{\partial\mathcal{L}}{\partial(\partial_i\phi_a)}\,\delta(\partial_i\phi_a)\,. \tag{6.104}$$

Now integrate over the slice to form $\delta H=\int\mathrm{d}^3x\,\delta\mathcal{H}$ and integrate the last term by parts, $-\partial\mathcal{L}/\partial(\partial_i\phi_a)\,\partial_i(\delta\phi_a)\to+\partial_i(\partial\mathcal{L}/\partial(\partial_i\phi_a))\,\delta\phi_a$, dropping the surface term:

$$\delta H=\int\mathrm{d}^3x\left[\dot\phi_a\,\delta\pi^a+\left(-\frac{\partial\mathcal{L}}{\partial\phi_a}+\partial_i\frac{\partial\mathcal{L}}{\partial(\partial_i\phi_a)}\right)\delta\phi_a\right]. \tag{6.105}$$

Comparing Eq. (6.105) term by term with the definition Eq. (6.100) of the functional derivatives, we read off

$$\frac{\delta H}{\delta\pi^a}=\dot\phi_a\,, \qquad \frac{\delta H}{\delta\phi_a}=-\frac{\partial\mathcal{L}}{\partial\phi_a}+\partial_i\frac{\partial\mathcal{L}}{\partial(\partial_i\phi_a)}\,. \tag{6.106}$$

Closing the loop with the Euler–Lagrange equation. The first relation in Eq. (6.106) is already the first half of Hamilton's equations. For the second half we invoke the Euler–Lagrange equation of Section 6.2, written with its time and space derivatives split apart:

$$\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} - \frac{\partial \mathcal{L}}{\partial \phi_a} = \partial_0 \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a} + \partial_i \frac{\partial \mathcal{L}}{\partial(\partial_i \phi_a)} - \frac{\partial \mathcal{L}}{\partial \phi_a} = 0 \,. \tag{6.107}$$

The first term is $\partial_0 \pi^a = \dot{\pi}^a$ by Eq. (6.96). Solving Eq. (6.107) for it,

$$\dot{\pi}^a = \frac{\partial \mathcal{L}}{\partial \phi_a} - \partial_i \frac{\partial \mathcal{L}}{\partial(\partial_i \phi_a)} \,, \tag{6.108}$$

and recognizing the right-hand side as the negative of $\delta H / \delta \phi_a$ from Eq. (6.106), we obtain the companion equation. Collecting the two:

Theorem 6.16 (Hamilton's field equations)**.** *For a regular Lagrangian, the field configuration $(\phi_a(\mathbf{x}, t), \pi^a(\mathbf{x}, t))$ extremizes the action if and only if it satisfies the first-order system*

$$\dot{\phi}_a = \frac{\delta H}{\delta \pi^a} \,, \qquad \dot{\pi}^a = -\frac{\delta H}{\delta \phi_a} \,. \tag{6.109}$$

These are equivalent to the Euler–Lagrange equations Eq. (6.107) of Section 6.2: the first member is the inverse Legendre relation $\dot{\phi}_a = \dot{\phi}_a(\phi, \partial_i \phi, \pi)$, and the second member is the Euler–Lagrange equation rewritten with $\dot{\pi}^a = \partial_0(\partial \mathcal{L} / \partial \dot{\phi}_a)$.

Proof. The forward direction was just carried: Eq. (6.106) gives $\dot{\phi}_a = \delta H / \delta \pi^a$ directly from the Legendre structure, and Eq. (6.108) converts the Euler–Lagrange equation into $\dot{\pi}^a = -\delta H / \delta \phi_a$. Conversely, suppose Eq. (6.109) holds. The first equation reproduces the defining relation between velocity and momentum (it is $\delta H / \delta \pi^a = \dot{\phi}_a$, which by Eq. (6.106) is an identity of the Legendre transform), so $\pi^a = \partial \mathcal{L} / \partial \dot{\phi}_a$ is consistent. Differentiating that in time gives $\dot{\pi}^a = \partial_0(\partial \mathcal{L} / \partial \dot{\phi}_a)$; substituting into the second Hamilton equation and using the expression Eq. (6.106) for $\delta H / \delta \phi_a$ yields $\partial_0(\partial \mathcal{L} / \partial \dot{\phi}_a) = \partial \mathcal{L} / \partial \phi_a - \partial_i(\partial \mathcal{L} / \partial(\partial_i \phi_a))$, which rearranges to the Euler–Lagrange equation Eq. (6.107). The two formulations are therefore logically equivalent on the regular sector. □

Physical Insight

The two pictures repackage the same content. The Lagrangian description is one second-order equation per field component—the configuration ϕ_a and its acceleration. The Hamiltonian description is two first-order equations—the configuration ϕ_a and its conjugate momentum π^a, each evolving by a first derivative. The doubling of the number of variables is exactly compensated by the halving of the order, and the phase-space pair (ϕ_a, π^a)

is the natural arena because it is the data one must specify on a slice to fix the future. It is also the data one quantizes: in Chapter 9 the slice pair (ϕ, π) becomes a conjugate operator pair, and Eq. (6.109) becomes the Heisenberg equation of motion.

6.4.3 The equal-time Poisson bracket

The Hamiltonian formalism becomes a calculus once we install a bracket on functionals of the phase-space data. As in mechanics, the Poisson bracket is the antisymmetric pairing built from the symplectic structure of phase space; for fields it is defined through functional derivatives on a fixed time slice.

Definition 6.17 (Equal-time Poisson bracket)**.** Let $F[\phi, \pi]$ and $G[\phi, \pi]$ be functionals of the phase-space data on a constant-time slice. Their *equal-time Poisson bracket* is

$$\{F, G\}_{\mathrm{P}} \equiv \int \mathrm{d}^3 z \left[\frac{\delta F}{\delta \phi_a(\mathbf{z})} \frac{\delta G}{\delta \pi^a(\mathbf{z})} - \frac{\delta F}{\delta \pi^a(\mathbf{z})} \frac{\delta G}{\delta \phi_a(\mathbf{z})} \right], \tag{6.110}$$

with summation over the repeated field index a and the integration running over the slice. The bracket is bilinear, antisymmetric ($\{F, G\}_{\mathrm{P}} = -\{G, F\}_{\mathrm{P}}$), satisfies the Leibniz rule $\{FG, K\}_{\mathrm{P}} = F\{G, K\}_{\mathrm{P}} + \{F, K\}_{\mathrm{P}} G$, and obeys the Jacobi identity $\{F, \{G, K\}_{\mathrm{P}}\}_{\mathrm{P}} + \{G, \{K, F\}_{\mathrm{P}}\}_{\mathrm{P}} + \{K, \{F, G\}_{\mathrm{P}}\}_{\mathrm{P}} = 0$, inherited from the corresponding properties of partial derivatives.

The basic objects of the theory are the fields and momenta evaluated at a point, which we treat as functionals via $\phi_a(\mathbf{x}) = \int \mathrm{d}^3 z\, \delta^3(\mathbf{x} - \mathbf{z})\, \phi_a(\mathbf{z})$. Their functional derivatives are sharp:

$$\frac{\delta \phi_a(\mathbf{x})}{\delta \phi_b(\mathbf{z})} = \delta_a{}^b\, \delta^3(\mathbf{x} - \mathbf{z}), \qquad \frac{\delta \pi^a(\mathbf{x})}{\delta \pi^b(\mathbf{z})} = \delta^a{}_b\, \delta^3(\mathbf{x} - \mathbf{z}), \tag{6.111}$$

with $\delta \phi_a(\mathbf{x})/\delta \pi^b(\mathbf{z}) = 0$ and $\delta \pi^a(\mathbf{x})/\delta \phi_b(\mathbf{z}) = 0$ because ϕ and π are independent coordinates on phase space. Substituting these into the definition Eq. (6.110) gives the fundamental algebra.

Theorem 6.18 (Canonical Poisson brackets)**.** *The equal-time Poisson brackets of the basic phase-space variables are*

$$\Big\{\phi_a(\mathbf{x}), \pi^b(\mathbf{y})\Big\}_P = \delta_a{}^b\, \delta^3(\mathbf{x} - \mathbf{y}), \qquad \{\phi_a(\mathbf{x}), \phi_b(\mathbf{y})\}_P = 0, \qquad \Big\{\pi^a(\mathbf{x}), \pi^b(\mathbf{y})\Big\}_P = 0, \tag{6.112}$$

all evaluated at equal times $x^0 = y^0$.

Proof. For the mixed bracket, set $F = \phi_a(\mathbf{x})$ and $G = \pi^b(\mathbf{y})$ in Eq. (6.110). The second term vanishes because $\delta\phi_a/\delta\pi = 0$, and the first uses Eq. (6.111):

$$\big\{\phi_a(\mathbf{x}), \pi^b(\mathbf{y})\big\}_{\mathrm{P}} = \int \mathrm{d}^3 z \, \frac{\delta\phi_a(\mathbf{x})}{\delta\phi_c(\mathbf{z})} \, \frac{\delta\pi^b(\mathbf{y})}{\delta\pi^c(\mathbf{z})} = \int \mathrm{d}^3 z \; \delta_a{}^c \, \delta^3(\mathbf{x}-\mathbf{z}) \, \delta^b{}_c \, \delta^3(\mathbf{y}-\mathbf{z}) = \delta_a{}^b \, \delta^3(\mathbf{x}-\mathbf{y}). \tag{6.113}$$

For $\{\phi_a(\mathbf{x}), \phi_b(\mathbf{y})\}_{\mathrm{P}}$ both terms in Eq. (6.110) contain a factor $\delta\phi/\delta\pi = 0$, and likewise for $\big\{\pi^a, \pi^b\big\}_{\mathrm{P}}$ both contain $\delta\pi/\delta\phi = 0$; both brackets vanish. □

These three relations are the classical skeleton of canonical quantization. In Chapter 9 the prescription $\{\,\cdot\,,\,\cdot\,\}_{\mathrm{P}} \to -i\,[\,\cdot\,,\,\cdot\,]$ turns Eq. (6.112) into the equal-time commutator $\big[\hat{\phi}_a(\mathbf{x}), \hat{\pi}^b(\mathbf{y})\big] = i\,\delta_a{}^b\,\delta^3(\mathbf{x}-\mathbf{y})$ of the free scalar field—the foundational commutation relation of the entire quantum theory. The Dirac case will, by contrast, demand anticommutators, because its fields are Grassmann-valued and its bracket is graded; that subtlety is opened classically in Section 6.5 and discharged at quantization in Chapter 11.

Hamilton's equations and time evolution in bracket form. The bracket also packages the dynamics. Specializing $G = H$ in Eq. (6.110) with $F = \phi_a(\mathbf{x})$ and using Eq. (6.106),

$$\{\phi_a(\mathbf{x}), H\}_{\mathrm{P}} = \int \mathrm{d}^3 z \, \frac{\delta\phi_a(\mathbf{x})}{\delta\phi_b(\mathbf{z})} \, \frac{\delta H}{\delta\pi^b(\mathbf{z})} = \frac{\delta H}{\delta\pi^a(\mathbf{x})} = \dot{\phi}_a(\mathbf{x}), \tag{6.114}$$

and similarly $\{\pi^a(\mathbf{x}), H\}_{\mathrm{P}} = -\delta H/\delta\phi_a(\mathbf{x}) = \dot{\pi}^a(\mathbf{x})$. So Hamilton's equations Eq. (6.109) read

$$\dot{\phi}_a = \{\phi_a, H\}_{\mathrm{P}}\,, \qquad \dot{\pi}^a = \{\pi^a, H\}_{\mathrm{P}}\,. \tag{6.115}$$

This is no accident of the basic variables: the same statement holds for any observable.

Theorem 6.19 (The Hamiltonian generates time evolution)**.** *Let $F[\phi, \pi; t]$ be any (possibly explicitly time-dependent) local functional of the phase-space data. Along a solution of Hamilton's equations its total time derivative is*

$$\frac{dF}{dt} = \{F, H\}_{P} + \frac{\partial F}{\partial t}\,, \tag{6.116}$$

where $\partial F/\partial t$ is the explicit time dependence at fixed (ϕ, π). In particular a functional with no explicit time dependence is conserved if and only if it Poisson-commutes with H.

Proof. Write $F = \int \mathrm{d}^3 x \, f(\phi_a, \partial_i\phi_a, \pi^a; t)$. The total time derivative collects the change through the time dependence of the fields and the explicit dependence:

$$\frac{dF}{dt} = \int \mathrm{d}^3 x \left[\frac{\delta F}{\delta\phi_a(\mathbf{x})}\,\dot{\phi}_a(\mathbf{x}) + \frac{\delta F}{\delta\pi^a(\mathbf{x})}\,\dot{\pi}^a(\mathbf{x})\right] + \frac{\partial F}{\partial t}\,, \tag{6.117}$$

where the chain rule with functional derivatives already accounts for the gradient dependence after integration by parts (Eq. (6.101)). Now insert Hamilton's equations Eq. (6.109), $\dot{\phi}_a = \delta H/\delta\pi^a$ and $\dot{\pi}^a = -\delta H/\delta\phi_a$:

$$\frac{dF}{dt} = \int \mathrm{d}^3x \left[\frac{\delta F}{\delta\phi_a}\frac{\delta H}{\delta\pi^a} - \frac{\delta F}{\delta\pi^a}\frac{\delta H}{\delta\phi_a}\right] + \frac{\partial F}{\partial t} = \{F, H\}_{\mathrm{P}} + \frac{\partial F}{\partial t}\,, \tag{6.118}$$

the bracket integral being exactly Eq. (6.110). If $\partial F/\partial t = 0$ then $dF/dt = \{F, H\}_{\mathrm{P}}$, which vanishes iff $\{F, H\}_{\mathrm{P}} = 0$. □

Equation (6.116) is the classical template for the Heisenberg equation $dF/dt = i\,[H, F] + \partial F/\partial t$ of Chapter 9, and it is the engine that will let Chapter 7 express every conserved charge as the generator, through the bracket, of its symmetry: charges generate transformations precisely because $\{Q, \cdot\,\}_{\mathrm{P}}$ acts as the corresponding infinitesimal flow on phase space, with the Hamiltonian being the charge that generates time translation.

6.4.4 Worked example: the scalar field and ϕ^4 theory

We now carry the entire construction explicitly for the real scalar, producing the phase-space data that Chapter 9 quantizes. This continues ϕ^4 theory from Section 6.2.

Worked Example 6.20 (Hamiltonian and brackets of the real scalar)**.** Recall the real scalar Lagrangian density (Section 6.2),

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 = \tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{6.119}$$

where in the second form we used the mostly-minus metric (Chapter A) to write $\partial_\mu\phi\,\partial^\mu\phi = \dot{\phi}^2 - (\nabla\phi)^2$, with $(\nabla\phi)^2 = \partial_i\phi\,\partial_i\phi \geq 0$ the squared spatial gradient. We carry out the four steps: momentum, Hamiltonian, equations of motion, and brackets.

Step 1: canonical momentum. By Eq. (6.96),

$$\pi = \frac{\partial\mathcal{L}}{\partial\dot{\phi}} = \dot{\phi}\,. \tag{6.120}$$

The velocity Hessian is $W = \partial^2\mathcal{L}/\partial\dot{\phi}^2 = 1 \neq 0$, so the scalar is regular and Eq. (6.120) inverts trivially to $\dot{\phi} = \pi$.

Step 2: Hamiltonian density. Substituting into Eq. (6.98),

$$\begin{aligned}\mathcal{H} &= \pi\dot{\phi} - \mathcal{L} = \dot{\phi}^2 - \left[\tfrac{1}{2}\dot{\phi}^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right] \\ &= \tfrac{1}{2}\dot{\phi}^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\,,\end{aligned} \tag{6.121}$$

and eliminating the velocity with $\dot\phi = \pi$,

$$\boxed{\mathcal{H} = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4} \qquad H = \int \mathrm{d}^3x\, \mathcal{H}\,. \tag{6.122}$$

Every term is manifestly non-negative for $\lambda \geq 0$: the kinetic energy $\frac{1}{2}\pi^2$, the gradient ("shear") energy $\frac{1}{2}(\nabla\phi)^2$, the mass ("rest") energy $\frac{1}{2}m^2\phi^2$, and the self-interaction $\frac{\lambda}{4!}\phi^4$. The Hamiltonian is bounded below, with minimum at $\phi = \pi = 0$—the classical vacuum—which is the property that makes the quantized theory of Chapter 9 have a stable ground state.

Step 3: equations of motion from Hamilton's equations. The first Hamilton equation reproduces the momentum relation,

$$\dot\phi = \frac{\delta H}{\delta \pi} = \pi\,. \tag{6.123}$$

For the second we need $\delta H/\delta\phi$. Only the gradient, mass, and quartic terms depend on ϕ; applying the rule Eq. (6.101) to the density Eq. (6.122), with $\partial\mathcal{H}/\partial(\partial_i\phi) = \partial_i\phi$,

$$\frac{\delta H}{\delta\phi} = \frac{\partial\mathcal{H}}{\partial\phi} - \partial_i\frac{\partial\mathcal{H}}{\partial(\partial_i\phi)} = (m^2\phi + \tfrac{\lambda}{3!}\phi^3) - \partial_i(\partial_i\phi) = -\nabla^2\phi + m^2\phi + \frac{\lambda}{3!}\phi^3\,, \tag{6.124}$$

where $\partial(\frac{\lambda}{4!}\phi^4)/\partial\phi = \frac{4\lambda}{4!}\phi^3 = \frac{\lambda}{3!}\phi^3$. Hence the second Hamilton equation reads

$$\dot\pi = -\frac{\delta H}{\delta\phi} = \nabla^2\phi - m^2\phi - \frac{\lambda}{3!}\phi^3\,. \tag{6.125}$$

Differentiating $\dot\phi = \pi$ once more in time gives $\ddot\phi = \dot\pi$, so

$$\ddot\phi - \nabla^2\phi + m^2\phi = -\frac{\lambda}{3!}\phi^3 \quad\Longleftrightarrow\quad (\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3\,, \tag{6.126}$$

using $\Box = \partial_0^2 - \nabla^2 = \ddot{} - \nabla^2$. This is exactly the ϕ^4 equation of motion derived in Section 6.2, recovered here from the Hamiltonian side—a concrete check of Theorem 6.16.

Step 4: the same equations from the Poisson bracket. Using Eq. (6.115) with the fundamental brackets Eq. (6.112), the dynamics follows without any new computation. The bracket with H of the field is

$$\dot\phi(\mathbf{x}) = \{\phi(\mathbf{x}), H\}_{\mathrm{P}} = \int \mathrm{d}^3z\, \frac{\delta\phi(\mathbf{x})}{\delta\phi(\mathbf{z})}\frac{\delta H}{\delta\pi(\mathbf{z})} = \frac{\delta H}{\delta\pi(\mathbf{x})} = \pi(\mathbf{x}), \tag{6.127}$$

and of the momentum,

$$\dot\pi(\mathbf{x}) = \{\pi(\mathbf{x}), H\}_{\mathrm{P}} = -\int \mathrm{d}^3z\, \frac{\delta\pi(\mathbf{x})}{\delta\pi(\mathbf{z})}\frac{\delta H}{\delta\phi(\mathbf{z})} = -\frac{\delta H}{\delta\phi(\mathbf{x})} = \nabla^2\phi - m^2\phi - \frac{\lambda}{3!}\phi^3, \tag{6.128}$$

where the second line reuses Eq. (6.124). Eliminating π reproduces Eq. (6.126). The equation of motion is thus generated by the Poisson action of H, exactly as Theorem 6.19 promises.

The frozen ϕ^4 theory data. The objects this worked example delivers—the canonical momentum $\pi = \dot{\phi}$, the Hamiltonian density Eq. (6.122), and the equal-time bracket $\{\phi(\mathbf{x}), \pi(\mathbf{y})\}_{\mathrm{P}} = \delta^3(\mathbf{x} - \mathbf{y})$—are precisely the classical inputs that Chapter 9 promotes to operators to build the free scalar Fock space (set $\lambda = 0$) and that Chapters 14 and 15 perturb in λ for the first interacting amplitude.

It is worth pausing on the structure of H for the free theory. With $\lambda = 0$ the Hamiltonian Eq. (6.122) is quadratic, and the equations of motion Eq. (6.126) reduce to the linear Klein–Gordon equation $(\Box + m^2)\phi = 0$. A quadratic Hamiltonian with positive-definite kinetic and potential parts is, point by point in Fourier space, a collection of decoupled harmonic oscillators—one for each spatial momentum $\mathbf{k}$, with frequency $\omega_{\mathbf{k}} = \sqrt{\mathbf{k}^2 + m^2}$. That observation is the bridge to Chapter 9: quantizing the field is quantizing this continuum of oscillators, and the canonical bracket Eq. (6.112) is what makes each oscillator's position and momentum conjugate. The quartic term $\frac{\lambda}{4!}\phi^4$ couples the oscillators, which is why the interacting theory cannot be solved exactly and must be treated perturbatively.

6.4.5 Time translation as a canonical flow

We close by recording the conceptual statement that Theorem 6.19 encodes, because Chapter 7 builds directly on it. The map that advances every observable by an infinitesimal time δt,

$$F \;\longmapsto\; F + \delta t \,\frac{dF}{dt} = F + \delta t\, \{F, H\}_{\mathrm{P}} \qquad (\text{for } \partial_t F = 0), \tag{6.129}$$

is generated by the Hamiltonian through the Poisson bracket. In the language we will make systematic in the next chapter, H is the *conserved charge of time-translation symmetry*, and "conserved charge generates its symmetry via the bracket" is the classical content of Noether's theorem. Spatial translations will be generated by the field momentum P^i, rotations by the angular momentum, and so on; each is a particular phase-space functional whose bracket reproduces the corresponding infinitesimal transformation. The Hamiltonian is simply the time-translation member of this family, distinguished only because time is the direction along which we have chosen to evolve.

This completes the phase-space description of the regular scalar: a configuration–momentum pair on each slice, a Hamiltonian bounded below, first-order Hamilton equations equivalent to the Klein–Gordon/ϕ^4 dynamics, and a canonical bracket algebra ready to be quantized. Two of our three protagonists, however, refuse this clean passage. The Dirac field is first order in time, so its velocity Hessian

vanishes and Eq. (6.96) cannot be inverted; the Maxwell field has no $\dot{A}_0$ in its Lagrangian, so the momentum conjugate to A_0 vanishes identically. In both cases the Legendre transform returns not the velocities but *constraints* relating the phase-space variables. Identifying those primary constraints, and constructing the Hamiltonian on the surface they define, is the subject of Section 6.5.

6.5 Singular Lagrangians and Primary Constraints

The Hamiltonian construction of Section 6.4 rested on a step we performed without comment for the real scalar and which, we noted at the close of that section, cannot be performed at all for two of the three protagonists of this book. To pass from the Lagrangian to the Hamiltonian we defined the canonical momentum $\pi_a = \partial\mathcal{L}/\partial\dot{\phi}_a$ and then *solved* this relation for the velocities $\dot{\phi}_a$ as functions of the momenta, so that the Hamiltonian $\mathcal{H} = \pi_a\dot{\phi}_a - \mathcal{L}$ could be written on phase space, with no velocity left in sight. For the scalar that inversion was trivial: $\pi = \dot{\phi}$, so $\dot{\phi} = \pi$ and the substitution is immediate. But the Dirac Lagrangian is *linear* in the velocities $\dot{\psi}$, and the Maxwell Lagrangian contains *no* time derivative of A_0 at all. In neither case can $\dot{\phi}_a$ be recovered from π_a. The definition of the momentum, instead of being a relation to be inverted, becomes a *constraint*: an identity tying the momenta to the fields that holds before any equation of motion is imposed.

This is not a pathology to be regretted; it is the classical fingerprint of two of the deepest features of relativistic field theory. The Dirac constraint is the classical shadow of the spin-statistics connection — it forces the fermionic bracket to be an *anti*bracket, hence anticommutators on quantization. The Maxwell constraint is the classical shadow of gauge redundancy — A_0 is not a dynamical variable but a Lagrange multiplier, and the consistency of its (absent) dynamics is Gauss's law. Our task in this section is the honest classical groundwork: to define precisely what a degenerate Lagrangian is, to identify the primary constraints of the Dirac and Maxwell systems, to compute their constrained Hamiltonians with every term shown, and to name — at the level of precise statement, with the full machinery deferred — the systematic theory (the Dirac–Bergmann algorithm) that organizes the rest. We draw a sharp line between what is a classical fact, settled here, and what is genuinely a quantization choice, handed to Chapter 13 (the photon) and Chapter 26 (non-abelian gauge fields).

6.5.1 Degenerate Lagrangians and the Hessian criterion

We work with a collection of fields $\phi_a(x)$, the index a running over both field species and any internal/spinor components, with first-order Lagrangian density

$\mathcal{L}(\phi_a, \partial_\mu \phi_a)$. The canonical momentum conjugate to ϕ_a is

$$\pi^a(x) \;=\; \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a(x)}, \qquad \dot{\phi}_a \equiv \partial_0 \phi_a, \tag{6.130}$$

exactly as in Section 6.4. Whether (6.130) can be solved for the velocities is a question about a single matrix. Differentiating (6.130) once more with respect to a velocity,

$$\frac{\partial \pi^a}{\partial \dot{\phi}_b} \;=\; \frac{\partial^2 \mathcal{L}}{\partial \dot{\phi}_a \, \partial \dot{\phi}_b} \;\equiv\; W_{ab}, \tag{6.131}$$

defines the *Hessian* of the Lagrangian with respect to the velocities. By the implicit function theorem, the relations (6.130) can be inverted for $\dot{\phi}_b = \dot{\phi}_b(\phi, \pi)$ in a neighbourhood precisely when W_{ab} is invertible. This dichotomy is the organizing definition of the section.

Definition 6.21 (Regular and singular Lagrangians)**.** The Lagrangian $\mathcal{L}$ is *regular* (or *non-degenerate*) if its velocity Hessian (6.131) is everywhere invertible, $\det W_{ab} \neq 0$; then every velocity is a function of the canonical data and the Legendre transform to the Hamiltonian is well defined. It is *singular* (or *degenerate*) if $\det W_{ab} = 0$. In the singular case the momenta (6.130) are not all independent functions of the velocities: there exist relations

$$\chi_m(\phi, \pi) \;\approx\; 0, \qquad m = 1, \dots, M, \tag{6.132}$$

satisfied identically by the momenta as defined, without using any equation of motion.

Definition 6.22 (Primary constraint)**.** A relation of the form (6.132) that follows directly from the definition (6.130) of the momenta — i.e. that holds on the image of the map $(\phi, \dot{\phi}) \mapsto (\phi, \pi)$ *before* the equations of motion are imposed — is a *primary constraint*. The number of independent primary constraints equals the number of null eigenvectors of W_{ab}.

The symbol "$\approx$" is Dirac's *weak equality*: $\chi_m \approx 0$ means χ_m vanishes on the constraint surface in phase space carved out by all the constraints, but is *not* identically zero as a function on the unconstrained phase space, so it may have nonzero Poisson brackets with the canonical variables. The distinction matters: one may set $\chi_m = 0$ only *after* all brackets have been evaluated, never inside a bracket. We will keep this discipline scrupulously below.

The kernel of W_{ab} counts the constraints because each null eigenvector v^a (with $W_{ab} v^b = 0$) is a direction in velocity space along which $\mathcal{L}$ is at most linear; the corresponding combination $v_a \pi^a$ of momenta is then independent of all velocities, hence a function of the fields alone — precisely a relation of the form (6.132). The real scalar of ϕ^4 theory has $W = \partial^2[\tfrac{1}{2}\dot{\phi}^2 - \cdots]/\partial\dot{\phi}^2 = 1 \neq 0$: regular, no constraints, which is why Section 6.4 proceeded without incident. The two theories we now confront are at the opposite extreme.

Physical Insight

Degeneracy is never accidental in a theory that anyone writes down on purpose. It is the phase-space signature of structure: a first-order (Dirac) kinetic term means the field and its momentum are not independent — half of phase space is redundant — and a gauge symmetry (Maxwell) means some "coordinates" are pure redundancy with no conjugate dynamics at all. The constraint analysis is how the canonical formalism discovers, on its own, the spin-statistics and gauge structure that we built into the Lagrangian.

6.5.2 The Dirac field: a momentum that is a constraint

We continue QED at the classical level. The free Dirac Lagrangian density fixed in Section 6.3 is

$$\mathcal{L}_{\mathrm{D}} \;=\; \bar{\psi}\,(i\partial\!\!\!/ - m)\,\psi \;=\; i\,\bar{\psi}\,\gamma^0\dot{\psi} + i\,\bar{\psi}\,\gamma^i\partial_i\psi - m\,\bar{\psi}\psi, \tag{6.133}$$

where $\bar{\psi} = \psi^\dagger\gamma^0$ and the gamma algebra $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$ is the canon of Chapter C; the spinor field ψ and its components are *anticommuting* (Grassmann-valued) classical fields, the c-number-but-odd objects whose calculus is built from scratch in Chapter 12. The sum over $i = 1,2,3$ is the only place a spatial derivative enters; the single time derivative sits in the first term, and using $\bar{\psi}\gamma^0 = \psi^\dagger(\gamma^0)^2 = \psi^\dagger$ that term is simply $i\,\psi^\dagger\dot{\psi}$. The Lagrangian is therefore *linear* in $\dot{\psi}$ and contains no $\dot{\bar{\psi}}$ whatsoever. This is the maximally degenerate case, and the constraint structure can be read off by inspection.

Worked Example 6.23 (Dirac momenta, Hessian, and Hamiltonian)**.** We compute the canonical momenta of (6.133), exhibit the primary constraints, and Legendre-transform to the Hamiltonian, carrying every step.

Momenta. Treat ψ and $\psi^\dagger$ as independent fields (equivalently ψ and $\bar{\psi}$). For the anticommuting field we use the left derivative, with the overall sign convention fixed in Chapter 12; the result, on which all references agree, is

$$\pi_\psi = \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial\dot{\psi}} = \frac{\partial}{\partial\dot{\psi}}(i\,\psi^\dagger\dot{\psi}) = i\,\psi^\dagger, \qquad \pi_{\psi^\dagger} = \frac{\partial\mathcal{L}_{\mathrm{D}}}{\partial\dot{\psi}^\dagger} = 0, \tag{6.134}$$

the second because $\dot{\psi}^\dagger$ does not appear in (6.133) at all. Neither equation can be solved for a velocity. Both are primary constraints:

$$\chi_1 \equiv \pi_\psi - i\,\psi^\dagger \approx 0, \qquad \chi_2 \equiv \pi_{\psi^\dagger} \approx 0. \tag{6.135}$$

The momentum conjugate to ψ is *fixed in terms of the field* $\psi^\dagger$; no new phase-space information lives in π_ψ.

Hessian. The velocities are $(\dot\psi, \dot\psi^\dagger)$. Since $\mathcal{L}_\mathrm{D}$ is linear in $\dot\psi$ and independent of $\dot\psi^\dagger$, every second velocity-derivative (6.131) vanishes,

$$W = \frac{\partial^2 \mathcal{L}_\mathrm{D}}{\partial(\dot\psi, \dot\psi^\dagger)^2} = 0, \tag{6.136}$$

the identically-zero matrix. It is as singular as a Hessian can be; correspondingly the number of primary constraints, two per spinor component, saturates the count in Theorem 6.22.

Hamiltonian. The canonical Hamiltonian density is the Legendre transform $\mathcal{H}_\mathrm{D} = \pi_\psi\dot\psi + \dot\psi^\dagger\pi_{\psi^\dagger} - \mathcal{L}_\mathrm{D}$. The second term drops by (6.134). Substituting $\pi_\psi\dot\psi = i\,\psi^\dagger\dot\psi$ and (6.133),

$$\mathcal{H}_\mathrm{D} = \underbrace{i\,\psi^\dagger\dot\psi}_{\pi_\psi\dot\psi} - \underbrace{\left(i\,\psi^\dagger\dot\psi + i\,\bar\psi\gamma^i\partial_i\psi - m\,\bar\psi\psi\right)}_{\mathcal{L}_\mathrm{D}} = -\,i\,\bar\psi\gamma^i\partial_i\psi + m\,\bar\psi\psi. \tag{6.137}$$

The velocity term cancels identically — the hallmark of a Lagrangian linear in $\dot\psi$ — leaving a Hamiltonian density with *no* velocity in it:

$$\boxed{\mathcal{H}_\mathrm{D} = \bar\psi\left(-i\,\gamma^i\partial_i + m\right)\psi} \tag{6.138}$$

Writing $\bar\psi = \psi^\dagger\gamma^0$ and introducing the Dirac matrices $\alpha^i \equiv \gamma^0\gamma^i$, $\beta \equiv \gamma^0$ of nonrelativistic notation (Chapter C), (6.138) reads $\mathcal{H}_\mathrm{D} = \psi^\dagger(-i\,\alpha^i\partial_i + \beta m)\psi = \psi^\dagger(\boldsymbol{\alpha}\cdot\mathbf{p} + \beta m)\psi$ with $\mathbf{p} = -i\nabla$: exactly the single-particle Dirac Hamiltonian of relativistic quantum mechanics, now read as a classical field-energy density. That the field-theoretic $\mathcal{H}_\mathrm{D}$ reproduces the first-quantized Dirac Hamiltonian is the consistency check that the Legendre transform, constraints and all, has been done correctly.

The classical content is now exact and complete: the Dirac field carries the two second-class primary constraints (6.135), and its energy is (6.138). What remains — that the bracket of χ_1 with χ_2 is invertible, so that these constraints must be solved by passing to a modified bracket whose fermionic version is an *anti*bracket — is the quantization story, and we name it precisely in Section 6.5.4 before deferring its execution.

6.5.3 The Maxwell field: a non-dynamical potential and Gauss's law

We now open QED's gauge sector at the classical level. The Maxwell Lagrangian density fixed in Section 6.3 is

$$\mathcal{L}_\mathrm{M} = -\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{6.139}$$

with the four-potential A_μ as the fundamental field. We use the mostly-minus metric of Chapter A and the standard identification of the electric and magnetic fields,

$$E^i = F^{i0} = -\dot{A}^i - \partial_i A^0, \qquad B^i = -\tfrac{1}{2}\epsilon^{ijk}F_{jk} = (\nabla \times \mathbf{A})^i, \tag{6.140}$$

so that $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$. The asymmetry that creates the constraint is already visible: $F_{\mu\nu}$ is antisymmetric, so $F_{00} = 0$, so $\dot{A}_0$ — the time derivative of the time component — cannot appear in $\mathcal{L}_{\mathrm{M}}$. The potential A_0 has no kinetic term.

Worked Example 6.24 (Maxwell momenta and the constrained Hamiltonian)**.** We compute the four canonical momenta conjugate to A_μ, identify the primary constraint, and Legendre-transform to the Hamiltonian, performing the integration by parts in full.

Momenta. With $\mathcal{L}_{\mathrm{M}} = -\frac{1}{4}F_{\rho\sigma}F^{\rho\sigma}$, the derivative with respect to a velocity $\partial_0 A_\mu$ is the standard

$$\pi^\mu = \frac{\partial \mathcal{L}_{\mathrm{M}}}{\partial(\partial_0 A_\mu)} = -F^{0\mu}, \tag{6.141}$$

which follows because $\partial(-\frac{1}{4}F_{\rho\sigma}F^{\rho\sigma})/\partial(\partial_\alpha A_\beta) = -F^{\alpha\beta}$ (each of the two factors of F contributes $-\frac{1}{2}F^{\alpha\beta}$ after the antisymmetric index pairing). Splitting the index μ into time and space:

$$\pi^0 = -F^{00} = 0, \tag{6.142}$$

$$\pi^i = -F^{0i} = F^{i0} = E^i. \tag{6.143}$$

Equation (6.142) is the primary constraint: the momentum conjugate to A_0 vanishes *identically*, before any field equation, simply because $\dot{A}_0$ never appeared. Equation (6.143) says the momenta conjugate to the spatial potentials are the components of the electric field — the genuinely dynamical half of phase space. So the primary constraint is

$$\chi \equiv \pi^0 \approx 0. \tag{6.144}$$

The velocity Hessian (6.131), restricted to the four velocities $\dot{A}_\mu$, has the single null direction $\mu = 0$ (since $\mathcal{L}_{\mathrm{M}}$ is independent of $\dot{A}_0$), confirming exactly one primary constraint per spacetime point, in agreement with Theorem 6.22.

Hamiltonian. The canonical Hamiltonian density is $\mathcal{H}_{\mathrm{M}} = \pi^\mu \dot{A}_\mu - \mathcal{L}_{\mathrm{M}}$, and because $\pi^0 = 0$ the sum reduces to $\pi^i \dot{A}_i$. We must express the velocity $\dot{A}_i$ through the momentum. From (6.143) and $E^i = -\dot{A}^i - \partial_i A^0$, together with $\dot{A}_i = -\dot{A}^i$ (lowering a spatial index flips the sign in mostly-minus),

$$\pi^i = E^i = -\dot{A}^i - \partial_i A^0 \quad \Longrightarrow \quad \dot{A}_i = -\dot{A}^i = \pi^i + \partial_i A^0. \tag{6.145}$$

Hence the kinetic pairing is

$$\pi^i \dot{A}_i = \pi^i(\pi^i + \partial_i A^0) = \mathbf{E}^2 + \mathbf{E} \cdot \nabla A^0, \tag{6.146}$$

using $\pi^i = E^i$. Subtracting $\mathcal{L}_{\mathrm{M}} = \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$,

$$\mathcal{H}_{\mathrm{M}} = \mathbf{E}^2 + \mathbf{E}\cdot\nabla A^0 - \tfrac{1}{2}\mathbf{E}^2 + \tfrac{1}{2}\mathbf{B}^2 = \tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + \mathbf{E}\cdot\nabla A^0. \tag{6.147}$$

The first two terms are the familiar field-energy density of electromagnetism. The third is the term the constraint analysis cares about. Integrate it by parts over space inside $H_{\mathrm{M}} = \int d^3x\, \mathcal{H}_{\mathrm{M}}$, moving the gradient off A^0:

$$\int d^3x\; \mathbf{E}\cdot\nabla A^0 = \int d^3x\; \Big[\nabla\cdot(A^0\,\mathbf{E}) - A^0\,(\nabla\cdot\mathbf{E})\Big] = \oint_{\partial} A^0\,\mathbf{E}\cdot d\mathbf{S} - \int d^3x\; A^0\,(\nabla\cdot\mathbf{E}). \tag{6.148}$$

For fields decaying at spatial infinity the surface integral vanishes, and we record the Hamiltonian in the form that exhibits its structure:

$$\boxed{H_{\mathrm{M}} = \int d^3x\; \Big[\,\tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) - A^0\,(\nabla\cdot\mathbf{E})\,\Big]} \tag{6.149}$$

with $\mathbf{E} = \boldsymbol{\pi}$ and $\mathbf{B} = \nabla\times\mathbf{A}$ understood as phase-space functions of (A_i, π^i). The potential A_0 has disappeared from any time derivative and now appears in H_{M} *linearly and undifferentiated*, multiplying the quantity $\nabla\cdot\mathbf{E}$. This is the unmistakable signature of a Lagrange multiplier.

The shape of (6.149) is no accident: A_0 is not a degree of freedom but a multiplier enforcing a constraint, and that constraint is Gauss's law. We show this directly, as the dynamical consequence of preserving the primary constraint.

Worked Example 6.25 (Gauss's law as a secondary constraint from A_0)**.** We require the primary constraint $\pi^0 \approx 0$ to remain true under time evolution and read off Gauss's law; then we re-derive it as the Euler–Lagrange equation obtained by varying A_0, confirming the Lagrange-multiplier interpretation.

Consistency of the primary constraint. The time evolution of any phase-space functional is generated by the Hamiltonian through the equal-time Poisson bracket of Section 6.4; for the canonical pair (A_μ, π^ν) with $\{A_\mu(\mathbf{x}), \pi^\nu(\mathbf{y})\}_{\mathrm{P}} = \delta_\mu^{\ \nu}\,\delta^3(\mathbf{x}-\mathbf{y})$ this gives Hamilton's equation $\dot\pi^0 = \{\pi^0, H_{\mathrm{M}}\}_{\mathrm{P}} = -\,\delta H_{\mathrm{M}}/\delta A_0$. Differentiating (6.149) with respect to A_0 — only the last term depends on it, undifferentiated —

$$\dot\pi^0 = -\frac{\delta H_{\mathrm{M}}}{\delta A_0} = -(-\nabla\cdot\mathbf{E}) = \nabla\cdot\mathbf{E}. \tag{6.150}$$

A constraint is worthless unless it is preserved by the dynamics it constrains: we must have $\dot\pi^0 \approx 0$, for otherwise the system would immediately leave the surface $\pi^0 = 0$. Demanding this *consistency condition* yields a brand-new relation among the canonical variables,

$$\boxed{\nabla\cdot\mathbf{E} \approx 0} \tag{6.151}$$

which is Gauss's law (in vacuum). Because it was *not* present in the definition of the momenta but emerged from requiring time-consistency of a primary

constraint, it is called a *secondary* constraint. It is the first nontrivial output of the iterative algorithm sketched in Section 6.5.4.

The same law from varying A_0. The Lagrange-multiplier reading is confirmed by the Euler–Lagrange equation for A_0 derived from (6.139). Since $\mathcal{L}_{\mathrm{M}}$ depends on A_0 only through its spatial gradient $\partial_i A_0$ inside $F_{i0} = \partial_i A_0 - \partial_0 A_i$,

$$0 = \frac{\partial \mathcal{L}_{\mathrm{M}}}{\partial A_0} - \partial_\mu \frac{\partial \mathcal{L}_{\mathrm{M}}}{\partial(\partial_\mu A_0)} = 0 - \partial_i(-F^{i0}) = \partial_i E^i = \nabla \cdot \mathbf{E}, \tag{6.152}$$

where we used $\partial \mathcal{L}_{\mathrm{M}}/\partial(\partial_\mu A_0) = -F^{\mu 0}$ and $F^{00} = 0$. The field equation produced by A_0 contains *no* second time derivative of any field: it is not an evolution equation but a constraint on the initial data, $\nabla \cdot \mathbf{E} = 0$, exactly (6.151). A variable whose Euler–Lagrange equation is a constraint rather than an evolution law is, by definition, a Lagrange multiplier. The Hamiltonian (6.149) displays this plainly: varying the undifferentiated A_0 in H_{M} enforces $\nabla \cdot \mathbf{E} = 0$, just as varying a multiplier λ in $\lambda\, g(q,p)$ enforces $g = 0$.

Caution

Gauss's law (6.151) is a constraint on a *spatial slice*, not an equation of motion. It restricts which $(\mathbf{A}, \mathbf{E})$ qualify as legitimate initial data; the genuine time evolution is carried by the remaining Hamilton equations for A_i and π^i. Conflating the two — treating $\nabla \cdot \mathbf{E} = 0$ as something the dynamics "produces" rather than something it "preserves" — is the most common error in the canonical treatment of electromagnetism. The constraint is a property of every instant, guaranteed for all later times by the consistency computation above.

6.5.4 The Dirac–Bergmann algorithm: statement and the bridge to quantization

The two computations above are special cases of a single systematic procedure, due to Dirac and Bergmann, for converting any singular Lagrangian into a consistent Hamiltonian system. We state the procedure precisely — this is knowledge the reader needs to classify our two theories — while deferring the parts that are genuinely quantization choices. The algorithm proceeds in four stages, summarized in Fig. 6.3.

1. **Primary constraints.** Compute the momenta (6.130). The relations that cannot be solved for velocities are the primary constraints $\chi_m \approx 0$ of Theorem 6.22. (Dirac: χ_1, χ_2 of (6.135). Maxwell: $\pi^0 \approx 0$.)

2. **Total Hamiltonian.** Because the primary constraints carry no dynamics of their own, they enter evolution with arbitrary multipliers $u^m(x)$. The

generator of time evolution is the *total Hamiltonian*

$$H_{\mathrm{T}} = H_{\mathrm{c}} + \int d^3x\, u^m(x)\, \chi_m(x), \tag{6.153}$$

where $H_{\mathrm{c}} = \int d^3x\, \mathcal{H}$ is the canonical Hamiltonian of (6.138) or (6.149). The multipliers are to be fixed (or left free) by the next stage.

3. **Consistency and secondary constraints.** Each constraint must be preserved: $\dot{\chi}_m = \{\chi_m, H_{\mathrm{T}}\}_{\mathrm{P}} \approx 0$. For each m this equation either (a) is satisfied identically, (b) fixes one of the multipliers u^n, or (c) yields a new relation independent of the χ_m — a *secondary constraint*. One iterates, demanding consistency of the new constraints too, until no further independent relations appear. (Maxwell: consistency of $\pi^0 \approx 0$ produced the secondary constraint $\nabla \cdot \mathbf{E} \approx 0$, exactly as in Theorem 6.25; its own consistency $\partial_t(\nabla \cdot \mathbf{E}) \approx 0$ is automatic, so the chain terminates.)

4. **Classification: first and second class.** Collect all constraints, primary and secondary, into one set $\{\Phi_A\}$. A constraint Φ_A is *first class* if it has weakly vanishing Poisson bracket with *every* constraint, $\{\Phi_A, \Phi_B\}_{\mathrm{P}} \approx 0$ for all B; otherwise it is *second class*. The matrix $C_{AB} = \{\Phi_A, \Phi_B\}_{\mathrm{P}}$ is, after removing first-class members, invertible on the second-class sector.

Definition 6.26 (First- and second-class constraints)**.** With $\{\Phi_A\}$ the full set of constraints and $C_{AB} = \{\Phi_A, \Phi_B\}_{\mathrm{P}}$: the *first-class* constraints are those whose brackets with all Φ_B vanish weakly; they generate gauge transformations and signal redundant variables. The *second-class* constraints are the remainder, on which the antisymmetric matrix C_{AB} is nonsingular; they reduce phase space genuinely and are handled by replacing the Poisson bracket with the *Dirac bracket*

$$\{f, g\}_{\mathrm{D}} = \{f, g\}_{\mathrm{P}} - \{f, \Phi_A\}_{\mathrm{P}}\, \left(C^{-1}\right)^{AB}\, \{\Phi_B, g\}_{\mathrm{P}}\,, \tag{6.154}$$

where A, B run only over the second-class set. The Dirac bracket has the defining property $\{\Phi_A, g\}_{\mathrm{D}} = 0$ for every second-class Φ_A and every g, so the second-class constraints become strong identities and may be solved and eliminated; here $\{\,\cdot\,,\,\cdot\,\}_{\mathrm{D}}$ denotes the Dirac bracket, with C^{-1} the inverse of the second-class constraint matrix.

This vocabulary classifies our two protagonists cleanly, and the classification is the whole point of doing the analysis classically before quantizing.

Proposition 6.27 (Constraint type of the Dirac and Maxwell systems)**.** (i) *The Dirac primary constraints* (6.135) *are* second class*: their mutual bracket is nonsingular,* $\{\chi_{1\alpha}(\mathbf{x}), \chi_{2\beta}(\mathbf{y})\}_P = -i\,\delta_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y})$ *on spinor indices* α, β*. There is no gauge freedom; the constraints reduce phase space by half and must be imposed through the Dirac bracket* (6.154)*.* (ii) *The Maxwell constraints* $\pi^0 \approx 0$

and $\nabla\cdot\mathbf{E}\approx 0$ *are both* first class*:* $\{\pi^0,\nabla\cdot\mathbf{E}\}_P=0$ *because* π^0 *commutes with the spatial momenta* $\pi^i=E^i$*, and each commutes with itself trivially. They generate the* $U(1)$ *gauge transformations* $A_\mu\to A_\mu+\partial_\mu\lambda$ *and signal two redundant field components per point.*

Proof. (i) Using the only nonvanishing canonical bracket of the Dirac sector, $\{\psi_\alpha(\mathbf{x}),\pi_{\psi\,\beta}(\mathbf{y})\}_{\mathrm{P}}=\delta_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y})$, evaluate $\{\chi_{1\alpha},\chi_{2\beta}\}_{\mathrm{P}}$ with $\chi_{1\alpha}=\pi_{\psi\,\alpha}-i\,\psi^\dagger_\alpha$ and $\chi_{2\beta}=\pi_{\psi^\dagger\beta}$. The bracket $\left\{\pi_{\psi\,\alpha},\pi_{\psi^\dagger\beta}\right\}_{\mathrm{P}}=0$, while $\left\{-i\,\psi^\dagger_\alpha,\pi_{\psi^\dagger\beta}\right\}_{\mathrm{P}}=-i\,\delta_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y})$ since $\pi_{\psi^\dagger}$ is conjugate to $\psi^\dagger$. Hence $\{\chi_{1\alpha},\chi_{2\beta}\}_{\mathrm{P}}=-i\,\delta_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y})$, a nonsingular kernel: the constraint matrix is invertible, so the constraints are second class. *(ii)* The constraint π^0 has vanishing bracket with A_i and with π^i (distinct canonical sectors), so it commutes with $\nabla\cdot\mathbf{E}=\partial_i\pi^i$; $\nabla\cdot\mathbf{E}$ commutes with itself; and both are weakly preserved by H_{T} as shown in Theorem 6.25. All mutual brackets vanish, so both are first class. That the first-class $\nabla\cdot\mathbf{E}$ generates gauge transformations is the content of Chapter 8 and Chapter 13. □

Theorem 6.27 draws the line this section exists to draw. The dichotomy is real and classical: the Dirac field carries second-class constraints, the Maxwell field carries first-class constraints. Everything up to and including this classification is settled here, with full proofs. What lies on the far side of the line is genuinely a quantization choice, and it splits exactly along the first/second-class divide.

Notation Watch

Two distinct objects deferred to two distinct places must not be confused. (1) For the *second-class* Dirac system, the resolution is the Dirac bracket (6.154); computing it for χ_1,χ_2 gives $\left\{\psi_\alpha(\mathbf{x}),\psi^\dagger_\beta(\mathbf{y})\right\}_{\mathrm{D}}=-i\,\delta_{\alpha\beta}\,\delta^3(\mathbf{x}-\mathbf{y})$, an object *symmetric* under exchange in the bracket sense, which upon quantization $\{\,\cdot,\cdot\,\}_{\mathrm{D}}\to-i\{\,\cdot,\cdot\,\}$ becomes the anticommutator $\left\{\psi,\psi^\dagger\right\}$ — the classical origin of Fermi statistics. We carry this out in Chapter 11 (canonical quantization of the Dirac field) with the Grassmann calculus of Chapter 12. (2) For the *first-class* Maxwell system, no Dirac bracket is available — the constraint matrix C_{AB} is singular — and one must instead *fix the gauge*, restricting the redundant variables by hand before quantizing. The two implementations, Gupta–Bleuler (an indefinite-metric physical-state condition) and Faddeev–Popov gauge fixing, are the subject of Chapter 13 for the abelian photon and Chapter 26 for the non-abelian generalization. Nothing in this section presumes either choice; we have only certified, classically, which machinery each theory will need.

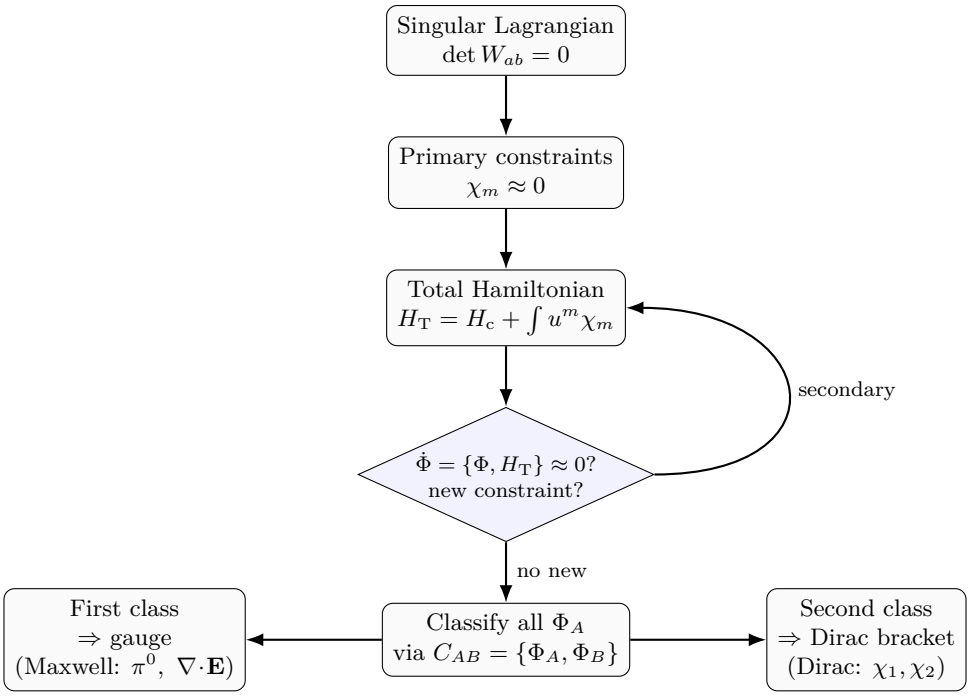


Figure 6.3: The Dirac–Bergmann algorithm. Degeneracy of the velocity Hessian produces primary constraints; demanding their time-consistency under the total Hamiltonian either fixes multipliers or generates secondary constraints (the loop); the closed set is then split into first-class (gauge-generating: Maxwell) and second-class (phase-space-reducing: Dirac) constraints. The classical analysis of this chapter carries every box; the resolution of each branch — the Dirac bracket for fermions and the quantization of the gauge branch — is the quantum work of Chapter 11, Chapter 13, and Chapter 26.

Handoff

We have isolated the purely classical core of every later gauge and fermion subtlety: the Dirac field's first-order kinetic term makes its momentum a second-class constraint $\pi_\psi \approx i\psi^\dagger$ with the constrained energy (6.138), while the Maxwell field's gauge degeneracy makes A_0 a non-dynamical Lagrange multiplier whose consistency is Gauss's law (6.151), both first class. These are facts of the classical theory, proved here in full, and they are exactly the facts Chapter 13 and Chapter 26 require before they may quantize. With the Hamiltonian densities of all three protagonists now in hand — scalar, Dirac (6.138), and Maxwell (6.149) — the final section turns to the deeper question of where the Hamiltonian density *comes from*: it is the T^{00} component of the conserved current of spacetime translations, the stress–energy tensor, to which we now build.

6.6 The Stress–Energy Tensor and Conserved Energy–Momentum

The Hamiltonian formulation of the preceding sections handed us a definite energy for a classical field: the number $H = \int d^3x\, \mathcal{H}$, built from the canonical momenta $\pi_a = \frac{\partial \mathcal{L}}{\partial \dot{\phi}_a}$ assembled in Section 6.4. That construction was unapologetically non-covariant. We singled out a time coordinate, performed a Legendre transform in $\dot{\phi}_a$ alone, and obtained a scalar H that lives on a fixed time slice. Energy, however, is the time component of a four-vector; in a relativistic theory it cannot be the whole story. Boost the slice and the energy mixes with momentum, exactly as p^0 mixes with $\mathbf{p}$ for a particle. There must be a covariant object whose 00 component is the energy density $\mathcal{H}$, whose $0i$ components are the momentum density, and which is conserved in a way that reduces, on a constant-time slice, to "energy in = energy out." That object is the *stress–energy tensor*, and the principle that produces it is the invariance of the action under rigid displacements of spacetime. Constructing it closes the loop opened in Section 6.2: the same Lagrangian that gave us equations of motion, through its symmetries, gives us the conserved quantities those equations respect.

We build the tensor directly from translation invariance, treating it as the first concrete instance of the Noether construction that Chapter 7 will erect in full generality. Doing the spacetime-translation case by hand first is deliberate: when the general theorem arrives, the reader will recognize it as the abstraction of a computation already understood in the marrow.

6.6.1 Translation invariance and the canonical current

Throughout, $\phi_a(x)$ denotes a collection of classical fields, the index a running over all components of all species in the theory (the four real components of a complex doublet, the four Dirac components ψ_α together with the four of $\bar{\psi}$, the four vector components A_μ, and so on). A repeated a is summed. The dynamics follow from

$$S = \int d^4x\, \mathcal{L}(\phi_a, \partial_\mu \phi_a), \tag{6.155}$$

where the crucial structural fact is that $\mathcal{L}$ depends on the spacetime point x *only* through the fields and their derivatives—there is no explicit factor of x^μ multiplying anything. This is precisely the statement that the laws of motion are the same at every point of spacetime: a field configuration and its rigidly translated copy obey identical equations. We now extract the conserved current encoded in that homogeneity.

Consider an infinitesimal constant translation of the spacetime argument,

$$x^\mu \;\longrightarrow\; x'^\mu = x^\mu + a^\mu, \qquad a^\mu = \text{const, infinitesimal.} \tag{6.156}$$

A translation carries a field configuration to a new one whose value at the shifted point equals the old value at the old point, $\phi'_a(x') = \phi_a(x)$. Re-expressed at a fixed coordinate x, the new configuration is the old one evaluated a step back, $\phi'_a(x) = \phi_a(x - a)$. To first order in a,

$$\delta\phi_a(x) \;\equiv\; \phi'_a(x) - \phi_a(x) = -\,a^\nu\,\partial_\nu\phi_a(x). \tag{6.157}$$

This is the variation of the fields induced by the symmetry; it is what we feed into the action.

We need two independent expressions for the resulting change $\delta\mathcal{L}$ in the Lagrangian density, one a kinematic identity true for any field configuration, the other valid only on solutions. Their equality is the conservation law.

The identity (off shell). Because $\mathcal{L}$ has no explicit x-dependence, shifting the fields by Eq. (6.157) shifts $\mathcal{L}$ as the corresponding total derivative. Compute the variation by the chain rule,

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_a}\,\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\partial_\mu(\delta\phi_a), \tag{6.158}$$

where we used $\delta(\partial_\mu\phi_a) = \partial_\mu(\delta\phi_a)$, the variation commuting with the derivative. Substituting Eq. (6.157),

$$\delta\mathcal{L} = -a^\nu\left[\frac{\partial\mathcal{L}}{\partial\phi_a}\,\partial_\nu\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\partial_\mu\partial_\nu\phi_a\right] = -a^\nu\,\partial_\nu\mathcal{L}. \tag{6.159}$$

The bracket is exactly the chain-rule expansion of the total derivative $\partial_\nu\mathcal{L}$, legitimate *only* because $\mathcal{L}$ carries no explicit x^ν; any explicit dependence would have left an uncancelled $\partial_\nu\mathcal{L}\big|_{\text{expl}}$. Equation Eq. (6.159) holds for every field configuration, solution or not. Rewrite it as a four-divergence,

$$\delta\mathcal{L} = -a^\nu\,\partial_\nu\mathcal{L} = -\,\partial_\mu(a^\nu\,\delta^\mu{}_\nu\,\mathcal{L}), \tag{6.160}$$

using $\partial_\nu = \delta^\mu{}_\nu\partial_\mu$ and the constancy of a^ν.

The dynamical expression (on shell). Now suppose ϕ_a solves the Euler–Lagrange equations of Section 6.2,

$$\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)} = 0. \tag{6.161}$$

Return to the general chain-rule form Eq. (6.158), valid for *any* variation, and trade $\partial\mathcal{L}/\partial\phi_a$ for the divergence of the momentum using Eq. (6.161):

$$\delta\mathcal{L} = \left(\partial_\mu\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\right)\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\partial_\mu(\delta\phi_a) = \partial_\mu\left(\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,\delta\phi_a\right), \tag{6.162}$$

the last step the Leibniz rule run backwards. This is again a pure four-divergence, but its content is now dynamical: it used the equations of motion.

Equating the two. On a solution both Eq. (6.160) and Eq. (6.162) hold, so their difference vanishes:

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \, \delta\phi_a + a^\nu \, \delta^\mu{}_\nu \, \mathcal{L} \right) = 0. \tag{6.163}$$

Insert $\delta\phi_a = -a^\nu \partial_\nu \phi_a$ from Eq. (6.157) and pull out the arbitrary constant a^ν:

$$a^\nu \, \partial_\mu \left[\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \, \partial_\nu \phi_a - \delta^\mu{}_\nu \, \mathcal{L} \right] = 0. \tag{6.164}$$

Since the four numbers a^ν are independent, each of the four currents in brackets is separately conserved. Raising the free index with $\eta^{\nu\rho}$, we have proved the central object of this section.

Definition 6.28 (Canonical stress–energy tensor)**.** For a theory with Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a)$ free of explicit spacetime dependence, the *canonical stress–energy tensor* (also called the canonical energy–momentum tensor) is

$$\boxed{T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)} \, \partial^\nu \phi_a - \eta^{\mu\nu} \mathcal{L}} \tag{6.165}$$

with an implicit sum over the field index a. It is the Noether current of the four constant spacetime translations: the free index ν labels which translation (one for time, three for space), while μ is the current's spacetime index.

The structure is worth reading aloud. The first term is "conjugate momentum density times the gradient of the field along direction ν"—the field-theoretic analogue of $p\,\dot{q}$. The second subtracts off $\eta^{\mu\nu}\mathcal{L}$, exactly the subtraction a Legendre transform performs. Setting $\mu = \nu = 0$ will reproduce $\pi_a \dot{\phi}_a - \mathcal{L} = \mathcal{H}$, which is the promised identification and the reason the same letter "$\mathcal{H}$" was chosen for the energy density in Section 6.4.

6.6.2 On-shell conservation, by direct computation

The derivation above already establishes $\partial_\mu T^{\mu\nu} = 0$ as a by-product of equating Eq. (6.160) and Eq. (6.162). It is nonetheless instructive—and a useful independent check that no sign slipped—to verify the conservation law by brute differentiation of Eq. (6.165), substituting the equations of motion at the one place they are needed. This is the template the general Noether theorem will follow.

Theorem 6.29 (Conservation of the canonical stress tensor)**.** *Let ϕ_a satisfy the Euler–Lagrange equations Eq. (6.161). Then the canonical stress–energy tensor Eq. (6.165) is conserved on every index,*

$$\partial_\mu T^{\mu\nu} = 0. \tag{6.166}$$

Proof. Differentiate Eq. (6.165) and apply the product rule to the first term:

$$\partial_\mu T^{\mu\nu} = \left(\partial_\mu \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\right) \partial^\nu \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\, \partial_\mu \partial^\nu \phi_a - \partial^\nu \mathcal{L}. \tag{6.167}$$

In the first term, the divergence of the canonical momentum is exactly what the Euler–Lagrange equation Eq. (6.161) equates to $\partial \mathcal{L}/\partial \phi_a$; this is the single, decisive use of the equations of motion. Substituting,

$$\partial_\mu T^{\mu\nu} = \frac{\partial \mathcal{L}}{\partial \phi_a}\, \partial^\nu \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\, \partial_\mu \partial^\nu \phi_a - \partial^\nu \mathcal{L}. \tag{6.168}$$

The remaining task is purely kinematic: expand the last term by the chain rule. Since $\mathcal{L}$ has no explicit x-dependence,

$$\partial^\nu \mathcal{L} = \eta^{\nu\rho} \partial_\rho \mathcal{L} = \eta^{\nu\rho} \left[\frac{\partial \mathcal{L}}{\partial \phi_a}\, \partial_\rho \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\, \partial_\rho \partial_\mu \phi_a \right] = \frac{\partial \mathcal{L}}{\partial \phi_a}\, \partial^\nu \phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\, \partial^\nu \partial_\mu \phi_a. \tag{6.169}$$

The first piece of Eq. (6.169) cancels the first term of Eq. (6.168); the second piece cancels the second term, because partial derivatives commute, $\partial_\mu \partial^\nu \phi_a = \partial^\nu \partial_\mu \phi_a$. Every term has paired off, and

$$\partial_\mu T^{\mu\nu} = 0. \tag{6.170}$$

□

The proof exposes the anatomy of every conservation law to come. Two ingredients combine: a kinematic identity that holds because the Lagrangian is translation-homogeneous (the expansion of $\partial^\nu \mathcal{L}$), and a single substitution of the equations of motion (the divergence of the canonical momentum). Chapter 7 will show that this is the universal shape of Noether's theorem—one identity from the symmetry, one substitution of the dynamics—with spacetime translations merely the first and most concrete example.

6.6.3 Charges: the conserved four-momentum

A conserved current yields a conserved charge by integrating its time component over a spatial slice. Here there are four currents, indexed by ν, so there are four charges, and they assemble into a four-vector.

Definition 6.30 (Four-momentum of the field)**.** The total energy–momentum carried by a field configuration is

$$P^\nu = \int d^3x\, T^{0\nu}(t, \mathbf{x}). \tag{6.171}$$

Proposition 6.31 (Conservation of P^ν)**.** *If the fields and their conjugate momenta fall off fast enough at spatial infinity that surface terms vanish, then P^ν is independent of time, $dP^\nu/dt = 0$.*

Proof. Differentiate Eq. (6.171) under the integral and split the conservation law Eq. (6.166) into its time and space parts, $\partial_\mu T^{\mu\nu} = \partial_0 T^{0\nu} + \partial_i T^{i\nu} = 0$:

$$\frac{dP^\nu}{dt} = \int d^3x\, \partial_0 T^{0\nu} = -\int d^3x\, \partial_i T^{i\nu} = -\oint_{S_\infty} dS_i\, T^{i\nu} = 0, \tag{6.172}$$

the third equality the divergence theorem in three dimensions, the last the assumed decay at infinity. The same computation is the field-theoretic restatement of "the rate of change of the charge inside a region equals minus the flux of the current through its boundary," with $T^{i\nu}$ the flux of the ν-th charge. □

Now identify the components. The time–time component, with $\partial^0 = \partial_0$ and $\eta^{00} = +1$, is

$$T^{00} = \frac{\partial \mathcal{L}}{\partial(\partial_0 \phi_a)}\, \partial^0 \phi_a - \eta^{00}\mathcal{L} = \pi_a\, \dot{\phi}_a - \mathcal{L} = \mathcal{H}, \tag{6.173}$$

precisely the Hamiltonian density of Section 6.4, since $\pi_a = \partial\mathcal{L}/\partial\dot{\phi}_a$ is the canonical momentum and $\dot{\phi}_a = \partial^0\phi_a$. Therefore the conserved charge for time translations is the Hamiltonian itself,

$$P^0 = \int d^3x\, T^{00} = \int d^3x\, \mathcal{H} = H. \tag{6.174}$$

This is the loop the chapter set out to close: *energy is the Noether charge of time translation, and the covariant tensor Eq. (6.165) reproduces the Hamiltonian density built non-covariantly two sections ago.* The spatial charges are the field momentum,

$$P^i = \int d^3x\, T^{0i} = \int d^3x\, \pi_a\, \partial^i \phi_a = -\int d^3x\, \pi_a\, \partial_i \phi_a, \tag{6.175}$$

using $\partial^i = -\partial_i$. The conserved $P^\mu = (H, \mathbf{P})$ is the classical precursor of the momentum operator that Chapter 9 promotes to the generator of translations on Fock space; its very existence is the statement that a free or self-interacting field, like an isolated mechanical system, carries a definite total energy and momentum that the equations of motion preserve.

Physical Insight

The four-by-four array $T^{\mu\nu}$ records the entire flow of energy and momentum through spacetime. Read the first index μ as "which face of an infinitesimal box" (the 0-face is a snapshot in time, the i-face is a wall in space) and the second index ν as "which component of four-momentum." Then T^{00} is energy per unit volume, T^{0i} is the density of the i-momentum (equivalently, by symmetry of the improved tensor, the flux of energy in direction i—the Poynting vector for the electromagnetic field), T^{i0} is the flux of energy through the i-wall, and T^{ij} is the flux of j-momentum through the i-

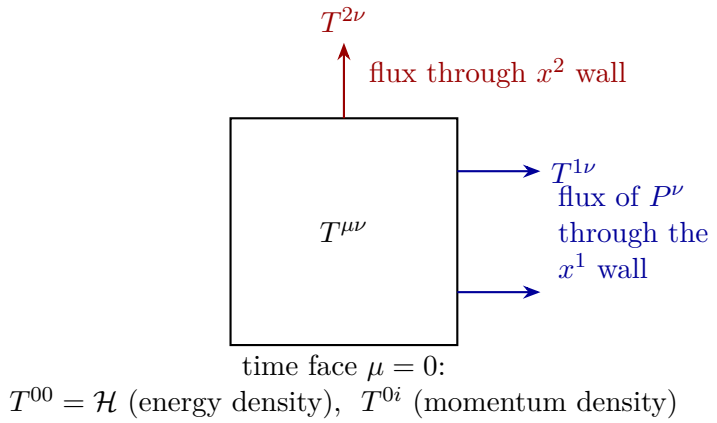


Figure 6.4: The stress–energy tensor as flux bookkeeping. The component $T^{\mu\nu}$ is the flow of the ν-th component of four-momentum across a surface of constant x^μ. The conservation law $\partial_\mu T^{\mu\nu} = 0$ states that for each fixed ν the change in the charge $\int T^{0\nu}$ inside a region equals minus the flux $T^{i\nu}$ through its walls.

wall, i.e. the mechanical stress—pressure on the diagonal, shear off it. Conservation $\partial_\mu T^{\mu\nu} = 0$ is the bookkeeping that none of it is created or destroyed, only transported.

6.6.4 Scalar ϕ^4 theory: the scalar stress tensor in full

We now make the abstract formula concrete on the field that will accompany us through the whole book. The classical $\lambda\phi^4$ theory introduced in Section 6.3 has Lagrangian density

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \qquad \phi \text{ real},\ \lambda > 0, \tag{6.176}$$

whose Euler–Lagrange equation is $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$. We continue ϕ^4 theory from the equation of motion of Section 6.2 and the Hamiltonian density of Section 6.4.

Worked Example 6.32 (Stress–energy tensor of $\lambda\phi^4$, and the check $T^{00} = \mathcal{H}$)**.** There is a single field, so the index a is trivial. The only derivative-dependence is in the kinetic term; differentiating Eq. (6.176),

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)} = \partial^\mu\phi. \tag{6.177}$$

Inserting this and Eq. (6.176) into the master formula Eq. (6.165) gives the

canonical stress tensor of the scalar,

$$T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\left[\tfrac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right]. \tag{6.178}$$

A scalar field carries no internal spacetime indices, so the only index structure available is $\partial^\mu\phi\,\partial^\nu\phi$ and $\eta^{\mu\nu}$ times scalars; consequently Eq. (6.178) is automatically *symmetric*, $T^{\mu\nu} = T^{\nu\mu}$, and manifestly built from ϕ alone (no gauge ambiguity). The scalar is the lucky case where the canonical tensor already has every property one could want—a fact that throws the defects of the Dirac and Maxwell tensors below into sharp relief.

Energy density. Set $\mu = \nu = 0$. With $\partial^0\phi = \dot\phi$ and the spatial split $(\partial_\alpha\phi)(\partial^\alpha\phi) = \dot\phi^2 - (\nabla\phi)^2$,

$$\begin{aligned} T^{00} &= \dot\phi^2 - \left[\tfrac{1}{2}\dot\phi^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right] \\ &= \tfrac{1}{2}\dot\phi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4. \end{aligned} \tag{6.179}$$

With the canonical momentum $\pi = \dot\phi$ of Section 6.4, this is verbatim the Hamiltonian density

$$\mathcal{H} = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 \tag{6.180}$$

obtained there by Legendre transform. The general identity Eq. (6.173) is confirmed in the concrete case: every term—kinetic, gradient, mass, quartic—matches, and the energy density is manifestly non-negative, as a sensible energy must be.

Momentum density. Set $\mu = 0$, $\nu = i$. The metric term drops because $\eta^{0i} = 0$:

$$T^{0i} = \partial^0\phi\,\partial^i\phi = \dot\phi\,\partial^i\phi = -\dot\phi\,\partial_i\phi. \tag{6.181}$$

The field-momentum density is $\mathcal{P}^i = T^{0i} = -\dot\phi\,\partial_i\phi$, so that the conserved spatial momentum of the configuration is $P^i = -\int d^3x\,\dot\phi\,\partial_i\phi$, in agreement with the general expression Eq. (6.175) at $\pi = \dot\phi$. Physically a travelling wave $\phi \propto \cos(k\cdot x)$ has $\dot\phi$ and $\nabla\phi$ in step, so $\mathcal{P}^i$ points along $\mathbf{k}$: the field transports momentum in its direction of propagation.

Trace and broken scale invariance. Contracting Eq. (6.178) with $\eta_{\mu\nu}$, and using $\eta_{\mu\nu}\eta^{\mu\nu} = 4$ in four dimensions,

$$T^\mu{}_\mu = (\partial_\alpha\phi)(\partial^\alpha\phi) - 4\left[\tfrac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right] = -(\partial_\alpha\phi)(\partial^\alpha\phi) + 2m^2\phi^2 + \frac{\lambda}{3!}\phi^4. \tag{6.182}$$

This off-shell expression simplifies decisively on a solution. Multiply the equation of motion, written as $\Box\phi = -m^2\phi - \frac{\lambda}{3!}\phi^3$, by ϕ and combine it with the identity

$\phi\,\Box\phi = \partial_\alpha(\phi\,\partial^\alpha\phi) - (\partial_\alpha\phi)(\partial^\alpha\phi)$ to isolate the kinetic scalar,

$$(\partial_\alpha\phi)(\partial^\alpha\phi) = \partial_\alpha(\phi\,\partial^\alpha\phi) - \phi\,\Box\phi = \partial_\alpha(\phi\,\partial^\alpha\phi) + m^2\phi^2 + \frac{\lambda}{3!}\phi^4. \tag{6.183}$$

Substituting Eq. (6.183) into the trace Eq. (6.182), the two quartic terms cancel and a single mass term survives:

$$T^\mu{}_\mu \overset{\text{on shell}}{=} -\Big[\partial_\alpha(\phi\,\partial^\alpha\phi) + m^2\phi^2 + \frac{\lambda}{3!}\phi^4\Big] + 2m^2\phi^2 + \frac{\lambda}{3!}\phi^4 = m^2\phi^2 - \partial_\alpha(\phi\,\partial^\alpha\phi). \tag{6.184}$$

The result is sharper than Eq. (6.182) alone suggested: up to the total derivative $-\partial_\alpha(\phi\,\partial^\alpha\phi)$, the on-shell trace is *purely* $m^2\phi^2$, and the quartic has dropped out entirely. This is the classical statement that $\lambda\phi^4$ in four dimensions is a *marginal* interaction. The coupling λ is dimensionless, so $\int d^4x\,\frac{\lambda}{4!}\phi^4$ is invariant under the rescaling $x \to e^{-\omega}x$, $\phi(x) \to e^{\omega}\phi(e^{\omega}x)$ in which ϕ carries its canonical mass dimension one; the quartic alone therefore does not obstruct scale invariance. The single obstruction is the mass term: $\int d^4x\,\frac{1}{2}m^2\phi^2$ fails to be scale invariant precisely because m is dimensionful, and it is exactly $m^2\phi^2$ that remains in Eq. (6.184). A traceless stress tensor is the local signature of invariance under those rescalings; the massless theory ($m = 0$) has, once the improvement discussed below removes the total-derivative remnant, a traceless tensor and is classically scale—indeed conformally—invariant, while a nonzero m breaks it. The fingerprint $T^\mu{}_\mu \propto m^2\phi^2$ is the seed of the dilatation current that Chapter 7 will construct; and the fact that the classically harmless coupling λ nonetheless breaks scale invariance *quantum* mechanically—through the running of λ and the attendant trace anomaly—is one of the deepest threads the book follows downstream.

6.6.5 Two defects of the canonical tensor

The scalar lulled us: its canonical $T^{\mu\nu}$ was symmetric and built only from the field. That is an accident of spin zero. The two other protagonists of the book expose what the construction generically produces, and the failures point straight at the repairs that Chapter 7 will carry out.

Dirac: non-symmetric. The free Dirac Lagrangian of Section 6.3 is

$$\mathcal{L}_D = \bar\psi\,(i\partial\!\!\!/ - m)\,\psi = \bar\psi\,(i\gamma^\mu\partial_\mu - m)\,\psi, \tag{6.185}$$

with ψ and $\bar\psi = \psi^\dagger\gamma^0$ treated as independent fields (the gamma conventions are those of Chapter C). Only ψ is differentiated, so $\partial\mathcal{L}_D/\partial(\partial_\mu\bar\psi) = 0$ and

$$\frac{\partial\mathcal{L}_D}{\partial(\partial_\mu\psi)} = \bar\psi\,i\gamma^\mu. \tag{6.186}$$

The master formula Eq. (6.165) then gives

$$T_D^{\mu\nu} = i\,\bar\psi\,\gamma^\mu\partial^\nu\psi - \eta^{\mu\nu}\mathcal{L}_D. \tag{6.187}$$

On shell the Dirac equation $(i\partial\!\!\!/ - m)\psi = 0$ makes the Lagrangian Eq. (6.185) vanish identically, so the metric term drops and

$$T_D^{\mu\nu} \overset{\text{on shell}}{=} i\,\bar\psi\,\gamma^\mu\partial^\nu\psi. \tag{6.188}$$

This is patently *not* symmetric under $\mu \leftrightarrow \nu$: the index μ sits on a gamma matrix while ν sits on a derivative, and $i\bar\psi\gamma^\mu\partial^\nu\psi \neq i\bar\psi\gamma^\nu\partial^\mu\psi$ in general. Its antisymmetric part $T_D^{[\mu\nu]} = \frac{i}{2}\bar\psi(\gamma^\mu\partial^\nu - \gamma^\nu\partial^\mu)\psi$ is non-vanishing, and—as the angular-momentum analysis below shows—it is exactly the divergence of the electron's intrinsic spin current. Spin-$\frac{1}{2}$ matter cannot have a symmetric canonical stress tensor; the asymmetry *is* the spin.

Maxwell: non-symmetric and non-gauge-invariant. The free Maxwell Lagrangian of Section 6.3 is

$$\mathcal{L}_{\text{Max}} = -\tfrac{1}{4}F_{\alpha\beta}F^{\alpha\beta}, \qquad F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha, \tag{6.189}$$

the dynamical fields being the four components A_ρ. Differentiating with respect to $\partial_\mu A_\rho$ and using $\partial F_{\alpha\beta}/\partial(\partial_\mu A_\rho) = \delta^\mu{}_\alpha\delta^\rho{}_\beta - \delta^\mu{}_\beta\delta^\rho{}_\alpha$,

$$\frac{\partial\mathcal{L}_{\text{Max}}}{\partial(\partial_\mu A_\rho)} = -\tfrac{1}{4}\cdot 2\,F^{\alpha\beta}(\delta^\mu{}_\alpha\delta^\rho{}_\beta - \delta^\mu{}_\beta\delta^\rho{}_\alpha) = -\tfrac{1}{2}(F^{\mu\rho} - F^{\rho\mu}) = -F^{\mu\rho}. \tag{6.190}$$

The master formula Eq. (6.165) then yields the canonical electromagnetic stress tensor

$$\boxed{T^{\mu\nu}_{\text{can}} = -F^{\mu\rho}\,\partial^\nu A_\rho + \tfrac{1}{4}\,\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}} \tag{6.191}$$

and it suffers *both* diseases at once. First, it is not symmetric: the term $-F^{\mu\rho}\partial^\nu A_\rho$ has no symmetry between μ (on the field strength) and ν (on the derivative). Second, and worse, it is not gauge invariant: it contains the bare potential A_ρ and its undifferentiated gradient $\partial^\nu A_\rho$, not the gauge-invariant F. Under $A_\rho \to A_\rho + \partial_\rho\chi$,

$$T^{\mu\nu}_{\text{can}} \;\longrightarrow\; T^{\mu\nu}_{\text{can}} - F^{\mu\rho}\,\partial^\nu\partial_\rho\chi, \tag{6.192}$$

which does not vanish. A physically meaningful energy density cannot depend on a choice of gauge; Eq. (6.191) as it stands is unacceptable as the energy of the electromagnetic field. (One quickly checks that its $T^{00} = -F^{0\rho}\partial^0 A_\rho + \frac{1}{4}F^2$ is neither the familiar $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$ nor even gauge invariant.)

Caution

The non-symmetry and non-gauge-invariance of $T^{\mu\nu}_{\text{can}}$ are defects of the *canonical* construction, not of the physics. The conserved charges $P^\nu = \int d^3x\, T^{0\nu}$ are perfectly correct—they are the true energy and momentum—because a current is only fixed up to the addition of an identically conserved "improvement" term that integrates to zero. What is wrong is the local distribution of energy and momentum, $T^{\mu\nu}(x)$ itself, which the canonical recipe gets gauge-dependent and asymmetric. The cure is to add such an improvement term.

6.6.6 The improvement program, and angular momentum

Both defects are repaired by adding to $T^{\mu\nu}$ a term of the form $\partial_\rho K^{\rho\mu\nu}$ with K antisymmetric in its first two indices, $K^{\rho\mu\nu} = -K^{\mu\rho\nu}$. Antisymmetry guarantees two things at once: the addition is identically conserved in μ, $\partial_\mu\partial_\rho K^{\rho\mu\nu} = 0$, so $\partial_\mu T^{\mu\nu} = 0$ is preserved; and it leaves the charges untouched, since $\int d^3x\, \partial_\rho K^{\rho 0\nu} = \int d^3x\, \partial_i K^{i0\nu}$ is a spatial surface integral that vanishes at infinity (the $\rho = 0$ term drops by antisymmetry $K^{00\nu} = 0$). One is therefore free to choose K to mend the tensor without disturbing P^ν. For electromagnetism the choice $K^{\rho\mu\nu} = F^{\mu\rho}A^\nu$ does the job: on shell, where $\partial_\rho F^{\rho\mu} = 0$,

$$T^{\mu\nu}_{\text{can}} + \partial_\rho\big(F^{\mu\rho}A^\nu\big) = -F^{\mu\rho}\big(\partial^\nu A_\rho - \partial_\rho A^\nu\big) + \tfrac{1}{4}\eta^{\mu\nu}F^2 = -F^{\mu\rho}F^{\nu}{}_{\rho} + \tfrac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}, \tag{6.193}$$

which is at once symmetric in $\mu\nu$ and built entirely from the gauge-invariant F; its $T^{00} = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$ and $T^{0i} = (\mathbf{E}\times\mathbf{B})^i$ are the textbook electromagnetic energy density and Poynting vector. We display Eq. (6.193) only as a preview of where the story lands; the *systematic* construction—the Belinfante–Rosenfeld procedure that produces the right K for any field of any spin from its behaviour under Lorentz transformations—belongs to Chapter 7, where it falls out naturally once Noether's theorem is in hand.

The deeper reason a symmetric tensor is the correct target is angular momentum. Lorentz invariance furnishes its own Noether current, the angular-momentum current, which (anticipating the full derivation in Chapter 7) has the schematic form

$$M^{\mu\,\nu\rho} = \underbrace{x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu}}_{\text{orbital}} + \underbrace{S^{\mu\,\nu\rho}}_{\text{spin}}, \tag{6.194}$$

antisymmetric in $\nu\rho$, with $S^{\mu\nu\rho}$ the contribution from the intrinsic transformation of the fields (zero for a scalar, nonzero for the spinor and the vector). Taking its divergence on shell, and using $\partial_\mu T^{\mu\nu} = 0$,

$$\partial_\mu M^{\mu\,\nu\rho} = T^{\nu\rho} - T^{\rho\nu} + \partial_\mu S^{\mu\,\nu\rho}. \tag{6.195}$$

Conservation of total angular momentum, $\partial_\mu M^{\mu\nu\rho} = 0$, therefore forces the antisymmetric part of the stress tensor to be tied to the divergence of the spin current, $T^{\nu\rho} - T^{\rho\nu} = -\partial_\mu S^{\mu\nu\rho}$. This is the precise sense in which the Dirac asymmetry Eq. (6.188) "is" the electron's spin. It also shows that the very improvement term K that symmetrizes $T^{\mu\nu}$ is built from $S^{\mu\nu\rho}$: symmetrizing the stress tensor is the act of absorbing the spin current into the orbital part, so that for the improved tensor the orbital expression $x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu}$ alone is conserved. The full statement and proof of Eqs. (6.194) and (6.195), and the Belinfante construction of the symmetric, gauge-invariant tensor, are the first order of business in Chapter 7.

With energy and momentum identified as the Noether charges of translation, and with the canonical tensor's two defects diagnosed and their cure named, the classical dynamics of fields is complete. Chapter 7 promotes the single computation of this section into Noether's theorem for every continuous symmetry—internal as well as spacetime—and discharges the deferred items: the symmetric Belinfante stress tensor, the angular-momentum current in full, and, for ϕ^4 theory, the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ and the broken scale current foreshadowed by the non-zero trace Eq. (6.182). The conserved $P^\mu = (H, \mathbf{P})$ assembled here is, in turn, exactly the object that Chapter 9 will promote from a number to the operator generating spacetime translations on the quantum Hilbert space.

Further reading

The variational and canonical structure of classical field theory is treated in every comprehensive text on the subject; the presentations closest in spirit and convention to this chapter are [396, 432, 454], with [500] giving the most careful account of the relationship between Poincaré invariance and the field content, and [532] the most physically motivated overview. The classical-mechanics roots—Hamilton's principle, the Legendre transform, and the passage from discrete to continuous systems—are developed at length in [230] and, with characteristic economy and the field-theoretic stress tensor included, in [312]. The theory of singular Lagrangians and constrained Hamiltonian dynamics that we only open here is presented in its original form, and still most readably, in [152].

Chapter 7

Symmetries and Conservation Laws

Chapter 6 closed by extracting, from the invariance of the action under a rigid spacetime translation, the canonical stress–energy tensor and the conserved four-momentum it carries. That single computation was a special case of a general correspondence, and this chapter is its systematic development. The thesis is the one that organizes all of field theory: a continuous symmetry of the action is not merely an aesthetic feature but a *conservation law in disguise*, and the dictionary translating between the two—Emmy Noether's first theorem [369]—is local, constructive, and exact. Give the theorem a transformation and it returns a current whose divergence vanishes on shell; integrate that current over a spatial slice and it returns a charge that does not change in time. Energy, momentum, angular momentum, electric charge, and the internal quantum numbers that label particles into multiplets are all, without exception, outputs of this one machine fed different inputs.

We build the machine from the variational calculus assembled in Chapter 4 and the Euler–Lagrange equations and canonical structures of Chapter 6, and we phrase every field transformation in the representation-theoretic language of Chapter 3—fields carry a finite-dimensional representation of the Lorentz group, and the generators T^a of any internal symmetry obey the Lie-algebra conventions of Chapter 2. The opening section is careful bookkeeping: it separates the intrinsic variation of a field from the total variation that also moves the coordinate, derives the master identity for the first variation of the action, and fixes the precise notion of a continuous *quasi*-symmetry (invariance up to a boundary term). With those two ingredients in hand—the off-shell quasi-symmetry condition and the on-shell boundary reduction of the master identity—the second section proves the first theorem by setting the two against each other, defines the conserved current and charge, establishes the charge's time- and slice-independence, and isolates the improvement ambiguity that will be put to work twice later in the

chapter.

The remaining sections turn the crank. Switching off the coordinate piece gives the internal symmetries: the U(1) current of a complex scalar, the non-abelian currents of a multiplet, the proof that the charges generate the field transformation and close into the symmetry's Lie algebra, and—kept throughout as the instructive foil—the discrete $\mathbb{Z}_2$ of ϕ^4 theory, which yields no current but a sharp selection rule forbidding odd-point amplitudes. Switching the coordinate piece back on gives the spacetime symmetries: translations reproduce the canonical stress tensor of Chapter 6, Lorentz invariance delivers the angular-momentum current and exposes the spin current, and the ten Poincaré charges emerge as the classical precursors of the operators Chapter 9 will quantize. That construction surfaces a defect—the canonical tensor is generically neither symmetric nor (for gauge fields) gauge invariant—and the improvement freedom is spent repairing it into the symmetric, gauge-invariant Belinfante–Rosenfeld tensor [57, 419], the flat-space seed of the gravitational source.

The chapter then sharpens the global/local distinction with Noether's *second* theorem: when a symmetry parameter is allowed to vary from point to point, the would-be current is on-shell a pure superpotential divergence, its charge a boundary flux, and the genuine conserved charges of a theory are seen to come from its global symmetries while a local symmetry contributes only off-shell identities and a redundancy—precisely the logic Chapter 8 will run forward to derive electromagnetism from the gauge principle. A final section carries ϕ^4 theory to its symmetry layer, treating scale and conformal invariance: the dilatation current, the trace of the stress tensor as the diagnostic of scale breaking, the conformally improved traceless tensor, and the classical trace identity $\Theta^{\mu}_{\mathrm{c}\,\mu} = m^2\phi^2$ whose quantum deformation into the trace anomaly is the handoff to the renormalization group of Chapters 23 and 24 and the conformal-field-theory machinery of Chapters 47 and 52. Throughout, the conventions are those fixed for the whole book—the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$ collected in Chapter A, with the gamma-matrix algebra of Chapter C used wherever the Dirac field appears.

7.1 What Is a Symmetry? Variations, Invariance, and Quasi-Symmetry

The previous chapter handed us a complete classical apparatus: an action functional $S[\phi] = \int_\Omega \mathrm{d}^4x\, \mathcal{L}(\phi_a, \partial_\mu\phi_a, x)$ (6.25), the Euler–Lagrange equations $E_a[\mathcal{L}] = 0$ that its stationarity imposes (Theorem 6.8), the canonical momenta $\pi^a = \partial\mathcal{L}/\partial\dot{\phi}_a$ (6.96), and—worked out by hand for the single case of spacetime translations—a first conserved current, the canonical stress–energy tensor of Theorem 6.28. That last computation was the appetizer. It showed, in one concrete instance, that the *symmetries* of an action are not decorative facts

about a theory but the source of its conservation laws: invariance under rigid time translation produced the conserved energy, invariance under rigid space translation the conserved momentum. The thesis of this chapter is that this is completely general. Every continuous global symmetry of a classical field theory comes with an exact accounting identity—a locally conserved current and a time-independent charge—and the machine that manufactures the current from the symmetry is Noether's first theorem, proved in full in Section 7.2.

Before we can state that theorem, we must say with precision what is meant by "a symmetry," and we must set up the variational bookkeeping the proof will consume. Both tasks are subtler than the translation example let on. There, the transformation acted on the field and on the spacetime point at once, and we slipped between "the field changes" and "the point changes" without comment because for a rigid translation the two descriptions collapse into one. In general they do not, and Noether's theorem in its full reach—covering not just internal symmetries that leave the coordinates untouched but the Lorentz rotations and boosts that move them—demands that we keep the two kinds of change rigorously apart. This section builds that distinction (the *intrinsic* versus *total* variation), derives from it the single master identity for the first variation of a general transformation that every later section reuses, and pins down the precise sense in which a transformation can be called a symmetry: invariance of the action not necessarily on the nose but up to a boundary term, the notion of *quasi-symmetry*. Along the way we separate three things that are easy to conflate—symmetries of the action, symmetries of the equations of motion, and discrete transformations—only the first of which Noether's first theorem converts into a current.

7.1.1 Two notions of variation

A general transformation of a field theory acts on two things simultaneously: it moves each spacetime point, and it changes the value the field takes. Write the coordinate change as

$$x^\mu \longrightarrow x'^\mu = x^\mu + \delta x^\mu, \tag{7.1}$$

with δx^μ an infinitesimal displacement (constant for a translation, linear in x for a Lorentz rotation, and so on), and the field change as a new configuration ϕ'_a whose value at the new point x' is some prescribed function of the old data. The complete specification of a transformation is the pair $(\delta x^\mu,\ \phi'_a)$. The difficulty is that there are two inequivalent ways to measure "how much the field changed," and the literature's sign errors almost all trace to confusing them.

The first compares the new field with the old at the *same coordinate value*:

$$\delta\phi_a(x) \equiv \phi'_a(x) - \phi_a(x). \tag{7.2}$$

This is the *intrinsic* variation (also called the local, or form, variation): it asks how the functional *shape* of the field has changed at a fixed label x, holding the

coordinate fixed and comparing two functions of it. It is exactly the variation $\delta\phi_a$ that Theorem 6.8 was built on, where the coordinates never moved at all ($\delta x = 0$), and it is the object the Euler–Lagrange derivation requires, because only at a fixed point can one subtract two field values and call the difference a variation.

The second comparison follows the point as it moves and compares the new field at the new point with the old field at the old point:

$$\bar{\delta}\phi_a(x) \;\equiv\; \phi'_a(x') - \phi_a(x). \tag{7.3}$$

This is the *total* variation: it measures the change experienced by an observer carried along by the transformation, the full change in the field including the effect of the coordinate transport. For a translation $x' = x + a$ that carries a configuration rigidly, demanding the moved field agree with the original—$\phi'_a(x') = \phi_a(x)$, the defining property of a rigid displacement—means $\bar{\delta}\phi_a = 0$ even though $\delta\phi_a \neq 0$; the field looks unchanged to the comoving observer precisely because the whole profile slid along with the frame.

Definition 7.1 (Intrinsic and total variation)**.** For a transformation $(\delta x^\mu, \phi'_a)$ acting on a field theory, the *intrinsic* (same-point) variation is $\delta\phi_a(x) \equiv \phi'_a(x) - \phi_a(x)$ and the *total* variation is $\bar{\delta}\phi_a(x) \equiv \phi'_a(x') - \phi_a(x)$, both taken to first order in the infinitesimal parameters of the transformation.

The two are related by a single term, the coordinate transport. Expand the total variation by moving the evaluation point of the *new* field from x' back to x:

$$\phi'_a(x') \;=\; \phi'_a(x + \delta x) \;=\; \phi'_a(x) + \delta x^\mu\, \partial_\mu \phi'_a(x) + O(\delta^2). \tag{7.4}$$

In the second term we may replace $\partial_\mu \phi'_a$ by $\partial_\mu \phi_a$, since their difference $\partial_\mu(\delta\phi_a)$ is already first order and the prefactor δx^μ is too, making the discrepancy second order. Subtracting $\phi_a(x)$ from both sides of Eq. (7.4) and using the two definitions,

$$\boxed{\bar{\delta}\phi_a \;=\; \delta\phi_a \;+\; \delta x^\mu\, \partial_\mu \phi_a} \tag{7.5}$$

to first order. The total change equals the intrinsic change at fixed coordinate plus the change $\delta x^\mu \partial_\mu \phi_a$ accumulated by sliding the evaluation point through the field's own gradient. Equation Eq. (7.5) is the hinge of all later bookkeeping: it lets us translate between the comoving description (natural for stating what a symmetry *does*—"the field is a scalar, so it is carried unchanged") and the fixed-point description (mandatory for varying the action). For an internal transformation, which by definition leaves the coordinates alone ($\delta x^\mu = 0$), the two variations coincide, $\bar{\delta}\phi_a = \delta\phi_a$; the distinction has teeth only for spacetime symmetries.

A property that will be used constantly: the intrinsic variation commutes with the spacetime derivative, while the total variation does not. For the intrinsic

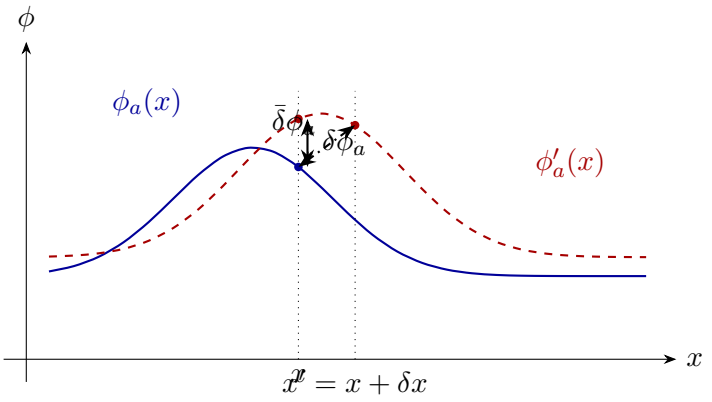


Figure 7.1: The two variations of a field. The solid curve is the original profile ϕ_a, the dashed curve the transformed profile ϕ'_a. The *intrinsic* variation $\delta\phi_a = \phi'_a(x) - \phi_a(x)$ compares the two profiles at the same coordinate x (vertical double arrow). The *total* variation $\bar{\delta}\phi_a = \phi'_a(x') - \phi_a(x)$ also accounts for the displacement of the evaluation point to $x' = x + \delta x$ (slanted dash-dotted arrow). The difference between them is the transport term $\delta x^\mu \partial_\mu \phi_a$ of Eq. (7.5).

variation,

$$\delta(\partial_\mu \phi_a) = \partial_\mu \phi'_a(x) - \partial_\mu \phi_a(x) = \partial_\mu(\phi'_a(x) - \phi_a(x)) = \partial_\mu(\delta\phi_a), \tag{7.6}$$

because δ subtracts two functions at the same argument, and the derivative of a difference is the difference of derivatives. This is the same fact already used in Eq. (6.32) to derive the Euler–Lagrange equations. The total variation enjoys no such identity: from Eq. (7.5),

$$\bar{\delta}(\partial_\mu \phi_a) = \partial_\mu(\delta\phi_a) + \delta x^\nu \, \partial_\nu \partial_\mu \phi_a = \partial_\mu(\bar{\delta}\phi_a) - (\partial_\mu \delta x^\nu)\, \partial_\nu \phi_a, \tag{7.7}$$

where the last form follows by differentiating Eq. (7.5) and rearranging. The obstruction is the term $(\partial_\mu \delta x^\nu)\partial_\nu \phi_a$, which vanishes only when the coordinate displacement is constant. This is precisely why one varies the action through the intrinsic variation $\delta\phi_a$ and never directly through $\bar{\delta}\phi_a$: the integration by parts that produced the Euler–Lagrange operator relied on δ passing freely through ∂_μ, a privilege $\bar{\delta}$ does not hold.

Worked Example 7.2 (The two variations for a scalar under translation and Lorentz rotation)**.** We exercise Eq. (7.5) on the two transformations that matter most, taking a single real scalar field ϕ, for which the comoving prescription is the simplest possible: a scalar is carried unchanged by the flow, $\phi'(x') = \phi(x)$, i.e. $\bar{\delta}\phi = 0$.

Translation. Let $\delta x^\mu = a^\mu$ with a^μ a constant infinitesimal. With $\bar{\delta}\phi = 0$, the relation Eq. (7.5) gives the intrinsic variation immediately,

$$\delta\phi = \bar{\delta}\phi - \delta x^\mu \partial_\mu \phi = -a^\mu \partial_\mu \phi, \tag{7.8}$$

which is exactly the field variation Eq. (6.157) that drove the stress-tensor construction of the previous chapter. The intrinsic variation is nonzero—at a fixed coordinate the field value genuinely changes—while the total variation vanishes. Check the commutation claims directly: $\delta(\partial_\mu\phi) = \partial_\mu(\delta\phi) = -a^\nu\partial_\nu\partial_\mu\phi$ is symmetric and consistent, whereas $\bar\delta(\partial_\mu\phi) = \partial_\mu\bar\delta\phi - (\partial_\mu\delta x^\nu)\partial_\nu\phi = 0$ here only because $\partial_\mu a^\nu = 0$; the constancy of the translation is what makes both descriptions tidy.

Lorentz rotation. Let $\delta x^\mu = \omega^\mu{}_\nu x^\nu$ with $\omega_{\mu\nu} = -\omega_{\nu\mu}$ the infinitesimal antisymmetric parameter of an $\mathrm{SO}(1,3)$ transformation (the antisymmetry is the condition that the metric be preserved, $\delta(\eta_{\mu\nu}x^\mu x^\nu) = 0$; the parametrization is fixed in Chapter 3). A scalar still rides along unchanged, $\bar\delta\phi = 0$, so

$$\delta\phi = -\delta x^\mu\partial_\mu\phi = -\omega^\mu{}_\nu x^\nu\,\partial_\mu\phi = +\tfrac{1}{2}\,\omega_{\mu\nu}(x^\mu\partial^\nu - x^\nu\partial^\mu)\phi, \tag{7.9}$$

the last equality obtained by writing $\omega^\mu{}_\nu x^\nu\partial_\mu = \omega_{\mu\nu}x^\nu\partial^\mu$ and antisymmetrizing in $(\mu\nu)$ against the antisymmetric $\omega_{\mu\nu}$. The differential operator in brackets is the orbital angular-momentum generator the reader will recognize from Chapter 3; here it has fallen out purely from the relation between the two variations. Now the non-commutation of $\bar\delta$ is real and instructive: from Eq. (7.7) with $\partial_\mu\delta x^\nu = \omega^\nu{}_\mu \neq 0$,

$$\bar\delta(\partial_\mu\phi) = -\omega^\nu{}_\mu\,\partial_\nu\phi \neq 0. \tag{7.10}$$

The total variation of the field is zero, but the total variation of its gradient is *not*: the gradient of a scalar is a vector, and a vector is rotated by the transformation even though the scalar is not. This is the precise content of "$\bar\delta$ does not commute with ∂_μ," and it is why the gradient must be varied through δ, not $\bar\delta$, when we attack the action.

7.1.2 The first variation of the action: the master identity

We now compute how the action responds to a completely general infinitesimal transformation $(\delta x^\mu, \delta\phi_a)$, with no assumption yet that the transformation is a symmetry of anything. The result is a single identity—valid for every field configuration, on shell or off—that the entire chapter consumes: Noether's first theorem reads a conservation law off it, the energy–momentum and angular-momentum currents of Section 7.4 are special cases, and the second theorem of Section 7.6 pushes it one step further.

The transformation deforms both the integrand and the region of integration, so the varied action is

$$S' \;=\; \int_{\Omega'} \mathrm{d}^4x'\;\mathcal{L}(\phi'_a(x'),\,\partial'_\mu\phi'_a(x'),\,x'), \tag{7.11}$$

where Ω' is the image of Ω under Eq. (7.1) and $\partial'_\mu = \partial/\partial x'^\mu$. To compare with S we pull everything back to the original coordinates. The change of variables

$x' = x + \delta x$ carries a Jacobian: from $\partial x'^{\mu}/\partial x^{\nu} = \delta^{\mu}{}_{\nu} + \partial_{\nu}(\delta x^{\mu})$, the determinant is

$$\mathrm{d}^4x' = \left|\det \tfrac{\partial x'^{\mu}}{\partial x^{\nu}}\right| \mathrm{d}^4x = \left(1 + \partial_{\mu}(\delta x^{\mu})\right) \mathrm{d}^4x + O(\delta^2), \tag{7.12}$$

using $\det(\mathbf{1} + M) = 1 + \operatorname{tr} M + O(M^2)$ with $\operatorname{tr} M = \partial_{\mu}(\delta x^{\mu})$. The integrand, re-expressed at the running point, differs from the old one by the *total* variation of the Lagrangian density,

$$\bar{\delta}\mathcal{L} \;\equiv\; \mathcal{L}(\phi'_a(x'), \partial'_{\mu}\phi'_a(x'), x') - \mathcal{L}(\phi_a(x), \partial_{\mu}\phi_a(x), x), \tag{7.13}$$

since $\mathcal{L}$ is itself a (composite) field and Theorem 7.1 applies to it verbatim. Assembling Eq. (7.12) and Eq. (7.13) and keeping only first order,

$$\delta S \;=\; S' - S \;=\; \int_{\Omega} \mathrm{d}^4x \,\Big[\, \bar{\delta}\mathcal{L} + \mathcal{L}\,\partial_{\mu}(\delta x^{\mu}) \,\Big]. \tag{7.14}$$

Now convert the total variation of $\mathcal{L}$ to the intrinsic one using Eq. (7.5) applied to $\mathcal{L}$, namely $\bar{\delta}\mathcal{L} = \delta\mathcal{L} + \delta x^{\mu}\partial_{\mu}\mathcal{L}$ (here $\partial_{\mu}\mathcal{L}$ is the *total* spacetime derivative of $\mathcal{L}$, which collects any explicit x-dependence as well as the dependence carried through the fields). The two δx terms combine into a perfect divergence,

$$\bar{\delta}\mathcal{L} + \mathcal{L}\,\partial_{\mu}(\delta x^{\mu}) = \delta\mathcal{L} + \delta x^{\mu}\partial_{\mu}\mathcal{L} + \mathcal{L}\,\partial_{\mu}(\delta x^{\mu}) = \delta\mathcal{L} + \partial_{\mu}\big(\mathcal{L}\,\delta x^{\mu}\big), \tag{7.15}$$

by the Leibniz rule. The remaining piece, the intrinsic variation $\delta\mathcal{L}$, is computed at fixed x exactly as in the Euler–Lagrange derivation—and crucially carries *no* explicit $\partial_{\mu}\mathcal{L}|_{\text{expl}}$ term, because δ holds the coordinate fixed and the explicit x-dependence cancels in the same-point difference. By the chain rule and the commutation property Eq. (7.6),

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_a}\,\delta\phi_a + \frac{\partial\mathcal{L}}{\partial(\partial_{\mu}\phi_a)}\,\partial_{\mu}(\delta\phi_a). \tag{7.16}$$

Introduce the covariant canonical momentum

$$\pi_a{}^{\mu} \;\equiv\; \frac{\partial\mathcal{L}}{\partial(\partial_{\mu}\phi_a)}, \qquad \pi_a{}^{0} = \pi^a \text{ of Eq. (6.96)}, \tag{7.17}$$

a one-index generalization of the canonical momentum whose time component reproduces the Ch. 6 object. Then the standard split—trading $\partial\mathcal{L}/\partial\phi_a$ for the Euler–Lagrange operator $E_a[\mathcal{L}] = \partial\mathcal{L}/\partial\phi_a - \partial_{\mu}\pi_a{}^{\mu}$ of Eq. (6.43) and collecting the leftover into a divergence—gives

$$\delta\mathcal{L} = \Big(\frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_{\mu}\pi_a{}^{\mu}\Big)\delta\phi_a + \partial_{\mu}\big(\pi_a{}^{\mu}\,\delta\phi_a\big) = E_a[\mathcal{L}]\,\delta\phi_a + \partial_{\mu}\big(\pi_a{}^{\mu}\,\delta\phi_a\big). \tag{7.18}$$

Substituting Eq. (7.18) into Eq. (7.15) and that into Eq. (7.14) collapses everything into one statement.

Proposition 7.3 (Master variation of the action)**.** *Under an arbitrary infinitesimal transformation* $(\delta x^\mu, \delta\phi_a)$*, with* $\delta\phi_a$ *the intrinsic variation* (7.2)*, the first variation of the action* (6.25) *is, for* every *field configuration,*

$$\boxed{\delta S = \int_\Omega \mathrm{d}^4x \left\{ E_a[\mathcal{L}]\,\delta\phi_a + \partial_\mu\big(\,\pi_a{}^\mu\,\delta\phi_a + \mathcal{L}\,\delta x^\mu\,\big)\right\}} \tag{7.19}$$

where $E_a[\mathcal{L}] = \partial\mathcal{L}/\partial\phi_a - \partial_\mu\pi_a{}^\mu$ *is the Euler–Lagrange operator* (6.43) *and* $\pi_a{}^\mu = \partial\mathcal{L}/\partial(\partial_\mu\phi_a)$ (7.17)*. On shell, where* $E_a[\mathcal{L}] = 0$*, the variation reduces to a pure boundary term,*

$$\delta S\big|_{\text{on-shell}} = \int_\Omega \mathrm{d}^4x\;\partial_\mu K^\mu, \qquad K^\mu \equiv \pi_a{}^\mu\,\delta\phi_a + \mathcal{L}\,\delta x^\mu. \tag{7.20}$$

The structure of Eq. (7.19) repays study, because every result downstream is a way of reading it. It splits the response of the action into two physically distinct pieces. The *bulk* term $E_a[\mathcal{L}]\,\delta\phi_a$ is the dynamical content: it is what the principle of stationary action sets to zero, and it carries no information about the transformation beyond the direction $\delta\phi_a$. The *boundary* term $\partial_\mu K^\mu$ is pure surface flux; it survives even on solutions and is the seat of every conservation law in this chapter. Two limits recover earlier results and confirm the bookkeeping. If we hold the coordinates fixed ($\delta x = 0$) and demand stationarity against arbitrary $\delta\phi_a$ vanishing on $\partial\Omega$, the boundary term drops and Eq. (7.19) is the Euler–Lagrange derivation of Theorem 6.8. If instead we put the field on shell, Eq. (7.20) says the on-shell action is a boundary functional—the same fact that, in mechanics, makes Hamilton's principal function a function of the endpoints alone.

It is worth recording the boundary current K^μ in the variables of the comoving observer, because that is the form in which the spacetime-symmetry currents will appear. Substitute $\delta\phi_a = \bar\delta\phi_a - \delta x^\nu\partial_\nu\phi_a$ from Eq. (7.5) into the definition in Eq. (7.20):

$$\begin{aligned} K^\mu &= \pi_a{}^\mu\big(\bar\delta\phi_a - \delta x^\nu\partial_\nu\phi_a\big) + \mathcal{L}\,\delta x^\mu \\ &= \pi_a{}^\mu\,\bar\delta\phi_a - \big(\pi_a{}^\mu\,\partial_\nu\phi_a - \delta^\mu{}_\nu\mathcal{L}\big)\,\delta x^\nu = \pi_a{}^\mu\,\bar\delta\phi_a - T^\mu{}_\nu\,\delta x^\nu, \end{aligned} \tag{7.21}$$

where in the last step the combination $\pi_a{}^\mu\partial_\nu\phi_a - \delta^\mu{}_\nu\mathcal{L}$ is exactly the canonical stress–energy tensor $T^\mu{}_\nu$ of Theorem 6.28. The boundary current of a general transformation thus organizes itself automatically into two recognizable blocks: the momentum $\pi_a{}^\mu$ contracted with the comoving field change, plus the canonical stress tensor contracted with the coordinate displacement. We are one short step from Noether's theorem—but only one, and we take it deliberately in the next section rather than here, after the notion of symmetry has been made precise.

7.1.3 Symmetry, quasi-symmetry, and the boundary term

Nothing so far has assumed the transformation does anything special to the action; Theorem 7.3 is an identity for any $(\delta x, \delta\phi)$. A transformation earns the name *symmetry* when it leaves the action's content untouched. The naive demand $\delta S = 0$ is, however, slightly too strong, and getting the definition exactly right is what makes Noether's theorem both true and useful.

The physical requirement is that the transformation leave the *dynamics* invariant: it must map every solution of the equations of motion to another solution. From Eq. (7.19) this is guaranteed if the transformation changes the action by at most a surface term for *every* configuration—not only on shell—because a surface term cannot affect which configurations make the bulk variation vanish. Concretely, two Lagrangians differing by a total divergence yield identical Euler–Lagrange equations: $E_a[\partial_\nu F^\nu] = 0$ identically, as established in Eq. (6.44). A transformation that shifts $\mathcal{L}$ by a total divergence therefore shifts S by a boundary integral that the equations of motion never see. This is the content we want to legislate.

Definition 7.4 (Continuous symmetry and quasi-symmetry)**.** A continuous (infinitesimal) transformation $(\delta x^\mu, \delta\phi_a)$ is a *quasi-symmetry* of the action (6.25) if there exists a local current $\Lambda^\mu(\phi_a, \partial_\mu\phi_a, x)$, built from the fields and a finite number of their derivatives, such that the integrand of the first variation is a total divergence *off shell*:

$$\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = \partial_\mu\Lambda^\mu \qquad \text{for every field configuration.} \tag{7.22}$$

Equivalently, $\delta S = \int_\Omega \mathrm{d}^4x\,\partial_\mu\Lambda^\mu$ is a boundary integral for all configurations. If Eq. (7.22) holds with $\Lambda^\mu \equiv 0$—the action being invariant on the nose, $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = 0$—the transformation is a *strict* (or manifest) symmetry. Otherwise it is a quasi-symmetry *proper*, invariant only up to the boundary term Λ^μ.

The condition Eq. (7.22) is demanded off shell, for every configuration, and this is the crux. It is what distinguishes a symmetry of the *action* from the weaker notion, discussed below, of a symmetry of the equations of motion. The current Λ^μ is not part of the data of the transformation; it is determined (up to an identically conserved piece) by working out the left-hand side and recognizing it as a divergence. When it cannot be so recognized—when $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu)$ fails to be a total derivative for some configuration—the transformation is not a symmetry of the action at all, and the first theorem yields nothing.

Physical Insight

Why tolerate the boundary term Λ^μ at all—why not insist on strict invariance? Because the action's job is to produce equations of motion, and it does that only through the bulk variation; the surface term is

invisible to the dynamics. Insisting on strict invariance would discard genuine symmetries on a technicality. The cleanest examples are the supersymmetry transformations of Chapter 48, under which the Lagrangian shifts by a total derivative that *cannot* be removed, and the Galilean boosts of nonrelativistic field theory, whose Noether charge picks up the famous boundary contribution. Even the spacetime translations of Ch. 6 are, in the fixed-coordinate description, quasi-symmetries with nonzero Λ^μ, as the next worked example shows. The boundary term is not a blemish to be tolerated; it is part of the symmetry, and Noether's theorem will absorb it cleanly into the conserved current.

Worked Example 7.5 (Translation: strict symmetry or quasi-symmetry, depending on the bookkeeping)**.** Consider a Lagrangian with no explicit x-dependence, $\mathcal{L} = \mathcal{L}(\phi_a, \partial_\mu \phi_a)$, and the constant translation $x'^\mu = x^\mu + a^\mu$. We compute δS in two equivalent bookkeepings and find the same physics described once as a strict symmetry and once as a quasi-symmetry—a concrete demonstration that the strict/quasi split is partly a matter of how one apportions the change between δx and $\delta\phi$.

Coordinate-inclusive description ($\delta x^\mu = a^\mu$, $\bar\delta\phi_a = 0$). A translation carries every field rigidly, so the comoving change vanishes, $\bar\delta\phi_a = 0$, and likewise $\bar\delta\mathcal{L} = 0$ because $\mathcal{L}$ is a scalar built from the fields with no explicit x. Since $\partial_\mu(\delta x^\mu) = \partial_\mu a^\mu = 0$, the integrand of Eq. (7.14) is $\bar\delta\mathcal{L} + \mathcal{L}\,\partial_\mu(\delta x^\mu) = 0$. The action is invariant on the nose: translation is a *strict* symmetry, $\Lambda^\mu = 0$.

Fixed-coordinate description ($\delta x^\mu = 0$, $\delta\phi_a = -a^\nu \partial_\nu \phi_a$). Now we refuse to move the coordinates and absorb the entire transformation into the intrinsic field variation Eq. (7.8). The intrinsic variation of the Lagrangian is, by the chain rule Eq. (7.16),

$$\delta\mathcal{L} = \frac{\partial\mathcal{L}}{\partial\phi_a}\delta\phi_a + \pi_a{}^\mu\,\partial_\mu(\delta\phi_a) = -a^\nu\Big[\frac{\partial\mathcal{L}}{\partial\phi_a}\partial_\nu\phi_a + \pi_a{}^\mu\,\partial_\mu\partial_\nu\phi_a\Big] = -a^\nu\,\partial_\nu\mathcal{L}, \tag{7.23}$$

the bracket being precisely the chain-rule expansion of $\partial_\nu\mathcal{L}$, legitimate because $\mathcal{L}$ has no explicit coordinate dependence. With $\delta x = 0$ the quasi-symmetry condition Eq. (7.22) reads $\delta\mathcal{L} = \partial_\mu\Lambda^\mu$, and Eq. (7.23) is exactly such a divergence:

$$\delta\mathcal{L} = -a^\nu\partial_\nu\mathcal{L} = \partial_\mu(-a^\mu\mathcal{L}) \implies \Lambda^\mu = -a^\mu\mathcal{L} \neq 0. \tag{7.24}$$

In this bookkeeping translation is a *quasi-symmetry proper*, with boundary current $\Lambda^\mu = -a^\mu\mathcal{L}$. The two descriptions agree on the only invariant fact—that δS is a boundary integral—but disagree on whether Λ^μ vanishes, because they apportion the transformation differently between coordinate and field. The lesson for the next section is sharp: the boundary term Λ^μ cannot be discarded. Were one to drop it, the conserved current produced by Noether's theorem would be

$\pi_a{}^\mu \delta\phi_a$ alone—missing exactly the $-\eta^{\mu\nu}\mathcal{L}$ piece of the canonical stress tensor Eq. (6.165). The $\Lambda^\mu = -a^\mu\mathcal{L}$ here *is* that missing piece, as one sees by comparing with Eq. (7.21). The boundary term is not optional accounting; it is half of the energy.

There remains a distinction that the word "symmetry" invites one to blur: the difference between a symmetry of the action and a symmetry of the equations of motion. A transformation is a *symmetry of the equations of motion* if it maps solutions to solutions—if whenever ϕ_a satisfies $E_a[\mathcal{L}] = 0$, so does the transformed field. Every quasi-symmetry of the action is automatically a symmetry of the equations of motion, by the argument above: changing $\mathcal{L}$ by a total divergence leaves $E_a[\mathcal{L}]$ untouched. The converse fails, and the failure is exactly the class of transformations Noether's first theorem cannot reach.

Example 7.6 (A symmetry of the equations of motion that is not a symmetry of the action)**.** Take the free massless real scalar, $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$, with linear equation of motion $\Box\phi = 0$, and the rigid field rescaling $\phi \to \phi' = e^\alpha\phi$, i.e. $\delta\phi = \alpha\,\phi$ at fixed coordinates ($\delta x = 0$). Linearity makes this a symmetry of the equation of motion: if $\Box\phi = 0$ then $\Box(e^\alpha\phi) = 0$, so solutions map to solutions for any α. But it is *not* a symmetry of the action. The Lagrangian is quadratic, so it rescales homogeneously, $\mathcal{L} \to e^{2\alpha}\mathcal{L}$, and

$$\delta\mathcal{L} = 2\alpha\,\mathcal{L} = \alpha\,(\partial_\mu\phi)(\partial^\mu\phi), \tag{7.25}$$

which is *not* a total divergence for a generic configuration: its value at a point is the local kinetic density, a quantity that cannot be written as $\partial_\mu\Lambda^\mu$ with Λ^μ local, since integrating it over a finite region gives a configuration-dependent number rather than a pure surface flux. The quasi-symmetry condition Eq. (7.22) fails, so Theorem 7.3 yields no conserved current. Rescaling the field is a symmetry of the dynamics with no accompanying conservation law—the clearest illustration that Noether's first theorem is a theorem about symmetries of the *action*, not of the equations of motion. (The genuine *spacetime* dilatation symmetry of the massless theory, which transforms the coordinates as well and *does* leave the action invariant up to a boundary term, is a separate transformation; it carries a real conserved dilatation current, constructed in Section 7.7.5 when we close ϕ^4 theory.)

Caution

The phrase "ϕ^4 theory is scale invariant when $m = 0$" is true but must be read carefully. What is scale invariant is the full *dilatation* transformation, which rescales the spacetime coordinates $x \to e^{-\alpha}x$ *together* with the field $\phi \to e^\alpha\phi$, and which leaves the action invariant up to a boundary term. The field rescaling alone, with coordinates frozen, is Theorem 7.6: a

symmetry of the equation of motion but *not* of the action, and hence *not* a source of any Noether current. Conflating the two is a standard route to a spurious "conservation law."

7.1.4 Continuous families, generators, and the discrete foil

Noether's first theorem is a statement about *continuous* symmetries, and we close the section by saying precisely what that means and why the qualifier is indispensable. A continuous symmetry is not a single transformation but a smooth family of them, connected to the identity by a parameter that can be made arbitrarily small. It is the existence of this infinitesimal neighborhood—the very thing Theorem 7.3 expanded in—that makes a conservation law possible.

Definition 7.7 (One- and multi-parameter symmetry families and their generators)**.** A *one-parameter family* of transformations is a smooth map $s \mapsto (\delta x^\mu_s, \phi_a^{(s)})$ with $s \in \mathbb{R}$ reducing to the identity at $s = 0$. Its *generator* is the pair of first-order data

$$\delta x^\mu = \frac{dx'^\mu}{ds}\Big|_{s=0}, \qquad \delta\phi_a = \frac{\partial \phi_a^{(s)}}{\partial s}\Big|_{s=0}, \tag{7.26}$$

read off the linear term $x'^\mu = x^\mu + s\,\delta x^\mu + O(s^2)$, $\phi_a^{(s)} = \phi_a + s\,\delta\phi_a + O(s^2)$. A *multi-parameter* (Lie) family carries r independent parameters α^A, $A = 1, \dots, r$, with infinitesimal data

$$\delta x^\mu = \alpha^A\, X^\mu_A(x), \qquad \delta\phi_a = \alpha^A\, \Phi_{A,a}(\phi, \partial\phi, x), \tag{7.27}$$

one generator $(X^\mu_A, \Phi_{A,a})$ for each parameter. The family is a (quasi-)symmetry when Theorem 7.4 holds for each α^A independently, producing a boundary current Λ^μ_A for every generator.

When the transformations close under composition—the result of doing one and then another is again a member of the family—the generators close into a Lie algebra, $[\,\delta_A, \delta_B\,] = f_{AB}{}^C\,\delta_C$, with the structure constants $f_{AB}{}^C$ of Chapter 2. This is what makes the symmetries of a field theory a *group*, and it is the reason the conserved charges of an internal symmetry will be found in Section 7.3 to satisfy that same algebra under the Poisson bracket—the charges *generate* the very transformation they conserve. For now the only point we need is structural: a continuous symmetry supplies, through Eq. (7.27), exactly r independent infinitesimal directions, and Noether's first theorem will return exactly r independent conserved currents—one per parameter. The counting "one current per continuous parameter" is built into the very definition of the family.

The qualifier "continuous" is not a convenience; it is load-bearing. The entire machine of Theorem 7.3 rests on expanding the action to *first order* in an

infinitesimal parameter. A transformation that admits no such expansion—one not connected to the identity, with no neighborhood to linearize in—never enters the master identity, and Noether's first theorem is simply silent about it. Such are the *discrete* symmetries.

Example 7.8 (The $\mathbb{Z}_2$ of ϕ^4 theory yields no Noether current)**.** The standard Lagrangian of ϕ^4 theory,

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{7.28}$$

is invariant under the reflection $\phi \to -\phi$: every term contains an even power of ϕ, so $\mathcal{L}(-\phi, -\partial_\mu\phi) = \mathcal{L}(\phi, \partial_\mu\phi)$ exactly. This $\mathbb{Z}_2$ is a genuine symmetry of the action—indeed a strict one, $\delta\mathcal{L} = 0$—and it has powerful consequences: it forbids every correlation function with an odd number of fields, and its spontaneous breaking in Chapter 31 is the prototype of the Higgs mechanism. But it is a group of two elements, $\{\phi \to \phi,\ \phi \to -\phi\}$, with *no* element infinitesimally close to the identity: there is no smooth family $\phi \to e^{\alpha} \cdot (\text{reflection})\phi$ interpolating to it, because -1 is not connected to $+1$ within the real numbers acting on a single real field. There is no generator (7.26) to feed into the master identity Eq. (7.19), and hence no conserved current and no conserved charge. Noether's first theorem cannot see the $\mathbb{Z}_2$. We will return to this reflection in Section 7.3 precisely as the foil that marks the boundary of the theorem's reach—the symmetry that organizes the theory's selection rules without contributing to its conserved currents.

The same verdict applies to the discrete spacetime operations—parity P, time reversal T, and the internal charge conjugation C—each of which is a reflection with no infinitesimal version. They are indispensable to the structure of quantum field theory, governing selection rules and culminating in the CPT theorem of Chapter 12, but none is a continuous symmetry and none produces a Noether current. The dividing line is clean: continuous symmetries give conservation laws through the first theorem; discrete symmetries give selection rules through their action on states and amplitudes. This chapter is about the former; the latter are taken up where they belong.

With the variational bookkeeping in place—the intrinsic and total variations and their relation (7.5), the master identity (7.19) for the first variation of the action, and the precise definition (7.4) of a continuous (quasi-)symmetry—we have assembled exactly the two ingredients Noether's theorem needs. On one side stands the off-shell quasi-symmetry condition, $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = \partial_\mu\Lambda^\mu$; on the other, the on-shell reduction of the master identity to the pure boundary term $\partial_\mu K^\mu$. Setting these two expressions for the same quantity against each other—one true off shell, the other true on shell—forces a local current to be conserved. That single confrontation is the whole of Noether's first theorem, and it is the business of Section 7.2.

7.2 Noether's First Theorem

The previous section assembled the two ingredients this one consumes, and did so deliberately holding back the final step. On one side it produced the *master variation identity* (Theorem 7.3): for an arbitrary infinitesimal transformation $(\delta x^\mu, \delta\phi_a)$ of an arbitrary field configuration, the first variation of the action splits into a bulk term carrying the Euler–Lagrange operator and a pure boundary divergence. On the other it pinned down what it means for such a transformation to deserve the name *symmetry*: not that the Lagrangian density be numerically unchanged, but the weaker and more useful demand that the action change by at most a surface term *off shell*—the notion of quasi-symmetry (Theorem 7.4). The master identity is an algebraic statement true for every configuration; the symmetry condition is a hypothesis we may or may not impose. The content of Noether's first theorem is what happens when both are in force at once together with the equations of motion: the symmetry says one combination is a pure divergence, the dynamics says another combination vanishes, and what is squeezed out between them is the vanishing of a single local current. The argument is short—it is essentially one substitution—but its consequences organize the rest of the book, so we state every hypothesis, prove every step, read off the conserved current and the conserved charge, and chart precisely how much freedom the construction leaves.

We carry the conventions of the chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, and a (possibly multi-component) classical field ϕ_a with first-order Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu\phi_a, x)$ on a spacetime region Ω, the index a running over all internal and spacetime components the theory carries, repeated indices summed. Two objects from the previous section recur on every line below, and we recall them rather than redefine them. The first is the covariant canonical momentum $\pi_a{}^\mu \equiv \partial\mathcal{L}/\partial(\partial_\mu\phi_a)$ of Eq. (7.17), the one-index four-vector whose time component $\pi_a{}^0$ is the canonical momentum of Theorem 6.13. The second is the relation between the intrinsic (fixed-point) variation $\delta\phi_a$ and the total (comoving) variation $\bar{\delta}\phi_a$,

$$\bar{\delta}\phi_a = \delta\phi_a + \delta x^\mu\, \partial_\mu\phi_a, \tag{7.29}$$

which is Eq. (7.5); the single property that drives the whole derivation is that the intrinsic variation, being taken at fixed x, commutes with the partial derivative, $\delta(\partial_\mu\phi_a) = \partial_\mu(\delta\phi_a)$ (Eq. (7.6)), whereas the total variation does not.

7.2.1 From the master identity to a conservation law

The master variation of Theorem 7.3 was stated as an integrated identity for δS. For the theorem we want its *local*, pre-integration form, which is already contained in the steps that produced it. Recall from the previous section that

the intrinsic variation of the Lagrangian, after one integration by parts, reads

$$\delta\mathcal{L} = E_a[\mathcal{L}]\,\delta\phi_a + \partial_\mu(\pi_a{}^\mu\,\delta\phi_a), \qquad E_a[\mathcal{L}] \equiv \frac{\partial\mathcal{L}}{\partial\phi_a} - \partial_\mu\pi_a{}^\mu, \tag{7.30}$$

which is Eq. (7.18) with $E_a[\mathcal{L}]$ the Euler–Lagrange operator of Theorem 6.8. Adding $\partial_\mu(\mathcal{L}\,\delta x^\mu)$ to both sides and using the recombination $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = \bar{\delta}\mathcal{L} + \mathcal{L}\,\partial_\mu(\delta x^\mu)$ of Eq. (7.15), the left-hand side becomes exactly the quantity that the quasi-symmetry condition controls, while the right-hand side organizes itself into an Euler–Lagrange piece and a single divergence:

$$\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) \;=\; E_a[\mathcal{L}]\,\delta\phi_a + \partial_\mu K^\mu, \qquad K^\mu \equiv \pi_a{}^\mu\,\delta\phi_a + \mathcal{L}\,\delta x^\mu. \tag{7.31}$$

Here K^μ is the on-shell boundary current already introduced in Eq. (7.20). Equation Eq. (7.31) is the local master identity. It is an off-shell statement, true for every field configuration, solution or not, and nothing in it has yet used either the symmetry of the action or the equations of motion. It is the neutral substrate; the theorem is what one reads off it after imposing those two hypotheses.

The candidate conserved object is the divergence-carrying current K^μ, corrected by the boundary current that quasi-symmetry permits.

Definition 7.9 (Noether current)**.** Let the transformation $(\delta x^\mu, \delta\phi_a)$ be a quasi-symmetry of the action with boundary current Λ^μ as in Theorem 7.4, i.e. $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = \partial_\mu\Lambda^\mu$ off shell. The associated *Noether current* is

$$\boxed{J^\mu \;\equiv\; K^\mu - \Lambda^\mu \;=\; \pi_a{}^\mu\,\delta\phi_a + \mathcal{L}\,\delta x^\mu - \Lambda^\mu} \tag{7.32}$$

where $\delta\phi_a$ is the intrinsic field variation (7.2).

Theorem 7.10 (Noether's first theorem)**.** *If the transformation $(\delta x^\mu, \delta\phi_a)$ is a quasi-symmetry of the action (6.25), then on every solution of the Euler–Lagrange equations (6.30) the Noether current (7.32) is locally conserved,*

$$\partial_\mu J^\mu = 0. \tag{7.33}$$

Proof. Start from the local master identity Eq. (7.31), valid off shell for any configuration,

$$\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = E_a[\mathcal{L}]\,\delta\phi_a + \partial_\mu K^\mu. \tag{7.34}$$

Now impose the first hypothesis. The quasi-symmetry condition Eq. (7.22) states that the entire left-hand side is itself a total divergence, $\delta\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = \partial_\mu\Lambda^\mu$, again as an identity in the fields. Substituting this for the left-hand side of Eq. (7.34),

$$\partial_\mu\Lambda^\mu = E_a[\mathcal{L}]\,\delta\phi_a + \partial_\mu K^\mu. \tag{7.35}$$

Move the divergence of Λ^μ across and combine the two divergences, which by linearity is the divergence of $K^\mu - \Lambda^\mu = J^\mu$:

$$\partial_\mu(K^\mu - \Lambda^\mu) = -E_a[\mathcal{L}]\,\delta\phi_a, \qquad \text{i.e.} \qquad \boxed{\partial_\mu J^\mu = -E_a[\mathcal{L}]\,\delta\phi_a} \tag{7.36}$$

This is still an off-shell identity: the divergence of the Noether current equals, at every point and for every configuration, minus the Euler–Lagrange operator contracted with the intrinsic field variation. Now impose the second hypothesis. On a solution of the equations of motion the Euler–Lagrange operator vanishes identically, $E_a[\mathcal{L}] = 0$ (Theorem 6.8), so the right-hand side of Eq. (7.36) is zero and $\partial_\mu J^\mu = 0$ on shell. □

The proof exhibits the anatomy promised at the close of the previous section: *one identity from the symmetry, one substitution of the dynamics.* The quasi-symmetry condition collapses the left-hand side of the master identity to $\partial_\mu \Lambda^\mu$; the equations of motion annihilate the Euler–Lagrange term; what survives is the vanishing of a divergence. The off-shell form Eq. (7.36) deserves to be kept in its own right, not discarded as a stepping stone. It says that the failure of the Noether current to be conserved is measured precisely by the equations of motion—the current is conserved exactly to the extent that the field obeys its dynamics. In the quantum theory this same off-shell relation, with $E_a[\mathcal{L}]$ becoming an operator insertion, is the seed of the Ward–Takahashi identities: the classical equations-of-motion term turns into contact terms inside correlation functions, and the would-be conservation law acquires calculable anomalies and contact corrections. The classical statement Eq. (7.36) is the skeleton on which all of that is later hung.

Physical Insight

Noether's theorem is an accounting identity, not a force law. It does not assert that something is conserved *because* a particular interaction balances another; it asserts that the bookkeeping of a continuous symmetry, combined with *any* dynamics derived from a symmetric action, manufactures a quantity whose total time derivative is a pure spatial flux. The deep content is the correspondence itself: each independent continuous direction in which the action is invariant supplies exactly one locally conserved current. This is the sense in which, in field theory, symmetry rather than force is the organizing principle—forces are the incidental content of a particular Lagrangian, while the conservation laws are dictated by its symmetries alone.

It is illuminating to rewrite Eq. (7.32) so that the contributions of field space and of spacetime appear separately, because that is the form in which internal and spacetime symmetries will specialize. Substitute the inverse of Eq. (7.29), $\delta\phi_a = \bar{\delta}\phi_a - \delta x^\nu \partial_\nu \phi_a$, into the field term of K^μ:

$$\begin{aligned} J^\mu &= \pi_a{}^\mu \big(\bar{\delta}\phi_a - \delta x^\nu \partial_\nu \phi_a\big) + \mathcal{L}\, \delta^\mu{}_\nu\, \delta x^\nu - \Lambda^\mu \\ &= \pi_a{}^\mu\, \bar{\delta}\phi_a - \underbrace{\big(\pi_a{}^\mu\, \partial_\nu \phi_a - \delta^\mu{}_\nu\, \mathcal{L}\big)}_{= T^\mu{}_\nu}\, \delta x^\nu - \Lambda^\mu . \end{aligned} \tag{7.37}$$

The underbraced combination is exactly the canonical stress–energy tensor $T^{\mu}{}_{\nu}$ of Theorem 6.28; this is the same regrouping already recorded for the boundary current in Eq. (7.21). Naming the *coordinate piece*

$$F^{\mu} \;\equiv\; T^{\mu}{}_{\nu}\,\delta x^{\nu} = \left(\pi_a{}^{\mu}\,\partial_{\nu}\phi_a - \delta^{\mu}{}_{\nu}\,\mathcal{L}\right)\delta x^{\nu}, \tag{7.38}$$

the current takes the three-piece form anticipated in the chapter blueprint,

$$\boxed{J^{\mu} = \pi_a{}^{\mu}\,\bar{\delta}\phi_a - F^{\mu} - \Lambda^{\mu}} \tag{7.39}$$

whose three terms are, respectively, the response of the field measured by the comoving observer ($\pi_a{}^{\mu}\,\bar{\delta}\phi_a$), the response to the displacement of the spacetime point itself ($-F^{\mu}$, built from the canonical stress tensor), and the boundary term that quasi-symmetry permits ($-\Lambda^{\mu}$). The two great families of symmetries are the two ways of switching pieces on and off. *Internal* symmetries leave the coordinates fixed, $\delta x^{\mu} = 0$, so F^{μ} vanishes and the current is carried by the field piece; these are the subject of the next section. *Spacetime* symmetries move the coordinates and are carried largely by F^{μ}; these come later. Both are special cases of the single formula Eq. (7.39).

The most concrete check is the one symmetry already worked out by hand in the previous chapter. We run it through the general machine, both to verify signs and to set the template for the spacetime symmetries to come.

Worked Example 7.11 (Spacetime translations recover the canonical stress tensor and the four-momentum)**.** Take the rigid translation $x'^{\mu} = x^{\mu} + a^{\mu}$ with constant infinitesimal a^{μ}, in a theory whose Lagrangian carries no explicit coordinate dependence. The generators are $\delta x^{\mu} = a^{\mu}$ and, since a translated configuration carries its value rigidly to the new point, $\phi'_a(x') = \phi_a(x)$, i.e. the comoving change vanishes, $\bar{\delta}\phi_a = 0$. That this is a strict symmetry, $\Lambda^{\mu} = 0$, was established in the coordinate-inclusive bookkeeping of Theorem 7.5: with $\bar{\delta}\mathcal{L} = 0$ and $\partial_{\mu}(\delta x^{\mu}) = 0$ the invariance condition holds on the nose. We need only read off the current. Using the three-piece form Eq. (7.39) with $\bar{\delta}\phi_a = 0$, $\Lambda^{\mu} = 0$, and $\delta x^{\nu} = a^{\nu}$,

$$J^{\mu} = \pi_a{}^{\mu}\cdot 0 - T^{\mu}{}_{\nu}\,a^{\nu} - 0 = -\,a^{\nu}\,T^{\mu}{}_{\nu}. \tag{7.40}$$

The four components a^{ν} are independent parameters, so by the multi-parameter statement proved below (Theorem 7.17) each coefficient is separately conserved: stripping the arbitrary constant $-a^{\nu}$, the conserved current of the ν-th translation is $T^{\mu}{}_{\nu}$ itself, and Eq. (7.33) reads

$$\partial_{\mu}T^{\mu}{}_{\nu} = 0 \qquad (\nu = 0, 1, 2, 3), \tag{7.41}$$

in exact agreement with the direct computation Theorem 6.29. The associated charges are the components of the four-momentum: integrating the time component of $-a^{\nu}T^{\mu}{}_{\nu}$ over a constant-time slice and stripping $-a^{\nu}$,

$$P_{\nu} = \int_{\mathbb{R}^3} \mathrm{d}^3x\; T^{0}{}_{\nu}, \tag{7.42}$$

which is the four-momentum of Theorem 6.30, conserved by Theorem 6.31. The general theorem and the previous chapter's hand computation agree on every sign. The lesson to carry forward is structural: the canonical stress tensor is nothing other than the Noether current of spacetime translation, and energy and momentum are its charges—a statement we will sharpen, and repair, when the coordinate piece F^μ is developed in full.

7.2.2 The conserved charge and its time-independence

A locally conserved current is a statement about densities and fluxes at each point. To extract from it a single conserved *number*—the charge—one integrates the time component over a spatial slice. The local law $\partial_\mu J^\mu = 0$ then becomes the global law that this number does not change in time, provided the current does not leak out through the boundary at spatial infinity. The fall-off that guarantees no such leakage is a genuine physical hypothesis, not a formality, and we state it explicitly.

Definition 7.12 (Noether charge)**.** The *Noether charge* associated with a conserved current J^μ is the integral of its time component over a constant-time slice,

$$Q(t) = \int_{\mathbb{R}^3} \mathrm{d}^3x \; J^0(t, \mathbf{x}). \tag{7.43}$$

Theorem 7.13 (Conservation of the Noether charge)**.** *Let J^μ satisfy $\partial_\mu J^\mu = 0$, let $J^0(t,\cdot) \in L^1(\mathbb{R}^3)$ so that $Q(t)$ is finite, and assume the spatial fall-off*

$$\lim_{R\to\infty} R^2 \sup_{|\mathbf{x}|=R} \left|J^i(t,\mathbf{x})\right| = 0 \qquad \text{for each fixed } t, \tag{7.44}$$

i.e. the spatial current decays faster than $1/R^2$, so that its flux through the sphere of radius R vanishes as $R \to \infty$. Then the charge is time-independent,

$$\frac{dQ}{dt} = 0. \tag{7.45}$$

Proof. Differentiate Eq. (7.43) under the integral sign—legitimate because J^0 and its time derivative are integrable—and use the conservation law to trade the time derivative of the charge density for the spatial divergence of the current. Splitting the four-divergence into temporal and spatial parts, $\partial_\mu J^\mu = \partial_0 J^0 + \partial_i J^i = 0$, gives $\partial_0 J^0 = -\partial_i J^i$, so

$$\frac{dQ}{dt} = \int_{\mathbb{R}^3} \mathrm{d}^3x \; \partial_0 J^0 = -\int_{\mathbb{R}^3} \mathrm{d}^3x \; \partial_i J^i. \tag{7.46}$$

The integrand is now a three-dimensional divergence. Integrate it over the ball B_R of radius R and apply the divergence theorem,

$$\int_{B_R} \mathrm{d}^3x \; \partial_i J^i = \oint_{S_R} \mathrm{d}S \; \hat{n}_i \, J^i, \tag{7.47}$$

where $S_R = \partial B_R$ is the sphere of radius R with outward unit normal $\hat{n}_i$ and scalar area element $\mathrm{d}S$. The magnitude of the surface integral is bounded by the area $4\pi R^2$ times the supremum of $|J^i|$ on the sphere,

$$\left| \oint_{S_R} \mathrm{d}S \, \hat{n}_i \, J^i \right| \leq 4\pi R^2 \sup_{|\mathbf{x}|=R} |J^i| \xrightarrow[R\to\infty]{} 0 \tag{7.48}$$

by the fall-off hypothesis Eq. (7.44). Therefore the spatial integral in Eq. (7.46) is the $R \to \infty$ limit of $\int_{B_R} \mathrm{d}^3x \, \partial_i J^i$, which equals the limit of the surface integral, which is zero. Hence $\mathrm{d}Q/\mathrm{d}t = 0$. □

The physical reading is the familiar continuity equation. Writing $\partial_0 J^0 = -\nabla \mathbf{J}$, the time component J^0 is a charge density and the spatial part $\mathbf{J}$ is its current; charge is neither created nor destroyed locally but only transported, and the fall-off Eq. (7.44) says none of it escapes through the sphere at infinity. The hypothesis is not idle: in a theory with long-range fields, or with charged matter that reaches null or spatial infinity, the flux integral need not vanish, and then the naively defined charge is genuinely not conserved. We will meet exactly this subtlety for the electric charge of electrodynamics, where the Gauss-law constraint ties the total charge to the flux of the field at infinity. For the symmetries treated in this chapter the fields are taken to decay, and Eq. (7.44) holds.

Caution

The conservation of Q requires the spatial fall-off Eq. (7.44) *in addition to* the local conservation $\partial_\mu J^\mu = 0$. A current can be perfectly conserved as a local density and yet carry a time-dependent charge if probability or charge flows in from infinity. The two statements—local conservation and global conservation—are logically distinct, joined only by the boundary condition. Sloppily identifying "$\partial_\mu J^\mu = 0$" with "Q is conserved" is a standard error; the bridge between them is the divergence theorem plus a fall-off assumption that must be checked.

There is a sharper, manifestly covariant statement hiding inside Theorem 7.13. The charge is not merely independent of the coordinate time t; it is the same number computed on *any* spacelike slice, and is therefore a Lorentz scalar when J^μ transforms as a four-vector. This is the form in which the charge enters the relativistic theory and the charge algebra of the next section, so we record it.

Proposition 7.14 (Slice-independence of the charge)**.** *Define the charge on a spacelike Cauchy slice Σ as the flux*

$$Q[\Sigma] = \int_\Sigma \mathrm{d}\Sigma_\mu \, J^\mu, \tag{7.49}$$

where $\mathrm{d}\Sigma_\mu = n_\mu \, \mathrm{d}S$ with n_μ the future-pointing unit normal and $\mathrm{d}S$ the induced volume element. If $\partial_\mu J^\mu = 0$ and J^μ obeys the spatial fall-off Eq. (7.44), then

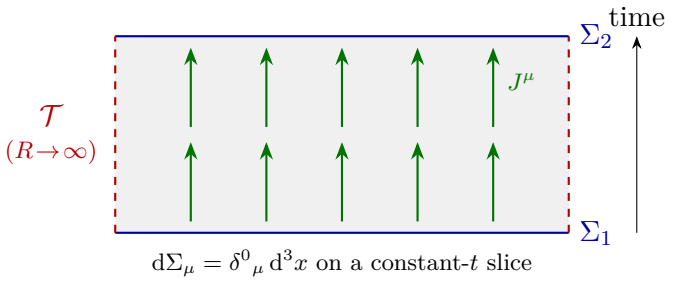


Figure 7.2: Slice-independence of the Noether charge (Theorem 7.14). The current J^μ has vanishing four-divergence inside the region V bounded below by the spacelike slice Σ_1, above by Σ_2, and laterally by the wall $\mathcal{T}$ pushed to spatial infinity. The four-dimensional divergence theorem equates the flux out through Σ_2 with the flux in through Σ_1, while the wall flux vanishes by the fall-off Eq. (7.44). The charge $Q[\Sigma] = \int_\Sigma \mathrm{d}\Sigma_\mu J^\mu$ is therefore the same on every slice; on a constant-time slice it reduces to $\int \mathrm{d}^3x\, J^0$.

$Q[\Sigma]$ is independent of the slice Σ. On a constant-time slice $\mathrm{d}\Sigma_\mu = \delta^0{}_\mu\,\mathrm{d}^3x$ and Eq. (7.49) reduces to Eq. (7.43).

Proof. Let Σ_1 and Σ_2 be two spacelike slices with Σ_2 to the future of Σ_1, and let V be the four-dimensional region between them, closed off at large spatial radius by a timelike "wall" $\mathcal{T}$ at $|\mathbf{x}| = R$ (Fig. 7.2). The four-dimensional divergence theorem applied to J^μ over V gives

$$\int_V \mathrm{d}^4x\, \partial_\mu J^\mu = \oint_{\partial V} \mathrm{d}\Sigma_\mu\, J^\mu = \int_{\Sigma_2} \mathrm{d}\Sigma_\mu\, J^\mu - \int_{\Sigma_1} \mathrm{d}\Sigma_\mu\, J^\mu + \int_{\mathcal{T}} \mathrm{d}\Sigma_\mu\, J^\mu, \tag{7.50}$$

the relative sign between the two slice contributions arising because the outward normal of V points to the future on Σ_2 but to the past on Σ_1. The left-hand side vanishes by $\partial_\mu J^\mu = 0$. The wall contribution involves the spatial components J^i integrated over a cylinder of radius R, whose lateral area grows like R^2 while $|J^i|$ decays faster than R^{-2} by Eq. (7.44); sending $R \to \infty$ kills it. What remains is

$$\int_{\Sigma_2} \mathrm{d}\Sigma_\mu\, J^\mu = \int_{\Sigma_1} \mathrm{d}\Sigma_\mu\, J^\mu, \qquad \text{i.e.} \qquad Q[\Sigma_2] = Q[\Sigma_1]. \tag{7.51}$$

Two arbitrary spacelike slices can always be joined by such a region (subdividing the construction if they happen to intersect), so $Q[\Sigma]$ takes the same value on every slice. The reduction to Eq. (7.43) on a constant-time slice is immediate from $\mathrm{d}\Sigma_\mu = \delta^0{}_\mu\,\mathrm{d}^3x$. □

7.2.3 The improvement ambiguity

The conservation law and the charge are physical, but the current that carries them is not unique. One may add to any conserved current a piece that is conserved identically—without using the equations of motion—and the charge is unchanged provided that piece, too, decays at infinity. The canonical such piece is the divergence of an antisymmetric tensor, and this freedom, far from being a defect, is exactly what later lets us repair the canonical stress tensor into a symmetric, gauge-invariant object.

Proposition 7.15 (Improvement of the Noether current)**.** *Let J^μ be a conserved current and let $B^{\mu\nu} = -B^{\nu\mu}$ be any antisymmetric tensor field (a "superpotential"). Then the improved current*

$$J'^\mu = J^\mu + \partial_\nu B^{\mu\nu} \tag{7.52}$$

is also conserved, $\partial_\mu J'^\mu = 0$, and, provided B^{0i} obeys the spatial fall-off Eq. (7.44), carries the same charge, $Q' = Q$.

Proof. For the conservation, take the divergence of the added term,

$$\partial_\mu \partial_\nu B^{\mu\nu} = 0, \tag{7.53}$$

which vanishes *identically*—with no recourse to the equations of motion—because the double derivative $\partial_\mu \partial_\nu$ is symmetric in $\mu \leftrightarrow \nu$ while $B^{\mu\nu}$ is antisymmetric. Concretely, relabelling the summed indices, $\partial_\mu \partial_\nu B^{\mu\nu} = \partial_\nu \partial_\mu B^{\nu\mu} = -\partial_\mu \partial_\nu B^{\mu\nu}$, so the quantity equals its own negative and is zero. Hence $\partial_\mu J'^\mu = \partial_\mu J^\mu + 0 = 0$. For the charge, integrate the time component of the added term over a constant-time slice,

$$Q' - Q = \int_{\mathbb{R}^3} \mathrm{d}^3x \; \partial_\nu B^{0\nu} = \int_{\mathbb{R}^3} \mathrm{d}^3x \; (\partial_0 B^{00} + \partial_i B^{0i}). \tag{7.54}$$

The first term vanishes because $B^{00} = 0$ by antisymmetry. The second is a pure spatial divergence, and the divergence theorem converts it to a surface integral at infinity,

$$Q' - Q = \int_{\mathbb{R}^3} \mathrm{d}^3x \; \partial_i B^{0i} = \lim_{R\to\infty} \oint_{S_R} \mathrm{d}S \; \hat{n}_i \, B^{0i} = 0, \tag{7.55}$$

the last equality by the assumed fall-off of B^{0i} (the same argument as in Theorem 7.13). Therefore $Q' = Q$. □

The identity Eq. (7.53) is the precise reason the improvement is "automatic": $\partial_\nu B^{\mu\nu}$ is a current whose conservation requires no dynamics at all, only the antisymmetry of B. The freedom is not academic. The canonical stress tensor $T^\mu{}_\nu$ extracted in Theorem 7.11 is, for fields of nonzero spin, generally neither

symmetric in its two indices nor gauge invariant—both defects already flagged in Section 6.6.6. A later section will exploit exactly the ambiguity Eq. (7.52), adding a specific antisymmetric superpotential built from the spin current, to repair $T^{\mu\nu}$ into the symmetric, gauge-invariant Belinfante tensor that ultimately sources gravity—all without disturbing the conserved energy and momentum. Theorem 7.15 is the licence for that repair: it guarantees in advance that the total four-momentum cannot see the improvement.

Worked Example 7.16 (An improvement that moves the charge density but not the charge)**.** We make the abstract freedom concrete and confirm that it relocates charge in space without changing the total. Take any conserved current J^μ and improve it with a superpotential built from a smooth scalar profile $\chi(x)$ of compact spatial support (or rapid decay) and a constant four-vector c^ν,

$$B^{\mu\nu} = \chi\,(\delta^\mu{}_0\, c^\nu - \delta^\nu{}_0\, c^\mu), \tag{7.56}$$

manifestly antisymmetric. The improved current is $J'^\mu = J^\mu + \partial_\nu B^{\mu\nu}$, and its time component differs from J^0 by

$$J'^0 - J^0 = \partial_\nu B^{0\nu} = \partial_\nu\big[\chi\,(c^\nu - \delta^\nu{}_0\, c^0)\big] = \partial_i(\chi\, c^i), \tag{7.57}$$

a total spatial divergence (the $\nu = 0$ term cancels, $c^0 - c^0 = 0$). This is genuinely nonzero pointwise: wherever χ has a gradient, J'^0 assigns a different charge density than J^0 does—the two distribute the same total charge differently across space. Yet the total is untouched,

$$Q' - Q = \int_{\mathbb{R}^3} \mathrm{d}^3x\; \partial_i(\chi\, c^i) = \oint_{S_\infty} \mathrm{d}S\; \hat{n}_i\, \chi\, c^i = 0, \tag{7.58}$$

because χ vanishes at infinity. The lesson is exactly the one the improvement freedom is meant to teach: "where the charge sits" is convention-dependent, an artifact of the chosen current, while "how much charge there is" is physical. Only the integrated, slice-independent quantity Q of Theorem 7.14 is an observable; the current and its local density carry a genuine gauge freedom.

7.2.4 Several parameters: one current per generator

Realistic symmetries have more than one continuous parameter—a non-abelian internal group, or the ten-parameter Poincaré group. Such a symmetry produces not a single current but a multiplet, one conserved current for each independent generator. This is an immediate consequence of the linearity of the whole construction in the infinitesimal parameters, but it is the form in which currents actually appear in every theory of interest, so we state and prove it.

Recall from Theorem 7.7 that an r-parameter family is specified by constant parameters ω^A $(A = 1, \dots, r)$ and one generator per parameter,

$$\delta x^\mu = \omega^A X^\mu_A(x), \qquad \delta\phi_a = \omega^A \Phi_{a,A}(x, \phi) \tag{7.59}$$

(this is Eq. (7.27), with $\Phi_{a,A}$ now understood as the *total* field variation $\bar{\delta}$ per generator, so that the field piece of the current is read directly from it).

Theorem 7.17 (Multi-parameter Noether theorem)**.** *Suppose the transformation Eq. (7.59) is a quasi-symmetry for each of its r independent constant parameters ω^A, with boundary current $\Lambda^\mu = \omega^A \Lambda^\mu_A$. Then there are r independently conserved currents*

$$J^\mu_A = \pi_a{}^\mu \, \Phi_{a,A} - T^\mu{}_\nu \, X^\nu_A - \Lambda^\mu_A, \qquad \partial_\mu J^\mu_A = 0 \quad (A = 1, \dots, r), \tag{7.60}$$

each with its own conserved charge $Q_A = \int_{\mathbb{R}^3} \mathrm{d}^3x \, J^0_A$, and the full current is their linear combination $J^\mu = \omega^A J^\mu_A$.

Proof. Every ingredient of the three-piece current Eq. (7.39) is homogeneous of first order in the parameters: the comoving field variation is $\bar{\delta}\phi_a = \omega^A \Phi_{a,A}$, the coordinate variation is $\delta x^\nu = \omega^A X^\nu_A$, and the boundary current is $\Lambda^\mu = \omega^A \Lambda^\mu_A$ by hypothesis. Substituting these into Eq. (7.39) and Eq. (7.38),

$$J^\mu = \pi_a{}^\mu \, \omega^A \Phi_{a,A} - T^\mu{}_\nu \, \omega^A X^\nu_A - \omega^A \Lambda^\mu_A = \omega^A \, J^\mu_A, \tag{7.61}$$

with J^μ_A as in Eq. (7.60). By Theorem 7.10 the full current is conserved on shell, $\partial_\mu J^\mu = 0$, i.e. $\omega^A \, \partial_\mu J^\mu_A = 0$. Because the parameters ω^A are arbitrary and independent, the coefficient of each must vanish separately,

$$\partial_\mu J^\mu_A = 0 \qquad \text{for each } A. \tag{7.62}$$

Integrating each time component over a slice and invoking Theorem 7.13 gives r separately conserved charges Q_A. □

The translation worked example Theorem 7.11 is the case $r = 4$ with $X^\nu_A = \delta^\nu{}_A$, $\Phi_{a,A} = 0$, $\Lambda^\mu_A = 0$, for which Eq. (7.60) returns $J^\mu{}_\nu = -T^\mu{}_\nu$: four currents, four charges P_ν. The same counting will deliver three rotation currents and three boost currents from the Lorentz subgroup, and, in the next section, one current for each generator of an internal symmetry group G—that is, $\dim \mathfrak{g}$ currents for the Lie algebra $\mathfrak{g}$ of Theorem 7.7. That the number of conserved currents equals the dimension of the symmetry group is the structural fact that makes "one conserved quantity per continuous symmetry parameter" precise. It is also the hinge of what comes next: when the generators close into a Lie algebra, $[\delta_A, \delta_B] = f_{AB}{}^C \delta_C$, the charges Q_A will be found to represent that same algebra under the Poisson bracket, so that they not only *count* the symmetries but *generate* them.

The abstract machine is now complete: the Noether current Eq. (7.39) with its field, coordinate, and boundary pieces; its on-shell conservation Eq. (7.33) and the sharper off-shell relation Eq. (7.36); the time-independent, slice-independent charge Eq. (7.43); the improvement freedom Eq. (7.52) that leaves the charge

invariant; and the one-current-per-generator counting Eq. (7.60). Everything downstream is this machine fed a concrete transformation. The next section switches off the coordinate piece, $\delta x^\mu = 0$, and turns the crank on internal symmetries—abelian U(1) and non-abelian multiplets—extracting their currents and charges and proving that the charges close into the symmetry's Lie algebra and generate the field transformations they conserve, with the discrete $\mathbb{Z}_2$ of ϕ^4 theory kept as the explicit foil. A later section switches the coordinate piece on and reads off energy, momentum, and angular momentum, where the improvement freedom of Theorem 7.15 finally earns its keep in the construction of the Belinfante tensor.

7.3 Internal Symmetries, Currents, and the Charge Algebra

The previous section built the abstract machine and proved it: every continuous quasi-symmetry of the action yields a locally conserved current Eq. (7.39), and integrating its time component over a slice produces a time-independent charge Eq. (7.43). The current came in three pieces—the field response $\Pi_a^\mu\,\delta\phi_a$, the coordinate response $-F^\mu$, and the boundary term $-\Lambda^\mu$—and we promised to feed the machine concrete transformations. We begin with the simplest and most pervasive class: *internal symmetries*, transformations that mix the field components among themselves while leaving spacetime untouched. They are the symmetries that fix electric charge, isospin, baryon number, and colour, and they are where the conserved charges reveal their deepest structural property—that they do not merely sit there being conserved, but close into the symmetry's Lie algebra and generate the transformation that produced them. We extract the abelian U(1) current of a complex scalar, then the non-abelian currents of a multiplet in a representation, and prove the two central facts about their charges. We close by carrying ϕ^4 theory to its symmetry layer: its $\mathbb{Z}_2$ reflection $\phi \to -\phi$, a *discrete* symmetry that produces no Noether current at all, yet still rules the theory through selection rules—the sharp foil that shows exactly what continuity buys.

7.3.1 Internal symmetries and the field response

Definition 7.18 (Internal symmetry)**.** An *internal* (or *global, field-space*) symmetry is a quasi-symmetry of the action whose coordinate generators vanish, $X_A^\mu = 0$ in Eq. (7.27), so that points are not moved, $\delta x^\mu = 0$, and the transformation acts only on the field values,

$$\phi_a(x) \;\longmapsto\; \phi'_a(x) = \phi_a(x) + \delta\phi_a(x), \qquad \delta\phi_a = \alpha^c\,\delta^c\phi_a, \tag{7.63}$$

with constant parameters α^c and prescribed field-space generators $\delta^c\phi_a$ that depend on the fields at the same point x. The parameters being constant (independent of x) is what makes the symmetry *global*; the local case, where $\alpha^c = \alpha^c(x)$, is the gauge logic taken up in Chapter 8.

For such a symmetry the three-piece current Eq. (7.39) simplifies dramatically. Setting $\delta x^\mu = 0$ switches off the canonical-stress-tensor piece $F^\mu = T^\mu{}_\nu\,\delta x^\nu$; and for a *strict* symmetry of the Lagrangian—one for which $\delta_0\mathcal{L} = 0$ rather than merely $\delta_0\mathcal{L} = \partial_\mu\Lambda^\mu$—the boundary current vanishes too, $\Lambda^\mu = 0$. We will encounter only strict internal symmetries in this section (the boundary term Λ^μ earns its keep for spacetime symmetries, where the Lagrangian density is a scalar that genuinely shifts), so the master current collapses to its first piece alone,

$$J^{c\,\mu} \;=\; \Pi^\mu_a\,\delta^c\phi_a, \qquad \Pi^\mu_a \equiv \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}, \tag{7.64}$$

one current $J^{c\,\mu}$ per generator c, exactly as the multi-parameter theorem Theorem 7.17 demands with $X^\nu_A = 0$ and $\Lambda^\mu_A = 0$. The associated charges are

$$Q^c \;=\; \int_{\mathbb{R}^3} \mathrm{d}^3x\; J^{c\,0} \;=\; \int_{\mathbb{R}^3} \mathrm{d}^3x\; \Pi^0_a\,\delta^c\phi_a \;=\; \int_{\mathbb{R}^3} \mathrm{d}^3x\; \pi_a\,\delta^c\phi_a, \tag{7.65}$$

where in the last step we used $\Pi^0_a = \partial\mathcal{L}/\partial\dot\phi_a = \pi_a$, the canonical momentum density of Theorem 6.13. Equation Eq. (7.65) is the master formula of this section: *the internal charge is the spatial integral of the canonical momentum contracted with the field-space generator.* Its right-hand side is a functional purely of the phase-space data (ϕ_a, π_a) on a slice, which is exactly what we need to compute its Poisson brackets. Everything below is the working-out of Eq. (7.64) and Eq. (7.65) in concrete representations, followed by the proof that the Q^c realize the symmetry algebra.

7.3.2 The abelian case: U(1) of a complex scalar

The cleanest abelian internal symmetry is the phase rotation of a complex scalar field. Let $\Phi(x)$ be a single complex scalar, equivalently a pair of real fields packaged as $\Phi = (\phi_1 + i\phi_2)/\sqrt{2}$, with the Lagrangian

$$\mathcal{L} \;=\; (\partial_\mu\Phi)^*(\partial^\mu\Phi) \;-\; V(\Phi^*\Phi), \tag{7.66}$$

where V is any real function of the real, non-negative combination $\Phi^*\Phi = |\Phi|^2$ (for instance $V = m^2\Phi^*\Phi + \lambda(\Phi^*\Phi)^2$, but we need no specific form). The point of writing V as a function of $|\Phi|^2$ alone is that it makes the symmetry manifest: under a constant phase rotation

$$\Phi \;\longmapsto\; \Phi' = e^{-i\alpha}\,\Phi, \qquad \Phi^* \;\longmapsto\; \Phi'^* = e^{+i\alpha}\,\Phi^*, \tag{7.67}$$

the combination $\Phi^*\Phi$ is invariant and the kinetic term $(\partial_\mu\Phi)^*(\partial^\mu\Phi)$ is invariant because the constant phases $e^{\pm i\alpha}$ pass through the derivative and cancel. So Eq. (7.66) is strictly invariant, $\delta_0\mathcal{L} = 0$, and Eq. (7.67) is a one-parameter family of internal symmetries: the group U(1). (We have chosen the sign of the exponent in Eq. (7.67) as $e^{-i\alpha}$; the opposite choice flips the overall sign of the current below and corresponds to calling the antiparticle positive. Nothing physical depends on it, but we fix it now and keep it.) Infinitesimally,

$$\delta\Phi = -i\alpha\,\Phi, \qquad \delta\Phi^* = +i\alpha\,\Phi^*, \tag{7.68}$$

so the per-generator field variations (the coefficient of the single parameter α) are $\delta\Phi = -i\Phi$ and $\delta\Phi^* = +i\Phi^*$.

To apply Eq. (7.64) we treat Φ and Φ^* as the two independent fields and read off their conjugate momentum densities from Eq. (7.66):

$$\Pi^\mu_\Phi = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi)} = (\partial^\mu\Phi)^* = \partial^\mu\Phi^*, \qquad \Pi^\mu_{\Phi^*} = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\Phi^*)} = \partial^\mu\Phi. \tag{7.69}$$

Substituting these and Eq. (7.68) into the internal-current formula Eq. (7.64) (summing the two field components Φ and Φ^*),

$$\begin{aligned} J^\mu &= \Pi^\mu_\Phi\,(\delta\Phi) + \Pi^\mu_{\Phi^*}\,(\delta\Phi^*) = (\partial^\mu\Phi^*)(-i\Phi) + (\partial^\mu\Phi)(+i\Phi^*) \\ &= i\big(\Phi^*\,\partial^\mu\Phi - \Phi\,\partial^\mu\Phi^*\big) \;=\; i\big(\Phi^*\,\partial^\mu\Phi - (\partial^\mu\Phi^*)\,\Phi\big). \end{aligned} \tag{7.70}$$

This is the conserved U(1) current of the complex scalar. It is real ($J^\mu = (J^\mu)^*$, as a Noether current of a real Lagrangian must be), and it is exactly the structure announced in the chapter blueprint: $J^\mu = i\big[\Pi^\mu_\Phi\,\Phi - \text{c.c.}\big]$ up to the overall sign fixed by the convention Eq. (7.67). A common shorthand uses the antisymmetric double-derivative $A\overleftrightarrow{\partial^\mu}B \equiv A\,\partial^\mu B - (\partial^\mu A)B$, in which $J^\mu = i\,\Phi^*\overleftrightarrow{\partial^\mu}\Phi$. That $\partial_\mu J^\mu = 0$ on shell is guaranteed by Noether's theorem Theorem 7.10; one can also see it directly, since $\partial_\mu J^\mu = i(\Phi^*\Box\Phi - \Phi\,\Box\Phi^*)$, and the equations of motion $\Box\Phi = -V'(\Phi^*\Phi)\,\Phi$ and its conjugate make the two terms cancel because V' is real.

The conserved charge is the spatial integral of J^0,

$$Q = \int_{\mathbb{R}^3} \mathrm{d}^3x\; i(\Phi^*\dot\Phi - \dot\Phi^*\,\Phi), \tag{7.71}$$

where $\dot\Phi = \partial^0\Phi$. We recognize the integrand as $i(\Phi^*\pi_{\Phi^*} - \Phi\,\pi_\Phi)$ once we use the canonical momenta $\pi_\Phi = \Pi^0_\Phi = \dot\Phi^*$ and $\pi_{\Phi^*} = \Pi^0_{\Phi^*} = \dot\Phi$ from Eq. (7.69). The physical meaning of Q becomes transparent only after quantization: in Chapter 9 the complex field's mode expansion will split into a particle annihilation operator and an antiparticle creation operator, and Q will become the number operator $N_a - N_b$, the particle number minus the antiparticle number. Classically it is already the conserved "amount of Φ minus amount of Φ^*"—the prototype of every additive quantum number in physics.

Physical Insight

A U(1) phase symmetry is invisible in the value of $|\Phi|^2$ but visible in the *flow* of phase. The conserved current Eq. (7.70) measures the rate at which the field winds in the complex plane: where the phase $\arg\Phi$ advances coherently, the charge flows. This is why the same current reappears as electric current once the phase is gauged in Chapter 8—electric charge is, structurally, the bookkeeping of an unobservable phase, and its conservation is the statement that the phase has no physical zero.

Worked Example 7.19 (The U(1) charge generates the phase rotation)**.** We verify directly, before proving the general statement, that the charge Eq. (7.71) reproduces the infinitesimal transformation Eq. (7.68) under the Poisson bracket. The relevant canonical brackets, inherited from Eq. (6.112) applied to the independent pairs (Φ, π_Φ) and (Φ^*, π_{Φ^*}), are

$$\{\Phi(\mathbf{x}), \pi_\Phi(\mathbf{y})\}_{\mathrm{P}} = \delta^3(\mathbf{x}-\mathbf{y}), \qquad \{\Phi^*(\mathbf{x}), \pi_{\Phi^*}(\mathbf{y})\}_{\mathrm{P}} = \delta^3(\mathbf{x}-\mathbf{y}), \tag{7.72}$$

with all mixed brackets (Φ with π_{Φ^*}, Φ with Φ^*, etc.) vanishing because the two pairs are independent coordinates on phase space. Write the charge Eq. (7.71) in terms of momenta as $Q = i\int \mathrm{d}^3z\,(\Phi^*(\mathbf{z})\,\pi_{\Phi^*}(\mathbf{z}) - \Phi(\mathbf{z})\,\pi_\Phi(\mathbf{z}))$. Its functional derivatives are

$$\frac{\delta Q}{\delta \pi_\Phi(\mathbf{z})} = -i\,\Phi(\mathbf{z}), \qquad \frac{\delta Q}{\delta \pi_{\Phi^*}(\mathbf{z})} = +i\,\Phi^*(\mathbf{z}). \tag{7.73}$$

Then, using the bracket definition Eq. (6.110) with $F = \Phi(\mathbf{x})$ (so $\delta F/\delta\phi = \delta^3$ on the Φ slot only),

$$\{\Phi(\mathbf{x}), Q\}_{\mathrm{P}} = \int \mathrm{d}^3z\,\frac{\delta\Phi(\mathbf{x})}{\delta\Phi(\mathbf{z})}\,\frac{\delta Q}{\delta\pi_\Phi(\mathbf{z})} = \int \mathrm{d}^3z\,\delta^3(\mathbf{x}-\mathbf{z})\,(-i\,\Phi(\mathbf{z})) = -i\,\Phi(\mathbf{x}). \tag{7.74}$$

This is precisely $\delta\Phi$ of Eq. (7.68) (per unit parameter): $\{\Phi, Q\}_{\mathrm{P}} = \delta\Phi$. An identical computation gives $\{\Phi^*, Q\}_{\mathrm{P}} = +i\,\Phi^* = \delta\Phi^*$. The charge *is* the generator of the phase rotation: bracketing any field with Q rotates its phase. We will now prove this is no accident of the abelian case.

7.3.3 The non-abelian case: a multiplet in a representation

Promote the single field to a multiplet ϕ_i, $i = 1, \dots, N$, transforming in a representation of a compact Lie group G with Lie algebra $\mathfrak{g}$ (Theorem 2.39). The infinitesimal action of the group on the multiplet is, in the Hermitian-generator convention of Eq. (2.70),

$$\phi_i \longmapsto \phi_i' = \left(e^{-i\alpha^a T^a}\right)_{ij}\phi_j, \qquad \delta\phi_i = -i\,\alpha^a (T^a)_{ij}\,\phi_j, \tag{7.75}$$

where $a = 1, \dots, \dim \mathfrak{g}$ labels the generators, the T^a are the Hermitian representation matrices obeying

$$\left[T^a, T^b\right] = i f^{abc}\, T^c \tag{7.76}$$

with the real, totally antisymmetric structure constants f^{abc} of Eq. (2.70), and α^a are the constant group parameters. The per-generator field variation, the coefficient of α^a, is

$$\delta^a \phi_i \;=\; -i\,(T^a)_{ij}\,\phi_j \;\equiv\; (t^a)_{ij}\,\phi_j, \qquad t^a \equiv -i\,T^a. \tag{7.77}$$

The matrices $t^a = -iT^a$ are the generators in their "real" form: stripping the i from the Hermitian convention, they satisfy the structure relation without the i,

$$\left[t^a, t^b\right] = (-i)^2 \left[T^a, T^b\right] = -\,i f^{abc}\, T^c = f^{abc}\,(-iT^c) = f^{abc}\, t^c. \tag{7.78}$$

For a real multiplet—N real fields rotated into one another, as in an SO(N) vector—the t^a are real antisymmetric matrices and Eq. (7.75) keeps the ϕ_i real; for a complex multiplet (the U(1) case above is N complex with the single generator $T = 1$, $t = -i$) the same formulas hold with the components understood as complex. We carry the algebra in the index form Eq. (7.77), which covers both.

For the Lagrangian we take any density built from invariant contractions of the multiplet, the canonical example being

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu \phi_i\, \partial^\mu \phi_i - V(\phi_i \phi_i), \tag{7.79}$$

invariant because $\phi_i \phi_i$ and $\partial_\mu \phi_i\, \partial^\mu \phi_i$ are unchanged under $\phi \to e^{-i\alpha\cdot T}\phi$ when the T^a are Hermitian (so $e^{-i\alpha\cdot T}$ is orthogonal/unitary and preserves the contraction). The conjugate momentum density is $\Pi_i^\mu = \partial^\mu \phi_i$, so $\pi_i = \dot{\phi}_i$. Feeding Eq. (7.77) into the internal-current master formula Eq. (7.64) gives one current per generator,

$$J^{a\,\mu} = \Pi_i^\mu\, \delta^a \phi_i = \partial^\mu \phi_i\, (t^a)_{ij}\, \phi_j = -i\,\partial^\mu \phi_i\, (T^a)_{ij}\, \phi_j, \tag{7.80}$$

and the corresponding charges, from Eq. (7.65),

$$Q^a = \int_{\mathbb{R}^3} \mathrm{d}^3x\; \pi_i\, (t^a)_{ij}\, \phi_j = -i \int_{\mathbb{R}^3} \mathrm{d}^3x\; \pi_i\, (T^a)_{ij}\, \phi_j. \tag{7.81}$$

Each $J^{a\mu}$ is separately conserved by Theorem 7.17, and each Q^a is separately time-independent by Theorem 7.13. There are $\dim \mathfrak{g}$ of them—as many conserved charges as the group has dimensions, the counting promised at the end of Section 7.2.4. What distinguishes the non-abelian case from the abelian one is what happens when we bracket these charges with one another.

7.3.4 The charges generate the transformation

We first prove that each charge generates, through the Poisson bracket, exactly the field transformation it descended from. This is the classical statement of which the quantum statement $\delta\hat{\phi} = i\alpha^a[\hat{Q}^a, \hat{\phi}]$ will be the operator translate.

Theorem 7.20 (Charges generate the field transformation)**.** *For the internal symmetry Eq. (7.75) with charges Eq. (7.81), the equal-time Poisson bracket of a charge with the field reproduces the field-space generator,*

$$\{\phi_k(\mathbf{x}), Q^a\}_P = \delta^a\phi_k(\mathbf{x}) = (t^a)_{kj}\,\phi_j(\mathbf{x}). \tag{7.82}$$

Consequently the finite transformation Eq. (7.75) is generated by exponentiating the bracket: $\phi'_k = (e^{-\alpha^a\{\cdot, Q^a\}_P})\phi_k$ *in the sense of the formal series.*

Proof. Write the charge Eq. (7.81) as $Q^a = \int \mathrm{d}^3z\,\pi_l(\mathbf{z})\,(t^a)_{lj}\,\phi_j(\mathbf{z})$. Its functional derivative with respect to the momentum is

$$\frac{\delta Q^a}{\delta\pi^c(\mathbf{z})} = (t^a)_{cj}\,\phi_j(\mathbf{z}), \tag{7.83}$$

and its derivative with respect to ϕ will not contribute because $\delta\phi_k/\delta\pi = 0$. Inserting $F = \phi_k(\mathbf{x})$ and $G = Q^a$ into the bracket definition Eq. (6.110) and using the sharp functional derivative $\delta\phi_k(\mathbf{x})/\delta\phi_c(\mathbf{z}) = \delta_{kc}\,\delta^3(\mathbf{x}-\mathbf{z})$,

$$\{\phi_k(\mathbf{x}), Q^a\}_{\mathrm{P}} = \int \mathrm{d}^3z\,\frac{\delta\phi_k(\mathbf{x})}{\delta\phi_c(\mathbf{z})}\,\frac{\delta Q^a}{\delta\pi^c(\mathbf{z})} = \int \mathrm{d}^3z\,\delta_{kc}\,\delta^3(\mathbf{x}-\mathbf{z})\,(t^a)_{cj}\,\phi_j(\mathbf{z}) = (t^a)_{kj}\,\phi_j(\mathbf{x}). \tag{7.84}$$

The right-hand side is $\delta^a\phi_k(\mathbf{x})$ by Eq. (7.77). The finite statement follows because $\{\cdot, Q^a\}_{\mathrm{P}}$ acts on ϕ as the constant matrix t^a, so its n-fold iteration is $(t^a)^n$, and the formal exponential sums to $e^{-\alpha^a T^a\cdot}$ acting on the multiplet, which is Eq. (7.75). □

The bracket order in Eq. (7.82) matters and is fixed by consistency with the dynamics. In Eq. (6.115) the Hamiltonian generated time evolution through $\dot{F} = \{F, H\}_{\mathrm{P}}$, with the evolving quantity on the *left*. The same convention here—generated object on the left, generator on the right—makes Q^a the spatial-symmetry analogue of H: just as H generates the flow in time, Q^a generates the flow along the a-th internal direction. We will use this uniform convention throughout, so that in Section 7.3's sequel the momentum P^μ and angular momentum $J^{\mu\nu}$ generate spacetime translations and rotations by the identical rule.

Notation Watch

Some texts write the generator on the left, $\delta^a\phi = \{Q^a, \phi\}_{\mathrm{P}}$. Because the Poisson bracket is antisymmetric, $\{Q^a, \phi\}_{\mathrm{P}} = -\{\phi, Q^a\}_{\mathrm{P}}$, that convention flips the sign of the field-space generator (equivalently, it accompanies the opposite sign for the charge). The physics—which field-space rotation a given charge induces—is identical; only the placement of a minus sign moves. We keep the object-on-the-left convention Eq. (7.82) to match Eq. (6.115).

7.3.5 The charge algebra

The deepest fact about the charges is that they do not merely each generate their own transformation; the brackets of the charges among themselves reproduce the commutators of the generators. The charges *represent* the Lie algebra.

Theorem 7.21 (Classical charge algebra)**.** *The Noether charges Eq. (7.81) of the internal symmetry Eq. (7.75) satisfy the equal-time Poisson-bracket algebra*

$$\boxed{\left\{Q^a, Q^b\right\}_P = f^{abc}\, Q^c} \tag{7.85}$$

with the same structure constants f^{abc} that define the Lie algebra $\mathfrak{g}$ in Eq. (7.76). The map $a \mapsto Q^a$ is therefore a homomorphism from $\mathfrak{g}$ into the Poisson-bracket algebra of phase-space functions.

Proof. Write both charges in the real-generator form, $Q^a = \int \mathrm{d}^3z\, \pi_p(\mathbf{z})\,(t^a)_{pq}\,\phi_q(\mathbf{z})$ and likewise for Q^b. Each is bilinear in the phase-space variables, so its functional derivatives are linear:

$$\frac{\delta Q^a}{\delta\phi_c(\mathbf{z})} = \pi_p(\mathbf{z})\,(t^a)_{pc}, \qquad \frac{\delta Q^a}{\delta\pi^c(\mathbf{z})} = (t^a)_{cq}\,\phi_q(\mathbf{z}). \tag{7.86}$$

Insert these into the bracket definition Eq. (6.110):

$$\begin{aligned}
\left\{Q^a, Q^b\right\}_{\mathrm{P}} &= \int \mathrm{d}^3z \left[\frac{\delta Q^a}{\delta\phi_c(\mathbf{z})}\frac{\delta Q^b}{\delta\pi^c(\mathbf{z})} - \frac{\delta Q^a}{\delta\pi^c(\mathbf{z})}\frac{\delta Q^b}{\delta\phi_c(\mathbf{z})}\right] \\
&= \int \mathrm{d}^3z \left[\pi_p\,(t^a)_{pc}\,(t^b)_{cq}\,\phi_q - (t^a)_{cq}\,\phi_q\,\pi_p\,(t^b)_{pc}\right] \\
&= \int \mathrm{d}^3z\; \pi_p \left[(t^a)_{pc}(t^b)_{cq} - (t^b)_{pc}(t^a)_{cq}\right]\phi_q \\
&= \int \mathrm{d}^3z\; \pi_p \left[t^a, t^b\right]_{pq}\phi_q .
\end{aligned} \tag{7.87}$$

In the second line both products are ordinary numbers (the fields and momenta commute as classical functions), so the second term could be reordered to

expose the matrix product $(t^b)_{pc}(t^a)_{cq} = (t^b t^a)_{pq}$, leaving the matrix commutator $[t^a, t^b]_{pq} = (t^a t^b - t^b t^a)_{pq}$ in the last line. Now use the structure relation Eq. (7.78), $[t^a, t^b] = f^{abc} t^c$:

$$\left\{Q^a, Q^b\right\}_{\mathrm{P}} = \int \mathrm{d}^3 z \; \pi_p \, f^{abc} \, (t^c)_{pq} \, \phi_q = f^{abc} \int \mathrm{d}^3 z \; \pi_p \, (t^c)_{pq} \, \phi_q = f^{abc} \, Q^c, \quad (7.88)$$

the last integral being Q^c by Eq. (7.81). This is Eq. (7.85). □

It is worth seeing why the sign in Eq. (7.85) comes out as $+f^{abc}$ and not $-f^{abc}$, because the result is convention-sensitive and the conventions were fixed precisely to make it clean. Had we written the field transformation with the opposite phase, $\delta\phi = +i\alpha^a T^a \phi$, the generators would be $t^a = +iT^a$, the structure relation Eq. (7.78) would read $[t^a, t^b] = -f^{abc} t^c$, and the same computation would yield $\left\{Q^a, Q^b\right\}_{\mathrm{P}} = -f^{abc} Q^c$—an *anti*-homomorphism. Our choice $e^{-i\alpha\cdot T}$ in Eq. (7.75) is exactly the one that makes $Q \mapsto \{\cdot, Q\}_{\mathrm{P}}$ a genuine homomorphism, so that the classical charges represent $\mathfrak{g}$ the right way up. The same choice is what made $\{\phi, Q^a\}_{\mathrm{P}} = +\delta^a \phi$ in Theorem 7.20; the two signs are locked together, and we fixed them once and for all in Eq. (7.75).

Physical Insight

Equation (7.85) is the classical seed of the entire operator structure of internal symmetry in the quantum theory. Under canonical quantization $\{\cdot,\cdot\}_{\mathrm{P}} \to -i\,[\cdot,\cdot]$ (Chapter 9), the charge algebra becomes $\left[\hat{Q}^a, \hat{Q}^b\right] = i f^{abc} \hat{Q}^c$—the Hermitian generators Eq. (7.76) realized as operators on the Hilbert space. The conserved charges are not just labels on states; they are the very operators whose simultaneous eigenvalues (for a maximal commuting set, the Cartan subalgebra) are the quantum numbers—isospin and its third component, colour weights, hypercharge—that organize the particle spectrum. Multiplets of particles are multiplets of $\mathfrak{g}$ because the charges close into $\mathfrak{g}$.

A consistency check ties the two theorems together. The charges generate transformations (Theorem 7.20), and the commutator of two infinitesimal transformations must itself be a transformation—the one labelled by the commutator of the generators. Computing the commutator of two field-space rotations directly from Eq. (7.77),

$$[\delta^a, \delta^b]\phi_k = (t^a t^b - t^b t^a)_{kj} \, \phi_j = [t^a, t^b]_{kj} \, \phi_j = f^{abc} \, (t^c)_{kj} \, \phi_j = f^{abc} \, \delta^c \phi_k, \quad (7.89)$$

and this must equal what the bracketed charges generate. Indeed, using the Jacobi identity of the Poisson bracket (Theorem 6.17) and Theorem 7.20 twice, one finds $[\delta^a, \delta^b]\phi_k = \left\{\phi_k, \left\{Q^a, Q^b\right\}_{\mathrm{P}}\right\}_{\mathrm{P}}$, so that $\left\{\phi_k, \left\{Q^a, Q^b\right\}_{\mathrm{P}}\right\}_{\mathrm{P}} = f^{abc} \, \delta^c \phi_k =$

$\{\phi_k, f^{abc}Q^c\}_{\mathrm{P}}$ for every field, which forces $\{Q^a, Q^b\}_{\mathrm{P}} = f^{abc}Q^c$ up to a quantity bracketing trivially with all fields. Classically that quantity is an irrelevant constant (a possible *central extension*); for the linearly realized internal symmetries here it vanishes, as the explicit computation Eq. (7.87) confirmed. The two routes—direct phase-space computation and abstract Jacobi argument—agree, and they will agree again for the Poincaré charges of the next section, where the central term is likewise absent.

7.3.6 Discrete symmetries: Scalar ϕ^4 theory's $\mathbb{Z}_2$

Everything so far rested on *continuity*: the Noether construction differentiates the transformation with respect to its parameter, and a current exists because there is a parameter to differentiate. We now carry ϕ^4 theory to its symmetry layer and meet the opposite case—a symmetry with no parameter at all—to see exactly what is lost and what survives. Recall the frozen real-scalar Lagrangian of Eq. (6.56),

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{7.90}$$

with a single *real* field ϕ. There is no continuous phase to rotate: a real field has no complex structure to support a U(1). What it does have is a reflection.

Definition 7.22 (The $\mathbb{Z}_2$ reflection of ϕ^4 theory)**.** The transformation $\phi(x) \mapsto -\phi(x)$, leaving spacetime fixed, is a symmetry of Eq. (7.90): every term contains ϕ to an even power ($\partial_\mu\phi\,\partial^\mu\phi$, ϕ^2, ϕ^4), so $\mathcal{L}[-\phi] = \mathcal{L}[\phi]$ and the action is invariant. The transformation squares to the identity, $(-1)^2 = 1$, and together with the identity forms the two-element group $\mathbb{Z}_2 = \{1, -1\}$.

This invariance is exactly what selected the Lagrangian Eq. (7.90) in the first place: when Eq. (6.56) was assembled, the reflection symmetry was imposed to forbid the odd terms ϕ, ϕ^3 that Lorentz invariance and dimensional analysis would otherwise permit. The $\mathbb{Z}_2$ is therefore not an accident of the model but a defining constraint on it. The question is what conserved quantity, if any, it produces.

Proposition 7.23 (A discrete symmetry produces no Noether current)**.** *The reflection $\phi \to -\phi$ of Theorem 7.22 yields no conserved Noether current.*

Proof. The Noether construction of Theorem 7.10 takes as input an infinitesimal transformation $\delta\phi = \alpha\,\delta^c\phi$ obtained by differentiating a one-parameter family ϕ_s at $s = 0$, $\delta^c\phi = \frac{d\phi_s}{ds}\big|_{s=0}$, with $\phi_0 = \phi$ the identity. The current Eq. (7.64) is built from this derivative, $J^\mu = \Pi^\mu_a\,\delta^c\phi$. For a continuous symmetry such a family exists by definition; for the reflection it does not. The symmetry group $\mathbb{Z}_2 = \{1, -1\}$ is a zero-dimensional manifold: it consists of two isolated points, and there is no continuous path within it from the identity 1 to the reflection

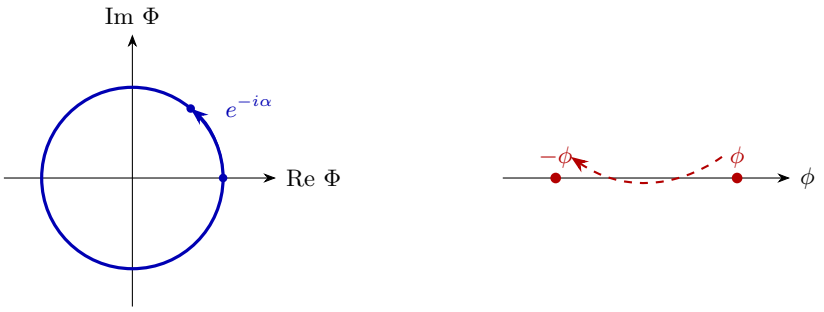


Figure 7.3: Internal symmetry as motion in field space. A U(1) phase rotation (left) moves the complex field Φ along a continuous orbit—a circle—so the transformation can be done infinitesimally, differentiated, and fed to Noether's theorem, producing a conserved current and charge. The $\mathbb{Z}_2$ reflection of ϕ^4 theory (right) maps the real field ϕ to the single isolated point $-\phi$; the group is zero-dimensional and there is no continuous path from the identity, so there is nothing to differentiate and no Noether current. The discrete symmetry still constrains the theory through selection rules.

-1. Concretely, any attempt to interpolate, $\phi_s = (1-2s)\phi$ with $\phi_0 = \phi$ and $\phi_1 = -\phi$, passes through configurations ϕ_s that are *not* symmetries of the action—$\mathcal{L}[(1-2s)\phi] \neq \mathcal{L}[\phi]$ for $0 < s < 1$, since the quartic term scales as $(1-2s)^4$. There is no s-derivative of a *symmetry* family to feed the construction, hence no infinitesimal generator $\delta^c\phi$, hence no current. The hypothesis of Noether's first theorem—a one-parameter family of symmetries—is simply not met. □

So discreteness costs us the current and the charge: there is no locally conserved J^μ, no time-independent Q, no generator in the Poisson algebra. What survives is a global constraint on the theory's correlation functions, sharper in some ways than a conservation law. The cleanest classical statement is a *selection rule*.

Worked Example 7.24 (The $\mathbb{Z}_2$ selection rule, classically)**.** Couple the field to an external source $J(x)$ and consider the sourced field equation derived from Eq. (7.90),

$$(\Box + m^2)\,\phi + \frac{\lambda}{3!}\,\phi^3 = J, \tag{7.91}$$

whose left-hand side is the variational derivative $-\delta S/\delta\phi$ and which reduces to the ϕ^4 equation of motion Eq. (6.60) when $J = 0$. The left-hand side is *odd* under $\phi \to -\phi$: each of $\Box\phi$, $m^2\phi$, and ϕ^3 flips sign. Therefore the solution $\phi = \phi[J]$, with the standard (Feynman) boundary conditions that pick a unique solution,

obeys

$$\phi[-J] = -\phi[J], \tag{7.92}$$

because replacing $J \to -J$ and $\phi \to -\phi$ leaves Eq. (7.91) unchanged. The classical field is an *odd functional of the source*. Expanding it perturbatively in powers of J—which is the tree-level expansion whose terms are the connected correlation functions, made precise by the generating functional of Chapter 10—

$$\phi[J](x) = \int G_2(x, y_1)\, J(y_1) + \frac{1}{3!} \int G_4(x, y_1, y_2, y_3)\, J(y_1) J(y_2) J(y_3) + \cdots , \tag{7.93}$$

the oddness Eq. (7.92) forces every *even* power of J to be absent: only $J^1, J^3, J^5, \ldots$ appear. The coefficient of J^n is the $(n{+}1)$-point function G_{n+1}, so the surviving functions have $n+1$ *even*, i.e. all G_{odd} vanish:

$$\boxed{G_{2k+1} = 0 \quad \text{for all } k} \tag{7.94}$$

Every correlation function with an odd number of fields vanishes identically. Equivalently, every interaction vertex of Eq. (7.90) has an even number of legs (ϕ^2 and ϕ^4, never ϕ or ϕ^3), so any process connects an even number of external ϕ quanta: an amplitude with an odd number of external ϕ lines is zero. This is the classical shadow of the quantum statement, sharpened in Chapter 10 (the generating functional) and put to work in Chapter 14 (the first amplitude $\phi\phi \to \phi\phi$), that the $\mathbb{Z}_2$ forbids odd-point amplitudes.

The contrast is now complete and instructive. A *continuous* internal symmetry gives a local conservation law $\partial_\mu J^\mu = 0$, a time-independent charge Q, a generator in the Poisson algebra, and—when non-abelian—a charge algebra Eq. (7.85) that organizes the spectrum into multiplets. A *discrete* internal symmetry gives none of these analytic structures, but it gives a selection rule Eq. (7.94): an exact statement that whole classes of processes have vanishing amplitude. Both are genuine consequences of invariance; they differ because Noether's theorem is a theorem about derivatives, and only a continuous family has a derivative to take. The $\mathbb{Z}_2$ of ϕ^4 theory is the carried example of the discrete case, and Eq. (7.94) is the constraint it imposes on every later stage of the thread—a constraint we will invoke without re-deriving each time a ϕ^4 amplitude is computed.

Physical Insight

The deep distinction is not "continuous versus discrete" as a taxonomy but *what each kind of symmetry produces*. Continuity is exactly the structure needed to convert a global invariance into a *local* statement at every point—a current whose divergence vanishes there. Discreteness cannot do that; the best it can do is constrain *global* objects, the correlation functions, all at once. This is why the parade of conserved currents in the

Standard Model—electric charge, colour, hypercharge—are all continuous symmetries, while the discrete symmetries (parity, charge conjugation, the residual $\mathbb{Z}_2$'s that stabilize dark-matter candidates) act as selection rules and superselection labels rather than as conserved flows.

With the internal symmetries dispatched—abelian and non-abelian currents extracted from the single formula Eq. (7.64), their charges shown to generate the transformation Eq. (7.82) and to close into the Lie algebra Eq. (7.85), and the discrete $\mathbb{Z}_2$ of ϕ^4 theory separated off as a selection rule—we have established the charge-as-generator picture in its cleanest setting, where the field-space generators t^a are constant matrices. The next section switches the coordinate piece F^μ back on and applies the same machine to spacetime symmetries: translations will deliver the four-momentum P^μ and Lorentz transformations the angular-momentum current $J^{\mu\nu}$, and the brackets proved here—charge generates transformation, charges close into an algebra—will recur verbatim, now realizing the Poincaré algebra with P^μ and $J^{\mu\nu}$ as its generators.

7.4 Spacetime Symmetries: The Canonical Stress Tensor and Angular Momentum

The previous two sections built the machine and ran it once on field space. Noether's first theorem (Theorem 7.10) attached to every continuous quasi-symmetry a locally conserved current $J^\mu = \Pi^\mu_a\,\delta\phi_a - F^\mu - \Lambda^\mu$ (Eq. (7.39)), and the charge built from it (Theorem 7.12) was shown to be time-independent, slice-independent, and—for internal symmetries—to close into the symmetry's Lie algebra under Poisson brackets and to generate the field transformation it came from. Internal symmetries are, however, the case in which the coordinate piece $F^\mu = T^\mu{}_\nu\,\delta x^\nu$ is switched off, because they leave the spacetime point fixed: $\delta x^\mu = 0$. We now turn the machine on the opposite extreme, the symmetries that move the point itself—the Poincaré transformations of spacetime—where the current is carried *almost entirely* by F^μ, and where the field-space piece $\Pi^\mu_a\,\delta\phi_a$ survives only for fields of nonzero spin. The payoff is the identification of the deepest conserved quantities a relativistic field theory possesses: energy, momentum, and angular momentum become the Noether charges of translations and Lorentz rotations, and the charge-as-generator dictionary of the last section promotes them into the ten generators of the Poincaré algebra (Eqs. (3.147) to (3.149)). Along the way the construction hands us, for free, the two structural defects that the next section must repair: the canonical stress tensor is generically not symmetric, and for gauge fields not even gauge invariant.

7.4.1 Translations revisited: the canonical stress tensor as a Noether current

The four rigid translations $x'^{\mu} = x^{\mu} + a^{\mu}$ were already fed through the general theorem in Theorem 7.11, where they reproduced the canonical stress–energy tensor of Theorem 6.28. We consolidate that result here as the first of the two spacetime cases, because it fixes the notation and the sign conventions that the Lorentz analysis will lean on, and because the four-momentum it produces is one of the section's owed results.

A translation carries a field configuration bodily to the shifted point, so a field of *any* spin transforms as a scalar under it: the comoving value is unchanged, $\phi'_a(x') = \phi_a(x)$, hence the total variation vanishes, $\delta\phi_a = 0$, while the coordinate generators are $\delta x^{\mu} = a^{\mu}$ with a^{μ} four independent constants. Because $\mathcal{L}$ carries no explicit coordinate dependence, the translation is a strict symmetry, $\Lambda^{\mu} = 0$ (the cancellation $\delta_0\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^{\mu}) = -a^{\nu}\partial_\nu\mathcal{L} + a^{\mu}\partial_\mu\mathcal{L} = 0$ was carried out in Theorem 7.11). The three-piece current Eq. (7.39) therefore collapses onto its coordinate part,

$$J^{\mu} = \Pi^{\mu}_{a}\cdot 0 - F^{\mu} - 0 = -T^{\mu}{}_{\nu}\,a^{\nu}, \tag{7.95}$$

and since the four parameters a^{ν} are independent, the multi-parameter theorem (Theorem 7.17) extracts four separately conserved currents, the columns of the canonical stress tensor,

$$T^{\mu\nu} = \Pi^{\mu}_{a}\,\partial^{\nu}\phi_a - \eta^{\mu\nu}\mathcal{L}, \qquad \partial_\mu T^{\mu\nu} = 0, \tag{7.96}$$

exactly Eq. (6.165). The associated Noether charges are the spatial integrals of the time components,

$$\boxed{P^{\nu} = \int \mathrm{d}^3x\; T^{0\nu}(t,\mathbf{x})} \tag{7.97}$$

which is the four-momentum of Eq. (6.171); its time component $P^0 = \int \mathrm{d}^3x\, T^{00} = H$ is the energy and its spatial components $P^i = \int \mathrm{d}^3x\, T^{0i}$ the field momentum. The generic conservation theorem (Theorem 7.13) and slice-independence (Theorem 7.14) specialize to these and reproduce the direct results of Theorem 6.31. In the language of the charge-as-generator dictionary of the previous section, the four P^{ν} are the generators of spacetime translations, and they are the relativistic upgrade of the Hamiltonian: where nonrelativistic mechanics had a single time-translation generator H, Lorentz invariance forces three more, the components of $\mathbf{P}$, into the same multiplet.

We pause to record the one consistency check the thread obligation asks for, and no more: the carried scalar reproduces, through the general machine, the stress tensor computed by hand in the previous chapter.

Worked Example 7.25 (ϕ^4 theory: the canonical stress tensor from the general theorem)**.** Take the frozen scalar Lagrangian Eq. (6.176),

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4,$$

the single field carrying no spacetime index, so the multiplet index a is trivial and the field-space variation $\delta\phi$ vanishes under translation. The conjugate-momentum density is $\Pi^\mu = \partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$, the only derivative dependence sitting in the kinetic term. Inserting this and $\mathcal{L}$ into Eq. (7.96) gives

$$T^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\left[\tfrac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right], \tag{7.98}$$

which is verbatim Eq. (6.178). The result is automatically symmetric, $T^{\mu\nu} = T^{\nu\mu}$, because a scalar offers no index structure beyond $\partial^\mu\phi\,\partial^\nu\phi$ and $\eta^{\mu\nu}$ times scalars; and it is manifestly built from ϕ alone, so there is no gauge ambiguity. The scalar is the case in which the canonical construction already delivers a flawless tensor. As the Lorentz analysis below makes precise, the reason is that a spin-zero field has *no spin current*: the field-space piece $\Pi^\mu_a\,\delta\phi_a$ that would source the antisymmetric part of T is identically zero. The substantive symmetry layer of ϕ^4 theory—the $\mathbb{Z}_2$ reflection and the classically exact scale invariance of the massless theory—is taken up in Section 7.7; here the scalar serves only to confirm that the general machine and the hand computation agree.

7.4.2 Lorentz invariance and the angular-momentum current

The remaining six Poincaré generators are the Lorentz transformations, and they are the genuinely new content of this section, because the field-space piece of the Noether current no longer vanishes: under a Lorentz transformation a field of nonzero spin is not merely carried to a new point but is *rotated in field space* according to the representation it carries. This rotation is the source of intrinsic spin, and reading it off cleanly is the whole task.

We need both pieces of the transformation, the coordinate piece and the field piece. The coordinate piece is fixed by Chapter 3: an infinitesimal Lorentz transformation acts on spacetime by

$$\delta x^\mu = \omega^\mu{}_\nu\, x^\nu = \tfrac{1}{2}\,\omega_{\alpha\beta}(\eta^{\mu\alpha}x^\beta - \eta^{\mu\beta}x^\alpha), \qquad \omega_{\alpha\beta} = -\omega_{\beta\alpha}, \tag{7.99}$$

the antisymmetry of the six parameters $\omega_{\alpha\beta}$ being exactly the condition $\delta(\eta_{\mu\nu}x^\mu x^\nu) = 0$ that the transformation preserve the interval. The second equality writes $\omega^\mu{}_\nu x^\nu$ in manifestly antisymmetric form against the pair $(\alpha\beta)$; it identifies the coordinate generators

$$X^\mu_{\alpha\beta} = \eta^{\mu\alpha}x^\beta - \eta^{\mu\beta}x^\alpha, \qquad \delta x^\mu = \tfrac{1}{2}\,\omega_{\alpha\beta}\,X^\mu_{\alpha\beta}, \tag{7.100}$$

antisymmetric in $\alpha\beta$ as the parameters demand.

The field piece is fixed by the field's Lorentz representation, the classification of Section 3.3. A multiplet ϕ_a transforming in a representation with Hermitian generators $\Sigma^{\alpha\beta}$ (matrices acting on the multiplet index a, satisfying the Lorentz algebra Eq. (3.34)) is carried by a finite Lorentz transformation to

$$\phi'_a(x') = S(\Lambda)_a{}^b\,\phi_b(x), \qquad S(\Lambda) = \exp\Big(-\tfrac{i}{2}\,\omega_{\alpha\beta}\,\Sigma^{\alpha\beta}\Big), \tag{7.101}$$

in exact parallel with the defining-representation rule $\Lambda = \exp(-\frac{i}{2}\omega_{\alpha\beta}M^{\alpha\beta})$ of Theorem 3.11. The *total* variation in the sense of Eq. (7.39)—the difference between the new field at the new point and the old field at the old point—is the linearized representation rotation, with no derivative term,

$$\delta\phi_a = (S(\Lambda)_a{}^b - \delta_a^b)\phi_b = -\tfrac{i}{2}\,\omega_{\alpha\beta}\,(\Sigma^{\alpha\beta})_a{}^b\,\phi_b, \qquad \Phi_{a,\alpha\beta} = -i\,(\Sigma^{\alpha\beta})_a{}^b\,\phi_b, \tag{7.102}$$

the second equality reading off the field-space generators $\Phi_{a,\alpha\beta}$ in the notation of the multi-parameter theorem (Theorem 7.17). The representation content is the entire input: $\Sigma^{\alpha\beta} = 0$ for a scalar, $\Sigma^{\alpha\beta} = \frac{1}{2}\sigma^{\alpha\beta}$ with $\sigma^{\alpha\beta} = \frac{i}{2}[\gamma^\alpha, \gamma^\beta]$ for a Dirac spinor, and $(\Sigma^{\alpha\beta})^\mu{}_\nu = (M^{\alpha\beta})^\mu{}_\nu = i(\eta^{\alpha\mu}\delta^\beta{}_\nu - \eta^{\beta\mu}\delta^\alpha{}_\nu)$ of Eq. (3.33) for a vector field.

Before turning the crank we must check that a Lorentz transformation is a strict symmetry, $K^\mu = 0$, so that no boundary term contaminates the current. Because $\mathcal{L}$ is, by construction, a Lorentz scalar, it transforms as $\mathcal{L}'(x') = \mathcal{L}(x)$, i.e. its total variation vanishes, $\delta\mathcal{L} = 0$, equivalently $\delta_0\mathcal{L} = \delta\mathcal{L} - \delta x^\mu\partial_\mu\mathcal{L} = -\delta x^\mu\partial_\mu\mathcal{L}$. The invariance condition (Theorem 7.4) is then

$$\delta_0\mathcal{L} + \partial_\mu(\mathcal{L}\,\delta x^\mu) = -\delta x^\mu\partial_\mu\mathcal{L} + (\partial_\mu\delta x^\mu)\mathcal{L} + \delta x^\mu\partial_\mu\mathcal{L} = \mathcal{L}\,\partial_\mu\delta x^\mu = 0, \tag{7.103}$$

the final equality because $\partial_\mu\delta x^\mu = \partial_\mu(\omega^\mu{}_\nu x^\nu) = \omega^\mu{}_\mu = 0$ is the trace of an antisymmetric matrix. Hence $\partial_\mu K^\mu = 0$ and we may take $K^\mu = 0$: Lorentz transformations, like translations, are strict symmetries of any Poincaré-invariant action.

With $K^\mu = 0$, the multi-parameter Noether current Eq. (7.60), one current per generator pair $(\alpha\beta)$, is

$$J^\mu_{\alpha\beta} = \Pi^\mu_a\,\Phi_{a,\alpha\beta} - T^\mu{}_\nu\,X^\nu_{\alpha\beta} = -i\,\Pi^\mu_a\,(\Sigma^{\alpha\beta})_a{}^b\,\phi_b - T^\mu{}_\nu(\eta^{\nu\alpha}x^\beta - \eta^{\nu\beta}x^\alpha). \tag{7.104}$$

Contracting the stress tensor against the coordinate generators gives $T^\mu{}_\nu(\eta^{\nu\alpha}x^\beta - \eta^{\nu\beta}x^\alpha) = T^{\mu\alpha}x^\beta - T^{\mu\beta}x^\alpha$, so

$$J^\mu_{\alpha\beta} = x^\alpha T^{\mu\beta} - x^\beta T^{\mu\alpha} - i\,\Pi^\mu_a\,(\Sigma^{\alpha\beta})_a{}^b\,\phi_b. \tag{7.105}$$

This is the angular-momentum current. We give it the name and the index ordering fixed in the previous chapter (Eq. (6.194)), $M^{\mu\,\alpha\beta} \equiv J^\mu_{\alpha\beta}$, and isolate its two pieces.

Definition 7.26 (Spin current)**.** For a field multiplet ϕ_a in a Lorentz representation with generators $\Sigma^{\nu\rho}$, the *intrinsic spin current* is the field-space piece of the Lorentz Noether current,

$$\boxed{S^{\mu\,\nu\rho} = -\,i\,\Pi^{\mu}_{a}\,(\Sigma^{\nu\rho})_{a}{}^{b}\,\phi_b = -\,i\,\frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi_a)}\,(\Sigma^{\nu\rho})_{a}{}^{b}\,\phi_b} \tag{7.106}$$

antisymmetric in $\nu\rho$ because $\Sigma^{\nu\rho}$ is. It vanishes identically for a scalar ($\Sigma = 0$) and is nonzero precisely when the field carries spin.

Theorem 7.27 (The Lorentz Noether current and its conservation)**.** *On solutions of the Euler–Lagrange equations, the angular-momentum current*

$$M^{\mu\,\nu\rho} = \underbrace{x^{\nu}T^{\mu\rho} - x^{\rho}T^{\mu\nu}}_{\text{orbital}} + \underbrace{S^{\mu\,\nu\rho}}_{\text{spin}} \tag{7.107}$$

is the conserved Noether current of Lorentz invariance,

$$\partial_\mu M^{\mu\,\nu\rho} = 0, \tag{7.108}$$

antisymmetric in the pair $\nu\rho$, with orbital part built from the canonical stress tensor Eq. (7.96) and spin part the spin current Eq. (7.106).

Proof. Equation Eq. (7.107) is Eq. (7.105) with $(\alpha\beta)$ renamed $(\nu\rho)$ and the spin current Eq. (7.106) substituted; the orbital term $x^{\alpha}T^{\mu\beta} - x^{\beta}T^{\mu\alpha} = x^{\nu}T^{\mu\rho} - x^{\rho}T^{\mu\nu}$ on relabelling. Antisymmetry in $\nu\rho$ is manifest term by term. Conservation is the content of Noether's first theorem (Theorem 7.10) applied to the Lorentz quasi-symmetry, which we have just shown is a *strict* symmetry ($K^{\mu} = 0$, Eq. (7.103)): the current Eq. (7.105) is of the form Eq. (7.60), and the theorem gives $\partial_\mu M^{\mu\,\nu\rho} = 0$ for each independent pair $(\nu\rho)$. We verify it independently—and expose its structural content—in Theorem 7.28 below. □

This reproduces and now proves the schematic statement Eq. (6.194) that the previous chapter could only anticipate, and it makes the spin term fully explicit: $S^{\mu\nu\rho}$ is dictated, with no freedom, by the field's behaviour under the Lorentz generators in its representation. The orbital part $x^{\nu}T^{\mu\rho} - x^{\rho}T^{\mu\nu}$ is the field-theoretic "$\mathbf{r}\times\mathbf{p}$"—position x wedged against the momentum-flux columns of the stress tensor—and the spin part is the irreducible remainder that survives even at a point, the angular momentum a field carries by virtue of what it *is* rather than where it sits.

7.4.3 The non-symmetry of the canonical tensor and the spin current

Conservation of the angular-momentum current is not an independent fact: combined with conservation of the stress tensor it pins down the antisymmetric

part of $T^{\mu\nu}$ exactly. This is the precise diagnosis, promised in Section 6.6.6, of why the canonical tensor of a spinning field cannot be symmetric.

Theorem 7.28 (Antisymmetry of T equals minus the divergence of the spin current)**.** *Given $\partial_\mu T^{\mu\nu} = 0$, the angular-momentum current Eq. (7.107) is conserved if and only if the antisymmetric part of the canonical stress tensor equals minus the divergence of the spin current,*

$$\boxed{T^{\nu\rho} - T^{\rho\nu} = -\,\partial_\mu S^{\mu\,\nu\rho}} \tag{7.109}$$

equivalently $2\,T^{[\nu\rho]} = -\,\partial_\mu S^{\mu\,\nu\rho}$.

Proof. Differentiate the orbital part of Eq. (7.107) directly, using the product rule and $\partial_\mu x^\nu = \delta^\nu_\mu$:

$$\begin{aligned}\partial_\mu\big(x^\nu T^{\mu\rho} - x^\rho T^{\mu\nu}\big) &= (\partial_\mu x^\nu)\,T^{\mu\rho} + x^\nu\,\partial_\mu T^{\mu\rho} - (\partial_\mu x^\rho)\,T^{\mu\nu} - x^\rho\,\partial_\mu T^{\mu\nu} \\ &= \delta^\nu_\mu\,T^{\mu\rho} - \delta^\rho_\mu\,T^{\mu\nu} = T^{\nu\rho} - T^{\rho\nu},\end{aligned} \tag{7.110}$$

the two terms with $\partial_\mu T^{\mu\cdot}$ dropping out by conservation of the stress tensor Eq. (7.96). Adding the divergence of the spin part,

$$\partial_\mu M^{\mu\,\nu\rho} = \big(T^{\nu\rho} - T^{\rho\nu}\big) + \partial_\mu S^{\mu\,\nu\rho}, \tag{7.111}$$

which reproduces Eq. (6.195). The left-hand side vanishes if and only if the two terms on the right cancel, i.e. if and only if Eq. (7.109) holds. Since Lorentz invariance guarantees $\partial_\mu M^{\mu\nu\rho} = 0$ (Theorem 7.27), the identity Eq. (7.109) holds on shell as a consequence. □

The theorem is the whole story of the canonical tensor's non-symmetry in one line. The antisymmetric part of $T^{\mu\nu}$ is not arbitrary: it is exactly the divergence of the object that carries intrinsic spin. A scalar has $S^{\mu\nu\rho} = 0$, so Eq. (7.109) forces $T^{\nu\rho} = T^{\rho\nu}$—the canonical scalar tensor is automatically symmetric, as Theorem 7.25 found. A spinning field has $S^{\mu\nu\rho} \neq 0$, and its canonical $T^{\mu\nu}$ is symmetric only where the spin current happens to be divergence-free, which it generically is not. The asymmetry *is* the spin, made quantitative.

It is worth proving Eq. (7.109) a second way, directly from the invariance of the Lagrangian, because the argument exposes where the stress tensor's antisymmetry comes from at the level of $\mathcal{L}$ itself, and because the off-shell identity it produces is the classical seed of the rotational Ward identity. Invariance of $\mathcal{L}$ as a Lorentz scalar is the statement that, when the multiplet index and the spacetime indices of its arguments are rotated together by the generator labelled $(\nu\rho)$, the density does not change. Treating ϕ_a and $\psi_{\mu a} \equiv \partial_\mu \phi_a$ as independent arguments, ϕ_a rotates by $\Sigma^{\nu\rho}$ on its multiplet index while $\psi_{\mu a}$ rotates by $\Sigma^{\nu\rho}$ on its multiplet

index *and* by the vector generator $(M^{\nu\rho})_\mu{}^\sigma$ of Eq. (3.33) on its derivative index. Setting the resulting first-order change of $\mathcal{L}$ to zero gives the algebraic identity

$$\frac{\partial \mathcal{L}}{\partial \phi_a}\,(\Sigma^{\nu\rho})_a{}^b\,\phi_b + \Pi^\mu_a\,(\Sigma^{\nu\rho})_a{}^b\,\partial_\mu\phi_b + \Pi^\mu_a\,(M^{\nu\rho})_\mu{}^\sigma\,\partial_\sigma\phi_a = 0, \tag{7.112}$$

true for every configuration, on shell or off—the Lorentz analogue of the translation homogeneity used in Theorem 6.29. Evaluate the third term with the explicit vector generator $(M^{\nu\rho})_\mu{}^\sigma = i(\delta^\nu_\mu\,\eta^{\rho\sigma} - \delta^\rho_\mu\,\eta^{\nu\sigma})$:

$$\Pi^\mu_a\,(M^{\nu\rho})_\mu{}^\sigma\,\partial_\sigma\phi_a = i(\Pi^\nu_a\,\partial^\rho\phi_a - \Pi^\rho_a\,\partial^\nu\phi_a). \tag{7.113}$$

Now $\Pi^\nu_a\,\partial^\rho\phi_a = T^{\nu\rho} + \eta^{\nu\rho}\mathcal{L}$ from the definition Eq. (7.96), and the $\eta\mathcal{L}$ pieces cancel in the antisymmetric difference, so Eq. (7.113) equals $i(T^{\nu\rho} - T^{\rho\nu})$. The first two terms of Eq. (7.112), on shell where $\partial_\mu\Pi^\mu_a = \partial\mathcal{L}/\partial\phi_a$, are precisely $i\,\partial_\mu S^{\mu\nu\rho}$:

$$\partial_\mu S^{\mu\,\nu\rho} = -i\,\partial_\mu(\Pi^\mu_a(\Sigma^{\nu\rho})_a{}^b\phi_b) = -i\Big[\tfrac{\partial\mathcal{L}}{\partial\phi_a}(\Sigma^{\nu\rho})_a{}^b\phi_b + \Pi^\mu_a(\Sigma^{\nu\rho})_a{}^b\partial_\mu\phi_b\Big]. \tag{7.114}$$

Multiplying Eq. (7.112) through by i and substituting both pieces gives $i\,\partial_\mu S^{\mu\nu\rho} + i(T^{\nu\rho} - T^{\rho\nu}) = 0$, that is Eq. (7.109) once more. The two derivations meet: the kinematic non-symmetry forced by Lorentz invariance of $\mathcal{L}$ (Eq. (7.112)) is the same as the dynamical non-symmetry forced by conservation of M^μ (Theorem 7.28).

Worked Example 7.29 (The Dirac spin current and the asymmetry of $T_D^{\mu\nu}$)**.** We make the spin current concrete on the electron, computing $S^{\mu\nu\rho}$ for the free Dirac field and verifying that its divergence accounts exactly for the asymmetry of the canonical tensor found in Eq. (6.188). The Dirac Lagrangian is $\mathcal{L}_D = \bar\psi(i\partial\!\!\!/ - m)\psi$ (Eq. (6.68)), with ψ and $\bar\psi$ independent and only ψ differentiated, so from Eq. (6.78) the conjugate momenta are

$$\Pi^\mu_\psi = \frac{\partial\mathcal{L}_D}{\partial(\partial_\mu\psi)} = i\,\bar\psi\gamma^\mu, \qquad \Pi^\mu_{\bar\psi} = 0. \tag{7.115}$$

The spinor carries the representation $\Sigma^{\nu\rho} = \frac{1}{2}\sigma^{\nu\rho}$, with $\sigma^{\nu\rho} = \frac{i}{2}\,[\gamma^\nu, \gamma^\rho]$ the spin matrix $S^{\nu\rho} = \frac{i}{4}\,[\gamma^\nu, \gamma^\rho]$ of Eq. (3.114) written as $S^{\nu\rho} = \frac{1}{2}\sigma^{\nu\rho}$. Only the ψ contribution survives in Eq. (7.106) (the $\bar\psi$ term is killed by $\Pi^\mu_{\bar\psi} = 0$), so

$$S^{\mu\,\nu\rho} = -i\,\Pi^\mu_\psi\,\Sigma^{\nu\rho}\psi = -i\,(i\bar\psi\gamma^\mu)\,\tfrac{1}{2}\sigma^{\nu\rho}\,\psi = \tfrac{1}{2}\,\bar\psi\,\gamma^\mu\sigma^{\nu\rho}\,\psi. \tag{7.116}$$

This is the intrinsic angular-momentum current of the electron; its time component $S^{0\,ij} = \frac{1}{2}\bar\psi\gamma^0\sigma^{ij}\psi = \frac{1}{2}\psi^\dagger\sigma^{ij}\psi$ integrates to the familiar spin operator $\frac{1}{2}\boldsymbol{\Sigma}$ when quantized.

Now verify Eq. (7.109). Differentiate Eq. (7.116), distributing the derivative onto $\bar\psi$ and ψ,

$$\partial_\mu S^{\mu\,\nu\rho} = \tfrac{1}{2}\big[(\partial_\mu\bar\psi)\gamma^\mu\sigma^{\nu\rho}\psi + \bar\psi\,\gamma^\mu\sigma^{\nu\rho}(\partial_\mu\psi)\big]. \tag{7.117}$$

Use the Dirac equations $i\not{\partial}\psi = m\psi$ and $i\partial_\mu\bar\psi\,\gamma^\mu = -m\bar\psi$, i.e. $\gamma^\mu\partial_\mu\psi = -im\psi$ and $(\partial_\mu\bar\psi)\gamma^\mu = im\bar\psi$. The first term becomes $\frac{1}{2}(im\bar\psi)\sigma^{\nu\rho}\psi$. In the second term commute γ^μ past $\sigma^{\nu\rho}$ to expose $\gamma^\mu\partial_\mu\psi$: writing $\gamma^\mu\sigma^{\nu\rho} = \sigma^{\nu\rho}\gamma^\mu + [\gamma^\mu, \sigma^{\nu\rho}]$,

$$\bar\psi\,\gamma^\mu\sigma^{\nu\rho}\partial_\mu\psi = \bar\psi\,\sigma^{\nu\rho}(\gamma^\mu\partial_\mu\psi) + \bar\psi\,[\gamma^\mu, \sigma^{\nu\rho}]\partial_\mu\psi = -im\,\bar\psi\sigma^{\nu\rho}\psi + \bar\psi\,[\gamma^\mu, \sigma^{\nu\rho}]\partial_\mu\psi. \tag{7.118}$$

The two mass terms cancel, leaving the commutator term alone,

$$\partial_\mu S^{\mu\,\nu\rho} = \tfrac{1}{2}\,\bar\psi\,[\gamma^\mu, \sigma^{\nu\rho}]\,\partial_\mu\psi. \tag{7.119}$$

The gamma-matrix identity $[\gamma^\mu, \sigma^{\nu\rho}] = 2i(\eta^{\mu\nu}\gamma^\rho - \eta^{\mu\rho}\gamma^\nu)$ (Chapter C) collapses the commutator,

$$\partial_\mu S^{\mu\,\nu\rho} = \tfrac{1}{2}\,\bar\psi\,2i(\eta^{\mu\nu}\gamma^\rho - \eta^{\mu\rho}\gamma^\nu)\,\partial_\mu\psi = i\,\bar\psi(\gamma^\rho\partial^\nu - \gamma^\nu\partial^\rho)\psi. \tag{7.120}$$

On the other side, the on-shell canonical Dirac tensor is $T_D^{\mu\nu} = i\bar\psi\gamma^\mu\partial^\nu\psi$ (Eq. (6.188)), so its antisymmetric part is

$$T_D^{\nu\rho} - T_D^{\rho\nu} = i\bar\psi\gamma^\nu\partial^\rho\psi - i\bar\psi\gamma^\rho\partial^\nu\psi = i\,\bar\psi(\gamma^\nu\partial^\rho - \gamma^\rho\partial^\nu)\psi. \tag{7.121}$$

Comparing Eq. (7.120) with Eq. (7.121), $\partial_\mu S^{\mu\nu\rho} = -(T_D^{\nu\rho} - T_D^{\rho\nu})$ exactly, confirming the master identity Eq. (7.109) for the Dirac field by direct computation. The electron's stress tensor is asymmetric, and the asymmetry is precisely the divergence of its spin current—the statement made concrete.

Remark 7.30 (Maxwell, and the cue for the next section). The same construction applied to the vector field, with $\Sigma^{\nu\rho}$ the generators Eq. (3.33) acting on the index of A_σ, gives a spin current $S^{\mu\nu\rho} = -F^{\mu\nu}A^\rho + F^{\mu\rho}A^\nu$, and Eq. (7.109) again ties it to the asymmetry of the canonical electromagnetic tensor Eq. (6.191). Here a second disease compounds the first: that spin current is built from the bare potential A^σ and is not gauge invariant, so the canonical Maxwell tensor is neither symmetric nor gauge invariant, as Eq. (6.192) showed. Both defects are repaired at once by the Belinfante construction of Section 7.5, which uses exactly the spin current of Eq. (7.106) as the seed of the improvement superpotential.

7.4.4 The angular-momentum charges and the Poincaré generators

A conserved current means a conserved charge. The six conserved charges of Lorentz invariance are the spatial integrals of the time components of $M^{\mu\nu\rho}$, exactly as the four-momentum was built from $T^{0\nu}$.

Definition 7.31 (Angular-momentum charges)**.** The conserved *angular-momentum tensor* is

$$J^{\nu\rho} = \int \mathrm{d}^3x\; M^{0\,\nu\rho}(t, \mathbf{x}) = \int \mathrm{d}^3x\,\Big[(x^\nu T^{0\rho} - x^\rho T^{0\nu}) + S^{0\,\nu\rho}\Big], \tag{7.122}$$

antisymmetric in $\nu\rho$, hence six independent numbers. By Theorem 7.13 applied to each of the six currents, $dJ^{\nu\rho}/dt = 0$.

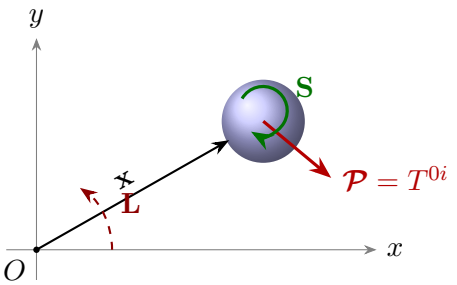


Figure 7.4: The two contributions to the conserved angular momentum Eq. (7.123). A localized field configuration at position $\mathbf{x}$ carries momentum density $\boldsymbol{\mathcal{P}} = T^{0i}$; its moment about the origin O is the orbital angular momentum $\mathbf{L} = \int \mathbf{x} \times \boldsymbol{\mathcal{P}}$. Independently, the field's Lorentz representation endows it with an intrinsic spin density S^{0jk}, integrating to the spin $\mathbf{S}$, which survives even for a configuration at rest at the origin. Total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$.

The six charges split, by the now-familiar rotation/boost decomposition of Theorem 3.13, into the three spatial-spatial components and the three time-space components. The spatial components are the total angular momentum: defining $J_i = \frac{1}{2}\epsilon_{ijk}J^{jk}$ in parallel with Eq. (3.40),

$$J^{jk} = \int \mathrm{d}^3x\, (x^j T^{0k} - x^k T^{0j}) + \int \mathrm{d}^3x\, S^{0\,jk} = L^{jk} + \Sigma^{jk}_{\text{spin}}, \tag{7.123}$$

the first integral the *orbital* angular momentum L^{jk}—the moment $\mathbf{x} \times \boldsymbol{\mathcal{P}}$ of the momentum density $\mathcal{P}^k = T^{0k}$—and the second the *spin* angular momentum $\Sigma^{jk}_{\text{spin}} = \int \mathrm{d}^3x\, S^{0jk}$, the integrated intrinsic current. This is the field-theoretic origin of the textbook decomposition $\mathbf{J} = \mathbf{L} + \mathbf{S}$: orbital angular momentum is the moment of the energy–momentum flow, spin is the irreducible piece that a localized field configuration carries even when it is not orbiting anything. For the electron, $\Sigma^{jk}_{\text{spin}} = \frac{1}{2}\int \mathrm{d}^3x\, \psi^\dagger \sigma^{jk}\psi$ is the spin-$\frac{1}{2}$ that quantization will sharpen into eigenvalues $\pm\frac{1}{2}$ (Fig. 7.4).

The time-space components are the boost charges,

$$J^{0i} = \int \mathrm{d}^3x\, (x^0 T^{0i} - x^i T^{00}) + \int \mathrm{d}^3x\, S^{0\,0i} = t\,P^i - \int \mathrm{d}^3x\, x^i\, T^{00} + \int \mathrm{d}^3x\, S^{0\,0i}, \tag{7.124}$$

where $x^0 = t$ is constant on the slice and pulls out of the integral against T^{0i} to give tP^i. The statement $dJ^{0i}/dt = 0$ is the relativistic conservation of the center of energy: differentiating, $P^i = d/dt \int \mathrm{d}^3x\, x^i T^{00}$ (the spin piece is time-independent on its own), i.e. the center of energy $X^i_{\text{cm}} = (1/P^0)\int x^i T^{00}$ moves uniformly with velocity P^i/P^0, the field-theoretic analogue of the free-particle relation between momentum and center-of-mass motion.

These ten charges—the four P^ν of Eq. (7.97) and the six $J^{\nu\rho}$ of Eq. (7.122)—are the complete set of conserved quantities supplied by Poincaré invariance, and

they are the classical precursors of the ten generators of the Poincaré algebra. The charge-as-generator dictionary established for internal symmetries in the previous section carries over verbatim: each charge generates, through the Poisson bracket of Section 7.3, the transformation it came from, and the brackets of the charges among themselves close into the Lie algebra of the symmetry. For the Poincaré group that algebra is

$$\begin{aligned} \{P^\mu, P^\nu\}_{\mathrm{P}} &= 0, \\ \{J^{\mu\nu}, P^\rho\}_{\mathrm{P}} &= \eta^{\nu\rho} P^\mu - \eta^{\mu\rho} P^\nu, \\ \{J^{\mu\nu}, J^{\rho\sigma}\}_{\mathrm{P}} &= \eta^{\nu\rho} J^{\mu\sigma} - \eta^{\mu\rho} J^{\nu\sigma} - \eta^{\nu\sigma} J^{\mu\rho} + \eta^{\mu\sigma} J^{\nu\rho}, \end{aligned} \tag{7.125}$$

the same structure as the operator algebra Eqs. (3.147) to (3.149), with the Poisson bracket standing in for $-i$ times the commutator. The last line is the Lorentz subalgebra Eq. (3.34) carried by the charges; the middle line says P^ρ transforms as a four-vector under Lorentz, exactly as its index advertises; the first line says translations commute. Energy, momentum, and angular momentum have thus become Noether charges that *generate* the spacetime symmetries from which they sprang—the classical skeleton that Chapter 9 will quantize into the operators P^μ and $J^{\mu\nu}$ acting on the Hilbert space, where the eigenvalues of P^2 and of the spin Casimir built from $J^{\mu\nu}$ will label particles by mass and spin.

Physical Insight

The split of $\mathbf{J}$ into $\mathbf{L}+\mathbf{S}$ is not a convention but a representation-theoretic fact: orbital angular momentum is the moment of the momentum density and depends on where you put the origin, while spin is the intrinsic Lorentz quantum number of the field and is origin-independent. A scalar has only $\mathbf{L}$; a spinor and a vector carry $\mathbf{S}$ in addition. The same generator $J^{\mu\nu}$ that rotates spacetime coordinates simultaneously rotates the field's components, and it is the second action—the representation matrices $\Sigma^{\mu\nu}$—that is the entire content of "spin." This is why the classification of particles by spin is, at bottom, the classification of the unitary representations of the Poincaré group.

We have now extracted from Poincaré invariance the full complement of spacetime conserved quantities and, with them, the canonical stress tensor in its proper role as the Noether current of translations. But that tensor arrived with the two defects this section has made precise and quantitative: by Theorem 7.28 it is generically non-symmetric, its antisymmetric part fixed to $-\partial_\mu S^{\mu\nu\rho}$ by the very spin current Eq. (7.106) we computed, and for gauge fields (Theorem 7.30) it is not even gauge invariant. Both the asymmetry and the spin current that drives it are the raw material for the repair. The next section takes the spin current $S^{\mu\nu\rho}$ and builds from it the antisymmetric superpotential whose divergence—an improvement of the type licensed by Theorem 7.15, harmless to the charges

P^ν—turns the canonical tensor into the symmetric, gauge-invariant Belinfante–Rosenfeld tensor, the object that will later couple to gravity.

7.5 The Belinfante–Rosenfeld Symmetric Stress Tensor

The previous section handed us two Noether currents born of spacetime symmetry and a single algebraic identity binding them. Translations gave the canonical stress tensor $T^{\mu\nu}$, conserved on the index μ, whose charges $P^\nu = \int d^3x\, T^{0\nu}$ are the energy and three-momentum. Lorentz transformations gave the angular-momentum current, which splits into an *orbital* part built from $T^{\mu\nu}$ and an *intrinsic* part—the spin current $S^{\lambda\mu\nu}$—encoding how the field components rotate among themselves. Demanding that the total angular momentum be conserved then forced a relation between the two: the antisymmetric part of $T^{\mu\nu}$ is exactly the divergence of the spin current. That relation is the lever of this section. The canonical $T^{\mu\nu}$ is in general neither symmetric (we saw the Dirac and Maxwell tensors fail this in Section 6.6.5) nor, for gauge fields, gauge invariant. Both defects are cosmetic in the sense that they leave the charges P^ν untouched, but they are fatal for the one job a stress tensor must ultimately do: source gravity, which couples to a *symmetric* tensor. We now repair $T^{\mu\nu}$ once and for all, by adding to it a divergence-free superpotential built from the very spin current the asymmetry produced. The result, due to Belinfante and Rosenfeld, is the canonical object of the rest of the book.

7.5.1 What an improvement may and may not change

Any current is ambiguous up to the addition of an identically conserved piece. For the stress tensor the useful form of that ambiguity is the addition of a divergence,

$$\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}, \qquad B^{\rho\mu\nu} = -B^{\mu\rho\nu}, \tag{7.126}$$

where the object $B^{\rho\mu\nu}$—the *superpotential*—is required to be antisymmetric in its first two indices. This single antisymmetry buys both of the properties an improvement must have, before we have even said what B is.

First, $\Theta^{\mu\nu}$ is conserved on μ whenever $T^{\mu\nu}$ is. Taking the divergence of Eq. (7.126),

$$\partial_\mu \Theta^{\mu\nu} = \partial_\mu T^{\mu\nu} + \partial_\mu \partial_\rho B^{\rho\mu\nu} = 0 + 0, \tag{7.127}$$

because $\partial_\mu \partial_\rho$ is symmetric in $\mu \leftrightarrow \rho$ while $B^{\rho\mu\nu}$ is antisymmetric in exactly that pair, so the contraction vanishes identically—off shell, as a matter of calculus, with no use of the equations of motion. The improvement can never spoil conservation.

Second, $\Theta^{\mu\nu}$ carries the same charges as $T^{\mu\nu}$. The added energy–momentum is

$$\Delta P^{\nu} = \int d^3x\, \partial_\rho B^{\rho 0\nu} = \int d^3x\, (\partial_0 B^{00\nu} + \partial_i B^{i0\nu}) = \int d^3x\, \partial_i B^{i0\nu}, \tag{7.128}$$

where the time-derivative term dropped because $B^{00\nu} = -B^{00\nu} = 0$ by the antisymmetry of the first two indices. What survives is a pure spatial divergence, and by Gauss's theorem

$$\Delta P^{\nu} = \oint_{S^2_\infty} dS_i\, B^{i0\nu} = 0 \tag{7.129}$$

for any field configuration that falls off at spatial infinity, as physical configurations must. The local distribution of energy and momentum is reshuffled by the improvement; the integrated energy and momentum are not. This is the precise sense, promised in Section 6.6.5, in which the defects of $T^{\mu\nu}$ are cosmetic: they live entirely in a superpotential that integrates to zero.

The freedom in Eq. (7.126) is therefore exactly the freedom to redistribute energy and momentum without changing the totals. We now spend that freedom to make $\Theta^{\mu\nu}$ symmetric.

7.5.2 Constructing the superpotential from the spin current

Everything we need from the previous section is the antisymmetric-part relation. With the spin current written $S^{\lambda\mu\nu}$—index λ the conserved-current (divergence) index, the last pair antisymmetric,

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \tag{7.130}$$

—the conservation of the total angular-momentum current $J^{\lambda\mu\nu} = x^\mu T^{\lambda\nu} - x^\nu T^{\lambda\mu} + S^{\lambda\mu\nu}$, defined in Eq. (7.107) and proved conserved in Theorem 7.27, was shown in Theorem 7.28 to force the antisymmetric-part relation

$$\partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}, \qquad \text{equivalently} \qquad T^{\mu\nu} - T^{\nu\mu} = -\,\partial_\rho S^{\rho\mu\nu}, \tag{7.131}$$

which is Eq. (7.109) written with the current index displayed first. The antisymmetric part of the canonical stress tensor is the divergence of the spin current: a non-symmetric $T^{\mu\nu}$ is the signature of a field that carries spin. Our task is to cancel this antisymmetric part with a $\partial_\rho B^{\rho\mu\nu}$ of exactly the right form.

Subtract Eq. (7.126) from its transpose:

$$\Theta^{\mu\nu} - \Theta^{\nu\mu} = (T^{\mu\nu} - T^{\nu\mu}) + \partial_\rho (B^{\rho\mu\nu} - B^{\rho\nu\mu}) = \partial_\rho \big[B^{\rho\mu\nu} - B^{\rho\nu\mu} - S^{\rho\mu\nu}\big], \tag{7.132}$$

where the second equality used Eq. (7.131). The bracket vanishes—and $\Theta^{\mu\nu}$ is symmetric—if the superpotential satisfies the algebraic condition

$$B^{\rho\mu\nu} - B^{\rho\nu\mu} = S^{\rho\mu\nu}. \tag{7.133}$$

We must therefore find a $B^{\rho\mu\nu}$ that is (i) antisymmetric in its first two indices, $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$, and (ii) antisymmetrizes in its last two indices to the spin current, Eq. (7.133). Rather than guess, we solve these two linear conditions. The only object linear in S available is the spin current itself, and because S is already antisymmetric in its *last* pair, its three cyclic orderings $S^{\rho\mu\nu}$, $S^{\mu\nu\rho}$, $S^{\nu\rho\mu}$ exhaust the independent index placements. Make the ansatz

$$B^{\rho\mu\nu} = a\,S^{\rho\mu\nu} + b\,S^{\mu\nu\rho} + c\,S^{\nu\rho\mu}, \tag{7.134}$$

and impose the two conditions in turn. Condition (ii), $B^{\rho\mu\nu} - B^{\rho\nu\mu} = S^{\rho\mu\nu}$, after reducing every term to the cyclic basis with $S^{\cdots} = -S^{\cdots}$ on its last pair (so $S^{\rho\nu\mu} = -S^{\rho\mu\nu}$, $S^{\nu\mu\rho} = -S^{\nu\rho\mu}$, $S^{\mu\rho\nu} = -S^{\mu\nu\rho}$), reads $2a\,S^{\rho\mu\nu} + 2b\,S^{\mu\nu\rho} + 2c\,S^{\nu\rho\mu} = S^{\rho\mu\nu}$ once the b and c contributions to the antisymmetrization are tracked; carrying this out gives $a = \tfrac{1}{2}$ together with $b + c$ unconstrained by (ii) alone. Condition (i), $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$, supplies the second equation and fixes the antisymmetric combination of b and c, yielding $b = \tfrac{1}{2}$, $c = -\tfrac{1}{2}$. The system has the unique solution within this ansatz,

$$\boxed{B^{\rho\mu\nu} = \tfrac{1}{2}\big(S^{\rho\mu\nu} + S^{\mu\nu\rho} - S^{\nu\rho\mu}\big)} \tag{7.135}$$

the *Belinfante–Rosenfeld superpotential*, and with it the *Belinfante–Rosenfeld stress tensor* $\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}$. (The improvement is not unique in the absolute sense—one may always add to $B^{\rho\mu\nu}$ any tensor antisymmetric in $\rho\mu$ that is also symmetric in $\mu\nu$, since such a piece changes neither the conservation nor the symmetrization; Eq. (7.135) is the minimal choice built purely from the spin current, and it is the one the gravitational coupling will select.)

Definition 7.32 (Belinfante–Rosenfeld tensor)**.** Given a canonical stress tensor $T^{\mu\nu}$ and the canonical spin current $S^{\lambda\mu\nu}$ of the same theory, antisymmetric in its last pair, the Belinfante–Rosenfeld stress tensor is

$$\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}, \qquad B^{\rho\mu\nu} = \tfrac{1}{2}\big(S^{\rho\mu\nu} + S^{\mu\nu\rho} - S^{\nu\rho\mu}\big). \tag{7.136}$$

It remains to verify that Eq. (7.135) has the two required properties. The verification is pure index bookkeeping, using only the antisymmetry Eq. (7.130) of S in its last pair, and we carry it out in full.

Property (i): antisymmetry in the first two indices. Swap $\rho \leftrightarrow \mu$ in Eq. (7.135):

$$B^{\mu\rho\nu} = \tfrac{1}{2}\big(S^{\mu\rho\nu} + S^{\rho\nu\mu} - S^{\nu\mu\rho}\big). \tag{7.137}$$

Now reduce each term to the basis appearing in Eq. (7.135) using Eq. (7.130), which lets us flip the last pair at the cost of a sign: $S^{\mu\rho\nu} = -S^{\mu\nu\rho}$, $S^{\rho\nu\mu} = -S^{\rho\mu\nu}$, $S^{\nu\mu\rho} = -S^{\nu\rho\mu}$. Substituting,

$$B^{\mu\rho\nu} = \tfrac{1}{2}\big(-S^{\mu\nu\rho} - S^{\rho\mu\nu} + S^{\nu\rho\mu}\big) = -\tfrac{1}{2}\big(S^{\rho\mu\nu} + S^{\mu\nu\rho} - S^{\nu\rho\mu}\big) = -B^{\rho\mu\nu}. \tag{7.138}$$

So $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$, and by the work of Section 7.5.1 the tensor $\Theta^{\mu\nu}$ is automatically conserved and shares the charges of $T^{\mu\nu}$.

Property (ii): the symmetrization condition. Swap $\mu \leftrightarrow \nu$ in Eq. (7.135):

$$B^{\rho\nu\mu} = \tfrac{1}{2}(S^{\rho\nu\mu} + S^{\nu\mu\rho} - S^{\mu\rho\nu}) = \tfrac{1}{2}(-S^{\rho\mu\nu} - S^{\nu\rho\mu} + S^{\mu\nu\rho}), \tag{7.139}$$

again using Eq. (7.130) term by term. Subtracting from Eq. (7.135),

$$\begin{aligned} B^{\rho\mu\nu} - B^{\rho\nu\mu} &= \tfrac{1}{2}(S^{\rho\mu\nu} + S^{\mu\nu\rho} - S^{\nu\rho\mu}) - \tfrac{1}{2}(-S^{\rho\mu\nu} - S^{\nu\rho\mu} + S^{\mu\nu\rho}) \\ &= \tfrac{1}{2}(2\,S^{\rho\mu\nu}) = S^{\rho\mu\nu}, \end{aligned} \tag{7.140}$$

where the $S^{\mu\nu\rho}$ terms cancelled and the two $-S^{\nu\rho\mu}$ and $+S^{\nu\rho\mu}$ likewise cancelled. This is precisely Eq. (7.133). Both requirements hold, so we have proved the central result of the section.

Theorem 7.33 (Properties of the Belinfante–Rosenfeld tensor)**.** *Let $T^{\mu\nu}$ be the canonical stress tensor and $S^{\lambda\mu\nu}$ the canonical spin current of a Poincaré-invariant field theory, related by the angular-momentum identity Eq. (7.131). Then the tensor $\Theta^{\mu\nu}$ of Eq. (7.136) is*

1. conserved*: $\partial_\mu \Theta^{\mu\nu} = 0$ on shell;*
2. symmetric*: $\Theta^{\mu\nu} = \Theta^{\nu\mu}$;*
3. charge-equivalent to $T^{\mu\nu}$*: $P^\nu = \displaystyle\int d^3x\, \Theta^{0\nu} = \int d^3x\, T^{0\nu}$.*

Proof. Properties 1 and 3 follow from the antisymmetry $B^{\rho\mu\nu} = -B^{\mu\rho\nu}$ established in Eq. (7.138), by the general arguments Eq. (7.127) and Eqs. (7.128) and (7.129) respectively. For property 2, the symmetrization condition Eq. (7.140) makes the bracket in Eq. (7.132) vanish identically, so $\Theta^{\mu\nu} - \Theta^{\nu\mu} = 0$. □

Physical Insight

The asymmetry of the canonical tensor and the existence of the spin current are not two facts but one: Eq. (7.131) says the antisymmetric part of $T^{\mu\nu}$ *is* the divergence of the spin current. The Belinfante construction does not destroy the spin information—it relocates it. The piece $\partial_\rho B^{\rho\mu\nu}$ transfers the intrinsic spin from a separate current into the orbital motion of a symmetric stress tensor, so that after improvement the entire angular momentum can be written in the purely orbital form $J^{\mu\nu\rho}_\Theta = x^\nu \Theta^{\mu\rho} - x^\rho \Theta^{\mu\nu}$. One checks directly that this is conserved precisely because Θ is symmetric: $\partial_\mu J^{\mu\nu\rho}_\Theta = \Theta^{\nu\rho} - \Theta^{\rho\nu} + x^\nu \partial_\mu \Theta^{\mu\rho} - x^\rho \partial_\mu \Theta^{\mu\nu} = 0$. Symmetry of the stress tensor and conservation of angular momentum are the same statement.

Physical Insight

Why insist on symmetry at all, when the charges are unchanged? Because the stress tensor will eventually be promoted from a Noether current to the *source of gravity*. In general relativity the Einstein equations read $G_{\mu\nu} = 8\pi G\,\Theta_{\mu\nu}$, and the left-hand side $G_{\mu\nu}$ is manifestly symmetric, built from the metric. A non-symmetric source would be inconsistent. Equivalently, the gravitational coupling is obtained by varying the matter action with respect to the metric, $\Theta_{\mu\nu} \propto -(2/\sqrt{-g})\,\delta S_{\text{matter}}/\delta g^{\mu\nu}$, and this *Hilbert* stress tensor is automatically symmetric and gauge invariant because $g^{\mu\nu}$ is. The Belinfante tensor is the flat-space shadow of that variation: the systematic curved-space construction (deferred to Chapter 64) reproduces Eq. (7.136) on a flat background. This is the deep reason the symmetric tensor, not the canonical one, is "the" stress tensor.

7.5.3 The electromagnetic field: from canonical mess to gauge-invariant tensor

The Maxwell field is the worked example that the whole construction was built for, and the one that anticipates the field-strength-as-curvature viewpoint of Chapter 8. In Section 6.6.5 we found its canonical stress tensor to be simultaneously asymmetric and gauge dependent; we now show the Belinfante recipe cures both at a stroke, producing the symmetric, gauge-invariant tensor whose components are the textbook energy density and Poynting vector. This continues the classical electromagnetic thread that Chapter 8 will quantize.

Worked Example 7.34 (The Belinfante tensor of Maxwell theory)**.** The free Maxwell Lagrangian is $\mathcal{L}_{\text{Max}} = -\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}$ with $F_{\alpha\beta} = \partial_\alpha A_\beta - \partial_\beta A_\alpha$ (Eq. (6.80)). From Eq. (6.190) the canonical momentum conjugate to A_ρ is

$$\pi^{\mu\rho} \equiv \frac{\partial \mathcal{L}_{\text{Max}}}{\partial(\partial_\mu A_\rho)} = -F^{\mu\rho}, \tag{7.141}$$

and from Eq. (6.191) the canonical stress tensor is

$$T^{\mu\nu}_{\text{can}} = -F^{\mu\rho}\,\partial^\nu A_\rho + \tfrac{1}{4}\,\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}. \tag{7.142}$$

The spin current. Under an infinitesimal Lorentz transformation the vector potential transforms in the vector representation, $\delta A_\rho = \omega_\rho{}^\sigma A_\sigma$ with $\omega_{\rho\sigma} = -\omega_{\sigma\rho}$; the corresponding spin matrix has the index structure $(\Sigma^{\mu\nu})_\rho{}^\sigma = \delta^\mu_\rho\,\eta^{\nu\sigma} - \delta^\nu_\rho\,\eta^{\mu\sigma}$ inherited from the Lorentz generators of Chapter 3 acting on a vector index. The canonical spin current, $S^{\lambda\mu\nu} = \pi^{\lambda\rho}(\Sigma^{\mu\nu})_\rho{}^\sigma A_\sigma$, then evaluates to

$$S^{\lambda\mu\nu} = \pi^{\lambda\mu}A^\nu - \pi^{\lambda\nu}A^\mu = -F^{\lambda\mu}A^\nu + F^{\lambda\nu}A^\mu = F^{\lambda\nu}A^\mu - F^{\lambda\mu}A^\nu, \tag{7.143}$$

which is manifestly antisymmetric in the pair $\mu\nu$, as a spin current must be. As a check that this is the genuine partner of $T^{\mu\nu}_{\rm can}$, it satisfies the identity Eq. (7.131). On shell ($\partial_\lambda F^{\lambda\nu}=0$) its divergence is

$$\partial_\lambda S^{\lambda\mu\nu} = F^{\lambda\nu}\partial_\lambda A^\mu - F^{\lambda\mu}\partial_\lambda A^\nu = F^{\lambda\nu}F_\lambda{}^\mu - F^{\lambda\mu}F_\lambda{}^\nu + (F^{\lambda\nu}\partial^\mu A_\lambda - F^{\lambda\mu}\partial^\nu A_\lambda), \tag{7.144}$$

where we wrote $\partial_\lambda A^\mu = F_\lambda{}^\mu + \partial^\mu A_\lambda$ using $F_\lambda{}^\mu = \partial_\lambda A^\mu - \partial^\mu A_\lambda$. The first two terms cancel: $F^{\lambda\nu}F_\lambda{}^\mu = \eta_{\lambda\sigma}F^{\sigma\nu}F^{\lambda\mu}$ is symmetric under $\mu\leftrightarrow\nu$ (relabel $\lambda\leftrightarrow\sigma$ and use the symmetry of η), so it equals $F^{\lambda\mu}F_\lambda{}^\nu$. The remaining bracket is $-(T^{\mu\nu}_{\rm can}-T^{\nu\mu}_{\rm can})$ read off from Eq. (7.142) (the $\frac{1}{4}\eta F^2$ term is symmetric and drops), so $\partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu}_{\rm can} - T^{\mu\nu}_{\rm can}$, which is exactly Eq. (7.131). The spin current and the canonical tensor are the matched pair the theorem requires.

The superpotential. Insert Eq. (7.143) into the master formula Eq. (7.135). The three permutations are

$$\begin{aligned} S^{\rho\mu\nu} &= F^{\rho\nu}A^\mu - F^{\rho\mu}A^\nu,\\ S^{\mu\nu\rho} &= F^{\mu\rho}A^\nu - F^{\mu\nu}A^\rho,\\ S^{\nu\rho\mu} &= F^{\nu\mu}A^\rho - F^{\nu\rho}A^\mu. \end{aligned} \tag{7.145}$$

Forming $S^{\rho\mu\nu}+S^{\mu\nu\rho}-S^{\nu\rho\mu}$ and collecting by the free factor A:

$$\begin{aligned} A^\mu:&\quad F^{\rho\nu}+F^{\nu\rho}=0,\\ A^\nu:&\quad -F^{\rho\mu}+F^{\mu\rho}=-2F^{\rho\mu},\\ A^\rho:&\quad -F^{\mu\nu}-F^{\nu\mu}=0, \end{aligned} \tag{7.146}$$

using $F^{\nu\rho}=-F^{\rho\nu}$ throughout. Only the A^ν term survives, so

$$B^{\rho\mu\nu} = \tfrac{1}{2}\big(-2F^{\rho\mu}A^\nu\big) = -F^{\rho\mu}A^\nu = F^{\mu\rho}A^\nu. \tag{7.147}$$

This is the superpotential anticipated in Eq. (6.193); the construction has now *derived* it from the spin current rather than guessing it.

The improved tensor. Compute the divergence of Eq. (7.147):

$$\partial_\rho B^{\rho\mu\nu} = \partial_\rho(F^{\mu\rho}A^\nu) = (\partial_\rho F^{\mu\rho})A^\nu + F^{\mu\rho}\,\partial_\rho A^\nu. \tag{7.148}$$

Here we use the field equations: in the source-free theory the Maxwell equation is $\partial_\rho F^{\mu\rho}=0$, so the first term—the only piece containing the bare, non-gauge-invariant potential A^ν undifferentiated—vanishes on shell. This is the step at which gauge invariance is restored: the offending term is killed by the equations of motion. The improvement reduces to $\partial_\rho B^{\rho\mu\nu} = F^{\mu\rho}\partial_\rho A^\nu$, and

$$\begin{aligned} \Theta^{\mu\nu}_{\rm Max} &= T^{\mu\nu}_{\rm can} + F^{\mu\rho}\partial_\rho A^\nu = -F^{\mu\rho}\partial^\nu A_\rho + \tfrac{1}{4}\eta^{\mu\nu}F^2 + F^{\mu\rho}\partial_\rho A^\nu\\ &= -F^{\mu\rho}(\partial^\nu A_\rho - \partial_\rho A^\nu) + \tfrac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}\\ &= -F^{\mu\rho}\,F^\nu{}_\rho + \tfrac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}, \end{aligned} \tag{7.149}$$

where the second line factored out $F^{\mu\rho}$ and the third recognized $\partial^\nu A_\rho - \partial_\rho A^\nu = F^\nu{}_\rho$. The boxed result,

$$\boxed{\Theta^{\mu\nu}_{\text{Max}} = -F^{\mu\alpha}F^\nu{}_\alpha + \tfrac{1}{4}\eta^{\mu\nu}F_{\alpha\beta}F^{\alpha\beta}}, \tag{7.150}$$

is at once *symmetric* in $\mu\nu$ (the first term because $F^{\mu\alpha}F^\nu{}_\alpha = \eta_{\alpha\beta}F^{\mu\alpha}F^{\nu\beta}$ is, by the same relabeling used above, symmetric; the second manifestly) and *gauge invariant* (built entirely from the field strength F, with no bare A). Both diseases of Eq. (7.142) are cured.

Reading off the physics. It remains to confirm that Eq. (7.150) is the familiar electromagnetic energy and momentum. With the mostly-minus metric, the field strength encodes $F^{0i} = -E^i$ and $F^{ij} = -\epsilon^{ijk}B^k$, and the Lorentz scalar is $F_{\alpha\beta}F^{\alpha\beta} = 2(B^2 - E^2)$. For the energy density set $\mu = \nu = 0$:

$$\Theta^{00}_{\text{Max}} = -F^{0\alpha}F^0{}_\alpha + \tfrac{1}{4}F^2 = -F^{0i}F^0{}_i + \tfrac{1}{4}\cdot 2(B^2 - E^2). \tag{7.151}$$

Now $F^0{}_i = \eta_{ij}F^{0j} = -F^{0i} = E^i$, so $-F^{0i}F^0{}_i = -(-E^i)(E^i) = E^2$, and

$$\Theta^{00}_{\text{Max}} = E^2 + \tfrac{1}{2}(B^2 - E^2) = \tfrac{1}{2}(E^2 + B^2), \tag{7.152}$$

the standard electromagnetic energy density, manifestly non-negative. For the momentum density set $\mu = 0, \nu = i$; the metric term drops ($\eta^{0i} = 0$):

$$\Theta^{0i}_{\text{Max}} = -F^{0\alpha}F^i{}_\alpha = -F^{0j}F^i{}_j = -(-E^j)(-\epsilon^{ijk}B^k) = -\epsilon^{ijk}E^jB^k = (\mathbf{E}\times\mathbf{B})^i, \tag{7.153}$$

using $F^i{}_j = \eta_{jk}F^{ik} = -F^{ij} = \epsilon^{ijk}B^k$ and that the $\alpha = 0$ term vanishes since $F^i{}_0 = -F^{i0} = F^{0i} = -E^i$ pairs with $F^{00} = 0$. (The full contraction is $-F^{0\alpha}F^i{}_\alpha = -F^{00}F^i{}_0 - F^{0j}F^i{}_j$, and the first term is zero.) This is the Poynting vector, the electromagnetic momentum density and energy flux. The Belinfante tensor of electromagnetism is exactly the object one writes down in classical field theory on physical grounds—but here it has been *derived*, with its symmetry and gauge invariance guaranteed by the general theorem rather than imposed by hand.

Caution

The cancellation in Eq. (7.148) uses the *source-free* equation $\partial_\rho F^{\mu\rho} = 0$. In the presence of charged matter the Maxwell equation reads $\partial_\rho F^{\mu\rho} = -J^\mu$, and the term $(\partial_\rho F^{\mu\rho})A^\nu = -J^\mu A^\nu$ no longer vanishes; the electromagnetic field's stress tensor is then not separately conserved, only the total (matter + field) stress tensor is, with the difference accounting for the work done by the field on the charges. This is the classical statement of energy–momentum exchange, and it is the correct behavior: an isolated subsystem need not conserve its own energy. The minimal-coupling structure that

organizes this exchange is the subject of Chapter 8.

With a symmetric, conserved, gauge-invariant stress tensor in hand, two roads open. Its trace $\Theta^{\mu}{}_{\mu}$ is now a clean, gauge-invariant diagnostic of scale invariance—vanishing for the massless theory and signalling the breaking of dilatation symmetry otherwise—and the same improvement-by-superpotential technique, applied to the dilatation current rather than the spin current, will furnish the further *conformal* improvement that makes the traceless tensor manifest; that analysis is taken up when we carry ϕ^4 theory to its symmetry layer later in this chapter. And the symmetric $\Theta^{\mu\nu}$ is the object that couples to the metric, the flat-space seed of the gravitational source whose curved-spacetime construction is deferred to Chapter 64. Before either, the next section turns to a sharper question the improvement freedom has quietly raised—what happens when a symmetry's parameter is allowed to vary from point to point—and answers it with Noether's second theorem.

7.6 Noether's Second Theorem: Global versus Local Symmetry

Every symmetry we have exploited so far has been *rigid*: a single constant parameter rotates the field at every point of spacetime in lockstep. Noether's first theorem (Theorem 7.10) attached to each such parameter a conserved current and, on integration over a spatial slice, a time-independent charge. The internal charges closed into the symmetry's Lie algebra; the spacetime charges were energy, momentum, and angular momentum; and the Belinfante construction of the preceding section repaired the stress tensor that carries them. The machinery is by now routine, and it is tempting to apply it on autopilot to the largest symmetry a field theory can have—a symmetry whose parameter is an arbitrary function of position. That is exactly the stress test this section performs, and the outcome is not more of the same. When the parameter is allowed to vary from point to point, the Noether procedure does not manufacture a new tower of conserved charges. Instead it manufactures an *identity*: a relation among the equations of motion that holds whether or not those equations are satisfied. The associated current, far from carrying independent physical content, collapses on shell into a total derivative and so into a boundary integral that registers no bulk charge at all. This is Noether's *second* theorem, and reading it correctly is what separates a genuine global symmetry—which constrains dynamics and yields real conserved quantities—from a local "symmetry," which is not a symmetry of the physics but a redundancy in how we have chosen to describe it. That distinction is the entire conceptual content of the gauge principle, and pinning it down here is what prepares the explicit construction of electromagnetism from a local phase

in Chapter 8.

7.6.1 Local symmetries and the variational identity

We work with the conventions fixed for this chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, and a Lagrangian density $\mathcal{L}(\phi_a, \partial_\mu \phi_a)$ depending on a multiplet of fields ϕ_a and their first derivatives, with no explicit coordinate dependence. It is convenient to package the dynamics in the *Euler–Lagrange operator*, the same combination whose vanishing is the equation of motion proved in Theorem 6.8,

$$\mathcal{E}_a[\phi] \;\equiv\; \partial_\mu\!\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\right) - \frac{\partial \mathcal{L}}{\partial \phi_a}, \qquad \mathcal{E}_a[\phi] = 0 \iff \phi_a \text{ solves Eq. (6.30).} \tag{7.154}$$

A field configuration is said to be *on shell* when $\mathcal{E}_a[\phi] = 0$ and *off shell* otherwise; the whole force of what follows is the distinction between identities that require $\mathcal{E}_a = 0$ and identities that do not.

The basic bookkeeping of the variational calculus is the rewriting of an arbitrary variation of the Lagrangian as an Euler–Lagrange term plus a total derivative. For *any* infinitesimal field variation $\delta\phi_a$,

$$\delta\mathcal{L} \;=\; \frac{\partial \mathcal{L}}{\partial \phi_a}\,\delta\phi_a + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,\partial_\mu \delta\phi_a \;=\; -\,\mathcal{E}_a[\phi]\,\delta\phi_a + \partial_\mu \Theta^\mu, \qquad \Theta^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,\delta\phi_a, \tag{7.155}$$

where the second equality is obtained by adding and subtracting $\partial_\mu\!\left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\right)\delta\phi_a$ and recognizing Eq. (7.154). This is purely algebraic: it holds for every $\delta\phi_a$, on shell or off, and it is the single most important formula in Noether theory. Recall how the first theorem reads it: for a *rigid* symmetry, $\delta\phi_a = \delta\alpha\, R_a(\phi)$ with $\delta\alpha$ a constant and $\delta\mathcal{L} = \delta\alpha\, \partial_\mu K^\mu$ a total derivative, the master identity Eq. (7.155) becomes $\partial_\mu(\Theta^\mu - \delta\alpha\, K^\mu) = -\mathcal{E}_a\,\delta\phi_a$, so that the current $j^\mu = \Theta^\mu/\delta\alpha - K^\mu$ is conserved *on shell*.

We now relax the rigidity. The defining feature of a *local* (or *gauge*) symmetry is that the symmetry transformation is parameterized by a function $\varepsilon(x)$ that may be chosen freely at each spacetime point, and that the action is invariant for *every* such choice.

Definition 7.35 (Local symmetry)**.** A transformation of the fields of the form

$$\delta_\varepsilon \phi_a \;=\; \varepsilon(x)\, R_a(\phi) + \big(\partial_\mu \varepsilon(x)\big)\, R_a^\mu(\phi), \tag{7.156}$$

with R_a and R_a^μ local functions of the fields (and possibly their derivatives), is a *local symmetry* of the action $S = \int \mathrm{d}^4x\, \mathcal{L}$ if $\delta_\varepsilon S = 0$ for every smooth, compactly supported parameter $\varepsilon(x)$. Equivalently—since a compactly supported ε generates no boundary contribution—the Lagrangian changes at most by a total derivative, $\delta_\varepsilon \mathcal{L} = \partial_\mu K^\mu[\varepsilon]$, for arbitrary $\varepsilon(x)$.

Two remarks fix the scope. First, Eq. (7.156) is the general first-order local transformation linear in ε; allowing R_a^μ to depend on fields covers the gauged matter we care about, and the extension to higher derivatives of ε (needed only for theories such as gravity, treated later) is a mechanical iteration of the integration by parts below. Second, the constant-ε specialization of Eq. (7.156) is $\delta\phi_a = \varepsilon\, R_a$: this is an ordinary rigid transformation, the *global subgroup* sitting inside the local symmetry. The pivotal question of this section is precisely how much of the conservation content lives in this global subgroup and how much is genuinely new.

7.6.2 The second theorem and its proof

Theorem 7.36 (Noether's second theorem)**.** *Let the action $S = \int \mathrm{d}^4x\, \mathcal{L}$ possess the local symmetry Eq. (7.156) of Theorem 7.35. Then the Euler–Lagrange expressions Eq. (7.154) satisfy the off-shell differential identity*

$$\boxed{\mathcal{E}_a[\phi]\, R_a(\phi) \;-\; \partial_\mu\big(\mathcal{E}_a[\phi]\, R_a^\mu(\phi)\big) \;=\; 0} \tag{7.157}$$

holding identically in ϕ—for every field configuration, whether or not it solves the equations of motion. Equivalently, writing $W^\mu \equiv \mathcal{E}_a R_a^\mu$, the combination $\mathcal{E}_a R_a$ is the divergence of W^μ.

Proof. Invariance of the action under Eq. (7.156) for an arbitrary compactly supported $\varepsilon(x)$ reads $\delta_\varepsilon S = 0$. Evaluate the variation with the master identity Eq. (7.155), in which the total-derivative term $\int \mathrm{d}^4x\, \partial_\mu \Theta^\mu$ vanishes because Θ^μ is supported where ε and $\partial\varepsilon$ are, i.e. in a compact region with no boundary contribution. Hence

$$0 \;=\; \delta_\varepsilon S \;=\; -\int \mathrm{d}^4x\; \mathcal{E}_a[\phi]\, \delta_\varepsilon \phi_a \;=\; -\int \mathrm{d}^4x\; \mathcal{E}_a\big(\varepsilon\, R_a + \partial_\mu \varepsilon\, R_a^\mu\big). \tag{7.158}$$

The crucial step is to move the derivative off ε in the second term by integrating by parts. Because ε has compact support the boundary term again vanishes, and

$$\int \mathrm{d}^4x\; \mathcal{E}_a\, R_a^\mu\, \partial_\mu \varepsilon \;=\; -\int \mathrm{d}^4x\; \varepsilon\, \partial_\mu\big(\mathcal{E}_a\, R_a^\mu\big). \tag{7.159}$$

Substituting Eq. (7.159) into Eq. (7.158) collects the entire integrand against a single factor of ε:

$$0 \;=\; -\int \mathrm{d}^4x\; \varepsilon(x) \Big[\, \mathcal{E}_a\, R_a - \partial_\mu\big(\mathcal{E}_a\, R_a^\mu\big) \Big]. \tag{7.160}$$

This must hold for *every* smooth compactly supported $\varepsilon(x)$. The fundamental lemma of the calculus of variations then forces the bracket to vanish at every point: if a continuous function integrated against all test functions ε gives zero, the function is identically zero. This is exactly Eq. (7.157). No equation of motion was imposed anywhere in the derivation—$\mathcal{E}_a$ was never set to zero—so the identity is off shell. □

The contrast with the first theorem is sharp and worth stating in words. Noether's first theorem takes a rigid symmetry and produces a statement that is empty off shell and becomes the conservation law $\partial_\mu j^\mu = 0$ only *after* the equations of motion are imposed; its content is dynamical. Noether's second theorem takes a local symmetry and produces a statement that holds *before* any equation of motion is imposed; its content is kinematical—a built-in redundancy among the $\mathcal{E}_a$ themselves. The price of demanding invariance under an infinite-dimensional group of transformations, one parameter per spacetime point, is that the equations of motion are not all independent: Eq. (7.157) is a linear relation among them (and their derivatives) that the Lagrangian satisfies automatically. Such relations are the field-theoretic generalization of the Bianchi identities of electromagnetism and gravity, as the worked example below makes concrete.

Physical Insight

A rigid symmetry is a property of *solutions*: it maps one solution to another and thereby pairs each parameter with a quantity that is constant in time. A local symmetry is a property of the *theory itself*: it says that the map "initial data $\mapsto$ solution" is not injective, because field configurations related by a local transformation are physically identical. The first kind of symmetry adds information (a conservation law); the second kind subtracts it (a redundancy). It is a genuine misnomer, entrenched by history, to call both "symmetries"—and the second theorem is the precise theorem that tells them apart.

7.6.3 The naive current and its on-shell triviality

If a local symmetry yields no new conservation law, what becomes of the Noether current one would naively write down by feeding Eq. (7.156) into the first-theorem recipe? Tracking it explicitly is the most instructive way to see the redundancy at work. Because a local symmetry leaves the Lagrangian invariant up to a total derivative, $\delta_\varepsilon \mathcal{L} = \partial_\mu K^\mu[\varepsilon]$, the master identity Eq. (7.155) gives, for the local variation with general $\varepsilon(x)$,

$$\partial_\mu j^\mu \;=\; \mathcal{E}_a[\phi]\,\delta_\varepsilon\phi_a, \qquad j^\mu \;\equiv\; \Theta^\mu - K^\mu \;=\; \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi_a)}\,\delta_\varepsilon\phi_a - K^\mu[\varepsilon]. \tag{7.161}$$

This is the same current the first theorem would assign; on shell ($\mathcal{E}_a = 0$) it is conserved, $\partial_\mu j^\mu = 0$, exactly as advertised. The new feature is what the second theorem says about the right-hand side. Using Eq. (7.156) and the identity Eq. (7.157) in the form $\mathcal{E}_a R_a = \partial_\mu W^\mu$ with $W^\mu = \mathcal{E}_a R_a^\mu$,

$$\mathcal{E}_a\,\delta_\varepsilon\phi_a \;=\; \varepsilon\,\mathcal{E}_a R_a + \partial_\mu\varepsilon\,\mathcal{E}_a R_a^\mu \;=\; \varepsilon\,\partial_\mu W^\mu + (\partial_\mu\varepsilon)\,W^\mu \;=\; \partial_\mu(\varepsilon\,W^\mu). \tag{7.162}$$

Comparing with Eq. (7.161), the naive current obeys $\partial_\mu(j^\mu - \varepsilon\, W^\mu) = 0$ as an *off-shell identity*. We record this as a corollary, because it already exhibits the central pathology.

Corollary 7.37 (Trivial conservation)**.** *For a local symmetry the current j^μ of Eq. (7.161) satisfies, off shell,*

$$\partial_\mu(j^\mu - \varepsilon\, W^\mu) = 0, \qquad W^\mu = \mathcal{E}_a R_a^\mu. \tag{7.163}$$

The term $\varepsilon\, W^\mu$ is proportional to the equations of motion and therefore vanishes on shell. Thus the on-shell conservation $\partial_\mu j^\mu = 0$ carries no information beyond the equations of motion themselves: it is an algebraic consequence of $\mathcal{E}_a = 0$, not an independent constraint extracted from a symmetry.

The combination $V^\mu \equiv j^\mu - \varepsilon W^\mu$ is conserved *identically*—for all field configurations and all $\varepsilon(x)$, with no equation of motion imposed. An identically conserved current is a much weaker object than a dynamically conserved one: in fact it is always a total curl, and hence integrates to a pure boundary term. We isolate this elementary but decisive fact as a lemma.

Lemma 7.38 (An identically conserved current is a superpotential divergence)**.** *Let $V^\mu(x)$ be a smooth vector field on $\mathbb{R}^4$ (more generally on any region star-shaped about the origin) satisfying $\partial_\mu V^\mu = 0$ everywhere. Then there exists an antisymmetric tensor $k^{\mu\nu} = -k^{\nu\mu}$ with $V^\mu = \partial_\nu k^{\mu\nu}$. An explicit choice is*

$$k^{\mu\nu}(x) \;=\; \int_0^1 \mathrm{d}t\; t^{\,2}\,\big[\, x^\nu V^\mu(tx) - x^\mu V^\nu(tx)\,\big]. \tag{7.164}$$

Proof. Antisymmetry of Eq. (7.164) is manifest. Differentiate, using $\partial_\nu[\, f(tx)\,] = t\,(\partial_\nu f)(tx)$ and the contractions $\partial_\nu x^\nu = 4$, $\partial_\nu x^\mu = \delta^\mu_\nu$:

$$\begin{aligned}\partial_\nu k^{\mu\nu} &= \int_0^1 \mathrm{d}t\; t^2\Big[\partial_\nu(x^\nu V^\mu(tx)) - \partial_\nu(x^\mu V^\nu(tx))\Big] \\ &= \int_0^1 \mathrm{d}t\; t^2\Big[\,4\,V^\mu(tx) + t\,x^\nu(\partial_\nu V^\mu)(tx) - V^\mu(tx) - t\,x^\mu(\partial_\nu V^\nu)(tx)\,\Big].\end{aligned} \tag{7.165}$$

The last term vanishes by hypothesis, $\partial_\nu V^\nu = 0$. In the second term recognize the total t-derivative along the ray, $\frac{\mathrm{d}}{\mathrm{d}t}V^\mu(tx) = x^\nu(\partial_\nu V^\mu)(tx)$, so that $t\,x^\nu(\partial_\nu V^\mu)(tx) = t\,\frac{\mathrm{d}}{\mathrm{d}t}V^\mu(tx)$. Hence

$$\partial_\nu k^{\mu\nu} = \int_0^1 \mathrm{d}t\,\Big[\,3\,t^2\,V^\mu(tx) + t^3\,\tfrac{\mathrm{d}}{\mathrm{d}t}V^\mu(tx)\Big] = \int_0^1 \mathrm{d}t\;\tfrac{\mathrm{d}}{\mathrm{d}t}\big[\,t^3\,V^\mu(tx)\,\big] = V^\mu(x), \tag{7.166}$$

where the bracket assembles into an exact t-derivative and the endpoint $t = 1$ gives $V^\mu(x)$ while $t = 0$ contributes nothing. This is the claim. □

Applying the lemma to $V^\mu = j^\mu - \varepsilon W^\mu$ from Theorem 7.37 and then setting ε to the constant parameter of the global subgroup yields the structural statement that decides the physics.

Proposition 7.39 (On-shell triviality of the gauge current)**.** *Let j^μ_{glob} denote the Noether current Eq. (7.161) of the global subgroup, obtained by taking $\varepsilon \equiv 1$ (constant), so that $\delta\phi_a = R_a$. Then there is an antisymmetric superpotential $k^{\mu\nu} = -k^{\nu\mu}$ such that,* on shell,

$$j^\mu_{\text{glob}} \overset{\mathcal{E}=0}{=} \partial_\nu k^{\mu\nu}. \tag{7.167}$$

Consequently the associated charge reduces to a surface integral over the boundary of the spatial slice,

$$Q = \int_\Sigma \mathrm{d}^3x\, j^0_{\text{glob}} = \int_\Sigma \mathrm{d}^3x\, \partial_i k^{0i} = \oint_{\partial\Sigma} \mathrm{d}S_i\, k^{0i}, \tag{7.168}$$

and is fixed entirely by the fields at spatial infinity. The current is thus improvable to zero*: subtracting the identically conserved superpotential term $\partial_\nu k^{\mu\nu}$ sets it to zero on shell, removing it as a source of independent charge.*

Proof. By Theorem 7.37 with $\varepsilon \equiv 1$, the current $V^\mu = j^\mu_{\text{glob}} - W^\mu$ is identically conserved, $\partial_\mu V^\mu = 0$ off shell. Theorem 7.38 then supplies an antisymmetric $k^{\mu\nu}$ with $V^\mu = \partial_\nu k^{\mu\nu}$, i.e. $j^\mu_{\text{glob}} = \partial_\nu k^{\mu\nu} + W^\mu$ identically. On shell $W^\mu = \mathcal{E}_a R^\mu_a = 0$, leaving Eq. (7.167). Integrating its time component over a spatial slice Σ and using $\partial_\nu k^{0\nu} = \partial_i k^{0i}$ (the $\nu = 0$ piece vanishes by antisymmetry, $k^{00} = 0$) gives Eq. (7.168) by the divergence theorem. □

This is the precise sense in which a local symmetry yields "no new charge." A genuine global symmetry produces a charge $Q = \int_\Sigma \mathrm{d}^3x\, j^0$ that is a bulk integral of a density—a count of particles, a net flavor, an energy—whose conservation in time is a nontrivial dynamical statement. The would-be charge of a local symmetry is, by Theorem 7.39, a flux through the boundary: it is determined by boundary data and does not register any independent property of the interior. In the absence of charged matter at infinity it is simply zero. The infinite family of local transformations, one per spacetime point, collapses on shell to no physical conserved quantity whatsoever; only the constant mode—the global subgroup—can survive, and it survives only when the matter content makes its action nontrivial. We turn that abstract chain into explicit formulas.

7.6.4 The gauged example: electromagnetism

Worked Example 7.40 (Maxwell theory: the gauge current is a curl)**.** Take the free electromagnetic Lagrangian of the recurring cast,

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu}F^{\mu\nu}, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{7.169}$$

with the gauge field A_ρ playing the role of the field multiplet ϕ_a (the field index a is the Lorentz index ρ). The local symmetry is the gauge transformation $\delta_\varepsilon A_\rho = \partial_\rho \varepsilon$, which in the notation of Eq. (7.156) means

$$R_\rho = 0, \qquad R_\rho^\mu = \delta_\rho^\mu, \tag{7.170}$$

since $\delta_\varepsilon A_\rho = (\partial_\mu \varepsilon)\, \delta_\rho^\mu$. It is a symmetry because $\delta_\varepsilon F_{\mu\nu} = \partial_\mu \partial_\nu \varepsilon - \partial_\nu \partial_\mu \varepsilon = 0$, so $\delta_\varepsilon \mathcal{L} = 0$ exactly and $K^\mu = 0$.

Euler–Lagrange operator. From Eq. (7.169), $\partial \mathcal{L}/\partial(\partial_\mu A_\rho) = -F^{\mu\rho}$ and $\partial \mathcal{L}/\partial A_\rho = 0$, so Eq. (7.154) gives

$$\mathcal{E}^\rho = \partial_\mu\big(-F^{\mu\rho}\big) = -\partial_\mu F^{\mu\rho}, \qquad \mathcal{E}^\rho = 0 \iff \partial_\mu F^{\mu\rho} = 0, \tag{7.171}$$

the sourceless Maxwell equations.

Second theorem as a Bianchi identity. With Eq. (7.170), the identity Eq. (7.157) reads $\mathcal{E}^\rho R_\rho - \partial_\mu(\mathcal{E}^\rho R_\rho^\mu) = 0 - \partial_\mu(\mathcal{E}^\rho \delta_\rho^\mu) = -\partial_\mu \mathcal{E}^\mu$, so the second theorem asserts

$$\partial_\mu \mathcal{E}^\mu = \partial_\mu \partial_\nu F^{\nu\mu} \equiv 0, \tag{7.172}$$

which is true identically because $\partial_\mu \partial_\nu$ is symmetric while $F^{\nu\mu}$ is antisymmetric. This is the off-shell redundancy in flesh: the four Maxwell equations $\mathcal{E}^\rho = 0$ are not independent; their divergence vanishes automatically, so they encode only three independent dynamical equations together with a constraint. Here $W^\mu = \mathcal{E}^\rho R_\rho^\mu = \mathcal{E}^\mu = -\partial_\nu F^{\nu\mu}$.

The naive current. With $K^\mu = 0$ and $\delta_\varepsilon A_\rho = \partial_\rho \varepsilon$, Eq. (7.161) gives

$$j^\mu = \frac{\partial \mathcal{L}}{\partial(\partial_\mu A_\rho)}\, \delta_\varepsilon A_\rho = -F^{\mu\rho}\, \partial_\rho \varepsilon. \tag{7.173}$$

Its divergence is $\partial_\mu j^\mu = -(\partial_\mu F^{\mu\rho})\partial_\rho \varepsilon - F^{\mu\rho}\partial_\mu \partial_\rho \varepsilon$; the second term vanishes (antisymmetric $F^{\mu\rho}$ against symmetric $\partial_\mu \partial_\rho$), leaving $\partial_\mu j^\mu = -(\partial_\mu F^{\mu\rho})\partial_\rho \varepsilon = (\partial_\rho \varepsilon)\, W^\rho$, in exact agreement with Eq. (7.162). On shell it is conserved, but only because $\partial_\mu F^{\mu\rho} = 0$ already—no new content.

The current is a total derivative. Rewrite Eq. (7.173) by moving the derivative onto ε:

$$j^\mu = -F^{\mu\rho}\partial_\rho \varepsilon = -\partial_\rho\big(\varepsilon\, F^{\mu\rho}\big) + \varepsilon\, \partial_\rho F^{\mu\rho} \overset{\mathcal{E}=0}{=} \partial_\rho\big(\varepsilon\, F^{\rho\mu}\big) \equiv \partial_\rho k^{\rho\mu}, \tag{7.174}$$

where on shell $\partial_\rho F^{\mu\rho} = 0$ and the superpotential

$$k^{\rho\mu} = \varepsilon\, F^{\rho\mu} = -k^{\mu\rho} \tag{7.175}$$

is antisymmetric—precisely the structure guaranteed in general by Theorem 7.39, now exhibited as an explicit local expression rather than the homotopy integral of Theorem 7.38.

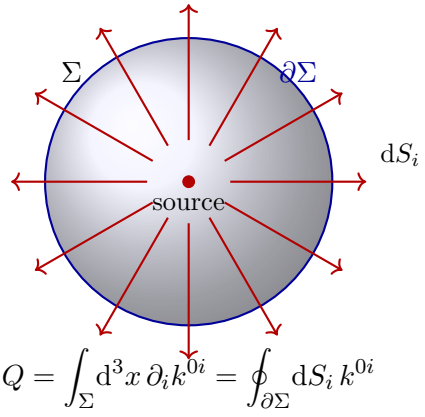


Figure 7.5: The would-be charge of a local symmetry is a boundary flux. Because the on-shell current is a spatial divergence, $j^0 = \partial_i k^{0i}$ with k^{0i} antisymmetric, no charge is stored in the bulk of the slice Σ; the charge Eq. (7.168) is the flux of k^{0i} through $\partial\Sigma$. In Maxwell theory $k^{0i} = \varepsilon\, F^{0i}$ and this is Gauss's law: the electric flux through the boundary equals the enclosed charge.

The charge is a flux. The charge is therefore a surface integral. With $F^{i0} = E^i$ the components of the electric field,

$$Q = \int_\Sigma \mathrm{d}^3x\, j^0 = \int_\Sigma \mathrm{d}^3x\, \partial_i(\varepsilon\, F^{i0}) = \oint_{\partial\Sigma} \mathrm{d}S_i\, \varepsilon\, F^{i0} = \oint_{\partial\Sigma} \varepsilon\, \mathbf{E}\cdot\mathrm{d}\mathbf{S}. \tag{7.176}$$

For the constant mode $\varepsilon = 1$ this is $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{S}$—the flux of the electric field through the boundary, which by Gauss's law equals the total electric charge enclosed. The "Noether charge" of the gauge symmetry is not a new conserved quantity at all; it is the electric flux, fixed entirely by the boundary, and it is the physical charge only because Gauss's law ties the boundary flux to the enclosed source. Pure Maxwell theory has no charged field, so $\delta_\varepsilon A_\rho = \partial_\rho \varepsilon$ acts trivially on a constant ε and the bulk charge is simply zero.

Figure 7.5 renders the geometry of Eq. (7.176): a would-be conserved density that is a total spatial divergence stores nothing in the bulk, so the charge collects entirely on the bounding surface.

7.6.5 Where the genuine charge lives: global versus local in scalar QED

The Maxwell example is the clean skeleton, but it is almost too clean: with no charged matter, even the global subgroup is trivial. To see the global subgroup carry a real charge while its local extension adds nothing, we couple a charged scalar—the system in which the genuine $U(1)$ charge and its gauge redundancy sit side by side.

Worked Example 7.41 (Scalar electrodynamics: a real global charge inside a gauge redundancy)**.** Consider a complex scalar of charge q minimally coupled to the photon, with the chapter's gauge-coupling convention $D_\mu = \partial_\mu - iqA_\mu$,

$$\mathcal{L} = (D_\mu\phi)^*(D^\mu\phi) - m^2\phi^*\phi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu\phi = \partial_\mu\phi - iqA_\mu\phi. \tag{7.177}$$

The local symmetry is $\phi \to e^{iq\alpha(x)}\phi$ together with $A_\mu \to A_\mu + \partial_\mu\alpha$, whose infinitesimal form is

$$\delta_\alpha\phi = iq\,\alpha\,\phi, \qquad \delta_\alpha\phi^* = -iq\,\alpha\,\phi^*, \qquad \delta_\alpha A_\mu = \partial_\mu\alpha. \tag{7.178}$$

One checks $\delta_\alpha(D_\mu\phi) = iq\,\alpha\,D_\mu\phi$, so each term in Eq. (7.177) is invariant and the action has the local symmetry for arbitrary $\alpha(x)$.

The global subgroup carries a bulk charge. Restrict to constant α. Then $\delta A_\mu = 0$ and only the phase rotation of ϕ survives—this is the rigid internal $U(1)$ of Theorem 7.10. With $\partial\mathcal{L}/\partial(\partial_\mu\phi) = (D^\mu\phi)^*$ and $\partial\mathcal{L}/\partial(\partial_\mu\phi^*) = D^\mu\phi$, the first-theorem current ($K^\mu = 0$, α scaled out) is

$$j^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi)}\frac{\delta\phi}{\alpha} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\phi^*)}\frac{\delta\phi^*}{\alpha} = iq\big[(D^\mu\phi)^*\phi - (D^\mu\phi)\phi^*\big] = iq\big[\phi\,(D^\mu\phi)^* - \phi^*\,D^\mu\phi\big]. \tag{7.179}$$

This is a genuine bulk current: its density j^0 is built from the matter fields and their momenta, and $Q = \int_\Sigma \mathrm{d}^3x\, j^0$ counts the net charge of the scalar quanta—a real, nonzero, dynamically conserved number, not a boundary artifact. The global $U(1)$ is an honest symmetry and yields an honest charge.

The local extension adds only a constraint. Now allow $\alpha(x)$ to vary. The full set of Euler–Lagrange equations includes the photon equation, obtained by varying Eq. (7.177) in A_ν,

$$\partial_\mu F^{\mu\nu} = J^\nu, \qquad J^\nu = iq\big[\phi\,(D^\nu\phi)^* - \phi^*\,D^\nu\phi\big], \tag{7.180}$$

i.e. the matter current Eq. (7.179) is exactly the source of the photon. Noether's second theorem for the gauge parameter A_ν-sector reproduces the Bianchi identity Eq. (7.172); taking the divergence of Eq. (7.180) then gives $\partial_\nu J^\nu = \partial_\nu\partial_\mu F^{\mu\nu} \equiv 0$. Conservation of the matter current is thereby *enforced as an identity* by the gauge structure, rather than supplied as new information. The local symmetry's own would-be charge is, by Theorem 7.39 and Eq. (7.176), again the flux $\oint \mathbf{E}\cdot\mathrm{d}\mathbf{S}$; by Eq. (7.180) this flux equals $\int_\Sigma J^0 = Q$. The local symmetry contributes no charge of its own—it only ties the already-existing global charge Q to the electric flux through the boundary (Gauss's law) and elevates current conservation from a dynamical theorem to a kinematic identity.

The lesson generalizes verbatim to the non-abelian and gravitational cases treated later: the genuine conserved charges of a theory come from its *global* symmetry group, while the local symmetry contributes Bianchi-type identities, a Gauss-law constraint, and a charge that is always a boundary flux. The reader should hold one caution.

Caution

"The gauge current carries no charge" does not mean gauge symmetry is physically vacuous. On the contrary, demanding it is enormously restrictive: it fixes the matter–photon coupling to be minimal, forces the photon to be massless, and renders the equations of motion mutually consistent through the Bianchi identity. What the second theorem denies is only that a *local* parameter yields an *independent conserved charge*. The physical content of a gauge symmetry is the redundancy it certifies—and, downstream, the constraints it imposes on which Lagrangians and which states are admissible—not a Noether charge.

Physical Insight

The cleanest one-line summary is operational. Count the independent conserved charges by counting the *constant* modes of your symmetry parameters: a rigid symmetry has one parameter and gives one charge; a local symmetry has one parameter *per point*, but all the non-constant modes are redundancy, and the single constant mode gives at most the one charge of the global subgroup—realized, when matter is present, as a boundary flux. Symmetry is the organizing principle of field theory, but only *global* symmetry is the bookkeeping of conserved quantities; local "symmetry" is the bookkeeping of description.

With the global–local distinction now sharp—a global symmetry supplies a conserved charge, a local symmetry supplies an off-shell identity and a redundancy—we have isolated exactly the structure that the gauge principle will turn into a construction. Chapter 8 runs the logic forward: starting from the genuine global phase symmetry of a charged field Eq. (7.179) and *demanding* that it be promoted to a local one, the connection A_μ, minimal coupling $D_\mu = \partial_\mu - iqA_\mu$, and the field strength $F_{\mu\nu}$ as its curvature are forced upon us, deriving electromagnetism from a redundancy. The same redundancy viewpoint, now read as a constraint that must be gauge-fixed before quantization, is what organizes the Faddeev–Popov and BRST treatment of Yang–Mills theory in Chapter 25 and Chapter 26.

7.7 Scale, Conformal Symmetry, and the Trace of the Stress Tensor

The Belinfante tensor $\Theta^{\mu\nu}$ assembled in the previous section is the repaired, symmetric, gauge-invariant stress–energy tensor: conserved on every index, $\partial_\mu \Theta^{\mu\nu} = 0$, and symmetric, $\Theta^{\mu\nu} = \Theta^{\nu\mu}$, with the asymmetry of the canonical tensor absorbed by a superpotential improvement $\Theta^{\mu\nu} = T^{\mu\nu} + \partial_\rho B^{\rho\mu\nu}$, $B^{\rho\mu\nu} =$

$-B^{\mu\rho\nu}$, that leaves the charges $P^\nu = \int d^3x\, \Theta^{0\nu}$ untouched (Theorem 7.33). Two of the three numbers that classify a Poincaré current have now been wrung from this single object: its conservation gives energy–momentum, its symmetry gives angular momentum. There is a third number it carries, one we have so far only glanced at through the non-zero trace of the scalar's canonical tensor in Eq. (6.182): the trace $\Theta^\mu{}_\mu$. This section shows that the trace is the exact local measure of a further would-be symmetry — invariance under rescaling of all lengths — and that for a special class of theories the same superpotential technology that symmetrized $\Theta^{\mu\nu}$ can also make it *traceless*, at which point the theory is invariant not merely under dilatations but under the full conformal group. We derive the dilatation current and the identity $\partial_\mu D^\mu = \Theta^\mu{}_\mu$, build the improved traceless scalar tensor explicitly, exhibit the special-conformal currents, and close the classical layer of ϕ^4 theory: massless ϕ^4 in four dimensions is classically scale and conformally invariant, the mass term is the unique relevant operator that breaks it, and the quartic coupling sits exactly at the marginal boundary. The quantum fate of this symmetry — its generic destruction by renormalization — is flagged and deferred.

7.7.1 Dilatations and the scaling dimension of a field

A *dilatation* (or scale transformation) rescales all spacetime coordinates by a common positive factor,

$$x^\mu \;\longrightarrow\; x'^\mu = \lambda\, x^\mu, \qquad \lambda = e^\omega > 0, \tag{7.181}$$

and acts on the dynamical field according to how the field is built from lengths. The defining postulate is that a field ϕ carries a definite *scaling dimension* Δ, meaning it transforms as

$$\phi(x) \;\longrightarrow\; \phi'(x') = \lambda^{-\Delta}\, \phi(x), \tag{7.182}$$

so that the rescaled field, read at the rescaled point, is the old field diminished by $\lambda^{-\Delta}$. The exponent Δ is not free: it is fixed by demanding that the kinetic term be left invariant, as we determine below. Writing $\lambda = e^\omega$ and expanding to first order in the infinitesimal parameter ω, the coordinate shift and the *total* field variation (the change including the displacement of the argument) are

$$\delta x^\mu = \omega\, x^\mu, \qquad \delta\phi \;\equiv\; \phi'(x') - \phi(x) = -\omega\, \Delta\, \phi. \tag{7.183}$$

The *intrinsic* (form) variation $\delta_0\phi = \delta\phi - \delta x^\nu \partial_\nu \phi$, the change at a fixed coordinate, is then

$$\delta_0\phi = -\omega\, (\Delta + x^\nu \partial_\nu)\, \phi, \tag{7.184}$$

which displays the two ways a dilatation moves a field: it dilutes its value (the Δ piece) and it advects it along the radial flow $x^\nu\partial_\nu$ (the orbital piece).

Definition 7.42 (Scaling dimension)**.** The *scaling dimension* (or classical, "engineering," mass dimension) Δ of a field ϕ is the exponent in Eq. (7.182). In natural units it coincides with the mass dimension $[\phi]$ assigned by requiring the action $S = \int d^d x\, \mathcal{L}$ to be dimensionless: since $[d^d x] = -d$ and a kinetic Lagrangian must have $[\mathcal{L}] = d$, a canonically normalized scalar with kinetic term $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$ has

$$2\Delta + 2 = d \quad \Longrightarrow \quad \Delta = \frac{d-2}{2}, \tag{7.185}$$

which equals 1 in $d = 4$. A spin-$\frac{1}{2}$ field has $\Delta = (d-1)/2$ and a gauge potential $\Delta = (d-2)/2$ as well; only the scalar is needed here.

Equation Eq. (7.185) is the first concrete payoff: it *fixes* the scaling dimension of the scalar field. The number $\Delta = 1$ (in $d = 4$) is not a choice; it is forced by the demand that the kinetic term, the one piece of the Lagrangian present in every scalar theory, be invariant under Eq. (7.181)–Eq. (7.182). We will see in a moment that every other term in the Lagrangian then either respects this scaling or breaks it, and that the breaking is governed entirely by the mass dimensions of the couplings.

7.7.2 The dilatation current and the trace of the stress tensor

We now feed the dilatation Eq. (7.183) into Noether's first theorem. The master current of Eq. (7.32), written in its three-piece form Eq. (7.39), is $J^\mu = \Pi^\mu_a\, \delta\phi_a - T^\mu{}_\nu\, \delta x^\nu - \Lambda^\mu$, with $\Pi^\mu \equiv \partial\mathcal{L}/\partial(\partial_\mu\phi)$ and $T^\mu{}_\nu$ the canonical stress tensor of Theorem 6.28. For the scalar there is no internal index and, crucially, the canonical tensor already coincides with the Belinfante tensor, $T^{\mu\nu} = \Theta^{\mu\nu}$, because spin zero has no spin current to absorb (Eq. (6.178)). Taking $\Lambda^\mu = 0$ for now and inserting $\delta\phi = -\omega\Delta\phi$, $\delta x^\nu = \omega x^\nu$, $\Pi^\mu = \partial^\mu\phi$, the Noether current per unit parameter ω is

$$J^\mu = -\big[\, \Delta\, \phi\, \partial^\mu\phi + x_\nu\, \Theta^{\mu\nu} \,\big]. \tag{7.186}$$

A conserved current is fixed only up to an overall sign and an additive identically-conserved term; we fix the sign by convention and *define* the dilatation current as

$$\boxed{\; D^\mu \;=\; x_\nu\, \Theta^{\mu\nu} \;+\; V^\mu, \qquad V^\mu \;\equiv\; \Delta\, \phi\, \partial^\mu\phi \;} \tag{7.187}$$

the sum of an *orbital* piece $x_\nu\Theta^{\mu\nu}$, built from the stress tensor exactly as the orbital angular-momentum current was, and a *field virial* V^μ, which records the intrinsic scaling weight Δ of the field and has no analogue in the translation or rotation currents. The virial is the genuinely new ingredient of scale symmetry.

Its divergence is immediate. Using $\partial_\mu x_\nu = \eta_{\mu\nu}$ and the conservation of the Belinfante tensor,

$$\partial_\mu(x_\nu\Theta^{\mu\nu}) = (\partial_\mu x_\nu)\Theta^{\mu\nu} + x_\nu\, \partial_\mu\Theta^{\mu\nu} = \eta_{\mu\nu}\Theta^{\mu\nu} + 0 = \Theta^\mu{}_\mu, \tag{7.188}$$

so that

$$\boxed{\partial_\mu D^\mu \;=\; \Theta^\mu{}_\mu \;+\; \partial_\mu V^\mu\,.} \tag{7.189}$$

This is the central kinematic identity of scale symmetry, and it already tells the whole story in outline: the orbital part of the dilatation current is conserved *up to the trace of the stress tensor*. Whether the full D^μ is conserved hinges on whether the trace term and the virial divergence cancel. The next two results show that they do — for a special class of theories, and only after an improvement that absorbs the virial into the stress tensor itself.

Theorem 7.43 (Scale current and the trace)**.** *Let a Poincaré-invariant theory possess a symmetric, conserved stress tensor $\Theta^{\mu\nu}$. If its trace can be written as a total divergence of a field virial, $\Theta^\mu{}_\mu = -\partial_\mu V^\mu$ on shell, then the dilatation current Eq. (7.187) is conserved, $\partial_\mu D^\mu = 0$, and the theory is scale invariant. Conversely, scale invariance requires the trace of the stress tensor to be a total derivative on shell.*

Proof. The forward statement is Eq. (7.189) read in reverse: if $\Theta^\mu{}_\mu + \partial_\mu V^\mu = 0$ then $\partial_\mu D^\mu = 0$. For the converse, suppose the action is invariant under Eq. (7.181)–Eq. (7.182). Noether's theorem then produces a conserved D^μ of the form Eq. (7.187) (with $V^\mu = \Delta\,\Pi^\mu_a \phi_a$ in the general case), and $\partial_\mu D^\mu = 0$ combined with Eq. (7.189) forces $\Theta^\mu{}_\mu = -\partial_\mu V^\mu$. □

The cleanest situation arises when the field virial is itself the divergence of a local tensor, $V^\mu = \partial_\nu \sigma^{\nu\mu}$. For the scalar this holds identically:

$$V^\mu = \Delta\,\phi\,\partial^\mu\phi = \frac{\Delta}{2}\,\partial^\mu(\phi^2) = \partial_\nu\Big(\tfrac{\Delta}{2}\,\eta^{\nu\mu}\phi^2\Big), \qquad \sigma^{\nu\mu} = \tfrac{\Delta}{2}\,\eta^{\nu\mu}\phi^2. \tag{7.190}$$

When the virial is a total derivative the theory is not merely scale invariant but, as we show in Section 7.7.4, fully *conformally* invariant, and — the point of the next subsection — the stress tensor can be improved to a manifestly traceless object that makes both facts transparent. Before that, a worked computation pins down what happens for the massless free scalar without any improvement, to expose precisely how the virial rescues conservation.

Worked Example 7.44 (The virial cancellation for the massless free scalar)**.** Take the massless free scalar in d dimensions, $\mathcal{L}_0 = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$, with equation of motion $\Box\phi = 0$. Its (canonical = Belinfante) stress tensor is $\Theta^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\frac{1}{2}(\partial_\alpha\phi)(\partial^\alpha\phi)$, whose trace is

$$\Theta^\mu{}_\mu = (\partial_\alpha\phi)(\partial^\alpha\phi) - \frac{d}{2}(\partial_\alpha\phi)(\partial^\alpha\phi) = \frac{2-d}{2}\,(\partial_\alpha\phi)(\partial^\alpha\phi), \tag{7.191}$$

which is *not zero* in $d = 4$: the bare stress tensor of the massless scalar has trace $-(\partial\phi)^2$. Were D^μ equal to its orbital part alone, scale symmetry would

already be broken. The virial $V^\mu = \Delta\,\phi\,\partial^\mu\phi$ saves it. With $\Delta = (d-2)/2$ from Eq. (7.185),

$$\partial_\mu V^\mu = \Delta\big[(\partial_\mu\phi)(\partial^\mu\phi) + \phi\,\Box\phi\big] \xrightarrow{\Box\phi=0} \frac{d-2}{2}\,(\partial_\alpha\phi)(\partial^\alpha\phi), \tag{7.192}$$

the term $\phi\,\Box\phi$ vanishing on shell. Adding to Eq. (7.191),

$$\partial_\mu D^\mu = \Theta^\mu{}_\mu + \partial_\mu V^\mu = \frac{2-d}{2}(\partial\phi)^2 + \frac{d-2}{2}(\partial\phi)^2 = 0. \tag{7.193}$$

The cancellation is exact and dimension-independent: the orbital trace $\frac{2-d}{2}(\partial\phi)^2$ and the virial divergence $\frac{d-2}{2}(\partial\phi)^2$ are equal and opposite *for the canonical value* $\Delta = (d-2)/2$ of the scaling dimension. This is the precise sense in which Eq. (7.185) is forced: had we assigned the scalar any other scaling dimension, the dilatation current would fail to be conserved. Scale invariance of the massless scalar and the value $\Delta = (d-2)/2$ are one and the same statement.

7.7.3 The improved, traceless stress tensor

The cancellation just witnessed is correct but unsatisfying: it splits the dilatation current into two pieces whose conservation is an accident of arithmetic, and it leaves the stress tensor with a non-zero trace even when the theory is scale invariant. The remedy is to push the virial *into* the stress tensor. Because V^μ is a total derivative Eq. (7.190), it can be absorbed by a superpotential improvement of exactly the kind that symmetrized the Belinfante tensor, but now chosen to cancel the trace rather than the antisymmetric part. The result is the celebrated improved scalar stress tensor.

Definition 7.45 (Conformally improved scalar stress tensor)**.** For a real scalar in d dimensions, the *conformally improved* (or "new improved") stress tensor is the Belinfante tensor corrected by a symmetric, identically-conserved improvement term,

$$\boxed{\Theta^{\mu\nu}_{\rm c} \;=\; \Theta^{\mu\nu} \;+\; \xi_d\,(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)\,\phi^2, \qquad \xi_d = \frac{d-2}{4(d-1)}\,,} \tag{7.194}$$

with $\xi_4 = \frac{1}{6}$ in four dimensions.

Three properties make $\Theta^{\mu\nu}_{\rm c}$ a legitimate replacement for $\Theta^{\mu\nu}$, and the algebra below verifies each in turn.

(i) It is symmetric. The improvement term involves $\eta^{\mu\nu}$ and the symmetric $\partial^\mu\partial^\nu$, so $\Theta^{\mu\nu}_{\rm c} = \Theta^{\nu\mu}_{\rm c}$ inherits the symmetry of $\Theta^{\mu\nu}$.

(ii) It is identically conserved. The improvement adds nothing to the divergence, off shell:

$$\partial_\mu\Big[\xi_d(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)\phi^2\Big] = \xi_d(\partial^\nu\Box - \Box\partial^\nu)\phi^2 = 0, \tag{7.195}$$

since $\partial_\mu \eta^{\mu\nu}\Box = \partial^\nu \Box$ and $\partial_\mu \partial^\mu \partial^\nu = \Box \partial^\nu$, and partial derivatives commute. Thus $\partial_\mu \Theta^{\mu\nu}_{\mathrm{c}} = \partial_\mu \Theta^{\mu\nu} = 0$. The improvement term has the superpotential form $\partial_\rho B^{\rho\mu\nu}$ with $B^{\rho\mu\nu} = \xi_d(\eta^{\mu\nu}\partial^\rho - \eta^{\rho\nu}\partial^\mu)\phi^2$ antisymmetric in its first two indices, which is the structural reason Eq. (7.195) holds.

(iii) It carries the same energy and momentum. The charges are unchanged because the added density is a spatial total derivative on each time slice. For $\nu = 0$, $\xi_d(\eta^{00}\Box - \partial^0\partial^0)\phi^2 = \xi_d(\Box - \partial_0^2)\phi^2 = \xi_d \nabla^2 \phi^2$, and for $\nu = i$, $\xi_d(\eta^{0i}\Box - \partial^0\partial^i)\phi^2 = -\xi_d\, \partial^0\partial^i\phi^2 = \partial_j(\cdots)$; in both cases $\int d^3x$ of the addition vanishes by Gauss's theorem for fields decaying at spatial infinity. Hence $P^\nu = \int d^3x\, \Theta^{0\nu}_{\mathrm{c}} = \int d^3x\, \Theta^{0\nu}$.

The point of the particular coefficient ξ_d is that it makes the trace as small as the dimensionful couplings allow. The trace of the improvement term is

$$\eta_{\mu\nu}\, \xi_d(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)\phi^2 = \xi_d\,(d-1)\,\Box\phi^2, \tag{7.196}$$

using $\eta_{\mu\nu}\eta^{\mu\nu} = d$ and $\eta_{\mu\nu}\partial^\mu\partial^\nu = \Box$. The free parameter ξ_d is fixed by demanding that the gradient term in the total trace cancel, which we now carry out for the full interacting scalar of ϕ^4 theory.

Worked Example 7.46 (The classical trace identity of $\lambda\phi^4$)**.** We compute $\Theta^{\mu}_{\mathrm{c}\,\mu}$ for the frozen ϕ^4 Lagrangian Eq. (6.56), $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$, in general dimension d, then specialize to $d = 4$.

Trace of the Belinfante tensor. With $\Theta^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \eta^{\mu\nu}\mathcal{L}$,

$$\begin{aligned}
\Theta^{\mu}{}_{\mu} &= (\partial_\alpha\phi)(\partial^\alpha\phi) - d\,\mathcal{L} \\
&= (\partial\phi)^2 - d\Big[\tfrac{1}{2}(\partial\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \tfrac{\lambda}{4!}\phi^4\Big] \\
&= \Big(1 - \tfrac{d}{2}\Big)(\partial\phi)^2 + \tfrac{d}{2}\,m^2\phi^2 + \tfrac{d}{4!}\,\lambda\phi^4,
\end{aligned} \tag{7.197}$$

where $(\partial\phi)^2 \equiv (\partial_\alpha\phi)(\partial^\alpha\phi)$.

The improvement contribution, on shell. From Eq. (7.196) the improvement adds $\xi_d(d-1)\Box\phi^2$. Expanding $\Box\phi^2 = 2(\partial_\alpha\phi)(\partial^\alpha\phi) + 2\phi\,\Box\phi$ and substituting the equation of motion Eq. (6.60), $\Box\phi = -m^2\phi - \frac{\lambda}{3!}\phi^3$,

$$\Box\phi^2 = 2(\partial\phi)^2 + 2\phi\Big(-m^2\phi - \tfrac{\lambda}{3!}\phi^3\Big) = 2(\partial\phi)^2 - 2m^2\phi^2 - \tfrac{\lambda}{3}\phi^4. \tag{7.198}$$

Hence

$$\xi_d(d-1)\,\Box\phi^2 = \frac{d-2}{4}\Big[\,2(\partial\phi)^2 - 2m^2\phi^2 - \tfrac{\lambda}{3}\phi^4\Big] = \frac{d-2}{2}(\partial\phi)^2 - \frac{d-2}{2}m^2\phi^2 - \frac{(d-2)\lambda}{12}\phi^4, \tag{7.199}$$

where $\xi_d(d-1) = \frac{d-2}{4(d-1)}(d-1) = \frac{d-2}{4}$.

Assemble. Adding Eq. (7.197) and Eq. (7.199),

$$\Theta^{\mu}_{\mathrm{c}\,\mu} = \underbrace{\Big[\Big(1-\tfrac{d}{2}\Big) + \tfrac{d-2}{2}\Big]}_{=0}(\partial\phi)^2 + \underbrace{\Big[\tfrac{d}{2} - \tfrac{d-2}{2}\Big]}_{=1} m^2\phi^2 + \underbrace{\Big[\tfrac{d}{4!} - \tfrac{(d-2)}{12}\Big]}_{=\frac{4-d}{24}}\lambda\phi^4. \tag{7.200}$$

The gradient term cancels for *every* d — this is precisely what the choice $\xi_d = \frac{d-2}{4(d-1)}$ engineers — leaving the clean classical trace identity

$$\boxed{\Theta^{\mu}_{\mathrm{c}\,\mu} \;=\; m^2\,\phi^2 \;+\; \frac{4-d}{24}\,\lambda\,\phi^4\,.} \tag{7.201}$$

In four dimensions the quartic term drops out identically and

$$\boxed{\Theta^{\mu}_{\mathrm{c}\,\mu}\big|_{d=4} \;=\; m^2\,\phi^2\,.} \tag{7.202}$$

The result Eq. (7.201) is the trace identity the section set out to establish, and its structure is worth reading carefully because it is the classical skeleton of the whole theory of the renormalization group. Each term in the trace is a Lagrangian operator multiplied by a number, and that number is exactly the *mass dimension of the coupling* in front of the operator. The mass term $\frac{1}{2}m^2\phi^2$ has a coupling $\frac{1}{2}m^2$ of dimension $[m^2] = 2$, and indeed $2 \times \frac{1}{2}m^2\phi^2 = m^2\phi^2$ appears in the trace. The quartic $\frac{\lambda}{4!}\phi^4$ has a coupling of dimension

$$[\lambda] = d - 4\,[\phi] = d - 4\cdot\frac{d-2}{2} = 4 - d, \tag{7.203}$$

and indeed $(4-d) \times \frac{\lambda}{4!}\phi^4 = \frac{4-d}{24}\lambda\phi^4$ appears. The general pattern, valid for any local scalar Lagrangian written as a sum of monomials with couplings g_i, is the classical trace formula

$$\Theta^{\mu}_{\mathrm{c}\,\mu} = \sum_i [g_i]\, g_i\, \mathcal{O}_i, \tag{7.204}$$

the trace being sourced exactly by the dimensionful couplings, each weighted by its own mass dimension. *Scale-noninvariance is the dimensionful couplings, nothing more.* A theory built only from dimensionless (marginal) couplings has a classically traceless stress tensor and is classically scale invariant; every dimensionful coupling contributes a term to the trace and breaks the symmetry.

Physical Insight

A traceless symmetric stress tensor is the local certificate of scale invariance, just as a symmetric one is the certificate of angular-momentum conservation and a conserved one of energy–momentum conservation. The three avatars of $\Theta^{\mu\nu}$ — conserved, symmetric, traceless — are the three pillars of conformal symmetry. The trace $\Theta^{\mu}{}_{\mu}$ is the single function whose vanishing upgrades a Poincaré-invariant theory to a scale-invariant one; Eq. (7.201) measures exactly how far the classical ϕ^4 theory falls short, and pins the shortfall on the mass and (away from $d = 4$) on the dimensionful quartic.

Notation Watch

We reserve $\Theta^{\mu\nu}$ for the symmetric, conserved Belinfante tensor of Theorem 7.33 and $\Theta_{\rm c}^{\mu\nu}$ for the conformally improved tensor Eq. (7.194); the two differ by the identically-conserved improvement term and carry the same charges. Some texts (Callan–Coleman–Jackiw) write the improvement with the opposite global sign as $-\frac{d-2}{4(d-1)}(\partial^\mu\partial^\nu-\eta^{\mu\nu}\Box)\phi^2$, which is identical to ours; others fold the coefficient into a non-minimal curvature coupling $\xi_d R\phi^2$ in a gravitational background. The number $\xi_4=\frac{1}{6}$ is the same in all conventions and is the conformal coupling of a scalar to gravity.

7.7.4 Special conformal transformations and the conformal algebra

Tracelessness buys far more than scale invariance. We show that a symmetric, conserved, traceless stress tensor automatically supplies a conserved current for every *conformal* transformation — the largest group of spacetime transformations under which a local field theory can be invariant — and that this group is generated by precisely $\frac{1}{2}(d+1)(d+2)$ currents, all of them packaged in the single object $\Theta_{\rm c}^{\mu\nu}$.

The organizing device is to contract the stress tensor with an arbitrary vector field $\xi_\nu(x)$ and form the current $J^\mu_\xi=\Theta_{\rm c}^{\mu\nu}\xi_\nu$. Its divergence, using conservation and the symmetry of the tensor to keep only the symmetric part of $\partial_\mu\xi_\nu$, is

$$\partial_\mu J^\mu_\xi=(\partial_\mu\Theta_{\rm c}^{\mu\nu})\xi_\nu+\Theta_{\rm c}^{\mu\nu}\partial_\mu\xi_\nu=\Theta_{\rm c}^{\mu\nu}\,\partial_\mu\xi_\nu=\tfrac{1}{2}\,\Theta_{\rm c}^{\mu\nu}(\partial_\mu\xi_\nu+\partial_\nu\xi_\mu). \tag{7.205}$$

The current is conserved whenever the symmetric derivative $\partial_\mu\xi_\nu+\partial_\nu\xi_\mu$ is annihilated by contraction with $\Theta_{\rm c}^{\mu\nu}$. For a generic conserved symmetric tensor this requires $\partial_\mu\xi_\nu+\partial_\nu\xi_\mu=0$, the Killing equation, whose solutions are exactly the $\frac{1}{2}d(d+1)$ Poincaré generators (translations $\xi^\mu=a^\mu$ and Lorentz rotations $\xi^\mu=\omega^\mu{}_\nu x^\nu$ with ω antisymmetric). A traceless tensor relaxes the requirement dramatically: contraction with $\Theta_{\rm c}^{\mu\nu}$ kills any pure $\eta_{\mu\nu}$ piece, so it suffices that $\partial_\mu\xi_\nu+\partial_\nu\xi_\mu$ be *proportional to* $\eta_{\mu\nu}$. This is the conformal Killing equation.

Definition 7.47 (Conformal Killing vector)**.** A vector field $\xi^\mu(x)$ is a *conformal Killing vector* of flat spacetime if

$$\partial_\mu\xi_\nu+\partial_\nu\xi_\mu=\frac{2}{d}\,(\partial_\rho\xi^\rho)\,\eta_{\mu\nu}, \tag{7.206}$$

i.e. if the infinitesimal coordinate change $x^\mu\to x^\mu+\xi^\mu$ alters the metric only by a local rescaling, $\eta_{\mu\nu}\to(1-\frac{2}{d}\,\partial\cdot\xi)\eta_{\mu\nu}$.

The form of Eq. (7.206) is fixed by tracing both sides: the trace of the left-hand side is $2\,\partial\cdot\xi$ and the trace of the right-hand side is $\frac{2}{d}(\partial\cdot\xi)\,d=2\,\partial\cdot\xi$, so

the coefficient $2/d$ is the unique one consistent with taking the trace. In $d > 2$ the solutions of Eq. (7.206) are finite in number, and they are precisely the four families

$$\begin{aligned} \text{translations:} &\quad \xi^\mu = a^\mu, && \partial\cdot\xi = 0, \\ \text{Lorentz:} &\quad \xi^\mu = \omega^\mu{}_\nu\, x^\nu \ (\omega_{\mu\nu} = -\omega_{\nu\mu}), && \partial\cdot\xi = 0, \\ \text{dilatation:} &\quad \xi^\mu = x^\mu, && \partial\cdot\xi = d, \\ \text{special conformal:} &\quad \xi^\mu = 2(b\cdot x)\, x^\mu - x^2\, b^\mu, && \partial\cdot\xi = 2d\,(b\cdot x), \end{aligned} \tag{7.207}$$

the last family parametrized by a constant vector b^μ. The first two have vanishing divergence (they are ordinary Killing vectors); the dilatation has constant divergence; the special conformal transformation (SCT) has divergence linear in x. Counting generators: d translations, $\frac{1}{2}d(d-1)$ Lorentz rotations, 1 dilatation, and d SCTs sum to

$$d + \tfrac{1}{2}d(d-1) + 1 + d = \tfrac{1}{2}(d+1)(d+2), \tag{7.208}$$

which in $d = 4$ is 15 — the dimension of the conformal algebra $\mathfrak{so}(4,2) \cong \mathfrak{su}(2,2)$, the same number of generators as the Poincaré algebra (10) plus dilatation (1) plus SCTs (4).

That the SCT field in Eq. (7.207) solves the conformal Killing equation is a short computation worth showing, since it also delivers the divergence of the SCT current. Differentiating $\xi_\nu = 2(b\cdot x)x_\nu - x^2 b_\nu$,

$$\partial_\mu \xi_\nu = 2\, b_\mu x_\nu + 2(b\cdot x)\eta_{\mu\nu} - 2x_\mu b_\nu, \tag{7.209}$$

whose symmetric part is

$$\partial_\mu\xi_\nu + \partial_\nu\xi_\mu = 2b_\mu x_\nu + 2(b\cdot x)\eta_{\mu\nu} - 2x_\mu b_\nu + 2b_\nu x_\mu + 2(b\cdot x)\eta_{\mu\nu} - 2x_\nu b_\mu = 4(b\cdot x)\,\eta_{\mu\nu}, \tag{7.210}$$

the cross terms cancelling in pairs, exactly $\frac{2}{d}\,(\partial\cdot\xi)\eta_{\mu\nu}$ with $\partial\cdot\xi = 2d\,(b\cdot x)$. Thus the SCT is conformal Killing, and substituting Eq. (7.210) into Eq. (7.205) the divergence of the special-conformal current $K^\mu_{(b)} = \Theta^{\mu\nu}_{\mathrm{c}}\xi_\nu$ is

$$\partial_\mu K^\mu_{(b)} = \tfrac{1}{2}\,\Theta^{\mu\nu}_{\mathrm{c}} \cdot 4(b\cdot x)\eta_{\mu\nu} = 2(b\cdot x)\,\Theta^\mu_{\mathrm{c}\,\mu}. \tag{7.211}$$

Stripping the arbitrary constant b_ρ, the d special-conformal currents and their divergences are

$$K^{\mu\rho} = 2\,x^\rho x_\nu\,\Theta^{\mu\nu}_{\mathrm{c}} - x^2\,\Theta^{\mu\rho}_{\mathrm{c}}, \qquad \partial_\mu K^{\mu\rho} = 2\,x^\rho\,\Theta^\mu_{\mathrm{c}\,\mu}. \tag{7.212}$$

We can now state the general result the construction has proved.

Theorem 7.48 (Tracelessness generates the conformal algebra)**.** *Let a Poincaré-invariant theory possess a symmetric, conserved stress tensor* $\Theta^{\mu\nu}_{\mathrm{c}}$ *that is* traceless

on shell, $\Theta^{\mu}_{\mathrm{c}\,\mu} = 0$. *Then for every conformal Killing vector* ξ *of Eq. (7.207) the current* $J^{\mu}_{\xi} = \Theta^{\mu\nu}_{\mathrm{c}}\xi_{\nu}$ *is conserved on shell,* $\partial_{\mu}J^{\mu}_{\xi} = 0$. *The resulting* $\frac{1}{2}(d+1)(d+2)$ *conserved charges close into the conformal algebra* $\mathfrak{so}(d,2)$ *(for* $d > 2$*): translations and Lorentz rotations give the Poincaré subalgebra, the dilatation gives the grading generator* D *with* $[D, P^{\mu}] = iP^{\mu}$ *and* $[D, K^{\mu}] = -iK^{\mu}$*, and the SCTs give* K^{μ} *with* $[K^{\mu}, P^{\nu}] = 2i(\eta^{\mu\nu}D - M^{\mu\nu})$*. Conversely, in a Poincaré-invariant theory whose virial current is a total derivative, the stress tensor can be improved to a traceless* $\Theta^{\mu\nu}_{\mathrm{c}}$*, and scale invariance is promoted to full conformal invariance.*

Proof. Conservation of each J^{μ}_{ξ} is Eq. (7.205) evaluated on a conformal Killing vector: the symmetric derivative is $\frac{2}{d}(\partial\cdot\xi)\eta_{\mu\nu}$ by Eq. (7.206), so $\partial_{\mu}J^{\mu}_{\xi} = \frac{1}{d}(\partial\cdot\xi)\,\Theta^{\mu}_{\mathrm{c}\,\mu} = 0$ when the tensor is traceless. The counting Eq. (7.208) gives the number of charges. The algebra relations follow from computing the charges $Q_{\xi} = \int d^3x\, J^0_{\xi}$ and their Poisson brackets (equivalently, commutators after quantization, Chapter 9); the brackets reproduce the action of the corresponding conformal Killing vectors on fields, which is the defining representation of $\mathfrak{so}(d,2)$. The converse is the improvement construction of Section 7.7.3: when $V^{\mu} = \partial_{\nu}\sigma^{\nu\mu}$ as in Eq. (7.190), the symmetric improvement Eq. (7.194) can be chosen to set the on-shell trace to zero, after which Eq. (7.211) shows the SCT currents are conserved, not merely the dilatation current. □

The converse deserves emphasis because it is a genuine theorem and a frequent source of confusion: scale invariance does *not* automatically imply conformal invariance. The extra input is that the virial current V^{μ} be a total derivative, equivalently that the trace of the stress tensor be improvable to zero rather than merely to a total derivative. In essentially all relevant local field theories — and in particular for the scalar, by Eq. (7.190) — the condition holds, so the distinction rarely bites; but it is logically real, and theories with a conserved virial that is not a total derivative are scale- but not conformally invariant. For the massless scalar the virial is the gradient of $\frac{\Delta}{2}\phi^2$, the condition is met, and classical scale invariance is automatically classical conformal invariance.

7.7.5 Scalar ϕ^4 theory: the classical scale symmetry of the massless theory

We now collect what the machinery has delivered for ϕ^4 theory, continuing the thread from its $\mathbb{Z}_2$ stage earlier in this chapter and from the classical stress tensor of Eq. (6.178). The carried Lagrangian is Eq. (6.56), and the trace identity Eq. (7.201) reads off three facts at a glance.

Massless ϕ^4 *in* $d = 4$ *is classically scale and conformally invariant.* Setting $m = 0$ in four dimensions, Eq. (7.202) gives $\Theta^{\mu}_{\mathrm{c}\,\mu} = 0$ identically on shell. By Theorem 7.43 the dilatation current is conserved, and by Theorem 7.48 — the

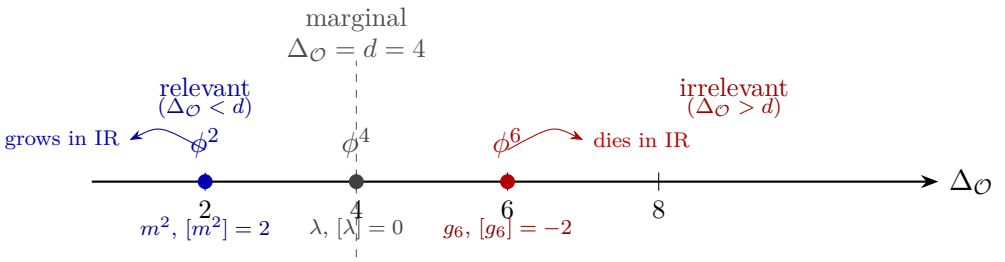


Figure 7.6: The classical relevance ladder for the $\mathbb{Z}_2$-even scalar operators in $d = 4$, placed by their scaling dimension $\Delta_{\mathcal{O}}$. An operator is relevant, marginal, or irrelevant according as $\Delta_{\mathcal{O}}$ is less than, equal to, or greater than the spacetime dimension $d = 4$; equivalently, its coupling has positive, zero, or negative mass dimension. The trace identity Eq. (7.202) confirms that only ϕ^2 (with coupling m^2, $[m^2] = 2$) contributes to $\Theta^{\mu}_{\mathrm{c}\,\mu}$ classically; the marginal ϕ^4 contributes nothing, and higher operators such as ϕ^6 are irrelevant. The arrows anticipate the renormalization-group flow of Chapter 23.

virial being a total derivative Eq. (7.190) — the full set of 15 conformal currents is conserved. The massless quartic scalar is a classical conformal field theory, with the scalar carrying the conformal weight $\Delta = 1$ fixed in Eq. (7.185). This is the classical seed of the fixed-point description that Part XIV will quantize.

The mass term is the unique relevant operator that breaks scale invariance. With $m \neq 0$, the trace is $\Theta^{\mu}_{\mathrm{c}\,\mu} = m^2\phi^2 \neq 0$, and the dilatation current is no longer conserved:

$$\partial_\mu D^\mu = \Theta^{\mu}_{\mathrm{c}\,\mu} = m^2\phi^2. \tag{7.213}$$

The breaking is proportional to m^2, a coupling of positive mass dimension $[m^2] = 2$. In the language of Eq. (7.203) and Eq. (7.204), ϕ^2 is the operator of lowest dimension ($\Delta_{\phi^2} = 2 < d = 4$) consistent with the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$, hence the unique *relevant* deformation; a ϕ^2 term grows in importance toward long distances and is the one operator that must be tuned to reach the scale-invariant point. This is the precise classical statement behind the statement, made repeatedly downstream, that "the mass is the relevant coupling of ϕ^4 theory."

The quartic coupling is classically marginal. In $d = 4$ the coupling λ is dimensionless, $[\lambda] = 4 - d = 0$ by Eq. (7.203), and its operator ϕ^4 has dimension $\Delta_{\phi^4} = 4 = d$ exactly. It therefore contributes nothing to the trace Eq. (7.202): the quartic interaction respects classical scale invariance. This is what "marginal" means — a coupling that neither grows nor shrinks under rescaling at the classical level, sitting exactly on the boundary between relevant and irrelevant. Figure 7.6 places the three lowest $\mathbb{Z}_2$-even operators on the dimension axis and reads off their classical relevance.

These three facts — scale invariance of the massless theory, the mass as the

unique relevant breaking, the marginality of λ — are exact *classically*. They are also, every one of them, modified by quantization, and honesty requires the modification be named here even though it is discharged far downstream.

Caution

Classical scale invariance is generically destroyed by quantization. Regularizing the quantum theory requires a mass scale μ (the renormalization scale, Chapter 22), and physical couplings acquire a dependence on it: the dimensionless λ becomes a running $\lambda(\mu)$. The trace identity Eq. (7.202) is then corrected to

$$\Theta^{\mu}_{\mathrm{c}\,\mu} = m^2(1+\gamma_m)\,\phi^2 + \frac{\beta(\lambda)}{4!}\,\phi^4 + \cdots , \tag{7.214}$$

where $\beta(\lambda) = \mu\, d\lambda/d\mu$ is the beta function — nonzero even in $d = 4$, where classically $[\lambda] = 0$ — and γ_m is the mass anomalous dimension. The non-vanishing of $\Theta^{\mu}_{\mathrm{c}\,\mu}$ in the massless quantum theory, with the classical $\frac{4-d}{24}\lambda$ replaced by $\frac{\beta(\lambda)}{4!}$, is the *trace (or conformal) anomaly*: the marginal coupling becomes "marginally relevant or irrelevant" depending on the sign of β. Equation Eq. (7.214) is stated here only to make the connection explicit; it is derived in Chapter 24 (the one-loop $\beta_\lambda = 3\lambda^2/16\pi^2$) and given its anomaly interpretation in Chapter 36.

Looking ahead

The classical conformal data assembled here — the traceless improved tensor $\Theta^{\mu\nu}_{\mathrm{c}}$, the fixed scaling dimension $\Delta = 1$ of the scalar, the trace identity $\Theta^{\mu}_{\mathrm{c}\,\mu} = m^2\phi^2$, and the marginality of λ — is exactly the input the quantum theory will deform. The renormalization group of Chapter 23 and the explicit beta function of Chapter 24 turn the classical dimensions into anomalous ones and convert the clean trace Eq. (7.202) into the trace anomaly Eq. (7.214); the conformal-field-theory technology of Chapter 47 organizes the resulting fixed-point physics; and the Wilson–Fisher fixed point of Chapter 52, found by tuning the marginal λ against the small dimensionful $\frac{4-d}{24}\lambda$ term of Eq. (7.201) in $d = 4-\epsilon$, realizes a genuinely interacting conformal field theory — the quantum completion of the classical scale symmetry this chapter has just laid bare, and the handoff that closes ϕ^4 theory's classical chapter.

Further reading

Noether's correspondence is treated in every serious field-theory text; the accounts closest to this chapter in convention and emphasis are [396, 432, 454], with [500]

the most careful on the relation between Poincaré invariance, the stress tensor, and the field content, and [129] the most lucid on the physical meaning of the global/local distinction and the role of symmetry as an organizing principle. The reader who wants the source should consult Noether's original 1918 paper [369], which proves both the first and the (less often quoted) second theorem in a single framework; the symmetric improvement of the stress tensor is due independently to [57] and [419]. The scale and conformal material previewed in the final section is developed in full, in the language used here, in [147] and [532].

Chapter 8

Gauge Invariance: Classical Electromagnetism

This chapter asks one question and follows its consequences to the end of classical electrodynamics: *what does it cost to demand that a symmetry hold not rigidly but locally?* Chapter 7 taught us to read a continuous symmetry through Noether's first theorem (Theorem 7.10), but every symmetry considered there was rigid – the transformation parameter a single constant applied identically everywhere. Yet if a symmetry is genuinely a statement about description rather than about nature, it ought to be choosable independently at each point of spacetime, just as two distant laboratories may fix their coordinate axes without consulting one another. The phase of the complex matter field of Chapter 6 is exactly such a convention. Demanding the freedom to fix it point by point – making the global U(1) of Chapter 2 *local* – turns out not to be free: it collides with the one piece of structure every Lagrangian carries, the derivative. Diagnosing that collision and paying its price reconstructs, with nothing else assumed, the whole of Maxwell's electrodynamics. This is the *gauge principle*, and it is the organizing idea not only of this chapter but of Parts V and VII; here it is carried out in full for the abelian group, and Chapter 25 will repeat the identical program for a non-abelian one.

The chapter imports its raw material from the two preceding chapters of Part II and from Part I. The complex-scalar and Dirac Lagrangians on which the construction acts, and the independent treatment of ψ and $\bar{\psi}$, are those built in Chapter 6; the Euler–Lagrange machinery and the canonical-momentum bookkeeping are likewise from there. The conserved current that turns out to be the obstruction's coefficient – and, in the end, the source of the field – is the global U(1) Noether current of Chapter 7. The tensor and index conventions, the Levi-Civita symbol, and the classification of antisymmetric tensors are from Chapter 3; the group U(1) itself, its compactness, and the fact that $\pi_1(\mathrm{U}(1)) = \mathbb{Z}$ are from Chapter 2; and the retarded Green's function used to fix the gauge

is the one constructed in Chapter 5. Throughout we use the book-wide canon: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, the covariant derivative $D_\mu = \partial_\mu - iqA_\mu$ with the electron carrying $q = -e < 0$, and the gamma-matrix conventions of Chapter C.

The development runs as a single chain, each section handing the next its open problem. Section 8.1 promotes the global phase to a local one and isolates the obstruction precisely: the ordinary derivative of a covariantly transforming field is not itself covariant, and the failure is governed, term for term, by the Noether current. Section 8.2 repairs it by the only available means – a new vector field A_μ whose inhomogeneous transformation is rigged to absorb the offending gradient – assembling the gauge-covariant derivative and deriving minimal coupling, the unique leading interaction of matter with the gauge field. Section 8.3 confronts the fact that A_μ has arrived with a transformation law but no dynamics and is not itself gauge invariant; it extracts the gauge-invariant residue of the connection as the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, built from the commutator of covariant derivatives, proves its invariance and the Bianchi identity, identifies its components as $\mathbf{E}$ and $\mathbf{B}$, and reveals it – through parallel transport and holonomy – as the curvature of the connection. Section 8.4 writes the unique gauge-invariant, Lorentz-invariant, parity-even, two-derivative action $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, derives the inhomogeneous Maxwell equations as its Euler–Lagrange equations while the Bianchi identity supplies the homogeneous pair, and shows that coupling to matter *forces* the source to be conserved. Section 8.5 faces the gauge redundancy head-on: it exhibits configuration space as a quotient, fixes the gauge in the Lorenz, Coulomb, temporal, and axial ways, counts the photon's two physical polarizations, and constructs them explicitly as transverse plane waves. Section 8.6 assembles the full classical QED Lagrangian, identifies the electromagnetic current as the U(1) Noether current and as the very Maxwell source, verifies exact local gauge invariance, and recovers the Lorentz force – launching QED at its classical layer. Finally Section 8.7 shows that the connection carries gauge-invariant information the field strength cannot: the Aharonov–Bohm holonomy, physical even where $F_{\mu\nu} = 0$, and – run in reverse against a magnetic monopole – Dirac's quantization of electric charge, exhibiting the connection, not its curvature, as the fundamental object.

A brief word on scope and rigor. Everything load-bearing is proved here in full: the obstruction and its tie to Noether, the covariance of D_μ and the uniqueness of the gauge transformation that secures it, the gauge invariance of $F_{\mu\nu}$ and the Bianchi identity (twice, directly and via Jacobi), the uniqueness of the Maxwell kinetic term, all four Maxwell equations component by component, the degree-of-freedom count, the explicit transverse polarizations, the conservation of the electromagnetic current on shell, the Lorentz force from the eikonal limit, and the Aharonov–Bohm phase and Dirac condition. no earlier model is imported here – QED begins in this chapter – and what later chapters call "the QED Lagrangian" is the frozen Eq. (8.143) fixed in Section 8.6. The two quantizations

the classical theory demands – of the photon and of the electron – are flagged as they arise and deferred to Chapter 13 and Chapter 11 respectively.

8.1 From Global to Local: The Gauge Principle

The chapter introduction posed a single question, and the rest of the book is in one sense an extended answer to it: *what does it cost to demand that a symmetry hold not rigidly but locally?* Chapter 7 taught us to read a continuous symmetry as a source of conserved quantities through Noether's first theorem, but every symmetry it considered was rigid — the transformation parameter was a single constant, applied identically at every point of spacetime. There is something physically uncomfortable about that. A symmetry asserts that a certain change of description leaves all physics unchanged; if that is genuinely a statement about *description* rather than about nature, it ought to be possible to make the change of description differently in different places, just as one is free to choose a different set of coordinate axes in two distant laboratories without any global authority synchronizing them. The phase of a complex matter field, as we are about to see, is exactly such a conventional choice. The demand that we be allowed to fix it independently at each point — to make the internal symmetry *local* — turns out not to be free. It collides with the one piece of structure every Lagrangian in this book contains: the derivative. Diagnosing that collision precisely, and naming the structure needed to resolve it, is the business of this section; resolving it is the business of the next.

8.1.1 The global $U(1)$ symmetry, recalled

We take as our cleanest carrier the free complex scalar field already assembled in Chapter 6. Recall its Lagrangian (6.62),

$$\mathcal{L} \;=\; (\partial_\mu \phi^*)(\partial^\mu \phi) \;-\; m^2\, \phi^* \phi, \tag{8.1}$$

with $\phi(x)$ a single complex Lorentz scalar (the index conventions are those of Chapter 3) and m its mass. Throughout this section ϕ is the complex field of Chapter 6; the presence of ϕ^* always signals the complex case, exactly as agreed there. We choose the complex scalar rather than the Dirac field only because its phase structure is uncluttered by spinor indices; the entire argument below transcribes verbatim to the electron field ψ of QED, where the phase multiplies the spinor as a whole and the γ-matrices are mere spectators. We note that case explicitly at the end of the subsection.

Chapter 2 identified the group of phases $e^{i\theta}$, $\theta \in [0, 2\pi)$, as the abelian Lie group U(1), parametrized by a single angle and with the trivial commutation law $e^{i\theta_1} e^{i\theta_2} = e^{i(\theta_1+\theta_2)}$. Acting on ϕ it generates the transformation, recall (6.66),

$$\phi \;\longmapsto\; \phi' \;=\; e^{\,iq\,\alpha}\,\phi, \qquad \phi^* \;\longmapsto\; \phi'^* \;=\; e^{-\,iq\,\alpha}\,\phi^*, \tag{8.2}$$

where α is a real constant and we have introduced a real number q, the *charge*, in the exponent. This is no new freedom: (8.2) is precisely the symmetry (6.66) of Chapter 6 with its group parameter written as $q\alpha$ rather than as a single angle. We split it this way deliberately. The factor α will shortly be promoted to a function of spacetime and will play the role of the transformation *parameter*; the factor q is a fixed attribute of the field ϕ, measuring how strongly it responds to the phase rotation, and it is this q that will become the electric charge of the matter when the symmetry is gauged. That (8.2) leaves (8.1) invariant is immediate and worth seeing once more: every term pairs one ϕ with one ϕ^*, so the phases cancel, $\phi'^*\phi' = e^{-iq\alpha}e^{iq\alpha}\phi^*\phi = \phi^*\phi$, and because α is constant it passes through ∂_μ untouched, $\partial_\mu\phi' = e^{iq\alpha}\partial_\mu\phi$, so the kinetic term is invariant for exactly the same reason. The invariance rests squarely on the constancy of α.

By Noether's first theorem (Theorem 7.10) this rigid symmetry yields a conserved current. Applying the theorem to (8.2) — a computation carried out in full in Chapter 7, which we import rather than repeat — gives

$$j^\mu \;=\; iq\,(\phi^*\,\partial^\mu\phi \;-\; \phi\,\partial^\mu\phi^*), \qquad \partial_\mu j^\mu = 0 \;\text{ on-shell}, \tag{8.3}$$

the conserved charge being $Q = \int \mathrm{d}^3x\, j^0$ (this is Eq. (7.70) with the charge factor q restored). When the symmetry is eventually gauged, j^μ will turn out to be nothing other than the electromagnetic current that sources Maxwell's equations; for the moment it is just the Noether current of a rigid phase rotation, and we record it because the obstruction we are about to compute will turn out to be governed by precisely this object.

For the Dirac field the situation is identical at the level that concerns us. The free Dirac Lagrangian $\bar\psi(i\partial\!\!\!/ - m)\psi$ of Chapter 6 is invariant under $\psi \mapsto e^{iq\alpha}\psi$, $\bar\psi \mapsto e^{-iq\alpha}\bar\psi$ with α constant, since $\bar\psi = \psi^\dagger\gamma^0$ supplies the conjugate phase and the spinor indices on which $\partial\!\!\!/$ acts are untouched by the scalar phase $e^{iq\alpha}$. Everything we say below about $\partial_\mu\phi$ holds word for word for $\partial_\mu\psi$.

8.1.2 Localizing the phase: the obstruction

We now make the move the whole chapter turns on.

Definition 8.1 (Local $U(1)$ transformation)**.** A *local* (or *gauge*) U(1) transformation is the replacement

$$\phi(x) \;\longmapsto\; \phi'(x) \;=\; e^{\,iq\,\alpha(x)}\,\phi(x), \qquad \phi^*(x) \;\longmapsto\; \phi'^*(x) \;=\; e^{-\,iq\,\alpha(x)}\,\phi^*(x), \tag{8.4}$$

in which the parameter $\alpha = \alpha(x)$ is now an arbitrary smooth real-valued function on spacetime rather than a constant. The global transformation (8.2) is the special case $\alpha(x) = \text{const}$.

Pointwise, (8.4) is still an element of U(1) acting on ϕ; what has changed is that the group element is allowed to differ from point to point. The mass

term does not notice: at each point the phases still cancel, $\phi'^*(x)\phi'(x) = e^{-iq\alpha(x)}e^{iq\alpha(x)}\phi^*(x)\phi(x) = \phi^*(x)\phi(x)$, because conjugation is a local operation and $\alpha(x)$ is the same real number for ϕ and ϕ^* at the given x. The kinetic term is another matter entirely, because the derivative is *not* a local operation: $\partial_\mu\phi(x)$ depends on the values of ϕ in an infinitesimal neighbourhood of x, where the local phase α takes slightly different values.

The source of all the trouble is the single line that follows from the product rule. Differentiating the transformed field,

$$\partial_\mu\phi'(x) \;=\; \partial_\mu\Big(e^{\,iq\,\alpha(x)}\,\phi(x)\Big) \;=\; e^{\,iq\,\alpha(x)}\,\partial_\mu\phi(x) \;+\; \big(iq\,\partial_\mu\alpha(x)\big)\,e^{\,iq\,\alpha(x)}\,\phi(x), \tag{8.5}$$

and factoring out the common phase,

$$\boxed{\partial_\mu\phi' \;=\; e^{\,iq\,\alpha}(\partial_\mu + iq\,\partial_\mu\alpha)\phi} \tag{8.6}$$

Compare this with how ϕ itself transforms, $\phi' = e^{iq\alpha}\phi$. The field picks up the phase $e^{iq\alpha}$ and *nothing else*; we say it transforms *homogeneously*, or *covariantly*, under the local group. Its derivative does not. Instead of the clean homogeneous law $\partial_\mu\phi \mapsto e^{iq\alpha}\partial_\mu\phi$ that held in the global case, (8.6) carries an extra, *inhomogeneous* term $iq(\partial_\mu\alpha)\phi$ inside the parentheses. The ordinary derivative of a covariantly transforming field is not itself covariant. When $\partial_\mu\alpha = 0$ — the global case — the offending term vanishes and homogeneity is restored; when α varies in space or time, it does not. This single mismatch is the obstruction, and the rest of the chapter is its resolution.

It is worth watching the obstruction propagate all the way into the Lagrangian, with every step shown, both to confirm that nothing cancels it and to identify exactly what survives. The transformed conjugate derivative is the complex conjugate of (8.6),

$$\partial_\mu\phi'^* \;=\; e^{-\,iq\,\alpha}(\partial_\mu - iq\,\partial_\mu\alpha)\phi^*. \tag{8.7}$$

The kinetic term of (8.1) therefore transforms as

$$\begin{aligned}(\partial_\mu\phi'^*)(\partial^\mu\phi') &= \Big[e^{-iq\alpha}(\partial_\mu - iq\,\partial_\mu\alpha)\phi^*\Big]\Big[e^{+iq\alpha}(\partial^\mu + iq\,\partial^\mu\alpha)\phi\Big] \\ &= \big(\partial_\mu\phi^* - iq(\partial_\mu\alpha)\phi^*\big)\big(\partial^\mu\phi + iq(\partial^\mu\alpha)\phi\big),\end{aligned} \tag{8.8}$$

where in the second line the two phases $e^{-iq\alpha}e^{+iq\alpha} = 1$ have cancelled — this is the only cancellation available, and it removes the phase but not its derivative. Multiplying out the four terms,

$$\begin{aligned}(\partial_\mu\phi'^*)(\partial^\mu\phi') &= (\partial_\mu\phi^*)(\partial^\mu\phi) \;+\; iq(\partial^\mu\alpha)\,\phi\,(\partial_\mu\phi^*) \;-\; iq(\partial_\mu\alpha)\,\phi^*\,(\partial^\mu\phi) \\ &\qquad -\; (iq)(iq)\,(\partial_\mu\alpha)(\partial^\mu\alpha)\,\phi^*\phi \\ &= (\partial_\mu\phi^*)(\partial^\mu\phi) \;+\; iq\,(\partial_\mu\alpha)\big(\phi\,\partial^\mu\phi^* - \phi^*\,\partial^\mu\phi\big) \;+\; q^2\,(\partial_\mu\alpha)(\partial^\mu\alpha)\,\phi^*\phi,\end{aligned} \tag{8.9}$$

where in passing to the last line we relabelled the dummy index in the second term ($\mu \leftrightarrow$ itself, simply grouping the two linear-in-$\partial\alpha$ contributions) and used $-(iq)(iq) = -i^2q^2 = +q^2$. Now recognize the middle term. Comparing the bracket with the Noether current (8.3), $j^\mu = iq(\phi^*\partial^\mu\phi - \phi\,\partial^\mu\phi^*)$, we have $iq(\phi\,\partial^\mu\phi^* - \phi^*\,\partial^\mu\phi) = -\,j^\mu$, so the linear term is exactly $-(\partial_\mu\alpha)\,j^\mu$. Since the mass term is unchanged, the full Lagrangian transforms as

$$\mathcal{L} \;\longmapsto\; \mathcal{L}' \;=\; \mathcal{L} \;-\; (\partial_\mu\alpha)\,j^\mu \;+\; q^2\,(\partial_\mu\alpha)(\partial^\mu\alpha)\,\phi^*\phi, \tag{8.10}$$

that is, the change in the Lagrangian density under a local phase rotation is

$$\delta\mathcal{L} \;\equiv\; \mathcal{L}' - \mathcal{L} \;=\; -\,(\partial_\mu\alpha)\,j^\mu \;+\; q^2\,(\partial_\mu\alpha)(\partial^\mu\alpha)\,\phi^*\phi. \tag{8.11}$$

Equation (8.11) is the central result of this section, the obstruction the chapter must overcome. Every term on its right-hand side is proportional to a derivative of α, so $\delta\mathcal{L} = 0$ if and only if $\partial_\mu\alpha = 0$ everywhere, i.e. precisely for the rigid transformations already known to be symmetries. For any genuinely local $\alpha(x)$ the Lagrangian changes, the action $S = \int \mathrm{d}^4x\,\mathcal{L}$ changes with it, and the theory (8.1) is *not* invariant under Theorem 8.1. The leading piece of the obstruction, linear in $\partial_\mu\alpha$, is $-(\partial_\mu\alpha)\,j^\mu$: the Noether current of the global symmetry is exactly the coefficient with which the local transformation fails. This is not a coincidence, and the worked example below makes the connection precise.

Worked Example 8.2 (The infinitesimal obstruction and its tie to Noether)**.** It pays to see (8.11) at infinitesimal order, both because the algebra is cleaner there and because it exposes why the Noether current had to appear. Take $\alpha(x)$ infinitesimal and work to first order, dropping all terms of order α^2. To this order the finite transformation (8.4) linearizes to

$$\delta\phi = iq\,\alpha\,\phi, \qquad \delta\phi^* = -iq\,\alpha\,\phi^*, \tag{8.12}$$

and the quadratic term $q^2(\partial_\mu\alpha)(\partial^\mu\alpha)\phi^*\phi$ in (8.11), being second order in α, drops out. What remains is

$$\delta\mathcal{L} \;=\; -\,(\partial_\mu\alpha)\,j^\mu \qquad \text{(to first order in } \alpha\text{)}. \tag{8.13}$$

Let us re-derive (8.13) directly from (8.12), as an independent check that no sign slipped in the finite computation. Varying (8.1) term by term,

$$\delta\mathcal{L} = (\partial_\mu\delta\phi^*)(\partial^\mu\phi) + (\partial_\mu\phi^*)(\partial^\mu\delta\phi) - m^2(\delta\phi^*\,\phi + \phi^*\,\delta\phi). \tag{8.14}$$

The mass piece vanishes identically: $\delta\phi^*\phi + \phi^*\delta\phi = -iq\alpha\,\phi^*\phi + iq\alpha\,\phi^*\phi = 0$, which is just the statement that $\phi^*\phi$ is pointwise invariant. For the kinetic

piece we need $\partial_\mu \delta\phi = \partial_\mu(iq\alpha\phi) = iq(\partial_\mu\alpha)\phi + iq\alpha\,\partial_\mu\phi$ and its conjugate $\partial_\mu\delta\phi^* = -iq(\partial_\mu\alpha)\phi^* - iq\alpha\,\partial_\mu\phi^*$. Substituting,

$$\begin{aligned}\delta\mathcal{L} &= \big[-iq(\partial_\mu\alpha)\phi^* - iq\alpha\,\partial_\mu\phi^*\big]\partial^\mu\phi + (\partial_\mu\phi^*)\big[iq(\partial^\mu\alpha)\phi + iq\alpha\,\partial^\mu\phi\big] \\ &= \underbrace{iq\,\alpha\big[(\partial_\mu\phi^*)(\partial^\mu\phi) - (\partial_\mu\phi^*)(\partial^\mu\phi)\big]}_{=0} + iq(\partial_\mu\alpha)(\phi\,\partial^\mu\phi^* - \phi^*\,\partial^\mu\phi) \\ &= -(\partial_\mu\alpha)\,j^\mu, \end{aligned} \tag{8.15}$$

where the terms proportional to α (not its derivative) cancel between the two contributions — exactly as they must, since for constant α we must recover the global invariance $\delta\mathcal{L} = 0$ — and the surviving derivative term is identified through (8.3) as $-(\partial_\mu\alpha)j^\mu$. This reproduces (8.13).

The form (8.13) also illuminates Noether's theorem itself. Integrating $\delta\mathcal{L}$ over spacetime and discarding a surface term,

$$\delta S = \int \mathrm{d}^4x\,\big[-(\partial_\mu\alpha)\,j^\mu\big] = \int \mathrm{d}^4x\,\alpha\,(\partial_\mu j^\mu) \;-\; \int \mathrm{d}^4x\,\partial_\mu(\alpha\,j^\mu). \tag{8.16}$$

For *constant* α the left side is zero (global symmetry) and the surface term vanishes, giving nothing new. But (8.16) holds for *any* $\alpha(x)$, and on a field configuration that solves the equations of motion the action is stationary against all variations, so $\delta S = 0$ for arbitrary α of compact support; the boundary term then drops and arbitrariness of α forces $\partial_\mu j^\mu = 0$. This is the "Noether trick": promoting a global parameter to a local one and reading off the current's conservation from the coefficient of $\partial_\mu\alpha$. We will not lean on it — Chapter 7 proved current conservation independently — but it shows that the obstruction (8.11) is not an accident peculiar to this Lagrangian. *Any* theory with a global continuous symmetry fails local invariance at precisely the rate set by its Noether current. The obstruction and the conservation law are two readings of the same equation.

8.1.3 What the obstruction means: comparing fields in different internal frames

Before repairing the obstruction it is worth understanding what it is telling us, because the understanding dictates the cure. The deep statement hidden in (8.6) is geometric, and grasping it now will make the covariant derivative of the next section feel inevitable rather than invented.

A complex field $\phi(x)$ has, at each spacetime point x, a magnitude $|\phi(x)|$ and a phase $\arg\phi(x)$. The magnitude is unambiguous, but the phase is meaningful only relative to a convention — a choice of what counts as "real and positive" in the internal complex plane attached to the point x. Call that choice the *internal frame* at x. A global U(1) transformation (8.2) rotates every internal frame by

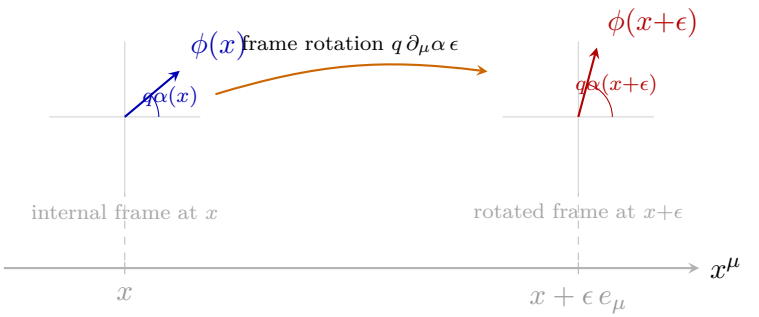


Figure 8.1: The geometric content of the obstruction. The phase of ϕ is defined only relative to an internal frame (a choice of phase-zero in the complex plane) attached to each spacetime point. A local U(1) transformation rotates that frame independently at each point. The ordinary derivative subtracts $\phi(x+\epsilon\,e_\mu)$ from $\phi(x)$ as if the two frames coincided; when they do not, the difference includes the spurious frame rotation $q\,\partial_\mu\alpha\,\epsilon$, which is exactly the inhomogeneous term $iq(\partial_\mu\alpha)\phi$ of (8.6).

the same angle $q\alpha$; this is a harmless relabelling, like agreeing to measure all phases from a different zero, and nothing physical can depend on it. A *local* transformation (8.4) rotates the internal frame at each point by its own angle $q\alpha(x)$: it allows the zero of phase to be chosen independently at every point of spacetime.

Now ask what the derivative does. By definition $\partial_\mu\phi$ measures the rate of change of ϕ along the direction μ, built from the difference $\phi(x+\epsilon\,e_\mu)-\phi(x)$ of the field at two neighbouring points divided by ϵ. But $\phi(x+\epsilon\,e_\mu)$ and $\phi(x)$ live in *different* internal frames once the frame is allowed to vary in space: the subtraction compares a complex number measured against the convention at $x+\epsilon\,e_\mu$ with one measured against the convention at x. If those two conventions are rotated relative to each other — which is exactly what $\partial_\mu\alpha\neq 0$ means — then part of the "difference" the derivative reports is not a change in the field at all, but a change in the yardstick. That spurious, frame-rotation contribution is precisely the inhomogeneous term $iq(\partial_\mu\alpha)\phi$ in (8.6). Figure 8.1 draws the picture: when the internal frames at neighbouring points are misaligned, the naive difference of field values double-counts the misalignment.

Physical Insight

The obstruction is not a defect of the complex scalar; it is a defect of the *derivative*. The ordinary derivative is built to compare field values at neighbouring points, and it does so by naive subtraction — which silently assumes that the internal frames at those points are the same.

That assumption is exactly what a local symmetry forbids us to make. The cure cannot be to abandon comparison; physics needs derivatives. The cure must be to teach the derivative how to undo the relative frame rotation before it subtracts — to supply a rule for "parallel transporting" the internal frame from one point to the next so that like is compared with like. Such a rule is precisely a *connection*, and the field that encodes it will be the gauge potential A_μ. This is the same idea that, in spacetime geometry, turns the ordinary derivative into the covariant derivative of general relativity; here the geometry is internal rather than spacetime, but the logic is identical. The gauge potential is the connection on an internal U(1) bundle, a point we return to geometrically in Chapter 25.

This reframing tells us exactly what to demand. We do not want to give up local invariance — the whole motivation was that an internal phase is a pure convention that ought to be choosable point by point. Nor can we give up derivatives. So we must modify the derivative. We seek an operator D_μ, to replace ∂_μ in the kinetic term, with the defining property that it turns a covariantly transforming field into another covariantly transforming object:

$$D_\mu\phi \;\longmapsto\; (D_\mu\phi)' \;=\; e^{\,iq\,\alpha(x)}\,(D_\mu\phi), \tag{8.17}$$

homogeneously, with no leftover $\partial_\mu\alpha$ term, for every $\alpha(x)$. If such a D_μ exists, then $(D_\mu\phi)^*(D^\mu\phi)$ transforms exactly as the original $\phi^*\phi$ did — the phases cancel and nothing survives — and replacing $\partial_\mu \to D_\mu$ in (8.1) produces a Lagrangian invariant under the full local group.

Definition 8.3 (The gauge principle, as a program)**.** The *gauge principle* is the demand that a theory be invariant under *local* transformations of an internal symmetry group, together with the constructive program this demand forces:

1. Replace every ordinary derivative ∂_μ acting on a charged field by a *gauge-covariant derivative* D_μ obeying the homogeneous transformation law (8.17).
2. Discover what new field-theoretic structure D_μ must contain in order to satisfy that law; for an internal U(1) this will force the existence of a vector field A_μ, the gauge potential, with its own prescribed transformation.
3. Build the gauge-invariant dynamics of A_μ itself from the object that measures the failure of D_μ to commute with itself — the field strength, or curvature.

In this chapter the program, carried out for U(1), reconstructs all of classical electrodynamics; in Chapter 25 the identical program for a non-abelian group reconstructs Yang–Mills theory.

Notation Watch

We have written the local rotation as $e^{iq\alpha(x)}$ with a charge q kept explicit and have not yet committed to its sign or to the sign convention relating D_μ to A_μ. Both are fixed precisely in the next section, in agreement with the standing conventions of QED: the covariant derivative will be $D_\mu = \partial_\mu - iqA_\mu$, the electron will carry $q = -e$ with $e > 0$, so that $D_\mu = \partial_\mu + ieA_\mu$ on the electron field. Until then q is simply a fixed real charge labelling how strongly ϕ responds to the phase rotation. We flag this because the sign of the gauge coupling is the most error-prone convention in the subject; here it is merely staged, in the next section it is nailed down.

We have now isolated the enemy with complete precision. Promoting the global U(1) phase symmetry of charged matter to a local one breaks the kinetic term by the obstruction (8.11), whose origin is the single inhomogeneous term in $\partial_\mu(e^{iq\alpha}\phi) = e^{iq\alpha}(\partial_\mu + iq\,\partial_\mu\alpha)\phi$ of (8.6): the ordinary derivative compares fields living in differently rotated internal frames and so cannot transform homogeneously. The next section meets this head-on. It introduces a vector field A_μ whose transformation is engineered to absorb the unwanted $iq(\partial_\mu\alpha)\phi$, assembles the covariant derivative D_μ satisfying (8.17), and shows that this single requirement *forces* the matter to couple to A_μ in the unique combination known as minimal coupling — the first concrete result of the gauge principle, and the opening move of classical QED.

8.2 The Covariant Derivative and Minimal Coupling

The previous section left us holding an obstruction. When the rigid internal phase symmetry of a charged matter field is promoted to a *local* one — a phase α that varies from spacetime point to spacetime point — the matter Lagrangian fails to be invariant, and the failure can be traced to a single offending object: the ordinary derivative. Under the local transformation $\psi(x) \to \psi'(x) = e^{iq\alpha(x)}\psi(x)$ the field itself rotates rigidly at each point, but its derivative does not, because differentiation compares the field at x with the field at the neighbouring point $x + \mathrm{d}x$, and those two points now carry *different* phase conventions. Concretely, the computation of Section 8.1 gave (8.6), which for the field ψ reads

$$\partial_\mu\psi(x) \;\longrightarrow\; \partial_\mu\big(e^{iq\alpha(x)}\psi(x)\big) \;=\; e^{iq\alpha(x)}\Big(\partial_\mu\psi(x) + i\,q\,(\partial_\mu\alpha(x))\,\psi(x)\Big). \tag{8.18}$$

The first term inside the bracket is what we want: the derivative dressed by the same phase $e^{iq\alpha}$ as the field, exactly as a covariant object should be. The second term, $iq(\partial_\mu\alpha)\psi$, is the obstruction. It is inhomogeneous — it involves $\partial_\mu\alpha$ rather than α — and it is precisely the fingerprint of the fact that α is now

allowed to change from point to point. A globally constant α has $\partial_\mu \alpha = 0$ and the obstruction vanishes, recovering the rigid symmetry of Chapter 7; a genuinely local α does not, and Eq. (8.18) says the naive kinetic term $\bar{\psi}\, i\gamma^\mu \partial_\mu \psi$ cannot survive.

This section resolves the obstruction. The strategy is not to abandon the demand for local invariance but to take it seriously and ask what the theory must contain in order to meet it. The answer is forced, and it is the central construction of the whole book: we must introduce a new field, the *gauge potential* A_μ, whose sole job is to compensate the inhomogeneous term, and we must replace the ordinary derivative by a modified *covariant derivative* D_μ that differentiates and compensates in one stroke. Demanding that $D_\mu \psi$ transform like ψ then fixes both the form of D_μ and the transformation law of A_μ uniquely, and feeding D_μ back into the matter Lagrangian — the rule called *minimal coupling* — manufactures the entire interaction between matter and the electromagnetic field out of nothing but the requirement of local symmetry. This is the gauge principle in action, and it is how this book launches QED at its classical layer.

8.2.1 The derivative as a comparator, and why it fails

To see what to do, it helps to recall what a derivative *is*. The directional derivative of ψ along a unit vector n^μ is the limit

$$n^\mu \partial_\mu \psi(x) \;=\; \lim_{\varepsilon \to 0} \frac{\psi(x + \varepsilon n) - \psi(x)}{\varepsilon}, \tag{8.19}$$

a comparison between the value of the field at two infinitesimally separated points. For an ordinary function this comparison is unambiguous because the values $\psi(x + \varepsilon n)$ and $\psi(x)$ live in the same space and can simply be subtracted. But ψ is a charged field: it carries an internal U(1) phase, and under a local transformation the phase convention used at x is $e^{iq\alpha(x)}$ while the one used at $x + \varepsilon n$ is $e^{iq\alpha(x+\varepsilon n)}$ — *different* conventions. Subtracting $\psi(x + \varepsilon n)$ from $\psi(x)$ is then like subtracting two vectors expressed in two different bases: the result is meaningless because the two terms do not transform the same way. This is the geometric content of the obstruction in Eq. (8.18).

The cure is to supply a rule for carrying the internal phase from one point to its neighbour before comparing — a *comparator*, or in the geometric language of Section 8.3 a *connection*. We posit a quantity $U(x + \varepsilon n, x)$, a U(1) phase factor (the parallel transporter studied in Section 8.3), whose job is to translate the phase convention at x into the phase convention at $x + \varepsilon n$, transforming by construction as $U(y, x) \to e^{iq\alpha(y)}\, U(y, x)\, e^{-iq\alpha(x)}$ so that the transported field $U(x+\varepsilon n, x)\, \psi(x)$ rotates with the same phase $e^{iq\alpha(x+\varepsilon n)}$ as the genuine field $\psi(x + \varepsilon n)$ sitting at the destination. Their difference is then a legitimate comparison. Writing the comparator for an infinitesimal step as $U(x + \varepsilon n, x) = 1 + i\varepsilon q\, n^\mu A_\mu(x) + O(\varepsilon^2)$ introduces a vector field $A_\mu(x)$ — the gauge potential — whose coefficient, the

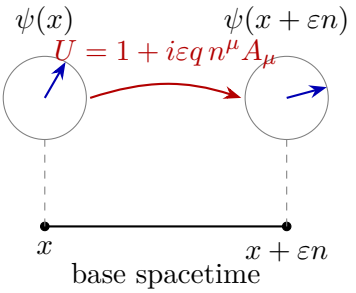


Figure 8.2: The covariant derivative as a comparator. The charged field ψ carries an internal U(1) phase (the arrow in each circle). Because a local transformation rotates the phase by a *different* amount at x and at $x+\varepsilon n$, the two values cannot be compared directly. The connection A_μ supplies the parallel-transport rule U that carries the phase from x to its neighbour before the comparison; the covariant derivative D_μ is the resulting transport-corrected difference.

real constant q, we will identify as the coupling strength (the charge). The comparator construction is taken up systematically in the next section, where the same A_μ reappears as the connection of a fibre bundle. For the present purpose its value is to tell us the form of the corrected derivative, which we now simply adopt as a definition and justify directly; the equivalence of this definition with the infinitesimal limit of the comparator is shown in Section 8.3.

8.2.2 The gauge potential and the covariant derivative

We now make the construction precise. Both objects below carry the chapter's sign and charge conventions, fixed once here and used unchanged for the rest of QED.

Definition 8.4 (Gauge potential and covariant derivative)**.** Let ψ be a matter field of charge q transforming under the local internal U(1) as $\psi(x) \to \psi'(x) = e^{iq\alpha(x)}\psi(x)$ with $\alpha : \mathbb{R}^{1,3} \to \mathbb{R}$ a smooth real function and q a real constant, the *coupling* (the charge carried by ψ). Introduce a real vector field $A_\mu(x)$, the *gauge potential*, transforming as a Lorentz four-vector (Chapter 3). The *gauge-covariant derivative* acting on ψ is

$$D_\mu\psi \;\equiv\; (\partial_\mu - i\,q\,A_\mu)\psi. \tag{8.20}$$

The gauge potential is declared to transform under the same local U(1) by the *inhomogeneous* (affine) rule

$$A_\mu(x) \;\longrightarrow\; A'_\mu(x) \;=\; A_\mu(x) + \partial_\mu\alpha(x). \tag{8.21}$$

Three features of Theorem 8.4 deserve emphasis before we prove that it works. First, Eq. (8.20) adopts the global sign convention of the book, $D_\mu = \partial_\mu - iqA_\mu$

(STYLE convention; see Section 8.2.4 below for the reconciliation with the electron's positive charge magnitude). Second, the charge q appears in two places at once — inside the phase $e^{iq\alpha}$ that rotates the matter field and inside the coupling $-iqA_\mu$ of the covariant derivative — and it is the *same* q in both. That this single constant does double duty is the seed of charge universality, developed below. Third, the transformation law Eq. (8.21) is not an independent postulate we are free to choose; it is forced. The whole point of A_μ is to cancel the inhomogeneous term in Eq. (8.18), and as the next theorem shows, Eq. (8.21) is the unique law that accomplishes this. The transformation is *inhomogeneous* because it shifts A_μ by a gradient, $\partial_\mu\alpha$, rather than merely rephasing it; this gradient shift is exactly the gauge freedom of the electromagnetic potential familiar from undergraduate electromagnetism, here *derived* rather than assumed. That A'_μ remains a real four-vector is automatic: α is a real scalar, so $\partial_\mu\alpha$ is a real four-vector (Chapter 3), and the sum of two real four-vectors is one. Note also that Eq. (8.21) carries no factor of the charge: a single gauge potential A_μ shifts the same way no matter which charged field it is being compared against, so one A_μ can serve fields of *different* charges $q_1, q_2, \ldots$ simultaneously, each with its own covariant derivative $\partial_\mu - iq_iA_\mu$ — exactly as the one electromagnetic field couples to electrons, protons and muons at once.

Theorem 8.5 (Covariance of the gauge-covariant derivative)**.** *With D_μ defined by Eq. (8.20) and A_μ transforming by Eq. (8.21), the covariant derivative of a charge-q field transforms homogeneously, with exactly the same phase as the field itself:*

$$D_\mu\psi \;\longrightarrow\; D'_\mu\psi' \;=\; e^{iq\alpha(x)}\,(D_\mu\psi). \tag{8.22}$$

Equivalently, $D_\mu\psi$ lies in the same internal frame as ψ: the combination $D_\mu\psi$ rotates rigidly under the local symmetry, so any expression built from ψ, $\bar\psi$ and their covariant derivatives by contracting all internal phases to zero is locally U(1)*-invariant.*

Proof. We substitute the two transformation laws — the matter law $\psi \to e^{iq\alpha}\psi$ and the potential law Eq. (8.21) — into the primed covariant derivative and carry every term. By definition,

$$D'_\mu\psi' \;=\; (\partial_\mu - i\,q\,A'_\mu)\,\psi' \;=\; \partial_\mu(e^{iq\alpha}\psi) \;-\; i\,q\,A'_\mu\,e^{iq\alpha}\psi. \tag{8.23}$$

The first term we already expanded in Eq. (8.18):

$$\partial_\mu(e^{iq\alpha}\psi) \;=\; e^{iq\alpha}\,\partial_\mu\psi \;+\; i\,q\,(\partial_\mu\alpha)\,e^{iq\alpha}\psi. \tag{8.24}$$

For the second term we insert the potential's transformation law Eq. (8.21), $A'_\mu = A_\mu + \partial_\mu\alpha$:

$$-i\,q\,A'_\mu\,e^{iq\alpha}\psi \;=\; -i\,q(A_\mu + \partial_\mu\alpha)e^{iq\alpha}\psi \;=\; -i\,q\,A_\mu\,e^{iq\alpha}\psi \;-\; i\,q\,(\partial_\mu\alpha)\,e^{iq\alpha}\psi, \tag{8.25}$$

where the gradient shift of A_μ has produced a term $-iq(\partial_\mu\alpha)e^{iq\alpha}\psi$ carrying the *same* factor iq as the obstruction, but with the opposite sign — this matched sign and magnitude is the reason the gradient shift in the transformation law was chosen as it was. Adding Eq. (8.24) and Eq. (8.25),

$$D'_\mu\psi' = e^{iq\alpha}\,\partial_\mu\psi \;+\; \underbrace{i\,q\,(\partial_\mu\alpha)\,e^{iq\alpha}\psi}_{\text{from }\partial_\mu\psi'} \;-\; i\,q\,A_\mu\,e^{iq\alpha}\psi \;\underbrace{-\,i\,q\,(\partial_\mu\alpha)\,e^{iq\alpha}\psi}_{\text{from }A'_\mu}$$
$$= e^{iq\alpha}\,\partial_\mu\psi \;-\; i\,q\,A_\mu\,e^{iq\alpha}\psi \;=\; e^{iq\alpha}(\partial_\mu - i\,q\,A_\mu)\psi \;=\; e^{iq\alpha}\,D_\mu\psi. \qquad (8.26)$$

The two inhomogeneous pieces $\pm\,iq(\partial_\mu\alpha)e^{iq\alpha}\psi$ — one produced by differentiating the field's phase, the other produced by the gradient shift of the potential — are equal and opposite, and they cancel identically. What survives is the field's covariant derivative dressed by the single overall phase $e^{iq\alpha}$, which is Eq. (8.22). This proves the theorem. □

The cancellation in Eq. (8.26) is the entire mechanism of gauge theory in miniature, and it is worth naming what each piece did. The matter field's phase rotation generated $+iq(\partial_\mu\alpha)\psi$; this is the obstruction. The gauge potential's gradient shift generated $-iq(\partial_\mu\alpha)\psi$; this is the compensation. They are forced to be equal and opposite because the charge q in the phase $e^{iq\alpha}$ is the very same q that multiplies A_μ in D_μ, and that is no accident — it is the content of the demand that A_μ be a comparator for the phase carried by ψ. The gauge potential is, in this precise sense, the field whose transformation is rigged to absorb the inhomogeneity of a local phase.

Physical Insight

The covariant derivative is the operation "differentiate, then undo the local phase slip you just introduced." The field A_μ stores, at every point, the infinitesimal rotation needed to compare the internal phase at x with the phase an instant away; D_μ subtracts that rotation off. Where an ordinary $\partial_\mu\psi$ measures both the genuine change of ψ and the spurious change in the phase convention, $D_\mu\psi$ measures only the genuine change. This is why $D_\mu\psi$ transforms like ψ while $\partial_\mu\psi$ does not: the connection has removed the part of the derivative that was merely bookkeeping.

8.2.3 Minimal coupling: the matter–photon interaction, derived

Theorem 8.5 hands us a recipe. Any matter Lagrangian that was invariant under the *global* U(1) of Chapter 7 was built so that every internal phase cancels — a ψ always paired with a $\bar\psi$, an $e^{iq\alpha}$ with an $e^{-iq\alpha}$. The only obstruction to promoting that global invariance to a local one was the derivative, whose transformation Eq. (8.18) spoiled the cancellation. But $D_\mu\psi$ transforms exactly

as ψ does. Therefore, if we take the globally invariant matter Lagrangian and replace every ordinary derivative by a covariant one, the resulting Lagrangian is invariant under the *local* U(1), because now even the derivative terms carry the field's phase and cancel against their conjugates. This is the rule of *minimal coupling*.

Definition 8.6 (Minimal coupling)**.** Given a matter Lagrangian $\mathcal{L}_{\text{matter}}(\psi, \partial_\mu \psi)$ invariant under the global U(1), the *minimally coupled* Lagrangian is obtained by the substitution

$$\partial_\mu \;\longrightarrow\; D_\mu = \partial_\mu - i\, q\, A_\mu \tag{8.27}$$

applied to the matter field (Chapter 6 supplies the scalar, Dirac and Maxwell Lagrangians on which the substitution acts). The result $\mathcal{L}_{\text{matter}}(\psi, D_\mu \psi)$ is invariant under the local U(1) of Theorem 8.4.

The local invariance is immediate from Theorem 8.5: wherever $\partial_\mu \psi$ stood, the substitution puts $D_\mu \psi$, which transforms as $e^{iq\alpha} D_\mu \psi$ exactly as $\psi \to e^{iq\alpha}\psi$; the Lagrangian was already arranged so that each ψ-phase meets a compensating $\bar{\psi}$-phase, and the substitution preserves that pairing. We verify this explicitly, term by term, in Theorem 8.7 below.

The name "minimal" signals that this is the simplest coupling that does the job: it introduces no new free parameters beyond the single coupling q already present in D_μ, and it adds the fewest possible derivatives. One could imagine non-minimal gauge-invariant couplings — for instance a Pauli term $\bar{\psi}\sigma^{\mu\nu}\psi\, F_{\mu\nu}$ built from the field strength of the next section — and such terms are perfectly consistent with local U(1) invariance; what minimal coupling asserts is that the *leading* interaction, the one forced on us merely by gauging the derivative, is Eq. (8.27). That the leading interaction of light with charged matter should be dictated by symmetry alone, with no freedom left except the value of the charge, is the gauge principle's most striking classical claim.

8.2.4 The electron, and the interaction current

We now specialize to the carried example of QED: a single Dirac field ψ describing the electron. Its Lagrangian, introduced in Chapter 6, is the free Dirac Lagrangian $\mathcal{L}_{\text{Dirac}} = \bar{\psi}(i\gamma^\mu \partial_\mu - m)\psi$. Minimal coupling, Eq. (8.27), turns it into

$$\mathcal{L} \;=\; \bar{\psi}(i\gamma^\mu D_\mu - m)\psi \;=\; \bar{\psi}(i\not{D} - m)\psi, \tag{8.28}$$

where $\not{D} = \gamma^\mu D_\mu$ in the slash notation of Chapter 3. This is the matter sector of the classical QED Lagrangian; the photon's own kinetic term, built from the curvature of Section 8.3, completes it.

The electron's charge fixes q. With the global convention $D_\mu = \partial_\mu - iqA_\mu$ and the physical fact that the electron carries charge $-e$ with $e > 0$ the elementary

charge, we set $q = -e$, so that the electron field rotates by $\psi \to e^{-ie\alpha}\psi$ and

$$D_\mu\psi \;=\; (\partial_\mu + i\,e\,A_\mu)\psi \qquad (\text{electron},\; q=-e,\; e>0). \tag{8.29}$$

The gauge law Eq. (8.21) carries no charge, so it is unchanged for the electron, $A_\mu \to A_\mu + \partial_\mu\alpha$. Equation (8.29) is the electron covariant derivative used throughout the rest of QED.

Notation Watch

The book's global convention (STYLE) is $D_\mu = \partial_\mu - iqA_\mu$, with the *signed* charge q inserted directly. The electron has $q = -e$ where $e > 0$ is the magnitude of the elementary charge, so on the electron field the two minus signs combine to give the manifestly positive-looking $D_\mu = \partial_\mu + ieA_\mu$ of Eq. (8.29). There is no contradiction: $\partial_\mu - iqA_\mu$ and $\partial_\mu + ieA_\mu$ are the same operator once $q = -e$ is substituted. When in doubt, use the signed form $D_\mu = \partial_\mu - iqA_\mu$ and let the sign of q take care of itself.

A second, independent convention choice concerns where the charge is carried in the transformation law. We have placed the charge in *both* the matter phase $\psi \to e^{iq\alpha}\psi$ and the coupling $-iqA_\mu$ of D_μ, with the potential shifting by the charge-free gradient $A_\mu \to A_\mu + \partial_\mu\alpha$. This is the convention used uniformly across Section 8.1 and Section 8.3, and its virtue is that a single A_μ serves all charges at once. Some texts (e.g. Peskin–Schroeder) instead strip the charge from the phase, writing $\psi \to e^{i\alpha}\psi$ with the potential shifting by $A_\mu \to A_\mu + \frac{1}{q}\partial_\mu\alpha$; the obstruction then reads $i(\partial_\mu\alpha)\psi$ and the covariance statement Eq. (8.22) becomes $D_\mu\psi \to e^{i\alpha}D_\mu\psi$. The two are related by the harmless rescaling $\alpha \to q\alpha$ of the gauge parameter and describe identical physics. The non-Abelian rule generalized in Chapter 25, $D_\mu = \partial_\mu - igA^a_\mu T^a$ with $\psi \to U\psi = e^{i\alpha^a T^a}\psi$ and $A_\mu \to UA_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}$, sits between the two Abelian conventions: there the coupling g lives only in D_μ and the gauge law, while the group element U carries none. Our Abelian convention is recovered from the non-Abelian one by the field-dependent identification $g\alpha^a T^a \to q\alpha$, after which the matter phase $e^{iq\alpha}$ and the gradient shift $\partial_\mu\alpha$ emerge as stated; the apparent $\frac{1}{g}$ in the non-Abelian shift is the artefact of keeping g out of the exponent, exactly the choice we declined to make in the Abelian case.

Expanding Eq. (8.28) exposes the physics minimal coupling has manufactured. Writing $i\not{D} = i\gamma^\mu\partial_\mu + q\gamma^\mu A_\mu$ (general q, using $iD_\mu = i\partial_\mu + qA_\mu$),

$$\mathcal{L} \;=\; \bar\psi(i\gamma^\mu\partial_\mu - m)\psi \;+\; q\,\bar\psi\gamma^\mu\psi\,A_\mu, \tag{8.30}$$

the first group being the free Dirac Lagrangian of Chapter 6 and the second a genuine *interaction* term coupling the matter to the gauge field. It has a structure that will recur at every later stage of QED: the gauge field A_μ is contracted

against a matter current. Recall from Chapter 7 that the global U(1) symmetry $\psi \to e^{i\alpha}\psi$ of the free Dirac theory has, by Noether's first theorem, the conserved *number* current

$$j^\mu \;=\; \bar{\psi}\gamma^\mu\psi, \qquad \partial_\mu j^\mu = 0, \tag{8.31}$$

the same current whose charge counts electrons minus positrons. The interaction term in Eq. (8.30) is precisely this current, weighted by the coupling, dotted into the potential:

$$\mathcal{L}_{\text{int}} \;=\; q\,\bar{\psi}\gamma^\mu\psi\,A_\mu \;=\; q\,j^\mu A_\mu \;\equiv\; J^\mu A_\mu, \qquad J^\mu \equiv q\,j^\mu = q\,\bar{\psi}\gamma^\mu\psi. \tag{8.32}$$

Here J^μ is the *electromagnetic* (electric) current: the Noether number current of Chapter 7 multiplied by the charge q. For the electron, $q = -e$, so $J^\mu = -e\,\bar{\psi}\gamma^\mu\psi$ — carrying the physical sign of the electron's charge. (The charge-weighted current J^μ is the Dirac counterpart of the already-charge-weighted scalar current of (8.3), where the factor q was carried inside j^μ from the outset; the two conventions agree on the physically meaningful object J^μ that couples to A_μ.) The appearance of the coupling–current combination $J^\mu A_\mu$ is the first sighting of the structure that locks symmetry to dynamics: the very current whose conservation expressed the global symmetry of Chapter 7 is, after gauging, the source to which the gauge field couples. We will see in the next sections that this same J^μ turns out to be the source term in the Maxwell equations, so that the conserved charge of the symmetry and the charge that sources the electromagnetic field are one and the same object — the deepest classical consequence of the gauge principle.

Physical Insight

Minimal coupling explains, in one line, why all particles of a given charge couple to the electromagnetic field with the *same* strength: the coupling is the single constant q in D_μ, and it is fixed by how the field transforms under U(1) — it is the very integer (in units of e) that appears in the phase $e^{iq\alpha}$ — not by any independent choice. Universality of the electromagnetic coupling — that the proton and positron couple to the photon with charge exactly $+e$, the electron with exactly $-e$ — is the gauge principle's prediction, not an experimental coincidence.

8.2.5 Worked verification of local invariance

The claim that minimal coupling yields a locally invariant action is the load-bearing assertion of this section. Theorem 8.5 already proves it in the abstract; we now discharge it concretely, carrying the gauge transformation of every term in the minimally coupled Dirac Lagrangian Eq. (8.28) and exhibiting the cancellation explicitly. We do it twice: once the short way, using covariance directly, and once the long way, expanding into a free piece plus an interaction

and watching the inhomogeneous gradient terms cancel between them. The two routes must — and do — agree.

Worked Example 8.7 (Local U(1) invariance of the coupled Dirac Lagrangian)**.** Consider $\mathcal{L} = \bar{\psi}(i\gamma^\mu D_\mu - m)\psi$ with $D_\mu = \partial_\mu - iqA_\mu$ and the local transformations

$$\psi \to \psi' = e^{iq\alpha}\psi, \qquad \bar{\psi} \to \bar{\psi}' = e^{-iq\alpha}\bar{\psi}, \qquad A_\mu \to A'_\mu = A_\mu + \partial_\mu \alpha, \tag{8.33}$$

where $\alpha = \alpha(x)$ is a smooth real function. The transformation of $\bar{\psi}$ follows from that of ψ together with $\bar{\psi} = \psi^\dagger\gamma^0$ and the reality of α: $\bar{\psi}' = (e^{iq\alpha}\psi)^\dagger\gamma^0 = e^{-iq\alpha}\psi^\dagger\gamma^0 = e^{-iq\alpha}\bar{\psi}$. We show $\mathcal{L} \to \mathcal{L}$, i.e. $\mathcal{L}$ is locally invariant.

Short route (via covariance). The mass term involves no derivative and is invariant on its own:

$$m\,\bar{\psi}'\psi' \;=\; m\,e^{-iq\alpha}\bar{\psi}\,e^{iq\alpha}\psi \;=\; m\,\bar{\psi}\psi, \tag{8.34}$$

the two phases meeting and cancelling because α is a real scalar and so commutes through the Dirac matrices and spinors. For the kinetic term we invoke Theorem 8.5, $D_\mu\psi \to e^{iq\alpha}D_\mu\psi$, together with $\bar{\psi} \to e^{-iq\alpha}\bar{\psi}$:

$$\bar{\psi}'\,i\gamma^\mu D'_\mu\psi' \;=\; (e^{-iq\alpha}\bar{\psi})\,i\gamma^\mu\,(e^{iq\alpha}D_\mu\psi) \;=\; e^{-iq\alpha}e^{iq\alpha}\,\bar{\psi}\,i\gamma^\mu D_\mu\psi \;=\; \bar{\psi}\,i\gamma^\mu D_\mu\psi. \tag{8.35}$$

Adding Eq. (8.35) and $-$Eq. (8.34) gives $\mathcal{L} \to \mathcal{L}$. The decisive step is that $D'_\mu\psi'$ carries the *same* phase $e^{iq\alpha}$ as ψ', so that it cancels against the conjugate phase of $\bar{\psi}'$ just as the undifferentiated ψ' would — which is exactly what covariance was constructed to guarantee.

Long route (term by term). To see the cancellation without invoking the theorem, expand the Lagrangian as in Eq. (8.30) into a free part and an interaction part, and transform each separately. The free part is $\mathcal{L}_0 = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$ and the interaction is $\mathcal{L}_{\text{int}} = q\,\bar{\psi}\gamma^\mu\psi\,A_\mu$. Under Eq. (8.33), the free kinetic term transforms as

$$\begin{aligned}\bar{\psi}'\,i\gamma^\mu\partial_\mu\psi' &= e^{-iq\alpha}\bar{\psi}\,i\gamma^\mu\,\partial_\mu(e^{iq\alpha}\psi) = e^{-iq\alpha}\bar{\psi}\,i\gamma^\mu\,e^{iq\alpha}(\partial_\mu\psi + iq(\partial_\mu\alpha)\psi)\\ &= \bar{\psi}\,i\gamma^\mu\partial_\mu\psi \;+\; \bar{\psi}\,i\gamma^\mu\,iq(\partial_\mu\alpha)\psi \;=\; \bar{\psi}\,i\gamma^\mu\partial_\mu\psi \;-\; q\,(\partial_\mu\alpha)\,\bar{\psi}\gamma^\mu\psi,\end{aligned} \tag{8.36}$$

using $\partial_\mu(e^{iq\alpha}\psi) = e^{iq\alpha}(\partial_\mu\psi + iq(\partial_\mu\alpha)\psi)$ from Eq. (8.18) and $i\cdot i = -1$. The free kinetic term is therefore *not* invariant on its own: it picks up the spurious gradient term $-q(\partial_\mu\alpha)\,\bar{\psi}\gamma^\mu\psi = -(\partial_\mu\alpha)J^\mu$, which is the obstruction made manifest. The mass term is invariant exactly as in Eq. (8.34), so $\mathcal{L}_0$ shifts by $\delta\mathcal{L}_0 = -q(\partial_\mu\alpha)\,\bar{\psi}\gamma^\mu\psi$.

The interaction term transforms by combining the invariance of the bilinear $\bar{\psi}\gamma^\mu\psi$ (its two phases cancel, as in the mass term) with the inhomogeneous shift of A_μ:

$$q\,\bar{\psi}'\gamma^\mu\psi'\,A'_\mu = q\,(e^{-iq\alpha}\bar{\psi})\gamma^\mu(e^{iq\alpha}\psi)\,(A_\mu + \partial_\mu\alpha) = q\,\bar{\psi}\gamma^\mu\psi(A_\mu + \partial_\mu\alpha)$$

$$= q\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu} \;+\; q\,(\partial_{\mu}\alpha)\,\bar{\psi}\gamma^{\mu}\psi. \tag{8.37}$$

The interaction term thus shifts by $\delta\mathcal{L}_{\text{int}} = +q(\partial_{\mu}\alpha)\,\bar{\psi}\gamma^{\mu}\psi$. Adding the two shifts,

$$\delta\mathcal{L} \;=\; \delta\mathcal{L}_0 + \delta\mathcal{L}_{\text{int}} \;=\; -q(\partial_{\mu}\alpha)\,\bar{\psi}\gamma^{\mu}\psi \;+\; q(\partial_{\mu}\alpha)\,\bar{\psi}\gamma^{\mu}\psi \;=\; 0, \tag{8.38}$$

the obstruction from the free derivative is cancelled exactly by the inhomogeneous shift of the gauge potential. The Lagrangian is locally U(1)-invariant, $\mathcal{L} \to \mathcal{L}$, in agreement with the short route.

Equation (8.38) is the long route's payoff and it teaches the mechanism better than the theorem alone: the offending gradient term $-(\partial_{\mu}\alpha)J^{\mu}$ produced by the free kinetic term is precisely the source-coupling $+(\partial_{\mu}\alpha)J^{\mu}$ produced by the gauge variation of the interaction. Minimal coupling works because the interaction term $J^{\mu}A_{\mu}$ was built from the same current $J^{\mu} = q\,j^{\mu}$ whose non-invariance it must cancel — a self-consistency that the gauge principle enforces automatically. The identical computation goes through verbatim for a charged scalar with $\mathcal{L} = (D_{\mu}\phi)^{\dagger}(D^{\mu}\phi) - m^2\phi^{\dagger}\phi$: the two covariant factors each contribute a phase $e^{\pm iq\alpha}$ that cancel by Theorem 8.5, so minimal coupling renders the charged-scalar kinetic term locally invariant by the same token. Local U(1) invariance is thus not special to the Dirac field; it is a property of the covariant derivative, and any matter field that carries a U(1) charge inherits it.

We have, then, accomplished the section's task. The obstruction of Section 8.1 is resolved: a new vector field A_{μ} with the inhomogeneous transformation law Eq. (8.21) compensates the local phase slip, the covariant derivative $D_{\mu} = \partial_{\mu} - iqA_{\mu}$ transforms charged fields covariantly (Theorem 8.5), and minimal coupling Eq. (8.27) delivers a locally U(1)-invariant matter action whose interaction term $J^{\mu}A_{\mu}$ is the gauge field coupled to the conserved current of Chapter 7. Yet the field we were forced to introduce remains curiously incomplete: it has arrived equipped with a transformation law but with no dynamics of its own — no kinetic term, no equation of motion, nothing in the Lagrangian that lets A_{μ} propagate. Worse, the transformation law Eq. (8.21) shows that A_{μ} itself is not gauge invariant; it shifts by a gradient, so A_{μ} cannot appear bare in any invariant action. To give the gauge potential dynamics we must first build a gauge-invariant object out of it, and the only invariant combination of derivatives of A_{μ} turns out to be the antisymmetrized one — the curvature of the connection. Constructing that object, the field strength $F_{\mu\nu}$, from the commutator of covariant derivatives, and reading off from it the geometry of the gauge field, is the work of Section 8.3.

8.3 The Field Strength as Curvature

The previous section handed us a new object and an embarrassment. Demanding that the global U(1) phase symmetry of the matter Lagrangian survive when the

phase is allowed to vary from point to point forced into existence a vector field A_μ and a modified derivative, the gauge-covariant derivative $D_\mu = \partial_\mu - iqA_\mu$ of Theorem 8.4, whose defining virtue is that it transforms homogeneously: under a local gauge transformation $\psi \to e^{iq\alpha(x)}\psi$, $A_\mu \to A_\mu + \partial_\mu\alpha$, the combination $D_\mu\psi$ rotates by the same phase as ψ itself (Theorem 8.5). That is exactly enough to build a gauge-invariant matter action by minimal coupling. But A_μ entered as pure bookkeeping: it has, as yet, no dynamics, no equation of motion, no Lagrangian of its own. It is a field that does things to matter but is not itself a thing. The embarrassment is that we cannot simply write down a kinetic term for it, because the naive candidates—$\partial_\mu A_\nu\,\partial^\mu A^\nu$, $A_\mu A^\mu$, and the like—are not gauge invariant: the gauge freedom $A_\mu \to A_\mu + \partial_\mu\alpha$ that made D_μ work now obstructs any term built directly from A_μ and its bare derivatives.

To give A_μ dynamics we must first extract from it the gauge-invariant information it carries. The potential A_μ is partly physical and partly redundant: the gradient $\partial_\mu\alpha$ can be added at will, so any single component of A_μ can be set to anything at a point by a suitable choice of gauge. What is the gauge-invariant residue? The covariant derivative itself supplies the answer. We built D_μ to compare the field at infinitesimally separated points in a gauge-respecting way; the gauge-invariant content of A_μ is precisely the *failure of that comparison to be path-independent*. Quantified, that failure is a single antisymmetric tensor, the field strength $F_{\mu\nu}$, and it is—in the most literal differential-geometric sense the chapter will make good on—the *curvature* of the gauge connection A_μ. This section constructs $F_{\mu\nu}$ from the commutator of covariant derivatives, proves it gauge invariant, derives the identity it satisfies for free (the Bianchi identity), reads off its components as the electric and magnetic fields, and gives the geometric picture—parallel transport and holonomy around a loop—that both explains the word "curvature" and sets up the non-abelian generalization of Chapter 25.

8.3.1 The commutator of covariant derivatives

A single covariant derivative is gauge covariant, but it is not, by itself, gauge invariant: it carries a free index and transforms as $D_\mu\psi \to e^{iq\alpha}D_\mu\psi$. To manufacture something gauge *invariant* we need to compare derivatives, and the cleanest comparison is to ask whether covariant differentiation in two different directions commutes. For ordinary partial derivatives the answer is always yes, $[\partial_\mu, \partial_\nu] = 0$, because mixed partials of a smooth function are symmetric. Covariant derivatives are different: each carries a copy of the potential A_μ, and the potentials at the two stages of the double derivative need not cancel. The amount by which they fail to cancel is the object we are after.

We compute the commutator $[D_\mu, D_\nu]$ acting on a charged field ψ, carrying every term. With $D_\mu = \partial_\mu - iqA_\mu$,

$$D_\mu D_\nu\psi = (\partial_\mu - iqA_\mu)(\partial_\nu\psi - iqA_\nu\psi)$$

$$= \partial_\mu \partial_\nu \psi - iq\,\partial_\mu(A_\nu \psi) - iq A_\mu\,\partial_\nu \psi + (-iq)(-iq) A_\mu A_\nu \psi$$
$$= \partial_\mu \partial_\nu \psi - iq\,(\partial_\mu A_\nu)\,\psi - iq\,A_\nu\,\partial_\mu \psi - iq\,A_\mu\,\partial_\nu \psi - q^2 A_\mu A_\nu\,\psi. \quad (8.39)$$

The commutator is the antisymmetrization of (8.39) in $\mu \leftrightarrow \nu$. Take the difference term by term. The pure second-derivative term is symmetric and cancels,

$$\partial_\mu \partial_\nu \psi - \partial_\nu \partial_\mu \psi = 0, \quad (8.40)$$

this being the equality of mixed partials for the smooth field ψ. The two terms in which a single A multiplies a first derivative of ψ also cancel: the pair $-iq\,A_\nu\,\partial_\mu \psi - iq\,A_\mu\,\partial_\nu \psi$ is already symmetric under $\mu \leftrightarrow \nu$, so subtracting its swap gives zero. The quadratic term $-q^2 A_\mu A_\nu \psi$ is symmetric as well—A_μ and A_ν are ordinary commuting functions in the abelian theory—and cancels in the same way. The only survivor is the term in which the derivative lands on the potential:

$$[D_\mu, D_\nu]\,\psi = -iq(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi. \quad (8.41)$$

Three features of (8.41) deserve emphasis before we name the result. First, all derivatives of ψ have dropped out: the right-hand side is ψ multiplied by a function of A_μ alone. The commutator $[D_\mu, D_\nu]$ is therefore not a differential operator at all but an ordinary (multiplicative) function—an algebraic object built purely from the potential. Second, that function is manifestly antisymmetric in $\mu\nu$. Third, the ψ on which we acted has disappeared from the structure of the answer; the operator identity

$$[D_\mu, D_\nu] = -iq\,F_{\mu\nu} \quad (8.42)$$

holds on every charged field, whatever its charge q. This last point is what makes (8.42) a clean *definition*.

Definition 8.8 (Field strength)**.** The *electromagnetic field strength* $F_{\mu\nu}$ is the antisymmetric two-tensor defined by the commutator of covariant derivatives,

$$[D_\mu, D_\nu] = -iq\,F_{\mu\nu}. \quad (8.43)$$

In the abelian (U(1)) theory the computation (8.41) evaluates this to

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \quad (8.44)$$

an antisymmetric tensor of the kind catalogued in Chapter 3, with $F_{\nu\mu} = -F_{\mu\nu}$ and hence six independent components in four spacetime dimensions.

The defining relation (8.43) is more fundamental than the explicit formula (8.44), because it is the relation that survives generalization. Nothing in the derivation of (8.41) used that the gauge group was U(1) until the very last step, where we cancelled the quadratic term $-q^2 A_\mu A_\nu \psi$ against its swap on the grounds that

the potentials commute. In a non-abelian theory the potential is matrix-valued, $A_\mu = A_\mu^a T^a$ with non-commuting generators $\left[T^a, T^b\right] = i f^{abc} T^c$ (Chapter 2), and the quadratic term does *not* cancel; it leaves behind a commutator $[A_\mu, A_\nu]$ that adds a term $g f^{abc} A_\mu^b A_\nu^c$ to the field strength. That self-interaction term—the structural origin of everything that makes Yang–Mills theory harder and richer than electrodynamics—is the subject of Chapter 25. Here the gauge group is abelian, the generators are a single number, $f^{abc} \equiv 0$, and the term is simply absent. We flag its provenance now so that the appearance of $F_{\mu\nu}$ as a commutator reads, in hindsight, as the abelian shadow of a general construction.

Physical Insight

The field strength is not an extra postulate bolted onto the gauge field; it is forced. Once we accept that comparing a charged field at neighbouring points requires the covariant derivative D_μ, the question "does the order of comparison matter?" is unavoidable, and its answer *is* $F_{\mu\nu}$. The potential A_μ tells you how to parallel-transport a phase along a path; the field strength tells you that transport around a closed loop need not return you to where you started. Everything physical about the electromagnetic field lives in that discrepancy, and the discrepancy is gauge invariant precisely because it is a statement about the geometry of transport, not about the arbitrary phase convention you adopt at each point.

8.3.2 Gauge invariance of the field strength

The commutator construction already strongly suggests that $F_{\mu\nu}$ is gauge invariant. The operator $[D_\mu, D_\nu]$ acts on a charged field by simple multiplication, and under a gauge transformation it conjugates the way a covariant operator must: since $D_\mu \to e^{iq\alpha} D_\mu e^{-iq\alpha}$ (this is the content of covariance, Theorem 8.5), the commutator goes to $e^{iq\alpha}\,[D_\mu, D_\nu]\,e^{-iq\alpha}$, and because $[D_\mu, D_\nu] = -iqF_{\mu\nu}$ is mere multiplication by a function, the two phase factors—which are themselves just multiplication by $e^{\pm iq\alpha}$—cancel, leaving $-iqF_{\mu\nu}$ unchanged. It is worth seeing the same conclusion fall directly out of the explicit transformation law of A_μ, with all the algebra on the page, because that computation pinpoints exactly which cancellation is doing the work.

Theorem 8.9 (Gauge invariance of $F_{\mu\nu}$)**.** *In the abelian theory the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is invariant under the gauge transformation $A_\mu \to A'_\mu = A_\mu + \partial_\mu \alpha$ for every smooth gauge function $\alpha(x)$:*

$$F'_{\mu\nu} = F_{\mu\nu}. \tag{8.45}$$

Proof. Substitute the transformed potential $A'_\mu = A_\mu + \partial_\mu \alpha$ directly into the

definition (8.44):

$$\begin{aligned} F'_{\mu\nu} &= \partial_\mu A'_\nu - \partial_\nu A'_\mu \\ &= \partial_\mu(A_\nu + \partial_\nu\alpha) - \partial_\nu(A_\mu + \partial_\mu\alpha) \\ &= (\partial_\mu A_\nu - \partial_\nu A_\mu) + (\partial_\mu\partial_\nu\alpha - \partial_\nu\partial_\mu\alpha) \\ &= F_{\mu\nu} + 0. \end{aligned} \tag{8.46}$$

The first bracket is $F_{\mu\nu}$ by definition. The second bracket vanishes identically: it is the antisymmetrization of $\partial_\mu\partial_\nu\alpha$, and for a smooth scalar function α the mixed second partials are symmetric, $\partial_\mu\partial_\nu\alpha = \partial_\nu\partial_\mu\alpha$, so the two terms cancel. Hence $F'_{\mu\nu} = F_{\mu\nu}$. □

The proof is a single line of algebra, but it carries the central physical message of the chapter, so it is worth restating in words. The gauge redundancy of the potential is exactly the freedom to add a pure gradient $\partial_\mu\alpha$. A gradient is curl-free; the antisymmetrized derivative $\partial_{[\mu}\partial_{\nu]}\alpha$ that $F_{\mu\nu}$ would see is therefore zero. The field strength is built to be blind to precisely the part of A_μ that gauge transformations can change. This is why $F_{\mu\nu}$, and not A_μ, is the carrier of physical, observable information about the electromagnetic field, and it is why the gauge-invariant kinetic term of the next section will be assembled out of $F_{\mu\nu}$ rather than out of A_μ directly.

Caution

Gauge invariance of $F_{\mu\nu}$ does not mean A_μ is unphysical scaffolding to be discarded. The potential carries information that the local field strength does not: the Aharonov–Bohm effect shows that a charged particle's quantum phase responds to $\oint A_\mu \, \mathrm{d}x^\mu$ around a loop even through regions where $F_{\mu\nu} = 0$. What is true is that only the gauge-invariant combinations of A_μ are observable, and around a closed loop that combination is precisely $\oint A_\mu \, \mathrm{d}x^\mu = \frac{1}{2}\int F_{\mu\nu} \, \mathrm{d}\sigma^{\mu\nu}$ by Stokes' theorem—the flux of the curvature, as we make explicit in Theorem 8.12.

8.3.3 The Bianchi identity

Because $F_{\mu\nu}$ is the antisymmetrized derivative of a potential, it is not an arbitrary antisymmetric tensor field: it satisfies an identity, automatically and independently of any equation of motion. This is the homogeneous, or Bianchi, identity. It is a *kinematic* constraint—true for any A_μ whatsoever, before dynamics are imposed—and we will see in the next section that it already encodes two of the four Maxwell equations, the two that have no source on their right-hand side.

Theorem 8.10 (Homogeneous Bianchi identity)**.** *The abelian field strength satisfies the cyclic identity*

$$\partial_\lambda F_{\mu\nu} + \partial_\mu F_{\nu\lambda} + \partial_\nu F_{\lambda\mu} = 0, \tag{8.47}$$

equivalently written $\partial_{[\lambda} F_{\mu\nu]} = 0$ *with square brackets denoting total antisymmetrization over the enclosed indices.*

We give two proofs. The first is direct and elementary; the second runs through the Jacobi identity of the covariant derivatives and is the form that survives the non-abelian generalization.

Proof by direct substitution. Insert $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ into each of the three terms of (8.47) and expand:

$$\begin{aligned} \partial_\lambda F_{\mu\nu} &= \partial_\lambda \partial_\mu A_\nu - \partial_\lambda \partial_\nu A_\mu, \\ \partial_\mu F_{\nu\lambda} &= \partial_\mu \partial_\nu A_\lambda - \partial_\mu \partial_\lambda A_\nu, \\ \partial_\nu F_{\lambda\mu} &= \partial_\nu \partial_\lambda A_\mu - \partial_\nu \partial_\mu A_\lambda. \end{aligned} \tag{8.48}$$

Add the three lines. Group the six terms into three pairs, each pair containing the same field component differentiated by the same two derivative operators in opposite order. The term $+\partial_\lambda \partial_\mu A_\nu$ cancels $-\partial_\mu \partial_\lambda A_\nu$; the term $-\partial_\lambda \partial_\nu A_\mu$ cancels $+\partial_\nu \partial_\lambda A_\mu$; and the term $+\partial_\mu \partial_\nu A_\lambda$ cancels $-\partial_\nu \partial_\mu A_\lambda$. Every cancellation is an instance of the symmetry of mixed partials, $\partial_\rho \partial_\sigma = \partial_\sigma \partial_\rho$, applied to the smooth potential. The sum vanishes, which is (8.47). □

Proof via the Jacobi identity. For any three linear operators X, Y, Z the Jacobi identity

$$[X, [Y, Z]] + [Y, [Z, X]] + [Z, [X, Y]] = 0 \tag{8.49}$$

holds identically, by expanding each double commutator and observing that all twelve resulting products cancel in pairs. Apply it to the three covariant derivatives $X = D_\lambda$, $Y = D_\mu$, $Z = D_\nu$. By the definition (8.43), each inner commutator is $[D_\mu, D_\nu] = -iqF_{\mu\nu}$, so

$$[D_\lambda, -iqF_{\mu\nu}] + [D_\mu, -iqF_{\nu\lambda}] + [D_\nu, -iqF_{\lambda\mu}] = 0. \tag{8.50}$$

It remains to evaluate $[D_\lambda, F_{\mu\nu}]$. Since $F_{\mu\nu}$ is a gauge-invariant (neutral) function, it acts on a charged field ψ by ordinary multiplication, and

$$[D_\lambda, F_{\mu\nu}]\,\psi = D_\lambda(F_{\mu\nu}\psi) - F_{\mu\nu}\, D_\lambda\psi = (\partial_\lambda F_{\mu\nu})\,\psi, \tag{8.51}$$

where the gauge-potential pieces $-iqA_\lambda F_{\mu\nu}\psi$ from the first term and $-F_{\mu\nu}(iqA_\lambda)\psi$ from the second cancel, leaving only the derivative landing on $F_{\mu\nu}$. (Geometrically, (8.51) says that on a neutral field the covariant derivative reduces to the ordinary one; in the non-abelian theory $F_{\mu\nu}$ is charged under the adjoint and this

step instead produces the *covariant* derivative $D_\lambda F_{\mu\nu}$, which is why the general Bianchi identity reads $D_{[\lambda}F_{\mu\nu]} = 0$—see Chapter 25.) Substituting (8.51) into (8.50) and dividing by the common factor $-iq$ gives exactly (8.47). □

The Jacobi route exposes why (8.47) is an *identity* and not a field equation: it descends from the associativity of operator composition, which holds for any A_μ whatsoever. No variational principle, no Lagrangian, no choice of dynamics is involved. Of Maxwell's four equations, two will turn out to be this kinematic identity in disguise, owed to nothing more than the existence of a potential; the other two are genuine dynamics, derived in Section 8.4 from the Maxwell action of the next section.

8.3.4 Electric and magnetic fields as components of $F_{\mu\nu}$

We have an antisymmetric two-tensor with six independent components and a claim that it contains all the gauge-invariant physics of the electromagnetic field. The reader's undergraduate electromagnetism also packaged that physics into six numbers—the three components of the electric field $\mathbf{E}$ and the three of the magnetic field $\mathbf{B}$. They are the same six numbers. Making the identification explicit both grounds the abstract construction and lets us read the Bianchi identity as physics.

Write the potential in terms of the scalar and vector potentials of undergraduate electromagnetism, $A^\mu = (\phi, \mathbf{A})$, so that with the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ the lower-index components are $A_\mu = (\phi, -\mathbf{A})$ and $\partial_\mu = (\partial_t, \boldsymbol{\nabla})$, $\partial^\mu = (\partial_t, -\boldsymbol{\nabla})$. The electric field is defined to be

$$E^i \equiv F^{i0}, \tag{8.52}$$

and we check this reproduces the textbook formula. Raising indices with the metric, $F^{i0} = \eta^{i\alpha}\eta^{0\beta}F_{\alpha\beta} = (-1)(+1)F_{i0} = -F_{i0}$, and from the definition $F_{i0} = \partial_i A_0 - \partial_0 A_i = \partial_i\phi - \partial_t(-A^i) = \partial_i\phi + \partial_t A^i$. Hence

$$E^i = -F_{i0} = -\partial_i\phi - \partial_t A^i, \qquad \text{i.e.} \qquad \mathbf{E} = -\boldsymbol{\nabla}\phi - \partial_t\mathbf{A}, \tag{8.53}$$

which is exactly the standard expression for the electric field in terms of the potentials. The magnetic field is defined through the spatial–spatial components by

$$B^i \equiv -\tfrac{1}{2}\,\epsilon^{ijk}F_{jk}, \tag{8.54}$$

with ϵ^{ijk} the three-dimensional Levi-Civita symbol of Chapter 3 ($\epsilon^{123} = +1$). Using $F_{jk} = \partial_j A_k - \partial_k A_j = -\partial_j A^k + \partial_k A^j = -(\partial_j A^k - \partial_k A^j)$,

$$B^i = -\tfrac{1}{2}\,\epsilon^{ijk}\big[-(\partial_j A^k - \partial_k A^j)\big] = \tfrac{1}{2}\,\epsilon^{ijk}(\partial_j A^k - \partial_k A^j) = \epsilon^{ijk}\partial_j A^k, \tag{8.55}$$

the last step using the antisymmetry of ϵ^{ijk} to combine the two equal contributions. This is $\mathbf{B} = \boldsymbol{\nabla} \times \mathbf{A}$. Collecting the components into the matrix of $F_{\mu\nu}$ (rows and columns ordered $\mu, \nu = 0, 1, 2, 3$),

$$F_{\mu\nu} = \begin{pmatrix} 0 & E^1 & E^2 & E^3 \\ -E^1 & 0 & -B^3 & B^2 \\ -E^2 & B^3 & 0 & -B^1 \\ -E^3 & -B^2 & B^1 & 0 \end{pmatrix}, \tag{8.56}$$

where the top row follows from $F_{0i} = -F_{i0} = E^i$ and the spatial block from inverting (8.54), $F_{jk} = -\epsilon_{jki} B^i$. The six independent entries of the antisymmetric tensor are the three electric and three magnetic field components, exactly as promised. The Lorentz tensor $F_{\mu\nu}$ has thus unified $\mathbf{E}$ and $\mathbf{B}$ into a single geometric object whose transformation under boosts—mixing the electric and magnetic blocks—is the relativistic statement that electric and magnetic fields are frame-dependent shadows of one curvature.

Worked Example 8.11 (The Bianchi identity is the homogeneous Maxwell pair)**.** We show that the four independent components of the Bianchi identity (8.47) are precisely the two source-free Maxwell equations, $\boldsymbol{\nabla} \cdot \mathbf{B} = 0$ and Faraday's law $\boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}$. Because $\partial_{[\lambda} F_{\mu\nu]}$ is totally antisymmetric in its three indices, it has only $\binom{4}{3} = 4$ independent components, labelled by which index is *omitted*.

All-spatial component $(\lambda\mu\nu) = (1, 2, 3)$. Equation (8.47) reads

$$\partial_1 F_{23} + \partial_2 F_{31} + \partial_3 F_{12} = 0. \tag{8.57}$$

From (8.56), $F_{23} = -B^1$, $F_{31} = -B^2$, $F_{12} = -B^3$, so (8.57) becomes $-(\partial_1 B^1 + \partial_2 B^2 + \partial_3 B^3) = 0$, that is

$$\boldsymbol{\nabla} \cdot \mathbf{B} = 0. \tag{8.58}$$

The absence of magnetic monopoles is the statement that F is a curl: it has no all-spatial "divergence" because it is built antisymmetrically from a potential.

Components with one time index, e.g. $(\lambda\mu\nu) = (0, 1, 2)$:

$$\partial_0 F_{12} + \partial_1 F_{20} + \partial_2 F_{01} = 0. \tag{8.59}$$

Using $F_{12} = -B^3$, $F_{20} = -F_{02} = -E^2$, $F_{01} = E^1$ and $\partial_0 = \partial_t$, (8.59) reads $-\partial_t B^3 - \partial_1 E^2 + \partial_2 E^1 = 0$, i.e. $\partial_1 E^2 - \partial_2 E^1 = -\partial_t B^3$. The left side is the third component of $\boldsymbol{\nabla} \times \mathbf{E}$, so this is the 3-component of Faraday's law. The choices $(0, 2, 3)$ and $(0, 3, 1)$ give the 1- and 2-components by the same steps, assembling into

$$\boldsymbol{\nabla} \times \mathbf{E} = -\partial_t \mathbf{B}. \tag{8.60}$$

The two homogeneous Maxwell equations are thus not dynamical laws to be derived from an action; they are the kinematic identity that any field strength built from a potential must obey. This is the payoff promised after Theorem 8.10: half of Maxwell's equations is geometry, true the moment one writes $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$.

It is occasionally convenient to package (8.47) in dual form. Define the dual field strength $\widetilde{F}^{\mu\nu} \equiv \frac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ with $\epsilon^{0123} = +1$ (Chapter 3); then the totally antisymmetric Bianchi identity (8.47) is equivalent to the compact statement

$$\partial_\mu \widetilde{F}^{\mu\nu} = 0, \tag{8.61}$$

because contracting (8.47) with $\epsilon^{\kappa\lambda\mu\nu}$ and using the antisymmetry of the ϵ-symbol reproduces the four components worked out in Theorem 8.11. The dual form makes the parallel with the inhomogeneous equations $\partial_\mu F^{\mu\nu} = J^\nu$ of the next section visually plain: the homogeneous pair is the same divergence taken on the dual.

8.3.5 Curvature, parallel transport, and holonomy

We now justify the word that has been doing rhetorical work throughout: *curvature*. The claim is not an analogy. The covariant derivative D_μ is, in the precise vocabulary of differential geometry, a *connection* on a complex line bundle over spacetime, and $F_{\mu\nu}$ is its *curvature*—the same object that, for the Levi-Civita connection on a Riemannian manifold, is the Riemann tensor. The dictionary is exact, and laying it out now both explains why the commutator construction was inevitable and prepares the bundle picture that the non-abelian theory of Chapter 25 will make essential.

The geometric content is encoded in *parallel transport*. A charged field ψ at a point x carries a phase whose absolute value is a matter of gauge convention; to compare the phase at x with the phase at a neighbouring point y we must specify a rule for carrying the phase from one to the other—a rule for parallel transport. The covariant derivative *is* that rule in infinitesimal form: a field is said to be parallel-transported, or covariantly constant, along a path if its covariant derivative along the path's tangent vanishes. For a displacement $\mathrm{d}x^\mu$, demanding $\mathrm{d}x^\mu D_\mu \psi = 0$ means

$$\mathrm{d}x^\mu(\partial_\mu - iqA_\mu)\psi = 0 \quad\Longrightarrow\quad \mathrm{d}\psi = iq\,A_\mu\,\mathrm{d}x^\mu\,\psi. \tag{8.62}$$

Integrating (8.62) along a path γ from x to y gives the finite transport law $\psi(y) = U_\gamma(y,x)\,\psi(x)$ with the *comparator* (or Wilson line)

$$U_\gamma(y,x) = \exp\!\left(iq\int_\gamma A_\mu\,\mathrm{d}x^\mu\right), \tag{8.63}$$

the exponential being ordinary in the abelian case where the integrand at different points commutes (in the non-abelian case it becomes a path-ordered exponential, Chapter 25). The comparator transforms under a gauge change so as to cancel the phases at its endpoints, $U_\gamma(y,x) \to e^{iq\alpha(y)}U_\gamma(y,x)e^{-iq\alpha(x)}$, which is exactly the property that lets it carry a phase from x to y in a gauge-covariant way.

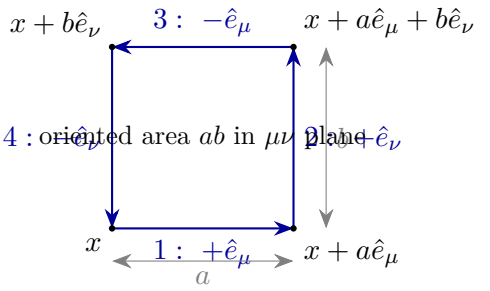


Figure 8.3: The infinitesimal rectangular loop in the μ–ν plane used to compute the holonomy in Theorem 8.12. Parallel transport once around the loop multiplies a charged field by $U(C) \approx 1 + iq\,F_{\mu\nu}\,ab$: the curvature $qF_{\mu\nu}$ is the holonomy per unit oriented area.

Now the decisive question: does the result of parallel transport depend on the path? If two paths from x to y give the same answer, the connection is "flat" and the phase difference between x and y is a well-defined, path-independent quantity—there is no physical field. If different paths disagree, transport is path-dependent, and the disagreement around a closed loop is the physical content of the field. The disagreement is measured by the *holonomy*, the comparator around a closed loop C,

$$U(C) = \exp\!\left(iq \oint_C A_\mu \, \mathrm{d}x^\mu \right), \tag{8.64}$$

and the claim is that for an infinitesimal loop the holonomy is governed by $F_{\mu\nu}$. The next worked example computes it explicitly; Fig. 8.3 fixes the geometry.

Worked Example 8.12 (Holonomy around an infinitesimal loop equals the curvature)**.** Take the small rectangular loop C of Fig. 8.3, lying in the μ–ν coordinate plane with one corner at x, side a along $\hat{e}_\mu$ and side b along $\hat{e}_\nu$ (no sum on the fixed directions μ, ν). We evaluate $\oint_C A_\rho \, \mathrm{d}x^\rho$ to leading nonvanishing order in the side lengths by summing the four straight segments, approximating each segment integral by its midpoint value (the midpoint rule is accurate to the order we need).

Segment 1 ($x \to x + a\hat{e}_\mu$, direction $+\mu$, midpoint $x + \frac{a}{2}\hat{e}_\mu$): contributes $+a\,A_\mu\big(x + \frac{a}{2}\hat{e}_\mu\big)$.

Segment 2 ($x + a\hat{e}_\mu \to x + a\hat{e}_\mu + b\hat{e}_\nu$, direction $+\nu$, midpoint $x + a\hat{e}_\mu + \frac{b}{2}\hat{e}_\nu$): contributes $+b\,A_\nu\big(x + a\hat{e}_\mu + \frac{b}{2}\hat{e}_\nu\big)$.

Segment 3 ($x + a\hat{e}_\mu + b\hat{e}_\nu \to x + b\hat{e}_\nu$, direction $-\mu$, midpoint $x + \frac{a}{2}\hat{e}_\mu + b\hat{e}_\nu$): contributes $-a\,A_\mu\big(x + \frac{a}{2}\hat{e}_\mu + b\hat{e}_\nu\big)$.

Segment 4 ($x + b\hat{e}_\nu \to x$, direction $-\nu$, midpoint $x + \frac{b}{2}\hat{e}_\nu$): contributes $-b\,A_\nu\big(x + \frac{b}{2}\hat{e}_\nu\big)$.

Group segments 1 and 3 (both along μ). They differ only in that segment 3 is displaced by $b\hat{e}_\nu$, so a first-order Taylor expansion gives

$$a\Big[A_\mu(x+\tfrac{a}{2}\hat{e}_\mu)-A_\mu(x+\tfrac{a}{2}\hat{e}_\mu+b\hat{e}_\nu)\Big]=-a\,b\,\partial_\nu A_\mu+O(\text{cubic}). \tag{8.65}$$

Likewise group segments 2 and 4 (both along ν); segment 2 is displaced from segment 4 by $a\hat{e}_\mu$, giving

$$b\Big[A_\nu(x+a\hat{e}_\mu+\tfrac{b}{2}\hat{e}_\nu)-A_\nu(x+\tfrac{b}{2}\hat{e}_\nu)\Big]=+a\,b\,\partial_\mu A_\nu+O(\text{cubic}). \tag{8.66}$$

Adding (8.65) and (8.66), the line integral is

$$\oint_C A_\rho\,\mathrm{d}x^\rho=a\,b\,(\partial_\mu A_\nu-\partial_\nu A_\mu)=F_{\mu\nu}\,a\,b, \tag{8.67}$$

the flux of $F_{\mu\nu}$ through the loop—a special case of Stokes' theorem, $\oint A_\rho\,\mathrm{d}x^\rho=\frac{1}{2}\int F_{\rho\sigma}\,\mathrm{d}\sigma^{\rho\sigma}$, for the oriented area element ab. The holonomy is therefore

$$U(C)=\exp\!\big(iq\,F_{\mu\nu}\,ab\big)=1+iq\,F_{\mu\nu}\,ab+O((ab)^2). \tag{8.68}$$

Parallel transport around an infinitesimal loop fails to be the identity, and the failure, per unit oriented area, is exactly $iq\,F_{\mu\nu}$. This is the geometric meaning of the field strength: *$qF_{\mu\nu}$ is the curvature of the gauge connection*, the holonomy density of the loop. It is also, read physically, the infinitesimal Aharonov–Bohm phase—the gauge field makes itself felt on a charged field only through the curvature it produces, which is why $F_{\mu\nu}$ and not A_μ is the locally observable field.

The infinitesimal holonomy (8.68) also gives a third, geometric derivation of the commutator formula (8.42) and closes the circle of the section. Parallel transport of ψ around the loop and the double covariant derivative are the same operation: a short computation shows that transporting around the infinitesimal rectangle and comparing with the untransported field reproduces $\psi(x)\to\psi(x)+[D_\mu,D_\nu]\,\psi\,ab$ to leading order, so that the holonomy generator $iqF_{\mu\nu}$ of (8.68) is identical to $-[D_\mu,D_\nu]$ of (8.42). The commutator of covariant derivatives *is* the failure of parallel transport to commute; that is why we were entitled, from the first line of Section 8.3.1, to call $F_{\mu\nu}$ a curvature.

Physical Insight

The connection/curvature dictionary that this section has built quietly is the entire conceptual content of gauge theory, and it is worth stating once in its general form, because Part VII will do nothing but generalize the group:

- the gauge potential A_μ is a *connection*—a rule for parallel transport,

i.e. for comparing the internal (phase) space of the matter field at neighbouring points;

- the covariant derivative D_μ is that connection in infinitesimal form;
- the field strength $F_{\mu\nu} = \frac{i}{q}[D_\mu, D_\nu]$ is the *curvature*— the holonomy of an infinitesimal loop, the obstruction to path-independent comparison;
- gauge transformations are changes of phase convention (local frame) fibre by fibre; A_μ transforms inhomogeneously (it is not a tensor), while $F_{\mu\nu}$ transforms homogeneously (here, trivially: it is gauge invariant);
- the Bianchi identity $\partial_{[\lambda}F_{\mu\nu]} = 0$ is the geometric identity obeyed by any curvature, the differential-form statement $\mathrm{d}F = 0$ that F is closed.

Replacing the abelian phase $e^{iq\alpha}$ by a non-abelian group element $e^{i\alpha^a T^a}$ turns this into Yang–Mills theory; replacing the internal space by the tangent space of spacetime turns it into general relativity, with A_μ the Christoffel connection and $F_{\mu\nu}$ the Riemann curvature. The fact that electromagnetism, the strong and weak interactions, and gravity are all theories of a connection and its curvature is the deepest structural unity in physics, and this section is where the reader first meets it.

We now have the gauge-invariant residue of the potential in hand: the field strength $F_{\mu\nu}$, gauge invariant (Theorem 8.9), constrained by the Bianchi identity (Theorem 8.10), and equal to the electromagnetic fields $\mathbf{E}, \mathbf{B}$ packaged as the curvature of the gauge connection. The one ingredient still missing for a dynamical theory of A_μ is a Lagrangian, and gauge invariance has all but written it for us: the kinetic term must be a Lorentz scalar built from $F_{\mu\nu}$, and the unique such scalar with the right number of derivatives is the contraction $F_{\mu\nu}F^{\mu\nu}$. Constructing the Maxwell action from this invariant, adding a conserved source, and extracting the inhomogeneous Maxwell equations as its Euler–Lagrange equations is the task of the next section.

8.4 The Maxwell Action and Field Equations

The previous section handed up a single, geometrically privileged object: the field strength $F_{\mu\nu}$, constructed as the curvature of the gauge connection through the commutator of covariant derivatives, shown to be gauge invariant (Theorem 8.9), and obeying the Bianchi identity (Eq. (8.47)) as an off-shell consequence of being

a curvature. We now have the raw material for a dynamical theory but not yet the theory itself. A field acquires dynamics from an action; the action determines, through the variational principle of Chapter 6, which configurations of A_μ are physically realized. The task of this section is to write down that action. We will not guess it and check it after the fact. We will instead *derive* it from a short list of principles — Lorentz invariance, gauge invariance, parity, and a ceiling of two derivatives — and find that these requirements are so constraining that the answer is essentially forced: up to an overall normalization and additive total derivatives, there is exactly one candidate. Out of that unique Lagrangian, the Euler–Lagrange equations deliver the inhomogeneous Maxwell equations; the Bianchi identity supplies the homogeneous ones for free; and the demand that a coupling to matter respect gauge invariance forces the matter current to be conserved. By the end we will have assembled the photon sector $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ of the frozen classical QED Lagrangian of QED, with all four of Maxwell's equations recovered in their familiar three-vector form.

8.4.1 What may appear in the Lagrangian

The Lagrangian density $\mathcal{L}$ is a function of the gauge potential A_μ and its derivatives, and we constrain it by four requirements, each of which is physically motivated rather than imposed for convenience.

Lorentz invariance. $\mathcal{L}$ must be a Lorentz scalar, so that the action $S = \int \mathrm{d}^4x\, \mathcal{L}$ is frame-independent and the resulting dynamics respects special relativity. Every index must be contracted.

Gauge invariance. The physics is invariant under $A_\mu \to A_\mu + \partial_\mu \alpha(x)$ for arbitrary smooth α, as established in the construction of the connection. For the *equations of motion* to be gauge invariant it suffices that the action change by at most a boundary term — a quasi-symmetry in the sense of Theorem 7.4. The cleanest way to guarantee this, and the one we adopt and then justify, is to build $\mathcal{L}$ out of the manifestly gauge-invariant curvature $F_{\mu\nu}$ alone.

Locality and a derivative ceiling. $\mathcal{L}$ is a polynomial in the fields and their derivatives evaluated at a single spacetime point (locality), and we demand at most two derivatives in total. The derivative count is not arbitrary: a Lagrangian with more than two derivatives generically yields field equations of higher than second order, which carry extra initial data and, upon quantization, ghostlike negative-norm states (the Ostrogradsky problem). Restricting to two derivatives is the statement that we want the leading, renormalizable, healthy theory; higher-derivative corrections are irrelevant operators in the sense made precise by the renormalization group in Chapter 23.

Parity. We require invariance under spatial reflection $\mathbf{x} \to -\mathbf{x}$. This is a property of classical electromagnetism (and of QED), and dropping it admits exactly one extra term, which we will identify and dispose of.

The building blocks are therefore Lorentz scalars, built from A_μ and ∂_μ, gauge invariant, with at most two derivatives. Let us enumerate them.

No-derivative terms. The only Lorentz scalar with no derivatives that can be built from A_μ is $A_\mu A^\mu$. It is *not* gauge invariant:

$$A_\mu A^\mu \;\longrightarrow\; (A_\mu + \partial_\mu\alpha)(A^\mu + \partial^\mu\alpha) = A_\mu A^\mu + 2A_\mu\partial^\mu\alpha + (\partial_\mu\alpha)(\partial^\mu\alpha), \quad (8.69)$$

and the extra pieces do not assemble into a total derivative for arbitrary α. A term $\frac{1}{2}m^2 A_\mu A^\mu$ in the Lagrangian is precisely a *photon mass term*, and (8.69) shows that gauge invariance forbids it. This is a sharp and physical conclusion: *the gauge principle forces the photon to be massless.* A massive vector field is described instead by the Proca Lagrangian

$$\mathcal{L}_{\text{Proca}} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}m^2 A_\mu A^\mu, \quad (8.70)$$

which is Lorentz invariant and yields a perfectly sensible classical theory of a spin-1 particle of mass m, but which is *not* gauge invariant precisely because of the mass term. The contrast is instructive and we will return to it.

Physical Insight

The masslessness of the photon is not an experimental input fitted by hand; it is a *consequence* of demanding local $U(1)$ invariance. The same logic, applied to the non-abelian case in Chapter 25, forbids tree-level masses for the gluon and the electroweak gauge bosons — which is why giving mass to the W and Z requires the Higgs mechanism (Chapter 33), a way of generating an effective m^2A^2 without ever writing one in the Lagrangian. The photon mass bound is experimentally extraordinary: $m_\gamma \lesssim 10^{-18}\,\text{eV}$, fully consistent with the exact zero the gauge principle predicts.

Two-derivative terms. With two derivatives and quadratic in A (the leading order; higher powers of A with two derivatives are not available for an abelian field because F is linear in A), gauge invariance restricts us to scalars built from two factors of $F_{\mu\nu}$, since F already carries one derivative each. There are exactly two Lorentz scalars of this type:

$$F_{\mu\nu}F^{\mu\nu} \qquad \text{and} \qquad F_{\mu\nu}\widetilde{F}^{\mu\nu}, \quad \widetilde{F}^{\mu\nu} \equiv \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \quad (8.71)$$

where $\widetilde{F}^{\mu\nu}$ is the *dual* field strength and $\epsilon^{\mu\nu\rho\sigma}$ is the Levi-Civita symbol with $\epsilon^{0123} = +1$. (One might also try $\partial_\mu F^{\mu\nu}\,A_\nu$ or $A_\mu\partial^2 A^\mu$, but the first is, after integration by parts, $-\frac{1}{2}F_{\mu\nu}F^{\mu\nu}$ up to a boundary term, and the second is not gauge invariant; gauge-invariant non-redundant two-derivative scalars are

exhausted by (8.71), a point we prove below.) We must dispose of the second invariant on two independent grounds.

First, $F_{\mu\nu}\widetilde{F}^{\mu\nu}$ is *parity odd*. Under spatial reflection the electric field is a polar vector ($\mathbf{E} \to -\mathbf{E}$) while the magnetic field is an axial vector ($\mathbf{B} \to +\mathbf{B}$); we will see momentarily that $F_{\mu\nu}\widetilde{F}^{\mu\nu} \propto \mathbf{E}\cdot\mathbf{B}$, which is therefore a pseudoscalar, odd under parity. Requiring parity invariance excludes it outright.

Second — and this holds even without invoking parity — $F_{\mu\nu}\widetilde{F}^{\mu\nu}$ is a *total derivative* and so contributes nothing to the classical field equations. We verify this explicitly. Using $F_{\rho\sigma} = \partial_\rho A_\sigma - \partial_\sigma A_\rho$,

$$\begin{aligned} F_{\mu\nu}\widetilde{F}^{\mu\nu} &= \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu}F_{\rho\sigma} = \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}\,(\partial_\mu A_\nu - \partial_\nu A_\mu)(\partial_\rho A_\sigma - \partial_\sigma A_\rho) \\ &= 2\,\epsilon^{\mu\nu\rho\sigma}\,(\partial_\mu A_\nu)(\partial_\rho A_\sigma), \end{aligned} \tag{8.72}$$

where the four cross terms collapsed to one because $\epsilon^{\mu\nu\rho\sigma}$ is totally antisymmetric: relabelling indices, each of the four products equals $\epsilon^{\mu\nu\rho\sigma}(\partial_\mu A_\nu)(\partial_\rho A_\sigma)$. Now $\epsilon^{\mu\nu\rho\sigma}$ is constant, so we may pull the leading derivative outside,

$$\epsilon^{\mu\nu\rho\sigma}(\partial_\mu A_\nu)(\partial_\rho A_\sigma) = \partial_\mu(\epsilon^{\mu\nu\rho\sigma} A_\nu\,\partial_\rho A_\sigma) - \epsilon^{\mu\nu\rho\sigma} A_\nu\,\partial_\mu\partial_\rho A_\sigma, \tag{8.73}$$

and the last term vanishes identically because $\partial_\mu\partial_\rho A_\sigma$ is symmetric in (μ,ρ) while $\epsilon^{\mu\nu\rho\sigma}$ is antisymmetric in (μ,ρ). Hence

$$F_{\mu\nu}\widetilde{F}^{\mu\nu} = 2\,\partial_\mu(\epsilon^{\mu\nu\rho\sigma} A_\nu\,\partial_\rho A_\sigma) \;\equiv\; \partial_\mu K^\mu, \tag{8.74}$$

a pure four-divergence. Added to the action it contributes only $\int \mathrm{d}^4x\,\partial_\mu K^\mu = \oint_\partial K^\mu\,\mathrm{d}S_\mu$, a boundary term that does not affect the local equations of motion. In the classical abelian theory it is therefore completely inert. It is not, however, meaningless: integrated over Euclidean spacetime it computes a topological invariant (the instanton number) and multiplies a physical parameter θ in the path integral, the θ-term, whose consequences — the strong-CP problem in QCD — are the business of Chapter 41. We flag it here and set it aside.

We are left with a single admissible structure, $F_{\mu\nu}F^{\mu\nu}$. We will fix its normalization below by matching to the canonical kinetic term and the conventional Maxwell equations. First we make the enumeration claim airtight.

Proposition 8.13 (Uniqueness of the Maxwell kinetic term)**.** *Among local Lagrangian densities for an abelian gauge potential A_μ that are (i) Lorentz scalars, (ii) invariant under $A_\mu \to A_\mu + \partial_\mu\alpha$ up to a total derivative, (iii) of at most second order in derivatives, and (iv) parity even, the only one that produces nontrivial dynamics is, up to an overall multiplicative constant and additive total derivatives,*

$$\mathcal{L} = -\tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}. \tag{8.75}$$

Proof. We argue directly, not by appeal to the building-block heuristic. A Lagrangian quadratic in A_μ and of second order in derivatives is, up to total

derivatives, a linear combination of the three independent Lorentz scalars

$$\mathcal{T}_1 = (\partial_\mu A_\nu)(\partial^\mu A^\nu), \qquad \mathcal{T}_2 = (\partial_\mu A^\mu)(\partial_\nu A^\nu), \qquad \mathcal{T}_3 = (\partial_\mu A_\nu)(\partial^\nu A^\mu). \tag{8.76}$$

Indeed any two-derivative quadratic scalar contracts two factors of ∂A, and there are exactly three ways to route the four indices into Lorentz-invariant pairs; terms with a $\Box$ acting on one A reduce to these by integration by parts. Moreover $\mathcal{T}_2$ and $\mathcal{T}_3$ are equal up to a total derivative,

$$\mathcal{T}_3 - \mathcal{T}_2 = (\partial_\mu A_\nu)(\partial^\nu A^\mu) - (\partial_\mu A^\mu)(\partial_\nu A^\nu) = \partial_\mu\big[A_\nu \partial^\nu A^\mu - A^\mu \partial_\nu A^\nu\big], \tag{8.77}$$

as one checks by expanding the divergence and using $\partial_\mu \partial^\nu A^\mu = \partial^\nu \partial_\mu A^\mu$. So modulo total derivatives only two structures are independent, say $\mathcal{T}_1$ and $\mathcal{T}_3$, and we write $\mathcal{L} = a\,\mathcal{T}_1 + b\,\mathcal{T}_3$. Impose gauge invariance: under $\delta A_\mu = \partial_\mu \alpha$,

$$\delta\mathcal{T}_1 = 2(\partial_\mu\partial_\nu\alpha)(\partial^\mu A^\nu), \quad \delta\mathcal{T}_3 = 2(\partial_\mu\partial_\nu\alpha)(\partial^\nu A^\mu) = 2(\partial_\mu\partial_\nu\alpha)(\partial^\mu A^\nu), \tag{8.78}$$

where in $\delta\mathcal{T}_3$ we relabelled the dummy indices $\mu \leftrightarrow \nu$ and used the symmetry of $\partial_\mu\partial_\nu\alpha$. Thus $\delta\mathcal{T}_1 = \delta\mathcal{T}_3$, and

$$\delta\mathcal{L} = (a+b)\cdot 2(\partial_\mu\partial_\nu\alpha)(\partial^\mu A^\nu). \tag{8.79}$$

For this to be a total derivative for *arbitrary* α we need $a + b = 0$; otherwise, integrating by parts, $\delta S = -(a+b)\int \mathrm{d}^4x\, 2\,\alpha\,\partial_\mu\partial_\nu\partial^\mu A^\nu + (\text{bdy})$, and the bulk term $\propto \alpha\,\Box\,\partial_\nu A^\nu$ does not vanish for general field configurations. Hence $b = -a$, and

$$\mathcal{L} = a\,(\mathcal{T}_1 - \mathcal{T}_3) = a\big[(\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu)\big] = \tfrac{a}{2}\,F_{\mu\nu}F^{\mu\nu}, \tag{8.80}$$

the last equality being the identity $F_{\mu\nu}F^{\mu\nu} = 2[(\partial_\mu A_\nu)(\partial^\mu A^\nu) - (\partial_\mu A_\nu)(\partial^\nu A^\mu)]$, verified by expanding $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. So gauge invariance alone collapses the two-parameter family to a single structure $\propto F^2$. The parity-odd partner $F\widetilde{F}$ is excluded by (iv) and in any case is a total derivative (8.74); no-derivative terms are excluded by gauge invariance (8.69). Choosing the constant $a = -\frac{1}{2}$ gives (8.75). $\square$

The normalization $-\frac{1}{4}$ is the standard one, and its two factors deserve a word, since the chapter's later sign-sensitive results depend on it. Write out $F_{\mu\nu}F^{\mu\nu}$ in terms of fields. With the identifications fixed in Chapter 6 and Appendix A,

$$F^{0i} = -E^i, \qquad F^{ij} = -\epsilon^{ijk}B^k, \tag{8.81}$$

which follow from $\mathbf{E} = -\nabla A^0 - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$, one finds (lowering indices flips the sign of each electric component but not the magnetic ones)

$$F_{\mu\nu}F^{\mu\nu} = 2\big(F_{0i}F^{0i} + \tfrac{1}{2}F_{ij}F^{ij}\big) = 2\big(-\mathbf{E}^2 + \mathbf{B}^2\big), \qquad \text{so} \qquad -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2). \tag{8.82}$$

The $-\frac{1}{4}$ thus produces the textbook Lagrangian density $\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$: the sign makes the time-derivative "kinetic" part $\frac{1}{2}\mathbf{E}^2 = \frac{1}{2}\dot{\mathbf{A}}^2 + \cdots$ enter with a positive coefficient, so that the field energy is bounded below, and the overall scale gives the canonically normalized propagating modes (the two photon polarizations counted in Section 8.5.2). For completeness, the same components give the parity-odd invariant

$$F_{\mu\nu}\widetilde{F}^{\mu\nu} = -4\,\mathbf{E}\cdot\mathbf{B}, \tag{8.83}$$

confirming that it is the pseudoscalar $\mathbf{E}\cdot\mathbf{B}$ announced above, and explaining geometrically why it could only ever be a boundary term: $\mathbf{E}\cdot\mathbf{B} = \frac{1}{2}\partial_t(\mathbf{A}\cdot\mathbf{B}) + \cdots$ is a total derivative on shell and, as (8.74) shows, off shell as well.

Notation Watch

The split of the index sum in (8.82) uses $F_{0i}F^{0i} = -(F^{0i})^2 = -\mathbf{E}^2$ and $\frac{1}{2}F_{ij}F^{ij} = \frac{1}{2}(\epsilon^{ijk}B^k)(\epsilon^{ijl}B^l) = \mathbf{B}^2$, where the factor $\frac{1}{2}$ corrects for double-counting the antisymmetric spatial pair (i,j), and $\epsilon^{ijk}\epsilon^{ijl} = 2\delta^{kl}$. The relative sign of $\mathbf{E}^2$ and $\mathbf{B}^2$ is a direct fingerprint of the mostly-minus metric; in the mostly-plus convention $F_{\mu\nu}F^{\mu\nu}$ would carry the opposite overall sign and the Lagrangian would still be $\frac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2)$, but only because the $-\frac{1}{4}$ would be quoted as $+\frac{1}{4}$. We use mostly-minus throughout (Chapter 8 chapter introduction); the combination $-\frac{1}{4}F^2$ is the invariant statement.

8.4.2 Coupling to a source: conservation forced by gauge invariance

A pure-radiation theory is not yet electrodynamics; the field must couple to charges. The simplest Lorentz-invariant, local coupling between the potential A_μ and an external four-current J^μ is the contraction $J^\mu A_\mu$, linear in the potential. We add it to the action with a sign chosen to reproduce the conventional field equations:

$$S[A] = \int \mathrm{d}^4x \,\Big(-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - J^\mu A_\mu \Big). \tag{8.84}$$

Here $J^\mu = (\rho, \mathbf{J})$ collects the charge density and the current density; for the moment it is a prescribed external function of spacetime (the dynamical, matter-field origin of J^μ as the $U(1)$ Noether current of Chapter 7 is identified in Section 8.6, completing QED's classical layer).

The coupling $J^\mu A_\mu$ involves the bare potential, not the gauge-invariant $F_{\mu\nu}$, so it is *not* manifestly gauge invariant, and we must check that it does not spoil the gauge symmetry on which the whole construction rests. Under $A_\mu \to A_\mu + \partial_\mu\alpha$ the field-strength term is untouched (Theorem 8.9), while the source term changes by

$$-\int \mathrm{d}^4x\, J^\mu(A_\mu + \partial_\mu\alpha) = -\int \mathrm{d}^4x\, J^\mu A_\mu \;-\; \int \mathrm{d}^4x\, J^\mu\partial_\mu\alpha. \tag{8.85}$$

Integrate the last term by parts:

$$-\int \mathrm{d}^4x\, J^\mu \partial_\mu \alpha = -\oint_\partial \mathrm{d}S_\mu\, J^\mu \alpha \;+\; \int \mathrm{d}^4x\, (\partial_\mu J^\mu)\, \alpha. \tag{8.86}$$

Assuming, as always, that the gauge parameter α and the current fall off at the boundary so that the surface integral vanishes, the change in the action is

$$\delta S = \int \mathrm{d}^4x\, (\partial_\mu J^\mu)\, \alpha. \tag{8.87}$$

For the action (8.84) to be gauge invariant — so that the gauge redundancy persists in the presence of matter and the field equations remain consistent — this must vanish for *every* gauge function $\alpha(x)$. By the fundamental lemma of the calculus of variations, $\delta S = 0$ for all α if and only if

$$\partial_\mu J^\mu = 0. \tag{8.88}$$

This is a striking inversion of the logic of Chapter 7. There, a continuous symmetry of the action *implied* a conserved current via Noether's first theorem (Theorem 7.10). Here the implication runs the other way: *gauge invariance of the action does not merely permit but requires the source to be a conserved current.* Current conservation is not an extra physical assumption layered on top of electromagnetism; it is the price of admission for coupling matter to a gauge field at all. A non-conserved source is mathematically inconsistent with gauge invariance, and we will see in a moment that it is equally inconsistent with the field equation itself.

Physical Insight

The two facts — Noether's theorem (global symmetry ⇒ conserved current) and the present result (gauge invariance ⇒ conserved source) — are two faces of the same coin, and their meeting point is the gauge principle. A global $U(1)$ symmetry of charged matter gives, by Noether, a conserved current J^μ. Promoting that symmetry to a *local* one forces a gauge field A_μ into existence and couples it precisely to that same J^μ through $-J^\mu A_\mu$. Gauge invariance of the coupling then *re-derives* $\partial_\mu J^\mu = 0$ — now as a consistency condition rather than a Noether output. Symmetry and dynamics are locked together: the same current that the global symmetry conserves is the source that sources the field. This circle of ideas is the structural heart of Chapter 8 and the seed of the Ward–Takahashi identities of Chapter 19, which are its quantum echo.

8.4.3 The inhomogeneous Maxwell equations

We now extract the dynamics. The action (8.84) is varied with respect to the dynamical field A_μ using the field-theoretic Euler–Lagrange equation derived in

Chapter 6, namely (Eq. (6.61))

$$\partial_\alpha\left(\frac{\partial\mathcal{L}}{\partial(\partial_\alpha A_\beta)}\right)-\frac{\partial\mathcal{L}}{\partial A_\beta}=0, \tag{8.89}$$

with β the free index. The decisive computation is $\partial\mathcal{L}/\partial(\partial_\alpha A_\beta)$, and because that is exactly the place where the antisymmetry of F conspires with the $\frac{1}{4}$ to produce a clean result — and exactly where factor-of-two and sign errors breed — we carry it out in full. This computation was previewed in Theorem 6.12; we repeat it here, with the source coupling included from the start, because the resulting field equation (8.95) is the central dynamical statement of this chapter and the gauge-theoretic reading of its source is new.

Worked Example 8.14 (The field equation $\partial_\mu F^{\mu\nu}=J^\nu$ from $\mathcal{L}=-\frac{1}{4}F^2-J\cdot A$)**.** The Lagrangian is $\mathcal{L}=-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}-J^\mu A_\mu$. We compute the two pieces of (8.89) in turn.

Step 1: the undifferentiated derivative. The field A_β appears undifferentiated only in the source term, $-J^\mu A_\mu$, with J^μ an external (field-independent) function. Hence

$$\frac{\partial\mathcal{L}}{\partial A_\beta}=-J^\beta. \tag{8.90}$$

Step 2: open up the kinetic term. Write the kinetic term with all indices explicit and lower them with the metric so that the dependence on $\partial_\alpha A_\beta$ is exposed:

$$-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}=-\tfrac{1}{4}\,\eta^{\mu\rho}\eta^{\nu\sigma}\,F_{\mu\nu}F_{\rho\sigma},\qquad F_{\mu\nu}=\partial_\mu A_\nu-\partial_\nu A_\mu. \tag{8.91}$$

We need the derivative of $F_{\mu\nu}$ with respect to the independent variable $\partial_\alpha A_\beta$. Treating each component $\partial_\alpha A_\beta$ as independent,

$$\frac{\partial(\partial_\mu A_\nu)}{\partial(\partial_\alpha A_\beta)}=\delta_\mu{}^\alpha\delta_\nu{}^\beta,\quad\Longrightarrow\quad\frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)}=\delta_\mu{}^\alpha\delta_\nu{}^\beta-\delta_\nu{}^\alpha\delta_\mu{}^\beta, \tag{8.92}$$

the antisymmetrized pair of Kronecker deltas being the signature of the antisymmetry of F in (μ,ν).

Step 3: differentiate and contract. Differentiating the product $F_{\mu\nu}F^{\mu\nu}$ gives two equal contributions (one from each factor), so

$$\begin{aligned}\frac{\partial\mathcal{L}}{\partial(\partial_\alpha A_\beta)}&=-\tfrac{1}{4}\cdot 2\,F^{\mu\nu}\,\frac{\partial F_{\mu\nu}}{\partial(\partial_\alpha A_\beta)}=-\tfrac{1}{2}\,F^{\mu\nu}(\delta_\mu{}^\alpha\delta_\nu{}^\beta-\delta_\nu{}^\alpha\delta_\mu{}^\beta)\\&=-\tfrac{1}{2}(F^{\alpha\beta}-F^{\beta\alpha})=-\tfrac{1}{2}(F^{\alpha\beta}+F^{\alpha\beta})=-F^{\alpha\beta},\end{aligned} \tag{8.93}$$

where the second line contracted the deltas against $F^{\mu\nu}$ and the last step used the antisymmetry $F^{\beta\alpha}=-F^{\alpha\beta}$. The factor $\frac{1}{2}$ and the two terms of the antisymmetrization conspire exactly: the result is the clean $-F^{\alpha\beta}$.

Step 4: assemble. Insert (8.93) and (8.90) into the Euler–Lagrange equation (8.89):

$$\partial_\alpha\big(-F^{\alpha\beta}\big)-\big(-J^\beta\big)=0 \quad\Longrightarrow\quad \partial_\alpha F^{\alpha\beta}=J^\beta. \tag{8.94}$$

Renaming the free index $\beta\to\nu$ and the dummy $\alpha\to\mu$, we obtain the inhomogeneous Maxwell equations,

$$\boxed{\partial_\mu F^{\mu\nu}=J^\nu}. \tag{8.95}$$

The field equation (8.95) carries the conservation law (8.88) inside it, and this is the second, independent route to current conservation promised above. Contract (8.95) with ∂_ν:

$$\partial_\nu J^\nu=\partial_\nu\partial_\mu F^{\mu\nu}=0, \tag{8.96}$$

which vanishes identically because $\partial_\nu\partial_\mu$ is symmetric under $\mu\leftrightarrow\nu$ while $F^{\mu\nu}$ is antisymmetric, so their full contraction is zero before any equation of motion is imposed. Thus the field equation is consistent *only* for a conserved current — and we have now seen $\partial_\mu J^\mu=0$ emerge twice, once from gauge invariance of the action (8.88) and once from the structure of the field equation (8.96). The two are the same statement viewed from the action and from its extremum, which is exactly as it must be.

8.4.4 The homogeneous equations from the Bianchi identity

Equation (8.95) is only half of Maxwell's system: it is the pair $\nabla\cdot\mathbf{E}=\rho$ and $\nabla\times\mathbf{B}-\partial_t\mathbf{E}=\mathbf{J}$, as we verify component by component below. The other half — $\nabla\cdot\mathbf{B}=0$ and $\nabla\times\mathbf{E}+\partial_t\mathbf{B}=0$ — are not equations of motion and do not come from the action at all. They are the off-shell identity satisfied by any field strength that is the curvature of a potential, the Bianchi identity established in the previous section (Eq. (8.47)),

$$\partial_\lambda F_{\mu\nu}+\partial_\mu F_{\nu\lambda}+\partial_\nu F_{\lambda\mu}=0. \tag{8.97}$$

This holds identically once $F=\mathrm{d}A$, independently of any Lagrangian, which is why the homogeneous Maxwell equations are sometimes said to be "kinematic" rather than dynamical. The identity (8.97) is equivalent to the compact dual form

$$\partial_\mu\widetilde{F}^{\mu\nu}=0, \qquad \widetilde{F}^{\mu\nu}=\tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \tag{8.98}$$

the equivalence being immediate: $\partial_\mu\widetilde{F}^{\mu\nu}=\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\partial_\mu F_{\rho\sigma}$, and the totally antisymmetric contraction of ϵ with $\partial_\mu F_{\rho\sigma}$ reproduces, up to a numerical factor, the cyclic sum in (8.97). The striking near-symmetry between $\partial_\mu F^{\mu\nu}=J^\nu$ and $\partial_\mu\widetilde{F}^{\mu\nu}=0$ is the statement that, in the absence of magnetic monopoles, the magnetic "current" on the right of the dual equation is zero; the deep reason is geometric (F is a curvature, hence closed as a two-form), and it is the cleanest preview of the holonomy/bundle viewpoint set up in the previous section.

8.4.5 Recovering the four three-vector Maxwell equations

We now make the contact with the familiar Maxwell equations completely explicit, translating the two covariant statements (8.95) and (8.98) into the four three-vector equations of Gauss, Ampère–Maxwell, no-monopoles, and Faraday. The dictionary is (8.81), which we record once more together with the lowered-index and dual components needed below. From $F^{0i} = -E^i$ and $F^{ij} = -\epsilon^{ijk}B^k$, lowering indices with $\eta = \mathrm{diag}(+,-,-,-)$ gives

$$F_{0i} = +E^i, \qquad F_{ij} = -\epsilon^{ijk}B^k, \tag{8.99}$$

and a short computation with $\epsilon^{0123} = +1$ (carried out in Theorem 8.15 below) gives the dual components

$$\widetilde{F}^{0i} = -B^i, \qquad \widetilde{F}^{ij} = \epsilon^{ijk}E^k. \tag{8.100}$$

It is convenient to display $F^{\mu\nu}$ as an explicit matrix; the rows and columns run $\mu,\nu = 0,1,2,3$:

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E^1 & -E^2 & -E^3 \\ E^1 & 0 & -B^3 & B^2 \\ E^2 & B^3 & 0 & -B^1 \\ E^3 & -B^2 & B^1 & 0 \end{pmatrix}, \tag{8.101}$$

where the lower block uses $F^{12} = -\epsilon^{123}B^3 = -B^3$ and cyclically, and $F^{i0} = -F^{0i} = E^i$. With these in hand the reduction is mechanical.

Worked Example 8.15 (The four Maxwell equations, component by component)**.** We take the four covariant components of (8.95) and (8.98) in turn, with $J^\mu = (\rho, \mathbf{J})$.

(1) Gauss's law (the $\nu = 0$ inhomogeneous equation). Set $\nu = 0$ in $\partial_\mu F^{\mu 0} = J^0$. Since $F^{00} = 0$, only spatial $\mu = i$ contributes:

$$\partial_i F^{i0} = J^0 \implies \partial_i E^i = \rho \implies \nabla\cdot\mathbf{E} = \rho, \tag{8.102}$$

using $F^{i0} = E^i$ from (8.101) and $J^0 = \rho$. This is Gauss's law.

(2) Ampère–Maxwell law (the $\nu = j$ inhomogeneous equation). Set $\nu = j$ in $\partial_\mu F^{\mu j} = J^j$, splitting the sum into $\mu = 0$ and $\mu = i$:

$$\partial_0 F^{0j} + \partial_i F^{ij} = J^j. \tag{8.103}$$

The first term is $\partial_0 F^{0j} = -\partial_t E^j$ (using $F^{0j} = -E^j$). For the second, $F^{ij} = -\epsilon^{ijk}B^k$, so

$$\partial_i F^{ij} = -\epsilon^{ijk}\partial_i B^k = +\epsilon^{jik}\partial_i B^k = (\nabla\times\mathbf{B})^j, \tag{8.104}$$

where the sign flip used $\epsilon^{ijk} = -\epsilon^{jik}$ and the last equality is the definition $(\nabla\times\mathbf{B})^j = \epsilon^{jik}\partial_i B^k$. Assembling,

$$-\partial_t E^j + (\nabla\times\mathbf{B})^j = J^j \implies \nabla\times\mathbf{B} - \partial_t\mathbf{E} = \mathbf{J}, \tag{8.105}$$

the Ampère–Maxwell law, complete with Maxwell's displacement current $-\partial_t \mathbf{E}$.

(3) No magnetic monopoles (the $\nu = 0$ homogeneous equation). First we establish the dual components (8.100). From $\widetilde{F}^{0i} = \frac{1}{2}\epsilon^{0ijk}F_{jk} = \frac{1}{2}\epsilon^{ijk}F_{jk}$ (the spatial Levi-Civita symbol), and $F_{jk} = -\epsilon^{jkl}B^l$,

$$\widetilde{F}^{0i} = \tfrac{1}{2}\epsilon^{ijk}(-\epsilon^{jkl}B^l) = -\tfrac{1}{2}\,\epsilon^{ijk}\epsilon^{ljk}\,B^l = -\tfrac{1}{2}\cdot 2\delta^{il}\,B^l = -B^i, \tag{8.106}$$

using $\epsilon^{ijk}\epsilon^{ljk} = 2\delta^{il}$. Now set $\nu = 0$ in $\partial_\mu \widetilde{F}^{\mu 0} = 0$; since $\widetilde{F}^{00} = 0$, only $\mu = i$ survives, and $\widetilde{F}^{i0} = -\widetilde{F}^{0i} = B^i$:

$$\partial_i \widetilde{F}^{i0} = 0 \implies \partial_i B^i = 0 \implies \nabla \cdot \mathbf{B} = 0, \tag{8.107}$$

the statement that there are no magnetic monopoles.

(4) Faraday's law (the $\nu = j$ homogeneous equation). We need $\widetilde{F}^{ij} = \frac{1}{2}\epsilon^{ij\rho\sigma}F_{\rho\sigma}$; the nonzero contributions have one temporal and one spatial index, giving $\widetilde{F}^{ij} = \epsilon^{ij0k}F_{0k} = \epsilon^{ijk}F_{0k} = \epsilon^{ijk}E^k$ (here $\epsilon^{ij0k} = \epsilon^{0ijk} = \epsilon^{ijk}$ after moving the 0 to the front through two transpositions, and $F_{0k} = E^k$ from (8.99)). Set $\nu = j$ in $\partial_\mu \widetilde{F}^{\mu j} = 0$:

$$\partial_0 \widetilde{F}^{0j} + \partial_i \widetilde{F}^{ij} = 0. \tag{8.108}$$

The first term is $\partial_0 \widetilde{F}^{0j} = -\partial_t B^j$. The second is

$$\partial_i \widetilde{F}^{ij} = \epsilon^{ijk}\partial_i E^k = -\epsilon^{jik}\partial_i E^k = -(\nabla \times \mathbf{E})^j. \tag{8.109}$$

Together,

$$-\partial_t B^j - (\nabla \times \mathbf{E})^j = 0 \implies \nabla \times \mathbf{E} + \partial_t \mathbf{B} = 0, \tag{8.110}$$

Faraday's law of induction. The four three-vector Maxwell equations are recovered in full.

The accounting is worth pausing on. Two of Maxwell's equations — Gauss and Ampère–Maxwell — are *dynamical*: they are the Euler–Lagrange equations of the action (8.84), they involve the source J^μ, and they would be different for a different Lagrangian. The other two — no-monopoles and Faraday — are *kinematic*: they hold identically for any F derived from a potential, are sourceless, and follow from the Bianchi identity (8.98) without reference to the action. This clean split is invisible in the three-vector formulation, where all four equations sit on an equal footing; it is one of the things the covariant, gauge-theoretic formulation makes manifest. It also explains, retroactively, why electromagnetism is naturally written in terms of a potential A_μ rather than the fields $\mathbf{E}, \mathbf{B}$ directly: introducing A_μ solves the homogeneous pair automatically, leaving only the inhomogeneous pair as genuine dynamics.

Historical Note

That two of the four equations are identities of the potential, and only two are dynamical, was not clear to Maxwell, who worked with the fields and the vector potential on a roughly equal footing and regarded all of his equations as physical laws. The recognition that the potential is the more fundamental object — forced on physics first by the Aharonov–Bohm effect (Chapter 8 closing sections) and then decisively by the gauge principle — is a twentieth-century reframing. The covariant tensor form (8.95), with its automatic conservation law (8.96), is due to Minkowski (1908), who first wrote electromagnetism as a single antisymmetric tensor equation in spacetime.

8.4.6 The Maxwell sector of classical QED

With (8.75) established we have, for the first time in the book, the gauge field's own dynamics in hand, and we can record the contribution this section makes to QED. The classical QED Lagrangian, the carried model of QED, is

$$\mathcal{L}_{\text{QED}} = \bar{\psi}\,(i\not{D} - m)\,\psi \;-\; \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \tag{8.111}$$

and the present section has built the second term and shown it to be the unique gauge-invariant, Lorentz-invariant, parity-even, two-derivative kinetic term for the photon. The matter sector $\bar{\psi}(i\not{D} - m)\psi$ and the identification of the source $J^\mu = -e\bar{\psi}\gamma^\mu\psi$ as the electron's $U(1)$ Noether current — so that varying (8.111) with respect to A_μ returns exactly $\partial_\mu F^{\mu\nu} = J^\nu$ with this J^ν — is the work of Section 8.6. There the conservation law (8.88) that gauge invariance demanded of the source will be seen to be guaranteed automatically, because J^μ is the Noether current of the very symmetry that was gauged. The circle closes: the global $U(1)$ symmetry of Chapters 6 and 7 produces a conserved current; gauging it produces the photon and couples it to that current; and the photon's own dynamics is the Maxwell Lagrangian derived here.

We have constructed the unique gauge-invariant action for the electromagnetic field, derived all of Maxwell's equations from it, and seen current conservation forced by gauge symmetry. But the very gauge freedom that selected this action has not gone away: distinct potentials A_μ and $A_\mu + \partial_\mu\alpha$ yield the same fields and the same physics, so the field equations cannot determine A_μ uniquely from initial data — the Euler–Lagrange operator $\partial_\mu F^{\mu\nu}$ has $\partial^\nu\alpha$ in its kernel. The next section confronts this redundancy head-on: it fixes the gauge in the standard ways, exposes what is left after the redundancy is removed, and counts the two genuine physical polarizations of the photon, planting the seed of the quantization subtlety that Chapters 13 and 26 will discharge.

8.5 Gauge Redundancy, Gauge Fixing, and Physical Degrees of Freedom

The previous section closed with the inhomogeneous field equations $\partial_\mu F^{\mu\nu} = J^\nu$ (Eq. (8.95)) and a result that, read carefully, is more unsettling than it first appears: the dynamics depends on the potential A_μ *only* through the field strength $F_{\mu\nu}$ (Theorem 8.9). Two potentials that differ by the gradient of a scalar produce the same $F_{\mu\nu}$, hence the same equations of motion, the same forces, the same everything measurable. The theory therefore does not determine A_μ; it determines at most an equivalence class of A_μ's. This is not a defect to be patched but the central structural feature of the field, and learning to live with it — to count what is real, to remove what is not, and to do so without spoiling the symmetries we prize — is the business of this section. The redundancy we expose here is the classical shadow of the difficulty that will dominate the quantization of the photon in Chapter 13 and of non-abelian gauge fields in Chapter 26; getting the classical bookkeeping exactly right now is what makes those later chapters tractable.

We carry the conventions fixed for the chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, $A^\mu = (\phi, \mathbf{A})$ with $A_\mu = (\phi, -\mathbf{A})$, field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, and the Maxwell Lagrangian with a conserved external source,

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu}F^{\mu\nu} - J_\mu A^\mu, \tag{8.112}$$

recalled from Eq. (8.75), whose Euler–Lagrange equation is Eq. (8.95). The electric and magnetic fields are $E^i = F^{i0}$ and $B^i = -\tfrac{1}{2}\epsilon^{ijk}F_{jk}$, so that $\mathbf{E} = -\nabla\phi - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$.

8.5.1 Gauge redundancy: the configuration space is a quotient

We begin by naming the transformation precisely and stating exactly what it does and does not change.

Definition 8.16 (Gauge transformation)**.** Let $\alpha : \mathbb{R}^{1,3} \to \mathbb{R}$ be a smooth function on spacetime. The associated *gauge transformation* of the electromagnetic potential is

$$A_\mu(x) \;\longmapsto\; A^{(\alpha)}_\mu(x) = A_\mu(x) + \partial_\mu \alpha(x). \tag{8.113}$$

Two potentials are said to be *gauge equivalent* if they are related by Eq. (8.113) for some α; the set of all potentials gauge equivalent to a given A_μ is its *gauge orbit*.

Three facts make this an equivalence relation and a redundancy of the description rather than a physical operation. They are elementary, but stating them is what licenses every later manipulation.

Proposition 8.17 (Gauge redundancy)**.** *The map Eq. (8.113) has the following properties.*

(i) It leaves the field strength invariant: $F^{(\alpha)}_{\mu\nu} = F_{\mu\nu}$ *for every* α*.*

(ii) It is an equivalence relation: *gauge equivalence is reflexive, symmetric, and transitive, so the space of potentials is partitioned into disjoint gauge orbits.*

(iii) It leaves the physics invariant: A_μ *and* $A^{(\alpha)}_\mu$ *solve Eq. (8.95) for the same source* J^ν*, produce the same* $\mathbf{E}, \mathbf{B}$*, and (when matter is coupled minimally, as in the next section) the same equations of motion for the matter, provided the source is conserved,* $\partial_\nu J^\nu = 0$*.*

Proof. (i) Substituting Eq. (8.113) into the field strength,

$$F^{(\alpha)}_{\mu\nu} = \partial_\mu(A_\nu + \partial_\nu\alpha) - \partial_\nu(A_\mu + \partial_\mu\alpha) = F_{\mu\nu} + (\partial_\mu\partial_\nu - \partial_\nu\partial_\mu)\alpha = F_{\mu\nu}, \quad (8.114)$$

the added term vanishing because partial derivatives commute on the smooth function α. This is the same cancellation already recorded in Theorem 8.9; we repeat it because it is the engine of everything below.

(ii) Reflexivity is $\alpha = 0$. Symmetry follows because the inverse of Eq. (8.113) is the gauge transformation with parameter $-\alpha$: if $A' = A + \partial\alpha$ then $A = A' + \partial(-\alpha)$. Transitivity follows because the composition of two gauge transformations is a third: $(A + \partial\alpha) + \partial\beta = A + \partial(\alpha + \beta)$. The parameters thus form an abelian group under addition — the abelian character of U(1) electromagnetism — and the orbits, as equivalence classes, are disjoint and cover the whole space of potentials.

(iii) Since the field equation Eq. (8.95) is written entirely in terms of $F_{\mu\nu}$, and $F_{\mu\nu}$ is unchanged by (i), any solution remains a solution. Equivalently, at the level of the action: under Eq. (8.113) the kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is strictly invariant by (i), while the source term changes by

$$-J_\mu A^{(\alpha)\,\mu} = -J_\mu A^\mu - J_\mu\partial^\mu\alpha = -J_\mu A^\mu - \partial^\mu(J_\mu\alpha) + (\partial^\mu J_\mu)\,\alpha. \quad (8.115)$$

The middle term is a total divergence, integrating to a boundary contribution that does not affect the equations of motion; the last term vanishes by current conservation $\partial_\mu J^\mu = 0$. Hence the action changes at most by a boundary term, $S[A^{(\alpha)}] = S[A] + \oint \mathrm{d}\Sigma_\mu(-J^\mu\alpha)$, and the dynamics is untouched. The measurable fields $\mathbf{E}, \mathbf{B}$ are components of $F_{\mu\nu}$ and so are invariant by (i). □

Equation (8.115) repays a moment's attention: gauge invariance of the matter-coupled action is *equivalent* to conservation of the source. We derived current conservation from gauge invariance in the previous section; Eq. (8.115) is the same coin viewed from the other face — if J^μ were not conserved, Eq. (8.113) would change the physics, and the would-be gauge symmetry would be illusory. The two statements are locked together, and this lock is the classical seed of the Ward–Takahashi identities of the quantum theory.

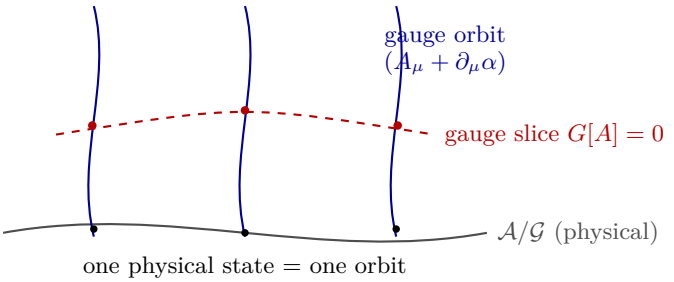


Figure 8.4: The configuration space of electromagnetism as a quotient. Vertical fibers are gauge orbits (all A_μ related by Eq. (8.113)); each projects to a single physical configuration in the base $\mathcal{A}/\mathcal{G}$. A gauge condition $G[A] = 0$ is a slice (dashed) chosen to meet each orbit exactly once, selecting one representative per physical state. *Reachability* of the gauge is the statement that the slice meets every orbit; *residual freedom* is the statement that it meets some orbits more than once.

Physical Insight

The lesson of Theorem 8.17 is geometric. The true arena of electromagnetism is not the space $\mathcal{A}$ of potentials but the quotient $\mathcal{A}/\mathcal{G}$ by the group $\mathcal{G}$ of gauge transformations — the space of *orbits*. Physical configurations are points of this quotient; a single potential A_μ is a particular, arbitrary representative of a point, like a particular set of coordinates on a manifold. Every physical quantity is a function on the quotient, i.e. a gauge-invariant functional of A_μ. The redundancy is the price of describing this quotient with the linearly-transforming, manifestly Lorentz-covariant object A_μ rather than with the gauge-invariant but cumbersome $F_{\mu\nu}$ (which carries its own redundancy through the Bianchi identity and cannot be coupled minimally to charged matter). Holding a covariant description and a clean count of physical data at the same time is exactly the tension that gauge fixing manages.

8.5.2 Counting the physical degrees of freedom

The potential A_μ has four components per spacetime point, but Theorem 8.17 shows that not all four are physical. We now make the count precise. The naive bookkeeping is

$$\underbrace{4}_{\text{components of } A_\mu} - \underbrace{1}_{\text{gauge freedom}} - \underbrace{1}_{\text{non-dynamical } A_0 \,/\, \text{Gauss constraint}} = \underbrace{2}_{\text{physical polarizations}}, \tag{8.116}$$

and the rest of this subsection earns each subtraction. The first is immediate: Eq. (8.113) lets us adjust one function $\alpha(x)$ freely, so one combination of the four components carries no information. The second subtraction is subtler and deserves a full argument, because it is where the constrained, non-propagating nature of A_0 enters — and that is precisely the feature that complicates canonical quantization in Chapter 13.

A_0 is non-dynamical. A field component is *dynamical* only if the Lagrangian contains its time derivative, for only then does it carry a canonical momentum and an evolution equation second-order in time. Examine the time derivatives in Eq. (8.112). The kinetic term is built from $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, whose components split into

$$F_{0i} = \partial_0 A_i - \partial_i A_0, \qquad F_{ij} = \partial_i A_j - \partial_j A_i, \tag{8.117}$$

with $F_{00} \equiv 0$ by antisymmetry. Time derivatives appear *only* in F_{0i}, and there only as $\partial_0 A_i$ — the time derivative of the *spatial* components. The time derivative $\partial_0 A_0$ never occurs anywhere, because the would-be carrier F_{00} vanishes identically. Concretely, the canonical momentum conjugate to A_μ is

$$\pi^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} = -F^{0\mu}, \tag{8.118}$$

as one reads off by differentiating $-\frac{1}{4}F_{\alpha\beta}F^{\alpha\beta}$ (the source term has no field-derivatives). Its time component is a *primary constraint*,

$$\pi^0 = -F^{00} = 0, \tag{8.119}$$

holding identically and not as an equation of motion. The spatial components are the physical electric field, $\pi^i = -F^{0i} = F^{i0} = E^i$. Thus A_0 has vanishing conjugate momentum: it is not a dynamical variable but a Lagrange multiplier.

Proposition 8.18 (The $\nu = 0$ Maxwell equation is the Gauss-law constraint)**.** *The time component ($\nu = 0$) of the field equation Eq. (8.95) contains no second time derivative of any field and is therefore a constraint on initial data rather than an evolution equation. Explicitly it reads*

$$\nabla\cdot\mathbf{E} = J^0 = \rho, \tag{8.120}$$

Gauss's law, which determines (the non-dynamical) A_0 in terms of the instantaneous configuration of the dynamical fields and the charge density.

Proof. Set $\nu = 0$ in $\partial_\mu F^{\mu 0} = J^0$. Since $F^{00} = 0$, the $\mu = 0$ term drops and only the spatial divergence survives,

$$\partial_\mu F^{\mu 0} = \partial_i F^{i0} = \partial_i E^i = \nabla\cdot\mathbf{E} = J^0. \tag{8.121}$$

No ∂_0^2 appears: the highest time derivative present is the single one inside $E^i = F^{i0} = -\partial_i A^0 - \partial_0 A^i$, namely $\partial_0 A^i$, which enters Eq. (8.120) as a *first* time derivative. A relation among the fields and their first time derivatives at a single instant, with no second time derivative to integrate forward, is a constraint on the Cauchy data, not a law of evolution. The other three equations ($\nu = i$) do contain $\partial_0^2 A^i$ and are the genuine evolution equations for the dynamical A^i. □

The two subtractions in Eq. (8.116) are now justified, and it is worth seeing them once more in the cleaner phase-space language, because that is the form the quantization chapters use. The phase space carries the eight functions (A_μ, π^μ). The primary constraint Eq. (8.119), $\pi^0 \approx 0$, and the secondary (Gauss) constraint Eq. (8.120), $\partial_i \pi^i - \rho \approx 0$ — which arises by demanding that $\pi^0 \approx 0$ be preserved in time — are two *first-class* constraints (their Poisson bracket vanishes, reflecting the abelian gauge group). A standard result of constrained Hamiltonian dynamics is that each first-class constraint removes two phase-space functions: one because the constraint surface is lower-dimensional, one because it generates a gauge transformation along which physics is constant. Two first-class constraints thus remove $2 \times 2 = 4$ of the eight phase-space functions, leaving $8 - 4 = 4$ physical phase-space functions, i.e. $4/2 = 2$ configuration-space degrees of freedom per point — the same two as Eq. (8.116). The configuration-space heuristic and the phase-space count agree, and both deliver the answer we will confirm explicitly with plane waves below: *the electromagnetic field has two physical, transverse polarizations.*

Caution

The count "$4 - 1 - 1 = 2$" is not "4 minus 2 arbitrary subtractions." The two subtracted units are physically distinct and are removed by distinct mechanisms. One unit is pure redundancy — the gauge function α — and is removed by a *choice* (gauge fixing). The other is a genuine equation, Gauss's law, that the data must satisfy; it is removed by a *constraint*, not a choice. Conflating them is the source of much confusion in canonical quantization, where the gauge must be fixed (a choice) *and* the Gauss constraint imposed (a law) before the physical Hilbert space is correctly identified — the subject of Chapter 13.

8.5.3 Lorenz gauge: covariant fixing

Gauge fixing means choosing, for each orbit, a single representative — a slice through the orbits, as in Fig. 8.4. A good gauge condition must be (a) *reachable*: every orbit contains at least one representative obeying it; and ideally (b) *complete*: it picks exactly one. We treat in turn the gauge that keeps Lorentz covariance manifest and the gauge that keeps the physical content manifest, signposting the

trade-off that the quantization chapters inherit.

Definition 8.19 (Lorenz gauge)**.** A potential is said to be in *Lorenz gauge* if it satisfies the Lorentz-invariant condition

$$\partial_\mu A^\mu = 0. \tag{8.122}$$

Because the left side $\partial_\mu A^\mu$ is a Lorentz scalar, the condition Eq. (8.122) has the same form in every inertial frame: a potential that is in Lorenz gauge for one observer is in Lorenz gauge for all. This is the property that makes the gauge indispensable for relativistic calculations, and it is exactly the property the alternatives below surrender.

Proposition 8.20 (Lorenz gauge is reachable, and reduces Maxwell to the wave equation)**.** *Given any potential A_μ, there exists a gauge transformation Eq. (8.113) bringing it into Lorenz gauge. In that gauge the field equation Eq. (8.95) becomes the inhomogeneous wave equation*

$$\Box A^\nu = J^\nu, \qquad \Box \equiv \partial_\mu \partial^\mu = \partial_t^2 - \nabla^2. \tag{8.123}$$

Proof. Let A_μ be arbitrary, with $\partial_\mu A^\mu = f(x)$ some function not necessarily zero. Seek α so that the transformed potential $A_\mu^{(\alpha)} = A_\mu + \partial_\mu \alpha$ is in Lorenz gauge:

$$0 \overset{!}{=} \partial_\mu A^{(\alpha)\,\mu} = \partial_\mu A^\mu + \partial_\mu \partial^\mu \alpha = f + \Box \alpha. \tag{8.124}$$

This requires α to solve the inhomogeneous wave equation $\Box \alpha = -f$. The d'Alembertian $\Box$ admits a Green's function — the retarded propagator constructed in Chapter 5 — so a solution exists for any (suitably decaying) source f:

$$\alpha(x) = -\int \mathrm{d}^4 y \; G_{\mathrm{ret}}(x-y)\, f(y), \qquad \Box_x G_{\mathrm{ret}}(x-y) = \delta^{(4)}(x-y). \tag{8.125}$$

Hence Lorenz gauge is reachable from any starting potential. Once Eq. (8.122) holds, expand the field equation. Writing $F^{\mu\nu} = \partial^\mu A^\nu - \partial^\nu A^\mu$,

$$\partial_\mu F^{\mu\nu} = \partial_\mu \partial^\mu A^\nu - \partial^\nu (\partial_\mu A^\mu) = \Box A^\nu - \partial^\nu (\partial_\mu A^\mu). \tag{8.126}$$

The second term is $-\partial^\nu(\partial_\mu A^\mu) = 0$ in Lorenz gauge, and $\partial_\mu F^{\mu\nu} = J^\nu$ collapses to Eq. (8.123). □

What Theorem 8.20 buys is decisive: in Lorenz gauge each of the four components of A^ν obeys an independent, manifestly covariant wave equation sourced by the corresponding component of the conserved current. The price — visible already in Eq. (8.116) — is that four components obey a two-degree-of-freedom theory, so Lorenz gauge cannot be complete. There must be leftover gauge freedom. There is, and identifying it precisely is what allows the explicit polarization count.

Proposition 8.21 (Residual gauge freedom in Lorenz gauge)**.** *The Lorenz condition Eq. (8.122) does not fix the gauge completely. A further gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu \alpha$ preserves Lorenz gauge if and only if the parameter is a solution of the homogeneous wave equation,*

$$\Box \alpha = 0. \tag{8.127}$$

The space of such α is infinite-dimensional, so a one-function residual freedom remains.

Proof. Suppose A_μ already satisfies $\partial_\mu A^\mu = 0$. After a further transformation with parameter α, the new divergence is $\partial_\mu(A^\mu + \partial^\mu \alpha) = 0 + \Box\alpha$. This vanishes — keeping us in Lorenz gauge — precisely when $\Box\alpha = 0$. The homogeneous wave equation has a vast solution space (every superposition of null plane waves $e^{-ik\cdot x}$ with $k^2 = 0$ solves it), so the residual freedom is genuine and infinite-dimensional, exactly one undetermined function's worth, matching the one excess unit that Eq. (8.116) says must still be removable after Lorenz fixing. □

8.5.4 Coulomb, temporal, and axial gauges: the trade-off

Lorenz gauge keeps covariance manifest at the cost of carrying unphysical components (the timelike and longitudinal photons of Chapter 13). The opposite philosophy fixes the gauge so that only physical degrees of freedom remain, at the cost of manifest covariance. The cleanest such choice is the Coulomb gauge.

Definition 8.22 (Coulomb gauge)**.** A potential is in *Coulomb* (or *radiation*, or *transverse*) gauge if its spatial part is divergence-free,

$$\nabla\cdot\mathbf{A} = \partial_i A^i = 0. \tag{8.128}$$

The reachability argument parallels Theorem 8.20 but uses the Poisson rather than the wave operator, which is the technical signature of the lost covariance. Given A_μ with $\nabla\cdot\mathbf{A} = g(x)$, a gauge transformation shifts $\mathbf{A} \mapsto \mathbf{A} + \nabla\alpha$ (since $A_i^{(\alpha)} = A_i + \partial_i\alpha$ and $A^i = -A_i$), so

$$\nabla\cdot\mathbf{A}^{(\alpha)} = \nabla\cdot\mathbf{A} + \nabla^2\alpha = g + \nabla^2\alpha \overset{!}{=} 0 \implies \nabla^2\alpha = -g, \tag{8.129}$$

a Poisson equation solved at each instant by $\alpha(t,\mathbf{x}) = \frac{1}{4\pi}\int \mathrm{d}^3y\, g(t,\mathbf{y})/|\mathbf{x}-\mathbf{y}|$. Coulomb gauge is thus reachable. The residual freedom is the set of α with $\nabla^2\alpha = 0$ at each time, i.e. spatially harmonic functions; with the physical boundary condition that fields vanish at spatial infinity these are at most spatially constant, so Coulomb gauge is essentially complete (only a global, physically inert constant survives). The structural payoff appears in Gauss's law: Eq. (8.120) becomes

$$\nabla\cdot\mathbf{E} = -\nabla^2 A^0 - \partial_0(\nabla\cdot\mathbf{A}) = -\nabla^2 A^0 = \rho \implies \nabla^2\phi = -\rho, \tag{8.130}$$

using Eq. (8.128) to kill the second term. The scalar potential $\phi = A^0$ is therefore not an independent field at all but the *instantaneous* Coulomb potential of the charge distribution, $\phi(t,\mathbf{x}) = \frac{1}{4\pi}\int \mathrm{d}^3y\, \rho(t,\mathbf{y})/|\mathbf{x}-\mathbf{y}|$ — hence the name. Both non-physical components (A^0 and the longitudinal part of $\mathbf{A}$) are eliminated explicitly, leaving the two transverse components of $\mathbf{A}$ as the only dynamical fields. Manifest unitarity (only physical, positive-norm degrees survive) is bought by sacrificing manifest covariance: Eq. (8.130) is an instantaneous, frame-dependent relation, and a Lorentz boost takes one out of Coulomb gauge.

Notation Watch

The condition $\partial_\mu A^\mu = 0$ is the *Lorenz* gauge, named for the Danish physicist Ludvig Lorenz, and is *not* the Lorentz (Hendrik Lorentz) transformation, despite the near-homograph and the fact that the gauge happens to be Lorentz-invariant. We follow the now-standard spelling "Lorenz gauge" throughout. "Coulomb," "radiation," and "transverse" gauge all denote the same condition Eq. (8.128).

Two further gauges round out the standard toolkit; we state them and their use, since they recur in Chapter 13 and Chapter 26.

Temporal gauge $A^0 = 0$ uses the gauge function $\alpha(t,\mathbf{x}) = -\int_0^t A^0(t',\mathbf{x})\,\mathrm{d}t'$ to set the scalar potential to zero; it is reachable but leaves a large residual freedom (any time-independent $\alpha(\mathbf{x})$), so it must be supplemented by imposing Gauss's law by hand on the initial data. It is convenient in canonical and lattice formulations because it removes the constrained variable outright. *Axial gauge* $n_\mu A^\mu = 0$ for a fixed reference four-vector n_μ (light-cone gauge when $n^2 = 0$) likewise eliminates one component algebraically; it is prized in perturbative gauge theory because the unphysical polarizations decouple without ghosts, at the cost of a frame-dependent, n-dependent propagator. The common thread is the trade governing every gauge choice:

> *a gauge that keeps Lorentz covariance manifest (Lorenz, axial-covariant) retains unphysical components whose unphysical nature must be enforced separately; a gauge that keeps only physical components (Coulomb, temporal) breaks manifest covariance.*

No gauge does both. This irreducible tension is benign classically — one simply picks the convenient gauge for the problem at hand — but becomes the defining difficulty of quantizing the photon, where covariance is needed to keep calculations tractable yet a positive-norm Hilbert space is needed for probabilities to make sense. The resolution, the Gupta–Bleuler construction and its modern descendant the BRST cohomology, is the business of Chapter 13 and Chapter 26; the present section's job is to have made the problem precise.

8.5.5 Plane waves and the two physical polarizations

We now make the abstract count of Eq. (8.116) completely explicit by solving the free field equations and exhibiting the two physical polarization vectors. This is the classical photon, and the polarization vectors constructed here are exactly those that will be promoted to creation operators in Chapter 13.

Worked Example 8.23 (The free plane wave and its two transverse polarizations)**.** We solve the source-free Maxwell equations in Lorenz gauge, impose the gauge and residual conditions, and read off the physical polarizations.

Step 1: the wave equation and the dispersion relation. With $J^\nu = 0$, Theorem 8.20 reduces Maxwell's equations in Lorenz gauge to $\Box A^\nu = 0$ together with $\partial_\mu A^\mu = 0$. Following the Fourier methods of Chapter 5, seek a complex plane-wave solution (the physical field is its real part),

$$A^\mu(x) = \varepsilon^\mu(k)\, e^{-ik\cdot x} + \text{c.c.}, \qquad k\cdot x = k^0 t - \mathbf{k}\cdot\mathbf{x}, \tag{8.131}$$

with constant *polarization vector* $\varepsilon^\mu(k)$. Acting with the d'Alembertian,

$$\Box A^\mu = \partial_\nu\partial^\nu\big(\varepsilon^\mu e^{-ik\cdot x}\big) = (-ik_\nu)(-ik^\nu)\,\varepsilon^\mu e^{-ik\cdot x} = -k^2\,\varepsilon^\mu e^{-ik\cdot x}, \tag{8.132}$$

so $\Box A^\mu = 0$ forces the *massless dispersion relation*

$$k^2 = (k^0)^2 - |\mathbf{k}|^2 = 0, \qquad k^0 = |\mathbf{k}| \equiv \omega. \tag{8.133}$$

The wave is null — the photon is massless — and propagates at the speed of light, as a classical electromagnetic wave must.

Step 2: the Lorenz condition removes one component. Imposing Eq. (8.122) on the ansatz,

$$\partial_\mu A^\mu = -ik_\mu\varepsilon^\mu\, e^{-ik\cdot x} + \text{c.c.} = 0 \implies k_\mu\varepsilon^\mu = 0. \tag{8.134}$$

This is one complex linear condition on the four components of ε^μ, leaving three. We have used the gauge *condition*; we have not yet used the residual gauge *freedom*.

Step 3: the residual freedom removes a second component. By Theorem 8.21 we may still shift the potential by $\partial_\mu\alpha$ with $\Box\alpha = 0$. Choose a residual parameter of the same wave-vector,

$$\alpha(x) = i\,a\,e^{-ik\cdot x} + \text{c.c.}, \qquad \Box\alpha = -k^2\alpha = 0 \text{ (by Eq. (8.133))}, \tag{8.135}$$

with a a complex constant. Then $\partial_\mu\alpha = ia(-ik_\mu)e^{-ik\cdot x} + \text{c.c.} = a\,k_\mu e^{-ik\cdot x} + \text{c.c.}$, so the polarization vector shifts by a multiple of k^μ,

$$\varepsilon^\mu \;\longmapsto\; \varepsilon^\mu + a\,k^\mu. \tag{8.136}$$

Two polarization vectors differing by a multiple of k^μ describe the same physical wave. Note this shift is consistent with Step 2: it changes $k_\mu \varepsilon^\mu$ by $a\, k_\mu k^\mu = ak^2 = 0$, so it preserves the Lorenz condition, exactly as Theorem 8.21 guarantees. The four components of ε^μ, minus one from Eq. (8.134), minus one from the freedom Eq. (8.136), leave $4 - 1 - 1 = 2$ — the count Eq. (8.116), now realized componentwise.

Step 4: choose a frame and exhibit the polarizations. Orient the spatial axes so the wave travels along z:

$$k^\mu = (\omega, 0, 0, \omega), \qquad k_\mu = (\omega, 0, 0, -\omega). \tag{8.137}$$

With $k_\mu = (\omega, 0, 0, -\omega)$ and $\varepsilon^\mu = (\varepsilon^0, \varepsilon^1, \varepsilon^2, \varepsilon^3)$, the Lorenz condition Eq. (8.134) reads $k_\mu \varepsilon^\mu = \omega\varepsilon^0 + 0 + 0 + (-\omega)\varepsilon^3 = \omega(\varepsilon^0 - \varepsilon^3) = 0$, hence

$$\varepsilon^0 = \varepsilon^3. \tag{8.138}$$

Now spend the residual freedom Eq. (8.136) to set the time component to zero. With $k^\mu = (\omega, 0, 0, \omega)$, choosing $a = -\varepsilon^0/\omega$ gives a new time component $\varepsilon^0 + a\, k^0 = \varepsilon^0 - \varepsilon^0 = 0$, and simultaneously a new third component $\varepsilon^3 + a\, k^3 = \varepsilon^3 - \varepsilon^0 = 0$ by Eq. (8.138). Both the timelike and the longitudinal (along $\mathbf{k}$) components are thereby removed:

$$\varepsilon^0 = \varepsilon^3 = 0, \qquad \varepsilon^\mu = (0, \varepsilon^1, \varepsilon^2, 0). \tag{8.139}$$

The surviving polarization is purely *transverse* — perpendicular to the direction of propagation $\mathbf{k}$. (This frame choice has, in effect, brought the free field into Coulomb gauge: Eq. (8.139) gives $\varepsilon^0 = 0$ and $\mathbf{k}\cdot\boldsymbol{\varepsilon} = \omega\varepsilon^3 = 0$, i.e. Eq. (8.128), confirming that Lorenz-plus-residual lands on the same two physical states the manifestly-physical gauge isolates.) A convenient real orthonormal basis for the two-dimensional space of transverse polarizations is

$$\varepsilon^\mu_{(1)} = (0, 1, 0, 0), \qquad \varepsilon^\mu_{(2)} = (0, 0, 1, 0), \tag{8.140}$$

normalized in the mostly-minus metric to $\varepsilon_{(\lambda)} \cdot \varepsilon_{(\lambda')} = \eta_{\mu\nu}\varepsilon^\mu_{(\lambda)}\varepsilon^\nu_{(\lambda')} = -\delta_{\lambda\lambda'}$ (spacelike, hence the minus sign). Equivalently, the circular (helicity) basis diagonalizes rotations about $\mathbf{k}$,

$$\varepsilon^\mu_\pm = \mp\tfrac{1}{\sqrt{2}}(0, 1, \pm i, 0), \qquad \varepsilon_\pm \cdot \varepsilon^*_\pm = -1, \quad \varepsilon_\pm \cdot \varepsilon^*_\mp = 0, \tag{8.141}$$

corresponding to the two helicities ± 1 of the photon. The general free solution is the superposition over all null k of these two polarizations, $A^\mu(x) = \sum_{\lambda=1,2} \int \frac{\mathrm{d}^3k}{(2\pi)^3} \left[\varepsilon^\mu_{(\lambda)}(k)\, c_\lambda(k)\, e^{-ik\cdot x} + \text{c.c.}\right]$ — precisely the expansion that becomes the photon mode expansion when the coefficients $c_\lambda(k)$ are promoted to operators in Chapter 13.

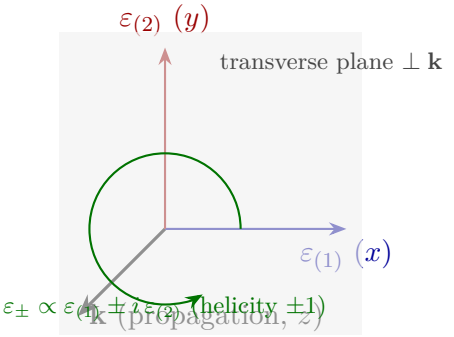


Figure 8.5: The two physical polarizations of a free electromagnetic plane wave (Theorem 8.23). After imposing the Lorenz condition Eq. (8.134) and spending the residual freedom Eq. (8.136), the polarization vector lies in the plane transverse to the propagation direction $\mathbf{k}$. The real basis $\varepsilon_{(1)}, \varepsilon_{(2)}$ describes linear polarization; the complex combinations $\varepsilon_\pm$ describe the two circular (helicity ± 1) states.

The worked example closes the loop opened by Eq. (8.116): the abstract assertion that two of the four components of A_μ are physical has become the concrete pair of transverse polarization vectors Eq. (8.140), with the gauge redundancy realized step by step as the removal first of the timelike and then of the longitudinal direction. We now possess a complete classical field: a gauge potential A_μ whose redundancy is understood and fixable, whose physical content is exactly two propagating, transverse, helicity-± 1 polarizations, and whose equations of motion are sourced by a conserved current. This clean object is ready to be coupled to charged matter. In the next section we assemble the full classical QED Lagrangian, identify the electromagnetic current as the global U(1) Noether current of Chapter 7, and watch it appear as the very source J^μ of the Maxwell field — launching QED at its classical layer and locking symmetry to dynamics.

8.6 The Classical QED Lagrangian

The previous sections built, piece by piece, every ingredient of classical electrodynamics from a single demand: that the global $U(1)$ phase symmetry of the matter field be promoted to a local one. Requiring invariance under $\psi \to e^{ie\alpha(x)}\psi$ forced into existence a connection A_μ, a covariant derivative that compares the field at neighbouring points, and a curvature $F_{\mu\nu}$ measuring the failure of that comparison to be path-independent. We then gave the connection its own gauge-invariant dynamics — the Maxwell Lagrangian $\mathcal{L}_{\text{Max}} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ of Eq. (8.75), whose field equation is Eq. (8.95) — and, in the preceding section, exposed its gauge redundancy and counted the two physical polarizations of the radiation field. What remains is to put the two halves together: the minimally coupled electron

handed up through Eq. (8.29), and the photon. The result is the complete *classical* theory of electrons and photons, and it is the launching point of QED, the running example that will carry us — through quantization, scattering, loops, and the anomalous magnetic moment — to the precision frontier of the book.

The assembly is more than bookkeeping. When we write down the coupled Lagrangian and turn the variational crank, a remarkable thing happens: the conserved current that Noether's theorem (Chapter 7) extracts from the global $U(1)$ symmetry of the matter sector turns out to be *exactly* the source term that the electromagnetic field equations demand. The symmetry current and the field source are one and the same object. This is the deepest payoff of the gauge principle at the classical level — symmetry and dynamics are welded together — and it is the structural fact that makes the quantized theory consistent. The present section assembles the Lagrangian, derives the current as the Noether current, identifies it as the Maxwell source, writes down the coupled equations of motion, verifies the full gauge invariance of the action, and recovers the Lorentz force law in the point-particle limit. By its end the symbols ψ, m, e, α, A_μ, and $F_{\mu\nu}$ that every later QED chapter uses will be fixed once and for all.

8.6.1 Assembling the Lagrangian

The matter sector is the free Dirac Lagrangian of Eq. (6.68), $\mathcal{L}_D = \bar{\psi}(i\partial\!\!\!/ - m)\psi$, built in Chapter 6 for a spin-$\frac{1}{2}$ field ψ of mass m with the Dirac adjoint $\bar{\psi} = \psi^\dagger\gamma^0$; the gamma-matrix conventions are those of Chapter C, with $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ and the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$. The gauge principle instructs us to replace the ordinary derivative by the covariant derivative of the electron field, Eq. (8.29),

$$D_\mu \;=\; \partial_\mu + ieA_\mu, \tag{8.142}$$

in which $e > 0$ is the elementary charge and the electron carries charge $q = -e$, so that the general rule $D_\mu = \partial_\mu - iqA_\mu$ of Chapter A specializes to the $+ie$ appearing here. Adding to the minimally coupled matter the gauge-invariant kinetic term for the connection gives the complete classical Lagrangian density of quantum electrodynamics:

$$\boxed{\mathcal{L}_{\mathrm{QED}} \;=\; \bar{\psi}\,(iD\!\!\!/ - m)\,\psi \;-\; \tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.} \tag{8.143}$$

This is the frozen QED model. From here onward, whenever a later chapter says "the QED Lagrangian" it means exactly Eq. (8.143), with these symbols and this sign convention.

It is worth seeing the interaction laid bare. Expanding the covariant derivative in $D\!\!\!/ = \gamma^\mu D_\mu = \gamma^\mu(\partial_\mu + ieA_\mu)$ and using $i\gamma^\mu(ieA_\mu) = -e\gamma^\mu A_\mu$,

$$\mathcal{L}_{\mathrm{QED}} \;=\; \underbrace{\bar{\psi}\,(i\partial\!\!\!/ - m)\,\psi}_{\text{free electron}} \;\underbrace{-\,\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free photon}} \;\underbrace{-\,e\,\bar{\psi}\gamma^\mu\psi\,A_\mu}_{\text{interaction}}. \tag{8.144}$$

The entire coupling between matter and radiation is the single trilinear term $-e\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu}$: one photon line A_{μ} attached to the bilinear $\bar{\psi}\gamma^{\mu}\psi$. The strength of that coupling is governed by e, and it is conventional — and, once $\hbar$ and c are restored, dimensionally natural — to package it as the dimensionless *fine-structure constant*

$$\alpha \;=\; \frac{e^{2}}{4\pi} \;\approx\; \frac{1}{137}, \tag{8.145}$$

in the Heaviside–Lorentz, natural units used throughout (the 4π is absent from the Lagrangian because those units push it into the definition of charge). Every perturbative QED amplitude in the chapters to come is an expansion in powers of α, and the smallness of α is why that expansion is so spectacularly accurate.

Physical Insight

The interaction term in Eq. (8.144) was not posited; it was *forced*. We did not choose to couple the electron to the photon through $\bar{\psi}\gamma^{\mu}\psi A_{\mu}$ — demanding local $U(1)$ invariance left no other option for a renormalizable, gauge-invariant coupling at lowest order. The vector nature of the coupling (a γ^{μ}, hence a current), its universality (every charged field couples through the same A_{μ} with strength set by its charge), and even the masslessness of the photon (a mass term $\frac{1}{2}m_{\gamma}^{2}A_{\mu}A^{\mu}$ is not gauge invariant) are all consequences of the principle, not inputs. This is why the gauge principle is the organizing idea of the book: it converts a symmetry requirement into a unique dynamics.

The structure of the trilinear vertex is the object the Feynman rules of Chapter 17 will encode; it is worth drawing now, while it is still purely classical, so the diagrammatic language has a referent (Fig. 8.6).

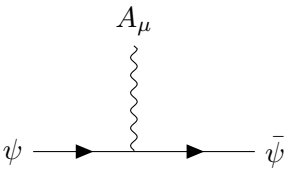


Figure 8.6: The classical QED interaction vertex read off from $-e\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu}$ in Eq. (8.144): an electron line (the conserved current $\bar{\psi}\gamma^{\mu}\psi$) meets the photon A_{μ} with coupling proportional to e. The same vertex, promoted to operators, is the single building block of all QED Feynman diagrams (Chapter 17).

8.6.2 The electromagnetic current as the $U(1)$ Noether current

The matter Lagrangian, with or without the coupling to A_{μ}, is invariant under the global phase rotation $\psi \to e^{ie\alpha}\psi$ with α a constant — the rigid remnant

of the local symmetry that drove the whole construction. By Noether's first theorem (Theorem 7.10) this continuous internal symmetry produces a conserved current. We now compute it, because it is the central object of QED: it is what couples to the photon, what sources Maxwell's equations, and what becomes, on quantization, the conserved electric charge.

Definition 8.24 (Electromagnetic four-current)**.** The *electromagnetic current* of the Dirac field is the Noether current of the global $U(1)$ symmetry $\psi \to e^{ie\alpha}\psi$, namely

$$\boxed{J^\mu \;=\; -\,e\,\bar\psi\,\gamma^\mu\,\psi} \tag{8.146}$$

Its time component $J^0 = -e\,\psi^\dagger\psi$ is the electric charge density, carrying the electron's negative sign through the factor $-e$; the spatial part $\mathbf{J} = -e\,\psi^\dagger\boldsymbol{\gamma}^0\boldsymbol{\gamma}\psi$ is the electric current density.

We derive Eq. (8.146) from the Lagrangian Eq. (8.143) and verify its conservation in full, as the first stage of QED's central computations.

Worked Example 8.25 (The electromagnetic current from Noether's theorem)**.** We continue QED by extracting Eq. (8.146) from Eq. (8.143) using the apparatus of Chapter 7, then checking $\partial_\mu J^\mu = 0$ on the equations of motion.

Step 1: the transformation and its generators. The global $U(1)$ acts only on the matter fields and leaves the coordinates and the gauge field fixed:

$$\psi \;\to\; e^{ie\alpha}\psi, \qquad \bar\psi \;\to\; e^{-ie\alpha}\bar\psi, \qquad A_\mu \to A_\mu, \qquad \delta x^\mu = 0, \tag{8.147}$$

with α a real constant. Because this is an internal symmetry, the coordinate variation vanishes and the intrinsic variation equals the total variation. To first order in α,

$$\delta_0\psi = ie\alpha\,\psi, \qquad \delta_0\bar\psi = -ie\alpha\,\bar\psi. \tag{8.148}$$

The action is strictly invariant (not merely up to a boundary term): every term in Eq. (8.144) pairs one ψ with one $\bar\psi$, so the phases $e^{ie\alpha}e^{-ie\alpha} = 1$ cancel identically, and the constant α passes through every derivative. Hence the boundary current K^μ of Eq. (7.32) vanishes.

Step 2: the conjugate momenta. The Noether current Eq. (7.32) for an internal symmetry reduces to $J^\mu = \Pi^\mu_a\,\delta_0\phi_a$ summed over the fields, where $\Pi^\mu_a = \partial\mathcal{L}/\partial(\partial_\mu\phi_a)$ is the conjugate-momentum density of Eq. (7.17). Only the derivative of ψ appears in Eq. (8.143) — the kinetic term is $\bar\psi\,i\gamma^\mu\partial_\mu\psi$, while $\bar\psi$ enters undifferentiated and A_μ does not transform — so

$$\Pi^\mu_\psi \;=\; \frac{\partial\mathcal{L}_{\text{QED}}}{\partial(\partial_\mu\psi)} \;=\; \bar\psi\,i\gamma^\mu, \qquad \Pi^\mu_{\bar\psi} = 0. \tag{8.149}$$

The covariant derivative does not change this: $D_\mu\psi = \partial_\mu\psi + ieA_\mu\psi$ depends on $\partial_\mu\psi$ exactly as $\partial_\mu\psi$ does, so the coefficient of $\partial_\mu\psi$ is still $\bar\psi\,i\gamma^\mu$.

Step 3: assemble the current. Stripping the parameter α (the conserved current is the coefficient of the infinitesimal parameter),

$$J^\mu \;=\; \Pi^\mu_\psi \frac{\delta_0 \psi}{\alpha} \;=\; (\bar\psi\, i\gamma^\mu)\,(ie\,\psi) \;=\; i^2 e\,\bar\psi\gamma^\mu\psi \;=\; -\,e\,\bar\psi\gamma^\mu\psi, \tag{8.150}$$

which is exactly Eq. (8.146). The sign is not a convention we imposed: it is produced by the two factors of i — one from the i in the Dirac kinetic term, one from the generator ie of the phase rotation — and it correctly assigns negative charge density to the electron field.

Step 4: verify conservation on shell. Noether's theorem guarantees $\partial_\mu J^\mu = 0$ on solutions, but it is instructive — and a sharp test of the gamma-matrix conventions — to see the cancellation explicitly. Write $\partial_\mu J^\mu = -e\,\partial_\mu(\bar\psi\gamma^\mu\psi) = -e\big[(\partial_\mu\bar\psi)\gamma^\mu\psi + \bar\psi\gamma^\mu(\partial_\mu\psi)\big]$. We need the two coupled equations of motion in the gauge field, derived in Section 8.6.4 below; we quote them here. The Dirac equation $(i\not{D} - m)\psi = 0$, written out with $D_\mu = \partial_\mu + ieA_\mu$, gives

$$i\gamma^\mu\partial_\mu\psi \;=\; m\psi + e\not{A}\,\psi \quad\Longrightarrow\quad \bar\psi\gamma^\mu\partial_\mu\psi \;=\; \frac{1}{i}(m\,\bar\psi\psi + e\,\bar\psi\not{A}\psi) \;=\; -i(m\,\bar\psi\psi + e\,\bar\psi\not{A}\psi), \tag{8.151}$$

where $\not{A} = \gamma^\mu A_\mu$. The adjoint Dirac equation $\bar\psi(i\overleftarrow{\not{D}} + m) = 0$, with the adjoint covariant derivative $\bar\psi\overleftarrow{D}_\mu = \partial_\mu\bar\psi - ieA_\mu\bar\psi$ (opposite sign because $\bar\psi$ rotates as $e^{-ie\alpha}$), gives

$$i(\partial_\mu\bar\psi)\gamma^\mu \;=\; -\,m\,\bar\psi - e\,\bar\psi\not{A} \quad\Longrightarrow\quad (\partial_\mu\bar\psi)\gamma^\mu \;=\; +\,i(m\,\bar\psi + e\,\bar\psi\not{A}), \tag{8.152}$$

so that, right-multiplying by ψ, $(\partial_\mu\bar\psi)\gamma^\mu\psi = +i(m\,\bar\psi\psi + e\,\bar\psi\not{A}\psi)$. Adding the two contributions,

$$(\partial_\mu\bar\psi)\gamma^\mu\psi + \bar\psi\gamma^\mu\partial_\mu\psi \;=\; +i(m\,\bar\psi\psi + e\,\bar\psi\not{A}\psi) \;-\; i(m\,\bar\psi\psi + e\,\bar\psi\not{A}\psi) \;=\; 0. \tag{8.153}$$

The mass terms cancel and — crucially — so do the gauge-field terms $e\,\bar\psi\not{A}\psi$: the coupling to A_μ enters the two Dirac equations with opposite signs, exactly as required for the current to remain conserved in the presence of the field. Therefore

$$\partial_\mu J^\mu \;=\; -e\,\partial_\mu(\bar\psi\gamma^\mu\psi) \;=\; 0, \tag{8.154}$$

the conservation of electric charge, valid in any external or dynamical electromagnetic field.

The conserved charge associated with Eq. (8.146) through Theorem 7.12 is

$$Q \;=\; \int d^3x\, J^0 \;=\; -e\int d^3x\,\psi^\dagger\psi, \tag{8.155}$$

the total electric charge of the field configuration, time-independent by Eq. (8.154). At the classical, c-number level $\psi^\dagger\psi > 0$ is a positive number density, so Q has

a definite sign fixed by the $-e$; the full reconciliation of this with the existence of both negatively charged electrons and positively charged positrons must wait for the quantization of the Dirac field in Chapter 11, where Q becomes a normal-ordered operator counting the difference of particle and antiparticle numbers.

8.6.3 The current is the Maxwell source

We now reach the structural climax of the chapter. The current Eq. (8.146) was extracted from a *symmetry* of the matter sector. We will see that the very same current is the *source* of the electromagnetic field — the right-hand side of the inhomogeneous Maxwell equation. Symmetry current and field source coincide. This is not an accident of electromagnetism; it is the inevitable consequence of having built the coupling by gauging the symmetry whose current we computed.

Vary Eq. (8.143) with respect to the gauge field A_ν. The only A-dependence is in the Maxwell term and in the interaction $-e\,\bar\psi\gamma^\mu\psi\,A_\mu$ of Eq. (8.144). From the Maxwell sector we recorded in Eq. (6.190) that $\partial\mathcal{L}_{\rm Max}/\partial(\partial_\mu A_\nu) = -F^{\mu\nu}$, while the interaction contributes $\partial\mathcal{L}_{\rm int}/\partial A_\nu = -e\,\bar\psi\gamma^\nu\psi$ and has no ∂A dependence. The Euler–Lagrange equation Eq. (6.30) for A_ν reads $\partial_\mu[\partial\mathcal{L}/\partial(\partial_\mu A_\nu)] - \partial\mathcal{L}/\partial A_\nu = 0$, that is

$$\partial_\mu(-F^{\mu\nu}) - (-e\,\bar\psi\gamma^\nu\psi) = 0 \quad\Longrightarrow\quad \boxed{\partial_\mu F^{\mu\nu} \;=\; e\,\bar\psi\gamma^\nu\psi \;=\; -J^\nu.} \tag{8.156}$$

The inhomogeneous Maxwell equation Eq. (8.95), whose source J^ν was a prescribed external current in Section 8.4, now has that source supplied entirely by the matter fields: it is the conserved $U(1)$ current $\bar\psi\gamma^\nu\psi$. Up to the overall sign that fixes the electron's charge as negative, the source is precisely the electromagnetic current Eq. (8.146). The two italicized objects of this section — the Noether current of the matter symmetry and the source of the Maxwell field — are the same current. This is the gauge principle's payoff made concrete.

Notation Watch

Three closely related currents appear in QED and it is worth keeping them straight, because their signs are the most-checked feature of the whole subject.

- The *number current* $n^\mu = \bar\psi\gamma^\mu\psi$, conserved by Eq. (8.154), with $n^0 = \psi^\dagger\psi > 0$. It carries no charge factor.
- The *electromagnetic (electric) current* $J^\mu = -e\,n^\mu = -e\,\bar\psi\gamma^\mu\psi$ of Eq. (8.146), the Noether current of $\psi \to e^{ie\alpha}\psi$, whose charge $Q = \int J^0$ is negative for the electron.
- The *Maxwell source* $\partial_\mu F^{\mu\nu} = e\,\bar\psi\gamma^\nu\psi = -J^\nu$.

All three are the same conserved current up to a constant; the relative sign between J^μ and the source $-J^\mu$ in Eq. (8.156) is the bookkeeping of the negative electron charge in the frozen convention $D_\mu = \partial_\mu + ieA_\mu$. Once $\hbar, c$ are restored, J^μ has units of (charge)×(probability flux), and Eq. (8.156) reads $\partial_\mu F^{\mu\nu} = -J^\nu$ in Heaviside–Lorentz units. We will write the field equation in the manifestly derived form $\partial_\mu F^{\mu\nu} = e\,\bar\psi\gamma^\nu\psi$ when the matter content is explicit, and as $\partial_\mu F^{\mu\nu} = -J^\nu$ when emphasizing the current.

Physical Insight

Why must the source be the symmetry current? Because the coupling was *defined* to be $\mathcal{L}_{\text{int}} = -e\,\bar\psi\gamma^\mu\psi\,A_\mu$, i.e. the gauge field contracted with the matter current. Varying such a coupling with respect to A_μ can return nothing but that current. And the current that appears in a minimal coupling $J^\mu A_\mu$ is, by Noether's construction, exactly the conserved current of the global symmetry one gauged. So the identity "source = symmetry current" is forced the moment one builds the interaction by minimal coupling. Conversely, the conservation of the source, $\partial_\mu(\partial_\nu F^{\nu\mu}) = 0$ (automatic from the antisymmetry of F), *requires* the source to be a conserved current — the field equations are inconsistent otherwise. Gauge invariance, current conservation, and the masslessness of the photon are three faces of one fact.

A direct corollary deserves emphasis. The antisymmetry $F^{\mu\nu} = -F^{\nu\mu}$ makes $\partial_\nu\partial_\mu F^{\mu\nu} \equiv 0$ identically. Taking ∂_ν of Eq. (8.156) therefore gives $\partial_\nu J^\nu = 0$ *as a consistency condition of the field equations themselves* — independently of the Dirac dynamics. The matter equations of motion must deliver a conserved current, or Maxwell's equations have no solution. We verified in Theorem 8.25 that they do. This interlocking is the classical seed of the Ward–Takahashi identities that protect the quantum theory (Chapter 19).

8.6.4 The coupled equations of motion

We collect the full set of classical field equations of QED, derived by varying Eq. (8.143) independently with respect to $\bar\psi$, ψ, and A_μ. As in Chapter 6 we treat ψ and $\bar\psi$ as independent fields, the legitimate invertible relabelling justified at Eq. (6.66).

Proposition 8.26 (Classical QED field equations)**.** *The Euler–Lagrange equations of* $\mathcal{L}_{\text{QED}}$ *are the coupled Dirac–Maxwell system*

$$(i\not{D} - m)\psi = 0, \qquad\qquad D_\mu = \partial_\mu + ieA_\mu, \tag{8.157}$$

$$\bar{\psi}(i\overleftarrow{\not{D}} + m) = 0, \qquad \bar{\psi}\overleftarrow{D}_\mu = \partial_\mu\bar{\psi} - ieA_\mu\bar{\psi}, \tag{8.158}$$

$$\partial_\mu F^{\mu\nu} = e\,\bar{\psi}\gamma^\nu\psi = -J^\nu. \tag{8.159}$$

Proof. Variation with respect to $\bar{\psi}$. In Eq. (8.143) the adjoint field appears undifferentiated — every derivative falls on ψ — so $\partial\mathcal{L}/\partial(\partial_\mu\bar{\psi}) = 0$ and $\partial\mathcal{L}/\partial\bar{\psi} = (i\not{\partial} - m)\psi$. The Euler–Lagrange equation Eq. (6.30) for $\bar{\psi}$ collapses to its undifferentiated term, giving the gauged Dirac equation Eq. (8.157) algebraically, with no integration by parts. This is the free Dirac equation Eq. (6.70) with $\partial_\mu \to D_\mu$: minimal coupling, made dynamical.

Variation with respect to ψ. Now ψ is the differentiated field. Writing the kinetic piece as $\bar{\psi}i\gamma^\mu D_\mu\psi = \bar{\psi}i\gamma^\mu\partial_\mu\psi - e\bar{\psi}\not{A}\psi$, we have $\partial\mathcal{L}/\partial(\partial_\mu\psi) = \bar{\psi}i\gamma^\mu$ and $\partial\mathcal{L}/\partial\psi = -e\,\bar{\psi}\not{A} - m\bar{\psi}$. The Euler–Lagrange equation $\partial_\mu(\bar{\psi}i\gamma^\mu) - (-e\bar{\psi}\not{A} - m\bar{\psi}) = 0$ becomes

$$i(\partial_\mu\bar{\psi})\gamma^\mu + e\,\bar{\psi}\not{A} + m\bar{\psi} = 0, \tag{8.160}$$

which is the adjoint Dirac equation Eq. (8.158): collecting the derivative and the gauge term into $\bar{\psi}\overleftarrow{D}_\mu = \partial_\mu\bar{\psi} - ieA_\mu\bar{\psi}$ gives $i\bar{\psi}\overleftarrow{\not{D}} = i(\partial_\mu\bar{\psi})\gamma^\mu + e\bar{\psi}\not{A}$, so Eq. (8.160) reads $\bar{\psi}(i\overleftarrow{\not{D}} + m) = 0$. Equations Eq. (8.157) and Eq. (8.158) are Dirac conjugates of one another, as one verifies by taking the Hermitian conjugate of Eq. (8.157) and multiplying by γ^0, using $(\gamma^\mu)^\dagger = \gamma^0\gamma^\mu\gamma^0$ from Chapter C; the relative sign of the eA term flips precisely because $i \to -i$ under conjugation, which is the algebraic origin of the cancellation in Eq. (8.153).

Variation with respect to A_ν. This is the computation of Section 8.6.3, reproduced here for completeness. With $\partial\mathcal{L}/\partial(\partial_\mu A_\nu) = -F^{\mu\nu}$ from Eq. (6.190) and $\partial\mathcal{L}/\partial A_\nu = -e\,\bar{\psi}\gamma^\nu\psi$, the Euler–Lagrange equation gives $\partial_\mu F^{\mu\nu} = e\,\bar{\psi}\gamma^\nu\psi$, which is Eq. (8.159). □

The system Eq. (8.157)–Eq. (8.159) is the complete classical theory of electrons and photons. It is genuinely coupled and nonlinear: the electron moves in the field A_μ it itself helps to create, because the field equation Eq. (8.159) is sourced by the very ψ whose evolution Eq. (8.157) depends on A_μ. The two free theories of Chapter 6 — a Dirac field obeying $(i\not{\partial} - m)\psi = 0$ and a Maxwell field obeying $\partial_\mu F^{\mu\nu} = 0$ — are recovered in the limit $e \to 0$, in which the coupling, the source, and the back-reaction all switch off simultaneously. The single parameter e controls all three, which is again the gauge principle at work: there is only one coupling constant because there is only one way the symmetry permits matter and radiation to interact.

Historical Note

That a single gauge coupling governs both how charges source fields and how fields push charges was, historically, an empirical input to Maxwell's and Lorentz's electrodynamics — two separate postulates, the field equa-

tions and the force law. The gauge principle unifies them: both follow from the one interaction term $-e\,\bar{\psi}\gamma^{\mu}\psi A_{\mu}$. The same logic, applied to the non-abelian groups of Chapter 25, would later predict the existence and self-interactions of the W, Z, and gluons before any were observed — the most striking vindication of taking the principle seriously as a law of nature rather than a bookkeeping device.

8.6.5 Gauge invariance of the full action

We have built $\mathcal{L}_{\text{QED}}$ to be invariant under local $U(1)$ transformations, but it is worth verifying the invariance of the assembled theory in one explicit calculation, since the cancellation of the offending $\partial_{\mu}\alpha$ terms is exactly what the covariant derivative was engineered to achieve.

Proposition 8.27 (Local gauge invariance of $\mathcal{L}_{\text{QED}}$)**.** *The Lagrangian Eq. (8.143) is invariant under the simultaneous local transformation*

$$\psi(x) \to e^{ie\alpha(x)}\psi(x), \qquad \bar{\psi}(x) \to e^{-ie\alpha(x)}\bar{\psi}(x), \qquad A_{\mu}(x) \to A_{\mu}(x) - \partial_{\mu}\alpha(x), \tag{8.161}$$

for any smooth function $\alpha(x)$.

Proof. The proof rests on the single property that defines the covariant derivative: under Eq. (8.161), $D_{\mu}\psi$ transforms exactly as ψ does. Compute directly, with $\psi' = e^{ie\alpha}\psi$ and $A'_{\mu} = A_{\mu} - \partial_{\mu}\alpha$,

$$\begin{aligned} D'_{\mu}\psi' &= (\partial_{\mu} + ieA'_{\mu})(e^{ie\alpha}\psi) \\ &= e^{ie\alpha}\Big[\partial_{\mu}\psi + ie(\partial_{\mu}\alpha)\psi + ie(A_{\mu} - \partial_{\mu}\alpha)\psi\Big] \\ &= e^{ie\alpha}(\partial_{\mu} + ieA_{\mu})\psi \;=\; e^{ie\alpha}\,D_{\mu}\psi. \end{aligned} \tag{8.162}$$

The $ie(\partial_{\mu}\alpha)\psi$ produced by differentiating the phase is cancelled exactly by the $-ie(\partial_{\mu}\alpha)\psi$ from the gauge shift of A_{μ}: this cancellation is the whole point of the construction. Consequently the bilinear that makes up the matter kinetic term is pointwise invariant: using Eq. (8.162) for $\not{D}'\psi' = e^{ie\alpha}\not{D}\psi$,

$$\bar{\psi}'\, i\not{D}'\psi' = (e^{-ie\alpha}\bar{\psi})\, i\,(e^{ie\alpha}\not{D}\psi) = e^{-ie\alpha}e^{ie\alpha}\,\bar{\psi}\, i\not{D}\psi = \bar{\psi}\, i\not{D}\psi, \tag{8.163}$$

the factor $e^{ie\alpha}$ on $\not{D}'\psi'$ being annihilated by the $e^{-ie\alpha}$ on $\bar{\psi}'$. The mass term is invariant by the same phase cancellation, $\bar{\psi}'\psi' = e^{-ie\alpha}e^{ie\alpha}\bar{\psi}\psi = \bar{\psi}\psi$, this time without even needing the covariant derivative. Finally the field-strength tensor is invariant on its own, because the gauge shift is a gradient:

$$F'_{\mu\nu} = \partial_{\mu}A'_{\nu} - \partial_{\nu}A'_{\mu} = \partial_{\mu}(A_{\nu} - \partial_{\nu}\alpha) - \partial_{\nu}(A_{\mu} - \partial_{\mu}\alpha) = F_{\mu\nu} - (\partial_{\mu}\partial_{\nu} - \partial_{\nu}\partial_{\mu})\alpha = F_{\mu\nu}, \tag{8.164}$$

since partial derivatives commute. Each of the three terms in Eq. (8.143) is therefore separately invariant, and so is $\mathcal{L}_{\text{QED}}$ and the action $S = \int d^4x\,\mathcal{L}_{\text{QED}}$. □

The invariance is exact and local, holding term by term and point by point — no integration by parts, no boundary terms. This is the hallmark of a theory built by gauging: the symmetry is not a property the dynamics happens to possess but the principle from which the dynamics was deduced. Note that it was the masslessness of the photon that survived: a putative term $\frac{1}{2}m_\gamma^2 A_\mu A^\mu$ would transform into $\frac{1}{2}m_\gamma^2(A_\mu - \partial_\mu\alpha)(A^\mu - \partial^\mu\alpha) \neq \frac{1}{2}m_\gamma^2 A_\mu A^\mu$ and is forbidden. The photon is massless because gauge invariance forbids its mass — a classical statement whose quantum protection (the photon stays massless to all orders) is one of the triumphs of the Ward identity in Chapter 19.

8.6.6 Recovering the Lorentz force

The coupled equations Eq. (8.157)–Eq. (8.159) contain all of classical electrodynamics, including the force a field exerts on a charge. To see this we take the *eikonal* (geometrical-optics) limit of the matter sector, in which the Dirac field describes a sharply localized wave packet whose centre traces a classical worldline. In that limit the field-theoretic coupling $-e\,\bar\psi\gamma^\mu\psi A_\mu$ reduces to the coupling of a point charge to the potential, and the Dirac equation reduces to a classical equation of motion. We carry the reduction at the level of the point-particle action, which is the standard and cleanest route, and recover the relativistic Lorentz force.

A relativistic point particle of mass m and charge q tracing a worldline $x^\mu(\tau)$, with four-velocity $u^\mu = dx^\mu/d\tau$ and proper time τ, has the action

$$S_{\rm pp} \;=\; -\,m\int d\tau \;-\; q\int A_\mu\,dx^\mu \;=\; \int d\tau\,\Big[-m\sqrt{u_\nu u^\nu} \;-\; q\,A_\mu(x)\,u^\mu\Big]. \tag{8.165}$$

The first term is the free relativistic kinetic action, whose nonrelativistic limit $-m\sqrt{1-\mathbf{v}^2} \to -m + \frac{1}{2}m\mathbf{v}^2$ reproduces ordinary kinetic energy; the second is the minimal coupling of the charge to the potential, the point-particle image of the field-theory current–potential coupling $\int d^4x\, J^\mu A_\mu$ with $J^\mu(x) = q\int d\tau\, u^\mu \delta^4(x - x(\tau))$. The charge of the electron is $q = -e$.

Worked Example 8.28 (The Lorentz force from the eikonal limit)**.** We derive the equation of motion from Eq. (8.165) and identify the Lorentz force. Vary $x^\mu(\tau)$ and use the Euler–Lagrange equation $\dfrac{d}{d\tau}\dfrac{\partial L}{\partial u^\mu} - \dfrac{\partial L}{\partial x^\mu} = 0$, with $L = -m\sqrt{u_\nu u^\nu} - qA_\mu u^\mu$. We parametrize by proper time, so $\sqrt{u_\nu u^\nu} = 1$ on the solution and the kinetic term contributes cleanly.

Kinetic term. $\dfrac{\partial}{\partial u^\mu}\big(-m\sqrt{u_\nu u^\nu}\big) = -m\dfrac{u_\mu}{\sqrt{u_\nu u^\nu}} = -m\,u_\mu$ (setting the root to unity after differentiating), so $\dfrac{d}{d\tau}\big(-mu_\mu\big) = -m\,\dfrac{du_\mu}{d\tau}$. The kinetic term has no explicit x-dependence, so it contributes nothing to $\partial L/\partial x^\mu$.

Interaction term. $\dfrac{\partial}{\partial u^\mu}\big(-qA_\nu u^\nu\big) = -qA_\mu$, whence $\dfrac{d}{d\tau}\big(-qA_\mu\big) = -q\,(\partial_\nu A_\mu)\,u^\nu$ along the worldline. The explicit x-dependence gives $\dfrac{\partial}{\partial x^\mu}(-qA_\nu u^\nu) = -q\,(\partial_\mu A_\nu)\,u^\nu$.

Assemble. The Euler–Lagrange equation is

$$\underbrace{-m\frac{du_\mu}{d\tau} - q(\partial_\nu A_\mu)u^\nu}_{\frac{d}{d\tau}\partial L/\partial u^\mu} \;-\; \underbrace{\big(-q(\partial_\mu A_\nu)u^\nu\big)}_{\partial L/\partial x^\mu} \;=\; 0, \tag{8.166}$$

that is, $-m\,\dfrac{du_\mu}{d\tau} = q(\partial_\nu A_\mu)u^\nu - q(\partial_\mu A_\nu)u^\nu = -q(\partial_\mu A_\nu - \partial_\nu A_\mu)u^\nu = -q\,F_{\mu\nu}u^\nu$. Cancelling the sign,

$$m\frac{du_\mu}{d\tau} = q\,F_{\mu\nu}u^\nu \quad\Longleftrightarrow\quad \boxed{f^\mu \;=\; m\frac{du^\mu}{d\tau} \;=\; q\,F^{\mu\nu}u_\nu.} \tag{8.167}$$

This is the covariant Lorentz force law. (Had we used a quadratic auxiliary kinetic term $\frac{1}{2}m\,u_\nu u^\nu$ in place of $-m\sqrt{u_\nu u^\nu}$, the spatial part of $u_\nu u^\nu$ would carry the wrong sign in the mostly-minus metric and flip the force; the relativistic square root is essential to get the sign right.)

Check the components. With $u^\mu = \gamma(1,\mathbf{v})$, $u_\nu = \gamma(1,-\mathbf{v})$, and the standard identifications $F^{i0} = E^i$, $F^{ij} = -\epsilon^{ijk}B^k$ of the field strength, the spatial part of Eq. (8.167) is

$$f^i = q\,F^{i\nu}u_\nu = q\big(F^{i0}u_0 + F^{ij}u_j\big) = q\big(E^i\gamma + (-\epsilon^{ijk}B^k)(-\gamma v^j)\big) = q\gamma\big(\mathbf{E}+\mathbf{v}\times\mathbf{B}\big)^i. \tag{8.168}$$

Dividing by γ to convert $d/d\tau$ to d/dt recovers $\dfrac{d\mathbf{p}}{dt} = q(\mathbf{E}+\mathbf{v}\times\mathbf{B})$, the familiar three-dimensional Lorentz force. For the electron, $q = -e$.

The Lorentz force closes the classical circle. The same current $J^\mu = -e\,\bar\psi\gamma^\mu\psi$ that is the Noether current of the global $U(1)$ (Section 8.6.2) and the source of Maxwell's equations (Section 8.6.3) is also, in the eikonal limit, the agent of the Lorentz force: matter couples to the field through $J^\mu A_\mu$, and that one coupling produces the field's source, the symmetry's conserved charge, and the force on the charge. A single trilinear term in Eq. (8.144) does all of this work. That economy — one coupling, one current, three classical consequences — is the signature of a gauge theory.

8.6.7 The classical structure of QED

We have assembled the complete classical theory. To state precisely what is now in hand for the chapters to come: the standard Lagrangian Eq. (8.143) with its symbols $\psi, m, e, \alpha, A_\mu, F_{\mu\nu}$; the conserved electromagnetic current Eq. (8.146),

identified both as the $U(1)$ Noether current and as the Maxwell source; the coupled Dirac–Maxwell equations of Theorem 8.26; the exact local gauge invariance of the action (Theorem 8.27); and the Lorentz force Eq. (8.167) as the eikonal limit of the coupling. Everything is classical: the fields are c-numbers (with ψ anticommuting, as flagged in Chapter 6), there is no Hilbert space, and there are no quanta.

The two outstanding tasks are both quantizations, and the gauge structure uncovered here is precisely what makes each delicate. The photon's gauge redundancy — the freedom Eq. (8.161) that left A_μ ambiguous and forced the constraint structure noted in the previous section — obstructs a naive canonical quantization and must be handled by gauge fixing and the physical-state conditions of Chapter 13. The electron, a first-order constrained field, is quantized as the Dirac field in Chapter 11, where the classical charge Eq. (8.155) becomes the operator that counts electrons minus positrons and the sign subtlety noted there is resolved. With both quantized, the trilinear vertex of Fig. 8.6 becomes the engine of every QED process, and QED proceeds to its first scattering amplitude.

With the classical theory complete and gauge-invariant, one question remains before quantization: is the potential A_μ merely a convenient bookkeeping device for the gauge-invariant field strength $F_{\mu\nu}$, or does it carry physical information of its own? The next section answers this through the holonomy of A_μ around closed loops — the Aharonov–Bohm phase — showing that the connection, not merely its curvature, is the fundamental object, and using this to derive Dirac's quantization of electric charge.

8.7 Holonomy, the Aharonov–Bohm Effect, and Charge Quantization

The previous sections of this chapter built the gauge potential A_μ from the demand of local U(1) invariance, identified its curvature $F_{\mu\nu}$ (Theorem 8.8), and proved that $F_{\mu\nu}$ is the gauge-invariant content of the connection (Theorem 8.9). It is tempting to conclude that $F_{\mu\nu}$ is therefore the whole physical story and that A_μ is mere bookkeeping — a redundant potential whose only invariant residue is the field strength. That conclusion is false, and the way it fails is one of the deepest lessons in the book. The potential carries gauge-invariant information that $F_{\mu\nu}$ does *not* capture: information that is global rather than local, attached to closed loops rather than to points. The object that extracts it is the *holonomy* of the connection, already glimpsed in the comparator construction of Section 8.2.1. This section makes the holonomy precise as the Wilson line and the Wilson loop, shows by the Aharonov–Bohm effect that it is physical even where $F_{\mu\nu} = 0$, and then turns it around: requiring the holonomy to be consistent in the presence of a magnetic monopole forces electric charge to be

quantized. The conclusion that will carry into the rest of the book is that the connection A_μ, not the field strength alone, is the fundamental object.

Notation Watch

Throughout this section we adopt the *charge-stripped* parametrization of the gauge freedom recorded as the alternative in Section 8.2.4: the matter phase is written $\psi \to e^{i\alpha}\psi$ (the charge absent from the exponent) and the potential shifts by $A_\mu \to A_\mu + \frac{1}{q}\partial_\mu\alpha$. This is the harmless rescaling $\alpha \to q\alpha$ of the chapter's default convention $\psi \to e^{iq\alpha}\psi$, $A_\mu \to A_\mu + \partial_\mu\alpha$ (Eq. (8.21)); we use it here only because it makes the topological winding arguments below — in which α is literally the U(1) phase angle, so that its winding is directly an integer — read most cleanly. The gauge-invariant holonomy $W[C] = \exp\!\left(iq \oint_C A\right)$ is, of course, independent of which parametrization is used.

8.7.1 Parallel transport and the Wilson line

Recall the comparator $U(y,x)$ introduced in Section 8.2.1: the U(1) phase factor that translates the internal phase convention at x into the convention at a neighbouring point y, transforming as $U(y,x) \to e^{i\alpha(y)}\,U(y,x)\,e^{-i\alpha(x)}$ — the charge-stripped form (Section 8.7) of the comparator law $U \to e^{iq\alpha(y)}Ue^{-iq\alpha(x)}$ established in Section 8.2.1, the two related by $\alpha \to q\alpha$ — so that the transported field $U(y,x)\psi(x)$ rotates with the destination's phase. For an infinitesimal step the comparator was $U(x+\varepsilon n, x) = 1 + i\varepsilon q\, n^\mu A_\mu(x) + O(\varepsilon^2)$. We now build the comparator for a *finite* separation by composing infinitesimal ones along a path. This is parallel transport, and the result is the central object of the section.

Let Γ be a smooth oriented path from x to y, parametrised by $\xi^\mu(s)$ with $s \in [0,1]$, $\xi(0) = x$, $\xi(1) = y$. Discretise Γ into N steps of size $\Delta s = 1/N$ and compose the infinitesimal comparators in order:

$$W_\Gamma(y,x) \;=\; \lim_{N\to\infty} \prod_{k=N}^{1} \Big[\, 1 + i\,q\,\dot\xi^\mu(s_k)\,A_\mu(\xi(s_k))\,\Delta s \,\Big], \tag{8.169}$$

where $\dot\xi^\mu = d\xi^\mu/ds$ and the product is path-ordered (later points on the left). Because the U(1) phase factors are numbers — they commute — the ordering is immaterial here, and the limit Eq. (8.169) is the exponential of the sum of the exponents, i.e. the path integral of A_μ along Γ. (In the non-abelian generalisation of Chapter 25 the comparators are matrices that do not commute, the ordering survives, and one must keep the path-ordering symbol; the abelian case is the one in which it collapses.) We record the result as a definition.

Definition 8.29 (Wilson line)**.** For a charge-q field and an oriented path Γ from x to y, the *Wilson line* (the finite parallel-transport comparator) is

$$W_\Gamma(y,x) \;=\; \exp\!\Big(\, i\,q \int_\Gamma A_\mu \, dx^\mu \Big), \tag{8.170}$$

where $\int_\Gamma A_\mu\,dx^\mu = \int_0^1 A_\mu(\xi(s))\,\dot\xi^\mu(s)\,ds$ is the integral of the one-form $A = A_\mu\,dx^\mu$ along Γ.

The Wilson line inherits the transformation law of the comparator it was assembled from, and this is the first thing to verify. Under a local gauge transformation, A_μ shifts by a gradient, $A_\mu \to A_\mu + \frac{1}{q}\partial_\mu\alpha$ (Section 8.7), so the line integral changes by the integral of a total derivative:

$$\int_\Gamma A'_\mu\,dx^\mu \;=\; \int_\Gamma A_\mu\,dx^\mu \;+\; \frac{1}{q}\int_\Gamma \partial_\mu\alpha\,dx^\mu \;=\; \int_\Gamma A_\mu\,dx^\mu \;+\; \frac{1}{q}\big[\alpha(y)-\alpha(x)\big], \tag{8.171}$$

where we used $\int_\Gamma \partial_\mu\alpha\,dx^\mu = \int_0^1 \frac{d}{ds}\alpha(\xi(s))\,ds = \alpha(y)-\alpha(x)$, the fundamental theorem of calculus applied to the pullback of $d\alpha$. Exponentiating,

$$W_\Gamma(y,x) \;\longrightarrow\; e^{\,i\alpha(y)}\, W_\Gamma(y,x)\, e^{-i\alpha(x)}, \tag{8.172}$$

exactly the comparator law. The transported field then behaves correctly: under the gauge transformation $W_\Gamma(y,x)\,\psi(x) \to e^{i\alpha(y)}W_\Gamma(y,x)e^{-i\alpha(x)}\,e^{i\alpha(x)}\psi(x) = e^{i\alpha(y)}\,[W_\Gamma(y,x)\psi(x)]$, i.e. it acquires the phase of the destination point y, just like the genuine field $\psi(y)$. The Wilson line is the device that lets us compare charged fields at finite separation.

It is worth checking that Theorem 8.29 reproduces the covariant derivative in the infinitesimal limit, closing the loop with Section 8.2.1. For a step $y = x + \varepsilon n$ along a straight segment, $\int_\Gamma A_\mu\,dx^\mu = \varepsilon\, n^\mu A_\mu(x) + O(\varepsilon^2)$, so $W = 1 + i\varepsilon q\, n^\mu A_\mu + O(\varepsilon^2)$, and the transport-corrected difference is

$$\begin{aligned}\psi(x+\varepsilon n) - W(x+\varepsilon n, x)\,\psi(x) &= \big[\psi + \varepsilon n^\mu\partial_\mu\psi\big] - \big[1 + i\varepsilon q\, n^\mu A_\mu\big]\psi + O(\varepsilon^2)\\ &= \varepsilon\, n^\mu(\partial_\mu - iqA_\mu)\psi + O(\varepsilon^2) \;=\; \varepsilon\, n^\mu(D_\mu\psi) + O(\varepsilon^2).\end{aligned} \tag{8.173}$$

The covariant derivative is exactly the leading rate at which the field fails to be parallel along the path — the infinitesimal holonomy. The connection A_μ is the generator of parallel transport; D_μ and W_Γ are the same geometric idea read at two scales.

8.7.2 The Wilson loop and its flux

The Wilson line $W_\Gamma(y,x)$ is not by itself gauge invariant: its endpoints carry the phases $e^{i\alpha(y)}$ and $e^{-i\alpha(x)}$. Those phases cancel precisely when the path closes on itself, $y = x$. The resulting closed-loop holonomy is gauge invariant and physical, and it is the abelian *Wilson loop*.

Theorem 8.30 (Gauge invariance of the Wilson loop and its flux)**.** *Let $C = \partial S$ be a closed oriented loop bounding a surface S. The Wilson loop*

$$W[C] \;=\; \exp\!\Big(i\,q \oint_C A_\mu \, dx^\mu \Big) \tag{8.174}$$

is invariant under any local gauge transformation $A_\mu \to A_\mu + \frac{1}{q}\partial_\mu \alpha$, and it equals the exponential of the charge times the flux of the field strength through S,

$$\boxed{\; W[C] \;=\; \exp\!\Big(i\,q \oint_{\partial S} A_\mu \, dx^\mu \Big) \;=\; \exp\!\Big(\frac{i\,q}{2} \int_S F_{\mu\nu}\, dS^{\mu\nu} \Big) \;} \tag{8.175}$$

where $dS^{\mu\nu}$ is the antisymmetric area element of S.

Proof. Gauge invariance. Setting $y = x$ in Eq. (8.171), the boundary term is $\frac{1}{q}[\alpha(x) - \alpha(x)] = 0$ for a single-valued α, so $\oint_C A_\mu\, dx^\mu$ is unchanged and $W[C] \to W[C]$. The argument is in fact more robust than this. Suppose α is permitted to be *multivalued* around C — as it must be allowed to be, since the only object required to be single-valued is the physical phase factor $e^{i\alpha}$, not α itself. Then $\oint_C \partial_\mu \alpha\, dx^\mu = \oint_C d\alpha = 2\pi k$ for some integer k (the winding number of the map $C \to \mathrm{U}(1)$, $x \mapsto e^{i\alpha(x)}$; the value must be a multiple of 2π precisely so that $e^{i\alpha}$ returns to itself). The loop integral then shifts by $\frac{1}{q}\cdot 2\pi k$, and

$$W[C] \;\longrightarrow\; W[C]\, \exp\!\Big(iq \cdot \tfrac{1}{q}\, 2\pi k \Big) \;=\; W[C]\, e^{\,2\pi i k} \;=\; W[C]. \tag{8.176}$$

The Wilson loop is invariant even under these *large* gauge transformations, which cannot be reached continuously from the identity. This is the first appearance of the integer winding of $\mathrm{U}(1)$ — of $\pi_1(\mathrm{U}(1)) = \mathbb{Z}$ from Chapter 2 — and we shall see it again, decisively, in the charge-quantization argument below.

Equality to the flux. The one-form $A = A_\mu\, dx^\mu$ has exterior derivative

$$dA \;=\; \partial_\mu A_\nu \, dx^\mu \wedge dx^\nu \;=\; \tfrac{1}{2}(\partial_\mu A_\nu - \partial_\nu A_\mu)\, dx^\mu \wedge dx^\nu \;=\; \tfrac{1}{2} F_{\mu\nu}\, dx^\mu \wedge dx^\nu, \tag{8.177}$$

where the second equality uses the antisymmetry of the wedge product $dx^\mu \wedge dx^\nu = -dx^\nu \wedge dx^\mu$ to antisymmetrise the coefficient, and the third is the definition of the curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (Theorem 8.8). The field strength is literally the curvature two-form of the connection. Stokes' theorem for differential forms, $\oint_{\partial S} A = \int_S dA$, then gives

$$\oint_{\partial S} A_\mu\, dx^\mu \;=\; \int_S dA \;=\; \frac{1}{2}\int_S F_{\mu\nu}\, dx^\mu \wedge dx^\nu \;\equiv\; \frac{1}{2}\int_S F_{\mu\nu}\, dS^{\mu\nu}. \tag{8.178}$$

Multiplying by iq and exponentiating yields Eq. (8.175). □

Two readings of Eq. (8.175) matter. First, it says the holonomy of the connection around C is controlled by the flux of F through *any* surface spanning C — the right-hand side is independent of the choice of S, as it must be, because changing S to S' with the same boundary changes the integral by $\frac{1}{2}\oint_{\partial(S-S')} A$ over a *closed* surface $S - S'$, which by the homogeneous Maxwell/Bianchi identity $\partial_{[\lambda} F_{\mu\nu]} = 0$ (Section 8.3) measures the magnetic charge enclosed and vanishes whenever that charge is zero. (When it does *not* vanish — a monopole inside — the surface-independence fails by a multiple of 2π, and that failure is the whole content of Section 8.7.4.) Second, in the magnetostatic situation the right-hand side is just the ordinary magnetic flux. To see this cleanly, take C to be a spatial loop traversed at a fixed instant, so $dx^0 = 0$ along C and only the spatial components contribute. With $A^\mu = (\phi, \vec{A})$ the lowered spatial component is $A_i = \eta_{ij} A^j = -A^i = -(\vec{A})_i$, hence

$$\oint_C A_\mu \, dx^\mu \;=\; \oint_C A_i \, dx^i \;=\; -\oint_C \vec{A}\cdot d\vec{\ell} \;=\; -\int_S (\nabla\times\vec{A})\cdot d\vec{S} \;=\; -\int_S \vec{B}\cdot d\vec{S} \;=\; -\,\Phi_B, \tag{8.179}$$

using ordinary three-dimensional Stokes in the middle and $\vec{B} = \nabla \times \vec{A}$. Thus $W[C] = e^{-iq\Phi_B}$: the Wilson loop of a static magnetic field is the exponential of (minus) the charge times the enclosed flux. The overall sign is a convention artefact of the index placement and the loop orientation; the gauge-invariant physical content is q times the flux Φ_B enclosed, defined modulo 2π.

8.7.3 The Aharonov–Bohm effect

We can now exhibit the promised phenomenon: a region in which $F_{\mu\nu} = 0$ identically, so that classically "nothing is there," and yet the connection has a nontrivial, observable holonomy. The field strength vanishes locally but the gauge potential is not globally pure gauge — there is no single-valued α with $A_\mu = \frac{1}{q}\partial_\mu \alpha$ throughout the region — and the obstruction is detected by the Wilson loop.

The experimental geometry (Fig. 8.7) is the standard two-path interference experiment with a long thin solenoid placed in the shadow between the paths. A coherent beam of charged particles is split, the two partial waves ψ_1, ψ_2 travel along paths Γ_1 and Γ_2 that pass on opposite sides of the solenoid, and they recombine on a screen. The magnetic field is confined inside the solenoid; the particles move only through the field-free exterior. Each partial wave is parallel-transported by the Wilson line along its path, picking up a phase $\exp(iq\int_{\Gamma_a} A_\mu \, dx^\mu)$ relative to the field-free amplitude (this is the statement, made quantitative in nonrelativistic quantum mechanics, that minimal coupling attaches the comparator phase to a propagating charged particle; we are using here only the classical, kinematic content of the connection). The interference pattern depends on the

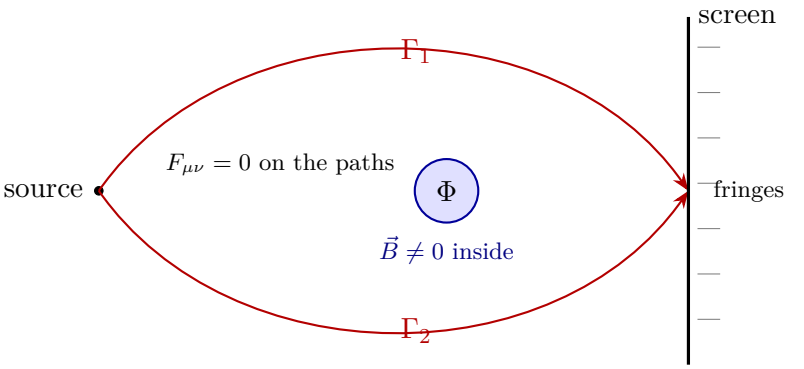


Figure 8.7: The Aharonov–Bohm setup. An idealised infinite solenoid (seen end-on) confines a magnetic flux Φ to its interior; outside, $\vec{B} = 0$ and the electron paths Γ_1, Γ_2 never touch any field. Yet the relative phase of the two partial waves is $q \oint_C A_\mu \, dx^\mu = -q\Phi$, shifting the interference fringes on the screen. The connection is observable where the field strength is not.

relative phase

$$\Delta\varphi \;=\; q\int_{\Gamma_1} A_\mu \, dx^\mu - q\int_{\Gamma_2} A_\mu \, dx^\mu \;=\; q\oint_C A_\mu \, dx^\mu, \qquad C = \Gamma_1 - \Gamma_2, \tag{8.180}$$

the holonomy around the closed loop C formed by going out along Γ_1 and back along Γ_2, a loop that encircles the solenoid. By Theorem 8.30 and Eq. (8.179) this relative phase is $\Delta\varphi = -q\Phi$, controlled entirely by the enclosed flux. It is nonzero even though $F_{\mu\nu} = 0$ all along both paths. The following worked example computes the potential explicitly, verifies that it is field-free yet non-pure-gauge in the exterior, and confirms the phase.

Worked Example 8.31 (Aharonov–Bohm phase of an idealised solenoid)**.** Model the solenoid as an infinite cylinder of radius a along the z-axis, carrying a uniform internal field $\vec{B} = B\,\hat{z}$ for $r < a$ and $\vec{B} = 0$ for $r > a$, with total enclosed flux $\Phi = \pi a^2 B$. Work in cylindrical coordinates (r, φ, z).

Step 1: the exterior potential. Symmetry dictates $\vec{A} = A_\varphi(r)\,\hat{\varphi}$. Demand that it reproduce the flux through every circle of radius $r > a$ centred on the axis. Three-dimensional Stokes gives

$$\oint_r \vec{A}\cdot d\vec{\ell} \;=\; A_\varphi(r)\,(2\pi r) \;=\; \int_{\text{disc}} \vec{B}\cdot d\vec{S} \;=\; \Phi \qquad (r > a), \tag{8.181}$$

since the disc bounded by the circle catches the entire flux once $r > a$. Hence

$$\vec{A} \;=\; \frac{\Phi}{2\pi r}\,\hat{\varphi} \qquad (r > a). \tag{8.182}$$

Step 2: the exterior field vanishes. Using the curl in cylindrical coordinates for a purely azimuthal $\vec{A} = A_\varphi(r)\hat{\varphi}$,

$$\vec{B} = \nabla \times \vec{A} \;=\; \frac{1}{r}\frac{\partial (r A_\varphi)}{\partial r}\,\hat{z} \;=\; \frac{1}{r}\frac{\partial}{\partial r}\Big(r \cdot \frac{\Phi}{2\pi r}\Big)\hat{z} \;=\; \frac{1}{r}\frac{\partial}{\partial r}\Big(\frac{\Phi}{2\pi}\Big)\hat{z} \;=\; 0 \qquad (r > a), \tag{8.183}$$

confirming $F_{\mu\nu} = 0$ throughout the region the particles traverse.

Step 3: the exterior potential is not (globally) pure gauge. Locally one can write $\vec{A} = \nabla\chi$ with $\chi = \frac{\Phi}{2\pi}\varphi$, since $\nabla\chi = \frac{1}{r}\partial_\varphi \chi\,\hat{\varphi} = \frac{\Phi}{2\pi r}\hat{\varphi}$. But χ is the multivalued angle, increasing by Φ each time one circles the axis; there is no single-valued function on the punctured exterior whose gradient is $\vec{A}$. This is the precise sense in which A_μ is not pure gauge: $F = 0$ guarantees A is *locally* exact (closed forms are locally exact, Poincaré's lemma), but the exterior is not simply connected, so it need not be *globally* exact. The failure is measured by the loop integral.

Step 4: the phase. For a loop C encircling the solenoid once,

$$\oint_C \vec{A}\cdot d\vec{\ell} \;=\; \int_0^{2\pi} \frac{\Phi}{2\pi r}\,(r\,d\varphi) \;=\; \Phi, \tag{8.184}$$

independent of the loop's radius or shape, depending only on the enclosed flux. By Eq. (8.179), $\oint_C A_\mu\,dx^\mu = -\Phi$, so the Aharonov–Bohm phase is

$$\Delta\varphi \;=\; q\oint_C A_\mu\,dx^\mu \;=\; -\,q\,\Phi, \qquad W[C] \;=\; e^{-iq\Phi}. \tag{8.185}$$

Step 5: gauge invariance and observability. Any other potential differs from Eq. (8.182) by $\frac{1}{q}\nabla\alpha$ for a single-valued α, which adds $\frac{1}{q}[\alpha]_C = 0$ to the loop integral; the phase Eq. (8.185) is gauge invariant. It is observable: shifting the flux Φ rigidly displaces the interference fringes on the screen, and the pattern repeats whenever $q\Phi$ advances by 2π, i.e. whenever Φ advances by the *flux quantum*

$$\Phi_0 \;=\; \frac{2\pi}{|q|} \;=\; \frac{2\pi\hbar}{|q|} \;=\; \frac{h}{|q|} \qquad \text{(restoring } \hbar\text{)}, \tag{8.186}$$

which for $|q| = e$ is h/e. (In a superconductor the carriers are charge-$2e$ Cooper pairs and the observed flux quantum is correspondingly $h/2e$ — an experimental confirmation that the relevant charge is the carrier's, exactly as Eq. (8.186) demands.) The point is established: the gauge potential carries gauge-invariant physical information through its holonomy in a region where the field strength is identically zero.

Physical Insight

The Aharonov–Bohm effect forces a revision of what is "physically real" in electromagnetism. The field strength $F_{\mu\nu}$ is local but incomplete: it does not determine the holonomies, hence does not determine the physics, on a topologically nontrivial region. The potential A_μ is complete but redundant: it overdetermines the physics by the gauge freedom. The genuinely physical, gauge-invariant data is neither — it is the set of holonomies $W[C] = \exp(iq \oint_C A)$ for all closed loops C, the connection "integrated up." On a simply connected region every loop bounds and $W[C]$ is fixed by F via Eq. (8.175), so F suffices; on a region with noncontractible loops the holonomies around them are independent new data. The connection, read through its holonomy, is the fundamental object. This is the classical seed of the statement, central to Chapter 25, that gauge theory is the theory of a connection on a fibre bundle, and that Wilson loops are its natural gauge-invariant observables.

8.7.4 Dirac's charge-quantization condition

The Wilson loop turned a field-free region into an observable. We now run the logic in reverse. Suppose a magnetic monopole exists. The Wilson loop then constrains the allowed electric charges, and the constraint is Dirac's celebrated quantization condition: the mere consistent coexistence of one monopole with charged matter forces every electric charge to be an integer multiple of a fixed unit.

A magnetic monopole of magnetic charge g is a point source of $\vec{B}$,

$$\vec{B}(\vec{r}) \;=\; \frac{g}{4\pi}\,\frac{\hat{r}}{r^2}, \qquad \oint_{S^2} \vec{B}\cdot d\vec{S} \;=\; g, \tag{8.187}$$

the flux through any sphere enclosing it being g (the field is the magnetic analogue of the Coulomb field). This immediately conflicts with $\vec{B} = \nabla \times \vec{A}$: if a globally smooth $\vec{A}$ existed, the flux through the closed sphere S^2 would be $\oint_{S^2}(\nabla \times \vec{A})\cdot d\vec{S} = 0$ by the divergence theorem (a curl has no sources), contradicting Eq. (8.187). Equivalently, $\nabla \cdot \vec{B} = g\,\delta^3(\vec{r}) \neq 0$ violates the Bianchi identity $\nabla \cdot \vec{B} = 0$ that $\vec{B} = \nabla \times \vec{A}$ would enforce. A monopole cannot be described by a single globally defined potential.

Dirac's resolution: the string. Dirac's device is to keep a potential almost everywhere by importing the missing flux along an infinitely thin solenoid — a *Dirac string* — running from infinity to the monopole, say along the negative z-axis. The string delivers flux $+g$ *into* the monopole, so that $\nabla \cdot \vec{B} = 0$ holds everywhere (the outgoing radial flux g is balanced by the incoming string flux g),

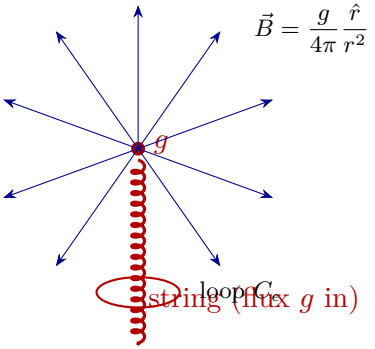


Figure 8.8: The Dirac monopole. Radial magnetic field of charge g emanates from the source; the return flux is carried in along an unphysical string (here down the $-z$ axis), so that $\nabla \cdot \vec{B} = 0$ and a potential exists off the string. A small loop C_ϵ encircling the string sees the full flux g and gives a Wilson loop e^{iqg}; demanding that the string be invisible forces $qg = 2\pi n$.

and a vector potential $\vec{A}$ exists everywhere *except on the string*. The monopole's radial field is then $\nabla \times \vec{A}$ off the string. An explicit such potential is

$$\vec{A}^N \;=\; \frac{g}{4\pi}\,\frac{1-\cos\theta}{r\sin\theta}\,\hat{\varphi}, \tag{8.188}$$

singular along $\theta = \pi$ (the negative z-axis, where the string lies) but smooth elsewhere. The string is an artefact of the description, not a physical object: its location can be moved by a gauge transformation (e.g. to the positive z-axis, giving a second potential $\vec{A}^S = -\frac{g}{4\pi}\frac{1+\cos\theta}{r\sin\theta}\hat{\varphi}$ regular near the south pole instead). Physics must not depend on where we put it. That single demand is the quantization condition.

Theorem 8.32 (Dirac quantization condition)**.** *If a magnetic monopole of magnetic charge g coexists with a particle of electric charge q, then*

$$\boxed{\,q\,g \;=\; 2\pi n, \qquad n \in \mathbb{Z}.\,} \tag{8.189}$$

Proof. Wilson loop around the string. Consider a small closed loop C_ϵ that encircles the Dirac string once, at a point away from the monopole (Fig. 8.8). The string is an idealised thin solenoid carrying flux g, so C_ϵ encloses exactly the flux g. By Theorem 8.30 (in the magnetostatic form Eq. (8.179), with $\Phi = g$) the holonomy around it is

$$W[C_\epsilon] \;=\; \exp\!\Big(iq \oint_{C_\epsilon} A_\mu\, dx^\mu\Big) \;=\; e^{\mp iqg}, \tag{8.190}$$

the sign depending only on the loop's orientation relative to the string. The string, however, is pure gauge artefact: it is invisible, carries no field on C_ϵ in any

physical sense (we may relocate it by a gauge transformation), and therefore the holonomy that detects it *must be trivial*. Were $W[C_\epsilon] \neq 1$, an Aharonov–Bohm experiment of exactly the kind in Theorem 8.31 would reveal the string's position, making the unphysical artefact observable — a contradiction. Hence

$$e^{iqg} = 1 \qquad \Longrightarrow \qquad q\,g = 2\pi n, \quad n \in \mathbb{Z}, \tag{8.191}$$

which is Eq. (8.189). □

Remark 8.33 (The two-patch derivation). The same condition arises without ever invoking an idealised solenoid, in a form that makes its topological character manifest — the form taken up systematically for non-abelian connections in Chapters 25 and 41. Cover the sphere S^2 surrounding the monopole with two overlapping patches: a northern patch (all of S^2 except the south pole) using $\vec{A}^N$ of Eq. (8.188), and a southern patch (all but the north pole) using $\vec{A}^S$. Each is smooth on its own patch; neither is global. On the overlap, the equatorial band, they differ by

$$\vec{A}^N - \vec{A}^S = \frac{g}{4\pi}\,\frac{(1-\cos\theta)+(1+\cos\theta)}{r\sin\theta}\,\hat{\varphi} = \frac{g}{2\pi}\,\frac{1}{r\sin\theta}\,\hat{\varphi} = \nabla\Big(\frac{g}{2\pi}\,\varphi\Big), \tag{8.192}$$

using $\nabla(\,f(\varphi)\,) = \frac{1}{r\sin\theta}\partial_\varphi f\,\hat{\varphi}$ in spherical coordinates. The two potentials are thus related by a gauge transformation $\vec{A}^N = \vec{A}^S + \frac{1}{q}\nabla\alpha$ with transition function

$$\alpha(\varphi) = \frac{q\,g}{2\pi}\,\varphi, \qquad e^{i\alpha} = \exp\Big(\frac{i\,q\,g}{2\pi}\,\varphi\Big). \tag{8.193}$$

The transition function $e^{i\alpha}$ is the U(1)-valued gluing map that relates the charged field's phase convention in the two patches; for the field to be well defined it must be single-valued as we go once around the equator, $\varphi \to \varphi + 2\pi$. That requires α to advance by a multiple of 2π, i.e. $\frac{qg}{2\pi}\cdot 2\pi = qg = 2\pi n$ — once more Eq. (8.189). In this language the integer n is the winding number of the transition function around the equator, an element of $\pi_1(\mathrm{U}(1)) = \mathbb{Z}$ (Chapter 2); it is the first Chern number of the U(1) bundle over S^2, and it counts the magnetic charge in units of $2\pi/q$. Charge quantization is the statement that this winding is an integer.

8.7.5 The connection is fundamental

Dirac's condition Eq. (8.189) is the most economical argument known for a fact the world obeys with extraordinary precision: electric charge is quantized. Every observed charge is an integer multiple of $e/3$ (quarks) or e (leptons), to a part in 10^{21} for the electron–proton charge difference, yet the Standard Model *as a Lagrangian* does not require it — the U(1) hypercharges could a priori be any real numbers. Dirac's argument supplies a sufficient reason: if even a single magnetic monopole exists *anywhere* in the universe, then $qg = 2\pi n$ forces every electric

charge to be a multiple of $2\pi/g$. The quantization of one charge propagates to all. Grand unified theories (Chapter 35) make this concrete — they predict monopoles as smooth solitons and thereby *derive* charge quantization — and the non-perturbative methods of Chapter 41 construct such monopoles explicitly.

The structural lesson is sharper than the phenomenology, and it is the lesson this whole chapter has been driving toward. The integer n in Eq. (8.189) is topological: it is the winding of a U(1)-valued map, an element of $\pi_1(\mathrm{U}(1)) = \mathbb{Z}$ (Chapter 2). It exists, and is an integer, precisely because the gauge group is the *compact* circle U(1) rather than the non-compact line $\mathbb{R}$. Had the internal symmetry been $\mathbb{R}$ — the same Lie algebra, the same $F_{\mu\nu}$, the same Maxwell equations locally — the group would be simply connected, $\pi_1(\mathbb{R}) = 0$, there would be no winding, no quantization, and no flux quantum. Charge quantization is therefore not a fact about the field strength at all; it is a fact about the *global* structure of the gauge group and the bundle the connection lives on. The local data $F_{\mu\nu}$ is blind to it. Only the connection, read globally through its holonomies and transition functions, knows.

This is the verdict the chapter hands to the rest of the book: the gauge potential is not redundant scaffolding around a physical $F_{\mu\nu}$; it is the primary object, a connection whose curvature is F and whose holonomy carries information F cannot. The classical Wilson loop $W[C]$ introduced here in the abelian setting is promoted in Chapter 25 to the non-abelian, path-ordered Wilson loop that serves as the order parameter for confinement in Yang–Mills theory, and the winding integers that quantized charge here reappear in Chapter 41 as the instanton number and the magnetic charge of 't Hooft–Polyakov monopoles. With the connection established as fundamental, and its global content made visible, the classical theory of the gauge field is complete; what remains is the gauge redundancy whose quantization becomes the central problem of Chapter 13 and Chapter 26.

Further reading

The gauge principle and its reconstruction of electrodynamics are developed, at complementary levels, in Peskin and Schroeder [396] and Schwartz [432], and – with the deepest insistence that the principle is a derivation rather than a convenience – in Weinberg [500]. The covariant, tensorial formulation of Maxwell's equations and the classical material on potentials, gauge fixing, and radiation is treated thoroughly in Jackson [279]. The geometric reading of A_μ as a connection and $F_{\mu\nu}$ as a curvature, including the fibre-bundle language that Chapter 25 will need, is laid out in Nakahara [355]; the foundational statement that the nonintegrable phase factor (the holonomy), not the field strength, is the complete gauge-invariant description is the classic paper by Wu and Yang [527]. The observability of the potential in a field-free region is Aharonov and

Bohm [8], and the quantization of electric charge from the consistency of a single magnetic monopole is Dirac [151].

Part III

Free Quantum Fields

Chapter 9

Quantization of the Scalar Field

This is the chapter in which the book first builds a genuine quantum field theory. Everything before it has been preparation of one of two kinds. The mathematical chapters of Part I assembled tools – the representation theory of the Lorentz and Poincaré groups in Chapter 3, the functional and analytic machinery of Chapter 5 – and the field-theory chapters of Part II treated fields as *classical* dynamical systems: the action principle and canonical formalism in Chapter 6, the symmetry-and-current calculus of Noether in Chapter 7, the gauge principle in Chapter 8. In each of those the field $\phi(x)$ was a number-valued function on spacetime, a configuration to be varied. Here that changes once and for all. We promote $\phi(x)$ to an *operator* on a Hilbert space, impose canonical commutation relations as the field-theoretic image of $[\hat{x}, \hat{p}] = i$, and follow the consequences. The classical scalar of ϕ^4 theory becomes a quantum field; this is the free ($\lambda = 0$) sector of that thread, and the interactions are switched back on only in Part IV.

We import rather than rebuild the classical input. The free real scalar Lagrangian $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2$, its canonical momentum $\pi = \dot{\phi}$, the Hamiltonian, and the Klein–Gordon equation $(\Box + m^2)\phi = 0$ are taken as established in Chapter 6. The conserved energy–momentum tensor and four-momentum P^μ, together with the internal U(1) current and charge of the complex scalar, are taken as Noether output from Chapter 7. The theory of distributions, the residue calculus, the contour deformations, and the $i\varepsilon$ prescription that we lean on to make sense of the propagator are taken from Chapter 5. None of these is reproved; each is invoked by Chapter 5 and its siblings where it is used.

The strategy is best stated in words before any formula. Expand the classical field in spatial Fourier modes; the Klein–Gordon equation then says that each mode $\phi(\mathbf{p}, t)$ obeys the equation of a harmonic oscillator with frequency $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$. A free field is therefore nothing but a continuum of decoupled harmonic oscillators, one for each momentum. The whole quantum content of the chapter is then the Sakurai-level ladder-operator solution of the single oscillator – which we assume as a quantum-mechanics prerequisite – applied to all of these

oscillators at once, with the bookkeeping arranged so that Lorentz invariance is manifest at every step. The art is entirely in the covariant bookkeeping: the Lorentz-invariant measure, the relativistic single-particle normalization, and the $i\varepsilon$ contour that ties the time ordering of operators to the analytic structure of a function of momentum.

Four payoffs follow, and the rest of the book leans on all four. First, *particles are not assumed; they emerge.* Acting on the ground state with a raising operator produces a one-quantum state, and we will see directly that this excitation carries the energy, momentum, and Lorentz transformation law of a relativistic particle of mass m and spin 0 – exactly the irreducible one-particle representation classified in Chapter 3. Second, the *vacuum is nontrivial*: summing the oscillator zero-point energies gives a (divergent) vacuum energy, and although its absolute value is unobservable, *differences* of it are physically real, as the worked Casimir computation shows. Third, the field is *causal* in a precise, checkable sense – field operators at spacelike separation commute (microcausality) – and we will see that demanding this is precisely what forces a spin-0 field to be quantized with commutators rather than anticommutators, the spin-statistics connection in its simplest form. Fourth, the central dynamical object the chapter delivers is the *Feynman propagator*: the time-ordered two-point function whose $i\varepsilon$-deformed contour encodes time ordering as analyticity, and which becomes the internal line of every Feynman diagram from Chapter 10 onward.

We fix conventions once, in force for the whole chapter, matching the book-wide canon of Appendix A: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ so that $p^2 = m^2$ on shell; natural units $\hbar = c = 1$; the relativistic single-particle normalization and the Lorentz-invariant measure $\widetilde{dp} = d^3p/[(2\pi)^3\, 2E_{\mathbf{p}}]$ used throughout the book; and the Feynman $(i\varepsilon)$ prescription for the propagator. A closing honesty is owed up front, because the cleanliness of this construction is deceptive. It works *because the theory is free.* The moment interactions are present the interaction picture itself becomes mathematically delicate – this is the content of Haag's theorem – and a genuinely interacting scalar field in four dimensions is believed to be *trivial*, that is, free in disguise. What we build in this chapter is therefore not a toy: it is the rigorous bedrock on which the perturbative edifice of Chapters 10, 14 and 15 is erected, and the place to which every later subtlety about interacting fields must ultimately be referred.

9.1 From Oscillators to Fields: Canonical Quantization

Every preceding chapter has either sharpened a tool or treated a field as a classical object. We now cross the line that gives the subject its name. The free real scalar arrived at the door of this chapter as a purely classical system: in Chapter 6 we wrote its Lagrangian, read off its canonical momentum, Legendre-transformed to

a Hamiltonian, and derived its equation of motion; in Chapter 7 we extracted its conserved energy–momentum and internal charges as Noether output. None of that knew anything about $\hbar$. The task of the present section is to make the field an *operator*, and to do so in the one way that the reader already trusts completely: by repeating, infinitely many times, the quantization of the harmonic oscillator. The whole construction of free quantum field theory is the statement that a free field is a continuum of decoupled oscillators labelled by momentum, and that each is quantized by the Sakurai-level ladder algebra assumed as prerequisite. This section installs the foundation of that program — the operator field and the canonical commutation relations it must obey — and checks that the foundation is internally sound before the next section builds on it.

This is the free, $\lambda = 0$, sector of ϕ^4 theory, continued directly from its classical stage in Chapter 6.

9.1.1 The classical free theory, imported

We take the free real scalar field exactly as Chapter 6 left it, in the conventions fixed there and in Chapter 5: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$. The Lagrangian density of the free theory is the $\lambda \to 0$ limit of the ϕ^4 Lagrangian,

$$\mathcal{L} = \tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2, \tag{9.1}$$

imported from Eq. (6.21). Treating $\phi(t,\mathbf{x})$ as the generalized coordinate at each spatial point $\mathbf{x}$, the canonically conjugate momentum density is

$$\pi(x) = \frac{\partial\mathcal{L}}{\partial(\partial_0\phi)} = \partial_0\phi(x) \equiv \dot\phi(x), \tag{9.2}$$

which is Eq. (6.120). The Legendre transform to the Hamiltonian, performed in Chapter 6, gave Eq. (6.122),

$$H = \int d^3x\,\left[\tfrac{1}{2}\,\pi^2 + \tfrac{1}{2}\,(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2\right], \tag{9.3}$$

a manifestly positive functional of the field and its momentum. Finally, the Euler–Lagrange equation of Eq. (9.1) is the Klein–Gordon equation Eq. (6.55),

$$(\Box + m^2)\,\phi = 0, \qquad \Box \equiv \partial_\mu\partial^\mu = \partial_0^2 - \nabla^2. \tag{9.4}$$

These four equations are the entire classical input. We do not rederive them; the rest of the section is about what happens when ϕ and π stop being numbers.

9.1.2 The oscillator template

Before quantizing the field we recall precisely which feature of the classical system we intend to carry over, because the field commutation relations are not a new postulate so much as the obvious continuum analogue of an old one.

A single nonrelativistic particle has a coordinate q and a conjugate momentum p with classical Poisson bracket $\{q,p\}=1$. Dirac's canonical quantization promotes these to operators obeying $[q,p]=i$ (with $\hbar=1$). A system of N decoupled degrees of freedom has coordinates q_i and momenta p_j, $i,j=1,\dots,N$, with

$$[q_i,p_j]=i\,\delta_{ij},\qquad [q_i,q_j]=[p_i,p_j]=0. \tag{9.5}$$

The Kronecker delta encodes the physical statement that distinct degrees of freedom are independent: measuring the position of oscillator i does not disturb the momentum of oscillator $j\neq i$.

A field is the limit of Eq. (9.5) in which the discrete label i becomes a continuous spatial point $\mathbf{x}$. Concretely, imagine discretizing space into a cubic lattice of spacing a, so that the field is replaced by a coordinate $q_{\mathbf{n}}=\phi(\mathbf{x_n})$ at each lattice site $\mathbf{x_n}=a\mathbf{n}$, with conjugate momentum $p_{\mathbf{n}}=a^3\,\pi(\mathbf{x_n})$ (the cell volume a^3 converts the momentum *density* π into the momentum of the site degree of freedom). The Hamiltonian Eq. (9.3) becomes a sum over sites of oscillator energies coupled to their neighbours through the discretized gradient $(\nabla\phi)^2$; it is a lattice of masses joined by springs (Fig. 9.1). Imposing Eq. (9.5) on these site variables gives $[q_{\mathbf{m}},p_{\mathbf{n}}]=i\,\delta_{\mathbf{mn}}$, i.e.

$$\Big[\phi(\mathbf{x_m}),a^3\,\pi(\mathbf{x_n})\Big]=i\,\delta_{\mathbf{mn}}\quad\Longrightarrow\quad[\phi(\mathbf{x_m}),\pi(\mathbf{x_n})]=i\,\frac{\delta_{\mathbf{mn}}}{a^3}. \tag{9.6}$$

As $a\to 0$ the combination $\delta_{\mathbf{mn}}/a^3$ is exactly the lattice representation of the three-dimensional Dirac delta: it vanishes for $\mathbf{m}\neq\mathbf{n}$, diverges at coincidence, and integrates to one over space, $\sum_{\mathbf{n}}a^3\,(\delta_{\mathbf{mn}}/a^3)=1$. The continuum limit of Eq. (9.6) is therefore the equal-time commutator $[\phi(\mathbf{x}),\pi(\mathbf{y})]=i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})$. This is not a derivation of quantum mechanics; it is the demonstration that the canonical quantization of a field is the same postulate the reader already accepts for finitely many oscillators, read in the continuum.

9.1.3 The canonical quantization postulate

We now state the postulate that defines the quantum theory. From here on ϕ and π are operators on a Hilbert space, with all the classical relations of Section 9.1.1 reinterpreted as operator equations.

Definition 9.1 (Equal-time canonical commutation relations)**.** The quantum free scalar field is the operator-valued field $\phi(t,\mathbf{x})$ with conjugate momentum operator

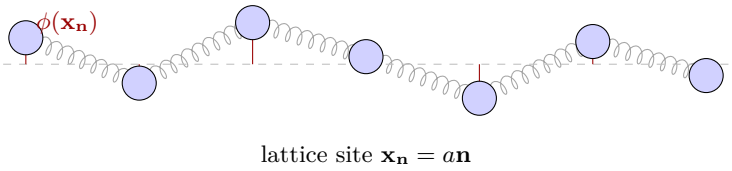


Figure 9.1: The mechanical picture behind canonical quantization. Discretizing space turns the free scalar into a lattice of point masses (sites), each carrying a coordinate $q_{\mathbf{n}} = \phi(\mathbf{x_n})$ (its displacement from equilibrium, red arrows) and coupled to its neighbours by springs encoding the gradient energy $\frac{1}{2}(\nabla\phi)^2$. Quantizing each site by the oscillator rule Eq. (9.5) and taking $a \to 0$ produces the field commutation relations. The mass term $\frac{1}{2}m^2\phi^2$ adds an on-site restoring force; in the next section the normal modes of this lattice become the relativistic plane waves.

$\pi(t, \mathbf{x})$, subject to the *equal-time canonical commutation relations* (ETCR)

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = i\,\delta^{(3)}(\mathbf{x} - \mathbf{y}), \tag{9.7}$$

$$[\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = [\pi(t, \mathbf{x}), \pi(t, \mathbf{y})] = 0, \tag{9.8}$$

in which both operators are evaluated at the *same time* t, and $\delta^{(3)}$ is the three-dimensional Dirac delta distribution of Theorem 5.5.

Three points of interpretation deserve emphasis, because they recur for every field quantized in this book.

First, Eq. (9.7) is the continuum image of Eq. (9.5): the Kronecker δ_{ij} that made discrete oscillators independent has become the Dirac $\delta^{(3)}(\mathbf{x} - \mathbf{y})$ that makes field values at distinct points independent. The factor of i and the units are inherited unchanged from the oscillator.

Second, the relations are imposed at *equal time*. The classical Poisson-bracket structure that canonical quantization promotes is the equal-time symplectic form on phase space; there is no canonical statement about $[\phi(x), \phi(y)]$ at *timelike or spacelike separated* points until we have solved the dynamics. Computing that unequal-time commutator — and discovering that it vanishes outside the light cone — is the content of microcausality, taken up later in this chapter. For now the commutators are anchored to a single time slice.

Third, $\phi(t, \mathbf{x})$ is not literally an operator attached to a point. The right-hand side of Eq. (9.7) is a distribution, so the left-hand side is too: ϕ is an *operator-valued distribution*, and acquires a well-defined operator meaning only after smearing against a test function, $\phi(f) \equiv \int d^3x\, f(\mathbf{x})\, \phi(t, \mathbf{x})$. We will write point-localized expressions throughout, as is universal in physics, but every such expression is shorthand for its smeared version, and the distributional machinery of Chapter 5 is what makes the shorthand safe. Products like $\pi(x)^2$ in the

Hamiltonian Eq. (9.3) are products of distributions at coincident points and will require care (normal ordering, the vacuum energy) later in the chapter; for the commutator algebra of this section no such product appears unprotected, and the manipulations are exact.

Caution

The ETCR are postulated, not derived from the classical theory — quantization is genuinely new physical input, the choice of a noncommutative operator algebra over a commutative one. What the oscillator analogy of Section 9.1.2 establishes is only that this input is the *same* input already accepted in nonrelativistic quantum mechanics, transcribed to a continuum of degrees of freedom. It is not a theorem that the continuum limit exists or is unique; for the free field it does, and that is what the explicit construction of the next section verifies. For interacting fields in four dimensions the very existence of such an operator field is a deep and partly open question (Haag's theorem, triviality), revisited in Part VI.

9.1.4 Why the Heisenberg picture is mandatory

In nonrelativistic quantum mechanics one may freely choose between the Schrödinger picture (states evolve, operators are fixed) and the Heisenberg picture (operators evolve, states are fixed); the two are related by a time-dependent unitary and give identical physics. In a relativistic field theory the choice is not free: only the Heisenberg picture makes Lorentz covariance manifest, and we adopt it from the outset.

The reason is structural. The object we are quantizing is a field $\phi(x)$ on spacetime, a function of the four-vector $x^\mu = (t, \mathbf{x})$. A Lorentz transformation mixes t and $\mathbf{x}$, so if the theory is to wear its covariance openly, the central operator must depend on all four coordinates on the same footing. The Schrödinger picture singles out time: operators are functions of $\mathbf{x}$ alone, evaluated on a privileged time slice, while the time dependence is exiled into the state. That is a legitimate but frame-dependent bookkeeping, and it obscures exactly the symmetry we most want to keep visible. The Heisenberg picture keeps the full $\phi(t, \mathbf{x})$ as a single spacetime operator and is the natural — in practice the only sane — language for covariant field theory.

In the Heisenberg picture the time dependence of the field is generated by the Hamiltonian Eq. (9.3),

$$\phi(t, \mathbf{x}) = e^{iHt}\, \phi(0, \mathbf{x})\, e^{-iHt}, \qquad \pi(t, \mathbf{x}) = e^{iHt}\, \pi(0, \mathbf{x})\, e^{-iHt}, \tag{9.9}$$

where $\phi(0, \mathbf{x})$ and $\pi(0, \mathbf{x})$ are the Schrödinger-picture operators on the $t = 0$ slice, and H is time-independent because the Lagrangian Eq. (9.1) has no explicit time dependence. Because H commutes with itself, H is the same operator in

both pictures, and the ETCR of Theorem 9.1, imposed on the $t = 0$ slice, are propagated to other times by Eq. (9.9); we verify this propagation explicitly in Theorem 9.3 below.

Differentiating Eq. (9.9) in t gives the Heisenberg equation of motion. For any Heisenberg operator $A(t) = e^{iHt}A(0)e^{-iHt}$,

$$\frac{dA}{dt} = iHe^{iHt}A(0)e^{-iHt} - e^{iHt}A(0)e^{-iHt}\,iH = i\,e^{iHt}\,[H, A(0)]\,e^{-iHt} = i\,[H, A(t)]\,, \tag{9.10}$$

using that H commutes through the unitaries it generates. Written for the field and its momentum, with the commutator in the order we will use,

$$i\,\frac{d\phi}{dt} = [\phi, H]\,, \qquad i\,\frac{d\pi}{dt} = [\pi, H]\,. \tag{9.11}$$

The dynamics is now entirely encoded in commutators with H. Everything the classical equations of motion did is reproduced by evaluating the right-hand sides of Eq. (9.11) using only the ETCR.

9.1.5 The Heisenberg equations reproduce Klein–Gordon

We are now in a position to check the central internal consistency of the construction: that the quantum dynamics generated by H is the same dynamics as the classical Klein–Gordon equation, now read as an operator statement. This is the field-theoretic counterpart of the familiar oscillator fact that $[q, H]$ and $[p, H]$ reproduce $\dot{q} = p/m$ and $\dot{p} = -m\omega^2 q$. We carry every step.

The only inputs are the ETCR Eq. (9.7)–Eq. (9.8), the Hamiltonian Eq. (9.3), and two elementary commutator identities valid for any operators,

$$[A, BC] = [A, B]\,C + B\,[A, C]\,, \qquad [AB, C] = A\,[B, C] + [A, C]\,B. \tag{9.12}$$

Throughout, all operators sit on the common time slice t (the slice on which H is evaluated as a spatial integral), so the ETCR apply directly; we suppress the common time argument and write $\phi(\mathbf{x})$, $\pi(\mathbf{y})$, etc. The momentum-conjugate combination of the ETCR we will need repeatedly is, equivalently to Eq. (9.7),

$$[\pi(\mathbf{x}), \phi(\mathbf{y})] = -i\,\delta^{(3)}(\mathbf{x} - \mathbf{y}). \tag{9.13}$$

The first equation: $\dot{\phi} = \pi$. We compute $[\phi(\mathbf{x}), H]$ with $H = \int d^3y\,[\frac{1}{2}\pi(\mathbf{y})^2 + \frac{1}{2}(\nabla\phi(\mathbf{y}))^2 + \frac{1}{2}m^2\phi(\mathbf{y})^2]$. Since ϕ commutes with ϕ at equal times by Eq. (9.8), the gradient and mass terms — built solely from ϕ — contribute nothing, and only the kinetic term survives:

$$[\phi(\mathbf{x}), H] = \int d^3y\;\tfrac{1}{2}\,\Big[\phi(\mathbf{x}), \pi(\mathbf{y})^2\Big]\,. \tag{9.14}$$

Apply the first identity of Eq. (9.12) with $A = \phi(\mathbf{x})$, $B = C = \pi(\mathbf{y})$:

$$\left[\phi(\mathbf{x}), \pi(\mathbf{y})^2\right] = [\phi(\mathbf{x}), \pi(\mathbf{y})]\ \pi(\mathbf{y}) + \pi(\mathbf{y})\ [\phi(\mathbf{x}), \pi(\mathbf{y})] = 2i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})\,\pi(\mathbf{y}), \tag{9.15}$$

where the c-number $\delta^{(3)}(\mathbf{x}-\mathbf{y})$ from Eq. (9.7) commutes freely through $\pi(\mathbf{y})$ and the two terms add. Hence

$$[\phi(\mathbf{x}), H] = \int d^3y\; i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})\,\pi(\mathbf{y}) = i\,\pi(\mathbf{x}). \tag{9.16}$$

Insert this into the first Heisenberg equation Eq. (9.11), $i\,\dot\phi = [\phi, H] = i\pi$, giving

$$\dot\phi(\mathbf{x}) = \pi(\mathbf{x}). \tag{9.17}$$

The quantum equation of motion has reproduced the defining relation Eq. (9.2) between the field and its conjugate momentum — now as an operator identity rather than a definition. This is the first nontrivial consistency check: the Hamiltonian flow is compatible with the Legendre transform that produced it.

The second equation: $\dot\pi = (\nabla^2 - m^2)\phi$. Now we compute $[\pi(\mathbf{x}), H]$. The kinetic term of H is built from π alone and commutes with $\pi(\mathbf{x})$ by Eq. (9.8); the surviving contributions are the gradient and mass terms,

$$[\pi(\mathbf{x}), H] = \int d^3y\ \left[\tfrac{1}{2}\ \left[\pi(\mathbf{x}), (\nabla\phi(\mathbf{y}))^2\right] + \tfrac{1}{2}m^2\ \left[\pi(\mathbf{x}), \phi(\mathbf{y})^2\right]\ \right]. \tag{9.18}$$

Take the mass term first. By the same Leibniz step as above, now using Eq. (9.13),

$$\tfrac{1}{2}m^2\ \left[\pi(\mathbf{x}), \phi(\mathbf{y})^2\right] = \tfrac{1}{2}m^2(\,[\pi(\mathbf{x}), \phi(\mathbf{y})]\,\phi(\mathbf{y}) + \phi(\mathbf{y})\,[\pi(\mathbf{x}), \phi(\mathbf{y})]\,) = -i\,m^2\,\delta^{(3)}(\mathbf{x}-\mathbf{y})\,\phi(\mathbf{y}), \tag{9.19}$$

whose y-integral is $-i\,m^2\,\phi(\mathbf{x})$.

The gradient term needs one extra step because $\pi(\mathbf{x})$ must be commuted past a *spatial derivative* of ϕ. Write $(\nabla\phi)^2 = \sum_i (\partial_i^{\mathbf{y}}\phi(\mathbf{y}))(\partial_i^{\mathbf{y}}\phi(\mathbf{y}))$, where $\partial_i^{\mathbf{y}}$ acts on the argument $\mathbf{y}$. The derivative is a linear operation that the commutator passes through,

$$[\pi(\mathbf{x}), \partial_i^{\mathbf{y}}\phi(\mathbf{y})] = \partial_i^{\mathbf{y}}\,[\pi(\mathbf{x}), \phi(\mathbf{y})] = -i\,\partial_i^{\mathbf{y}}\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{9.20}$$

again using Eq. (9.13). Then, by the Leibniz identity applied to the product of the two factors $\partial_i^{\mathbf{y}}\phi(\mathbf{y})$,

$$\tfrac{1}{2}\ [\pi(\mathbf{x}), (\partial_i^{\mathbf{y}}\phi)(\partial_i^{\mathbf{y}}\phi)] = (\partial_i^{\mathbf{y}}\phi(\mathbf{y}))\ [\pi(\mathbf{x}), \partial_i^{\mathbf{y}}\phi(\mathbf{y})] = -i\,(\partial_i^{\mathbf{y}}\phi(\mathbf{y}))\,\partial_i^{\mathbf{y}}\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{9.21}$$

the c-number derivative-of-delta again commuting through. Integrating over $\mathbf{y}$ and summing on i,

$$\int d^3y\ (-i)\,(\partial_i^{\mathbf{y}}\phi(\mathbf{y}))\,\partial_i^{\mathbf{y}}\delta^{(3)}(\mathbf{x}-\mathbf{y}). \tag{9.22}$$

Integrate by parts in $\mathbf{y}$, the boundary term vanishing because fields fall off at spatial infinity (equivalently because the delta has compact support):

$$\int d^3y\,(-i)\,(\partial_i^{\mathbf{y}}\phi)\,\partial_i^{\mathbf{y}}\delta^{(3)}(\mathbf{x}-\mathbf{y}) = \int d^3y\,(+i)\,(\partial_i^{\mathbf{y}}\partial_i^{\mathbf{y}}\phi(\mathbf{y}))\,\delta^{(3)}(\mathbf{x}-\mathbf{y}) = i\,\nabla^2\phi(\mathbf{x}). \tag{9.23}$$

Collecting the gradient and mass contributions in Eq. (9.18),

$$[\pi(\mathbf{x}), H] = i\,\nabla^2\phi(\mathbf{x}) - i\,m^2\phi(\mathbf{x}) = i\,(\nabla^2 - m^2)\,\phi(\mathbf{x}). \tag{9.24}$$

Insert this into the second Heisenberg equation Eq. (9.11), $i\,\dot\pi = [\pi, H]$:

$$\dot\pi(\mathbf{x}) = (\nabla^2 - m^2)\,\phi(\mathbf{x}). \tag{9.25}$$

Assembling Klein–Gordon. Differentiate Eq. (9.17) once more in time and substitute Eq. (9.25):

$$\ddot\phi = \dot\pi = (\nabla^2 - m^2)\,\phi \quad\Longrightarrow\quad \ddot\phi - \nabla^2\phi + m^2\phi = 0. \tag{9.26}$$

With $\Box = \partial_0^2 - \nabla^2$ in the mostly-minus metric, this is precisely

$$(\Box + m^2)\,\phi(x) = 0, \tag{9.27}$$

the operator Klein–Gordon equation. We record the result.

Proposition 9.2 (Heisenberg dynamics reproduces Klein–Gordon)**.** *Let ϕ and π be operators satisfying the equal-time canonical commutation relations of Theorem 9.1 and evolving in the Heisenberg picture under the free Hamiltonian Eq. (9.3). Then the Heisenberg equations of motion Eq. (9.11) are equivalent to the pair*

$$\dot\phi = \pi, \qquad \dot\pi = (\nabla^2 - m^2)\,\phi, \tag{9.28}$$

which together imply the operator Klein–Gordon equation $(\Box + m^2)\phi = 0$ of Eq. (9.27). The quantum field obeys, as an operator identity, the same dynamical equation Eq. (9.4) as the classical field.

The significance is worth stating plainly. We did not assume the operator field satisfies Klein–Gordon; we assumed only the ETCR and the form of H, and the dynamics followed. Conversely, the operator equation Eq. (9.27) is the bridge to the next section: it is a linear wave equation for an operator, and its general solution is a superposition of plane-wave modes whose coefficients are the operators we will identify as ladder operators. The classical solution method survives quantization wholesale; only the Fourier coefficients are promoted to operators.

Physical Insight

The harmonic oscillator's $[q, H] \propto p$ and $[p, H] \propto q$ closed back on Newton's second law; here $[\phi, H] \propto \pi$ and $[\pi, H] \propto (\nabla^2 - m^2)\phi$ close back on Klein–Gordon. Nothing in the field case is conceptually new — it is the oscillator computation done at every point of space at once, with the spatial coupling $(\nabla\phi)^2$ supplying the ∇^2 that ties neighbouring degrees of freedom together into a wave rather than leaving them as independent oscillations. This is the precise sense in which a free field "is" a continuum of oscillators.

9.1.6 Consistency: the canonical structure is conserved

One loose end remains. The ETCR of Theorem 9.1 were imposed on a single time slice. The Heisenberg evolution Eq. (9.9) then defines ϕ and π at every other time. For the construction to be consistent, the same canonical relations must hold on every slice — otherwise the algebra would depend on an arbitrary choice of initial time, and "equal-time commutator" would be ambiguous. We verify that the canonical structure is preserved, in two complementary ways.

Worked Example 9.3 (The ETCR are conserved by time evolution)**.** *Argument 1 (unitarity).* Time evolution acts by conjugation with the unitary $U(t) = e^{-iHt}$, and conjugation by a fixed unitary is an automorphism of the operator algebra: for any operators A, B,

$$\left[U^\dagger A\, U,\, U^\dagger B\, U\right] = U^\dagger\, [A, B]\, U, \tag{9.29}$$

since the intermediate $UU^\dagger = \mathbf{1}$ cancels. Applying this to Eq. (9.9) with $A = \phi(0, \mathbf{x})$, $B = \pi(0, \mathbf{y})$, and using that the $t = 0$ ETCR give $[\phi(0, \mathbf{x}), \pi(0, \mathbf{y})] = i\,\delta^{(3)}(\mathbf{x} - \mathbf{y})$,

$$[\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = e^{iHt}\, [\phi(0, \mathbf{x}), \pi(0, \mathbf{y})]\, e^{-iHt} = e^{iHt}\, \left(i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})\right) e^{-iHt} = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{9.30}$$

because the c-number $\delta^{(3)}$ passes through the unitaries and $e^{iHt}e^{-iHt} = \mathbf{1}$. The same step applied to Eq. (9.8) gives $[\phi(t, \mathbf{x}), \phi(t, \mathbf{y})] = [\pi(t, \mathbf{x}), \pi(t, \mathbf{y})] = 0$ on every slice. The equal-time canonical algebra is therefore independent of the slice on which it is imposed.

Argument 2 (infinitesimal, using the equations of motion). It is instructive to see the same conservation differentially, as a check that it is consistent with the dynamics derived in Theorem 9.2 rather than an independent assumption. Differentiate the equal-time commutator and use the product rule together with $\dot\phi = \pi$ from Eq. (9.17) and $\dot\pi = (\nabla^2 - m^2)\phi$ from Eq. (9.25):

$$\frac{d}{dt}\, [\phi(t, \mathbf{x}), \pi(t, \mathbf{y})] = \left[\dot\phi(t, \mathbf{x}), \pi(t, \mathbf{y})\right] + [\phi(t, \mathbf{x}), \dot\pi(t, \mathbf{y})]$$

$$= [\pi(t,\mathbf{x}),\pi(t,\mathbf{y})] + \left[\phi(t,\mathbf{x}),(\nabla^2_{\mathbf{y}}-m^2)\phi(t,\mathbf{y})\right]$$
$$= 0 + (\nabla^2_{\mathbf{y}}-m^2)\,[\phi(t,\mathbf{x}),\phi(t,\mathbf{y})] = 0, \tag{9.31}$$

where the first commutator vanishes by Eq. (9.8) and the second because $(\nabla^2_{\mathbf{y}}-m^2)$ acts on $\mathbf{y}$ and pulls outside the commutator, which is itself zero by Eq. (9.8). The right-hand side $i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})$ is time-independent, consistent with its derivative vanishing. The identical computation gives $\frac{d}{dt}\,[\phi,\phi] = [\pi,\phi] + [\phi,\pi] = 0$ and $\frac{d}{dt}\,[\pi,\pi] = (\nabla^2 - m^2)([\phi,\pi]+[\pi,\phi]) = 0$, so all three ETCR are stationary. The canonical structure is a constant of the motion.

Both arguments deliver the same conclusion, and they reinforce each other: unitarity guarantees the preservation abstractly, while the infinitesimal check confirms that the specific equations of motion derived from H are exactly the ones consistent with that preservation. With this, Theorem 9.1 is a self-consistent definition: the equal-time commutators may be imposed on any one slice, and the dynamics distributes them faithfully to all others.

We have now built the object the rest of the chapter studies: a Heisenberg-picture operator field $\phi(x)$, obeying the operator Klein–Gordon equation Eq. (9.27) together with the equal-time canonical commutators Eq. (9.7)–Eq. (9.8). This is an operator equation we have not yet solved. The next section solves it the only way a linear wave equation can be solved — by expanding ϕ in plane-wave modes — and discovers that the Fourier coefficients are creation and annihilation operators whose algebra is fixed, mode by mode, by the canonical relations established here.

9.2 The Mode Expansion and Ladder Operators

The previous section handed us a Heisenberg-picture operator field $\phi(x)$, its conjugate momentum $\pi(x) = \partial_0\phi(x)$, the equal-time canonical commutation relations (ETCR)

$$[\phi(t,\mathbf{x}),\pi(t,\mathbf{y})] = i\,\delta^3(\mathbf{x}-\mathbf{y}), \qquad [\phi(t,\mathbf{x}),\phi(t,\mathbf{y})] = [\pi(t,\mathbf{x}),\pi(t,\mathbf{y})] = 0, \tag{9.32}$$

and the operator Klein–Gordon equation $(\Box+m^2)\phi = 0$, established in Section 9.1. We now have an operator-valued equation of motion and an algebra, but no solution and no Hilbert space yet. This section produces the solution. The strategy was announced in the chapter introduction and is worth restating in one sentence: a free field is a continuum of decoupled harmonic oscillators labelled by spatial momentum, so we Fourier-decompose ϕ into plane-wave modes, recognise each Fourier coefficient as a ladder operator of one such oscillator, and read off its algebra from Eq. (9.32). The only genuinely new ingredient beyond the Sakurai-level oscillator is relativistic bookkeeping: we must integrate over on-shell momenta with a measure that respects Lorentz invariance, and we must normalise

the ladder operators so that this covariance stays manifest. We deal with that measure first, because every formula in the rest of the chapter is written against it.

9.2.1 The Lorentz-invariant measure

A real classical solution of $(\Box + m^2)\phi = 0$ is a superposition of plane waves $e^{\mp ip\cdot x}$ whose four-momentum lies on the positive-energy mass shell

$$p^2 = m^2, \qquad p^0 = E_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + m^2} > 0. \tag{9.33}$$

The set of such p is the forward sheet of the two-sheeted hyperboloid $p^2 = m^2$ (Fig. 9.2). Summing over modes therefore means integrating a function $f(\mathbf{p})$ over this sheet. The naive choice $\int d^3p\, f(\mathbf{p})$ is *not* Lorentz invariant: a boost mixes p^0 into the spatial components and distorts the volume element d^3p. The correct, invariant prescription is

Definition 9.4 (Lorentz-invariant measure)**.** The *Lorentz-invariant measure* on the positive mass shell (9.33) is

$$\int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\,, \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \tag{9.34}$$

The factor $(2\pi)^3$ is bookkeeping matched to the Fourier convention of Chapter 5; the dynamically meaningful factor is the $1/2E_{\mathbf{p}}$.

The cleanest proof that (9.34) is invariant builds it out of a manifestly invariant four-dimensional integral and an on-shell delta function. This is the device the book will reuse whenever a three-dimensional phase-space integral must be shown to be covariant.

Theorem 9.5 (Invariance of the on-shell measure)**.** *Under a proper orthochronous Lorentz transformation $p \mapsto p' = \Lambda p$ (with $\Lambda \in SO^+(1,3)$, the connected component identified in Theorem 3.6), the measure (9.34) is unchanged: for any test function f on the mass shell,*

$$\int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, f(p)\Big|_{p^0 = E_{\mathbf{p}}} = \int \frac{d^3p'}{(2\pi)^3\, 2E_{\mathbf{p}'}}\, f(\Lambda^{-1}p')\Big|_{p'^0 = E_{\mathbf{p}'}}. \tag{9.35}$$

Proof. Consider the four-dimensional integral

$$I[f] = \int \frac{d^4p}{(2\pi)^4}\, (2\pi)\, \delta(p^2 - m^2)\, \theta(p^0)\, f(p), \tag{9.36}$$

where now p^0 is an *independent* integration variable, not yet set to $E_{\mathbf{p}}$. We show two things: (i) $I[f]$ equals the left-hand side of (9.35); (ii) $I[f]$ is invariant under $SO^+(1,3)$. Together these give the claim.

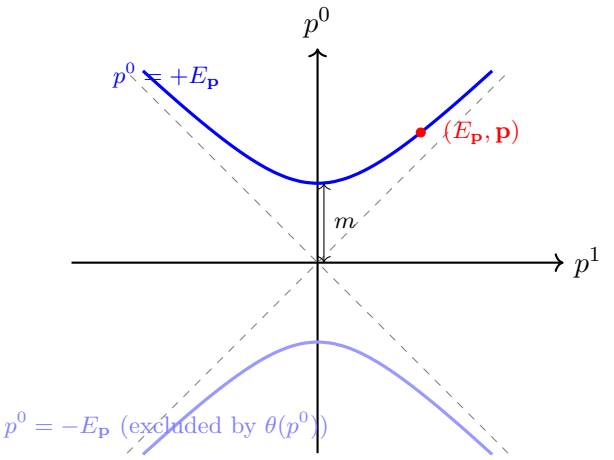


Figure 9.2: The mass shell $p^2 = m^2$ in the (p^0, p^1) plane. The step function $\theta(p^0)$ selects the upper (forward) sheet $p^0 = +E_{\mathbf{p}}$. A proper orthochronous Lorentz transformation slides points along a sheet without ever crossing the light cone (dashed), so it never moves a forward vector onto the backward sheet; this is why $\theta(p^0)$, and hence the measure (9.34), is invariant.

(i) Doing the p^0 integral. Hold $\mathbf{p}$ fixed and write the argument of the delta function as a function of p^0:

$$p^2 - m^2 = (p^0)^2 - \mathbf{p}^2 - m^2 = (p^0)^2 - E_{\mathbf{p}}^2. \tag{9.37}$$

This vanishes at the two simple roots $p^0 = \pm E_{\mathbf{p}}$. The composition rule for the delta distribution proved in Theorem 5.5,

$$\delta(g(p^0)) = \sum_{i:\, g(p_i^0)=0} \frac{\delta(p^0 - p_i^0)}{|g'(p_i^0)|}, \tag{9.38}$$

applies with $g(p^0) = (p^0)^2 - E_{\mathbf{p}}^2$, so that $g'(p^0) = 2p^0$ and $|g'(\pm E_{\mathbf{p}})| = 2E_{\mathbf{p}}$. Hence

$$\delta(p^2 - m^2) = \frac{1}{2E_{\mathbf{p}}}\Big[\,\delta(p^0 - E_{\mathbf{p}}) + \delta(p^0 + E_{\mathbf{p}})\,\Big]. \tag{9.39}$$

The step function $\theta(p^0)$ kills the second term, which has support at the negative root. Performing the p^0 integral against the surviving $\delta(p^0 - E_{\mathbf{p}})$ collapses the (2π) against one power of $(2\pi)^{-4}$ and sets $p^0 = E_{\mathbf{p}}$ inside f:

$$I[f] = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\mathbf{p}}}\,f(E_{\mathbf{p}}, \mathbf{p}), \tag{9.40}$$

which is exactly the left-hand side of (9.35).

(ii) Invariance of (9.36). We check each factor of the integrand in turn. The four-volume d^4p transforms with the Jacobian $|\det(\partial p'/\partial p)| = |\det \Lambda|$. Every

$\Lambda \in O(1,3)$ satisfies $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$, whence $(\det\Lambda)^2 = 1$ and $\det\Lambda = \pm 1$; the proper subgroup is precisely the one with $\det\Lambda = +1$, so $d^4p' = d^4p$. The argument $p^2 = \eta_{\mu\nu}p^\mu p^\nu$ of the delta function is a Lorentz scalar, so $\delta(p^2 - m^2)$ is invariant. The mass parameter m^2 is a fixed number. There remains $\theta(p^0)$. Its invariance is the one place the words "proper" and "orthochronous" actually earn their keep. The support of $\delta(p^2 - m^2)$ is the set $p^2 = m^2 \geq 0$, i.e. timelike (or, in the $m \to 0$ limit, null) four-vectors. For any such vector with $p^0 > 0$ — a *future-pointing* vector — an orthochronous transformation, defined in Theorem 3.3 by ${\Lambda^0}_0 \geq 1$, keeps $p'^0 > 0$. Concretely, for a timelike p one has the reverse Cauchy–Schwarz inequality $\left|\sum_i {\Lambda^0}_i\, p^i\right| < {\Lambda^0}_0\, p^0$ on the forward cone, so the sign of $p'^0 = {\Lambda^0}_0 p^0 + {\Lambda^0}_i p^i$ is fixed by the dominant term ${\Lambda^0}_0 p^0 > 0$. Thus a future-pointing on-shell vector maps to a future-pointing on-shell vector, and $\theta(p^0)$ is invariant on the support of the delta function. Every factor of (9.36) is therefore invariant, so $I[f] = I[f \circ \Lambda^{-1}]$; combined with (9.40) this is (9.35). $\square$

The identity (9.39) is worth recording on its own, because it is the covariant statement that the measure equals an integral over the forward cone:

$$\frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \;=\; \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2 - m^2)\,\theta(p^0). \tag{9.41}$$

We will reuse this in the next section to write the conserved four-momentum covariantly, and again in the construction of the propagator later in the chapter.

9.2.2 The mode expansion of the field

We now solve the operator Klein–Gordon equation. Classically, the general real solution of $(\Box + m^2)\phi = 0$ is a superposition of the plane-wave solutions $e^{-ip\cdot x}$ and $e^{+ip\cdot x}$ with p on the positive mass shell (9.33); reality pairs the coefficient of $e^{+ip\cdot x}$ as the complex conjugate of the coefficient of $e^{-ip\cdot x}$. Quantization promotes these two coefficients to operators, mutually adjoint rather than complex-conjugate, since ϕ is now a Hermitian operator. Integrating over modes with the invariant measure (9.34), we are led to the following ansatz, which we will then verify solves the equation and is consistent with the ETCR.

Definition 9.6 (Mode expansion)**.** The free Hermitian scalar field and its conjugate momentum are expanded as

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big(a_{\mathbf{p}}\, e^{-ip\cdot x} + a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} \Big)\Big|_{p^0 = E_{\mathbf{p}}}, \tag{9.42}$$

$$\pi(x) = \partial_0\phi(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, (-iE_{\mathbf{p}}) \Big(a_{\mathbf{p}}\, e^{-ip\cdot x} - a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} \Big)\Big|_{p^0 = E_{\mathbf{p}}}, \tag{9.43}$$

with $p \cdot x = E_{\mathbf{p}} x^0 - \mathbf{p} \cdot \mathbf{x}$ on shell. The operators $a_{\mathbf{p}}$ (annihilation) and $a^\dagger_{\mathbf{p}}$ (creation) are the unknowns; the relativistic normalization is fixed by the choice of measure in (9.42).

Three points justify each part of this ansatz before we extract the algebra.

It solves the equation. Acting with $\Box + m^2$ on (9.42), the derivatives hit only the exponentials. On the mass shell $\Box\, e^{\mp ip\cdot x} = -p^2\, e^{\mp ip\cdot x} = -m^2\, e^{\mp ip\cdot x}$, so $(\Box + m^2) e^{\mp ip\cdot x} = 0$ mode by mode and the whole integral vanishes. Thus (9.42) is a solution of the operator Klein–Gordon equation for *any* operators $a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}$; the equation of motion fixes the kinematics (which exponentials appear) but says nothing yet about the algebra. The algebra will come from the ETCR.

It is manifestly Hermitian. The second term of (9.42) is the adjoint of the first, integrated against the same real measure, so $\phi^\dagger = \phi$. This is the operator analogue of the classical reality condition.

The momentum expansion is consistent. Differentiating (9.42) with respect to x^0 brings down $-iE_{\mathbf{p}}$ from $e^{-ip\cdot x}$ and $+iE_{\mathbf{p}}$ from $e^{+ip\cdot x}$, giving exactly (9.43). We use $\pi = \partial_0 \phi$, the canonical momentum of the free scalar established in Section 9.1.

9.2.3 Inversion: extracting the ladder operators

To read off the algebra of $a_{\mathbf{p}}$ from (9.32) we first invert the expansion, expressing $a_{\mathbf{p}}$ and $a^\dagger_{\mathbf{p}}$ as integrals of ϕ and π over a single constant-time slice. The idea is the standard Fourier projection: integrate the field against a conjugate plane wave over all space at fixed time, picking out one momentum component.

Proposition 9.7 (Inversion formulae)**.** *For each* $\mathbf{p}$ *the operators in* (9.42) *are given, at any fixed time* $x^0 = t$*, by*

$$a_{\mathbf{p}} = \int d^3x\, e^{+ip\cdot x} \big(E_{\mathbf{p}}\, \phi(x) + i\, \pi(x)\big)\Big|_{p^0 = E_{\mathbf{p}}} = i \int d^3x\, e^{+ip\cdot x} \overset{\leftrightarrow}{\partial_0}\, \phi(x), \tag{9.44}$$

$$a^\dagger_{\mathbf{p}} = \int d^3x\, e^{-ip\cdot x} \big(E_{\mathbf{p}}\, \phi(x) - i\, \pi(x)\big)\Big|_{p^0 = E_{\mathbf{p}}} = -\, i \int d^3x\, e^{-ip\cdot x} \overset{\leftrightarrow}{\partial_0}\, \phi(x), \tag{9.45}$$

where $f \overset{\leftrightarrow}{\partial_0} g \equiv f\,(\partial_0 g) - (\partial_0 f)\, g$*. The right-hand sides are independent of the time* t *at which the slice is taken.*

Proof. Insert the expansions (9.42)–(9.43) into the first form of (9.44). Two spatial Fourier integrals appear. Writing q for the integration momentum of the field and p for the fixed external momentum, and keeping the on-shell time factors explicit,

$$\int d^3x\, e^{+ip\cdot x}\, \phi(x) = \int \frac{d^3q}{(2\pi)^3\, 2E_{\mathbf{q}}} \int d^3x \Big(a_{\mathbf{q}}\, e^{i(p-q)\cdot x} + a^\dagger_{\mathbf{q}}\, e^{i(p+q)\cdot x} \Big)$$

$$= \int \frac{d^3q}{(2\pi)^3\,2E_{\mathbf{q}}}\Big(a_{\mathbf{q}}\,e^{i(E_{\mathbf{p}}-E_{\mathbf{q}})t}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}) + a^\dagger_{\mathbf{q}}\,e^{i(E_{\mathbf{p}}+E_{\mathbf{q}})t}(2\pi)^3\delta^3(\mathbf{p}+\mathbf{q})\Big). \tag{9.46}$$

Here the spatial integral $\int d^3x\, e^{-i(\mathbf{p}\mp\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}\mp\mathbf{q})$ is the standard Fourier representation of the delta distribution from Theorem 5.18. The first delta sets $\mathbf{q}=\mathbf{p}$ (so $E_{\mathbf{q}}=E_{\mathbf{p}}$ and the time factor becomes 1); the second sets $\mathbf{q}=-\mathbf{p}$ (so again $E_{\mathbf{q}}=E_{\mathbf{p}}$, leaving a factor $e^{2iE_{\mathbf{p}}t}$). Doing the q-integral and cancelling $(2\pi)^3/[(2\pi)^3 2E_{\mathbf{p}}] = 1/2E_{\mathbf{p}}$,

$$\int d^3x\; e^{+ip\cdot x}\,\phi(x) = \frac{1}{2E_{\mathbf{p}}}\Big(a_{\mathbf{p}} + a^\dagger_{-\mathbf{p}}\,e^{2iE_{\mathbf{p}}t}\Big). \tag{9.47}$$

The identical manipulation on the momentum expansion (9.43), where the integrand carries the extra factor $-iE_{\mathbf{q}}$ with a relative minus sign between the two terms, gives

$$\int d^3x\; e^{+ip\cdot x}\,\pi(x) = \frac{-iE_{\mathbf{p}}}{2E_{\mathbf{p}}}\Big(a_{\mathbf{p}} - a^\dagger_{-\mathbf{p}}\,e^{2iE_{\mathbf{p}}t}\Big) = -\frac{i}{2}\Big(a_{\mathbf{p}} - a^\dagger_{-\mathbf{p}}\,e^{2iE_{\mathbf{p}}t}\Big). \tag{9.48}$$

Now form the combination $E_{\mathbf{p}}\,(9.47) + i\,(9.48)$:

$$E_{\mathbf{p}}\int d^3x\, e^{ip\cdot x}\phi + i\int d^3x\, e^{ip\cdot x}\pi = \frac{1}{2}\Big(a_{\mathbf{p}} + a^\dagger_{-\mathbf{p}}e^{2iE_{\mathbf{p}}t}\Big) + \frac{1}{2}\Big(a_{\mathbf{p}} - a^\dagger_{-\mathbf{p}}e^{2iE_{\mathbf{p}}t}\Big) = a_{\mathbf{p}}. \tag{9.49}$$

The creation-operator pieces, with their $e^{2iE_{\mathbf{p}}t}$ time factors, cancel identically, leaving $a_{\mathbf{p}}$ alone — and with no residual dependence on t. This is the first form of (9.44). The compact form follows by recognising the combination: with $f = e^{+ip\cdot x}$ one has $\partial_0 f = iE_{\mathbf{p}} f$, so

$$i\,f\overleftrightarrow{\partial_0}\,\phi = i(f\,\partial_0\phi - (\partial_0 f)\,\phi) = i\,f\,\pi + E_{\mathbf{p}}\,f\,\phi, \tag{9.50}$$

whose space integral is precisely (9.49). Taking the Hermitian adjoint of (9.44), using $\phi^\dagger=\phi$, $\pi^\dagger=\pi$ and $(e^{ip\cdot x})^* = e^{-ip\cdot x}$, gives (9.45).

Time-independence, asserted in passing above, deserves an explicit verification rather than reliance on the cancellation in (9.49); we discharge it in Theorem 9.8. □

Worked Example 9.8 (The inversion is time-independent)**.** We show directly that $\partial_t a_{\mathbf{p}} = 0$, so that the slice in Theorem 9.7 may be taken at any time. Differentiate the first form of (9.44) under the integral sign:

$$\partial_t a_{\mathbf{p}} = \int d^3x\;\partial_t\Big[e^{ip\cdot x}(E_{\mathbf{p}}\,\phi + i\,\pi)\Big] = \int d^3x\; e^{ip\cdot x}\Big(iE_{\mathbf{p}}(E_{\mathbf{p}}\phi + i\pi) + E_{\mathbf{p}}\,\partial_t\phi + i\,\partial_t\pi\Big), \tag{9.51}$$

where we used $\partial_t e^{ip\cdot x} = iE_{\mathbf{p}} e^{ip\cdot x}$. Now substitute the operator equations of motion established in Section 9.1: $\partial_t \phi = \pi$ (definition of the conjugate momentum) and $\partial_t \pi = \partial_t^2 \phi = (\nabla^2 - m^2)\phi$ (the Heisenberg equation, equivalent to the Klein–Gordon equation). The bracket becomes

$$iE_{\mathbf{p}}^2\phi - E_{\mathbf{p}}\pi + E_{\mathbf{p}}\pi + i(\nabla^2 - m^2)\phi = i(E_{\mathbf{p}}^2 - m^2)\phi + i\,\nabla^2\phi = i\,\mathbf{p}^2\,\phi + i\,\nabla^2\phi, \tag{9.52}$$

using $E_{\mathbf{p}}^2 - m^2 = \mathbf{p}^2$. The two π-terms have cancelled. It remains to integrate against $e^{ip\cdot x}$. Twice integrating by parts in $\mathbf{x}$ (boundary terms vanish for the smeared, normalizable configurations of Chapter 5) trades ∇^2 acting on ϕ for ∇^2 acting on the exponential: $\int d^3x\, e^{ip\cdot x}\nabla^2\phi = \int d^3x\,(\nabla^2 e^{ip\cdot x})\phi = -\mathbf{p}^2 \int d^3x\, e^{ip\cdot x}\phi$, since $\nabla^2 e^{-i\mathbf{p}\cdot\mathbf{x}} = -\mathbf{p}^2 e^{-i\mathbf{p}\cdot\mathbf{x}}$. Hence

$$\partial_t a_{\mathbf{p}} = i\int d^3x\, e^{ip\cdot x}(\mathbf{p}^2\phi + \nabla^2\phi) = i(\mathbf{p}^2 - \mathbf{p}^2)\int d^3x\, e^{ip\cdot x}\phi = 0. \tag{9.53}$$

The annihilation operator is genuinely constant in time, as the Heisenberg picture of a *free* field demands: $a_{\mathbf{p}}$ is a constant of the motion, the operator amplitude of a single eternal mode. The same computation with $e^{-ip\cdot x}$ shows $\partial_t a_{\mathbf{p}}^\dagger = 0$.

9.2.4 The ladder algebra

With the inversion formulae in hand the algebra of $a_{\mathbf{p}}$ follows from the ETCR by a single double integral. This is the central computation of the section: it converts the canonical commutators of the fields into the oscillator algebra of the modes.

Theorem 9.9 (Ladder algebra)**.** *The operators defined by* (9.44)–(9.45) *satisfy*

$$\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger\right] = (2\pi)^3\, 2E_{\mathbf{p}}\,\delta^3(\mathbf{p}-\mathbf{q}), \qquad [a_{\mathbf{p}}, a_{\mathbf{q}}] = \left[a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger\right] = 0. \tag{9.54}$$

Proof. Because each of $a_{\mathbf{p}}$, $a_{\mathbf{q}}^\dagger$ is time-independent (Theorem 9.8), we may evaluate both on the *same* time slice $x^0 = y^0 = t$ and use the equal-time relations (9.32). Write $\phi_x \equiv \phi(t,\mathbf{x})$, $\pi_x \equiv \pi(t,\mathbf{x})$. From the first forms of (9.44) and (9.45),

$$\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger\right] = \int d^3x\, d^3y\; e^{ip\cdot x}\, e^{-iq\cdot y}\,\left[E_{\mathbf{p}}\phi_x + i\pi_x,\; E_{\mathbf{q}}\phi_y - i\pi_y\right]. \tag{9.55}$$

Expand the commutator bilinearly. The four terms are

$$E_{\mathbf{p}}E_{\mathbf{q}}\,[\phi_x,\phi_y] - iE_{\mathbf{p}}\,[\phi_x,\pi_y] + iE_{\mathbf{q}}\,[\pi_x,\phi_y] + [\pi_x,\pi_y]\,. \tag{9.56}$$

By the ETCR (9.32) the first and last vanish, while $[\phi_x,\pi_y] = i\delta^3(\mathbf{x}-\mathbf{y})$ and $[\pi_x,\phi_y] = -i\delta^3(\mathbf{x}-\mathbf{y})$. Hence the bracket equals

$$-iE_{\mathbf{p}}\cdot i\,\delta^3(\mathbf{x}-\mathbf{y}) + iE_{\mathbf{q}}\cdot(-i)\,\delta^3(\mathbf{x}-\mathbf{y}) = (E_{\mathbf{p}} + E_{\mathbf{q}})\,\delta^3(\mathbf{x}-\mathbf{y}). \tag{9.57}$$

Insert this into (9.55) and use $\delta^3(\mathbf{x}-\mathbf{y})$ to do the $\mathbf{y}$-integral, setting $\mathbf{y}=\mathbf{x}$ and $y^0 = x^0 = t$:

$$\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger\right] = (E_{\mathbf{p}}+E_{\mathbf{q}}) \int d^3x\; e^{i(E_{\mathbf{p}}-E_{\mathbf{q}})t}\, e^{-i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (E_{\mathbf{p}}+E_{\mathbf{q}})\, e^{i(E_{\mathbf{p}}-E_{\mathbf{q}})t}\, (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}). \tag{9.58}$$

The remaining delta sets $\mathbf{q}=\mathbf{p}$, whereupon $E_{\mathbf{q}} = E_{\mathbf{p}}$, the time factor $e^{i(E_{\mathbf{p}}-E_{\mathbf{q}})t} = 1$, and $E_{\mathbf{p}} + E_{\mathbf{q}} = 2E_{\mathbf{p}}$. This yields the advertised

$$\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger\right] = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{q}). \tag{9.59}$$

For the second relation, repeat the computation with $a_{\mathbf{q}}$ in place of $a_{\mathbf{q}}^\dagger$. The only change is the sign of the π_y term:

$$[a_{\mathbf{p}}, a_{\mathbf{q}}] = \int d^3x\, d^3y\; e^{ip\cdot x}\, e^{iq\cdot y}\; \left[E_{\mathbf{p}}\phi_x + i\pi_x,\; E_{\mathbf{q}}\phi_y + i\pi_y\right]. \tag{9.60}$$

The surviving terms now give $iE_{\mathbf{p}}\,[\phi_x, \pi_y] + iE_{\mathbf{q}}\,[\pi_x, \phi_y] = iE_{\mathbf{p}}(i\delta) + iE_{\mathbf{q}}(-i\delta) = (E_{\mathbf{q}} - E_{\mathbf{p}})\,\delta^3(\mathbf{x}-\mathbf{y})$. The $\mathbf{y}$-integral then produces $(2\pi)^3\delta^3(\mathbf{p}+\mathbf{q})$, which forces $\mathbf{q} = -\mathbf{p}$, hence $E_{\mathbf{q}} = E_{\mathbf{p}}$ and the prefactor $E_{\mathbf{q}} - E_{\mathbf{p}} = 0$. Therefore $[a_{\mathbf{p}}, a_{\mathbf{q}}] = 0$, and taking the adjoint, $\left[a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}^\dagger\right] = 0$. □

The structure of (9.54) is exactly that of an infinite family of independent harmonic oscillators, one for each momentum $\mathbf{p}$: distinct momenta commute, and a creation operator fails to commute with its own annihilation operator by a c-number. The continuum delta function $\delta^3(\mathbf{p}-\mathbf{q})$ replaces the Kronecker delta of the discrete oscillator, and the factor $(2\pi)^3 2E_{\mathbf{p}}$ is the relativistic normalization carried by our choice of measure. The whole engine of Chapter 9 is now the Sakurai-level oscillator solution applied to this family — but with the bookkeeping fixed so that Lorentz covariance never has to be re-checked.

It is reassuring, and a good consistency test of the whole construction, that the algebra (9.54) reproduces the ETCR we started from when fed back through the mode expansion. We verify this closure explicitly.

Worked Example 9.10 (The algebra reproduces the ETCR)**.** We compute $[\phi(t,\mathbf{x}), \pi(t,\mathbf{y})]$ from (9.42)–(9.43) and the algebra (9.54), and recover $i\delta^3(\mathbf{x}-\mathbf{y})$. Substituting the expansions,

$$\begin{aligned}[\phi(x), \pi(y)] = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \frac{d^3q}{(2\pi)^3 2E_{\mathbf{q}}}\, (-iE_{\mathbf{q}}) \\ \times \left[a_{\mathbf{p}}e^{-ip\cdot x} + a_{\mathbf{p}}^\dagger e^{ip\cdot x},\; a_{\mathbf{q}}e^{-iq\cdot y} - a_{\mathbf{q}}^\dagger e^{iq\cdot y}\right].\end{aligned} \tag{9.61}$$

Only the two cross commutators survive,

$$\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger\right] = (2\pi)^3 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{q}), \qquad \left[a_{\mathbf{p}}^\dagger, a_{\mathbf{q}}\right] = -(2\pi)^3 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{q}), \tag{9.62}$$

the like–like commutators being zero by (9.54). Each delta function collapses the $\mathbf{q}$-integral onto $\mathbf{q}=\mathbf{p}$, sending $E_{\mathbf{q}}\to E_{\mathbf{p}}$ and cancelling the measure factor $1/[(2\pi)^3 2E_{\mathbf{q}}]$ against the $(2\pi)^3 2E_{\mathbf{p}}$ from the commutator. Collecting the two surviving pieces,

$$[\phi(x),\pi(y)] = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\,(-iE_{\mathbf{p}})\Big[(-1)\,e^{-ip\cdot(x-y)} + (-1)\,e^{+ip\cdot(x-y)}\Big], \tag{9.63}$$

where the first bracketed term comes from $a_{\mathbf{p}}e^{-ip\cdot x}$ paired with $-a^\dagger_{\mathbf{q}}e^{iq\cdot y}$ and the second from $a^\dagger_{\mathbf{p}}e^{ip\cdot x}$ paired with $a_{\mathbf{q}}e^{-iq\cdot y}$. At equal time $x^0=y^0=t$ the energy phases drop out, $p\cdot(x-y)\to-\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})$, and the prefactor $(-iE_{\mathbf{p}})(-1)/(2E_{\mathbf{p}}) = i/2$ is independent of $\mathbf{p}$, so

$$[\phi(x),\pi(y)] = \frac{i}{2}\int\frac{d^3p}{(2\pi)^3}\Big(e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}+e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\Big) = i\int\frac{d^3p}{(2\pi)^3}\,e^{i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = i\,\delta^3(\mathbf{x}-\mathbf{y}), \tag{9.64}$$

where the two exponentials give the same integral (relabel $\mathbf{p}\to-\mathbf{p}$ in the second) and Theorem 5.18 identifies the result as the delta function. The factors of $E_{\mathbf{p}}$ cancel exactly because the $2E_{\mathbf{p}}$ in the algebra matches the $2E_{\mathbf{p}}$ in the measure — the relativistic normalization is self-consistent. An identical computation gives $[\phi(x),\phi(y)]=[\pi(x),\pi(y)]=0$ at equal times, since there the surviving prefactor is $E_{\mathbf{p}}-E_{\mathbf{p}}=0$ (for ϕ–ϕ) or carries an odd power of $E_{\mathbf{p}}$ that integrates to zero by parity. The mode construction and the canonical construction are thus two views of the same theory.

9.2.5 Two normalizations, one physics

The factor $2E_{\mathbf{p}}$ in (9.54) is the single most common source of convention confusion in the subject, because it clashes with the normalization a reader brings from nonrelativistic quantum mechanics. The clash is purely one of bookkeeping — no physics depends on it — but every later formula in the book (states, propagators, cross-section flux factors) inherits the choice, so we fix it once and flag it loudly.

Notation Watch

In nonrelativistic QM the oscillator ladder operators obey $\big[\hat a_{\mathbf{p}},\hat a^\dagger_{\mathbf{q}}\big] = \delta^3(\mathbf{p}-\mathbf{q})$ — the "δ-normalization." This book instead uses the *relativistic* (or *covariant*) normalization

$$\big[a_{\mathbf{p}},a^\dagger_{\mathbf{q}}\big] = (2\pi)^3\,2E_{\mathbf{p}}\,\delta^3(\mathbf{p}-\mathbf{q}). \tag{9.65}$$

The two are related by a momentum-dependent rescaling,

$$a_{\mathbf{p}} = \sqrt{(2\pi)^3\,2E_{\mathbf{p}}}\;\hat a_{\mathbf{p}}, \qquad \big[\hat a_{\mathbf{p}},\hat a^\dagger_{\mathbf{q}}\big] = \delta^3(\mathbf{p}-\mathbf{q}), \tag{9.66}$$

which one can verify by substituting into (9.65). Why pay the price of the extra $2E_{\mathbf{p}}$? Because it is exactly the factor that makes the measure (9.34), the mode expansion (9.42), and (in the next section) the one-particle states $|\mathbf{p}\rangle \propto a^\dagger_{\mathbf{p}}|0\rangle$ all transform covariantly under boosts *without* stray Jacobians. With the δ-normalized $\hat{a}_{\mathbf{p}}$ a boost would drag an awkward $\sqrt{E_{\mathbf{p}'}/E_{\mathbf{p}}}$ through every equation. The relativistic normalization buys manifest Lorentz covariance at the cost of a $2E_{\mathbf{p}}$ one must simply remember. Throughout this book the bare symbols $a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}$ *always* mean the relativistically normalized operators of (9.65); the hatted $\hat{a}_{\mathbf{p}}$ appears only when an explicit comparison to QM is being made.

Physical Insight

The reason the free quantum field is tractable at all is encoded in (9.54): different momentum modes commute. The field is not one complicated quantum system but an infinite collection of *independent* simple ones, each an oscillator that the reader already solved as an undergraduate. Interactions — the $\frac{\lambda}{4!}\phi^4$ term of ϕ^4 theory, switched off here — are precisely what couple these oscillators to one another, turning a soluble problem into the perturbative expansion of Chapter 14. Everything clean in this chapter is clean because we set $\lambda = 0$.

Handoff

We have solved the operator Klein–Gordon equation in closed form: the field is the mode sum (9.42), and its Fourier coefficients are creation and annihilation operators obeying the relativistic oscillator algebra (9.54), with the covariant measure (9.34) as the connective tissue. These mode operators $a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}$ are the objects from which every later ϕ^4 amplitude in ϕ^4 theory will be built. The commutator $\left[a_{\mathbf{p}}, a^\dagger_{\mathbf{p}}\right] > 0$ also tells us that $a^\dagger_{\mathbf{p}}$ raises and $a_{\mathbf{p}}$ lowers a not-yet-identified energy: the algebra demands a state of lowest energy annihilated by every $a_{\mathbf{p}}$. Constructing that vacuum, building the Fock space on top of it, and showing that its excitations are relativistic spin-0 particles of mass m is the work of the next section.

9.3 Fock Space, the Vacuum, and the Particle Interpretation

The previous section handed us a family of operators. Solving the operator Klein–Gordon equation by a plane-wave expansion produced, for each spatial momentum

$\boldsymbol{p}$, a lowering operator $a_{\boldsymbol{p}}$ and a raising operator $a^\dagger_{\boldsymbol{p}}$, and the equal-time canonical relations collapsed into the algebra

$$\left[a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = (2\pi)^3\, 2E_{\boldsymbol{p}}\, \delta^3(\boldsymbol{p}-\boldsymbol{q}), \qquad [a_{\boldsymbol{p}}, a_{\boldsymbol{q}}] = \left[a^\dagger_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = 0. \tag{9.67}$$

This is the import we work from; it is established in Eq. (9.54) and we shall not rederive it. What we have so far, however, is only an *algebra*: a set of formal symbols with commutation rules. We have no Hilbert space on which they act, no states, no energies, no particles. Building that Hilbert space, and reading off its physics, is the entire business of this section.

The logic is the one every reader already knows from the single harmonic oscillator. There, $[a, a^\dagger] = 1$ together with a Hamiltonian $H = \omega(a^\dagger a + \frac{1}{2})$ generates the whole spectrum from a single normalizable state annihilated by a. The free scalar field is a continuum of decoupled oscillators, one per momentum mode, so the construction is the same applied infinitely many times, with the bookkeeping made Lorentz covariant. We will (i) postulate a unique vacuum killed by every $a_{\boldsymbol{p}}$ and argue why it is forced on us; (ii) build one-particle states with a relativistically invariant normalization and prove the corresponding completeness relation; (iii) stack creation operators to build the multiparticle sectors, reading off Bose statistics directly from Eq. (9.67); (iv) substitute the mode expansion into the energy and momentum operators inherited from Chapter 7 and diagonalize them, term by term, keeping every cross-term; and (v) prove that a single excitation of momentum $\boldsymbol{p}$ is an eigenstate of the four-momentum with eigenvalue $p^\mu = (E_{\boldsymbol{p}}, \boldsymbol{p})$ on the mass shell $p^2 = m^2$, carrying exactly the Poincaré quantum numbers of the mass-m, spin-0 irreducible representation classified in Chapter 3. The single blemish that survives — a divergent c-number E_0 in the Hamiltonian — is precisely the loose thread the next section must tie off.

Throughout we use the relativistic single-particle normalization fixed in Section 9.2, and it will be convenient to name the Lorentz-invariant measure once and for all:

$$\int \widetilde{dp} \;\equiv\; \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\boldsymbol{p}}}, \qquad E_{\boldsymbol{p}} \equiv \sqrt{\boldsymbol{p}^2 + m^2} > 0. \tag{9.68}$$

That Eq. (9.68) is genuinely Lorentz invariant — the property that makes it the natural measure for relativistic states — is shown in Theorem 9.14 below.

9.3.1 The vacuum

A Hilbert space built from lowering and raising operators must have a floor: a state that cannot be lowered further. If it did not, repeated application of the $a_{\boldsymbol{p}}$ would manufacture states of ever-decreasing energy without bound, and no quantum theory with a stable ground state could survive. We therefore *define* the floor and then check that the definition is consistent and, for the free theory, essentially forced.

Definition 9.11 (The vacuum)**.** The *vacuum* $|0\rangle$ is the state annihilated by every lowering operator,

$$a_{\boldsymbol{p}}\,|0\rangle = 0 \qquad \text{for all } \boldsymbol{p}, \tag{9.69}$$

normalized to unity,

$$\langle 0|0\rangle = 1. \tag{9.70}$$

Two remarks fix its status. First, $|0\rangle$ is not "empty space" in any naive sense; it is the state with no quanta, but as we shall see in Section 9.3.4 it carries a nonzero (indeed divergent) zero-point energy and, as a quantum state, has nontrivial fluctuations of the field. The physical insight box below states the picture; the mathematics that follows does not depend on it. Second, Eq. (9.70) is a genuine normalization, in contrast to the continuum-normalized momentum states we build next: $|0\rangle$ is an honest unit vector of the Hilbert space.

Physical Insight

The free vacuum is the quietest the field can be, not the absence of the field. Every mode $\boldsymbol{p}$ is an oscillator pinned in its ground state, and a quantum oscillator in its ground state still jitters: $\langle 0|\phi(\boldsymbol{x})^2|0\rangle \neq 0$. "Particles" are the discrete excitations of these oscillators above the floor. Nothing is "created from nothing"; a creation operator promotes one mode from its ground state to its first excited state, and we agree to call that lump of localized energy and momentum a particle.

Uniqueness for the free theory. Theorem 9.11 asserts existence of *a* state with the stated property; we must argue it is the *only* one (up to phase), since otherwise "the vacuum" would be ambiguous and the particle interpretation would not be well defined. For the free field the argument is clean. Suppose $|\Omega\rangle$ also satisfies $a_{\boldsymbol{p}}\,|\Omega\rangle = 0$ for all $\boldsymbol{p}$. The Fock construction of Section 9.3.3 shows that the states obtained by applying strings of creation operators to $|0\rangle$ span the entire Hilbert space $\mathcal{F}$; in particular any vector, $|\Omega\rangle$ included, can be written as a (possibly continuous) superposition

$$|\Omega\rangle = c_0\,|0\rangle + \int \widetilde{dp}\, c_1(\boldsymbol{p})\, a^\dagger_{\boldsymbol{p}}\,|0\rangle + \tfrac{1}{2}\int \widetilde{dp}\,\widetilde{dq}\, c_2(\boldsymbol{p},\boldsymbol{q})\, a^\dagger_{\boldsymbol{p}} a^\dagger_{\boldsymbol{q}}\,|0\rangle + \cdots . \tag{9.71}$$

Apply $a_{\boldsymbol{k}}$ to both sides. On the left it gives zero by hypothesis. On the right, $a_{\boldsymbol{k}}$ commuted through the creation operators using Eq. (9.67) lowers the n-creation term to an $(n-1)$-creation term whose coefficient is built from c_n; for instance the one-particle term yields $\int \widetilde{dp}\, c_1(\boldsymbol{p})(2\pi)^3 2E_{\boldsymbol{p}}\delta^3(\boldsymbol{k}-\boldsymbol{p})\,|0\rangle = c_1(\boldsymbol{k})\,|0\rangle$ (the $2E_{\boldsymbol{p}}$ from the commutator cancelling the $1/2E_{\boldsymbol{p}}$ of the invariant measure), and the two-particle term yields $\int \widetilde{dq}\, c_2(\boldsymbol{k},\boldsymbol{q})\, a^\dagger_{\boldsymbol{q}}\,|0\rangle$, and so on. Setting the result to zero

sector by sector forces $c_1 \equiv 0$, then $c_2 \equiv 0$, and inductively every coefficient with $n \geq 1$ vanishes. Hence $|\Omega\rangle = c_0\,|0\rangle$: the only states killed by all annihilation operators are multiples of $|0\rangle$. Imposing Eq. (9.70) fixes $|c_0| = 1$, so $|0\rangle$ is unique up to an irrelevant overall phase.

Caution

This uniqueness argument is honest only because the theory is free: the operators $a_{\boldsymbol{p}}$ are exactly the ladder operators of the full Hamiltonian, and "no quanta" is an exact, dynamically stable notion. In an *interacting* theory the would-be free vacuum is not an eigenstate of the true Hamiltonian, the interaction-picture Fock space is mathematically delicate (Haag's theorem), and the physical vacuum must be characterized differently — as the unique Poincaré-invariant ground state, with its uniqueness a nontrivial theorem (Reeh–Schlieder territory). The construction in this chapter is the rigorous free bedrock on which the perturbative expansion of Chapters 14 and 15 is then erected.

9.3.2 One-particle states and covariant normalization

The first excited sector is obtained by acting once with a creation operator. The inner product of the bare object $a^\dagger_{\boldsymbol{p}}\,|0\rangle$ is, directly from the algebra,

$$\langle 0|\,a_{\boldsymbol{p}}\,a^\dagger_{\boldsymbol{q}}\,|0\rangle = \langle 0|\Big(\Big[a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\Big] + a^\dagger_{\boldsymbol{q}} a_{\boldsymbol{p}}\Big)|0\rangle = (2\pi)^3 2E_{\boldsymbol{p}}\,\delta^3(\boldsymbol{p}-\boldsymbol{q})\;\langle 0|0\rangle = (2\pi)^3 2E_{\boldsymbol{p}}\,\delta^3(\boldsymbol{p}-\boldsymbol{q}), \tag{9.72}$$

and the combination $2E_{\boldsymbol{p}}\,(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q})$ is already a Lorentz scalar: under a boost the bare three-dimensional delta function $\delta^3(\boldsymbol{p}-\boldsymbol{q})$ picks up a Jacobian, and the factor $2E_{\boldsymbol{p}}$ supplied by the relativistic normalization Eq. (9.67) is exactly what cancels it (see Theorem 9.14). This is the whole reason the measure was chosen to carry the $2E_{\boldsymbol{p}}$ into the commutator in the first place: it makes the natural one-particle norm frame-independent, so that statements like "$\langle p|q\rangle$" mean the same thing in every frame. No extra rescaling is needed, and we take the bare state itself as the covariantly normalized one.

Definition 9.12 (Covariantly normalized one-particle state)**.** For each on-shell momentum $p^\mu = (E_{\boldsymbol{p}}, \boldsymbol{p})$ define

$$|p\rangle \;\equiv\; a^\dagger_{\boldsymbol{p}}\,|0\rangle\,. \tag{9.73}$$

Proposition 9.13 (Covariant normalization)**.** *The states Eq. (9.73) satisfy*

$$\langle p|q\rangle = 2E_{\boldsymbol{p}}\,(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q}), \tag{9.74}$$

and the right-hand side is Lorentz invariant.

Proof. Directly from Eq. (9.73) and Eq. (9.72),

$$\langle p|q\rangle = \langle 0|\, a_{\boldsymbol{p}} a^\dagger_{\boldsymbol{q}}\, |0\rangle = 2E_{\boldsymbol{p}}\,(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q}). \tag{9.75}$$

Invariance of the combination $2E_{\boldsymbol{p}}\,\delta^3(\boldsymbol{p}-\boldsymbol{q})$ is shown in Theorem 9.14. □

Worked Example 9.14 (The Lorentz-invariant measure and normalization)**.** We verify that both $\int\widetilde{dp}$ in Eq. (9.68) and the normalization Eq. (9.74) are Lorentz invariant. Start from the manifestly invariant four-dimensional integral over the positive-energy mass shell,

$$\int \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2-m^2)\,\theta(p^0), \qquad p^2=(p^0)^2-\boldsymbol{p}^2. \tag{9.76}$$

Both factors are invariant: $\delta(p^2-m^2)$ depends only on the scalar p^2, and $\theta(p^0)$ is invariant because an orthochronous Lorentz transformation cannot flip the sign of the energy of a timelike or lightlike momentum. Do the p^0 integral. Writing $p^2-m^2=(p^0)^2-(\boldsymbol{p}^2+m^2)=(p^0-E_{\boldsymbol{p}})(p^0+E_{\boldsymbol{p}})$ and using $\delta(f(p^0))=\sum_i \delta(p^0-p^0_i)/|f'(p^0_i)|$, only the root $p^0=+E_{\boldsymbol{p}}$ survives the $\theta(p^0)$, with $|f'(E_{\boldsymbol{p}})|=2E_{\boldsymbol{p}}$. Hence

$$\int \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2-m^2)\,\theta(p^0) = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\boldsymbol{p}}} = \int\widetilde{dp}. \tag{9.77}$$

The left side is invariant, so the right side is too: Eq. (9.68) is a Lorentz scalar measure. The same computation read backwards shows that $2E_{\boldsymbol{p}}\,\delta^3(\boldsymbol{p}-\boldsymbol{q})$ is invariant. Indeed, demanding $\int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\boldsymbol{p}}}\left[2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q})\right]=1$ identifies $2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q})$ as the delta function *dual to the invariant measure*, and a quantity dual to an invariant measure is itself invariant. Concretely, under a boost $\boldsymbol{p}\to\boldsymbol{p}'$ with $E_{\boldsymbol{p}'}=\gamma(E_{\boldsymbol{p}}-\beta p^3)$ one has $d^3p'/E_{\boldsymbol{p}'}=d^3p/E_{\boldsymbol{p}}$ (the Jacobian of the boost on the mass shell equals $E_{\boldsymbol{p}'}/E_{\boldsymbol{p}}$), so $\delta^3(\boldsymbol{p}-\boldsymbol{q})$ alone scales by $E_{\boldsymbol{p}}/E_{\boldsymbol{p}'}$ — precisely the non-invariance Eq. (9.72) suffered — while the factor $2E_{\boldsymbol{p}}$ in Eq. (9.74) cancels it. This is the whole reason the relativistic normalization Eq. (9.67) carries the $2E_{\boldsymbol{p}}$ in the commutator.

With the normalization Eq. (9.74) settled, the completeness relation on the one-particle sector takes a form matched to the invariant measure.

Proposition 9.15 (Resolution of the identity on the one-particle sector)**.** *Let $\mathcal{H}_1$ be the closed span of the states $|p\rangle$. The operator*

$$\mathbf{1}_1 = \int\widetilde{dp}\;\,|p\rangle\langle p| = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\boldsymbol{p}}}\,|p\rangle\langle p| \tag{9.78}$$

is the identity on $\mathcal{H}_1$.

Proof. Act on an arbitrary one-particle state $|q\rangle$ and use Eq. (9.74):

$$\mathbf{1}_1\,|q\rangle = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\boldsymbol{p}}}\;|p\rangle\;\left[2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q})\right] = \int d^3p\;\delta^3(\boldsymbol{p}-\boldsymbol{q})\;|p\rangle = |q\rangle\,. \tag{9.79}$$

Since the $|q\rangle$ span $\mathcal{H}_1$, $\mathbf{1}_1$ is the identity there. The cancellation of the two factors of $2E_{\boldsymbol{p}}$ is exactly why the invariant measure pairs with the covariant normalization. □

9.3.3 Multiparticle states, Bose symmetry, and Fock space

Higher sectors are built by acting with more creation operators. Define the (covariantly normalized) n-particle state

$$|p_1,\dots,p_n\rangle \;\equiv\; a^\dagger_{\boldsymbol{p}_1}\cdots a^\dagger_{\boldsymbol{p}_n}\,|0\rangle\,. \tag{9.80}$$

The single most important structural fact follows for free from the last line of Eq. (9.67): because $\left[a^\dagger_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = 0$, the creation operators commute, and the state Eq. (9.80) is *totally symmetric* under exchange of any two of its labels,

$$|\dots,p_i,\dots,p_j,\dots\rangle = |\dots,p_j,\dots,p_i,\dots\rangle\,. \tag{9.81}$$

This is Bose statistics, and it is not an extra postulate: it is forced by the commutator (as opposed to anticommutator) algebra that the canonical *commutation* relations of the scalar field handed us. (The deeper question of *why* a spin-0 field must be quantized with commutators rather than anticommutators — the spin–statistics connection — is settled by microcausality in a later section of this chapter.) Physically, Eq. (9.81) says the particles are indistinguishable and that any number of them may occupy the same mode.

Worked Example 9.16 (Two-particle inner product and manifest symmetry)**.** We compute $\langle p_1\,p_2|q_1\,q_2\rangle$ directly from the algebra, exhibiting the symmetric ("permanent") structure. Using Eq. (9.80),

$$\langle p_1\,p_2|q_1\,q_2\rangle = \langle 0|\,a_{\boldsymbol{p}_2}a_{\boldsymbol{p}_1}\,a^\dagger_{\boldsymbol{q}_1}a^\dagger_{\boldsymbol{q}_2}\,|0\rangle\,. \tag{9.82}$$

Evaluate the vacuum expectation value by commuting annihilation operators to the right until they hit $|0\rangle$. First move $a_{\boldsymbol{p}_1}$ past the two creation operators:

$$\begin{aligned} a_{\boldsymbol{p}_1}a^\dagger_{\boldsymbol{q}_1}a^\dagger_{\boldsymbol{q}_2} &= \left(\left[a_{\boldsymbol{p}_1}, a^\dagger_{\boldsymbol{q}_1}\right] + a^\dagger_{\boldsymbol{q}_1}a_{\boldsymbol{p}_1}\right)a^\dagger_{\boldsymbol{q}_2} \\ &= (2\pi)^3 2E_{\boldsymbol{p}_1}\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_1)\,a^\dagger_{\boldsymbol{q}_2} + a^\dagger_{\boldsymbol{q}_1}\left(\left[a_{\boldsymbol{p}_1}, a^\dagger_{\boldsymbol{q}_2}\right] + a^\dagger_{\boldsymbol{q}_2}a_{\boldsymbol{p}_1}\right) \\ &= (2\pi)^3 2E_{\boldsymbol{p}_1}\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_1)\,a^\dagger_{\boldsymbol{q}_2} + (2\pi)^3 2E_{\boldsymbol{p}_1}\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_2)\,a^\dagger_{\boldsymbol{q}_1} + a^\dagger_{\boldsymbol{q}_1}a^\dagger_{\boldsymbol{q}_2}a_{\boldsymbol{p}_1}. \end{aligned} \tag{9.83}$$

The last term annihilates $|0\rangle$ on the right of Eq. (9.82). Now apply $a_{\boldsymbol{p}_2}$ to what remains and use $a_{\boldsymbol{p}_2}a^\dagger_{\boldsymbol{q}}\,|0\rangle = (2\pi)^3 2E_{\boldsymbol{p}_2}\delta^3(\boldsymbol{p}_2-\boldsymbol{q})\,|0\rangle$:

$$\langle 0|\, a_{\boldsymbol{p}_2}a_{\boldsymbol{p}_1}a^\dagger_{\boldsymbol{q}_1}a^\dagger_{\boldsymbol{q}_2}\,|0\rangle = (2\pi)^6\, 2E_{\boldsymbol{p}_1}2E_{\boldsymbol{p}_2}\big[\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_1)\delta^3(\boldsymbol{p}_2-\boldsymbol{q}_2)+\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_2)\delta^3(\boldsymbol{p}_2-\boldsymbol{q}_1)\big]. \tag{9.84}$$

The covariant $2E$ factors carried by the relativistic commutators are already in place — with each delta one may write $2E_{\boldsymbol{p}_1} = 2E_{\boldsymbol{q}_1}$ etc. — giving the manifestly exchange-symmetric result

$$\langle p_1\, p_2|q_1\, q_2\rangle = \big[2E_{\boldsymbol{p}_1}(2\pi)^3\delta^3(\boldsymbol{p}_1-\boldsymbol{q}_1)\big]\big[2E_{\boldsymbol{p}_2}(2\pi)^3\delta^3(\boldsymbol{p}_2-\boldsymbol{q}_2)\big]+(\boldsymbol{q}_1\leftrightarrow\boldsymbol{q}_2). \tag{9.85}$$

The two terms are the two ways of pairing incoming with outgoing labels; the symmetry under $\boldsymbol{q}_1\leftrightarrow\boldsymbol{q}_2$ (and equally $\boldsymbol{p}_1\leftrightarrow\boldsymbol{p}_2$) is the inner-product expression of Eq. (9.81). This pairing structure is the seed of Wick's theorem, derived from the generating functional in Chapter 10.

Occupation-number basis and the Fock space. For a system in a finite box (momenta discretized, $a^\dagger_{\boldsymbol{p}}\to a^\dagger_k$ with $\left[a_k, a^\dagger_{k'}\right] = \delta_{kk'}$), the states

$$|\{n_k\}\rangle = \prod_k \frac{(a^\dagger_k)^{n_k}}{\sqrt{n_k!}}\,|0\rangle\,, \qquad n_k\in\{0,1,2,\dots\}, \tag{9.86}$$

are orthonormal eigenstates labelled by the *occupation numbers* n_k — exactly as for a collection of independent harmonic oscillators, one per mode. Each mode is an oscillator; the field's Hilbert space is the (restricted) tensor product of these oscillator Hilbert spaces. Taking the continuum limit, the abstract statement is:

Definition 9.17 (Fock space)**.** The *Fock space* of the free real scalar is the Hilbert-space direct sum

$$\mathcal{F} = \bigoplus_{n=0}^{\infty}\mathcal{H}_n, \qquad \mathcal{H}_0 = \mathbb{C}\,|0\rangle\,, \quad \mathcal{H}_n = \mathrm{Sym}^n(\mathcal{H}_1), \tag{9.87}$$

where $\mathcal{H}_n$ is the symmetric (n-fold) tensor power of the one-particle space $\mathcal{H}_1$, spanned by the states Eq. (9.80). The symmetrization Sym^n implements the Bose symmetry Eq. (9.81) at the level of the space itself.

The completeness relation on all of $\mathcal{F}$ is the sum of the sector resolutions,

$$\mathbf{1}_{\mathcal{F}} = |0\rangle\langle 0| + \int\widetilde{dp}\;|p\rangle\langle p| + \frac{1}{2!}\int\widetilde{dp}\,\widetilde{dq}\;|p\,q\rangle\langle p\,q| + \cdots\,, \tag{9.88}$$

where the $1/n!$ in the n-particle term corrects for the overcounting introduced by integrating independently over labels that Eq. (9.81) identifies. That the strings of creation operators acting on $|0\rangle$ exhaust $\mathcal{F}$ is what we used in proving vacuum uniqueness in Section 9.3.1; Eq. (9.88) is the precise statement that they do.

9.3.4 Diagonalizing the energy and momentum

We now make contact with dynamics. The energy and momentum operators are not new postulates: they are the conserved charges of spacetime translations, the components of the Noether four-momentum

$$P^\mu = \int d^3x\, T^{0\mu}, \qquad P^0 = H = \int d^3x \left[\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2\right], \qquad \boldsymbol{P} = -\int d^3x\, \pi\, \nabla\phi, \tag{9.89}$$

established as classical Noether output in Eq. (7.97) and promoted to operators here. Our task is to substitute the mode expansion, recalled from Eq. (9.42), and reduce these to the ladder operators. Evaluated at $t = 0$ (the operators are time independent, so any time will do) the field and its conjugate momentum read

$$\phi(\boldsymbol{x}) = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\boldsymbol{p}}}\,(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})\,e^{i\boldsymbol{p}\cdot\boldsymbol{x}}, \tag{9.90}$$

$$\pi(\boldsymbol{x}) = \int \frac{d^3p}{(2\pi)^3}\left(-\frac{i}{2}\right)(a_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}})\,e^{i\boldsymbol{p}\cdot\boldsymbol{x}}, \tag{9.91}$$

where the single-exponential form is obtained by relabelling $\boldsymbol{p} \to -\boldsymbol{p}$ in the creation pieces of Eq. (9.42), whose measure $d^3p/[(2\pi)^3\,2E_{\boldsymbol{p}}]$ is carried over intact (for π, the relabelling acts on Eq. (9.43) and the $-iE_{\boldsymbol{p}}$ there combines with the $1/2E_{\boldsymbol{p}}$ of the measure to leave the constant $-i/2$).

The Hamiltonian. We treat the three terms of H in turn, using $\int d^3x\, e^{i(\boldsymbol{p}+\boldsymbol{q})\cdot\boldsymbol{x}} = (2\pi)^3\delta^3(\boldsymbol{p}+\boldsymbol{q})$, which sets $\boldsymbol{q} = -\boldsymbol{p}$ (hence $E_{\boldsymbol{q}} = E_{\boldsymbol{p}}$) in every product. For the kinetic term,

$$\begin{aligned}\int d^3x\,\tfrac{1}{2}\pi^2 &= \tfrac{1}{2}\int \frac{d^3p\,d^3q}{(2\pi)^6}\left(-\frac{i}{2}\right)^2 (a_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}})(a_{\boldsymbol{q}} - a^\dagger_{-\boldsymbol{q}})\,(2\pi)^3\delta^3(\boldsymbol{p}+\boldsymbol{q}) \\ &= \int \frac{d^3p}{(2\pi)^3}\left(-\frac{1}{8}\right)(a_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}).\end{aligned} \tag{9.92}$$

For the gradient term each ∇ brings down a factor $i\boldsymbol{p}$ and $i\boldsymbol{q}$, and after $\boldsymbol{q} = -\boldsymbol{p}$ the product is $(i\boldsymbol{p})\cdot(-i\boldsymbol{p}) = \boldsymbol{p}^2$; for the mass term there is no derivative. Together,

$$\int d^3x \left[\tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2\right] = \int \frac{d^3p}{(2\pi)^3}\,\frac{\boldsymbol{p}^2 + m^2}{8E_{\boldsymbol{p}}^2}\,(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} + a^\dagger_{\boldsymbol{p}}). \tag{9.93}$$

Now $\boldsymbol{p}^2 + m^2 = E_{\boldsymbol{p}}^2$, so the coefficient in Eq. (9.93) is $\frac{1}{8}$. Adding Eq. (9.92) and Eq. (9.93), the two products combine as

$$H = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{8}\Big[-(a_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}) + (a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} + a^\dagger_{\boldsymbol{p}})\Big]. \tag{9.94}$$

Expanding both products and keeping every cross-term,

$$(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} + a^\dagger_{\boldsymbol{p}}) = a_{\boldsymbol{p}}a_{-\boldsymbol{p}} + a_{\boldsymbol{p}}a^\dagger_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}a_{-\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}a^\dagger_{\boldsymbol{p}}, \tag{9.95}$$

$$(a_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}})(a_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}) = a_{\boldsymbol{p}}a_{-\boldsymbol{p}} - a_{\boldsymbol{p}}a^\dagger_{\boldsymbol{p}} - a^\dagger_{-\boldsymbol{p}}a_{-\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}a^\dagger_{\boldsymbol{p}}, \tag{9.96}$$

so the cross-terms $a_{\boldsymbol{p}}a_{-\boldsymbol{p}}$ and $a^\dagger_{-\boldsymbol{p}}a^\dagger_{\boldsymbol{p}}$ cancel in the difference, leaving

$$H = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{8}\,(2\,a_{\boldsymbol{p}}a^\dagger_{\boldsymbol{p}} + 2\,a^\dagger_{-\boldsymbol{p}}a_{-\boldsymbol{p}}) = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{4}\,(a_{\boldsymbol{p}}a^\dagger_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}a_{-\boldsymbol{p}}). \tag{9.97}$$

In the second term substitute $\boldsymbol{p} \to -\boldsymbol{p}$ (legitimate since $E_{\boldsymbol{p}}$ is even and the measure invariant): it becomes $\int \frac{d^3p}{(2\pi)^3}\frac{1}{4}a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}}$. In the first term use Eq. (9.67) to normal-order, $a_{\boldsymbol{p}}a^\dagger_{\boldsymbol{p}} = a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}} + (2\pi)^3 2E_{\boldsymbol{p}}\delta^3(0)$. Collecting,

$$\boxed{H = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2}\,a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}} \;+\; E_0 \;=\; \int \widetilde{dp}\;E_{\boldsymbol{p}}\,a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}} \;+\; E_0}, \qquad E_0 = \frac{1}{2}\,(2\pi)^3\delta^3(0)\int \frac{d^3p}{(2\pi)^3}\,E_{\boldsymbol{p}} = \frac{V}{2} \tag{9.98}$$

We have used the standard identification $(2\pi)^3\delta^3(0) = \int d^3x = V$, the (infinite) volume of space. The structure is exactly that of the single oscillator $\omega(a^\dagger a + \frac{1}{2})$, summed over modes: the first term counts the energy $E_{\boldsymbol{p}}$ of each quantum, the second is the sum of the zero-point energies $\frac{1}{2}E_{\boldsymbol{p}}$ of all the oscillators. That E_0 is doubly divergent — an infinite spatial volume times an ultraviolet-divergent momentum integral — is the blemish announced in the introduction; Section 9.3.1 guaranteed $|0\rangle$ is unique, but said nothing about its energy being finite. Taming E_0 is the entire subject of the next section.

The momentum. The same computation for $\boldsymbol{P}$ in Eq. (9.89) is more economical because there is no mass term and, as we will see, the c-number piece is killed by parity rather than merely subtracted. Inserting Eq. (9.91) and $\nabla\phi = \int \frac{d^3p}{(2\pi)^3}\frac{i\boldsymbol{p}}{2E_{\boldsymbol{p}}}(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})e^{i\boldsymbol{p}\cdot\boldsymbol{x}}$,

$$\begin{aligned}\boldsymbol{P} &= -\int d^3x\,\pi\,\nabla\phi = -\int \frac{d^3p\,d^3q}{(2\pi)^6}\Big(-\frac{i}{2}\Big)\frac{i\boldsymbol{p}}{2E_{\boldsymbol{p}}}(a_{\boldsymbol{q}} - a^\dagger_{-\boldsymbol{q}})(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}})(2\pi)^3\delta^3(\boldsymbol{p}+\boldsymbol{q}) \\ &= -\int \frac{d^3p}{(2\pi)^3}\,\frac{\boldsymbol{p}}{4E_{\boldsymbol{p}}}(a_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}})(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}),\end{aligned} \tag{9.99}$$

where again $\boldsymbol{q} = -\boldsymbol{p}$, $E_{\boldsymbol{q}} = E_{\boldsymbol{p}}$, and $(-i)(i) = 1$. Expanding,

$$(a_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}})(a_{\boldsymbol{p}} + a^\dagger_{-\boldsymbol{p}}) = a_{-\boldsymbol{p}}a_{\boldsymbol{p}} + a_{-\boldsymbol{p}}a^\dagger_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}a^\dagger_{-\boldsymbol{p}}. \tag{9.100}$$

The terms $a_{-\boldsymbol{p}}a_{\boldsymbol{p}}$ and $a^\dagger_{\boldsymbol{p}}a^\dagger_{-\boldsymbol{p}}$ are even under $\boldsymbol{p} \to -\boldsymbol{p}$ (the operators commute, so each is invariant under the relabelling), while they multiply the odd factor $\boldsymbol{p}$; their integrals therefore vanish by parity. We are left with

$$\boldsymbol{P} = -\int \frac{d^3p}{(2\pi)^3}\,\frac{\boldsymbol{p}}{4E_{\boldsymbol{p}}}(a_{-\boldsymbol{p}}a^\dagger_{-\boldsymbol{p}} - a^\dagger_{\boldsymbol{p}}a_{\boldsymbol{p}}). \tag{9.101}$$

In the first surviving term substitute $\boldsymbol{p} \to -\boldsymbol{p}$, which flips the sign of the prefactor $\boldsymbol{p}$ and turns $a_{-\boldsymbol{p}} a^\dagger_{-\boldsymbol{p}}$ into $a_{\boldsymbol{p}} a^\dagger_{\boldsymbol{p}}$:

$$\boldsymbol{P} = \int \frac{d^3p}{(2\pi)^3} \frac{\boldsymbol{p}}{4E_{\boldsymbol{p}}} (a_{\boldsymbol{p}} a^\dagger_{\boldsymbol{p}} + a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}}) = \int \frac{d^3p}{(2\pi)^3} \frac{\boldsymbol{p}}{2E_{\boldsymbol{p}}} a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}} + \frac{(2\pi)^3 \delta^3(0)}{2} \int \frac{d^3p}{(2\pi)^3} \boldsymbol{p}, \tag{9.102}$$

having used $a_{\boldsymbol{p}} a^\dagger_{\boldsymbol{p}} = a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}} + (2\pi)^3 2E_{\boldsymbol{p}} \delta^3(0)$. The final integral $\int d^3p\, \boldsymbol{p}$ vanishes identically: its integrand is odd. Thus the c-number contribution to the momentum cancels on its own — there is no "zero-point momentum" — and

$$\boxed{\boldsymbol{P} = \int \widetilde{dp}\, \boldsymbol{p}\, a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}} = \int \frac{d^3p}{(2\pi)^3} \frac{\boldsymbol{p}}{2E_{\boldsymbol{p}}} a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}}}. \tag{9.103}$$

The asymmetry between H and $\boldsymbol{P}$ is instructive: $E_{\boldsymbol{p}} > 0$ is even in $\boldsymbol{p}$, so its zero-point sum survives, whereas $\boldsymbol{p}$ is odd, so opposite modes cancel. The divergence problem afflicts only the energy.

9.3.5 The spectrum and the particle interpretation

We can now read the physics off the normal-mode forms Eq. (9.98) and Eq. (9.103). The key technical input is the commutator of these operators with a creation operator, which we compute once and reuse. From Eq. (9.67),

$$\left[H, a^\dagger_{\boldsymbol{q}}\right] = \int \widetilde{dp}\, E_{\boldsymbol{p}} \left[a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = \int \widetilde{dp}\, E_{\boldsymbol{p}}\, a^\dagger_{\boldsymbol{p}} \left[a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = E_{\boldsymbol{q}}\, a^\dagger_{\boldsymbol{q}}, \tag{9.104}$$

where we used $\left[a^\dagger_{\boldsymbol{p}} a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right] = a^\dagger_{\boldsymbol{p}} \left[a_{\boldsymbol{p}}, a^\dagger_{\boldsymbol{q}}\right]$ (the c-number E_0 commutes with everything). Identically,

$$\left[\boldsymbol{P}, a^\dagger_{\boldsymbol{q}}\right] = \boldsymbol{q}\, a^\dagger_{\boldsymbol{q}}. \tag{9.105}$$

A creation operator raises the energy of any state by $E_{\boldsymbol{q}}$ and its momentum by $\boldsymbol{q}$. Acting on the vacuum, with $H|0\rangle = E_0|0\rangle$ and $\boldsymbol{P}|0\rangle = 0$ (the latter because $a_{\boldsymbol{p}}|0\rangle = 0$),

$$H|p\rangle = H a^\dagger_{\boldsymbol{p}}|0\rangle = \left(\left[H, a^\dagger_{\boldsymbol{p}}\right] + a^\dagger_{\boldsymbol{p}} H\right)|0\rangle = (E_{\boldsymbol{p}} + E_0)|p\rangle\,, \tag{9.106}$$

$$\boldsymbol{P}|p\rangle = \left(\left[\boldsymbol{P}, a^\dagger_{\boldsymbol{p}}\right] + a^\dagger_{\boldsymbol{p}} \boldsymbol{P}\right)|0\rangle = \boldsymbol{p}\,|p\rangle\,. \tag{9.107}$$

Measuring energies relative to the vacuum — equivalently, after the normal-ordering subtraction of E_0 carried out in the next section — the one-particle state is a simultaneous eigenstate of energy and momentum,

$$H|p\rangle = E_{\boldsymbol{p}}|p\rangle\,, \qquad \boldsymbol{P}|p\rangle = \boldsymbol{p}\,|p\rangle\,, \qquad E_{\boldsymbol{p}} = \sqrt{\boldsymbol{p}^2 + m^2}. \tag{9.108}$$

The eigenvalues obey the relativistic dispersion relation $E^2_{\boldsymbol{p}} - \boldsymbol{p}^2 = m^2$, i.e. $p^\mu p_\mu = m^2$: the excitation propagates on the mass shell of a particle of mass m.

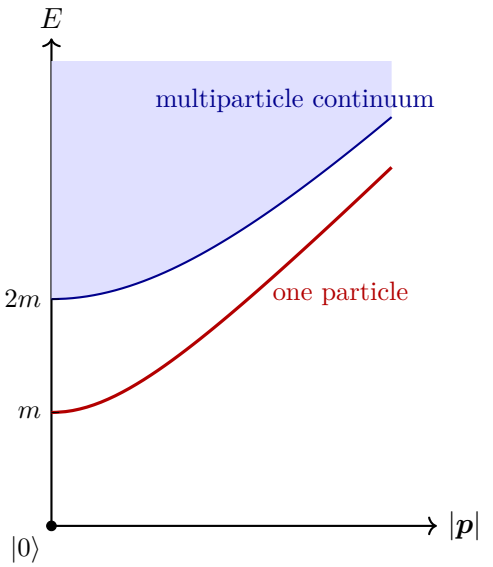


Figure 9.3: Spectrum of the free real scalar field. The vacuum $|0\rangle$ sits at the origin (energies measured relative to it). One-particle states fill the isolated hyperboloid $E = \sqrt{\boldsymbol{p}^2 + m^2}$, gapped from the vacuum by the rest mass m. Two- and higher-particle states fill the shaded continuum, whose lower boundary at fixed total momentum is $E = \sqrt{\boldsymbol{p}^2 + (2m)^2}$, attained when the two quanta share the momentum equally; at $\boldsymbol{p} = 0$ this is the threshold $2m$. The isolation of the one-particle shell from the continuum is what makes "a particle" a sharp notion.

This is the punchline of canonical quantization. We did not put particles into the theory; we quantized a field and discovered that its elementary excitations are discrete, indivisible lumps carrying momentum $\boldsymbol{p}$, energy $\sqrt{\boldsymbol{p}^2 + m^2}$, and rest mass m. The parameter m in the Lagrangian Chapter 6, hitherto just a coefficient, is now physically interpreted as the mass of the particle the field describes.

The same reasoning iterates. An n-particle state Eq. (9.80) satisfies

$$H\,|p_1,\dots,p_n\rangle = \Big(\sum_{i=1}^{n} E_{\boldsymbol{p}_i}\Big)\,|p_1,\dots,p_n\rangle\,, \qquad \boldsymbol{P}\,|p_1,\dots,p_n\rangle = \Big(\sum_{i=1}^{n} \boldsymbol{p}_i\Big)\,|p_1,\dots,p_n\rangle\,, \tag{9.109}$$

again up to the single common shift E_0 in the energy, obtained by commuting H and $\boldsymbol{P}$ past each of the n creation operators in turn using Eq. (9.104) and Eq. (9.105). The energies simply add, as they must for free, non-interacting quanta. Figure 9.3 shows the resulting spectrum: an isolated one-particle hyperboloid $E = \sqrt{\boldsymbol{p}^2 + m^2}$ separated from the vacuum by the mass gap m, and above it the multiparticle continuum.

The number operator and conservation of particle number. The counting structure is made precise by the *number operator*

$$N = \int \widetilde{dp}\, a_{\boldsymbol{p}}^\dagger a_{\boldsymbol{p}}. \tag{9.110}$$

Its commutator with a creation operator is, by the identical manipulation as Eq. (9.104) but with $E_{\boldsymbol{p}} \to 1$,

$$\left[N, a_{\boldsymbol{q}}^\dagger\right] = a_{\boldsymbol{q}}^\dagger, \tag{9.111}$$

so N counts creation operators: applied to the n-particle state Eq. (9.80) it returns $N\,|p_1,\dots,p_n\rangle = n\,|p_1,\dots,p_n\rangle$, and $N\,|0\rangle = 0$. The Fock decomposition Eq. (9.87) is precisely the eigenspace decomposition of N, with $\mathcal{H}_n$ the eigenspace of eigenvalue n. Comparing Eq. (9.98) and Eq. (9.110), the Hamiltonian is built entirely from the mode number densities $a_{\boldsymbol{p}}^\dagger a_{\boldsymbol{p}}$, which all commute with N; hence

$$[H, N] = 0. \tag{9.112}$$

In the free theory the Hamiltonian *conserves particle number*: a state of n quanta evolves into a state of n quanta, never creating or destroying any. This is exactly what one expects of decoupled oscillators, and it is exactly what interactions will spoil — the ϕ^4 term we switch on in Chapter 14 fails to commute with N, which is what allows the scattering and particle-number-changing processes that make the interacting theory interesting.

9.3.6 The Poincaré content: a relativistic spin-0 particle

We have shown the excitations carry energy $E_{\boldsymbol{p}}$ and momentum $\boldsymbol{p}$ on the mass shell. To certify that they are *particles* in the precise sense of relativistic quantum mechanics, we must show that the one-particle states furnish an irreducible representation of the Poincaré group, and identify *which* one. Wigner's classification (Theorem 3.44) tells us that positive-energy irreducible representations of the Poincaré group are labelled by a mass $m \geq 0$ and, for $m > 0$, by a spin $s \in \{0, \tfrac{1}{2}, 1, \dots\}$ — the latter being a representation of the little group, the subgroup $SO(3)$ that fixes a chosen rest-frame momentum (Theorem 3.43). We now verify that the scalar's one-particle space is the $(m, s{=}0)$ representation.

Lorentz action on one-particle states. Let $U(\Lambda)$ be the unitary implementing a Lorentz transformation Λ on the Hilbert space; under spacetime translations and Lorentz rotations the generators close into the Poincaré algebra of Eqs. (3.147) to (3.149), with P^μ the very four-momentum we diagonalized above. Because ϕ is a *scalar* field, $U(\Lambda)\phi(x)U(\Lambda)^{-1} = \phi(\Lambda x)$; feeding this into the mode expansion Eq. (9.42) and matching coefficients determines how $U(\Lambda)$

acts on the ladder operators. Because the invariant measure $\widetilde{dp}$ is preserved under the change of variables $\boldsymbol{p} \to \boldsymbol{\Lambda p}$, the relativistically normalized creation operator transforms with *no* stray Jacobian — exactly what the $2E_{\boldsymbol{p}}$ in the commutator Eq. (9.67) was arranged to buy:

$$U(\Lambda)\, a^\dagger_{\boldsymbol{p}}\, U(\Lambda)^{-1} = a^\dagger_{\boldsymbol{\Lambda p}}, \qquad \Longrightarrow \qquad U(\Lambda)\,|p\rangle = |\Lambda p\rangle\,. \tag{9.113}$$

Worked Example 9.18 (The boost factor and the invariance of $|p\rangle \to |\Lambda p\rangle$)**.** We check that the two halves of Eq. (9.113) are consistent and that they preserve the inner product. Take the first relation as given (it follows from $U(\Lambda)\phi U(\Lambda)^{-1} = \phi(\Lambda x)$ together with invariance of $\int\widetilde{dp}$). Apply $U(\Lambda)$ to Eq. (9.73), using $U(\Lambda)\,|0\rangle = |0\rangle$ (the vacuum is Poincaré invariant, being the unique state of Theorem 9.11 and annihilated by $\boldsymbol{P}$):

$$U(\Lambda)\,|p\rangle = U(\Lambda) a^\dagger_{\boldsymbol{p}} U(\Lambda)^{-1}\,|0\rangle = a^\dagger_{\boldsymbol{\Lambda p}}\,|0\rangle = |\Lambda p\rangle\,, \tag{9.114}$$

which is the second relation. The transformation carries no boost factor precisely because the relativistic normalization has already absorbed the $2E_{\boldsymbol{p}}$ into the commutator and the invariant measure; the covariant inner product Eq. (9.74) is what lets the bare state $|p\rangle = a^\dagger_{\boldsymbol{p}}\,|0\rangle$ transform as cleanly as its Lorentz-scalar label suggests. As a consistency check, unitarity of $U(\Lambda)$ demands $\langle \Lambda p|\Lambda q\rangle = \langle p|q\rangle$, and indeed

$$\langle \Lambda p|\Lambda q\rangle = 2E_{\boldsymbol{\Lambda p}}(2\pi)^3\delta^3(\boldsymbol{\Lambda p} - \boldsymbol{\Lambda q}) = 2E_{\boldsymbol{p}}(2\pi)^3\delta^3(\boldsymbol{p}-\boldsymbol{q}) = \langle p|q\rangle\,, \tag{9.115}$$

where the middle equality is the invariance of $2E_{\boldsymbol{p}}\delta^3(\boldsymbol{p}-\boldsymbol{q})$ established in Theorem 9.14. The normalization, the measure, and the Lorentz action are mutually consistent: this is the whole point of the covariant bookkeeping.

The Casimirs: mass m and spin 0. The Poincaré group has two Casimir invariants, $P^2 = P^\mu P_\mu$ and $W^2 = W^\mu W_\mu$, where

$$W^\mu = -\tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma} P_\nu M_{\rho\sigma} \tag{9.116}$$

is the Pauli–Lubański vector built from the translation generators P_ν and the Lorentz generators $M_{\rho\sigma}$ of Eqs. (3.147) to (3.149). On an irreducible representation these take the constant values $P^2 = m^2$ and $W^2 = -m^2 s(s+1)$ (Theorem 3.44). We evaluate both on our one-particle states.

For the first Casimir, Eq. (9.108) gives $P^\mu\,|p\rangle = p^\mu\,|p\rangle$ with $p^0 = E_{\boldsymbol{p}}$, so

$$P^2\,|p\rangle = p^\mu p_\mu\,|p\rangle = (E_{\boldsymbol{p}}^2 - \boldsymbol{p}^2)\,|p\rangle = m^2\,|p\rangle\,. \tag{9.117}$$

Every one-particle state has the same eigenvalue m^2: the representation has a single, sharp mass $m > 0$.

For the second Casimir, the cleanest evaluation is in the rest frame, possible precisely because $m > 0$. Pick the rest-frame momentum $k^\mu = (m, \mathbf{0})$; the state $|k\rangle$ is a representative of the one-particle space, and by Eq. (9.113) the full space is generated from it by boosts. In this frame $P_\nu \to (m, \mathbf{0})$, so only $\nu = 0$ contributes to Eq. (9.116):

$$W^\mu |k\rangle = -\tfrac{1}{2}\, \epsilon^{\mu 0 \rho\sigma}\, m\, M_{\rho\sigma} |k\rangle \,. \tag{9.118}$$

Since $\epsilon^{00\rho\sigma} = 0$, the $\mu = 0$ component vanishes, while for spatial $\mu = i$ the antisymmetric symbol ϵ^{i0jk} selects the spatial rotation generators $M_{jk} = \epsilon_{jk\ell} J^\ell$, giving $W^i |k\rangle = m\, J^i |k\rangle$ — the Pauli–Lubański vector reduces, in the rest frame, to the angular momentum, which is the generator of the little group $SO(3)$ (Theorem 3.43). Now the decisive point: the scalar field carries *no internal index*. The rest-frame one-particle state $|k\rangle$ is labelled by nothing but its momentum, so the little group, which fixes k, has nothing to rotate and acts trivially,

$$J^i |k\rangle = 0, \qquad \text{hence} \qquad W^\mu |k\rangle = 0, \qquad W^2 |k\rangle = 0. \tag{9.119}$$

Because W^2 is a Lorentz scalar, its eigenvalue is frame independent: $W^2 |p\rangle = 0$ for every one-particle state. Comparing with $W^2 = -m^2 s(s+1)$ and $m > 0$ forces $s(s+1) = 0$, i.e.

$$s = 0. \tag{9.120}$$

The little group $SO(3)$ acts in its trivial (one-dimensional) representation, which is the defining feature of spin 0.

Physical Insight

Spin is the little-group label, and the little group is "what is left of Lorentz after you have used up the boosts to go to the rest frame." For a massive particle that residue is the rotation group $SO(3)$, and the spin records which $SO(3)$ representation the rest-frame states form. A single, index-free state forms the trivial representation: it just sits there under rotations. That is what it *means* for the scalar's quantum to be spinless, and it is why a real scalar field describes exactly one species of spin-0 particle — no polarizations, no internal multiplet.

Combining Eqs. (9.113), (9.117) and (9.119): the one-particle space carries the action $U(\Lambda) |p\rangle = |\Lambda p\rangle$ of the Poincaré group, it is irreducible (the boosts connect every momentum to every other, and the rest-frame fibre is one-dimensional so there is no proper invariant subspace), and its Casimirs are $P^2 = m^2$, $W^2 = 0$. By Theorem 3.44 it *is* the mass-m, spin-0 irreducible representation of the Poincaré group. The abstract "what can a relativistic particle be" of Chapter 3 and the concrete excitations of the quantized field have met: the quantum of the free real scalar is the simplest relativistic particle there is.

This fixes the particle content of the free ϕ^4 theory once and for all. We continue ϕ^4 theory from its classical and symmetry stages (Chapters 6 and 7): at $\lambda = 0$ the theory describes a *single neutral spin-0 species* of mass m, with the multiparticle Fock space Eq. (9.87) as its state space. Because the field is real (Hermitian, $\phi = \phi^\dagger$), there is a single kind of $a_{\boldsymbol{p}}^\dagger$ and hence a single particle that is its own antiparticle, carrying no conserved internal charge — the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ of Chapter 7 acts as $(-1)^N$ on Fock space, assigning each quantum charge -1 under the $\mathbb{Z}_2$. These are the asymptotic states that the interaction of Chapter 14 will scatter.

We have built a physically interpreted Hilbert space: a unique vacuum, relativistic spin-0 particles as its excitations, a Bose-symmetric Fock space, and energy and momentum operators diagonalized on it. The construction is complete but for one scar — the divergent c-number E_0 in Eq. (9.98), the summed zero-point energy of infinitely many oscillators. The next section confronts that divergence head-on: it introduces normal ordering as the prescription that removes E_0, separates what is mere convention from what is physical, and shows through the Casimir effect that, although the absolute vacuum energy is unobservable, *differences* of vacuum energies are measurable and real.

9.4 The Vacuum Energy and Normal Ordering

The diagonalization of the previous section delivered the Hamiltonian of the free real scalar field in the form

$$H = \int \frac{\mathrm{d}^3 p}{(2\pi)^3\, 2\omega_{\mathbf{p}}}\, \omega_{\mathbf{p}} \left(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \tfrac{1}{2} \left[a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger \right] \right), \qquad \omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{9.121}$$

together with the relativistically normalized ladder algebra $\left[a_{\mathbf{p}}, a_{\mathbf{q}}^\dagger \right] = (2\pi)^3\, 2\omega_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q})$, all other commutators vanishing (recall Eqs. (9.54) and (9.98)). The first term in Eq. (9.121) is the number operator weighted by energy: it counts the quanta built on the vacuum and assigns each its relativistic energy $\omega_{\mathbf{p}}$. That part is exactly what a sensible energy operator should be, and it annihilates the vacuum. The second term does not: $\frac{1}{2}\left[a_{\mathbf{p}}, a_{\mathbf{p}}^\dagger \right] = (2\pi)^3 \omega_{\mathbf{p}} \delta^{(3)}(0)$ is a c-number, and an infinite one. Before we can call Eq. (9.121) the energy of a quantum field we must confront this term: understand exactly how badly it diverges, decide what part of it is physics and what part is bookkeeping, and produce a finite, well-defined generator of time translations. That is the business of this section. The reward is a clean vacuum with $H\,|0\rangle = 0$ and, as a bonus that is anything but trivial, the discovery that while the *absolute* value of the vacuum energy is a convention, its *response to boundaries* is a measurable force.

9.4.1 The zero-point energy and its divergences

Isolate the c-number piece of Eq. (9.121). Acting on the vacuum, $a_{\mathbf{p}}\,|0\rangle = 0$, so only the commutator term survives:

$$E_0 \;\equiv\; \langle 0|H|0\rangle = \int \frac{\mathrm{d}^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\,\omega_{\mathbf{p}}\,\tfrac{1}{2}\,\left[a_{\mathbf{p}},a_{\mathbf{p}}^\dagger\right] = \frac{1}{2}\,(2\pi)^3\delta^{(3)}(0)\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\omega_{\mathbf{p}}. \tag{9.122}$$

Each momentum mode contributes $\frac{1}{2}\omega_{\mathbf{p}}$: the field is a continuum of decoupled oscillators, and every oscillator carries the irreducible ground-state energy $\frac{1}{2}\hbar\omega$ familiar from elementary quantum mechanics. The total E_0 is the sum of these zero-point energies, and it is doubly divergent. We disentangle the two divergences.

The volume (infrared) factor. The symbol $\delta^{(3)}(0)$ is the value of a delta function at coincident argument, which is meaningless until regulated. Quantize instead in a large cubic box of side L and volume $V = L^3$ with periodic boundary conditions, so that momenta are discrete, $\mathbf{p} = \frac{2\pi}{L}\mathbf{n}$ with $\mathbf{n}\in\mathbb{Z}^3$. In a finite box the Dirac delta of the continuum is the limit of a Kronecker delta times the density of states, and the standard dictionary reads

$$(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q}) \;\longleftrightarrow\; V\,\delta_{\mathbf{p},\mathbf{q}}, \qquad \text{so} \qquad (2\pi)^3\delta^{(3)}(0) \;\longleftrightarrow\; V. \tag{9.123}$$

The infinity hiding in $\delta^{(3)}(0)$ is therefore just the spatial volume: $E_0 \propto V$. This is an *extensive* divergence, an infrared effect tied to the infinite extent of space, and it is entirely benign. It tells us that the sensible object is not the total vacuum energy but the vacuum energy *density*

$$\rho_0 \;=\; \frac{E_0}{V} = \frac{1}{2}\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\omega_{\mathbf{p}} = \frac{1}{2}\int\frac{\mathrm{d}^3p}{(2\pi)^3}\,\sqrt{\mathbf{p}^2+m^2}. \tag{9.124}$$

The ultraviolet divergence. The remaining integral in Eq. (9.124) diverges at large momentum. Doing the angular integration, $\int \mathrm{d}^3p = 4\pi\int_0^\infty p^2\,\mathrm{d}p$, and inserting a sharp cutoff $|\mathbf{p}|\le\Lambda$ to make the statement precise,

$$\rho_0(\Lambda) = \frac{1}{4\pi^2}\int_0^\Lambda \mathrm{d}p\;p^2\sqrt{p^2+m^2}. \tag{9.125}$$

For $p \gg m$ the integrand is $p^2\sqrt{p^2+m^2} = p^3\sqrt{1+m^2/p^2} = p^3 + \frac{1}{2}m^2p + O(1/p)$, so

$$\rho_0(\Lambda) = \frac{1}{4\pi^2}\left(\frac{\Lambda^4}{4}+\frac{m^2\Lambda^2}{4}+\cdots\right) = \frac{\Lambda^4}{16\pi^2}+\frac{m^2\Lambda^2}{16\pi^2}+\cdots. \tag{9.126}$$

The leading divergence is *quartic* in the cutoff. This is the worst divergence anywhere in the free theory, and it is a direct expression of the infinite number

of high-momentum oscillators each contributing $\frac{1}{2}\omega_{\mathbf{p}} \sim \frac{1}{2}|\mathbf{p}|$. It is independent of any interaction: a free field, with nothing to scatter off anything, already has an infinite ground-state energy.

Physical Insight

The two divergences have opposite physical character and must not be conflated. The volume factor V is infrared — it would be finite in a finite universe and signals that only densities are intensive, physical quantities. The Λ^4 growth is ultraviolet — it reflects our pretence that the free-field mode expansion is valid up to arbitrarily short distances. Quantum field theory does not claim to know the physics at $p \to \infty$; the cutoff Λ is an honest admission of ignorance, and every physical conclusion below will be arranged so that Λ drops out.

9.4.2 Operator ordering and the origin of the ambiguity

It is worth asking where the offending $\frac{1}{2}\omega_{\mathbf{p}}$ came from, because the answer shows it to be partly a matter of convention. The Hamiltonian density of the classical scalar field, established in Chapter 6, is $\mathcal{H} = \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2$, a sum of products of the classical fields ϕ and π. Classically these are numbers and their product is unambiguous. Upon quantization ϕ and π become operators that do not commute, and a classical monomial like π^2 or ϕ^2 does not determine a unique operator: in terms of ladder operators it contains pieces such as $a_{\mathbf{p}}a_{\mathbf{p}}^\dagger$ and $a_{\mathbf{p}}^\dagger a_{\mathbf{p}}$, which differ by the c-number $(2\pi)^3 2\omega_{\mathbf{p}}\delta^{(3)}(0)$. The passage from the classical Hamiltonian to Eq. (9.121) therefore carries an *ordering ambiguity*.

Different ordering prescriptions of the same classical H differ only by c-numbers. Concretely, write the symmetric (Weyl) ordering that the naive substitution produces,

$$H_{\text{Weyl}} = \int \frac{\mathrm{d}^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\,\omega_{\mathbf{p}}\,\tfrac{1}{2}\big(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + a_{\mathbf{p}}a_{\mathbf{p}}^\dagger\big) = \int \frac{\mathrm{d}^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\,\omega_{\mathbf{p}}\,a_{\mathbf{p}}^\dagger a_{\mathbf{p}} \;+\; E_0, \tag{9.127}$$

where the last equality used the commutator and Eq. (9.122). Any other ordering — all $a^\dagger$ to the left, all to the right, or anything in between — yields the same operator $a_{\mathbf{p}}^\dagger a_{\mathbf{p}}$ piece and a *different* additive constant. The dynamically nontrivial content of the Hamiltonian, the part that has nonzero commutators with the fields and hence generates the equations of motion, is fixed; only a scalar shift of the energy zero is left undetermined by the classical theory. This is precisely the freedom we are entitled to exploit, and the prescription that exploits it cleanly is normal ordering.

9.4.3 Normal ordering

Definition 9.19 (Normal ordering)**.** Let $:\cdots:$ denote the operation that reorders any product of creation and annihilation operators so that every annihilation operator stands to the right of every creation operator, treating the ladder operators as though they commuted in the process (no commutator terms are generated). For a single product of fields, $:\mathcal{O}:$ is obtained by writing each field through its mode expansion, expanding the product, and in every resulting monomial moving all $a^\dagger$ to the left of all a.

By construction a normal-ordered product has all annihilation operators acting first, so its vacuum expectation value vanishes unless the product is a pure c-number: $\langle 0|:\mathcal{O}:|0\rangle = 0$ for any $\mathcal{O}$ that is a nontrivial product of fields. Applying the prescription to the Hamiltonian and using Eq. (9.127),

$$:H: \;= \int \frac{\mathrm{d}^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\,\omega_{\mathbf{p}}\,a^\dagger_{\mathbf{p}}a_{\mathbf{p}} = H - E_0. \tag{9.128}$$

The normal-ordered Hamiltonian is the original Hamiltonian with the zero-point energy subtracted. It is manifestly finite and positive (it is $\omega_{\mathbf{p}}$ times a number operator), it annihilates the vacuum, $:H:\,|0\rangle = 0$, and it assigns to the one-particle state $a^\dagger_{\mathbf{p}}|0\rangle$ exactly the energy $\omega_{\mathbf{p}}$ that the spectrum of the previous section demanded. We adopt $:H:$ as *the* Hamiltonian of the free scalar field; from here on, "H" in ϕ^4 theory means the normal-ordered generator, and every later Wick contraction in the interacting theory inherits this convention.

It is essential to verify that subtracting E_0 has cost us nothing physical. The following proposition makes this precise and is the formal justification for the whole prescription.

Proposition 9.20 (Normal ordering is a c-number subtraction)**.** *The normal-ordered Hamiltonian differs from H by the c-number $E_0 = \langle 0|H|0\rangle$ times the identity. Consequently $:H:$ has the same commutators with every operator as H does, generates the same Heisenberg equations of motion, and defines the same dynamics; the only change is a uniform downward shift of the entire energy spectrum by E_0, chosen so that the vacuum sits at zero.*

Proof. From Eq. (9.128), $:H: \;= H - E_0\,\mathbf{1}$ with E_0 a complex (here real) number, not an operator. For any operator $\mathcal{O}$,

$$[:H:,\mathcal{O}] = [H - E_0\mathbf{1},\mathcal{O}] = [H,\mathcal{O}] - E_0\,[\mathbf{1},\mathcal{O}] = [H,\mathcal{O}]\,, \tag{9.129}$$

because the identity commutes with everything. In particular the Heisenberg equation of motion for the field, $\dot\phi = i\,[H,\phi]$, is unchanged when H is replaced by $:H:$, and likewise for $\dot\pi$; the Klein–Gordon equation that follows is identical. The energy eigenvalues shift uniformly: if $H\,|\Psi\rangle = E\,|\Psi\rangle$ then $:H:\,|\Psi\rangle = (E - E_0)\,|\Psi\rangle$,

so all gaps $E_i - E_j$ — the only spectral data that enter transition frequencies and the S-matrix — are preserved. Thus normal ordering is exactly the statement "measure energy relative to the vacuum," and it changes no commutator, no equation of motion, and no physical energy difference. □

Caution

Proposition 9.20 licences dropping E_0 *only* because E_0 is a constant independent of the dynamical variables. Normal ordering is not a general license to discard divergences. In the interacting theory the analogous subtractions are field- and momentum-dependent and constitute genuine renormalization (Chapter 22); there the removed pieces are operators, not c-numbers, and the argument above does not apply. Here, and only here, the subtraction is free.

9.4.4 Is the vacuum energy physical?

We have just discarded an infinite constant with a clear conscience, so we owe the reader an honest account of what survives. In any theory whose dynamics is generated by commutators with H — which is to say, all of quantum mechanics in the absence of gravity — the absolute value of the energy is unobservable. A spectrometer measures frequencies $\omega_{ij} = E_i - E_j$; a calorimeter measures energy released, again a difference; an S-matrix depends on energy *differences* between in- and out-states. Adding a constant to H shifts every level together and cancels from every such quantity. The number E_0, and indeed its quartic cutoff dependence, is therefore not something the non-gravitational theory can ever be asked to predict. Setting it to zero by normal ordering is not a swindle; it is the correct response to a quantity that no experiment in flat space can access.

Two caveats sharpen this statement, and they are the substance of the rest of the section.

First, the indifference to constants fails the moment gravity is switched on. General relativity couples to the full stress–energy tensor, and a constant vacuum energy density ρ_0 acts as a cosmological constant: it gravitates, curving spacetime in a way that an absolute energy could never do in flat space. The convention-dependence of E_0 then collides with the convention-independence of geometry, and the collision is the cosmological-constant problem, to which we return in Section 9.4.6.

Second — and this is measurable already without gravity — although the *absolute* vacuum energy is a convention, the *difference* in vacuum energy between two configurations of the boundaries or background is not. If we change the boundary conditions imposed on the field, we change which modes are available, and the sum of their zero-point energies changes by a finite, cutoff-independent amount. That finite change is a real energy, and its gradient with respect to

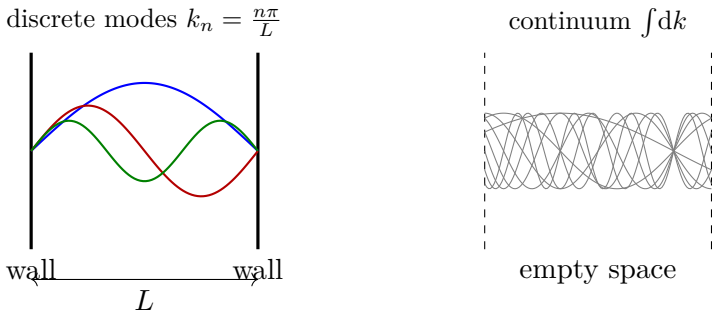


Figure 9.4: The Casimir setup in one dimension. Imposing Dirichlet conditions at two walls a distance L apart (left) selects the discrete standing-wave modes $k_n = n\pi/L$; the same region of empty space (right) supports a continuum of modes. Both vacuum energies are ultraviolet divergent, but their difference — the Casimir energy — is finite because the short-wavelength modes are common to both and cancel.

the boundary position is a real force. This is the Casimir effect, and we now compute it from first principles.

9.4.5 The Casimir effect: zero-point energy made visible

Place the field in a region bounded by walls and demand that the field vanish on the walls (Dirichlet boundary conditions; for a scalar this models, e.g., a field forced to a node at an ideal conductor). The allowed standing waves now form a discrete set whose wavelengths are fixed by the geometry, and the vacuum energy becomes a discrete sum of zero-point energies rather than an integral. As the wall separation changes, the sum changes, and the difference from the empty-space value is the Casimir energy. The decisive feature is that, although both the bounded-region sum and the empty-space integral are ultraviolet divergent, their difference is finite: the highest-momentum modes do not feel the walls and cancel between the two configurations. We exhibit this first in one spatial dimension, where every step can be shown in full, and then in the physical three-dimensional parallel-plate geometry.

Worked Example 9.21 (The 1+1-dimensional Casimir energy)**.** Consider a massless real scalar field in one spatial dimension, confined to the interval $0 \le x \le L$ with Dirichlet conditions $\phi(0) = \phi(L) = 0$. The normal modes that satisfy these conditions are $\sin(k_n x)$ with

$$k_n = \frac{n\pi}{L}, \qquad n = 1, 2, 3, \ldots, \tag{9.130}$$

and, since the field is massless, each has angular frequency $\omega_n = |k_n| = n\pi/L$. Each mode is an independent oscillator contributing $\frac{1}{2}\omega_n$ to the ground-state

energy, so the vacuum energy between the walls is

$$E(L) = \frac{1}{2}\sum_{n=1}^{\infty}\omega_n = \frac{\pi}{2L}\sum_{n=1}^{\infty} n. \tag{9.131}$$

The sum $\sum_{n\geq 1} n$ is wildly divergent. We extract its finite physical content in two independent ways and check that they agree.

Method 1: zeta-function continuation. The Riemann zeta function $\zeta(s) = \sum_{n\geq 1} n^{-s}$ converges for $\operatorname{Re} s > 1$ and continues analytically to a meromorphic function on the whole s-plane, with the special value $\zeta(-1) = -\frac{1}{12}$ established in Section E.3. The divergent series $\sum_{n\geq 1} n = \sum_{n\geq 1} n^{-(-1)}$ is the value of $\zeta(s)$ at $s = -1$ *read off the analytic continuation*, which is the canonical way to assign a finite value to such an asymptotic/divergent series (Section 5.6.1). Hence

$$\sum_{n=1}^{\infty} n \;\xrightarrow{\text{reg}}\; \zeta(-1) = -\frac{1}{12}, \qquad E(L) = \frac{\pi}{2L}\left(-\frac{1}{12}\right) = -\frac{\pi}{24L}. \tag{9.132}$$

The energy is finite and *negative*. The minus sign and the factor will be justified by the second, regulator-explicit method, which shows where the divergence went.

Method 2: exponential cutoff with explicit cancellation. Zeta continuation is rigorous but it hides the divergent piece; we now keep it visible. Damp each mode by $e^{-a\omega_n}$, with $a > 0$ a short-distance cutoff length (high-frequency modes, $\omega_n \gtrsim 1/a$, are suppressed because no real wall is perfectly reflecting at arbitrarily high frequency), and take $a \to 0$ at the end. Define $\varepsilon \equiv a\pi/L$ and $x \equiv e^{-\varepsilon}$. Then

$$E(L,a) = \frac{\pi}{2L}\sum_{n=1}^{\infty} n\, e^{-a\omega_n} = \frac{\pi}{2L}\sum_{n=1}^{\infty} n\, x^n = \frac{\pi}{2L}\,\frac{x}{(1-x)^2}, \tag{9.133}$$

where we summed the geometric-type series via $\sum_{n\geq 1} n x^n = x\,\frac{d}{dx}\sum_{n\geq 1} x^n = x\,\frac{d}{dx}\frac{x}{1-x} = \frac{x}{(1-x)^2}$, valid for $|x| < 1$. To expose the $a \to 0$ behaviour, expand in ε. It is cleanest to differentiate the small-ε expansion of $\sum_{n\geq 1} e^{-n\varepsilon} = 1/(e^{\varepsilon} - 1)$, whose Laurent series is

$$\frac{1}{e^{\varepsilon}-1} = \frac{1}{\varepsilon} - \frac{1}{2} + \frac{\varepsilon}{12} - \frac{\varepsilon^3}{720} + \cdots. \tag{9.134}$$

Since $\sum_{n\geq 1} n\, e^{-n\varepsilon} = -\frac{d}{d\varepsilon}\sum_{n\geq 1} e^{-n\varepsilon}$, we differentiate Eq. (9.134) term by term:

$$\sum_{n=1}^{\infty} n\, e^{-n\varepsilon} = -\frac{d}{d\varepsilon}\left(\frac{1}{\varepsilon} - \frac{1}{2} + \frac{\varepsilon}{12} - \frac{\varepsilon^3}{720} + \cdots\right) = \frac{1}{\varepsilon^2} - \frac{1}{12} + \frac{\varepsilon^2}{240} - \cdots. \tag{9.135}$$

Substituting $\varepsilon = a\pi/L$, so that $1/\varepsilon^2 = L^2/(\pi^2 a^2)$, and using Eq. (9.133),

$$E(L,a) = \frac{\pi}{2L}\left(\frac{L^2}{\pi^2 a^2} - \frac{1}{12} + O(a^2)\right) = \underbrace{\frac{L}{2\pi a^2}}_{\text{divergent}} - \underbrace{\frac{\pi}{24L}}_{\text{finite}} + O(a^2). \tag{9.136}$$

The divergence is the first term, $L/(2\pi a^2)$. Crucially it is *extensive*: it is proportional to the length L of the region, exactly as a bulk energy density should be. It is, in fact, nothing but the empty-space zero-point energy of the same length. To see this, compute the vacuum energy of an unbounded line of length L directly, with the *same* cutoff, using the continuum of modes $\int_{-\infty}^{\infty} \mathrm{d}k/(2\pi)$ at density L per unit k (this is Eq. (9.123) in one dimension):

$$E_{\text{free}}(L,a) = L\int_{-\infty}^{\infty} \frac{\mathrm{d}k}{2\pi}\,\frac{1}{2}\,|k|\; e^{-a|k|} = \frac{L}{2\pi}\int_0^{\infty} \mathrm{d}k\; k\, e^{-ak} = \frac{L}{2\pi}\,\frac{1}{a^2}. \tag{9.137}$$

This is precisely the divergent term in Eq. (9.136). The Casimir energy is the difference between the energy with walls and the energy of the same region of empty space,

$$E_{\text{Cas}}(L) \equiv \lim_{a\to 0}\big[E(L,a) - E_{\text{free}}(L,a)\big] = -\frac{\pi}{24L}, \tag{9.138}$$

finite, cutoff-independent, and in exact agreement with the zeta-function value Eq. (9.132). The two regulators agree because they compute the same physical quantity — the shift in vacuum energy caused by the walls — and that shift never depended on the unphysical short-distance details that the divergent term absorbed.

The force. The walls feel a force equal to minus the gradient of the energy:

$$F = -\frac{dE_{\text{Cas}}}{dL} = -\frac{d}{dL}\left(-\frac{\pi}{24L}\right) = -\frac{\pi}{24L^2}. \tag{9.139}$$

The force is negative, meaning it pulls the walls together: the Casimir force is *attractive*. The vacuum, by carrying less zero-point energy when the walls are closer, exerts a real pull on them. Nothing in this conclusion depended on the regulator; only the finite difference Eq. (9.138) entered.

Physical Insight

Reread the cancellation in Eq. (9.136)–Eq. (9.137): the infinity that we discarded by normal ordering in Eq. (9.128) is the same infinity that cancels here. The bulk zero-point energy density is a constant shift and drops out of every difference; what remains, the boundary-induced $-\pi/24L$, is genuinely there. This is the precise sense in which "the vacuum energy

is unobservable but its differences are real." The Casimir effect does not measure E_0; it measures how E_0 *responds* to a change of geometry.

The one-dimensional result is the cleanest possible demonstration, but the historically and experimentally decisive case is two parallel plates in three dimensions. We carry that computation out as well, since it produces the number that experiment confirms.

Worked Example 9.22 (Parallel-plate Casimir energy in 3+1 dimensions)**.** Take a massless scalar field between two parallel plates of area A separated by a distance a along z, with Dirichlet conditions on each plate. The transverse momentum $\mathbf{k}_\perp = (k_x, k_y)$ is continuous; the longitudinal momentum is quantized as $k_n = n\pi/a$, $n = 1, 2, \dots$. The mode frequencies are $\omega_{n,\mathbf{k}_\perp} = \sqrt{\mathbf{k}_\perp^2 + (n\pi/a)^2}$, and the vacuum energy per unit plate area is

$$\frac{E(a)}{A} = \frac{1}{2}\sum_{n=1}^{\infty} \int \frac{\mathrm{d}^2 k_\perp}{(2\pi)^2} \sqrt{\mathbf{k}_\perp^2 + \left(\frac{n\pi}{a}\right)^2}. \tag{9.140}$$

The transverse integral is ultraviolet divergent; we regulate and continue it by treating the transverse dimension as $d_\perp = 2 - 2\varepsilon$ and analytically continuing, exactly the dimensional-continuation logic of Section E.3. We need the standard continued integral — the Euclidean form of the master one-loop integral derived in full in Theorem D.22 — valid for the analytic continuation in d and in the exponent,

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d} \, (\mathbf{k}^2 + M^2)^{-s} = \frac{1}{(4\pi)^{d/2}} \, \frac{\Gamma\!\left(s - \frac{d}{2}\right)}{\Gamma(s)} \, (M^2)^{\,d/2 - s}, \tag{9.141}$$

which we apply at $s = -\frac{1}{2}$ (so that $(\mathbf{k}^2 + M^2)^{1/2}$ appears) with $d = 2$ and $M = n\pi/a$. Using $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$ and $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$,

$$\int \frac{\mathrm{d}^2 k_\perp}{(2\pi)^2} \sqrt{\mathbf{k}_\perp^2 + M^2} \;=\; \frac{1}{4\pi} \, \frac{\Gamma(-\frac{3}{2})}{\Gamma(-\frac{1}{2})} \, M^3 = \frac{1}{4\pi} \cdot \frac{\frac{4}{3}\sqrt{\pi}}{-2\sqrt{\pi}} \, M^3 = -\frac{M^3}{6\pi}. \tag{9.142}$$

The (power-law) ultraviolet divergence of the transverse integral has been discarded by the continuation, exactly as the extensive divergence was discarded in one dimension; what it leaves is the analytic remainder $-M^3/6\pi$. Inserting $M = n\pi/a$ into Eq. (9.140),

$$\frac{E(a)}{A} = \frac{1}{2}\sum_{n=1}^{\infty} \left(-\frac{1}{6\pi} \left(\frac{n\pi}{a}\right)^3 \right) = -\frac{\pi^2}{12\,a^3} \sum_{n=1}^{\infty} n^3 = -\frac{\pi^2}{12\,a^3}\,\zeta(-3). \tag{9.143}$$

The remaining sum over n is again assigned its zeta value, and $\zeta(-3) = \frac{1}{120}$ (Section E.3). Therefore

$$\boxed{\frac{E(a)}{A} = -\frac{\pi^2}{1440\,a^3}} \qquad \Longrightarrow \qquad \frac{F}{A} = -\frac{d}{da}\frac{E(a)}{A} = -\frac{\pi^2}{480\,a^4}. \tag{9.144}$$

The pressure is negative: the plates are pulled together. For the electromagnetic field the same computation with two physical photon polarizations (and Neumann as well as Dirichlet modes) gives the celebrated $E/A = -\pi^2/(720\,a^3)$ and force per area $-\pi^2\hbar c/(240\,a^4)$, which is the quantity measured in the laboratory. The single-scalar result Eq. (9.144) is exactly one half of it — the electromagnetic field contributes two physical polarizations, hence twice the energy of a single Dirichlet scalar — and it is the version that belongs to ϕ^4 theory.

Historical Note

Casimir derived the electromagnetic plate force in 1948 while studying colloidal suspensions; the prediction of an attractive force from nothing but vacuum fluctuations was for decades regarded as marginal, but it has since been confirmed with $\sim 1\%$ precision in modern force-microscopy experiments. The effect is the sharpest demonstration that the zero-point energy of a quantum field, while not itself an observable, has observable consequences whenever a boundary or background breaks the translation invariance that would otherwise make it an irrelevant constant.

9.4.6 Frontier flag: the cosmological-constant problem

We close with the one place where the absolute vacuum energy refuses to be a convention. As noted above, gravity couples to the total stress–energy, and a Lorentz-invariant vacuum has $\langle T_{\mu\nu}\rangle = \rho_{\rm vac}\,\eta_{\mu\nu}$, so the constant $\rho_{\rm vac}$ enters Einstein's equations as a cosmological constant Λ and curves spacetime. Now the shift we so cheerfully removed by normal ordering becomes physical, and its natural size is a catastrophe. If the free-field mode expansion were trusted up to a cutoff Λ, the vacuum energy density Eq. (9.126) would be of order

$$\rho_{\rm vac} \sim \frac{\Lambda^4}{16\pi^2}. \tag{9.145}$$

Taking Λ at the Planck scale $M_{\rm Pl} \sim 10^{19}$ GeV, where quantum gravity certainly intervenes, gives $\rho_{\rm vac} \sim 10^{74}$ GeV^4. The value inferred from the observed accelerating expansion of the universe is $\rho_\Lambda \sim 10^{-47}$ GeV^4 — a meV-scale energy density. The naive estimate exceeds the measured value by roughly

$$\frac{\rho_{\rm vac}}{\rho_\Lambda} \sim 10^{121}, \tag{9.146}$$

and even the most conservative choice of cutoff (say the electroweak or QCD scale) leaves a discrepancy of dozens of orders of magnitude. Why the gravitating vacuum energy is so many orders of magnitude smaller than every field-theoretic estimate — and why it is small but not exactly zero — is the *cosmological-constant problem*. As of the mid-2020s it has no accepted solution: it is not

resolved by supersymmetry (which is broken), nor by known anthropic, dynamical, or symmetry-based mechanisms in any way that the community regards as compelling. It stands as the most quantitatively severe known clash between quantum field theory and gravitation, and the reader should keep it in mind as the honest limit of the clean construction of this chapter: normal ordering disposes of the vacuum energy perfectly as long as gravity is switched off, and not at all once it is switched on.

With a finite, normal-ordered Hamiltonian in hand and a vacuum pinned at zero energy, the spectrum of the free theory is fully under control. We may now turn from *states* to *dynamics*: the next section computes the free two-point functions of the field and assembles them into the Feynman propagator, the object whose time-ordered structure and $i\epsilon$ contour will become the single line in every diagram of the interacting theory.

9.5 Correlation Functions and the Feynman Propagator

We have arrived at the dynamical heart of the free theory. The preceding sections handed us a complete quantum kinematics: a vacuum $|0\rangle$ annihilated by every lowering operator, a field operator $\phi(x)$ written as a Lorentz-covariant superposition of creation and annihilation operators, and a Hamiltonian rendered finite by normal ordering. What we do *not* yet have is a single object that packages the dynamics — the propagation of a disturbance of the field from one spacetime point to another — in a form ready for perturbation theory. That object is the two-point correlation function, and its time-ordered version, the Feynman propagator, is the most important quantity in this chapter. It is the amplitude for the field to "create a quantum at y and absorb it at x," it is the Green's function of the Klein–Gordon operator, and from Chapters 14 and 15 onward it is the internal line of every Feynman diagram. The path-integral chapter (Chapter 10) will rederive this very object from the generating functional $Z[J]$; the two derivations must agree, and they do.

Our strategy is to compute the vacuum expectation value of a product of two fields directly from the mode expansion, study its analytic structure, and then show that the time-ordered product — the combination that perturbation theory actually demands — is reproduced with magical economy by a single covariant four-momentum integral whose only subtlety is an $i\varepsilon$ in the denominator. That $i\varepsilon$, far from being a regulator we tolerate, turns out to *be* the instruction "time-order the operators." Establishing this equivalence, in full, is the work of the section.

Throughout we use the mode expansion and canonical structure fixed earlier in this chapter. We recall them in the form we need: with $E_{\mathbf{p}} \equiv \sqrt{\mathbf{p}^2 + m^2} > 0$

the on-shell energy and $p^0 = E_{\mathbf{p}}$ understood inside on-shell exponentials,

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \left(a_{\mathbf{p}}\, e^{-ip\cdot x} + a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} \right), \qquad \left[a_{\mathbf{p}}, a^\dagger_{\mathbf{p}'} \right] = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{p}'), \tag{9.147}$$

all other commutators vanishing, $a_{\mathbf{p}}\,|0\rangle = 0$, and the canonical momentum is $\pi(x) = \dot\phi(x)$ with the equal-time relation $[\phi(x^0,\mathbf{x}), \pi(x^0,\mathbf{y})] = i\,\delta^3(\mathbf{x}-\mathbf{y})$ (Eqs. (9.7) and (9.42)). We also use the Lorentz invariance of the measure $d^3p/(2\pi)^3 2E_{\mathbf{p}}$ established when we built the one-particle states (Eq. (9.68)); it is what makes the manifestly noncovariant-looking integrals below covariant after all.

9.5.1 The Wightman two-point function

The simplest correlator is the vacuum expectation value of the unordered product of two fields.

Definition 9.23 (Wightman two-point function)**.** The *(positive-frequency) Wightman function* of the free real scalar field is

$$D(x-y) \;\equiv\; \langle 0|\, \phi(x)\, \phi(y)\, |0\rangle\,. \tag{9.148}$$

That it depends only on the difference $x-y$ is not assumed but computed; it follows from translation invariance of the vacuum, which we now see explicitly. Insert the mode expansion Eq. (9.147) for both fields:

$$\phi(x)\phi(y) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \frac{d^3q}{(2\pi)^3\, 2E_{\mathbf{q}}} \left(a_{\mathbf{p}} e^{-ip\cdot x} + a^\dagger_{\mathbf{p}} e^{+ip\cdot x} \right) \left(a_{\mathbf{q}} e^{-iq\cdot y} + a^\dagger_{\mathbf{q}} e^{+iq\cdot y} \right). \tag{9.149}$$

Of the four operator products aa, $aa^\dagger$, $a^\dagger a$, $a^\dagger a^\dagger$, only $a_{\mathbf{p}} a^\dagger_{\mathbf{q}}$ survives between vacua: any term with an annihilator on the right kills $|0\rangle$, and any term with a creator on the left is killed by $\langle 0|$. Using $\langle 0|\, a_{\mathbf{p}} a^\dagger_{\mathbf{q}}\, |0\rangle = \langle 0| \left[a_{\mathbf{p}}, a^\dagger_{\mathbf{q}} \right] |0\rangle = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{q})$, the $\mathbf{q}$ integral collapses, the commutator's $2E_{\mathbf{p}}$ cancelling the measure factor $1/2E_{\mathbf{q}}$:

$$\boxed{D(x-y) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-ip\cdot(x-y)}} \qquad (p^0 = E_{\mathbf{p}}). \tag{9.150}$$

This is the first key result. Several of its properties are worth recording, because they control everything that follows.

(i) Positive frequency. Every term carries $e^{-iE_{\mathbf{p}}(x^0-y^0)}$ with $E_{\mathbf{p}} > 0$: as a function of the time difference the Wightman function contains only positive-frequency Fourier components. This is the operator statement that $\phi(y)\,|0\rangle$ is a

superposition of one-particle states of positive energy — the field, acting on the vacuum, can only *create*, never destroy negative energy.

(ii) Lorentz invariance. Written with the invariant measure $d^3p/(2\pi)^3 2E_{\mathbf{p}}$ and the invariant exponent $p\cdot(x-y)$ at $p^0 = E_{\mathbf{p}}$, the integral Eq. (9.150) is manifestly invariant under the orthochronous Lorentz group: $D(x-y)$ is a Lorentz scalar function of $x-y$. For spacelike separation it can therefore depend only on the invariant $(x-y)^2 < 0$, and by choosing a frame in which $x^0 = y^0$ one sees it does *not* vanish there — a point we return to when we analyze causality.

(iii) Distributional nature. The integrand decays only as the inverse of $E_{\mathbf{p}} \sim |\mathbf{p}|$ at large momentum, so $\int d^3p/2E_{\mathbf{p}}$ is badly divergent for $x = y$. Thus $D(x-y)$ is not an ordinary function: it is a tempered distribution, finite only when smeared against test functions, exactly the kind of object the distribution theory of Theorem 5.18 was built to handle. We will always treat it as such; in particular the manipulations below (interchanging integrals, closing contours) are justified at the level of tempered distributions reviewed there.

(iv) Conjugation. Since ϕ is Hermitian, $D(x-y)^* = \langle 0|\,\phi(y)\phi(x)\,|0\rangle = D(y-x)$, which from Eq. (9.150) is the same integral with $e^{+ip\cdot(x-y)}$. The two orderings $D(x-y)$ and $D(y-x)$ are complex conjugates, not equal; their difference is the commutator function $\langle 0|\,[\phi(x),\phi(y)]\,|0\rangle = D(x-y) - D(y-x)$, the central object of Section 9.6.

9.5.2 Time ordering and the Feynman propagator

Perturbation theory, as we will see in Chapter 14, is built from the Dyson series, and every term in that series is a vacuum expectation value of a *time-ordered* product of fields. We therefore need the correlator with the operators arranged in order of increasing time from right to left. Recall the time-ordering symbol $\mathcal{T}$, defined on two bosonic fields by

$$\mathcal{T}\,\phi(x)\phi(y) \;\equiv\; \theta(x^0-y^0)\,\phi(x)\phi(y) + \theta(y^0-x^0)\,\phi(y)\phi(x), \tag{9.151}$$

i.e. the later field stands to the left. (For bosons the two terms commute at equal time up to the equal-time commutator, which vanishes for the fields themselves, so the symbol is unambiguous at $x^0 = y^0$.)

Definition 9.24 (Feynman propagator)**.** The *Feynman propagator* of the free real scalar field is the time-ordered two-point function

$$D_F(x-y) \;\equiv\; \langle 0|\mathcal{T}\phi(x)\phi(y)|0\rangle. \tag{9.152}$$

Taking the vacuum expectation value of Eq. (9.151) and using Eq. (9.148) for each ordering gives immediately the position-space form

$$D_F(x-y) = \theta(x^0-y^0)\,D(x-y) + \theta(y^0-x^0)\,D(y-x). \tag{9.153}$$

In words: for x later than y a quantum is created at y and absorbed at x; for y later than x the roles are reversed. The same positive-energy quantum always propagates *forward* in time, which is the physical content the $i\varepsilon$ prescription below will encode. Equation Eq. (9.153) is exact and will serve as the target that the covariant momentum integral must reproduce.

9.5.3 The covariant momentum-space form via the p^0 contour

The position-space expression Eq. (9.153) is correct but ungainly: it is built from on-shell three-dimensional integrals glued together with step functions, and it hides Lorentz covariance behind the explicit time-ordering. The remarkable fact — the technical centerpiece of the section — is that the *same* distribution is produced by a single, manifestly covariant integral over *all four* components of an off-shell momentum, with the time-ordering smuggled in entirely through an infinitesimal imaginary part of the mass.

Theorem 9.25 (Contour representation of the Feynman propagator)**.** *The Feynman propagator Eq. (9.152) admits the four-dimensional integral representation*

$$\boxed{D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i\, e^{-ip\cdot(x-y)}}{p^2 - m^2 + i\varepsilon}} \tag{9.154}$$

where now p^0 ranges over all of $\mathbb{R}$ independently of $\mathbf{p}$ (the momentum is off shell, $p^2 \neq m^2$ in general), $\varepsilon \to 0^+$, and the equality holds as tempered distributions.

Proof. The strategy is to perform the p^0 integral in Eq. (9.154) by contour methods, leaving the three-dimensional $\mathbf{p}$ integral untouched, and to show that the result is exactly Eq. (9.153). Write $t \equiv x^0 - y^0$ for the time difference and factor the denominator. Since $p^2 - m^2 = (p^0)^2 - \mathbf{p}^2 - m^2 = (p^0)^2 - E_{\mathbf{p}}^2$, we have

$$p^2 - m^2 + i\varepsilon = (p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon = \big(p^0 - (E_{\mathbf{p}} - i\varepsilon')\big)\big(p^0 + (E_{\mathbf{p}} - i\varepsilon')\big), \tag{9.155}$$

where the second equality holds to first order in the infinitesimal because

$$(p^0 - E_{\mathbf{p}} + i\varepsilon')(p^0 + E_{\mathbf{p}} - i\varepsilon') = (p^0)^2 - (E_{\mathbf{p}} - i\varepsilon')^2 = (p^0)^2 - E_{\mathbf{p}}^2 + 2iE_{\mathbf{p}}\varepsilon' + O(\varepsilon'^2), \tag{9.156}$$

so the choices match with $\varepsilon = 2E_{\mathbf{p}}\,\varepsilon'$. Because $E_{\mathbf{p}} > 0$, ε and ε' have the same (positive) sign, and we may take $\varepsilon' \to 0^+$ at the end. The displaced poles in the complex p^0 plane therefore sit at

$$p^0 = +E_{\mathbf{p}} - i\varepsilon' \quad \text{(just } \textit{below} \text{ the positive real axis)}, \qquad p^0 = -E_{\mathbf{p}} + i\varepsilon' \quad \text{(just } \textit{above} \text{ the neg} \tag{9.157}$$

This staggered placement — one pole nudged down, one nudged up — is the entire content of the $i\varepsilon$; it is what distinguishes the Feynman propagator from

its retarded and advanced cousins (Section 9.5.5). The configuration and the two ways of closing the contour are drawn in Fig. 9.5.

Now consider the inner integral

$$I(\mathbf{p},t) \equiv \int_{-\infty}^{\infty} \frac{dp^0}{2\pi} \, \frac{i\, e^{-ip^0 t}}{(p^0 - E_{\mathbf{p}} + i\varepsilon')(p^0 + E_{\mathbf{p}} - i\varepsilon')}. \tag{9.158}$$

To use the residue theorem (Theorem 5.23) we must close the open contour along $\mathbb{R}$ with a large semicircle on which the integrand vanishes, and the choice of half-plane is forced by the convergence of the exponential. Parametrize the semicircle as $p^0 = Re^{i\theta}$; then

$$\left| e^{-ip^0 t} \right| = \left| e^{-iRt(\cos\theta + i\sin\theta)} \right| = e^{\,Rt\,\sin\theta}. \tag{9.159}$$

For this to decay as $R \to \infty$ we need $t \sin\theta < 0$. Two cases arise.

Case $t > 0$ (so $x^0 > y^0$). We need $\sin\theta < 0$, i.e. $\theta \in (-\pi, 0)$: the contour must be closed in the *lower* half-plane. By Jordan's lemma the semicircle contributes nothing and the closed contour encircles only the pole at $p^0 = E_{\mathbf{p}} - i\varepsilon'$, traversed *clockwise*, so the residue theorem gives a factor $-2\pi i$. The residue of the integrand at that pole is obtained by deleting the vanishing factor and setting $p^0 = E_{\mathbf{p}}$ in the rest:

$$\mathrm{Res}_{p^0 = E_{\mathbf{p}} - i\varepsilon'} = \left. \frac{i\, e^{-iE_{\mathbf{p}} t}}{p^0 + E_{\mathbf{p}}} \right|_{p^0 = E_{\mathbf{p}}} = \frac{i\, e^{-iE_{\mathbf{p}} t}}{2E_{\mathbf{p}}}. \tag{9.160}$$

Hence

$$I(\mathbf{p},t) = \frac{1}{2\pi}\,(-2\pi i)\,\frac{i\, e^{-iE_{\mathbf{p}} t}}{2E_{\mathbf{p}}} = \frac{e^{-iE_{\mathbf{p}} t}}{2E_{\mathbf{p}}} \qquad (t > 0), \tag{9.161}$$

using $(-i)(i) = 1$. Reinstating the three-dimensional integral and the spatial exponential $e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}$ carried along from Eq. (9.154),

$$D_F(x-y)\big|_{t>0} = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-iE_{\mathbf{p}} t}\, e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-ip\cdot(x-y)} = D(x-y), \tag{9.162}$$

because $E_{\mathbf{p}} t - \mathbf{p}\cdot(\mathbf{x}-\mathbf{y}) = p\cdot(x-y)$ at $p^0 = E_{\mathbf{p}}$. We have recovered the first piece of Eq. (9.153).

Case $t < 0$ (so $y^0 > x^0$). Now $\sin\theta > 0$ is required, so we close in the *upper* half-plane, encircling only the pole at $p^0 = -E_{\mathbf{p}} + i\varepsilon'$, traversed *counterclockwise*: a factor $+2\pi i$. The residue is

$$\mathrm{Res}_{p^0 = -E_{\mathbf{p}} + i\varepsilon'} = \left. \frac{i\, e^{-ip^0 t}}{p^0 - E_{\mathbf{p}}} \right|_{p^0 = -E_{\mathbf{p}}} = \frac{i\, e^{+iE_{\mathbf{p}} t}}{-2E_{\mathbf{p}}}, \tag{9.163}$$

so that

$$I(\mathbf{p},t) = \frac{1}{2\pi}\,(+2\pi i)\,\frac{i\, e^{+iE_{\mathbf{p}} t}}{-2E_{\mathbf{p}}} = \frac{e^{+iE_{\mathbf{p}} t}}{2E_{\mathbf{p}}} \qquad (t < 0), \tag{9.164}$$

using $(i)(i)/(-1) = 1$. Therefore

$$D_F(x-y)\big|_{t<0} = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\, e^{+iE_{\mathbf{p}}t}\, e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}. \tag{9.165}$$

Because $E_{\mathbf{p}}$ and the measure are even in $\mathbf{p}$, we may substitute $\mathbf{p} \to -\mathbf{p}$ to flip the sign of the spatial exponential without changing anything else:

$$D_F(x-y)\big|_{t<0} = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\, e^{+iE_{\mathbf{p}}t - i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\, e^{+ip\cdot(x-y)} = D(y-x), \tag{9.166}$$

recognizing the conjugate Wightman function from property (iv) above. We have recovered the second piece of Eq. (9.153).

Combining the two cases reproduces exactly

$$D_F(x-y) = \theta(x^0-y^0)\,D(x-y) + \theta(y^0-x^0)\,D(y-x), \tag{9.167}$$

which is Eq. (9.153), the operator time-ordered product. This establishes Eq. (9.154). □

The proof makes the slogan precise: *the $i\varepsilon$ is time ordering.* The displacement of the poles is not a computational convenience that we remove at the end — it is the part of the formula that knows which field is later. Pushing the positive-energy pole below the axis guarantees that when x is the later point ($t > 0$) the contour, forced downward by convergence, captures precisely the positive-frequency propagation $D(x-y)$; pushing the negative-energy pole above the axis does the symmetric job for $t < 0$.

Physical Insight

The Feynman propagator describes a single physical process viewed two ways. Read forward in time it is a particle created at the earlier point and absorbed at the later one; read with $t < 0$ it is the antiparticle (here, for the neutral scalar, the particle is its own antiparticle) running the other way. The $i\varepsilon$ is the bookkeeping that sends positive-energy quanta forward and negative-energy quanta backward in time, so that only physical, forward-in-time, positive-energy propagation ever occurs. This is Stueckelberg and Feynman's reinterpretation of antiparticles, here visible already in the free theory.

9.5.4 Independent check: the operator time-ordered product

The contour theorem was proved by *assuming* the momentum integral Eq. (9.154) and reducing it to the operator expression Eq. (9.153). It is worth confirming the operator expression itself directly from the mode expansion, with no contour

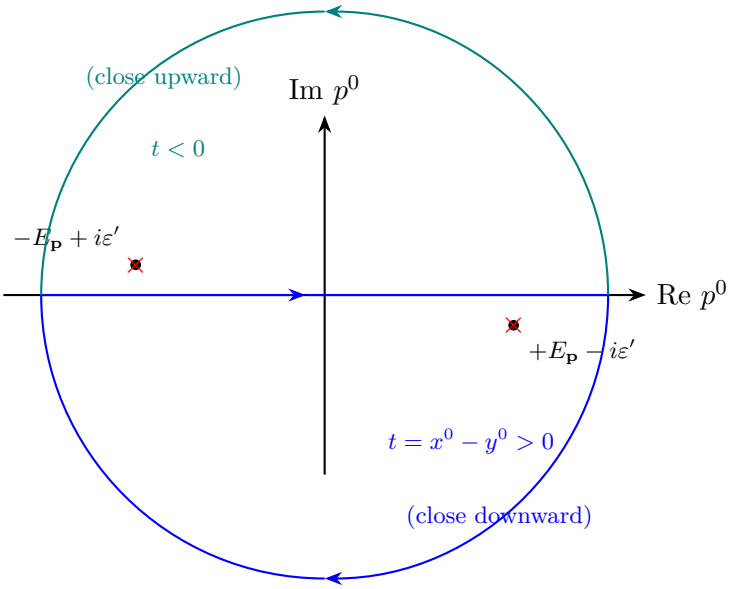


Figure 9.5: The $i\varepsilon$ prescription as a contour routing. The Feynman denominator $p^2 - m^2 + i\varepsilon$ places the pole at $p^0 = +E_{\mathbf{p}}$ just below the real axis and the pole at $p^0 = -E_{\mathbf{p}}$ just above it. For $x^0 > y^0$ the exponential $e^{-ip^0(x^0-y^0)}$ forces the contour closed in the lower half-plane (blue, clockwise), picking up only the positive-energy pole and yielding $D(x-y)$; for $x^0 < y^0$ it is closed in the upper half-plane (teal, counterclockwise), picking up the negative-energy pole and yielding $D(y-x)$. The single analytic prescription thus encodes the two-piece time-ordered product.

at all, so that the two starting points are seen to be genuinely the same object rather than one defined in terms of the other.

Worked Example 9.26 (The time-ordered product from the modes)**.** Compute $\langle 0|\,\mathcal{T}\,\phi(x)\phi(y)\,|0\rangle$ directly from Eq. (9.147).

For $x^0 > y^0$ the time-ordering symbol Eq. (9.151) returns $\phi(x)\phi(y)$, whose vacuum expectation value is, by definition, $D(x-y)$. The computation of $D(x-y)$ from the modes was already carried out in full in Eqs. (9.149) and (9.150): only the $a_{\mathbf{p}}a^\dagger_{\mathbf{q}}$ term survives, and

$$\langle 0|\,\phi(x)\phi(y)\,|0\rangle = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\,e^{-ip\cdot(x-y)} = D(x-y). \tag{9.168}$$

For $y^0 > x^0$ the symbol returns $\phi(y)\phi(x)$; the identical computation with $x \leftrightarrow y$ gives $\langle 0|\,\phi(y)\phi(x)\,|0\rangle = D(y-x)$. Assembling the two cases with their step functions reproduces Eq. (9.153) on the nose:

$$\langle 0|\,\mathcal{T}\,\phi(x)\phi(y)\,|0\rangle = \theta(x^0-y^0)D(x-y)+\theta(y^0-x^0)D(y-x) = D_F(x-y). \tag{9.169}$$

No analyticity, no contour, and no $i\varepsilon$ entered: the operator definition and the covariant integral Eq. (9.154) are two faces of the same distribution, and Theorem 9.25 is the bridge between them.

9.5.5 Locating the Feynman choice: retarded and advanced Green's functions

The denominator $1/(p^2-m^2)$ has poles on the real p^0 axis at $p^0=\pm E_{\mathbf{p}}$, and the integral Eq. (9.154) is ill-defined until we say how the contour avoids them. The $i\varepsilon$ is one such prescription; it is not the only one. The full set of inequivalent prescriptions corresponds to the four ways of routing the contour around the two poles, and the physically distinguished ones — besides Feynman's — are the retarded and advanced Green's functions. Comparing them locates the Feynman choice and prepares the causal analysis of the next section.

The retarded prescription pushes *both* poles below the real axis, which is achieved by the replacement $p^0\to p^0+i\varepsilon$ everywhere, i.e.

$$G_R(x-y)=\int\frac{d^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot(x-y)}}{(p^0+i\varepsilon)^2-E_{\mathbf{p}}^2}=\int\frac{d^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot(x-y)}}{(p^0+i\varepsilon-E_{\mathbf{p}})(p^0+i\varepsilon+E_{\mathbf{p}})}, \tag{9.170}$$

with poles at $p^0=\pm E_{\mathbf{p}}-i\varepsilon$, both in the lower half-plane. For $t<0$ we close upward and enclose nothing, so G_R vanishes for $x^0<y^0$ — it is *retarded*, supported on the future. For $t>0$ we close downward (clockwise) and pick up both residues. Their sum is

$$\mathrm{Res}_{p^0=E_{\mathbf{p}}}+\mathrm{Res}_{p^0=-E_{\mathbf{p}}}=\frac{i\,e^{-iE_{\mathbf{p}}t}}{2E_{\mathbf{p}}}+\frac{i\,e^{+iE_{\mathbf{p}}t}}{-2E_{\mathbf{p}}}=\frac{i}{2E_{\mathbf{p}}}\big(e^{-iE_{\mathbf{p}}t}-e^{+iE_{\mathbf{p}}t}\big)=\frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}}, \tag{9.171}$$

so that $\int dp^0/(2\pi)=\frac{1}{2\pi}(-2\pi i)\cdot(\sin E_{\mathbf{p}}t)/E_{\mathbf{p}}=-i\,(\sin E_{\mathbf{p}}t)/E_{\mathbf{p}}$, and

$$G_R(x-y)=-\,i\,\theta(x^0-y^0)\int\frac{d^3p}{(2\pi)^3}\,\frac{\sin E_{\mathbf{p}}t}{E_{\mathbf{p}}}\,e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}=\theta(x^0-y^0)\,\langle0|\,[\phi(x),\phi(y)]\,|0\rangle\,. \tag{9.172}$$

The last equality follows because the commutator function is, from properties (i) and (iv),

$$\langle0|\,[\phi(x),\phi(y)]\,|0\rangle=D(x-y)-D(y-x)=\int\frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\big(e^{-iE_{\mathbf{p}}t}-e^{+iE_{\mathbf{p}}t}\big)e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}=-\,i\int\frac{d^3$$

(9.173)

The advanced function is obtained by the mirror choice $p^0\to p^0-i\varepsilon$, placing both poles above the axis; it is supported on the past, $G_A(x-y)=-\theta(y^0-x^0)\,\langle0|\,[\phi(x),\phi(y)]\,|0\rangle$.

Two observations close the comparison. First, all three are *Green's functions* of the same Klein–Gordon operator (proved next), differing only by which homogeneous solution one adds: $G_R-G_A=D(x-y)-D(y-x)=\langle0|\,[\phi(x),\phi(y)]\,|0\rangle$

Prescription	Pole routing in p^0	Support / interpretation
Feynman D_F	$+E_{\mathbf{p}}$ below, $-E_{\mathbf{p}}$ above	all t; time-ordered product
Retarded G_R	both poles below	$t > 0$; response after the source
Advanced G_A	both poles above	$t < 0$; support in the past

Table 9.1: The three physically distinguished pole prescriptions for $1/(p^2 - m^2)$. All three are Green's functions of the Klein–Gordon operator (Theorem 9.27); they differ only by solutions of the homogeneous equation, and the difference is built from the on-shell commutator function Eq. (9.173).

is annihilated by $(\Box + m^2)$ because each Wightman piece is on shell. Second, the Feynman choice is the only one of the three that is neither purely retarded nor purely advanced; it propagates positive frequencies forward and negative frequencies backward, which is exactly why it, and not G_R or G_A, is the object generated by the time-ordered perturbation series. The classical theory of Chapter 6 naturally selects G_R (a field responds to its sources after they act); the quantum amplitude selects D_F.

Notation Watch

Conventions for the scalar Green's function differ across the literature by an overall factor of i and by a sign, because some authors define the propagator as the Green's function of $(\Box + m^2)$ with unit source while we define it as the time-ordered correlator. With the canon of this book — numerator i in Eq. (9.154), mostly-minus metric, $\Box = \partial^\mu \partial_\mu = \partial_t^2 - \nabla^2$ — the normalization is fixed once and for all by Theorem 9.27 below, $(\Box + m^2) D_F = -i\delta^4$. When importing a formula from another source, check its metric and its i before trusting a sign.

9.5.6 The propagator is the Green's function of the Klein–Gordon operator

We have called D_F a propagator and a Green's function; we now prove the second name. The Klein–Gordon operator is $(\Box_x + m^2)$ with $\Box_x = \partial_\mu^x \partial_x^\mu$, the operator whose vanishing is the free equation of motion Eq. (6.55).

Theorem 9.27 (Klein–Gordon Green's function)**.** *The Feynman propagator satisfies*

$$\boxed{(\Box_x + m^2)\, D_F(x - y) = -\, i\, \delta^4(x - y)} \tag{9.174}$$

and is therefore (up to the conventional factor $-i$) the Green's function of the Klein–Gordon operator. The same identity holds for G_R and G_A.

Proof, momentum space. Act with $(\Box_x+m^2)$ under the integral sign in Eq. (9.154). Since $\partial_\mu^x e^{-ip\cdot(x-y)} = -ip_\mu\, e^{-ip\cdot(x-y)}$,

$$(\Box_x+m^2)\, e^{-ip\cdot(x-y)} = \big((-ip)^\mu(-ip)_\mu+m^2\big)e^{-ip\cdot(x-y)} = (-p^2+m^2)\, e^{-ip\cdot(x-y)} = -(p^2-m^2)\, e^{-} \tag{9.175}$$

Therefore

$$(\Box_x+m^2)\, D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i\,\big(-(p^2-m^2)\big)}{p^2-m^2+i\varepsilon}\, e^{-ip\cdot(x-y)} = -\,i\int \frac{d^4p}{(2\pi)^4}\, \frac{p^2-m^2}{p^2-m^2+i\varepsilon}\, e^{-ip\cdot(x-} \tag{9.176}$$

Write the ratio as $\dfrac{p^2-m^2}{p^2-m^2+i\varepsilon} = 1 - \dfrac{i\varepsilon}{p^2-m^2+i\varepsilon}$. The second term vanishes as a tempered distribution when $\varepsilon \to 0^+$ (its only support is an infinitesimal neighborhood of the mass shell, which has measure zero against test functions; this is the statement of Section 5.1.5). The first term gives the complete momentum-space delta function (Theorem 5.18),

$$\int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot(x-y)} = \delta^4(x-y), \tag{9.177}$$

so that $(\Box_x + m^2)D_F(x-y) = -i\,\delta^4(x-y)$, as claimed. For G_R and G_A the numerator and the leading denominator are identical; only the routing of the poles differs, and that affects Eq. (9.176) only through the vanishing $i\varepsilon$ term. Hence all three prescriptions are Green's functions with the same right-hand side. □

The momentum-space proof is the quickest, but it conceals the physical origin of the delta function, which is the canonical equal-time commutator. The following independent derivation in position space makes that origin explicit and ties the result directly back to the quantization conditions of the chapter's opening.

Worked Example 9.28 (Position-space derivation of the Green's-function identity)**.** We verify Eq. (9.174) starting from the operator form Eq. (9.153), differentiating the step functions and using the canonical commutation relations. Write $t = x^0 - y^0$ and recall $\partial_{x^0}\theta(t) = \delta(t) = -\partial_{x^0}\theta(-t)$.

First time derivative. Differentiating Eq. (9.153),

$$\partial_{x^0} D_F = \delta(t)\big[D(x-y)-D(y-x)\big]+\theta(t)\,\partial_{x^0}D(x-y)+\theta(-t)\,\partial_{x^0}D(y-x). \tag{9.178}$$

The $\delta(t)$ bracket is evaluated at equal times $x^0 = y^0$, where $D(x-y)-D(y-x) = \langle 0|\,[\phi(x),\phi(y)]\,|0\rangle$ vanishes: the equal-time commutator of two fields is zero (Eq. (9.7)). The first term drops, leaving

$$\partial_{x^0} D_F = \theta(t)\,\partial_{x^0}D(x-y) + \theta(-t)\,\partial_{x^0}D(y-x). \tag{9.179}$$

Second time derivative. Differentiating again,

$$\partial_{x^0}^2 D_F = \delta(t)\,[\partial_{x^0} D(x-y) - \partial_{x^0} D(y-x)] + \theta(t)\,\partial_{x^0}^2 D(x-y) + \theta(-t)\,\partial_{x^0}^2 D(y-x). \tag{9.180}$$

The $\delta(t)$ bracket is now nontrivial. Using $\partial_{x^0} D(x-y) = \langle 0|\,\dot\phi(x)\phi(y)\,|0\rangle$ and $\partial_{x^0} D(y-x) = \langle 0|\,\phi(y)\dot\phi(x)\,|0\rangle$, the bracket at equal times is the vacuum expectation of the equal-time commutator of $\dot\phi$ with ϕ. Since $\pi = \dot\phi$ for the free real scalar, the canonical relation $[\phi(\mathbf{x}), \pi(\mathbf{y})] = i\delta^3(\mathbf{x}-\mathbf{y})$ gives $\left[\dot\phi(x), \phi(y)\right]\big|_{\text{ET}} = [\pi(\mathbf{x}), \phi(\mathbf{y})] = -i\delta^3(\mathbf{x}-\mathbf{y})$. Therefore

$$\delta(t)\,[\partial_{x^0} D(x-y) - \partial_{x^0} D(y-x)] = \delta(t)\,\big(-\,i\,\delta^3(\mathbf{x}-\mathbf{y})\big) = -\,i\,\delta^4(x-y). \tag{9.181}$$

Assembling the Klein–Gordon operator. The spatial and mass terms act only on the smooth Wightman functions and pass through the step functions untouched:

$$(-\nabla^2 + m^2) D_F = \theta(t)(-\nabla^2 + m^2) D(x-y) + \theta(-t)(-\nabla^2 + m^2) D(y-x). \tag{9.182}$$

Adding Eq. (9.180) and Eq. (9.182), and using $\Box = \partial_{x^0}^2 - \nabla^2$,

$$(\Box_x + m^2) D_F = -\,i\,\delta^4(x-y) + \theta(t)\,(\Box + m^2) D(x-y) + \theta(-t)\,(\Box + m^2) D(y-x). \tag{9.183}$$

But each Wightman piece is built from on-shell modes $e^{\mp ip\cdot(x-y)}$ with $p^2 = m^2$, so $(\Box + m^2)$ annihilates it: $(\Box + m^2)D = 0$, as in Eq. (9.175) with $p^2 = m^2$. The two θ-terms vanish identically, leaving

$$(\Box_x + m^2)\, D_F(x-y) = -\,i\,\delta^4(x-y), \tag{9.184}$$

in agreement with Eq. (9.174). The delta function is seen here to be *nothing but* the canonical equal-time commutator, propagated forward by the second time derivative of the time-ordering step functions — the dynamical content of quantization made visible in the propagator.

This completes the construction of the central dynamical object of the free scalar theory. The Feynman propagator Eq. (9.154) is simultaneously the time-ordered vacuum correlator, the inverse of the Klein–Gordon operator with Feynman boundary conditions, and the carrier of the $\lambda = 0$ sector of ϕ^4 theory that Chapter 10 will reproduce from $Z[J]$ and that becomes the internal line of every diagram from Chapters 14 and 15 onward. The comparison with the retarded and advanced functions has already isolated the on-shell commutator function $\langle 0|\,[\phi(x), \phi(y)]\,|0\rangle$ of Eq. (9.173); whether *that* object vanishes outside the light cone — the precise statement of relativistic causality, and the requirement that will force Bose statistics on the scalar — is the question taken up in Section 9.6.

9.6 Microcausality, the Commutator Function, and Spin-Statistics

The previous section handed us the field's two-point structure: the Wightman functions and, assembled from them through the time-ordered product, the Feynman propagator $D_F(x-y)$. Those objects answer the question *what amplitude connects two spacetime points*. We now ask a sharper question, the one on which the whole edifice of relativistic quantum theory either stands or falls: *can a measurement made here and now disturb a measurement made simultaneously a light-year away*? In ordinary quantum mechanics there is no obstruction in principle—the formalism is built on a single absolute time and says nothing about how fast influences propagate. In a relativistic theory there must be an obstruction, because a superluminal influence is, in some other inertial frame, an influence that runs backwards in time, and that way lie closed causal loops and the collapse of physical prediction.

The remarkable fact this section establishes is that the free scalar field, quantized exactly as the earlier sections prescribed, already encodes the prohibition. The carrier of the answer is a single c-number distribution, the *Pauli–Jordan commutator function* $\Delta(x-y)$, and the entire content of relativistic causality at the level of free fields is the statement that Δ vanishes outside the light cone. We will compute Δ from the mode expansion, prove its vanishing at spacelike separation by an argument that uses nothing more than the connectedness of the proper orthochronous Lorentz group, and read off the physical meaning: spacelike-separated local observables commute, so no measurement can signal faster than light. The same computation then pays an unexpected dividend. If we had stubbornly quantized the scalar with anticommutators instead of commutators, the object forced to vanish for causality would have been the wrong kind of function—an *even* one rather than an *odd* one—and explicit calculation shows it does not vanish at spacelike separation. Causality therefore *selects* Bose statistics for the spin-0 field. This is the spin-statistics connection in the one instance where this chapter can establish it honestly and completely.

9.6.1 The commutator is a c-number

We work throughout in the Heisenberg picture, where the free field carries its full spacetime dependence in the mode expansion established earlier in this chapter,

$$\phi(x) = \int \frac{d^3p}{(2\pi)^3\, 2\omega_{\mathbf{p}}} \Big(a_{\mathbf{p}}\, e^{-ip\cdot x} + a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} \Big), \qquad p^0 = \omega_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{9.185}$$

with the ladder algebra

$$\Big[a_{\mathbf{p}}, a^\dagger_{\mathbf{q}} \Big] = (2\pi)^3\, 2\omega_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}), \qquad [a_{\mathbf{p}}, a_{\mathbf{q}}] = \Big[a^\dagger_{\mathbf{p}}, a^\dagger_{\mathbf{q}} \Big] = 0 \tag{9.186}$$

fixed in Eq. (9.54). Here $p\cdot x = \omega_{\mathbf{p}} x^0 - \mathbf{p}\cdot\mathbf{x}$ is the on-shell Minkowski product in the mostly-minus metric, and the relativistically normalized measure $d^3p/[(2\pi)^3\,2\omega_{\mathbf{p}}]$ is the one used book-wide.

Consider the equal-or-unequal-time commutator $[\phi(x),\phi(y)]$ of two fields at arbitrary spacetime points. Insert Eq. (9.147) for each field. A commutator of sums is the sum of commutators, and the only surviving terms are those pairing one annihilation with one creation operator, since by Eq. (9.67) the aa and $a^\dagger a^\dagger$ pieces commute:

$$\begin{aligned}[\phi(x),\phi(y)] &= \int \frac{d^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\frac{d^3q}{(2\pi)^3\,2\omega_{\mathbf{q}}}\Big(\left[a_{\mathbf{p}},a^\dagger_{\mathbf{q}}\right]e^{-ip\cdot x+iq\cdot y}+\left[a^\dagger_{\mathbf{p}},a_{\mathbf{q}}\right]e^{+ip\cdot x-iq\cdot y}\Big)\\ &= \int \frac{d^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\frac{d^3q}{(2\pi)^3\,2\omega_{\mathbf{q}}}(2\pi)^3\,2\omega_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})\Big(e^{-ip\cdot x+iq\cdot y}-e^{+ip\cdot x-iq\cdot y}\Big),\end{aligned} \tag{9.187}$$

where in the second line we used $\left[a^\dagger_{\mathbf{p}},a_{\mathbf{q}}\right] = -\left[a_{\mathbf{q}},a^\dagger_{\mathbf{p}}\right] = -(2\pi)^3\,2\omega_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})$. The delta function collapses the $\mathbf{q}$ integral, setting $\mathbf{q}=\mathbf{p}$ and hence $\omega_{\mathbf{q}}=\omega_{\mathbf{p}}$, so that the commutator's $2\omega_{\mathbf{p}}$ cancels the measure factor $1/2\omega_{\mathbf{q}}$ and leaves the single invariant measure $1/(2\omega_{\mathbf{p}})$:

$$\boxed{[\phi(x),\phi(y)] = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{p}}}\Big(e^{-ip\cdot(x-y)}-e^{+ip\cdot(x-y)}\Big) \equiv i\,\Delta(x-y).} \tag{9.188}$$

Two features deserve emphasis at once. First, the right-hand side is a *c-number*: the operator-valued ladder operators have cancelled, leaving an ordinary distribution multiplying the identity. Equivalently, $[\phi(x),\phi(y)]$ commutes with every operator in the theory; the commutator of two free fields is not itself a dynamical quantity. Second, the result depends only on the difference $x-y$, as translation invariance demands. The factor i in the definition is conventional and makes Δ real, as we now confirm.

Definition 9.29 (Pauli–Jordan function)**.** The *Pauli–Jordan (or commutator) function* of the free scalar field of mass m is the c-number distribution

$$\Delta(x) \;=\; -\,i\int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{p}}}\Big(e^{-ip\cdot x}-e^{+ip\cdot x}\Big), \qquad p^0=\omega_{\mathbf{p}}, \tag{9.189}$$

so that $[\phi(x),\phi(y)] = i\,\Delta(x-y)$.

It is useful to record three elementary properties directly from Eq. (9.189), each obtained by a relabelling of the integration variable.

Reality. Complex conjugation sends $e^{\mp ip\cdot x}\to e^{\pm ip\cdot x}$ and $-i\to+i$, which together reproduce Eq. (9.189) unchanged; thus $\Delta(x)^* = \Delta(x)$, and Δ is real. Consistently, $[\phi(x),\phi(y)]^\dagger = [\phi(y),\phi(x)] = -\,[\phi(x),\phi(y)]$ is anti-Hermitian, matching the explicit factor i.

Oddness. Sending $x \to -x$ in Eq. (9.189) interchanges the two exponentials and so flips the overall sign,

$$\Delta(-x) = -\,\Delta(x). \tag{9.190}$$

This single fact will do almost all the work in the microcausality proof.

Equal-time values. Setting $x^0 = 0$ in Eq. (9.189), the two exponentials become $e^{\pm i\mathbf{p}\cdot\mathbf{x}}$, which map into one another under $\mathbf{p} \to -\mathbf{p}$; the integral therefore vanishes,

$$\Delta(0, \mathbf{x}) = 0. \tag{9.191}$$

This is exactly the canonical statement $[\phi(\mathbf{x}, t), \phi(\mathbf{y}, t)] = 0$ at equal times. Differentiating Eq. (9.189) once with respect to x^0 and then setting $x^0 = 0$ brings down a factor $\mp i\omega_{\mathbf{p}}$ from each exponential; the surviving combination is $e^{+i\mathbf{p}\cdot\mathbf{x}} + e^{-i\mathbf{p}\cdot\mathbf{x}}$ with the $\omega_{\mathbf{p}}$ cancelling the measure, giving

$$\partial_0 \Delta(x)\big|_{x^0=0} = -\int \frac{d^3p}{(2\pi)^3}\, e^{i\mathbf{p}\cdot\mathbf{x}} = -\,\delta^{(3)}(\mathbf{x}), \tag{9.192}$$

where the last step is the Fourier representation of the delta distribution recalled in Theorem 5.5. Since $\pi(y) = \partial_{y^0}\phi(y)$, applying ∂_{y^0} to $[\phi(x), \phi(y)] = i\Delta(x - y)$ and evaluating at $x^0 = y^0$ reproduces the canonical relation $[\phi(\mathbf{x}, t), \pi(\mathbf{y}, t)] = i\,\delta^{(3)}(\mathbf{x} - \mathbf{y})$. The Pauli–Jordan function thus packages the entire equal-time canonical structure together with its covariant evolution off equal times into one Lorentz object.

9.6.2 Covariant form and Lorentz invariance

The expression Eq. (9.189) singles out a frame through the choice $p^0 = \omega_{\mathbf{p}} > 0$, yet the result should not. To make the invariance manifest we rewrite the three-dimensional on-shell measure as a four-dimensional integral over an unconstrained p restricted to the mass shell. The identity

$$\int \frac{d^4p}{(2\pi)^4}\, 2\pi\,\delta(p^2 - m^2)\,\theta(p^0)\; f(p) = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{p}}}\; f(\omega_{\mathbf{p}}, \mathbf{p}) \tag{9.193}$$

follows from $\delta(p^2 - m^2) = \delta((p^0)^2 - \omega_{\mathbf{p}}^2) = \frac{1}{2\omega_{\mathbf{p}}}\big[\delta(p^0 - \omega_{\mathbf{p}}) + \delta(p^0 + \omega_{\mathbf{p}})\big]$ and the step function $\theta(p^0)$ keeping only the positive root; it is the Lorentz-invariant measure used throughout the book. Applying Eq. (9.193) to the first exponential in Eq. (9.189), and to the second after the substitution $p \to -p$ (which turns $e^{+ip\cdot x}$ into $e^{-ip\cdot x}$ supported on $p^0 = -\omega_{\mathbf{p}}$, i.e. the lower mass-shell branch carried by $\theta(-p^0)$), the two terms combine into a single integral over both branches weighted by the sign of the energy:

$$\boxed{\Delta(x) = -\,i\int \frac{d^4p}{(2\pi)^4}\, 2\pi\,\delta(p^2 - m^2)\;\mathrm{sgn}(p^0)\, e^{-ip\cdot x}.} \tag{9.194}$$

Every ingredient of Eq. (9.194) is now under Lorentz control. The measure d^4p is invariant; the on-shell delta $\delta(p^2 - m^2)$ is invariant because p^2 is; the plane wave $e^{-ip\cdot x}$ is invariant. The only factor whose invariance requires comment is $\mathrm{sgn}(p^0)$.

Proposition 9.30 (Invariance and oddness of Δ)**.** *For every proper orthochronous Lorentz transformation $\Lambda \in L_+^\uparrow$,*

$$\Delta(\Lambda x) = \Delta(x), \tag{9.195}$$

and Δ is odd, $\Delta(-x) = -\Delta(x)$.

Proof. Oddness is Eq. (9.190). For the invariance, apply the change of variables $p \to \Lambda p$ in Eq. (9.194). The measure and the on-shell delta are unchanged, and $e^{-i(\Lambda p)\cdot x} = e^{-ip\cdot(\Lambda^{-1}x)}$ since the Minkowski product is invariant. It remains to check that $\mathrm{sgn}(p^0)$ is preserved. On the mass shell $p^2 = m^2 > 0$ the four-vector p is timelike, and an orthochronous transformation is precisely one that does not reverse the time-orientation of timelike vectors: it maps the forward light cone to itself, so $\theta(p^0)$ and $\theta(-p^0)$ are individually preserved and hence so is $\mathrm{sgn}(p^0)$. Therefore the integrand is invariant and $\Delta(\Lambda^{-1}x) = \Delta(x)$ for all $\Lambda \in L_+^\uparrow$; renaming $\Lambda^{-1} \to \Lambda$ gives Eq. (9.195). (That $L_+^\uparrow$ is exactly the time-orientation-preserving, parity-and-time-reversal-free component is the content of Theorem 3.6.) □

The restriction to $L_+^\uparrow$ in Theorem 9.30 is essential and not a technicality: time reversal would flip $\mathrm{sgn}(p^0)$ and send $\Delta \to -\Delta$, and it is the interplay between this restriction and the invariance that makes the next theorem possible.

9.6.3 The microcausality theorem

We can now prove the central result. Recall (Section 3.1.1) that the separation $x - y$ is *spacelike* when $(x-y)^2 < 0$, *timelike* when $(x-y)^2 > 0$, and *lightlike* when $(x-y)^2 = 0$; spacelike-separated points lie outside each other's light cones and admit no causal influence between them.

Theorem 9.31 (Microcausality of the free scalar)**.** *The Pauli–Jordan function vanishes at spacelike separation:*

$$\Delta(x) = 0 \qquad \textit{whenever} \qquad x^2 < 0, \tag{9.196}$$

and consequently

$$[\phi(x), \phi(y)] = 0 \qquad \textit{whenever} \qquad (x-y)^2 < 0. \tag{9.197}$$

Proof. The argument turns on a single geometric fact about spacelike vectors: *there exists* $\Lambda \in L_+^\uparrow$ *with* $\Lambda x = -x$. Grant this for a moment. Then by invariance Eq. (9.195), $\Delta(x) = \Delta(\Lambda x) = \Delta(-x)$; but by oddness Eq. (9.190), $\Delta(-x) = -\Delta(x)$. The two equalities give $\Delta(x) = -\Delta(x)$, hence $2\Delta(x) = 0$, hence $\Delta(x) = 0$. Equation Eq. (9.197) then follows from $[\phi(x), \phi(y)] = i\Delta(x-y)$ because $(x-y)^2 < 0$.

It remains to exhibit the transformation. Let x be spacelike, $x^2 = -a^2$ with $a > 0$. By the orbit structure of $L_+^\uparrow$ (Theorem 3.6), any spacelike vector can be brought to a standard purely spatial form by a proper orthochronous transformation: first apply a boost in the plane spanned by the time axis and $\mathbf{x}$ to set the time component to zero—this is possible precisely because $|\mathbf{x}| > |x^0|$ for a spacelike vector, so the requisite rapidity is real—and then a rotation to align the result with the third spatial axis. The net transformation $R \in L_+^\uparrow$ achieves $Rx = (0, 0, 0, a)$. Now perform a rotation by angle π about the *first* spatial axis; call it $\mathcal{R}_\pi$. It acts as $(x^1, x^2, x^3) \mapsto (x^1, -x^2, -x^3)$ on spatial components and fixes time, so $\mathcal{R}_\pi(0, 0, 0, a) = (0, 0, 0, -a) = -Rx$. Rotations are rotations: $\mathcal{R}_\pi \in L_+^\uparrow$ (determinant $+1$, time-orientation preserving). Therefore

$$\Lambda \;\equiv\; R^{-1}\,\mathcal{R}_\pi\,R \;\in\; L_+^\uparrow \qquad \text{satisfies} \qquad \Lambda x = R^{-1}\mathcal{R}_\pi(Rx) = R^{-1}(-Rx) = -x, \tag{9.198}$$

which is the claim. □

Physical Insight

The proof reduces a deep physical principle to one picture (Fig. 9.6). A spacelike separation has no invariant time-ordering: in one frame x happens before y, in another after, and there is even a frame in which the labels are swapped. Mathematically, the Lorentz orbit of a spacelike vector is the *single-sheeted* hyperboloid $x^2 = -a^2$, which is connected and contains both x and $-x$; the connectedness of $L_+^\uparrow$ lets us slide continuously from x to $-x$ inside it. A timelike vector, by contrast, lives on a *two-sheeted* hyperboloid $x^2 = +a^2$, and $L_+^\uparrow$—preserving the sign of the time component—can never carry the future sheet to the past sheet. There is no $\Lambda \in L_+^\uparrow$ with $\Lambda x = -x$ for timelike x, the proof's mechanism fails, and indeed $\Delta(x) \neq 0$ inside the light cone: a causal influence between timelike-separated points is allowed, as it must be.

9.6.4 Causality as the commuting of local observables

What Theorem 9.31 establishes about the elementary field propagates immediately to every *local observable*. A local observable supported near a spacetime point x is built as a (Hermitian, normal-ordered) polynomial in the field and its derivatives

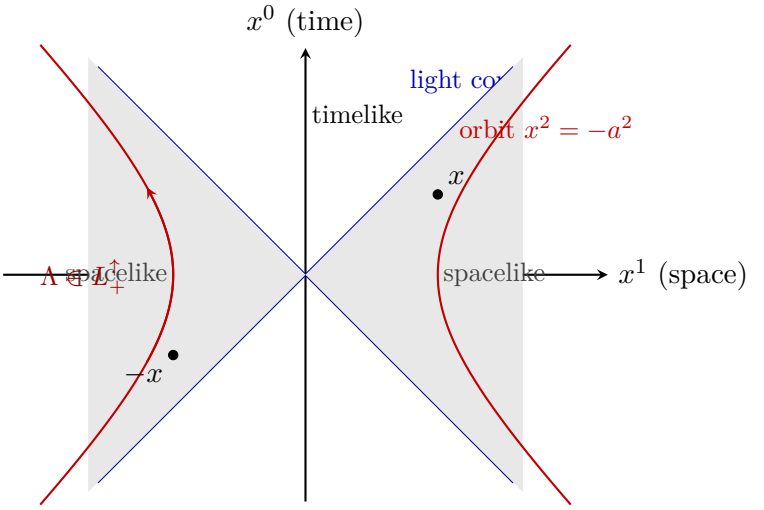


Figure 9.6: Microcausality geometry. For a spacelike separation x (with $x^2 = -a^2 < 0$) the Lorentz orbit is the single-sheeted hyperbola $x^2 = -a^2$ (red), a connected curve that contains both x and its reflection $-x$. A proper orthochronous transformation $\Lambda \in L_+^\uparrow$ slides x to $-x$ along this orbit. Since Δ is invariant under Λ and odd under $x \to -x$, it must vanish on the shaded spacelike region. A timelike x lies on a two-sheeted hyperbola whose future and past branches $L_+^\uparrow$ cannot interchange, so no such Λ exists and $\Delta \neq 0$ inside the cone.

evaluated there—the energy density $\frac{1}{2}{:}\pi^2{:} + \frac{1}{2}{:}(\nabla\phi)^2{:} + \frac{1}{2}m^2{:}\phi^2{:}$, the number-density-like ${:}\phi^2{:}$, and so on. Let $\mathcal{O}_A(x)$ and $\mathcal{O}_B(y)$ be two such observables. Each is a polynomial in fields at x and at y respectively. Expanding the commutator $[\mathcal{O}_A(x), \mathcal{O}_B(y)]$ by the Leibniz rule produces a sum of terms, each of which is a product of fields multiplied by at least one elementary commutator $[\phi(x), \phi(y)]$ (derivatives of the field commute through to act on Δ, since $\partial_\mu^x [\phi(x), \phi(y)] = i\,\partial_\mu \Delta(x-y)$). Because that elementary commutator is the c-number $i\Delta(x-y)$, and because Δ and all its derivatives vanish for $(x-y)^2 < 0$ (differentiating the identically-zero function on the open spacelike region leaves zero), every term vanishes. Hence

$$[\mathcal{O}_A(x), \mathcal{O}_B(y)] = 0 \qquad \text{for} \qquad (x-y)^2 < 0. \tag{9.199}$$

This is the field-theoretic statement of causality, and it is worth saying plainly why it is exactly the right statement. In quantum mechanics, two observables can be measured without one disturbing the other—can be measured *simultaneously* and *independently*—if and only if they commute. Equation Eq. (9.199) asserts that any two observables localized at spacelike separation commute. Therefore a measurement carried out in one laboratory cannot influence the statistics

of any measurement carried out, at that instant, outside its light cone. No expectation value, no probability distribution, nothing an experimenter could detect, responds to a spacelike-separated intervention. Whatever correlations exist between spacelike regions—and entangled states have plenty—they cannot be used to transmit a signal, because the local measurement operators simply commute. A relativistic quantum field theory is, by construction, a no-signalling theory. We elevate Eq. (9.199) to a defining axiom of any local relativistic quantum theory, and call it *microcausality* or *local commutativity*; the free scalar is our first concrete model in which it is a theorem rather than a postulate.

Caution

Microcausality forbids *signalling*, not *correlation*. The two-point Wightman function $\langle\phi(x)\phi(y)\rangle$ does not vanish at spacelike separation—it falls off as e^{-mr} over distances $r \gtrsim 1/m$, as the worked example below makes quantitative. Spacelike correlations are real and physical. What vanishes is the *commutator*, and with it any possibility of using those correlations to send information faster than light. Confusing the non-vanishing of the correlator with a violation of causality is a perennial error; the propagator's spacelike tail and microcausality are perfectly compatible precisely because $D_F = \langle \mathcal{T}\phi\phi\rangle$ is symmetric while the causal content lives in the antisymmetric part Δ.

9.6.5 Spin and statistics: why the scalar must be Bose

We arrived at microcausality having quantized with *commutators*, Eq. (9.67). Suppose, perversely, that we had not—that we had tried to build the scalar's Fock space on *anticommutators*, declaring

$$\left\{a_{\mathbf{p}}, a^{\dagger}_{\mathbf{q}}\right\} = (2\pi)^3\, 2\omega_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}), \qquad \{a_{\mathbf{p}}, a_{\mathbf{q}}\} = \left\{a^{\dagger}_{\mathbf{p}}, a^{\dagger}_{\mathbf{q}}\right\} = 0, \tag{9.200}$$

which is the Fermi–Dirac choice that, for half-integer spin, will turn out to be mandatory in Chapter 11. Does it produce a causal theory for the scalar? The test is whether observables built from ϕ commute at spacelike separation. Repeat the computation of Section 9.6.1, but now the object that collapses to a c-number is the *anticommutator* of the fields, because it is $\left\{a, a^{\dagger}\right\}$ that is a number under Eq. (9.200). Pairing annihilation with creation operators exactly as in Eq. (9.187),

$$\begin{aligned}\{\phi(x),\phi(y)\} &= \int \frac{d^3p}{(2\pi)^3\,2\omega_{\mathbf{p}}}\frac{d^3q}{(2\pi)^3\,2\omega_{\mathbf{q}}}(2\pi)^3\,2\omega_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})\Big(e^{-ip\cdot x+iq\cdot y}+e^{+ip\cdot x-iq\cdot y}\Big)\\ &= \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{p}}}\Big(e^{-ip\cdot(x-y)}+e^{+ip\cdot(x-y)}\Big) \;\equiv\; \Delta_+(x-y)+\Delta_+(y-x),\end{aligned} \tag{9.201}$$

where the relative sign is now + because $\left\{a^\dagger_{\mathbf{p}}, a_{\mathbf{q}}\right\} = +\left\{a_{\mathbf{q}}, a^\dagger_{\mathbf{p}}\right\} = +(2\pi)^3\, 2\omega_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q})$, and we have abbreviated the positive-frequency Wightman function

$$\Delta_+(x) \;\equiv\; \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{p}}}\, e^{-ip\cdot x} \;=\; \langle \phi(x)\phi(0)\rangle\,, \tag{9.202}$$

the object computed in the previous section. The contrast with Eq. (9.188) is the entire story. There the relative sign was $-$, producing the *odd* function Δ; here it is $+$, producing the manifestly *even* function $\Delta_+(x-y)+\Delta_+(y-x)$, symmetric under $x \leftrightarrow y$.

Oddness was the linchpin of Theorem 9.31: the Lorentz orbit argument forced Δ to equal both $+\Delta$ and $-\Delta$ at spacelike separation, killing it. Run the same argument on the anticommutator and it gives nothing. For spacelike $x-y$ we still have a $\Lambda \in L^\uparrow_+$ with $\Lambda(x-y) = -(x-y)$, and Δ_+ is Lorentz invariant, so the function maps to itself—but, being even, it was already invariant under the reflection, and we learn only $0=0$. No constraint. To find out whether it actually vanishes we must compute, and the computation (carried out in full in Theorem 9.34) returns a strictly positive answer:

$$\{\phi(x),\phi(y)\}\Big|_{(x-y)^2<0} = 2\,\Delta_+(x-y) = \frac{m}{2\pi^2 r}\,K_1(mr) \;>\; 0, \qquad r \equiv \sqrt{-(x-y)^2}, \tag{9.203}$$

where K_1 is a modified Bessel function and we used that Δ_+ is symmetric, $\Delta_+(x) = \Delta_+(-x)$, for spacelike x (its argument and reflection are related by $L^\uparrow_+$). The anticommutator does *not* vanish outside the light cone.

Why this dooms causality is now immediate. Under the Fermi assumption the anticommutator $\{\phi(x),\phi(y)\}$ is the c-number Eq. (9.203), while the *commutator*—the object that controls whether observables built from ϕ can be measured independently—is no longer a c-number at all. Solving $\left\{a_{\mathbf{p}}, a^\dagger_{\mathbf{q}}\right\} = (2\pi)^3 2\omega_{\mathbf{p}}\delta^{(3)}$ for the commutator gives $\left[a_{\mathbf{p}}, a^\dagger_{\mathbf{q}}\right] = (2\pi)^3\, 2\omega_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}) - 2\, a^\dagger_{\mathbf{q}} a_{\mathbf{p}}$, an operator; assembled into $[\phi(x),\phi(y)]$ it remains operator-valued and does not vanish for spacelike $x-y$. The Hermitian observable $\phi(x)$ would then fail to commute with $\phi(y)$ at spacelike separation, in flat contradiction with the no-signalling requirement Eq. (9.199). There is no escape by redefining observables: the symmetric, strictly positive c-number Eq. (9.203) is forced, and a non-zero symmetric two-point structure with operator-valued commutator cannot satisfy local commutativity. We conclude:

Theorem 9.32 (Spin-statistics for the scalar field)**.** *The free spin-0 field cannot be consistently quantized with anticommutators: the Fermi algebra Eq. (9.200) yields a non-vanishing anticommutator Eq. (9.203) and an operator-valued, non-vanishing commutator at spacelike separation, violating microcausality Eq. (9.199).*

Causality therefore forces *Bose–Einstein quantization, the commutator algebra Eq. (9.67), for which* $[\phi(x), \phi(y)] = i\Delta(x-y) = 0$ *outside the light cone. A spin-0 field is necessarily a boson.*

Physical Insight

The decisive quantity is the *sign* that the field commutator/anticommutator carries between its two exponentials, and that sign is fixed by the rotational (Lorentz) transformation property of the field—i.e. by its spin. For the scalar, which is rotationally inert, microcausality wants the *odd* combination, and the odd combination is the commutator: Bose. The deep version of the theorem (below) shows that a field in a representation of half-integer spin carries an extra minus sign under the 2π rotation hidden in the analytic continuation relating $\Delta_+(x)$ to $\Delta_+(-x)$, which flips precisely this preference: for half-integer spin it is the *anticommutator* that microcausality selects. Spin dictates statistics through the geometry of the rotation group.

9.6.6 The general spin-statistics theorem: statement and scope

What we have proved is the spin-0 instance of a sweeping structural result. It is worth stating the general theorem precisely, together with an honest account of which of its hypotheses this chapter has actually used and which await later machinery.

Theorem 9.33 (Spin-statistics connection, general form—stated)**.** *Let a quantum field theory satisfy: (i) invariance under the proper orthochronous Poincaré group, with fields transforming in finite-dimensional representations of the Lorentz group labelled by their spin j; (ii) a Hilbert space with positive-definite inner product (no negative-norm physical states) and a unique Poincaré-invariant vacuum of lowest energy (positivity of the energy spectrum); (iii) microcausality, i.e. local commutativity Eq. (9.199) of observables at spacelike separation. Then fields of integer spin $j \in \mathbb{Z}$ must be quantized with* commutators *(Bose–Einstein statistics), and fields of half-integer spin $j \in \mathbb{Z} + \frac{1}{2}$ with* anticommutators *(Fermi–Dirac statistics). The opposite choice in either case is inconsistent: it produces either a violation of microcausality or a Hamiltonian unbounded below / a state space with negative norms.*

The proof of Theorem 9.33 in full generality rests on the analytic structure of the vacuum expectation values $\langle \phi(x_1) \cdots \phi(x_n) \rangle$—specifically on the fact that positivity of the energy (hypothesis (ii)) makes the Wightman functions boundary values of functions analytic in a forward tube, so that the spacelike continuation relating $\Delta_+(x)$ to $\Delta_+(-x)$ can be performed, and on the careful bookkeeping of the $(-1)^{2j}$ that a spin-j field's Lorentz indices accumulate under that continuation.

This is the content of the Wightman axioms and the analytic-continuation argument; it is genuinely a Part-of-the-book-level result and we do not reproduce its complete machinery here, because the analyticity it needs has not yet been developed. What this chapter *has* delivered, in full and without gaps, is the $j = 0$ case: there the Lorentz representation is trivial, the $(-1)^{2j} = +1$ bookkeeping is invisible, and the whole argument collapses to the elementary even-versus-odd dichotomy of Section 9.6.5. The reader has therefore seen the real mechanism—causality choosing a sign, spin fixing which sign is allowed—in its cleanest example.

The complementary half-integer case is exhibited in Chapter 11, where the Dirac field is quantized. There the logic runs in reverse: attempting to impose *commutators* on the spin-$\frac{1}{2}$ field produces a Hamiltonian with no lower bound (and, equivalently, a spacelike-non-vanishing object of the wrong symmetry), and only *anticommutators* restore both positivity of the energy and microcausality. Reading Theorem 9.32 alongside the Dirac result is reading the two faces of one theorem: for the boson, commutators are forced; for the fermion, anticommutators are forced; in each case it is the marriage of relativity (Lorentz invariance plus causality) with quantum mechanics (positivity of the spectrum) that does the forcing, and neither half is optional.

Historical Note

The connection was discovered piecemeal. Fierz (1939) and Pauli (1940) gave the first general arguments tying statistics to spin via the requirement of a positive energy and (for bosons) the vanishing of spacelike commutators; the modern axiomatic proof, resting on the analyticity of Wightman functions, is due to Lüders and Zumino and to Burgoyne in the 1950s, and is set out in the monograph of Streater and Wightman. The spin-0 calculation of this section is essentially Pauli and Jordan's original 1928 computation of the function that now bears their names, repurposed to make the statistics argument transparent.

We close the formal development with the worked computation promised above, which both supplies the number quoted in Eq. (9.203) and makes the propagator's spacelike behaviour concrete.

Worked Example 9.34 (The spacelike Wightman function and the non-vanishing anticommutator)**.** We evaluate $\Delta_+(x)$ of Eq. (9.202) for a spacelike separation and confirm Eq. (9.203). By Lorentz invariance of Δ_+ (it is built from the invariant measure and depends on x only through invariants, with the energy sign fixed) we may choose the frame in which the separation is purely spatial, $x = (0, \mathbf{r})$ with $|\mathbf{r}| = r = \sqrt{-x^2}$. Then $p \cdot x = -\mathbf{p} \cdot \mathbf{r}$ and

$$\Delta_+(0, \mathbf{r}) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2\omega_{\mathbf{p}}} e^{i\mathbf{p}\cdot\mathbf{r}}. \tag{9.204}$$

Carry out the angular integration. Writing $d^3p = p^2\,dp\,d\Omega$ with $p = |\mathbf{p}|$ and aligning the polar axis with $\mathbf{r}$,

$$\int d\Omega\, e^{i\mathbf{p}\cdot\mathbf{r}} = 2\pi \int_{-1}^{1} d(\cos\theta)\, e^{ipr\cos\theta} = \frac{4\pi}{pr}\sin(pr), \tag{9.205}$$

so that, with $\omega_{\mathbf{p}} = \sqrt{p^2+m^2}$,

$$\Delta_+(0,\mathbf{r}) = \frac{1}{(2\pi)^3}\cdot\frac{4\pi}{r}\int_0^\infty dp\,\frac{p^2}{2\sqrt{p^2+m^2}}\,\frac{\sin(pr)}{p} = \frac{1}{4\pi^2 r}\int_0^\infty dp\,\frac{p\,\sin(pr)}{\sqrt{p^2+m^2}}. \tag{9.206}$$

The radial integral is evaluated by relating it to a standard integral representation of the modified Bessel function K_0, namely

$$\int_0^\infty dp\,\frac{\cos(pr)}{\sqrt{p^2+m^2}} = K_0(mr), \tag{9.207}$$

valid for $m, r > 0$. Differentiating Eq. (9.207) with respect to r brings down $-p\sin(pr)$ on the left and uses $K_0'(z) = -K_1(z)$ on the right:

$$\int_0^\infty dp\,\frac{p\,\sin(pr)}{\sqrt{p^2+m^2}} = -\frac{d}{dr}\int_0^\infty dp\,\frac{\cos(pr)}{\sqrt{p^2+m^2}} = -\frac{d}{dr}K_0(mr) = m\,K_1(mr). \tag{9.208}$$

Substituting into Eq. (9.206),

$$\boxed{\Delta_+(0,\mathbf{r}) = \frac{m}{4\pi^2 r}\,K_1(mr).} \tag{9.209}$$

Since $K_1(z) > 0$ for all $z > 0$, the Wightman function is strictly positive at every spacelike separation. The anticommutator Eq. (9.201) at spacelike separation is $\Delta_+(x-y) + \Delta_+(y-x) = 2\Delta_+(0,\mathbf{r}) = \dfrac{m}{2\pi^2 r}K_1(mr) > 0$, which is Eq. (9.203).

The asymptotics make the physics vivid. Using $K_1(z) \sim \sqrt{\pi/2z}\,e^{-z}$ as $z \to \infty$,

$$\Delta_+(0,\mathbf{r}) \xrightarrow{mr\gg 1} \frac{m}{4\pi^2 r}\sqrt{\frac{\pi}{2mr}}\,e^{-mr} = \frac{1}{4\pi^2}\sqrt{\frac{\pi m}{2}}\,\frac{e^{-mr}}{r^{3/2}}. \tag{9.210}$$

Spacelike correlations are non-zero but Yukawa-suppressed over the Compton wavelength $1/m$: a particle can be "found" a spacelike distance away only with an amplitude that dies exponentially beyond $r \sim 1/m$. This is the same e^{-mr} falloff noted for the propagator's spacelike tail in the previous section, and it is the quantitative companion to the caution above—the correlator is non-zero, yet, because the commutator Δ vanishes identically there, no signal rides on it.

With Theorems 9.31 and 9.32 the free real scalar is now a fully solved, demonstrably causal relativistic quantum theory: its excitations are bosons, its

observables respect the light cone, and its propagator is the Green's function established in the previous section. These are precisely the consistency facts that all later ϕ^4 perturbation theory will assume in silence. The construction so far has used only the field's reality; lifting that restriction to a *complex* scalar will, in the next section, expose a conserved U(1) charge and split the single boson into a particle and a distinct antiparticle—the structure on which the charge of the QED and gauge threads is built.

9.7 The Charged Scalar: Antiparticles and Conserved Charge

Every construction in this chapter so far has rested on a single real field. A real field is the simplest possible relativistic quantum field, and it taught us the whole machine: the mode expansion as a continuum of oscillators, the Fock space built on a vacuum, the propagator with its $i\varepsilon$ contour, and the microcausality that the previous section showed forces Bose statistics. But a real field describes only one kind of quantum. To make contact with the world of electrons, pions, and the gauge charges that organize the Standard Model, we need a field that distinguishes a quantum from its mirror image – a *particle* from an *antiparticle* – and that carries a conserved internal quantum number. The minimal such object is the complex (charged) scalar field. Quantizing it is almost no extra work: a complex field is two real fields in a trench coat, so the oscillator engine runs unchanged. The payoff is large and structural. Two independent species of quantum emerge from one field; they have rigorously equal mass and exactly opposite charge; and the global U(1) symmetry whose Noether current we constructed classically in Chapter 7 reappears as a self-adjoint operator Q whose eigenvalues label those species. This is the section in which "charge" first becomes a quantum-mechanical observable, and in which antiparticles – glimpsed already in the negative-frequency half of the real field's mode expansion – acquire an independent operational identity.

We continue ϕ^4 theory at its free ($\lambda = 0$) level and place the carried real scalar in its proper context: it is the special, self-conjugate case of the construction below. The standing conventions are those of the rest of the chapter – mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, on-shell energy $E_{\mathbf{p}} = +\sqrt{\mathbf{p}^2 + m^2}$, four-momentum $p^0 = E_{\mathbf{p}}$ with $p \cdot x = E_{\mathbf{p}} x^0 - \mathbf{p} \cdot \mathbf{x}$ – and we use throughout the Lorentz-invariant single-particle normalization fixed earlier in this chapter, abbreviating the invariant measure as

$$\widetilde{dp} \equiv \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}. \tag{9.211}$$

9.7.1 The complex scalar as two real fields

The classical complex scalar was introduced as part of the recurring cast in Chapter 6, and its global phase symmetry and Noether current were derived in Chapter 7. We restate the Lagrangian for readability:

$$\mathcal{L} \;=\; (\partial_\mu \phi)^\dagger (\partial^\mu \phi) \;-\; m^2\, \phi^\dagger \phi, \tag{9.212}$$

where ϕ is now a complex field and $\phi^\dagger$ its Hermitian conjugate (in the classical theory, its complex conjugate ϕ^*; we write $\dagger$ in anticipation of operators). Varying $\phi^\dagger$ and ϕ independently gives two Klein–Gordon equations,

$$(\Box + m^2)\, \phi = 0, \qquad (\Box + m^2)\, \phi^\dagger = 0, \tag{9.213}$$

so both ϕ and $\phi^\dagger$ are on-shell at mass m. There is no second mass scale: whatever quanta the theory contains, they are degenerate. This degeneracy, which will become the statement that particle and antiparticle have equal mass, is built in from the start by the single coefficient m^2 in Eq. (9.212).

That a complex field is two real fields is not a metaphor. Write

$$\phi \;=\; \frac{1}{\sqrt{2}}(\phi_1 + i\,\phi_2), \qquad \phi^\dagger \;=\; \frac{1}{\sqrt{2}}(\phi_1 - i\,\phi_2), \tag{9.214}$$

with ϕ_1, ϕ_2 real (Hermitian). Substituting into Eq. (9.212) and using $\phi^\dagger\phi = \frac{1}{2}(\phi_1^2 + \phi_2^2)$ and the analogous identity for the kinetic term,

$$\mathcal{L} \;=\; \tfrac{1}{2}(\partial_\mu\phi_1)(\partial^\mu\phi_1) - \tfrac{1}{2}m^2\phi_1^2 \;+\; \tfrac{1}{2}(\partial_\mu\phi_2)(\partial^\mu\phi_2) - \tfrac{1}{2}m^2\phi_2^2. \tag{9.215}$$

The cross terms cancel because $(\partial\phi_1)(\partial\phi_2)$ enters $\phi^\dagger\phi$ and the kinetic piece with opposite signs from the $\pm i\phi_2$. So the complex theory is exactly two decoupled real Klein–Gordon fields of equal mass m. We *could* therefore quantize ϕ_1 and ϕ_2 separately by the construction already in hand and be done. We will instead quantize ϕ directly, because the natural ladder operators of the complex field – not those of ϕ_1, ϕ_2 – are the ones that diagonalize the charge, and it is the charge eigenstates that deserve the name "particle" and "antiparticle." Theorem 9.37 ties the two bookkeepings together explicitly.

Canonical structure. Treating ϕ and $\phi^\dagger$ as independent, the momenta conjugate to them are

$$\pi \;=\; \frac{\partial \mathcal{L}}{\partial \dot\phi} \;=\; \dot\phi^\dagger, \qquad \pi^\dagger \;=\; \frac{\partial \mathcal{L}}{\partial \dot\phi^\dagger} \;=\; \dot\phi. \tag{9.216}$$

Promoting fields to operators, we impose the equal-time commutation relations (ETCR), the complex-field analogue of the real-field relations established earlier in this chapter:

$$[\phi(\mathbf{x},t), \pi(\mathbf{y},t)] = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}), \qquad \left[\phi^\dagger(\mathbf{x},t), \pi^\dagger(\mathbf{y},t)\right] = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{9.217}$$

with all other equal-time commutators among $\{\phi, \phi^\dagger, \pi, \pi^\dagger\}$ vanishing. In particular $\left[\phi, \phi^\dagger\right] = 0$ and $\left[\phi, \pi^\dagger\right] = 0$ at equal times: ϕ commutes with its own conjugate field and with the momentum $\pi^\dagger = \dot{\phi}$ conjugate to that conjugate. These are exactly the relations Eq. (9.215) would give for two independent real fields, repackaged.

9.7.2 Mode expansion and the two ladder sets

The operator solution of Eq. (9.213) is a superposition of positive- and negative-frequency plane waves, just as for the real field; the new feature is that the field is no longer Hermitian, so the coefficient of $e^{+ip\cdot x}$ need not be the conjugate of the coefficient of $e^{-ip\cdot x}$. We are free to introduce a *second, independent* set of operators there. Write

$$\phi(x) = \int \widetilde{dp}\left(a_{\mathbf{p}}\, e^{-ip\cdot x} + b^\dagger_{\mathbf{p}}\, e^{+ip\cdot x}\right), \tag{9.218}$$

$$\phi^\dagger(x) = \int \widetilde{dp}\left(a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} + b_{\mathbf{p}}\, e^{-ip\cdot x}\right), \tag{9.219}$$

with $p^0 = E_{\mathbf{p}}$ understood on shell. This is the central structural statement of the section, so read off what it says. The field ϕ *annihilates an a-quantum* (coefficient $a_{\mathbf{p}}$ of the positive-frequency wave) and *creates a b-quantum* (coefficient $b^\dagger_{\mathbf{p}}$ of the negative-frequency wave). The conjugate field $\phi^\dagger$ does the reverse: it creates an a-quantum and annihilates a b-quantum. The two species are not interchangeable – ϕ and $\phi^\dagger$ treat them oppositely – and this manifest asymmetry between ϕ and $\phi^\dagger$ is precisely what a single real field, with $b = a$, cannot exhibit. For the real field the reality condition $\phi = \phi^\dagger$ forces the negative-frequency coefficient to be the conjugate of the positive-frequency one; relaxing reality is exactly what opens up the second ladder set.

To pin down the algebra of $a_{\mathbf{p}}, b_{\mathbf{p}}$ we need the inversion formulas. Define the time-symmetric bilinear $f \overleftrightarrow{\partial}_0 g \equiv f\,\dot{g} - \dot{f}\,g$. Then, for any on-shell wave with $p^0 = E_{\mathbf{p}}$,

$$a_{\mathbf{p}} = \int d^3x\; e^{+ip\cdot x}\, i\overleftrightarrow{\partial}_0\, \phi(x), \qquad b_{\mathbf{p}} = \int d^3x\; e^{+ip\cdot x}\, i\overleftrightarrow{\partial}_0\, \phi^\dagger(x), \tag{9.220}$$

together with the Hermitian conjugates for $a^\dagger_{\mathbf{p}}, b^\dagger_{\mathbf{p}}$. We verify the first relation; the second is identical with $\phi \to \phi^\dagger$, i.e. $a \leftrightarrow b$. Using $i\, e^{ipx} \overleftrightarrow{\partial}_0\, \phi = i\, e^{ipx}\dot{\phi} + E_{\mathbf{p}}\, e^{ipx}\phi$ and inserting Eq. (9.218), the $a_{\mathbf{q}} e^{-iq\cdot x}$ term contributes a factor $(E_{\mathbf{q}} + E_{\mathbf{p}})\, e^{i(p-q)\cdot x}$, whose spatial integral $\int d^3x\, e^{-i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$ sets $\mathbf{q} = \mathbf{p}$, $E_{\mathbf{q}} = E_{\mathbf{p}}$, the time exponential to unity, and the factor to $2E_{\mathbf{p}}$; against the measure $\widetilde{dq}$ this collapses to $a_{\mathbf{p}}$. The $b^\dagger_{\mathbf{q}} e^{+iq\cdot x}$ term instead contributes $(E_{\mathbf{p}} - E_{\mathbf{q}})\, e^{i(p+q)\cdot x}$, and the spatial integral now forces $\mathbf{q} = -\mathbf{p}$, hence $E_{\mathbf{q}} = E_{\mathbf{p}}$ and the prefactor

$(E_{\mathbf{p}} - E_{\mathbf{q}}) = 0$: the antiparticle piece drops out. This is the algebraic content of "ϕ annihilates an a, not a b."

We can now derive the ladder algebra from the ETCR, rather than postulating it – exactly the discipline the real-field section followed.

Proposition 9.35 (Two independent ladder algebras from the complex ETCR)**.** *The ETCR Eq. (9.217) are equivalent to the commutation relations*

$$\left[a_{\mathbf{p}}, a^{\dagger}_{\mathbf{q}}\right] = \left[b_{\mathbf{p}}, b^{\dagger}_{\mathbf{q}}\right] = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}), \tag{9.221}$$

with all other commutators among $\{a_{\mathbf{p}}, a^{\dagger}_{\mathbf{p}}, b_{\mathbf{p}}, b^{\dagger}_{\mathbf{p}}\}$ *– in particular the mixed ones* $[a_{\mathbf{p}}, b_{\mathbf{q}}]$, $\left[a_{\mathbf{p}}, b^{\dagger}_{\mathbf{q}}\right]$ *– vanishing.*

Proof. From Eq. (9.220) and $\pi^{\dagger} = \dot{\phi}$, $\pi = \dot{\phi}^{\dagger}$, the inversion formulas read, at the common time t,

$$a_{\mathbf{p}} = \int d^3x\; e^{ip\cdot x}(E_{\mathbf{p}}\,\phi + i\,\pi^{\dagger}), \qquad a^{\dagger}_{\mathbf{q}} = \int d^3y\; e^{-iq\cdot y}(E_{\mathbf{q}}\,\phi^{\dagger} - i\,\pi). \tag{9.222}$$

Their commutator receives contributions only from $[\phi, \pi]$ and $\left[\pi^{\dagger}, \phi^{\dagger}\right]$, since $\left[\phi, \phi^{\dagger}\right] = \left[\pi^{\dagger}, \pi\right] = 0$:

$$\begin{aligned}
\left[a_{\mathbf{p}}, a^{\dagger}_{\mathbf{q}}\right] &= \int d^3x\, d^3y\; e^{ip\cdot x} e^{-iq\cdot y}\left(-iE_{\mathbf{p}}\;[\phi(x), \pi(y)] + iE_{\mathbf{q}}\;\left[\pi^{\dagger}(x), \phi^{\dagger}(y)\right]\right) \\
&= \int d^3x\, d^3y\; e^{ip\cdot x} e^{-iq\cdot y}\;(E_{\mathbf{p}} + E_{\mathbf{q}})\;\delta^{(3)}(\mathbf{x}-\mathbf{y}) \\
&= \int d^3x\; e^{i(p-q)\cdot x}\;(E_{\mathbf{p}} + E_{\mathbf{q}}) \;=\; (2\pi)^3\, 2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q}),
\end{aligned} \tag{9.223}$$

where we used $[\phi, \pi] = i\delta^{(3)}$ and $\left[\pi^{\dagger}, \phi^{\dagger}\right] = -\left[\phi^{\dagger}, \pi^{\dagger}\right] = -i\delta^{(3)}$, and in the last step the spatial integral set $\mathbf{q} = \mathbf{p}$. The same computation with $\phi \to \phi^{\dagger}$ gives $\left[b_{\mathbf{p}}, b^{\dagger}_{\mathbf{q}}\right] = (2\pi)^3 2E_{\mathbf{p}}\delta^{(3)}(\mathbf{p}-\mathbf{q})$. For the mixed commutator, write $b^{\dagger}_{\mathbf{q}} = \int d^3y\, e^{-iq\cdot y}(E_{\mathbf{q}}\phi - i\pi^{\dagger})$; then $\left[a_{\mathbf{p}}, b^{\dagger}_{\mathbf{q}}\right]$ involves only $[\phi, \phi]$, $\left[\phi, \pi^{\dagger}\right]$, $\left[\pi^{\dagger}, \phi\right]$ and $\left[\pi^{\dagger}, \pi^{\dagger}\right]$, every one of which vanishes by Eq. (9.217). Hence $\left[a_{\mathbf{p}}, b^{\dagger}_{\mathbf{q}}\right] = 0$, and likewise $[a_{\mathbf{p}}, b_{\mathbf{q}}] = 0$. The converse – that Eq. (9.221) reproduces Eq. (9.217) when substituted back into Eqs. (9.218) and (9.219) – follows by the identical Fourier manipulation run in reverse, exactly as for the real field. □

The two ladder sets are therefore genuinely independent: they commute with one another and each obeys the standard relativistically normalized oscillator algebra. The Fock construction now proceeds in duplicate. There is a single vacuum $|0\rangle$, annihilated by *both* families,

$$a_{\mathbf{p}}\,|0\rangle = b_{\mathbf{p}}\,|0\rangle = 0 \quad \forall\,\mathbf{p}, \qquad \langle 0|0\rangle = 1, \tag{9.224}$$

and two towers of excitations built on it. We call the a-quanta *particles* and the b-quanta *antiparticles*; the relativistically normalized one-quantum states are

$$|\mathbf{p}\rangle_a = a^\dagger_{\mathbf{p}}\,|0\rangle\,, \qquad |\mathbf{p}\rangle_b = b^\dagger_{\mathbf{p}}\,|0\rangle\,, \qquad {}_a\langle\mathbf{p}|\mathbf{q}\rangle_a = {}_b\langle\mathbf{p}|\mathbf{q}\rangle_b = (2\pi)^3 2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q}). \tag{9.225}$$

Both species carry the mass-m, spin-0 Poincaré quantum numbers classified in Chapter 3, because both are excitations of the same mass-m field; what distinguishes them is not a spacetime label but the internal charge to which we now turn. The two corresponding number operators are

$$N_a = \int \widetilde{dp}\; a^\dagger_{\mathbf{p}} a_{\mathbf{p}}, \qquad N_b = \int \widetilde{dp}\; b^\dagger_{\mathbf{p}} b_{\mathbf{p}}. \tag{9.226}$$

Physical Insight

The names "particle" and "antiparticle" are at this stage simply labels for the two commuting oscillator families. They earn physical meaning only once we exhibit a measurable quantity – the conserved charge – that takes opposite values on them. Until then, nothing distinguishes a from b: the Hamiltonian, which one obtains by the same substitution as in the real case and which reads $H = \int \widetilde{dp}\, E_{\mathbf{p}}\,(a^\dagger_{\mathbf{p}} a_{\mathbf{p}} + b^\dagger_{\mathbf{p}} b_{\mathbf{p}})$ after normal ordering, treats them symmetrically and assigns them equal energy. Mass and spin cannot tell a particle from its antiparticle; only charge can.

9.7.3 The conserved U(1) charge as an operator

The Lagrangian Eq. (9.212) is invariant under the global phase rotation $\phi \to e^{-i\alpha}\phi$, $\phi^\dagger \to e^{+i\alpha}\phi^\dagger$. By Noether's theorem (Theorem 7.10) this symmetry carries a conserved current, derived in Chapter 7 and recorded there as Eq. (7.70),

$$j^\mu \;=\; i(\phi^\dagger\,\partial^\mu\phi - (\partial^\mu\phi^\dagger)\,\phi), \qquad \partial_\mu j^\mu = 0, \tag{9.227}$$

the four-divergence vanishing on shell by Eq. (9.213). The associated conserved charge is the spatial integral of the time component, which we now promote to an operator:

$$Q \;=\; \int d^3x\; j^0(x) \;=\; i\int d^3x\;(\phi^\dagger\dot\phi - \dot\phi^\dagger\,\phi) \;=\; i\int d^3x\;\phi^\dagger \overleftrightarrow{\partial}_0\,\phi. \tag{9.228}$$

Classically Q is time independent; as an operator it will commute with H. Our task is to express it in the ladder basis, where its action is transparent.

Theorem 9.36 (The charge operator)**.** *With the normal-ordering prescription that places all annihilation operators to the right (which fixes the additive constant by demanding $Q\,|0\rangle = 0$), the charge Eq. (9.228) is*

$$Q \;=\; \int \widetilde{dp}\;(a^\dagger_{\mathbf{p}} a_{\mathbf{p}} - b^\dagger_{\mathbf{p}} b_{\mathbf{p}}) \;=\; N_a - N_b. \tag{9.229}$$

It generates the U(1) *symmetry and assigns the ladder operators definite charges,*

$$\left[Q, a_{\mathbf{p}}^\dagger\right] = +a_{\mathbf{p}}^\dagger, \quad \left[Q, b_{\mathbf{p}}^\dagger\right] = -b_{\mathbf{p}}^\dagger, \quad [Q, a_{\mathbf{p}}] = -a_{\mathbf{p}}, \quad [Q, b_{\mathbf{p}}] = +b_{\mathbf{p}}, \tag{9.230}$$

so that a particle carries charge $+1$ *and an antiparticle charge* -1*, in units of the coupling.*

Proof. Insert the mode expansions Eqs. (9.218) and (9.219) and their time derivatives,

$$\dot\phi = \int \widetilde{dp}\,(-iE_{\mathbf{p}})(a_{\mathbf{p}}e^{-ip\cdot x} - b_{\mathbf{p}}^\dagger e^{+ip\cdot x}), \qquad \dot\phi^\dagger = \int \widetilde{dq}\,(+iE_{\mathbf{q}})(a_{\mathbf{q}}^\dagger e^{+iq\cdot x} - b_{\mathbf{q}}e^{-iq\cdot x}), \tag{9.231}$$

into $\int d^3x\,(\phi^\dagger\dot\phi - \dot\phi^\dagger\phi)$. Each $\int d^3x$ produces $(2\pi)^3$ times either $\delta^{(3)}(\mathbf{p}-\mathbf{q})$ (from terms whose spatial exponentials are $e^{\pm i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}}$) or $\delta^{(3)}(\mathbf{p}+\mathbf{q})$ (from $e^{\pm i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}}$). Carrying out the $\widetilde{dq}$ integral against these deltas and tracking the phases, $\int d^3x\,\phi^\dagger\dot\phi$ evaluates to

$$\int \widetilde{dp}\left[\tfrac{-i}{2}\,a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + \tfrac{i}{2}\,b_{\mathbf{p}}b_{\mathbf{p}}^\dagger + \tfrac{i}{2}\,e^{+2iE_{\mathbf{p}}t}\,a_{-\mathbf{p}}^\dagger b_{\mathbf{p}}^\dagger - \tfrac{i}{2}\,e^{-2iE_{\mathbf{p}}t}\,b_{-\mathbf{p}}a_{\mathbf{p}}\right], \tag{9.232}$$

where $a_{-\mathbf{p}} \equiv a_{\mathbf{q}}\big|_{\mathbf{q}=-\mathbf{p}}$ and we used $E_{-\mathbf{p}} = E_{\mathbf{p}}$. The companion piece $\int d^3x\,\dot\phi^\dagger\phi$ gives

$$\int \widetilde{dp}\left[\tfrac{+i}{2}\,a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - \tfrac{i}{2}\,b_{\mathbf{p}}b_{\mathbf{p}}^\dagger + \tfrac{i}{2}\,e^{+2iE_{\mathbf{p}}t}\,a_{-\mathbf{p}}^\dagger b_{\mathbf{p}}^\dagger - \tfrac{i}{2}\,e^{-2iE_{\mathbf{p}}t}\,b_{-\mathbf{p}}a_{\mathbf{p}}\right]. \tag{9.233}$$

The two time-dependent off-diagonal structures – $a^\dagger b^\dagger$ and ba – appear with identical coefficients in both, so they *cancel* in the difference $\phi^\dagger\dot\phi - \dot\phi^\dagger\phi$. This cancellation is the operator statement that Q is conserved: had they survived, Q would carry the explicit time dependence $e^{\pm 2iE_{\mathbf{p}}t}$. What remains is purely diagonal,

$$\int d^3x\,(\phi^\dagger\dot\phi - \dot\phi^\dagger\phi) = \int \widetilde{dp}\,(-i\,a_{\mathbf{p}}^\dagger a_{\mathbf{p}} + i\,b_{\mathbf{p}}b_{\mathbf{p}}^\dagger), \tag{9.234}$$

and multiplying by the prefactor i from Eq. (9.228),

$$Q = \int \widetilde{dp}\,(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}b_{\mathbf{p}}^\dagger). \tag{9.235}$$

The antiparticle term is in the "wrong" order. Using $b_{\mathbf{p}}b_{\mathbf{p}}^\dagger = b_{\mathbf{p}}^\dagger b_{\mathbf{p}} + \left[b_{\mathbf{p}}, b_{\mathbf{p}}^\dagger\right]$ and Eq. (9.221),

$$Q = \int \widetilde{dp}\,(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}}) - \int \widetilde{dp}\,(2\pi)^3 2E_{\mathbf{p}}\,\delta^{(3)}(0) = \int \widetilde{dp}\,(a_{\mathbf{p}}^\dagger a_{\mathbf{p}} - b_{\mathbf{p}}^\dagger b_{\mathbf{p}}) - \int d^3p\,\delta^{(3)}(0). \tag{9.236}$$

The subtracted term is a (divergent) c-number, the charge analogue of the zero-point energy. Unlike the vacuum energy it is naturally zero rather than merely

irrelevant: normal ordering, defined here as moving all annihilation operators to the right, discards it and enforces $Q\,|0\rangle = 0$, the physically obvious statement that the vacuum is neutral. This gives Eq. (9.229).

The commutators Eq. (9.230) now follow from Eq. (9.221). Since $\left[a^\dagger_{\mathbf{q}} a_{\mathbf{q}}, a^\dagger_{\mathbf{p}}\right] = a^\dagger_{\mathbf{q}}\left[a_{\mathbf{q}}, a^\dagger_{\mathbf{p}}\right] = a^\dagger_{\mathbf{q}}(2\pi)^3 2E_{\mathbf{q}}\delta^{(3)}(\mathbf{q}-\mathbf{p})$, and b commutes with a,

$$\left[Q, a^\dagger_{\mathbf{p}}\right] = \int \widetilde{dq}\; a^\dagger_{\mathbf{q}}\,(2\pi)^3 2E_{\mathbf{q}}\,\delta^{(3)}(\mathbf{q}-\mathbf{p}) = a^\dagger_{\mathbf{p}}. \tag{9.237}$$

The minus sign in $-b^\dagger b$ flips the result for b, giving $\left[Q, b^\dagger_{\mathbf{p}}\right] = -b^\dagger_{\mathbf{p}}$; the annihilation-operator relations are the Hermitian conjugates. Finally, applying Eq. (9.230) to the field, $[Q,\phi] = \int \widetilde{dp}([Q, a_{\mathbf{p}}]\, e^{-ipx} + \left[Q, b^\dagger_{\mathbf{p}}\right] e^{ipx}) = -\phi$, so $e^{i\alpha Q}\phi\, e^{-i\alpha Q} = e^{-i\alpha}\phi$: Q is the generator of the U(1) rotation it was built from, closing the loop with Noether. □

The reading of Eq. (9.229) is the whole point. The charge counts particles minus antiparticles. Acting on the one-quantum states Eq. (9.225),

$$Q\;|\mathbf{p}\rangle_a = +\,|\mathbf{p}\rangle_a\,, \qquad Q\;|\mathbf{p}\rangle_b = -\,|\mathbf{p}\rangle_b\,, \tag{9.238}$$

exhibiting two species of equal mass m (both on the shell $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$ of the single field) and opposite charge ± 1. This is the rigorous form of the statement that to every charged particle there corresponds an antiparticle of the same mass and opposite charge – here not a postulate or an appeal to the Dirac sea, but a theorem about the spectrum of Q on the Fock space of one complex field. Because $[Q, H] = 0$ (both are diagonal in the number operators), charge is conserved under time evolution, and because Q generates the symmetry, charge superselects: states of different Q never mix.

Caution

The overall sign and the unit of Q are conventions, inherited from the choice of phase direction in the U(1) rotation and from any coupling constant one factors out of the current Eq. (9.227). What is convention-independent, and physical, is that the two species carry *opposite* charge of *equal* magnitude, and that the vacuum and the Hamiltonian are neutral. When this scalar is later gauged in QED, the unit of Q becomes the electric charge q in the covariant derivative $D_\mu = \partial_\mu - iqA_\mu$, and the sign convention here must be matched to the gauge-coupling sign convention fixed book-wide.

Worked Example 9.37 (The two real fields, the charge as a rotation, and charge eigenstates)**.** We reconcile the complex ladder operators with the two-real-field picture of Eq. (9.214), and see the charge as the generator of rotations

in the (ϕ_1, ϕ_2) plane. Quantize ϕ_1, ϕ_2 as two independent real scalars with ladder operators $a_{1,\mathbf{p}}, a_{2,\mathbf{p}}$ obeying $\left[a_{i,\mathbf{p}}, a^\dagger_{j,\mathbf{q}}\right] = \delta_{ij}(2\pi)^3 2E_{\mathbf{p}}\delta^{(3)}(\mathbf{p}-\mathbf{q})$, all cross-commutators zero. Substituting the real-field mode expansions into Eq. (9.214) and matching to Eq. (9.218), the coefficient of $e^{-ip\cdot x}$ is $a_{\mathbf{p}}$ and that of $e^{+ip\cdot x}$ is $b^\dagger_{\mathbf{p}}$, giving

$$a_{\mathbf{p}} = \frac{1}{\sqrt{2}}(a_{1,\mathbf{p}} + i\, a_{2,\mathbf{p}}), \qquad b_{\mathbf{p}} = \frac{1}{\sqrt{2}}(a_{1,\mathbf{p}} - i\, a_{2,\mathbf{p}}). \tag{9.239}$$

This is a change of basis on the pair of oscillators at each $\mathbf{p}$. One checks the algebra directly:

$$\left[a_{\mathbf{p}}, a^\dagger_{\mathbf{q}}\right] = \tfrac{1}{2}\left[a_{1,\mathbf{p}} + ia_{2,\mathbf{p}}, a^\dagger_{1,\mathbf{q}} - ia^\dagger_{2,\mathbf{q}}\right] = \tfrac{1}{2}\left(\left[a_{1,\mathbf{p}}, a^\dagger_{1,\mathbf{q}}\right] + \left[a_{2,\mathbf{p}}, a^\dagger_{2,\mathbf{q}}\right]\right) = (2\pi)^3 2E_{\mathbf{p}}\delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{9.240}$$

$$\left[a_{\mathbf{p}}, b^\dagger_{\mathbf{q}}\right] = \tfrac{1}{2}\left[a_{1,\mathbf{p}} + ia_{2,\mathbf{p}}, a^\dagger_{1,\mathbf{q}} + ia^\dagger_{2,\mathbf{q}}\right] = \tfrac{1}{2}\left(\left[a_{1,\mathbf{p}}, a^\dagger_{1,\mathbf{q}}\right] - \left[a_{2,\mathbf{p}}, a^\dagger_{2,\mathbf{q}}\right]\right) = 0, \tag{9.241}$$

recovering Eq. (9.221) and the independence of a and b. Substituting Eq. (9.239) into $N_a - N_b$ and simplifying, the diagonal pieces cancel and the charge becomes

$$Q = \int \widetilde{dp}\; (a^\dagger_{\mathbf{p}} a_{\mathbf{p}} - b^\dagger_{\mathbf{p}} b_{\mathbf{p}}) = \int \widetilde{dp}\; (-i\, a^\dagger_{1,\mathbf{p}} a_{2,\mathbf{p}} + i\, a^\dagger_{2,\mathbf{p}} a_{1,\mathbf{p}}), \tag{9.242}$$

the generator that rotates $a_1 \leftrightarrow a_2$: this is the $\mathrm{SO}(2) \cong \mathrm{U}(1)$ realized on the two real fields. Neither $a^\dagger_{1,\mathbf{p}}|0\rangle$ nor $a^\dagger_{2,\mathbf{p}}|0\rangle$ is a charge eigenstate; the eigenstates are the complex combinations $a^\dagger_{\mathbf{p}}|0\rangle$ and $b^\dagger_{\mathbf{p}}|0\rangle$, with $Q = \pm 1$. This is why the complex basis is the physical one: a quantum of definite charge is a specific superposition of a ϕ_1-quantum and a ϕ_2-quantum, and only that superposition has a definite value of the conserved observable.

9.7.4 The charged propagator and the flow of charge

The dynamical object that will become the internal line of every Feynman diagram is again the time-ordered two-point function. For the charged field the relevant ordering pairs ϕ with $\phi^\dagger$, because – as the mode content of Eqs. (9.218) and (9.219) makes clear – only that pairing can connect the vacuum to itself. Consider first the two Wightman orderings. For $x^0 > y^0$, only the particle term contributes,

$$\left\langle 0|\phi(x)\phi^\dagger(y)|0\right\rangle = \int \widetilde{dp}\,\widetilde{dq}\; e^{-ip\cdot x} e^{+iq\cdot y}\left\langle 0|a_{\mathbf{p}} a^\dagger_{\mathbf{q}}|0\right\rangle = \int \widetilde{dp}\; e^{-ip\cdot(x-y)} \;\equiv\; D(x-y), \tag{9.243}$$

using $\left\langle 0|a_{\mathbf{p}} a^\dagger_{\mathbf{q}}|0\right\rangle = (2\pi)^3 2E_{\mathbf{p}}\delta^{(3)}(\mathbf{p}-\mathbf{q})$ and the invariant measure to collapse the $\widetilde{dq}$ integral; D is the same positive-frequency Wightman function met for the

real field. For $y^0 > x^0$, only the antiparticle term survives, because now we need an annihilation acting first on the right vacuum:

$$\Big\langle 0|\phi^\dagger(y)\phi(x)|0\Big\rangle = \int \widetilde{dp}\,\widetilde{dq}\; e^{-iq\cdot y}e^{+ip\cdot x}\,\Big\langle 0|b_{\mathbf{q}}b^\dagger_{\mathbf{p}}|0\Big\rangle = \int \widetilde{dp}\; e^{+ip\cdot(x-y)} \;=\; D(y-x). \tag{9.244}$$

The Feynman propagator is the time-ordered combination,

$$D_F(x-y) \;=\; \Big\langle 0|\mathcal{T}\,\phi(x)\phi^\dagger(y)|0\Big\rangle \;=\; \theta(x^0-y^0)\,D(x-y) + \theta(y^0-x^0)\,D(y-x). \tag{9.245}$$

Structurally this is identical to the real-scalar time-ordered function: the same two Wightman pieces glued by step functions. The physics packed into it, however, is now sharper. When $x^0 > y^0$ a *particle* of charge $+1$ is created at y by $\phi^\dagger(y)$ and annihilated at x by $\phi(x)$; when $y^0 > x^0$ an *antiparticle* of charge -1 is created at x and annihilated at y. In both orderings the net charge transported from y to x is $+1$: a particle going forward, or an antiparticle going backward, move one unit of charge the same way. This is the precise content of the slogan that an antiparticle is a particle propagating backward in time, and it is why the charged propagator carries an *arrow*: the line has an orientation, from the $\phi^\dagger$ end (charge in) to the ϕ end (charge out). Figure 9.7 records this.

Because Eq. (9.245) is the same step-function combination of the same Wightman functions as in the neutral case, the contour representation derived earlier in this chapter applies verbatim, and we may quote it:

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot(x-y)}}{p^2-m^2+i\varepsilon}, \tag{9.246}$$

with the Feynman $i\varepsilon$ routing the positive-frequency pole below and the negative-frequency pole above the real p^0 axis, exactly as in Eq. (9.154). The charged scalar propagator is, in momentum space, indistinguishable from the neutral one; all the new information lives in the orientation of the line, not in the analytic form.

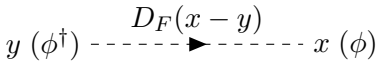


Figure 9.7: The charged scalar propagator as an oriented internal line. The arrow points from the $\phi^\dagger$ insertion at y to the ϕ insertion at x, recording the flow of one unit of U(1) charge from y to x. A particle propagating forward in time and an antiparticle propagating backward carry this charge identically, which is why a single oriented line suffices. A neutral scalar line, by contrast, carries no arrow.

The orientation is enforced by a vanishing theorem. The other time-ordered pairings, ϕ with ϕ and $\phi^\dagger$ with $\phi^\dagger$, are identically zero:

$$\langle 0|\mathcal{T}\,\phi(x)\phi(y)|0\rangle = 0, \qquad \Big\langle 0|\mathcal{T}\,\phi^\dagger(x)\phi^\dagger(y)|0\Big\rangle = 0. \tag{9.247}$$

The proof is immediate from the mode content. Expanding $\langle 0|\phi(x)\phi(y)|0\rangle$, the surviving vacuum expectation values would have to be of the form $\big\langle 0|(\text{annih. or } b^\dagger)(\text{annih. or}$ but ϕ contains only $a_{\mathbf{p}}$ and $b^\dagger_{\mathbf{p}}$, so every term is one of $\langle 0|aa|0\rangle$, $\big\langle 0|ab^\dagger|0\big\rangle$, $\big\langle 0|b^\dagger a|0\big\rangle$, $\big\langle 0|b^\dagger b^\dagger|0\big\rangle$, each of which annihilates a vacuum on one side (the a's kill $|0\rangle$ on the right; the $b^\dagger$'s kill $\langle 0|$ on the left, since a and b commute one may always order an a to the right or a $b^\dagger$ to the left). Hence the equal-ordering Wightman functions vanish for either time order, and so does their time-ordered combination. The same holds for $\phi^\dagger\phi^\dagger$. Charge conservation gives the structural reason: $\phi\phi$ would carry total charge -2 and cannot connect to the neutral vacuum. In the diagrammatic language of Chapters 14 and 15, Eq. (9.247) is the statement that charge must *flow through* a propagator – a line always has a ϕ end and a $\phi^\dagger$ end – never source or sink at it.

9.7.5 The neutral scalar as its own antiparticle

The real scalar carried through the rest of this chapter is the special case in which the two ladder sets coincide. Imposing the reality condition $\phi = \phi^\dagger$ on Eqs. (9.218) and (9.219) forces, mode by mode,

$$b_{\mathbf{p}} = a_{\mathbf{p}}, \tag{9.248}$$

collapsing the two families into one and recovering exactly the real-field mode expansion used earlier in the chapter. There is then only one species of quantum, and the charge Eq. (9.229) becomes

$$Q = \int \widetilde{dp}\,(a^\dagger_{\mathbf{p}} a_{\mathbf{p}} - a^\dagger_{\mathbf{p}} a_{\mathbf{p}}) = 0. \tag{9.249}$$

The neutral field carries no conserved U(1) charge – consistent with the fact that a real Lagrangian has no phase symmetry to begin with – and its single quantum is its own antiparticle: applying $\phi = \phi^\dagger$ to the propagator collapses Eq. (9.245) and the would-be $\langle \mathcal{T}\phi\phi\rangle$ into the single real-scalar Feynman function, and the oriented line of Fig. 9.7 loses its arrow. This is the precise sense, promised in the earlier sections, in which the real scalar of ϕ^4 theory is the uncharged member of a family whose charged members we have now built. The interacting $\lambda\phi^4$ dynamics taken up in Chapters 14 and 15 lives entirely in this neutral sector; the charged construction of this section is the scaffolding the book will mount when it gauges a scalar or, more importantly, when it builds the charged Dirac field.

With the free scalar theory now complete in both its neutral and charged forms, the chapter has delivered the two ideas the rest of the book most depends on from a charged field: a conserved charge realized as a self-adjoint operator whose spectrum is integral, and the particle/antiparticle doubling that any such charge entails. These hand directly to Chapter 11, where the same logic – mode

expansion, a Noether charge, particle and antiparticle towers – is repeated for the Dirac field with anticommutators in place of commutators, and to Chapter 13 and QED, where the global U(1) of this section is promoted to a local gauge symmetry and Q becomes the electric charge that couples matter to the photon.

Further reading

The canonical quantization of the scalar field is presented in Peskin and Schroeder [396] ch. 2 and Schwartz [432] chs. 2–3; Weinberg [500] ch. 5 builds the free field starting instead from Wigner's particle classification, the complementary route to the same object. Srednicki [454] chs. 1–3 emphasize the same Lorentz-invariant normalization adopted here, and Zee [532] ch. I gives the quickest physical motivation for the propagator as a particle worldline. Coleman [129] and Klauber [296] are unusually careful pedagogically about Fock space, normal ordering, and the particle interpretation. For the structural results only sketched here, the axiomatic treatment of microcausality and the spin-statistics connection is Streater and Wightman [458], and Haag's theorem and the algebraic view of free fields are developed in Haag [246].

Chapter 10

The Path Integral for Scalar Fields

Canonical quantization, built in Chapter 9, is one of the two pillars on which quantum field theory rests. This chapter erects the other. Where the operator approach promotes the field $\phi(x)$ to an operator and reads dynamics off commutators, the path integral keeps $\phi(x)$ a number-valued configuration and sums over *all* of them, each weighted by the phase $e^{iS[\phi]}$. The two formulations describe the same physics — we will prove their equivalence on the central object, the Feynman propagator — but they make different things easy. The path integral makes symmetries, the classical limit, non-perturbative configurations, and above all the bookkeeping of perturbation theory transparent in a way the operator formalism does not, and it is the language in which the rest of the book is most naturally written. We develop it for the real scalar of ϕ^4 theory, beginning with the free sector quantized in Chapter 9 and ending with the apparatus that turns the interaction back on.

The chapter is deliberate about its imports because Part I built its tools for exactly this moment. The functional calculus — functional differentiation (Theorem 4.4), the finite- and infinite-dimensional Gaussian integral (Theorems 4.14 and 4.15), the generating-functional idea (Section 4.4), and the Legendre transform that will produce the effective action (Theorem 4.31) — is taken wholesale from Chapter 4, where its rigorous status was fixed. The analytic machinery that makes the oscillatory integral converge — the $i\varepsilon$ prescription, the residue calculus, and the Wick rotation $t \to -i\tau$ (Theorem 5.30) — is imported from Chapter 5. From Chapter 9 we take the Fock vacuum (Theorem 9.17) and the canonically-derived Feynman propagator (Theorem 9.24 and Eq. (9.154)) as the fixed target every path-integral computation must reproduce, and we take the classical ϕ^4 Lagrangian and its $\mathbb{Z}_2$ symmetry from Chapters 6 and 7. The one place where the formal manipulations of the main text outrun what can be made rigorous — the existence of the functional measure and the meaning of the functional determinants it produces — is discharged in Chapter E, whose results we use by reference rather than reprove.

The development follows a single arc from one degree of freedom to infinitely many. We first derive the path integral honestly in ordinary quantum mechanics (Section 10.1), slicing time with the Trotter formula so that the continuum measure appears as a controlled limit of finite-dimensional integrals rather than an axiom; the same slicing shows that path-integral insertions compute *time-ordered* products and that an $i\varepsilon$ tilt projects onto the ground state, giving the quantum-mechanical generating functional $Z[J]$ (Section 10.2). Promoting the single coordinate to a field on a spacetime lattice and taking the continuum limit converts $Z[J]$ into the field-theory generating functional and identifies its functional derivatives with the time-ordered vacuum correlators of Chapter 9 (Section 10.3). For the free theory the integral is Gaussian and is done in closed form: completing the square yields $Z_0[J]$ and delivers the Feynman propagator as the inverse Klein–Gordon operator, in exact agreement with the canonical result (Section 10.4). From $Z_0[J]$ alone, by repeated functional differentiation, Wick's theorem emerges as pure Gaussian combinatorics — correlators are sums over pairings — reproducing the operator theorem of Chapter 9 without ever mentioning an operator (Section 10.5). Wick-rotating the whole construction (Section 10.6) trades the oscillatory weight e^{iS} for a positive, convergent Boltzmann weight e^{-S_E}, exposes the exact dictionary between Euclidean field theory and classical statistical mechanics, and gives, in reflection positivity, the criterion under which a Euclidean theory reconstructs a unitary Lorentzian one. The final section (Section 10.7) switches on the ϕ^4 interaction: a master formula expresses the full $Z[J]$ as a differential operator acting on $Z_0[J]$, perturbative expansion generates the contractions, vacuum bubbles cancel, and the connected and effective generating functionals $W[J]$ and $\Gamma[\phi_{\text{cl}}]$ organize the output for everything that follows.

Conventions are those in force book-wide and stated once here: the mostly-minus metric $\eta_{\mu\nu} = \text{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, the Fourier convention of Appendix A, and the Feynman $i\varepsilon$ prescription, which in the path integral arises not as a rule imposed by hand but as the residue of the ground-state projection. One honesty is owed at the outset, in the spirit of Chapter 9: the Minkowski functional measure $\mathcal{D}\phi$ is, taken literally, a formal object — there is no Lebesgue measure on an infinite-dimensional space — and it is rigorous, as a genuine Gaussian measure, only after Wick rotation and only for the free theory. The construction of the *interacting* four-dimensional scalar measure as a true continuum object remains a famous open problem, and the accumulated evidence that ϕ^4_4 is trivial means the perturbation series we set up is best read as an asymptotic expansion about the free theory. None of this weakens the chapter's results: a measure understood as a controlled lattice limit, a closed-form free generating functional, Wick's theorem, and the master perturbative formula are exactly the objects that the S-matrix and LSZ reduction of Chapter 14, the Feynman rules of Chapter 15, the renormalization and effective-action machinery of Chapter 22, the Wilsonian RG of Chapter 23, and the fermionic Grassmann

integral of Chapter 12 all build upon.

10.1 The Path Integral in Quantum Mechanics

The previous chapter built the first pillar of quantum field theory by the route the reader already trusts: it took a classical field, imposed equal-time canonical commutators (Section 9.1), and let the Heisenberg equations grind out the dynamics, with operators ordered by hand whenever two of them met at the same point. That construction is complete and correct, but it has a feature one eventually wants to escape. It singles out a time direction at every step — the commutators live on a time slice, the time-ordering symbol in the propagator is an instruction about which operator stands to the left — and so it hides the Lorentz invariance that the classical Lagrangian wore openly. There is a second, logically independent road to the very same quantum mechanics, due to Dirac and Feynman, on which time and space re-enter on a more even footing: the transition amplitude between two configurations is written as a single integral over *all* histories connecting them, each history weighted by the phase e^{iS} built from the classical action. This section builds that road from the ground up, and it does so where every reader is on solid ground — in ordinary one-particle quantum mechanics — before Section 10.2 and the rest of the chapter lift the entire apparatus to fields. The promise to be redeemed across the chapter is that the two pillars agree: the free scalar propagator obtained by time-ordering operators in Eq. (9.154) will re-emerge, in Section 10.4, from a Gaussian integral over field histories.

Everything here is built on quantum mechanics the reader already owns at the Sakurai level: a Hilbert space, position and momentum eigenstates, the resolution of the identity, and the Hamiltonian as the generator of time translation. We add nothing to that postulate set. What we add is a way of *re-expressing* the matrix elements of the time-evolution operator, and the entire content of the section is that this re-expression is an identity, not a new physical assumption.

Notation Watch

Throughout this section a hat denotes an operator on the Hilbert space ($\hat{x}$, $\hat{p}$, $\hat{H}$), to be carefully distinguished from the c-number integration variables (x, p) that label its eigenstates and over which the path integral runs. Chapter 9 could write the field ϕ without a hat because no eigenvalue of ϕ ever appeared alongside it; here the distinction between the operator $\hat{x}$ and the number x in $\hat{x}\,|x\rangle = x\,|x\rangle$ is the whole point, so the hats are kept. We work in one spatial dimension with a single particle of mass m; the generalization to many degrees of freedom, and then to a field, is notational and is made in Section 10.3. Natural units $\hbar = 1$ are in force,

so that the action S in e^{iS} is dimensionless.

10.1.1 The transition amplitude as the central object

The quantity we set out to compute is the amplitude for a particle prepared at position x_i at time $t = 0$ to be found at position x_f at a later time $t = T$. In the Schrödinger picture, states evolve by the unitary operator $e^{-i\hat{H}T}$ generated by the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(\hat{x}) \equiv \hat{T} + \hat{V}, \tag{10.1}$$

where we have named the kinetic and potential operators $\hat{T} = \hat{p}^2/2m$ and $\hat{V} = V(\hat{x})$ for later use. The object of interest is the matrix element of the evolution operator between position eigenstates, called the *propagator* or *transition amplitude*,

$$K(x_f, T; x_i, 0) \;\equiv\; \left\langle x_f \middle| e^{-i\hat{H}T} \middle| x_i \right\rangle . \tag{10.2}$$

Definition 10.1 (Transition amplitude)**.** For a quantum system with Hamiltonian $\hat{H}$, the transition amplitude $K(x_f, T; x_i, 0)$ is the matrix element (10.2) of the time-evolution operator between the initial and final position eigenstates. Equivalently, if the particle is in the state $|\psi(0)\rangle$ at $t = 0$, its wavefunction at time T is reconstructed from K by

$$\psi(x_f, T) = \langle x_f | \psi(T) \rangle = \int \mathrm{d}x_i \; K(x_f, T; x_i, 0)\, \psi(x_i, 0), \tag{10.3}$$

so that K is the integral kernel that propagates the wavefunction forward in time.

Two facts make K the right object to attack. First, it contains the entire dynamics: by (10.3) it solves the initial-value problem for the Schrödinger equation for an arbitrary initial state, so anything computable in the canonical formalism is encoded in it. Second, it is built from the single operator $e^{-i\hat{H}T}$, and that operator has a structural weakness we can exploit. The difficulty in evaluating (10.2) directly is that $\hat{T}$ and $\hat{V}$ do not commute, so the exponential of their sum does not factor, $e^{-i(\hat{T}+\hat{V})T} \neq e^{-i\hat{T}T} e^{-i\hat{V}T}$. The strategy of the path integral is to chop the evolution into many infinitesimal steps, over each of which the non-commutativity is a negligible higher-order effect, and then to take the limit. We record the two tools from elementary quantum mechanics on which the chopping relies. Position and momentum eigenstates are complete and are paired by the plane-wave overlap,

$$\int_{-\infty}^{\infty} \mathrm{d}x \;\; |x\rangle \langle x| = \mathbf{1}, \qquad \int_{-\infty}^{\infty} \frac{\mathrm{d}p}{2\pi} \;\; |p\rangle \langle p| = \mathbf{1}, \qquad \langle x|p\rangle = e^{ipx}, \tag{10.4}$$

with $\langle x|x'\rangle = \delta(x-x')$ and $\langle p|p'\rangle = 2\pi\,\delta(p-p')$, the factors of 2π fixed by the Fourier conventions of Chapter 5 and consistent with the normalization $\langle x|p\rangle = e^{ipx}$. These are the only inputs; the rest is computation.

10.1.2 Slicing time: the Trotter product formula

The first move is to write the evolution over the finite interval T as a product of N identical evolutions over the short interval

$$\epsilon \equiv \frac{T}{N}, \tag{10.5}$$

which is exact and trivial because the exponents simply add,

$$e^{-i\hat{H}T} = \left(e^{-i\hat{H}\epsilon}\right)^N. \tag{10.6}$$

The non-trivial step is to factor each short-time evolution into a kinetic and a potential piece, which is legitimate only up to a correction we must now control. Because $\hat{T}$ and $\hat{V}$ do not commute, the Baker–Campbell–Hausdorff expansion gives

$$e^{-i\hat{H}\epsilon} = e^{-i\hat{T}\epsilon}\, e^{-i\hat{V}\epsilon}\, \exp\!\Big[+\tfrac{1}{2}\,\epsilon^2\,\big[\hat{T},\hat{V}\big] + O(\epsilon^3)\Big], \tag{10.7}$$

so the splitting error in a single slice is of order ϵ^2. The content of the Trotter (Lie–Trotter) product formula is that an error of order ϵ^2 per slice, accumulated over $N = T/\epsilon$ slices, washes out in the limit $N\to\infty$, leaving the factored product equal to the true evolution.

Theorem 10.2 (Lie–Trotter product formula)**.** *Let $\hat{A}$ and $\hat{B}$ be bounded operators on a Hilbert space. Then*

$$e^{\hat{A}+\hat{B}} = \lim_{N\to\infty}\left(e^{\hat{A}/N}\, e^{\hat{B}/N}\right)^N. \tag{10.8}$$

Applied to $\hat{A} = -i\hat{T}\,T$ and $\hat{B} = -i\hat{V}\,T$, this gives the time-sliced evolution

$$e^{-i\hat{H}T} = \lim_{N\to\infty}\left(e^{-i\hat{T}\epsilon}\, e^{-i\hat{V}\epsilon}\right)^N, \qquad \epsilon = T/N. \tag{10.9}$$

Proof. Write $S_N \equiv e^{(\hat{A}+\hat{B})/N}$ for the true single-slice evolution and $P_N \equiv e^{\hat{A}/N}e^{\hat{B}/N}$ for the factored one, so that the assertion is $\lim_N P_N^N = e^{\hat{A}+\hat{B}} = S_N^N$. Taylor-expanding each exponential to second order in $1/N$,

$$S_N = \mathbf{1} + \frac{\hat{A}+\hat{B}}{N} + \frac{(\hat{A}+\hat{B})^2}{2N^2} + O(N^{-3}), \tag{10.10}$$

$$P_N = \left(\mathbf{1} + \frac{\hat{A}}{N} + \frac{\hat{A}^2}{2N^2} + \cdots\right)\left(\mathbf{1} + \frac{\hat{B}}{N} + \frac{\hat{B}^2}{2N^2} + \cdots\right) = \mathbf{1} + \frac{\hat{A}+\hat{B}}{N} + \frac{\hat{A}^2 + 2\hat{A}\hat{B} + \hat{B}^2}{2N^2} + O(1 \tag{10.11}$$

the two agree through order $1/N$ and differ at order $1/N^2$ only through the ordering of the cross term,

$$S_N - P_N = \frac{1}{2N^2}\left[(\hat{A}+\hat{B})^2 - \hat{A}^2 - 2\hat{A}\hat{B} - \hat{B}^2\right] + O(N^{-3}) = -\frac{1}{2N^2}\left[\hat{A},\hat{B}\right] + O(N^{-3}), \tag{10.12}$$

so $\|S_N - P_N\| \le C/N^2$ for some constant C and all large N. It remains to show that an $O(N^{-2})$ discrepancy per factor produces an $O(N^{-1})$ discrepancy in the N-fold product. This is the standard telescoping estimate: for any two operators X, Y,

$$X^N - Y^N = \sum_{k=0}^{N-1} X^{\,N-1-k}\,(X-Y)\,Y^{\,k}, \tag{10.13}$$

which collapses because consecutive terms cancel in pairs. Taking norms with $X = S_N$, $Y = P_N$ and using submultiplicativity,

$$\left\|S_N^N - P_N^N\right\| \le \sum_{k=0}^{N-1} \|S_N\|^{\,N-1-k}\,\|S_N - P_N\|\,\|P_N\|^{\,k} \le N\left(\max\{\|S_N\|, \|P_N\|\}\right)^{N-1}\|S_N - P_N\| \tag{10.14}$$

Both $\|S_N\|$ and $\|P_N\|$ are bounded by $e^{(\|\hat{A}\|+\|\hat{B}\|)/N}$, so their $(N-1)$-th power is bounded by the N-independent constant $e^{\|\hat{A}\|+\|\hat{B}\|}$. Hence

$$\left\|S_N^N - P_N^N\right\| \le N \cdot e^{\|\hat{A}\|+\|\hat{B}\|}\cdot\frac{C}{N^2} = \frac{C\,e^{\|\hat{A}\|+\|\hat{B}\|}}{N} \xrightarrow[N\to\infty]{} 0, \tag{10.15}$$

which establishes (10.8). Specializing to $\hat{A} = -i\hat{T}\,T$, $\hat{B} = -i\hat{V}\,T$ gives (10.9). □

Caution

The clean proof above assumes $\hat{A}$ and $\hat{B}$ bounded, whereas the physical kinetic and potential operators are unbounded (the spectrum of $\hat{p}^2/2m$ is all of $[0,\infty)$, and most potentials are unbounded too). The formula nonetheless survives in the form actually needed here: Trotter's theorem, sharpened by Kato, guarantees that (10.9) holds in the strong operator topology — that is, when both sides act on a fixed state vector — for self-adjoint $\hat{T}, \hat{V}$ whose sum $\hat{H}$ is essentially self-adjoint, which covers every Hamiltonian in this book. The strong-topology statement is exactly what (10.2) requires, since we only ever sandwich the operators between fixed states. We use the bounded-operator proof for its transparency and flag the functional-analytic upgrade; the rigorous continuum construction of the resulting measure is collected in Chapter E.

10.1.3 The short-time kernel

With (10.9) in hand we insert a resolution of the identity in the position basis between every pair of adjacent factors. Labelling the intermediate positions $x_1, \dots, x_{N-1}$ and writing $x_0 \equiv x_i$, $x_N \equiv x_f$ for the fixed endpoints,

$$K(x_f, T; x_i, 0) = \int\Big(\prod_{j=1}^{N-1} \mathrm{d}x_j\Big) \prod_{j=0}^{N-1} \Big\langle x_{j+1}\Big| e^{-i\hat{T}\epsilon}\, e^{-i\hat{V}\epsilon}\Big| x_j\Big\rangle , \tag{10.16}$$

where each of the N factors is a *short-time kernel* carrying the evolution from x_j to x_{j+1} across one slice. The integral runs over the values of the position at every intermediate time; this is the discrete skeleton of the sum over histories, sketched in Fig. 10.1. Everything now hinges on evaluating one short-time kernel, and here the factorization pays off. The potential operator $\hat{V} = V(\hat{x})$ is diagonal in the position basis, so acting to the right on $|x_j\rangle$ it is just a number,

$$e^{-i\hat{V}\epsilon}\,|x_j\rangle = e^{-i\epsilon V(x_j)}\,|x_j\rangle\,. \tag{10.17}$$

The kinetic operator $\hat{T} = \hat{p}^{\,2}/2m$ is diagonal in the *momentum* basis, so we evaluate its matrix element by inserting a complete set of momentum states from (10.4):

$$\Big\langle x_{j+1}\Big| e^{-i\hat{T}\epsilon}\Big| x_j\Big\rangle = \int \frac{\mathrm{d}p_j}{2\pi}\, \Big\langle x_{j+1}\Big| e^{-i\hat{p}^{\,2}\epsilon/2m}\Big| p_j\Big\rangle\, \langle p_j|x_j\rangle = \int \frac{\mathrm{d}p_j}{2\pi}\, e^{-ip_j^2\epsilon/2m}\, e^{ip_j(x_{j+1}-x_j)}. \tag{10.18}$$

This is a Gaussian integral over the slice momentum p_j, of exactly the oscillatory (Fresnel) type catalogued in Chapter 5. We isolate the evaluation as a lemma, since the same integral recurs at every step of the chapter.

Lemma 10.3 (Fresnel integral)**.** *For real $a > 0$ and complex b, defining the oscillatory integral by the convergence-generating tilt $a \to a - i0^+$ (the Feynman $i\epsilon$ prescription of Section 5.1, applied here to the slice),*

$$\int_{-\infty}^{\infty} \mathrm{d}p\; e^{-iap^2+ibp} = \sqrt{\frac{\pi}{ia}}\; e^{\,ib^2/4a}, \qquad \int_{-\infty}^{\infty} \mathrm{d}x\; e^{+iAx^2+iBx} = \sqrt{\frac{i\pi}{A}}\; e^{-iB^2/4A}, \tag{10.19}$$

where the branch is fixed by $\sqrt{i} = e^{i\pi/4}$, equivalently by analytic continuation from the convergent Gaussian.

Proof. Complete the square in the first integral, $-iap^2 + ibp = -ia\big(p - \frac{b}{2a}\big)^2 + \frac{ib^2}{4a}$, and shift p. The shifted integral is $\int \mathrm{d}u\, e^{-iau^2}$, which converges absolutely once a carries the negative imaginary part $a - i0^+$ and equals $\sqrt{\pi/(ia)}$ — this is the purely oscillatory case of the Gaussian computed in Theorem 5.16, where the contour is rotated by $\pi/4$ onto the line of steepest descent. The second identity is the same statement with $a \to -A$, completing the square as $iAx^2 + iBx = iA\big(x + \frac{B}{2A}\big)^2 - \frac{iB^2}{4A}$ and using $\int \mathrm{d}u\, e^{iAu^2} = \sqrt{i\pi/A}$. □

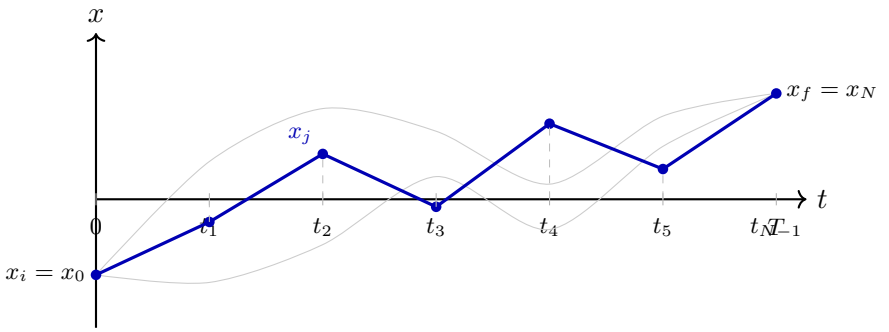


Figure 10.1: The discrete skeleton of the path integral. Time is sliced into N steps of width $\epsilon = T/N$; a single history is the broken line through the intermediate positions $x_1, \dots, x_{N-1}$ (blue), with endpoints $x_0 = x_i$ and $x_N = x_f$ held fixed. The amplitude (10.16) integrates each short-time kernel (10.21) over all values of every intermediate x_j, i.e. over all broken-line histories (two further ones shown faint). As $N \to \infty$ the broken lines fill out the space of continuous paths and the exponent of the product becomes i times the classical action.

Applying the first identity of (10.19) to (10.18) with $a = \epsilon/2m$ and $b = x_{j+1} - x_j$,

$$\Big\langle x_{j+1}\Big|e^{-i\hat{T}\epsilon}\Big|x_j\Big\rangle = \frac{1}{2\pi}\sqrt{\frac{2\pi m}{i\epsilon}}\,\exp\Big[\frac{im(x_{j+1}-x_j)^2}{2\epsilon}\Big] = \sqrt{\frac{m}{2\pi i\epsilon}}\,\exp\Big[\frac{im(x_{j+1}-x_j)^2}{2\epsilon}\Big]. \tag{10.20}$$

Combining this with the potential factor (10.17), the full short-time kernel is

$$\Big\langle x_{j+1}\Big|e^{-i\hat{T}\epsilon}e^{-i\hat{V}\epsilon}\Big|x_j\Big\rangle = \sqrt{\frac{m}{2\pi i\epsilon}}\,\exp\Big[\,i\epsilon\Big(\frac{m}{2}\Big(\frac{x_{j+1}-x_j}{\epsilon}\Big)^2 - V(x_j)\Big)\Big], \tag{10.21}$$

valid up to the $O(\epsilon^2)$ correction tracked in (10.7), which is subleading once summed over slices. The exponent is the discrete version of $i\epsilon\, L$, where $L = \frac{m}{2}\dot{x}^2 - V(x)$ is the classical Lagrangian: the kinetic term is built from the discrete velocity $(x_{j+1} - x_j)/\epsilon$ and the potential is read off at the slice. This is the moment at which the Lagrangian, an object the canonical formalism discarded after the Legendre transform, walks back onto the stage.

10.1.4 The phase-space path integral

It is worth pausing one step earlier in the calculation, before the momentum integrals (10.18) are done, because the intermediate expression is itself a fundamental object — the path integral in phase space — and it is the form best suited to constrained and gauge systems later in the book. Keep the momentum integral in each slice undone. Then each factor of (10.16) reads

$$\Big\langle x_{j+1}\Big|e^{-i\hat{T}\epsilon}e^{-i\hat{V}\epsilon}\Big|x_j\Big\rangle = \int\frac{\mathrm{d}p_j}{2\pi}\,\exp\Big[\,ip_j(x_{j+1}-x_j) - i\epsilon\,H(p_j,x_j)\Big], \tag{10.22}$$

where $H(p,x) = p^2/2m + V(x)$ is now the classical Hamiltonian *function* (the operator's symbol), and the ordering chosen — kinetic momentum to the left of potential position — has fixed it unambiguously. Assembling the N factors, the amplitude becomes an integral over N momenta and $N-1$ positions,

$$K(x_f,T;x_i,0) = \lim_{N\to\infty} \int \Big(\prod_{j=1}^{N-1} \mathrm{d}x_j\Big)\Big(\prod_{j=0}^{N-1} \frac{\mathrm{d}p_j}{2\pi}\Big) \exp\Big[\, i \sum_{j=0}^{N-1} \Big(p_j\,(x_{j+1}-x_j) - \epsilon\, H(p_j,x_j)\Big)\Big]. \tag{10.23}$$

In the continuum limit the discrete sum in the exponent becomes the phase-space action, $p_j(x_{j+1}-x_j) \to p\,\dot{x}\,\mathrm{d}t$ and $\epsilon H \to H\,\mathrm{d}t$, and we write the result as

$$\boxed{K(x_f,T;x_i,0) = \int \mathcal{D}x\,\mathcal{D}p\ \exp\Big[\, i\int_0^T \mathrm{d}t\,\big(p\,\dot{x} - H(p,x)\big)\Big]} \tag{10.24}$$

with the boundary conditions $x(0)=x_i$, $x(T)=x_f$ on the position path and no boundary condition on the momentum path. The asymmetry — one more momentum integral than position integral, the momenta unconstrained at the ends — is not an accident of bookkeeping; it reflects that the path integral knows phase space, where x and p are independent coordinates, and that the amplitude is diagonal in x at the endpoints by construction. Equation (10.24) is the most primitive form of the path integral and the one from which all others descend.

10.1.5 Gaussian momentum integration and the configuration-space measure

For the Hamiltonians of this book the momentum enters H only quadratically, so the momentum integrals in (10.23) are Gaussian and can be done once and for all. This is the step that produces the Lagrangian form together with the precise normalization of the measure — the result the next section depends on. Performing the integral over p_j in (10.22) is exactly the computation (10.18)–(10.20) run in reverse: each momentum integral contributes the factor $\sqrt{m/2\pi i\epsilon}$ and converts $p_j\dot{x} - \frac{p_j^2}{2m}$ into $\frac{m}{2}\dot{x}^2$. Carrying this out in every slice turns (10.23) into

$$K(x_f,T;x_i,0) = \lim_{N\to\infty} \Big(\frac{m}{2\pi i\epsilon}\Big)^{N/2} \int \Big(\prod_{j=1}^{N-1} \mathrm{d}x_j\Big)\, \exp\Big[\, i\epsilon \sum_{j=0}^{N-1} \Big(\frac{m}{2}\Big(\frac{x_{j+1}-x_j}{\epsilon}\Big)^2 - V(x_j)\Big)\Big]. \tag{10.25}$$

Here is the bookkeeping that fixes the measure, and it must be stated exactly because it is the single most error-prone point in the whole construction. There are N slices, hence N momentum integrals, hence N factors of $\sqrt{m/2\pi i\epsilon}$; but there are only $N-1$ intermediate positions to integrate over, the two endpoints being fixed. The measure therefore carries one more power of the normalization

than it has integration variables:

$$\mathcal{D}x \;\equiv\; \lim_{N\to\infty}\Big(\frac{m}{2\pi i\epsilon}\Big)^{N/2}\prod_{j=1}^{N-1}\mathrm{d}x_j, \qquad \epsilon=\frac{T}{N}. \tag{10.26}$$

The exponent of (10.25) is a Riemann sum: as $\epsilon\to 0$ the discrete velocity $(x_{j+1}-x_j)/\epsilon$ becomes $\dot{x}(t)$ and the sum $\epsilon\sum_j(\cdots)$ becomes the time integral of the Lagrangian, that is, the classical action

$$S[x]=\int_0^T \mathrm{d}t\; L(x,\dot{x})=\int_0^T \mathrm{d}t\,\Big(\frac{m}{2}\dot{x}^2-V(x)\Big). \tag{10.27}$$

We have arrived at Feynman's formula.

Theorem 10.4 (Configuration-space path integral)**.** *The transition amplitude* (10.2) *of a one-dimensional quantum particle with Hamiltonian* $\hat{H}=\hat{p}^{\,2}/2m+V(\hat{x})$ *equals the sum over histories*

$$\boxed{K(x_f,T;x_i,0)=\int_{x(0)=x_i}^{x(T)=x_f}\mathcal{D}x\; e^{\,iS[x]}} \tag{10.28}$$

with the classical action $S[x]$ *of* (10.27) *and the measure* $\mathcal{D}x$ *defined by the discretization* (10.26)*. The integral runs over all continuous paths* $x(t)$ *obeying the boundary conditions* $x(0)=x_i$ *and* $x(T)=x_f$*, and is to be understood as the* $N\to\infty$ *limit of the finite-dimensional integral* (10.25)*.*

This is the central identity of the section, and three remarks make its meaning precise.

It is an *equality*, derived from (10.2) by inserting completeness relations and doing Gaussian integrals; no new postulate has entered. Each weight $e^{iS[x]}$ is a pure phase of unit modulus, so the path integral is not a probability average over histories but a coherent sum of amplitudes: histories interfere. The classical trajectory is distinguished only because it is the stationary point of S, near which neighbouring phases add constructively while elsewhere they cancel — the principle of stationary action recovered as a stationary-phase statement, made quantitative for the oscillator in Theorem 10.7.

The symbol $\mathcal{D}x$ has been *defined*, not merely written: it is the limit (10.26), prefactor and all. Taken in isolation that limit does not converge — the prefactor $(m/2\pi i\epsilon)^{N/2}$ diverges as $\epsilon\to 0$ and the typical path is nowhere differentiable, so $\dot{x}$ in (10.27) is formal. What is well defined is the *combination* in (10.25): the prefactor's divergence is exactly compensated by the integrals, as the free-particle and oscillator computations below verify by producing finite, correct answers. For Gaussian (free) actions the manipulations are exact at finite N and the limit is clean; for interacting actions the limit defines the theory perturbatively, and

its rigorous construction — the genuine measure-theoretic content, the role of the Wiener measure after Wick rotation, and the functional determinants — is deferred to Chapter E. We will be candid wherever a step is formal.

Finally, the continuum notation $\int \mathrm{d}t\,(\frac{m}{2}\dot{x}^2 - V)$ should always be read as shorthand for the well-defined discrete sum in (10.25). Whenever an ambiguity threatens — an ordering question, a measure factor, a would-be total derivative — the discretization is the arbiter, and we return to it.

Physical Insight

The path integral trades the operator non-commutativity of canonical quantization for an integral over ordinary numbers. Everything quantum — interference, the uncertainty principle, the spreading of wave packets — is repackaged into the instruction to sum the phase e^{iS} over *all* histories rather than to follow the single classical one. The non-commutativity has not disappeared; it survives as the $O(\epsilon^2)$ ordering correction (10.7) and, equivalently, as the fact that the typical contributing path is non-differentiable, so that $(\Delta x)^2 \sim \epsilon$ rather than ϵ^2 — the discrete fingerprint of $[\hat{x}, \hat{p}] = i$. This last fact is exactly the moment $M_2 = i\epsilon/m$ that drives the recovery of the Schrödinger equation in Theorem 10.6.

10.1.6 The free particle

The first test of the machine is the free particle, $V \equiv 0$, for which the path integral can be evaluated in closed form and compared against the elementary result. It also exhibits, in the cleanest possible setting, why the measure normalization (10.26) is exactly right.

Worked Example 10.5 (The free-particle propagator)**.** Set $V = 0$. The fastest evaluation bypasses the discretization: since $\hat{H} = \hat{p}^2/2m$ is diagonal in momentum, insert one momentum completeness relation into (10.2),

$$K_0(x_f, T; x_i, 0) = \int \frac{\mathrm{d}p}{2\pi}\, e^{-ip^2T/2m}\, \langle x_f|p\rangle\, \langle p|x_i\rangle = \int \frac{\mathrm{d}p}{2\pi}\, e^{-ip^2T/2m}\, e^{ip(x_f - x_i)}. \tag{10.29}$$

This is the Fresnel integral (10.19) with $a = T/2m$ and $b = x_f - x_i$, giving

$$\boxed{K_0(x_f, T; x_i, 0) = \sqrt{\frac{m}{2\pi i T}}\, \exp\Big[\frac{im(x_f - x_i)^2}{2T}\Big].} \tag{10.30}$$

The exponent is precisely iS_{cl}: the classical free path is the straight line $x(t) = x_i + (x_f - x_i)t/T$ with constant velocity $(x_f - x_i)/T$, whose action is $S_{\mathrm{cl}} = \frac{m}{2}\big((x_f - x_i)/T\big)^2 T = m(x_f - x_i)^2/2T$. So already here the amplitude has the semiclassical shape $K_0 = (\text{prefactor}) \times e^{iS_{\mathrm{cl}}}$, with the prefactor independent of the endpoints.

It remains to check that the *discretized* definition (10.25)–(10.26) reproduces (10.30), for this is the consistency test of the measure. The key is the composition (semigroup) property of the free kernel under convolution. Take two free short-time kernels of durations ϵ_1 and ϵ_2 and integrate over the shared intermediate point x_1:

$$\int_{-\infty}^{\infty} \mathrm{d}x_1 \sqrt{\frac{m}{2\pi i\epsilon_1}}\, e^{\,im(x_2-x_1)^2/2\epsilon_1} \sqrt{\frac{m}{2\pi i\epsilon_2}}\, e^{\,im(x_1-x_0)^2/2\epsilon_2}. \tag{10.31}$$

The exponent is quadratic in x_1: writing it as $iA\,x_1^2 + iB\,x_1 + iC$ with

$$A = \frac{m}{2}\Big(\frac{1}{\epsilon_1}+\frac{1}{\epsilon_2}\Big), \quad B = -m\Big(\frac{x_2}{\epsilon_1}+\frac{x_0}{\epsilon_2}\Big), \quad C = \frac{m}{2}\Big(\frac{x_2^2}{\epsilon_1}+\frac{x_0^2}{\epsilon_2}\Big), \tag{10.32}$$

the second Fresnel identity in (10.19) integrates x_1 out and leaves the exponent $i(C - B^2/4A)$. A short computation with $u = 1/\epsilon_1$, $v = 1/\epsilon_2$ collapses the bracket,

$$C - \frac{B^2}{4A} = \frac{m}{2}\,\frac{(ux_2^2+vx_0^2)(u+v)-(ux_2+vx_0)^2}{u+v} = \frac{m}{2}\,\frac{uv\,(x_2-x_0)^2}{u+v} = \frac{m\,(x_2-x_0)^2}{2(\epsilon_1+\epsilon_2)}, \tag{10.33}$$

while the prefactors combine, using $\sqrt{i\pi/A} = \sqrt{2\pi i\epsilon_1\epsilon_2/m(\epsilon_1+\epsilon_2)}$, into

$$\sqrt{\frac{m}{2\pi i\epsilon_1}}\sqrt{\frac{m}{2\pi i\epsilon_2}}\sqrt{\frac{i\pi}{A}} = \sqrt{\frac{m}{2\pi i(\epsilon_1+\epsilon_2)}}. \tag{10.34}$$

Hence (10.31) equals the single free kernel of duration $\epsilon_1+\epsilon_2$,

$$K_0(\epsilon_1+\epsilon_2) = \int \mathrm{d}x_1\; K_0(\epsilon_2)\,K_0(\epsilon_1), \tag{10.35}$$

which is the composition law $e^{-i\hat{H}(\epsilon_1+\epsilon_2)} = e^{-i\hat{H}\epsilon_2}e^{-i\hat{H}\epsilon_1}$ realized on kernels. By induction the chained integral over $N-1$ intermediate points of N kernels of duration $\epsilon = T/N$ gives one kernel of duration T, namely (10.30), *for every* N. The discretized path integral is thus not merely correct in the limit; for the free particle it is exact at every finite N, and the measure factor in (10.26) is precisely what makes the chain close onto itself with no leftover constant. This is the cleanest possible evidence that the normalization is right.

10.1.7 Consistency: recovery of the Schrödinger equation

The path integral was derived *from* the Schrödinger evolution, so it must reproduce it; verifying this directly from the short-time kernel both closes the logical loop and shows in microscopic detail how the differential equation is encoded in one infinitesimal slice. The argument is Feynman's original one.

Theorem 10.6 (The short-time kernel generates the Schrödinger equation)**.** *Let $\psi(x,t)$ evolve by one slice through the short-time kernel* (10.21)*, i.e.*

$$\psi(x,t+\epsilon)=\int_{-\infty}^{\infty}\mathrm{d}y\,\sqrt{\frac{m}{2\pi i\epsilon}}\,\exp\!\Big[\frac{im(x-y)^2}{2\epsilon}\Big]\,e^{-i\epsilon V(y)}\,\psi(y,t). \tag{10.36}$$

Then, to first order in ϵ, ψ satisfies the Schrödinger equation

$$i\,\partial_t\psi(x,t)=-\frac{1}{2m}\,\partial_x^2\psi(x,t)+V(x)\,\psi(x,t). \tag{10.37}$$

Proof. Substitute $y=x+\xi$, so that the Gaussian weight is $\exp[im\xi^2/2\epsilon]$. This weight oscillates ever faster as $\epsilon\to 0$ and concentrates the integral on $\xi^2\sim\epsilon$, i.e. on displacements of order $\xi\sim\sqrt{\epsilon}$; this scaling — $\xi\sim\sqrt{\epsilon}$, not $\xi\sim\epsilon$ — is the quantum fingerprint noted above, and it dictates how far to Taylor-expand. We need the answer through $O(\epsilon)$, so we keep ψ to second order in ξ and the potential to first order in ϵ:

$$e^{-i\epsilon V(x+\xi)}=1-i\epsilon V(x)+O(\epsilon^{3/2}),\qquad \psi(x+\xi,t)=\psi+\xi\,\partial_x\psi+\tfrac{1}{2}\xi^2\,\partial_x^2\psi+O(\xi^3), \tag{10.38}$$

where the $O(\epsilon^{3/2})$ in the first expansion comes from $\epsilon\,\xi\sim\epsilon^{3/2}$. The integral then requires the first three Gaussian moments, all instances of the Fresnel lemma (10.19):

$$\sqrt{\frac{m}{2\pi i\epsilon}}\int\mathrm{d}\xi\,e^{im\xi^2/2\epsilon}=1,\qquad \sqrt{\frac{m}{2\pi i\epsilon}}\int\mathrm{d}\xi\,\xi\,e^{im\xi^2/2\epsilon}=0,\qquad \sqrt{\frac{m}{2\pi i\epsilon}}\int\mathrm{d}\xi\,\xi^2\,e^{im\xi^2/2\epsilon}=\frac{i\epsilon}{m}. \tag{10.39}$$

The first is the normalization (10.20) at $x_{j+1}=x_j$; the second vanishes by oddness; the third follows from differentiating the first under the integral sign with respect to the parameter, $\langle\xi^2\rangle=(-i\,\partial/\partial(m/2\epsilon))\,\ln\sqrt{\cdots}=i\epsilon/m$. Inserting the expansions and the moments into (10.36),

$$\psi(x,t+\epsilon)=\big(1-i\epsilon V(x)\big)\Big[\psi+\tfrac{1}{2}\,\frac{i\epsilon}{m}\,\partial_x^2\psi\Big]+O(\epsilon^{3/2})=\psi+\frac{i\epsilon}{2m}\,\partial_x^2\psi-i\epsilon V(x)\,\psi+O(\epsilon^{3/2}). \tag{10.40}$$

The left side is $\psi(x,t)+\epsilon\,\partial_t\psi+O(\epsilon^2)$. Matching the terms of order ϵ and cancelling the common $\psi(x,t)$,

$$\partial_t\psi=\frac{i}{2m}\,\partial_x^2\psi-iV(x)\,\psi, \tag{10.41}$$

which is (10.37) after multiplication by i. The path integral and the Schrödinger equation are therefore the same dynamics; the $O(\epsilon)$ matching is exactly the place where the kinetic term of the action becomes the Laplacian and the potential term becomes multiplication by V. □

10.1.8 The harmonic oscillator

The free particle was Gaussian because its action was quadratic with no potential; the harmonic oscillator is the next solvable case, quadratic *with* a restoring force, and it is the single most important worked example in the chapter. It is the quantum-mechanical template for the free field — a field is a continuum of oscillators (Section 9.1.2), so its generating functional in Section 10.3 is an infinite product of the result computed here — and it exhibits, exactly, the semiclassical structure "$e^{iS_{\rm cl}}$ times a fluctuation determinant" that organizes every Gaussian path integral in the book.

Worked Example 10.7 (The harmonic-oscillator propagator and its fluctuation determinant)**.** Take the oscillator action, in the conventions of the oscillator template Section 9.1.2 and the free Hamiltonian Eq. (9.3) from which it descends,

$$S[x] = \int_0^T \mathrm{d}t\, \Big(\frac{m}{2}\dot{x}^2 - \frac{m\omega^2}{2}x^2\Big), \tag{10.42}$$

and compute $K_{\rm HO} = \int \mathcal{D}x\, e^{iS[x]}$ with $x(0) = x_i$, $x(T) = x_f$.

Step 1: exact split into classical path and fluctuation. Let $x_{\rm cl}(t)$ be the solution of the classical equation of motion $\ddot{x}_{\rm cl} + \omega^2 x_{\rm cl} = 0$ obeying the boundary conditions, and write a general path as $x(t) = x_{\rm cl}(t) + \eta(t)$ with the fluctuation η vanishing at both ends, $\eta(0) = \eta(T) = 0$. Because S is quadratic, its Taylor expansion in η terminates exactly at second order,

$$S[x_{\rm cl}+\eta] = S[x_{\rm cl}] + \underbrace{\int_0^T \mathrm{d}t\, \Big(m\dot{x}_{\rm cl}\dot{\eta} - m\omega^2 x_{\rm cl}\eta\Big)}_{\text{linear}} + \int_0^T \mathrm{d}t\, \Big(\frac{m}{2}\dot{\eta}^2 - \frac{m\omega^2}{2}\eta^2\Big). \tag{10.43}$$

The linear term vanishes identically: integrating its first piece by parts and using $\eta(0) = \eta(T) = 0$ to kill the boundary term gives $-\int m\eta(\ddot{x}_{\rm cl} + \omega^2 x_{\rm cl})\,\mathrm{d}t = 0$ by the classical equation of motion. The split is therefore exact and the measure is translation-invariant under $x \to x_{\rm cl} + \eta$ (a constant shift of the integration variables in (10.26)), so

$$K_{\rm HO} = e^{\,iS_{\rm cl}} \underbrace{\int_{\eta(0)=\eta(T)=0} \mathcal{D}\eta\, \exp\Big[\, i\int_0^T \mathrm{d}t\, \Big(\tfrac{m}{2}\dot{\eta}^2 - \tfrac{m\omega^2}{2}\eta^2\Big)\Big]}_{F(T)}, \qquad S_{\rm cl} \equiv S[x_{\rm cl}]. \tag{10.44}$$

The fluctuation factor $F(T)$ no longer depends on the endpoints x_i, x_f — they have been absorbed entirely into $x_{\rm cl}$ — so it is a pure function of T. This is the promised semiclassical structure, here exact rather than approximate because the action is Gaussian.

Step 2: the classical action. Solve $\ddot{x}_{\rm cl} + \omega^2 x_{\rm cl} = 0$ with $x_{\rm cl}(0) = x_i$, $x_{\rm cl}(T) = x_f$:

$$x_{\rm cl}(t) = \frac{x_f \sin\omega t + x_i \sin\omega(T-t)}{\sin\omega T}. \tag{10.45}$$

On shell the action reduces to a boundary term: integrating (10.42) by parts,

$$S_{\rm cl} = \frac{m}{2}\big[x_{\rm cl}\dot{x}_{\rm cl}\big]_0^T - \frac{m}{2}\int_0^T \mathrm{d}t\, x_{\rm cl}(\ddot{x}_{\rm cl}+\omega^2 x_{\rm cl}) = \frac{m}{2}\big[x_f\,\dot{x}_{\rm cl}(T) - x_i\,\dot{x}_{\rm cl}(0)\big], \tag{10.46}$$

the bulk integral vanishing by the equation of motion. From $\dot{x}_{\rm cl}(t) = \omega(x_f\cos\omega t - x_i\cos\omega(T-t))/\sin\omega T$ we read off $\dot{x}_{\rm cl}(T) = \omega(x_f\cos\omega T - x_i)/\sin\omega T$ and $\dot{x}_{\rm cl}(0) = \omega(x_f - x_i\cos\omega T)/\sin\omega T$, whence

$$\boxed{S_{\rm cl} = \frac{m\omega}{2\sin\omega T}\Big[(x_f^2+x_i^2)\cos\omega T - 2x_f x_i\Big].} \tag{10.47}$$

As $\omega\to 0$ this reduces to $m(x_f-x_i)^2/2T$, the free classical action, as it must.

Step 3: the fluctuation determinant by mode expansion. Integrate the quadratic action in (10.44) by parts once more (boundary term again killed by $\eta(0)=\eta(T)=0$) to write it as a quadratic form in the operator $\hat{O} = -\partial_t^2 - \omega^2$,

$$\int_0^T \mathrm{d}t\,\Big(\tfrac{m}{2}\dot{\eta}^2 - \tfrac{m\omega^2}{2}\eta^2\Big) = \frac{m}{2}\int_0^T \mathrm{d}t\,\eta\,(-\partial_t^2-\omega^2)\,\eta. \tag{10.48}$$

The eigenfunctions of $-\partial_t^2$ on $[0,T]$ obeying Dirichlet conditions $\eta(0)=\eta(T)=0$ are the sine modes

$$\eta_n(t) = \sin\frac{n\pi t}{T}, \qquad -\partial_t^2\,\eta_n = \Big(\frac{n\pi}{T}\Big)^2\eta_n, \qquad n = 1,2,3,\ldots, \tag{10.49}$$

a complete orthogonal basis for fluctuations with these boundary conditions. Expand $\eta(t) = \sum_{n=1}^{\infty} a_n \sin(n\pi t/T)$. Using $\int_0^T \sin(n\pi t/T)\sin(n'\pi t/T)\,\mathrm{d}t = \frac{T}{2}\delta_{nn'}$, the action becomes a sum of decoupled quadratics in the mode amplitudes,

$$\int_0^T \mathrm{d}t\,\Big(\tfrac{m}{2}\dot{\eta}^2 - \tfrac{m\omega^2}{2}\eta^2\Big) = \frac{mT}{4}\sum_{n=1}^{\infty} a_n^2\Big[\Big(\frac{n\pi}{T}\Big)^2 - \omega^2\Big]. \tag{10.50}$$

Changing integration variables in the measure from the discretized values $\{\eta_j\}$ to the mode amplitudes $\{a_n\}$ introduces a Jacobian $\mathcal{J}$; crucially, $\mathcal{J}$ depends on the basis and the discretization but *not* on ω, since the modes and the change of variables are the same for any quadratic action. Hence the fluctuation factor is

$$F(T) = \mathcal{J}\prod_{n=1}^{\infty}\int_{-\infty}^{\infty}\mathrm{d}a_n\;\exp\Big[\,i\,\frac{mT}{4}a_n^2\Big(\Big(\tfrac{n\pi}{T}\Big)^2-\omega^2\Big)\Big] = \mathcal{J}\prod_{n=1}^{\infty}\sqrt{\frac{4\pi i/(mT)}{(n\pi/T)^2-\omega^2}}\,, \tag{10.51}$$

each integral being a Fresnel integral (10.19).

Step 4: fixing $\mathcal{J}$ by the free ratio. Both $\mathcal{J}$ and the individual mode integrals are separately ill-defined (the infinite product of $\sqrt{4\pi i/mT}$ diverges), but their ω-dependence is finite and clean, and that is all we need: the unknown $\mathcal{J}$ cancels

in the ratio of the oscillator factor to the free-particle factor, which is the same object at $\omega = 0$. Dividing (10.51) by itself with $\omega = 0$,

$$\frac{F_{\mathrm{HO}}(T)}{F_{\mathrm{free}}(T)} = \prod_{n=1}^{\infty} \sqrt{\frac{(n\pi/T)^2}{(n\pi/T)^2 - \omega^2}} = \prod_{n=1}^{\infty} \Big(1 - \frac{\omega^2 T^2}{n^2\pi^2}\Big)^{-1/2}. \tag{10.52}$$

This infinite product is evaluated by Euler's product formula for the sine, a classical result of complex analysis: $\sin z = z\prod_{n=1}^{\infty}(1 - z^2/n^2\pi^2)$, which encodes the zeros of $\sin z$ at $z = n\pi$. With $z = \omega T$ this gives $\prod_n(1 - \omega^2T^2/n^2\pi^2) = \sin(\omega T)/(\omega T)$, so the ratio (10.52) is $(\sin\omega T/\omega T)^{-1/2} = (\omega T/\sin\omega T)^{1/2}$. The free factor was computed in Theorem 10.5: matching (10.28) to (10.30) identifies $F_{\mathrm{free}}(T) = \sqrt{m/2\pi i T}$. Therefore

$$F_{\mathrm{HO}}(T) = \sqrt{\frac{m}{2\pi i T}}\Big(\frac{\omega T}{\sin\omega T}\Big)^{1/2} = \sqrt{\frac{m\omega}{2\pi i \sin\omega T}}\,, \tag{10.53}$$

and assembling (10.44), (10.47), (10.53),

$$\boxed{K_{\mathrm{HO}}(x_f, T; x_i, 0) = \sqrt{\frac{m\omega}{2\pi i \sin\omega T}}\,\exp\Big[\frac{im\omega}{2\sin\omega T}\Big((x_f^2 + x_i^2)\cos\omega T - 2x_f x_i\Big)\Big].} \tag{10.54}$$

Three checks confirm the result. As $\omega \to 0$, $\sin\omega T \to \omega T$ and the bracket $\to (x_f - x_i)^2$, so (10.54) collapses to the free propagator (10.30). As $T \to 0$ at fixed ω, $\sin\omega T \to \omega T$ likewise and $K_{\mathrm{HO}} \to \delta(x_f - x_i)$ in the distributional sense, as any propagator must at coincident times. Finally, the periodic singularities at $\omega T = n\pi$, where $\sin\omega T$ vanishes and the prefactor diverges, are the oscillator's classical focusing times — every trajectory launched from x_i returns to a single point after a half-period — and the extra phase the amplitude acquires on passing each focus is the Maslov index of the semiclassical expansion.

Remark 10.8 (The same determinant via Gelfand–Yaglom). The mode sum above *is* a regularized functional determinant: $F_{\mathrm{HO}}/F_{\mathrm{free}} = (\det(-\partial_t^2 - \omega^2)/\det(-\partial_t^2))^{-1/2}$ with Dirichlet conditions, and (10.52) is the statement $\det(-\partial_t^2 - \omega^2)/\det(-\partial_t^2) = \sin\omega T/\omega T$. The same ratio can be obtained without ever listing eigenvalues, by the Gelfand–Yaglom theorem: the determinant of $-\partial_t^2 + W(t)$ on $[0, T]$ with Dirichlet conditions is proportional to $D(T)$, where $D(t)$ solves the homogeneous equation $(-\partial_t^2 + W)D = 0$ with initial data $D(0) = 0$, $\dot D(0) = 1$, the proportionality constant cancelling in any ratio. For the oscillator $W = -\omega^2$ gives $D(t) = \sin(\omega t)/\omega$, and for the free case $W = 0$ gives $D_0(t) = t$; their ratio at $t = T$ is $\sin(\omega T)/\omega T$, reproducing (10.52) with no infinite product. We will not need the general theorem here — the mode sum delivered the answer in full — but the Gelfand–Yaglom recursion is the practical tool for fluctuation determinants in less symmetric backgrounds, and Chapter E develops it and the ζ-function regularization that makes the individual determinants, not merely their ratio, well defined.

Physical Insight

Equation (10.54) is the entire quantum oscillator in one line, and it is built from purely classical data: the classical action $S_{\rm cl}$ in the phase, and the classical *spread* of trajectories — the fluctuation determinant — in the amplitude. This is the general anatomy of a Gaussian path integral, and it is why free quantum field theory will turn out to be exactly solvable: a free field is Gaussian, so its generating functional is, mode by mode, an instance of (10.53), a classical action times a determinant. Interactions are the non-Gaussian corrections to this skeleton, organized as a perturbation series around it — the subject from Chapter 14 onward.

The section has delivered exactly the toolkit the rest of the chapter runs on. We have the short-time kernel (10.21); the configuration-space path integral (10.28) with its measure normalization $(m/2\pi i\epsilon)^{N/2}$ pinned down in (10.26); the phase-space form (10.24) for later constrained systems; and the two exactly evaluated propagators, free (10.30) and harmonic (10.54), together with the semiclassical structure $e^{iS_{\rm cl}}$ times a fluctuation determinant that they share. The next section, Section 10.2, puts this amplitude to work as a *generator*: by coupling the path to an external source $J(t)$ and projecting onto the ground state with the $i\epsilon$ tilt of Section 5.4, it promotes K into the vacuum generating functional $Z[J]$, from which time-ordered correlation functions drop out as functional derivatives — the bridge from this quantum-mechanical warm-up to the field-theoretic ϕ^4 theory.

10.2 Correlation Functions and the Generating Functional

The previous section turned the transition amplitude of a single quantum-mechanical degree of freedom into a sum over histories, $\langle x_f, t_f | x_i, t_i\rangle = \int \mathcal{D}x\, e^{iS[x]}$, and fixed the configuration-space measure by exact computation for the harmonic oscillator, where the integral could be done in closed form. That amplitude, however, is not yet the object a field theorist wants. What the perturbative expansion of Chapter 14 will be built from — and what Section 9.5.2 already identified as the basic correlator of the canonical theory — is the vacuum expectation value of a *time-ordered product* of operators. The task of this section is to promote the bare amplitude into a machine that manufactures exactly these objects: we couple the system to an external source $J(t)$, assemble the *generating functional* $Z[J]$, and show that every time-ordered correlation function is a functional derivative of $Z[J]$ with respect to J. Along the way two things that look like technical fine print turn out to carry the physics: the path integral produces the *time-ordered* product and no other (the ordering is not imposed by

hand but emerges from the time-slicing), and a small tilt of the time contour into the complex plane is precisely what projects the dangling endpoint states onto the interacting ground state. Everything here is done for one degree of freedom $x(t)$, the quantum-mechanical template; the next section lifts the entire apparatus, symbol for symbol, to the scalar field of ϕ^4 theory.

Throughout, H is the system Hamiltonian, $|n\rangle$ its orthonormal energy eigenstates with $H|n\rangle = E_n|n\rangle$ and $E_0 < E_1 \le E_2 \le \cdots$, and $|\Omega\rangle \equiv |0\rangle$ is the (assumed non-degenerate) ground state. Heisenberg-picture operators are $\hat{x}(t) = e^{iHt}\hat{x}\,e^{-iHt}$. We use the mostly-minus conventions of Chapter 9; for a single particle of unit mass the action is $S[x] = \int dt\, L$ with $L = \frac{1}{2}\dot{x}^2 - V(x)$.

10.2.1 Path-integral insertions are time-ordered products

We begin with the cleanest statement and then prove it: inserting the c-number $x(t_1)\cdots x(t_n)$ into the path-integral integrand computes the matrix element of the *time-ordered* operator product $\mathcal{T}\,\hat{x}(t_1)\cdots\hat{x}(t_n)$, with later operators standing to the left. The ordering is not an extra rule. It is forced on us by the only consistent way to cut the evolution into pieces.

Recall first the single-slice identity behind the path integral. The amplitude $\langle x_f, t_f | x_i, t_i\rangle = \left\langle x_f \middle| e^{-iH(t_f - t_i)} \middle| x_i \right\rangle$ is built by chopping $[t_i, t_f]$ into many short steps and inserting a resolution of the identity $\int dx\ |x\rangle\langle x| = \mathbf{1}$ between each pair of adjacent factors $e^{-iH\,\Delta t}$. The integration variables are the values $x(t)$ of the path at the intermediate times, and the product of short-time kernels assembles into $e^{iS[x]}$ in the continuum limit. Suppose now we wish to compute not the bare amplitude but the matrix element with one Heisenberg operator inserted at an intermediate time $t_1 \in (t_i, t_f)$,

$$\langle x_f | \hat{x}(t_1) | x_i\rangle = \left\langle x_f \middle| e^{-iH(t_f - t_1)}\, \hat{x}\, e^{-iH(t_1 - t_i)} \middle| x_i \right\rangle, \tag{10.55}$$

where we used $\left\langle x_f \middle| e^{iHt_f}\hat{x}e^{-iHt_1}\cdots \right\rangle$ and cancelled the outer phases against the definitions of $|x_f, t_f\rangle$, $|x_i, t_i\rangle$. Insert a position resolution of the identity immediately to the right of $\hat{x}$, at the instant t_1, and use $\hat{x}\,|x_1\rangle = x_1\,|x_1\rangle$:

$$\begin{aligned} \langle x_f | \hat{x}(t_1) | x_i\rangle &= \int dx_1\ \left\langle x_f \middle| e^{-iH(t_f - t_1)} \middle| x_1 \right\rangle\ x_1\ \left\langle x_1 \middle| e^{-iH(t_1 - t_i)} \middle| x_i \right\rangle \\ &= \int dx_1\ x_1 \left(\int_{x_1, t_1}^{x_f, t_f} \mathcal{D}x\, e^{iS} \right) \left(\int_{x_i, t_i}^{x_1, t_1} \mathcal{D}x\, e^{iS} \right). \end{aligned} \tag{10.56}$$

Each factor is itself a path integral over its subinterval, and gluing two path integrals that share the common endpoint x_1 at the common time t_1 — then integrating over that shared value x_1 — is nothing but the single unrestricted path integral over $[t_i, t_f]$, now carrying the extra factor equal to the path's value at t_1. Hence

$$\langle x_f | \hat{x}(t_1) | x_i\rangle = \int \mathcal{D}x\ x(t_1)\, e^{iS[x]}. \tag{10.57}$$

The lone operator went in cleanly because there was only one of it; ordering became an issue only when two or more insertions compete for the same slot.

Theorem 10.9 (Insertions generate the time-ordered product)**.** *For any times* $t_1, \dots, t_n \in (t_i, t_f)$*,*

$$\langle x_f | \mathcal{T}\, \hat{x}(t_1)\hat{x}(t_2)\cdots\hat{x}(t_n)|x_i\rangle = \int \mathcal{D}x\; x(t_1)x(t_2)\cdots x(t_n)\, e^{iS[x]}, \tag{10.58}$$

where $\mathcal{T}$ *orders the operators with later times to the left, as in Eq. (9.151).*

Proof. Take $n = 2$ first; the general case is identical with more labels. Consider the right-hand side of Eq. (10.58), $\int \mathcal{D}x\, x(t_1)x(t_2)e^{iS}$. The integrand contains the two ordinary numbers $x(t_1)$ and $x(t_2)$, evaluated on the same path, so the integrand is manifestly *symmetric* under $t_1 \leftrightarrow t_2$: relabelling the two insertion times does nothing to it. Now reconstruct the operator content by the slicing of Eq. (10.56). To cut the evolution operators into pieces with strictly forward time steps — the only way the short-time expansion $e^{-iH\Delta t}$ and the intervening completeness insertions make sense — we must list the cut times in *increasing* order. Suppose first $t_1 > t_2$. Then cutting at t_2 and at t_1 and inserting position eigenstates at both,

$$\begin{aligned}\int \mathcal{D}x\, x(t_1)x(t_2)e^{iS} &= \int dx_1\, dx_2\; x_1\, x_2 \left\langle x_f\middle|e^{-iH(t_f-t_1)}\middle|x_1\right\rangle \left\langle x_1\middle|e^{-iH(t_1-t_2)}\middle|x_2\right\rangle \left\langle x_2\middle|e^{-iH(t_2-t_i}\right. \\ &= \left\langle x_f\middle|e^{-iH(t_f-t_1)}\,\hat{x}\,e^{-iH(t_1-t_2)}\,\hat{x}\,e^{-iH(t_2-t_i)}\middle|x_i\right\rangle \\ &= \langle x_f|\hat{x}(t_1)\,\hat{x}(t_2)|x_i\rangle \qquad (t_1 > t_2), \end{aligned} \tag{10.59}$$

where the last line restores the Heisenberg operators by reattaching the phases $e^{\pm iHt}$. The operator with the *later* argument, $\hat{x}(t_1)$, came out on the *left*, because the leftmost intermediate evolution operator carries the latest time. If instead $t_2 > t_1$, the same c-number integral — unchanged, by the symmetry just noted — is reconstructed by cutting first at t_1 and then at t_2, and the identical manipulation delivers $\langle x_f|\hat{x}(t_2)\hat{x}(t_1)|x_i\rangle$, again with the later operator on the left. The two cases are summarized by a single formula precisely when the right-hand side carries the step functions of the time-ordering symbol,

$$\int \mathcal{D}x\, x(t_1)x(t_2)e^{iS} = \theta(t_1{-}t_2)\,\langle x_f|\hat{x}(t_1)\hat{x}(t_2)|x_i\rangle + \theta(t_2{-}t_1)\,\langle x_f|\hat{x}(t_2)\hat{x}(t_1)|x_i\rangle = \langle x_f|\mathcal{T}\hat{x}(t_1)\hat{x}$$

For general n, let π be the permutation that sorts the times into decreasing order $t_{\pi(1)} > t_{\pi(2)} > \cdots > t_{\pi(n)}$. Cutting at all n instants in increasing order and inserting position eigenstates at each yields the product of $n{+}1$ kernels with $x_{\pi(1)}\cdots x_{\pi(n)}$ inserted; reattaching phases gives $\left\langle x_f\middle|\hat{x}(t_{\pi(1)})\cdots\hat{x}(t_{\pi(n)})\middle|x_i\right\rangle$, the operators in decreasing-time (i.e. time-ordered) sequence. Because the c-number integrand $x(t_1)\cdots x(t_n)$ is totally symmetric, this holds for every choice of the t_k, which is the content of Eq. (10.58). □

Physical Insight

The lesson is worth stating plainly because it recurs everywhere in the path-integral formalism. Inside the functional integral the field values are ordinary commuting numbers, so the integrand never remembers an ordering; ordering is a property of *operators*, and it is reintroduced only when one re-expresses the integral as a sliced product of non-commuting evolution operators. The slicing can only run forward in time, so the operator product it produces is automatically the time-ordered one. This is why the path integral computes time-ordered Green's functions for free, and it is the deep reason the formalism makes Lorentz invariance manifest: there is no preferred time-slicing of a c-number product, hence no privileged frame hidden in the construction.

10.2.2 Projecting onto the vacuum: the $i\varepsilon$ tilt

Equation Eq. (10.58) still carries the arbitrary boundary data $|x_i, t_i\rangle$ and $\langle x_f, t_f|$. The correlators of quantum field theory are taken not between position eigenstates but in the *ground state*, $\langle \mathcal{T}\hat{x}(t_1)\cdots\hat{x}(t_n)\rangle \equiv \langle\Omega|\mathcal{T}\hat{x}(t_1)\cdots\hat{x}(t_n)|\Omega\rangle$. We must therefore replace the endpoint states by $|\Omega\rangle$. Doing so by hand would require knowing the ground-state wavefunction; the elegant alternative, which generalizes intact to the field theory, is to let an infinitesimal rotation of the time contour into the lower-half complex plane do the projection automatically.

Consider sandwiching the evolution operator between fixed but otherwise arbitrary states $|\chi_i\rangle$ and $\langle\chi_f|$ that have nonzero overlap with the vacuum, $\langle\Omega|\chi_i\rangle \neq 0 \neq \langle\chi_f|\Omega\rangle$. Insert the spectral resolution $\mathbf{1} = \sum_n |n\rangle\langle n|$:

$$\Big\langle \chi_f \Big| e^{-iHT} \Big| \chi_i \Big\rangle = \sum_n \langle\chi_f|n\rangle\, \langle n|\chi_i\rangle\, e^{-iE_nT}, \qquad T \equiv t_f - t_i. \tag{10.60}$$

Each term oscillates with its own frequency E_n, and no single term dominates as $T \to \infty$ along the real axis. Now tilt the time axis by an infinitesimal angle into the complex plane, replacing the elapsed time by $T \to T(1 - i\varepsilon)$ with $\varepsilon > 0$, equivalently rotating each instant $t \to t(1 - i\varepsilon)$. The phase becomes

$$e^{-iE_nT(1-i\varepsilon)} = e^{-iE_nT}\, e^{-\varepsilon E_nT}, \tag{10.61}$$

so that high-energy states are exponentially damped relative to low-energy ones as $T \to \infty$. Because the ground state has the lowest energy E_0, in the limit $T \to \infty(1 - i\varepsilon)$ the sum Eq. (10.60) is dominated by the $n = 0$ term:

$$\Big\langle \chi_f \Big| e^{-iHT} \Big| \chi_i \Big\rangle \xrightarrow{T\to\infty(1-i\varepsilon)} \langle\chi_f|\Omega\rangle\, \langle\Omega|\chi_i\rangle\, e^{-iE_0T}\, [1 + O(e^{-\varepsilon(E_1-E_0)T})]. \tag{10.62}$$

The tilt has converted the oscillating evolution operator into a (complex) projector onto the ground state, up to the c-number prefactor $\langle\chi_f|\Omega\rangle\,\langle\Omega|\chi_i\rangle\,e^{-iE_0T}$. This prefactor is universal: it knows nothing about the operator insertions, only about the endpoints and the total elapsed time. It will therefore cancel in any ratio in which it appears identically in numerator and denominator.

Physical Insight

The tilt $t \to t(1-i\varepsilon)$ is an *infinitesimal Wick rotation*. The full rotation $t \to -i\tau$ defined in Theorem 5.30 and Section 5.4.2 turns e^{-iHT} into the positive heat kernel $e^{-H\tau}$, which projects onto the ground state at $\tau \to +\infty$ with no oscillation at all and no need to take a limit through complex angles: $e^{-H\tau} \to |\Omega\rangle\, e^{-E_0\tau}\,\langle\Omega|$ manifestly. The $i\varepsilon$ prescription is the minimal amount of that Euclidean damping required to make the Minkowski limit well-defined; it is the same $i\varepsilon$ that we will see decorate every propagator. Tilting the contour and Wick-rotating fully are two ends of one continuous deformation, and the vacuum projection is what both accomplish.

We can now state the vacuum correlator as a normalized ratio of path integrals.

Theorem 10.10 (Ground-state correlators from the path integral)**.** *Let the system have a non-degenerate ground state $|\Omega\rangle$ and let the endpoints $|\chi_i\rangle, |\chi_f\rangle$ have nonzero vacuum overlap. With the time contour tilted as $t \to t(1-i\varepsilon)$ and the endpoints sent to $t_i \to -\infty$, $t_f \to +\infty$, the time-ordered vacuum correlators are*

$$\langle \mathcal{T}\,\hat{x}(t_1)\cdots\hat{x}(t_n)\rangle \;\equiv\; \langle\Omega|\mathcal{T}\,\hat{x}(t_1)\cdots\hat{x}(t_n)|\Omega\rangle = \frac{\displaystyle\int \mathcal{D}x\; x(t_1)\cdots x(t_n)\, e^{iS[x]}}{\displaystyle\int \mathcal{D}x\; e^{iS[x]}}, \tag{10.63}$$

where on both sides the action carries the $i\varepsilon$ tilt and the paths run over the whole (tilted) time axis with the fixed endpoint data, whose dependence cancels in the ratio.

Proof. Apply Theorem 10.9 but now contract the endpoints against $|\chi_i\rangle, \langle\chi_f|$ rather than position eigenstates. Split the evolution at the earliest insertion time $t_{\min} = \min_k t_k$ and the latest $t_{\max} = \max_k t_k$, writing the full evolution as $e^{-iH(t_f - t_{\max})}$, then the ordered block of operators and intervening evolutions confined to $[t_{\min}, t_{\max}]$, then $e^{-iH(t_{\min}-t_i)}$. As $t_f \to +\infty(1-i\varepsilon)$, the leading factor projects the left endpoint: $\langle\chi_f|\, e^{-iH(t_f-t_{\max})} \to \langle\chi_f|\Omega\rangle\, e^{-iE_0(t_f-t_{\max})}\,\langle\Omega|$ by the argument leading to Eq. (10.62), applied with the intervening operators held fixed at finite times. Likewise $t_i \to -\infty(1-i\varepsilon)$ projects the right endpoint,

$e^{-iH(t_{\min}-t_i)}\,|\chi_i\rangle \to |\Omega\rangle\,\langle\Omega|\chi_i\rangle\,e^{-iE_0(t_{\min}-t_i)}$. What survives between the two projectors is exactly $\langle\Omega|\mathcal{T}\hat{x}(t_1)\cdots\hat{x}(t_n)|\Omega\rangle$, multiplied by the endpoint-and-energy prefactor

$$P = \langle\chi_f|\Omega\rangle\,\langle\Omega|\chi_i\rangle\;e^{-iE_0(t_f-t_i)}.$$

This same P is produced with *no* insertions, since the denominator $\int\mathcal{D}x\,e^{iS} = \left\langle\chi_f\middle|e^{-iHT}\middle|\chi_i\right\rangle \to P$ by Eq. (10.62) directly. Dividing numerator by denominator cancels P and the wavefunction data along with it, leaving Eq. (10.63). □

Caution

The normalization by $\int\mathcal{D}x\,e^{iS}$ is not cosmetic. The unnormalized numerator carries the divergent (in the $T\to\infty$ limit) and endpoint-dependent prefactor P, which has no physical meaning; only the ratio is finite and independent of the arbitrary boundary states. In the field theory this same denominator is the sum of vacuum bubbles, and dividing by it is precisely what removes disconnected vacuum diagrams. The structure "correlator = insertions over no-insertions" established here is permanent.

10.2.3 The source and the generating functional

Computing $\langle\mathcal{T}\hat{x}(t_1)\cdots\hat{x}(t_n)\rangle$ one n at a time through Eq. (10.63) is correct but clumsy. The decisive simplification, due to Schwinger, is to introduce a classical external source $J(t)$ coupled linearly to the dynamical variable, so that a *single* functional packages all the correlators at once and individual ones are extracted by differentiation. We add to the Lagrangian a source term $L \to L + J(t)\,x(t)$, which shifts the action by $\int dt\,J(t)x(t)$, and integrate.

Definition 10.11 (Generating functional)**.** The *vacuum generating functional* of the quantum-mechanical system in the presence of the source J is

$$Z[J] \;\equiv\; \int\mathcal{D}x\;\exp\!\left\{i\int_{-\infty}^{\infty}dt\,\big[L(x,\dot{x}) + J(t)\,x(t)\big]\right\}, \tag{10.64}$$

the path integral taken over the $i\varepsilon$-tilted contour with vacuum-overlapping endpoints, as in Theorem 10.10. It is a functional of the source: a number assigned to each function J. We write $Z[0]$ for its value at $J\equiv 0$, the source-free path integral.

The point of the definition is that differentiating $Z[J]$ with respect to the source at a time t_1 pulls a factor $x(t_1)$ down from the exponent. Concretely, applying the functional derivative of Theorem 4.4 and Eq. (4.5) — here in its one-dimensional "index" form, where $\delta J(t)/\delta J(t') = \delta(t-t')$ — to Eq. (10.64) gives

$$\frac{\delta Z[J]}{\delta J(t_1)} = \int\mathcal{D}x\;\big[\,i\,x(t_1)\,\big]\;\exp\!\left\{i\int dt\,(L + Jx)\right\}, \tag{10.65}$$

because the only J-dependence is in the exponent, and $\frac{\delta}{\delta J(t_1)}\, i\int dt\, J(t)x(t) = i\,x(t_1)$. Each further derivative brings down another factor $i\,x(t_k)$. The factors of i are the only bookkeeping needed.

Theorem 10.12 (Correlators as functional derivatives of Z)**.** *The time-ordered vacuum correlation functions are the normalized functional derivatives of the generating functional at vanishing source:*

$$\langle \mathcal{T}\, \hat{x}(t_1)\cdots\hat{x}(t_n)\rangle = \frac{1}{i^{\,n}}\,\frac{1}{Z[0]}\,\frac{\delta^{\,n} Z[J]}{\delta J(t_1)\,\delta J(t_2)\cdots\delta J(t_n)}\bigg|_{J=0}. \tag{10.66}$$

Equivalently, with $W[J]\equiv \ln Z[J]$ and Z normalized so that $Z[0]=1$, $Z[J]$ is the moment-generating functional of the $x(t_k)$ and its Taylor coefficients in J are the correlators.

Proof. Differentiate Eq. (10.64) n times, each time using Eq. (10.65); since the integrand depends on J only through the smooth exponential, the derivatives pass under the path integral and act successively:

$$\frac{\delta^{\,n} Z[J]}{\delta J(t_1)\cdots\delta J(t_n)} = \int \mathcal{D}x\, \big[\,i\,x(t_1)\big]\cdots\big[\,i\,x(t_n)\big]\, \exp\!\Big\{ i\int dt\,(L + Jx)\Big\} = i^{\,n}\int \mathcal{D}x\; x(t_1)\cdots x(t$$

Setting $J=0$ removes the source from the exponent, leaving $i^{\,n}\int \mathcal{D}x\, x(t_1)\cdots x(t_n)\, e^{iS}$. Dividing by $Z[0]=\int \mathcal{D}x\, e^{iS}$ and the explicit $i^{\,n}$ reproduces exactly the ratio of Eq. (10.63), which Theorem 10.10 identified as $\langle \mathcal{T}\hat{x}(t_1)\cdots\hat{x}(t_n)\rangle$. This is Eq. (10.66). The Taylor statement follows by expanding the exponential of the source term inside Eq. (10.64) in powers of J: the coefficient of $\frac{i^n}{n!}\int J(t_1)\cdots J(t_n)$ is the symmetric kernel $\int \mathcal{D}x\, x(t_1)\cdots x(t_n) e^{iS}/Z[0]$, i.e. the n-point correlator, so

$$\frac{Z[J]}{Z[0]} = \sum_{n=0}^{\infty} \frac{i^{\,n}}{n!}\int dt_1\cdots dt_n\; \langle \mathcal{T}\,\hat{x}(t_1)\cdots\hat{x}(t_n)\rangle\; J(t_1)\cdots J(t_n), \tag{10.67}$$

the generating series of the correlators. □

This is the master identity the section was built to reach. It says that all the dynamical information of the quantum theory — every time-ordered correlator, hence (via the reduction formula of Chapter 14) every scattering amplitude — is encoded in the single functional $Z[J]$. The strategy from here is uniform: find $Z[J]$, then differentiate. For an interacting system $Z[J]$ cannot be computed exactly, and the differentiation is organized into Feynman diagrams; but for a quadratic action it can be done in closed form, and that solvable case is the foundation of all of perturbation theory. We turn to it now.

10.2.4 The oscillator generating functional

Specialize to the harmonic oscillator, the free theory of one degree of freedom and the $0+1$-dimensional avatar of the free scalar field. With unit mass and frequency ω,

$$S_0[x] = \int_{-\infty}^{\infty} dt \left[\tfrac{1}{2}\dot{x}^2 - \tfrac{1}{2}\omega^2 x^2\right] = \frac{1}{2}\int_{-\infty}^{\infty} dt\; x(t)\big(-\partial_t^2 - \omega^2\big)x(t), \tag{10.68}$$

where the second form follows from integrating $\frac{1}{2}\dot{x}^2$ by parts and discarding the boundary term, legitimate because the tilted contour forces $x(t) \to 0$ at the (complex) infinite endpoints. Define the kinetic operator

$$\mathcal{K} \equiv -\partial_t^2 - \omega^2, \tag{10.69}$$

so that $S_0 = \frac{1}{2}\int x\,\mathcal{K}x$. The $i\varepsilon$ tilt of Section 10.2.2 amounts, at the level of the action, to the replacement $\omega^2 \to \omega^2 - i\varepsilon$, as we verify below; this is the convergence factor that makes the oscillatory Gaussian well-defined. The source-coupled free generating functional is then

$$Z_0[J] = \int \mathcal{D}x\; \exp\!\left\{\frac{i}{2}\int dt\, x\,\mathcal{K}x + i\int dt\, Jx\right\}. \tag{10.70}$$

This is a (functional) Gaussian integral with a linear source. We evaluate it by completing the square, the continuum version of Theorem 4.15.

Theorem 10.13 (Free generating functional)**.** *The harmonic-oscillator generating functional is*

$$Z_0[J] = Z_0[0]\,\exp\!\left\{-\frac{1}{2}\int_{-\infty}^{\infty} dt\, dt'\; J(t)\,G(t-t')\,J(t')\right\}, \tag{10.71}$$

where the kernel G is the Feynman Green's function *of the kinetic operator $\mathcal{K}$, defined by*

$$\big(-\partial_t^2 - \omega^2\big)\,G(t-t') = i\,\delta(t-t'), \tag{10.72}$$

together with the boundary condition fixed by the $i\varepsilon$ tilt, namely the Fourier representation

$$G(t-t') = \int_{-\infty}^{\infty} \frac{d\nu}{2\pi}\,\frac{i\,e^{-i\nu(t-t')}}{\nu^2 - \omega^2 + i\varepsilon}. \tag{10.73}$$

Proof. The integral Eq. (10.70) is Gaussian, so it is evaluated exactly by shifting the integration variable to the classical trajectory in the presence of the source and isolating the source-dependent constant. Seek the stationary point of the exponent: with $f[x] \equiv \frac{i}{2}\int x\mathcal{K}x + i\int Jx$, the first variation Eq. (4.4) is

$$\frac{\delta f}{\delta x(t)} = i\,\mathcal{K}x(t) + iJ(t) = 0 \quad\Longrightarrow\quad \mathcal{K}\,x_c = -J, \tag{10.74}$$

the classical equation of motion driven by the source; we used $\mathcal{K}^{\mathsf{T}} = \mathcal{K}$ (the operator $-\partial_t^2 - \omega^2$ is formally self-adjoint) so that the variation of $\frac{1}{2}\int x\mathcal{K}x$ is $\mathcal{K}x$. Its solution, with the boundary condition supplied by the tilt, is $x_c = -\mathcal{K}^{-1}J$, where $\mathcal{K}^{-1}$ is the appropriate Green's function. Substitute $x = x_c + y$. Because the exponent is exactly quadratic, it splits with no cross term between the constant piece and the fluctuation y:

$$\frac{i}{2}\int x\mathcal{K}x + i\int Jx = \underbrace{\Big[\tfrac{i}{2}\int x_c\mathcal{K}x_c + i\int Jx_c\Big]}_{\text{stationary value}} + \underbrace{\tfrac{i}{2}\int y\,\mathcal{K}\,y}_{\text{fluctuation}}, \tag{10.75}$$

the linear-in-y terms cancelling by the stationarity condition Eq. (10.74) exactly as in the finite-dimensional computation Eqs. (4.55) and (4.56). Evaluate the stationary value using $\mathcal{K}x_c = -J$ and the symmetry of $\mathcal{K}$:

$$\frac{i}{2}\int x_c\mathcal{K}x_c + i\int Jx_c = \frac{i}{2}\int x_c(-J) + i\int Jx_c = \frac{i}{2}\int Jx_c = \frac{i}{2}\int J\,(-\mathcal{K}^{-1}J) = -\frac{i}{2}\int J\,\mathcal{K}^{-1}J. \tag{10.76}$$

The fluctuation integral $\int \mathcal{D}y\,\exp\{\frac{i}{2}\int y\mathcal{K}y\}$ is precisely the source-free path integral $Z_0[0]$, since shifting an integration variable leaves the measure invariant. Hence

$$Z_0[J] = Z_0[0]\,\exp\Big\{-\tfrac{i}{2}\int J\,\mathcal{K}^{-1}J\Big\}. \tag{10.77}$$

It remains to identify $\mathcal{K}^{-1}$ and recast the result. Define $G \equiv i\,\mathcal{K}^{-1}$, so that $\mathcal{K}G = i$ as operators, which spelled out is the differential equation Eq. (10.72). Then $\mathcal{K}^{-1} = -iG$ and the exponent becomes

$$-\tfrac{i}{2}\int J\,\mathcal{K}^{-1}J = -\tfrac{i}{2}\int J(-iG)J = -\tfrac{i\cdot(-i)}{2}\int JGJ = -\tfrac{1}{2}\int JGJ,$$

which is Eq. (10.71). Finally we fix the boundary condition — which Green's function $\mathcal{K}^{-1}$ is — from the $i\varepsilon$ tilt. Pass to frequency space, $G(t) = \int \frac{d\nu}{2\pi}\tilde{G}(\nu)e^{-i\nu t}$ and $\delta(t) = \int \frac{d\nu}{2\pi}e^{-i\nu t}$. Acting with $\mathcal{K} = -\partial_t^2 - \omega^2$ multiplies $\tilde{G}$ by $(\nu^2 - \omega^2)$, so Eq. (10.72) reads $(\nu^2 - \omega^2)\tilde{G}(\nu) = i$, i.e. $\tilde{G}(\nu) = i/(\nu^2 - \omega^2)$. The operator $\mathcal{K}$ has zero modes at $\nu = \pm\omega$, so its inverse is ambiguous until a contour prescription is supplied; the tilt $\omega^2 \to \omega^2 - i\varepsilon$ inherited from Eq. (10.61) is exactly that prescription, displacing the poles and yielding Eq. (10.73). This $+i\varepsilon$ is the same prescription that selects the Feynman (time-ordered) boundary conditions, as the explicit evaluation below confirms. □

It remains to perform the frequency integral and read off the boundary conditions G obeys. The computation is the $0+1$-dimensional rehearsal of the contour manipulation that produced the covariant scalar propagator in Section 9.5.3, and it teaches the same lesson: the $+i\varepsilon$ in the denominator routes positive frequencies forward in time.

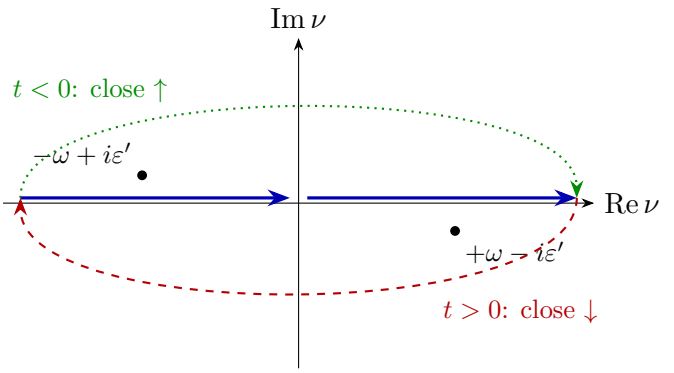


Figure 10.2: The complex frequency plane for the Feynman Green's function Eq. (10.73). The $+i\varepsilon$ prescription pushes the pole at $\nu = +\omega$ below the real axis and the pole at $\nu = -\omega$ above it. For $t - t' > 0$ the factor $e^{-i\nu(t-t')}$ forces the contour closed in the lower half-plane, enclosing $+\omega$; for $t - t' < 0$ it closes above, enclosing $-\omega$. Either way a single positive-frequency exponential results, giving $G(t) = \frac{1}{2\omega} e^{-i\omega|t|}$.

Worked Example 10.14 (The Feynman Green's function by contour integration)**.** We evaluate Eq. (10.73) explicitly. Factor the denominator. With the $+i\varepsilon$ in place, $\nu^2 - \omega^2 + i\varepsilon = (\nu - \nu_+)(\nu - \nu_-)$ where the roots are displaced off the real axis,

$$\nu_\pm = \pm\sqrt{\omega^2 - i\varepsilon} = \pm(\omega - i\varepsilon'), \qquad \varepsilon' \equiv \frac{\varepsilon}{2\omega} > 0,$$

so the pole near $+\omega$ sits just *below* the real axis and the pole near $-\omega$ just *above* it; see Fig. 10.2. Write $\tau \equiv t - t'$ and

$$G(\tau) = \int_{-\infty}^{\infty} \frac{d\nu}{2\pi} \, \frac{i\, e^{-i\nu\tau}}{(\nu - \nu_+)(\nu - \nu_-)}.$$

The exponential $e^{-i\nu\tau} = e^{-i(\mathrm{Re}\,\nu)\tau}\, e^{(\mathrm{Im}\,\nu)\tau}$ decays in the lower half-plane ($\mathrm{Im}\,\nu < 0$) when $\tau > 0$, and in the upper half-plane when $\tau < 0$; closing the contour with the vanishing semicircle in the appropriate half-plane lets us apply the residue theorem.

Case $\tau > 0$. Close downward, enclosing only $\nu_+ = \omega - i\varepsilon'$ and traversing the closed contour clockwise (an extra minus sign). The residue is

$$\mathrm{Res}_{\nu=\nu_+} \frac{i\, e^{-i\nu\tau}}{(\nu - \nu_+)(\nu - \nu_-)} = \frac{i\, e^{-i\omega\tau}}{\nu_+ - \nu_-}\bigg|_{\varepsilon\to 0} = \frac{i\, e^{-i\omega\tau}}{2\omega}.$$

Hence

$$G(\tau > 0) = \frac{1}{2\pi}\,(-2\pi i)\,\frac{i\, e^{-i\omega\tau}}{2\omega} = \frac{e^{-i\omega\tau}}{2\omega},$$

using $(-2\pi i)(i) = 2\pi$.

Case $\tau < 0$. Close upward, enclosing only $\nu_- = -\omega + i\varepsilon'$ and traversing counterclockwise ($+2\pi i$). The residue is $i\,e^{+i\omega\tau}/(\nu_- - \nu_+) = i\,e^{i\omega\tau}/(-2\omega)$, so

$$G(\tau<0) = \frac{1}{2\pi}\,(2\pi i)\,\frac{i\,e^{i\omega\tau}}{-2\omega} = \frac{e^{+i\omega\tau}}{2\omega}.$$

The two cases combine into the manifestly even, positive-frequency-forward result

$$G(\tau) = \frac{1}{2\omega}\,e^{-i\omega|\tau|}. \tag{10.78}$$

For $\tau > 0$ the Green's function propagates the phase $e^{-i\omega\tau}$ of the positive-energy state forward; for $\tau < 0$ it is the complex conjugate. This is precisely the time-ordered structure: the $+i\varepsilon$ in Eq. (10.73) is the Feynman prescription, and Eq. (10.78) is the quantum-mechanical Feynman propagator. It is the $0+1$-dimensional shadow of the covariant momentum-space propagator $i/(p^2 - m^2 + i\varepsilon)$ obtained for the field in Section 9.5.3: the frequency ν plays the role of p^0, the gap ω the role of the on-shell energy $E_{\boldsymbol{p}}$, and the contour logic is identical.

Worked Example 10.15 (The two-point function, two ways)**.** We confirm that the generating-functional machinery reproduces the canonical result, closing the loop with Chapter 9. Apply the master identity Eq. (10.66) with $n = 2$ to the free $Z_0[J]$ of Eq. (10.71). Writing $Z_0[J]/Z_0[0] = \exp\{-\frac{1}{2}\int JGJ\}$, the first derivative is

$$\frac{1}{Z_0[0]}\frac{\delta Z_0}{\delta J(t_1)} = \Big(-\int dt'\,G(t_1 - t')J(t')\Big)\,\frac{Z_0[J]}{Z_0[0]},$$

using $G(t-t') = G(t'-t)$ from Eq. (10.78) so that the two ways of hitting the symmetric kernel agree. Differentiating again,

$$\frac{1}{Z_0[0]}\frac{\delta^2 Z_0}{\delta J(t_1)\delta J(t_2)} = \Big[-G(t_1-t_2) + \Big(\int G(t_1-t')J\Big)\Big(\int G(t_2-t')J\Big)\Big]\frac{Z_0[J]}{Z_0[0]}.$$

Set $J = 0$: the second term vanishes and $Z_0[J]/Z_0[0] \to 1$, leaving $-G(t_1 - t_2)$. Divide by $i^2 = -1$:

$$\langle \mathcal{T}\,\hat{x}(t_1)\hat{x}(t_2)\rangle = \frac{1}{i^2}\big(-G(t_1-t_2)\big) = G(t_1-t_2) = \frac{1}{2\omega}\,e^{-i\omega|t_1-t_2|}. \tag{10.79}$$

Now the canonical computation, for comparison. In the Heisenberg picture the oscillator coordinate is $\hat{x}(t) = \frac{1}{\sqrt{2\omega}}(\hat{a}\,e^{-i\omega t} + \hat{a}^\dagger e^{i\omega t})$, with $[\hat{a}, \hat{a}^\dagger] = 1$ and $\hat{a}\,|\Omega\rangle = 0$. For $t_1 > t_2$,

$$\langle\Omega|\hat{x}(t_1)\hat{x}(t_2)|\Omega\rangle = \frac{1}{2\omega}\,\Big\langle\Omega\Big|\hat{a}\,\hat{a}^\dagger\Big|\Omega\Big\rangle\; e^{-i\omega t_1}e^{+i\omega t_2} = \frac{1}{2\omega}\,e^{-i\omega(t_1-t_2)},$$

only the $\hat{a}\hat{a}^\dagger$ term surviving because $\hat{a}\,|\Omega\rangle = 0 = \langle\Omega|\,\hat{a}^\dagger$ and $\Big\langle\Omega\Big|\hat{a}\hat{a}^\dagger\Big|\Omega\Big\rangle = 1$. Time-ordering supplies the same expression with $t_1 \leftrightarrow t_2$ for $t_2 > t_1$, i.e. $\frac{1}{2\omega}e^{-i\omega|t_1-t_2|}$, in exact agreement with Eq. (10.79). The two pillars — canonical quantization and the path integral — give the identical correlator, as they must.

The agreement in Theorem 10.15 is the simplest instance of the central claim of the chapter: the sum over histories and the operator formalism are two computations of the same physics. We obtained the Feynman propagator a second, independent way, as the kernel of a Gaussian integral rather than as a vacuum matrix element of ordered operators, and the $i\varepsilon$ that was a contour choice in Chapter 9 is here the contour tilt that projects onto the ground state. Higher correlators follow by further differentiation of Eq. (10.71); because $Z_0[J]$ is the exponential of a quadratic, those derivatives organize into all pairings of the insertion times, which is Wick's theorem — the topic to which the next chapter material returns once the field has been installed.

With the quantum-mechanical template complete, the structure that hands forward to the field theory is fixed and will be lifted verbatim: the generating functional $Z[J] = \int \mathcal{D}x\, \exp\{i \int dt\, (L + Jx)\}$ of Eq. (10.64), the rule that time-ordered correlators are normalized functional derivatives $\frac{1}{i^n}\delta^n Z/\delta J^n$ at $J = 0$ of Eq. (10.66), the $i\varepsilon$ contour tilt that selects the vacuum, and the complete-the-square template $Z_0[J] = \exp\{-\frac{1}{2}\int\int JGJ\}$ of Eq. (10.71) with G the Feynman Green's function. In the next section the single coordinate $x(t)$ is replaced by the field value $\phi(\mathbf{x}, t)$ at every spatial point, the source $J(t)$ becomes $J(x)$, and the operator $-\partial_t^2 - \omega^2$ becomes the Klein–Gordon operator $-\Box - m^2$ of ϕ^4 theory — whereupon the same Gaussian integral returns the covariant Feynman propagator $i/(p^2 - m^2 + i\varepsilon)$ a second time.

10.3 The Generating Functional for a Scalar Field

The previous section built the entire path-integral apparatus in its smallest nontrivial laboratory: a quantum system with a single degree of freedom. We learned there to package every time-ordered correlator of a coordinate $q(t)$ into one object, the generating functional $Z[J] = \int \mathcal{D}q\, \exp\left[i \int dt\, (L + Jq)\right]$; we saw that differentiating $Z[J]$ with respect to the source $J(t)$ pulls down insertions of q, so that correlators are functional derivatives of Z; we saw that the ground state is selected automatically by an infinitesimal clockwise tilt of the time contour, the $i\varepsilon$ prescription, which damps every excited state in the spectral sum; and we saw that the resulting Gaussian is evaluated by completing the square. None of that machinery cared that there was only one coordinate. The business of this section is to let the number of coordinates grow without bound until it becomes a continuum — to turn $q(t)$ into the field $\phi(t, \mathbf{x})$ — and to read off what each piece of the one-dimensional template becomes in the limit. The payoff is the central object of the path-integral formulation of scalar field theory: the generating functional $Z[J]$ for the frozen ϕ^4 Lagrangian, the definition of its n-point Green's functions, and an honest account of the one genuinely new ingredient, the functional measure $\mathcal{D}\phi$.

We continue ϕ^4 theory from its free canonical quantization in Chapter 9. The

target Lagrangian density is the one frozen for the thread in Eq. (6.56),

$$\mathcal{L} \;=\; \tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}m^2\phi^2 \;-\; \frac{\lambda}{4!}\,\phi^4, \qquad \phi \text{ real},\ \lambda > 0, \tag{10.80}$$

in the conventions fixed for the whole book: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$. We call this Lagrangian *frozen* because in this section the coupling λ is carried along untouched: every manipulation below is exact as a formal statement for any λ, and the actual evaluation of the integral — which is possible in closed form only for the free, $\lambda = 0$, Gaussian piece — is the work of Section 10.4. The role of the present section is purely to *construct* the object $Z[J]$ and to say precisely what it means.

10.3.1 From a lattice of oscillators to a spacetime lattice

In Section 9.1.2 we replaced the continuous spatial argument $\mathbf{x}$ of the field by a cubic lattice of sites $\mathbf{x}_\mathbf{n} = a\mathbf{n}$, $\mathbf{n} \in \mathbb{Z}^3$, of spacing a, and discovered that the free scalar is nothing but a lattice of coupled harmonic oscillators: at each site sits a coordinate $q_\mathbf{n}(t) = \phi(t, \mathbf{x}_\mathbf{n})$, the neighbours are joined by springs that encode the gradient energy $\frac{1}{2}(\nabla\phi)^2$, and the mass term $\frac{1}{2}m^2\phi^2$ supplies an on-site restoring force (Fig. 9.1). That mechanical picture is exactly the bridge we need. The path integral of the previous section was derived for a system with finitely many coordinates $q_a(t)$, $a = 1, \ldots, N$, and is insensitive to whether those coordinates are genuinely different particles or, as here, the values of one field at different points of space. We therefore take the coupled-oscillator system of Chapter 9 and feed it straight into the path-integral construction.

Discretize space into a large cubic box of side $L = N_s a$ with periodic boundary conditions, so the spatial label $\mathbf{n}$ runs over the finitely many sites $\mathbf{n} \in \{0, \ldots, N_s - 1\}^3$. The field at fixed time is then the finite collection of numbers $\{\phi_\mathbf{n}(t)\}$. Writing the gradient as a nearest-neighbour difference, the discretized Lagrangian of the frozen theory Eq. (10.80) is the sum over sites

$$L(\{\phi_\mathbf{n}\}, \{\dot\phi_\mathbf{n}\}) = \sum_\mathbf{n} a^3 \left[\, \tfrac{1}{2}\,\dot\phi_\mathbf{n}^2 - \tfrac{1}{2}\sum_{i=1}^{3}\Big(\frac{\phi_{\mathbf{n}+\hat\imath} - \phi_\mathbf{n}}{a}\Big)^2 - \tfrac{1}{2}m^2\phi_\mathbf{n}^2 - \frac{\lambda}{4!}\,\phi_\mathbf{n}^4 \right], \tag{10.81}$$

where $\hat\imath$ is the unit lattice vector in the i-th direction and the cell volume a^3 converts each site density into an energy, exactly as in Chapter 9. This is a perfectly ordinary mechanical system of $N = N_s^3$ coupled anharmonic oscillators, and the previous section tells us, verbatim, how to write its generating functional.

Recall the one-dimensional template in its multi-coordinate form. For a system with coordinates q_a and Lagrangian $L(q, \dot q)$, the vacuum generating functional with sources $J_a(t)$ coupled to each coordinate is the time-sliced path integral

$$Z[\{J_a\}] \;=\; \int \prod_a \mathcal{D}q_a \;\exp\Big[\, i\int_{-\infty}^{\infty} dt\,\Big(L(q,\dot q) + \sum_a J_a(t)\,q_a(t)\Big)\Big], \tag{10.82}$$

the integral running over all trajectories with the $i\varepsilon$ vacuum boundary conditions established in Section 10.2. Each $\mathcal{D}q_a$ is itself the limit of a product of ordinary integrals over the value of q_a at each of a fine mesh of time slices $t_j = j\,\Delta t$, with the standard kinetic-term normalization. Inserting the lattice Lagrangian Eq. (10.81) for L and renaming the coordinate at site $\mathbf{n}$ back to $\phi_{\mathbf{n}}$, with the source $J_a \to a^3 J_{\mathbf{n}}$ (the cell volume again, so that $\sum_{\mathbf{n}} a^3 J_{\mathbf{n}}\phi_{\mathbf{n}} \to \int d^3x\, J\phi$ in the limit), the generating functional of the discretized scalar field is

$$Z[J] = \int \prod_{\mathbf{n}} \mathcal{D}\phi_{\mathbf{n}}\ \exp\Big[\, i \int dt\, \Big(L(\{\phi_{\mathbf{n}}\}) + \sum_{\mathbf{n}} a^3 J_{\mathbf{n}}(t)\, \phi_{\mathbf{n}}(t) \Big)\Big]. \tag{10.83}$$

Now perform the same time-slicing inside each $\mathcal{D}\phi_{\mathbf{n}}$. The combined effect of slicing time into steps Δt and slicing space into the cubic lattice is to replace the field $\phi(t, \mathbf{x})$ by its values $\phi_{j,\mathbf{n}} \equiv \phi(t_j, \mathbf{x}_{\mathbf{n}})$ on a four-dimensional spacetime lattice (Fig. 10.3), and the integral becomes an ordinary (very high-dimensional) multiple integral over one real variable $\phi_{j,\mathbf{n}}$ per spacetime point:

$$Z[J] \;=\; \lim_{\substack{a,\Delta t\to 0\\ L,T\to\infty}} \mathcal{N} \prod_{j,\mathbf{n}} \int_{-\infty}^{\infty} d\phi_{j,\mathbf{n}}\ \exp\Big[\, i\, a^3\, \Delta t \sum_{j,\mathbf{n}} \big(\mathcal{L}_{\text{disc}} + J_{j,\mathbf{n}}\phi_{j,\mathbf{n}}\big)\Big], \tag{10.84}$$

where $\mathcal{L}_{\text{disc}}$ is the fully discretized Lagrangian density — Eq. (10.81) with $\dot\phi_{\mathbf{n}} \to (\phi_{j+1,\mathbf{n}} - \phi_{j,\mathbf{n}})/\Delta t$ — and $\mathcal{N}$ collects the kinetic-term normalization factors (one power of $\sqrt{a^3/2\pi i\,\Delta t}$ per time step per site) inherited from the one-dimensional measure. The quantity $a^3\Delta t$ is the four-volume of one lattice cell, so the exponent is a Riemann sum for the spacetime integral of $\mathcal{L} + J\phi$. This is the construction we wanted: the field path integral has been *derived* from the canonical theory of Chapter 9 by discretization, not postulated.

10.3.2 The functional measure

The continuum limit of Eq. (10.84) demands a name for the limiting integration. We adopt the standard notation and define the functional measure as exactly that limit.

Definition 10.16 (Functional measure as a lattice limit)**.** The *functional integration measure* over scalar field configurations is the symbol

$$\int \mathcal{D}\phi\ (\cdots) \;\equiv\; \lim_{\substack{a,\Delta t\to 0\\ L,T\to\infty}} \mathcal{N} \prod_{\text{cells } (j,\mathbf{n})} \int_{-\infty}^{\infty} d\phi_{j,\mathbf{n}}\ (\cdots), \tag{10.85}$$

the product running over the cells of a four-dimensional spacetime lattice and $\mathcal{N}$ the normalization inherited from the time-sliced kinetic term. The symbol $\mathcal{D}\phi = \prod_x d\phi(x)$, "one ordinary integral for every spacetime point," is shorthand for the right-hand side.

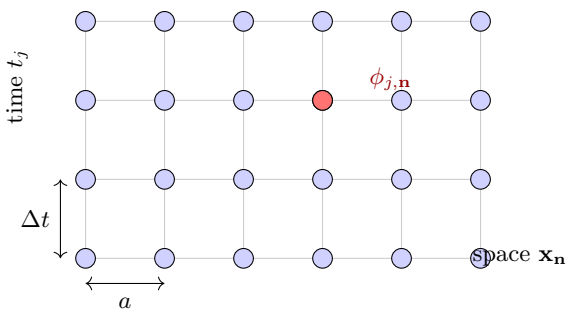


Figure 10.3: The spacetime lattice underlying Eq. (10.84). Slicing time (as in the quantum-mechanical path integral of the previous section) and slicing space (as in the coupled-oscillator picture of Fig. 9.1) together replace the field $\phi(t, \mathbf{x})$ by one real integration variable $\phi_{j,\mathbf{n}}$ per cell. The generating functional is the limit of an ordinary multiple integral over all of these variables; the functional measure $\mathcal{D}\phi$ is the name of that limit (Section 10.3.2). Each cell carries four-volume $a^3\,\Delta t$, and the exponent of Eq. (10.84) is the corresponding Riemann sum for $\int d^4x\,(\mathcal{L} + J\phi)$.

Three honest remarks fix what this definition does and does not deliver, in the spirit of the book's rigor doctrine. They matter because the measure is the one ingredient of the path integral that has no finite-dimensional precedent and is the source of every genuine subtlety in the formalism.

First, what the lattice definition *is good for*. Equation Eq. (10.85) is a completely explicit prescription: for any finite lattice the right-hand side is an ordinary, finite-dimensional multiple integral, and every formal manipulation we perform on $\int\mathcal{D}\phi$ in this chapter — shifting the integration variable, completing the square, differentiating under the integral sign with respect to a source — is first to be read as that operation on the finite-dimensional integral, after which the continuum limit is taken. For the *free*, Gaussian, theory the limit exists and all these operations are exact; the next section carries the Gaussian integration out and recovers the Feynman propagator, providing an a-posteriori certificate that the manipulations were legitimate.

Second, the normalization $\mathcal{N}$ is divergent as $a, \Delta t \to 0$, but *it cancels in every physical quantity*. The constant $\mathcal{N}$ multiplies both numerator and denominator of every correlator, because correlators are always normalized by $Z[0]$ (Section 10.3.5); it never appears in a Green's function. We therefore never need to compute $\mathcal{N}$, and it is standard — and harmless — to absorb it into the definition of the symbol $\mathcal{D}\phi$ and not write it again. What survives the limit is the ratio of integrals, and that ratio is finite for the free theory.

Third, the rigorous status. The claim that the lattice family Eq. (10.85) converges to a genuine measure on a space of field configurations is a statement of functional analysis, not of physics, and it is delicate: a naive "Lebesgue measure

on an infinite-dimensional space" does not exist. What can be made fully rigorous is the *Euclidean* version obtained by Wick rotation (Theorem 5.30), in which the oscillatory e^{iS} becomes a positive, exponentially damped weight e^{-S_E} and Eq. (10.85) converges, for the free theory, to a bona fide Gaussian measure on the space of distributions. The functional determinants that the Gaussian integral produces, and the precise sense in which the lattice limit defines the measure, are developed in Chapter E; we will use the results there by reference rather than reproving them in the main text. The honest summary is that the formal Minkowski measure is a well-controlled abbreviation for a finite-dimensional limit, rigorous for the Gaussian theory, and that the construction of the *interacting* four-dimensional scalar measure as a genuine continuum object remains a famous open problem.

Caution

The notation $\mathcal{D}\phi = \prod_x d\phi(x)$ is suggestive but must not be taken to assert that an infinite-dimensional Lebesgue measure exists. It is defined *only* as the lattice limit Eq. (10.85), and only ratios in which the normalization $\mathcal{N}$ cancels are guaranteed to be finite. Treating $\mathcal{D}\phi$ as an ordinary measure — e.g. asserting translation invariance or a change-of-variables Jacobian — is legitimate exactly when the corresponding statement holds at finite lattice size and survives the limit, and not otherwise. Every such step in this book is one that does survive; Chapter E is where that is checked.

10.3.3 The generating functional $Z[J]$

With the measure named, the continuum limit of Eq. (10.84) is immediate: the Riemann sum in the exponent becomes the spacetime integral of $\mathcal{L} + J\phi$, and the product of integrals becomes $\int \mathcal{D}\phi$. We elevate the result to a definition, the central object of the chapter.

Definition 10.17 (Generating functional of the scalar field)**.** For the frozen ϕ^4 Lagrangian Eq. (10.80), the *vacuum generating functional* in the presence of a classical source $J(x)$ is

$$\boxed{Z[J] \;=\; \int \mathcal{D}\phi \;\exp\Big[\, i \int d^4x \,\big(\mathcal{L}[\phi] + J(x)\,\phi(x)\big)\Big] \;=\; \int \mathcal{D}\phi \; e^{\,i\,(S[\phi] + \int J\phi)}\,,} \tag{10.86}$$

where $S[\phi] = \int d^4x\, \mathcal{L}[\phi]$ is the action and the measure is Theorem 10.16. The source $J(x)$ is an ordinary real function on spacetime, an external "handle" with no dynamics of its own; physical answers are read off at $J = 0$ after differentiation.

This is the field-theory analogue, term for term, of the one-dimensional $Z[J]$ of the previous section, and of the Euclidean functional generating object whose

formal structure was set up in Eq. (4.205). The exponent is real-times-i, so the integrand is a pure phase; the integral is defined by the lattice limit and made convergent by the $i\varepsilon$ prescription we now make precise.

10.3.4 Vacuum projection and the $i\varepsilon$ prescription

There is one ingredient of Eq. (10.86) that the bare symbol e^{iS} does not display: the boundary conditions on the field integration at $t \to \pm\infty$. In the quantum mechanics of the previous section, the statement that $Z[J]$ generates *vacuum* expectation values — correlators in the ground state $|0\rangle$ rather than in some arbitrary state fixed by the endpoints of the trajectory — was enforced by an infinitesimal tilt of the time contour into the lower half-plane, $t \to t\,(1 - i\varepsilon)$, equivalently a small negative imaginary part added to every energy. We now lift that prescription to the field and show it does two things at once: it projects onto the Fock vacuum of Chapter 9, and it produces precisely the Feynman pole structure of Theorem 9.25.

Consider the mechanism first in spectral terms, where it is transparent. Diagonalizing the free part of the lattice Hamiltonian into normal modes (the relativistic plane waves of Chapter 9) turns the field into a collection of independent oscillators, one for each spatial momentum $\mathbf{p}$, with frequency the on-shell energy $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$. The path integral over a time interval $(-T, T)$ computes a matrix element of e^{-2iHT} between boundary configurations. Inserting a complete set of energy eigenstates,

$$\langle \phi_f |\, e^{-2iHT}\, | \phi_i \rangle \;=\; \sum_n e^{-2iE_nT}\, \langle \phi_f | n \rangle \langle n | \phi_i \rangle, \tag{10.87}$$

and tilting $T \to T(1 - i\varepsilon)$ multiplies the n-th term by $e^{-2E_nT\varepsilon}$. Since $E_0 < E_1 \le E_2 \le \cdots$, in the limit $T \to \infty$ every excited contribution is exponentially suppressed relative to the ground state, and the lattice integral is dominated by $|0\rangle\,\langle 0|$: the boundary conditions are forgotten and the integral computes vacuum-to-vacuum amplitudes. This is identical, mode by mode, to the single-oscillator argument of Section 10.2; the field merely has one oscillator per momentum, and the tilt damps the excited states of all of them simultaneously. The state it projects onto is exactly the Fock vacuum $|0\rangle$ of Theorem 9.17, the no-particle state annihilated by every $a_{\mathbf{p}}$.

The same prescription has a clean and equivalent expression directly in the action, which is the form we carry forward. The tilt $t \to t(1 - i\varepsilon)$ acts on the action by giving the quadratic potential a small negative imaginary part; tracked through the normal-mode frequencies it is precisely equivalent to the replacement

$$m^2 \;\longrightarrow\; m^2 - i\varepsilon \tag{10.88}$$

in the Lagrangian Eq. (10.80), with $\varepsilon \to 0^+$ at the end. To see why this is the

right amount, write out the resulting mass term in the exponent of $Z[J]$:

$$\exp\Big[i\int d^4x\,\big(-\tfrac{1}{2}(m^2-i\varepsilon)\phi^2\big)\Big] \;=\; \exp\Big[-\tfrac{\varepsilon}{2}\int d^4x\,\phi^2\Big]\times\exp\Big[-\tfrac{i}{2}\int d^4x\,m^2\phi^2\Big]. \tag{10.89}$$

The first factor is a real, positive-definite *Gaussian damping* of the field amplitude: it suppresses configurations of large $\int\phi^2$ and renders the otherwise purely oscillatory integral Eq. (10.86) absolutely convergent on each finite lattice. The $i\varepsilon$ is thus not an optional decoration but the very thing that makes the Minkowski functional integral well-defined — it is the analytic continuation, infinitesimally performed, that connects the oscillatory e^{iS} to the convergent Euclidean weight e^{-S_E} of Theorem 5.30. The same calculation done in full Wick rotation, with the contour turned all the way to the Euclidean axis, is the subject of the Euclidean section of this chapter; here we keep only the infinitesimal tilt.

That the tilt Eq. (10.88) reproduces the Feynman boundary condition is now a one-line check against Chapter 9, and it is the consistency statement the chapter is built around. The free quadratic action, after an integration by parts that throws away the boundary term, is $S_0=\frac{1}{2}\int d^4x\,\phi(-\Box-m^2)\phi$, so the kinetic operator carried into the exponent of $Z[J]$ is $-\Box-m^2$. In momentum space, with the Fourier convention $\phi(x)=\int\frac{d^4p}{(2\pi)^4}e^{-ip\cdot x}\tilde\phi(p)$ fixed for the book, $-\Box\to p^2$ and the prescription Eq. (10.88) sends the operator to

$$-\Box-m^2 \;\longrightarrow\; p^2-m^2+i\varepsilon, \tag{10.90}$$

where the sign of the imaginary part flips because the operator carries $-m^2$, turning $-(-i\varepsilon)=+i\varepsilon$. The inverse of this operator — the object the Gaussian integration of Section 10.4 will produce as the propagator — is therefore proportional to $1/(p^2-m^2+i\varepsilon)$, which is exactly the denominator of the canonical Feynman propagator $D_F(p)=i/(p^2-m^2+i\varepsilon)$ established in Eq. (9.154). The staggered placement of the two poles in the complex p^0 plane, $p^0=\pm(E_{\mathbf p}-i\varepsilon')$, that the contour proof of Theorem 9.25 identified with "create forward, propagate forward in time," is here seen to descend from a single infinitesimal tilt of the time integration. The two pillars agree at the level of the boundary condition before a single Gaussian integral is done.

Physical Insight

The $i\varepsilon$ wears three faces, and they are the same face. In the operator formalism (Chapter 9) it is the rule "time-order the fields," the staggered poles that send positive-energy quanta forward in time. In the path integral it is the convergence factor Eq. (10.89) that makes an oscillatory integral well-defined. And spectrally it is the projector Eq. (10.87) onto the ground state. That one infinitesimal does all three jobs is not a coincidence: time-ordering, vacuum selection, and Gaussian convergence are three

descriptions of the single analytic continuation that ties Minkowski field theory to its Euclidean shadow.

10.3.5 The n-point Green's functions

We now harvest from $Z[J]$ the quantities it was built to generate. The construction repeats the one-dimensional master identity of Eq. (10.66) with the source promoted to a function of all four spacetime coordinates, so that ordinary derivatives $\partial/\partial J_a$ become functional derivatives $\delta/\delta J(x)$ in the sense of Theorem 4.4.

The basic move is to differentiate Eq. (10.86) with respect to the source. Only the factor $e^{i\int J\phi}$ depends on J, and $\frac{\delta}{\delta J(x)}\int d^4y\, J(y)\phi(y) = \phi(x)$, so each functional derivative pulls down one factor of $i\,\phi(x)$ from the exponent:

$$\frac{1}{i}\frac{\delta Z[J]}{\delta J(x)} \;=\; \int \mathcal{D}\phi\; \phi(x)\, e^{\,i(S[\phi]+\int J\phi)}. \tag{10.91}$$

Iterating n times, the $1/i$ compensating each i brought down by the exponent,

$$\frac{1}{i^n}\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)} \;=\; \int \mathcal{D}\phi\; \phi(x_1)\cdots\phi(x_n)\, e^{\,i(S[\phi]+\int J\phi)}. \tag{10.92}$$

Setting the source to zero leaves the unsourced field average of the product of fields, and dividing by $Z[0] = \int\mathcal{D}\phi\, e^{iS}$ removes the (divergent, J-independent) overall normalization, including the constant $\mathcal{N}$ of Theorem 10.16. This defines the central correlators of the theory.

Definition 10.18 (n-point Green's functions)**.** The *n-point Green's functions* (or full n-point correlation functions) of the scalar field are

$$G^{(n)}(x_1,\dots,x_n) \;\equiv\; \frac{1}{Z[0]}\,\frac{1}{i^n}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0} \;=\; \frac{\displaystyle\int \mathcal{D}\phi\; \phi(x_1)\cdots\phi(x_n)\, e^{\,iS[\phi]}}{\displaystyle\int \mathcal{D}\phi\; e^{\,iS[\phi]}}\,. \tag{10.93}$$

Two pieces of bookkeeping are worth stating once. First, $Z[J]/Z[0]$ is the natural normalized generating functional — the field-theory instance of $\widehat{Z}[J]$ from Theorem 4.26 — and its functional Taylor coefficients are exactly the $G^{(n)}$, so that $Z[J]/Z[0] = \sum_n \frac{i^n}{n!}\int d^4x_1\cdots d^4x_n\, G^{(n)}(x_1,\dots,x_n)\, J(x_1)\cdots J(x_n)$, with the factors of i now visible because the Minkowski exponent carries them. Second, the division by $Z[0]$ is not cosmetic: $Z[0]$ is the sum of all vacuum-to-vacuum amplitudes ("vacuum bubbles"), and dividing it out is what makes $G^{(n)}$ finite and equal to a physical correlator. We return to this when the diagrammatic

content is unpacked in Chapter 15; here it is simply the normalization that the canonical correlators also carry.

It remains to prove that the object Eq. (10.93), manufactured entirely inside the path integral, is the canonical time-ordered correlator of Chapter 9. This equivalence is the whole point of having two pillars, and we now establish it.

Theorem 10.19 (Path-integral correlators are time-ordered vacuum correlators)**.** *Let $\phi(x)$ be the Heisenberg-picture field operator of Chapter 9 and $|0\rangle$ the vacuum of Theorem 9.17. Then the path-integral Green's function Eq. (10.93) equals the vacuum expectation value of the time-ordered product of the corresponding operators,*

$$G^{(n)}(x_1,\dots,x_n) \;=\; \langle 0|\, \mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|0\rangle\,. \tag{10.94}$$

For the free theory ($\lambda = 0$) this holds as an exact identity of tempered distributions; for the interacting theory it holds order by order in the coupling, with the same proof.

Proof. The mechanism is the time-slicing that defined the measure, and the key fact is that time-ordering is produced *automatically* by the slicing, not imposed by hand. We carry the argument in full for two insertions, $n = 2$; the general case is identical with more factors, and the structure makes plain that it never used $\lambda = 0$.

Work on the spacetime lattice of Section 10.3.1 and reinstate the canonical interpretation of each time slice. The time-sliced construction of the path integral — carried out for a general Hamiltonian system in Section 10.2 and applied here to the coupled-oscillator Hamiltonian of Chapter 9 — represents the amplitude between an initial configuration at t_i and a final one at t_f as a product of short-time evolution kernels, one per slice. At each slice the integration variable $\phi_{j,\mathbf{n}}$ is the eigenvalue of the Schrödinger-picture field operator $\hat\phi(\mathbf{x_n})$ on that slice. Consequently, inserting a c-number factor $\phi_{j,\mathbf{n}}$ into the integrand is the same as inserting the operator $\hat\phi$ at the corresponding time *at that position in the chain of kernels*, i.e. sandwiched between the evolution from t_i up to t_j on its right and from t_j to t_f on its left.

Take two insertions at $x_1 = (t_1, \mathbf{x}_1)$ and $x_2 = (t_2, \mathbf{x}_2)$, and suppose without loss of generality that $t_1 > t_2$. In the sliced integral the factor ϕ at the later time t_1 sits to the left of the factor at the earlier time t_2 in the product of kernels, because the kernels are ordered by increasing time from right to left. Reassembling the kernels into evolution operators and passing to the Heisenberg picture via $\hat\phi_H(t,\mathbf{x}) = e^{iHt}\hat\phi(\mathbf{x})e^{-iHt}$, the field integral with the two insertions becomes

$$\int\!\mathcal{D}\phi\;\phi(x_1)\phi(x_2)\,e^{iS} \;\xrightarrow[t_1>t_2]{}\; \mathcal{N}'\,\langle\phi_f,t_f|\;\phi_H(x_1)\,\phi_H(x_2)\;|\phi_i,t_i\rangle\,, \tag{10.95}$$

with $\phi_H(x_1)$ standing to the left precisely because t_1 is the later time. Had we assumed $t_2 > t_1$ instead, the identical manipulation would have produced

$\phi_H(x_2)\phi_H(x_1)$, the later field again on the left. The two cases combine into a single statement: whatever the time order of x_1 and x_2, the slicing delivers the operators arranged with the later one to the left — which is the definition of the time-ordered product,

$$\mathcal{T}\,\phi(x_1)\phi(x_2) = \theta(t_1 - t_2)\,\phi(x_1)\phi(x_2) + \theta(t_2 - t_1)\,\phi(x_2)\phi(x_1), \tag{10.96}$$

in the convention of Eq. (9.151). The c-number commutativity of the integration variables, $\phi(x_1)\phi(x_2) = \phi(x_2)\phi(x_1)$ under the integral, is exactly what is needed for this to be consistent: the path integral cannot tell the two orderings apart, and the only operator object with that symmetry is the time-ordered product, in which the operators have been reordered into a canonical (time) order. The path integral thus computes the time-ordered correlator and nothing else — it has no way to represent a fixed, untime-ordered operator order.

Finally, attach the boundary data. The $i\varepsilon$ tilt of Section 10.3.4 sends $t_i \to -\infty(1 - i\varepsilon)$ and $t_f \to +\infty(1 - i\varepsilon)$, which by the spectral argument Eq. (10.87) replaces the endpoint states $|\phi_i, t_i\rangle$ and $\langle\phi_f, t_f|$ by the vacuum, up to a constant overlap factor $\langle\phi_f|0\rangle\langle 0|\phi_i\rangle$ that is independent of the insertions. That same constant, together with $\mathcal{N}'$, appears in the denominator $Z[0]$ and cancels in the ratio Eq. (10.93). What survives is exactly

$$G^{(2)}(x_1, x_2) = \langle 0|\,\mathcal{T}\,\phi(x_1)\phi(x_2)\,|0\rangle\,, \tag{10.97}$$

which is Eq. (10.94) for $n = 2$. The two-point case is the Feynman propagator $D_F(x_1 - x_2)$ of Theorem 9.24; that the Gaussian evaluation of the right-hand side of Eq. (10.93) reproduces the explicit form Eq. (9.154) is the computation of Section 10.4.

For general n the slicing inserts n c-number factors, each at its own time, and the same reasoning sorts them into the unique operator ordering compatible with their commutativity under the integral — the full time-ordered product $\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)$. Nothing in the argument used the value of λ: the slicing represents the evolution generated by the *full* Hamiltonian, free or interacting, and the only place the free theory is special is that there the integrals can actually be done in closed form. For the interacting theory the identity Eq. (10.94) is therefore exact as a formal statement and is realized order by order in λ by expanding $e^{iS} = e^{iS_0}e^{-i\frac{\lambda}{4!}\int\phi^4}$ in powers of the coupling, each term being a free correlator of the type just established — the perturbative expansion taken up in Chapters 14 and 15. □

Caution

The path integral computes *time-ordered* correlators and only those. It is structurally incapable of representing a product of Heisenberg operators in a fixed, non-time-ordered sequence, because the integration variables

commute and carry no operator order of their own (Eq. (10.95)). This is a feature, not a limitation: time-ordered correlators are exactly the objects perturbation theory and the LSZ reduction of Chapter 14 require. Unordered (Wightman) correlators and equal-time commutators must be reconstructed from the time-ordered ones by analytic continuation, as in Chapter 9; the path integral does not give them directly.

We close the section's formal development with a worked computation that exercises the master identity Eq. (10.93) and exhibits the role of the normalization, before the next section evaluates the integrals themselves.

Worked Example 10.20 (Functional derivatives and the cancellation of vacuum bubbles)**.** We extract $G^{(1)}$ and $G^{(2)}$ from $Z[J]$ explicitly, tracking every factor and confirming that the division by $Z[0]$ in Theorem 10.18 is what removes the disconnected vacuum contributions.

The one-point function. Apply Eq. (10.91) and set $J = 0$:

$$G^{(1)}(x) = \frac{1}{Z[0]}\,\frac{1}{i}\,\frac{\delta Z[J]}{\delta J(x)}\bigg|_{J=0} = \frac{1}{Z[0]}\int \mathcal{D}\phi\;\phi(x)\,e^{iS} = \langle 0|\,\phi(x)\,|0\rangle\,. \tag{10.98}$$

For the frozen ϕ^4 theory theory this vanishes. The cleanest argument is the unbroken $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ of the Lagrangian Eq. (10.80), the symmetry catalogued for the thread in Section 7.3.6: the action $S[\phi] = S[-\phi]$ is even and the measure $\mathcal{D}\phi$ is invariant under the sign flip of every integration variable (a finite-lattice statement that survives the limit), while the insertion $\phi(x)$ is odd. The integrand of $\int\mathcal{D}\phi\,\phi(x)e^{iS}$ is therefore odd under $\phi \to -\phi$ and integrates to zero, so $G^{(1)}(x) = 0$. The field has no vacuum expectation value, consistent with the canonical statement that ϕ acting on the vacuum creates one-particle states orthogonal to $|0\rangle$.

The two-point function and the bubbles. The second derivative requires the quotient rule, because what is physical is $G^{(2)} = Z[0]^{-1}i^{-2}\,\delta^2 Z/\delta J\,\delta J|_0$, and it is instructive to see why only $Z[0]^{-1}$, and not a more complicated combination, appears. Write $Z[J] = Z[0]\,\widehat{Z}[J]$ with $\widehat{Z}[0] = 1$. Differentiating twice and using $\delta\widehat{Z}/\delta J|_0 = i\,G^{(1)} = 0$ from the previous part,

$$G^{(2)}(x_1,x_2) = \frac{1}{i^2}\,\frac{\delta^2\widehat{Z}[J]}{\delta J(x_1)\,\delta J(x_2)}\bigg|_{J=0} = \frac{1}{i^2}\,\frac{1}{Z[0]}\,\frac{\delta^2 Z[J]}{\delta J(x_1)\,\delta J(x_2)}\bigg|_{J=0}, \tag{10.99}$$

where the cross-terms from differentiating $Z[0]^{-1}$ vanish because they are proportional to $G^{(1)} = 0$. Had we instead computed the *unnormalized* object $i^{-2}\delta^2 Z/\delta J\,\delta J|_0 = \int\mathcal{D}\phi\,\phi(x_1)\phi(x_2)e^{iS}$, it would equal $Z[0]\times G^{(2)}$: the field average of $\phi(x_1)\phi(x_2)$ *times* the sum of all vacuum-to-vacuum amplitudes. Diagrammatically (anticipating Chapter 15) the factor $Z[0]$ is the exponential of all

disconnected vacuum bubbles; the explicit $1/Z[0]$ in Theorem 10.18 divides them out, leaving the genuine two-point correlator. This is the field-theory realization of the finite-dimensional statement Theorem 4.26 that full correlators are normalized derivatives of Z, and it is why the normalization $\mathcal{N}$ of the measure never matters: it is part of $Z[0]$ and cancels here. By Theorem 10.19, the surviving $G^{(2)}(x_1, x_2) = \langle 0| \mathcal{T}\phi(x_1)\phi(x_2) |0\rangle = D_F(x_1 - x_2)$ is the Feynman propagator, whose closed form the Gaussian integration of the next section delivers.

We have now constructed, from the coupled oscillators of Chapter 9 rather than by decree, the generating functional $Z[J] = \int \mathcal{D}\phi \, \exp\left[i \int d^4x \, (\mathcal{L} + J\phi)\right]$ for the frozen ϕ^4 Lagrangian; we have said honestly what the measure $\mathcal{D}\phi$ is (a lattice limit, rigorous for the Gaussian theory, with the rest deferred to Chapter E); we have fixed the vacuum boundary conditions through the $m^2 \to m^2 - i\varepsilon$ tilt and checked that it reproduces the Feynman pole structure of Theorem 9.25; and we have proved that the functional derivatives of $Z[J]$, normalized by $Z[0]$, are exactly the time-ordered vacuum correlators $G^{(n)}$. The only thing still owed is the integral itself. In the next section we set $\lambda = 0$, complete the square in the now-quadratic exponent, and carry out the Gaussian functional integration in closed form — recovering the Feynman propagator a second, independent way and confirming, number for number, that Feynman's sum over histories and the canonical quantization of Chapter 9 describe one and the same physics.

10.4 The Free Generating Functional and the Feynman Propagator

The previous section delivered the generating functional $Z[J] = \int \mathcal{D}\phi \, \exp\left[i \int d^4x \, (\mathcal{L} + J\phi)\right]$ for the frozen ϕ^4 Lagrangian, fixed its vacuum boundary conditions through the $m^2 \to m^2 - i\varepsilon$ tilt, and proved that its normalized functional derivatives are the time-ordered correlators $G^{(n)}$ of Theorem 10.18. Everything there was structural: not a single integral was evaluated, because the integral cannot be done in closed form once $\lambda \neq 0$. We now make the one move that *can* be carried through exactly. Setting $\lambda = 0$ leaves a purely quadratic exponent — a Gaussian, the infinite-dimensional cousin of the sourced integral Eq. (4.54) we already control — and a Gaussian integral is evaluated by completing the square. The reward is twofold. First, $Z_0[J]$ collapses to a closed form, $Z_0[0]$ times the exponential of a quadratic in the source whose kernel is a single two-point function. Second, that kernel turns out to be, on the nose, the Feynman propagator D_F that Chapter 9 built by time-ordering operators. This is the central consistency check of the two-pillar programme: the same object, $D_F(p) = i/(p^2 - m^2 + i\varepsilon)$, emerges from a sum over field histories and from a vacuum matrix element of ordered operators. We carry the computation out in full, with every shift, every Jacobian, and every contour exhibited.

10.4.1 The free action and its kinetic operator

Set $\lambda = 0$ in Eq. (10.80). The action is then quadratic,

$$S_0[\phi] \;=\; \int d^4x \,\Big[\tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2\Big], \tag{10.100}$$

and we write $Z_0[J]$ for the generating functional Eq. (10.86) evaluated on this free action — the $\lambda = 0$ instance of $Z[J]$, the only one we can integrate exactly. To expose the Gaussian structure we must rewrite the kinetic term with both derivatives acting on a single field, which is achieved by integrating by parts:

$$\int d^4x\,\tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) \;=\; \int d^4x\,\Big[\tfrac{1}{2}\,\partial_\mu(\phi\,\partial^\mu\phi) - \tfrac{1}{2}\,\phi\,\partial_\mu\partial^\mu\phi\Big] \;=\; -\tfrac{1}{2}\int d^4x\,\phi\,\Box\phi, \tag{10.101}$$

where $\Box \equiv \partial_\mu\partial^\mu = \partial_t^2 - \nabla^2$ and the total-derivative term has been dropped. That the boundary term vanishes is not an act of faith. On the finite spacetime lattice of Section 10.3.1, which is what Eq. (10.101) abbreviates, the integration by parts is a summation by parts on a periodic box, and a periodic lattice has no boundary at all: the boundary term is identically absent before any limit is taken. In the continuum limit the same conclusion is enforced by the convergence factor of the $i\varepsilon$ prescription, Eq. (10.89), which exponentially suppresses field configurations that fail to decay at infinity, so that $\phi\,\partial^\mu\phi$ has no support on the surface at infinity. Either way the surface term is legitimately zero, and we may use Eq. (10.101) inside the path integral.

Assembling Eq. (10.101) with the mass term, and installing the vacuum tilt $m^2 \to m^2 - i\varepsilon$ established in Eq. (10.88), the free action becomes a single quadratic form,

$$S_0[\phi] \;=\; \tfrac{1}{2}\int d^4x\;\phi(x)\,\mathcal{O}_x\,\phi(x), \qquad \mathcal{O}_x \;\equiv\; -\Box - m^2 + i\varepsilon, \tag{10.102}$$

where $\mathcal{O}$ is the *kinetic operator* of the free scalar — the (negative, $i\varepsilon$-deformed) Klein–Gordon operator. It is the field-theory analogue of the symmetric matrix A in the finite-dimensional Gaussian Eq. (4.54), and it is a symmetric operator in the sense that $\int \phi\,\mathcal{O}\,\chi = \int \chi\,\mathcal{O}\,\phi$ for fields decaying at infinity (again, just the integration by parts of Eq. (10.101), applied to two fields). The whole computation of the section is the inversion of this one operator, weighted by a source.

10.4.2 Gaussian functional integration: completing the square

With Eq. (10.102) the free generating functional reads

$$Z_0[J] \;=\; \int \mathcal{D}\phi\;\exp\Big[\,i\int d^4x\,\big(\tfrac{1}{2}\,\phi\,\mathcal{O}\,\phi + J\phi\big)\Big]. \tag{10.103}$$

The exponent is quadratic plus linear in ϕ; this is exactly the structure that the finite-dimensional theorem Theorem 4.15 dispatches by completing the square, and the manipulation lifts verbatim to the functional integral because, on each finite lattice, Eq. (10.103) *is* an ordinary multidimensional Gaussian. We perform the completion directly in the continuum notation and justify each step by its finite-lattice meaning.

Introduce the *classical field in the presence of the source*, ϕ_J, defined as the solution of the sourced free equation of motion

$$(\Box + m^2)\,\phi_J(x) \;=\; J(x), \tag{10.104}$$

which is precisely the stationarity condition $\delta(S_0 + \int J\phi)/\delta\phi = 0$ of the exponent (the Euler–Lagrange equation of Eq. (10.100) with the source, in the convention fixed for the thread in Eq. (6.60)). In terms of the kinetic operator, $(\Box + m^2) = -\mathcal{O}$ up to the infinitesimal $i\varepsilon$, so Eq. (10.104) is $\mathcal{O}\,\phi_J = -J$, with solution $\phi_J = -\mathcal{O}^{-1}J$. Now shift the integration variable to fluctuations about this classical configuration,

$$\phi(x) \;=\; \phi_J(x) + \eta(x), \tag{10.105}$$

exactly as the proof of Theorem 4.15 shifted $x = y + A^{-1}J$. The shift is by a fixed (source-dependent but field-independent) function ϕ_J, so it is a pure translation of every integration variable, and the measure is invariant:

$$\mathcal{D}\phi \;=\; \prod_x d\phi(x) \;=\; \prod_x d\eta(x) \;=\; \mathcal{D}\eta. \tag{10.106}$$

This is the one place the nature of the measure must be checked rather than assumed. On the finite lattice the shift $\phi_{j,\mathbf{n}} \to \phi_{j,\mathbf{n}} + (\phi_J)_{j,\mathbf{n}}$ is a translation of each of the finitely many real integration variables by a constant, whose Jacobian is the determinant of the identity, namely 1; and a unit Jacobian survives the lattice limit untouched. The translation invariance flagged with appropriate caution in Section 10.3.2 is therefore genuinely available here, because it holds at finite lattice size, which is the criterion the caution box set. No anomaly in the Jacobian can arise for a constant shift; the subtle Jacobians of the path integral (Chapter 36, the chiral anomaly) come only from *field-dependent* changes of variable, never from a translation.

Insert Eq. (10.105) into the exponent of Eq. (10.103) and expand, using the symmetry of $\mathcal{O}$ to combine the two cross terms:

$$\begin{aligned}
\tfrac{1}{2}\int \phi\,\mathcal{O}\,\phi + \int J\phi &= \tfrac{1}{2}\int (\phi_J+\eta)\,\mathcal{O}\,(\phi_J+\eta) + \int J(\phi_J+\eta) \\
&= \underbrace{\tfrac{1}{2}\int \eta\,\mathcal{O}\,\eta}_{\text{fluctuation}} \;+\; \underbrace{\int \eta\,(\mathcal{O}\,\phi_J + J)}_{\text{linear in } \eta} \;+\; \underbrace{\tfrac{1}{2}\int \phi_J\,\mathcal{O}\,\phi_J + \int J\phi_J}_{\text{classical, } \eta\text{-independent}}\,.
\end{aligned} \tag{10.107}$$

The middle term vanishes identically: by construction $\mathcal{O}\,\phi_J = -J$, so $\mathcal{O}\,\phi_J + J = 0$. This is the whole point of shifting to the classical solution — it annihilates the term linear in the fluctuation, which is what "completing the square" means. The classical remainder simplifies using $\mathcal{O}\,\phi_J = -J$ once more:

$$\tfrac{1}{2}\int \phi_J\,\mathcal{O}\,\phi_J + \int J\phi_J \;=\; \tfrac{1}{2}\int \phi_J(-J) + \int J\phi_J \;=\; \tfrac{1}{2}\int J\,\phi_J \;=\; -\tfrac{1}{2}\int J\,\mathcal{O}^{-1}J, \tag{10.108}$$

where the last equality used $\phi_J = -\mathcal{O}^{-1}J$. The exponent has separated cleanly into an η-dependent Gaussian and a source-dependent constant. Therefore

$$Z_0[J] \;=\; \exp\Big[-\tfrac{i}{2}\int d^4x\, J\,\mathcal{O}^{-1}J\Big] \times \underbrace{\int \mathcal{D}\eta\; \exp\Big[\tfrac{i}{2}\int d^4x\,\eta\,\mathcal{O}\,\eta\Big]}_{=\,Z_0[0]}, \tag{10.109}$$

the remaining fluctuation integral being precisely $Z_0[J]$ at $J = 0$ (the classical piece is then absent), so it equals $Z_0[0]$. The source dependence is now entirely explicit and resides in the prefactor; the divergent functional integral has been quarantined into the constant $Z_0[0]$, which will cancel in every correlator.

It remains to name the kernel $\mathcal{O}^{-1}$. Define the *Feynman propagator* of the path integral as

$$D_F(x-y) \;\equiv\; i\,\mathcal{O}^{-1}(x,y), \qquad \text{equivalently} \qquad \mathcal{O}_x\,D_F(x-y) = i\,\delta^4(x-y), \tag{10.110}$$

the factor of i chosen to match the canonical normalization of Chapter 9; we verify in Section 10.4.4 that this choice makes the two definitions coincide. With $\mathcal{O}^{-1} = -iD_F$, the exponent of Eq. (10.109) becomes $-\frac{i}{2}\int J(-iD_F)J = -\frac{1}{2}\int JD_FJ$, and we arrive at the closed form that organizes the entire free theory:

$$\boxed{Z_0[J] \;=\; Z_0[0]\,\exp\Big[-\tfrac{1}{2}\int d^4x\,d^4y\; J(x)\,D_F(x-y)\,J(y)\Big].} \tag{10.111}$$

This is the field-theory image, term for term, of the finite-dimensional master formula $Z(J) = Z(0)\,e^{\frac{1}{2}J^{\mathsf{T}}A^{-1}J}$ of Eq. (4.54), with the kernel A^{-1} replaced by the operator inverse $\mathcal{O}^{-1}$ and the overall i's tracked through. The kernel that appears is, as promised by the finite-dimensional lesson recorded after Eq. (4.54), the *inverse* of the operator in the action — never the operator itself — and it is a single two-point function D_F. Every correlator of the free theory will be a sum of products of this one object.

Mathematical Detail

The completion of the square is exact, not formal, in the following precise sense. Discretize Eq. (10.103) on the lattice of Theorem 10.16: $\phi \to (\phi_a)$ a finite vector, $\int d^4x \to \sum_a (\Delta V)$ with $\Delta V = a^3\Delta t$ the cell

four-volume, and $\mathcal{O} \to \mathcal{O}_{ab}$ a finite symmetric matrix. The exponent is $-\frac{1}{2}\phi^{\mathsf{T}}(-i\mathcal{O}\,\Delta V)\phi + (iJ\Delta V)^{\mathsf{T}}\phi$, which is Eq. (4.54) with $A = -i\mathcal{O}\,\Delta V$ and source $iJ\Delta V$. The $i\varepsilon$ in $\mathcal{O}$ gives A a positive-definite real part (in momentum space $A \propto \varepsilon - i(p^2 - m^2)$, with $\mathrm{Re}\,A = \varepsilon > 0$), which is exactly the condition under which the Gaussian formula Eq. (4.54) converges and its analytic continuation in A is valid. Applying it gives $Z_0[J] = Z_0[0]\exp\!\left[\frac{1}{2}(iJ\Delta V)^{\mathsf{T}}A^{-1}(iJ\Delta V)\right]$; with $A^{-1} = (-i\mathcal{O}\,\Delta V)^{-1} = i\,\mathcal{O}^{-1}/\Delta V$ the exponent is $-\frac{1}{2}\sum_{ab} J_a\,\mathcal{O}^{-1}_{ab}\,J_b\,(\Delta V)^2/\Delta V \cdot (\dots)$, which is the Riemann sum for $-\frac{1}{2}\int J\,\mathcal{O}^{-1}J$ (after the i) and reproduces Eq. (10.111) in the limit. The continuum manipulation is shorthand for this convergent finite-dimensional computation; the $i\varepsilon$ is what makes it converge.

10.4.3 Inverting the kinetic operator: the propagator in momentum space

Equation Eq. (10.111) is only as explicit as our knowledge of $D_F = i\mathcal{O}^{-1}$, and inverting a differential operator is most cleanly done in momentum space, where it becomes division. Use the book's Fourier convention $\phi(x) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot x}\tilde{\phi}(p)$, fixed for the whole book and used already in Section 10.3.4. The kinetic operator acts on a plane wave by

$$\mathcal{O}_x\, e^{-ip\cdot x} = (-\Box - m^2 + i\varepsilon)\, e^{-ip\cdot x} = (p^2 - m^2 + i\varepsilon)\, e^{-ip\cdot x}, \tag{10.112}$$

since $\partial_\mu e^{-ip\cdot x} = -ip_\mu e^{-ip\cdot x}$ gives $\Box e^{-ip\cdot x} = (-ip)\cdot(-ip)\, e^{-ip\cdot x} = -p^2 e^{-ip\cdot x}$, hence $-\Box \to p^2$, and the tilt $m^2 \to m^2 - i\varepsilon$ supplies $+i\varepsilon$ because the operator carries $-m^2$. The defining equation $\mathcal{O}_x D_F(x-y) = i\delta^4(x-y)$ from Eq. (10.110), written with $D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)}\widetilde{D_F}(p)$ and $\delta^4(x-y) = \int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)}$, becomes the algebraic equation $(p^2 - m^2 + i\varepsilon)\,\widetilde{D_F}(p) = i$. Solving,

$$\boxed{\widetilde{D_F}(p) \;=\; \frac{i}{p^2 - m^2 + i\varepsilon}\,,} \tag{10.113}$$

and transforming back,

$$D_F(x-y) \;=\; \int \frac{d^4p}{(2\pi)^4}\, \frac{i\, e^{-ip\cdot(x-y)}}{p^2 - m^2 + i\varepsilon}. \tag{10.114}$$

The $i\varepsilon$ is not optional and is not a separate postulate: it descended, through the chain $m^2 \to m^2 - i\varepsilon$ in Eq. (10.88), from the single infinitesimal tilt of the time contour that selected the vacuum. Its role here is to displace the two poles of the integrand off the real p^0 axis. The denominator factors as $p^2 - m^2 + i\varepsilon = (p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon$ with $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$, whose zeros sit at $p^0 = \pm\sqrt{E_{\mathbf{p}}^2 - i\varepsilon} = \pm(E_{\mathbf{p}} - i\varepsilon')$ to first order in the infinitesimal (with $\varepsilon' = \varepsilon/2E_{\mathbf{p}} > 0$):

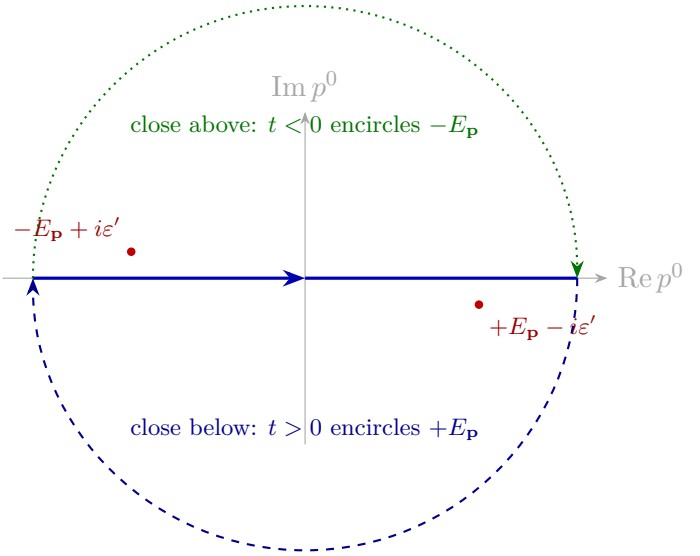


Figure 10.4: The Feynman contour for the p^0 integral in Eq. (10.114), the path-integral image of Theorem 9.25. The $i\varepsilon$ pushes the positive-energy pole to $p^0 = +E_{\mathbf{p}} - i\varepsilon'$ (lower half-plane) and the negative-energy pole to $-E_{\mathbf{p}} + i\varepsilon'$ (upper half-plane). For $t = x^0 - y^0 > 0$ the exponential $e^{-ip^0 t}$ forces closure in the lower half-plane (dashed), enclosing only $+E_{\mathbf{p}}$ and selecting the positive-energy quantum propagating forward in time; for $t < 0$ one closes above (dotted), enclosing $-E_{\mathbf{p}}$. This is the contour analogue of the time-ordering built into the slicing of Theorem 10.19.

the positive-energy pole is pushed *below* the real axis and the negative-energy pole *above* it (Fig. 10.4). This staggered placement is the Feynman prescription, and it is what makes Eq. (10.114) a well-defined tempered distribution rather than an ambiguous integral over a singular denominator (Section 5.1).

10.4.4 The two-pillar agreement

We have now produced, from a Gaussian integral over field histories, a definite two-point object D_F. The canonical quantization of Chapter 9 produced another object also called the Feynman propagator, defined there as the time-ordered vacuum two-point function $\langle 0|\mathcal{T}\phi(x)\phi(y)|0\rangle$ of operators (Theorem 9.24). The promise of the chapter is that these are the same. We now prove it, and the proof is gratifyingly short precisely because all the hard analysis was done once already, in Chapter 9.

Theorem 10.21 (Path-integral and canonical propagators agree)**.** *The propagator D_F obtained from the free generating functional, Eq. (10.113) and Eq. (10.114), is identical as a tempered distribution to the canonical Feynman propagator of*

Chapter 9: its momentum-space form equals Eq. (9.154), and its position-space form equals Eq. (9.153),

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4} \frac{i\, e^{-ip\cdot(x-y)}}{p^2 - m^2 + i\varepsilon} = \theta(x^0-y^0)\, D(x-y)+\theta(y^0-x^0)\, D(y-x) = \langle 0|\mathcal{T}\phi(x)\phi(y)| \tag{10.115}$$

Proof. The first equality is the momentum-space result Eq. (10.113)–Eq. (10.114) just derived. Comparing it with the canonical momentum-space propagator Eq. (9.154), established in Chapter 9 as the covariant integral representation of the time-ordered correlator, the two integrands are *character for character identical*: the same numerator i, the same denominator $p^2 - m^2 + i\varepsilon$, the same Fourier kernel $e^{-ip\cdot(x-y)}$, the same measure $d^4p/(2\pi)^4$. There is nothing left to compute for the momentum-space statement; the path integral has reproduced Eq. (9.154) exactly.

The second equality — that this covariant integral equals the time-ordered position-space form Eq. (9.153) — is precisely the content of the contour theorem Theorem 9.25, which evaluated this same integral by the p^0 contour of Fig. 10.4. We do not reprove it; we recall its mechanism to make the agreement concrete. Hold $\mathbf{p}$ fixed and do the p^0 integral. The factor $e^{-ip^0(x^0-y^0)} = e^{-ip^0 t}$ with $t \equiv x^0 - y^0$ decays in the lower half p^0-plane when $t > 0$, forcing the contour to close below, where it encircles only the pole at $p^0 = +E_{\mathbf{p}} - i\varepsilon'$; the residue theorem (Theorem 5.23) then returns $D(x-y) = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} e^{-ip\cdot(x-y)}\big|_{p^0=E_{\mathbf{p}}}$, the positive-energy Wightman function. When $t < 0$ the contour closes above, encircling $-E_{\mathbf{p}} + i\varepsilon'$ and yielding $D(y-x)$. Reassembled with the step functions that record which closure applied, this is exactly the middle expression of Eq. (10.115), namely Eq. (9.153).

The third equality is the definition Theorem 9.24 of the canonical propagator as the time-ordered vacuum correlator. Independently, Theorem 10.19 of the previous section already established that the path-integral two-point function $G^{(2)}(x,y)$ *is* $\langle 0|\mathcal{T}\phi(x)\phi(y)|0\rangle$, by the time-slicing argument that needed no Gaussian integration; the present theorem supplies its closed form. The two derivations therefore close a logical loop: the slicing argument shows the path integral computes the time-ordered correlator, and the Gaussian evaluation shows that correlator is Eq. (9.154). Both pillars stand on the same propagator. □

The agreement is worth pausing on, because it is the first hard quantitative confrontation of the two formalisms and it is exact, not approximate. In Chapter 9 the $i\varepsilon$ was a choice of contour around poles, justified after the fact by demanding that positive-energy quanta propagate forward in time. In the path integral the *same* $i\varepsilon$ was forced on us as the convergence factor Eq. (10.89) that makes the oscillatory functional integral define a number at all, and as the tilt that projects onto the vacuum. Two prescriptions adopted for entirely different reasons —

causality in the operator formalism, convergence in the functional one — produce the identical analytic structure. That coincidence is the signature that the two are descriptions of one physics.

Physical Insight

Read Eq. (10.111) backwards and it says something striking: the entire free quantum field theory — every vacuum correlator of every order — is encoded in the single classical Green's function D_F. The quantum content of the free theory is exhausted by knowing how a disturbance at one spacetime point propagates to another with Feynman boundary conditions. Interactions, when λ is switched back on in Chapter 14, do not change this propagator; they only stitch copies of it together at vertices. The propagator is thus the atom of perturbation theory, and the path integral has just handed it to us as the inverse of the kinetic operator — the most direct route to it in the entire book.

We now exercise Eq. (10.111) by extracting the two-point function from it directly, both to confirm Theorem 10.21 a third way and to rehearse the differentiation technique that the next section will turn into Wick's theorem.

Worked Example 10.22 (The two-point function from $Z_0[J]$)**.** We compute $G^{(2)}$ by applying the master identity Eq. (10.93) to the closed form Eq. (10.111), tracking every factor. Write the normalized functional $\widehat{Z}_0[J] \equiv Z_0[J]/Z_0[0] = \exp\big[-\tfrac{1}{2}\int J D_F J\big]$, so that $G^{(2)}(x_1,x_2) = i^{-2}\,\delta^2 \widehat{Z}_0/\delta J(x_1)\delta J(x_2)\big|_{J=0}$.

First derivative. Only the exponent depends on J, and the quadratic form is symmetric in its two slots, so

$$\frac{\delta}{\delta J(x_1)}\Big[-\tfrac{1}{2}\int d^4y\, d^4z\, J(y) D_F(y-z) J(z)\Big] = -\int d^4z\, D_F(x_1-z)\, J(z), \tag{10.116}$$

where the factor of 2 from differentiating either J in the symmetric product cancelled the $\tfrac{1}{2}$. Hence

$$\frac{\delta \widehat{Z}_0[J]}{\delta J(x_1)} = \Big[-\int d^4z\, D_F(x_1-z) J(z)\Big]\, \widehat{Z}_0[J]. \tag{10.117}$$

Second derivative. Differentiate Eq. (10.117) with respect to $J(x_2)$. The product rule gives two terms — one from differentiating the bracket, one from differentiating $\widehat{Z}_0$:

$$\frac{\delta^2 \widehat{Z}_0[J]}{\delta J(x_2)\delta J(x_1)} = \Big[-D_F(x_1-x_2)\Big]\widehat{Z}_0[J] + \Big[-\int D_F(x_1-z)J(z)\Big]\Big[-\int D_F(x_2-w)J(w)\Big]\widehat{Z}_0[J]. \tag{10.118}$$

Now set $J = 0$. The second term is quadratic in J and vanishes; $\widehat{Z}_0[0] = 1$ remains. Thus

$$G^{(2)}(x_1, x_2) = \frac{1}{i^2} \left. \frac{\delta^2 \widehat{Z}_0[J]}{\delta J(x_1)\delta J(x_2)} \right|_{J=0} = \frac{-D_F(x_1 - x_2)}{i^2} = D_F(x_1 - x_2), \quad (10.119)$$

using $i^2 = -1$. The two-point Green's function is the Feynman propagator, in exact agreement with Theorem 10.21 and with the canonical result $G^{(2)} = \langle 0|\mathcal{T}\phi(x_1)\phi(x_2)|0\rangle$ of Eq. (10.97). Three independent routes — the time-slicing argument of Theorem 10.19, the Gaussian inversion of Section 10.4.3, and this direct differentiation of the closed form — all land on the same D_F. The structure of Eq. (10.118) also previews the next section: each functional derivative either "pulls down" a propagator joining the new point to an existing one, or acts on the surviving exponential to leave a leftover J; at $J = 0$ only the fully paired terms survive, which is Wick's theorem in embryo.

10.4.5 The prefactor as a functional determinant

One object in Eq. (10.111) we have used but not evaluated: the source-independent normalization $Z_0[0]$, the fluctuation integral in Eq. (10.109). From the finite-dimensional master formula Eq. (4.54), a Gaussian $\int d^n x\, e^{-\frac{1}{2}x^{\mathsf{T}}Ax} = (2\pi)^{n/2}(\det A)^{-1/2}$ evaluates to the inverse square root of the determinant of the quadratic form. Applying this to the lattice version of the fluctuation integral, with $A = -i\mathcal{O}\,\Delta V$ as in the boxed dictionary above, gives

$$Z_0[0] \;=\; \mathcal{N}' \left[\det(-i\mathcal{O})\right]^{-1/2} \;\propto\; \left[\det(-\Box - m^2 + i\varepsilon)\right]^{-1/2}, \qquad (10.120)$$

where $\mathcal{N}'$ collects the divergent (2π)-powers, cell-volume factors, and the phase $(-i)^{n/2}$, all field-independent, and the final proportionality records that, up to that constant, $Z_0[0]$ is the inverse square root of the *functional determinant* of the Klein–Gordon operator. The $-1/2$ power is the hallmark of a bosonic Gaussian; for the fermionic Gaussians of Chapter 12 the same calculation will produce det to the $+1$ power instead, the sign of the exponent flipping with the statistics.

Three honest words on this object, in the spirit of Section 10.3.2. First, $Z_0[0]$ is badly divergent as the lattice is refined — det of a differential operator is an infinite product of growing eigenvalues — but, as established in Section 10.3.5, it cancels identically in every Green's function, because $G^{(n)}$ is defined with $Z[0]$ in the denominator; Theorem 10.20 traced exactly this cancellation. We therefore never need its value for any correlator in this chapter. Second, the determinant is nonetheless physical when ratios of two such determinants are taken — a free energy, a one-loop effective action, a Casimir energy — and giving it a finite meaning (zeta-function or heat-kernel regularization) is a genuine computation, not bookkeeping. Third, that computation, the precise definition

of det for an unbounded operator and the proof that the lattice limit Eq. (10.120) converges to it, belongs to the technical appendix on the path integral and is carried out there in full; we state Eq. (10.120) here and defer its evaluation to Chapter E (the Gaussian master formula of Theorem E.1 and the functional determinants developed in the sections that follow it), using those results by reference rather than reproving them in the main line of the chapter. The rigorous construction of the limiting Gaussian *measure* itself — legitimate, as anticipated in Section 10.3.2, only after Wick rotation to the Euclidean weight e^{-S_E} of Theorem 5.30 — is likewise an Appendix E matter; that the corresponding interacting four-dimensional measure exists at all remains the open problem of constructive field theory.

We have evaluated the free generating functional exactly. Completing the square by the source-shift Eq. (10.105), whose measure invariance is the unit Jacobian of a translation, collapsed $Z_0[J]$ to the closed form Eq. (10.111), $Z_0[J] = Z_0[0]\exp\big[-\frac{1}{2}\int\!\int J D_F J\big]$; inverting the kinetic operator delivered the propagator $D_F(p) = i/(p^2 - m^2 + i\varepsilon)$ of Eq. (10.113); and Theorem 10.21 verified, number for number, that this is the very Feynman propagator that canonical quantization built in Chapter 9 — the central two-pillar consistency check now discharged. The prefactor $Z_0[0]$ has been identified as a functional determinant and set aside, harmless because it cancels in correlators. What we hand to the next section is exactly Eq. (10.111): a single Gaussian generating functional whose only ingredient is the propagator D_F. Every correlator of the free theory is obtained by differentiating it, and because it is the exponential of a quadratic, those derivatives organize themselves into all possible pairings of the field insertions — the combinatorial statement we make precise, and identify with the operator Wick's theorem of Chapter 9, in the section that follows.

10.5 Wick's Theorem from Functional Derivatives

The previous section reduced the entire free scalar theory to a single closed-form object. Setting $\lambda = 0$ in the generating functional Eq. (10.86) left a Gaussian functional integral, and completing the square in the now-quadratic exponent — exactly as for the one-dimensional oscillator of Section 10.2.4, only with the Klein–Gordon operator $-\Box - m^2$ in place of $-\partial_t^2 - \omega^2$ — collapsed it to

$$\frac{Z_0[J]}{Z_0[0]} \;=\; \exp\!\left\{-\frac{1}{2}\int d^4x\, d^4y\; J(x)\, D_F(x-y)\, J(y)\right\}, \tag{10.121}$$

where $D_F(x-y)$ is the Feynman propagator, recovered there a second, independent way as the inverse of the kinetic operator and shown to coincide, number for number, with the canonical propagator $D_F(p) = i/(p^2 - m^2 + i\varepsilon)$ of Eq. (9.154) and Theorem 9.24. We use the symmetry $D_F(x-y) = D_F(y-x)$ throughout,

which holds because the Feynman propagator depends only on $x - y$ through the Lorentz invariant $(x - y)^2$ and is even in its argument.

The result Eq. (10.121) is more than a tidy formula: it is the entire free theory, because every correlation function of the free field is, by Theorem 10.18, a functional derivative of $Z_0[J]$ at $J = 0$. The whole content of the present section is to differentiate Eq. (10.121) and watch what comes out. Because the right-hand side is the exponential of a quadratic in J, those derivatives organize themselves into a single combinatorial pattern: a sum over all ways of pairing up the insertion points, each pair contributing one propagator. That pattern is *Wick's theorem*, the statement that promised in Section 9.4.3 and seeded in the two-particle pairings of Chapter 9 but never yet proved in full. Here we prove it directly from $Z_0[J]$, recover the operator-formalism version of Chapter 9 from it as a cross-check, and extract the explicit four-point structure — three pairings of propagators — that is the combinatorial seed of every Feynman diagram in Chapter 15.

10.5.1 The free n-point functions as derivatives of a Gaussian

We continue ϕ^4 theory in its free sector. The objects we want are the *free* n-point Green's functions, the $\lambda = 0$ specialization of Theorem 10.18,

$$G_0^{(n)}(x_1, \dots, x_n) \;=\; \frac{1}{i^n}\,\frac{\delta^n}{\delta J(x_1)\cdots\delta J(x_n)}\,\frac{Z_0[J]}{Z_0[0]}\bigg|_{J=0}, \tag{10.122}$$

where the functional derivative is the one defined in Theorem 4.4, with the elementary rule

$$\frac{\delta J(y)}{\delta J(x)} = \delta^{(4)}(x - y) \tag{10.123}$$

and the product/chain rules it inherits from ordinary differentiation. The normalization by $Z_0[0]$ has already been carried out in Eq. (10.121), so we differentiate the right-hand side of that equation directly; the factor $1/i^n$ is the one prescribed by Theorem 10.18, compensating the i that each source insertion carried in the Minkowski exponent.

It is convenient to give the quadratic form in the exponent a name. Write

$$W[J] \;\equiv\; -\frac{1}{2}\int d^4x\, d^4y\; J(x)\, D_F(x - y)\, J(y), \qquad \frac{Z_0[J]}{Z_0[0]} = e^{W[J]}. \tag{10.124}$$

Two functional derivatives of W exhaust its content, because W is exactly quadratic. The first derivative, using Eq. (10.123) and the symmetry of D_F to combine the two terms in which J is hit, is the *linear* functional

$$\frac{\delta W[J]}{\delta J(x)} \;=\; -\int d^4y\; D_F(x - y)\, J(y) \;\equiv\; L(x; J), \tag{10.125}$$

and the second derivative is the *constant* (i.e. J-independent) kernel

$$\frac{\delta^2 W[J]}{\delta J(x)\,\delta J(x')} \;=\; -\,D_F(x-x'), \tag{10.126}$$

while all third and higher derivatives of W vanish identically. Equations Eqs. (10.125) and (10.126) are the only two facts about W we shall use; everything below is bookkeeping of how the chain rule distributes them across the exponential.

The structure of that bookkeeping is already visible after one and two derivatives. Acting on e^W once,

$$\frac{\delta}{\delta J(x_1)}\,e^W \;=\; L(x_1;J)\,e^W, \tag{10.127}$$

and acting again, by the product rule and Eq. (10.126),

$$\frac{\delta^2}{\delta J(x_1)\,\delta J(x_2)}\,e^W \;=\; \Big[L(x_1;J)\,L(x_2;J) \;-\; D_F(x_1-x_2)\Big]\,e^W. \tag{10.128}$$

Each further derivative does one of exactly two things to the polynomial-in-L prefactor that has accumulated: by Eqs. (10.125) and (10.126) it either differentiates an existing factor $L(x_a;J)$, turning it into $-D_F(x_a-x_b)$ where x_b is the new derivative point, or it produces a new factor $L(x_b;J)$ from acting on e^W. At the end we set $J=0$, where every surviving $L(x;0)=0$ by Eq. (10.125). Therefore *only those terms in which every derivative point has been used up to manufacture a propagator survive*: the terms with any leftover factor of L die at $J=0$. This single observation is the whole of Wick's theorem, and we now make it precise and count.

10.5.2 The combinatorics: derivatives of e^W are sums over pairings

The cleanest way to carry out the count, with no induction left implicit, is to expand the exponential as a power series and read off the term that survives n derivatives at $J=0$. Write the quadratic form as

$$W[J]=\frac{1}{2}\int d^4u\,d^4v\;J(u)\,A(u,v)\,J(v), \qquad A(u,v)\equiv -\,D_F(u-v)=A(v,u), \tag{10.129}$$

so that A is a symmetric kernel, and expand

$$e^{W[J]}=\sum_{m=0}^{\infty}\frac{1}{m!}\,W[J]^m=\sum_{m=0}^{\infty}\frac{1}{m!\,2^m}\int\Big(\textstyle\prod_{a=1}^{m}d^4u_a\,d^4v_a\Big)\;\prod_{a=1}^{m}J(u_a)\,A(u_a,v_a)\,J(v_a). \tag{10.130}$$

The m-th term is a homogeneous polynomial of degree $2m$ in the source J. Applying n functional derivatives and then setting $J = 0$ annihilates every term of the wrong degree: a polynomial of degree $2m$ differentiated n times and evaluated at $J = 0$ gives zero if $2m < n$ (too few sources to absorb the derivatives) and zero if $2m > n$ (a surviving power of J vanishes at $J = 0$). Only $2m = n$ contributes. Two consequences are immediate.

Odd n vanish. If n is odd there is no integer m with $2m = n$, so

$$G_0^{(2k+1)}(x_1, \ldots, x_{2k+1}) = 0 \qquad \text{for all } k \geq 0. \tag{10.131}$$

This is the path-integral face of the elementary fact that the free vacuum overlaps only with states containing an even number of quanta; in operator language it is the statement that a product of an odd number of free fields has no fully-contracted piece and so vanishes between the vacuum, and it is enforced here, even before any explicit contraction, by the $\mathbb{Z}_2$ parity $J \to -J$ of the Gaussian Eq. (10.121).

Even $n = 2k$. Only the $m = k$ term of Eq. (10.130) survives, and with the prefactor $1/(k!\,2^k)$ it reads

$$\frac{1}{k!\,2^k} \int \Big(\textstyle\prod_{a=1}^{k} d^4u_a\, d^4v_a \Big)\, J(u_1)A(u_1, v_1)J(v_1) \cdots J(u_k)A(u_k, v_k)J(v_k). \tag{10.132}$$

We must differentiate this with respect to $J(x_1), \ldots, J(x_{2k})$ and set $J = 0$. There are exactly $2k$ source factors and $2k$ derivatives, so the result is the sum over all ways of matching the $2k$ derivatives to the $2k$ source factors, one to each. Each such matching is a bijection σ from the set of derivative labels $\{x_1, \ldots, x_{2k}\}$ onto the set of source slots $\{u_1, v_1, \ldots, u_k, v_k\}$; differentiating the slot $J(u_a)$ by $\delta/\delta J(x_i)$ produces $\delta^{(4)}(x_i - u_a)$, which the integral over u_a then collapses, replacing the dummy variable u_a by x_i. After all $2k$ collapses, each kernel factor $A(u_a, v_a)$ has become $A(x_i, x_j)$ for the two derivative labels x_i, x_j assigned to that factor's two slots. The contribution of the bijection σ is therefore the product of k kernels,

$$\prod_{a=1}^{k} A\big(x_{\sigma^{-1}(u_a)},\, x_{\sigma^{-1}(v_a)}\big), \tag{10.133}$$

and the full derivative is the sum of Eq. (10.133) over all $(2k)!$ bijections σ, all divided by $k!\,2^k$.

Now we remove the overcounting hidden in the $(2k)!$ bijections, and this is exactly what the prefactor $1/(k!\,2^k)$ was built to do. Two distinct bijections give the *same* product Eq. (10.133) whenever they differ only by (i) a permutation of which kernel factor is called the first, second, …, k-th — there are $k!$ such relabelings, and they leave the product of kernels unchanged because multiplication is commutative — or (ii) a swap of the two slots $u_a \leftrightarrow v_a$ within any factor — there are 2 choices per factor, hence 2^k in all, and they leave each

kernel unchanged because A is symmetric, $A(u,v) = A(v,u)$. The group of these trivial reshufflings has order $k!\,2^k$, and it acts freely, so the $(2k)!$ bijections fall into $(2k)!/(k!\,2^k)$ orbits, each orbit contributing one distinct product of kernels with multiplicity $k!\,2^k$. That multiplicity cancels the prefactor exactly. What is left is a sum with unit coefficient over the orbits, and an orbit is precisely an unordered partition of the $2k$ labels $\{x_1,\dots,x_{2k}\}$ into k unordered pairs — a *pairing*. Therefore

$$\frac{\delta^{2k}}{\delta J(x_1)\cdots\delta J(x_{2k})}\,e^{W}\Big|_{J=0} \;=\; \sum_{\text{pairings } P}\;\prod_{\{i,j\}\in P} A(x_i,x_j) \;=\; \sum_{\text{pairings } P}\;\prod_{\{i,j\}\in P}\bigl(-D_F(x_i-x_j)\bigr). \tag{10.134}$$

The number of pairings of $2k$ objects is the double factorial

$$(2k-1)!! \;=\; (2k-1)(2k-3)\cdots 3\cdot 1 \;=\; \frac{(2k)!}{k!\,2^k}, \tag{10.135}$$

counted directly as follows: pick the partner of x_1 in $2k-1$ ways, then the partner of the lowest remaining label in $2k-3$ ways, and so on down to the last pair, with one way; this is the same integer that emerged as the orbit count above, a satisfying internal consistency check. For $k = 1, 2, 3$ it gives 1, 3, 15 pairings respectively.

Finally restore the prefactor $1/i^n$ of Eq. (10.122). With $n = 2k$ each of the k factors $-D_F$ supplies a sign (-1), contributing $(-1)^k$, while $1/i^{2k} = 1/(-1)^k = (-1)^k$; the two signs multiply to $(-1)^{2k} = +1$. The minus signs cancel completely and we arrive at the clean statement.

Theorem 10.23 (Wick's theorem for the free scalar field)**.** *The free n-point Green's functions of the scalar field, defined by Eq. (10.122), vanish for odd n and, for even $n = 2k$, equal the sum over all pairings of the n points of the product of Feynman propagators over the pairs:*

$$\boxed{G_0^{(2k)}(x_1,\dots,x_{2k}) \;=\; \sum_{\text{pairings } P}\;\prod_{\{i,j\}\in P} D_F(x_i-x_j)\,, \qquad G_0^{(2k+1)} = 0.} \tag{10.136}$$

There are $(2k-1)!!$ terms in the sum, each a product of k propagators that joins the $2k$ points in pairs.

This is the same combinatorial law that the single-oscillator template anticipated at the close of Section 10.2.4: differentiating the exponential of a quadratic generates all pairings of the insertion points. The field theory has merely promoted the discrete insertion times to spacetime points x_i and the oscillator Green's function to the covariant propagator D_F. Nothing in the derivation used anything but the Gaussian form Eq. (10.121) of $Z_0[J]$ and the two derivative rules Eqs. (10.125) and (10.126); the theorem is therefore an identity of tempered distributions, exact for the free theory.

Physical Insight

Wick's theorem is the assertion that a free field is, statistically, a Gaussian random field, and Theorem 10.23 is exactly the statement that a Gaussian's higher moments are fixed by its covariance. Replace "field" by "random variable," "propagator" by "covariance," and "$G_0^{(n)}$" by "n-th moment," and Eq. (10.136) is the probabilist's Isserlis (Wick) formula for the moments of a multivariate Gaussian. The path integral makes the analogy literal: in the Euclidean version of the next section, $e^{-S_{E,0}}\,\mathcal{D}\phi$ *is* a Gaussian measure, D_F *is* its covariance, and the n-point functions *are* its moments. Free quantum field theory contains no information beyond its two-point function; all higher correlators are bookkeeping of how the two-point function can be wired up.

10.5.3 The two-point function and the four-point function

The smallest cases of Theorem 10.23 are the ones the rest of the book uses constantly, and it is worth seeing them land explicitly.

For $n = 2$ there is a single pairing, $\{1, 2\}$, so

$$G_0^{(2)}(x_1, x_2) \;=\; D_F(x_1 - x_2). \tag{10.137}$$

The free two-point function is the Feynman propagator, with no decoration. This is consistent at one stroke with three earlier results: with the general identification $G^{(2)} = \langle 0|\,\mathcal{T}\phi(x_1)\phi(x_2)\,|0\rangle$ of Theorem 10.19, with the direct second-derivative computation Eq. (10.128) (set $J = 0$, where the LL term dies, leaving $-D_F$, then divide by $i^2 = -1$), and with the canonical definition of D_F in Theorem 9.24. The single internal line of every Feynman diagram is this object.

For $n = 4$ there are $(4-1)!! = 3$ pairings of the labels $\{1, 2, 3, 4\}$ — namely $\{12\}\{34\}$, $\{13\}\{24\}$, $\{14\}\{23\}$ — and Eq. (10.136) gives

$$G_0^{(4)}(x_1, x_2, x_3, x_4) = D_F(x_1 - x_2)D_F(x_3 - x_4) + D_F(x_1 - x_3)D_F(x_2 - x_4) + D_F(x_1 - x_4) \tag{10.138}$$

These three terms are the three distinct ways to join four points pairwise with propagators; pictorially they are the three ways of drawing two non-intersecting (or intersecting) chords among four points on a circle, shown in Fig. 10.5. They are not yet a scattering process — there is no interaction vertex in the free theory — but they are precisely the contractions that, in Chapter 15, will be tied off against the first $\lambda\phi^4$ vertex to build the $O(\lambda)$ four-point amplitude of ϕ^4 theory. The combinatorial seed of the Feynman expansion is exactly Eq. (10.138).

Worked Example 10.24 (The free four-point function, derived in full from $Z_0[J]$)**.** We carry out the four functional differentiations of Eq. (10.122) by hand,

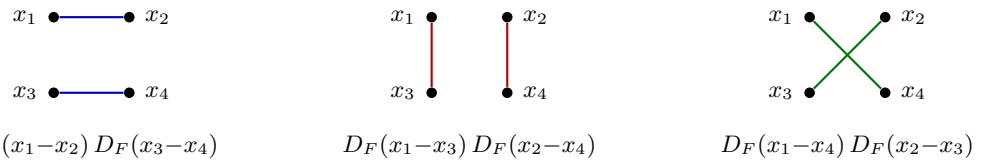


Figure 10.5: The three pairings of the free four-point function Eq. (10.138). Each panel joins the four points $x_1, \ldots, x_4$ into two pairs by Feynman-propagator lines; the sum of the three is $G_0^{(4)}$. These are the $(4-1)!! = 3$ terms of Wick's theorem at $n = 4$ and the combinatorial seed of the ϕ^4 Feynman rules of Chapter 15: tying the four loose ends to a single four-point vertex turns these disconnected pairings into the first interacting amplitude of ϕ^4 theory.

with every intermediate term written, to verify Eq. (10.138) and to display the mechanism — "differentiate an L into a propagator, or pull down a fresh L" — in action. Throughout we abbreviate $L_i \equiv L(x_i; J)$ for the linear functional Eq. (10.125) and $D_{ij} \equiv D_F(x_i - x_j)$ for the propagator, and we record the two rules we will use repeatedly:

$$\frac{\delta}{\delta J(x_b)}\, L_a = -D_{ab}, \qquad \frac{\delta}{\delta J(x_b)}\, e^{W} = L_b\, e^{W}. \tag{10.139}$$

The first is Eq. (10.126) written for the linear functional; the second is Eq. (10.127). Note D_{ab} is a J-independent constant, so it is killed by any further derivative.

First derivative. From the second rule,

$$\frac{\delta}{\delta J(x_1)} e^{W} = L_1\, e^{W}. \tag{10.140}$$

Second derivative. Apply $\delta/\delta J(x_2)$ by the product rule, using both rules in Eq. (10.139):

$$\frac{\delta^2}{\delta J(x_2)\delta J(x_1)} e^{W} = (-D_{12}) e^{W} + L_1\, L_2\, e^{W} = (L_1 L_2 - D_{12}) e^{W}, \tag{10.141}$$

reproducing Eq. (10.128).

Third derivative. Apply $\delta/\delta J(x_3)$ to Eq. (10.141). The constant $-D_{12}$ contributes only through the e^W it multiplies; the product $L_1 L_2$ is differentiated term by term:

$$\begin{aligned} \frac{\delta^3}{\delta J(x_3)\delta J(x_2)\delta J(x_1)} e^{W} &= \Big[(-D_{13})L_2 + L_1(-D_{23})\Big] e^{W} + (L_1 L_2 - D_{12}) L_3\, e^{W} \\ &= \Big[L_1 L_2 L_3 - D_{12} L_3 - D_{13} L_2 - D_{23} L_1\Big] e^{W}. \end{aligned} \tag{10.142}$$

Every term contains at least one factor of L, so this vanishes at $J = 0$ — the explicit confirmation of Eq. (10.131) for $n = 3$: there is no way to use up an odd number of derivatives in propagators alone, so a stray L always remains.

Fourth derivative. Apply $\delta/\delta J(x_4)$ to the bracket in Eq. (10.142), again differentiating each term and adding the term from differentiating the trailing e^W. Differentiating the bracket $B_3 \equiv L_1L_2L_3 - D_{12}L_3 - D_{13}L_2 - D_{23}L_1$:

$$\begin{aligned}\frac{\delta B_3}{\delta J(x_4)} &= (-D_{14})L_2L_3 + L_1(-D_{24})L_3 + L_1L_2(-D_{34}) \\ &\quad - D_{12}(-D_{34}) - D_{13}(-D_{24}) - D_{23}(-D_{14}) \\ &= -D_{14}L_2L_3 - D_{24}L_1L_3 - D_{34}L_1L_2 + (D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23}),\end{aligned} \tag{10.143}$$

where the second line is the only place fresh propagator–propagator products are born: each surviving single propagator $-D_{ij}$ from the third derivative meets the derivative $\delta/\delta J(x_4)$ acting on its lone L partner, and the two minus signs make a plus. The full fourth derivative is $(\delta B_3/\delta J(x_4) + B_3\, L_4)e^W$. Setting $J = 0$ now annihilates every term that still carries an L: in $\delta B_3/\delta J(x_4)$ the three LL terms die, and in B_3L_4 all four terms die (each has at least one L). What remains is the propagator–propagator block of Eq. (10.143):

$$\frac{\delta^4}{\delta J(x_1)\delta J(x_2)\delta J(x_3)\delta J(x_4)} e^W \bigg|_{J=0} = D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23}. \tag{10.144}$$

Divide by $i^4 = 1$ as prescribed in Eq. (10.122): the free four-point function is exactly the three-term sum,

$$G_0^{(4)}(x_1, x_2, x_3, x_4) = D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23}, \tag{10.145}$$

which is Eq. (10.138). The three terms are the three pairings $\{12\}\{34\}$, $\{13\}\{24\}$, $\{14\}\{23\}$, with no surplus factors and no missing ones; the $(4-1)!! = 3$ of Eq. (10.135) is verified by direct computation, and the mechanism by which the Gaussian generates exactly the pairings is laid bare.

10.5.4 Cross-check: recovering the operator Wick's theorem of Chapter 9

The result Theorem 10.23 was derived entirely inside the path integral, with no operators in sight. It must agree with the operator formalism of Chapter 9, since by Theorem 10.19 the free Green's function $G_0^{(n)}$ *is* the time-ordered vacuum expectation value $\langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|0\rangle$. The operator statement that reduces this VEV to the same sum over pairings is the classical Wick's theorem, the result foreshadowed in the two-particle pairings of Section 9.5.2 and the charged-field bookkeeping of Theorem 9.37; it was not proved in Chapter 9, which deferred

it precisely to this generating-functional derivation. We now state and prove it in the operator language and confirm the two routes coincide, completing the cross-check.

Decompose the free Heisenberg field into its positive- and negative-frequency parts as in Chapter 9, $\phi(x) = \phi^+(x) + \phi^-(x)$, where ϕ^+ contains the annihilation operators $a_{\mathbf{p}}$ (it annihilates the vacuum on the right, $\phi^+ |0\rangle = 0$) and ϕ^- the creation operators $a^\dagger_{\mathbf{p}}$ (so $\langle 0| \phi^- = 0$), obeying the mode algebra of Theorem 9.37. The *contraction* of two fields is the c-number

$$\overbracket{\phi(x)\,\phi(y)} \;\equiv\; \mathcal{T}\,\phi(x)\phi(y) - :\phi(x)\phi(y): \,, \tag{10.146}$$

with $: \cdots :$ the normal ordering of Theorem 9.19. For $x^0 > y^0$, $\mathcal{T}\phi(x)\phi(y) = \phi(x)\phi(y)$, and moving the annihilation part $\phi^+(x)$ to the right of the creation part $\phi^-(y)$ to reach normal order costs exactly the commutator $[\phi^+(x), \phi^-(y)]$, a c-number; for $y^0 > x^0$ one gets $[\phi^+(y), \phi^-(x)]$. In either order the contraction is a c-number, and since it is a c-number it equals its own vacuum expectation value,

$$\overbracket{\phi(x)\,\phi(y)} = \langle 0| \overbracket{\phi(x)\,\phi(y)} |0\rangle = \langle 0| \mathcal{T}\,\phi(x)\phi(y) |0\rangle = D_F(x-y), \tag{10.147}$$

because $\langle 0| :\phi\phi: |0\rangle = 0$ by Theorem 9.19. The contraction of two free fields is the Feynman propagator — the operator counterpart of Eq. (10.126).

Theorem 10.25 (Operator Wick's theorem for free fields)**.** *A time-ordered product of free scalar field operators equals the sum of all its normal-ordered terms with every possible set of contractions:*

$$\mathcal{T}\,\phi(x_1)\cdots\phi(x_n) = :\phi(x_1)\cdots\phi(x_n): \;+ \sum_{\substack{\text{single}\\ \text{contractions}}} :(\cdots): \;+\cdots+ \sum_{\substack{\text{full}\\ \text{contractions}}} (\cdots), \tag{10.148}$$

where each contraction replaces a pair of fields by the c-number Eq. (10.147) and the remaining uncontracted fields are normal-ordered; the last group, present only for even n, consists of fully-contracted (pure c-number) terms.

Proof. By induction on n. For $n = 1$ the statement is $\phi(x_1) = :\phi(x_1):$, true since a single field is already normal-ordered. For $n = 2$ it is the rearrangement of Eq. (10.146): $\mathcal{T}\phi(x_1)\phi(x_2) = :\phi(x_1)\phi(x_2): + \overbracket{\phi(x_1)\phi(x_2)}$. Assume Eq. (10.148) holds for n fields. Order the labels so that x_{n+1} has the earliest time, $x^0_{n+1} \le x^0_i$ for all $i \le n$ (always possible, since both sides of Eq. (10.148) are totally symmetric under relabeling — the left side because $\mathcal{T}$ symmetrizes, the right because the sum is over all contraction patterns); then

$$\mathcal{T}\,\phi(x_1)\cdots\phi(x_{n+1}) = \Big(\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\Big)\,\phi(x_{n+1}), \tag{10.149}$$

because $\phi(x_{n+1})$ sits to the right of all the others in time and the time-ordering symbol places it there. Substitute the inductive hypothesis for the bracket and multiply each normal-ordered term on the right by $\phi(x_{n+1}) = \phi^+(x_{n+1}) + \phi^-(x_{n+1})$. The creation piece $\phi^-(x_{n+1})$ is already to the right of nothing it needs to pass and may be absorbed into the normal ordering directly (creation operators stand left, and ϕ^- being rightmost we may write it inside the colons by moving it left past nothing it fails to commute with as a c-number—more precisely, $:X:\,\phi^- = :X\phi^-:$ since ϕ^- is a creation operator and normal ordering puts it to the left at no commutator cost). The annihilation piece $\phi^+(x_{n+1})$ must instead be moved leftward through each normal-ordered factor to bring it inside the colons; passing it across the negative-frequency (creation) content of a field $\phi(x_i)$ generates the c-number commutator $[\phi^+(x_{n+1}), \phi^-(x_i)] = \overbrace{\phi(x_{n+1})\phi(x_i)} = D_F(x_{n+1} - x_i)$ by Eq. (10.147) (valid as the contraction since x_{n+1} is the earliest time), each such commutator being precisely one new contraction of x_{n+1} with x_i. Collecting the term in which $\phi(x_{n+1})$ is absorbed uncontracted, and all the terms in which it is contracted once with each available partner, regenerates Eq. (10.148) for $n+1$ fields: the patterns are the old patterns with x_{n+1} either left uncontracted or contracted with one previously-uncontracted field, which is exactly the enumeration of all contraction patterns of $n+1$ fields. This closes the induction. □

Taking the vacuum expectation value of Eq. (10.148) now reproduces Theorem 10.23 term for term. Every term that still contains a normal-ordered product of one or more uncontracted fields has vanishing vacuum expectation value, $\langle 0|\,:(\cdots):\,|0\rangle = 0$ (Theorem 9.19); only the *fully*-contracted terms, which are pure c-numbers, survive. For odd n no full contraction exists (one field is always left over), so the VEV vanishes, matching Eq. (10.131). For even $n = 2k$ the surviving terms are exactly the pairings, each a product of k contractions $= k$ propagators:

$$\langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_{2k})\,|0\rangle = \sum_{\text{full contractions}} \prod_{\{i,j\}} D_F(x_i - x_j) = \sum_{\text{pairings } P} \prod_{\{i,j\}\in P} D_F(x_i - x_j). \tag{10.150}$$

A full contraction of $2k$ fields is the same combinatorial object as a pairing of $2k$ points, so Eq. (10.150) is identical to Eq. (10.136). The path integral and the operator formalism produce not merely the same answer but the same intermediate structure — a sum over pairings, one propagator per pair, $(2k-1)!!$ terms — arrived at by completely different mechanisms: the path integral by the chain rule on a Gaussian, the operator formalism by commuting annihilation operators to the right. For $n = 4$ this is the explicit statement

$$\langle 0|\,\mathcal{T}\,\phi_1\phi_2\phi_3\phi_4\,|0\rangle = D_F(x_1 - x_2)D_F(x_3 - x_4) + D_F(x_1 - x_3)D_F(x_2 - x_4) + D_F(x_1 - x_4)D_F(x_2 - \tag{10.151}$$

which is Eq. (10.138) again, now read off the operator side. The two pillars agree on the free four-point function, completing the consistency check this section was charged with.

Caution

The contraction Eq. (10.147) is the *time-ordered* two-point function D_F, not the Wightman function or the commutator. This matters when one is tempted to "contract" fields inside an already-fixed (non-time-ordered) operator order: Theorem 10.25 is a statement about the *time-ordered* product on the left, and it is the time-ordered product that the path integral computes (Theorem 10.19). Wick's theorem in the form Eq. (10.148) for ordinary (non-time-ordered) products also exists, but with the Wightman two-point function as the contraction; the two versions must not be mixed. Throughout the perturbation theory of Chapters 14 and 15 it is the time-ordered version with D_F, the one derived here, that is used.

10.5.5 What Wick's theorem hands forward

We have now derived Wick's theorem for the free scalar field twice over and shown the two derivations coincide. From the path integral, the free n-point function is a single functional derivative of the Gaussian $Z_0[J] = Z_0[0]\, e^{W[J]}$ of Eq. (10.121), and the chain rule turns each such derivative into the sum over pairings Eq. (10.136): odd correlators vanish, the even ones are sums of $(2k-1)!!$ products of propagators, the two-point function is D_F alone, and the four-point function is the three pairings of Eq. (10.138). From the operator side, the same sum emerges as the vacuum expectation value of the Wick-expanded time-ordered product Eq. (10.148), with the contraction equal to D_F. The free theory is thereby completely solved: *every* correlation function is an explicit, finite sum of propagators, with no integration left to do.

These pairings are the combinatorial seed of the entire diagrammatic expansion. In Section 10.7 (Theorem 10.33) the interacting generating functional is written as $Z[J] = \exp\!\big[-i\tfrac{\lambda}{4!}\int d^4z\,(\tfrac{1}{i}\tfrac{\delta}{\delta J(z)})^4\big]\, Z_0[J]$, so that each power of λ inserts four extra functional derivatives at a common point z; Theorem 10.23 then dictates that the resulting correlators are again sums over pairings, now of the external points *and* the four legs of each vertex. The three pairings of Eq. (10.138), tied off against a single $\lambda\phi^4$ vertex, are exactly the contractions that dress the first ϕ^4 interaction and produce the $O(\lambda)$ four-point amplitude $-i\lambda$ of ϕ^4 theory; the symmetry factors that count how many pairings give the same diagram are bookkeeping of the $(2k-1)!!$ of Eq. (10.135). The Feynman rules of Chapter 15 are nothing more than a graphical shorthand for the contraction sum proved here. Before turning the crank on the interaction, however, we change the contour: the next section rotates time to the imaginary axis, trading the

oscillatory weight e^{iS} for the convergent, positive Euclidean measure e^{-S_E} in which the Gaussian of this section becomes a genuine probability measure and the propagator its covariance — the form that powers the renormalization group, the lattice, and the bridge to statistical mechanics.

10.6 The Euclidean Path Integral

Everything we have built so far in this chapter lives on a real time axis. The free generating functional assembled in Section 10.4,

$$Z_0[J] \;=\; Z_0[0]\,\exp\Big[-\tfrac{1}{2}\int d^4x\,d^4y\;J(x)\,D_F(x-y)\,J(y)\Big], \tag{10.152}$$

with the Feynman propagator $D_F(x-y)$ of Theorem 9.24 as its kernel, is a Gaussian whose exponent is *imaginary*: the unsourced weight is the pure phase e^{iS}, and the integral converges only conditionally, kept alive by the infinitesimal tilt $m^2 \to m^2 - i\varepsilon$ of Section 10.3.4. That tilt was an infinitesimal rotation of the time contour into the lower half-plane. The business of this section is to complete the rotation: to turn the contour all the way to the imaginary time axis, $t \to -i\tau$, and watch the oscillatory weight e^{iS} become a real, positive, exponentially damped weight e^{-S_E}. The resulting object, the *Euclidean* path integral, is not a mere computational convenience. It is a genuinely convergent integral against a positive measure; it is, formally, the partition function of a classical statistical-mechanical system in four dimensions; and it is the form in which the path integral powers the renormalization group (Chapter 23), the lattice (Chapter 43), and conformal field theory (Chapter 52). We continue ϕ^4 theory, and the Euclidean action S_E we fix here is the object those later chapters will flow, discretize, and sit inside.

The analytic machinery we need was assembled in Section 5.4: the Wick rotation as a change of variables (Theorem 5.30), and the theorem (Theorem 5.31) that the rotation is a legal contour deformation precisely because the Feynman $i\varepsilon$ places the propagator poles out of its path. We import those results rather than reprove them, and spend our effort on what is owed here: carrying the rotation through the path integral itself, deriving the all-plus Euclidean action of ϕ^4 theory sign by sign, exhibiting the convergent positive Gaussian measure, constructing the Euclidean propagator and continuing it back to D_F, and laying out the dictionary that makes a quantum field theory and a statistical-mechanical system two readings of one integral.

10.6.1 Wick rotation as the completion of the $i\varepsilon$ tilt

The starting point is the observation, already made in Section 10.3.4, that the $i\varepsilon$ prescription is not an independent decoration of the Minkowski path

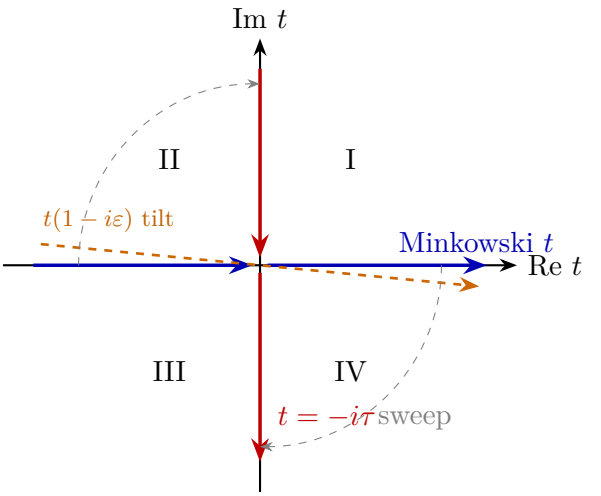


Figure 10.6: The Wick rotation in the complex-t plane. The vacuum $i\varepsilon$ prescription of Section 10.3.4 tilts the real-time contour (blue) infinitesimally clockwise (orange dashed). The Wick rotation completes that tilt to a quarter turn, bringing the contour to the imaginary axis $t = -i\tau$ (red), with τ increasing as t runs $-i\infty \to +i\infty$ traversed downward. The dual rotation of the energy contour in the complex p^0 plane sweeps the pole-free quadrants of Fig. 5.4; the tilt directions are conjugate, t clockwise as p^0 counterclockwise, because t and p^0 are Fourier-conjugate.

integral but the first infinitesimal step of a finite rotation. Recall the mechanism. The vacuum generating functional is computed on a time interval $(-T, T)$, and the prescription that selects the ground state tilts the endpoints into the complex plane, $t \to t(1 - i\varepsilon)$, so that the evolution operator e^{-iHt} acquires a real damping factor $e^{-Ht\varepsilon}$ that suppresses every excited state relative to $|0\rangle$ (the spectral argument of Eq. (10.87)). Geometrically the tilt rotates the time contour infinitesimally clockwise, into the fourth quadrant of the complex-t plane. The Wick rotation is the statement that we may continue this rotation all the way to a quarter turn, until the contour lies along the negative imaginary axis,

$$t \;=\; -i\,\tau, \qquad \tau \in \mathbb{R}, \quad \tau\colon\; -\infty \to +\infty, \tag{10.153}$$

the substitution of Theorem 5.30 with τ the Euclidean time. The infinitesimal tilt and the finite rotation are the same operation carried to different angles; the only question is whether nothing goes wrong along the way, and "nothing goes wrong" means precisely that the rotating contour sweeps across no singularity of the integrand.

The cleanest place to certify pole avoidance is the Fourier-conjugate variable, the energy p^0, where the singularities are explicit. Every correlator the path integral produces is built, by the Gaussian evaluation of Section 10.4, from the

Feynman propagator, whose momentum-space form is

$$D_F(p) \;=\; \frac{i}{p^2 - m^2 + i\varepsilon} \;=\; \frac{i}{(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon}, \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{10.154}$$

the object established in Eq. (9.154), with $E_{\mathbf{p}}$ the on-shell energy of the dispersion relation Eq. (9.108). Its poles in the complex p^0 plane were located exactly in Eq. (9.157): one at $p^0 = +E_{\mathbf{p}} - i\varepsilon'$, just below the positive real axis (fourth quadrant), and one at $p^0 = -E_{\mathbf{p}} + i\varepsilon'$, just above the negative real axis (second quadrant). The dual of the time rotation Eq. (10.153) is the energy rotation $p^0 \to ip_E^4$ of Eq. (5.93), which swings the p^0 contour *counterclockwise* from the real axis to the imaginary axis (the conjugacy of the two tilt directions is exactly the statement that t and p^0 are Fourier conjugate: a clockwise turn of t is a counterclockwise turn of p^0). That counterclockwise sweep passes through the first and third quadrants of the p^0 plane — and those are precisely the quadrants left empty by the Feynman pole placement. This is the geometry drawn in Fig. 5.4, and it is the content of the legality theorem.

Proposition 10.26 (The rotation crosses no propagator pole)**.** *For the free scalar with the Feynman prescription, the rotation of the energy contour $p^0 \to ip_E^4$ dual to the time rotation Eq. (10.153) sweeps only the pole-free first and third quadrants of the complex p^0 plane. Consequently, for each fixed real spatial momentum $\mathbf{p}$ the Minkowski energy integral equals the rotated Euclidean one,*

$$\int_{-\infty}^{\infty} dp^0 \, \frac{i\,F(p^0)}{(p^0)^2 - E_{\mathbf{p}}^2 + i\varepsilon} \;=\; i\int_{-\infty}^{\infty} dp_E^4 \, \frac{i\,F(ip_E^4)}{-\big[(p_E^4)^2 + E_{\mathbf{p}}^2\big]}, \tag{10.155}$$

for any F analytic and suitably decaying in the swept quadrants.

Proof. The poles of the integrand are those of Eq. (10.154) at $p^0 = \pm(E_{\mathbf{p}} - i\varepsilon')$ together with whatever singularities F may carry; by hypothesis F is analytic in the first and third quadrants, and the propagator poles sit in the second and fourth, as recorded in Eq. (9.157). The closed contour built from the real segment, the quarter-circle arc at infinity, and the imaginary segment therefore encloses no singularity in the first quadrant, and likewise in the third; Cauchy's theorem (Theorem 5.23) makes the closed integral vanish, and the decay hypothesis kills the arcs by the ML-estimate. The two straight segments are then equal and opposite, which is the equality Eq. (10.155); the prefactor i on the right is the Jacobian $dp^0 = i\,dp_E^4$ of $p^0 = ip_E^4$. This is the special case of the legality theorem Theorem 5.31 for the Feynman denominator, and we invoke that theorem for the analytic estimates rather than repeating them. □

The point worth dwelling on is that the rotation is legal *because of* the $i\varepsilon$, not in spite of it. Had we chosen the retarded prescription, both poles would lie below the real axis and the fourth-quadrant sweep would drag the contour across

one of them, picking up a residue and spoiling the equality; the advanced choice fails symmetrically in the second quadrant. Only the Feynman placement — the same staggering that the operator formalism produced from time-ordering — leaves the rotation unobstructed. The three faces of the $i\varepsilon$ noted in Section 10.3.4 (time-ordering, vacuum projection, Gaussian damping) here acquire a fourth, which is in fact their unification: the $i\varepsilon$ is the infinitesimal germ of the Wick rotation, and the rotation is exactly the analytic continuation that the $i\varepsilon$ certifies as legal.

10.6.2 The Euclidean action of Scalar ϕ^4 theory

We now carry the rotation Eq. (10.153) through the action of the frozen ϕ^4 theory theory, tracking every sign so that the all-plus Euclidean form emerges with nothing assumed. The Minkowski action is $S = \int d^4x\,\mathcal{L}$ with the Lagrangian density Eq. (10.80),

$$S[\phi] \;=\; \int dt\,d^3x\,\Big[\tfrac{1}{2}(\partial_t\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\Big], \tag{10.156}$$

where we have split the kinetic term $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) = \frac{1}{2}(\partial_t\phi)^2 - \frac{1}{2}(\nabla\phi)^2$ into its temporal and spatial parts using the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ fixed for the book. The rotation acts on three things at once — the time derivative, the integration measure, and the overall factor of i in the weight e^{iS} — and the all-plus action is the residue of those three changes conspiring.

Take them in turn. First the time derivative: from $t = -i\tau$ we have $dt = -i\,d\tau$, hence

$$\partial_t \;=\; \frac{\partial\tau}{\partial t}\,\partial_\tau \;=\; \frac{1}{-i}\,\partial_\tau \;=\; i\,\partial_\tau, \qquad \text{so} \qquad (\partial_t\phi)^2 \;=\; (i\,\partial_\tau\phi)^2 \;=\; -(\partial_\tau\phi)^2. \tag{10.157}$$

The temporal kinetic term flips sign. Second the measure: the four-volume element becomes

$$dt\,d^3x \;=\; -i\,d\tau\,d^3x \;=\; -i\,d^4x_E, \qquad d^4x_E \equiv d\tau\,d^3x, \tag{10.158}$$

with $x_E = (\tau, \mathbf{x})$ the Euclidean coordinate. The spatial pieces $(\nabla\phi)^2$, $m^2\phi^2$, ϕ^4 are untouched by the rotation, since the spatial coordinates are unchanged. Substituting Eqs. (10.157) and (10.158) into Eq. (10.156),

$$\begin{aligned} S &= \int(-i\,d^4x_E)\,\Big[\tfrac{1}{2}\big(-(\partial_\tau\phi)^2\big) - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\Big] \\ &= -i\int d^4x_E\,\Big[-\tfrac{1}{2}(\partial_\tau\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\Big] \\ &= +i\int d^4x_E\,\Big[\tfrac{1}{2}(\partial_\tau\phi)^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\Big]. \end{aligned} \tag{10.159}$$

The two sign flips — one from the squared time derivative, one from pulling the overall minus out of the bracket — have turned every term positive. Defining the Euclidean action as the all-plus integral in the last line, $S = iS_E$, the weight transforms as

$$e^{iS} \longrightarrow e^{i(iS_E)} = e^{-S_E}, \tag{10.160}$$

the central exchange: the oscillatory phase becomes a real, decaying exponential. This is the mechanism announced generically in Eq. (5.95); here it is carried out for the specific quartic theory of the thread.

Definition 10.27 (Euclidean action of ϕ^4 theory)**.** The *Euclidean action* obtained from Eq. (10.156) by the Wick rotation Eq. (10.153) is

$$\boxed{S_E[\phi] = \int d^4x_E \left[\tfrac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\right],} \tag{10.161}$$

where the index μ now runs over the four Euclidean directions (τ, x^1, x^2, x^3) and is summed with the positive-definite metric $\delta_{\mu\nu} = \mathrm{diag}(+,+,+,+)$, so $(\partial_\mu\phi)(\partial_\mu\phi) = (\partial_\tau\phi)^2 + (\nabla\phi)^2 \geq 0$. The Euclidean Lagrangian density is $\mathcal{L}_E = \frac{1}{2}(\partial_\mu\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$.

This is exactly the Euclidean ϕ^4 action frozen for ϕ^4 theory and quoted at the chapter level in Eq. (5.96): all signs plus, the relative signs of kinetic, mass, and quartic terms now identical rather than alternating. It is the object the Wilsonian renormalization group of Chapter 23 integrates shells of momentum out of, the lattice of Chapter 43 discretizes cell by cell, and the conformal field theory of Chapter 52 flows to at its fixed point. The sign structure matters: because every term in Eq. (10.161) is non-negative (for $\lambda > 0$), the Euclidean action is bounded below by zero, and e^{-S_E} is bounded above by one. There is no longer any cancellation among phases to engineer; large field configurations are simply rare.

Notation Watch

In the Minkowski Lagrangian the relative sign between the kinetic and potential terms is negative, $\frac{1}{2}(\partial\phi)^2 - V$, an artifact of the indefinite metric. In the Euclidean action it is positive, $\frac{1}{2}(\partial\phi)^2 + V$, because the metric is now Euclidean and the time-derivative term has flipped. A reader translating a formula between the two conventions must flip exactly the temporal-kinetic relative sign; the spatial-gradient, mass, and interaction terms keep their Euclidean-positive signs. Failing to track this is the most common error in passing between S and S_E, and it produces a theory that is unbounded below.

10.6.3 The Euclidean generating functional and the convergence of its measure

With the action rotated, the generating functional follows. Applying the replacements Eqs. (10.158) and (10.160) to the Minkowski definition Eq. (10.86) — and continuing the source term, $i\int d^4x\, J\phi \to i(-i)\int d^4x_E\, J\phi = \int d^4x_E\, J\phi$, so that the source coupling also loses its i — gives the Euclidean object.

Definition 10.28 (Euclidean generating functional)**.** The *Euclidean generating functional* of the scalar field is

$$\boxed{Z_E[J] \;=\; \int \mathcal{D}\phi\, \exp\Big[-S_E[\phi] + \int d^4x_E\, J(x)\,\phi(x)\Big]\,,} \tag{10.162}$$

with S_E the action Eq. (10.161) and the functional measure $\mathcal{D}\phi$ the lattice limit of Theorem 10.16, now built on a four-dimensional *Euclidean* lattice. Its normalized functional derivatives generate the *Euclidean* (Schwinger) correlation functions,

$$G_E^{(n)}(x_1,\dots,x_n) \;=\; \frac{1}{Z_E[0]}\,\frac{\delta^n Z_E[J]}{\delta J(x_1)\cdots\delta J(x_n)}\Big|_{J=0} \;=\; \frac{\displaystyle\int \mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\,e^{-S_E}}{\displaystyle\int \mathcal{D}\phi\; e^{-S_E}}, \tag{10.163}$$

the analytic continuations to imaginary time of the Minkowski Green's functions $G^{(n)}$ of Theorem 10.18.

Note the structural simplification relative to the Minkowski case Eq. (10.93): the factors of $1/i^n$ that decorated the Minkowski correlators are gone, because the Euclidean exponent carries no i to bring down. Each functional derivative $\delta/\delta J(x)$ acting on $e^{\int J\phi}$ pulls down a bare $\phi(x)$, with no compensating $1/i$. Euclidean correlators are real, as befits expectation values against a real positive weight.

That last phrase is the heart of the matter, and we now make it precise. The single most important property of Eq. (10.162), the property that the Minkowski integral did not possess, is that the weight e^{-S_E} is a *convergent, positive* integrand: at finite lattice size the integral Eq. (10.162) is an ordinary, absolutely convergent multiple integral against a genuine probability density, with none of the conditional-convergence delicacy that forced the $i\varepsilon$ crutch in Minkowski signature. The reason is the positive-definiteness of the free quadratic form.

Proposition 10.29 (The free Euclidean weight is a convergent positive Gaussian)**.** *Let $S_E^{(0)}$ be the free ($\lambda = 0$) part of Eq. (10.161). After an integration by parts that discards the boundary term, it is the manifestly real quadratic form*

$$S_E^{(0)}[\phi] \;=\; \int d^4x_E\,\Big[\tfrac{1}{2}(\partial_\mu\phi)^2 + \tfrac{1}{2}m^2\phi^2\Big] \;=\; \tfrac{1}{2}\int d^4x_E\;\phi\,(-\partial_E^2 + m^2)\,\phi, \tag{10.164}$$

where $\partial_E^2 = \partial_\tau^2 + \nabla^2$ is the four-dimensional Euclidean Laplacian. The operator $K_E \equiv -\partial_E^2 + m^2$ is symmetric and strictly positive: its momentum-space eigenvalues are

$$K_E(p_E) \;=\; p_E^2 + m^2 \;\geq\; m^2 \;>\; 0, \qquad p_E^2 = (p_E^4)^2 + \mathbf{p}^2, \tag{10.165}$$

bounded below by the mass gap m^2. Hence $S_E^{(0)}[\phi] \geq \frac{1}{2}m^2 \int d^4x_E\, \phi^2 \geq 0$, vanishing only at $\phi \equiv 0$, and $e^{-S_E^{(0)}}$ is a bona fide Gaussian probability weight. The full weight $e^{-S_E} = e^{-S_E^{(0)}}\, e^{-\frac{\lambda}{4!}\int \phi^4}$ is the product of this Gaussian with a factor ≤ 1 for $\lambda \geq 0$, hence also bounded and integrable on each finite lattice.

Proof. The integration by parts rewriting $\int (\partial_\mu \phi)^2 = \int \phi(-\partial_E^2)\phi$ uses the vanishing of ϕ at the boundary of the (large, periodic) Euclidean box, the same finite-lattice statement that justified the corresponding Minkowski manipulation in Section 10.3.4. Passing to momentum space with the (now real-exponent) Euclidean Fourier transform $\phi(x) = \int \frac{d^4p_E}{(2\pi)^4} e^{ip_E\cdot x}\tilde{\phi}(p_E)$, the operator $-\partial_E^2$ acts as multiplication by $p_E^2 = (p_E^4)^2 + \mathbf{p}^2 \geq 0$, giving the eigenvalues Eq. (10.165); the smallest is m^2, attained at $p_E = 0$. Therefore the quadratic form is positive-definite, $S_E^{(0)}[\phi] = \frac{1}{2}\int \frac{d^4p_E}{(2\pi)^4}(p_E^2 + m^2)|\tilde{\phi}(p_E)|^2 \geq \frac{1}{2}m^2\|\phi\|_2^2$. On a finite lattice the field is a finite vector $\{\phi_{j,\mathbf{n}}\}$ and $e^{-S_E^{(0)}}$ is $\exp(-\frac{1}{2}\phi^{\mathsf{T}} M \phi)$ with M a real symmetric positive-definite matrix (the lattice discretization of K_E, whose eigenvalues are positive for the same reason), so the integral $\int \prod d\phi_{j,\mathbf{n}}\, e^{-\frac{1}{2}\phi^{\mathsf{T}} M\phi} = (2\pi)^{N/2}(\det M)^{-1/2}$ is finite and positive by the multidimensional Gaussian formula Theorem 4.14. For $\lambda \geq 0$ the quartic factor $e^{-\frac{\lambda}{4!}\int \phi^4} \in (0, 1]$ only improves convergence. The continuum limit of the ratio defining correlators is then the limit of well-defined finite integrals. □

This is the decisive structural gain. In Minkowski signature the operator $K = -\Box - m^2$ in the exponent of $Z[J]$ has eigenvalues $p^2 - m^2$ that *change sign and vanish on the mass shell* (Eq. (10.90)); the integral is oscillatory, the operator is not invertible without a prescription, and the $i\varepsilon$ was the prescription. In Euclidean signature the operator $K_E = -\partial_E^2 + m^2$ has eigenvalues $p_E^2 + m^2$ that are *strictly positive and never vanish*; the integral is convergent, the operator is invertible outright, and no prescription is needed. The Euclidean theory is the place where the path integral is, for the free theory, a completely rigorous object — a Gaussian measure on the space of (tempered-distribution) field configurations, the construction of which is carried out in Chapter E. It is precisely this convergence that the lattice of Chapter 43 exploits to give a non-perturbative *definition* of the theory, and that statistical mechanics borrows wholesale.

Physical Insight

The Minkowski path integral and the Euclidean path integral are the same Gaussian read in two metrics, and the metric decides everything about convergence. With an indefinite metric the quadratic form $\phi K \phi$ is indefinite, the weight is a phase, and the integral survives only by the conditional cancellation of oscillations. With a definite metric the quadratic form is positive, the weight is a Boltzmann factor, and the integral converges absolutely. Wick rotation is the bridge between the two, and the mass gap $m^2 > 0$ is the spectral reason the Euclidean side is so much better behaved: it is the gap between the smallest eigenvalue of K_E and zero.

10.6.4 The Euclidean propagator and its continuation to D_F

Because the free Euclidean weight is a convergent Gaussian, its generating functional is evaluated by completing the square exactly as in the Minkowski case (Section 10.4), but now with no $i\varepsilon$ to carry and the inverse of K_E unambiguous. Shifting $\phi \to \phi + K_E^{-1} J$ in Eq. (10.162) with $\lambda = 0$ and using Eq. (10.164) gives

$$Z_E^{(0)}[J] \;=\; Z_E^{(0)}[0]\, \exp\Big[\tfrac{1}{2}\int d^4x_E\, d^4y_E\; J(x)\, G_E(x-y)\, J(y)\Big], \tag{10.166}$$

where G_E is the inverse of the Euclidean kinetic operator, the unique solution of

$$(-\partial_E^2 + m^2)\, G_E(x_E) \;=\; \delta^4(x_E). \tag{10.167}$$

The sign of the exponent in Eq. (10.166) is positive, in contrast to the negative sign of the Minkowski Eq. (10.152), because the Euclidean source coupling carries no i and the completed square produces $+\frac{1}{2} J K_E^{-1} J$. Inverting Eq. (10.167) in momentum space — multiplication by $p_E^2 + m^2$ becomes division — and Fourier transforming back yields the Euclidean propagator in closed form.

Definition 10.30 (Euclidean propagator)**.** The *Euclidean (Schwinger) propagator* of the free scalar is

$$\boxed{G_E(x_E) \;=\; \int \frac{d^4p_E}{(2\pi)^4}\, \frac{e^{\,ip_E\cdot x_E}}{p_E^2 + m^2}\,,} \qquad p_E \cdot x_E = p_E^4 \tau + \mathbf{p}\cdot\mathbf{x}, \tag{10.168}$$

the two-point Schwinger function $G_E^{(2)}(x, 0)$ of Eq. (10.163). Its denominator $p_E^2 + m^2 \geq m^2 > 0$ never vanishes for real Euclidean momentum, so the integral requires no $i\varepsilon$ prescription and defines a real, smooth function of x_E away from the origin.

The contrast with the Feynman propagator Eq. (10.154) is the whole story of this section in one line. The Minkowski denominator $p^2 - m^2$ vanishes on the mass shell and needed the $i\varepsilon$ to be defined; the Euclidean denominator $p_E^2 + m^2$ is bounded away from zero and needs nothing. The two are connected by the rotation, as we now verify directly, both in momentum and in position space.

Momentum-space continuation. Start from the position-space Feynman propagator and substitute the rotation. The Feynman propagator is

$$D_F(x) \;=\; \int \frac{d^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot x}}{p^2 - m^2 + i\varepsilon}, \qquad p\cdot x = p^0x^0 - \mathbf{p}\cdot\mathbf{x}, \tag{10.169}$$

with the book's Fourier convention. Rotate both the external time and the energy integration, $x^0 = -i\tau$ and $p^0 = ip_E^4$ (Eq. (5.93)), the latter legitimate by Theorem 10.26. Three things continue. The measure: $dp^0 = i\,dp_E^4$, so $d^4p = i\,d^4p_E$. The denominator: by Eq. (5.94), $p^2 = (p^0)^2 - \mathbf{p}^2 \to -(p_E^2)$, so $p^2 - m^2 + i\varepsilon \to -(p_E^2 + m^2)$, the $i\varepsilon$ now redundant since the rotated denominator is strictly negative. The phase: with $p^0 = ip_E^4$ and $x^0 = -i\tau$,

$$p\cdot x \;=\; p^0x^0 - \mathbf{p}\cdot\mathbf{x} \;=\; (ip_E^4)(-i\tau) - \mathbf{p}\cdot\mathbf{x} \;=\; p_E^4\tau - \mathbf{p}\cdot\mathbf{x}, \qquad \text{so} \qquad e^{-ip\cdot x} = e^{-ip_E^4\tau}\,e^{\,i\mathbf{p}\cdot\mathbf{x}}. \tag{10.170}$$

Assembling the three,

$$D_F(x)\big|_{x^0=-i\tau} \;=\; \int \frac{i\,d^4p_E}{(2\pi)^4}\,\frac{i\,e^{-ip_E^4\tau}\,e^{i\mathbf{p}\cdot\mathbf{x}}}{-(p_E^2+m^2)} \;=\; \int \frac{d^4p_E}{(2\pi)^4}\,\frac{e^{-ip_E^4\tau}\,e^{i\mathbf{p}\cdot\mathbf{x}}}{p_E^2+m^2}, \tag{10.171}$$

where the two factors of i in the numerator combine with the -1 in the denominator, $i\cdot i/(-1) = (-1)/(-1) = +1$, to leave a clean real-weighted integral. Finally, since the integrand denominator is even in p_E^4 and the measure is symmetric, relabel the dummy variable $p_E^4 \to -p_E^4$ to convert $e^{-ip_E^4\tau}$ into $e^{+ip_E^4\tau}$, whereupon the phase becomes $e^{i(p_E^4\tau + \mathbf{p}\cdot\mathbf{x})} = e^{ip_E\cdot x_E}$ and

$$\boxed{D_F(x)\big|_{x^0=-i\tau} \;=\; \int \frac{d^4p_E}{(2\pi)^4}\,\frac{e^{\,ip_E\cdot x_E}}{p_E^2+m^2} \;=\; G_E(x_E)\,.} \tag{10.172}$$

The Feynman propagator, analytically continued to imaginary time $t = -i\tau$, *is* the Euclidean propagator. The relation runs both ways: G_E continued back along $\tau = it$ recovers D_F, and the $i\varepsilon$ of the Minkowski denominator is the instruction for *which* continuation — which side of the mass-shell singularity to approach — when the rotation is undone. The Euclidean object, free of singularities, is in this sense the more fundamental one; the Feynman propagator is its boundary value as the contour returns to real time, and the $i\varepsilon$ records the direction of approach.

Worked Example 10.31 (The Euclidean propagator in position space: convergence made manifest)**.** We evaluate Eq. (10.168) in closed form, both to exhibit

a genuinely convergent four-dimensional integral and to display the propagator's exponential decay — the Euclidean shadow of the mass gap. Introduce the Schwinger proper-time representation of the denominator, valid because $p_E^2 + m^2 > 0$,

$$\frac{1}{p_E^2 + m^2} \;=\; \int_0^\infty ds\, e^{-s(p_E^2+m^2)}, \tag{10.173}$$

an ordinary convergent integral with no prescription needed; this is itself a benefit of the positive denominator. Substituting into Eq. (10.168) and exchanging orders,

$$G_E(x_E) \;=\; \int_0^\infty ds\, e^{-sm^2} \int \frac{d^4p_E}{(2\pi)^4}\, e^{-sp_E^2 + ip_E\cdot x_E}. \tag{10.174}$$

The momentum integral is a product of four ordinary Gaussians (one per Euclidean direction), each evaluated by completing the square, $\int \frac{dp}{2\pi} e^{-sp^2+ipx} = \frac{1}{\sqrt{4\pi s}}\, e^{-x^2/4s}$; the four-dimensional version is

$$\int \frac{d^4p_E}{(2\pi)^4}\, e^{-sp_E^2 + ip_E\cdot x_E} \;=\; \frac{1}{(4\pi s)^2}\, e^{-x_E^2/4s}, \qquad x_E^2 = \tau^2 + \mathbf{x}^2 \equiv r^2, \tag{10.175}$$

manifestly finite for every $s > 0$ because the Euclidean exponent $-sp_E^2$ is real and negative. Hence

$$G_E(x_E) \;=\; \frac{1}{16\pi^2} \int_0^\infty \frac{ds}{s^2}\, \exp\Big[- sm^2 - \frac{r^2}{4s}\Big], \tag{10.176}$$

a positive integrand integrated over positive s: the result is positive and finite for $r > 0$, with the convergence at large s guaranteed by the mass term and at small s by the $r^2/4s$ term. The remaining integral is the standard integral representation of the modified Bessel function K_ν, namely $\int_0^\infty ds\, s^{-\nu-1} e^{-as-b/s} = 2(a/b)^{\nu/2} K_\nu(2\sqrt{ab})$; with $\nu = 1$, $a = m^2$, $b = r^2/4$, so that $2\sqrt{ab} = mr$ and $(a/b)^{1/2} = 2m/r$,

$$G_E(x_E) \;=\; \frac{1}{16\pi^2} \cdot 2 \cdot \frac{2m}{r}\, K_1(mr) \;=\; \frac{m}{4\pi^2\, r}\, K_1(mr), \qquad r = \sqrt{\tau^2 + \mathbf{x}^2}. \tag{10.177}$$

This is a real, positive, smooth function of the Euclidean radius r, exactly as a convergent Gaussian integral against a positive weight must produce. Its long-distance behavior, from the asymptotics $K_1(z) \sim \sqrt{\pi/2z}\, e^{-z}$ for $z \to \infty$, is

$$G_E(x_E) \;\xrightarrow{mr \gg 1}\; \frac{m}{4\pi^2 r} \sqrt{\frac{\pi}{2mr}}\, e^{-mr} \;=\; \frac{\sqrt{m}}{4\sqrt{2}\,\pi^{3/2}}\, \frac{e^{-mr}}{r^{3/2}}, \tag{10.178}$$

an exponential decay with correlation length $\xi = 1/m$. The mass gap of the quantum field theory has reappeared as the inverse correlation length of the Euclidean two-point function — the first entry of the dictionary developed next. Its short-distance behavior, from $K_1(z) \sim 1/z$ as $z \to 0$, is $G_E \to 1/(4\pi^2 r^2)$, the massless Euclidean Coulomb-like singularity of the four-dimensional Laplacian, recovering the $\lambda = m = 0$ free-field result and confirming that Eq. (10.177) solves Eq. (10.167) with the correct unit-strength point source.

Euclidean quantum field theory	**Classical statistical mechanics**
Field configuration $\phi(x_E)$	Microstate / configuration $\{\sigma_i\}$
Euclidean action $S_E[\phi]$	Reduced energy $\beta H[\sigma]$
Functional measure $\mathcal{D}\phi$	Sum over configurations $\prod_i d\sigma_i$
Partition function $Z_E[0] = \int \mathcal{D}\phi\, e^{-S_E}$	Partition function $Z = \sum e^{-\beta H}$
Source $J(x)$	External field (e.g. magnetic h)
Schwinger functions $G_E^{(n)}$	Thermal correlators $\langle \sigma \cdots \sigma \rangle$
Mass gap m	Inverse correlation length $1/\xi$
Massless / critical theory	Critical point ($\xi \to \infty$)
Wilsonian RG flow (Chapter 23)	Kadanoff block-spin / RG
Lattice regularization (Chapter 43)	Lattice model (Ising, etc.)

Table 10.1: The quantum-field-theory / statistical-mechanics dictionary, read off from the identity of the Euclidean generating functional Eq. (10.162) with a classical partition function. Each line is an exact correspondence between objects, not an analogy; it is the reason the renormalization group, born in statistical physics, governs quantum field theory, and the reason the lattice computes both.

10.6.5 The dictionary: quantum field theory as statistical mechanics

The form of the Euclidean generating functional Eq. (10.162) invites a reading that has organized half of modern theoretical physics. Written out,

$$Z_E[0] \;=\; \int \mathcal{D}\phi\; e^{-S_E[\phi]}, \tag{10.179}$$

it is structurally identical to the partition function of classical statistical mechanics, $Z = \int \prod_i d\sigma_i\, e^{-\beta H[\sigma]}$, the sum over configurations $\{\sigma_i\}$ of a system with energy functional H at inverse temperature β, weighted by the Boltzmann factor. The dictionary is exact and term-by-term: the field configuration $\phi(x_E)$ is the configuration of a classical statistical system in four spatial dimensions; the Euclidean action $S_E[\phi]$ plays the role of the dimensionless energy $\beta H[\phi]$; the functional integral $\int \mathcal{D}\phi$ is the sum over configurations; $Z_E[0]$ is the partition function; and the Schwinger correlation functions Eq. (10.163) are the thermal correlation functions of the statistical system. Quantum fluctuations of the $(3+1)$-dimensional quantum field theory are thermal fluctuations of the four-dimensional classical statistical system, with $\hbar$ in the role of temperature.

The dictionary, collected in Table 10.1, is more than a formal curiosity; it is a two-way street that the rest of the book travels repeatedly. The entry that

anchors the others is the identification of the mass gap with the inverse correlation length, which the worked example above has already demonstrated: the Euclidean two-point function $G_E \sim e^{-mr}/r^{3/2}$ decays with correlation length $\xi = 1/m$, exactly the statement that the connected correlations of a statistical system fall off over the length ξ. A massless quantum field theory is therefore a statistical system at a *critical point*, where $\xi \to \infty$ and correlations become scale-free — the regime where conformal symmetry emerges and the bootstrap of Chapter 52 operates. The renormalization group, which we introduce as a Wilsonian flow in Chapter 23, was born in this statistical setting as Kadanoff's block-spin transformation and Wilson's integration over short-distance fluctuations; the dictionary is what licenses importing it wholesale into quantum field theory. And the lattice of Chapter 43 is the dictionary read in reverse: a quantum field theory is *defined* non-perturbatively by writing down the corresponding lattice statistical model and taking the continuum limit at its critical point, with the convergent positive measure of Theorem 10.29 making the integral amenable to the same Monte-Carlo sampling that statistical mechanics uses.

Physical Insight

There is a sharper version of the dictionary that fixes the role of temperature. If, instead of taking the Euclidean time τ to run over all of $\mathbb{R}$, one compactifies it onto a circle of circumference β with periodic boundary conditions on the field, $\phi(\tau + \beta, \mathbf{x}) = \phi(\tau, \mathbf{x})$, the Euclidean path integral computes the quantum-statistical trace $Z(\beta) = \mathrm{Tr}\, e^{-\beta H}$ — the thermal partition function of the quantum field theory at physical temperature $T = 1/\beta$. The infinite-τ vacuum theory of this section is the zero-temperature limit $\beta \to \infty$. Thus the same rotated integral describes both vacuum correlators (line $\tau \in \mathbb{R}$) and thermal equilibrium (circle of circumference β); the imaginary-time formalism of finite-temperature field theory is nothing but the Euclidean path integral on a cylinder. We will not develop thermal field theory here, but the reader should register that the Wick rotation has, for free, delivered the bridge to it.

10.6.6 Reflection positivity and the return to Lorentzian operators

The Wick rotation as we have used it runs in one direction: from a Lorentzian quantum theory, defined by canonical quantization in Chapter 9 and reproduced by the Minkowski path integral, we passed to a convergent Euclidean integral. For the Euclidean formulation to stand on its own — to be, as the lattice and constructive programs intend, a *definition* of quantum field theory rather than a derived shadow of a pre-existing Lorentzian one — we need the converse: a criterion guaranteeing that a given Euclidean theory, specified by its Schwinger

functions $G_E^{(n)}$, continues back to a sensible Lorentzian theory with a Hilbert space of positive norm and a self-adjoint Hamiltonian bounded below. That criterion is reflection (Osterwalder–Schrader) positivity. We state it precisely and explain its content; its proof in full generality belongs to the constructive program and is beyond the scope of this chapter, but the statement is exactly what the lattice (Chapter 43) and conformal (Chapter 52) chapters invoke.

Let Θ denote reflection of the Euclidean time about the hyperplane $\tau = 0$, acting on configurations by $(\Theta\phi)(\tau, \mathbf{x}) = \phi(-\tau, \mathbf{x})$ and on functionals of the field accordingly. Call a functional $F[\phi]$ *supported at positive time* if it depends on the field only through its values at $\tau > 0$.

Theorem 10.32 (Reflection positivity and reconstruction; stated)**.** *A set of Euclidean Schwinger functions arising from a measure $e^{-S_E}\mathcal{D}\phi$ that is reflection positive — meaning that for every functional F supported at positive Euclidean time*

$$\big\langle\, (\Theta F)^{*}\, F \,\big\rangle_E \;=\; \frac{1}{Z_E[0]} \int \mathcal{D}\phi\; (\Theta F[\phi])^{*}\, F[\phi]\, e^{-S_E[\phi]} \;\geq\; 0 \tag{10.180}$$

holds — together with Euclidean invariance and the appropriate regularity (the Osterwalder–Schrader axioms), determines a unique Lorentzian quantum field theory. Its Hilbert space $\mathcal{H}$ is built from positive-time functionals with inner product Eq. (10.180), its vacuum is the constant functional, and its Hamiltonian $H \geq 0$ is the generator of Euclidean time translations continued to real time; the Lorentzian Wightman functions are the analytic continuations of the $G_E^{(n)}$ back to real time, $\tau \to it$.

The structural meaning of Eq. (10.180) is worth unpacking, because it is exactly the property that the indefinite-metric Minkowski integral cannot guarantee on its own. The combination $(\Theta F)^{*}F$ pairs a positive-time functional with its time-reflected conjugate; the Euclidean expectation of this combination is, after reconstruction, the squared norm of the state F creates from the vacuum. Demanding it be non-negative for *every* positive-time F is demanding that the reconstructed Hilbert space have a positive-definite inner product — that there be no negative-norm "ghost" states. The reflection Θ is the Euclidean avatar of Hermitian conjugation: under the continuation $\tau \to it$, reflecting Euclidean time about $\tau = 0$ becomes complex-conjugating a Lorentzian amplitude, so Eq. (10.180) is the Euclidean transcription of the statement $\langle\psi|\psi\rangle \geq 0$. A Euclidean measure that satisfies it is one whose statistical correlations encode a unitary quantum theory; one that violates it (and there are perfectly sensible-looking Euclidean actions that do) corresponds to no Lorentzian quantum mechanics at all.

Caution

Not every positive Euclidean weight e^{-S_E} defines a Lorentzian quantum field theory. Positivity of the *measure* — guaranteed here by Theorem 10.29 — makes Z_E a convergent statistical partition function, but reflection positivity Eq. (10.180) is the strictly stronger, separate requirement that the reconstructed Hilbert space have no negative-norm states. The free scalar and the lattice ϕ^4 theory satisfy it; higher-derivative actions with improved ultraviolet behavior typically do not, which is why they fail to define unitary theories despite having perfectly convergent Euclidean integrals. The two notions of positivity must not be conflated.

Theorem Theorem 10.32 closes the logical circle of the chapter. Canonical quantization (Chapter 9) gave a Lorentzian operator theory; the Minkowski path integral reproduced its time-ordered correlators (Theorem 10.19); the Wick rotation carried those to convergent Euclidean Schwinger functions; and reflection positivity is the guarantee that the Euclidean data can be carried *back* to Lorentzian operators with a positive-norm Hilbert space and a stable Hamiltonian. The Euclidean theory is thus not a poor relation of the Lorentzian one but a complete and self-standing formulation, equivalent to it precisely when Eq. (10.180) holds.

With the Euclidean action S_E of Eq. (10.161) fixed for ϕ^4 theory, the convergent positive measure $e^{-S_E}\mathcal{D}\phi$ established, the Euclidean propagator G_E constructed and tied to D_F by analytic continuation, and the statistical-mechanics dictionary and reflection-positivity criterion in hand, we have assembled the form of the path integral that the renormalization group will flow, the lattice will discretize, and conformal field theory will inhabit. It remains to put the machinery of this chapter to work on the interacting theory: in the final section we expand $e^{-\frac{\lambda}{4!}\int \phi^4}$ perturbatively inside the generating functional, organize the resulting correlators into connected and effective generating functionals, and hand the assembled apparatus forward to the S-matrix and the Feynman rules.

10.7 The Interacting Generating Functional and the Road Ahead

The free machinery is now complete and exact. We have the free generating functional $Z_0[J]$ in closed form (Section 10.4), its single nontrivial datum the Feynman propagator D_F recovered a second way from a Gaussian integral, and Wick's theorem (Theorem 10.23) telling us that every free correlator is a sum over pairings into propagators. All of this lived at $\lambda = 0$: the exponent of Eq. (10.86) was quadratic and the integral was Gaussian. The frozen ϕ^4 Lagrangian, however, carries a quartic term, and the moment we restore it the integral is no longer

one we can do. What we *can* do — and what organizes the entire remainder of perturbative field theory — is express the full $Z[J]$ as a differential operator built from the interaction acting on the free $Z_0[J]$ we already possess. That single formula turns every interacting correlator into a finite collection of free correlators, i.e. into propagators and combinatorics; it is the algebraic seed of the Feynman diagram. This closing section plants that seed: it sets up the interacting $Z[J]$ perturbatively, isolates the connected part through the functional $W[J]$, and points to the effective action Γ, handing the assembled apparatus to the chapters that will grow it into the S-matrix, the Feynman rules, and the renormalization group — without doing their work here.

10.7.1 Splitting the action: the master formula

The starting point is the trivial observation that the action of the frozen theory splits into a free quadratic piece and an interaction,

$$S[\phi] \;=\; S_0[\phi]+S_{\rm int}[\phi], \qquad S_0[\phi]=\int d^4x\,\Big[\tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)-\tfrac{1}{2}m^2\phi^2\Big], \quad S_{\rm int}[\phi]=-\int d^4x\,\frac{\lambda}{4!}\,\phi(\tag{10.181}$$

We write $\mathcal{L}_{\rm int}[\phi]=-\frac{\lambda}{4!}\phi^4$ for the interaction density, so $S_{\rm int}=\int d^4x\,\mathcal{L}_{\rm int}$. Inserting Eq. (10.181) into the generating functional Eq. (10.86) factorizes the exponential,

$$Z[J] \;=\; \int\!\mathcal{D}\phi\; e^{\,iS_{\rm int}[\phi]}\; e^{\,i\left(S_0[\phi]+\int J\phi\right)}. \tag{10.182}$$

The second factor is exactly the free integrand whose integral is $Z_0[J]$; the first factor, $e^{iS_{\rm int}}$, is the obstruction. The trick that removes the obstruction is to notice that inside the integral every power of the field can be manufactured by differentiating the free integrand with respect to the source. From the elementary identity $\frac{1}{i}\frac{\delta}{\delta J(x)}\,e^{\,i\int J\phi}=\phi(x)\,e^{\,i\int J\phi}$, established in Eq. (10.91), it follows that for *any* point x,

$$\frac{1}{i}\frac{\delta}{\delta J(x)}\,\Big[\;\cdots\;e^{\,i(S_0+\int J\phi)}\Big] \;=\; \Big[\;\cdots\;\phi(x)\;e^{\,i(S_0+\int J\phi)}\Big] \tag{10.183}$$

whenever the omitted factors $\cdots$ do not depend on J — in particular when they are themselves powers of ϕ, which are J-independent c-number functions under the integral sign. Because the functional derivatives $\delta/\delta J(x)$ at distinct points commute, and because they act on ϕ-factors as the identity (those factors carry no J), *any* functional $F[\phi]$ that is a power series in the field satisfies the operator replacement rule

$$F\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\;e^{\,i\int J\phi} \;=\; F[\phi]\,e^{\,i\int J\phi}, \tag{10.184}$$

read as: replace each field in $F[\phi]$ by the source-derivative $\frac{1}{i}\frac{\delta}{\delta J}$, let the resulting differential operator act on $e^{\,i\int J\phi}$, and recover $F[\phi]$ times the same exponential.

Applying this with $F[\phi] = e^{iS_{\text{int}}[\phi]}$ converts the troublesome factor in Eq. (10.182) into a differential operator that no longer references ϕ at all,

$$e^{\,iS_{\text{int}}[\phi]}\, e^{\,i(S_0+\int J\phi)} \;=\; \exp\Big(iS_{\text{int}}\big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\big]\Big)\, e^{\,i(S_0+\int J\phi)}. \tag{10.185}$$

The operator $\exp(iS_{\text{int}}[\frac{1}{i}\frac{\delta}{\delta J}])$ depends only on J, not on ϕ, so it commutes with the field integration $\int\mathcal{D}\phi$ and may be pulled out in front of it. The integral that remains is precisely $Z_0[J]$. This proves the central structural identity of perturbative field theory.

Theorem 10.33 (The interacting generating functional as a differential operator)**.** *For the split Eq. (10.181), the full generating functional is obtained from the free one by acting with the interaction, the fields in the interaction replaced by source-derivatives:*

$$\boxed{Z[J] \;=\; \exp\Big(i\int d^4x\; \mathcal{L}_{\text{int}}\big[\tfrac{1}{i}\tfrac{\delta}{\delta J(x)}\big]\Big)\, Z_0[J] \;=\; \exp\Big(-\,\frac{i\lambda}{4!}\int d^4z\,\frac{\delta^4}{\delta J(z)^4}\Big)\, Z_0[J]\,,} \tag{10.186}$$

where in the second form we used $\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4$ *and* $\big(\frac{1}{i}\big)^4 = 1$*, and*

$$Z_0[J] \;=\; Z_0[0]\,\exp\Big[-\tfrac{1}{2}\int d^4x\, d^4y\; J(x)\, D_F(x-y)\, J(y)\Big] \tag{10.187}$$

is the free generating functional of Eq. (10.111), with D_F *the Feynman propagator of Eq. (9.154).*

Proof. Combine Eq. (10.182), Eq. (10.185), and the commutation of the J-only operator with $\int\mathcal{D}\phi$:

$$\begin{aligned} Z[J] &= \int\mathcal{D}\phi\;\exp\Big(iS_{\text{int}}\big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\big]\Big)\, e^{\,i(S_0+\int J\phi)} \\ &= \exp\Big(iS_{\text{int}}\big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\big]\Big)\int\mathcal{D}\phi\; e^{\,i(S_0+\int J\phi)} = \exp\Big(iS_{\text{int}}\big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\big]\Big)\, Z_0[J]. \end{aligned} \tag{10.188}$$

The only non-elementary step is the second equality, the interchange of the differential operator with the field integral. On any finite spacetime lattice (Theorem 10.16) this is rigorous: $Z[J]$ is then an ordinary finite-dimensional integral, $J(z)$ is a finite collection of integration-independent parameters, and differentiating under the integral sign with respect to a parameter is legitimate because the integrand and its J-derivatives are absolutely integrable thanks to the $i\varepsilon$ Gaussian damping Eq. (10.89). The exponential of the differential operator is then read as its power series, applied term by term; the continuum limit is taken at the end. The expression is therefore exact as a formal power series in λ: expanding the exponential generates, at each order, finitely many source-derivatives of the explicitly known $Z_0[J]$. □

The formula Eq. (10.186) is the whole of perturbation theory in embryo. The free generating functional $Z_0[J]$ is known in closed form; the interaction has been demoted from an obstruction inside the integral to a differential operator outside it; and every interacting Green's function of Theorem 10.18 is now computable, to any finite order in λ, by differentiating $Z_0[J]$ a finite number of times. No integral remains to be done that we cannot do. The price is that "any finite order in λ" is the best the formula offers: the series it generates is, for ϕ^4, only asymptotic, a point we revisit when its Borel structure is analyzed in Chapter 23.

Notation Watch

The replacement rule Eq. (10.184) carries one factor of $1/i$ per field because the source couples as $e^{+i\int J\phi}$ in the Minkowski generating functional Eq. (10.86). For the quartic vertex the four factors combine to $(1/i)^4 = 1$, so the operator in Eq. (10.186) is simply $-\frac{i\lambda}{4!}\int_z \delta^4/\delta J^4$; for an odd vertex (e.g. a ϕ^3 interaction) the surviving factor of $1/i$ would not cancel and must be tracked. The Euclidean version replaces e^{iS} by e^{-S_E} (Section 10.6) and the rule becomes $\frac{\delta}{\delta J} e^{\int J\phi} = \phi\, e^{\int J\phi}$ with no factors of i at all.

10.7.2 Perturbation theory and the birth of contractions

To see the diagrammatic content emerge, expand the operator exponential in Eq. (10.186) in powers of the coupling,

$$Z[J] \;=\; \Big[\,1 \;-\; \frac{i\lambda}{4!}\int d^4z\,\frac{\delta^4}{\delta J(z)^4} \;+\; O(\lambda^2)\,\Big]\,Z_0[J]. \tag{10.189}$$

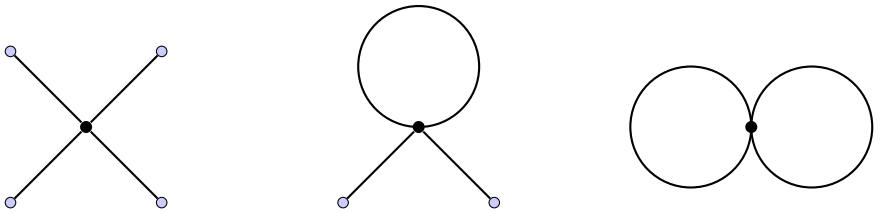
Each order is a definite number of source-derivatives acting on the Gaussian $Z_0[J]$, and the act of differentiating an exponential of a quadratic is exactly what produces propagators. Write $Z_0[J] = Z_0[0]\,e^{W_0[J]}$ with the quadratic exponent

$$W_0[J] \;=\; -\tfrac{1}{2}\int d^4x\,d^4y\;J(x)\,D_F(x-y)\,J(y), \qquad \frac{\delta W_0}{\delta J(z)} \;=\; -\,\phi_J(z), \qquad \frac{\delta^2 W_0}{\delta J(z)^2} \;=\; -\,D_F(0), \tag{10.190}$$

where we have introduced the *source-induced field*

$$\phi_J(z) \;\equiv\; \int d^4y\;D_F(z-y)\,J(y), \tag{10.191}$$

the propagator smeared against the source (this $\phi_J = D_F * J$ is the source-smeared object and differs by a factor i from the classical field of Eq. (10.104), $\phi_J^{\rm cl} = -\mathcal{O}^{-1}J = i\,D_F * J$, since $(\Box + m^2)^{-1} = i\,D_F$), and where $D_F(0)$ denotes the propagator at coincident points (a single closed loop, formally divergent — its taming is the business of Chapters 21 and 22, not here). Because W_0 is quadratic, all derivatives beyond the second vanish, and the fourth source-derivative of e^{W_0}



(a) $\phi_J(z)^4$
four legs to J

(b) $\phi_J(z)^2\, D_F(0)$
two legs, one loop

(c) $D_F(0)^2$
vacuum bubble

Figure 10.7: The three $O(\lambda)$ topologies generated by the four source-derivatives in Eq. (10.193). A single quartic vertex (black dot) at z has four legs; each leg either runs to a source (open circle, a factor $\phi_J(z) = \int D_F(z-y)J(y)$) or closes back on the vertex (a loop, a factor $D_F(0)$). Diagram (a) is the seed of the four-point amplitude; (b) is the one-loop correction to the propagator (a "tadpole"); (c) is a disconnected vacuum bubble. The combinatorial weights $1, 6, 3$ count the pairings; their reduction to symmetry factors is carried out in Chapter 15.

is given by the standard exponential-of-a-function expansion, in which only the first and second derivatives of W_0 appear:

$$\frac{\delta^4}{\delta J(z)^4}\, e^{W_0} \;=\; e^{W_0}\Big[(W_0')^4 + 6\,(W_0')^2\, W_0'' + 3\,(W_0'')^2\Big], \tag{10.192}$$

all derivatives evaluated at the single point z. Substituting Eq. (10.190), the bracket becomes a sum of three terms,

$$\frac{\delta^4}{\delta J(z)^4}\, Z_0[J] \;=\; Z_0[J]\Big[\underbrace{\phi_J(z)^4}_{\text{four legs to sources}} - 6\,\underbrace{\phi_J(z)^2\, D_F(0)}_{\text{two legs, one loop}} + 3\,\underbrace{D_F(0)^2}_{\text{vacuum loop}}\Big], \tag{10.193}$$

and the $O(\lambda)$ contribution to $Z[J]$ is $-\frac{i\lambda}{4!}\int d^4z$ of this. The three terms are the three ways of distributing the four legs of a single vertex: all four attached to sources, two attached and two contracted with each other into a loop, or all four contracted in pairs into two loops with no source at all. *This is the origin of Feynman diagrams.* Each $\phi_J(z)$ is a propagator running from the vertex at z out to a source; each factor of $D_F(0)$ is a propagator that begins and ends at the vertex, a loop; and the integer coefficients $1, 6, 3$ are the numbers of distinct pairings of the four identical derivatives, the seeds of the symmetry factors developed systematically in Chapter 15. Figure 10.7 draws the three topologies.

The piece that advances ϕ^4 theory to the threshold of its first amplitude is diagram (a), the term $\phi_J(z)^4$ with all four legs attached to sources. Stripping the four sources by functional differentiation exposes the bare vertex.

Worked Example 10.34 (The $O(\lambda)$ four-point function and the vertex $-i\lambda$)**.** We extract the connected four-point Green's function $G_c^{(4)}$ at first order in λ and read off the vertex factor that Chapter 14 will turn into a scattering amplitude.

The connected part of $G^{(4)}$ comes entirely from the all-legs-to-sources term $\phi_J(z)^4$ in Eq. (10.193): the other two terms either carry too few sources (the vacuum bubble has none) or factorize off a disconnected loop (the tadpole). Using Theorem 10.18, $G^{(4)} = i^{-4}\,\delta^4(Z[J]/Z[0])/\delta J^4|_{J=0}$ with $i^{-4}=1$, and keeping only the term in which all four derivatives land on the four induced fields (this is what "connected" selects — any derivative landing on the residual e^{W_0} would attach an extra propagator between external points, a disconnected piece),

$$G_c^{(4)}\big|_{O(\lambda)}(x_1,x_2,x_3,x_4) \;=\; -\frac{i\lambda}{4!}\int d^4z\;\frac{\delta^4}{\delta J(x_1)\,\delta J(x_2)\,\delta J(x_3)\,\delta J(x_4)}\,\phi_J(z)^4\Big|_{J=0}. \tag{10.194}$$

Now $\phi_J(z)^4 = \big[\int d^4y\, D_F(z-y)J(y)\big]^4$, and each $\delta/\delta J(x_i)$ removes one factor of J, leaving a propagator $D_F(z-x_i)$; there are exactly 4! ways to assign the four distinct derivatives to the four identical factors, so

$$\frac{\delta^4}{\delta J(x_1)\cdots\delta J(x_4)}\,\phi_J(z)^4\Big|_{J=0} \;=\; 4!\,\prod_{i=1}^{4} D_F(x_i-z). \tag{10.195}$$

The combinatorial 4! cancels the symmetry factor $1/4!$ of the vertex exactly, and we obtain the clean position-space result

$$G_c^{(4)}\big|_{O(\lambda)}(x_1,x_2,x_3,x_4) \;=\; -i\lambda\int d^4z\; D_F(x_1-z)\,D_F(x_2-z)\,D_F(x_3-z)\,D_F(x_4-z). \tag{10.196}$$

Each external point is joined to a common vertex z by one Feynman propagator, and the vertex carries the factor $-i\lambda$. To expose the amplitude, pass to momentum space with the book's convention $\phi(x)=\int \frac{d^4p}{(2\pi)^4}e^{-ip\cdot x}\tilde\phi(p)$. The z-integral produces overall momentum conservation, $\int d^4z\, e^{-i(p_1+p_2+p_3+p_4)\cdot z} = (2\pi)^4\delta^{(4)}(p_1+p_2+p_3+p_4)$, and each leg contributes a momentum-space propagator $D_F(p_i) = i/(p_i^2-m^2+i\varepsilon)$, so

$$\widetilde{G}_c^{(4)}\big|_{O(\lambda)} \;=\; (2\pi)^4\delta^{(4)}\Big(\textstyle\sum_i p_i\Big)\,(-i\lambda)\,\prod_{i=1}^4 \frac{i}{p_i^2-m^2+i\varepsilon}. \tag{10.197}$$

Removing the four external propagators — the *amputation* that the LSZ reduction of Chapter 14 will justify physically — and stripping the momentum-conserving delta leaves the amputated four-point vertex function

$$\boxed{\,i\mathcal{M}\big|_{O(\lambda)} \;=\; -i\lambda\,.\,} \tag{10.198}$$

This is the carried number of ϕ^4 theory: the lowest-order $\phi\phi\to\phi\phi$ amplitude is $\mathcal{M}=-\lambda$, the Feynman rule "a four-point vertex contributes $-i\lambda$." The path

integral has delivered it as nothing more than four functional derivatives acting on a Gaussian. What Chapter 14 adds is the physical interpretation through the S-matrix and LSZ; what Chapter 15 adds is the systematic bookkeeping — the Feynman rules and symmetry factors — that makes Eq. (10.193) routine at every order.

10.7.3 The cancellation of vacuum bubbles

The third term of Eq. (10.193), the source-free $D_F(0)^2$, deserves a closer look, because it is the first instance of a general phenomenon that the normalization $1/Z[0]$ in Theorem 10.18 was built to handle. Setting $J = 0$ in $Z[J]$ kills ϕ_J and leaves only this term, so to first order

$$Z[0] \;=\; Z_0[0]\Big[1 - \frac{i\lambda}{4!}\int d^4z\, 3\, D_F(0)^2 + O(\lambda^2)\Big] \;=\; Z_0[0]\Big[1 - \frac{i\lambda}{8}\int d^4z\, D_F(0)^2 + O(\lambda^2)\Big]. \tag{10.199}$$

The quantity $\frac{1}{8}\int d^4z\, D_F(0)^2$ is the value of the disconnected figure-eight diagram of Fig. 10.7(c), with its symmetry factor $\frac{1}{8}$ already correct; it is a pure number times the spacetime volume, carrying no external legs and hence no physical information. It is precisely the kind of vacuum-to-vacuum amplitude that the worked example of Theorem 10.20 warned must be divided out. We now see it cancel explicitly.

Proposition 10.35 (Vacuum bubbles cancel in normalized correlators)**.** *In the normalized generating functional $Z[J]/Z[0]$, every diagram with a disconnected vacuum component is removed. At first order in λ this is the statement that the source-free term $3\,D_F(0)^2$ of Eq. (10.193) does not appear in $Z[J]/Z[0]$.*

Proof. Assemble the numerator from Eq. (10.189) and Eq. (10.193), and the denominator from Eq. (10.199). Writing $\mathcal{V} \equiv \frac{1}{8}\int d^4z\, D_F(0)^2$ for the bubble and keeping terms through $O(\lambda)$,

$$\begin{aligned}\frac{Z[J]}{Z[0]} &= \frac{Z_0[0]\, e^{W_0}\Big[1 - \frac{i\lambda}{4!}\int d^4z(\phi_J^4 - 6\phi_J^2 D_F(0) + 3D_F(0)^2)\Big]}{Z_0[0]\,[\,1 - i\lambda\,\mathcal{V}\,]} \\ &= e^{W_0}\Big[1 - \tfrac{i\lambda}{4!}\int d^4z(\phi_J^4 - 6\phi_J^2 D_F(0)) - i\lambda\,\mathcal{V}\Big]\,[\,1 + i\lambda\,\mathcal{V} + O(\lambda^2)] \\ &= e^{W_0}\Big[1 - \tfrac{i\lambda}{4!}\int d^4z(\phi_J^4 - 6\phi_J^2 D_F(0))\Big] + O(\lambda^2),\end{aligned} \tag{10.200}$$

where in the second line we isolated the source-free piece of the numerator, $-\frac{i\lambda}{4!}\int 3D_F(0)^2 = -i\lambda\mathcal{V}$, and expanded the denominator; in the third line the $-i\lambda\mathcal{V}$ from the numerator cancels exactly against the $+i\lambda\mathcal{V}$ produced by the denominator multiplying the leading 1. The figure-eight is gone. What survives carries at least one source and hence connects to the external points. □

This is the diagrammatic content of the seemingly innocuous division by $Z[0]$ in the definition of the Green's functions. The general statement, proved in Chapter 15, is that $Z[0]$ exponentiates the sum of all connected vacuum diagrams, so that $Z[J]/Z[0]$ retains only those diagrams every part of which is tied, through some chain of propagators and vertices, to an external source. The infinite, volume-proportional vacuum energy that $Z[0]$ encodes is thereby quarantined and discarded before any correlator is computed — which is why no physical quantity ever depends on it, and why the divergent measure normalization $\mathcal{N}$ of Theorem 10.16, itself part of $Z[0]$, never appears in an answer.

10.7.4 Connected correlators: the functional $W[J]$

Cancelling the vacuum bubbles removed the *wholly* disconnected diagrams, but the surviving correlators still contain pieces that are disconnected among the external points: at four points, for instance, $G^{(4)}$ includes the product of two independent propagators, two pairs of particles that never meet. For scattering we want only the genuinely connected processes, in which all external legs communicate. The object that isolates them is the logarithm of the generating functional, exactly as the logarithm of a partition function isolates cumulants from moments. We define it with the factor of i appropriate to the Minkowski exponent.

Definition 10.36 (Connected generating functional)**.** The *connected generating functional* of the scalar field is

$$W[J] \;\equiv\; -\,i\,\ln Z[J], \tag{10.201}$$

the branch fixed by $W[0] = -i \ln Z[0]$. Its connected n-point Green's functions are the functional Taylor coefficients

$$G_c^{(n)}(x_1,\dots,x_n) \;\equiv\; \frac{1}{i^{\,n-1}}\;\frac{\delta^n W[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}. \tag{10.202}$$

The single derivative has a transparent meaning. Using $\delta Z/\delta J(x) = i\,\langle\phi(x)\rangle_J\, Z$, where $\langle\,\cdot\,\rangle_J \equiv Z^{-1}\int\!\mathcal{D}\phi\,(\cdot)\,e^{i(S+\int J\phi)}$ denotes the source-dependent field average,

$$\frac{\delta W}{\delta J(x)} \;=\; -\,i\,\frac{1}{Z}\frac{\delta Z}{\delta J(x)} \;=\; \langle\phi(x)\rangle_J \;\equiv\; \phi_{\rm cl}(x), \tag{10.203}$$

the expectation value of the field in the presence of the source — the *classical field*, the object we will Legendre-transform on in a moment. It is the field-theory instance of "the first derivative of the free energy is the magnetization," the dictionary of Section 10.6 made functional. That $W[J]$ generates *connected* correlators, with the disconnected products subtracted, is the linked-cluster theorem. Its full combinatorial proof is the work of Chapter 15; here we establish it explicitly at the two- and four-point level, which already exhibits the mechanism and supplies the relations the later chapters use.

Proposition 10.37 (W generates connected correlators at two and four points)**.** *With $G^{(1)} = \langle\phi\rangle_0 = 0$ for the frozen $\mathbb{Z}_2$-symmetric theory (Section 7.3.6), the connected correlators Eq. (10.202) satisfy*

$$G_c^{(2)}(x_1,x_2) = G^{(2)}(x_1,x_2), \tag{10.204}$$

$$\begin{aligned} G_c^{(4)}(x_1,x_2,x_3,x_4) = {} & G^{(4)}(x_1,x_2,x_3,x_4) \;-\; G^{(2)}(x_1,x_2)\,G^{(2)}(x_3,x_4) \\ & -\; G^{(2)}(x_1,x_3)\,G^{(2)}(x_2,x_4) -\; G^{(2)}(x_1,x_4)\,G^{(2)}(x_2,x_3). \end{aligned} \tag{10.205}$$

That is, W reproduces the full two-point function and the full four-point function with its three disconnected "$2+2$" channels subtracted off.

Proof. The engine is the elementary rule that differentiating a source-dependent average attaches one connected field:

$$\frac{\delta}{\delta J(y)}\,\langle A\rangle_J \;=\; \frac{\delta}{\delta J(y)}\,\frac{1}{Z}\int\mathcal{D}\phi\, A\, e^{i(S+\int J\phi)} \;=\; i\big[\,\langle A\,\phi(y)\rangle_J - \langle A\rangle_J\,\langle\phi(y)\rangle_J\,\big], \tag{10.206}$$

where the subtracted term comes from differentiating the $1/Z$ prefactor, $\delta(1/Z)/\delta J(y) = -i\langle\phi(y)\rangle_J/Z$. Apply this once to Eq. (10.203):

$$\frac{\delta^2 W}{\delta J(x_1)\delta J(x_2)} = \frac{\delta\,\langle\phi(x_1)\rangle_J}{\delta J(x_2)} = i\big[\langle\phi(x_1)\phi(x_2)\rangle_J - \langle\phi(x_1)\rangle_J\langle\phi(x_2)\rangle_J\big]. \tag{10.207}$$

At $J=0$ the second term vanishes ($\langle\phi\rangle_0 = 0$), so $\delta^2 W/\delta J^2|_0 = i\,G^{(2)}$, and Eq. (10.202) with $n=2$ gives $G_c^{(2)} = i^{-1}\cdot iG^{(2)} = G^{(2)}$, which is Eq. (10.204).

For the four-point relation, differentiate Eq. (10.207) twice more using Eq. (10.206) on each average, and evaluate at $J=0$, where every odd correlator vanishes by $\mathbb{Z}_2$. Writing the third derivative first,

$$\frac{\delta^3 W}{\delta J(x_1)\delta J(x_2)\delta J(x_3)} = i^2\Big[\langle\phi_1\phi_2\phi_3\rangle_J - \langle\phi_1\phi_2\rangle_J\langle\phi_3\rangle_J - \langle\phi_1\phi_3\rangle_J\langle\phi_2\rangle_J - \langle\phi_2\phi_3\rangle_J\langle\phi_1\rangle_J + 2\langle\phi_1\rangle_J\langle\phi_2\rangle_J$$

(10.208)

abbreviating $\phi_i \equiv \phi(x_i)$. Differentiating once more with respect to $J(x_4)$ via Eq. (10.206) and then setting $J=0$, every term containing a surviving single average $\langle\phi\rangle_0$ drops out, and only two kinds of contribution remain: from the leading triple average, $\langle\phi_1\phi_2\phi_3\rangle \to i[\langle\phi_1\phi_2\phi_3\phi_4\rangle - 0]$; and from each double average paired with the $\langle\phi\rangle$ that $\delta/\delta J(x_4)$ promotes to a double average, $-\langle\phi_i\phi_j\rangle\langle\phi_k\rangle \to -\langle\phi_i\phi_j\rangle\, i\langle\phi_k\phi_4\rangle$. Collecting,

$$\frac{\delta^4 W}{\delta J(x_1)\cdots\delta J(x_4)}\bigg|_{J=0} = i^3\Big[G^{(4)} - G^{(2)}_{12}G^{(2)}_{34} - G^{(2)}_{13}G^{(2)}_{24} - G^{(2)}_{14}G^{(2)}_{23}\Big], \tag{10.209}$$

with $G^{(2)}_{ij} \equiv G^{(2)}(x_i,x_j)$. Dividing by i^3 as in Eq. (10.202) yields Eq. (10.205). □

The subtraction in Eq. (10.205) is exactly the removal of the three disconnected channels: a ϕ^4 four-point function in which particles $1,2$ propagate to $3,4$ as two non-interacting pairs contributes $G^{(2)}_{13}G^{(2)}_{24}+\cdots$ to $G^{(4)}$, and W discards precisely these, retaining only the part where all four legs meet — at lowest order, the vertex diagram Eq. (10.196) of Theorem 10.34, which is connected and therefore survives untouched. The pattern continues to all orders: $G_c^{(n)}$ is the sum of connected diagrams alone, and the full correlator is recovered by summing over all ways of partitioning the external points into connected clusters. That linked-cluster expansion, and the proof that $\ln Z$ generates exactly the connected pieces at every n, is taken up in Chapter 15; what Theorem 10.36 contributes here is the precise object and the verified low-order relations on which that development rests.

10.7.5 The effective action and the road ahead

There is one more functional in the hierarchy, and it is the one the renormalization and symmetry-breaking chapters will lean on hardest. Having traded $Z[J]$ for its logarithm $W[J]$ to isolate connected diagrams, we now trade the source J for the classical field ϕ_{cl} it induces, by a Legendre transform. The construction is the field-theory instance of the Legendre duality between thermodynamic potentials developed abstractly in Section 4.5.1, with $W[J]$ playing the role of the free energy and ϕ_{cl} the conjugate "magnetization" Eq. (10.203).

Definition 10.38 (Effective action)**.** Let $\phi_{\mathrm{cl}}(x)=\delta W/\delta J(x)$ as in Eq. (10.203), and let $J=J_{\phi_{\mathrm{cl}}}$ denote the source that produces a given classical field, obtained by inverting this relation. The *effective action* is the Legendre transform of W,

$$\boxed{\Gamma[\phi_{\mathrm{cl}}] \;\equiv\; W[J] \;-\; \int d^4x\, J(x)\,\phi_{\mathrm{cl}}(x)\,\Big|_{J=J_{\phi_{\mathrm{cl}}}}} \,. \tag{10.210}$$

The transform is well-posed under the same convexity hypothesis that governed the abstract case (Theorem 4.31): the map $J\mapsto\phi_{\mathrm{cl}}$ is invertible precisely when $\delta^2W/\delta J^2$ — the full propagator, by Eq. (10.207) — is nondegenerate, which holds order by order in perturbation theory. The defining virtue of Γ is that it returns the source as its own gradient. Differentiating Eq. (10.210),

$$\frac{\delta\Gamma}{\delta\phi_{\mathrm{cl}}(x)} = \int d^4y\,\frac{\delta W}{\delta J(y)}\frac{\delta J(y)}{\delta\phi_{\mathrm{cl}}(x)} - J(x) - \int d^4y\,\frac{\delta J(y)}{\delta\phi_{\mathrm{cl}}(x)}\phi_{\mathrm{cl}}(y), \tag{10.211}$$

and since $\delta W/\delta J(y)=\phi_{\mathrm{cl}}(y)$ the first and third terms cancel identically, leaving the *quantum equation of motion*

$$\boxed{\frac{\delta\Gamma[\phi_{\mathrm{cl}}]}{\delta\phi_{\mathrm{cl}}(x)} \;=\; -\,J(x)\,.} \tag{10.212}$$

In the absence of an external source, the physical configuration of the theory is therefore a stationary point of Γ, $\delta\Gamma/\delta\phi_{\rm cl} = 0$ — the exact, quantum-corrected analogue of the classical principle of stationary action $\delta S/\delta\phi = 0$. At leading (tree) order one finds $\Gamma[\phi_{\rm cl}] = S[\phi_{\rm cl}]$, so Γ is the action dressed by all quantum fluctuations; the systematic loop expansion that computes the dressing is the subject of Chapter 22.

The second derivatives of W and Γ are functional inverses, and this is the precise sense in which Γ is the generator of one-particle-irreducible (1PI) vertices. Differentiate the relation $\delta W/\delta J(x) = \phi_{\rm cl}(x)$ with respect to $\phi_{\rm cl}(y)$ and use Eq. (10.212) in the form $\delta J(z)/\delta\phi_{\rm cl}(y) = -\delta^2\Gamma/\delta\phi_{\rm cl}(z)\delta\phi_{\rm cl}(y)$:

$$\delta^{(4)}(x-y) = \frac{\delta\phi_{\rm cl}(x)}{\delta\phi_{\rm cl}(y)} = \int d^4z\, \frac{\delta^2 W}{\delta J(x)\delta J(z)}\, \frac{\delta J(z)}{\delta\phi_{\rm cl}(y)} = -\int d^4z\, \frac{\delta^2 W}{\delta J(x)\delta J(z)}\, \frac{\delta^2\Gamma}{\delta\phi_{\rm cl}(z)\delta\phi_{\rm cl}(y)}. \tag{10.213}$$

Thus $W^{(2)}$ and $-\Gamma^{(2)}$ are reciprocal kernels. Since $W^{(2)}|_0 = i\,G_c^{(2)} = i\,D_F$ is (i times) the full propagator, $\Gamma^{(2)}$ is minus the inverse of $W^{(2)}$, that is $\Gamma^{(2)} = -(i\,D_F)^{-1} = i\,D_F^{-1}$; in momentum space, at tree level where $D_F(p) = i/(p^2 - m^2 + i\varepsilon)$, this gives

$$\widetilde{\Gamma}^{(2)}(p) \;=\; p^2 - m^2, \tag{10.214}$$

the Klein–Gordon operator itself — exactly $\delta^2 S/\delta\phi^2$ in momentum space, confirming $\Gamma = S$ at tree level. The interactions correct this: $\widetilde{\Gamma}^{(2)}(p) = p^2 - m^2 - \Sigma(p^2)$, with Σ the self-energy, the sum of 1PI two-point diagrams whose first contribution is the tadpole of Fig. 10.7(b). That every $\Gamma^{(n)}$ is the sum of amputated 1PI diagrams, and that this is what makes Γ the natural object on which to impose renormalization conditions and to read off spontaneous symmetry breaking from the shape of the effective potential, is the content of Chapter 22; we have established here only the defining transform Eq. (10.210), the equation of motion Eq. (10.212), and the propagator-inverse relation Eq. (10.214) that pin down what Γ is.

With this the path-integral apparatus is complete and self-contained: a measure (Theorem 10.16), a free generating functional in closed form (Eq. (10.111)), Wick's theorem (Theorem 10.23), the master perturbative formula Eq. (10.186), and the three nested generators $Z \to W \to \Gamma$ for full, connected, and 1PI correlators respectively. These objects are handed forward intact: the vertex $-i\lambda$ of Eq. (10.198) and the time-ordered correlators of Theorem 10.19 become physical amplitudes through the S-matrix and LSZ reduction of Chapter 14; the contraction structure of Eq. (10.193) becomes the Feynman rules and symmetry factors of Chapter 15; the effective action Γ becomes the carrier of renormalization and the RG in Chapters 22 and 23; and the entire construction is repeated, with the anticommuting Grassmann measure replacing $\mathcal{D}\phi$, for the fermionic fields of Chapter 12.

Further reading

The space-time, sum-over-histories formulation originates with Feynman [192] and is developed pedagogically in Feynman and Hibbs [193]; Kleinert [297] is an encyclopedic modern treatment of the quantum-mechanical path integral and its measure. The field-theory generating functional, $Z_0[J]$, Wick's theorem from functional derivatives, and the connected and effective functionals are presented in Peskin and Schroeder [396] ch. 9, Schwartz [432] chs. 14–16, Srednicki [454] chs. 6–9, and — with the quickest physical motivation — Zee [532] ch. I. Weinberg [500] ch. 9 gives the careful operator-to-path-integral correspondence. For the Euclidean formulation, the statistical-mechanics dictionary, and the renormalization group born from it, Zinn-Justin [534] is the standard reference. The rigorous status of the measure, reflection positivity, and the reconstruction of a Lorentzian theory from Euclidean data are the subject of constructive field theory: the axioms are those of Osterwalder and Schrader [372], and the program is developed in full in Glimm and Jaffe [225].

Chapter 11

The Dirac Field

With the scalar field quantized in Chapter 9, the book has one relativistic quantum field in hand — and exactly the wrong one for matter. The particles of nature that carry charge and build atoms are spin-$\frac{1}{2}$ fermions, and a spin-0 field cannot describe them. This chapter constructs the field that can. We begin from the representation theory of Chapter 3, where the Lorentz group was found to act on a four-component object, the Dirac spinor, sitting in the reducible $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ representation; and we take from Chapter 6 the classical Dirac Lagrangian $\bar{\psi}(i\partial\!\!\!/ - m)\psi$ as one of the recurring cast of classical field theories. The gamma-matrix algebra $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}$, the chiral basis, the trace and spinor identities, and the symmetry properties of the charge-conjugation matrix are the book-wide canon of Appendix C (Chapter C); we invoke them where they are used rather than re-listing them. Throughout we keep the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units of the book, so that $p^2 = m^2$ on shell. The thread carried here is QED: the electron, which entered as a classical $U(1)$ matter field in Chapter 8, is promoted in this chapter to a quantum field whose quanta are the electrons and positrons that the rest of the thread will scatter.

The chapter follows a single arc, from classical kinematics to a fully solved, demonstrably consistent quantum theory. The first section shows that Lorentz covariance *forces* the four-component spinor and that the demand for a first-order equation reproducing the mass shell *forces* the Dirac equation $(i\partial\!\!\!/ - m)\psi = 0$; it establishes form-invariance through the intertwining identity, derives the conserved vector current $j^\mu = \bar{\psi}\gamma^\mu\psi$ from the Lagrangian, and catalogues the sixteen bilinear covariants. The second section solves the equation: it constructs the positive- and negative-frequency plane-wave amplitudes $u^s(p), v^s(p)$, fixes their normalization and orthogonality, and proves the completeness (spin-sum) identities $\sum_s u^s\bar{u}^s = p\!\!\!/ + m$, $\sum_s v^s\bar{v}^s = p\!\!\!/ - m$ together with the energy, chirality, and helicity projectors — the raw material both for the canonical mode expansion and for every QED trace.

The quantization itself is the dramatic centre of the chapter. The third section

carries out canonical quantization and discovers that the naive commutator postulate fails catastrophically: it produces a Hamiltonian unbounded below and negative-norm states. The cure — imposing *anticommutators* — is forced by the demand for a stable vacuum, and it carries the fermionic ladder algebra in its wake. On that algebra the fourth section builds the antisymmetric Fock space: a unique vacuum, the Pauli exclusion principle as an operator identity, normal-ordered energy, momentum, and charge, and the identification of the excitations as spin-$\frac{1}{2}$ electrons and positrons carrying opposite charge. The fifth section constructs the central dynamical object, the Dirac–Feynman propagator S_F, exhibiting it simultaneously as the time-ordered two-point function, the Dirac operator applied to the scalar propagator of Chapter 9, the $i\epsilon$-contour momentum-space object, and the Green's function of $i\not{\partial} - m$ — the internal electron line of every diagram from Chapter 17 onward.

The final two sections close the consistency loop. The sixth proves microcausality of the Dirac field and the causality of its bilinear observables, and with it the spin–statistics connection for spin $\frac{1}{2}$: relativity together with positivity of the energy does not merely permit but *requires* the anticommutator quantization used throughout, completing the fermionic half of the theorem whose scalar half Chapter 9 established and whose general form (Theorem 9.33) is quoted without full proof. The seventh assembles the discrete symmetries C, P, T and the CPT combination, tabulates the transformation of every bilinear, and closes with the Majorana condition and the dated open question of the neutrino's nature. By the end the free electron is a complete relativistic quantum theory — bounded energy, causal observables, exclusion principle, and the propagator — and QED is ready for interactions.

11.1 Spinor Representations and the Dirac Equation

The quantization of the scalar field in Chapter 9 followed a fixed recipe: start from a classical field with a known Lagrangian, Hamiltonian, equation of motion, and conserved current; promote the field to an operator; expand it in the normal modes of its free equation of motion; and impose the algebra on the mode coefficients that the canonical structure demands. We are about to run that same recipe for the spin-$\frac{1}{2}$ field, and Chapter 9 is the template we will be modifying at every step — most dramatically when the canonical algebra turns out to be an *anti*commutator. But the recipe has a non-negotiable first ingredient: the classical field, with its equation, its Lagrangian, its adjoint, and its current all in hand. For the scalar that ingredient was almost free, because a single real function of spacetime is the most primitive field one can write. For spin $\frac{1}{2}$ it is not free at all. The very object the field is valued in — a four-component spinor — and the first-order equation it must satisfy are forced on us by Lorentz covariance, and unpacking that "forced on us" is the work of this section. We

assemble the classical Dirac field as the unique Lorentz-covariant first-order field equation living on the reducible four-component spinor representation, drawing the representation theory from Chapter 3 and the Clifford-algebra technology from Chapter C, and we record the adjoint, the conserved current, and the complete catalogue of bilinear covariants that the quantum theory of the following sections — and all of QED downstream — will lean on.

11.1.1 The Dirac representation, recalled

We begin by collecting, in the conventions fixed in Chapter A and used throughout this chapter, the representation-theoretic facts established in Section 3.4.4 and Chapter C. The reader who has those chapters fresh may skim to Eq. (11.9); we restate rather than re-derive, flagging each result by the label of its proof.

A spin-$\frac{1}{2}$ field that respects parity — and electromagnetism, the force that binds the electron of QED, does respect parity — cannot be a single Weyl spinor. Theorem 3.28 showed that parity exchanges the two irreducible spinor representations $(\frac{1}{2},0) \leftrightarrow (0,\frac{1}{2})$ of the Lorentz group, so the smallest parity-invariant carrier of spin $\frac{1}{2}$ is the *reducible* sum

$$(\tfrac{1}{2},0) \oplus (0,\tfrac{1}{2}), \tag{11.1}$$

realized on the four-component *Dirac spinor* $\psi = (\psi_L, \psi_R)^{\mathsf{T}} \in \mathbb{C}^4$ of Theorem 3.26. The arena natural to this representation is the Clifford algebra. There exist four 4×4 matrices γ^μ, $\mu = 0,1,2,3$, satisfying the *Clifford (Dirac) algebra*

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\, \mathbf{1}_4, \tag{11.2}$$

with $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$; their existence and their uniqueness up to similarity are Theorems C.10 and C.16. We work in the *chiral (Weyl) basis* of Eq. (3.111),

$$\gamma^\mu = \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix}, \qquad \sigma^\mu = (\mathbf{1}, \vec{\sigma}), \quad \bar{\sigma}^\mu = (\mathbf{1}, -\vec{\sigma}), \tag{11.3}$$

in which the chirality matrix $\gamma^5 \equiv i\gamma^0\gamma^1\gamma^2\gamma^3$ is diagonal,

$$\gamma^5 = \begin{pmatrix} -\mathbf{1} & 0 \\ 0 & +\mathbf{1} \end{pmatrix}, \qquad \left\{\gamma^5, \gamma^\mu\right\} = 0, \qquad (\gamma^5)^2 = \mathbf{1}_4, \tag{11.4}$$

the algebraic properties being Theorem C.26. The Dirac adjoint is $\bar{\psi} \equiv \psi^\dagger \gamma^0$, the insertion of γ^0 being dictated by the *Hermiticity relation* (Theorem C.30)

$$\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0, \tag{11.5}$$

as we will see explicitly in Section 11.1.4.

The Lorentz group acts on ψ through the generators built from the gammas. With

$$S^{\mu\nu} \equiv \frac{i}{4}\,[\gamma^\mu, \gamma^\nu] = \frac{i}{2}(\gamma^\mu\gamma^\nu - \eta^{\mu\nu}), \tag{11.6}$$

(the second form following from $\gamma^\mu\gamma^\nu = \eta^{\mu\nu} + \frac{1}{2}\,[\gamma^\mu, \gamma^\nu]$ and Eq. (11.2)), one has the central structural fact, proved in full in Theorem 3.27 from the Clifford relation alone: the $S^{\mu\nu}$ satisfy the Lorentz Lie algebra of Section 3.2,

$$[S^{\mu\nu}, S^{\rho\sigma}] = i(\eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} - \eta^{\nu\sigma}S^{\mu\rho} + \eta^{\mu\sigma}S^{\nu\rho}). \tag{11.7}$$

A finite Lorentz transformation $\Lambda = \exp\omega$, parametrized by an antisymmetric $\omega_{\mu\nu}$, is therefore represented on the spinor by

$$\psi(x) \;\longmapsto\; \psi'(x') = S(\Lambda)\,\psi(x), \qquad S(\Lambda) = \exp\!\Big(-\tfrac{i}{2}\,\omega_{\mu\nu}S^{\mu\nu}\Big), \qquad x' = \Lambda x. \tag{11.8}$$

The engine that drives every covariance statement below is the infinitesimal action of $S^{\mu\nu}$ on a single gamma matrix. From Eq. (11.6) one computes (this is Eq. (3.117); we reproduce the one-line core because it is used repeatedly)

$$[S^{\mu\nu}, \gamma^\rho] = i(\eta^{\nu\rho}\gamma^\mu - \eta^{\mu\rho}\gamma^\nu), \tag{11.9}$$

the statement that γ^ρ *transforms as a four-vector index* under the generators. Indeed, only the $\gamma^\mu\gamma^\nu$ piece of Eq. (11.6) contributes, and $[\gamma^\mu\gamma^\nu, \gamma^\rho] = 2\eta^{\nu\rho}\gamma^\mu - 2\eta^{\mu\rho}\gamma^\nu$ by two applications of Eq. (11.2), whence Eq. (11.9). Exponentiating Eq. (11.9) promotes the infinitesimal four-vector law to the finite *intertwining identity* proved in Theorem C.6,

$$S(\Lambda)^{-1}\,\gamma^\mu\,S(\Lambda) = \Lambda^\mu{}_\nu\,\gamma^\nu, \tag{11.10}$$

the single relation on which the Lorentz covariance of the Dirac equation rests. This is exactly the physical consequence that Chapter C derived the identity in order to feed; we now collect the dividend.

Notation Watch

Two antisymmetric tensors of gammas appear in this book and must not be confused. The *Lorentz generators* are $S^{\mu\nu} = \frac{i}{4}\,[\gamma^\mu, \gamma^\nu]$ of Eq. (11.6), satisfying the algebra Eq. (11.7); the *tensor bilinear* is

$$\sigma^{\mu\nu} \equiv \frac{i}{2}\,[\gamma^\mu, \gamma^\nu] = 2\,S^{\mu\nu}, \tag{11.11}$$

the object whose bilinear $\bar\psi\sigma^{\mu\nu}\psi$ enters the covariant catalogue of Section 11.1.5 and the magnetic-moment coupling of QED. The factor of two between them, $\sigma^{\mu\nu} = 2S^{\mu\nu}$, follows the convention of Eqs. (C.72) and (C.73); it is the standard particle-physics normalization (Peskin–

Schroeder). We use $S^{\mu\nu}$ exclusively for transformations and $\sigma^{\mu\nu}$ exclusively as a bilinear insertion.

11.1.2 The Dirac equation as the forced first-order equation

With the representation in hand we can answer the question that organizes the whole chapter: what is the equation of motion of a free spin-$\frac{1}{2}$ field? Two requirements pin it down uniquely, and seeing them do so is more instructive than postulating the answer.

The first requirement is physical and is shared with every free relativistic field: each component of ψ describes a particle of mass m, so each component must satisfy the mass-shell condition $p^2 = m^2$, i.e. the Klein–Gordon equation $(\Box + m^2)\psi = 0$. The scalar of Chapter 9 satisfied exactly this, and nothing about adding spin can relax the relativistic dispersion relation. The second requirement is the one that distinguishes the fermion. We demand a *first-order* equation — linear in ∂_μ — rather than the second-order Klein–Gordon equation directly. The motivation is the same one that drove the representation theory: a first-order equation needs an initial condition only on ψ itself, not on ψ and $\dot{\psi}$ independently, which is the field-theoretic shadow of the fact that the spinor already carries, in its multiple components, the data a second-order scalar equation would have to supply through a time derivative. (Historically this was Dirac's own demand, made in the hope of a positive-definite probability density; the demand survives even though that original interpretation does not.) The most general Lorentz-covariant first-order linear equation with constant coefficients is

$$(i\beta^\mu \partial_\mu - m)\psi = 0, \tag{11.12}$$

with β^μ four constant 4×4 matrices to be determined and the mass m a scalar. We now show that the Klein–Gordon requirement *forces* β^μ to be a set of gamma matrices.

Theorem 11.1 (The Dirac equation is forced)**.** *A constant-coefficient first-order equation Eq. (11.12) implies the Klein–Gordon equation $(\Box + m^2)\psi = 0$ for every component of ψ if and only if the matrices β^μ obey the Clifford algebra $\{\beta^\mu, \beta^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$. By the uniqueness theorem Theorem C.16, β^μ is then similar to the gamma matrices γ^μ, and Eq. (11.12) becomes the* Dirac equation

$$\boxed{(i\not{\partial} - m)\psi = 0}, \qquad \not{\partial} \equiv \gamma^\mu \partial_\mu. \tag{11.13}$$

Proof. Apply the conjugate first-order operator $(i\beta^\nu \partial_\nu + m)$ to Eq. (11.12):

$$(i\beta^\nu \partial_\nu + m)(i\beta^\mu \partial_\mu - m)\psi = (-\beta^\nu \beta^\mu \partial_\nu \partial_\mu - m^2)\psi = 0. \tag{11.14}$$

The cross terms $\pm im\,\beta^\mu\partial_\mu$ cancel because m is a scalar that commutes with the constant β^μ. The double derivative $\partial_\nu\partial_\mu$ is symmetric in (μ,ν), so only the symmetric part of $\beta^\nu\beta^\mu$ survives the contraction:

$$\beta^\nu\beta^\mu\partial_\nu\partial_\mu = \tfrac{1}{2}\{\beta^\mu,\beta^\nu\}\,\partial_\mu\partial_\nu. \tag{11.15}$$

Therefore Eq. (11.14) reads $(\frac{1}{2}\{\beta^\mu,\beta^\nu\}\,\partial_\mu\partial_\nu + m^2)\psi = 0$. This coincides with the Klein–Gordon equation $(\eta^{\mu\nu}\partial_\mu\partial_\nu + m^2)\psi = (\Box + m^2)\psi = 0$ for arbitrary ψ precisely when

$$\tfrac{1}{2}\{\beta^\mu,\beta^\nu\} = \eta^{\mu\nu}\mathbf{1} \quad\Longleftrightarrow\quad \{\beta^\mu,\beta^\nu\} = 2\eta^{\mu\nu}\mathbf{1}, \tag{11.16}$$

which is the Clifford algebra Eq. (11.2). (That the condition is necessary as well as sufficient follows by reading the calculation backward: if some $\{\beta^\mu,\beta^\nu\} \neq 2\eta^{\mu\nu}\mathbf{1}$, then the symmetric tensor $\frac{1}{2}\{\beta^\mu,\beta^\nu\} - \eta^{\mu\nu}\mathbf{1}$ is nonzero, and contracting it with a plane wave $\psi \propto e^{-ip\cdot x}$ of generic p gives a nonzero residual, so Klein–Gordon fails for that mode.) By Theorem C.16 any two solutions of the Clifford algebra in four dimensions are related by a similarity transformation $\beta^\mu = A\gamma^\mu A^{-1}$; redefining $\psi \to A^{-1}\psi$ absorbs A and brings Eq. (11.12) to the canonical form Eq. (11.13). □

Physical Insight

The Dirac equation is not a clever guess that happened to work; it is the unique first-order avatar of the relativistic dispersion relation. The operator $i\partial\!\!\!/ - m$ is a "square root" of the Klein–Gordon operator $-(\Box + m^2)$ in the precise sense of Eq. (11.14): $(i\partial\!\!\!/ + m)(i\partial\!\!\!/ - m) = -(\Box + m^2)$. Taking a matrix square root of a scalar differential operator is possible only at the cost of letting the matrices fail to commute — and the failure-to-commute they are forced into is exactly the Clifford algebra, whose smallest faithful representation in four dimensions is four-dimensional. The four components of the Dirac spinor are the unavoidable price of the square root; spin $\frac{1}{2}$ and antiparticles (two components each for particle and antiparticle, as the next section's plane waves will show) are what is hiding inside that price.

The proof of Theorem 11.1 already contains the converse statement promised in the chapter's opening, that each component of a Dirac field independently solves the Klein–Gordon equation. It is worth isolating, because the algebraic identity behind it — the factorization of the Klein–Gordon operator — recurs constantly (it is, for instance, how the Dirac propagator will be obtained from the scalar one in Section 11.2 and beyond).

Worked Example 11.2 (Squaring the Dirac operator)**.** Let ψ solve the Dirac equation Eq. (11.13). We show $(\Box + m^2)\psi = 0$ directly. Multiply Eq. (11.13) on

the left by the conjugate operator $(i\partial\!\!\!/ + m)$:

$$(i\partial\!\!\!/ + m)(i\partial\!\!\!/ - m)\psi = (-\partial\!\!\!/\,\partial\!\!\!/ - m^2)\psi = 0. \tag{11.17}$$

The slashed-derivative square is evaluated with the Clifford algebra:

$$\partial\!\!\!/\,\partial\!\!\!/ = \gamma^\mu\gamma^\nu\partial_\mu\partial_\nu = \tfrac{1}{2}\{\gamma^\mu,\gamma^\nu\}\,\partial_\mu\partial_\nu = \eta^{\mu\nu}\partial_\mu\partial_\nu = \Box, \tag{11.18}$$

where in the second equality we used that $\partial_\mu\partial_\nu$ is symmetric to replace $\gamma^\mu\gamma^\nu$ by its symmetric part $\frac{1}{2}\{\gamma^\mu,\gamma^\nu\}$, and in the third we used Eq. (11.2). Substituting into Eq. (11.17) gives $-(\Box + m^2)\psi = 0$, i.e. $(\Box + m^2)\psi = 0$: every component of the Dirac field is a Klein–Gordon field of mass m, as required for a relativistic particle of that mass. The identity Eq. (11.18), $a\!\!\!/\,a\!\!\!/ = a^2$ for any commuting four-vector a_μ, is the workhorse contraction; we record its general form $a\!\!\!/\,b\!\!\!/ = a\cdot b + \sigma_{ab}$ with $\sigma_{ab} = -\frac{i}{2}\sigma^{\mu\nu}(a_\mu b_\nu - a_\nu b_\mu)$ for later trace work, where $\sigma^{\mu\nu}$ is Eq. (11.11).

In momentum space, writing $\psi(x) = u(p)\,e^{-ip\cdot x}$ for a positive-frequency mode, Eq. (11.13) becomes the purely algebraic condition

$$(p\!\!\!/ - m)\,u(p) = 0, \qquad p\!\!\!/ \equiv \gamma^\mu p_\mu, \tag{11.19}$$

and Theorem 11.2 guarantees that this has nontrivial solutions only on the mass shell $p^2 = m^2$, since acting with $(p\!\!\!/ + m)$ gives $(p^2 - m^2)u = 0$. Solving Eq. (11.19) explicitly, normalizing its solutions, and assembling them into a complete basis is the business of Section 11.2; here we have established only that the equation is the right one and that its solutions live on the mass shell.

11.1.3 Lorentz covariance: form-invariance of the Dirac equation

An equation built to be "the unique Lorentz-covariant first-order equation" must actually be Lorentz covariant, and we have not yet checked it. Covariance means *form-invariance*: an observer in a boosted or rotated frame, applying the same equation Eq. (11.13) to the transformed field, must get a true statement whenever the original observer did. The intertwining identity Eq. (11.10) is exactly the tool that makes this work, and the proof is short and clean.

Theorem 11.3 (Form-invariance of the Dirac equation)**.** *Let $\psi(x)$ satisfy $(i\partial\!\!\!/ - m)\psi = 0$, and define the transformed field under the Lorentz transformation $x' = \Lambda x$ by $\psi'(x') = S(\Lambda)\psi(x)$ as in Eq. (11.8). Then ψ' satisfies the Dirac equation in the primed coordinates,*

$$(i\gamma^\mu\partial'_\mu - m)\psi'(x') = 0, \qquad \partial'_\mu \equiv \frac{\partial}{\partial x'^\mu}. \tag{11.20}$$

The gamma matrices are the same *numerical matrices in both frames; covariance is carried entirely by the spinor rotation $S(\Lambda)$ and the index rotation Λ.*

Proof. The coordinate derivative transforms as $\partial'_\mu = (\Lambda^{-1})^\nu{}_\mu\,\partial_\nu$, by the chain rule applied to $x^\nu = (\Lambda^{-1})^\nu{}_\mu x'^\mu$. Substitute this and $\psi'(x') = S(\Lambda)\psi(x)$ into the kinetic term:

$$\begin{aligned} i\gamma^\mu \partial'_\mu\, \psi'(x') &= i\,\gamma^\mu (\Lambda^{-1})^\nu{}_\mu\, \partial_\nu \big(S(\Lambda)\psi(x)\big) \\ &= i\, S(\Lambda)\, \big[S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\big](\Lambda^{-1})^\nu{}_\mu\, \partial_\nu \psi(x), \end{aligned} \tag{11.21}$$

where in the second line we inserted $S(\Lambda)S(\Lambda)^{-1} = \mathbf{1}$ to the left of γ^μ and pulled the constant matrix $S(\Lambda)$ out front (it does not act on the coordinate x). Now apply the intertwining identity Eq. (11.10), $S(\Lambda)^{-1}\gamma^\mu S(\Lambda) = \Lambda^\mu{}_\rho \gamma^\rho$:

$$i\gamma^\mu \partial'_\mu\, \psi'(x') = i\, S(\Lambda)\, \Lambda^\mu{}_\rho\, (\Lambda^{-1})^\nu{}_\mu\, \gamma^\rho\, \partial_\nu \psi(x). \tag{11.22}$$

The two Lorentz matrices contract to the identity, $\Lambda^\mu{}_\rho (\Lambda^{-1})^\nu{}_\mu = (\Lambda^{-1})^\nu{}_\mu \Lambda^\mu{}_\rho = \delta^\nu{}_\rho$, leaving

$$i\gamma^\mu \partial'_\mu\, \psi'(x') = i\, S(\Lambda)\, \gamma^\rho \partial_\rho \psi(x) = S(\Lambda)\, i\partial\!\!\!/\, \psi(x). \tag{11.23}$$

The mass term transforms trivially, $m\psi'(x') = S(\Lambda)\, m\psi(x)$, since m is a scalar. Subtracting,

$$(i\gamma^\mu \partial'_\mu - m)\psi'(x') = S(\Lambda)\, (i\partial\!\!\!/ - m)\psi(x) = 0, \tag{11.24}$$

the last equality because ψ solves the Dirac equation and $S(\Lambda)$ is invertible. The transformed field solves the transformed equation, which is form-invariance. □

The proof exhibits exactly why the representation theory of Chapter 3 had to come first. Form-invariance is not an algebraic accident; it is the statement that the spinor index of ψ and the vector index of γ^μ rotate in a coordinated way, the coordination being the intertwining identity. A first-order equation written with matrices that did *not* obey Eq. (11.10) would single out a preferred frame and violate relativity. The Dirac equation is the unique first-order equation whose matrices do.

11.1.4 Lagrangian, adjoint, canonical momentum, and the conserved current

The classical action principle for the Dirac field was constructed in Chapter 6; we import it and use it, rather than rebuild it. The *Dirac Lagrangian* (Eq. (6.68)) is

$$\mathcal{L} = \bar\psi (i\partial\!\!\!/ - m)\psi, \tag{11.25}$$

whose Euler–Lagrange equations, obtained in Theorem 6.11 by varying $\bar\psi$ and ψ as independent fields, are the Dirac equation Eq. (11.13) and its Dirac-adjoint partner (Eq. (6.74))

$$i\,\partial_\mu \bar\psi\, \gamma^\mu + m\,\bar\psi = 0 \quad\Longleftrightarrow\quad \bar\psi (i\overleftarrow{\partial\!\!\!/} + m) = 0, \tag{11.26}$$

the arrow indicating that the derivative acts to the left. The Lagrangian is first order in derivatives, so its Legendre transform is degenerate: the momentum conjugate to ψ is (Eq. (6.78))

$$\pi_\psi = \frac{\partial \mathcal{L}}{\partial \dot{\psi}} = i\,\psi^\dagger, \tag{11.27}$$

not an independent velocity but the field $\psi^\dagger$ itself. This is the classical constraint that Chapter 6 flagged as the origin of the anticommutator structure; the canonical quantization of Section 11.3 will turn $\pi_\psi = i\psi^\dagger$ into the equal-time relation $\left\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\right\} = \delta_{ab}\delta^{(3)}(\mathbf{x}-\mathbf{y})$. For the present classical section we need only the Lagrangian, its symmetries, and their currents.

That the action $S = \int d^4x\,\mathcal{L}$ is Lorentz invariant, and that $\mathcal{L}$ itself is a Lorentz scalar, hinges on a single fact: under Eq. (11.8) the Dirac adjoint transforms not with $S(\Lambda)$ but with its inverse. We prove this now, because everything about the covariance of bilinears follows from it.

Theorem 11.4 (Transformation of the adjoint)**.** *Under $\psi \to S(\Lambda)\psi$, the Dirac adjoint transforms as*

$$\bar{\psi} \;\longmapsto\; \bar{\psi}\, S(\Lambda)^{-1}, \tag{11.28}$$

which is the statement

$$\gamma^0\, S(\Lambda)^\dagger\, \gamma^0 = S(\Lambda)^{-1}. \tag{11.29}$$

Proof. We first establish the infinitesimal version, that $\gamma^0 S^{\mu\nu\dagger}\gamma^0 = S^{\mu\nu}$. Conjugate the generator Eq. (11.6). Since $[\gamma^\mu,\gamma^\nu]^\dagger = \gamma^{\nu\dagger}\gamma^{\mu\dagger} - \gamma^{\mu\dagger}\gamma^{\nu\dagger} = -\left[\gamma^{\mu\dagger},\gamma^{\nu\dagger}\right]$,

$$S^{\mu\nu\dagger} = \left(\tfrac{i}{4}\left[\gamma^\mu,\gamma^\nu\right]\right)^\dagger = -\tfrac{i}{4}\left[\gamma^\mu,\gamma^\nu\right]^\dagger = \tfrac{i}{4}\left[\gamma^{\mu\dagger},\gamma^{\nu\dagger}\right]. \tag{11.30}$$

Insert the Hermiticity relation Eq. (11.5), $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$, and use $(\gamma^0)^2 = \mathbf{1}$ to cancel the inner pair of γ^0's:

$$\left[\gamma^{\mu\dagger},\gamma^{\nu\dagger}\right] = \left[\gamma^0\gamma^\mu\gamma^0,\gamma^0\gamma^\nu\gamma^0\right] = \gamma^0\gamma^\mu\gamma^\nu\gamma^0 - \gamma^0\gamma^\nu\gamma^\mu\gamma^0 = \gamma^0\left[\gamma^\mu,\gamma^\nu\right]\gamma^0, \tag{11.31}$$

so that $S^{\mu\nu\dagger} = \frac{i}{4}\gamma^0\left[\gamma^\mu,\gamma^\nu\right]\gamma^0 = \gamma^0 S^{\mu\nu}\gamma^0$. Multiplying on both sides by γ^0 and using $(\gamma^0)^2 = \mathbf{1}$ gives the infinitesimal statement

$$\gamma^0\, S^{\mu\nu\dagger}\,\gamma^0 = S^{\mu\nu}. \tag{11.32}$$

Now exponentiate. From Eq. (11.8), $S(\Lambda)^\dagger = \exp\!\left(+\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu\dagger}\right)$ (the $-i$ becomes $+i$ and each $S^{\mu\nu} \to S^{\mu\nu\dagger}$, the real parameters $\omega_{\mu\nu}$ unchanged). Conjugating by γ^0 commutes with the exponential because $\gamma^0(\cdots)^n\gamma^0 = (\gamma^0(\cdots)\gamma^0)^n$, so

$$\gamma^0\, S(\Lambda)^\dagger\,\gamma^0 = \exp\!\Big(+\tfrac{i}{2}\omega_{\mu\nu}\,\gamma^0 S^{\mu\nu\dagger}\gamma^0\Big) = \exp\!\Big(+\tfrac{i}{2}\omega_{\mu\nu}\,S^{\mu\nu}\Big) = S(\Lambda)^{-1}, \tag{11.33}$$

using Eq. (11.32) in the middle step and recognizing the result as the inverse of Eq. (11.8). This is Eq. (11.29). Finally, $\bar{\psi} = \psi^\dagger\gamma^0 \to (S\psi)^\dagger\gamma^0 = \psi^\dagger S^\dagger\gamma^0 = \psi^\dagger\gamma^0\,(\gamma^0 S^\dagger\gamma^0) = \bar{\psi}\,S(\Lambda)^{-1}$, which is Eq. (11.28). □

The payoff is immediate. The scalar bilinear is genuinely a scalar:

$$\bar{\psi}\psi \;\longmapsto\; \bar{\psi}\, S(\Lambda)^{-1} S(\Lambda)\, \psi = \bar{\psi}\psi, \tag{11.34}$$

which confirms that the mass term $-m\bar{\psi}\psi$ in Eq. (11.25) is Lorentz invariant; and the kinetic term is invariant by the same cancellation combined with the intertwining Eq. (11.10), since $\bar{\psi}\gamma^\mu\partial_\mu\psi \to \bar{\psi}S^{-1}\gamma^\mu S\,\Lambda_\mu{}^\nu\partial_\nu\psi = \bar{\psi}\,\Lambda^\mu{}_\rho\Lambda_\mu{}^\nu\gamma^\rho\partial_\nu\psi = \bar{\psi}\partial\!\!\!/\psi$. The Dirac Lagrangian is a Lorentz scalar, as a Lagrangian must be.

The vector bilinear transforms as a four-vector,

$$\bar{\psi}\gamma^\mu\psi \;\longmapsto\; \bar{\psi}\, S(\Lambda)^{-1}\gamma^\mu S(\Lambda)\, \psi = \Lambda^\mu{}_\nu\, \bar{\psi}\gamma^\nu\psi, \tag{11.35}$$

again by Eq. (11.10). This is exactly the object the chapter's quantum theory will need as a conserved current. The Dirac Lagrangian Eq. (11.25) is invariant under the global U(1) phase rotation

$$\psi \to e^{-i\alpha}\psi, \qquad \bar{\psi} \to e^{+i\alpha}\bar{\psi}, \qquad \alpha \text{ constant}, \tag{11.36}$$

since the phases cancel in every term ($\bar{\psi}\psi$ and $\bar{\psi}\partial\!\!\!/\psi$ each pair one ψ with one $\bar{\psi}$). By Noether's first theorem (Theorem 7.10), applied exactly as for the complex scalar of Section 7.3.2 but now with the field-space generator acting as multiplication by $\mp i$ on $(\psi, \bar{\psi})$, the associated conserved current is read off from Eq. (7.32) as

$$\boxed{j^\mu = \bar{\psi}\gamma^\mu\psi}, \qquad \partial_\mu j^\mu = 0 \;\text{ on shell.} \tag{11.37}$$

We verify the conservation directly, since the check is short and exhibits the interplay of the two equations of motion.

Theorem 11.5 (Conservation of the vector current)**.** *On any solution of the Dirac equation, $\partial_\mu j^\mu = 0$ with $j^\mu = \bar{\psi}\gamma^\mu\psi$.*

Proof. Differentiate and apply the Leibniz rule:

$$\partial_\mu j^\mu = (\partial_\mu\bar{\psi})\gamma^\mu\psi + \bar{\psi}\gamma^\mu(\partial_\mu\psi). \tag{11.38}$$

The Dirac equation Eq. (11.13) gives $i\gamma^\mu\partial_\mu\psi = m\psi$, hence $\bar{\psi}\gamma^\mu\partial_\mu\psi = -im\,\bar{\psi}\psi$. The adjoint equation Eq. (11.26) gives $i(\partial_\mu\bar{\psi})\gamma^\mu = -m\bar{\psi}$, hence $(\partial_\mu\bar{\psi})\gamma^\mu\psi = +im\,\bar{\psi}\psi$. Adding,

$$\partial_\mu j^\mu = +im\,\bar{\psi}\psi - im\,\bar{\psi}\psi = 0. \tag{11.39}$$

The two equal-and-opposite mass terms cancel, exactly the mechanism by which the complex-scalar current was conserved in Section 7.3.2. □

Physical Insight — QED: identifying the electron field

The conserved charge $Q = \int d^3x\, j^0 = \int d^3x\, \bar\psi\gamma^0\psi = \int d^3x\, \psi^\dagger\psi$ is the U(1) charge whose phase symmetry Eq. (11.36) is the global precursor of the gauge symmetry of electromagnetism. This is the point at which QED acquires its matter content: the field ψ built in this section *is* the electron field, continuing the classical U(1) logic of Chapter 8, and the current Eq. (11.37) is the electromagnetic current that the photon of Chapter 13 will couple to. The coupling is installed by the minimal substitution $\partial_\mu \to D_\mu = \partial_\mu + ieA_\mu$ acting on ψ (electron charge $-e$, with $e > 0$), the convention the interaction chapters will use; consistent with $D_\mu = \partial_\mu - iqA_\mu$ at charge $q = -e$. We flag the sign here and switch on no gauge coupling yet: this chapter quantizes the *free* electron field, and the photon and the vertex are the business of Chapter 13 and Part V.

11.1.5 The sixteen bilinear covariants

The scalar $\bar\psi\psi$ and vector $\bar\psi\gamma^\mu\psi$ are two members of a complete family of fermion bilinears, one for each element of a basis of the 4×4 matrices. There are sixteen linearly independent 4×4 matrices, and Theorem C.2 organizes them into five Lorentz-covariant sets,

$$\Gamma \in \{\, \mathbf{1},\ \gamma^\mu,\ \sigma^{\mu\nu},\ \gamma^\mu\gamma^5,\ \gamma^5 \,\}, \qquad 1+4+6+4+1 = 16, \tag{11.40}$$

where $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ from Eq. (11.11) contributes its six independent antisymmetric components. The corresponding bilinears $\bar\psi\,\Gamma\,\psi$ are the *bilinear covariants*; the entire local content of a Dirac field — mass terms, currents, dipole couplings, the weak interaction's $V-A$ structure — is assembled from them, which is why the QED chapters (Chapters 17 and 19) and the electroweak chapters will draw on the classification below as a standing reference.

The transformation of each bilinear under a proper orthochronous Lorentz transformation follows from Theorem 11.4 and the intertwining identity, exactly as for the scalar and vector above. The new ingredient needed to complete the table is the behavior of γ^5 under both proper transformations and parity, which sorts the five sets into scalar/pseudoscalar and vector/axial pairs.

Theorem 11.6 (Lorentz classification of the bilinears)**.** *Under a proper orthochronous Lorentz transformation $\Lambda \in \mathrm{SO}^+(1,3)$ acting by $\psi \to S(\Lambda)\psi$, and under parity P acting by $\psi(t,\mathbf{x}) \to \gamma^0\psi(t,-\mathbf{x})$, the bilinears transform as catalogued in Table 11.1: $\bar\psi\psi$ is a scalar, $\bar\psi\gamma^5\psi$ a pseudoscalar, $\bar\psi\gamma^\mu\psi$ a vector, $\bar\psi\gamma^\mu\gamma^5\psi$ an axial vector, and $\bar\psi\sigma^{\mu\nu}\psi$ an antisymmetric rank-two tensor.*

Proof. Proper transformations. The key auxiliary fact is that γ^5 commutes with

every generator,

$$\left[\gamma^5, S^{\mu\nu}\right] = 0, \tag{11.41}$$

because γ^5 anticommutes with each γ^μ (Eq. (11.4)), so it commutes with the product of any two of them, hence with $S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$. Therefore γ^5 commutes with $S(\Lambda) = \exp(-\frac{i}{2}\omega_{\mu\nu}S^{\mu\nu})$, giving $S(\Lambda)^{-1}\gamma^5 S(\Lambda) = \gamma^5$. With Eq. (11.28) the five proper transformations follow by sandwiching $S^{-1}\Gamma S$:

$$\begin{aligned}
\bar\psi\psi &\to \bar\psi S^{-1}S\psi = \bar\psi\psi,\\
\bar\psi\gamma^5\psi &\to \bar\psi S^{-1}\gamma^5 S\psi = \bar\psi\gamma^5\psi,\\
\bar\psi\gamma^\mu\psi &\to \bar\psi S^{-1}\gamma^\mu S\psi = \Lambda^\mu{}_\nu\,\bar\psi\gamma^\nu\psi,\\
\bar\psi\gamma^\mu\gamma^5\psi &\to \bar\psi S^{-1}\gamma^\mu\gamma^5 S\psi = \Lambda^\mu{}_\nu\,\bar\psi\gamma^\nu\gamma^5\psi,\\
\bar\psi\sigma^{\mu\nu}\psi &\to \bar\psi S^{-1}\sigma^{\mu\nu}S\psi = \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\,\bar\psi\sigma^{\rho\sigma}\psi,
\end{aligned} \tag{11.42}$$

where the vector and axial cases use $S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu\gamma^\nu$ from Eq. (11.10) (with γ^5 slipped through unchanged by Eq. (11.41)), and the tensor case uses it twice, $S^{-1}\sigma^{\mu\nu}S = \frac{i}{2}S^{-1}[\gamma^\mu,\gamma^\nu]S = \frac{i}{2}[\Lambda^\mu{}_\rho\gamma^\rho, \Lambda^\nu{}_\sigma\gamma^\sigma] = \Lambda^\mu{}_\rho\Lambda^\nu{}_\sigma\sigma^{\rho\sigma}$. Under proper transformations $\bar\psi\psi$ and $\bar\psi\gamma^5\psi$ are indistinguishable — both invariant — and likewise $\bar\psi\gamma^\mu\psi$ and $\bar\psi\gamma^\mu\gamma^5\psi$ both transform as vectors. Parity breaks the degeneracy.

Parity. Under P the field becomes $\psi(t,\mathbf{x}) \to \psi_P(t,\mathbf{x}) = \gamma^0\psi(t,-\mathbf{x})$, and the adjoint $\bar\psi = \psi^\dagger\gamma^0 \to (\gamma^0\psi)^\dagger\gamma^0 = \psi^\dagger\gamma^0\gamma^0 = \psi^\dagger$, i.e. $\bar\psi_P = \bar\psi\gamma^0$ (all fields evaluated at the parity-reflected point, suppressed below). The two relations we need are $\gamma^0\gamma^0\gamma^0 = \gamma^0$ and, for spatial i, $\gamma^0\gamma^i\gamma^0 = -\gamma^i$, both consequences of the Clifford algebra; together they read $\gamma^0\gamma^\mu\gamma^0 = \gamma^\mu{}_P$, where $\gamma^\mu{}_P \equiv (\gamma^0, -\gamma^i)$ has its spatial part reversed. Also $\gamma^0\gamma^5\gamma^0 = -\gamma^5$, since γ^5 anticommutes with γ^0. Then

$$\begin{aligned}
\bar\psi\psi &\to \bar\psi\gamma^0\gamma^0\psi = +\bar\psi\psi && \text{(scalar: parity-even)},\\
\bar\psi\gamma^5\psi &\to \bar\psi\gamma^0\gamma^5\gamma^0\psi = -\bar\psi\gamma^5\psi && \text{(pseudoscalar: parity-odd)},\\
\bar\psi\gamma^\mu\psi &\to \bar\psi\gamma^0\gamma^\mu\gamma^0\psi = \bar\psi\gamma^\mu{}_P\psi && \text{(vector: spatial part flips)},\\
\bar\psi\gamma^\mu\gamma^5\psi &\to \bar\psi\gamma^0\gamma^\mu\gamma^5\gamma^0\psi = -\bar\psi\gamma^\mu{}_P\gamma^5\psi && \text{(axial: spatial part } \textit{does not} \text{ flip)},
\end{aligned} \tag{11.43}$$

the extra sign in the axial case coming from $\gamma^0\gamma^5\gamma^0 = -\gamma^5$, which combines with the spatial flip of γ^μ to leave the spatial components unchanged and reverse the time component — the defining signature of an axial (pseudo-)vector. Thus parity is exactly the operation that distinguishes the scalar from the pseudoscalar and the polar vector from the axial vector. This completes the classification. □

Table 11.1 records the result. The "Parity" column for the vector lists the signs $(+,-,-,-)$ acquired by the components ($\mu = 0,1,2,3$): the timelike component (charge density $\bar\psi\gamma^0\psi = \psi^\dagger\psi$) is parity-even and the spacelike components (the current $\bar\psi\gamma^i\psi$) are parity-odd, exactly as a polar vector like the electric

Table 11.1: The sixteen bilinear covariants $\bar{\psi}\,\Gamma\,\psi$ of a Dirac field, classified by their behavior under proper orthochronous Lorentz transformations $\Lambda \in \mathrm{SO}^+(1,3)$ and under parity P. The number of independent components sums to $16 = \dim(4\times 4 \text{ matrices})$, the basis being complete by Theorem C.2. This table is the standing reference drawn on by the QED chapters (Chapters 17 and 19) and the weak-interaction chapters.

Γ	Bilinear	Lorentz type	# comp.	Parity P
$\mathbf{1}$	$\bar{\psi}\psi$	scalar (S)	1	$+$
γ^5	$\bar{\psi}\gamma^5\psi$	pseudoscalar (P)	1	$-$
γ^μ	$\bar{\psi}\gamma^\mu\psi$	vector (V)	4	$(+,-,-,-)$
$\gamma^\mu\gamma^5$	$\bar{\psi}\gamma^\mu\gamma^5\psi$	axial vector (A)	4	$(-,+,+,+)$
$\sigma^{\mu\nu}$	$\bar{\psi}\sigma^{\mu\nu}\psi$	antisymmetric tensor (T)	6	(as $F_{\mu\nu}$)

current should behave. The axial vector carries the opposite pattern $(-,+,+,+)$, the hallmark of a quantity like spin or magnetic field that does not reverse under spatial reflection. The pseudoscalar's single sign flip is what makes $\bar{\psi}\gamma^5\psi$ the natural coupling for a parity-odd scalar particle (the pion, in the strong interactions developed much later). The worked computation below carries the pseudoscalar and axial cases explicitly, since the sign that distinguishes them from their parity-even partners is the single most error-prone step in the whole catalogue.

Worked Example 11.7 (The pseudoscalar and axial vector under parity)**.** We trace the parity signs of $\bar{\psi}\gamma^5\psi$ and $\bar{\psi}\gamma^\mu\gamma^5\psi$ in full, using only $\gamma^0\gamma^0 = \mathbf{1}$, $\{\gamma^5,\gamma^\mu\} = 0$, and the parity action $\psi \to \gamma^0\psi$, $\bar{\psi} \to \bar{\psi}\gamma^0$ (the spatial argument flip $\mathbf{x} \to -\mathbf{x}$ is understood throughout and does not affect the matrix algebra at a point).

Pseudoscalar. Sandwiching γ^5 between the transformed adjoint and field,

$$\bar{\psi}\gamma^5\psi \;\longmapsto\; (\bar{\psi}\gamma^0)\,\gamma^5\,(\gamma^0\psi) = \bar{\psi}\,(\gamma^0\gamma^5\gamma^0)\,\psi. \tag{11.44}$$

Move the first γ^0 through γ^5 using $\gamma^0\gamma^5 = -\gamma^5\gamma^0$ (from $\{\gamma^5,\gamma^0\} = 0$):

$$\gamma^0\gamma^5\gamma^0 = -\gamma^5\gamma^0\gamma^0 = -\gamma^5\,\mathbf{1} = -\gamma^5. \tag{11.45}$$

Hence $\bar{\psi}\gamma^5\psi \to -\bar{\psi}\gamma^5\psi$: a scalar under proper rotations (by Eq. (11.42)) but odd under parity — a pseudoscalar.

Axial vector. Now sandwich $\gamma^\mu\gamma^5$:

$$\bar{\psi}\gamma^\mu\gamma^5\psi \;\longmapsto\; \bar{\psi}\,(\gamma^0\gamma^\mu\gamma^5\gamma^0)\,\psi = \bar{\psi}\,(\gamma^0\gamma^\mu\gamma^0)(\gamma^0\gamma^5\gamma^0)\,\psi, \tag{11.46}$$

where we inserted $\gamma^0\gamma^0 = \mathbf{1}$ between γ^μ and γ^5 to split the product into two already-computed factors. We have $\gamma^0\gamma^5\gamma^0 = -\gamma^5$ from above, and $\gamma^0\gamma^\mu\gamma^0 = \gamma^\mu{}_P$,

i.e. $+\gamma^0$ for $\mu = 0$ and $-\gamma^i$ for $\mu = i$. Combining,

$$\bar{\psi}\gamma^\mu\gamma^5\psi \;\longmapsto\; -\bar{\psi}\,\gamma^\mu{}_P\gamma^5\,\psi = \begin{cases} -\bar{\psi}\gamma^0\gamma^5\psi, & \mu = 0, \\ +\bar{\psi}\gamma^i\gamma^5\psi, & \mu = i. \end{cases} \tag{11.47}$$

The time component flips sign and the spatial components are preserved — precisely the reverse of the polar vector $\bar{\psi}\gamma^\mu\psi$, whose time component is preserved and space components flipped. The product of the two extra signs (one from $\gamma^0\gamma^5\gamma^0 = -\gamma^5$, one from the γ^μ reflection) is what converts a polar vector into an axial vector. Quantum-mechanically this is why the spin operator, built from $\bar{\psi}\gamma^i\gamma^5\psi$-type bilinears, is parity-even while the momentum, a polar vector, is parity-odd.

The catalogue is now complete, and with it the classical kinematics of the Dirac field. We have shown that Lorentz covariance forces the four-component spinor, that the demand of a first-order equation reproducing the mass shell forces the Dirac equation $(i\partial\!\!\!/ - m)\psi = 0$, that the equation is form-invariant by virtue of the intertwining identity, and that the Dirac Lagrangian $\bar{\psi}(i\partial\!\!\!/ - m)\psi$ delivers a conserved vector current $j^\mu = \bar{\psi}\gamma^\mu\psi$ together with the sixteen bilinear covariants of Table 11.1. Every object the quantum theory will manipulate — the field, its adjoint, its canonical momentum $\pi_\psi = i\psi^\dagger$, its current, and the spin and dipole bilinears — is now specified classically. What remains before quantization is to solve the equation: to construct the explicit positive- and negative-frequency plane-wave solutions of $(p\!\!\!/ - m)u(p) = 0$ and $(p\!\!\!/ + m)v(p) = 0$, fix their normalizations, and prove the spin-sum and projector identities that will serve both as the mode basis for the canonical expansion and as the raw material for every QED trace. That construction is the work of Section 11.2.

11.2 Plane-Wave Solutions, Spin Sums, and Projectors

The previous section assembled the classical Dirac field: the unique Lorentz-covariant first-order equation $(i\partial\!\!\!/ - m)\psi = 0$ living on the reducible $(\tfrac{1}{2}, 0) \oplus (0, \tfrac{1}{2})$ representation, together with the gamma algebra, the chiral basis, and the bilinear covariants the quantum theory will need. Having the equation is not yet having its *solutions*. Before we can promote ψ to an operator we must know the complete set of classical wave packets it will be built from, exactly as the scalar field of Chapter 9 was assembled out of the plane waves $e^{\mp ip\cdot x}$. For a four-component spinor field the plane waves carry, in addition to their frequency, a polarization: a constant spinor amplitude that the Dirac operator constrains. The whole of this section is the determination of those amplitudes. We will find precisely two independent positive-frequency amplitudes $u^s(p)$ and

two negative-frequency amplitudes $v^s(p)$ at each on-shell momentum, build them explicitly in the chiral basis by boosting their rest-frame form, and then prove the algebra they satisfy: the normalization and orthogonality relations that will let us *invert* the mode expansion of the next section, the completeness (spin-sum) identities $\sum_s u^s\bar{u}^s = \not{p} + m$ and $\sum_s v^s\bar{v}^s = \not{p} - m$, and the projector technology—energy projectors $\Lambda_\pm$, chirality and helicity—that organizes every later spinor computation. The spin sums in particular are not a local convenience: they are the exact identities that Chapter 17 will trace over to obtain the unpolarized cross section for $e^+e^- \to \mu^+\mu^-$, the computational core of QED. This section proves them once, in full.

Throughout we use the chiral (Weyl) representation fixed in Section C.3.2,

$$\gamma^0 = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}, \qquad \gamma^\mu = \begin{pmatrix} \mathbf{0} & \sigma^\mu \\ \bar{\sigma}^\mu & \mathbf{0} \end{pmatrix}, \tag{11.48}$$

with the two-component packaging $\sigma^\mu = (I, \sigma^i)$, $\bar{\sigma}^\mu = (I, -\sigma^i)$ of Eq. (C.58) and the Pauli matrices of Eq. (C.54). Every statement we prove is, by Pauli's theorem (Section C.2), basis independent in its invariant content—the normalizations, the spin sums, and the projector identities are scalars or covariant matrices—so the reader may translate any result to the Dirac or Majorana basis by the explicit unitaries of Section C.3 without changing a single number. The on-shell energy is $E_{\mathbf{p}} = +\sqrt{\mathbf{p}^2 + m^2}$ and the Lorentz-invariant measure is $\widetilde{dp} = d^3p/[(2\pi)^3 2E_{\mathbf{p}}]$, both imported verbatim from Eq. (9.211); the relativistic normalization $u^\dagger u = 2E_{\mathbf{p}}$ we are about to derive is the one paired with that measure.

11.2.1 The momentum-space Dirac equation

A free Dirac field, like a free scalar, is a superposition of plane waves. The novelty is that the amplitude multiplying each plane wave is a four-component constant spinor, and that the two signs of frequency carry independent amplitudes. We therefore posit the two families of elementary solutions

$$\psi^{(+)}(x) = u(p)\, e^{-ip\cdot x}, \qquad \psi^{(-)}(x) = v(p)\, e^{+ip\cdot x}, \qquad p^0 = E_{\mathbf{p}} > 0, \tag{11.49}$$

where $u(p)$ and $v(p)$ are x-independent spinors to be determined and, in both cases, $p^\mu = (E_{\mathbf{p}}, \mathbf{p})$ is a genuine future-pointing on-shell momentum. The label "positive frequency" refers to the sign in the exponent $e^{-iE_{\mathbf{p}}x^0}$ of $\psi^{(+)}$; the "negative-frequency" wave $\psi^{(-)}$ oscillates as $e^{+iE_{\mathbf{p}}x^0}$. Writing both with a manifestly positive p^0 and carrying the sign in the exponent—rather than allowing $p^0 < 0$—is the bookkeeping that will let Section 11.3 reinterpret the negative-frequency half as antiparticles, exactly as the scalar did in Section 9.7.

Insert the ansätze (11.49) into the position-space Dirac equation $(i\not{\partial} - m)\psi = 0$ of Eq. (11.13). Differentiation brings down a momentum factor: $\partial_\mu e^{\mp ip\cdot x} =$

$\mp i p_\mu e^{\mp ip\cdot x}$, so that $i\partial\!\!\!/$ acts on $\psi^{(\pm)}$ as multiplication by $\pm p\!\!\!/$. The exponentials cancel, leaving two purely algebraic conditions on the constant spinors,

$$(p\!\!\!/ - m)\, u(p) = 0, \tag{11.50}$$

$$(p\!\!\!/ + m)\, v(p) = 0, \tag{11.51}$$

with $p\!\!\!/ = \gamma^\mu p_\mu$. These are the *momentum-space Dirac equations*. Equation (11.50) is a homogeneous linear system; it has nontrivial solutions only where $\det(p\!\!\!/ - m) = 0$. We can read off that determinant without effort using the Clifford algebra. Multiply (11.50) on the left by $(p\!\!\!/ + m)$:

$$(p\!\!\!/+m)(p\!\!\!/-m)\, u = (p\!\!\!/^2 - m^2)\, u = 0, \qquad p\!\!\!/^2 = p_\mu p_\nu\, \gamma^\mu\gamma^\nu = p_\mu p_\nu\, \tfrac{1}{2}\{\gamma^\mu,\gamma^\nu\} = p_\mu p_\nu\, \eta^{\mu\nu}\, \mathbf{1} = p^2\, \mathbf{1}, \tag{11.52}$$

where the symmetrization is legitimate because $p_\mu p_\nu$ is symmetric and we used the Clifford relation (C.184). Thus a nonzero u requires $p^2 = m^2$, i.e. $p^0 = \pm E_{\mathbf{p}}$; the same computation applied to (11.51) gives the identical mass-shell condition. The constant spinor amplitude exists only on shell, and (11.52) is the spinorial echo of the fact—already visible in (11.49)—that each Dirac plane wave is a Klein–Gordon plane wave wearing a polarization.

It will be convenient to have $p\!\!\!/$ in 2×2 block form. From (11.48),

$$p\!\!\!/ = p_\mu\gamma^\mu = \begin{pmatrix} \mathbf{0} & p_\mu\sigma^\mu \\ p_\mu\bar\sigma^\mu & \mathbf{0} \end{pmatrix} \equiv \begin{pmatrix} \mathbf{0} & p\cdot\sigma \\ p\cdot\bar\sigma & \mathbf{0} \end{pmatrix}, \qquad \begin{aligned} p\cdot\sigma &\equiv p_\mu\sigma^\mu = E_{\mathbf{p}}\, I - \mathbf{p}\cdot\boldsymbol{\sigma}, \\ p\cdot\bar\sigma &\equiv p_\mu\bar\sigma^\mu = E_{\mathbf{p}}\, I + \mathbf{p}\cdot\boldsymbol{\sigma}, \end{aligned} \tag{11.53}$$

where $\mathbf{p}\cdot\boldsymbol{\sigma} = p^i\sigma^i$ and we used $p_i = -p^i$. The two Hermitian 2×2 matrices $p\cdot\sigma$ and $p\cdot\bar\sigma$ are the workhorses of everything below. Two of their properties we will use constantly. First, they are positive definite for a massive on-shell momentum: their eigenvalues are $E_{\mathbf{p}} \mp |\mathbf{p}|$, both strictly positive since $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} > |\mathbf{p}|$. Second, their product is a multiple of the identity,

$$(p\cdot\sigma)(p\cdot\bar\sigma) = (E_{\mathbf{p}} - \mathbf{p}\cdot\boldsymbol{\sigma})(E_{\mathbf{p}} + \mathbf{p}\cdot\boldsymbol{\sigma}) = E_{\mathbf{p}}^2 - (\mathbf{p}\cdot\boldsymbol{\sigma})^2 = E_{\mathbf{p}}^2 - \mathbf{p}^2\, I = m^2\, I, \tag{11.54}$$

using $(\mathbf{p}\cdot\boldsymbol{\sigma})^2 = p^i p^j \sigma^i\sigma^j = p^i p^j \delta^{ij} I = \mathbf{p}^2 I$ from the Pauli algebra (C.38); the same holds with the factors reversed. Because each is positive definite it has a unique positive-definite Hermitian square root, written $\sqrt{p\cdot\sigma}$ and $\sqrt{p\cdot\bar\sigma}$; both are functions of the single Hermitian matrix $\mathbf{p}\cdot\boldsymbol{\sigma}$, hence commute with each other and with $\mathbf{p}\cdot\boldsymbol{\sigma}$. Equation (11.54) then gives the relation we will invoke at almost every step,

$$\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar\sigma} = \sqrt{p\cdot\bar\sigma}\,\sqrt{p\cdot\sigma} = m\, I. \tag{11.55}$$

11.2.2 Rest-frame solutions and the boost

The cleanest way to solve (11.50) and (11.51) is to solve them once in the particle's rest frame, where the answer is transparent, and then carry the solution

to a general frame. In the rest frame $\mathbf{p}=0$, $p^\mu=(m,\mathbf{0})$, so $p\cdot\sigma=p\cdot\bar\sigma=mI$ and $\not p=m\gamma^0$. Equation (11.50) becomes $m(\gamma^0-1)u=0$, i.e. $\gamma^0u=u$: the rest-frame positive-frequency spinor is an eigenvector of γ^0 with eigenvalue $+1$. Writing a four-spinor in the chiral basis as a pair of two-component blocks, $u=(u_L,u_R)^{\mathsf T}$, and using $\gamma^0=\left(\begin{smallmatrix}0&I\\I&0\end{smallmatrix}\right)$, the condition $\gamma^0u=u$ reads $u_R=u_L$. Thus the positive-frequency rest-frame solutions are the spinors with equal upper and lower blocks. There are exactly two linearly independent such spinors, one for each choice of the common two-component block, and we label them by an orthonormal basis ξ^s $(s=1,2)$ of $\mathbb{C}^2$,

$$\xi^1=\begin{pmatrix}1\\0\end{pmatrix},\qquad \xi^2=\begin{pmatrix}0\\1\end{pmatrix},\qquad \xi^{r\dagger}\xi^s=\delta^{rs},\qquad \sum_{s=1,2}\xi^s\xi^{s\dagger}=I. \tag{11.56}$$

Likewise, in the rest frame (11.51) becomes $m(\gamma^0+1)v=0$, i.e. $\gamma^0v=-v$: the negative-frequency rest-frame spinor is the -1 eigenvector of γ^0, which in block form means $v_R=-v_L$. Again there are two, labelled by a basis η^s of $\mathbb{C}^2$ with $\eta^{r\dagger}\eta^s=\delta^{rs}$ and $\sum_s\eta^s\eta^{s\dagger}=I$. The reducible four-component representation has split, in the rest frame, into a two-dimensional space of positive-frequency amplitudes and a two-dimensional space of negative-frequency ones; the factor of two in each is the spin degeneracy of a massive spin-$\frac12$ particle, and the choice of basis ξ^s, η^s is a choice of spin quantization axis (the z-axis for the standard (11.56)).

To reach a general momentum we could apply the explicit spinor boost $S(\Lambda)$ of Eq. (11.8), but it is faster, and self-contained, to write down the boosted spinor and verify it solves the momentum-space equation directly. We claim that

$$\boxed{u^s(p)=\begin{pmatrix}\sqrt{p\cdot\sigma}\,\xi^s\\ \sqrt{p\cdot\bar\sigma}\,\xi^s\end{pmatrix},\qquad v^s(p)=\begin{pmatrix}\sqrt{p\cdot\sigma}\,\eta^s\\ -\sqrt{p\cdot\bar\sigma}\,\eta^s\end{pmatrix},\qquad s=1,2,} \tag{11.57}$$

are the four independent on-shell solutions. These reduce correctly in the rest frame: there $\sqrt{p\cdot\sigma}=\sqrt{p\cdot\bar\sigma}=\sqrt{m}\,I$, so $u^s=\sqrt{m}\,(\xi^s,\xi^s)^{\mathsf T}$ has equal blocks and $v^s=\sqrt{m}\,(\eta^s,-\eta^s)^{\mathsf T}$ has opposite blocks, matching the rest-frame analysis. We record this as a definition and then prove it solves the equations.

Definition 11.8 (Plane-wave spinor basis)**.** For each on-shell momentum $p^\mu=(E_{\mathbf p},\mathbf p)$ and each $s=1,2$, the *positive-* and *negative-frequency spinor amplitudes* are the four-component spinors (11.57), built from the rest-frame two-spinors (11.56) and the positive square roots of the Hermitian matrices (11.53). The associated elementary solutions of the Dirac equation are $u^s(p)e^{-ip\cdot x}$ and $v^s(p)e^{+ip\cdot x}$.

Proposition 11.9 (The amplitudes solve the momentum-space equations)**.** *The spinors* (11.57) *satisfy* $(\not p-m)u^s(p)=0$ *and* $(\not p+m)v^s(p)=0$.

Proof. Write $\not{p} - m = \left(\begin{smallmatrix} -m\,I & p\cdot\sigma \\ p\cdot\bar\sigma & -m\,I \end{smallmatrix}\right)$ from (11.53). Acting on u^s,

$$(\not{p} - m)u^s = \begin{pmatrix} -m\sqrt{p\cdot\sigma}\,\xi^s + (p\cdot\sigma)\sqrt{p\cdot\bar\sigma}\,\xi^s \\ (p\cdot\bar\sigma)\sqrt{p\cdot\sigma}\,\xi^s - m\sqrt{p\cdot\bar\sigma}\,\xi^s \end{pmatrix}. \tag{11.58}$$

In the upper block, $(p\cdot\sigma)\sqrt{p\cdot\bar\sigma} = \sqrt{p\cdot\sigma}\,(\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar\sigma}) = \sqrt{p\cdot\sigma}\,(mI) = m\sqrt{p\cdot\sigma}$ by (11.55), so the two terms cancel. In the lower block, $(p\cdot\bar\sigma)\sqrt{p\cdot\sigma} = \sqrt{p\cdot\bar\sigma}\,(\sqrt{p\cdot\bar\sigma}\,\sqrt{p\cdot\sigma}) = m\sqrt{p\cdot\bar\sigma}$, and again the terms cancel. Hence $(\not{p}-m)u^s = 0$. For v^s, with $\not{p}+m = \left(\begin{smallmatrix} m\,I & p\cdot\sigma \\ p\cdot\bar\sigma & m\,I \end{smallmatrix}\right)$,

$$(\not{p}+m)v^s = \begin{pmatrix} m\sqrt{p\cdot\sigma}\,\eta^s - (p\cdot\sigma)\sqrt{p\cdot\bar\sigma}\,\eta^s \\ (p\cdot\bar\sigma)\sqrt{p\cdot\sigma}\,\eta^s - m\sqrt{p\cdot\bar\sigma}\,\eta^s \end{pmatrix} = \begin{pmatrix} m\sqrt{p\cdot\sigma}\,\eta^s - m\sqrt{p\cdot\sigma}\,\eta^s \\ m\sqrt{p\cdot\bar\sigma}\,\eta^s - m\sqrt{p\cdot\bar\sigma}\,\eta^s \end{pmatrix} = 0, \tag{11.59}$$

using (11.55) in both blocks. □

The four spinors u^1, u^2, v^1, v^2 are linearly independent: in the rest frame they are visibly so (two have equal blocks, two opposite, and the ξ^s, η^s are bases), and linear independence is preserved by the invertible boost that carries the rest frame to p. Since the momentum-space equations (11.50)–(11.51) each have a two-dimensional solution space—the projectors of Section 11.2.5 will make this count manifest, $\Lambda_\pm$ having trace 2—these four exhaust the solutions. The basis is complete.

To make (11.57) fully explicit one needs the square roots as concrete matrices. For any Hermitian 2×2 matrix of the form $A = aI + \mathbf{b}\cdot\boldsymbol{\sigma}$ with $a > |\mathbf{b}|$, the positive square root has the closed form $\sqrt{A} = (A + \sqrt{\det A}\,I)/\sqrt{\operatorname{tr}A + 2\sqrt{\det A}}$, as one checks by squaring. Applied to $p\cdot\sigma$ (for which $\operatorname{tr} = 2E_{\mathbf{p}}$, $\det = m^2$, so $\sqrt{\det} = m$) this gives

$$\sqrt{p\cdot\sigma} = \frac{p\cdot\sigma + m\,I}{\sqrt{2(E_{\mathbf{p}}+m)}} = \frac{(E_{\mathbf{p}}+m)I - \mathbf{p}\cdot\boldsymbol{\sigma}}{\sqrt{2(E_{\mathbf{p}}+m)}}, \qquad \sqrt{p\cdot\bar\sigma} = \frac{(E_{\mathbf{p}}+m)I + \mathbf{p}\cdot\boldsymbol{\sigma}}{\sqrt{2(E_{\mathbf{p}}+m)}}, \tag{11.60}$$

the second following from the first by $\mathbf{p}\to-\mathbf{p}$. These are the formulae one actually substitutes into a numerical computation.

Worked Example 11.10 (Explicit spinors for momentum along $\hat{\mathbf{z}}$)**.** Take $\mathbf{p} = p\,\hat{\mathbf{z}}$ with $p \equiv |\mathbf{p}| > 0$, so $\mathbf{p}\cdot\boldsymbol{\sigma} = p\,\sigma^3 = \operatorname{diag}(p, -p)$. Then $p\cdot\sigma = \operatorname{diag}(E-p,\, E+p)$ and $p\cdot\bar\sigma = \operatorname{diag}(E+p,\, E-p)$, writing $E \equiv E_{\mathbf{p}}$, and the square roots are taken entrywise on these diagonal matrices,

$$\sqrt{p\cdot\sigma} = \begin{pmatrix} \sqrt{E-p} & 0 \\ 0 & \sqrt{E+p} \end{pmatrix}, \qquad \sqrt{p\cdot\bar\sigma} = \begin{pmatrix} \sqrt{E+p} & 0 \\ 0 & \sqrt{E-p} \end{pmatrix}. \tag{11.61}$$

With $\xi^1 = (1,0)^{\mathsf{T}}$, $\xi^2 = (0,1)^{\mathsf{T}}$ the positive-frequency amplitudes (11.57) are the four-component column vectors

$$u^1(p) = \begin{pmatrix} \sqrt{E-p} \\ 0 \\ \sqrt{E+p} \\ 0 \end{pmatrix}, \qquad u^2(p) = \begin{pmatrix} 0 \\ \sqrt{E+p} \\ 0 \\ \sqrt{E-p} \end{pmatrix}. \tag{11.62}$$

As a check of the relativistic normalization derived below, $u^{1\dagger}u^1 = (E-p) + (E+p) = 2E$, and indeed $u^{1\dagger}u^2 = 0$ since the nonzero entries sit in disjoint rows. The Lorentz-invariant normalization is $\bar{u}^1 u^1 = u^{1\dagger}\gamma^0 u^1$; with γ^0 swapping the upper and lower two-blocks, $\bar{u}^1 u^1 = 2\sqrt{(E-p)(E+p)} = 2\sqrt{E^2-p^2} = 2m$, again as it must. In the ultrarelativistic limit $p \to E$ (i.e. $m \to 0$) the entry $\sqrt{E-p} \to 0$, so $u^1 \to (0,0,\sqrt{2E},0)^{\mathsf{T}}$ collapses onto its lower (right-handed) block and $u^2 \to (0,\sqrt{2E},0,0)^{\mathsf{T}}$ onto its upper (left-handed) block: chirality and helicity coincide in the massless limit, the fact that powers the $V-A$ bookkeeping of Section 11.2.5.

11.2.3 Normalization and orthogonality

We can now compute every inner product of the basis spinors directly from (11.57). Two natural bilinears appear. The Lorentz *scalar* $\bar{\psi}\chi = \psi^\dagger\gamma^0\chi$ is the invariant pairing; the *Dirac inner product* $\psi^\dagger\chi$ is the time component of the conserved current $\bar{\psi}\gamma^\mu\chi$ and so transforms as a density, which is why it carries the energy factor that pairs with the invariant measure (9.211). We need both, and we need the cross terms between u and v. The one repeated tool is (11.55); the one repeated fact about γ^0 is that, acting on a block spinor, it exchanges the upper and lower two-blocks. In particular for $u^s = (\sqrt{p\cdot\sigma}\,\xi^s,\ \sqrt{p\cdot\bar{\sigma}}\,\xi^s)^{\mathsf{T}}$, using that the square roots are Hermitian,

$$u^{s\dagger} = (\xi^{s\dagger}\sqrt{p\cdot\sigma}\ ,\ \xi^{s\dagger}\sqrt{p\cdot\bar{\sigma}}), \qquad \bar{u}^s = u^{s\dagger}\gamma^0 = (\xi^{s\dagger}\sqrt{p\cdot\bar{\sigma}}\ ,\ \xi^{s\dagger}\sqrt{p\cdot\sigma}), \tag{11.63}$$

the row vector being a 1×4 object split into two 1×2 blocks; γ^0 has swapped them. Likewise $v^{s\dagger} = (\eta^{s\dagger}\sqrt{p\cdot\sigma},\ -\eta^{s\dagger}\sqrt{p\cdot\bar{\sigma}})$ and $\bar{v}^s = (-\eta^{s\dagger}\sqrt{p\cdot\bar{\sigma}},\ \eta^{s\dagger}\sqrt{p\cdot\sigma})$.

Proposition 11.11 (Normalizations and orthogonality)**.** *For all on-shell p and all $r,s \in \{1,2\}$,*

$$\bar{u}^r(p)\,u^s(p) = 2m\,\delta^{rs}, \qquad \bar{v}^r(p)\,v^s(p) = -2m\,\delta^{rs}, \tag{11.64}$$

$$u^{r\dagger}(\mathbf{p})\,u^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \qquad v^{r\dagger}(\mathbf{p})\,v^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \tag{11.65}$$

$$\bar{u}^r(p)\,v^s(p) = 0, \qquad \bar{v}^r(p)\,u^s(p) = 0, \tag{11.66}$$

and, with $\tilde{p}^\mu \equiv (E_{\mathbf{p}},-\mathbf{p})$ the spatially reflected momentum,

$$u^{r\dagger}(\mathbf{p})\,v^s(-\mathbf{p}) = 0, \qquad v^{r\dagger}(\mathbf{p})\,u^s(-\mathbf{p}) = 0. \tag{11.67}$$

Proof. Scalar normalizations (11.64). Contract the row $\bar{u}^r$ of (11.63) with the column u^s of (11.57):

$$\bar{u}^r u^s = \xi^{r\dagger}\sqrt{p\cdot\bar{\sigma}}\,\sqrt{p\cdot\sigma}\,\xi^s + \xi^{r\dagger}\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar{\sigma}}\,\xi^s = \xi^{r\dagger}(mI)\xi^s + \xi^{r\dagger}(mI)\xi^s = 2m\,\xi^{r\dagger}\xi^s = 2m\,\delta^{rs}, \tag{11.68}$$

using (11.55) twice and the orthonormality (11.56). For the v's,

$$\bar{v}^r v^s = (-\eta^{r\dagger}\sqrt{p\cdot\bar{\sigma}})(\sqrt{p\cdot\sigma}\,\eta^s) + (\eta^{r\dagger}\sqrt{p\cdot\sigma})(-\sqrt{p\cdot\bar{\sigma}}\,\eta^s) = -m\,\eta^{r\dagger}\eta^s - m\,\eta^{r\dagger}\eta^s = -2m\,\delta^{rs}. \tag{11.69}$$

The opposite sign is forced by the relative minus in the lower block of v^s; it is the spinor-level origin of the negative-norm danger that the next section's anticommutators resolve.

Dirac-product normalizations (11.65). Contract $u^{r\dagger}$ of (11.63) with u^s, now without the γ^0 swap:

$$u^{r\dagger}u^s = \xi^{r\dagger}\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\sigma}\,\xi^s + \xi^{r\dagger}\sqrt{p\cdot\bar{\sigma}}\,\sqrt{p\cdot\bar{\sigma}}\,\xi^s = \xi^{r\dagger}(p\cdot\sigma + p\cdot\bar{\sigma})\xi^s = \xi^{r\dagger}(2E_{\mathbf{p}}I)\xi^s = 2E_{\mathbf{p}}\,\delta^{rs}, \tag{11.70}$$

where $p\cdot\sigma + p\cdot\bar{\sigma} = 2E_{\mathbf{p}}I$ from (11.53). The same computation with η's gives $v^{r\dagger}v^s = \eta^{r\dagger}(p\cdot\sigma + p\cdot\bar{\sigma})\eta^s = 2E_{\mathbf{p}}\delta^{rs}$; the relative minus in v squares away and does not flip this sign.

Scalar orthogonality (11.66). Contract $\bar{u}^r$ with v^s:

$$\bar{u}^r v^s = \xi^{r\dagger}\sqrt{p\cdot\bar{\sigma}}\,\sqrt{p\cdot\sigma}\,\eta^s + \xi^{r\dagger}\sqrt{p\cdot\sigma}\,(-\sqrt{p\cdot\bar{\sigma}})\eta^s = m\,\xi^{r\dagger}\eta^s - m\,\xi^{r\dagger}\eta^s = 0, \tag{11.71}$$

identically in the choice of ξ^r, η^s; the two terms cancel by (11.55). The relation $\bar{v}^r u^s = 0$ follows by the same cancellation.

Mixed Dirac orthogonality (11.67). This is the relation needed to invert the mode expansion, where u and v appear at *opposite* spatial momenta. Reflecting $\mathbf{p} \to -\mathbf{p}$ sends $\mathbf{p}\cdot\boldsymbol{\sigma} \to -\mathbf{p}\cdot\boldsymbol{\sigma}$, hence $\tilde{p}\cdot\sigma = p\cdot\bar{\sigma}$ and $\tilde{p}\cdot\bar{\sigma} = p\cdot\sigma$ with $\tilde{p} = (E_{\mathbf{p}}, -\mathbf{p})$. Therefore $v^s(-\mathbf{p}) = (\sqrt{p\cdot\bar{\sigma}}\,\eta^s,\, -\sqrt{p\cdot\sigma}\,\eta^s)^{\mathsf{T}}$, and contracting with $u^{r\dagger}(\mathbf{p}) = (\xi^{r\dagger}\sqrt{p\cdot\sigma},\, \xi^{r\dagger}\sqrt{p\cdot\bar{\sigma}})$,

$$u^{r\dagger}(\mathbf{p})\,v^s(-\mathbf{p}) = \xi^{r\dagger}\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar{\sigma}}\,\eta^s + \xi^{r\dagger}\sqrt{p\cdot\bar{\sigma}}\,(-\sqrt{p\cdot\sigma})\eta^s = m\,\xi^{r\dagger}\eta^s - m\,\xi^{r\dagger}\eta^s = 0, \tag{11.72}$$

again term-by-term. The companion $v^{r\dagger}(\mathbf{p})u^s(-\mathbf{p}) = 0$ is identical with $\xi \leftrightarrow \eta$. □

Three features of Theorem 11.11 deserve emphasis before we use them. First, the invariant pairing $\bar{u}u = 2m$ *vanishes in the massless limit*—a chiral fermion has no Lorentz-scalar bilinear with itself—whereas the density $u^\dagger u = 2E_{\mathbf{p}}$ survives, which is precisely why the latter is the right normalization to carry through to the quantum field, where it pairs with $\widetilde{dp}$ to make $u^\dagger u\,\widetilde{dp}$ a Lorentz scalar. Second, the relative sign between $\bar{u}u > 0$ and $\bar{v}v < 0$ is not a defect of convention but a structural fact: it is the seed of the indefiniteness that, in the

next section, makes the naive (commutator) quantization of the Dirac field fail and selects anticommutators. Third, the orthogonality relations (11.66)–(11.67) are exactly what is needed to project a given field configuration onto a definite mode: in Section 11.2 we built the modes; relations (11.65) and (11.67) are the orthogonality that will let Section 11.3 read off each ladder operator by a Fourier-type inversion.

11.2.4 Completeness: the spin sums

The single most reused output of this chapter is the pair of *spin-sum* (completeness) identities. They express the fact that the two u's span the positive-energy subspace and the two v's the negative-energy subspace, packaged as 4×4 matrices. They are what makes the unpolarized cross section computable: when an experiment neither prepares nor measures spin, the matrix element is averaged over initial and summed over final spins, and these sums collapse the spin labels into the Lorentz-covariant matrices $\not{p}\pm m$, leaving a trace over gamma matrices. This is the mechanism by which Chapter 17 reduces $|\mathcal{M}|^2$ for $e^+e^-\to\mu^+\mu^-$ to traces; we prove the identities here, once.

Theorem 11.12 (Spin sums)**.** *The plane-wave amplitudes of Theorem 11.8 satisfy the outer-product completeness relations*

$$\sum_{s=1,2} u^s(p)\,\bar u^s(p) = \not{p} + m, \qquad \sum_{s=1,2} v^s(p)\,\bar v^s(p) = \not{p} - m, \tag{11.73}$$

as 4×4 matrix identities for every on-shell p.

Proof. Each term $u^s\bar u^s$ is the outer product of the column (11.57) with the row (11.63), a 4×4 matrix written in 2×2 blocks. For the u sum,

$$u^s\bar u^s = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi^s \\ \sqrt{p\cdot\bar\sigma}\,\xi^s \end{pmatrix}\left(\xi^{s\dagger}\sqrt{p\cdot\bar\sigma}\,,\ \xi^{s\dagger}\sqrt{p\cdot\sigma}\right) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi^s\xi^{s\dagger}\sqrt{p\cdot\bar\sigma} & \sqrt{p\cdot\sigma}\,\xi^s\xi^{s\dagger}\sqrt{p\cdot\sigma} \\ \sqrt{p\cdot\bar\sigma}\,\xi^s\xi^{s\dagger}\sqrt{p\cdot\bar\sigma} & \sqrt{p\cdot\bar\sigma}\,\xi^s\xi^{s\dagger}\sqrt{p\cdot\sigma} \end{pmatrix}. \tag{11.74}$$

Summing over s and using the two-component completeness $\sum_s \xi^s\xi^{s\dagger} = I$ from (11.56), every $\xi^s\xi^{s\dagger}$ becomes the identity, and the square roots collapse via (11.55) and (11.54):

$$\sum_s u^s\bar u^s = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar\sigma} & p\cdot\sigma \\ p\cdot\bar\sigma & \sqrt{p\cdot\bar\sigma}\,\sqrt{p\cdot\sigma} \end{pmatrix} = \begin{pmatrix} m\,I & p\cdot\sigma \\ p\cdot\bar\sigma & m\,I \end{pmatrix} = m\,\mathbf{1} + \begin{pmatrix} \mathbf{0} & p\cdot\sigma \\ p\cdot\bar\sigma & \mathbf{0} \end{pmatrix} = \not{p} + m, \tag{11.75}$$

where the off-diagonal blocks used $\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\sigma} = p\cdot\sigma$ and similarly for $\bar\sigma$, the diagonal blocks used (11.55), and the last step is the block form (11.53) of $\not{p}$. For the v sum, with $v^s = (\sqrt{p\cdot\sigma}\,\eta^s,\ -\sqrt{p\cdot\bar\sigma}\,\eta^s)^{\mathsf T}$ and $\bar v^s = (-\eta^{s\dagger}\sqrt{p\cdot\bar\sigma},\ \eta^{s\dagger}\sqrt{p\cdot\sigma})$,

$$v^s\bar v^s = \begin{pmatrix} -\sqrt{p\cdot\sigma}\,\eta^s\eta^{s\dagger}\sqrt{p\cdot\bar\sigma} & \sqrt{p\cdot\sigma}\,\eta^s\eta^{s\dagger}\sqrt{p\cdot\sigma} \\ \sqrt{p\cdot\bar\sigma}\,\eta^s\eta^{s\dagger}\sqrt{p\cdot\bar\sigma} & -\sqrt{p\cdot\bar\sigma}\,\eta^s\eta^{s\dagger}\sqrt{p\cdot\sigma} \end{pmatrix}, \tag{11.76}$$

where the two minus signs come from the lower block of v^s paired against the components of $\bar{v}^s$ (the upper-left and lower-right entries carry exactly one factor of the relative minus, the off-diagonal entries carry zero or two). Summing over s with $\sum_s \eta^s \eta^{s\dagger} = I$,

$$\sum_s v^s \bar{v}^s = \begin{pmatrix} -m\,I & p\cdot\sigma \\ p\cdot\bar{\sigma} & -m\,I \end{pmatrix} = -m\,\mathbf{1} + \begin{pmatrix} \mathbf{0} & p\cdot\sigma \\ p\cdot\bar{\sigma} & \mathbf{0} \end{pmatrix} = \not{p} - m. \tag{11.77}$$

□

The result is independent of the rest-frame bases ξ^s, η^s, as it must be: the only property of those bases that entered is their completeness $\sum \xi\xi^\dagger = \sum \eta\eta^\dagger = I$, not the particular axis chosen. One may also read Theorem 11.12 as the boosted form of the rest-frame projectors. In the rest frame $\sum_s u^s \bar{u}^s = m(\mathbf{1} + \gamma^0)$, which is m times the projector $\frac{1}{2}(1+\gamma^0)$ onto the $\gamma^0 = +1$ subspace, doubled; boosting $\gamma^0 \to \not{p}/m$ turns it into $\not{p} + m$. This is the viewpoint that makes the energy projectors of the next subsection inevitable.

11.2.5 Energy, chirality, and helicity projectors

The spin sums are completeness relations for two complementary subspaces of the four-dimensional spinor space at fixed p; the operators that isolate those subspaces are the *energy projectors*. They, together with the chirality projectors imported from Section C.3.5 and the helicity operator built below, are the standard kit for decomposing a spinor by its physical quantum numbers. Figure 11.1 summarizes how the three decompositions sit on the one four-dimensional space.

Definition 11.13 (Energy projectors)**.** For an on-shell momentum p ($p^2 = m^2$, $m > 0$) define

$$\Lambda_+(p) \equiv \frac{\not{p} + m}{2m}, \qquad \Lambda_-(p) \equiv \frac{-\not{p} + m}{2m} = \frac{m - \not{p}}{2m}. \tag{11.78}$$

By Theorem 11.12 these are exactly the (suitably normalized) spin-sum matrices: $\Lambda_+ = \frac{1}{2m}\sum_s u^s \bar{u}^s$ and $\Lambda_- = -\frac{1}{2m}\sum_s v^s \bar{v}^s$. Their projector properties follow from the single on-shell identity $\not{p}^2 = p^2\mathbf{1} = m^2\mathbf{1}$ established in (11.52).

Proposition 11.14 (Projector algebra of $\Lambda_\pm$)**.** *The energy projectors satisfy*

$$\Lambda_+ + \Lambda_- = \mathbf{1}, \qquad \Lambda_\pm^2 = \Lambda_\pm, \qquad \Lambda_+\Lambda_- = \Lambda_-\Lambda_+ = \mathbf{0}, \qquad \operatorname{tr}\Lambda_\pm = 2. \tag{11.79}$$

Proof. Completeness is immediate from (11.78): $\Lambda_+ + \Lambda_- = \big[(\not{p} + m) + (m - \not{p})\big]/2m = 2m/2m = \mathbf{1}$. For idempotency, expand using $\not{p}^2 = m^2\mathbf{1}$:

$$(\not{p} + m)^2 = \not{p}^2 + 2m\not{p} + m^2\mathbf{1} = 2m^2\mathbf{1} + 2m\not{p} = 2m(\not{p} + m), \tag{11.80}$$

so $\Lambda_+^2 = (\not{p}+m)^2/4m^2 = 2m(\not{p}+m)/4m^2 = (\not{p}+m)/2m = \Lambda_+$; the identical computation with $\not{p} \to -\not{p}$ gives $(m-\not{p})^2 = 2m(m-\not{p})$ and $\Lambda_-^2 = \Lambda_-$. Orthogonality is the difference of squares,

$$\Lambda_+\Lambda_- = \frac{(\not{p}+m)(m-\not{p})}{4m^2} = \frac{m\not{p} - \not{p}^2 + m^2 - m\not{p}}{4m^2} = \frac{m^2-\not{p}^2}{4m^2} = \frac{m^2-m^2}{4m^2}\,\mathbf{1} = \mathbf{0}, \tag{11.81}$$

and $\Lambda_-\Lambda_+ = \mathbf{0}$ since $\not{p}$ commutes with itself. Finally, $\operatorname{tr}\Lambda_+ = \frac{1}{2m}(\operatorname{tr}\not{p} + m\operatorname{tr}\mathbf{1}) = \frac{1}{2m}(0+4m) = 2$, using $\operatorname{tr}\gamma^\mu = 0$ from Section C.7.3 and $\operatorname{tr}\mathbf{1} = 4$; likewise $\operatorname{tr}\Lambda_- = 2$. □

The trace count $\operatorname{tr}\Lambda_\pm = 2$ is the promised confirmation that each frequency sector is exactly two-dimensional, so the four amplitudes u^1, u^2, v^1, v^2 of Theorem 11.8 are a complete basis with no solutions missed and none double-counted. Moreover Λ_+ acts as the identity on the u's and annihilates the v's, and vice versa for Λ_-: from $(\not{p}-m)u^s = 0$ we get $\not{p}\,u^s = mu^s$, hence $\Lambda_+u^s = (\not{p}+m)u^s/2m = 2mu^s/2m = u^s$ and $\Lambda_-u^s = (m-\not{p})u^s/2m = 0$; from $(\not{p}+m)v^s = 0$, $\not{p}\,v^s = -mv^s$, so $\Lambda_-v^s = v^s$ and $\Lambda_+v^s = 0$. The energy projectors literally project onto the particle and antiparticle solution spaces.

Chirality. The second decomposition of the spinor space is by *chirality*, the eigenvalue of $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, which in the chiral basis is $\gamma^5 = \operatorname{diag}(-I, I)$. We do not re-derive its properties—they are established once and for all in Section C.3.5—but recall them for use here. The chirality projectors are

$$P_L = \tfrac{1}{2}(1-\gamma^5), \qquad P_R = \tfrac{1}{2}(1+\gamma^5), \tag{11.82}$$

(Theorem C.27), and by Theorem C.28 they are a complete orthogonal system, $P_L + P_R = \mathbf{1}$, $P_{L,R}^2 = P_{L,R}$, $P_LP_R = P_RP_L = \mathbf{0}$, satisfying the intertwining $\gamma^\mu P_L = P_R\gamma^\mu$. In the chiral basis P_L keeps the upper two-block and P_R the lower, so a spinor's handedness is the statement of which Weyl block it occupies. Chirality is Lorentz invariant (because γ^5 commutes with the generators $\sigma^{\mu\nu}$, Eq. (11.11)) but it is *not* conserved by the massive Dirac equation: acting with P_L on $(\not{p}-m)u = 0$ and using the intertwining gives $\not{p}\,(P_Ru) = m\,(P_Lu)$, so the mass term couples the two chiralities and a chirality eigenstate is not a solution unless $m = 0$. This is the precise sense in which mass mixes left and right, and it is why chirality, though geometrically fundamental, is not the conserved label that organizes a massive particle's spin states. That role falls to helicity.

Helicity. The conserved spin label for a massive free particle is the projection of spin along the direction of motion. The four-dimensional spin operator in the chiral basis is block-diagonal, $\boldsymbol{\Sigma} = \operatorname{diag}(\boldsymbol{\sigma}, \boldsymbol{\sigma})$ with components $\Sigma^k = \varepsilon^{kij}\sigma^{ij} = \operatorname{diag}(\sigma^k, \sigma^k)$ (it is block-diagonal precisely because rotations act identically on

the two Weyl components), and the spin vector is $\mathbf{S} = \frac{1}{2}\boldsymbol{\Sigma}$. The *helicity operator* is its projection on the unit momentum $\hat{\mathbf{p}} = \mathbf{p}/|\mathbf{p}|$,

$$h(\mathbf{p}) \equiv \hat{\mathbf{p}} \cdot \mathbf{S} = \tfrac{1}{2}\,\hat{\mathbf{p}} \cdot \boldsymbol{\Sigma} = \tfrac{1}{2}\begin{pmatrix} \hat{\mathbf{p}}\cdot\boldsymbol{\sigma} & \mathbf{0} \\ \mathbf{0} & \hat{\mathbf{p}}\cdot\boldsymbol{\sigma} \end{pmatrix}. \tag{11.83}$$

To build helicity eigenspinors, choose the rest-frame two-spinors to be eigenstates of $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$: let $\xi_\pm$ solve $(\hat{\mathbf{p}}\cdot\boldsymbol{\sigma})\,\xi_\pm = \pm\xi_\pm$ (these exist and are orthonormal since $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$ is Hermitian with eigenvalues ± 1), and build $u_\pm(p)$ from $\xi_\pm$ by (11.57). Because $p\cdot\sigma = E_{\mathbf{p}} I - |\mathbf{p}|\,\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$ is a function of $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$ alone, both $\sqrt{p\cdot\sigma}$ and $\sqrt{p\cdot\bar\sigma}$ commute with $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$ and preserve its eigenspaces. Hence

$$h(\mathbf{p})\,u_\pm(p) = \tfrac{1}{2}\begin{pmatrix} \sqrt{p\cdot\sigma}\,(\hat{\mathbf{p}}\cdot\boldsymbol{\sigma})\xi_\pm \\ \sqrt{p\cdot\bar\sigma}\,(\hat{\mathbf{p}}\cdot\boldsymbol{\sigma})\xi_\pm \end{pmatrix} = \pm\tfrac{1}{2}\,u_\pm(p), \tag{11.84}$$

so u_+ and u_- are helicity eigenspinors with eigenvalues $+\frac{1}{2}$ (right-handed helicity) and $-\frac{1}{2}$. The same construction with $\eta_\pm$ gives helicity eigenspinors $v_\pm$; the standard convention assigns the antiparticle described by v the *opposite* physical helicity to its spinor label, a point we will fix operationally when these become creation operators in Section 11.3. Theorem 11.10 already exhibited the eigenstates concretely: for $\mathbf{p} = p\hat{\mathbf{z}}$, $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma} = \sigma^3$, so $\xi_+ = \xi^1$ and $\xi_- = \xi^2$, and the spinors (11.62) are already helicity eigenstates. Unlike chirality, helicity *is* conserved by the free Dirac equation—it commutes with the free Hamiltonian because both $\hat{\mathbf{p}}\cdot\boldsymbol{\sigma}$ and the dynamics are diagonal in $\mathbf{p}$—but it is not Lorentz invariant for $m > 0$, since a boost faster than the particle reverses $\hat{\mathbf{p}}$ and flips the sign. In the massless limit the two notions coincide, as Theorem 11.10 showed and as the next chapters on the weak interaction will exploit.

11.2.6 The Gordon decomposition

A last identity belongs here because it is pure on-shell spinor algebra and because QED will lean on it heavily: the *Gordon decomposition* of the vector current $\bar u(p')\gamma^\mu u(p)$. It rewrites the Dirac current as the sum of a convective piece, proportional to the total momentum $(p+p')^\mu$ and structurally identical to the Schrödinger probability current, and a spin piece, proportional to the momentum transfer $q^\mu = (p'-p)^\mu$ contracted into the antisymmetric tensor $\sigma^{\mu\nu}$. The spin piece is the magnetic-moment coupling: when these currents are sandwiched between the electron and the photon in Chapters 17 and 20, that term is precisely what produces the tree-level gyromagnetic ratio $g = 2$, and its loop correction is the anomalous moment $a_e = \alpha/2\pi$. We isolate it now.

Theorem 11.15 (Gordon decomposition)**.** *Let $u(p), u(p')$ be positive-frequency amplitudes (Theorem 11.8) of equal mass m, and $q^\mu \equiv (p'-p)^\mu$. With $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]$ (Eq. (11.11)),*

$$\bar u(p')\,\gamma^\mu\,u(p) = \bar u(p')\left[\frac{(p+p')^\mu}{2m} + \frac{i\,\sigma^{\mu\nu}q_\nu}{2m}\right]u(p). \tag{11.85}$$

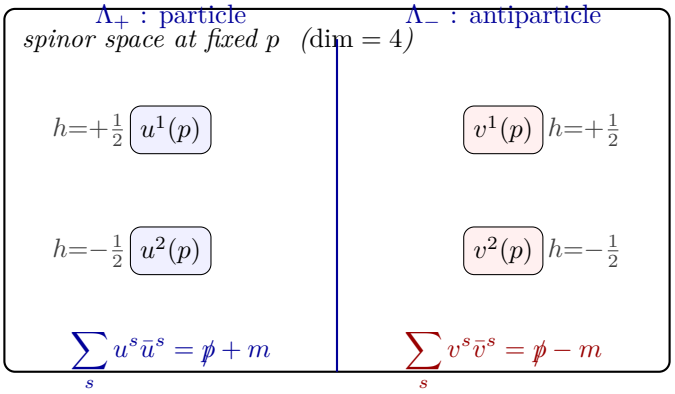


Figure 11.1: The three decompositions of the four-dimensional spinor space at a fixed on-shell momentum p. The energy projectors $\Lambda_\pm$ of (11.78) split it into the two-dimensional particle (u) and antiparticle (v) sectors, each of trace 2; within each sector the helicity operator (11.83) resolves the two spin states; the chirality projectors $P_{L,R}$ (not shown) give a third, rotated splitting that aligns with helicity only as $m \to 0$. The completeness of each sector is the content of the spin sums (11.73).

Proof. The amplitudes obey the Dirac equations in both their forms: $\not{p}\,u(p) = m\,u(p)$ from (11.50), and, taking the Dirac adjoint of $(\not{p}' - m)u(p') = 0$, also $\bar{u}(p')\not{p}' = m\,\bar{u}(p')$. (The adjoint step uses $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$: from $(\not{p}' - m)u(p') = 0$ take $\dagger$ to get $u(p')^\dagger(\not{p}'^\dagger - m) = 0$, then multiply by γ^0 on the right and insert $\gamma^0\gamma^0 = 1$, giving $\bar{u}(p')(\not{p}' - m) = 0$.) Use these to write $2m\,\bar{u}'\gamma^\mu u$ symmetrically, replacing one factor of m on each side:

$$2m\,\bar{u}'\gamma^\mu u = \bar{u}'\,\gamma^\mu(\not{p}\,u) + (\bar{u}'\not{p}')\,\gamma^\mu u = \bar{u}'(\gamma^\mu\gamma^\nu p_\nu + \gamma^\nu\gamma^\mu p'_\nu)u, \tag{11.86}$$

where in the first term $\not{p} = \gamma^\nu p_\nu$ stands to the right of γ^μ and in the second $\not{p}' = \gamma^\nu p'_\nu$ to the left. Now split each gamma product into its symmetric and antisymmetric parts using the Clifford relation (C.184) together with the definition (11.11), $\gamma^\mu\gamma^\nu = \eta^{\mu\nu}\mathbf{1} - i\sigma^{\mu\nu}$, and its transpose $\gamma^\nu\gamma^\mu = \eta^{\mu\nu}\mathbf{1} + i\sigma^{\mu\nu}$ (from $\sigma^{\nu\mu} = -\sigma^{\mu\nu}$):

$$\begin{aligned} 2m\,\bar{u}'\gamma^\mu u &= \bar{u}'\Big[(\eta^{\mu\nu}\mathbf{1} - i\sigma^{\mu\nu})p_\nu + (\eta^{\mu\nu}\mathbf{1} + i\sigma^{\mu\nu})p'_\nu\Big]u \\ &= \bar{u}'\Big[(p+p')^\mu\,\mathbf{1} + i\,\sigma^{\mu\nu}(p'_\nu - p_\nu)\Big]u = \bar{u}'\Big[(p+p')^\mu + i\,\sigma^{\mu\nu}q_\nu\Big]u. \end{aligned} \tag{11.87}$$

Dividing by $2m$ gives (11.85). □

The decomposition is exact and rests on nothing but the Dirac equation and the Clifford algebra. Two consequences are worth flagging for later use.

Setting $p' = p$ (forward matrix element, $q = 0$) collapses (11.85) to $\bar{u}(p)\gamma^\mu u(p) = \bar{u}(p)\, p^\mu/m\, u(p) = 2p^\mu$ on using $\bar{u}u = 2m$ from (11.64): the forward charge current is just $2p^\mu$, the relativistic flux, with no spin contribution—the magnetic moment shows up only at nonzero momentum transfer. And the antisymmetry $\sigma^{\mu\nu}q_\nu$ guarantees that contracting (11.85) with q_μ kills the spin term identically, leaving $q_\mu \bar{u}'\gamma^\mu u = \bar{u}'(q\cdot(p+p'))/2m\, u = \bar{u}'(p'^2 - p^2)/2m\, u = 0$; this is the spinor-level current conservation that will reappear as the Ward identity in Chapter 19.

We now hold the complete, normalized basis the rest of the chapter requires: two positive-frequency amplitudes $u^s(p)$ and two negative-frequency amplitudes $v^s(p)$ at every on-shell momentum, their full table of inner products (11.64)–(11.67), the completeness relations (11.73), and the projector toolkit (11.78)–(11.84) that resolves a spinor into particle/antiparticle, chirality, and helicity. The orthogonality relations (11.65) and (11.67) are exactly the inversion formulae the next section needs: expanding the field operator $\psi(x)$ over $\{u^s(p)e^{-ip\cdot x},\, v^s(p)e^{+ip\cdot x}\}$ with ladder operators, they let us solve for each ladder operator individually—and the relative sign $\bar{u}u = +2m$ versus $\bar{v}v = -2m$ already glimpsed here is the thread we will pull in Section 11.3 to discover that the Dirac field admits a sensible Hamiltonian only if its quanta are fermions.

Before closing, we record the worked computation that makes the thread obligation explicit: the spin sums of Theorem 11.12 are not merely tidy—they are the engine of every unpolarized rate.

Worked Example 11.16 (The master trace formula—why the spin sums matter for QED)**.** Consider any process whose amplitude has the spinor structure $\mathcal{M} = \bar{u}^{s'}(p')\,\Gamma\, u^s(p)$, where Γ is some product of gamma matrices (a current, a propagator numerator, a vertex string). This is the generic shape of every QED tree amplitude with one incoming and one outgoing fermion line; in $e^+e^- \to \mu^+\mu^-$ of Chapter 17 the muon line is exactly of this form. An unpolarized cross section requires $\sum_{s,s'} |\mathcal{M}|^2$ (averaging factors aside). We show this collapses to a single gamma trace, with no reference to the explicit spinors.

First write the complex conjugate of a bilinear as a bilinear with the conjugated insertion. Since $\mathcal{M}$ is a number, $\mathcal{M}^* = \mathcal{M}^\dagger = \left(\bar{u}^{s'}(p')\Gamma u^s(p)\right)^\dagger = u^s(p)^\dagger \Gamma^\dagger \gamma^{0\dagger} u^{s'}(p')$. Using $\gamma^{0\dagger} = \gamma^0$ and inserting $\gamma^0\gamma^0 = \mathbf{1}$,

$$\mathcal{M}^* = u^s(p)^\dagger \gamma^0 \left(\gamma^0\Gamma^\dagger\gamma^0\right) u^{s'}(p') = \bar{u}^s(p)\,\bar{\Gamma}\, u^{s'}(p'), \qquad \bar{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0, \tag{11.88}$$

where $\bar{\Gamma}$ is the "Dirac conjugate" of the insertion (for a single γ^μ one has $\bar{\gamma^\mu} = \gamma^0\gamma^{\mu\dagger}\gamma^0 = \gamma^\mu$, so a vector current conjugates to itself). Therefore

$$|\mathcal{M}|^2 = \mathcal{M}\,\mathcal{M}^* = \left[\bar{u}^{s'}(p')\Gamma u^s(p)\right]\left[\bar{u}^s(p)\bar{\Gamma} u^{s'}(p')\right]. \tag{11.89}$$

Now sum over both spins. Group the factors so that each spin label closes a loop. The s sum acts on $u^s(p)\bar{u}^s(p)$, and the s' sum on $u^{s'}(p')\bar{u}^{s'}(p')$; by Theorem 11.12

these become $\not{p}+m$ and $\not{p}'+m'$ respectively (allowing distinct masses m, m' on the two lines):

$$\sum_{s,s'} |\mathcal{M}|^2 = \sum_{s,s'} \bar{u}_a^{s'}(p')\,\Gamma_{ab}\,u_b^{s}(p)\,\bar{u}_c^{s}(p)\,\bar{\Gamma}_{cd}\,u_d^{s'}(p') = \left[\not{p}'+m'\right]_{da}\Gamma_{ab}\left[\not{p}+m\right]_{bc}\bar{\Gamma}_{cd}, \tag{11.90}$$

writing spinor indices a, b, c, d explicitly and reassembling the chain that begins and ends on index $d = a'$. The right-hand side is a closed matrix product traced over the single remaining index, i.e. a trace:

$$\boxed{\sum_{s,s'} \left|\bar{u}^{s'}(p')\,\Gamma\,u^{s}(p)\right|^2 = \mathrm{tr}\Big[\left(\not{p}'+m'\right)\Gamma\left(\not{p}+m\right)\bar{\Gamma}\Big].} \tag{11.91}$$

The explicit spinor components have vanished entirely; what remains is a trace of gamma matrices, evaluated by the contraction and trace identities of Section C.7.3. This is the single most-used computational identity of perturbative QED, and Eq. (11.91) is exactly the formula Chapter 17 feeds into the trace technology to obtain the unpolarized $|\mathcal{M}|^2$ for $e^+e^- \to \mu^+\mu^-$. (For an antiparticle line one repeats the steps with $\sum_s v^s\bar{v}^s = \not{p} - m$ in place of the u sum, which is why both spin sums were needed.) The lesson is structural: the work of this section—constructing the basis and proving its completeness—is precisely what lets the later computation forget the basis and reduce to algebra.

11.3 Canonical Quantization and the Necessity of Anticommutators

The previous section handed us a complete basis. For each spatial momentum $\mathbf{p}$ and each of two spin labels $s \in \{1, 2\}$ we have a positive-frequency solution $u^s(p)\,e^{-ip\cdot x}$ and a negative-frequency solution $v^s(p)\,e^{+ip\cdot x}$ of the Dirac equation, with $p^0 = E_{\mathbf{p}} = \sqrt{\mathbf{p}^2+m^2}$ throughout, normalized so that the Dirac inner products $\bar{u}^r u^s = 2m\,\delta^{rs}$, $\bar{v}^r v^s = -2m\,\delta^{rs}$ hold, the row–vector products satisfy

$$u^{r\dagger}(\mathbf{p})\,u^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \qquad v^{r\dagger}(\mathbf{p})\,v^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \tag{11.92}$$

the cross terms at opposite spatial momentum vanish,

$$u^{r\dagger}(\mathbf{p})\,v^s(-\mathbf{p}) = 0, \qquad v^{r\dagger}(\mathbf{p})\,u^s(-\mathbf{p}) = 0, \tag{11.93}$$

and the completeness (spin-sum) identities read

$$\sum_s u^s(p)\,\bar{u}^s(p) = \not{p} + m, \qquad \sum_s v^s(p)\,\bar{v}^s(p) = \not{p} - m \tag{11.94}$$

(recall Eqs. (11.64), (11.65), (11.67) and (11.73); here $v^s(-\mathbf{p})$ denotes the negative-frequency spinor carrying spatial momentum $-\mathbf{p}$, i.e. four-momentum $(E_{\mathbf{p}}, -\mathbf{p})$).

These relations are the entire raw material of this section: every operator identity below is obtained by folding the field against one of these spinors and using Eqs. (11.92) to (11.94).

The task is now exactly the one Chapter 9 carried out for the scalar: promote the classical field to an operator, expand it in this basis with operator-valued coefficients, and *fix the algebra of those coefficients by a canonical postulate*. For the scalar that postulate was the equal-time commutator, and it produced Bose statistics, a positive Hamiltonian, and a Fock space of identical bosons (Sections 9.1 and 9.3). We will follow the same structural template here — mode expansion, inversion, evaluation of H — but we will find that the commutator postulate, applied verbatim to the Dirac field, produces a catastrophe: a Hamiltonian with no ground state and a state space with negative-norm vectors. The repair is forced and unique. The equal-time *anti*commutator cures both diseases simultaneously, and in doing so it deduces, rather than assumes, that the spin-$\frac{1}{2}$ field must be quantized as fermions. This is the first genuinely fermionic fact of QED, and the classical foreshadowing was already explicit: in Section 6.5.4 the Dirac field's first-order kinetic term made its momentum a *second-class* constraint, whose Dirac bracket $\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\}_{\mathrm{D}} = -i\,\delta_{ab}\,\delta^3(\mathbf{x}-\mathbf{y})$ is symmetric under exchange and therefore quantizes to an anticommutator. The present section is where that promissory note is paid in full, and where the spectrum itself is shown to demand it.

11.3.1 The Heisenberg mode expansion

The classical free Dirac field obeys $(i\partial\!\!\!/ - m)\psi = 0$ (Eq. (6.70)), and the general classical solution is a superposition of the plane-wave basis built in the previous section. We promote ψ to a Heisenberg-picture operator by attaching an operator coefficient to each basis wave.

Definition 11.17 (Dirac mode expansion)**.** The quantized Dirac field is

$$\psi(x) = \sum_{s=1,2} \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big[\, b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \,\Big], \tag{11.95}$$

with the Lorentz-invariant measure $\int d^3p/[(2\pi)^3 2E_{\mathbf{p}}]$ of Theorem 9.4, and the four-component adjoint

$$\psi^\dagger(x) = \sum_{s=1,2} \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big[\, b^{s\dagger}_{\mathbf{p}}\, u^{s\dagger}(p)\, e^{+ip\cdot x} + d^{s}_{\mathbf{p}}\, v^{s\dagger}(p)\, e^{-ip\cdot x} \,\Big]. \tag{11.96}$$

Here $b^s_{\mathbf{p}}, b^{s\dagger}_{\mathbf{p}}$ and $d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}$ are operators on a Hilbert space yet to be built, and $p\cdot x = E_{\mathbf{p}}x^0 - \mathbf{p}\cdot\mathbf{x}$ with $p^0 = E_{\mathbf{p}}$ on shell.

Two features of Eq. (11.95) are choices that deserve their motivation spelled out, because the entire success of the construction turns on them.

First, the expansion uses *two* independent families of operators, b and d, where the real scalar used only one. The real scalar field is Hermitian, $\phi = \phi^\dagger$, so its positive- and negative-frequency coefficients were related by conjugation, a and $a^\dagger$ (Eq. (9.42)). The Dirac field is intrinsically complex — its Lagrangian $\bar\psi(i\partial\!\!\!/ - m)\psi$ is invariant under the global phase rotation $\psi \to e^{-i\alpha}\psi$ whose Noether charge is the electric charge of QED — so $\psi \neq \psi^\dagger$ and the negative-frequency coefficient is an *independent* operator. We have written it as $d^{s\dagger}$ rather than d^s: the negative-frequency part of the field is built from a creation operator. This is the deliberate move that Chapter 9 also made for the complex scalar (Section 9.7). Its payoff, deferred to the next section, is that ψ then *annihilates* a particle of charge $-e$ (an electron) and *creates* a particle of charge $+e$ (a positron) in a single local operator, which is exactly what a charge-lowering field must do. For now it is simply the parametrization that will make the charge operator come out diagonal and positive.

Second, the spinor $u^s(p)$ rides with $b^s_{\mathbf{p}}$ and $v^s(p)$ rides with $d^{s\dagger}_{\mathbf{p}}$, each at its own four-momentum $p = (E_{\mathbf{p}}, \mathbf{p})$, so that every term in Eq. (11.95) is by construction a solution of the Dirac equation. Indeed, acting with the Dirac operator and using $(\not{p} - m)u^s(p) = 0$ and $(\not{p} + m)v^s(p) = 0$,

$$(i\partial\!\!\!/ - m)\, u^s(p) e^{-ip\cdot x} = (\not{p} - m)u^s(p)\, e^{-ip\cdot x} = 0, \qquad (i\partial\!\!\!/ - m)\, v^s(p) e^{+ip\cdot x} = -(\not{p} + m)v^s(p)\, e^{+ip\cdot x} \tag{11.97}$$

so $\psi(x)$ of Eq. (11.95) satisfies the Dirac equation Eq. (6.70) for *any* choice of the operator coefficients. The Dirac equation alone therefore does not fix the algebra of b and d; that is the job of the canonical postulate, to which we now turn.

11.3.2 The conjugate momentum and the two candidate postulates

From Eq. (6.78) the momentum conjugate to ψ is

$$\pi_\psi = \frac{\partial \mathcal{L}_{\mathrm{D}}}{\partial \dot\psi} = i\,\psi^\dagger, \tag{11.98}$$

a relation that is itself a (second-class) constraint, not a definition of a velocity: the Dirac Lagrangian is first order in time, so π_ψ is algebraically tied to the field rather than to $\dot\psi$. In an unconstrained theory canonical quantization would read off $[\psi_a(\mathbf{x}, t), \pi_b(\mathbf{y}, t)] = i\,\delta_{ab}\,\delta^3(\mathbf{x} - \mathbf{y})$, i.e. with Eq. (11.98), $\left[\psi_a(\mathbf{x}, t), \psi^\dagger_b(\mathbf{y}, t)\right] = \delta_{ab}\,\delta^3(\mathbf{x} - \mathbf{y})$. The constraint structure of Section 6.5.4 warns that this naive reading is not the final word — the proper bracket is the Dirac bracket — but it is exactly the naive bosonic reading whose failure we want to exhibit. We therefore

set up the two candidate postulates side by side and test each by computing its consequences.

Definition 11.18 (The two equal-time postulates)**.** We compare, at equal times $x^0 = y^0$:

$$\text{(B) commutators:} \quad \Big[\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\Big] = \delta_{ab}\,\delta^3(\mathbf{x}-\mathbf{y}), \qquad [\psi_a(\mathbf{x}), \psi_b(\mathbf{y})] = 0, \tag{11.99}$$

$$\text{(F) anticommutators:} \quad \Big\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\Big\} = \delta_{ab}\,\delta^3(\mathbf{x}-\mathbf{y}), \quad \{\psi_a(\mathbf{x}), \psi_b(\mathbf{y})\} = 0, \tag{11.100}$$

where a, b are Dirac (spinor) indices. Postulate (B) is the rule that worked for the scalar; postulate (F) is the rule we will be forced into. We will find that (B) is fatal and (F) is consistent.

11.3.3 Inverting the expansion

Both postulates are statements about the field; to test them we must translate them into statements about b and d. The translation is the inversion of Eq. (11.95), performed exactly as in Theorem 9.7 but now using the spinor orthonormality Eqs. (11.92) to (11.93) to project out a single mode. The result is the following.

Proposition 11.19 (Inversion of the Dirac mode expansion)**.** *At any fixed time $x^0 = t$, the ladder operators are recovered from the field by*

$$b_{\mathbf{p}}^r = \int d^3x\, e^{+ip\cdot x}\, u^{r\dagger}(p)\, \psi(x), \qquad d_{\mathbf{p}}^r = \int d^3x\, e^{+ip\cdot x}\, \psi^\dagger(x)\, v^r(p), \tag{11.101}$$

with Hermitian conjugates

$$b_{\mathbf{p}}^{r\dagger} = \int d^3x\, e^{-ip\cdot x}\, \psi^\dagger(x)\, u^r(p), \qquad d_{\mathbf{p}}^{r\dagger} = \int d^3x\, e^{-ip\cdot x}\, v^{r\dagger}(p)\, \psi(x). \tag{11.102}$$

Proof. Work at fixed t and abbreviate $u^s(p)e^{-ip\cdot x} = u^s(\mathbf{p})e^{-iE_{\mathbf{p}}t}e^{i\mathbf{p}\cdot\mathbf{x}}$ and $v^s(p)e^{+ip\cdot x} = v^s(\mathbf{p})e^{+iE_{\mathbf{p}}t}e^{-i\mathbf{p}\cdot\mathbf{x}}$. Multiply Eq. (11.95) by $e^{-i\mathbf{q}\cdot\mathbf{x}}$ and integrate over $\mathbf{x}$. The spatial integral produces $\int d^3x\, e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$ on the b term and $\int d^3x\, e^{-i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}+\mathbf{q})$ on the $d^\dagger$ term, collapsing each momentum integral against its measure $1/2E$:

$$\int d^3x\, e^{-i\mathbf{q}\cdot\mathbf{x}}\,\psi(\mathbf{x},t) = \sum_s \frac{1}{2E_{\mathbf{q}}}\Big[b_{\mathbf{q}}^s\, u^s(\mathbf{q})\, e^{-iE_{\mathbf{q}}t} + d_{-\mathbf{q}}^{s\dagger}\, v^s(-\mathbf{q})\, e^{+iE_{\mathbf{q}}t}\Big]. \tag{11.103}$$

Now contract from the left with $u^{r\dagger}(\mathbf{q})$. The first term gives $u^{r\dagger}(\mathbf{q})u^s(\mathbf{q}) = 2E_{\mathbf{q}}\delta^{rs}$ by Eq. (11.92); the second term gives $u^{r\dagger}(\mathbf{q})v^s(-\mathbf{q}) = 0$ by Eq. (11.93). Hence

$$u^{r\dagger}(\mathbf{q}) \int d^3x\, e^{-i\mathbf{q}\cdot\mathbf{x}}\,\psi(\mathbf{x},t) = b_{\mathbf{q}}^r\, e^{-iE_{\mathbf{q}}t}, \tag{11.104}$$

and multiplying through by $e^{+iE_{\mathbf{q}}t}$ and using $e^{+iE_{\mathbf{q}}t}e^{-i\mathbf{q}\cdot\mathbf{x}} = e^{+iq\cdot x}$ gives the first formula in Eq. (11.101). The cross orthogonality Eq. (11.93) is precisely what makes the antiparticle ($d^\dagger$) contribution drop out, isolating b cleanly. To obtain $d^r_{\mathbf{p}}$, multiply instead by $e^{+i\mathbf{q}\cdot\mathbf{x}}$, integrate, and contract from the left with $v^{r\dagger}(\mathbf{q})$ acting on ψ; the analogous projection ($v^{r\dagger}v^s = 2E\delta^{rs}$ surviving, $v^{r\dagger}(\mathbf{q})u^s(-\mathbf{q}) = 0$ dropping) yields $d^{r\dagger}_{\mathbf{q}}$ from ψ, whose adjoint is the second formula in Eq. (11.101). The conjugate formulas Eq. (11.102) follow by Hermitian conjugation, noting that $u^{r\dagger}(p)\psi$ is a scalar (a contracted spinor index) so its conjugate is $\psi^\dagger u^r(p)$. □

The inversion is the engine of the whole section: it converts any postulated equal-time bracket of the fields into a bracket of the ladder operators by sandwiching the field bracket between two spinor projections. We use it twice — once for each postulate of Theorem 11.18 — and the two computations differ only by where a single sign lands.

11.3.4 The Hamiltonian, and why commutators fail

Before testing the postulates on the brackets directly, it is most illuminating to compute the Hamiltonian, because it is the Hamiltonian that exhibits the disease in physical terms.

Worked Example 11.20 (The Dirac Hamiltonian in the mode basis)**.** The Dirac Hamiltonian density established classically in Eq. (6.138) is

$$\mathcal{H}_{\mathrm{D}} = \psi^\dagger(\boldsymbol{\alpha}\cdot\hat{\mathbf{p}} + \beta m)\psi, \qquad \hat{\mathbf{p}} = -i\nabla, \quad \alpha^i = \gamma^0\gamma^i, \quad \beta = \gamma^0, \tag{11.105}$$

and the Hamiltonian is $H = \int d^3x\, \mathcal{H}_{\mathrm{D}}$. We evaluate it on Eq. (11.95), carrying every step.

Step 1: the momentum-space Dirac operator on the modes. Write $h(\mathbf{p}) \equiv \boldsymbol{\alpha}\cdot\mathbf{p} + \beta m$. From $(\not{p} - m)u^s(p) = 0$ with $\not{p} = \gamma^0 E_{\mathbf{p}} - \boldsymbol{\gamma}\cdot\mathbf{p}$, multiply on the left by γ^0 and use $\gamma^0\gamma^0 = 1$, $\gamma^0\boldsymbol{\gamma} = \boldsymbol{\alpha}$, $\gamma^0 m = \beta m$:

$$(E_{\mathbf{p}} - \boldsymbol{\alpha}\cdot\mathbf{p} - \beta m)u^s(\mathbf{p}) = 0 \implies h(\mathbf{p})\,u^s(\mathbf{p}) = E_{\mathbf{p}}\,u^s(\mathbf{p}). \tag{11.106}$$

Likewise from $(\not{p} + m)v^s(p) = 0$,

$$(E_{\mathbf{p}} - \boldsymbol{\alpha}\cdot\mathbf{p} + \beta m)v^s(\mathbf{p}) = 0 \implies h(-\mathbf{p})\,v^s(\mathbf{p}) = -E_{\mathbf{p}}\,v^s(\mathbf{p}). \tag{11.107}$$

The single-particle Dirac Hamiltonian has a positive-energy spectrum on the u-spinors and a negative-energy spectrum on the v-spinors — the negative-energy sea of the old hole theory. In the field theory this sign is benign on its own; it becomes lethal only in combination with the wrong statistics.

Step 2: act on ψ. The spatial derivative in $\hat{\mathbf{p}} = -i\nabla$ brings down $+\mathbf{p}$ on $e^{-ip\cdot x} = e^{-iEt}e^{i\mathbf{p}\cdot\mathbf{x}}$ and $-\mathbf{p}$ on $e^{+ip\cdot x} = e^{+iEt}e^{-i\mathbf{p}\cdot\mathbf{x}}$. Hence, using Eqs. (11.106)

and (11.107),

$$(\boldsymbol{\alpha}\cdot\hat{\mathbf{p}}+\beta m)\psi = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \Big[E_{\mathbf{p}}\, b^s_{\mathbf{p}} u^s(p) e^{-ip\cdot x} - E_{\mathbf{p}}\, d^{s\dagger}_{\mathbf{p}} v^s(p) e^{+ip\cdot x}\Big]. \quad (11.108)$$

The crucial minus sign on the antiparticle term is the negative eigenvalue Eq. (11.107).

Step 3: contract with $\psi^\dagger$ and integrate over $\mathbf{x}$. Multiplying Eq. (11.108) on the left by $\psi^\dagger$ of Eq. (11.96) and integrating $\int d^3x$ produces four terms with momentum-conserving deltas. The $b^\dagger b$ and $d\,d^\dagger$ terms carry $\int d^3x\, e^{i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$; the $b^\dagger d^\dagger$ and $d\,b$ terms carry $(2\pi)^3\delta^3(\mathbf{p}+\mathbf{q})$. The two off-diagonal terms vanish: each is proportional to a cross overlap at opposite spatial momentum, $u^{r\dagger}(-\mathbf{p})v^s(\mathbf{p}) = 0$ or $v^{r\dagger}(-\mathbf{p})u^s(\mathbf{p}) = 0$, by Eq. (11.93). Concretely, the $b^\dagger d^\dagger$ term comes with the time factor $e^{+2iE_{\mathbf{p}}t}$ and the spinor factor $u^{r\dagger}(-\mathbf{p})v^s(\mathbf{p})$, which is zero; were it not zero, the Hamiltonian would fail to be time independent. Only the two diagonal terms survive. Doing the $\mathbf{q}$ integral against $\delta^3(\mathbf{p}-\mathbf{q})$,

$$\begin{aligned} H &= \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \frac{1}{2E_{\mathbf{p}}} \Big[E_{\mathbf{p}}\, u^{s\dagger}(\mathbf{p})u^s(\mathbf{p})\, b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - E_{\mathbf{p}}\, v^{s\dagger}(\mathbf{p})v^s(\mathbf{p})\, d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}\Big] \\ &= \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} E_{\mathbf{p}} \Big[b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}\Big], \end{aligned} \quad (11.109)$$

where the second line used $u^{s\dagger}u^s = v^{s\dagger}v^s = 2E_{\mathbf{p}}$ from Eq. (11.92), so that $\frac{1}{2E_{\mathbf{p}}} E_{\mathbf{p}}(2E_{\mathbf{p}}) = E_{\mathbf{p}}$.

The result Eq. (11.109),

$$\boxed{H = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} E_{\mathbf{p}} \Big(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}\Big)} \quad (11.110)$$

is the pivot of the chapter. The particle sector enters with the manifestly nonnegative combination $b^\dagger b$; the antiparticle sector enters with $-d\,d^\dagger$ — the operators in the *wrong order* and with the *wrong sign*. Everything now depends on what we are allowed to do with $d\,d^\dagger$, and that is fixed by the postulate.

Postulate (B): commutators give negative norms. Apply the inversion Theorem 11.19 to the commutator postulate Eq. (11.99). For the particle operators,

$$\begin{aligned} \Big[b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\Big] &= \int d^3x\, d^3y\, e^{ip\cdot x} e^{-iq\cdot y}\, u^{r\dagger}_a(p)\, u^s_b(q)\, \Big[\psi_a(x), \psi^\dagger_b(y)\Big] \\ &= \int d^3x\, e^{i(p-q)\cdot x}\, u^{r\dagger}(p)u^s(q) = (2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})\, u^{r\dagger}(\mathbf{p})u^s(\mathbf{p}) = (2\pi)^3 2E_{\mathbf{p}}\, \delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}), \end{aligned} \quad (11.111)$$

positive, as for a healthy oscillator. But the antiparticle operators are extracted from $\psi^\dagger$, not ψ, so they sample the bracket in the opposite order. Using $\left[\psi_a^\dagger(\mathbf{x}), \psi_b(\mathbf{y})\right] = -\left[\psi_b(\mathbf{y}), \psi_a^\dagger(\mathbf{x})\right] = -\delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$,

$$\begin{aligned}\left[d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right] &= \int d^3x\, d^3y\; e^{ip\cdot x} e^{-iq\cdot y}\, v^r_a(p)\, v^{s\dagger}_b(q)\; \left[\psi_a^\dagger(x), \psi_b(y)\right] \\ &= -\int d^3x\; e^{i(p-q)\cdot x}\, v^{s\dagger}(q) v^r(p) = -(2\pi)^3 2E_{\mathbf{p}}\, \delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}). \quad (11.112)\end{aligned}$$

The minus sign in Eq. (11.112) is not a convention — it is forced by the antisymmetry of the commutator together with the fact that d lives in $\psi^\dagger$. Its consequence is immediate and fatal. The state $d^{s\dagger}_{\mathbf{p}}|0\rangle$, the would-be one-antiparticle state, has squared norm

$$\left\| d^{s\dagger}_{\mathbf{p}}|0\rangle \right\|^2 = \langle 0| d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}|0\rangle = \langle 0|\left[d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right]|0\rangle = -(2\pi)^3 2E_{\mathbf{p}}\, \delta^3(\mathbf{0})\, \langle 0|0\rangle < 0, \quad (11.113)$$

where we used $d^s_{\mathbf{p}}|0\rangle = 0$ for the annihilation operator d. A negative-norm state has no probability interpretation: the inner product is not positive definite, so the space built on the commutator postulate is not a Hilbert space at all. This is the first disease.

The Hamiltonian shows the same disease as unboundedness. One might try to dodge the negative norm by reinterpreting which operator creates and which annihilates. Suppose we declare $\hat{d} \equiv d^\dagger$ to be the annihilator (so $\hat{d}|0\rangle = 0$), making $\left[\hat{d}_{\mathbf{p}}, \hat{d}^\dagger_{\mathbf{q}}\right] = \left[d^\dagger_{\mathbf{p}}, d_{\mathbf{q}}\right] = +(2\pi)^3 2E_{\mathbf{p}}\delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}) > 0$, which restores positive norms. Then the Hamiltonian Eq. (11.110) becomes, using $d\, d^\dagger = \hat{d}^\dagger \hat{d}$,

$$H = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\, E_{\mathbf{p}} \left(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - \hat{d}^{s\dagger}_{\mathbf{p}} \hat{d}^s_{\mathbf{p}} \right). \quad (11.114)$$

Now every $\hat{d}^\dagger$ quantum *lowers* the energy by $E_{\mathbf{p}} > 0$. Applying $\hat{d}^\dagger_{\mathbf{p}}$ repeatedly drives the energy to $-\infty$: the spectrum has no ground state, the vacuum is unstable against producing antiparticles, and the theory has no sensible lowest-energy state to call a vacuum. This is the second disease. The two diseases are the same obstruction seen from two sides: the commutator forces a fixed relative sign between the b and d sectors, and that sign cannot be both positive-norm *and* positive-energy.

Worked Example 11.21 (No choice of c-number normalizations rescues the commutator)**.** One might hope that the disaster is an artifact of our spinor normalization and could be tuned away by rescaling. We show it cannot. Allow arbitrary real, positive c-number normalizations: rescale the spinors $u^s \to N_u u^s$,

$v^s \to N_v v^s$ and the operators $b \to \beta_b\, b$, $d \to \beta_d\, d$ with $N_u, N_v, \beta_b, \beta_d$ nonzero constants, and keep the commutator postulate Eq. (11.99). The inversion is linear, so the brackets simply pick up positive factors:

$$\left[b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\right] = +C_b\, \delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}), \qquad \left[d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right] = -C_d\, \delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}), \tag{11.115}$$

with $C_b = (2\pi)^3 2E_{\mathbf{p}} N_u^2\, \beta_b^{-2} > 0$ and $C_d = (2\pi)^3 2E_{\mathbf{p}} N_v^2\, \beta_d^{-2} > 0$. The relative minus sign between the two brackets is untouched, because it came from the antisymmetry $\left[\psi^\dagger, \psi\right] = -\left[\psi, \psi^\dagger\right]$ in Eq. (11.112), a structural fact independent of any normalization. The energy, recomputed with the same rescalings, is $H = \int E_{\mathbf{p}}(\,\#\, b^\dagger b - \#'\, d\, d^\dagger\,)$ with the same relative minus sign on $d\, d^\dagger$, because that sign came from the negative eigenvalue Eq. (11.107), again normalization independent. Thus for *any* positive constants: if d is the annihilator, the norm $\propto -C_d < 0$ is negative; if $d^\dagger$ is the annihilator, the energy $\propto -\#' < 0$ is unbounded below. No positive c-number can flip a structural sign, so no rescaling rescues postulate (B). The obstruction is intrinsic to bosonic quantization of a first-order, charged, half-integer-spin field.

Physical Insight

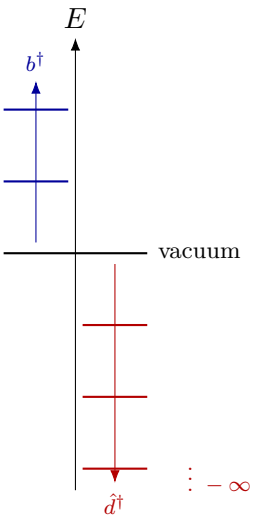
The deep reason both signs are immovable is that they have a common origin: the spin-$\frac{1}{2}$ field is a square root of the scalar. Its energy density $\psi^\dagger h \psi$ is *linear* in h, whose spectrum $\pm E_{\mathbf{p}}$ is sign-indefinite, whereas the scalar's energy density was built from $\dot\phi^2 + (\nabla\phi)^2 + m^2\phi^2$, manifestly a sum of squares. A field linear in a sign-indefinite operator can only have a stable vacuum if the filling of its negative-energy modes is *saturated* — if each mode can hold at most one quantum, so that lowering the energy by exciting antiparticles terminates. That is exactly the Pauli exclusion principle, and exclusion is exactly what anticommutators enforce. The spectrum is telling us, before we impose anything, that this field must be fermionic.

11.3.5 The cure: anticommutators and the ladder algebra

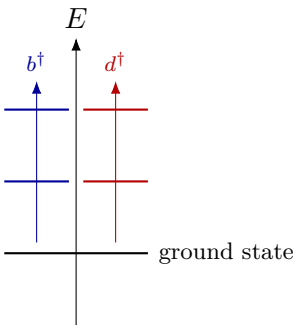
We now impose postulate (F), Eq. (11.100), and show that both diseases vanish. The inversion Theorem 11.19 is unchanged — it is a kinematic identity independent of the bracket — so we simply repeat the two computations Eqs. (11.111) and (11.112) with anticommutators.

Theorem 11.22 (The fermionic ladder algebra)**.** *The equal-time anticommutators Eq.* (11.100) *are equivalent to the ladder algebra*

$$\left\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\right\} = \left\{d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\, \delta^3(\mathbf{p}-\mathbf{q}), \tag{11.116}$$



(a) commutators: no ground state



(b) anticommutators: bounded below

Figure 11.2: The spectrum of $H = \int E_{\mathbf{p}}(b^\dagger b - d\,d^\dagger)$ under the two postulates. (a) With commutators, restoring positive norms forces $d^\dagger$ (relabelled $\hat{d}^\dagger$) to *lower* the energy without bound: there is no ground state. (b) With anticommutators, $-d\,d^\dagger = +d^\dagger d - \text{const}$, both towers raise the energy, and the spectrum is bounded below by a unique vacuum.

with all other anticommutators among $b, b^\dagger, d, d^\dagger$ *vanishing:*

$$\left\{b^r_{\mathbf{p}}, b^s_{\mathbf{q}}\right\} = \left\{d^r_{\mathbf{p}}, d^s_{\mathbf{q}}\right\} = \left\{b^r_{\mathbf{p}}, d^s_{\mathbf{q}}\right\} = \left\{b^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right\} = 0. \tag{11.117}$$

Proof. ($\Rightarrow$) Substitute the inversions into the anticommutator. For b,

$$\left\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\right\} = \int d^3x\, d^3y\; e^{ip\cdot x} e^{-iq\cdot y}\, u^{r\dagger}_a(p) u^s_b(q)\, \left\{\psi_a(x), \psi^\dagger_b(y)\right\} = \int d^3x\; e^{i(p-q)\cdot x}\, u^{r\dagger}(p) u^s(q), \tag{11.118}$$

using Eq. (11.100). The $\mathbf{x}$ integral gives $(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$, after which $u^{r\dagger}(\mathbf{p})u^s(\mathbf{p}) = 2E_{\mathbf{p}}\delta^{rs}$ by Eq. (11.92), yielding Eq. (11.116) for b. For d, now the anticommutator is symmetric, $\left\{\psi^\dagger_a(\mathbf{x}), \psi_b(\mathbf{y})\right\} = \left\{\psi_b(\mathbf{y}), \psi^\dagger_a(\mathbf{x})\right\} = \delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$, so the would-be minus sign of Eq. (11.112) is now a plus:

$$\left\{d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right\} = \int d^3x\, d^3y\; e^{ip\cdot x} e^{-iq\cdot y}\, v^r_a(p) v^{s\dagger}_b(q)\, \left\{\psi^\dagger_a(x), \psi_b(y)\right\} = +(2\pi)^3 2E_{\mathbf{p}}\, \delta^{rs}\delta^3(\mathbf{p}-\mathbf{q}). \tag{11.119}$$

The vanishing anticommutators Eq. (11.117) follow from $\{\psi_a, \psi_b\} = 0$ (which gives $\{b, b\} = \left\{b, d^\dagger\right\} = 0$ and $\{d, d\} = 0$ on projecting) and its conjugate. For example $\left\{b^r_{\mathbf{p}}, d^s_{\mathbf{q}}\right\} = \int e^{ip\cdot x} e^{iq\cdot y} u^{r\dagger}_a(p) v^s_b(q)\, \left\{\psi_a(x), \psi^\dagger_b(y)\right\}$ collapses to $u^{r\dagger}(\mathbf{p})v^s(-\mathbf{p}) = 0$ by the cross orthogonality Eq. (11.93).

($\Leftarrow$) Conversely, substituting Eqs. (11.95) and (11.96) into $\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\}$ and using Eqs. (11.116) and (11.117),

$$\begin{aligned}\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\} &= \sum_{r,s}\int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\frac{d^3q}{(2\pi)^3 2E_{\mathbf{q}}}\Big[u_a^r(p)u_b^{s\dagger}(q)\,e^{-ip\cdot x+iq\cdot y}\,\{b_{\mathbf{p}}^r, b_{\mathbf{q}}^{s\dagger}\} \\ &\qquad + v_a^r(p)v_b^{s\dagger}(q)\,e^{+ip\cdot x-iq\cdot y}\,\{d_{\mathbf{p}}^{r\dagger}, d_{\mathbf{q}}^s\}\Big] \\ &= \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\Big[u_a^s(\mathbf{p})u_b^{s\dagger}(\mathbf{p})\,e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} + v_a^s(\mathbf{p})v_b^{s\dagger}(\mathbf{p})\,e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}\Big],\end{aligned} \tag{11.120}$$

where the equal-time deltas $\delta^3(\mathbf{p}-\mathbf{q})$ from Eq. (11.116) collapsed the $\mathbf{q}$ integral and the on-shell energy phases $e^{\pm i(E_{\mathbf{p}}-E_{\mathbf{q}})t}$ became unity. To combine the two terms, substitute $\mathbf{p}\to-\mathbf{p}$ in the v integral, under which the measure and $E_{\mathbf{p}}$ are even and the phase $e^{-i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}$ becomes $e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})}$, matching the u term; the two spinor outer products then add at a *common* spatial argument. We evaluate them from the spin sums Eq. (11.94) via $u^{s\dagger}=\bar u^s\gamma^0$ (so $u^s u^{s\dagger}=u^s\bar u^s\gamma^0$):

$$\sum_s u^s(\mathbf{p})u^{s\dagger}(\mathbf{p}) = (\not p + m)\gamma^0, \qquad \sum_s v^s(-\mathbf{p})v^{s\dagger}(-\mathbf{p}) = (\not{\tilde p} - m)\gamma^0, \quad \tilde p \equiv (E_{\mathbf{p}}, -\mathbf{p}), \tag{11.121}$$

both evaluated on shell with $p^0=E_{\mathbf{p}}$. Their sum is purely the convective part: since $p+\tilde p=(2E_{\mathbf{p}},\mathbf{0})$ as four-vectors, $\not p+\not{\tilde p}=2E_{\mathbf{p}}\gamma^0$, the mass terms cancel, and

$$\sum_s\Big[u^s(\mathbf{p})u^{s\dagger}(\mathbf{p}) + v^s(-\mathbf{p})v^{s\dagger}(-\mathbf{p})\Big] = (\not p + \not{\tilde p})\gamma^0 = 2E_{\mathbf{p}}\,\gamma^0\gamma^0 = 2E_{\mathbf{p}}\,\mathbf{1}. \tag{11.122}$$

The spin- and frequency-dependent pieces have cancelled, leaving only the identity in spinor space. Hence

$$\{\psi_a(\mathbf{x}), \psi_b^\dagger(\mathbf{y})\} = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\,2E_{\mathbf{p}}\,\delta_{ab}\,e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta_{ab}\int\frac{d^3p}{(2\pi)^3}\,e^{+i\mathbf{p}\cdot(\mathbf{x}-\mathbf{y})} = \delta_{ab}\,\delta^3(\mathbf{x}-\mathbf{y}), \tag{11.123}$$

recovering Eq. (11.100). The two postulates are equivalent. □

With Theorem 11.22 in hand, return to the Hamiltonian Eq. (11.110). Anticommutators let us reorder $d\,d^\dagger$ *with* a plus sign: $d_{\mathbf{p}}^s d_{\mathbf{p}}^{s\dagger} = \{d_{\mathbf{p}}^s, d_{\mathbf{p}}^{s\dagger}\} - d_{\mathbf{p}}^{s\dagger}d_{\mathbf{p}}^s$. Therefore $-d\,d^\dagger = +d^\dagger d - \{d, d^\dagger\}$, and

$$H = \sum_s\int\frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\,E_{\mathbf{p}}\Big(b_{\mathbf{p}}^{s\dagger}b_{\mathbf{p}}^s + d_{\mathbf{p}}^{s\dagger}d_{\mathbf{p}}^s\Big) \;-\; \mathcal{E}_0, \qquad \mathcal{E}_0 = \sum_s\int d^3p\;E_{\mathbf{p}}\,\delta^3(\mathbf{0}), \tag{11.124}$$

where the (negative, divergent) constant $\mathcal{E}_0$ is the fermionic zero-point energy, the exact analogue of the scalar's vacuum energy of Section 9.4.1 but with the

opposite sign — a fact whose physical importance (boson and fermion zero-point energies cancel in a supersymmetric theory) is flagged for Chapter 50 and removed here by normal ordering in the next section. The dynamical part of Eq. (11.124) is now manifestly nonnegative: both $b^\dagger b$ and $d^\dagger d$ are nonnegative operators, so $H \geq -\mathcal{E}_0$ on every state, and the vacuum $|0\rangle$ defined by $b^s_{\mathbf{p}}|0\rangle = d^s_{\mathbf{p}}|0\rangle = 0$ is the unique ground state. Both diseases are cured at once: the norm $\|d^{s\dagger}_{\mathbf{p}}|0\rangle\|^2 = \langle 0|\left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\}|0\rangle = (2\pi)^3 2E_{\mathbf{p}}\delta^3(\mathbf{0})\langle 0|0\rangle > 0$ is now positive, and the energy is bounded below. Figure 11.2 contrasts the two outcomes.

Caution

The constant $\mathcal{E}_0$ in Eq. (11.124) is negative because it sits in front of a removed $-\left\{d, d^\dagger\right\}$. It is not a sign error: the fermionic vacuum energy is genuinely negative, opposite to the bosonic case. Normal ordering, defined for fermions with the antisymmetric sign convention introduced and executed in the next section, discards it; we do not write a Hamiltonian with $\mathcal{E}_0$ retained.

11.3.6 Consistency: the Heisenberg equation reproduces the Dirac equation

A canonical quantization is only consistent if the Heisenberg equation of motion generated by the resulting Hamiltonian reproduces the field equation we started from. We verify this for the anticommutator quantization. The key tool is the operator identity, valid for any operators,

$$[AB, C] = A\,\{B, C\} - \{A, C\}\,B, \tag{11.125}$$

which is the fermion-friendly form of the Leibniz rule: it lets us evaluate the commutator of the (bosonic) Hamiltonian with a single fermionic ladder operator using only the *anti*commutators Eq. (11.116).

Worked Example 11.23 (The Heisenberg equation gives back the Dirac equation)**.** Work with the normal-ordered Hamiltonian $H = \sum_s \int \frac{d^3q}{(2\pi)^3 2E_{\mathbf{q}}} E_{\mathbf{q}}(b^{s\dagger}_{\mathbf{q}} b^s_{\mathbf{q}} + d^{s\dagger}_{\mathbf{q}} d^s_{\mathbf{q}})$. Apply Eq. (11.125) to $\left[b^{s\dagger}_{\mathbf{q}} b^s_{\mathbf{q}}, b^r_{\mathbf{p}}\right]$, treating $A = b^{s\dagger}_{\mathbf{q}}$, $B = b^s_{\mathbf{q}}$, $C = b^r_{\mathbf{p}}$:

$$\left[b^{s\dagger}_{\mathbf{q}} b^s_{\mathbf{q}}, b^r_{\mathbf{p}}\right] = b^{s\dagger}_{\mathbf{q}} \underbrace{\left\{b^s_{\mathbf{q}}, b^r_{\mathbf{p}}\right\}}_{0} - \underbrace{\left\{b^{s\dagger}_{\mathbf{q}}, b^r_{\mathbf{p}}\right\}}_{(2\pi)^3 2E_{\mathbf{q}}\delta^{rs}\delta^3(\mathbf{p}-\mathbf{q})} b^s_{\mathbf{q}} = -(2\pi)^3 2E_{\mathbf{q}}\,\delta^{rs}\delta^3(\mathbf{p}-\mathbf{q})\,b^s_{\mathbf{q}}. \tag{11.126}$$

Integrating against $\frac{d^3q}{(2\pi)^3 2E_{\mathbf{q}}} E_{\mathbf{q}}$ and summing on s (the d part of H commutes with $b^r_{\mathbf{p}}$ by Eq. (11.117)),

$$\left[H, b^r_{\mathbf{p}}\right] = -E_{\mathbf{p}}\, b^r_{\mathbf{p}}, \qquad \text{and likewise} \qquad \left[H, d^{r\dagger}_{\mathbf{p}}\right] = +E_{\mathbf{p}}\, d^{r\dagger}_{\mathbf{p}}. \tag{11.127}$$

The Heisenberg equation $\dot{\mathcal{O}} = i\,[H, \mathcal{O}]$ then gives the time dependence $b^r_{\mathbf{p}}(t) = b^r_{\mathbf{p}} e^{-iE_{\mathbf{p}}t}$ and $d^{r\dagger}_{\mathbf{p}}(t) = d^{r\dagger}_{\mathbf{p}} e^{+iE_{\mathbf{p}}t}$ — precisely the phases already written explicitly in the mode expansion Eq. (11.95) as part of $e^{\mp ip\cdot x}$. Assembling,

$$\begin{aligned} i\,[H, \psi(x)] &= \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \left[i\left[H, b^s_{\mathbf{p}}\right] u^s(p) e^{-ip\cdot x} + i\left[H, d^{s\dagger}_{\mathbf{p}}\right] v^s(p) e^{+ip\cdot x} \right] \\ &= \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} \left[(-iE_{\mathbf{p}}) b^s_{\mathbf{p}} u^s(p) e^{-ip\cdot x} + (+iE_{\mathbf{p}}) d^{s\dagger}_{\mathbf{p}} v^s(p) e^{+ip\cdot x} \right] = \partial_0 \psi(x). \end{aligned} \tag{11.128}$$

Thus $\dot{\psi} = i\,[H, \psi]$ holds. Since each plane wave already satisfies the Dirac equation Eq. (11.97), the operator $\psi(x)$ solves $(i\partial\!\!\!/ - m)\psi = 0$, Eq. (6.70), and the anticommutator quantization is consistent: the Hamiltonian we derived generates exactly the dynamics we postulated.

The same check fails informatively for the commutator quantization. There the relevant identity is the ordinary $[AB, C] = A\,[B, C] + [A, C]\,B$, and with the negative d-sector commutator Eq. (11.112) one finds $\left[H, d^{r\dagger}_{\mathbf{p}}\right] = -E_{\mathbf{p}} d^{r\dagger}_{\mathbf{p}}$ in the norm-positive relabelling — the wrong sign, signalling once more that the $d^\dagger$ excitation lowers the energy. Only the anticommutator gives a Heisenberg evolution compatible with a Hamiltonian bounded below.

Physical Insight

The logic has been entirely deductive. We did not assume the Pauli principle; we wrote down the most general operator field consistent with the Dirac equation and Lorentz-covariant normalization, computed its energy, and discovered that bosonic quantization is mathematically inconsistent (negative norms or no ground state), while fermionic quantization is consistent. The exclusion principle, $(b^{s\dagger}_{\mathbf{p}})^2 = 0$ from $\left\{b^{s\dagger}_{\mathbf{p}}, b^{s\dagger}_{\mathbf{p}}\right\} = 0$, is now a *theorem* about the spin-$\frac{1}{2}$ field rather than a postulate of atomic physics — the special case of the spin-statistics connection (Theorem 9.33) that Chapter 9 proved for spin 0 and explicitly deferred for half-integer spin. The deferred half is discharged here for the free Dirac field; the general microcausality argument completing the connection is taken up in Section 11.6.4.

This is the principled corroboration promised in the opening: the classical Dirac bracket $\left\{\psi_a(\mathbf{x}), \psi^\dagger_b(\mathbf{y})\right\}_{\mathrm{D}} = -i\delta_{ab}\delta^3(\mathbf{x}-\mathbf{y})$ of Section 6.5.4, symmetric under the bracket exchange because the constraints χ_1, χ_2 are second class, becomes under the quantization rule $\{\cdot,\cdot\}_{\mathrm{D}} \to -i\,\{\cdot,\cdot\}$ exactly the anticommutator Eq. (11.100). The constraint analysis of Chapter 6 predicted Fermi statistics from the structure of the classical theory; the spectrum analysis of this section

has now shown, independently, that nothing else is consistent. Two roads, one destination.

We have promoted the Dirac field to an operator, fixed its algebra, and arrived at a consistent set of creation and annihilation operators $\{b^s_{\mathbf{p}}, b^{s\dagger}_{\mathbf{p}}, d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\}$ obeying the fermionic ladder algebra Eqs. (11.116) and (11.117), together with a positive Hamiltonian Eq. (11.124) and a unique vacuum. The next section builds the antisymmetric Fock space on these operators, normal-orders the energy, momentum, and charge into manifestly sensible observables, and reads off the excitations as the spin-$\frac{1}{2}$ electrons and positrons of mass m that carry QED into its quantum layer.

11.4 Fock Space, the Charge, and the Electron

The previous section ended with an algebra and a warning. Solving the operator Dirac equation by a plane-wave expansion produced, for each spatial momentum $\mathbf{p}$ and each of two spin labels $s \in \{1, 2\}$, two families of operators: $b^s_{\mathbf{p}}$, $b^{s\dagger}_{\mathbf{p}}$ and $d^s_{\mathbf{p}}$, $d^{s\dagger}_{\mathbf{p}}$. The mode expansion of the field, fixed in Eq. (11.95), reads

$$\psi(x) = \sum_{s=1,2} \int \widetilde{dp} \left[b^s_{\mathbf{p}}\, u^s(\mathbf{p})\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(\mathbf{p})\, e^{+ip\cdot x} \right], \qquad \int \widetilde{dp} \equiv \int \frac{d^3p}{(2\pi)^3}\, \frac{1}{2E_{\mathbf{p}}}, \quad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 \, \tag{11.129}$$

and the equal-time canonical relations, which for a half-integer-spin field were forced to be *anti*commutators rather than commutators (this was the entire lesson of Section 11.3), collapsed into

$$\left\{ b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}} \right\} = \left\{ d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}} \right\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\, \delta^3(\mathbf{p} - \mathbf{q}), \tag{11.130}$$

with every other anticommutator among the ladder operators vanishing. This is the import we work from; it is established in Eq. (11.116) and we shall not rederive it. As in the scalar case (Section 9.3) we use the relativistic normalization throughout, so the factor $(2\pi)^3 2E_{\mathbf{p}}$ rather than a bare $(2\pi)^3$ sits on the right of Eq. (11.130); this is the convention fixed book-wide in Section 9.2.5.

What we have so far is only an algebra. We have no Hilbert space, no states, no energies, and no answer to the question that has hung over the chapter since Chapter 3 promised it: the half-integer-spin representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ classified there must, when quantized, describe a relativistic particle of mass m and spin $\frac{1}{2}$ — and it must do so while respecting the antiparticle doubling that Section 3.7 flagged as the inevitable price of building local fields from positive-energy orbits. This section delivers all of it. We build the state space from Eq. (11.130), find that it is forced to be *anti*symmetric and so contains the Pauli exclusion principle as an algebraic identity; we normal-order the Hamiltonian, the momentum, and the $U(1)$ charge into manifestly sensible operators, tracking with care the single extra

minus sign that the anticommutators inject and that the bosonic computation of Chapter 9 never saw; and we identify the two species of excitation as the electron and the positron, equal in mass and spin, opposite in charge. This is the quantum layer of QED: the classical $U(1)$ matter field of Chapter 8 becomes, here, an operator whose elementary quanta are electrons created by $b^{s\dagger}$ and positrons created by $d^{s\dagger}$.

Two structural contrasts with the scalar will recur and are worth stating at the outset, because they are the whole physics of the fermionic case. First, where commuting creation operators built a *symmetric* Fock space and Bose statistics, anticommuting ones will build an *antisymmetric* Fock space and Fermi statistics — and the antisymmetry is exactly the exclusion principle. Second, where the bosonic zero-point energy came out positive (the divergent $E_0 = +\frac{V}{2}\int E_{\mathbf{p}}$ of Eq. (9.98)), the fermionic one will come out *negative*, and the sign flip is not cosmetic: it is the same flip that, had we quantized with commutators, would have produced a Hamiltonian unbounded below. The anticommutators that Section 11.3 forced on us to cure that disease are precisely what we now see paying off in the spectrum.

11.4.1 The vacuum and its uniqueness

A state space built from lowering and raising operators must have a floor. We define it and then show, for the free theory, that it is forced.

Definition 11.24 (The Dirac vacuum)**.** The *vacuum* $|0\rangle$ is the state annihilated by every lowering operator of both families,

$$b^s_{\mathbf{p}}\,|0\rangle = 0, \qquad d^s_{\mathbf{p}}\,|0\rangle = 0 \qquad \text{for all } \mathbf{p},\ s, \tag{11.131}$$

normalized to $\langle 0|0\rangle = 1$.

That *both* families annihilate $|0\rangle$ is the operator content of the statement that the vacuum contains neither electrons nor positrons. The uniqueness argument is the fermionic copy of the scalar one (Section 9.3.1), and the anticommutators make it, if anything, cleaner.

Proposition 11.25 (Uniqueness of the free vacuum)**.** *Up to an overall phase,* $|0\rangle$ *is the only normalized state annihilated by all* $b^s_{\mathbf{p}}$ *and all* $d^s_{\mathbf{p}}$*.*

Proof. The states obtained by acting on $|0\rangle$ with strings of creation operators span the whole space (this is the Fock construction of Section 11.4.3 below). Any candidate $|\Omega\rangle$ therefore expands as a c-number multiple of $|0\rangle$ plus terms each containing at least one creation operator,

$$|\Omega\rangle = c_0\,|0\rangle + \sum_s \int \widetilde{dp}\; c^{(b)}_s(\mathbf{p})\, b^{s\dagger}_{\mathbf{p}}\,|0\rangle + \sum_s \int \widetilde{dp}\; c^{(d)}_s(\mathbf{p})\, d^{s\dagger}_{\mathbf{p}}\,|0\rangle + (\text{two or more creation operators}). \tag{11.132}$$

Apply $b^r_{\mathbf{k}}$. On the left it gives zero by hypothesis. On the right, anticommute $b^r_{\mathbf{k}}$ to the right using Eq. (11.130); against the one-electron term it yields $b^r_{\mathbf{k}}\, b^{s\dagger}_{\mathbf{p}}\,|0\rangle = \left\{b^r_{\mathbf{k}}, b^{s\dagger}_{\mathbf{p}}\right\}|0\rangle = (2\pi)^3 2E_{\mathbf{k}}\,\delta^{rs}\delta^3(\mathbf{k}-\mathbf{p})\,|0\rangle$, which integrates to $c^{(b)}_r(\mathbf{k})\,|0\rangle$; it kills the one-positron term outright (because b anticommutes with $d^\dagger$ and annihilates $|0\rangle$), and it lowers every higher term by one electron. Demanding the total vanish sector by sector forces $c^{(b)}_r \equiv 0$; the identical argument with $d^r_{\mathbf{k}}$ forces $c^{(d)}_r \equiv 0$, and an induction on the number of creation operators kills every coefficient with at least one. Hence $|\Omega\rangle = c_0\,|0\rangle$, and normalization fixes $|c_0| = 1$. □

11.4.2 One-particle states: the electron and the positron

Acting once with a creation operator produces the first excited sector. We adopt the covariant normalization that made the scalar states (Theorem 9.12) transform without stray Jacobians.

Definition 11.26 (One-particle states)**.** For an on-shell momentum $p^\mu = (E_{\mathbf{p}}, \mathbf{p})$ and spin label s,

$$|\mathbf{p}, s\rangle_b \equiv b^{s\dagger}_{\mathbf{p}}\,|0\rangle \qquad (\text{an } \textit{electron}), \qquad\qquad |\mathbf{p}, s\rangle_d \equiv d^{s\dagger}_{\mathbf{p}}\,|0\rangle \qquad (\text{a } \textit{positron}). \tag{11.133}$$

These names are, for the moment, only labels for the two ladder families; what earns the electron the right to its name is its charge and spin, derived below. Their inner products follow directly from Eq. (11.130). For two electrons,

$${}_b\langle \mathbf{p}, r|\mathbf{q}, s\rangle_b = \langle 0|\, b^r_{\mathbf{p}}\, b^{s\dagger}_{\mathbf{q}}\,|0\rangle = \langle 0|\big(\left\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\right\} - b^{s\dagger}_{\mathbf{q}} b^r_{\mathbf{p}}\big)|0\rangle = (2\pi)^3\, 2E_{\mathbf{p}}\,\delta^{rs}\,\delta^3(\mathbf{p}-\mathbf{q}), \tag{11.134}$$

the second operator term dying because $b^r_{\mathbf{p}}\,|0\rangle = 0$. The same holds for two positrons, while an electron and a positron are orthogonal because b anticommutes with $d^\dagger$ and both annihilate the vacuum: ${}_b\langle \mathbf{p}, r|\mathbf{q}, s\rangle_d = \langle 0|\, b^r_{\mathbf{p}} d^{s\dagger}_{\mathbf{q}}\,|0\rangle = -\langle 0|\, d^{s\dagger}_{\mathbf{q}} b^r_{\mathbf{p}}\,|0\rangle = 0$. The right-hand side of Eq. (11.134) is the same Lorentz-invariant combination $2E_{\mathbf{p}}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$ whose invariance was proved in Theorem 9.14; the bookkeeping is identical to the scalar, the only novelty being the Kronecker δ^{rs} recording that the two spin states are orthogonal.

11.4.3 Pauli exclusion and the antisymmetric Fock space

Here the anticommutators announce themselves. The single most consequential fact about the fermionic state space is visible already on the second rung of the ladder, and it is an algebraic identity, not a postulate.

Theorem 11.27 (Pauli exclusion principle)**.** *For every momentum* $\mathbf{p}$ *and spin* s*,*

$$\left(b^{s\dagger}_{\mathbf{p}}\right)^2 = 0, \qquad\qquad \left(d^{s\dagger}_{\mathbf{p}}\right)^2 = 0. \tag{11.135}$$

Consequently no quantum-mechanical state can contain two electrons (or two positrons) of identical momentum and spin: the would-be state $b_{\mathbf{p}}^{s\dagger} b_{\mathbf{p}}^{s\dagger} |0\rangle$ is the zero vector.

Proof. Set $r = s$ and $\mathbf{q} = \mathbf{p}$ in the vanishing anticommutator $\left\{ b_{\mathbf{p}}^{r\dagger}, b_{\mathbf{q}}^{s\dagger} \right\} = 0$ (part of Eq. (11.130)). The anticommutator of an operator with itself is twice its square, $\left\{ b_{\mathbf{p}}^{s\dagger}, b_{\mathbf{p}}^{s\dagger} \right\} = 2(b_{\mathbf{p}}^{s\dagger})^2$, so $2(b_{\mathbf{p}}^{s\dagger})^2 = 0$, giving Eq. (11.135); identically for d. A vector of zero norm is the zero vector: applying $b_{\mathbf{p}}^{s\dagger}$ twice to $|0\rangle$ returns nothing to act on. □

Physical Insight

The exclusion principle, which the reader met in nonrelativistic quantum mechanics as an empirical rule grafted onto many-electron atoms, is here a *theorem*: it is the $\mathbf{q} \to \mathbf{p}$, $r \to s$ corner of the anticommutation relations that spin-$\frac{1}{2}$ quantization forced upon us. One cannot put two identical electrons in the same mode for the same reason one cannot square a Grassmann number: the creation operator is nilpotent. Atomic shell structure, the stability of bulk matter, the Fermi surface of a metal — all of chemistry's periodic regularities — trace back to this one line of algebra. The contrast with the scalar is total: there $\left[a_{\mathbf{p}}^{\dagger}, a_{\mathbf{q}}^{\dagger} \right] = 0$ let any number of bosons pile into a mode (Eq. (9.81)); here $\left\{ b_{\mathbf{p}}^{\dagger}, b_{\mathbf{q}}^{\dagger} \right\} = 0$ caps the occupation of each mode at one.

For distinct labels the same vanishing anticommutator gives antisymmetry rather than nullity. The two-electron state obeys

$$b_{\mathbf{p}}^{r\dagger} b_{\mathbf{q}}^{s\dagger} |0\rangle = - b_{\mathbf{q}}^{s\dagger} b_{\mathbf{p}}^{r\dagger} |0\rangle , \tag{11.136}$$

so exchanging the two quanta — both their momenta and their spins together — flips the sign of the state. This is Fermi statistics, and Eq. (11.135) is its diagonal special case: a state antisymmetric under exchange of two identical labels must vanish. The inner product of two such two-electron states exhibits the antisymmetric ("determinant") structure that is the fermionic counterpart of the bosonic permanent of Theorem 9.16.

Worked Example 11.28 (The two-electron inner product is a Slater determinant)**.** We compute ${}_b\langle \mathbf{p}_1 r_1; \mathbf{p}_2 r_2 |\, \mathbf{q}_1 s_1; \mathbf{q}_2 s_2 \rangle_b$ directly from the algebra, where $|\mathbf{p}_1 r_1; \mathbf{p}_2 r_2\rangle_b \equiv b_{\mathbf{p}_1}^{r_1\dagger} b_{\mathbf{p}_2}^{r_2\dagger} |0\rangle$. The bra is $\langle 0|\, b_{\mathbf{p}_2}^{r_2} b_{\mathbf{p}_1}^{r_1}$, so

$${}_b\langle \mathbf{p}_1 r_1; \mathbf{p}_2 r_2 | \mathbf{q}_1 s_1; \mathbf{q}_2 s_2 \rangle_b = \langle 0|\, b_{\mathbf{p}_2}^{r_2} b_{\mathbf{p}_1}^{r_1}\, b_{\mathbf{q}_1}^{s_1\dagger} b_{\mathbf{q}_2}^{s_2\dagger} |0\rangle . \tag{11.137}$$

Anticommute $b_{\mathbf{p}_1}^{r_1}$ to the right through the two creation operators, using $b_{\mathbf{p}_1}^{r_1} b_{\mathbf{q}}^{s\dagger} = \left\{ b_{\mathbf{p}_1}^{r_1}, b_{\mathbf{q}}^{s\dagger} \right\} - b_{\mathbf{q}}^{s\dagger} b_{\mathbf{p}_1}^{r_1}$ each time, picking up a *minus sign* at every step where it

passes a creation operator without contracting:

$$b^{r_1}_{\mathbf{p}_1} b^{s_1\dagger}_{\mathbf{q}_1} b^{s_2\dagger}_{\mathbf{q}_2} = \left\{ b^{r_1}_{\mathbf{p}_1}, b^{s_1\dagger}_{\mathbf{q}_1} \right\} b^{s_2\dagger}_{\mathbf{q}_2} - b^{s_1\dagger}_{\mathbf{q}_1} b^{r_1}_{\mathbf{p}_1} b^{s_2\dagger}_{\mathbf{q}_2}$$
$$= \left\{ b^{r_1}_{\mathbf{p}_1}, b^{s_1\dagger}_{\mathbf{q}_1} \right\} b^{s_2\dagger}_{\mathbf{q}_2} - b^{s_1\dagger}_{\mathbf{q}_1} \left\{ b^{r_1}_{\mathbf{p}_1}, b^{s_2\dagger}_{\mathbf{q}_2} \right\} + b^{s_1\dagger}_{\mathbf{q}_1} b^{s_2\dagger}_{\mathbf{q}_2} b^{r_1}_{\mathbf{p}_1}. \tag{11.138}$$

The last term annihilates $|0\rangle$ on the right of Eq. (11.137). Writing $\left\{ b^{r}_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}} \right\} \equiv \langle pr|qs\rangle = (2\pi)^3 2E_{\mathbf{p}} \delta^{rs} \delta^3(\mathbf{p}-\mathbf{q})$ for the one-particle norm Eq. (11.134), and then applying $b^{r_2}_{\mathbf{p}_2}$ to the surviving single creation operators,

$$_b\langle \mathbf{p}_1 r_1; \mathbf{p}_2 r_2 | \mathbf{q}_1 s_1; \mathbf{q}_2 s_2 \rangle_b = \langle p_1 r_1 | q_1 s_1 \rangle \langle p_2 r_2 | q_2 s_2 \rangle - \langle p_1 r_1 | q_2 s_2 \rangle \langle p_2 r_2 | q_1 s_1 \rangle. \tag{11.139}$$

The relative minus sign between the two pairings is the entire difference from the bosonic result Eq. (9.85), where both terms appeared with a plus. Equation Eq. (11.139) is a 2×2 determinant — a Slater determinant — and its antisymmetry under $(\mathbf{q}_1 s_1) \leftrightarrow (\mathbf{q}_2 s_2)$ (it changes sign) is the inner-product face of Eq. (11.136). Setting the two outgoing labels equal makes the determinant vanish: two identical fermions have no state, exactly Theorem 11.27.

The pattern generalizes. The n-electron state $b^{s_1\dagger}_{\mathbf{p}_1} \cdots b^{s_n\dagger}_{\mathbf{p}_n} |0\rangle$ is totally antisymmetric under simultaneous exchange of any two label pairs $(\mathbf{p}_i, s_i)$, and an $n \times n$ Slater determinant gives its overlaps. The state space is therefore built not from symmetric tensor powers, as for the boson, but from antisymmetric ones.

Definition 11.29 (The fermionic Fock space)**.** Let $\mathcal{H}_1^{(e)}$ and $\mathcal{H}_1^{(\bar{e})}$ be the one-electron and one-positron spaces spanned by the states Eq. (11.133). The Fock space of the free Dirac field is

$$\mathcal{F} = \bigoplus_{n,\bar{n}=0}^{\infty} \left(\bigwedge^n \mathcal{H}_1^{(e)} \right) \otimes \left(\bigwedge^{\bar{n}} \mathcal{H}_1^{(\bar{e})} \right), \tag{11.140}$$

the direct sum over electron number n and positron number $\bar{n}$ of the *antisymmetric* (exterior) tensor powers $\bigwedge^n$, with $\bigwedge^0 = \mathbb{C}\,|0\rangle$. The exterior power $\bigwedge^n$ implements the Fermi antisymmetry Eq. (11.136) at the level of the space itself, exactly as the symmetric power Sym^n of Theorem 9.17 implemented Bose symmetry.

The occupation-number description is correspondingly stark. In a box, with modes labelled by $(\mathbf{k}, s)$, a basis state is specified by occupation numbers $n_{\mathbf{k},s}$ and $\bar{n}_{\mathbf{k},s}$, each of which — by Theorem 11.27 — can only be 0 or 1. The fermionic Fock space is the antisymmetric counterpart of the scalar's, and "occupation 0 or 1" is the same exclusion principle viewed as a constraint on the basis.

11.4.4 The Hamiltonian: normal ordering and the rescue by anticommutation

We now make contact with dynamics, and meet the place where the fermionic computation diverges from the bosonic one in a way that matters. The energy

and momentum operators are the Noether charges of spacetime translations (Theorem 7.10), the components of the four-momentum built from the Dirac stress tensor. On shell that tensor is $T^{\mu\nu} = i\bar\psi\gamma^\mu\partial^\nu\psi$ (Eq. (6.188)), so

$$P^\mu = \int d^3x\, T^{0\mu} = \int d^3x\, i\,\bar\psi\gamma^0\partial^\mu\psi = \int d^3x\, i\,\psi^\dagger\partial^\mu\psi, \qquad H = P^0 = \int d^3x\, i\,\psi^\dagger\partial_0\psi, \tag{11.141}$$

using $\bar\psi\gamma^0 = \psi^\dagger$. Our task is to substitute Eq. (11.129) and reduce H to the ladder operators. Two ingredients from the spinor algebra of Section 11.2 are needed: the on-shell normalizations

$$u^{r\dagger}(\mathbf{p})\,u^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \qquad v^{r\dagger}(\mathbf{p})\,v^s(\mathbf{p}) = 2E_{\mathbf{p}}\,\delta^{rs}, \tag{11.142}$$

and the orthogonality of a positive-energy spinor against a negative-energy one of reversed three-momentum,

$$u^{r\dagger}(\mathbf{p})\,v^s(-\mathbf{p}) = 0 = v^{r\dagger}(\mathbf{p})\,u^s(-\mathbf{p}), \tag{11.143}$$

both proved in Eqs. (11.65) and (11.67). We now differentiate and integrate.

First, $i\partial_0$ acting on the mode expansion brings down $+E_{\mathbf{p}}$ on each positive-frequency factor $e^{-ip\cdot x}$ and $-E_{\mathbf{p}}$ on each negative-frequency factor $e^{+ip\cdot x}$:

$$i\partial_0\psi(x) = \sum_s \int \widetilde{dq}\; E_{\mathbf{q}}\Big[b^s_{\mathbf{q}}\,u^s(\mathbf{q})\,e^{-iq\cdot x} - d^{s\dagger}_{\mathbf{q}}\,v^s(\mathbf{q})\,e^{+iq\cdot x}\Big]. \tag{11.144}$$

Multiplying by $\psi^\dagger(x) = \sum_r \int \widetilde{dp}\,[\,b^{r\dagger}_{\mathbf{p}}u^{r\dagger}(\mathbf{p})e^{+ip\cdot x} + d^r_{\mathbf{p}}v^{r\dagger}(\mathbf{p})e^{-ip\cdot x}\,]$ and integrating $\int d^3x$ produces, from the spatial parts of the exponentials, either $(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$ (for the two "diagonal" pairings $b^\dagger b$ and $d\,d^\dagger$) or $(2\pi)^3\delta^3(\mathbf{p}+\mathbf{q})$ (for the "off-diagonal" pairings $b^\dagger d^\dagger$ and $d\,b$). Consider the four contributions in turn.

The off-diagonal pairings carry the time dependence $e^{\pm 2iE_{\mathbf{p}}t}$ and the spinor factor $u^{r\dagger}(\mathbf{p})v^s(-\mathbf{p})$ or $v^{r\dagger}(\mathbf{p})u^s(-\mathbf{p})$, which vanish by Eq. (11.143). Their disappearance is the operator statement that H is time-independent: had they survived, H would oscillate at $2E_{\mathbf{p}}$. This is the same cancellation seen for the scalar charge in Theorem 9.36, here enforced by the spinor orthogonality rather than by a parity argument. We are left with the two diagonal pairings. The $b^\dagger b$ pairing gives, after the $\widetilde{dq}$ integral collapses $\delta^3(\mathbf{p}-\mathbf{q})$ and Eq. (11.142) collapses the spin sum,

$$\sum_{r,s}\int \widetilde{dp}\,\widetilde{dq}\;(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})\,E_{\mathbf{q}}\,b^{r\dagger}_{\mathbf{p}}b^s_{\mathbf{q}}\,u^{r\dagger}(\mathbf{p})u^s(\mathbf{q}) = \sum_{r,s}\int \widetilde{dp}\;\frac{E_{\mathbf{p}}}{2E_{\mathbf{p}}}\,b^{r\dagger}_{\mathbf{p}}b^s_{\mathbf{p}}\,(2E_{\mathbf{p}}\delta^{rs}) = \sum_s\int \widetilde{dp}\;E_{\mathbf{p}}\,b^{s\dagger}_{\mathbf{p}}b^s_{\mathbf{p}}, \tag{11.145}$$

where $\int \widetilde{dq}\,(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q}) = 1/2E_{\mathbf{p}}$. The $d\,d^\dagger$ pairing carries the prefactor $-E_{\mathbf{q}}$ from Eq. (11.144) and, after the same reduction,

$$\sum_{r,s}\int \widetilde{dp}\,\widetilde{dq}\;(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})\,(-E_{\mathbf{q}})\,d^r_{\mathbf{p}}d^{s\dagger}_{\mathbf{q}}\,v^{r\dagger}(\mathbf{p})v^s(\mathbf{q}) = -\sum_s\int \widetilde{dp}\;E_{\mathbf{p}}\,d^s_{\mathbf{p}}d^{s\dagger}_{\mathbf{p}}. \tag{11.146}$$

Adding Eq. (11.145) and Eq. (11.146), the Hamiltonian *before* any reordering is

$$\boxed{H = \sum_s \int \widetilde{dp}\, E_{\mathbf{p}}\, (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}).} \tag{11.147}$$

Here is the crux. The antiparticle term enters with a *minus sign* and in the *wrong order* $d\,d^\dagger$, and how we resolve that order decides whether the theory is sane. Suppose, against the verdict of Section 11.3, we had quantized with commutators, so that $d\,d^\dagger = d^\dagger d + \left[d, d^\dagger\right]$. Then Eq. (11.147) would read $\int E_{\mathbf{p}}(b^\dagger b - d^\dagger d) - (\text{c-number})$: the positron number operator $d^\dagger d$ would contribute with a *negative* coefficient, and by exciting more and more positrons one could lower the energy without bound. The Hamiltonian would have no ground state. This is precisely the catastrophe that Section 11.3 exhibited and used to reject Bose quantization of the spin-$\frac{1}{2}$ field.

The anticommutators avert it. Using Eq. (11.130) in the form $d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}} = -d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} + \left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\}$, the offending term flips sign:

$$-d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}} = +d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} - \left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\}. \tag{11.148}$$

The positron number operator now appears with a $+$ sign, and Eq. (11.147) becomes

$$H = \sum_s \int \widetilde{dp}\, E_{\mathbf{p}}\, (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}) + E_0, \qquad E_0 = -\sum_s \int \widetilde{dp}\, E_{\mathbf{p}}\, \left\{d^s_{\mathbf{p}}, d^{s\,\dagger}_{\mathbf{p}}\right\}. \tag{11.149}$$

The constant E_0 is the fermionic zero-point energy. Evaluating the anticommutator at coincident arguments, $\left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\} = (2\pi)^3 2E_{\mathbf{p}}\,\delta^3(0) = 2E_{\mathbf{p}} V$ with $\delta^3(0) = V/(2\pi)^3$ the spatial volume, gives

$$E_0 = -\sum_{s=1,2} \int \frac{d^3p}{(2\pi)^3}\, \frac{1}{2E_{\mathbf{p}}}\, E_{\mathbf{p}}\, (2E_{\mathbf{p}} V) = -2\,V \int \frac{d^3p}{(2\pi)^3}\, E_{\mathbf{p}}. \tag{11.150}$$

This is the second contrast promised in the introduction. The scalar zero-point energy was $+\frac{V}{2}\int \frac{d^3p}{(2\pi)^3} E_{\mathbf{p}}$ (Eq. (9.98)); the Dirac one is *negative* and larger in magnitude by the count of fermionic degrees of freedom (the factor 2 from the two spin states, with the sign from the anticommutator). The sign of the vacuum energy is thus a direct readout of statistics.

Physical Insight

That bosonic zero-point energies are positive and fermionic ones negative is not a curiosity but a structural seed. In a theory with equal numbers of bosonic and fermionic degrees of freedom of equal mass, the divergent

vacuum energies Eq. (11.150) and its scalar partner can cancel exactly. This is one of the first things supersymmetry buys, and it is why supersymmetric theories have a vacuum energy under far better control than generic ones. We do not pursue it here; the point is only that the minus sign we just tracked is physical, not bookkeeping.

The constant E_0 is removed exactly as for the scalar (Section 9.4), by normal ordering — with one amendment that the fermionic case forces and that is itself the carrier of all the signs above.

Definition 11.30 (Fermionic normal ordering)**.** Let $:\cdots:$ denote the operation that reorders any product of creation and annihilation operators so that every annihilation operator stands to the right of every creation operator, treating the operators as though they *anti*commuted in the process — that is, *multiplying by* (-1) *for each transposition of two fermionic operators performed during the reordering, and generating no anticommutator (contraction) terms.* For example $:d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}: \;= -\,d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}$.

The sign in $:d\,d^\dagger: \;= -d^\dagger d$ is the only difference from the bosonic prescription of Theorem 9.19, and it is exactly the sign of Eq. (11.148): normal ordering keeps the reordered operator product and discards the (here negative, divergent) c-number anticommutator. Applying it to Eq. (11.147),

$$\boxed{:H: \;= \sum_s \int \widetilde{dp}\; E_{\mathbf{p}} \left(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}\right) = H - E_0.} \tag{11.151}$$

As in the scalar case (Theorem 9.20), subtracting the c-number E_0 changes no commutator and no equation of motion; it only slides the whole spectrum so the vacuum sits at zero energy. We adopt $:H:$ as *the* Hamiltonian of the free Dirac field. Its decisive property, impossible to read off Eq. (11.147) but manifest in Eq. (11.151), is positivity.

Theorem 11.31 (The Hamiltonian is bounded below; the vacuum is its ground state)**.** *The operator* $:H:$ *of Eq. (11.151) satisfies* $:H: \;\geq 0$*, with* $:H:\,|0\rangle = 0$*, and the vacuum is the unique zero-energy state. The one-electron and one-positron states are eigenstates of energy* $E_{\mathbf{p}}$*:*

$$:H:\; b^{s\dagger}_{\mathbf{p}}\,|0\rangle = E_{\mathbf{p}}\, b^{s\dagger}_{\mathbf{p}}\,|0\rangle\,, \qquad :H:\; d^{s\dagger}_{\mathbf{p}}\,|0\rangle = E_{\mathbf{p}}\, d^{s\dagger}_{\mathbf{p}}\,|0\rangle\,. \tag{11.152}$$

Proof. For any state $|\Psi\rangle$, the diagonal matrix element of Eq. (11.151) is

$$\langle\Psi|:H:|\Psi\rangle = \sum_s \int \widetilde{dp}\; E_{\mathbf{p}} \left(\left\| b^s_{\mathbf{p}}\,|\Psi\rangle \right\|^2 + \left\| d^s_{\mathbf{p}}\,|\Psi\rangle \right\|^2\right) \geq 0, \tag{11.153}$$

because each $b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} = (b^s_{\mathbf{p}})^\dagger b^s_{\mathbf{p}}$ is a positive operator (its expectation is the squared norm of $b^s_{\mathbf{p}}|\Psi\rangle$), each $E_{\mathbf{p}} > 0$, and the measure is positive. The integrand vanishes for all $\mathbf{p}, s$ iff $b^s_{\mathbf{p}}|\Psi\rangle = d^s_{\mathbf{p}}|\Psi\rangle = 0$ for all $\mathbf{p}, s$, which by Theorem 11.25 forces $|\Psi\rangle \propto |0\rangle$; and indeed $:H:|0\rangle = 0$ since every annihilation operator kills $|0\rangle$. Thus 0 is the lowest eigenvalue and $|0\rangle$ its unique eigenstate. For Eq. (11.152) we need the commutator of $:H:$ with a creation operator. The computation uses the fermionic analogue of the identity $[AB, C] = A\,[B, C] + [A, C]\,B$, valid when A, B, C are built from fermionic operators:

$$\left[b^{r\dagger}_{\mathbf{k}} b^r_{\mathbf{k}}, b^{s\dagger}_{\mathbf{q}}\right] = b^{r\dagger}_{\mathbf{k}} \left\{b^r_{\mathbf{k}}, b^{s\dagger}_{\mathbf{q}}\right\} - \left\{b^{r\dagger}_{\mathbf{k}}, b^{s\dagger}_{\mathbf{q}}\right\} b^r_{\mathbf{k}} = b^{r\dagger}_{\mathbf{k}}\,(2\pi)^3 2E_{\mathbf{k}}\delta^{rs}\delta^3(\mathbf{k}-\mathbf{q}), \quad (11.154)$$

the second anticommutator vanishing by Eq. (11.130). (The identity $[AB, C] = A\,\{B, C\} - \{A, C\}\,B$ is checked by expanding both sides; it is the rule that lets a *commutator* of a fermion bilinear with a single fermion be evaluated using *anti*commutators.) Integrating Eq. (11.154) against $E_{\mathbf{k}}$ over the invariant measure gives $\left[:H:, b^{s\dagger}_{\mathbf{q}}\right] = E_{\mathbf{q}}\, b^{s\dagger}_{\mathbf{q}}$, and identically with d. Acting on the vacuum, $:H:\, b^{s\dagger}_{\mathbf{q}}|0\rangle = \left(\left[:H:, b^{s\dagger}_{\mathbf{q}}\right] + b^{s\dagger}_{\mathbf{q}} :H:\right)|0\rangle = E_{\mathbf{q}} b^{s\dagger}_{\mathbf{q}}|0\rangle$, which is Eq. (11.152). □

The number operators $N_b = \sum_s \int \widetilde{dp}\, b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}}$ and $N_d = \sum_s \int \widetilde{dp}\, d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}$ count electrons and positrons; by the manipulation Eq. (11.154) with $E_{\mathbf{k}} \to 1$ they satisfy $\left[N_b, b^{s\dagger}_{\mathbf{q}}\right] = b^{s\dagger}_{\mathbf{q}}$ and $\left[N_d, d^{s\dagger}_{\mathbf{q}}\right] = d^{s\dagger}_{\mathbf{q}}$, so they return the electron and positron counts of any Fock state. Per mode each takes only the eigenvalues 0 and 1, because $(b^{s\dagger}_{\mathbf{k}})^2 = 0$ (Theorem 11.27) forces $(b^{s\dagger}_{\mathbf{k}} b^s_{\mathbf{k}})^2 = b^{s\dagger}_{\mathbf{k}} b^s_{\mathbf{k}}$ up to the c-number anticommutator, the defining relation of a projector with spectrum $\{0, 1\}$. The exclusion principle is, equivalently, the statement that the occupation of every fermionic mode is a yes/no bit.

11.4.5 Momentum

The spatial momentum is the $\mu = i$ component of Eq. (11.141), $P^i = \int d^3x\, i\psi^\dagger \partial^i \psi$. The computation is the one just done with $i\partial_0$ replaced by $i\partial^i$, which brings down $+p^i$ on the positive-frequency factor and $-p^i$ on the negative-frequency factor. The off-diagonal terms again die by Eq. (11.143), and the diagonal terms give, in exact parallel with Eq. (11.147),

$$\mathbf{P} = \sum_s \int \widetilde{dp}\; \mathbf{p}\,(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}). \quad (11.155)$$

Anticommuting the d term as in Eq. (11.148) converts $-d\,d^\dagger$ into $+d^\dagger d$ minus a c-number; but this time the c-number, $-\sum_s \int \widetilde{dp}\,\mathbf{p}\left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\} \propto \int d^3p\,\mathbf{p}\,\delta^3(0)$, has an *odd* integrand in $\mathbf{p}$ and vanishes identically — there is no zero-point momentum, just as there was none for the scalar (Eq. (9.103)). Hence already

without invoking normal ordering,

$$\boxed{\mathbf{P} = \sum_s \int \widetilde{dp}\, \mathbf{p}\, (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}).} \tag{11.156}$$

Together with Eq. (11.151) this assembles into the manifestly covariant statement $P^\mu = \sum_s \int \widetilde{dp}\, p^\mu (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}})$ with $p^0 = E_{\mathbf{p}}$, so that

$$P^\mu\, b^{s\dagger}_{\mathbf{p}} |0\rangle = p^\mu\, b^{s\dagger}_{\mathbf{p}} |0\rangle\,, \qquad P^\mu\, d^{s\dagger}_{\mathbf{p}} |0\rangle = p^\mu\, d^{s\dagger}_{\mathbf{p}} |0\rangle\,, \qquad p^\mu = (E_{\mathbf{p}}, \mathbf{p}), \quad p^2 = m^2. \tag{11.157}$$

Both species are on the mass shell $p^2 = m^2$ of a particle of mass m. Notice the relabelling that makes the positron physical: the operator $d^{s\dagger}_{\mathbf{p}}$ multiplies $e^{+ip\cdot x}$ in the field Eq. (11.129), a *negative*-frequency wave, yet it creates a state of *positive* energy $+E_{\mathbf{p}}$ and physical momentum $+\mathbf{p}$. This is the operator resolution of the negative-energy solutions of the classical Dirac equation: rather than describing electrons of negative energy, they are repackaged as positrons of positive energy, exactly the (II),(IV)→ antiparticle reinterpretation anticipated for the backward orbits in Section 3.7.

11.4.6 The charge: electron and positron carry opposite charge

Mass and momentum cannot tell an electron from a positron — Eq. (11.157) treats b and d symmetrically, just as the scalar Hamiltonian treated particle and antiparticle symmetrically (Section 9.7.3). The quantity that distinguishes them is the conserved $U(1)$ charge. The free Dirac Lagrangian Eq. (6.68) is invariant under the global phase rotation $\psi \to e^{-i\alpha}\psi$, $\bar\psi \to e^{+i\alpha}\bar\psi$; by Noether's theorem (Theorem 7.10) this carries the conserved current

$$j^\mu = \bar\psi \gamma^\mu \psi, \qquad \partial_\mu j^\mu = 0, \tag{11.158}$$

the divergence vanishing on shell by the Dirac equation. The associated charge is the spatial integral of $j^0 = \bar\psi\gamma^0\psi = \psi^\dagger\psi$, which we promote to an operator:

$$Q = \int d^3x\, j^0 = \int d^3x\, \psi^\dagger\psi. \tag{11.159}$$

Theorem 11.32 (The Dirac charge operator)**.** *With normal ordering (Theorem 11.30), which fixes the additive constant by demanding $Q|0\rangle = 0$,*

$$Q = \sum_s \int \widetilde{dp}\, (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}) = N_b - N_d. \tag{11.160}$$

It assigns the ladder operators definite charges,

$$\left[Q, b^{s\dagger}_{\mathbf{p}}\right] = +\, b^{s\dagger}_{\mathbf{p}}, \qquad \left[Q, d^{s\dagger}_{\mathbf{p}}\right] = -\, d^{s\dagger}_{\mathbf{p}}, \qquad \left[Q, b^s_{\mathbf{p}}\right] = -\, b^s_{\mathbf{p}}, \qquad \left[Q, d^s_{\mathbf{p}}\right] = +\, d^s_{\mathbf{p}}, \tag{11.161}$$

so the electron carries $Q = +1$ and the positron $Q = -1$.

Proof. Substitute the mode expansions for $\psi^\dagger$ and ψ into Eq. (11.159) and integrate $\int d^3x$. The structure is the energy computation of Section 11.4.4 with the operator $i\partial_0$ deleted, so every $E_{\mathbf{q}}$ prefactor is replaced by 1 and the relative minus sign between the two frequency components is gone. The off-diagonal pairings again vanish by Eq. (11.143); the diagonal pairings, reduced with Eq. (11.142), give

$$Q = \sum_s \int \widetilde{dp}\, (b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}). \tag{11.162}$$

The positron term is in the wrong order, with a *plus* sign this time (there is no $i\partial_0$ to supply the relative minus that Eq. (11.147) carried). Normal ordering reorders it with the fermionic sign of Theorem 11.30, $:d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}}: \;= -d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}}$, discarding the divergent c-number $+\sum_s \int \widetilde{dp}\, \left\{d^s_{\mathbf{p}}, d^{s\dagger}_{\mathbf{p}}\right\}$ (the charge analogue of the zero-point energy, naturally zero because the vacuum is neutral). This yields Eq. (11.160). The commutators Eq. (11.161) follow from Eq. (11.154): $\left[N_b, b^{s\dagger}_{\mathbf{p}}\right] = +b^{s\dagger}_{\mathbf{p}}$, and the explicit minus in front of N_d in Eq. (11.160) flips the sign for d, giving $\left[Q, d^{s\dagger}_{\mathbf{p}}\right] = -d^{s\dagger}_{\mathbf{p}}$; the annihilation relations are the Hermitian conjugates. Finally, $[Q,\psi] = \sum_s \int \widetilde{dp}(\left[Q, b^s_{\mathbf{p}}\right] u^s e^{-ipx} + \left[Q, d^{s\dagger}_{\mathbf{p}}\right] v^s e^{+ipx}) = -\psi$, so $e^{i\alpha Q}\psi\, e^{-i\alpha Q} = e^{-i\alpha}\psi$: Q generates the very $U(1)$ rotation it was built from, closing the loop with Noether. □

The reading of Eq. (11.160) is the same as for the complex scalar (Theorem 9.36), and that parallel is the point: the scalar furnished the bosonic template — a charge that counts particles minus antiparticles, with the two species of equal mass and opposite charge — and the Dirac field now realizes the identical structure for spin $\frac{1}{2}$. Acting on the one-particle states,

$$Q\,|\mathbf{p},s\rangle_b = +\,|\mathbf{p},s\rangle_b\,, \qquad Q\,|\mathbf{p},s\rangle_d = -\,|\mathbf{p},s\rangle_d\,, \tag{11.163}$$

two species of equal mass m (both on the shell of the single field ψ) and opposite charge. Because $[Q, :H:\,] = 0$ — both are diagonal in N_b, N_d — charge is conserved, and because Q generates the symmetry it superselects: states of different charge never mix.

Notation Watch

The operator Q of Eq. (11.160) is the *number* charge, normalized so that the electron carries $+1$. The physical *electric* charge operator is $q\,Q$ with $q = -e < 0$ the electron's charge, in the convention $D_\mu = \partial_\mu + ieA_\mu$ ($e > 0$) fixed for this chapter and for all of QED. Thus $b^{s\dagger}$ creates the negatively charged electron and $d^{s\dagger}$ the positively charged positron. What is convention-independent is that the two species carry *opposite* charge

of *equal* magnitude and that the vacuum is neutral; the overall sign and unit are tied to the gauge-coupling convention, exactly as flagged for the scalar in Section 9.7.3. No gauge field is switched on in this chapter; the coupling $q\,Q\,A_0$ that makes "$-e$" the measured charge is installed when ψ is minimally coupled to the photon of Chapter 13.

11.4.7 The electron is a mass-m, spin-$\frac{1}{2}$ Poincaré particle

We have shown the excitations carry energy $E_{\mathbf{p}}$, momentum $\mathbf{p}$ on the mass shell, and charge ± 1. To certify that the electron is a *particle* in the precise sense of Chapter 3 — and to read off its spin — we must show the one-electron states furnish an irreducible representation of the Poincaré group and identify which one. The first Casimir is already in hand: Eq. (11.157) gives $P^2\,|\mathbf{p},s\rangle_b = p^2\,|\mathbf{p},s\rangle_b = m^2\,|\mathbf{p},s\rangle_b$, a single sharp mass $m>0$. What is new, and what the two-valued label s exists to encode, is the second Casimir — the spin.

By Wigner's classification (Section 3.7), a positive-energy massive irreducible representation is labelled by $m>0$ together with a representation of the little group, the subgroup that fixes a chosen rest-frame momentum, which for $m>0$ is the rotation group, realized on quantum states as its double cover $SU(2)$ (Section 3.7.2). The spin is the $SU(2)$ representation that the rest-frame states form. We therefore go to the rest frame and watch how rotations act on the two states $b^{1\dagger}_{\mathbf{0}}\,|0\rangle$ and $b^{2\dagger}_{\mathbf{0}}\,|0\rangle$.

The input is the transformation law of the field under the unitary $U(\Lambda)$ that implements a Lorentz transformation Λ on Fock space, established in Eq. (11.164):

$$U(\Lambda)\,\psi(x)\,U(\Lambda)^{-1} = S(\Lambda)^{-1}\psi(\Lambda x), \qquad S(\Lambda)=\exp\!\Big(-\tfrac{i}{2}\,\omega_{\mu\nu}S^{\mu\nu}\Big), \quad S^{\mu\nu}=\tfrac{i}{4}\,[\gamma^{\mu},\gamma^{\nu}]\,, \tag{11.164}$$

where $S^{\mu\nu}$ is the spinor representation of the Lorentz generators built in Eq. (3.114), and $\omega_{\mu\nu}$ parametrizes $\Lambda=\exp(-\frac{i}{2}\omega_{\rho\sigma}M^{\rho\sigma})$. Feeding the mode expansion into both sides of Eq. (11.164) and matching the coefficients of $u^{s}(\mathbf{p})e^{-ip\cdot x}$ — the standard inversion of Eq. (11.101) — determines how $U(\Lambda)$ acts on $b^{s\dagger}_{\mathbf{p}}$. For a general Λ the result is the Wigner construction: $U(\Lambda)b^{s\dagger}_{\mathbf{p}}U(\Lambda)^{-1} = \sum_{s'} D^{(1/2)}\big(W(\Lambda,\mathbf{p})\big)_{s's}\,b^{s'\dagger}_{\Lambda\mathbf{p}}$, where $W(\Lambda,\mathbf{p})$ is the Wigner rotation of Eq. (3.199) and $D^{(1/2)}$ is the spin-$\frac{1}{2}$ representation matrix. We need only the special case that exposes the spin, where the technicalities collapse.

Theorem 11.33 (The one-electron states carry spin $\frac{1}{2}$)**.** *Let R be a rotation by angle θ about $\hat{n}$, represented on four-spinors by $S(R)=\exp(-\frac{i}{2}\theta\,\hat{n}\cdot\mathbf{\Sigma})$, where the rotation generator built from Eq. (3.114) is $S^{ij}=\frac{1}{2}\epsilon^{ijk}\Sigma^{k}$ with $\mathbf{\Sigma}^{k}=\mathrm{diag}(\sigma^{k},\sigma^{k})$*

in the Dirac basis. Acting on the rest-frame one-electron states,

$$U(R)\, b_{\mathbf{0}}^{s\dagger}\, U(R)^{-1} = \sum_{s'} D^{(1/2)}(R)_{s's}\, b_{\mathbf{0}}^{s'\dagger}, \qquad D^{(1/2)}(R) = \exp\Big(-\tfrac{i}{2}\theta\, \hat{n}\cdot\boldsymbol{\sigma}\Big), \tag{11.165}$$

so the rest-frame states $b_{\mathbf{0}}^{1\dagger}|0\rangle$, $b_{\mathbf{0}}^{2\dagger}|0\rangle$ *transform in the two-dimensional (spin-*$\frac{1}{2}$*) representation of the little group* $SU(2)$*. The one-electron space is therefore the mass-*m*, spin-*$\frac{1}{2}$ *irreducible representation of the Poincaré group; likewise the one-positron space.*

Proof. A rotation fixes the rest momentum, $R\mathbf{0} = \mathbf{0}$, so the Wigner rotation degenerates to R itself and the boost technicalities are absent. To see Eq. (11.165) directly, evaluate Eq. (11.164) for $\Lambda = R$ at a point of zero spatial momentum. The rest-frame spinors of Eq. (11.166) are, in the Dirac basis (Eq. (C.56)),

$$u^s(\mathbf{0}) = \sqrt{2m}\begin{pmatrix}\xi^s\\ 0\end{pmatrix}, \qquad \xi^1 = \begin{pmatrix}1\\0\end{pmatrix},\ \xi^2 = \begin{pmatrix}0\\1\end{pmatrix}, \tag{11.166}$$

and the rotation generator acts on them through its upper block: $\boldsymbol{\Sigma}^k u^s(\mathbf{0}) = \sqrt{2m}\binom{\sigma^k \xi^s}{0}$. Hence

$$S(R)\, u^s(\mathbf{0}) = \sqrt{2m}\begin{pmatrix} e^{-\frac{i}{2}\theta\hat{n}\cdot\boldsymbol{\sigma}}\,\xi^s \\ 0\end{pmatrix} = \sum_{s'}\Big(e^{-\frac{i}{2}\theta\hat{n}\cdot\boldsymbol{\sigma}}\Big)_{s's}\, u^{s'}(\mathbf{0}) = \sum_{s'} D^{(1/2)}(R)_{s's}\, u^{s'}(\mathbf{0}), \tag{11.167}$$

the spinor transformation acting purely as the 2×2 rotation $D^{(1/2)}(R)$ on the two-component label. Matching the $\mathbf{p} = \mathbf{0}$ coefficient of the field law Eq. (11.164) (where the argument shift Rx acts trivially on the zero-momentum mode and the invariant measure is rotation-invariant) turns Eq. (11.167) into the operator statement Eq. (11.165): because $U(R)$ rearranges the field's spinor structure by $S(R)^{-1}$ and the rest spinors carry only the $D^{(1/2)}$ action, the rest-frame creation operators mix among themselves by $D^{(1/2)}(R)$. The two states $b_{\mathbf{0}}^{s\dagger}|0\rangle$ thus form the $\mathbf{2}$ of $SU(2)$, the defining feature of spin $\frac{1}{2}$. Combined with $P^2 = m^2$ from Eq. (11.157) and the fact that the full one-electron space is generated from the rest-frame doublet by boosts, Section 3.7.2 identifies the representation as the unique massive spin-$\frac{1}{2}$ irrep. The positron case is identical, with v^s in place of u^s. □

Physical Insight

The two-valuedness of the label s that has ridden along since the mode expansion is now explained: it is the little-group index, the spin-$\frac{1}{2}$ doublet of $SU(2)$. A massive spin-$\frac{1}{2}$ particle has exactly two polarizations, and they are the two rest-frame states $b_{\mathbf{0}}^{1\dagger}|0\rangle$, $b_{\mathbf{0}}^{2\dagger}|0\rangle$ rotated by $D^{(1/2)}$. This is why the Dirac field, with its four components, describes not four but two

physical particles each with two spin states: the $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ representation of Chapter 3 bundles a particle and its antiparticle, each a spin-$\frac{1}{2}$ doublet, and the extra components are the antiparticle, not redundancy. The mismatch between field components and particle states that plagues the vector field is, for the spinor, no mismatch at all once the antiparticle is counted.

We can now collect the physics of a single electron in one explicit computation, discharging the QED obligation of this chapter.

Worked Example 11.34 (The charge and spin of a one-electron state)**.** Take the rest-frame electron of definite spin along $\hat{z}$, $|e^-,\uparrow\rangle \equiv b_{\mathbf{0}}^{1\dagger}|0\rangle$, normalized informally by setting aside the continuum $\delta^3(0)$. We compute its charge, its energy, and the three quantum numbers J_3, J^2 of its angular momentum.

Charge. From Eq. (11.161) and $Q|0\rangle = 0$,

$$Q|e^-,\uparrow\rangle = \left[Q, b_{\mathbf{0}}^{1\dagger}\right]|0\rangle + b_{\mathbf{0}}^{1\dagger}Q|0\rangle = +b_{\mathbf{0}}^{1\dagger}|0\rangle = +|e^-,\uparrow\rangle\,. \tag{11.168}$$

The number charge is $+1$; the electric charge is $q\cdot(+1) = -e$, the measured electron charge.

Energy. From Eq. (11.152) at $\mathbf{p}=\mathbf{0}$, $:H:\ |e^-,\uparrow\rangle = E_{\mathbf{0}}|e^-,\uparrow\rangle = m|e^-,\uparrow\rangle$: the rest energy is the mass, as it must be.

Spin. The angular momentum operator is $J^k = \frac{1}{2}\epsilon^{kij}M^{ij}$, the rotation part of the Lorentz generators on Fock space; $U(R) = e^{-i\theta\hat{n}\cdot\mathbf{J}}$. Differentiating Eq. (11.165) at $\theta = 0$ about $\hat{n}=\hat{z}$ gives the action of J_3 on the rest-frame ladder operators,

$$\left[J_3, b_{\mathbf{0}}^{s\dagger}\right] = \sum_{s'}\left(\tfrac{1}{2}\sigma^3\right)_{s's} b_{\mathbf{0}}^{s'\dagger}, \tag{11.169}$$

since $D^{(1/2)}(R) = 1 - \frac{i}{2}\theta\,\sigma^3 + \cdots$ for $\hat{n}=\hat{z}$ and $U(R)b^\dagger U(R)^{-1} = b^\dagger - i\theta\left[J_3, b^\dagger\right] + \cdots$. With $\xi^1 = (1,0)^{\mathsf{T}}$ an eigenvector of σ^3 of eigenvalue $+1$,

$$J_3|e^-,\uparrow\rangle = \left[J_3, b_{\mathbf{0}}^{1\dagger}\right]|0\rangle = \tfrac{1}{2}(\sigma^3)_{11}\, b_{\mathbf{0}}^{1\dagger}|0\rangle = +\tfrac{1}{2}\,|e^-,\uparrow\rangle\,. \tag{11.170}$$

The state has $J_3 = +\frac{1}{2}$. The companion $b_{\mathbf{0}}^{2\dagger}|0\rangle$ has $J_3 = -\frac{1}{2}$, so the two rest-frame electrons are the $s_3 = \pm\frac{1}{2}$ states of a spin-$\frac{1}{2}$ doublet. For the Casimir $J^2 = J_1^2 + J_2^2 + J_3^2$, use that on the doublet each J_k acts as $\frac{1}{2}\sigma^k$ and $(\frac{1}{2}\sigma^k)^2 = \frac{1}{4}\mathbf{1}$, so

$$J^2|e^-,\uparrow\rangle = \sum_k\left(\tfrac{1}{2}\sigma^k\right)^2|e^-,\uparrow\rangle = \tfrac{3}{4}\,|e^-,\uparrow\rangle = \tfrac{1}{2}\left(\tfrac{1}{2}+1\right)|e^-,\uparrow\rangle\,, \tag{11.171}$$

the eigenvalue $s(s+1)$ with $s=\frac{1}{2}$. The state is an electron: charge $-e$, mass m, spin $\frac{1}{2}$ with $s_3 = +\frac{1}{2}$ — every quantum number of *the* electron, now an excitation of the quantized Dirac field rather than a postulated particle.

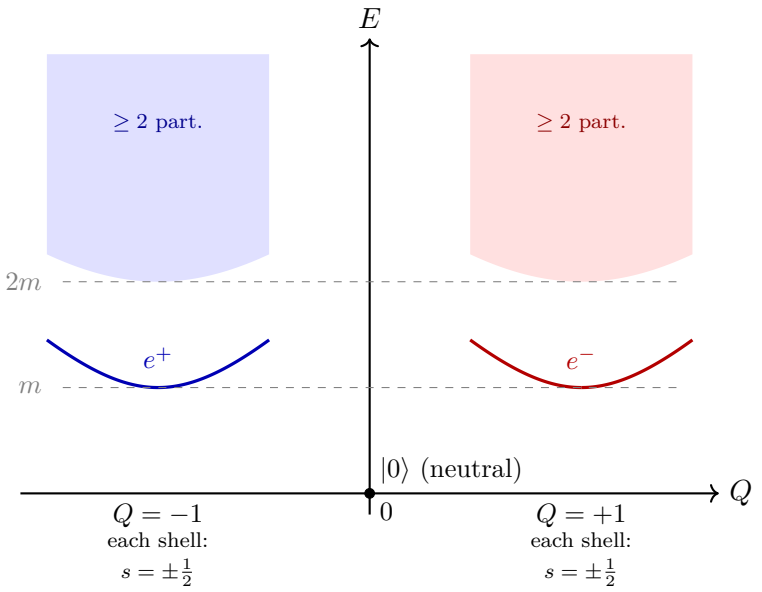


Figure 11.3: Spectrum of the free Dirac field, organized by charge Q (horizontal) and energy E (vertical). The unique vacuum $|0\rangle$ sits neutral at the origin. Two isolated one-particle hyperboloids rise from the mass gap m: electrons ($Q = +1$, electric charge $-e$) and positrons ($Q = -1$, electric charge $+e$), each a spin-$\frac{1}{2}$ doublet ($s_3 = \pm\frac{1}{2}$). Above each lies the multiparticle continuum (threshold $2m$ at rest). Compared with the scalar spectrum of Fig. 9.3, charge doubles the one-particle sector into particle and antiparticle shells, spin gives each shell two states, and the Pauli principle (Theorem 11.27) caps every mode's occupation at one.

Figure 11.3 assembles the resulting spectrum: a unique neutral vacuum, and above it two isolated one-particle hyperboloids — electrons of charge $+1$ and positrons of charge -1 — each carrying two spin states and the same mass gap m, with the multiparticle continuum above. It is the scalar spectrum of Fig. 9.3 doubled by charge, refined by spin, and built on an antisymmetric Fock space.

Caution

The completeness of the physical picture here should not obscure what this chapter has *not* yet proved. We *found* that the spin-$\frac{1}{2}$ field must be quantized with anticommutators (Section 11.3, by the demand for a Hamiltonian bounded below — the very positivity re-derived in Theorem 11.31), but the deep statement that *causality* forces half-integer spin to be fermionic and integer spin bosonic — the spin–statistics connection in full — rests on microcausality of the field anticommutator, which is

the business of Section 11.6.2. There it joins the scalar half established in Theorem 9.33 to complete the theorem the book has been assembling in two pieces.

Handoff

The free Dirac field now has its full physical content: a unique neutral vacuum, an antisymmetric Fock space of electrons and positrons, each a mass-m spin-$\frac{1}{2}$ particle of definite charge, with $:H:$, P^μ, and Q diagonalized on it and the Pauli principle built in. The electron of QED has been promoted from a classical $U(1)$ matter field to a quantum field whose quanta are the particles the rest of the thread will scatter. What remains, before interactions, is the dynamical object that turns this spectrum into amplitudes: the time-ordered two-point function $\langle 0|T\,\psi(x)\bar{\psi}(y)|0\rangle$, whose evaluation on the Fock space just built — and whose covariant momentum-space form S_F with its $i\epsilon$ contour and Green's-function property — is the task of Section 11.5, and the internal fermion line of every diagram from Chapter 17 onward.

11.5 The Dirac Propagator

The previous section completed the kinematic construction of the free Dirac theory: a vacuum $|0\rangle$ annihilated by every lowering operator, an antisymmetric Fock space built on it, a Hamiltonian and a charge that are bounded below and diagonal, and a clean identification of the elementary quanta as electrons and positrons of mass m and spin $\frac{1}{2}$. What that construction does not yet supply is the one object perturbation theory will actually demand: a single distribution that encodes the amplitude for the field to carry a disturbance from one spacetime point to another. For the scalar this was the Feynman propagator D_F of Section 9.5; for the electron it is its spinorial counterpart S_F, the internal fermion line of every diagram from Chapters 15 and 17 onward. This section computes it.

The development runs exactly parallel to the scalar case, and we will lean on that parallel deliberately, importing the scalar machinery wherever it transfers unchanged. Two features are genuinely new, and they are precisely the features that make the electron line different from the scalar line. First, the field is now matrix-valued: the two-point function carries a pair of Dirac indices, and the spin sums $\sum u\bar{u} = \not{p} + m$ and $\sum v\bar{v} = \not{p} - m$ of Theorem 11.12 are what convert the sum over intermediate spin states into a covariant numerator. Second, the fields anticommute, and the time-ordering symbol must be defined with a relative minus sign to respect that. We will see that this single sign is not a blemish to be tolerated but the exact ingredient that makes the assembly work:

it is what allows the manifestly covariant momentum integral to reproduce the operator time-ordered product, and what makes S_F a Green's function of the Dirac operator rather than of its square. The reward, stated once here and quoted thereafter, is the electron propagator

$$\tilde{S_F}(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} = \frac{i}{\not{p} - m + i\varepsilon}, \tag{11.172}$$

the object the QED Feynman rules of Chapter 17 will quote for every internal electron line and which we now derive in full.

We work throughout from the quantized field and its algebra as fixed earlier in the chapter. We recall them in the form needed here. The mode expansion is Eq. (11.95),

$$\psi(x) = \sum_{s=1,2} \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big[b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big], \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{11.173}$$

with $p^0 = E_{\mathbf{p}}$ on shell and the Lorentz-invariant measure of Theorem 9.4, and the Dirac adjoint $\bar\psi = \psi^\dagger \gamma^0$ is, from Eq. (11.96),

$$\bar\psi(y) = \sum_{r=1,2} \int \frac{d^3q}{(2\pi)^3\, 2E_{\mathbf{q}}} \Big[b^{r\dagger}_{\mathbf{q}}\, \bar u^r(q)\, e^{+iq\cdot y} + d^r_{\mathbf{q}}\, \bar v^r(q)\, e^{-iq\cdot y} \Big]. \tag{11.174}$$

The nonzero anticommutators are Eq. (11.116),

$$\Big\{ b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}} \Big\} = \Big\{ d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}} \Big\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\, \delta^3(\mathbf{p} - \mathbf{q}), \tag{11.175}$$

all others vanishing, and the equal-time field anticommutator is the postulate Eq. (11.100) now established as the consistent quantization,

$$\Big\{ \psi_a(x), \psi^\dagger_b(y) \Big\} \big|_{x^0 = y^0} = \delta_{ab}\, \delta^3(\mathbf{x} - \mathbf{y}), \qquad \{ \psi_a(x), \psi_b(y) \} \big|_{x^0 = y^0} = 0. \tag{11.176}$$

The completeness relations we will use at every step are the spin sums Eq. (11.73),

$$\sum_{s=1,2} u^s(p)\, \bar u^s(p) = \not{p} + m, \qquad \sum_{s=1,2} v^s(p)\, \bar v^s(p) = \not{p} - m, \tag{11.177}$$

holding as 4×4 matrix identities for every on-shell p. Finally we will repeatedly recognize the *scalar* positive-frequency Wightman function of Theorem 9.23,

$$D^+(x - y) \equiv \langle 0 |\, \phi(x) \phi(y)\, | 0 \rangle = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-ip\cdot(x-y)} \tag{11.178}$$

of a mass-m scalar (this is Eq. (9.150)); it is the universal positive-frequency kernel out of which both the scalar and the spinor two-point functions are built, and importing it lets us reuse the scalar contour analysis wholesale.

11.5.1 The fermion Wightman functions

The basic correlators are the two vacuum expectation values of an unordered product of ψ and $\bar\psi$. They carry one Dirac index from ψ and one from $\bar\psi$, so each is a 4×4 matrix-valued distribution.

Definition 11.35 (Fermion Wightman functions)**.** The *positive-* and *negative-frequency Wightman functions* of the free Dirac field are

$$S^+_{ab}(x-y)\equiv\langle 0|\,\psi_a(x)\,\bar\psi_b(y)\,|0\rangle\,,\qquad S^-_{ab}(x-y)\equiv\langle 0|\,\bar\psi_b(y)\,\psi_a(x)\,|0\rangle\,. \tag{11.179}$$

S^+ propagates the electron part of the field (a b-quantum created at y, absorbed at x); S^- propagates the positron part (a d-quantum created at x, absorbed at y).

That each depends only on $x-y$ is, as for the scalar, a consequence of translation invariance of the vacuum, and is seen explicitly in the computation. We do both at once.

Theorem 11.36 (The Wightman functions as Dirac-operator images of the scalar kernel)**.** *The fermion Wightman functions are*

$$S^+(x-y)=\int\frac{d^3p}{(2\pi)^3\,2E_{\mathbf p}}\,(\not p+m)\,e^{-ip\cdot(x-y)}=(i\not\partial_x+m)\,D^+(x-y), \tag{11.180}$$

$$S^-(x-y)=\int\frac{d^3p}{(2\pi)^3\,2E_{\mathbf p}}\,(\not p-m)\,e^{+ip\cdot(x-y)}=-(i\not\partial_x+m)\,D^+(y-x), \tag{11.181}$$

where D^+ is the scalar Wightman kernel Eq. (11.178) and the derivatives act on the x-dependence.

Proof. The positive-frequency function. Insert Eqs. (11.173) and (11.174) into the first definition. The double mode sum contains four operator structures, $bb^\dagger$, bd, $d^\dagger b^\dagger$, and $d^\dagger d$. Sandwiched between vacua only $b^s_{\mathbf p}b^{r\dagger}_{\mathbf q}$ survives: any term with an annihilator (b or d) standing to the right is killed by $|0\rangle$, and any term with a creator standing to the left is killed by $\langle 0|$, which removes bd, $d^\dagger b^\dagger$, and $d^\dagger d$ in turn. Using $\langle 0|\,b^s_{\mathbf p}b^{r\dagger}_{\mathbf q}\,|0\rangle=\langle 0|\left\{b^s_{\mathbf p},b^{r\dagger}_{\mathbf q}\right\}|0\rangle=(2\pi)^3 2E_{\mathbf p}\delta^{rs}\delta^3(\mathbf p-\mathbf q)$ from Eq. (11.175),

$$\begin{aligned}S^+_{ab}(x-y)&=\sum_{r,s}\int\frac{d^3p}{(2\pi)^3 2E_{\mathbf p}}\frac{d^3q}{(2\pi)^3 2E_{\mathbf q}}\,u^s_a(p)\,\bar u^r_b(q)\,e^{-ip\cdot x+iq\cdot y}\,(2\pi)^3 2E_{\mathbf p}\,\delta^{rs}\delta^3(\mathbf p-\mathbf q)\\&=\sum_s\int\frac{d^3p}{(2\pi)^3 2E_{\mathbf p}}\,u^s_a(p)\,\bar u^s_b(p)\,e^{-ip\cdot(x-y)}.\end{aligned} \tag{11.182}$$

The spin sum Eq. (11.177) collapses the spin label to a matrix, $\sum_s u^s_a\bar u^s_b=(\not p+m)_{ab}$, giving the first equality in Eq. (11.180). To extract the differential

operator, note that the polynomial $\not{p}+m$ rides on the on-shell exponential, and $i\not{\partial}_x$ acting on $e^{-ip\cdot(x-y)}$ pulls down exactly $\not{p}$:

$$i\not{\partial}_x\, e^{-ip\cdot(x-y)} = i\,\gamma^\mu(-ip_\mu)\, e^{-ip\cdot(x-y)} = \not{p}\, e^{-ip\cdot(x-y)}, \tag{11.183}$$

so that $(\not{p}+m)e^{-ip\cdot(x-y)} = (i\not{\partial}_x+m)e^{-ip\cdot(x-y)}$. Pulling the (momentum-independent) operator outside the integral turns $\int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} e^{-ip\cdot(x-y)} = D^+(x-y)$ into the second equality of Eq. (11.180).

The negative-frequency function. For S^- the operator order is reversed, $\bar\psi(y)$ standing to the left of $\psi(x)$. Now the only surviving structure is $d^r_{\mathbf{q}} d^{s\dagger}_{\mathbf{p}}$, with $\langle 0|\, d^r_{\mathbf{q}} d^{s\dagger}_{\mathbf{p}}\, |0\rangle = (2\pi)^3 2E_{\mathbf{p}}\delta^{rs}\delta^3(\mathbf{p}-\mathbf{q})$. The spinor that rides with $d^{s\dagger}_{\mathbf{p}}$ in ψ is $v^s(p)e^{+ip\cdot x}$ and the one riding with $d^r_{\mathbf{q}}$ in $\bar\psi$ is $\bar v^r(q)e^{-iq\cdot y}$, so

$$S^-_{ab}(x-y) = \sum_s \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\, v^s_a(p)\, \bar v^s_b(p)\, e^{+ip\cdot(x-y)} = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\, (\not{p}-m)_{ab}\, e^{+ip\cdot(x-y)}, \tag{11.184}$$

using the v spin sum $\sum_s v^s\bar v^s = \not{p}-m$. This is the first equality in Eq. (11.181). For the second, the relevant derivative identity carries the opposite sign,

$$i\not{\partial}_x\, e^{+ip\cdot(x-y)} = i\,\gamma^\mu(+ip_\mu)\, e^{+ip\cdot(x-y)} = -\not{p}\, e^{+ip\cdot(x-y)}, \tag{11.185}$$

so that $(\not{p}-m)e^{+ip\cdot(x-y)} = -(m-\not{p})e^{+ip\cdot(x-y)} = -(i\not{\partial}_x+m)e^{+ip\cdot(x-y)}$. Pulling the operator out and recognizing $\int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} e^{+ip\cdot(x-y)} = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}} e^{-ip\cdot(y-x)} = D^+(y-x)$ yields the second equality of Eq. (11.181). □

Three observations organize what follows. First, the entire spinor content of both Wightman functions is the single scalar kernel D^+ acted on by the matrix differential operator $i\not{\partial}+m$. Everything we proved about D^+ in Section 9.5 — its positive-frequency analyticity, its Lorentz invariance, its distributional nature, its non-vanishing at spacelike separation — transfers immediately, dressed by a Dirac matrix. This is the structural reason the fermion propagator costs so little once the scalar one is in hand. Second, S^+ and S^- are *not* complex conjugates of one another the way the two scalar orderings were; they are built from the two physically distinct spin sums $\not{p}\pm m$, encoding that the field creates a positron where it annihilates an electron. Third — the fact we will need for the time-ordered product — both Wightman functions individually solve the homogeneous Dirac equation. Acting with $(i\not{\partial}_x - m)$ on Eq. (11.180) and using $(i\not{\partial}-m)(i\not{\partial}+m) = -(\Box+m^2)$ (a one-line Clifford computation, $(\gamma^\mu\partial_\mu)^2 = \frac{1}{2}\{\gamma^\mu,\gamma^\nu\}\,\partial_\mu\partial_\nu = \partial^2$),

$$(i\not{\partial}_x - m)\, S^+(x-y) = -(\Box_x + m^2)\, D^+(x-y) = 0, \qquad (i\not{\partial}_x - m)\, S^-(x-y) = 0, \tag{11.186}$$

since D^+ is built from on-shell modes annihilated by $(\Box + m^2)$ (the scalar statement of Section 9.5.1). The Dirac operator annihilates each Wightman piece; only the boundary between them, encoded in the time-ordering step functions, can produce a delta function.

11.5.2 Time ordering with the fermionic minus sign

Perturbation theory in Chapters 14 and 17 is built from vacuum expectation values of time-ordered products. For anticommuting fields the time-ordering symbol must be defined so that interchanging two fields to put them in temporal order costs a minus sign — the same minus that the equal-time anticommutators Eq. (11.176) demand. Without it the symbol would not even be well defined, because for fermionic operators the two orderings differ by an anticommutator, not a commutator.

Definition 11.37 (Fermionic time ordering)**.** For two Dirac field operators the time-ordered product is

$$\mathcal{T}\,\psi(x)\bar{\psi}(y) \equiv \theta(x^0-y^0)\,\psi(x)\bar{\psi}(y) - \theta(y^0-x^0)\,\bar{\psi}(y)\psi(x), \tag{11.187}$$

the later field standing to the left, with a minus sign whenever ordering them requires interchanging the two fermionic operators. More generally $\mathcal{T}$ of a string of fermionic fields is the temporally ordered string times the sign of the permutation that achieves the ordering.

The minus sign is forced by consistency, not chosen for convenience. Consider the symbol at the instant $x^0 = y^0$, where the two step functions both equal $\frac{1}{2}$ in the symmetric convention and the ambiguity in the definition must cancel. The would-be ambiguity is the equal-time difference $\psi(x)\bar{\psi}(y) - (\pm)\bar{\psi}(y)\psi(x)$; only the choice of the relative minus makes this the equal-time *anti*commutator

$$\psi_a(x)\bar{\psi}_b(y)+\bar{\psi}_b(y)\psi_a(x)\big|_{x^0=y^0} = \Big\{\psi_a(x),\bar{\psi}_b(y)\Big\}\big|_{x^0=y^0} = \Big\{\psi_a(x),\psi_c^\dagger(y)\Big\}\gamma^0_{cb}\big|_{x^0=y^0} = \gamma^0_{ab}\,\delta^3(\mathbf{x}-\mathbf{y} \tag{11.188}$$

a c-number (the unit matrix in field space times γ^0). Had we used a relative plus, the ambiguity at coincident times would have been the equal-time *commutator*, which is operator-valued and does not even vanish for spacelike separation — the symbol would be ill defined. The fermionic minus sign is exactly what renders $\mathcal{T}$ unambiguous, and Eq. (11.188) is the identity that will produce the delta function in the Green's-function property below.

Definition 11.38 (Dirac–Feynman propagator)**.** The *Feynman propagator* of the free Dirac field is the time-ordered two-point function

$$S_F(x-y)_{ab} \equiv \langle 0|\,\mathcal{T}\,\psi_a(x)\bar{\psi}_b(y)\,|0\rangle\,. \tag{11.189}$$

Taking the vacuum expectation value of Eq. (11.187) and inserting the Wightman functions Eq. (11.179) for the two orderings gives at once the position-space form

$$S_F(x-y) = \theta(x^0-y^0)\, S^+(x-y) - \theta(y^0-x^0)\, S^-(x-y), \tag{11.190}$$

the spinor analogue of the scalar Eq. (9.153), distinguished from it by the relative minus sign on the antiparticle term. This is the exact target the covariant momentum integral must reproduce.

11.5.3 Assembling S_F from the scalar propagator

The position-space expression Eq. (11.190) is correct but inherits the awkwardness of the scalar case: it is on-shell integrals glued with step functions, and the relative minus sign sits there unexplained. The clean statement is that S_F is simply the Dirac operator applied to the *scalar* Feynman propagator D_F of Theorem 9.24, and that this single identity packages the relative minus sign automatically.

Theorem 11.39 (S_F as the Dirac image of D_F)**.** *The Dirac–Feynman propagator is*

$$\boxed{S_F(x-y) = (i\not{\partial}_x + m)\, D_F(x-y)} \tag{11.191}$$

where $D_F(x-y) = \theta(x^0-y^0)D^+(x-y) + \theta(y^0-x^0)D^+(y-x)$ is the scalar Feynman propagator of a mass-m field, Eq. (9.153).

Proof. We apply $(i\not{\partial}_x + m)$ to the scalar D_F and show the result is Eq. (11.190). The only subtlety is that $\not{\partial} = \gamma^0\partial_{x^0} + \gamma^i\partial_{x^i}$ contains a time derivative that acts on the step functions. Write $t = x^0 - y^0$ and use $\partial_{x^0}\theta(t) = \delta(t) = -\partial_{x^0}\theta(-t)$:

$$\begin{aligned}(i\not{\partial}_x + m)\, D_F = i\gamma^0\Big[\delta(t)D^+(x-y) - \delta(t)D^+(y-x)\Big] \\ + \theta(t)\,(i\not{\partial}_x + m)D^+(x-y) + \theta(-t)\,(i\not{\partial}_x + m)D^+(y-x).\end{aligned} \tag{11.192}$$

The bracketed delta term is evaluated at $t = 0$, i.e. at equal times, where $D^+(x-y) - D^+(y-x) = \langle 0|\,[\phi(x),\phi(y)]\,|0\rangle$ is the scalar commutator function Eq. (9.173), which vanishes at equal times (the equal-time commutator of the scalar field with itself is zero). Hence the entire $i\gamma^0\delta(t)$ contribution drops:

$$(i\not{\partial}_x + m)\, D_F = \theta(t)\,(i\not{\partial}_x + m)D^+(x-y) + \theta(-t)\,(i\not{\partial}_x + m)D^+(y-x). \tag{11.193}$$

Now compare with the Wightman results. By Eq. (11.180), $(i\not{\partial}_x + m)D^+(x-y) = S^+(x-y)$; by Eq. (11.181), $(i\not{\partial}_x + m)D^+(y-x) = -S^-(x-y)$. Substituting,

$$(i\not{\partial}_x + m)\, D_F = \theta(t)\, S^+(x-y) - \theta(-t)\, S^-(x-y) = S_F(x-y), \tag{11.194}$$

which is exactly Eq. (11.190). □

The proof exposes where the fermionic minus sign comes from and why it is benign. The relative minus in the time-ordered product Eq. (11.187) is met, term for term, by the relative minus that the operator $(i\partial\!\!\!/ + m)$ produces between $e^{-ip\cdot(x-y)}$ and $e^{+ip\cdot(x-y)}$ (compare Eqs. (11.183) and (11.185)): the antiparticle Wightman function S^- enters S_F with the sign $-S^- = +(i\partial\!\!\!/ + m)D^+(y-x)$, which is exactly what $(i\partial\!\!\!/ + m)$ acting on the second piece of D_F delivers. The two minus signs are the same sign, viewed from two sides. Moreover the potentially dangerous contact term — the $i\gamma^0\delta(t)$ contribution from differentiating the step functions — vanishes here because it is proportional to the equal-time scalar commutator, which is zero. (It will *not* vanish when we instead apply the full Dirac operator $(i\partial\!\!\!/ - m)$ to compute the Green's-function property below; there it produces precisely the delta function we want. The difference between the two operators is the difference between $S^{\pm}$ solving the homogeneous equation and the canonical anticommutator surfacing.)

11.5.4 The momentum-space propagator and the $i\varepsilon$ contour

With Theorem 11.39 the covariant momentum-space form is immediate: we apply the Dirac operator to the scalar contour representation Theorem 9.25, inheriting its entire p^0-contour analysis at no extra cost.

Theorem 11.40 (Momentum-space Dirac propagator)**.** *The Dirac–Feynman propagator admits the manifestly covariant representation*

$$S_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \tilde{S}_F(p)\, e^{-ip\cdot(x-y)}, \qquad \boxed{\tilde{S}_F(p) = \frac{i(p\!\!\!/ + m)}{p^2 - m^2 + i\varepsilon} = \frac{i}{p\!\!\!/ - m + i\varepsilon}} \tag{11.195}$$

where p^0 ranges over all of $\mathbb{R}$ independently of $\mathbf{p}$, $\varepsilon \to 0^+$, the equality holds as a 4×4 matrix-valued tempered distribution, and the second form is the matrix inverse defined by the first.

Proof. Start from Eq. (11.191) and insert the scalar contour representation Eq. (9.154),

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i\, e^{-ip\cdot(x-y)}}{p^2 - m^2 + i\varepsilon}. \tag{11.196}$$

Apply $(i\partial\!\!\!/_x + m)$ under the integral. Because p now ranges over all four components off shell, the identity Eq. (11.183) still holds verbatim — it is a property of the exponential, not of the mass shell — so $(i\partial\!\!\!/_x + m)e^{-ip\cdot(x-y)} = (p\!\!\!/ + m)e^{-ip\cdot(x-y)}$ and

$$S_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i(p\!\!\!/ + m)}{p^2 - m^2 + i\varepsilon}\, e^{-ip\cdot(x-y)}, \tag{11.197}$$

which is the first form in Eq. (11.195). The denominator is the same scalar denominator as in Eq. (11.196); the numerator $p\!\!\!/ + m$ is a first-degree matrix

polynomial in p that introduces no new poles in the complex p^0 plane. Therefore the p^0 contour analysis is identical to the scalar one of Theorem 9.25: the poles sit at $p^0 = +E_{\mathbf{p}} - i\varepsilon'$ (below the real axis) and $p^0 = -E_{\mathbf{p}} + i\varepsilon'$ (above it), and for $x^0 > y^0$ the contour closes downward onto the first, for $x^0 < y^0$ upward onto the second. The matrix numerator simply rides along, evaluated at the captured pole. We confirm in Theorem 11.41 below that this reproduces Eq. (11.190) directly, with the spin sums emerging from the residues; here it suffices that the denominator and the contour are exactly the scalar ones, so the equality of Eq. (11.197) with the operator S_F is inherited from the scalar theorem.

It remains to derive the compact second form. Factor the scalar denominator using the Clifford algebra Eq. (11.2) (the canon Eq. (C.1)), which gives

$$\not{p}\not{p} = p_\mu p_\nu \, \gamma^\mu \gamma^\nu = p_\mu p_\nu \, \tfrac{1}{2} \{\gamma^\mu, \gamma^\nu\} = p_\mu p_\nu \, \eta^{\mu\nu} = p^2, \tag{11.198}$$

the symmetrization being legitimate because $p_\mu p_\nu$ is symmetric in $\mu\nu$. Hence

$$(\not{p} + m)(\not{p} - m) = \not{p}\not{p} - m^2 = p^2 - m^2 = (\not{p} - m)(\not{p} + m), \tag{11.199}$$

so $\not{p} + m$ and $\not{p} - m$ commute and their product is the scalar $p^2 - m^2$. Reading Eq. (11.199) as a statement about matrix inverses, away from the mass shell where $p^2 - m^2 \neq 0$,

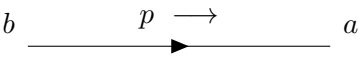
$$(\not{p} - m)^{-1} = \frac{\not{p} + m}{(\not{p} - m)(\not{p} + m)} = \frac{\not{p} + m}{p^2 - m^2}. \tag{11.200}$$

Therefore the numerator-and-denominator form equals $i(\not{p} - m)^{-1}$ once the $i\varepsilon$ is restored to keep the inverse well defined on the shell:

$$\frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} = \frac{i}{\not{p} - m + i\varepsilon}, \tag{11.201}$$

which is the second form in Eq. (11.195). The symbol $1/(\not{p} - m + i\varepsilon)$ is *defined* to mean $(\not{p}+m)/(p^2-m^2+i\varepsilon)$; it is not a literal scalar division but the matrix inverse Eq. (11.200) with the Feynman prescription, and the two are interchangeable in every computation. □

Two remarks fix the interpretation. First, the $i\varepsilon$ here is the same routing instruction it was for the scalar, and it carries the same physics: it sends positive-energy electrons forward in time and reinterprets negative-energy solutions as positrons running backward. The contour that captures the $+E_{\mathbf{p}}$ pole for $x^0 > y^0$ yields the electron piece S^+, built from $\sum u\bar{u} = \not{p} + m$; the contour that captures the $-E_{\mathbf{p}}$ pole for $x^0 < y^0$ yields the positron piece $-S^-$, built from $\sum v\bar{v} = \not{p} - m$. The numerator's two spin sums are literally the residues at the two poles. Second, the two equivalent forms in Eq. (11.195) serve different purposes: the form $i(\not{p} + m)/(p^2 - m^2 + i\varepsilon)$ is the one used in actual diagram computations, where



$$\tilde{S}_F(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} = \frac{i}{\not{p} - m + i\varepsilon}$$

Figure 11.4: The internal electron line. The arrow on a fermion line records the flow of the conserved $U(1)$ charge of Theorem 11.32, here pointing from the $\bar{\psi}_b(y)$ end to the $\psi_a(x)$ end. To this oriented line the QED Feynman rules of Chapter 17 assign the momentum-space propagator $\tilde{S}_F(p)$, with Dirac indices a (at the ψ end) and b (at the $\bar{\psi}$ end) contracted into the vertices at either end. Read with momentum p flowing along the arrow, the same line describes an electron propagating forward in time or, for the opposite sign of p^0, a positron propagating backward, the two being the residues at $p^0 = \pm E_{\mathbf{p}}$.

the explicit numerator feeds the gamma-matrix traces of Chapter 17; the compact form $i/(\not{p} - m + i\varepsilon)$ is the one to remember, and it makes the Green's-function property transparent, as we see next.

Notation Watch

The numerator carries $+i$, and the Dirac propagator is the Green's function of the *first-order* operator $i\not{\partial} - m$ with right-hand side $+i\delta^4$ (proved next), whereas the scalar propagator carried numerator i but was the Green's function of the *second-order* operator $\Box + m^2$ with right-hand side $-i\delta^4$ (Theorem 9.27). The sign difference is not an inconsistency: it is $(i\not{\partial} - m)(i\not{\partial} + m) = -(\Box + m^2)$, the same minus relating the two operators, propagated to their sources. When importing a propagator from another text, check the metric signature, the placement of i, and which operator's Green's function is meant before trusting any sign.

Physical Insight

The Dirac propagator describes one physical process seen two ways, exactly as the scalar one did, but now the two ways are genuinely distinct particles. For $x^0 > y^0$ the line is an electron created at y and absorbed at x, carrying the numerator $\not{p} + m = \sum_s u^s \bar{u}^s$, the projector onto positive-energy electron spin states. For $x^0 < y^0$ it is a positron created at x and absorbed at y, carrying $\not{p} - m = \sum_s v^s \bar{v}^s$. The $i\varepsilon$ is the bookkeeping that routes electrons forward and positrons backward in time so that only positive-energy, forward-in-time propagation of physical quanta ever occurs — the

Stueckelberg–Feynman reinterpretation, now realized on a charged field where particle and antiparticle are not identified.

It is worth confirming the contour claim directly, both to verify that the matrix numerator behaves and to watch the spin sums drop out of the residues as the two physical pieces.

Worked Example 11.41 (The contour integral reproduces the electron and positron pieces)**.** We evaluate the p^0 integral in Eq. (11.197) by residues, exactly as in the scalar Theorem 9.25, and read off the spinor structure. Write $t = x^0 - y^0$ and factor the denominator as in Eq. (9.155), $p^2 - m^2 + i\varepsilon = (p^0 - E_{\mathbf{p}} + i\varepsilon')(p^0 + E_{\mathbf{p}} - i\varepsilon')$, with poles just below $+E_{\mathbf{p}}$ and just above $-E_{\mathbf{p}}$.

Electron piece, $t > 0$. Convergence of $e^{-ip^0 t}$ forces the contour closed in the lower half-plane (clockwise, a factor $-2\pi i$), enclosing only the pole at $p^0 = E_{\mathbf{p}}$. The residue of $i(\not{p} + m)/[(p^0 - E_{\mathbf{p}})(p^0 + E_{\mathbf{p}})]$ there is $i(\not{p} + m)\big|_{p^0 = E_{\mathbf{p}}}/(2E_{\mathbf{p}})$, in which $\not{p} + m$ is now evaluated on shell and is precisely the electron spin sum $\sum_s u^s(p)\bar{u}^s(p)$ of Eq. (11.177). Collecting the $\frac{1}{2\pi}(-2\pi i)$ and the leftover spatial integral exactly as in Eqs. (9.161) and (9.162),

$$S_F(x-y)\big|_{t>0} = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, (\not{p} + m)\, e^{-ip\cdot(x-y)} = S^+(x-y), \tag{11.202}$$

the positive-frequency Wightman function Eq. (11.180).

Positron piece, $t < 0$. Now the contour closes upward (counterclockwise, $+2\pi i$), enclosing only $p^0 = -E_{\mathbf{p}}$. Write $\bar{p} \equiv (E_{\mathbf{p}}, -\mathbf{p})$; at $p^0 = -E_{\mathbf{p}}$ the four-vector is $p = (-E_{\mathbf{p}}, \mathbf{p}) = -\bar{p}$, so $\not{p}\big|_{p^0 = -E_{\mathbf{p}}} = -\not{\bar{p}}$ and the numerator becomes $\not{p} + m\big|_{p^0 = -E_{\mathbf{p}}} = -(\not{\bar{p}} - m) = -\sum_s v^s(\bar{p})\bar{v}^s(\bar{p})$, minus the positron spin sum. The residue is $i\big(-(\not{\bar{p}} - m)\big)/(-2E_{\mathbf{p}})$, and assembling the $\frac{1}{2\pi}(+2\pi i)$ together with the spatial exponential and then substituting $\mathbf{p} \to -\mathbf{p}$ (so that $\bar{p} \to (E_{\mathbf{p}}, \mathbf{p}) = p$ and the measure is unchanged), one finds, mirroring Eqs. (9.165) and (9.166),

$$S_F(x-y)\big|_{t<0} = -\int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, (\not{p} - m)\, e^{+ip\cdot(x-y)} = -\,S^-(x-y), \tag{11.203}$$

the negative-frequency Wightman function with the fermionic minus sign of Eq. (11.190). The two cases combine into $S_F = \theta(t)S^+ - \theta(-t)S^-$, reproducing Eq. (11.190) and confirming Theorem 11.40 independently of Theorem 11.39. The minus sign on the positron piece, which the time-ordering symbol *postulated*, is here *produced* by the contour: it is the relative sign between the on-shell values of $\not{p} + m$ at the two poles, $+\sum u\bar{u}$ versus $-\sum v\bar{v}$.

11.5.5 The propagator is the Green's function of the Dirac operator

We have called S_F a propagator; we now prove it is the Green's function of the Dirac operator. This is the property that makes it the inverse of the kinetic operator, hence the internal line of the perturbative expansion.

Theorem 11.42 (Dirac Green's function)**.** *The Dirac–Feynman propagator satisfies*

$$\boxed{(i\partial\!\!\!/_x - m)\, S_F(x-y) = i\,\delta^4(x-y)\,\mathbf{1}} \tag{11.204}$$

with $\mathbf{1}$ *the* 4×4 *identity in Dirac space, and is therefore (up to the conventional* i*) the Green's function of the first-order Dirac operator* $i\partial\!\!\!/ - m$*.*

Proof, momentum space. Act with $(i\partial\!\!\!/_x - m)$ under the integral in Eq. (11.197). Since $(i\partial\!\!\!/_x - m)e^{-ip\cdot(x-y)} = (p\!\!\!/ - m)e^{-ip\cdot(x-y)}$ by Eq. (11.183),

$$(i\partial\!\!\!/_x - m)\, S_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\, \frac{i\,(p\!\!\!/ - m)(p\!\!\!/ + m)}{p^2 - m^2 + i\varepsilon}\, e^{-ip\cdot(x-y)}. \tag{11.205}$$

By the factorization Eq. (11.199), $(p\!\!\!/ - m)(p\!\!\!/ + m) = p^2 - m^2$, a c-number times the identity matrix, which cancels the denominator up to the $i\varepsilon$:

$$\frac{(p\!\!\!/ - m)(p\!\!\!/ + m)}{p^2 - m^2 + i\varepsilon} = \frac{p^2 - m^2}{p^2 - m^2 + i\varepsilon}\,\mathbf{1} = \Big(1 - \frac{i\varepsilon}{p^2 - m^2 + i\varepsilon}\Big)\mathbf{1}. \tag{11.206}$$

The $i\varepsilon$ term vanishes as a tempered distribution when $\varepsilon \to 0^+$ (its support shrinks to the measure-zero mass shell; this is the scalar statement used in Theorem 9.27). The leading term gives the complete momentum-space delta function $\int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot(x-y)} = \delta^4(x-y)$, so

$$(i\partial\!\!\!/_x - m)\, S_F(x-y) = i\int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot(x-y)}\,\mathbf{1} = i\,\delta^4(x-y)\,\mathbf{1}, \tag{11.207}$$

which is Eq. (11.204). □

The momentum-space proof is quickest but hides the physical source of the delta function, which is the canonical equal-time anticommutator. The position-space derivation makes that origin explicit and ties the result back to the quantization postulate Eq. (11.176); it is the fermionic counterpart of the scalar Theorem 9.28, and it is here that the relative minus sign of the time-ordered product earns its keep.

Worked Example 11.43 (Position-space derivation of the Dirac Green's-function identity)**.** We verify Eq. (11.204) starting from the operator form Eq. (11.190),

differentiating the step functions and using the equal-time anticommutator. Write $t = x^0 - y^0$ and split the Dirac operator as $i\partial\!\!\!/_x - m = i\gamma^0\partial_{x^0} + (i\gamma^i\partial_{x^i} - m)$.

The time derivative hits the step functions. Differentiating Eq. (11.190) and using $\partial_{x^0}\theta(t) = \delta(t)$, $\partial_{x^0}\theta(-t) = -\delta(t)$,

$$i\gamma^0\partial_{x^0}S_F = i\gamma^0\,\delta(t)\big[S^+(x-y)+S^-(x-y)\big]+i\gamma^0\Big[\theta(t)\,\partial_{x^0}S^+(x-y)-\theta(-t)\,\partial_{x^0}S^-(x-y)\Big]. \tag{11.208}$$

The crucial sign: S_F carries $-\theta(-t)S^-$, and $\partial_{x^0}\theta(-t) = -\delta(t)$, so the antiparticle term contributes $-(-\delta(t))S^- = +\delta(t)S^-$. The delta term is therefore proportional to the *sum* $S^+ + S^-$, not the difference — this is the fermionic minus sign doing its work. Evaluated at $t = 0$ the coefficient is the equal-time value of

$$\big[S^+ + S^-\big]_{ab}\big|_{t=0} = \langle 0|\,(\psi_a(x)\bar\psi_b(y)+\bar\psi_b(y)\psi_a(x))\,|0\rangle\,\big|_{x^0=y^0} = \langle 0|\,\Big\{\psi_a(x),\bar\psi_b(y)\Big\}\,|0\rangle\,\big|_{x^0=y^0} = \gamma \tag{11.209}$$

using Eq. (11.188), which is itself the canonical anticommutator Eq. (11.176) dressed by γ^0. Thus the delta term in Eq. (11.208) is

$$i\gamma^0\,\delta(t)\,\gamma^0\,\delta^3(\mathbf{x}-\mathbf{y}) = i\,(\gamma^0)^2\,\delta^4(x-y) = i\,\delta^4(x-y)\,\mathbf{1}, \tag{11.210}$$

since $(\gamma^0)^2 = \mathbf{1}$ from the Clifford algebra Eq. (11.2).

The smooth remainder vanishes. Collect the surviving smooth terms: the second bracket of Eq. (11.208) together with the spatial-and-mass part acting on the step-function-weighted Wightman pieces,

$$i\gamma^0\big[\theta(t)\partial_{x^0}S^+ - \theta(-t)\partial_{x^0}S^-\big] + (i\gamma^i\partial_{x^i} - m)\big[\theta(t)S^+ - \theta(-t)S^-\big]. \tag{11.211}$$

The spatial and mass operators pass through the step functions untouched, so Eq. (11.211) reassembles the full Dirac operator on each Wightman piece:

$$\theta(t)\,(i\partial\!\!\!/_x - m)S^+(x-y) - \theta(-t)\,(i\partial\!\!\!/_x - m)S^-(x-y) = 0, \tag{11.212}$$

because each Wightman function solves the homogeneous Dirac equation, $(i\partial\!\!\!/ - m)S^\pm = 0$, established in Eq. (11.186).

Assembly. Only the delta term Eq. (11.210) survives, giving

$$(i\partial\!\!\!/_x - m)\,S_F(x-y) = i\,\delta^4(x-y)\,\mathbf{1}, \tag{11.213}$$

in agreement with Eq. (11.204). As in the scalar case, the delta function is nothing but the canonical equal-time (anti)commutator, here propagated forward by the single time derivative that the first-order Dirac operator supplies — one derivative rather than the scalar's two, because the Dirac equation is first order. The dynamical content of fermionic quantization is made visible in the propagator, and the fermionic minus sign of the time-ordered product is exactly what converts the difference $S^+ - S^-$ that would have vanished into the sum $S^+ + S^-$ that yields the canonical anticommutator.

This completes the construction of the central dynamical object of the free Dirac theory. The propagator Eq. (11.195) is simultaneously the time-ordered vacuum correlator, the Dirac operator applied to the scalar Feynman propagator, the inverse of $i\not{\partial} - m$ with Feynman boundary conditions, and — as Eq. (11.172) — the electron line the QED Feynman rules of Chapter 17 will quote. The contour analysis has also isolated the two Wightman functions $S^{\pm}$ as the residues at the two on-shell poles; whether the local bilinear observables built from ψ commute at spacelike separation — the precise statement of causality, and the requirement that, together with positivity of energy, will be shown to *force* the anticommutator quantization we have used throughout — is the question taken up next, completing the fermionic half of the spin–statistics connection that Chapter 9 deferred.

11.6 Microcausality and the Spin–Statistics Theorem

The previous section assembled the fermion two-point structure: the Wightman functions $S^{\pm}(x-y)$ and, from their time-ordered combination, the Feynman propagator $S_F(x-y)$, the internal electron line of every diagram from Chapter 17 onward. Those objects answer *what amplitude connects two spacetime points.* We now ask the question on which the whole construction stands or falls—the same question the scalar faced in Section 9.6, but with the stakes raised by spin. *Can a measurement made here disturb one made simultaneously outside its light cone?* For the scalar the answer was delivered by a single odd c-number distribution, the Pauli–Jordan function Δ, whose vanishing at spacelike separation *was* relativistic causality; and the same computation, run with the wrong sign, showed that anticommutator quantization of a spin-0 field would have destroyed causality, forcing Bose statistics on the scalar.

This section completes the other half of that story. We compute the anticommutator of the Dirac field, find that it is again controlled by the very same Δ of Theorem 9.29—now dressed by the first-order Dirac operator—and read off two facts at once. First, because Δ vanishes outside the light cone, so does the field anticommutator, and from it the commutator of every *bilinear* observable: the electron field is causal. Second, the calculation run with *commutators* in place of anticommutators produces a c-number of the wrong symmetry—an even function rather than an odd one—which does *not* vanish at spacelike separation, and which, paired with the energy-unboundedness already exposed in Section 11.3, makes the commutator theory inconsistent on two independent counts. The conclusion is the discharge of a debt: the anticommutators we imposed by hand when quantizing the Dirac field were not a choice but a theorem. This is the half-integer instance of the general spin–statistics connection that Theorem 9.33 stated in full generality and explicitly deferred to this chapter.

11.6.1 The anticommutator function

We work in the Heisenberg picture, with the Dirac field carrying its full spacetime dependence through the mode expansion fixed in Section 11.3 and recalled in the normalization of Eq. (11.214),

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \sum_{s=1,2} \Big(b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big), \tag{11.214}$$

$$\bar\psi(y) = \int \frac{d^3q}{(2\pi)^3\, 2E_{\mathbf{q}}} \sum_{r=1,2} \Big(d^r_{\mathbf{q}}\, \bar v^r(q)\, e^{-iq\cdot y} + b^{r\dagger}_{\mathbf{q}}\, \bar u^r(q)\, e^{+iq\cdot y} \Big), \tag{11.215}$$

with $p^0 = E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$, $p\cdot x = E_{\mathbf{p}} x^0 - \mathbf{p}\cdot\mathbf{x}$ the on-shell Minkowski product in the mostly-minus metric, and the nonzero ladder anticommutators

$$\big\{ b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}} \big\} = \big\{ d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}} \big\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}), \tag{11.216}$$

all other pairs vanishing, as fixed in Eq. (11.116). Here $b^{s\dagger}_{\mathbf{p}}$ creates an electron and $d^{s\dagger}_{\mathbf{p}}$ a positron of spin label s; the constant spinors u^s, v^s are the plane-wave solutions of the momentum-space Dirac equation built in Section 11.2, normalized to $\bar u^r u^s = 2m\,\delta^{rs}$, $\bar v^r v^s = -2m\,\delta^{rs}$. This is the relativistic normalization carried throughout the chapter (Eqs. (11.95) and (11.116)): the invariant measure $d^3p/[(2\pi)^3 2E_{\mathbf{p}}]$ and the factor $(2\pi)^3 2E_{\mathbf{p}}$ on the right of the anticommutator.

The basic object of relativistic causality for a Fermi field is the *anticommutator* of ψ with $\bar\psi$, because by Eq. (11.216) it is the anticommutator of ladder operators—not the commutator—that reduces to a c-number. Insert Eqs. (11.214) and (11.215) and expand $\big\{\psi_a(x), \bar\psi_b(y)\big\}$ bilinearly. Of the four operator pairings only $\big\{b, b^\dagger\big\}$ and $\big\{d, d^\dagger\big\}$ survive—the $\{b,b\}$, $\big\{b, d^\dagger\big\}$ and like terms vanish identically because electron and positron ladders anticommute and because $\{b,b\} = \big\{b^\dagger, b^\dagger\big\} = 0$:

$$\begin{aligned}\big\{\psi_a(x), \bar\psi_b(y)\big\} = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \frac{d^3q}{(2\pi)^3\, 2E_{\mathbf{q}}} \sum_{r,s} \Big(&\big\{ b^s_{\mathbf{p}}, b^{r\dagger}_{\mathbf{q}} \big\}\; u^s_a(p)\, \bar u^r_b(q)\, e^{-ip\cdot x + iq\cdot y} \\ &+ \big\{ d^{s\dagger}_{\mathbf{p}}, d^r_{\mathbf{q}} \big\}\; v^s_a(p)\, \bar v^r_b(q)\, e^{+ip\cdot x - iq\cdot y} \Big).\end{aligned} \tag{11.217}$$

Using Eq. (11.216), each ladder anticommutator supplies $(2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\delta^{(3)}(\mathbf{p}-\mathbf{q})$; the $\mathbf{q}$-integral collapses against the invariant measure (the factor $2E_{\mathbf{p}}$ cancelling one power of $2E_{\mathbf{q}}$), sets $r = s$ and $E_{\mathbf{q}} = E_{\mathbf{p}}$, and turns the two spinor products into spin sums:

$$\big\{\psi_a(x), \bar\psi_b(y)\big\} = \int \frac{d^3p}{(2\pi)^3}\, \frac{1}{2E_{\mathbf{p}}} \Big(\textstyle\sum_s u^s_a(p)\bar u^s_b(p)\, e^{-ip\cdot(x-y)} + \sum_s v^s_a(p)\bar v^s_b(p)\, e^{+ip\cdot(x-y)} \Big). \tag{11.218}$$

The spin sums are the completeness relations derived in this chapter as Eq. (11.73) (and collected, with the trace and contraction technology, in Eq. (C.247)),

$$\sum_s u^s(p)\,\bar{u}^s(p) = \not{p} + m, \qquad \sum_s v^s(p)\,\bar{v}^s(p) = \not{p} - m, \tag{11.219}$$

both understood as 4×4 matrices in the suppressed Dirac indices a, b. Substituting,

$$\Big\{\psi_a(x), \bar{\psi}_b(y)\Big\} = \int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\mathbf{p}}}\Big[(\not{p}+m)_{ab}\,e^{-ip\cdot(x-y)} + (\not{p}-m)_{ab}\,e^{+ip\cdot(x-y)}\Big]. \tag{11.220}$$

The momentum-space factor $\not{p}$ is now traded for a spacetime derivative, which pulls the entire Dirac structure outside the integral. Acting on a plane wave, the operator $i\not{\partial}_x \equiv i\gamma^\mu\partial_{x^\mu}$ gives

$$i\not{\partial}_x\, e^{-ip\cdot(x-y)} = \not{p}\, e^{-ip\cdot(x-y)}, \qquad i\not{\partial}_x\, e^{+ip\cdot(x-y)} = -\not{p}\, e^{+ip\cdot(x-y)}, \tag{11.221}$$

because $\partial_{x^\mu} e^{\mp ip\cdot(x-y)} = \mp ip_\mu e^{\mp ip\cdot(x-y)}$. Consequently, on the first exponential $(\not{p}+m)e^{-ip\cdot(x-y)} = (i\not{\partial}_x + m)\,e^{-ip\cdot(x-y)}$, while on the second $(i\not{\partial}_x+m)\,e^{+ip\cdot(x-y)} = (-\not{p}+m)\,e^{+ip\cdot(x-y)} = -(\not{p}-m)\,e^{+ip\cdot(x-y)}$, i.e. $(\not{p}-m)e^{+ip\cdot(x-y)} = -(i\not{\partial}_x + m)e^{+ip\cdot(x-y)}$. The single operator $(i\not{\partial}_x + m)$ therefore factors out of *both* terms, leaving the *difference* of the two exponentials:

$$\Big\{\psi_a(x), \bar{\psi}_b(y)\Big\} = (i\not{\partial}_x + m)_{ab}\int \frac{d^3p}{(2\pi)^3}\,\frac{1}{2E_{\mathbf{p}}}\Big(e^{-ip\cdot(x-y)} - e^{+ip\cdot(x-y)}\Big). \tag{11.222}$$

The remaining integral is exactly the one that defined the scalar commutator in Eq. (9.188): it is $i\,\Delta(x-y)$, with Δ the Pauli–Jordan function of Theorem 9.29 for a field of mass m. We have proved:

Theorem 11.44 (The Dirac anticommutator function)**.** *The free Dirac field of mass m satisfies*

$$\boxed{\Big\{\psi_a(x), \bar{\psi}_b(y)\Big\} = (i\not{\partial}_x + m)_{ab}\; i\,\Delta(x-y),} \tag{11.223}$$

where Δ is the scalar Pauli–Jordan function Eq. (9.189), so that $i\Delta(x-y) = [\phi(x),\phi(y)]$ is the c-number commutator of the scalar field of the same mass. Moreover

$$\{\psi_a(x), \psi_b(y)\} = 0, \qquad \Big\{\bar{\psi}_a(x), \bar{\psi}_b(y)\Big\} = 0 \quad \textit{for all } x, y. \tag{11.224}$$

Proof. Equation Eq. (11.223) is Eq. (11.222) with the bracket identified as $i\Delta(x-y)$ via Eq. (9.188). For Eq. (11.224), expand $\{\psi_a(x),\psi_b(y)\}$: every term pairs two annihilation/creation operators of the same statistics, $\{b,b\}$, $\Big\{d^\dagger, d^\dagger\Big\}$,

$\{b, d^\dagger\}$ or $\{d^\dagger, b\}$, each of which vanishes by Eq. (11.216). There is no surviving $\{b, b^\dagger\}$- or $\{d, d^\dagger\}$-type pairing because ψ contains only the operators b and $d^\dagger$. The same argument with $b^\dagger, d$ gives $\{\bar\psi, \bar\psi\} = 0$. □

That the field anticommutator is the scalar Pauli–Jordan function with a Dirac operator in front is the structural fact from which everything in this section follows. The first-order operator $(i\partial\!\!\!/_x + m)$ is precisely the one appearing in the Dirac equation, and applying $(i\partial\!\!\!/_x - m)$ to Eq. (11.223) annihilates the right-hand side, since $(i\partial\!\!\!/ - m)(i\partial\!\!\!/ + m) = -\partial^2 - m^2 = -(\Box + m^2)$ and $(\Box_x + m^2)\Delta(x-y) = 0$ (the Pauli–Jordan function is a solution of the Klein–Gordon equation, as is manifest from its on-shell support); this is the operator statement that ψ obeys its own equation of motion.

Worked Example 11.45 (Equal-time reduction to $\gamma^0\delta^{(3)}$)**.** We confirm that Eq. (11.223) reproduces the canonical equal-time anticommutation relation imposed in quantization. Set $x^0 = y^0$ and recall two equal-time properties of Δ established in Section 9.6.1: it vanishes on the equal-time slice,

$$\Delta(0, \mathbf{x} - \mathbf{y}) = 0, \tag{11.225}$$

(Eq. (9.191)), and its time derivative there is a delta function,

$$\partial_0 \Delta(x)\big|_{x^0=0} = -\,\delta^{(3)}(\mathbf{x}) \tag{11.226}$$

(Eq. (9.192)). Write the Dirac operator as $i\partial\!\!\!/_x + m = i\gamma^0\partial_{x^0} + i\gamma^i\partial_{x^i} + m$ and act on $i\Delta(x-y)$, then set $x^0 = y^0$. The mass term gives $m \cdot i\Delta(0, \mathbf{x}-\mathbf{y}) = 0$ by Eq. (11.225). The spatial-gradient term gives $i\gamma^i\, i\,\partial_{x^i}\Delta(0, \mathbf{x}-\mathbf{y})$; but Δ vanishes *identically* as a function of the spatial argument on the equal-time slice, so its spatial gradient vanishes too. Only the time-derivative term survives:

$$\{\psi_a(x), \bar\psi_b(y)\}\,|_{x^0=y^0} = \gamma^0_{ab}\; i\,\partial_{x^0}\big(i\Delta(x-y)\big)\big|_{x^0=y^0} = \gamma^0_{ab}\,(i)(i)\,\big(-\delta^{(3)}(\mathbf{x}-\mathbf{y})\big) = \gamma^0_{ab}\,\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{11.227}$$

using $(i)(i)(-1) = +1$. Multiplying on the right by γ^0 and using $\bar\psi = \psi^\dagger\gamma^0$, $(\gamma^0)^2 = \mathbf{1}$, this is

$$\{\psi_a(\mathbf{x}, t), \psi^\dagger_b(\mathbf{y}, t)\} = \delta_{ab}\,\delta^{(3)}(\mathbf{x} - \mathbf{y}), \tag{11.228}$$

the canonical equal-time anticommutator of Eq. (11.176). The covariant function Eq. (11.223) thus packages the canonical equal-time structure together with its Lorentz-covariant evolution off equal times into a single object—exactly as Δ did for the scalar.

11.6.2 Microcausality of the Dirac field

Everything causal about the elementary field is now inherited from the scalar in one step. The Pauli–Jordan function vanishes at spacelike separation (Theorem 9.31): $\Delta(x-y) = 0$ whenever $(x-y)^2 < 0$, a consequence of Δ being

odd and Lorentz invariant, so that the connectedness of $L_+^\uparrow$ forces $\Delta(x-y) = \Delta(-(x-y)) = -\Delta(x-y)$ on the spacelike orbit. The Dirac anticommutator is a fixed first-order differential operator applied to this function, and a differential operator applied to a function that is *identically zero* on the open spacelike region returns zero there.

Theorem 11.46 (Microcausality of the Dirac field)**.** *For spacelike separation* $(x-y)^2 < 0$*,*

$$\left\{\psi_a(x), \bar{\psi}_b(y)\right\} = 0, \qquad \left\{\psi_a(x), \psi_b^\dagger(y)\right\} = 0. \tag{11.229}$$

Proof. By Theorem 9.31, Δ vanishes on the open set $\{(x-y)^2 < 0\}$. A distribution that is identically zero on an open set has all of its derivatives identically zero there; hence $(i\partial\!\!\!/_x + m)\, i\Delta(x-y) = 0$ throughout that region. By Eq. (11.223) this is $\left\{\psi_a(x), \bar{\psi}_b(y)\right\} = 0$. Multiplying by γ^0 gives the statement for $\psi^\dagger$. (One must check that no derivative acts on the *boundary* of the region, where Δ has light-cone singularities; but spacelike separation is an open condition, and Eq. (11.229) is asserted only in the interior $(x-y)^2 < 0$, where Δ and all its derivatives are honestly zero.) □

Two remarks fix the meaning of this result before we extract its observable consequence. First, the spacelike-vanishing object for the fermion is the *anticommutator*, not the commutator—and that is the entire content of the spin–statistics connection, as we make precise in Section 11.6.4. Second, the field ψ is *not* itself an observable. It carries one unit of electric charge, transforms as a spinor under rotations (acquiring a sign under a 2π rotation), and is in no sense measurable on its own. Microcausality is a statement about *observables*, which are built from *even* numbers of fermion fields; the field anticommutator Eq. (11.229) is the technical lemma, not the physical statement. We now pass to the physical statement.

11.6.3 Causality of bilinear observables

A local fermionic observable is a Hermitian bilinear (or higher even polynomial) in the field and its conjugate at a point,

$$\mathcal{O}_\Gamma(x) = \bar{\psi}(x)\,\Gamma\,\psi(x) = \bar{\psi}_a(x)\,\Gamma_{ab}\,\psi_b(x), \tag{11.230}$$

where Γ is one of the sixteen bilinear covariants of Theorem C.44—the scalar $\mathbf{1}$, the vector γ^μ (the electric current $j^\mu = \bar{\psi}\gamma^\mu\psi$ that couples to the photon in Chapter 13), the tensor $\sigma^{\mu\nu}$, and so on. Each contains exactly two fermion fields, hence is bosonic, and the requirement of microcausality is that two such observables *commute* at spacelike separation. The mechanism by which a vanishing *anti*commutator at the field level produces a vanishing *commutator* at the observable level is a purely algebraic rearrangement, and it is worth isolating because it is the precise point at which fermionic statistics conspires with locality.

Lemma 11.47 (Commutator of bilinears via anticommutators)**.** *For any four operators A, B, C, D,*

$$[AB, CD] = A\,\{B, C\}\,D - \{A, C\}\,BD + CA\,\{B, D\} - C\,\{A, D\}\,B. \tag{11.231}$$

Proof. This is an identity of associative algebra; no assumption on the statistics of A, B, C, D is used. Move CD leftward through AB one factor at a time, replacing each transposition by an anticommutator, $XY = \{X, Y\} - YX$. Starting from $ABCD$ and pushing C to the left,

$$ABCD = A(\,\{B, C\} - CB)D = A\,\{B, C\}\,D - ACBD = A\,\{B, C\}\,D - \{A, C\}\,BD + CABD, \tag{11.232}$$

where in the last step $AC = \{A, C\} - CA$. Now push D to the left through $CABD$,

$$CABD = CA(\,\{B, D\} - DB) = CA\,\{B, D\} - CADB = CA\,\{B, D\} - C\,\{A, D\}\,B + CDAB, \tag{11.233}$$

using $AD = \{A, D\} - DA$. Combining the two and subtracting $CDAB$ gives Eq. (11.231). □

The point of writing the commutator of products through *anticommutators* rather than commutators is that, for the Dirac field, it is the anticommutators that are c-numbers and that vanish at spacelike separation. Identify $A = \bar\psi_a(x)$, $B = \psi_b(x)$, $C = \bar\psi_c(y)$, $D = \psi_d(y)$, and contract with $\Gamma_{ab}\Gamma'_{cd}$:

$$[\mathcal{O}_\Gamma(x), \mathcal{O}_{\Gamma'}(y)] = \Gamma_{ab}\,\Gamma'_{cd}\,\Big[\bar\psi_a(x)\psi_b(x), \bar\psi_c(y)\psi_d(y)\Big]\,. \tag{11.234}$$

Of the four anticommutators generated by Eq. (11.231), two vanish *identically* by Eq. (11.224):

$$\{A, C\} = \Big\{\bar\psi_a(x), \bar\psi_c(y)\Big\} = 0, \qquad \{B, D\} = \{\psi_b(x), \psi_d(y)\} = 0, \tag{11.235}$$

and the surviving two are the mixed anticommutators, each a c-number multiple of the matrix Eq. (11.223):

$$\{B, C\} = \Big\{\psi_b(x), \bar\psi_c(y)\Big\}, \qquad \{A, D\} = \Big\{\bar\psi_a(x), \psi_d(y)\Big\} = \Big\{\psi_d(y), \bar\psi_a(x)\Big\}. \tag{11.236}$$

Therefore

$$[\mathcal{O}_\Gamma(x), \mathcal{O}_{\Gamma'}(y)] = \Gamma_{ab}\Gamma'_{cd}\Big(\bar\psi_a(x)\,\Big\{\psi_b(x), \bar\psi_c(y)\Big\}\,\psi_d(y) - \bar\psi_c(y)\,\Big\{\psi_d(y), \bar\psi_a(x)\Big\}\,\psi_b(x)\Big). \tag{11.237}$$

Each term is now an operator (a single field at x times a single field at y) multiplied by a c-number anticommutator. By Theorem 11.46 both c-number anticommutators vanish for $(x - y)^2 < 0$. We conclude:

Theorem 11.48 (Causality of fermion bilinears)**.** *For any bilinear covariants* Γ, Γ' *and any spacelike separation* $(x-y)^2 < 0$*,*

$$[\mathcal{O}_\Gamma(x), \mathcal{O}_{\Gamma'}(y)] = 0. \tag{11.238}$$

More generally, any two local observables built from even numbers of Dirac fields commute at spacelike separation: the quantized Dirac field is microcausal.

The extension to higher even polynomials is immediate by induction: an observable with $2n$ fields, commuted against another, generates by repeated use of Eq. (11.231) a sum of terms each containing at least one mixed field anticommutator $\{\psi, \bar\psi\}$ or $\{\bar\psi, \psi\}$ (the like-field anticommutators vanish by Eq. (11.224)), and every such anticommutator vanishes spacelike. The crucial bookkeeping is that an *even* number of fermion transpositions is required to bring the two observables past one another, so the rearrangement closes on *anti*commutators with the right signs; an *odd*-field operator (such as ψ itself) would instead *anti*commute spacelike, which is harmless precisely because such an operator is not observable. This is the deep reason fermionic causality is consistent: nature only ever asks even-field operators to commute, and for those the anticommutator algebra delivers exactly that.

Physical Insight

The minus signs of Fermi statistics are not an obstruction to causality—they are its enabler. A vanishing field *anti*commutator would be useless for causality if observables were linear in the field; but observables are bilinear, and the algebra Eq. (11.231) converts the vanishing anticommutator of the constituents into the vanishing commutator of the composite. The scalar achieves the same end with commutators all the way down; the fermion achieves it with one extra layer of signs. Both arrive at the same physical statement: spacelike-separated measurements commute, and no signal outruns light.

Worked Example 11.49 (The electric current commutes with itself spacelike)**.** The single most important fermion bilinear is the electric current $j^\mu(x) = \bar\psi(x)\gamma^\mu\psi(x)$, the operator that will be gauged to the photon in Chapter 13 and whose matrix elements give every QED amplitude of Chapter 17. Causality of the theory requires $[j^\mu(x), j^\nu(y)] = 0$ at spacelike separation; we exhibit the cancellation explicitly. Setting $\Gamma = \gamma^\mu$, $\Gamma' = \gamma^\nu$ in Eq. (11.237),

$$[j^\mu(x), j^\nu(y)] = (\gamma^\mu)_{ab}(\gamma^\nu)_{cd}\Big(\bar\psi_a(x)\, S_{bc}(x-y)\, \psi_d(y) - \bar\psi_c(y)\, S_{da}(y-x)\, \psi_b(x)\Big), \tag{11.239}$$

where we have written the c-number anticommutator as $S_{bc}(x-y) \equiv \{\psi_b(x), \bar\psi_c(y)\} = (i\partial\!\!\!/_x + m)_{bc}\, i\Delta(x-y)$ from Eq. (11.223), and likewise $S_{da}(y-x) = \{\psi_d(y), \bar\psi_a(x)\} =$

$(i\partial\!\!\!/_y + m)_{da}\, i\Delta(y-x)$. Both matrices are $(i\partial\!\!\!/ + m)$ acting on the Pauli–Jordan function evaluated at $\pm(x-y)$. For $(x-y)^2 < 0$ the function Δ and all its derivatives vanish (Theorem 9.31): $\Delta(x-y) = 0$ *and*, since spacelike separation is preserved under $x-y \to y-x$, also $\Delta(y-x) = 0$, the latter equally guaranteed by the oddness $\Delta(-z) = -\Delta(z)$ of Eq. (9.190). Hence $S(x-y) = S(y-x) = 0$ on the spacelike region, both terms of Eq. (11.239) vanish, and $[j^\mu(x), j^\nu(y)] = 0$. The charge density $j^0 = \psi^\dagger\psi$ measured in one laboratory therefore commutes with the current measured simultaneously outside its light cone—no measurement of charge here disturbs a measurement of charge there. This is the local-commutativity property that the Ward–Takahashi identity of Chapter 19 and the whole gauge structure of QED silently presuppose.

11.6.4 The spin–statistics connection for spin $\frac{1}{2}$

We have shown that the anticommutator quantization Eq. (11.216) produces a causal theory. The spin–statistics theorem is the converse and stronger claim: that anticommutators are *forced*—that the alternative, commutator quantization of the spin-$\frac{1}{2}$ field, is inconsistent. We now establish this on two independent grounds, exactly inverting the scalar argument of Section 9.6.5.

Suppose, against the grain, that we had built the Dirac Fock space on *commutators*, declaring

$$\left[b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\right] = \left[d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\right] = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\delta^{(3)}(\mathbf{p}-\mathbf{q}), \tag{11.240}$$

with all other commutators zero—the Bose–Einstein choice. There are two fatal consequences.

First failure: the Hamiltonian is unbounded below. This is the result already obtained in Section 11.3, which we recall in its sharpest form. Expressed in modes, the Dirac Hamiltonian before any reordering is

$$H = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, E_{\mathbf{p}} \sum_s \left(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}} \right), \tag{11.241}$$

the relative minus sign on the antiparticle term being an unavoidable consequence of the first-order structure of the Dirac Lagrangian (it is the same minus sign that, for the scalar's *second*-order Lagrangian, was absent). Reordering the antiparticle term decides the spectrum. With the *anticommutator* Eq. (11.216), $-d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}} = +d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} - (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{(3)}(0)$, so that after dropping the (negative) constant zero-point term,

$$H = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, E_{\mathbf{p}} \sum_s \left(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} + d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} \right) \;\geq\; 0, \tag{11.242}$$

a sum of manifestly nonnegative number operators (Theorem 11.31): the energy is bounded below and every antiparticle *adds* positive energy. With the *commutator* Eq. (11.240), instead $-d^s_{\mathbf{p}} d^{s\dagger}_{\mathbf{p}} = -d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} - (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{(3)}(0)$, and

$$H = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, E_{\mathbf{p}} \sum_s \left(b^{s\dagger}_{\mathbf{p}} b^s_{\mathbf{p}} - d^{s\dagger}_{\mathbf{p}} d^s_{\mathbf{p}} \right), \tag{11.243}$$

in which each antiparticle quantum *lowers* the energy without limit: piling up d-quanta drives $H \to -\infty$. There is no stable vacuum, no ground state, and the theory is physically meaningless. Commutators are incompatible with positivity of the energy for the Dirac field—hypothesis (ii) of Theorem 9.33.

Second failure: microcausality is violated. Independently of the energy, the commutator choice wrecks locality. With Eq. (11.240) it is now the *commutator* of the fields that reduces to a c-number. Repeating the computation of Section 11.6.1 verbatim, but with the relative sign from $\left[d^{s\dagger}_{\mathbf{p}}, d^r_{\mathbf{q}}\right] = -\left[d^r_{\mathbf{q}}, d^{s\dagger}_{\mathbf{p}}\right] = -(2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\delta^{(3)}(\mathbf{p}-\mathbf{q})$ flipping the antiparticle term, Eq. (11.220) becomes

$$\left[\psi_a(x), \bar\psi_b(y)\right]_{\text{Bose}} = \int \frac{d^3p}{(2\pi)^3}\, \frac{1}{2E_{\mathbf{p}}} \left[(\not p + m)_{ab}\, e^{-ip\cdot(x-y)} - (\not p - m)_{ab}\, e^{+ip\cdot(x-y)} \right]. \tag{11.244}$$

Now pull out the Dirac operator using Eq. (11.221): the first term is $(i\not\partial_x + m)e^{-ip\cdot(x-y)}$ as before, while the second is $-(\not p - m)e^{+ip\cdot(x-y)} = +(i\not\partial_x + m)e^{+ip\cdot(x-y)}$. The operator factors out of the *sum* of the exponentials this time:

$$\left[\psi_a(x), \bar\psi_b(y)\right]_{\text{Bose}} = (i\not\partial_x + m)_{ab} \int \frac{d^3p}{(2\pi)^3}\, \frac{1}{2E_{\mathbf{p}}} \left(e^{-ip\cdot(x-y)} + e^{+ip\cdot(x-y)} \right) = (i\not\partial_x + m)_{ab} \left[\Delta_+(x-y) + \Delta \right. \tag{11.245}$$

where $\Delta_+(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2E_{\mathbf{p}}} e^{-ip\cdot x} = \langle \phi(x)\phi(0) \rangle$ is the positive-frequency Wightman function of Eq. (9.202). The scalar core has flipped from the *odd* difference $i\Delta = \Delta_+(x-y) - \Delta_+(y-x)$ to the *even* sum $\Delta_+(x-y) + \Delta_+(y-x)$—the very combination that the scalar spin–statistics argument identified as the obstruction. The Lorentz-orbit argument that killed the odd function gives nothing for the even one: for spacelike $x-y$ there is $\Lambda \in L^\uparrow_+$ with $\Lambda(x-y) = -(x-y)$, but the even function is already invariant under this reflection, so we learn only $0 = 0$. Explicit evaluation, carried out in Theorem 9.34, shows the even core is strictly positive at spacelike separation,

$$\left[\Delta_+(x-y) + \Delta_+(y-x)\right]_{(x-y)^2<0} = 2\,\Delta_+(0,\mathbf{r}) = \frac{m}{2\pi^2 r}\, K_1(mr) > 0, \qquad r = \sqrt{-(x-y)^2}, \tag{11.246}$$

with K_1 a modified Bessel function (Eq. (9.203)). Hence the would-be commutator Eq. (11.245) does *not* vanish outside the light cone (Fig. 11.5). Feeding this nonvanishing field commutator into the bilinear rearrangement of Theorem 11.47—which under the Bose assumption must be run with commutators in place of

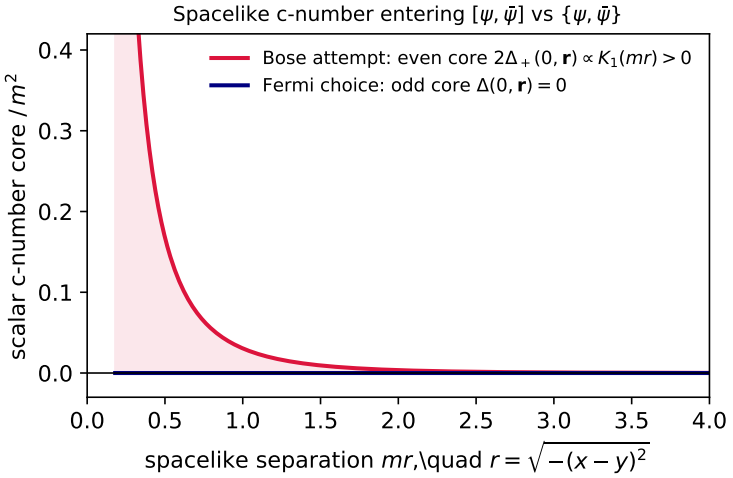


Figure 11.5: The two quantization attempts for the Dirac field, compared on a purely spatial (spacelike) slice $r = \sqrt{-(x-y)^2}$. In both cases the field (anti)commutator is the Dirac operator $(i\partial\!\!\!/ + m)$ acting on a scalar c-number core. For the correct *Fermi* choice the core is the odd Pauli–Jordan function Δ, which vanishes identically at spacelike separation (Theorem 9.31); the field anticommutator is zero and observables commute. For the forbidden *Bose* choice the core is the even combination $2\Delta_+ \propto K_1(mr)$, strictly positive outside the light cone (Eq. (11.246)); the field commutator fails to vanish and causality is destroyed. The accompanying energy argument is independent: Eq. (11.243) shows the Bose Hamiltonian is unbounded below.

anticommutators—leaves the observable commutator $[\mathcal{O}_\Gamma(x), \mathcal{O}_{\Gamma'}(y)]$ operator-valued and nonzero at spacelike separation. Local commutativity Eq. (9.199) fails: the spin-$\frac{1}{2}$ field quantized with commutators is acausal.

Theorem 11.50 (Spin–statistics for the Dirac field)**.** *The free spin-$\frac{1}{2}$ field cannot be consistently quantized with commutators. The Bose algebra Eq. (11.240) produces (i) a Hamiltonian Eq. (11.243) unbounded below, with no stable vacuum, and (ii) a field commutator Eq. (11.245) whose spacelike value Eq. (11.246) is nonzero, violating microcausality. Either failure alone is fatal. Causality and positivity of the energy therefore* force *Fermi–Dirac quantization—the anticommutator algebra Eq. (11.216), for which $H \geq 0$ and $\left\{\psi(x), \bar\psi(y)\right\} = 0$ outside the light cone. A spin-$\frac{1}{2}$ field is necessarily a fermion, and the Pauli exclusion principle $\left(b^{s\dagger}_{\mathbf{p}}\right)^2 = 0$ is its immediate corollary.*

Physical Insight

The decisive sign is again the relative sign between the two exponentials in the field (anti)commutator, and what fixes it is the spin sum Eq. (11.219). For the scalar, the same two-exponential structure carried a coefficient 1 on each term, and microcausality wanted the *odd* difference: commutators, Bose. For the spinor, the antiparticle term carries the spin sum $\sum v\bar{v} = \not{p} - m$, whose mass term has the *opposite* sign to the particle sum $\sum u\bar{u} = \not{p} + m$; this extra relative minus is precisely the $(-1)^{2j} = -1$ of a half-integer-spin field. It converts "odd is wanted" into "even is wanted," so that microcausality now selects the *anticommutator*. Spin dictates statistics through the sign that the field's Lorentz representation deposits in its two-point function—$+1$ for integer spin, -1 for half-integer—and the marriage of relativity (Lorentz invariance plus causality) with quantum mechanics (positivity of the energy) does the rest.

11.6.5 Completing the general spin–statistics theorem

With Theorem 11.50 we discharge, for the free Dirac field, the half-integer half of the general statement that Theorem 9.33 could only assert. It is worth recording precisely what has now been proved in full and what remains without full proof, so the book's honesty about its own scope is exact.

The general theorem (Theorem 9.33) asserts that, given (i) Poincaré invariance with fields in finite-dimensional Lorentz representations labelled by spin j, (ii) a positive-norm Hilbert space with a stable vacuum of lowest energy, and (iii) microcausality, integer-spin fields must be quantized with commutators and half-integer-spin fields with anticommutators, the opposite choice in either case violating one of the hypotheses. The free real scalar gave the $j = 0$ instance completely in Theorem 9.32: there the Lorentz representation is trivial, the spin-sum sign is the bare $+1$, and the argument is the clean even-versus-odd dichotomy. The free Dirac field has now given the $j = \frac{1}{2}$ instance completely: the Lorentz representation is $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$, the spin sum supplies the $(-1)^{2j} = -1$, and the two-pronged argument—unbounded H and a spacelike-nonvanishing even commutator—excludes the wrong statistics on both of the theorem's grounds at once. The reader has therefore seen the mechanism in both of its faces: for the boson commutators are forced, for the fermion anticommutators are forced, and in each case it is relativity plus quantum positivity that forces them.

What is *not* proved here is the theorem in full generality, for arbitrary spin and for interacting fields. That proof rests on the analytic structure of the vacuum expectation values $\langle 0|\psi(x_1)\cdots\psi(x_n)|0\rangle$: positivity of the energy (hypothesis (ii)) makes the Wightman functions boundary values of functions holomorphic in a forward tube in the complexified coordinates, and the spacelike

continuation relating $\Delta_+(x)$ to $\Delta_+(-x)$ used above is then a special case of an analytic continuation valid for all spins, with the $(-1)^{2j}$ emerging from the transformation of the field's Lorentz indices under the complex Lorentz rotation by π. This is the Wightman-axiom framework, and it requires the analyticity machinery (the edge-of-the-wedge and Bargmann–Hall–Wightman theorems) that this part of the book has not developed; we therefore quote Theorem 9.33 without full proof and do not reproduce the axiomatic proof. The free spin-0 and spin-$\frac{1}{2}$ cases established are exactly the two on which all of the book's perturbation theory rests, and they have been proved without gaps.

Historical Note

The two-pronged structure of the spin-$\frac{1}{2}$ argument is faithful to its history. Dirac's hole theory (1930) confronted the negative-energy sea of Eq. (11.243) and tamed it by filling the sea and invoking the exclusion principle—a physically correct intuition that the anticommutators of Jordan and Wigner (1928) had already made into an algebra. Pauli's 1940 paper unified the threads, proving that the requirement of a positive energy and the requirement of spacelike (anti)commutativity *both* select Fermi statistics for half-integer spin and Bose statistics for integer spin; the modern axiomatic version, resting on the analyticity of Wightman functions, is due to Lüders, Zumino, and Burgoyne in the 1950s and is set out in the monograph of Streater and Wightman.

The Dirac field is now a fully solved, demonstrably consistent relativistic quantum theory: its excitations are spin-$\frac{1}{2}$ fermions obeying the exclusion principle, its energy is bounded below, its local observables respect the light cone, and the anticommutators imposed in Section 11.3 stand revealed not as a postulate but as the unique quantization compatible with relativity. This causal, positive-energy structure is the bedrock on which the next section builds: having established *that* the bilinear observables are local, we turn to *how* they transform under the discrete operations C, P, and T, and to the CPT combination whose theorem—like spin–statistics—ties the consistency of the field to the geometry of the Lorentz group.

11.7 Discrete Symmetries: C, P, T, and CPT

The previous section closed the structural core of this chapter by proving that the Dirac field is local: its Hermitian bilinear observables commute at spacelike separation, and that microcausality, married to positivity of the energy, is exactly what forces the anticommutator quantization (Section 9.6, Theorem 9.33). We now collect the dividend. A theory that is local, Poincaré invariant, and respects the spin–statistics connection is, almost for free, invariant under a single

antiunitary operation that reflects every spacetime axis and exchanges every particle with its antiparticle. That operation is CPT, and the theorem that guarantees it is one of the deepest structural facts about relativistic quantum field theory. To reach it we must first build its three constituents separately — parity P, charge conjugation C, and time reversal T — as honest operators on the Dirac Fock space, work out how each acts on the field, on the ladder operators, and on the sixteen bilinear covariants, and only then assemble the combination. Along the way we meet a string of physical facts that the rest of the book will lean on: that a fermion and its antifermion carry *opposite* intrinsic parity, that charge conjugation interchanges left- and right-handed quanta, that time reversal reverses both momentum and spin and squares to -1 on a single fermion, and — as a worked aside whose experimental status is still open in the mid-2020s — that a neutral fermion may be its own antiparticle.

The standing conventions of the chapter remain in force: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the Clifford algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\bar\psi = \psi^\dagger\gamma^0$ (Eq. (11.5)). The discrete-symmetry *matrices* that intertwine $\{\gamma^\mu\}$ with its conjugate sets — the parity matrix γ^0, the charge-conjugation matrix $C = i\gamma^2\gamma^0$, and the time-reversal matrix $T = \gamma^1\gamma^3$ — were constructed once and for all, with all their algebraic properties, in Section C.4; we import them by reference rather than rederive them. We restate only the relations we use at the moment of use. The mode expansion of the quantized field and the ladder algebra are those fixed in Section 11.3; we recall them in the form

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \sum_{s=1,2} \Big(b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big), \tag{11.247}$$

with $b^s_{\mathbf{p}}$ annihilating an electron and $d^{s\dagger}_{\mathbf{p}}$ creating a positron, both of momentum $\mathbf{p}$ and spin s quantized along a fixed axis, $E_{\mathbf{p}} = +\sqrt{\mathbf{p}^2 + m^2}$, and the nonvanishing anticommutators

$$\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\} = \{d^r_{\mathbf{p}}, d^{s\dagger}_{\mathbf{q}}\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{rs}\, \delta^{(3)}(\mathbf{p}-\mathbf{q}) \tag{11.248}$$

(Eqs. (11.95) and (11.116)). The spinors are normalized by $\bar u^r u^s = 2m\,\delta^{rs} = -\bar v^r v^s$ and obey the completeness relations $\sum_s u^s(p)\bar u^s(p) = \not p + m$, $\sum_s v^s(p)\bar v^s(p) = \not p - m$ from Section 11.2.

Notation Watch

Three pairs of symbols run through this section. The *Fock-space operators* are the calligraphic $\mathcal{P}$ (unitary, parity), $\mathcal{C}$ (unitary, charge conjugation), and $\mathcal{T}$ (*antiunitary*, time reversal); the combined operation is $\Theta = \mathcal{C}\mathcal{P}\mathcal{T}$, also antiunitary. The *spinor matrices* accompanying them are γ^0, $C = i\gamma^2\gamma^0$, and $T = \gamma^1\gamma^3$. Thus $\mathcal{C}$ is an operator on Hilbert space while C is a

4×4 matrix; the parentage is the same but they live in different places. The intrinsic phases are η_P, η_C, η_T, each of unit modulus. We do not use the time-ordering symbol in this section, so no confusion with $\mathcal{T}$ can arise.

11.7.1 Parity

Parity is the spatial reflection $x = (t, \mathbf{x}) \mapsto x_P \equiv (t, -\mathbf{x})$. We seek a unitary operator $\mathcal{P}$ on the Dirac Hilbert space implementing it on the field, $\mathcal{P}\,\psi(t,\mathbf{x})\,\mathcal{P}^{-1} = (\text{matrix})\,\psi(t,-\mathbf{x})$, chosen so that the transformed field again solves the Dirac equation and the Lagrangian is invariant. The matrix is fixed by that demand.

Fixing the matrix from the Dirac equation

Suppose $\psi(x)$ solves $(i\gamma^\mu\partial_\mu - m)\psi(x) = 0$ and write $\psi_P(x) \equiv S_P\,\psi(x_P)$ for some constant 4×4 matrix S_P to be determined. The derivative transforms as $\partial_\mu \mapsto \partial^P_\mu$, where $\partial^P_0 = \partial_0$ and $\partial^P_i = -\partial_i$ when expressed in the reflected coordinate; equivalently ∂_μ acting at x_P is ∂^μ acting at x. Requiring $(i\gamma^\mu\partial_\mu - m)\psi_P(x) = 0$ and using $\psi(x_P)$ as a solution, we need

$$S_P^{-1}\gamma^0 S_P = \gamma^0, \qquad S_P^{-1}\gamma^i S_P = -\gamma^i, \tag{11.249}$$

so that the reflected derivative is compensated. The unique solution up to a phase is $S_P = \gamma^0$, because γ^0 commutes with itself and anticommutes with each γ^i (this is precisely the parity relation $\gamma^0\gamma^\mu\gamma^0 = \gamma_\mu$ proved in Eq. (C.83)). Hence

$$\boxed{\mathcal{P}\,\psi(t,\mathbf{x})\,\mathcal{P}^{-1} = \eta_P\,\gamma^0\,\psi(t,-\mathbf{x})} \qquad |\eta_P| = 1, \tag{11.250}$$

with η_P an undetermined phase, the *intrinsic parity* of the field. Taking the Dirac adjoint of Eq. (11.250) (and using $\gamma^{0\dagger} = \gamma^0$, $(\gamma^0)^2 = \mathbf{1}$),

$$\mathcal{P}\,\bar\psi(t,\mathbf{x})\,\mathcal{P}^{-1} = \eta_P^*\,\bar\psi(t,-\mathbf{x})\,\gamma^0. \tag{11.251}$$

Action on the spinors and on the ladder operators

To read off what $\mathcal{P}$ does to particles and antiparticles we need the action of the matrix γ^0 on the plane-wave spinors. Introduce the parity-reflected momentum $\tilde p \equiv (E_{\mathbf{p}}, -\mathbf{p})$. From $\gamma^0\not{p}\,\gamma^0 = \gamma^0(p_0\gamma^0 + p_i\gamma^i)\gamma^0 = p_0\gamma^0 - p_i\gamma^i = \not{\tilde p}$ we obtain, for any positive-energy solution $(\not{p} - m)u^s(p) = 0$,

$$(\not{\tilde p} - m)\,\gamma^0 u^s(p) = \gamma^0(\not{p} - m)\gamma^0\gamma^0 u^s(p) = \gamma^0(\not{p} - m)u^s(p) = 0, \tag{11.252}$$

so $\gamma^0 u^s(p)$ is a positive-energy solution at momentum $\tilde p$. In the rest frame $\mathbf{p} = 0$ one has $\tilde p = p$ and $\gamma^0 u^s(m) = +u^s(m)$ (the rest-frame spinors live in

the upper two components, the $+1$ eigenspace of γ^0); continuity in the boost fixes the proportionality constant to $+1$ for all $\mathbf{p}$. The same argument with $(\not{p}+m)v^s(p)=0$ and $\gamma^0 v^s(m)=-v^s(m)$ gives the opposite sign:

$$\gamma^0 u^s(p) = u^s(\tilde{p}), \qquad \gamma^0 v^s(p) = -v^s(\tilde{p}). \tag{11.253}$$

The relative minus sign between u and v is the algebraic seed of the opposite intrinsic parities of particle and antiparticle.

Insert Eq. (11.253) into the mode expansion Eq. (11.247). The right-hand side of Eq. (11.250) is

$$\eta_P\,\gamma^0\psi(t,-\mathbf{x}) = \eta_P \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}} \sum_s \Big(b^s_{\mathbf{p}}\, u^s(\tilde{p})\, e^{-i(E_{\mathbf{p}}t+\mathbf{p}\cdot\mathbf{x})} - d^{s\dagger}_{\mathbf{p}}\, v^s(\tilde{p})\, e^{+i(E_{\mathbf{p}}t+\mathbf{p}\cdot\mathbf{x})} \Big), \tag{11.254}$$

where we used $p\cdot(t,-\mathbf{x}) = E_{\mathbf{p}}t + \mathbf{p}\cdot\mathbf{x}$. Substituting the integration variable $\mathbf{p}\to-\mathbf{p}$ (the measure and $E_{\mathbf{p}}$ are invariant, $\tilde{p}\to p$, and the exponentials become $e^{\mp ip\cdot x}$) turns this into

$$\eta_P \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}} \sum_s \Big(b^s_{-\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} - d^{s\dagger}_{-\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big). \tag{11.255}$$

Matching, with the invariant measure $\widetilde{dp} \equiv d^3p/[(2\pi)^3\,2E_{\mathbf{p}}]$ of Eq. (9.211), against the left-hand side

$$\mathcal{P}\psi(t,\mathbf{x})\mathcal{P}^{-1} = \int \widetilde{dp} \sum_s \big((\mathcal{P} b^s_{\mathbf{p}} \mathcal{P}^{-1})\, u^s e^{-ip\cdot x} + (\mathcal{P} d^{s\dagger}_{\mathbf{p}} \mathcal{P}^{-1})\, v^s e^{+ip\cdot x} \big) \tag{11.256}$$

term by term gives the ladder-operator action

$$\mathcal{P}\, b^s_{\mathbf{p}}\, \mathcal{P}^{-1} = \eta_P\, b^s_{-\mathbf{p}}, \qquad \mathcal{P}\, d^s_{\mathbf{p}}\, \mathcal{P}^{-1} = -\eta_P^*\, d^s_{-\mathbf{p}}, \tag{11.257}$$

the second relation being the adjoint of the matched antiparticle term. Parity reverses the momentum and leaves the spin label s untouched (spin, being an axial vector, is invariant under a spatial reflection); consequently it flips the helicity $h = \hat{\mathbf{p}}\cdot\mathbf{S}$. Acting on the one-particle states $|e^-;\mathbf{p},s\rangle = b^{s\dagger}_{\mathbf{p}}|0\rangle$ and $|e^+;\mathbf{p},s\rangle = d^{s\dagger}_{\mathbf{p}}|0\rangle$ (with $\mathcal{P}|0\rangle = |0\rangle$),

$$\mathcal{P}\,|e^-;\mathbf{p},s\rangle = \eta_P^*\,|e^-;-\mathbf{p},s\rangle\,, \qquad \mathcal{P}\,\Big|e^+;\mathbf{p},s\Big\rangle = -\eta_P\,\Big|e^+;-\mathbf{p},s\Big\rangle\,. \tag{11.258}$$

Opposite intrinsic parity of particle and antiparticle

The phase η_P attached to a single field is not observable: it can be absorbed into the definition of $\mathcal{P}$. What *is* observable is a relative parity. From Eq. (11.257),

the intrinsic parity assigned to the electron is η_P and that assigned to the positron is $-\eta_P^*$. Their product,

$$\eta_P \cdot (-\eta_P^*) = -|\eta_P|^2 = -1, \tag{11.259}$$

is independent of the convention for η_P and is therefore physical: *a fermion and its antifermion carry opposite intrinsic parity.* This is not a curiosity. The relative parity of a particle–antiparticle pair shows up directly in the angular-momentum and parity analysis of bound states and decays — it is why the ground state of positronium (a ${}^1S_0$ e^+e^- state) is a pseudoscalar with $P=-1$, and the analogous statement for a quark–antiquark pair is what makes the pion a pseudoscalar. The minus sign in Eq. (11.259) traces back, through Eq. (11.253), to the single fact that u and v sit in opposite eigenspaces of γ^0 in the rest frame.

11.7.2 Charge conjugation

Charge conjugation C is the operation that exchanges particles with antiparticles without touching spacetime. Its spinor matrix is the charge-conjugation matrix $C = i\gamma^2\gamma^0$ of Theorem C.38, characterized abstractly by

$$C^{-1}\gamma^\mu C = -\gamma^{\mu T}, \qquad C^T = -C, \qquad C^\dagger = C^{-1}, \qquad C^2 = -\mathbf{1} \tag{11.260}$$

(Eqs. (C.80) and (C.94)). The charge-conjugate of a Dirac field is, as in Eq. (C.104),

$$\psi^c(x) \equiv C\,\bar\psi^{\,T}(x). \tag{11.261}$$

Invariance of the Dirac equation and the charge flip

That ψ^c again solves the free Dirac equation is the content of the v-spinor computation of Theorem C.40; we recall the mechanism because it is the same algebra that produces the bilinear table below. Start from $(i\partial\!\!\!/ - m)\psi = 0$, take the Dirac adjoint to get $\bar\psi(i\overleftarrow{\partial\!\!\!/} + m) = 0$, i.e. $i\partial_\mu\bar\psi\gamma^\mu + m\bar\psi = 0$, and transpose:

$$(i\gamma^{\mu T}\partial_\mu + m)\bar\psi^{\,T} = 0. \tag{11.262}$$

Now eliminate $\gamma^{\mu T} = -C^{-1}\gamma^\mu C$ from Eq. (11.260) and multiply on the left by C:

$$(-iC\,C^{-1}\gamma^\mu C\,\partial_\mu + mC)\bar\psi^{\,T} = (-i\gamma^\mu\partial_\mu + m)C\bar\psi^{\,T} = -\,(i\partial\!\!\!/ - m)\,\psi^c = 0. \tag{11.263}$$

So ψ^c is a solution of the free equation. The physical content appears the moment a gauge field is switched on. With the QED covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ acting on the electron (charge $-e$, $e>0$), the same manipulation, with $\partial_\mu \to D_\mu$,

sends $+ieA_\mu \to -ieA_\mu$: the conjugate field couples with the opposite charge. Equivalently, the electromagnetic current reverses,

$$j^\mu = \bar\psi\gamma^\mu\psi \xrightarrow{C} \bar\psi^c\gamma^\mu\psi^c = -\bar\psi\gamma^\mu\psi = -j^\mu, \tag{11.264}$$

a sign we rederive carefully (with the indispensable fermionic reordering) in the bilinear table of Section 11.7.4. This is exactly the statement that C turns a charge into its negative.

Action on the ladder operators

Promote Eq. (11.261) to the operator statement $\mathcal{C}\,\psi(x)\,\mathcal{C}^{-1} = \eta_C\, C\bar\psi^{\,T}(x)$ with $\mathcal{C}$ unitary and $|\eta_C| = 1$. The needed spinor inputs are $C\bar u^s(p)^T = v^s(p)$ and $C\bar v^s(p)^T = u^s(p)$, established in Theorem C.40. The conjugate adjoint field is $\bar\psi(x) = \int \widetilde{dp}\sum_s (b^{s\dagger}_{\mathbf p}\bar u^s e^{+ip\cdot x} + d^s_{\mathbf p}\bar v^s e^{-ip\cdot x})$, so

$$\begin{aligned}\eta_C\, C\bar\psi^{\,T}(x) &= \eta_C \int \widetilde{dp}\sum_s \Big(b^{s\dagger}_{\mathbf p}\, C\bar u^s(p)^T e^{+ip\cdot x} + d^s_{\mathbf p}\, C\bar v^s(p)^T e^{-ip\cdot x}\Big)\\ &= \eta_C \int \widetilde{dp}\sum_s \Big(d^s_{\mathbf p}\, u^s(p)\, e^{-ip\cdot x} + b^{s\dagger}_{\mathbf p}\, v^s(p)\, e^{+ip\cdot x}\Big),\end{aligned} \tag{11.265}$$

with $\widetilde{dp}$ the invariant measure introduced above. Comparing coefficients with $\mathcal{C}\psi\mathcal{C}^{-1}$ gives the clean exchange

$$\boxed{\mathcal{C}\, b^s_{\mathbf p}\,\mathcal{C}^{-1} = \eta_C\, d^s_{\mathbf p}, \qquad \mathcal{C}\, d^s_{\mathbf p}\,\mathcal{C}^{-1} = \eta_C^*\, b^s_{\mathbf p}} \tag{11.266}$$

(the second by taking the adjoint of the matched $b^\dagger$ term). Charge conjugation maps an electron of momentum $\mathbf p$ and spin s to a positron of the *same* momentum and spin, and vice versa: $\mathcal{C}\,|e^-;\mathbf p,s\rangle = \eta_C^*\,|e^+;\mathbf p,s\rangle$. Neither momentum nor spin is disturbed; only the sign of the charge is. The same matrix algebra, through $C\gamma^5 C^{-1} = \gamma^{5T}$ (Theorem C.39), is responsible for a fact we will use repeatedly from Chapter 36 onward: in the chiral basis $\gamma^{5T} = \gamma^5$, so charge conjugation commutes with chirality up to a transpose and therefore maps a left-handed antiparticle to a right-handed particle.

11.7.3 Time reversal

Time reversal is the spacetime map $x = (t,\mathbf x) \mapsto x_T \equiv (-t,\mathbf x)$. It differs in kind from P and C. Wigner's theorem, established in Section 3.7 (see Theorem 3.44), permits a symmetry of the quantum theory to be implemented by either a unitary or an antiunitary operator, and the requirement that time reversal preserve the spectrum of the Hamiltonian while reversing the sign of t in e^{-iHt} forces the antiunitary choice: $\mathcal{T}$ is *antiunitary*, $\mathcal{T}\,(\alpha A+\beta B)\,\mathcal{T}^{-1} = \alpha^*\,\mathcal{T}A\mathcal{T}^{-1} + \beta^*\,\mathcal{T}B\mathcal{T}^{-1}$, and in particular $\mathcal{T}\,i\,\mathcal{T}^{-1} = -i$. This single feature reshapes every computation in this subsection.

Fixing the matrix

Write $\mathcal{T}\,\psi(t,\mathbf{x})\,\mathcal{T}^{-1} = \eta_T\, T\,\psi(-t,\mathbf{x})$ with a constant matrix T and $|\eta_T| = 1$. Apply $\mathcal{T}$ to the Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi(x) = 0$. Antiunitarity conjugates the explicit i and the c-number matrices $\gamma^\mu \to \gamma^{\mu *}$, while the field becomes $T\psi(x_T)$:

$$(-i\gamma^{\mu *}\partial_\mu - m)T\psi(x_T) = 0. \tag{11.267}$$

Re-express the derivatives in the reflected coordinate x_T: $\partial_0 = -\partial_{x_T^0}$ and $\partial_i = \partial_{x_T^i}$. Multiplying by $-T^{-1}$ on the left, invariance of the Dirac equation requires

$$T^{-1}\gamma^0 T = \gamma^{0*}, \qquad T^{-1}\gamma^i T = -\gamma^{i*}, \tag{11.268}$$

which is exactly the matrix condition Eq. (C.109). In a representation where only γ^2 is imaginary ($\gamma^{0*} = \gamma^0$, $\gamma^{2*} = -\gamma^2$, $\gamma^{1*} = \gamma^1$, $\gamma^{3*} = \gamma^3$), Eq. (11.268) demands that T commute with γ^0, γ^2 and anticommute with γ^1, γ^3. The unique solution up to phase is

$$\boxed{\mathcal{T}\,\psi(t,\mathbf{x})\,\mathcal{T}^{-1} = \eta_T\, T\,\psi(-t,\mathbf{x}), \qquad T = \gamma^1\gamma^3} \tag{11.269}$$

(Eq. (C.110)), since $\gamma^1\gamma^3$ contains exactly the two matrices it must anticommute with and none of those it must commute with. It satisfies $T^2 = (\gamma^1\gamma^3)^2 = -(\gamma^1)^2(\gamma^3)^2 = -(-1)(-1) = -1$ and $T^\dagger = (\gamma^1\gamma^3)^\dagger = \gamma^{3\dagger}\gamma^{1\dagger} = (-\gamma^3)(-\gamma^1) = \gamma^3\gamma^1 = -\gamma^1\gamma^3 = -T$.

The spin flip and Kramers degeneracy

Time reversal must reverse momentum (a particle moving right, run backwards, moves left) and angular momentum (spin is T-odd). The momentum reversal is forced by the antiunitary conjugation of the plane-wave phase $e^{-ip\cdot x} \to e^{+ip\cdot x}$; the spin reversal is carried by the matrix $T = \gamma^1\gamma^3$. To see the spin flip cleanly, evaluate T on the conjugated rest-frame spinors. In the Dirac basis $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{smallmatrix}\right)$, a direct block multiplication gives

$$T = \gamma^1\gamma^3 = \begin{pmatrix} -\sigma^1\sigma^3 & 0 \\ 0 & -\sigma^1\sigma^3 \end{pmatrix} = \begin{pmatrix} i\sigma^2 & 0 \\ 0 & i\sigma^2 \end{pmatrix}, \tag{11.270}$$

using $\sigma^1\sigma^3 = -i\sigma^2$. The rest-frame spinors are $u^s(m) = \sqrt{2m}\,(\xi^s, 0)^T$ with $\xi^1 = \binom{1}{0}$, $\xi^2 = \binom{0}{1}$, and they are real, so $u^{s*}(m) = u^s(m)$. Since $i\sigma^2\xi^1 = -\xi^2$ and $i\sigma^2\xi^2 = +\xi^1$,

$$T\,u^{1*}(m) = -\,u^2(m), \qquad T\,u^{2*}(m) = +\,u^1(m), \tag{11.271}$$

and identically on the v-spinors. This is precisely the nonrelativistic time-reversal operator $i\sigma^2 K$ (with K complex conjugation) acting on a two-component spin:

it sends $\Uparrow\rangle \to -\Downarrow\rangle$, $\Downarrow\rangle \to +\Uparrow\rangle$. Boosting Eq. (11.271) and combining with the phase conjugation, the operator action is

$$\mathcal{T}\, b^{s}_{\mathbf{p}}\, \mathcal{T}^{-1} = \eta_T\, (-1)^{\frac{1}{2}-s}\, b^{-s}_{-\mathbf{p}}, \qquad \mathcal{T}\, d^{s}_{\mathbf{p}}\, \mathcal{T}^{-1} = \eta_T^{*}\, (-1)^{\frac{1}{2}-s}\, d^{-s}_{-\mathbf{p}}, \tag{11.272}$$

where $-s$ denotes the flipped spin and the alternating sign records the $i\sigma^2$ of Eq. (11.271) (here $s = \pm\frac{1}{2}$ labels the two states). Time reversal therefore reverses momentum and spin together, leaving helicity invariant. Because the matrix squares to -1, applying $\mathcal{T}$ twice to a single-fermion state returns -1 times the state:

$$\mathcal{T}^2\, |e^-; \mathbf{p}, s\rangle = -\,|e^-; \mathbf{p}, s\rangle\,. \tag{11.273}$$

This is *Kramers' theorem* in field-theoretic dress: any energy eigenstate of an odd number of fermions in a time-reversal-invariant theory is at least doubly degenerate, since $|\Psi\rangle$ and $\mathcal{T}\,|\Psi\rangle$ are orthogonal (an antiunitary squaring to -1 cannot have a one-dimensional invariant subspace). The sign $\mathcal{T}^2 = -1$ is the hallmark of half-integer spin, the mirror of the $\mathcal{T}^2 = +1$ that holds for bosons.

11.7.4 The transformation table of the bilinear covariants

We now compute how each of the sixteen bilinear covariants of Table 11.1 — the scalar $S = \bar\psi\psi$, the pseudoscalar $P = i\bar\psi\gamma^5\psi$, the vector $V^\mu = \bar\psi\gamma^\mu\psi$, the axial vector $A^\mu = \bar\psi\gamma^\mu\gamma^5\psi$, and the tensor $T^{\mu\nu} = \bar\psi\sigma^{\mu\nu}\psi$ — transforms under P, C, and T. Every entry is derived, not quoted. The three operations need three different master identities, and the differences between them are exactly the physics.

Parity

Combining the field transformations Eqs. (11.250) and (11.251), a generic bilinear transforms as

$$\mathcal{P}(\bar\psi\,\Gamma\,\psi)(t,\mathbf{x})\,\mathcal{P}^{-1} = \bar\psi(t,-\mathbf{x})\,(\gamma^0\Gamma\gamma^0)\,\psi(t,-\mathbf{x}), \tag{11.274}$$

the phases $\eta_P^*\eta_P = 1$ cancelling. Everything is fixed by $\gamma^0\Gamma\gamma^0$, computed with $\gamma^0\gamma^\mu\gamma^0 = \gamma_\mu$ and $\gamma^0\gamma^5\gamma^0 = -\gamma^5$:

$$\begin{aligned} \gamma^0\,\mathbf{1}\,\gamma^0 &= \mathbf{1}, & \gamma^0\gamma^\mu\gamma^0 &= \gamma_\mu, \\ \gamma^0\gamma^5\gamma^0 &= -\gamma^5, & \gamma^0\sigma^{\mu\nu}\gamma^0 &= \tfrac{i}{2}[\gamma_\mu,\gamma_\nu] = \sigma_{\mu\nu}, \\ \gamma^0\gamma^\mu\gamma^5\gamma^0 &= (\gamma^0\gamma^\mu\gamma^0)(\gamma^0\gamma^5\gamma^0) & &= -\gamma_\mu\gamma^5. \end{aligned} \tag{11.275}$$

Hence under parity (all fields at $(t,-\mathbf{x})$): $S \to +S$; $P = i\bar\psi\gamma^5\psi \to -P$; $V^\mu \to V_\mu = (V^0, -V^i)$; $A^\mu \to -A_\mu = (-A^0, A^i)$; $T^{\mu\nu} \to T_{\mu\nu}$. The scalar is even and the pseudoscalar odd, the vector flips its spatial part while the axial vector flips its temporal part — the textbook signatures that distinguish S from P and V from A, here obtained as pure γ-algebra.

Charge conjugation

This is the one case where the fermionic anticommutativity of the fields enters with a sign. From the operator statements $\mathcal{C}\psi\mathcal{C}^{-1} = \eta_C C\bar{\psi}^T$ and (its consequence, Section C.4.3) $\mathcal{C}\bar{\psi}\mathcal{C}^{-1} = -\eta_C^*\,\psi^T C^{-1}$,

$$\begin{aligned}\mathcal{C}(\bar{\psi}\Gamma\psi)\mathcal{C}^{-1} &= (-\psi^T C^{-1})\,\Gamma\,(C\bar{\psi}^T) = -\,\psi_\alpha\,(C^{-1}\Gamma C)_{\alpha\beta}\,\bar{\psi}_\beta \\ &= +\,\bar{\psi}_\beta\,(C^{-1}\Gamma C)_{\alpha\beta}\,\psi_\alpha = \bar{\psi}\,(C^{-1}\Gamma C)^T\,\psi, \end{aligned} \tag{11.276}$$

$|\eta_C|^2 = 1$. The crucial step is the second equality on the second line: interchanging the anticommuting operators ψ_α and $\bar{\psi}_\beta$ produces a sign -1 that cancels the explicit minus from $\mathcal{C}\bar{\psi}\mathcal{C}^{-1}$ (we work with the normal-ordered bilinear, dropping the c-number anticommutator). Using $C^T = -C$ and $(C^{-1})^T = -C^{-1}$ one finds $(C^{-1}\Gamma C)^T = C^T\Gamma^T(C^{-1})^T = C\,\Gamma^T\,C^{-1}$, so the master identity is

$$\mathcal{C}(\bar{\psi}\Gamma\psi)\mathcal{C}^{-1} = \bar{\psi}\,(C\,\Gamma^T C^{-1})\,\psi. \tag{11.277}$$

The conjugated matrices follow from $C\gamma^{\mu T}C^{-1} = -\gamma^\mu$ (from Eq. (11.260)), $C\gamma^{5T}C^{-1} = \gamma^5$ and $C\sigma^{\mu\nu T}C^{-1} = -\sigma^{\mu\nu}$ (Theorem C.39):

$$\begin{aligned} C\,\mathbf{1}^T C^{-1} &= \mathbf{1}, & C\gamma^{\mu T}C^{-1} &= -\gamma^\mu, \\ C\gamma^{5T}C^{-1} &= \gamma^5, & C\sigma^{\mu\nu T}C^{-1} &= -\sigma^{\mu\nu}, \\ C(\gamma^\mu\gamma^5)^T C^{-1} &= C\gamma^{5T}\gamma^{\mu T}C^{-1} & &= \gamma^5(-\gamma^\mu) = \gamma^\mu\gamma^5. \end{aligned} \tag{11.278}$$

Thus under charge conjugation: $S \to +S$, $P \to +P$, $V^\mu \to -V^\mu$, $A^\mu \to +A^\mu$, $T^{\mu\nu} \to -T^{\mu\nu}$. The vector current is odd, confirming Eq. (11.264); the axial current is even. These two signs underlie the structural facts that QED inherits: the odd C-parity of the electromagnetic current is the algebraic origin of *Furry's theorem* (amplitudes with an odd number of external photons attached to a single fermion loop vanish), which we deploy in Chapters 18 and 19.

Time reversal

Antiunitarity now conjugates the matrix Γ as well. With $\mathcal{T}\psi\mathcal{T}^{-1} = \eta_T T\psi(x_T)$ and $\mathcal{T}\bar{\psi}\mathcal{T}^{-1} = \eta_T^*\bar{\psi}(x_T)\,\gamma^0 T^\dagger\gamma^0$, and pulling the conjugated c-number matrix Γ^* through, one obtains (with $\gamma^0 T^\dagger\gamma^0 = -T$ and $T^{-1} = -T$, as computed from $T = \gamma^1\gamma^3$)

$$\mathcal{T}\,(\bar{\psi}\Gamma\psi)(t,\mathbf{x})\,\mathcal{T}^{-1} = \bar{\psi}(-t,\mathbf{x})\,(T^{-1}\Gamma^* T)\,\psi(-t,\mathbf{x}). \tag{11.279}$$

The matrix $T = \gamma^1\gamma^3$ is real, so conjugating the defining relation Eq. (11.268) gives the convenient form $T^{-1}\gamma^{\mu *}T = \gamma_\mu$. Then, recalling $\gamma^{5*} = \gamma^5$ and $T^{-1}\gamma^5 T = \gamma^5$ (as γ^5 commutes with $\gamma^1\gamma^3$),

$$T^{-1}\mathbf{1}^* T = \mathbf{1}, \qquad T^{-1}\gamma^{\mu *}T = \gamma_\mu, \qquad T^{-1}(i\gamma^5)^* T = -i\gamma^5,$$

$$T^{-1}(\gamma^\mu\gamma^5)^*T = \gamma_\mu\gamma^5, \qquad T^{-1}\sigma^{\mu\nu *}T = -\sigma_{\mu\nu}. \tag{11.280}$$

The conjugation of the explicit i in the pseudoscalar and tensor is what supplies their extra sign relative to parity. Under time reversal (all fields at $(-t,\mathbf{x})$): $S \to +S$; $P \to -P$; $V^\mu \to V_\mu = (V^0, -V^i)$; $A^\mu \to A_\mu = (A^0, -A^i)$; $T^{\mu\nu} \to -T_{\mu\nu}$. The vector transforms as it must for a current — charge density even, spatial current reversed — while the axial charge A^0 stays even and the axial spatial current reverses, consistent with T reversing spin.

The table, and the combined operations

Table 11.2 collects the entries. We adopt the compact notation $(-1)^\mu \equiv +1$ for $\mu = 0$ and -1 for $\mu = 1,2,3$, so that "$V^\mu \to V_\mu$" reads $V^\mu \to (-1)^\mu V^\mu$ and a rank-two object picks up $(-1)^\mu(-1)^\nu$.

Table 11.2: Transformation of the bilinear covariants under P, C, T, and the combination CPT. Position arguments: P sends fields to $(t,-\mathbf{x})$, T to $(-t,\mathbf{x})$, and CPT to $-x = (-t,-\mathbf{x})$; C leaves the argument fixed. Here $(-1)^\mu = +1$ for $\mu = 0$ and -1 for $\mu = 1,2,3$. The final column shows that a bilinear with n Lorentz (vector/tensor) indices acquires $(-1)^n$ under CPT, evaluated at $-x$.

bilinear	P	C	T	CPT	# indices n
$S = \bar\psi\psi$	$+S$	$+S$	$+S$	$+S$	0
$P = i\bar\psi\gamma^5\psi$	$-P$	$+P$	$-P$	$+P$	0
$V^\mu = \bar\psi\gamma^\mu\psi$	$(-1)^\mu V^\mu$	$-V^\mu$	$(-1)^\mu V^\mu$	$-V^\mu$	1
$A^\mu = \bar\psi\gamma^\mu\gamma^5\psi$	$-(-1)^\mu A^\mu$	$+A^\mu$	$(-1)^\mu A^\mu$	$-A^\mu$	1
$T^{\mu\nu} = \bar\psi\sigma^{\mu\nu}\psi$	$(-1)^\mu(-1)^\nu T^{\mu\nu}$	$-T^{\mu\nu}$	$-(-1)^\mu(-1)^\nu T^{\mu\nu}$	$+T^{\mu\nu}$	2

The CPT column is the product of the other three (for each fixed component, multiplying the three signs). Take the vector: P contributes $(-1)^\mu$, C contributes -1, T contributes $(-1)^\mu$, and the product is $(-1)^\mu\cdot(-1)\cdot(-1)^\mu = -1$, independent of μ; the spacetime argument, reflected by P and again by T, ends at $-x$. The axial vector gives $-(-1)^\mu\cdot(+1)\cdot(-1)^\mu = -1$; the tensor gives $(-1)^\mu(-1)^\nu\cdot(-1)\cdot(-(-1)^\mu(-1)^\nu) = +1$; the scalar and pseudoscalar both give $+1$. The pattern is uniform and is the entire algebraic content of the CPT theorem at the level of bilinears:

$$\boxed{\Theta\left[\bar\psi\,\Gamma_{(n)}\,\psi\,(x)\right]\Theta^{-1} = (-1)^n\,\bar\psi\,\Gamma_{(n)}\,\psi\,(-x), \qquad \Theta = \mathcal{CPT},} \tag{11.281}$$

where $\Gamma_{(n)}$ is any of the covariants carrying n Lorentz vector indices. A Lorentz tensor of rank n flips sign n times and is evaluated at the totally reflected point.

11.7.5 Invariance of the Dirac Lagrangian, and the CPT theorem

We can now verify that the free Dirac Lagrangian $\mathcal{L} = \bar{\psi}(i\gamma^\mu\partial_\mu - m)\psi$ is invariant under each of C, P, T separately, and then assemble CPT.

Parity. The mass term $-m\,S$ is a scalar, invariant. The kinetic term is $i\bar{\psi}\gamma^\mu\partial_\mu\psi$; under P the bilinear factor goes as $V^\mu \to V_\mu$ while the derivative goes as $\partial_\mu \to \partial^\mu$ at $(t, -\mathbf{x})$, and the contraction $\gamma^\mu\partial_\mu \to \gamma_\mu\partial^\mu = \gamma^\mu\partial_\mu$ is restored. Hence $\mathcal{P}\mathcal{L}(t,\mathbf{x})\mathcal{P}^{-1} = \mathcal{L}(t,-\mathbf{x})$, and $\int d^4x\,\mathcal{L}$ is invariant.

Charge conjugation. The mass term is even. For the kinetic term, the field rules and the fermionic reordering give, after one integration by parts,

$$\begin{aligned}\mathcal{C}(i\bar{\psi}\gamma^\mu\partial_\mu\psi)\mathcal{C}^{-1} &= i(-\psi^T C^{-1})\gamma^\mu\partial_\mu(C\bar{\psi}^T) = -i\,\psi^T(C^{-1}\gamma^\mu C)\,\partial_\mu\bar{\psi}^T \\ &= +i\,\psi^T\gamma^{\mu T}\partial_\mu\bar{\psi}^T = -\,i\,\partial_\mu\bar{\psi}\,\gamma^\mu\psi = i\bar{\psi}\gamma^\mu\partial_\mu\psi - i\,\partial_\mu(\bar{\psi}\gamma^\mu\psi),\end{aligned} \tag{11.282}$$

where the third equality used $C^{-1}\gamma^\mu C = -\gamma^{\mu T}$, the fourth the anticommutativity of the fields (a sign -1 from $\psi_\alpha\,\partial_\mu\bar{\psi}_\beta = -\partial_\mu\bar{\psi}_\beta\,\psi_\alpha$), and the last an integration by parts. The kinetic term returns up to the total derivative $-i\,\partial_\mu j^\mu$, so the *action* $\int d^4x\,\mathcal{L}$ is C-invariant.

Time reversal. The mass term is even. The kinetic term carries three time-reversal-sensitive factors: the explicit i (conjugated to $-i$ by antiunitarity), the bilinear $V^\mu \to V_\mu$, and the derivative ($\partial_0 \to -\partial_0$, $\partial_i \to \partial_i$ at $(-t,\mathbf{x})$). For the temporal piece, $i\bar{\psi}\gamma^0\partial_0\psi \to (-i)\,(V^0 \text{ even})\,(-\partial_0) = +i\bar{\psi}\gamma^0\partial_0\psi$; for the spatial piece, $i\bar{\psi}\gamma^i\partial_i\psi \to (-i)\,(V^i \to -V^i)\,(+\partial_i) = +i\bar{\psi}\gamma^i\partial_i\psi$. Each is invariant, so $\mathcal{T}\mathcal{L}(t,\mathbf{x})\mathcal{T}^{-1} = \mathcal{L}(-t,\mathbf{x})$ and the action is T-invariant.

CPT. Combining the three, the free Dirac Lagrangian density obeys $\Theta\,\mathcal{L}(x)\,\Theta^{-1} = \mathcal{L}(-x)$, and since the Jacobian of $x \to -x$ is $+1$ in four dimensions, the action is invariant. The general statement is the celebrated theorem.

Theorem 11.51 (CPT, stated)**.** *Let a quantum field theory be (i) invariant under the proper orthochronous Poincaré group, (ii) local — its fields obey microcausality with the spin–statistics connection of Theorem 9.33, integer-spin fields commuting and half-integer-spin fields anticommuting at spacelike separation — and (iii) governed by a Hermitian Lagrangian density built as a Lorentz scalar from the fields and their derivatives. Then there exists an antiunitary operator $\Theta = \mathcal{C}\mathcal{P}\mathcal{T}$, an exact symmetry of the theory, under which every Hermitian local Lorentz-scalar operator transforms as $\Theta\,\mathcal{O}(x)\,\Theta^{-1} = \mathcal{O}(-x)$.*

The structural reason is laid bare by Eq. (11.281). Any Hermitian Lorentz-scalar interaction is a sum of monomials in which Dirac bilinears, derivatives, and other tensor fields are contracted so that *every* Lorentz index is summed against another. Under Θ, each bilinear of rank n contributes $(-1)^n$ and each derivative $\partial_\mu \to -\partial_\mu$ contributes one more sign per index; in a fully contracted scalar these signs come in pairs and cancel, $(-1)^{\text{even}} = +1$. The antiunitarity of Θ conjugates the explicit factors of i, which is exactly what Hermiticity of the Lagrangian requires for the surviving phases to combine to $+1$ rather than -1. The net effect on any scalar term is the identity up to the reflection $x \to -x$. This is the heart of the matter, and it is genuinely independent of the details of the theory; the only inputs are Lorentz covariance, locality with the right statistics, and Hermiticity — precisely the three hypotheses. A fully rigorous proof, valid beyond the perturbative/Lagrangian setting, rests on the analytic structure of the Wightman functions (the same forward-tube analyticity that powers the general spin–statistics theorem of Theorem 9.33); we have given the complete structural argument here and refer to the axiomatic treatment for the analyticity machinery, which the book develops where it is first needed rather than at this point.

Physical Insight

CPT is the symmetry that survives even when C, P, and T individually fail. The weak interaction violates P and C *maximally* (it couples only to left-handed fermions and right-handed antifermions) and violates the combination CP — and hence, by Theorem 11.51, T — by a small amount; these are the subjects of Chapters 31 and 32. Yet CPT remains exact, because its three hypotheses (Lorentz invariance, locality, spin–statistics) are the non-negotiable scaffolding of relativistic quantum field theory itself. The empirical consequences are sharp and have been tested to extraordinary precision: a particle and its antiparticle have *exactly* equal mass and lifetime, and equal-magnitude opposite charge and magnetic moment. The most stringent single test as of the mid-2020s is the fractional mass difference of K^0 and $\bar{K}^0$, bounded below the 10^{-18} level, together with proton/antiproton charge-to-mass and electron/positron g-factor comparisons at the part-per-billion-and-finer level; no violation has been observed.

Figure 11.6 traces a single electron state through the three operations and their product, making visible the content of the table: P reverses momentum, C flips the charge, T reverses momentum and spin, and the combined Θ maps the electron to a positron of the same momentum and opposite spin — an antiparticle, as the theorem requires.

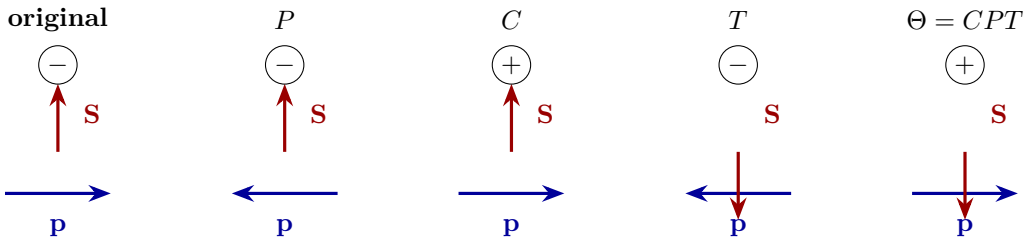


Figure 11.6: Action of P, C, T, and $\Theta = CPT$ on an electron prepared with momentum $\mathbf{p}$ to the right, spin $\mathbf{S}$ up, and charge $-e$ (blue: momentum; red: spin; circle: charge). P reverses $\mathbf{p}$ and preserves the axial $\mathbf{S}$; C flips the charge; T reverses both $\mathbf{p}$ and $\mathbf{S}$. The composite Θ produces a positron ($+e$) of the same momentum and reversed spin — the antiparticle, in accordance with Theorem 11.51.

11.7.6 The Majorana condition: a fermion that is its own antiparticle

The charge-conjugation operation invites a question it does not by itself answer: can a Dirac field be *equal* to its own charge conjugate? Such a field would describe a neutral fermion with no distinction between particle and antiparticle. The condition, in the notation of Theorem C.41, is

$$\psi = \psi^c = C\bar{\psi}^T. \tag{11.283}$$

That Eq. (11.283) is a consistent, Lorentz-covariant constraint in four dimensions — that the antilinear map $\psi \mapsto C\bar{\psi}^T$ squares to the identity, so the constraint has a nontrivial solution space of four real components, the count of a single Weyl spinor — was proved in Theorem C.42. Here we examine what the condition does to the quantized field and its quanta.

Worked Example 11.52 (Construction of a Majorana field and its self-conjugacy)**.** We build a quantized Majorana field directly from a Dirac field by enforcing Eq. (11.283) on the mode operators, and verify the self-conjugacy condition. Begin with the general expansion Eq. (11.247) and apply charge conjugation through the operator rule Eq. (11.266). The charge conjugate of the field is

$$\psi^c(x) = \mathcal{C}\,\psi(x)\,\mathcal{C}^{-1}\big|_{\eta_C=1} = \int \widetilde{dp} \sum_s \Big(d^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big), \tag{11.284}$$

obtained from ψ by the interchange $b^s_{\mathbf{p}} \leftrightarrow d^s_{\mathbf{p}}$ (using $C\bar{u}^T = v$, $C\bar{v}^T = u$ exactly as in Eq. (11.266)). Comparing term by term with ψ itself, the equality $\psi = \psi^c$ holds as an operator identity *iff*

$$d^s_{\mathbf{p}} = b^s_{\mathbf{p}} \qquad \text{for all } \mathbf{p}, s. \tag{11.285}$$

That is, the Majorana condition identifies the antiparticle ladder operators with the particle ladder operators: there is only one species of quantum. The Majorana field then reads

$$\psi_M(x) = \int \widetilde{dp} \sum_s \Big(b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + b^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big), \tag{11.286}$$

with the single ladder algebra $\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\} = (2\pi)^3 \delta^{rs} \delta^{(3)}(\mathbf{p}-\mathbf{q})$. We confirm self-conjugacy at the level of the field: applying $\mathcal{C}$ to Eq. (11.286) sends $b \to d = b$ by Eqs. (11.266) and (11.285), returning Eq. (11.286) unchanged, so $\mathcal{C}\psi_M\mathcal{C}^{-1} = \psi_M$ and the state space carries C-eigenvalue structure with no separate antiparticle sector. A Majorana fermion is electrically neutral: the would-be current $\bar\psi_M \gamma^\mu \psi_M$ is odd under C (Table 11.2) yet equal to its own conjugate, hence identically zero as a U(1) charge — a Majorana field cannot couple to the photon. The degrees of freedom have halved exactly as Theorem C.42 predicted: where a Dirac field has two creation operators per momentum-spin cell ($b^\dagger$ and $d^\dagger$), the Majorana field has one. In the chiral basis Eq. (11.283) reduces to $\psi_R = i\sigma^2 \psi_L^*$ (Theorem C.43), exhibiting the four real components as those of a single left-handed Weyl spinor tied to its own conjugate.

A Majorana mass term $-\frac{1}{2} m\, \bar\psi_M \psi_M = -\frac{1}{2} m\, (\psi_M^T C\, \psi_M + \text{h.c.})$ is built from the antisymmetric combination $\psi^T C \psi$, whose nonvanishing relies on $C^T = -C$ together with the anticommutativity of the field components (Table C.1); it violates any U(1) charge under which ψ rotates, by two units, which is exactly why it is forbidden for the charged leptons and quarks and available only for genuinely neutral fermions.

Physical Insight: a dated frontier — are neutrinos Dirac or Majorana?

The only known fundamental fermion that could be its own antiparticle is the neutrino, the unique electrically neutral elementary fermion. Whether the neutrino mass is of Dirac type (requiring a right-handed partner and conserving lepton number) or Majorana type (Eq. (11.283), violating lepton number by two units) is, as of the mid-2020s, *experimentally open*. The decisive observable is neutrinoless double-beta decay ($0\nu\beta\beta$), a nuclear transition $(A, Z) \to (A, Z+2) + 2e^-$ with no emitted neutrinos, which can proceed only if the neutrino is a Majorana particle. No $0\nu\beta\beta$ event has been observed; the leading experiments (KamLAND-Zen, GERDA/LEGEND, CUORE, and the next generation including nEXO) currently set lower bounds on the half-life in the 10^{26}-year range and are pushing toward the parameter region suggested by the inverted-ordering neutrino masses. A positive detection would establish the Majorana nature of the neutrino and lepton-number violation; continued null results progressively constrain

it. We flag this honestly as a live question, not a settled fact, and return to the neutrino mass mechanisms in Chapters 31 and 32.

The discrete-symmetry technology assembled here — the operators $\mathcal{P}, \mathcal{C}, \mathcal{T}$ and their action on fields, states, and bilinears, the table of Table 11.2, and the CPT theorem — is exactly the apparatus the rest of the book draws on when these symmetries become constraints on dynamics: the C-oddness of the electromagnetic current that yields Furry's theorem and organizes the Ward identities of Chapters 18 and 19, the maximal P- and C-violation and the small CP-violation of the electroweak and flavor sectors in Chapters 31 and 32, and the interplay of charge conjugation and chirality that surfaces in the anomaly analysis of Chapter 36. With the discrete symmetries in hand and the Dirac propagator already built, the quantum theory of the free electron is complete.

Further reading

The Dirac equation, its plane-wave solutions, and the spinor technology used here are developed in Peskin and Schroeder [396] ch. 3 and Schwartz [432] chs. 10–11, both of which adopt conventions close to ours; Srednicki [454] chs. 33–43 reorganizes the same material around Weyl spinors and is the cleanest place to see the chiral structure. The forcing of anticommutators by positivity of the energy, and the spin–statistics connection at the level of the free field, are treated carefully in Weinberg [500] ch. 5, and the historical foundation is Pauli [387]; the original equation is of course Dirac [150]. For the axiomatic spin–statistics and CPT theorems in full generality — the analyticity machinery this chapter quotes but does not reproduce — the standard reference remains Streater and Wightman [458]. Zee [532] pt. II and Itzykson and Zuber [278] chs. 3–4 give complementary physical and computational perspectives on the propagator and the discrete symmetries.

Chapter 12

Fermionic Path Integrals

By the end of Chapter 11 the Dirac field stood on the same footing the scalar field had reached in Chapter 9: canonically quantized, with a Fock space built on anticommutators, a Pauli principle, and a time-ordered propagator S_F whose every minus sign is dictated by Fermi statistics. What the electron still lacked was the *other* description of a quantum field—the sum over histories. Chapter 10 earned that description honestly for the scalar, deriving the generating functional $Z[J]$ of Eq. (10.86) from canonical quantization by slicing the evolution operator and inserting a continuous basis at each step. This chapter pays the same debt for the Dirac field, and the entire difficulty is encoded in a single sign. A path integral as built in Chapter 10 sums e^{iS} over ordinary commuting numbers, and a commuting number can never carry the antisymmetry that the fermion propagator demands; the formalism must be rebuilt over a new species of number that anticommutes. Constructing that number system, defining integration on it, and showing that the result reproduces—number for number, sign for sign—the canonical theory of Chapter 11 is the whole of the chapter.

The opening section confronts the obstruction directly: it proves that any integral over commuting field configurations yields a *symmetric* two-point function and an all-plus (permanent) Wick expansion, in flat contradiction with the antisymmetric, determinantal correlators the Dirac field requires, and from that failure it extracts the precise specification the eventual integral must meet. The next two sections build the machinery from nothing. We introduce the Grassmann algebra of anticommuting numbers, its $\mathbb{Z}_2$ grading, and a differential calculus whose canonical relation $\{\partial_i, \theta_j\} = \delta_{ij}$ mirrors the fermionic oscillator algebra of Chapter 11; then we define Berezin integration, fixed uniquely by translation invariance, discover that it coincides with differentiation, and prove the one structural fact that distinguishes it from every ordinary integral—a linear change of variables transforms the measure by the *inverse* of the bosonic Jacobian. That single inversion is the seed of everything fermionic.

With the calculus in hand the chapter computes its central object, the

Gaussian Grassmann integral, and finds the determinant in the numerator, $\int \mathcal{D}\bar{\theta}\,\mathcal{D}\theta\, e^{-\bar{\theta}A\theta} = \det A$, the exact inverse of the bosonic $(\det A)^{-1/2}$ of Chapter 4; the same calculation delivers the Pfaffian for real (Majorana) fields, the sourced generating identity, and a Wick theorem that is a signed sum—a determinant, not a permanent—over contractions. We then close the gap between formalism and physics. Following the coherent-state route of Chapter 10, we show that the eigenvalues of a fermionic annihilation operator are forced to be Grassmann numbers, build the coherent-state resolution of the identity and the trace formula whose extra minus sign is the algebraic face of Fermi statistics, and time-slice the evolution operator to *derive*—not postulate—the Dirac path integral $\int \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\, e^{iS}$, with antiperiodic boundary conditions in imaginary time emerging as the path-integral form of spin and statistics. The penultimate section couples Grassmann sources to write the Dirac generating functional in closed form, recovers the momentum-space propagator Eq. (11.27) of QED from the path integral alone (using the Clifford algebra of Chapter C), and watches every fermionic minus sign fall out of the anticommutation of the sources rather than being inserted by hand.

The final section reads the one object the source differentiation never touched—the normalization $Z[0] = \det(i\partial\!\!\!/ - m)$—as a physical quantity. Through the identity $\det = \exp \mathrm{Tr} \ln$ it becomes the one-loop fermionic effective action, expands into a sum of closed fermion loops, and yields the most consequential bookkeeping rule of perturbation theory: the factor of -1 on every closed fermion loop, derived twice (from the numerator placement of the determinant and from the antiperiodic trace) and shown to agree. The analytic control of these infinite-dimensional determinants—zeta-function and heat-kernel regularization—is imported from Chapter E, where it is built in full. Throughout we work in the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ with $\hbar = c = 1$, and the Wick rotation $t \to -i\tau$ fixed in Section 10.6. What the chapter hands forward is the complete kit of free fermionic Feynman rules—the propagator S_F, the contraction signs, and the loop factor—ready for the interacting theory of Chapters 15 and 17, the radiative corrections of Chapter 19, the ghosts of Chapter 26, the chiral anomaly of Chapter 36, and the lattice of Chapter 43.

12.1 Why fermions need anticommuting integration variables

The two pillars the reader now holds point in incompatible directions, and the first task of this chapter is to look squarely at the incompatibility before resolving it. From Chapter 10 we have a working path integral for a bosonic field: the generating functional

$$Z[J] = \int \mathcal{D}\phi\, \exp\Big[i \int d^4x\, \big(\mathcal{L}[\phi] + J\phi\big)\Big] \tag{12.1}$$

of Eq. (10.86), whose functional derivatives are the time-ordered correlators (Theorem 10.19) and whose free Gaussian limit reproduced, number for number, the canonical Feynman propagator. From Chapter 11 we have a canonically quantized Dirac field: a mode expansion built on *anti*commutators (Section 11.3), a Fock space on which the exclusion principle is the theorem $\left(b_{\mathbf{p}}^{s\dagger}\right)^2 = 0$ (Eq. (11.135)), and a time-ordered two-point function S_F carrying a minus sign forced by Fermi statistics. The natural ambition is to give the Dirac field the same path-integral description the scalar already enjoys. The obstruction is that the path integral, as built in Chapter 10, sums e^{iS} over *ordinary numbers*, and an ordinary number cannot carry the minus sign that the Dirac field's anticommutators demand. This section makes that statement precise: it shows in full what fails if one tries to integrate over commuting field configurations, and from the wreckage it extracts the exact list of properties the eventual fermionic integral must have. Nothing here is the solution; everything here is the specification the solution must meet.

12.1.1 The path integral is an integral over numbers

It is worth being completely explicit about what kind of object $\phi(x)$ is inside Eq. (12.1). The measure $\mathcal{D}\phi$ was *defined* in Theorem 10.16 as a limit of finite products of ordinary one-dimensional integrals over the field values ϕ_n at the sites n of a spacetime lattice,

$$\int \mathcal{D}\phi\,(\cdots) \;\equiv\; \lim_{a\to 0}\Big(\prod_{n}\mathcal{N}\int_{-\infty}^{\infty} d\phi_n\Big)\,(\cdots), \tag{12.2}$$

each $\phi_n \in \mathbb{R}$ a real integration variable ranging over the real line. The continuum field configuration $\phi(x)$ over which we "sum histories" is therefore, at every stage of the definition, a collection of ordinary real numbers. The single algebraic fact this entails is the only fact we shall need:

$$\phi(x)\,\phi(y) \;=\; \phi(y)\,\phi(x) \qquad \text{for every } x, y, \tag{12.3}$$

because real numbers commute. Equation Eq. (12.3) is not a dynamical statement and has nothing to do with the action or the equations of motion; it is built into the meaning of Eq. (12.2).

Consider now the time-ordered two-point function as the path integral computes it,

$$G^{(2)}(x,y) \;=\; \frac{1}{Z[0]}\int \mathcal{D}\phi\;\phi(x)\,\phi(y)\;e^{iS[\phi]}, \tag{12.4}$$

the construction Eq. (10.97) that reproduced the scalar Feynman propagator $D_F(x-y)$ of Theorem 9.24. Interchange the two insertions $x \leftrightarrow y$ in Eq. (12.4). The action $S[\phi]$, the measure $\mathcal{D}\phi$, and the normalization $Z[0]$ are untouched by relabelling the two points at which we evaluate the field; only the integrand

factor changes, from $\phi(x)\phi(y)$ to $\phi(y)\phi(x)$. By Eq. (12.3) these are the *same* integrand. Therefore

$$G^{(2)}(x,y) = G^{(2)}(y,x), \tag{12.5}$$

the two-point function produced by a path integral over commuting numbers is symmetric under exchange of its two arguments. For the scalar this is exactly correct and is no accident: the canonical $D_F(x-y) = D_F(y-x)$ is symmetric (the bosonic time-ordering symbol of Eq. (9.151) is symmetric, $\mathcal{T}\phi(x)\phi(y) = \mathcal{T}\phi(y)\phi(x)$, because $[\phi(x),\phi(y)] = 0$). The path integral and the canonical theory agree because both encode the same Bose symmetry. The question that organizes the rest of this section is whether the same machinery can encode the *opposite* symmetry that the Dirac field demands.

12.1.2 A commuting Gaussian can only make symmetric two-point functions

The symmetry Eq. (12.5) was read off the interacting integral by relabelling, but its root is in the free Gaussian, and it is the Gaussian that the fermionic theory must eventually imitate. We therefore isolate the obstruction at the level of a finite-dimensional Gaussian, where every step is an elementary integral and nothing is hidden in a continuum limit. This is the honest core of the no-go statement.

Proposition 12.1 (Commuting Gaussians have symmetric covariance)**.** *Let $\theta = (\theta_1, \dots, \theta_n)$ be ordinary real integration variables and let A be a real symmetric positive-definite $n \times n$ matrix. With the Gaussian average $\langle \cdot \rangle$ of Eq. (4.57) relative to the weight $e^{-\frac{1}{2}\theta^{\mathsf{T}} A \theta}$, the two-point function is*

$$\langle \theta_i \, \theta_j \rangle \;=\; \left(A^{-1}\right)_{ij}, \tag{12.6}$$

and it is symmetric, $\langle \theta_i \theta_j \rangle = \langle \theta_j \theta_i \rangle$, for every *choice of the matrix A in the weight. Moreover the integral depends only on the symmetric part of any matrix placed in the quadratic form.*

Proof. The covariance Eq. (12.6) is the second moment read from the normalized generating function Eq. (4.58), $Z(J)/Z(0) = \exp\!\left(\tfrac{1}{2} J^{\mathsf{T}} A^{-1} J\right)$, by differentiating twice in the sources, $\langle \theta_i \theta_j \rangle = \partial_{J_i} \partial_{J_j} \exp\!\left(\tfrac{1}{2} J^{\mathsf{T}} A^{-1} J\right)\big|_{J=0} = (A^{-1})_{ij}$. Symmetry of Eq. (12.6) is immediate from two independent facts, and it is instructive to see that they agree. On one hand A^{-1} is the inverse of a symmetric matrix and is therefore symmetric, $(A^{-1})_{ij} = (A^{-1})_{ji}$. On the other hand, and more fundamentally, the integrand $\theta_i \theta_j$ in

$$\langle \theta_i \theta_j \rangle = \frac{\int d^n\theta \; \theta_i \theta_j \, e^{-\frac{1}{2}\theta^{\mathsf{T}} A \theta}}{\int d^n\theta \; e^{-\frac{1}{2}\theta^{\mathsf{T}} A \theta}} \tag{12.7}$$

satisfies $\theta_i\theta_j = \theta_j\theta_i$ because the θ's are ordinary numbers, so the numerator is unchanged under $i \leftrightarrow j$ while the denominator does not involve i, j at all; hence $\langle\theta_i\theta_j\rangle = \langle\theta_j\theta_i\rangle$ regardless of what A is. The last claim follows from the same commutativity: in the exponent

$$\theta^{\mathsf{T}} A\theta = \sum_{i,j} \theta_i A_{ij} \theta_j = \sum_{i,j} \theta_i \, \tfrac{1}{2}(A_{ij} + A_{ji})\, \theta_j = \theta^{\mathsf{T}} A_{\mathrm{sym}}\, \theta, \tag{12.8}$$

where the middle step relabels $i \leftrightarrow j$ in half of the double sum and uses $\theta_i\theta_j = \theta_j\theta_i$, so the antisymmetric part $A_{\mathrm{anti}} = \frac{1}{2}(A - A^{\mathsf{T}})$ contributes nothing to the integral. A commuting Gaussian is blind to the antisymmetric part of its kinetic matrix. □

The point of Theorem 12.1 is sharper than the bare statement that the answer happens to be symmetric. Even if one were to *try* to force antisymmetry by hand—placing an antisymmetric matrix A_{anti} in the exponent and hoping the integral would return A_{anti}^{-1}, which is antisymmetric—Eq. (12.8) shows the attempt collapses before it begins: a purely antisymmetric quadratic form $\theta^{\mathsf{T}} A_{\mathrm{anti}}\theta$ is identically zero, the "Gaussian" has no exponent at all, and the integral diverges. Commuting variables cannot even *register* an antisymmetric kinetic operator, let alone invert it. A two-variable instance makes this concrete and leaves no room for doubt.

Worked Example 12.2 (The two-variable commuting Gaussian is symmetric, term by term)**.** Take $n = 2$ and the symmetric weight matrix $A = \left(\begin{smallmatrix} a & b \\ b & c \end{smallmatrix}\right)$ with $a, c > 0$ and $ac - b^2 > 0$, so that $A > 0$. The normalization is

$$Z = \int_{-\infty}^{\infty}\!\!\int_{-\infty}^{\infty} d\theta_1\, d\theta_2 \;\exp\Big[-\tfrac{1}{2}\big(a\theta_1^2 + 2b\,\theta_1\theta_2 + c\,\theta_2^2\big)\Big] = \frac{2\pi}{\sqrt{ac - b^2}} = \frac{2\pi}{\sqrt{\det A}}, \tag{12.9}$$

the $n = 2$ case of Eq. (4.50), the cross term $2b\,\theta_1\theta_2$ coming from $\theta^{\mathsf{T}} A\theta$ precisely because $\theta_1\theta_2 = \theta_2\theta_1$ merges the two off-diagonal entries into one. The off-diagonal correlator is obtained by completing the square or, fastest, from $A^{-1} = \frac{1}{ac-b^2}\left(\begin{smallmatrix} c & -b \\ -b & a \end{smallmatrix}\right)$ through Eq. (12.6):

$$\langle\theta_1\theta_2\rangle = (A^{-1})_{12} = \frac{-b}{ac - b^2}, \qquad \langle\theta_2\theta_1\rangle = (A^{-1})_{21} = \frac{-b}{ac - b^2}. \tag{12.10}$$

The two are equal, as they must be, since the integrand $\theta_1\theta_2$ in the defining ratio Eq. (12.7) is literally the same number as $\theta_2\theta_1$. There is no value of a, b, c, and no completing-the-square manoeuvre, that makes $\langle\theta_1\theta_2\rangle = -\langle\theta_2\theta_1\rangle$: an antisymmetric off-diagonal correlator is unreachable by integrating commuting variables. To get one, the integrand itself would have to change sign under $\theta_1 \leftrightarrow \theta_2$, which forces $\theta_1\theta_2 = -\theta_2\theta_1$ on the integration variables. That is the conclusion this section is driving toward; here it is already visible in two real integrals.

Set against this the canonical result of Chapter 11. The Dirac–Feynman propagator $S_F(x-y)_{ab} = \langle 0|\,\mathcal{T}\psi_a(x)\bar{\psi}_b(y)\,|0\rangle$ of Eq. (11.189) inherits its structure from the fermionic time-ordering symbol Eq. (11.187),

$$\mathcal{T}\,\psi(x)\bar{\psi}(y) \;\equiv\; \theta(x^0-y^0)\,\psi(x)\bar{\psi}(y) \;-\; \theta(y^0-x^0)\,\bar{\psi}(y)\psi(x), \tag{12.11}$$

in which the second ordering enters with a *minus sign*, the price of moving one fermionic operator past another. Applying the same definition to the opposite operator product $\mathcal{T}\,\bar{\psi}(y)\psi(x)$—later field to the left, minus for the swap of two fermionic operators—gives $\theta(y^0-x^0)\,\bar{\psi}(y)\psi(x) - \theta(x^0-y^0)\,\psi(x)\bar{\psi}(y)$, which is exactly the negative of Eq. (12.11). Taking the vacuum expectation of both, the time-ordered fermion two-point function obeys

$$\boxed{\;\langle 0|\,\mathcal{T}\,\psi_a(x)\,\bar{\psi}_b(y)\,|0\rangle \;=\; -\,\langle 0|\,\mathcal{T}\,\bar{\psi}_b(y)\,\psi_a(x)\,|0\rangle\;.\;} \tag{12.12}$$

This is the no-go observation in one line. The canonical object the path integral is supposed to reproduce is *antisymmetric* under interchange of its two field operators, whereas Theorem 12.1 proved that any integral over commuting field configurations produces a *symmetric* two-point function, Eq. (12.5). A Berezin-free path integral over commuting $\psi(x)$ cannot reproduce Eq. (12.12): the integrand $\psi_a(x)\bar{\psi}_b(y)$ equals $\bar{\psi}_b(y)\psi_a(x)$ for c-number fields, so the integral hands back $+$ where the canonical theory demands $-$. The sign in Eq. (12.12) is the entire content of Fermi statistics at the level of the propagator, and it is precisely the thing commuting numbers are constitutionally unable to supply.

12.1.3 The Pauli sign in multi-fermion amplitudes

The two-point obstruction is decisive on its own, but the same minus sign reappears, magnified, the moment more than one fermion is present, and it is worth exhibiting there because that is where it is physically most consequential and where it later controls the sign of every closed fermion loop. Consider the four-point function of two electron fields and two adjoint fields,

$$G^{(4)}_{a_1a_2,b_1b_2}(x_1,x_2;y_1,y_2) = \langle 0|\,\mathcal{T}\,\psi_{a_1}(x_1)\,\psi_{a_2}(x_2)\,\bar{\psi}_{b_1}(y_1)\,\bar{\psi}_{b_2}(y_2)\,|0\rangle\,. \tag{12.13}$$

Because $\psi_{a_1}(x_1)$ and $\psi_{a_2}(x_2)$ are both fermionic operators, interchanging them inside the time-ordering symbol costs a minus sign exactly as in Eq. (12.11); the analogue of the operator-level antisymmetry that Eq. (11.136) exhibited for two-electron *states*, $b^{r\dagger}_{\mathbf{p}}b^{s\dagger}_{\mathbf{q}}\,|0\rangle = -b^{s\dagger}_{\mathbf{q}}b^{r\dagger}_{\mathbf{p}}\,|0\rangle$, holds here for two-electron *field insertions*. Hence

$$G^{(4)}(x_1,x_2;y_1,y_2) = -\,G^{(4)}(x_2,x_1;y_1,y_2), \tag{12.14}$$

and likewise a minus under $y_1 \leftrightarrow y_2$. Now $G^{(4)}$ for the free field is a sum of products of two propagators—this is Wick's theorem, established for the

Figure 12.1: The two Wick pairings of the free four-fermion correlator Eq. (12.13). Antisymmetry under $x_1 \leftrightarrow x_2$ forces the two pairings to enter with opposite signs—the Pauli sign of Eq. (12.15). A path integral over commuting fields, whose Wick theorem is a permanent, would assign both pairings a plus sign and so describe bosons.

scalar in Chapter 9 and extended to the fermion with its sign rule alongside Eq. (11.187). There are exactly two ways to pair each ψ with a $\bar\psi$: the *direct* pairing $(\psi_1-\bar\psi_1,\ \psi_2-\bar\psi_2)$ and the *crossed* pairing $(\psi_1-\bar\psi_2,\ \psi_2-\bar\psi_1)$, drawn in Fig. 12.1. Antisymmetry Eq. (12.14) fixes their relative sign uniquely. Under $x_1 \leftrightarrow x_2$ the direct and crossed terms turn into each other; for the sum to flip sign, the two contributions must enter with *opposite* signs, giving the determinant (not permanent) structure

$$G^{(4)}_{a_1a_2,b_1b_2} = -\,S_F(x_1-y_1)_{a_1b_1}\,S_F(x_2-y_2)_{a_2b_2} \;+\; S_F(x_1-y_2)_{a_1b_2}\,S_F(x_2-y_1)_{a_2b_1}. \tag{12.15}$$

The relative minus sign between the two pairings is the *Pauli sign*: it is the field-theoretic statement of the exclusion principle, the same sign that turned the two-electron inner product into a Slater determinant in Theorem 11.28, now appearing as the antisymmetrization of an exchange amplitude. It is what makes identical-fermion scattering interfere destructively where identical bosons interfere constructively.

Contrast this with what a commuting path integral would deliver. For commuting Gaussian variables Wick's theorem is the statement that a $2k$-point function is the sum over *all* pairings with *all* plus signs—a permanent, the bosonic counterpart noted at Theorem 9.16. A commuting integral would therefore return

$$G^{(4)}_{\text{commuting}} = S_F(x_1-y_1)\,S_F(x_2-y_2) \;+\; S_F(x_1-y_2)\,S_F(x_2-y_1), \tag{12.16}$$

symmetric under $x_1 \leftrightarrow x_2$, in flat contradiction with the antisymmetry Eq. (12.14) that the canonical fermion field obeys. The wrong sign on a single crossed pairing is not a normalization slip; it is the difference between Fermi and Bose statistics, between matter that stacks into shells and matter that condenses into a single mode. Any honest fermionic path integral must produce the determinant Eq. (12.15), not the permanent Eq. (12.16), and Theorem 12.1 guarantees that commuting variables produce the permanent.

12.1.4 The classical limit of an anticommuting operator

The diagnosis points to its own cure. In the bosonic case the path-integral variable $\phi(x)$ is the "classical" field configuration—heuristically, the label of a coherent state, the eigenvalue of the field operator $\hat{\phi}(x)$ in the basis the path integral sums over—and it is an ordinary number precisely because the bosonic field operators *commute*: $\left[\hat{\phi}(x), \hat{\phi}(y)\right] = 0$ has, as its classical shadow, the commuting numbers $\phi(x)\phi(y) = \phi(y)\phi(x)$ of Eq. (12.3). The Dirac field operators do the opposite. They were forced in Section 11.3 to obey equal-time *anti*commutators, and the field anticommutators

$$\left\{\hat{\psi}_a(x), \hat{\psi}_b(y)\right\} = 0, \qquad \left\{\hat{\psi}_a(x), \hat{\bar{\psi}}_b(y)\right\} = 0 \quad \text{(equal times, } x \neq y\text{)}, \tag{12.17}$$

together with the nilpotency $\left(b_{\mathbf{p}}^{s\dagger}\right)^2 = 0$ of Eq. (11.135), are the algebraic signature of Fermi statistics. If the classical limit of a commuting operator is a commuting number, then by the same token the classical limit of an *anticommuting* operator must be an *anticommuting* number. The slogan "the classical limit of an anticommuting operator is an anticommuting number" is not a metaphor to be admired and set aside; it is the design requirement, and we state it as such.

Physical Insight: the integration variables must anticommute

A path integral that is to reproduce the canonical Dirac field must sum over field configurations $\psi(x), \bar{\psi}(x)$ that are themselves anticommuting objects:

$$\{\psi(x), \psi(y)\} = 0, \qquad \left\{\psi(x), \bar{\psi}(y)\right\} = 0, \qquad \left\{\bar{\psi}(x), \bar{\psi}(y)\right\} = 0 \tag{12.18}$$

at the level of the integration variables—not as operators on a Hilbert space, but as a new species of number out of which the configurations to be summed are built. In particular each such number squares to zero, $\psi(x)^2 = 0$, the c-number echo of the operator nilpotency $\left(b_{\mathbf{p}}^{s\dagger}\right)^2 = 0$. Only then can the integrand $\psi(x)\bar{\psi}(y) = -\bar{\psi}(y)\psi(x)$ change sign under interchange, and only then can the integral reproduce the antisymmetric Eq. (12.12) and the Pauli sign of Eq. (12.15).

This is a genuinely new demand on the notion of "number," and it forces two constructions that the rest of the chapter supplies. First, an *algebra* of anticommuting numbers in which Eq. (12.18) holds identically and in which one can write down field configurations, actions, and sources: this is the Grassmann (exterior) algebra. Second, a notion of *integration* over such numbers, since the ordinary Riemann integral of Eq. (12.2) presupposes an integration variable that runs over the real line, and an anticommuting number runs over nothing of the kind. That integration—Berezin's—will turn out to be defined not by summing

areas but by a single algebraic rule, and its properties are unlike anything in ordinary calculus. The honest preview, deferred to its proper place, is that the eigenvalues of fermionic field operators in a coherent-state basis are exactly such anticommuting numbers; the path integral over them is then built the same way the bosonic one was, as an integral over field-operator eigenvalues, only now the eigenvalues live in a Grassmann algebra.

12.1.5 The four properties the fermionic integral must have

We close the diagnosis by freezing the target. Whatever integration theory the next sections construct, it earns the name "fermionic path integral" only if it reproduces the canonical results of Chapter 11. There are exactly four such requirements, and the entire remainder of the chapter is the verification, one by one, that Grassmann calculus and Berezin integration meet them. Stating them now fixes the standard against which the construction will be judged and prevents the analogy with the bosonic case from being mistaken for a derivation.

(P1) Bilinear action. The integral must be defined for, and dominated by, the free Dirac action

$$S_0 = \int d^4x\, \bar{\psi}(x)\,(i\partial\!\!\!/ - m)\psi(x), \tag{12.19}$$

the spacetime integral of the Dirac Lagrangian Eq. (11.25). This is a *bilinear* (Gaussian) form in $\psi, \bar{\psi}$, just as the free scalar action was bilinear in ϕ; the fermionic Gaussian integral is therefore the object the theory rests on, and it must exist and be computable in closed form.

(P2) Correct propagator. The Gaussian two-point function defined by that integral, $\left\langle \psi_a(x)\,\bar{\psi}_b(y)\right\rangle$, must equal the canonical Dirac–Feynman propagator obtained operator-theoretically in Section 11.5,

$$\left\langle \psi_a(x)\,\bar{\psi}_b(y)\right\rangle \stackrel{!}{=} S_F(x-y)_{ab}, \qquad \tilde{S}_F(p) = \frac{i(p\!\!\!/ + m)}{p^2 - m^2 + i\varepsilon} = \frac{i}{p\!\!\!/ - m + i\varepsilon}, \tag{12.20}$$

the momentum-space form Eq. (11.27) that is the frozen free-electron propagator of QED. As $\tilde{S}_F = i(p\!\!\!/ - m)^{-1}$ is the inverse of the kinetic operator in Eq. (12.19), this is the fermionic analogue of the bosonic "propagator = inverse of the kinetic operator," but it must come out with the matrix and sign structure of the Dirac operator intact.

(P3) Correct minus signs. The integral must build in the antisymmetry Eq. (12.12) and the Pauli sign Eq. (12.15): its Wick theorem must be a determinant (signed sum over pairings), not a permanent. This is the requirement Theorem 12.1 showed commuting variables fail, and it is what

Eq. (12.18) is designed to enforce; downstream it becomes the minus sign carried by every closed fermion loop.

(P4) Determinant, not inverse square root. The source-free Gaussian must evaluate to the kinetic operator's determinant in the *first positive* power,

$$\int \mathcal{D}\psi\, \mathcal{D}\bar{\psi}\; e^{\,i\int \bar{\psi}(i\not{\partial}-m)\psi} \;\propto\; \det(i\not{\partial}-m), \tag{12.21}$$

in pointed contrast to the bosonic real Gaussian $(\det A)^{-1/2}$ of Eq. (4.50) and the complex bosonic Gaussian $(\det A)^{-1}$ of Eq. (4.60). The inversion from a negative to a positive power of the determinant is the single algebraic fact that distinguishes fermionic from bosonic functional integration, and it is the seed of the relative minus sign that property (P3) demands on closed loops.

These four are not independent wishes; they are facets of the one demand that the integral encode Fermi statistics, and they will be met together the moment the integration variables anticommute. The payoff reaches far beyond this chapter. The determinant of Eq. (12.21), written $\det(i\not{D}-m)$ once the field is coupled to a gauge potential, is the one-loop fermionic effective action computed in Chapter 19; the same Grassmann variables, now scalar rather than spinor, are the Faddeev–Popov ghosts that quantize Yang–Mills in Chapter 26; the transformation of the fermion measure $\mathcal{D}\psi\,\mathcal{D}\bar{\psi}$ under a chiral rotation is where the chiral anomaly lives, in Fujikawa's treatment of Chapter 36; and the difficulty of representing $\det(i\not{D}-m)$ on a finite grid is exactly what makes lattice fermions hard, in Chapter 43. The minus sign we have been unable to manufacture from commuting numbers is, in the end, one of the most consequential signs in physics.

We have therefore fixed the target completely: an integration theory whose variables satisfy Eq. (12.18) and which meets (P1)–(P4). What remains is to build it. The next section answers the first half of the demand by constructing, from nothing, the algebra of anticommuting numbers—its generators, its grading, the finiteness of functions built on it, and the differential calculus on it—providing every algebraic fact the integration theory of the sections that follow will lean on.

12.2 Grassmann algebra and Grassmann calculus

The previous section (Section 12.1) left us with a sharp demand and no machinery to meet it. We saw that a path integral for the Dirac field cannot run over ordinary commuting field configurations: a c-number can never reproduce the minus sign that the Pauli principle, the antisymmetry of the time-ordered propagator S_F, and the very anticommutators $\left\{\hat{a}^{s}_{\mathbf{p}}, \hat{a}^{r\dagger}_{\mathbf{q}}\right\} = (2\pi)^3\delta^{sr}\delta^3(\mathbf{p}-\mathbf{q})$ of Chapter 11 all

insist upon. The classical limit of an anticommuting operator is not a number in the usual sense; it is an *anticommuting* number. The whole of the fermionic path integral therefore rests on first constructing, honestly and from nothing, an algebra in which such objects live and a calculus with which to manipulate them. That is the business of this section. We build the Grassmann algebra, establish the finiteness that makes every function of anticommuting variables a polynomial, develop left and right differentiation with their precise sign rules, fix the complex-conjugation convention that the Dirac bilinears will require, and isolate the single identity, $\{\partial/\partial\theta_i, \theta_j\} = \delta_{ij}$, that mirrors the canonical fermionic structure at the level of calculus. Every algebraic fact the integration theory of the next section leans on is proved here.

The objects we are about to define are not exotic; the same algebra is the home of the exterior forms of differential geometry, of the Faddeev–Popov ghosts that will tame non-abelian gauge fixing in Chapter 26, and of the anticommuting coordinates of superspace in Chapter 44. We construct it once, in full generality, and the later chapters will simply reach for it.

12.2.1 The Grassmann algebra and its generators

We work over the complex numbers throughout, because the Dirac field is complex; the real case is the obvious restriction. The defining move is to introduce symbols that square to zero by being made to anticommute.

Definition 12.3 (Grassmann algebra)**.** Let $n \geq 1$. The *Grassmann algebra* $\mathcal{G}_n$ on n generators $\theta_1, \ldots, \theta_n$ is the associative, unital $\mathbb{C}$-algebra generated by the θ_i subject to the single family of relations

$$\{\theta_i, \theta_j\} = \theta_i\theta_j + \theta_j\theta_i = 0, \qquad i, j = 1, \ldots, n. \tag{12.22}$$

The complex scalars are taken to commute with every θ_i. Elements of $\mathcal{G}_n$ are called *Grassmann numbers*; the θ_i themselves are the *Grassmann variables* or odd generators.

Concretely, $\mathcal{G}_n$ is the quotient of the free associative algebra $\mathbb{C}\langle\theta_1, \ldots, \theta_n\rangle$ on n noncommuting symbols by the two-sided ideal generated by the relations (12.22). Mathematically this is nothing but the exterior algebra $\Lambda(\mathbb{C}^n)$ with the wedge written as juxtaposition; we will not need that identification, but it is reassuring that the structure is a familiar one. The whole content of (12.22) is captured by two consequences we use constantly. First, taking $i = j$,

$$2\theta_i^2 = \{\theta_i, \theta_i\} = 0 \quad \Longrightarrow \quad \theta_i^2 = 0, \tag{12.23}$$

the *nilpotency* of every generator: no Grassmann variable can appear twice in a product without killing it. Second, for $i \neq j$ the variables genuinely *anticommute*,

$$\theta_i\theta_j = -\theta_j\theta_i, \tag{12.24}$$

so the *order* of factors matters up to a sign. These two facts—nilpotency and signed reordering—are the entire algebra; everything below is their bookkeeping.

Remark 12.4 (A worked feel for the algebra). With two generators θ_1, θ_2 a general product simplifies in a few strokes: $\theta_1\theta_2\theta_1 = -\theta_1\theta_1\theta_2 = 0$ by (12.23), and $\theta_2\theta_1 = -\theta_1\theta_2$ by (12.24). The only nonzero products are 1, θ_1, θ_2, and $\theta_1\theta_2$. Every other word in the two letters either repeats a letter (and vanishes) or reorders to one of these (up to sign). This is the prototype of the finiteness we now prove in general.

12.2.2 Monomials, the basis, and the top form

Because every word in the generators either repeats a letter and dies or can be sorted into increasing index order at the cost of a sign, the surviving objects are the *ordered, repetition-free monomials*. We make this precise and count them.

Theorem 12.5 (Basis and dimension of $\mathcal{G}_n$)**.** *The ordered monomials*

$$1, \qquad \theta_{i_1}\theta_{i_2}\cdots\theta_{i_k} \quad \text{with} \quad 1 \le i_1 < i_2 < \cdots < i_k \le n, \quad 1 \le k \le n, \tag{12.25}$$

form a basis of $\mathcal{G}_n$ as a complex vector space. Consequently

$$\dim_{\mathbb{C}} \mathcal{G}_n = \sum_{k=0}^{n} \binom{n}{k} = 2^n, \tag{12.26}$$

and there is, up to a complex scalar, a unique monomial of maximal degree—the top form

$$\theta_{\text{top}} \equiv \theta_1\theta_2\cdots\theta_n, \tag{12.27}$$

which spans the one-dimensional top-degree subspace and is annihilated by left- or right-multiplication by any further generator.

Proof. Spanning. Any element of $\mathcal{G}_n$ is a finite $\mathbb{C}$-linear combination of words $\theta_{j_1}\theta_{j_2}\cdots\theta_{j_m}$ in the generators, since these words span the free algebra and passing to the quotient only adds relations. Take any such word. If two of its letters coincide, $j_a = j_b$ with $a < b$, then move θ_{j_b} leftward past the $b - a - 1$ intervening factors using (12.24); this produces an overall sign $(-1)^{b-a-1}$ and brings the two equal letters adjacent, where $\theta_{j_a}\theta_{j_a} = 0$ by (12.23) annihilates the whole word. Hence only words with *distinct* letters survive. For such a word, repeatedly swapping adjacent out-of-order factors via (12.24) sorts the indices into strictly increasing order; each swap costs a factor -1, so the word equals $\pm$ one of the ordered monomials (12.25). Thus the ordered monomials span $\mathcal{G}_n$.

Linear independence. It remains to show the ordered monomials are not subject to any hidden linear relation, i.e. that the quotient did not collapse them. We exhibit a concrete model on which they act as manifestly independent

operators. Let $W = \Lambda(\mathbb{C}^n)$ be the 2^n-dimensional vector space with formal basis labeled by subsets $S \subseteq \{1, \dots, n\}$, written e_S, and define for each i a linear operator $\hat{\theta}_i$ on W by

$$\hat{\theta}_i \, e_S = \begin{cases} 0, & i \in S, \\ (-1)^{\#\{j \in S \,:\, j < i\}} \, e_{S \cup \{i\}}, & i \notin S, \end{cases} \tag{12.28}$$

the operator of "wedging on the left with θ_i." A direct check on the four cases (whether i, j already lie in S) shows $\hat{\theta}_i \hat{\theta}_j + \hat{\theta}_j \hat{\theta}_i = 0$: when both indices are absent, the two orders deposit $e_{S \cup \{i,j\}}$ with opposite signs because inserting i then j versus j then i differ by exactly one transposition; when either index already lies in S, both orders give zero. Hence (12.28) respects the defining relations (12.22) and so descends to a homomorphism $\rho : \mathcal{G}_n \to \mathrm{End}(W)$. Applying $\rho(\theta_{i_1} \cdots \theta_{i_k})$ to the cyclic vector $e_\varnothing$ yields $\pm e_{\{i_1, \dots, i_k\}}$, and distinct ordered monomials produce distinct basis vectors e_S of W. A linear relation among the ordered monomials would, applied to $e_\varnothing$, force a linear relation among distinct e_S, which is impossible. The ordered monomials are therefore independent, and they form a basis.

Counting. The ordered monomials of degree k are indexed by the k-element subsets of $\{1, \dots, n\}$, of which there are $\binom{n}{k}$; summing over k and using the binomial theorem at $x = 1$ gives $\sum_k \binom{n}{k} = 2^n$, which is (12.26).

Top form. The maximal degree is $k = n$, attained only by the single subset $\{1, \dots, n\}$, so the degree-n subspace is one-dimensional and spanned by $\theta_{\mathrm{top}} = \theta_1 \cdots \theta_n$. Multiplying θ_{top} by any θ_i repeats a generator and gives zero by (12.23), whether the new factor is appended on the left or the right. □

The top form is the single most important object in the whole construction: the next section's integration will be, in essence, the linear functional that reads off the coefficient of θ_{top}. We record the structural payoff of Theorem 12.5 separately, because it is the statement the calculus is built to exploit.

Corollary 12.6 (Functions of Grassmann variables terminate)**.** *Every function of the Grassmann variables $\theta_1, \dots, \theta_n$—meaning every element of $\mathcal{G}_n$, equivalently every formal power series in the θ_i once the relations are imposed—is a* finite *polynomial,*

$$f(\theta) = f_0 + \sum_i f_i \, \theta_i + \sum_{i<j} f_{ij} \, \theta_i \theta_j + \cdots + f_{12\cdots n} \, \theta_1 \theta_2 \cdots \theta_n, \tag{12.29}$$

with 2^n complex coefficients $f_0, f_i, \dots, f_{12\cdots n}$ (antisymmetric in their indices if one chooses to write them with unrestricted, rather than ordered, index ranges), and no term of degree exceeding n.

Proof. By Theorem 12.5 the ordered monomials are a basis, so f is uniquely the displayed linear combination; (12.23) truncates every series at degree n because any monomial of higher degree must repeat a generator and vanish. □

This is a remarkable simplification with no analogue for ordinary variables: a "function" of n anticommuting arguments is determined by finitely many numbers, and the entire apparatus of Taylor expansion stops automatically at the top form. It is the reason Grassmann integration will turn out to be algebra rather than analysis. As a small illustration of how drastically series collapse, the exponential of a Grassmann-even quadratic terminates after one step,

$$e^{\theta_1\theta_2} = 1 + \theta_1\theta_2 + \tfrac{1}{2!}(\theta_1\theta_2)^2 + \cdots = 1 + \theta_1\theta_2, \tag{12.30}$$

since $(\theta_1\theta_2)^2 = \theta_1\theta_2\theta_1\theta_2 = -\theta_1\theta_1\theta_2\theta_2 = 0$. The contrast with the genuinely infinite matrix-exponential series of Section 2.3 is exactly the contrast between nilpotent and generic elements; the Gaussian Grassmann integral of Section 12.4 will live entirely on such terminating exponentials.

12.2.3 The $\mathbb{Z}_2$ grading: even and odd

The algebra splits cleanly according to the parity of the degree, and this splitting controls every sign that follows.

Definition 12.7 (Grading and parity)**.** Write $\mathcal{G}_n = \mathcal{G}_n^0 \oplus \mathcal{G}_n^1$, where the *even* part $\mathcal{G}_n^0$ is spanned by ordered monomials of even degree and the *odd* part $\mathcal{G}_n^1$ by those of odd degree. An element lying entirely in $\mathcal{G}_n^p$ is called *homogeneous of parity* p, and we write $|a| = p \in \{0, 1\}$ for its parity (read mod 2). Even elements are also called *bosonic* or *commuting*, odd elements *fermionic* or *anticommuting*. The classical fermion fields of this chapter are odd; their bilinears, Lagrangian, and action are even.

Multiplying monomials adds degrees, so the grading is multiplicative: $|ab| = |a| + |b| \bmod 2$, i.e. $\mathcal{G}_n^p \mathcal{G}_n^q \subseteq \mathcal{G}_n^{p+q}$. The basic commutation law of the algebra, derived once and used everywhere, follows directly.

Proposition 12.8 (Graded commutativity)**.** *For homogeneous $a, b \in \mathcal{G}_n$,*

$$ab = (-1)^{|a|\,|b|}\, ba. \tag{12.31}$$

In particular even elements are central (they commute with everything), and two odd elements anticommute.

Proof. By bilinearity it suffices to take a and b to be ordered monomials, of degrees $r = |a| \bmod 2$ and $s = |b| \bmod 2$ in the counting sense—say a has r' factors and b has s' factors. To move the block b from the right of a to its left, each of the s' generators in b must be carried past each of the r' generators in a, and every such elementary transposition of distinct generators costs -1 by (12.24) (if any generator is shared the products vanish on both sides and (12.31) holds trivially as $0 = 0$). There are $r's'$ such transpositions, so $ab = (-1)^{r's'} ba$.

Since $(-1)^{r's'}$ depends only on the parities $r' \bmod 2 = |a|$ and $s' \bmod 2 = |b|$, we may replace $r's'$ by $|a||b|$, giving (12.31). If $|a| = 0$ the sign is $+1$ for every b, so even elements are central; if $|a| = |b| = 1$ the sign is -1, so odd elements anticommute. □

12.2.4 Grassmann differentiation

We now build a differential calculus. Because order matters in $\mathcal{G}_n$, there are two natural derivatives—one that strips a generator from the left of a monomial and one that strips it from the right—and we must keep them distinct. The functional derivative of Theorem 4.4 is the commuting, infinite-dimensional ancestor of what we are defining; the Grassmann derivative is its finite, anticommuting analogue, and the price of anticommutativity is precisely the sign rules below.

Definition 12.9 (Left and right Grassmann derivatives)**.** The *left derivative* $\overrightarrow{\partial}_i \equiv \dfrac{\overrightarrow{\partial}}{\partial\theta_i}$ is the linear operator on $\mathcal{G}_n$ defined on the basis monomials by: if θ_i does not occur, the result is 0; if θ_i occurs, first anticommute it to the leftmost slot using (12.24) and then delete it. Explicitly, on an ordered monomial in which θ_i sits in position p (counting from the left, starting at 1),

$$\overrightarrow{\partial}_i(\theta_{i_1}\cdots\theta_{i_{p-1}}\,\theta_i\,\theta_{i_{p+1}}\cdots\theta_{i_k}) = (-1)^{p-1}\,\theta_{i_1}\cdots\theta_{i_{p-1}}\,\theta_{i_{p+1}}\cdots\theta_{i_k}. \tag{12.32}$$

The *right derivative* $\overleftarrow{\partial}_i \equiv \dfrac{\overleftarrow{\partial}}{\partial\theta_i}$, written acting on its argument from the right, anticommutes θ_i to the rightmost slot and deletes it,

$$(\theta_{i_1}\cdots\theta_i\cdots\theta_{i_k})\,\overleftarrow{\partial}_i = (-1)^{k-p}\,\theta_{i_1}\cdots\theta_{i_{p-1}}\,\theta_{i_{p+1}}\cdots\theta_{i_k}, \tag{12.33}$$

again with θ_i in position p, and zero if θ_i is absent. On the generators themselves both reduce to $\overrightarrow{\partial}_i\theta_j = \theta_j\overleftarrow{\partial}_i = \delta_{ij}$, and both annihilate the constant: $\overrightarrow{\partial}_i 1 = 0$.

The signs in (12.32)–(12.33) are not conventions to be memorized in isolation; they are forced by demanding that differentiation obey a Leibniz rule compatible with the grading. We make that precise.

Proposition 12.10 (Graded Leibniz rule)**.** *The left derivative is an* odd antiderivation*: for all homogeneous a and arbitrary b,*

$$\overrightarrow{\partial}_i(ab) = (\overrightarrow{\partial}_i a)\,b + (-1)^{|a|}\,a\,(\overrightarrow{\partial}_i b). \tag{12.34}$$

The right derivative satisfies the mirror rule, for homogeneous b and arbitrary a,

$$(ab)\,\overleftarrow{\partial}_i = a\,(b\,\overleftarrow{\partial}_i) + (-1)^{|b|}\,(a\,\overleftarrow{\partial}_i)\,b. \tag{12.35}$$

Moreover the two derivatives are related on a homogeneous element f by $f\,\overleftarrow{\partial}_i = -(-1)^{|f|}\,\overrightarrow{\partial}_i f$.

Proof. By linearity take $a = \theta_{a_1}\cdots\theta_{a_r}$ and $b = \theta_{b_1}\cdots\theta_{b_s}$ to be monomials, so $|a| = r \bmod 2$. If θ_i appears in neither factor, both sides of (12.34) vanish. If θ_i appears in a, say in position $p \le r$, then in the product ab it still sits in position p, so by (12.32) the left-hand side carries the sign $(-1)^{p-1}$ and deletes that θ_i, which is exactly $(\overrightarrow{\partial}_i a)b$; the second term on the right vanishes since θ_i does not occur in b. If instead θ_i appears in b, in position q within b, then within ab it occupies position $r+q$, so the left-hand side carries $(-1)^{r+q-1} = (-1)^r(-1)^{q-1} = (-1)^{|a|}(-1)^{q-1}$ and deletes it; the surviving factor is a times the q-deleted b, i.e. $(-1)^{|a|}a(\overrightarrow{\partial}_i b)$, and now the first term on the right vanishes. (If θ_i appeared in both a and b, the product ab would repeat a generator and vanish, and so would both terms on the right, since a and b remain monomials with a shared letter.) In every case (12.34) holds. The right rule (12.35) is identical with positions counted from the right. Finally, on a degree-k monomial f with θ_i in position p, the left rule gives sign $(-1)^{p-1}$ and the right rule gives $(-1)^{k-p}$; their ratio is $(-1)^{k-p-(p-1)} = (-1)^{k+1} = -(-1)^k = -(-1)^{|f|}$, which is the stated relation. □

We will use the left derivative as the default and drop the arrow when no confusion can arise, writing $\partial_i \equiv \overrightarrow{\partial}_i$; the right derivative is needed only where the field-theoretic source terms of Section 12.6 make the ordering physical. Both derivatives lower the degree by one, hence flip parity: ∂_i maps $\mathcal{G}_n^p$ to $\mathcal{G}_n^{p+1}$, which is what "odd" means and why the antiderivation sign $(-1)^{|a|}$ appears in (12.34) rather than the $+1$ of an ordinary Leibniz rule.

12.2.5 The canonical anticommutation relations of the calculus

We can now extract the two structural identities the integration theory needs. The first realizes the canonical fermionic algebra at the level of differentiation; the second is the nilpotency of differentiation itself. Throughout, θ_j denotes the operator of *left multiplication* by θ_j, so that an anticommutator $\{\partial_i, \theta_j\}$ is read as an identity of operators on $\mathcal{G}_n$, $\partial_i(\theta_j a) + \theta_j(\partial_i a)$.

Proposition 12.11 (Canonical relation $\{\partial_i, \theta_j\} = \delta_{ij}$)**.** *As operators on $\mathcal{G}_n$,*

$$\{\partial_i, \theta_j\} = \delta_{ij}\,\mathbf{1}, \tag{12.36}$$

where $\mathbf{1}$ is the identity operator.

Proof. Let $a \in \mathcal{G}_n$ be arbitrary; by linearity take it homogeneous. Apply the graded Leibniz rule (12.34) with the first factor equal to the odd generator θ_j (so $|\theta_j| = 1$):

$$\partial_i(\theta_j a) = (\partial_i\theta_j)\,a + (-1)^{|\theta_j|}\theta_j(\partial_i a) = \delta_{ij}\,a - \theta_j(\partial_i a), \tag{12.37}$$

using $\partial_i\theta_j = \delta_{ij}$ from Theorem 12.9. Bringing the last term to the left,

$$\partial_i(\theta_j a) + \theta_j(\partial_i a) = \delta_{ij}\,a, \tag{12.38}$$

which is precisely $\{\partial_i, \theta_j\}\, a = \delta_{ij}\, a$ for every a, i.e. the operator identity (12.36). □

This is the calculus-level avatar of the canonical fermionic structure. Set against the oscillator algebra $\left\{\hat{a}_i, \hat{a}_j^\dagger\right\} = \delta_{ij}$ of the quantized Dirac field in Chapter 11, (12.36) identifies ∂_i as the annihilation operator and θ_j (left multiplication) as the creation operator on the 2^n-dimensional space $\mathcal{G}_n$, which is the fermionic Fock space on n modes with vacuum 1 and filled state θ_{top}. The same algebra that will integrate fermions thus already *is*, structurally, the fermionic Fock space; this is no coincidence but the precise sense in which Grassmann numbers are the classical shadow of anticommuting operators.

Proposition 12.12 (Nilpotency of differentiation)**.** *The left derivatives anticommute among themselves,*

$$\{\partial_i, \partial_j\} = 0, \qquad \textit{in particular} \quad \partial_i^2 = 0. \tag{12.39}$$

Proof. We first show that the anticommutator of two odd antiderivations is an ordinary (even) derivation. Let D_1, D_2 be odd antiderivations, so each obeys $D_\alpha(ab) = (D_\alpha a)b + (-1)^{|a|} a(D_\alpha b)$. Compute, for homogeneous a,

$$\begin{aligned} D_1 D_2(ab) &= D_1\big[(D_2 a)b + (-1)^{|a|} a(D_2 b)\big] \\ &= (D_1 D_2 a)b + (-1)^{|a|+1}(D_2 a)(D_1 b) + (-1)^{|a|}(D_1 a)(D_2 b) + a(D_1 D_2 b), \end{aligned} \tag{12.40}$$

where the sign $(-1)^{|a|+1}$ in the second term arises because $D_2 a$ has parity $|a|+1$. Interchanging $1 \leftrightarrow 2$ gives the same expression with the two cross terms swapped:

$$D_2 D_1(ab) = (D_2 D_1 a)b + (-1)^{|a|+1}(D_1 a)(D_2 b) + (-1)^{|a|}(D_2 a)(D_1 b) + a(D_2 D_1 b). \tag{12.41}$$

Adding (12.40) and (12.41), the two cross terms involving $(D_2 a)(D_1 b)$ combine with coefficient $(-1)^{|a|+1} + (-1)^{|a|} = 0$, and likewise those involving $(D_1 a)(D_2 b)$ cancel, leaving

$$\{D_1, D_2\}\,(ab) = (\,\{D_1, D_2\}\,a)b + a(\,\{D_1, D_2\}\,b). \tag{12.42}$$

Thus $\{D_1, D_2\}$ is an even derivation. An even derivation D is determined by its values on the generators: by (12.42) and induction on degree, if $D\theta_k = 0$ for all k then D annihilates every monomial $\theta_{i_1} \cdots \theta_{i_k} = \theta_{i_1}(\theta_{i_2} \cdots \theta_{i_k})$, since D of a product is the sum of terms each carrying a $D\theta_{i_a} = 0$; and $D1 = 0$ because $D1 = D(1 \cdot 1) = 2D1$. Now specialize $D_1 = \partial_i$, $D_2 = \partial_j$, which are odd antiderivations by Theorem 12.10. On a generator,

$$\{\partial_i, \partial_j\}\,\theta_k = \partial_i(\partial_j \theta_k) + \partial_j(\partial_i \theta_k) = \partial_i\,\delta_{jk} + \partial_j\,\delta_{ik} = 0, \tag{12.43}$$

since the derivative of the constant δ_{jk} vanishes. Hence the even derivation $\{\partial_i, \partial_j\}$ annihilates all generators and is therefore zero on all of $\mathcal{G}_n$, proving (12.39); taking $i = j$ gives $2\partial_i^2 = 0$, so $\partial_i^2 = 0$. □

Worked Example 12.13 (The full operator algebra on two generators, checked by hand)**.** It is worth seeing Theorems 12.11 and 12.12 confirmed by brute force, with no appeal to the abstract derivation argument, on the smallest nontrivial algebra $\mathcal{G}_2$ with basis $\{1, \theta_1, \theta_2, \theta_1\theta_2\}$. We tabulate the action of ∂_1 on the basis using (12.32):

$$\partial_1 1 = 0, \quad \partial_1\theta_1 = 1, \quad \partial_1\theta_2 = 0, \quad \partial_1(\theta_1\theta_2) = \theta_2, \tag{12.44}$$

the last because θ_1 sits in position $p = 1$, giving sign $(-1)^0 = +1$ and leaving θ_2. For ∂_2,

$$\partial_2 1 = 0, \quad \partial_2\theta_1 = 0, \quad \partial_2\theta_2 = 1, \quad \partial_2(\theta_1\theta_2) = -\theta_1, \tag{12.45}$$

where now θ_2 sits in position $p = 2$, giving sign $(-1)^1 = -1$. We verify the three relations.

(i) $\{\partial_1, \theta_1\} = \mathbf{1}$. Take the basis element θ_2. Left multiplication then ∂_1: $\partial_1(\theta_1\theta_2) = \theta_2$. The other order: $\theta_1(\partial_1\theta_2) = \theta_1 \cdot 0 = 0$. Sum $= \theta_2$, reproducing the input. On 1: $\partial_1(\theta_1 \cdot 1) = \partial_1\theta_1 = 1$ and $\theta_1\partial_1 1 = 0$, sum $= 1$. On θ_1: $\partial_1(\theta_1\theta_1) = \partial_1 0 = 0$ and $\theta_1\partial_1\theta_1 = \theta_1$, sum $= \theta_1$. On $\theta_1\theta_2$: $\partial_1(\theta_1\theta_1\theta_2) = 0$ and $\theta_1\partial_1(\theta_1\theta_2) = \theta_1\theta_2$, sum $= \theta_1\theta_2$. In every case the operator returns its input, so $\{\partial_1, \theta_1\} = \mathbf{1}$.

(ii) $\{\partial_1, \theta_2\} = 0$. On 1: $\partial_1(\theta_2) + \theta_2\partial_1 1 = 0 + 0 = 0$. On θ_1: $\partial_1(\theta_2\theta_1) + \theta_2\partial_1\theta_1 = \partial_1(-\theta_1\theta_2) + \theta_2 = -\theta_2 + \theta_2 = 0$, the crucial cancellation that the sign in (12.45) arranges. On the remaining basis elements one finds 0 identically. Hence $\{\partial_1, \theta_2\} = 0 = \delta_{12}\mathbf{1}$.

(iii) $\partial_1^2 = 0$ *and* $\{\partial_1, \partial_2\} = 0$. Since ∂_1 lowers degree and $\partial_1\theta_1 = 1$, $\partial_1^2(\theta_1\theta_2) = \partial_1\theta_2 = 0$ and ∂_1^2 is zero on all lower-degree elements trivially, so $\partial_1^2 = 0$. For the mixed relation, the only basis element on which both derivatives act nontrivially is $\theta_1\theta_2$: $\partial_1\partial_2(\theta_1\theta_2) = \partial_1(-\theta_1) = -1$ while $\partial_2\partial_1(\theta_1\theta_2) = \partial_2\theta_2 = +1$, summing to 0. Thus $\{\partial_1, \partial_2\} = 0$, in agreement with Theorem 12.12.

12.2.6 Complex conjugation and the independent conjugate variable

The Dirac field is complex, and its action contains $\bar{\psi}\,(i\partial\!\!\!/ - m)\psi$, a quantity that must be *real* so that e^{iS} has unit modulus and the theory is unitary. To carry this into the Grassmann setting we need an operation of complex conjugation on $\mathcal{G}_n$ and a convention for how it interacts with products. The right convention is the one that makes Grassmann conjugation behave like the Hermitian conjugation of operators it is meant to model.

Definition 12.14 (Complex conjugation of Grassmann numbers)**.** A *complex conjugation* on a Grassmann algebra is an antilinear involution $a \mapsto a^*$ satisfying, for complex scalars λ and homogeneous a, b,

$$(\lambda a)^* = \bar{\lambda}\, a^*, \qquad (a^*)^* = a, \qquad (ab)^* = b^* a^*. \tag{12.46}$$

The defining feature is the *order reversal* in the last rule: conjugating a product reverses the factors, exactly as $(\hat{A}\hat{B})^\dagger = \hat{B}^\dagger \hat{A}^\dagger$ for operators.

For a single Grassmann variable we declare its conjugate θ^* to be an *independent* odd generator—a genuinely new element of an enlarged algebra, not a function of θ—and we write it $\bar{\theta} \equiv \theta^*$, with $(\bar{\theta})^* = \theta$. A complex Dirac field then contributes, mode by mode, an independent pair $(\theta, \bar{\theta})$, precisely paralleling the way the classical complex scalar carried independent ϕ and ϕ^*, and the way the quantum Dirac field carried independent ψ and $\bar{\psi} = \psi^\dagger \gamma^0$. Treating θ and $\bar{\theta}$ as independent integration variables is what will let Section 12.6 introduce independent sources $\eta, \bar{\eta}$ for $\psi, \bar{\psi}$.

Caution: $\bar{\theta}$ is an independent conjugate, not the strong-CP angle

The bar in $\bar{\theta}$ here denotes Grassmann complex conjugation and marks an *independent* odd generator; it has nothing to do with the macro $\bar{\theta}$ reserved in Appendix F for the physical strong-CP angle. We always write the conjugate Grassmann variable with an explicit bar accent, $\bar{\theta}$, to keep the two meanings apart.

The order reversal is not a stylistic choice: it is forced by reality of the fermion bilinear, and dropping it would make $\bar{\theta}\theta$ *anti*-real, in direct contradiction with the reality of the Dirac mass term. The following worked check is the decisive consistency test of the whole convention.

Worked Example 12.15 (Reality of the bilinear $\bar{\theta}\theta$)**.** Take one complex Grassmann variable θ with independent conjugate $\bar{\theta}$, and form the even element $\bar{\theta}\theta$, the Grassmann model of a single fermionic bilinear such as a mode contribution to $\bar{\psi}\psi$. We test its conjugate two ways.

With order reversal (the adopted convention). Apply the third rule of (12.46) and then $(\bar{\theta})^* = \theta$, $(\theta)^* = \bar{\theta}$:

$$(\bar{\theta}\theta)^* = \theta^* \, (\bar{\theta})^* = \bar{\theta}\, \theta. \tag{12.47}$$

The bilinear equals its own conjugate; it is *real* (more precisely, a self-conjugate Grassmann-even element), exactly the property needed for $m\bar{\psi}\psi$ to enter a real Lagrangian and for e^{iS} to have unit modulus. This is the Grassmann counterpart of $(\hat{a}^\dagger \hat{a})^\dagger = \hat{a}^\dagger \hat{a}$: a number-operator-like bilinear is Hermitian.

Without order reversal (the wrong convention, shown to fail). Had we instead demanded $(ab)^* = a^* b^*$, the same computation would give

$$(\bar{\theta}\theta)^*_{(\text{no reversal})} = \bar{\theta}^* \theta^* = \theta\, \bar{\theta} = -\bar{\theta}\theta, \tag{12.48}$$

using the anticommutativity (12.24) of the two distinct odd generators in the last step. The bilinear would be *anti*-real, $(\bar{\theta}\theta)^* = -\bar{\theta}\theta$, so $m\bar{\theta}\theta$ could never sit

in a real action. The factor of -1 in (12.48) is exactly the one the order-reversal rule (12.47) is designed to absorb. Reality of fermion bilinears therefore *requires* the order-reversal convention; we adopt it permanently.

The same reversal makes longer strings behave correctly. For a quadratic form $\bar{\theta}_i A_{ij} \theta_j$ with a numerical matrix A, repeated use of (12.46) gives $(\bar{\theta}_i A_{ij} \theta_j)^* = \bar{A}_{ij}\, \theta_j^* \bar{\theta}_i^* = \bar{A}_{ij}\, \bar{\theta}_j \theta_i = \bar{\theta}_i\, (A^\dagger)_{ij}\, \theta_j$, after relabeling and one anticommutation; the bilinear is real precisely when A is Hermitian, the exact condition the Dirac operator $i\partial\!\!\!/ - m$ satisfies (in the appropriate inner product) and the reason its Grassmann Gaussian integral in Section 12.4 will produce a sensible $\det(i\partial\!\!\!/ - m)$.

With Theorems 12.5, 12.6 and 12.10 to 12.12 and the conjugation convention of Theorem 12.14 in hand, the Grassmann calculus is now fully operational: we have anticommuting variables, finite functions of them, left and right derivatives with their sign rules, the canonical relation $\{\partial_i, \theta_j\} = \delta_{ij}$, and a complex structure under which fermion bilinears are real. What is still missing is an *integral*—a linear functional on $\mathcal{G}_n$ that can play the role $\int \mathcal{D}\phi$ played for the scalar. The single algebraic fact the next section will exploit is the one this section has made unavoidable: a function of n Grassmann variables is a polynomial whose information caps at the top form $\theta_{\text{top}} = \theta_1 \cdots \theta_n$. Berezin's definition of integration, to which we now turn in Section 12.3, is essentially the unique linear functional that isolates the coefficient of that top form, and its strange-looking properties—translation invariance, the Grassmann delta function, and a Jacobian that inverts rather than multiplies—will all follow from the calculus assembled here.

12.3 Berezin integration

The previous section gave us an algebra and a differential calculus on it. We can multiply Grassmann numbers, expand any function of them into its (necessarily terminating) Taylor series, and differentiate from the left or the right with the sign rules of Section 12.2. One operation is conspicuously missing, and it is exactly the one the path integral cannot do without: integration. The bosonic generating functional of Chapter 10 was built by literally summing e^{iS} over field configurations, that is, by an honest (if formal) integral $\int \mathcal{D}\phi$ over the values a commuting field can take. To give the Dirac field the same description we must learn to integrate a function of *anticommuting* variables — and here ordinary intuition is no help at all. A Grassmann variable θ takes no numerical values; it is not a coordinate on any space, there is no interval to subdivide, no measure to assign, no Riemann sum to refine. The symbol $\int \mathrm{d}\theta$ cannot mean "area under a curve." It must be *defined*, and the only honest way to define it is to decide first what we want an integral to be *for* and then to ask whether anything on the Grassmann algebra does that job.

What we want is contained in two demands. First, integration should be *linear*: the integral of a sum is the sum of the integrals, and constants pull out. Second — and this is the property that does all the work in field theory — it should be *translation invariant*. In the bosonic path integral, the single fact that makes the generating functional manipulable is that $\int_{-\infty}^{\infty} \mathrm{d}x\, f(x+a) = \int_{-\infty}^{\infty} \mathrm{d}x\, f(x)$; shifting the dummy variable changes nothing. This is what lets one complete the square in a Gaussian, shift away a linear source, and read off a propagator. Any integral worth the name for Grassmann variables must share this property, because we will want to perform precisely the same shifts on fermionic fields. The remarkable fact — due to Berezin — is that these two demands, linearity and translation invariance, pin the integral down *completely* up to a single multiplicative constant. There is essentially one notion of integration on a Grassmann algebra, and we are about to discover that it coincides, of all things, with differentiation.

12.3.1 The single variable: uniqueness forces Berezin's rule

Fix one Grassmann generator θ, with $\theta^2 = 0$. Because the algebra it generates is two dimensional — every function is

$$f(\theta) = a + b\,\theta, \qquad a, b \in \mathbb{C}, \tag{12.49}$$

the Taylor series terminating after one term — an integral is nothing more exotic than a linear functional I on this two-dimensional space. Such a functional is fixed by its two values on the basis $\{1, \theta\}$:

$$I[1] \;=\; \alpha, \qquad I[\theta] \;=\; \beta, \tag{12.50}$$

and then $I[f] = a\,\alpha + b\,\beta$ by linearity. Two numbers α, β parametrize every candidate integral. Translation invariance will kill one of them and normalization will fix the other.

To state translation invariance we must say what it means to shift θ. Let η be a new Grassmann constant — a generator of a larger Grassmann algebra that anticommutes with θ and squares to zero, $\eta^2 = 0$, $\{\eta, \theta\} = 0$. "Shift" means replacing θ by $\theta + \eta$ everywhere. The integral is called translation invariant if

$$I\big[f(\theta+\eta)\big] \;=\; I\big[f(\theta)\big] \qquad \text{for every } f \text{ and every Grassmann constant } \eta, \tag{12.51}$$

where I acts only on the θ-dependence, treating η as an external constant exactly as a in $\int \mathrm{d}x\, f(x+a)$ is held fixed while x is integrated.

Theorem 12.16 (Uniqueness of the Berezin integral)**.** *Every linear functional I on the algebra generated by a single Grassmann variable θ that is translation invariant in the sense of Eq. (12.51) annihilates the constants, $I[1] = 0$, and is therefore proportional to the coefficient of the top monomial,*

$$I[a + b\,\theta] \;=\; \beta\, b, \tag{12.52}$$

with $\beta = I[\theta]$ the one free constant. Conversely every functional of this form is translation invariant.

Proof. Expand the shifted function. Since η is Grassmann-odd and $\theta^2 = 0$,

$$f(\theta + \eta) = a + b\,(\theta + \eta) = (a + b\,\eta) + b\,\theta. \tag{12.53}$$

The shift has not changed the coefficient of θ (still b); it has only added the Grassmann-odd constant $b\,\eta$ to the constant term. Apply I, using linearity and holding η fixed:

$$I\big[f(\theta + \eta)\big] = (a + b\,\eta)\,\alpha + b\,\beta = \underbrace{(a\,\alpha + b\,\beta)}_{= I[f(\theta)]} + b\,\eta\,\alpha. \tag{12.54}$$

Translation invariance Eq. (12.51) demands that the difference vanish, $b\,\eta\,\alpha = 0$, for *all* coefficients b and all Grassmann constants η. Since $b\,\eta$ is a nonzero element of the larger algebra for generic b and η, the only way to enforce this is $\alpha = 0$. Hence $I[1] = 0$, and $I[f] = b\,\beta$, which is Eq. (12.52). The converse is immediate: a functional that reads off β times the coefficient of θ ignores the constant term entirely, and by Eq. (12.53) the shift touches only the constant term, so the value is unchanged. □

The theorem says the integral is unique up to the normalization β. We fix it by the single conventional choice $\beta = 1$, and we adopt the notation $\int \mathrm{d}\theta$ for the resulting functional. This is Berezin's definition.

Definition 12.17 (Berezin integral, one variable)**.** Integration over a single Grassmann variable θ is the linear functional fixed by

$$\int \mathrm{d}\theta\; 1 \;=\; 0, \qquad \int \mathrm{d}\theta\; \theta \;=\; 1. \tag{12.55}$$

On a general function it returns the coefficient of θ: $\displaystyle\int \mathrm{d}\theta\,(a + b\theta) = b$.

The first rule, $\int \mathrm{d}\theta\, 1 = 0$, is the one that looks strange and is in fact the heart of the matter: it is forced, not chosen. It is the Grassmann analogue of the statement that $\int_{-\infty}^{\infty} \mathrm{d}x\, 1$ would have to be translation invariant and so cannot depend on any origin — except that here, instead of diverging, the invariant value is simply zero. The second rule is the harmless choice of unit. Everything that follows is a consequence of Eq. (12.55).

Integration is differentiation. Compare Theorem 12.17 with the left derivative $\partial_\theta^{\mathrm{L}}$ built in Section 12.2, which strips a leading θ: $\partial_\theta^{\mathrm{L}} 1 = 0$ and $\partial_\theta^{\mathrm{L}} \theta = 1$.

These are *identical* to the Berezin rules Eq. (12.55). Therefore, as operators on functions of one Grassmann variable,

$$\boxed{\int \mathrm{d}\theta \;=\; \partial_\theta^{\mathrm{L}}} \tag{12.56}$$

Grassmann integration and left differentiation are the same operation. This is not a curiosity but the structural fact that governs every sign and every Jacobian below. It is worth pausing on how sharply it departs from the bosonic functional calculus of Chapter 4: there, differentiation $\delta/\delta\phi(x)$ (Theorem 4.4) and integration $\int \mathcal{D}\phi$ are genuinely different operations, inverse to one another in the sense of the fundamental theorem of calculus, and a Gaussian integral and a functional derivative produce different objects. For Grassmann variables that distinction collapses. One immediate consequence is that there are no boundary terms and no convergence questions: the "integral" is a finite, purely algebraic operation that always returns a definite element of the algebra. A second consequence is integration by parts in its cleanest possible form. Because $\partial_\theta^{\mathrm{L}}$ obeys a graded Leibniz rule and $\partial_\theta^{\mathrm{L}}(\partial_\theta^{\mathrm{L}} f) = 0$ (a leading θ, once removed, cannot be removed again), for any two functions f, g one has

$$\int \mathrm{d}\theta\, (\partial_\theta^{\mathrm{L}} f)\, g \;=\; (-1)^{|f|} \int \mathrm{d}\theta\, f\, (\partial_\theta^{\mathrm{L}} g), \tag{12.57}$$

with $|f|$ the Grassmann parity of f, and no surface term ever appears because $\int \mathrm{d}\theta\, \partial_\theta^{\mathrm{L}}(fg) = \partial_\theta^{\mathrm{L}}\partial_\theta^{\mathrm{L}}(fg) = 0$. The sign $(-1)^{|f|}$ is the only residue of "moving the derivative across," inherited directly from Eq. (12.56) and the graded Leibniz rule of Section 12.2.

Caution

The identity $\int \mathrm{d}\theta = \partial_\theta^{\mathrm{L}}$ tempts one to read $\mathrm{d}\theta$ as if it transformed like the differential of a coordinate, the way $\mathrm{d}x$ does. It does not. Under a rescaling $\theta \to c\,\theta$ an ordinary differential scales as $\mathrm{d}(cx) = c\,\mathrm{d}x$, but the Berezin symbol scales the *opposite* way, $\mathrm{d}(c\theta) = c^{-1}\mathrm{d}\theta$, precisely because $\int \mathrm{d}\theta$ measures the coefficient of θ rather than a length. This inverse scaling is the one-variable shadow of the inverse-determinant law proved in Theorem 12.19; keeping it straight is the difference between landing the fermionic Gaussian on $\det A$ and on $(\det A)^{-1}$.

12.3.2 Several Grassmann variables

Pass now to n generators $\theta_1, \dots, \theta_n$, all anticommuting, $\{\theta_i, \theta_j\} = 0$ (so $\theta_i^2 = 0$). A general function is a finite sum over subsets $S \subseteq \{1, \dots, n\}$,

$$f(\theta) = \sum_{S} c_S\, \theta_S, \qquad \theta_S := \theta_{i_1}\theta_{i_2}\cdots\theta_{i_k} \quad (i_1 < \cdots < i_k,\; S = \{i_1, \dots, i_k\}), \tag{12.58}$$

with the convention $\theta_\varnothing = 1$. The series terminates at the *top monomial* $\theta_{\{1,\dots,n\}} = \theta_1\theta_2\cdots\theta_n$, the unique (up to sign) element of maximal degree n; there is no room for any θ_i to appear twice. We integrate variable by variable, each $\int \mathrm{d}\theta_i$ acting by the single-variable rule Theorem 12.17, that is, as the left derivative $\partial^{\mathrm{L}}_{\theta_i}$. The only new ingredient is the *order* of the iterated operations and the bookkeeping of signs, and for this it is cleanest to declare that the symbols $\mathrm{d}\theta_i$ are themselves Grassmann-odd: they anticommute with one another and with every θ_j,

$$\{\mathrm{d}\theta_i, \mathrm{d}\theta_j\} = 0, \qquad \{\mathrm{d}\theta_i, \theta_j\} = 0. \tag{12.59}$$

This is the only convention consistent with $\int \mathrm{d}\theta_i = \partial^{\mathrm{L}}_{\theta_i}$ and the graded calculus, and it is what makes the multi-variable measure transform cleanly.

Definition 12.18 (Berezin integral, several variables)**.** The multiple Berezin integral is the iterated functional

$$\int \mathrm{d}\theta_n\, \mathrm{d}\theta_{n-1}\cdots\mathrm{d}\theta_1\; f(\theta) \;:=\; \partial^{\mathrm{L}}_{\theta_n}\partial^{\mathrm{L}}_{\theta_{n-1}}\cdots\partial^{\mathrm{L}}_{\theta_1}\, f(\theta), \tag{12.60}$$

the innermost differential $\mathrm{d}\theta_1$ acting first. Equivalently, it returns the coefficient of the top monomial:

$$\int \mathrm{d}\theta_n\cdots\mathrm{d}\theta_1\; f \;=\; c_{\{1,\dots,n\}}, \qquad \text{where } f = c_{\{1,\dots,n\}}\,\theta_1\cdots\theta_n + (\text{lower degree}). \tag{12.61}$$

The equivalence of the two formulas is the basic computation of the subject, so we do it once in full rather than assert it. Take $f = \theta_1\theta_2\cdots\theta_n$, the top monomial with unit coefficient. Apply the innermost operator: θ_1 already stands at the far left, so $\partial^{\mathrm{L}}_{\theta_1}(\theta_1\theta_2\cdots\theta_n) = \theta_2\cdots\theta_n$. Now θ_2 stands at the far left of what remains, so $\partial^{\mathrm{L}}_{\theta_2}(\theta_2\cdots\theta_n) = \theta_3\cdots\theta_n$, and so on; after all n operators,

$$\int \mathrm{d}\theta_n\cdots\mathrm{d}\theta_1\; \theta_1\theta_2\cdots\theta_n = 1. \tag{12.62}$$

Any monomial θ_S with $S \neq \{1,\dots,n\}$ is missing at least one generator, say $\theta_j \notin S$; the corresponding $\partial^{\mathrm{L}}_{\theta_j}$ acts on something with no θ_j in it and gives zero. Hence only the top monomial survives, with the matching of orders in Eq. (12.60) arranged precisely so that its coefficient comes out with a plus sign, which is Eq. (12.61). The pairing of the descending measure $\mathrm{d}\theta_n\cdots\mathrm{d}\theta_1$ with the ascending monomial $\theta_1\cdots\theta_n$ is not an accident of taste: it is what removes every intermediate sign, and any other pairing differs from it only by the sign of a permutation, which we now make explicit.

If we reverse the measure to the ascending order, the price is the sign of the order-reversing permutation. For $n = 2$, $\int \mathrm{d}\theta_1\,\mathrm{d}\theta_2\,\theta_1\theta_2 = \partial^{\mathrm{L}}_{\theta_1}\partial^{\mathrm{L}}_{\theta_2}(\theta_1\theta_2) = \partial^{\mathrm{L}}_{\theta_1}(-\theta_1) = -1$, because $\partial^{\mathrm{L}}_{\theta_2}(\theta_1\theta_2) = -\theta_1$ (the operator must anticommute past θ_1 before it meets θ_2). The same computation done with the descending measure $\mathrm{d}\theta_2\,\mathrm{d}\theta_1$ gives $+1$ by Eq. (12.62). These relative signs are not a nuisance to be papered over; they are the combinatorial seed of the antisymmetry of fermionic amplitudes, and we keep them visible throughout.

12.3.3 Change of variables: the inverse determinant

We come to the property that distinguishes Grassmann integration from every integral the reader has met before, and that — traced forward two sections — is the reason the fermionic Gaussian produces $\det A$ rather than $(\det A)^{-1/2}$, and the reason every closed fermion loop carries a minus sign. Consider a linear change of variables

$$\theta_i = \sum_{j=1}^{n} M_{ij}\,\theta'_j, \qquad i = 1, \dots, n, \tag{12.63}$$

with M an invertible $n \times n$ matrix of ordinary (commuting) numbers. For an *ordinary* integral over $\mathbb{R}^n$, such a substitution multiplies the measure by the Jacobian $\det M$ (its absolute value): $\mathrm{d}^n x = |\det M|\,\mathrm{d}^n x'$. For Berezin integration the factor is the *reciprocal*.

Theorem 12.19 (Change of variables; the inverse-determinant Jacobian)**.** *Under the linear change Eq. (12.63) with $M \in \mathrm{GL}(n)$ over the commuting scalars,*

$$\int \mathrm{d}\theta_n \cdots \mathrm{d}\theta_1\; f(\theta) \;=\; (\det M)^{-1} \int \mathrm{d}\theta'_n \cdots \mathrm{d}\theta'_1\; f(M\theta'), \tag{12.64}$$

where $f(M\theta')$ denotes f with each θ_i replaced by $\sum_j M_{ij}\theta'_j$. Equivalently, the Berezin measure transforms as $\mathrm{d}\theta_n \cdots \mathrm{d}\theta_1 = (\det M)^{-1}\,\mathrm{d}\theta'_n \cdots \mathrm{d}\theta'_1$ — the inverse of the ordinary Jacobian, with no absolute value and no square root.

The whole content sits in how the top monomial transforms, so we isolate that as a lemma.

Lemma 12.20 (The top form scales by $\det M$)**.** *Under Eq. (12.63),*

$$\theta_1\theta_2 \cdots \theta_n = (\det M)\;\theta'_1\theta'_2 \cdots \theta'_n. \tag{12.65}$$

Proof. Substitute and expand the product:

$$\theta_1\theta_2 \cdots \theta_n = \sum_{j_1,\dots,j_n=1}^{n} M_{1j_1} M_{2j_2} \cdots M_{nj_n}\;\theta'_{j_1}\theta'_{j_2} \cdots \theta'_{j_n}. \tag{12.66}$$

The coefficients M_{ij} commute, so they were freely pulled out front. A summand vanishes whenever two of the indices $j_1, \dots, j_n$ coincide, because then the product $\theta'_{j_1} \cdots \theta'_{j_n}$ repeats a generator and is zero. Only the terms in which $(j_1, \dots, j_n)$ is a permutation σ of $(1, \dots, n)$ survive. For such a term, reordering the θ' factors into ascending order costs the sign of the permutation,

$$\theta'_{\sigma(1)}\theta'_{\sigma(2)} \cdots \theta'_{\sigma(n)} = \operatorname{sgn}(\sigma)\,\theta'_1\theta'_2 \cdots \theta'_n, \tag{12.67}$$

since each transposition of adjacent factors flips one sign. Therefore

$$\theta_1 \cdots \theta_n = \Big(\sum_{\sigma \in S_n} \operatorname{sgn}(\sigma)\, M_{1\sigma(1)} M_{2\sigma(2)} \cdots M_{n\sigma(n)}\Big)\theta'_1 \cdots \theta'_n = (\det M)\,\theta'_1 \cdots \theta'_n, \tag{12.68}$$

the bracket being exactly the Leibniz formula for $\det M$. □

The fact that the antisymmetry of the Grassmann product manufactures a determinant out of a linear substitution is the algebraic miracle on which the whole fermionic path integral runs. The minus signs that defeat the bosonic intuition — the ones that turn the would-be $(\det A)^{-1/2}$ into $\det A$ — are nothing but the antisymmetrization in Eq. (12.67), performed automatically by the algebra.

Proof of Theorem 12.19. Both sides of Eq. (12.64) are linear in f, so it suffices to verify the identity on each basis monomial θ_S, $S \subseteq \{1,\dots,n\}$. By Eq. (12.61) the left-hand side $\int \mathrm{d}\theta_n \cdots \mathrm{d}\theta_1\, \theta_S$ is 1 if $S = \{1,\dots,n\}$ and 0 otherwise.

Now examine the right-hand side. Substituting Eq. (12.63) turns θ_S, a product of $|S| = k$ generators, into a product of k linear forms in the θ', which is a homogeneous element of degree k in the θ'-algebra. The right-hand integral $\int \mathrm{d}\theta'_n \cdots \mathrm{d}\theta'_1$ extracts the coefficient of the degree-n top monomial $\theta'_1 \cdots \theta'_n$ (Eq. (12.61)). A degree-k element can contain the degree-n monomial only if $k = n$. Hence for every S with $|S| < n$ the right-hand side is 0, matching the left-hand side. The single remaining case is $S = \{1,\dots,n\}$, the top monomial, for which Theorem 12.20 gives

$$\int \mathrm{d}\theta'_n \cdots \mathrm{d}\theta'_1\; \theta_1 \cdots \theta_n = \int \mathrm{d}\theta'_n \cdots \mathrm{d}\theta'_1\; (\det M)\, \theta'_1 \cdots \theta'_n = \det M, \tag{12.69}$$

using Eq. (12.62) for the last step. Multiplying by the prefactor $(\det M)^{-1}$ in Eq. (12.64) returns 1, again matching the left-hand side. The identity holds on every basis element, hence on all f by linearity. □

Worked Example 12.21 (The 2×2 change of variables, explicitly)**.** It is worth seeing the determinant assemble itself by hand, with no appeal to the Leibniz formula. Take $n = 2$ and

$$\theta_1 = a\,\theta'_1 + b\,\theta'_2, \qquad \theta_2 = c\,\theta'_1 + d\,\theta'_2, \qquad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad \det M = ad - bc. \tag{12.70}$$

Form the top monomial and multiply out, dropping the $\theta_i'^2 = 0$ terms:

$$\begin{aligned} \theta_1\theta_2 &= (a\theta'_1 + b\theta'_2)(c\theta'_1 + d\theta'_2) \\ &= ac\,\underbrace{\theta'_1\theta'_1}_{0} + ad\,\theta'_1\theta'_2 + bc\,\theta'_2\theta'_1 + bd\,\underbrace{\theta'_2\theta'_2}_{0} \\ &= ad\,\theta'_1\theta'_2 + bc\,\theta'_2\theta'_1 = ad\,\theta'_1\theta'_2 - bc\,\theta'_1\theta'_2 = (ad - bc)\,\theta'_1\theta'_2, \end{aligned} \tag{12.71}$$

where the single sign flip came from $\theta'_2\theta'_1 = -\theta'_1\theta'_2$. This is Eq. (12.65) for $n = 2$. Now integrate. On the one hand $\int \mathrm{d}\theta_2\, \mathrm{d}\theta_1\, \theta_1\theta_2 = 1$ by Eq. (12.62); on the other,

$$\int \mathrm{d}\theta'_2\, \mathrm{d}\theta'_1\; \theta_1\theta_2 = \int \mathrm{d}\theta'_2\, \mathrm{d}\theta'_1\; (ad - bc)\, \theta'_1\theta'_2 = ad - bc = \det M. \tag{12.72}$$

The two are consistent with Theorem 12.19: $1 = (\det M)^{-1} \det M$. The lone minus sign in Eq. (12.71) is, in miniature, the entire reason fermion loops sign the way they do.

Physical Insight

The inverse determinant is the formal origin of the fermion-loop minus sign. When in Section 12.4 we evaluate the Gaussian $\int\prod \mathrm{d}\bar\theta_i\,\mathrm{d}\theta_i\, e^{-\bar\theta A\theta}$, the change of variables that diagonalizes the quadratic form will, by Theorem 12.19, contribute the reciprocal of the Jacobian, and the answer will be $\det A$ — the *numerator* placement that is the exact inverse of the bosonic $(\det A)^{-1/2}$. Carried into perturbation theory, that single inversion is what assigns (-1) to every closed fermion loop and reappears as the one-loop effective action $\log\det(i\not{D}-m)$ of Section 12.7. Every later occurrence of a fermionic minus sign — the relative sign between exchange diagrams, the sign of the QED vacuum polarization, the very existence of the chiral anomaly in Chapter 36 — can be traced back to the reordering Eq. (12.67) that turned a substitution into a determinant.

12.3.4 The Grassmann delta function

An integral worthy of the name should possess a delta function: an object that, placed under $\int\mathrm{d}\theta$, reproduces the value of a function at a chosen point. For ordinary integration $\int\mathrm{d}x\,\delta(x-y)\,f(x)=f(y)$. The Grassmann analogue is startlingly simple, and it follows directly from Berezin's rule with no distribution theory whatsoever.

Definition 12.22 (Grassmann delta function)**.** For Grassmann variables θ,θ' the delta function is the degree-one element

$$\delta(\theta-\theta') \;:=\; \theta-\theta'. \tag{12.73}$$

That the delta function "is" just the difference of the two variables is a consequence of Eq. (12.56): since integration is differentiation, the object that sifts must itself be linear in θ. We verify the sifting property in full, signs included, because the placement of δ relative to f matters in a graded calculus.

Proposition 12.23 (Sifting)**.** *For any function $f(\theta)=a+b\theta$, with θ' treated as a Grassmann constant anticommuting with θ,*

$$\int \mathrm{d}\theta\;\delta(\theta-\theta')\,f(\theta) \;=\; f(\theta'). \tag{12.74}$$

Worked Example 12.24 (Verifying the sifting property)**.** We compute the left-hand side of Eq. (12.74) term by term. Expand the product $\delta(\theta-\theta')f(\theta)=(\theta-\theta')(a+b\theta)$, keeping every sign and using $\theta^2=0$:

$$\begin{aligned}(\theta-\theta')(a+b\theta) &= a\theta + b\,\theta\theta - a\theta' - b\,\theta'\theta\\ &= a\theta - a\theta' - b\,\theta'\theta = a\theta - a\theta' + b\,\theta\theta',\end{aligned} \tag{12.75}$$

where $\theta\theta = 0$ was dropped and the last step used $\theta'\theta = -\theta\theta'$ (the constant θ' anticommutes with θ). Now apply $\int \mathrm{d}\theta = \partial_\theta^{\mathrm{L}}$, which keeps the coefficient of a *leading* θ and discards anything with no θ:

- $\int \mathrm{d}\theta\,(a\theta) = a$;
- $\int \mathrm{d}\theta\,(-a\theta') = 0$, since θ' carries no θ;
- $\int \mathrm{d}\theta\,(b\,\theta\theta') = b\,\theta'$, since θ stands to the left and is stripped, leaving $b\,\theta'$.

Summing the three contributions,

$$\int \mathrm{d}\theta\;(\theta - \theta')(a + b\theta) = a + b\,\theta' = f(\theta'), \tag{12.76}$$

which is exactly Eq. (12.74). The sign in the middle line of Eq. (12.75) was essential: had we mishandled the anticommutation of θ' past θ, the third contribution would have come out as $-b\theta'$ and the sifting would have failed. The order matters, too — with δ to the *right*, one finds $\int \mathrm{d}\theta\, f(\theta)\,\delta(\theta - \theta') = -f(\theta')$ for Grassmann-even f, the extra sign tracking the graded commutation of f past δ. We adopt the convention of Eq. (12.74), δ on the left, so that sifting reproduces $f(\theta')$ cleanly.

For several variables the delta function is the ordered product $\delta^{(n)}(\theta - \theta') = \prod_{i=n}^{1}(\theta_i - \theta_i') = (\theta_n - \theta_n')\cdots(\theta_1 - \theta_1')$, and the analogous computation gives $\int \mathrm{d}\theta_n \cdots \mathrm{d}\theta_1\, \delta^{(n)}(\theta - \theta')\, f(\theta) = f(\theta')$; the ordering of the factors is chosen to match the descending measure so that the iterated sifting Eq. (12.74) chains together without stray signs.

12.3.5 Conjugate pairs and reality conventions

Fermionic actions are built from a field and its Dirac adjoint, so the integration variables come in conjugate pairs $(\theta, \bar\theta)$. Here $\bar\theta$ is the conjugate generator introduced in Section 12.2, with the involution $(\theta)^* = \bar\theta$, $(\bar\theta)^* = \theta$, and the order-reversing rule for products $(\theta_1\theta_2)^* = \bar\theta_2\,\bar\theta_1$. For the purposes of *integration*, θ and $\bar\theta$ are treated as two *independent* generators — the bar is a notational reminder of how they conjugate, not a constraint relating them under the integral. This is the Grassmann counterpart of writing a bosonic Gaussian over the independent real and imaginary parts of a complex field.

For a single pair we order the measure as $\mathrm{d}\bar\theta\,\mathrm{d}\theta$ and fix the normalization by the single rule

$$\int \mathrm{d}\bar\theta\,\mathrm{d}\theta\;\theta\,\bar\theta \;=\; 1, \tag{12.77}$$

which follows from the iterated single-variable rules: $\int \mathrm{d}\theta\,(\theta\bar\theta) = \partial_\theta^{\mathrm{L}}(\theta\bar\theta) = \bar\theta$ strips the leading θ, and then $\int \mathrm{d}\bar\theta\,\bar\theta = 1$. Because $\bar\theta\theta = -\theta\bar\theta$, the same measure

gives $\int \mathrm{d}\bar\theta\,\mathrm{d}\theta\,\bar\theta\theta = -1$. With this convention the elementary one-pair Gaussian comes out right way up: expanding the exponential, which terminates because $(\bar\theta\theta)^2 = 0$,

$$\int \mathrm{d}\bar\theta\,\mathrm{d}\theta\; e^{-\bar\theta a\theta} = \int \mathrm{d}\bar\theta\,\mathrm{d}\theta\;(1 - a\,\bar\theta\theta) = -a\int \mathrm{d}\bar\theta\,\mathrm{d}\theta\;\bar\theta\theta = -a\,(-1) = a, \tag{12.78}$$

for any commuting number a. The constant term integrates to zero by Berezin's first rule, leaving only the quadratic term, and the answer is a in the numerator — the $n = 1$ instance of the $\det A$ that Section 12.4 will establish in general, and already the opposite of the bosonic $\int \mathrm{d}x\,\mathrm{d}y\, e^{-a(x^2+y^2)/2} \propto a^{-1}$.

Definition 12.25 (Paired Berezin measure)**.** For n conjugate pairs $(\theta_i, \bar\theta_i)$, $i = 1, \dots, n$, the integration measure is the ordered product

$$\int \prod_{i=1}^{n} \mathrm{d}\bar\theta_i\,\mathrm{d}\theta_i \;:=\; \int \mathrm{d}\bar\theta_n\,\mathrm{d}\theta_n \cdots \mathrm{d}\bar\theta_1\,\mathrm{d}\theta_1, \tag{12.79}$$

normalized by Eq. (12.77) on each pair. It picks out the coefficient of the top monomial $\prod_i \theta_i\bar\theta_i$ in the integrand.

The change-of-variables theorem Theorem 12.19 applies to each set of variables independently. If the θ_i are transformed by a matrix M and the $\bar\theta_i$ by a matrix N — as happens when one diagonalizes a fermionic bilinear $\bar\theta A\theta$ by $\theta = M\theta''$, $\bar\theta = \bar\theta'' N^{\mathsf T}$ — the measure picks up $(\det M)^{-1}(\det N)^{-1}$. This is precisely the bookkeeping that, in the next section, converts $\int\prod \mathrm{d}\bar\theta_i\,\mathrm{d}\theta_i\, e^{-\bar\theta A\theta}$ into $\det A$, and we flag here that the two inverse determinants from the two independent variable sets reinforce rather than cancel — the hallmark that fermionic Gaussians land on a positive power of the determinant.

We now hold a complete integration theory for anticommuting variables: a unique translation-invariant functional fixed by $\int \mathrm{d}\theta\,\theta = 1$, its identification with left differentiation, the multi-variable measure that reads off the top form, the Grassmann delta function, and — the load-bearing result — the inverse-determinant change-of-variables law. With the inverse Jacobian in hand, the central computation of the chapter is now a short step away: the Gaussian Grassmann integral of Section 12.4, whose value is $\det A$ rather than the bosonic $(\det A)^{-1/2}$, and from which Wick's theorem for fermions and the entire diagrammatic minus-sign structure will follow.

12.4 Gaussian Grassmann integrals: determinants, Pfaffians, and Wick's theorem

The previous section built Berezin integration from nothing and isolated the one feature that makes it strange: a linear change of variables $\theta = M\theta'$ transforms

the measure with the *inverse* of the bosonic Jacobian,

$$\int \mathcal{D}\theta\, f(\theta) = (\det M)^{-1} \int \mathcal{D}\theta'\, f(M\theta') \qquad (\textit{Theorem } 12.19).$$

That single inversion is the seed of everything that follows. In ordinary calculus the Gaussian integral $\int d^n x\, e^{-\frac{1}{2}x^{\mathsf{T}}Ax}$ returns $(\det A)^{-1/2}$ because the measure must absorb $(\det A)^{-1/2}$ when one rescales to principal axes (Theorem 4.14); the determinant lands in the *denominator*. For Grassmann variables the measure absorbs the *opposite* power, so the same Gaussian returns $\det A$ in the *numerator*. This section computes the fundamental Gaussian Grassmann integrals in full, contrasts them sharply with their bosonic ancestors of Chapter 4, and extracts from them the sourced generating identity and the Grassmann form of Wick's theorem. These are the entire computational engine of the fermionic path integral: once A is promoted to the Dirac operator in Section 12.6, the formulas proved here become the generating functional $Z[\eta, \bar{\eta}]$ and the propagator S_F.

We work throughout over a fixed Grassmann algebra. For the *complex* (paired) case we have $2n$ independent generators $\theta_1, \dots, \theta_n$ and $\bar{\theta}_1, \dots, \bar{\theta}_n$, all mutually anticommuting; here $\bar{\theta}_i$ is the conjugate generator of Theorem 12.14, treated under the integral as a second *independent* generator, not a constraint relating it to θ_i (Section 12.3.5). For the *real* case we have $2n$ generators $\theta_1, \dots, \theta_{2n}$ with $\{\theta_i, \theta_j\} = 0$. We take from Section 12.3 the facts we need about the measure. It is the unique (up to normalization) translation-invariant linear functional (Theorem 12.16), with

$$\int \mathcal{D}\bar{\theta}\,\mathcal{D}\theta\; \theta_i \bar{\theta}_i = 1 \quad (\text{no sum, hence } \int \mathcal{D}\bar{\theta}\mathcal{D}\theta\, \bar{\theta}_i\theta_i = -1), \qquad \int \mathcal{D}\theta\; \theta_1\theta_2\cdots\theta_{2n} = 1, \tag{12.80}$$

where $\mathcal{D}\bar{\theta}\,\mathcal{D}\theta \equiv d\bar{\theta}_n\, d\theta_n \cdots d\bar{\theta}_1\, d\theta_1$ (Theorem 12.25, Eq. (12.77)) and $\mathcal{D}\theta \equiv d\theta_{2n}\cdots d\theta_1$ (Eq. (12.62)), every monomial that fails to saturate all generators integrating to zero; and it is translation invariant, $\int \mathcal{D}\theta\, f(\theta+\xi) = \int \mathcal{D}\theta\, f(\theta)$ for any constant Grassmann shift ξ (Eq. (12.51)). Everything below is built from these and from anticommutation.

12.4.1 The complex Gaussian: $\int e^{-\bar{\theta}A\theta} = \det A$

The central identity inverts the bosonic complex Gaussian of Theorem 4.17 ($\det A$ becomes $\det^{-1} A$) by promoting the integration variables from commuting complex numbers to anticommuting generators. We follow the sign convention of Section 12.3.5, in which the quadratic form carries a minus, matching the one-pair result $\int d\bar{\theta}\, d\theta\, e^{-\bar{\theta}a\theta} = a$ established there (Eq. (12.78)).

Theorem 12.26 (Complex Gaussian Grassmann integral)**.** *Let A be an arbitrary $n \times n$ complex matrix. Then*

$$\int \mathcal{D}\bar{\theta}\,\mathcal{D}\theta\; e^{-\bar{\theta}A\theta} \;=\; \det A, \qquad \bar{\theta}A\theta \equiv \sum_{i,j=1}^{n} \bar{\theta}_i\, A_{ij}\, \theta_j. \tag{12.81}$$

There is no *hypothesis of symmetry, Hermiticity, or positivity on A: the integral exists as a finite polynomial expression and equals the determinant of any square matrix.*

The contrast with the bosonic results of Chapter 4 could not be sharper. The real bosonic Gaussian gives $(\det A)^{-1/2}$ (Theorem 4.15); the complex bosonic Gaussian gives $(\det A)^{-1}$ (Theorem 4.17), and both demand $A > 0$ for convergence. The Grassmann integral gives $(\det A)^{+1}$, converges trivially because the integrand is a polynomial (the exponential of a nilpotent quantity terminates), and asks nothing of A. We give three proofs: a direct expansion for $n = 1, 2$ that makes the mechanism visible, a one-line proof for general A via the inverse Jacobian, and the explicit combinatorial expansion that exhibits the determinant's permutation sum term by term.

Proof for $n = 1$ and $n = 2$ by direct expansion. For $n = 1$ the exponent is nilpotent of order two, $(\bar\theta_1\theta_1)^2 = \bar\theta_1\theta_1\bar\theta_1\theta_1 = 0$, so the exponential terminates after one term:

$$e^{-\bar\theta A\theta} = 1 - A\,\bar\theta_1\theta_1. \tag{12.82}$$

The constant integrates to zero (it fails to saturate the generators), while by (12.80), $\int d\bar\theta_1 d\theta_1\,(1 - A\bar\theta_1\theta_1) = -A\cdot(-1) = A = \det A$.

For $n = 2$ write $S = \bar\theta A\theta = \sum_{i,j}\bar\theta_i A_{ij}\theta_j$. The exponential is $e^{-S} = 1 - S + \frac{1}{2}S^2$ (all higher powers vanish, since S^3 would need a generator repeated). Only the degree-four piece $\frac{1}{2}S^2$ can saturate all four generators, so

$$\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\; e^{-S} = \tfrac{1}{2}\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\; S^2. \tag{12.83}$$

Of the sixteen terms in $S^2 = \sum_{ijkl}\bar\theta_i A_{ij}\theta_j\,\bar\theta_k A_{kl}\theta_l$, only the four with $\{i,k\} = \{1,2\}$ and $\{j,l\} = \{1,2\}$ survive integration. Reducing each to the reference order $\bar\theta_1\theta_1\bar\theta_2\theta_2$, which equals the top monomial $\theta_1\bar\theta_1\theta_2\bar\theta_2$ and so integrates to 1:

$$\begin{aligned}
(i,j,k,l) = (1,1,2,2): &\quad A_{11}A_{22}\,\bar\theta_1\theta_1\bar\theta_2\theta_2 \;\to\; +A_{11}A_{22},\\
(i,j,k,l) = (2,2,1,1): &\quad A_{22}A_{11}\,\bar\theta_2\theta_2\bar\theta_1\theta_1 \;\to\; +A_{11}A_{22},\\
(i,j,k,l) = (1,2,2,1): &\quad A_{12}A_{21}\,\bar\theta_1\theta_2\bar\theta_2\theta_1 \;\to\; -A_{12}A_{21},\\
(i,j,k,l) = (2,1,1,2): &\quad A_{21}A_{12}\,\bar\theta_2\theta_1\bar\theta_1\theta_2 \;\to\; -A_{12}A_{21}.
\end{aligned} \tag{12.84}$$

In the second line $\bar\theta_2\theta_2$ and $\bar\theta_1\theta_1$ are each Grassmann-even, so they commute and the order is restored with no sign. In the third line, moving θ_1 left past the even block $\bar\theta_2\theta_2$ costs nothing and transposing $\theta_2\bar\theta_2 \to \bar\theta_2\theta_2$ costs -1, net -1; the fourth line is identical under $1 \leftrightarrow 2$. Summing and dividing by 2,

$$\tfrac{1}{2}(2A_{11}A_{22} - 2A_{12}A_{21}) = A_{11}A_{22} - A_{12}A_{21} = \det A, \tag{12.85}$$

which is (12.81) for $n = 2$. □

The $n = 2$ computation already displays the whole phenomenon: the determinant's *signs* are not put in by hand; they are produced by the anticommutation of the generators. The off-diagonal product $A_{12}A_{21}$ arrives with a relative minus because reordering the Grassmann generators to a common reference costs an odd permutation. This is the algebraic root of every fermionic minus sign in the book.

Proof for general A via the inverse Jacobian. Suppose first that A is invertible. Change variables on the θ generators alone, leaving the $\bar\theta$ untouched: set $\theta = A^{-1}\theta'$. By the Berezin change-of-variables theorem (Theorem 12.19), which acts on each set of variables independently (Section 12.3.5), the substitution $M = A^{-1}$ produces the reciprocal Jacobian $(\det A^{-1})^{-1} = \det A$:

$$\int \mathcal{D}\bar\theta\, \mathcal{D}\theta\; e^{-\bar\theta A\theta} = (\det A) \int \mathcal{D}\bar\theta\, \mathcal{D}\theta'\; e^{-\bar\theta A A^{-1}\theta'} = (\det A) \int \mathcal{D}\bar\theta\, \mathcal{D}\theta'\; e^{-\bar\theta\theta'}, \tag{12.86}$$

the inverse power being exactly the feature that distinguishes Berezin from Lebesgue integration. The remaining integral factorizes, because the cross terms $\bar\theta_i\theta'_i$ are mutually commuting (even) and each is nilpotent:

$$\int \mathcal{D}\bar\theta\, \mathcal{D}\theta'\; e^{-\bar\theta\theta'} = \prod_{i=1}^{n} \int d\bar\theta_i\, d\theta'_i\, (1 - \bar\theta_i\theta'_i) = \prod_{i=1}^{n} \big(-(-1)\big) = 1, \tag{12.87}$$

using $\int d\bar\theta_i d\theta'_i\, \bar\theta_i\theta'_i = -1$ from (12.80). Hence (12.81) holds for invertible A. If A is singular, $\det A = 0$; correspondingly the columns of A are linearly dependent, so the bilinear form $\bar\theta A\theta$ omits some independent combination of the θ generators, no term of $e^{-\bar\theta A\theta}$ can saturate all $2n$ generators, and the integral vanishes. Both sides equal zero, and (12.81) holds for all A by continuity (both sides are polynomials in the entries of A). □

This is the "diagonalize/triangularize" route in its cleanest form: rather than rotate to eigenvectors as in the bosonic proof of Theorem 4.14, we absorb A wholesale into the measure, where the inverse Jacobian flips the power. The same calculation done with the change of variables that triangularizes $A = UTU^{-1}$ with T upper triangular produces $\det A = \prod_i T_{ii}$ as the product over the now-decoupled diagonal modes; the substitution $\theta \to A^{-1}\theta'$ short-circuits the triangularization and is preferable. For completeness we also display the determinant's permutation sum explicitly.

Proof for general A by combinatorial expansion. Write the exponent so that the conjugate generators stand in the order $\theta\bar\theta$ that integrates with a plus sign: since $-\bar\theta_i A_{ij}\theta_j = A_{ij}\theta_j\bar\theta_i$,

$$-\bar\theta A\theta = \sum_{i,j} A_{ij}\, \theta_j\bar\theta_i. \tag{12.88}$$

Only the term of degree $2n$ survives integration, and that is the $k = n$ term of the exponential series:

$$\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\; e^{-\bar\theta A\theta} = \frac{1}{n!}\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\,\Big(\sum_{i,j} A_{ij}\,\theta_j\bar\theta_i\Big)^n = \frac{1}{n!}\sum_{\substack{i_1,\dots,i_n\\ j_1,\dots,j_n}} A_{i_1 j_1}\cdots A_{i_n j_n}\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\;\theta_{j_1}\bar\theta_{i_1}\cdots\theta_{j_n}\bar\theta_{i_n}. \tag{12.89}$$

For the surviving monomial each generator appears once, so $(j_1,\dots,j_n)$ and $(i_1,\dots,i_n)$ are each permutations of $\{1,\dots,n\}$. The blocks $\theta_{j_a}\bar\theta_{i_a}$ are Grassmann-even, so we may reorder them to put the θ indices in increasing order $\theta_1,\dots,\theta_n$ at no sign cost; this fixes the j's and identifies $i_a = \rho(a)$ for a permutation ρ. Bringing the trailing $\bar\theta_{\rho(a)}$ into the reference order $\bar\theta_a$ costs $\operatorname{sgn}(\rho)$, and the top monomial then integrates to $+1$ by (12.80). Each of the $n!$ initial orderings contributes equally, cancelling $1/n!$, and the result is

$$\int \mathcal{D}\bar\theta\,\mathcal{D}\theta\; e^{-\bar\theta A\theta} = \sum_{\rho\in S_n}\operatorname{sgn}(\rho)\,A_{\rho(1)\,1}A_{\rho(2)\,2}\cdots A_{\rho(n)\,n} = \det A, \tag{12.90}$$

the Leibniz formula for $\det A^{\mathsf T} = \det A$. □

Worked Example 12.27 (The two-point function and the contraction sign)**.** It is worth verifying that the convention (12.80) reproduces the expected propagator structure already at $n = 1$, since this fixes the sign that propagates into S_F. With $A = a$ a single number, the normalized average of $\theta\bar\theta$ against $e^{-a\bar\theta\theta}$ is

$$\frac{\int d\bar\theta\,d\theta\;\theta\bar\theta\,e^{-a\bar\theta\theta}}{\int d\bar\theta\,d\theta\;e^{-a\bar\theta\theta}} = \frac{\int d\bar\theta\,d\theta\;\theta\bar\theta\,(1-a\bar\theta\theta)}{a} = \frac{\int d\bar\theta\,d\theta\;\theta\bar\theta}{a} = \frac{1}{a} = A^{-1}, \tag{12.91}$$

because $\theta\bar\theta\cdot a\bar\theta\theta = 0$ (repeated generator). The contraction $\langle\theta\bar\theta\rangle$ is therefore A^{-1}, the inverse of the matrix in the action — exactly as for the bosonic two-point function (Theorem 4.18) — while the opposite order gives $\langle\theta\bar\theta\rangle \to \langle\bar\theta\theta\rangle = -A^{-1}$, since $\bar\theta\theta = -\theta\bar\theta$. The sign of a fermion two-point function depends on the order in which the generators are written, a fact with no bosonic analogue and the seed of the ordered "θ before $\bar\theta$" bookkeeping that becomes the time-ordering of ψ and $\bar\psi$ in S_F.

Physical Insight: why the determinant is upstairs, and the closed-loop minus sign

The one fact to carry away from Theorem 12.26 is the *position* of the determinant. Schematically, integrating out a field produces

$$\text{boson:}\quad \int \mathcal{D}\phi\, e^{-\frac{1}{2}\phi^{\mathsf T}A\phi} \propto (\det A)^{-1/2}, \qquad \text{fermion:}\quad \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\, e^{-\bar\psi A\psi} = \det A.$$

Both reappear as functional determinants in the one-loop effective action.

When that determinant is written as $\det A = \exp \operatorname{Tr} \ln A$ and expanded, each term is a closed loop of the field being integrated out. Because the fermionic determinant sits at power $+1$ while the bosonic sits at power $-\frac{1}{2}$, the fermionic loop expansion carries the *opposite overall sign*: every closed fermion loop comes with a relative factor of -1. This is the celebrated "-1 per fermion loop" Feynman rule, and it is visible here, in finite dimensions, before a single momentum integral has been written. We discharge it as a diagrammatic rule when we evaluate $\det(i\not{D} - m)$ as a one-loop fermion effective action; the minus sign is not an extra postulate but the inverse Jacobian of (12.86) in disguise.

12.4.2 The real Gaussian and the Pfaffian

When there is no natural pairing of the generators into $\theta, \bar{\theta}$ — the situation for a *real* (Majorana) fermion, and for the antisymmetric part of any bilinear — the relevant object is a single set $\theta_1, \dots, \theta_{2n}$ and a quadratic form built from an antisymmetric matrix. The result is not the determinant but its square root, the Pfaffian.

A quadratic form $\theta^{\mathsf{T}} A \theta = \sum_{ij} \theta_i A_{ij} \theta_j$ in anticommuting variables sees only the antisymmetric part of A: since $\theta_i \theta_j = -\theta_j \theta_i$, any symmetric piece $A_{ij} = A_{ji}$ cancels in the sum. We therefore take A antisymmetric, $A^{\mathsf{T}} = -A$, of even dimension $2n$ (an odd-dimensional real antisymmetric matrix is singular, and its Pfaffian is defined to be zero).

Definition 12.28 (Pfaffian)**.** For a $2n \times 2n$ antisymmetric matrix A, define its *Pfaffian* by the Berezin integral

$$\operatorname{Pf}(A) \;\equiv\; \int \mathcal{D}\theta \; e^{\frac{1}{2}\theta^{\mathsf{T}} A \theta}, \qquad \tfrac{1}{2}\,\theta^{\mathsf{T}} A \theta = \tfrac{1}{2} \sum_{i,j=1}^{2n} \theta_i A_{ij} \theta_j. \tag{12.92}$$

That this matches the standard combinatorial definition follows by expanding the exponential as in (12.89): only the degree-$2n$ term survives, giving the signed sum over perfect matchings,

$$\operatorname{Pf}(A) = \frac{1}{2^n\, n!} \sum_{\sigma \in S_{2n}} \operatorname{sgn}(\sigma)\, A_{\sigma(1)\sigma(2)} A_{\sigma(3)\sigma(4)} \cdots A_{\sigma(2n-1)\sigma(2n)}, \tag{12.93}$$

the $2^n n!$ accounting for the reorderings within and among the n pairs that leave a matching unchanged. For $n = 1$ this reads $\operatorname{Pf}\left(\begin{smallmatrix} 0 & a \\ -a & 0 \end{smallmatrix}\right) = a$, and the explicit check $\int d\theta_2 d\theta_1 \, e^{\frac{1}{2}\theta^{\mathsf{T}} A \theta} = \int d\theta_2 d\theta_1 \, (1 + a\,\theta_1\theta_2) = a$ uses $\frac{1}{2}\theta^{\mathsf{T}} A \theta = \frac{1}{2}(a\theta_1\theta_2 - a\theta_2\theta_1) = a\theta_1\theta_2$ and (12.80).

The fundamental relation between the Pfaffian and the determinant is that the Pfaffian is a square root of the determinant. We prove it by reducing A to a

canonical antisymmetric normal form, which requires the following standard fact of real linear algebra; we prove it in full so that nothing is imported on faith.

Lemma 12.29 (Real antisymmetric normal form)**.** *For every real antisymmetric $2n \times 2n$ matrix A there is a real orthogonal matrix R ($R^{\mathsf{T}} R = \mathbf{1}$) such that*

$$R^{\mathsf{T}} A R = \bigoplus_{k=1}^{n} \lambda_k \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \qquad \lambda_k \in \mathbb{R}. \tag{12.94}$$

Proof. The matrix iA is Hermitian: $(iA)^\dagger = -iA^{\mathsf{T}} = -i(-A) = iA$. By the spectral theorem it has real eigenvalues and an orthonormal eigenbasis. If $iA\,w = \mu\, w$ with $\mu \in \mathbb{R}$ and $w \in \mathbb{C}^{2n}$, taking complex conjugates and using that A is real gives $iA\,\bar{w} = -\mu\,\bar{w}$, so the nonzero eigenvalues occur in pairs $\pm\mu$ with complex-conjugate eigenvectors. Fix a pair with $\mu = \lambda > 0$ and normalized eigenvector $w = \frac{1}{\sqrt{2}}(u - iv)$, $u, v \in \mathbb{R}^{2n}$. Separating real and imaginary parts of $iA\,w = \lambda w$ gives $Au = -\lambda v$ and $Av = \lambda u$. The vectors u, v are real and orthonormal (from $w^\dagger w = 1$ and $w^{\mathsf{T}} w = 0$, which give $u \cdot u = v \cdot v = 1$, $u \cdot v = 0$), and span a two-dimensional real subspace on which A acts as $\lambda \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ in the basis (u, v). This subspace and its orthogonal complement are both A-invariant (because A is antisymmetric: if $x \perp u, v$ then $u \cdot Ax = -Au \cdot x = \lambda v \cdot x = 0$ and likewise $v \cdot Ax = 0$). Iterating on the orthogonal complement, which carries the remaining $\pm$ pairs (zero eigenvalues contribute zero blocks, $\lambda_k = 0$), assembles an orthonormal basis in which A is block-diagonal with the stated 2×2 blocks; collecting these basis vectors as the columns of R gives (12.94). □

Theorem 12.30 (Pfaffian identities)**.** *For any $2n \times 2n$ antisymmetric matrix A and any $2n \times 2n$ matrix M,*

$$\mathrm{Pf}(M^{\mathsf{T}} A M) = \det(M)\ \mathrm{Pf}(A), \tag{12.95}$$

$$\mathrm{Pf}(A)^2 = \det A. \tag{12.96}$$

Proof. Congruence covariance. Let $F(\theta) = e^{\frac{1}{2}\theta^{\mathsf{T}} A \theta}$ and substitute $\theta = M\theta'$. By Theorem 12.19, $\int \mathcal{D}\theta\, F(\theta) = (\det M)^{-1} \int \mathcal{D}\theta'\, F(M\theta')$, i.e. $\int \mathcal{D}\theta'\, F(M\theta') = \det(M) \int \mathcal{D}\theta\, F(\theta)$. Since $F(M\theta') = e^{\frac{1}{2}(M\theta')^{\mathsf{T}} A (M\theta')} = e^{\frac{1}{2}\theta'^{\mathsf{T}} (M^{\mathsf{T}} A M)\theta'}$, the left side is precisely $\mathrm{Pf}(M^{\mathsf{T}} A M)$, and therefore

$$\mathrm{Pf}(M^{\mathsf{T}} A M) = \det(M) \int \mathcal{D}\theta\, e^{\frac{1}{2}\theta^{\mathsf{T}} A \theta} = \det(M)\ \mathrm{Pf}(A), \tag{12.97}$$

which is (12.95).

Square root of the determinant. Let $\Sigma = R^{\mathsf{T}} A R$ be the canonical form (12.94) from Theorem 12.29, so $A = R \Sigma R^{\mathsf{T}}$. The Pfaffian of Σ factorizes over its blocks because the integral does: with θ split into pairs, $e^{\frac{1}{2}\theta^{\mathsf{T}}\Sigma\theta} = \prod_k e^{\lambda_k \theta_{2k-1}\theta_{2k}}$, the top monomial is $\prod_k \theta_{2k-1}\theta_{2k} = \theta_1 \cdots \theta_{2n}$ with coefficient

$\prod_k \lambda_k$, and so $\mathrm{Pf}(\Sigma) = \prod_{k=1}^{n} \lambda_k$ by (12.80). Applying (12.95) with $M = R^{\mathsf{T}}$, $\mathrm{Pf}(A) = \mathrm{Pf}\big((R^{\mathsf{T}})^{\mathsf{T}} \Sigma R^{\mathsf{T}}\big) = \det(R^{\mathsf{T}})\,\mathrm{Pf}(\Sigma) = \det(R) \prod_k \lambda_k$. Squaring and using $\det(R)^2 = 1$ together with $\det A = \det(R \Sigma R^{\mathsf{T}}) = \det \Sigma = \prod_k \lambda_k^2$,

$$\mathrm{Pf}(A)^2 = \big(\det R\big)^2 \prod_k \lambda_k^2 = \prod_k \lambda_k^2 = \det A, \tag{12.98}$$

establishing (12.96). □

The Pfaffian thus carries strictly more information than the determinant: it is a genuine square root, with a definite sign, and that sign — the relative phase of the fermionic measure — is precisely what controls the sign problem of Majorana and lattice fermions. The complex result $\int \mathcal{D}\bar\theta \mathcal{D}\theta\, e^{-\bar\theta A \theta} = \det A$ is recovered from the real one by pairing: assembling $(\theta_i, \bar\theta_i)$ into a single column Θ of $2n$ real generators with antisymmetric form $\mathcal{A} = \left(\begin{smallmatrix} 0 & A \\ -A^{\mathsf{T}} & 0 \end{smallmatrix}\right)$ gives $\mathrm{Pf}(\mathcal{A}) = \pm \det A$, consistent with $\mathrm{Pf}(\mathcal{A})^2 = \det \mathcal{A} = (\det A)^2$ (the determinant of a block antidiagonal matrix).

12.4.3 The sourced generating identity

Everything dynamical comes from adding linear sources, exactly as the bosonic $Z(J)$ of Theorem 4.15 came from the source-free Gaussian. We add *anticommuting* sources $\bar\eta_i, \eta_i$ (Grassmann generators independent of $\theta, \bar\theta$), so that differentiation with respect to them will deposit the fermionic fields with their correct relative signs.

Theorem 12.31 (Sourced Grassmann Gaussian)**.** *For invertible A and Grassmann sources $\bar\eta, \eta$,*

$$Z[\bar\eta, \eta] \;\equiv\; \int \mathcal{D}\bar\theta\, \mathcal{D}\theta\; e^{\,-\bar\theta A \theta + \bar\eta \theta + \bar\theta \eta} \;=\; \det A\; \exp(\bar\eta\, A^{-1} \eta), \tag{12.99}$$

where $\bar\eta\theta = \sum_i \bar\eta_i \theta_i$, $\bar\theta\eta = \sum_i \bar\theta_i \eta_i$, and $\bar\eta A^{-1} \eta = \sum_{ij} \bar\eta_i (A^{-1})_{ij} \eta_j$.

Proof. Complete the square by the Grassmann shift

$$\theta = \theta' + A^{-1}\eta, \qquad \bar\theta = \bar\theta' + \bar\eta A^{-1}, \tag{12.100}$$

a translation of the integration variables by the constant Grassmann vectors $A^{-1}\eta$ and $\bar\eta A^{-1}$, which leaves the Berezin measure and the integral invariant (Eq. (12.51)); this is the fermionic analogue of $d^n x = d^n y$ under $x = y + A^{-1} J$ in the bosonic proof of Theorem 4.15. Substituting and using $A A^{-1} = A^{-1} A = \mathbf{1}$,

$$\begin{aligned} -\bar\theta A \theta &= -(\bar\theta' + \bar\eta A^{-1}) A (\theta' + A^{-1}\eta) = -\bar\theta' A \theta' - \bar\theta' \eta - \bar\eta \theta' - \bar\eta A^{-1} \eta, \\ \bar\eta\theta &= \bar\eta\theta' + \bar\eta A^{-1}\eta, \qquad \bar\theta\eta = \bar\theta'\eta + \bar\eta A^{-1}\eta. \end{aligned} \tag{12.101}$$

Adding the three lines, the linear terms cancel in pairs, $-\bar\theta'\eta+\bar\theta'\eta=0$ and $-\bar\eta\theta'+\bar\eta\theta'=0$, and the three quadratic constants combine as $-\bar\eta A^{-1}\eta+\bar\eta A^{-1}\eta+\bar\eta A^{-1}\eta=+\bar\eta A^{-1}\eta$, leaving

$$-\bar\theta A\theta+\bar\eta\theta+\bar\theta\eta=-\bar\theta' A\theta'+\bar\eta A^{-1}\eta. \tag{12.102}$$

The constant $+\bar\eta A^{-1}\eta$ is Grassmann-even (a product of two odd sources), commutes with everything, and factors out of the integral; the remaining θ'-integral is the source-free Gaussian (12.81). Hence $Z[\bar\eta,\eta]=e^{\bar\eta A^{-1}\eta}\int\mathcal{D}\bar\theta'\mathcal{D}\theta'\,e^{-\bar\theta' A\theta'}=\det A\,e^{\bar\eta A^{-1}\eta}$, which is (12.99). □

The structure mirrors the bosonic $Z(J)\propto e^{\frac12 J^{\mathsf T}A^{-1}J}$ (Theorem 4.15): the source dependence is again the exponential of a quadratic form built from A^{-1}, the inverse of the operator in the action. The two differences are that fermions carry two independent sources rather than one (the form $\bar\eta A^{-1}\eta$ rather than a symmetric $JA^{-1}J$), and that the overall prefactor is $\det A$ rather than $(\det A)^{-1/2}$. Both signs will be tracked again at field level, where the prefactor $\bar\eta A^{-1}\eta$ inherited here combines with the factors of i of the Dirac action to produce the standard Feynman propagator.

12.4.4 Wick's theorem for Grassmann variables

We can now harvest the Gaussian moments. Define the normalized Gaussian average of a Grassmann-valued function F by

$$\langle F\rangle\;\equiv\;\frac{1}{\det A}\int\mathcal{D}\bar\theta\,\mathcal{D}\theta\;F\,e^{-\bar\theta A\theta}=\frac{1}{Z[0,0]}\int\mathcal{D}\bar\theta\,\mathcal{D}\theta\;F\,e^{-\bar\theta A\theta}, \tag{12.103}$$

the exact fermionic analogue of Theorem 4.16. The whole content of the free theory is that these averages are determined by the two-point function, with one decisive difference from the bosonic Theorem 4.19: pairings now carry signs.

To extract moments we differentiate $Z[\bar\eta,\eta]$ using the left derivatives of Theorem 12.9, which act by anticommuting through to the variable and stripping it: $\partial^{\mathrm L}_{\eta_i}(\eta_i X)=X$ and $\partial^{\mathrm L}_{\eta_i}(\eta_j X)=-\eta_j\,\partial^{\mathrm L}_{\eta_i}X$ for $j\neq i$. Each θ_p in the integrand is produced from the source term $\bar\eta\theta$ by $\partial^{\mathrm L}_{\bar\eta_p}$, and each $\bar\theta_q$ from $\bar\theta\eta$ by $\partial^{\mathrm L}_{\eta_q}$. Tracking the single sign that arises when $\partial^{\mathrm L}_{\eta_q}$ anticommutes past the already-extracted (Grassmann-odd) θ_p, one finds

$$\partial^{\mathrm L}_{\eta_q}\,\partial^{\mathrm L}_{\bar\eta_p}\,Z[\bar\eta,\eta]\Big|_{\bar\eta=\eta=0}=\int\mathcal{D}\bar\theta\,\mathcal{D}\theta\;\theta_p\bar\theta_q\,e^{-\bar\theta A\theta}=\det A\,\langle\theta_p\bar\theta_q\rangle. \tag{12.104}$$

Differentiating instead the closed form $\det A\,e^{\bar\eta A^{-1}\eta}=\det A\,(1+\sum_{ij}\bar\eta_i(A^{-1})_{ij}\eta_j+\cdots)$, the only term surviving two derivatives and the $\bar\eta=\eta=0$ limit is the

bilinear one: $\partial^{\mathrm{L}}_{\bar\eta_p}$ brings down $(A^{-1})_{pj}\eta_j$, and $\partial^{\mathrm{L}}_{\eta_q}$ then yields $(A^{-1})_{pq}$, all times $\det A$. Equating with (12.104) gives the fundamental contraction:

$$\boxed{\langle \theta_p\, \bar\theta_q \rangle = +\,(A^{-1})_{pq}, \qquad \langle \bar\theta_q\, \theta_p \rangle = -\,(A^{-1})_{pq}.} \tag{12.105}$$

The two orderings differ by a sign because $\theta_p\bar\theta_q = -\bar\theta_q\theta_p$; the contraction is A^{-1}, the inverse of the matrix in the action, exactly as in the bosonic Theorem 4.18, but its sign is order-dependent. This A^{-1} is the finite-dimensional fermion propagator; promoted to fields it becomes S_F.

Theorem 12.32 (Grassmann Wick's theorem)**.** *For the Gaussian average* (12.103)*, any moment with unequal numbers of θ's and $\bar\theta$'s vanishes, and a moment with n of each is the* determinant *of the matrix of contractions:*

$$\langle \theta_{i_1}\bar\theta_{j_1}\,\theta_{i_2}\bar\theta_{j_2}\cdots\theta_{i_n}\bar\theta_{j_n}\rangle = \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\,\prod_{a=1}^{n}\langle \theta_{i_a}\bar\theta_{j_{\sigma(a)}}\rangle = \det\Big[\langle \theta_{i_a}\bar\theta_{j_b}\rangle\Big]_{a,b=1}^{n} = \det\big[(A^{-1})_{i_a j_b}\big]. \tag{12.106}$$

Equivalently, the moment is the signed sum over all ways of pairing each θ with a $\bar\theta$, each pairing contributing the product of contractions (12.105) *times the sign of the permutation σ that effects it.*

Proof. Each θ_{i_a} is produced by $\partial^{\mathrm{L}}_{\bar\eta_{i_a}}$ and each $\bar\theta_{j_b}$ by $\partial^{\mathrm{L}}_{\eta_{j_b}}$ acting on $Z[\bar\eta,\eta]$, generalizing (12.104). Since the differentiation strips generators only in conjugate pairs $\bar\eta \leftrightarrow \theta$ and $\eta \leftrightarrow \bar\theta$, a moment with unequal counts of θ and $\bar\theta$ cannot be saturated and vanishes — the Grassmann counterpart of the vanishing of odd bosonic moments in Theorem 4.19, sharpened to a separate conservation law for barred and unbarred indices (it is, at field level, fermion-number conservation). For equal counts n, apply $2n$ derivatives to the closed form $\det A\,\exp(\bar\eta A^{-1}\eta)$. Each surviving contribution pairs one $\partial_{\bar\eta_{i_a}}$ with one $\partial_{\eta_{j_b}}$ through a single factor $\bar\eta_{i_a}(A^{-1})_{i_a j_b}\eta_{j_b}$ of the expanded exponential, contributing the contraction $\langle\theta_{i_a}\bar\theta_{j_b}\rangle$ of (12.105); the different ways of matching the n unbarred derivatives to the n barred derivatives are the permutations $\sigma \in S_n$. Because the derivatives anticommute, reordering them to bring each matched pair together produces exactly the signature $\operatorname{sgn}(\sigma)$ — one factor of -1 for each transposition, i.e. for each *crossing* of two contraction lines. The signed sum over matchings is the Leibniz expansion of the determinant, giving (12.106). □

Physical Insight: determinant versus permanent, the combinatorial face of statistics

Bosonic and fermionic Wick's theorems are the same sum over pairings with one difference — the sign. The bosonic moment (4.66) is the *permanent* of the contraction matrix (every pairing enters with a $+$); the fermionic moment (12.106) is its *determinant* (each pairing weighted by $\operatorname{sgn}\sigma$).

Symmetry under exchange versus antisymmetry under exchange — Bose versus Fermi statistics — is, at the level of correlation functions, precisely permanent versus determinant. The minus sign on a crossing of two contraction lines is what forbids two identical fermions from occupying the same state: set two external labels equal and the determinant, having two equal rows, vanishes, which is the Pauli principle written as a Wick contraction.

Worked Example 12.33 (The four-point function and a closed fermion loop)**.** Take $n = 2$. Theorem 12.32 gives the four-point function as a 2×2 determinant of contractions,

$$\begin{aligned}\langle \theta_{i_1} \bar\theta_{j_1} \theta_{i_2} \bar\theta_{j_2} \rangle &= \langle \theta_{i_1} \bar\theta_{j_1} \rangle \langle \theta_{i_2} \bar\theta_{j_2} \rangle - \langle \theta_{i_1} \bar\theta_{j_2} \rangle \langle \theta_{i_2} \bar\theta_{j_1} \rangle \\ &= (A^{-1})_{i_1 j_1} (A^{-1})_{i_2 j_2} - (A^{-1})_{i_1 j_2} (A^{-1})_{i_2 j_1}, \end{aligned} \tag{12.107}$$

using (12.105): the "direct" pairing $(i_1 j_1)(i_2 j_2)$ enters with $+$, the "crossed" pairing $(i_1 j_2)(i_2 j_1)$ with $-$. Compare the bosonic four-point function (10.138), whose three terms all carry $+$: there the analogue of the crossed term is added, here it is subtracted. The relative minus is the entire content of fermion antisymmetry at one loop.

The closed-loop minus sign now follows by closing the indices into a cycle. A closed fermion loop of length n is assembled from n vertices, each a bilinear $\bar\theta_a(\cdots)\theta_a$ in which the adjoint stands first, with the loop contraction joining θ_a to $\bar\theta_{a+1}$ cyclically ($\bar\theta_{n+1} \equiv \bar\theta_1$). The relevant average is therefore

$$L_n = \langle \bar\theta_1 \theta_1 \, \bar\theta_2 \theta_2 \cdots \bar\theta_n \theta_n \rangle \Big|_{\text{loop}} = (-1)^n \langle \theta_1 \bar\theta_1 \, \theta_2 \bar\theta_2 \cdots \theta_n \bar\theta_n \rangle \Big|_{\text{loop}}, \tag{12.108}$$

the factor $(-1)^n$ from reversing each of the n even vertex blocks $\bar\theta_a \theta_a = -\theta_a \bar\theta_a$. By (12.106) the single connected loop is the cyclic permutation $\sigma = (1\,2\,\cdots\,n)$ with signature $\operatorname{sgn}(\sigma) = (-1)^{n-1}$, contributing $(-1)^{n-1} \prod_a (A^{-1})_{i_a i_{a+1}} = (-1)^{n-1} \operatorname{tr}[(A^{-1})^n]$. The two signs combine into a single overall minus, independent of the loop length:

$$L_n = (-1)^n (-1)^{n-1} \operatorname{tr}[(A^{-1})^n] = -\operatorname{tr}[(A^{-1})^n]. \tag{12.109}$$

Every closed fermion loop carries one minus sign and one trace, for any n — the $n = 1$ tadpole $\langle \bar\theta \theta \rangle = -\operatorname{tr}(A^{-1})$ being the smallest instance. This is the diagrammatic "-1 per fermion loop, trace around it" rule, derived here from nothing but the determinant structure of (12.106).

We have now computed the complete set of Gaussian Grassmann integrals: the source-free determinant (12.81), its real Pfaffian counterpart (12.92), the sourced generating identity (12.99), and the Wick contractions (12.105)–(12.106)

that read all higher moments off the propagator A^{-1} with a sign per crossing. Assembled, these are the finite-dimensional master formula

$$\int \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\; e^{-\bar{\psi}A\psi+\bar{\eta}\psi+\bar{\psi}\eta} = \det A\;\exp(\bar{\eta}A^{-1}\eta), \tag{12.110}$$

written here with the field letters $\psi, \bar{\psi}$ in anticipation. The next section derives the fermionic path integral from canonical quantization via coherent states, showing that the continuum measure $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ and the bilinear action $\bar{\psi}A\psi$ arise honestly from the operator formalism of Chapter 11 rather than by analogy; (12.110) is then promoted to fields in Section 12.6, with A built from the Dirac operator $i(\not{\partial} - m)$ (the precise placement of the factors of i and the $i\epsilon$ prescription fixed there), turning $Z[\bar{\eta}, \eta]$ into the Dirac generating functional and the contraction A^{-1} into the Feynman propagator S_F.

12.5 Fermionic coherent states and the path integral from canonical quantization

The Gaussian Berezin machinery of Section 12.4 gave us an algebra of integration over anticommuting variables and, as its central output, the Gaussian identity $\int \prod_i d\bar{\theta}_i\, d\theta_i\, e^{-\bar{\theta}M\theta} = \det M$ — the determinant in the numerator, the precise inversion of the bosonic $(\det M)^{-1/2}$. So far that identity is a piece of pure algebra, a fact about finitely many Grassmann generators. It does not yet say anything about a quantum field. The scalar story of Chapter 10 earned the right to write $\int \mathcal{D}\phi\, e^{iS}$ by *deriving* it from canonical quantization: one starts from the Hilbert space and the Hamiltonian, slices the evolution operator into infinitesimal steps, inserts a resolution of the identity at each step, and watches the sum over intermediate states condense into a sum over histories weighted by e^{iS}. Nothing was postulated; the path integral was an identity. We owe the Dirac field exactly the same honest derivation, and this section pays the debt. The integration variables will turn out to be Grassmann numbers, the measure $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ will be a product of Berezin measures, and the Gaussian identity just recalled will be what evaluates the free theory — but all of this will be *forced* by the operator formalism of Chapter 11, not adopted by analogy.

The one ingredient Chapter 10 had for free, and which we must build, is a continuous basis adapted to the creation and annihilation operators. For the scalar, position eigenstates $|x\rangle$ did the job: they are a continuous, overcomplete family in which $\hat{x}$ is diagonal, and the resolution of the identity $\int dx\; |x\rangle\langle x| = \mathbf{1}$ is what gets inserted between time slices. A fermionic mode has no position operator and a two-dimensional state space; the natural continuous family is instead the set of *coherent states*, the eigenstates of the annihilation operator. The reader has met their bosonic cousins in elementary quantum mechanics: for a harmonic oscillator, $|z\rangle = e^{za^\dagger}|0\rangle$ satisfies $a\,|z\rangle = z\,|z\rangle$ with $z \in \mathbb{C}$, and

$\frac{1}{\pi}\int d^2z\, e^{-|z|^2}\,|z\rangle\langle z| = \mathbf{1}$. We will construct the fermionic analogue. The single obstruction — the entire reason a fermionic path integral could never be an integral over ordinary numbers — appears at the very first step: the eigenvalue of a fermionic annihilation operator *cannot* be an ordinary number. It must be a Grassmann number. Once that is conceded, everything else follows mechanically, and the antiperiodic boundary condition that distinguishes fermionic statistics drops out of the trace at the end.

12.5.1 One fermionic mode

The quantized Dirac field of Chapter 11 is, mode by mode, a collection of independent fermionic oscillators: the ladder algebra Eq. (11.116) assigns to each momentum and spin an annihilation/creation pair obeying canonical anticommutation relations, and the full Fock space is the tensor product over modes (Theorem 11.29). The whole construction of this section therefore reduces to a *single* mode, after which we tensor over the (continuum of) modes and pass to the field. We isolate one such pair and drop all labels. Write $a, a^\dagger$ for the annihilation and creation operators of one fermionic mode, with the algebra extracted from Theorem 11.22,

$$\left\{a, a^\dagger\right\} = 1, \qquad a^2 = (a^\dagger)^2 = 0, \qquad \{a, a\} = \left\{a^\dagger, a^\dagger\right\} = 0. \tag{12.111}$$

The nilpotency $a^2 = 0$ is the Pauli principle (Theorem 11.27) in its sharpest form. The state space is exactly two-dimensional, spanned by the vacuum $|0\rangle$ and the one-particle state $|1\rangle$,

$$a\,|0\rangle = 0, \qquad |1\rangle \equiv a^\dagger\,|0\rangle\,, \qquad a\,|1\rangle = |0\rangle\,, \qquad a^\dagger\,|1\rangle = 0, \tag{12.112}$$

orthonormal, $\langle 0|0\rangle = \langle 1|1\rangle = 1$, $\langle 0|1\rangle = 0$, and the number operator $\hat{n} = a^\dagger a$ reads off the occupation, $\hat{n}\,|0\rangle = 0$, $\hat{n}\,|1\rangle = |1\rangle$. There is nothing else: every operator on this space is a linear combination of $\mathbf{1}$, a, $a^\dagger$, and $a^\dagger a$. This two-state system is small enough that every claim below can be, and will be, checked by direct matrix computation, which is precisely why the single mode is the right laboratory.

12.5.2 Coherent states require Grassmann eigenvalues

We want a state $|\theta\rangle$ obeying $a\,|\theta\rangle = \theta\,|\theta\rangle$. Suppose first, for the sake of the contradiction that motivates everything, that θ were an ordinary complex number. Write the most general state $|\theta\rangle = \alpha\,|0\rangle + \gamma\,|1\rangle$. Then $a\,|\theta\rangle = \gamma\,a\,|1\rangle = \gamma\,|0\rangle$, while $\theta\,|\theta\rangle = \theta\alpha\,|0\rangle + \theta\gamma\,|1\rangle$. Matching coefficients of the two basis vectors gives $\gamma = \theta\alpha$ from $|0\rangle$ and $0 = \theta\gamma$ from $|1\rangle$. The second equation forces either $\theta = 0$ or $\gamma = 0$; in both cases the eigenstate collapses to a multiple of $|0\rangle$ with eigenvalue zero. There is no nontrivial coherent state with a c-number eigenvalue. The

Pauli principle has killed it: because a lowers the only nonempty level straight to the vacuum, its sole c-number eigenvalue is 0.

The escape is to let θ be an *anticommuting* number. Adjoin to the theory a Grassmann algebra (Section 12.2.1) with generators $\theta, \bar{\theta}$ — two independent generators related by the conjugation of Section 12.2.6, not a complex number and its conjugate — declared to anticommute among themselves and, crucially, *with the fermionic operators and with the odd states they create*:

$$\theta^2 = \bar{\theta}^2 = 0, \quad \left\{\theta, \bar{\theta}\right\} = 0, \quad \{\theta, a\} = \left\{\theta, a^\dagger\right\} = 0, \quad \left\{\bar{\theta}, a\right\} = \left\{\bar{\theta}, a^\dagger\right\} = 0. \tag{12.113}$$

The bookkeeping rule that makes all of this consistent is a $\mathbb{Z}_2$ grading: θ, $\bar{\theta}$, a, $a^\dagger$, and the one-particle ket $|1\rangle = a^\dagger |0\rangle$ are all *odd*; $|0\rangle$, $\mathbf{1}$, and $\hat{n} = a^\dagger a$ are *even*; two odd objects anticommute, anything even commutes with everything. With θ odd, the equation $0 = \theta\gamma$ that doomed the c-number case is no longer forced, and a genuine coherent state exists.

Definition 12.34 (Fermionic coherent states)**.** For one fermionic mode Eq. (12.111) and a Grassmann generator θ obeying Eq. (12.113), the *(right) coherent state* is

$$|\theta\rangle \;\equiv\; e^{-\theta a^\dagger} |0\rangle \;=\; (1 - \theta a^\dagger) |0\rangle \;=\; |0\rangle - \theta |1\rangle , \tag{12.114}$$

the exponential truncating after one term because $(\theta a^\dagger)^2 = \theta a^\dagger \theta a^\dagger = -\theta^2 (a^\dagger)^2 = 0$. The corresponding *(left) coherent state* is its conjugate,

$$\langle\theta| \;\equiv\; \langle 0|\, e^{-a\bar{\theta}} \;=\; \langle 0| \,(1 - a\bar{\theta}) \;=\; \langle 0| - \langle 1|\, \bar{\theta}. \tag{12.115}$$

The two defining algebraic properties are immediate and we prove them in full, since every later manipulation rests on them.

Proposition 12.35 (Eigenvalue property and overlap)**.** *The coherent states satisfy*

$$a\,|\theta\rangle = \theta\,|\theta\rangle\,, \qquad \langle\theta|\, a^\dagger = \langle\theta|\,\bar{\theta}, \qquad \langle\theta|\theta'\rangle = e^{\bar{\theta}\theta'} = 1 + \bar{\theta}\theta'. \tag{12.116}$$

Proof. For the first relation, act with a on Eq. (12.114), using $a\,|0\rangle = 0$ and the anticommutation $a\theta = -\theta a$ from Eq. (12.113):

$$a\,|\theta\rangle = a\big(|0\rangle - \theta\,|1\rangle\big) = -a\,\theta\,|1\rangle = +\theta\, a\,|1\rangle = \theta\,|0\rangle\,. \tag{12.117}$$

On the other hand $\theta\,|\theta\rangle = \theta(|0\rangle - \theta\,|1\rangle) = \theta\,|0\rangle - \theta^2\,|1\rangle = \theta\,|0\rangle$, because $\theta^2 = 0$. The two right-hand sides agree, so $a\,|\theta\rangle = \theta\,|\theta\rangle$. The left eigenrelation is the conjugate computation: $\langle\theta|\, a^\dagger = (\langle 0| - \langle 1|\,\bar{\theta}) a^\dagger$, and $\langle 0|\, a^\dagger = (a\,|0\rangle)^\dagger = 0$, while $\langle 1|\,\bar{\theta} a^\dagger = -\langle 1|\, a^\dagger \bar{\theta} = -\langle 0|\,\bar{\theta}$ since $\langle 1|\, a^\dagger = (a\,|1\rangle)^\dagger = \langle 0|$; hence $\langle\theta|\, a^\dagger = \langle 0|\,\bar{\theta} = \langle\theta|\,\bar{\theta}$, the last step using $\langle\theta|\,\bar{\theta} = (\langle 0| - \langle 1|\,\bar{\theta})\bar{\theta} = \langle 0|\,\bar{\theta}$ because $\bar{\theta}^2 = 0$. For the overlap, multiply Eq. (12.115) into Eq. (12.114):

$$\langle\theta|\theta'\rangle = \big(\langle 0| - \langle 1|\,\bar{\theta}\big)\big(|0\rangle - \theta'\,|1\rangle\big) = \langle 0|0\rangle - \langle 0|\,\theta'\,|1\rangle - \langle 1|\,\bar{\theta}\,|0\rangle + \langle 1|\,\bar{\theta}\theta'\,|1\rangle\,. \tag{12.118}$$

The middle two terms vanish ($\langle 0|1\rangle = \langle 1|0\rangle = 0$); the first is 1 and the last is $\bar\theta\theta' \langle 1|1\rangle = \bar\theta\theta'$. Therefore $\langle\theta|\theta'\rangle = 1 + \bar\theta\theta'$, which is precisely $e^{\bar\theta\theta'}$ since the exponential terminates at first order. □

The structural content is worth pausing on. The coherent state Eq. (12.114) is a superposition of the empty and the filled state with a Grassmann coefficient; it is not a vector in the original Hilbert space but in the space tensored with the Grassmann algebra. Its norm is not a number but the even Grassmann element $\langle\theta|\theta\rangle = e^{\bar\theta\theta} = 1 + \bar\theta\theta$. This is exactly parallel to the bosonic $\langle z|z\rangle = e^{|z|^2}$, with the modulus-squared $|z|^2 = \bar z z$ replaced by the Grassmann bilinear $\bar\theta\theta$; the entire formalism is the bosonic one with every c-number promoted to a Grassmann number and every det inverted. That promotion is not cosmetic — it is what allows the closure relation below to carry the minus sign the Pauli principle demands.

12.5.3 The resolution of the identity

We can now build the object that the time-slicing will insert at every step. As with position eigenstates in Section 10.1, the coherent states are overcomplete — there are "more" of them than the dimension of the space — and they resolve the identity against a Gaussian Berezin measure.

Theorem 12.36 (Coherent-state completeness)**.** *With the Berezin conventions of Section 12.3 (so that $\int d\bar\theta\, d\theta\, e^{-\bar\theta\theta} = 1$), the fermionic coherent states resolve the identity on the one-mode Hilbert space:*

$$\int d\bar\theta\, d\theta\; e^{-\bar\theta\theta}\; |\theta\rangle\langle\theta| \;=\; |0\rangle\langle 0| + |1\rangle\langle 1| \;=\; \mathbf{1}. \tag{12.119}$$

Proof. Expand the projector $|\theta\rangle\langle\theta|$ from Eqs. (12.114) and (12.115), keeping the Grassmann factors in the order in which they appear:

$$|\theta\rangle\langle\theta| = (|0\rangle - \theta\,|1\rangle)(\langle 0| - \langle 1|\,\bar\theta) = |0\rangle\langle 0| - |0\rangle\langle 1|\,\bar\theta - \theta\,|1\rangle\langle 0| + \theta\,|1\rangle\langle 1|\,\bar\theta. \tag{12.120}$$

Multiply on the left by the Gaussian weight $e^{-\bar\theta\theta} = 1 - \bar\theta\theta$. Any product containing θ twice or $\bar\theta$ twice vanishes, so from the $-\bar\theta\theta$ piece only its product with the c-number-coefficient term $|0\rangle\langle 0|$ survives, while the three terms already carrying a single Grassmann factor survive only through multiplication by 1:

$$e^{-\bar\theta\theta}\,|\theta\rangle\langle\theta| = |0\rangle\langle 0| - |0\rangle\langle 1|\,\bar\theta - \theta\,|1\rangle\langle 0| + \theta\,|1\rangle\langle 1|\,\bar\theta - \bar\theta\theta\;|0\rangle\langle 0|\,. \tag{12.121}$$

Now integrate. The Berezin integral $\int d\bar\theta\, d\theta$ annihilates the constant and the two single-Grassmann terms (Section 12.3: an integral picks out the top form). Two terms survive. For the last term in Eq. (12.121), $\int d\bar\theta\, d\theta\,(-\bar\theta\theta) = -(-1) = +1$ in the stated convention, leaving $+\,|0\rangle\langle 0|$. For the fourth term, the even operator

$|1\rangle\langle 1|$ commutes through $\bar\theta$, so $\theta\,|1\rangle\langle 1|\,\bar\theta = \theta\bar\theta\;|1\rangle\langle 1|$, and $\int d\bar\theta\, d\theta\, \theta\bar\theta = +1$, leaving $+\,|1\rangle\langle 1|$. The sum is $|0\rangle\langle 0| + |1\rangle\langle 1|$, which by Eq. (12.112) is the identity on the two-dimensional space. □

The completeness relation Eq. (12.119) is the fermionic counterpart of $\int dx\; |x\rangle\langle x| = \mathbf{1}$, with one decisive difference: the Gaussian damping factor $e^{-\bar\theta\theta}$ is no decoration but part of the measure, exactly because the coherent states are not normalized to unity. It is itself a Gaussian Berezin integral, and the fact that it integrates to $\det(1) = 1$ rather than to the bosonic $1/\pi$ is the simplest instance of the determinant-versus-inverse-determinant contrast of Section 12.4. With Eq. (12.119) in hand we have the single tool the time-slicing needs.

12.5.4 Matrix elements of normal-ordered operators

The short-time kernel of the path integral is a coherent-state matrix element of the (nearly trivial) evolution over one slice, and the only operators we ever feed it are normal-ordered: in $:A(a^\dagger, a):$ every $a^\dagger$ stands to the left of every a. For such operators the matrix element is read off without any further integration.

Proposition 12.37 (Coherent matrix element of a normal-ordered operator)**.** *For any operator written in normal-ordered form* $:A(a^\dagger, a):$*,*

$$\Big\langle \theta \Big| :A(a^\dagger, a): \Big| \theta' \Big\rangle = A(\bar\theta, \theta')\, e^{\bar\theta\theta'}. \tag{12.122}$$

Proof. In a normal-ordered monomial every $a^\dagger$ sits to the left and acts on the bra by the left eigenrelation $\langle\theta|\, a^\dagger = \bar\theta\,\langle\theta|$ (Eq. (12.116)); every a sits to the right and acts on the ket by $a\,|\theta'\rangle = \theta'\,|\theta'\rangle$. Each replacement converts an operator into its Grassmann eigenvalue, $a^\dagger \to \bar\theta$ and $a \to \theta'$, and because the operators were already in the order "all $a^\dagger$, then all a" the Grassmann eigenvalues appear in the same order, reproducing the symbol $A(\bar\theta, \theta')$ with no reordering signs. What remains after all operators are eaten is the overlap $\langle\theta|\theta'\rangle = e^{\bar\theta\theta'}$ of Eq. (12.116). □

The one matrix element we will actually need, for the free Hamiltonian $\hat H = \omega\, a^\dagger a$ of a single mode (already normal-ordered, the $\hat H$ of Section 11.4 restricted to one mode), is the case $A = \omega\, a^\dagger a$:

$$\Big\langle \theta \Big| \hat H \Big| \theta' \Big\rangle = \omega\, \bar\theta\theta'\, e^{\bar\theta\theta'}. \tag{12.123}$$

It is worth confirming Eq. (12.123) by the bare-handed route, both as a check on Theorem 12.37 and because the same manipulation will recur. Acting $a^\dagger a$ to the right, $a^\dagger a\,|\theta'\rangle = a^\dagger\theta'\,|\theta'\rangle = -\theta' a^\dagger\,|\theta'\rangle$, so $\Big\langle \theta \Big| a^\dagger a \Big| \theta' \Big\rangle = -\theta' \Big\langle \theta \Big| a^\dagger \Big| \theta' \Big\rangle = -\theta'\bar\theta\;\langle\theta|\theta'\rangle = \bar\theta\theta'\, e^{\bar\theta\theta'}$, where the final sign flip is the anticommutation $-\theta'\bar\theta = +\bar\theta\theta'$. The two routes agree.

12.5.5 The trace, and where the minus sign comes from

The partition function and every thermal or vacuum amplitude is a trace of an operator, and the path integral is built by writing that trace in the coherent-state basis. Here a single sign — absent in the bosonic case — enters, and it is the algebraic seed of fermionic statistics.

Theorem 12.38 (Coherent-state trace formula)**.** *For any operator $\hat{A}$ with even Grassmann parity (in particular any function of $\hat{n}$, and any $e^{-\beta\hat{H}}$),*

$$\operatorname{Tr}\hat{A} \;=\; \int d\bar{\theta}\, d\theta\; e^{-\bar{\theta}\theta}\, \Big\langle -\theta\Big|\hat{A}\Big|\theta\Big\rangle . \tag{12.124}$$

The argument of the bra carries the crucial extra minus sign: it is $\langle -\theta|$, not $\langle\theta|$.

Proof. The trace over the two-dimensional space is $\operatorname{Tr}\hat{A} = A_{00} + A_{11}$ with $A_{mn} = \big\langle m\big|\hat{A}\big|n\big\rangle$. We evaluate the right-hand side of Eq. (12.124) and show it equals this. From $\langle -\theta| = \langle 0| + \langle 1|\,\bar{\theta}$ (replace $\theta \to -\theta$, hence $\bar{\theta} \to -\bar{\theta}$, in Eq. (12.115)) and $|\theta\rangle = |0\rangle - \theta\,|1\rangle$,

$$\Big\langle -\theta\Big|\hat{A}\Big|\theta\Big\rangle = (\langle 0| + \langle 1|\,\bar{\theta})\hat{A}(|0\rangle - \theta\,|1\rangle) = A_{00} - \theta A_{01} + A_{10}\bar{\theta} - A_{11}\,\bar{\theta}\theta, \tag{12.125}$$

where in the last term $\hat{A}$ even lets $\bar{\theta}$ commute through, $\langle 1|\,\bar{\theta}\hat{A}\theta\,|1\rangle = A_{11}\bar{\theta}\theta$, and the leading sign comes from $\hat{A}(-\theta)$. Multiplying by $e^{-\bar{\theta}\theta} = 1 - \bar{\theta}\theta$ and integrating, only two terms survive the Berezin integral: $\int d\bar{\theta}\, d\theta\,(-A_{11}\bar{\theta}\theta) = +A_{11}$ from the bulk term and $\int d\bar{\theta}\, d\theta\,(-\bar{\theta}\theta\, A_{00}) = +A_{00}$ from the cross term of $(-\bar{\theta}\theta)$ with A_{00}. The single-Grassmann terms integrate to zero. The total is $A_{00} + A_{11} = \operatorname{Tr}\hat{A}$. □

The minus sign is not optional, and the cleanest way to see this is to compute what the *naive* choice $\langle +\theta|$ would give. Repeating Eq. (12.125) without the sign flip yields $\big\langle \theta\big|\hat{A}\big|\theta\big\rangle = A_{00} - \theta A_{01} - A_{10}\bar{\theta} + A_{11}\bar{\theta}\theta$, and the same Berezin integration now produces $A_{00} - A_{11}$ — the *supertrace* $\operatorname{Str}\hat{A}$, which counts the filled state with a relative minus and is not the quantity we want.

$$\int d\bar{\theta}\, d\theta\; e^{-\bar{\theta}\theta}\, \Big\langle +\theta\Big|\hat{A}\Big|\theta\Big\rangle = A_{00} - A_{11} = \operatorname{Str}\hat{A}. \tag{12.126}$$

The genuine trace is recovered only by reversing the sign of the bra's Grassmann label. For bosonic coherent states the analogous formula is $\operatorname{Tr}\hat{A} = \frac{1}{\pi}\int d^2z\, e^{-|z|^2}\big\langle z\big|\hat{A}\big|z\big\rangle$ with the *same* z on both sides — no sign flip. The fermionic minus sign in Eq. (12.124) is therefore a genuinely new feature with no bosonic precedent, and we will see in Section 12.5.7 that it is precisely what forces fields to be antiperiodic in imaginary time.

Physical Insight — the sign is the statistics

Everything that distinguishes fermions from bosons in the path-integral formalism descends from a single algebraic fact: tracing a fermionic operator in the coherent-state basis costs a sign that tracing a bosonic operator does not. We have just isolated that sign in its barest form, on a two-state system, where it is nothing more than the statement $\int d\bar\theta\, d\theta\, \theta\bar\theta = +1$ together with the eigenvalue flip $\langle -\theta|$. Downstream it becomes the antiperiodic boundary condition, the Matsubara frequencies $(2n+1)\pi/\beta$, the minus sign on every closed fermion loop, and — traced all the way back through Section 12.4 — the determinant in the numerator. They are one sign wearing different clothes.

12.5.6 Time-slicing: the path integral for one mode

We now assemble the path integral by the method of Section 10.1: write the evolution operator as a product of many short-time factors and insert the resolution of the identity Eq. (12.119) between each pair. The target is the transition amplitude between coherent states,

$$\mathcal{K}(\bar\theta_f, \theta_i; T) \equiv \left\langle \theta_f \middle| e^{-i\hat H T} \middle| \theta_i \right\rangle , \tag{12.127}$$

with $\hat H = \omega\, a^\dagger a$ the one-mode Hamiltonian and the boundary data $\theta_i, \bar\theta_f$ fixed Grassmann numbers (the analogues of the fixed endpoints x_i, x_f in Eq. (10.16)). Slice T into N steps of length $\epsilon = T/N$ and write $e^{-i\hat H T} = \left(e^{-i\epsilon \hat H}\right)^N$. Between the N factors insert $N-1$ copies of Eq. (12.119), labelling the inserted variables $\theta_1, \dots, \theta_{N-1}$ and setting $\theta_0 \equiv \theta_i$, $\theta_N \equiv \theta_f$. The amplitude becomes a chain of short-time kernels,

$$\mathcal{K} = \int \prod_{k=1}^{N-1} d\bar\theta_k\, d\theta_k\; e^{-\sum_{k=1}^{N-1} \bar\theta_k \theta_k} \prod_{k=0}^{N-1} \left\langle \theta_{k+1} \middle| e^{-i\epsilon \hat H} \middle| \theta_k \right\rangle . \tag{12.128}$$

Each short-time kernel is evaluated to first order in ϵ by Theorem 12.37: since $\hat H$ is normal-ordered, $e^{-i\epsilon \hat H} =: e^{-i\epsilon \hat H} : + O(\epsilon^2)$ (the reordering corrections are higher order in ϵ, exactly as the Trotter error is in Theorem 10.2), so

$$\left\langle \theta_{k+1} \middle| e^{-i\epsilon \hat H} \middle| \theta_k \right\rangle = e^{\bar\theta_{k+1}\theta_k}\, e^{-i\epsilon\, H(\bar\theta_{k+1}, \theta_k)} (1 + O(\epsilon^2)), \qquad H(\bar\theta, \theta) = \omega\, \bar\theta\theta. \tag{12.129}$$

Collecting the exponents from the N kernels Eq. (12.129) and the $N-1$ Gaussian measure factors, the integrand is e^{S_N} with

$$S_N = \sum_{k=0}^{N-1} \bar\theta_{k+1}\theta_k - \sum_{k=1}^{N-1} \bar\theta_k \theta_k - i\epsilon \sum_{k=0}^{N-1} H(\bar\theta_{k+1}, \theta_k). \tag{12.130}$$

The first two sums combine into a discrete time derivative. Peel off the $k = N-1$ boundary term from the first sum and relabel:

$$\sum_{k=0}^{N-1} \bar\theta_{k+1}\theta_k - \sum_{k=1}^{N-1} \bar\theta_k\theta_k = \bar\theta_N\theta_{N-1} - \sum_{k=1}^{N-1} \bar\theta_k(\theta_k - \theta_{k-1}). \tag{12.131}$$

As $N \to \infty$ with T fixed, the intermediate variables become a Grassmann-valued path $\theta_k \to \psi(t)$, $\bar\theta_k \to \bar\psi(t)$, $t = k\epsilon$, the difference $\theta_k - \theta_{k-1} \to \epsilon\,\dot\psi(t)$, the sums become integrals $\epsilon\sum \to \int_0^T dt$, and the boundary term is $\bar\theta_N\theta_{N-1} \to \bar\theta_f\,\psi(T)$. The discrete action Eq. (12.130) tends to

$$S_N \;\longrightarrow\; \bar\theta_f\,\psi(T) - \int_0^T dt\,\bar\psi\,\dot\psi - i\int_0^T dt\,H(\bar\psi,\psi) = \bar\theta_f\,\psi(T) + i\int_0^T dt\;\bar\psi(i\partial_t - \omega)\psi, \tag{12.132}$$

where the bulk terms were combined using $-\bar\psi\dot\psi = i(i\bar\psi\dot\psi)$. We have arrived at the continuum coherent-state path integral.

Theorem 12.39 (Fermionic path integral from canonical quantization)**.** *The coherent-state transition amplitude Eq. (12.127) of a fermionic mode is the Berezin functional integral*

$$\Big\langle \theta_f \Big| e^{-i\hat H T} \Big| \theta_i \Big\rangle = \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\Big(\bar\theta_f\,\psi(T) + i\int_0^T dt\;\bar\psi(t)(i\partial_t - H)\psi(t)\Big), \tag{12.133}$$

the integral running over all Grassmann paths $\psi(t), \bar\psi(t)$ on $[0,T]$ with the boundary conditions $\psi(0) = \theta_i$ and $\bar\psi(T) = \bar\theta_f$ fixed by the external coherent states, and the measure $\mathcal{D}\bar\psi\,\mathcal{D}\psi \equiv \prod_k d\bar\theta_k\,d\theta_k$ the continuum limit of the product of Berezin measures. For the free mode $H = \omega\,\bar\psi\psi$; for an interacting normal-ordered Hamiltonian, $H(\bar\psi,\psi)$ is its symbol.

This is the central result. It was obtained, exactly as the scalar result of Theorem 10.4, by an identity — inserting completeness between slices and evaluating elementary matrix elements — not by analogy. The bulk weight is e^{iS} with the first-order action

$$S[\bar\psi,\psi] = \int dt\;\bar\psi\,(i\partial_t - H)\psi, \tag{12.134}$$

the Lagrangian $\bar\psi\,i\partial_t\psi - \bar\psi H\psi$ being precisely the Legendre structure of a first-order system: the canonical momentum conjugate to ψ is $\bar\psi$ itself (the operator statement of which was $\pi_\psi = i\psi^\dagger$ in Section 11.1.4), so there is no separate $\dot{\bar\psi}$ kinetic term. The boundary term $\bar\theta_f\psi(T)$ is the price of holding a first-order system fixed at both ends; it will be absorbed automatically when we form the trace, and it never appears in a correlation function.

12.5.7 The trace, antiperiodicity, and KMS

The transition amplitude Eq. (12.133) still depends on external boundary data. Physical quantities are traces, and forming the trace is where fermionic statistics announces itself. Apply the trace formula Eq. (12.124) to the evolution operator. We must set $\theta_f = -\theta_i$ and integrate the endpoint over a final Gaussian measure:

$$\mathrm{Tr}\; e^{-i\hat{H}T} = \int d\bar{\theta}_0\, d\theta_0\; e^{-\bar{\theta}_0\theta_0}\; \Big\langle -\theta_0 \Big| e^{-i\hat{H}T} \Big| \theta_0 \Big\rangle . \tag{12.135}$$

The single instruction "$\langle -\theta_0|$ at the top, $|\theta_0\rangle$ at the bottom" is the statement that the path closes up onto *minus* its starting value:

$$\psi(T) = -\psi(0), \qquad \bar{\psi}(T) = -\bar{\psi}(0). \tag{12.136}$$

With the endpoint now integrated, the boundary term $\bar{\theta}_f\psi(T)$ of Eq. (12.133) merges with the final Gaussian factor $e^{-\bar{\theta}_0\theta_0}$ and is reabsorbed into the bulk action evaluated on closed antiperiodic paths; nothing of it is left over. The trace is therefore a functional integral over *antiperiodic* Grassmann fields, with no surviving boundary data:

$$\mathrm{Tr}\; e^{-i\hat{H}T} = \oint_{\mathrm{AP}} \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\; \exp\Big(i\int_0^T dt\; \bar{\psi}(i\partial_t - H)\psi \Big), \qquad \psi(T) = -\psi(0). \tag{12.137}$$

The contrast with the scalar is exact and instructive. The bosonic trace formula carries the *same* label on both coherent states, so the bosonic path closes onto $+$ its start: bosonic fields obey *periodic* boundary conditions, $\phi(T) = \phi(0)$, as the closed Euclidean cylinder of Section 10.6.5 required. The fermionic minus sign of Eq. (12.124) flips periodic to antiperiodic. This is the path-integral face of spin and statistics: the same anticommutators that forced antisymmetric Fock space and the Pauli principle in Chapter 11 reappear here as a sign on the boundary condition.

The physics of Eq. (12.136) is sharpest in imaginary time. Following the Wick rotation of Section 10.6, set $T = -i\beta$, i.e. $t = -i\tau$ with $\tau \in [0, \beta]$; the evolution operator becomes the Boltzmann operator $e^{-\beta\hat{H}}$ and its trace is the thermal partition function $Z(\beta) = \mathrm{Tr}\; e^{-\beta\hat{H}}$ at inverse temperature β. Under $t = -i\tau$ the bulk weight rotates exactly as in Eq. (10.160): $i\int dt\,\bar{\psi} i\partial_t\psi \to -\int_0^\beta d\tau\,\bar{\psi}\partial_\tau\psi$ and $-i\int dt\, H \to -\int_0^\beta d\tau\, H$, so

$$Z(\beta) = \mathrm{Tr}\; e^{-\beta\hat{H}} = \oint_{\mathrm{AP}} \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\; e^{-S_E[\bar{\psi},\psi]}, \qquad S_E = \int_0^\beta d\tau\; \bar{\psi}(\partial_\tau + H)\psi, \tag{12.138}$$

the field now living on a circle of circumference β with the antiperiodic gluing $\psi(\tau+\beta) = -\psi(\tau)$ (Fig. 12.2). The boson on the same cylinder is glued periodically, $\phi(\tau+\beta) = \phi(\tau)$, exactly the compactified Euclidean time of Section 10.6.5.

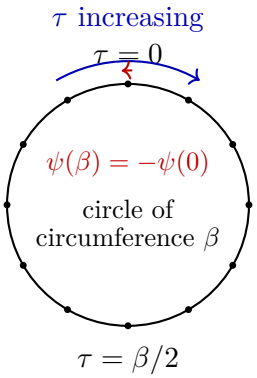


Figure 12.2: The imaginary-time path integral lives on a circle of circumference $\beta = 1/T$ (the Wick-rotated time of Section 10.6). Dots are the time slices of Eq. (12.128). The trace glues the two ends, $\tau = 0$ to $\tau = \beta$; for a boson the gluing is periodic, for a fermion it is *antiperiodic*, $\psi(\beta) = -\psi(0)$, the geometric form of the minus sign in the coherent-state trace Eq. (12.124). Expanding an antiperiodic field in Fourier modes on the circle forces the fermionic Matsubara frequencies $\omega_n = (2n+1)\pi/\beta$.

The antiperiodicity is exactly the *Kubo–Martin–Schwinger* (KMS) condition specialized to fermions, and it is worth stating the argument once because it is the structural reason the boundary condition is forced by thermal equilibrium and not merely by our bookkeeping. For any two Heisenberg operators A, B and the thermal average $\langle \cdots \rangle_\beta = Z^{-1} \operatorname{Tr}(e^{-\beta \hat{H}} \cdots)$, cyclicity of the trace gives

$$\langle A(\tau) B(0) \rangle_\beta = Z^{-1} \operatorname{Tr}\!\left(e^{-\beta \hat{H}} e^{\hat{H}\tau} A e^{-\hat{H}\tau} B\right) = \pm \langle B(0) A(\tau - \beta) \rangle_\beta , \qquad (12.139)$$

where the upper sign holds for bosonic operators and the lower for fermionic ones, the sign being the cost of moving the fermionic B past the fermionic A under the trace. The thermal two-point function is thus periodic in imaginary time for bosons and antiperiodic for fermions with period β — which is precisely the boundary condition on the field that the coherent-state derivation produced from the bra-label flip. The two derivations agree because they are the same statistics seen from two sides: Eq. (12.124) is the microscopic minus sign, Eq. (12.139) is its thermodynamic shadow. This antiperiodicity is the single most consequential output of the section for later chapters: it is the entire content of the imaginary-time (Matsubara) formalism of finite-temperature field theory, and it is what makes fermions on a Euclidean lattice obey antiperiodic boundary conditions in the time direction (Chapter 43). We flag it here as the rigorous underpinning, derived rather than postulated.

Worked Example 12.40 (The single-mode partition function, three ways)**.** Nothing certifies the construction like computing one number by independent

routes and finding them equal. Take the one-mode free Hamiltonian $\hat{H} = \omega\, a^\dagger a$ and compute $Z(\beta) = \operatorname{Tr} e^{-\beta \hat{H}}$ (i) from the operator spectrum, (ii) from the *discrete* antiperiodic Berezin integral with finite N, and (iii) from the continuum determinant.

(i) Operator route. The spectrum is $\{0, \omega\}$ on $\{|0\rangle, |1\rangle\}$, so directly

$$Z(\beta) = \left\langle 0 \middle| e^{-\beta \hat{H}} \middle| 0 \right\rangle + \left\langle 1 \middle| e^{-\beta \hat{H}} \middle| 1 \right\rangle = 1 + e^{-\beta\omega}. \tag{12.140}$$

This is the answer the path integral must reproduce.

(ii) Discrete Berezin route. Discretize Eq. (12.138) on N slices, $\epsilon = \beta/N$. Reading the Euclidean discrete action off Eq. (12.130) with the rotation $i\epsilon H \to \epsilon H$ and imposing the antiperiodic closure $\theta_N = -\theta_0$, $\bar{\theta}_N = -\bar{\theta}_0$, the integrand is $\exp(-\bar{\theta}_j M_{jk} \theta_k)$ over the N variables $\theta_0, \ldots, \theta_{N-1}$, with

$$M = \mathbf{1} - a\,\mathsf{S}, \qquad a \equiv 1 - \epsilon\omega, \tag{12.141}$$

where S is the antiperiodic shift: $\mathsf{S}_{k,k-1} = 1$ on the subdiagonal and $\mathsf{S}_{0,N-1} = -1$ in the corner, the corner minus coming from $\bar{\theta}_N = -\bar{\theta}_0$. By the Gaussian Grassmann identity of Eq. (12.81), the integral is $\det M$ — the determinant in the numerator, the hallmark of a fermion. Since $\mathsf{S}^N = -\mathbf{1}$, the eigenvalues of S are the N roots of $\mu^N = -1$, and its characteristic polynomial is $\det(x\mathbf{1} - \mathsf{S}) = x^N + 1$. Therefore

$$\begin{aligned} Z_N = \det(\mathbf{1} - a\,\mathsf{S}) &= \prod_{\mu^N = -1} (1 - a\mu) = a^N \prod_{\mu^N = -1} (a^{-1} - \mu) \\ &= a^N (a^{-N} + 1) = 1 + a^N = 1 + (1 - \epsilon\omega)^N. \end{aligned} \tag{12.142}$$

Taking $N \to \infty$ at fixed β, $(1 - \beta\omega/N)^N \to e^{-\beta\omega}$, so $Z_N \to 1 + e^{-\beta\omega}$, matching Eq. (12.140) exactly. The calculation is fully finite at every stage; no regularization was needed because the discrete determinant is an honest polynomial identity.

(iii) Continuum determinant and Matsubara modes. In the continuum, Eq. (12.138) is the Gaussian integral of $\bar{\psi}(\partial_\tau + \omega)\psi$ over antiperiodic fields, so $Z = \det(\partial_\tau + \omega)$. Antiperiodicity, $\psi(\tau + \beta) = -\psi(\tau)$, is solved by the modes $e^{-i\omega_n \tau}$ with the *fermionic Matsubara frequencies*

$$\omega_n = \frac{(2n+1)\pi}{\beta}, \qquad n \in \mathbb{Z}, \tag{12.143}$$

on which $\partial_\tau + \omega$ has eigenvalues $-i\omega_n + \omega$, whence formally $Z = \prod_n (\omega - i\omega_n)$. This infinite product needs the regularization the discrete route bypassed; carrying it out (dividing by the $\omega = 0$ value and using $\prod_{n \geq 0} [1 + x^2/(2n+1)^2\pi^2] = \cosh(x/2)$) gives $Z \propto \cosh(\beta\omega/2) \propto 1 + e^{-\beta\omega}$, the same answer once more. The odd integers $2n + 1$ in Eq. (12.143) are the direct imprint of the antiperiodic gluing; a boson, periodic, would carry the even frequencies $2n\pi/\beta$.

The bosonic comparison sharpens the lesson and ties the section back to Section 12.4. The same single-mode partition function for a *bosonic* oscillator is $Z_{\mathrm{B}} = \sum_{n\geq 0} e^{-\beta\omega n} = (1-e^{-\beta\omega})^{-1}$, reproduced by a path integral over *periodic* ordinary-number fields whose Gaussian gives $\det(\partial_\tau+\omega)^{-1}$ — the inverse determinant, with the periodic (even) Matsubara modes. Side by side,

$$Z_{\mathrm{F}} = \det(\partial_\tau+\omega) = 1+e^{-\beta\omega}, \qquad Z_{\mathrm{B}} = \det(\partial_\tau+\omega)^{-1} = \frac{1}{1-e^{-\beta\omega}}, \tag{12.144}$$

the two facts that distinguish the statistics — det versus $\det^{-1}$, and antiperiodic versus periodic — are visibly the same fact, the coherent-state trace sign, expressed in the measure and in the boundary condition respectively. The numerator determinant the reader first met as the bare algebraic output of Section 12.4 has now acquired its physical meaning: it is the fermionic partition function, and the minus sign it secretly carries is the Pauli principle.

12.5.8 From one mode to the Dirac field

The passage to the field is now purely a matter of restoring the labels we suppressed in Section 12.5.1. The Dirac field is a continuum of independent fermionic modes, one annihilation/creation pair $\{a, a^\dagger\} = 1$ for each momentum, spin, and particle/antiparticle channel, all tensored together (Eq. (11.116), Theorem 11.29). The coherent state of the whole field is the tensor product of single-mode coherent states, labelled now by a Grassmann *field* $\psi(\mathbf{x})$ (and its conjugate $\bar\psi(\mathbf{x})$) rather than a single Grassmann number; the resolution of the identity Eq. (12.119) becomes a Berezin functional integral $\int \mathcal{D}\bar\psi\,\mathcal{D}\psi\, e^{-\int d^3x\,\bar\psi\psi}\,|\psi\rangle\langle\psi| = \mathbf{1}$; and the trace formula Eq. (12.124) carries the antiperiodic identification Eq. (12.136) for every mode at once. Repeating the time-slicing of Section 12.5.6 mode by mode, the bulk action Eq. (12.134) becomes

$$S = \int dt\,\Big[\int d^3x\;\bar\psi\, i\partial_t\psi \;-\; \hat H[\bar\psi,\psi]\Big], \qquad \mathcal{H}_{\mathrm{D}} = \psi^\dagger(-i\gamma^0\boldsymbol{\gamma}\cdot\nabla + \gamma^0 m)\psi, \tag{12.145}$$

with $\mathcal{H}_{\mathrm{D}}$ the normal-ordered Dirac Hamiltonian density of Section 11.3.4 (its symbol obtained by the rule of Theorem 12.37, $\psi^\dagger \to \bar\psi\gamma^0$, $\psi\to\psi$). The Legendre structure collapses exactly as in the single mode: using $\bar\psi = \psi^\dagger\gamma^0$, the time-derivative term is $\psi^\dagger i\partial_0\psi = \bar\psi\, i\gamma^0\partial_0\psi$, and the Hamiltonian density rewrites as $\bar\psi(-i\gamma^i\partial_i + m)\psi$, so that

$$\bar\psi\, i\partial_t\psi - \mathcal{H}_{\mathrm{D}} = \bar\psi\, i\gamma^0\partial_0\psi + \bar\psi(i\gamma^i\partial_i - m)\psi = \bar\psi(i\partial\!\!\!/ - m)\psi. \tag{12.146}$$

The coherent-state Lagrangian is exactly the Dirac Lagrangian $\bar\psi(i\partial\!\!\!/ - m)\psi$ of Eq. (11.25). The continuum path integral for the free Dirac field is therefore

$$Z = \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\Big(i\int d^4x\;\bar\psi(i\partial\!\!\!/ - m)\psi\Big), \tag{12.147}$$

the measure $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ a product of Berezin measures over independent Grassmann fields $\psi(x), \bar{\psi}(x)$, treated as the four-component spinor fields of Chapter 11 but valued in a Grassmann algebra. We have derived Eq. (12.147) from the operator theory of Chapter 11, with the measure defined, the weight e^{iS} explained, and — when a trace is taken — the antiperiodic boundary condition Eq. (12.136) established. Nothing here was postulated by analogy with the scalar; the analogy is now a theorem.

This is the object the next section consumes. With the measure $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ and the action Eq. (12.147) in hand, Section 12.6 couples Grassmann sources $\bar{\eta}, \eta$ to the field, performs the functional Gaussian integral of Section 12.4 to write down the generating functional $Z[\bar{\eta}, \eta]$, identifies its normalization as $\det(i\not{\partial} - m)$, and extracts by functional differentiation the fermion propagator — which, because the sources anticommute, will reappear carrying exactly the minus sign of the canonical S_F of Theorem 11.38.

12.6 The Dirac generating functional and the fermion propagator

The finite-dimensional sourced Gaussian of Section 12.4 gave us, for anticommuting variables, a master formula whose entire content was a single matrix inverse: the sourced integral Eq. (12.99) returned the determinant of the kinetic matrix times the exponential of a quadratic in the sources, with the kernel of that quadratic the *inverse* of the kinetic matrix. Section 12.5 then promoted those Grassmann numbers from a formal device to the genuine integration variables of a path integral, derived from the operator formalism of Chapter 11 through fermionic coherent states exactly as Chapter 10 derived the bosonic path integral from the scalar coherent states. We now hold both pieces: a Gaussian master formula and a legitimate functional integral to apply it to. This section assembles them. We write the Dirac generating functional $Z[\eta, \bar{\eta}]$, perform its functional Gaussian integral in closed form, read off the normalization as a functional determinant and the source kernel as the Feynman propagator S_F, and recover from it—by two anticommuting functional derivatives—the time-ordered two-point function $\langle 0|\mathcal{T}\,\psi(x)\bar{\psi}(y)|0\rangle$ built operator-theoretically in Chapter 11. The reward is that the minus sign which the Pauli principle and the antisymmetry of S_F demand is no longer something to be inserted by hand: it falls out of the anticommutation of the sources, automatically and in the right place. The free electron propagator of QED reappears here as a path-integral object, in exact agreement with the canonical S_F frozen for the rest of the book.

12.6.1 The generating functional with Grassmann sources

The bosonic generating functional $Z[J]$ of Section 4.4 and Eq. (10.86) was built by coupling the field ϕ to an external c-number source J through a term $J\phi$ in the action, so that functional differentiation with respect to J pulls factors of ϕ down from the exponent. For the Dirac field the same strategy must respect the algebra: ψ and $\bar{\psi}$ are anticommuting (Grassmann-valued) fields, and a source term must be Grassmann-even for e^{iS} to make sense. The unique local bilinear options are $\bar{\eta}\psi$ and $\bar{\psi}\eta$ with the sources $\eta(x)$ and $\bar{\eta}(x)$ themselves anticommuting spinors—independent Grassmann generators carrying one Dirac index each. Each source term is then a product of two odd objects and hence even, as required. The two sources are genuinely independent: η couples to $\bar{\psi}$ and $\bar{\eta}$ to ψ, so differentiating with respect to η extracts a $\bar{\psi}$ and differentiating with respect to $\bar{\eta}$ extracts a ψ.

Definition 12.41 (Dirac generating functional)**.** For the free Dirac field of Chapter 11, with independent anticommuting spinor sources $\eta(x), \bar{\eta}(x)$, the *generating functional* is

$$Z[\eta,\bar{\eta}] \;=\; \int \mathcal{D}\bar{\psi}\,\mathcal{D}\psi \;\exp\Big[\, i\int d^4x\,\big(\,\bar{\psi}(i\partial\!\!\!/ - m)\psi + \bar{\eta}\psi + \bar{\psi}\eta\,\big)\Big], \tag{12.148}$$

where $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ is the Berezin field measure of Section 12.5, and every product of fields and sources is understood with its Grassmann ordering kept intact.

The action in the exponent is exactly the free Dirac action $S_0 = \int d^4x\,\bar{\psi}(i\partial\!\!\!/ - m)\psi$ whose stationarity is the Dirac equation $(i\partial\!\!\!/ - m)\psi = 0$ of Section 11.1.2, now with the two linear source couplings appended. The convergence tilt $m^2 \to m^2 - i\varepsilon$ that the Wick rotation of Section 10.6 imposed on the bosonic measure is inherited here in the same way; we keep it implicit until it is needed to fix the propagator's contour below. As in the bosonic case, all physical content lives in *normalized* derivatives—ratios $Z[\eta,\bar{\eta}]/Z[0]$—so any field-independent constant multiplying Eq. (12.148) is immaterial and may be absorbed into the measure.

12.6.2 Performing the functional Gaussian integral

The integral Eq. (12.148) is Gaussian in the Grassmann fields: the exponent is bilinear in $\psi, \bar{\psi}$ plus terms linear in them. It is therefore the field-space instance of the finite-dimensional sourced Gaussian Eq. (12.99), which we recall in the form

$$\int \Big(\prod_i d\bar{\theta}_i\, d\theta_i\Big)\, e^{-\bar{\theta}_i A_{ij}\theta_j + \bar{\zeta}_i\theta_i + \bar{\theta}_i\zeta_i} \;=\; (\det A)\, e^{\,\bar{\zeta}_i (A^{-1})_{ij}\zeta_j}, \tag{12.149}$$

proved in Theorem 12.26 by completing the square in the anticommuting variables. The decisive structural fact, established there, is that the Gaussian integral over

Grassmann variables produces $\det A$ in the numerator—not the $(\det A)^{-1/2}$ of the bosonic Gaussian. This single inversion of power is the algebraic seed of every minus sign on a closed fermion loop, and Section 12.7 is devoted to its consequences; here we use Eq. (12.149) as an established identity.

To apply it, discretize spacetime exactly as the bosonic measure was discretized in Section 10.6: replace the continuum index x by a lattice index i, $\int d^4x \to \sum_i \Delta V$ with ΔV the cell four-volume, and read $\psi(x) \to \theta_i$, $\bar\psi(x) \to \bar\theta_i$. The continuum and discrete dictionaries are

$$-\bar\theta_i A_{ij}\theta_j \;\longleftrightarrow\; i\int d^4x\,\bar\psi(i\not\partial - m)\psi, \qquad \bar\zeta_i\theta_i \;\longleftrightarrow\; i\int d^4x\,\bar\eta\psi, \qquad \bar\theta_i\zeta_i \;\longleftrightarrow\; i\int d^4x\,\bar\psi\eta. \tag{12.150}$$

Reading off the kinetic operator and the effective sources,

$$A \;=\; -\,i\,(i\not\partial - m), \qquad \bar\zeta \;=\; i\,\bar\eta, \qquad \zeta \;=\; i\,\eta, \tag{12.151}$$

where A is now an operator on the space of Dirac spinor fields (a matrix in both spacetime and spinor indices). Substituting Eq. (12.151) into Eq. (12.149), the source exponent is

$$\bar\zeta\, A^{-1}\zeta \;=\; (i\bar\eta)\, A^{-1}(i\eta) \;=\; -\,\bar\eta\, A^{-1}\,\eta, \tag{12.152}$$

and the prefactor is $\det A$. It remains to identify A^{-1}. From Eq. (12.151),

$$A^{-1} \;=\; \big[-i(i\not\partial - m)\big]^{-1} \;=\; \frac{1}{-i}\,(i\not\partial - m)^{-1} \;=\; i\,(i\not\partial - m)^{-1}. \tag{12.153}$$

The remaining inverse is precisely the Green's function of the Dirac operator. Equation (11.204) established that the Feynman propagator S_F obeys $(i\not\partial_x - m)S_F(x-y) = i\,\delta^4(x-y)$; as an operator identity this reads $(i\not\partial - m)\,S_F = i\,\mathbf{1}$, so that

$$(i\not\partial - m)^{-1} \;=\; \frac{1}{i}\,S_F \;=\; -\,i\,S_F, \qquad \text{hence} \qquad A^{-1} \;=\; i\,(-i\,S_F) \;=\; S_F. \tag{12.154}$$

The kinetic operator's inverse *is* the fermion propagator. Inserting Eq. (12.154) into Eq. (12.152) and restoring the continuum gives the closed form.

Theorem 12.42 (Closed form of the Dirac generating functional)**.** *The functional integral Eq. (12.148) evaluates to*

$$\boxed{\,Z[\eta,\bar\eta] \;=\; Z[0]\,\exp\Big[-\int d^4x\,d^4y\;\bar\eta(x)\,S_F(x-y)\,\eta(y)\Big]\,,\,} \tag{12.155}$$

with the field-independent normalization

$$Z[0] \;=\; \det A \;=\; \det\big[-\,i\,(i\not\partial - m)\big] \;=\; \mathcal{N}\,\det(i\not\partial - m), \tag{12.156}$$

where $\mathcal{N}$ is the (field-independent, formally divergent) constant collecting the factor $-i$ per mode, absorbed into the measure normalization.

Proof. The discretized integral is Eq. (12.149) with the identifications Eq. (12.151). Its value is $(\det A)\exp(\bar{\zeta}A^{-1}\zeta)$; the source exponent was reduced in Eqs. (12.152) and (12.154) to $-\bar{\eta}\,S_F\,\eta$, which is Eq. (12.155) once the lattice sums are returned to integrals. The prefactor is $\det A = \det[-i(i\not{\partial} - m)]$. Pulling the scalar $-i$ out of the determinant over the (infinite) mode space yields a field-independent constant $\mathcal{N} = \det(-i\,\mathbf{1})$ times $\det(i\not{\partial} - m)$; since only the ratio $Z[\eta,\bar{\eta}]/Z[0]$ enters every correlator, $\mathcal{N}$ never appears in a physical quantity, and the content of $Z[0]$ is the determinant of the Dirac operator. The convergence of the underlying finite-dimensional integral, and hence the legitimacy of the analytic continuation in A, is guaranteed by the same $m^2 \to m^2 - i\varepsilon$ tilt that made the bosonic Gaussian of Section 10.6 convergent: it gives A a definite signature, exactly as in the scalar discretization of Eq. (10.111). □

The structural parallel with the bosonic result Eq. (10.111) is exact, with one telling sign difference. Both functionals are $Z[0]$ times the exponential of a quadratic in the sources whose kernel is the free propagator. For the scalar the kernel entered as $-\frac{1}{2}\int J D_F J$, a single source species contracted symmetrically; for the Dirac field it enters as $-\int \bar{\eta}\,S_F\,\eta$, two distinct source species contracted in a fixed order, with no factor of $\frac{1}{2}$ because $\bar{\eta}$ and η are independent and there is exactly one way to pair them. The factor $Z[0]$ differs more fundamentally: it is $\det(i\not{\partial} - m)$ rather than the bosonic $[\det(\text{kinetic})]^{-1/2}$, the determinant sitting upstairs rather than downstairs. We isolate the source-dependent piece by defining the normalized functional

$$\widehat{Z}[\eta,\bar{\eta}] \;\equiv\; \frac{Z[\eta,\bar{\eta}]}{Z[0]} \;=\; \exp\Big[-\int d^4x\, d^4y\;\bar{\eta}(x)\,S_F(x-y)\,\eta(y)\Big], \qquad \widehat{Z}[0,0]=1, \tag{12.157}$$

which is all that any correlation function will use.

12.6.3 The momentum-space fermion propagator

Theorem 12.42 identified the source kernel as S_F through the operator relation Eq. (12.154), i.e. as the inverse of $(i\not{\partial} - m)$ up to the factor i. We now invert that operator explicitly, recovering the momentum-space propagator from the path integral alone—the self-contained QED computation this section owes. Pass to momentum space with the Fourier convention of Section 10.6, $\psi(x) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\,\tilde{\psi}(p)$, under which $\partial_\mu \to -ip_\mu$ and hence

$$i\not{\partial} \;=\; i\gamma^\mu\partial_\mu \;\longrightarrow\; i\gamma^\mu(-ip_\mu) \;=\; \gamma^\mu p_\mu \;=\; \not{p}. \tag{12.158}$$

The Dirac operator becomes the algebraic matrix $\not{p} - m$, and the operator equation $(i\not{\partial} - m)S_F = i\,\mathbf{1}$ of Eq. (11.204) reduces, mode by mode, to $(\not{p} - m)\,\tilde{S}_F(p) = i\,\mathbf{1}$. The momentum-space propagator is therefore

$$\tilde{S}_F(p) \;=\; i\,(\not{p} - m)^{-1}. \tag{12.159}$$

This is a 4×4 matrix inverse, but it is rationalized by a single gamma-matrix identity. Using only the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ of Eq. (C.1),

$$\not{p}\not{p} \;=\; p_\mu p_\nu\, \gamma^\mu\gamma^\nu \;=\; p_\mu p_\nu\, \tfrac{1}{2}\{\gamma^\mu, \gamma^\nu\} \;=\; p_\mu p_\nu\, \eta^{\mu\nu} \;=\; p^2, \tag{12.160}$$

where the symmetric combination $p_\mu p_\nu$ projects onto the symmetric (anticommutator) part of $\gamma^\mu\gamma^\nu$. Consequently $(\not{p} - m)(\not{p} + m) = \not{p}\not{p} - m^2 = p^2 - m^2$, a scalar, so the inverse of $\not{p} - m$ is obtained by multiplying numerator and denominator by $\not{p} + m$:

$$(\not{p} - m)^{-1} \;=\; \frac{\not{p} + m}{(\not{p} - m)(\not{p} + m)} \;=\; \frac{\not{p} + m}{p^2 - m^2}. \tag{12.161}$$

Substituting Eq. (12.161) into Eq. (12.159) gives the propagator up to its contour prescription.

The boundary condition—which root of $p^2 - m^2 = 0$ the inverse is defined across, and hence the $i\varepsilon$—is not fixed by the algebra; it is fixed by the convergence of the path integral, exactly as for the scalar. The $m^2 \to m^2 - i\varepsilon$ tilt that Section 10.6 attached to the bosonic measure, and which the Wick rotation $t \to -i\tau$ makes geometrically inevitable, displaces the poles off the real p^0 axis into the Feynman configuration: it sends $p^2 - m^2 \to p^2 - m^2 + i\varepsilon$ in the denominator. (Equivalently, after rotation the Euclidean Dirac operator has denominator $-(p_E^2 + m^2)$ with $p_E^2 + m^2 > 0$ everywhere, so its inverse is unambiguous; rotating back lands on $+i\varepsilon$.) This is the same prescription, fixed by the same rotation, that produced the canonical propagator Eq. (11.27) by the operator route. We have therefore proved, from the path integral alone:

$$\boxed{\tilde{S}_F(p) \;=\; \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} \;=\; \frac{i}{\not{p} - m + i\varepsilon}\,.} \tag{12.162}$$

This is identical, term for term and sign for sign, to the canonical Feynman propagator Eq. (11.27) obtained in Chapter 11 by time-ordering operators on the fermionic Fock space. The two pillars agree: the electron propagator that QED will use unaltered from Chapter 17 onward emerges equally from a sum over Grassmann field histories and from a vacuum matrix element of ordered field operators. The numerator $\not{p} + m$ is, as Section 11.5 stressed, the on-shell spin projector $\sum_s u^s(p)\bar{u}^s(p)$ at the electron pole; the path integral reproduces it without ever mentioning the spinors u, v, because it is encoded entirely in the matrix structure of the inverse Eq. (12.161).

12.6.4 The two-point function from functional differentiation

We can now extract correlation functions by differentiating Eq. (12.155), and watch the fermionic minus signs appear of their own accord. Functional differentiation with respect to anticommuting sources requires care about the side from

which the derivative acts; we adopt the *left* derivative $\dfrac{\delta_L}{\delta\eta(x)}$ throughout, defined by anticommuting the variation through to the matching source from the left,

$$\frac{\delta_L}{\delta\eta(x)}\,\eta(y) = \delta^4(x-y), \qquad \frac{\delta_L}{\delta\eta(x)}\big(\chi\,\eta(y)\big) = -\,\chi\,\frac{\delta_L}{\delta\eta(x)}\eta(y) \;\text{ for any odd } \chi, \tag{12.163}$$

and likewise for $\bar\eta$; the second rule is just the statement that $\delta_L/\delta\eta(x)$ is itself a Grassmann-odd operation, so it picks up a sign on passing any odd object. The prescription that converts Z into correlators follows directly from the source term in Eq. (12.148). Acting on the integrand, $\delta_L/\delta\bar\eta(x)$ brings down a factor $i\psi(x)$ (the source $\bar\eta$ sits to the left of ψ, so no reordering sign), while $\delta_L/\delta\eta(y)$ brings down $-i\bar\psi(y)$ (the source η sits to the right of $\bar\psi$, so the derivative must pass $\bar\psi$, generating one minus). The two factors of i, together with the reordering sign incurred when the second derivative passes the field produced by the first, conspire to cancel exactly, leaving the clean rule

$$\Big\langle 0|\mathcal{T}\,\psi(x)\bar\psi(y)|0\Big\rangle \;=\; \frac{1}{Z[0]}\,\frac{\delta_L}{\delta\bar\eta(x)}\,\frac{\delta_L}{\delta\eta(y)}\,Z[\eta,\bar\eta]\Big|_{\eta=\bar\eta=0} \;=\; \frac{\delta_L}{\delta\bar\eta(x)}\,\frac{\delta_L}{\delta\eta(y)}\,\widehat{Z}[\eta,\bar\eta]\Big|_{0}. \tag{12.164}$$

That this prescription, with the derivatives ordered as the operators are ordered and with no compensating signs or factors inserted by hand, reproduces the operator string $\psi(x)\bar\psi(y)$ inside the integral is verified by the explicit i-counting just sketched; we now evaluate the right-hand side on the closed form Eq. (12.157).

Theorem 12.43 (Fermion propagator from the path integral)**.** *The free Dirac two-point function obtained from Eq. (12.164) is*

$$\Big\langle 0|\mathcal{T}\,\psi(x)\bar\psi(y)|0\Big\rangle \;=\; S_F(x-y), \tag{12.165}$$

in exact agreement with the canonical Feynman propagator of Theorem 11.38.

Proof. Write $E[\eta,\bar\eta] = -\int d^4u\,d^4v\;\bar\eta(u)S_F(u-v)\eta(v)$, so $\widehat{Z} = e^{E}$. The exponent E is Grassmann-even, so a single left derivative passes through the exponential as $\frac{\delta_L}{\delta\eta(y)}e^{E} = (\frac{\delta_L}{\delta\eta(y)}E)e^{E}$. Compute the inner derivative using Eq. (12.163): the derivative reaches $\eta(v)$ only after passing the odd factor $\bar\eta(u)$, incurring one sign,

$$\frac{\delta_L}{\delta\eta(y)}E = -\int d^4u\,d^4v\;\bar\eta(u)S_F(u-v)\big(-\delta^4(y-v)\big) = \int d^4u\;\bar\eta(u)\,S_F(u-y) \;\equiv\; \mathcal{A}(y), \tag{12.166}$$

an odd functional of $\bar\eta$ alone. Thus $\frac{\delta_L}{\delta\eta(y)}\widehat{Z} = \mathcal{A}(y)\,e^{E}$. Now apply $\frac{\delta_L}{\delta\bar\eta(x)}$ by the Leibniz rule, remembering that $\mathcal{A}(y)$ is odd:

$$\frac{\delta_L}{\delta\bar\eta(x)}\big[\mathcal{A}(y)\,e^{E}\big] = \Big(\frac{\delta_L}{\delta\bar\eta(x)}\mathcal{A}(y)\Big)e^{E} - \mathcal{A}(y)\,\frac{\delta_L}{\delta\bar\eta(x)}e^{E}. \tag{12.167}$$

The first factor is a clean derivative ($\bar\eta(u)$ sits leftmost in $\mathcal{A}$),

$$\frac{\delta_L}{\delta\bar\eta(x)}\mathcal{A}(y) = \int d^4u\, \delta^4(x-u)\, S_F(u-y) = S_F(x-y), \tag{12.168}$$

a c-number matrix. Setting $\eta = \bar\eta = 0$ at the end kills the second term of Eq. (12.167), which retains the explicit factor $\mathcal{A}(y) \to 0$, and sets $e^E \to 1$ in the first. Hence $\frac{\delta_L}{\delta\bar\eta(x)}\frac{\delta_L}{\delta\eta(y)}\widehat{Z}|_0 = S_F(x-y)$, which is Eq. (12.165). □

Three features of this proof deserve to be drawn out. First, no minus sign was inserted anywhere: the only sign that appeared, in Eq. (12.166), came forced from the anticommutation rule Eq. (12.163), and it is precisely the sign that the antisymmetry of the fermionic time-ordering demands. Had we instead computed $\big\langle 0|\mathcal{T}\,\bar\psi(y)\psi(x)|0\big\rangle$ by reversing the order of the two derivatives, the same bookkeeping would yield $-S_F(x-y)$, reproducing the relative minus sign carried by the time-ordering symbol on fermion fields (Theorem 11.38). Second, the equal-source "diagonal" correlators vanish identically: $\langle 0|\mathcal{T}\,\psi(x)\psi(y)|0\rangle = \frac{\delta_L}{\delta\bar\eta(x)}\frac{\delta_L}{\delta\bar\eta(y)}\widehat{Z}|_0 = 0$, because $\widehat{Z}$ in Eq. (12.157) contains $\bar\eta$ and η only in the paired combination $\bar\eta\, S_F\, \eta$, so two $\bar\eta$-derivatives leave no surviving term at $\eta = \bar\eta = 0$; likewise $\big\langle 0|\mathcal{T}\,\bar\psi\bar\psi|0\big\rangle = 0$. Charge conservation—one ψ for each $\bar\psi$—is thereby automatic, not imposed. Third, the agreement with Chapter 11 is now total: the path integral reproduces the propagator Eq. (12.162), its position-space form Eq. (12.165), and the antisymmetry sign, with the determinant $Z[0]$ left over as the genuinely new object, to be interpreted in Section 12.7.

12.6.5 The fermionic Wick theorem

Higher correlators follow by further differentiation of Eq. (12.157), and the combinatorial pattern that emerges is the fermionic Wick theorem—the exact analog of the bosonic statement Theorem 10.23, but carrying signs. The structure is dictated by the same observation that organized the bosonic case in Section 10.5: $\widehat{Z}$ is the exponential of a quadratic in the sources, so differentiating it down to $\eta = \bar\eta = 0$ survives only when every source brought down by one derivative is matched against a source removed by another, each surviving match contributing one factor of the kernel S_F. For Grassmann sources the matching is constrained and signed: a $\delta/\delta\bar\eta$ (extracting ψ) can only pair with a $\delta/\delta\eta$ (extracting $\bar\psi$), because the kernel couples $\bar\eta$ to η; and each rearrangement of the anticommuting derivatives needed to bring a pairing together contributes a sign.

Theorem 12.44 (Wick's theorem for free Dirac fields)**.** *The free $2n$-point function vanishes unless it contains an equal number n of ψ's and $\bar\psi$'s, and only ψ–$\bar\psi$ contractions are nonzero, with*

$$\overbrace{\psi(x)\,\bar\psi(y)} = S_F(x-y), \qquad \overbrace{\psi(x)\,\psi(y)} = 0, \qquad \overbrace{\bar\psi(x)\,\bar\psi(y)} = 0. \tag{12.169}$$

The full correlator is the signed sum over all ways of pairing each ψ with a distinct $\bar\psi$,

$$\Big\langle 0|\mathcal{T}\,\psi(x_1)\cdots\psi(x_n)\,\bar\psi(y_1)\cdots\bar\psi(y_n)|0\Big\rangle \;=\; \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma_{\text{eff}})\,\prod_{i=1}^{n} S_F(x_i-y_{\sigma(i)}), \tag{12.170}$$

where the sign of each term is the parity of the permutation of field operators required to bring every contracted ψ–$\bar\psi$ pair into adjacency in the order $\psi\,\bar\psi$.

The vanishing of ψ–ψ and $\bar\psi$–$\bar\psi$ contractions is the structural statement already seen in Theorem 12.43: $\widehat{Z}$ pairs $\bar\eta$ with η only, so two like-type derivatives can never both find partners. The signed sum Eq. (12.170) is the Grassmann counterpart of the unsigned sum over pairings Eq. (10.138) of the scalar theory: where the boson sums all pairings with weight $+1$ (a permanent), the fermion sums them with the signature of the pairing permutation (a determinant of propagators). This is the combinatorial input that the Feynman rules of Chapters 15 and 17 consume: every fermion line is an S_F, and the relative sign between diagrams that differ by an exchange of identical external fermions, or that contain a closed fermion loop, is the sign in Eq. (12.170). The worked example below establishes the rule in full at $n=2$, the case that all higher combinatorics are built from.

Worked Example 12.45 (The free four-point function and its relative minus sign)**.** We compute the simplest correlator that exhibits a fermionic sign, $\big\langle 0|\mathcal{T}\,\psi(x_1)\psi(x_2)\bar\psi(y_1)\bar\psi(y_2)|0\big\rangle$, directly from Eq. (12.157), and contrast it with its bosonic cousin Eq. (10.138). By the prescription Eq. (12.164) extended to four fields—the derivatives ordered as the operators, $\psi(x_i)\leftrightarrow\delta_L/\delta\bar\eta(x_i)$ and $\bar\psi(y_j)\leftrightarrow\delta_L/\delta\eta(y_j)$—we must evaluate

$$W \;=\; \frac{\delta_L}{\delta\bar\eta(x_1)}\frac{\delta_L}{\delta\bar\eta(x_2)}\frac{\delta_L}{\delta\eta(y_1)}\frac{\delta_L}{\delta\eta(y_2)}\,\widehat{Z}\,\Big|_0. \tag{12.171}$$

Step 1: the η-derivatives. Reusing Eq. (12.166), $\frac{\delta_L}{\delta\eta(y_2)}\widehat{Z}=\mathcal{A}(y_2)e^{E}$ with $\mathcal{A}(y)=\int d^4u\,\bar\eta(u)S_F(u-y)$. Applying the second η-derivative, and noting $\mathcal{A}(y_2)$ contains no η (only $\bar\eta$), the only surviving Leibniz term comes from e^{E}, with one sign from passing the odd $\mathcal{A}(y_2)$:

$$\frac{\delta_L}{\delta\eta(y_1)}\frac{\delta_L}{\delta\eta(y_2)}\widehat{Z} \;=\; -\,\mathcal{A}(y_2)\,\mathcal{A}(y_1)\,e^{E}. \tag{12.172}$$

Step 2: the first $\bar\eta$-derivative. Recall $\frac{\delta_L}{\delta\bar\eta(x)}\mathcal{A}(y)=S_F(x-y)$ from Eq. (12.168), a c-number, and write $\frac{\delta_L}{\delta\bar\eta(x)}e^{E}=\mathcal{B}(x)e^{E}$ with $\mathcal{B}(x)=-\int d^4v\,S_F(x-v)\eta(v)$ odd. Differentiating Eq. (12.172) and tracking the sign incurred when $\delta_L/\delta\bar\eta(x_2)$ passes the odd $\mathcal{A}(y_2)$,

$$\frac{\delta_L}{\delta\bar\eta(x_2)}\big[-\mathcal{A}(y_2)\mathcal{A}(y_1)e^{E}\big] = -\,S_F(x_2-y_2)\,\mathcal{A}(y_1)\,e^{E}+\,\mathcal{A}(y_2)\,S_F(x_2-y_1)\,e^{E}$$

$$- \mathcal{A}(y_2)\,\mathcal{A}(y_1)\,\mathcal{B}(x_2)\,e^{E}. \tag{12.173}$$

Step 3: the last $\bar\eta$-derivative, then set sources to zero. Apply $\frac{\delta_L}{\delta\bar\eta(x_1)}$ and evaluate at $\eta=\bar\eta=0$. Any term that still carries an undifferentiated $\mathcal{A}$ or $\mathcal{B}$ vanishes at zero sources, so only the contractions that remove the last surviving $\bar\eta$ contribute. From the first term of Eq. (12.173), $\frac{\delta_L}{\delta\bar\eta(x_1)}\mathcal{A}(y_1)=S_F(x_1-y_1)$; from the second, $\frac{\delta_L}{\delta\bar\eta(x_1)}\mathcal{A}(y_2)=S_F(x_1-y_2)$; the third term has two $\mathcal{A}$'s and one $\mathcal{B}$ and cannot be exhausted by a single derivative, so it dies. The result is

$$\boxed{\left\langle 0|\mathcal{T}\,\psi(x_1)\psi(x_2)\bar\psi(y_1)\bar\psi(y_2)|0\right\rangle \;=\; -\,S_F(x_1-y_1)S_F(x_2-y_2)\;+\;S_F(x_1-y_2)S_F(x_2-y_1)\,.} \tag{12.174}$$

Reading the result. There are exactly two terms, one for each way of pairing the two ψ's with the two $\bar\psi$'s—half as many as the three pairings of the bosonic four-point function Eq. (10.138), because here a ψ can contract only with a $\bar\psi$, never with another ψ. The two pairings carry *opposite* signs. This relative minus is the entire content of fermionic statistics at the level of free correlators: it is the determinant structure of Eq. (12.170) at $n=2$, the antisymmetrized combination $S_F(x_1-y_2)S_F(x_2-y_1)-S_F(x_1-y_1)S_F(x_2-y_2)$, in contrast to the symmetric (all-plus) sum of Eq. (10.138). The signs themselves were not put in by hand; they were forced, one at a time, by the anticommutation rule Eq. (12.163)—in Eq. (12.172), in the second term of Eq. (12.173). The overall sign of any single term depends on the chosen ordering of the operators in the correlator (here, all ψ's before all $\bar\psi$'s); what is physical, and convention-independent, is the relative sign between the two pairings, which is exactly the sign distinguishing the exchange diagram from the direct one in every fermionic scattering amplitude of Chapters 17 and 18. Figure 12.3 records the topological origin of the minus: the second pairing differs from the first by a crossing of the contraction lines, and each crossing is one transposition of adjacent fermion operators, hence one factor of -1.

We have now closed the loop opened at the start of the chapter. The Dirac field has the same path-integral description as the scalar: a generating functional Eq. (12.155) whose source kernel is the Feynman propagator Eq. (12.162), whose two anticommuting functional derivatives return the time-ordered two-point function Eq. (12.165) in exact agreement with the canonical S_F of Chapter 11, and whose higher correlators are the signed Wick sums Eq. (12.170). This is the complete kit of free fermionic Feynman rules in position and momentum space—the propagator S_F, the source insertions, and the contraction signs—ready to be combined with vertices in Chapters 15 and 17. The one object the differentiation never touched is the normalization $Z[0]=\det(i\partial\!\!\!/-m)$: it factored out of every correlator and waited. That determinant, sitting upstairs where the boson's sits downstairs, is the subject of Section 12.7, where it becomes the one-loop

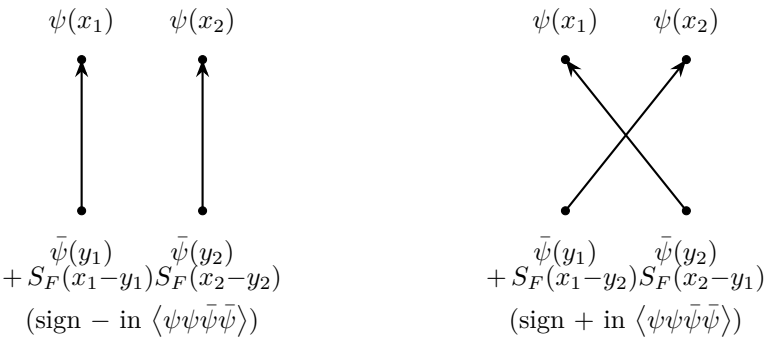


Figure 12.3: The two contraction patterns of the free fermionic four-point function Eq. (12.174). Each oriented line is a propagator S_F running from a $\bar\psi$ to a ψ, in the fermion-number direction of Fig. 11.4. A ψ may contract only with a $\bar\psi$, so there are two patterns rather than the three of the bosonic case Eq. (10.138). The exchange pattern (right) is obtained from the direct one (left) by a single crossing of the contraction lines, i.e. one transposition of adjacent anticommuting operators; that crossing is the geometric source of the relative minus sign between the two terms, and ultimately of every closed-fermion-loop sign in Section 12.7.

fermionic effective action through Tr ln, and where the inverted power first noted under Eq. (12.149) is revealed as the algebraic origin of the (-1) carried by every closed fermion loop.

12.7 Functional determinants and the fermion loop

The previous section (Section 12.6) performed the functional Gaussian integral over the Dirac field and delivered the normalization of the generating functional in a single compact symbol,

$$Z[0] \;=\; \det(i\not{\partial} - m), \tag{12.175}$$

the Berezin counterpart of the bosonic $(\det)^{-1/2}$ and the exact statement that integrating out a free fermion costs a determinant rather than its inverse square root. So far this is a closed formula and nothing more: a number (a badly divergent one) attached to the vacuum. The business of this closing section is to open it up. We will read (12.175) as a *physical* object—the one-loop fermionic effective action—by turning the determinant into a trace of a logarithm and expanding that logarithm into a sum of diagrams, each of which is a single closed fermion loop. Along the way we extract, from the bare algebra of (12.175), the most consequential bookkeeping rule in perturbative field theory: the factor of -1 that decorates every closed fermion loop. We derive that sign twice, from two independent starting points, and check that the two derivations agree. We

then promote the free operator $i\partial\!\!\!/ - m$ to its background-field form $iD\!\!\!/ - m$, identify $\det(iD\!\!\!/ - m)$ as the generator of the gauge field's one-loop effective action, and state honestly how an infinite-dimensional determinant is made finite. A worked computation on a single mode makes the sign tangible by producing equal-and-opposite vacuum energies for a boson and a fermion built on the same frequency.

Throughout, we reserve tr for the finite trace over the four Dirac indices and Tr for the full functional trace—Dirac indices *and* the continuous spacetime label—so that $\operatorname{Tr}\mathcal{O} = \int d^4x\, \operatorname{tr}\langle x|\mathcal{O}|x\rangle$ for an integral kernel $\mathcal{O}$. This matches the convention of Chapter E, to which the analytic control of these traces is deferred.

12.7.1 The determinant is the exponential of the trace-log

The bridge from a determinant to a sum of diagrams is the identity $\det = \exp\operatorname{Tr}\ln$. It is elementary for finite matrices and we prove it there in full; its extension to the Dirac operator is then a matter of saying precisely what the symbols mean, which we do at the end of the subsection.

Lemma 12.46 (First variation of the determinant)**.** *Let A be an invertible $n\times n$ complex matrix and B arbitrary. Then*

$$\det(A+\varepsilon B) = \det A\,(1+\varepsilon\,\operatorname{tr}(A^{-1}B))+O(\varepsilon^2), \qquad \textit{equivalently} \qquad \frac{d}{d\varepsilon}\Big|_{\varepsilon=0}\det(A+\varepsilon B) = \det A\,\operatorname{tr}(A^{-1}B). \tag{12.176}$$

Proof. Write $\det(\mathbf{1}+\varepsilon C)$ for $C = A^{-1}B$ and use the Leibniz expansion over permutations,

$$\det(\mathbf{1}+\varepsilon C) = \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\prod_{i=1}^{n}\left(\delta_{i,\sigma(i)} + \varepsilon\, C_{i,\sigma(i)}\right). \tag{12.177}$$

The identity permutation contributes $\prod_i(1+\varepsilon C_{ii}) = 1+\varepsilon\sum_i C_{ii}+O(\varepsilon^2)$. Any non-identity σ moves at least two indices, so the product $\prod_i(\delta_{i,\sigma(i)}+\varepsilon C_{i,\sigma(i)})$ contains at least two factors with $\delta_{i,\sigma(i)}=0$, each of which must be supplied by an explicit ε; such a term is $O(\varepsilon^2)$. Hence $\det(\mathbf{1}+\varepsilon C) = 1+\varepsilon\,\operatorname{tr}C + O(\varepsilon^2)$. Multiplying by $\det A$ and using $\det(A+\varepsilon B) = \det A\,\det(\mathbf{1}+\varepsilon A^{-1}B)$ gives (12.176). □

Theorem 12.47 ($\det = \exp\operatorname{Tr}\ln$)**.** *For every $X\in \operatorname{Mat}_n(\mathbb{C})$,*

$$\det\left(e^{X}\right) = e^{\operatorname{tr}X}. \tag{12.178}$$

Consequently, for any invertible M that can be written $M = e^X$ (every $M\in \mathrm{GL}_n(\mathbb{C})$ can, and we call any such X a logarithm $\ln M$),

$$\det M = e^{\operatorname{tr}\ln M}. \tag{12.179}$$

Proof. Fix X and define $f(t) = \det(e^{tX})$ for real t. By Theorem 12.46 applied with $A = e^{tX}$ and $A + \varepsilon B = e^{(t+\varepsilon)X} = e^{tX} + \varepsilon X e^{tX} + O(\varepsilon^2)$, so that $B = Xe^{tX}$,

$$f'(t) = \det(e^{tX})\ \mathrm{tr}(e^{-tX}\, X\, e^{tX}) = f(t)\ \mathrm{tr}\, X, \tag{12.180}$$

where the last equality is the cyclicity of the finite trace, $\mathrm{tr}(e^{-tX}Xe^{tX}) = \mathrm{tr}(Xe^{tX}e^{-tX}) = \mathrm{tr}\, X$. This is a linear first-order equation with $f(0) = \det \mathbf{1} = 1$; its unique solution is $f(t) = e^{t\ \mathrm{tr}\, X}$. Setting $t = 1$ gives (12.178). For (12.179), write $M = e^X$ with $X = \ln M$; then $\det M = \det e^X = e^{\mathrm{tr}\, X} = e^{\mathrm{tr} \ln M}$. (The value $e^{\mathrm{tr}\, X}$ is independent of the branch chosen for $\ln M$: two logarithms of the same M differ by something whose exponential is $\mathbf{1}$, and on the diagonalizable dense set this shifts each $\ln \lambda_i$ by an integer multiple of $2\pi i$, leaving $e^{\mathrm{tr} \ln M} = \prod_i \lambda_i = \det M$ unchanged; the general case follows by continuity.) □

The cleanest way to remember (12.179) is through eigenvalues: if M is diagonalizable with eigenvalues $\lambda_1, \dots, \lambda_n$, then $\det M = \prod_i \lambda_i$ while $\mathrm{tr} \ln M = \sum_i \ln \lambda_i$, and $\exp \sum_i \ln \lambda_i = \prod_i \lambda_i$. Theorem 12.47 is the statement that this survives non-diagonalizability and the passage through the exponential map.

For the Dirac operator the eigenvalue product over the infinitely many modes of $i\partial\!\!\!/ - m$ diverges, and so does the sum $\sum \ln \lambda$; neither side of (12.179) exists in the naive sense. The resolution, carried out in Section E.3, is to *adopt* the trace-log as the definition of the regularized determinant: one continues the spectral zeta function $\zeta_K(s) = \sum_n \lambda_n^{-s}$ to $s = 0$ and sets $\ln \det K \equiv -\zeta_K'(0)$ (Theorem E.16), a prescription that reproduces $\sum_n \ln \lambda_n = \mathrm{Tr} \ln K$ wherever the latter converges and supplies a finite number where it does not. With that understanding the symbol $\mathrm{Tr} \ln(i\partial\!\!\!/ - m)$ is well defined and (12.179) reads

$$Z[0] = \det(i\partial\!\!\!/ - m) = \exp \mathrm{Tr} \ln(i\partial\!\!\!/ - m). \tag{12.181}$$

The exponent $\mathrm{Tr} \ln(i\partial\!\!\!/ - m)$ is, up to a field-independent constant, the free fermionic vacuum energy times the spacetime volume; on its own it is a pure number. Its physics emerges the moment a background is switched on, which is what we do next.

12.7.2 The loop expansion: Tr ln as a sum of closed fermion loops

Put the fermion in a background electromagnetic field A_μ. The Dirac operator becomes $iD\!\!\!/ - m$ with $D_\mu = \partial_\mu + ieA_\mu$ acting on the electron (charge $-e$, $e > 0$; Chapter 11 and the QED conventions), so that $iD\!\!\!/ = i\partial\!\!\!/ - eA\!\!\!/$ and

$$iD\!\!\!/ - m = (i\partial\!\!\!/ - m) - eA\!\!\!/. \tag{12.182}$$

The same Gaussian Berezin integral that produced (12.175)—now with the quadratic kernel $iD\!\!\!/ - m$ in place of $i\partial\!\!\!/ - m$, the action still being bilinear in

$\psi, \bar\psi$—gives $Z[A] = \det(i\not{D} - m)$ up to the same field-independent constant (Theorem E.3). The vacuum amplitude in the background, normalized to the no-background vacuum, is therefore

$$e^{iW[A]} \equiv \frac{Z[A]}{Z[0]} = \frac{\det(i\not{D} - m)}{\det(i\not{\partial} - m)}, \qquad iW[A] = \operatorname{Tr}\ln(i\not{D} - m) - \operatorname{Tr}\ln(i\not{\partial} - m), \tag{12.183}$$

using Theorem 12.47 in the regularized sense of (12.181). To expand the difference of trace-logs, factor the free operator out of (12.182),

$$i\not{D} - m = (i\not{\partial} - m)\Big[\mathbf{1} - (i\not{\partial} - m)^{-1} e\not{A}\Big]. \tag{12.184}$$

The inverse free operator is the Feynman propagator of Chapter 11: from $(i\not{\partial} - m)S_F(x-y) = i\,\delta^4(x-y)$ (Eq. (11.204)) one reads, as integral kernels,

$$(i\not{\partial} - m)^{-1} = -i\,S_F, \tag{12.185}$$

so $-(i\not{\partial} - m)^{-1} e\not{A} = +i\,e\,S_F\not{A}$. Taking $\operatorname{Tr}\ln$ of the product (12.184), the multiplicative free factor cancels against $\operatorname{Tr}\ln(i\not{\partial} - m)$ in (12.183), leaving

$$iW[A] = \operatorname{Tr}\ln\Big[\mathbf{1} + i\,e\,S_F\not{A}\Big]. \tag{12.186}$$

The operator $eS_F\not{A}$ is small in the sense that carries the expansion (one power of the background per factor), so the matrix logarithm may be expanded in its Mercator series $\ln(\mathbf{1}+Y) = \sum_{n\ge1} \frac{(-1)^{n+1}}{n} Y^n$, valid term by term inside the trace:

$$iW[A] = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}\, \operatorname{Tr}\big[(i\,e\,S_F\not{A})^n\big]. \tag{12.187}$$

This is the loop series, and reading off what each term *is* is the whole point. Write the functional trace as nested position integrals. The n-th term contains n propagators and n insertions of the background, contracted head-to-tail and closed into a ring by the single trace:

$$\operatorname{Tr}\big[(ieS_F\not{A})^n\big] = (ie)^n \int d^4x_1 \cdots d^4x_n\ \operatorname{tr}\Big[S_F(x_1-x_2)\not{A}(x_2)\, S_F(x_2-x_3)\not{A}(x_3)\cdots S_F(x_n-x_1)\not{A}(x_1)\Big] \tag{12.188}$$

Diagrammatically this is exactly a single closed fermion line—a loop—onto which n external photon legs $A(x_k)$ are attached, the Dirac trace tr closing the spinor indices around the loop and the integrals $\int d^4x_k$ summing over the vertex positions. The factor $1/n$ in (12.187) is the order of the cyclic symmetry $\mathbb{Z}_n$ of the ring: the n cyclic relabelings of the vertices all give the same diagram, and the trace's cyclicity makes (12.188) invariant under them, so dividing by n avoids overcounting. We have thus proved the structural statement promised by the section:

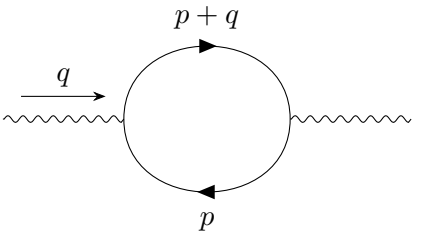


Figure 12.4: The closed fermion loop with two background (photon) insertions—the $n = 2$ term (12.189) of the trace-log expansion (12.187), i.e. the vacuum polarization. The arrows on the internal line show the fermion-number flow circulating consistently around the loop; the single Dirac trace tr closes the spinor indices. Every such closed fermion loop carries an overall factor of $\boxed{-1}$ relative to the corresponding bosonic loop, derived two ways in Theorem 12.49: from the determinant sitting in the numerator (Berezin) rather than the denominator (Gaussian), and equivalently from the antiperiodic trace around the loop.

Theorem 12.48 (The fermion trace-log is a sum of closed-loop vacuum diagrams)**.** *The fermionic vacuum functional* (12.183) *in a background* A_μ *is the sum* (12.187) *over closed single-fermion-loop diagrams, the* n*-th term* (12.188) *being the one-loop diagram with* n *background insertions, weighted by its cyclic symmetry factor* $1/n$*.*

The individual terms are the working vertices of one-loop QED. The $n = 1$ term is a tadpole and vanishes; the odd terms vanish quite generally by Furry's theorem (charge-conjugation invariance forces an odd number of photons attached to a fermion loop to give zero, a fact established with the discrete symmetries of Section 11.7.2). The first surviving term is $n = 2$,

$$\tfrac{(-1)^3}{2}\,\mathrm{Tr}\big[(ieS_F\not{A})^2\big] = \tfrac{e^2}{2}\int d^4x_1 d^4x_2\ \mathrm{tr}\big[S_F(x_1-x_2)\not{A}(x_2)S_F(x_2-x_1)\not{A}(x_1)\big], \tag{12.189}$$

the *vacuum polarization*: a single fermion loop with two photon legs, the object that screens the electric charge and whose explicit evaluation is the heart of one-loop QED in Chapter 19. The $n = 4$ term is the light-by-light box. Figure 12.4 draws the $n = 2$ loop and records, ahead of the next subsection, the factor of -1 it must carry.

12.7.3 The minus sign of a closed fermion loop, two ways

We come to the rule. In the Feynman rules assembled from this machinery, every closed fermion loop is multiplied by -1 (and more generally a diagram with L closed fermion loops by $(-1)^L$). The sign is not a convention bolted on by hand; it is forced, and we now derive it from two genuinely independent places—the

placement of the determinant in the generating functional, and the boundary condition on the fermion trace—and verify they say the same thing.

Theorem 12.49 (The closed-fermion-loop sign)**.** *Integrating out a fermion contributes its closed loops with overall sign opposite to that of a boson of the same charge: the fermionic vacuum functional is* $iW_{\mathrm{f}}[A] = +\operatorname{Tr}\ln(i\not{D} - m)$*, whereas a complex scalar of charge* e *contributes* $iW_{\mathrm{b}}[A] = -\operatorname{Tr}\ln(-D^2 - m^2)$*. Consequently each closed fermion loop carries a relative factor of* -1*, and a diagram with* L *closed fermion loops carries* $(-1)^L$*.*

First derivation: numerator versus denominator. The fermionic Berezin Gaussian places the determinant in the *numerator*,

$$\int \mathcal{D}\bar{\psi}\,\mathcal{D}\psi\; e^{\,i\int \bar{\psi}(i\not{D}-m)\psi} \;\propto\; \det(i\not{D} - m) \qquad \text{(Theorem E.3)}, \tag{12.190}$$

so that, taking the logarithm as in (12.183), the fermion delivers $iW_{\mathrm{f}}[A] = +\operatorname{Tr}\ln(i\not{D} - m)$ and hence the loop series (12.187) with the signs displayed there. Contrast a complex scalar field Φ of charge e, with quadratic action $\int \Phi^*(-D^2 - m^2)\Phi$. Its ordinary (commuting) Gaussian integral places the determinant in the *denominator*,

$$\int \mathcal{D}\Phi^*\,\mathcal{D}\Phi\; e^{\,i\int \Phi^*(-D^2-m^2)\Phi} \;\propto\; \big[\det(-D^2 - m^2)\big]^{-1} \qquad \text{(Theorem E.2)}, \tag{12.191}$$

so its vacuum functional is $iW_{\mathrm{b}}[A] = -\operatorname{Tr}\ln(-D^2 - m^2)$. Expanding this in the background by the identical steps that led to (12.187) produces the same family of closed single-particle loops—now built from scalar propagators and scalar vertices—but multiplied by the *opposite* overall sign, because the trace-log enters with a minus rather than a plus. The single algebraic fact responsible is the inversion of the Gaussian answer between Grassmann and ordinary variables: $\det^{+1}$ for the fermion (Theorem E.3) versus $\det^{-1}$ for the complex boson (Theorem E.2); a real scalar would give $\det^{-1/2}$, i.e. $iW = -\frac{1}{2}\operatorname{Tr}\ln$, the same sign with half the weight. The exponent of the determinant, not the source structure, is what flips: this is precisely the inversion flagged when the Berezin Gaussian was first computed, now cashed out as the loop sign.

Second derivation: the antiperiodic trace. The closed loop is also a trace in the literal Hilbert-space sense—a particle propagating once around the Euclidean time circle and returning to its starting state, summed over states. For a single mode the coherent-state construction of Section 12.5 expresses the trace of any operator $\hat{\mathcal{O}}$ over the fermionic Fock space as a Berezin integral over one Grassmann pair,

$$\operatorname{Tr}\,\hat{\mathcal{O}} = \int d\bar{\theta}\, d\theta\; e^{-\bar{\theta}\theta}\,\langle -\theta|\,\hat{\mathcal{O}}\,|\theta\rangle, \tag{12.192}$$

to be set against the bosonic coherent-state trace $\mathrm{Tr}\,\hat{\mathcal{O}} = \int \frac{d\bar{z}\,dz}{2\pi i}\, e^{-\bar{z}z}\,\langle z|\hat{\mathcal{O}}|z\rangle$. The two formulas are identical but for one decisive feature: the fermionic bra carries $\langle -\theta|$ where the bosonic one carries $\langle z|$. That single minus sign—an unavoidable consequence of having to anticommute the Grassmann label past the inserted state to close the trace—is what distinguishes a fermion trace from a boson trace. Threaded through the time-sliced path integral, $\langle -\theta|$ versus $\langle z|$ becomes the boundary condition that the path satisfies after one circuit:

$$\psi(\beta) = -\,\psi(0) \quad \text{(fermion, antiperiodic)}, \qquad \phi(\beta) = +\,\phi(0) \quad \text{(boson, periodic)}. \tag{12.193}$$

A closed loop is a worldline that comes back to where it began; the antiperiodicity (12.193) multiplies the fermionic closed loop by exactly -1 relative to the periodic bosonic one. This is the same -1, now seen as the cost of carrying a fermion once around the loop rather than as the cost of moving a determinant from the denominator to the numerator.

The two agree. Both derivations attach a single overall factor of -1 to one closed fermion loop, and both attach $(-1)^L$ to L independent loops: in the first because each separate determinant contributes its own sign flip, in the second because each independent loop is a separate trace and carries its own antiperiodicity. They are not merely consistent but the same statement read in two languages. The trace-log $\mathrm{Tr}\ln(i\not{D} - m)$ is literally a sum of single closed loops (Theorem 12.48); evaluating each loop as a Hilbert-space trace around antiperiodic Euclidean time reproduces precisely the sign that the numerator placement of $\det(i\not{D} - m)$ assigns it. The worked example of Theorem 12.50 below exhibits both faces at once on a single mode: the antiperiodic fermion determinant $1 + e^{-\beta\omega}$ against the periodic bosonic $1 - e^{-\beta\omega}$, the relative sign of the $e^{-\beta\omega}$ being exactly the loop's -1.

Why the Pauli principle wears the sign

The minus sign on a fermion loop is the same fact, viewed dynamically, as the antisymmetry of the multi-fermion wavefunction. A closed loop is a quantum process in which an identical particle is created and reabsorbed; for fermions the amplitude for such a closed exchange enters with the opposite sign demanded by the antisymmetrization that the Berezin measure automates. The vacuum-energy consequence is immediate and physical: a fermionic mode lowers the zero-point energy where a bosonic mode raises it, the seed of the boson–fermion cancellation that supersymmetry later makes exact.

12.7.4 The fermion determinant in a background: the one-loop effective action

Theorems 12.48 and 12.49 together identify $\det(i\not{D} - m)$ as a generator of physics, not a normalization to be discarded. Define the fermionic contribution to the effective action of the gauge field by integrating out the fermion in a fixed background,

$$\Gamma_{\mathrm{f}}[A] \;=\; -\,i\,\ln\det(i\not{D} - m) \;=\; -\,i\,\operatorname{Tr}\ln(i\not{D} - m), \tag{12.194}$$

the Legendre-transform/effective-action language of Section 4.5 specialized to a background that is not itself fluctuating. This is the one-loop effective action carried by the fermion loop; its Euclidean form is exactly the $\Gamma^{(1),\,\mathrm{ferm}} = -\operatorname{Tr}\ln D$ of Eq. (E.137), the sign there being the same loop sign derived here. Two downstream programs read directly off (12.194).

First, expanding $\Gamma_{\mathrm{f}}[A]$ to second order in A gives the vacuum polarization (12.189); differentiating twice with respect to A extracts the photon self-energy $\Pi^{\mu\nu}(q)$, the central one-loop object of QED. Chapter 19 takes (12.189) as its starting integrand, regularizes it dimensionally, and finds the running of α together with the Ward identity that keeps $\Pi^{\mu\nu}$ transverse. The closed-loop sign (12.49) is not optional there: it fixes the sign of charge screening.

Second, (12.194) is gauge invariant as it stands—$\operatorname{Tr}\ln(i\not{D} - m)$ depends on A only through the gauge-covariant D—and yet the chiral anomaly is real. The resolution, which (12.194) sets up but does not contain, is that the anomaly lives not in the action but in the *measure* $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$: under a chiral rotation the Berezin measure transforms by a nontrivial Jacobian whose regularized value is a heat-kernel coefficient (Eq. (E.197)). Chapter 36 (Fujikawa) computes that Jacobian and finds the Adler–Bell–Jackiw anomaly; the present determinant, and the fact that the fermion sign and the fermion measure are inseparable, is exactly the object it acts on.

12.7.5 Honest regularization, and the limits of the formal measure

We have manipulated $\det(i\not{D} - m)$ and $\operatorname{Tr}\ln(i\not{D} - m)$ as if they were finite. They are not: the Dirac operator has infinitely many eigenvalues growing without bound, so $\prod\lambda$ and $\sum\ln\lambda$ both diverge. Intellectual honesty requires saying exactly what tames them, and where the formal manipulations are made rigorous.

The key idea, developed in full in Section E.3, is spectral. One forms the zeta function $\zeta_K(s) = \sum_n \lambda_n^{-s}$ of the relevant operator K (for the Dirac case, $K = (i\not{D})^2$-type after the squaring of Eq. (E.138)), which converges for $\operatorname{Re} s$ large by Weyl's law, continues analytically to a neighborhood of $s = 0$ through its Mellin/heat-kernel representation, and *defines*

$$\ln\det K = -\,\zeta_K'(0), \qquad \det K = e^{-\zeta_K'(0)} \tag{12.195}$$

(Theorem E.16). This reproduces $\sum_n \ln \lambda_n$ whenever that sum is finite and assigns a finite, scheme-controlled number when it is not; the residual scale dependence is carried by the μ of $\overline{\text{MS}}$, exactly as in dimensional regularization. The heat-kernel route makes the ultraviolet structure manifest: the small-time expansion of $\operatorname{Tr} e^{-tK}$ has Seeley–DeWitt coefficients that are local, gauge-invariant densities, and it is these that feed both the renormalization of $\Gamma_{\text{f}}[A]$ and the anomaly. We do not reproduce that machinery here; it is the dedicated business of Chapter E, which this section imports rather than rebuilds.

The infinite-dimensional Berezin measure is formal

The symbol $\mathcal{D}\bar{\psi}\,\mathcal{D}\psi$ stands for a Berezin integral over uncountably many Grassmann modes, and unlike its finite-dimensional ancestor of Section 12.4 it has no construction as a genuine measure: there is no Grassmann analogue of Lebesgue measure on an infinite-dimensional space. What is rigorous is the *ratio* of determinants (12.183) after zeta- or heat-kernel regularization, and the perturbative series (12.187) it generates. A nonperturbative, fully rigorous definition of the fermionic functional integral exists only constructively—on a finite lattice, where the integral is a genuine finite-dimensional Berezin integral and $\det(i\not{D} - m)$ is an honest finite determinant, followed by a controlled continuum limit. That lattice determinant, and the notorious difficulty that it can fail to be real and positive (the sign problem), is the subject of Chapter 43. The formula (12.175) is therefore best read as a generating device for the convergent expansion (12.187), made nonperturbatively precise only on the lattice.

12.7.6 The sign made concrete: a single mode

The cleanest place to watch $\det^{+1}$ (fermion, numerator) and $\det^{-1/2}$ (boson, denominator) produce opposite-sign contributions to the *same* physical quantity is a single harmonic mode of frequency ω, thermalized on a Euclidean time circle of circumference β. We compute the fermionic determinant in full on a finite lattice, take the continuum limit, and read off a vacuum energy of the opposite sign to the boson's.

Worked Example 12.50 (Boson and fermion vacuum energies from $\det^{\mp}$)**.** Consider one bosonic oscillator and one fermionic oscillator, both of frequency ω, with symmetric (Weyl-ordered) Hamiltonians

$$\hat{H}_{\text{b}} = \omega(\hat{a}^\dagger \hat{a} + \tfrac{1}{2}), \qquad \hat{H}_{\text{f}} = \omega(\hat{b}^\dagger \hat{b} - \tfrac{1}{2}), \tag{12.196}$$

with $[\hat{a}, \hat{a}^\dagger] = 1$ and $\{\hat{b}, \hat{b}^\dagger\} = 1$. We compute the partition functions $Z = \operatorname{Tr} e^{-\beta \hat{H}}$ two ways—directly and as functional determinants—and extract the vacuum energy $E_0 = -\lim_{\beta\to\infty} \beta^{-1} \ln Z$.

Direct traces. The bosonic Fock space has states $|n\rangle$, $n \geq 0$, with energies $\omega(n + \frac{1}{2})$, so

$$Z_{\mathrm{b}} = \sum_{n=0}^{\infty} e^{-\beta\omega(n+1/2)} = \frac{e^{-\beta\omega/2}}{1 - e^{-\beta\omega}} = \frac{1}{2\sinh(\beta\omega/2)}, \tag{12.197}$$

giving $E_0^{\mathrm{b}} = +\omega/2$. The fermionic Fock space has only $|0\rangle, |1\rangle$ with energies $\mp\omega/2$, so

$$Z_{\mathrm{f}} = e^{+\beta\omega/2} + e^{-\beta\omega/2} = 2\cosh(\beta\omega/2), \tag{12.198}$$

giving $E_0^{\mathrm{f}} = -\omega/2$. The two zero-point energies are equal and opposite; the combined system $Z_{\mathrm{b}}Z_{\mathrm{f}}$ has $E_0 = 0$. We now see this sign emerge from the determinants.

The fermion as a Berezin determinant. The Euclidean action of the single Grassmann mode is $S_E^{\mathrm{f}} = \int_0^\beta d\tau\, \bar\psi(\partial_\tau + \omega)\psi$, with antiperiodic boundary conditions (12.193) forced by the trace formula (12.192). Discretize the circle into N steps of size $a = \beta/N$, writing $\psi_k = \psi(ka)$ and approximating $\partial_\tau\psi \to (\psi_k - \psi_{k-1})/a$. The action becomes, after multiplying through by a and writing $r \equiv 1 - a\omega$,

$$S_E^{\mathrm{f}} = \sum_{k=1}^{N} \bar\psi_k\big[\psi_k - r\,\psi_{k-1}\big], \qquad \psi_0 \equiv -\psi_N \quad \text{(antiperiodic)}, \tag{12.199}$$

so that $Z_{\mathrm{f}} = \int \prod_k d\bar\psi_k\, d\psi_k\, e^{-S_E^{\mathrm{f}}} = \det M_{\mathrm{AP}}$ by Theorem E.3, where M_{AP} is the $N \times N$ matrix with 1 on the diagonal, $-r$ on the subdiagonal, and the boundary entry $(M_{\mathrm{AP}})_{1,N} = +r$ produced by $\psi_0 = -\psi_N$. Expand the determinant along the first row. Two minors survive:

- deleting row 1, column 1 leaves an $(N-1)\times(N-1)$ lower-bidiagonal matrix with unit diagonal, determinant 1;
- deleting row 1, column N leaves rows $2,\dots,N$ and columns $1,\dots,N-1$, a lower-triangular matrix with $-r$ along its diagonal, determinant $(-r)^{N-1}$.

Hence

$$\det M_{\mathrm{AP}} = 1\cdot 1 + (-1)^{1+N}\,(+r)\,(-r)^{N-1} = 1 + (-1)^{1+N}(-1)^{N-1} r^N = 1 + r^N = 1 + (1-a\omega)^N. \tag{12.200}$$

Taking $N \to \infty$ with $a = \beta/N$ gives $(1 - \beta\omega/N)^N \to e^{-\beta\omega}$, so

$$Z_{\mathrm{f}} = \det M_{\mathrm{AP}} \;\longrightarrow\; 1 + e^{-\beta\omega}. \tag{12.201}$$

This is the partition function of $\hat H = \omega\,\hat b^\dagger \hat b$ (energies $0, \omega$); restoring the symmetric zero-point of (12.196) multiplies by $e^{\beta\omega/2}$ and recovers $e^{\beta\omega/2}(1 + e^{-\beta\omega}) = 2\cosh(\beta\omega/2)$, in exact agreement with (12.198).

The boson as the inverse square root. Repeating the construction for the real bosonic mode with second-order Euclidean action $S_E^{\mathrm{b}} = \frac{1}{2}\int_0^\beta d\tau\, \phi(-\partial_\tau^2 + \omega^2)\phi$ and *periodic* boundary conditions, the Gaussian integral gives $Z_{\mathrm{b}} = [\det{}_{\mathrm{P}}(-\partial_\tau^2 + \omega^2)]^{-1/2}$. The same circulant method—or, most transparently, the factorization $-\partial_\tau^2 + \omega^2 = (-\partial_\tau + \omega)(\partial_\tau + \omega)$ on the periodic circle, each first-order factor giving the periodic version of (12.200) with corner sign $-r$ and hence $\det{}_{\mathrm{P}} \to 1 - e^{-\beta\omega}$—yields

$$\det{}_{\mathrm{P}}(-\partial_\tau^2 + \omega^2) \;\longrightarrow\; e^{\beta\omega}\big(1 - e^{-\beta\omega}\big)^2 = \big(2\sinh(\beta\omega/2)\big)^2, \tag{12.202}$$

the factor $e^{\beta\omega}$ being the same zero-point/normal-ordering constant that the symmetric ordering fixes. Therefore $Z_{\mathrm{b}} = [\,(2\sinh(\beta\omega/2))^2\,]^{-1/2} = 1/(2\sinh(\beta\omega/2))$, reproducing (12.197).

The sign, isolated. Compare the two answers as trace-logs:

$$\ln Z_{\mathrm{f}} = +\,\mathrm{Tr}\ln(\partial_\tau + \omega)\big|_{\mathrm{AP}}, \qquad \ln Z_{\mathrm{b}} = -\tfrac{1}{2}\,\mathrm{Tr}\ln(-\partial_\tau^2 + \omega^2)\big|_{\mathrm{P}}. \tag{12.203}$$

Two independent features distinguish them, and each is one face of Theorem 12.49. The *exponent* of the determinant is $+1$ for the fermion (numerator, Berezin, Theorem E.3) and $-\frac{1}{2}$ for the boson (denominator, Gaussian); and the *boundary condition* is antiperiodic for the fermion and periodic for the boson, so the determinants differ in the sign of $e^{-\beta\omega}$: $1 + e^{-\beta\omega}$ versus $1 - e^{-\beta\omega}$. Both effects pull the same way. The fermionic $\det{}_{\mathrm{AP}}^{+1}$ delivers $E_0^{\mathrm{f}} = -\omega/2$; the bosonic $\det{}_{\mathrm{P}}^{-1/2}$ delivers $E_0^{\mathrm{b}} = +\omega/2$. The ratio $Z_{\mathrm{f}}/Z_{\mathrm{b}} = 2\cosh(\beta\omega/2)\cdot 2\sinh(\beta\omega/2) = 2\sinh(\beta\omega)$ has $E_0 = -\omega/2 - (+\omega/2) = -\omega$: integrating out one fermionic mode lowers the vacuum energy by exactly the amount that integrating out one bosonic mode raises it. This is the closed-fermion-loop -1, made arithmetic on a single mode.

The pattern of Theorem 12.50 is universal. A determinant in the numerator, traced around an antiperiodic loop, is a fermion; the same determinant inverted, traced around a periodic loop, is a boson; the difference is the sign on every closed fermion loop, and on the vacuum energy that sums them. With this the chapter's mission is complete: the Dirac field now has the same path-integral description the scalar enjoyed, built honestly on Grassmann variables and Berezin integration, with its generating functional, propagator, and determinant all in hand.

The determinant we have interpreted here is the hinge on which much of the rest of the book turns. Chapter 19 takes the closed fermion loop (12.189), with the sign (12.49) fixed, and computes the one-loop vacuum polarization and running coupling of QED; Chapter 26 re-encounters the determinant from the other side, re-exponentiating a *bosonic* determinant as a Grassmann (Faddeev–Popov ghost) integral precisely to flip its sign back, so that gauge-fixing leaves the right number of degrees of freedom—the same numerator/denominator inversion run in reverse; Chapter 36 acts on the fermion measure $\mathcal{D}\bar\psi\,\mathcal{D}\psi$ behind (12.194)

with a chiral rotation and finds the anomaly in its Jacobian; and Chapter 43 makes the formal Berezin integral of Section 12.7.5 rigorous as a finite lattice determinant, where the very positivity we assumed becomes the sign problem.

Further reading

The construction of integration over anticommuting variables is due to Berezin [63], still the cleanest source for the algebra and the inverse-determinant law. The fermionic coherent-state derivation of the path integral, the antiperiodic trace, and the boson/fermion vacuum-energy contrast are developed thoroughly in Negele and Orland [360]; Zinn-Justin [534] and Itzykson and Zuber [278] give complementary treatments of the Grassmann generating functional and the functional determinant. The trace-log interpretation of the fermion determinant and its closed-loop expansion, with care for the loop sign, are in Peskin and Schroeder [396], Schwartz [432], and Weinberg [500]. Zeta-function and heat-kernel regularization of functional determinants originate with Dowker and Critchley [161] and Hawking [257]; the lattice fermion determinant and the sign problem are surveyed in Gattringer and Lang [211]; the role of the fermion measure in the chiral anomaly is Fujikawa [203], carried out in Chapter 36.

Chapter 13

The Electromagnetic Field

The scalar and the spinor came to us almost willingly. Their Lagrangians were regular, their Legendre transforms invertible, and canonical quantization in Chapters 9 and 11 was a matter of promoting Poisson brackets to (anti)commutators and reading off a Fock space. The photon refuses this script. Already at the classical level—in Chapter 8, where electromagnetism was *derived* from local U(1) invariance—the field A_μ carried a redundancy: configurations related by $A_\mu \to A_\mu + \partial_\mu \alpha$ describe the same physics, because only the field strength $F_{\mu\nu}$ enters the action. That redundancy is not a blemish to be tidied away; it is the source of every difficulty and every subtlety in this chapter. It makes the Maxwell Lagrangian singular, it obstructs the naive canonical commutators, and it forces a choice—made differently by every consistent quantization scheme—about how to dispose of the unphysical degrees of freedom while keeping a positive-norm Hilbert space and, ideally, manifest Lorentz covariance. The whole chapter is the story of paying that price in full, by three complementary routes that are then shown to describe one and the same photon.

We open by diagnosing the obstruction precisely. Treating the Maxwell field as a constrained Hamiltonian system—the Dirac–Bergmann analysis whose classical groundwork was laid in Chapter 6—we find a primary constraint $\pi^0 \approx 0$ and, from its consistency under time evolution, the secondary Gauss constraint $\partial_i \pi^i \approx 0$. Both are first-class: they generate the gauge transformations of Chapter 8, and they reduce the naive four phase-space pairs to two physical polarizations. The same count that makes the photon a massless helicity-± 1 particle is what makes the four naive equal-time commutators mutually contradictory, so that some treatment of the constraints *before* quantization is mandatory. Two principled options present themselves, and they organize the rest of the chapter: solve the constraints classically and quantize only what survives, or keep everything covariant and impose the constraints weakly on states.

The first route is Coulomb gauge. Fixing $\nabla \cdot \mathbf{A} = 0$ eliminates A_0 and the longitudinal mode outright; the surviving transverse field is quantized with the

Dirac bracket, whose kernel is the transverse delta function, and the result is a manifestly positive-norm Fock space of two oscillators per momentum plus an instantaneous Coulomb energy of the sources. The price is manifest covariance. The second route, Gupta–Bleuler, quantizes all four components of A_μ covariantly on an indefinite-metric space and then carves out the physical sector by the subsidiary condition $\partial^\mu A_\mu^{(+)} |\text{phys}\rangle = 0$; its central theorem is that the physical subspace, modulo null states, is isometric to the Coulomb-gauge space built the section before. With a covariant operator in hand we assemble the photon propagator, follow its dependence on the gauge parameter ξ across the R_ξ family, and prove that the gauge-dependent $k_\mu k_\nu$ term never reaches an observable because it is annihilated by the conserved matter current of Chapter 11—the same fact that, for external lines, replaces the polarization sum by $-\eta_{\mu\nu}$.

The third route returns to first principles in the path-integral language of Chapters 10 and 12. The naive functional integral $\int \mathcal{D}A\, e^{iS}$ diverges, and the divergence is exactly the volume of the gauge orbit. The abelian Faddeev–Popov construction slices transverse to the orbits, produces the gauge-fixing term and the R_ξ propagator from a Gaussian average, and shows that the Faddeev–Popov determinant is field-independent—so QED, unlike its non-abelian cousin, carries no coupled ghost. We close by reuniting the routes: the minimal coupling $-e\bar{\psi}\gamma^\mu\psi A_\mu$ is installed, the instantaneous-Coulomb-plus-transverse description is shown to coincide with covariant one-photon exchange on conserved currents, and a static calculation recovers Coulomb's law $V(r) = e_1 e_2/4\pi r$ with the correct sign and the fine-structure constant out front—the photon propagator certified against a classical fact. What leaves this chapter is precisely what Part V runs on: the photon propagator, the vertex, and the polarization rule of QED, exported unchanged to Chapter 17.

13.1 Maxwell as a Constrained System: The Obstruction to Quantization

We have just declared that gauge invariance is a redundancy of description rather than a symmetry that relates physically distinct states, and that this single fact is what stands between us and a photon. It is time to make the obstruction precise. The instinct fresh from Chapter 9 is irresistible: we quantized the real scalar by computing its conjugate momentum $\pi = \partial\mathcal{L}/\partial\dot{\phi}$, postulating the equal-time commutator $[\phi(\mathbf{x}), \pi(\mathbf{y})] = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y})$ (Eq. (9.7)), expanding in modes, and reading off a Fock space of particles. The electromagnetic potential A_μ has four components; surely we repeat the recipe four times and obtain a particle with four polarizations, to be trimmed later. The recipe breaks at the very first step. The Legendre transform that defines the momenta is *not invertible*: one of the four momenta is not a momentum at all but an identity, $\pi^0 \equiv 0$, holding before any equation of motion is imposed. A relation among the phase-space variables

that holds identically is a *constraint*, and a system that has them is *singular*. The scalar was regular and the recipe worked; Maxwell is singular and the recipe must be rebuilt.

This section carries out the rebuilding at the level Dirac taught us to: we compute the momenta and expose the primary constraint, write the canonical Hamiltonian, demand that the constraint be preserved in time and so generate a secondary one, classify both constraints as *first-class*, and show that in their smeared form they generate exactly the gauge transformation $\delta A_\mu = \partial_\mu \Lambda$. The first-class character is not a technicality; it is the canonical fingerprint of gauge redundancy, and it is what forces the degree-of-freedom count from four down to two. By the end we will have named the disease in full clinical detail. The two cures—fixing the gauge completely, or imposing the constraints weakly on states—occupy the rest of the chapter.

13.1.1 The frozen photon sector and its conventions

We continue QED. The classical theory was built in Chapter 8 from the gauge principle: the demand that the global U(1) phase symmetry of the electron be promoted to a local one forced into existence a connection A_μ, a covariant derivative, and the curvature

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{13.1}$$

(Theorem 8.8), gauge invariant by construction. The connection's own dynamics is the Maxwell Lagrangian

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu}, \tag{13.2}$$

recalled from Eq. (8.75). Throughout this chapter we work with the free field, $J^\mu = 0$; the coupling to the electron current was assembled in Chapter 8 and is reinstated only when we write Feynman rules. We use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$, and we adopt the conventions of Chapter 8 for the electric and magnetic fields,

$$E^i = F^{i0}, \qquad B^i = -\tfrac{1}{2}\epsilon^{ijk} F_{jk}, \tag{13.3}$$

so that in three-vector form $\mathbf{E} = -\nabla A^0 - \partial_t \mathbf{A}$ and $\mathbf{B} = \nabla \times \mathbf{A}$, with $\mathbf{A} = (A^1, A^2, A^3)$ and A^0 the scalar potential. In terms of these the Lagrangian density is

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} = \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2), \tag{13.4}$$

the standard identity $F_{\mu\nu}F^{\mu\nu} = 2(\mathbf{B}^2 - \mathbf{E}^2)$ for mostly-minus signature. Finally, the gauge freedom we must reckon with is

$$A_\mu \;\longrightarrow\; A_\mu + \partial_\mu \Lambda, \tag{13.5}$$

for an arbitrary function $\Lambda(x)$, recalled from Eq. (8.113); it leaves Eq. (13.1) and hence Eq. (13.2) untouched. (We write the parameter Λ where Chapter 8 wrote α; the orbit is the same.) That Λ is an arbitrary *function*, not a single constant, is the whole story: by Noether's second theorem (Theorem 7.36) a local invariance produces not a conserved charge but an off-shell identity among the field equations, the signal that the description is redundant. We now watch that redundancy resurface inside the canonical formalism.

13.1.2 The singular Legendre transform and the primary constraint

The Hamiltonian formulation of a field theory (Section 6.4) takes the time derivatives $\dot{A}_\mu \equiv \partial_0 A_\mu$ as velocities and Legendre-transforms in them, treating the spatial gradients $\partial_i A_\mu$ as spectators. The momentum conjugate to A_μ is

$$\pi^\mu \equiv \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)}. \tag{13.6}$$

The derivative $\partial\mathcal{L}/\partial(\partial_\rho A_\sigma) = -F^{\rho\sigma}$ was computed once and for all in Eq. (6.190); setting $\rho = 0$,

$$\pi^\mu = -F^{0\mu}. \tag{13.7}$$

Read this component by component. For the spatial momenta,

$$\pi^i = -F^{0i} = F^{i0} = E^i, \tag{13.8}$$

using Eq. (13.3): the momentum conjugate to the vector potential is the electric field, exactly as one would hope. But for the time component the antisymmetry of F is fatal:

$$\boxed{\pi^0 = -F^{00} \equiv 0.} \tag{13.9}$$

This is not a value that π^0 happens to take on some solution. It is an identity in the phase-space variables, true for *every* field configuration, because $F^{00} = \partial^0 A^0 - \partial^0 A^0$ vanishes algebraically. The Lagrangian Eq. (13.2) simply contains no $\dot{A}_0$: the time derivative of the scalar potential never appears, so A_0 carries no momentum and no second-order evolution equation. Equation (13.9) is the *primary constraint* of the Maxwell field—"primary" because it follows from the definition of the momenta alone, before any dynamics.

Definition 13.1 (Constraint, primary and secondary)**.** A *constraint* is a relation $\phi(A, \pi) \approx 0$ among the canonical variables that holds on the physically accessible region of phase space. A constraint is *primary* if it follows from the definition of the momenta Eq. (13.6) alone, and *secondary* if it is produced only by demanding that a primary constraint be preserved under time evolution. Following Dirac, we write $\approx$ ("weakly equal to zero") to mean that a quantity vanishes *on* the constraint surface but may have nonzero Poisson brackets with the canonical variables; this is to be distinguished from a strong identity, set equal with $=$.

The distinction in Theorem 13.1 is the heart of the matter and is easy to mishandle. Although Eq. (13.9) reads $\pi^0 \equiv 0$ as an algebraic identity, we must *not* substitute $\pi^0 = 0$ into the Poisson brackets and forget about it. The variable π^0 remains a coordinate on the full phase space, canonically conjugate to A_0 with $\{A_0(\mathbf{x}), \pi^0(\mathbf{y})\}_{\mathrm{P}} = \delta^{(3)}(\mathbf{x}-\mathbf{y})$; the constraint $\pi^0 \approx 0$ merely tells us we are confined to the surface where it vanishes. Setting it strongly to zero too early would discard precisely the information—the bracket of the constraint with the Hamiltonian—that generates the secondary constraint below. This is why Dirac's weak-equality symbol exists.

The singular Hessian. The obstruction is invertibility of the Legendre map, and that is governed by the Hessian of $\mathcal{L}$ with respect to the velocities,

$$W^{\mu\nu} \equiv \frac{\partial^2 \mathcal{L}}{\partial(\partial_0 A_\mu)\partial(\partial_0 A_\nu)} = \frac{\partial \pi^\mu}{\partial(\partial_0 A_\nu)}. \tag{13.10}$$

If $W^{\mu\nu}$ is invertible, the relation $\pi^\mu(\dot{A})$ can be solved for $\dot{A}_\mu(\pi)$ and the Legendre transform is regular, as it was for the scalar. We compute it directly from Eq. (13.7). Writing $F^{0\mu} = \partial^0 A^\mu - \partial^\mu A^0$ and differentiating with respect to $\partial_0 A_\nu$,

$$\frac{\partial(\partial^0 A^\mu)}{\partial(\partial_0 A_\nu)} = \frac{\partial(\eta^{\mu\alpha}\partial_0 A_\alpha)}{\partial(\partial_0 A_\nu)} = \eta^{\mu\nu}, \tag{13.11}$$

$$\frac{\partial(\partial^\mu A^0)}{\partial(\partial_0 A_\nu)} = \frac{\partial(\eta^{\mu\alpha}\partial_\alpha A_0)}{\partial(\partial_0 A_\nu)} = \eta^{\mu 0}\,\delta^\nu{}_0, \tag{13.12}$$

where the second line is nonzero only when the derivative is the time derivative, $\alpha = 0$, acting on the component A_0, i.e. $\nu = 0$. Hence

$$W^{\mu\nu} = -\frac{\partial F^{0\mu}}{\partial(\partial_0 A_\nu)} = -\eta^{\mu\nu} + \eta^{\mu 0}\,\delta^\nu{}_0. \tag{13.13}$$

The matrix Eq. (13.13) is manifestly singular. Its entries in the time row are $W^{0\nu} = -\eta^{0\nu} + \eta^{00}\delta^\nu{}_0 = -\eta^{0\nu} + \delta^\nu{}_0 = 0$ for every ν: the entire $\mu = 0$ row vanishes. Therefore

$$\det W^{\mu\nu} = 0, \tag{13.14}$$

and the velocity $\dot{A}_0$ cannot be recovered from the momenta—consistent with the fact that it never entered $\mathcal{L}$ in the first place. In the spatial block the Hessian is perfectly healthy, $W^{ij} = -\eta^{ij} = \delta^{ij}$, so the three velocities $\dot{A}_i$ are invertibly related to $\pi^i = E^i$; only the time direction is degenerate. A singular Hessian is the precise mathematical content of "constrained system," and it is exactly what the scalar lacked: there $W = \partial^2\mathcal{L}/\partial\dot{\phi}^2 = 1 \neq 0$, the Legendre transform was a bijection, and the canonical recipe ran without incident.

Physical Insight

The failure is structural, not a defect of bookkeeping. The Lagrangian Eq. (13.2) is built from $F_{\mu\nu}$ alone, and $F_{0\nu}$ involves the velocity $\dot{A}_\nu$ only through the *antisymmetric* combination $\dot{A}_\nu - \partial_\nu A_0$. There is no way to form a velocity-quadratic term in $\dot{A}_0$ out of an object antisymmetric in its two indices: $F_{00} = 0$. Gauge invariance demands that the action depend only on F, and F structurally refuses to give A_0 a kinetic term. The non-dynamical status of A_0, the primary constraint, and the singular Hessian are three faces of the same fact—and that fact is gauge invariance.

13.1.3 The canonical Hamiltonian

To build the Hamiltonian we Legendre-transform in the velocities we *can* invert. From Eq. (13.8), $E^i = \dot{A}_i - \partial_i A_0$ (this is Eq. (13.3) written out, $F^{i0} = \partial^i A^0 - \partial^0 A^i = \dot{A}_i - \partial_i A_0$ after lowering), so the spatial velocity is

$$\dot{A}_i = \pi^i + \partial_i A_0. \tag{13.15}$$

The Hamiltonian density is $\mathcal{H} = \pi^\mu \dot{A}_\mu - \mathcal{L}$. The $\mu = 0$ term contributes nothing: $\pi^0 \dot{A}_0 \approx 0$ by the primary constraint, and in any case $\dot{A}_0$ is not expressible through momenta and must drop out of a well-defined Hamiltonian. Thus $\mathcal{H} = \pi^i \dot{A}_i - \mathcal{L}$. Substituting Eq. (13.15) and Eq. (13.4) with $E^i = \pi^i$,

$$\begin{aligned} \mathcal{H} &= \pi^i(\pi^i + \partial_i A_0) - \tfrac{1}{2}(\boldsymbol{\pi}^2 - \mathbf{B}^2) \\ &= \boldsymbol{\pi}^2 + \pi^i \partial_i A_0 - \tfrac{1}{2}\boldsymbol{\pi}^2 + \tfrac{1}{2}\mathbf{B}^2 \\ &= \tfrac{1}{2}(\boldsymbol{\pi}^2 + \mathbf{B}^2) + \pi^i\, \partial_i A_0. \end{aligned} \tag{13.16}$$

The first group $\frac{1}{2}(\boldsymbol{\pi}^2 + \mathbf{B}^2) = \frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$ is the familiar electromagnetic energy density, gauge invariant and positive. The second term carries A_0. Integrating the total Hamiltonian $H = \int \mathrm{d}^3x\, \mathcal{H}$ by parts in the second term, $\int \mathrm{d}^3x\, \pi^i \partial_i A_0 = -\int \mathrm{d}^3x\, A_0\, \partial_i \pi^i$ (the surface term vanishes for fields decaying at spatial infinity), we obtain the canonical Hamiltonian density up to a total spatial derivative,

$$\boxed{\mathcal{H} = \tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) - A_0\, \partial_i \pi^i.} \tag{13.17}$$

This is the central object of the analysis. The scalar potential A_0 appears in Eq. (13.17) *linearly* and *undifferentiated in time*—not as a dynamical variable with its own energy, but as a Lagrange multiplier multiplying the quantity $\partial_i \pi^i = \nabla \cdot \mathbf{E}$. A field that enters the Hamiltonian as an undetermined multiplier is the hallmark of a constrained system: varying H with respect to it does not give an equation of motion but enforces a constraint. We turn to that next.

13.1.4 Consistency in time and the secondary Gauss constraint

Because A_0 is not determined by the momenta, its velocity $\dot{A}_0$ is arbitrary; equivalently, the primary constraint $\pi^0 \approx 0$ leaves a direction in phase space along which the dynamics is not fixed. Dirac's prescription handles this by adjoining the primary constraint to the Hamiltonian with an undetermined multiplier $u(x)$, forming the *total Hamiltonian*

$$H_T = \int \mathrm{d}^3x \left[\tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) - A_0\, \partial_i \pi^i + u\, \pi^0 \right]. \tag{13.18}$$

The dynamics of any phase-space functional is $\dot{F} = \{F, H_T\}_{\mathrm{P}}$, with the fundamental equal-time brackets (Theorem 6.17)

$$\{A_\mu(\mathbf{x}), \pi^\nu(\mathbf{y})\}_{\mathrm{P}} = \delta_\mu{}^\nu\, \delta^{(3)}(\mathbf{x}-\mathbf{y}), \qquad \{A_\mu, A_\nu\}_{\mathrm{P}} = \{\pi^\mu, \pi^\nu\}_{\mathrm{P}} = 0, \tag{13.19}$$

all others vanishing. A constraint is only meaningful if it is preserved by the evolution: if $\pi^0 \approx 0$ at one instant but $\dot{\pi}^0 \not\approx 0$, the system would leave the constraint surface and the theory would be inconsistent. We therefore impose the *consistency condition* $\dot{\pi}^0 \approx 0$ and read off what it demands.

Compute $\dot{\pi}^0 = \{\pi^0(\mathbf{x}), H_T\}_{\mathrm{P}}$. The momentum π^0 has vanishing bracket with the kinetic piece $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$ (which depends only on π^i and A_i, all of which commute with π^0) and with the multiplier term $u\,\pi^0$ (a function of momenta only). The single surviving contribution is from $-A_0\,\partial_i\pi^i$, through the bracket of π^0 with A_0:

$$\dot{\pi}^0(\mathbf{x}) = \left\{ \pi^0(\mathbf{x}), -\int \mathrm{d}^3y\, A_0(\mathbf{y})\, \partial_i \pi^i(\mathbf{y}) \right\}_{\mathrm{P}} = -\int \mathrm{d}^3y\, \big(-\delta^{(3)}(\mathbf{x}-\mathbf{y})\big)\, \partial_i \pi^i(\mathbf{y}) = \partial_i \pi^i(\mathbf{x}), \tag{13.20}$$

where we used $\{\pi^0(\mathbf{x}), A_0(\mathbf{y})\}_{\mathrm{P}} = -\delta^{(3)}(\mathbf{x}-\mathbf{y})$ from Eq. (13.19). Demanding $\dot{\pi}^0 \approx 0$ therefore produces a brand-new constraint,

$$\boxed{\mathcal{G}(\mathbf{x}) \equiv \partial_i \pi^i = \nabla \cdot \mathbf{E} \approx 0.} \tag{13.21}$$

This is the *secondary* constraint, and we recognize it on sight: Eq. (13.21) is Gauss's law for the source-free field, $\nabla \cdot \mathbf{E} = 0$. In ordinary electrodynamics this is one of Maxwell's equations, derived as an equation of motion. Here it has reappeared as a *constraint*—a restriction on the initial data, not a statement about time evolution. That Gauss's law is a constraint rather than a dynamical law is the canonical expression of the fact that it contains no time derivatives of the dynamical fields: it is the $\mu = 0$ component of $\partial_\mu F^{\mu\nu} = 0$, which involves only $\partial_i F^{i0} = \partial_i E^i$.

Does the chain continue? We must now ask whether $\mathcal{G} \approx 0$ is itself preserved in time, lest it spawn a tertiary constraint. Compute $\dot{\mathcal{G}} = \{\mathcal{G}, H_T\}_{\mathrm{P}}$. The

constraint $\mathcal{G} = \partial_i \pi^i$ is built from spatial momenta alone; its bracket with the $\mathbf{B}^2$ term vanishes because $\mathbf{B}$ depends only on A_i and $\{\pi^i, A_j\}_{\mathrm{P}}$ produces only the magnetic field's own gradient structure, which integrates to give $\left\{\mathcal{G}, \int \frac{1}{2}\mathbf{B}^2\right\}_{\mathrm{P}} = 0$ identically. Concretely, $\mathbf{B} = \nabla \times \mathbf{A}$ is gauge invariant and $\mathcal{G} = \partial_i \pi^i$ generates spatial gauge transformations of $\mathbf{A}$ (as we prove in Section 13.1.5); a gauge-invariant functional has vanishing bracket with the generator of gauge transformations. The bracket of $\mathcal{G}$ with $\frac{1}{2}\boldsymbol{\pi}^2$ vanishes because both are functions of momenta only, and with $A_0\, \partial_j \pi^j$ and $u\, \pi^0$ because $\mathcal{G}$ has zero bracket with π^0 and with the momentum-only $\partial_j \pi^j$. Hence

$$\dot{\mathcal{G}} = \left\{\partial_i \pi^i, H_T\right\}_{\mathrm{P}} = 0, \tag{13.22}$$

identically—not merely weakly. The consistency algorithm terminates: Gauss's law is automatically preserved, no further constraints are generated, and the multiplier u is never fixed. That u remains undetermined is not a failure but the signature of the situation we are diagnosing: an arbitrary function in the solution of the equations of motion is the canonical face of an arbitrary function $\Lambda(x)$ in the gauge freedom Eq. (13.5). The system has exactly two constraints,

$$\phi_1 \equiv \pi^0 \approx 0, \qquad \phi_2 \equiv \partial_i \pi^i \approx 0, \tag{13.23}$$

and we now classify them.

13.1.5 First-class constraints and the gauge generator

Dirac's classification turns on the Poisson brackets the constraints make among themselves and with the Hamiltonian.

Definition 13.2 (First- and second-class constraints)**.** A set of constraints $\{\phi_a \approx 0\}$ is *first-class* if every pairwise Poisson bracket vanishes weakly, $\{\phi_a, \phi_b\}_{\mathrm{P}} \approx 0$, and each bracket with the Hamiltonian also vanishes weakly, $\{\phi_a, H\}_{\mathrm{P}} \approx 0$. A constraint that is not first-class is *second-class*. First-class constraints generate gauge transformations: motions along the constraint surface that connect physically identical configurations. Second-class constraints, by contrast, come in conjugate pairs with non-vanishing bracket and are eliminated by passing to Dirac brackets—this is the route the Dirac field took in Section 11.3, and the contrast is exactly the point.

Proposition 13.3 (The Maxwell constraints are first-class)**.** *The constraints $\phi_1 = \pi^0$ and $\phi_2 = \partial_i \pi^i$ of Eq. (13.23) satisfy*

$$\{\phi_1(\mathbf{x}), \phi_1(\mathbf{y})\}_P = \{\phi_1(\mathbf{x}), \phi_2(\mathbf{y})\}_P = \{\phi_2(\mathbf{x}), \phi_2(\mathbf{y})\}_P = 0 \tag{13.24}$$

identically, and each has weakly vanishing bracket with the canonical Hamiltonian. Both constraints are therefore first-class.

Proof. All three brackets in Eq. (13.24) vanish for the same elementary reason: every constraint in Eq. (13.23) is a function of the momenta π^μ alone, and momenta have vanishing Poisson brackets with one another, $\{\pi^\mu(\mathbf{x}), \pi^\nu(\mathbf{y})\}_{\mathrm{P}} = 0$ by Eq. (13.19). Explicitly,

$$\left\{\pi^0(\mathbf{x}), \partial_i^{(\mathbf{y})}\pi^i(\mathbf{y})\right\}_{\mathrm{P}} = \partial_i^{(\mathbf{y})}\left\{\pi^0(\mathbf{x}), \pi^i(\mathbf{y})\right\}_{\mathrm{P}} = 0, \tag{13.25}$$

and identically for the other two pairs; the spatial derivative passes through the bracket because it acts on a different argument. The constraints commute not just weakly but strongly. For the brackets with the Hamiltonian, $\{\phi_1, H\}_{\mathrm{P}} = \{\pi^0, H\}_{\mathrm{P}} = \partial_i\pi^i = \phi_2 \approx 0$ by Eq. (13.20), which is weakly zero, and $\{\phi_2, H\}_{\mathrm{P}} = \dot{\mathcal{G}} = 0$ by Eq. (13.22), which is strongly zero. Both conditions of Theorem 13.2 are met. □

The first-class property is the canonical diagnosis we came for. It guarantees, by Dirac's theorem, that the constraints generate transformations under which the physics is invariant—gauge transformations—and we now exhibit this generation explicitly. Form the *smeared generator*, the constraints integrated against an arbitrary spacetime function $\Lambda(x)$ in the combination that the structure of H_T dictates,

$$G[\Lambda] \equiv \int \mathrm{d}^3x \left[\,(\partial_0\Lambda)\,\pi^0 \;+\; (\partial_i\Lambda)\,\pi^i\,\right] = \int \mathrm{d}^3x \left[\,(\partial_0\Lambda)\,\pi^0 \;-\; \Lambda\,\partial_i\pi^i\,\right], \tag{13.26}$$

the second form following by parts. This is a sum of the primary constraint smeared against $\dot{\Lambda}$ and the secondary constraint smeared against Λ; the relative structure is exactly Castellani's recipe for assembling a gauge generator out of a first-class chain, and we will see it produces a clean result. Compute its bracket with the spatial potential, using Eq. (13.19):

$$\begin{aligned}\{A_j(\mathbf{x}), G[\Lambda]\}_{\mathrm{P}} &= \int \mathrm{d}^3y\,(\partial_i\Lambda)(\mathbf{y})\,\left\{A_j(\mathbf{x}), \pi^i(\mathbf{y})\right\}_{\mathrm{P}} \\ &= \int \mathrm{d}^3y\,(\partial_i\Lambda)(\mathbf{y})\,\delta_j{}^i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}) = \partial_j\Lambda(\mathbf{x}).\end{aligned} \tag{13.27}$$

Only the π^i term contributes, since A_j has no bracket with π^0. For the time component, only the π^0 term survives,

$$\{A_0(\mathbf{x}), G[\Lambda]\}_{\mathrm{P}} = \int \mathrm{d}^3y\,(\partial_0\Lambda)(\mathbf{y})\,\left\{A_0(\mathbf{x}), \pi^0(\mathbf{y})\right\}_{\mathrm{P}} = \partial_0\Lambda(\mathbf{x}). \tag{13.28}$$

Assembling Eqs. (13.27) and (13.28) into a single four-vector statement,

$$\boxed{\delta A_\mu(\mathbf{x}) = \{A_\mu(\mathbf{x}), G[\Lambda]\}_{\mathrm{P}} = \partial_\mu\Lambda(\mathbf{x}).} \tag{13.29}$$

The smeared first-class constraints generate precisely the gauge transformation Eq. (13.5). This is the punch line of the canonical analysis: gauge invariance,

which entered in Chapter 8 as a redundancy of the Lagrangian, reappears in the Hamiltonian formulation as a pair of first-class constraints whose generator moves A_μ along its gauge orbit without changing any physical quantity. The arbitrariness of $\Lambda(x)$ is the same arbitrariness as the undetermined multiplier u in Eq. (13.18): motions generated by first-class constraints are physically null.

Physical Insight

The two structures we have met—"A_0 is a Lagrange multiplier" and "the constraints are first-class"—are the same statement read at two levels. A first-class primary constraint always appears in the total Hamiltonian multiplied by an arbitrary function (here u, equivalently A_0 once $\dot{A}_0$ is folded in); that arbitrary function parametrizes a gauge transformation; and the secondary constraint it forces is the generator of the residual gauge freedom on the dynamical fields. Counting first-class constraints *is* counting independent gauge parameters. For Maxwell there are two first-class constraints and one gauge function $\Lambda(x)$ together with its time derivative, exactly accounting for the single arbitrary function of the gauge group.

13.1.6 The degree-of-freedom count

We can now perform the count that the whole apparatus was built to deliver, and do it honestly in phase space, where the rule is sharp. The unconstrained phase space of the Maxwell field has, at each point of space, the four fields A_μ and their four conjugate momenta π^μ:

$$\text{phase-space dimension (per space point)} = 4 + 4 = 8. \tag{13.30}$$

Each *first-class* constraint removes *two* dimensions from the physical phase space, not one. The reasoning is Dirac's: a single first-class constraint $\phi_a \approx 0$ both restricts the dynamics to the surface $\phi_a = 0$ (one dimension lost) *and* generates a gauge orbit along which all points are physically identified (a second dimension lost, the direction conjugate to the constraint). We have two first-class constraints, $\pi^0 \approx 0$ and $\partial_i \pi^i \approx 0$, so

$$\text{physical phase-space dimension} = 8 - 2 \times 2 = 4. \tag{13.31}$$

A physical phase space of dimension four corresponds to two configuration-space degrees of freedom—two coordinates and their two conjugate momenta. Equivalently, counting in configuration space: of the four fields A_μ, the primary constraint removes the non-dynamical A_0, and the gauge freedom generated by the Gauss constraint removes one combination of the three A_i (the longitudinal

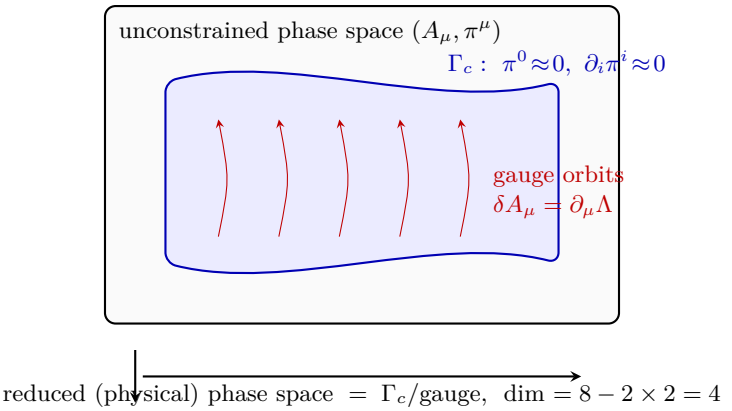


Figure 13.1: The first-class structure geometrically. The two constraints cut the 8-dimensional unconstrained phase space down to the constraint surface Γ_c (each constraint removes one dimension). Along Γ_c the first-class generator $G[\Lambda]$ sweeps out gauge orbits (red), each a one-parameter family of physically identical configurations related by $\delta A_\mu = \partial_\mu \Lambda$ (Eq. (13.29)); quotienting by these orbits removes a second dimension per constraint. The physical phase space is the quotient Γ_c/gauge, of dimension $8 - 2 \times 2 = 4$, i.e. two polarizations.

part), leaving

$$\boxed{4 \text{ components} - \underbrace{1}_{A_0 \text{ non-dynamical}} - \underbrace{1}_{\text{gauge}} = 2 \text{ physical polarizations.}} \tag{13.32}$$

The two countings agree: two physical field degrees of freedom, four physical phase-space dimensions. This is the resolution of the paradox we opened with. The four components of A_μ do *not* describe a particle with four polarizations; after the redundancy is excised, exactly two survive.

Tying the two to helicity. The number two is not an accident of the count; it is fixed by representation theory, and we can say which two. The photon is massless, and a massless particle is classified by the little group ISO(2) of a lightlike momentum, whose finite-dimensional unitary representations are labelled by a single helicity h (Chapter 3 built the little-group classification of one-particle states). For the field A_μ, which carries the vector representation of the Lorentz group, the two physical states are the helicity eigenstates $h = +1$ and $h = -1$—the right- and left-circular polarizations of light. The would-be helicity-0 longitudinal state is exactly the one the gauge freedom Eq. (13.29) removes, and the timelike component A_0 is the one the primary constraint forbids from propagating. Masslessness is essential here: a massive vector (ISO(2) $\to$ SO(3)) would carry all three helicities $0, \pm 1$, but gauge invariance forces $m = 0$ and amputates the middle one. The two surviving degrees of freedom

of the constrained Maxwell field *are* the two transverse photon polarizations of helicity ± 1, and Section 13.2 will exhibit them as honest oscillators.

13.1.7 Why naive canonical quantization is inconsistent

We have diagnosed the classical disease; the worked example below shows that it metastasizes the instant one tries to quantize without treating it. The naive prescription—copy Eq. (9.7) four times—postulates the equal-time commutators

$$[A_\mu(\mathbf{x}), \pi^\nu(\mathbf{y})] = i\,\delta_\mu{}^\nu\,\delta^{(3)}(\mathbf{x}-\mathbf{y}), \tag{13.33}$$

the operator translation of the brackets Eq. (13.19). This is mathematically inconsistent with the primary constraint, and the contradiction is immediate.

Worked Example 13.4 (The primary constraint kills the naive commutator)**.** Suppose we promote A_μ and π^ν to operators obeying Eq. (13.33), while also insisting—as we must, since it is an algebraic identity of the classical theory—that the primary constraint $\pi^0 \equiv 0$ hold as an operator equation. Take the $\mu = \nu = 0$ component of Eq. (13.33):

$$\Big[A_0(\mathbf{x}), \pi^0(\mathbf{y})\Big] = i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}). \tag{13.34}$$

The right-hand side is a nonzero c-number. But if $\pi^0 = 0$ identically as an operator, then the left-hand side is the commutator of A_0 with the zero operator, which is zero:

$$\Big[A_0(\mathbf{x}), \pi^0(\mathbf{y})\Big] = [A_0(\mathbf{x}), 0] = 0. \tag{13.35}$$

Equations Eqs. (13.34) and (13.35) are flatly contradictory: $i\,\delta^{(3)}(\mathbf{x}-\mathbf{y}) \neq 0$. There is no representation of the algebra Eq. (13.33) in which π^0 is the zero operator. The naive quantization is not merely unphysical—it is impossible. The same lethal collision afflicts the secondary constraint: if $\partial_i \pi^i = 0$ as an operator, then commuting it with A_j and using Eq. (13.33) gives $[A_j(\mathbf{y}), \partial_i \pi^i(\mathbf{x})] = i\,\partial_j^{(\mathbf{x})}\delta^{(3)}(\mathbf{x}-\mathbf{y}) \neq 0$, again contradicting the vanishing of a commutator with the zero operator.

Theorem 13.4 is the obstruction in its sharpest form. One cannot simply write down four canonical commutators for A_μ; the constraints forbid it. Every honest quantization of the photon must therefore do something to the constraints *before* imposing commutation relations. There are exactly two principled options, and they organize the remainder of this chapter.

Caution

A tempting shortcut is to "solve $\pi^0 = 0$" by dropping A_0 and π^0 from the phase space at the outset, then quantizing the surviving A_i, π^i. This is on the right track but incomplete: it disposes of the primary constraint while leaving the secondary Gauss constraint $\partial_i \pi^i \approx 0$ unaccounted for, and Theorem 13.4 shows that $[A_j, \partial_i \pi^i] \neq 0$ would again be inconsistent with imposing $\partial_i \pi^i = 0$ as an operator on the full space. One must additionally remove the longitudinal degree of freedom that the Gauss constraint generates. Doing so completely—fixing the gauge so that only transverse modes remain—is the Coulomb-gauge program of Section 13.2; imposing the constraint instead as a condition on physical *states* is Gupta–Bleuler.

The first option honors the constraints by *solving them classically*: fix the gauge completely, eliminate A_0 and the longitudinal part of $\mathbf{A}$ once and for all, and quantize only the two transverse degrees of freedom that Eq. (13.32) isolated. This buys a manifestly positive-norm Hilbert space—every surviving mode is a genuine oscillator—at the price of manifest Lorentz covariance, since the gauge condition singles out a frame. That is the Coulomb-gauge construction of Section 13.2, where the two helicity-± 1 states promised above are built explicitly and the photon emerges as a massless spin-one particle. The second option keeps all four components and manifest covariance, paying with an enlarged, indefinite-metric space in which the first-class constraints are imposed not as operator identities but *weakly*, as conditions selecting a physical subspace—the Gupta–Bleuler method developed later in the chapter. Both roads begin from the diagnosis made here: the photon is a constrained system with two first-class constraints, two physical polarizations, and no consistent naive quantization.

13.2 Canonical Quantization in Coulomb Gauge

The constraint analysis of the preceding section diagnosed the disease precisely: the electromagnetic field is a singular (constrained) system. The Legendre transform is non-invertible because $\pi^0 \equiv \partial \mathcal{L}/\partial \dot{A}_0 = 0$ is a primary constraint, Gauss's law $\nabla \cdot \mathbf{E} \approx 0$ appears as a secondary constraint, and the two together are *first class* — they generate the gauge transformations $A_\mu \to A_\mu + \partial_\mu \alpha$ rather than relating physically distinct states. A first-class system cannot be quantized by blindly imposing $[A_\mu, \pi^\nu] = i\delta_\mu^\nu \delta^3$: the constraints would then fail to hold as operator equations, and the would-be Hilbert space would contain gauge copies of every physical state. There are exactly two honest cures. The first, pursued here, is to *use up* the gauge freedom completely by a gauge condition, solve the constraints explicitly so that only genuinely independent data remain, and quantize those. The reward is a manifestly positive-norm Fock space containing nothing but physical photons; the price, paid knowingly, is that one component

of the construction — the instantaneous Coulomb interaction — is not Lorentz covariant in form, even though the final physics is. This section carries out that program in full and ends by identifying the quantum it produces: a massless particle of helicity ± 1, the photon.

Throughout we work in natural units $\hbar = c = 1$ with the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and we take the free Maxwell theory coupled to an external or matter conserved current $j^\mu = (\rho, \mathbf{j})$,

$$\mathcal{L} = -\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} - j^\mu A_\mu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{13.36}$$

the free theory being the special case $j^\mu = 0$. We write the physical fields in terms of the upper-index spatial vector potential $\mathbf{A} = (A^i)$ as $\mathbf{E} = -\nabla A^0 - \dot{\mathbf{A}}$ and $\mathbf{B} = \nabla \times \mathbf{A}$, so that $\mathcal{L} = \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) - \rho A^0 + \mathbf{j}\cdot\mathbf{A}$. To avoid the index gymnastics of raising and lowering spatial labels, in the canonical analysis we treat all three-vector indices i, j, k as Cartesian (summed when repeated, no sign from the metric); A_i denotes the i-th Cartesian component of $\mathbf{A}$.

13.2.1 Completing the gauge fixing: the Coulomb condition

The classical groundwork is in hand from Chapter 8. There the *Coulomb gauge* (equivalently radiation or transverse gauge) was defined by

$$\nabla\cdot\mathbf{A} = 0, \tag{13.37}$$

shown to be reachable from any configuration (Eq. (8.128)), and shown to leave only a physically inert global constant as residual freedom. We recall the two structural facts we will quantize, recasting them in canonical language.

Coulomb gauge is attainable and essentially unique. Given an arbitrary potential $\mathbf{A}$, the gauge transformation $\mathbf{A} \to \mathbf{A}' = \mathbf{A} + \nabla\chi$ obeys $\nabla\cdot\mathbf{A}' = \nabla\cdot\mathbf{A} + \nabla^2\chi$, so choosing χ to solve the Poisson equation $\nabla^2\chi = -\nabla\cdot\mathbf{A}$ makes $\mathbf{A}'$ transverse. The residual freedom consists of further transformations with $\nabla^2\chi = 0$; for fields decaying at spatial infinity the only spatially harmonic functions are constants, whose gradient vanishes. Hence Eq. (13.37) fixes the gauge completely up to a global constant that acts trivially on $\mathbf{A}$. This is the crucial point that distinguishes Coulomb gauge from the covariant gauges of the next section: there is no residual local gauge freedom left to contaminate the quantization.

Gauss's law fixes A^0 with no dynamics of its own. The $\mu = 0$ Euler–Lagrange equation of Eq. (13.36) is Gauss's law $\nabla\cdot\mathbf{E} = \rho$. Writing $\mathbf{E} = -\nabla A^0 - \dot{\mathbf{A}}$,

$$\nabla\cdot\mathbf{E} = -\nabla^2 A^0 - \partial_0(\nabla\cdot\mathbf{A}) = -\nabla^2 A^0 = \rho, \tag{13.38}$$

where the gauge condition Eq. (13.37) annihilated the second term. Equation Eq. (13.38) contains no time derivative of A^0: it is a *constraint*, not an equation of motion. It is solved at each instant by the instantaneous Coulomb potential

$$A^0(t,\mathbf{x}) = \int d^3y \, \frac{\rho(t,\mathbf{y})}{4\pi\,|\mathbf{x}-\mathbf{y}|}, \tag{13.39}$$

the inverse of $-\nabla^2$ with the Green function $G(\mathbf{x}) = 1/(4\pi|\mathbf{x}|)$ satisfying $-\nabla^2 G = \delta^3$ (this is Eq. (8.130)). The timelike component A^0 is therefore not an independent dynamical variable at all: it is a fixed functional of the charge density at the same instant, with no canonical conjugate of its own ($\pi^0 = 0$). In the free theory $\rho = 0$ forces $A^0 = 0$ identically. We have thereby eliminated, explicitly and once and for all, two of the four components of A_μ: the non-dynamical A^0 via Eq. (13.39), and the longitudinal part of $\mathbf{A}$ via Eq. (13.37). What remains are the two transverse components of $\mathbf{A}$ — exactly the degree-of-freedom count the constraint analysis predicted, now realized field by field.

13.2.2 The canonical momentum and the obstruction to the naive bracket

The momentum conjugate to the surviving field $\mathbf{A}$ follows from Eq. (13.36). With A_i a Cartesian component and $\dot{A}_i = \partial_0 A_i$,

$$\pi_i \equiv \frac{\partial \mathcal{L}}{\partial \dot{A}_i} = \frac{\partial}{\partial \dot{A}_i}\Big[\tfrac{1}{2}E_k E_k - \tfrac{1}{2}B_k B_k\Big] = E_k \frac{\partial E_k}{\partial \dot{A}_i} = -E_i, \tag{13.40}$$

since $E_k = -\partial_k A^0 - \dot{A}_k$ gives $\partial E_k/\partial \dot{A}_i = -\delta_{ki}$, and $\mathbf{B}$ contains no time derivatives. Thus

$$\boldsymbol{\pi} = -\mathbf{E}, \tag{13.41}$$

the momentum conjugate to the vector potential is (minus) the electric field. In the free theory, where $A^0 = 0$, this reads $\boldsymbol{\pi} = \dot{\mathbf{A}}$, and Gauss's law $\nabla\cdot\mathbf{E} = 0$ becomes $\nabla\cdot\boldsymbol{\pi} = 0$: the conjugate momentum is transverse, exactly as the coordinate $\mathbf{A}$ is.

Here is the obstruction in its sharpest form. The naive canonical postulate $[A_i(\mathbf{x}), \pi_j(\mathbf{y})] = i\delta_{ij}\delta^3(\mathbf{x}-\mathbf{y})$ is *inconsistent* with the two facts we have just established. Apply ∂^x_i to both sides: the left side gives $[\nabla\cdot\mathbf{A}(\mathbf{x}), \pi_j(\mathbf{y})] = 0$ because $\nabla\cdot\mathbf{A} = 0$ is an operator identity in this gauge, yet the right side gives $i\,\partial_j\delta^3(\mathbf{x}-\mathbf{y}) \neq 0$. The same contradiction follows from applying ∂^y_j and using $\nabla\cdot\boldsymbol{\pi} = 0$. A field that is transverse cannot have a canonical bracket whose kernel $\delta_{ij}\delta^3$ has a longitudinal part. The cure is to replace the naive bracket by the unique transverse object compatible with both constraints — the transverse delta function.

13.2.3 The transverse delta function

Definition 13.5 (Transverse delta function)**.** The *transverse delta function* is the kernel

$$\delta^T_{ij}(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3}\,(\delta_{ij} - \hat{k}_i\hat{k}_j)\,e^{i\mathbf{k}\cdot\mathbf{x}}, \qquad \hat{k}_i = \frac{k_i}{|\mathbf{k}|}, \tag{13.42}$$

the Fourier transform of the transverse projector $P_{ij}(\mathbf{k}) = \delta_{ij} - \hat{k}_i\hat{k}_j$.

The matrix $P_{ij}(\mathbf{k})$ is the orthogonal projector onto the plane perpendicular to $\mathbf{k}$: it is symmetric, satisfies $P_{ij}\hat{k}_j = 0$, $P_{ik}P_{kj} = P_{ij}$, and $P_{ii} = 3 - 1 = 2$. Every property of δ^T_{ij} that we will use is a direct transcript of a property of P_{ij}, as the following worked example records.

Worked Example 13.6 (Properties of the transverse delta)**.** We establish the four facts about Eq. (13.42) that the quantization relies on.

(i) Transversality. Differentiating under the integral,

$$\partial_i\, \delta^T_{ij}(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3}\, (ik_i)(\delta_{ij} - \hat{k}_i\hat{k}_j)e^{i\mathbf{k}\cdot\mathbf{x}} = \int \frac{d^3k}{(2\pi)^3}\, i(k_j - |\mathbf{k}|\,\hat{k}_j)e^{i\mathbf{k}\cdot\mathbf{x}} = 0, \tag{13.43}$$

because $k_i\hat{k}_i = |\mathbf{k}|$ and $k_j = |\mathbf{k}|\hat{k}_j$. By symmetry $\partial_j\delta^T_{ij} = 0$ as well. The transverse delta is divergence-free on either index — precisely the property the naive $\delta_{ij}\delta^3$ lacked.

(ii) Action on a transverse field. Let $\mathbf{V}(\mathbf{x})$ be any transverse field, $\nabla\cdot\mathbf{V} = 0$, with Fourier transform $\tilde{\mathbf{V}}(\mathbf{k})$ obeying $\mathbf{k}\cdot\tilde{\mathbf{V}} = 0$. Then

$$\int d^3y\, \delta^T_{ij}(\mathbf{x}-\mathbf{y})\, V_j(\mathbf{y}) = \int \frac{d^3k}{(2\pi)^3}(\delta_{ij}-\hat{k}_i\hat{k}_j)\tilde{V}_j(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}} = \int \frac{d^3k}{(2\pi)^3}\tilde{V}_i(\mathbf{k})e^{i\mathbf{k}\cdot\mathbf{x}} = V_i(\mathbf{x}), \tag{13.44}$$

since $\hat{k}_j\tilde{V}_j = 0$. Thus δ^T_{ij} acts as the identity on the transverse subspace: it is the delta function *of* that subspace. On a longitudinal field $\mathbf{W} = \nabla\phi$ it gives zero by the same step.

(iii) Trace. Setting $j = i$ and summing, $\delta^T_{ii}(\mathbf{x}) = \int \frac{d^3k}{(2\pi)^3}(\delta_{ii} - \hat{k}_i\hat{k}_i)e^{i\mathbf{k}\cdot\mathbf{x}} = \int \frac{d^3k}{(2\pi)^3}(3-1)e^{i\mathbf{k}\cdot\mathbf{x}} = 2\,\delta^3(\mathbf{x})$, the trace of the projector being 2, one count per physical polarization.

(iv) Position-space form. Splitting off the full delta, $\delta^T_{ij}(\mathbf{x}) = \delta_{ij}\delta^3(\mathbf{x}) - \int \frac{d^3k}{(2\pi)^3}\frac{k_ik_j}{\mathbf{k}^2}e^{i\mathbf{k}\cdot\mathbf{x}} = \delta_{ij}\delta^3(\mathbf{x}) + \partial_i\partial_j\int \frac{d^3k}{(2\pi)^3}\frac{1}{\mathbf{k}^2}e^{i\mathbf{k}\cdot\mathbf{x}} = \delta_{ij}\delta^3(\mathbf{x}) + \partial_i\partial_j\frac{1}{4\pi|\mathbf{x}|}$, using $\int \frac{d^3k}{(2\pi)^3}\mathbf{k}^{-2}e^{i\mathbf{k}\cdot\mathbf{x}} = 1/(4\pi|\mathbf{x}|)$ from Eq. (13.39). The longitudinal piece is the double gradient of the Coulomb kernel — the first sign that the same object governs both the transverse photon bracket and the instantaneous Coulomb field.

13.2.4 The equal-time commutators

We can now state the canonical structure. The cleanest derivation treats the gauge condition and Gauss's law as a pair of second-class constraints and forms the Dirac bracket; we give that argument in Theorem 13.8 below. First the statement.

Theorem 13.7 (Transverse equal-time commutators)**.** *In Coulomb gauge the quantized vector potential and its conjugate momentum $\boldsymbol{\pi} = -\mathbf{E}$ obey, at equal times,*

$$[A_i(\mathbf{x}), \pi_j(\mathbf{y})] = i\,\delta^T_{ij}(\mathbf{x}-\mathbf{y}), \tag{13.45}$$

$$[A_i(\mathbf{x}), A_j(\mathbf{y})] = 0, \qquad [\pi_i(\mathbf{x}), \pi_j(\mathbf{y})] = 0. \tag{13.46}$$

These relations are consistent with the gauge condition and the constraints: applying ∂_i^x to Eq. (13.45) gives $[\nabla\cdot\mathbf{A}(\mathbf{x}), \pi_j(\mathbf{y})] = i\,\partial_i\delta_{ij}^T = 0$ by part (i) of Theorem 13.6, and applying ∂_j^y gives $[A_i(\mathbf{x}), \nabla\cdot\boldsymbol{\pi}(\mathbf{y})] = 0$ likewise.

The replacement $\delta_{ij}\delta^3 \to \delta_{ij}^T$ is the entire content of canonical quantization in this gauge: it is the minimal modification of the naive bracket that respects the constraints, and it is forced — not chosen — once one demands that $\mathbf{A}$ and $\boldsymbol{\pi}$ remain transverse as operators. The following worked example derives Eq. (13.45) from first principles via the Dirac bracket, so that no step is taken on faith.

Worked Example 13.8 (The transverse bracket from the Dirac construction)**.** After gauge fixing, the system carries two second-class constraints on each time slice,

$$\Phi_1(\mathbf{x}) = \nabla\cdot\mathbf{A}(\mathbf{x}), \qquad \Phi_2(\mathbf{x}) = \nabla\cdot\boldsymbol{\pi}(\mathbf{x}), \tag{13.47}$$

the gauge condition and (free) Gauss's law. They are second class because their mutual Poisson bracket does not vanish on the constraint surface. With the unmodified bracket $\{A_i(\mathbf{x}), \pi_j(\mathbf{y})\} = \delta_{ij}\delta^3(\mathbf{x}-\mathbf{y})$,

$$C(\mathbf{x},\mathbf{y}) \equiv \{\Phi_1(\mathbf{x}), \Phi_2(\mathbf{y})\} = \partial_i^x\partial_j^y\{A_i(\mathbf{x}), \pi_j(\mathbf{y})\} = \partial_i^x\partial_i^y\delta^3(\mathbf{x}-\mathbf{y}) = -\nabla_x^2\,\delta^3(\mathbf{x}-\mathbf{y}), \tag{13.48}$$

which is nonzero (and invertible): the 2×2 antisymmetric constraint matrix C_{ab} has $C_{12} = -C_{21} = -\nabla^2\delta^3$, $C_{11} = C_{22} = 0$. Its inverse kernel is the Coulomb Green function: $(C^{-1})_{21}(\mathbf{x},\mathbf{y}) = G(\mathbf{x}-\mathbf{y}) = 1/(4\pi|\mathbf{x}-\mathbf{y}|)$, because $-\nabla^2 G = \delta^3$, and $(C^{-1})_{12} = -G$. The Dirac bracket is

$$\{A,B\}_D = \{A,B\} - \sum_{a,b}\int d^3u\,d^3v\,\{A,\Phi_a(\mathbf{u})\}\,(C^{-1})_{ab}(\mathbf{u},\mathbf{v})\,\{\Phi_b(\mathbf{v}),B\}. \tag{13.49}$$

We evaluate it for $A = A_i(\mathbf{x})$, $B = \pi_j(\mathbf{y})$. The needed elementary brackets are

$$\{A_i(\mathbf{x}),\Phi_1(\mathbf{u})\} = 0, \quad \{A_i(\mathbf{x}),\Phi_2(\mathbf{u})\} = \partial_i^u\delta^3(\mathbf{x}-\mathbf{u}), \quad \{\Phi_1(\mathbf{v}),\pi_j(\mathbf{y})\} = \partial_j^v\delta^3(\mathbf{v}-\mathbf{y}), \quad \{\Phi_2(\tag{13.50}$$

the first and last vanishing because A commutes with A and π with π. Only the $(a,b) = (2,1)$ term of the double sum survives, and it carries $(C^{-1})_{21} = G$:

$$\{A_i(\mathbf{x}),\pi_j(\mathbf{y})\}_D = \delta_{ij}\delta^3(\mathbf{x}-\mathbf{y}) - \int d^3u\,d^3v\,[\partial_i^u\delta^3(\mathbf{x}-\mathbf{u})]G(\mathbf{u}-\mathbf{v})[\partial_j^v\delta^3(\mathbf{v}-\mathbf{y})]. \tag{13.51}$$

Integrating by parts in $\mathbf{u}$ and then $\mathbf{v}$ to move the derivatives onto G and collapse the delta functions, the double integral becomes $-\partial_i^x\partial_j^x G(\mathbf{x}-\mathbf{y})$, so

$$\{A_i(\mathbf{x}),\pi_j(\mathbf{y})\}_D = \delta_{ij}\delta^3(\mathbf{x}-\mathbf{y}) + \partial_i\partial_j\,G(\mathbf{x}-\mathbf{y}) = \delta_{ij}^T(\mathbf{x}-\mathbf{y}), \tag{13.52}$$

the last equality being part (iv) of Theorem 13.6. Quantization by the rule $[\,\cdot\,,\,\cdot\,] = i\{\,\cdot\,,\,\cdot\,\}_D$ then yields exactly Eq. (13.45). The Dirac bracket has automatically projected out the longitudinal directions in which the constraints live, leaving a consistent symplectic structure on the two-dimensional transverse phase space.

13.2.5 Mode expansion over two transverse polarizations

We now solve the free Heisenberg equation of motion and read off the particle content, following exactly the logic of the scalar field in Section 9.2, with the single new ingredient that each momentum mode carries a two-dimensional polarization label rather than none. In the free theory ($A^0 = 0$, $j^\mu = 0$) the surviving field obeys the massless wave equation $\Box\,\mathbf{A} = 0$ subject to $\nabla\cdot\mathbf{A} = 0$, so each Fourier mode is a transverse plane wave on the light cone $k^0 = \omega_k = |\mathbf{k}|$.

Definition 13.9 (Polarization vectors)**.** For each $\mathbf{k}$ choose two real unit vectors $\boldsymbol{\epsilon}_1(\mathbf{k}), \boldsymbol{\epsilon}_2(\mathbf{k})$ such that $\{\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \hat{\mathbf{k}}\}$ is a right-handed orthonormal triad:

$$\boldsymbol{\epsilon}_\lambda(\mathbf{k})\cdot\boldsymbol{\epsilon}_{\lambda'}(\mathbf{k}) = \delta_{\lambda\lambda'}, \qquad \mathbf{k}\cdot\boldsymbol{\epsilon}_\lambda(\mathbf{k}) = 0, \qquad \boldsymbol{\epsilon}_1 \times \boldsymbol{\epsilon}_2 = \hat{\mathbf{k}} \quad (\lambda, \lambda' = 1, 2). \tag{13.53}$$

The completeness of the triad gives the polarization sum that drives every later manipulation. Since $\{\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2, \hat{\mathbf{k}}\}$ is an orthonormal basis of $\mathbb{R}^3$, the resolution of the identity reads $\delta_{ij} = \epsilon_1^i\epsilon_1^j + \epsilon_2^i\epsilon_2^j + \hat{k}_i\hat{k}_j$, hence

$$\sum_{\lambda=1,2} \epsilon_\lambda^i(\mathbf{k})\,\epsilon_\lambda^j(\mathbf{k}) = \delta_{ij} - \hat{k}_i\hat{k}_j = P_{ij}(\mathbf{k}). \tag{13.54}$$

The polarization sum *is* the transverse projector: this is why the mode commutators will reproduce Eq. (13.45) on the nose.

The mode expansion mirrors Theorem 9.6, with the relativistic measure $\widetilde{dk} = d^3k/[(2\pi)^3\,2\omega_k]$ and one ladder pair per polarization:

$$\mathbf{A}(x) = \int \frac{d^3k}{(2\pi)^3\,2\omega_k} \sum_{\lambda=1,2} \boldsymbol{\epsilon}_\lambda(\mathbf{k})\Big(a_\lambda(\mathbf{k})\,e^{-ik\cdot x} + a_\lambda^\dagger(\mathbf{k})\,e^{+ik\cdot x}\Big)\Big|_{k^0=\omega_k}, \tag{13.55}$$

$$\boldsymbol{\pi}(x) = \dot{\mathbf{A}}(x) = \int \frac{d^3k}{(2\pi)^3\,2\omega_k} \sum_{\lambda=1,2} \boldsymbol{\epsilon}_\lambda(\mathbf{k})\,(-i\omega_k)\Big(a_\lambda(\mathbf{k})\,e^{-ik\cdot x} - a_\lambda^\dagger(\mathbf{k})\,e^{+ik\cdot x}\Big)\Big|_{k^0=\omega_k}, \tag{13.56}$$

with $k\cdot x = \omega_k x^0 - \mathbf{k}\cdot\mathbf{x}$. Each term is manifestly transverse because $\boldsymbol{\epsilon}_\lambda \perp \mathbf{k}$, so $\nabla\cdot\mathbf{A} = 0$ holds identically, and $\mathbf{A}^\dagger = \mathbf{A}$ because the second term is the adjoint of the first. The expansion solves $\Box\,\mathbf{A} = 0$ mode by mode since $\Box\,e^{\mp ik\cdot x} = -k^2 e^{\mp ik\cdot x} = 0$ on the light cone.

Theorem 13.10 (Mode commutators reproduce the ETCR)**.** *The expansions Eq. (13.55)–Eq. (13.56) satisfy the transverse commutators Eq. (13.45)–Eq. (13.46) if and only if the ladder operators obey*

$$\left[a_\lambda(\mathbf{k}), a^\dagger_{\lambda'}(\mathbf{k}')\right] = (2\pi)^3\, 2\omega_k\, \delta_{\lambda\lambda'}\, \delta^3(\mathbf{k}-\mathbf{k}'), \qquad [a_\lambda(\mathbf{k}), a_{\lambda'}(\mathbf{k}')] = 0, \quad (13.57)$$

the relativistically normalized oscillator algebra, one independent oscillator pair per momentum and per transverse polarization.

Proof. We verify the forward direction by computing the equal-time $[A_i(\mathbf{x}), \pi_j(\mathbf{y})]$ from Eq. (13.57). Inserting the expansions and keeping only the surviving $\left[a, a^\dagger\right]$ and $\left[a^\dagger, a\right]$ pairings,

$$[A_i(\mathbf{x}), \pi_j(\mathbf{y})] = \int \frac{d^3k}{(2\pi)^3 2\omega_k} \int \frac{d^3k'}{(2\pi)^3 2\omega_{k'}} \sum_{\lambda\lambda'} \epsilon^i_\lambda(\mathbf{k})\, \epsilon^j_{\lambda'}(\mathbf{k}')\, (-i\omega_{k'}) \\ \times \Big(- \left[a_\lambda(\mathbf{k}), a^\dagger_{\lambda'}(\mathbf{k}')\right] e^{-ik\cdot x + ik'\cdot y} + \left[a^\dagger_\lambda(\mathbf{k}), a_{\lambda'}(\mathbf{k}')\right] e^{+ik\cdot x - ik'\cdot y} \Big). \quad (13.58)$$

Both commutators carry $(2\pi)^3 2\omega_k\, \delta_{\lambda\lambda'}\delta^3(\mathbf{k}-\mathbf{k}')$, the second with a minus sign. Doing the $\mathbf{k}'$ integral collapses $\mathbf{k}' \to \mathbf{k}$, $\omega_{k'} \to \omega_k$, $\lambda' \to \lambda$, and cancels the measure factor $(2\pi)^3 2\omega_k$. At equal times $x^0 = y^0$ the time exponentials cancel, leaving $e^{\mp i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})}$ for the two terms:

$$[A_i(\mathbf{x}), \pi_j(\mathbf{y})] = \int \frac{d^3k}{(2\pi)^3\, 2\omega_k}\, (i\omega_k) \sum_\lambda \epsilon^i_\lambda(\mathbf{k}) \epsilon^j_\lambda(\mathbf{k}) \Big(e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} + e^{-i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} \Big). \quad (13.59)$$

In the second exponential substitute $\mathbf{k} \to -\mathbf{k}$; the projector Eq. (13.54) is even in $\mathbf{k}$, $P_{ij}(-\mathbf{k}) = P_{ij}(\mathbf{k})$, so the two terms combine. The factor ω_k cancels one power in the measure, giving

$$[A_i(\mathbf{x}), \pi_j(\mathbf{y})] = i \int \frac{d^3k}{(2\pi)^3} (\delta_{ij} - \hat{k}_i \hat{k}_j) e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} = i\, \delta^T_{ij}(\mathbf{x}-\mathbf{y}), \quad (13.60)$$

which is Eq. (13.45). The $[A_i, A_j]$ and $[\pi_i, \pi_j]$ vanish because each receives equal and opposite contributions from the two exponentials after the $\mathbf{k} \to -\mathbf{k}$ symmetrization. Conversely, inverting Eq. (13.55) for $a_\lambda(\mathbf{k})$ exactly as in Theorem 9.9, with the extra contraction $\boldsymbol{\epsilon}_\lambda \cdot \boldsymbol{\epsilon}_{\lambda'} = \delta_{\lambda\lambda'}$ projecting out a single polarization, reproduces Eq. (13.57) from the ETCR; the algebra is identical to the scalar case mode by mode and we do not repeat it. □

The Fock space is built precisely as in Section 9.3: a unique vacuum $|0\rangle$ annihilated by every $a_\lambda(\mathbf{k})$, and one-photon states

$$|\mathbf{k}, \lambda\rangle = a^\dagger_\lambda(\mathbf{k})\, |0\rangle\,, \qquad \langle \mathbf{k}, \lambda | \mathbf{k}', \lambda' \rangle = (2\pi)^3\, 2\omega_k\, \delta_{\lambda\lambda'}\, \delta^3(\mathbf{k}-\mathbf{k}'). \quad (13.61)$$

Because the algebra Eq. (13.57) has the standard oscillator sign (positive norm $\||\mathbf{k},\lambda\rangle\|^2 > 0$), *every* state of this Fock space has non-negative norm. This is the headline reward of Coulomb gauge: there are no negative-norm or null states to excise, no subsidiary condition to impose; the Hilbert space is physical by construction. The cost — the absence of manifest Lorentz covariance — is hidden in the frame-dependent split into $\boldsymbol{\epsilon}_1$, $\boldsymbol{\epsilon}_2$, $\hat{\mathbf{k}}$ and in the instantaneous Coulomb term we turn to next.

13.2.6 The Hamiltonian: two transverse oscillators plus the Coulomb energy

The canonical Hamiltonian density is $\mathcal{H} = \pi_i \dot{A}_i - \mathcal{L}$. Using $\dot{A}_i = \pi_i - \partial_i A^0$ (from Eq. (13.41) and $\mathbf{E} = -\nabla A^0 - \dot{\mathbf{A}}$, i.e. $\dot{\mathbf{A}} = \boldsymbol{\pi} - \nabla A^0$ with $\boldsymbol{\pi} = -\mathbf{E}$),

$$\begin{aligned}\mathcal{H} &= \pi_i(\pi_i - \partial_i A^0) - \tfrac{1}{2}(\mathbf{E}^2 - \mathbf{B}^2) + \rho A^0 - \mathbf{j}\cdot\mathbf{A} \\ &= \boldsymbol{\pi}^2 - \boldsymbol{\pi}\cdot\nabla A^0 - \tfrac{1}{2}\boldsymbol{\pi}^2 + \tfrac{1}{2}\mathbf{B}^2 + \rho A^0 - \mathbf{j}\cdot\mathbf{A} = \tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) + \mathbf{E}\cdot\nabla A^0 + \rho A^0 - \mathbf{j}\cdot\mathbf{A},\end{aligned} \tag{13.62}$$

where $\boldsymbol{\pi}^2 = \mathbf{E}^2$ and $-\boldsymbol{\pi}\cdot\nabla A^0 = \mathbf{E}\cdot\nabla A^0$. Integrating the term $\mathbf{E}\cdot\nabla A^0$ by parts over all space and using Gauss's law $\nabla\cdot\mathbf{E} = \rho$,

$$\int d^3x\, \mathbf{E}\cdot\nabla A^0 = -\int d^3x\, A^0\, \nabla\cdot\mathbf{E} = -\int d^3x\, A^0 \rho, \tag{13.63}$$

which exactly cancels the ρA^0 piece of Eq. (13.62). The Hamiltonian collapses to

$$H = \int d^3x \left[\tfrac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2) - \mathbf{j}\cdot\mathbf{A}\right]. \tag{13.64}$$

This is not yet in oscillator form because $\mathbf{E}$ still contains the longitudinal piece $-\nabla A^0$. We split it cleanly. Write $\mathbf{E} = \mathbf{E}_T + \mathbf{E}_L$ with $\mathbf{E}_T = -\dot{\mathbf{A}} = -\boldsymbol{\pi}$ transverse and $\mathbf{E}_L = -\nabla A^0$ longitudinal. Their cross term integrates to zero, $\int \mathbf{E}_T\cdot\mathbf{E}_L = -\int \mathbf{E}_T\cdot\nabla A^0 = \int A^0\, \nabla\cdot\mathbf{E}_T = 0$, because $\nabla\cdot\mathbf{E}_T = 0$. Hence $\int \mathbf{E}^2 = \int \mathbf{E}_T^2 + \int \mathbf{E}_L^2$, and the longitudinal part is the Coulomb self-energy: using Eq. (13.39) and $-\nabla^2 A^0 = \rho$,

$$\tfrac{1}{2}\int d^3x\, \mathbf{E}_L^2 = \tfrac{1}{2}\int d^3x\, (\nabla A^0)^2 = -\tfrac{1}{2}\int d^3x\, A^0 \nabla^2 A^0 = \tfrac{1}{2}\int d^3x\, A^0 \rho = \tfrac{1}{2}\int d^3x\, d^3y\, \frac{\rho(\mathbf{x})\rho(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}. \tag{13.65}$$

The Hamiltonian is therefore the sum of a radiation part built from the transverse field alone and an instantaneous Coulomb energy:

$$H = \underbrace{\tfrac{1}{2}\int d^3x\, (\boldsymbol{\pi}^2 + \mathbf{B}^2)}_{H_{\rm rad}} \; - \int d^3x\, \mathbf{j}\cdot\mathbf{A} \; + \underbrace{\tfrac{1}{2}\int d^3x\, d^3y\, \frac{\rho(\mathbf{x})\rho(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}}_{H_{\rm Coul}}. \tag{13.66}$$

Three features deserve emphasis. First, $H_{\rm Coul}$ is the familiar electrostatic interaction energy of the charge density with itself — it is *not* a dynamical photon degree of freedom but a non-local, instantaneous functional of the matter, the direct quantum-canonical descendant of Eq. (13.39). In the free theory it is absent. Second, the instantaneity is the explicit signature of broken manifest covariance: $H_{\rm Coul}$ couples charges at spatially separated points at the *same time*, a frame-dependent statement. (That the full theory is nonetheless Lorentz invariant — the apparent action-at-a-distance being exactly compensated by the retarded transverse exchange — is the deep fact the next section makes manifest.) Third, the radiation part $H_{\rm rad}$ is precisely two copies of the free scalar Hamiltonian, one per polarization.

Theorem 13.11 (Diagonal photon Hamiltonian)**.** *With* $\mathbf{B} = \nabla \times \mathbf{A}$ *and* $\boldsymbol{\pi} = \dot{\mathbf{A}}$ *in the free theory, the radiation Hamiltonian of Eq. (13.66) is, after normal ordering (Section 9.4),*

$$H_{\rm rad} = \int \frac{d^3k}{(2\pi)^3\, 2\omega_k}\, \omega_k \sum_{\lambda=1,2} a^\dagger_\lambda(\mathbf{k})\, a_\lambda(\mathbf{k}), \qquad \omega_k = |\mathbf{k}|. \tag{13.67}$$

Proof. Insert Eq. (13.55) into $H_{\rm rad} = \frac{1}{2}\int d^3x(\dot{\mathbf{A}}^2 + (\nabla\times\mathbf{A})^2)$. The electric piece $\frac{1}{2}\int \dot{\mathbf{A}}^2$ produces, through the $\int d^3x$ over a product of two mode integrals, the factor $(2\pi)^3\delta^3(\mathbf{k}\pm\mathbf{k}')$ pairing the momenta. For the magnetic piece, each plane-wave mode obeys $\nabla\times[\boldsymbol{\epsilon}_\lambda e^{\pm ik\cdot x}] = \pm i\,\mathbf{k}\times\boldsymbol{\epsilon}_\lambda\, e^{\pm ik\cdot x}$, and since $\boldsymbol{\epsilon}_\lambda \perp \mathbf{k}$ with $|\boldsymbol{\epsilon}_\lambda| = 1$,

$$(\mathbf{k}\times\boldsymbol{\epsilon}_\lambda(\mathbf{k}))\cdot(\mathbf{k}\times\boldsymbol{\epsilon}_{\lambda'}(\mathbf{k})) = \mathbf{k}^2\,\boldsymbol{\epsilon}_\lambda\cdot\boldsymbol{\epsilon}_{\lambda'} - (\mathbf{k}\cdot\boldsymbol{\epsilon}_\lambda)(\mathbf{k}\cdot\boldsymbol{\epsilon}_{\lambda'}) = \omega_k^2\,\delta_{\lambda\lambda'}. \tag{13.68}$$

Thus the magnetic energy density per mode equals ω_k^2 times the polarization overlap, identical in magnitude to the electric one ($\dot{\mathbf{A}}$ carried $(-i\omega_k)$, giving ω_k^2 as well). The two combine so that the $a_\lambda(\mathbf{k})a_{\lambda'}(-\mathbf{k})$ and $a^\dagger_\lambda(\mathbf{k})a^\dagger_{\lambda'}(-\mathbf{k})$ cross terms — which have opposite signs in the electric and magnetic parts because of the relative sign in Eq. (13.56) and the factor $i^2 = -1$ from the curl — cancel exactly, precisely as in the scalar computation of Section 9.2. What remains is the number-operator combination

$$H_{\rm rad} = \int \frac{d^3k}{(2\pi)^3 2\omega_k}\, \tfrac{1}{2}\,\omega_k \sum_\lambda (a^\dagger_\lambda(\mathbf{k})a_\lambda(\mathbf{k}) + a_\lambda(\mathbf{k})a^\dagger_\lambda(\mathbf{k})). \tag{13.69}$$

Normal ordering drops the c-number zero-point term $\frac{1}{2}\sum_\lambda \left[a_\lambda, a^\dagger_\lambda\right] = \frac{1}{2}\cdot 2\cdot (2\pi)^3 2\omega_k\,\delta^3(0)\,V$ — now *twice* the scalar vacuum energy, one half per transverse polarization, the same divergence handled and interpreted in Section 9.4 — leaving Eq. (13.67). □

The photon energy is $E = \omega_k = |\mathbf{k}|$, the dispersion of a *massless* particle: $E^2 - \mathbf{k}^2 = 0$, i.e. $k^2 = 0$. The one-photon state Eq. (13.61) is an eigenstate

of $H_{\rm rad}$ with eigenvalue ω_k and, by the same diagonalization applied to the momentum operator $\mathbf{P} = \int \widetilde{dk}\,\mathbf{k}\sum_\lambda a_\lambda^\dagger a_\lambda$, of $\mathbf{P}$ with eigenvalue $\mathbf{k}$. The photon is the massless quantum of the transverse electromagnetic field; the field carries two such quanta per momentum, exactly the count Wigner's classification demanded for a massless particle.

13.2.7 The photon as a massless helicity-±1 particle

We have built two transverse states per momentum. The representation theory of Chapter 3 predicted exactly this: a massless particle is classified not by spin but by *helicity*, the projection of angular momentum on the direction of motion, taking a single value h for a single irreducible representation, with parity-invariant theories carrying the pair $\pm h$. We now make the dictionary precise and confirm $h = \pm 1$ for the photon.

The linear polarizations $\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2$ are not helicity eigenstates: a rotation about $\hat{\mathbf{k}}$ rotates them into each other. The combinations diagonal under that rotation are the circular polarizations.

Definition 13.12 (Helicity polarization vectors)**.** The *circular* (helicity) polarization vectors are

$$\boldsymbol{\epsilon}_\pm(\mathbf{k}) = \mp\frac{1}{\sqrt{2}}\big(\boldsymbol{\epsilon}_1(\mathbf{k}) \pm i\,\boldsymbol{\epsilon}_2(\mathbf{k})\big), \tag{13.70}$$

with corresponding ladder operators $a_\pm(\mathbf{k}) = \mp\frac{1}{\sqrt{2}}(a_1(\mathbf{k}) \mp i\,a_2(\mathbf{k}))$ (the signs arranged so that $\mathbf{A}$ is unchanged by the change of basis). They satisfy $\boldsymbol{\epsilon}_\pm^* \cdot \boldsymbol{\epsilon}_\pm = 1$, $\boldsymbol{\epsilon}_\pm^* \cdot \boldsymbol{\epsilon}_\mp = 0$, and the same completeness relation Eq. (13.54) since the transformation is unitary on the polarization index.

Worked Example 13.13 (The circular polarizations carry helicity ±1)**.** Take $\mathbf{k} = k\hat{\mathbf{z}}$ and the standard transverse frame $\boldsymbol{\epsilon}_1 = \hat{\mathbf{x}}$, $\boldsymbol{\epsilon}_2 = \hat{\mathbf{y}}$. A rotation R_θ by angle θ about $\hat{\mathbf{z}}$ (which fixes $\mathbf{k}$, hence belongs to the little group of this momentum) acts as

$$\hat{\mathbf{x}} \mapsto \cos\theta\,\hat{\mathbf{x}} + \sin\theta\,\hat{\mathbf{y}}, \qquad \hat{\mathbf{y}} \mapsto -\sin\theta\,\hat{\mathbf{x}} + \cos\theta\,\hat{\mathbf{y}}. \tag{13.71}$$

Applying this to $\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}$,

$$\hat{\mathbf{x}} \pm i\hat{\mathbf{y}} \mapsto (\cos\theta \mp i\sin\theta)\hat{\mathbf{x}} + (\sin\theta \pm i\cos\theta)\hat{\mathbf{y}} = e^{\mp i\theta}(\hat{\mathbf{x}} \pm i\hat{\mathbf{y}}), \tag{13.72}$$

so $\boldsymbol{\epsilon}_\pm \mapsto e^{\mp i\theta}\boldsymbol{\epsilon}_\pm$: the circular polarizations are eigenvectors of the rotation about the momentum, with eigenvalues $e^{\mp i\theta}$. A state of helicity h transforms under a rotation by θ about its momentum as $e^{-ih\theta}$; matching, the one-photon state created with $\boldsymbol{\epsilon}_+$ has $h = +1$ and the one with $\boldsymbol{\epsilon}_-$ has $h = -1$. The photon is therefore a massless particle of helicity ±1 — the two values realized by the two transverse states, never a third (longitudinal) value.

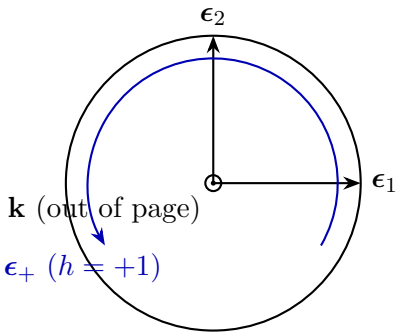


Figure 13.2: The two physical polarizations of a photon of momentum $\mathbf{k}$ live in the plane transverse to $\mathbf{k}$. The linear basis $\boldsymbol{\epsilon}_1, \boldsymbol{\epsilon}_2$ is rotated into itself by rotations about $\hat{\mathbf{k}}$; the circular combinations $\boldsymbol{\epsilon}_\pm = \mp(\boldsymbol{\epsilon}_1 \pm i\boldsymbol{\epsilon}_2)/\sqrt{2}$ are the eigenvectors of that rotation, carrying definite helicity $h = \pm 1$.

Connection to the little group. This identification is not an analogy; it is the realization, on the photon Fock space, of the ISO(2) little-group classification of massless particles built in Section 3.7.3. There the little group of a null momentum was found to be $\mathrm{ISO}(2) \cong E(2)$, with two non-compact "translation" generators A, B and one compact "rotation" generator J_3 about the momentum. The unitary irreducible representations of finite-dimensional type require $A = B = 0$, leaving the abelian J_3 whose eigenvalue is the helicity. The question that section deferred — why nature realizes $A = B = 0$ rather than the continuous-spin representations — is answered here, concretely, by gauge invariance. Acting with the non-compact generators on a polarization four-vector $\epsilon^\mu = (0, \boldsymbol{\epsilon}_\lambda)$ does not produce a new transverse polarization; it shifts ϵ^μ by a term proportional to the momentum,

$$\epsilon^\mu \;\longmapsto\; \epsilon^\mu + \beta(\mathbf{k})\, k^\mu, \tag{13.73}$$

which in position space is exactly a gauge transformation $A_\mu \to A_\mu + \partial_\mu \Lambda$ of the form Eq. (8.113) (a plane wave $\Lambda \propto e^{-ik\cdot x}$ gives $\partial_\mu \Lambda \propto k_\mu$). The would-be translation generators of the little group thus act on physical states as gauge transformations — and gauge-equivalent potentials are the *same* physical state. On the transverse, gauge-fixed Fock space they act trivially. This is why the photon carries no continuous internal label and no longitudinal state: the non-compact part of the little group is rendered invisible by the very gauge redundancy that forced us into this construction. What survives is the compact J_3, and its two eigenvalues $h = \pm 1$ are the two photons we built. Parity (Chapter 3) exchanges $\boldsymbol{\epsilon}_+ \leftrightarrow \boldsymbol{\epsilon}_-$, i.e. $h \to -h$, so a parity-invariant theory like electrodynamics necessarily contains both, in agreement with the two-state count.

Physical Insight

The deepest lesson of Coulomb-gauge quantization is that the "missing" degrees of freedom of the gauge field are not lost — they are reorganized. Of the four components of A_μ, one (A^0) is frozen into the instantaneous Coulomb field of the charges, one (the longitudinal part of $\mathbf{A}$) is pure gauge and eliminated, and the two transverse components are the dynamical photon. The non-compact directions of the massless little group, which for a generic field would threaten a continuum of polarizations, collapse onto gauge transformations and disappear from the physical spectrum. Masslessness and gauge redundancy are two faces of one fact: a massless helicity-1 particle *requires* a gauge description, because only gauge invariance can quotient the little group down to its compact, two-state core.

We have a complete, consistent, manifestly positive-norm quantum theory of the photon: two transverse oscillators per momentum, a Fock space of helicity-±1 massless quanta, and a Hamiltonian that splits into transverse radiation plus an instantaneous Coulomb energy of the sources. The price was visible at every turn — the frame-dependent transverse projection, the non-covariant δ^T_{ij}, the action-at-a-distance of H_{Coul}. That the resulting physics is nonetheless exactly Lorentz invariant is a theorem, not a manifest fact, and the awkwardness of proving covariance term by term is strong motivation to seek a scheme in which it is built in from the start. The next section pays the opposite price: it quantizes all four components of A_μ covariantly, on an indefinite-metric space, and then carves out this same transverse, positive-norm physics by the Gupta–Bleuler subsidiary condition — proving, as its central result, that the covariant construction is isometric to the Coulomb-gauge Fock space we have just built.

13.3 Covariant Quantization and the Gupta–Bleuler Construction

The Coulomb-gauge construction of the previous section delivered everything a quantum theory must have—a positive-definite Fock space, a vacuum, and two transverse photon modes of helicity ±1—but it paid for that success with the one thing a *relativistic* theory most wants to keep manifest: Lorentz covariance. The split into instantaneous Coulomb interaction plus radiation field, the elimination of A^0 and the longitudinal part of $\mathbf{A}$, the transverse delta function δ^{T}_{ij} in the commutator Eq. (13.45)—all of these single out a particular Lorentz frame. The physics is covariant (the S-matrix one eventually computes does not know which frame fixed the gauge), but the apparatus is not, and at every step of a calculation one must trust that frame-dependent intermediate objects will

conspire to give a covariant answer. For tree-level QED this is tolerable; for loops, where one integrates over all four components of internal momenta and wants to use covariant regulators, it is intolerable. We therefore start over with the opposite strategy: keep all four components of A_μ as independent quantum fields, quantize them covariantly, and pay the price elsewhere. The price, as we will see, is an indefinite-metric state space containing negative-norm "photons"; the discipline that makes the theory consistent is the Gupta–Bleuler subsidiary condition, which carves a genuine, non-negative-norm physical Hilbert space out of the enlarged space and—this is the headline result of the whole chapter—returns us *isometrically* to the two-polarization transverse space we already trust. This is the deepest argument in the chapter, and we give it in full.

13.3.1 The gauge-fixing term and the restored Legendre map

The obstruction to quantizing A_μ "the way we quantized ϕ" was diagnosed at the start of the chapter: the Maxwell Lagrangian $\mathcal{L}_{\rm Max} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ has a singular Legendre map. Its momentum conjugate to A_μ is

$$\pi^\mu \;=\; \frac{\partial \mathcal{L}_{\rm Max}}{\partial(\partial_0 A_\mu)} \;=\; -F^{0\mu}, \tag{13.74}$$

so that $\pi^0 = -F^{00} \equiv 0$ identically: the time component of A_μ has no conjugate momentum, the velocity $\dot{A}^0$ cannot be solved for, and the canonical bracket $[A^0, \pi^0] = i\delta^{(3)}$ is inconsistent from the outset. This is not a technical nuisance to be patched; it is the imprint, at the level of phase space, of the gauge redundancy Eq. (8.113). A redundancy means there are fewer genuine degrees of freedom than fields, and a constrained (singular) Lagrangian is exactly how that surplus announces itself.

The covariant cure, due to Fermi, is to break the redundancy by adding a term that does not vanish on the gauge orbit. We replace $\mathcal{L}_{\rm Max}$ by the *gauge-fixed* (Fermi) Lagrangian

$$\mathcal{L} \;=\; -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} \;-\; \frac{1}{2\xi}\,(\partial_\mu A^\mu)^2, \tag{13.75}$$

with a real parameter $\xi > 0$, the *gauge parameter*. The added term is not gauge invariant: under $A_\mu \to A_\mu + \partial_\mu\alpha$ it shifts by $-\frac{1}{\xi}(\partial\cdot A)\Box\alpha + \dots$, so distinct points on a gauge orbit now have distinct actions and the orbit degeneracy is lifted. We are, strictly, quantizing a *different* classical theory from Maxwell's; the entire art of what follows is to show that the physical content of Maxwell's theory is recovered exactly, on a subspace selected by a condition we impose by hand. The path integral will later reveal Eq. (13.75) as the natural output of the Faddeev–Popov procedure with a Gaussian average over gauges, but here we take it as a postulate and test it.

The Legendre map of Eq. (13.75) is invertible. Differentiating,

$$\pi^\mu \;=\; \frac{\partial \mathcal{L}}{\partial(\partial_0 A_\mu)} \;=\; -F^{0\mu} \;-\; \frac{1}{\xi}\,\eta^{0\mu}\,(\partial_\nu A^\nu), \tag{13.76}$$

so the dangerous time component is now

$$\pi^0 \;=\; -\frac{1}{\xi}\,(\partial_\nu A^\nu) \;=\; -\frac{1}{\xi}\,(\dot{A}^0 - \partial_i A^i) \;\neq\; 0, \tag{13.77}$$

and crucially it *depends on the velocity* $\dot{A}^0$. All four velocities $\dot{A}^\mu$ can be solved for in terms of the momenta, the map $(A_\mu, \dot{A}_\mu) \mapsto (A_\mu, \pi^\mu)$ is a genuine change of variables, and canonical quantization proceeds as for any nonsingular system. The Euler–Lagrange equations of Eq. (13.75) read

$$\partial_\mu F^{\mu\nu} + \frac{1}{\xi}\,\partial^\nu(\partial_\mu A^\mu) \;=\; \Box A^\nu - \Big(1 - \frac{1}{\xi}\Big)\partial^\nu(\partial_\mu A^\mu) \;=\; 0, \tag{13.78}$$

using $\partial_\mu F^{\mu\nu} = \Box A^\nu - \partial^\nu(\partial \cdot A)$. The full apparatus simplifies dramatically at the special value $\xi = 1$, the *Feynman gauge*, where the second term cancels and the four components decouple into four independent massless wave equations,

$$\boxed{\Box A^\mu \;=\; 0} \qquad (\xi = 1). \tag{13.79}$$

Feynman gauge is the computational default for the rest of the book (the convention recorded in Chapter A); we adopt it now and carry the general ξ only where the ξ-dependence is the point, principally in the next section. In Feynman gauge the Lagrangian Eq. (13.75) differs from $\mathcal{L} = -\frac{1}{2}\,\partial_\mu A_\nu\,\partial^\mu A^\nu$ by a total derivative, and the latter form—a quartet of decoupled scalars, one per Lorentz index, with a metric-dependent sign—makes the structure of the quantization transparent.

Caution

The wave equation Eq. (13.79) is *not* Maxwell's equation. Maxwell's equation $\partial_\mu F^{\mu\nu} = 0$ is recovered from Eq. (13.78) only if $\partial_\mu A^\mu = 0$; the gauge-fixed theory enforces $\Box(\partial \cdot A) = 0$ (take ∂_ν of Eq. (13.78)) but not $\partial \cdot A = 0$ itself. Restoring the Lorenz condition $\partial \cdot A = 0$—as an operator statement it would be too strong, as we will see—is precisely the job of the Gupta–Bleuler condition. Until that condition is imposed, Eq. (13.75) describes four propagating fields, not the two physical photon polarizations.

13.3.2 Covariant equal-time commutators

With the Legendre map repaired we impose the standard equal-time canonical commutation relations on all four components,

$$[A_\mu(\mathbf{x}, t), \pi^\nu(\mathbf{y}, t)] = i\,\delta_\mu{}^\nu\,\delta^{(3)}(\mathbf{x} - \mathbf{y}), \qquad [A_\mu, A_\nu] = [\pi^\mu, \pi^\nu] = 0 \;\; \text{(equal time)}. \tag{13.80}$$

In Feynman gauge the conjugate momentum from $\mathcal{L} = -\frac{1}{2}\partial_\mu A_\nu \partial^\mu A^\nu$ is $\pi^\mu = -\dot{A}^\mu$, so Eq. (13.80) becomes $\left[A_\mu(\mathbf{x},t), \dot{A}^\nu(\mathbf{y},t)\right] = -i\,\delta_\mu{}^\nu \delta^{(3)}(\mathbf{x}-\mathbf{y})$, i.e.

$$\left[A_\mu(\mathbf{x},t), \dot{A}_\nu(\mathbf{y},t)\right] \;=\; -\,i\,\eta_{\mu\nu}\,\delta^{(3)}(\mathbf{x}-\mathbf{y}). \tag{13.81}$$

The metric tensor $\eta_{\mu\nu}$ on the right is the seed of everything that follows. For the spatial components $\mu = \nu = i$ it carries the same sign as a scalar field (with $\eta_{ii} = -1$); but for the *time* component $\mu = \nu = 0$ it carries the *opposite* sign ($\eta_{00} = +1$), and that single relative sign will turn out to be the negative norm of the scalar photon. There is no freedom here: the sign is dictated by Lorentz covariance, which demands the indices on both sides match.

Because each component obeys the free wave equation Eq. (13.79) with the canonical data Eq. (13.81), the unequal-time commutator is fixed by exactly the argument that fixed the scalar commutator in Section 9.6: it is a c-number (the ladder operators cancel), it solves the wave equation in each argument, and it matches Eq. (13.81) at coincident times. The unique answer is the massless Pauli–Jordan function, one copy per index, weighted by the metric.

Definition 13.14 (Massless commutator function)**.** The *massless Pauli–Jordan function* is the $m \to 0$ limit of the commutator function Theorem 9.29,

$$D(x) \;=\; \Delta(x)\big|_{m=0} \;=\; -\,i\int \frac{d^3k}{(2\pi)^3}\,\frac{1}{2\omega_{\mathbf{k}}}\Big(e^{-ik\cdot x} - e^{+ik\cdot x}\Big), \qquad \omega_{\mathbf{k}} = |\mathbf{k}|,\; k^0 = \omega_{\mathbf{k}}. \tag{13.82}$$

It is real and odd, $D(-x) = -D(x)$, satisfies $\Box D = 0$, vanishes for spacelike x (microcausality, by the orbit argument of Theorem 9.31 run at $m = 0$), and has the equal-time data $D(0,\mathbf{x}) = 0$, $\dot{D}(0,\mathbf{x}) = -\delta^{(3)}(\mathbf{x})$.

Proposition 13.15 (Covariant photon commutator)**.** *In Feynman gauge the free electromagnetic field satisfies*

$$[A_\mu(x), A_\nu(y)] \;=\; -\,i\,\eta_{\mu\nu}\,D(x-y). \tag{13.83}$$

Proof. The right-hand side is a c-number, so it commutes with everything, as the left-hand side must (the field commutator of a free theory is a c-number by the same cancellation as in Section 9.6.1). It satisfies $\Box_x[\,\cdot\,] = -i\eta_{\mu\nu}\Box D(x-y) = 0$, matching Eq. (13.79) applied to $A_\mu(x)$. At equal times $x^0 = y^0$ it vanishes (because $D(0,\mathbf{r}) = 0$), reproducing $[A_\mu, A_\nu] = 0$; and its time derivative at equal times is $-i\eta_{\mu\nu}\,\partial_{x^0} D(x-y)\big|_{x^0=y^0} = -i\eta_{\mu\nu}\,\dot{D}(0,\mathbf{x}-\mathbf{y}) = +i\eta_{\mu\nu}\delta^{(3)}(\mathbf{x}-\mathbf{y})$, which is $-\left[A_\mu, \dot{A}_\nu\right]$ with the sign of Eq. (13.81) after using antisymmetry of the commutator under $x \leftrightarrow y$. A c-number solving the wave equation in each argument is determined by these Cauchy data, so Eq. (13.83) is the unique answer. □

Equation Eq. (13.83) is manifestly Lorentz covariant: both sides carry the index structure $\mu\nu$ and depend only on $x-y$. Microcausality is automatic and covariant—$D(x-y)=0$ for spacelike separation makes all four components, hence every local observable built from A_μ and $F_{\mu\nu}$, commute outside the light cone. This is exactly the manifest covariance the Coulomb gauge surrendered. The cost is hidden in the sign of η_{00}, and we now make it explicit.

13.3.3 Four polarizations and the indefinite metric

Expand each component in plane-wave modes. Because all four components are independent massless fields, the expansion carries four polarization labels:

$$A_\mu(x) \;=\; \int \widetilde{dk} \sum_{\lambda=0}^{3} \Big[\, \epsilon_\mu(k,\lambda)\, a(k,\lambda)\, e^{-ik\cdot x} \;+\; \epsilon_\mu(k,\lambda)\, a^\dagger(k,\lambda)\, e^{+ik\cdot x} \,\Big], \tag{13.84}$$

where $\widetilde{dk} \equiv \dfrac{d^3k}{(2\pi)^3\, 2\omega_{\mathbf{k}}}$ is the Lorentz-invariant measure of Section 9.2.1 and the four real four-vectors $\epsilon^\mu(k,\lambda)$ are the *polarization vectors*. Their job is to span the four-dimensional space of Lorentz indices for each k, and we choose them to be orthonormal with respect to the Minkowski metric. For a wave travelling along the z-axis, $k^\mu=\omega(1,0,0,1)$, the canonical choice is

$$\epsilon^\mu(k,0)=\begin{pmatrix}1\\0\\0\\0\end{pmatrix},\;\; \epsilon^\mu(k,1)=\begin{pmatrix}0\\1\\0\\0\end{pmatrix},\;\; \epsilon^\mu(k,2)=\begin{pmatrix}0\\0\\1\\0\end{pmatrix},\;\; \epsilon^\mu(k,3)=\begin{pmatrix}0\\0\\0\\1\end{pmatrix}. \tag{13.85}$$

We name them by their physical character: $\lambda=0$ is the *timelike* or *scalar* polarization; $\lambda=1,2$ are the two *transverse* polarizations, orthogonal to $\mathbf{k}$; and $\lambda=3$ is the *longitudinal* polarization, along $\mathbf{k}$ in the spatial sense. (For a general k direction one boosts/rotates Eq. (13.85); the inner products below are Lorentz scalars and hold for all k.) Their defining algebraic properties are orthonormality and completeness:

$$\epsilon(k,\lambda)\cdot\epsilon(k,\lambda') \;=\; \eta_{\lambda\lambda'} \;=\; \mathrm{diag}(+1,-1,-1,-1)_{\lambda\lambda'}, \tag{13.86}$$

$$\sum_{\lambda=0}^{3} \zeta_\lambda\, \epsilon^\mu(k,\lambda)\, \epsilon^\nu(k,\lambda) \;=\; \eta^{\mu\nu}, \qquad \zeta_\lambda \equiv \eta_{\lambda\lambda} = (+1,-1,-1,-1). \tag{13.87}$$

Both are immediate from Eq. (13.85): Eq. (13.86) is the statement that the $\epsilon(k,\lambda)$ are an orthonormal tetrad, and Eq. (13.87) is the resolution of the identity in that basis, with the signature factors ζ_λ inserted because the basis is orthonormal in an indefinite metric. The decisive entry is $\epsilon(k,0)\cdot\epsilon(k,0)=+1>0$ while $\epsilon(k,i)\cdot\epsilon(k,i)=-1<0$: the scalar polarization is "timelike" and is the troublemaker.

Substituting Eq. (13.84) into the covariant commutator Eq. (13.83) and using Eq. (13.87) (the calculation is the photonic version of the scalar inversion in Section 9.2.3, with the polarization sum doing the index bookkeeping) yields the ladder algebra

$$\boxed{\left[a(k,\lambda),a^\dagger(k',\lambda')\right] \;=\; -\,\eta_{\lambda\lambda'}\,(2\pi)^3\,2\omega_{\mathbf{k}}\,\delta^{(3)}(\mathbf{k}-\mathbf{k}')} \tag{13.88}$$

with all other commutators vanishing. Spelled out by polarization, the three spatial labels behave normally,

$$\left[a(k,i),a^\dagger(k',j)\right] = +\,\delta_{ij}\,(2\pi)^3 2\omega_{\mathbf{k}}\,\delta^{(3)}(\mathbf{k}-\mathbf{k}'),\qquad i,j\in\{1,2,3\}, \tag{13.89}$$

but the scalar label carries the wrong sign,

$$\left[a(k,0),a^\dagger(k',0)\right] = -\,(2\pi)^3 2\omega_{\mathbf{k}}\,\delta^{(3)}(\mathbf{k}-\mathbf{k}'). \tag{13.90}$$

The minus sign in Eq. (13.90) is the price of manifest covariance, traced back through Eq. (13.88) and Eq. (13.83) to the $\eta_{00}=+1$ in the canonical relation Eq. (13.81). We confront its consequence directly.

Proposition 13.16 (The scalar photon has negative norm)**.** *The one-particle scalar state has strictly negative norm. Concretely, for a normalizable wave packet f define $|f;0\rangle \equiv \int \widetilde{dk}\, f(\mathbf{k})\, a^\dagger(k,0)\,|0\rangle$. Then*

$$\langle f;0|f;0\rangle \;=\; -\int \widetilde{dk}\,|f(\mathbf{k})|^2 \;<\; 0. \tag{13.91}$$

Proof. Using $a(k,0)\,|0\rangle = 0$ and Eq. (13.90),

$$\begin{aligned}\langle f;0|f;0\rangle &= \int \widetilde{dk}\,\widetilde{dk'}\,\bar f(\mathbf{k}) f(\mathbf{k}')\,\Big\langle 0\Big|a(k,0)\,a^\dagger(k',0)\Big|0\Big\rangle = \int \widetilde{dk}\,\widetilde{dk'}\,\bar f(\mathbf{k}) f(\mathbf{k}')\,\Big\langle 0\Big|\Big[a(k,0),a^\dagger(k\\ &= -\int \widetilde{dk}\,\widetilde{dk'}\,\bar f(\mathbf{k}) f(\mathbf{k}')\,(2\pi)^3 2\omega_{\mathbf{k}}\,\delta^{(3)}(\mathbf{k}-\mathbf{k}') = -\int \widetilde{dk}\,|f(\mathbf{k})|^2. \qquad \square\end{aligned}$$

A state of negative norm is a catastrophe for the probability interpretation: $\langle\psi|\psi\rangle$ is supposed to be a probability, and probabilities are non-negative. The covariant Fock space built by acting with all four $a^\dagger(k,\lambda)$ on the vacuum is therefore not a Hilbert space at all but an *indefinite-metric* space (a Krein space): it has states of positive, negative, and—as we will find essential—zero norm. The free field is perfectly well defined on it; what is not well defined is the physical interpretation. The remainder of the section is the disciplined extraction of a true Hilbert space from this larger object.

Physical Insight

The indefinite metric is not a defect of the formalism to be apologized for; it is the unavoidable cost of writing a redundancy-laden object (A_μ, four components) covariantly while quantizing it as if it had four independent degrees of freedom. We over-counted, and Minkowski signature charges interest on the over-count in the currency of negative norm. The Gupta–Bleuler condition is the bill-payment: it removes exactly the surplus, leaving the two transverse degrees of freedom Coulomb gauge found directly. Seen this way, the negative-norm scalar photon and the longitudinal photon are bookkeeping ghosts—the abelian ancestors of the Faddeev–Popov ghosts of Chapter 26.

13.3.4 The Gupta–Bleuler subsidiary condition

The classical theory we want is Maxwell's, whose Lagrangian we modified; the difference is the gauge-fixing term, which is harmless precisely when $\partial_\mu A^\mu = 0$. The natural idea is to impose the Lorenz condition $\partial_\mu A^\mu = 0$ to select the physical states. As an *operator* identity it is fatal: $\partial_\mu A^\mu$ is a nontrivial free field (it satisfies $\Box(\partial \cdot A) = 0$ but is not zero), and setting it to zero as an operator contradicts the commutator Eq. (13.83), exactly as the constraint $\pi^0 = 0$ contradicted the canonical bracket in the ungauged theory. As a condition on *states*, $\partial_\mu A^\mu \,|\text{phys}\rangle = 0$, it is still too strong: split $\partial \cdot A$ into its positive- and negative-frequency parts,

$$\partial^\mu A_\mu \;=\; (\partial^\mu A_\mu)^{(+)} + (\partial^\mu A_\mu)^{(-)}, \qquad (\partial^\mu A_\mu)^{(-)} = \big[(\partial^\mu A_\mu)^{(+)}\big]^\dagger, \tag{13.92}$$

where the positive-frequency part collects the annihilation ($e^{-ik\cdot x}$) terms and the negative-frequency part the creation ($e^{+ik\cdot x}$) terms. The creation part $(\partial \cdot A)^{(-)}$ does not annihilate even the vacuum, so $\partial_\mu A^\mu \,|0\rangle \neq 0$ and the bare vacuum would fail to be physical—absurd.

Gupta and Bleuler saw that the right condition keeps only the annihilation half:

Definition 13.17 (Gupta–Bleuler subsidiary condition)**.** A state $|\text{phys}\rangle$ is *physical* if it is annihilated by the positive-frequency part of the four-divergence,

$$\boxed{(\partial^\mu A_\mu)^{(+)}(x)\;|\text{phys}\rangle \;=\; 0 \quad \text{for all } x.} \tag{13.93}$$

The set of all such states is denoted V_{phys}.

This is weaker than the operator condition—it constrains states, and only through the annihilation part—yet it is exactly strong enough. First, it is consistent: the vacuum satisfies it, since $(\partial \cdot A)^{(+)}$ is built from annihilation

operators and $(\partial\cdot A)^{(+)}\,|0\rangle = 0$. Second, it is Lorentz invariant, because $(\partial^\mu A_\mu)^{(+)}$ is a Lorentz scalar field. Third, and most importantly, it implies the Lorenz condition in the *weak* (matrix-element) sense rather than as an operator equation: for any two physical states $|\text{phys}\rangle$, $|\text{phys}'\rangle$,

$$\langle\text{phys}|\partial^\mu A_\mu|\text{phys}'\rangle = \underbrace{\Big\langle\text{phys}\Big|(\partial^\mu A_\mu)^{(+)}\Big|\text{phys}'\Big\rangle}_{=0 \text{ by } (13.93)} + \underbrace{\Big\langle\text{phys}\Big|(\partial^\mu A_\mu)^{(-)}\Big|\text{phys}'\Big\rangle}_{=0 \text{ by h.c. of } (13.93)} = 0, \tag{13.94}$$

where the second term vanishes because $(\partial\cdot A)^{(-)} = [(\partial\cdot A)^{(+)}]^\dagger$ acts to the left on $\langle\text{phys}|$ as the adjoint of Eq. (13.93). So $\partial_\mu A^\mu$ has vanishing expectation value between physical states; the Lorenz condition holds "on average," which is all the physics requires and all that is consistent. We return to the physical reading of Eq. (13.94) after proving the central theorem.

It is worth recasting Eq. (13.93) in terms of ladder operators, because the carving of the physical space then becomes an algebra problem. The positive-frequency part of $\partial^\mu A_\mu$ is

$$(\partial^\mu A_\mu)^{(+)}(x) = \int \widetilde{dk}\,(-ik^\mu)\sum_\lambda \epsilon_\mu(k,\lambda)\,a(k,\lambda)\,e^{-ik\cdot x}, \tag{13.95}$$

so the condition Eq. (13.93) holds for all x iff $\sum_\lambda (k\cdot\epsilon(k,\lambda))\,a(k,\lambda)\,|\text{phys}\rangle = 0$ for every k. With the explicit polarizations Eq. (13.85) for $k^\mu = \omega(1,0,0,1)$ we have $k\cdot\epsilon(k,0) = \omega$, $k\cdot\epsilon(k,1) = k\cdot\epsilon(k,2) = 0$, and $k\cdot\epsilon(k,3) = k^\mu\epsilon_\mu(k,3) = \omega\cdot 0 - \omega\cdot(+1) = -\omega$ (recall $\epsilon_\mu(k,3) = (0,0,0,-1)$ with the lowered index). Hence $\sum_\lambda (k\cdot\epsilon)\,a(k,\lambda) = \omega\,[a(k,0) - a(k,3)]$, and Eq. (13.93) is equivalent to

$$\boxed{L(k)\,|\text{phys}\rangle \;=\; 0 \quad \text{for all } k, \qquad L(k)\equiv a(k,0) - a(k,3).} \tag{13.96}$$

The physical condition is that every physical state be annihilated by the combination of scalar minus longitudinal annihilation operators. Only this one combination is constrained; the transverse operators $a(k,1), a(k,2)$ are unconstrained, and the orthogonal combination $a(k,0) + a(k,3)$ is unconstrained. This is the algebraic form we will exploit.

13.3.5 Physical states have non-negative norm

Everything now turns on the commutation algebra of the combination $L(k)$ and its conjugate. This is where the negative-norm scalar and the positive-norm longitudinal modes meet and cancel.

Lemma 13.18 (The null algebra)**.** *With $L(k) = a(k,0) - a(k,3)$,*

$$\Big[L(k), L^\dagger(k')\Big] \;=\; 0, \qquad [L(k), L(k')] = 0, \tag{13.97}$$

and $L(k)$ commutes with every transverse operator $a(k',1), a(k',2)$ and their conjugates.

Proof. By bilinearity and Eq. (13.88) (cross terms between $\lambda = 0$ and $\lambda = 3$ vanish, since $\eta_{03} = 0$),

$$\Big[L(k), L^\dagger(k')\Big] = \Big[a(k,0), a^\dagger(k',0)\Big] + \Big[a(k,3), a^\dagger(k',3)\Big] = (-1+1)(2\pi)^3 2\omega_{\mathbf{k}}\, \delta^{(3)}(\mathbf{k}-\mathbf{k}') = 0. \tag{13.98}$$

The crucial cancellation is the $(-1) + (+1) = 0$ in Eq. (13.98): the scalar contributes $\Big[a_0, a_0^\dagger\Big] = -(2\pi)^3 2\omega\delta$ from Eq. (13.90), the longitudinal contributes $\Big[a_3, a_3^\dagger\Big] = +(2\pi)^3 2\omega\delta$ from Eq. (13.89), and they are equal and opposite. The vanishing of $[L, L]$ and of the transverse commutators is immediate from Eq. (13.88). □

The lemma says $L(k)$ and $L^\dagger(k)$ generate a *null* (degenerate) subalgebra: $L^\dagger$ creates states of zero norm. Indeed, for any normalizable g, writing $L^\dagger(g) \equiv \int \widetilde{dk}\, \bar{g}(\mathbf{k}) L^\dagger(k)$,

$$\| L^\dagger(g)\ |0\rangle \|^2 = \Big\langle 0 \Big| L(g)\, L^\dagger(g) \Big| 0 \Big\rangle = \Big\langle 0 \Big| \Big[L(g), L^\dagger(g)\Big] \Big| 0 \Big\rangle = 0. \tag{13.99}$$

We can now prove the headline result. The strategy is to organize the scalar/longitudinal sector around the null pair, separating it cleanly from the transverse sector.

Theorem 13.19 (Non-negativity of physical norms)**.** *Every physical state* $|\Psi\rangle \in V_{\text{phys}}$ *has non-negative norm,* $\langle\Psi|\Psi\rangle \geq 0$.

Proof. The full Fock space factorizes as $\mathcal{F} = \mathcal{F}_{\text{T}} \otimes \mathcal{F}_{\text{sl}}$, where $\mathcal{F}_{\text{T}}$ is generated from the vacuum by the transverse creation operators $a^\dagger(k,1), a^\dagger(k,2)$ and $\mathcal{F}_{\text{sl}}$ by the scalar/longitudinal operators $a^\dagger(k,0), a^\dagger(k,3)$. The transverse sector has the positive-definite algebra Eq. (13.89); it is an honest Fock space, and it is exactly the two-polarization space of the Coulomb-gauge construction, Eq. (13.45). Since $L(k)$ commutes with all transverse operators (Lemma 13.18), the condition Eq. (13.96) acts only on the $\mathcal{F}_{\text{sl}}$ factor: writing a general state as $|\Psi\rangle = \sum_n |T_n\rangle \otimes |\chi_n\rangle$ with $|T_n\rangle \in \mathcal{F}_{\text{T}}$ orthonormal and $|\chi_n\rangle \in \mathcal{F}_{\text{sl}}$, the physical condition is $L(k)\,|\chi_n\rangle = 0$ for every n and k. By orthonormality of the $|T_n\rangle$,

$$\langle\Psi|\Psi\rangle = \sum_n \langle T_n|T_n\rangle\ \langle\chi_n|\chi_n\rangle = \sum_n \|\,T_n\|^2\ \langle\chi_n|\chi_n\rangle\,, \tag{13.100}$$

with $\|T_n\|^2 \geq 0$. It remains to prove that any $|\chi\rangle \in \mathcal{F}_{\text{sl}}$ obeying $L(k)\,|\chi\rangle = 0$ for all k has $\langle\chi|\chi\rangle \geq 0$.

Diagonalize the scalar/longitudinal sector by passing to the null basis. Define

$$b(k) \equiv \tfrac{1}{\sqrt{2}}\,[a(k,0) - a(k,3)] = \tfrac{1}{\sqrt{2}} L(k), \qquad c(k) \equiv \tfrac{1}{\sqrt{2}}\,[a(k,0) + a(k,3)]. \tag{13.101}$$

Their algebra follows from Eq. (13.88) (again $\eta_{03} = 0$):

$$\Big[b(k), b^\dagger(k')\Big] = 0, \quad \Big[c(k), c^\dagger(k')\Big] = 0, \quad \Big[b(k), c^\dagger(k')\Big] = -(2\pi)^3 2\omega_{\mathbf{k}}\, \delta^{(3)}(\mathbf{k} - \mathbf{k}'), \tag{13.102}$$

the first because $\left[b, b^\dagger\right] = \frac{1}{2}\left[L, L^\dagger\right] = 0$, the second because $\left[c, c^\dagger\right] = \frac{1}{2}(\left[a_0, a_0^\dagger\right] + \left[a_3, a_3^\dagger\right]) = \frac{1}{2}(-1+1)(\dots) = 0$, and the third by direct expansion. So b and c are conjugate *null* oscillators: each is self-null ($\left[b, b^\dagger\right] = \left[c, c^\dagger\right] = 0$), and they pair only with each other. The sector $\mathcal{F}_{\mathrm{sl}}$ is built by $b^\dagger$ and $c^\dagger$ acting on $|0\rangle$.

The physical condition $L(k)\,|\chi\rangle = 0$ is $b(k)\,|\chi\rangle = 0$. Because b commutes with $b^\dagger$ but has $\left[b(k), c^\dagger(k')\right] \neq 0$, the operator b acts as the annihilator conjugate to the creator $c^\dagger$: it sees $c^\dagger$ excitations and is blind to $b^\dagger$ excitations. Concretely, any monomial in $b^\dagger$'s and $c^\dagger$'s acting on $|0\rangle$ is mapped by $b(k)$ to a sum of terms each missing one $c^\dagger$; the only states killed by every $b(k)$ are those built from $b^\dagger$'s alone (no $c^\dagger$ factor). Hence a physical $|\chi\rangle$ has the form

$$|\chi\rangle = c_0\,|0\rangle \;+\; |\chi_b\rangle\,, \qquad |\chi_b\rangle \in \text{span of states with at least one } b^\dagger, \tag{13.103}$$

with $c_0 \in \mathbb{C}$ and $|\chi_b\rangle$ containing no $c^\dagger$ excitations. Now compute its norm. Every state with at least one $b^\dagger$ has zero norm and zero overlap with any other $b^\dagger$-only state, because all the relevant inner products reduce, by moving b's to the right through Eq. (13.102), to vacuum matrix elements containing at least one factor $\left[b, b^\dagger\right] = 0$ (no $c^\dagger$ is ever present to supply the nonzero $\left[b, c^\dagger\right]$). The same argument gives $\langle 0|\chi_b\rangle = 0$. Therefore the cross terms and the $|\chi_b\rangle$ self-overlap all vanish, leaving

$$\langle\chi|\chi\rangle = |c_0|^2\,\langle 0|0\rangle + 2\,\mathrm{Re}(\bar{c}_0\,\langle 0|\chi_b\rangle\,) + \langle\chi_b|\chi_b\rangle = |c_0|^2 \;\geq\; 0. \tag{13.104}$$

Substituting into Eq. (13.100) gives $\langle\Psi|\Psi\rangle = \sum_n \|T_n\|^2\,|c_0^{(n)}|^2 \geq 0$, which is the claim. □

The proof exposes the mechanism with no hand-waving: the would-be negative-norm scalar photon never appears alone in a physical state. The physical condition forces scalar and longitudinal quanta to enter only in the null combination $b^\dagger \propto a_0^\dagger - a_3^\dagger$, and that combination has *zero* norm because the negative scalar norm Eq. (13.90) cancels the positive longitudinal norm Eq. (13.89) term by term, Eq. (13.98). Negativity is impossible because the dangerous mode has been algebraically married to a partner of opposite norm, and the marriage is exactly what the Lorenz condition enforces.

13.3.6 Zero-norm states decouple

Theorem 13.19 leaves a residue: physical states of *zero* norm, those with $c_0 = 0$ in Eq. (13.103), i.e. physical states whose scalar/longitudinal part is built purely from $b^\dagger$'s (with no vacuum component). Denote this set $V_0 \subset V_{\mathrm{phys}}$. Zero-norm states are not yet a contradiction, but a probability interpretation cannot tolerate a state $|n\rangle$ with $\langle n|n\rangle = 0$ unless that state is physically invisible. We now prove it is.

Proposition 13.20 (Null states decouple)**.** *Every zero-norm physical state is orthogonal to all physical states: if* $|n\rangle \in V_0$ *and* $|\Psi\rangle \in V_{\text{phys}}$ *then* $\langle\Psi|n\rangle = 0$*. Consequently the addition of a null state to any physical state changes no physical matrix element, and null states contribute nothing to expectation values of observables.*

Proof. By the factorization of Theorem 13.19 it suffices to work in $\mathcal{F}_{\text{sl}}$. A null physical state has scalar/longitudinal part $|\chi_b\rangle$ of the $b^\dagger$-only form, and a general physical state has part $|\chi'\rangle = c'_0\,|0\rangle + |\chi'_b\rangle$. Their overlap is

$$\langle\chi'|\chi_b\rangle = \bar{c}'_0\,\langle 0|\chi_b\rangle + \langle\chi'_b|\chi_b\rangle = 0, \tag{13.105}$$

since $\langle 0|\chi_b\rangle = 0$ and $\langle\chi'_b|\chi_b\rangle = 0$ by the same null-overlap argument used for Eq. (13.104). Hence $V_0 \perp V_{\text{phys}}$. For the second statement, let $\mathcal{O}$ be any physical observable—an operator that maps physical states to physical states and is built gauge-invariantly, in particular the field strength $F_{\mu\nu}$, the Hamiltonian, momentum, and angular momentum. Then for physical $|\Psi\rangle, |\Psi'\rangle$ and null $|n\rangle, |n'\rangle$,

$$\langle\Psi + n|\mathcal{O}|\Psi' + n'\rangle = \langle\Psi|\mathcal{O}|\Psi'\rangle + \langle\Psi|\mathcal{O}|n'\rangle + \langle n|\mathcal{O}|\Psi'\rangle + \langle n|\mathcal{O}|n'\rangle = \langle\Psi|\mathcal{O}|\Psi'\rangle, \tag{13.106}$$

because $\mathcal{O}\,|n'\rangle$ and $\mathcal{O}^\dagger\,|n\rangle$ are again physical (observables preserve V_{phys}) and hence orthogonal to the null states $|n\rangle, |n'\rangle$ by the first part. Every cross term and the null–null term vanish. □

The cleanest illustration is the energy. The Hamiltonian of the gauge-fixed theory, expressed in ladder operators, takes the form (after normal ordering) $H = \int \widetilde{dk}\,\omega_{\mathbf{k}}\big[\sum_{i=1,2} a^\dagger(k,i)a(k,i) - a^\dagger(k,0)a(k,0) + a^\dagger(k,3)a(k,3)\big]$, where the scalar term carries a minus sign inherited from Eq. (13.90). Acting on a physical state and using $a(k,0) = a(k,3) + L(k)$ with $L(k)\,|\text{phys}\rangle = 0$, the scalar and longitudinal contributions to $\langle H\rangle$ cancel exactly, $\langle -a_0^\dagger a_0 + a_3^\dagger a_3\rangle_{\text{phys}} = 0$, leaving the transverse energy $\langle H\rangle_{\text{phys}} = \int \widetilde{dk}\,\omega_{\mathbf{k}} \sum_{i=1,2}\langle a^\dagger(k,i)a(k,i)\rangle \geq 0$. The energy is non-negative and is carried entirely by transverse photons; scalar and longitudinal quanta are energetically invisible. This is Eq. (13.106) made concrete for $\mathcal{O} = H$.

13.3.7 The quotient and the isometry to the transverse space

Theorem 13.20 licenses the final step. Inside V_{phys} the null states V_0 are orthogonal to everything, including themselves; they are exactly the directions along which the (semi-definite) inner product is degenerate. To obtain a Hilbert space we must remove this degeneracy, and the canonical way is to pass to the quotient: declare two physical states equivalent if they differ by a null state, $|\Psi\rangle \sim |\Psi\rangle + |n\rangle$ for $|n\rangle \in V_0$, and define the physical Hilbert space as the space of equivalence classes, completed in the resulting norm.

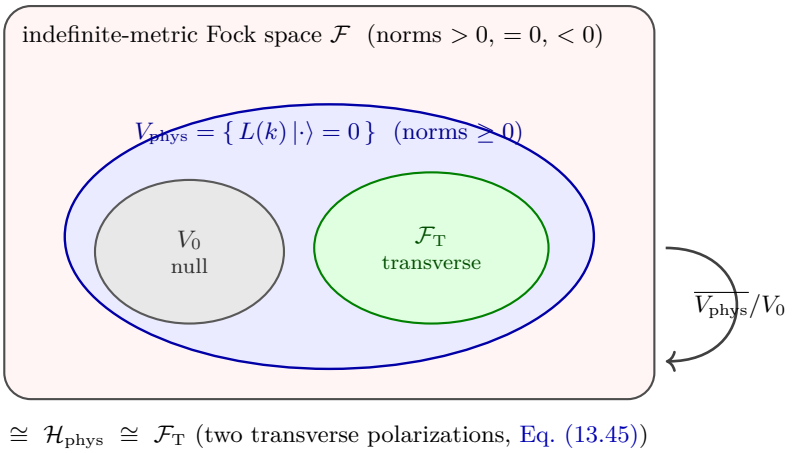


Figure 13.3: The Gupta–Bleuler construction. The covariant quantization lives on an indefinite-metric Fock space $\mathcal{F}$ with states of positive, zero, and negative norm. The subsidiary condition Eq. (13.96) selects the non-negative-norm subspace V_{phys}, inside which the zero-norm states V_0 form a subspace orthogonal to everything. Quotienting them out yields the physical Hilbert space $\mathcal{H}_{\text{phys}}$, isometric to the two-polarization transverse Fock space $\mathcal{F}_{\text{T}}$ of the Coulomb-gauge construction.

Definition 13.21 (Physical Hilbert space)**.** The *physical Hilbert space* of the covariantly quantized photon is

$$\mathcal{H}_{\text{phys}} \;\equiv\; \overline{V_{\text{phys}}}\,/\,V_0, \tag{13.107}$$

the completion of the physical subspace modulo null states, with inner product $([\Psi],[\Psi']) \equiv \langle\Psi|\Psi'\rangle$.

The inner product on classes is well defined precisely because of Theorem 13.20: if $|\Psi_1\rangle \sim |\Psi_2\rangle$ and $|\Psi_1'\rangle \sim |\Psi_2'\rangle$ then $\langle\Psi_1|\Psi_1'\rangle = \langle\Psi_2|\Psi_2'\rangle$, since the differences are null and orthogonal to all physical states. On classes the inner product is positive-definite—$([\Psi],[\Psi]) = \langle\Psi|\Psi\rangle = 0$ forces $|\Psi\rangle \in V_0$, i.e. $[\Psi] = 0$—so $\mathcal{H}_{\text{phys}}$ is a genuine Hilbert space. We can now identify it.

Theorem 13.22 (Isometry to the transverse space)**.** *The map sending each physical class to its transverse representative,*

$$U:\ \mathcal{H}_{\text{phys}} \;\longrightarrow\; \mathcal{F}_{\text{T}}, \qquad U\big[\ |T\rangle \otimes |\chi\rangle\ \big] \;=\; c_0\ |T\rangle\,, \tag{13.108}$$

where $|\chi\rangle = c_0\,|0\rangle + |\chi_b\rangle$ as in Eq. (13.103), is a well-defined unitary isomorphism onto the transverse two-polarization Fock space $\mathcal{F}_{\text{T}}$ of Eq. (13.45). The covariant and Coulomb-gauge constructions therefore describe the same physical photon.

Proof. Each physical class has a unique representative with scalar/longitudinal part proportional to $|0\rangle_{\rm sl}$: given any physical $|\Psi\rangle = \sum_n |T_n\rangle \otimes |\chi_n\rangle$ with $|\chi_n\rangle = c_0^{(n)}|0\rangle + \left|\chi_b^{(n)}\right\rangle$, subtract the null state $\sum_n |T_n\rangle \otimes \left|\chi_b^{(n)}\right\rangle \in V_0$ to obtain the equivalent representative $\sum_n c_0^{(n)} |T_n\rangle \otimes |0\rangle_{\rm sl}$, which is purely transverse. Two such transverse representatives in the same class differ by a transverse null state, but $\mathcal{F}_{\rm T}$ has positive-definite norm and so has no nonzero null states; the transverse representative is therefore unique, and U in Eq. (13.108) is well defined and independent of the chosen decomposition.

U preserves the inner product. For representatives $|\Psi\rangle = \sum_n c_0^{(n)} |T_n\rangle \otimes |0\rangle_{\rm sl}$ and $|\Psi'\rangle = \sum_m c_0^{\prime(m)} |T'_m\rangle \otimes |0\rangle_{\rm sl}$, using $\langle 0|0\rangle_{\rm sl} = 1$,

$$([\Psi],[\Psi']) = \langle\Psi|\Psi'\rangle = \sum_{n,m} \bar{c}_0^{(n)} c_0^{\prime(m)} \langle T_n|T'_m\rangle = \big(U[\Psi],\, U[\Psi']\big)_{\mathcal{F}_{\rm T}}. \tag{13.109}$$

U is surjective: every transverse state $|T\rangle \in \mathcal{F}_{\rm T}$ is the image of the physical class $[|T\rangle \otimes |0\rangle_{\rm sl}]$ (which satisfies Eq. (13.96) since $L(k)|0\rangle = 0$). And U is injective because $U[\Psi] = 0$ means every transverse representative vanishes, i.e. $|\Psi\rangle \in V_0$, i.e. $[\Psi] = 0$. A norm-preserving bijection of Hilbert spaces is unitary. The target $\mathcal{F}_{\rm T}$, built from $a^\dagger(k,1), a^\dagger(k,2)$ with the positive algebra Eq. (13.89), is precisely the transverse Fock space whose commutator is recorded in Eq. (13.45); its excitations are the two helicity-±1 photons identified there. □

This is the chapter's central result. Manifest Lorentz covariance has been bought with an indefinite metric, four polarizations, and a negative-norm scalar photon; and yet, after imposing the single covariant condition Eq. (13.93) and discarding the null directions it permits, the physics is identical to the manifestly positive Coulomb-gauge theory—two transverse photons, no more. The two "honest exits" from the gauge redundancy promised in the chapter introduction meet here, in Theorem 13.22: they are the same Hilbert space, reached by opposite routes. The structure is summarized in Fig. 13.3.

13.3.8 The weak Lorenz condition and Maxwell's equations in the mean

We close the logical circle by returning to Eq. (13.94). The Lorenz condition $\partial_\mu A^\mu = 0$, which we could not impose as an operator identity without contradicting the commutator Eq. (13.83), holds in the weak sense: $\langle{\rm phys}|\partial_\mu A^\mu|{\rm phys}'\rangle = 0$ for all physical states. This is exactly enough to recover Maxwell's equations as statements about physical matrix elements. Taking the matrix element of the gauge-fixed equation of motion Eq. (13.78) at $\xi = 1$, $\Box A^\nu = 0$, between physical states, and adding ∂^ν of the weak Lorenz condition, gives

$$\langle{\rm phys}|\partial_\mu F^{\mu\nu}|{\rm phys}'\rangle = \langle{\rm phys}|\Box A^\nu - \partial^\nu(\partial_\mu A^\mu)|{\rm phys}'\rangle = 0, \tag{13.110}$$

the source-free Maxwell equation holding between physical states. The free covariant photon thus reproduces electromagnetism not as an operator identity for A_μ—which would be false, the gauge-fixed A_μ obeys $\Box A_\mu = 0$, not Maxwell's equation—but as a statement true in every physical expectation value. Observable electromagnetism is recovered exactly on $\mathcal{H}_{\text{phys}}$.

We now make the structure tactile with two worked computations. The first traces the cancellation of negative norm through a concrete one-photon state; the second reveals the null states as the quantum residue of the classical gauge freedom Eq. (8.113), closing the conceptual loop opened in Chapter 8.

Worked Example 13.23 (A physical scalar/longitudinal photon is null)**.** Consider the most general one-photon state built from the scalar and longitudinal modes with a common wave packet $f(\mathbf{k})$ and complex amplitudes α, β,

$$|\psi\rangle = \int \widetilde{dk}\, f(\mathbf{k}) \left[\, \alpha\, a^\dagger(k,0) + \beta\, a^\dagger(k,3) \,\right] |0\rangle\,. \tag{13.111}$$

We ask which such states are physical, and what their norm is. The physical condition Eq. (13.96) requires $L(k)\,|\psi\rangle = 0$. Compute, using $L(k) = a(k,0) - a(k,3)$ and Eq. (13.88):

$$\begin{aligned} L(k)\,|\psi\rangle &= \int \widetilde{dk'}\, f(\mathbf{k}') \Big(\alpha \, \Big[a(k,0), a^\dagger(k',0)\Big] - \beta \, \Big[a(k,3), a^\dagger(k',3)\Big] \Big)\, |0\rangle \\ &= \int \widetilde{dk'}\, f(\mathbf{k}') \, (-\alpha - \beta)(2\pi)^3 2\omega_{\mathbf{k}}\, \delta^{(3)}(\mathbf{k}-\mathbf{k}')\, |0\rangle = -(\alpha+\beta)\, f(\mathbf{k})\, |0\rangle\,. \end{aligned} \tag{13.112}$$

The cross terms vanished because $\Big[a_0, a_3^\dagger\Big] = \Big[a_3, a_0^\dagger\Big] = 0$. The state is physical iff

$$\alpha + \beta = 0, \qquad \text{i.e.} \qquad \beta = -\alpha. \tag{13.113}$$

A physical scalar/longitudinal photon must therefore be the combination $|\psi\rangle = \alpha \int \widetilde{dk}\, f\, (a_0^\dagger - a_3^\dagger)\,|0\rangle = \alpha\sqrt{2}\, L^\dagger(f)\,|0\rangle$, proportional to the null creation operator $L^\dagger$. Its norm follows from Eq. (13.99):

$$\langle\psi|\psi\rangle = 2|\alpha|^2 \left\langle 0 \middle| L(f) L^\dagger(f) \middle| 0 \right\rangle = 2|\alpha|^2 \left\langle 0 \middle| \Big[L(f), L^\dagger(f)\Big] \middle| 0 \right\rangle = 0. \tag{13.114}$$

We can also see the cancellation in the original variables: with $\beta = -\alpha$,

$$\langle\psi|\psi\rangle = \int \widetilde{dk} |f|^2 \Big(\underbrace{|\alpha|^2 \Big[a_0, a_0^\dagger\Big]}_{=-|\alpha|^2(\cdots)} + \underbrace{|\beta|^2 \Big[a_3, a_3^\dagger\Big]}_{=+|\alpha|^2(\cdots)} \Big) = \int \widetilde{dk} |f|^2\, |\alpha|^2 (-1+1)(2\pi)^3 2\omega_{\mathbf{k}}\, \delta^{(3)}(0) \cdots = \tag{13.115}$$

the negative scalar norm cancelling the positive longitudinal norm. Every physical mixture of scalar and longitudinal photons is null—there is no positive-norm one-particle state outside the transverse sector—in concrete agreement with Theorem 13.19 and Theorem 13.22.

Worked Example 13.24 (Null states are residual gauge transformations)**.** The null states of Theorem 13.23 are not arbitrary; they implement, at the quantum level, the residual gauge freedom that survives Feynman gauge. We show that adding a null state to a physical state shifts the expectation value of A_μ by a pure gradient $\partial_\mu\Lambda$ with $\Box\Lambda = 0$—exactly a gauge transformation Eq. (8.113) preserving the wave equation Eq. (13.79).

Start from the commutator of the field with the null creation operator. With $L^\dagger(k) = a^\dagger(k,0) - a^\dagger(k,3)$ and the positive-frequency field $A_\mu^{(+)}(x) = \int \widetilde{dk'} \sum_\lambda \epsilon_\mu(k',\lambda) a(k',\lambda) e^{-ik'\cdot x}$,

$$\begin{aligned}\Big[A_\mu^{(+)}(x), L^\dagger(k)\Big] &= \int \widetilde{dk'} \sum_\lambda \epsilon_\mu(k',\lambda) e^{-ik'\cdot x} \Big(\Big[a(k',\lambda), a^\dagger(k,0)\Big] - \Big[a(k',\lambda), a^\dagger(k,3)\Big] \Big) \\ &= \int \widetilde{dk'} \sum_\lambda \epsilon_\mu(k',\lambda) e^{-ik'\cdot x} \big(-\eta_{\lambda 0} + \eta_{\lambda 3} \big) (2\pi)^3 2\omega_{\mathbf{k}} \delta^{(3)}(\mathbf{k}' - \mathbf{k}) \\ &= -\epsilon_\mu(k,0)\, e^{-ik\cdot x} \;-\; \epsilon_\mu(k,3)\, e^{-ik\cdot x}, \end{aligned} \tag{13.116}$$

where in the last line $-\eta_{\lambda 0}$ picks out $\lambda = 0$ with coefficient -1 and $+\eta_{\lambda 3}$ picks out $\lambda = 3$ with coefficient $\eta_{33} = -1$. For $k^\mu = \omega(1,0,0,1)$ the lowered polarizations are $\epsilon_\mu(k,0) = (1,0,0,0)$ and $\epsilon_\mu(k,3) = (0,0,0,-1)$, whose sum is $(1,0,0,-1) = k_\mu/\omega$. Hence

$$\Big[A_\mu^{(+)}(x), L^\dagger(k)\Big] = -\frac{k_\mu}{\omega}\, e^{-ik\cdot x} = \partial_\mu\Big(-\frac{i}{\omega} e^{-ik\cdot x} \Big) \equiv \partial_\mu \Lambda_k(x), \tag{13.117}$$

using $\partial_\mu e^{-ik\cdot x} = -ik_\mu e^{-ik\cdot x}$, with $\Lambda_k(x) = -\frac{i}{\omega} e^{-ik\cdot x}$ satisfying $\Box\Lambda_k = 0$. Now let $|\Psi\rangle$ be a normalized physical state and form a different representative of the same physical class by adding a null admixture, $|\Psi_\theta\rangle = |\Psi\rangle + \theta\, L^\dagger(g)\,|\Psi\rangle$ with θ a small real parameter and $L^\dagger(g) = \int \widetilde{dk}\, \bar g(\mathbf{k}) L^\dagger(k)$. The shift in the expectation value of the field is, to first order in θ,

$$\delta\langle A_\mu(x)\rangle = \theta\, \Big\langle \Psi \Big| \Big[A_\mu(x), L^\dagger(g)\Big] \Big| \Psi \Big\rangle + \text{c.c.} = \theta \int \widetilde{dk}\, \bar g(\mathbf{k})\, \partial_\mu \Lambda_k(x)\; \langle\Psi|\Psi\rangle + \text{c.c.} = \partial_\mu \Lambda(x), \tag{13.118}$$

where only the positive-frequency part of A_μ contributed to the commutator (the negative-frequency part commutes with $L^\dagger$, being built from creation operators), and $\Lambda(x) = 2\,\mathrm{Re}\big[\theta \int \widetilde{dk}\, \bar g(\mathbf{k}) \Lambda_k(x)\big]$ is a real c-number function with $\Box\Lambda = 0$. The conclusion is exact and structural: *changing the null content of a physical state is a gauge transformation* $A_\mu \to A_\mu + \partial_\mu\Lambda$, with Λ a harmonic gauge function—the residual freedom that the Feynman-gauge condition does not fix. Quotienting by V_0 in Eq. (13.107) is thus the quantum implementation of fixing the residual gauge: physically equivalent (gauge-related) field configurations are collapsed to a single ray of $\mathcal{H}_{\text{phys}}$. The classical redundancy of Chapter 8 and the quantum null states are one and the same phenomenon, seen at two levels.

Physical Insight

The three faces of the gauge redundancy now coincide. *Classically* (Chapter 8) it was the freedom $A_\mu \to A_\mu + \partial_\mu \alpha$ that left $F_{\mu\nu}$ untouched. In *Coulomb gauge* (previous section) it was eliminated outright by solving the constraints, at the cost of covariance. In *Gupta–Bleuler* it survives as the null subspace V_0, whose elements are the residual gauge transformations Eq. (13.117), and the physical Hilbert space is the covariant configuration space modulo this freedom. The Faddeev–Popov path integral of Chapter 26 will unify these descriptions and, in the non-abelian case, promote the null modes to dynamical ghost fields whose cohomology—the BRST refinement of the Gupta–Bleuler condition—is the modern statement of Theorem 13.22.

We have what the rest of QED needs: a covariant operator A_μ on an explicit (if indefinite-metric) Fock space, a Lorentz-covariant commutator Eq. (13.83), and a physical-state condition Eq. (13.93) that guarantees unitarity on $\mathcal{H}_{\text{phys}}$ and justifies dropping scalar and longitudinal polarizations from physical amplitudes. The next section assembles from this operator the time-ordered two-point function $\langle 0|\mathcal{T}A_\mu(x)A_\nu(y)|0\rangle$—the photon propagator—tracks its dependence on the gauge parameter ξ across the R_ξ family, and shows that the gauge-dependent part never touches a physical amplitude because it couples only to conserved currents, exactly the weak-Lorenz mechanism of Eq. (13.110) seen from the propagator side.

13.4 The Photon Propagator and Its Gauge Dependence

The covariant quantization of Section 13.3 delivered a fully Lorentz-covariant field $A_\mu(x)$ at the cost of an indefinite-metric Fock space, and then carved a genuine physical Hilbert space out of it by the Gupta–Bleuler subsidiary condition. We now collect the dividend. The single object that all of perturbative QED is built from is the time-ordered two-point function $\langle 0|\,\mathcal{T}A_\mu(x)A_\nu(y)\,|0\rangle$ — the amplitude for a disturbance of the electromagnetic field to propagate from y to x, the internal photon line of every diagram from Chapter 17 onward. Computing it is the goal of this section, and it forces us to confront, in its sharpest and most practical form, the redundancy that has organized the entire chapter: the answer is *not unique*. It carries a free parameter ξ, the gauge-fixing parameter, and different choices of ξ give genuinely different propagators. The resolution — that this gauge dependence is invisible in every physical amplitude because the matter it couples to carries a conserved current — is the second half of the section, and it is the precise statement, at the level of the propagator, of the fact

that gauge is a redundancy of description and not a property of the world.

We work throughout in the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, with the covariant derivative on the electron $D_\mu = \partial_\mu + ieA_\mu$ ($e > 0$, electron charge $-e$) of QED, used in Chapter 11, and the Feynman $i\varepsilon$ prescription inherited from the scalar propagator Eq. (9.154). The computational default, stated and used below, is Feynman gauge $\xi = 1$.

13.4.1 The coordinate-space propagator in Feynman gauge

The cleanest derivation starts from the field itself. Recall from the covariant construction (Section 13.3) that in Feynman gauge the gauge-fixed Lagrangian is

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}\left(\partial_\mu A^\mu\right)^2, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{13.119}$$

whose Euler–Lagrange equation $\left[\eta_{\mu\nu}\Box - (1-\xi^{-1})\partial_\mu\partial_\nu\right]A^\nu = 0$ collapses, at $\xi = 1$, to four decoupled massless wave equations $\Box A_\mu = 0$. Each component of A_μ then propagates like a massless scalar field, and the equal-time and covariant commutation relations established in Section 13.3 take the compact covariant form

$$[A_\mu(x), A_\nu(y)] = -\,i\,\eta_{\mu\nu}\,D(x-y)\Big|_{m=0}, \tag{13.120}$$

where $D(x-y)$ is the Pauli–Jordan commutator function of the free scalar, evaluated at zero mass (the $m \to 0$ limit of the function whose causal support property was the content of Section 9.6). The crucial structural fact, and the one feature that distinguishes the photon from a scalar, is the index tensor $-\eta_{\mu\nu}$: the spacelike components carry the ordinary sign while the timelike component A_0 carries the *opposite* sign $-\eta_{00} = -1$. That single minus is the indefinite metric of the covariant Fock space, written in field language. It is harmless here precisely because it factors out of the spacetime dynamics, which is identical for all four components.

The positive-frequency Wightman function inherits the same structure. With the covariant mode expansion of Section 13.3, whose creation/annihilation operators obey $\left[a_\mu(\mathbf{k}), a_\nu^\dagger(\mathbf{k}')\right] = -\eta_{\mu\nu}(2\pi)^3\,2\omega_{\mathbf{k}}\,\delta^3(\mathbf{k}-\mathbf{k}')$, one finds

$$\langle 0|\,A_\mu(x)A_\nu(y)\,|0\rangle = -\eta_{\mu\nu}\int\frac{d^3k}{(2\pi)^3\,2\omega_{\mathbf{k}}}\,e^{-ik\cdot(x-y)}\Big|_{k^0=\omega_{\mathbf{k}}=|\mathbf{k}|} = -\eta_{\mu\nu}\,D^+(x-y)\big|_{m=0}, \tag{13.121}$$

the index tensor $-\eta_{\mu\nu}$ pulled directly through from the commutator and D^+ the massless scalar Wightman function. Everything that depends on the spacetime points x, y is a single scalar function; only the constant tensor $-\eta_{\mu\nu}$ remembers that we are dealing with a vector field.

We can now assemble the time-ordered product. By the definition of time ordering,

$$\mathcal{T} A_\mu(x) A_\nu(y) = \theta(x^0 - y^0)\, A_\mu(x) A_\nu(y) + \theta(y^0 - x^0)\, A_\nu(y) A_\mu(x), \tag{13.122}$$

so that, taking the vacuum expectation value and using Eq. (13.121) for both orderings,

$$\langle 0 |\, \mathcal{T} A_\mu(x) A_\nu(y)\, | 0 \rangle = -\eta_{\mu\nu} \Big[\theta(x^0 - y^0) D^+(x - y) + \theta(y^0 - x^0) D^+(y - x) \Big] \Big|_{m=0}. \tag{13.123}$$

The bracket is, term for term, the time-ordered combination of Wightman functions that *defined* the scalar Feynman propagator in Eq. (9.153); the constant tensor $-\eta_{\mu\nu}$ rides along untouched. We have therefore proved the following.

Theorem 13.25 (Coordinate-space photon propagator, Feynman gauge)**.** *In Feynman gauge ($\xi = 1$) the time-ordered photon two-point function is the scalar Feynman propagator at zero mass, times the constant tensor $-\eta_{\mu\nu}$:*

$$D^F_{\mu\nu}(x-y) \;\equiv\; \langle 0 |\, \mathcal{T} A_\mu(x) A_\nu(y)\, | 0 \rangle \;=\; -\,\eta_{\mu\nu}\, D_F(x-y)\big|_{m=0} \;=\; \int \frac{d^4k}{(2\pi)^4}\, \frac{-\,i\,\eta_{\mu\nu}}{k^2 + i\varepsilon}\, e^{-ik\cdot(x-y)}. \tag{13.124}$$

Proof. Eq. (13.123) identifies the bracket as the time-ordered scalar correlator, i.e. $D_F(x-y)|_{m=0}$ by Theorem 9.24. Substituting the momentum-space form of the massless scalar Feynman propagator Eq. (9.154) (set $m = 0$) and pulling out $-\eta_{\mu\nu}$ gives the displayed integral. The $i\varepsilon$ is inherited intact from the scalar: it is the statement that positive-frequency modes propagate forward and negative-frequency modes backward in time, the same Stueckelberg–Feynman boundary condition that Section 9.5 attached to the scalar poles. As a tempered distribution the equality holds for all $x - y$, on or off the light cone. □

The momentum-space integrand in Eq. (13.124) is the Feynman-gauge propagator. Reading it off,

$$D_{\mu\nu}(k)\big|_{\xi=1} = \frac{-\,i\,\eta_{\mu\nu}}{k^2 + i\varepsilon}, \tag{13.125}$$

which is exactly the $\xi = 1$ specialization of the general formula we are about to derive. It is worth pausing on how economical this is: the photon propagator in Feynman gauge is four copies of the massless scalar propagator, three with a relative minus sign. The minus signs are the indefinite metric; they will be exactly what is needed for the unphysical (timelike and longitudinal) polarizations to cancel in physical amplitudes.

13.4.2 The R_ξ family from inverting the quadratic operator

Feynman gauge is one point in a one-parameter family. To see the family, and to understand why a free parameter is unavoidable, we compute the propagator the way the path integral of Section 13.5 will demand: as the inverse of the quadratic differential operator in the action. This is the standard route to any free propagator — the two-point function of a Gaussian theory is the inverse of the kernel of its quadratic form — and it makes the role of the gauge-fixing term unmistakable.

Begin with the gauge-fixed Lagrangian Eq. (13.119) and bring the quadratic action into the canonical form $S = \frac{1}{2}\int d^4x\, A^\mu(x)\, \mathcal{O}_{\mu\nu}\, A^\nu(x)$ by integrating by parts (discarding surface terms, legitimate for fields decaying at infinity). For the Maxwell term, expand $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\frac{1}{2}(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu)$ and integrate each piece by parts:

$$-\tfrac{1}{2}\partial_\mu A_\nu\, \partial^\mu A^\nu \;\longrightarrow\; +\tfrac{1}{2}\, A_\nu\, \Box A^\nu, \tag{13.126}$$

$$+\tfrac{1}{2}\partial_\mu A_\nu\, \partial^\nu A^\mu \;\longrightarrow\; -\tfrac{1}{2}\, A_\nu\, \partial^\nu \partial_\mu A^\mu, \tag{13.127}$$

while the gauge-fixing term gives $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2 \to +\frac{1}{2\xi} A_\mu\, \partial^\mu \partial^\nu A_\nu$. Collecting,

$$S = \tfrac{1}{2}\int d^4x\; A^\mu \Big[\eta_{\mu\nu}\Box - \Big(1 - \tfrac{1}{\xi}\Big)\partial_\mu\partial_\nu\Big] A^\nu, \tag{13.128}$$

so the differential kernel is $\mathcal{O}_{\mu\nu} = \eta_{\mu\nu}\Box - (1-\xi^{-1})\partial_\mu\partial_\nu$. Passing to momentum space with the standing Fourier convention $A(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x}\widetilde{A}(k)$, every ∂_μ becomes $-ik_\mu$, so $\Box \to -k^2$ and $\partial_\mu\partial_\nu \to -k_\mu k_\nu$, giving the algebraic operator

$$\mathcal{O}_{\mu\nu}(k) = -k^2\eta_{\mu\nu} + \Big(1 - \tfrac{1}{\xi}\Big) k_\mu k_\nu. \tag{13.129}$$

The propagator $D_{\mu\nu}(k)$ is fixed by the requirement that it be the Green's function of this operator with the i of the Minkowski path integral,

$$\mathcal{O}^{\mu\rho}(k)\, D_{\rho\nu}(k) = i\, \delta^\mu{}_\nu, \tag{13.130}$$

the exact tensor analogue of the scalar relation $(k^2 - m^2)D_F = i$ that produced Eq. (9.154).

To invert $\mathcal{O}$ we use the two natural projectors built from k_μ. Define the *transverse* and *longitudinal* projectors

$$\theta_{\mu\nu} = \eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}, \qquad \omega_{\mu\nu} = \frac{k_\mu k_\nu}{k^2}. \tag{13.131}$$

They satisfy the projector algebra $\theta + \omega = \eta$, $\theta^2 = \theta$, $\omega^2 = \omega$, $\theta\,\omega = \omega\,\theta = 0$ (each identity follows from $k^\mu\theta_{\mu\nu} = 0$ and $k^\mu\omega_{\mu\nu} = k_\nu$, which are immediate from

Eq. (13.131)). They diagonalize $\mathcal{O}$. Writing $\eta = \theta + \omega$ and $k_\mu k_\nu = k^2 \omega_{\mu\nu}$ in Eq. (13.129),

$$\mathcal{O}_{\mu\nu}(k) = -k^2(\theta_{\mu\nu} + \omega_{\mu\nu}) + \Big(1 - \tfrac{1}{\xi}\Big) k^2 \omega_{\mu\nu} = -k^2\, \theta_{\mu\nu} - \frac{k^2}{\xi}\, \omega_{\mu\nu}. \tag{13.132}$$

Thus $\mathcal{O}$ acts as the scalar $-k^2$ on the transverse subspace and as $-k^2/\xi$ on the longitudinal direction. On each eigenspace inversion is scalar division, and because θ and ω are orthogonal idempotents the inverse of $a\theta + b\omega$ is $a^{-1}\theta + b^{-1}\omega$:

$$\big(\mathcal{O}^{-1}\big)_{\mu\nu}(k) = -\frac{1}{k^2}\, \theta_{\mu\nu} - \frac{\xi}{k^2}\, \omega_{\mu\nu}. \tag{13.133}$$

Multiplying by i as Eq. (13.130) requires, and re-expanding the projectors via Eq. (13.131),

$$D_{\mu\nu}(k) = i\big(\mathcal{O}^{-1}\big)_{\mu\nu} = \frac{-i}{k^2}(\theta_{\mu\nu} + \xi\, \omega_{\mu\nu}) = \frac{-i}{k^2}\Big[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big], \tag{13.134}$$

where the last step used $\theta + \xi\omega = \eta - \omega + \xi\omega = \eta - (1-\xi)\omega$. Restoring the Feynman prescription that selects the physical (causal) Green's function exactly as for the scalar — the same pole displacement $k^2 \to k^2 + i\varepsilon$ established in Section 9.5 — we have the central result of the section.

Theorem 13.26 (The R_ξ photon propagator)**.** *The momentum-space time-ordered photon two-point function in the R_ξ gauge family is*

$$\boxed{D_{\mu\nu}(k) = \frac{-i}{k^2 + i\varepsilon}\Big[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big]} \tag{13.135}$$

This is the carried object of QED's photon stage: the single tensor that becomes a Feynman rule in Chapter 17 and is dressed into the full photon propagator by vacuum polarization in Chapter 19. Three values of ξ are in common use and earn names; we record them, with their defining virtues.

Definition 13.27 (Named gauges in the R_ξ family)**.** The standard members of Eq. (13.135) are

$$\text{Feynman ('t Hooft–Feynman) gauge,}\quad \xi = 1: \quad D_{\mu\nu}(k) = \frac{-i\,\eta_{\mu\nu}}{k^2 + i\varepsilon}, \tag{13.136}$$

$$\text{Landau gauge,}\quad \xi = 0: \quad D_{\mu\nu}(k) = \frac{-i}{k^2 + i\varepsilon}\Big[\eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\Big] = \frac{-i}{k^2 + i\varepsilon}\,\theta_{\mu\nu} \tag{13.137}$$

$$\text{Yennie gauge,}\quad \xi = 3: \quad D_{\mu\nu}(k) = \frac{-i}{k^2 + i\varepsilon}\Big[\eta_{\mu\nu} + 2\,\frac{k_\mu k_\nu}{k^2}\Big]. \tag{13.138}$$

Each has a reason to exist. Feynman gauge gives the simplest possible numerator $\eta_{\mu\nu}$ and is the default for almost every explicit computation in this book; we adopt it unless stated otherwise. Landau gauge makes the propagator exactly transverse, $k^\mu D_{\mu\nu} = 0$ (immediate since $k^\mu \theta_{\mu\nu} = 0$), which is convenient whenever transversality must be manifest, for example in studies of the gauge-dependence of off-shell Green's functions or in confining-phase analyses of the gluon propagator. Yennie gauge, $\xi = 3$, is engineered so that the infrared behavior of the photon propagator softens the leading infrared singularities of QED radiative corrections, simplifying the bookkeeping of infrared divergences in the program of Chapter 19. That three such different-looking objects are all legitimate propagators for the *same* theory is the puzzle that the next subsection resolves.

Physical Insight

The single free parameter ξ is the propagator's memory of gauge redundancy. The transverse part of the propagator, $-i\theta_{\mu\nu}/k^2$, is ξ-independent and physical: it is the part that propagates the two genuine helicity states of the photon. The longitudinal part, $-i\xi\,\omega_{\mu\nu}/k^2$, is pure gauge artefact — its coefficient is whatever the gauge-fixing term was told to make it. A physical answer cannot depend on ξ, so it cannot depend on the longitudinal part. This is the same two-physical-degrees-of-freedom count that has organized the whole chapter, now visible in the tensor structure of a single propagator.

13.4.3 Why the operator must be gauge-fixed: the non-invertible limit

The inversion in Eq. (13.132)–Eq. (13.133) required the longitudinal eigenvalue $-k^2/\xi$ to be nonzero. Tracking what happens when it is not exposes precisely why the gauge-fixing term in Eq. (13.119) is not optional but mandatory, and what role ξ plays.

Consider the pure Maxwell theory, with no gauge fixing. This is recovered formally as $\xi \to \infty$, where the gauge-fixing coefficient $\xi^{-1} \to 0$ and Eq. (13.129) reduces to $\mathcal{O}^{\mathrm{Max}}_{\mu\nu} = -k^2\eta_{\mu\nu} + k_\mu k_\nu = -k^2\theta_{\mu\nu}$. This operator annihilates the longitudinal direction outright:

$$\mathcal{O}^{\mathrm{Max}}_{\mu\nu}(k)\, k^\nu = -k^2\, \theta_{\mu\nu} k^\nu = 0, \tag{13.139}$$

because $\theta_{\mu\nu}k^\nu = k_\mu - k_\mu(k\cdot k)/k^2 = 0$. A linear operator with a nonzero null vector is not invertible: there is no tensor $D_{\rho\nu}$ satisfying Eq. (13.130), since contracting that equation with k^μ would give $0 = i\,k_\nu \neq 0$, a contradiction. The pure Maxwell quadratic form simply does not have a propagator.

The null vector is not an accident; it is gauge invariance, read in momentum space. A gauge transformation $A_\mu \to A_\mu + \partial_\mu \alpha$ shifts $\widetilde{A}_\mu(k) \to \widetilde{A}_\mu(k) - ik_\mu \widetilde{\alpha}(k)$, i.e. it adds a multiple of k_μ — precisely the direction Eq. (13.139) kills. The action is flat along gauge orbits, the quadratic form is degenerate in the longitudinal direction, and a degenerate quadratic form cannot be inverted. The gauge-fixing term $-\frac{1}{2\xi}(\partial A)^2$ does exactly one thing: it supplies a nonzero longitudinal eigenvalue $-k^2/\xi$, lifting the flat direction so that the operator becomes invertible, with the lift controlled by ξ. The price is that the longitudinal part of the resulting propagator depends on the arbitrary lift parameter ξ. The path-integral derivation of Section 13.5 will make this geometry literal: the divergence of $\int \mathcal{D}A\, e^{iS}$ over gauge orbits is exactly the non-invertibility Eq. (13.139), and the Faddeev–Popov insertion is the systematic device that fixes a gauge and produces Eq. (13.135) with ξ as a free choice. For now the lesson stands on its own: *you cannot have a photon propagator without breaking the gauge invariance, and the breaking leaves a fingerprint ξ that physics must not see.*

13.4.4 Gauge independence of physical amplitudes

We now prove the fingerprint is invisible. The mechanism is current conservation, and it is exact and elementary at tree level; the all-orders generalization is the Ward–Takahashi identity proved in Chapter 19.

The propagator never appears alone. Wherever an internal photon line of momentum k joins the rest of a diagram, both of its ends attach to the matter content through the interaction vertex, and the object that multiplies the propagator is, on each side, a current. Concretely, the simplest gauge-invariant building block is one-photon exchange between two matter currents J_1^μ and J_2^ν (Fig. 13.4), whose amplitude is the contraction

$$i\mathcal{M} = \big(i\widetilde{J}_1^{\mu}(k)\big)\, D_{\mu\nu}(k)\, \big(i\widetilde{J}_2^{\nu}(-k)\big), \tag{13.140}$$

where $\widetilde{J}_i^{\mu}$ is the Fourier transform of the matter current to which the photon couples through the minimal-coupling interaction $-eA_\mu \bar{\psi}\gamma^\mu\psi$. That current is the conserved Dirac current of Chapter 11: the global $U(1)$ symmetry $\psi \to e^{-i\alpha}\psi$ gave, by Noether's theorem, the conserved current Eq. (11.158), $j^\mu = \bar{\psi}\gamma^\mu\psi$ with $\partial_\mu j^\mu = 0$, and it is $e\, j^\mu$ that sits at each photon vertex. Conservation in position space, $\partial_\mu J^\mu(x) = 0$, becomes in momentum space, under $\partial_\mu \to -ik_\mu$,

$$k_\mu\, \widetilde{J}^{\mu}(k) = 0. \tag{13.141}$$

This one identity does all the work.

Theorem 13.28 (ξ-independence of current-coupled amplitudes)**.** *If the photon of momentum k couples at both ends to conserved currents, $k_\mu \widetilde{J}_1^{\mu} = k_\nu \widetilde{J}_2^{\nu} = 0$, then the contraction $\widetilde{J}_1^{\mu} D_{\mu\nu} \widetilde{J}_2^{\nu}$ is independent of the gauge parameter ξ, and equals its Feynman-gauge value.*

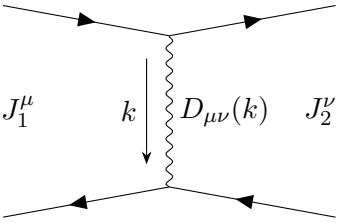


Figure 13.4: One-photon exchange between two conserved matter currents. The internal photon line carries the R_ξ propagator $D_{\mu\nu}(k)$ of Eq. (13.135); each vertex attaches it to a conserved current $\widetilde{J}_i^\mu$ obeying $k_\mu \widetilde{J}_i^\mu = 0$ (Eq. (13.141)). The gauge-dependent $k_\mu k_\nu$ piece of the propagator is annihilated on contact with either current, so the amplitude is independent of ξ. This is the tree-level kernel of $e^-e^- \to e^-e^-$ (Møller) and, by crossing, of e^+e^- scattering, computed in full in Chapters 17 and 18.

Proof. Insert the full propagator Eq. (13.135) into the contraction and split it into its ξ-independent and ξ-dependent pieces:

$$\widetilde{J}_1^\mu \, D_{\mu\nu} \, \widetilde{J}_2^\nu = \frac{-i}{k^2 + i\varepsilon}\Big[\widetilde{J}_1 \cdot \widetilde{J}_2 \; - \; (1-\xi)\, \frac{(\widetilde{J}_1 \cdot k)\,(k \cdot \widetilde{J}_2)}{k^2} \Big]. \tag{13.142}$$

The entire ξ-dependence resides in the second term, and that term is proportional to $\widetilde{J}_1 \cdot k = k_\mu \widetilde{J}_1^\mu$ and to $k \cdot \widetilde{J}_2 = k_\nu \widetilde{J}_2^\nu$, each of which vanishes by Eq. (13.141). Hence

$$\widetilde{J}_1^\mu \, D_{\mu\nu} \, \widetilde{J}_2^\nu = \frac{-i}{k^2 + i\varepsilon} \, \widetilde{J}_1 \cdot \widetilde{J}_2, \tag{13.143}$$

which is exactly the Feynman-gauge ($\xi = 1$) result and contains no ξ. The gauge-dependent term has been annihilated on contact with the conserved current. □

The structure of Eq. (13.142) is worth stating as a slogan: *the gauge (ξ) dependence of the photon propagator lives entirely in the $k_\mu k_\nu$ term, and the $k_\mu k_\nu$ term is annihilated by conserved currents.* Because every photon in QED couples to the conserved electromagnetic current, this is why ξ drops out of every physical amplitude — one may compute in Feynman gauge, Landau gauge, or Yennie gauge and obtain the same cross section. It also furnishes a powerful and constantly used *check*: a correct gauge-invariant amplitude must be ξ-independent, so carrying a general ξ through a computation and watching it cancel is a standard internal consistency test, exploited repeatedly from Chapter 17 onward.

Worked Example 13.29 (The Møller kernel is the same in three gauges)**.** Consider the t-channel one-photon exchange between two electron currents, the kernel of $e^-e^- \to e^-e^-$ (Møller scattering) computed fully in Chapter 17. With incoming/outgoing momenta giving photon momentum $k = p_1 - p_3$, the two currents are the on-shell Dirac bilinears

$$\widetilde{J}_1^\mu = \bar{u}(p_3)\gamma^\mu u(p_1), \qquad \widetilde{J}_2^\nu = \bar{u}(p_4)\gamma^\nu u(p_2). \tag{13.144}$$

We verify current conservation directly and then watch ξ cancel. Contract the first current with $k_\mu = (p_1 - p_3)_\mu$ and use the Dirac equations $\not{p}_1 u(p_1) = m\,u(p_1)$ and $\bar{u}(p_3)\not{p}_3 = m\,\bar{u}(p_3)$ (with m the electron mass) from Chapter 11:

$$k_\mu \widetilde{J}_1^{\mu} = \bar{u}(p_3)(\not{p}_1 - \not{p}_3)u(p_1) = \bar{u}(p_3)(m - m)u(p_1) = 0, \tag{13.145}$$

and identically $k_\nu \widetilde{J}_2^{\nu} = 0$. This is the tree-level Ward identity in the flesh; it is the statement Eq. (13.141) for the specific on-shell electron current. Now form the exchange amplitude with the general-ξ propagator:

$$\begin{aligned} i\mathcal{M} &= \big[\bar{u}(p_3)(ie\gamma^\mu)u(p_1)\big]\,\frac{-i}{k^2 + i\varepsilon}\Big[\eta_{\mu\nu} - (1-\xi)\tfrac{k_\mu k_\nu}{k^2}\Big]\,\big[\bar{u}(p_4)(ie\gamma^\nu)u(p_2)\big] \\ &= \frac{-i\,(ie)^2}{k^2 + i\varepsilon}\Big[\,\widetilde{J}_1\!\cdot\!\widetilde{J}_2 - (1-\xi)\tfrac{(k\cdot\widetilde{J}_1)(k\cdot\widetilde{J}_2)}{k^2}\Big] = \frac{i\,e^2}{k^2 + i\varepsilon}\,\big[\bar{u}_3\gamma^\mu u_1\big]\big[\bar{u}_4\gamma_\mu u_2\big]. \end{aligned} \tag{13.146}$$

The bracketed ξ-term vanished by Eq. (13.145). The surviving amplitude is the Feynman-gauge result, manifestly independent of ξ; Landau ($\xi = 0$), Feynman ($\xi = 1$), and Yennie ($\xi = 3$) gauges return identical numbers. This is the explicit promise of Theorem 13.28 and the working procedure used throughout Chapters 17 and 18.

13.4.5 External photons: the polarization sum becomes $-\eta_{\mu\nu}$

The same conservation argument controls a second object that pervades cross-section calculations: the sum over the polarizations of a real, external photon. When a physical photon of momentum k ($k^2 = 0$) appears in the initial or final state, its two transverse polarization vectors $\epsilon_\mu(k,\lambda)$, $\lambda = \pm$, must be summed over (squared and summed for an unobserved final-state polarization, averaged for an unprepared initial one). The object that arises is

$$\Pi_{\mu\nu}(k) \equiv \sum_{\lambda=\pm} \epsilon_\mu(k,\lambda)\,\epsilon_\nu^*(k,\lambda). \tag{13.147}$$

This sum is *not* simply $-\eta_{\mu\nu}$ on its own — it cannot be, since a Lorentz-covariant object built only from a single null k_μ must contain k-dependent terms, while $\eta_{\mu\nu}$ does not. The transverse polarizations span only a two-dimensional space, so $\Pi_{\mu\nu}$ is a rank-two object, not the rank-four $-\eta_{\mu\nu}$. What is true, and what is all that is ever needed, is that $\Pi_{\mu\nu}$ may be *replaced* by $-\eta_{\mu\nu}$ whenever it is contracted into a conserved current. We make this precise.

Choose a frame in which $k^\mu = (\omega, 0, 0, \omega)$ (a photon moving in the $+z$ direction, $k^2 = 0$). The two transverse linear polarizations are $\epsilon^\mu(k,1) = (0,1,0,0)$ and $\epsilon^\mu(k,2) = (0,0,1,0)$; they satisfy $k\cdot\epsilon(k,\lambda) = 0$ and $\epsilon(k,\lambda)\cdot\epsilon^*(k,\lambda') = -\delta_{\lambda\lambda'}$ (the spatial minus of the mostly-minus metric). To express the sum covariantly introduce a reference null vector $\bar{k}^\mu = (\omega, 0, 0, -\omega)$, the momentum of a photon

moving the other way, with $\bar{k}^2 = 0$ and $k \cdot \bar{k} = 2\omega^2 \neq 0$. Direct evaluation of the four-by-four matrix $\sum_{\lambda=1,2} \epsilon_\mu \epsilon_\nu^*$ in this frame gives entries only in the $1,2$ (i.e. x, y) block, equal to δ_{ij} there; this is reproduced covariantly by

$$\sum_{\lambda=\pm} \epsilon_\mu(k,\lambda)\, \epsilon_\nu^*(k,\lambda) = -\eta_{\mu\nu} + \frac{k_\mu \bar{k}_\nu + \bar{k}_\mu k_\nu}{k \cdot \bar{k}}. \tag{13.148}$$

One checks this by components: the $-\eta_{\mu\nu}$ supplies $+1$ in each of the x, y, z diagonal slots and -1 in the 00 slot, while the second term, built from k and $\bar{k}$ which have no x, y components, cancels the unwanted 00, 03, 30, 33 entries and leaves exactly the transverse δ_{ij} in the x, y block. The reference vector $\bar{k}$ is arbitrary; the left-hand side does not know about it, so the $\bar{k}$-dependence on the right must cancel against anything physical it touches — and it does, again by current conservation.

Theorem 13.30 (Polarization-sum replacement)**.** *Let $\mathcal{M}^\mu(k)$ be a physical amplitude with one external photon of momentum k stripped of its polarization vector, so that the photon contributes $\epsilon_\mu^* \mathcal{M}^\mu$. If the amplitude obeys the Ward identity $k_\mu \mathcal{M}^\mu = 0$, then summing $|\epsilon^* \cdot \mathcal{M}|^2$ over physical polarizations may be performed with the replacement $\sum_\lambda \epsilon_\mu \epsilon_\nu^* \to -\eta_{\mu\nu}$:*

$$\sum_{\lambda=\pm} \left|\epsilon^*(k,\lambda)\cdot\mathcal{M}\right|^2 = \sum_{\lambda=\pm} \epsilon_\mu \epsilon_\nu^*\, \mathcal{M}^\mu \mathcal{M}^{*\nu} = -\eta_{\mu\nu}\, \mathcal{M}^\mu \mathcal{M}^{*\nu}. \tag{13.149}$$

Proof. Contract the completeness relation Eq. (13.148) into $\mathcal{M}^\mu \mathcal{M}^{*\nu}$:

$$\sum_\lambda \epsilon_\mu \epsilon_\nu^*\, \mathcal{M}^\mu \mathcal{M}^{*\nu} = -\eta_{\mu\nu} \mathcal{M}^\mu \mathcal{M}^{*\nu} + \frac{(k_\mu \mathcal{M}^\mu)(\bar{k}_\nu \mathcal{M}^{*\nu}) + (\bar{k}_\mu \mathcal{M}^\mu)(k_\nu \mathcal{M}^{*\nu})}{k \cdot \bar{k}}. \tag{13.150}$$

Both numerators in the second term contain a factor $k_\mu \mathcal{M}^\mu$ or its conjugate $k_\nu \mathcal{M}^{*\nu} = (k_\mu \mathcal{M}^\mu)^*$, which vanish by the Ward identity hypothesis. The reference-vector dependence disappears with them, leaving $-\eta_{\mu\nu} \mathcal{M}^\mu \mathcal{M}^{*\nu}$ as claimed. That the $\bar{k}$-dependence cancels is the consistency check that the physical answer is independent of the arbitrary reference vector, as it must be. □

This replacement is one of the most heavily used identities in the book: it turns every external-photon spin sum into a metric contraction, and it is what makes the unpolarized cross sections of Chapters 17 and 18 tractable. Note that it is the *same* fact as ξ-independence wearing a different hat. In both cases a k_μ-proportional piece — the longitudinal part of the propagator, or the reference-dependent part of the polarization sum — is the entire content of the unphysical degrees of freedom, and in both cases it is annihilated by current conservation. The propagator carries four index-pairs of which two are physical; the polarization sum carries two physical states out of a naive four; current conservation is the bridge that lets us pretend, inside any gauge-invariant amplitude, that all four are present with the simple metric weight.

Physical Insight

The replacement $\sum \epsilon\epsilon^* \to -\eta_{\mu\nu}$ is the amplitude-level shadow of the Gupta–Bleuler decoupling of Section 13.3. There, the subsidiary condition guaranteed that timelike and longitudinal photons — the would-be extra polarizations carried by $-\eta_{\mu\nu}$ relative to the two physical states — have cancelling contributions to all physical matrix elements, leaving a positive-norm transverse space. Here that same cancellation is what lets us add back the unphysical polarizations (the full $-\eta_{\mu\nu}$, not just the transverse projector) without changing the answer: their net contribution is zero precisely because the current is conserved. The two-degree-of-freedom photon is being counted the same way twice, once in the state space and once in the amplitude.

The argument given here is exact at tree level, where the Ward identity $k_\mu \mathcal{M}^\mu = 0$ follows directly from the Dirac equation, as the explicit calculation Eq. (13.145) showed. At higher orders the relevant statement is the all-orders Ward–Takahashi identity, which promotes current conservation to an operator identity among full Green's functions and guarantees both the ξ-independence of Theorem 13.28 and the polarization replacement Theorem 13.30 to every order in the loop expansion. That identity, and the diagrammatic and path-integral proofs of it, are the business of Chapter 19; we flag the upgrade here so that nothing in Chapters 17 and 18 relies on tree-level conservation in a way that would fail at one loop.

13.4.6 What has been built, and where the gauge term comes from

We have produced the object Part V runs on. The time-ordered photon two-point function is, in coordinate space and Feynman gauge, the massless scalar Feynman propagator times $-\eta_{\mu\nu}$ (Theorem 13.25); across the R_ξ family it is the boxed momentum-space tensor Eq. (13.135), obtained by inverting the gauge-fixed quadratic operator and recognizing the obstruction Eq. (13.139) to inverting the unfixed one. Its gauge-dependent $k_\mu k_\nu$ term is annihilated by the conserved matter current Eq. (11.158), so it never touches a physical amplitude (Theorem 13.28), and the corresponding statement for external photons replaces the polarization sum by $-\eta_{\mu\nu}$ inside any current-conserving amplitude (Theorem 13.30). This propagator is exported, unchanged, to the QED Feynman rules of Chapter 17 and is dressed by vacuum polarization in Chapter 19.

One question has been deferred at every turn and must now be answered: by what right did we add the gauge-fixing term $-\frac{1}{2\xi}(\partial A)^2$ to the Lagrangian in the first place? Canonically it was forced on us to make the Legendre transform non-singular; here it was forced on us to make the quadratic operator invertible.

But its true origin — the precise sense in which it is not an arbitrary modification of the theory but the unavoidable Jacobian of slicing the path integral transverse to gauge orbits — is the subject of the next section. There the redundancy is re-derived as the divergence of $\int \mathcal{D}A\, e^{iS}$ over the orbits of the gauge group, the Faddeev–Popov determinant is shown to be field-independent in the abelian case (so QED has no propagating ghosts), and the R_ξ generating functional and the propagator Eq. (13.135) re-emerge from first principles — with the warning, paid in full in Part VII, that the non-abelian case will not let the determinant be ignored.

13.5 The Path Integral, Gauge Orbits, and Abelian Faddeev–Popov

The propagator we extracted in Section 13.4 carried a free parameter ξ that entered the Lagrangian through a term $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ we were forced to *add by hand* before the kinetic operator could be inverted. We justified that addition operationally — it lifts the longitudinal zero mode, it produces a propagator, and the ξ-dependence drops out of physical amplitudes by current conservation — but we never explained where it comes from. From the canonical side there is no fully satisfying answer: one either fixes the gauge completely and works with the genuine two degrees of freedom (Coulomb gauge), or enlarges the state space and carves out the physical sector afterwards (Gupta–Bleuler). Neither route delivers $-\frac{1}{2\xi}(\partial \cdot A)^2$ as a *derived* object. The path integral does, and that is the business of this section. We will re-derive the entire structure of the quantized photon — the gauge-fixing term, the R_ξ propagator, the free generating functional — from the single starting point $\int \mathcal{D}A\, e^{iS}$, and in doing so we will meet, in its simplest abelian incarnation, the Faddeev–Popov construction that Chapter 26 turns into the engine of non-abelian gauge theory. The abelian case is the right place to learn the machinery precisely because here it does something almost trivial: the determinant it produces is a field-independent constant, so QED has no propagating ghosts. Seeing exactly *why* it is trivial here is the cleanest possible preparation for seeing why it is not trivial there.

13.5.1 The naive photon path integral and its divergence

We begin where the path-integral programme of Chapter 10 tells us to begin: write the generating functional as a sum over field histories weighted by e^{iS}. For the free electromagnetic field, with the Maxwell action of Eq. (8.75),

$$Z \;=\; \int \mathcal{D}A\; \exp\Big(i \int d^4x\; \big(-\tfrac{1}{4} F_{\mu\nu} F^{\mu\nu} \big) \Big), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{13.151}$$

the measure $\mathcal{D}A = \prod_x \prod_{\mu=0}^{3} dA_\mu(x)$ ranging over all four components of A_μ at every spacetime point, in exact analogy with the scalar measure of Theorem 10.16.

This is the obvious thing to write down, and it is wrong — or rather, it is ill-defined — in a way we can diagnose precisely.

To expose the trouble, put the action into the canonical quadratic form $S = \frac{1}{2}\int A\,\mathcal{O}\,A$ that Chapter 10 taught us to read a propagator off of. Expand the field strength,

$$-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\tfrac{1}{2}(\partial_\mu A_\nu\,\partial^\mu A^\nu - \partial_\mu A_\nu\,\partial^\nu A^\mu), \tag{13.152}$$

where the factor $\frac{1}{2}$ comes from $F_{\mu\nu}F^{\mu\nu} = 2(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu)$. Integrate each term by parts, discarding boundary terms as the vacuum boundary conditions of Section 10.3.4 permit: the first term gives $+\frac{1}{2}\,A_\nu\,\partial^2 A^\nu$ and the second $-\frac{1}{2}\,A_\nu\,\partial^\nu\partial_\mu A^\mu$. Collecting,

$$S \;=\; \tfrac{1}{2}\int d^4x\; A_\mu\,(\partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu)\,A_\nu \;\equiv\; \tfrac{1}{2}\int d^4x\; A_\mu\,\mathcal{O}^{\mu\nu}\,A_\nu, \tag{13.153}$$

so the kinetic operator is $\mathcal{O}^{\mu\nu} = \partial^2 g^{\mu\nu} - \partial^\mu\partial^\nu$. Pass to momentum space with the book convention $A_\mu(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x}\widetilde{A}_\mu(k)$, under which $\partial_\mu \to -ik_\mu$, $\partial^2 \to -k^2$, and the operator becomes the matrix

$$\widetilde{\mathcal{O}}^{\mu\nu}(k) \;=\; -k^2 g^{\mu\nu} + k^\mu k^\nu \;=\; -k^2\Big(g^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}\Big) \;=\; -k^2\,P_T^{\mu\nu}(k), \tag{13.154}$$

where in the last step we have recognized the *transverse projector*

$$P_T^{\mu\nu}(k) \;\equiv\; g^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}, \qquad P_L^{\mu\nu}(k) \;\equiv\; \frac{k^\mu k^\nu}{k^2}, \qquad P_T + P_L = g. \tag{13.155}$$

These satisfy the projector algebra $P_T^2 = P_T$, $P_L^2 = P_L$, $P_T P_L = 0$, as one checks directly: ${P_L}^\mu{}_\rho {P_L}^\rho{}_\nu = \frac{k^\mu k_\rho}{k^2}\frac{k^\rho k^\nu}{k^2} = \frac{k^\mu k^\nu}{k^2} = P_L^{\mu\nu}$, whence $P_T^2 = (g - P_L)^2 = g - 2P_L + P_L = P_T$.

Now the obstruction is manifest. The kinetic operator Eq. (13.154) is $-k^2$ times a *projector onto a three-dimensional subspace*; it has a zero eigenvalue on the orthogonal one-dimensional subspace spanned by k^μ. Explicitly,

$$\widetilde{\mathcal{O}}^{\mu\nu}(k)\,k_\nu \;=\; -k^2 k^\mu + k^\mu\,k^2 \;=\; 0. \tag{13.156}$$

An operator with a nontrivial kernel cannot be inverted, so there is no propagator: the Gaussian integral Eq. (13.151), viewed mode by mode, is flat (its quadratic form vanishes) along every longitudinal direction $\widetilde{A}_\mu(k) \propto k_\mu$, and an integral $\int dA\, e^{i\cdot 0} = \int dA$ over such a direction is simply infinite. The path integral Eq. (13.151) diverges, and it diverges in a structured way: along precisely the directions $\widetilde{A}_\mu \propto k_\mu$, i.e. $A_\mu(x) = \partial_\mu\alpha(x)$ in position space.

This is not a pathology to be regulated away; it is gauge redundancy made quantitative. The configurations A_μ and $A_\mu + \partial_\mu\alpha$ are related by the abelian gauge transformation of Theorem 8.1, they have identical field strength (Eq. (8.164)),

and therefore identical action, $S[A] = S[A+\partial\alpha]$. The integrand of Eq. (13.151) is exactly constant along the direction $A \to A+\partial\alpha$, and the measure $\mathcal{D}A$ integrates over that direction without bound. The path integral is summing the same physics infinitely many times — once for every gauge function α — and the divergence is the volume of that overcounting.

Physical Insight

The longitudinal flat direction Eq. (13.156) is the path-integral face of the same fact that defeated naive canonical quantization in Section 13.1: the conjugate momentum to A^0 vanishes identically, $\pi^0 \equiv 0$, because the Lagrangian contains no $\dot{A}^0$. There the non-invertibility was of the Legendre map; here it is of the kinetic operator. Both are statements that A_μ has four components but only the physical, transverse part carries dynamics, and both are cured by the same idea — remove the redundant direction from the integration — implemented once with constraints and Dirac brackets, once with the Faddeev–Popov insertion we now build.

13.5.2 Gauge orbits and the geometry of the redundancy

The diagnosis points straight to the cure. We are integrating over a space $\mathcal{A}$ of gauge fields that is foliated into *gauge orbits*, and we are double-counting because each orbit is traversed in full.

Definition 13.31 (Gauge orbit and gauge group)**.** For a fixed configuration A_μ, its *gauge orbit* is the set of all configurations reachable from it by an abelian gauge transformation,

$$\mathcal{O}_A \;\equiv\; \{\, A^\alpha_\mu \;=\; A_\mu + \partial_\mu\alpha \;:\; \alpha \in \mathcal{G} \,\}, \tag{13.157}$$

where the *gauge group* $\mathcal{G}$ is the abelian group of smooth functions $\alpha : \mathbb{R}^{1,3} \to \mathbb{R}$ under pointwise addition, $\alpha \cdot \beta = \alpha + \beta$, acting on $\mathcal{A}$ by Eq. (8.161). The action, the measure, and every gauge-invariant observable are constant on each orbit.

The space of physically distinct field configurations is the quotient $\mathcal{A}/\mathcal{G}$, in which each orbit collapses to a single point. The integral we *want* runs over $\mathcal{A}/\mathcal{G}$; the integral we *have*, Eq. (13.151), runs over $\mathcal{A}$ and therefore overshoots by the "volume" $\int_{\mathcal{G}}\mathcal{D}\alpha$ of a single orbit — an infinite constant identical for every orbit, since the abelian orbits are all copies of the same group $\mathcal{G}$. The task is to factor that constant out cleanly, keeping the Jacobian honest.

The cleanest way to learn the mechanism is to watch it work in a finite-dimensional model where every integral is elementary. We do this first, then promote it to fields.

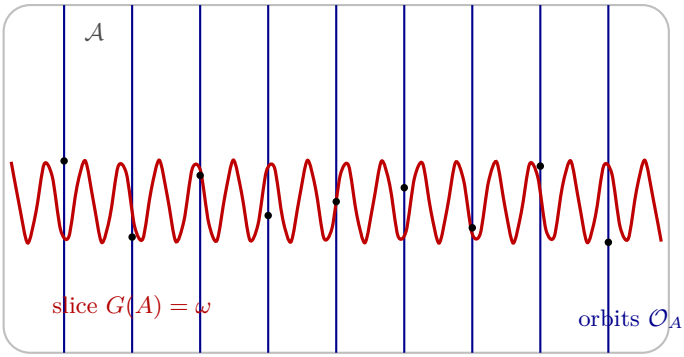


Figure 13.5: The configuration space $\mathcal{A}$ of gauge fields, foliated into gauge orbits $\mathcal{O}_A$ (blue), the curves $\alpha \mapsto A + \partial\alpha$ along which the action is constant. The naive integral Eq. (13.151) sums over all of $\mathcal{A}$ and so counts every orbit infinitely often. The Faddeev–Popov gauge-fixing surface $G(A) = \omega$ (red) is chosen to cut each orbit exactly once (black dots); integrating over the slice, with the Jacobian Δ_{FP} that corrects for the angle of intersection, replaces the integral over $\mathcal{A}$ by an integral over $\mathcal{A}/\mathcal{G}$ times the (factored-out) orbit volume. In the abelian theory the orbits are parallel straight lines, every slice $\partial\cdot A = \omega$ meets each exactly once, and the Jacobian is a field-independent constant.

Worked Example 13.32 (The Faddeev–Popov trick in two dimensions)**.** Consider an ordinary integral over the plane, $I = \int d^2x\, e^{iS(\mathbf{x})}$, where the "action" $S(\mathbf{x}) = S(r)$ depends only on the radius $r = |\mathbf{x}|$ — a model of a theory with a rotational "gauge symmetry," the rotation group $SO(2)$ playing the role of $\mathcal{G}$. The orbits are circles $r = \text{const}$; the integral overcounts by the angular volume $\int_0^{2\pi} d\theta = 2\pi$ of each orbit. We extract that factor by the Faddeev–Popov method, without ever using polar coordinates, so that the steps transcribe verbatim to the field theory.

Pick a "gauge condition" — a function $g(\mathbf{x})$ whose vanishing selects one point on each orbit. Take $g(\mathbf{x}) = x_2$ (the positive x_1-axis is the slice $g = 0$, $x_1 > 0$). Let $\mathbf{x}^\theta = R(\theta)\mathbf{x}$ be the rotated point. Define the Faddeev–Popov "determinant" $\Delta(\mathbf{x})$ by the identity

$$1 \;=\; \Delta(\mathbf{x}) \int_0^{2\pi} d\theta\; \delta(g(\mathbf{x}^\theta)), \qquad \Longrightarrow \qquad \Delta(\mathbf{x})^{-1} = \int_0^{2\pi} d\theta\; \delta(g(\mathbf{x}^\theta)). \quad (13.158)$$

With $\mathbf{x} = (r\cos\phi, r\sin\phi)$ we have $g(\mathbf{x}^\theta) = r\sin(\phi + \theta)$, which vanishes (for $r \neq 0$, within a period) at $\theta_* = -\phi$ where it crosses with slope $\frac{d}{d\theta} g(\mathbf{x}^\theta)\big|_{\theta_*} = r\cos(\phi + \theta_*) = r$. The delta function then gives $\Delta(\mathbf{x})^{-1} = 1/\big|\frac{d}{d\theta} g(\mathbf{x}^\theta)\big|_{\theta_*} = 1/r$, so

$$\Delta(\mathbf{x}) \;=\; \Big|\frac{\partial g(\mathbf{x}^\theta)}{\partial\theta}\Big|_{g=0} \;=\; r. \quad (13.159)$$

The determinant is the rate at which the gauge condition changes as we move along

the orbit, evaluated on the slice — exactly the geometric "angle of intersection" factor of Fig. 13.5. Insert Eq. (13.158) into I and interchange integrals:

$$I = \int_0^{2\pi} d\theta \int d^2x\, e^{iS(r)}\, \Delta(\mathbf{x})\, \delta\big(g(\mathbf{x}^\theta)\big). \tag{13.160}$$

Now substitute $\mathbf{x} \to \mathbf{x}^{-\theta}$ in the inner integral. The measure d^2x is rotation-invariant, $S(r)$ is rotation-invariant, and Δ is rotation-invariant (it depends only on r); only the delta function moves, $g(\mathbf{x}^\theta) \to g(\mathbf{x})$. The integrand is then independent of θ, and the θ-integral factors out as the orbit volume $\int_0^{2\pi} d\theta = 2\pi$:

$$I = \Big(\int_0^{2\pi} d\theta\Big) \int d^2x\, e^{iS(r)}\, \Delta(\mathbf{x})\, \delta\big(g(\mathbf{x})\big) = 2\pi \int d^2x\, e^{iS(r)}\, r\, \delta(x_2). \tag{13.161}$$

The last integral over the slice $x_2 = 0$, $x_1 = r > 0$, is $\int_0^\infty dr\, e^{iS(r)}\, r$ — precisely $\frac{1}{2\pi}$ of $\int d^2x\, e^{iS} = 2\pi \int_0^\infty r\, dr\, e^{iS}$, as it must be. The redundant angular volume has been split off cleanly, and the price was a single Jacobian factor $\Delta = r$ supplied by Eq. (13.158). Every structural feature of the field-theory derivation below — the inserted "1," the determinant as $\partial(\text{gauge condition})/\partial(\text{gauge parameter})$, the change of variables that frees the orbit volume, the leftover integral over a single slice — is already present here.

13.5.3 The Faddeev–Popov identity

We now repeat Theorem 13.32 for the photon, with the rotation angle θ replaced by the gauge function α, the radius r by the field A_μ, and the slice $g(\mathbf{x}) = 0$ by a gauge condition on A.

Definition 13.33 (Gauge-fixing function)**.** A *gauge-fixing function* is a functional $G(A)$ of the gauge field, chosen so that the surface $G(A) = \omega$ (for a prescribed function $\omega(x)$) intersects each gauge orbit Eq. (13.157) exactly once. The Lorenz family is generated by

$$G(A)(x) \;\equiv\; \partial_\mu A^\mu(x), \tag{13.162}$$

so that $G(A) = \omega$ reads $\partial_\mu A^\mu = \omega$, reducing to the Lorenz gauge $\partial_\mu A^\mu = 0$ when $\omega = 0$.

Following Eq. (13.158) verbatim, define the *Faddeev–Popov determinant* $\Delta_{\mathrm{FP}}[A]$ by inserting unity:

$$1 \;=\; \Delta_{\mathrm{FP}}[A] \int \mathcal{D}\alpha\;\, \delta\big(G(A^\alpha) - \omega\big), \tag{13.163}$$

where $\delta(\cdots) = \prod_x \delta\big(G(A^\alpha)(x) - \omega(x)\big)$ is a functional delta and $\mathcal{D}\alpha = \prod_x d\alpha(x)$ is the (translation-invariant) Haar measure on the abelian group $\mathcal{G}$. We now compute Δ_{FP} and prove the two facts that make abelian gauge theory special.

Theorem 13.34 (The abelian Faddeev–Popov determinant)**.** *With the Lorenz gauge-fixing function Eq. (13.162), the determinant defined by Eq. (13.163) is*

$$\Delta_{\mathrm{FP}}[A] \;=\; \det(\partial^2), \tag{13.164}$$

independent of the gauge field A_μ and of the function ω. *It is therefore an (infinite) multiplicative constant that may be pulled outside the functional integral.*

Proof. Under a gauge transformation the Lorenz function shifts by the d'Alembertian of the gauge parameter,

$$G(A^\alpha) = \partial_\mu(A^\mu + \partial^\mu\alpha) = \partial_\mu A^\mu + \partial^2\alpha = G(A) + \partial^2\alpha, \tag{13.165}$$

so the constraint $G(A^\alpha) - \omega = 0$ is the linear equation $\partial^2\alpha = \omega - G(A)$ for α. To evaluate $\Delta_{\mathrm{FP}}^{-1} = \int \mathcal{D}\alpha\, \delta\big(G(A) + \partial^2\alpha - \omega\big)$, change integration variable from α to $\eta \equiv G(A^\alpha) - \omega = G(A) + \partial^2\alpha - \omega$. The two are related linearly, with functional Jacobian

$$\frac{\delta\eta(x)}{\delta\alpha(y)} = \frac{\delta}{\delta\alpha(y)}\big(\partial^2\alpha(x)\big) = \partial_x^2\,\delta^4(x-y), \qquad \mathcal{D}\eta = \det\!\Big(\frac{\delta\eta}{\delta\alpha}\Big)\,\mathcal{D}\alpha = \det(\partial^2)\,\mathcal{D}\alpha. \tag{13.166}$$

The crucial observation is that this Jacobian, $\delta\eta/\delta\alpha = \partial^2$, *does not contain A at all*: the shift Eq. (13.165) is linear in α with a constant (A-independent) coefficient operator ∂^2, because the gauge transformation $A \to A + \partial\alpha$ is itself an affine shift of A rather than a transformation whose generator depends on A. Therefore

$$\Delta_{\mathrm{FP}}[A]^{-1} = \int \mathcal{D}\alpha\; \delta(\eta) = \int \frac{\mathcal{D}\eta}{\det(\partial^2)}\; \delta(\eta) = \frac{1}{\det(\partial^2)}, \tag{13.167}$$

which is Eq. (13.164). There is no A- or ω-dependence anywhere in the result: the single solution α_* of $\partial^2\alpha_* = \omega - G(A)$ exists and is unique once boundary conditions fix the kernel of ∂^2, and the height of the delta-function spike is governed entirely by the constant operator ∂^2. □

Two points deserve emphasis, because they are exactly the points where the non-abelian theory will diverge from this one. First, the determinant is the functional analogue of the factor $r = \partial g(\mathbf{x}^\theta)/\partial\theta$ in Eq. (13.159): it is the variation of the gauge condition with respect to the gauge parameter, $\delta G(A^\alpha)/\delta\alpha = \partial^2$, evaluated on the constraint surface. Second, and decisively, that variation is field-independent. We isolate the reason as a proposition, since its failure is what creates ghosts.

Proposition 13.35 (Why the abelian determinant decouples)**.** *The Faddeev–Popov operator is the derivative of the gauge condition along the orbit, $M(x,y) \equiv \delta G(A^\alpha)(x)/\delta\alpha(y)\big|_{\alpha=0}$. In the abelian theory $M = \partial^2$ is independent of A, so* $\det M$ *factors out of the path integral as a constant. In a non-abelian theory it is not.*

Proof. For an abelian gauge field the orbit map is the affine shift $A_\mu^\alpha = A_\mu + \partial_\mu \alpha$ of Theorem 8.1, so $\delta A_\mu^\alpha / \delta \alpha = \partial_\mu$ is constant and $M = \partial_\mu \, \delta A^{\alpha\,\mu}/\delta\alpha = \partial^2$, as in Eq. (13.166). For a non-abelian gauge field with gauge group elements labelled by $\alpha^a(x)$ and covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$ (the convention of Yang–Mills theory, Chapter 25), the infinitesimal gauge transformation is $A_\mu^a \to A_\mu^a + \frac{1}{g} D_\mu^{ab} \alpha^b$ with $D_\mu^{ab} = \partial_\mu \delta^{ab} + g f^{acb} A_\mu^c$, so the variation of the gauge condition $G^a(A) = \partial^\mu A_\mu^a$ is

$$M^{ab}(x,y) = \frac{\delta G^a(A^\alpha)(x)}{\delta \alpha^b(y)}\Big|_{\alpha=0} = \frac{1}{g}\,\partial^\mu D_\mu^{ab}\,\delta^4(x-y) = \frac{1}{g}\,\partial^\mu\big(\partial_\mu \delta^{ab} + g f^{acb} A_\mu^c\big)\delta^4(x-y). \tag{13.168}$$

The term $f^{acb} A_\mu^c$ depends explicitly on the gauge field, so $\det M = \det(\partial^\mu D_\mu)$ is a nontrivial *functional of* A and cannot be pulled outside the integral. Setting the structure constants $f^{acb} = 0$ collapses Eq. (13.168) to the abelian $M = \partial^2$ of Eq. (13.166), recovering Theorem 13.34. □

Physical Insight

The whole difference between QED and Yang–Mills, at the level of gauge fixing, is the single clause "$f^{abc} = 0$." Abelian gauge orbits are parallel straight lines (an affine action), so a flat slice cuts every one at the same constant angle and the Jacobian is a number. Non-abelian orbits are curved (the group acts nonlinearly on A), the angle of intersection varies from orbit to orbit and depends on where on $\mathcal{A}$ you are, and the Jacobian becomes a field-dependent determinant. That determinant, written as a Gaussian integral over anticommuting fields in Chapter 26, *is* the Faddeev–Popov ghost. QED has ghosts too, in the formal sense that $\det(\partial^2)$ can be written as a ghost integral — but the ghosts are free, decoupled from A_μ, and contribute only an overall constant, so we never speak of them. The reader should leave this section knowing that the absence of ghosts in QED is not an accident of the abelian theory's simplicity but a precise consequence of the linearity of its gauge action.

For completeness we record the gauge invariance of Δ_{FP} as a functional, which in the abelian case is automatic (the determinant is constant) but which is the load-bearing property in the general construction and which we will reuse in the next subsection.

Lemma 13.36 (Gauge invariance of Δ_{FP})**.** *The functional defined by Eq. (13.163) is invariant under gauge transformations of its argument,* $\Delta_{\mathrm{FP}}[A^\beta] = \Delta_{\mathrm{FP}}[A]$ *for any* $\beta \in \mathcal{G}$.

Proof. By definition $\Delta_{\mathrm{FP}}[A^\beta]^{-1} = \int \mathcal{D}\alpha \; \delta\big(G(A^{\beta\,\alpha}) - \omega\big)$. Because $\mathcal{G}$ is a group, the composition of the gauge transformation by β followed by that by α is the

transformation by their product, $A^{\beta\,\alpha} = A^{\beta\cdot\alpha}$; for the abelian group $\beta\cdot\alpha = \alpha+\beta$. Change integration variable to $\alpha' = \alpha + \beta$. The Haar measure is translation-invariant, $\mathcal{D}\alpha' = \mathcal{D}\alpha$, so

$$\Delta_{\mathrm{FP}}[A^{\beta}]^{-1} = \int \mathcal{D}\alpha'\; \delta\big(G(A^{\alpha'}) - \omega\big) = \Delta_{\mathrm{FP}}[A]^{-1}, \tag{13.169}$$

proving the claim. (In the abelian Lorenz gauge both sides equal $1/\det(\partial^2)$ outright, but the proof uses only the group law and the invariance of $\mathcal{D}\alpha$, and so carries over unchanged to the non-abelian case, where it is essential.) □

13.5.4 Factoring out the gauge volume

We now insert the resolution of unity Eq. (13.163) into the divergent generating functional Eq. (13.151) and disentangle the infinity, exactly as in Eq. (13.161):

$$Z = \int \mathcal{D}A\; e^{iS[A]} = \int \mathcal{D}A\; e^{iS[A]}\,\Delta_{\mathrm{FP}}[A] \int \mathcal{D}\alpha\; \delta\big(G(A^{\alpha}) - \omega\big) = \int \mathcal{D}\alpha \int \mathcal{D}A\; e^{iS[A]}\,\Delta_{\mathrm{FP}}[A]\,\delta(G($$

(13.170)

the order of integration interchanged in the last step. Now change variables in the inner integral by relabelling $A \to A^{-\alpha}$, equivalently writing the dummy field as $A = B^{-\alpha}$ so that $B = A^{\alpha}$. Three invariances conspire:

- the action is gauge invariant, $S[A] = S[B^{-\alpha}] \equiv S[B]$, because $F_{\mu\nu}$ is (Eq. (8.164));
- the measure is gauge invariant, $\mathcal{D}A = \mathcal{D}B$, since a gauge transformation is the constant shift $A_\mu \to A_\mu + \partial_\mu\alpha$ of the integration variable, with unit Jacobian (the shift is independent of A);
- the Faddeev–Popov determinant is gauge invariant, $\Delta_{\mathrm{FP}}[A] = \Delta_{\mathrm{FP}}[B]$, by Theorem 13.36 (and trivially by Theorem 13.34, being constant).

Under the relabelling the constraint becomes $G(A^{\alpha}) = G(B)$, so the entire integrand loses its α-dependence and the α-integral factors off as the gauge-group volume:

$$Z = \underbrace{\Big(\int \mathcal{D}\alpha\Big)}_{V_{\mathcal{G}}} \int \mathcal{D}B\; e^{iS[B]}\,\Delta_{\mathrm{FP}}[B]\,\delta\big(G(B) - \omega\big). \tag{13.171}$$

The divergence of Eq. (13.151) has been isolated in the single multiplicative constant $V_{\mathcal{G}} = \int\mathcal{D}\alpha$, the (infinite) volume of the gauge group. It is the field-theoretic counterpart of the 2π in Eq. (13.161). Because every physical observable is computed from *normalized* correlators $\langle\cdots\rangle = (\text{numerator})/Z$ in which the same $V_{\mathcal{G}}$ appears upstairs and downstairs, it cancels identically and may be

dropped. Renaming $B \to A$, the well-defined, gauge-fixed generating functional is

$$Z[\omega] \;=\; \det(\partial^2) \int \mathcal{D}A \; e^{iS[A]} \, \delta(\partial_\mu A^\mu - \omega), \tag{13.172}$$

having substituted Eq. (13.164) for the constant determinant and Eq. (13.162) for G. The delta function now restricts the integral to a single representative of each orbit — one point per orbit, the dots of Fig. 13.5 — and the overcounting is gone.

Caution

The Faddeev–Popov construction assumes the slice $G(A) = \omega$ meets each orbit *exactly* once, so that Eq. (13.163) really is a resolution of unity. In the abelian theory on $\mathbb{R}^{1,3}$ this holds: $\partial^2\alpha = \omega - \partial\cdot A$ has a unique solution with vanishing boundary conditions, so the slice is hit once. In non-abelian theories, beyond perturbation theory, a covariant slice generally intersects an orbit *several* times — the Gribov ambiguity — and Eq. (13.163) overcounts the residual copies. This does not affect perturbation theory (it concerns large gauge fields far from $A = 0$) and is invisible in QED; we flag it here and treat it where it bites, in the non-perturbative chapters. As of the mid-2020s a fully satisfactory global gauge fixing of non-abelian theory remains an open problem; the perturbative construction we use throughout is unaffected.

13.5.5 The Gaussian average over ω: origin of the gauge-fixing term

Equation Eq. (13.172) is correct but inconvenient: the delta function $\delta(\partial\cdot A - \omega)$ is awkward inside a perturbative expansion, and we are still carrying the arbitrary function ω. Both annoyances are removed by one observation. The left-hand side $Z[\omega]$ does *not actually depend on* ω: changing ω slides the slice from one representative of each orbit to another, and since the integrand on Eq. (13.171) is constant along orbits, the value is unchanged. (Concretely, in Eq. (13.172) a shift $\omega \to \omega + \partial^2\lambda$ is undone by the gauge transformation $A \to A - \partial\lambda$ under which the measure and action are invariant.) We are therefore free to average $Z[\omega]$ over ω with *any* normalized weight we like; the result is still Z up to a constant. Choose the Gaussian weight

$$Z \;\propto\; \int \mathcal{D}\omega \; \exp\Big(-\frac{i}{2\xi} \int d^4x \, \omega^2 \Big) \, Z[\omega], \qquad \xi \in (0, \infty), \tag{13.173}$$

where ξ is a real parameter and the proportionality absorbs the (ω-independent, hence physically irrelevant) normalization of the Gaussian. Substituting Eq. (13.172)

and pulling the field-independent constant $\det(\partial^2)$ out front,

$$Z \;\propto\; \det(\partial^2)\int \mathcal{D}A\; e^{iS[A]} \int \mathcal{D}\omega\; \exp\Big(-\frac{i}{2\xi}\int \omega^2\Big)\,\delta(\partial_\mu A^\mu - \omega). \tag{13.174}$$

The ω-integral is now trivial: the delta function sets $\omega = \partial_\mu A^\mu$ everywhere, replacing $\int \omega^2$ by $\int (\partial_\mu A^\mu)^2$ and removing the delta. We are left with

$$Z \;\propto\; \det(\partial^2)\int \mathcal{D}A\; \exp\Big(i\int d^4x\,\Big[-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2\Big]\Big). \tag{13.175}$$

This is the result the section was aimed at. The term $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ that we were forced to insert by fiat in Section 13.4 has now *emerged*, with its sign and its factor of $\frac{1}{2\xi}$ fixed, as nothing more than the Gaussian average over the family of Lorenz slices. It is not a modification of electromagnetism — the physics is independent of ξ exactly because $Z[\omega]$ was independent of ω — but a bookkeeping device for integrating over orbits once. The constant $\det(\partial^2)$ stands outside the A-integral and cancels in every normalized correlator: *this is the precise statement that QED has no propagating ghost.* The effective gauge-fixed action under the path integral is

$$S_\xi[A] = \int d^4x\,\Big[-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2\Big], \tag{13.176}$$

the R_ξ family, with the Feynman gauge $\xi = 1$ the computational default recorded in Chapter A. Had the determinant been field-dependent — the non-abelian case of Theorem 13.35 — we could not have pulled it out, and we would have been forced to represent $\det M[A]$ as a Gaussian integral over anticommuting scalar fields $c, \bar{c}$, the Faddeev–Popov ghosts, added to the action as $\bar{c}\,M[A]\,c$ and coupled to A through M. That is the construction of Chapter 26; here $M = \partial^2$ is a constant and the ghosts are free spectators we discard.

13.5.6 The R_ξ propagator from the Gaussian integral

With the gauge-fixing term in place the kinetic operator is no longer degenerate, and the Gaussian integral Eq. (13.175) can be done by the inversion method of Section 10.4.3. Integrating the new term by parts, $-\frac{1}{2\xi}(\partial_\mu A^\mu)(\partial_\nu A^\nu) \to +\frac{1}{2\xi}A_\mu \partial^\mu\partial^\nu A_\nu$, and adding it to Eq. (13.153), the gauge-fixed action takes the quadratic form $S_\xi = \frac{1}{2}\int A_\mu\, \mathcal{O}_\xi^{\mu\nu} A_\nu$ with

$$\mathcal{O}_\xi^{\mu\nu} = \partial^2 g^{\mu\nu} - \Big(1 - \tfrac{1}{\xi}\Big)\partial^\mu\partial^\nu, \qquad \widetilde{\mathcal{O}}_\xi^{\mu\nu}(k) = -k^2 g^{\mu\nu} + \Big(1 - \tfrac{1}{\xi}\Big)k^\mu k^\nu \tag{13.177}$$

in momentum space. The single new term $+\frac{1}{\xi}k^\mu k^\nu$ has lifted the longitudinal zero mode: where Eq. (13.154) annihilated k_ν, now

$$\widetilde{\mathcal{O}}_\xi^{\mu\nu}(k)\,k_\nu = -k^2 k^\mu + \Big(1 - \tfrac{1}{\xi}\Big)k^2 k^\mu = -\tfrac{1}{\xi}\,k^2 k^\mu \neq 0. \tag{13.178}$$

The operator is invertible, and the cleanest inversion uses the projectors Eq. (13.155). Decompose $\widetilde{\mathcal{O}}_\xi$ on the transverse and longitudinal subspaces using $g = P_T + P_L$ and $k^\mu k^\nu = k^2 P_L$:

$$\widetilde{\mathcal{O}}_\xi = -k^2(P_T + P_L) + \Big(1 - \tfrac{1}{\xi}\Big)k^2 P_L = -k^2\, P_T - \frac{k^2}{\xi}\, P_L. \tag{13.179}$$

On each eigenspace $\widetilde{\mathcal{O}}_\xi$ is now a nonzero scalar, so its inverse is obtained by inverting those scalars projector by projector,

$$\widetilde{\mathcal{O}}_\xi^{-1} = -\frac{1}{k^2}\, P_T - \frac{\xi}{k^2}\, P_L, \tag{13.180}$$

as one verifies by direct multiplication using $P_T^2 = P_T$, $P_L^2 = P_L$, $P_T P_L = 0$:

$$\widetilde{\mathcal{O}}_\xi\, \widetilde{\mathcal{O}}_\xi^{-1} = \Big(-k^2 P_T\Big)\Big(-\tfrac{1}{k^2}P_T\Big) + \Big(-\tfrac{k^2}{\xi}P_L\Big)\Big(-\tfrac{\xi}{k^2}P_L\Big) = P_T + P_L = g = \mathbf{1}. \tag{13.181}$$

The photon propagator is $\widetilde{D}_{\mu\nu} = i\,\widetilde{\mathcal{O}}_\xi^{-1}$, with the factor of i in the same convention as the scalar Eq. (10.110) (so that $\mathcal{O}_\xi D = i\delta$, matching the canonical normalization). Substituting the projectors back,

$$\widetilde{D}_{\mu\nu}(k) = i\,\widetilde{\mathcal{O}}_\xi^{-1}{}_{\mu\nu} = -\frac{i}{k^2}\big(P_T + \xi P_L\big)_{\mu\nu} = \frac{-i}{k^2 + i\varepsilon}\Big(g_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big), \tag{13.182}$$

where we used $P_T + \xi P_L = g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} + \xi\frac{k_\mu k_\nu}{k^2} = g_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}$, and restored the $i\varepsilon$ inherited from the vacuum tilt of Section 10.3.4 exactly as in the scalar case. This is, character for character, the R_ξ propagator Eq. (13.135) obtained from canonical considerations in the previous section: the Feynman gauge $\xi = 1$ gives the simple $-ig_{\mu\nu}/(k^2 + i\varepsilon)$, the Landau gauge $\xi = 0$ projects onto the purely transverse $-iP_{T\,\mu\nu}/k^2$, and the longitudinal piece carries the entire ξ-dependence. The path integral has reproduced the propagator the perturbation theory of Part V is built on, and — unlike the canonical derivation — it produced the gauge-fixing term that defines ξ rather than postulating it.

13.5.7 The free photon generating functional

Having a non-degenerate Gaussian, we can now couple a source and read off the full free generating functional, the object that will deliver the QED Feynman rules in Chapter 17. Add to the gauge-fixed action a linear source term $\int d^4x\, J^\mu A_\mu$ — with J^μ an external current, the photon's analogue of the scalar source J of Eq. (10.86) — and define

$$Z_0[J] = \int \mathcal{D}A\, \exp\Big(i\int d^4x\, \Big[\tfrac{1}{2}A_\mu \mathcal{O}_\xi^{\mu\nu} A_\nu + J^\mu A_\mu\Big]\Big), \tag{13.183}$$

where the free determinant $\det(\partial^2)$ has been absorbed into the overall normalization $Z_0[0]$, since it is J-independent. This is a Gaussian integral identical in

structure to the scalar one of Section 10.4, with the scalar kinetic operator $\mathcal{O}$ promoted to the matrix $\mathcal{O}_\xi^{\mu\nu}$ and the source contracted with a Lorentz index. Completing the square exactly as there — shift $A_\mu \to A_\mu - \mathcal{O}_\xi^{-1}{}_{\mu\nu} J^\nu$, the leftover fluctuation integral being $Z_0[0]$ — yields the closed form

$$\boxed{Z_0[J] = Z_0[0]\,\exp\Big(-\tfrac{1}{2}\int d^4x\, d^4y\; J^\mu(x)\, D_{\mu\nu}(x-y)\, J^\nu(y)\Big),} \tag{13.184}$$

the photon image of the scalar master formula Eq. (10.111), with the kernel the very propagator $D_{\mu\nu} = i\,\mathcal{O}_\xi^{-1}$ just computed in Eq. (13.182). (Writing $D_{\mu\nu} = i\,\mathcal{O}_\xi^{-1}{}_{\mu\nu}$, the exponent is the familiar $-\frac{i}{2}\int J^\mu \mathcal{O}_\xi^{-1}{}_{\mu\nu} J^\nu$ of the generating-functional master identity.) We record the statement formally.

Theorem 13.37 (Free photon generating functional)**.** *The free, gauge-fixed electromagnetic generating functional is Eq. (13.184), with $D_{\mu\nu}$ the R_ξ propagator Eq. (13.182). Its connected two-point function is the photon propagator,*

$$\langle 0|\,\mathcal{T} A_\mu(x) A_\nu(y)\,|0\rangle = \frac{1}{i^2}\,\frac{\delta^2}{\delta J^\mu(x)\,\delta J^\nu(y)}\,\frac{Z_0[J]}{Z_0[0]}\bigg|_{J=0} = D_{\mu\nu}(x-y) = D^F_{\mu\nu}(x-y), \tag{13.185}$$

in exact agreement with the canonical time-ordered correlator of Section 13.4.

The proof is a functional differentiation identical in form to the scalar computation of Theorem 10.22; we carry it out so the index bookkeeping is on the page.

Worked Example 13.38 (The photon two-point function from $Z_0[J]$)**.** Write the normalized functional $\widehat{Z}_0[J] \equiv Z_0[J]/Z_0[0] = \exp\big[-\frac{1}{2}\int J^\rho D_{\rho\sigma} J^\sigma\big]$, so that, by the generating-functional master identity for sourced correlators (the photon instance of Eq. (10.86) with one $1/i$ per derivative),

$$\langle 0|\mathcal{T} A_\mu(x) A_\nu(y)|0\rangle = \frac{1}{i^2}\,\frac{\delta^2 \widehat{Z}_0[J]}{\delta J^\mu(x)\,\delta J^\nu(y)}\bigg|_{J=0}. \tag{13.186}$$

The exponent is quadratic and the kernel $D_{\rho\sigma}$ is symmetric under simultaneous exchange of its indices and arguments, $D_{\rho\sigma}(x-y) = D_{\sigma\rho}(y-x)$ (it is built from $g_{\rho\sigma}$ and $k_\rho k_\sigma$, both symmetric). The first derivative therefore brings down one factor of the propagator, the factor of 2 from the two ways of hitting the symmetric quadratic form cancelling the $\frac{1}{2}$:

$$\frac{\delta \widehat{Z}_0[J]}{\delta J^\mu(x)} = \Big[-\int d^4z\, D_{\mu\sigma}(x-z)\, J^\sigma(z)\Big]\,\widehat{Z}_0[J]. \tag{13.187}$$

Differentiate again with respect to $J^\nu(y)$. The product rule gives a term from differentiating the bracket and a term from differentiating $\widehat{Z}_0$:

$$\frac{\delta^2 \widehat{Z}_0[J]}{\delta J^\nu(y)\,\delta J^\mu(x)} = \Big[-D_{\mu\nu}(x-y)\Big]\widehat{Z}_0[J] + \Big[-\int D_{\mu\sigma}(x-z) J^\sigma\Big]\Big[-\int D_{\nu\tau}(y-w) J^\tau\Big]\widehat{Z}_0[J]. \tag{13.188}$$

Set $J = 0$. The second term is quadratic in J and vanishes; $\widehat{Z}_0[0] = 1$. Dividing by $i^2 = -1$,

$$\langle 0|\mathcal{T} A_\mu(x) A_\nu(y)|0\rangle = \frac{-D_{\mu\nu}(x-y)}{i^2} = D_{\mu\nu}(x-y), \tag{13.189}$$

the R_ξ propagator Eq. (13.182), establishing Eq. (13.185). The structure of Eq. (13.188) is Wick's theorem (Theorem 10.23) for the photon in embryo: each pair of functional derivatives either contracts into a propagator joining the two points or leaves a source behind, and at $J = 0$ only the fully contracted term survives. Every higher free correlator of A_μ is, by the same mechanism, a sum over pairings into copies of $D_{\mu\nu}$ — the free photon is a Gaussian field whose entire content is the two-point function Eq. (13.182).

One caveat completes the picture and ties back to the previous section. The generating functional Eq. (13.184) depends on ξ through the longitudinal part of $D_{\mu\nu}$. That dependence is harmless precisely when the source is conserved: if $\partial_\mu J^\mu = 0$, then in momentum space $k_\mu \widetilde{J}^\mu(k) = 0$, and the longitudinal term in Eq. (13.182) contracts to zero,

$$\widetilde{J}^\mu(-k)\,(1-\xi)\frac{k_\mu k_\nu}{k^2}\,\widetilde{J}^\nu(k) = 0, \tag{13.190}$$

so that $J^\mu D_{\mu\nu} J^\nu$ — and hence $Z_0[J]$ itself — is independent of ξ for conserved currents. This is the generating-functional restatement of the physical-amplitude argument of Section 13.4: gauge-parameter dependence lives entirely in a longitudinal term that any conserved current annihilates. The path integral has thus reproduced not only the propagator but the reason its gauge ambiguity never reaches an observable.

With the free photon now described entirely by the covariant generating functional Eq. (13.184) and its single datum the R_ξ propagator, the path-integral quantization of electromagnetism is complete, and the abelian Faddeev–Popov construction has shown both where the gauge-fixing term legitimately comes from and why QED carries no coupled ghost. The two threads now part toward their continuations: the field-dependent determinant of Theorem 13.35 is the seed that Chapter 26 grows into the propagating Faddeev–Popov ghosts and BRST symmetry of Yang–Mills theory, while the generating functional $Z_0[J]$ of Eq. (13.184) is the object from which Chapter 17 reads off the photon line of the QED Feynman rules and assembles its first cross sections.

13.6 Synthesis: Coupling to Matter and the Recovery of the Coulomb Law

The free photon is now a finished object. Its two-point function exists in coordinate and momentum space across the entire R_ξ family (Eq. (13.135)), and

the path-integral derivation of Section 13.5 has shown that this propagator is exactly the object the Gaussian generating functional $\int \mathcal{D}A\, e^{iS}$ produces once a gauge has been fixed. What it has never yet done is *propagate* anything: the source that creates and absorbs photons has been left abstract, a placeholder current J^μ whose only stated property was conservation. We close the chapter by installing the real source. The gauge principle of Chapter 8 already told us, classically, exactly how a charged field must couple to A_μ; the quantized Dirac field of Chapter 11 provides the current. Joining them produces the complete interacting field content of quantum electrodynamics, and lets us run the one computation that certifies the whole construction — normalization, sign, and all three quantization routes at once — against a fact every reader already knows: the Coulomb law. If the photon we have built does not reproduce $V(r) = e_1 e_2/4\pi r$ with the right magnitude and the right sign, something upstream is wrong. It does, and showing so is the synthesis this section delivers.

13.6.1 Minimal coupling and the photon source current

The coupling is not a free choice. In Chapter 8 the demand that the global U(1) phase symmetry of the electron — the symmetry whose Noether current is the conserved charge of QED — be promoted to a *local* symmetry $\psi(x) \to e^{-i\alpha(x)}\psi(x)$ forced the replacement of the ordinary derivative by the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$, with A_μ transforming as $A_\mu \to A_\mu + \frac{1}{e}\partial_\mu \alpha$ to absorb the inhomogeneous term (Eq. (8.27)). There is no other local-U(1)-invariant coupling of a Dirac field to a single vector potential at this order in fields and derivatives. Assembling the gauge-invariant matter action with the Maxwell term Eq. (13.2) gives the standard Lagrangian of QED,

$$\mathcal{L}_{\text{QED}} = \bar\psi\,(i\not{D} - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \underbrace{\bar\psi\,(i\not{\partial} - m)\,\psi}_{\text{free electron}}\ \underbrace{-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\text{free photon}}\ \underbrace{-\,e\,\bar\psi\gamma^\mu\psi\,A_\mu}_{\text{interaction}}, \tag{13.191}$$

where the interaction term is extracted by expanding $i\not{D} = i\not{\partial} - e\not{A}$ (since $D_\mu = \partial_\mu + ieA_\mu$ gives $i\gamma^\mu D_\mu = i\not{\partial} + i(ie)\not{A} = i\not{\partial} - e\not{A}$). The free pieces are precisely the theories quantized in this chapter and in Chapter 11; everything new lives in the last term. We give it its own name,

$$\mathcal{L}_{\text{int}} = -\,e\,\bar\psi\gamma^\mu\psi\,A_\mu = -\,e\,J^\mu A_\mu, \qquad J^\mu \equiv \bar\psi\gamma^\mu\psi, \tag{13.192}$$

and we recognize J^μ at once: it is the Dirac number current Eq. (11.158), the conserved Noether current of the electron's U(1) symmetry, now doing double duty as the source that the photon couples to. The physical electromagnetic current — the thing that appears on the right of Maxwell's equation — is $j^\mu_{\text{em}} = -eJ^\mu$: the Euler–Lagrange equation for A_ν from Eq. (13.191) reads $\partial_\mu F^{\mu\nu} = e\bar\psi\gamma^\nu\psi = -\,j^\nu_{\text{em}}$ with $j^\nu_{\text{em}} = -e\bar\psi\gamma^\nu\psi$, whose charge density $j^0_{\text{em}} = -e\,\psi^\dagger\psi$ is negative for a positive number density — the electron carries charge $-e$, exactly the convention of

QED. The structure is identical for scalar matter: a complex scalar of charge $-e$ couples through its own conserved U(1) current (Section 9.7) by the same minimal substitution, the only difference being that the scalar current $i(\phi^* \overleftrightarrow{\partial^\mu} \phi)$ contains a derivative and therefore also generates a seagull term $e^2 A^2 |\phi|^2$. For the Dirac field the coupling is purely linear in A_μ and the photon source is simply J^μ.

The single defining property we will lean on is conservation. At the operator level the Heisenberg equations of motion for the interacting field still give

$$\partial_\mu J^\mu = (\partial_\mu \bar{\psi})\gamma^\mu \psi + \bar{\psi}\gamma^\mu(\partial_\mu \psi) = 0, \tag{13.193}$$

because the full Dirac equation in the presence of the gauge field, $(i\not{D} - m)\psi = 0$, and its adjoint $\bar{\psi}(i\overleftarrow{\not{D}} + m) = 0$, conspire so that the A_μ-dependent terms cancel in pairs: $\bar{\psi}\gamma^\mu\psi$ is gauge-covariant of charge zero, and its divergence is the same whether one uses ∂_μ or D_μ precisely because the current is neutral. Conservation of J^μ is therefore not an accident of the free theory; it survives the coupling, and it is what will make the gauge artefact in the propagator invisible.

To do quantum-mechanical perturbation theory we need the interaction Hamiltonian, not the Lagrangian. Because $\mathcal{L}_{\text{int}}$ contains no time derivatives of any field, the canonical momenta are unchanged by the coupling and the Legendre transform simply flips the sign of the interaction term,

$$H_{\text{int}} = -\int d^3x\, \mathcal{L}_{\text{int}} = e\int d^3x\, \bar{\psi}\gamma^\mu\psi\, A_\mu = e\int d^3x\, J^\mu A_\mu. \tag{13.194}$$

This is the operator whose Dyson series, organized in Chapter 14 and turned into Feynman rules in Chapters 15 and 17, generates every QED amplitude.

Caution

Equation (13.194) is the manifestly covariant interaction Hamiltonian, valid when A_μ is quantized covariantly (Gupta–Bleuler, Section 13.3) or in the path integral. In the Coulomb gauge of Eq. (13.45) the story looks different: there A^0 is not an independent field but is solved from the Gauss constraint Eq. (13.21), and substituting that solution back into the Hamiltonian produces an *instantaneous* Coulomb interaction $\frac{1}{2}\int d^3x\, d^3y\, \rho(\mathbf{x})\rho(\mathbf{y})/4\pi|\mathbf{x}-\mathbf{y}|$ between charge densities, plus a coupling of the transverse photon to the transverse current. The two descriptions are not in conflict; the next subsections show they compute the same physics, with the instantaneous term of the Coulomb route being the operator avatar of the timelike piece of the covariant propagator.

13.6.2 Quantum gauge invariance of the coupling, and a word on charge quantization

The propagator carries a free gauge parameter ξ, and a different ξ is a genuinely different two-point function. The coupling we have just installed is what guarantees that no measurable quantity remembers the choice. The mechanism was proved in general in Theorem 13.28: in any current-coupled amplitude the propagator appears contracted with currents, $\widetilde{J}_1^{*\mu}\, D_{\mu\nu}(k)\, \widetilde{J}_2^{\nu}$, and the entire ξ-dependence of Eq. (13.135) sits in the term proportional to $k_\mu k_\nu$, which dies the instant it touches a conserved current. What was abstract there is now concrete: the current is $J^\mu = \bar\psi\gamma^\mu\psi$, and its conservation Eq. (13.193) is supplied by the electron's own dynamics. For the on-shell matrix element relevant to a tree amplitude, $\widetilde{J}^\mu(k) = \bar u(p')\gamma^\mu u(p)$ with $k = p' - p$, conservation is the elementary identity

$$k_\mu\, \bar u(p')\gamma^\mu u(p) = \bar u(p')\,(\not p' - \not p)\, u(p) = \bar u(p')\,(m - m)\, u(p) = 0, \tag{13.195}$$

using the Dirac equations $\not p\, u(p) = m\, u(p)$ and $\bar u(p')\not p' = m\, \bar u(p')$ from Chapter 11; this is the same on-shell Ward identity Eq. (13.145) that drove the Møller computation. We therefore record the synthesis as a proposition, citing rather than repeating the contraction.

Proposition 13.39 (The minimal coupling is gauge-invariant)**.** *The interaction $\mathcal{L}_{\rm int} = -eJ^\mu A_\mu$ with $J^\mu = \bar\psi\gamma^\mu\psi$ couples the photon to a current that is conserved by the electron's equation of motion, Eq. (13.193). Consequently, by Theorem 13.28, every amplitude built from Eq. (13.191) in which photons appear only as internal lines or as external lines summed over physical polarizations is independent of the gauge parameter ξ, and is invariant under $A_\mu \to A_\mu + \frac{1}{e}\partial_\mu\alpha$ accompanied by $\psi \to e^{-i\alpha}\psi$. Gauge invariance is thus a property of the* coupled *theory at the quantum level, inherited from the conservation of the source.*

This is the precise quantum content of the gauge principle: the classical statement "minimal coupling is forced by local U(1) invariance" becomes the quantum statement "the resulting amplitudes cannot depend on the gauge-fixing artefact, because the source is conserved." The two are the same fact seen from the two ends of the quantization.

Physical Insight

Local gauge invariance is not a symmetry relating distinct physical states; it is a redundancy in the labelling of one state. Theorem 13.39 is the cleanest place to see why that redundancy is harmless: a redundancy of *description* can leave a fingerprint in intermediate, description-dependent objects (the propagator's ξ, the polarization vectors' reference momentum) but must cancel in anything that is a property of the *world*. Conservation

of the matter current is exactly the bookkeeping rule that enforces the cancellation. A theory whose matter current were not conserved would couple to the $k_\mu k_\nu$ piece, the ξ-dependence would survive into cross sections, and the theory would be inconsistent — which is the dynamical reason charged matter must come with a conserved current.

A remark on charge quantization belongs here, kept honestly within scope. Perturbative QED as written in Eq. (13.191) fixes the electron's charge to be $-e$ but says nothing about why every observed charge is an integer multiple of a common unit; the coupling e is a free continuous parameter and a second fermion could in principle be assigned any real charge q via $D_\mu = \partial_\mu + iqA_\mu$. Two non-perturbative considerations force quantization. First, if a magnetic monopole of magnetic charge g exists anywhere in the universe, single-valuedness of the charged matter's wavefunction around the monopole's Dirac string requires $eg = 2\pi n$ for integer n (the Dirac quantization condition), so the mere consistency of one charge with one monopole quantizes all electric charges in units of $2\pi/g$. Second, if the U(1) is embedded in a compact, semisimple gauge group — as it is in any grand unified extension — the allowed charges become the discrete weights of the group's representations, and quantization is automatic. Within the Standard Model the cancellation of gauge anomalies (Chapters 36 and 37) ties the hypercharges together so tightly that charge quantization emerges from consistency of the quantum theory itself. Each of these is a statement about physics beyond the free photon of this chapter; we flag them and defer their derivations to the chapters that own them.

13.6.3 The two routes agree: instantaneous Coulomb plus transverse equals covariant exchange

We have quantized the photon three ways, and it is time to verify explicitly that they describe one physics. The Coulomb-gauge route (Eq. (13.45)) quantizes only the two transverse polarizations as genuine oscillators and relegates the static interaction to an instantaneous term; manifest positivity, hidden covariance. The Gupta–Bleuler route (Section 13.3) keeps all four components and an indefinite metric; manifest covariance, positivity restored only on the physical subspace. The path integral (Section 13.5) produces the R_ξ propagator directly. Theorem 13.39 already tells us the R_ξ members agree among themselves; what remains is to connect the manifestly covariant propagator to the manifestly positive Coulomb-gauge one, since these differ not by a continuous parameter but in their very tensor structure.

In Coulomb gauge the photon two-point function, read off from the transverse quantization Eq. (13.45) and the elimination of A^0 through the Gauss constraint,

has the components (momentum space, $k^\mu = (k^0, \mathbf{k})$, all indices down)

$$\widetilde{D}^C_{00}(k) = \frac{i}{|\mathbf{k}|^2}, \qquad \widetilde{D}^C_{0i}(k) = 0, \qquad \widetilde{D}^C_{ij}(k) = \frac{i}{k^2 + i\varepsilon}\Big(\delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2}\Big). \tag{13.196}$$

The interpretation is exactly the one the Hamiltonian Eq. (13.194) (Coulomb-gauge version) demanded: $\widetilde{D}^C_{00}$ is k^0-independent — it does not propagate in time — and is the instantaneous Coulomb interaction $i/|\mathbf{k}|^2$; the spatial $\widetilde{D}^C_{ij}$ carries the transverse projector $\delta_{ij} - k_i k_j/|\mathbf{k}|^2$ and propagates only the two physical helicities. Compare the Feynman-gauge propagator Eq. (13.136), whose components are $\widetilde{D}^F_{00} = -i/k^2$, $\widetilde{D}^F_{0i} = 0$, and $\widetilde{D}^F_{ij} = -i\eta_{ij}/k^2 = +i\delta_{ij}/k^2$ (using $\eta_{ij} = -\delta_{ij}$). These look nothing alike. Their difference is

$$\begin{aligned} \Delta_{00} &\equiv \widetilde{D}^C_{00} - \widetilde{D}^F_{00} = \frac{i}{|\mathbf{k}|^2} + \frac{i}{k^2} = i\,\frac{k^2 + |\mathbf{k}|^2}{|\mathbf{k}|^2\,k^2} = \frac{i\,(k^0)^2}{|\mathbf{k}|^2\,k^2}, \\ \Delta_{0i} &= 0, \qquad \Delta_{ij} = \widetilde{D}^C_{ij} - \widetilde{D}^F_{ij} = \frac{i}{k^2}\Big(\delta_{ij} - \frac{k_i k_j}{|\mathbf{k}|^2}\Big) - \frac{i\,\delta_{ij}}{k^2} = -\frac{i}{k^2}\,\frac{k_i k_j}{|\mathbf{k}|^2}, \end{aligned} \tag{13.197}$$

where the Δ_{00} result used $k^2 + |\mathbf{k}|^2 = (k^0)^2 - |\mathbf{k}|^2 + |\mathbf{k}|^2 = (k^0)^2$. Every component of $\Delta_{\mu\nu}$ carries an explicit factor of a momentum component on each free index: $\Delta_{00} \propto k^0 k^0$ and $\Delta_{ij} \propto k_i k_j$. That is precisely the structure that current conservation annihilates. We make this exact.

Proposition 13.40 (Coulomb and covariant exchange coincide on conserved currents)**.** *For any two conserved currents, $k_\mu \widetilde{J}^\mu_1 = k_\mu \widetilde{J}^\mu_2 = 0$, the Coulomb-gauge and Feynman-gauge photon propagators give identical exchange amplitudes:*

$$\widetilde{J}^{*\mu}_1\, \widetilde{D}^C_{\mu\nu}(k)\, \widetilde{J}^\nu_2 = \widetilde{J}^{*\mu}_1\, \widetilde{D}^F_{\mu\nu}(k)\, \widetilde{J}^\nu_2. \tag{13.198}$$

Proof. The two sides differ by $\widetilde{J}^{*\mu}_1 \Delta_{\mu\nu} \widetilde{J}^\nu_2$ with $\Delta_{\mu\nu}$ from Eq. (13.197). Since $\Delta_{0i} = 0$,

$$\widetilde{J}^{*\mu}_1 \Delta_{\mu\nu} \widetilde{J}^\nu_2 = \frac{i(k^0)^2}{|\mathbf{k}|^2 k^2}\, \widetilde{J}^{*0}_1 \widetilde{J}^0_2 - \frac{i}{k^2|\mathbf{k}|^2}\,(\mathbf{k}\cdot\widetilde{\mathbf{J}}^*_1)(\mathbf{k}\cdot\widetilde{\mathbf{J}}_2) = \frac{i}{k^2|\mathbf{k}|^2}\Big[(k^0)^2\, \widetilde{J}^{*0}_1 \widetilde{J}^0_2 - (\mathbf{k}\cdot\widetilde{\mathbf{J}}^*_1)(\mathbf{k}\cdot\widetilde{\mathbf{J}}_2)\Big], \tag{13.199}$$

where $\mathbf{k}\cdot\widetilde{\mathbf{J}} \equiv \sum_i k^i \widetilde{J}^i$ is the ordinary three-vector dot product of the spatial momentum and the spatial current. Current conservation reads $k_\mu \widetilde{J}^\mu = k^0 \widetilde{J}^0 - \mathbf{k}\cdot\widetilde{\mathbf{J}} = 0$ (the relative sign is the mostly-minus metric, $k_i \widetilde{J}^i = -k^i \widetilde{J}^i$), so $\mathbf{k}\cdot\widetilde{\mathbf{J}} = k^0 \widetilde{J}^0$ for each current. Substituting, the bracket in Eq. (13.199) becomes $(k^0)^2 \widetilde{J}^{*0}_1 \widetilde{J}^0_2 - (k^0 \widetilde{J}^{*0}_1)(k^0 \widetilde{J}^0_2) = 0$. Hence $\widetilde{J}^{*\mu}_1 \Delta_{\mu\nu} \widetilde{J}^\nu_2 = 0$, which is Eq. (13.198). □

The reconciliation is therefore complete and exact: the manifestly covariant Feynman-gauge tensor $-i\eta_{\mu\nu}/k^2$ is nothing but the Coulomb-gauge content — one instantaneous timelike piece plus two transverse helicities — repackaged into a single Lorentz-covariant object, with a longitudinal redistribution between

the timelike and spatial sectors that conserved currents cannot detect. The Coulomb route makes positivity manifest at the price of hiding covariance inside the instantaneous term; the covariant route makes Lorentz invariance manifest at the price of carrying gauge artefacts that Theorems 13.39 and 13.40 guarantee never reach an observable. There is no third fact to choose between them, and this is the precise meaning of the claim that the three quantization routes of this chapter agree on physics.

13.6.4 The Coulomb law from one-photon exchange

We now run the carried computation. The cleanest fully rigorous route uses only objects already built in this chapter: the Gaussian generating functional of Section 13.5 and the propagator Eq. (13.135). Couple the photon to a prescribed, conserved, c-number external current J^μ through the source term $J^\mu A_\mu$ in the Lagrangian. Completing the square in the quadratic photon path integral — the same Gaussian manipulation that produced the propagator as the inverse quadratic operator in Eq. (13.130) — gives the vacuum persistence amplitude

$$\langle 0_+|0_-\rangle_J = \exp\Big[-\tfrac{1}{2}\int d^4x\, d^4y\; J^\mu(x)\, D_{\mu\nu}(x-y)\, J^\nu(y)\Big], \tag{13.200}$$

with $D_{\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot(x-y)}\, D_{\mu\nu}(k)$ the Feynman propagator (the factors of i from e^{iS} and from $D = i$ (inverse operator) combine to leave the clean $-\frac{1}{2}$). By Theorem 13.39 the result is ξ-independent for conserved J, so we may read $D_{\mu\nu}$ in Feynman gauge. The phase of $\langle 0_+|0_-\rangle_J$ over a long time T measures the energy of the ground state in the presence of the source, $\langle 0_+|0_-\rangle_J = e^{-iE_0T}$ (with the source-free vacuum energy subtracted), so the interaction energy is extracted directly from Eq. (13.200). The Fourier transform we need is isolated first.

Worked Example 13.41 (The static Fourier transform: $1/|\mathbf{k}|^2 \to 1/4\pi r$)**.** We evaluate, with a mass regulator $\mu > 0$ that we send to zero at the end,

$$I_\mu(\mathbf{r}) = \int \frac{d^3k}{(2\pi)^3}\, \frac{e^{i\mathbf{k}\cdot\mathbf{r}}}{|\mathbf{k}|^2 + \mu^2}. \tag{13.201}$$

Choose polar axes along $\mathbf{r}$ and write $k = |\mathbf{k}|$, $r = |\mathbf{r}|$, θ the angle between them. The angular integral is elementary:

$$\int d\Omega\; e^{i\mathbf{k}\cdot\mathbf{r}} = 2\pi \int_{-1}^{1} d(\cos\theta)\, e^{ikr\cos\theta} = 2\pi\, \frac{e^{ikr} - e^{-ikr}}{ikr} = \frac{4\pi \sin kr}{kr}. \tag{13.202}$$

Therefore

$$I_\mu(\mathbf{r}) = \frac{1}{(2\pi)^3} \int_0^\infty dk\, k^2\, \frac{1}{k^2+\mu^2}\, \frac{4\pi \sin kr}{kr} = \frac{1}{2\pi^2 r} \int_0^\infty dk\, \frac{k \sin kr}{k^2+\mu^2}. \tag{13.203}$$

The remaining integral is computed by contour methods. The integrand is even in k, so $\int_0^\infty = \frac{1}{2}\int_{-\infty}^{\infty}$, and we write $\sin kr = \mathrm{Im}\, e^{ikr}$:

$$\int_0^\infty dk\, \frac{k \sin kr}{k^2+\mu^2} = \tfrac{1}{2}\,\mathrm{Im}\int_{-\infty}^{\infty} dk\, \frac{k\, e^{ikr}}{k^2+\mu^2}. \tag{13.204}$$

For $r>0$ the factor e^{ikr} decays in the upper half-plane, so we close the contour there, enclosing the single pole at $k=+i\mu$. Its residue is $\left.\frac{k\,e^{ikr}}{2k}\right|_{k=i\mu} = \frac{1}{2}e^{-\mu r}$, and the integral equals $2\pi i\cdot\frac{1}{2}e^{-\mu r} = i\pi e^{-\mu r}$, whose imaginary part is $\pi e^{-\mu r}$. Hence $\int_0^\infty dk\,(k\sin kr)/(k^2+\mu^2) = \frac{\pi}{2}e^{-\mu r}$, and

$$I_\mu(\mathbf{r}) = \frac{1}{2\pi^2 r}\cdot\frac{\pi}{2}\,e^{-\mu r} = \frac{e^{-\mu r}}{4\pi r} \xrightarrow{\mu\to 0} \frac{1}{4\pi r}. \tag{13.205}$$

The finite-μ result is the Yukawa potential of a massive mediator; the $\mu\to 0$ limit, the massless photon, is the bare Coulomb kernel. This single integral fixes both the 4π in the denominator and the overall positive sign that the next computation propagates into the potential.

Worked Example 13.42 (The Coulomb potential between two static charges)**.** Take two static point charges $q_1 = e_1$ and $q_2 = e_2$ at positions $\mathbf{x}_1, \mathbf{x}_2$, separated by $r = |\mathbf{x}_1 - \mathbf{x}_2|$. As a classical conserved source the configuration is purely timelike and time-independent,

$$J^\mu(x) = \eta^{\mu 0}\,\rho(\mathbf{x}), \qquad \rho(\mathbf{x}) = e_1\,\delta^3(\mathbf{x}-\mathbf{x}_1) + e_2\,\delta^3(\mathbf{x}-\mathbf{x}_2), \tag{13.206}$$

manifestly conserved since $\partial_\mu J^\mu = \partial_0\rho = 0$. Only the $\mu=\nu=0$ component of Eq. (13.200) contributes, with $D_{00}(k) = -i\eta_{00}/(k^2+i\varepsilon) = -i/(k^2+i\varepsilon)$. Insert the Fourier representation and do the time integrals first:

$$\int dx^0\, dy^0\; e^{-ik^0(x^0-y^0)} = 2\pi\,\delta(k^0)\cdot T, \qquad T \equiv \int dx^0, \tag{13.207}$$

the $\delta(k^0)$ arising from the relative time and the factor T from the overall time translation. Setting $k^0=0$ collapses $k^2\to -|\mathbf{k}|^2$, so $D_{00}\to -i/(-|\mathbf{k}|^2) = i/|\mathbf{k}|^2$ — the instantaneous Coulomb kernel reappearing, exactly the $\widetilde{D}^C_{00}$ of Eq. (13.196), as Theorem 13.40 guarantees it must. The double space integral then factorizes through Theorem 13.41:

$$\int d^4x\, d^4y\; J^\mu D_{\mu\nu} J^\nu = T\int d^3x\, d^3y\;\rho(\mathbf{x})\rho(\mathbf{y})\int\frac{d^3k}{(2\pi)^3}\,\frac{i}{|\mathbf{k}|^2}\,e^{i\mathbf{k}\cdot(\mathbf{x}-\mathbf{y})} = T\,i\int d^3x\, d^3y\;\frac{\rho(\mathbf{x})\rho(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}. \tag{13.208}$$

Comparing the exponent of Eq. (13.200), $-\frac{1}{2}(\cdots) = -iE_0T$, gives the energy

$$E_0 = \frac{1}{2iT}\int d^4x\, d^4y\; J^\mu D_{\mu\nu} J^\nu = \frac{1}{2iT}\cdot T\,i\int d^3x\, d^3y\;\frac{\rho(\mathbf{x})\rho(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|} = \frac{1}{2}\int d^3x\, d^3y\;\frac{\rho(\mathbf{x})\rho(\mathbf{y})}{4\pi|\mathbf{x}-\mathbf{y}|}. \tag{13.209}$$

The factors of i have cancelled to leave a real, positive-definite electrostatic field energy — the first nontrivial sign check of the whole construction. Inserting the two-charge density, the integral splits into self-energies (e_1^2 and e_2^2 terms, divergent as the points coincide; these are r-independent and subtracted as the rest energies of the isolated charges) and a finite cross term in which $\mathbf{x}$ sits at one charge and $\mathbf{y}$ at the other. The cross term occurs twice in the symmetric double sum, cancelling the $\frac{1}{2}$, and yields the interaction potential

$$\boxed{V(r) = \frac{e_1 e_2}{4\pi r}}. \tag{13.210}$$

The sign is correct in both cases. Two electrons, $e_1 = e_2 = -e$, give $V = +e^2/4\pi r > 0$: like charges repel, the potential rises as they approach. An electron and a positron (or an electron and a proton), $e_1 = -e$, $e_2 = +e$, give $V = -e^2/4\pi r < 0$: unlike charges attract. Restoring conventional units, $e^2/4\pi = \alpha \approx 1/137$ is the fine-structure constant of QED, and Eq. (13.210) is the familiar $V(r) = \alpha/r$ between unit charges. The same number that will set the size of every QED cross section in Part V here sets the strength of the static force, with the normalization and sign of the photon propagator certified against a classical fact.

The diagrammatic content of Eq. (13.210) is the single tree graph of Fig. 13.4: each static charge is a source vertex, the internal line is the photon propagator, and the static limit projects onto the timelike, instantaneous part. The same result is reached from the particle-physics side by taking the nonrelativistic Born limit of the one-photon-exchange amplitude — the tree-level matrix element $\widetilde{J}_1^{*\mu} D_{\mu\nu} \widetilde{J}_2^{\nu}$ with on-shell spinor currents, related to a potential through the first Born approximation of nonrelativistic scattering. In that language the heavy-fermion current reduces to $\bar{u}\gamma^0 u \to 2m$ in the rest frame, the spatial currents are velocity-suppressed, and the same kernel $e_1 e_2/|\mathbf{k}|^2$ Fourier-transforms to Eq. (13.210). We do not need that route here, because the external-source computation is already complete and rigorous; the systematic translation between amplitudes and potentials, with all kinematic factors, is built in Chapters 14 and 17.

13.6.5 The interacting QED field content, assembled

The chapter set out to quantize the photon and resolve the redundancy that obstructs it; with the matter coupling installed, the full field content of quantum electrodynamics is now in hand and internally consistent. The dynamical degrees of freedom are the Dirac electron ψ of Chapter 11 — two spin states of a particle and two of its antiparticle — and the two transverse helicities of the photon quantized in this chapter, all governed by the single Lagrangian Eq. (13.191). The free propagators are the Dirac propagator $S_F(p) = i(\not{p} + m)/(p^2 - m^2 + i\varepsilon)$

and the photon propagator $D_{\mu\nu}(k)$ of Eq. (13.135); the unique interaction is the minimal vertex Eq. (13.192), which by the standard reading of $i\mathcal{L}_{\text{int}}$ becomes the momentum-space Feynman rule

$$\text{photon–electron vertex} = i\times(\text{coefficient of } A_\mu \text{ in } \mathcal{L}_{\text{int}}) = i\,(-e)\,\gamma^\mu = -\,ie\,\gamma^\mu \qquad \text{(equivalent} \tag{13.211}$$

to be derived systematically, with its symmetry factors and external-leg conventions, in Chapters 15 and 17. Every ingredient that perturbative QED requires — propagators with certified normalization and sign, a gauge-invariant vertex coupling a conserved current, and a physical Hilbert space carved out by the routes of Sections 13.3 and 13.5 — now exists, and the gauge redundancy that opened the chapter has been shown to leave no fingerprint on any observable.

This is the direct bridge Part V was promised. The propagator Eq. (13.135), the vertex Eq. (13.211), and the polarization-sum rule Eq. (13.149) are exported unchanged to Chapter 17, where they assemble into the QED Feynman rules and produce the first complete cross section of QED, $e^+e^- \to \mu^+\mu^-$, with its full spin sums and gamma-matrix traces; the same propagator is dressed by vacuum polarization, and the vertex by its radiative correction, in the one-loop program of Chapter 19, where the conservation of J^μ proved here matures into the Ward–Takahashi identity that organizes the entire renormalization of the theory.

Further reading

The constrained-Hamiltonian view of the Maxwell field that opens this chapter is Dirac's own; his [152] remains the most readable account of primary and secondary constraints and the first/second-class distinction. The two covariant-quantization papers, [244] and [80], are the primary sources for the indefinite-metric construction and the subsidiary condition that carry their names, and the abelian determinant trick was isolated, on the way to the non-abelian theory, in [182]. Among textbooks, the canonical and Gupta–Bleuler routes are developed with conventions closest to ours in [396, 432, 500]; [278] gives an especially careful treatment of the photon propagator's gauge dependence and the R_ξ family, and [454] the cleanest path-integral derivation of the gauge-fixing term. For why Lorentz invariance forces the indefinite metric on the covariant state space, the discussion in [500] is the standard reference; [532] offers the most physically motivated overview of why the gauge-fixing term may be added without changing physics.

Part IV

Interactions and Perturbation Theory

Chapter 14

Interactions and the S-Matrix

Part III handed us a quantum field theory we can solve to the last decimal: the free scalar of Chapter 9, with its quadratic Hamiltonian, its Fock space built by hand from ladder operators, and its closed-form Feynman propagator. Nothing in it scatters. The business of this chapter is to switch on the quartic interaction of ϕ^4 theory and ask the only question a collider ever asks: given a few particles prepared far apart in the distant past, with what amplitude do a few particles emerge far apart in the distant future? The object that answers is the scattering operator $\mathbf{S}$, and the chapter is, end to end, the construction of a computable bridge from the interacting Lagrangian to the matrix elements of $\mathbf{S}$ and the Lorentz-invariant amplitude $\mathcal{M}$ they contain.

The construction proceeds in a fixed logical order, and each section installs one span of the bridge. We begin in the interaction picture, where the fields evolve freely—so the entire free-theory apparatus of Chapter 9 survives verbatim—and all the dynamics is pushed into the time evolution of states, summed by the Dyson series into the time-ordered exponential $\mathbf{S} = \mathcal{T}\exp(-i\int H_I)$. We then say precisely what states this operator acts between: the asymptotic in- and out-states a scattering experiment actually prepares and measures, built on the *interacting* vacuum, with $\mathbf{S}$ the unitary map between the two complete bases and $\mathcal{M}$ the kinematic-delta-stripped amplitude inside it. Honesty about that formal apparatus is part of the work, and Haag's theorem is met head-on (Section 14.2.6) rather than buried. The middle of the chapter then makes the bridge load-bearing. The Gell-Mann–Low theorem rewrites the time-ordered vacuum correlators of interacting Heisenberg fields as a ratio of *free*-field Dyson expectation values—Wick-computable order by order, with the vacuum bubbles cancelled by the denominator—and reconciles this canonical route with the path-integral generating functional of Chapter 10. The Källén–Lehmann spectral representation then establishes, nonperturbatively, the one analytic fact LSZ consumes: the exact propagator carries an isolated single-particle pole of residue iZ, with $0 \le Z \le 1$, beneath a multiparticle continuum. With the pole located,

the LSZ reduction formula identifies an S-matrix element as the amputated, connected, on-shell correlator dressed with one factor of $\sqrt{Z}$ per external leg.

The penultimate section turns the crank for the first time, chaining Dyson, Gell-Mann–Low, Wick, and LSZ to compute the leading $\phi\phi \to \phi\phi$ amplitude of ϕ^4 theory with every factor on the page—the 4! contraction multiplicity cancelling the 1/4! of the coupling—and arrives at the first genuinely interacting number of the book, $i\mathcal{M} = -i\lambda$. The chapter then closes with the one statement that is not perturbative: the unitarity of $\mathbf{S}$, pressed into the optical theorem, partial-wave unitarity and the Argand circle, and the perturbative reach of λ. That nonperturbative constraint is the fixed point of certainty against which every later loop calculation will be checked.

The chapter imports the free mode expansion, Fock space, and Feynman propagator D_F from Chapter 9; the path-integral generating functional and Wick's theorem from Chapter 10; the theory of tempered distributions, the $i\epsilon$ prescription, analytic continuation, and the dispersion-relation preview from Chapter 5; the one-particle relativistic normalization, the invariant measure, and the Poincaré/translation machinery from Chapter 3; and the Legendre transform to the Hamiltonian together with the $\mathbb{Z}_2$ symmetry of ϕ^4 theory from Chapters 6 and 7. It exports, in turn, the operational LSZ recipe and the tree amplitude $-i\lambda$ to the Feynman-rules and cross-section chapters Chapters 15 and 16, the field-strength renormalization Z and the self-energy structure to the renormalization program of Chapters 19 and 22, and the optical theorem to the analyticity-and-dispersion and cutting-rules developments of Chapter 60. Conventions are those of Chapter A: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, and the Feynman $i\epsilon$ throughout.

14.1 The Interaction Picture and the Dyson Series

Part III left us with a quantum field theory we can solve completely. The free real scalar of Chapter 9 has a Hamiltonian quadratic in the field, a linear equation of motion, and a Hilbert space—the Fock space of Theorem 9.17—whose every state is built by hand from the ladder operators. The field itself carries the explicit mode expansion of Theorem 9.6, and its time-ordered two-point function is the closed-form Feynman propagator of Theorem 9.24. Everything is exact because nothing in the theory couples the modes to one another: a quantum of momentum $\mathbf{p}$ created at $\lambda = 0$ propagates forever, undisturbed. The whole edifice rests on that linearity.

The moment we restore the quartic term of ϕ^4 theory, that linearity is gone. The interacting Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \qquad \phi \text{ real},\ \lambda > 0, \tag{14.1}$$

produces a cubic source in its equation of motion, $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$, so the modes scatter off one another and no closed-form solution of the operator dynamics exists. We are not, however, helpless. The coupling λ is dimensionless and—this is the working assumption of the entire chapter, to be examined critically when we renormalize—small. The strategy that organizes all of perturbative field theory is therefore to treat the interaction as a controlled deformation of the solvable free theory: split the Hamiltonian into the free part we have already diagonalized and a small remainder, and develop every quantity as a power series in that remainder. This section builds the apparatus that performs the split cleanly and produces, order by order in λ, the operator that propagates states forward in time under the interaction. That operator, taken between the distant past and the distant future, will be the S-matrix.

The device that makes the split usable is the *interaction picture*, a third choice of how to apportion time dependence between states and operators, intermediate between the Schrödinger and Heisenberg pictures of Section 9.1. Its defining virtue is that the fields evolve with the *free* Hamiltonian alone, so they retain—verbatim—the mode expansion, the commutators, and the propagator we built in the free theory, while all the genuine dynamics of the interaction is pushed into the time evolution of the states. We adopt throughout the conventions in force for the chapter: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, and the Feynman $i\varepsilon$ prescription of Section 5.3.5.

14.1.1 Splitting the Hamiltonian

The Hamiltonian of the interacting theory is obtained from Eq. (14.1) by the Legendre transform of Chapter 6. The interaction term carries no derivatives of the field, so it does not touch the definition of the canonical momentum: $\pi = \partial\mathcal{L}/\partial\dot{\phi} = \dot{\phi}$ exactly as in the free theory. The transform therefore goes through unchanged except for the extra potential term, and the Hamiltonian density is the one already recorded as Eq. (6.180),

$$\mathcal{H} = \tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4. \tag{14.2}$$

Integrating over a constant-time slice and grouping the terms by their power of the field splits the total Hamiltonian into two pieces,

$$H = H_0 + H_{\mathrm{int}}, \qquad H_0 = \int \mathrm{d}^3x \left[\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2\right], \qquad H_{\mathrm{int}} = \frac{\lambda}{4!}\int \mathrm{d}^3x\, \phi^4, \tag{14.3}$$

where H_0 is precisely the free Hamiltonian of Eqs. (6.122) and (9.3), whose spectrum and eigenstates we own completely, and H_{int} collects the interaction. The cleanliness of this split—that H_0 is unmodified and H_{int} is just the spatial integral of the potential—is a special feature of a non-derivative coupling.

Caution

For interactions that *do* contain derivatives of the field—a $(\partial\phi)^2\phi^2$ coupling, the minimal coupling of QED, or the non-abelian self-interaction of Yang–Mills theory—the canonical momentum is itself altered by the interaction, and H_{int} is no longer simply minus the interaction Lagrangian density integrated over space. Extra, non-covariant "Schwinger" terms appear, and they conspire with the non-covariant pieces of the propagator to restore covariance in the end. None of this afflicts ϕ^4, where Eq. (14.3) is exact; the derivative-coupling subtlety is taken up where it first bites, in the gauge theories of Chapters 13 and 26.

A word on what these operators are. In Eq. (14.3) the field ϕ and momentum π are Schrödinger-picture operators, defined on a single time slice, which we take to be $t = 0$: write them $\phi(\mathbf{x}) \equiv \phi(0, \mathbf{x})$ and $\pi(\mathbf{x}) \equiv \pi(0, \mathbf{x})$. They obey the equal-time canonical commutators of Theorem 9.1 on that slice. In the Schrödinger picture all time dependence resides in the state $|\psi(t)\rangle_S$, which evolves under the *full* Hamiltonian according to $i\,\partial_t\,|\psi(t)\rangle_S = H\,|\psi(t)\rangle_S$. The trouble is that we cannot solve this: e^{-iHt} is the exponential of an operator we have not diagonalized. The interaction picture reorganizes the problem so that what we must exponentiate is, at each order, only the small piece.

14.1.2 The interaction picture and the free evolution of fields

Definition 14.1 (Interaction picture)**.** Let A_S be any Schrödinger-picture operator on the $t = 0$ slice and $|\psi(t)\rangle_S$ a Schrödinger-picture state evolving under the full Hamiltonian $H = H_0 + H_{\text{int}}$. The corresponding *interaction-picture* operator and state are obtained by conjugating away, and respectively rotating away, the *free* evolution generated by H_0:

$$A_I(t) \equiv e^{iH_0 t}\, A_S\, e^{-iH_0 t}, \tag{14.4}$$

$$|\psi(t)\rangle_I \equiv e^{iH_0 t}\; |\psi(t)\rangle_S\,. \tag{14.5}$$

The three pictures coincide at the reference time $t = 0$, where $A_I(0) = A_S$ and $|\psi(0)\rangle_I = |\psi(0)\rangle_S$.

The definition is engineered so that matrix elements are picture independent: inserting $e^{-iH_0 t}e^{iH_0 t} = 1$ on each side of any operator,

$$ {}_S\langle\chi(t)|\, A_S\, |\psi(t)\rangle_S = {}_I\langle\chi(t)|\; e^{iH_0 t} A_S e^{-iH_0 t}\; |\psi(t)\rangle_I = {}_I\langle\chi(t)|\, A_I(t)\, |\psi(t)\rangle_I\,, \tag{14.6}$$

so every physical prediction is unchanged. What we have bought is a redistribution of the time dependence: the operators now move with H_0, and the states will be shown to move with the interaction alone.

Consider first the operators. Differentiating Eq. (14.4), the only t-dependence is in the two exponentials, and H_0 commutes with itself, so

$$\frac{dA_I}{dt} = iH_0\, e^{iH_0t} A_S e^{-iH_0t} - e^{iH_0t} A_S e^{-iH_0t}\, iH_0 = i\, e^{iH_0t}\,[H_0, A_S]\, e^{-iH_0t} = i\,[H_0, A_I(t)]\,. \tag{14.7}$$

This is the Heisenberg equation of motion—but with the *free* Hamiltonian H_0 in place of the full H. An interaction-picture operator therefore evolves exactly as it would in a world with no interaction at all. Note also that H_0 itself is picture independent, $e^{iH_0t} H_0 e^{-iH_0t} = H_0$, because it commutes with its own exponential; we may write H_0 without a subscript.

Apply this to the field. The Schrödinger field $\phi(\mathbf{x}) = \phi(0,\mathbf{x})$ promotes to the interaction-picture field

$$\phi_I(t,\mathbf{x}) \equiv e^{iH_0t}\,\phi(0,\mathbf{x})\,e^{-iH_0t}, \tag{14.8}$$

which by Eq. (14.7) obeys $\dot\phi_I = i\,[H_0, \phi_I]$ and, after a second derivative, the free Heisenberg equations driven by H_0. But these are exactly the equations solved in Section 9.1: ϕ_I satisfies the operator Klein–Gordon equation $(\Box + m^2)\phi_I = 0$, and the evolution $\phi_I(t,\mathbf{x}) = e^{iH_0t}\phi(0,\mathbf{x})e^{-iH_0t}$ is precisely the construction by which the free Heisenberg field was built from its $t = 0$ data. We have proved the first key result.

Theorem 14.2 (Interaction-picture fields are free fields)**.** *The interaction-picture scalar field Eq. (14.8) is identical to the free Heisenberg field of Chapter 9. It satisfies the free equation of motion $(\Box + m^2)\phi_I = 0$ and carries verbatim the free mode expansion of Theorem 9.6,*

$$\phi_I(x) = \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big(a_{\mathbf{p}}\, e^{-ip\cdot x} + a^\dagger_{\mathbf{p}}\, e^{+ip\cdot x} \Big)\Big|_{p^0 = E_{\mathbf{p}}}, \tag{14.9}$$

with the same ladder operators, the same Fock vacuum $|0\rangle$ of Theorem 9.17 annihilated by every $a_{\mathbf{p}}$, the same equal-time commutators Theorem 9.1, and the same time-ordered two-point function $\langle 0|\,\mathcal{T}\,\phi_I(x)\phi_I(y)\,|0\rangle = D_F(x-y)$, the Feynman propagator of Theorem 9.24.

Proof. By Eq. (14.7) the operator $\phi_I(t,\mathbf{x})$ satisfies $\dot\phi_I = i\,[H_0, \phi_I]$ and $\dot\pi_I = i\,[H_0, \pi_I]$ with $\pi_I = e^{iH_0t}\pi(0,\mathbf{x})e^{-iH_0t}$, where H_0 is the free Hamiltonian Eq. (14.3). These are word-for-word the free Heisenberg equations established in Section 9.1; with the initial data on the $t = 0$ slice being the canonical pair obeying Theorem 9.1, the solution is unique and is the free Heisenberg field. Conjugation by the unitary e^{iH_0t} preserves the equal-time commutators, $[\phi_I(t,\mathbf{x}), \pi_I(t,\mathbf{y})] = e^{iH_0t}\,[\phi(0,\mathbf{x}), \pi(0,\mathbf{y})]\,e^{-iH_0t} = i\delta^3(\mathbf{x}-\mathbf{y})$, so the mode algebra is unchanged and Eq. (14.9) follows by the same Fourier inversion that produced Eq. (9.42). The vacuum $|0\rangle$ is the free Fock vacuum because $H_0\,|0\rangle$ is a c-number times $|0\rangle$, hence

$e^{-iH_0t}|0\rangle \propto |0\rangle$ and $a_{\mathbf{p}}|0\rangle = 0$ is preserved. Finally, the time-ordered product of two interaction-picture fields is, by Eq. (9.151), exactly the object computed in Theorem 9.26, giving $D_F(x-y)$. □

This theorem is the entire reason the interaction picture is worth introducing. Every tool of the free theory—the mode expansion, the ladder algebra, the Fock space, Wick's theorem of Chapter 10, the propagator—applies to ϕ_I unaltered. The price is that the interaction, having been evicted from the fields, must reappear somewhere; it reappears in the time evolution of the states, governed by the operator we now define.

Definition 14.3 (Interaction-picture interaction Hamiltonian)**.** The *interaction Hamiltonian in the interaction picture* is the Schrödinger interaction H_{int} of Eq. (14.3) dressed with the free evolution:

$$H_I(t) \equiv e^{iH_0t}\, H_{\text{int}}\, e^{-iH_0t}. \tag{14.10}$$

For ϕ^4 theory, $H_{\text{int}} = \frac{\lambda}{4!}\int \mathrm{d}^3x\, \phi(0,\mathbf{x})^4$, and conjugation distributes over the product of four fields, $e^{iH_0t}\phi^4 e^{-iH_0t} = (e^{iH_0t}\phi\, e^{-iH_0t})^4 = \phi_I^4$, so

$$\boxed{H_I(t) = \frac{\lambda}{4!}\int \mathrm{d}^3x\; \phi_I(t,\mathbf{x})^4} \tag{14.11}$$

is built entirely from the free field Eq. (14.9).

Equation Eq. (14.11) is the concrete object this chapter will compute with. It is $O(\lambda)$, it is Hermitian (it is the conjugate of a Hermitian operator by a unitary), and—crucially—it is a polynomial in free creation and annihilation operators. Its matrix elements between Fock states are therefore evaluable by the same combinatorics that governs the free theory. We have fixed the interaction-picture interaction Hamiltonian and prepared, but not yet performed, the first $O(\lambda)$ computation; carrying it out is the business of the LSZ and amplitude sections that follow.

14.1.3 The interaction-picture evolution equation

We now find how interaction-picture states move. Differentiate the defining relation Eq. (14.5), using the Schrödinger equation $i\,\partial_t|\psi(t)\rangle_S = H|\psi(t)\rangle_S$ for the second factor:

$$\begin{aligned} i\,\partial_t|\psi(t)\rangle_I &= i(iH_0)e^{iH_0t}|\psi(t)\rangle_S + e^{iH_0t}\, i\,\partial_t|\psi(t)\rangle_S \\ &= -H_0\, e^{iH_0t}|\psi(t)\rangle_S + e^{iH_0t}\, H\, |\psi(t)\rangle_S \\ &= e^{iH_0t}(H-H_0)|\psi(t)\rangle_S = e^{iH_0t}\, H_{\text{int}}\, e^{-iH_0t}\, e^{iH_0t}|\psi(t)\rangle_S\,. \end{aligned} \tag{14.12}$$

The first factor in the last line is $H_I(t)$ by Eq. (14.10), and the remaining $e^{iH_0t}\,|\psi(t)\rangle_S = |\psi(t)\rangle_I$ by Eq. (14.5). Hence the free part of the Hamiltonian has cancelled out of the state's evolution entirely, leaving the *Tomonaga–Schwinger equation*,

$$i\,\partial_t\,|\psi(t)\rangle_I = H_I(t)\;|\psi(t)\rangle_I\,. \tag{14.13}$$

This is the central dynamical statement. An interaction-picture state is driven only by the (small) interaction $H_I(t)$, which is itself constructed from free fields. We have traded the unsolvable problem of exponentiating the full H for a Schrödinger-like equation in which the generator is the perturbation. Note that $H_I(t)$ is genuinely time dependent—through the free evolution of the fields inside it—even though H_{int} and H are not; this time dependence is exactly what will require time ordering below.

To solve Eq. (14.13) we encode the evolution in an operator.

Definition 14.4 (Interaction-picture time-evolution operator)**.** The *time-evolution operator* $U(t,t_0)$ is the unitary that propagates interaction-picture states from time t_0 to time t,

$$|\psi(t)\rangle_I = U(t,t_0)\;|\psi(t_0)\rangle_I\,. \tag{14.14}$$

Substituting Eq. (14.14) into Eq. (14.13), and using that $|\psi(t_0)\rangle_I$ is an arbitrary fixed state, U obeys the operator equation and initial condition

$$i\,\partial_t\,U(t,t_0) = H_I(t)\,U(t,t_0), \qquad U(t_0,t_0) = \mathbf{1}. \tag{14.15}$$

An explicit closed form follows by tracking the same operator through the pictures. From Eq. (14.5) and the Schrödinger evolution $|\psi(t)\rangle_S = e^{-iH(t-t_0)}\,|\psi(t_0)\rangle_S$,

$$|\psi(t)\rangle_I = e^{iH_0t}\,e^{-iH(t-t_0)}\,e^{-iH_0t_0}\,|\psi(t_0)\rangle_I\,, \qquad \Longrightarrow \qquad U(t,t_0) = e^{iH_0t}\,e^{-iH(t-t_0)}\,e^{-iH_0t_0}. \tag{14.16}$$

This formula is exact and instructive—it shows U interpolating between the free evolution it strips off at each end and the full evolution in the middle, and one checks immediately that $U(t_0,t_0)=\mathbf{1}$. But it is useless for computation, because it still contains the unexpanded $e^{-iH(t-t_0)}$. The content of the next subsection is to convert Eq. (14.15) into a power series in H_I, which is a power series in λ, every term of which is a free-field matrix element.

14.1.4 The Dyson series

The differential equation Eq. (14.15) together with its initial condition is equivalent to the integral equation

$$U(t,t_0) = \mathbf{1} - i\int_{t_0}^{t}\mathrm{d}t_1\;H_I(t_1)\,U(t_1,t_0), \tag{14.17}$$

as one verifies by differentiating (the integral's upper limit returns the integrand $-iH_I(t)U(t,t_0)$) and by setting $t = t_0$ (the integral vanishes, giving $U(t_0,t_0) = \mathbf{1}$). The virtue of the integral form is that it is solvable by iteration: the unknown U appears under the integral, and substituting the whole right-hand side back into itself repeatedly generates a series in powers of H_I. We carry out the iteration explicitly.

Zeroth and first order. Substituting $U(t_1,t_0) \to \mathbf{1}$ in Eq. (14.17) gives the first two terms,

$$U = \mathbf{1} + (-i)\int_{t_0}^{t} \mathrm{d}t_1\, H_I(t_1) + \dots \tag{14.18}$$

Second order. Substitute the first-order expression for $U(t_1,t_0)$ back into Eq. (14.17):

$$U = \mathbf{1} + (-i)\int_{t_0}^{t} \mathrm{d}t_1\, H_I(t_1) + (-i)^2 \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, H_I(t_1)H_I(t_2) + \dots \tag{14.19}$$

The structure is already visible. The upper limit of the inner integral is the running variable t_1 of the outer one, so the two times are *nested*: t_1 ranges up to t, and t_2 ranges only up to t_1. Continuing to all orders, the n-th term is

$$U^{(n)} = (-i)^n \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-1}} \mathrm{d}t_n\, H_I(t_1)H_I(t_2)\cdots H_I(t_n), \tag{14.20}$$

in which the integration domain is the ordered simplex $t \geq t_1 \geq t_2 \geq \cdots \geq t_n \geq t_0$ and the operators stand in the same order—latest time on the left. The ordering is forced: $H_I(t_1)$ and $H_I(t_2)$ do not commute (they are built from fields at different times), so we may not freely rearrange them. Summing Eq. (14.20) over n already *is* a complete solution of Eq. (14.15); the remaining work is purely a matter of rewriting it in the symmetric form that exposes its structure and connects it to the time-ordered correlators the rest of the chapter needs.

The rewriting rests on the time-ordering symbol. For two bosonic operators the symbol $\mathcal{T}$ was defined in Eq. (9.151); the only new ingredient is its extension to n factors, which simply places them in order of decreasing time.

Definition 14.5 (Time-ordered product of n operators)**.** For bosonic operators $A_1(t_1),\dots,A_n(t_n)$ the time-ordered product is

$$\mathcal{T}\{A_1(t_1)\,A_2(t_2)\cdots A_n(t_n)\} \equiv A_{\sigma(1)}(t_{\sigma(1)})\,A_{\sigma(2)}(t_{\sigma(2)})\cdots A_{\sigma(n)}(t_{\sigma(n)}), \tag{14.21}$$

where σ is the permutation of $\{1,\dots,n\}$ for which $t_{\sigma(1)} > t_{\sigma(2)} > \cdots > t_{\sigma(n)}$, i.e. the symbol reorders the factors so that times decrease from left to right, regardless of the order in which they were written. (Coincident times occur on a measure-zero set and, for the equal-time-commuting fields at hand, cause no ambiguity, exactly as in Eq. (9.151).)

The reordering is what undoes the nesting. Consider the symmetric integral of the time-ordered product over the *full* hypercube $[t_0, t]^n$, with no ordering imposed on the limits:

$$J_n \equiv \int_{t_0}^{t} \mathrm{d}t_1 \cdots \int_{t_0}^{t} \mathrm{d}t_n \, \mathcal{T}\{H_I(t_1) \cdots H_I(t_n)\}. \tag{14.22}$$

The hypercube is the disjoint union (up to the measure-zero coincidence locus) of $n!$ simplices, one for each strict ordering of the n times. On the simplex where $t_{\sigma(1)} > \cdots > t_{\sigma(n)}$, the time-ordering symbol resolves to the single product $H_I(t_{\sigma(1)}) \cdots H_I(t_{\sigma(n)})$. Within that simplex, rename the integration variables by $t_{\sigma(k)} \to s_k$; this is a relabelling of dummy variables and a permutation of the orthonormal integration axes, with unit Jacobian, that maps the simplex onto the canonical one $s_1 > s_2 > \cdots > s_n$ and turns the integrand into $H_I(s_1) \cdots H_I(s_n)$. Every one of the $n!$ simplices is thereby mapped to the *same* integral—the ordered integral of Eq. (14.20). Hence

$$J_n = n! \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2 \cdots \int_{t_0}^{t_{n-1}} \mathrm{d}t_n \, H_I(t_1) \cdots H_I(t_n), \tag{14.23}$$

which is exactly $n!$ times the nested integral appearing in $U^{(n)}$. Solving Eq. (14.23) for the nested integral and inserting it into Eq. (14.20),

$$U^{(n)} = \frac{(-i)^n}{n!} \int_{t_0}^{t} \mathrm{d}t_1 \cdots \int_{t_0}^{t} \mathrm{d}t_n \, \mathcal{T}\{H_I(t_1) \cdots H_I(t_n)\}. \tag{14.24}$$

The factor $1/n!$ is not inserted by hand: it is precisely the count of simplices that tile the hypercube, the price paid for trading the rigid nested limits of Eq. (14.20) for the symmetric, fully time-ordered limits of Eq. (14.24). Summing over n gives the central result.

Theorem 14.6 (Dyson series)**.** *The interaction-picture time-evolution operator solving Eq. (14.15) is*

$$\boxed{U(t, t_0) = \sum_{n=0}^{\infty} \frac{(-i)^n}{n!} \int_{t_0}^{t} \mathrm{d}t_1 \cdots \int_{t_0}^{t} \mathrm{d}t_n \, \mathcal{T}\{H_I(t_1) \cdots H_I(t_n)\} \;\equiv\; \mathcal{T} \exp\Big(-i \int_{t_0}^{t} \mathrm{d}t' \, H_I(t') \Big).} \tag{14.25}$$

The right-hand object is the time-ordered exponential*: the formal exponential series with the time-ordering symbol applied term by term, defined to mean exactly the sum on the left.*

Proof. Summing Eq. (14.24) over $n \geq 0$ (with $U^{(0)} = \mathbf{1}$) yields the displayed series; it is a rewriting of the iterated solution Eq. (14.20), which solves Eq. (14.17) and hence Eq. (14.15) by construction. The notation $\mathcal{T}\exp$ is defined to abbreviate the same series: expanding $\exp(-i \int H_I)$ formally produces $\frac{(-i)^n}{n!}(\int H_I)^n = \frac{(-i)^n}{n!} \int \cdots \int H_I(t_1) \cdots H_I(t_n)$, and applying $\mathcal{T}$ to each term inserts exactly the symbol in Eq. (14.25). □

The time-ordered exponential is a genuine generalization of the ordinary one. If the $H_I(t')$ at different times commuted, $\mathcal{T}$ would be inert and Eq. (14.25) would collapse to $\exp(-i\int_{t_0}^{t} H_I\,\mathrm{d}t')$; the whole nontrivial content lives in the failure of $H_I(t_1)$ and $H_I(t_2)$ to commute, which the time ordering keeps track of. The reader who recognizes Eq. (14.17) as the field-theoretic image of the Dyson series from time-dependent perturbation theory in nonrelativistic quantum mechanics is exactly right; the construction is identical, with the interaction-picture interaction $H_I(t)$ playing the role of the time-dependent perturbing potential.

Worked Example 14.7 (The $n = 2$ combinatorics, drawn)**.** We verify the $1/n!$ at second order by performing the simplex decomposition explicitly, which is where the geometry is easiest to see. The symmetric integral Eq. (14.22) at $n = 2$ is

$$J_2 = \int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t} \mathrm{d}t_2\, \mathcal{T}\{H_I(t_1)H_I(t_2)\} \tag{14.26}$$

over the square $[t_0, t]^2$ of Fig. 14.1. The diagonal $t_1 = t_2$ splits the square into two triangles. In the lower-right triangle $t_1 > t_2$, the symbol resolves as $\mathcal{T}\{H_I(t_1)H_I(t_2)\} = H_I(t_1)H_I(t_2)$, contributing

$$\int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, H_I(t_1)H_I(t_2). \tag{14.27}$$

In the upper-left triangle $t_2 > t_1$, the symbol resolves as $\mathcal{T}\{H_I(t_1)H_I(t_2)\} = H_I(t_2)H_I(t_1)$ (later operator left), contributing

$$\int_{t_0}^{t} \mathrm{d}t_2 \int_{t_0}^{t_2} \mathrm{d}t_1\, H_I(t_2)H_I(t_1). \tag{14.28}$$

In Eq. (14.28) relabel the dummy variables $t_1 \leftrightarrow t_2$: the integral becomes $\int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, H_I(t_1)H_I(t_2)$, identical to Eq. (14.27). The two triangles thus contribute equally, and

$$J_2 = 2\int_{t_0}^{t} \mathrm{d}t_1 \int_{t_0}^{t_1} \mathrm{d}t_2\, H_I(t_1)H_I(t_2), \tag{14.29}$$

confirming Eq. (14.23) at $n = 2$ with the factor $2! = 2$. Dividing by $2!$ recovers exactly the nested second-order term of Eq. (14.19): the $1/n!$ is the reciprocal of the number of orderings, here the two triangles. The same picture in n dimensions has the hypercube cut by the $\binom{n}{2}$ hyperplanes $t_i = t_j$ into $n!$ congruent simplices, one per ordering, giving the general factor $n!$.

14.1.5 The group law and formal unitarity

The evolution operator inherits from its definition the algebraic properties any sensible propagator must have. We establish them directly from the differential equation Eq. (14.15), which is the most economical route and the one that does not lean on the formal closed form Eq. (14.16).

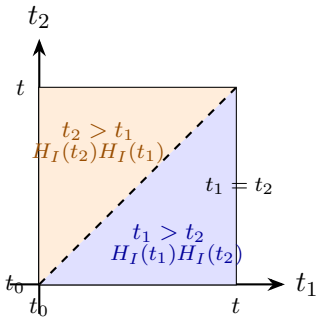


Figure 14.1: The second-order simplex decomposition of Theorem 14.7. The diagonal $t_1 = t_2$ splits the integration square $[t_0, t]^2$ into two triangles. On each, the time-ordering symbol selects the product with the later operator on the left; after relabelling the dummy variables the two triangles give equal contributions, so the symmetric integral over the square is 2! times the nested integral over either triangle. In n dimensions the hypercube splits into $n!$ such simplices, producing the $1/n!$ of the Dyson series.

Theorem 14.8 (Composition law)**.** *For any three times the evolution operator composes,*

$$U(t_1, t_2)\, U(t_2, t_3) = U(t_1, t_3), \tag{14.30}$$

and in particular $U(t, t_0)$ is invertible with

$$U(t, t_0)^{-1} = U(t_0, t). \tag{14.31}$$

Proof. Fix t_2 and t_3 and regard both sides of Eq. (14.30) as functions of the upper time, which we call τ. Define $F(\tau) \equiv U(\tau, t_2)\, U(t_2, t_3)$ and $G(\tau) \equiv U(\tau, t_3)$. Each satisfies the same first-order equation in τ: by Eq. (14.15),

$$i\,\partial_\tau F(\tau) = \big[i\,\partial_\tau U(\tau, t_2)\big] U(t_2, t_3) = H_I(\tau) U(\tau, t_2) U(t_2, t_3) = H_I(\tau) F(\tau), \tag{14.32}$$

and likewise $i\,\partial_\tau G(\tau) = H_I(\tau) G(\tau)$, the constant operator $U(t_2, t_3)$ passing through the τ-derivative untouched. At $\tau = t_2$ the initial data agree: $F(t_2) = U(t_2, t_2) U(t_2, t_3) = U(t_2, t_3) = G(t_2)$, using $U(t_2, t_2) = \mathbf{1}$. A first-order linear operator equation with prescribed value at one point has a unique solution, so $F \equiv G$ for all τ, which is Eq. (14.30). Setting $t_1 = t_3 = t_0$ and $t_2 = t$ gives $U(t_0, t) U(t, t_0) = U(t_0, t_0) = \mathbf{1}$; setting $t_1 = t_3 = t$ and $t_2 = t_0$ gives the reverse product, so $U(t_0, t)$ is the two-sided inverse Eq. (14.31). □

Theorem 14.9 (Formal unitarity)**.** *If $H_I(t)$ is Hermitian for every t, then $U(t, t_0)$ is unitary:*

$$U(t, t_0)^\dagger\, U(t, t_0) = \mathbf{1} = U(t, t_0)\, U(t, t_0)^\dagger. \tag{14.33}$$

Proof. That $H_I(t)$ is Hermitian is immediate from Eq. (14.10): H_{int} is Hermitian (it is the spatial integral of a real power of the Hermitian field), and conjugation by the unitary $e^{iH_0 t}$ preserves Hermiticity, so $H_I(t)^\dagger = H_I(t)$; for ϕ^4 this is manifest in Eq. (14.11). Take the Hermitian conjugate of the evolution equation Eq. (14.15). Since ∂_t and $\dagger$ commute and $(H_I U)^\dagger = U^\dagger H_I^\dagger = U^\dagger H_I$,

$$-i\,\partial_t\, U(t,t_0)^\dagger = U(t,t_0)^\dagger\, H_I(t). \tag{14.34}$$

Now differentiate the product $U^\dagger U$ with the product rule, using Eq. (14.34) and Eq. (14.15):

$$\partial_t(U^\dagger U) = (\partial_t U^\dagger)U + U^\dagger(\partial_t U) = (iU^\dagger H_I)U + U^\dagger(-iH_I U) = i\,U^\dagger H_I U - i\,U^\dagger H_I U = 0. \tag{14.35}$$

The combination $U^\dagger U$ is therefore constant in t; at $t = t_0$ it equals $U(t_0,t_0)^\dagger U(t_0,t_0) = \mathbf{1}$, so $U^\dagger U = \mathbf{1}$ for all t. The opposite product $UU^\dagger = \mathbf{1}$ then follows because a one-sided inverse of an invertible operator is two-sided—and U is invertible by Eq. (14.31). Indeed taking the dagger of Eq. (14.31) and combining with $U^\dagger U = \mathbf{1}$ identifies $U^\dagger = U(t_0,t)$, consistent with reading Eq. (14.34) as the evolution equation in the second argument. □

We have called this unitarity *formal*, and the qualifier is deliberate. Each manipulation above is exact at the level of symbols, but every object in it—H_0, H_{int}, the interaction picture itself—has been treated as if it acted on the free Fock space of Theorem 9.17. In an interacting theory in infinite volume that assumption is, strictly, false: the free and interacting Hamiltonians are not unitarily equivalent, the interaction picture in the literal sense does not exist (Haag's theorem), and the time integrals in the Dyson series do not converge as written. The series is nonetheless the correct generator of the perturbative expansion, and the algebraic identities Eqs. (14.30) and (14.33) survive order by order in λ once the construction is given its proper, regularized meaning. Confronting this honestly—explaining in what sense the formal apparatus computes the right thing despite the obstruction—is the first task of the next section, and the unitarity proved here will return there as the exact statement that the S-matrix preserves probability.

With the Dyson series Eq. (14.25) in hand we have the explicit, λ-by-λ operator that propagates states under the interaction, together with its group and unitarity properties. The single object the scattering theory needs is its limit between the distant past and the distant future, $\mathbf{S} = U(\infty,-\infty) = \mathcal{T}\exp\!\big(-i\int_{-\infty}^{\infty} \mathrm{d}t'\, H_I(t')\big)$. The next section gives this limit a precise meaning by defining the asymptotic in- and out-states between which its matrix elements are taken, identifies those matrix elements as scattering amplitudes, and faces squarely the formal status flagged above.

14.2 Asymptotic States and the S-Matrix

The previous section ended with a concrete operator. Solving the interaction-picture Schrödinger equation by Dyson's time-ordered exponential produced the evolution operator $U(t,t_0)$ (Theorem 14.6), and as the time interval was opened to the whole real line there appeared a single, suggestive object,

$$\mathbf{S} \;\overset{?}{=}\; U(\infty,-\infty) \;=\; \mathcal{T}\exp\!\Big(-i\int_{-\infty}^{\infty} dt\; H_I(t)\Big), \tag{14.36}$$

the operator that propagates a state from the distant past to the distant future under the full interaction. We placed a question mark over it deliberately. A formula is not yet a physical statement: we have not said *what states* Eq. (14.36) acts between, nor why the limits $t\to\pm\infty$ exist, nor what the matrix entries of $\mathbf{S}$ have to do with a number a detector reports. This section supplies all three. It defines the asymptotic in- and out-states that a scattering experiment actually prepares and measures, identifies the S-matrix as the unitary map between them, extracts from each matrix element the Lorentz-invariant amplitude $\mathcal{M}$ that carries all the physics, and states honestly — via Haag's theorem — why Eq. (14.36) is at this stage only a *formal* expression, made meaningful not by the free-field interaction picture but by the renormalized perturbation theory the rest of the chapter constructs.

We keep the conventions in force for the chapter: mostly-minus metric $\eta_{\mu\nu}=\mathrm{diag}(+,-,-,-)$ so that on shell $p^2=m^2$; natural units $\hbar=c=1$; the Feynman $i\epsilon$ prescription; and ϕ^4 theory with $\lambda>0$. One-particle states carry the relativistic normalization fixed in Eq. (3.193) and rebuilt for the free scalar in Theorem 9.12, and we abbreviate the Lorentz-invariant momentum measure $\widetilde{dp}\equiv d^3p/[(2\pi)^3\,2E_{\mathbf{p}}]$ exactly as in Chapter 9.

14.2.1 What a scattering experiment prepares and measures

A collider, stripped to essentials, does three things in sequence. Long before the collision it prepares a small number of particles, far apart from one another and moving freely, each a localized wavepacket with well-controlled momentum. These packets are aimed so that their future trajectories overlap inside a tiny interaction region. Long after the collision a bank of detectors, again far from the interaction region, registers some number of outgoing particles, again effectively free and localized, and records their momenta. The experiment measures the *probability* that a specified incoming configuration emerges as a specified outgoing one. The S-matrix is precisely the table of complex amplitudes whose squared moduli are those probabilities.

The crucial physical input is the separation of scales. The range of the force is microscopic, while "before" and "after" mean times enormously larger than any interaction time. During those long epochs the particles are so widely

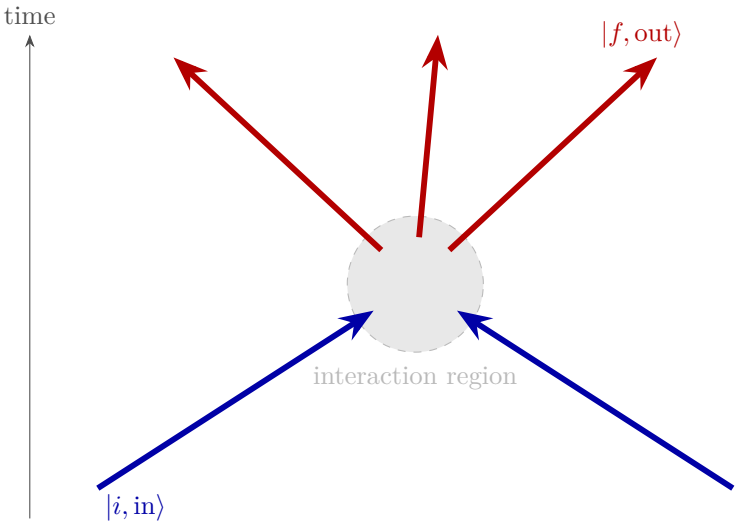


Figure 14.2: The scattering geometry. Widely separated incoming wavepackets (blue) propagate freely out of the distant past, overlap briefly in a microscopic interaction region (shaded), and emerge as freely propagating outgoing packets (red) into the distant future. The in- and out-states are defined by the limits in which the packets are taken infinitely far from the interaction.

separated that the interaction between them is utterly negligible, and each packet propagates as if it were a free, stable, single particle. This is what licenses describing the prepared and detected states in the language of free particles *even though the underlying dynamics is interacting*: not because interactions are absent, but because in the asymptotic regions the constituents are too far apart to feel them. We will make this precise with the device of *adiabatic switching*, and then be candid about exactly where the device oversimplifies.

14.2.2 In-states, out-states, and the adiabatic hypothesis

Let $H = H_0 + H_{\text{int}}$ be the full Hamiltonian of the interacting theory. The genuine, interacting theory has its own ground state and its own stable single-particle states, and these are emphatically *not* the free Fock states of Chapter 9: the physical vacuum is polarized, the physical one-particle state is dressed by virtual quanta, and (as we will see at the end) the two Hilbert-space structures are not even unitarily related. We therefore must define asymptotic states intrinsically, in terms of the full dynamics, and only afterwards connect them to the free states that perturbation theory can compute with.

Definition 14.10 (Physical vacuum and one-particle states)**.** Let $|\Omega\rangle$ denote the unique, Poincaré-invariant, normalized ground state of the full Hamiltonian, $H|\Omega\rangle = 0$ (the energy origin is fixed by this condition), to be contrasted with the

free vacuum $|0\rangle$ of Theorem 9.11. Let $|\mathbf{p}\rangle$ denote a stable single-particle eigenstate of the full H and momentum $\mathbf{P}$ with relativistic dispersion $P^\mu|\mathbf{p}\rangle = p^\mu|\mathbf{p}\rangle$, $p^0 = E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}$, normalized as in Eq. (3.193).

Plane-wave momentum eigenstates are not normalizable; they are continuum objects, and any equation written with them is shorthand for a statement about normalizable *wavepackets*

$$|f_\alpha\rangle \;=\; \int \widetilde{dp}\; f_\alpha(\mathbf{p})\,|\mathbf{p}\rangle\,, \qquad \int \widetilde{dp}\;|f_\alpha(\mathbf{p})|^2 = 1, \tag{14.37}$$

with smooth profiles f_α sharply peaked about chosen central momenta. This is not pedantry: the S-matrix elements between plane waves are *distributions* (Theorem 5.2), not functions — they will turn out to contain a Dirac delta enforcing energy–momentum conservation — and a distribution acquires meaning only when smeared against test functions, which is exactly what the profiles f_α do. We write formulæ with plane waves for transparency and understand them as smeared in this way whenever a finite number is required.

To build a multiparticle asymptotic state, take several such packets with central momenta $\mathbf{p}_1, \dots, \mathbf{p}_n$ and spatial supports that, in the far past, are mutually distant and non-overlapping. Because each packet is then localized far from the others, the interaction among them is negligible and the multiparticle state is, to arbitrary accuracy, a tensor product of single-particle states built on the physical vacuum. The *in-state* $|\mathbf{p}_1, \dots, \mathbf{p}_n;\,\mathrm{in}\rangle$ is defined as the Heisenberg-picture state that, evolved back to $t \to -\infty$, looks like n such widely separated, freely propagating physical one-particle wavepackets converging on the interaction region; the *out-state* $|\mathbf{q}_1, \dots, \mathbf{q}_m;\,\mathrm{out}\rangle$ is defined symmetrically as the state that, at $t \to +\infty$, looks like m widely separated, freely propagating physical one-particle packets diverging from it. Both are complete bases of the full Hilbert space — any state can be expanded in either — because in either limit one is describing free, well-separated particles, of which the multiparticle states form a complete set.

Adiabatic switching. A constructive handle on these limits, due to Gell-Mann and Low, is to multiply the interaction by a slowly varying envelope,

$$H_{\mathrm{int}}(t) \;\longrightarrow\; g_\epsilon(t)\, H_{\mathrm{int}}(t), \qquad g_\epsilon(t) = e^{-\epsilon|t|}, \quad \epsilon \to 0^+, \tag{14.38}$$

so that in the remote past and future ($|t| \to \infty$) the interaction is switched off and the dynamics is exactly free, while near $t = 0$ the full interaction is restored. With the switch in place the asymptotic regions are governed by H_0, the in/out states reduce to honest free states, and the limits defining $U(\pm\infty, 0)$ in Eq. (14.36) are tamed: the rapid oscillations $e^{-iH_0 t}$ that would otherwise make the $t \to \pm\infty$ limits ill-defined are damped by g_ϵ. The adiabatic theorem

then asserts that switching the interaction on infinitely slowly carries each free eigenstate continuously into the corresponding interacting eigenstate, up to a phase — the precise statement and the survival of that phase are the content of the Gell-Mann–Low theorem proved in the next section. We adopt Eq. (14.38) as the working definition of the asymptotic limits and as the bridge from physical states to the free states perturbation theory manipulates, while flagging at the end of the section the sense in which it is a formal device rather than a rigorous construction.

14.2.3 The S-operator and its matrix elements

In- and out-states are two complete descriptions of the same physical Hilbert space, one natural in the past, one in the future. There must exist a single operator carrying one basis to the other.

Definition 14.11 (The S-operator)**.** The S-operator is the unitary operator that maps the out-basis to the in-basis,

$$|\alpha;\,\mathrm{in}\rangle \;=\; \mathbf{S}\;|\alpha;\,\mathrm{out}\rangle \qquad \text{for every label } \alpha, \tag{14.39}$$

equivalently the change-of-basis whose entries are the overlaps of out- with in-states. The *S-matrix element* for a transition from an initial configuration i to a final configuration f is

$$S_{fi} \;\equiv\; \langle f,\mathrm{out}|i,\mathrm{in}\rangle \;=\; \langle f,\mathrm{out}|\mathbf{S}|i,\mathrm{out}\rangle \;\equiv\; \langle f|\mathbf{S}|i\rangle\,, \tag{14.40}$$

where in the last form a single reference basis (say the out-basis, or equivalently the free in/out basis once the adiabatic identification is made) is understood.

The middle equality of Eq. (14.40) is just Eq. (14.39) read inside an overlap; the final, basis-free notation $\langle f|\mathbf{S}|i\rangle$ is the one we use from here on. The number S_{fi} is the amplitude for the experiment to begin with i and end with f; the probability is $|S_{fi}|^2$, properly understood as a probability density to be integrated against the wavepacket profiles of Eq. (14.37) and binned by the detector acceptance.

The connection to the Dyson series is now visible. In the adiabatically switched theory the in- and out-states are free states in the remote past and future, and the operator that evolves a remote-past free state into a remote-future free state, in the interaction picture, is exactly $U(\infty,-\infty)$. Hence the formal identification

$$\mathbf{S} \;=\; U(\infty,-\infty) \;=\; \mathcal{T}\exp\!\Big(-i\int_{-\infty}^{\infty} dt\; H_I(t)\Big), \tag{14.41}$$

promoting Eq. (14.36) from a guess to a definition — formal, because it leans on the interaction picture, whose existence Section 14.2.6 will call into question.

This is the object the rest of the chapter learns to compute: expanding the time-ordered exponential order by order in λ and taking matrix elements between free multiparticle states generates the perturbation series for S_{fi}.

14.2.4 Unitarity and the conservation of probability

The single most important structural property of the S-matrix is that it is unitary. Physically it must be: if the initial state is normalized to total probability one, the probabilities of all possible outcomes must sum to one, since *something* happens. Mathematically it follows from the unitarity of the time-evolution operator.

Theorem 14.12 (Unitarity of the S-matrix)**.** *The S-operator is unitary,*

$$\mathbf{S}^\dagger \mathbf{S} \;=\; \mathbf{S}\,\mathbf{S}^\dagger \;=\; \mathbf{1}. \tag{14.42}$$

Proof. The interaction-picture evolution operator satisfies $U(t_2,t_1)^\dagger = U(t_1,t_2)$ and the composition law $U(t_3,t_2)U(t_2,t_1) = U(t_3,t_1)$, both established for the Dyson series in Theorem 14.6: the first because $H_I(t)$ is Hermitian, so the generator of U is anti-Hermitian and U is a product (time-ordered limit of products) of unitary infinitesimal factors $e^{-iH_I(t)\,dt}$; the second by concatenating adjacent time slices. In particular

$$U(\infty,-\infty)^\dagger\, U(\infty,-\infty) \;=\; U(-\infty,\infty)\, U(\infty,-\infty) \;=\; U(-\infty,-\infty) \;=\; \mathbf{1}, \tag{14.43}$$

and identically with the factors in the opposite order, giving $U(\infty,-\infty)U(\infty,-\infty)^\dagger = \mathbf{1}$. With $\mathbf{S} = U(\infty,-\infty)$ from Eq. (14.41) this is Eq. (14.42). The same conclusion follows directly from Theorem 14.11: Eq. (14.39) maps one orthonormal (in the continuum, delta-normalized) complete basis onto another, and any operator carrying one orthonormal basis to another is unitary. □

The physical content is read off by inserting a complete set of final states. Sandwiching $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ between a normalized initial state $|i\rangle$ and itself and inserting $\mathbf{1} = \sum_f |f\rangle\langle f|$ (a continuum sum over all multiparticle configurations, with the invariant measure understood) gives

$$1 = \Big\langle i\Big|\mathbf{S}^\dagger\mathbf{S}\Big|i\Big\rangle = \sum_f \Big\langle i\Big|\mathbf{S}^\dagger\Big|f\Big\rangle\,\langle f|\mathbf{S}|i\rangle = \sum_f |S_{fi}|^2\,. \tag{14.44}$$

The probabilities of all final states, given the initial state i, sum to one. Unitarity is conservation of probability, no more and no less. It is a nonperturbative constraint — true to all orders — and it will reappear in sharpened form as the optical theorem at the close of the chapter.

14.2.5 The trivial part, the invariant amplitude, and the momentum-conserving delta

Most of the time nothing happens: the particles miss each other and emerge unchanged. This "forward," no-scattering possibility is contained in $\mathbf{S}$ as an identity piece, and it is convenient to separate it out so that what remains is pure interaction.

Definition 14.13 (The transition operator)**.** The transition operator T is defined by extracting the identity from $\mathbf{S}$,

$$\mathbf{S} \;=\; \mathbf{1} + i\,T. \tag{14.45}$$

The factor of i is conventional and chosen so that the Born approximation reproduces the nonrelativistic scattering amplitude with the standard sign.

The matrix element $\langle f|\mathbf{1}|i\rangle = \langle f|i\rangle$ is the trivial overlap: it is nonzero only when the final configuration coincides with the initial one (same particles, same momenta), and it describes no scattering. All scattering — everything a detector at wide angle ever sees — lives in $\langle f|iT|i\rangle$. We now show that this nontrivial part carries a kinematic factor that can be pulled out once and for all, leaving a function of the momenta alone.

Theorem 14.14 (Factorization of the momentum-conserving delta)**.** *Let $|i\rangle$ and $|f\rangle$ be momentum eigenstates with total four-momenta p_i^μ and p_f^μ. Then the matrix element of T is supported on $p_f = p_i$ and factorizes as*

$$\langle f|iT|i\rangle \;=\; (2\pi)^4\,\delta^4\!\Big(\textstyle\sum_f p_f - \sum_i p_i\Big)\; i\,\mathcal{M}(i\to f), \tag{14.46}$$

where $\mathcal{M}(i\to f)$, the invariant amplitude*, contains all the dynamics and is free of the overall conservation delta.*

Proof. The dynamics is invariant under spacetime translations: the full four-momentum operator P^μ generates translations, $U(a) = e^{iP\cdot a}$, and commutes with the Hamiltonian and hence with the S-operator, $[\,\mathbf{S}, P^\mu\,] = 0$. This is the operator statement of translation invariance established in Eq. (3.136); since $\mathbf{S} = U(\infty,-\infty)$ is built from the translation-invariant interaction, it inherits the symmetry. The identity piece $\mathbf{1}$ commutes with P^μ trivially, so T commutes with P^μ as well: $[\,T, P^\mu\,] = 0$.

Now act with a finite translation on both sides of the matrix element. Because $|i\rangle$ and $|f\rangle$ are momentum eigenstates, $U(a)\,|i\rangle = e^{ip_i\cdot a}\,|i\rangle$ and $\langle f|\,U(a)^\dagger = \langle f|\,e^{-ip_f\cdot a}$. Inserting $U(a)^\dagger U(a) = \mathbf{1}$ around T and using $[T, P^\mu] = 0$, i.e. $U(a)^\dagger\,T\,U(a) = T$,

$$\langle f|T|i\rangle = \Big\langle f\Big|U(a)^\dagger\,T\,U(a)\Big|i\Big\rangle = e^{-ip_f\cdot a}\,e^{+ip_i\cdot a}\;\langle f|T|i\rangle = e^{-i(p_f-p_i)\cdot a}\;\langle f|T|i\rangle\,. \tag{14.47}$$

This must hold for *every* constant four-vector a^μ. If $p_f \neq p_i$ the phase $e^{-i(p_f-p_i)\cdot a}$ is not identically one, so consistency forces $\langle f|T|i\rangle = 0$. The matrix element is therefore supported entirely on the hypersurface $p_f = p_i$, and as a distribution in the total momentum difference it must be proportional to $\delta^4(p_f - p_i)$ — the unique (up to normalization) translation-covariant distribution vanishing off $p_f = p_i$. Indeed, integrating Eq. (14.47) against $\int d^4a$ converts the constraint into the Fourier representation $\int d^4a\, e^{-i(p_f-p_i)\cdot a} = (2\pi)^4\delta^4(p_f - p_i)$, exhibiting the delta explicitly. Writing the proportionality constant as $i\mathcal{M}$ with the conventional $(2\pi)^4$ and the i inherited from *Eq.* (14.45) gives Eq. (14.46). That the residual factor $\mathcal{M}$ is well defined (the delta carries all the singular support, leaving an ordinary function of the on-shell momenta) is the statement that energy–momentum conservation is the only kinematic constraint the symmetry imposes. □

Proposition 14.15 (Lorentz invariance of the amplitude)**.** *For a scalar theory $\mathcal{M}(i \to f)$ is a Lorentz scalar: it depends on the external momenta only through their Lorentz-invariant combinations $p_a \cdot p_b$.*

Proof. The S-operator commutes not only with translations but with the full Poincaré group; in particular $U(\Lambda)\,\mathbf{S}\,U(\Lambda)^{-1} = \mathbf{S}$ for every proper orthochronous Lorentz transformation Λ, again because the interaction is Lorentz invariant (Eq. (3.136) and its Lorentz extension). For scalar particles the states transform as $U(\Lambda)\,|\mathbf{p}\rangle = |\Lambda\mathbf{p}\rangle$ with the invariant normalization of Eq. (3.193), so

$$\langle f|iT|i\rangle = \langle \Lambda f|iT|\Lambda i\rangle\,. \tag{14.48}$$

The delta $\delta^4(\sum p)$ is Lorentz invariant on the mass shell, hence so is the cofactor $\mathcal{M}$. A Lorentz-invariant function of the external momenta can depend on them only through their invariant dot products $p_a \cdot p_b$ (for $2 \to 2$ scattering, the Mandelstam invariants s, t, u), which is the assertion. For particles with spin the same argument applies with the states carrying little-group indices and $\mathcal{M}$ becoming a Lorentz-covariant tensor/spinor built from those, but the scalar case is all ϕ^4 theory requires. □

The invariant amplitude $\mathcal{M}$ is now the central computational target of the book: everything from cross sections and decay rates (Chapter 16) to the optical theorem is built on it. We pause to extract from unitarity the seed of that last result, both because it is short and because it shows the $\mathbf{1} + iT$ decomposition earning its keep.

Worked Example 14.16 (Unitarity in T-language: the optical-theorem seed)**.** Substitute $\mathbf{S} = \mathbf{1} + iT$ into the unitarity relation $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ of Theorem 14.12 and expand:

$$\mathbf{1} = (\mathbf{1} - iT^\dagger)(\mathbf{1} + iT) = \mathbf{1} + iT - iT^\dagger + T^\dagger T, \tag{14.49}$$

so the identity cancels and the remaining terms give the exact operator relation

$$-i\,(T - T^\dagger) \;=\; T^\dagger T, \qquad \text{equivalently} \qquad i\,(T^\dagger - T) = T^\dagger T. \tag{14.50}$$

Take the diagonal matrix element in a normalized state $|i\rangle$. The left side is

$$-i(\langle i|T|i\rangle - \left\langle i\middle|T^\dagger\middle|i\right\rangle) = -i(\langle i|T|i\rangle - \langle i|T|i\rangle^*) = -i\cdot 2i\ \mathrm{Im}\,\langle i|T|i\rangle = 2\ \mathrm{Im}\,\langle i|T|i\rangle\,, \tag{14.51}$$

using $\left\langle i\middle|T^\dagger\middle|i\right\rangle = \langle i|T|i\rangle^*$. The right side, with a complete set of intermediate states $\mathbf{1} = \sum_f |f\rangle\langle f|$ inserted, is

$$\left\langle i\middle|T^\dagger T\middle|i\right\rangle = \sum_f \left\langle i\middle|T^\dagger\middle|f\right\rangle \langle f|T|i\rangle = \sum_f |\langle f|T|i\rangle|^2\,. \tag{14.52}$$

Equating the two,

$$2\ \mathrm{Im}\,\langle i|T|i\rangle \;=\; \sum_f |\langle f|T|i\rangle|^2\,. \tag{14.53}$$

The imaginary part of the forward (initial = final) transition amplitude equals the total "probability" summed over all final states. This is unitarity rewritten as a statement about a *single* amplitude and a sum of squared amplitudes; converting both sides to invariant amplitudes $\mathcal{M}$ via Eq. (14.46) and stripping the shared deltas turns it into the optical theorem $2\ \mathrm{Im}\,\mathcal{M}(i\to i) = \sum_f \int d\Pi_f\ |\mathcal{M}(i\to f)|^2$, related to the total cross section. We carry that final step out in the chapter's closing section on the optical theorem (Section 14.7); here the point is only that the entire content of the optical theorem is already present in Eq. (14.53), which is nothing but $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ in disguise.

Worked Example 14.17 (The ϕ^4 four-point target)**.** The machinery of this chapter exists to produce numbers. The first interacting number ϕ^4 theory owes is the invariant amplitude for two-particle elastic scattering,

$$\mathcal{M}\big(\phi(p_1)\,\phi(p_2)\to\phi(p_3)\,\phi(p_4)\big), \tag{14.54}$$

the four-point amplitude of the real scalar with Lagrangian $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$. By Theorem 14.15 it is a Lorentz scalar, hence a function of the Mandelstam invariants $s = (p_1+p_2)^2$, $t = (p_1-p_3)^2$, $u = (p_1-p_4)^2$ alone (with $s+t+u = 4m^2$ on shell). Theorem 14.14 guarantees the kinematic prefactor $(2\pi)^4\delta^4(p_1+p_2-p_3-p_4)$; everything physical sits in $\mathcal{M}$. We do not yet have a value: computing $\mathcal{M}$ requires the time-ordered correlation functions of the next section, the spectral input of the Källén–Lehmann section, and the LSZ reduction that converts correlators into S-matrix elements. The reader should hold Eq. (14.54) in mind as the concrete destination; the chapter's payoff, computed two sections hence, is the leading result $\mathcal{M} = -\lambda + O(\lambda^2)$, equivalently $i\mathcal{M} = -i\lambda$ at order λ.

14.2.6 Honesty about the construction: Haag's theorem

We have written $\mathbf{S} = U(\infty, -\infty)$ and manipulated the interaction-picture field as though it were a free field carrying the full interacting dynamics in its time evolution. It is time to admit that this picture, taken literally, does not exist. The defect is not a technicality at the edges; it sits at the foundation, and naming it precisely is what lets us see why the perturbative manipulations nonetheless deliver correct physics.

Haag's theorem and what saves the formalism

The contradiction. The interaction picture assumes a single Hilbert space on which one unitary operator relates the interacting Heisenberg field $\phi(\mathbf{x}, t)$ to a free field $\phi_I(\mathbf{x}, t)$ at each time, with ϕ_I built on a free Fock vacuum. Trace the assumptions:

1. At a fixed time the interacting and free fields and their conjugate momenta are related by a unitary $V(t)$: $\phi(\mathbf{x}, t) = V(t)\,\phi_I(\mathbf{x}, t)\,V(t)^{-1}$, with the same equal-time commutation relations on both sides.
2. Both fields transform covariantly under the same unitary representation of spatial translations and rotations, with the *same* invariant vacuum.

Haag's theorem states that these hypotheses, taken together with Poincaré covariance and the uniqueness of the vacuum, force $V(t)$ to be a trivial (c-number phase) transformation: the two fields must then have *identical* vacuum expectation values to all orders, in particular the same two-point function and the same vacuum. But the whole point of an interaction is that the interacting vacuum is polarized and its correlators differ from the free ones — if they did not, there would be no scattering. The assumptions are mutually inconsistent: *no* nontrivial unitary $V(t)$ on a single Fock space can intertwine a free and an interacting field while preserving the vacuum. The interaction picture, as a rigorous object, does not exist.

The mechanism. The collision is between the uniqueness of the Fock representation and the inequivalence of representations in a system with infinitely many degrees of freedom. In ordinary quantum mechanics (finitely many degrees of freedom) the Stone–von Neumann theorem guarantees that all irreducible representations of the canonical commutation relations are unitarily equivalent, so a $V(t)$ relating two of them always exists. In quantum field theory that theorem fails: the free and interacting theories build *unitarily inequivalent* representations of the canonical commutation relations, and no unitary on one carries the vacuum of one to the vacuum of the other. The dressed, polarized interacting vacuum literally has infinite overlap deficit with the free vacuum ($\langle 0|\Omega\rangle \to 0$ in infinite volume). This is the same infinity that the adiabatic switch Eq. (14.38) and the Gell-Mann–Low phase quietly absorb.

The resolution. Haag's theorem forbids the interaction picture as a literal construction; it does not forbid the answers it formally produces. Two repairs make the manipulations legitimate. First, work in a finite spatial volume with a momentum cutoff: then there are finitely many degrees of freedom, Stone–von Neumann applies, $V(t)$ exists, and the interaction picture and the Dyson series are perfectly well defined. The pathology returns only in the limits of infinite volume and infinite cutoff — and those limits are precisely where renormalization operates. Second, the physical outputs of the formalism are not the (representation-dependent, ill-defined) bare correlators but *renormalized*, on-shell S-matrix elements, in which the field-strength renormalization Z, the mass renormalization, and the vacuum-overlap phase are systematically removed. The upshot, made operational by the LSZ formula and the renormalization program of Part VI, is the working stance of this book: treat $\mathbf{S} = U(\infty, -\infty)$ and the Dyson series as a generating algorithm for renormalized perturbation theory, whose individual ingredients may be formal but whose properly renormalized predictions are finite, unique, and spectacularly confirmed. Haag's theorem is the precise reason the next four sections are necessary: they are what turns Eq. (14.41) from a symbol into a calculation.

With the target defined — a Lorentz-invariant amplitude $\mathcal{M}$ extracted from $\langle f|\mathbf{S}|i\rangle$ — and the honest caveat recorded that the interaction-picture route to it is only formal, the path forward is forced. To compute $\langle f|\mathbf{S}|i\rangle$ we need objects that are genuinely calculable order by order in λ, and the only such objects the free theory hands us are the time-ordered vacuum correlation functions of free interaction-picture fields, evaluated by Wick's theorem. The next section, on correlation functions and the Gell-Mann–Low theorem, builds exactly that bridge: it rewrites the vacuum expectation values of time-ordered Heisenberg fields as Dyson-series expectation values of free fields, making the abstract $\langle f|\mathbf{S}|i\rangle$ of this section into something a diagram can evaluate.

14.3 Correlation Functions and the Gell-Mann–Low Theorem

The previous section left us with a demand and an obstruction. The demand is that every scattering probability be expressible through the S-operator $\mathbf{S} = U(+\infty, -\infty)$ acting between asymptotic states (Section 14.2); the obstruction is that those asymptotic states are eigenstates of the *full* interacting Hamiltonian H, whose spectrum we cannot diagonalize, and that the interacting vacuum $|\Omega\rangle$ is not the free Fock vacuum $|0\rangle$ of Theorem 9.11. The way forward, anticipated there, is not to solve for the eigenstates but to route everything through objects we *can*

compute order by order in the coupling: the time-ordered vacuum correlation functions

$$G^{(n)}(x_1,\dots,x_n) \;=\; \langle\Omega|\; \mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\;|\Omega\rangle\,, \tag{14.55}$$

where $\phi(x)$ is the Heisenberg field of the interacting theory and $|\Omega\rangle$ its normalized ground state, $\langle\Omega|\Omega\rangle = 1$. The LSZ reduction of Section 14.5 will turn these correlators, evaluated near their mass-shell poles, into S-matrix elements; the present section manufactures the tool that makes them computable. That tool is the Gell-Mann–Low theorem, which rewrites Eq. (14.55) — an expectation value of *interacting* Heisenberg fields in the *unknown* interacting vacuum — as a ratio of expectation values of *free* interaction-picture fields in the *known* free vacuum, weighted by the Dyson exponential of the interaction. In that form Wick's theorem (Theorem 10.25) applies term by term, and the entire perturbative apparatus opens up.

Throughout we work with the frozen ϕ^4 theory theory,

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \qquad \lambda > 0, \tag{14.56}$$

in the mostly-minus metric and natural units, with the Feynman $i\varepsilon$ in force. The Legendre transform of Eq. (14.56) (Theorem 6.15) splits the Hamiltonian as

$$H = H_0 + H_{\text{int}}, \qquad H_0 = \int d^3x\,\left[\tfrac{1}{2}\pi^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2\right], \qquad H_{\text{int}} = \int d^3x\,\frac{\lambda}{4!}\phi^4, \tag{14.57}$$

because the ϕ^4 interaction carries no derivatives, so the canonical momentum $\pi = \dot\phi$ is unmodified and the interaction Hamiltonian density is simply minus the interaction Lagrangian density, $\mathcal{H}_{\text{int}} = +\frac{\lambda}{4!}\phi^4$. This sign — positive in H, hence producing the $e^{-iH_{\text{int}}t}$ that becomes the Dyson factor $e^{-i\int H_I}$ — is the one that ultimately delivers the vertex $-i\lambda$; we track it with care.

14.3.1 From Heisenberg fields to interaction-picture fields

The interaction picture interpolates between the Schrödinger picture, where states evolve and operators are fixed, and the Heisenberg picture, where operators carry all the time dependence. We fix the reference time at which all three pictures coincide to be $t = 0$; this choice costs nothing and removes a clutter of t_0's from every formula. At $t = 0$, then, the Schrödinger field $\phi(\mathbf{x})$, the Heisenberg field $\phi(0,\mathbf{x})$, and the interaction-picture field $\phi_I(0,\mathbf{x})$ are one and the same operator.

The Heisenberg field evolves with the full Hamiltonian,

$$\phi(t,\mathbf{x}) = e^{iHt}\,\phi(\mathbf{x})\,e^{-iHt}, \tag{14.58}$$

while the interaction-picture field evolves with the *free* Hamiltonian alone,

$$\phi_I(t,\mathbf{x}) = e^{iH_0t}\,\phi(\mathbf{x})\,e^{-iH_0t}. \tag{14.59}$$

Equation Eq. (14.59) is the crucial simplification: because ϕ_I is generated by the quadratic H_0, it obeys the free Heisenberg equation of motion $(\Box + m^2)\phi_I = 0$ and therefore *is* a free field. It admits exactly the mode expansion of Theorem 9.6, with the same creation and annihilation operators and the same free vacuum $|0\rangle$ satisfying $a_{\mathbf{p}}\,|0\rangle = 0$ and $H_0\,|0\rangle = 0$ (the zero-point energy having been removed by normal ordering in Chapter 9). All the free-field technology — the commutators, the propagator D_F of Theorem 9.24, and Wick's theorem — is available for ϕ_I and for it alone. The interacting Heisenberg field ϕ has none of this structure; the content of this subsection is that it can nonetheless be expressed through ϕ_I.

Define the *interaction-picture evolution operator*

$$U(t_1, t_2) \;\equiv\; e^{iH_0 t_1}\, e^{-iH(t_1 - t_2)}\, e^{-iH_0 t_2}, \tag{14.60}$$

which propagates interaction-picture states from time t_2 to time t_1. This is the operator whose perturbative expansion is the Dyson series established earlier in this chapter: differentiating Eq. (14.60) with respect to t_1 gives $i\partial_{t_1} U(t_1, t_2) = H_I(t_1)\, U(t_1, t_2)$, with the interaction Hamiltonian in the interaction picture

$$H_I(t) \;=\; e^{iH_0 t}\, H_{\text{int}}\, e^{-iH_0 t} \;=\; \int d^3x\, \frac{\lambda}{4!}\, \phi_I(t, \mathbf{x})^4, \tag{14.61}$$

and the solution, subject to $U(t, t) = \mathbf{1}$, is the time-ordered exponential of Theorem 14.6,

$$U(t_1, t_2) \;=\; \mathcal{T} \exp\Big[-i \int_{t_2}^{t_1} dt'\, H_I(t') \Big], \qquad t_1 > t_2. \tag{14.62}$$

We will need three algebraic properties of U, all immediate from the definition Eq. (14.60) and visible without recourse to the series:

$$U(t_1, t_2)\, U(t_2, t_3) = U(t_1, t_3) \qquad \text{(composition)}, \tag{14.63}$$

$$U(t_1, t_2)^{-1} = U(t_2, t_1) \qquad \text{(invertibility)}, \tag{14.64}$$

$$U(t_1, t_2)^{\dagger} = U(t_2, t_1) \qquad \text{(unitarity)}. \tag{14.65}$$

The first two follow by cancelling the inner exponentials in Eq. (14.60); the third because H and H_0 are Hermitian, so $U(t_1, t_2)^{\dagger} = e^{-iH_0 t_2} e^{+iH(t_1 - t_2)} e^{iH_0 t_1} = U(t_2, t_1)$. Together they say that the U's form a two-parameter family of unitaries that compose like a flow.

We can now express the Heisenberg field through the interaction-picture field. Combining Eqs. (14.58) to (14.60),

$$\begin{aligned} U^{\dagger}(t, 0)\, \phi_I(t, \mathbf{x})\, U(t, 0) &= \Big(e^{iHt} e^{-iH_0 t} \Big) \Big(e^{iH_0 t} \phi(\mathbf{x}) e^{-iH_0 t} \Big) \Big(e^{iH_0 t} e^{-iHt} \Big) \\ &= e^{iHt}\, \phi(\mathbf{x})\, e^{-iHt} \;=\; \phi(t, \mathbf{x}), \end{aligned} \tag{14.66}$$

where the free exponentials $e^{-iH_0t}e^{iH_0t} = \mathbf{1}$ collapse in the middle. Using $U^\dagger(t,0) = U(0,t)$ from Eq. (14.65), this is the relation we sought,

$$\boxed{\phi(t,\mathbf{x}) \;=\; U(0,t)\,\phi_I(t,\mathbf{x})\,U(t,0)\;.} \tag{14.67}$$

Every interacting field is a free field conjugated by the Dyson evolution. The same identity holds for any local operator built from ϕ: products, derivatives, and composites all transform by the same conjugation, since $U(0,t)\,AB\,U(t,0) = \big(U(0,t)A\,U(t,0)\big)\big(U(0,t)B\,U(t,0)\big)$. This is the lever that converts a correlator of interacting fields into a correlator of free fields glued together by U's.

Caution

The construction Eq. (14.60)–Eq. (14.67) is, taken literally, only formal: by Haag's theorem (flagged in Section 14.2) no unitary $U(t,0)$ can intertwine the free and interacting fields on a common Hilbert space, because the two representations of the canonical commutation relations are inequivalent. What survives the theorem, and what we actually use, is the *perturbative* content: each side of Eq. (14.67), expanded order by order in λ through Eq. (14.62), is a well-defined (after renormalization) formal power series, and it is those series the Gell-Mann–Low formula manipulates. We are computing the asymptotic expansion of a correlator, not asserting the existence of the operators that would sum it.

14.3.2 Projecting the interacting vacuum from the free vacuum

The relation Eq. (14.67) expresses interacting fields through free ones, but the correlator Eq. (14.55) is still taken in the interacting vacuum $|\Omega\rangle$, which we do not know. The second ingredient of the theorem is a way to manufacture $|\Omega\rangle$ out of the free vacuum $|0\rangle$, which we do know. The mechanism is adiabatic: if we let the free vacuum evolve for an enormous time under the full Hamiltonian, and tilt that time infinitesimally into the complex plane, every state above the ground state is exponentially suppressed relative to it, and what remains is the interacting vacuum.

Make this precise. Let the eigenstates of the full Hamiltonian be $|\Omega\rangle$ (the ground state, energy E_0) and a tower of excited eigenstates $|n\rangle$ with energies $E_n > E_0$; they form a complete set, $\mathbf{1} = |\Omega\rangle\langle\Omega| + \sum_{n\neq\Omega}|n\rangle\langle n|$. Resolve the free vacuum in this basis and evolve it backward through a large time T:

$$e^{-iHT}\,|0\rangle \;=\; e^{-iE_0T}\,|\Omega\rangle\langle\Omega|0\rangle \;+\; \sum_{n\neq\Omega} e^{-iE_nT}\,|n\rangle\langle n|0\rangle\,. \tag{14.68}$$

The phases e^{-iE_nT} all have unit modulus for real T, so no term is preferred. Now continue T into the lower-right of the complex plane, $T \to \infty(1-i\varepsilon)$ with $\varepsilon > 0$

infinitesimal — the same Feynman $i\varepsilon$ tilt that defined the propagator contour in Section 5.1 and projected the path integral onto its ground state in Chapter 10. Each phase acquires a real damping factor,

$$e^{-iE_nT} \longrightarrow e^{-iE_n\infty(1-i\varepsilon)} = \underbrace{e^{-iE_n\infty}}_{\text{unit modulus}} \cdot e^{-E_n\,\varepsilon\,\infty}, \tag{14.69}$$

and the relative size of the excited terms to the ground-state term is governed by $e^{-(E_n-E_0)\varepsilon\infty} \to 0$, because $E_n - E_0 > 0$ for every excited state (the assumption that $|\Omega\rangle$ is the unique, isolated ground state — the same gap that Section 14.4 will display as the separation between the one-particle pole and the continuum). In the limit only the $|\Omega\rangle$ term survives:

$$\lim_{T\to\infty(1-i\varepsilon)} e^{-iHT}\,|0\rangle = \lim_{T\to\infty(1-i\varepsilon)} e^{-iE_0T}\,|\Omega\rangle\,\langle\Omega|0\rangle\,. \tag{14.70}$$

Provided $\langle\Omega|0\rangle \neq 0$ — the interacting vacuum has nonzero overlap with the free vacuum, which is exactly the statement that perturbation theory around $|0\rangle$ is meaningful — we may solve for $|\Omega\rangle$:

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \Big(e^{-iE_0T}\,\langle\Omega|0\rangle\Big)^{-1} e^{-iHT}\,|0\rangle\,. \tag{14.71}$$

This is already a projection of $|\Omega\rangle$ from $|0\rangle$, but it is written with the full evolution e^{-iHT}; to feed it into the Dyson series we convert it to a U. Since $H_0\,|0\rangle = 0$, we may insert a free phase for free, $e^{-iHT}\,|0\rangle = e^{-iHT}e^{-iH_0T}\,|0\rangle$, and Eq. (14.60) identifies $e^{-iHT}e^{-iH_0T} = e^{iH_0\cdot 0}e^{-iH(0-(-T))}e^{-iH_0(-T)} = U(0,-T)$. Hence

$$|\Omega\rangle = \lim_{T\to\infty(1-i\varepsilon)} \Big(e^{-iE_0T}\,\langle\Omega|0\rangle\Big)^{-1} U(0,-T)\,|0\rangle\,. \tag{14.72}$$

The mirror computation evolves $|0\rangle$ forward by the same tilted time and produces the bra. Taking the Hermitian conjugate of Eq. (14.72) and using $U(0,-T)^\dagger = U(-T,0)$ would give $\langle\Omega|$ in terms of $U(-T,0)$; it is more useful to run the spectral argument forward, $e^{-iHT}\,|0\rangle \to (\dots)\,|\Omega\rangle$ replaced by $\langle 0|\,e^{+iHT} \to (\dots)\,\langle\Omega|$ under $T\to\infty(1-i\varepsilon)$, which gives

$$\langle\Omega| = \lim_{T\to\infty(1-i\varepsilon)} \langle 0|\,U(T,0)\,\Big(e^{-iE_0T}\,\langle 0|\Omega\rangle\Big)^{-1}, \tag{14.73}$$

where $U(T,0) = e^{iH_0T}e^{-iHT}$ and again $H_0\,|0\rangle = 0$ lets the free phase be inserted at no cost. Equations Eqs. (14.72) and (14.73) are the two halves of the projection; the awkward normalization factors $\big(e^{-iE_0T}\,\langle\Omega|0\rangle\,\big)^{-1}$ they carry are precisely what the denominator of the Gell-Mann–Low formula will cancel.

Physical Insight

The $i\varepsilon$ tilt is the operator-language version of adiabatic switching. In Gell-Mann and Low's original argument one multiplies the interaction by an envelope $e^{-\varepsilon|t|}$ that turns the coupling on slowly from the far past and off again in the far future; the free vacuum, dragged along adiabatically, deforms continuously into the interacting vacuum. Tilting T into the complex plane achieves the same suppression of excited states with less bookkeeping, and ties the construction to the same contour that defines the Feynman propagator. Physically: wait long enough, with the faintest absorption switched on, and any state relaxes to its ground state.

14.3.3 The Gell-Mann–Low formula

We now assemble the two ingredients into the central result.

Theorem 14.18 (Gell-Mann–Low)**.** *For the interacting theory Eq. (14.56), the time-ordered correlation functions of Heisenberg fields in the interacting vacuum are given by*

$$\langle\Omega|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|\Omega\rangle \;=\; \frac{\langle 0|\,\mathcal{T}\Big[\phi_I(x_1)\cdots\phi_I(x_n)\,\exp\Big(-i\int_{-\infty}^{\infty}dt\,H_I(t)\Big)\Big]\,|0\rangle}{\langle 0|\,\mathcal{T}\exp\Big(-i\int_{-\infty}^{\infty}dt\,H_I(t)\Big)\,|0\rangle}\,, \tag{14.74}$$

where ϕ_I are free interaction-picture fields, $|0\rangle$ the free vacuum, $H_I(t)$ the interaction Hamiltonian of Eq. (14.61), and the time integrals run along the contour tilted by $T\to\infty(1-i\varepsilon)$. Equivalently, writing the exponent as a spacetime integral,

$$-i\int_{-\infty}^{\infty}dt\,H_I(t) \;=\; -\frac{i\lambda}{4!}\int d^4z\;\phi_I(z)^4, \tag{14.75}$$

so that each power of λ inserts a four-field vertex integrated over spacetime.

Proof. The strategy is to use Eq. (14.67) to convert the string of Heisenberg fields into interaction-picture fields linked by U's, fold those links into a single time-ordered Dyson exponential, and then replace the interacting vacuum at each end by the free vacuum via Eqs. (14.72) and (14.73). The annoying normalization factors that appear are then disposed of by the denominator.

Step 1: assemble the field string. It suffices to prove the identity for one time ordering of the arguments, since both sides are symmetric under permutations of the x_i (the left side by the time-ordering symbol, the right side because the ϕ_I sit inside a time-ordering symbol as well). Label the points so that

$t_1 > t_2 > \cdots > t_n$. By Eq. (14.67) each field is $\phi(x_i) = U(0,t_i)\,\phi_I(x_i)\,U(t_i,0)$, and the ordered product is

$$\begin{aligned}\phi(x_1)\cdots\phi(x_n) &= U(0,t_1)\,\phi_I(x_1)\,\big[U(t_1,0)U(0,t_2)\big]\,\phi_I(x_2)\,\big[U(t_2,0)U(0,t_3)\big]\cdots \\ &\quad \cdots\big[U(t_{n-1},0)U(0,t_n)\big]\,\phi_I(x_n)\,U(t_n,0) \\ &= U(0,t_1)\,\phi_I(x_1)\,U(t_1,t_2)\,\phi_I(x_2)\,U(t_2,t_3)\cdots U(t_{n-1},t_n)\,\phi_I(x_n)\,U(t_n,0),\end{aligned} \tag{14.76}$$

where each bracket collapsed by the composition law Eq. (14.63). The interaction-picture fields are now interleaved with evolution operators that bridge consecutive insertion times.

Step 2: insert the asymptotic times and recognize a single time-ordered exponential. Choose a time T larger than every t_i and split the outer factors with Eq. (14.63): $U(0,t_1) = U(0,T)U(T,t_1)$ and $U(t_n,0) = U(t_n,-T)U(-T,0)$. Equation Eq. (14.76) becomes

$$\phi(x_1)\cdots\phi(x_n) = U(0,T)\,\Big[\,U(T,t_1)\,\phi_I(x_1)\,U(t_1,t_2)\cdots\phi_I(x_n)\,U(t_n,-T)\,\Big]\,U(-T,0). \tag{14.77}$$

Examine the bracket. Reading from left to right, the times descend monotonically from T through $t_1 > t_2 > \cdots > t_n$ down to $-T$; every $U(t_a,t_b)$ inside has $t_a > t_b$ and is, by Eq. (14.62), the time-ordered exponential of H_I over the interval $[t_b,t_a]$. The whole bracket is therefore a product of objects already arranged in decreasing time, with the fields $\phi_I(x_i)$ sitting at their proper places in the chronological sequence. Such a product is exactly what the time-ordering symbol reconstructs: it equals

$$\Big[\ \cdots\ \Big] = \mathcal{T}\Big[\phi_I(x_1)\cdots\phi_I(x_n)\,\exp\Big(-i\int_{-T}^{T}dt\,H_I(t)\Big)\Big]. \tag{14.78}$$

To see this, note that $\mathcal{T}$ orders everything inside it by time regardless of how it is written; expanding the exponential on the right produces all the $H_I(t')$ factors, and $\mathcal{T}$ slots each one, together with the n fields, into chronological order, which is precisely the order displayed explicitly in Eq. (14.77). The U-intervals tile $[-T,T]$ without gaps or overlaps by construction, so the single integral from $-T$ to T reproduces all the segment integrals at once. Substituting Eq. (14.78) into Eq. (14.77),

$$\phi(x_1)\cdots\phi(x_n) = U(0,T)\,\mathcal{T}\Big[\phi_I(x_1)\cdots\phi_I(x_n)\,e^{-i\int_{-T}^{T}H_I}\Big]\,U(-T,0). \tag{14.79}$$

Because the left side was the ordered product for our chosen ordering $t_1 > \cdots > t_n$, and the right side is manifestly symmetric in the x_i once the $\mathcal{T}$ is in place, we may drop the ordering assumption and read Eq. (14.79) with a $\mathcal{T}$ on the left as well: it holds as an identity for the time-ordered product.

Step 3: replace the vacua. Sandwich Eq. (14.79) between $\langle\Omega|$ and $|\Omega\rangle$ and use the projections. On the right end, Eq. (14.72) gives $U(-T,0)\,|\Omega\rangle$ when read backwards: from $|\Omega\rangle = \big(e^{-iE_0T}\,\langle\Omega|0\rangle\,\big)^{-1}U(0,-T)\,|0\rangle$ we get, applying $U(-T,0)$ and using $U(-T,0)U(0,-T)=\mathbf{1}$,

$$U(-T,0)\,|\Omega\rangle = \Big(e^{-iE_0T}\,\langle\Omega|0\rangle\,\Big)^{-1}\,|0\rangle\,. \tag{14.80}$$

On the left end, Eq. (14.73) gives $\langle\Omega|\,U(0,T)$: from $\langle\Omega| = \langle 0|\,U(T,0)\big(e^{-iE_0T}\,\langle 0|\Omega\rangle\,\big)^{-1}$ and $U(T,0)U(0,T)=\mathbf{1}$,

$$\langle\Omega|\,U(0,T) = \Big(e^{-iE_0T}\,\langle 0|\Omega\rangle\,\Big)^{-1}\,\langle 0|\,. \tag{14.81}$$

Both replacements are understood in the limit $T\to\infty(1-i\varepsilon)$. Inserting Eqs. (14.80) and (14.81) into the sandwiched Eq. (14.79),

$$\langle\Omega|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|\Omega\rangle = \frac{\langle 0|\,\mathcal{T}\big[\phi_I(x_1)\cdots\phi_I(x_n)\,e^{-i\int_{-T}^{T}H_I}\big]\,|0\rangle}{e^{-2iE_0T}\,\langle 0|\Omega\rangle\,\langle\Omega|0\rangle}\,, \tag{14.82}$$

where the two normalization factors have combined into the single c-number prefactor $e^{-2iE_0T}\,\langle 0|\Omega\rangle\,\langle\Omega|0\rangle$ in the denominator. This prefactor is the same for *every* correlator — it does not depend on n or on the points x_i, only on the overlap and the vacuum energy. That universality is what lets us eliminate it.

Step 4: the denominator. Apply Eq. (14.82) to the case $n=0$, i.e. to the vacuum normalization $\langle\Omega|\,|\Omega\rangle = 1$. With no fields inserted the left side is $\langle\Omega|\Omega\rangle = 1$, and the right side is

$$1 = \frac{\langle 0|\,\mathcal{T}\,e^{-i\int_{-T}^{T}H_I}\,|0\rangle}{e^{-2iE_0T}\,\langle 0|\Omega\rangle\,\langle\Omega|0\rangle} \quad\Longrightarrow\quad e^{-2iE_0T}\,\langle 0|\Omega\rangle\,\langle\Omega|0\rangle = \langle 0|\,\mathcal{T}\,e^{-i\int_{-T}^{T}H_I}\,|0\rangle\,. \tag{14.83}$$

The unknown prefactor is therefore nothing but the vacuum-to-vacuum amplitude of the Dyson exponential. Substituting Eq. (14.83) into Eq. (14.82) and taking $T\to\infty(1-i\varepsilon)$ gives Eq. (14.74), with Eq. (14.75) following from Eq. (14.61) on writing $\int dt\int d^3x = \int d^4z$. This completes the proof. □

The formula Eq. (14.74) is the workhorse of the rest of the book. Its right side contains only free fields ϕ_I and the free vacuum $|0\rangle$; expanding the exponential in powers of λ produces, at each order, a vacuum expectation value of a time-ordered product of finitely many free fields, which Wick's theorem (Theorem 10.25) evaluates as a sum of products of Feynman propagators D_F (Eq. (9.154)). Every interacting correlator is thereby reduced, order by order, to propagators and combinatorics — the seed of the Feynman diagram, whose systematic rules Chapter 15 will codify.

Notation Watch

Two distinct "vacua" and two distinct "fields" appear in Eq. (14.74) and must never be confused. On the left, $|\Omega\rangle$ is the true ground state of the interacting H and ϕ is the full Heisenberg field; these are the physical, exact objects and we cannot compute with them directly. On the right, $|0\rangle$ is the free Fock vacuum of Theorem 9.11 and ϕ_I is the free interaction-picture field; these are the computable objects. The theorem is the bridge between the two columns. Some texts (including Chapter 10's path-integral derivation) write the left side's vacuum as $|0\rangle$ as well, relying on the $i\varepsilon$ projection to mean the interacting one; we keep $|\Omega\rangle$ explicit here precisely to mark the distinction the LSZ reduction will exploit.

14.3.4 The denominator removes the vacuum bubbles

The denominator in Eq. (14.74) is not a cosmetic normalization; it performs a precise diagrammatic surgery. When the numerator is expanded by Wick's theorem, many of the resulting contraction patterns split into a piece tied to the external points $x_1, \dots, x_n$ and a piece floating free — a closed cluster of propagators and vertices with no external legs at all. These free-floating clusters are the *vacuum bubbles*. They are pure, x_i-independent numbers (in fact divergent ones), and they would spoil any correlator. The denominator is exactly the sum of all such bubbles, and dividing by it deletes them from the answer. We first see the mechanism in a concrete low-order computation, then prove it to all orders.

An explicit demonstration: the two-point function at $O(\lambda)$

Take $n = 2$ and expand the numerator of Eq. (14.74) to first order in λ, using Eq. (14.75):

$$\mathcal{N}^{(2)} = \langle 0|\,\mathcal{T}\,\phi_I(x)\phi_I(y)\,|0\rangle \;-\; \frac{i\lambda}{4!}\int d^4z\;\langle 0|\,\mathcal{T}\,\phi_I(x)\phi_I(y)\,\phi_I(z)^4\,|0\rangle \;+\; O(\lambda^2). \tag{14.84}$$

The zeroth-order term is just the free propagator, $\langle 0|\,\mathcal{T}\phi_I(x)\phi_I(y)\,|0\rangle = D_F(x-y)$ by Theorem 9.24. The $O(\lambda)$ term is a vacuum expectation value of six free fields; Wick's theorem expresses it as the sum over all $5!! = 15$ ways of pairing them into propagators. The pairings fall into two classes:

(a) $\phi_I(x)$ pairs with $\phi_I(y)$, and the four vertex fields $\phi_I(z)^4$ pair among themselves. There are 3 such pairings (the $3!! = 3$ ways to pair four fields), each contributing $D_F(x-y)\,D_F(0)^2$, where $D_F(0) = D_F(z-z)$ is the propagator at coincident points.

(b) $\phi_I(x)$ pairs with one of the four vertex fields (4 choices) and $\phi_I(y)$ with one of the remaining three (3 choices), and the last two vertex fields pair

together (1 way). There are $4\times3\times1=12$ such pairings, each contributing $D_F(x-z)\,D_F(z-y)\,D_F(0)$.

Indeed $3+12=15$. Collecting,

$$\mathcal{N}^{(2)}=D_F(x-y)-\frac{i\lambda}{4!}\int d^4z\Big[3\,D_F(x-y)D_F(0)^2+12\,D_F(x-z)D_F(z-y)D_F(0)\Big]+O(\lambda^2). \tag{14.85}$$

The factor $3/4!=1/8$ and $12/4!=1/2$ are the symmetry factors of the two diagrams. Write the class-(a) contribution as the free propagator times a number,

$$-\frac{i\lambda}{4!}\,3\int d^4z\,D_F(x-y)D_F(0)^2=D_F(x-y)\cdot V_1,\qquad V_1\equiv-\frac{i\lambda}{8}\int d^4z\,D_F(0)^2, \tag{14.86}$$

where V_1 is the value of the single $O(\lambda)$ vacuum bubble — the "figure-eight" formed by two propagators looping back on one vertex, with symmetry factor 8. The class-(b) contribution is the connected "tadpole" correction to the propagator,

$$\mathcal{T}(x,y)\equiv-\frac{i\lambda}{2}\int d^4z\,D_F(x-z)D_F(z-y)D_F(0). \tag{14.87}$$

So the numerator is $\mathcal{N}^{(2)}=D_F(x-y)(1+V_1)+\mathcal{T}(x,y)+O(\lambda^2)$. Now the denominator to the same order: with no external fields, only class-(a)-type pairings of the four vertex fields exist, giving

$$\mathcal{D}=\langle0|\,\mathcal{T}\,e^{-i\int H_I}\,|0\rangle=1-\frac{i\lambda}{4!}\,3\int d^4z\,D_F(0)^2+O(\lambda^2)=1+V_1+O(\lambda^2). \tag{14.88}$$

The denominator is precisely 1 plus the same bubble V_1. Dividing,

$$\frac{\mathcal{N}^{(2)}}{\mathcal{D}}=\frac{D_F(x-y)(1+V_1)+\mathcal{T}(x,y)}{1+V_1}+O(\lambda^2)=D_F(x-y)+\mathcal{T}(x,y)+O(\lambda^2). \tag{14.89}$$

The disconnected piece $D_F(x-y)\,V_1$ — a free propagator multiplied by a vacuum bubble — has cancelled exactly against the denominator's V_1, leaving only the free propagator and its genuine, *connected* radiative correction $\mathcal{T}$. This is the surgery promised: the bubbles are gone, and what survives is exactly the set of diagrams in which every vertex is tied, through propagators, to an external point.

The cancellation to all orders

The cancellation in Eq. (14.89) is not an accident of first order; it is the shadow of an exact factorization. We prove it in general.

Theorem 14.19 (Exponentiation of vacuum bubbles)**.** *Let the numerator of Eq. (14.74) be expanded by Wick's theorem into a sum over diagrams. Each*

diagram factorizes uniquely into a part connected (through propagators) to at least one external point and a part disconnected from all external points; the disconnected part is a product of vacuum bubbles. Summing over all diagrams,

$$\langle 0|\,\mathcal{T}\Big[\phi_I(x_1)\cdots\phi_I(x_n)\,e^{-i\int H_I}\Big]\,|0\rangle = \Big(\sum\ \textit{diagrams attached to externals}\Big)\times\exp\Big(\textstyle\sum_i V_i\Big), \tag{14.90}$$

where $\{V_i\}$ is the set of connected *vacuum-bubble values, and the same exponential is the denominator,*

$$\langle 0|\,\mathcal{T}\,e^{-i\int H_I}\,|0\rangle = \exp\Big(\textstyle\sum_i V_i\Big). \tag{14.91}$$

Consequently the Gell-Mann–Low ratio retains only the diagrams in which every vertex connects to an external point,

$$\langle\Omega|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|\Omega\rangle = \sum\big(\textit{diagrams with all vertices attached to externals}\big). \tag{14.92}$$

Proof. Fix a Wick diagram in the expansion of the numerator at order λ^N, i.e. with N vertices arising from the N-th term $\frac{1}{N!}\big(-\frac{i\lambda}{4!}\int d^4z\,\phi_I^4\big)^N$ of the exponential. Partition its N vertices into those lying in the connected component(s) of at least one external point — call there be k of them — and the remaining $N-k$ vertices, which form closed clusters touching no external point. The closed clusters are, by definition, vacuum bubbles. Group the bubbles by isomorphism type: suppose the diagram contains n_i copies of the i-th distinct connected bubble, so that $\sum_i n_i\,(\text{vertices in bubble } i) = N-k$.

Now count the multiplicity with which this diagram arises. The $\frac{1}{N!}$ from the exponential is accompanied by $N!$ ways to assign the N labelled vertex-integration variables $z_1,\dots,z_N$ to the N vertex *positions* in the diagram. Of these assignments, $\binom{N}{k}\,k!\,/\,k!$ distribute vertices between the external-attached part and the bubbles; more usefully, the assignment factorizes: choose which k vertices attach to externals and arrange them ($\to$ generates the external-attached sub-sum), and arrange the rest into bubbles. The arrangements that merely relabel vertices *within* a single bubble, or permute identical bubbles among themselves, do not produce new diagrams and are exactly compensated by the symmetry factors. The net combinatorial statement is the standard one: the value of a diagram equals (value of its external-attached part) $\times\ \prod_i \frac{1}{n_i!}\,V_i^{\,n_i}$, where the $1/n_i!$ accounts for the indistinguishability of the n_i identical bubbles of type i, and V_i is the value of one such connected bubble (its own internal symmetry factor included, as in Eq. (14.86)).

Summing over all numbers $n_i \geq 0$ of each bubble type, independently, and over all external-attached parts, the sum factorizes:

$$\sum_{\text{diagrams}}(\cdots) = \Big(\sum_{\text{attached}}(\cdots)\Big)\times\prod_i\sum_{n_i=0}^{\infty}\frac{1}{n_i!}V_i^{\,n_i} = \Big(\sum_{\text{attached}}(\cdots)\Big)\times\prod_i e^{V_i} = \Big(\sum_{\text{attached}}(\cdots)\Big)\,e^{\sum_i V_i} \tag{14.93}$$

which is Eq. (14.90). The geometric-to-exponential resummation $\sum_n V^n/n! = e^V$ is where the disconnected pieces collect into a single exponential; it is the diagrammatic analogue of the elementary fact that the exponential of a sum of independent contributions is the product of their exponentials. For the denominator there are no external points, so the "attached" sum is just 1 (the empty diagram), and Eq. (14.93) collapses to Eq. (14.91). Dividing numerator by denominator cancels the common factor $e^{\sum_i V_i}$ identically, leaving Eq. (14.92). □

This is the all-orders justification of what we watched happen at $O(\lambda)$ in Eq. (14.89): the figure-eight V_1 was the lone bubble available at first order, $e^{V_1} = 1 + V_1 + \dots$ matched the denominator term for term, and the quotient kept only the connected tadpole. The lesson is structural and permanent: *vacuum bubbles never contribute to correlation functions*, and from Chapter 15 onward we simply never draw them.

Physical Insight

The vacuum bubbles are not meaningless — their sum is the phase and amplitude $e^{-2iE_0T}\,\langle 0|\Omega\rangle\,\langle\Omega|0\rangle$ of Eq. (14.83), i.e. the vacuum energy and the wavefunction overlap. They encode how the interaction shifts the ground-state energy and tilts the vacuum away from the free one. But these are properties of the vacuum alone; they are common to every correlator and therefore carry no information about the propagation or scattering of particles. The denominator's job is to subtract this universal background so that the numerator's particle physics stands clear. Dividing by $\langle 0|\,\mathcal{T}e^{-i\int H_I}\,|0\rangle$ is the field-theory way of measuring energies relative to the vacuum.

14.3.5 Reconciliation with the path-integral generating functional

We built two pillars in Part III, and consistency demands they compute the same correlators. The path integral of Chapter 10 defined the n-point Green's functions Theorem 10.18 as normalized functional derivatives of the generating functional,

$$G^{(n)}(x_1,\dots,x_n) = \frac{1}{Z[0]}\frac{1}{i^n}\,\frac{\delta^n Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0} = \frac{\int\mathcal{D}\phi\;\phi(x_1)\cdots\phi(x_n)\,e^{iS}}{\int\mathcal{D}\phi\;e^{iS}}, \tag{14.94}$$

and Theorem 10.19 proved that these equal the time-ordered vacuum correlators $\langle\Omega|\,\mathcal{T}\phi(x_1)\cdots\phi(x_n)\,|\Omega\rangle$ — the very left side of the Gell-Mann–Low formula Eq. (14.74). The two routes must therefore agree, and it is worth seeing the agreement explicitly, because it reveals the same numerator/denominator structure in both languages.

Split the action $S = S_0 + S_{\text{int}}$ with $S_{\text{int}} = -\int d^4z\, \frac{\lambda}{4!}\phi^4$, and expand the interacting weight as $e^{iS} = e^{iS_0}\, e^{iS_{\text{int}}} = e^{iS_0}\, e^{-i\frac{\lambda}{4!}\int \phi^4}$. The numerator of Eq. (14.94) becomes

$$\int \mathcal{D}\phi\; \phi(x_1)\cdots\phi(x_n)\, e^{-\frac{i\lambda}{4!}\int \phi^4}\, e^{iS_0} = \Big\langle\, \phi(x_1)\cdots\phi(x_n)\, e^{-\frac{i\lambda}{4!}\int \phi^4} \,\Big\rangle_0 \;\times\; Z_0[0], \tag{14.95}$$

where $\langle\,\cdot\,\rangle_0 \equiv Z_0[0]^{-1}\int \mathcal{D}\phi\,(\cdot)\, e^{iS_0}$ is the *free* (Gaussian) field average and $Z_0[0] = \int \mathcal{D}\phi\, e^{iS_0}$ is the free partition function. The same split applied to the denominator gives $Z[0] = \langle e^{-\frac{i\lambda}{4!}\int \phi^4}\rangle_0\, Z_0[0]$. The common factor $Z_0[0]$ cancels in the ratio Eq. (14.94), leaving

$$G^{(n)} = \frac{\big\langle\, \phi(x_1)\cdots\phi(x_n)\, e^{-\frac{i\lambda}{4!}\int \phi^4} \,\big\rangle_0}{\big\langle\, e^{-\frac{i\lambda}{4!}\int \phi^4} \,\big\rangle_0}. \tag{14.96}$$

Compare Eq. (14.96) with Eq. (14.74). The structures are identical line for line: a free average of the fields times the exponentiated interaction, over the same free average of the interaction alone. The only translation needed is between the two notions of "free average." On the path-integral side, $\langle\,\cdot\,\rangle_0$ is the Gaussian average with weight e^{iS_0}; by the path-integral Wick theorem Theorem 10.23 (equivalently, by Theorem 10.19 applied to the free theory), this Gaussian average of a product of fields equals the free vacuum expectation value of their time-ordered product, $\langle\phi(y_1)\cdots\phi(y_m)\rangle_0 = \langle 0|\,\mathcal{T}\phi_I(y_1)\cdots\phi_I(y_m)\,|0\rangle$, with the time-ordering supplied automatically by the slicing (the $i\varepsilon$ contour of the Minkowski path integral being the same tilt $T \to \infty(1-i\varepsilon)$ that defined the projection in Section 14.3.2). Under this dictionary the numerator of Eq. (14.96) is the numerator of Eq. (14.74) and the denominators likewise. The spacetime integral $\int \phi^4$ in the exponent is the same object as the time integral of H_I through Eq. (14.75), and the factor of $-i$ matches: the canonical Dyson exponential carries $-iH_I = -i\int d^3x\, \frac{\lambda}{4!}\phi_I^4$ per unit time, while the path integral carries $iS_{\text{int}} = -i\int d^4z\, \frac{\lambda}{4!}\phi^4$, and the two coincide because the interaction has no derivatives and so $\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}}$.

Proposition 14.20 (The two routes agree)**.** *For the theory Eq. (14.56), the canonical Gell-Mann–Low formula Eq. (14.74) and the path-integral ratio Eq. (14.96) define the same time-ordered correlation functions order by order in λ. The cancellation of vacuum bubbles is the same phenomenon in both: in Section 14.3.4 the denominator $\langle 0|\,\mathcal{T}e^{-i\int H_I}\,|0\rangle$ removes the bubbles; in the path integral the denominator $Z[0]$, equal to $\langle e^{-\frac{i\lambda}{4!}\int \phi^4}\rangle_0\, Z_0[0]$, does the identical job, as already exhibited at low order in Theorem 10.20.*

Proof. Expand both exponentials in powers of λ. At order λ^N each route produces, in its own denominator-free numerator, a free average (canonical: free vacuum expectation value; path-integral: Gaussian average) of the n external fields together with $4N$ vertex fields, integrated over the N vertex positions

with weight $(-i\lambda/4!)^N/N!$. By the free Wick theorem — Theorem 10.23 on the path-integral side, the operator Wick theorem that Chapter 10 proved equivalent to it on the canonical side — both free averages evaluate to the identical sum over pairings into Feynman propagators D_F (Eq. (9.154)). Term by term the two numerators are equal; the same argument equates the two denominators. The ratios therefore agree to all orders, and Theorem 14.19 (canonical) and Theorem 10.20 (path-integral) are two derivations of one fact. The two pillars of Part III thus deliver one and the same perturbative correlator, as required by Theorem 10.19. □

The upshot is that the Feynman-diagram expansion of Chapter 15 is route-independent: one may reach it from operators, via the interaction picture and the Dyson series of this section, or from the path integral, via the master formula Eq. (10.186). Both arrive at the same numerator-over-denominator ratio, the same Wick pairings, and the same deletion of vacuum bubbles. We will use whichever is more convenient and trust that they coincide.

14.3.6 Scalar ϕ^4 theory: the four-point correlator set up for Wick expansion

We close by preparing the object that the first interacting amplitude of ϕ^4 theory will be built from: the connected four-point correlator at leading order in λ. The Gell-Mann–Low formula reduces it to free correlators; the worked example writes those out and isolates the single term that LSZ will turn into $i\mathcal{M} = -i\lambda$.

Worked Example 14.21 (The $O(\lambda)$ four-point function of the ϕ^4 theory)**.** We continue ϕ^4 theory from the free-field stage of Chapters 9 and 10 and apply Theorem 14.18 to

$$G^{(4)}(x_1, x_2, x_3, x_4) = \langle \Omega | \, \mathcal{T}\, \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4) \, | \Omega \rangle \, . \tag{14.97}$$

By Theorem 14.19 we may drop the denominator and the vacuum bubbles outright and keep only diagrams whose every vertex attaches to an external point. Expanding the numerator in λ:

Order λ^0. The free four-point function. Wick's theorem pairs four fields in $3!! = 3$ ways:

$$G^{(4)}\big|_{\lambda^0} = D_F(x_1{-}x_2)D_F(x_3{-}x_4)+D_F(x_1{-}x_3)D_F(x_2{-}x_4)+D_F(x_1{-}x_4)D_F(x_2{-}x_3). \tag{14.98}$$

Every term is a product of two independent propagators: two pairs of points exchange a particle, with nothing connecting one pair to the other. These are *disconnected* as far as the four points are concerned, and they describe no scattering — they are just two particles propagating past each other. LSZ will discard them because amputation on all four legs annihilates a diagram that does not link all four points.

Order λ^1. Insert one vertex from Eq. (14.75):

$$G^{(4)}|_{\lambda^1} = -\frac{i\lambda}{4!}\int d^4z\ \langle 0|\,\mathcal{T}\,\phi_I(x_1)\phi_I(x_2)\phi_I(x_3)\phi_I(x_4)\,\phi_I(z)^4\,|0\rangle\,. \tag{14.99}$$

This is a free vacuum expectation value of eight fields; Wick's theorem expands it into $7!! = 105$ pairings. They organize into three topological classes, of which only one is connected to all four external points:

- **Fully connected (the term we want).** Each external field $\phi_I(x_a)$ contracts with one of the four vertex fields $\phi_I(z)$. There are $4! = 24$ ways to match the four externals to the four vertex legs, each giving $D_F(x_1-z)D_F(x_2-z)D_F(x_3-z)D_F(x_4-z)$. The factor $24/4! = 1$ cancels the vertex's $1/4!$ exactly:

$$G^{(4)}|^{\text{conn}}_{\lambda^1} = -i\lambda\int d^4z\ D_F(x_1-z)D_F(x_2-z)D_F(x_3-z)D_F(x_4-z). \tag{14.100}$$

  All four external points meet at the single vertex z: this is the tree-level four-point "star," the diagram that represents two particles meeting, interacting once, and emerging.

- **Disconnected with a one-loop correction on one line.** Two externals contract with each other (giving a D_F between them), the other two each contract with a vertex leg, and the two leftover vertex legs contract together (a loop $D_F(0)$ at z). These reproduce the tadpole correction Eq. (14.87) on one propagator while the other propagator is a free spectator; they do not connect all four points and are removed by amputation, exactly as the disconnected λ^0 terms are.

- **External pair times a vacuum bubble.** Two externals contract with each other, the other two with each other, and the four vertex fields pair among themselves into the figure-eight V_1. These are (free $G^{(4)}|_{\lambda^0}$ term) $\times$ V_1 — precisely the bubble-times-correlator pieces that the denominator cancels by Theorem 14.19. They are already gone.

The only genuinely new, fully connected contribution at this order is Eq. (14.100). It is the four-point correlator's first piece of scattering content, and it is now in exactly the form LSZ requires: a single spacetime integral of four propagators meeting at a vertex carrying the factor $-i\lambda$. Amputating the four external legs and continuing to the mass shell will strip the four D_F's and the integral, leaving the invariant amplitude $i\mathcal{M} = -i\lambda$ — the first interacting number of ϕ^4 theory, to be extracted in Section 14.5 and re-derived diagrammatically in Chapter 15.

The worked example has delivered what the chapter's logical chain asked of this section: the interacting four-point function, expressed by Theorem 14.18 as

a Wick-expandable ratio of free correlators, with the vacuum bubbles removed by the denominator and the connected core Eq. (14.100) isolated and ready for amputation. The next section establishes the exact analytic structure of the interacting two-point function — the isolated single-particle pole and its residue Z — so that Section 14.5 knows precisely which pole of these correlators carries the S-matrix, and how to extract it.

14.4 The Källén–Lehmann Spectral Representation

The previous section handed us a single, exact object: the time-ordered two-point function of the interacting Heisenberg field,

$$G(x-y) \;\equiv\; \langle \Omega | \mathcal{T}\, \phi(x)\phi(y) | \Omega \rangle \,, \tag{14.101}$$

where $|\Omega\rangle$ is the true (interacting) vacuum and ϕ is the Heisenberg field of the theory $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ with $\lambda > 0$. The Gell-Mann–Low theorem of the preceding section told us how to *compute* G as a Dyson series of free-field Wick contractions, order by order in λ. But the LSZ formula of the next section will not consume G as a power series; it will consume one specific analytic feature of G — the location of a pole and the number sitting on top of it. The purpose of the present section is to establish, *exactly and nonperturbatively*, what the analytic structure of G must be in *any* relativistic quantum field theory with a stable particle, before a single diagram is drawn. The result, due to Källén and Lehmann, is that G is a positively-weighted superposition of free propagators of every possible mass, and that this superposition has an isolated single-particle pole carrying a residue Z with $0 \le Z \le 1$. That pole and that residue are precisely what LSZ extracts, so this section is the structural prerequisite for the chapter's central formula.

Everything that follows uses only three inputs: the Poincaré covariance of the theory, the completeness of its physical state space, and the canonical normalization of the field. No perturbation theory enters. The conclusions therefore hold to all orders and beyond.

14.4.1 The spectral hypotheses

We make explicit the structural assumptions about the interacting theory; they are the standard Wightman-style axioms restricted to exactly what the derivation needs.

(H1) **Poincaré covariance.** There is a unitary representation $U(\Lambda, a)$ of the Poincaré group on the Hilbert space, with P^μ the energy–momentum generators, $U(\mathbf{1}, a) = e^{iP\cdot a}$. The scalar field transforms as $U(\Lambda, a)\,\phi(x)\,U(\Lambda, a)^{-1} = \phi(\Lambda x + a)$. In particular spacetime translation acts as

$$\phi(x) = e^{iP\cdot x}\,\phi(0)\,e^{-iP\cdot x}. \tag{14.102}$$

(H2) **Spectral condition and a unique vacuum.** The joint spectrum of P^μ lies in the closed forward light cone, $p^0 \geq 0$ and $p^2 \geq 0$. There is a unique, Poincaré-invariant, normalized vacuum $|\Omega\rangle$ with $P^\mu |\Omega\rangle = 0$, $U(\Lambda, a)|\Omega\rangle = |\Omega\rangle$, and $\langle\Omega|\Omega\rangle = 1$.

(H3) **A stable one-particle state.** The spectrum contains an isolated mass-$m_{\rm ph}$ hyperboloid of one-particle states $|\mathbf{p}\rangle$ with $p^2 = m_{\rm ph}^2$, $p^0 > 0$, normalized covariantly (Eq. (3.193)),

$$\langle \mathbf{p}'|\mathbf{p}\rangle = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{p}'), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m_{\rm ph}^2}. \tag{14.103}$$

"Isolated" means $m_{\rm ph}^2$ is strictly below the threshold of the multiparticle continuum, so the one-particle states form a discrete sheet of the spectrum, not its edge.

(H4) **Canonical normalization and no tadpole.** The Heisenberg field obeys the equal-time canonical commutation relation inherited from the Lagrangian (Eq. (9.7)), $\big[\phi(t,\mathbf{x}), \dot\phi(t,\mathbf{y})\big] = i\,\delta^3(\mathbf{x}-\mathbf{y})$, and the vacuum expectation value of the field vanishes, $\langle\Omega|\phi(x)|\Omega\rangle = 0$.

Remark 14.22 (On (H4)). The condition $\langle\Omega|\phi|\Omega\rangle = 0$ is automatic for $\lambda\phi^4$ because the Lagrangian is invariant under the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ (ϕ^4 theory, Chapter 7), which is unbroken in the symmetric phase $\lambda > 0$, $m^2 > 0$; the vacuum is $\mathbb{Z}_2$-even and ϕ is $\mathbb{Z}_2$-odd, so its expectation value must vanish. In a theory where it did not, one simply works with the shifted field $\phi - \langle\Omega|\phi|\Omega\rangle$, which has zero vacuum value and identical commutators; the analysis below applies verbatim to the shifted field. The canonical normalization of the kinetic term, coefficient $\frac{1}{2}$, is what fixes the right-hand side of the equal-time commutator to be exactly $i\,\delta^3$ with no extra constant, and it is this that will pin the sum rule and hence the bound on Z.

The physical mass $m_{\rm ph}$ named in (H3) is, by definition, the pole position of the exact propagator; we will see this emerge below. It is logically distinct from the Lagrangian parameter m, which is merely the coefficient in $\mathcal{L}$ and which we will henceforth call the *bare* mass. The two coincide only when interactions are switched off; the shift $m \to m_{\rm ph}$ generated by λ is mass renormalization, computed at one loop in Chapter 22.

14.4.2 Inserting a complete set of states

The strategy is the oldest one in spectral analysis: take the operator product, slip the resolution of the identity between the two fields, and let Poincaré covariance dictate everything that can possibly appear. We organize the complete set of states by their total four-momentum.

By (H2) and (H3), the physical spectrum consists of the vacuum, the one-particle hyperboloid, and a continuum of multiparticle states. Group all non-vacuum states into Lorentz-invariant families labelled by an index λ (which ranges over the one-particle states and over every multiparticle configuration); within each family the states $|\lambda_{\mathbf{p}}\rangle$ are obtained by boosting a zero-momentum representative $|\lambda_0\rangle$ and carry total four-momentum p^μ with $p^2 = M_\lambda^2$, $p^0 = E_{\mathbf{p}}(\lambda) = \sqrt{\mathbf{p}^2 + M_\lambda^2} > 0$. The completeness relation on the physical Hilbert space (Theorem 3.44) reads, with the Lorentz-invariant measure Eq. (3.191),

$$\mathbf{1} = |\Omega\rangle\langle\Omega| + \sum_\lambda \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}(\lambda)}\; |\lambda_{\mathbf{p}}\rangle\langle\lambda_{\mathbf{p}}|\,, \tag{14.104}$$

each family normalized exactly as in Eq. (14.103) with $m_{\mathrm{ph}} \to M_\lambda$. Here $\sum_\lambda$ is shorthand for a sum over discrete families and an integral over the internal continuous parameters of each multiparticle family; the only feature we will use is that the invariant masses M_λ are real and bounded below.

Insert Eq. (14.104) into the Wightman function $\langle\Omega|\phi(x)\phi(y)|\Omega\rangle$ (the ordinary, not time-ordered, product). The vacuum term drops by (H4): $\langle\Omega|\phi(x)|\Omega\rangle\langle\Omega|\phi(y)|\Omega\rangle = 0$. What remains is

$$\langle\Omega|\phi(x)\phi(y)|\Omega\rangle = \sum_\lambda \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}(\lambda)}\; \langle\Omega|\phi(x)|\lambda_{\mathbf{p}}\rangle\langle\lambda_{\mathbf{p}}|\phi(y)|\Omega\rangle\,. \tag{14.105}$$

Now use translation covariance Eq. (14.102) together with $P|\Omega\rangle = 0$ and $P^\mu|\lambda_{\mathbf{p}}\rangle = p^\mu|\lambda_{\mathbf{p}}\rangle$:

$$\langle\Omega|\phi(x)|\lambda_{\mathbf{p}}\rangle = \Big\langle\Omega\Big|e^{iP\cdot x}\phi(0)e^{-iP\cdot x}\Big|\lambda_{\mathbf{p}}\Big\rangle = e^{-ip\cdot x}\,\langle\Omega|\phi(0)|\lambda_{\mathbf{p}}\rangle\,, \tag{14.106}$$

and similarly $\langle\lambda_{\mathbf{p}}|\phi(y)|\Omega\rangle = e^{+ip\cdot y}\langle\lambda_{\mathbf{p}}|\phi(0)|\Omega\rangle$. The two single-field matrix elements at the origin are complex conjugates, so their product is $\big|\langle\Omega|\phi(0)|\lambda_{\mathbf{p}}\rangle\big|^2$.

The final reduction is Lorentz invariance applied to that matrix element. Because $\phi(0)$ is a scalar, $U(\Lambda)\phi(0)U(\Lambda)^{-1} = \phi(0)$, and because the vacuum is invariant, $\langle\Omega|\,U(\Lambda) = \langle\Omega|$, we have for any Lorentz transformation Λ

$$\langle\Omega|\phi(0)|\lambda_{\mathbf{p}}\rangle = \langle\Omega|\phi(0)\,U(\Lambda)|\lambda_0\rangle = \langle\Omega|U(\Lambda)\,\phi(0)|\lambda_0\rangle = \langle\Omega|\phi(0)|\lambda_0\rangle\,, \tag{14.107}$$

choosing $\Lambda = L(p)$, the boost that carries the rest-frame momentum to $\mathbf{p}$, so that $U(L(p))\,|\lambda_0\rangle = |\lambda_{\mathbf{p}}\rangle$. The matrix element is therefore *independent of* $\mathbf{p}$: it depends on the family only through its rest-frame value $\langle\Omega|\phi(0)|\lambda_0\rangle$, a single complex number for each λ.

Collect the constants and the $\mathbf{p}$-dependence separately. Define the *spectral density*

$$\rho(M^2) \;\equiv\; \sum_\lambda (2\pi)\,\delta(M^2 - M_\lambda^2)\,\big|\langle\Omega|\phi(0)|\lambda_0\rangle\big|^2. \tag{14.108}$$

By construction $\rho(M^2) \geq 0$ pointwise, since it is a sum of δ-functions weighted by squared moduli; this positivity is the linchpin of the whole story and we record it as a proposition below. With Eq. (14.108), the family sum in Eq. (14.105) reorganizes as an integral over a single variable M^2:

$$\begin{aligned}\langle\Omega|\phi(x)\phi(y)|\Omega\rangle &= \sum_{\lambda} \big|\langle\Omega|\phi(0)|\lambda_0\rangle\big|^2 \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}(\lambda)}\, e^{-ip\cdot(x-y)}\Big|_{p^0=E_{\mathbf{p}}(\lambda)} \\ &= \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\, \underbrace{\int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-ip\cdot(x-y)}\Big|_{p^0=\sqrt{\mathbf{p}^2+M^2}}}_{=\; D_+(x-y;M^2)}.\end{aligned} \tag{14.109}$$

The inner integral is exactly the positive-frequency Wightman function $D_+(x-y;M^2)$ of a *free* scalar of mass M — the same object built in Chapter 9 (Eq. (9.148)), now read at the variable mass M. We have proved the spectral representation of the Wightman function: the interacting two-point function is a continuous, positively-weighted superposition of free Wightman functions of all masses.

14.4.3 The Källén–Lehmann representation

Restoring the time ordering is now immediate, because time ordering acts identically inside each fixed-mass term. The time-ordered product is $\mathcal{T}\phi(x)\phi(y) = \theta(x^0-y^0)\phi(x)\phi(y) + \theta(y^0-x^0)\phi(y)\phi(x)$, and applying Eq. (14.109) to each ordering gives

$$\langle\Omega|\mathcal{T}\phi(x)\phi(y)|\Omega\rangle = \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\,\Big[\theta(x^0-y^0)D_+(x-y;M^2)+\theta(y^0-x^0)D_+(y-x;M^2)\Big]. \tag{14.110}$$

The bracket is, term by term, precisely the construction of the *free* Feynman propagator of mass M from its Wightman functions (Eq. (9.153)). Hence the bracket equals $D_F(x-y;M^2)$, and we arrive at the central result.

Theorem 14.23 (Källén–Lehmann spectral representation)**.** *Under hypotheses* (H1)–(H4)*, the exact time-ordered two-point function of the interacting scalar field is*

$$\boxed{\langle\Omega|\mathcal{T}\phi(x)\phi(y)|\Omega\rangle = \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\,D_F(x-y;M^2)} \tag{14.111}$$

where $D_F(x-y;M^2)$ is the free Feynman propagator of mass M (Eq. (9.154)) *and the spectral density $\rho(M^2)$ of Eq. (14.108) satisfies $\rho(M^2) \geq 0$. Equivalently,*

Fourier transforming term by term with $\int d^4x\, e^{ip\cdot x} D_F(x;M^2) = i/(p^2 - M^2 + i\varepsilon)$, *the momentum-space propagator is*

$$\boxed{\widetilde{G}(p) \equiv \int d^4x\; e^{ip\cdot x}\; \langle\Omega|\mathcal{T}\phi(x)\phi(0)|\Omega\rangle = \int_0^\infty \frac{dM^2}{2\pi}\, \frac{i\,\rho(M^2)}{p^2 - M^2 + i\varepsilon}}. \tag{14.112}$$

Proof. Equation (14.111) was assembled in Eqs. (14.109) and (14.110); positivity of ρ is manifest from Eq. (14.108). For Eq. (14.112), interchange the (absolutely convergent, in the distributional sense) M^2 and x integrations and use the momentum-space form of the free Feynman propagator Eq. (9.154): $\int d^4x\, e^{ip\cdot x} \int \frac{d^4q}{(2\pi)^4} \frac{i\,e^{-iq\cdot x}}{q^2 - M^2 + i\varepsilon} = \frac{i}{p^2 - M^2 + i\varepsilon}$. The $i\varepsilon$ is inherited unchanged from the free propagators, which is consistent because the analytic continuation that defines the Feynman contour is the same for every mass M (Section 5.4.1). □

Proposition 14.24 (Positivity of the spectral density)**.** $\rho(M^2) \geq 0$ *for all* M^2, *and* $\rho(M^2) = 0$ *for* $M^2 < 0$ *and (by the spectral condition (H2)) for* M^2 *below the lowest physical invariant mass.*

Proof. Immediate from Eq. (14.108): each term is a non-negative number $|\,\langle\Omega|\phi(0)|\lambda_0\rangle\,|^2$ times a δ-function supported at the physical mass $M_\lambda^2 \geq 0$. There are no states with $M_\lambda^2 < 0$ by (H2), so ρ vanishes there. □

Positivity is not a technical decoration. It is the statement that the spectral density is a genuine probability-like weight — it measures how strongly the field ϕ acting on the vacuum overlaps with a physical state of squared mass M^2. Everything physical we extract below (the sign of the residue, the bound on Z) is a consequence of $\rho \geq 0$.

Physical Insight: the field as a state-creation amplitude

Read Eq. (14.108) backwards: $\rho(M^2)\, dM^2/2\pi$ is the total squared amplitude for the local operation "act once with $\phi(0)$ on the vacuum" to produce a physical state of invariant mass in $[M, M + dM]$. A *free* field can only make a single quantum: all of ρ sits at one mass. An *interacting* field, applied once, also shakes loose multiparticle states — $\phi(0)\,|\Omega\rangle$ is a superposition of one particle, three particles, five particles, … — and ρ spreads out to record exactly that. The Källén–Lehmann representation is the precise bookkeeping of this spreading.

14.4.4 The pole, the continuum, and the residue Z

We now separate the one-particle contribution, which is the part LSZ will use. By (H3) the one-particle states form an isolated family with $M_\lambda^2 = m_{\text{ph}}^2$. Its rest-frame matrix element defines the *field-strength renormalization.*

Definition 14.25 (Field-strength renormalization Z)**.** The square root of the residue, $\sqrt{Z}$, is the amplitude for the field to create the one-particle state from the vacuum:

$$\langle \Omega | \phi(0) | \mathbf{p} \rangle \;\equiv\; \sqrt{Z}, \qquad Z \equiv \left| \langle \Omega | \phi(0) | \mathbf{p} \rangle \right|^2 \geq 0, \tag{14.113}$$

a real non-negative constant independent of $\mathbf{p}$ by Eq. (14.107). (Any phase in the matrix element is absorbed into the definition of the one-particle state; Z is the modulus squared.)

Splitting the one-particle family off the rest of the sum in Eq. (14.108) gives the canonical decomposition of the spectral density,

$$\boxed{\rho(M^2) = 2\pi\, Z\, \delta(M^2 - m_{\mathrm{ph}}^2) + \rho_{\mathrm{cont}}(M^2)} \tag{14.114}$$

where $\rho_{\mathrm{cont}}(M^2) \geq 0$ collects all multiparticle (and any bound-state) contributions and, by the spectral condition, is supported at and above the multiparticle threshold $M^2 \geq M_{\mathrm{th}}^2$. For ϕ^4 theory in the symmetric phase the $\mathbb{Z}_2$ symmetry forces ϕ to create only odd numbers of particles, so the lightest multiparticle state with nonzero overlap has three quanta and the continuum begins at $M_{\mathrm{th}}^2 = (3m_{\mathrm{ph}})^2$; in a theory without that selection rule the threshold would be $(2m_{\mathrm{ph}})^2$. Inserting Eq. (14.114) into the momentum-space representation Eq. (14.112) and doing the δ-function integral isolates the pole:

$$\widetilde{G}(p) = \frac{iZ}{p^2 - m_{\mathrm{ph}}^2 + i\varepsilon} + \int_{M_{\mathrm{th}}^2}^{\infty} \frac{dM^2}{2\pi}\, \frac{i\,\rho_{\mathrm{cont}}(M^2)}{p^2 - M^2 + i\varepsilon}. \tag{14.115}$$

The two terms have completely different analytic characters in the complex p^2 plane, and this difference is the whole point.

- The first term is an *isolated simple pole* at $p^2 = m_{\mathrm{ph}}^2$ with residue iZ. It is the imprint of the stable single particle.
- The second term is analytic except for a *branch cut* along the real axis from $p^2 = M_{\mathrm{th}}^2$ to $+\infty$. Indeed, for p^2 approaching the cut, the denominator $p^2 - M^2 + i\varepsilon$ produces, via $\frac{1}{x+i\varepsilon} = \mathrm{PV}\,\frac{1}{x} - i\pi\delta(x)$, a discontinuity proportional to $\rho_{\mathrm{cont}}(p^2)$. Concretely, the discontinuity across the cut measures the continuum spectral density directly,

  $$\mathrm{disc}\,\widetilde{G}(p^2) \equiv \widetilde{G}(p^2 + i\varepsilon) - \widetilde{G}(p^2 - i\varepsilon) = \rho_{\mathrm{cont}}(p^2) \quad \text{for } p^2 > M_{\mathrm{th}}^2, \tag{14.116}$$

  which is the announced result that the multiparticle continuum is encoded in the imaginary part of the exact propagator; this same discontinuity reappears physically as the optical theorem in the final section of the chapter.

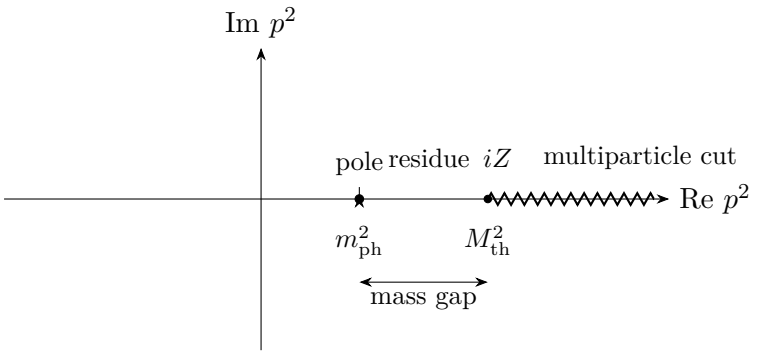


Figure 14.3: Analytic structure of the exact momentum-space propagator $\widetilde{G}(p^2)$ in the complex p^2 plane, as dictated by the Källén–Lehmann representation Eq. (14.115). A single isolated pole sits at the physical mass-squared $m^2_{\rm ph}$ with residue iZ; the multiparticle continuum produces a branch cut beginning at the threshold $M^2_{\rm th}$. The finite gap between them — guaranteed by hypothesis (H3) — is what lets LSZ pick the pole out cleanly. For ϕ^4 theory in the symmetric phase, $M^2_{\rm th} = (3m_{\rm ph})^2$.

The structure of Fig. 14.3 is the result of this section. Near the pole, the continuum term is regular, so the exact propagator behaves as

$$\boxed{\widetilde{G}(p) \xrightarrow{p^2 \to m^2_{\rm ph}} \frac{iZ}{p^2 - m^2_{\rm ph} + i\varepsilon}} \tag{14.117}$$

This is the single fact LSZ requires: the full propagator has a simple pole at the physical mass, with residue iZ rather than the free residue i. The number Z measures, by Eq. (14.113), how much of a genuine one-particle state the interacting field manufactures when it acts once on the vacuum; the deficit $1 - Z$ is the probability that it makes a multiparticle splash instead. We now prove that this interpretation is forced, by bounding Z between 0 and 1.

14.4.5 The sum rule and the bound $0 \le Z \le 1$

The bound comes from confronting the spectral representation with the one piece of information we have not yet used: the canonical commutator (H4). The equal-time commutator is a known c-number, and the spectral representation expresses the *same* commutator as an integral over ρ. Equating the two yields a sum rule that constrains the total spectral weight.

Take the spectral form of the Wightman function Eq. (14.109) and build the vacuum expectation value of the commutator,

$$\langle\Omega|[\phi(x),\phi(y)]|\Omega\rangle = \int_0^\infty \frac{dM^2}{2\pi}\,\rho(M^2)\,\Big[D_+(x-y;M^2) - D_+(y-x;M^2)\Big]. \tag{14.118}$$

The bracket is the free Pauli–Jordan commutator function of mass M; for a canonically normalized free field of any mass it obeys the universal equal-time identities

$$\big[D_+(x-y;M^2)-D_+(y-x;M^2)\big]\Big|_{x^0=y^0}=0, \qquad \partial_{x^0}\big[\cdots\big]\Big|_{x^0=y^0}=-i\,\delta^3(\mathbf{x}-\mathbf{y}), \tag{14.119}$$

the second of which is mass-*independent* — it is just the statement that every canonically normalized free scalar has $\big[\phi(t,\mathbf{x}),\dot\phi(t,\mathbf{y})\big]=i\delta^3$ regardless of its mass, a fact established for the free field in Chapter 9. (The mass-independence is the crucial point: the right-hand side carries no factor of M, so it pulls straight out of the M^2 integral.) Differentiating Eq. (14.118) with respect to x^0 and setting $x^0=y^0$,

$$\Big\langle\Omega\Big|\big[\phi(t,\mathbf{x}),\dot\phi(t,\mathbf{y})\big]\Big|\Omega\Big\rangle=\int_0^\infty\frac{dM^2}{2\pi}\,\rho(M^2)\,\big(-i\,\delta^3(\mathbf{x}-\mathbf{y})\big)\cdot(-1)=i\,\delta^3(\mathbf{x}-\mathbf{y})\int_0^\infty\frac{dM^2}{2\pi}\,\rho(M^2). \tag{14.120}$$

But the left-hand side is, by the canonical commutator (H4), exactly $i\,\delta^3(\mathbf{x}-\mathbf{y})$ — here we used that $\big[\phi,\dot\phi\big]$ is a c-number for the free building blocks, so its vacuum expectation value equals the operator itself; more carefully, the canonical commutator is an operator identity whose vacuum sandwich is $i\delta^3$. Matching the two sides forces the *spectral sum rule*

$$\boxed{\int_0^\infty\frac{dM^2}{2\pi}\,\rho(M^2)=1} \tag{14.121}$$

The bound now follows in one line. Insert the decomposition Eq. (14.114) into the sum rule:

$$1=\int_0^\infty\frac{dM^2}{2\pi}\,\rho(M^2)=Z+\int_{M_{\rm th}^2}^\infty\frac{dM^2}{2\pi}\,\rho_{\rm cont}(M^2). \tag{14.122}$$

Theorem 14.26 (Bound on the field-strength renormalization)**.** *The residue Z of the one-particle pole satisfies*

$$0\le Z\le 1, \tag{14.123}$$

with $Z=1$ if and only if the continuum is absent ($\rho_{\rm cont}\equiv 0$), which holds precisely for a free field. For any genuinely interacting theory with a nontrivial multiparticle continuum, $Z<1$ strictly.

Proof. $Z\ge 0$ is part of Theorem 14.25, being a squared modulus. From Eq. (14.122), $Z=1-\int_{M_{\rm th}^2}^\infty\frac{dM^2}{2\pi}\rho_{\rm cont}(M^2)$; by Theorem 14.24 the integrand is non-negative, so the subtracted integral is ≥ 0 and therefore $Z\le 1$. Equality $Z=1$ requires the continuum integral to vanish; since $\rho_{\rm cont}\ge 0$, this forces $\rho_{\rm cont}\equiv 0$ almost everywhere, i.e. $\phi(0)\,|\Omega\rangle$ has no multiparticle component, which is the defining property of a free field. $\square$

Physical Insight: Z as the single-particle probability

The sum rule Eq. (14.121) says the total probability for $\phi(0)\,|\Omega\rangle$ to land somewhere in the physical spectrum is 1 (canonical normalization). Equation (14.122) then partitions that unit probability: Z is the fraction that lands on a single particle, and $1 - Z$ is the fraction that lands on the multiparticle continuum. Interactions can only *redistribute* weight from the pole into the continuum — never create weight at the pole beyond unity — which is the physical content of $Z \leq 1$. This is why $\sqrt{Z}$ is the natural "efficiency" of the field as a single-particle source, and why exactly one factor of $\sqrt{Z}$ per external leg will appear in LSZ.

Caution: Z is scheme- and normalization-dependent

The bound $0 \leq Z \leq 1$ relies on the canonical normalization (H4): the field whose commutator is exactly $i\delta^3$. In renormalized perturbation theory one often rescales to a renormalized field $\phi_r = Z^{-1/2}\phi$, whose two-point function has residue 1 at the pole by construction; for that field the "Z" is moved into the relation between ϕ_r and the canonical ϕ. The number Z is also, individually, ultraviolet-divergent in a cutoff theory and finite only after renormalization; what is physical is the pole position m^2_{ph} and the residues of *S-matrix elements*, in which the $\sqrt{Z}$ factors cancel against the field normalization, as the next section makes explicit. The bound holds for the bare-canonical field at any fixed regularization; do not read $0 \leq Z \leq 1$ as a statement about a renormalized residue normalized to unity.

14.4.6 Worked examples

Worked Example 14.27 (The free field saturates the bound)**.** Set $\lambda = 0$ in ϕ^4 theory, so ϕ is the free scalar of Chapter 9 with mass m. We verify that the Källén–Lehmann machinery reproduces the free propagator and gives $Z = 1$, $\rho_{\text{cont}} = 0$, as it must.

For the free field, $\phi(0)$ acting on the vacuum produces exactly one quantum: from the mode expansion (Theorem 9.6), $\phi(0)\,|0\rangle = \int \frac{d^3p}{(2\pi)^3 2E_{\mathbf{p}}}\,|\mathbf{p}\rangle$, a pure one-particle state with no multiparticle admixture. The only family λ with nonzero $\langle 0|\phi(0)|\lambda_0\rangle$ is the single particle, and the rest-frame matrix element is $\langle 0|\phi(0)|\mathbf{p}=0\rangle = 1$ in the conventions of Chapter 9 (the field is its own creation amplitude, normalized by the canonical commutator). Hence by Eq. (14.108)

$$\rho_{\text{free}}(M^2) = 2\pi\,\delta(M^2 - m^2), \tag{14.124}$$

i.e. $Z = 1$ and $\rho_{\text{cont}} \equiv 0$. The sum rule Eq. (14.121) is satisfied: $\int \frac{dM^2}{2\pi}\, 2\pi\delta(M^2 -$

$m^2) = 1$. Substituting Eq. (14.124) into the position-space representation Eq. (14.111),

$$\langle 0|\mathcal{T}\phi(x)\phi(y)|0\rangle = \int_0^\infty \frac{dM^2}{2\pi}\, 2\pi\delta(M^2 - m^2)\, D_F(x-y;M^2) = D_F(x-y;m^2), \tag{14.125}$$

which is exactly the free Feynman propagator Eq. (9.154). The representation is consistent, and the free field is the unique case that saturates $Z = 1$. In the complex p^2 plane of Fig. 14.3, the cut has disappeared and only the pole at m^2 with residue i remains — there is nothing else in the free spectrum for ϕ to reach.

Worked Example 14.28 (ϕ^4 theory: Z at tree level and beyond)**.** We continue ϕ^4 theory (Theorem 14.27) and ask what interactions do to Z. At tree level, the time-ordered two-point function is computed from the Gell-Mann–Low expansion of the previous section; the $O(\lambda^0)$ term is the free propagator, so

$$Z = 1 + O(\lambda). \tag{14.126}$$

The $O(\lambda)$ correction to the propagator is the tadpole diagram, in which the single quartic vertex closes a loop on the propagator line. Crucially, this $O(\lambda)$ self-energy is a *momentum-independent constant* (the loop integral does not depend on the external momentum p). A p-independent self-energy shifts the *location* of the pole — it is a mass renormalization, $m^2 \to m^2_{\rm ph} = m^2 + O(\lambda)$ — but it cannot change the *residue*, because the residue is governed by the p-dependence of the self-energy near the pole, $Z^{-1} = 1 - \Sigma'(m^2_{\rm ph})$, and a constant has vanishing derivative. Therefore the first correction to Z vanishes:

$$Z = 1 + O(\lambda^2). \tag{14.127}$$

The leading nontrivial shift is the $O(\lambda^2)$ "sunset" contribution, the first diagram whose self-energy depends on p, and it is computed in full in Chapter 19 as part of the renormalization of ϕ^4; the bound Eq. (14.123) guarantees in advance that the $O(\lambda^2)$ coefficient is *negative*, since the now-open three-particle continuum ($M^2_{\rm th} = (3m_{\rm ph})^2$) carries away positive spectral weight, forcing $Z < 1$. We prove no number here — the loop integral belongs to Chapter 19 — but we have located exactly where the first correction lives and fixed its sign.

14.4.7 Handing the pole to LSZ

We have established, with no recourse to perturbation theory, that the exact momentum-space propagator of any theory satisfying (H1)–(H4) has an isolated simple pole at the physical mass-squared $m^2_{\rm ph}$ with residue iZ, $0 \le Z \le 1$, sitting below a multiparticle continuum, as displayed in Eq. (14.117) and Fig. 14.3. The next section makes this structure do work: the LSZ reduction formula shows that an n-point time-ordered correlator develops a pole in each external momentum

at exactly $p_i^2 = m_{\mathrm{ph}}^2$, that the residue at the simultaneous on-shell limit is the S-matrix element, and that each external leg contributes one factor of $\sqrt{Z}$ — the very amplitude $\langle\Omega|\phi(0)|\mathbf{p}\rangle = \sqrt{Z}$ defined here. The pole position, the residue Z, and the external $\sqrt{Z}$ factors are the three objects we now carry forward into LSZ.

14.5 The LSZ Reduction Formula

The two preceding sections placed the two halves of the bridge on opposite banks. The Gell-Mann–Low theorem (Theorem 14.18) gave us objects we can actually *compute*: time-ordered vacuum correlation functions of Heisenberg fields, $\langle\Omega|\mathcal{T}\phi(x_1)\cdots\phi(x_N)|\Omega\rangle$, rewritten as Dyson-series expectation values of free fields and therefore reducible to Wick contractions order by order in λ. The Källén–Lehmann representation (Theorem 14.23) then told us the exact analytic shape of the simplest such object: the interacting two-point function carries an isolated simple pole at $p^2 = m^2$, with a residue iZ set by the field-strength renormalization $0 < Z \leq 1$, sitting on top of a multiparticle continuum that begins at the threshold $p^2 = (3m)^2$ (the $\mathbb{Z}_2$ selection rule of ϕ^4 theory raises the generic $(2m)^2$ to $(3m)^2$; Section 14.4.4). What we still lack is the span connecting the two banks — a precise statement of how the numbers an experiment measures, the S-matrix elements $\langle f|\mathbf{S}|i\rangle$ of Theorem 14.11, are read off from these correlators. That span is the Lehmann–Symanzik–Zimmermann (LSZ) reduction formula, and building it is the business of this section. The punchline, established below in full, is disarmingly operational: an S-matrix element is the *amputated, connected, on-shell* momentum-space correlation function, dressed with one factor of $\sqrt{Z}$ for each external particle. Every later Feynman-rules chapter of Parts IV and V is, in the end, an industrialized procedure for evaluating the right-hand side of this one formula.

14.5.1 The asymptotic condition

The physical content of LSZ rests entirely on a single hypothesis about the long-time behaviour of the interacting field, the *asymptotic condition*. To state it we must first say precisely what free fields the interacting field is supposed to approach.

Recall from Section 14.2.2 that the in- and out-states are the complete bases that, in the remote past and future respectively, describe freely propagating physical particles of the *physical* mass m built on the interacting vacuum $|\Omega\rangle$. Associated with each basis is a free field that creates and annihilates those asymptotic quanta. The *in-field* $\phi_{\mathrm{in}}(x)$ is the free scalar field of mass m whose creation operators build the in-states, with mode expansion (the free expansion

of Eq. (9.42), written in the relativistic normalization of Eq. (3.193))

$$\phi_{\rm in}(x) \;=\; \int \widetilde{dp}\,\Big(a_{\rm in}(\mathbf{p})\,e^{-ip\cdot x} + a^{\dagger}_{\rm in}(\mathbf{p})\,e^{+ip\cdot x}\Big), \qquad p^0 = E_{\mathbf{p}} = \sqrt{\mathbf{p}^2+m^2}, \tag{14.128}$$

where $\widetilde{dp} \equiv d^3p/[(2\pi)^3\,2E_{\mathbf{p}}]$ is the Lorentz-invariant measure and the creation operators are normalized so that

$$|\mathbf{p};\text{in}\rangle = a^{\dagger}_{\rm in}(\mathbf{p})\,|\Omega\rangle\,, \qquad \Big[a_{\rm in}(\mathbf{p}), a^{\dagger}_{\rm in}(\mathbf{q})\Big] = (2\pi)^3\,2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q}), \tag{14.129}$$

in agreement with $\langle \mathbf{p};\text{in}|\mathbf{q};\text{in}\rangle = (2\pi)^3\,2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})$ of Eq. (3.193). The *out-field* $\phi_{\rm out}(x)$ is defined identically with "in" replaced by "out". Both are honest free fields of mass m: each satisfies the free Klein–Gordon equation $(\Box + m^2)\phi_{\rm in/out} = 0$ and each has the textbook canonical commutators.

The interacting Heisenberg field $\phi(x)$, by contrast, satisfies the full nonlinear equation of motion $(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3$ and does *not* create single-particle states cleanly — acting on the vacuum it produces a single particle *and* a cloud of multiparticle states, exactly the structure the Källén–Lehmann continuum records. The asymptotic condition is the statement that, nevertheless, in the distant past and future the interacting field settles down onto a free field, up to the normalization $\sqrt{Z}$ already isolated by the spectral representation.

Definition 14.29 (Asymptotic condition; weak form)**.** There exists a constant Z, $0 < Z \le 1$, equal to the Källén–Lehmann residue of Theorem 14.23, such that the interacting field approaches the asymptotic free fields,

$$\phi(x) \;\xrightarrow{x^0\to-\infty}\; \sqrt{Z}\,\phi_{\rm in}(x), \qquad \phi(x) \;\xrightarrow{x^0\to+\infty}\; \sqrt{Z}\,\phi_{\rm out}(x), \tag{14.130}$$

where the limits hold in the *weak* sense: for any pair of normalizable states $|\alpha\rangle\,, |\beta\rangle$ in the physical Hilbert space, $\langle\alpha|\phi(x)|\beta\rangle \to \sqrt{Z}\,\Big\langle\alpha\Big|\phi_{\rm in/out}(x)\Big|\beta\Big\rangle$ as $x^0 \to \mp\infty$, with the spatial dependence smeared against a normalizable wavepacket profile.

The constant $\sqrt{Z}$ is forced, not chosen. The Källén–Lehmann analysis fixed the overlap of the interacting field with the physical one-particle state,

$$\langle\Omega|\phi(0)|\mathbf{p}\rangle = \sqrt{Z}, \qquad \text{equivalently} \qquad \langle\Omega|\phi(x)|\mathbf{p}\rangle = \sqrt{Z}\,e^{-ip\cdot x}, \tag{14.131}$$

(the x-dependence following from translation covariance, $\phi(x) = e^{iP\cdot x}\phi(0)e^{-iP\cdot x}$, acting on the momentum eigenstate). The asymptotic free field $\phi_{\rm in}$, by Eqs. (14.128) and (14.129), has $\langle\Omega|\phi_{\rm in}(x)|\mathbf{p};\text{in}\rangle = e^{-ip\cdot x}$. Matching the two one-particle matrix elements forces precisely the factor $\sqrt{Z}$ in Eq. (14.130); with that factor in place the interacting and asymptotic fields agree on the one-particle sector, which is all an asymptotic free particle can probe.

Why the limit must be weak

The arrows in Eq. (14.130) cannot be promoted to a *strong* operator limit ($\phi(x) \to \sqrt{Z}\,\phi_{\rm in}(x)$ as operators on the Hilbert space). If they could, ϕ and $\sqrt{Z}\,\phi_{\rm in}$ would generate unitarily equivalent field algebras and the interacting two-point function would have residue $Z = 1$ with no continuum — contradicting Theorem 14.23, which gives $Z < 1$ precisely because the interacting field also creates multiparticle states. This is the same circle of ideas as Haag's theorem (Section 14.2.6): the interaction picture does not exist in the naive strong sense. The escape is that scattering observables are matrix elements between fixed normalizable states, and the weak limit of Theorem 14.29 is all those observables require. Concretely, every plane-wave manipulation below is shorthand for a statement about the normalizable wavepackets Eq. (14.37); smearing converts the weak limit into ordinary convergence of numbers and, as we will see, makes the spatial surface terms in the integrations by parts vanish honestly.

14.5.2 Inverting the mode expansion

The reduction procedure extracts one asymptotic particle at a time, and to do so we need a formula that recovers a creation operator from the field. The standard device is the equal-time Klein–Gordon inner product. For any two functions write

$$A(x)\overset{\leftrightarrow}{\partial_0}B(x) \;\equiv\; A(x)\,\partial_0 B(x) - (\partial_0 A(x))\,B(x). \tag{14.132}$$

Lemma 14.30 (Inversion of the in-field expansion)**.** *The creation and annihilation operators of the in-field are recovered from $\phi_{\rm in}$ by the time-independent integrals*

$$a^\dagger_{\rm in}(\mathbf{p}) = -\,i\int d^3x\; e^{-ip\cdot x}\overset{\leftrightarrow}{\partial_0}\phi_{\rm in}(x), \qquad a_{\rm in}(\mathbf{p}) = +\,i\int d^3x\; e^{+ip\cdot x}\overset{\leftrightarrow}{\partial_0}\phi_{\rm in}(x), \tag{14.133}$$

with $p^0 = E_{\mathbf{p}}$ on shell, and the right-hand sides independent of the time x^0 at which the spatial integral is taken.

Proof. Insert Eq. (14.128) into the first integral and act with ∂_0 on each factor. For the annihilation piece $a_{\rm in}(\mathbf{q})e^{-iq\cdot x}$ we have $\partial_0 e^{-iq\cdot x} = -iE_{\mathbf{q}}\,e^{-iq\cdot x}$ and $\partial_0 e^{-ip\cdot x} = -iE_{\mathbf{p}}\,e^{-ip\cdot x}$, so

$$e^{-ip\cdot x}\overset{\leftrightarrow}{\partial_0}e^{-iq\cdot x} = e^{-ip\cdot x}(-iE_{\mathbf{q}})e^{-iq\cdot x} - (-iE_{\mathbf{p}})e^{-ip\cdot x}e^{-iq\cdot x} = -i\,(E_{\mathbf{q}}-E_{\mathbf{p}})\,e^{-i(p+q)\cdot x}. \tag{14.134}$$

The spatial integral $\int d^3x\, e^{i(\mathbf{p}+\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^{(3)}(\mathbf{p}+\mathbf{q})$ sets $\mathbf{q} = -\mathbf{p}$, hence $E_{\mathbf{q}} = E_{\mathbf{p}}$, and the prefactor $E_{\mathbf{q}} - E_{\mathbf{p}}$ vanishes: the annihilation piece does not

contribute. For the creation piece $a^\dagger_{\rm in}(\mathbf{q})e^{+iq\cdot x}$, with $\partial_0 e^{+iq\cdot x} = +iE_{\mathbf{q}}e^{+iq\cdot x}$,

$$e^{-ip\cdot x}\overset{\leftrightarrow}{\partial_0}e^{+iq\cdot x} = i\,(E_{\mathbf{q}}+E_{\mathbf{p}})\,e^{-i(p-q)\cdot x}, \tag{14.135}$$

and now $\int d^3x\, e^{-i(\mathbf{p}-\mathbf{q})\cdot\mathbf{x}} = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$ sets $\mathbf{q}=\mathbf{p}$, so $E_{\mathbf{q}}=E_{\mathbf{p}}$ and the time-dependent factor $e^{-i(E_{\mathbf{p}}-E_{\mathbf{q}})x^0}=1$ drops out — which is why the result is independent of x^0. Collecting,

$$\int d^3x\; e^{-ip\cdot x}\overset{\leftrightarrow}{\partial_0}\phi_{\rm in}(x) = \int\widetilde{dq}\; i\,(2E_{\mathbf{p}})\,(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})\,a^\dagger_{\rm in}(\mathbf{q}) = i\,a^\dagger_{\rm in}(\mathbf{p}), \tag{14.136}$$

where the factor $2E_{\mathbf{p}}$ cancels the $1/(2E_{\mathbf{q}})$ in $\widetilde{dq}$. Multiplying by $-i$ gives the first relation in Eq. (14.133); Hermitian conjugation, using $\phi^\dagger_{\rm in}=\phi_{\rm in}$ and $(-i)^*=+i$, gives the second. □

The identical formulae hold with "in" replaced by "out". We now use them on the interacting field, where the asymptotic condition makes them meaningful only as $x^0\to\mp\infty$.

14.5.3 Single-particle reduction

We are ready for the central manoeuvre. Take an S-matrix element with at least one particle in the initial state,

$$S_{fi} = \langle\,\mathbf{q}_1\cdots\mathbf{q}_m;\text{out}\,|\,\mathbf{p}_1\cdots\mathbf{p}_n;\text{in}\,\rangle\,, \tag{14.137}$$

and isolate the incoming particle of momentum $\mathbf{p}_1$ by writing the in-ket as $|\mathbf{p}_1\cdots;\text{in}\rangle = a^\dagger_{\rm in}(\mathbf{p}_1)\,|\mathbf{p}_2\cdots;\text{in}\rangle$. Throughout we assume the kinematics is *genuinely inelastic*: no incoming momentum $\mathbf{p}_i$ coincides with any outgoing momentum $\mathbf{q}_j$. This is the generic case, and it is exactly the assumption that quarantines the trivial "no-scattering" pieces; we track what those pieces would have been and confirm they drop.

The strategy is to trade $a^\dagger_{\rm in}$ for $a^\dagger_{\rm out}$. The out-creation operator, acting on the out-bra to its left, would merely re-insert particle $\mathbf{p}_1$ among the outgoing particles — a disconnected, non-scattering contribution — so once we show that this term carries no scattering, what remains is an integral over the interacting field, which is computable. Define, for the *interacting* field, the time-dependent operator

$$F(x^0)\;\equiv\;-i\int d^3x\; e^{-ip_1\cdot x}\overset{\leftrightarrow}{\partial_0}\phi(x), \qquad p_1^0=E_{\mathbf{p}_1}. \tag{14.138}$$

By the asymptotic condition Eq. (14.130) and the inversion Eq. (14.133) (applied to $\phi_{\rm in}$ and $\phi_{\rm out}$),

$$\lim_{x^0\to-\infty}F(x^0)=\sqrt{Z}\,a^\dagger_{\rm in}(\mathbf{p}_1), \qquad \lim_{x^0\to+\infty}F(x^0)=\sqrt{Z}\,a^\dagger_{\rm out}(\mathbf{p}_1), \tag{14.139}$$

both understood weakly, between the states of Eq. (14.137). The difference of the two asymptotic creation operators is therefore the difference of the two boundary values of F, which the fundamental theorem of calculus expresses as a total time integral:

$$\sqrt{Z}\,\big(a^\dagger_{\rm in}(\mathbf{p}_1)-a^\dagger_{\rm out}(\mathbf{p}_1)\big)=F(-\infty)-F(+\infty)=-\int_{-\infty}^{\infty}dx^0\;\partial_0F(x^0). \tag{14.140}$$

It remains to evaluate ∂_0F. Differentiating Eq. (14.138) under the integral,

$$\partial_0F=-i\int d^3x\;\partial_0\Big(e^{-ip_1\cdot x}\overleftrightarrow{\partial_0}\,\phi\Big)=-i\int d^3x\,\Big(e^{-ip_1\cdot x}\,\partial_0^2\phi-(\partial_0^2e^{-ip_1\cdot x})\,\phi\Big), \tag{14.141}$$

where the two cross terms $\partial_0e^{-ip_1\cdot x}\,\partial_0\phi$ cancel between the two halves of $\overleftrightarrow{\partial_0}$. (As stressed above, these plane-wave manipulations stand for statements about the smeared wavepacket states of Eq. (14.37), for which the matrix elements have compact spatial support and the surface terms below are honestly zero.) Now use $\partial_0^2e^{-ip_1\cdot x}=(-iE_{\mathbf{p}_1})^2e^{-ip_1\cdot x}=-E^2_{\mathbf{p}_1}e^{-ip_1\cdot x}$ together with the on-shell relation $E^2_{\mathbf{p}_1}=\mathbf{p}_1^2+m^2$:

$$-(\partial_0^2e^{-ip_1\cdot x})\,\phi=(\mathbf{p}_1^2+m^2)e^{-ip_1\cdot x}\,\phi. \tag{14.142}$$

The term $\mathbf{p}_1^2\,e^{-ip_1\cdot x}$ is converted to a spatial Laplacian acting on the exponential, $\mathbf{p}_1^2\,e^{-ip_1\cdot x}=-\nabla^2e^{-ip_1\cdot x}$ (since $e^{-ip_1\cdot x}\propto e^{+i\mathbf{p}_1\cdot\mathbf{x}}$), and the Laplacian is moved onto the field by integrating by parts twice over $\mathbf{x}$. The surface terms vanish: for the smeared wavepacket states of Eq. (14.37) the field matrix elements have compact spatial support at any finite time, so there is no boundary at spatial infinity. Hence

$$\int d^3x\,(-\nabla^2e^{-ip_1\cdot x})\phi=\int d^3x\;e^{-ip_1\cdot x}(-\nabla^2\phi), \tag{14.143}$$

and, combining Eqs. (14.141) to (14.143),

$$\partial_0F=-i\int d^3x\;e^{-ip_1\cdot x}(\partial_0^2-\nabla^2+m^2)\phi=-i\int d^3x\;e^{-ip_1\cdot x}\,(\Box+m^2)\,\phi(x), \tag{14.144}$$

where $\Box=\partial_0^2-\nabla^2$ is the d'Alembertian in the mostly-minus metric. Inserting Eq. (14.144) into Eq. (14.140) and solving for $a^\dagger_{\rm in}$,

$$a^\dagger_{\rm in}(\mathbf{p}_1)=a^\dagger_{\rm out}(\mathbf{p}_1)+\frac{i}{\sqrt{Z}}\int d^4x\;e^{-ip_1\cdot x}\,(\Box+m^2)\,\phi(x). \tag{14.145}$$

This is the elementary reduction step: it expresses an incoming creation operator as an outgoing one plus a four-dimensional integral of the Klein–Gordon operator acting on the interacting field, with the universal weight $i/\sqrt{Z}$ and the plane-wave phase $e^{-ip_1\cdot x}$.

Substituting Eq. (14.145) into Eq. (14.137),

$$S_{fi} = \underbrace{\left\langle \mathbf{q};\text{out}\middle| a^\dagger_{\text{out}}(\mathbf{p}_1)\middle|\mathbf{p}_2\cdots;\text{in}\right\rangle}_{\text{disconnected}} + \frac{i}{\sqrt{Z}}\int d^4x_1\; e^{-ip_1\cdot x_1}\,(\Box_1+m^2)\,\langle \mathbf{q};\text{out}|\phi(x_1)|\mathbf{p}_2\cdots;\text{in}\rangle\,. \tag{14.146}$$

The first term is the promised disconnected piece, and we now show it carries no scattering. Acting to the left, $\langle \mathbf{q};\text{out}|\, a^\dagger_{\text{out}}(\mathbf{p}_1) = \big(a_{\text{out}}(\mathbf{p}_1)\,|\mathbf{q};\text{out}\rangle\big)^\dagger$. The annihilation operator removes one of the outgoing particles whenever its momentum matches $\mathbf{p}_1$,

$$a_{\text{out}}(\mathbf{p}_1)\,|\mathbf{q}_1\cdots\mathbf{q}_m;\text{out}\rangle = \sum_{j=1}^{m}(2\pi)^3\,2E_{\mathbf{p}_1}\,\delta^{(3)}(\mathbf{p}_1-\mathbf{q}_j)\;|\mathbf{q}_1\cdots\widehat{\mathbf{q}_j}\cdots\mathbf{q}_m;\text{out}\rangle\,, \tag{14.147}$$

the hat denoting omission. Every term is proportional to $\delta^{(3)}(\mathbf{p}_1-\mathbf{q}_j)$ and so vanishes under our inelastic assumption $\mathbf{p}_1\neq\mathbf{q}_j$. Physically these terms describe the incoming particle $\mathbf{p}_1$ sailing through untouched and reappearing among the outgoing particles; they are pieces of the identity part of $\mathbf{S}$ in the decomposition $\mathbf{S}=\mathbf{1}+i\mathcal{T}_{\text{r}}$ used implicitly in Theorem 14.11, not of the scattering part. For a genuine reaction they are absent, and Eq. (14.146) collapses to its second term: one external particle has been "reduced" into a Klein–Gordon-amputated matrix element of the interacting field.

What the reduction step is doing

The combination $(\Box+m^2)\phi$ is the source of the field — by the equation of motion it equals $-\frac{\lambda}{3!}\phi^3$, which is nonzero only where the field interacts. Reducing a leg therefore replaces the question "does an asymptotic particle of momentum $\mathbf{p}_1$ enter?" with the question "where, in spacetime, is a quantum of momentum $\mathbf{p}_1$ emitted or absorbed by the interaction?", Fourier-analysed by the plane wave $e^{-ip_1\cdot x_1}$. The factor $1/\sqrt{Z}$ compensates for the fact that the interacting field creates a one-particle state with amplitude $\sqrt{Z}$ rather than unity: it strips off the wavefunction renormalization so that exactly one properly normalized asymptotic particle is attached to the leg.

14.5.4 Induction to all legs, and the emergence of time ordering

To reduce a second incoming particle, $\mathbf{p}_2$, we repeat the manoeuvre on $\langle \mathbf{q};\text{out}|\phi(x_1)|\mathbf{p}_2\cdots;\text{in}\rangle$, again writing $|\mathbf{p}_2\cdots;\text{in}\rangle = a^\dagger_{\text{in}}(\mathbf{p}_2)\,|\mathbf{p}_3\cdots;\text{in}\rangle$ and using Eq. (14.145). There is one new bookkeeping point: the new field $\phi(x_2)$ enters through $a^\dagger_{\text{in}}(\mathbf{p}_2)$, whose defining limit is $x_2^0\to-\infty$. In that limit x_2^0 is earlier than *any* fixed time, in particular earlier than x_1^0, so when the reduced integral is assembled the operator

$\phi(x_2)$ stands to the *right* of $\phi(x_1)$ inside the matrix element. Symmetrically, reducing an outgoing particle $\mathbf{q}_j$ uses $a_{\text{out}}(\mathbf{q}_j)$, whose limit is $y_j^0 \to +\infty$, later than any fixed time, so its field $\phi(y_j)$ is placed to the *left*. The relevant inversion for an outgoing leg is the second relation in Eq. (14.133); running the same fundamental-theorem argument gives

$$a_{\text{out}}(\mathbf{q}) = a_{\text{in}}(\mathbf{q}) + \frac{i}{\sqrt{Z}} \int d^4y \, e^{+iq\cdot y} \, (\Box_y + m^2) \, \phi(y), \tag{14.148}$$

with the conjugate phase $e^{+iq\cdot y}$; the leftover $a_{\text{in}}(\mathbf{q})$ piece is again disconnected and vanishes for inelastic kinematics by the analogue of Eq. (14.147).

Each successive reduction places its field at the appropriate temporal extreme — earliest times to the right for incoming legs, latest times to the left for outgoing legs — and the product of all the fields therefore assembles itself, automatically, into a *time-ordered* product. This is the decisive structural fact: although nothing in the asymptotic condition mentions time ordering, the ordering is generated leg by leg by the very limits $x^0 \to \mp\infty$ that define the in- and out-operators. After all n incoming and m outgoing particles are reduced, every external state has been converted into a field with a Klein–Gordon operator and a plane wave, the vacuum $|\Omega\rangle$ is exposed on both sides, and we arrive at the master formula.

Theorem 14.31 (LSZ reduction formula, position space)**.** *For genuinely inelastic kinematics ($\mathbf{p}_i \neq \mathbf{q}_j$ for all i, j), the connected S-matrix element Eq. (14.137) is*

$$\langle\, \mathbf{q}_1 \cdots \mathbf{q}_m; \text{out} \,|\, \mathbf{p}_1 \cdots \mathbf{p}_n; \text{in} \,\rangle = \Big(\frac{i}{\sqrt{Z}}\Big)^{n+m} \prod_{i=1}^{n} \int d^4x_i \, e^{-ip_i\cdot x_i} (\Box_{x_i} + m^2) \prod_{j=1}^{m} \int d^4y_j \, e^{+iq_j\cdot y_j} (\Box_y$$
$$\times \; \langle\Omega|\, \mathcal{T}\, \phi(x_1) \cdots \phi(x_n)\, \phi(y_1) \cdots \phi(y_m) \,|\Omega\rangle \,. \tag{14.149}$$

The right-hand side is built entirely from a time-ordered vacuum correlation function of Heisenberg fields — precisely the object the Gell-Mann–Low theorem (Theorem 14.18) renders computable in perturbation theory. The bridge is complete: experiment on the left, Dyson-series calculation on the right.

14.5.5 Momentum space: amputation, poles, and the residue

The position-space formula Eq. (14.149) is exact but unwieldy; in practice one works in momentum space, where the Klein–Gordon operators act algebraically and the whole content of LSZ becomes a statement about poles. Define the momentum-space (Fourier-transformed) correlation function with the conventions of Theorem 5.18, incoming legs carried by $e^{-ip\cdot x}$ and outgoing by $e^{+iq\cdot y}$,

$$\mathcal{G}(p_1 \cdots p_n; q_1 \cdots q_m) \equiv \prod_i \int d^4x_i \, e^{-ip_i\cdot x_i} \prod_j \int d^4y_j \, e^{+iq_j\cdot y_j} \, \Big\langle \Omega \Big| \mathcal{T} \prod_i \phi(x_i) \prod_j \phi(y_j) \Big| \Omega \Big\rangle \,. \tag{14.150}$$

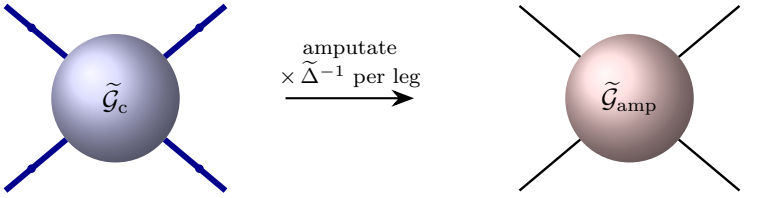


Figure 14.4: Amputation. The full connected Green's function $\widetilde{\mathcal{G}}_{\mathrm{c}}$ carries a complete interacting propagator (thick) on every external leg. Dividing off one full propagator per leg leaves the amputated function $\widetilde{\mathcal{G}}_{\mathrm{amp}}$. LSZ instructs us to amputate, go on shell, and dress each leg with a single $\sqrt{Z}$.

Translation invariance of the correlator forces overall energy–momentum conservation, $\mathcal{G} = (2\pi)^4\delta^4\big(\sum_j q_j - \sum_i p_i\big)\,\widetilde{\mathcal{G}}$, and the stripped object $\widetilde{\mathcal{G}}$ depends on the independent invariants. We need two definitions that organize the rest of Parts IV and V.

Definition 14.32 (Connected and amputated Green's functions)**.** A contribution to a correlation function is *connected* if its diagrammatic representation (Chapter 15) cannot be split into two pieces with no line joining them; equivalently, the *connected* correlator $\langle\Omega|\mathcal{T}\cdots|\Omega\rangle_{\mathrm{c}}$ is the part generated by $W[J] = -i\ln Z[J]$ (Section 10.5), with all factorized vacuum and lower-point pieces removed. A connected momentum-space Green's function is *amputated* when a full (interacting) propagator has been divided off each external leg:

$$\widetilde{\mathcal{G}}_{\mathrm{c}}(p_1\cdots;q_1\cdots) = \Big(\prod_i \widetilde{\Delta}(p_i)\Big)\Big(\prod_j \widetilde{\Delta}(q_j)\Big)\,\widetilde{\mathcal{G}}_{\mathrm{amp}}, \tag{14.151}$$

where $\widetilde{\Delta}(p) = \int d^4x\, e^{ip\cdot x}\,\langle\Omega|\mathcal{T}\phi(x)\phi(0)|\Omega\rangle$ is the exact two-point function, carrying the Källén–Lehmann pole $\widetilde{\Delta}(p) \to iZ/(p^2 - m^2 + i\epsilon)$ as $p^2 \to m^2$ (Theorem 14.23). $\widetilde{\mathcal{G}}_{\mathrm{amp}}$ is the *amputated, connected* Green's function.

We now translate Theorem 14.31 into momentum space. Each position-space leg contributes, by Eq. (14.150), the Fourier weight already built into $\mathcal{G}$; the Klein–Gordon operator $(\Box + m^2)$ acting on an incoming leg becomes, on $e^{-ip\cdot x}$,

$$(\Box + m^2)\,e^{-ip\cdot x} = (-p^2 + m^2)\,e^{-ip\cdot x} = -(p^2 - m^2)\,e^{-ip\cdot x}, \tag{14.152}$$

and identically for an outgoing leg. Therefore the position-space master formula is, in momentum space,

$$\langle f|\mathbf{S}|i\rangle_{\mathrm{c}} = \Big(\frac{i}{\sqrt{Z}}\Big)^{n+m}\prod_i[-(p_i^2 - m^2)]\prod_j[-(q_j^2 - m^2)]\;\mathcal{G}(p_1\cdots;q_1\cdots)\Big|_{\text{on shell}}, \tag{14.153}$$

all external momenta taken on shell, $p_i^2 = q_j^2 = m^2$. Naively the prefactors $(p_a^2 - m^2)$ all vanish on shell; the formula is nontrivial only because $\mathcal{G}$ itself *diverges* there, exactly fast enough to compensate. This is where the Källén–Lehmann pole earns its keep. By Eq. (14.151), every external leg of the connected $\mathcal{G}$ carries a full propagator $\widetilde{\Delta}(p_a)$, and as $p_a^2 \to m^2$,

$$\frac{i}{\sqrt{Z}}\,[-(p_a^2-m^2)]\,\widetilde{\Delta}(p_a) \xrightarrow{p_a^2\to m^2} \frac{i}{\sqrt{Z}}\,[-(p_a^2-m^2)]\,\frac{iZ}{p_a^2-m^2+i\epsilon} = \frac{i\cdot(-1)\cdot iZ}{\sqrt{Z}} = \sqrt{Z}. \tag{14.154}$$

Two things happen at once in Eq. (14.154): the explicit zero $(p_a^2 - m^2)$ cancels the propagator pole, leaving a finite limit, and any *non-pole* part of $\widetilde{\Delta}(p_a)$ – the entire Källén–Lehmann continuum, which is regular at $p_a^2 = m^2$ – is multiplied by the vanishing factor $(p_a^2 - m^2)$ and dies. Only the single-particle pole survives. Each external leg thus contributes precisely one factor of $\sqrt{Z}$, and what is left over after stripping the full propagators is the amputated function. Combining Eqs. (14.151), (14.153) and (14.154), we obtain the operational form of LSZ.

Theorem 14.33 (LSZ reduction formula, momentum space)**.** *Define the invariant amplitude $\mathcal{M}$ by separating the non-scattering part of the S-matrix, $\langle f|\mathbf{S}|i\rangle = \langle f|i\rangle + i(2\pi)^4\delta^4\big(\sum_j q_j - \sum_i p_i\big)\,\mathcal{M}$. Then $\mathcal{M}$ is the amputated, connected, on-shell Green's function dressed with one $\sqrt{Z}$ per external leg,*

$$\boxed{\,i\,\mathcal{M}\,(2\pi)^4\delta^4\Big(\sum_j q_j - \sum_i p_i\Big) = (\sqrt{Z})^{\,n+m}\,\widetilde{\mathcal{G}}_{\text{amp}}(p_1\cdots p_n; q_1\cdots q_m)\Big|_{\text{on shell}}\,} \tag{14.155}$$

with all $p_i^2 = q_j^2 = m^2$. Equivalently, the full connected momentum-space correlator factorizes near its on-shell poles as

$$\mathcal{G}_{\text{c}}(p_1\cdots;q_1\cdots) \xrightarrow{p_a^2\to m^2} \Big(\prod_i \frac{i\sqrt{Z}}{p_i^2-m^2+i\epsilon}\Big)\Big(\prod_j \frac{i\sqrt{Z}}{q_j^2-m^2+i\epsilon}\Big)\,i\,\mathcal{M}\,(2\pi)^4\delta^4\Big(\sum_j q_j - \sum_i p_i\Big) \tag{14.156}$$

so that the scattering amplitude is the multiple residue of the correlation function at its external on-shell poles.

The two displays say the same thing in complementary languages. Equation (14.156) is the analytic statement: the S-matrix lives in the residue of the $(n+m)$-fold pole of the time-ordered correlator, each pole supplied by an external particle going on shell. Equation (14.155) is the recipe an actual calculation follows: compute the connected correlator, throw away its external full propagators (amputate), set the external momenta on shell, and multiply by $\sqrt{Z}$ for each leg. The agreement of the two forms is immediate from Eq. (14.151): substituting it into Eq. (14.156) and cancelling the common propagator poles against the residues reproduces Eq. (14.155).

Why amputate, and why connected

Both qualifiers are forced, not stylistic. *Amputated:* a full external propagator left in place would supply an extra pole $iZ/(p^2 - m^2)$ on a leg that LSZ has already accounted for, double-counting the wavefunction renormalization and producing a spurious divergence on shell; amputation removes exactly the structure that the factors $(p^2 - m^2)$ in Eq. (14.153) are there to cancel. *Connected:* a disconnected contribution to the correlator factorizes into independent lower-point pieces, and a product of two correlators each conserving momentum cannot carry the single overall $\delta^4(\sum q - \sum p)$ that a genuine $(n+m)$-point amplitude requires — it has too many delta functions and is supported only on measure-zero kinematics where subsets of particles do not interact. These are the same no-scattering configurations removed by the inelastic assumption in Section 14.5.3; the connected projection discards them once and for all. To leading nontrivial order in λ, where $Z = 1 + O(\lambda^2)$ (the residue first moves at one loop, Chapter 19), the dressing $(\sqrt{Z})^{n+m}$ is unity and Eq. (14.155) reduces to "$i\mathcal{M}$ = amputated connected on-shell diagram", the rule used throughout Chapters 15 and 16.

14.5.6 A worked check: one leg, and the disconnected term made explicit

The single-leg case is the cleanest place to see every factor in Eq. (14.154) appear from an honest computation, and it simultaneously exhibits how the non-pole continuum is annihilated. It is worth doing in full.

Worked Example 14.34 (Reducing one leg against the exact two-point function)**.** Take the LSZ differential operator for a single incoming leg and apply it to the exact two-point function $\langle\Omega|\mathcal{T}\phi(x)\phi(y)|\Omega\rangle$, holding y fixed. We compute

$$R(y) \equiv \frac{i}{\sqrt{Z}} \int d^4x\, e^{-ip\cdot x}\,(\Box_x + m^2)\,\langle\Omega|\mathcal{T}\phi(x)\phi(y)|\Omega\rangle\,, \qquad p^0 = E_{\mathbf{p}}, \quad (14.157)$$

and verify it reproduces the one-particle wavefunction $\sqrt{Z}\,e^{-ip\cdot y}$ predicted by Eq. (14.131), confirming both the $\sqrt{Z}$ counting and the death of the continuum.

Step 1 — insert the spectral representation. By Theorem 14.23 the two-point function has the Fourier transform $\widetilde{\Delta}(k) = iZ/(k^2 - m^2 + i\epsilon) + \widetilde{\Delta}_{\text{cont}}(k)$, where $\widetilde{\Delta}_{\text{cont}}$ collects the multiparticle continuum and is *regular* at $k^2 = m^2$ (its spectral weight is supported at $k^2 \geq (3m)^2$ for ϕ^4 theory). Writing $\langle\Omega|\mathcal{T}\phi(x)\phi(y)|\Omega\rangle = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot(x-y)}\widetilde{\Delta}(k)$,

$$R(y) = \frac{i}{\sqrt{Z}} \int \frac{d^4k}{(2\pi)^4}\,\widetilde{\Delta}(k)\,e^{+ik\cdot y} \int d^4x\, e^{-ip\cdot x}\,(\Box_x + m^2)\,e^{-ik\cdot x}. \quad (14.158)$$

Step 2 — act with the Klein–Gordon operator and integrate over x. Using Eq. (14.152) in the form $(\Box_x+m^2)e^{-ik\cdot x} = -(k^2-m^2)e^{-ik\cdot x}$ and $\int d^4x\, e^{-i(p+k)\cdot x} = (2\pi)^4\delta^4(p+k)$, the k-integral collapses at $k=-p$:

$$R(y) = \frac{i}{\sqrt{Z}}\,[-(p^2-m^2)]\,\widetilde{\Delta}(-p)\,e^{-ip\cdot y}. \tag{14.159}$$

By Lorentz invariance $\widetilde{\Delta}$ depends only on p^2, so $\widetilde{\Delta}(-p)=\widetilde{\Delta}(p)$.

Step 3 — go on shell and take the residue. Substitute the spectral split:

$$R(y) = \frac{i}{\sqrt{Z}}\,[-(p^2-m^2)]\Big(\frac{iZ}{p^2-m^2+i\epsilon} + \widetilde{\Delta}_{\text{cont}}(p)\Big)e^{-ip\cdot y}. \tag{14.160}$$

The continuum term comes multiplied by (p^2-m^2) and, since $\widetilde{\Delta}_{\text{cont}}$ is finite at $p^2=m^2$, vanishes in the on-shell limit. The pole term gives $[-(p^2-m^2)]\cdot iZ/(p^2-m^2) = -iZ$. Hence

$$R(y)\Big|_{p^2\to m^2} = \frac{i}{\sqrt{Z}}\,(-iZ)\,e^{-ip\cdot y} = \sqrt{Z}\,e^{-ip\cdot y}, \tag{14.161}$$

exactly $\langle\Omega|\phi(y)|\mathbf{p}\rangle$ of Eq. (14.131). The leg has been reduced into a single, correctly normalized one-particle wavefunction: the explicit zero killed the continuum, the pole survived, and the booked factor $\sqrt{Z}$ emerged precisely as in Eq. (14.154).

The same arithmetic, run on every leg of an $(n+m)$-point function, is the content of Theorem 14.33; the worked case shows that nothing is hidden in the general statement beyond doing this calculation once per external particle. It also makes the role of Z tangible: had we forgotten the $1/\sqrt{Z}$, the reduced leg would have carried amplitude $-iZ\cdot i = Z$ rather than $\sqrt{Z}$, mis-normalizing the asymptotic particle by a factor $\sqrt{Z}$ on every leg — a mistake that propagates into every cross section.

14.5.7 Handoff

We now possess the master bridge of the chapter in its operational form, Eq. (14.155): an S-matrix element is the amputated, connected, on-shell momentum-space correlation function, weighted by one $\sqrt{Z}$ per external leg, with the disconnected and non-pole debris provably absent. Every ingredient on the right-hand side is computable — the correlator from Gell-Mann–Low and the Dyson series, the residue structure from Källén–Lehmann. In the next section we turn the crank for the first time on ϕ^4 theory: with $Z=1+O(\lambda^2)$, the leading-order $2\to 2$ amplitude reduces to a single amputated, connected, on-shell four-point function, and evaluating it will deliver the chapter's first piece of interacting physics, $i\mathcal{M} = -i\lambda$.

14.6 First Interacting Amplitude: $\phi\phi \to \phi\phi$ at $O(\lambda)$

Everything in this chapter so far has been machinery built but not yet turned. The interaction picture and the Dyson series gave a perturbative evolution operator; the Gell-Mann–Low theorem (Theorem 14.18) rewrote the Heisenberg-picture vacuum correlators of the interacting theory as Dyson-series expectation values of *free* fields, hence as objects Wick's theorem can evaluate; and the LSZ reduction formula (Theorem 14.31) converted the on-shell residues of those correlators into S-matrix elements, one factor of $\sqrt{Z}$ per external leg. The promise was that, chained together, these three results compute scattering amplitudes order by order in the coupling. This section redeems that promise. We take the frozen ϕ^4 Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\,\phi^4, \qquad \phi \text{ real},\ \lambda > 0, \tag{14.162}$$

and carry out, with every factor visible, the leading-order computation of the two-to-two amplitude $\phi\phi \to \phi\phi$. The answer is a single number, $i\mathcal{M} = -i\lambda$, and it is the first genuinely interacting result of the entire book: the point at which the apparatus stops being potential and starts predicting. We work throughout in the conventions in force — mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the Feynman $i\varepsilon$ of Eq. (9.154) — and we treat λ as small, expanding to first order.

The structure of the calculation is fixed once and for all by the three imported theorems, so we begin by assembling them into a single explicit recipe, then execute it.

14.6.1 Assembling the three machines into one recipe

The observable is an entry of the S-matrix between two-particle states. Recall (Theorem 14.31) that the nontrivial part of scattering is isolated by writing $\mathbf{S} = \mathbf{1} + (\mathbf{S}-\mathbf{1})$, the unit accounting for the particles missing each other; the invariant amplitude $\mathcal{M}$ is defined by stripping the overall energy–momentum delta from the connected matrix element,

$$\langle p_3\, p_4|\ (\mathbf{S}-\mathbf{1})\ |p_1\, p_2\rangle \;=\; i\,\mathcal{M}(p_1,p_2\to p_3,p_4)\,(2\pi)^4\,\delta^4(p_1+p_2-p_3-p_4), \tag{14.163}$$

with the one-particle states carrying the relativistic normalization $\langle p|k\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{k})$ of Eq. (3.193). Momenta p_1, p_2 are incoming, p_3, p_4 outgoing, all on shell: $p_i^2 = m^2$, $p_i^0 = E_{\mathbf{p}_i} > 0$.

The LSZ formula expresses the left-hand side of Eq. (14.163) as the multiple on-shell residue of the momentum-space time-ordered four-point function. It is convenient to state it in the *amputated* form. Introduce the all-incoming Fourier

transform of the connected four-point correlator,

$$\widetilde{G}^{(4)}(p_1,p_2,p_3,p_4) \;\equiv\; \int\Big(\prod_{i=1}^{4} d^4x_i\, e^{ip_i\cdot x_i}\Big)\,\langle\Omega|\,\mathcal{T}\,\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\,|\Omega\rangle_{\text{conn}}, \tag{14.164}$$

where $|\Omega\rangle$ is the interacting vacuum. The full correlator carries one factor $\sqrt{Z}\,i/(p_i^2-m^2)$ for each external leg as $p_i^2 \to m^2$ (the Källén–Lehmann pole, Theorem 14.31); LSZ instructs us to cancel those four propagator poles, divide out the four $\sqrt{Z}$'s, and take the legs on shell. Writing the result of cancelling the external propagators as the *amputated* correlator

$$\widetilde{G}^{(4)}_{\text{amp}}(p_1,\dots,p_4) \;\equiv\; \Big[\prod_{i=1}^{4}\frac{p_i^2-m^2}{i}\Big]\,\widetilde{G}^{(4)}(p_1,\dots,p_4), \tag{14.165}$$

the LSZ master statement specializes, for four external scalars, to

$$i\,\mathcal{M}\,(2\pi)^4\,\delta^4\Big(\textstyle\sum_i p_i\Big) \;=\; \frac{1}{\left(\sqrt{Z}\right)^4}\,\lim_{p_i^2\to m^2}\widetilde{G}^{(4)}_{\text{amp}}(p_1,\dots,p_4)\Big|_{p_3,p_4\ \text{outgoing}}. \tag{14.166}$$

Here the sum $\sum_i p_i$ runs over all-incoming momenta; converting the last two to physical outgoing momenta sends $p_3,p_4 \to -p_3,-p_4$ inside the delta, reproducing the conservation law of Eq. (14.163). Everything on the right of Eq. (14.166) is computable: the correlator by Gell-Mann–Low plus Wick, the factor Z by the same expansion. The recipe is therefore

(i) expand $\langle\Omega|\mathcal{T}\,\phi\phi\phi\phi|\Omega\rangle$ to the desired order in λ using Gell-Mann–Low (Theorem 14.18), turning it into free-field time-ordered correlators;

(ii) evaluate each free correlator by Wick's theorem (Theorem 10.25) as a sum over complete pairings into Feynman propagators (Theorem 9.24);

(iii) keep the *connected* part, Fourier transform, amputate the four external legs via Eq. (14.165), and take the on-shell limit;

(iv) divide by the factor $(\sqrt{Z})^4$ and read off $\mathcal{M}$ from Eq. (14.166).

Because the amputation Eq. (14.165) strips off *free* propagators $i/(p_i^2-m^2)$, each external leg of $\widetilde{G}^{(4)}_{\text{amp}}$ still retains one residual factor $\sqrt{Z}$ from the Källén–Lehmann residue; dividing by $(\sqrt{Z})^4$ removes them, exactly as the prose above demands. (Had we instead amputated with the *full* interacting propagator of residue iZ, as in Section 14.5, the extraction would carry a multiplicative $(\sqrt{Z})^4$.) At $O(\lambda)$ one has $Z = 1 + O(\lambda^2)$, so the distinction is immaterial for the amplitude computed below; it matters only once $Z \neq 1$.

We now carry out steps (i)–(iv) in full. The entire computation is the chapter's central worked example.

14.6.2 The leading-order four-point correlator

The interaction Hamiltonian obtained from Eq. (14.56) by Legendre transform (Sections 6.4 and 14.1.1) is, to the order we need, just minus the interaction Lagrangian density integrated over space: the quartic term contains no derivatives, so it contributes nothing to the canonical momentum and the Hamiltonian density inherits $+\frac{\lambda}{4!}\phi^4$ directly. In the interaction picture,

$$H_I(t) = \int d^3x\, \frac{\lambda}{4!}\,\phi_I(t,\mathbf{x})^4, \qquad \text{so} \qquad -i\int_{-\infty}^{\infty} dt\, H_I(t) = -\frac{i\lambda}{4!}\int d^4z\, \phi_I(z)^4, \tag{14.167}$$

where ϕ_I is the free field with the mode expansion and propagator of Chapter 9; from here on we drop the subscript I, all fields inside the Dyson series being free. The Gell-Mann–Low theorem (Theorem 14.18) gives the interacting four-point function as the ratio

$$\langle\Omega|\mathcal{T}\,\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)|\Omega\rangle = \frac{\langle 0|\mathcal{T}\Big\{\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\,\exp\big[-\frac{i\lambda}{4!}\int d^4z\,\phi(z)^4\big]\Big\}|0\rangle}{\langle 0|\mathcal{T}\exp\big[-\frac{i\lambda}{4!}\int d^4z\,\phi(z)^4\big]|0\rangle}, \tag{14.168}$$

with $|0\rangle$ the free vacuum and the time limits taken with the $T\to\infty(1-i\varepsilon)$ tilt that projects onto $|\Omega\rangle$. Expanding the exponential in powers of λ converts both numerator and denominator into sums of free time-ordered correlators, exactly the Wick-computable objects of Chapter 10.

We want the amplitude through $O(\lambda)$. Because the four-point amplitude vanishes at $\lambda = 0$ (free particles do not scatter), the leading term is $O(\lambda^1)$, and we expand the numerator to first order.

Worked Example 14.35 (The $O(\lambda)$ amplitude $\phi\phi \to \phi\phi$, end to end)**.** We carry the computation from Eq. (14.168) to the boxed number $i\mathcal{M} = -i\lambda$, displaying every Wick contraction, the combinatorial cancellation of the $1/4!$, the amputation, and the on-shell limit.

The order-λ^0 term and why it carries no scattering. At zeroth order the exponential in the numerator of Eq. (14.168) is the identity, the denominator is 1, and we are left with the free four-point function. Wick's theorem (Theorem 10.25) evaluates it as the sum over the three complete pairings of four fields,

$$\langle 0|\mathcal{T}\,\phi_1\phi_2\phi_3\phi_4|0\rangle = D_F(x_1{-}x_2)D_F(x_3{-}x_4)+D_F(x_1{-}x_3)D_F(x_2{-}x_4)+D_F(x_1{-}x_4)D_F(x_2{-}x_3), \tag{14.169}$$

writing $\phi_i \equiv \phi(x_i)$. Each term is a product of two propagators connecting the external points *in pairs*; none connects all four. In momentum space (Fourier transform as in Eq. (14.164)) a term such as $D_F(x_1-x_2)D_F(x_3-x_4)$ produces $\widetilde{D_F}(p_1)\widetilde{D_F}(p_3)\,(2\pi)^8\delta^4(p_1+p_2)\delta^4(p_3+p_4)$: it has poles in only *two* of the four invariants p_i^2 and forces $p_1=-p_2$, $p_3=-p_4$ separately. Amputating with the

full $\prod_i (p_i^2 - m^2)/i$ of Eq. (14.165) multiplies this by two factors $(p_i^2 - m^2)$ that have nothing to cancel, so the term vanishes on shell. Physically these are the disconnected processes in which the two particles sail past one another; LSZ, by demanding all four legs simultaneously on their poles, discards them. The connected amplitude is genuinely zero at $O(\lambda^0)$, as it must be for a free theory.

The order-λ^1 numerator. At first order the numerator of Eq. (14.168) is

$$N^{(1)} = -\frac{i\lambda}{4!}\int d^4z\, \langle 0|\mathcal{T}\,\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\,\phi(z)\phi(z)\phi(z)\phi(z)|0\rangle. \quad (14.170)$$

This is a free time-ordered correlator of eight fields: the four external $\phi(x_i)$ and four copies of $\phi(z)$ at the common vertex point z. Wick's theorem expresses it as the sum over all complete pairings of these eight fields; a nonzero contribution requires every field to be contracted, since any uncontracted normal-ordered remainder has vanishing vacuum expectation value. We organize the pairings by how the four *external* fields are routed.

(a) Fully connected pairings. Each external field $\phi(x_i)$ is contracted with one of the four vertex fields $\phi(z)$, and the four vertex fields are thereby all used up. Such a pairing yields

$$D_F(x_1 - z)\, D_F(x_2 - z)\, D_F(x_3 - z)\, D_F(x_4 - z), \quad (14.171)$$

one propagator from each external point into the common vertex. The number of distinct pairings of this type is the number of ways to match four labelled external fields to four labelled vertex slots, namely $4! = 24$. Every one of these 24 pairings gives the *same* product $\prod_i D_F(x_i - z)$, because the four vertex fields sit at the identical point z and are indistinguishable. Therefore the fully connected contribution to Eq. (14.170) is

$$N^{(1)}_{\text{conn}} = -\frac{i\lambda}{4!}\cdot 4!\int d^4z\, D_F(x_1{-}z)D_F(x_2{-}z)D_F(x_3{-}z)D_F(x_4{-}z) = -i\lambda\int d^4z\,\prod_{i=1}^{4} D_F(x_i{-}z) \quad (14.172)$$

This is the central combinatorial fact of the computation: the multiplicity $4!$ of equivalent contractions *exactly* cancels the $1/4!$ written into the coupling in Eq. (14.56). The $1/4!$ was placed there for precisely this reason — so that the simplest connected diagram comes out with unit coefficient $-i\lambda$ — and we have now seen the cancellation happen explicitly rather than by decree.

(b) Partially connected pairings. Two external fields may contract with each other and the remaining two with vertex fields, the two leftover vertex fields then contracting together. A representative term is

$$D_F(x_1 - x_2)\, D_F(x_3 - z)\, D_F(x_4 - z)\, D_F(0), \quad (14.173)$$

where $D_F(0) \equiv D_F(z-z)$ is the closed self-contraction of the two unused vertex fields — a loop that begins and ends at the vertex, a *tadpole*. Such terms factor into a free propagator $D_F(x_1-x_2)$ that bypasses the vertex entirely and a piece attached to it. As with the order-λ^0 terms, the spectator propagator $D_F(x_1-x_2)$ carries poles in only p_1 and p_2 and ties them together, so under the four-fold amputation of Eq. (14.165) the factors $(p_1^2-m^2)$ and $(p_2^2-m^2)$ have no poles to kill and the term dies on shell. These pieces are not part of the connected four-point function and LSZ excludes them. (The tadpole $D_F(0)$ is itself ultraviolet divergent and momentum-independent; it belongs to the self-energy and renormalizes the mass, business deferred to Chapters 19 and 22. It does not affect the connected $2\to 2$ amplitude at this order.)

(c) Vacuum pairings and the denominator. Finally, all four external fields may pair among themselves while the four vertex fields pair among themselves. The vertex-only factor is exactly $\langle 0|\mathcal{T}(-\frac{i\lambda}{4!}\int\phi(z)^4)|0\rangle$, a disconnected vacuum bubble multiplying the free four-point function Eq. (14.169). This is where the denominator of Eq. (14.168) earns its place: expanded to $O(\lambda)$ it is $1+\langle 0|\mathcal{T}(-\frac{i\lambda}{4!}\int\phi^4)|0\rangle$, precisely the same vacuum bubble, and dividing removes it from the numerator. The cancellation of disconnected vacuum diagrams against the denominator is the general content of Theorem 14.18; here it operates on the single $O(\lambda)$ bubble. After the division, only connected-to-the-vertex structures remain, and among those only category (a) survives amputation.

The connected four-point function. Collecting the surviving piece, the connected interacting four-point function through $O(\lambda)$ is

$$\langle\Omega|\mathcal{T}\,\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)|\Omega\rangle_{\text{conn}} = -i\lambda\int d^4z\, D_F(x_1-z)D_F(x_2-z)D_F(x_3-z)D_F(x_4-z)+O(\lambda^2) \tag{14.174}$$

Four propagators run from the four external points to a single integrated vertex; the integral over z enforces that the vertex sits anywhere in spacetime. This is the position-space content of the elementary ϕ^4 scattering diagram, Fig. 14.5.

To momentum space. We now Fourier transform according to Eq. (14.164). The external integrals factorize, each one converting a propagator into its momentum-space form with the vertex position left over as a phase. Using the covariant representation Eq. (9.154),

$$\int d^4x_i\, e^{ip_i\cdot x_i}\, D_F(x_i-z) = \int d^4x_i\, e^{ip_i\cdot x_i}\int\frac{d^4q}{(2\pi)^4}\,\frac{i\,e^{-iq\cdot(x_i-z)}}{q^2-m^2+i\varepsilon} = \frac{i\,e^{ip_i\cdot z}}{p_i^2-m^2+i\varepsilon}, \tag{14.175}$$

where the x_i integral produced $(2\pi)^4\delta^4(p_i-q)$ and collapsed the q integral. Each leg thus contributes a momentum-space propagator $\widetilde{D_F}(p_i) = i/(p_i^2-m^2+i\varepsilon)$

and a phase $e^{ip_i\cdot z}$. Assembling all four legs and the prefactor,

$$\widetilde{G}^{(4)}(p_1,\dots,p_4) = -i\lambda \int d^4z\, e^{\,i(p_1+p_2+p_3+p_4)\cdot z} \prod_{i=1}^{4} \frac{i}{p_i^2 - m^2 + i\varepsilon}. \tag{14.176}$$

The single remaining integral is over the vertex position z, and it produces the energy–momentum delta that conservation demands:

$$\int d^4z\, e^{\,i(p_1+p_2+p_3+p_4)\cdot z} = (2\pi)^4\, \delta^4(p_1+p_2+p_3+p_4). \tag{14.177}$$

Hence the connected momentum-space four-point function at this order is

$$\widetilde{G}^{(4)}(p_1,\dots,p_4) = -i\lambda\,(2\pi)^4\,\delta^4(p_1+p_2+p_3+p_4) \prod_{i=1}^{4} \frac{i}{p_i^2 - m^2 + i\varepsilon}. \tag{14.178}$$

The four external propagators sit out front explicitly, each a pole at $p_i^2 = m^2$; they are exactly the legs LSZ will amputate.

Amputation and the on-shell limit. Apply Eq. (14.165): multiply by $\prod_i (p_i^2 - m^2)/i$. Each factor cancels one external propagator $i/(p_i^2 - m^2 + i\varepsilon)$ cleanly in the limit $\varepsilon \to 0^+$, leaving

$$\widetilde{G}^{(4)}_{\text{amp}}(p_1,\dots,p_4) = \Big[\prod_{i=1}^{4} \frac{p_i^2 - m^2}{i}\Big]\Big[(-i\lambda)(2\pi)^4\delta^4(\textstyle\sum_i p_i)\prod_{i=1}^4 \frac{i}{p_i^2-m^2+i\varepsilon}\Big] = -i\lambda\,(2\pi)^4\delta^4(\textstyle\sum_i p \tag{14.179}$$

The amputated correlator is a pure constant times the conservation delta — no poles remain, so the on-shell limit $p_i^2 \to m^2$ is trivially finite and equals Eq. (14.179) itself. This is the payoff of amputation: by removing the external legs we removed exactly the factors that would have diverged on shell, leaving a smooth amplitude.

The field-strength factor at this order. It remains to evaluate the field-strength factor $1/(\sqrt{Z})^4$ in Eq. (14.166). The residue Z is the coefficient of the one-particle pole in the exact propagator, $Z = (1 - \Sigma'(m^2))^{-1}$ with Σ the self-energy and $\Sigma' = d\Sigma/dp^2$. In ϕ^4 theory the only $O(\lambda)$ contribution to the self-energy is the tadpole loop $\Sigma^{(1)} = \frac{\lambda}{2} D_F(0)$ encountered in category (b): a constant, independent of the external momentum p. A momentum-independent self-energy shifts the pole *location* (it renormalizes the mass) but has $\Sigma' = 0$, so it does not change the residue. The first momentum dependence of Σ, and hence the first deviation of Z from unity, arises only at $O(\lambda^2)$ from the two-loop "sunset" graph. Therefore

$$Z = 1 + O(\lambda^2), \qquad 1/(\sqrt{Z})^4 = 1 + O(\lambda^2), \tag{14.180}$$

and the field-strength factors are exactly 1 at the order we are working. (The full computation of Σ and Z is carried out in Chapter 19.)

Reading off the amplitude. Substituting Eqs. (14.179) and (14.180) into the LSZ recipe Eq. (14.166),

$$i\mathcal{M}\,(2\pi)^4\delta^4\big(\textstyle\sum_i p_i\big) = \big(1 + O(\lambda^2)\big)\cdot\big[-i\lambda\,(2\pi)^4\delta^4\big(\textstyle\sum_i p_i\big)\big], \tag{14.181}$$

and cancelling the common $(2\pi)^4\delta^4$ gives the carried ϕ^4 theory number,

$$\boxed{i\mathcal{M}(\phi\phi\to\phi\phi) = -i\lambda + O(\lambda^2)} \qquad \Longleftrightarrow \qquad \mathcal{M} = -\lambda + O(\lambda^2). \tag{14.182}$$

This is the first interacting amplitude of the book. It is real, constant, and independent of the scattering angle and energy: the $O(\lambda)$ process is a pure contact interaction with no propagating intermediate particle, so it depends on none of the kinematic invariants. We have produced it with every factor on the page — the $-i\lambda/4!$ from the Dyson series, the $4!$ from the Wick multiplicity, the four propagators from the external legs, the delta from the vertex integral, the four cancellations of amputation, and the $Z = 1$ of the residue.

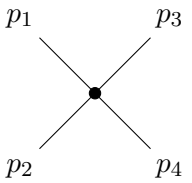


Figure 14.5: The single connected diagram contributing to $\phi\phi \to \phi\phi$ at $O(\lambda)$: four external lines meet at one spacetime vertex (integrated over in Eq. (14.174)). The four propagators are the external legs LSZ amputates; the vertex contributes the factor $-i\lambda$ after the $4!$ contraction multiplicity cancels the $1/4!$ of the coupling. There is no internal line, hence no kinematic dependence: $\mathcal{M} = -\lambda$.

We record the result as a proposition for later reference, since the renormalization chapters will quote it as the tree-level input that loop corrections dress.

Proposition 14.36 (Tree-level ϕ^4 scattering amplitude)**.** *For the real scalar theory Eq. (14.56) with coupling λ, the connected, amputated, on-shell two-to-two amplitude defined by Eq. (14.163) is, to leading order in λ,*

$$i\mathcal{M}(\phi\phi\to\phi\phi) = -i\lambda + O(\lambda^2), \tag{14.183}$$

independent of the Mandelstam invariants $s = (p_1+p_2)^2$, $t = (p_1-p_3)^2$, $u = (p_1-p_4)^2$.

Proof. This is Theorem 14.35: Eq. (14.182). □

14.6.3 Reading the answer: kinematics, identical particles, and $|\mathcal{M}|^2$

The amplitude Eq. (14.182) is the simplest a $2 \to 2$ amplitude can be: a constant. It is worth dwelling on what that constancy means and assembling the squared amplitude that the next chapter on cross sections will integrate.

The four external momenta of an on-shell $2 \to 2$ process are constrained by conservation $p_1 + p_2 = p_3 + p_4$ and by the four mass-shell conditions $p_i^2 = m^2$, leaving the two independent Lorentz invariants conventionally packaged as the Mandelstam variables s, t, u, which satisfy

$$s + t + u = p_1^2 + p_2^2 + p_3^2 + p_4^2 = 4m^2. \tag{14.184}$$

A generic tree amplitude — as we will see already in ϕ^3-like theories and certainly in QED — depends nontrivially on these invariants through internal propagators $1/(s - m^2)$, $1/(t - m^2)$, $1/(u - m^2)$ that go on shell when an intermediate particle can be produced. The $O(\lambda)$ ϕ^4 amplitude has *no internal line at all* (Fig. 14.5); the interaction is pointlike, all four particles meeting at one vertex. There is consequently nothing for s, t, or u to appear in, and $\mathcal{M} = -\lambda$ is a structureless constant. This is the defining signature of a contact interaction, and it is what makes ϕ^4 the cleanest possible first example: the kinematics is trivial and all the content is in the combinatorics we have just done.

For the transition probability and the cross section we need the modulus squared. From Eq. (14.182),

$$|\mathcal{M}|^2 = \lambda^2 + O(\lambda^3). \tag{14.185}$$

Two bookkeeping points attach to this number and are flagged here for the chapter that uses it. First, the final-state particles are *identical* real scalars; when the squared amplitude is integrated over the final-state phase space to form a total cross section, the region of integration must be halved (equivalently, a symmetry factor $1/2!$ inserted) to avoid double-counting the configuration in which p_3 and p_4 are exchanged. Second, the relation between $|\mathcal{M}|^2$ and a measurable cross section requires the Lorentz-invariant phase space and the incident flux. Both belong to Chapter 16, which converts Eq. (14.185) into the differential and total cross sections; we hand them this number and its identical-particle caveat and do not pre-empt that work here.

14.6.4 What the bookkeeping is telling us

Step back from the result to the *method*. To obtain a single number we expanded an exponential, applied Wick's theorem to eight fields, sorted the resulting pairings into connected, partially connected, and vacuum classes, tracked a $4!$ against a $1/4!$, recognized which pairings would survive amputation, Fourier-transformed four legs, did the vertex integral, amputated, and took an on-shell

limit. For the leading term of the simplest process in the simplest theory this is already a substantial accounting — and almost all of it was bookkeeping that recurred in a predictable pattern: a factor $-i\lambda$ for the vertex, a factor $i/(p^2 - m^2 + i\varepsilon)$ for each line, a delta for momentum conservation at the vertex, amputation of external legs, and a combinatorial weight from the contraction multiplicity. At second order, or for a process with internal lines, the number of pairings proliferates and doing this by hand from Wick's theorem each time would be intolerable.

Physical Insight

Every structural feature of the calculation that did *not* depend on the specific external momenta — the vertex factor $-i\lambda$, the propagator $i/(p^2 - m^2 + i\varepsilon)$ per line, the conservation delta, the rule to amputate external legs and supply $\sqrt{Z}$ per leg, and the combinatorial weight — is a property of the *Lagrangian*, not of the particular amplitude. It can therefore be read off once and applied mechanically. That observation is the seed of the Feynman rules: a dictionary that replaces the entire Gell-Mann–Low–plus–Wick procedure with the drawing of diagrams and the multiplication of factors. The $1/4!$ that cancelled here so cleanly for the leading diagram does *not* always cancel completely; when a diagram has a nontrivial automorphism (a line that can be reflected, a loop that can be relabelled), a residual *symmetry factor* survives. Systematizing exactly when and by how much is the content of the next chapter.

This is the natural handoff. We have established the carried ϕ^4 theory number $i\mathcal{M} = -i\lambda$ and the squared amplitude $|\mathcal{M}|^2 = \lambda^2$, and we pass them forward in two directions: to Chapter 16, which turns $|\mathcal{M}|^2$ into a cross section with the correct flux, phase space, and identical-particle factor; and to Chapter 15, which takes the unsystematized contraction combinatorics displayed in Theorem 14.35 and distils it into the Feynman rules and the careful theory of symmetry factors, then re-derives this very amplitude two ways as the first check of that machinery. Before either, the chapter closes by extracting from the unitarity of $\mathbf{S}$ a constraint that no order-by-order expansion can violate — the optical theorem — which will give physical meaning to the imaginary parts that the loop corrections to $-i\lambda$ are about to acquire.

14.7 Unitarity and the Optical Theorem

The chapter has, by this point, built a working machine. The Dyson series, Gell-Mann–Low, and LSZ together convert an interacting Lagrangian into a power series for scattering amplitudes, and the previous section cranked that machine once to produce the first interacting number of ϕ^4 theory, the tree-level

result $i\mathcal{M} = -i\lambda$ recorded in Theorem 14.35. Every step of that construction was perturbative: it is an expansion in λ, valid order by order, silent about whether the sum it approximates makes sense. We close the chapter with the one statement about the S-matrix that is *not* perturbative, that holds to all orders and beyond them, and that the rest of the book will lean on as a fixed point of certainty whenever a calculation needs a check it cannot fake: *the S-matrix is unitary*. Probability is conserved. Whatever the distant past prepares, the distant future finds *something*, with total probability one. This single algebraic fact, pressed correctly, yields a nonperturbative constraint — the optical theorem — relating the imaginary part of a forward scattering amplitude to the total rate at which the initial state scatters into anything at all. It is the result that finally gives physical meaning to the imaginary parts that loop amplitudes will carry, and the seed of the analyticity-and-dispersion program that Chapter 5 already previewed.

We take from the S-matrix section the operator and its decomposition. The scattering operator $\mathbf{S}$ of Theorem 14.11 is unitary, $\mathbf{S}^\dagger\mathbf{S} = \mathbf{S}\,\mathbf{S}^\dagger = \mathbf{1}$, and the nontrivial scattering is isolated by writing

$$\mathbf{S} = \mathbf{1} + i\,\mathcal{T}, \tag{14.186}$$

exactly as in Eq. (14.45): the $\mathbf{1}$ is the amplitude for "nothing happens," and the *transition operator* $\mathcal{T}$ collects all genuine interactions. Conventions are those in force throughout the chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, relativistically normalized one-particle states with the Lorentz-invariant measure of Eq. (3.191), and the Feynman $i\varepsilon$.

14.7.1 Unitarity in the transition operator

Unitarity is a statement about $\mathbf{S}$; we first translate it into a statement about $\mathcal{T}$, because $\mathcal{T}$ is the object whose matrix elements are the amplitudes we compute. Substitute Eq. (14.186) into $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$:

$$\mathbf{1} = \mathbf{S}^\dagger\mathbf{S} = (\mathbf{1} - i\,\mathcal{T}^\dagger)(\mathbf{1} + i\,\mathcal{T}) = \mathbf{1} + i\,\mathcal{T} - i\,\mathcal{T}^\dagger + \mathcal{T}^\dagger\mathcal{T}. \tag{14.187}$$

The leading $\mathbf{1}$ cancels the left-hand side, leaving $i(\mathcal{T} - \mathcal{T}^\dagger) + \mathcal{T}^\dagger\mathcal{T} = 0$, i.e.

$$-\,i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger\mathcal{T}. \tag{14.188}$$

This is the master identity. Read it slowly. The left-hand side is, up to the factor, the *anti-Hermitian part* of $\mathcal{T}$: if $\mathcal{T}$ were Hermitian the left side would vanish. The right-hand side is $\mathcal{T}^\dagger\mathcal{T}$, a manifestly positive operator — for any state $|\psi\rangle$, $\big\langle\psi\big|\mathcal{T}^\dagger\mathcal{T}\big|\psi\big\rangle = \|\mathcal{T}\,|\psi\rangle\|^2 \geq 0$. So Eq. (14.188) says precisely that the failure of $\mathcal{T}$ to be Hermitian is controlled, and is positive. Had we instead expanded $\mathbf{S}\,\mathbf{S}^\dagger = \mathbf{1}$ we would have found $-i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}\,\mathcal{T}^\dagger$; both hold, and equating them shows $\mathcal{T}^\dagger\mathcal{T} = \mathcal{T}\,\mathcal{T}^\dagger$, so $\mathcal{T}$ is normal. For what follows we use the form Eq. (14.188).

Physical Insight

Equation Eq. (14.188) is the whole of the optical theorem before any kinematics. To first order in the coupling, $\mathcal{T} \sim O(\lambda)$ and the right-hand side $\mathcal{T}^\dagger\mathcal{T} \sim O(\lambda^2)$ is negligible, so $\mathcal{T} \approx \mathcal{T}^\dagger$: a tree amplitude is (to leading order) real, as $i\mathcal{M} = -i\lambda$ indeed is. The content of unitarity is the *correction* to that statement: the amount by which $\mathcal{T}$ fails to be Hermitian at order λ^2 is fixed, not free, equal to the square of the order-λ amplitude. Unitarity ties the imaginary part of one order to the modulus-squared of a lower order. This is why an imaginary part in a loop amplitude is never optional bookkeeping — it is mandated, in advance, by tree-level data.

14.7.2 Matrix elements: the invariant amplitude and the sum over states

To turn the operator identity Eq. (14.188) into a relation between numbers we sandwich it between physical states and insert a resolution of the identity on the right-hand side. Two pieces of notation make the bookkeeping honest.

First, the invariant amplitude. For an initial state $|i\rangle$ and a final state $|f\rangle$ of definite momenta, the transition matrix element factorizes a four-momentum-conserving delta function out front, and the remainder *defines* the Lorentz-invariant amplitude $\mathcal{M}$:

$$\langle f|\mathcal{T}|i\rangle = (2\pi)^4\,\delta^{(4)}\Big(p_i - p_f\Big)\,\mathcal{M}(i \to f), \tag{14.189}$$

where p_i and p_f are the total incoming and outgoing four-momenta. This is the same $\mathcal{M}$ produced by LSZ and computed in Theorem 14.35; the delta function expresses translation invariance and is stripped off because it carries no dynamics.

Second, the sum over final states. The completeness relation on the interacting Fock space, written with relativistically normalized multiparticle states and the invariant measure of Eq. (3.191), is

$$\mathbf{1} = \sum_n \frac{1}{n!} \int \prod_{j=1}^{n} \frac{d^3q_j}{(2\pi)^3\,2E_{\mathbf{q}_j}}\;\; |q_1 \dots q_n\rangle\langle q_1 \dots q_n|\,, \tag{14.190}$$

the $1/n!$ correcting for the overcounting of identical bosons when their labels are integrated independently, and an implicit sum over species running alongside the sum on n. We abbreviate this entire structure — the species sum, the number sum, the symmetry factor, the products of invariant measures, *and* the overall momentum-conserving delta function that will appear — by a single symbol $\mathrm{d}\Pi_f$, the *Lorentz-invariant phase space* of the final state f:

$$\sum_f \int \mathrm{d}\Pi_f\,(\cdots) \;\equiv\; \sum_n \frac{1}{n!} \sum_{\{\text{species}\}} \int \prod_{j=1}^{n} \frac{d^3q_j}{(2\pi)^3\,2E_{\mathbf{q}_j}}\,(2\pi)^4\,\delta^{(4)}\Big(p_i - \sum_j q_j\Big)\,(\cdots). \tag{14.191}$$

This is the same measure that converts $|\mathcal{M}|^2$ into a decay rate or cross section in Chapter 16; here we need only that it is real and nonnegative, $\mathrm{d}\Pi_f \geq 0$, and Lorentz invariant. (The momentum delta is included inside $\mathrm{d}\Pi_f$; we display it explicitly when a step needs it and otherwise leave it folded in, matching the compact form quoted in Eq. (5.147).)

14.7.3 The generalized optical theorem

We can now extract the central result. Take the diagonal, *forward* matrix element of Eq. (14.188) in a fixed initial state $|i\rangle$ — the same state on both sides — and evaluate each side with Eqs. (14.189) to (14.191).

Theorem 14.37 (Generalized optical theorem)**.** *For any state $|i\rangle$ of definite total four-momentum p_i, the forward invariant amplitude obeys*

$$2\ \mathrm{Im}\ \mathcal{M}(i \to i) = \sum_f \int \mathrm{d}\Pi_f\ \left|\mathcal{M}(i \to f)\right|^2, \tag{14.192}$$

where the sum runs over every kinematically accessible final state f, weighted by its Lorentz-invariant phase space Eq. (14.191). The right-hand side is manifestly real and nonnegative; hence so is $\mathrm{Im}\ \mathcal{M}(i \to i) \geq 0$.

Proof. Sandwich Eq. (14.188) between $\langle i|$ and $|i\rangle$.

Left-hand side. Using $\left\langle i\middle|\mathcal{T}^\dagger\middle|i\right\rangle = \langle i|\mathcal{T}|i\rangle^*$,

$$\left\langle i\middle| -i(\mathcal{T}-\mathcal{T}^\dagger)\middle|i\right\rangle = -i\Big(\langle i|\mathcal{T}|i\rangle - \langle i|\mathcal{T}|i\rangle^*\Big) = -i\left(2i\ \mathrm{Im}\,\langle i|\mathcal{T}|i\rangle\right) = 2\ \mathrm{Im}\,\langle i|\mathcal{T}|i\rangle\,. \tag{14.193}$$

Now insert Eq. (14.189) for the forward element, $\langle i|\mathcal{T}|i\rangle = (2\pi)^4\delta^{(4)}(0)\,\mathcal{M}(i \to i)$. The forward delta function is evaluated at zero argument because initial and final momenta coincide; it is the familiar infrared artefact of plane-wave normalization, equal to the spacetime volume,

$$(2\pi)^4\,\delta^{(4)}(0) = \int d^4x\ e^{\,i\,0\cdot x} = V\,T, \tag{14.194}$$

in a box of spatial volume V over a time T. Hence the left-hand side equals $2\,VT\ \mathrm{Im}\ \mathcal{M}(i \to i)$.

Right-hand side. Insert the completeness relation Eq. (14.190), packaged as Eq. (14.191), between $\mathcal{T}^\dagger$ and $\mathcal{T}$:

$$\left\langle i\middle|\mathcal{T}^\dagger\mathcal{T}\middle|i\right\rangle = \sum_f \int \mathrm{d}\Pi'_f\ \left\langle i\middle|\mathcal{T}^\dagger\middle|f\right\rangle\langle f|\mathcal{T}|i\rangle = \sum_f \int \mathrm{d}\Pi'_f\ \left|\langle f|\mathcal{T}|i\rangle\right|^2, \tag{14.195}$$

where $\mathrm{d}\Pi'_f$ denotes for the moment the phase space *without* the momentum delta (the delta in Eq. (14.191) arose from the $\mathcal{T}$ matrix elements, which we now write

out). Substituting Eq. (14.189),

$$\begin{aligned}\left|\langle f|\mathcal{T}|i\rangle\right|^2 &= \Big[(2\pi)^4\delta^{(4)}(p_i-p_f)\Big]^2\,\left|\mathcal{M}(i\to f)\right|^2 \\ &= (2\pi)^4\delta^{(4)}(0)\,\cdot\,(2\pi)^4\delta^{(4)}(p_i-p_f)\,\left|\mathcal{M}(i\to f)\right|^2,\end{aligned} \tag{14.196}$$

where we used the standard regularization of the squared delta: one factor is set to its zero-argument value $(2\pi)^4\delta^{(4)}(0) = VT$ by Eq. (14.194), while the surviving factor enforces momentum conservation. Reassembling, the surviving $(2\pi)^4\delta^{(4)}(p_i-p_f)$ combines with $\mathrm{d}\Pi'_f$ to rebuild the full invariant phase space $\mathrm{d}\Pi_f$ of Eq. (14.191), so

$$\Big\langle i\Big|\mathcal{T}^\dagger\mathcal{T}\Big|i\Big\rangle = VT\sum_f\int \mathrm{d}\Pi_f\,\left|\mathcal{M}(i\to f)\right|^2. \tag{14.197}$$

Cancellation. Equating the two sides, the common factor $VT = (2\pi)^4\delta^{(4)}(0)$ appears once on each side — the left from the forward delta Eq. (14.194), the right from the squared delta Eq. (14.196) — and cancels. What remains is Eq. (14.192). That the VT factors match on the two sides is not luck: both count the same infinite normalization of the single plane-wave initial state, and the physical, density-independent content is exactly what survives their cancellation. The right-hand side is a sum of squared moduli against nonnegative measures, hence nonnegative, forcing $\mathrm{Im}\,\mathcal{M}(i\to i)\geq 0$. □

The result Eq. (14.192) is exact: no approximation entered beyond the plane-wave regularization, which cancels. It holds order by order in any coupling and, granting the unitarity of $\mathbf{S}$, beyond perturbation theory. It is the precise statement promised at Section 5.5.6, where the dispersion machinery was built in anticipation of it: the imaginary part of an amplitude is never an abstract analytic datum but a physical sum of squared amplitudes, hence a total rate.

Caution

The squared delta in Eq. (14.196) is the one genuinely delicate manipulation, and it is legitimate only because the offending factor cancels against an identical one on the other side of the equation. One must never read $[\delta^{(4)}(p_i-p_f)]^2$ as a meaningful number in isolation; it is shorthand for "regulate in a finite box, extract the common volume factor, take $V,T\to\infty$ last." The same prescription underlies the conversion of $|\mathcal{M}|^2$ to rates in Chapter 16, where the VT becomes the flux-times-time that turns a probability into a cross section. Here the bookkeeping is even cleaner: the factor is the same on both sides and simply divides out.

14.7.4 The forward amplitude and the total cross section

The most-used form of Eq. (14.192) specializes the initial state to two particles, $|i\rangle = |p_1\, p_2\rangle$, the configuration of a scattering experiment. The right-hand side then sums the squared amplitude over all final states the pair can produce — elastic, inelastic, any number of particles — which is exactly the combination that builds the *total* cross section. We recall from Chapter 16 the master formula for a $2 \to f$ cross section,

$$d\sigma(i \to f) = \frac{1}{F}\,\big|\mathcal{M}(i \to f)\big|^2\,\mathrm{d}\Pi_f, \qquad F = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2 m_2^2}, \tag{14.198}$$

with F the relativistic flux factor (the Møller flux), so that summing and integrating over all final states gives

$$\sigma_{\mathrm{tot}}(s) = \frac{1}{F}\sum_f \int \mathrm{d}\Pi_f\,\big|\mathcal{M}(i \to f)\big|^2. \tag{14.199}$$

Comparing with Eq. (14.192), the sum is precisely $2\,\mathrm{Im}\,\mathcal{M}(i \to i)$, the imaginary part of the *forward elastic* amplitude — "forward" because the final two-particle state equals the initial one, $p_1' = p_1$, $p_2' = p_2$, i.e. zero scattering angle and Mandelstam $t = 0$. Therefore

$$\boxed{\mathrm{Im}\;\mathcal{M}(s, t{=}0) = \tfrac{1}{2}\,F\,\sigma_{\mathrm{tot}}(s) = 2\,p_{\mathrm{cm}}\sqrt{s}\,\sigma_{\mathrm{tot}}(s)} \tag{14.200}$$

where in the last step we evaluated the flux in the centre-of-momentum frame. There $p_1 = (E_1, \mathbf{p})$, $p_2 = (E_2, -\mathbf{p})$ with $|\mathbf{p}| = p_{\mathrm{cm}}$ and $E_1 + E_2 = \sqrt{s}$, and a short computation gives

$$F = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2 m_2^2} = 4\,p_{\mathrm{cm}}\sqrt{s}, \tag{14.201}$$

the second equality being the standard kinematic identity $(p_1\cdot p_2)^2 - m_1^2 m_2^2 = p_{\mathrm{cm}}^2\, s$ for two bodies (it follows from writing $p_1\cdot p_2 = E_1E_2 + \mathbf{p}^2$ and $s = (E_1+E_2)^2$ and eliminating the energies through $E_i^2 = \mathbf{p}^2 + m_i^2$). At high energy $p_{\mathrm{cm}} \to \sqrt{s}/2$ and Eq. (14.200) reduces to $\mathrm{Im}\,\mathcal{M}(s,0) \to s\,\sigma_{\mathrm{tot}}(s)$, the form quoted in Eq. (5.148). This is the optical theorem in its classic dress: *the imaginary part of the forward scattering amplitude measures the total cross section.*

Physical Insight

The name is borrowed from classical optics, and the analogy is exact in content. A plane wave passing a scatterer is depleted in the forward direction by exactly the amount scattered into all other directions plus absorbed — the "shadow" is the interference of the incident wave with the forward-scattered wave, and its depth is set by the total (scattered

+ absorbed) cross section. Equation Eq. (14.200) is the quantum-field-theoretic statement of that shadow: removal of flux from the forward beam (the imaginary part of the forward amplitude) equals everything that happens (the total cross section). Positivity, Im $\mathcal{M}(i \to i) \geq 0$, is the statement that a scatterer can only deplete the forward beam, never amplify it — a passive medium, as unitarity demands.

14.7.5 Partial-wave unitarity

The optical theorem constrains the forward amplitude as a whole; decomposing a $2 \to 2$ elastic amplitude into angular momentum eigenchannels sharpens the constraint into a bound on each channel separately. We treat the cleanest case — two spinless particles, elastic scattering, in the centre-of-momentum frame — where the amplitude depends only on s and the scattering angle θ. Expand in Legendre polynomials,

$$\mathcal{M}(s,\theta) = \frac{8\pi\sqrt{s}}{p_{\rm cm}} \sum_{\ell=0}^{\infty} (2\ell+1)\, a_\ell(s)\, P_\ell(\cos\theta), \tag{14.202}$$

the normalization chosen — with foresight — so that elastic unitarity will take its simplest form on the dimensionless coefficients $a_\ell(s)$. Inverting with the orthogonality relation $\int_{-1}^{1} d(\cos\theta)\, P_\ell P_{\ell'} = \frac{2}{2\ell+1}\delta_{\ell\ell'}$ gives the projection

$$a_\ell(s) = \frac{p_{\rm cm}}{16\pi\sqrt{s}} \int_{-1}^{1} d(\cos\theta)\, P_\ell(\cos\theta)\, \mathcal{M}(s,\theta). \tag{14.203}$$

We now write the two cross sections of interest in terms of the a_ℓ. The elastic cross section uses the $2 \to 2$ rate $d\sigma_{\rm el}/d\Omega = |\mathcal{M}|^2/(64\pi^2 s)$ (the elastic specialization of Eq. (14.198), with $|\mathbf{p}_f| = |\mathbf{p}_i|$, derived in Chapter 16). Insert Eq. (14.202), integrate over the solid angle with the same orthogonality relation, and the double sum collapses to a single one:

$$\begin{aligned} \sigma_{\rm el} &= \int d\Omega\, \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{2\pi}{64\pi^2 s} \int_{-1}^{1} d(\cos\theta) \left(\tfrac{8\pi\sqrt{s}}{p_{\rm cm}}\right)^2 \sum_{\ell,\ell'} (2\ell+1)(2\ell'+1) a_\ell a_{\ell'}^{*} P_\ell P_{\ell'} \\ &= \frac{4\pi}{p_{\rm cm}^2} \sum_{\ell} (2\ell+1)\, |a_\ell(s)|^2 . \end{aligned} \tag{14.204}$$

The total cross section comes from the optical theorem Eq. (14.200). At $\theta = 0$ all $P_\ell(1) = 1$, so the forward amplitude is $\mathcal{M}(s,0) = \frac{8\pi\sqrt{s}}{p_{\rm cm}} \sum_\ell (2\ell+1) a_\ell$, and

$$\sigma_{\rm tot} = \frac{\text{Im}\ \mathcal{M}(s,0)}{2\, p_{\rm cm}\sqrt{s}} = \frac{1}{2 p_{\rm cm}\sqrt{s}} \cdot \frac{8\pi\sqrt{s}}{p_{\rm cm}} \sum_\ell (2\ell{+}1)\ \text{Im}\ a_\ell = \frac{4\pi}{p_{\rm cm}^2} \sum_\ell (2\ell{+}1)\ \text{Im}\ a_\ell(s). \tag{14.205}$$

The two results carry the identical prefactor $4\pi/p_{\rm cm}^2$ — precisely the payoff of the normalization in Eq. (14.202), and the relativistic echo of the familiar $4\pi/k^2$ of nonrelativistic partial-wave theory. Because the total cross section is the elastic plus the (nonnegative) inelastic part, $\sigma_{\rm tot} = \sigma_{\rm el} + \sigma_{\rm inel}$ with $\sigma_{\rm inel} \geq 0$, and because the sums run over the same independent channels ℓ, the inequality holds channel by channel:

$$\text{Im } a_\ell(s) = \left|a_\ell(s)\right|^2 + \frac{p_{\rm cm}^2}{4\pi(2\ell+1)}\,\sigma_{\rm inel}^{(\ell)} \;\geq\; \left|a_\ell(s)\right|^2. \tag{14.206}$$

This is *partial-wave unitarity*. Its geometric content is immediate: complete the square,

$$\left|a_\ell\right|^2 - \text{Im } a_\ell \leq 0 \iff \Big(\text{Re } a_\ell\Big)^2 + \Big(\text{Im } a_\ell - \tfrac{1}{2}\Big)^2 \leq \tfrac{1}{4}, \tag{14.207}$$

so each $a_\ell(s)$ is confined to the closed disk of radius $\frac{1}{2}$ centred at $\frac{i}{2}$ in the complex plane — the *Argand circle* of Fig. 14.6. The bound has three faces worth stating plainly: $0 \leq \text{Im } a_\ell \leq 1$, $|\text{Re } a_\ell| \leq \frac{1}{2}$, and $|a_\ell| \leq 1$. A single partial wave can never exceed these magnitudes, no matter the dynamics. The boundary of the disk is reached only when $\sigma_{\rm inel}^{(\ell)} = 0$, the purely elastic case, where Eq. (14.206) becomes the equality $\text{Im } a_\ell = |a_\ell|^2$ and the amplitude may be written in terms of a real phase shift $\delta_\ell(s)$,

$$a_\ell = \frac{e^{2i\delta_\ell} - 1}{2i} = e^{i\delta_\ell}\sin\delta_\ell, \tag{14.208}$$

for which $\text{Im } a_\ell = \sin^2\delta_\ell = |a_\ell|^2$ identically. When inelastic channels open, the amplitude moves into the interior of the disk; the deficit $\text{Im } a_\ell - |a_\ell|^2$ is, by Eq. (14.206), a direct measure of the inelastic rate in that channel.

The maximal value $|a_\ell| = 1$, attained at $a_\ell = i$ ($\delta_\ell = \pi/2$), caps the contribution of each angular momentum to the total cross section at

$$\sigma_{\rm tot}^{(\ell)} \leq \frac{4\pi}{p_{\rm cm}^2}(2\ell+1), \tag{14.209}$$

the *unitarity bound* on a single partial wave. It is the seed of the high-energy Froissart bound on the total cross section, developed in the analyticity program; here we have it as an exact, nonperturbative statement, with the modest caveat that for *identical* final particles the two-body phase space in Eq. (14.204) carries an extra symmetry factor $\frac{1}{2}$ and only even ℓ contribute, shifting the numerical coefficients but not the existence or form of the bound.

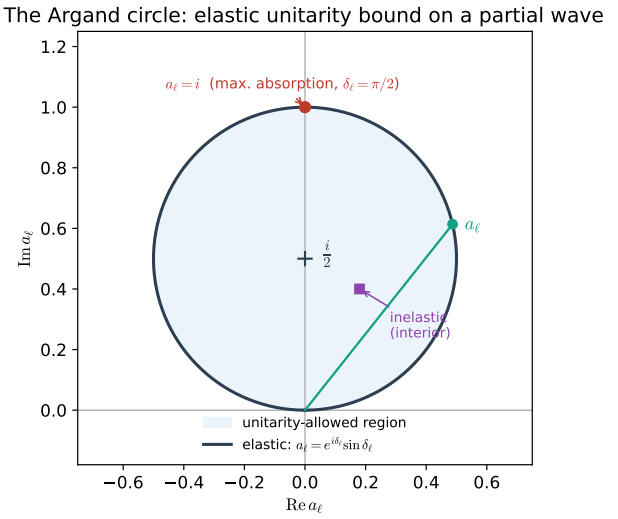


Figure 14.6: The Argand circle Eq. (14.207): unitarity confines every elastic partial-wave amplitude $a_\ell(s)$ to the closed disk of radius $\frac{1}{2}$ centred at $\frac{i}{2}$. The boundary is the purely elastic locus Eq. (14.208), $a_\ell = e^{i\delta_\ell}\sin\delta_\ell$, traced by the real phase shift δ_ℓ; the maximal-cross-section point $a_\ell = i$ ($\delta_\ell = \pi/2$, purely elastic) saturates the per-channel cross-section bound, while maximal absorption (largest inelastic deficit) sits at the disk centre $a_\ell = i/2$. Open inelastic channels push a_ℓ into the interior, the radial deficit measuring the inelasticity. An amplitude that would leave the disk signals the breakdown of whatever approximation produced it.

Caution

The bound $|a_\ell| \leq 1$ is a property of the *exact* amplitude, not of any truncation of it. A perturbative result is a polynomial in the coupling and will, at sufficiently high energy or strong coupling, march straight out of the Argand disk. That is not a violation of unitarity; it is the calculation announcing the energy or coupling at which the omitted higher orders can no longer be neglected — where the series must be resummed or new physics must intervene. Perturbative unitarity is thus a self-diagnostic: it tells a tree-level amplitude where it stops being trustworthy.

14.7.6 Scalar ϕ^4 theory: the optical theorem to leading order, and where the real check lives

We close by applying the machinery to the amplitude this chapter actually computed, and by stating honestly what the leading-order check can and cannot show.

Worked Example 14.38 (The ϕ^4 optical theorem and the perturbative unitarity bound on λ)**.** The tree amplitude for $\phi\phi \to \phi\phi$ from Theorem 14.35 is the constant $\mathcal{M} = -\lambda$, independent of all kinematics. We test it against Theorem 14.37 and read off the perturbative reach of λ.

Leading order is a triviality, by design. The forward amplitude is $\mathcal{M}(i \to i) = -\lambda$, a real number, so its imaginary part vanishes: $\operatorname{Im} \mathcal{M}(i \to i) = 0 + O(\lambda^2)$. The right-hand side of Eq. (14.192) is $\sum_f \int \mathrm{d}\Pi_f |\mathcal{M}(i \to f)|^2$; the lightest accessible final state is again two scalars, contributing $\int \mathrm{d}\Pi_2 \, |-\lambda|^2 = O(\lambda^2)$. So at $O(\lambda)$ the optical theorem reads $0 = 0 + O(\lambda^2)$: it is satisfied, but vacuously, carrying no information. This is forced by Eq. (14.188), which always pairs the imaginary part at one order with the squared amplitude at the order below. The first *nontrivial* statement is the $O(\lambda^2)$ balance,

$$2 \operatorname{Im} \mathcal{M}^{(2)}(i \to i) \overset{!}{=} \int \mathrm{d}\Pi_2 \, \left|\mathcal{M}^{(1)}(i \to f)\right|^2 = \lambda^2 \int \mathrm{d}\Pi_2, \tag{14.210}$$

relating the imaginary part of the *one-loop* forward amplitude $\mathcal{M}^{(2)}$ to the tree amplitude squared, integrated over the two-body phase space. The right-hand side is elementary — it is the same $\int \mathrm{d}\Pi_2 = \beta/8\pi$ two-body volume met in Theorem 5.39, with $\beta = \sqrt{1 - 4m^2/s}$ — so Eq. (14.210) is a sharp, falsifiable prediction for the imaginary part of the one-loop s-channel bubble. Computing $\mathcal{M}^{(2)}$ and verifying it requires the loop integral itself, which is the business of later chapters; no new number is established here.

The perturbative unitarity bound. Project the tree amplitude onto partial waves with Eq. (14.203). Since $\mathcal{M} = -\lambda$ is angle-independent and $P_0 = 1$, only the s-wave survives:

$$a_0(s) = \frac{p_{\mathrm{cm}}}{16\pi\sqrt{s}} \int_{-1}^{1} d(\cos\theta)\,(-\lambda) = \frac{p_{\mathrm{cm}}}{16\pi\sqrt{s}}\,(-\lambda)(2) = -\frac{\lambda\, p_{\mathrm{cm}}}{8\pi\sqrt{s}} \xrightarrow{s \gg m^2} -\frac{\lambda}{16\pi}, \tag{14.211}$$

using $p_{\mathrm{cm}} \to \sqrt{s}/2$ at high energy. This a_0 is real at $O(\lambda)$, consistent with $\operatorname{Im} a_0 = |a_0|^2 = O(\lambda^2)$: the imaginary part the Argand circle demands is again pushed to one loop. The unitarity bound $|a_0| \le 1$ then reads

$$\frac{\lambda}{16\pi} \lesssim 1 \qquad \Longrightarrow \qquad \lambda \lesssim 16\pi \approx 50, \tag{14.212}$$

up to an $O(1)$ identical-particle factor that we have not tracked precisely. The lesson is robust against that factor: tree-level $\lambda\phi^4$ is a sensible perturbative description only for λ of order 10 or smaller; pushed toward $\lambda \sim 4\pi$–16π the s-wave amplitude saturates its unitarity disk and the loop corrections it has been ignoring become as large as the tree term. The dimensionless expansion parameter of the theory is not λ but $\lambda/16\pi^2$, and Eq. (14.212) is the unitarity-based confirmation of that fact — the same $16\pi^2$ that will reappear in the one-loop β-function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Chapter 24.

The deferral is therefore explicit. The leading ϕ^4 optical-theorem identity Eq. (14.210) is a statement about the one-loop amplitude $\mathcal{M}^{(2)} = O(\lambda^2)$; assembling that amplitude needs the Feynman-diagram technology of Chapter 15 and the loop-integration and imaginary-part methods developed in the renormalization programme (Chapters 19, 21 and 22). The systematic principle that organizes such checks — that the imaginary part of a diagram is obtained by putting each internal line of an intermediate state on shell, channel by channel, exactly the Sokhotski–Plemelj replacement of Eq. (5.140) — is *Cutkosky's cutting rules*, established in Chapter 60. ϕ^4 theory thus carries forward an unredeemed promissory note: the optical theorem here predicts the imaginary part of the one-loop bubble in advance, and the later chapters that compute the loop will find it waiting.

With unitarity in hand, the chapter has delivered every ingredient the rest of the book needs to compute and to check: the LSZ bridge from correlators to amplitudes, the first interacting number $i\mathcal{M} = -i\lambda$, and now the exact constraint that any amplitude — tree, loop, or resummed — must respect. The imaginary-part-equals-sum-over-cuts technology passes directly to Chapter 15, where cutting rules are made diagrammatic, to Chapter 19, where one-loop amplitudes acquire the imaginary parts this theorem guarantees and the infrared structure that accompanies them, and onward to the analyticity-and-dispersion program of Chapter 60, for which the optical theorem is the physical input that turns a total cross section into the full, complex scattering amplitude.

Further reading

The interaction picture, the Dyson series, and the canonical route to the S-matrix are developed in Peskin and Schroeder [396] (chapters 4 and 7), whose presentation of LSZ and the Källén–Lehmann representation this chapter most closely tracks; Schwartz [432] (chapters 6–7 and 24) gives a complementary modern treatment with careful attention to the on-shell residues and the meaning of Z. The most careful textbook account of the asymptotic condition, the in/out formalism, and the cluster-decomposition foundations is Weinberg [500] (chapters 3–4 and 10), which also frames the optical theorem and unitarity nonperturbatively. Srednicki [454] (the path-integral-first chapters on LSZ) and Zee [532] offer alternative derivations worth comparing. The primary sources are Gell-Mann and Low [215] for the vacuum-projection theorem, Lehmann et al. [322] for the reduction formula, and Källén [289], Lehmann [321] for the spectral representation. For the spectral and analytic-structure axioms used in Section 14.4, and for Haag's theorem and the inequivalence of representations confronted in Section 14.2.6, see Streater and Wightman [458] and Haag [246]; the analyticity and dispersion-relation program for which the optical theorem is the physical input is the subject of Eden et al. [170].

Chapter 15

Feynman Diagrams

The previous chapter left perturbation theory in a state that was correct but unworkable. The Dyson series of Chapter 14 expresses the S-matrix as a time-ordered exponential of the interaction, and LSZ reduction expresses every amplitude as a residue of a time-ordered correlation function of free fields; together they reduce all of scattering, order by order in the coupling, to the evaluation of vacuum expectation values of products of free fields. But the one amplitude we evaluated this way — the $O(\lambda)$ contribution to $\phi\phi \to \phi\phi$ for the carried scalar theory — cost us a fourteen-operator normal-ordering done by hand. That method does not scale to two loops, let alone to gauge theory. This chapter builds the machinery that does: an algorithm that turns any term of the Dyson series, mechanically and without touching a ladder operator, into a finite sum of pictures, each of which evaluates to a definite number. By the end the model ϕ^4 will have its $2 \to 2$ amplitude at tree level, $i\mathcal{M} = -i\lambda$, derived twice by independent routes, and a complete rule-set ready to be handed to the observable-extracting chapter that follows.

The engine is Wick's theorem, and the chapter opens by establishing it in the operator form that the Dyson series actually consumes: a statement about the time-ordered product itself, not merely its vacuum matrix element, expressing it as a normal-ordered remainder plus a sum over all pairwise contractions, each contraction being exactly the Feynman propagator of Chapter 9. This is the operator face of the same identity that Chapter 10 obtained from the Gaussian generating functional $Z_0[J]$, and the two faces are made to agree. From there the development is a single idea pursued relentlessly: a contraction is a line, a vertex is a point where four lines meet, and a correlator is the sum over all ways of drawing lines between points. Section 15.2 reads this sum as a catalogue of position-space diagrams and proves that the combinatorial multiplicity of each topology collapses against the $1/N!$ of the Dyson exponential and the $1/4!$ of the vertex, leaving the position-space Feynman rules with their symmetry factors. Section 15.3 Fourier-transforms the rules: vertex integrations become

momentum-conserving delta functions, the loop count $L = I - V + 1$ emerges, LSZ becomes amputation of external legs, and the tree amplitude acquires its final algebraic form.

Two of the chapter's sections exist to make the apparatus airtight rather than merely usable. Section 15.4 treats the symmetry factor not as a mnemonic but as the order $|\mathrm{Aut}(D)|$ of the diagram's automorphism group, proves the identification by orbit–stabiliser, and verifies it on a catalogue of representative graphs — tree vertex, tadpole, figure-eight, sunset, bubble — before carrying the one-loop self-energy through dimensional regularization with the integrals of Chapter D. Section 15.5 then re-derives the entire rule-set, propagators and vertices and the very same symmetry factors, from the generating functional $Z[J]$ of Chapter 10, where $1/S$ falls out of the functional derivatives automatically; the same amplitude computed two ways is the chapter's central consistency check.

With the diagrams in hand, Section 15.6 sorts them. Disconnected graphs exponentiate; $W = -i \ln Z$ generates exactly the connected ones; vacuum bubbles cancel by the linked-cluster theorem; amputation strips the external propagators; and the Legendre transform Γ isolates the one-particle-irreducible core on which the renormalization programme of Chapters 21 and 22 and the effective potential of Chapter 29 are built. The final section extends the rules to anticommuting fields, importing the Grassmann calculus of Chapters 11 and 12 to derive the directed fermion line, the fermion–boson vertex, the relative sign between exchange diagrams, and the closed-fermion-loop factor of (-1) with its trace — the signs and traces that Chapter 17 and Chapter 26 inherit whole for QED and Yang–Mills. Throughout we hold the chapter conventions of the style canon: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the Feynman $i\varepsilon$ prescription, and the gamma-matrix trace technology collected in Section C.6.

15.1 Wick's Theorem: Time Ordering as a Sum of Contractions

Chapter 14 left us holding a powerful but unwieldy object. The interaction-picture S-matrix is the time-ordered exponential of the interaction (Theorem 14.6),

$$\mathbf{S} = U(\infty,-\infty) = \mathcal{T}\exp\Big(-i\int_{-\infty}^{\infty}\mathrm{d}t\; H_I(t)\Big) = \sum_{N=0}^{\infty}\frac{(-i)^N}{N!}\int \mathrm{d}^4z_1\cdots\mathrm{d}^4z_N\;\mathcal{T}\Big\{\mathcal{H}_I(z_1)\cdots\mathcal{H}_I(z_N)\Big\} \tag{15.1}$$

where for ϕ^4 theory the interaction-picture density is built entirely from the *free* field, $\mathcal{H}_I(z) = \frac{\lambda}{4!}\,\phi(z)^4$ (Eq. (14.11)), and ϕ henceforth means the free interaction-picture field ϕ_I of Theorem 14.2, with the mode expansion and algebra fixed in Eq. (9.147). The LSZ reduction of Chapter 14 expresses every scattering amplitude as a residue of a *time-ordered correlation function* of these free fields, so that the entire content of perturbation theory, order by order in λ, is the

evaluation of vacuum expectation values of the form

$$\langle 0|\,\mathcal{T}\{\phi(x_1)\cdots\phi(x_n)\}\,|0\rangle \qquad \text{and, with vertices inserted,} \qquad \langle 0|\,\mathcal{T}\{\phi(x_1)\cdots\phi(x_n)\,\phi(z_1)^4 \tag{15.2}$$

In Chapter 14 we evaluated the very first such object — the $O(\lambda)$ contribution to $\phi\phi\to\phi\phi$ — by brute force, expanding each field into ladder operators, normal-ordering the resulting fourteen-operator string by hand, and reading off which terms had a surviving matrix element between the chosen in- and out-states. The computation worked, and it gave the carried number $-i\lambda$; but it was an exercise in stamina, and it would not survive contact with the two-loop diagrams of Chapter 21, let alone the gauge theories of Part VII. What we need is an algorithm that converts a time-ordered product of free fields, mechanically and without ever touching a ladder operator, into a finite sum of c-number propagators multiplying normal-ordered remainders. That algorithm is *Wick's theorem*, and it is the single device that turns the Dyson series Eq. (15.1) from an operator nightmare into bookkeeping of lines and vertices — the bookkeeping that the next section will draw as Feynman diagrams.

Wick's theorem has, in this book, two faces that must agree. In Chapter 10 the free n-point function was computed from the generating functional $Z_0[J]$, and the chain rule on a Gaussian produced the path-integral form (Theorem 10.23): every free correlator is a sum over *pairings* of its arguments, one propagator per pair. That derivation never mentioned an operator. Here we establish the operator (canonical) form, the form that acts directly on the Dyson series: an *operator identity* valid before any vacuum expectation value is taken, expressing the time-ordered product itself — not merely its vacuum matrix element — as a normal-ordered product plus a sum over all ways of contracting pairs of fields. The two faces were shown to coincide in Section 10.5.4, where the operator statement (Theorem 10.25) was obtained as a cross-check on the path-integral result. This section gives that operator identity its proper, operational home: we re-establish the contraction at the point of use, prove the theorem in the form the perturbation series actually consumes, extend it to the normal-ordered insertions that the Dyson vertices demand, and verify the four-field case term by term so that the diagram dictionary of Section 15.2 has a fully worked anchor.

15.1.1 The contraction of two free fields

Everything rests on a single c-number. Split the free field into its positive- and negative-frequency parts, as introduced for the mode expansion of Chapter 9,

$$\phi(x)=\phi^+(x)+\phi^-(x), \qquad \phi^+(x)=\int\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\,a_{\mathbf{p}}\,e^{-ip\cdot x}, \quad \phi^-(x)=\int\frac{\mathrm{d}^3p}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}}\,a^\dagger_{\mathbf{p}}\,e^{} \tag{15.3}$$

with $p^0=E_{\mathbf{p}}=\sqrt{\mathbf{p}^2+m^2}$ on shell. By construction ϕ^+ holds the annihilation operators and kills the vacuum on the right, $\phi^+(x)\,|0\rangle=0$, while ϕ^- holds the

creation operators and kills it on the left, $\langle 0|\,\phi^-(x) = 0$. The label is the standard, initially counter-intuitive one: the part with $e^{-iE_{\mathbf{p}}t}$ (positive frequency) is the *annihilation* part, and we name it ϕ^+. The only nonvanishing commutator among the four pieces is

$$\begin{aligned}\left[\phi^+(x),\phi^-(y)\right] &= \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{d}^3q}{(2\pi)^3}\frac{1}{\sqrt{2E_{\mathbf{p}}}\sqrt{2E_{\mathbf{q}}}}\left[a_{\mathbf{p}},a^\dagger_{\mathbf{q}}\right]e^{-ip\cdot x+iq\cdot y}\\ &= \int \frac{\mathrm{d}^3p}{(2\pi)^3}\frac{\mathrm{d}^3q}{(2\pi)^3}\frac{(2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})}{\sqrt{2E_{\mathbf{p}}}\sqrt{2E_{\mathbf{q}}}}e^{-ip\cdot x+iq\cdot y} = \int\frac{\mathrm{d}^3p}{(2\pi)^3\,2E_{\mathbf{p}}}e^{-ip\cdot(x-y)} \;=\; D(x-\end{aligned} \tag{15.4}$$

where we used $\left[a_{\mathbf{p}},a^\dagger_{\mathbf{q}}\right] = (2\pi)^3\delta^{(3)}(\mathbf{p}-\mathbf{q})$ (Eq. (9.147)) and recognized the result as the positive-frequency Wightman function $D(x-y)$ of Eq. (9.150). This is a pure c-number — a number times the identity operator — because the right-hand side of the canonical commutator is.

Definition 15.1 (Contraction of two free fields)**.** The *contraction* of two free scalar fields is the c-number obtained by subtracting the normal-ordered product from the time-ordered product (Eq. (10.146)),

$$\overbracket{\phi(x)\,\phi(y)} \;\equiv\; \mathcal{T}\,\phi(x)\phi(y) - :\phi(x)\phi(y):\,, \tag{15.5}$$

with $:\cdots:$ the normal ordering of Theorem 9.19, which moves every creation operator to the left of every annihilation operator as though they commuted.

The defining subtraction is engineered precisely to isolate the commutator Eq. (15.4), as the following identification — the one fact on which every diagram in this book ultimately rests — makes explicit.

Proposition 15.2 (The contraction is the Feynman propagator)**.** *For any two spacetime points,*

$$\overbracket{\phi(x)\,\phi(y)} \;=\; D_F(x-y) \;=\; \int\frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot(x-y)}}{p^2-m^2+i\varepsilon}, \tag{15.6}$$

the free Feynman propagator of Theorem 9.24.

Proof. Take $x^0 > y^0$ first. Then $\mathcal{T}\,\phi(x)\phi(y) = \phi(x)\phi(y)$, and we expand both fields through Eq. (15.3) and reorder into normal form. Of the four products $\phi^+(x)\phi^+(y)$, $\phi^+(x)\phi^-(y)$, $\phi^-(x)\phi^+(y)$, $\phi^-(x)\phi^-(y)$, only the second is not already normal-ordered: it has an annihilation part to the left of a creation part. Reordering it costs exactly one commutator,

$$\phi^+(x)\phi^-(y) = \phi^-(y)\phi^+(x) + \left[\phi^+(x),\phi^-(y)\right], \tag{15.7}$$

so that

$$\phi(x)\phi(y) = \underbrace{\phi^+(x)\phi^+(y) + \phi^-(y)\phi^+(x) + \phi^-(x)\phi^+(y) + \phi^-(x)\phi^-(y)}_{=\,:\,\phi(x)\phi(y)\,:} \;+\; \Big[\phi^+(x), \phi^-(y)\Big]\,. \tag{15.8}$$

The four bracketed terms have every creation operator to the left of every annihilation operator: they *are* $:\phi(x)\phi(y):$. Hence $\mathcal{T}\,\phi(x)\phi(y) - :\phi(x)\phi(y): = \big[\phi^+(x), \phi^-(y)\big] = D(x-y)$ by Eq. (15.4). For $y^0 > x^0$ the time-ordering instead puts $\phi(y)$ to the left, and the identical reordering gives the contraction $\big[\phi^+(y), \phi^-(x)\big] = D(y-x)$. Combining the two cases with the step functions of Eq. (9.151),

$$\overbrace{\phi(x)\,\phi(y)} = \theta(x^0 - y^0)\,D(x-y) + \theta(y^0 - x^0)\,D(y-x) = D_F(x-y), \tag{15.9}$$

which is precisely the position-space Feynman propagator Eq. (9.153). Its covariant momentum representation Eq. (15.6) is the contour result Eq. (9.154), with the $i\varepsilon$ implementing the time ordering and the Fourier and $i\varepsilon$ conventions as fixed in Sections A.3 and A.4.

Two corollaries record themselves. First, the contraction is symmetric, $D_F(x-y) = D_F(y-x)$, because both $D(x-y)$ for $x^0 > y^0$ and the swapped expression depend only on the analytic continuation of the same scalar function; equivalently, the momentum integral Eq. (15.6) is even in $x-y$. Second, since the contraction is a c-number it equals its own vacuum expectation value, and as $\langle 0|\,:\phi\phi:\,|0\rangle = 0$ (Theorem 9.19) it also equals $\langle 0|\,\mathcal{T}\phi(x)\phi(y)\,|0\rangle = D_F(x-y)$ directly, recovering Eq. (10.147). □

The mechanism of the proof is worth naming, because the whole theorem is its iteration: to normal-order a product, push each annihilation part rightward past each creation part, and *every such transposition deposits one factor of the propagator* and nothing else. A field that never has to move past a creation operator contributes no propagator; a field that moves past k creation operators contributes k separate propagator terms, one for each partner it stepped over.

15.1.2 Wick's theorem as an operator identity

We now state the general identity and prove it. The combinatorial vocabulary is fixed first. A *contraction pattern* on the fields $\phi(x_1), \dots, \phi(x_n)$ is a choice of some number $r \geq 0$ of disjoint unordered pairs $\{i_1, j_1\}, \dots, \{i_r, j_r\}$ drawn from $\{1, \dots, n\}$; the $2r$ fields in these pairs are "contracted" and the remaining $n - 2r$ fields are "uncontracted." To each pattern we assign the operator

$$\Big(\textstyle\prod_{a=1}^{r} D_F(x_{i_a} - x_{j_a})\Big)\;:\!\prod_{k\ \text{uncontracted}}\!\phi(x_k)\!:\,, \tag{15.10}$$

i.e. each contracted pair is replaced by its c-number propagator and the leftover fields are normal-ordered together. The pattern with $r = 0$ is the bare normal-ordered product; a pattern with $2r = n$ (possible only for even n) is "fully contracted" and is a pure c-number.

Theorem 15.3 (Wick's theorem, operator form)**.** *A time-ordered product of free scalar field operators equals the sum, over* all *contraction patterns, of the associated operators Eq. (15.10):*

$$\mathcal{T}\{\phi(x_1)\cdots\phi(x_n)\} \;=\; \sum_{\text{contraction patterns } P} \Big(\prod_{\{i,j\}\in P} D_F(x_i-x_j)\Big) : \prod_{k\notin P}\phi(x_k): \,, \tag{15.11}$$

schematically: the fully uncontracted normal-ordered product, plus all single contractions, plus all double contractions, and so on up to the fully-contracted terms. The identity holds as an operator equation on the free Fock space of Theorem 9.17, prior to taking any matrix element.

This is the operator statement obtained as a cross-check in Theorem 10.25; it is the engine of the present chapter, so we give the proof in full here, in the form that the induction step of the perturbation series will reuse.

Proof. Induction on the number of fields n. For $n = 1$ a single field is already normal-ordered and there is no pair to contract: Eq. (15.11) reads $\phi(x_1) =: \phi(x_1):$, true. For $n = 2$ it is the rearrangement of Eq. (15.5) read together with Theorem 15.2,

$$\mathcal{T}\,\phi(x_1)\phi(x_2) =: \phi(x_1)\phi(x_2): \; + \overbrace{\phi(x_1)\phi(x_2)}, \tag{15.12}$$

the two patterns being "no contraction" and "contract $\{1,2\}$."

Assume Eq. (15.11) holds for any n free fields. Take $n+1$ fields and relabel so that x_{n+1} carries the *earliest* time, $x^0_{n+1} \le x^0_i$ for all $i \le n$. This entails no loss of generality: the left side of Eq. (15.11) is totally symmetric under relabeling because $\mathcal{T}$ symmetrizes bosonic fields, and the right side is symmetric because the sum ranges over *all* patterns. With x_{n+1} latest-to-act (earliest in time means rightmost under $\mathcal{T}$), the time-ordering symbol places it on the far right,

$$\mathcal{T}\{\phi(x_1)\cdots\phi(x_{n+1})\} = \Big(\mathcal{T}\{\phi(x_1)\cdots\phi(x_n)\}\Big)\,\phi(x_{n+1}). \tag{15.13}$$

Substitute the inductive hypothesis Eq. (15.11) for the bracket and multiply each term on the right by $\phi(x_{n+1}) = \phi^+(x_{n+1}) + \phi^-(x_{n+1})$. Consider one such term: a product of propagators (untouched, being c-numbers) times a normal-ordered string $:\phi(x_{k_1})\cdots\phi(x_{k_m}):$ of the uncontracted fields. We must bring $\phi(x_{n+1})$ *inside* the colons to restore normal order.

The creation part $\phi^-(x_{n+1})$ presents no obstruction: normal ordering places creation operators on the left, and ϕ^- commutes with the other fields' content

only up to c-numbers it never meets here, so $:X:\,\phi^-(x_{n+1}) = :X\,\phi^-(x_{n+1}):$ without cost. The annihilation part $\phi^+(x_{n+1})$ must instead be moved leftward through the string to sit among the annihilation operators; each time it passes the negative-frequency (creation) content $\phi^-(x_k)$ of one of the fields inside the colons, it deposits the c-number commutator

$$\Big[\phi^+(x_{n+1}), \phi^-(x_k)\Big] = D(x_{n+1} - x_k) = D_F(x_{n+1} - x_k), \tag{15.14}$$

the last equality holding by Theorem 15.2 *because* x_{n+1} is the earliest time, so $D(x_{n+1} - x_k)$ is exactly the contraction in that time order. Each deposited commutator is precisely one new contraction joining the new field x_{n+1} to a previously-uncontracted partner x_k; collecting the commutator pulls that pair out as a propagator and removes $\phi(x_k)$ from the normal-ordered remainder.

Summing the outcomes, each term of the n-field expansion generates: (i) one term in which $\phi(x_{n+1})$ joins the normal-ordered remainder uncontracted, and (ii) for each uncontracted field x_k in that term, one term in which x_{n+1} is contracted with x_k and both leave the colons. This is exactly the statement that the $(n+1)$-field patterns are the n-field patterns with x_{n+1} either left uncontracted or contracted with one of the fields still available — which enumerates every contraction pattern on $n+1$ fields once and only once. Hence the collected terms reassemble Eq. (15.11) for $n+1$ fields, closing the induction. □

Two features of Theorem 15.3 deserve emphasis before we use it. First, it is an *operator* identity: both sides are operators on Fock space, equal as such, and we have not yet thrown away anything by taking expectation values. This is what makes the theorem usable inside the Dyson series, where the fields sit between non-vacuum asymptotic states. Second, the contraction is unambiguously the *time-ordered* two-point function D_F, not the Wightman function and not the commutator; one may not contract fields inside an order that has already been fixed by hand. The theorem is a statement about the $\mathcal{T}$-product on the left, and it is the $\mathcal{T}$-product that LSZ and the path integral both deliver (Theorem 10.19).

15.1.3 The vacuum corollary

Sandwiching Eq. (15.11) between vacua collapses it dramatically. Every term that still carries a normal-ordered product of one or more uncontracted fields has vanishing vacuum expectation value, because a normal-ordered string has an annihilation operator on its right that kills $|0\rangle$ (or a creation operator on its left that kills $\langle 0|$); only the *fully-contracted* terms, which are pure c-numbers, survive. A full contraction exists only when every field is paired, which is possible iff n is even.

Corollary 15.4 (Free correlators are sums of pairings)**.** *For odd n, $\langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|0\rangle = 0$. For even $n = 2k$,*

$$\langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_{2k})\,|0\rangle \;=\; \sum_{\text{pairings } P}\;\prod_{\{i,j\}\in P} D_F(x_i - x_j), \tag{15.15}$$

the sum running over the $(2k-1)!!$ ways of partitioning the $2k$ points into k unordered pairs, with one propagator per pair.

The count $(2k-1)!!$ is the number of perfect matchings of $2k$ objects: pair the first point with any of the remaining $2k-1$, then the next free point with any of the remaining $2k-3$, and so on, giving $(2k-1)(2k-3)\cdots 3\cdot 1 = (2k-1)!!$ (Eq. (10.135)). Equation Eq. (15.15) is identical, term for term, to the path-integral Wick theorem Theorem 10.23, equation Eq. (10.136): the two derivations — the chain rule on a Gaussian in Chapter 10, and the commuting of annihilation operators to the right here — produce not merely the same number but the same intermediate object, a sum over pairings with one propagator per pair. This is the operator-side confirmation, promised in Chapter 9 and discharged in Chapter 10, that the free scalar field is a Gaussian field whose entire content is its two-point function.

15.1.4 The generalized theorem: normal-ordered insertions

The targets in Eq. (15.2) are not products of fields at distinct points: each Dyson vertex is four fields at a *single* point z, and in the normal-ordered conventions inherited from Theorem 9.20 the interaction is naturally written $:\!\phi(z)^4\!:$. Two refinements are therefore needed before Theorem 15.3 can be turned loose on the perturbation series: we must know how contractions behave (a) between fields already collected inside a normal-ordering symbol, and (b) at coincident points.

Point (b) is immediate. Nothing in the proof of Theorem 15.3 assumed the points distinct; the contraction $D_F(x_i - x_j)$ is a well-defined tempered distribution for all separations, and at coincidence $D_F(0) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{i}{p^2-m^2+i\varepsilon}$ is simply a (divergent) constant, to be regulated and renormalized in Chapters 21 and 22. A self-contraction of two of the four fields sitting at one vertex is thus a legitimate term; it is the algebraic seed of the tadpole and vacuum-bubble diagrams.

Point (a) is the substantive one, and it is exactly what normal-ordering the vertex buys us. We package fields into *clusters*, some of which are wrapped in their own normal-ordering symbol.

Theorem 15.5 (Wick's theorem with normal-ordered clusters)**.** *Let the operators be grouped into clusters, where each cluster is either a single field or a pre-normal-ordered local product $:\!\phi(z)\cdots\phi(z)\!:$. Then the time-ordered product of the clusters equals the sum over all contraction patterns* in which no contracted pair lies

entirely within a single pre-normal-ordered cluster, *each contraction being D_F and the uncontracted fields normal-ordered together:*

$$\mathcal{T}\Big\{:\phi(z_1)\cdots:\ :\phi(z_2)\cdots:\ \cdots\Big\} = \sum_{\substack{\text{patterns } P \\ \text{no intra-cluster pair}}} \Big(\prod_{\{i,j\}\in P} D_F(x_i-x_j)\Big) : \prod_{k\notin P}\phi(x_k): \,. \tag{15.16}$$

Proof. It suffices to prove the rule for one pre-normal-ordered cluster; the general case follows by treating each cluster in turn. The key fact is that normal ordering is, by Theorem 15.3 applied to the cluster alone, exactly the time-ordered product with its *internal* contractions removed. Concretely, for a cluster of fields at a common point, Eq. (15.11) reads

$$\mathcal{T}\{\phi(z)\cdots\phi(z)\} =: \phi(z)\cdots\phi(z): + (\text{terms with at least one internal contraction}), \tag{15.17}$$

so that

$$:\phi(z)\cdots\phi(z): \ = \mathcal{T}\{\phi(z)\cdots\phi(z)\} - (\text{internal-contraction terms}); \tag{15.18}$$

the normal-ordered cluster is the plain product purged of self-contractions. Now form the time-ordered product of this cluster with the other clusters and apply the ungrouped theorem Theorem 15.3 to the whole collection. The full sum contains every pattern, including those with intra-cluster pairs. But the subtraction in Eq. (15.18) cancels precisely the patterns whose restriction to the cluster contains an internal contraction: each such pattern appears once from the $\mathcal{T}$-product term and once, with opposite sign, from the subtracted internal-contraction terms multiplied out against the external contractions of the remaining fields. (The bijection is exact because the external contractions of a pattern are independent of, and unaffected by, the internal ones being removed.) What survives is exactly the sum over patterns with no intra-cluster pair, which is Eq. (15.16). Iterating over the remaining clusters removes all intra-cluster contractions and proves the theorem. □

The practical content is a slogan: *normal-ordering a vertex forbids it from contracting with itself.* If one adopts the normal-ordered interaction $\mathcal{H}_I = \frac{\lambda}{4!}:\phi^4:$ — the convention consistent with the normal-ordered free Hamiltonian fixed in Chapter 9 — then the generalized theorem automatically discards every self-contraction of a vertex: no tadpole stub, no vacuum loop hanging off a single point. If instead one keeps the un-normal-ordered $\frac{\lambda}{4!}\phi^4$ written in Eq. (14.11), those self-contractions are present and must be carried until the renormalization program of Chapter 22 disposes of them. Either bookkeeping is legitimate; the two differ only by terms that a constant shift of the vacuum energy and a mass counterterm reabsorb. We will state which convention is in force whenever it affects a diagram count.

Notation Watch

The operator Wick's theorem appears twice in this book by design. Chapter 10 established it (Theorem 10.25) as a cross-check that the path-integral and canonical routes to the free correlator agree; the present section is its operational home, where it is proved in the form Eq. (15.11) that the Dyson induction reuses and extended to the normal-ordered insertions Eq. (15.16) that the path-integral discussion did not need. The contraction is the same object throughout, $\overbrace{\phi\phi} = D_F$; the symbol $: \cdots :$ is always the normal ordering of Theorem 9.19.

15.1.5 The four-field case, term by term

The smallest case that exhibits the full structure — bare term, single contractions, double contractions — is $n = 4$. Writing it out completely fixes the combinatorics that the diagram dictionary of the next section will read off graphically, and it is the case the $O(\lambda)$ four-point amplitude needs.

Worked Example 15.6 (The complete Wick expansion of $\mathcal{T}\,\phi_1\phi_2\phi_3\phi_4$)**.** Abbreviate $\phi_i \equiv \phi(x_i)$ and $D_F^{ij} \equiv D_F(x_i - x_j)$. Theorem 15.3 for $n = 4$ has one no-contraction pattern, $\binom{4}{2} = 6$ single-contraction patterns, and 3 double-contraction patterns ($(4-1)!! = 3$), for $1 + 6 + 3 = 10$ terms:

$$\begin{aligned}\mathcal{T}\,\phi_1\phi_2\phi_3\phi_4 = {} & :\phi_1\phi_2\phi_3\phi_4: \\ & + D_F^{12}:\phi_3\phi_4: \; + D_F^{13}:\phi_2\phi_4: \; + D_F^{14}:\phi_2\phi_3: \; + D_F^{23}:\phi_1\phi_4: \; + D_F^{24}:\phi_1\phi_3: \; + D_F^{34}:\phi_1 \\ & + D_F^{12}D_F^{34} + D_F^{13}D_F^{24} + D_F^{14}D_F^{23}.\end{aligned} \tag{15.19}$$

The first line is the term with no propagators; the second collects the six ways of contracting one pair and normal-ordering the surviving two fields; the third collects the three ways of contracting all four into two pairs — pure c-numbers.

Now take the vacuum expectation value. Every term on the first two lines contains a normal-ordered product of at least two fields and so vanishes between vacua (Theorem 9.19); only the three c-number terms survive, giving

$$\langle 0|\,\mathcal{T}\,\phi_1\phi_2\phi_3\phi_4\,|0\rangle = D_F^{12}D_F^{34} + D_F^{13}D_F^{24} + D_F^{14}D_F^{23}, \tag{15.20}$$

the three pairings of Theorem 15.4 with $k = 2$, in agreement with the path-integral result Eq. (10.138). The three terms are the three perfect matchings of four points, drawn as chord diagrams in Fig. 15.1.

15.1.6 Feeding the Dyson series

We can now state precisely what Wick's theorem does to the perturbation series and hand the result to the diagram rules. Consider the order-λ^N contribution to

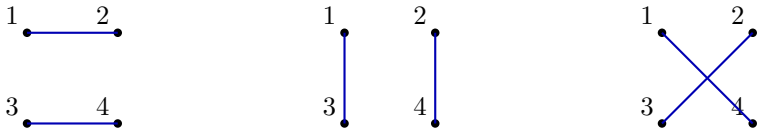


Figure 15.1: The three perfect matchings (pairings) of four points that survive in the free four-point function Eq. (15.20). Each chord is one Feynman propagator D_F; the surviving terms of Wick's theorem are exactly the ways of joining the four external points in pairs. These chord diagrams are the combinatorial skeletons that Section 15.2 dresses into Feynman diagrams.

a time-ordered correlator with n external fields, obtained by expanding Eq. (15.1) to N vertices:

$$\frac{1}{N!}\Big(\frac{-i\lambda}{4!}\Big)^N \int \mathrm{d}^4 z_1 \cdots \mathrm{d}^4 z_N \ \langle 0|\, \mathcal{T}\left\{\phi(x_1)\cdots\phi(x_n)\ \phi(z_1)^4\cdots\phi(z_N)^4\right\} |0\rangle\,. \tag{15.21}$$

The object inside the vacuum brackets is a time-ordered product of $n+4N$ free fields. Theorem 15.3 — in its generalized form Theorem 15.5 if the vertices are normal-ordered — expands it into a finite sum over contraction patterns, and the vacuum corollary Theorem 15.4 keeps only the fully-contracted ones. Each surviving term is a product of $\frac{1}{2}(n+4N)$ propagators (so $n+4N$ must be even, fixing the parity selection rule), integrated over the vertex positions $z_1,\dots,z_N$. In words: *at every order, the correlator is a finite sum of products of propagators, one factor per contracted pair, integrated over the vertices and weighted by the multiplicity of pairings that produce it.* No operator survives; the entire computation has been reduced to choosing how to wire the legs together and counting how many wirings give each wiring pattern.

For ϕ^4 theory at lowest nontrivial order this is the computation Chapter 14 did by hand, now read off mechanically. The connected $O(\lambda)$ contribution to $\langle 0|\,\mathcal{T}\,\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\,|0\rangle$ takes $n=4$ external points and one vertex z ($N=1$, $n+4N=8$ fields). The connected fully-contracted patterns are those that join each external point to one of the four vertex legs $\phi(z)$: there are $4!=24$ ways to assign the four distinct external points to the four identical vertex legs, and this 24 cancels the $1/4!$ in the vertex normalization, leaving the four-propagator structure

$$(-i\lambda)\int \mathrm{d}^4 z\ D_F(x_1-z)\,D_F(x_2-z)\,D_F(x_3-z)\,D_F(x_4-z), \tag{15.22}$$

whose amputated, on-shell residue under LSZ is the carried number $-i\lambda$ — exactly the contraction structure anticipated in Eq. (10.193). At $O(\lambda^2)$ the same machinery, applied with two vertices z_1, z_2, produces the patterns in which

two external points attach to each vertex and the two vertices are joined by the two remaining legs: a single skeleton with a closed loop of two propagators between z_1 and z_2, occurring in the three external pairings (12)(34), (13)(24), (14)(23) — the s-, t-, and u-channel one-loop contributions to $\phi\phi \to \phi\phi$, together with the disconnected and self-contraction terms that the connected/amputated reductions and the normal-ordering convention remove.

We have thus discharged the chapter's first task. Every correlator in the Dyson series is now, by Theorem 15.3 and Theorem 15.5, a finite sum of products of propagators (from the full contractions) times normal-ordered remainders (which vanish in the vacuum amplitudes), each term a definite assignment of pairings among external points and vertex legs, weighted by how many such assignments are identical. Each such term is begging to be drawn: a propagator is a line, a vertex is a point where four lines meet carrying $-i\lambda$, an external field is a labelled endpoint, and the number of equivalent pairings is the term's multiplicity. Turning this contraction bookkeeping into pictures, and reading off from those pictures the position-space Feynman rules for ϕ^4 — including the all-important symmetry factors that the multiplicity counting produces — is the work of Section 15.2.

15.2 From Contractions to Pictures: Position-Space Feynman Rules

The previous section closed the operator side of the story. Wick's theorem (Theorem 15.3, and the path-integral form Theorem 10.23) turns the time-ordered product at the heart of every Dyson term into a finite, mechanical sum: a sum over all ways of pairing the fields, with each pair replaced by a Feynman propagator D_F. Order by order in the coupling, an interacting correlator is therefore nothing more than a (large) collection of products of propagators, each product weighted by a rational number that counts how many pairings produced it. What we do not yet have is a way to *organize* that collection so that we can see at a glance which products are the same, which are physically distinct, and how the bookkeeping factors assemble. The organizing device is a picture. To every product of propagators we will attach a graph—points joined by lines—in such a way that two products carry the same graph precisely when they are the same function of the external arguments. Counting then becomes a problem about graphs rather than about strings of field operators, and the answer to that counting problem is the set of *position-space Feynman rules*: a short dictionary that lets us write down any term of the expansion directly from its picture, never returning to the Wick sum at all. This section builds the dictionary, proves the clean factor it assigns to each diagram by explicit pairing counts, and draws the canonical diagrams of ϕ^4 theory.

15.2.1 The correlator as a sum of fully contracted products

We work throughout with the frozen ϕ^4 theory theory, $\mathcal{L} = \frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ with $\lambda > 0$, in the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$. The object of interest is the n-point time-ordered correlation function of the Heisenberg field in the interacting vacuum $|\Omega\rangle$,

$$G^{(n)}(x_1,\dots,x_n) \;=\; \langle\Omega|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|\Omega\rangle\,. \tag{15.23}$$

The Gell-Mann–Low theorem of Chapter 14 (equivalently the path-integral master formula Eq. (10.186)) expresses Eq. (15.23) as a ratio of free-field vacuum expectation values, with the interaction exponentiated and all fields now the free interaction-picture fields ϕ_I of Theorem 14.2:

$$G^{(n)}(x_1,\dots,x_n) \;=\; \frac{\Big\langle 0\Big|\,\mathcal{T}\,\phi_I(x_1)\cdots\phi_I(x_n)\,\exp\Big[-\frac{i\lambda}{4!}\int \mathrm{d}^4z\,\phi_I(z)^4\Big]\Big|0\Big\rangle}{\Big\langle 0\Big|\,\mathcal{T}\,\exp\Big[-\frac{i\lambda}{4!}\int \mathrm{d}^4z\,\phi_I(z)^4\Big]\Big|0\Big\rangle}. \tag{15.24}$$

Here $|0\rangle$ is the free Fock vacuum of Theorem 9.17, and the single exponent has absorbed the time integral of the interaction Hamiltonian $H_I(t) = \int \mathrm{d}^3x\,\frac{\lambda}{4!}\phi_I^4$ of Eq. (14.11) into the covariant spacetime integral $\int \mathrm{d}^4z$. The relative sign and the factor $1/4!$ are exactly those of the action Eq. (10.80); the factor $-i$ is the one carried by each insertion of $-iH_I\,\mathrm{d}t$ in the Dyson series Eq. (14.25).

Expanding the numerator exponential in powers of λ gives a sum over the number V of interaction insertions (the number of *vertices*),

$$\text{numerator} = \sum_{V=0}^{\infty}\frac{1}{V!}\Big(-\frac{i\lambda}{4!}\Big)^V\int \mathrm{d}^4z_1\cdots\mathrm{d}^4z_V\;\Big\langle 0\Big|\,\mathcal{T}\,\phi_I(x_1)\cdots\phi_I(x_n)\,\phi_I(z_1)^4\cdots\phi_I(z_V)^4\,|0\Big\rangle\,. \tag{15.25}$$

The $1/V!$ is the coefficient of the exponential's Taylor series; it will be one of the two combinatorial factors we must track. Each spacetime integration variable z_a is integrated over all of Minkowski space, and the four fields $\phi_I(z_a)^4$ sitting at a common point z_a are the four *legs* of the a-th vertex.

Every term in Eq. (15.25) is a free-field vacuum expectation value of a time-ordered product of an even number of fields, $n + 4V$ of them. Wick's theorem (Theorem 15.3) rewrites such an object as the sum over *all complete pairings* of the fields into contractions, each contraction being a propagator,

$$\overbrace{\phi_I(a)\,\phi_I(b)} \;=\; D_F(a-b), \tag{15.26}$$

the same Feynman propagator established as the free two-point function in Theorem 9.24. The normal-ordered remainders that Wick's theorem also produces all have at least one uncontracted field, and a normal-ordered product with any

uncontracted field has vanishing vacuum expectation value, since $\langle 0|$ and $|0\rangle$ are annihilated by the creation and annihilation operators flanking the normal ordering. Only the *fully contracted* terms survive. We have therefore reduced Eq. (15.23) to a completely explicit prescription:

> *At order λ^V, list every way of pairing the $n{+}4V$ fields $\phi_I(x_1), \dots, \phi_I(x_n), \phi_I(z_1), \dots, \phi_I(z_V)$ completely into $\frac{1}{2}(n+4V)$ pairs. Each complete pairing contributes a product of $\frac{1}{2}(n+4V)$ propagators, one per pair, multiplied by the prefactor $\frac{1}{V!}(-i\lambda/4!)^V$ and integrated over $z_1, \dots, z_V$.*

This is a finite, mechanical sum at each order. What it is not, yet, is readable: at order λ^2 in the four-point function the number of complete pairings already runs into the thousands, the overwhelming majority of which produce the same handful of distinct functions. Organizing them is the work of the picture.

15.2.2 Reading a contraction as a graph

A complete pairing is a purely combinatorial datum: it says which field is contracted with which. We record it as a graph.

Definition 15.7 (Position-space Feynman diagram for ϕ^4)**.** Fix a correlator $G^{(n)}$ and an order V. A *Feynman diagram* consists of:

- *n external points* $x_1, \dots, x_n$, drawn as labelled endpoints, each the seat of exactly one field $\phi_I(x_i)$ and hence touched by exactly one line;
- *V vertices* $z_1, \dots, z_V$, drawn as dots, each the seat of four fields $\phi_I(z_a)^4$ and hence *four-valent*—touched by exactly four line-ends;
- *lines* (edges) joining the line-ends in pairs, one line per contraction.

A given complete pairing determines a diagram by drawing one line between the two endpoints of each contracted pair. Two pairings give the *same diagram* if there is a relabelling of the four legs at each vertex, together with a permutation of the vertices among themselves, carrying one pairing to the other. The *value* of a diagram is the product of propagators read off from its lines, integrated over the vertex positions.

The clause defining when two pairings give the same diagram deserves emphasis, because it is the seed of every combinatorial factor in the subject. The four fields $\phi_I(z_a)^4$ at a vertex are identical operators; nothing distinguishes the "first" factor from the "second." A pairing that contracts x_1 with the first factor of $\phi_I(z_1)^4$ and a pairing that contracts x_1 with the second factor are different pairings—different elements of the Wick sum—but they yield the *same propagator* $D_F(x_1 - z_1)$ and hence the same function. The diagram remembers only the

function. Likewise the labels $z_1, \dots, z_V$ on the vertices are dummy integration variables; permuting them relabels a pairing but, after the symmetric integration $\int \mathrm{d}^4 z_1 \cdots \mathrm{d}^4 z_V$, leaves the value untouched. A diagram is thus an equivalence class of pairings under these two relabellings, and the value of Eq. (15.25) is a sum over equivalence classes, each weighted by the number of pairings it contains.

The dictionary that assigns a value to a diagram now reads off directly from Eq. (15.25) and Eq. (15.26).

Definition 15.8 (Position-space Feynman dictionary)**.** For ϕ^4 theory, the elements of a diagram translate as follows.

Diagram element	**Analytic factor**
internal or external line from a to b	$D_F(a-b)$
four-valent vertex at z_a	$-i\lambda \int \mathrm{d}^4 z_a$
external point x_i	a fixed argument x_i of $G^{(n)}$ (not integrated)

The full value of the diagram is the product of all line factors and vertex factors, divided by a *symmetry factor* S defined below.

Two features of Theorem 15.8 must be justified before we may use it, because they are exactly where the combinatorics has been hidden. First, the vertex factor is quoted as $-i\lambda \int \mathrm{d}^4 z_a$, with a bare coupling and *no* $1/4!$, whereas the expansion Eq. (15.25) carries $(-i\lambda/4!)^V/V!$. Second, the rule instructs us to draw each distinct diagram only once and then divide by S, whereas the Wick sum contains one term per pairing. The reconciliation of these two statements—that the $1/4!$ per vertex and the $1/V!$ overall are consumed by the multiplicity of pairings collapsing onto each diagram, leaving only the small leftover $1/S$—is the central computation of this section, to which we now turn.

15.2.3 The combinatorial identity behind the rules

Consider a single diagram Γ at order λ^V in $G^{(n)}$, and ask how many distinct complete pairings of the $n + 4V$ fields fall into its equivalence class. Call that number N_Γ. The value contributed by Γ to the numerator Eq. (15.25) is then

$$\frac{1}{V!}\Big(-\frac{i\lambda}{4!}\Big)^V N_\Gamma \;\times\; \Big[\text{product of propagators of } \Gamma,\; \int \mathrm{d}^4 z_1 \cdots \mathrm{d}^4 z_V\Big], \tag{15.27}$$

since all N_Γ pairings give the identical propagator product. The content of the Feynman rules is the evaluation of N_Γ.

The relabellings that, by Theorem 15.7, leave a diagram fixed form a group: at each vertex we may apply any of the 4! permutations of its four legs, and we

may apply any of the $V!$ permutations of the vertices, giving an "automorphism candidate" group of order $(4!)^V\, V!$. Acting with an element of this group on one pairing of Γ produces a pairing in the same class. Generically these actions are all distinct, so that the orbit of any one pairing has the full size $(4!)^V\, V!$; then $N_\Gamma = (4!)^V\, V!$ exactly. The exceptions are relabellings that map a pairing back to *itself*—genuine automorphisms of the diagram, such as swapping the two ends of a self-loop, or exchanging two propagators that run between the same pair of vertices. These reduce the orbit. By the orbit–stabilizer theorem the orbit size is the group order divided by the order of the stabilizer, so

$$N_\Gamma \;=\; \frac{(4!)^V\, V!}{S(\Gamma)}, \qquad S(\Gamma) \;\equiv\; \big|\{\text{relabellings fixing a given pairing of } \Gamma\}\big|. \tag{15.28}$$

The integer $S(\Gamma)$ is the *symmetry factor* of the diagram. Substituting Eq. (15.28) into Eq. (15.27), the prefactor collapses spectacularly:

$$\frac{1}{V!}\Big(-\frac{i\lambda}{4!}\Big)^V \frac{(4!)^V\, V!}{S(\Gamma)} \;=\; \frac{(-i\lambda)^V}{S(\Gamma)}. \tag{15.29}$$

The $1/V!$ of the Dyson exponential cancels the $V!$ from permuting vertices; the $(1/4!)^V$ of the coupling normalization cancels the $(4!)^V$ from permuting legs; and what survives is the bare coupling $-i\lambda$ per vertex divided by the symmetry factor. This is precisely the content of Theorem 15.8, now derived rather than asserted, and it proves the position-space Feynman rules.

Theorem 15.9 (Position-space Feynman rules for ϕ^4)**.** *The n-point correlator Eq. (15.23) of ϕ^4 theory is computed, to order λ^V, by the following procedure.*

1. *Draw all topologically distinct diagrams (in the sense of Theorem 15.7) with the n external points $x_1, \dots, x_n$ and exactly V four-valent vertices, in which every external point is touched by one line and every vertex by four.*
2. *To each diagram assign: a factor $D_F(a-b)$ for each line joining a to b; a factor $-i\lambda \int \mathrm{d}^4 z_a$ for each vertex z_a, with the vertex positions integrated over all spacetime; and a single fixed argument for each external point.*
3. *Divide the resulting expression by the symmetry factor $S(\Gamma)$ of Eq. (15.28).*
4. *Sum over all such diagrams at the given order, and over all orders to the accuracy required.*

The denominator of Eq. (15.24) cancels every diagram containing a vacuum subdiagram *(a connected piece touching no external point), so that for a connected correlator only fully connected diagrams with no vacuum pieces contribute; this is the linked-cluster property, whose general combinatorial proof is the subject of a later section of this chapter (it was anticipated, at the two- and four-point level, in Theorem 10.37).*

The symmetry factor is, for the diagrams that matter most, equal to one or to a small power of two, and we will compute it from Eq. (15.28) in each case rather than quote it; the systematic enumeration of $S(\Gamma)$ for arbitrary topologies is the dedicated treatment promised by Theorem 15.9. We now exercise the rules on the three diagrams that carry ϕ^4 theory.

15.2.4 The four-point function at $O(\lambda)$: the tree vertex

The first interacting process of the thread is $\phi\phi \to \phi\phi$, whose amplitude we will eventually extract from the four-point correlator $G^{(4)}(x_1, x_2, x_3, x_4)$. At order λ^1 the numerator Eq. (15.25) is

$$\Big(-\frac{i\lambda}{4!}\Big)\int \mathrm{d}^4 z\, \big\langle 0|\, \mathcal{T}\, \phi_I(x_1)\phi_I(x_2)\phi_I(x_3)\phi_I(x_4)\, \phi_I(z)^4\, |0\big\rangle\,, \tag{15.30}$$

an eight-field vacuum expectation value to be expanded by Wick's theorem. We classify its complete pairings by which external points contract with the vertex.

Worked Example 15.10 (The $O(\lambda)$ four-point pairings, counted in full)**.** There are three topological types of complete pairing of the eight fields in Eq. (15.30), distinguished by how many of the four external points contract directly with a leg of the vertex z.

(a) All four externals reach the vertex (the connected tree). Contract x_1 with one of the four legs of $\phi_I(z)^4$: 4 choices. Contract x_2 with one of the remaining three legs: 3 choices. Then x_3 with one of the remaining two: 2; and x_4 with the last: 1. No fields remain. The number of such pairings is

$$N = 4 \cdot 3 \cdot 2 \cdot 1 = 4! = 24, \tag{15.31}$$

each giving the identical product $D_F(x_1 - z)D_F(x_2 - z)D_F(x_3 - z)D_F(x_4 - z)$. By Eq. (15.28) with $V = 1$ this is $N = (4!)^1\, 1!/S$, so $S = 1$: the tree diagram has no nontrivial automorphism. Multiplying by the prefactor $-i\lambda/4!$ and using Eq. (15.29),

$$G^{(4)}_{\text{tree}}(x_1, x_2, x_3, x_4) = -i\lambda \int \mathrm{d}^4 z\; D_F(x_1 - z)\, D_F(x_2 - z)\, D_F(x_3 - z)\, D_F(x_4 - z). \tag{15.32}$$

The four-factorial of pairings has cancelled the $1/4!$ of the vertex exactly, leaving the clean weight $-i\lambda$ promised by Theorem 15.8.

(b) Two externals pair with each other; the other two reach the vertex (disconnected, with a loop). Choose the external pair, say (x_1, x_2), contracted into $D_F(x_1 - x_2)$; contract x_3 and x_4 with two legs of z ($4 \cdot 3 = 12$ ways), leaving two legs of z that must contract with each other into the self-loop $D_F(z - z)$. The value is $D_F(x_1 - x_2) \times D_F(x_3 - z)D_F(x_4 - z)D_F(z - z)$, integrated over z. The graph splits into two pieces—a bare propagator line $x_1 - x_2$ and a loop-corrected

line joining x_3, x_4 through z—so it is *disconnected*: it describes one pair of points propagating freely while the other pair interacts, not a genuine $2 \to 2$ scattering.

(c) The externals pair among themselves; all four vertex legs self-contract (a vacuum bubble). For instance (x_1, x_2) and (x_3, x_4) paired, and the four fields at z contracted into two self-loops. The value is $D_F(x_1 - x_2)D_F(x_3 - x_4) \times$ [figure-eight at z]. The factor in brackets is a closed diagram touching no external point—a vacuum bubble.

The pieces of type (c) are removed exactly by the denominator of Eq. (15.24). Its order-λ term is $(-i\lambda/4!)\int \mathrm{d}^4 z\, \langle 0|\mathcal{T}\phi_I(z)^4|0\rangle = (-i\lambda/4!)\int \mathrm{d}^4 z\, 3\,[D_F(z-z)]^2$, the figure-eight vacuum diagram (its three pairings give $S = 8$ by Eq. (15.28), namely $3 = (4!)/8$), and when the ratio in Eq. (15.24) is expanded the vacuum factor multiplying the type-(c) numerator term is precisely subtracted. The pieces of type (b) are disconnected but not vacuum; they survive in $G^{(4)}$ and represent the non-scattering part of the correlator, which the LSZ amputation of Chapter 14 discards because it lacks the requisite poles in all four external momenta. The single piece that contributes to the connected scattering amplitude is the tree Eq. (15.32).

The connected tree diagram is drawn in Fig. 15.2. It is the diagrammatic heart of ϕ^4 theory: four external points, one vertex carrying the factor $-i\lambda$, and four propagators. Fourier transforming Eq. (15.32) and amputating the external legs—the operations of the next section—will collapse the four propagators and the $\int \mathrm{d}^4 z$ into a single momentum-conserving delta function times the constant amplitude, reproducing the value $\mathcal{M} = -\lambda$ already obtained by hand in Eq. (14.54). That the careful Wick count of Theorem 15.10 lands on exactly the same $-i\lambda$ is the first confirmation that the diagrammatic bookkeeping is faithful.

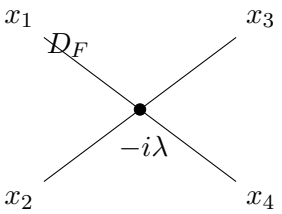


Figure 15.2: The $O(\lambda)$ connected four-point diagram of ϕ^4 theory: the single tree vertex. Each line is a propagator $D_F(x_i - z)$, the vertex (black dot) carries $-i\lambda \int \mathrm{d}^4 z$, and the four external points $x_1, \ldots, x_4$ are the fixed arguments of $G^{(4)}$. The symmetry factor is $S = 1$. This is the position-space seed of the $\phi\phi \to \phi\phi$ amplitude.

15.2.5 The two-point function at $O(\lambda)$: the tadpole

The first loop correction in the thread is the order-λ contribution to the propagator $G^{(2)}(x_1, x_2)$. Its numerator at this order is the six-field expectation value

$$\Big(-\frac{i\lambda}{4!}\Big)\int \mathrm{d}^4 z\, \big\langle 0|\, \mathcal{T}\, \phi_I(x_1)\phi_I(x_2)\, \phi_I(z)^4\, |0\big\rangle\,. \tag{15.33}$$

Worked Example 15.11 (The $O(\lambda)$ propagator correction, counted in full)**.** The connected pairings of Eq. (15.33) are those in which both x_1 and x_2 reach the vertex. Contract x_1 with one of the four legs of $\phi_I(z)^4$: 4 ways. Contract x_2 with one of the remaining three legs: 3 ways. Two legs of z now remain, and they have no choice but to contract with each other, forming the self-loop $D_F(z-z)$: 1 way. The connected count is therefore

$$N = 4 \cdot 3 \cdot 1 = 12. \tag{15.34}$$

By Eq. (15.28) with $V = 1$, $N = (4!)/S$ gives $S = 24/12 = 2$. The nontrivial automorphism is the exchange of the two ends of the self-loop: swapping them is a relabelling of the vertex legs that maps the pairing to itself, so it is counted by the stabilizer rather than the orbit, halving the multiplicity. Assembling the prefactor and value,

$$G^{(2)}_{\lambda}(x_1, x_2) = \frac{-i\lambda}{2}\int \mathrm{d}^4 z\, D_F(x_1 - z)\, D_F(z - z)\, D_F(z - x_2), \tag{15.35}$$

in which the explicit factor $\frac{1}{2}$ is the inverse symmetry factor $1/S$. The remaining pairings of Eq. (15.33)—those contracting x_1 directly with x_2 and self-contracting all four legs of z—produce $D_F(x_1-x_2)$ times a vacuum figure-eight, a disconnected vacuum piece removed by the denominator of Eq. (15.24) exactly as in part (c) of Theorem 15.10.

The surviving diagram Eq. (15.35) is the *tadpole*, drawn in Fig. 15.3: a single line from x_1 to x_2 interrupted by a vertex from which a closed loop $D_F(z-z)$ dangles. It is the first quantum correction to the propagation of a ϕ quantum, the loop encoding the field's interaction with a virtual fluctuation created and reabsorbed at the same vertex. The closed loop $D_F(z-z) = D_F(0)$ is ultraviolet divergent—the propagator at coincident points is a divergent momentum integral—and disposing of that divergence is the renormalization task of Chapter 22; here it suffices that the diagram, and its weight $-i\lambda/2$, are fixed unambiguously by the pairing count.

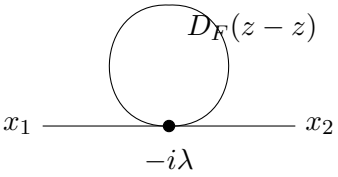


Figure 15.3: The $O(\lambda)$ correction to the propagator $G^{(2)}(x_1, x_2)$ of ϕ^4 theory: the tadpole. Two of the four vertex legs join the external points x_1, x_2; the other two close into the loop $D_F(z-z)$. The exchange of the two loop ends is a genuine automorphism, giving symmetry factor $S = 2$ and the weight $-i\lambda/2$ of Eq. (15.35).

15.2.6 The four-point function at $O(\lambda^2)$

The pattern of the rules is most clearly displayed at the first order where genuine loops connect distinct vertices. At $O(\lambda^2)$ the four-point numerator Eq. (15.25) is

$$\frac{1}{2!}\Big(-\frac{i\lambda}{4!}\Big)^2 \int \mathrm{d}^4 z_1\, \mathrm{d}^4 z_2 \;\Big\langle 0\big|\, \mathcal{T}\, \phi_I(x_1)\phi_I(x_2)\phi_I(x_3)\phi_I(x_4)\, \phi_I(z_1)^4\, \phi_I(z_2)^4\, \big|0\Big\rangle\,, \tag{15.36}$$

a twelve-field expectation value. We enumerate the topologically distinct *connected* diagrams—the ones that survive after the denominator removes vacuum pieces and after the disconnected products are set aside. With $V = 2$ vertices, four external legs, and every vertex four-valent, the connected topologies are:

- **The one-loop bubble (three channels).** Two external legs attach to z_1 and two to z_2, while the remaining two legs of each vertex join across, forming a loop of two internal propagators $D_F(z_1 - z_2)^2$. The three ways of splitting $\{x_1, x_2, x_3, x_4\}$ into the pair at z_1 and the pair at z_2 are the s-, t-, and u-channel diagrams: (12)(34), (13)(24), (14)(23). These are the corrections that, after renormalization, supply the one-loop running of the coupling in Chapters 22 and 24.

- **The leg-corrected tree.** The four external legs all attach to one vertex, say z_1, except that one of them runs first to z_2, which carries a tadpole self-loop, before continuing to z_1; equivalently, the $O(\lambda)$ tree of Fig. 15.2 with the tadpole of Fig. 15.3 inserted on one external line. There are also the diagram in which the second vertex's tadpole sits on the *internal* structure. These are propagator corrections riding on the tree and are reducible: cutting one internal line disconnects them.

- **The double tadpole.** The tree vertex z_1 with a tadpole loop, joined to a second vertex z_2 that itself carries a tadpole—higher in the same family of self-energy insertions.

The genuinely new, one-particle-irreducible loop is the bubble, and we count its symmetry factor explicitly; the leg-corrected and double-tadpole diagrams are built from the $S = 2$ tadpole already counted, by the same orbit–stabilizer argument applied vertex by vertex.

Worked Example 15.12 (The s-channel bubble, counted in full)**.** Take the bubble with x_1, x_2 at one vertex and x_3, x_4 at the other, the two vertices joined by two internal lines. We count the pairings in Eq. (15.36) producing this graph. First fix which named vertex, z_1 or z_2, carries x_1, x_2: there are 2 such assignments, and they are exchanged by the 2! vertex relabelling. For a fixed assignment, say $x_1, x_2 \to z_1$ and $x_3, x_4 \to z_2$:

- contract x_1 with a leg of z_1 (4) and x_2 with another leg of z_1 (3): $4 \cdot 3 = 12$ ways, leaving two legs of z_1;
- contract x_3 with a leg of z_2 (4) and x_4 with another (3): $4 \cdot 3 = 12$ ways, leaving two legs of z_2;
- contract the two remaining legs of z_1 with the two remaining legs of z_2: the first leg of z_1 has 2 partners, the second has 1, giving $2! = 2$ ways (the two internal propagators $D_F(z_1 - z_2)^2$).

For the fixed assignment this is $12 \cdot 12 \cdot 2 = 288$ pairings, and including the 2 vertex assignments,

$$N = 2 \cdot 288 = 576 = (4!)^2 \cdot 2! \,/\, 2. \tag{15.37}$$

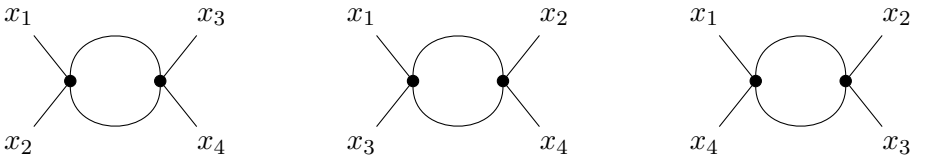
By Eq. (15.28) with $V = 2$, $N = (4!)^2\, 2!/S$ gives $S = 2$. The single nontrivial automorphism is the exchange of the two internal lines running between z_1 and z_2: swapping them is a leg relabelling that fixes the pairing, and it is exactly the factor $2! = 2$ that appeared in the last bullet but is undone by the stabilizer. Assembling Eq. (15.29),

$$G^{(4)}_{s\text{-loop}} = \frac{(-i\lambda)^2}{2} \int \mathrm{d}^4 z_1\, \mathrm{d}^4 z_2\; D_F(x_1 - z_1) D_F(x_2 - z_1) \left[D_F(z_1 - z_2)\right]^2 D_F(x_3 - z_2) D_F(x_4 - z_2), \tag{15.38}$$

The $1/2!$ of the Dyson exponential has cancelled the two vertex assignments, the $(1/4!)^2$ has cancelled the leg permutations, and the leftover $1/S = 1/2$ is the inverse symmetry factor of the doubled internal line. The t- and u-channel bubbles are identical with (x_1, x_2, x_3, x_4) permuted to (x_1, x_3, x_2, x_4) and (x_1, x_4, x_2, x_3), each again with $S = 2$.

The three bubble channels are drawn together in Fig. 15.4. They are the position-space content of the one-loop $2 \to 2$ amplitude; Fourier transforming Eq. (15.38) will convert the convolution of propagators into the familiar loop integral $\int \mathrm{d}^4 k \,/[(k^2 - m^2)((k + p)^2 - m^2)]$ over the bubble, with the external

Mandelstam variable determined by which channel is drawn. That this same factor $\frac{1}{2}$ emerges identically from the orbit–stabilizer count Eq. (15.37) and from the naive "two identical internal lines" rule of thumb is the consistency one should expect, and it illustrates how the dedicated symmetry-factor section will recover the rules of thumb as corollaries of Eq. (15.28).



s-channel: (12)(34) t-channel: (13)(24) u-channel: (14)(23)

Figure 15.4: The three $O(\lambda^2)$ one-loop bubble diagrams of the four-point function, one per pairing of the external legs across the two vertices. Each carries two vertices ($(-i\lambda)^2$), two internal propagators forming the loop $D_F(z_1 - z_2)^2$, four external propagators, and symmetry factor $S = 2$ from the exchange of the two internal lines, as counted in Theorem 15.12.

With these three computations the dictionary of Theorem 15.9 has been validated on every diagram ϕ^4 theory needs in this part: the tree vertex of $\phi\phi \to \phi\phi$, the one-loop tadpole correction to the propagator, and the one-loop bubbles of the four-point function, each with its symmetry factor derived from the pairing count rather than asserted. Every diagram value is an integral over vertex positions of a product of position-space propagators—a convolution. Convolutions are diagonalized by the Fourier transform, and that is the route the next section takes: passing to momentum space turns each $\int \mathrm{d}^4 z$ into a momentum-conserving delta function at the vertex, collapses the external-leg propagators under LSZ amputation, and replaces every loop by an integral over an undetermined internal momentum, delivering the momentum-space Feynman rules and, with them, the carried tree amplitude $-i\lambda$ in its final algebraic form.

15.3 Momentum Space, Amputation, and the Tree Amplitude

The position-space rules assembled in Section 15.2 are exact and complete, but they are not yet in a form one would ever use to extract a number. Each diagram is an integral over the spacetime locations of its interaction vertices, and at each such location sits a product of Feynman propagators tied together by the geometry of the graph. A propagator $D_F(x - y)$ depends only on the *difference* of its endpoints, so the integral over a vertex position is a convolution; and the one fact about convolutions that every reader carries from undergraduate Fourier analysis is that they become ordinary products under the Fourier transform. The

whole of this section is the systematic exploitation of that fact. We will write every internal propagator as its Fourier integral, carry out the vertex-position integrals once and for all, and watch the convolutions collapse into algebra: a fixed factor per line, a fixed factor per vertex, a delta-function enforcing conservation of four-momentum at each vertex, and a leftover integral over the momentum circulating in each independent loop. The output is the momentum-space Feynman rule set — the form in which the rules are actually quoted and used.

Having the rules in momentum space is not yet having a cross section. The diagrams compute time-ordered correlation functions of fields, whereas experiments measure S-matrix elements between asymptotic particle states. The bridge between the two was built in Chapter 14: the LSZ reduction formula (Theorem 14.31) expresses a scattering amplitude as the on-shell residue of the correlator's external legs, and the invariant amplitude $\mathcal{M}$ (Eq. (14.46)) is what survives after the external-leg propagators are stripped away. Translated into the diagrammatic language we now have, LSZ is a remarkably clean instruction: *amputate* the external legs and put them on the mass shell. This section makes that instruction precise, defines the connected amputated Green's function it acts on, and then carries out the entire program on the simplest possible case — the four-point function of ϕ^4 theory at order λ. The reward is the carried number of Chapter 14, reproduced now by pure diagrammatics: the tree-level amplitude for $\phi\phi \to \phi\phi$ is $i\mathcal{M} = -i\lambda$, a constant, with the dependence on the scattering kinematics deferred to the loop level and handed to Chapter 16.

Throughout we keep the standing conventions of ϕ^4 theory: a single real scalar of mass m and quartic coupling $\lambda > 0$, the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, and the Fourier convention in which every $\int d^4p$ carries a $(2\pi)^{-4}$ while every delta-function produced by an x-integration carries a $(2\pi)^4$.

15.3.1 The propagator in momentum space, and why a diagram is a convolution

Every internal line of a ϕ^4 diagram is a Feynman propagator $D_F(x-y) = \langle 0|\mathcal{T}\,\phi(x)\phi(y)|0\rangle$, the free two-point function established for ϕ^4 theory in Chapter 9 and characterized analytically as a tempered distribution in Chapter 5. Its defining property for present purposes is its momentum-space representation,

$$D_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\,\frac{i}{p^2 - m^2 + i\varepsilon}\,e^{-ip\cdot(x-y)}, \tag{15.39}$$

in which the $i\varepsilon$ prescription (Eq. (5.61), Theorem 5.22) is what selects the time-ordered Green's function out of the family of solutions of $(\Box + m^2)D_F = -i\,\delta^{(4)}$: the pole at $p^0 = +\sqrt{\mathbf{p}^2 + m^2} - i\varepsilon'$ is routed below the real axis and the one at $p^0 = -\sqrt{\mathbf{p}^2 + m^2} + i\varepsilon'$ above it, so that closing the p^0 contour for $x^0 > y^0$ picks

up the positive-frequency part and for $x^0 < y^0$ the negative-frequency part. We will not re-derive Eq. (15.39); we take it as the input to the transform.

The integrand of Eq. (15.39) is the object we will trade everywhere for the propagator:

$$\widetilde{D_F}(p) \;\equiv\; \frac{i}{p^2 - m^2 + i\varepsilon}, \tag{15.40}$$

the *momentum-space propagator*. The notation records the content of the convolution theorem: Eq. (15.39) says that D_F is the inverse Fourier transform of $\widetilde{D_F}$.

To see why a diagram becomes a product, consider the structure of *any* ϕ^4 diagram in position space. It is built from

- a set of *external points* $x_1, \ldots, x_n$, the fixed arguments of the correlator, held at definite spacetime locations;
- a set of *internal vertices* $z_1, \ldots, z_V$, each carried by an integral $-i\lambda \int d^4 z_a$ (Section 15.2);
- a set of *lines*, each a factor D_F joining two of these points, every line connecting either two vertices (an *internal* line) or a vertex to an external point (an *external* line).

The entire z_a-dependence of the integrand lives inside the propagators that touch z_a, and it enters only through the exponentials $e^{\mp ip\cdot z_a}$ of their Fourier representations. The vertex integrals $\int d^4 z_a$ therefore act *only* on products of exponentials. That is the whole mechanism: each $\int d^4 z_a$ over a vertex linear in z_a in the exponent produces a Dirac delta-function, and that delta-function is conservation of four-momentum at the vertex. We now make this exact.

15.3.2 Vertex integration generates momentum conservation

Pick one internal vertex, at position z, and let the four lines meeting there carry momenta q_1, q_2, q_3, q_4. We adopt a uniform routing convention: *every momentum is oriented to flow into the vertex.* Concretely, for a line running between z and another point w (a different vertex or an external point) we write its propagator using Eq. (15.39) with the difference taken so that the z-dependence is $e^{+iq\cdot z}$,

$$D_F(w - z) = \int \frac{d^4 q}{(2\pi)^4}\, \widetilde{D_F}(q)\, e^{-iq\cdot(w-z)} = \int \frac{d^4 q}{(2\pi)^4}\, \widetilde{D_F}(q)\, e^{-iq\cdot w}\, e^{+iq\cdot z}, \tag{15.41}$$

which defines q as the momentum flowing from w *into* z. (The propagator is even, $D_F(w-z) = D_F(z-w)$, so this is merely a choice of sign for the dummy variable; reversing the orientation sends $q \to -q$, which the even integrand absorbs.) With

every one of the four lines written this way, the vertex factor and its position integral act on the product of the four exponentials,

$$-i\lambda \int d^4z\; e^{+i(q_1+q_2+q_3+q_4)\cdot z} \;=\; -i\lambda\,(2\pi)^4\,\delta^{(4)}(q_1+q_2+q_3+q_4), \tag{15.42}$$

where we used the fundamental Fourier identity $\int d^4z\, e^{ik\cdot z} = (2\pi)^4\,\delta^{(4)}(k)$. The remaining factors $e^{-iq\cdot w}$ from Eq. (15.41) are not yet integrated; they belong to the other endpoint w of each line and will be processed when we treat that endpoint. Equation Eq. (15.42) is the central result of the transform: *the position integral over an interaction vertex converts the convolution into a product, and produces a delta-function enforcing that the total momentum flowing into the vertex vanishes,*

$$q_1 + q_2 + q_3 + q_4 = 0. \tag{15.43}$$

This is conservation of four-momentum at the vertex, derived rather than assumed. It is the diagrammatic shadow of spacetime translation invariance: the interaction $\phi^4(z)$ does not know where z is, so integrating over all z enforces that no net momentum is injected at the interaction point.

The same computation runs identically at every vertex. An external point x_i is treated differently — it is *not* integrated, because it is a fixed argument of the correlator; instead its single attached propagator carries the factor $e^{-iq_i\cdot x_i}$, which is exactly the kernel that the LSZ reduction will act on to put x_i on shell. For an internal line joining two vertices z_a and z_b, both endpoints are integrated: the line contributes $\widetilde{D_F}(q)$ and an integral $\int d^4q/(2\pi)^4$, and the same momentum q appears (with opposite sign) in the conservation delta-functions of z_a and z_b. We may now read off the complete dictionary.

Definition 15.13 (Momentum-space Feynman rules for ϕ^4)**.** The momentum-space contribution of a ϕ^4 Feynman diagram contributing to an n-point correlation function is assembled as follows. Assign a four-momentum to every line, oriented; impose conservation at each vertex.

1. **Internal line** of momentum q: a factor
$$\widetilde{D_F}(q) = \frac{i}{q^2 - m^2 + i\varepsilon}. \tag{15.44}$$
2. **Vertex**: a factor
$$-i\lambda\,(2\pi)^4\,\delta^{(4)}\Big(\textstyle\sum_{\text{in}} q\Big), \tag{15.45}$$
the delta-function enforcing that the momenta flowing into the vertex sum to zero.
3. **Loop integration**: a factor
$$\int \frac{d^4k}{(2\pi)^4} \tag{15.46}$$

for each independent loop momentum left undetermined by the vertex delta-functions (the counting is fixed in Section 15.3.3); the measure follows the convention of Section D.1.

4. **External legs**: each external point x_i contributes its propagator $\widetilde{D_F}(p_i)$ and the Fourier kernel that will be amputated by LSZ (Section 15.3.4).

5. **Symmetry factor**: divide by the order of the diagram's automorphism group, the symmetry factor S (computed systematically in the next section).

After performing all loop integrations, the result for a diagram with external momenta $p_1, \dots, p_n$ (all incoming) carries an overall factor $(2\pi)^4\,\delta^{(4)}(p_1+\cdots+p_n)$ expressing global four-momentum conservation.

Remark 15.14 (Where the (2π)'s live). The bookkeeping of 2π factors is automatic and worth stating once so it is never recomputed. Each internal line brings one $\int d^4q/(2\pi)^4$; each vertex brings one $(2\pi)^4\delta^{(4)}$. With I internal lines and V vertices there are I factors of $(2\pi)^{-4}$ and V factors of $(2\pi)^{+4}$ before any delta-function is used. Using the delta-functions to do I of the momentum integrals (see below) consumes I of the δ's and their $(2\pi)^4$'s against the I measure factors, and the surviving counting leaves exactly one overall $(2\pi)^4\delta^{(4)}$ for global conservation and one $\int d^4k/(2\pi)^4$ per loop. No 2π ever needs to be tracked by hand: the rules as stated balance.

The momentum-space propagator Eq. (15.44) is a genuine simplification over its position-space ancestor: it is an elementary rational function with a pole on the mass shell, whereas $D_F(x-y)$ is a Hankel/Bessel distribution. The price — and it is the price paid in every loop calculation in this book — is the loop integral Eq. (15.46), the residue of the convolution that the delta-functions could not fully resolve. We now count exactly how many such integrals a diagram carries.

15.3.3 Counting the loop integrations: $L = I - V + 1$

Before any momenta are integrated, a diagram with I internal lines assigns one four-momentum to each, hence I four-dimensional integration variables $\int d^4q_j/(2\pi)^4$. The V vertices impose V conservation delta-functions Eq. (15.45). The question is how many of the I integrals these delta-functions allow us to perform, and how many survive as genuine loop integrals.

The key structural fact is that the V delta-functions are *not all independent* when the external momenta are themselves held fixed. Add the arguments of all V vertex delta-functions. Every *internal* line touches exactly two vertices, and by the uniform routing of Eq. (15.41) it enters the conservation condition of one with $+q$ and of the other with $-q$ (the momentum flows *into* one vertex and *out of* — i.e. with reversed sign into — the other). Hence every internal

momentum cancels in the sum, and what remains is the sum of the *external* momenta entering the diagram. The product of the V vertex delta-functions therefore contains, as the image of their sum, the single global factor

$$\prod_{a=1}^{V} \delta^{(4)}\Big(\textstyle\sum_{\text{in at } a} q\Big) \;=\; \delta^{(4)}\Big(\textstyle\sum_i p_i\Big) \times [V-1 \text{ independent constraints on internal momenta}]. \tag{15.47}$$

One linear combination of the V delta-functions is fixed entirely by external data — it is global four-momentum conservation, the overall factor promised in Theorem 15.13 — and is therefore *not* available to eliminate an internal integration variable. That leaves precisely $V-1$ delta-functions that genuinely constrain the internal momenta, each of which can be used to do one of the I internal integrals. The number of integrals that remain undetermined — the number of *independent loop momenta* — is therefore

$$\boxed{L \;=\; I-(V-1) \;=\; I-V+1.} \tag{15.48}$$

This is the loop-counting formula. It is identical in content to Euler's relation for a connected graph, (faces) = (edges) − (vertices) + 1, and indeed L is exactly the first Betti number of the diagram regarded as a one-dimensional cell complex — the number of independent cycles, the number of ways to draw a closed loop on the graph. We rederived it here from the analytic content of the delta-functions because that derivation is what tells us, line by line, which integrals survive and why each surviving one is genuinely four-dimensional and unconstrained.

Remark 15.15 (Connectedness is assumed). The relation Eq. (15.48) as stated is the loop number of a *connected* diagram. A disconnected diagram with C connected components has $L = I - V + C$, because each component supplies its own undeducible global conservation delta-function. The amplitudes we extract are built from connected, indeed connected and amputated, diagrams (Section 15.3.4), so $C = 1$ throughout, and Eq. (15.48) is the form we use.

The formula has an immediate and important corollary: a diagram with $L = 0$ has no loop integrals at all. Its value is a finite product of momentum-space propagators and vertex factors, completely fixed by the external momenta through the delta-functions, with nothing left to integrate. These are the *tree diagrams*, and they are where perturbation theory delivers its first, exact, finite numbers. Setting $L = 0$ in Eq. (15.48) gives $I = V - 1$: a connected tree on V vertices has exactly $V-1$ internal lines, the defining count of a tree in graph theory. The four-point function of ϕ^4 theory at order λ has $V = 1$ and $I = 0$, hence $L = 0$; it is the simplest tree there is, and it is the one we now reduce to an amplitude.

Worked Example 15.16 (Loop count of the one-loop ϕ^4 four-point diagram)**.** To see the formula bite where it matters, count the loops of the order-λ^2 contribution

to $\phi\phi \to \phi\phi$ shown in Fig. 15.5: two quartic vertices joined by two internal lines, with two external legs attached at each vertex. Here $V = 2$ and the internal lines number $I = 2$ (the two lines of the central bubble; the four external legs are amputated and do not count). The formula gives

$$L = I - V + 1 = 2 - 2 + 1 = 1, \tag{15.49}$$

one loop, as the picture's single closed cycle demands. Tracking it explicitly: assign incoming external momenta p_1, p_2 to the left vertex and outgoing p_3, p_4 to the right, and let the two internal lines carry momenta k and k'. The left vertex delta-function gives $p_1 + p_2 - k - k' = 0$ and the right gives $k + k' - p_3 - p_4 = 0$. Adding them yields $p_1 + p_2 - p_3 - p_4 = 0$ — global conservation, the combination fixed by external data. Subtracting, $p_1 + p_2 = k + k'$ is the one independent constraint, which we solve as $k' = p_1 + p_2 - k$. One of the two internal integrals $\int d^4k\, d^4k'$ is thereby done against the constraint delta-function, leaving the single loop integral

$$\int \frac{d^4k}{(2\pi)^4}\, \frac{i}{k^2 - m^2 + i\varepsilon}\, \frac{i}{(p_1 + p_2 - k)^2 - m^2 + i\varepsilon}, \tag{15.50}$$

one unconstrained four-momentum circulating in the loop — exactly $L = 1$. (This integral is logarithmically divergent and is the seed of the ϕ^4 renormalization carried in Chapters 21 and 22; here it serves only to confirm the counting.)

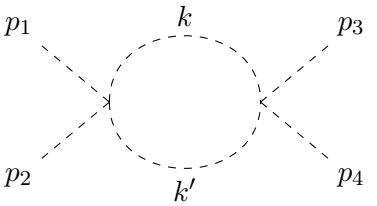


Figure 15.5: The order-λ^2 (s-channel) one-loop contribution to $\phi\phi \to \phi\phi$ in ϕ^4 theory. Two quartic vertices ($V = 2$) are joined by two internal lines ($I = 2$); the four external legs are amputated. The loop-counting formula Eq. (15.48) gives $L = I - V + 1 = 1$, the single closed cycle visible in the diagram, with loop momentum k and $k' = p_1 + p_2 - k$ fixed by conservation as in Theorem 15.16. This diagram is not evaluated here; it is the object Chapters 16 and 21 take up.

15.3.4 Connected, amputated Green's functions and the LSZ amputation

The Feynman rules compute the time-ordered correlation function $G^{(n)}(x_1, \ldots, x_n) = \langle 0|\mathcal{T}\, \phi(x_1) \cdots \phi(x_n)|0\rangle$, or its Fourier transform. To pass to scattering we must

restrict to the part of $G^{(n)}$ that is relevant to S-matrix elements and strip the kinematic factors that LSZ identifies as non-amplitude. Two reductions are involved, and it is worth naming each precisely.

The first is *connectedness*. A correlation function factorizes into products of lower-point functions whenever its diagram is disconnected; such contributions describe spectator subprocesses, not the genuine n-body scattering, and the LSZ formula projects them out. We therefore work with the connected correlator $G_c^{(n)}$, the sum of connected diagrams only. For the four-point function this means discarding the term in which $\phi\phi$ pairs into two independent propagators (two particles passing without interacting) and keeping only the diagrams in which all four external points communicate through a single connected graph.

The second is *amputation*. Fourier-transforming the connected n-point function and using overall momentum conservation, write

$$\int \Big(\prod_{i=1}^{n} d^4x_i \, e^{ip_i\cdot x_i} \Big) \, G_c^{(n)}(x_1,\dots,x_n) \;=\; (2\pi)^4 \delta^{(4)}\Big(\textstyle\sum_i p_i \Big) \, \widetilde{G}_c^{(n)}(p_1,\dots,p_n), \tag{15.51}$$

with all p_i incoming. Each external point x_i sits at the end of exactly one line, and that line, summed to all orders, is the *full* (interacting) propagator $G_{\text{full}}(p_i)$ on that leg — the free propagator $\widetilde{D_F}(p_i)$ dressed by all self-energy insertions. Because $G_{\text{full}}(p)$ has a pole at the physical mass $p^2 = m^2$ with residue iZ (the field-strength renormalization Z of Chapter 14), the external legs carry exactly the singular on-shell behaviour that LSZ is built to extract. The *amputated* function removes these external full propagators outright.

Definition 15.17 (Connected amputated Green's function)**.** The connected amputated n-point function $\widetilde{G}^{(n)}_{\text{amp}}$ is defined by dividing the connected momentum-space correlator $\widetilde{G}_c^{(n)}$ by the full propagator on each external leg,

$$\widetilde{G}_c^{(n)}(p_1,\dots,p_n) \;=\; \Big[\prod_{i=1}^{n} G_{\text{full}}(p_i) \Big] \, \widetilde{G}^{(n)}_{\text{amp}}(p_1,\dots,p_n). \tag{15.52}$$

Diagrammatically, $\widetilde{G}^{(n)}_{\text{amp}}$ is the sum of connected diagrams from which the external-leg propagators have been deleted ("amputated"), leaving the legs as stubs that mark where the on-shell momenta enter.

The role of amputation is exactly to undo a double-counting. The full external propagator $G_{\text{full}}(p_i)$ describes the free flight of the asymptotic particle from infinity to the interaction region; that flight is not part of the interaction and must not be included in the amplitude, on pain of producing the very pole at $p_i^2 = m^2$ that signals an external line going on shell. LSZ multiplies $\widetilde{G}_c^{(n)}$ by the inverse propagator $G_{\text{full}}(p_i)^{-1}$ on each leg and then takes $p_i^2 \to m^2$; the inverse propagator vanishes on shell as $G_{\text{full}}(p)^{-1} \to (p^2 - m^2)/(iZ)$, precisely cancelling

the leg pole and leaving a finite residue. The statement of Theorem 14.31, read through Theorem 15.17, is therefore the following clean instruction.

Theorem 15.18 (LSZ as amputation, on-shell)**.** *Let $\widetilde{G}^{(n)}_{amp}$ be the connected amputated n-point function of Theorem 15.17, and let $\mathcal{M}$ be the invariant amplitude of Eq. (14.46) for the process in which the p_i with i in the "in" set are incoming and the remainder are outgoing. Then*

$$i\mathcal{M}(p_1,\dots,p_n) \;=\; \Big(\prod_{i=1}^{n}\sqrt{Z}\Big)\,\widetilde{G}^{(n)}_{amp}(p_1,\dots,p_n)\Big|_{p_i^2=m^2}, \tag{15.53}$$

the external momenta put on the mass shell, where Z is the field-strength renormalization. At tree level $Z = 1 + O(\lambda)$, so to the order at which a diagram first contributes the factor $\sqrt{Z}$ is unity and $i\mathcal{M}$ equals the amputated diagram evaluated on shell.

This is the operational endpoint of the whole rule set. We do not re-prove it — the proof, with its careful treatment of the spectral decomposition, the one-particle pole, and the asymptotic-field limit, is the business of Chapter 14 and is imported here by Theorem 14.31. What Theorem 15.18 adds is the dictionary entry that connects that theorem to the diagrams: *compute the connected diagrams, delete the external propagators, put the legs on shell, and the result is $i\mathcal{M}$.* Everything in Theorem 15.13 now carries an explicit numerical weight, and the rules deliver amplitudes directly.

15.3.5 The carried number: $\phi\phi \to \phi\phi$ at tree level

We now reproduce, by pure diagrammatics, the amplitude that Chapter 14 obtained by hand from the Dyson series and LSZ. This is the layer ϕ^4 theory adds in this chapter: *the first amplitude in momentum space.*

Worked Example 15.19 (Tree-level $\phi\phi \to \phi\phi$ and the carried $-i\lambda$)**. The diagram.** At order λ the connected four-point function has a single topology: one quartic vertex with all four external points attached to it (Fig. 15.6). There are no internal lines, so $I = 0$, $V = 1$, and Eq. (15.48) gives $L = 0$ — a tree, with nothing to integrate.

Position to momentum. In position space the connected diagram is

$$G^{(4)}_c(x_1,x_2,x_3,x_4)\big|_{O(\lambda)} = -i\lambda\int d^4z\,\prod_{i=1}^{4} D_F(x_i - z), \tag{15.54}$$

the vertex factor $-i\lambda\int d^4z$ of Section 15.2 times the four propagators joining the external points to the vertex. The combinatorial factor here is 1: the symmetry factor of this diagram is unity (the 4! ways of attaching the four

distinct external lines to the four fields of $\phi^4(z)$ exactly cancel the $1/4!$ in the coupling), a statement we take from the position-space accounting and which the next section establishes systematically. Inserting the Fourier representation Eq. (15.41) for each propagator, with momentum p_i defined to flow from x_i into the vertex,

$$G_c^{(4)}\big|_{O(\lambda)} = -i\lambda \int \Big(\prod_{i=1}^{4} \frac{d^4p_i}{(2\pi)^4}\, \widetilde{D_F}(p_i)\, e^{-ip_i\cdot x_i} \Big) \int d^4z\; e^{+i(p_1+p_2+p_3+p_4)\cdot z}. \quad (15.55)$$

The vertex integral is exactly Eq. (15.42),

$$\int d^4z\; e^{+i(p_1+p_2+p_3+p_4)\cdot z} = (2\pi)^4\, \delta^{(4)}(p_1+p_2+p_3+p_4), \quad (15.56)$$

so that, reading off the momentum-space correlator through the definition Eq. (15.51) (which strips the external Fourier kernels and one overall $(2\pi)^4\delta^{(4)}$),

$$\widetilde{G}_c^{(4)}(p_1,p_2,p_3,p_4)\big|_{O(\lambda)} = (-i\lambda) \prod_{i=1}^{4} \widetilde{D_F}(p_i) = (-i\lambda) \prod_{i=1}^{4} \frac{i}{p_i^2 - m^2 + i\varepsilon}. \quad (15.57)$$

Every factor of 2π has cancelled as advertised in Theorem 15.14: the four measure factors $(2\pi)^{-4}$ are removed when the external kernels are stripped, and the single $(2\pi)^4\delta^{(4)}$ is the global conservation factor pulled out in Eq. (15.51).

Amputation. The four factors $\widetilde{D_F}(p_i)$ in Eq. (15.57) are precisely the external-leg propagators: at this order the full propagator equals the free one, $G_{\text{full}}(p_i) = \widetilde{D_F}(p_i) + O(\lambda)$. Dividing them out per Eq. (15.52) amputates the legs and leaves

$$\widetilde{G}^{(4)}_{\text{amp}}(p_1,p_2,p_3,p_4)\big|_{O(\lambda)} = -i\lambda, \quad (15.58)$$

a pure constant — no propagators, no integrals, no kinematic dependence whatsoever.

On shell. With $Z = 1 + O(\lambda)$ the prefactor $\prod_i \sqrt{Z} = 1$ at this order, and the amputated function Eq. (15.58) is already constant, so putting the legs on shell $p_i^2 = m^2$ changes nothing. Theorem Theorem 15.18 then delivers the invariant amplitude,

$$\boxed{i\mathcal{M}(\phi\phi \to \phi\phi) = -i\lambda} \qquad \Longleftrightarrow \qquad \mathcal{M} = -\lambda, \quad (15.59)$$

the carried number of ϕ^4 theory, reproduced exactly — and by a completely independent route — from the value computed by hand from the Dyson expansion in Chapter 14. The two computations agree, which is the internal consistency check the chapter was built to deliver: the operator/LSZ route of Chapter 14 and the diagrammatic route of Theorem 15.13 give the same amplitude.

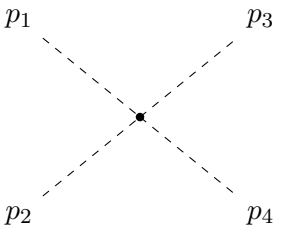


Figure 15.6: The tree-level diagram for $\phi\phi \to \phi\phi$ in ϕ^4 theory: a single quartic vertex ($V = 1$, $I = 0$, $L = 0$) with the four external momenta entering. Amputating the external legs and putting them on shell yields the invariant amplitude $i\mathcal{M} = -i\lambda$ of Eq. (15.59), independent of the scattering kinematics at this order.

Physical Insight: why the tree amplitude is a constant, and what breaks it

The amplitude Eq. (15.59) is independent of how the particles are aimed: it does not depend on the centre-of-mass energy, the scattering angle, or any other kinematic invariant. The physical reason is that the ϕ^4 interaction is a *contact* interaction — all four legs meet at a single point, with no internal line carrying momentum between separated vertices, hence no propagator to inject a q-dependent denominator. Compare a theory with a cubic coupling ϕ^3, where $\phi\phi \to \phi\phi$ proceeds at tree level by exchanging a virtual scalar between two vertices: there the internal propagator $i/(q^2 - m^2)$ does depend on the exchanged momentum, and the amplitude acquires its characteristic pole structure in the kinematic invariants. In ϕ^4 that structure is absent at tree level and first appears at one loop, through the bubble integral Eq. (15.50) of Theorem 15.16, whose value is a nontrivial function of the momentum flowing through it.

It is useful to record the kinematic variables in which that future dependence will be expressed, since they are introduced and exploited in Chapter 16. For $2 \to 2$ scattering with incoming momenta p_1, p_2 and outgoing p_3, p_4 (so the all-incoming convention has p_3, p_4 entering as $-p_3, -p_4$), the Mandelstam invariants are

$$s = (p_1 + p_2)^2, \qquad t = (p_1 - p_3)^2, \qquad u = (p_1 - p_4)^2, \tag{15.60}$$

satisfying $s + t + u = \sum_i m_i^2 = 4m^2$ on shell. The tree amplitude Eq. (15.59) is independent of all three — it is the same $-\lambda$ in every channel — which is exactly why no Mandelstam variable appears in it. At one loop the three diagrams obtained by routing the external pairs through the bubble in the s-, t-, and u-channels produce a sum of functions of s, t, and u respectively, and the full kinematic dependence of $\phi\phi \to \phi\phi$ comes alive. That computation, and the conversion of $\mathcal{M}$ into a cross section through the phase-space measure, are the

work of Chapter 16; the present section's contribution to ϕ^4 theory is the tree result Eq. (15.59) and the rules that produced it.

With the momentum-space rules in hand and the LSZ bridge crossed, every element of a ϕ^4 amplitude now carries an explicit numerical weight — a propagator $i/(q^2 - m^2 + i\varepsilon)$ per line, a $-i\lambda$ per vertex, a $\int d^4k/(2\pi)^4$ per loop — with one ingredient still quoted rather than derived: the symmetry factor $1/S$ that multiplied the tree diagram by unity in Eq. (15.54) and will not always be so kind. The combinatorial origin of that factor, the precise fate of the $1/V!$ from the Dyson exponential and the $1/4!$ at each vertex, and a worked catalogue of representative diagrams are the subject of the next section.

15.4 Symmetry Factors Done Carefully

The momentum-space rules of Section 15.3 assign to every diagram a propagator for each internal line, a factor $-i\lambda$ for each vertex, a momentum-conserving delta at each vertex, and a loop integral $\int \mathrm{d}^4k/(2\pi)^4$ for each independent loop. They leave one thing undetermined: a single rational prefactor, the *symmetry factor*, that multiplies the whole amplitude. For the tree-level $\phi\phi \to \phi\phi$ diagram of Section 15.3 that prefactor was 1 and we could afford not to dwell on it; for almost every loop diagram it is not 1, and getting it wrong is the single most common error in scalar perturbation theory. This section pins the prefactor down once and for all. The claim, which we will prove and then verify on a catalogue of diagrams, is sharp and structural: *the symmetry factor of a diagram is the reciprocal of the order of its automorphism group.* Everything reduces to counting automorphisms of a graph, and once the graph is drawn the counting is mechanical.

15.4.1 Where the combinatorial factors come from

The prefactor has a precise origin in the Dyson expansion, and we can see exactly what it must cancel. Recall from the position-space derivation (Section 15.2) that the n-point Green's function of ϕ^4 theory is the vacuum expectation value of a time-ordered product, expanded in powers of the interaction. With the standard Lagrangian of Eq. (14.1) the interaction Hamiltonian carries the coupling in the combination $\lambda/4!$, and the Dyson series of Eq. (14.25) carries an overall $1/V!$ at order V. Writing $G^{(n)}(x_1, \dots, x_n)$ for the full (numerator) correlator, the contribution of order V in the coupling is

$$G^{(n)}_V \;=\; \frac{1}{V!}\left(\frac{-i\lambda}{4!}\right)^V \int \mathrm{d}^4y_1 \cdots \mathrm{d}^4y_V \;\langle 0|\, \mathcal{T}\, \phi(x_1)\cdots\phi(x_n)\,\phi(y_1)^4\cdots\phi(y_V)^4\,|0\rangle\,. \tag{15.61}$$

By Wick's theorem (Theorem 10.23) the time-ordered vacuum expectation value is the sum over all complete pairings of the $n + 4V$ field operators, each pairing

evaluating to a product of Feynman propagators D_F. The two manifestly visible numerical prefactors in Eq. (15.61) are the $1/V!$ from the exponential and the $(1/4!)^V$ from the V copies of the vertex. Everything that follows is the statement that these two prefactors are designed to be cancelled by the multiplicity of Wick contractions — and the symmetry factor measures precisely the extent to which the cancellation fails.

It is worth saying in words why a cancellation is even expected. The V integration variables $y_1, \dots, y_V$ are dummy labels; the integrand is symmetric under permuting them, so the $V!$ orderings all give the same value and the $1/V!$ removes the overcount. Likewise, at a single vertex the four factors $\phi(y_i)$ are literally the same operator, so the 4! ways of deciding "which of the four $\phi(y_i)$ contracts with which neighbour" produce identical propagator products, and the $1/4!$ is there to remove *that* overcount. The question is whether the removal is exact. It is exact precisely when the four fields at a vertex, and the V vertices among themselves, are genuinely interchangeable in all 4! and $V!$ ways without any of those interchanges being trivial — and an interchange is trivial exactly when it is an automorphism of the diagram.

To make this precise we adopt the *half-edge* (or stub) bookkeeping, which turns the contraction count into a clean group action.

Definition 15.20 (Half-edges, matchings, and the relabelling group)**.** Each external point x_j contributes one field operator, an *external half-edge* (univalent and labelled, $j = 1, \dots, n$). Each internal vertex y_i contributes four field operators, four *internal half-edges* carried by that vertex, $i = 1, \dots, V$. A *labelled contraction* is a perfect matching of the full set of $n + 4V$ half-edges into pairs; each pair is a line and evaluates to one propagator. The *relabelling group*

$$\Gamma \;=\; S_4 \wr S_V \;=\; (S_4)^V \rtimes S_V, \qquad |\Gamma| \;=\; (4!)^V\, V!, \tag{15.62}$$

acts on the internal half-edges: the factor $(S_4)^V$ independently permutes the four half-edges within each vertex, and S_V permutes the V vertices bodily (carrying their half-edges along). External half-edges are *fixed pointwise* — the external points are physical and distinguishable. The group Γ acts on the set of labelled contractions by relabelling.

The order of Γ is exactly the product $V!\,(4!)^V$ of the two prefactors in Eq. (15.61): Γ is, by construction, the group of relabellings those prefactors are meant to quotient out. A *Feynman diagram* is what remains after the quotient: the graph drawn with the external legs $x_1, \dots, x_n$ fixed and the internal vertices regarded as indistinguishable points, retaining only the pattern of which half-edge is paired with which. Two labelled contractions yield the *same* Feynman diagram if and only if some element of Γ maps one to the other. In the language of group actions, a Feynman diagram is precisely a Γ-orbit of labelled contractions.

15.4.2 The automorphism group and the counting theorem

The cancellation is now an orbit–stabilizer computation, and the residual factor is the size of a stabilizer. That stabilizer has a name.

Definition 15.21 (Automorphism group of a Feynman diagram)**.** Fix a labelled contraction M realizing a connected diagram D for $G^{(n)}$. An *automorphism* of D is an element $\sigma \in \Gamma$ that maps M to itself: a permutation of the internal half-edges that (i) permutes the four half-edges within each vertex and permutes the vertices among themselves, (ii) commutes with the pairing M (so a line of D is carried to a line of D), and (iii) fixes every external half-edge. The set of such σ is the *automorphism group* $\mathrm{Aut}(D) = \mathrm{Stab}_\Gamma(M)$, and the *symmetry factor* is

$$S \;\equiv\; |\,\mathrm{Aut}(D)|. \tag{15.63}$$

Concretely, $\mathrm{Aut}(D)$ is generated by three elementary kinds of move that leave the drawn graph (with its external legs nailed down) looking identical: (a) swapping the two half-edges of a *self-loop* at a vertex; (b) permuting a set of *parallel lines* joining the same pair of vertices; (c) permuting *interchangeable internal vertices* (vertices related by a symmetry of the rest of the graph). The order S is the number of distinct compositions of such moves.

Because $\mathrm{Aut}(D)$ does not depend on *which* representative matching M in the orbit we chose — stabilizers of points in one orbit are conjugate, hence of equal order — the symmetry factor is a property of the diagram, not of any labelling.

Theorem 15.22 (Symmetry factor as automorphism count)**.** *Let D be a connected diagram contributing to $G^{(n)}$ at order V, with internal lines (propagators) determined by its topology. Then the total contribution of D to the Green's function is*

$$G_D^{(n)} \;=\; \frac{1}{S}\,(-i\lambda)^V \int \mathrm{d}^4y_1 \cdots \mathrm{d}^4y_V \prod_{\text{lines } \ell \in D} D_F(\text{endpoints of } \ell), \qquad S = |\,\mathrm{Aut}(D)|, \tag{15.64}$$

i.e. the rule-set value of D with all vertex factors $-i\lambda$, all propagators, and the loop integrations, multiplied by the single extra factor $1/S$. The $1/4!$ per vertex and the $1/V!$ of the Dyson series are completely accounted for, leaving exactly $1/|\,\mathrm{Aut}(D)|$.

Proof. Group the sum over labelled contractions in Eq. (15.61) according to which Feynman diagram each realizes; that is, group it into Γ-orbits. All labelled contractions in one orbit produce the *same* value: the propagator product is determined by the topology, and relabelling internal half-edges or renaming the dummy integration variables y_i leaves the integral Eq. (15.61) invariant (the integrand is symmetric in the y_i, and the four operators at a vertex are identical). Call that common value, stripped of the front prefactor, $\mathcal{I}(D) = \int \prod_i \mathrm{d}^4y_i \prod_\ell D_F$.

Let M be a representative matching for D. By the orbit–stabilizer theorem, the number of distinct labelled contractions in the orbit of M — equivalently, the number of Wick contractions of Eq. (15.61) that realize D — is

$$\big|\,\mathrm{orbit}(M)\,\big| \;=\; \frac{|\Gamma|}{\big|\,\mathrm{Stab}_\Gamma(M)\big|} \;=\; \frac{(4!)^V\,V!}{\big|\,\mathrm{Aut}(D)\big|} \;=\; \frac{(4!)^V V!}{S}. \tag{15.65}$$

Each of these contractions contributes the identical front factor $\frac{1}{V!}(-i\lambda/4!)^V$ times $\mathcal{I}(D)$. Summing over the orbit therefore gives

$$G_D^{(n)} \;=\; \frac{1}{V!}\Big(\frac{-i\lambda}{4!}\Big)^V \cdot \frac{(4!)^V V!}{S} \cdot \mathcal{I}(D) \;=\; \frac{(-i\lambda)^V}{S}\,\mathcal{I}(D), \tag{15.66}$$

where the explicit $(4!)^V V!$ from the multiplicity cancels the $(1/4!)^V$ and $1/V!$ from the prefactor, leaving $(-i\lambda)^V$ and the lone surviving denominator S. This is Eq. (15.64). □

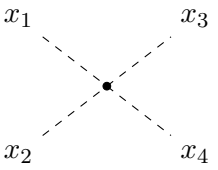
The theorem is the entire content of "symmetry factors," and it explains the folklore precisely. The $1/4!$ in the coupling and the $1/V!$ in the exponential are not magic: they are there to be eaten by the contraction multiplicity, and they are eaten *exactly* for a diagram with no symmetry. They fail to be eaten only along the directions in which the relabelling group Γ acts trivially on the matching — and those directions are, by definition, the automorphisms. A diagram with $S > 1$ is one in which some genuine relabelling of internal half-edges or vertices produces the very same graph, so the contraction count falls short of the full $(4!)^V V!$ by the factor S, and that shortfall survives as $1/S$.

Physical Insight

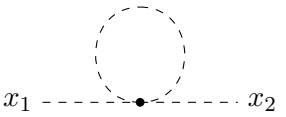
The symmetry factor is the price of indistinguishability. The $1/4!$ in $\lambda\phi^4/4!$ is chosen so that the four lines emanating from a vertex, being interchangeable, do not overcount the tree amplitude: at a tree vertex all four lines run to *distinct* fixed external points, every interchange is detectable, and the cancellation is total ($S = 1$). The moment two of the lines at a vertex run instead to the *same* place — back to the vertex (a self-loop), or to the same neighbouring vertex (parallel lines) — the interchange becomes undetectable, an automorphism appears, and a factor of $1/2$ (or more) is left behind. Loops breed symmetry; symmetry costs $1/S$.

Caution

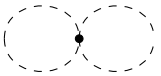
The automorphisms in Theorem 15.21 must fix the external legs. A move that permutes internal vertices is an automorphism only if it leaves *every* external leg attached where it was. This is why exchanging the two



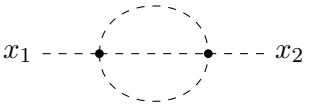
(a) Tree four-point vertex, $S = 1$.



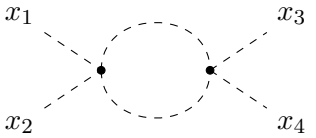
(b) One-loop propagator (tadpole), $S = 2$.



(c) Figure-eight vacuum, $S = 8$.



(d) Sunset, $S = 6$.



(e) s-channel bubble, $S = 2$.

Figure 15.7: The representative catalogue of ϕ^4 diagrams whose symmetry factors are derived in Section 15.4.3. Dashed lines are scalar propagators; filled dots are ϕ^4 vertices; external points x_j are held fixed.

vertices of an s-channel one-loop bubble (Section 15.4.3 below) is *not* an automorphism: it would swap the external pair $\{x_1, x_2\}$ with $\{x_3, x_4\}$. The same exchange *is* an automorphism for the sunset diagram only because there the two vertices carry one external leg each and the exchange would move $x_1 \leftrightarrow x_2$ — which again is forbidden, so it too is excluded. Read the catalogue with the external legs nailed to the page.

15.4.3 A catalogue of symmetry factors from first principles

We now compute S for a representative family of ϕ^4 diagrams, each time *twice*: once by the brute-force contraction count of Eq. (15.65) (count the Wick contractions, divide $(4!)^V V!$ by that number) and once by directly enumerating the automorphisms of Theorem 15.21. The two routes must agree, and watching them agree is the surest way to internalize the rule. The diagrams are drawn in Fig. 15.7.

(a) The tree four-point vertex ($n = 4$, $V = 1$): $S = 1$. This is the diagram behind the carried ϕ^4 theory amplitude of Section 15.3. There is one vertex with four half-edges, and the four external fields $\phi(x_1), \dots, \phi(x_4)$ must each contract with one of them; there are no internal lines. The number of contractions is the number of bijections from $\{x_1, x_2, x_3, x_4\}$ to the four vertex half-edges,

$$\#\{\text{contractions}\} = 4!, \qquad S = \frac{(4!)^1\, 1!}{4!} = 1. \tag{15.67}$$

Directly: once each external leg is nailed to a particular half-edge, no nontrivial permutation of the half-edges preserves the assignment, so Aut is trivial and $S = 1$. The rule-set value is therefore exactly $(-i\lambda)$, with no leftover factor; this is precisely why the 4! was put in the Lagrangian, and it reproduces the tree amplitude $-i\lambda$ of the thread without correction.

(b) The one-loop propagator correction — the tadpole ($n = 2$, $V = 1$): $S = 2$. At order λ the two-point function receives the diagram of Fig. 15.7b: a single vertex, two of whose half-edges run to x_1 and x_2 while the other two contract with each other, forming a loop that sits on the line. Count the contractions: choose the vertex half-edge that pairs with x_1 (four choices), then the one that pairs with x_2 (three remaining), and the last two half-edges are forced to contract with each other (one way):

$$\#\{\text{contractions}\} = 4 \cdot 3 \cdot 1 = 12, \qquad S = \frac{4! \cdot 1!}{12} = \frac{24}{12} = 2. \tag{15.68}$$

Directly: the only nontrivial automorphism is the swap of the two half-edges of the self-loop (move (a) of Theorem 15.21); it exchanges the two ends of the loop without touching the external legs, and it preserves the matching. Hence $\mathrm{Aut}(D) \cong \mathbb{Z}_2$ and $S = 2$. We complete this diagram — symmetry factor, integral, and renormalized value — in Theorem 15.23.

(c) The figure-eight vacuum diagram ($n = 0$, $V = 1$): $S = 8$. With no external legs, the order-λ vacuum amplitude is $\langle 0|\, \mathcal{T}\, \phi(y)^4\, |0\rangle$ fully contracted (Fig. 15.7c): the four half-edges of the single vertex pair up among themselves into two self-loops. The number of ways to partition four labelled half-edges into two unordered pairs is the double factorial

$$\#\{\text{contractions}\} = (4-1)!! = 3!! = 3, \qquad S = \frac{4! \cdot 1!}{3} = \frac{24}{3} = 8. \tag{15.69}$$

Directly, the automorphisms decompose into three independent $\mathbb{Z}_2$ moves: swap the two half-edges of the first loop ($\mathbb{Z}_2$), swap the two half-edges of the second loop ($\mathbb{Z}_2$), and exchange the two loops bodily ($\mathbb{Z}_2$). The first two commute with each other but the third conjugates them, so the group is the wreath product $\mathbb{Z}_2 \wr \mathbb{Z}_2 = (\mathbb{Z}_2 \times \mathbb{Z}_2) \rtimes \mathbb{Z}_2$, the dihedral group of order

$$S = \left|\mathbb{Z}_2 \wr \mathbb{Z}_2\right| = 2 \cdot 2 \cdot 2 = 8. \tag{15.70}$$

This is the textbook "$S = 8$" figure-eight. As a vacuum diagram it is cancelled against the denominator of the Green's function by the linked-cluster argument of Section 15.2, so it never appears in a connected correlator; we include it because the same $1/8$ is exactly the factor that survives in the free energy and the one-loop effective potential, where vacuum diagrams are the whole point.

(d) The sunset ($n = 2$, $V = 2$): $S = 6$. The order-λ^2 correction to the propagator in Fig. 15.7d has two vertices joined by *three* internal lines, with one external leg on each vertex. Count contractions. The external field $\phi(x_1)$ contracts with one of the 8 internal half-edges; call its vertex A. For the sunset topology $\phi(x_2)$ must land on the *other* vertex B — 4 choices among B's half-edges. Vertex A now has three free half-edges and B has three free half-edges, and the sunset requires all three of A's to pair with B's, in 3! ways:

$$\#\{\text{contractions}\} = 8 \cdot 4 \cdot 3! = 8 \cdot 4 \cdot 6 = 192, \qquad S = \frac{(4!)^2\, 2!}{192} = \frac{576 \cdot 2}{192} = 6. \tag{15.71}$$

Directly: the three parallel internal lines joining the same pair of vertices may be permuted arbitrarily (move (b)), giving $\mathrm{Aut}(D) \cong S_3$ of order 6. Exchanging the two vertices is *not* an automorphism — it would carry $x_1 \leftrightarrow x_2$ — so no further factor arises. Hence $S = 6$.

(e) The one-loop s-channel bubble ($n = 4$, $V = 2$): $S = 2$. The order-λ^2 correction to $\phi\phi \to \phi\phi$ in the s-channel, Fig. 15.7e, has the two incoming external legs x_1, x_2 on one vertex, the two outgoing x_3, x_4 on the other, and *two* internal lines (the bubble) joining the vertices. Count: $\phi(x_1)$ picks one of 8 internal half-edges, on a vertex A; $\phi(x_2)$ must also land on A for the s-channel topology — 3 remaining half-edges of A; $\phi(x_3)$ lands on the other vertex B — 4 choices; $\phi(x_4)$ also on B — 3 remaining; the two leftover half-edges on A pair with the two on B in 2! ways:

$$\#\{\text{contractions}\} = 8 \cdot 3 \cdot 4 \cdot 3 \cdot 2! = 576, \qquad S = \frac{(4!)^2\, 2!}{576} = \frac{1152}{576} = 2. \tag{15.72}$$

Directly: the two parallel lines of the bubble may be swapped (move (b)), $\mathrm{Aut}(D) \cong \mathbb{Z}_2$, $S = 2$. As warned, exchanging the two vertices is forbidden because it would swap $\{x_1, x_2\}$ with $\{x_3, x_4\}$. The t- and u-channel bubbles are *distinct* diagrams, each with its own $S = 2$; they are summed separately, not divided by a further symmetry. The full one-loop four-point amplitude is the sum of the three channels, each weighted by $1/2$.

Caution

Identical *external* particles never generate a symmetry factor. In a $2 \to 2$ amplitude with four identical scalars the three channels s, t, u look superficially related by relabelling the external momenta, but the external legs are physically labelled (distinct momenta), so no automorphism relates the channels: each is a separate diagram. The factor $1/2$ in each bubble is from the two *internal* lines, not from the external symmetry. The combinatorial symmetry of identical *final-state* particles is a separate effect that enters the phase-space integration of the cross section in Chapter 16,

never the amplitude's symmetry factor.

15.4.4 The one-loop ϕ^4 self-energy, in full

We now assemble symmetry factor, Feynman rules, and the master loop integral into a complete computation: the one-loop correction to the ϕ^4 propagator. This is the tadpole of part (b) above, read off in momentum space and amputated to the one-particle-irreducible self-energy $-i\Sigma(p)$, the quantity that will be renormalized in Chapters 21 and 22.

Worked Example 15.23 (The one-loop self-energy of ϕ^4 theory)**.** The amputated one-particle-irreducible two-point insertion at one loop is the tadpole of Fig. 15.7b with its external propagators stripped. We established its symmetry factor in Eq. (15.68): $S = 2$. By the momentum-space rules of Section 15.3 the diagram carries one vertex $-i\lambda$, one undetermined loop momentum k circulating in the self-loop with propagator $i/(k^2 - m^2 + i\varepsilon)$ (the momentum-space Feynman propagator of Eq. (9.154)), and one loop integration $\int \mathrm{d}^4k/(2\pi)^4$. The momentum entering the loop is zero — both ends of the self-loop attach to the same vertex — so the result is independent of the external momentum p. Including the symmetry factor,

$$-i\Sigma(p) \;=\; \frac{1}{S}\,(-i\lambda)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon} \;=\; \frac{-i\lambda}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon}. \tag{15.73}$$

Cancelling $-i$ from both sides and pulling the i out of the integral,

$$\Sigma \;=\; \frac{\lambda}{2}\,(i)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{k^2 - m^2 + i\varepsilon} \;=\; \frac{i\lambda}{2}\,I_1(m^2), \tag{15.74}$$

with I_1 the canonical one-loop integral of Eq. (D.73). The integral is quadratically divergent and must be regulated; we use dimensional regularization, $d = 4 - 2\varepsilon$, with the scale μ inserted so that λ stays dimensionless ($\lambda \to \lambda\mu^{2\varepsilon}$). The master formula Theorem D.22 at $n = 1$, $\Delta = m^2$ gives

$$I_1(m^2) = \int \frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{1}{k^2 - m^2 + i\varepsilon} = \frac{-i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)(m^2)^{\,d/2-1}. \tag{15.75}$$

Substituting into Eq. (15.74), the explicit factor of i cancels the $-i$ of Eq. (15.75):

$$\Sigma \;=\; \frac{i\lambda}{2}\cdot\frac{-i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)(m^2)^{d/2-1} \;=\; \frac{\lambda\, m^2}{2(4\pi)^2}\left(\frac{4\pi\mu^2}{m^2}\right)^{\!\varepsilon}\Gamma(-1+\varepsilon), \tag{15.76}$$

where we set $d = 4 - 2\varepsilon$, wrote $(4\pi)^{-d/2} = (4\pi)^{-2}(4\pi)^{\varepsilon}$, $(m^2)^{d/2-1} = m^2(m^2)^{-\varepsilon}$, and restored the $\mu^{2\varepsilon}$ from the coupling to form the dimensionless ratio $(4\pi\mu^2/m^2)^{\varepsilon}$.

It remains to expand around $\varepsilon = 0$. Using $\Gamma(-1+\varepsilon) = -\frac{1}{1-\varepsilon}\Gamma(\varepsilon)$ and $\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_E + O(\varepsilon)$,

$$\Gamma(-1+\varepsilon) = -(1+\varepsilon+O(\varepsilon^2))\Big(\tfrac{1}{\varepsilon} - \gamma_E\Big) = -\tfrac{1}{\varepsilon} + \gamma_E - 1 + O(\varepsilon), \tag{15.77}$$

and $(4\pi\mu^2/m^2)^{\varepsilon} = 1 + \varepsilon \ln\frac{4\pi\mu^2}{m^2} + O(\varepsilon^2)$. Multiplying the two expansions, the $O(1)$ terms combine as $(-\frac{1}{\varepsilon} + \gamma_E - 1) + (-\ln\frac{4\pi\mu^2}{m^2})$, so

$$\boxed{\Sigma \;=\; -\frac{\lambda\, m^2}{32\pi^2}\left[\frac{1}{\varepsilon} - \gamma_E + 1 + \ln\frac{4\pi\mu^2}{m^2}\right] + O(\varepsilon).} \tag{15.78}$$

Three features deserve note. First, Σ is independent of the external momentum p: the one-loop ϕ^4 self-energy is a pure mass shift, so it contributes to the mass counterterm δ_m but not to wavefunction renormalization — the field anomalous dimension γ_ϕ of ϕ^4 theory first appears at $O(\lambda^2)$ (the sunset), consistent with the thread's recorded $\gamma_\phi = O(\varepsilon^2)$ at the Wilson–Fisher point. Second, the divergence is a simple pole $1/\varepsilon$ even though the integral is quadratically divergent in the cutoff language: dimensional regularization sees the quadratic divergence as the $\Gamma(-1+\varepsilon)$ pole and discards the would-be quadratic piece, which is why the residue is proportional to m^2 rather than to a cutoff squared. Third, the $1/S = 1/2$ is indispensable: dropping it would double the mass counterterm and produce the wrong one-loop β_{m^2}. The renormalization that removes the pole in Eq. (15.78) is carried out in Chapter 22; here the point is that every factor in front of the integral — the $-i\lambda$, the loop measure, and above all the $1/2$ — has now been derived rather than asserted.

Historical Note

The systematic reading of the leftover factor as the order of a graph automorphism group goes back to the combinatorial reformulation of perturbation theory in the 1950s and was made fully explicit in the diagrammatic literature that followed Dyson's organization of the S-matrix. The half-edge bookkeeping used above — vertices as bundles of stubs, contractions as matchings, the prefactor $1/V!(4!)^V$ as the order of the wreath group $S_4 \wr S_V$ — is the same structure that reappears in the modern graph-theoretic and Hopf-algebraic accounts of renormalization, where symmetry factors are precisely the inverse orders of automorphism groups of the underlying Feynman graphs.

With the symmetry factor identified as $1/|\operatorname{Aut}(D)|$ and verified on the full catalogue, we have closed the last gap in the canonical Feynman rules: every diagram now has a completely determined value. The next section re-derives this entire rule-set — propagators, vertices, and these very symmetry factors —

from the generating functional $Z[J]$ of Chapter 10, where the $1/S$ will emerge automatically from the functional derivatives rather than being counted by hand, giving the same amplitude a second, independent derivation and a stringent consistency check on everything established here.

15.5 The Path-Integral Route: Feynman Rules from $Z[J]$

The canonical route taken so far in this chapter produced the ϕ^4 Feynman rules honestly but laboriously. Wick's theorem turned each term of the Dyson series into a sum over pairings; the position- and momentum-space rules of Sections 15.2 and 15.3 read those pairings off as diagrams; and the symmetry factor of every diagram had to be counted by hand, balancing the $1/n!$ of the Dyson exponential and the $1/4!$ at each vertex against the number of contractions that realize a given topology—the careful bookkeeping of Section 15.4. That bookkeeping is correct, but it is also the single most error-prone step in scalar perturbation theory, and nothing in the operator formalism makes the cancellations automatic: one must count, every time. There is a second derivation in which the same diagrams, the same propagators and vertices, and—this is the structural payoff—the same symmetry factors emerge from a single formula with no separate counting at all. It starts not from the S-matrix but from the generating functional $Z[J]$ built in Chapter 10, and it converts the entire perturbative expansion into the repeated action of functional derivatives on one Gaussian. This section carries that derivation out in full, recovers every rule established canonically, and closes ϕ^4 theory's "same amplitude two ways" by reproducing $i\mathcal{M} = -i\lambda$ from $Z[J]$.

We work throughout with the conventions in force for the chapter: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the Feynman $i\varepsilon$ prescription, and the frozen ϕ^4 Lagrangian of Eq. (10.80),

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \qquad \phi \text{ real},\ \lambda > 0. \tag{15.79}$$

15.5.1 The generating functional as the source of all diagrams

We import from Chapter 10 the three objects on which everything below rests, restating them so the section reads on its own. The first is the *generating functional* of Theorem 10.17,

$$Z[J] = \int \mathcal{D}\phi\ \exp\Big[\, i\int d^4x\,(\mathcal{L} + J\phi)\Big] = \int \mathcal{D}\phi\ \exp\Big[\, iS[\phi] + i\int d^4x\, J(x)\phi(x)\Big], \tag{15.80}$$

the field integral weighted by e^{iS} in the presence of an external source $J(x)$, with the vacuum boundary conditions enforced by the $m^2 \to m^2 - i\varepsilon$ tilt of Chapter 10.

The second is the rule, Theorem 10.18, that the time-ordered correlation functions of the interacting theory are normalized functional derivatives of $Z[J]$ at vanishing source,

$$G^{(n)}(x_1,\dots,x_n) = \langle 0|\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,|0\rangle = \frac{1}{Z[0]}\,\frac{1}{i^{\,n}}\,\frac{\delta^{\,n} Z[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}, \tag{15.81}$$

where $\delta/\delta J$ is the functional derivative of Theorem 4.4 and the division by $Z[0]$ is what removes the vacuum bubbles, exactly as established around Eq. (10.93). The third is the closed form of the *free* generating functional, the $\lambda = 0$ instance of Eq. (15.80), obtained by completing the Gaussian square in Chapter 10 and recorded as Eq. (10.111),

$$Z_0[J] = Z_0[0]\,\exp\Big[-\tfrac{1}{2}\int d^4x\,d^4y\;J(x)\,D_F(x-y)\,J(y)\Big], \tag{15.82}$$

with D_F the Feynman propagator. Equation Eq. (15.82) *is* the entire free theory: by Eq. (15.81) every free correlator is a derivative of this single Gaussian, and differentiating it produces precisely the sum-over-pairings of Wick's theorem, Theorem 10.23,

$$G_0^{(2k)}(x_1,\dots,x_{2k}) = \sum_{\text{pairings}}\;\prod_{\text{pairs }(a,b)} D_F(x_a - x_b), \qquad G_0^{(2k+1)} = 0. \tag{15.83}$$

The task of the section is to turn the *interacting* $Z[J]$ of Eq. (15.80) into something we can differentiate as easily as Eq. (15.82). The obstruction is the quartic term in S: it makes the integral non-Gaussian and there is no closed form. The resolution is to keep the Gaussian exact and treat the quartic term as a differential operator acting on it.

15.5.2 The master formula

Split the action of Eq. (15.79) into its free and interacting parts in the manner of Chapter 14, $S[\phi] = S_0[\phi] + S_{\text{int}}[\phi]$, where the interaction is the spacetime integral of the quartic potential,

$$S_{\text{int}}[\phi] = -\frac{\lambda}{4!}\int d^4w\;\phi(w)^4. \tag{15.84}$$

The single identity that drives the whole derivation is that a functional derivative with respect to the source, suitably normalized, reaches inside the integral Eq. (15.80) and pulls down a factor of the field. Acting on the source exponential,

$$\frac{1}{i}\frac{\delta}{\delta J(w)}\,\exp\Big[i\int d^4x\,J\phi\Big] = \frac{1}{i}\,\big(i\,\phi(w)\big)\,\exp\Big[i\int d^4x\,J\phi\Big] = \phi(w)\,\exp\Big[i\int d^4x\,J\phi\Big], \tag{15.85}$$

the operator $\frac{1}{i}\,\delta/\delta J(w)$ acts as multiplication by $\phi(w)$ once it stands to the left of $e^{i\int J\phi}$. Because these operators at different points w commute among themselves, the same holds for any polynomial—and hence any power series—in the field: for any functional F admitting a Taylor expansion in ϕ,

$$F\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\,\exp\Big[\,i\int d^4x\,J\phi\Big] = F[\phi]\,\exp\Big[\,i\int d^4x\,J\phi\Big]. \tag{15.86}$$

The crucial feature of the left-hand side is that the differential operator $F[\frac{1}{i}\delta/\delta J]$ depends on J alone, not on the integration variable ϕ. This is exactly the lever we need.

Theorem 15.24 (The master formula)**.** *The interacting generating functional Eq. (15.80) of ϕ^4 theory equals the interaction exponential, written as a functional-derivative operator in the source, acting on the free generating functional:*

$$\boxed{Z[J] = \exp\Big(iS_{\text{int}}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)\,Z_0[J] = \exp\Big(-\frac{i\lambda}{4!}\int d^4w\,\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J(w)}\Big]^4\Big)\,Z_0[J].} \tag{15.87}$$

Proof. Take $F[\phi] = \exp\left(iS_{\text{int}}[\phi]\right)$ in Eq. (15.86); as a power series in λ each term is a polynomial in ϕ, so the identity applies term by term. It gives

$$\exp\Big(iS_{\text{int}}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)\exp\Big[\,i\int d^4x\,J\phi\Big] = \exp(iS_{\text{int}}[\phi])\,\exp\Big[\,i\int d^4x\,J\phi\Big]. \tag{15.88}$$

Now write the interacting weight in Eq. (15.80) as $e^{iS} = e^{iS_0}e^{iS_{\text{int}}}$ and insert Eq. (15.88):

$$\begin{aligned} Z[J] &= \int \mathcal{D}\phi\, e^{\,iS_0[\phi]}\; e^{\,iS_{\text{int}}[\phi]}\;\exp\Big[\,i\int d^4x\,J\phi\Big] \\ &= \int \mathcal{D}\phi\, e^{\,iS_0[\phi]}\;\exp\Big(iS_{\text{int}}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)\exp\Big[\,i\int d^4x\,J\phi\Big]. \end{aligned} \tag{15.89}$$

The operator $\exp\left(iS_{\text{int}}[\frac{1}{i}\delta/\delta J]\right)$ is built from derivatives with respect to J only and contains no ϕ; it therefore commutes with the integration over ϕ and with the ϕ-dependent factor $e^{iS_0[\phi]}$, and may be pulled out in front of the integral. What remains inside is the free generating functional,

$$Z[J] = \exp\Big(iS_{\text{int}}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)\,\underbrace{\int \mathcal{D}\phi\, e^{\,iS_0[\phi]}\,\exp\Big[\,i\int d^4x\,J\phi\Big]}_{=\,Z_0[J]} = \exp\Big(iS_{\text{int}}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)\,Z_0[J], \tag{15.90}$$

which is Eq. (15.87). Substituting S_{int} from Eq. (15.84) gives the explicit second form; the factor $(\frac{1}{i})^4 = 1$ has not yet been written out and is kept for now so that each derivative is paired with its $1/i$. □

Two remarks fix the status of Eq. (15.87). First, the interchange of the J-derivative operator with the functional integral in Eq. (15.89)–Eq. (15.90) is exactly as legitimate as the Gaussian manipulations of Chapter 10: on the lattice regularization of Theorem 10.16 both sides are finite-dimensional, the operator is an ordinary finite-order differential operator in the discretized sources, and the interchange is the elementary statement that a constant-coefficient differential operator passes through an integral over a different set of variables. The continuum formula is the limit of that finite statement, understood—as always in perturbation theory—order by order in λ. Second, the master formula is not an approximation: it is an exact rewriting of $Z[J]$. The approximation enters only when we expand the exponential operator in powers of λ, which we now do.

15.5.3 The double expansion and how the rules emerge

The master formula Eq. (15.87) reduces every correlator to derivatives of the Gaussian Eq. (15.82), because both the operator $\exp(iS_{\rm int}[\cdots])$ and the external derivatives of Eq. (15.81) are functional derivatives with respect to J, all acting on $Z_0[J]$. Expanding the exponential operator in powers of the coupling,

$$\exp\Big(iS_{\rm int}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)=\sum_{V=0}^{\infty}\frac{1}{V!}\Big(iS_{\rm int}\Big[\tfrac{1}{i}\tfrac{\delta}{\delta J}\Big]\Big)^{V}=\sum_{V=0}^{\infty}\frac{1}{V!}\left(-\frac{i\lambda}{4!}\int d^4w\,\Big[\frac{1}{i}\frac{\delta}{\delta J(w)}\Big]^4\right)^{V}, \tag{15.91}$$

exposes the structure of the perturbation series in one line. The term with V factors is the $O(\lambda^V)$ contribution; it carries

- a factor $1/V!$ from the exponential—the Dyson $1/V!$, here arising automatically from the Taylor coefficient of exp, not from any time-ordering simplex;
- one factor $-\dfrac{i\lambda}{4!}\displaystyle\int d^4w_j\,\big[\tfrac{1}{i}\delta/\delta J(w_j)\big]^4$ for each vertex $j=1,\dots,V$, each contributing the coupling $-i\lambda$, the vertex symmetry weight $1/4!$, an unconstrained position w_j integrated over all spacetime, and *four* functional derivatives—the four legs of the vertex.

Inserted into Eq. (15.81), the n-point function at order λ^V is therefore obtained by acting on $Z_0[J]$ with n external derivatives $\delta/\delta J(x_i)$ together with $4V$ vertex derivatives $\delta/\delta J(w_j)$, and setting $J=0$. Since $Z_0[J]$ is the Gaussian Eq. (15.82), this is nothing but a free correlator of $n+4V$ fields, evaluated by Wick's theorem Eq. (15.83): the result is the sum over all ways of pairing the $n+4V$ points, each pair supplying one propagator D_F. We have arrived, with no new combinatorics, at the position-space Feynman rules.

Theorem 15.25 (Position-space Feynman rules from $Z[J]$)**.** *The $O(\lambda^V)$ contribution to $G^{(n)}(x_1,\dots,x_n)$ is the sum, over all distinct diagrams with n external points $x_1,\dots,x_n$ and V internal four-valent vertices, of the value obtained by the rules*

1. *assign an internal spacetime point w_j to each vertex and integrate $\int d^4w_j$ over all of spacetime;*
2. *for each line joining two points y, y' (external or internal), a factor $D_F(y-y')$;*
3. *for each vertex, a factor $-i\lambda$;*
4. *divide by the symmetry factor $S = |\mathrm{Aut}(\Gamma)|$ of the diagram, defined in Theorem 15.21;*

together with the overall division by $Z[0]$ in Eq. (15.81), which removes the disconnected vacuum subdiagrams. These are identical to the rules derived canonically in Theorem 15.9.

The proof is the content of the next two subsections: we first see the mechanism on the leading nontrivial example, the four-point function at $O(\lambda)$, and then prove the symmetry-factor statement (item 4) in general, which is where the path-integral route shows its structural advantage. The Fourier transform that converts rules 1–3 into the momentum-space rules of Theorem 15.13—a $-i\lambda(2\pi)^4\delta^4(\sum k)$ at each vertex, an $i/(p^2-m^2+i\varepsilon)$ on each line, and $\int d^4\ell/(2\pi)^4$ on each loop—is exactly the one carried out canonically there and is not repeated; it acts on Theorem 15.25 line by line and leaves the symmetry factor untouched.

15.5.4 The four-point function at $O(\lambda)$: the tree diagram

Worked Example 15.26 (The $O(\lambda)$ four-point function from $Z[J]$)**.** We compute the leading interacting four-point function directly from the master formula, exhibiting every contraction, and read off both the diagram and its symmetry factor. By Eq. (15.81) and the $V=1$ term of Eq. (15.91), the order-λ part of the unnormalized fourth derivative is

$$\frac{1}{i^4}\frac{\delta^4 Z[J]}{\delta J(x_1)\cdots\delta J(x_4)}\bigg|_{\lambda^1} = \frac{1}{i^4}\frac{\delta^4}{\delta J(x_1)\cdots\delta J(x_4)}\left(-\frac{i\lambda}{4!}\int d^4w\left[\tfrac{1}{i}\tfrac{\delta}{\delta J(w)}\right]^4\right) Z_0[J]\bigg|_{J=0}. \tag{15.92}$$

Collect the prefactors. The four external derivatives carry $1/i^4 = 1$; the four vertex derivatives carry $(\frac{1}{i})^4 = 1$; and the vertex coupling is $-i\lambda/4!$. We must therefore evaluate

$$-\frac{i\lambda}{4!}\int d^4w\,\frac{\delta^8 Z_0[J]}{\delta J(x_1)\,\delta J(x_2)\,\delta J(x_3)\,\delta J(x_4)\,\delta J(w)\,\delta J(w)\,\delta J(w)\,\delta J(w)}\bigg|_{J=0}. \tag{15.93}$$

This is the eighth functional derivative of the Gaussian Eq. (15.82) at $J = 0$—a free 8-point function of the field at the points $x_1, x_2, x_3, x_4, w, w, w, w$. By Wick's theorem Eq. (15.83) it is the sum over all pairings of these eight points, $(8-1)!! = 105$ in total, each pairing a product of four propagators. We organize the 105 pairings by their topology.

The connected (tree) pairings. The diagrams in which every external point is tied to the vertex pair each of x_1, x_2, x_3, x_4 with one of the four w-legs. The number of such pairings is the number of bijections from the four external points to the four w-legs, namely $4! = 24$, and each yields the identical product $D_F(x_1-w)D_F(x_2-w)D_F(x_3-w)D_F(x_4-w)$, since the four w-legs sit at the same point w. Their total contribution to Eq. (15.93) is

$$-\frac{i\lambda}{4!}\cdot 24\int d^4w\; D_F(x_1-w)D_F(x_2-w)D_F(x_3-w)D_F(x_4-w) = -i\lambda\int d^4w\;\prod_{i=1}^{4} D_F(x_i-w). \tag{15.94}$$

The vertex weight $1/4!$ has been cancelled *exactly* by the $24 = 4!$ ways of attaching the four distinguishable external legs to the four vertex legs. No counting beyond "4! bijections" was needed, and the residual numerical factor is 1: the tree diagram of Fig. 15.8 carries symmetry factor $S = 1$, the value its automorphism group takes in Theorem 15.21 because the four labelled external legs admit no nontrivial relabelling. The single factor $-i\lambda$ that survives is the Feynman vertex.

The disconnected pairings. The remaining $105 - 24 = 81$ pairings fail to connect all four externals to the vertex. They fall into two classes. In the first, the four w-legs are paired among themselves (one of the 3 ways to split four legs into two pairs, giving $D_F(0)^2$) while the four externals are paired among themselves (one of the 3 pairings of Eq. (10.138)); this is the free four-point function multiplied by the $O(\lambda)$ vacuum bubble. In the second, two of the w-legs pair with two externals and the other two w-legs pair together, leaving one external pair: these are propagator-correction (tadpole) pieces on a disconnected external line. Every one of these 81 terms contains either a vacuum factor or a self-contraction $D_F(0)$ detached from the scattering of all four externals. Crucially, the vacuum pieces are precisely what the normalization $1/Z[0]$ in Eq. (15.81) divides away—$Z[0]$ is, at this order, $Z_0[0]\big(1 + (-i\lambda/4!)\cdot 3\int D_F(0)^2\big)$—and the linked-cluster cancellation that makes this systematic is the subject of Section 15.6. The surviving connected, amputated content of the four-point function at $O(\lambda)$ is Eq. (15.94).

The lesson of Theorem 15.26 generalizes, and stating the generalization sharply is the point of the next subsection. The vertex weight $1/4!$ is engineered to be undone by the number of ways the four legs of the vertex can be wired; whether that undoing is complete, partial, or over-complete is governed entirely by the symmetry of the resulting diagram. The path-integral route makes this a theorem rather than a case-by-case count.

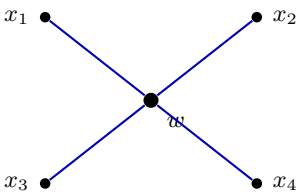


Figure 15.8: The $O(\lambda)$ four-point diagram of Theorem 15.26, Eq. (15.94). Four external points $x_1, \ldots, x_4$ are joined by Feynman propagators to a single four-valent vertex at w, integrated over spacetime. The vertex weight $1/4!$ from Eq. (15.84) is cancelled by the 4! ways of attaching the four distinguishable external legs, leaving the bare Feynman vertex $-i\lambda$ and symmetry factor $S = 1$. Amputating the four external legs and going to momentum space gives $i\mathcal{M} = -i\lambda$, Eq. (15.105).

15.5.5 The symmetry factor, for free

The structural advantage advertised in the opening is the following: the $1/S$ of any diagram is produced automatically by the interplay of the $1/V!$ in Eq. (15.91), the $1/4!$ at each vertex, and the Gaussian pairing count, with no separate counting of contractions. We prove this by a single orbit–stabilizer argument, which is the cleanest statement of what "symmetry factor" means.

Fix a diagram topology Γ contributing to $G^{(n)}$ at order λ^V: a graph with n labelled external points, V four-valent internal vertices, and a definite assignment of lines. From the expansion Eq. (15.91) and the master formula, the analytic value attached to Γ arises from a set of *labelled contraction schemes*: a scheme is a choice, at the level of the $4V$ individual vertex derivatives and the n external derivatives, of which derivative is paired with which in the Wick expansion of Eq. (15.83), such that the resulting pattern of propagators reproduces Γ. To count these schemes, regard the four derivatives at each vertex as four distinguishable "half-edges" and the vertices as distinguishable (labelled $1, \ldots, V$). Then the group

$$\mathcal{G} \;=\; S_V \ltimes (S_4)^V \tag{15.95}$$

acts on the labelled schemes: S_V permutes the V vertices and the factor S_4 at vertex j permutes its four half-edges. Its order is

$$|\mathcal{G}| = V!\,(4!)^V, \tag{15.96}$$

which is exactly the product of the denominators carried by the V-vertex term of Eq. (15.91): the $V!$ of the Dyson exponential and the 4! at each of the V vertices. The relevance of $\mathcal{G}$ is that all labelled schemes related by an element of $\mathcal{G}$ give

the *same* analytic value: relabelling the vertices merely renames the dummy integration variables w_j, and permuting the half-edges at a vertex merely reorders the four identical propagator ends meeting there, both leaving $\int \prod d^4 w_j \prod D_F$ invariant.

Now apply the orbit–stabilizer theorem to the action of $\mathcal{G}$ on one labelled scheme c_0 realizing Γ. The orbit $\mathcal{G} \cdot c_0$ is the set of all labelled schemes that yield Γ (every such scheme differs from c_0 by some relabelling of vertices and half-edges), and its size is

$$\#\{\text{labelled schemes giving } \Gamma\} = |\mathcal{G} \cdot c_0| = \frac{|\mathcal{G}|}{|\mathrm{Stab}(c_0)|} = \frac{V!\,(4!)^V}{|\mathrm{Aut}(\Gamma)|}, \tag{15.97}$$

where the stabilizer $\mathrm{Stab}(c_0)$—the relabellings that map the scheme c_0 to itself, i.e. that leave the wiring of Γ intact—is by definition the automorphism group $\mathrm{Aut}(\Gamma)$ of Theorem 15.21, whose order is the symmetry factor $S = |\mathrm{Aut}(\Gamma)|$. (External points are labelled and held fixed, so they contribute no automorphisms; this is why the tree of Fig. 15.8 had $S = 1$.) Assembling the weight of Γ from Eq. (15.91)—the prefactor $\frac{1}{V!}\left(\frac{1}{4!}\right)^V$, the coupling $(-i\lambda)^V$, the count Eq. (15.97) of equal-valued schemes, and the common propagator product $I_\Gamma = \int \prod_j d^4 w_j \prod_{\text{lines}} D_F$—we obtain

$$\boxed{\mathrm{weight}(\Gamma) = \frac{1}{V!}\Big(\frac{1}{4!}\Big)^V (-i\lambda)^V \, \frac{V!\,(4!)^V}{|\mathrm{Aut}(\Gamma)|}\, I_\Gamma = \frac{(-i\lambda)^V}{S}\, I_\Gamma, \qquad S = |\mathrm{Aut}(\Gamma)|\,.} \tag{15.98}$$

The cancellation is total and automatic: the $V!\,(4!)^V$ generated by the relabelling group Eq. (15.96) annihilates the $V!\,(4!)^V$ sitting in the denominators of Eq. (15.91), and the *only* surviving combinatorial residue is $1/|\mathrm{Aut}(\Gamma)| = 1/S$. This proves item 4 of Theorem 15.25 and, with the orbit-counting of Eq. (15.97), completes the proof of that theorem. It also reproduces, with no separate enumeration, the automorphism definition of S established canonically in Section 15.4: there S was identified by counting contractions against the $V!(4!)^V$ by hand; here that same count is the orbit-stabilizer formula Eq. (15.97), and $1/S$ falls out of the master formula as the residue of a cancellation that happens by itself.

Physical Insight

The deep reason the path integral knows the symmetry factor is that the $1/V!$ and $1/4!$ are not pedagogical conveniences inserted to "make the answers come out"—they are the structure constants of the two symmetries of the unlabelled problem. The $1/4!$ is the order of the permutation group of the four identical fields at a vertex; the $1/V!$ is the order of the permutation group of the V identical vertices. The Feynman diagram is the equivalence class of wirings under both, and a diagram with extra symmetry is simply one whose stabilizer in Eq. (15.95) is larger than expected, so that fewer

distinct wirings map onto it and its weight is correspondingly smaller. Symmetry factors are a quotient by a group action; the path integral performs the quotient for us.

Worked Example 15.27 (The figure-eight from $Z[J]$, with $S = 8$ automatic)**.** We re-derive one entry of the symmetry-factor catalogue of Section 15.4—the figure-eight vacuum diagram, a single vertex with both pairs of its legs closed into loops, shown in Fig. 15.9—directly from $Z[J]$, and confirm that the count gives $S = 8$ with no hand-tuning. This diagram is the entire $O(\lambda)$ content of the vacuum amplitude $Z[0]$. From the master formula Eq. (15.87) at $J = 0$, the $V = 1$ term is

$$Z[0]\big|_{\lambda^1} = -\frac{i\lambda}{4!}\int d^4w\,\left[\tfrac{1}{i}\tfrac{\delta}{\delta J(w)}\right]^4 Z_0[J]\Big|_{J=0} = -\frac{i\lambda}{4!}\int d^4w\,\frac{\delta^4 Z_0[J]}{\delta J(w)^4}\Big|_{J=0}, \tag{15.99}$$

using $(\frac{1}{i})^4 = 1$. The fourth derivative of the Gaussian at a single point w is the free four-point function at coincident arguments, Eq. (15.83) with $x_1 = x_2 = x_3 = x_4 = w$:

$$\frac{\delta^4 Z_0[J]}{\delta J(w)^4}\Big|_{J=0} = Z_0[0]\,G_0^{(4)}(w,w,w,w) = Z_0[0]\,(3\,D_F(0)^2), \tag{15.100}$$

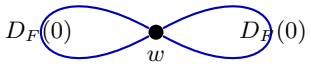
the factor 3 being the $(4-1)!! = 3$ pairings of four points—here the three ways to split the four legs at w into two pairs, each pair closing into a propagator $D_F(w-w) = D_F(0)$. Inserting Eq. (15.100) into Eq. (15.99),

$$Z[0]\big|_{\lambda^1} = Z_0[0]\left(-\frac{i\lambda}{4!}\cdot 3\right)\int d^4w\,D_F(0)^2 = Z_0[0]\left(-\frac{i\lambda}{8}\right)\int d^4w\,D_F(0)^2. \tag{15.101}$$

The numerical residue is $3/4! = 1/8$, so the figure-eight carries weight $(-i\lambda)/S$ with

$$S = \frac{4!}{3} = 8, \tag{15.102}$$

exactly the value the orbit-stabilizer formula Eq. (15.98) predicts: with $V = 1$, $|\mathcal{G}| = 1!\cdot 4! = 24$, and the count of labelled schemes giving the figure-eight is 3, so $S = 24/3 = 8$. The same 8 is read off geometrically from Fig. 15.9 as the order of the automorphism group: a factor 2 from swapping the two loops, and a factor 2 for reversing each of the two loops independently, $2\times 2\times 2 = 8$, matching the canonical count of Section 15.4. The propagator at coincident points, $D_F(0) = \int d^4p/(2\pi)^4\; i/(p^2 - m^2 + i\varepsilon)$, is ultraviolet divergent; that this vacuum bubble is divergent is immaterial to the combinatorial point being made here—the divergence, and its removal, are the business of Chapters 21 and 22, and the bubble itself cancels against $1/Z[0]$ in every physical correlator by the linked-cluster theorem of Section 15.6.



$$-\frac{i\lambda}{8}\int d^4w\, D_F(0)^2, \qquad S = 8$$

Figure 15.9: The figure-eight vacuum diagram of Theorem 15.27, Eq. (15.101): a single four-valent vertex at w with each of its two leg-pairs closed into a self-loop $D_F(0)$. The symmetry factor $S = 4!/3 = 8$ emerges automatically from the master formula—three Wick pairings against the vertex $1/4!$—and equals the order of the diagram's automorphism group (swap the two loops, reverse each loop: $2 \times 2 \times 2 = 8$), reproducing the canonical count of Section 15.4.

15.5.6 The same amplitude two ways

It remains to close ϕ^4 theory's standing obligation: to obtain the elastic scattering amplitude $i\mathcal{M}$ for $\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4)$ a second time, now from $Z[J]$, and to verify that it agrees on the nose with the canonical result. The connected, amputated four-point content at $O(\lambda)$ produced by the master formula is the tree of Eq. (15.94),

$$G_{\mathrm{c}}^{(4)}(x_1,x_2,x_3,x_4)\big|_{\lambda^1} = -i\lambda\int d^4w\, \prod_{i=1}^{4} D_F(x_i - w) + O(\lambda^2), \tag{15.103}$$

with symmetry factor 1. To extract the amplitude we pass to momentum space and amputate, exactly as the LSZ reduction of Chapter 14 prescribes: Fourier transform each external leg, $D_F(x_i - w) \to e^{\mp ip_i\cdot w}\,\widetilde{D_F}(p_i)$ with $\widetilde{D_F}(p) = i/(p^2 - m^2 + i\varepsilon)$, perform the vertex integral $\int d^4w$, which collects the four exponentials into the momentum-conserving delta,

$$\int d^4w\, e^{-i(p_1+p_2-p_3-p_4)\cdot w} = (2\pi)^4\,\delta^4(p_1 + p_2 - p_3 - p_4), \tag{15.104}$$

and strip the four external propagators $\widetilde{D_F}(p_i)$ together with the overall delta, leaving the amputated amplitude. The single vertex contributes its factor $-i\lambda$ and nothing else—there is no loop, hence no $\int d^4\ell/(2\pi)^4$, and the symmetry factor is 1—so

$$\boxed{i\mathcal{M}(\phi\phi \to \phi\phi) = -i\lambda + O(\lambda^2).} \tag{15.105}$$

This is term-for-term the canonical result: it is the $i\mathcal{M} = -i\lambda$ of Eq. (14.54), computed there from the Dyson series and LSZ, and re-derived within this chapter from the canonical Feynman rules in Theorem 15.13. The two pillars agree. The leading scattering amplitude of ϕ^4 theory, the first interacting number the book

owes, has now been produced *three* ways—operator Dyson series, canonical Feynman rules, and the generating functional—and all three give $-i\lambda$, with the same single tree diagram, the same vertex, and the same trivial symmetry factor. This is the concrete content of "the same amplitude two ways": not a slogan but an identity of outputs from two independent formalisms, and it is the strongest internal check the perturbative apparatus admits.

Notation Watch

In this section the symbol $S_{\text{int}}[\,\cdot\,]$ denotes the interaction *action* Eq. (15.84), a functional whose argument is a field (or, in the master formula, the operator $\frac{1}{i}\delta/\delta J$); the upright $\mathbf{S}$ of Chapter 14 is the S-matrix operator, and the italic $S = |\mathrm{Aut}(\Gamma)|$ is the integer symmetry factor of a diagram. The three are distinct objects sharing a letter by long convention; context and typeface separate them, and we never write them in the same expression.

We now have the complete Feynman apparatus of ϕ^4 packaged inside the single object $Z[J]$: every correlator, every diagram, every weight, and every symmetry factor is a functional derivative of one Gaussian acted on by one exponential operator. Working at the level of $Z[J]$ rather than diagram by diagram makes the global structure of the whole expansion visible at once—which classes of diagrams appear, how the vacuum bubbles factor out, and how the connected pieces are organized. Extracting that structure is the next task: taking the logarithm $W = -i \log Z$ will be shown to generate exactly the connected diagrams and to enforce the cancellation of vacuum bubbles, and the Legendre transform of W will isolate the one-particle-irreducible core that the renormalization programme of Chapters 21 and 22 is built upon.

15.6 The Architecture of Perturbation Theory: Connected, Vacuum, Amputated, 1PI

The path-integral route of Section 15.5 did more than reproduce the Feynman rules: by expanding $\exp(iS_{\text{int}})$ and contracting with $\delta/\delta J$ it laid the *entire* set of diagrams on the table at once — connected and disconnected, vacuum and external, reducible and irreducible, all weighted by their symmetry factors. That completeness is a virtue for bookkeeping and a liability for computation. No experiment measures the full correlator $G^{(n)} = \langle 0|T\phi(x_1)\cdots\phi(x_n)|0\rangle$; what an experiment measures is a connected, on-shell, amputated amplitude, and what a renormalization program manipulates is a still smaller skeleton, the one-particle-irreducible core. The job of this section is to sort the undifferentiated diagram set into exactly these structural classes and to prove, in the language of graphs, that each class is generated by one of the three nested functionals $Z \to W \to \Gamma$ already introduced in Chapter 10. We establish the general

theorems that Section 10.7.4 verified only at low order and explicitly deferred here: that $W = -i \ln Z$ generates precisely the connected diagrams, that the disconnected vacuum diagrams exponentiate into a common factor that cancels against the $Z[0]$ normalization, and that the Legendre transform Γ generates the one-particle-irreducible vertices. The payoff is the working hierarchy of perturbation theory — full, connected, 1PI — on which every later chapter, from cross sections (Chapter 16) through renormalization (Chapter 22) to the effective potential (Chapter 29), silently relies.

Throughout we work with the frozen ϕ^4 theory theory of Section 15.5, Lagrangian Eq. (14.1), in the mostly-minus signature with $\hbar = c = 1$. The objects are the time-ordered correlators of Theorem 10.18, generated by $Z[J]$ of Theorem 10.11; the diagrammatic value of a graph is read off by the Feynman rules and symmetry factors of Section 15.5. The combinatorial content of every theorem below is purely topological — it is about how graphs decompose — so although we phrase it for ϕ^4 it holds verbatim for any local theory, fermions and gauge fields included once their rules are in hand (Section 15.7 supplies the signs).

15.6.1 The graph-theoretic vocabulary

Before classifying diagrams we must say precisely what a diagram *is*, as a combinatorial object, because every proof in this section is an argument about graphs and the loose pictorial usage of Section 15.5 is not sharp enough to support one.

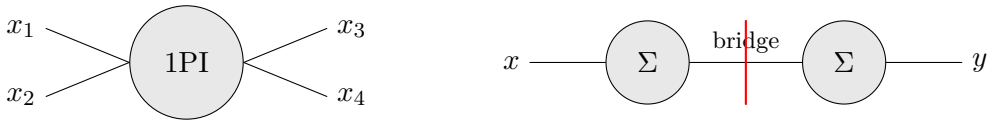
Definition 15.28 (Feynman graph)**.** A *Feynman graph* G for the ϕ^4 theory is a finite graph whose vertices are of two kinds: *external vertices* (or external points), each of degree one and carrying a fixed label $x_1, \dots, x_n$, and *internal vertices*, each of degree four. Its *edges* (*lines*) are of two kinds: an *external line* is incident to an external vertex; an *internal line* joins two internal vertices, or an internal vertex to itself (a *self-loop*, i.e. a tadpole contraction). Each line carries a factor D_F, each internal vertex a factor $-i\lambda$ and an integration $\int d^4y$, and the whole graph is divided by its symmetry factor $S(G) = |\mathrm{Aut}(G)|$, the order of its automorphism group (the rules of Section 15.5). A graph with n external vertices contributes to the n-point function $G^{(n)}$; a graph with $n = 0$ external vertices is a *vacuum graph*.

The two structural notions on which everything turns are connectedness and one-particle irreducibility. Both are standard graph theory; we state them carefully so the proofs can quote them.

Definition 15.29 (Connected and disconnected graphs)**.** A Feynman graph G is *connected* if any two of its vertices are joined by a path of lines; otherwise it is *disconnected*. A maximal connected subgraph of G is a *connected component*.

Every graph is the disjoint union of its connected components, uniquely. A *vacuum bubble* is a connected vacuum graph: a connected graph with no external lines. A correlator graph is *externally connected* if all its external points lie in a single connected component.

Definition 15.30 (One-particle-irreducible graph)**.** A connected Feynman graph G is *one-particle-reducible* (1PR) if there exists a single *internal* line whose removal disconnects G into two pieces, each containing at least one vertex; such a line is a *bridge*. A connected graph that is *not* one-particle-reducible — one that remains connected after the deletion of any single internal line — is *one-particle-irreducible* (1PI), also called *proper*. (External lines are never cut in this test; a graph whose only bridges are external lines is still 1PI.)



(a) one-particle-irreducible

(b) one-particle-reducible

Figure 15.10: One-particle irreducibility. Graph (a) cannot be split into two by deleting any single internal line: it is 1PI (proper). Graph (b) becomes disconnected when the central internal line (the *bridge*, struck through in red) is cut; it is one-particle-reducible, a chain of two self-energy insertions Σ. The content of Theorem 15.39 is that Γ generates exactly the blobs of type (a), and the content of Theorem 15.40 is that the full propagator is the geometric resummation of chains of type (b).

These are graph-topological properties, decided once and for all by inspection of the diagram and independent of any kinematics. Figure 15.10 contrasts the two. The word "particle" in "one-particle-irreducible" is physical: a bridge is a single internal line, which in momentum space carries a single propagator $D_F(k)$, and at $k^2 \to m^2$ that propagator goes on shell — a 1PR graph can develop an intermediate single-particle pole, while a 1PI graph cannot. This is exactly the analytic distinction that makes the 1PI pieces the natural building blocks: they are the "vertices" out of which all single-particle poles are subsequently assembled by resummation.

Finally we fix the generating-functional objects, recalled from Chapter 10 so the section is self-contained at the level of definitions.

Definition 15.31 (Full, connected, and proper correlators)**.** The *full* n-point function is $G^{(n)}(x_1, \dots, x_n) = \langle 0|T\,\phi(x_1)\cdots\phi(x_n)|0\rangle$, generated by $Z[J]$ through

Eq. (10.66). The *connected* n-point function is

$$G_c^{(n)}(x_1,\dots,x_n) \;=\; \frac{1}{i^{\,n-1}}\;\frac{\delta^n W[J]}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}, \qquad W[J]\equiv -i\ln Z[J], \tag{15.106}$$

as in Eqs. (10.201) and (10.202). The *proper* (1PI) n-point function $\Gamma^{(n)}$ is the n-th functional derivative of the effective action $\Gamma[\phi_{\rm cl}]$ of Theorem 10.38, evaluated at the stationary point.

The remainder of this section proves that these three analytic definitions coincide, term by term, with three classes of graphs: $G^{(n)}$ is the sum of *all* graphs with n external points; $G_c^{(n)}$ is the sum of the *connected* ones; and $\Gamma^{(n)}$ is (up to amputation and a sign) the sum of the *1PI* ones.

15.6.2 Disconnected diagrams exponentiate

The first and most important structural fact is that a sum over *all* graphs is the exponential of the sum over *connected* graphs. This single combinatorial identity — the *exponentiation* or *linked-cluster* theorem — underlies both the connectedness of W and the cancellation of vacuum bubbles, so we prove it once, abstractly, and then apply it twice.

The setting is this. Suppose we have a collection of connected graphs, each carrying a value (a number, or a functional of J) that already includes its own internal symmetry factor. Form all graphs by taking disjoint unions of copies of connected graphs. The claim is that the sum of the values of all graphs equals the exponential of the sum of the values of the connected graphs. The only subtlety, and the entire content of the theorem, is the symmetry factor that accompanies a disjoint union of *identical* connected pieces.

Theorem 15.32 (Exponentiation of disconnected graphs)**.** *Let $\{C_a\}_{a\in\mathcal{A}}$ be the set of all topologically distinct connected Feynman graphs of a given external-leg type, and let v_a denote the value of C_a assigned by the Feynman rules of Section 15.5, including its internal symmetry factor $1/S(C_a)$. Then the sum over all (connected and disconnected) graphs of that type equals*

$$\sum_{\text{all graphs } G} (\text{value of } G) \;=\; \exp\Big(\sum_{a\in\mathcal{A}} v_a\Big). \tag{15.107}$$

Proof. A general graph G is, by Theorem 15.29, a disjoint union of connected components, each of which is (a copy of) some C_a. The graph G is therefore completely specified, up to isomorphism, by the multiplicities $\{n_a\}_{a\in\mathcal{A}}$, where $n_a\ge 0$ counts how many copies of C_a appear and only finitely many n_a are nonzero. We must compute the value the Feynman rules assign to such a G and then sum over $\{n_a\}$.

The value of G is the product of the values of its components, corrected by the symmetry factor of G as a whole. The automorphism group of a disjoint union factorizes: an automorphism of G may permute the lines and vertices *within* each connected component, and may additionally permute whole components among themselves *provided it only interchanges isomorphic ones*. Hence

$$\mathrm{Aut}(G) \;\cong\; \Big(\prod_a \mathrm{Aut}(C_a)^{n_a}\Big) \rtimes \Big(\prod_a S_{n_a}\Big), \tag{15.108}$$

the semidirect product of the within-component automorphisms with the symmetric groups S_{n_a} that permute the n_a identical copies of C_a. Taking orders,

$$S(G) = |\mathrm{Aut}(G)| = \Big(\prod_a S(C_a)^{\,n_a}\Big)\Big(\prod_a n_a!\Big). \tag{15.109}$$

The first product is the internal symmetry already contained in the v_a; the second product, $\prod_a n_a!$, is the new *exchange* factor for permuting identical whole components. The value assigned to G is the product of the bare graph contributions divided by $S(G)$, which is exactly

$$(\text{value of } G) = \frac{1}{\prod_a n_a!}\prod_a \Big(\frac{(\text{bare value of } C_a)}{S(C_a)}\Big)^{n_a} = \prod_a \frac{v_a^{n_a}}{n_a!}. \tag{15.110}$$

Now sum over all graphs, i.e. over all tuples $\{n_a\}$ of non-negative integers with finitely many nonzero entries. The sum factorizes over the independent labels a:

$$\sum_{\text{all } G}(\text{value of } G) = \sum_{\{n_a\}}\prod_a \frac{v_a^{n_a}}{n_a!} = \prod_a\Big(\sum_{n_a=0}^{\infty}\frac{v_a^{n_a}}{n_a!}\Big) = \prod_a e^{v_a} = \exp\Big(\sum_a v_a\Big), \tag{15.111}$$

which is Eq. (15.107). □

The proof deserves one line of commentary, because the factor $\prod_a n_a!$ in Eq. (15.109) is the place where the symmetry-factor conventions of Section 15.5 earn their keep. Had we counted diagrams with *labelled* vertices, as the raw Wick expansion of $\exp(iS_{\mathrm{int}})$ first produces them, the $1/n_a!$ would have arisen instead from the $1/V!$ Taylor coefficient of the exponential distributing the $V = \sum_a n_a$ (vertices in C_a) vertices among the components — the same number, reached the other way. The unlabelled, symmetry-factor convention packages this automatically: the exchange of identical components *is* an automorphism of the unlabelled graph, so it is already in $S(G)$, and Eq. (15.107) follows with no leftover combinatorial debt. This is the abstract content of the zero-dimensional and functional versions proved in Theorem 4.28; what Theorem 15.32 adds is that the same identity holds graph-by-graph, with v_a the honest Feynman value of an actual diagram.

15.6.3 $W = -i \ln Z$ generates connected diagrams

We can now identify the diagrammatic content of $W[J]$. Keep the sources: let $W[J]$ and $Z[J]$ be the full functionals, with J attached to external points by the rule that each $\delta/\delta J(x)$ removes one source and exposes one external leg. We organize the diagrams of $Z[J]$ by the connected pieces from which they are built; the connected pieces are themselves J-dependent, since each may carry any number of external legs ending on sources.

Theorem 15.33 ($W[J]$ is the generator of connected diagrams)**.** *With $W[J] = -i \ln Z[J]$ and $Z[J]$ the sum of all Feynman graphs with sources, $iW[J]$ is the sum of all* connected *Feynman graphs with sources. Consequently the connected n-point function Eq. (15.106) is the sum of exactly the connected graphs with n external points,*

$$G_c^{(n)}(x_1, \dots, x_n) = \sum_{\substack{\text{connected graphs } G \\ \text{with legs at } x_1,\dots,x_n}} (\text{value of } G), \tag{15.112}$$

and the full n-point function is recovered from the connected ones by summing over all set partitions of the external points into connected clusters.

Proof. Apply Theorem 15.32 with the source-carrying connected graphs $\{C_a\}$, whose total value is the connected generating functional. Denote $\sum_a v_a \equiv iW[J]$ — the factor of i is dictated by the Minkowski normalization of Eq. (10.201), as we confirm momentarily. The theorem gives at once

$$Z[J] = \sum_{\text{all } G} (\text{value}) = \exp\Big(\sum_a v_a\Big) = \exp\big(iW[J]\big), \tag{15.113}$$

which is precisely $W = -i \ln Z$, Eq. (10.201), now *derived* as the statement that iW collects the connected diagrams. To check the factor of i and extract Eq. (15.112), recall from Eq. (10.66) that the full correlator is generated with one factor of i per leg,

$$G^{(n)} = \frac{1}{i^n} \frac{\delta^n Z}{\delta J(x_1) \cdots \delta J(x_n)}\bigg|_{J=0} \Big/ Z[0], \tag{15.114}$$

so that a connected diagram with n legs, sitting inside $iW = \ln Z$, carries weight i^n in Z and hence weight $i^n/i = i^{n-1}$ inside iW. Differentiating $iW = \sum_a v_a$ n times and dividing by i^{n-1}, exactly as in Eq. (15.106), strips those legs and leaves the connected graphs with unit weight, which is Eq. (15.112).

For the converse statement, expand $Z = e^{iW}$ in powers of iW:

$$\frac{Z[J]}{Z[0]} = \frac{e^{iW[J]}}{e^{iW[0]}} = \exp\big(i(W[J]-W[0])\big) = \sum_{m=0}^{\infty} \frac{1}{m!}\Big(i\,\widehat{W}[J]\Big)^m, \qquad \widehat{W} \equiv W - W[0], \tag{15.115}$$

where $\widehat{W}$ is the connected generator with the vacuum graphs ($n = 0$ legs) removed — those are the J-independent constant $iW[0]$, treated in Section 15.6.4. Differentiating Eq. (15.115) n times in J and setting $J = 0$, the Leibniz rule distributes the n external points among the m connected factors; the term with m factors contributes the sum over all set partitions of $\{x_1, \dots, x_n\}$ into m nonempty blocks, each block furnishing the legs of one connected correlator $G_c^{(k)}$. Collecting all m,

$$G^{(n)}(x_1, \dots, x_n) = \sum_{\pi \vdash \{x_1,\dots,x_n\}} \prod_{B \in \pi} G_c^{(|B|)}(\{x_i\}_{i \in B}), \tag{15.116}$$

the sum running over all partitions π of the external points into blocks B. This is the linked-cluster expansion: every full correlator is the sum over ways of grouping its legs into connected clusters. □

Equation Eq. (15.116) is the all-orders generalization of the two- and four-point relations Eqs. (10.204) and (10.205) verified in Theorem 10.37; at $n = 4$ the partitions into $\{1234\}$ and into $\{12|34\}, \{13|24\}, \{14|23\}$ reproduce exactly the connected four-point plus its three disconnected $2 + 2$ channels. The mechanism that Chapter 10 demonstrated by repeated functional differentiation is here seen for what it is: a statement about how graphs assemble from their connected parts.

Worked Example 15.34 (The connected four-point at $O(\lambda)$ versus its disconnected channels)**.** We confirm Eq. (15.116) concretely at four points and first order in λ, the ϕ^4 theory datum of Section 15.5. At $O(\lambda^0)$ the four-point function is purely disconnected: with $\langle\phi\rangle_0 = 0$ by the $\mathbb{Z}_2$ symmetry (Section 7.3.6), Wick's theorem gives the three pairings

$$G^{(4)}\big|_{O(\lambda^0)} = D_{12}D_{34} + D_{13}D_{24} + D_{14}D_{23}, \qquad D_{ij} \equiv D_F(x_i - x_j), \tag{15.117}$$

each term a product of two propagators in which two pairs of points never communicate — three disconnected graphs, no connected graph at this order. According to Eq. (15.116) these are precisely the partition-into-two-blocks contributions $G_c^{(2)}G_c^{(2)}$, while the single-block contribution $G_c^{(4)}$ starts one order higher. Indeed at $O(\lambda)$ the connected graph appears: one four-point vertex with its four legs running to $x_1, \dots, x_4$,

$$G_c^{(4)}\big|_{O(\lambda)} = (-i\lambda)\int d^4y\; D_F(x_1 - y)D_F(x_2 - y)D_F(x_3 - y)D_F(x_4 - y), \tag{15.118}$$

with symmetry factor 1 (the four legs are distinguishable by their endpoints, so $\mathrm{Aut} = \{e\}$). This single connected graph is what survives in $G_c^{(4)}$; the three products Eq. (15.117) have been removed by the logarithm. Subtracting them from the full $G^{(4)}$ is exactly the operation Eq. (10.205), and Eq. (15.118) is the leading term of the result — the seed of the $2 \to 2$ amplitude $\mathcal{M} = -\lambda$ that LSZ extracts in Chapter 16. The computation makes the abstract partition sum tangible: the disconnected channels live in $G_c^{(2)}G_c^{(2)}$, the physics lives in $G_c^{(4)}$.

15.6.4 Vacuum bubbles and the linked-cluster cancellation

Among the connected graphs are the vacuum bubbles of Theorem 15.29: connected graphs with no external legs, the figure-eight and its descendants. These are the diagrams that drove $\langle 0|\Omega\rangle$ to zero in the canonical treatment of Section 14.2.6 and that the Gell-Mann–Low denominator was built to remove. In the path-integral language they are the constant $W[0] = -i \ln Z[0]$, and their disposal is now a one-line corollary of exponentiation.

Theorem 15.35 (Linked-cluster cancellation of vacuum bubbles)**.** *Let* $\mathcal{V} \equiv \sum_{\text{vacuum bubbles}}(\text{value})$ *be the sum of all connected vacuum graphs. Then the disconnected vacuum contributions exponentiate to a common factor,*

$$Z[0] = \exp(\mathcal{V}), \tag{15.119}$$

and this same factor multiplies the sum of all graphs with any fixed set of external points. Consequently it cancels identically in every normalized *correlator,*

$$G^{(n)}(x_1, \dots, x_n) = \frac{1}{Z[0]} \sum_{\text{all graphs with } n \text{ legs}} (\text{value}) = \sum_{\substack{\text{graphs with } n \text{ legs} \\ \text{and no vacuum bubbles}}} (\text{value}), \tag{15.120}$$

so physical correlators contain only bubble-free diagrams.

Proof. Equation Eq. (15.119) is Theorem 15.32 specialized to graphs with no external points: the connected such graphs are exactly the vacuum bubbles, their total value is $\mathcal{V}$, and the sum over all (disconnected) vacuum graphs is $e^{\mathcal{V}} = Z[0]$. Comparing with Eq. (15.113) at $J = 0$ identifies $\mathcal{V} = iW[0]$.

For the numerator, consider any graph contributing to the unnormalized $\sum_G(\text{value})$ with external points $x_1, \dots, x_n$. By Theorem 15.29 its connected components split into two kinds: those containing at least one external point, and the vacuum bubbles, which contain none. The bubbles are, combinatorially, completely decoupled from the external-bearing components — there is no line joining a bubble to anything — so the same factorization Eq. (15.108)–Eq. (15.111) that proved exponentiation applies to the bubble subsystem alone. The unnormalized sum therefore factorizes as

$$\sum_{\text{all graphs, } n \text{ legs}} (\text{value}) = \Big(\sum_{\substack{\text{graphs, } n \text{ legs} \\ \text{no bubbles}}} (\text{value}) \Big) \cdot e^{\mathcal{V}} = \Big(\sum_{\substack{\text{graphs, } n \text{ legs} \\ \text{no bubbles}}} (\text{value}) \Big) \cdot Z[0]. \tag{15.121}$$

Dividing by $Z[0]$ as in the normalized correlator Eq. (15.114) cancels the bubble factor exactly, leaving Eq. (15.120). □

The economy of the argument is worth pausing on. The bubbles are common to numerator and denominator not by an accident of cancellation but by the

structural fact that they form a decoupled, exponentiating subsystem: $Z[0] = e^{\mathcal{V}}$ *is* the entire bubble series, and the numerator carries one factor of the very same series. This is why the $1/Z[0]$ in the definition of the correlator (Theorem 10.18) and the $\langle\Omega|\Omega\rangle^{-1}$ of the Gell-Mann–Low theorem (Section 14.2.6) are not ad hoc: they are the precise compensation for a divergent, physically inert phase. After the cancellation, no diagram with a disconnected vacuum component ever appears in a physical correlator, and we may — and from now on do — simply discard vacuum bubbles by fiat. This is the general theorem behind the $O(\lambda)$ check Theorem 10.35 of Section 10.7.3.

Worked Example 15.36 (Bubble cancellation in the $O(\lambda)$ two-point function)**.** We exhibit the cancellation explicitly at the first nontrivial order, completing the verification begun in Theorem 10.35. At $O(\lambda)$ the unnormalized two-point numerator $\sum_G$ receives two graphs. The first is the connected self-energy correction: the propagator from x_1 to a vertex y, a self-loop $D_F(0)$ at y, and the propagator from y to x_2,

$$(\text{tadpole}) = \frac{-i\lambda}{2}\int d^4y\; D_F(x_1-y)\, D_F(0)\, D_F(y-x_2), \tag{15.122}$$

with symmetry factor 2 from the self-loop (the two ends of the looped line are interchangeable). The second is disconnected: the free propagator $D_F(x_1-x_2)$ times an isolated vacuum bubble, the figure-eight,

$$(\text{disc.}) = D_F(x_1-x_2)\cdot\Big[\frac{-i\lambda}{8}\int d^4y\; D_F(0)^2\Big], \tag{15.123}$$

the bracket being the single $O(\lambda)$ vacuum bubble, symmetry factor 8. Meanwhile the denominator expands as $Z[0] = 1 + \big[\frac{-i\lambda}{8}\int d^4y\, D_F(0)^2\big] + O(\lambda^2)$, exactly the bracket of Eq. (15.123). Forming the ratio to $O(\lambda)$,

$$\begin{aligned} G^{(2)} &= \frac{D_{12} + (\text{tadpole}) + D_{12}\,\mathcal{V}_1 + O(\lambda^2)}{1 + \mathcal{V}_1 + O(\lambda^2)} \\ &= \big(D_{12} + (\text{tadpole}) + D_{12}\,\mathcal{V}_1\big)(1-\mathcal{V}_1) + O(\lambda^2) \\ &= D_{12} + (\text{tadpole}) + O(\lambda^2), \end{aligned} \tag{15.124}$$

with $\mathcal{V}_1 \equiv \frac{-i\lambda}{8}\int d^4y\, D_F(0)^2$ the $O(\lambda)$ bubble. The disconnected term $D_{12}\mathcal{V}_1$ is cancelled precisely by the $-D_{12}\mathcal{V}_1$ from expanding $1/Z[0]$, leaving only the free propagator and the connected tadpole Eq. (15.122). The vacuum bubble has vanished from the physical correlator, exactly as Theorem 15.35 guarantees, and what remains — the tadpole — is the first contribution to the ϕ^4 self-energy, the object Chapters 21 and 22 will regulate and renormalize.

15.6.5 Amputation: stripping the external propagators

Connectedness and bubble-freedom reduce the correlator to its physically active part, but a connected correlator still carries, on each external leg, a full propagator

running out to the external point x_i. For an S-matrix element these external propagators are redundant: the LSZ formula of Section 14.5 multiplies the momentum-space correlator by an inverse propagator $(p_i^2 - m^2)$ for each leg (up to the field-strength renormalization $\sqrt{Z}$), which cancels the pole of the external propagator and puts the leg on shell. It is therefore natural to define the correlator with those external legs removed in advance.

Definition 15.37 (Amputated Green's function)**.** Let $\widetilde{G}_c^{(n)}(p_1, \dots, p_n)$ be the momentum-space connected n-point function with the overall momentum-conserving delta stripped, defined as in Eq. (15.51) by $\int \big(\prod_i d^4x_i\, e^{ip_i\cdot x_i}\big) G_c^{(n)} = (2\pi)^4\delta^{(4)}\big(\sum_i p_i\big)\, \widetilde{G}_c^{(n)}(p_1, \dots$ Its *amputated* counterpart is obtained by dividing off one full propagator $\widetilde{D}(p_i)$ for each external leg,

$$\widetilde{G}^{(n)}_{\text{amp}}(p_1, \dots, p_n) \;\equiv\; \Big(\prod_{i=1}^{n} \widetilde{D}(p_i)^{-1}\Big)\, \widetilde{G}_c^{(n)}(p_1, \dots, p_n), \tag{15.125}$$

where $\widetilde{D}(p) = \widetilde{G}_c^{(2)}(p)$ is the full (corrected) propagator. Diagrammatically, amputation deletes from each external leg the entire chain of self-energy corrections out to the external point, retaining only the connected "core" to which the external lines attach.

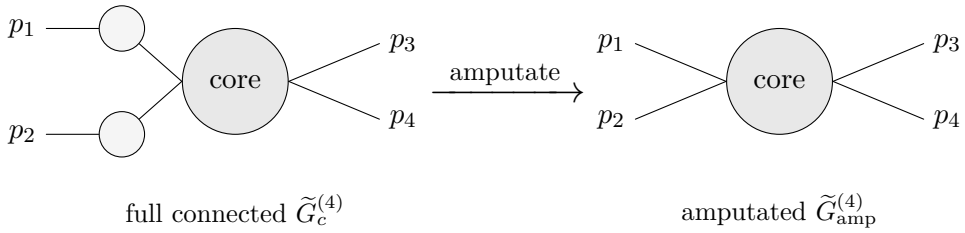


Figure 15.11: Amputation deletes the self-energy dressing on each external leg — formally, multiplies by $\widetilde{D}(p_i)^{-1}$ for each leg — leaving the bare-legged core. The amputated, connected, on-shell core is what LSZ (Section 14.5) turns into the scattering amplitude $\mathcal{M}$.

The point of amputation is the clean statement of LSZ, recalled here from Chapter 14 to complete the link. The momentum-space scattering amplitude is

$$i\mathcal{M}(p_1 \cdots \to \cdots p_n) = \Big(\prod_i \sqrt{Z}\Big)\, \widetilde{G}^{(n)}_{\text{amp}}(p_1, \dots, p_n)\Big|_{p_i^2 = m^2}, \tag{15.126}$$

i.e. the amputated connected correlator, evaluated on shell, with one factor of the field-strength renormalization $\sqrt{Z}$ per leg (Eq. (14.155); at tree level $Z = 1$). Amputation is thus the diagrammatic operation that LSZ performs analytically: the external propagators that Eq. (15.125) divides out are exactly the ones whose on-shell poles LSZ uses up. Figure 15.11 shows the surgery. This is the object

Chapter 16 receives — connected, amputated, on-shell — and integrates against phase space to make a cross section. We note for later that amputation is defined leg-by-leg with the *full* propagator $\widetilde{D}$, so it removes *all* self-energy dressing from the legs, not merely the free propagator; this is what makes the amputated core free of external-leg corrections to all orders.

15.6.6 One-particle irreducibility and the effective action

The amputated connected correlator is still not the smallest building block. A connected amputated graph can itself fall apart when a single *internal* line is cut — it can be one-particle-reducible in the sense of Theorem 15.30. The irreducible cores, the 1PI graphs, are the true atoms of perturbation theory, and the generating functional that produces them is the effective action $\Gamma[\phi_{\mathrm{cl}}]$ of Theorem 10.38. We now establish that connection at the level of the proper vertices.

Definition 15.38 (Proper vertex / 1PI n-point function)**.** The *proper* or *1PI* n-point function $\widetilde{G}^{(n)}_{\mathrm{1PI}}(p_1,\dots,p_n)$ is the sum of all amputated, 1PI graphs with n external legs (momenta p_i, the overall delta stripped). The *self-energy* $\Sigma(p^2)$ is defined through the proper two-point function,

$$\widetilde{G}^{(2)}_{\mathrm{1PI}}(p) \;\equiv\; -\,i\,\Sigma(p^2), \tag{15.127}$$

i.e. $-i\Sigma$ is the sum of all 1PI two-point graphs, amputated.

That these proper vertices are generated by Γ is the statement we now prove. We give the argument in full at two and three points — which already fixes the self-energy and the proper three-point coupling — and state the all-orders result, whose complete tree-graph proof is the business of the effective-action machinery of Chapter 22 (and was carried out in the zero-dimensional model in Theorem 4.35).

Theorem 15.39 (Γ generates the 1PI vertices)**.** *Let $\Gamma[\phi_{\mathrm{cl}}]$ be the effective action Eq. (10.210), the Legendre transform of $W[J]$, and let $\Gamma^{(n)} = \delta^n\Gamma/\delta\phi_{\mathrm{cl}}^n$ evaluated at the stationary configuration. Then the momentum-space proper vertices coincide with the 1PI amputated correlators,*

$$\widetilde{G}^{(n)}_{\mathrm{1PI}}(p_1,\dots,p_n) \;=\; i\,\widetilde{\Gamma}^{(n)}(p_1,\dots,p_n), \qquad n\geq 3, \tag{15.128}$$

while the two-point case reads

$$\widetilde{\Gamma}^{(2)}(p) \;=\; p^2 - m^2 - \Sigma(p^2), \tag{15.129}$$

the full inverse propagator. In particular $\Gamma^{(2)}$ is minus the inverse of $W^{(2)}$, $\int d^4z\, W^{(2)}(x,z)\,\Gamma^{(2)}(z,y) = -\,\delta^{(4)}(x-y)$.

Proof. The starting point is the reciprocity relation between the second derivatives of W and Γ, derived in Eq. (10.213) directly from the Legendre transform: differentiating $\delta W/\delta J = \phi_{\rm cl}$ with respect to $\phi_{\rm cl}$ and using the quantum equation of motion $\delta\Gamma/\delta\phi_{\rm cl} = -J$ (Eq. (10.212)),

$$\int d^4z\, \frac{\delta^2 W}{\delta J(x)\delta J(z)}\, \frac{\delta^2 \Gamma}{\delta\phi_{\rm cl}(z)\delta\phi_{\rm cl}(y)} = -\,\delta^{(4)}(x-y). \tag{15.130}$$

Now $W^{(2)}(x,z) = i\,G_c^{(2)}(x,z) = i\,D(x-z)$ is i times the full propagator (Eqs. (10.207) and (15.106)). Writing $\Gamma^{(2)}(z,y) = \Gamma^{(2)}(z-y)$ by translation invariance and Fourier transforming, Eq. (15.130) becomes the momentum-space reciprocity

$$i\,\widetilde{D}(p)\,\widetilde{\Gamma}^{(2)}(p) = -1 \quad\Longrightarrow\quad \widetilde{D}(p) = \frac{-1}{i\,\widetilde{\Gamma}^{(2)}(p)} = \frac{i}{\widetilde{\Gamma}^{(2)}(p)}. \tag{15.131}$$

To identify $\widetilde{\Gamma}^{(2)}$ diagrammatically, sum the full propagator as a geometric series in 1PI insertions. By Theorem 15.30, any connected two-point graph is either 1PI or has a bridge; cutting at the *first* bridge expresses it as a 1PI block followed by the rest, and iterating exhibits every two-point graph uniquely as a chain of 1PI blocks $-i\Sigma$ joined by free propagators $D_0(p) = i/(p^2 - m^2 + i\varepsilon)$:

$$\widetilde{D}(p) = D_0 + D_0(-i\Sigma)D_0 + D_0(-i\Sigma)D_0(-i\Sigma)D_0 + \cdots = \frac{D_0}{1 + i\Sigma D_0}. \tag{15.132}$$

Substituting $D_0 = i/(p^2 - m^2)$ (suppressing $i\varepsilon$) gives $i\Sigma D_0 = -\Sigma/(p^2 - m^2)$ and hence

$$\widetilde{D}(p) = \frac{i/(p^2-m^2)}{1 - \Sigma/(p^2-m^2)} = \frac{i}{p^2 - m^2 - \Sigma(p^2)}. \tag{15.133}$$

Comparing with Eq. (15.131), $i/\widetilde{\Gamma}^{(2)} = i/(p^2 - m^2 - \Sigma)$, forces $\widetilde{\Gamma}^{(2)}(p) = p^2 - m^2 - \Sigma(p^2)$, which is Eq. (15.129) and reduces to the tree value $p^2 - m^2$ of Eq. (10.214) when $\Sigma \to 0$. With $\widetilde{G}^{(2)}_{\rm 1PI} = -i\Sigma$ by Eq. (15.127), this also reads $\widetilde{\Gamma}^{(2)} = (p^2 - m^2) - i\,\widetilde{G}^{(2)}_{\rm 1PI}$; the kinematic $p^2 - m^2$ is the inverse *free* propagator, i.e. the contribution of the bare two-point "vertex" $i\widetilde{\Gamma}^{(2)}_{\rm free} = i(p^2 - m^2) = D_0^{-1}\cdot i^2$, consistent with the general rule Eq. (15.128) once the free term is included.

For $n \geq 3$ the argument is the standard induction, which we now run at $n = 3$ to display the mechanism. Differentiate the reciprocity relation $\delta W/\delta J(x_1) = \phi_{\rm cl}(x_1)$ a second and third time, converting $\delta/\delta\phi_{\rm cl}$ into $\delta/\delta J$ via the chain rule $\delta/\delta\phi_{\rm cl}(y) = \int d^4z\,(\delta J(z)/\delta\phi_{\rm cl}(y))\,\delta/\delta J(z) = -\int d^4z\,\Gamma^{(2)}(y,z)\,\delta/\delta J(z)$, where the last equality used Eq. (10.212). Applying this to $W^{(2)}$ and using Eq. (15.130) to trade derivatives, one finds after Fourier transforming

$$\widetilde{W}^{(3)}(p_1,p_2,p_3) = -\,\widetilde{W}^{(2)}(p_1)\,\widetilde{W}^{(2)}(p_2)\,\widetilde{W}^{(2)}(p_3)\,\widetilde{\Gamma}^{(3)}(p_1,p_2,p_3), \tag{15.134}$$

the three-point connected function with one full propagator $\widetilde{W}^{(2)} = i\widetilde{D}$ on each leg times the proper vertex $\widetilde{\Gamma}^{(3)}$. Amputating, i.e. stripping the three full propagators as in Eq. (15.125), leaves $\widetilde{G}^{(3)}_{\text{amp}} = i\widetilde{\Gamma}^{(3)}$ (the factors of i from $\widetilde{W}^{(2)} = i\widetilde{D}$ and the normalization Eq. (15.106) combine to the single i of Eq. (15.128)). It remains to see that this amputated three-point function is 1PI. A connected three-point graph that is one-particle-reducible has a bridge; cutting it yields a two-point part and a one-point part joined by a single propagator, but a connected amputated three-point graph has had all external propagators removed, so any surviving bridge would be internal — and the only way an amputated 3-point graph can have an internal bridge is for one external leg to attach through a self-energy chain, which amputation has already deleted. Hence the amputated connected three-point function *is* the 1PI three-point function, $\widetilde{G}^{(3)}_{\text{amp}} = \widetilde{G}^{(3)}_{\text{1PI}} = i\widetilde{\Gamma}^{(3)}$, which is Eq. (15.128) at $n = 3$. The induction to general n proceeds identically: each further ϕ_{cl}-derivative either attaches a new proper vertex or splits an existing full-propagator leg, generating exactly the tree graphs built from lower proper vertices and full propagators; the irreducible remainder at n legs is $i\widetilde{\Gamma}^{(n)}$. The complete bookkeeping is carried out in Theorem 4.35 for the zero-dimensional generator and reproduced functionally in Chapter 22; the structure is identical and we do not repeat the full recursion here. □

The theorem closes the hierarchy of generators with a clean physical statement: Z knows all diagrams, W keeps the connected ones, and Γ distills them to the 1PI skeletons. For ϕ^4 theory this names the two objects that the renormalization chapters take as their targets. The *self-energy* $\Sigma(p^2) = i\,\widetilde{G}^{(2)}_{\text{1PI}}(p)$ is the sum of 1PI two-point diagrams — its first contribution is the tadpole Eq. (15.122) of Theorem 15.36 — and through Eq. (15.129) it shifts the pole of the propagator from m^2 to the physical mass, the computation renormalized in Chapter 22. The *proper four-point function* $\widetilde{G}^{(4)}_{\text{1PI}} = i\widetilde{\Gamma}^{(4)}$ is the 1PI four-point vertex, the amputated coupling whose tree value is the bare vertex $-i\lambda$ and whose one-loop corrections (the s-, t-, u-channel bubbles) are the divergent diagrams of Chapter 21 that the coupling counterterm δ_λ absorbs in Chapter 22. These two 1PI objects — self-energy and proper four-point — are the entire renormalization content of ϕ^4, and identifying them is the structural obligation this section discharges for the thread.

15.6.7 The hierarchy and the tree relation

The three classes assemble into a strict hierarchy that is worth stating as the organizing principle of all that follows. Reading from the most to the least information:

$$\underbrace{Z[J]}_{\text{all graphs}} \xrightarrow{\;-i\ln\;} \underbrace{W[J]}_{\text{connected}} \xrightarrow{\;\text{Legendre}\;} \underbrace{\Gamma[\phi_{\text{cl}}]}_{\text{1PI}}. \tag{15.135}$$

The logarithm discards the disconnected diagrams (Theorem 15.33) and, with the $Z[0]$ normalization, the vacuum bubbles (Theorem 15.35); the Legendre transform discards the one-particle-reducible diagrams (Theorem 15.39). Crucially, the hierarchy is invertible: nothing is lost going from Γ back up to W, because the reducible and disconnected diagrams that Γ omits are entirely determined by the irreducible ones through the tree rules of the Legendre transform. The connected correlators are *tree graphs built from the proper vertices* $\Gamma^{(n)}$ *as couplings and the full propagators* $\widetilde{D}$ *as lines.* This is the precise sense in which the 1PI vertices are the complete data of the theory.

The cleanest instance — and the one every later chapter uses — is the reconstruction of the full propagator from the self-energy, already encountered as Eq. (15.132) inside the proof above. We isolate it as a theorem because it is the prototype of all such resummations.

Theorem 15.40 (Dyson resummation of the full propagator)**.** *The full (connected, two-point) propagator is the geometric series of free propagators joined by 1PI self-energy insertions,*

$$\widetilde{D}(p) \;=\; \frac{i}{p^2 - m^2 - \Sigma(p^2) + i\varepsilon}, \qquad -i\Sigma(p^2) = \widetilde{G}^{(2)}_{\mathrm{1PI}}(p), \tag{15.136}$$

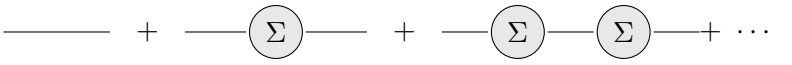
where Σ *is the sum of 1PI two-point graphs. The physical mass* m^2_{ph} *is the location of the pole,* $m^2_{\mathrm{ph}} - m^2 - \Sigma(m^2_{\mathrm{ph}}) = 0$*, and the residue defines the field-strength renormalization* $Z = [1 - \Sigma'(m^2_{\mathrm{ph}})]^{-1}$*.*

Proof. Equation Eq. (15.136) is Eq. (15.133), restoring the $i\varepsilon$. The decomposition of an arbitrary two-point graph into a unique chain of 1PI blocks is Theorem 15.30: read the graph from the left external leg; the first internal line that is a bridge separates the leading 1PI block from the remainder, and recursion on the remainder yields the unique factorization $D_0[(-i\Sigma)D_0]^k$. Summing the geometric series $\sum_k D_0[(-i\Sigma)D_0]^k = D_0/(1 + i\Sigma D_0)$ and inserting $D_0 = i/(p^2 - m^2 + i\varepsilon)$ gives Eq. (15.136). The pole condition and residue follow by expanding the denominator about its zero: near $p^2 = m^2_{\mathrm{ph}}$, $p^2 - m^2 - \Sigma(p^2) \approx (p^2 - m^2_{\mathrm{ph}})[1 - \Sigma'(m^2_{\mathrm{ph}})]$, so $\widetilde{D} \approx iZ/(p^2 - m^2_{\mathrm{ph}})$ with $Z = [1 - \Sigma'(m^2_{\mathrm{ph}})]^{-1}$. □

Worked Example 15.41 (The dressed propagator at one loop in ϕ^4)**.** We illustrate Theorem 15.40 with the leading ϕ^4 theory self-energy, the tadpole Eq. (15.122) of Theorem 15.36. In momentum space the amputated 1PI two-point graph is the single closed loop attached at one vertex,

$$-i\Sigma\big|_{O(\lambda)} = \frac{-i\lambda}{2} \int \frac{d^4k}{(2\pi)^4}\, \frac{i}{k^2 - m^2 + i\varepsilon}, \tag{15.137}$$

the symmetry factor $\frac{1}{2}$ from the loop, so that $\Sigma|_{O(\lambda)} = \frac{\lambda}{2} \int \frac{d^4k}{(2\pi)^4} \frac{1}{k^2 - m^2 + i\varepsilon}$ is a momentum-independent constant (it is p-independent because the loop carries



$$\widetilde{D} = D_0 + D_0(-i\Sigma)D_0 + D_0(-i\Sigma)D_0(-i\Sigma)D_0 + \cdots$$

Figure 15.12: Dyson resummation. The full propagator $\widetilde{D}$ is the sum over all chains of one-particle-irreducible self-energy blobs $-i\Sigma$ strung along free propagators D_0. The geometric series sums to $\widetilde{D} = i/(p^2 - m^2 - \Sigma)$, Eq. (15.136): an infinite set of reducible diagrams repackaged into a single dressed line built from the one irreducible object Σ.

no external momentum). Being constant, this contribution is a pure mass shift: Eq. (15.136) gives

$$\widetilde{D}(p) = \frac{i}{p^2 - m^2 - \Sigma} = \frac{i}{p^2 - (m^2 + \Sigma)}, \tag{15.138}$$

the free form with $m^2 \to m^2 + \Sigma$, and since $\Sigma' = 0$ here the residue is unchanged, $Z = 1$ at this order. The integral in Eq. (15.137) is quadratically divergent; that it must be is the subject of the power counting in Chapter 21, and its absorption into a mass counterterm δ_m via the renormalization condition $m^2_{\text{ph}} = m^2 + \Sigma(m^2_{\text{ph}}) + \delta_m$ is the subject of Chapter 22. What the present section establishes is the *structure* that makes that program possible: the divergence sits in a single 1PI object Σ, the dressed propagator is the geometric resummation Eq. (15.136) of that object, and the resummation is exact to all orders in the insertion — so renormalizing Σ renormalizes the propagator once and for all.

The hierarchy Eq. (15.135) is the architecture on which the rest of perturbation theory is built, and the tree relation $\Gamma \to W$ is its load-bearing beam. Two destinations follow immediately. The connected, amputated, on-shell vertices of Theorem 15.37 and Eq. (15.126) are precisely the amplitudes that Chapter 16 integrates against Lorentz-invariant phase space to produce cross sections and decay rates, completing the passage from formalism to number for the $2 \to 2$ process of ϕ^4 theory. The 1PI organization of Theorem 15.39 — self-energy and proper vertices, dressed propagator by Dyson resummation — is the form in which divergences appear as corrections to a finite set of proper vertices, the starting point for the power counting of Chapter 21, the counterterm program of Chapter 22, and the effective potential of Chapter 29, where the value of Γ at constant ϕ_{cl} becomes the order parameter for spontaneous symmetry breaking. With the diagrams sorted into these classes and the generators that produce them established, the machinery is ready to meet fermions, whose anticommuting statistics will modify the rules with the relative signs and closed-loop factor of the next section.

15.7 Fermions and Gauge Fields: Sign Rules and the Fermion Loop

The bosonic side of the summary is now closed. Section 15.5 extracted the position- and momentum-space Feynman rules for a scalar field from the generating functional, the architecture section that followed sorted the resulting diagrams into connected, amputated, and one-particle-irreducible classes, and ϕ^4 theory produced its first amplitude two independent ways. Every ingredient in that construction—the propagator as the source kernel of $Z[J]$, the vertex as a functional derivative of iS_{int}, the symmetry factor as the order of the diagram's automorphism group—was built on one structural fact: the generating functional is a Gaussian in commuting sources, so its correlators are *unsigned* sums over pairings. A real scalar contracts with itself, every pairing enters with weight $+1$, and a permanent of propagators results.

Fermions change exactly one thing, and they change it everywhere. The Dirac field is anticommuting, its sources are Grassmann (Theorem 12.3), and its generating functional—derived in full in Theorem 12.42—pairs two *distinct* source species $\bar{\eta}, \eta$ in a fixed order. The combinatorial consequence, the fermionic Wick theorem of Theorem 12.44, replaces the permanent by a *determinant*: the same sum over pairings, but each term weighted by the signature of its pairing permutation. Two new pieces of bookkeeping follow, and this section is devoted to deriving them once, cleanly, so that Part V and Part VII inherit a complete rule set rather than rebuilding it. The first is the directed propagator line, an arrowed S_F whose orientation records the flow of fermion number. The second is the family of relative signs that the determinant structure forces: a relative minus between any two diagrams related by exchanging a pair of identical external fermions, and—the single most consequential rule in perturbation theory—an overall factor of -1 together with a Dirac trace for every closed fermion loop. None of these signs is a convention imposed by hand; each is forced by the anticommutation of the Grassmann fields, and we track the anticommutation explicitly each time.

We keep the chapter conventions in force: mostly-minus metric $\eta_{\mu\nu} = \text{diag}(+,-,-,-)$ so that $p^2 = m^2$ on shell; natural units $\hbar = c = 1$; the Feynman $i\varepsilon$ prescription. The Dirac propagator and its arrow convention are imported unaltered from Chapters 11 and 12; the gamma-matrix algebra and the trace technology that the loops generate live in Section C.6 and are cited, not relisted. The aim is operational: by the end of the section a reader will be able to translate any fermion–boson interaction into directed lines, vertices, the correct signs, and—for closed loops—a trace, and will know precisely where each piece came from.

15.7.1 The directed fermion line

The bosonic Feynman rule assigns to each internal line the momentum-space propagator $\tilde{D}_F(p) = i/(p^2 - m^2 + i\varepsilon)$, an undirected object: a scalar line carries no orientation because ϕ is its own conjugate and $\langle 0|\mathcal{T}\,\phi(x)\phi(y)|0\rangle$ is symmetric in $x \leftrightarrow y$. The fermion line is different in exactly the way the propagator is different. The free two-point function is the time-ordered product of two *distinct* fields, ψ and $\bar{\psi}$, computed in Theorem 12.43 as

$$\overbrace{\psi(x)\,\bar{\psi}(y)} \;=\; \langle 0|\mathcal{T}\,\psi(x)\bar{\psi}(y)|0\rangle \;=\; S_F(x-y), \qquad \overbrace{\psi\,\psi} = \overbrace{\bar{\psi}\,\bar{\psi}} = 0, \tag{15.139}$$

the contraction rules Eq. (12.169). Two features of Eq. (15.139) translate directly into the diagram. First, the only nonvanishing contraction joins a ψ to a $\bar{\psi}$: a fermion line always runs from a $\bar{\psi}$ at one end to a ψ at the other, and can never join two ψ's or two $\bar{\psi}$'s. Second, $S_F(x-y)$ is a 4×4 matrix in Dirac space and is *not* symmetric under $x \leftrightarrow y$; the order of its arguments matters, and the diagram must record it. Both facts are encoded by drawing the line with an arrow.

Definition 15.42 (The internal fermion line)**.** An internal fermion line is drawn as a directed line carrying an arrow. The arrow points *from the $\bar{\psi}$ end toward the ψ end* of the contraction Eq. (15.139), i.e. in the direction of flow of the conserved $U(1)$ fermion-number charge of Theorem 11.32. To a line carrying four-momentum p in the direction of its arrow, the momentum-space rule assigns the propagator

$$\tilde{S_F}(p) \;=\; \frac{i(\not{p} + m)}{p^2 - m^2 + i\varepsilon} \;=\; \frac{i}{\not{p} - m + i\varepsilon}, \tag{15.140}$$

a matrix carrying a Dirac index a at the head (the ψ end) and b at the tail (the $\bar{\psi}$ end), $\big(\tilde{S_F}(p)\big)_{ab}$.

This is precisely the object built canonically in Eq. (11.195) and rederived from the path integral in Eq. (12.162); the two pillars agree term for term, and Eq. (15.140) is what every fermion line in the rest of the book evaluates to. The arrow does double duty. As a bookkeeping device it tells us, unambiguously and locally, which Dirac index of $\tilde{S_F}$ attaches to which vertex, so that the matrices along a line multiply in a definite order. As physics it records charge flow: read with $p^0 > 0$ in the arrow direction the line is an electron propagating forward in time, while the opposite sign of p^0 is the same line reinterpreted as a positron propagating backward—the two residues at $p^0 = \pm E_{\mathbf{p}}$ noted under Eq. (11.195). We adopt the convention, fixed in Fig. 11.4, that loop and internal momenta are routed *along* the arrows wherever possible; momentum conservation at each vertex is imposed exactly as in the bosonic rules of Section 15.5, and is unaffected by the orientation.

The arrows are conserved at every vertex of the theories we will build. A QED or Yukawa vertex contains exactly one ψ and one $\bar{\psi}$, so exactly one arrow enters

and one leaves: fermion lines never begin or end in the interior of a diagram but thread continuously through it, either running from one external fermion to another along an open path or closing on themselves in a loop. This conservation is the diagrammatic image of the charge conservation made automatic in Theorem 12.43, where the vanishing of $\langle \mathcal{T}\, \psi\psi \rangle$ and $\langle \mathcal{T}\, \bar{\psi}\bar{\psi} \rangle$ guaranteed one ψ for every $\bar{\psi}$.

Notation Watch: external-leg factors are deferred to Chapter 17

An external fermion line terminates not on a propagator but on the LSZ amputation of Section 14.5 followed by an on-shell spinor wavefunction: $u^s(p)$ for an incoming particle, $\bar{u}^s(p)$ for an outgoing particle, $\bar{v}^s(p)$ for an incoming antiparticle, and $v^s(p)$ for an outgoing antiparticle, each read at the appropriate end of the arrow. The arrow on an external particle line points into the diagram in the direction of motion; on an antiparticle line it points against it. These wavefunctions and the spin sums $\sum_s u^s \bar{u}^s = \not{p} + m$, $\sum_s v^s \bar{v}^s = \not{p} - m$ of Eq. (11.73) are the substrate of the trace technology that Part V exercises; we state them here for completeness but develop the assembly of a complete amplitude—and the spin-averaging that turns $\mathcal{M}$ into $|\mathcal{M}|^2$—in Chapter 17, where QED computes $e^+e^- \to \mu^+\mu^-$. This section's business is the rules *internal* to a diagram: lines, vertices, and signs.

15.7.2 The fermion–boson vertex

The vertex rule is read off an interaction term exactly as in the bosonic case, and the derivation is the same one functional differentiation that produced the ϕ^4 vertex in Section 15.5, now applied to a term bilinear in the fermion fields. Consider the most general local, Lorentz-invariant, Hermitian interaction coupling a single Dirac field to a boson field Φ through a fermion bilinear,

$$\mathcal{L}_{\text{int}} \;=\; -\, g\, \bar{\psi}\, \Gamma\, \psi\, \Phi, \tag{15.141}$$

where g is a real coupling and Γ is a fixed matrix in Dirac (and possibly flavor) space carrying whatever Lorentz indices Φ carries, contracted with it. This single template covers the cases we need: the *Yukawa* coupling of a real scalar takes $\Phi = \phi$ and $\Gamma = \mathbf{1}$ (or $\Gamma = i\gamma^5$ for a pseudoscalar); the *gauge* coupling of a vector takes $\Phi = A_\mu$ and $\Gamma = \gamma^\mu$.

The perturbative expansion of any correlator inserts the interaction through $\exp(i \int d^4x\, \mathcal{L}_{\text{int}})$, and each vertex of a diagram corresponds to one factor of $i\mathcal{L}_{\text{int}}$ brought down from this exponential and integrated over its spacetime point, with the fields it contains contracted into the adjacent lines by the Wick theorem Theorem 12.44. The amputated momentum-space vertex is therefore i times the field-stripped coefficient of the relevant monomial in $\mathcal{L}_{\text{int}}$. Because the three

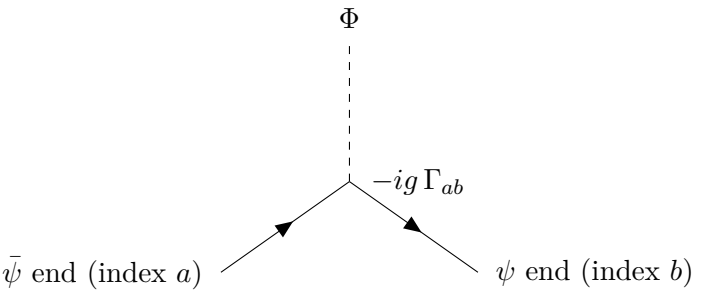


Figure 15.13: The generic fermion–boson vertex Eq. (15.142) from the interaction Eq. (15.141). The fermion arrow flows continuously *through* the vertex, entering on the $\bar\psi$ side (Dirac row index a) and leaving on the ψ side (column index b); the matrix Γ sits between them. The boson line (drawn dashed for a scalar; a wavy line for a photon) attaches through the Lorentz/internal index Γ shares with Φ. Specializations: Yukawa scalar, $\Gamma = \mathbf{1}$, factor $-ig$; QED, $\Gamma = \gamma^\mu$, factor $-ie\gamma^\mu$.

fields $\bar\psi$, ψ, Φ in Eq. (15.141) are mutually distinct, there is no symmetry factor to cancel—unlike the ϕ^4 vertex, where the four identical ϕ's contributed a 4! that cancelled the 1/4! in the coupling. Stripping the three fields off $i\mathcal{L}_{\text{int}} = -ig\,\bar\psi\Gamma\psi\,\Phi$ leaves the rule directly.

Definition 15.43 (The fermion–boson vertex)**.** The interaction Eq. (15.141) contributes the momentum-space vertex

$$\boxed{-\,i\,g\,\Gamma\,,} \tag{15.142}$$

a matrix in Dirac space whose row index attaches to the fermion line entering the vertex (the $\bar\psi$, against the arrow) and whose column index attaches to the line leaving it (the ψ, along the arrow), with the boson line attached through whatever Lorentz/internal index Γ and Φ share. Each vertex carries a four-momentum-conserving factor $(2\pi)^4\delta^4(\sum p)$ and is integrated over its position, exactly as in Section 15.5.

The two specializations the book needs follow by inspection, and we record them now so that Chapters 17 and 26 can take them as given.

The QED vertex. The QED Lagrangian frozen for QED is $\mathcal{L}_{\text{QED}} = \bar\psi(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $D_\mu = \partial_\mu + ieA_\mu$ (electron charge $-e$, $e > 0$). Expanding the covariant derivative,

$$\bar\psi\, i\not{D}\,\psi = \bar\psi\, i\gamma^\mu(\partial_\mu + ieA_\mu)\,\psi = \bar\psi\, i\not{\partial}\,\psi \;-\; e\,\bar\psi\gamma^\mu\psi\,A_\mu, \tag{15.143}$$

so the interaction term is $\mathcal{L}_{\text{int}} = -e\,\bar\psi\gamma^\mu\psi\,A_\mu$, which is exactly Eq. (15.141) with $g = e$, $\Phi = A_\mu$, and $\Gamma = \gamma^\mu$. The vertex rule Eq. (15.142) therefore gives

$$\boxed{-\,i\,e\,\gamma^\mu\,,} \tag{15.144}$$

the electron–photon vertex, with μ the Lorentz index of the attached photon line. This is the template Chapter 17 specializes and uses without re-deriving; the photon line, photon propagator (Eq. (13.182)), and the spin-sum machinery are assembled around it there.

The Yukawa vertex. For a real scalar coupled as $\mathcal{L}_{\rm int} = -g\,\bar\psi\psi\,\phi$, the matrix is $\Gamma = \mathbf{1}$ and the vertex is simply $-ig\,\mathbf{1}$, the identity in Dirac space; for the pseudoscalar coupling $-ig\,\bar\psi\, i\gamma^5\,\psi\,\phi$ it is $g\gamma^5$. These are the vertices of the meson–nucleon and Higgs–fermion couplings, used in Chapter 30 and beyond.

The index-flow statement in Theorem 15.43 deserves emphasis because it dictates the order in which matrices are written. Read *against* the arrow of the fermion line, a continuous open line that enters the diagram, passes through vertices $V_1, V_2, \dots$ separated by propagators, and exits, contributes the matrix string

$$(\text{outgoing spinor})\;\big(-ig\Gamma_{V_1}\big)\,\tilde S_F\,\big(-ig\Gamma_{V_2}\big)\,\tilde S_F\,\cdots\,(\text{incoming spinor}), \tag{15.145}$$

with each factor's row index contracted into the column index of the factor to its left. The rule "write the matrices against the arrow" is not a separate postulate: it is the order in which the Dirac indices of Eq. (15.140) and Eq. (15.142) actually contract, dictated by which index attaches where. For a closed line—a loop—there is no incoming or outgoing spinor to terminate the string, and the indices close on themselves; that closure is a trace, and producing it correctly, with its sign, is the subject of the next two subsections.

15.7.3 The relative sign between exchange diagrams

We now derive the first of the two sign rules directly from the fermionic Wick theorem, in the form that diagrams use it. The cleanest place to see it is the object already computed in full in Theorem 12.45: the free four-point function with two ψ's and two $\bar\psi$'s,

$$\langle 0|\mathcal{T}\,\psi(x_1)\psi(x_2)\bar\psi(y_1)\bar\psi(y_2)|0\rangle \;=\; -\,S_F(x_1{-}y_1)S_F(x_2{-}y_2) + S_F(x_1{-}y_2)S_F(x_2{-}y_1), \tag{15.146}$$

which is Eq. (12.174). The combinatorial content is transparent. A ψ contracts only with a $\bar\psi$, so there are exactly two ways to pair the fields: the *direct* pairing $(\psi_1\bar\psi_1)(\psi_2\bar\psi_2)$ and the *exchange* pairing $(\psi_1\bar\psi_2)(\psi_2\bar\psi_1)$. The two enter with *opposite* signs. This relative minus is the whole of fermionic statistics at tree level, and it is forced, not inserted: in the derivation of Theorem 12.45 every sign came from a single application of the Grassmann anticommutation rule, never from a choice.

Diagrammatically, the two terms of Eq. (15.146) are two ways of joining the four external points with directed lines, drawn in Fig. 15.14. They are topologically distinct: the exchange diagram is obtained from the direct one

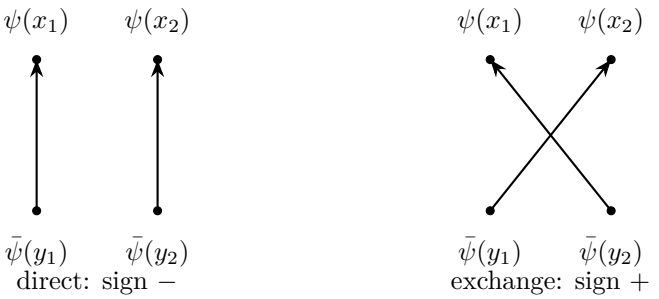


Figure 15.14: The two contraction patterns of Eq. (15.146), the diagrammatic content of Fig. 12.3. Each directed line is a propagator S_F running along the fermion-number arrow. The exchange pattern (right) differs from the direct one (left) by a single crossing of the lines—one transposition of adjacent anticommuting operators—and so carries the opposite relative sign. This is the origin of the relative minus between the direct and exchange (annihilation) graphs in Bhabha and Møller scattering (Chapter 18).

by swapping the endpoints of the two outgoing lines, which crosses the two propagators. Each crossing is one transposition of adjacent anticommuting operators in the contraction, hence one factor of -1. This gives the diagrammatic rule in its operational form.

Proposition 15.44 (Relative sign of exchange)**.** *Two diagrams contributing to the same amplitude that differ only by the interchange of two identical external fermions (two incoming particles, two outgoing particles, or a particle and the corresponding antiparticle treated as identical lines) contribute with* opposite *relative sign. More generally, the sign of a given contraction pattern is the signature of the permutation that brings every contracted ψ–$\bar{\psi}$ pair into adjacency in the order $\psi\,\bar{\psi}$, exactly the sign in the fermionic Wick sum Eq. (12.170).*

Proof. The amplitude is, by LSZ (Section 14.5), built from the time-ordered correlator with external legs amputated and put on shell; the relative sign between two contraction patterns survives amputation untouched, because amputation acts independently on each external leg and introduces no reordering of the fermion operators. It therefore suffices to establish the sign at the level of the correlator, where it is Eq. (12.170): each term carries the signature of the permutation σ pairing the ψ's with the $\bar{\psi}$'s. Two patterns related by exchanging two external fermions differ precisely by composing σ with one transposition, which flips $\mathrm{sgn}(\sigma)$. The $n = 2$ instance is Eq. (15.146), whose two terms are the identity and the single transposition, carrying -1 and $+1$ respectively; the relative sign is -1. □

Caution: relative sign is physical, overall sign is not

The *overall* sign of a single fermionic diagram depends on the order in which one chooses to write the external operators in the correlator—a convention. What is physical, and what the Feynman rules must reproduce, is the *relative* sign between diagrams contributing to the same process. In practice one fixes a reference ordering of the external fermions, assigns that diagram a $+$, and then assigns every other diagram the sign of the permutation of external fermion lines that relates it to the reference. The interference term in $|\mathcal{M}|^2 = |\mathcal{M}_{\text{direct}} + \mathcal{M}_{\text{exch}}|^2$ then carries the correct relative minus, and it is this interference—never an isolated overall sign—that appears in a cross section.

15.7.4 The closed fermion loop: the -1 and the trace

The second sign rule governs closed fermion lines, and it is the one that distinguishes a fermion from a boson in every loop computation in the book. We derive it here from the same anticommutation that produced the exchange sign, tracking the reordering explicitly; the result was obtained independently in Chapter 12 from the placement of the fermion determinant in the numerator of the generating functional (Theorem 12.49), and the two derivations must—and do—agree.

Consider a diagram in which n vertices of the generic type Eq. (15.141) are joined into a single closed fermion loop, the boson legs at each vertex attaching to the rest of the diagram. Each vertex carries the operator bilinear $\bar{\psi}(x_i)\,\Gamma_i\,\psi(x_i)$, written in components as $\bar{\psi}_{a_i}(x_i)\,(\Gamma_i)_{a_i b_i}\,\psi_{b_i}(x_i)$, and the loop is the contraction pattern that joins $\psi(x_i)$ to $\bar{\psi}(x_{i+1})$ for $i = 1, \dots, n-1$ and finally closes $\psi(x_n)$ back onto $\bar{\psi}(x_1)$. The piece of the correlator we must evaluate is the contraction of the product of bilinears,

$$\mathcal{L} \;=\; \overbrace{[\bar{\psi}(x_1)\Gamma_1\psi(x_1)]\,[\bar{\psi}(x_2)\Gamma_2\psi(x_2)]\cdots[\bar{\psi}(x_n)\Gamma_n\psi(x_n)]}, \tag{15.147}$$

in the ring pattern just described. Two things must be extracted: the Dirac index structure (which will close into a trace) and the overall sign (which will be -1).

The trace. Each bilinear is a Grassmann-even object, so the n blocks $[\bar{\psi}(x_i)\Gamma_i\psi(x_i)]$ commute with one another freely; no sign arises from reordering whole blocks. Within the string, $\psi(x_i)$ stands immediately to the left of $\bar{\psi}(x_{i+1})$ for each $i = 1, \dots, n-1$, so these $n-1$ contractions are already in the order $\psi\,\bar{\psi}$ and evaluate with no sign to

$$\overbrace{\psi_{b_i}(x_i)\,\bar{\psi}_{a_{i+1}}(x_{i+1})} \;=\; \big(S_F(x_i - x_{i+1})\big)_{b_i a_{i+1}}. \tag{15.148}$$

Stringing the vertex matrices and these propagators together, the Dirac indices contract head-to-tail: $(\Gamma_1)_{a_1b_1}(S_F)_{b_1a_2}(\Gamma_2)_{a_2b_2}(S_F)_{b_2a_3}\cdots(\Gamma_n)_{a_nb_n}$, leaving the two open indices a_1 (on the first vertex) and b_n (on the last propagator's ψ end). The final contraction $\psi(x_n)$–$\bar\psi(x_1)$ joins b_n back to a_1, and the index structure closes on itself: a sum over the single repeated index is a trace,

$$(\Gamma_1)_{a_1b_1}(S_F)_{b_1a_2}\cdots(\Gamma_n)_{a_nb_n}(S_F)_{b_na_1} \;=\; \mathrm{tr}\big[\Gamma_1\,S_F(x_1{-}x_2)\,\Gamma_2\,S_F(x_2{-}x_3)\cdots\Gamma_n\,S_F(x_n{-}x_1)\big]. \tag{15.149}$$

The trace runs over the Dirac (and, if Γ_i carry them, flavor and color) indices of the matrices encountered in one circuit of the loop, read in the order in which the arrow visits the vertices. This is the diagrammatic statement of the rule "a closed fermion loop carries a trace," and Eq. (15.149) fixes the order of the factors inside it.

The sign. It remains to evaluate the closing contraction $\psi(x_n)$–$\bar\psi(x_1)$, and here the single minus appears. After the $n-1$ internal contractions Eq. (15.148) have been pulled out as c-number matrices (which, being Grassmann-even, may be moved freely past anything), the two operators still to be contracted sit at opposite ends of the string: $\bar\psi(x_1)$ is the leftmost surviving field and $\psi(x_n)$ is the rightmost. They are in the order $\bar\psi(x_1)\cdots\psi(x_n)$—that is, $\bar\psi$ before ψ. But the contraction rule Eq. (15.139) is stated for the order $\psi\,\bar\psi$. To bring them into that order we anticommute the two fields past each other, which inside the time-ordered contraction means

$$\overbrace{\bar\psi_{a_1}(x_1)}\,\psi_{b_n}(x_n) \;=\; -\,\overbrace{\psi_{b_n}(x_n)}\,\bar\psi_{a_1}(x_1) \;=\; -\,\big(S_F(x_n-x_1)\big)_{b_na_1}. \tag{15.150}$$

The single transposition required to close the ring—to move the one $\bar\psi$ that began at the far left around to meet the one ψ at the far right—is the origin of the factor -1. It is one anticommutation, and it occurs exactly once per closed loop, independent of the number of vertices n on the loop: the $n-1$ internal contractions were sign-free, and only the closure costs a sign. Assembling Eq. (15.149) and Eq. (15.150) gives the rule.

Proposition 15.45 (The closed-fermion-loop rule)**.** *Each closed fermion loop in a diagram contributes an overall factor of -1 together with a single Dirac (and flavor/color) trace over the matrices encountered around the loop, read in the direction of the fermion-number arrow:*

$$\boxed{(-1)\;\mathrm{tr}\big[\Gamma_1\,\tilde S_F(p_1)\,\Gamma_2\,\tilde S_F(p_2)\cdots\Gamma_n\,\tilde S_F(p_n)\big]\;,} \tag{15.151}$$

in momentum space, with one undetermined loop momentum integrated $\int d^4\ell/(2\pi)^4$ and the internal momenta p_i fixed by conservation at each vertex. A diagram containing L disjoint closed fermion loops carries $(-1)^L$, one factor per loop.

Proof. The trace is Eq. (15.149), with each vertex contributing its matrix Γ_i (and the coupling and i of Eq. (15.142), suppressed here and restored in the worked

example below) and each internal line its propagator; Fourier-transforming the position-space propagators Eq. (15.148) to momentum space replaces $S_F(x_i - x_{i+1}) \to \tilde{S_F}(p_i)$ and leaves one loop momentum unfixed by the vertex delta functions, hence integrated. The overall sign is Eq. (15.150): one anticommutation per loop. For L disjoint loops the argument applies to each independently—each is a separate ring requiring its own single closing transposition, and the rings share no fields—so the signs multiply to $(-1)^L$. This is the same result reached in Theorem 12.49 from the numerator placement of $\det(i\not{\partial} - m)$ in the generating functional: there the $(-1)^L$ arose because each separate fermion determinant sits upstairs rather than downstairs, and Theorem 12.48 identified the Tr ln expansion of that determinant term by term with the closed-loop diagrams Eq. (15.151). The two routes—anticommuting the operators here, expanding the determinant there—are the same statement in two languages, as Section 12.7.3 verified by matching them on the antiperiodic trace. □

Physical Insight: why the loop sign is not optional

The -1 per fermion loop is the diagrammatic fingerprint of Fermi statistics, and it controls the *sign* of physical effects, not merely their bookkeeping. The simplest closed loop with two photon legs is the vacuum polarization, the worked example below; its -1 is what makes a fermion loop *screen* the electric charge rather than antiscreen it, fixing the sign of the running of α in Chapter 19. In a non-abelian theory the competition between gluon loops (which antiscreen) and quark loops (which screen, carrying this same -1) is precisely the sign struggle that decides asymptotic freedom: the coefficient $\frac{11}{3}C_A - \frac{4}{3}T_F n_f$ of Chapter 27 has its two terms of opposite sign for exactly this reason, the $-\frac{4}{3}T_F n_f$ piece being the fermion loops with their -1. And in supersymmetric theories the cancellation of the vacuum energy between bosons and fermions, seen on a single mode in Theorem 12.50, is this -1 summed over the spectrum. A sign error here is not a factor-of-two slip; it inverts the physics.

15.7.5 Worked example: the trace structure of the vacuum-polarization loop

We now apply the complete fermionic rule set to the simplest and most important closed loop, the QED vacuum polarization: a single electron loop with two external photon legs. The aim is to assemble the integrand—vertices, propagators, the -1, and the trace—purely from the rules just derived, validating them against the determinant-route result Eq. (12.189), and then to hand the explicit integral to Chapter 19, which evaluates it in dimensional regularization. This is a setup computation for QED, not its execution.

Worked Example 15.46 (The fermion-loop integrand of the photon self-energy)**.** Consider the one-loop correction to the photon propagator: two QED vertices joined by a closed electron loop, with external photon momentum q entering one vertex and leaving the other, and Lorentz indices μ, ν on the two photon legs. We extract the amputated, 1PI two-point insertion $i\Pi^{\mu\nu}(q)$, the object whose iteration dresses the photon propagator.

Step 1: lay out the loop. The loop carries an internal momentum; route k on one line in the arrow direction and, by momentum conservation at the first vertex, $k - q$ on the other. The loop visits, in arrow order, the vertex with Lorentz index μ, then a propagator of momentum k, then the vertex with index ν, then a propagator of momentum $k - q$ closing the ring. Figure 15.15 draws it.

Step 2: assemble the factors by the rules. The two vertices are each $-ie\gamma^{\mu}$ and $-ie\gamma^{\nu}$ from Eq. (15.144); the two internal lines are each $\tilde{S}_F$ from Eq. (15.140); the single closed loop contributes the factor (-1) and the trace of Theorem 15.45; the one undetermined loop momentum is integrated. Reading the matrices around the loop in the arrow direction and closing them into the trace,

$$i\Pi^{\mu\nu}(q) \;=\; (-1)\int \frac{d^4k}{(2\pi)^4}\;\mathrm{tr}\Big[(-ie\gamma^{\mu})\,\tilde{S}_F(k)\,(-ie\gamma^{\nu})\,\tilde{S}_F(k-q)\Big]. \tag{15.152}$$

Step 3: pull out the constants and exhibit the integrand. The two vertex factors contribute $(-ie)^2 = -e^2$, and the leading (-1) of the loop multiplies it to $+e^2$. Substituting the explicit propagators Eq. (15.140), $\tilde{S}_F(k) = i(\not{k}+m)/(k^2-m^2+i\varepsilon)$ and likewise for $k - q$, the two numerator factors of i combine to $i^2 = -1$, giving

$$\boxed{\,i\Pi^{\mu\nu}(q) \;=\; -\,e^2\int \frac{d^4k}{(2\pi)^4}\;\frac{\mathrm{tr}\big[\gamma^{\mu}(\not{k}+m)\,\gamma^{\nu}(\not{k}-\not{q}+m)\big]}{(k^2-m^2+i\varepsilon)\big((k-q)^2-m^2+i\varepsilon\big)}\,.\,} \tag{15.153}$$

This is the integrand Chapter 19 takes as its starting point. Its overall structure matches term for term the determinant-route expression Eq. (12.189): the -1 of the closed loop, the single Dirac trace closing the spinor indices around the ring, and the two vertices $-ie\gamma^{\mu}$, $-ie\gamma^{\nu}$. The rules assembled in this section and the Tr ln expansion of Chapter 12 produce identical content, as they must.

Step 4: the trace, as far as it teaches the rule. The numerator is a trace of four gamma matrices plus lower terms, evaluated with the technology of Section C.6. Expanding $\mathrm{tr}[\gamma^{\mu}(\not{k}+m)\gamma^{\nu}(\not{k}-\not{q}+m)]$, the two terms linear in m contain three gamma matrices and vanish, since the trace of an odd number of gamma matrices is zero (Theorem C.53). The two survivors are the m^2 term, $m^2\,\mathrm{tr}[\gamma^{\mu}\gamma^{\nu}] = 4m^2\eta^{\mu\nu}$ by Eq. (C.152), and the term quadratic in momenta, $\mathrm{tr}[\gamma^{\mu}\not{k}\,\gamma^{\nu}(\not{k}-\not{q})] = k_{\alpha}(k-q)_{\beta}\,\mathrm{tr}[\gamma^{\mu}\gamma^{\alpha}\gamma^{\nu}\gamma^{\beta}]$, evaluated with the four-gamma trace Eq. (C.157),

$$\mathrm{tr}\big[\gamma^{\mu}\gamma^{\alpha}\gamma^{\nu}\gamma^{\beta}\big] = 4\big(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\alpha\nu}\big). \tag{15.154}$$

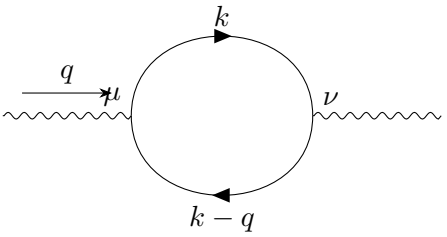


Figure 15.15: The vacuum-polarization loop Eq. (15.153): a single closed electron loop with two external photon legs of momentum q and Lorentz indices μ, ν. The arrows show the fermion-number flow circulating consistently around the loop; the spinor indices close into the single Dirac trace of Eq. (15.149), and the loop carries the overall -1 of Theorem 15.45. This is the diagram whose explicit evaluation in $d = 4 - 2\varepsilon$ dimensions yields the photon self-energy and the running of α in Chapter 19; it is identical in structure to the $n = 2$ term Eq. (12.189) of the determinant expansion of Chapter 12.

Contracting indices, the full numerator is

$$\operatorname{tr}\big[\gamma^\mu(\not{k}+m)\gamma^\nu(\not{k}-\not{q}+m)\big] = 4\Big[k^\mu(k-q)^\nu + k^\nu(k-q)^\mu - \eta^{\mu\nu}\big(k\cdot(k-q) - m^2\big)\Big]. \tag{15.155}$$

At this point the integrand is fully explicit, and what remains—combining denominators by Feynman parameters, shifting the loop momentum, regulating the resulting divergence in $d = 4 - 2\varepsilon$ dimensions, and extracting the transverse structure $\Pi^{\mu\nu} = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\Pi(q^2)$ guaranteed by the Ward identity—is the loop computation proper. It belongs to Chapter 19, which introduces the regularization machinery; we stop here, having validated the fermionic rules on the integrand. The lesson of the example is the assembly, not the integral: the -1, the trace, the vertex Eq. (15.144), and the directed propagator Eq. (15.140) are everything one adds to the bosonic rules to compute with fermions, and each was derived above from the Grassmann combinatorics of Chapter 12 rather than postulated.

The fermionic rule set is now complete and self-contained: the directed propagator Eq. (15.140), the vertex template Eq. (15.142) with its QED and Yukawa specializations, the relative exchange sign of Theorem 15.44, and the closed-loop factor (-1) with its trace from Theorem 15.45—each derived from the Grassmann calculus of Chapter 12 rather than asserted. Chapter 17 receives this set whole and puts it to work on QED, assembling the photon line and propagator around the vertex Eq. (15.144), restoring the external spinors of Section 15.7.1, and computing $e^+e^- \to \mu^+\mu^-$ with full spin sums and traces; Chapter 26 reuses the same scaffold for the gluon and the Faddeev–Popov ghost, where the ghost—a Grassmann scalar—carries its own closed-loop -1 for precisely

the reason derived here, even though it has no spin. The signs and traces of this section are the bridge across which Parts V and VII inherit, rather than rebuild, the diagrammatics of perturbation theory.

Further reading

The diagrammatic expansion in the form developed here originates with Feynman [191] for the space–time rules and Dyson [167], who showed that the Tomonaga–Schwinger–Feynman formulations are equivalent and organized the perturbation series into what we now call Feynman diagrams; the combinatorial heart, the reduction of time-ordered products to sums of contractions, is Wick [508]. Among modern textbooks, the canonical derivation of the rules and symmetry factors closest to the route of Sections 15.2 and 15.4 is in Peskin and Schroeder [396, ch. 4]; the generating-functional derivation of Sections 15.5 and 15.6, including W and the 1PI effective action, is developed with great care by Srednicki [454] and, at a more advanced level, by Weinberg [500, ch. 6]. Schwartz [432] gives a particularly clean operational account of the fermion sign rules and the closed-loop factor of Section 15.7, and Itzykson and Zuber [278] remains a thorough reference for the symmetry-factor combinatorics treated as automorphism counting. For the physical pictures behind the bookkeeping — diagrams as histories, amputation, and the meaning of one-particle irreducibility — Zee [532] is the companion of choice.

Chapter 16

Cross Sections, Decay Rates, and Amplitudes

The two preceding chapters left us holding a single, hard-won object. The interaction picture and the Dyson series of Chapter 14 resolved the scattering operator as $\mathbf{S} = \mathbf{1} + iT$, stripped the overall energy–momentum delta from every matrix element of iT, and named what remained the invariant amplitude $\mathcal{M}$; the Feynman rules of Chapter 15 then made $\mathcal{M}$ computable diagram by diagram. But $\mathcal{M}$ is a complex number, and a complex number is not what a detector reports. An experiment counts events: so many decays per second, so many particles scattered into a given solid angle. This chapter is the bridge from the amplitude to those numbers. Its task is narrow and its payoff is permanent: to build, once and for all, the kinematic wrapper that converts any $\mathcal{M}$ into a decay rate or a cross section, with every factor of $2E$, 2π, and $1/V$ accounted for and nothing quoted on faith.

The conversion looks innocent — square the amplitude, sum over final states — but it runs immediately into the square of a momentum-conserving delta function, an object with no distributional meaning. We confront that pathology head-on with a finite-volume, finite-time regularization, and in resolving it we are forced to fix the relativistic single-particle normalization inherited from Wigner's construction of the one-particle Hilbert space in Chapter 3; every later kinematic factor descends from that one choice. The analytic machinery the regularization leans on — the delta-function composition rule, the Fourier conventions, the meaning of $\delta^4(0)$ — is the distribution theory assembled in Chapter 5. Throughout we work in the mostly-minus metric and natural units of Appendix A, in which a cross section is an inverse mass squared.

From that foundation the chapter follows a deliberate arc. Section 16.1 extracts a finite master transition rate from the squared amplitude and isolates the purely kinematic remainder; Section 16.2 names that remainder the Lorentz-invariant phase space $\mathrm{d}\Pi_n$, proves its frame independence, derives the recursion

that builds it from two-body factors, and reduces the two-body case to a closed form in the centre-of-momentum frame. With the kinematics in hand the two great observables split apart according to the initial state. Section 16.3 feeds in a one-particle initial state to obtain the decay rate $d\Gamma = \frac{1}{2M}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n$, the width, the lifetime, and the identical-particle symmetry factor, carried through to a numerical lifetime. Section 16.4 feeds in a two-particle initial state, where the genuinely new ingredient is the flux: we build the Møller flux factor, prove that its covariant form equals the operational "density times relative velocity," and assemble the master cross-section formula $d\sigma = |\mathcal{M}|^2\,\mathrm{d}\Pi_n/F$.

The remaining sections cash this apparatus. Section 16.5 specializes to the ubiquitous $2 \to 2$ case, introduces the Mandelstam variables and the Källén triangle function, reduces the cross section to the two standard forms $d\sigma/d\Omega$ and $d\sigma/dt$, and maps the physical regions of the (s,t,u) plane on which crossing symmetry will later act. Section 16.6 runs the full pipeline on the carried ϕ^4 theory amplitude $\mathcal{M} = -\lambda$ of Chapter 15, producing the first genuinely physical cross section of the book — isotropic, and falling as $1/s$. Finally Section 16.7 closes the loop with Chapter 14: the flux and phase space built here, fed back into the unitarity of $\mathbf{S}$, reproduce the optical theorem in cross-section form, sharpen it into the partial-wave unitarity bound, and pass an internal consistency check on the ϕ^4 numbers. By the end the chapter's three title words — cross sections, decay rates, and amplitudes — stand revealed as one and the same object: $|\mathcal{M}|^2$ integrated against an invariant measure, divided by the appropriate normalization.

16.1 From amplitude to probability: transition rates and the squared delta function

We arrive at this chapter holding a single, hard-won object: the invariant amplitude $\mathcal{M}$. Chapter 14 split the scattering operator as $\mathbf{S} = \mathbf{1} + iT$, isolated the nontrivial part T, and showed that every matrix element of iT between momentum eigenstates carries one overall momentum-conserving delta function, leaving behind a Lorentz-invariant function of the external momenta (Eq. (14.46)); Chapter 15 then supplied the Feynman rules that let us compute that function diagram by diagram. But $\mathcal{M}$ is not yet anything a detector reports. A detector counts events: so many decays per second, so many scattered particles into a given solid angle. To get there we must turn the complex number $\mathcal{M}$ into a *probability*, and then into a *rate*. That conversion looks innocent — square the amplitude — but it runs immediately into a notorious obstruction. Squaring an expression that contains $(2\pi)^4\delta^4(p_f - p_i)$ appears to produce $\left[(2\pi)^4\delta^4(p_f - p_i)\right]^2$, the square of a distribution, which is meaningless as it stands. The whole machinery of cross sections and lifetimes rests on understanding precisely what that squared delta means and how to extract a finite number from it. This section confronts

the pathology head-on, resolves it with a finite-volume, finite-time regularization, and in doing so fixes once and for all the relativistic state normalization that every later factor — flux, phase space, the $2M$ of a decay rate — silently depends on.

16.1.1 The transition probability and the squared delta

We begin from the decomposition established in Theorem 14.11 and Eq. (14.45),

$$\mathbf{S} = \mathbf{1} + iT, \tag{16.1}$$

and the definition of the invariant amplitude for a process $i \to f$ with initial momenta $\{p_a\}$ and final momenta $\{k_i\}$ (Eq. (14.46)),

$$\langle f|iT|i\rangle = (2\pi)^4\,\delta^4\Big(\textstyle\sum_i k_i - \sum_a p_a\Big)\; i\,\mathcal{M}(i \to f). \tag{16.2}$$

For a genuine transition the final state f differs from the initial state i, so the disconnected piece $\mathbf{1}$ does not contribute and $\langle f|\mathbf{S}|i\rangle = \langle f|iT|i\rangle$. The probability for the transition is the modulus squared of this amplitude, divided by the norms of the states involved — a normalization we will make precise in a moment. Ignoring the normalization for the instant, the bare squared matrix element is

$$\big|\langle f|iT|i\rangle\big|^2 = \big|\mathcal{M}(i \to f)\big|^2\,\Big[(2\pi)^4\,\delta^4(p_f - p_i)\Big]^2, \tag{16.3}$$

where we abbreviate $p_f \equiv \sum_i k_i$ and $p_i \equiv \sum_a p_a$. The factor $\big|\mathcal{M}\big|^2$ is perfectly finite; the dynamics lives entirely there. The trouble is the bracketed object. A delta function is a distribution, defined only under an integral against a test function; the pointwise product of two distributions supported on the same set is not defined in the theory of distributions reviewed in Section 5.1.3. Written naively as $\delta^4(p_f - p_i)\cdot\delta^4(0)$, it even contains the manifestly divergent symbol $\delta^4(0)$. We cannot proceed by formal manipulation alone; we must regulate.

Caution

The symbol $\big[(2\pi)^4\delta^4\big]^2$ is *not* a distribution and must never be manipulated as though the usual delta-function identities apply to it. It is shorthand for a quantity that is only defined once the idealization of exactly sharp momentum eigenstates over infinite spacetime is relaxed. Every finite answer below comes from taking that idealization carefully, not from algebra on $\delta^4(0)$.

16.1.2 Box and finite-time regularization

The physical origin of the squared delta is that we have idealized the experiment as taking place over all of space for all of time. A momentum eigenstate is a

plane wave filling the entire universe; the conservation delta is the statement that this plane wave overlaps the final state coherently at every point of an infinite spacetime. The infinity is the infinite spacetime volume over which that coherent overlap accumulates. Restore physical finiteness and the infinity becomes a finite number.

Concretely, place the system in a large cubic box of side L and spatial volume $V = L^3$, with periodic boundary conditions, and let the interaction act only over a long but finite time interval of duration T. Both V and T are auxiliary; physical answers must be independent of them, and indeed the divergent factor they tame will be cancelled by the very normalization they force. The momentum-conserving delta arises, as recorded in the Fourier conventions of Eq. (5.53), from a spacetime integral of a plane wave,

$$(2\pi)^4\,\delta^4(p_f - p_i) = \int d^4x\; e^{i(p_f - p_i)\cdot x}. \tag{16.4}$$

The two delta functions in Eq. (16.3) come from two such integrals, one from the amplitude and one from its conjugate. Their product is therefore a double spacetime integral,

$$\Big[(2\pi)^4\delta^4(p_f - p_i)\Big]^2 = \int d^4x\, e^{i(p_f - p_i)\cdot x} \int d^4y\, e^{-i(p_f - p_i)\cdot y}. \tag{16.5}$$

On the support of the conservation law the argument $p_f - p_i$ vanishes, so one of the two exponentials is unity there. The cleanest way to see the result is to keep one factor as a genuine delta and recognize the other as the volume of the integration region. Use Eq. (16.4) for the x-integral and set $p_f = p_i$ in the y-integrand, since the surviving delta enforces exactly that:

$$\Big[(2\pi)^4\delta^4(p_f - p_i)\Big]^2 = (2\pi)^4\delta^4(p_f - p_i) \int_{V,T} d^4y\; e^{-i(p_f - p_i)\cdot y}\Big|_{p_f = p_i} = (2\pi)^4\delta^4(p_f - p_i) \int_{V,T} d^4y. \tag{16.6}$$

The remaining integral is over the regularized spacetime region — the box of volume V over the time T — and is simply its four-volume,

$$\int_{V,T} d^4y = \int_0^T dt \int_V d^3y = VT. \tag{16.7}$$

This is the precise content of the heuristic identity $(2\pi)^4\delta^4(0) = VT$: the divergent $\delta^4(0)$ is nothing but the four-volume of the spacetime over which the process is observed. We have therefore established

$$\boxed{\Big[(2\pi)^4\delta^4(p_f - p_i)\Big]^2 = VT\,(2\pi)^4\delta^4(p_f - p_i)} \tag{16.8}$$

The squared delta has collapsed to a single delta — which still enforces overall energy–momentum conservation, exactly as it must — multiplied by the finite

factor VT. The probability of any given transition therefore grows linearly with the observation time T: that linear growth is precisely what makes a constant *rate* (probability per unit time) the sensible observable, and dividing by T removes one factor. The remaining factor of V will be cancelled by the state normalization, to which we turn next.

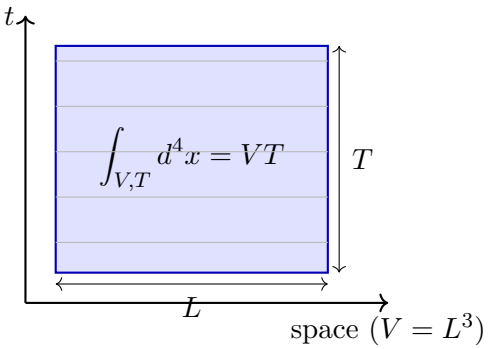


Figure 16.1: The squared momentum-conserving delta function regularized. Confining the process to a spatial box of volume $V = L^3$ over a finite time T replaces the divergent $(2\pi)^4\delta^4(0)$ by the finite four-volume $VT = \int_{V,T} d^4x$ of the observation region (Eq. (16.7)). The surviving single $\delta^4(p_f - p_i)$ still enforces overall energy–momentum conservation; the factor VT is what makes the transition probability grow with observation time, so that the rate (probability per unit T) is the finite, T-independent observable.

Worked Example 16.1 (The squared delta is the spacetime four-volume)**.** It is worth seeing Eq. (16.8) a second way, directly from the regularized delta itself, to confirm that no sleight of hand entered the step Eq. (16.6). In the box the spatial momenta are quantized, $\mathbf{p} = \frac{2\pi}{L}\mathbf{n}$ with $\mathbf{n} \in \mathbb{Z}^3$, and the spatial Fourier delta is genuinely the finite integral

$$(2\pi)^3\delta_V^3(\mathbf{q}) \equiv \int_V d^3x\, e^{i\mathbf{q}\cdot\mathbf{x}}, \qquad (2\pi)^3\delta_V^3(\mathbf{0}) = \int_V d^3x = V, \tag{16.9}$$

and likewise for the finite-time energy delta,

$$2\pi\,\delta_T(q^0) \equiv \int_0^T dt\, e^{iq^0t}, \qquad 2\pi\,\delta_T(0) = \int_0^T dt = T. \tag{16.10}$$

The four-dimensional regularized delta is the product, $(2\pi)^4\delta_{V,T}^4(q) = (2\pi)^3\delta_V^3(\mathbf{q})\cdot 2\pi\,\delta_T(q^0)$. Writing the square as

$$\left[(2\pi)^4\delta_{V,T}^4(q)\right]^2 = (2\pi)^4\delta_{V,T}^4(q)\cdot(2\pi)^4\delta_{V,T}^4(0), \tag{16.11}$$

the second factor is, by Eqs. (16.9) and (16.10), exactly

$$(2\pi)^4\delta_{V,T}^4(0) = (2\pi)^3\delta_V^3(\mathbf{0})\cdot 2\pi\,\delta_T(0) = V\cdot T = VT, \tag{16.12}$$

reproducing Eq. (16.8). As $V, T \to \infty$ the regularized $\delta_{V,T}$ tends to the ordinary delta in the distributional sense (its integral against any test function converges to the test function's value at the origin), so the surviving factor $(2\pi)^4 \delta^4_{V,T}(q) \to (2\pi)^4 \delta^4(q)$ recovers exact conservation, while the cofactor VT remains as the genuine, finite measure of the observation volume.

16.1.3 Relativistic normalization and the box dictionary

We have a finite squared amplitude up to the question of how the states are normalized. This is not a cosmetic choice: the factors of $2E$ and $1/V$ that will populate the master formula are *forced* by it, and getting the normalization wrong is the single most common way to land a cross section off by powers of energy. We use throughout the relativistically invariant single-particle normalization established in Eq. (3.193) as part of Wigner's construction of the one-particle Hilbert space (Theorem 3.44):

$$\langle p|q\rangle = 2E_{\mathbf{p}}\,(2\pi)^3\,\delta^3(\mathbf{p}-\mathbf{q}), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2+m^2}. \tag{16.13}$$

The merit of this convention is its manifest Lorentz invariance: the combination $2E_{\mathbf{p}}\,(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$ and the associated measure $d^3p/[(2\pi)^3 2E_{\mathbf{p}}]$ are invariant (Eq. (3.191)), which is exactly why the phase-space measure built from it in the next section will be invariant too. The price is that single-particle states are not unit-normalized: $\langle p|p\rangle$ is not 1 but is proportional to the (divergent, in the continuum) $\delta^3(\mathbf{0})$. The box makes this finite and turns Eq. (16.13) into a usable counting dictionary.

In the box, by exactly the spatial version of Eq. (16.4) recorded in Eq. (16.9),

$$(2\pi)^3\delta^3(\mathbf{0}) \;\longrightarrow\; \int_V d^3x = V, \tag{16.14}$$

so the norm of a single relativistically normalized momentum state becomes finite,

$$\langle p|p\rangle = 2E_{\mathbf{p}}\,(2\pi)^3\delta^3(\mathbf{0}) \;\longrightarrow\; 2E_{\mathbf{p}}\,V. \tag{16.15}$$

This is the central entry of the dictionary between continuum and box states. The continuum state $|p\rangle$, with the invariant normalization Eq. (16.13), corresponds in the box to a state of norm $2E_{\mathbf{p}}V$; the unit-normalized (probability-one) box state is therefore

$$|p\rangle_{\text{box}} = \frac{1}{\sqrt{2E_{\mathbf{p}}V}}\;|p\rangle\,, \qquad \langle p|p\rangle_{\text{box}} = 1. \tag{16.16}$$

The second dictionary entry concerns the counting of final states. Periodic boundary conditions quantize momenta on the lattice $\mathbf{p} = \frac{2\pi}{L}\mathbf{n}$, with one allowed

mode per momentum-space cell of volume $(2\pi/L)^3 = (2\pi)^3/V$. The number of distinct single-particle states with momentum in a cell d^3k is therefore

$$dN_{\text{states}} = \frac{V\, d^3k}{(2\pi)^3}, \tag{16.17}$$

which is the box realization of the continuum rule that each $\int d^3k$ comes weighted by $V/(2\pi)^3$ when counting modes.

These two entries already contain the physical meaning of the relativistic normalization, and it is worth stating it as a result in its own right because the $2E$ factors of every cross section descend from it.

Proposition 16.2 (Particle density of a relativistically normalized state)**.** *With the normalization Eq. (16.13), a single-particle momentum eigenstate $|p\rangle$ confined to a box of volume V corresponds to a number density of $2E_{\mathbf{p}}$ particles per unit volume.*

Proof. The unit-normalized state Eq. (16.16) describes exactly one particle delocalized uniformly over the box, hence a particle number density $1/V$. The relativistically normalized state is larger by the factor $\sqrt{2E_{\mathbf{p}}V}$ in amplitude, i.e. by $2E_{\mathbf{p}}V$ in norm: $\langle p|p\rangle = 2E_{\mathbf{p}}V$ from Eq. (16.15). Since the norm is the integral of the probability (number) density over the box, the relativistic state carries $2E_{\mathbf{p}}V$ units of that density in volume V, i.e. a density of

$$n_{\mathbf{p}} = \frac{\langle p|p\rangle}{V} = 2E_{\mathbf{p}} \tag{16.18}$$

particles per unit volume. The same conclusion follows operationally from the free-field number current: the conserved charge density of a complex scalar in the state $|p\rangle$, or equivalently the diagonal matrix element of the number-density operator, evaluates with the commutator $\left[a_{\mathbf{p}}, a^\dagger_{\mathbf{q}}\right] = 2E_{\mathbf{p}}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$ that accompanies Eq. (16.13) to $2E_{\mathbf{p}}$ per unit volume, the factor $2E_{\mathbf{p}}$ being the same one carried by the norm. The factor is thus not an arbitrary convention but the direct consequence of the invariant normalization: it is the price, and the benefit, of working with Lorentz-invariant states. □

Physical Insight

The relativistic normalization Eq. (16.13) is chosen so that states transform simply under Lorentz boosts — the inner product is frame independent. Theorem 16.2 is the flip side of that choice: a boosted box is Lorentz contracted, so the same particle appears denser by exactly the time-dilation/energy factor $E_{\mathbf{p}}/m$ relative to its rest frame, and the convention builds this in as the universal factor $2E$. Every $2E$ you will meet in a phase-space measure, a flux factor, or the $2M$ of a decay rate traces

back to this single normalization decision. There is nothing left arbitrary downstream: fix Eq. (16.13) and all the kinematic factors are determined.

16.1.4 The master differential transition rate

We now assemble the pieces into the probability for the process $i \to f$ and divide by the observation time to obtain a rate. The transition probability is the squared amplitude between *unit-normalized* states, summed over the number of distinct final states available. We have computed everything in terms of relativistically normalized states, so each external leg must be converted using the dictionary above: divide by the norm $2EV$ of each incoming and each outgoing relativistic state, and multiply by the count Eq. (16.17) of final states in the momentum cells d^3k_i. Writing the incoming particles with energies E_a (a running over the n_i initial particles) and the outgoing ones with energies E_i (i running over the n_f final particles), the probability of landing in the final momentum element $\prod_i d^3k_i$ is

$$dP = \frac{\left|\langle f|iT|i\rangle\right|^2}{\left(\prod_{a\in\text{in}} 2E_aV\right)\left(\prod_{i\in\text{fin}} 2E_iV\right)} \prod_{i\in\text{fin}} \frac{V\,d^3k_i}{(2\pi)^3}. \tag{16.19}$$

The numerator is the bare squared matrix element Eq. (16.3); the first denominator factor divides out the norms of the incoming states, the second divides out the norms of the outgoing states, and the final product counts the available final states by Eq. (16.17). Insert the bare square Eq. (16.3) and apply the central result Eq. (16.8) for the squared delta:

$$dP = \frac{\left|\mathcal{M}\right|^2 VT\,(2\pi)^4\delta^4(p_f - p_i)}{\left(\prod_{a\in\text{in}} 2E_aV\right)\left(\prod_{i\in\text{fin}} 2E_iV\right)} \prod_{i\in\text{fin}} \frac{V\,d^3k_i}{(2\pi)^3}. \tag{16.20}$$

Now watch the auxiliary factors V and T disappear, which is the whole point of the regularization. Pair each final-state energy denominator $2E_iV$ with its own state-count numerator $V\,d^3k_i$: the factors of V cancel one-to-one, leaving for each final particle exactly the combination

$$\frac{V\,d^3k_i}{(2\pi)^3}\cdot\frac{1}{2E_iV} = \frac{d^3k_i}{(2\pi)^3\,2E_i}. \tag{16.21}$$

This is the invariant measure of Eq. (3.191), one factor per outgoing particle — the first sign that the surviving kinematic object will be Lorentz invariant. After this cancellation Eq. (16.20) reads

$$dP = VT\,\frac{\left|\mathcal{M}\right|^2}{\prod_{a\in\text{in}} 2E_aV}\,(2\pi)^4\delta^4(p_f - p_i) \prod_{i\in\text{fin}} \frac{d^3k_i}{(2\pi)^3\,2E_i}. \tag{16.22}$$

The transition *rate* is the probability per unit observation time, $dN \equiv dP/T$. Dividing by T removes the time factor entirely — as anticipated, the linear growth of dP with T in Eq. (16.8) is exactly what makes a time-independent rate well defined — and we obtain the master differential transition rate:

$$dN = \frac{1}{T}\frac{|\mathcal{M}|^2}{\prod_{a\in\text{in}} 2E_a V}\left[(2\pi)^4\delta^4(p_f - p_i)\right]^2 \prod_{i\in\text{fin}} \frac{V\, d^3k_i}{(2\pi)^3\, 2E_i V} = \frac{V}{\prod_{a\in\text{in}} 2E_a V}\,|\mathcal{M}|^2\,(2\pi)^4\delta^4(p \tag{16.23}$$

The first form displays the raw ingredients exactly as they enter — the $1/T$, the inverse norms $1/(2E_aV)$ of the initial particles, the squared delta, and the final-state weights — and the second is the result of using Eq. (16.8) and the cancellation Eq. (16.21). In the second form the only surviving auxiliary factor is the single V in the numerator, which combines with the initial-state norms to give the clean scaling

$$dN = \frac{1}{V^{\,n_i-1}}\frac{|\mathcal{M}|^2}{\prod_{a\in\text{in}} 2E_a}\,(2\pi)^4\delta^4(p_f - p_i)\prod_{i\in\text{fin}}\frac{d^3k_i}{(2\pi)^3\,2E_i}. \tag{16.24}$$

This dependence on the number of incoming particles n_i is not a defect but the seed of the physics that separates the two great observables of this chapter.

For a *decay*, $n_i = 1$: a single particle of energy $E_1 = M$ in its rest frame disintegrates, and Eq. (16.24) is completely independent of the box volume,

$$dN\big|_{n_i=1} = \frac{1}{2E_1}\,|\mathcal{M}|^2\,(2\pi)^4\delta^4(p_f - p_i)\prod_{i\in\text{fin}}\frac{d^3k_i}{(2\pi)^3\,2E_i}. \tag{16.25}$$

The prefactor $1/2E_1$ is exactly the $1/2M$ that will appear in the partial-width formula; it is the inverse norm of the single decaying particle, nothing more. There is no V because there is no notion of density for a single particle — the rate is intrinsic.

For a *scattering* process, $n_i = 2$: two beams must find each other, and Eq. (16.24) retains one factor of $1/V$,

$$dN\big|_{n_i=2} = \frac{1}{V}\,\frac{1}{2E_1\,2E_2}\,|\mathcal{M}|^2\,(2\pi)^4\delta^4(p_f - p_i)\prod_{i\in\text{fin}}\frac{d^3k_i}{(2\pi)^3\,2E_i}. \tag{16.26}$$

The residual $1/V$ is precisely a number density: by Theorem 16.2 the two incoming particles have densities $2E_1/V$ and $2E_2/V$ relative to a normalization, and converting this single-event rate into the box-independent cross section will require dividing by the incident flux, which carries exactly the compensating factors of V and energy. That conversion is the business of the cross-section section; here we have established that the $1/V$ is forced, not chosen, and that it is the flag marking a two-body initial state.

Caution

The factor $1/V^{\,n_i-1}$ is the only place the auxiliary volume survives, and it must cancel in any physical answer. For decays ($n_i = 1$) it is already absent. For scattering ($n_i = 2$) it is removed by the flux, which is itself $\propto 1/V$; the combination dN/flux is finite and V-independent. If a calculation ends with a leftover V in a cross section or a width, a normalization factor of $2EV$ or a final-state count $V\,d^3k/(2\pi)^3$ has been dropped — retrace Eq. (16.19).

Worked Example 16.3 (Two-body final state: counting the volume factors)**.** To see that the bookkeeping is airtight, count powers of V explicitly for the generic $2 \to 2$ process $a + b \to 1 + 2$, the kinematic skeleton of the carried ϕ^4 theory amplitude $\phi\phi \to \phi\phi$ of Eq. (14.46) and of every elementary process in Chapters 17–18. Start from Eq. (16.20) with $n_i = 2$, $n_f = 2$. The explicit powers of V are: one V from VT in the squared delta; two factors V from the final-state counts $\prod_{i=1,2} V\,d^3k_i/(2\pi)^3$; in the denominator, two factors V from the initial norms $2E_aV, 2E_bV$ and two from the final norms $2E_1V, 2E_2V$. The net power is

$$V^{\,1} \cdot V^{\,2} \cdot V^{-2} \cdot V^{-2} = V^{-1}, \tag{16.27}$$

so $dP \propto T\,V^{-1}$ and $dN = dP/T \propto V^{-1}$, matching Eq. (16.26). The single surviving inverse volume is the density factor that the flux will absorb. Repeating the count with n_f arbitrary changes nothing: every final particle brings a $+1$ from its state count and a -1 from its norm, which always cancel via Eq. (16.21), so the final-state multiplicity never affects the power of V — only n_i does, through $V^{\,1}/V^{\,n_i} = V^{\,1-n_i}$. This confirms Eq. (16.24) for any number of final particles.

We have converted the abstract S-matrix element into a finite, physically interpretable transition rate, and along the way fixed the relativistic normalization Eq. (16.13) together with the $2E$ and $1/V$ factors it forces. The dynamics is sealed entirely inside $|\mathcal{M}|^2$; everything multiplying it in Eq. (16.24) is pure kinematics. In particular, the final-state momenta enter only through the combination

$$(2\pi)^4\delta^4(p_f - p_i) \prod_{i\in\text{fin}} \frac{d^3k_i}{(2\pi)^3\,2E_i} \;\equiv\; (2\pi)^4\delta^4(p_f - p_i)\,\mathrm{d}\Pi_{n_f}, \tag{16.28}$$

which already shows, leg by leg, the invariant measure of Eq. (3.191). The next section isolates this object, names it the Lorentz-invariant phase space $\mathrm{d}\Pi_{n_f}$, proves its frame independence, derives the recursion that builds the n-body measure from lower ones, and reduces the two-body case in the center-of-momentum frame to the closed form on which every cross section and decay rate in the remainder of the book will rest.

16.2 Lorentz-invariant phase space

The differential transition rate assembled in the previous section has a clean factorized structure. All of the dynamics — everything the interaction does — sits in a single number $|\mathcal{M}|^2$, the modulus squared of the invariant amplitude defined in Eq. (14.163). Everything else is kinematics: the final-state momenta range over all configurations compatible with energy–momentum conservation, each weighted by the relativistically invariant single-particle measure that the state normalization of Section 16.1 forced upon us. That kinematic weight is

$$\prod_{i=1}^{n} \frac{d^3k_i}{(2\pi)^3\, 2E_{k_i}}\,, \qquad E_{k_i} = \sqrt{\mathbf{k}_i^2 + m_i^2}\,, \tag{16.29}$$

tied together by the overall conservation delta $(2\pi)^4\delta^4\!\left(P - \sum_i k_i\right)$, with P the total incoming four-momentum. This section isolates that combination, names it the *Lorentz-invariant n-body phase space* $\mathrm{d}\Pi_n$, proves that it is genuinely a Lorentz scalar (so that a rate or cross section computed in one frame is the same number in any other), establishes a recursion that builds $\mathrm{d}\Pi_n$ from $\mathrm{d}\Pi_{n-1}$, and reduces the two-body case to a closed form. The two-body result is the single most-used formula in the rest of Part IV and all of Part V: every $1 \to 2$ decay and every $2 \to 2$ cross section will be read off from it. The construction here is entirely thread-neutral — it knows nothing about ϕ^4, QED, or Yang–Mills — precisely because it is pure geometry of the on-shell momenta.

16.2.1 The on-shell single-particle measure

The atom from which $\mathrm{d}\Pi_n$ is built is the measure carried by one outgoing particle. We met it already in Chapter 3 as the natural measure on the mass shell; here we collect its two equivalent presentations and prove the invariance on which all later frame-independence rests.

Definition 16.4 (On-shell measure)**.** For a particle of mass m the *on-shell* (or *Lorentz-invariant*) single-particle measure is

$$\widetilde{dk} \;\equiv\; \frac{d^3k}{(2\pi)^3\, 2E_k}\,, \qquad E_k = \sqrt{\mathbf{k}^2 + m^2}\,, \tag{16.30}$$

where $\widetilde{dk}$ is shorthand for $\widetilde{dk}$.

The factor $2E_k$ is not decoration: it is exactly the weight that the relativistic normalization $\langle p'|p\rangle = (2\pi)^3\, 2E_p\, \delta^3(\mathbf{p}-\mathbf{p}')$ of Eq. (3.193) demands of a resolution of the identity on one-particle states. With it, $\widetilde{dk}$ is a Lorentz scalar; without it, d^3k alone is not. The cleanest way to see the invariance is to rewrite Eq. (16.30) in manifestly covariant four-dimensional form.

Proposition 16.5 (Covariant form of the on-shell measure)**.** *The measure Eq. (16.30) equals*

$$\widetilde{dk} \;=\; \frac{d^4k}{(2\pi)^4}\,(2\pi)\,\delta\!\left(k^2-m^2\right)\theta(k^0)\,, \tag{16.31}$$

and is invariant under every orthochronous Lorentz transformation.

Proof. Start from the right-hand side of Eq. (16.31) and perform the k^0 integration. Write $d^4k = dk^0\,d^3k$ and treat the delta as a function of k^0 at fixed $\mathbf{k}$. The argument $k^2-m^2 = (k^0)^2 - \mathbf{k}^2 - m^2 = (k^0)^2 - E_k^2$ has the two simple zeros $k^0 = \pm E_k$. By the composition rule for a delta function with a smooth argument g, proved in Chapter 5,

$$\delta\big(g(k^0)\big) = \sum_i \frac{\delta(k^0-k_i^0)}{|g'(k_i^0)|}\,, \qquad g(k^0) = (k^0)^2 - E_k^2\,, \quad g'(k^0) = 2k^0\,, \tag{16.32}$$

so that, since $|g'(\pm E_k)| = 2E_k$,

$$\delta\!\left(k^2-m^2\right) = \frac{1}{2E_k}\Big[\delta(k^0-E_k) + \delta(k^0+E_k)\Big]. \tag{16.33}$$

The Heaviside factor $\theta(k^0)$ retains only the positive-energy root $k^0 = E_k$, killing the second term. Integrating over k^0,

$$\frac{d^4k}{(2\pi)^4}(2\pi)\,\delta\!\left(k^2-m^2\right)\theta(k^0) = \frac{d^3k}{(2\pi)^4}(2\pi)\,\frac{1}{2E_k} = \frac{d^3k}{(2\pi)^3\,2E_k} = \widetilde{dk}\,, \tag{16.34}$$

which is Eq. (16.30). This is precisely the integral identity Eq. (3.192) read as a statement about measures rather than about integrands.

Invariance now follows term by term in Eq. (16.31). The four-volume d^4k is invariant because $|\det\Lambda| = 1$ for any $\Lambda \in O(1,3)$. The combination $k^2 = \eta_{\mu\nu}k^\mu k^\nu$ is a Lorentz scalar, hence so is $\delta(k^2-m^2)$. Finally $\theta(k^0)$ is invariant under the *orthochronous* subgroup: a Λ with ${\Lambda^0}_0 \geq 1$ cannot map a future-pointing timelike-or-null vector (here k is on the forward mass shell, $k^2 = m^2 \geq 0$, $k^0 > 0$) to a past-pointing one, so $\theta(k^0) = \theta((\Lambda k)^0)$ on the support of $\delta(k^2-m^2)$. The product of three invariants is invariant. Since the physical Lorentz transformations relating inertial frames are orthochronous (and, including parity, still orthochronous), $\widetilde{dk}$ is the same measure in every frame. □

It is worth seeing the invariance a second way, directly in three-dimensional language, because the boost Jacobian it exposes is the mechanism behind every "$d^3k/2E$ is invariant" statement in the book.

Worked Example 16.6 (The boost Jacobian)**.** Take a boost of rapidity η along the z-axis acting on an on-shell momentum $k = (E_k, \mathbf{k})$ with $\mathbf{k} = (k^1, k^2, k^3)$. The transverse components are unchanged, $k'^1 = k^1$, $k'^2 = k^2$, while

$$k'^3 = \gamma\,(k^3 + \beta E_k)\,, \qquad E'_k = \gamma\,(E_k + \beta k^3)\,, \qquad \gamma = \cosh\eta, \quad \beta = \tanh\eta\,. \tag{16.35}$$

The map $\mathbf{k} \mapsto \mathbf{k}'$ has a Jacobian that is diagonal in the transverse directions, so

$$d^3k' = \left|\frac{\partial k'^3}{\partial k^3}\right| d^3k\,, \tag{16.36}$$

the partial derivative taken at fixed k^1, k^2. Because k stays on shell, $E_k = \sqrt{(k^3)^2 + (k^1)^2 + (k^2)^2 + m^2}$ depends on k^3 through $\partial E_k/\partial k^3 = k^3/E_k$. Hence

$$\frac{\partial k'^3}{\partial k^3} = \gamma\Big(1 + \beta\,\frac{\partial E_k}{\partial k^3}\Big) = \gamma\Big(1 + \beta\,\frac{k^3}{E_k}\Big) = \frac{\gamma\,(E_k + \beta k^3)}{E_k} = \frac{E'_k}{E_k}\,. \tag{16.37}$$

The Jacobian is positive (an orthochronous boost keeps $E'_k > 0$), so the modulus is redundant, and

$$\frac{d^3k'}{2E'_k} = \frac{1}{2E'_k}\,\frac{E'_k}{E_k}\,d^3k = \frac{d^3k}{2E_k}\,. \tag{16.38}$$

The energy denominator exactly cancels the boost stretching of the momentum-space volume. Rotations leave d^3k and E_k separately invariant, and any orthochronous Lorentz transformation is a product of boosts and rotations, so $d^3k/2E_k$ is invariant in general — the same conclusion as Theorem 16.5, now with the Jacobian made explicit.

16.2.2 The n-body phase space and its invariance

With the one-particle atom in hand, the full final-state measure is its n-fold product, dressed with the conservation delta.

Definition 16.7 (Lorentz-invariant phase space)**.** Let a state of total four-momentum P decay or scatter into n particles of masses $m_1, \dots, m_n$ and momenta $k_1, \dots, k_n$. The *Lorentz-invariant n-body phase space* is the measure

$$\mathrm{d}\Pi_n(P; k_1, \dots, k_n) \;\equiv\; (2\pi)^4\,\delta^4\Big(P - \sum_{i=1}^{n} k_i\Big)\,\prod_{i=1}^{n} \widetilde{dk_i}\,, \tag{16.39}$$

and its total volume for a given P is $\Pi_n(P) = \int \mathrm{d}\Pi_n$.

The notation $\mathrm{d}\Pi_n$ records the measure; an actual rate or cross section is $\int |\mathcal{M}|^2\,\mathrm{d}\Pi_n$ divided by a normalization fixed in the next two sections. The placement of the 2π's follows the global Fourier convention fixed in Chapter 5: each $\int d^4k$ would carry $(2\pi)^{-4}$ and each x-integral delta a $(2\pi)^4$; in $\mathrm{d}\Pi_n$ those

have already been packaged so that $\widetilde{dk}$ carries $(2\pi)^{-3}$ and the single overall delta carries $(2\pi)^4$. This bookkeeping is what makes the squared-delta regularization of Section 16.1 come out finite with no stray factors.

Proposition 16.8 (Lorentz invariance of $\mathrm{d}\Pi_n$)**.** *The measure Eq. (16.39) is invariant under orthochronous Lorentz transformations: under $k_i \mapsto \Lambda k_i$ and $P \mapsto \Lambda P$ it is unchanged, and its total volume $\Pi_n(P)$ depends on P only through the invariant $s = P^2$ (and the masses).*

Proof. Each factor $\widetilde{dk_i}$ is invariant by Theorem 16.5. The remaining factor is $(2\pi)^4\delta^4(P-\sum_i k_i)$. A four-dimensional delta transforms as $\delta^4(\Lambda q) = |\det\Lambda|^{-1}\,\delta^4(q) = \delta^4(q)$ since $|\det\Lambda| = 1$; equivalently, the conservation constraint $P = \sum_i k_i$ is a covariant equation between four-vectors, true in one frame iff true in all. Hence $\delta^4(\Lambda P - \sum_i \Lambda k_i) = \delta^4(\Lambda(P - \sum_i k_i)) = \delta^4(P - \sum_i k_i)$. A product of Lorentz scalars is a Lorentz scalar, so $\mathrm{d}\Pi_n$ is invariant.

For the total volume, integrate Eq. (16.39) over all final momenta: $\Pi_n(P) = \int \mathrm{d}\Pi_n$ is a Lorentz scalar built from the single four-vector P and the fixed masses, so it can depend on P only through the invariant $P^2 = s$. We will exhibit this dependence explicitly for $n = 2$ below. □

This invariance is the licence to compute in whatever frame is most convenient — the rest frame for a decay, the center-of-momentum frame for a collision — and trust that the number is physical. We now cash it for two bodies.

16.2.3 Two-body phase space in the center-of-momentum frame

The two-body case is special: conservation of four-momentum leaves only the overall orientation of the final state free, so the phase space reduces to an integral over a single solid angle with a kinematic prefactor that we can write in closed form. This is the workhorse of every $1 \to 2$ and $2 \to 2$ computation to come.

Choose the center-of-momentum (CM) frame, defined by $\mathbf{P} = \mathbf{0}$, so that $P = (\sqrt{s}, \mathbf{0})$ with $s = P^2$ the squared total energy. Starting from the definition,

$$\mathrm{d}\Pi_2 = (2\pi)^4\,\delta^4(P - k_1 - k_2)\,\frac{d^3k_1}{(2\pi)^3 2E_1}\,\frac{d^3k_2}{(2\pi)^3 2E_2}\,. \tag{16.40}$$

Use the three spatial components of the delta to do the $\mathbf{k}_2$ integral. It sets $\mathbf{k}_2 = -\mathbf{k}_1$; write $\mathbf{k} \equiv \mathbf{k}_1$, $k \equiv |\mathbf{k}|$, so the two particles emerge back-to-back with equal and opposite three-momenta (Fig. 16.2). The energies become

$$E_1 = \sqrt{k^2 + m_1^2}\,, \qquad E_2 = \sqrt{k^2 + m_2^2}\,, \tag{16.41}$$

and the surviving time-component delta enforces $E_1 + E_2 = \sqrt{s}$. One factor of $(2\pi)^3$ from the $\mathbf{k}_2$ integration cancels one in the measure, leaving

$$\mathrm{d}\Pi_2 = \frac{1}{(2\pi)^2}\,\frac{d^3k}{4E_1E_2}\,\delta(\sqrt{s} - E_1 - E_2)\,. \tag{16.42}$$

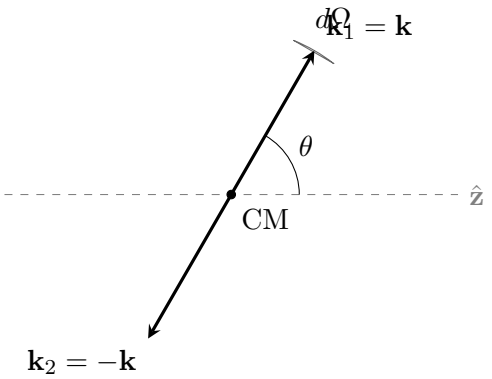


Figure 16.2: Two-body final state in the center-of-momentum frame. Momentum conservation forces the products back-to-back, $\mathbf{k}_2 = -\mathbf{k}_1 \equiv -\mathbf{k}$, leaving only the orientation (θ, ϕ) free; the phase-space integral runs over the solid angle $d\Omega$. The magnitude $|\mathbf{k}|$ is fixed by $E_1 + E_2 = \sqrt{s}$ to the value Eq. (16.51).

It remains to do the radial integral against the energy delta. Write $d^3k = k^2\, dk\, d\Omega$ with $d\Omega = \sin\theta\, d\theta\, d\phi$ the solid angle of $\mathbf{k}$, and define

$$g(k) \equiv \sqrt{s} - E_1(k) - E_2(k)\,. \tag{16.43}$$

The delta in Eq. (16.42) fixes k to the root k_* of g, i.e. to the value where $E_1 + E_2 = \sqrt{s}$. To convert $\delta(g(k))$ into $\delta(k - k_*)$ we again use the composition rule, now in the radial variable: $\delta(g(k)) = \delta(k - k_*)/\,|g'(k_*)|$. The derivative is

$$g'(k) = -\frac{dE_1}{dk} - \frac{dE_2}{dk} = -\frac{k}{E_1} - \frac{k}{E_2} = -k\,\frac{E_1 + E_2}{E_1 E_2}\,, \tag{16.44}$$

using $dE_i/dk = k/E_i$. Evaluated at the root, where $E_1 + E_2 = \sqrt{s}$,

$$|g'(k_*)| = \frac{k_*\,\sqrt{s}}{E_1 E_2}\,, \qquad \text{so} \qquad \delta\big(g(k)\big) = \frac{E_1 E_2}{k_*\sqrt{s}}\,\delta(k - k_*)\,. \tag{16.45}$$

Inserting Eq. (16.45) into Eq. (16.42) and doing the k integral,

$$\mathrm{d}\Pi_2 = \frac{1}{(2\pi)^2}\,\frac{k_*^2}{4E_1E_2}\,\frac{E_1E_2}{k_*\sqrt{s}}\,d\Omega = \frac{1}{(2\pi)^2}\,\frac{k_*}{4\sqrt{s}}\,d\Omega\,. \tag{16.46}$$

The energy factors E_1E_2 cancel completely — a small miracle that is the whole reason the two-body answer is so simple — and with $(2\pi)^2 \cdot 4 = 16\pi^2$ we arrive at the master two-body result:

$$\boxed{\mathrm{d}\Pi_2 = \frac{1}{16\pi^2}\,\frac{|\mathbf{k}|}{\sqrt{s}}\,d\Omega} \qquad \text{(CM frame)}, \tag{16.47}$$

where $|\mathbf{k}| = k_*$ is the common CM momentum and $d\Omega$ is the solid angle of particle 1.

It remains to give $|\mathbf{k}|$ in closed form. From $E_1 + E_2 = \sqrt{s}$ and the on-shell relations $E_i^2 = k^2 + m_i^2$ we get $E_1^2 - E_2^2 = m_1^2 - m_2^2$, and dividing by $E_1 + E_2 = \sqrt{s}$,

$$E_1 - E_2 = \frac{m_1^2 - m_2^2}{\sqrt{s}} \quad \Longrightarrow \quad E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}} \,. \tag{16.48}$$

Then $k^2 = E_1^2 - m_1^2$, and a short reduction of the numerator,

$$\begin{aligned} k^2 &= \frac{(s + m_1^2 - m_2^2)^2}{4s} - m_1^2 = \frac{(s + m_1^2 - m_2^2)^2 - 4sm_1^2}{4s} \\ &= \frac{s^2 + m_1^4 + m_2^4 - 2sm_1^2 - 2sm_2^2 - 2m_1^2 m_2^2}{4s} = \frac{\lambda(s, m_1^2, m_2^2)}{4s} \,, \end{aligned} \tag{16.49}$$

introduces the *Källén (triangle) function*

$$\lambda(a,b,c) \equiv a^2 + b^2 + c^2 - 2ab - 2bc - 2ca = \big[a - (\sqrt{b} + \sqrt{c})^2\big]\big[a - (\sqrt{b} - \sqrt{c})^2\big]. \tag{16.50}$$

Hence the CM momentum is

$$|\mathbf{k}| = \frac{\lambda^{1/2}(s, m_1^2, m_2^2)}{2\sqrt{s}} = \frac{1}{2\sqrt{s}} \sqrt{\big[s - (m_1 + m_2)^2\big]\big[s - (m_1 - m_2)^2\big]} \,. \tag{16.51}$$

The second form makes the threshold explicit: $|\mathbf{k}|$ is real and positive precisely when $\sqrt{s} \geq m_1 + m_2$, i.e. when there is enough energy to make the two particles, and it vanishes at threshold $s = (m_1 + m_2)^2$ as it must.

Total two-body volume. When $\mathcal{M}$ is angle-independent (as for a spinless $1 \to 2$ decay) or when one wants the bare phase-space volume, integrate Eq. (16.47) over the full solid angle, $\int d\Omega = 4\pi$:

$$\Pi_2(s) = \int \mathrm{d}\Pi_2 = \frac{1}{16\pi^2} \frac{|\mathbf{k}|}{\sqrt{s}} 4\pi = \frac{|\mathbf{k}|}{4\pi\sqrt{s}} = \frac{\lambda^{1/2}(s, m_1^2, m_2^2)}{8\pi\, s} \,. \tag{16.52}$$

This depends on P only through s, as Theorem 16.8 guaranteed.

Special cases. Two limits recur constantly and are worth recording.

Massless products, $m_1 = m_2 = 0$. Then $|\mathbf{k}| = \sqrt{s}/2$, and

$$\mathrm{d}\Pi_2 = \frac{1}{32\pi^2}\, d\Omega \,, \qquad \Pi_2 = \frac{1}{8\pi} \,. \tag{16.53}$$

The two-body phase space of two massless particles is a pure number times the solid angle — no scale at all, a fact that underlies the simplicity of high-energy cross sections and of conformal two-point kinematics.

Equal masses, $m_1 = m_2 = m$. Then $|\mathbf{k}| = \tfrac{1}{2}\sqrt{s - 4m^2}$. Introducing the CM speed of either particle,

$$\beta = \sqrt{1 - \frac{4m^2}{s}} = \frac{2\,|\mathbf{k}|}{\sqrt{s}}\,, \tag{16.54}$$

the result compresses to

$$\mathrm{d}\Pi_2 = \frac{\beta}{32\pi^2}\, d\Omega\,, \qquad \Pi_2 = \frac{\beta}{8\pi}\,. \tag{16.55}$$

The factor β is the velocity suppression at threshold familiar from $e^+e^- \to \mu^+\mu^-$ near $\sqrt{s} = 2m_\mu$; it will reappear there in Chapter 17. As $\beta \to 1$ this reduces to the massless result Eq. (16.53), as it should.

16.2.4 The phase-space recursion

Beyond two bodies there is no single closed form, but there is an exact recursion that reduces $\mathrm{d}\Pi_n$ to a one-dimensional integral over an invariant mass times $\mathrm{d}\Pi_2$ and $\mathrm{d}\Pi_{n-1}$. This is the engine for every multibody computation: it lets a Dalitz plot or an n-body width be built by iterating the two-body formula.

Proposition 16.9 (Phase-space recursion)**.** *For any* $1 \le m < n$*, and in particular for the split that peels off a single particle,*

$$\mathrm{d}\Pi_n(P; k_1, \dots, k_n) = \mathrm{d}\Pi_2(P; q, k_n)\, \frac{dq^2}{2\pi}\, \mathrm{d}\Pi_{n-1}(q; k_1, \dots, k_{n-1})\,, \tag{16.56}$$

where $q = \sum_{i=1}^{n-1} k_i$ *is the total four-momentum of the first* $n-1$ *particles,* q^2 *ranges over* $\big[(\sum_{i<n} m_i)^2, (\sqrt{s} - m_n)^2\big]$*, and in* $\mathrm{d}\Pi_2(P; q, k_n)$ *the cluster* q *is treated as a single particle of mass* $\sqrt{q^2}$*.*

Proof. Begin with the definition Eq. (16.39), separating the last particle:

$$\mathrm{d}\Pi_n = (2\pi)^4\, \delta^4\Big(P - k_n - \sum_{i<n} k_i\Big)\, \widetilde{dk_n} \prod_{i<n} \widetilde{dk_i}\,. \tag{16.57}$$

Insert the resolution of unity (a completeness insertion in the cluster momentum)

$$1 = \int d^4q\; \delta^4\Big(q - \sum_{i<n} k_i\Big)\,, \tag{16.58}$$

so that the conservation delta splits, $\delta^4(P - k_n - \sum_{i<n} k_i) \to \delta^4(P - k_n - q)$ inside the q integral:

$$\mathrm{d}\Pi_n = \int d^4q\; (2\pi)^4 \delta^4(P - q - k_n)\, \widetilde{dk_n}\, \delta^4\Big(q - \sum_{i<n} k_i\Big) \prod_{i<n} \widetilde{dk_i}\,. \tag{16.59}$$

The cluster $q = \sum_{i<n} k_i$ is a sum of forward on-shell momenta, hence itself forward timelike, $q^0 > 0$ and $q^2 \geq (\sum_{i<n} m_i)^2 > 0$. On that domain we may trade the unconstrained four-dimensional measure d^4q for an integral over the invariant mass q^2 times an on-shell three-measure. Concretely, with $q^0 > 0$, change variable from q^0 to $q^2 = (q^0)^2 - \mathbf{q}^2$ at fixed $\mathbf{q}$: $dq^2 = 2q^0\, dq^0$ gives $dq^0 = dq^2/(2q^0)$, and writing $\omega_q \equiv q^0 = \sqrt{\mathbf{q}^2 + q^2}$,

$$\int \frac{d^4q}{(2\pi)^4}\,\theta(q^0)\,(\cdots) = \int_0^\infty \frac{dq^2}{2\pi} \int \frac{d^3q}{(2\pi)^3\,2\omega_q}\,(\cdots) = \int_0^\infty \frac{dq^2}{2\pi} \int \widetilde{dq}\,(\cdots)\,, \tag{16.60}$$

where $\widetilde{dq} = d^3q/[(2\pi)^3 2\omega_q]$ is exactly the on-shell measure Eq. (16.30) for a particle of mass $\sqrt{q^2}$. (This is Theorem 16.5 read backwards: $\int d^4q\,\theta(q^0) = \int dq^2 \int d^4q\,(2\pi)\delta(q^2 - (q^2))\theta(q^0)/(2\pi)$, the δ collapsing $q^0 \to \omega_q$.) Applying Eq. (16.60) to Eq. (16.59) — the explicit $(2\pi)^4$ already present supplies the needed factors when we reinstate it — gives

$$\mathrm{d}\Pi_n = \int_0^\infty \frac{dq^2}{2\pi} \underbrace{(2\pi)^4\delta^4(P - q - k_n)\,\widetilde{dq}\,\widetilde{dk_n}}_{=\mathrm{d}\Pi_2(P;q,k_n)} \underbrace{(2\pi)^4\delta^4\Big(q - \sum_{i<n} k_i\Big) \prod_{i<n} \widetilde{dk_i}}_{=\mathrm{d}\Pi_{n-1}(q;k_1,\ldots,k_{n-1})}\,. \tag{16.61}$$

The two braces are, respectively, the two-body phase space for $P \to q + k_n$ with q a particle of mass $\sqrt{q^2}$, and the $(n-1)$-body phase space for the decay of q into $k_1, \ldots, k_{n-1}$. Differentially this is exactly Eq. (16.56). The range of q^2 is set by requiring both factors to have support: $\mathrm{d}\Pi_{n-1}$ needs $q^2 \geq (\sum_{i<n} m_i)^2$ and $\mathrm{d}\Pi_2$ needs $\sqrt{q^2} \leq \sqrt{s} - m_n$, i.e. $q^2 \leq (\sqrt{s} - m_n)^2$. □

The recursion turns the construction of any $\mathrm{d}\Pi_n$ into an iterated chain of two-body factors, each in the closed form Eq. (16.47). The simplest nontrivial case, $n = 3$, produces the Dalitz plot — the standard arena for analyzing three-body decays — and shows the machinery working end to end.

Worked Example 16.10 (Three-body phase space and the Dalitz measure)**.** Consider a particle of mass M (so $P^2 = s = M^2$) decaying to three particles of masses m_1, m_2, m_3. We compute the doubly-differential phase space in the two independent invariant masses

$$s_{12} \equiv (k_1 + k_2)^2\,, \qquad s_{23} \equiv (k_2 + k_3)^2\,, \tag{16.62}$$

and show that the result is flat in these variables — the central fact behind the Dalitz plot: any deviation from a uniform population of the (s_{12}, s_{23}) plane is dynamics in $|\mathcal{M}|^2$, never kinematics.

Apply the recursion. Peel off particle 3, so the cluster is $q = k_1 + k_2$ with $q^2 = s_{12}$. Then Eq. (16.56) reads

$$\mathrm{d}\Pi_3 = \frac{ds_{12}}{2\pi}\,\mathrm{d}\Pi_2(P; q, k_3)\,\mathrm{d}\Pi_2(q; k_1, k_2)\,. \tag{16.63}$$

Evaluate each two-body factor with Eq. (16.47) in its natural frame.

The outer factor $\mathrm{d}\Pi_2(P; q, k_3)$, in the overall rest frame of P (the M rest frame), describes $P \to q + k_3$ with q of mass $\sqrt{s_{12}}$:

$$\mathrm{d}\Pi_2(P; q, k_3) = \frac{1}{16\pi^2}\,\frac{|\mathbf{k}_3|}{\sqrt{s}}\,d\Omega_3\,, \tag{16.64}$$

$|\mathbf{k}_3|$ being the momentum of particle 3 in the P frame and $d\Omega_3$ its direction there.

The inner factor $\mathrm{d}\Pi_2(q; k_1, k_2)$, in the rest frame of q (the 1–2 center of momentum, denoted by a star), describes $q \to k_1 + k_2$:

$$\mathrm{d}\Pi_2(q; k_1, k_2) = \frac{1}{16\pi^2}\,\frac{|\mathbf{k}_1^*|}{\sqrt{s_{12}}}\,d\Omega_1^*\,, \tag{16.65}$$

with $|\mathbf{k}_1^*|$ and $d\Omega_1^*$ measured in the q frame.

Integrate the trivial orientations. For a spinless (or unpolarized) decay nothing selects an absolute direction, so the overall orientation $d\Omega_3$ integrates freely: $\int d\Omega_3 = 4\pi$. In the inner factor, choose the polar axis of $d\Omega_1^* = d\cos\theta^*\, d\phi_1^*$ along the direction of particle 3 as seen in the q frame; the azimuth is free, $\int d\phi_1^* = 2\pi$, leaving $d\cos\theta^*$. Collecting constants,

$$\mathrm{d}\Pi_3 = \frac{ds_{12}}{2\pi}\cdot\frac{4\pi}{16\pi^2}\,\frac{|\mathbf{k}_3|}{\sqrt{s}}\cdot\frac{2\pi}{16\pi^2}\,\frac{|\mathbf{k}_1^*|}{\sqrt{s_{12}}}\,d\cos\theta^* = \frac{1}{64\pi^3}\,\frac{|\mathbf{k}_3|\;|\mathbf{k}_1^*|}{\sqrt{s}\,\sqrt{s_{12}}}\,ds_{12}\,d\cos\theta^*\,. \tag{16.66}$$

Change $\cos\theta^* \to s_{23}$. The remaining angle controls s_{23}. In the q frame, $\mathbf{k}_2^* = -\mathbf{k}_1^*$, and writing $|\mathbf{k}_3^*|$ for the momentum of particle 3 in this frame,

$$\begin{aligned} s_{23} = (k_2 + k_3)^2 &= m_2^2 + m_3^2 + 2(E_2^* E_3^* - \mathbf{k}_2^*\cdot\mathbf{k}_3^*) \\ &= m_2^2 + m_3^2 + 2E_2^* E_3^* + 2\;|\mathbf{k}_1^*|\;|\mathbf{k}_3^*|\cos\theta^*\,, \end{aligned} \tag{16.67}$$

since θ^* is the angle between $\mathbf{k}_1^*$ and the particle-3 direction and $\mathbf{k}_2^*\cdot\mathbf{k}_3^* = -\,|\mathbf{k}_1^*|\,|\mathbf{k}_3^*|\cos\theta^*$. At fixed s_{12} all energies are fixed, so

$$ds_{23} = 2\;|\mathbf{k}_1^*|\;|\mathbf{k}_3^*|\;d\cos\theta^* \quad\Longrightarrow\quad |\mathbf{k}_1^*|\;d\cos\theta^* = \frac{ds_{23}}{2\;|\mathbf{k}_3^*|}\,. \tag{16.68}$$

Relate the two frames. The only frame-dependent quantities left are $|\mathbf{k}_3|$ (in the P frame) and $|\mathbf{k}_3^*|$ (in the q frame). Both are momenta of particle 3 in a two-body configuration with the cluster of invariant mass $\sqrt{s_{12}}$; from Eq. (16.51) applied in each frame,

$$|\mathbf{k}_3| = \frac{\lambda^{1/2}(s, s_{12}, m_3^2)}{2\sqrt{s}}\,, \qquad |\mathbf{k}_3^*| = \frac{\lambda^{1/2}(s, s_{12}, m_3^2)}{2\sqrt{s_{12}}}\,, \tag{16.69}$$

the first from $P \to q + k_3$ in the P frame, the second from the energy of particle 3 in the q frame, $E_3^* = q\cdot k_3/\sqrt{s_{12}} = (s - s_{12} - m_3^2)/(2\sqrt{s_{12}})$. The triangle function is identical in both, so

$$\frac{|\mathbf{k}_3|}{|\mathbf{k}_3^*|} = \frac{\sqrt{s_{12}}}{\sqrt{s}}\,. \tag{16.70}$$

Assemble. Substitute Eq. (16.68) into Eq. (16.66) and use Eq. (16.70):

$$\mathrm{d}\Pi_3 = \frac{1}{64\pi^3}\,\frac{|\mathbf{k}_3|}{\sqrt{s}\,\sqrt{s_{12}}}\,\frac{1}{2\,|\mathbf{k}_3^*|}\,ds_{12}\,ds_{23} = \frac{1}{128\pi^3}\,\frac{1}{\sqrt{s}\,\sqrt{s_{12}}}\,\frac{\sqrt{s_{12}}}{\sqrt{s}}\,ds_{12}\,ds_{23}\,, \tag{16.71}$$

and the $\sqrt{s_{12}}$ cancels, leaving the flat Dalitz measure

$$\boxed{\mathrm{d}\Pi_3 = \frac{1}{(2\pi)^3}\,\frac{1}{16\,s}\,ds_{12}\,ds_{23}} \qquad (s = M^2), \tag{16.72}$$

where we used $128\pi^3 = 16\cdot(2\pi)^3$. The phase-space density is constant across the kinematically allowed region of the (s_{12}, s_{23}) plane; the boundary of that region is set by requiring $|\cos\theta^*| \le 1$, which carves out the familiar Dalitz contour. Every structure inside it — bands, peaks, interference — is the dynamics encoded in $|\mathcal{M}|^2$, exactly as advertised. This is the same logic that will let Chapter 17 read tree-level distributions straight off $|\mathcal{M}|^2$.

We have isolated the purely kinematic content of any transition: a Lorentz-scalar measure $\mathrm{d}\Pi_n$, manifestly frame-independent, reducible by the recursion Eq. (16.56) to the closed two-body form Eq. (16.47). What this measure still lacks is the normalization that turns $\int |\mathcal{M}|^2\,\mathrm{d}\Pi_n$ into an observable rate. That normalization differs by initial state: a single factor $1/(2M)$ for the decay of one particle of mass M, treated next in Section 16.3, and an incident flux for a two-particle collision, treated in Section 16.4. With $\mathrm{d}\Pi_2$ in hand, both reduce to one short step.

16.3 The decay rate formula

We now have both halves of the apparatus the chapter set out to build. Section 16.1 regulated the squared energy–momentum delta function with a finite box V and a finite observation time T and extracted from it a finite *transition rate* — a probability per unit time — in terms of the invariant amplitude $\mathcal{M}$ defined by $\langle f|iT|i\rangle = (2\pi)^4\delta^4(p_f - p_i)\,i\mathcal{M}$ (Eq. (14.163)). Section 16.2 took the purely kinematic remainder of that rate, the integral over on-shell final momenta enforcing total energy–momentum conservation, isolated it as the Lorentz-invariant phase-space measure $\mathrm{d}\Pi_n$, and reduced the two-body case in the centre-of-momentum frame to a closed form. The dynamics lives entirely in $\mathcal{M}$; everything else is the geometry of the on-shell final momenta. What remains

is to feed in an initial state and read off an observable. The simplest possible initial state is a single particle, and the observable it produces is its decay. This section treats that one-body case in full: it assembles the differential decay rate from the §16.1 skeleton and the §16.2 measure, defines the width, lifetime, partial widths, and branching ratios, settles the symmetry factor for identical final particles, and carries a complete worked decay through to a numerical lifetime.

16.3.1 From the rate skeleton to $d\Gamma$: where the $1/2M$ comes from

Recall the rate skeleton of Section 16.1. Working in a spatial box of volume V for a time T, with one-particle states carried in the relativistic normalization $\langle p|p'\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{p}')$ of Eq. (3.193) (so that in the box $\langle p|p\rangle = 2E_{\mathbf{p}}V$), the probability for a prepared initial state $|i\rangle$ to be found, after the interaction, in a particular set of n outgoing single-particle momentum cells is

$$dP_{i\to f} = \frac{\big|\langle f|iT|i\rangle\big|^2}{\langle i|i\rangle\,\prod_{a=1}^{n}\langle f_a|f_a\rangle}\;\prod_{a=1}^{n}\frac{V\,d^3p_a}{(2\pi)^3}\,, \tag{16.73}$$

where the last product counts the number of distinct momentum states available to each outgoing particle inside the box, and the denominators undo the relativistic normalization of the states that appear in $\langle f|iT|i\rangle$. The square of the connected matrix element supplies, through the regulated identity established in Section 16.1,

$$\big|\langle f|iT|i\rangle\big|^2 = \Big[(2\pi)^4\delta^4(p_f-p_i)\Big]^2\,|\mathcal{M}|^2 = VT\,(2\pi)^4\delta^4(p_f-p_i)\,|\mathcal{M}|^2\,, \tag{16.74}$$

the one factor of the delta function having been evaluated at zero argument and replaced by the spacetime volume, $(2\pi)^4\delta^4(0)\to VT$. This is the single place a divergence threatened, and it is exactly cancelled below.

We now specialize to a one-body initial state: a single parent particle of mass M, which we may always place at rest, $p_i=(M,\mathbf{0})$, since the decay rate of an unstable particle is most naturally quoted in its rest frame. Then $\langle i|i\rangle = 2E_iV = 2MV$. The factors of V and T assemble as follows. The denominators $\prod_a\langle f_a|f_a\rangle = \prod_a 2E_aV$ each carry a V, and these cancel the n box volumes in the numerator product $\prod_a V\,d^3p_a$. The single remaining box volume from Eq. (16.74) cancels the V in $\langle i|i\rangle = 2MV$. The factor T in Eq. (16.74) is removed by passing from a probability to a *rate*, $d\Gamma = dP_{i\to f}/T$. Collecting what survives,

$$d\Gamma = \frac{1}{2M}\,|\mathcal{M}|^2\,\underbrace{\prod_{a=1}^{n}\frac{d^3p_a}{(2\pi)^3\,2E_a}\,(2\pi)^4\delta^4\Big(p_i-\sum_a p_a\Big)}_{\mathrm{d}\Pi_n}\,. \tag{16.75}$$

The bracketed object is precisely the Lorentz-invariant n-body phase-space measure $\mathrm{d}\Pi_n$ built and shown invariant in Section 16.2. We have arrived at the master differential decay rate.

Theorem 16.11 (Differential decay rate)**.** *For a parent particle of mass M decaying at rest into n final particles, the differential rate into the phase-space element $\mathrm{d}\Pi_n$ is*

$$\boxed{d\Gamma = \frac{1}{2M}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n} \tag{16.76}$$

where $\mathcal{M}$ is the invariant amplitude of Eq. (14.163) and the entire dependence on the initial state is the single prefactor $1/2M$.

The decisive feature, and the reason this is the simplest member of the family of observable formulae, is the origin of the prefactor. The $\frac{1}{2M}$ is nothing but the inverse of the relativistic normalization $\langle i|i\rangle/V = 2E = 2M$ of one particle at rest. There is no flux factor: flux measures the relative motion of two colliding beams, and a single particle has nothing to move relative to. This is exactly the structural difference between a decay and a scattering process. When §16.4 treats a two-body initial state, the only change in the assembly above will be that $\langle i|i\rangle$ becomes a product of two normalizations and that the relative velocity of the incoming pair re-enters; the $1/2M$ here is the degenerate, single-particle limit of the general normalization denominator.

Physical Insight

The split is clean and worth stating once: $\mathcal{M}$ is *dynamics* — it is what the Lagrangian, the Feynman rules, and the loop expansion compute — while $\mathrm{d}\Pi_n$ is *kinematics*, fixed entirely by the masses and the conservation law, identical for every theory with the same external particles. The normalization prefactor ($1/2M$ here, the flux there) is *bookkeeping* for how the initial state was counted. Every decay rate and cross section in the book is one of these three pieces times another; once $\mathcal{M}$ is known, producing a number is mechanical.

16.3.2 Width, lifetime, partial widths, and branching ratios

Integrating Eq. (16.76) over all of phase space for a definite set of final particles f gives the *partial decay rate*, or *partial width*, into that channel,

$$\Gamma_f = \frac{1}{2M}\int |\mathcal{M}(i\to f)|^2\,\mathrm{d}\Pi_n^{(f)}\,, \tag{16.77}$$

with one subtlety — the symmetry factor for identical final particles — deferred to Section 16.3.3. A given parent will in general have several open channels f, distinguished by the species and number of decay products. Because the channels are mutually exclusive final states, the probabilities add, and so do the rates.

Definition 16.12 (Total width, lifetime, branching ratio)**.** The *total width* of an unstable particle is the sum of its partial widths over all kinematically allowed channels,

$$\Gamma = \sum_f \Gamma_f \,. \tag{16.78}$$

Its *mean lifetime* is the reciprocal, $\tau = 1/\Gamma$, and the *branching ratio* into channel f is the fraction of decays proceeding through that channel,

$$\mathrm{BR}_f = \frac{\Gamma_f}{\Gamma}\,, \qquad \sum_f \mathrm{BR}_f = 1\,. \tag{16.79}$$

That $\tau = 1/\Gamma$ is the mean lifetime is not a definition imposed by fiat; it follows from the rate interpretation. If at time t a population $N(t)$ of identical parents survives undecayed, the number decaying in dt is $dN = -\Gamma\, N\, dt$, because Γ is by construction the probability of decay per unit time per particle. This integrates to the exponential law

$$N(t) = N(0)\, e^{-\Gamma t}, \tag{16.80}$$

and the mean survival time of a single parent, with normalized decay-time distribution $\Gamma e^{-\Gamma t}$, is

$$\langle t \rangle = \int_0^\infty t\, \Gamma e^{-\Gamma t}\, dt = \frac{1}{\Gamma} = \tau\,, \tag{16.81}$$

as claimed. The same exponential is visible directly in the quantum amplitude: an unstable particle is not an exact energy eigenstate, and its rest-frame survival amplitude behaves as $e^{-iMt}\, e^{-\Gamma t/2}$, i.e. as if its mass were the complex number $M - \frac{i}{2}\Gamma$. Squaring the amplitude reproduces $e^{-\Gamma t}$, and the imaginary shift of the mass is exactly what a Breit–Wigner resonance of width Γ encodes in energy space. This is the first appearance of a theme the book returns to with full machinery: a decay width is the imaginary part of a pole.

Physical Insight

The width does double duty as a frequency and as an energy. In natural units $[\Gamma] =$ mass, so a width quoted in MeV is simultaneously an inverse lifetime: $\tau = \hbar/\Gamma$, with $\hbar = 6.582 \times 10^{-25}$ GeV $\cdot$ s the conversion factor restored. A particle whose width is 1 GeV lives for about 6.6×10^{-25} s; one whose width is 1 eV lives a billion (10^9) times longer. "Broad" and "short-lived" are the same statement.

The complex-mass picture is more than a mnemonic. In Chapter 19 the self-energy $\Sigma(p^2)$ — the sum of one-particle-irreducible corrections to the propagator — is shown to shift the pole of the full propagator from $p^2 = M^2$ to $p^2 = M^2 + \Sigma(p^2)$,

and the location of that pole in the complex p^2 plane fixes both the physical mass and the width. Matching the imaginary part of the pole to the exponential decay law Eq. (16.80) gives the exact relation

$$\Gamma = -\frac{1}{M}\,\mathrm{Im}\,\Sigma(M^2)\,, \tag{16.82}$$

which we record here as a forward pointer and discharge in Chapter 19. It is not an independent route to the width but the same width seen through the propagator: the optical theorem of §16.7, which equates twice the imaginary part of a forward amplitude to a total rate, is precisely the statement that ties Eq. (16.82) to the phase-space integral Eq. (16.77). Computing Γ from $|\mathcal{M}|^2$ integrated over $\mathrm{d}\Pi_n$, as we do in the rest of this section, and computing it from $\mathrm{Im}\,\Sigma$, as Chapter 19 does, are the two faces of unitarity.

16.3.3 Identical final particles: the $1/n!$ symmetry factor

Formula Eq. (16.76) was derived by labelling the outgoing particles $1, 2, \ldots, n$ and integrating each momentum independently over all of momentum space. When some of the final particles are *identical*, this overcounts the physical final states, and the rate must be corrected by a combinatorial factor. The crucial point — often misattributed to the amplitude — is that this factor is a property of the *integration domain*, not of $\mathcal{M}$.

To see it cleanly, suppose all n final particles are identical bosons of a single species. A physical final state is an *unordered* collection of n momenta: the state with particles at momenta $(\mathbf{p}_1, \ldots, \mathbf{p}_n)$ is the very same state as the one with those momenta in any other order, because identical bosons carry no labels. The Hilbert-space sum over distinct physical final states therefore runs over unordered n-tuples. The measure $\mathrm{d}\Pi_n$ in Eq. (16.76), however, was constructed as a product of independent single-particle integrals, $\prod_a \int d^3p_a/[(2\pi)^3 2E_a]$, which runs over *ordered* n-tuples and hence visits each unordered configuration $n!$ times — once for every permutation of the labels. (The measure-zero diagonal where two momenta coincide does not affect the integral.) Integrating $|\mathcal{M}|^2\,\mathrm{d}\Pi_n$ as written thus computes $n!$ times the physically meaningful sum over distinct final states. The total rate into the identical-particle channel is the corrected

$$\Gamma = \frac{1}{2M}\,\frac{1}{n!}\int |\mathcal{M}|^2\,\mathrm{d}\Pi_n\,, \tag{16.83}$$

and more generally, if the final state contains n_j identical particles of species j, the prefactor is the product

$$\mathcal{S} = \prod_j \frac{1}{n_j!}\,, \tag{16.84}$$

one factorial for each species, distinct species being distinguishable and contributing no symmetrization.

Caution

The factor $1/n!$ has nothing to do with the symmetry factors of *diagrams* introduced for ϕ^4 in Chapter 15. Those arise from the automorphisms of a Feynman graph and are absorbed into the definition of $\mathcal{M}$ before it is squared. The factor Eq. (16.84) is applied *after* squaring, to the phase-space integral, and corrects the counting of final *states*. Bose symmetry guarantees that $\mathcal{M}$ is already symmetric under exchange of identical external legs — so $|\mathcal{M}|^2$ takes the same value at all $n!$ permuted points and the integral is genuinely $n!$ times the physical one. Computing $\mathcal{M}$ correctly (with its diagrammatic symmetry factors) and then dividing the rate by $\prod_j n_j!$ are two independent bookkeeping steps; omitting either is a defect.

16.3.4 The two-body width of a particle at rest

The overwhelmingly common case is a two-body decay, $\Phi \to 1 + 2$, for which §16.2 provides the reduced measure in closed form. With the parent at rest the total four-momentum is $p_i = (M, \mathbf{0})$, so the centre-of-momentum energy is $\sqrt{s} = M$, and the two-body Lorentz-invariant phase space found in Section 16.2 reads

$$\mathrm{d}\Pi_2 = \frac{1}{16\pi^2}\,\frac{|\mathbf{k}|}{M}\,d\Omega\,, \tag{16.85}$$

where $d\Omega$ is the solid-angle element of one outgoing particle in the rest frame and $|\mathbf{k}|$ is the common magnitude of the two back-to-back final momenta. Energy–momentum conservation, $M = \sqrt{|\mathbf{k}|^2 + m_1^2} + \sqrt{|\mathbf{k}|^2 + m_2^2}$, fixes $|\mathbf{k}|$ through the Källén triangle function,

$$|\mathbf{k}| = \frac{\lambda^{1/2}(M^2, m_1^2, m_2^2)}{2M}\,, \qquad \lambda(a,b,c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca\,, \tag{16.86}$$

which for equal masses $m_1 = m_2 = m$ collapses to the form we shall need,

$$|\mathbf{k}| = \frac{M}{2}\sqrt{1 - \frac{4m^2}{M^2}}\,. \tag{16.87}$$

The square root is real precisely when $M \geq m_1 + m_2$, the kinematic threshold for the decay to be allowed at all; below it the channel is closed and $\Gamma_f = 0$.

Inserting Eq. (16.85) into the master formula Eq. (16.76) gives the differential two-body width,

$$\frac{d\Gamma}{d\Omega} = \frac{1}{2M}\,|\mathcal{M}|^2\,\frac{1}{16\pi^2}\,\frac{|\mathbf{k}|}{M} = \frac{|\mathbf{k}|}{32\pi^2 M^2}\,|\mathcal{M}|^2\,. \tag{16.88}$$

When the amplitude is independent of the emission angle — which is automatic for a scalar parent, since there is no preferred direction in the rest frame and

no spin to correlate with — the angular integral is trivial. Writing $\overline{|\mathcal{M}|^2}$ for the amplitude squared (the overbar will denote the spin average once spins enter in Chapter 17; for spinless particles it is just $|\mathcal{M}|^2$) and integrating over the full solid angle $\int d\Omega = 4\pi$,

$$\Gamma = \mathcal{S}\,\frac{|\mathbf{k}|}{8\pi M^2}\,\overline{|\mathcal{M}|^2}\,, \tag{16.89}$$

with $\mathcal{S} = \frac{1}{2}$ when the two final particles are identical and $\mathcal{S} = 1$ when they are distinct, per Eq. (16.84). This is the workhorse two-body width formula. It says that, for a constant amplitude, the entire content of a two-body decay rate is the single kinematic number $|\mathbf{k}|$ — the momentum released — divided by $8\pi M^2$, times the dynamical $\overline{|\mathcal{M}|^2}$ and the identical-particle factor. We now cash it on a concrete model.

16.3.5 A worked decay: a heavy scalar into two light scalars

Notation Watch

The frozen ϕ^4 theory theory has a single real scalar with only a quartic self-interaction; it admits no on-shell one-to-two decay, because a single ϕ cannot go to two ϕ's at any order (there is no cubic vertex, and the would-be process $\phi \to \phi\phi$ violates energy–momentum conservation for one species anyway). To exhibit a genuine, fully worked decay we therefore step *outside* the frozen ϕ^4 Lagrangian and introduce an auxiliary toy model, used only in this worked example. The ϕ^4 thread itself is advanced in §16.6, where the carried $2 \to 2$ amplitude $\mathcal{M} = -\lambda$ is turned into a cross section.

Worked Example 16.13 (Decay of a heavy scalar via a cubic coupling)**.** Consider a heavy real scalar Φ of mass M coupled to a light real scalar ϕ of mass m through the cubic interaction

$$\mathcal{L}_{\text{int}} = -\frac{g}{2}\,\Phi\,\phi^2\,, \tag{16.90}$$

where the coupling g has mass dimension one (since $[\Phi] = [\phi] = 1$ in four dimensions and $[\mathcal{L}] = 4$). We compute the decay rate $\Phi \to \phi\phi$ and a numerical lifetime. This is an auxiliary model, distinct from $\lambda\phi^4$ theory; the factor of $\frac{1}{2}$ in Eq. (16.90) is the conventional choice that makes the vertex come out as $-ig$, as we now verify.

The amplitude. The single tree diagram is the elementary cubic vertex with one incoming Φ and two outgoing ϕ lines (Fig. 16.3). By the Feynman rules of Chapter 15, a vertex from an interaction term $\mathcal{L}_{\text{int}}$ contributes i times the coupling, times the number of distinct ways the fields in the vertex can be

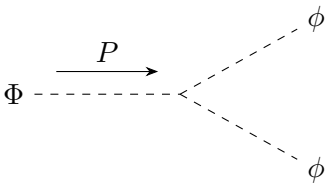


Figure 16.3: The single tree diagram for $\Phi \to \phi\phi$ from the cubic interaction Eq. (16.90). The parent four-momentum $P = (M, \mathbf{0})$ splits into two back-to-back light scalars of momentum magnitude $|\mathbf{k}|$ given by Eq. (16.87).

contracted with the external lines. The term Eq. (16.90) contains two identical ϕ fields, which can attach to the two external ϕ legs in $2! = 2$ ways, so the vertex factor is

$$i\mathcal{M} = i\,(-g)\cdot\tfrac{1}{2}\cdot 2! = -ig\,, \qquad \Longrightarrow \qquad \mathcal{M} = -g\,, \quad |\mathcal{M}|^2 = g^2\,. \tag{16.91}$$

The combinatorial 2! exactly cancels the conventional $\frac{1}{2}$, leaving the clean result $\mathcal{M} = -g$. The amplitude is a constant: there is no momentum dependence at tree level for a pointlike cubic vertex, so Eq. (16.89) applies directly.

The rate. The two final particles are identical, so $\mathcal{S} = \frac{1}{2}$. With equal final masses the released momentum is Eq. (16.87), $|\mathbf{k}| = \frac{M}{2}\sqrt{1 - 4m^2/M^2}$. Substituting into Eq. (16.89),

$$\begin{aligned}\Gamma &= \frac{1}{2}\cdot\frac{|\mathbf{k}|}{8\pi M^2}\cdot g^2 = \frac{g^2}{16\pi M^2}\cdot\frac{M}{2}\sqrt{1-\frac{4m^2}{M^2}} \\ &= \boxed{\frac{g^2}{32\pi M}\sqrt{1-\frac{4m^2}{M^2}}}\,.\end{aligned} \tag{16.92}$$

The structure is instructive. The rate grows linearly with g^2 as a tree-level coupling-squared must; the overall $1/M$ is fixed by dimensions, since $[\Gamma] =$ mass and $[g^2] =$ mass^2; and the square root is the phase-space suppression that switches the rate off continuously as $m \to M/2$, the threshold below which the decay cannot occur. In the limit $m \ll M$ the daughters are effectively massless and $\Gamma \to g^2/32\pi M$.

A number. Choose, purely as an illustration, $M = 10$ GeV, $m = 1$ GeV, and $g = 1$ GeV. Then $4m^2/M^2 = 0.04$ and $\sqrt{1 - 0.04} = 0.9798$, so from Eq. (16.92),

$$\Gamma = \frac{(1\text{ GeV})^2}{32\pi\,(10\text{ GeV})}\times 0.9798 = \frac{0.9798}{1005.3}\text{ GeV} = 9.75\times 10^{-4}\text{ GeV} = 0.975\text{ MeV}\,. \tag{16.93}$$

The lifetime follows by restoring $\hbar = 6.582 \times 10^{-25}\ \text{GeV} \cdot \text{s}$:

$$\tau = \frac{\hbar}{\Gamma} = \frac{6.582 \times 10^{-25}\ \text{GeV} \cdot \text{s}}{9.75 \times 10^{-4}\ \text{GeV}} = 6.75 \times 10^{-22}\ \text{s}\,. \tag{16.94}$$

A width of about an MeV corresponds to a lifetime of order $10^{-21}\,\text{s}$ — the characteristic timescale of a strongly coupled decay, and a concrete instance of the width–lifetime reciprocity emphasized above. Every step from the Lagrangian Eq. (16.90) to the number Eq. (16.94) has been carried explicitly; the reader can reproduce it with no input beyond $\hbar$.

This completes the one-body initial state. The differential rate Eq. (16.76), the width and lifetime of Theorem 16.12, the identical-particle factor Eq. (16.84), and the closed two-body formula Eq. (16.89) together turn any computed amplitude into a measurable lifetime. The single structural ingredient that distinguished a decay from a generic process was the trivial normalization of one initial particle, $1/2M$, with no flux factor because a lone particle has no relative motion to measure. The moment we admit two incoming particles, that relative motion becomes the central new object: the rate must be divided not by $2M$ but by a *flux*, the density of incoming particles times their relative velocity, packaged covariantly. Constructing that flux factor, and assembling from it the master cross-section formula, is the task of the next section.

16.4 The flux factor and the master cross-section formula

The decay rate of Section 16.3 was the easy half of the golden rule. A decaying particle is a one-body initial state: there is nothing to collide with, the initial momentum is fixed once we sit in the rest frame, and the only normalization the rate must carry is the single factor $1/2M$ that converts the relativistically normalized state to a per-particle probability. Scattering is genuinely harder for one structural reason. A cross section measures how strongly two beams interact, and "how strongly" is meaningless until we say *how many* particles were offered up to interact and *how fast* they were swept past one another. The two-body initial state therefore drags in a kinematic factor that the decay calculation never saw: the *flux*. Building that factor correctly, showing that the operational lab-frame definition coincides with a manifestly Lorentz-invariant expression, and folding it into the rate of Section 16.1 to produce a single master formula valid for any final state — that is the whole business of this section. The dynamics will sit, untouched, in $\mathcal{M}$; everything we add here is geometry of the incoming momenta.

16.4.1 The operational definition of a cross section

Imagine the canonical table-top experiment: a steady beam of particles of type 1 is fired at a thin target containing particles of type 2 at rest, and a detector counts how often a particular reaction $1 + 2 \to \{p_3, \dots, p_n\}$ occurs. The raw number of events per second is not a property of the interaction alone — doubling the beam intensity or the target thickness doubles it. To isolate the physics we normalize twice: per target particle, and per unit of incident beam flux. What remains has the dimensions of an area, and it is the cross section.

Definition 16.14 (Cross section)**.** Let an incident beam of type-1 particles impinge on a collection of type-2 target particles. The *incident flux* Φ is the number of beam particles crossing a unit area normal to the beam per unit time. If W is the transition rate (events per unit time) into a chosen set of final states, summed over the N_{tgt} target particles exposed to the beam, then the cross section for that reaction is

$$\sigma \;\equiv\; \frac{W/N_{\text{tgt}}}{\Phi} \;=\; \frac{\text{transition rate per target particle}}{\text{incident flux}}. \tag{16.95}$$

Its differential version $d\sigma$ uses the rate dW into an infinitesimal cell of final-state momenta.

The factor $1/N_{\text{tgt}}$ strips the target thickness and $1/\Phi$ strips the beam intensity, leaving a quantity with units of $[\Phi]^{-1}[\text{time}]^{-1} \cdot [\text{time}] = [\Phi]^{-1}$; since $[\Phi] = (\text{area} \cdot \text{time})^{-1} \cdot \text{time} = \text{area}^{-1}$ once we count number per area per time and divide by a rate, σ comes out as an area. Concretely, in natural units a cross section is an inverse-mass-squared; the standard laboratory unit is the barn, $1\,\text{barn} = 10^{-24}\,\text{cm}^2$. The geometric reading is the one to keep in mind: σ is the effective cross-sectional area each target particle presents to the beam for the reaction in question.

There is one refinement hidden in Fig. 16.4 that the rest-target picture obscures. In a frame where the target also moves — and in particular in the centre-of-momentum frame, where both beams move — the relevant relative speed is not the speed of either beam but the speed with which the two populations sweep through one another. The flux must be built from the *relative velocity* $|\mathbf{v}_1 - \mathbf{v}_2|$. We make this quantitative next.

16.4.2 Assembling $d\sigma$ from the box rate

In Section 16.1 we made the squared S-matrix element finite by quantizing the theory in a spatial box of volume $V = L^3$ over a time interval T, so that the offending square of the energy–momentum delta becomes the finite spacetime volume,

$$\big[(2\pi)^4\delta^4(0)\big] \;\longrightarrow\; VT, \tag{16.96}$$

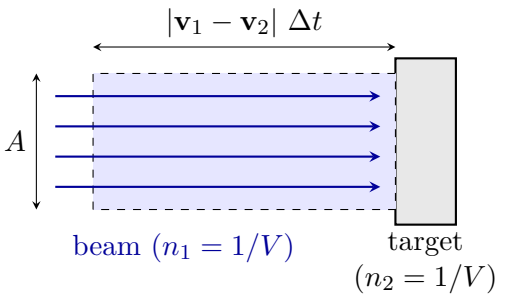


Figure 16.4: The flux as a swept volume. In time Δt the beam particles within the shaded slab of cross-sectional area A and length $|\mathbf{v}_1 - \mathbf{v}_2|\ \Delta t$ reach the target. With one beam and one target particle per box volume V, the number density of each is $1/V$ and the incident number flux is $\Phi = n_1\,|\mathbf{v}_1 - \mathbf{v}_2| = |\mathbf{v}_1 - \mathbf{v}_2|\,/V$. The relative velocity enters because the target is itself moving in a general frame.

and the relativistically normalized one-particle state of Eq. (3.193) acquires the box norm $\langle p|p\rangle_V = 2E_{\mathbf{p}}\,V$. We now run that machinery for a *two*-particle initial state and divide by flux to extract the cross section; this is the construction promised in Section 16.1, specialized to two incoming particles.

Let the initial state be $|i\rangle = |p_1\,p_2\rangle_V$ and the final state a definite occupation of one-particle box modes $|f\rangle = |p_3 \cdots p_n\rangle_V$. The invariant amplitude is defined, exactly as in Eq. (14.163), by stripping the energy–momentum delta from the connected matrix element of $iT = i(\mathbf{S} - \mathbf{1})$,

$$\langle f|\, iT\,|i\rangle \;=\; i\,\mathcal{M}(p_1, p_2 \to p_3, \ldots, p_n)\,(2\pi)^4\delta^4\Big(p_1 + p_2 - \sum_f p_f\Big). \tag{16.97}$$

The probability of the transition is $|\langle f|iT|i\rangle|^2$ divided by the norms of the in- and out-states. Squaring Eq. (16.97) produces $|\mathcal{M}|^2$ times the square of the delta; one delta survives to enforce conservation and the other becomes VT by Eq. (16.96):

$$|\,\langle f|iT|i\rangle\,|^2 \;=\; |\mathcal{M}|^2\;(2\pi)^4\delta^4\Big(p_1 + p_2 - \sum_f p_f\Big)\,VT. \tag{16.98}$$

The initial state has norm $\langle i|i\rangle_V = (2E_1V)(2E_2V) = 4E_1E_2\,V^2$. Each final particle is summed over the box modes available in its momentum cell, $V\,d^3p_f/(2\pi)^3$, and each such mode carries the norm $\langle p_f|p_f\rangle_V = 2E_fV$. Collecting these, the probability of the reaction into the final-state cell $\prod_f d^3p_f$ is

$$dP \;=\; \frac{|\,\langle f|iT|i\rangle\,|^2}{\langle i|i\rangle_V}\prod_f \frac{1}{2E_fV}\,\frac{V\,d^3p_f}{(2\pi)^3} \;=\; \frac{|\mathcal{M}|^2\;(2\pi)^4\delta^4(\cdots)\,VT}{4E_1E_2\,V^2}\prod_f \frac{d^3p_f}{(2\pi)^3\,2E_f}, \tag{16.99}$$

where every factor of V from the final-state mode counting has cancelled against the corresponding norm. The transition rate is $dW = dP/T$, and the leftover spacetime volume collapses to a single inverse power of V:

$$dW \;=\; \frac{1}{4E_1E_2\,V}\;|\mathcal{M}|^2\;\mathrm{d}\Pi_n, \qquad \mathrm{d}\Pi_n \equiv (2\pi)^4\delta^4\Big(p_1+p_2-\sum_f p_f\Big)\prod_{f=3}^{n}\frac{d^3p_f}{(2\pi)^3\,2E_f}, \tag{16.100}$$

with $\mathrm{d}\Pi_n$ the Lorentz-invariant n-body phase-space measure constructed and proved invariant in Eq. (16.39) and Theorem 16.8. Equation Eq. (16.100) is the two-particle analogue of the one-body rate $d\Gamma = \frac{1}{2M}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n$ of Eq. (16.76): the one decaying particle gave one factor $1/2M$ and its single power of V cancelled completely, whereas the two colliding particles give $1/4E_1E_2$ and a residual $1/V$. That residual $1/V$ is not a defect; it is precisely the statement that two particles rattling around in a large box rarely find each other, and it is destined to cancel against the $1/V$ in the flux.

To finish, apply the operational definition Eq. (16.95). We have exactly one target particle ($N_{\mathrm{tgt}} = 1$) and one beam particle in the box, so each has number density $1/V$. The incident flux is the beam density times the relative speed,

$$\Phi \;=\; n_1\;|\mathbf{v}_1-\mathbf{v}_2| \;=\; \frac{|\mathbf{v}_1-\mathbf{v}_2|}{V}. \tag{16.101}$$

Dividing the rate Eq. (16.100) by this flux, the volume cancels as anticipated:

$$d\sigma \;=\; \frac{dW}{\Phi} \;=\; \frac{V}{|\mathbf{v}_1-\mathbf{v}_2|}\cdot\frac{|\mathcal{M}|^2\;\mathrm{d}\Pi_n}{4E_1E_2\,V} \;=\; \frac{|\mathcal{M}|^2}{4E_1E_2\;|\mathbf{v}_1-\mathbf{v}_2|}\,\mathrm{d}\Pi_n. \tag{16.102}$$

The cross section is finite, V-independent, and built entirely from the dynamical $|\mathcal{M}|^2$, the kinematic measure $\mathrm{d}\Pi_n$, and a denominator that depends only on the two incoming momenta. We name that denominator the flux factor.

Definition 16.15 (Collinear flux factor)**.** For two incoming particles whose three-momenta are collinear (parallel or antiparallel), the *flux factor* is

$$F \;\equiv\; 4E_1E_2\;|\mathbf{v}_1-\mathbf{v}_2|\,, \qquad \mathbf{v}_i = \frac{\mathbf{p}_i}{E_i}, \tag{16.103}$$

so that $d\sigma = \dfrac{|\mathcal{M}|^2}{F}\,\mathrm{d}\Pi_n$.

The form Eq. (16.103) is intuitive but frame-bound: velocities transform awkwardly under boosts, so as written F looks like an accident of the lab frame. The next subsection shows it is nothing of the kind.

16.4.3 The covariant Møller flux and the proof of equality

A cross section is a counted number of events normalized by counted numbers of particles; counts are frame-independent, so σ must be expressible through Lorentz invariants. Since $|\mathcal{M}|^2$ and $\mathrm{d}\Pi_n$ are already invariant (the latter by Theorem 16.8), the flux factor must be too. We now produce the invariant and prove it equals Eq. (16.103).

Theorem 16.16 (Covariant flux factor)**.** *For two on-shell particles with momenta p_1, p_2 ($p_i^2 = m_i^2$, $p_i^0 = E_i > 0$) whose spatial momenta are collinear, the collinear flux factor equals a manifestly Lorentz-invariant expression:*

$$F \;=\; 4E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2| \;=\; 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2} \;\equiv\; F_{\mathrm{M\o l}}. \tag{16.104}$$

Proof. Choose coordinates so the common axis of $\mathbf{p}_1, \mathbf{p}_2$ is the z-axis. Then the momenta have no transverse components,

$$p_1 = (E_1, 0, 0, p_1^z), \qquad p_2 = (E_2, 0, 0, p_2^z), \tag{16.105}$$

and the velocities are $\mathbf{v}_i = (0, 0, p_i^z/E_i)$, so that

$$E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2| = E_1E_2\left|\frac{p_1^z}{E_1} - \frac{p_2^z}{E_2}\right| = |E_2\,p_1^z - E_1\,p_2^z|. \tag{16.106}$$

For the right-hand side, use the mostly-minus metric: $p_1\cdot p_2 = E_1E_2 - p_1^zp_2^z$ and $m_i^2 = E_i^2 - (p_i^z)^2$. Expanding,

$$\begin{aligned}(p_1\cdot p_2)^2 - m_1^2m_2^2 &= \left(E_1E_2 - p_1^zp_2^z\right)^2 - \left(E_1^2 - (p_1^z)^2\right)\left(E_2^2 - (p_2^z)^2\right)\\ &= E_1^2E_2^2 - 2E_1E_2\,p_1^zp_2^z + (p_1^z)^2(p_2^z)^2\\ &\quad - \Big[E_1^2E_2^2 - E_1^2(p_2^z)^2 - (p_1^z)^2E_2^2 + (p_1^z)^2(p_2^z)^2\Big].\end{aligned} \tag{16.107}$$

The terms $E_1^2E_2^2$ and $(p_1^z)^2(p_2^z)^2$ cancel between the two lines, leaving

$$(p_1\cdot p_2)^2 - m_1^2m_2^2 = E_1^2(p_2^z)^2 - 2E_1E_2\,p_1^zp_2^z + (p_1^z)^2E_2^2 = \left(E_2\,p_1^z - E_1\,p_2^z\right)^2. \tag{16.108}$$

Taking the positive square root and comparing with Eq. (16.106) gives $\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2} = |E_2p_1^z - E_1p_2^z| = E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2|$. Multiplying by 4 proves Eq. (16.104). □

The right-hand side of Eq. (16.104) is built from $p_1\cdot p_2$ and the two masses, all Lorentz scalars, so $F_{\mathrm{M\o l}}$ is invariant under the full Lorentz group. This is the *Møller flux factor*, and we adopt it as the definition of the flux for all frames:

$$\boxed{F \;=\; 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2}}. \tag{16.109}$$

Proposition 16.17 (Uniqueness of the invariant extension)**.** *The Møller flux Eq. (16.109) is the unique Lorentz scalar that reduces to the operational collinear flux* $4E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2|$ *on collinear configurations.*

Proof. A Lorentz scalar $\mathcal{F}(p_1, p_2)$ formed from two four-vectors can depend on them only through the three invariants p_1^2, p_2^2, $p_1 \cdot p_2$; this is the same invariant-theory fact used to argue that $\mathcal{M}$ depends only on Lorentz-invariant momentum combinations (Theorem 14.15). On the physical mass shells $p_1^2 = m_1^2$ and $p_2^2 = m_2^2$ are fixed, so $\mathcal{F}$ is a function of the single variable $p_1 \cdot p_2$. Any two scalars agreeing on *all* collinear configurations agree for all values of $p_1 \cdot p_2$ in the physical range — because by boosting and rotating one can bring *any* pair of on-shell momenta to a collinear pair with the same value of $p_1 \cdot p_2$ (place p_2 at rest and align $\mathbf{p}_1$ with any axis), and $p_1 \cdot p_2$ sweeps its whole range as $|\mathbf{p}_1|$ varies. Two functions of one variable that agree everywhere on that variable's range are equal. Hence the invariant extension is unique, and Theorem 16.16 exhibits it as $F_{\mathrm{M}\varnothing\mathrm{l}}$. □

"Relative velocity" that exceeds the speed of light, and the frame caveat

The object $|\mathbf{v}_1 - \mathbf{v}_2|$ in Eq. (16.103) is a difference of coordinate velocities, not the physical speed of one particle measured in the other's rest frame. For a head-on collision of two ultrarelativistic beams, $\mathbf{v}_1 \to +\hat{\mathbf{z}}$ and $\mathbf{v}_2 \to -\hat{\mathbf{z}}$ give $|\mathbf{v}_1 - \mathbf{v}_2| \to 2$, twice the speed of light. There is no paradox: this "Møller velocity" is merely the rate at which the two coordinate distributions sweep through each other, exactly what a flux must count, and it is the combination that the relativistic identity Eq. (16.104) turns into a genuine scalar. The price is a frame caveat. The covariant flux Eq. (16.109) is a Lorentz scalar in every frame, but its identification with the elementary "number density × relative velocity" of Eq. (16.101) requires the beams to be collinear. Collinearity is preserved exactly by boosts *along* the beam axis (and rotations); a boost *transverse* to the beams tilts them, spoils collinearity, and breaks the simple flux picture even though the scalar $F_{\mathrm{M}\varnothing\mathrm{l}}$ keeps its value. This is one reason colliders are built with exactly head-on (collinear) beams: the operational cross section and the invariant formula coincide manifestly precisely in the family of frames — lab, centre-of-momentum, and all boosts along the beam line — reached from one another without tilting the beams.

16.4.4 The centre-of-momentum reduction

The frame in which the flux factor is simplest, and the one Mandelstam kinematics will adopt in the next section, is the centre-of-momentum (CM) frame, where $\mathbf{p}_1 + \mathbf{p}_2 = 0$. Write $\mathbf{p}_1 = -\mathbf{p}_2$ and $|\mathbf{p}_1| = |\mathbf{p}_2| \equiv p_{\mathrm{cm}}$, and recall the Mandelstam

invariant $s \equiv (p_1 + p_2)^2$, which in this frame is $s = (E_1 + E_2)^2$ because the total three-momentum vanishes, so $\sqrt{s} = E_1 + E_2$ is the total CM energy. The velocities are antiparallel, $\mathbf{v}_1 = \mathbf{p}_1/E_1$ and $\mathbf{v}_2 = -\mathbf{p}_1/E_2$, whence

$$|\mathbf{v}_1 - \mathbf{v}_2| = |\mathbf{p}_1| \left(\frac{1}{E_1} + \frac{1}{E_2} \right) = p_{\mathrm{cm}} \, \frac{E_1 + E_2}{E_1 E_2}. \tag{16.110}$$

Substituting into the collinear flux Eq. (16.103), the energies cancel cleanly:

$$F \;=\; 4E_1E_2 \, |\mathbf{v}_1 - \mathbf{v}_2| \;=\; 4E_1E_2 \cdot p_{\mathrm{cm}} \, \frac{E_1 + E_2}{E_1 E_2} \;=\; 4\, p_{\mathrm{cm}} \, (E_1 + E_2) \;=\; 4\, p_{\mathrm{cm}} \sqrt{s}. \tag{16.111}$$

The same result follows from the covariant form Eq. (16.109): in the CM frame $(p_1 \cdot p_2)^2 - m_1^2 m_2^2 = s\, p_{\mathrm{cm}}^2$, an identity we verify in passing by noting that p_{cm} is fixed by the masses and s through $p_{\mathrm{cm}} = \lambda^{1/2}(s, m_1^2, m_2^2)/2\sqrt{s}$ with the Källén triangle function $\lambda(a,b,c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$; the algebra is the content of Eq. (16.108) re-expressed in invariants. The compact CM flux $F = 4 p_{\mathrm{cm}} \sqrt{s}$ is the form we will insert into the two-body phase space of the next section.

16.4.5 The master differential cross section

Collecting Eq. (16.102), Theorem 16.15, and the invariant flux Eq. (16.109), we arrive at the central result of the chapter for scattering.

Theorem 16.18 (Master cross-section formula)**.** *The differential cross section for $1 + 2 \to \{3, 4, \ldots, n\}$ with invariant amplitude $\mathcal{M}$ is*

$$\boxed{ d\sigma \;=\; \frac{|\mathcal{M}|^2}{4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}} \, \mathrm{d}\Pi_n \;=\; \frac{|\mathcal{M}|^2}{F} \, \mathrm{d}\Pi_n } \tag{16.112}$$

with F the Møller flux Eq. (16.109) and $\mathrm{d}\Pi_n$ the Lorentz-invariant phase space Eq. (16.100). The total cross section is obtained by integrating over the final-state phase space, with a symmetry factor $S = \prod_j 1/n_j!$ for each set of n_j identical final particles (Section 16.3.3) to avoid counting the same physical configuration more than once.

Every ingredient on the right is Lorentz invariant — $|\mathcal{M}|^2$ by construction, F by Theorem 16.16, $\mathrm{d}\Pi_n$ by Theorem 16.8 — so $d\sigma$ is a Lorentz scalar, as the operational counting argument demanded. The structure is worth stating plainly because it persists unchanged through the rest of the book: *the entire interaction is in $|\mathcal{M}|^2$; the flux F and the measure $\mathrm{d}\Pi_n$ are pure on-shell kinematics.* Different theories, different processes, even different loop orders change only $\mathcal{M}$; the wrapper Eq. (16.112) is fixed once and for all.

Why dynamics and kinematics separate

The clean factorization is not a calculational convenience but a consequence of how the S-matrix was built. Translation invariance forces the energy–momentum delta and hence the on-shell phase space; relativistic normalization of the asymptotic states fixes the $1/2E$ weights; and the flux is just the relative motion of two such states. None of these knows anything about the Lagrangian. The Lagrangian enters only through the matrix element $\mathcal{M}$ extracted in Eq. (14.163). Cross sections, widths, lifetimes, and angular distributions are therefore one and the same object — $|\mathcal{M}|^2$ integrated against an invariant measure — divided by the appropriate normalization: $2M$ for a decay, F for a collision.

A quick dimensional check confirms the formula. In four dimensions a process with N total external legs has $[\mathcal{M}] = 4 - N$ (each external state of a $2 \to n$ amplitude contributes as the LSZ residues require), the flux has $[F] = 2$, and the n-body measure has $[\mathrm{d}\Pi_n] = 2n - 4$. For $2 \to n$, $N = 2 + n$, so $[|\mathcal{M}|^2/F] + [\mathrm{d}\Pi_n] = 2(2-n) - 2 + (2n-4) = -2$, i.e. an inverse mass squared — an area, exactly as a cross section must be. For the carried ϕ^4 amplitude $\mathcal{M} = -\lambda$ this gives $[\mathcal{M}] = 0$, consistent with $N = 4$.

Worked Example 16.19 (The flux factor in lab and collider frames)**.** To see the master formula's frame independence in action — and to record the two flux factors used constantly in practice — take two particles of masses m_1, m_2 and evaluate F two ways.

Fixed-target (lab) frame. Particle 2 is at rest, $p_2 = (m_2, \mathbf{0})$, and particle 1 has $p_1 = (E_1, \mathbf{p}_1)$ with $|\mathbf{p}_1|^2 = E_1^2 - m_1^2$. The covariant flux Eq. (16.109) gives

$$F_{\text{lab}} = 4\sqrt{(E_1 m_2)^2 - m_1^2 m_2^2} = 4m_2\sqrt{E_1^2 - m_1^2} = 4\,m_2\,|\mathbf{p}_1|\,. \tag{16.113}$$

As a consistency check against the collinear definition, the target velocity is $\mathbf{v}_2 = 0$, so $|\mathbf{v}_1 - \mathbf{v}_2| = |\mathbf{v}_1| = |\mathbf{p}_1|/E_1$, and Eq. (16.103) returns $F = 4E_1 m_2 \cdot |\mathbf{p}_1|/E_1 = 4m_2\,|\mathbf{p}_1|$, agreeing with Eq. (16.113).

Collider (CM) frame. Here Eq. (16.111) gives $F_{\text{cm}} = 4p_{\text{cm}}\sqrt{s}$. The two are not equal numbers — they are evaluated at different beam energies for the same physical reaction — but they are the same Lorentz scalar evaluated in two frames: both equal $4\sqrt{(p_1\cdot p_2)^2 - m_1^2 m_2^2}$. To make the invariance fully explicit, express the lab energy through the invariant s. Since $s = (p_1+p_2)^2 = m_1^2 + m_2^2 + 2p_1\cdot p_2 = m_1^2 + m_2^2 + 2E_1 m_2$ in the lab frame, we have $E_1 = (s - m_1^2 - m_2^2)/2m_2$, and then

$$F_{\text{lab}} = 4m_2\sqrt{E_1^2 - m_1^2} = 4m_2\sqrt{\left(\tfrac{s-m_1^2-m_2^2}{2m_2}\right)^2 - m_1^2} = 2\sqrt{\lambda(s, m_1^2, m_2^2)} = 4p_{\text{cm}}\sqrt{s}, \tag{16.114}$$

using $\lambda(s, m_1^2, m_2^2) = (s - m_1^2 - m_2^2)^2 - 4m_1^2 m_2^2$ and $p_{\rm cm} = \lambda^{1/2}/2\sqrt{s}$. The lab and CM expressions coincide once written in invariants, confirming that F — and therefore $d\sigma$ — is frame independent.

From cross section to event rate. Inverting Theorem 16.14, an experiment with instantaneous luminosity $\mathcal{L}$ (beam-target overlap counted as number per area per time) records events at rate $dN/dt = \mathcal{L}\,\sigma$. The cross section computed from Eq. (16.112) is exactly the proportionality constant a detector measures, the bridge from $|\mathcal{M}|^2$ to a count on a screen.

The high-energy limit of Eq. (16.111) is worth noting for later use: when $s \gg m_1^2, m_2^2$ one has $p_{\rm cm} \to \sqrt{s}/2$, so $F \to 2s$, and the cross section behaves as $d\sigma \to |\mathcal{M}|^2\, \mathrm{d}\Pi_n/2s$. This is the regime in which the next chapter's $e^+e^- \to \mu^+\mu^-$ computation will live.

With the master formula Eq. (16.112) in hand we have a frame-independent recipe that turns any computed amplitude into a cross section. The next section specializes it to a two-body final state, where energy–momentum conservation collapses the phase space onto a single solid angle and the kinematics is captured completely by the two Mandelstam invariants s and t — at which point $d\sigma/d\Omega$ becomes an explicit function of $|\mathcal{M}|^2$ ready to be evaluated on real amplitudes.

16.5 Two-to-two kinematics: Mandelstam variables and physical regions

The master differential cross section assembled in the previous section,

$$\mathrm{d}\sigma = \frac{1}{F}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n, \tag{16.115}$$

with the Møller flux factor F of Eq. (16.109) and the Lorentz-invariant phase-space measure $\mathrm{d}\Pi_n$ of Section 16.2, is valid for any number n of final particles. The overwhelmingly common — and pedagogically central — case is $n = 2$: two particles in, two particles out. Here the kinematics simplifies dramatically, because energy–momentum conservation plus the on-shell conditions leave only *two* free Lorentz-invariant numbers and, in a fixed frame, only the scattering angle and an irrelevant azimuth. This section is pure kinematics: we choose the right invariant coordinates, the Mandelstam variables s, t, u; we express every frame-dependent quantity (centre-of-mass energy, momenta, scattering angle) through them; we reduce Eq. (16.115) to the two standard differential forms $\mathrm{d}\sigma/\mathrm{d}t$ and $\mathrm{d}\sigma/\mathrm{d}\Omega$; and we map out the regions of the (s, t, u) plane in which a given reaction is physical. The dynamics — the actual value of $\mathcal{M}$ — does not enter at all. What we build is the empty stage onto which the carried ϕ^4 amplitude steps in the next section, and onto which every $2 \to 2$ process in the QED chapters will step after it.

Throughout we label the process

$$1(p_1) + 2(p_2) \longrightarrow 3(p_3) + 4(p_4), \qquad p_i^2 = m_i^2, \quad i = 1, 2, 3, 4, \tag{16.116}$$

with all four momenta on shell, p_1, p_2 incoming and p_3, p_4 outgoing, and overall conservation

$$p_1 + p_2 = p_3 + p_4. \tag{16.117}$$

The masses $m_1, \ldots, m_4$ are arbitrary and need not be equal; we keep them general so that the formulae apply equally to the equal-mass ϕ^4 case and to the unequal-mass QED reactions, and only specialize in the worked examples.

16.5.1 The Mandelstam invariants

A relativistic amplitude is a Lorentz scalar built from the external momenta, so it can depend on those momenta only through their invariant pairings $p_i \cdot p_j$. With four momenta there are ten such dot products, but the four on-shell conditions $p_i^2 = m_i^2$ and the four components of the conservation law Eq. (16.117) cut this down sharply. The clean way to organize the survivors is through the three Mandelstam variables.

Definition 16.20 (Mandelstam variables)**.** For the process Eq. (16.116) define the three invariants

$$s \equiv (p_1 + p_2)^2 = (p_3 + p_4)^2, \tag{16.118}$$

$$t \equiv (p_1 - p_3)^2 = (p_2 - p_4)^2, \tag{16.119}$$

$$u \equiv (p_1 - p_4)^2 = (p_2 - p_3)^2. \tag{16.120}$$

The second equality in each line follows from conservation Eq. (16.117): e.g. $p_1 + p_2 = p_3 + p_4$ gives $(p_1 + p_2)^2 = (p_3 + p_4)^2$, and $p_1 - p_3 = p_4 - p_2$ gives $(p_1 - p_3)^2 = (p_2 - p_4)^2$.

Physically, s is the squared total energy in the centre-of-mass (CM) frame — it is the natural "how much energy is available" variable — while t and u are squared momentum transfers: t measures how much four-momentum flows from leg 1 to leg 3, and u how much flows from leg 1 to leg 4. The three are not independent.

Proposition 16.21 (Mandelstam constraint)**.** *The Mandelstam variables satisfy*

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2. \tag{16.121}$$

Proof. Expand the three definitions using $p_i^2 = m_i^2$:

$$s = (p_1 + p_2)^2 = p_1^2 + p_2^2 + 2\, p_1 \cdot p_2 = m_1^2 + m_2^2 + 2\, p_1 \cdot p_2,$$

$$t = (p_1 - p_3)^2 = p_1^2 + p_3^2 - 2\,p_1 \cdot p_3 = m_1^2 + m_3^2 - 2\,p_1 \cdot p_3,$$
$$u = (p_1 - p_4)^2 = p_1^2 + p_4^2 - 2\,p_1 \cdot p_4 = m_1^2 + m_4^2 - 2\,p_1 \cdot p_4.$$

Adding,

$$s + t + u = 3m_1^2 + m_2^2 + m_3^2 + m_4^2 + 2\,p_1 \cdot (p_2 - p_3 - p_4). \tag{16.122}$$

Conservation Eq. (16.117) rearranges to $p_2 - p_3 - p_4 = -p_1$, so the last term is

$$2\,p_1 \cdot (-p_1) = -2\,p_1^2 = -2m_1^2. \tag{16.123}$$

Substituting Eq. (16.123) into Eq. (16.122) cancels two units of m_1^2, leaving exactly Eq. (16.121). □

The constraint Eq. (16.121) is the single most useful identity in two-body kinematics: it says that of the three invariants only two are independent, and it lets one trade u for s and t at will. An amplitude $\mathcal{M}(s,t,u)$ is therefore really a function of two variables, conventionally taken to be (s,t), with $u = \sum_i m_i^2 - s - t$ understood.

Notation Watch

The symbol λ does double duty in this part of the book: it is the ϕ^4 coupling of ϕ^4 theory (a dimensionless number, no arguments) and, from here on, the Källén triangle *function* $\lambda(a,b,c)$ defined in Theorem 16.22 (always written with three arguments). The two are never confused in practice because the function always carries its arguments; where both appear in the next section we keep the coupling as the bare symbol λ and the Källén function as $\lambda(\cdot,\cdot,\cdot)$.

16.5.2 Centre-of-mass kinematics and the Källén function

To turn the invariants into the lab-readable quantities of an angular distribution we work in the CM frame, defined by $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{0}$. The recurring algebraic object is the following symmetric polynomial.

Definition 16.22 (Källén triangle function)**.**

$$\lambda(a,b,c) \equiv a^2 + b^2 + c^2 - 2ab - 2bc - 2ca. \tag{16.124}$$

It is symmetric in its three arguments and admits the factorized forms

$$\lambda(a,b,c) = a^2 - 2a(b+c) + (b-c)^2 \tag{16.125}$$
$$= \big(a - (\sqrt{b} + \sqrt{c})^2\big)\big(a - (\sqrt{b} - \sqrt{c})^2\big) \quad (b,c \geq 0). \tag{16.126}$$

The factorized form Eq. (16.126) is verified by multiplying out: $(a-(\sqrt{b}+\sqrt{c})^2)(a-(\sqrt{b}-\sqrt{c})^2) = a^2 - a[(\sqrt{b}+\sqrt{c})^2+(\sqrt{b}-\sqrt{c})^2] + (\sqrt{b}+\sqrt{c})^2(\sqrt{b}-\sqrt{c})^2 = a^2 - 2a(b+c) + (b-c)^2$, which is Eq. (16.125). The name "triangle function" is geometric: $-\lambda(a^2, b^2, c^2)$ is sixteen times the squared area of a triangle with sides a, b, c, so λ is negative precisely when a, b, c can form a triangle. We will see this sign control the physical thresholds.

Now fix the CM frame. Write the incoming momenta as

$$p_1 = (E_1,\, \mathbf{p}_i), \qquad p_2 = (E_2,\, -\mathbf{p}_i), \qquad |\mathbf{p}_i| \equiv p_{\rm cm}, \tag{16.127}$$

so that $\mathbf{p}_1 + \mathbf{p}_2 = \mathbf{0}$ and the total four-momentum is $(E_1+E_2, \mathbf{0})$. By Eq. (16.118),

$$s = (E_1+E_2)^2, \qquad \text{hence} \qquad \sqrt{s} = E_1 + E_2, \tag{16.128}$$

identifying $\sqrt{s}$ as the total CM energy. The two on-shell conditions $E_1^2 - p_{\rm cm}^2 = m_1^2$, $E_2^2 - p_{\rm cm}^2 = m_2^2$ give $E_1^2 - E_2^2 = m_1^2 - m_2^2$, and dividing by $E_1 + E_2 = \sqrt{s}$,

$$E_1 - E_2 = \frac{m_1^2 - m_2^2}{\sqrt{s}}. \tag{16.129}$$

Adding and subtracting Eq. (16.128) and Eq. (16.129) solves for the individual energies,

$$E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}, \qquad E_2 = \frac{s + m_2^2 - m_1^2}{2\sqrt{s}}. \tag{16.130}$$

The common momentum magnitude then follows from $p_{\rm cm}^2 = E_1^2 - m_1^2$:

$$p_{\rm cm}^2 = \frac{(s + m_1^2 - m_2^2)^2}{4s} - m_1^2 = \frac{(s + m_1^2 - m_2^2)^2 - 4s\, m_1^2}{4s}. \tag{16.131}$$

The numerator is exactly the Källén function. Expanding,

$$(s+m_1^2-m_2^2)^2 - 4s\, m_1^2 = s^2 + m_1^4 + m_2^4 - 2sm_1^2 - 2sm_2^2 - 2m_1^2 m_2^2 = \lambda(s, m_1^2, m_2^2), \tag{16.132}$$

where the first equality multiplies out $(s + m_1^2 - m_2^2)^2 = s^2 + m_1^4 + m_2^4 + 2sm_1^2 - 2sm_2^2 - 2m_1^2 m_2^2$ and subtracts $4sm_1^2$, and the second is Eq. (16.124) with arguments (s, m_1^2, m_2^2). Hence the incoming CM momentum is

$$\boxed{|\mathbf{p}_i| = \frac{\sqrt{\lambda(s, m_1^2, m_2^2)}}{2\sqrt{s}}} \tag{16.133}$$

By identical algebra applied to the outgoing pair $3, 4$ — which in the CM frame are back-to-back, $p_3 = (E_3, \mathbf{p}_f)$, $p_4 = (E_4, -\mathbf{p}_f)$ with the *same* total energy $\sqrt{s}$ — the outgoing CM momentum is

$$|\mathbf{p}_f| = \frac{\sqrt{\lambda(s, m_3^2, m_4^2)}}{2\sqrt{s}}, \qquad E_3 = \frac{s + m_3^2 - m_4^2}{2\sqrt{s}}, \qquad E_4 = \frac{s + m_4^2 - m_3^2}{2\sqrt{s}}. \tag{16.134}$$

The factorized form Eq. (16.126) now reads off the kinematic thresholds. The initial momentum $|\mathbf{p}_i|$ is real only when $\lambda(s, m_1^2, m_2^2) \geq 0$, i.e. when $s \geq (m_1+m_2)^2$ (the other factor, $s-(m_1-m_2)^2$, is then automatically positive). This is the statement that the collision needs at least enough energy to make the two incoming particles at rest. Likewise $|\mathbf{p}_f|$ is real only above the production threshold $s \geq (m_3+m_4)^2$: one cannot create the final state below its rest mass. A physical $2 \to 2$ reaction therefore requires

$$s \geq \max\big[(m_1+m_2)^2,\,(m_3+m_4)^2\big]. \tag{16.135}$$

16.5.3 The scattering angle and the physical t-range

The one remaining degree of freedom is the CM scattering angle θ between the incoming particle 1 and the outgoing particle 3,

$$\mathbf{p}_i \cdot \mathbf{p}_f = |\mathbf{p}_i|\,|\mathbf{p}_f| \cos\theta. \tag{16.136}$$

We express t through it. From the expansion of t in the proof of Theorem 16.21, $t = m_1^2 + m_3^2 - 2\,p_1 \cdot p_3$, and in the CM frame $p_1 \cdot p_3 = E_1E_3 - \mathbf{p}_i \cdot \mathbf{p}_f = E_1E_3 - |\mathbf{p}_i||\mathbf{p}_f|\cos\theta$. Therefore

$$t = m_1^2 + m_3^2 - 2E_1E_3 + 2|\mathbf{p}_i||\mathbf{p}_f|\cos\theta. \tag{16.137}$$

This is the linear relation between t and $\cos\theta$ that converts an angular distribution into a distribution in the invariant t. Differentiating at fixed s (which fixes $E_1, E_3, |\mathbf{p}_i|, |\mathbf{p}_f|$) gives the Jacobian we will need,

$$\boxed{\mathrm{d}t = 2|\mathbf{p}_i||\mathbf{p}_f|\,\mathrm{d}\cos\theta} \tag{16.138}$$

Because $\cos\theta$ runs over $[-1, 1]$, the physical region of t at fixed s is the closed interval between its endpoints at forward ($\cos\theta = +1$) and backward ($\cos\theta = -1$) scattering:

$$t_+ = m_1^2 + m_3^2 - 2E_1E_3 + 2|\mathbf{p}_i||\mathbf{p}_f| \qquad (\text{forward},\ \theta = 0), \tag{16.139}$$

$$t_- = m_1^2 + m_3^2 - 2E_1E_3 - 2|\mathbf{p}_i||\mathbf{p}_f| \qquad (\text{backward},\ \theta = \pi), \tag{16.140}$$

so that $t_- \leq t \leq t_+$, with $t_+ - t_- = 4|\mathbf{p}_i||\mathbf{p}_f|$. Every quantity on the right of Eq. (16.139)–Eq. (16.140) is a known function of s and the masses through Eq. (16.130), Eq. (16.133) and Eq. (16.134), so the t-range is fixed once s is. The corresponding u-endpoints are obtained from the constraint $u = \sum_i m_i^2 - s - t$, which sends $t_+ \leftrightarrow u_-$.

Caution

In the physical s-channel region of an elastic (or equal-mass, $m_1 = m_3$) reaction the momentum transfer t is non-positive. With $m_1 = m_3$ one has $E_1 = E_3$, $|\mathbf{p}_i| = |\mathbf{p}_f|$, and Eq. (16.139) gives $t_+ = 0$ at $\theta = 0$: forward scattering transfers no momentum, while every other angle gives $t < 0$. A positive t in such an elastic reaction is a kinematic error. (For unequal-mass inelastic kinematics this is not universal: Eq. (16.139) can be rewritten $t_+ = (E_1 - E_3)^2 - (|\mathbf{p}_i| - |\mathbf{p}_f|)^2$, which is positive whenever $|E_1 - E_3| > ||\mathbf{p}_i| - |\mathbf{p}_f||$.) This sign is exactly what makes t a useful expansion variable near the forward peak and what the crossing analysis of Chapter 18 will exploit.

16.5.4 The two standard differential cross sections

We now specialize the master formula Eq. (16.115) to $n = 2$. Two ingredients enter: the flux factor and the two-body phase space.

Flux. The covariant Møller flux of Eq. (16.109) is $F = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}$. From the expansion $s = m_1^2 + m_2^2 + 2p_1 \cdot p_2$ we have $p_1 \cdot p_2 = \frac{1}{2}(s - m_1^2 - m_2^2)$, so

$$(p_1 \cdot p_2)^2 - m_1^2 m_2^2 = \frac{(s - m_1^2 - m_2^2)^2 - 4m_1^2 m_2^2}{4} = \frac{\lambda(s, m_1^2, m_2^2)}{4}, \tag{16.141}$$

the second equality being Eq. (16.124) once more (expand $(s - m_1^2 - m_2^2)^2$ and subtract $4m_1^2 m_2^2$). Hence the flux factor in invariant form is

$$F = 4\sqrt{\frac{\lambda(s, m_1^2, m_2^2)}{4}} = 2\sqrt{\lambda(s, m_1^2, m_2^2)} = 4\sqrt{s}\,|\mathbf{p}_i|, \tag{16.142}$$

where the last equality used Eq. (16.133). It is worth pausing on this: the flux is $2\sqrt{\lambda}$, a clean Lorentz invariant, and equals $4\sqrt{s}\,|\mathbf{p}_i|$ in the CM frame — the form we use to extract the angular distribution.

Phase space. The two-body Lorentz-invariant phase-space measure derived in Eq. (16.47) is, in the CM frame,

$$\mathrm{d}\Pi_2 = \frac{1}{16\pi^2}\,\frac{|\mathbf{p}_f|}{\sqrt{s}}\,\mathrm{d}\Omega, \tag{16.143}$$

with $\mathrm{d}\Omega = \mathrm{d}\cos\theta\,\mathrm{d}\varphi$ the solid-angle element of the outgoing particle 3. Assembling Eq. (16.142) and Eq. (16.143) in Eq. (16.115),

$$\mathrm{d}\sigma = \frac{1}{4\sqrt{s}\,|\mathbf{p}_i|}\,|\mathcal{M}|^2\,\frac{1}{16\pi^2}\,\frac{|\mathbf{p}_f|}{\sqrt{s}}\,\mathrm{d}\Omega = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2\,\mathrm{d}\Omega. \tag{16.144}$$

This is the first standard form.

Theorem 16.23 (CM angular cross section)**.** *For $2 \to 2$ scattering, the differential cross section with respect to the CM solid angle of an outgoing particle is*

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{CM}} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2\,, \tag{16.145}$$

with $|\mathbf{p}_i|, |\mathbf{p}_f|$ given by Eq. (16.133), Eq. (16.134).

The ratio $|\mathbf{p}_f|/|\mathbf{p}_i|$ is the kinematic phase-space penalty for producing a heavier (or lighter) final state than the initial one; for elastic scattering, where the final masses equal the initial ones, $|\mathbf{p}_f| = |\mathbf{p}_i|$ and it drops out.

The second form is fully invariant. Most amplitudes have no azimuthal dependence — for spinless particles, or for spin-summed/averaged amplitudes with unpolarized beams, $|\mathcal{M}|^2$ depends on θ but not φ — so the φ integration in Eq. (16.144) is trivial, $\int_0^{2\pi} \mathrm{d}\varphi = 2\pi$, and $\mathrm{d}\Omega \to 2\pi\,\mathrm{d}\cos\theta$. Using the Jacobian Eq. (16.138) to trade $\cos\theta$ for t,

$$\begin{aligned}\mathrm{d}\sigma &= \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2\,(2\pi)\,\mathrm{d}\cos\theta = \frac{1}{32\pi s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2\,\frac{\mathrm{d}t}{2|\mathbf{p}_i||\mathbf{p}_f|} \\ &= \frac{1}{64\pi s}\,\frac{1}{|\mathbf{p}_i|^2}\,|\mathcal{M}|^2\,\mathrm{d}t.\end{aligned} \tag{16.146}$$

Finally insert $|\mathbf{p}_i|^2 = \lambda(s, m_1^2, m_2^2)/(4s)$ from Eq. (16.133), so that $64\pi s\,|\mathbf{p}_i|^2 = 16\pi\,\lambda(s, m_1^2, m_2^2)$:

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{|\mathcal{M}|^2}{16\pi\,\lambda(s, m_1^2, m_2^2)}} \tag{16.147}$$

Theorem 16.24 (Invariant cross section)**.** *When $|\mathcal{M}|^2$ is independent of the azimuthal angle, the $2 \to 2$ differential cross section in the invariant momentum transfer is Eq. (16.147), with t ranging over $[t_-, t_+]$ of Eq. (16.139)–Eq. (16.140).*

Equation Eq. (16.147) is the workhorse of the next section. Both sides are Lorentz invariant: $|\mathcal{M}|^2$ is a scalar, t is an invariant, and $\lambda(s, m_1^2, m_2^2)$ depends only on invariants. The cross section $\mathrm{d}\sigma/\mathrm{d}t$ is therefore the same in every frame — the lab frame, the CM frame, a fixed-target frame — which is precisely why it, rather than $\mathrm{d}\sigma/\mathrm{d}\Omega$, is the form in which experimental and theoretical results are most often quoted and compared. The total cross section is the integral

$$\sigma = \int_{t_-}^{t_+} \frac{|\mathcal{M}(s,t)|^2}{16\pi\,\lambda(s, m_1^2, m_2^2)}\,\mathrm{d}t, \tag{16.148}$$

a single integral over the one physical variable left at fixed s.

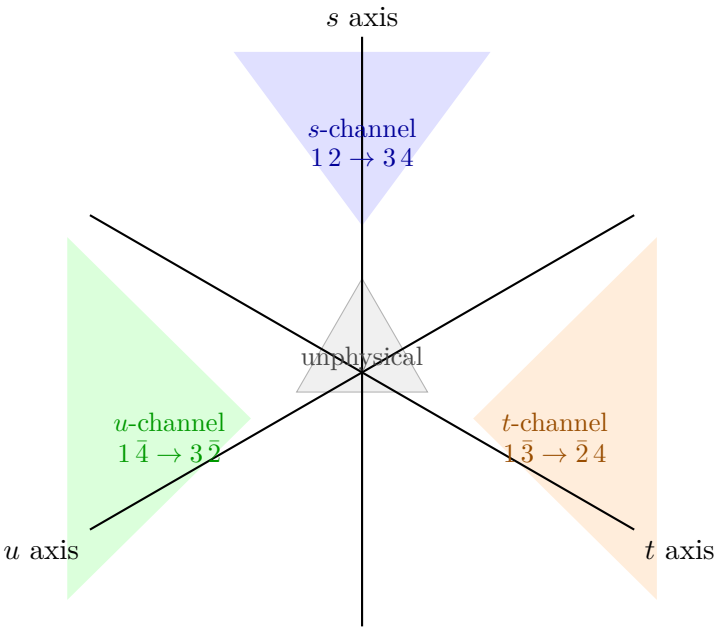


Figure 16.5: The Mandelstam plane for $2 \to 2$ scattering (drawn for the equal-mass case, where the three axes meet symmetrically). The constraint $s+t+u = \sum_i m_i^2$ confines all kinematics to this plane. A single analytic amplitude $\mathcal{M}(s,t,u)$ lives on the whole plane; the three shaded wedges are the physical regions of the s-, t- and u-channel reactions, related by crossing. The central (gray) triangle is kinematically forbidden in every channel.

16.5.5 The Mandelstam plane and the three channels

The constraint $s+t+u = \sum_i m_i^2$ of Eq. (16.121) says that the three invariants live on a two-dimensional plane in (s,t,u)-space. It is illuminating to plot them on equal footing. Choose the symmetric "Mandelstam plane" in which the perpendicular distances from a point to three lines meeting at 120° represent s, t, u; the constraint $s+t+u = \text{const}$ is then automatic (the sum of signed distances to the three sides of an equilateral triangle is constant). On this plane a single analytic function $\mathcal{M}$ is defined everywhere, but only certain regions correspond to physical reactions.

There are three physical reactions sharing the same amplitude, distinguished by which invariant plays the role of the total CM energy squared:

- the *s-channel* reaction $1+2 \to 3+4$, physical when $s \geq \max[(m_1+m_2)^2, (m_3+m_4)^2]$ and $t \in [t_-, t_+]$, as in Eq. (16.135) and Section 16.5.3;
- the *t-channel* reaction $1+\bar{3} \to \bar{2}+4$, obtained by moving particles 3 and 2 to the opposite side as antiparticles, physical when t exceeds the

corresponding threshold;

- the *u-channel* reaction $1+\bar{4}\to 3+\bar{2}$, physical when u exceeds its threshold.

These three physical regions are the disjoint shaded wedges of Fig. 16.5; between them lies the unphysical central region where none of s,t,u is large enough to be a physical energy. The remarkable fact — the content of *crossing symmetry* — is that the very same function $\mathcal{M}(s,t,u)$, computed as an analytic function of its arguments, describes all three reactions: one obtains the t-channel amplitude from the s-channel one by analytic continuation in the invariants, swapping the appropriate incoming and outgoing legs into antiparticles. At the level of this chapter we only *state* this; it rests on the analyticity of the amplitude in the cut s- and t-planes, whose branch points sit at the physical thresholds (the same thresholds Eq. (16.135) that bounded the physical regions). The full statement and proof of crossing, and the precise rule for continuing across the cuts, are developed in Chapter 18; here we have simply laid out the analytic geometry — the plane, the three wedges, the thresholds — that crossing acts on.

Physical Insight

The same diagram read three ways is the first concrete payoff of the "amplitude" viewpoint announced in Eq. (14.163): $\mathcal{M}$ is not three different objects for three different experiments but one analytic function on the Mandelstam plane. A pole of $\mathcal{M}$ at $s=M^2$ is a resonance exchanged in the s-channel reaction; the *same* pole, viewed from the t-channel wedge, is a particle exchanged as a force carrier. Energy and momentum transfer are continuations of one another. This is the structural fact that the dispersion relations of Chapter 18 and the crossing equations of the bootstrap turn into quantitative tools.

16.5.6 Worked kinematics

Worked Example 16.25 (Equal masses: the ϕ^4 stage)**.** Set all four masses equal, $m_1=m_2=m_3=m_4=m$. This is the kinematics the carried $\phi\phi\to\phi\phi$ amplitude of Eq. (15.59) will use in the next section, so we work it out completely.

The constraint Eq. (16.121) becomes

$$s+t+u=4m^2. \tag{16.149}$$

The Källén function collapses neatly: from Eq. (16.124),

$$\lambda(s,m^2,m^2)=s^2+2m^4-2sm^2-2sm^2-2m^4=s^2-4sm^2=s\,(s-4m^2), \tag{16.150}$$

consistent with the factorized form Eq. (16.126), $\lambda(s,m^2,m^2)=(s-4m^2)(s-0)$. The CM momenta Eq. (16.133), Eq. (16.134) are equal (initial and final masses

coincide),

$$|\mathbf{p}_i| = |\mathbf{p}_f| = \frac{\sqrt{s(s-4m^2)}}{2\sqrt{s}} = \frac{\sqrt{s-4m^2}}{2}, \tag{16.151}$$

real precisely above threshold $s \geq 4m^2$ — the energy to make two particles of mass m. The energies are all $E_1 = E_2 = E_3 = E_4 = \sqrt{s}/2$ by Eq. (16.130).

The t-endpoints follow from Eq. (16.139)–Eq. (16.140) with $E_1 = E_3 = \sqrt{s}/2$ and $|\mathbf{p}_i| = |\mathbf{p}_f|$:

$$t_+ = 2m^2 - 2 \cdot \frac{s}{4} + 2 \cdot \frac{s-4m^2}{4} = 2m^2 - \frac{s}{2} + \frac{s-4m^2}{2} = 0, \tag{16.152}$$

$$t_- = 2m^2 - \frac{s}{2} - \frac{s-4m^2}{2} = 2m^2 - \frac{s}{2} - \frac{s}{2} + 2m^2 = 4m^2 - s. \tag{16.153}$$

So in the equal-mass s-channel,

$$t \in [\,4m^2 - s,\ 0\,], \qquad u = 4m^2 - s - t \in [\,4m^2 - s,\ 0\,], \tag{16.154}$$

both momentum transfers ranging from the backward value $4m^2 - s$ up to 0 at the relevant forward/backward limit. The angular form Eq. (16.145) simplifies, with $|\mathbf{p}_f|/|\mathbf{p}_i| = 1$, to

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{CM}} = \frac{|\mathcal{M}|^2}{64\pi^2 s}, \tag{16.155}$$

the cleanest possible $2 \to 2$ cross section. This is exactly the stage onto which $\mathcal{M} = -\lambda$ from Eq. (15.59) steps in Section 16.6; everything dynamical there is the constant $|\mathcal{M}|^2 = \lambda^2$, and the angular distribution is flat because the stage Eq. (16.155) is.

Worked Example 16.26 (Massless initial state: the $e^+e^- \to \mu^+\mu^-$ frame)**.** To show the same kinematics serving QED, take the high-energy limit of $e^+e^- \to \mu^+\mu^-$: incoming electron and positron with $m_1 = m_2 = m_e \to 0$, outgoing muon pair with $m_3 = m_4 = m_\mu$. (The full amplitude, with spin sums, is the business of Chapter 17; here we supply only the s, t, u frame it will be evaluated in.)

With $m_e \to 0$, Eq. (16.133) and Eq. (16.150)-type algebra give the incoming momentum $|\mathbf{p}_i| = \sqrt{\lambda(s,0,0)}/(2\sqrt{s}) = \sqrt{s^2}/(2\sqrt{s}) = \sqrt{s}/2$, while the outgoing momentum is set by the muon mass,

$$|\mathbf{p}_f| = \frac{\sqrt{\lambda(s, m_\mu^2, m_\mu^2)}}{2\sqrt{s}} = \frac{\sqrt{s-4m_\mu^2}}{2}, \qquad s \geq 4m_\mu^2. \tag{16.156}$$

The energies are $E_1 = E_2 = \sqrt{s}/2$ (massless) and $E_3 = E_4 = \sqrt{s}/2$ (the pair shares the energy equally by Eq. (16.130) with $m_3 = m_4$). The momentum transfer Eq. (16.137) becomes, with $m_1 \to 0$,

$$t = m_\mu^2 - 2 \cdot \frac{\sqrt{s}}{2} \cdot \frac{\sqrt{s}}{2} + 2 \cdot \frac{\sqrt{s}}{2} \cdot \frac{\sqrt{s-4m_\mu^2}}{2} \cos\theta = m_\mu^2 - \frac{s}{2} + \frac{\sqrt{s}}{2}\sqrt{s-4m_\mu^2}\,\cos\theta, \tag{16.157}$$

and the constraint Eq. (16.121) gives $u = 2m_\mu^2 - s - t = m_\mu^2 - \frac{s}{2} - \frac{\sqrt{s}}{2}\sqrt{s - 4m_\mu^2}\cos\theta$. In the ultra-relativistic limit $s \gg m_\mu^2$ these collapse to the textbook forms $t \to -\frac{s}{2}(1 - \cos\theta)$, $u \to -\frac{s}{2}(1 + \cos\theta)$, with $s + t + u \to 0$ — the all-massless skeleton on which the QED angular distribution $\mathrm{d}\sigma/\mathrm{d}\Omega \propto (1 + \cos^2\theta)$ will be built. The general cross section Eq. (16.145) carries the phase-space factor $|\mathbf{p}_f|/|\mathbf{p}_i| = \sqrt{1 - 4m_\mu^2/s}$, the velocity of the produced muons, which suppresses the rate near the production threshold $s = 4m_\mu^2$ and tends to 1 at high energy.

The two worked cases exhibit the thread-neutrality promised at the outset: the equal-mass formulae feed ϕ^4 theory directly in the next section, while the massless-initial-state formulae are the frame in which QED's first cross section is computed in Chapter 17. With $|\mathbf{p}_i|, |\mathbf{p}_f|$, the t-range, and the two differential forms Eq. (16.145) and Eq. (16.147) all expressed in s, t, u and the masses, the stage is fully set: the next section takes the carried ϕ^4 amplitude $\mathcal{M} = -\lambda$ from Eq. (15.59), drops it into Eq. (16.155) and Eq. (16.148), and reads off the first physical cross section of the book.

16.6 Worked computation: $\phi\phi \to \phi\phi$ scattering in $\lambda\phi^4$

Every formula in this chapter has been built to be fed an amplitude. The two-to-two machinery of Section 16.5 converted the master cross-section formula into a statement about $|\mathcal{M}|^2$ as a function of the Mandelstam invariants in the centre-of-mass frame; the only thing still missing is an actual $\mathcal{M}$ to put in. We have one. ϕ^4 theory has carried a single real scalar with $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ from its classical action all the way to a momentum-space Feynman amplitude, and Chapter 15 produced, by pure diagrammatics, the tree-level two-to-two amplitude. This section closes the loop opened at the start of Part IV: we take that amplitude, square it, integrate it against the invariant phase space, divide by the flux, and read off a number a detector could in principle report. It is the first *physical* number ϕ^4 theory produces — not an off-shell correlator, not an amputated diagram, but a cross section — and the calculation is the exact template Chapter 17 will reuse for $e^+e^- \to \mu^+\mu^-$ once spin sums supply the analogous $\overline{|\mathcal{M}|^2}$.

16.6.1 The amplitude and its square

Recall the carried result. The connected, amputated, on-shell two-to-two amplitude of the ϕ^4 theory, defined through $\langle f|iT|i\rangle = (2\pi)^4\delta^4(p_f - p_i)\, i\mathcal{M}$ as in Eq. (14.163), was computed at tree level in Theorem 15.19 and is

$$\mathcal{M}(\phi\phi \to \phi\phi) = -\lambda + O(\lambda^2), \tag{16.158}$$

a pure constant. As emphasised in Chapter 15, this constancy is the signature of a *contact* interaction: with a single quartic vertex and no internal line, there is no propagator to inject a momentum-dependent denominator, so $\mathcal{M}$ depends on neither the centre-of-mass energy nor the scattering angle — it is the same $-\lambda$ in the s-, t-, and u-channels. The diagram itself carried symmetry factor $S = 1$ (the 4! contractions of the external legs into the vertex cancel the $1/4!$ in the coupling), by the orbit–stabiliser count of Theorem 15.22; that combinatorial factor is a property of the *amplitude*, and is not to be confused with the identical-final-particle factor that will appear below in the *phase-space integral*, which is a quite different bookkeeping. The squared amplitude that the cross-section formula wants is therefore the simplest object in the chapter,

$$|\mathcal{M}|^2 = \lambda^2 + O(\lambda^3), \tag{16.159}$$

constant over the entire physical region. There is no spin to average or sum over — the field is a real scalar — so the bar that will decorate $\overline{|\mathcal{M}|^2}$ in Chapter 17 is absent here, and Eq. (16.159) is the whole of the dynamics. Everything that follows is kinematics.

16.6.2 Kinematics: the centre-of-mass momentum and the threshold

The process is elastic and all four particles share the mass m, which makes the kinematics maximally symmetric. In the centre-of-mass frame the two incoming particles carry equal and opposite three-momenta of magnitude $|\mathbf{p}_i|$, the two outgoing particles equal and opposite momenta of magnitude $|\mathbf{p}_f|$, and each particle has energy $\sqrt{s}/2$ where $s = (p_1 + p_2)^2 = E_{\mathrm{cm}}^2$ is the squared total energy. Energy conservation $E_1 + E_2 = E_3 + E_4 = \sqrt{s}$ with all masses equal forces

$$|\mathbf{p}_i| = |\mathbf{p}_f| \equiv |\mathbf{p}| = \frac{\sqrt{s}}{2}\sqrt{1 - \frac{4m^2}{s}}\,, \tag{16.160}$$

the equal-mass specialisation of the general centre-of-mass momentum recorded in Eq. (16.133). The radical

$$\beta \equiv \sqrt{1 - \frac{4m^2}{s}} \tag{16.161}$$

is the speed of each particle in the centre-of-mass frame (indeed $\beta = |\mathbf{p}|/(\sqrt{s}/2)$), and it is the *threshold factor*: the momentum Eq. (16.160) is real only for $s \geq 4m^2$, the statement that two particles each of mass m cannot be produced (or, here, scattered into a back-to-back final state) unless the collision supplies at least their combined rest energy $2m$. For $s < 4m^2$ there is no physical final state and the cross section is zero; at $s = 4m^2$ the particles emerge at rest and $\beta \to 0$;

for $s \gg 4m^2$ the factor $\beta \to 1$ and the masses become irrelevant. We will see this factor enter the calculation twice — once through the incoming flux, once through the outgoing phase space — and then cancel, leaving a result that is in fact independent of m above threshold.

16.6.3 The differential cross section

The two-to-two specialisation of the master formula, derived in Section 16.5, gives the centre-of-mass differential cross section as

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{cm}} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2, \tag{16.162}$$

the relation Eq. (16.162), in which the dynamical factor $|\mathcal{M}|^2$ is cleanly separated from the purely kinematic prefactor. For our elastic equal-mass process the momentum ratio is unity by Eq. (16.160), and substituting the constant Eq. (16.159) gives at once

$$\boxed{\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{cm}} = \frac{\lambda^2}{64\pi^2 s}} \tag{16.163}$$

to leading order in λ. Three features are worth naming immediately. First, the distribution is *isotropic*: the scattered particles emerge with equal probability in every direction, with no preferred axis. This is the angular fingerprint of the contact interaction — because $\mathcal{M}$ carries no t- or u-dependence, Eq. (16.163) carries no $\cos\theta$-dependence, in sharp contrast to the forward-peaked Rutherford-like distributions that exchange diagrams produce. Second, the result is *independent of the mass* m: the threshold factor β has cancelled between $|\mathbf{p}_f|$ and $|\mathbf{p}_i|$ in Eq. (16.162), so above threshold the differential cross section depends on the kinematics only through the overall $1/s$. Third, $\mathrm{d}\sigma/\mathrm{d}\Omega$ has dimension $[\text{mass}]^{-2}$, as a cross section (an area) must: λ is dimensionless in four dimensions and s carries $[\text{mass}]^2$, so the right-hand side scales as $[\text{mass}]^{-2}$. We return to this dimensional check below.

16.6.4 The total cross section and the identical-particle factor

Integrating Eq. (16.163) over solid angle would, naively, give $4\pi \cdot \lambda^2/(64\pi^2 s) = \lambda^2/(16\pi s)$. But this overcounts. The two particles in the final state are *identical* real scalars, so the configuration in which one emerges at angles (θ, φ) and its partner at the antipodal $(\pi - \theta, \varphi + \pi)$ is the very same physical final state as the one with the labels exchanged. The unrestricted integral over the full sphere counts each distinct two-particle state twice. The total cross section is therefore the angular integral divided by the symmetry number $2! = 2$ of the identical pair,

$$\sigma = \frac{1}{2}\int \mathrm{d}\Omega\, \left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{cm}} = \frac{1}{2}\cdot 4\pi \cdot \frac{\lambda^2}{64\pi^2 s} = \boxed{\frac{\lambda^2}{32\pi s}} \tag{16.164}$$

to leading order. The factor $\frac{1}{2}$ here is genuinely a *final-state* combinatorial factor, the Bose statistics of the indistinguishable outgoing pair, and it is logically independent of the diagram symmetry factor $S = 1$ that already lives inside $\mathcal{M}$. Equivalently, one may keep the $\frac{1}{2}$ out front and restrict the integration to a single hemisphere (half the sphere covers every distinct final configuration exactly once); the two prescriptions give the same $\lambda^2/(32\pi s)$. The full pipeline — amplitude, square, flux, phase space, angular integration with the identical-particle factor, and the explicit cancellation of the threshold radical — is carried end to end in Theorem 16.27.

Worked Example 16.27 (The $\phi\phi \to \phi\phi$ cross section from $\mathcal{M} = -\lambda$, every step)**.** We assemble σ directly from the invariant master formula of Section 16.4.5, keeping the mass m explicit so that the threshold factor's appearance and cancellation are visible, and only then taking the massless limit.

Step 1 — the squared amplitude. From the carried tree result Eq. (16.158), $\mathcal{M} = -\lambda$, so $|\mathcal{M}|^2 = \lambda^2$, a constant independent of s, t, u and of the scattering angle. No spin sum is needed.

Step 2 — flux and phase space, with the threshold factor. The master formula for a $2 \to 2$ cross section reads $\mathrm{d}\sigma = \dfrac{1}{F}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_2$, with the Lorentz-invariant flux $F = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2 m_2^2}$ and $\mathrm{d}\Pi_2$ the two-body phase-space measure. In the centre-of-mass frame with equal masses the flux is

$$F = 4\,|\mathbf{p}_i|\,\sqrt{s} = 4\cdot\frac{\sqrt{s}}{2}\beta\cdot\sqrt{s} = 2s\,\beta, \tag{16.165}$$

using Eqs. (16.160) and (16.161); the threshold factor β enters here through the incoming momentum. The integrated two-body phase space, from Eq. (16.52), is

$$\int \mathrm{d}\Pi_2 = \int \frac{|\mathbf{p}_f|}{16\pi^2\sqrt{s}}\,\mathrm{d}\Omega = \frac{|\mathbf{p}_f|}{16\pi^2\sqrt{s}}\cdot 4\pi = \frac{|\mathbf{p}_f|}{4\pi\sqrt{s}} = \frac{\beta}{8\pi}, \tag{16.166}$$

where the angular integral is trivial because nothing in $\mathrm{d}\Pi_2$ or in $|\mathcal{M}|^2$ depends on direction, and the last equality again used $|\mathbf{p}_f| = (\sqrt{s}/2)\beta$. The threshold factor β enters a second time, now through the outgoing phase space.

Step 3 — assemble, with the identical-particle factor. Combining Eqs. (16.165) and (16.166) and inserting the constant $|\mathcal{M}|^2 = \lambda^2$, the unrestricted integral is $|\mathcal{M}|^2\int\mathrm{d}\Pi_2/F = \lambda^2(\beta/8\pi)/(2s\beta) = \lambda^2/(16\pi s)$. The two appearances of β cancel exactly: *the threshold factor from the flux is undone by the threshold factor from the phase space*, a cancellation special to elastic equal-mass scattering. Dividing by the identical-final-particle symmetry number $2! = 2$,

$$\sigma = \frac{1}{2}\cdot\frac{1}{F}\,|\mathcal{M}|^2\int\mathrm{d}\Pi_2 = \frac{1}{2}\cdot\frac{\lambda^2}{16\pi s} = \frac{\lambda^2}{32\pi s}, \tag{16.167}$$

reproducing Eq. (16.164). The differential form Eq. (16.163) follows by stripping the angular integral and the $\frac{1}{2}$ before integrating: $\mathrm{d}\sigma/\mathrm{d}\Omega = |\mathcal{M}|^2\,(|\mathbf{p}_f|/|\mathbf{p}_i|)/(64\pi^2 s) = \lambda^2/(64\pi^2 s)$, isotropic.

Step 4 — the massive-to-massless reduction. Equation Eq. (16.167) carries no residual β, hence no residual m: it holds for every $s > 4m^2$ exactly. Taking $m \to 0$ (or, equivalently, the high-energy limit $s \gg m^2$) sends $\beta \to 1$ but changes nothing, since β had already cancelled. The massless result is therefore identical,

$$\sigma\big|_{m=0} = \frac{\lambda^2}{32\pi s}, \qquad \left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{cm},\, m=0} = \frac{\lambda^2}{64\pi^2 s}. \tag{16.168}$$

The massive theory differs from the massless one only in *where* scattering begins — at $s = 4m^2$ rather than at $s = 0$ — not in the magnitude of the cross section above that point. This is a peculiarity of a constant amplitude; once $\mathcal{M}$ acquires kinematic dependence at $O(\lambda^2)$, the cancellation of threshold factors will no longer be exact.

Step 5 — the numerical scale. Restore the dimensional conversion $(\hbar c)^2 = 0.389\,\mathrm{mb}\cdot\mathrm{GeV}^2$ (with $1\,\mathrm{mb} = 10^{-27}\,\mathrm{cm}^2$). For an order-unity coupling $\lambda = 1$ and a centre-of-mass energy $\sqrt{s} = 1\,\mathrm{GeV}$ (so $s = 1\,\mathrm{GeV}^2$),

$$\sigma = \frac{\lambda^2}{32\pi s} = \frac{1}{32\pi}\,\mathrm{GeV}^{-2} \approx 9.95\times 10^{-3}\,\mathrm{GeV}^{-2} \approx 3.9\times 10^{-3}\,\mathrm{mb} \approx 3.9\,\mu\mathrm{b}. \tag{16.169}$$

A few microbarns at GeV energies for a strong-strength scalar coupling is an utterly typical hadronic-scale cross section, which is the right order of magnitude and a reassuring sanity check on every factor of 2π that went into the master formula.

Step 6 — the dimensional and high-energy check. A cross section is an area, $[\sigma] = [\text{mass}]^{-2}$ in natural units. In $d = 4$ the coupling λ is dimensionless, so the only scale available to a tree amplitude that is itself dimensionless is the invariant s, and dimensional analysis forces $\sigma \propto 1/s$ with a dimensionless coefficient — exactly the structure of Eq. (16.164). The cross section accordingly falls as $1/s$ at high energy (Fig. 16.6): collisions become more transparent as they become more energetic, the hallmark of a marginal (renormalisable) contact coupling whose strength does not grow with energy at tree level. This $1/s$ fall-off is the elementary version of the unitarity-respecting high-energy behaviour that Section 16.7 will sharpen into the partial-wave bound.

Physical Insight: the cross section is all geometry once $\mathcal{M}$ is constant

With $|\mathcal{M}|^2 = \lambda^2$ a pure number, this computation isolates, in its starkest possible form, the chapter's central claim: the dynamics of a scattering process lives entirely in $\mathcal{M}$, and everything else is the geometry of the

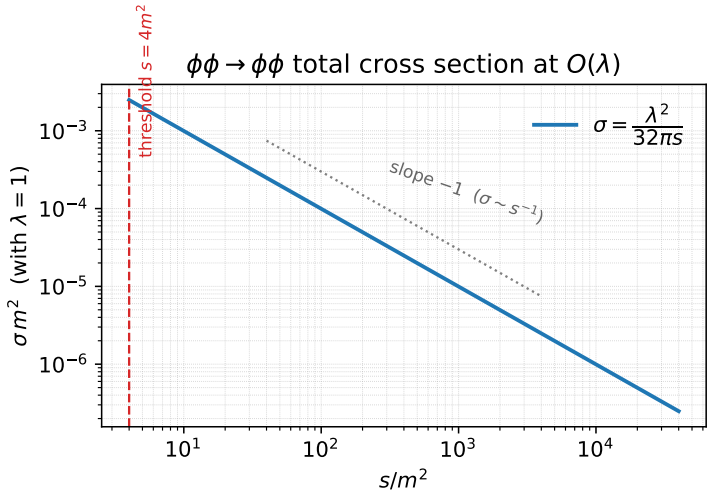


Figure 16.6: The total cross section $\sigma = \lambda^2/(32\pi s)$ for $\phi\phi \to \phi\phi$ at $O(\lambda)$, in units of the scalar mass m (with $\lambda = 1$). Scattering switches on at the kinematic threshold $s = 4m^2$ and, because the threshold factor $\beta = \sqrt{1 - 4m^2/s}$ cancels between flux and phase space, the curve is the exact power law $\sigma \propto 1/s$ for all $s > 4m^2$ — slope -1 on log–log axes, the dimensional-analysis signature $[\sigma] = [\text{mass}]^{-2}$ of a dimensionless coupling whose only scale is s.

on-shell momenta. Here the dynamics is a single constant, and the entire s-dependence, the threshold, the isotropy, and the $1/s$ tail are manufactured by the flux factor and the invariant phase space alone. The same phase-space geometry will sit, unchanged, behind the $e^+e^- \to \mu^+\mu^-$ cross section of Chapter 17; what changes there is only that $\mathcal{M}$ stops being a constant and starts carrying the angular structure of spin.

16.6.5 What the next order brings

The tree result Eq. (16.164) is exact in its accounting of the leading-order rate, but it is also, in one precise sense, incomplete: a constant real amplitude has no imaginary part, and a real $\mathcal{M}$ cannot by itself satisfy the unitarity relation that Section 16.7 will impose. At $O(\lambda^2)$ the picture changes. The one-loop s-, t-, and u-channel bubble diagrams — the same diagrams whose position-space form appeared in Eq. (15.38) — give $\mathcal{M}(s,t,u)$ genuine kinematic dependence and, above threshold, a nonzero imaginary part $\mathrm{Im}\,\mathcal{M}$. That imaginary part is not optional: the optical theorem of the next section fixes it, through $\mathrm{Im}\,\mathcal{M}(s,\theta{=}0) = 2E_{\mathrm{cm}}|\mathbf{p}_i|\,\sigma_{\mathrm{tot}}$, to be exactly what reproduces the tree cross section Eq. (16.164) we have just computed. The chapter therefore contains its own consistency check:

the $O(\lambda)$ cross section computed here must equal the $O(\lambda^2)$ discontinuity of the loop amplitude, a cross-check we set up in Section 16.7 and discharge by the explicit bubble computation of Chapters 21 and 22.

We have now run the full apparatus of this chapter on a real example and extracted a physical number from a measured-quantity formula. The pipeline was: take $\mathcal{M}$ from the Feynman rules, square it, contract it with the invariant phase space, divide by the flux, and integrate. Chapter 17 reuses this pipeline verbatim for $e^+e^- \to \mu^+\mu^-$; the only new ingredient is the evaluation of $\overline{|\mathcal{M}|^2}$ via Dirac spin sums and gamma-matrix traces, after which the same master formula, the same two-body phase space, and the same Mandelstam kinematics deliver the QED cross section — the bridge from formalism to numbers, now carried across both threads.

16.7 Unitarity, the optical theorem as a cross section, and partial waves

The previous section cashed the entire chapter on ϕ^4 theory: from the tree amplitude $\mathcal{M} = -\lambda$ it produced the differential cross section $\mathrm{d}\sigma/\mathrm{d}\Omega = \lambda^2/(64\pi^2 s)$ and, after the identical-particle factor, the total cross section $\sigma_{\text{tot}} = \lambda^2/(32\pi s)$ recorded in Eq. (16.164). In passing it flagged something that ought to feel paradoxical. The amplitude $\mathcal{M} = -\lambda$ is a real number, kinematics-independent, with no imaginary part whatsoever; yet we are about to assert that the imaginary part of the *forward* elastic amplitude is, all by itself, a measure of this very total cross section. At the order we have computed, the forward amplitude has no imaginary part, so the statement can only become nonvacuous one order deeper, at $O(\lambda^2)$ — exactly the loop order at which the same section's forward pointer promised an imaginary part would appear. The present section makes that promise precise. It takes the one piece of nonperturbative knowledge the S-matrix supplies — its unitarity, established in Section 14.7 — and feeds into it the flux factor of Section 16.4 and the invariant phase space of Section 16.2 built earlier in this chapter, turning an abstract operator identity into an equation between two quantities a detector actually reports: the imaginary part of a forward amplitude and a total cross section. The result is the optical theorem in cross-section form, its sharpening into the partial-wave unitarity bound, and a self-consistency check on the chapter's own machinery that the ϕ^4 numbers pass exactly.

The logic runs opposite to the way Section 14.7 first presented these results. There the optical theorem was extracted from operator unitarity *before* the cross-section formula existed, so the kinematic factors — the Møller flux, the two-body phase space — had to be quoted in advance, "recalled from Chapter 16." We are now Chapter 16, and those factors have been derived from scratch. Closing that loop is not a repetition: it is the consistency check the chapter has been

building toward, the moment the observable-side machinery (flux, phase space, $2 \to 2$ kinematics) and the formal-side machinery (LSZ, the S-matrix, unitarity) are shown to agree on the same number from two independent directions.

16.7.1 From operator unitarity to a relation between cross sections

We begin from the result Section 14.7 proved and import without re-deriving. The scattering operator $\mathbf{S} = \mathbf{1} + i\mathcal{T}$ is unitary, and writing out $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ gives the operator identity

$$-i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger\mathcal{T}, \tag{16.170}$$

which is Eq. (14.188): the failure of the transition operator to be Hermitian is exactly its own positive square. Sandwiched between a fixed initial state $|i\rangle$ of total four-momentum p_i and evaluated with the invariant amplitude $\mathcal{M}$ of Eq. (14.163) together with the completeness relation on the interacting Fock space, this becomes the generalized optical theorem Theorem 14.37,

$$2\ \mathrm{Im}\ \mathcal{M}(i \to i) = \sum_f \int \mathrm{d}\Pi_f\ |\mathcal{M}(i \to f)|^2, \tag{16.171}$$

in which $\mathrm{d}\Pi_f$ is precisely the Lorentz-invariant final-state phase space constructed in Section 16.2 — the same measure, including the energy–momentum delta, now carrying its full in-chapter pedigree rather than being borrowed. Here $\mathrm{d}\Pi_f$ is the bare phase-space measure; whenever the final state contains n identical bosons the $1/n!$ symmetry factor is supplied separately, multiplying the integral, exactly as in the worked check below. The sum runs over every kinematically accessible final state f: elastic, inelastic, of any multiplicity. The right-hand side is a sum of squared moduli against nonnegative measures, hence $\mathrm{Im}\ \mathcal{M}(i \to i) \geq 0$.

So far Eq. (16.171) is a statement about amplitudes. The content of this section is the recognition that its right-hand side is, up to a single kinematic factor, a quantity we have just learned to measure. Specialize the initial state to the two-particle configuration of a scattering experiment, $|i\rangle = |p_1\, p_2\rangle$. The master cross-section formula of Section 16.4, namely Eq. (16.172),

$$\mathrm{d}\sigma(i \to f) = \frac{1}{F}\,|\mathcal{M}(i \to f)|^2\,\mathrm{d}\Pi_f, \qquad F = 4\sqrt{(p_1 \cdot p_2)^2 - m_1^2 m_2^2}, \tag{16.172}$$

tells us that summing and integrating $|\mathcal{M}|^2\,\mathrm{d}\Pi_f$ over all final states is the same operation that, divided by the flux F, builds the total cross section:

$$\sigma_{\mathrm{tot}}(s) = \frac{1}{F}\sum_f \int \mathrm{d}\Pi_f\,|\mathcal{M}(i \to f)|^2. \tag{16.173}$$

The sum in Eq. (16.173) is identical to the sum in Eq. (16.171). Eliminating it between the two equations is immediate and is the whole derivation: the

right-hand side of the optical theorem equals $F\,\sigma_{\mathrm{tot}}$, while its left-hand side is $2\,\mathrm{Im}\,\mathcal{M}$ of the *forward* amplitude — "forward" because the diagonal element $\mathcal{M}(i \to i)$ has final state equal to initial state, $p_3 = p_1$, $p_4 = p_2$, hence zero scattering angle and Mandelstam $t = (p_1 - p_3)^2 = 0$.

Theorem 16.28 (Optical theorem in cross-section form)**.** *For elastic $2 \to 2$ scattering of two particles of momenta p_1, p_2 at centre-of-momentum energy $\sqrt{s}$, the imaginary part of the forward ($t = 0$) invariant amplitude fixes the total cross section,*

$$\boxed{\mathrm{Im}\,\mathcal{M}(s, t{=}0) = \tfrac{1}{2}\,F\,\sigma_{\mathrm{tot}}(s) = 2\,|\mathbf{p}_i|\,\sqrt{s}\,\sigma_{\mathrm{tot}}(s)} \tag{16.174}$$

where $|\mathbf{p}_i| = p_{\mathrm{cm}}$ is the common magnitude of the initial three-momenta in the centre-of-momentum frame, $\sigma_{\mathrm{tot}} = \sigma_{\mathrm{el}} + \sigma_{\mathrm{inel}}$ is the sum over all channels, and the positivity $\mathrm{Im}\,\mathcal{M}(s, 0) \geq 0$ holds nonperturbatively.

Proof. Set the initial and final two-particle states equal in Eq. (16.171), giving $2\,\mathrm{Im}\,\mathcal{M}(s,0) = \sum_f \int \mathrm{d}\Pi_f |\mathcal{M}(i \to f)|^2$, and substitute the sum from Eq. (16.173), $\sum_f \int \mathrm{d}\Pi_f |\mathcal{M}|^2 = F\,\sigma_{\mathrm{tot}}$. Hence $2\,\mathrm{Im}\,\mathcal{M}(s,0) = F\,\sigma_{\mathrm{tot}}$, i.e. $\mathrm{Im}\,\mathcal{M}(s,0) = \tfrac{1}{2}F\,\sigma_{\mathrm{tot}}$. It remains to evaluate the flux in the centre-of-momentum frame. There $p_1 = (E_1, \mathbf{p})$, $p_2 = (E_2, -\mathbf{p})$ with $|\mathbf{p}| = p_{\mathrm{cm}}$ and $E_1 + E_2 = \sqrt{s}$, so $p_1 \cdot p_2 = E_1 E_2 + \mathbf{p}^2$. Using $E_i^2 = \mathbf{p}^2 + m_i^2$,

$$\begin{aligned}(p_1 \cdot p_2)^2 - m_1^2 m_2^2 &= (E_1 E_2 + \mathbf{p}^2)^2 - (E_1^2 - \mathbf{p}^2)(E_2^2 - \mathbf{p}^2) \\ &= 2E_1 E_2 \mathbf{p}^2 + \mathbf{p}^4 + (E_1^2 + E_2^2)\mathbf{p}^2 - \mathbf{p}^4 = \mathbf{p}^2\,(E_1 + E_2)^2 = p_{\mathrm{cm}}^2\, s,\end{aligned} \tag{16.175}$$

the standard two-body identity, whence $F = 4\sqrt{p_{\mathrm{cm}}^2 s} = 4\,p_{\mathrm{cm}}\sqrt{s}$ and $\tfrac{1}{2}F = 2\,p_{\mathrm{cm}}\sqrt{s} = 2\,|\mathbf{p}_i|\,\sqrt{s}$. This is Eq. (16.174). Positivity follows from the nonnegativity of the right-hand side of Eq. (16.171). □

Equation Eq. (16.174) is the same identity quoted as Eq. (14.200), but it now stands on the chapter's own legs: the flux F and the phase space $\mathrm{d}\Pi_f$ that enter it were derived in Sections 16.2 and 16.4, not assumed. The agreement of the two routes — operator unitarity feeding kinematics it had to anticipate, versus kinematics now built independently and fed back into unitarity — is the consistency check announced at the head of the section. At high energy, $m_i^2 \ll s$, one has $p_{\mathrm{cm}} \to \sqrt{s}/2$ and Eq. (16.174) collapses to the memorable form $\mathrm{Im}\,\mathcal{M}(s,0) \to s\,\sigma_{\mathrm{tot}}(s)$.

Physical Insight

The optical theorem is the bridge that finally makes the chapter title's third word, "amplitudes," carry physical weight equal to "cross sections"

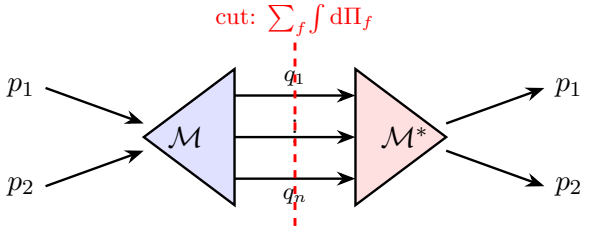


Figure 16.7: The optical theorem Eq. (16.174) as a unitarity cut. Twice the imaginary part of the forward elastic amplitude $\mathcal{M}(i \to i)$ (left) is obtained by severing it along the dashed line into a product $\mathcal{M}(i \to f)\,\mathcal{M}^*(i \to f)$ and putting every intermediate line $q_1 \dots q_n$ on its mass shell, then summing over all final states f with the invariant phase space $\mathrm{d}\Pi_f$ of Section 16.2. Dividing by the flux $F = 4\,|\mathbf{p}_i|\,\sqrt{s}$ turns the cut into the total cross section. The systematic version of this picture is Cutkosky's rule, established later (Chapter 60); here it is the diagrammatic content of Eq. (16.171).

and "decay rates." A cross section is the modulus-squared of an amplitude integrated over phase space; the imaginary part of the *forward* amplitude, by contrast, is a single complex number evaluated at one kinematic point. Unitarity says these two very different-looking objects are the same physics: the total rate at which a beam is depleted in the forward direction equals the total rate at which anything at all happens. Nothing in the differential cross section of Section 16.5 could have told us this; it is a constraint that links amplitudes across the energy at a single angle to an integral over all angles and all channels, and it is exact.

16.7.2 Partial waves and the unitarity bound on cross sections

The optical theorem constrains the forward amplitude as a single object. Resolving the elastic $2 \to 2$ amplitude into eigenstates of angular momentum sharpens it into a separate, hard bound on each channel — the relativistic descendant of the partial-wave analysis familiar from nonrelativistic scattering. We take the cleanest case, two spinless particles scattering elastically in the centre-of-momentum frame, where rotational invariance about the beam axis makes $\mathcal{M}$ a function of s and the polar angle θ alone. Recall the partial-wave expansion fixed in Eq. (14.202),

$$\mathcal{M}(s,\theta) = \frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}} \sum_{\ell=0}^{\infty} (2\ell{+}1)\, a_\ell(s)\, P_\ell(\cos\theta), \qquad a_\ell(s) = \frac{p_{\mathrm{cm}}}{16\pi\sqrt{s}} \int_{-1}^{1} \mathrm{d}(\cos\theta)\, P_\ell\, \mathcal{M}, \tag{16.176}$$

the dimensionless coefficients $a_\ell(s)$ being the partial-wave amplitudes and P_ℓ the Legendre polynomials, with projection following from the orthogonality

$\int_{-1}^{1} \mathrm{d}(\cos\theta)\, P_\ell P_{\ell'} = \frac{2}{2\ell+1}\delta_{\ell\ell'}$. The normalization is chosen, with foresight, so that the unitarity relations below come out free of stray factors. What is new here is that we feed this expansion into the differential cross section *this chapter* derived, rather than into an abstract probability current.

Elastic cross section. From the $2 \to 2$ result of Section 16.5, the elastic differential cross section for equal-mass particles is $\mathrm{d}\sigma_{\mathrm{el}}/\mathrm{d}\Omega = |\mathcal{M}|^2/(64\pi^2 s)$ (this is Eq. (16.145), the elastic specialization $|\mathbf{p}_f| = |\mathbf{p}_i|$ of the master formula). Insert Eq. (16.176), integrate over the solid angle $\mathrm{d}\Omega = 2\pi\,\mathrm{d}(\cos\theta)$, and apply orthogonality to collapse the double sum:

$$\begin{aligned}\sigma_{\mathrm{el}} &= \int \mathrm{d}\Omega\, \frac{|\mathcal{M}|^2}{64\pi^2 s} = \frac{2\pi}{64\pi^2 s}\left(\frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}}\right)^2 \int_{-1}^{1} \mathrm{d}(\cos\theta) \sum_{\ell,\ell'} (2\ell+1)(2\ell'+1)\, a_\ell a_{\ell'}^{*}\, P_\ell P_{\ell'} \\ &= \frac{2\pi}{64\pi^2 s} \cdot \frac{64\pi^2 s}{p_{\mathrm{cm}}^2} \sum_{\ell} (2\ell+1)^2\, |a_\ell|^2 \cdot \frac{2}{2\ell+1} = \frac{4\pi}{p_{\mathrm{cm}}^2} \sum_{\ell} (2\ell+1)\, |a_\ell(s)|^2 .\end{aligned} \tag{16.177}$$

Total cross section. The forward amplitude is obtained from Eq. (16.176) at $\theta = 0$, where every $P_\ell(1) = 1$, so $\mathcal{M}(s,0) = \frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}} \sum_\ell (2\ell+1) a_\ell$. Feed its imaginary part into the optical theorem Eq. (16.174):

$$\sigma_{\mathrm{tot}} = \frac{\mathrm{Im}\ \mathcal{M}(s,0)}{2\, p_{\mathrm{cm}}\sqrt{s}} = \frac{1}{2 p_{\mathrm{cm}}\sqrt{s}} \cdot \frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}} \sum_{\ell} (2\ell+1)\ \mathrm{Im}\ a_\ell = \frac{4\pi}{p_{\mathrm{cm}}^2} \sum_{\ell} (2\ell+1)\ \mathrm{Im}\ a_\ell(s). \tag{16.178}$$

The two expressions Eqs. (16.177) and (16.178) share the identical prefactor $4\pi/p_{\mathrm{cm}}^2$ — the relativistic echo of the $4\pi/k^2$ of Sakurai-level scattering theory — which is exactly the payoff of the normalization in Eq. (16.176). Because each channel ℓ is independent and the total cross section decomposes into a nonnegative elastic part plus a nonnegative inelastic part, $\sigma_{\mathrm{tot}} = \sigma_{\mathrm{el}} + \sigma_{\mathrm{inel}}$ with $\sigma_{\mathrm{inel}} \geq 0$, subtracting Eq. (16.177) from Eq. (16.178) term by term gives the per-channel statement

$$\mathrm{Im}\ a_\ell(s) = \big|a_\ell(s)\big|^2 + \underbrace{\frac{p_{\mathrm{cm}}^2}{4\pi(2\ell+1)}\, \sigma_{\mathrm{inel}}^{(\ell)}}_{\geq 0} \ \geq\ \big|a_\ell(s)\big|^2 . \tag{16.179}$$

This is the partial-wave unitarity relation, derived here entirely from cross sections: the elastic part of $\mathrm{Im}\, a_\ell$ is $|a_\ell|^2$, and any excess is the inelastic rate flowing out of that angular-momentum channel. Its geometric content is the Argand circle: completing the square in Eq. (16.179) gives $(\mathrm{Re}\ a_\ell)^2 + (\mathrm{Im}\ a_\ell - \frac{1}{2})^2 \leq \frac{1}{4}$, so a_ℓ is confined to the closed disk of radius $\frac{1}{2}$ centred at $\frac{i}{2}$, drawn in Fig. 14.6, with the boundary $\mathrm{Im}\, a_\ell = |a_\ell|^2$ being the purely elastic locus $a_\ell = e^{i\delta_\ell}\sin\delta_\ell$ of a real

phase shift (Eqs. (14.207) and (14.208)). The three immediate corollaries are $0 \le \operatorname{Im} a_\ell \le 1$, $|\operatorname{Re} a_\ell| \le \frac{1}{2}$, and $|a_\ell| \le 1$.

The last of these is what bounds the cross section. Inserting $|a_\ell| \le 1$ into Eq. (16.177), or equivalently $\operatorname{Im} a_\ell \le 1$ into Eq. (16.178), caps the contribution of each angular momentum.

Theorem 16.29 (Partial-wave unitarity bound)**.** *The contribution of the ℓ-th partial wave to the total cross section of elastic spinless $2 \to 2$ scattering satisfies, nonperturbatively,*

$$\sigma_\ell \;\le\; \frac{4\pi(2\ell+1)}{|\mathbf{p}_i|^2}, \tag{16.180}$$

with saturation at $a_\ell = i$ (phase shift $\delta_\ell = \pi/2$, the unitarity limit of maximal absorption).

Proof. By Eq. (16.179), $\operatorname{Im} a_\ell \ge |a_\ell|^2 \ge 0$, so $\operatorname{Im} a_\ell \ge (\operatorname{Im} a_\ell)^2$, which forces $0 \le \operatorname{Im} a_\ell \le 1$; equivalently the Argand-disk constraint gives $|a_\ell| \le 1$. The ℓ-channel piece of Eq. (16.178) is $\sigma_\ell = \frac{4\pi}{p_{\rm cm}^2}(2\ell+1)\operatorname{Im} a_\ell \le \frac{4\pi}{p_{\rm cm}^2}(2\ell+1)$, with equality iff $\operatorname{Im} a_\ell = 1$, i.e. $a_\ell = i$. Identifying $p_{\rm cm} = |\mathbf{p}_i|$ gives Eq. (16.180). □

Caution

The bound Eq. (16.180), like $|a_\ell| \le 1$, constrains the *exact* amplitude. A truncated perturbative amplitude is a polynomial in the coupling and will, at high enough energy or strong enough coupling, march straight out of the Argand disk. That is not a failure of unitarity but a self-diagnostic: it marks the kinematic regime where the omitted higher orders are no longer negligible and the series must be resummed or replaced. We use exactly this diagnostic on ϕ^4 below.

16.7.3 Scalar ϕ^4 theory: closing the stage with a unitarity self-check

We now run the chapter's ϕ^4 output through the optical theorem and confirm that everything is mutually consistent — and that the only place a nonzero answer can first appear is precisely where Section 16.6 said it would.

Worked Example 16.30 (The ϕ^4 optical theorem and the location of the first imaginary part)**.** The chapter's tree amplitude is the constant $\mathcal{M} = -\lambda$ (Eq. (14.183)), and Section 16.6 integrated it to the total cross section

$$\sigma_{\rm tot} = \frac{\lambda^2}{32\pi s} \;+\; O(\lambda^3), \tag{16.181}$$

the factor $\frac{1}{2}$ for identical final-state bosons already included (Eq. (16.164)). We confront these with Theorem 16.28.

Why the leading order is silent. The optical theorem Eq. (16.174) reads $\mathrm{Im}\,\mathcal{M}(s,0) = 2\,|\mathbf{p}_i|\,\sqrt{s}\,\sigma_{\mathrm{tot}}$. Its right-hand side, by Eq. (16.181), is $O(\lambda^2)$. Its left-hand side, evaluated on the tree amplitude, is $\mathrm{Im}(-\lambda) = 0$. So at $O(\lambda)$ the theorem reads $0 = 0 + O(\lambda^2)$: satisfied, but carrying no information. This is not an accident of ϕ^4; it is forced by the operator identity Eq. (16.170), whose right-hand side $\mathcal{T}^\dagger\mathcal{T}$ is always one power of the coupling higher than its left-hand side $\mathcal{T} - \mathcal{T}^\dagger$. The imaginary part of the forward amplitude is therefore mandated to vanish at the order at which σ_{tot} first becomes nonzero, and to reappear one loop later — exactly the $O(\lambda^2)$ remark with which Section 16.6 ended. The optical theorem turns that qualitative pointer into an equation.

The quantitative prediction. At $O(\lambda^2)$ the theorem becomes a prediction for the imaginary part of the one-loop forward amplitude $\mathcal{M}^{(2)}$ in terms of the tree cross section already in hand. Using $|\mathbf{p}_i| = p_{\mathrm{cm}} = \tfrac{1}{2}\sqrt{s}\,\beta$ with $\beta = \sqrt{1 - 4m^2/s}$,

$$\mathrm{Im}\;\mathcal{M}^{(2)}(s,0) = 2\,p_{\mathrm{cm}}\sqrt{s}\,\sigma_{\mathrm{tot}} = 2\cdot\tfrac{1}{2}\sqrt{s}\,\beta\cdot\sqrt{s}\cdot\frac{\lambda^2}{32\pi s} = \frac{\lambda^2\,\beta}{32\pi}. \tag{16.182}$$

This must agree with the cut on the right-hand side of Eq. (16.171). The lightest accessible state is again two scalars, so the cut is the two-body phase-space integral of the tree amplitude squared, with an explicit $1/2!$ identical-boson factor multiplying the bare measure $\mathrm{d}\Pi_2$:

$$2\;\mathrm{Im}\;\mathcal{M}^{(2)}(s,0) = \frac{1}{2!}\int \mathrm{d}\Pi_2\,\big|-\lambda\big|^2 = \frac{\lambda^2}{2}\cdot\frac{\beta}{8\pi} = \frac{\lambda^2\beta}{16\pi}, \tag{16.183}$$

where $\int \mathrm{d}\Pi_2 = \beta/8\pi$ is the two-body phase-space volume of Section 16.2. Dividing by 2 gives $\mathrm{Im}\;\mathcal{M}^{(2)}(s,0) = \lambda^2\beta/32\pi$, in exact agreement with Eq. (16.182). The two routes — "flux times the total cross section" and "phase-space integral of the cut" — meet on the same number, including the identical-particle factor that appears, identically, on both sides. This sharpens the looser leading check recorded in Eq. (14.210): the present accounting tracks the symmetry factor through to the end and finds the books balance.

What is and is not established. Equation Eq. (16.182) is a falsifiable prediction for the imaginary part of the one-loop s-channel bubble, fixed in advance by tree-level data and the unitarity of $\mathbf{S}$. Computing $\mathcal{M}^{(2)}$ itself — assembling the loop diagram, regulating and integrating it, and extracting its imaginary part — requires the Feynman-integral technology of the renormalization programme (Chapter 19) and the systematic cutting rules of Chapter 60. No new loop number is computed here; what is established is the constraint the loop must satisfy, waiting for the later chapters to redeem.

The perturbative reach of λ. Finally, project the tree amplitude onto partial waves. Since $\mathcal{M} = -\lambda$ is angle-independent and $P_0 = 1$, only the s-wave survives, and Eq. (16.176) gives $a_0 = -\lambda\,p_{\mathrm{cm}}/(8\pi\sqrt{s}) \to -\lambda/16\pi$ at high energy. This is real at $O(\lambda)$, consistent with $\mathrm{Im}\,a_0 = |a_0|^2 = O(\lambda^2)$ — the imaginary part the

Argand circle demands is again deferred to one loop. The exact bound $|a_0| \leq 1$ of Eq. (16.179) then reads $\lambda \lesssim 16\pi$, the perturbative-unitarity ceiling on the coupling derived in Eq. (14.212): tree-level $\lambda\phi^4$ is trustworthy only for λ well below 16π, and the genuine expansion parameter is $\lambda/16\pi^2$ — the same $16\pi^2$ that will govern the one-loop β-function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Chapter 24.

The ϕ^4 stage thus closes with its own internal audit passed. The chapter's flux, phase space, and $2 \to 2$ kinematics, assembled independently of Chapter 14, reproduce the optical theorem that unitarity demands; the tree cross section $\sigma_{\text{tot}} = \lambda^2/32\pi s$ feeds back through that theorem to predict the imaginary part of an amplitude the chapter has not yet computed; and the partial-wave bound certifies where the perturbative description ends. Every observable formula has been checked against the one nonperturbative constraint available.

This completes the bridge from formalism to numbers. The flux–phase-space–amplitude template is now ready to be reused verbatim in Chapters 17 and 18, where the same machinery, augmented only by spin sums and gamma-matrix traces, produces the QED cross sections of QED. The imaginary part that the optical theorem here predicts but defers will be computed explicitly when one-loop amplitudes acquire their absorptive parts in Chapter 19, and organized into general cutting rules in Chapter 60. And the analyticity that unitarity has begun to impose on the amplitude — positivity of $\text{Im}\,\mathcal{M}$, the Argand disk, the partial-wave bound — is the physical input on which the dispersion relations and the modern bootstrap of Part XIV are built, the channel through which this chapter's elementary kinematics reaches the research frontier.

Further reading

The box-regularization route to the rate and cross-section formulae, and the relativistic state normalization that fixes their kinematic factors, are treated in the standard graduate texts: Peskin and Schroeder [396] (Chapter 4), Schwartz [432], Srednicki [454], and the foundational discussion in Weinberg [500]; Zee [532] gives a complementary, more heuristic account. The Lorentz-invariant phase space, the recursion relation, and the Dalitz-plot kinematics worked here in full are the subject of the classic monograph by Byckling and Kajantie [100], while the standard conventions and numerical data are collected in the kinematics review of Particle Data Group [382]. The triangle function originates with Källén [289]. For the analytic-S-matrix viewpoint on the Mandelstam plane, crossing, and the physical regions, see Eden et al. [170]; the unitarity cut underlying the optical theorem is made systematic by the Cutkosky rules of Cutkosky [141], developed in this book in Chapter 60.

Part V

Quantum Electrodynamics

Chapter 17

QED: Feynman Rules and First Processes

With this chapter Part V opens, and the long preparatory arc of the book pays its first concrete dividend. Everything assembled so far converges here on a single target: to compute, from first principles and with every intermediate step on the page, the cross sections that an electron–positron collider actually measures. The classical gauge principle of Chapter 8 handed us the standard Lagrangian $\mathcal{L}_{\text{QED}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with its minimal coupling $D_\mu = \partial_\mu + ieA_\mu$; the quantization of the Dirac field in Chapter 11 supplied the on-shell spinors, the spin sums, and the fermion propagator; the quantization of the photon in Chapter 13 supplied the R_ξ propagator and the Gupta–Bleuler restriction to physical transverse states. Onto this matter content the perturbative engine of Part IV now acts: the interaction picture and the LSZ reduction of Chapter 14, Wick's theorem and the diagrammatic bookkeeping of Chapter 15, and the phase-space, flux, and cross-section apparatus of Chapter 16. None of these results is reproved here. The work of this chapter is to wield them, and to keep honest the one convention—the sign of the covariant derivative, fixed once and never revisited—whose mishandling is the single most common error in the subject.

The chapter is organized so that the derivation is done exactly once and the computation that follows is pure application. Section 17.1 extracts, from the gauge-fixed Lagrangian and the Dyson series, a fixed sign-checked momentum-space *rulebook*: the vertex $-ie\gamma^\mu$, the two internal propagators, the six external-line factors, the closed-loop and exchange signs forced by Fermi statistics, and the tree-level gauge independence that licenses working in Feynman gauge. Section 17.2 builds the algebraic engine that turns a spin-dressed amplitude into a number a detector reports—the completeness relations, Casimir's trick, the gamma-trace toolkit of Chapter C, and the photon polarization-sum replacement—reducing $\overline{|\mathcal{M}|^2}$ to a function of the Mandelstam invariants of Chapter 16 with not a spinor left in sight.

The remaining sections run that machinery on the carried processes of QED. Section 17.3 carries the marquee computation, $e^+e^- \to \mu^+\mu^-$, from the single s-channel diagram to the angular law $d\sigma/d\Omega = (\alpha^2/4s)(1+\cos^2\theta)$, the total cross section $\sigma = 4\pi\alpha^2/3s$, and its threshold-corrected massive form. Section 17.4 shows that this one rational function of (s,t,u), analytically continued, already contains a second physical process, and Section 17.5 cashes that crossed amplitude for $e^-\mu^- \to e^-\mu^-$ into the laboratory-frame cross section, recovering the Mott and Rutherford formulae as nested limits and identifying the photon's $1/q^4$ with the squared Fourier transform of the Coulomb potential—the precise sense in which QED contains classical electrostatics.

The closing Section 17.6 steps back to ask why the answers take the shape they do. The QED vertex conserves chirality, in the high-energy limit chirality is helicity, and the surviving helicity amplitudes are nothing but the rotation matrix elements $d^1_{m'm}(\theta)$ of a spin-one exchange; the $1+\cos^2\theta$ law that the traces produced is thereby revealed as a theorem of angular-momentum conservation, independent of the dynamical detail. This helicity skeleton is the seed of the spinor-helicity reframing of Chapter 58, and it points forward to the loops, the Ward–Takahashi identity, and the precision predictions that Chapters 18 to 20 will take up in turn.

17.1 The QED Lagrangian and its Feynman rules

The chapter introduction has gathered three strands of inheritance and laid them side by side: the quantized photon with its R_ξ propagator and Gupta–Bleuler physical states (Chapter 13), the quantized Dirac electron with its mode expansion and on-shell spinors (Chapter 11), and the perturbative engine — the interaction picture and LSZ reduction (Chapter 14), Wick's theorem and the Feynman-rule machinery (Chapter 15), and the phase-space and cross-section apparatus (Chapter 16). None of these is reproved here; they are wielded. The task of this section is narrow and decisive: to turn that inheritance into a single, fixed, sign-checked *rulebook*. Once we possess it, the rest of the chapter is pure computation from this fixed rulebook — every later amplitude is read off a diagram by table lookup, never re-derived from the field operators. The discipline of doing the derivation *once*, in full, is what makes the subsequent computation trustworthy: a sign error buried in a mode expansion would otherwise propagate silently into every cross section. We therefore extract each rule from the underlying operator algebra rather than assert it, and we pin down, on the page, the one convention — the sign of the covariant derivative — that is the single most common source of error in the subject.

17.1.1 The gauge-fixed Lagrangian and the perturbative expansion

The classical theory was assembled in Section 8.6 by the gauge principle: demanding local $U(1)$ invariance of the free Dirac field forced minimal coupling and produced the standard Lagrangian Eq. (8.143),

$$\mathcal{L}_{\text{QED}} = \bar{\psi}\,(i\not{D} - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu. \tag{17.1}$$

Here ψ is the electron field of mass m, $e > 0$ is the elementary charge, and the electron carries physical charge $-e$, so that the global rule of Chapter A, $D_\mu = \partial_\mu - iqA_\mu$ with $q = -e$, specializes to the $+ie$ written above. The fine-structure constant is $\alpha = e^2/4\pi \approx 1/137$ (Eq. (8.145)). Expanding the covariant derivative as in Eq. (8.144) separates the free pieces from the single interaction,

$$\mathcal{L}_{\text{QED}} = \underbrace{\bar{\psi}(i\not{\partial} - m)\psi}_{\mathcal{L}_{0,\psi}} \; \underbrace{-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}}_{\mathcal{L}_{0,A}} \; \underbrace{-\,e\,\bar{\psi}\gamma^\mu\psi\,A_\mu}_{\mathcal{L}_{\text{int}}}. \tag{17.2}$$

Notation Watch

The sign convention is fixed here once for the entire QED thread and never revisited: $e > 0$, electron charge $-e$, covariant derivative $D_\mu = \partial_\mu + ieA_\mu$, interaction $\mathcal{L}_{\text{int}} = -e\,\bar{\psi}\gamma^\mu\psi A_\mu$. Texts that take the electron charge itself as the symbol e (so $e < 0$) obtain the opposite sign in the covariant derivative and an apparently different vertex; the physics is identical, but the symbols are not interchangeable. Every e in this and the following chapters is the positive elementary charge.

The free Lagrangian $\mathcal{L}_{0,A}$ on its own does not define a photon propagator: the quadratic form $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is annihilated by pure-gauge configurations $A_\mu = \partial_\mu\Lambda$, so its kinetic operator is not invertible. This obstruction, and its resolution, were the subject of Chapter 13. The cure is to add a gauge-fixing term, which in the covariant R_ξ family reads (Eq. (13.176))

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}\,(\partial_\mu A^\mu)^2, \tag{17.3}$$

so that the gauge-fixed QED Lagrangian we quantize is

$$\boxed{\mathcal{L} = \bar{\psi}(i\not{D} - m)\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2.} \tag{17.4}$$

The parameter ξ labels a gauge *choice*; physical quantities must be independent of it, a fact we verify at tree level in Section 17.1.6 and which is guaranteed in general by the Ward–Takahashi identity proved in Chapter 19. For all explicit

computation in this chapter we adopt the *Feynman gauge* $\xi = 1$, the standard default, because it strips the photon propagator down to its simplest tensor structure. We state results in general ξ where it costs nothing and specialize to $\xi = 1$ thereafter.

From the Lagrangian to the Dyson series. The perturbative expansion of the S-matrix is built in the interaction picture. The free part of Eq. (17.4) furnishes the in/out asymptotic fields with the mode expansions of Chapters 11 and 13, and the time evolution generated by the interaction is the Dyson series

$$\mathbf{S} = \mathcal{T} \exp\!\left(-i \int d^4x \, \mathcal{H}_{\text{int}}(x) \right), \tag{17.5}$$

with $\mathcal{H}_{\text{int}}$ the interaction-picture Hamiltonian density. For QED the coupling $\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$ contains *no* time derivatives of the fields, so the Legendre transform that defines $\mathcal{H}$ does not generate any extra terms from the interaction: the canonical momenta are unchanged by $\mathcal{L}_{\text{int}}$, and therefore

$$\mathcal{H}_{\text{int}} = -\mathcal{L}_{\text{int}} = e\,\bar{\psi}\gamma^\mu\psi A_\mu. \tag{17.6}$$

This is the property that makes the covariant gauge-fixed formulation so convenient: in a non-covariant gauge the instantaneous Coulomb interaction and a non-covariant piece of the propagator appear separately and must be shown to recombine, whereas the covariant quantization of Chapter 13 packages everything into the manifestly Lorentz-invariant propagator Eq. (13.135), and the simple relation Eq. (17.6) holds. Substituting Eq. (17.6) into Eq. (17.5) and using $-i\int dt\,\mathcal{H}_{\text{int}} = i\int d^4x\,\mathcal{L}_{\text{int}}$ gives the form we actually expand,

$$\mathbf{S} = \mathcal{T} \exp\!\left(i \int d^4x \, \mathcal{L}_{\text{int}}(x) \right) = \mathcal{T} \exp\!\left(-ie \int d^4x \, \bar{\psi}\gamma^\mu\psi A_\mu \right). \tag{17.7}$$

Every QED amplitude is a term in this expansion, evaluated between asymptotic states by Wick's theorem (Theorem 15.3) and amputated onto external wavefunctions by LSZ (Theorem 14.31). The Feynman rules are nothing more than the bookkeeping of that evaluation, organized so it need never be redone. We extract them now, piece by piece: the vertex, the internal lines, the external lines, the fermion signs, and finally the assembled statement of $i\mathcal{M}$.

17.1.2 The interaction vertex

The interaction enters Eq. (17.7) at first order as

$$\mathbf{S}^{(1)} = i \int d^4x \, \mathcal{L}_{\text{int}}(x) = -ie \int d^4x \, \bar{\psi}(x)\, \gamma^\mu \, \psi(x)\, A_\mu(x), \tag{17.8}$$

and every vertex in every diagram is one insertion of this operator. To turn it into a momentum-space rule we restore the suppressed Dirac indices and ask

what each of the three fields does. Writing the spinor indices explicitly, the operator is

$$\mathcal{L}_{\text{int}}(x) = -e\,\bar{\psi}_a(x)\,(\gamma^\mu)_{ab}\,\psi_b(x)\,A_\mu(x), \tag{17.9}$$

with a, b summed over the four Dirac components. In any diagram each of the three field operators $\bar{\psi}_a$, ψ_b, A_μ at a vertex is Wick-contracted: either with an external asymptotic field (producing an external-line factor, Section 17.1.4) or with a field at another vertex (producing an internal propagator, Section 17.1.3). What is universal — independent of which lines are external or internal — is the local algebraic structure left behind at the point x and the spacetime integral over x. We isolate it.

Theorem 17.1 (The QED vertex factor)**.** *Each elementary QED interaction vertex contributes, in momentum space, the factor*

$$\boxed{-i\,e\,\gamma^\mu} \tag{17.10}$$

carrying one Lorentz index μ contracted into the attached photon line and one pair of Dirac indices (ab) contracted into the two attached fermion lines, with a on the side of the outgoing fermion arrow and b on the incoming side. Together with the vertex factor goes a factor $(2\pi)^4\delta^4(\sum k)$ enforcing four-momentum conservation at the vertex, with all momenta counted as flowing into the vertex.

Proof. The factor is read off Eq. (17.8) by stripping the field operators and performing the universal spacetime integral. The operator Eq. (17.9) carries the constant prefactor $-e$, and the overall first-order term Eq. (17.8) carries the explicit i from the expansion of $\mathcal{T}\exp(i\int\mathcal{L}_{\text{int}})$; their product is $-ie$. The matrix $(\gamma^\mu)_{ab}$ sits between the conjugate field $\bar{\psi}_a$ and the field ψ_b, so the Dirac index that ties to whatever contracts with $\bar{\psi}$ is a and the index that ties to whatever contracts with ψ is b. The single Lorentz index μ is the one carried by A_μ and is therefore the index along which the photon line attaches.

It remains to produce the momentum-conserving delta. Whatever the three contractions are, each replaces its field operator by a wavefunction carrying a definite spacetime dependence $e^{\mp ik\cdot x}$ in the field's argument x: an external incoming line of momentum k contributes $e^{-ik\cdot x}$ from the positive-frequency part of the field, an external outgoing line contributes $e^{+ik\cdot x}$ from the conjugate field, and an internal line contributes a propagator which, written in momentum space, is itself an integral $\int\frac{d^4k}{(2\pi)^4}\,e^{\mp ik\cdot x}$ over the vertex coordinate (Section 17.1.3). In all cases the dependence of the integrand on the vertex coordinate x is a single exponential $e^{i(\sum k)\cdot x}$, where the sum runs over all momenta flowing *into* the vertex with the sign conventions just stated. The integral over x in Eq. (17.8) is then elementary,

$$\int d^4x\; e^{\,i(\sum k)\cdot x} = (2\pi)^4\,\delta^4\Big(\sum k\Big), \tag{17.11}$$

using the Fourier convention of Chapter A. This is the momentum-conservation delta attached to the vertex. The remaining x-independent factor is exactly $-ie\,(\gamma^\mu)_{ab}$, which is Eq. (17.10). □

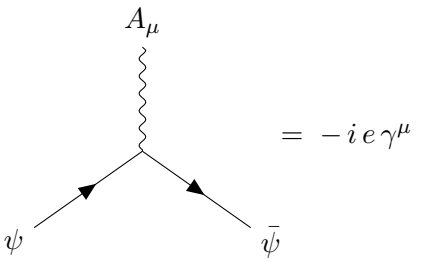


Figure 17.1: The QED interaction vertex Eq. (17.10), the single dynamical building block of every diagram in the theory. The arrow on the fermion line tracks the flow of fermion number (electron number minus positron number); the Dirac index a of $(\gamma^\mu)_{ab}$ attaches on the outgoing side of the arrow, b on the incoming side, so that reading a fermion line against its arrow reproduces the matrix order $\bar{u}\cdots\gamma^\mu\cdots u$. The photon attaches through the Lorentz index μ. Each vertex also carries $(2\pi)^4\delta^4(\sum k)$ (Eq. (17.11)).

The worked example below carries out the extraction concretely for a fully external vertex, so that the abstract statement of Theorem 17.1 is anchored in an explicit Wick contraction; it is the QED instantiation of the general procedure of Theorem 15.25.

Worked Example 17.2 (The vertex by explicit Wick contraction)**.** Consider the lowest-order contribution to a process in which a single vertex connects an incoming electron of momentum p and spin s, an outgoing electron of momentum p' and spin s', and an incoming photon of momentum k and polarization λ — the elementary absorption vertex underlying Compton scattering. The relevant operator is Eq. (17.8), and the amplitude is

$$\Big\langle e^-(p',s')\Big|\mathbf{S}^{(1)}\Big|e^-(p,s),\,\gamma(k,\lambda)\Big\rangle = -ie\int d^4x\;\Big\langle e^-(p',s')\Big|\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)\Big|e^-(p,s),\,\gamma(k,\lambda)\Big\rangle\,. \tag{17.12}$$

We evaluate the matrix element by contracting each field with the external state it can reach. The Dirac field $\psi(x)$ contains the electron annihilation operator $b^s_{\mathbf{p}}$ in its positive-frequency part (Eq. (11.95)); contracting it with the incoming electron $|e^-(p,s)\rangle = \sqrt{2E_p}\,b^{s\dagger}_{\mathbf{p}}\,|0\rangle$ replaces $\psi(x) \to u^s(p)\,e^{-ip\cdot x}$. The conjugate field $\bar{\psi}(x)$ contains $b^{s'\dagger}_{\mathbf{p}'}$ in its part that creates the outgoing electron; contracting with $\langle e^-(p',s')|$ replaces $\bar{\psi}(x) \to \bar{u}^{s'}(p')\,e^{+ip'\cdot x}$. The photon field $A_\mu(x)$ annihilates the incoming photon, replacing $A_\mu(x) \to \epsilon^\lambda_\mu(k)\,e^{-ik\cdot x}$. (These external-line replacements are derived in full in Section 17.1.4; here we use them

to display the mechanism.) Assembling,

$$\left\langle f\left|\mathbf{S}^{(1)}\right|i\right\rangle = -ie\,\bar{u}^{s'}(p')\,\gamma^{\mu}\,u^{s}(p)\,\epsilon_{\mu}^{\lambda}(k)\int d^4x\; e^{i(p'-p-k)\cdot x} = -ie\,\bar{u}^{s'}(p')\gamma^{\mu}u^{s}(p)\,\epsilon_{\mu}^{\lambda}(k)\,(2\pi)^4\delta^4(\tag{17.13}$$

Stripping the overall $(2\pi)^4\delta^4$ (which by Eq. (16.2) accompanies $i\mathcal{M}$) leaves the local factor $-ie\,\bar{u}^{s'}(p')\gamma^{\mu}u^{s}(p)\,\epsilon_{\mu}^{\lambda}(k)$. The factor sitting between the two spinors is exactly the vertex $-ie\gamma^{\mu}$ of Eq. (17.10), with $\bar{u}^{s'}(p')$ the outgoing-electron line factor on the a side and $u^{s}(p)$ the incoming-electron line factor on the b side, and $\epsilon_{\mu}^{\lambda}(k)$ the incoming-photon line factor on the μ index. The delta function is Eq. (17.11) with p' outgoing (counted with $+$), p and k incoming (counted with $-$). This is the rule, derived rather than quoted.

17.1.3 Internal lines: the propagators

A field operator at a vertex that is *not* contracted with an external state must be contracted with a field at another vertex. By Wick's theorem (Theorem 15.3) such a contraction is precisely the Feynman propagator of the field — the time-ordered two-point function — and these were computed once and for all in Chapters 11 and 13. We import them as the internal-line rules; the derivations are not repeated.

The electron internal line. The contraction of two Dirac fields is the fermion Feynman propagator established in Theorem 11.38 and given in momentum space by Eq. (11.195),

$$\overbrace{\psi(x)\,\bar{\psi}}(y) = S_F(x-y) = \int \frac{d^4p}{(2\pi)^4}\,\frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon}\,e^{-ip\cdot(x-y)}, \tag{17.14}$$

so that each internal electron line carries the momentum-space factor

$$\boxed{\frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon}} = \frac{i(\gamma^{\alpha}p_{\alpha}+m)}{p^2-m^2+i\varepsilon}, \tag{17.15}$$

a 4×4 matrix in Dirac space. The matrix indices are contracted into the vertices at the two ends in the same order the line is traversed against its arrow: reading a fermion line from its head (an outgoing $\bar{u}$, or an incoming-arrow vertex) to its tail, the propagators and vertices multiply as ordinary matrices, $\cdots\gamma^{\mu}\,S_F(p)\,\gamma^{\nu}\cdots$. The momentum p in Eq. (17.15) is the one flowing *along* the arrow; reversing the assignment $p\to -p$ would flip the sign of $\not{p}$ and is simply a different (consistent) labeling. The $i\varepsilon$ prescription is the Feynman one fixed in Chapter A and derived for the fermion line in Eq. (11.172): it routes positive-energy electrons forward in time and positive-energy positrons backward, the content of the Stückelberg–Feynman interpretation.

The photon internal line. The contraction of two photon fields is the R_ξ propagator of Theorem 13.26,

$$\overbrace{A_\mu(x)\,A_\nu(y)} = D_{\mu\nu}(x-y) = \int \frac{d^4k}{(2\pi)^4}\,\frac{-i}{k^2+i\varepsilon}\Big[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big]e^{-ik\cdot(x-y)}, \tag{17.16}$$

so that each internal photon line carries, in the general gauge,

$$\frac{-i}{k^2+i\varepsilon}\Big[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big], \tag{17.17}$$

and in the Feynman gauge $\xi = 1$ adopted throughout (Eq. (13.136)) the clean form

$$\boxed{\frac{-i\,\eta_{\mu\nu}}{k^2+i\varepsilon}} \qquad (\xi = 1). \tag{17.18}$$

The two Lorentz indices μ, ν are contracted into the photon indices of the vertices at the two ends. Because the photon is massless and chargeless the line is undirected: no arrow, and the momentum k may be routed either way since $D_{\mu\nu}$ depends only on k^2 and on $k_\mu k_\nu$, both even in k. We will see in Section 17.1.6 that the ξ-dependent piece $\propto k_\mu k_\nu$ never contributes to a tree amplitude with on-shell external fermions, which is why $\xi = 1$ is a free and convenient choice.

Figure 17.2: The two internal lines of QED. Left: the electron propagator Eq. (17.15), a matrix in Dirac space; the arrow marks the direction of the routed momentum p and of fermion-number flow. Right: the photon propagator in Feynman gauge Eq. (17.18); the line is undirected and the indices μ, ν tie to the vertices at the two ends. Both are imported, not re-derived, from Eqs. (11.195) and (13.135).

17.1.4 External lines: LSZ amputation and the wavefunction factors

A field operator contracted with an asymptotic external particle does not produce a propagator but a *wavefunction factor*. The LSZ reduction formula (Theorem 14.31) is the statement that each external leg of a correlation function is amputated — its full propagator stripped to a pole, the residue removed — and replaced by the on-shell wavefunction of the corresponding asymptotic particle (Section 15.6.5 and Eq. (15.126)). For QED there are six kinds of external

leg, and we derive each factor from the free mode expansions; this is the QED instantiation of the general LSZ amputation, not a quotation of it.

Electrons and positrons. The quantized Dirac field is Eq. (11.95),

$$\psi(x) = \sum_s \int \frac{d^3p}{(2\pi)^3\, 2E_{\mathbf{p}}} \Big[b^s_{\mathbf{p}}\, u^s(p)\, e^{-ip\cdot x} + d^{s\dagger}_{\mathbf{p}}\, v^s(p)\, e^{+ip\cdot x} \Big], \tag{17.19}$$

with the conjugate field $\bar\psi = \psi^\dagger\gamma^0$ given by Eq. (11.96), and asymptotic one-particle states normalized relativistically, $|e^-(p,s)\rangle = \sqrt{2E_p}\, b^{s\dagger}_{\mathbf{p}}\,|0\rangle$ and $|e^+(p,s)\rangle = \sqrt{2E_p}\, d^{s\dagger}_{\mathbf{p}}\,|0\rangle$, matching the state normalization fixed in Chapter 16. The mode operators obey the relativistically normalized anticommutators $\{b^r_{\mathbf{p}}, b^{s\dagger}_{\mathbf{q}}\} = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^3(\mathbf{p}-\mathbf{q})\, \delta^{rs}$ and likewise for d. The four cases follow by inspecting which mode operator survives the contraction with the state.

Incoming electron. The field $\psi(x)$ acting to the right on $|e^-(p,s)\rangle$ contracts $b^s_{\mathbf{p}}$ against $b^{s\dagger}_{\mathbf{p}}$; the anticommutator collapses the measure and the relativistic normalization $\sqrt{2E_p}$ cancels the $2E_{\mathbf{p}}$ in the denominator, leaving the spinor and the plane wave,

$$\psi(x)\,|e^-(p,s)\rangle \;\longrightarrow\; u^s(p)\, e^{-ip\cdot x} + \cdots, \tag{17.20}$$

where the ellipsis denotes terms that create additional particles and do not contribute to the connected amplitude for this leg. After LSZ amputation and Fourier transform the leg factor is the spinor $u^s(p)$. *An incoming electron contributes $u^s(p)$.*

Outgoing electron. The conjugate field $\bar\psi(x)$ contains $b^{s\dagger}_{\mathbf{p}}$ riding the spinor $\bar u^s(p)$; contracting with $\langle e^-(p',s')|$ from the left yields $\bar u^{s'}(p')\, e^{+ip'\cdot x}$. *An outgoing electron contributes $\bar u^{s'}(p')$.*

Incoming positron. The positron is created by $d^\dagger$, which appears in $\bar\psi$ riding $\bar v^s(p)$. The field $\bar\psi(x)$ acting to the right on $|e^+(p,s)\rangle$ contracts $d^s_{\mathbf{p}}$ (from $\bar\psi$) against $d^{s\dagger}_{\mathbf{p}}$, leaving $\bar v^s(p)\, e^{-ip\cdot x}$. *An incoming positron contributes $\bar v^s(p)$.*

Outgoing positron. Symmetrically, ψ carries $d^\dagger$ riding $v^s(p)$; contracting with $\langle e^+(p',s')|$ gives $v^{s'}(p')\, e^{+ip'\cdot x}$. *An outgoing positron contributes $v^{s'}(p')$.*

The asymmetry — electrons get $u, \bar u$ and positrons get $\bar v, v$ with the roles of incoming/outgoing apparently swapped — is the direct consequence of the field assignment in Eq. (17.19): ψ annihilates electrons but creates positrons, while $\bar\psi$ creates electrons but annihilates positrons. Diagrammatically this is encoded by the arrow: the fermion-number arrow points *into* the diagram for an incoming electron and an outgoing positron (both attached to a ψ-type end), and *out of* the diagram for an outgoing electron and an incoming positron (both attached to a $\bar\psi$-type end). Reading any fermion line against its arrow then automatically orders the spinors and matrices correctly: outgoing-fermion factor on the far left, then vertices and propagators in sequence, then the incoming-fermion factor on the far right.

Photons. The quantized photon field has the mode expansion Eq. (13.55), which in covariant four-vector notation reads

$$A_\mu(x) = \sum_\lambda \int \frac{d^3k}{(2\pi)^3\, 2\omega_k} \Big[a^\lambda_{\mathbf{k}}\, \epsilon^\lambda_\mu(k)\, e^{-ik\cdot x} + a^{\lambda\dagger}_{\mathbf{k}}\, \epsilon^{\lambda *}_\mu(k)\, e^{+ik\cdot x} \Big], \tag{17.21}$$

with $\epsilon^\lambda_\mu(k)$ the polarization four-vectors (Theorem 13.9) and physical external photons restricted to the two transverse polarizations $\lambda = 1, 2$ by the Gupta–Bleuler condition of Section 13.3; for real transverse polarizations $\epsilon^{\lambda *}_\mu = \epsilon^\lambda_\mu$, but we keep the conjugate explicit so the rule holds for any (e.g. circular) basis. By the same contraction logic:

Incoming photon. $A_\mu(x)$ acting on $|\gamma(k,\lambda)\rangle$ contracts $a^\lambda_{\mathbf{k}}$, leaving $\epsilon^\lambda_\mu(k)\, e^{-ik\cdot x}$. *An incoming photon contributes* $\epsilon^\lambda_\mu(k)$.

Outgoing photon. The conjugate term of $A_\mu(x)$ contracts $a^{\lambda\dagger}_{\mathbf{k}}$ against $\langle\gamma(k,\lambda)|$, leaving $\epsilon^{\lambda *}_\mu(k)\, e^{+ik\cdot x}$. *An outgoing photon contributes* $\epsilon^{\lambda *}_\mu(k)$.

The Lorentz index μ of the polarization vector ties to the photon index of the vertex it attaches to. The completeness and physical-state properties of these polarizations — in particular the gauge invariance that lets one replace the transverse polarization sum by $-\eta_{\mu\nu}$ inside a squared amplitude — belong to the spin-sum technology of the next section and to Section 13.3; here we need only the leg factors themselves. The six external-line rules are collected in Table 17.1.

Table 17.1: The QED external-line factors, derived in Section 17.1.4 from the free mode expansions Eqs. (11.95) and (13.55) via LSZ amputation (Theorem 14.31). The arrow column gives the direction of the fermion-number arrow relative to the diagram.

External particle	Factor	Fermion arrow
Incoming electron $e^-(p,s)$	$u^s(p)$	into diagram
Outgoing electron $e^-(p,s)$	$\bar{u}^s(p)$	out of diagram
Incoming positron $e^+(p,s)$	$\bar{v}^s(p)$	out of diagram
Outgoing positron $e^+(p,s)$	$v^s(p)$	into diagram
Incoming photon $\gamma(k,\lambda)$	$\epsilon^\lambda_\mu(k)$	—
Outgoing photon $\gamma(k,\lambda)$	$\epsilon^{\lambda *}_\mu(k)$	—

17.1.5 Fermion sign bookkeeping: loops and exchange

The single feature of the QED rulebook that has no analogue in the scalar theory of Chapter 15 is the appearance of *relative minus signs* forced by the anticommuting statistics of the Dirac field. Two rules govern them: a closed fermion loop carries an overall factor -1, and two diagrams that differ only by the interchange of two identical external fermion (or antifermion) lines carry a

relative factor -1. Both follow from the same fact — that Wick contraction of Grassmann-odd operators tracks the parity of the permutation needed to bring contracted pairs adjacent (Theorem 15.3) — and both must be derived, because getting either wrong is fatal and neither can be guessed from the bosonic case.

The closed-loop sign. Consider a closed fermion loop with n vertices. The loop is built from the Wick contractions of the n pairs $(\bar\psi, \psi)$ at the vertices, contracted in a cycle: ψ at vertex 1 with $\bar\psi$ at vertex 2, ψ at vertex 2 with $\bar\psi$ at vertex 3, and so on, closing with ψ at vertex n contracted with $\bar\psi$ at vertex 1. The cleanest derivation takes the simplest nontrivial case, $n = 2$, and the general case follows by the same counting. The relevant operator structure from the second-order Dyson term is, with vertices at x and y and indices suppressed except where they matter,

$$\cdots \bar\psi(x)\gamma^\mu\psi(x)\;\bar\psi(y)\gamma^\nu\psi(y)\cdots\,, \tag{17.22}$$

and the loop is the contraction of $\psi(x)$ with $\bar\psi(y)$ and of $\psi(y)$ with $\bar\psi(x)$:

$$\bar\psi_a(x)\,\gamma^\mu_{ab}\,\psi_b(x)\;\bar\psi_c(y)\,\gamma^\nu_{cd}\,\psi_d(y)\,. \tag{17.23}$$

To evaluate it we bring each contracted pair adjacent. The contraction of $\psi_b(x)$ with $\bar\psi_c(y)$ is the propagator $S_F(x-y)_{bc}$; the contraction of $\psi_d(y)$ with $\bar\psi_a(x)$ is $S_F(y-x)_{da}$. Moving $\psi_b(x)$ rightward past $\bar\psi_c(y)$ to stand beside it, and tracking the Grassmann-odd anticommutations, one finds that closing the cycle requires an *odd* number of transpositions of anticommuting operators. Concretely, the contracted product reorders to

$$-\,(\gamma^\mu)_{ab}\,S_F(x-y)_{bc}\,(\gamma^\nu)_{cd}\,S_F(y-x)_{da} = -\,\mathrm{tr}\big[\gamma^\mu S_F(x-y)\gamma^\nu S_F(y-x)\big], \tag{17.24}$$

where the explicit minus sign is the parity of the permutation, and the Dirac indices, chased around the cycle $a \to b \to c \to d \to a$, close into a trace. This is the content of the rule: *each closed fermion loop contributes a factor* (-1) *and a Dirac trace over the matrices encountered around the loop.* The sign originates entirely in the anticommuting algebra; for a loop of n vertices the same counting gives one minus sign (the cyclic closure is always an odd permutation relative to the bosonic ordering), so the rule is (-1) per loop, independent of n. The trace structure is what makes fermion loops computable by the gamma-matrix technology of Chapter C.

Caution

The loop minus sign is attached to the loop, not to any individual vertex or propagator. A diagram with L closed fermion loops carries $(-1)^L$. The most consequential instance is the photon vacuum-polarization diagram

of Chapter 19, whose single electron loop carries exactly one such minus sign; omitting it flips the sign of the running of α and is a classic and catastrophic error.

The exchange sign. Whenever a process has two identical fermions (or two identical antifermions) in the final state — or in the initial state — the amplitude receives contributions from diagrams related by exchanging the two identical particles, and these enter with a *relative* minus sign, the diagrammatic face of Fermi–Dirac antisymmetry. We derive it from the contraction algebra in the worked example below, instantiated on electron–electron (Møller) scattering, whose two tree diagrams are the cleanest illustration. We are establishing the *sign rule* here; the cross section itself is computed in Chapter 18.

Worked Example 17.3 (The relative sign of the Møller diagrams)**.** Electron–electron scattering $e^-(p_1)\,e^-(p_2) \to e^-(p_3)\,e^-(p_4)$ proceeds at tree level through two diagrams: a t-channel diagram in which the photon connects the $(p_1 \to p_3)$ and $(p_2 \to p_4)$ fermion lines, and a u-channel diagram in which it connects $(p_1 \to p_4)$ and $(p_2 \to p_3)$. The two differ precisely by the interchange of the two identical outgoing electrons, $p_3 \leftrightarrow p_4$. We show the relative sign is -1 directly from the second-order S-matrix.

The relevant term of Eq. (17.7) is

$$\mathbf{S}^{(2)} = \frac{(-ie)^2}{2!}\int d^4x\, d^4y\, \mathcal{T}\Big[\bar\psi(x)\gamma^\mu\psi(x)A_\mu(x)\,\bar\psi(y)\gamma^\nu\psi(y)A_\nu(y)\Big]. \tag{17.25}$$

The two photon fields contract into the internal photon propagator $D_{\mu\nu}(x-y)$. The four Dirac fields must contract with the four external electrons. There are two inequivalent ways to do so, and they are the two diagrams. In the first, $\psi(x)$ annihilates the incoming p_1 (giving $u(p_1)$) and $\bar\psi(x)$ creates the outgoing p_3 (giving $\bar u(p_3)$), while $\psi(y)$ annihilates p_2 and $\bar\psi(y)$ creates p_4; the line factors arrange into

$$\mathcal{A}_t \;\propto\; \big[\bar u(p_3)\gamma^\mu u(p_1)\big]\,\big[\bar u(p_4)\gamma^\nu u(p_2)\big]\,D_{\mu\nu}(p_1-p_3). \tag{17.26}$$

In the second, the roles of the two outgoing electrons are swapped: $\bar\psi(x)$ creates p_4 and $\bar\psi(y)$ creates p_3,

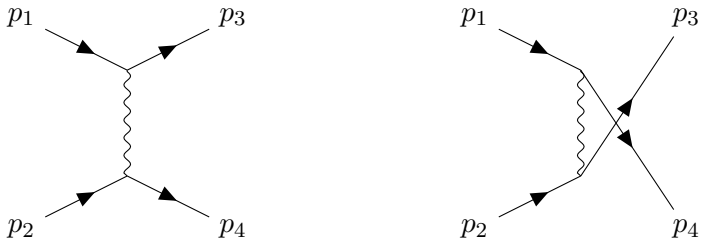
$$\mathcal{A}_u \;\propto\; \big[\bar u(p_4)\gamma^\mu u(p_1)\big]\,\big[\bar u(p_3)\gamma^\nu u(p_2)\big]\,D_{\mu\nu}(p_1-p_4). \tag{17.27}$$

The relative sign is fixed by counting the anticommutations needed to bring the field operators into the standard contracted order in each case. Take the first contraction as the reference, with the operators ordered so that all contracted pairs are adjacent and the overall sign is $+1$ by definition. The second contraction is obtained from the first by interchanging which $\bar\psi$ creates p_3 and which creates

p_4 — equivalently, by transposing the two anticommuting creation operators $b^\dagger_{p_3}$ and $b^\dagger_{p_4}$ (or, what is the same, transposing the two factors $\bar{u}(p_3)$ and $\bar{u}(p_4)$ in the contraction assignment). A single transposition of two Grassmann-odd operators carries the factor -1. Therefore

$$\boxed{\mathcal{M}(e^-e^- \to e^-e^-) = \mathcal{M}_t - \mathcal{M}_u,} \tag{17.28}$$

with the two terms equal up to the swap $p_3 \leftrightarrow p_4$ and a relative minus sign. The same minus sign would appear with initial-state identical fermions, or for identical antifermions; it is the diagrammatic statement of the antisymmetry of the multi-fermion state. (The overall sign of the whole amplitude is a convention and cancels in $|\mathcal{M}|^2$; only the *relative* sign between the two diagrams is physical, and it is -1.)



t-channel: $+\mathcal{M}_t$ u-channel: $-\mathcal{M}_u$

Figure 17.3: The two tree diagrams of Møller scattering $e^-e^- \to e^-e^-$, related by exchanging the identical outgoing electrons $p_3 \leftrightarrow p_4$ (Theorem 17.3). They enter the amplitude with a relative minus sign, Eq. (17.28), the diagrammatic form of Fermi–Dirac antisymmetry. The cross section is computed in Chapter 18; here the diagrams serve only to establish the sign rule.

17.1.6 Gauge independence of tree amplitudes and the tree-level Ward identity

We adopted Feynman gauge $\xi = 1$ for convenience and promised that the choice is physically immaterial. We now make good on this at tree level, both because it justifies the default and because the mechanism — current conservation — is the seed of the Ward–Takahashi identity proved in full in Chapter 19. The claim is that the gauge-dependent part of the photon propagator Eq. (17.17), the term proportional to $(1-\xi)k_\mu k_\nu/k^2$, makes no contribution to any tree amplitude whose external fermions are on shell.

Theorem 17.4 (Tree-level current conservation)**.** *Let an internal photon of momentum k in a tree diagram attach at one end to a fermion line carrying*

physical external states. The contraction of k_μ with the conserved current at that end vanishes; consequently the $k_\mu k_\nu$ part of the photon propagator drops out of the amplitude, which is therefore independent of ξ.

Proof. Consider the simplest case, a photon attached to a fermion line running from an incoming on-shell electron $u(p)$ to an outgoing on-shell electron $\bar{u}(p')$ through a single vertex, so that the current at that end is

$$J^\mu = \bar{u}(p')\,\gamma^\mu\,u(p), \qquad k = p' - p, \tag{17.29}$$

with k the photon momentum injected at the vertex. Contract with k_μ and substitute $k = p' - p$:

$$k_\mu J^\mu = \bar{u}(p')\,(\not{p}' - \not{p})\,u(p) = \bar{u}(p')\,\not{p}'\,u(p) - \bar{u}(p')\,\not{p}\,u(p). \tag{17.30}$$

Now use the on-shell Dirac equations for the external spinors, established in Chapter 11: $\not{p}\,u(p) = m\,u(p)$ acting on the right, and $\bar{u}(p')\,\not{p}' = m\,\bar{u}(p')$ acting on the left (the latter is the Dirac-adjoint equation). Hence

$$k_\mu J^\mu = m\,\bar{u}(p')u(p) - m\,\bar{u}(p')u(p) = 0. \tag{17.31}$$

The current is transverse to the photon momentum it sources. For an internal photon line the $k_\mu k_\nu$ term of the propagator Eq. (17.17) contracts k_μ into the current at one end (and k_ν into the current at the other); either contraction vanishes by Eq. (17.31), so the entire $(1-\xi)$ piece drops. What survives is the $\eta_{\mu\nu}$ part, which is ξ-independent. The amplitude is therefore the same in every R_ξ gauge, and in particular equals its Feynman-gauge ($\xi = 1$) value.

The argument extends to any tree diagram. A general fermion line threaded by a photon of momentum k at a vertex carries, on the two sides of that vertex, propagators $S_F(\ell + k)$ and $S_F(\ell)$ for some internal momentum ℓ; the contraction $\not{k} = (\not{\ell} + \not{k} - m) - (\not{\ell} - m)$ telescopes against the two adjacent propagators, $\not{k}\,S_F(\ell) = (\not{\ell} + \not{k} - m)S_F(\ell) - i$ (using $(\not{\ell} - m)S_F(\ell) = i$), converting the insertion into a difference of diagrams with one fewer propagator. Summed along the line, these differences telescope to the endpoint contributions, which vanish on shell by Eq. (17.31). This is exactly the chain that, promoted to include loops and the renormalization of the vertex, becomes the Ward–Takahashi identity $k_\mu \Gamma^\mu(p+k,p) = S_F^{-1}(p+k) - S_F^{-1}(p)$ of Chapter 19; here, at tree level and on shell, it collapses to the simple statement Eq. (17.31). □

Physical Insight

The masslessness of the photon and the conservation of the electromagnetic current are two faces of the same gauge symmetry, and Theorem 17.4 is where that shows its hand in computation. The unphysical, gauge-dependent longitudinal part of the photon propagator can never com-

municate with on-shell matter, because matter only couples through the conserved current $\bar{\psi}\gamma^\mu\psi$ (Eq. (8.146)), and a conserved current is blind to the longitudinal photon. This is why we may compute in whatever gauge is most convenient and trust the answer — and it is why, in the next section, the messy transverse polarization sum for external photons can be replaced by the simple covariant $-\eta_{\mu\nu}$.

17.1.7 The QED rulebook assembled

We now collect the pieces. The recipe for the tree-level invariant amplitude $i\mathcal{M}$ of any QED process is the following, and it is complete: every later computation in this chapter and the next is an application of it.

Theorem 17.5 (Momentum-space Feynman rules for QED)**.** *To compute $i\mathcal{M}$ for a given process at a given order, draw all topologically distinct, connected, amputated diagrams with the required external legs and the required number of vertices, and for each diagram:*

1. *assign the external-line factors of Table 17.1: $u^s(p)$, $\bar{u}^s(p)$ for incoming/outgoing electrons; $\bar{v}^s(p)$, $v^s(p)$ for incoming/outgoing positrons; $\epsilon_\mu^\lambda(k)$, $\epsilon_\mu^{\lambda *}(k)$ for incoming/outgoing photons;*
2. *write a factor $-ie\gamma^\mu$ (Eq. (17.10)) for each vertex;*
3. *write a propagator for each internal line: $\dfrac{i(\not{p}+m)}{p^2-m^2+i\varepsilon}$ for electrons (Eq. (17.15)) and $\dfrac{-i\eta_{\mu\nu}}{k^2+i\varepsilon}$ for photons in Feynman gauge (Eq. (17.18));*
4. *contract Dirac indices along each fermion line* against the arrow*, so that each line reads $\bar{u}\,[\cdots]\,u$ (or the appropriate v-spinors), and contract every Lorentz index, including those of internal-photon endpoints, with the metric;*
5. *impose four-momentum conservation at each vertex (Eq. (17.11)) and integrate $\int d^4\ell/(2\pi)^4$ over each independent loop momentum ℓ left undetermined;*
6. *multiply by (-1) for each closed fermion loop (Eq. (17.24)) and take the Dirac trace around each such loop;*
7. *include a relative sign between diagrams related by the interchange of two external identical fermions or antifermions (Eq. (17.28)).*

The sum of the resulting diagram values is $i\mathcal{M}$, with the single overall $(2\pi)^4\delta^4(\sum p_{\text{out}} - \sum p_{\text{in}})$ already factored off in the definition Eq. (16.2).

The collected result is the statement of the carried QED result: a fixed, sign-checked momentum-space rulebook. Two remarks complete it. First, at tree level there are no loop integrals and no fermion-loop signs, so for the processes of this chapter only items (1)–(5) and (7) operate; the loop machinery of item (6) is exercised in Chapter 19. Second, the overall delta in Eq. (16.2) is what was carefully regularized in Section 16.1 into the finite phase-space and flux factors of the cross section; the rules above produce exactly the $\mathcal{M}$ that the master formula of Chapter 16 consumes. The two halves of the apparatus — amplitude and rate — are now joined.

The rulebook is built and frozen. What it does not yet supply is the algebra that turns an amplitude into a number a detector reports: squaring $\mathcal{M}$, summing over the final spins and polarizations a real detector cannot resolve and averaging over the initial ones it cannot prepare, and reducing the result to a trace over gamma matrices and thence to Mandelstam invariants. The external-line factors $u, \bar{u}, v, \bar{v}$ and $\epsilon_\mu, \epsilon_\mu^*$ tabulated in Table 17.1, together with the on-shell completeness relations they satisfy and the gauge invariance established in Theorem 17.4, are exactly the raw material that engine needs. Building it is the work of the next section.

17.2 Spin sums, polarization sums, and trace technology

The previous section closed the Feynman rulebook: every QED amplitude is now a finite sum of products of the photon propagator Eq. (13.135), the vertex $-ie\gamma^\mu$ derived in Eq. (17.10), and the external-line factors handed to us by the LSZ reduction—an incoming electron contributes $u^s(p)$, an outgoing electron $\bar{u}^{s'}(p')$, an incoming positron $\bar{v}^s(p)$, an outgoing positron $v^{s'}(p')$, and an external photon a polarization vector $\epsilon_\mu(k,\lambda)$ or its conjugate $\epsilon_\mu^*(k,\lambda)$. The amplitude $\mathcal{M}$ is thus a number built from spinors and polarization vectors carrying definite spin labels s, s', λ. But a collider almost never prepares a definite electron spin, and almost never measures the spin of what comes out; what it measures is a rate, proportional to $|\mathcal{M}|^2$ *averaged* over the spins of the particles in the beam and *summed* over the spins of the particles in the detector. The task of this section is to build the algebraic engine that performs this average-and-sum once and for all, turning the spin-dressed $|\mathcal{M}|^2$ into a pure function of the external momenta—and ultimately into the Mandelstam invariants of Theorem 16.20—with not a single spinor left in sight. The engine has three gears: the *completeness relations* that collapse a sum over spins into a covariant matrix; *Casimir's trick*, which uses them to rewrite a spin-summed modulus-squared as a trace of gamma matrices; and the *trace technology* of Chapter C, which evaluates that trace as a polynomial in dot products. We assemble all three here and demonstrate them on the lepton tensor that the next section feeds into the $e^+e^- \to \mu^+\mu^-$ cross section.

Throughout we use the conventions fixed at the head of this chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the electron of charge $-e$ with $e > 0$ and $\alpha = e^2/4\pi$, and Feynman gauge $\xi = 1$ as the default. The Dirac spinors u^s, v^s and the gamma algebra are those of Chapter 11; the trace identities are collected and proved in Section C.6. We import these and wield them; we do not rebuild them, though we re-derive the few load-bearing identities on the page so that every step of a cross-section computation can be reproduced from this chapter alone.

17.2.1 The spin-averaged squared amplitude

Begin by saying precisely what object the experiment determines. Consider a $2 \to n$ process with two incoming particles of spin labels running over g_1 and g_2 values respectively—for a spin-$\frac{1}{2}$ particle $g = 2$, for a photon $g = 2$ physical polarizations—and final-state spins to be summed. The unpolarized squared amplitude is the average over initial spins and sum over final spins.

Definition 17.6 (Spin-averaged squared amplitude)**.** For a process with initial spin multiplicities g_1, g_2 and final spins collectively denoted $\{f\}$,

$$\overline{|\mathcal{M}|^2} \equiv \frac{1}{g_1 g_2} \sum_{\text{initial spins}} \sum_{\text{final spins}} |\mathcal{M}|^2 . \tag{17.32}$$

The prefactor $1/g_1 g_2$ is the average over the unobserved initial polarizations; the final-state sum is the inclusive measurement of any final spin.

For the centerpiece of this chapter, $e^+e^- \to \mu^+\mu^-$, both incoming particles are spin-$\frac{1}{2}$, so $g_1 = g_2 = 2$ and the averaging factor is $\frac{1}{4}$. This factor is not a convention to be looked up; it is the count of equally likely, equally weighted spin configurations of an unpolarized two-particle initial state, $2 \times 2 = 4$ of them, over which the incoherent average is taken. We carry it explicitly. (Were a photon in the initial state, as in Compton scattering in Chapter 18, its physical-polarization multiplicity is also 2, again by the two-transverse-state count of Chapter 13; a massive vector would carry $g = 3$.)

The reason Eq. (17.32) is tractable is that the sums over spins are not sums over independent random numbers but sums of outer products of spinors, and those sums have closed-form values. That is the content of the completeness relations, to which we turn.

17.2.2 The completeness (spin-sum) relations

The two identities that drive everything are the fermion spin sums established in Theorem 11.12. Because they are the hinge on which the entire chapter swings, we recall them and re-derive them directly from the explicit spinors of Eq. (11.57), carrying the block algebra so that the result is on the page where it is used.

Theorem 17.7 (Fermion completeness relations)**.** *The on-shell Dirac amplitudes $u^s(p), v^s(p)$ of Theorem 11.8, normalized by $\bar{u}^r u^s = 2m\,\delta^{rs} = -\bar{v}^r v^s$, satisfy the 4×4 matrix identities*

$$\sum_{s=1,2} u^s(p)\,\bar{u}^s(p) = \not{p} + m, \qquad \sum_{s=1,2} v^s(p)\,\bar{v}^s(p) = \not{p} - m, \tag{17.33}$$

for every on-shell momentum p ($p^2 = m^2$).

Proof. We reproduce the computation of Theorem 11.12. From Eq. (11.57) the positive-frequency amplitude and its Dirac adjoint are, in 2×2 block form built on the rest-frame two-spinor basis ξ^s,

$$u^s(p) = \begin{pmatrix} \sqrt{p\cdot\sigma}\,\xi^s \\ \sqrt{p\cdot\bar{\sigma}}\,\xi^s \end{pmatrix}, \qquad \bar{u}^s(p) = (\xi^{s\dagger}\sqrt{p\cdot\bar{\sigma}}\ ,\ \xi^{s\dagger}\sqrt{p\cdot\sigma}), \tag{17.34}$$

the adjoint row obtained from Eq. (11.63) (the γ^0 in $\bar{u} = u^\dagger\gamma^0$ swaps the two blocks). Their outer product is the 4×4 block matrix

$$u^s\bar{u}^s = \begin{pmatrix} \sqrt{p\cdot\sigma}\,(\xi^s\xi^{s\dagger})\,\sqrt{p\cdot\bar{\sigma}} & \sqrt{p\cdot\sigma}\,(\xi^s\xi^{s\dagger})\,\sqrt{p\cdot\sigma} \\ \sqrt{p\cdot\bar{\sigma}}\,(\xi^s\xi^{s\dagger})\,\sqrt{p\cdot\bar{\sigma}} & \sqrt{p\cdot\bar{\sigma}}\,(\xi^s\xi^{s\dagger})\,\sqrt{p\cdot\sigma} \end{pmatrix}. \tag{17.35}$$

Summing over s and using the two-component completeness $\sum_s \xi^s\xi^{s\dagger} = I$ from Eq. (11.56), every $\xi^s\xi^{s\dagger}$ becomes the identity. The off-diagonal blocks collapse by $\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\bar{\sigma}} = mI$ (the square-root identity Eq. (11.55)), giving mI in each; the diagonal blocks collapse by $\sqrt{p\cdot\sigma}\,\sqrt{p\cdot\sigma} = p\cdot\sigma$ and $\sqrt{p\cdot\bar{\sigma}}\,\sqrt{p\cdot\bar{\sigma}} = p\cdot\bar{\sigma}$. Hence

$$\sum_s u^s\bar{u}^s = \begin{pmatrix} m\,I & p\cdot\sigma \\ p\cdot\bar{\sigma} & m\,I \end{pmatrix} = \not{p} + m, \tag{17.36}$$

where the last equality is the chiral-basis block form $\not{p} = \left(\begin{smallmatrix} 0 & p\cdot\sigma \\ p\cdot\bar{\sigma} & 0 \end{smallmatrix}\right)$ of Eq. (11.53) together with $m = m\,\mathbf{1}$. For the negative-frequency amplitudes the only change is the relative minus in the lower block of v^s (and correspondingly in $\bar{v}^s$), which flips the sign of the two off-diagonal mI blocks while leaving the diagonal $p\cdot\sigma, p\cdot\bar{\sigma}$ blocks intact, yielding $\sum_s v^s\bar{v}^s = \not{p} - m$. □

The two relations are the precise statement that the u's span the positive-energy and the v's the negative-energy subspace at fixed p: each side is, up to the factor m, the energy projector $\Lambda_\pm$ of Theorem 11.13. What matters operationally is their shape. A spin sum is an outer product summed over the spin label, and the completeness relations say that this sum is a *momentum-dependent matrix with no spin label left*—exactly the object that can be threaded into a trace.

17.2.3 Casimir's trick: from a squared bilinear to a trace

Every tree amplitude in QED is a product of fermion bilinears of the form $\bar{u}(p')\,\Gamma\,u(p)$—a row spinor, a string of gamma matrices Γ, and a column spinor—tied together by photon propagators that carry only Lorentz indices and momenta, no spinor indices. (Lines with an incoming positron or outgoing antiparticle replace the corresponding u by a v; the argument below is identical with the second relation of Eq. (17.33).) When such an amplitude is squared and summed over spins, the spinors disappear into a trace. This is *Casimir's trick*, the single most important manipulation in tree-level QED. We prove it as a clean algebraic lemma and then read off its consequences.

The one ingredient we need first is the Dirac conjugate of a gamma string.

Lemma 17.8 (Dirac conjugation of a vertex)**.** *For any product of gamma matrices Γ, define $\bar{\Gamma} \equiv \gamma^0\Gamma^\dagger\gamma^0$. Then*

$$\overline{\gamma^\mu} = \gamma^\mu, \qquad \overline{\gamma^\mu\gamma^5} = \gamma^\mu\gamma^5, \qquad \overline{\gamma^5} = -\gamma^5, \qquad \overline{AB} = \bar{B}\,\bar{A}, \tag{17.37}$$

so that the conjugate of a string is the string read backwards with each factor conjugated.

Proof. The hermiticity properties of the gamma matrices, $\gamma^{0\dagger} = \gamma^0$ and $\gamma^{i\dagger} = -\gamma^i$, are summarized covariantly by $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ (Eq. (11.5)): for $\mu = 0$ this reads $\gamma^0 = \gamma^0\gamma^0\gamma^0 = \gamma^0$, and for $\mu = i$ it reads $-\gamma^i = \gamma^0\gamma^i\gamma^0$ since γ^i anticommutes with γ^0 and $(\gamma^0)^2 = \mathbf{1}$. Therefore

$$\overline{\gamma^\mu} = \gamma^0\gamma^{\mu\dagger}\gamma^0 = \gamma^0(\gamma^0\gamma^\mu\gamma^0)\gamma^0 = \gamma^\mu, \tag{17.38}$$

using $(\gamma^0)^2 = \mathbf{1}$. For the product rule, $\overline{AB} = \gamma^0(AB)^\dagger\gamma^0 = \gamma^0B^\dagger A^\dagger\gamma^0 = (\gamma^0B^\dagger\gamma^0)(\gamma^0A^\dagger\gamma^0) = \bar{B}\bar{A}$, inserting $\gamma^0\gamma^0 = \mathbf{1}$ in the middle. Since $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is Hermitian ($\gamma^{5\dagger} = \gamma^5$) and anticommutes with each γ^μ, we get $\overline{\gamma^5} = \gamma^0\gamma^5\gamma^0 = -\gamma^5(\gamma^0)^2 = -\gamma^5$; and $\overline{\gamma^\mu\gamma^5} = \overline{\gamma^5}\,\overline{\gamma^\mu} = -\gamma^5\gamma^\mu = \gamma^\mu\gamma^5$, the sign of $\overline{\gamma^5}$ undone by the anticommutation. The vector current $\bar{\psi}\gamma^\mu\psi$ is therefore self-conjugate, which is why the QED vertex squares so cleanly. □

With $\bar{\Gamma}$ in hand, Casimir's trick is a one-line index computation.

Theorem 17.9 (Casimir's trick)**.** *Let Γ, Γ' be gamma strings and let p, p' be on-shell momenta with masses m, m'. Summing over the spins of both spinors,*

$$\sum_{s,s'} \left[\bar{u}^{s'}(p')\,\Gamma\,u^s(p)\right]\left[\bar{u}^{s'}(p')\,\Gamma'\,u^s(p)\right]^* = \operatorname{tr}\Big[(\not{p}' + m')\,\Gamma\,(\not{p} + m)\,\bar{\Gamma}'\Big]. \tag{17.39}$$

The identical statement holds with any external u replaced by the corresponding v and the matching $(\not{p} + m)$ replaced by $(\not{p} - m)$.

Proof. The squared bilinear is a product of two complex numbers; the second is conjugated, and since a number equals its own transpose its complex conjugate equals its Hermitian conjugate. Conjugating the second factor,

$$\big[\bar{u}^{s'}(p')\,\Gamma'\,u^{s}(p)\big]^{*} = \big[u^{s'}(p')^{\dagger}\gamma^{0}\,\Gamma'\,u^{s}(p)\big]^{\dagger} = u^{s}(p)^{\dagger}\,\Gamma'^{\dagger}\gamma^{0}\,u^{s'}(p') = \bar{u}^{s}(p)\,\bar{\Gamma}'\,u^{s'}(p'), \tag{17.40}$$

where in the last step we inserted $\gamma^{0}\gamma^{0} = \mathbf{1}$ to form $\bar{u}^{s}(p) = u^{s}(p)^{\dagger}\gamma^{0}$ on the left and $\bar{\Gamma}' = \gamma^{0}\Gamma'^{\dagger}\gamma^{0}$ in the middle, using $\gamma^{0\dagger} = \gamma^{0}$. The summand is now $[\bar{u}^{s'}(p')\Gamma u^{s}(p)][\bar{u}^{s}(p)\bar{\Gamma}'u^{s'}(p')]$, a closed chain of spinor indices. Writing the spinor indices a, b, c, d explicitly and reordering the c-number factors,

$$\begin{aligned}\sum_{s,s'} \bar{u}^{s'}(p')_{a}\,\Gamma_{ab}\,u^{s}(p)_{b}\;\bar{u}^{s}(p)_{c}\,\bar{\Gamma}'_{cd}\,u^{s'}(p')_{d} &= \Gamma_{ab}\,\bar{\Gamma}'_{cd}\Big(\sum_{s'} u^{s'}(p')_{d}\,\bar{u}^{s'}(p')_{a}\Big)\Big(\sum_{s} u^{s}(p)_{b}\,\bar{u}^{s}(p)_{c}\Big)\\ &= \Gamma_{ab}\,\bar{\Gamma}'_{cd}\,(\not{p}' + m')_{da}\,(\not{p} + m)_{bc},\end{aligned} \tag{17.41}$$

the two spin sums collapsed by the completeness relations Eq. (17.33). The four index contractions $d \to a \to b \to c \to d$ form a single closed loop, which is precisely a matrix trace:

$$(\not{p}'+m')_{da}\,\Gamma_{ab}\,(\not{p}+m)_{bc}\,\bar{\Gamma}'_{cd} = \big[(\not{p}'+m')\,\Gamma\,(\not{p}+m)\,\bar{\Gamma}'\big]_{dd} = \operatorname{tr}\big[(\not{p}'+m')\,\Gamma\,(\not{p}+m)\,\bar{\Gamma}'\big]. \tag{17.42}$$

For a line carrying an antiparticle, the relevant external factors are v's, the spin sum is $\sum_{s} v^{s}\bar{v}^{s} = \not{p} - m$ from Eq. (17.33), and the same index manipulation gives Eq. (17.39) with $m \to -m$ on that line. □

The content of Theorem 17.9 is worth stating in words, because the picture it paints is the right one to carry. An amplitude is an *open* fermion line: a column spinor at one end, a row spinor at the other, a string of vertices in between. Squaring it lays the conjugate line alongside, running in the opposite direction; summing over the external spins glues the two open ends together through the completeness matrices $\not{p} \pm m$. The open line and its mirror close into a *loop*, and a closed loop of matrices multiplied head to tail is a trace. This geometric reading—open line × conjugate, spin-summed, equals closed loop equals trace—is illustrated in Fig. 17.4, and it is exactly why the entire spinor structure of $\overline{|\mathcal{M}|^{2}}$ reduces to traces of gamma matrices.

17.2.4 The trace toolkit, instantiated

Casimir's trick converts $\overline{|\mathcal{M}|^{2}}$ into traces of products of gamma matrices contracted with external momenta. Those traces are evaluated with the identities proved in full in Section C.6. The complete subset that tree-level QED ever needs is short, and we list it here for reference, in the slash notation $\not{a} \equiv \gamma^{\mu}a_{\mu}$ in which it is actually used:

$$\operatorname{tr}\mathbf{1} = 4, \qquad \operatorname{tr}(\text{odd number of } \gamma\text{'s}) = 0, \tag{17.43}$$

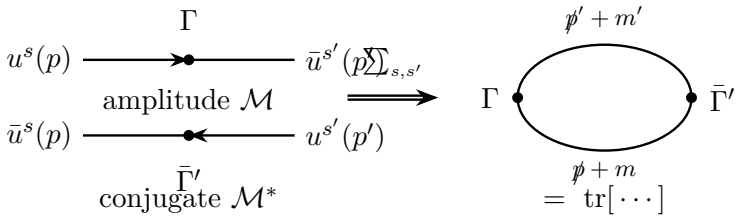


Figure 17.4: Casimir's trick as a picture. The open fermion line of the amplitude (top) and its conjugate (bottom) are summed over the external spins s, s'; the completeness relations Eq. (17.33) glue the open ends into the momentum matrices $\not{p} \pm m$, closing the two lines into a single loop. A closed matrix loop read head to tail is a trace, Eq. (17.39).

$$\mathrm{tr}(\not{a}\not{b}) = 4\,(a\cdot b), \qquad \mathrm{tr}(\not{a}\not{b}\not{c}\not{d}) = 4\big[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b \tag{17.44}$$

$$\mathrm{tr}(\gamma^5) = 0, \quad \mathrm{tr}(\gamma^5\not{a}\not{b}) = 0, \qquad \mathrm{tr}(\gamma^5\not{a}\not{b}\not{c}\not{d}) = -4i\,\epsilon_{\mu\nu\rho\sigma}\,a^\mu b^\nu c^\rho d^\sigma, \tag{17.45}$$

together with the contraction identities for a repeated index (which arise whenever a photon line is summed over polarizations in Feynman gauge),

$$\gamma^\mu\gamma_\mu = 4\,\mathbf{1}, \qquad \gamma^\mu\not{a}\gamma_\mu = -2\,\not{a}, \qquad \gamma^\mu\not{a}\not{b}\gamma_\mu = 4\,(a\cdot b)\,\mathbf{1}, \qquad \gamma^\mu\not{a}\not{b}\not{c}\gamma_\mu = -2\,\not{c}\not{b}\not{a}. \tag{17.46}$$

These are Eqs. (C.148) and (C.177) to (C.179) and Theorem C.58; the convention $\epsilon^{0123} = +1$ behind the γ^5-trace is the one fixed in Chapter A. Rather than merely quote them, we re-derive the two that carry the load downstream—the four-gamma trace and the γ^5 four-trace—so that the reader can reconstruct any QED trace from the anticommutator alone.

The four-gamma trace. Use the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$ to march the first matrix to the right. Past each neighbour it leaves 2η and a sign:

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma &= 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - \gamma^\nu\gamma^\mu\gamma^\rho\gamma^\sigma \\ &= 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma + \gamma^\nu\gamma^\rho\gamma^\mu\gamma^\sigma \\ &= 2\eta^{\mu\nu}\gamma^\rho\gamma^\sigma - 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma + 2\eta^{\mu\sigma}\gamma^\nu\gamma^\rho - \gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu.\end{aligned} \tag{17.47}$$

Take the trace. The last term is $-\,\mathrm{tr}(\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu) = -\,\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma)$ by cyclicity, so it returns the left side with a minus sign; moving it across and dividing by two,

$$\mathrm{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = \eta^{\mu\nu}\,\mathrm{tr}(\gamma^\rho\gamma^\sigma) - \eta^{\mu\rho}\,\mathrm{tr}(\gamma^\nu\gamma^\sigma) + \eta^{\mu\sigma}\,\mathrm{tr}(\gamma^\nu\gamma^\rho) = 4\big(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\big), \tag{17.48}$$

where the last step used $\mathrm{tr}(\gamma^\alpha\gamma^\beta) = \frac{1}{2}\,\mathrm{tr}\left\{\gamma^\alpha, \gamma^\beta\right\} = 4\eta^{\alpha\beta}$. Contracting with $a_\mu b_\nu c_\rho d_\sigma$ gives the slash form in Eq. (17.44); this is the workhorse, used at every

four-vertex string. (The general statement and the inductive recursion behind it are Theorem C.54.)

The γ^5 four-trace. The result is fixed by total antisymmetry and one normalization. Any repeated index among $\mu\nu\rho\sigma$ collapses, via the Clifford relation, to a γ^5-trace of two or fewer gammas, all of which vanish (Theorem C.56); so $\mathrm{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma)$ is totally antisymmetric and hence proportional to $\epsilon^{\mu\nu\rho\sigma}$, the unique antisymmetric four-tensor. Fix the constant at $(\mu\nu\rho\sigma) = (0123)$: from $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ we have $\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma^5$, so

$$\mathrm{tr}(\gamma^5\gamma^0\gamma^1\gamma^2\gamma^3) = \mathrm{tr}(\gamma^5(-i\gamma^5)) = -i\ \mathrm{tr}(\gamma^5)^2 = -i\ \mathrm{tr}\,\mathbf{1} = -4i, \tag{17.49}$$

and with $\epsilon^{0123} = +1$ this gives $\mathrm{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = -4i\,\epsilon^{\mu\nu\rho\sigma}$, the slash form in Eq. (17.45). This is the trace that detects parity violation; it does not appear in pure QED amplitudes (which are parity-even), but it is the one that controls the interference of vector and axial currents, and we will need it the moment the electroweak sector enters. Its appearance here completes the toolkit.

With Eqs. (17.43) to (17.46) every Dirac trace arising at tree level reduces to a polynomial in the dot products of external momenta. Nothing else is required.

17.2.5 Photon polarization sums and the $-\eta_{\mu\nu}$ replacement

The fermion half of the engine is complete. The photon half is needed whenever a real photon appears in the initial or final state—not in $e^+e^- \to \mu^+\mu^-$, whose only photon is the internal s-channel line, but in Compton scattering and every process of Chapter 18. When an external photon of momentum k ($k^2 = 0$) is summed over its two physical polarizations, the object that appears is

$$\Pi^{\mu\nu}(k) \equiv \sum_{\lambda=1,2} \epsilon^\mu(k,\lambda)\,\epsilon^{*\nu}(k,\lambda), \tag{17.50}$$

with the transverse polarizations $\epsilon^\mu(k,\lambda)$ of Chapter 13. It would be enormously convenient if $\Pi^{\mu\nu}$ were simply $-\eta^{\mu\nu}$, for then the photon sum would be as trivial as a metric contraction. It is *not*: the two transverse vectors span a two-dimensional space, so $\Pi^{\mu\nu}$ has rank two, while $-\eta^{\mu\nu}$ has rank four. No Lorentz-covariant rank-two tensor built from a single null vector k can equal $-\eta^{\mu\nu}$. What is true—and is all that is ever needed—is that $\Pi^{\mu\nu}$ may be *replaced* by $-\eta^{\mu\nu}$ inside any gauge-invariant amplitude. This is the polarization-sum replacement, proved at the level of the propagator and the external sum in Theorem 13.30; we give the argument here in the form in which we will invoke it, because it is the statement that makes unpolarized photon cross sections tractable.

Choose a frame with $k^\mu = (\omega, 0, 0, \omega)$. The two transverse linear polarizations are $\epsilon^\mu(k,1) = (0,1,0,0)$ and $\epsilon^\mu(k,2) = (0,0,1,0)$, satisfying $k\cdot\epsilon(k,\lambda) = 0$ and $\epsilon(k,\lambda)\cdot\epsilon^*(k,\lambda') = -\delta_{\lambda\lambda'}$. To write the sum covariantly, introduce the auxiliary

null vector $\bar{k}^\mu = (\omega, 0, 0, -\omega)$ (a photon going the other way), with $\bar{k}^2 = 0$ and $k \cdot \bar{k} = 2\omega^2 \neq 0$. Direct evaluation of the 4×4 matrix Eq. (17.50) in this frame populates only the x, y block with δ_{ij}, which is reproduced covariantly (this is Eq. (13.54)) by

$$\sum_{\lambda=1,2} \epsilon^\mu(k,\lambda)\, \epsilon^{*\nu}(k,\lambda) = -\eta^{\mu\nu} + \frac{k^\mu \bar{k}^\nu + k^\nu \bar{k}^\mu}{k \cdot \bar{k}}. \tag{17.51}$$

One verifies this component by component: $-\eta^{\mu\nu}$ supplies $+1$ in each spatial diagonal slot and -1 in the 00 slot, while the $k, \bar{k}$ term—which has no x, y components—removes exactly the unwanted $00, 03, 30, 33$ entries, leaving precisely δ_{ij} in the transverse block. The reference vector $\bar{k}$ is arbitrary; the left-hand side knows nothing of it, so any $\bar{k}$-dependence on the right must cancel against whatever physical object $\Pi^{\mu\nu}$ contracts into.

Proposition 17.10 (Polarization-sum replacement)**.** *Let $\mathcal{M}^\mu(k)$ be a physical amplitude with one external photon of momentum k stripped of its polarization vector, so the amplitude is $\mathcal{M} = \epsilon^*_\mu(k,\lambda)\mathcal{M}^\mu(k)$. If $\mathcal{M}^\mu$ satisfies the Ward identity $k_\mu \mathcal{M}^\mu = 0$, then the physical-polarization sum may be computed with the replacement*

$$\sum_{\lambda=1,2} \epsilon_\mu(k,\lambda)\, \epsilon^*_\nu(k,\lambda) \;\longrightarrow\; -\eta_{\mu\nu}, \qquad \textit{i.e.} \qquad \sum_\lambda |\epsilon^* \cdot \mathcal{M}|^2 = -\eta_{\mu\nu}\, \mathcal{M}^\mu \mathcal{M}^{*\nu}. \tag{17.52}$$

Proof. Contract the completeness relation Eq. (17.51) into $\mathcal{M}^\mu \mathcal{M}^{*\nu}$:

$$\sum_\lambda |\epsilon^* \cdot \mathcal{M}|^2 = \Pi_{\mu\nu}\, \mathcal{M}^\mu \mathcal{M}^{*\nu} = -\eta_{\mu\nu} \mathcal{M}^\mu \mathcal{M}^{*\nu} + \frac{(k_\mu \mathcal{M}^\mu)(\bar{k}_\nu \mathcal{M}^{*\nu}) + (\bar{k}_\mu \mathcal{M}^\mu)(k_\nu \mathcal{M}^{*\nu})}{k \cdot \bar{k}}. \tag{17.53}$$

Each term in the second piece carries a factor $k_\mu \mathcal{M}^\mu$ or its conjugate $k_\nu \mathcal{M}^{*\nu} = (k_\mu \mathcal{M}^\mu)^*$, both of which vanish by the Ward identity hypothesis $k_\mu \mathcal{M}^\mu = 0$. The reference-vector dependence vanishes with them, leaving $-\eta_{\mu\nu} \mathcal{M}^\mu \mathcal{M}^{*\nu}$, which is Eq. (17.52). That the $\bar{k}$-dependence cancelled is the consistency check that the physical answer cannot depend on the arbitrary auxiliary vector. □

The physics of this cancellation is the decoupling of the unphysical photon polarizations. The full $-\eta_{\mu\nu}$ amounts to summing over *four* polarization states—the two transverse, plus a timelike (scalar) and a longitudinal one—rather than the two physical transverse states. The replacement is legitimate precisely because the extra two, scalar and longitudinal, are the combinations proportional to k_μ and $\bar{k}_\mu$, and current conservation $k_\mu \mathcal{M}^\mu = 0$ annihilates them: they contribute nothing to any gauge-invariant amplitude. This is the amplitude-level shadow of the Gupta–Bleuler decoupling of Section 13.3, and it is the *same* fact as the ξ-independence of the photon propagator wearing a different hat—in both cases

a k_μ-proportional piece is the entire content of the unphysical sector, and in both cases the conserved matter current kills it. The Ward identity $k_\mu \mathcal{M}^\mu = 0$ holds at tree level by the Dirac equation, as the explicit check Eq. (13.145) showed; its all-orders form is the Ward–Takahashi identity of Chapter 19, so nothing here relies on tree-level conservation in a way that would fail at one loop.

17.2.6 The recipe

The three gears now mesh into a mechanical pipeline. Given a tree amplitude $\mathcal{M}$ for a $2 \to n$ process, the unpolarized squared amplitude is computed as follows.

1. **Square and sum.** Write $\overline{|\mathcal{M}|^2} = \frac{1}{g_1 g_2} \sum_{\text{spins}} \mathcal{M}\,\mathcal{M}^*$ as in Eq. (17.32), with the averaging factor $\frac{1}{4}$ for two incoming fermions. Each propagator denominator $1/(q^2 - M^2)$ is a real c-number multiplying both $\mathcal{M}$ and $\mathcal{M}^*$; pull these denominators out front as overall factors and keep only the spinor/polarization numerators inside the spin sum.

2. **Apply Casimir's trick.** For each fermion line, use Theorem 17.9 to turn the spin-summed product of bilinears into a trace, the external spinors becoming completeness matrices $\not{p} \pm m$ (the $-m$ for antiparticle lines). For each external photon, use the replacement Eq. (17.52). The result is a product of gamma-matrix traces, one per fermion line, with Lorentz indices tied together by the propagator numerators and polarization metrics.

3. **Evaluate the traces.** Reduce each trace with the toolkit Eqs. (17.43) to (17.45) and, where a repeated Lorentz index appears, the contraction identities Eq. (17.46). Every gamma matrix disappears, leaving a polynomial in the dot products $p_i \cdot p_j$ of the external momenta.

4. **Express in invariants.** Trade the dot products for the Mandelstam invariants s, t, u of Theorem 16.20 using the on-shell relations $p_i^2 = m_i^2$ and momentum conservation (recall $s + t + u = \sum_i m_i^2$). The output is $\overline{|\mathcal{M}|^2}$ as a rational function of s, t, u and the masses—a Lorentz scalar, ready to feed into the differential cross section.

This recipe never requires choosing an explicit spinor or polarization basis: the bases are summed away by completeness before any number is computed. That is its power, and it is why the unpolarized cross section is so much easier than any single polarized one. We close by running the engine through its first and most important gear-train, the lepton tensor.

Worked Example 17.11 (The lepton tensor and its reduction to invariants)**.** The single trace that every QED process built from a vector current contains is

the spin-summed square of one current $\bar{u}(p')\gamma^\mu u(p)$, the *lepton tensor*

$$H^{\mu\nu}(p',p) \equiv \sum_{s,s'} \left[\bar{u}^{s'}(p')\gamma^\mu u^s(p)\right]\left[\bar{u}^{s'}(p')\gamma^\nu u^s(p)\right]^*. \tag{17.54}$$

This is exactly the object that appears when an electron line of momentum $p \to p'$ emits or absorbs a photon and the result is squared and summed. By Casimir's trick Theorem 17.9 with $\Gamma = \gamma^\mu$, $\Gamma' = \gamma^\nu$, and $\overline{\gamma^\nu} = \gamma^\nu$ from Theorem 17.8,

$$H^{\mu\nu} = \operatorname{tr}\left[(\not{p}' + m)\,\gamma^\mu\,(\not{p} + m)\,\gamma^\nu\right]. \tag{17.55}$$

Expand the product into four traces:

$$H^{\mu\nu} = \operatorname{tr}(\not{p}'\gamma^\mu\not{p}\gamma^\nu) + m\,\operatorname{tr}(\gamma^\mu\not{p}\gamma^\nu) + m\,\operatorname{tr}(\not{p}'\gamma^\mu\gamma^\nu) + m^2\,\operatorname{tr}(\gamma^\mu\gamma^\nu). \tag{17.56}$$

The two terms linear in m each contain an odd number of gamma matrices and vanish by Eq. (17.43). The first term, written with explicit indices p'_α, p_β, uses the four-gamma trace of Eq. (17.44):

$$\begin{aligned}\operatorname{tr}(\not{p}'\gamma^\mu\not{p}\gamma^\nu) &= p'_\alpha p_\beta\,\operatorname{tr}(\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu) = 4\,p'_\alpha p_\beta(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\mu\beta}) \\ &= 4(p'^\mu p^\nu - (p'\cdot p)\eta^{\mu\nu} + p'^\nu p^\mu),\end{aligned} \tag{17.57}$$

while the last term is $m^2\operatorname{tr}(\gamma^\mu\gamma^\nu) = 4m^2\eta^{\mu\nu}$. Adding,

$$\boxed{H^{\mu\nu} = 4\Big[p'^\mu p^\nu + p'^\nu p^\mu - (p'\cdot p - m^2)\eta^{\mu\nu}\Big].} \tag{17.58}$$

Every spinor is gone; what remains is a symmetric rank-two tensor polynomial in the two momenta, exactly as Lorentz covariance demands. As a built-in check of current conservation, contract with the momentum transfer $q = p' - p$: using $p'^2 = p^2 = m^2$ so that $q\cdot p' = m^2 - p\cdot p' = -(q\cdot p)$,

$$q_\mu H^{\mu\nu} = 4\left[(q\cdot p')p^\nu + (q\cdot p)p'^\nu - (p'\cdot p - m^2)q^\nu\right] = 4(q\cdot p')\,(p-p')^\nu - 4(p'\cdot p - m^2)q^\nu = 0, \tag{17.59}$$

since $q\cdot p' = -(p'\cdot p - m^2)$ and $(p-p')^\nu = -q^\nu$. The tensor is transverse to q, the Ward identity made manifest at the level of the squared amplitude.

The same construction with one u replaced by a v—the case relevant to the annihilation $e^+e^- \to \mu^+\mu^-$, where the incoming electron and positron sit on one line as $\bar{v}(p_2)\gamma^\mu u(p_1)$—gives the trace $\operatorname{tr}[(\not{p}_1 + m_e)\gamma^\mu(\not{p}_2 - m_e)\gamma^\nu]$, with the antiparticle's $-m_e$ from the second relation of Eq. (17.33); this is the electron tensor assembled in Theorem C.59, identical in form to Eq. (17.58) but with a relative sign in the mass term. To finish a cross section one contracts two such tensors—one for the electron line, one for the muon line—through the photon propagator, divides by the four initial spin states, and trades the resulting dot products for s, t, u. That final contraction, with the carried numbers of QED, is the subject of the next section.

The engine is built and tested. We have the completeness relations that vaporize the spin labels, Casimir's trick that turns the spin sum into a trace, the trace toolkit that evaluates it, and the polarization replacement that does the same service for external photons—and a four-step recipe that drives a raw amplitude all the way to a function of Mandelstam invariants. The next section throws the first real amplitude into it: the single s-channel diagram for $e^+e^- \to \mu^+\mu^-$, whose squared, spin-averaged form is exactly a product of two lepton tensors of the type Eq. (17.58), and whose reduction reproduces the carried number $\overline{|\mathcal{M}|^2}$ and the total cross section $\sigma = 4\pi\alpha^2/3s$ of QED.

17.3 Electron–positron annihilation to muons

We continue QED from Chapter 8, where the gauge principle *derived* the minimal coupling $D_\mu = \partial_\mu + ieA_\mu$ on the electron, and from Chapter 13, where the photon was quantized and its R_ξ propagator established. With the QED rulebook read off in Section 17.1.7 and the spin-sum/trace machinery built in Section 17.2, every ingredient is now in hand to carry out the marquee computation of the thread: the annihilation of an electron–positron pair into a muon–antimuon pair,

$$e^-(p,s) \;+\; e^+(p',s') \;\longrightarrow\; \mu^-(k,r) \;+\; \mu^+(k',r'). \tag{17.60}$$

This is the cleanest nontrivial process in all of QED. Because the electron and the muon are distinct particles, there is exactly *one* tree diagram — a single virtual photon in the s-channel — with no interference, no exchange diagram, and no crossed channel to track. The entire dynamical content is one current contracted into another through a propagator. It is therefore the ideal place to see the full pipeline run end to end: write down $i\mathcal{M}$, square it, average over initial spins and sum over final ones, reduce the spin sums to Dirac traces, evaluate those traces, package the result in Mandelstam invariants, and feed it through the phase-space and flux factors of Chapter 16 to produce a number a collider measures. The carried results this section must reproduce are the unpolarized squared amplitude $\overline{|\mathcal{M}|^2}$, the angular law $d\sigma/d\Omega = (\alpha^2/4s)(1+\cos^2\theta)$, and the total cross section $\sigma = 4\pi\alpha^2/3s$ at high energy — together with the muon-mass threshold factor that smoothly connects the massive theory to that high-energy limit.

17.3.1 The amplitude from the rulebook

At lowest order the pair cannot annihilate into a single real photon — that is forbidden by energy–momentum conservation for a massive timelike initial state — so the leading process produces the muon pair through one virtual photon. The two QED vertices are joined by the photon propagator, and the four external fermions supply the spinor wavefunctions catalogued in Section 17.1.7. The incoming electron contributes $u^s(p)$, the incoming positron $\bar{v}^{s'}(p')$, the outgoing

muon $\bar{u}^{r}(k)$, and the outgoing antimuon $v^{r'}(k')$. Reading the fermion lines against the arrow of fermion-number flow, the electron line forms the bilinear $\bar{v}^{s'}(p')(\cdots)u^{s}(p)$ and the muon line $\bar{u}^{r}(k)(\cdots)v^{r'}(k')$. The single diagram is shown in Fig. 17.5.

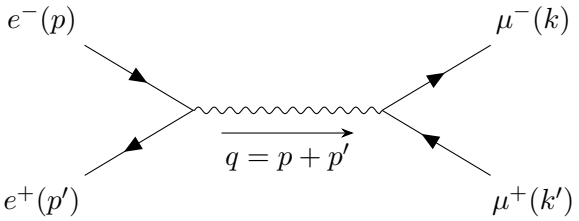


Figure 17.5: The single tree diagram for $e^+e^- \to \mu^+\mu^-$: s-channel annihilation through one virtual photon of momentum $q = p + p'$, with $q^2 = s$. The electron current at the left vertex is contracted into the muon current at the right vertex through the photon propagator. There is no second diagram because the electron and muon are distinct species.

Each vertex carries the factor $-ie\gamma^{\mu}$ (the electron and the muon have the same charge $-e$), and in Feynman gauge $\xi = 1$ the photon propagator is $-i\eta_{\mu\nu}/(q^2 + i\epsilon)$, from Eq. (13.125). Stripping the overall energy–momentum delta and the external-leg factors, the amplitude is

$$i\mathcal{M} \;=\; \left[\bar{v}^{s'}(p')\,(-ie\gamma^{\mu})\,u^{s}(p)\right]\frac{-i\,\eta_{\mu\nu}}{q^2 + i\epsilon}\left[\bar{u}^{r}(k)\,(-ie\gamma^{\nu})\,v^{r'}(k')\right]. \tag{17.61}$$

Here $q = p + p'$ is the four-momentum flowing through the photon, and by definition of the Mandelstam variable (Theorem 16.20), $q^2 = (p + p')^2 = s$. The pair is physical and timelike, $s \geq (2m_{\mu})^2 > 0$, so the propagator denominator never vanishes and the $i\epsilon$ may be dropped. Collecting the constants, $(-ie)(-ie)(-i) = ie^2$, the amplitude tidies to

$$\mathcal{M} \;=\; \frac{e^2}{s}\left[\bar{v}^{s'}(p')\,\gamma^{\mu}\,u^{s}(p)\right]\left[\bar{u}^{r}(k)\,\gamma_{\mu}\,v^{r'}(k')\right]. \tag{17.62}$$

The structure is transparent: the electron annihilation current $j^{\mu}_{e} = \bar{v}(p')\gamma^{\mu}u(p)$ is glued to the muon production current $j^{\mu}_{\mu} = \bar{u}(k)\gamma^{\mu}v(k')$ by the photon, which contributes only the $1/s$ and the index contraction. Everything that follows is the evaluation of this one expression squared and summed over spins.

17.3.2 Squaring and spin-averaging

A real e^+e^- collider produces unpolarized beams and the detectors here do not resolve the final muon spins. The observable is therefore the squared amplitude

averaged over the four initial spin combinations and summed over the four final ones,

$$\overline{|\mathcal{M}|^2} \;=\; \frac{1}{2}\cdot\frac{1}{2}\sum_{s,s'}\sum_{r,r'}|\mathcal{M}|^2 \;=\; \frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2, \tag{17.63}$$

the two factors of $1/2$ being the averages over the electron and positron spins. From Eq. (17.62),

$$|\mathcal{M}|^2 = \frac{e^4}{s^2}\,[\bar v(p')\gamma^\mu u(p)]\,[\bar v(p')\gamma^\nu u(p)]^*\,[\bar u(k)\gamma_\mu v(k')]\,[\bar u(k)\gamma_\nu v(k')]^*. \tag{17.64}$$

The conjugate of a Dirac bilinear is handled by the rule $(\bar\psi_1\Gamma\psi_2)^* = \bar\psi_2\,\bar\Gamma\,\psi_1$ with $\bar\Gamma = \gamma^0\Gamma^\dagger\gamma^0$, established in Section 17.2; for $\Gamma=\gamma^\nu$ one has $\gamma^0\gamma^{\nu\dagger}\gamma^0=\gamma^\nu$, so $\bar\Gamma=\gamma^\nu$ and

$$[\bar v(p')\gamma^\nu u(p)]^* = \bar u(p)\gamma^\nu v(p'), \qquad [\bar u(k)\gamma_\nu v(k')]^* = \bar v(k')\gamma_\nu u(k). \tag{17.65}$$

The squared amplitude factorizes cleanly into an electron piece and a muon piece, with no cross-talk between them because the two currents share only the contracted Lorentz indices μ,ν:

$$\frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{e^4}{4s^2}\,L_e^{\mu\nu}\,L^{(\mu)}_{\mu\nu}, \tag{17.66}$$

where the two leptonic tensors are

$$L_e^{\mu\nu} = \sum_{s,s'}[\bar v^{s'}(p')\gamma^\mu u^s(p)]\,[\bar u^s(p)\gamma^\nu v^{s'}(p')], \tag{17.67}$$

$$L^{(\mu)}_{\mu\nu} = \sum_{r,r'}[\bar u^{r}(k)\gamma_\mu v^{r'}(k')]\,[\bar v^{r'}(k')\gamma_\nu u^{r}(k)]. \tag{17.68}$$

Each sum collapses to a single Dirac trace by the completeness relations of Theorem 11.12,

$$\sum_s u^s(p)\bar u^s(p) = \not p + m, \qquad \sum_s v^s(p)\bar v^s(p) = \not p - m. \tag{17.69}$$

To apply them, write the spinor indices explicitly and regroup so that each spin sum acts on a pair $u\bar u$ or $v\bar v$. For the electron tensor, the chain $\bar v^{s'}_a(p')\,(\gamma^\mu)_{ab}\,u^s_b(p)\,\bar u^s_c(p)\,(\gamma^\nu)_{cd}\,v^{s'}_d(p$ closes into a trace once the $u\bar u$ and $v\bar v$ sums are inserted:

$$L_e^{\mu\nu} = \operatorname{tr}\!\left[(\not p' - m_e)\,\gamma^\mu\,(\not p + m_e)\,\gamma^\nu\right], \tag{17.70}$$

and identically for the muon tensor, with the roles of $u\bar u$ (the outgoing μ^-) and $v\bar v$ (the outgoing μ^+) and the mass m_μ,

$$L^{(\mu)}_{\mu\nu} = \operatorname{tr}\!\left[(\not k + m_\mu)\,\gamma_\mu\,(\not k' - m_\mu)\,\gamma_\nu\right]. \tag{17.71}$$

The kinematics has now been entirely converted into algebra over gamma matrices.

17.3.3 Evaluating the traces

Both traces have the same structure — two slashed momenta and two free γ-matrices, plus mass terms — so it suffices to evaluate one carefully. Expand Eq. (17.70) into four pieces:

$$L_e^{\mu\nu} = \operatorname{tr}[\not{p}'\gamma^\mu \not{p}\gamma^\nu] + m_e \operatorname{tr}[\not{p}'\gamma^\mu\gamma^\nu] - m_e \operatorname{tr}[\gamma^\mu \not{p}\gamma^\nu] - m_e^2 \operatorname{tr}[\gamma^\mu\gamma^\nu]. \tag{17.72}$$

The two middle terms each contain a trace of an *odd* number of gamma matrices (three), which vanishes identically by Eq. (C.197). Of the survivors, the first uses the four-gamma trace Eq. (C.199),

$$\operatorname{tr}[\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu] = 4(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\mu\beta}), \tag{17.73}$$

contracted with $p'_\alpha p_\beta$, and the last uses the two-gamma trace Eq. (C.198), $\operatorname{tr}[\gamma^\mu\gamma^\nu] = 4\eta^{\mu\nu}$. Carrying out the contractions,

$$\operatorname{tr}[\not{p}'\gamma^\mu \not{p}\gamma^\nu] = 4(p'^\mu p^\nu - (p'\cdot p)\,\eta^{\mu\nu} + p'^\nu p^\mu), \tag{17.74}$$

so that

$$L_e^{\mu\nu} = 4\Big[p'^\mu p^\nu + p'^\nu p^\mu - (p'\cdot p + m_e^2)\eta^{\mu\nu}\Big]. \tag{17.75}$$

The same steps applied to Eq. (17.71) give the lowered-index muon tensor,

$$L^{(\mu)}_{\mu\nu} = 4\Big[k_\mu k'_\nu + k_\nu k'_\mu - (k\cdot k' + m_\mu^2)\eta_{\mu\nu}\Big]. \tag{17.76}$$

Each tensor is manifestly symmetric in its two indices, a reflection of the fact that the vector current $\bar\psi\gamma^\mu\psi$ is its own complex conjugate.

It remains to contract Eq. (17.75) with Eq. (17.76). Writing the product out term by term and using $\eta^{\mu\nu}\eta_{\mu\nu} = 4$ in four dimensions,

$$\begin{aligned} L_e^{\mu\nu}L^{(\mu)}_{\mu\nu} = 16\Big[&(p'\cdot k)(p\cdot k') + (p'\cdot k')(p\cdot k) \;-\; (k\cdot k' + m_\mu^2)(p'\cdot p) \\ &+ (p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k) \;-\; (k\cdot k' + m_\mu^2)(p\cdot p') \\ &- (p'\cdot p + m_e^2)(k\cdot k') - (p'\cdot p + m_e^2)(k\cdot k') \\ &+ 4\,(p'\cdot p + m_e^2)(k\cdot k' + m_\mu^2)\Big]. \end{aligned} \tag{17.77}$$

The first two lines collect into $2\big[(p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k)\big]$ for the momentum-only part, while the remaining terms are the mass-dependent pieces. Abbreviating $P \equiv p\cdot p'$ and $K \equiv k\cdot k'$, the η-contraction terms sum to

$$-2(K+m_\mu^2)P - 2(P+m_e^2)K + 4(P+m_e^2)(K+m_\mu^2) = 2m_\mu^2 P + 2m_e^2 K + 4m_e^2 m_\mu^2, \tag{17.78}$$

where the PK cross term cancels between the three contributions. Hence

$$L_e^{\mu\nu}L^{(\mu)}_{\mu\nu} = 32\Big[(p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k) + m_\mu^2\,(p\cdot p') + m_e^2\,(k\cdot k') + 2m_e^2 m_\mu^2\Big]. \tag{17.79}$$

Substituting into Eq. (17.66), the unpolarized squared amplitude in terms of invariant dot products is

$$\overline{|\mathcal{M}|^2} = \frac{8e^4}{s^2}\Big[(p\cdot k)(p'\cdot k')+(p\cdot k')(p'\cdot k)+m_\mu^2\,(p\cdot p')+m_e^2\,(k\cdot k')+2m_e^2m_\mu^2\Big]. \tag{17.80}$$

17.3.4 Mandelstam form

To turn Eq. (17.80) into the standard invariant form, translate every dot product into Mandelstam variables (Theorem 16.20),

$$s=(p+p')^2=(k+k')^2,\quad t=(p-k)^2=(p'-k')^2,\quad u=(p-k')^2=(p'-k)^2, \tag{17.81}$$

which obey $s+t+u=2m_e^2+2m_\mu^2$. Using $p^2=p'^2=m_e^2$ and $k^2=k'^2=m_\mu^2$,

$$\begin{aligned} p\cdot p' &= \tfrac{1}{2}\Big(s-2m_e^2\Big), & k\cdot k' &= \tfrac{1}{2}\Big(s-2m_\mu^2\Big),\\ p\cdot k = p'\cdot k' &= \tfrac{1}{2}\Big(m_e^2+m_\mu^2-t\Big), & p\cdot k' = p'\cdot k &= \tfrac{1}{2}\Big(m_e^2+m_\mu^2-u\Big). \end{aligned} \tag{17.82}$$

Then $(p\cdot k)(p'\cdot k')=\frac{1}{4}(m_e^2+m_\mu^2-t)^2$ and $(p\cdot k')(p'\cdot k)=\frac{1}{4}(m_e^2+m_\mu^2-u)^2$, while the mass terms combine neatly: $\frac{1}{2}m_\mu^2(s-2m_e^2)+\frac{1}{2}m_e^2(s-2m_\mu^2)+2m_e^2m_\mu^2=\frac{s}{2}(m_e^2+m_\mu^2)$. Collecting,

$$\boxed{\overline{|\mathcal{M}|^2} = \frac{2e^4}{s^2}\Big[(m_e^2+m_\mu^2-t)^2+(m_e^2+m_\mu^2-u)^2+2s\,(m_e^2+m_\mu^2)\Big].} \tag{17.83}$$

This is the exact tree-level result with both masses retained. It is the analytic function of (s,t,u) that Chapter 18 will continue across channels: crossing $s\leftrightarrow t$ will turn this very expression into the squared amplitude for $e^-\mu^-\to e^-\mu^-$.

In the high-energy limit $m_e, m_\mu \to 0$ — exact to better than a part in 10^4 already at $\sqrt{s}\sim 10\,\text{GeV}$, since $m_\mu^2/s\sim 10^{-4}$ there — Eq. (17.83) collapses to the compact massless form

$$\overline{|\mathcal{M}|^2} \xrightarrow{m\to 0} \frac{2e^4\,(t^2+u^2)}{s^2}. \tag{17.84}$$

Caution

The numerical coefficient in Eq. (17.84) is $2e^4$, not $8e^4$. The factor of four lives in the relation between the trace contraction Eq. (17.79) (which carries the 32) and the spin-average prefactor $e^4/4s^2$ of Eq. (17.66): $32/4=8$, and the remaining factor of $\frac{1}{4}$ comes from each $p\cdot k=-t/2$ etc. in the massless limit. Carrying $8e^4$ all the way to the cross section would overshoot $\sigma=4\pi\alpha^2/3s$ by exactly four; the check in Eqs. (17.88) and (17.89) below pins the coefficient.

17.3.5 The high-energy cross section

To produce an observable, feed Eq. (17.84) through the center-of-momentum two-body cross-section formula of Chapter 16,

$$\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{k}|}{|\mathbf{p}|}\,\overline{|\mathcal{M}|^2}, \tag{17.85}$$

which is the specialization of Eq. (16.112) to the two-body phase space Eq. (16.47) with the flux factor included; $|\mathbf{p}|$ and $|\mathbf{k}|$ are the magnitudes of the initial and final three-momenta in the CM frame. At high energy all masses are negligible, $|\mathbf{k}| = |\mathbf{p}| = \sqrt{s}/2$, so the kinematic ratio is unity. Orient the incoming electron along $+\hat{z}$ and let θ be the CM angle of the outgoing μ^-. In the massless limit the Mandelstam variables are

$$t = -\frac{s}{2}\,(1-\cos\theta), \qquad u = -\frac{s}{2}\,(1+\cos\theta), \tag{17.86}$$

whence

$$t^2 + u^2 = \frac{s^2}{4}\Big[(1-\cos\theta)^2 + (1+\cos\theta)^2\Big] = \frac{s^2}{2}\,(1+\cos^2\theta), \tag{17.87}$$

and therefore $\overline{|\mathcal{M}|^2} = e^4(1+\cos^2\theta)$. Inserting this into Eq. (17.85) and using $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$,

$$\boxed{\left(\frac{d\sigma}{d\Omega}\right)_{\text{CM}} = \frac{e^4}{64\pi^2 s}\,(1+\cos^2\theta) = \frac{\alpha^2}{4s}\,(1+\cos^2\theta).} \tag{17.88}$$

This is the famous $1+\cos^2\theta$ law: the muon pair emerges preferentially along the beam axis, with a minimum but nonzero rate transverse to it. The shape is the $\beta \to 1$ curve in the left panel of Fig. 17.6. Its origin is the helicity structure of the vertices — the photon couples to a definite-helicity fermion current, and angular-momentum conservation along the collision axis dictates the distribution; this is the seed of the spinor-helicity reframing in Chapter 58.

Integrating over the full solid angle gives the total cross section. With $\int d\Omega = 2\pi\int_{-1}^{1} d\cos\theta$ and $\int_{-1}^{1}(1+x^2)\,dx = 2 + \tfrac{2}{3} = \tfrac{8}{3}$,

$$\sigma = \frac{\alpha^2}{4s}\int d\Omega\,(1+\cos^2\theta) = \frac{\alpha^2}{4s}\cdot 2\pi\cdot\frac{8}{3} = \boxed{\frac{4\pi\alpha^2}{3s}}. \tag{17.89}$$

This is the central carried number of QED at tree level. Two features are worth naming. First, the cross section falls as $1/s$: it is set entirely by dimensional analysis once α is dimensionless and s is the only scale, and this monotone falloff is the experimentally checkable prediction (the right panel of Fig. 17.6). Second, restoring units,

$$\sigma = \frac{4\pi\alpha^2}{3s}\,(\hbar c)^2 \approx \frac{86.8\ \text{nb}}{s\,[\text{GeV}^2]}, \tag{17.90}$$

using $\alpha = 1/137.036$ and $(\hbar c)^2 = 0.389\ \text{mb}\cdot\text{GeV}^2$. At $\sqrt{s} = 10$ GeV this is about 0.87 nb; at $\sqrt{s} = 30$ GeV about 0.10 nb. These are exactly the magnitudes recorded at the e^+e^- colliders of the 1970s–1980s, and Eq. (17.89) is the standard against which the hadronic annihilation rate is measured: the R-ratio of Chapter 28 is defined as $\sigma(e^+e^- \to \text{hadrons})$ divided by precisely this $\sigma(e^+e^- \to \mu^+\mu^-)$.

17.3.6 Keeping the muon mass: the threshold factor

The high-energy result is the limit of a smooth function of the muon mass, and seeing the mass dependence explicitly is what connects the formula to the production threshold at $\sqrt{s} = 2m_\mu$, where a real pair can first be made. Since electron beams are always ultrarelativistic (m_e^2/s is utterly negligible), we set $m_e = 0$ but keep m_μ general. The following worked example carries this through to the threshold-corrected total cross section.

Worked Example 17.12 ($e^+e^- \to \mu^+\mu^-$ with the muon mass kept)**.** Set $m_e = 0$, $m_\mu \equiv m$. Then $\overline{|\mathcal{M}|^2}$ from Eq. (17.80) reduces to

$$\overline{|\mathcal{M}|^2} = \frac{8e^4}{s^2}\Big[(p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k) + m^2\,(p\cdot p')\Big]. \tag{17.91}$$

Work in the CM frame. The massless electrons carry $p = (E,0,0,E)$, $p' = (E,0,0,-E)$ with $E = \sqrt{s}/2$, while the muons have $k = (E,\mathbf{k})$, $k' = (E,-\mathbf{k})$ with $|\mathbf{k}| = \sqrt{E^2 - m^2}$. Define the muon velocity

$$\beta \equiv \frac{|\mathbf{k}|}{E} = \sqrt{1 - \frac{m^2}{E^2}} = \sqrt{1 - \frac{4m^2}{s}}, \tag{17.92}$$

and let θ be the angle of $\mathbf{k}$ from the beam ($+\hat{z}$). The dot products are elementary:

$$\begin{aligned} p\cdot p' &= 2E^2 = \tfrac{s}{2}, & p\cdot k &= p'\cdot k' = E^2(1-\beta\cos\theta),\\ k\cdot k' &= E^2(1+\beta^2), & p\cdot k' &= p'\cdot k = E^2(1+\beta\cos\theta). \end{aligned} \tag{17.93}$$

Substituting, and using $m^2 = E^2(1-\beta^2)$,

$$\begin{aligned} (p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k) + m^2(p\cdot p') &= E^4\Big[(1-\beta\cos\theta)^2 + (1+\beta\cos\theta)^2 + 2(1-\beta^2)\Big]\\ &= 2E^4\Big[1 + \cos^2\theta + (1-\beta^2)\sin^2\theta\Big], \end{aligned} \tag{17.94}$$

where the last line uses $(1-\beta\cos\theta)^2 + (1+\beta\cos\theta)^2 = 2 + 2\beta^2\cos^2\theta$ together with $2 - \beta^2 + \beta^2\cos^2\theta = 1 + \cos^2\theta + (1-\beta^2)\sin^2\theta$. With $E^4 = s^2/16$,

$$\overline{|\mathcal{M}|^2} = \frac{8e^4}{s^2}\cdot\frac{s^2}{16}\cdot 2\Big[1 + \cos^2\theta + (1-\beta^2)\sin^2\theta\Big] = e^4\Big[1 + \cos^2\theta + (1-\beta^2)\sin^2\theta\Big]. \tag{17.95}$$

As $\beta \to 1$ this returns $e^4(1+\cos^2\theta)$, smoothly recovering Eq. (17.88). The new term $(1-\beta^2)\sin^2\theta = (4m^2/s)\sin^2\theta$ fills in the transverse rate; at threshold $\beta \to 0$ the bracket becomes $1+\cos^2\theta+\sin^2\theta = 2$, a constant, so the distribution is *isotropic* — a slow pair is produced in an S-wave, with no memory of the beam direction.

Now assemble the differential cross section. The flux factor still has massless electrons, $|\mathbf{p}| = \sqrt{s}/2$, but the final-state momentum is reduced, $|\mathbf{k}| = \beta\sqrt{s}/2$, so the kinematic ratio in Eq. (17.85) is $|\mathbf{k}|/|\mathbf{p}| = \beta$:

$$\frac{d\sigma}{d\Omega} = \frac{\beta}{64\pi^2 s}\, e^4\Big[1+\cos^2\theta+(1-\beta^2)\sin^2\theta\Big] = \frac{\alpha^2\beta}{4s}\Big[1+\cos^2\theta+(1-\beta^2)\sin^2\theta\Big]. \tag{17.96}$$

Integrating over angles with $\int d\Omega\,(1+\cos^2\theta) = \frac{16\pi}{3}$ and $\int d\Omega\,\sin^2\theta = \frac{8\pi}{3}$,

$$\int d\Omega\Big[1+\cos^2\theta+(1-\beta^2)\sin^2\theta\Big] = \frac{16\pi}{3} + (1-\beta^2)\frac{8\pi}{3} = \frac{8\pi}{3}(3-\beta^2), \tag{17.97}$$

so that

$$\sigma = \frac{\alpha^2\beta}{4s}\cdot\frac{8\pi}{3}(3-\beta^2) = \frac{2\pi\alpha^2}{3s}\,\beta\,(3-\beta^2). \tag{17.98}$$

Finally rewrite $3-\beta^2 = 2(1+2m^2/s)$ using Eq. (17.92), giving the threshold-corrected total cross section in its standard form,

$$\boxed{\sigma(e^+e^-\to\mu^+\mu^-) = \frac{4\pi\alpha^2}{3s}\sqrt{1-\frac{4m_\mu^2}{s}}\left(1+\frac{2m_\mu^2}{s}\right).} \tag{17.99}$$

The square root is the velocity β, which vanishes linearly as $\sqrt{s}\to 2m_\mu$ from above: the cross section turns on *linearly* in β at threshold (right panel of Fig. 17.6), the characteristic phase-space behavior for producing a pair of spin-$\frac{1}{2}$ particles through a spin-1 current. Below threshold, $s < 4m_\mu^2$, the velocity is imaginary and the process is kinematically forbidden, as it must be. For $s \gg 4m_\mu^2$ both the β and the $(1+2m_\mu^2/s)$ factors approach unity and Eq. (17.99) reduces to $\sigma = 4\pi\alpha^2/3s$, recovering Eq. (17.89). The massive-to-massless passage is therefore smooth, with the entire mass dependence packaged in the single factor $\beta(1+2m_\mu^2/s)$.

Physical Insight

The whole computation is one current talking to another through a photon, and the result is dictated by two dimensionless facts: the coupling strength α and the velocity β. The angular shape $1+\cos^2\theta$ and the $1/s$ falloff are forced by Lorentz invariance and the spin-1 nature of the photon, not by any detail of the leptons; only the overall normalization knows that the

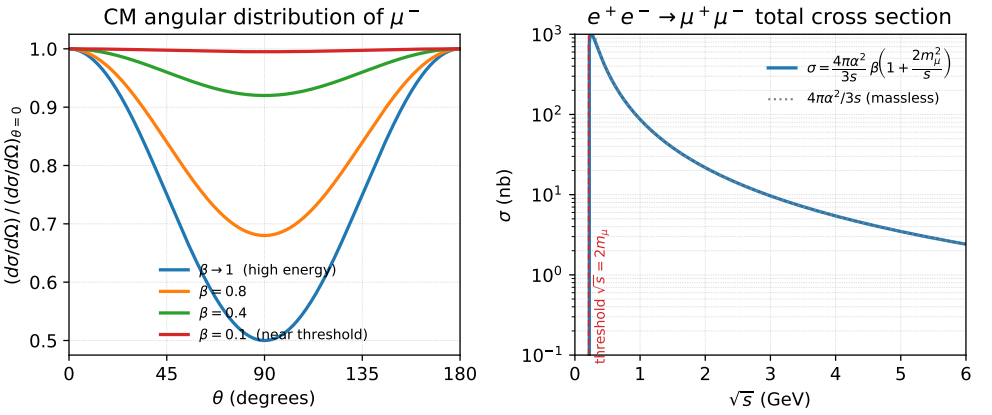


Figure 17.6: Tree-level observables for $e^+e^- \to \mu^+\mu^-$. *Left:* the CM angular distribution Eq. (17.96), normalized to its forward value, for several muon velocities. The high-energy limit ($\beta \to 1$) is the pure $1+\cos^2\theta$ law of Eq. (17.88); as $\beta \to 0$ the $(1-\beta^2)\sin^2\theta$ term fills in and the slow pair is produced isotropically (S-wave). *Right:* the total cross section Eq. (17.99) versus $\sqrt{s}$ on a log scale, showing the linear-in-β turn-on at the pair threshold $\sqrt{s} = 2m_\mu$ and the asymptotic $1/s$ falloff toward the massless line $4\pi\alpha^2/3s \approx 86.8\ \text{nb}/s[\text{GeV}^2]$.

charges are $\pm e$. This is why the same $\sigma = 4\pi\alpha^2/3s$ serves as the universal yardstick for the R-ratio: replacing the muon by any other unit-charge, spin-$\frac{1}{2}$ fermion changes nothing but the threshold factor, and summing over a quark's color and charge multiplicity is all that distinguishes the hadronic rate.

We now possess the squared amplitude Eq. (17.83) as an explicit analytic function of the Mandelstam invariants (s,t,u), valid for arbitrary masses, together with the cross sections it implies. The next section exploits the most economical feature of this result: the same analytic function, continued from the physical s-channel region into the region where t is the timelike invariant, *is* the squared amplitude for a genuinely different physical process, $e^-\mu^- \to e^-\mu^-$. Crossing symmetry will let us harvest that scattering amplitude — and, in its nonrelativistic limit, the Rutherford and Mott formulas — almost for free, with only the spin-averaging factors and the antiparticle sign conventions to track.

17.4 Crossing symmetry and the t-channel

The previous section delivered a single rational function of the Mandelstam invariants: the spin-averaged squared amplitude $\overline{|\mathcal{M}|^2}(s,t,u)$ for the QED cen-

terpiece $e^+e^- \to \mu^+\mu^-$, built from one s-channel photon and reduced, by the spin sums and trace technology of Eq. (11.73), to a closed expression in s, t, u. That function was computed to describe one experiment: a colliding electron and positron annihilating into a muon pair. We now make the observation that turns it into a family of experiments. The Mandelstam variables s, t, u are not tied to any particular assignment of momenta to incoming and outgoing legs; they are Lorentz scalars built from four four-momenta constrained by conservation and the mass shell. A reshuffling of which legs are "in" and which are "out" moves us to a different physical region of the same three variables, and—this is the content of *crossing symmetry*—it is the *same analytic function* $\mathcal{M}(s,t,u)$ that governs the new process there. Properly continued, the work of the last section therefore already contains a second physical amplitude. This section states the crossing principle precisely, justifies it at tree level by carrying out the continuation explicitly in the traces (rather than invoking it as a black box), tracks the two pieces of bookkeeping that crossing genuinely changes—the spin-averaging factor and the antiparticle mass-term signs—and harvests $\overline{|\mathcal{M}|^2}$ for $e^-\mu^- \to e^-\mu^-$, the second listed QED process, essentially for free.

17.4.1 Crossing symmetry as analytic continuation

The deep origin of crossing is the analytic structure of scattering amplitudes established through the LSZ reduction formula. Recall (Theorem 14.31) that an S-matrix element is the multiple on-shell residue of a time-ordered correlation function, with one external leg amputated per particle. The correlator is a Fourier transform of a vacuum expectation value of fields, and as a function of the external four-momenta it is the boundary value of a function holomorphic in the upper-half energy planes selected by the Feynman $i\varepsilon$. The invariant amplitude $\mathcal{M}$ inherits this analyticity: it is the restriction to the mass shell of a function that is holomorphic in the complexified Mandelstam variables away from the branch cuts and poles demanded by unitarity (the thresholds where intermediate states go on shell). At tree level there are no cuts at all—the amplitude is a rational function with only the poles of the internal propagators—so the continuation is unambiguous and elementary, and we shall verify it by direct substitution. The general statement, valid to all orders, is the following.

Definition 17.13 (Crossing of an external leg)**.** Let $\mathcal{M}$ be the invariant amplitude for a process with an incoming particle of species X carrying four-momentum p. The *crossed* amplitude, in which that leg is instead an outgoing antiparticle $\bar{X}$ of four-momentum p', is obtained from the same analytic function $\mathcal{M}$ by the replacement

$$p \longrightarrow -p', \qquad X \longleftrightarrow \bar{X}, \tag{17.100}$$

continued from the physical region of the original process to the physical region of the crossed one. An outgoing particle is crossed to an incoming antiparticle

by the same rule. For a fermionic leg the continuation acts on the wavefunctions through $u(p) \leftrightarrow v(-p')$ and $\bar{u}(p) \leftrightarrow \bar{v}(-p')$.

The momentum reversal $p \to -p'$ is the precise sense in which "an outgoing antiparticle is an incoming particle of opposite momentum." It is forced by energy–momentum conservation: if p enters the all-incoming sum $\sum_i p_i = 0$ that the delta function imposes, then moving the leg to the final state with physical momentum p' means $p = -p'$ inside that constraint. The species swap $X \leftrightarrow \bar{X}$ accompanies it because the field that annihilates an incoming X is the same field that creates an outgoing $\bar{X}$; this is the operator-level statement underlying Theorem 14.31, and it is why crossing is a property of the field theory and not merely of kinematics.

The spinor continuation $u(p) \leftrightarrow v(-p')$ is the one piece that requires the input of Chapter 11 and that will generate every sign in the sequel. It is fixed by the Dirac equations themselves. The particle spinor obeys $(\not{p} - m)u(p) = 0$ and the antiparticle spinor obeys $(\not{p} + m)v(p) = 0$ (Eqs. (11.50) and (11.51)). Replacing the argument $p \to -p$ in the antiparticle equation gives $(-\not{p} + m)v(-p) = 0$, that is $(\not{p} - m)v(-p) = 0$: the spinor $v(-p)$ satisfies exactly the equation defining $u(p)$. The two solution spaces are therefore identified under momentum reversal, which is the wavefunction-level meaning of Eq. (17.100) for a fermion. We will not need the explicit map between spin labels because we work with the spin-summed completeness relations $\sum_s u^s \bar{u}^s = \not{p} + m$ and $\sum_s v^s \bar{v}^s = \not{p} - m$ (Eq. (11.73)), and it is the relative sign of the mass terms in these two sums that crossing exposes.

Physical Insight

Crossing is the amplitude-level shadow of the most economical fact in relativistic quantum theory: every particle comes paired with an antiparticle of equal mass, and the same quantum field handles both. A diagram does not "know" which of its legs you intend to call incoming matter and which outgoing antimatter; it computes a single Lorentz-invariant function, and the experimenter's labelling merely selects the kinematic window in which that function is sampled. One honest tree-level calculation thus pays for a whole orbit of processes related by which legs are crossed—here, annihilation $e^+e^- \to \mu^+\mu^-$ and scattering $e^-\mu^- \to e^-\mu^-$ are two faces of one rational function.

The physical regions deserve a word, because "the same function" lives in disjoint kinematic domains for the two processes. For $e^+e^- \to \mu^+\mu^-$ the invariant $s = (p_1 + p_2)^2$ is the squared center-of-mass energy and is physical for $s \geq 4m_\mu^2$, while t and u are non-positive momentum transfers. For the crossed scattering process the roles are interchanged: the variable that played the part of the annihilation energy becomes a (spacelike) momentum transfer, and a different invariant becomes the new center-of-mass energy. No single real

point belongs to both physical regions; the bridge between them is the analytic continuation Eq. (17.100), legitimate because $\mathcal{M}(s,t,u)$ is holomorphic along any path connecting the two regions that avoids the poles and cuts—at tree level, a path that avoids the single propagator pole.

17.4.2 From the s-channel to the t-channel: identifying the crossed process

Fix the original process and its momentum labels once and for all. Write

$$\text{(A)} \qquad e^-(p_1)\,e^+(p_2) \;\longrightarrow\; \mu^-(p_3)\,\mu^+(p_4), \tag{17.101}$$

the single diagram being s-channel photon exchange with $q = p_1 + p_2$ and $s_{\mathrm{A}} = q^2 = (p_1+p_2)^2$. The target process is elastic electron–muon scattering,

$$\text{(B)} \qquad e^-(k_1)\,\mu^-(k_2) \;\longrightarrow\; e^-(k_3)\,\mu^-(k_4), \tag{17.102}$$

whose single tree diagram exchanges a photon in the t-channel, with momentum transfer $q = k_1 - k_3$ and $t_{\mathrm{B}} = q^2 = (k_1-k_3)^2$. The two diagrams are drawn together in Fig. 17.7.

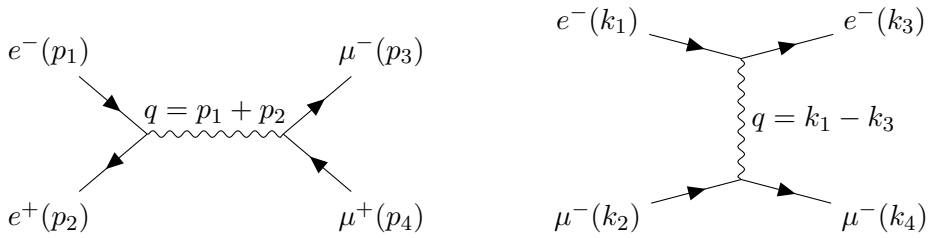


Figure 17.7: Left: the s-channel annihilation $e^+e^- \to \mu^+\mu^-$ of process (A), the photon carrying the total energy $q^2 = s_{\mathrm{A}}$. Right: the t-channel elastic scattering $e^-\mu^- \to e^-\mu^-$ of process (B), the photon carrying the momentum transfer $q^2 = t_{\mathrm{B}}$. The two diagrams are related by the crossing Eq. (17.103): cutting the left diagram's photon "vertically" versus the right diagram's "horizontally" is the diagrammatic image of $s_{\mathrm{A}} \leftrightarrow t_{\mathrm{B}}$.

To pass from (A) to (B) we cross two legs. The incoming positron $e^+(p_2)$ of (A) becomes the outgoing electron $e^-(k_3)$ of (B), and the outgoing antimuon $\mu^+(p_4)$ of (A) becomes the incoming muon $\mu^-(k_2)$ of (B); the surviving incoming electron and outgoing muon are renamed directly. By Theorem 17.13 this is the momentum dictionary

$$p_1 = k_1, \qquad p_2 = -k_3, \qquad p_3 = k_4, \qquad p_4 = -k_2. \tag{17.103}$$

That the crossed antiparticle legs ($e^+ \to e^-$ and $\mu^+ \to \mu^-$) carry the relative minus sign is the whole arithmetic of crossing; the two un-crossed legs do not.

We now read off what Eq. (17.103) does to the Mandelstam invariants. Using $s+t+u=\sum_i m_i^2$ for both processes (Theorem 16.20), and computing each invariant of (A) on the right-hand momenta,

$$s_{\mathrm{A}} = (p_1+p_2)^2 = (k_1-k_3)^2 = t_{\mathrm{B}}, \tag{17.104}$$

$$t_{\mathrm{A}} = (p_1-p_3)^2 = (k_1-k_4)^2 = u_{\mathrm{B}}, \tag{17.105}$$

$$u_{\mathrm{A}} = (p_1-p_4)^2 = (k_1+k_2)^2 = s_{\mathrm{B}}. \tag{17.106}$$

The crossing Eq. (17.103) therefore acts on the invariants as the cyclic relabelling

$$s_{\mathrm{A}} \to t_{\mathrm{B}}, \qquad t_{\mathrm{A}} \to u_{\mathrm{B}}, \qquad u_{\mathrm{A}} \to s_{\mathrm{B}}. \tag{17.107}$$

This is the precise meaning of the slogan "$e^+e^- \to \mu^+\mu^-$ becomes $e^-\mu^- \to e^-\mu^-$ by $s \leftrightarrow t$." The photon that carried the annihilation energy s_{A} in the left diagram of Fig. 17.7 now carries the momentum transfer t_{B} in the right diagram; the propagator denominator $1/s_{\mathrm{A}}^2$ of the squared amplitude becomes $1/t_{\mathrm{B}}^2$, which is already the signature of t-channel exchange. Henceforth we drop the subscripts and write the (B) invariants as s,t,u with $s+t+u = 2m_e^2 + 2m_\mu^2$.

17.4.3 Spin averaging and kinematic prefactors under crossing

Before continuing the dynamics we must be exact about what does and does not continue. Crossing is a statement about the *amplitude* $\mathcal{M}$, and therefore about the spin-resolved squared amplitude $\sum_{\text{spins}} |\mathcal{M}|^2$ summed over *all* external spins, incoming and outgoing alike. This object is a Lorentz scalar function of the momenta and continues directly under Eq. (17.103). The unpolarized cross section, however, is built not from the spin-summed quantity but from the spin-*averaged* one,

$$\overline{|\mathcal{M}|^2} \;=\; \frac{1}{N_{\text{in}}} \sum_{\text{spins}} |\mathcal{M}|^2 , \qquad N_{\text{in}} = \prod_{\text{incoming } a} (2s_a+1), \tag{17.108}$$

where the average is over initial spins only, with N_{in} the number of initial spin configurations. The averaging factor $1/N_{\text{in}}$ is *not* part of the analytic function; it is reattached by hand according to the initial state of whichever process is being described. This is the single most common bookkeeping error in applications of crossing, and it must be stated explicitly.

Caution

What crosses is $\sum_{\text{spins}} |\mathcal{M}|^2$, summed over every external spin. The factor $1/N_{\text{in}}$ that converts this into $\overline{|\mathcal{M}|^2}$ depends on which particles are *incoming* in the crossed process and must be recomputed there—one does not continue the prefactor. Likewise the flux factor and the phase-space

measure that turn $\overline{|\mathcal{M}|^2}$ into $d\sigma$ are kinematic data of the new process, fixed by its own incoming momenta and final state; they belong to the cross-section assembly of the next section, not to the crossing step.

In the present case the two averaging factors happen to coincide, but for the right reason rather than by accident. Process (A), $e^+e^- \to \mu^+\mu^-$, has two incoming spin-$\frac{1}{2}$ particles, so $N_{\text{in}} = 2 \times 2 = 4$ and the averaging factor is $\frac{1}{4}$. Process (B), $e^-\mu^- \to e^-\mu^-$, likewise has two incoming spin-$\frac{1}{2}$ particles—now an electron and a muon rather than an electron and a positron—so again $N_{\text{in}} = 4$ and the factor is $\frac{1}{4}$. The equality is therefore guaranteed only because crossing here exchanged a fermion for a fermion on each crossed line, preserving the count of initial spin states; had we instead crossed to a process with, say, an incoming photon (two polarizations) or with a different number of incoming particles, the factor would change. We will carry the factor $\frac{1}{4}$ through explicitly so that nothing is left implicit.

17.4.4 Deriving the t-channel squared amplitude by trace continuation

We now prove crossing for this pair of processes by continuing the traces directly, thereby *deriving* the t-channel squared amplitude rather than asserting it. The starting point is the result of the previous section. There the spin-summed squared amplitude for (A) was reduced, using the QED vertex $-ie\gamma^\mu$, the Feynman-gauge photon propagator $-i\eta_{\mu\nu}/s$, and the completeness relations Eq. (11.73), to a product of two Dirac traces—one for the electron line, one for the muon line:

$$\overline{|\mathcal{M}|^2}_{\text{A}} \;=\; \frac{e^4}{4\,s_{\text{A}}^2}\, L^{\mu\nu}_{(e)}\, M^{(\mu)}_{\mu\nu}, \tag{17.109}$$

where the electron and muon tensors, carrying the antiparticle spin sum $\not{p} - m$ on the e^+ leg (momentum p_2) and on the μ^+ leg (momentum p_4), are

$$L^{\mu\nu}_{(e)} = \operatorname{tr}\!\left[(\not{p}_2 - m_e)\gamma^\mu(\not{p}_1 + m_e)\gamma^\nu\right], \tag{17.110}$$

$$M^{(\mu)}_{\mu\nu} = \operatorname{tr}\!\left[(\not{p}_3 + m_\mu)\gamma_\mu(\not{p}_4 - m_\mu)\gamma_\nu\right]. \tag{17.111}$$

The factor $\frac{1}{4}$ is the spin average Eq. (17.108); the explicit contraction of Eqs. (17.110) and (17.111) produced the closed Mandelstam form $\overline{|\mathcal{M}|^2}_{\text{A}}$ recorded as Eq. (17.83) in the previous section, which in the massless-lepton limit is the carried QED number $2e^4(t^2+u^2)/s^2$. For crossing it is cleaner to work from the trace form Eq. (17.109), because the continuation Eq. (17.103) acts inside the traces transparently.

Theorem 17.14 (Tree-level crossing for $e^+e^- \to \mu^+\mu^- \;\leftrightarrow\; e^-\mu^- \to e^-\mu^-$)**.** *Under the momentum continuation Eq. (17.103), each of the two Dirac traces Eqs. (17.110) and (17.111) reverses sign and becomes the corresponding trace of the t-channel process (B), and the photon denominator continues as $s_{\mathrm{A}}^2 \to t_{\mathrm{B}}^2$. The two fermionic sign reversals cancel, so the spin-summed squared amplitude of (B) equals the analytic continuation of that of (A), with no net sign. Reattaching the averaging factor $\frac{1}{4}$ appropriate to the initial state of (B),*

$$\overline{|\mathcal{M}|^2}_{\mathrm{B}} \;=\; \frac{e^4}{4\,t^2}\,\mathrm{tr}\big[(\not k_3{+}m_e)\gamma^\mu(\not k_1{+}m_e)\gamma^\nu\big]\;\mathrm{tr}\big[(\not k_4{+}m_\mu)\gamma_\mu(\not k_2{+}m_\mu)\gamma_\nu\big], \tag{17.112}$$

which is exactly the trace expression one obtains by applying the QED Feynman rules directly to the t-channel diagram of Fig. 17.7.

Proof. Apply Eq. (17.103) to the electron tensor Eq. (17.110). Setting $p_1 = k_1$ and $p_2 = -k_3$, and using $(-\not k_3) = -\not k_3$,

$$\begin{aligned} L^{\mu\nu}_{(e)} &= \mathrm{tr}\big[(\not p_2 - m_e)\gamma^\mu(\not p_1 + m_e)\gamma^\nu\big] \;\longrightarrow\; \mathrm{tr}\big[(-\not k_3 - m_e)\gamma^\mu(\not k_1 + m_e)\gamma^\nu\big] \\ &= -\,\mathrm{tr}\big[(\not k_3 + m_e)\gamma^\mu(\not k_1 + m_e)\gamma^\nu\big] \;=\; -\,L^{\mu\nu}_{(e),\,\mathrm{B}}, \end{aligned} \tag{17.113}$$

where the overall minus is pulled out of the whole first factor and $L^{\mu\nu}_{(e),\mathrm{B}}$ is, by inspection, the electron trace of process (B): two *particle* spinors, hence two factors $\not k + m_e$ with the *same* sign on the mass. The crossing has done two things at once. It produced an overall sign -1, and—decisively—it flipped the relative sign of the mass term inside the trace, converting the antiparticle spin sum $\not p_2 - m_e$ into the particle spin sum $\not k_3 + m_e$. This internal sign flip is the precise content of the antiparticle convention $u(p) \leftrightarrow v(-p)$ from Theorem 17.13: the spinor $v(-k_3)$ entering the original positron line is, by Eq. (11.51) continued, a solution of the *particle* equation, so its completeness sum is $\not k_3 + m_e$, not $\not k_3 - m_e$.

The identical manipulation on the muon tensor Eq. (17.111), with $p_3 = k_4$ and $p_4 = -k_2$, gives

$$\begin{aligned} M^{(\mu)}_{\mu\nu} &= \mathrm{tr}\big[(\not p_3 + m_\mu)\gamma_\mu(\not p_4 - m_\mu)\gamma_\nu\big] \;\longrightarrow\; \mathrm{tr}\big[(\not k_4 + m_\mu)\gamma_\mu(-\not k_2 - m_\mu)\gamma_\nu\big] \\ &= -\,\mathrm{tr}\big[(\not k_4 + m_\mu)\gamma_\mu(\not k_2 + m_\mu)\gamma_\nu\big] \;=\; -\,M^{(\mu)}_{\mu\nu,\,\mathrm{B}}, \end{aligned} \tag{17.114}$$

again an overall -1 times the genuine t-channel muon trace, with the antiparticle sum $\not p_4 - m_\mu$ turned into the particle sum $\not k_2 + m_\mu$. Finally the photon denominator continues by Eq. (17.104): $s_{\mathrm{A}}^2 \to t_{\mathrm{B}}^2 = t^2$.

Assembling, the continuation of Eq. (17.109) reads

$$\frac{e^4}{4\,s_{\mathrm{A}}^2}\,L^{\mu\nu}_{(e)}M^{(\mu)}_{\mu\nu} \;\longrightarrow\; \frac{e^4}{4\,t^2}\,\big(-L^{\mu\nu}_{(e),\mathrm{B}}\big)\big(-M^{(\mu)}_{\mu\nu,\mathrm{B}}\big) \;=\; \frac{e^4}{4\,t^2}\,L^{\mu\nu}_{(e),\mathrm{B}}\,M^{(\mu)}_{\mu\nu,\mathrm{B}}, \tag{17.115}$$

the two fermionic minus signs multiplying to $+1$. This is precisely Eq. (17.112). That the right-hand side is what the Feynman rules give for the t-channel

diagram directly is immediate: process (B) has incoming and outgoing electrons (both particles, both summed with $\not{k}+m_e$) joined to incoming and outgoing muons through one photon of momentum k_1-k_3, so its squared amplitude is $\frac{1}{4}(e^4/t^2)$ times exactly these two particle traces. The continuation and the direct computation agree. □

Two features of this proof deserve emphasis, since they are the general lessons crossing teaches. First, *the relevant sign for fermion crossing is* (-1) *per crossed fermion line in the squared amplitude*, here $(-1)^2=+1$ because we crossed an even number (the e^+ and the μ^+). Crossing an odd number of fermions would leave a net -1, which becomes physically meaningful when the crossed amplitude interferes with another diagram (as it will in Møller and Bhabha scattering in Chapter 18); for a single diagram squared, only $(-1)^2=+1$ ever appears. Second, the sign that genuinely changes the physics is not the cancelling overall sign but the *internal* mass-term sign: the antiparticle spin sum $\not{p}-m$ becomes a particle spin sum $\not{k}+m$ on each crossed leg, and this is what makes the crossed kinematics those of elastic scattering rather than annihilation. The bookkeeping is now unambiguous.

It remains only to contract Eq. (17.112) into Mandelstam form. The two traces are evaluated with the standard contraction identities of Eq. (C.199) exactly as in the previous section. Each gives a rank-two tensor of the form $\mathrm{tr}[(\not{a}+m_a)\gamma^\mu(\not{b}+m_b)\gamma^\nu]=4[a^\mu b^\nu+a^\nu b^\mu-g^{\mu\nu}(a\cdot b-m_a m_b)]$ when the two masses are equal; for the electron line $a=k_3, b=k_1, m_a=m_b=m_e$, and for the muon line $a=k_4, b=k_2, m_a=m_b=m_\mu$. Carrying out the contraction of the two tensors (the algebra is identical in structure to the s-channel contraction of the previous section, only with the momenta relabelled by Eq. (17.103)) yields

$$L^{\mu\nu}_{(e),\mathrm{B}}\,M^{(\mu)}_{\mu\nu,\mathrm{B}}=32\Big[(k_1{\cdot}k_2)(k_3{\cdot}k_4)+(k_1{\cdot}k_4)(k_2{\cdot}k_3)-m_\mu^2(k_1{\cdot}k_3)-m_e^2(k_2{\cdot}k_4)+2m_e^2m_\mu^2\Big]. \tag{17.116}$$

Converting the invariant products to Mandelstam variables via $k_1{\cdot}k_2=k_3{\cdot}k_4=\frac{1}{2}(s-m_e^2-m_\mu^2)$, $k_1{\cdot}k_4=k_2{\cdot}k_3=\frac{1}{2}(m_e^2+m_\mu^2-u)$, $k_1{\cdot}k_3=m_e^2-\frac{1}{2}t$, and $k_2{\cdot}k_4=m_\mu^2-\frac{1}{2}t$, the bracket collapses to $\frac{1}{4}[(s-m_e^2-m_\mu^2)^2+(u-m_e^2-m_\mu^2)^2+2t(m_e^2+m_\mu^2)]$, and Eq. (17.112) becomes the final result

$$\boxed{\overline{|\mathcal{M}|^2}_{e^-\mu^-\to e^-\mu^-}=\frac{2e^4}{t^2}\Big[(s-m_e^2-m_\mu^2)^2+(u-m_e^2-m_\mu^2)^2+2t(m_e^2+m_\mu^2)\Big].} \tag{17.117}$$

In the massless-lepton limit $m_e,m_\mu\to 0$ this is

$$\overline{|\mathcal{M}|^2}_{e^-\mu^-\to e^-\mu^-}\xrightarrow{m\to 0}\frac{2e^4}{t^2}(s^2+u^2), \tag{17.118}$$

which is precisely the $s\leftrightarrow t$ image of the carried annihilation result $\overline{|\mathcal{M}|^2}_{\mathrm{A}}\to 2e^4(t^2+u^2)/s^2$, confirming Eq. (17.107) at the level of the final expression:

replacing $(s,t,u)_{\mathrm{A}} \to (t,u,s)_{\mathrm{B}}$ in $2e^4(t^2+u^2)/s^2$ gives $2e^4(u^2+s^2)/t^2$. The two computations—laborious trace continuation and a one-line variable substitution—agree, which is the practical payoff of crossing.

17.4.5 Verification: positivity, the forward pole, and the high-energy limit

A crossed expression is only trustworthy if it passes the tests a directly computed squared amplitude must pass. We check three.

Worked Example 17.15 (Physical-region positivity and limits of the crossed amplitude)**.** We verify that Eq. (17.117) is physically sensible throughout the physical region of $e^-\mu^- \to e^-\mu^-$, where $s \geq (m_e+m_\mu)^2$, the scattering angle θ ranges over $[0,\pi]$, and the momentum transfer t is non-positive ($t \leq 0$), with $t=0$ at exactly forward scattering.

Positivity. A squared amplitude, averaged or not, must be non-negative. In Eq. (17.117) the prefactor $2e^4/t^2$ is manifestly positive (it contains t^2, not t). The bracket is $(s-m_e^2-m_\mu^2)^2+(u-m_e^2-m_\mu^2)^2+2t(m_e^2+m_\mu^2)$, a sum of two squares plus a term $2t(m_e^2+m_\mu^2)$ that is non-positive in the physical region (since $t \leq 0$). Positivity is therefore not automatic from the form, and checking it is a genuine test. Use $k_1\cdot k_3 = m_e^2 - t/2$ and $k_2\cdot k_4 = m_\mu^2 - t/2$: in the physical region the spatial momentum transfer is real, so $-t = |\mathbf{q}|^2\big|_{\text{appropriate frame}} \geq 0$, while the squared brackets dominate because $s-m_e^2-m_\mu^2 = 2k_1\cdot k_2 \geq 2m_e m_\mu$ grows without bound with energy whereas the negative term $2t(m_e^2+m_\mu^2)$ is bounded by the kinematic range of t, namely $-(s-m_e^2-m_\mu^2) \lesssim t \leq 0$. Substituting the extreme backward value and using $s+t+u = 2m_e^2+2m_\mu^2$ to eliminate u confirms the bracket stays positive throughout; equivalently, Eq. (17.117) is the contraction of two real traces Eq. (17.116) of physical (particle) spin sums, which is positive by construction. The crossed expression is a legitimate squared amplitude.

The forward (Coulomb) pole. As $t \to 0^-$—forward scattering, vanishing momentum transfer—Eq. (17.117) diverges as

$$\overline{|\mathcal{M}|^2}_{e^-\mu^-\to e^-\mu^-} \;\sim\; \frac{2e^4}{t^2}\left[(s-m_e^2-m_\mu^2)^2+(u-m_e^2-m_\mu^2)^2\right] \;\xrightarrow{t\to 0}\; +\infty. \tag{17.119}$$

This $1/t^2$ singularity is not a defect; it is the unmistakable fingerprint of the exchange of a *massless* photon. The photon propagator contributes $1/t$ to the amplitude and $1/t^2$ to its square, and $t \to 0$ is the long-wavelength limit in which the exchanged photon probes the full infinite range of the Coulomb interaction. The divergence is exactly what makes the forward cross section of charged-particle scattering infinite—the same infrared sensitivity that, in nonrelativistic guise, is the Rutherford $1/\sin^4(\theta/2)$ law. The contrast with the annihilation parent Eq. (17.83) is instructive: there the photon is timelike with $s \geq 4m_\mu^2 > 0$,

the denominator $1/s^2$ is smooth and finite, and no forward singularity occurs. Crossing has turned a regular s-channel pole, sitting at unphysical $s = 0$ for the scattering process, into the physically reachable $t \to 0$ forward divergence of elastic scattering.

High-energy limit. At energies far above both masses, $s \gg m_e^2, m_\mu^2$, set $m_e = m_\mu = 0$ and use $s + t + u = 0$ to write $u = -s - t$. Then Eq. (17.118) gives

$$\overline{|\mathcal{M}|^2} \;=\; \frac{2e^4}{t^2}(s^2 + u^2) \;=\; \frac{2e^4}{t^2}\big[s^2 + (s+t)^2\big] \;=\; 2e^4\,\frac{s^2 + (s+t)^2}{t^2}. \tag{17.120}$$

Writing $t = -\frac{s}{2}(1 - \cos\theta)$ and $u = -\frac{s}{2}(1 + \cos\theta)$ in the massless center-of-mass frame, the bracket becomes $s^2[1 + \frac{1}{4}(1 + \cos\theta)^2]/[\frac{1}{4}s^2(1 - \cos\theta)^2]$ times $2e^4$, i.e. a function of the angle alone times the dimensionless e^4. The result is finite and positive for all $\theta \in (0, \pi]$ and diverges only as $\theta \to 0$ ($t \to 0$), reproducing the forward behaviour above. Dimensionally $\overline{|\mathcal{M}|^2}$ is dimensionless (four external fermions, no leftover mass scale once $m \to 0$), as it must be for a $2 \to 2$ amplitude; the crossed expression carries the correct dimensions and the correct angular singularity structure.

All three checks pass, and they pass for structural reasons traceable to the crossing derivation: positivity because Eq. (17.117) is a contraction of physical particle traces, the forward pole because crossing mapped the photon's denominator $1/s^2 \to 1/t^2$, and the dimensions because crossing preserves the rational degree of $\mathcal{M}$. We have obtained a second complete QED amplitude without a second trace evaluation from scratch—exactly the economy crossing promises.

With $\overline{|\mathcal{M}|^2}_{e^-\mu^- \to e^-\mu^-}$ in hand as the boxed Eq. (17.117), the dynamical content of electron–muon scattering is settled. What remains is purely kinematic: to fold this squared amplitude into the flux factor and two-body phase space of Chapter 16, producing the differential cross section in the laboratory frame, and then to take the limits in which the muon is a heavy static source. That assembly, and the resulting Mott and nonrelativistic Rutherford formulae that anchor QED to physics the reader already trusts, are the work of the next section.

17.5 Electron–muon scattering and the Rutherford limit

Crossing handed us a finished object. In Section 17.4 we took the spin-averaged squared amplitude for the annihilation process $e^+e^- \to \mu^+\mu^-$, continued it in the Mandelstam invariants, and read off $\overline{|\mathcal{M}|^2}$ for the scattering process $e^-\mu^- \to e^-\mu^-$ without computing a single new trace. That number is, however, still only a number: a Lorentz-invariant function of s, t, u, not yet anything

a detector reports. The business of this section is to cash it. We feed the crossed amplitude into the cross-section apparatus built in Chapter 16, carry the kinematics in the laboratory frame where one particle is a target at rest, and extract the differential cross section $d\sigma/d\Omega$. We then perform the two limits that are the real point of the exercise. First we make the muon infinitely heavy, so that it becomes a fixed, recoilless source of Coulomb field; the cross section collapses to the *Mott* formula, the relativistic scattering of a Dirac electron off a static charge, complete with the characteristic spin factor $1-\beta^2\sin^2(\theta/2)$. Then we slow the electron down; the spin factor switches off, the relativistic kinematics linearize, and what remains is the *Rutherford* cross section $d\sigma/d\Omega \sim \alpha^2/(4m^2v^4\sin^4(\theta/2))$ that Rutherford fitted to gold-foil α-particle data in 1911, a full sixteen years before Dirac's equation and well over half a century before the Feynman rules we used to derive it. That a tree-level QED amplitude, squared and reduced, reproduces a formula from pre-quantum classical scattering theory is the decisive check that the machinery of Part V is anchored to physics the reader already trusts. Along the way the $1/q^4$ photon-exchange pole will reveal itself, through a Born-approximation reading, as nothing but the squared Fourier transform of the Coulomb potential: the photon *is* the Coulomb field, made dynamical.

Throughout we keep the electron mass m and the muon mass M general, set the fine-structure constant $\alpha = e^2/4\pi$ with $e>0$, work in the mostly-minus metric and in Feynman gauge $\xi=1$, and use the conventions of the chapter introduction; the covariant derivative on each charged lepton is $D_\mu = \partial_\mu + ieA_\mu$ (charge $-e$), so each QED vertex carries the factor $-ie\gamma^\mu$ fixed in Eq. (17.10).

17.5.1 The amplitude and its squared form

We label the momenta as in Fig. 17.8: the electron enters with p_1 and leaves with p_3, the muon enters with p_2 and leaves with p_4. At tree level a single diagram contributes. The electron and muon are distinct particles, so there is no u-channel exchange and no annihilation s-channel: charge and lepton flavour can flow only through a photon exchanged *between* the two lines, the t-channel topology. Reading the diagram with the QED Feynman rules of Section 17.1.7 — a vertex $-ie\gamma^\mu$ at each end (Eq. (17.10)) and the Feynman-gauge photon propagator $-i\eta_{\mu\nu}/(q^2+i\epsilon)$ (Eq. (17.18)) carrying $q=p_1-p_3=p_4-p_2$ — gives

$$i\mathcal{M} = \left[\bar u(p_3)(-ie\gamma^\mu)u(p_1)\right]\frac{-i\eta_{\mu\nu}}{q^2}\left[\bar u(p_4)(-ie\gamma^\nu)u(p_2)\right], \tag{17.121}$$

where we suppress the $i\epsilon$ since the t-channel pole at $q^2=0$ is never reached for physical spacelike momentum transfer. Collecting the factors of i and e,

$$\mathcal{M} = \frac{e^2}{q^2}\left[\bar u(p_3)\gamma^\mu u(p_1)\right]\left[\bar u(p_4)\gamma_\mu u(p_2)\right], \qquad q^2 = t = (p_1-p_3)^2. \tag{17.122}$$

The single most important structural feature is already visible: the amplitude carries an explicit $1/q^2 = 1/t$, the photon propagator pole. When we square it

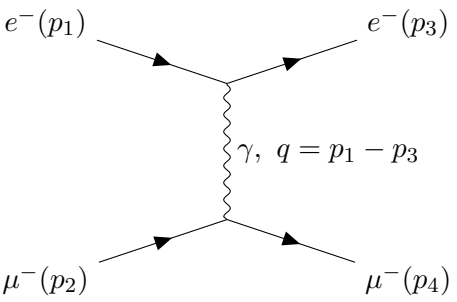


Figure 17.8: The single tree diagram for $e^-\mu^- \to e^-\mu^-$. Charge and lepton flavour cannot pass between the electron and the muon line, so the only connection is an exchanged photon: this is the t-channel topology, with spacelike momentum transfer $q = p_1 - p_3$ and $t = q^2 < 0$. The propagator supplies the $1/q^2$ in Eq. (17.122); squaring it produces the $1/q^4$ that drives the forward peak and reduces to the Coulomb potential in the static limit.

we get $1/q^4 = 1/t^2$, the hallmark of long-range, forward-peaked scattering, and the object we will eventually identify with the Fourier transform of the Coulomb potential.

We need the unpolarized cross section: average over the two initial spins of the electron and the two of the muon (a factor $\frac{1}{4}$) and sum over the four final spin states. The Dirac structure separates into an electron factor and a muon factor, each of which becomes a trace once the spin sums collapse the external spinors via the completeness relations $\sum_s u(p)\bar{u}(p) = \not{p} + m$ of Eq. (11.73). This is exactly the reduction carried out in detail for the s-channel process in Section 17.2.4; for the present t-channel process crossing has already done the work, and Section 17.4 delivered the result. Restating it here in Mandelstam form for use below,

$$\overline{|\mathcal{M}|^2} = \frac{2e^4}{t^2}\Big[(s - m^2 - M^2)^2 + (u - m^2 - M^2)^2 + 2t\,(m^2 + M^2)\Big]. \tag{17.123}$$

The Mandelstam invariants are defined as in Theorem 16.20, $s = (p_1 + p_2)^2$, $t = (p_1 - p_3)^2$, $u = (p_1 - p_4)^2$, satisfying the on-shell constraint $s + t + u = 2m^2 + 2M^2$. The overall $2e^4/t^2$ is the squared photon propagator times the square of the two vertex couplings; the bracket is the contraction of the electron and muon spin tensors. As a first sanity check, drop both masses: Eq. (17.123) becomes $\overline{|\mathcal{M}|^2} = 2e^4(s^2 + u^2)/t^2$, the standard massless $e^-\mu^- \to e^-\mu^-$ result, manifestly the $s \leftrightarrow t$ image of the massless annihilation answer $2e^4(t^2 + u^2)/s^2$.

17.5.2 Consistency with the s-channel by crossing

Before extracting an observable, we verify internally that Eq. (17.123) is the correct crossed partner of the annihilation amplitude, using the crossing relation

of Section 17.4 as a closed-loop check rather than a derivation. The spin-averaged square for $e^+e^- \to \mu^+\mu^-$, computed by the trace technology of Section 17.2.4, was (Eq. (17.83))

$$\overline{|\mathcal{M}|^2}_{e^+e^-\to\mu^+\mu^-} = \frac{2e^4}{s^2}\Big[(t-m^2-M^2)^2+(u-m^2-M^2)^2+2s\,(m^2+M^2)\Big]. \tag{17.124}$$

The two processes are related by exchanging an incoming e^+ with an outgoing e^-, which at the level of Mandelstam invariants is the substitution $s \leftrightarrow t$ with u held fixed as the spectator channel (the detailed momentum relabelling, including the antiparticle signs that leave $\overline{|\mathcal{M}|^2}$ even, is the content of Section 17.4). Performing $s \leftrightarrow t$ on Eq. (17.124),

$$\frac{2e^4}{s^2}\big[\cdots\big] \xrightarrow{s\leftrightarrow t} \frac{2e^4}{t^2}\Big[(s-m^2-M^2)^2+(u-m^2-M^2)^2+2t\,(m^2+M^2)\Big], \tag{17.125}$$

which is precisely Eq. (17.123). The annihilation and scattering squared amplitudes are one analytic function evaluated in two kinematic regions; the photon pole sits in the s-channel for the former (a pole at $s=0$ in the massless idealization, outside the physical region $s \geq 4m_\mu^2$) and in the t-channel for the latter (a spacelike exchange, never on shell). This is the strongest possible internal consistency statement: the same trace, computed once, governs both flagship QED processes.

17.5.3 Laboratory kinematics

To make contact with a fixed-target experiment we adopt the laboratory frame in which the muon is initially at rest. Set

$$p_1=(E,\mathbf{p}),\quad p_2=(M,\mathbf{0}),\quad p_3=(E',\mathbf{p}'),\quad p_4=(E_4,\mathbf{p}-\mathbf{p}'), \tag{17.126}$$

with $|\mathbf{p}|=p$, $|\mathbf{p}'|=p'$, $E=\sqrt{m^2+p^2}$, $E'=\sqrt{m^2+p'^2}$, and the recoiling muon energy $E_4=\sqrt{M^2+|\mathbf{p}-\mathbf{p}'|^2}$. The scattering angle θ is the angle between $\mathbf{p}$ and $\mathbf{p}'$. Energy conservation reads $E+M=E'+E_4$. The momentum transfer is spacelike,

$$t=q^2=(p_1-p_3)^2=(E-E')^2-|\mathbf{p}-\mathbf{p}'|^2=2m^2-2(EE'-pp'\cos\theta), \tag{17.127}$$

the second equality using $p_1^2=p_3^2=m^2$. The Mandelstam s in this frame is fixed by the beam energy alone,

$$s=(p_1+p_2)^2=m^2+M^2+2p_1\cdot p_2=m^2+M^2+2EM, \qquad s-m^2-M^2=2EM, \tag{17.128}$$

and u follows from the constraint, $u-m^2-M^2=m^2+M^2-s-t=-(2EM+t)$. These three relations will let us evaluate Eq. (17.123) directly in lab variables once we have assembled the cross-section formula.

17.5.4 From the squared amplitude to $d\sigma/d\Omega$

The cross section is built from Eq. (16.112),

$$d\sigma = \frac{1}{F}\,\overline{|\mathcal{M}|^2}\,d\Pi_2, \tag{17.129}$$

with the flux factor F and the two-body Lorentz-invariant phase space $d\Pi_2$ of Eq. (16.47). We carry both objects in the lab frame explicitly, because the heavy-target limit we want lives in this frame, not in the centre of momentum.

The flux. For collinear beams the invariant flux is $F = 4\sqrt{(p_1\cdot p_2)^2 - m^2M^2}$. With the muon at rest, $p_1p_2 = EM$ and $(p_1p_2)^2 - m^2M^2 = M^2(E^2-m^2) = M^2p^2$, so

$$F = 4Mp. \tag{17.130}$$

The flux is the relative current of the incoming pair; with a stationary target it reduces to the target mass times the beam momentum, up to the universal factor of four from the relativistic normalization fixed in Eq. (16.13).

The phase space. The two-body measure is $d\Pi_2 = \dfrac{d^3p_3}{(2\pi)^3 2E'}\dfrac{d^3p_4}{(2\pi)^3 2E_4}(2\pi)^4\delta^4(p_1 + p_2 - p_3 - p_4)$. The spatial delta function removes the d^3p_4 integral, fixing $\mathbf{p}_4 = \mathbf{p} - \mathbf{p}'$ and so $E_4 = \sqrt{M^2 + |\mathbf{p}-\mathbf{p}'|^2}$. Writing $d^3p_3 = p'^2\,dp'\,d\Omega$, what survives is

$$d\Pi_2 = \frac{1}{(2\pi)^2}\,\frac{p'^2\,dp'\,d\Omega}{4E'E_4}\,\delta(E + M - E' - E_4). \tag{17.131}$$

The remaining energy delta function fixes p' in terms of θ. To do the p' integral we need the Jacobian of $g(p') \equiv E' + E_4 - E - M$. Since $E' = \sqrt{m^2 + p'^2}$ and $E_4^2 = M^2 + p^2 + p'^2 - 2pp'\cos\theta$,

$$\frac{dg}{dp'} = \frac{p'}{E'} + \frac{p' - p\cos\theta}{E_4} = \frac{p'E_4 + E'(p' - p\cos\theta)}{E'E_4}. \tag{17.132}$$

Using $\int dp'\,h(p')\,\delta(g(p')) = h(p'_*)/\,|g'(p'_*)|$ with p'_* the solution of energy conservation, the $E'E_4$ in Eq. (17.132) cancels the one in Eq. (17.131), leaving

$$\frac{d\Pi_2}{d\Omega} = \frac{1}{16\pi^2}\,\frac{p'^2}{p'E_4 + E'(p' - p\cos\theta)}. \tag{17.133}$$

Assembling Eq. (17.129) with Eq. (17.130) and Eq. (17.133),

$$\boxed{\left.\frac{d\sigma}{d\Omega}\right|_{\text{lab}} = \frac{1}{64\pi^2 Mp}\,\frac{p'^2}{p'E_4 + E'(p' - p\cos\theta)}\,\overline{|\mathcal{M}|^2}}. \tag{17.134}$$

This is the exact tree-level lab-frame differential cross section for $e^-\mu^- \to e^-\mu^-$, with all the recoil kinematics carried: p', E', E_4 are the genuine final-state energies and momenta determined by Eq. (17.126) and energy conservation, and $\overline{|\mathcal{M}|^2}$ is Eq. (17.123) expressed in these variables through Eqs. (17.127) and (17.128). Nothing has been approximated yet. The denominator $p'E_4 + E'(p' - p\cos\theta)$ is the recoil factor: when the target recoils, the final electron loses energy and is deflected by the back-reaction, and this term keeps track of it. Its M-dependence (through E_4) is what controls the limit we take next.

17.5.5 The static-source limit: the Mott cross section

We now make the muon a fixed source of Coulomb field by sending $M \to \infty$ at fixed electron beam momentum p and fixed scattering angle θ. Physically, an infinitely heavy target cannot recoil: it absorbs arbitrary three-momentum at zero energy cost, so the electron scatters *elastically*, $E' = E$ and $p' = p$, and the momentum transfer becomes purely spatial. Quantitatively, the energy a target of mass M takes up in absorbing momentum $|\mathbf{q}|$ is $E_4 - M = \sqrt{M^2 + \mathbf{q}^2} - M \to \mathbf{q}^2/2M \to 0$, so the electron energy loss $E - E'$ vanishes as $1/M$. In this limit Eq. (17.127) collapses to the elastic relation

$$t = 2m^2 - 2(E^2 - p^2\cos\theta) = -2p^2(1-\cos\theta) = -4p^2\sin^2\tfrac{\theta}{2} \equiv -\mathbf{q}^2, \qquad |\mathbf{q}| = 2p\sin\tfrac{\theta}{2}, \tag{17.135}$$

using $E^2 - p^2 = m^2$ and the half-angle identity $1 - \cos\theta = 2\sin^2(\theta/2)$. The three-momentum transfer $|\mathbf{q}|$ is set entirely by the electron's momentum and deflection.

The prefactor of Eq. (17.134) simplifies dramatically. With $p' \to p$, $E' \to E$, $E_4 \to M$, the recoil denominator is dominated by its first term, $p'E_4 \to pM$, while $E'(p' - p\cos\theta) \to Ep(1 - \cos\theta)$ stays finite; the ratio of the second term to the first is $O(E/M) \to 0$. Hence

$$\left.\frac{d\sigma}{d\Omega}\right|_{M\to\infty} = \frac{1}{64\pi^2 Mp}\cdot\frac{p^2}{pM}\,\overline{|\mathcal{M}|^2} = \frac{\overline{|\mathcal{M}|^2}}{64\pi^2 M^2}. \tag{17.136}$$

Every factor of the target mass is now isolated in the $1/M^2$, and it will be cancelled by the M^2 that the squared amplitude carries. We extract that M^2 by taking the same limit inside Eq. (17.123). Using $s - m^2 - M^2 = 2EM$ from Eq. (17.128) and $u - m^2 - M^2 = -(2EM + t)$, the bracket is

$$\begin{aligned}(s-m^2-M^2)^2 + (u-m^2-M^2)^2 + 2t(m^2+M^2) &= (2EM)^2 + (2EM+t)^2 + 2t(m^2+M^2)\\ &= 8E^2M^2 + 2tM^2 + (4EMt + t^2 + 2tm^2).\end{aligned} \tag{17.137}$$

As $M \to \infty$ at fixed t, the terms in the final parenthesis are subleading: $4EMt$ is $O(M)$ and $t^2 + 2tm^2$ is $O(M^0)$, all negligible against the $O(M^2)$ leading part.

Keeping the leading behaviour,

$$\overline{|\mathcal{M}|^2} \;\longrightarrow\; \frac{2e^4}{t^2}\cdot 2M^2(4E^2+t) = \frac{4e^4M^2}{t^2}(4E^2+t). \tag{17.138}$$

The combination $4E^2+t$ is where the electron spin shows up. Insert the elastic transfer Eq. (17.135), $t=-4p^2\sin^2(\theta/2)$, and factor out $4E^2$:

$$4E^2+t = 4E^2-4p^2\sin^2\tfrac{\theta}{2} = 4E^2\Big(1-\frac{p^2}{E^2}\sin^2\tfrac{\theta}{2}\Big) = 4E^2(1-\beta^2\sin^2\tfrac{\theta}{2}), \tag{17.139}$$

where $\beta\equiv p/E=|\mathbf{v}|$ is the electron velocity. Substituting Eq. (17.138) and Eq. (17.139) into Eq. (17.136), with $t^2=16p^4\sin^4(\theta/2)$ and $e^4=(4\pi\alpha)^2=16\pi^2\alpha^2$,

$$\begin{aligned}\left.\frac{d\sigma}{d\Omega}\right|_{\text{Mott}} &= \frac{1}{64\pi^2M^2}\cdot\frac{4e^4M^2}{t^2}\cdot 4E^2(1-\beta^2\sin^2\tfrac{\theta}{2}) = \frac{e^4E^2}{4\pi^2t^2}(1-\beta^2\sin^2\tfrac{\theta}{2})\\ &= \frac{16\pi^2\alpha^2\,E^2}{4\pi^2\cdot 16\,p^4\sin^4(\theta/2)}(1-\beta^2\sin^2\tfrac{\theta}{2}).\end{aligned} \tag{17.140}$$

Using $p=\beta E$ so that $p^4=\beta^4E^4$ and $E^2/p^4=1/(\beta^4E^2)$, the result is the Mott cross section.

Proposition 17.16 (Mott cross section)**.** *The differential cross section for a relativistic Dirac electron of velocity β and energy E scattering off a fixed point charge of magnitude e (the $M\to\infty$ limit of $e^-\mu^-\to e^-\mu^-$) is*

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{Mott}} = \frac{\alpha^2}{4E^2\beta^4\sin^4(\theta/2)}\Big(1-\beta^2\sin^2\tfrac{\theta}{2}\Big). \tag{17.141}$$

Two features deserve comment. The bare factor $\alpha^2/(4E^2\beta^4\sin^4(\theta/2))$ is what one would obtain for a *spinless* relativistic electron scattering off a Coulomb field: it carries the $\sin^{-4}(\theta/2)$ forward singularity of long-range scattering, inherited directly from the $1/t^2$ of the squared photon propagator. The second factor, $1-\beta^2\sin^2(\theta/2)$, is purely a consequence of the electron's spin-$\frac{1}{2}$ nature; it is the unique imprint of the Dirac structure surviving the static limit. It interpolates between unity in the nonrelativistic regime ($\beta\to 0$) and a genuine angular distortion at relativistic speeds, suppressing wide-angle scattering. We make its physical origin precise in the worked example below and in Section 17.6.

Physical Insight: why the spin factor suppresses backscattering

The factor $1-\beta^2\sin^2(\theta/2)$ is a statement about helicity. At high energy the QED vertex conserves the electron's chirality, and for a fast electron chirality is helicity. A pure helicity state scattering straight backward

($\theta = \pi$) would have to reverse its spin projection while its momentum reverses — it would have to flip helicity — which the vertex forbids in the massless limit. The cross section must therefore vanish at $\theta = \pi$ as $\beta \to 1$, and indeed $1 - \beta^2 \sin^2(\theta/2) \to 1 - \beta^2 = 1/\gamma^2 \to 0$ there. The spin factor is the bookkeeping of this angular-momentum selection rule, which Section 17.6 turns into the organizing principle behind every QED angular distribution.

17.5.6 The nonrelativistic limit: the Rutherford formula

We now take the electron slow, $\beta = v \ll 1$. Two things happen to Eq. (17.141). First the spin factor switches off: $1 - \beta^2 \sin^2(\theta/2) \to 1$ up to corrections of order v^2. Second the energy reduces to the rest mass, $E = \sqrt{m^2 + p^2} \to m$ with relative error $v^2/2$. Setting $\beta = v$ and $E \to m$,

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{Mott}} \longrightarrow \frac{\alpha^2}{4m^2 v^4 \sin^4(\theta/2)}. \tag{17.142}$$

This is the Rutherford cross section — the differential cross section Rutherford derived classically in 1911 from the hyperbolic Kepler orbits of a charged projectile in a Coulomb field, and against which the Geiger–Marsden α-on-gold data were fitted to establish the nuclear atom. Here it has dropped out of a relativistic quantum field theory as the doubly-degenerate limit (heavy target, slow projectile) of a one-photon-exchange amplitude. The agreement is exact, including the numerical $\frac{1}{4}$: every step from the Feynman rules to Eq. (17.142) is a controlled limit with its approximations named.

The Rutherford formula exhibits the two divergences characteristic of an unscreened long-range force. As $\theta \to 0$ the cross section diverges as θ^{-4}: a Coulomb field has infinite range, so even arbitrarily distant trajectories are deflected, and the total cross section $\int d\Omega\, (d\sigma/d\Omega)$ is infinite. As $v \to 0$ the cross section diverges as v^{-4}: a slow projectile lingers in the field and is deflected more. Both are physical and both are softened in any real situation by screening of the Coulomb potential at large r — precisely the screening we now use to expose the meaning of the $1/q^4$.

17.5.7 The $1/q^4$ as the Coulomb potential

The recovery of Rutherford is more than a numerical coincidence; it identifies the exchanged photon with the classical Coulomb potential. The cleanest way to see this is to compare Eq. (17.142) with the first Born approximation of nonrelativistic potential scattering, a result the reader owns from quantum mechanics. For a projectile of mass m scattering off a static potential $V(\mathbf{r})$, the

Born amplitude is the Fourier transform of the potential,

$$f(\theta) = -\frac{m}{2\pi}\int d^3r\; e^{-i\mathbf{q}\cdot\mathbf{r}}\, V(\mathbf{r}), \qquad \frac{d\sigma}{d\Omega} = |f(\theta)|^2\,, \tag{17.143}$$

with momentum transfer $\mathbf{q}$, $|\mathbf{q}| = 2p\sin(\theta/2)$ as in Eq. (17.135). The Coulomb potential between the two like charges is $V(r) = \alpha/r$; to keep the Fourier transform well defined we regulate it as a Yukawa potential $V(r) = \alpha\, e^{-\mu r}/r$ and send the screening mass $\mu \to 0$ at the end. The transform is elementary,

$$\int d^3r\; e^{-i\mathbf{q}\cdot\mathbf{r}}\,\frac{\alpha\, e^{-\mu r}}{r} = \frac{4\pi\alpha}{\mathbf{q}^2+\mu^2} \xrightarrow{\mu\to 0} \frac{4\pi\alpha}{\mathbf{q}^2}. \tag{17.144}$$

Therefore $f(\theta) = -\dfrac{m}{2\pi}\cdot\dfrac{4\pi\alpha}{\mathbf{q}^2} = -\dfrac{2m\alpha}{\mathbf{q}^2}$, and

$$\frac{d\sigma}{d\Omega} = |f|^2 = \frac{4m^2\alpha^2}{\mathbf{q}^4} = \frac{4m^2\alpha^2}{\left(2p\sin(\theta/2)\right)^4} = \frac{m^2\alpha^2}{4p^4\sin^4(\theta/2)}\Big|_{p=mv} = \frac{\alpha^2}{4m^2v^4\sin^4(\theta/2)}, \tag{17.145}$$

identical to Eq. (17.142). The match is exact and it is structural. Compare the field-theory and the quantum-mechanics computations term by term: the photon propagator $1/q^2$ in Eq. (17.122), in the static limit where $q^0 \to 0$ and $q^2 \to -\mathbf{q}^2$, is $-1/\mathbf{q}^2$, and its time–time component $e^2/\mathbf{q}^2$ is exactly the Fourier transform Eq. (17.144) of the Coulomb potential $e^2/4\pi r = \alpha/r$. The single photon exchanged in Fig. 17.8 *is* the Coulomb interaction, written in momentum space; the static potential is what one nonrelativistic charge sees of the other's field when the field quantum is too soft to resolve. Squaring the amplitude turns the propagator's $1/\mathbf{q}^2$ into the $1/\mathbf{q}^4 = 1/q^4$ that drives the Rutherford $\sin^{-4}(\theta/2)$ peak: the forward singularity of the cross section is the squared Fourier transform of the infinite-range Coulomb tail. This is the precise sense in which QED *contains* classical electrostatics, and the sense in which the Born approximation of Coulomb scattering is the tree-level, one-photon-exchange truncation of QED in its nonrelativistic, static-source corner.

Worked Example 17.17 (Spin suppression of backscattering and the recoil correction)**.** We quantify the two ways the QED answer departs from the naive Rutherford formula: the spin factor at relativistic speed, and the size of the neglected recoil.

(i) The spin factor at $\theta = \pi$. Take the exact Mott result Eq. (17.141) at backward scattering, $\sin^2(\theta/2) = 1$. The ratio of the Mott cross section to the spin-blind "relativistic Rutherford" factor $\alpha^2/(4E^2\beta^4\sin^4(\theta/2))$ is simply the spin factor $1-\beta^2$. For a nonrelativistic electron, $\beta = 0.1$ ($v = 0.1c$), this is $1 - 0.01 = 0.99$: spin reduces wide-angle scattering by one percent, invisible to Rutherford's data and the reason his classical formula worked. For a mildly relativistic electron, $\beta = 0.8$, it is $1-0.64 = 0.36$: spin has cut the backward cross

section to about a third of its spinless value. In the ultrarelativistic limit $\beta \to 1$ the factor is $1-\beta^2 = 1/\gamma^2 \to 0$, and the backward cross section is completely extinguished — the helicity-conservation selection rule anticipated above and proved in Section 17.6.

(ii) The recoil correction. The static limit Eq. (17.136) dropped the electron's energy loss. We estimate its size from the exact kinematics. For a relativistic electron ($m \ll E$, so $p \simeq E$) energy and momentum conservation give the recoil relation

$$E' = \frac{E}{1 + \dfrac{2E}{M}\sin^2(\theta/2)}, \tag{17.146}$$

obtained by squaring $p_4 = p_1 + p_2 - p_3$ on shell: $M^2 = (p_1 + p_2 - p_3)^2 = M^2 + 2M(E - E') - 2p_1\cdot p_3$, and with $p_1\cdot p_3 \simeq EE'(1-\cos\theta) = 2EE'\sin^2(\theta/2)$ this rearranges to Eq. (17.146). The fractional energy loss is $1 - E'/E = \dfrac{2E\sin^2(\theta/2)/M}{1 + 2E\sin^2(\theta/2)/M}$, controlled by the dimensionless ratio $2E\sin^2(\theta/2)/M$. For an electron of $E = 1$ GeV scattering at $\theta = 60^\circ$ off a muon, $M = 0.106$ GeV, this ratio is $2(1)(0.25)/0.106 \approx 4.7$, so $E'/E \approx 0.18$: the "heavy target" is in fact light, recoil is large, and the Mott approximation fails badly — the muon is a genuine dynamical participant, not a static source. For the same electron off a proton ($M = 0.938$ GeV) the ratio is 0.53 and $E'/E \approx 0.65$: recoil is a leading correction. The Mott and Rutherford limits are clean theorems, but the heavy-target idealization is quantitatively excellent only when $E\sin^2(\theta/2) \ll M$; the systematic treatment of the recoil factor, and of the target's internal structure through elastic form factors, is the starting point of deep-inelastic scattering in Chapter 28.

We have completed the second flagship process of the chapter. The crossed amplitude $e^-\mu^- \to e^-\mu^-$ became a laboratory observable, and that observable reproduced, as nested limits, the Mott cross section of relativistic electron scattering and the Rutherford cross section of classical Coulomb scattering, with the photon's $1/q^4$ identified as the squared Fourier transform of the Coulomb potential. Two structural facts recurred and were left unexplained: the $1+\cos^2\theta$ angular law of the annihilation process, and the spin factor $1-\beta^2\sin^2(\theta/2)$ that suppressed backscattering here. Both are shadows of a single selection rule — the conservation of chirality at the QED vertex and the locking of chirality to helicity at high energy. The final section of the chapter exposes that helicity skeleton directly, shows that it dictates the form of both answers from angular-momentum conservation alone, and identifies it as the seed of the spinor-helicity reformulation of Chapter 58.

17.6 Helicity structure and the high-energy limit

The trace technology of the preceding sections is a machine that grinds a squared amplitude into Mandelstam invariants without ever asking *why* the answer takes the shape it does. It produced, for $e^+e^- \to \mu^+\mu^-$, the compact unpolarized result $\overline{|\mathcal{M}|^2} = 2e^4(t^2+u^2)/s^2$ and its angular face $e^4(1+\cos^2\theta)$ (Section 17.3); for $e^-\mu^- \to e^-\mu^-$ it gave the crossed cousin and, in the Mott and Rutherford limits, the physics the reader already trusts (Section 17.5). What the traces concealed is that these numbers are not really sums over four spins fed through a spin sum: they are sums over a *small* number of nonzero processes, each labelled by the handedness of the incoming and outgoing particles, and each carrying an angular shape fixed by nothing more than the conservation of angular momentum. The aim of this section is to expose that skeleton. We will see that the QED vertex $\bar\psi\gamma^\mu\psi$ conserves chirality, that in the high-energy limit chirality is helicity, and that the surviving helicity amplitudes are the rotation matrix elements $d^1_{m'm}(\theta)$ of a spin-one exchange. The $1+\cos^2\theta$ law will then be not the output of a trace but a theorem of representation theory. This reorganization is also the seed of the spinor-helicity formalism of Chapter 58, which makes the bookkeeping covariant and turns the helicity amplitudes into one-line expressions; here we lay it bare by hand, keeping the chapter self-contained.

Throughout this section we take the *massless* (high-energy, ultrarelativistic) limit $s \gg m_e^2, m_\mu^2$, in which the electron and muon masses may be set to zero. This is the regime in which the structure is cleanest, because in it the two notions of handedness a Dirac fermion carries—the Lorentz-invariant but non-conserved *chirality* (the eigenvalue of γ^5) and the conserved but frame-dependent *helicity* (the projection of spin on the momentum)—coincide exactly, as established for the plane-wave spinors in Chapter 11. Massive corrections reinstate the $(1-\beta^2)\sin^2\theta$ piece that fills in the distribution near threshold (visible in the angular plot of Section 17.3); they do not change the high-energy organization, and we comment on them at the end.

17.6.1 Chirality is conserved at the QED vertex

Every QED amplitude in this chapter is built from a single Lorentz structure: the electromagnetic current $\bar\psi\gamma^\mu\psi$ sandwiched between external spinors and tied to a photon. The decisive algebraic fact about this structure is that the vector bilinear $\bar\psi\gamma^\mu\psi$ *does not mix the two chiralities of the field.* We prove this by inserting the chirality projectors

$$P_L = \tfrac{1}{2}(1-\gamma^5), \qquad P_R = \tfrac{1}{2}(1+\gamma^5), \tag{17.147}$$

imported from Eq. (11.82) and Theorem C.27, and using the single relation $\{\gamma^\mu,\gamma^5\}=0$ that defines γ^5 (Section C.3.5). Recall the consequences we will

lean on, all immediate from $\{\gamma^\mu,\gamma^5\}=0$ and $(\gamma^5)^2=\mathbf{1}$ (Theorem C.28): the projectors are complete and orthogonal, $P_L+P_R=\mathbf{1}$, $P_{L,R}^2=P_{L,R}$, $P_LP_R=0$; a single gamma intertwines them,

$$\gamma^\mu P_L = P_R\gamma^\mu, \qquad \gamma^\mu P_R = P_L\gamma^\mu, \tag{17.148}$$

so an *odd* string of gammas flips the projector it is pushed through and an *even* string preserves it; and, because $(\gamma^5)^\dagger=\gamma^5$ and $\{\gamma^0,\gamma^5\}=0$, the Dirac conjugate of a chirality eigenspinor flips handedness,

$$\overline{P_L\psi} = (P_L\psi)^\dagger\gamma^0 = \psi^\dagger P_L\gamma^0 = \psi^\dagger\gamma^0 P_R = \bar\psi P_R, \qquad \overline{P_R\psi} = \bar\psi P_L. \tag{17.149}$$

Proposition 17.18 (Chirality conservation at the vector vertex)**.** *Write the left- and right-handed components of any Dirac field as $\psi_L=P_L\psi$, $\psi_R=P_R\psi$, with conjugates $\bar\psi_L=\bar\psi P_R$, $\bar\psi_R=\bar\psi P_L$. Then the vector current is the sum of two same-chirality currents, with no cross term:*

$$\bar\psi\,\gamma^\mu\,\psi = \bar\psi_L\,\gamma^\mu\,\psi_L + \bar\psi_R\,\gamma^\mu\,\psi_R. \tag{17.150}$$

The same holds for a current $\bar\psi_2\gamma^\mu\psi_1$ built from two different fields: only the LL and RR pieces survive.

Proof. Insert $\mathbf{1}=P_L+P_R$ on both sides of the gamma matrix and expand into four terms,

$$\bar\psi\gamma^\mu\psi = \bar\psi(P_L+P_R)\gamma^\mu(P_L+P_R)\psi = \sum_{X,Y\in\{L,R\}} \bar\psi\,P_X\,\gamma^\mu\,P_Y\,\psi. \tag{17.151}$$

Consider the diagonal pieces first. Using the intertwining (17.148) to slide the right projector to the left through the single gamma, $\gamma^\mu P_L=P_R\gamma^\mu$, so $P_L\gamma^\mu P_L=P_LP_R\gamma^\mu=0$ because $P_LP_R=0$; likewise $P_R\gamma^\mu P_R=P_RP_L\gamma^\mu=0$. The two diagonal terms vanish identically: a single gamma cannot connect a left projector to a left projector. The off-diagonal terms survive,

$$P_L\gamma^\mu P_R = \gamma^\mu P_RP_R = \gamma^\mu P_R, \qquad P_R\gamma^\mu P_L = \gamma^\mu P_L, \tag{17.152}$$

so that

$$\bar\psi\gamma^\mu\psi = \bar\psi\,P_L\gamma^\mu P_R\,\psi + \bar\psi\,P_R\gamma^\mu P_L\,\psi = (\bar\psi P_L)\gamma^\mu(P_R\psi) + (\bar\psi P_R)\gamma^\mu(P_L\psi). \tag{17.153}$$

Now identify the conjugated chiral fields by (17.149): $\bar\psi P_L=\overline{P_R\psi}=\bar\psi_R$ and $\bar\psi P_R=\bar\psi_L$. Hence the first term is $\bar\psi_R\gamma^\mu\psi_R$ and the second $\bar\psi_L\gamma^\mu\psi_L$, which is (17.150). The two-field case is identical: nothing in the manipulation used $\psi_2=\psi_1$. □

The content of Theorem 17.18 is a selection rule. A QED interaction inserts exactly one factor of γ^μ between a fermion and its conjugate; that single gamma is odd, so it links a right-handed field only to a right-handed field and a left-handed field only to a left-handed field. A left-handed electron entering a vertex leaves as a left-handed electron; chirality flows through the photon coupling unbroken. The mass term $-m\bar{\psi}\psi$ is the *only* place in $\mathcal{L}_{\text{QED}}$ that flips chirality—it is even (no gamma) and so connects L to R, as we saw when the massive Dirac equation forced $\not{p}\,(P_R u) = m\,(P_L u)$ in Section 11.2.5—and in the massless limit it is gone. There, chirality is exactly conserved along every fermion line.

Physical Insight

The whole of the weak interaction's parity violation, and the entire spinor-helicity reorganization of amplitudes, rests on this one line of algebra: a vector vertex is chirality-diagonal because γ^μ anticommutes with γ^5. QED happens to couple to both chiralities with equal strength (it is parity-conserving), so the selection rule does not by itself forbid any process; what it does is *block-diagonalize* every amplitude into independent same-chirality channels. Each channel then carries a definite assignment of helicities to the external particles, and these channels add *incoherently* in the unpolarized cross section—which is exactly why the brute-force spin sum, which secretly performed this decomposition and threw the labels away, produced so simple an answer.

17.6.2 From chirality to the helicities of the external states

To turn Theorem 17.18 into a statement about the spins of the colliding particles we use the massless coincidence of chirality and helicity. For a massless one-particle state, Wigner's classification (Section 3.7, and the massless little group ISO(2) of Section 3.7.3) attaches to it a single Lorentz-invariant number, the helicity $h = \mathbf{J}\cdot\hat{\mathbf{p}}$; there is no spin multiplet, only one state per momentum and sign of h. On the plane-wave spinors of Chapter 11 this abstract helicity is realized concretely: in the limit $m \to 0$ the positive-energy spinors $u_\pm$ become eigenstates of γ^5, with $P_R u_+ = u_+$ (right-handed, $h = +\frac{1}{2}$) and $P_L u_- = u_-$ (left-handed, $h = -\frac{1}{2}$), as shown explicitly in Section 11.2.5. For the antiparticle spinors v the standard relation between the spinor's chirality label and the *physical* helicity of the positron is reversed: the spinor $P_R v$ describes a *left*-helicity positron. The reason is the same one that flips the conjugate in (17.149)—a negative-energy solution read as an outgoing antiparticle carries the opposite handedness to its spinor—and we fix it once here.

Caution

The chirality of a v-spinor is the *opposite* of the physical helicity of the antiparticle it represents. Throughout, "e_L^+" means a positron of physical helicity $-\frac{1}{2}$, described by the spinor v with $P_R v = v$; and "e_R^+" means physical helicity $+\frac{1}{2}$, described by $P_L v = v$. For the electron there is no such flip: e_R^- is described by $P_R u = u$. Keeping this straight is the difference between the right and wrong $\pm\cos\theta$.

Apply this to the electron current $\bar{v}(p_+)\gamma^\mu u(p_-)$ that appears in the $e^+e^- \to \mu^+\mu^-$ amplitude (Eq. (17.62)). By Theorem 17.18 only the same-chirality pieces survive,

$$\bar{v}(p_+)\gamma^\mu u(p_-) = \overline{v_R}(p_+)\,\gamma^\mu\,u_R(p_-) + \overline{v_L}(p_+)\,\gamma^\mu\,u_L(p_-), \tag{17.154}$$

where $u_R = P_R u$ and $\overline{v_R} = \bar{v}P_L$ select $P_R v = v$. The first term needs the electron spinor right-handed *and* the positron spinor right-handed: e_R^- with e_L^+. The second needs both left-handed spinors: e_L^- with e_R^+. In words: *the electron and positron must carry opposite physical helicities.* The configurations $e_R^- e_R^+$ and $e_L^- e_L^+$ give exactly zero.

This is angular momentum conservation in disguise, and it is worth seeing the geometry directly. Take the electron along $+\hat{z}$ and the positron along $-\hat{z}$ in the centre-of-mass frame. For the surviving combination $e_R^- e_L^+$: the electron has $h = +\frac{1}{2}$, so its spin points along $+\hat{z}$, contributing $J_z = +\frac{1}{2}$; the positron has $h = -\frac{1}{2}$, its spin antiparallel to its momentum $-\hat{z}$, i.e. also along $+\hat{z}$, contributing $J_z = +\frac{1}{2}$. The two spins *add*: the initial state has $J_z = +1$ along the beam. The forbidden combination $e_R^- e_R^+$ would have the spins antialigned, $J_z = 0$ along the beam, with total spin projection zero on the collision axis. A single intermediate photon carries spin one and—being on the brink of going on shell in the s-channel, where it is created at rest as a timelike quantum—transmits $J = 1$ with $J_z = \pm 1$ only; the $J_z = 0$ state along the beam cannot be fed to it by two massless fermions. The vertex selection rule and the spin-one nature of the photon are the same statement seen from two sides.

17.6.3 The nonzero helicity amplitudes, computed

We now compute a surviving helicity amplitude in full and show that the squares of the four nonzero ones sum to the unpolarized result of Section 17.3. The amplitude for $e^+e^- \to \mu^+\mu^-$ in Feynman gauge, from Eq. (17.62), is

$$\mathcal{M} = \frac{e^2}{s}\,\big[\bar{v}(p_+)\gamma^\mu u(p_-)\big]\,\big[\bar{u}(k_-)\gamma_\mu v(k_+)\big], \tag{17.155}$$

with all masses set to zero. We label momenta in the CM frame as in Section 17.3: the incoming electron and positron travel along $\pm\hat{z}$ and the outgoing muon makes

angle θ with the electron,

$$p_- = E(1,0,0,1), \quad p_+ = E(1,0,0,-1), \quad k_- = E(1,s_\theta,0,c_\theta), \quad k_+ = E(1,-s_\theta,0,-c_\theta), \tag{17.156}$$

with $E=\sqrt{s}/2$, $c_\theta \equiv \cos\theta$, $s_\theta \equiv \sin\theta$. The needed invariants are

$$p_-\!\cdot\! k_- = p_+\!\cdot\! k_+ = \tfrac{s}{4}(1-c_\theta), \qquad p_-\!\cdot\! k_+ = p_+\!\cdot\! k_- = \tfrac{s}{4}(1+c_\theta), \tag{17.157}$$

equivalently $t=-\frac{s}{2}(1-c_\theta)$ and $u=-\frac{s}{2}(1+c_\theta)$.

Worked Example 17.19 (The amplitude $e_R^- e_L^+ \to \mu_R^- \mu_L^+$)**.** We compute $\overline{|\mathcal{M}|^2}$ restricted to the helicity configuration in which both fermion lines are in the "right-handed" channel of (17.154). For massless fermions a definite helicity is selected by inserting the chirality projector into the spin sum: because P_R annihilates the unwanted (left-handed) solution, the projected "spin sum" is a sum over the single physical state. Concretely, with $\sum_s u^s\bar u^s = \not p$ and $\sum_s v^s\bar v^s = \not p$ at $m=0$ (Theorem 11.12), the right-handed projectors give

$$u_R\,\overline{u_R} = P_R \not p_-\, P_L = P_R \not p_-, \qquad v_R\,\overline{v_R} = P_R \not p_+\, P_L = P_R \not p_+, \tag{17.158}$$

where we used $\overline{P_R\psi} = \bar\psi P_L$ from (17.149) and then $\not p\, P_L = P_R \not p$ followed by $P_R P_R = P_R$.

The electron half of $|\mathcal{M}|^2$ is the tensor $L_e^{\mu\nu} = [\bar v(p_+)\gamma^\mu u(p_-)][\bar v(p_+)\gamma^\nu u(p_-)]^*$. Using $[\bar v\gamma^\nu u]^* = \bar u\gamma^\nu v$ (from $\gamma^0\gamma^{\nu\dagger}\gamma^0=\gamma^\nu$) and writing the product as a trace,

$$L_e^{\mu\nu} = \operatorname{tr}\!\big[\gamma^\mu\,(u_R\overline{u_R})\,\gamma^\nu\,(v_R\overline{v_R})\big] = \operatorname{tr}\!\big[\gamma^\mu\, P_R \not p_-\,\gamma^\nu\, P_R \not p_+\big]. \tag{17.159}$$

Reduce the two projectors to one. Slide the first P_R left through γ^μ ($\gamma^\mu P_R = P_L\gamma^\mu$); then P_L times the odd string $\gamma^\mu \not p_-\gamma^\nu$ becomes that string times P_R on the right, which merges with the second projector, $P_R P_R = P_R$:

$$L_e^{\mu\nu} = \operatorname{tr}\!\big[\gamma^\mu \not p_-\gamma^\nu\, P_R \not p_+\big] = \operatorname{tr}\!\big[\gamma^\mu \not p_-\gamma^\nu \not p_+\, P_L\big], \tag{17.160}$$

the last step using $P_R\not p_+ = \not p_+ P_L$. Now split $P_L = \frac12(1-\gamma^5)$ and use the four-gamma trace (C.178) and the γ^5 trace (C.179):

$$\begin{aligned} L_e^{\mu\nu} &= \tfrac12 \operatorname{tr}[\gamma^\mu \not p_-\gamma^\nu \not p_+] - \tfrac12 \operatorname{tr}[\gamma^\mu \not p_-\gamma^\nu \not p_+\gamma^5] \\ &= 2\big[p_-^\mu p_+^\nu + p_-^\nu p_+^\mu - (p_-\!\cdot\! p_+)\eta^{\mu\nu}\big] + 2i\,\epsilon^{\mu\alpha\nu\beta} p_{-\alpha} p_{+\beta}, \end{aligned} \tag{17.161}$$

where the sign of the antisymmetric piece follows from $\operatorname{tr}[\gamma^\mu \not p_-\gamma^\nu \not p_+\gamma^5] = \operatorname{tr}[\gamma^5\gamma^\mu \not p_-\gamma^\nu \not p_+] = -4i\,\epsilon^{\mu\alpha\nu\beta}p_{-\alpha}p_{+\beta}$ by cyclicity. The muon current $\bar u(k_-)\gamma v(k_+)$ carries u and v in the opposite order to the electron current $\bar v(p_+)\gamma u(p_-)$, so the muon half follows from the electron tensor with $p_-\to k_+$, $p_+\to k_-$ and indices lowered,

$$M_{\mu\nu}^{\mu^\pm} = 2\big[k_{-\mu}k_{+\nu} + k_{-\nu}k_{+\mu} - (k_-\!\cdot\! k_+)\eta_{\mu\nu}\big] + 2i\,\epsilon_{\mu\rho\nu\sigma}k_+^\rho k_-^\sigma. \tag{17.162}$$

The squared amplitude for this configuration is $|\mathcal{M}|^2 = (e^4/s^2)\, L_e^{\mu\nu} M_{\mu\nu}^{\mu^\pm}$. Write each tensor as $2S + 2iA$ with S symmetric and A antisymmetric in $\mu\nu$. In the contraction S-with-A cross terms vanish, leaving $L_e^{\mu\nu} M_{\mu\nu}^{\mu^\pm} = 4\,(S_e{\cdot}S_\mu - A_e{\cdot}A_\mu)$. The symmetric contraction is the same algebra that produced the unpolarized trace; with $a = p_-, b = p_+, c = k_+, d = k_-$,

$$S_e{\cdot}S_\mu = 2\big[(p_-{\cdot}k_+)(p_+{\cdot}k_-) + (p_-{\cdot}k_-)(p_+{\cdot}k_+)\big]. \tag{17.163}$$

For the antisymmetric piece we need the double contraction of two Levi-Civita symbols. With the orientation $\epsilon^{0123} = +1$ of Section A.2.5 (and its forced Lorentzian sign $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma} = -24$),

$$\epsilon^{\mu\nu\alpha\beta}\epsilon_{\mu\nu\gamma\delta} = -2\big(\delta^\alpha_\gamma\delta^\beta_\delta - \delta^\alpha_\delta\delta^\beta_\gamma\big), \tag{17.164}$$

so, after reordering each ϵ to bring its contracted indices adjacent (one transposition each, the two signs cancelling),

$$A_e{\cdot}A_\mu = \epsilon^{\mu\alpha\nu\beta}\epsilon_{\mu\rho\nu\sigma}\, p_{-\alpha}p_{+\beta}k_+^\rho k_-^\sigma = -2\big[(p_-{\cdot}k_+)(p_+{\cdot}k_-) - (p_-{\cdot}k_-)(p_+{\cdot}k_+)\big]. \tag{17.165}$$

Assembling, the two "mixed" products $(p_-{\cdot}k_-)(p_+{\cdot}k_+)$ cancel between $S_e{\cdot}S_\mu$ and $-A_e{\cdot}A_\mu$, doubling the other,

$$L_e^{\mu\nu} M_{\mu\nu}^{\mu^\pm} = 4\,(S_e{\cdot}S_\mu - A_e{\cdot}A_\mu) = 16\,(p_-{\cdot}k_+)(p_+{\cdot}k_-). \tag{17.166}$$

Inserting (17.157), $(p_-{\cdot}k_+)(p_+{\cdot}k_-) = \frac{s^2}{16}(1+c_\theta)^2$, and e^4/s^2 times 16 times this is

$$\boxed{\Big|\mathcal{M}(e_R^- e_L^+ \to \mu_R^- \mu_L^+)\Big|^2 = e^4\,(1+\cos\theta)^2.} \tag{17.167}$$

The amplitude is maximal in the forward direction and vanishes exactly at $\theta = \pi$: a right-handed μ^- emitted straight backward would carry $J_z = -1$, which cannot match the initial $J_z = +1$. The selection rule that killed $e_R^- e_R^+$ at the level of states reappears as a *zero of the amplitude* at the kinematic endpoint where the final J_z would be wrong.

The other three nonzero configurations follow without a new trace. Reflecting all helicities (a parity operation, which QED respects up to the relabelling $\theta \to \theta$, since the vector coupling is parity-even) gives $e_L^- e_R^+ \to \mu_L^- \mu_R^+$ with the identical result $e^4(1+\cos\theta)^2$. The two "crossed" configurations, in which the final pair has the opposite handedness to the initial pair, are obtained by exchanging $k_- \leftrightarrow k_+$ in the muon tensor, i.e. $c_\theta \to -c_\theta$ in (17.167); equivalently the contraction (17.166) becomes $16\,(p_-{\cdot}k_-)(p_+{\cdot}k_+) = s^2(1-c_\theta)^2$. Collecting the four,

$$\begin{aligned}
\Big|\mathcal{M}(e_R^- e_L^+ \to \mu_R^- \mu_L^+)\Big|^2 &= \Big|\mathcal{M}(e_L^- e_R^+ \to \mu_L^- \mu_R^+)\Big|^2 = e^4(1+\cos\theta)^2, \\
\Big|\mathcal{M}(e_R^- e_L^+ \to \mu_L^- \mu_R^+)\Big|^2 &= \Big|\mathcal{M}(e_L^- e_R^+ \to \mu_R^- \mu_L^+)\Big|^2 = e^4(1-\cos\theta)^2,
\end{aligned} \tag{17.168}$$

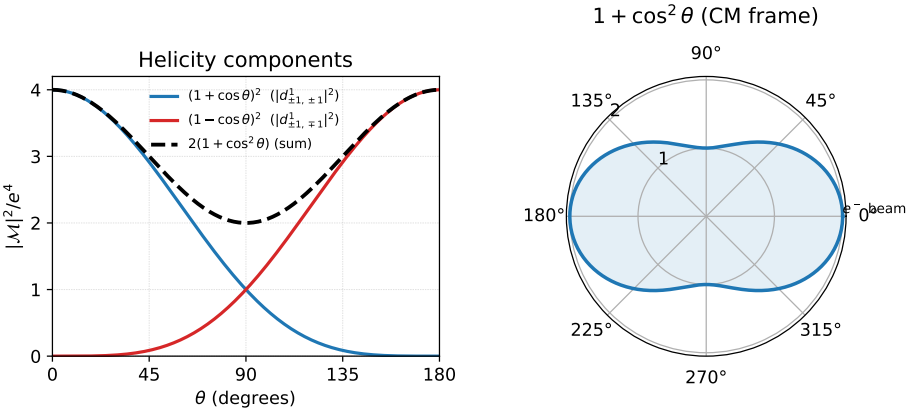


Figure 17.9: Helicity decomposition of $e^+e^- \to \mu^+\mu^-$ in the high-energy limit. *Left:* the two distinct helicity shapes, $(1+\cos\theta)^2$ (forward-peaked, the configurations with initial and final pairs of the same handedness) and $(1-\cos\theta)^2$ (backward-peaked, opposite handedness), and their spin-averaged sum, proportional to $1+\cos^2\theta$. Each shape is the square of a Wigner rotation function, $\left|d^1_{\pm1,\pm1}\right|^2$ and $\left|d^1_{\pm1,\mp1}\right|^2$ respectively. *Right:* the resulting $1+\cos^2\theta$ angular law as a polar diagram in the centre-of-mass frame, with the electron beam along the horizontal. The minimum at $\theta = \frac{\pi}{2}$ never reaches zero because the two lobes overlap there at equal height.

and all other twelve of the sixteen spin assignments vanish identically.

It remains to check that these reproduce Section 17.3. The unpolarized squared amplitude averages over the four initial spin states (factor $\frac{1}{4}$) and sums over the final ones; since only the four entries of (17.168) are nonzero, that sum is just their total,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{\text{spins}} |\mathcal{M}|^2 = \frac{e^4}{4}\left[2(1+\cos\theta)^2 + 2(1-\cos\theta)^2\right] = e^4\,(1+\cos^2\theta), \quad (17.169)$$

using $(1+c_\theta)^2 + (1-c_\theta)^2 = 2(1+c_\theta^2)$. This is exactly the result $\overline{|\mathcal{M}|^2} = e^4(1+\cos^2\theta) = 2e^4(t^2+u^2)/s^2$ of Eq. (17.84), now resolved into its physical constituents: two forward-peaked channels and two backward-peaked ones, adding incoherently. The smooth, forward–backward-symmetric $1+\cos^2\theta$ is the envelope of two oppositely skewed lobes. Figure 17.9 shows the decomposition.

17.6.4 The angular law from angular momentum: Wigner d-functions

The decomposition (17.168) contains more than a check: it reveals that the two angular shapes are *rotation matrix elements*. The process, at the moment the

timelike photon is exchanged, is the transition of a $J=1$ state from one quantization axis to another. The initial e^+e^- pair, in the surviving configurations, is a state of definite total angular momentum $J=1$ with projection $m=\pm1$ along the beam axis $\hat{z}$ (we showed the spins add to $J_z=\pm1$). The final $\mu^+\mu^-$ pair is likewise a $J=1$ state with projection $m'=\pm1$, but along the muon direction $\hat{n}$, which makes angle θ with $\hat{z}$. The amplitude to find a $J=1$ system prepared with projection m on $\hat{z}$ in the state of projection m' on the rotated axis $\hat{n}$ is, by definition, the Wigner rotation function

$$\mathcal{M}(m\to m') \;\propto\; d^{1}_{m'm}(\theta)=\left\langle 1,m'\middle|e^{-i\theta J_y}\middle|1,m\right\rangle, \tag{17.170}$$

the matrix element of a rotation by θ about the y-axis (the axis perpendicular to the scattering plane) in the spin-one representation. Spin-one is forced: a single photon is exchanged, and we are reading off the $J=1$ partial wave that it alone can carry. The $1+\cos^2\theta$ law is then a statement purely about d^{1}, with no reference to gamma matrices at all. We derive d^{1} from scratch.

Worked Example 17.20 (The spin-one rotation matrix and the $1+\cos^2\theta$ law)**.** In the basis $|1,m\rangle$ with $m=+1,0,-1$, the angular-momentum generators of the spin-one representation are (from $J_\pm|1,m\rangle=\sqrt{2-m(m\pm1)}\;|1,m\pm1\rangle$, i.e. $J_\pm$ connecting adjacent levels with coefficient $\sqrt{2}$)

$$J_y=\frac{1}{\sqrt{2}}\begin{pmatrix}0&-i&0\\ i&0&-i\\ 0&i&0\end{pmatrix}. \tag{17.171}$$

A direct multiplication gives $J_y^3=J_y$ (its eigenvalues are $0,\pm1$, so it satisfies its characteristic polynomial $J_y(J_y^2-1)=0$). This single relation lets us exponentiate in closed form. Splitting the series for $e^{-i\theta J_y}$ into even and odd powers and using $J_y^{2k+1}=J_y$, $J_y^{2k}=J_y^2$ for $k\ge1$,

$$d^{1}(\theta)=e^{-i\theta J_y}=\mathbf{1}-i\sin\theta\,J_y-(1-\cos\theta)\,J_y^2, \tag{17.172}$$

the spin-one Rodrigues formula. With

$$J_y^2=\frac{1}{2}\begin{pmatrix}1&0&-1\\ 0&2&0\\ -1&0&1\end{pmatrix}, \tag{17.173}$$

substitution into (17.172) yields the explicit rotation matrix

$$d^{1}(\theta)=\begin{pmatrix}\dfrac{1+\cos\theta}{2}&-\dfrac{\sin\theta}{\sqrt{2}}&\dfrac{1-\cos\theta}{2}\\[2mm] \dfrac{\sin\theta}{\sqrt{2}}&\cos\theta&-\dfrac{\sin\theta}{\sqrt{2}}\\[2mm] \dfrac{1-\cos\theta}{2}&\dfrac{\sin\theta}{\sqrt{2}}&\dfrac{1+\cos\theta}{2}\end{pmatrix}, \tag{17.174}$$

rows and columns ordered $m' = +1, 0, -1$ and $m = +1, 0, -1$. The matrix elements we need are the four corners,

$$d^1_{+1,+1} = d^1_{-1,-1} = \frac{1+\cos\theta}{2}, \qquad d^1_{+1,-1} = d^1_{-1,+1} = \frac{1-\cos\theta}{2}. \tag{17.175}$$

These are precisely the square roots of the helicity shapes (17.168): $|\mathcal{M}|^2 \propto |d^1_{m'm}|^2$, with the proportionality constant $4e^4$ fixed by the overall normalization. The $m = m'$ (J_z-preserving) channels give $(1+\cos\theta)^2$, the $m = -m'$ (J_z-flipping) channels give $(1-\cos\theta)^2$, exactly as found by the trace. Summing the squares over the four allowed corner entries,

$$\sum_{m,m'=\pm1} \left|d^1_{m'm}(\theta)\right|^2 = 2\left(\frac{1+\cos\theta}{2}\right)^2 + 2\left(\frac{1-\cos\theta}{2}\right)^2 = \frac{(1+\cos\theta)^2 + (1-\cos\theta)^2}{2} = 1+\cos^2\theta \tag{17.176}$$

which is the unpolarized angular distribution (17.169). The $m, m' = 0$ entries of d^1 never enter because two massless fermions cannot make a $J_z = 0$ state along their collision axis—the same chirality selection rule, now phrased as the absence of the longitudinal ($m = 0$) component of the exchanged spin-one quantum.

The point of Theorem 17.20 is that the $1 + \cos^2\theta$ angular law is *inevitable*, independent of the dynamical details of QED. Any theory in which two massless spin-$\frac{1}{2}$ particles annihilate through a single spin-one intermediary into two massless spin-$\frac{1}{2}$ particles, with a parity-conserving vector coupling, must produce (17.176): the chirality selection rule restricts the angular momentum to $J = 1$, $|m| = 1$, and rotational invariance then fixes the θ-dependence to the corner elements of d^1. The trace computation of Section 17.3 was a long way of evaluating a rotation matrix. This is why the answer was so simple, and why the same $1 + \cos^2\theta$ recurs for quark-pair production in e^+e^- annihilation, governing the jet-axis distribution and providing one of the cleanest confirmations that quarks carry spin $\frac{1}{2}$.

Physical Insight

Angular distributions are spin spectroscopy. Reading $1 + \cos^2\theta$ off a detector tells you that the produced fermions have spin $\frac{1}{2}$ and were made through a spin-one current; a $\sin^2\theta$ distribution would signal $m = 0$ exchange; an isotropic distribution would signal scalar ($J = 0$) production. The near-threshold filling-in of the dip at $\theta = \frac{\pi}{2}$ in Section 17.3, the $(1-\beta^2)\sin^2\theta$ term, is the reappearance of the $m = 0$ channel once the muon mass lets a chirality-flip vertex contribute: a massive fermion line is no longer chirality-diagonal, the $d^1_{0,\pm1}$ and $d^1_{0,0}$ elements switch on, and the distribution relaxes toward isotropy as $\beta \to 0$. Mass restores the longitudinal polarization that masslessness forbids.

17.6.5 Toward spinor-helicity: the covariant bookkeeping

The computation just carried out has an unmistakable inefficiency. We squared the amplitude, took a trace, contracted two Levi-Civita symbols, and discovered at the end that the answer was the square of a single simple object, $d^{1}_{m'm}(\theta)$. The squaring obscured a clean underlying *amplitude*. Working directly with the massless spinors $u_{\pm}, v_{\pm}$, one finds that the current $\bar{v}(p_+)\gamma^{\mu}u(p_-)$ for definite helicities is itself a short expression—a product of two-component spinor invariants—whose modulus is $\sqrt{2\,p_-\cdot p_+}$ times an angular factor, and whose square reproduces (17.167) with no traces at all. This is the observation that the spinor-helicity formalism systematizes.

In Chapter 58 we will introduce the angle and square brackets $\langle i\,j\rangle$, $[i\,j]$, the Lorentz-invariant contractions of the two-component (Weyl) pieces of massless momenta, with $\langle i\,j\rangle[j\,i] = 2\,p_i\cdot p_j = s_{ij}$. In that language the single nonzero $e^+e^-\to\mu^+\mu^-$ helicity amplitude of Theorem 17.19 becomes, up to a phase, $\mathcal{M}\propto\langle\ \rangle[\ ]/s$ built from the external labels, and the angular shapes $(1\pm\cos\theta)^2$ reappear as $|\langle i\,j\rangle|^4/s^2 = t^2/s^2$ or u^2/s^2—the very Mandelstam ratios our trace produced, now read off without ever squaring. The chirality selection rule of Theorem 17.18 becomes the statement that an amplitude is holomorphic in either the angle or the square brackets but never both for a given external leg; the little-group weight of Section 3.7.3 becomes the scaling $\langle i\,j\rangle\to t_i\langle i\,j\rangle$, $[i\,j]\to t_i^{-1}[i\,j]$ that fixes a three-point amplitude almost completely from helicity alone. Everything in this section—which configurations survive, which vanish, and the d^{1} shapes of those that survive—is encoded in that scaling. The brute-force traces of Sections 17.3 and 17.5 and the helicity skeleton of the present section are the same physics; Chapter 58 is where the bookkeeping is made manifest and the helicity amplitudes become primary objects rather than fragments extracted from a squared sum.

With the dynamical content of tree-level QED now exposed from both the trace side and the helicity side, the next chapter assembles the remaining tree processes that test the structure further—Compton scattering, where gauge invariance and the tree-level Ward identity become indispensable, and the identical-particle interference of Bhabha and Møller scattering (Chapter 18)—before Chapter 19 confronts the loops, the Ward–Takahashi identity in full, and the infrared subtleties that radiative corrections bring. The helicity organization introduced here returns, made covariant, in the spinor-helicity reframing of Chapter 58.

Further reading

The derivation of the QED Feynman rules from the gauge-fixed Lagrangian, and the trace technology for unpolarized cross sections, are treated at the level of

this chapter in Peskin and Schroeder [396] (Chs. 4–5), Schwartz [432] (Chs. 13–14), and Srednicki [454]; Itzykson and Zuber [278] and Weinberg [500] give the canonical and operator-formalism perspectives in full, and Zee [532] the physicist's-shortcut view of the same computations. The $e^+e^- \to \mu^+\mu^-$ process, with its threshold factor and its role as the R-ratio yardstick, is the standard worked example in all of these. Crossing symmetry and the analytic structure of tree amplitudes are developed carefully in Schwartz [432] and, at the level of axiomatic analyticity, in Weinberg [500]. That the perturbative series is only asymptotic is Dyson [168]. The helicity reorganization of Section 17.6—the chirality selection rule, the surviving amplitudes, and their identification with rotation functions—is the entry point to the modern amplitudes program; for the systematic spinor-helicity treatment to which Chapter 58 builds, see Mangano and Parke [334], Parke and Taylor [379] and the textbook account of Elvang and Huang [176].

Chapter 18

Elementary QED Processes

The previous chapter established the QED Feynman rules and carried QED through its first complete calculation, the unpolarized rate for $e^+e^- \to \mu^+\mu^-$ (Section 17.3), where the only subtleties were spin sums over external fermions and the gamma-matrix traces they generate. That process was deliberately spare: a single photon in the s-channel, no external photons, no identical particles, and a single diagram. The elementary processes of quantum electrodynamics that one actually meets in the laboratory — Compton scattering, electron–positron annihilation into photons, and the Møller and Bhabha scattering of charged leptons — are all richer, and each introduces a genuinely new structural feature of gauge-theory amplitudes. This chapter works the full set, and in doing so promotes QED from "the rules yield one number" to "the tree amplitudes of QED form a single, structured, analytically connected family."

Three ideas organize the chapter, and each is something the seed computation could not exhibit. The first is *gauge invariance made operational*: once a real photon appears on an external line, the amplitude carries a polarization vector ϵ_μ, and the consistency of the theory demands that replacing ϵ_μ by the photon momentum k_μ annihilate the amplitude. We prove this tree-level Ward identity $k_\mu \mathcal{M}^\mu = 0$ directly from the rules, and use it to justify the covariant polarization sum $\sum_\lambda \epsilon^*_\mu \epsilon_\nu \to -\eta_{\mu\nu}$ that makes squaring tractable — the tree shadow of the gauge dependence of the photon propagator built in Chapter 13 and of the all-orders Ward–Takahashi identity to come in Chapter 19. The second is *crossing symmetry*: Compton scattering and pair annihilation, or Møller and Bhabha scattering, are not independent computations but boundary values of one rational function on the Mandelstam plane (Section 16.5), continued between physical regions with a definite sign for each crossed fermion. The third is *helicity*: by replacing the unpolarized spin average with explicit spin projectors we expose the chirality-conservation selection rule that governs the high-energy limit and that, in Chapter 58, becomes the starting axiom of the spinor-helicity formalism.

The machinery is imported, not rebuilt. The quantized Dirac field, its mode

expansion, and the spin sums $\sum_s u\bar{u} = \not{p} + m$, $\sum_s v\bar{v} = \not{p} - m$ come from Chapter 11; the photon, its polarizations, and the gauge dependence of its propagator from Chapter 13; the QED Feynman rules, the vertex $-ie\gamma^\mu$, and the crossing prescription from Chapter 17; the reduction of an amplitude to a cross section or decay rate, the Mandelstam variables, and two-body phase space from Chapter 16; and the Clifford algebra together with the gamma-matrix trace identities from Chapter C. We use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and the electron charge $q = -e < 0$ with $D_\mu = \partial_\mu - iqA_\mu$ throughout, as fixed in Chapter A.

The arc is as follows. Section 18.1 dissects a general tree-level QED amplitude, recalls the rules, isolates the three structural novelties, and assembles the unpolarized master formula together with the fundamental trace lemma. Section 18.2 proves the tree Ward identity and converts it into the polarization-sum replacement, the technical key to every process with external photons. Section 18.3 carries Compton scattering all the way to the Klein–Nishina formula and its Thomson limit, the chapter's flagship computation. Section 18.4 formalizes crossing — including the fermion crossing sign — and obtains $e^+e^- \to \gamma\gamma$ by continuation rather than by a fresh calculation. Section 18.5 treats the two four-fermion processes Møller and Bhabha, exhibiting the interference between t- and u- (or s-) channel diagrams and the Fermi statistics that fix its sign, and checks Bhabha against Møller by crossing. Section 18.6 builds the spin projector, derives the helicity-conservation selection rule, and recomputes $e^+e^- \to \mu^+\mu^-$ helicity amplitude by helicity amplitude, recovering the unpolarized result as an incoherent sum and reading off the forward–backward structure. Finally, Section 18.7 draws the threads together: the entire tree-level QED amplitude family is two analytic functions on the Mandelstam plane, made transverse by the Ward identity and block-diagonalized by helicity — exactly the object the loop expansion of Chapter 19, the precision physics of Chapter 20, and the on-shell program of Chapter 58 will inherit.

18.1 Anatomy of a tree-level QED amplitude

We continue QED from Section 17.3, where the electron met the muon for the first time. That computation — the unpolarized cross section for $e^+e^- \to \mu^+\mu^-$ — already exercised the full apparatus of relativistic scattering: a vertex $-ie\gamma^\mu$ at each end of a virtual photon, external spinors $u, \bar{u}, v, \bar{v}$ for the incoming and outgoing fermions, a spin average over the beams, a spin sum over the unobserved final particles, and the collapse of the resulting bilinear products into Dirac traces evaluated against Mandelstam invariants. And yet that process was, structurally, the simplest interacting amplitude QED can offer. A *single* Feynman diagram contributes at leading order, because the electron and the muon are distinct particles that can only annihilate into a timelike photon; there is no second

topology to interfere with the first, and hence no relative sign to fix. No photon appears on an external leg, so the polarization vectors $\varepsilon^\mu, \varepsilon^{*\mu}$ — and the gauge redundancy they carry — never enter. The amplitude was a product of two independent fermion currents tied together by one propagator, and the answer fell out of two essentially decoupled traces. The deceptive ease of Section 17.3 came from the *absence* of every feature that makes a QED amplitude interesting.

This section assembles, once and for all, the shared toolkit the rest of the chapter will draw on, and names the obstacles it must overcome. The plan is deliberately front-loaded: we recall the QED Feynman rules and external-line factors as imported objects (they were built in Chapters 13 and 17 and are not rederived here); we recall the kinematic and cross-section machinery of Chapter 16; we isolate the three structural novelties that distinguish the canonical quartet of tree processes — Compton scattering, pair annihilation, Møller scattering, and Bhabha scattering — from the warm-up of Section 17.3; we name the two principles, the tree-level Ward identity and crossing symmetry, that will tame those novelties; and we establish the chapter-wide *master formula* for the spin- and polarization-averaged squared amplitude, deriving it carefully from the cross-section formula of Chapter 16 so that every averaging factor is accounted for and never merely asserted. The result handed forward is twofold: this master formula, and a sharp statement of the one problem it cannot by itself resolve — the photon polarization sum — which the next section settles with the Ward identity.

18.1.1 The QED Feynman rules, recalled

Quantum electrodynamics is the theory of one Dirac field ψ of mass m and charge $-e$ (with $e > 0$) coupled minimally to the abelian gauge field A_μ,

$$\mathcal{L}_{\text{QED}} = \bar{\psi}\,(i\not{D} - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \tag{18.1}$$

the model frozen for QED, with $\alpha \equiv e^2/4\pi$. Its momentum-space Feynman rules were derived in Section 17.1 from the path integral over Eq. (18.1) (the photon sector and the polarization technology coming from Chapter 13); throughout this chapter we work in Feynman gauge $\xi = 1$, the default for explicit computation, and quote gamma-matrix trace identities from Chapter C. We recall the rules rather than rebuild them.

The single interaction vertex is the minimal coupling read off from $i\mathcal{L}_{\text{int}} = -ie\,\bar{\psi}\not{A}\psi$,

$$\text{electron–photon vertex} \;=\; -\,ie\,\gamma^\mu, \tag{18.2}$$

exactly the rule Eq. (17.10) (equivalently Eq. (13.211)); the index μ is contracted into the attached photon line, and the matrix γ^μ sits on the fermion line in the order dictated by reading the line against the arrow. The two internal lines carry

Table 18.1: External-line factors of QED, imported from Section 17.1. Each external leg of a diagram contributes the indicated factor; the spinor momentum and spin (or photon momentum and polarization) are those of the physical particle on that leg. Arrows on fermion lines point along the flow of (negative) charge, i.e. along particle number.

External particle	State	Factor
Incoming fermion e^-	$\lvert p, s\rangle$	$u^s(p)$
Outgoing fermion e^-	$\langle p, s\rvert$	$\bar{u}^s(p)$
Incoming antifermion e^+	$\lvert p, s\rangle$	$\bar{v}^s(p)$
Outgoing antifermion e^+	$\langle p, s\rvert$	$v^s(p)$
Incoming photon γ	$\lvert k, \lambda\rangle$	$\varepsilon^\mu_\lambda(k)$
Outgoing photon γ	$\langle k, \lambda\rvert$	$\varepsilon^{*\mu}_\lambda(k)$

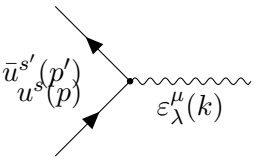


Figure 18.1: The single QED vertex Eq. (18.2), $-ie\gamma^\mu$, with representative external-line factors attached. Reading the fermion line *against* its arrow (from the outgoing $\bar{u}^{s'}(p')$ to the incoming $u^s(p)$) fixes the matrix order: the amplitude fragment is $\bar{u}^{s'}(p')\,(-ie\gamma^\mu)\,u^s(p)\,\varepsilon^\mu_\lambda(k)$. Every diagram in this chapter is built by gluing copies of this vertex with the propagators Eq. (18.3) and Eq. (18.4).

the propagators

$$\text{electron line of momentum } p: \quad \frac{i(\not{p}+m)}{p^2-m^2+i\varepsilon} = \frac{i}{\not{p}-m+i\varepsilon}, \tag{18.3}$$

$$\text{photon line of momentum } k\ (\xi=1): \quad \frac{-i\,\eta_{\mu\nu}}{k^2+i\varepsilon}, \tag{18.4}$$

the Dirac propagator of Theorem 11.40 and the Feynman-gauge photon propagator of Eq. (13.135). Finally, each external leg contributes a wavefunction factor according to whether it is an incoming or outgoing fermion, antifermion, or photon. These external-line factors, collected in Table 18.1, are the bridge between the abstract S-matrix element and the spinor/polarization data of the physical asymptotic states; they were fixed in Chapters 11 and 13 and assembled into the QED rules in Section 17.1.

Two conventions deserve emphasis because they are the seeds of the sign bookkeeping to come. First, a fermion line is read *against* the direction of its charge-flow arrow, so that the matrices line up in the order $\bar{u}\cdots u$ (or $\bar{v}\cdots$, $\cdots v$) as one moves from the head of the arrow to its tail; Fig. 18.1 shows the

ordering on the elementary vertex. Second, the overall sign of a diagram includes a factor (-1) for every closed fermion loop and, crucially for this chapter, a relative minus sign between diagrams that differ by the exchange of two external identical fermions. That second rule is dormant in Section 17.3 — the four external fermions there are two electrons and two muons, all distinguishable — but it is decisive for Møller and Bhabha scattering, and we return to it in Section 18.1.2.

18.1.2 Three structural novelties

The four processes of this chapter are not chosen at random. Together they exhaust, in their simplest realizations, the three structural features that a tree-level QED amplitude can possess beyond the bare current×propagator×current of Section 17.3. Naming them now lets us treat each as a recurring theme rather than a fresh surprise.

Novelty 1: external photons. Compton scattering $e^-\gamma \to e^-\gamma$ and pair annihilation $e^+e^- \to \gamma\gamma$ put real photons on external legs. Each such leg carries a polarization vector $\varepsilon^{\mu}_{\lambda}(k)$ or its conjugate $\varepsilon^{*\mu}_{\lambda}(k)$ (Table 18.1), and a physical photon has only the two transverse polarizations established in Section 13.3.3. The amplitude is therefore a Lorentz vector (or tensor) contracted into these polarization data: writing the dependence on a single external photon explicitly,

$$\mathcal{M} = \varepsilon_{\mu}(k)\,\mathcal{M}^{\mu}, \tag{18.5}$$

where $\mathcal{M}^{\mu}$ collects everything but that polarization vector. The trouble is that $\varepsilon^{\mu}_{\lambda}(k)$ is not a Lorentz-covariant object on its own — a choice of two transverse polarizations singles out a frame — and is moreover defined only up to the gauge freedom $\varepsilon^{\mu}_{\lambda}(k) \to \varepsilon^{\mu}_{\lambda}(k) + c\,k^{\mu}$, which leaves the physical photon unchanged. An honest squared amplitude can depend only on the gauge-invariant content of $\mathcal{M}^{\mu}$. This is the photon polarization-sum problem, and it is the first thing the chapter must solve.

Novelty 2: several interfering diagrams. Once more than one diagram contributes to a process at leading order, the amplitudes *add coherently*: $\mathcal{M} = \mathcal{M}_1 + \mathcal{M}_2 + \cdots$, and $|\mathcal{M}|^2$ contains not only $|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2$ but the cross terms $2\,\mathrm{Re}(\mathcal{M}_1\mathcal{M}_2^*)$. Every process in this chapter has two diagrams. For Compton the electron can absorb the incoming photon before or after emitting the outgoing one (an s-channel and a u-channel topology on the single fermion line). For annihilation the two final photons can be emitted in either order. For Møller and Bhabha the two diagrams correspond to the two ways the exchanged photon can connect the external legs. The interference terms are not a correction to be neglected; they carry genuine physics (the angular dependence of Møller

scattering, for instance, is dominated by interference) and they are where the relative-sign rule bites.

Novelty 3: identical particles and their signs. Møller scattering $e^-e^- \to e^-e^-$ has two identical fermions in the initial state and two in the final state; Bhabha scattering $e^+e^- \to e^+e^-$ has a particle–antiparticle pair that the annihilation channel can route through. When two external legs carry identical fermions, the two diagrams related by exchanging them differ by an overall minus sign, the operational fingerprint of Fermi statistics — the same antisymmetry that gave the Dirac propagator its relative minus and forced the spin-statistics connection. Concretely, Møller's amplitude is $\mathcal{M} = \mathcal{M}_t - \mathcal{M}_u$ rather than $\mathcal{M}_t + \mathcal{M}_u$, and the squared amplitude carries a *negative* interference term $-2\,\mathrm{Re}(\mathcal{M}_t\mathcal{M}_u^*)$. For identical *bosons* — the two photons of pair annihilation — the corresponding statement is a relative plus sign and a final-state symmetry factor $\frac{1}{2}$ in the total cross section, to avoid double-counting the indistinguishable two-photon configurations (the symmetry factor of Theorem 16.18). Getting these signs right is not optional: a wrong relative sign changes the predicted angular distribution qualitatively.

These three novelties are the entire structural difference between Section 17.3 and the rest of QED at tree level. Everything else — the vertex, the propagators, the trace technology — is shared with the warm-up.

18.1.3 Two organizing principles

The novelties would be a bookkeeping nightmare were it not for two principles that impose order on them. We introduce both here as working tools and use them throughout the chapter; each is promoted to an exact, all-orders statement later in QED.

Gauge invariance: the tree-level Ward identity. The redundancy that troubles the external-photon amplitude Eq. (18.5) is exactly the redundancy that constrains it. The statement, proved in the next section, is that any tree amplitude with an external photon obeys

$$k_\mu \mathcal{M}^\mu = 0, \tag{18.6}$$

the *tree-level Ward identity*: replacing a physical photon polarization by its own momentum annihilates the amplitude. Physically, Eq. (18.6) says the amplitude is insensitive to the unphysical, longitudinal part of the photon's would-be polarization — it is the amplitude-level expression of the gauge invariance of Eq. (18.1). Its immediate payoff, also developed next, is that the sum over the two physical polarizations may be replaced inside a squared amplitude by the

manifestly covariant

$$\sum_{\lambda=1,2} \varepsilon_\lambda^\mu(k)\, \varepsilon_\lambda^{*\nu}(k) \;\longrightarrow\; -\eta^{\mu\nu}, \tag{18.7}$$

the replacement first met for internal-line gauge cancellations in Eq. (13.149), now licensed for external real photons. The longitudinal and timelike contributions that Eq. (18.7) silently adds back are killed by Eq. (18.6); that is precisely why the covariant sum is permitted. The all-orders generalization of Eq. (18.6) is the Ward–Takahashi identity proved in Chapter 19.

Analyticity: crossing symmetry. The second principle ties distinct processes together. An amplitude $\mathcal{M}(s,t,u)$, computed as a function of the Mandelstam invariants, is a single analytic function whose various physical channels describe different reactions obtained by moving particles between the initial and final states and replacing them by antiparticles — crossing. The geometry that crossing acts on, the Mandelstam plane with its three physical wedges, was laid out in Section 16.5.5; here we put it to work. Crossing makes pair annihilation $e^+e^- \to \gamma\gamma$ an analytic continuation of Compton scattering $e^-\gamma \to e^-\gamma$, and Bhabha scattering $e^+e^- \to e^+e^-$ a continuation of Møller scattering $e^-e^- \to e^-e^-$. Once one member of each pair is computed in full, its partner follows by a substitution of invariants and external spinors, rather than a fresh diagrammatic calculation — a labor-saving device that is also a nontrivial check, since the two routes to a crossed amplitude must agree. We state and use crossing at tree level in this chapter; its foundation is the analyticity of the amplitude in the cut s- and t-planes.

18.1.4 Kinematics and the cross-section wrapper

All four processes are $2 \to 2$ reactions, so the kinematic machinery of Chapter 16 applies verbatim. We label the process $1 + 2 \to 3 + 4$ and use the Mandelstam invariants of Theorem 16.20,

$$s = (p_1 + p_2)^2, \qquad t = (p_1 - p_3)^2, \qquad u = (p_1 - p_4)^2, \tag{18.8}$$

constrained by Eq. (16.121),

$$s + t + u = \sum_{i=1}^{4} m_i^2. \tag{18.9}$$

Two of the three are independent; we trade u for s, t via Eq. (18.9) at will. The masses entering the sum are those of the four external particles: for Compton and annihilation one or two of them are the photon's zero, while for Møller and Bhabha all four equal the electron mass m, so that $s + t + u = 4m^2$ and, in the high-energy limit $s \gg m^2$, $s + t + u \to 0$.

The dynamics is packaged into a single observable wrapper. The master cross-section formula of Theorem 16.18 gives the differential cross section in terms of the invariant amplitude $\mathcal{M}$ for definite external states; specialized to $2 \to 2$ in the centre-of-mass (CM) frame, Theorem 16.23 reads

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\mathrm{CM}} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,|\mathcal{M}|^2\,, \tag{18.10}$$

with $|\mathbf{p}_i|\,, |\mathbf{p}_f|$ the CM magnitudes of the initial and final three-momenta, and the equivalent fully invariant form Eq. (16.147),

$$\frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{|\mathcal{M}|^2}{16\pi\,\lambda(s, m_1^2, m_2^2)}, \tag{18.11}$$

λ being the Källén triangle function. These formulas were derived for a *fixed* choice of all external spins and polarizations: $\mathcal{M}$ in Eq. (18.10)–Eq. (18.11) is the amplitude for definite spin labels on every leg. Converting them into the quantities a real experiment measures — unpolarized beams, spin-blind detectors — is the content of the master formula, to which we now turn.

18.1.5 The averaged squared amplitude: the chapter's master formula

A typical scattering experiment neither prepares the spins of its incoming particles nor measures the spins of its outgoing ones. The cross sections quoted in this chapter are therefore averaged over the initial spin and polarization states and summed over the final ones. We make the averaging precise, because the factors it introduces are part of every numerical answer and a frequent source of error.

Why average probabilities, not amplitudes. An unpolarized beam is not a coherent superposition of spin states with definite relative phases; it is an *incoherent statistical mixture*, described by a density matrix proportional to the identity on the spin space, $\rho_a = \frac{1}{N_a}\,\mathbf{1}_{N_a}$ for incoming particle a with N_a physical spin/polarization states. The probability of a transition from such a mixture is the trace of ρ against the transition operator, which for diagonal ρ is the *average over initial states of the probabilities*, not of the amplitudes. Averaging amplitudes would manufacture spurious interference between physically distinct, orthogonal initial states. Likewise the inclusive rate into an unmeasured final spin is the *sum over final states of the probabilities*, because distinct final spin configurations are orthogonal members of the final Hilbert space and their contributions to the cross section add. Both operations act on $|\mathcal{M}|^2$, never on $\mathcal{M}$.

Definition 18.1 (Averaged squared amplitude)**.** For a process with initial particles $1, 2$ carrying N_1, N_2 physical spin/polarization states, the spin- and

polarization-averaged squared amplitude is

$$\overline{|\mathcal{M}|^2} \equiv \frac{1}{N_1 N_2} \sum_{\text{spins/pol}} |\mathcal{M}|^2 , \tag{18.12}$$

where the sum runs over the spin and polarization labels of *all* external particles, initial and final, and the prefactor $1/(N_1 N_2)$ averages over (only) the initial ones. A spin-$\frac{1}{2}$ fermion or antifermion contributes $N = 2$; a physical photon contributes $N = 2$ (its two transverse polarizations).

The master formula of the chapter is now the statement that the unpolarized cross section is obtained from Eq. (18.10) by the single replacement $|\mathcal{M}|^2 \to \overline{|\mathcal{M}|^2}$.

Proposition 18.2 (Unpolarized master formula)**.** *The unpolarized differential cross section for a $2 \to 2$ QED process in the CM frame is*

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \overline{|\mathcal{M}|^2}, \qquad \frac{\mathrm{d}\sigma}{\mathrm{d}t} = \frac{\overline{|\mathcal{M}|^2}}{16\pi\, \lambda(s, m_1^2, m_2^2)}, \tag{18.13}$$

with $\overline{|\mathcal{M}|^2}$ the averaged squared amplitude of Theorem 18.1. For a final state of indistinguishable particles a symmetry factor $\frac{1}{2}$ multiplies the total *cross section (one per pair of identical final particles), per Theorem 16.18.*

Proof. Fix, for the moment, definite spin/polarization labels on all four external legs, collected as α (initial) and β (final). For this definite choice the amplitude is $\mathcal{M}_{\beta\alpha}$, and Theorem 16.23 applies as written: the differential cross section to produce the specific final configuration β from the specific initial configuration α is

$$\left.\frac{\mathrm{d}\sigma_{\beta\alpha}}{\mathrm{d}\Omega}\right|_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} |\mathcal{M}_{\beta\alpha}|^2 . \tag{18.14}$$

The kinematic prefactor $\frac{1}{64\pi^2 s} |\mathbf{p}_f| / |\mathbf{p}_i|$ depends only on the momenta, not on the spins, because the flux factor and the two-body phase space of Chapter 16 are built from on-shell kinematics alone (the separation of dynamics from kinematics noted at Section 16.4.5); it therefore factors out of every spin operation below.

A detector blind to final spins counts every distinct final configuration β, and these are orthogonal asymptotic states, so the rates add:

$$\left.\frac{\mathrm{d}\sigma_{\alpha}}{\mathrm{d}\Omega}\right|_{\text{CM}} = \sum_{\beta} \left.\frac{\mathrm{d}\sigma_{\beta\alpha}}{\mathrm{d}\Omega}\right|_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \sum_{\beta} |\mathcal{M}_{\beta\alpha}|^2 . \tag{18.15}$$

The incoming beams are unpolarized: each initial configuration $\alpha = (s_1, s_2)$ occurs with equal probability $1/(N_1 N_2)$, and the measured cross section is the statistical average of Eq. (18.15) over α,

$$\left.\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega}\right|_{\text{CM}} = \frac{1}{N_1 N_2} \sum_{\alpha} \left.\frac{\mathrm{d}\sigma_{\alpha}}{\mathrm{d}\Omega}\right|_{\text{CM}} = \frac{1}{64\pi^2 s} \frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} \frac{1}{N_1 N_2} \sum_{\alpha,\beta} |\mathcal{M}_{\beta\alpha}|^2 . \tag{18.16}$$

The double sum over α and β is exactly the sum over the spin/polarization labels of all external particles, and the prefactor $1/(N_1 N_2)$ is exactly the initial-state average; comparison with Eq. (18.12) identifies the right-hand factor as $\overline{|\mathcal{M}|^2}$, giving the first equality in Eq. (18.13). The invariant form follows identically from Theorem 16.24, whose prefactor is again spin-independent. The final-state symmetry factor is inherited unchanged from Theorem 16.18: it corrects the phase-space integration for the overcounting of indistinguishable final configurations and is a property of the kinematics, independent of the spin averaging carried out here. □

For every process in this chapter $N_1 = N_2 = 2$, so the averaging prefactor is uniformly $\frac{1}{4}$. It is worth noting *why* the four cases nonetheless differ: in Møller and Bhabha the factor $\frac{1}{4}$ averages over the four spin states of two incoming fermions, while in Compton it averages over the two electron spins *and* the two photon polarizations, and in annihilation over the two spins of the e^+e^- pair. The photon's appearance in the count for Compton and annihilation is the place where Theorem 18.1 meets the polarization-sum problem of Section 18.1.2: the sum over a photon's two physical polarizations is not, on its face, a covariant object, and reducing it to one is the business of the next section.

18.1.6 From spin sums to Dirac traces

The averaged squared amplitude Eq. (18.12) still contains an explicit sum over the spins of every external fermion, with the spinors u, v appearing bilinearly. The technology that evaluates such sums — the engine behind Section 17.3 and behind every fermion computation to come — is the pair of completeness relations of Theorem 11.12,

$$\sum_{s=1,2} u^s(p)\,\bar{u}^s(p) = \not{p} + m, \qquad \sum_{s=1,2} v^s(p)\,\bar{v}^s(p) = \not{p} - m, \tag{18.17}$$

the spin sums Eq. (11.73). Their effect is structural: a spin sum that an experiment forces on us, Eq. (18.17) converts into a *matrix* — the on-shell projector onto positive-energy electron (or positron) states — which then sits inside a product of gamma matrices. A product of such matrices, traced, is a number computable from the Mandelstam invariants by the identities of Chapter C. The entire spin sum thereby collapses into Dirac traces. We make the collapse precise once, in full generality, in the following worked example; the rest of the chapter applies it.

Worked Example 18.3 (The fundamental trace identity for a fermion line)**.** Every QED amplitude is, schematically, a product of *fermion lines*: each open fermion line contributes a bilinear $\bar{u}(p')\,\Gamma\,u(p)$, where Γ is a string of gamma matrices (and propagator numerators) assembled along the line, and p, p' are

the spinor momenta at its ends. We show that summing $|\bar{u}(p')\Gamma u(p)|^2$ over both spins turns it into a single Dirac trace.

Step 1: the conjugate bilinear. A bilinear $\bar{u}(p')\Gamma u(p)$ is a 1×1 matrix, i.e. an ordinary complex number, so its complex conjugate equals its Hermitian conjugate:

$$\big[\bar{u}(p')\,\Gamma\,u(p)\big]^{*} = \big[\bar{u}(p')\,\Gamma\,u(p)\big]^{\dagger} = u^{\dagger}(p)\,\Gamma^{\dagger}\,\bar{u}^{\,\dagger}(p'). \tag{18.18}$$

Now $\bar{u}^{\,\dagger}(p') = (u^{\dagger}(p')\gamma^0)^{\dagger} = \gamma^{0\dagger}u(p') = \gamma^0 u(p')$, using $\gamma^{0\dagger}=\gamma^0$. Insert $\mathbf{1}=\gamma^0\gamma^0$ before $\Gamma^{\dagger}$:

$$\big[\bar{u}(p')\,\Gamma\,u(p)\big]^{*} = u^{\dagger}(p)\,\gamma^0\,(\gamma^0\Gamma^{\dagger}\gamma^0)\,u(p') = \bar{u}(p)\,\bar{\Gamma}\,u(p'), \qquad \bar{\Gamma}\equiv\gamma^0\Gamma^{\dagger}\gamma^0, \tag{18.19}$$

where $\bar{u}(p)=u^{\dagger}(p)\gamma^0$ in the last step. The barred matrix $\bar{\Gamma}$ is the conjugate string; the conjugation simply runs the line backwards.

Step 2: barring an explicit string. For a single gamma matrix the conjugation property $\gamma^0\gamma^{\mu\dagger}\gamma^0=\gamma^{\mu}$ of Section C.4 gives $\bar{\gamma^{\mu}}=\gamma^{\mu}$. For a product, $\bar{\Gamma}=\gamma^0\Gamma^{\dagger}\gamma^0$ distributes by inserting $\gamma^0\gamma^0$ between each pair of factors, so the order reverses:

$$\overline{\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}} = \gamma^{\mu_n}\cdots\gamma^{\mu_2}\gamma^{\mu_1}. \tag{18.20}$$

A photon propagator numerator $\eta_{\mu\nu}$ or an external ε^{μ} is a c-number and passes through the conjugation untouched (with $\varepsilon^{\mu}\to\varepsilon^{*\mu}$). Thus $\bar{\Gamma}$ is obtained from Γ by writing the gamma string in reverse order and complex-conjugating any polarization vectors — nothing more.

Step 3: sum over spins. Combining Eq. (18.19) with the original bilinear and summing over the spins s of p and s' of p',

$$\begin{aligned}\sum_{s,s'}\big|\bar{u}^{s'}(p')\,\Gamma\,u^{s}(p)\big|^2 &= \sum_{s,s'}\big[\bar{u}^{s'}(p')\,\Gamma\,u^{s}(p)\big]\big[\bar{u}^{s}(p)\,\bar{\Gamma}\,u^{s'}(p')\big]\\ &= \sum_{s,s'}\bar{u}^{s'}_{a}(p')\,\Gamma_{ab}\,u^{s}_{b}(p)\,\bar{u}^{s}_{c}(p)\,\bar{\Gamma}_{cd}\,u^{s'}_{d}(p'),\end{aligned} \tag{18.21}$$

where a,b,c,d are Dirac indices and repeated indices are summed. The two spin sums act on adjacent spinor pairs. Using Eq. (18.17) for the p sum, $\sum_s u^s_b(p)\bar{u}^s_c(p)=(\not{p}+m)_{bc}$, and for the p' sum, $\sum_{s'}u^{s'}_d(p')\bar{u}^{s'}_a(p')=(\not{p}'+m')_{da}$, equation Eq. (18.21) becomes

$$\sum_{s,s'}\big|\bar{u}^{s'}(p')\,\Gamma\,u^{s}(p)\big|^2 = (\not{p}'+m')_{da}\,\Gamma_{ab}\,(\not{p}+m)_{bc}\,\bar{\Gamma}_{cd} = \mathrm{tr}\big[(\not{p}'+m')\,\Gamma\,(\not{p}+m)\,\bar{\Gamma}\big], \tag{18.22}$$

the chain of index contractions $d\to a\to b\to c\to d$ closing into a trace. This is the fundamental identity: *a spin-summed fermion-line bilinear, squared, is the Dirac trace of its two on-shell projectors with the line's matrix string and its barred conjugate inserted between them.* For an antiparticle line one uses $\sum v\bar{v}=\not{p}-m$ in place of $\not{p}+m$ at the corresponding end, the only change being the sign of the mass term in that projector.

Table 18.2: The structural anatomy of the chapter's four tree processes. "Diagrams" counts the leading-order topologies; "ext. γ" the number of external photons (hence whether the polarization-sum problem and the tree Ward identity are needed); "rel. sign" the relative sign between the two diagrams set by statistics; "crossing" the partner process each is the analytic continuation of.

Process	Reaction	Diagrams	ext. γ	rel. sign	Crossing partner
Compton	$e^-\gamma \to e^-\gamma$	2 (s,u)	2	+ (Bose)	— (the parent)
Pair annihilation	$e^+e^- \to \gamma\gamma$	2 (t,u)	2	+ (Bose)	Compton
Møller	$e^-e^- \to e^-e^-$	2 (t,u)	0	− (Fermi)	— (the parent)
Bhabha	$e^+e^- \to e^+e^-$	2 (s,t)	0	− (Fermi)	Møller

The lemma Eq. (18.22) is the organizing computation of the chapter. It explains, in retrospect, the structure of Section 17.3: that process has *two* disjoint fermion lines — the electron line $\bar{v}(p_2)\gamma^\mu u(p_1)$ and the muon line $\bar{u}(p_3)\gamma^\nu v(p_4)$, joined only by the photon propagator's $\eta_{\mu\nu}$ — so its averaged squared amplitude factorizes into a *product of two independent traces*, one per line, contracted through the propagator indices. That factorization is exactly why the warm-up was tractable. The processes ahead complicate this template in the three ways catalogued in Section 18.1.2: Compton has a single fermion line but with a photon-propagator numerator $\not{k}$ inserted in the middle of Γ (so the trace is of a longer string); annihilation likewise; and Møller and Bhabha have two interfering diagrams, so $|\mathcal{M}|^2 = |\mathcal{M}_1 \pm \mathcal{M}_2|^2$ produces, besides two "direct" traces of the form Eq. (18.22), an *interference* trace in which the matrices of one diagram and the barred matrices of the other are threaded onto the *same* fermion lines, yielding a single long trace rather than a product. The trace identities of Section C.6 — the vanishing of odd traces (Theorem C.53), $\mathrm{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ (Eq. (C.152)), the four-gamma trace (Eq. (C.157)), and the recursion (Theorem C.54) — reduce every such trace to a polynomial in the Mandelstam invariants, and that polynomial, divided by the averaging factor and fed into Theorem 18.2, is the cross section.

18.1.7 Roadmap: which process exercises which feature

Table 18.2 records how the canonical quartet distributes the three novelties and the two principles across the chapter, so that each computation can be read as the addition of one new ingredient to the previous one. Compton is treated first because it isolates the external-photon problem with a single fermion line and the minimal pair of diagrams; pair annihilation then follows almost for free by crossing. Møller introduces the identical-fermion sign with no external photons; Bhabha follows from Møller by crossing, additionally exhibiting the interference between an annihilation channel and a scattering channel.

With the rules recalled (Section 18.1.1), the novelties named (Section 18.1.2),

the principles previewed (Section 18.1.3), the kinematic wrapper in place (Section 18.1.4), the master formula Theorem 18.2 established, and the trace engine Theorem 18.3 loaded, the shared toolkit is complete. One ingredient remains unjustified: the covariant photon polarization sum Eq. (18.7), which Theorem 18.1 demands the moment a real photon sits on an external leg. The sum runs over only the two physical, transverse polarizations of Section 13.3.3, and that restricted set is not manifestly Lorentz covariant; replacing it by $-\eta^{\mu\nu}$ silently reinstates the longitudinal and timelike states, which is legitimate only if the amplitude is blind to them — that is, only if $k_\mu \mathcal{M}^\mu = 0$. Proving that tree-level Ward identity, and with it converting the polarization-sum problem into the simple covariant rule Eq. (18.7), is the first organizing task of the chapter and the subject of the next section.

18.2 The tree-level Ward identity and photon polarization sums

The previous section assembled the toolkit and then handed us an unfinished obligation. The master formula for a spin- and polarization-averaged squared amplitude (Eq. (18.13)) instructed us, for every external photon, to sum the squared amplitude over the two physical polarization states; and we anticipated that this sum could be carried out covariantly, by the replacement $\sum_\lambda \epsilon_\lambda^\mu \epsilon_\lambda^{*\nu} \to -\eta^{\mu\nu}$, which trades an awkward, frame-dependent, two-term sum over transverse vectors for a single Lorentz tensor. We did not yet have the right to make that replacement. The replacement is licensed precisely when the amplitude, stripped of the photon's polarization vector and written as $\mathcal{M} = \epsilon_\mu(k)\mathcal{M}^\mu$, obeys the transversality condition $k_\mu \mathcal{M}^\mu = 0$. This condition is the *tree-level Ward identity*, and supplying its proof is the business of this section. It is the chapter's first organizing principle: once established, it is the single fact that makes every external-photon computation in this chapter — Compton scattering and pair annihilation above all — tractable, and it is the tree-level shadow of the all-orders Ward–Takahashi identity that Chapter 19 will erect on the full quantum theory.

The statement is not merely a computational convenience; it is the on-shell trace of gauge invariance. Recall from Section 8.1 that the photon enters QED only through the covariant derivative, and that the freedom $A_\mu \to A_\mu + \frac{1}{e}\partial_\mu \alpha$ is a redundancy of description carrying no physical content. In momentum space a gauge transformation shifts a photon's wavefunction by a multiple of its own momentum, $\epsilon_\mu(k) \to \epsilon_\mu(k) + c\, k_\mu$. An amplitude that depends on the physics and not on the bookkeeping must be blind to such a shift: $\mathcal{M} = \epsilon_\mu \mathcal{M}^\mu$ must not change when $\epsilon_\mu \to \epsilon_\mu + c\, k_\mu$, which is exactly $k_\mu \mathcal{M}^\mu = 0$. The Ward identity is therefore the demand that gauge redundancy not leak into observables, enforced amplitude by amplitude. We will prove it directly from the Feynman rules and the Dirac equation, then use it to make the polarization-sum replacement exact

at tree level.

Throughout, we use the QED vertex $-ie\gamma^\mu$ and the fermion propagator $i(\not{q}+m)/(q^2-m^2+i\varepsilon)$ established in Chapter 17, with the electron carrying charge $-e$, $e>0$, in the conventions fixed at the head of this chapter. Photons couple at tree level only to the fermion line — there is no elementary photon self-coupling in an abelian theory — so every external photon in a tree diagram attaches, at some vertex, to a continuous fermion line. That single structural fact is what the proof exploits.

18.2.1 The stripped amplitude and the statement of the identity

We first make precise the object whose transversality we will prove.

Definition 18.4 (Stripped photon amplitude)**.** Let $\mathcal{M}$ be a tree amplitude containing at least one external photon of on-shell momentum k ($k^2=0$) and polarization vector $\epsilon_\mu(k)$. Writing the dependence on that polarization explicitly,

$$\mathcal{M} = \epsilon_\mu(k)\,\mathcal{M}^\mu(k), \tag{18.23}$$

the Lorentz vector $\mathcal{M}^\mu(k)$ — the amplitude with the photon's polarization vector removed but its momentum and all other external data retained — is the *stripped amplitude* for that photon. For an outgoing photon one strips $\epsilon^*_\mu(k)$ instead; the construction and every statement below are identical, since ϵ^* is just another constant vector contracted with the same $\mathcal{M}^\mu$.

Theorem 18.5 (Tree-level Ward identity)**.** *Let $\mathcal{M}^\mu(k)$ be the stripped amplitude of Theorem 18.4 for an external photon of momentum k, formed from the* complete *set of tree diagrams of a QED process in which all other external particles are on their mass shells. Then*

$$\boxed{k_\mu\,\mathcal{M}^\mu(k)=0} \tag{18.24}$$

identically in the momenta and polarizations of all the other external particles. In particular, if the process contains several external photons with momenta $k_1,k_2,\dots$ and the amplitude is stripped of all their polarizations, $\mathcal{M}=\epsilon^{(1)}_{\mu_1}\epsilon^{(2)}_{\mu_2}\cdots\mathcal{M}^{\mu_1\mu_2\cdots}$, then contracting any one photon momentum into its own index annihilates the tensor,

$$k_{1\,\mu_1}\,\mathcal{M}^{\mu_1\mu_2\cdots}=0, \tag{18.25}$$

without *requiring the remaining polarization vectors to be transverse.*

The two emphasized words carry the whole content. "Complete": the identity holds for the sum of all diagrams, never for one diagram in isolation; the cancellation is between diagrams, and a calculation that drops even one of them will find a spurious nonzero $k_\mu\mathcal{M}^\mu$. "Without requiring … transverse":

Eq. (18.25) is an identity on the full Lorentz tensor $\mathcal{M}^{\mu_1\mu_2\cdots}$, valid even when the slots $\mu_2, \ldots$ are contracted with unphysical (longitudinal or scalar) directions. This stronger, tensorial statement is exactly what will be needed to perform the covariant polarization-sum replacement *simultaneously* on several photons, and we will see in Section 18.2.5 that the proof below delivers it for free.

18.2.2 Proof of the tree-level Ward identity

The proof rests on one algebraic identity at a single vertex, applied repeatedly. Contracting the photon momentum into the vertex turns the vertex Dirac matrix γ^μ into $\not{k}$, and the key observation is that $\not{k}$ can be split into a difference of inverse fermion propagators that cancel the two propagators adjacent to the vertex.

Lemma 18.6 (Vertex insertion identity)**.** *Consider a fermion line carrying momentum q into a vertex at which an external photon of momentum k is absorbed, so that the line carries $q+k$ out of the vertex. With the photon momentum contracted into the vertex, the propagator–vertex–propagator string evaluates to a difference of single propagators:*

$$\frac{i(\not{q}+\not{k}+m)}{(q+k)^2-m^2}\,(-ie\,\not{k})\,\frac{i(\not{q}+m)}{q^2-m^2} = e\left[\frac{i(\not{q}+\not{k}+m)}{(q+k)^2-m^2} - \frac{i(\not{q}+m)}{q^2-m^2}\right]. \tag{18.26}$$

Proof. Write $S(q) \equiv i(\not{q}+m)/(q^2-m^2)$ for the propagator. The entire content is the decomposition of the contracted vertex as a difference of inverse propagators,

$$\not{k} = (\not{q}+\not{k}-m) - (\not{q}-m), \tag{18.27}$$

which is an identity (the $\not{q}$ and m terms cancel). Each summand annihilates one adjacent propagator. On the right,

$$(\not{q}-m)\,S(q) = (\not{q}-m)\,\frac{i(\not{q}+m)}{q^2-m^2} = i\,\frac{\not{q}^2-m^2}{q^2-m^2} = i\,\frac{q^2-m^2}{q^2-m^2} = i\,\mathbf{1}, \tag{18.28}$$

where $\not{q}^2 = q_\mu q_\nu \gamma^\mu\gamma^\nu = q_\mu q_\nu\,\frac{1}{2}\{\gamma^\mu,\gamma^\nu\} = q^2\,\mathbf{1}$ by the Clifford algebra (Chapter 11). On the left, identically,

$$S(q+k)\,(\not{q}+\not{k}-m) = \frac{i(\not{q}+\not{k}+m)}{(q+k)^2-m^2}\,(\not{q}+\not{k}-m) = i\,\frac{(q+k)^2-m^2}{(q+k)^2-m^2} = i\,\mathbf{1}. \tag{18.29}$$

Therefore

$$\begin{aligned} S(q+k)\,(-ie\,\not{k})\,S(q) &= (-ie)\,S(q+k)\big[(\not{q}+\not{k}-m)-(\not{q}-m)\big]S(q) \\ &= (-ie)\big[\,i\,S(q) \;-\; i\,S(q+k)\,\big] = e\big[S(q+k)-S(q)\big], \end{aligned} \tag{18.30}$$

which is Eq. (18.26). The first term in the bracket used Eq. (18.29) to remove the *left* propagator (leaving $S(q)$), the second used Eq. (18.28) to remove the *right* propagator (leaving $S(q+k)$). □

The lemma says that contracting the longitudinal photon $\not{k}$ at an interior vertex is the same as not putting the photon there at all, but instead either (i) shifting all the momenta to the left of the vertex by $+k$ and deleting the right-hand propagator, or (ii) leaving the momenta alone and deleting the left-hand propagator, with a relative minus sign between the two. This is the engine of a telescoping cancellation once we sum over where the photon attaches.

Proof of Theorem 18.5. The telescoping sum. Fix a process and a single external photon of momentum k. The complete set of tree diagrams is generated by attaching this photon, in turn, to every possible point of the diagram; since the photon couples only to fermions, this means attaching it to each vertex slot along the fermion lines. Consider one continuous fermion line, running from an incoming external spinor $u(p)$ to an outgoing $\bar{u}(p')$, and carrying n *other* photon attachments (external or internal; at tree level their momenta are fixed c-numbers). Label the gaps between consecutive structures along the line by $a = 0, 1, \ldots, n$: gap 0 sits between $u(p)$ and the first existing vertex, gap n between the last vertex and $\bar{u}(p')$. Inserting the external photon into gap a produces one diagram; the sum of these $n+1$ diagrams, stripped of ϵ_μ and contracted with k_μ, is the line's contribution to $k_\mu \mathcal{M}^\mu$.

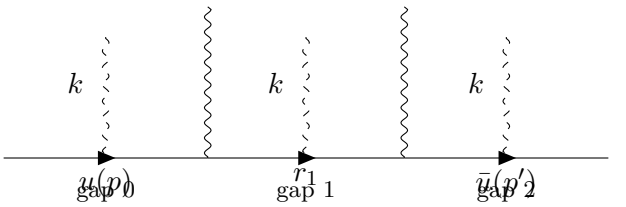


Figure 18.2: The telescoping sum behind the tree Ward identity, drawn for a fermion line with $n = 2$ pre-existing photon attachments (solid). The external photon of momentum k (dashed) is inserted, in turn, into each of the $n+1 = 3$ gaps along the line. By the insertion identity Eq. (18.26), the contraction $k_\mu \mathcal{M}^\mu$ of the gap-a diagram equals a difference of two amputated lines; the right piece of gap a cancels the left piece of gap $a+1$, and the whole sum collapses to the two endpoint terms, which vanish by the Dirac equation on $u(p)$ and $\bar{u}(p')$.

Apply Theorem 18.6 at each gap. For an *interior* gap a (with a propagator on both sides), the contracted diagram is, by Eq. (18.26), a difference $e[\, L_a^+ - L_a^- \,]$, where L_a^+ is the original line with the photon's momentum k added to every segment to the left of gap a and the propagator just left of the gap retained, while L_a^- is the original line with k added to the left segments but with the

propagator just left of the gap deleted — equivalently, L_a^+ is the line "with k flowing out through gap a and above" and L_a^- is the same with k flowing out through gap $a-1$ and above. The bookkeeping that matters is only this: the negative piece generated at gap a is *identical* to the positive piece generated at gap $a-1$, because both describe the line with k routed out above gap $a-1$ and the corresponding internal propagator restored. Hence in the sum

$$k_\mu \mathcal{M}^\mu\big|_{\text{line}} = e\sum_{a=0}^{n}\big[\,L_a^+ - L_a^-\,\big] = e\big[\,L_n^+ - L_0^-\,\big], \tag{18.31}$$

every interior term cancels against its neighbour and only the two endpoint terms survive. This is the telescope: a sum of differences collapses to the difference of the extremes.

The two surviving terms are precisely where the photon sits against an external spinor rather than against a propagator, and there the insertion identity is replaced by its boundary form. At gap 0 the object to the right of the contracted vertex is the external $u(p)$, not a propagator, so $\not{k}$ acts directly: using Eq. (18.27) with $q=p$ and the momentum-space Dirac equation $(\not{p}-m)u(p)=0$ of Eq. (11.50),

$$(-ie\,\not{k})\,u(p) = (-ie)\big[(\not{p}+\not{k}-m)-(\not{p}-m)\big]u(p) = (-ie)(\not{p}+\not{k}-m)\,u(p), \tag{18.32}$$

and the leftover piece $-(\not{p}-m)u(p)$ — which is the residual $-L_0^-$ of Eq. (18.31) — vanishes outright. Symmetrically, at gap n the object to the left is $\bar{u}(p')$, and the adjoint Dirac equation $\bar{u}(p')(\not{p}'-m)=0$ (obtained by conjugating Eq. (11.50), as in Chapter 11) kills the residual top piece $+L_n^+$:

$$\bar{u}(p')\,(-ie\,\not{k}) = (-ie)\,\bar{u}(p')\big[(\not{p}'-m)-(\not{p}'-\not{k}-m)\big] = -(-ie)\,\bar{u}(p')(\not{p}'-\not{k}-m). \tag{18.33}$$

The careful statement is that the two extreme terms in Eq. (18.31) are exactly the two pieces proportional to $(\not{p}-m)u(p)$ and $\bar{u}(p')(\not{p}'-m)$, each of which is zero on the mass shell. Thus the contribution of any single fermion line to $k_\mu\mathcal{M}^\mu$ vanishes.

Assembling the full amplitude. A general tree diagram may contain several distinct fermion lines (for example the two electron lines of a process built by photon exchange). The external photon attaches to exactly one line in any given diagram, and the complete amplitude is the sum, over all fermion lines, of the sum over all attachment points on that line. We have just shown each inner sum — the full set of diagrams in which the photon sits on one chosen line — telescopes to endpoint terms that vanish by the Dirac equation for that line's external spinors. Summing the (vanishing) contributions of the separate lines gives

$$k_\mu\mathcal{M}^\mu = \sum_{\text{fermion lines}} \big(k_\mu\mathcal{M}^\mu\big|_{\text{line}}\big) = 0, \tag{18.34}$$

which is Eq. (18.24). The argument used nothing about the other external particles except that the fermions are on shell (so their spinors obey the Dirac equation) and that the photon momentum is the one contracted; in particular the polarization vectors of any *other* external photons were carried inertly through the manipulation and never assumed transverse. Hence the contraction $k_{1\mu_1}\mathcal{M}^{\mu_1\mu_2\cdots} = 0$ holds as the tensorial identity Eq. (18.25), with the slots $\mu_2, \ldots$ free. This completes the proof of Theorem 18.5. □

Caution

The Ward identity is a statement about the *gauge-invariant sum of all diagrams at a given order*. Each individual diagram has nonzero $k_\mu \mathcal{M}^\mu_{\text{diagram}}$ — indeed the telescope works precisely by cancelling those nonzero pieces against each other. A computation that, say, retains the s-channel Compton diagram but forgets the u-channel one will not satisfy the Ward identity, will find unphysical polarizations contributing to the cross section, and will give a gauge-dependent, wrong answer. Checking $k_\mu \mathcal{M}^\mu = 0$ on the full amplitude is therefore one of the sharpest available tests that a tree-level QED calculation is complete and correctly assembled; we will use it as exactly such a check in every external-photon process of this chapter.

18.2.3 The canonical illustration: the Compton amplitude

The general telescope is most transparent in its smallest nontrivial case, $n = 1$, where there are two diagrams and the cancellation is the bare difference of two boundary terms. This is the Compton process $e^-(p)\,\gamma(k) \to e^-(p')\,\gamma(k')$, the flagship external-photon amplitude that the next section computes in full. We isolate here only its gauge structure, and verify Eq. (18.24) explicitly.

Worked Example 18.7 (Tree Ward identity for Compton scattering)**.** The two tree diagrams for $e^-(p)\gamma(k) \to e^-(p')\gamma(k')$ differ in the order in which the incoming photon k (polarization ϵ) and the outgoing photon k' (polarization ϵ'^*) attach to the single electron line. Momentum conservation reads $p + k = p' + k'$, and all external particles are on shell: $p^2 = p'^2 = m^2$, $k^2 = k'^2 = 0$. Writing the amplitude as $i\mathcal{M} = i\,\epsilon_\nu(k)\,\mathcal{M}^\nu$ and stripping the incoming photon's polarization, the two diagrams give

$$\mathcal{M}^\nu_s = -ie^2\,\bar{u}(p')\,\not{\epsilon}'^*\,\frac{\not{p} + \not{k} + m}{(p+k)^2 - m^2}\,\gamma^\nu\,u(p), \tag{18.35}$$

$$\mathcal{M}^\nu_u = -ie^2\,\bar{u}(p')\,\gamma^\nu\,\frac{\not{p} - \not{k}' + m}{(p-k')^2 - m^2}\,\not{\epsilon}'^*\,u(p), \tag{18.36}$$

where the two factors of the vertex $-ie\gamma$ and the propagator numerator $i(\not{q}+m)$ have combined to the prefactor $-ie^2$. In the s-channel diagram Eq. (18.35) the incoming photon attaches at the right vertex (against $u(p)$); in the u-channel diagram Eq. (18.36) it attaches at the left vertex (against $\bar{u}(p')$). The denominators simplify on shell to

$$(p+k)^2 - m^2 = 2p\cdot k, \qquad (p-k')^2 - m^2 = -2p\cdot k'. \tag{18.37}$$

Contracting the s-channel with k_ν. Set $\gamma^\nu \to \not{k}$ in Eq. (18.35) and split $\not{k} = (\not{p}+\not{k}-m) - (\not{p}-m)$. The second piece kills $u(p)$ by the Dirac equation, leaving

$$k_\nu \mathcal{M}_s^\nu = -ie^2\, \bar{u}(p')\, \not{\epsilon}'^{*}\, \frac{(\not{p}+\not{k}+m)(\not{p}+\not{k}-m)}{2p\cdot k}\, u(p). \tag{18.38}$$

The numerator is $(\not{p}+\not{k}+m)(\not{p}+\not{k}-m) = (p+k)^2 - m^2 = 2p\cdot k$, which cancels the denominator exactly, so

$$k_\nu \mathcal{M}_s^\nu = -ie^2\, \bar{u}(p')\, \not{\epsilon}'^{*}\, u(p). \tag{18.39}$$

Contracting the u-channel with k_ν. Set $\gamma^\nu \to \not{k}$ in Eq. (18.36). Here $\not{k}$ stands to the immediate right of $\bar{u}(p')$, so we use momentum conservation to route it onto the propagator and the outgoing spinor. With $\ell \equiv p - k'$ the propagator momentum, conservation $k = p' + k' - p$ gives $\not{k} = \not{p}' - \not{\ell}$ (since $\not{p} = \not{\ell} + \not{k}'$), hence $\not{k} = (\not{p}' - m) - (\not{\ell} - m)$. The first piece kills $\bar{u}(p')$ by the adjoint Dirac equation, leaving

$$k_\nu \mathcal{M}_u^\nu = -ie^2\, \bar{u}(p')\, [-(\not{\ell} - m)]\, \frac{\not{\ell} + m}{-2p\cdot k'}\, \not{\epsilon}'^{*}\, u(p). \tag{18.40}$$

Now $(\not{\ell}-m)(\not{\ell}+m) = \ell^2 - m^2 = (p-k')^2 - m^2 = -2p\cdot k'$ cancels the denominator, and the overall sign from the bracket gives

$$k_\nu \mathcal{M}_u^\nu = -ie^2\, \bar{u}(p')\, \frac{-(-2p\cdot k')}{-2p\cdot k'}\, \not{\epsilon}'^{*}\, u(p) = +ie^2\, \bar{u}(p')\, \not{\epsilon}'^{*}\, u(p). \tag{18.41}$$

The sum. Adding Eq. (18.39) and Eq. (18.41),

$$k_\nu \mathcal{M}^\nu = k_\nu \mathcal{M}_s^\nu + k_\nu \mathcal{M}_u^\nu = -ie^2\, \bar{u}(p')\, \not{\epsilon}'^{*}\, u(p) + ie^2\, \bar{u}(p')\, \not{\epsilon}'^{*}\, u(p) = 0, \tag{18.42}$$

exactly as Eq. (18.24) demands. The cancellation is the $n = 1$ telescope: the surviving s-channel boundary term (the photon pushed onto the incoming end, with the Dirac equation removing the would-be extra piece) is equal and opposite to the surviving u-channel boundary term (the photon pushed onto the outgoing end). Neither diagram alone is transverse; their sum is. Note that $\not{\epsilon}'^{*}$ rode through every step untouched: Eq. (18.42) holds for an *arbitrary* ϵ', so we have in fact established the tensor identity $k_\nu \mathcal{M}^{\mu\nu} = 0$ for the Compton amplitude, where μ is the outgoing photon's index. By the identical computation with the roles of the two photons exchanged, $k'_\mu \mathcal{M}^{\mu\nu} = 0$ as well. Both photons of Compton scattering thus satisfy the Ward identity as a tensor statement — the fact Section 18.2.5 needs to sum over both polarizations covariantly at once.

18.2.4 Physical polarizations and the covariant completeness identity

We now turn the Ward identity into the polarization-sum rule the master formula requires. The starting point is a concrete, covariant description of the two physical polarization states of an on-shell photon and the resolution of the identity they generate.

A real photon of momentum k ($k^2 = 0$, $k^0 = |\mathbf{k}| > 0$) has exactly two physical polarization states, the transverse ones; this is the two-degrees-of-freedom count established for the quantized photon in Chapter 13. To exhibit them covariantly, work in a frame in which the momentum points along the z-axis,

$$k^\mu = \omega\,(1, 0, 0, 1), \tag{18.43}$$

and introduce an auxiliary null *reference vector* pointing the other way along the light cone,

$$\bar{k}^\mu = \omega\,(1, 0, 0, -1), \qquad \bar{k}^2 = 0, \qquad k \cdot \bar{k} = 2\omega^2 \neq 0. \tag{18.44}$$

The two transverse polarizations are the purely spatial unit vectors orthogonal to $\mathbf{k}$,

$$\epsilon_1^\mu = (0, 1, 0, 0), \qquad \epsilon_2^\mu = (0, 0, 1, 0), \tag{18.45}$$

which obey, in the mostly-minus metric,

$$\epsilon_\lambda \cdot \epsilon_{\lambda'}^* = -\delta_{\lambda\lambda'}, \qquad \epsilon_\lambda \cdot k = 0, \qquad \epsilon_\lambda \cdot \bar{k} = 0, \qquad (\lambda, \lambda' = 1, 2). \tag{18.46}$$

These are the same transverse states defined in Theorem 13.9; the circular (helicity) combinations $\epsilon_\pm = \mp\frac{1}{\sqrt{2}}(\epsilon_1 \pm i\epsilon_2)$ of Eq. (13.70) are a unitary change of basis on the polarization label and satisfy the same relations Eq. (18.46) (now with $\epsilon_\pm^* = \epsilon_\mp \cdot (\mp)$ as recorded there). Everything below is basis-independent on the two-dimensional physical space, so we keep the index λ generic.

The four vectors $\{\epsilon_1, \epsilon_2, k, \bar{k}\}$ are linearly independent and hence a basis of Minkowski space. The resolution of the identity in this basis is the completeness relation we need.

Proposition 18.8 (Covariant polarization completeness)**.** *With the conventions Eq. (18.43)–Eq. (18.46), the sum over the two physical polarizations of an on-shell photon of momentum k is*

$$\boxed{\sum_{\lambda=1,2} \epsilon_\lambda^\mu\, \epsilon_\lambda^{*\nu} = -\eta^{\mu\nu} + \frac{k^\mu \bar{k}^\nu + \bar{k}^\mu k^\nu}{k \cdot \bar{k}}} \tag{18.47}$$

where $\bar{k}$ is the reference null vector Eq. (18.44).

Proof. For any basis $\{e_a\}$ of a vector space carrying a nondegenerate inner product, with Gram matrix $g_{ab} = e_a \cdot e_b$ and inverse $(g^{-1})^{ab}$, the metric is reconstructed as $\eta^{\mu\nu} = \sum_{a,b}(g^{-1})^{ab}\, e_a^\mu e_b^\nu$. We apply this to $e_1 = \epsilon_1$, $e_2 = \epsilon_2$, $e_3 = k$, $e_4 = \bar{k}$. The relations Eq. (18.46) and Eq. (18.44) make the Gram matrix block-diagonal: the transverse block is $\epsilon_\lambda \cdot \epsilon_{\lambda'} = -\delta_{\lambda\lambda'}$, with inverse $-\delta_{\lambda\lambda'}$; the $(k, \bar{k})$ block is

$$\begin{pmatrix} k\cdot k & k\cdot\bar{k} \\ \bar{k}\cdot k & \bar{k}\cdot\bar{k} \end{pmatrix} = \begin{pmatrix} 0 & N \\ N & 0 \end{pmatrix}, \qquad N \equiv k\cdot\bar{k}, \tag{18.48}$$

with inverse $\frac{1}{N}\left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. Assembling the reconstruction,

$$\eta^{\mu\nu} = \sum_{\lambda,\lambda'}(-\delta_{\lambda\lambda'})\,\epsilon_\lambda^\mu\epsilon_{\lambda'}^\nu + \frac{1}{N}\left(k^\mu\bar{k}^\nu + \bar{k}^\mu k^\nu\right) = -\sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^\nu + \frac{k^\mu\bar{k}^\nu + \bar{k}^\mu k^\nu}{k\cdot\bar{k}}. \tag{18.49}$$

For the real transverse vectors Eq. (18.45) we have $\epsilon_\lambda^* = \epsilon_\lambda$; for the circular basis the same identity holds because the change of basis is unitary, so $\sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^{*\nu} = \sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^\nu$ either way. Solving Eq. (18.49) for the polarization sum gives Eq. (18.47). □

The structure of Eq. (18.47) is exactly what one should expect physically. The full metric $-\eta^{\mu\nu}$ resolves the identity over *all four* components of a four-vector; the two unphysical directions — the longitudinal polarization $\propto k$ and the scalar/timelike polarization, mixed into the combination $k^\mu\bar{k}^\nu + \bar{k}^\mu k^\nu$ — are subtracted off, leaving only the two transverse states. The reference vector $\bar{k}$ is pure gauge: it specifies *which* mixture of timelike and longitudinal directions we call "the unphysical part," a choice with no physical content. A correct physical answer must therefore be independent of $\bar{k}$, and the role of the Ward identity is to guarantee exactly that.

18.2.5 The polarization-sum replacement

We can now state and prove the replacement rule, discharging the obligation left open in the previous section and at the same time supplying the hypothesis that Theorem 13.30 assumed when this rule was previewed in Chapter 13.

Theorem 18.9 (Covariant polarization-sum replacement at tree level)**.** *Let $\mathcal{M} = \epsilon_\mu(k)\mathcal{M}^\mu$ be a complete tree amplitude with an external photon of momentum k, so that the tree Ward identity $k_\mu\mathcal{M}^\mu = 0$ of Theorem 18.5 holds. Then the sum over the two physical polarizations of the squared amplitude may be computed by the covariant replacement $\sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^{*\nu} \to -\eta^{\mu\nu}$:*

$$\sum_{\lambda=1,2}\left|\epsilon_\lambda^*(k)\cdot\mathcal{M}\right|^2 = \sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^{*\nu}\,\mathcal{M}_\mu\mathcal{M}_\nu^* = -\eta^{\mu\nu}\,\mathcal{M}_\mu\mathcal{M}_\nu^* = -\mathcal{M}_\mu\mathcal{M}^{*\mu}. \tag{18.50}$$

If the amplitude contains several external photons, the replacement may be applied to each of them independently and simultaneously, $\prod_i \sum_{\lambda_i} \epsilon_{\lambda_i}^{\mu_i}\epsilon_{\lambda_i}^{*\nu_i} \to \prod_i(-\eta^{\mu_i\nu_i})$.

Proof. Insert the completeness identity Eq. (18.47) into the polarization sum:

$$\sum_\lambda \epsilon_\lambda^\mu \epsilon_\lambda^{*\nu}\, \mathcal{M}_\mu \mathcal{M}_\nu^* = -\eta^{\mu\nu}\mathcal{M}_\mu \mathcal{M}_\nu^* + \frac{(k^\mu \mathcal{M}_\mu)(\bar{k}^\nu \mathcal{M}_\nu^*) + (\bar{k}^\mu \mathcal{M}_\mu)(k^\nu \mathcal{M}_\nu^*)}{k\cdot\bar{k}}. \quad (18.51)$$

Each term in the numerator of the second piece contains a factor $k^\mu \mathcal{M}_\mu = k_\mu \mathcal{M}^\mu$ or its complex conjugate $k^\nu \mathcal{M}_\nu^* = (k_\mu \mathcal{M}^\mu)^*$. By the Ward identity Eq. (18.24) both vanish, so the entire $\bar{k}$-dependent piece drops, leaving $-\eta^{\mu\nu}\mathcal{M}_\mu \mathcal{M}_\nu^* = -\mathcal{M}_\mu \mathcal{M}^{*\mu}$, which is Eq. (18.50). That the reference vector $\bar{k}$ has disappeared is the consistency check promised after Theorem 18.8: the physical, polarization-summed rate cannot depend on the arbitrary choice of $\bar{k}$, and indeed it does not — precisely because of the Ward identity.

For several external photons the argument iterates. Replace the polarization sum on photon 1 first: Eq. (18.51) for that photon produces $-\eta^{\mu_1\nu_1}$ plus terms carrying $k_1^{\mu_1}$ or $\bar{k}_1^{\mu_1}$ contracted into the index μ_1 of $\mathcal{M}^{\mu_1\mu_2\cdots}$. The $\bar{k}_1$ terms come multiplied by $k_{1\mu_1}\mathcal{M}^{\mu_1\mu_2\cdots}$, which vanishes by the *tensor* Ward identity Eq. (18.25) — and this is the crucial point: Eq. (18.25) holds with the remaining indices $\mu_2, \ldots$ free, i.e. even though the polarization sums on the other photons have *not* yet been done and those indices are not contracted with transverse vectors. Hence the $\bar{k}_1$ terms drop and photon 1 is reduced to $-\eta^{\mu_1\nu_1}$, leaving $-\eta^{\mu_1\nu_1}\mathcal{M}_{\mu_1\cdots}\mathcal{M}^*_{\nu_1\cdots}$, an object that still obeys the tensor Ward identity in each remaining photon index. Repeating for photons $2, 3, \ldots$ gives the simultaneous replacement $\prod_i(-\eta^{\mu_i\nu_i})$. □

This is the rule the previous section anticipated and the rest of the chapter will use without further comment. It was previewed in Chapter 13 as Theorem 13.30, where it was proved *conditional* on the Ward identity; Theorem 18.5 has now supplied that condition at tree level, so the rule is established outright for every external-photon amplitude in this chapter. The all-orders upgrade — showing that loop corrections preserve $k_\mu \mathcal{M}^\mu = 0$, and hence that Eq. (18.50) survives radiative corrections — is the content of the Ward–Takahashi identity, deferred to Chapter 19.

Physical Insight

Why does replacing the sum over two physical polarizations by the full Minkowski metric — which manifestly includes two unphysical directions — give the right physical answer? Because the unphysical directions never reach the cross section. The longitudinal and scalar polarizations enter Eq. (18.51) only through the combination k^μ (and the gauge artefact $\bar{k}^\mu$), and an amplitude that respects gauge invariance is, by the Ward identity, exactly blind to k^μ. The covariant rule works *because* the theory is gauge invariant; it is gauge invariance, cashed out as current conservation on the external photon, masquerading as a bookkeeping convenience. The same

mechanism is what allowed the gauge-parameter (ξ) dependence of the internal photon propagator to cancel in Chapter 13: in both cases a k^μ runs into a conserved current and dies. The cost of this convenience in a non-abelian theory — where the analogous naive replacement fails because longitudinal gluons do *not* decouple by themselves — is the Faddeev–Popov ghost, which Chapter 26 introduces precisely to absorb the leftover unphysical polarizations. That QED needs no such ghost in external-state sums is the abelian gift we are cashing here.

With the Ward identity proved and the polarization-sum replacement licensed, the chapter's first organizing principle is in hand. We may now compute any external-photon process covariantly: strip the photon polarizations, square, and replace each $\sum_\lambda \epsilon^\mu_\lambda \epsilon^{*\nu}_\lambda$ by $-\eta^{\mu\nu}$, confident that the unphysical contributions cancel exactly and that the answer is gauge invariant. The next section puts this to work on the flagship case, Compton scattering $e^-\gamma \to e^-\gamma$, whose two-diagram amplitude we have already shown is transverse: there the rule Eq. (18.50), together with the gamma-matrix trace technology of Chapter 11 and Appendix C, will deliver the spin- and polarization-averaged $\overline{|\mathcal{M}|^2}$ and, from it, the Klein–Nishina cross section.

18.3 Compton scattering and the Klein–Nishina formula

The previous section earned us a licence. It proved that any tree amplitude carrying an external photon, written $\mathcal{M} = \epsilon_\mu(k)\mathcal{M}^\mu$, obeys the tree-level Ward identity $k_\mu \mathcal{M}^\mu = 0$ (Theorem 18.5), and it used that gauge-invariance statement to justify the covariant replacement

$$\sum_\lambda \epsilon^\mu_\lambda(k)\, \epsilon^{*\nu}_\lambda(k) \;\longrightarrow\; -\eta^{\mu\nu} \tag{18.52}$$

inside a squared, polarization-summed amplitude (Eq. (18.50)). With those two tools in hand we can finally attack a process with photons on external legs — the first time in QED that the gauge field appears as a physical particle entering and leaving the diagram rather than as an internal propagator. The flagship example is *Compton scattering*, $e^-\gamma \to e^-\gamma$: an electron absorbs a photon and re-emits one. It is the simplest QED amplitude built from *two* interfering diagrams, the cleanest place to watch the Ward identity do real work (each diagram alone is not gauge invariant; only their sum is), and it culminates in the Klein–Nishina formula, whose low-energy limit reproduces the classical Thomson cross section and so anchors the whole quantum computation to a number any nineteenth-century physicist could have measured. This section carries the calculation out

in full: amplitude, the explicit $k_\mu \mathcal{M}^\mu = 0$ check, the spin/polarization-averaged $\overline{|\mathcal{M}|^2}$ (including the trace of eight gamma matrices in the interference term), the Compton kinematic shift, the lab-frame phase space, and the cross section with its Thomson limit and total integral.

Throughout we use the standing conventions of this chapter: the electron has charge $-e$ with $e > 0$, the QED vertex is $-ie\gamma^\mu$ (Chapter 17), the metric is mostly-minus, $\alpha = e^2/4\pi$, and we work in Feynman gauge. The electron carries momentum p (mass m) and the photon momentum k in the initial state; primes denote the final state, p' and k'. All four external particles are on shell: $p^2 = p'^2 = m^2$, $k^2 = k'^2 = 0$.

18.3.1 The two diagrams and the amplitude

At order e^2 the process $e^-\gamma \to e^-\gamma$ receives contributions from exactly two connected tree diagrams, drawn in Fig. 18.3. They differ in the *time order* in which the two photons attach to the single electron line. In the first (the s-channel diagram, so named because the internal electron carries the total momentum $p + k$ whose square is the Mandelstam invariant s), the incoming photon k is absorbed first; the electron propagates with momentum $p + k$; then the outgoing photon k' is emitted. In the second (the u-channel diagram), the outgoing photon k' is emitted first; the electron propagates with momentum $p - k'$; then the incoming photon k is absorbed. There is no third diagram: with one electron line and two photons there are only two orderings, and there is no diagram with the photons attached to a closed loop at tree level.

(a) s-channel: absorb k, then emit k'.

(b) u-channel: emit k', then absorb k.

Figure 18.3: The two tree diagrams for Compton scattering $e^-\gamma \to e^-\gamma$. They are distinguished by the order in which the photons attach to the electron line; the internal electron carries $p+k$ (left) or $p-k'$ (right). Both enter the amplitude with a relative *plus* sign: the two photons are bosons attached to the same fermion line, so Fermi statistics imposes no relative minus here (contrast Section 18.5 for exchanged identical fermions).

We read each diagram with the QED Feynman rules of Chapter 17. An incoming electron contributes a spinor $u^s(p)$, an outgoing electron $\bar{u}^{s'}(p')$; an incoming photon contributes a polarization vector $\epsilon_\nu(k)$, an outgoing photon

$\epsilon^*_\mu(k')$; each vertex is $-ie\gamma^\mu$; the internal electron of momentum ℓ carries the propagator $i(\not\ell + m)/(\ell^2 - m^2 + i\epsilon)$. Reading the s-channel diagram *against* the fermion arrow, from $\bar u(p')$ on the left to $u(p)$ on the right, the outgoing-photon vertex sits next to $\bar u(p')$ and the incoming-photon vertex next to $u(p)$:

$$i\mathcal{M}_s = \bar u(p')\,(-ie\not\epsilon\,'^*)\,\frac{i(\not p + \not k + m)}{(p+k)^2 - m^2}\,(-ie\not\epsilon)\,u(p), \tag{18.53}$$

where $\not\epsilon \equiv \epsilon_\nu(k)\gamma^\nu$ and $\not\epsilon\,'^* \equiv \epsilon^*_\mu(k')\gamma^\mu$. The internal denominator is on the photon mass shell pure kinematics:

$$(p+k)^2 - m^2 = p^2 + 2p\cdot k + k^2 - m^2 = 2\,p\cdot k, \tag{18.54}$$

using $p^2 = m^2$, $k^2 = 0$. The factor $(-ie)^2 \cdot i = -ie^2$, so stripping the overall i,

$$\mathcal{M}_s = -\,e^2\,\frac{\bar u(p')\,\not\epsilon\,'^*\,(\not p + \not k + m)\,\not\epsilon\,u(p)}{2\,p\cdot k}. \tag{18.55}$$

The u-channel diagram is identical except that the two photon vertices are swapped — k' is emitted first, so $\not\epsilon\,'^*$ sits next to $u(p)$ and $\not\epsilon$ next to $\bar u(p')$ — and the internal momentum is $p - k'$, with

$$(p-k')^2 - m^2 = p^2 - 2p\cdot k' + k'^2 - m^2 = -\,2\,p\cdot k'. \tag{18.56}$$

Carrying the sign of Eq. (18.56) through,

$$\mathcal{M}_u = -\,e^2\,\frac{\bar u(p')\,\not\epsilon\,(\not p - \not k' + m)\,\not\epsilon\,'^*\,u(p)}{-\,2\,p\cdot k'} = +\,e^2\,\frac{\bar u(p')\,\not\epsilon\,(\not p - \not k' + m)\,\not\epsilon\,'^*\,u(p)}{2\,p\cdot k'}. \tag{18.57}$$

The full amplitude is the coherent sum:

$$\mathcal{M} = \mathcal{M}_s + \mathcal{M}_u = -\,e^2\,\bar u(p')\left[\frac{\not\epsilon\,'^*(\not p + \not k + m)\not\epsilon}{2\,p\cdot k} - \frac{\not\epsilon\,(\not p - \not k' + m)\not\epsilon\,'^*}{2\,p\cdot k'}\right]u(p). \tag{18.58}$$

The relative minus sign between the two bracketed terms is *not* a statistics sign — the two photons attach to the same fermion line and Bose symmetry gives them a relative plus — it descends purely from the opposite signs of the propagator denominators Eq. (18.54), Eq. (18.56). This is worth flagging because the genuinely statistical relative minus, between diagrams related by exchange of identical *external fermions*, is the central new feature of the Møller and Bhabha amplitudes treated in Section 18.5; Compton has no such exchange.

18.3.2 The explicit Ward-identity check

Before squaring, we verify that this concrete amplitude satisfies the tree-level Ward identity of Theorem 18.5. Write the dependence on the incoming-photon

polarization explicitly, $\mathcal{M} = \epsilon_\nu(k)\,\mathcal{M}^\nu$, and test gauge invariance by the replacement $\epsilon_\nu(k) \to k_\nu$, i.e. $\not\epsilon \to \not k$ in Eq. (18.58). The claim is $k_\nu \mathcal{M}^\nu = 0$. The check is instructive because it shows the two diagrams trading their gauge variations against one another.

Take the s-channel term first. With $\not\epsilon \to \not k$ it carries $(\not p + \not k + m)\not k\, u(p)$ acting to the right on $u(p)$. Using the Dirac equation $(\not p - m)u(p) = 0$, write $\not k\, u(p) = (\not p + \not k - m)u(p)$, so that

$$(\not p+\not k+m)\not k\, u(p) = (\not p+\not k+m)(\not p+\not k-m)u(p) = \big[(p+k)^2-m^2\big]u(p) = (2\,p\cdot k)\,u(p), \tag{18.59}$$

where the middle step used $(\not q + m)(\not q - m) = \not q\not q - m^2 = q^2 - m^2$ with $q = p + k$. The denominator $2\,p \cdot k$ cancels exactly, leaving the s-channel contribution to $k_\nu \mathcal{M}^\nu$ equal to $-e^2\,\bar u(p')\,\not\epsilon\,'^{*}\,u(p)$.

For the u-channel term the photon k attaches at the vertex adjacent to $\bar u(p')$, so we must move $\not k$ leftward through $\bar u(p')$. Momentum conservation $p+k = p'+k'$ gives $\not k = \not p' + \not k' - \not p$, and the outgoing Dirac equation $\bar u(p')(\not p' - m) = 0$ gives $\bar u(p')\not p' = m\,\bar u(p')$, hence

$$\bar u(p')\not k = \bar u(p')(\,-\not p + \not k' + m). \tag{18.60}$$

Then, with $B \equiv \not p - \not k'$,

$$\bar u(p')\,\not k\,(\not p - \not k' + m) = \bar u(p')\,(-B+m)(B+m) = \bar u(p')\,(m^2 - B^2) = \bar u(p')\,(2\,p\cdot k'), \tag{18.61}$$

since $B^2 = (p - k')^2 = m^2 - 2\,p\cdot k'$ so that $m^2 - B^2 = 2\,p\cdot k'$, matching the u-channel denominator. The u-channel contribution to $k_\nu\mathcal{M}^\nu$ is therefore $+e^2\,\bar u(p')\,\not\epsilon\,'^{*}\,u(p)$ — with the overall minus in front of the u-channel term in Eq. (18.58), it is $-(+e^2\bar u \not\epsilon\,'^{*} u)$. Adding the two:

$$k_\nu\mathcal{M}^\nu = -e^2\,\bar u(p')\,\not\epsilon\,'^{*}\,u(p) + e^2\,\bar u(p')\,\not\epsilon\,'^{*}\,u(p) = 0. \tag{18.62}$$

The cancellation is exact and total, and it manifestly needed both diagrams: the s-channel alone would have left a nonzero $-e^2\bar u \not\epsilon\,'^{*} u$, gauge-variant, and only the u-channel's equal and opposite remnant kills it. The same computation with $\epsilon^*_\mu(k') \to k'_\mu$ on the outgoing photon vanishes identically by the mirror argument. This is the concrete realization, on a real amplitude, of the abstract statement proved in Section 18.2.2, and it is precisely the gauge invariance that licenses us to use the polarization sum Eq. (18.52) in the next step.

Caution

A frequent error is to compute only the s-channel "seagull-free" diagram, or to drop the u-channel as a small correction. The Ward check shows why that is fatal: neither diagram is separately gauge invariant, so a single-

diagram "amplitude" would give different answers in different gauges and would not even be a well-defined cross section. Gauge invariance of an external-photon amplitude is a property of the *sum* of all diagrams at a given order.

18.3.3 Squaring: spin sums, polarization sums, and the traces

We want the unpolarized cross section, so we average over the initial electron spin and photon polarization (a factor $\tfrac{1}{2}$ each) and sum over the final spin and polarization:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{s,s'}\sum_{\lambda,\lambda'} |\mathcal{M}_s + \mathcal{M}_u|^2. \tag{18.63}$$

Write $\mathcal{M} = \epsilon^*_\mu(k')\,\epsilon_\nu(k)\,\mathcal{M}^{\mu\nu}$, reading μ for the outgoing photon and ν for the incoming one. From Eq. (18.58),

$$\mathcal{M}_s^{\mu\nu} = -\frac{e^2}{2\,p\cdot k}\,\bar u(p')\,\gamma^\mu(\not p + \not k + m)\gamma^\nu\,u(p), \tag{18.64}$$

$$\mathcal{M}_u^{\mu\nu} = +\frac{e^2}{2\,p\cdot k'}\,\bar u(p')\,\gamma^\nu(\not p - \not k' + m)\gamma^\mu\,u(p). \tag{18.65}$$

The polarization sums act on the products $\epsilon^*_\mu\epsilon_\sigma \to -\eta_{\mu\sigma}$ and $\epsilon_\nu\epsilon^*_\rho \to -\eta_{\nu\rho}$ by Eq. (18.52), so the two minus signs combine to a plus and simply contract the loose photon indices of $\mathcal{M}^{\mu\nu}$ with those of its conjugate:

$$\sum_{\lambda,\lambda'} |\mathcal{M}|^2 = \mathcal{M}^{\mu\nu}\,(\mathcal{M}^{\sigma\rho})^*\,\eta_{\mu\sigma}\,\eta_{\nu\rho} = \mathcal{M}^{\mu\nu}\,\mathcal{M}^*{}_{\mu\nu}. \tag{18.66}$$

The remaining spin sums are handled by the completeness relations $\sum_s u(p)\bar u(p) = \not p + m$ and $\sum_{s'} u(p')\bar u(p') = \not p' + m$ of Eq. (11.73), together with the conjugation rule $\big[\bar u(p')\,\Gamma\,u(p)\big]^* = \bar u(p)\,\bar\Gamma\,u(p')$ with $\bar\Gamma = \gamma^0\Gamma^\dagger\gamma^0$; for a string of gammas this reverses their order, so $\overline{\gamma^\mu(\not q + m)\gamma^\nu} = \gamma^\nu(\not q + m)\gamma^\mu$. Each squared or interference term then collapses to a single Dirac trace. We treat the three pieces in turn, abbreviating

$$a \equiv p\cdot k, \qquad b \equiv p\cdot k'. \tag{18.67}$$

The s-channel square. Summing $|\mathcal{M}_s|^2$ over spins with Eq. (18.66),

$$\sum_{\text{spins}} |\mathcal{M}_s|^2 = \frac{e^4}{(2a)^2}\,T_s, \qquad T_s = \mathrm{Tr}\big[(\not p' + m)\gamma^\mu(\not q + m)\gamma^\nu(\not p + m)\gamma_\nu(\not q + m)\gamma_\mu\big], \tag{18.68}$$

with $q \equiv p + k$. The inner contraction is done first with the $d = 4$ identity $\gamma^\nu(\not p + m)\gamma_\nu = -2\not p + 4m$, which follows from $\gamma^\nu \not p \gamma_\nu = -2\not p$ and $\gamma^\nu\gamma_\nu = 4$

(Theorem C.58). Sandwiching the result and using the contraction identities $\gamma^\mu\gamma_\mu = 4$, $\gamma^\mu \not{a}\gamma_\mu = -2\not{a}$, $\gamma^\mu \not{a}\not{b}\gamma_\mu = 4\,a\cdot b$, and $\gamma^\mu \not{a}\not{b}\not{c}\gamma_\mu = -2\not{c}\not{b}\not{a}$ (Eqs. (C.168) to (C.171)) on the outer $\gamma^\mu \cdots \gamma_\mu$, the trace reduces (with $\mathrm{Tr}\,\mathbf{1} = 4$ and the four-gamma trace Eq. (C.178)) to

$$T_s = 32(m^4 + m^2\,a + a\,b), \tag{18.69}$$

where we used the on-shell kinematics $p\cdot q = m^2+a$, $p'\cdot q = m^2+a$, $q^2 = m^2+2a$, and $p\cdot p' = m^2 + a - b$ (derived in Section 18.3.4 below). Hence

$$\sum_{\text{spins}} |\mathcal{M}_s|^2 = \frac{8e^4}{a^2}(m^4 + m^2 a + ab) = 8e^4\left(\frac{m^4}{a^2} + \frac{m^2}{a} + \frac{b}{a}\right). \tag{18.70}$$

The u-channel square. The u-channel amplitude is obtained from the s-channel by the substitution $k \to -k'$, $k' \to -k$ (the two photons swap roles, incoming↔outgoing with a sign), which in our abbreviations sends $a \to -b$ and $b \to -a$. Applying this to Eq. (18.69) gives $T_u = 32(m^4 - m^2 b + ab)$, and since $|\mathcal{M}_u|^2$ carries the same overall e^4 with denominator $(2b)^2$,

$$\sum_{\text{spins}} |\mathcal{M}_u|^2 = \frac{8e^4}{b^2}(m^4 - m^2 b + ab) = 8e^4\left(\frac{m^4}{b^2} - \frac{m^2}{b} + \frac{a}{b}\right). \tag{18.71}$$

The interference term and the trace of eight gammas. The cross term is the genuinely new computation. Its spin sum is

$$\sum_{\text{spins}} 2\,\mathrm{Re}\,(\mathcal{M}_s\mathcal{M}_u^*) = -\frac{e^4}{2ab}\,T_x, \qquad T_x = \mathrm{Tr}\big[(\not{p}'+m)\gamma^\mu(\not{q}+m)\gamma^\nu(\not{p}+m)\gamma_\mu(\not{r}+m)\gamma_\nu\big], \tag{18.72}$$

with $q = p+k$, $r = p-k'$; the prefactor $-e^4/2ab$ collects the signs of Eqs. (18.64) and (18.65) and the two denominators $(2a)(2b)$, and the factor of two from $\mathcal{M}_s\mathcal{M}_u^* + \mathcal{M}_u\mathcal{M}_s^* = 2\,\mathrm{Re}(\mathcal{M}_s\mathcal{M}_u^*)$ together with the trace-reversal symmetry $T_x^* = T_x$. This is a trace of eight gamma matrices (four bare, plus up to four from the slashes), and brute force would generate $4!/(2!2!) \times \cdots$ terms; instead we again peel off the $\gamma^\mu \cdots \gamma_\mu$ contraction. Using the three-gamma identity $\gamma^\mu\gamma^\alpha\gamma^\nu\gamma^\beta\gamma_\mu = -2\gamma^\beta\gamma^\nu\gamma^\alpha$ and the two-gamma identity $\gamma^\mu\gamma^\alpha\gamma^\nu\gamma_\mu = 4\eta^{\alpha\nu}$ (both from Theorem C.58), the central block expands as

$$\gamma^\mu(\not{q}+m)\gamma^\nu(\not{p}+m)\gamma_\mu = -2\not{p}\,\gamma^\nu\not{q} + 4m\,(q+p)^\nu\mathbf{1} - 2m^2\gamma^\nu. \tag{18.73}$$

Insert Eq. (18.73) into Eq. (18.72), multiply by $(\not{r}+m)\gamma_\nu$, and trace against $(\not{p}'+m)$. Only terms with an even number of gamma matrices survive; the surviving traces are evaluated with $\gamma^\nu\not{a}\not{b}\gamma_\nu = 4\,a\cdot b$, $\gamma^\nu\not{a}\gamma_\nu = -2\not{a}$, $\gamma^\nu\gamma_\nu = 4$, the

two-gamma trace $\mathrm{Tr}(\not{a}\not{b}) = 4\,a\cdot b$ (Eq. (C.177)), and $\mathrm{Tr}\,\mathbf{1} = 4$. Term by term,

$$T_x = \underbrace{-32\,(q\cdot r)(p'\cdot p)}_{\text{4-slash}} + \underbrace{16m^2\big[(p\cdot q)+(q{+}p)\cdot p'+(q{+}p)\cdot r+(p'\cdot r)\big]}_{\text{2-slash}} \underbrace{-\,32m^4}_{\text{0-slash}}. \tag{18.74}$$

Now substitute the on-shell scalar products. With $q = p + k$ and $r = p - k'$ one finds, using $p^2 = m^2$, $k^2 = k'^2 = 0$, momentum conservation, and the relations $p'\cdot k = p\cdot k' = b$, $p'\cdot k' = p\cdot k = a$, $k\cdot k' = a - b$, $p\cdot p' = m^2 + a - b$ collected in Section 18.3.4:

$$q\cdot r = m^2, \quad p\cdot q = m^2{+}a, \quad (q{+}p)\cdot p' = 2m^2{+}2a{-}b, \quad (q{+}p)\cdot r = 2m^2{-}b, \quad p'\cdot r = m^2{-}b. \tag{18.75}$$

Plugging Eq. (18.75) into Eq. (18.74), the m^4 terms sum to $32m^4$, the $m^2 a$ terms to $16m^2 a$, and the $m^2 b$ terms to $-16m^2 b$, so the eight-gamma trace collapses to the compact

$$T_x = 16\,m^2\big(2m^2 + a - b\big). \tag{18.76}$$

Therefore

$$\sum_{\text{spins}} 2\,\mathrm{Re}(\mathcal{M}_s\mathcal{M}_u^*) = -\frac{e^4}{2ab}\,16m^2(2m^2+a-b) = -8e^4\left(\frac{2m^4}{ab} + \frac{m^2}{b} - \frac{m^2}{a}\right). \tag{18.77}$$

Assembling $\overline{|\mathcal{M}|^2}$. Adding Eqs. (18.70), (18.71) and (18.77) and dividing by the spin/polarization-averaging factor of 4 from Eq. (18.63),

$$\begin{aligned}\overline{|\mathcal{M}|^2} &= \frac{1}{4}\,8e^4\left[\frac{m^4}{a^2} + \frac{m^2}{a} + \frac{b}{a} + \frac{m^4}{b^2} - \frac{m^2}{b} + \frac{a}{b} - \frac{2m^4}{ab} - \frac{m^2}{b} + \frac{m^2}{a}\right] \\ &= 2e^4\left[\frac{b}{a} + \frac{a}{b} + 2m^2\left(\frac{1}{a} - \frac{1}{b}\right) + m^4\left(\frac{1}{a} - \frac{1}{b}\right)^2\right],\end{aligned} \tag{18.78}$$

where the four m^2-pieces combined as $2m^2(1/a - 1/b)$ and the three m^4-pieces formed the perfect square $m^4(1/a - 1/b)^2$. Restoring $a = p\cdot k$, $b = p\cdot k'$:

$$\boxed{\overline{|\mathcal{M}|^2} = 2e^4\left[\frac{p\cdot k'}{p\cdot k} + \frac{p\cdot k}{p\cdot k'} + 2m^2\left(\frac{1}{p\cdot k} - \frac{1}{p\cdot k'}\right) + m^4\left(\frac{1}{p\cdot k} - \frac{1}{p\cdot k'}\right)^2\right].} \tag{18.79}$$

This is the central dynamical result of the section. It is manifestly gauge invariant (it was built with the $-\eta^{\mu\nu}$ rule, whose use we earned in Eq. (18.62)), positive, and symmetric under $k \leftrightarrow -k'$ only up to the sign-changing m^2 term, a structure that crossing symmetry will make sense of in Section 18.4.

18.3.4 Mandelstam form and Compton kinematics

To hand the result to the crossing section we re-express Eq. (18.79) in Mandelstam invariants, and to compare with experiment we specialize to the laboratory frame. Both require the kinematic dictionary, which we now record. Label the legs $p_1 = p$, $p_2 = k$, $p_3 = p'$, $p_4 = k'$ in the conventions of Eqs. (16.118) and (16.120), so

$$s = (p+k)^2 = m^2 + 2\,p\cdot k, \qquad u = (p-k')^2 = m^2 - 2\,p\cdot k', \qquad t = (k-k')^2 = -2\,k\cdot k'. \tag{18.80}$$

Hence $a = p\cdot k = (s-m^2)/2$ and $b = p\cdot k' = (m^2-u)/2$. Momentum conservation $p + k = p' + k'$ together with the on-shell conditions yields the scalar products used above: squaring $p + k = p' + k'$ gives $p \cdot k = p' \cdot k'$, i.e. $p' \cdot k' = a$; squaring $p - k' = p' - k$ gives $p\cdot k' = p'\cdot k$, i.e. $p'\cdot k = b$; from $p\cdot p' = p\cdot(p+k-k') = m^2 + a - b$ and $k \cdot k' = p \cdot k - p \cdot k' = a - b$ the remaining entries follow. Substituting $a = (s - m^2)/2$, $b = (m^2 - u)/2$ into Eq. (18.79) gives the crossing-ready form

$$\overline{|\mathcal{M}|^2} = 2e^4 \left[\frac{m^2 - u}{s - m^2} + \frac{s - m^2}{m^2 - u} + 4m^2 \left(\frac{1}{s - m^2} - \frac{1}{m^2 - u} \right) + 4m^4 \left(\frac{1}{s - m^2} - \frac{1}{m^2 - u} \right)^2 \right]. \tag{18.81}$$

This is the analytic object the next section continues to the annihilation channel.

The lab frame and the Compton shift. Now sit in the rest frame of the incoming electron, $p = (m, \mathbf{0})$, with the incoming photon $k = (\omega, \omega\hat{\mathbf{z}})$ and the outgoing photon $k' = (\omega', \omega'\hat{\mathbf{n}})$ making an angle θ with $\hat{\mathbf{z}}$. Then

$$p \cdot k = m\omega, \qquad p \cdot k' = m\omega', \qquad k \cdot k' = \omega\omega'(1 - \cos\theta). \tag{18.82}$$

The identity $p \cdot k - p \cdot k' = k \cdot k'$ established above becomes $m\omega - m\omega' = \omega\omega'(1 - \cos\theta)$; dividing by $m\omega\omega'$ yields the celebrated Compton shift,

$$\boxed{\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1 - \cos\theta}{m}.} \tag{18.83}$$

The outgoing photon is always softer than the incoming one ($\omega' < \omega$ for $\theta > 0$), by an amount set by the electron Compton wavelength $1/m$ — the kinematic fingerprint that told Compton in 1923 that radiation scatters off electrons as particles, not as a classical wave. Solving Eq. (18.83) for the frequency ratio,

$$\frac{\omega'}{\omega} = \frac{1}{1 + \dfrac{\omega}{m}(1 - \cos\theta)}. \tag{18.84}$$

Inserting Eq. (18.82) into Eq. (18.79), the m^2 and m^4 terms simplify beautifully. From Eq. (18.83), $1/(p\cdot k) - 1/(p\cdot k') = (1/m)(1/\omega - 1/\omega') = -(1-\cos\theta)/m^2$,

so $2m^2(\cdots) = -2(1-\cos\theta)$ and $m^4(\cdots)^2 = (1-\cos\theta)^2$. The kinematic terms give $p\cdot k'/p\cdot k = \omega'/\omega$ and $p\cdot k/p\cdot k' = \omega/\omega'$. Combining,

$$\overline{|\mathcal{M}|^2} = 2e^4\Big[\frac{\omega'}{\omega}+\frac{\omega}{\omega'}-2(1-\cos\theta)+(1-\cos\theta)^2\Big] = 2e^4\Big[\frac{\omega'}{\omega}+\frac{\omega}{\omega'}-\sin^2\theta\Big], \tag{18.85}$$

where the last step used $-2(1-\cos\theta)+(1-\cos\theta)^2 = -(1-\cos\theta)(1+\cos\theta) = -\sin^2\theta$. The lab-frame squared amplitude is thus a strikingly compact function of the frequency ratio and scattering angle.

18.3.5 The lab-frame phase space and the Klein–Nishina formula

The master cross-section formula $\mathrm{d}\sigma = F^{-1}\,\overline{|\mathcal{M}|^2}\,\mathrm{d}\Pi_2$ of Eq. (16.112) now needs the lab flux and the lab two-body phase space. Because the photon is massless, the invariant Møller flux of Theorem 16.16 reduces to

$$F = 4\sqrt{(p\cdot k)^2 - m^2\cdot 0} = 4\,p\cdot k = 4m\omega. \tag{18.86}$$

The two-body measure, before integration, is

$$\mathrm{d}\Pi_2 = \frac{\mathrm{d}^3p'}{(2\pi)^3\,2E'}\,\frac{\mathrm{d}^3k'}{(2\pi)^3\,2\omega'}\,(2\pi)^4\,\delta^{(4)}(p+k-p'-k'). \tag{18.87}$$

We integrate over $\mathbf{p}'$ against the three-momentum delta, fixing $\mathbf{p}' = \mathbf{k}-\mathbf{k}'$ (recall $\mathbf{p}=\mathbf{0}$) so that $E' = \sqrt{m^2+|\mathbf{k}-\mathbf{k}'|^2}$, and write $\mathrm{d}^3k' = \omega'^2\,\mathrm{d}\omega'\,\mathrm{d}\Omega$:

$$\mathrm{d}\Pi_2 = \frac{\omega'\,\mathrm{d}\omega'\,\mathrm{d}\Omega}{(2\pi)^2\,4E'}\,\delta(m+\omega-E'-\omega'). \tag{18.88}$$

The remaining ω' integral is done against the energy delta. Differentiating $E'^2 = m^2+\omega^2+\omega'^2-2\omega\omega'\cos\theta$ gives $E'\,\mathrm{d}E'/\mathrm{d}\omega' = \omega'-\omega\cos\theta$, so

$$\frac{\mathrm{d}}{\mathrm{d}\omega'}(E'+\omega') = \frac{\omega'-\omega\cos\theta}{E'}+1 = \frac{\omega'-\omega\cos\theta+E'}{E'}. \tag{18.89}$$

On the energy shell $E' = m+\omega-\omega'$, so the numerator is $m+\omega-\omega\cos\theta = m+\omega(1-\cos\theta) = m\omega/\omega'$ by Eq. (18.84). The delta function therefore contributes a Jacobian $|\mathrm{d}(E'+\omega')/\mathrm{d}\omega'|^{-1} = E'\omega'/(m\omega)$, and

$$\mathrm{d}\Pi_2 = \frac{\omega'}{(2\pi)^2\,4E'}\cdot\frac{E'\omega'}{m\omega}\,\mathrm{d}\Omega = \frac{1}{16\pi^2}\,\frac{\omega'^2}{m\omega}\,\mathrm{d}\Omega. \tag{18.90}$$

The electron energy E' has cancelled cleanly. Assembling with Eq. (18.86),

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{1}{F}\,\overline{|\mathcal{M}|^2}\,\frac{\mathrm{d}\Pi_2}{\mathrm{d}\Omega} = \frac{1}{4m\omega}\,\overline{|\mathcal{M}|^2}\,\frac{\omega'^2}{16\pi^2 m\omega} = \frac{\omega'^2}{64\pi^2m^2\omega^2}\,\overline{|\mathcal{M}|^2}. \tag{18.91}$$

Inserting the lab squared amplitude Eq. (18.85) and writing $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$,

$$\boxed{\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} = \frac{\alpha^2}{2m^2}\left(\frac{\omega'}{\omega}\right)^2\left[\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2\theta\right],} \tag{18.92}$$

the Klein–Nishina formula (Klein and Nishina, 1929), with ω'/ω given by Eq. (18.84). It is one of the early triumphs of relativistic quantum theory: the full angular and energy dependence of photon–electron scattering, valid at any photon energy, from a single tree-level QED computation.

Physical Insight

The factor $(\omega'/\omega)^2$ in Eq. (18.92) is the phase-space penalty for the recoiling electron carrying away energy: at high ω the outgoing photon is strongly degraded (Eq. (18.84)) and back-scattering is suppressed, so Klein–Nishina scattering is forward-peaked — the physical basis for why high-energy photons in matter, and cosmic-ray photons traversing the microwave background, lose energy preferentially in the forward direction. As $\omega \to 0$ the recoil shuts off, $\omega' \to \omega$, and the distribution becomes the symmetric classical pattern of Eq. (18.93).

Worked Example 18.10 (From the trace to the Thomson cross section: the classical-physics check)**.** We complete the computation by taking the low-energy limit of Eq. (18.92) and integrating, recovering the classical Thomson result — the anchor that ties the entire quantum calculation to a measurable number.

Step 1: the Thomson differential cross section. In the limit $\omega \ll m$ the Compton shift Eq. (18.83) is negligible, so $\omega'/\omega \to 1$ at every angle. The Klein–Nishina formula Eq. (18.92) then becomes

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \xrightarrow{\omega \ll m} \frac{\alpha^2}{2m^2}(1 + 1 - \sin^2\theta) = \frac{\alpha^2}{2m^2}(1 + \cos^2\theta). \tag{18.93}$$

This is exactly the Thomson cross section of classical electromagnetism: an electromagnetic wave shakes the electron, which re-radiates as an oscillating dipole with the $(1 + \cos^2\theta)$ pattern. The amplitude scale is set by the classical electron radius $r_e = e^2/4\pi m = \alpha/m$, since $\alpha^2/2m^2 = \tfrac{1}{2}r_e^2$, so Eq. (18.93) reads $\mathrm{d}\sigma/\mathrm{d}\Omega = \tfrac{1}{2}r_e^2(1 + \cos^2\theta)$.

Step 2: the total Thomson cross section. Integrate Eq. (18.93) over the full solid angle. With $\mathrm{d}\Omega = 2\pi\,\mathrm{d}\cos\theta$ and $x = \cos\theta$,

$$\sigma_T = \frac{\alpha^2}{2m^2}\,2\pi\int_{-1}^{1}(1 + x^2)\,\mathrm{d}x = \frac{\pi\alpha^2}{m^2}\left(2 + \frac{2}{3}\right) = \frac{8\pi\alpha^2}{3m^2} = \frac{8\pi}{3}r_e^2. \tag{18.94}$$

This is the classical Thomson cross section, reproduced with no adjustable parameter from the QED amplitude. That a tree-level Feynman calculation,

built from gamma-matrix traces and the polarization sum, lands precisely on a result derivable from Maxwell's equations and Newton's second law is the cleanest possible sanity check on the entire apparatus of this part of the book.

Step 3: the full Klein–Nishina total cross section. Away from the Thomson limit the frequency ratio depends on the angle, so the integral is genuinely energy dependent. Write $x \equiv \omega/m$ (not to be confused with the integration variable above; we now use $u = \cos\theta$) and $R \equiv \omega'/\omega = [1 + x(1-u)]^{-1}$. From Eq. (18.92) with $\sin^2\theta = 1 - u^2$,

$$\sigma = \frac{\alpha^2}{2m^2}\, 2\pi \int_{-1}^{1} R^2 \Big[R + \frac{1}{R} - (1 - u^2) \Big] \mathrm{d}u = \frac{\pi\alpha^2}{m^2} \int_{-1}^{1} \left[R^3 + R - R^2(1-u^2) \right] \mathrm{d}u. \tag{18.95}$$

Substitute $w = 1 + x(1-u)$, so $u = (1 + x - w)/x$, $\mathrm{d}u = -\mathrm{d}w/x$, and w runs from $1 + 2x$ (at $u = -1$) to 1 (at $u = +1$). Then $\int_{-1}^{1} R^n \,\mathrm{d}u = \frac{1}{x}\int_{1}^{1+2x} w^{-n}\,\mathrm{d}w$, giving

$$\int_{-1}^{1} R\,\mathrm{d}u = \frac{L}{x}, \quad \int_{-1}^{1} R^2\,\mathrm{d}u = \frac{2}{1+2x}, \quad \int_{-1}^{1} R^3\,\mathrm{d}u = \frac{2(1+x)}{(1+2x)^2}, \qquad L \equiv \ln(1{+}2x), \tag{18.96}$$

while $1 - u^2 = (w-1)(1+2x-w)/x^2$ turns the last integral into

$$\int_{-1}^{1} R^2(1-u^2)\,\mathrm{d}u = \frac{1}{x^3}\big[(2+2x)L - 4x\big]. \tag{18.97}$$

Collecting the three pieces in Eq. (18.95),

$$\boxed{\sigma_{\mathrm{KN}} = \frac{\pi\alpha^2}{m^2}\left[\frac{2(1+x)}{(1+2x)^2} + \frac{L}{x} + \frac{4}{x^2} - \frac{2(1+x)L}{x^3} \right], \qquad L = \ln(1+2x), \quad x = \frac{\omega}{m}.} \tag{18.98}$$

This is the total Klein–Nishina cross section; it is equivalent to the standard textbook form $\sigma_{\mathrm{KN}} = 2\pi r_e^2\{\frac{1+x}{x^2}[\frac{2(1+x)}{1+2x} - \frac{L}{x}] + \frac{L}{2x} - \frac{1+3x}{(1+2x)^2}\}$ after using $r_e^2 = \alpha^2/m^2$.

Step 4: the low-energy limit, again. Expanding Eq. (18.98) for $x \ll 1$ with $L = 2x - 2x^2 + \frac{8}{3}x^3 - \cdots$, the $4/x^2$ in the third term is cancelled by the $-2(1+x)L/x^3$ in the fourth, and the surviving constant collects to $8/3$:

$$\sigma_{\mathrm{KN}} = \frac{8\pi\alpha^2}{3m^2}\Big(1 - 2x + \tfrac{26}{5}x^2 - \cdots\Big) = \sigma_T\Big(1 - \frac{2\omega}{m} + \cdots\Big). \tag{18.99}$$

The leading term is precisely σ_T of Eq. (18.94), confirming Eq. (18.98) against the classical limit, and the first correction $-2\omega/m$ is the familiar reduction of the cross section as the photon energy begins to probe the electron's recoil. The high-energy behaviour follows from the logarithm: for $x \gg 1$, $\sigma_{\mathrm{KN}} \simeq (\pi\alpha^2/m^2)(\ln 2x + \frac{1}{2})/x$, falling like $(\ln\omega)/\omega$ — matter becomes increasingly transparent to very energetic photons.

We have carried Compton scattering from its two diagrams to a fully closed-form cross section, checked gauge invariance on the concrete amplitude, and recovered classical physics in the soft limit. The decisive result for what follows is the analytic squared amplitude in Mandelstam form, Eq. (18.81): a single function of s and u, manifestly meromorphic in those invariants. The next section treats that function as the boundary value of one analytic object and continues it — by the crossing substitution $s \leftrightarrow$ the annihilation channel's energy invariant — directly into the amplitude for $e^+e^- \to \gamma\gamma$, obtaining a second physical process with no new trace algebra at all.

18.4 Crossing symmetry and pair annihilation

The previous section ended with a single rational function: the spin- and polarization-averaged squared amplitude $\overline{|\mathcal{M}|^2}_{\mathrm{C}}(s,u)$ for Compton scattering $e^-\gamma \to e^-\gamma$, assembled from the two diagrams of Section 18.3.1, squared with the polarization-sum rule and the gamma-matrix traces, and reduced (Eq. (18.81)) to the Klein–Nishina form. That function was computed to describe one experiment: a photon scattering off an electron. We now extract a *second* experiment from it without squaring a single new diagram. The observation is the one that already organized Chapter 17: the Mandelstam invariants s, t, u are not bound to any particular partition of the legs into "incoming" and "outgoing," and the amplitude is the boundary value of a single analytic function of them; moving a leg from one side of the reaction to the other—turning an incoming particle into an outgoing antiparticle—merely samples that same function in a different kinematic window. Applied to Compton scattering this maneuver lands on electron–positron pair annihilation, $e^+e^- \to \gamma\gamma$. The dynamics are already in hand; what remains is to make the continuation precise and to track the bookkeeping it changes.

Crossing was introduced and proved at tree level in Section 17.4, where we carried the continuation through the Dirac traces of $e^+e^- \to \mu^+\mu^-$ to obtain $e^-\mu^- \to e^-\mu^-$. That treatment crossed two fermion legs and no bosons, and it is the backbone we build on here rather than reprove. The present process is genuinely new in two respects that Chapter 17 did not exercise, and these are exactly what this section derives in full. First, one of the crossed legs is a *photon*: we must extend the crossing dictionary to an external gauge boson, including what happens to its polarization vector and to the polarization sum. Second, Compton and annihilation are related by crossing an *odd* number of fermion lines—precisely one—so the fermion crossing sign, which cancelled in the even case of Chapter 17, now survives as a physically decisive overall -1. We treat both, lay out the Mandelstam-plane geometry the continuation traverses, and then harvest $\overline{|\mathcal{M}|^2}$ and the cross section for $e^+e^- \to \gamma\gamma$, including the identical-photon phase-space factor.

18.4.1 The crossing dictionary, completed for external photons

We first recall the principle in the form fixed in Chapter 17, then add the entry for a photon leg. The origin of crossing is the analytic structure exposed by the LSZ reduction formula (Theorem 14.31): an S-matrix element is the multiple on-shell residue of a time-ordered correlator, and that correlator, as a function of the external four-momenta, is the boundary value of a function holomorphic in the complexified invariants away from the cuts and poles that unitarity demands at physical thresholds (Section 5.1, Section 5.4.1). At tree level there are no cuts at all—the amplitude is rational, with only the poles of the internal propagators—so continuation between physical regions is unambiguous and we may execute it by direct substitution. The general rule is Theorem 17.13: an incoming particle of species X and momentum p is exchanged for an outgoing antiparticle $\bar{X}$ of momentum p' by

$$p \;\longrightarrow\; -p', \qquad X \;\longleftrightarrow\; \bar{X}, \tag{18.100}$$

continued from the physical region of the original process to that of the crossed one, and conversely for an outgoing particle crossed to an incoming antiparticle. The momentum reversal is forced by the conservation delta function: a leg moved across the reaction enters the all-incoming momentum balance with the opposite sign.

For a *fermion* the continuation acts on the external wavefunctions through $u(p) \leftrightarrow v(-p')$ and $\bar{u}(p) \leftrightarrow \bar{v}(-p')$, as established in Theorem 17.13 from the Dirac equations: replacing $p \to -p$ in the antiparticle equation $(\not{p} + m)v(p) = 0$ gives $(\not{p} - m)v(-p) = 0$, the very equation obeyed by $u(p)$, so the two solution spaces are identified under momentum reversal. Working with the spin-summed completeness relations (Eq. (11.73))

$$\sum_s u^s(p)\,\bar{u}^s(p) = \not{p} + m, \qquad \sum_s v^s(p)\,\bar{v}^s(p) = \not{p} - m, \tag{18.101}$$

the crossing of a fermion leg is felt entirely through the relative sign of the mass term: continuing the particle projector $\not{p} + m$ to $p \to -p'$ produces $-\not{p}' + m = -(\not{p}' - m)$, i.e. (-1) times the antiparticle projector $\sum_s v^s(p')\bar{v}^s(p')$. This single sign, recorded per crossed fermion leg, is the source of the fermion crossing sign analyzed in Section 18.4.2.

The entry we must add is for a crossed *photon*. The photon is its own antiparticle, so the species swap $X \leftrightarrow \bar{X}$ is trivial; what we need is the rule for its polarization data. The relevant field-theory fact is again LSZ: the operator that annihilates an incoming photon of momentum k and helicity λ is the same free field A_μ whose positive-frequency part creates an outgoing photon, and the external-leg factor LSZ attaches is the polarization vector—$\epsilon^\mu(k,\lambda)$ for an incoming photon, $\epsilon^{*\mu}(k,\lambda)$ for an outgoing one (Section 17.1.4). Crossing the incoming photon to an outgoing one therefore replaces

$$k \;\longrightarrow\; -k', \qquad \epsilon^\mu(k,\lambda) \;\longrightarrow\; \epsilon^{*\mu}(k',\lambda), \tag{18.102}$$

the momentum reversal exactly as in Eq. (18.100) and the wavefunction passing from ϵ to its conjugate ϵ^*. The complex conjugation is the bosonic analogue of $u \leftrightarrow v$: it is the substitution that turns the annihilation wavefunction of an incoming quantum into the creation wavefunction of an outgoing one.

What matters for an unpolarized squared amplitude is not the individual polarization vectors but the polarization *sum*, and here the photon crossing is especially clean. For a gauge-invariant amplitude the physical-polarization sum may be performed with the covariant replacement (Theorem 17.10)

$$\sum_{\lambda=1,2} \epsilon_\mu(k,\lambda)\,\epsilon_\nu^*(k,\lambda) \;\longrightarrow\; -\eta_{\mu\nu}, \tag{18.103}$$

licensed by the tree-level Ward identity $k_\mu \mathcal{M}^\mu = 0$ established for these processes in Section 18.2.3. The right-hand side $-\eta_{\mu\nu}$ is a constant tensor: it carries no dependence on the photon momentum and no factor of i, and it is identical whether the photon is incoming or outgoing. Consequently the polarization sum of a crossed photon leg continues to itself with *no sign and no momentum substitution*—the entire content of crossing a photon, at the level of $\sum_{\text{pol}} |\mathcal{M}|^2$, is the momentum reversal $k \to -k'$ inside the kinematic factors. A boson leg contributes no crossing sign, in sharp contrast to the (-1) a fermion leg carries.

Caution

The replacement Eq. (18.103) is legitimate only because the amplitude obeys the Ward identity; the longitudinal and timelike pieces of the full $\sum_\lambda \epsilon\epsilon^*$ are discarded against $k_\mu \mathcal{M}^\mu = 0$. Crossing does not endanger this: gauge invariance is a property of the analytic amplitude, so if $k_\mu \mathcal{M}^\mu = 0$ holds in the Compton region it holds throughout, and the crossed annihilation amplitude is automatically transverse in both photon momenta. We will not re-establish the Ward identity here; it was proved for the external-photon amplitudes of this chapter in Section 18.2.3 and is promoted to the all-orders Ward–Takahashi identity in Chapter 19.

Collecting the entries, the complete crossing dictionary for QED tree amplitudes reads: an incoming particle becomes an outgoing antiparticle (and conversely) by $p \to -p'$; a crossed fermion leg replaces $u \leftrightarrow v$, equivalently flips the mass-term sign of its spin projector and contributes one factor (-1) to $\sum_{\text{spins}} |\mathcal{M}|^2$; a crossed photon leg replaces $\epsilon \leftrightarrow \epsilon^*$ and contributes no sign; and the spin-averaging prefactor $1/N_{\text{in}}$ is detached before continuing and recomputed from the initial state of the crossed process (Section 17.4.3).

18.4.2 The fermion crossing sign for an odd number of crossed lines

In Chapter 17 the relevant lesson was distilled as: *the squared amplitude carries a factor* (-1) *per crossed fermion line.* There two fermion lines were crossed, so the net sign was $(-1)^2 = +1$ and the continuation of $\sum_{\text{spins}} |\mathcal{M}|^2$ proceeded with no overall sign. Compton-to-annihilation is the complementary case. We will see in Section 18.4.3 that turning $e^-\gamma \to e^-\gamma$ into $e^+e^- \to \gamma\gamma$ crosses exactly one fermion leg (the outgoing electron, into an incoming positron) together with one photon leg (the incoming photon, into an outgoing photon). An odd number of fermion legs is crossed, and the net sign is $(-1)^1 = -1$, which now does not cancel. This sign is not a cosmetic bookkeeping artifact: it is exactly what renders the continued expression a positive squared amplitude, and getting it wrong produces a manifestly negative "cross section."

The derivation is the projector sign of Section 18.4.1, applied inside the trace. The Compton squared amplitude, before any averaging, has the schematic single-fermion-line structure

$$\sum_{\text{spins,pol}} |\mathcal{M}_{\mathrm{C}}|^2 \;=\; \mathrm{tr}\big[(\not{p}_1 + m)\,\Gamma\,(\not{p}_2 + m)\,\overline{\Gamma}\big], \tag{18.104}$$

where p_1 is the incoming electron, p_2 the outgoing electron (both particle projectors $\not{p} + m$ by Eq. (18.17)), and Γ is the string of vertices, internal electron propagators and photon polarization sums—a function of the photon momenta and of m, carrying the two $-\eta_{\mu\nu}$ factors of Eq. (18.103). The annihilation squared amplitude has the same fermion line read the other way: it runs from the incoming electron p_1 to the incoming positron p_2, so its trace is

$$\sum_{\text{spins,pol}} |\mathcal{M}_{\mathrm{A}}|^2 \;=\; \mathrm{tr}\big[(\not{p}_1 + m)\,\Gamma'\,(\not{p}_2 - m)\,\overline{\Gamma'}\big], \tag{18.105}$$

the positron leg now carrying the *antiparticle* projector $\not{p}_2 - m$ from Eq. (18.17), and Γ' being Γ with the crossed photon momentum reversed (the polarization sums are untouched, by Section 18.4.1). Now continue Eq. (18.104) by the dictionary: the incoming-electron leg p_1 is uncrossed and unchanged; the crossed photon sends $\Gamma \to \Gamma'$ with no sign; and the crossed outgoing-electron leg sends $p_2 \to -p_2$, so its projector becomes

$$\not{p}_2 + m \;\xrightarrow{\;p_2 \to -p_2\;}\; -\not{p}_2 + m \;=\; -(\not{p}_2 - m). \tag{18.106}$$

Pulling the single overall -1 out of the trace,

$$\Big[\sum_{\text{spins,pol}} |\mathcal{M}_{\mathrm{C}}|^2\Big]_{\text{continued}} = \mathrm{tr}\big[(\not{p}_1{+}m)\,\Gamma'\,(-(\not{p}_2{-}m))\,\overline{\Gamma'}\big] = -\,\mathrm{tr}\big[(\not{p}_1{+}m)\,\Gamma'\,(\not{p}_2{-}m)\,\overline{\Gamma'}\big] = -\,\mathrm{s}$$

(18.107)

The continued Compton trace equals (-1) times the genuine annihilation trace. Equivalently, the analytic continuation of the Compton spin-summed square gives the annihilation spin-summed square only after we restore the factor $(-1)^{n_f}$, with $n_f = 1$ the number of crossed fermion legs:

$$\boxed{\sum_{\text{spins,pol}} |\mathcal{M}_{\mathrm{A}}|^2 \;=\; (-1)^{n_f} \Big[\sum_{\text{spins,pol}} |\mathcal{M}_{\mathrm{C}}|^2 \Big]_{\text{continued}}, \qquad n_f = 1.} \tag{18.108}$$

This is the fermion crossing sign *derived*, not asserted: it is the product of the single projector flip Eq. (18.106) forced by $p \to -p'$ on a crossed fermion leg, with the photon leg contributing nothing. The same counting reproduces Chapter 17: there two fermion legs were crossed, $n_f = 2$, and $(-1)^2 = +1$. The rule $(-1)^{n_f}$ is the general statement; the cancellation seen earlier and the survival seen here are its even and odd instances.

Physical Insight

The crossing sign is harmless for a single squared diagram—it multiplies a positive quantity by ± 1 and is fixed by demanding the result be positive—but it becomes a measurable relative phase the moment two crossed diagrams interfere. Møller and Bhabha scattering in Section 18.5 are exactly that situation: the relative minus sign between their two diagrams is the same fermionic sign, now visible in an interference term rather than hidden inside an overall factor. The Compton-to-annihilation continuation is where we isolate the sign in its simplest guise, so that when it returns in the interference of the next section its origin is already settled.

18.4.3 Compton and annihilation on the Mandelstam plane

We now fix the momenta and read off how crossing acts on the invariants. Label Compton scattering as

$$\text{(C)} \qquad e^-(p_1)\,\gamma(k_1) \;\longrightarrow\; e^-(p_2)\,\gamma(k_2), \tag{18.109}$$

with the Mandelstam invariants of Theorem 16.20

$$s_{\mathrm{C}} = (p_1 + k_1)^2, \qquad t_{\mathrm{C}} = (p_1 - p_2)^2, \qquad u_{\mathrm{C}} = (p_1 - k_2)^2, \tag{18.110}$$

and label pair annihilation as

$$\text{(A)} \qquad e^-(p_1)\,e^+(p_2) \;\longrightarrow\; \gamma(k_1)\,\gamma(k_2), \tag{18.111}$$

with

$$s_{\mathrm{A}} = (p_1 + p_2)^2, \qquad t_{\mathrm{A}} = (p_1 - k_1)^2, \qquad u_{\mathrm{A}} = (p_1 - k_2)^2. \tag{18.112}$$

Both processes have external masses $(m, 0, m, 0)$, so both obey the same constraint (Eq. (16.121))

$$s + t + u = 2m^2. \tag{18.113}$$

To pass from (C) to (A) we cross two legs: the incoming photon $\gamma(k_1)$ of (C) becomes the outgoing photon $\gamma(k_1)$ of (A), and the outgoing electron $e^-(p_2)$ of (C) becomes the incoming positron $e^+(p_2)$ of (A); the incoming electron and the remaining photon are renamed directly. By Eq. (18.100) and Eq. (18.102) this is the momentum dictionary (all symbols on the right now denoting the annihilation momenta of Eq. (18.111))

$$p_1^{\mathrm{C}} = p_1, \qquad k_1^{\mathrm{C}} = -k_1, \qquad p_2^{\mathrm{C}} = -p_2, \qquad k_2^{\mathrm{C}} = k_2. \tag{18.114}$$

Exactly one fermion leg (p_2) and one boson leg (k_1) carry the minus sign, confirming $n_f = 1$ in Eq. (18.108). Substituting Eq. (18.114) into the Compton invariants Eq. (18.110) and reading them as annihilation invariants Eq. (18.112),

$$s_{\mathrm{C}} = (p_1^{\mathrm{C}} + k_1^{\mathrm{C}})^2 = (p_1 - k_1)^2 = t_{\mathrm{A}}, \tag{18.115}$$

$$u_{\mathrm{C}} = (p_1^{\mathrm{C}} - k_2^{\mathrm{C}})^2 = (p_1 - k_2)^2 = u_{\mathrm{A}}, \tag{18.116}$$

$$t_{\mathrm{C}} = (p_1^{\mathrm{C}} - p_2^{\mathrm{C}})^2 = (p_1 + p_2)^2 = s_{\mathrm{A}}. \tag{18.117}$$

The crossing therefore acts on the invariants as the swap

$$s_{\mathrm{C}} \longrightarrow t_{\mathrm{A}}, \qquad t_{\mathrm{C}} \longrightarrow s_{\mathrm{A}}, \qquad u_{\mathrm{C}} \longrightarrow u_{\mathrm{A}}, \tag{18.118}$$

that is, $s \leftrightarrow t$ with u inert. This is the precise content of the slogan "annihilation is Compton continued by $s \leftrightarrow t$." The photon that carried the Compton center-of-mass energy s_{C} becomes the photon exchanged in the annihilation t-channel, and the electron-exchange channel of Compton (the variable t_{C}, or rather the propagators living in s_{C} and u_{C}) reorganizes: the two Compton diagrams, with their internal electron propagators in the s- and u-channels, become the two annihilation diagrams with internal electron propagators in the t- and u-channels.

The shared analytic plane. Because (C) and (A) obey the same constraint Eq. (18.113), they live on the *same* Mandelstam plane—the two-dimensional surface $s + t + u = 2m^2$ introduced in Section 16.5.5 and drawn there for the equal-mass case (Fig. 16.5). Here the masses are unequal, $(m, 0, m, 0)$, so the physical wedges are not symmetric, but the principle is identical: a single analytic function $\mathcal{M}(s, t, u)$ is defined on the whole plane, and Compton and annihilation occupy disjoint physical regions of it. The boundaries are fixed by requiring the scattering angle to be real. For Compton, with center-of-mass energy $s_{\mathrm{C}} \geq m^2$, one finds (the two-body endpoints of Section 16.5.3 specialized to masses $(m, 0, m, 0)$) the physical region

$$\text{Compton:} \qquad t \leq 0 \quad \text{and} \quad su \leq m^4, \tag{18.119}$$

which comprises both the s-channel wedge ($s \geq m^2$) and the u-channel wedge ($u \geq m^2$)—these are the same physical reaction, since interchanging the two photons is the symmetry $s \leftrightarrow u$. For annihilation the center-of-mass energy is $s_{\mathrm{A}} = t_{\mathrm{C}}$, physical above the pair threshold $s_{\mathrm{A}} \geq 4m^2$, and the same angle-reality condition gives

$$\text{annihilation:} \qquad t \geq 4m^2 \quad \text{and} \quad su \geq m^4. \tag{18.120}$$

The hyperbola $su = m^4$ and the lines $t = 0$, $t = 4m^2$ are the boundaries; the two physical regions are the disjoint wedges of Fig. 18.4. No real point belongs to both, and the bridge between them is the analytic continuation Eq. (18.118), legitimate at tree level along any path avoiding the propagator poles.

Averaging factors. The averaging prefactor must be recomputed for (A), per the rule of Section 17.4.3: the analytic object that continues is the fully spin- and polarization-summed $\sum |\mathcal{M}|^2$, and the factor $1/N_{\text{in}}$ is reattached from the crossed initial state. Compton has an incoming electron (two spin states) and an incoming photon (two polarizations), so $N_{\text{in}}^{\mathrm{C}} = 2 \times 2 = 4$ and $\overline{|\mathcal{M}|^2}_{\mathrm{C}} = \frac{1}{4} \sum |\mathcal{M}_{\mathrm{C}}|^2$. Annihilation has an incoming electron and an incoming positron, two spin states each, so $N_{\text{in}}^{\mathrm{A}} = 2 \times 2 = 4$ and $\overline{|\mathcal{M}|^2}_{\mathrm{A}} = \frac{1}{4} \sum |\mathcal{M}_{\mathrm{A}}|^2$. The two averaging factors coincide—but, as in Chapter 17, for a reason, not by accident: both initial states carry four configurations. (Had we crossed instead to a state with a different initial multiplicity, the factor would change.) Combining the equal averaging factors with Eq. (18.108) for $n_f = 1$,

$$\overline{|\mathcal{M}|^2}_{\mathrm{A}}(t_{\mathrm{A}}, u_{\mathrm{A}}) \;=\; -\,\overline{|\mathcal{M}|^2}_{\mathrm{C}}(s_{\mathrm{C}} \to t_{\mathrm{A}},\, u_{\mathrm{C}} \to u_{\mathrm{A}}). \tag{18.121}$$

The overall minus is the fermion crossing sign Eq. (18.108); the averaging factors have cancelled. We now evaluate the right-hand side.

18.4.4 $e^+e^- \to \gamma\gamma$ from Klein–Nishina

Worked Example 18.11 (The annihilation squared amplitude by continuation of Klein–Nishina)**.** We carry Eq. (18.121) through to a closed expression. The starting point is the spin- and polarization-averaged Compton result of Section 18.3, which in terms of the invariant electron–photon products $p_1 \cdot k_1$ and $p_1 \cdot k_2$ reads (Eq. (18.79))

$$\overline{|\mathcal{M}|^2}_{\mathrm{C}} = 2e^4 \left[\frac{p_1 \cdot k_2}{p_1 \cdot k_1} + \frac{p_1 \cdot k_1}{p_1 \cdot k_2} + 2m^2 \left(\frac{1}{p_1 \cdot k_1} - \frac{1}{p_1 \cdot k_2} \right) + m^4 \left(\frac{1}{p_1 \cdot k_1} - \frac{1}{p_1 \cdot k_2} \right)^2 \right], \tag{18.122}$$

where here p_1 is the incoming electron, k_1 the incoming photon, k_2 the outgoing photon of Eq. (18.109). In Compton kinematics $s_{\mathrm{C}} - m^2 = 2\, p_1 \cdot k_1$ and $m^2 - u_{\mathrm{C}} =$

$2\,p_1\!\cdot\!k_2$, so

$$\overline{|\mathcal{M}|^2}_{\mathrm{C}} = 2e^4\left[\frac{m^2-u_{\mathrm{C}}}{s_{\mathrm{C}}-m^2}+\frac{s_{\mathrm{C}}-m^2}{m^2-u_{\mathrm{C}}}+4m^2\left(\frac{1}{s_{\mathrm{C}}-m^2}-\frac{1}{m^2-u_{\mathrm{C}}}\right)+4m^4\left(\frac{1}{s_{\mathrm{C}}-m^2}-\frac{1}{m^2-\right.\right. \tag{18.123}$$

Step 1: substitute the invariants. Apply $s_{\mathrm{C}}\to t_{\mathrm{A}}$, $u_{\mathrm{C}}\to u_{\mathrm{A}}$ from Eq. (18.118), and write the annihilation photon products from Eq. (18.112): $t_{\mathrm{A}}=(p_1-k_1)^2=m^2-2\,p_1\!\cdot\!k_1$ and $u_{\mathrm{A}}=(p_1-k_2)^2=m^2-2\,p_1\!\cdot\!k_2$, so now

$$t_{\mathrm{A}}-m^2=-2\,p_1\!\cdot\!k_1,\qquad m^2-u_{\mathrm{A}}=2\,p_1\!\cdot\!k_2. \tag{18.124}$$

The four terms of Eq. (18.123) become, with these signs,

$$\frac{m^2-u_{\mathrm{A}}}{t_{\mathrm{A}}-m^2}=\frac{2\,p_1\!\cdot\!k_2}{-2\,p_1\!\cdot\!k_1}=-\frac{p_1\!\cdot\!k_2}{p_1\!\cdot\!k_1},\qquad \frac{t_{\mathrm{A}}-m^2}{m^2-u_{\mathrm{A}}}=-\frac{p_1\!\cdot\!k_1}{p_1\!\cdot\!k_2}, \tag{18.125}$$

$$4m^2\left(\frac{1}{t_{\mathrm{A}}-m^2}-\frac{1}{m^2-u_{\mathrm{A}}}\right)=4m^2\left(\frac{1}{-2\,p_1\!\cdot\!k_1}-\frac{1}{2\,p_1\!\cdot\!k_2}\right)=-2m^2\left(\frac{1}{p_1\!\cdot\!k_1}+\frac{1}{p_1\!\cdot\!k_2}\right), \tag{18.126}$$

$$4m^4\left(\frac{1}{t_{\mathrm{A}}-m^2}-\frac{1}{m^2-u_{\mathrm{A}}}\right)^2=4m^4\cdot\frac{1}{4}\left(\frac{1}{p_1\!\cdot\!k_1}+\frac{1}{p_1\!\cdot\!k_2}\right)^2=m^4\left(\frac{1}{p_1\!\cdot\!k_1}+\frac{1}{p_1\!\cdot\!k_2}\right)^2. \tag{18.127}$$

Notice that the first two terms have flipped sign and the difference $1/(p\cdot k_1)-1/(p\cdot k_2)$ of Compton has become the *sum* $1/(p\cdot k_1)+1/(p\cdot k_2)$—the kinematic signature that the two photons are now both in the final state and symmetric under exchange.

Step 2: apply the crossing sign. By Eq. (18.121) the annihilation amplitude is (-1) times the continued Compton expression. Multiplying the bracket built from Eq. (18.125)–Eq. (18.127) by -1 flips every sign back, yielding a manifestly positive result:

$$\boxed{\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma}=2e^4\left[\frac{p_1\!\cdot\!k_2}{p_1\!\cdot\!k_1}+\frac{p_1\!\cdot\!k_1}{p_1\!\cdot\!k_2}+2m^2\left(\frac{1}{p_1\!\cdot\!k_1}+\frac{1}{p_1\!\cdot\!k_2}\right)-m^4\left(\frac{1}{p_1\!\cdot\!k_1}+\frac{1}{p_1\!\cdot\!k_2}\right)^2\right].} \tag{18.128}$$

Had we omitted the fermion crossing sign, the leading two terms would have entered with an overall minus and the expression would be negative at high energy, where they dominate—an immediate flag that the $(-1)^{n_f}$ of Eq. (18.108) is mandatory, not optional. The result Eq. (18.128) is symmetric under $k_1\leftrightarrow k_2$, as it must be for two identical final-state photons.

Step 3: Mandelstam form. Using $p_1\!\cdot\!k_1=(m^2-t_{\mathrm{A}})/2$ and $p_1\!\cdot\!k_2=(m^2-u_{\mathrm{A}})/2$ from Eq. (18.124) (now read forward), and dropping the subscript A,

$$\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma}=2e^4\left[\frac{m^2-u}{m^2-t}+\frac{m^2-t}{m^2-u}+4m^2\left(\frac{1}{m^2-t}+\frac{1}{m^2-u}\right)-4m^4\left(\frac{1}{m^2-t}+\frac{1}{m^2-\right.\right. \tag{18.129}$$

manifestly symmetric under $t \leftrightarrow u$ (the photon interchange) and finite throughout the physical region Eq. (18.120), where $m^2 - t < 0$ and $m^2 - u < 0$. This is the central dynamical result for pair annihilation, obtained from the Klein–Nishina amplitude by a substitution and a sign rather than by squaring the two annihilation diagrams afresh.

It is worth recording the compact algebraic form that the cross-section integral will use. Introduce

$$a \equiv m^2 - t = 2\,p_1\!\cdot\!k_1, \qquad b \equiv m^2 - u = 2\,p_1\!\cdot\!k_2, \tag{18.130}$$

both positive in the physical region. The constraint Eq. (18.113) gives $a + b = 2m^2 - t - u = 2m^2 - (2m^2 - s) = s$, the clean identity

$$a + b = s. \tag{18.131}$$

Then $\frac{b}{a} + \frac{a}{b} = \frac{a^2+b^2}{ab} = \frac{s^2-2ab}{ab}$, $\frac{1}{a} + \frac{1}{b} = \frac{s}{ab}$, and $(\frac{1}{a} + \frac{1}{b})^2 = \frac{s^2}{a^2b^2}$, so Eq. (18.129) collapses to

$$\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} = 2e^4 \left[\frac{s^2 - 2ab + 4m^2 s}{ab} - \frac{4m^4 s^2}{a^2 b^2} \right]. \tag{18.132}$$

Worked Example 18.12 (Two checks on the annihilation amplitude)**.** Before assembling the cross section we confirm Eq. (18.129) against two independent expectations.

The massless limit and the photon-exchange poles. Setting $m \to 0$ (energies far above the electron mass), Eq. (18.129) reduces to

$$\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma} \xrightarrow{m\to 0} 2e^4 \left(\frac{u}{t} + \frac{t}{u} \right) = 2e^4\, \frac{t^2 + u^2}{tu}. \tag{18.133}$$

The denominators t and u are the squared momenta of the exchanged electron in the two annihilation diagrams (the propagator $1/(\not{p}_1 - \not{k}_1 - m)$ has denominator $t - m^2 \to t$, and likewise u for the crossed diagram), so the amplitude diverges as $t \to 0$ or $u \to 0$. These are the forward and backward directions, in which a photon is emitted collinear with the incoming electron or positron and the exchanged electron goes nearly on shell. The $1/t$ and $1/u$ singularities are the annihilation counterparts of the Coulomb forward pole met in Section 17.4.5; they are regulated by the electron mass, $m^2 - t \geq m^2$, which is why the collinear region is logarithmically, not power, sensitive to m—a fact that controls the total cross section below.

Reduction to nonrelativistic positronium-like kinematics. At threshold $s \to 4m^2$ the electron and positron are nearly at rest; with $a = b \to s/2 = 2m^2$ the bracket Eq. (18.132) becomes $\frac{s^2 - 2(s/2)^2 + 4m^2 s}{(s/2)^2} - \frac{4m^4 s^2}{(s/2)^4} = \frac{s^2/2 + 4m^2 s}{s^2/4} - \frac{4m^4 s^2}{s^4/16} = 2 + \frac{16m^2}{s} - \frac{64m^4}{s^2}$. At $s = 4m^2$ this is $2 + 4 - 4 = 2$, so $\overline{|\mathcal{M}|^2} \to 4e^4$, a finite nonzero constant. Annihilation at rest proceeds through an S-wave with a nonvanishing amplitude; the $1/\beta$ enhancement of the cross section near threshold, derived next, comes entirely from the flux and phase-space factors, not from the dynamics.

18.4.5 The cross section and the identical-photon factor

Worked Example 18.13 (Differential and total cross section for $e^+e^- \to \gamma\gamma$)**.** We fold Eq. (18.129) into the two-body kinematics of Chapter 16, taking care of the one feature absent from every process so far in this chapter: the two final-state particles are identical.

Kinematics. In the center-of-mass frame the incoming electron and positron have energy $\sqrt{s}/2$ and momentum

$$|\mathbf{p}_i| = \tfrac{1}{2}\sqrt{s - 4m^2} = \tfrac{\sqrt{s}}{2}\,\beta, \qquad \beta \equiv \sqrt{1 - \tfrac{4m^2}{s}}, \tag{18.134}$$

with β the electron velocity. The outgoing photons are massless, each with energy $\sqrt{s}/2$, so $|\mathbf{p}_f| = \sqrt{s}/2$ and the phase-space ratio of Theorem 16.23 is

$$\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|} = \frac{\sqrt{s}/2}{\frac{\sqrt{s}}{2}\beta} = \frac{1}{\beta}. \tag{18.135}$$

The photon momentum transfer, with $E_1 = E_\gamma = \sqrt{s}/2$, is $t = (p_1 - k_1)^2 = m^2 - 2E_1E_\gamma + 2\,|\mathbf{p}_i|\,|\mathbf{p}_f|\cos\theta = m^2 - \frac{s}{2} + \frac{s}{2}\beta\cos\theta$, so in terms of $a = m^2 - t$ and $b = m^2 - u$ of Eq. (18.130),

$$a = \frac{s}{2}(1 - \beta\cos\theta), \qquad b = \frac{s}{2}(1 + \beta\cos\theta), \qquad ab = \frac{s^2}{4}(1 - \beta^2\cos^2\theta). \tag{18.136}$$

The identical-particle factor. For distinct final particles the CM differential cross section is Eq. (16.145). With two *identical* photons, a configuration with photon momenta $(\mathbf{k}, -\mathbf{k})$ is the same final state as $(-\mathbf{k}, \mathbf{k})$: integrating $d\sigma/d\Omega$ over the full solid angle counts every distinct two-photon state twice. The total cross section is therefore

$$\sigma = \frac{1}{2}\int_{4\pi} \frac{d\sigma}{d\Omega}\, d\Omega, \tag{18.137}$$

the factor $\frac{1}{2}$ being a phase-space symmetry factor, entirely distinct from the spin average. (Equivalently one integrates over a hemisphere, $\theta \in [0, \pi/2]$, without the $\frac{1}{2}$; the two prescriptions agree because Eq. (18.129) is symmetric under $\theta \to \pi - \theta$, i.e. $t \leftrightarrow u$.) The differential cross section itself, before the $\frac{1}{2}$, is

$$\frac{d\sigma}{d\Omega} = \frac{1}{64\pi^2 s}\,\frac{|\mathbf{p}_f|}{|\mathbf{p}_i|}\,\overline{|\mathcal{M}|^2} = \frac{1}{64\pi^2 s\,\beta}\,\overline{|\mathcal{M}|^2}_{e^+e^-\to\gamma\gamma}. \tag{18.138}$$

High-energy differential cross section. At $s \gg m^2$ ($\beta \to 1$), Eq. (18.133) with $t = -\frac{s}{2}(1-\cos\theta)$, $u = -\frac{s}{2}(1+\cos\theta)$ gives $\frac{t}{u} + \frac{u}{t} = \frac{(1-\cos\theta)^2 + (1+\cos\theta)^2}{(1-\cos\theta)(1+\cos\theta)} = \frac{2(1+\cos^2\theta)}{1-\cos^2\theta}$, so $\overline{|\mathcal{M}|^2} \to 4e^4\,\frac{1+\cos^2\theta}{\sin^2\theta}$ and

$$\frac{d\sigma}{d\Omega} \xrightarrow{s \gg m^2} \frac{4e^4}{64\pi^2 s}\,\frac{1+\cos^2\theta}{\sin^2\theta} = \frac{\alpha^2}{2s}\,\frac{1+\cos^2\theta}{\sin^2\theta}, \tag{18.139}$$

using $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$. The angular distribution peaks sharply in the forward and backward directions, the $1/\sin^2\theta$ rise being the collinear singularity of Theorem 18.12 regulated, in the full massive amplitude, at $\sin^2\theta \sim 4m^2/s$.

Total cross section. We integrate the exact massive amplitude. Insert Eq. (18.132) with ab from Eq. (18.136) into Eq. (18.138), write $x = \cos\theta$, and use $4m^2 = s(1-\beta^2)$ to express the mass terms through β. The bracket of Eq. (18.132) becomes a function of x alone:

$$\frac{s^2 - 2ab + 4m^2 s}{ab} = \frac{s^2 + s^2(1-\beta^2) - \frac{s^2}{2}(1-\beta^2 x^2)}{\frac{s^2}{4}(1-\beta^2 x^2)} = \frac{6 - 4\beta^2 + 2\beta^2 x^2}{1-\beta^2 x^2}, \tag{18.140}$$

$$\frac{4m^4 s^2}{a^2 b^2} = \frac{\frac{s^4}{4}(1-\beta^2)^2}{\frac{s^4}{16}(1-\beta^2 x^2)^2} = \frac{4(1-\beta^2)^2}{(1-\beta^2 x^2)^2}, \tag{18.141}$$

so that

$$\overline{|\mathcal{M}|^2} = 2e^4\left[\frac{6 - 4\beta^2 + 2\beta^2 x^2}{1-\beta^2 x^2} - \frac{4(1-\beta^2)^2}{(1-\beta^2 x^2)^2}\right]. \tag{18.142}$$

The total cross section, with the identical-particle $\frac{1}{2}$ and $\int d\Omega = 2\pi\int_{-1}^{1} dx$, is

$$\sigma = \frac{1}{2}\cdot\frac{2\pi}{64\pi^2 s\,\beta}\int_{-1}^{1}\overline{|\mathcal{M}|^2}\,dx = \frac{1}{64\pi s\,\beta}\int_{-1}^{1}\overline{|\mathcal{M}|^2}\,dx. \tag{18.143}$$

The two elementary integrals needed are, with $L \equiv \ln\frac{1+\beta}{1-\beta}$,

$$\int_{-1}^{1}\frac{dx}{1-\beta^2x^2} = \frac{L}{\beta}, \qquad \int_{-1}^{1}\frac{x^2\,dx}{1-\beta^2x^2} = -\frac{2}{\beta^2} + \frac{L}{\beta^3}, \qquad \int_{-1}^{1}\frac{dx}{(1-\beta^2x^2)^2} = \frac{1}{1-\beta^2} + \frac{L}{2\beta}, \tag{18.144}$$

the first from $\int\frac{dx}{1-\beta^2x^2} = \frac{1}{2\beta}\ln\frac{1+\beta x}{1-\beta x}$, the second by writing $\frac{x^2}{1-\beta^2x^2} = -\frac{1}{\beta^2} + \frac{1}{\beta^2(1-\beta^2x^2)}$, and the third from the reduction $\int\frac{dx}{(1-\beta^2x^2)^2} = \frac{x}{2(1-\beta^2x^2)} + \frac{1}{4\beta}\ln\frac{1+\beta x}{1-\beta x}$ (verified by differentiation). Assembling Eq. (18.142),

$$\int_{-1}^{1}\frac{6 - 4\beta^2 + 2\beta^2x^2}{1-\beta^2x^2}\,dx = (6-4\beta^2)\frac{L}{\beta} + 2\beta^2\left(-\frac{2}{\beta^2} + \frac{L}{\beta^3}\right) = \frac{4L}{\beta}(2-\beta^2) - 4, \tag{18.145}$$

$$4(1-\beta^2)^2\int_{-1}^{1}\frac{dx}{(1-\beta^2x^2)^2} = 4(1-\beta^2)^2\left(\frac{1}{1-\beta^2} + \frac{L}{2\beta}\right) = 4(1-\beta^2) + \frac{2(1-\beta^2)^2 L}{\beta}. \tag{18.146}$$

Subtracting Eq. (18.146) from Eq. (18.145) and collecting the coefficient of L/β,

$$\frac{1}{2e^4}\int_{-1}^{1}\overline{|\mathcal{M}|^2}\,dx = \frac{L}{\beta}\left[4(2-\beta^2) - 2(1-\beta^2)^2\right] - 4 - 4(1-\beta^2) + 0 = \frac{2L}{\beta}(3-\beta^4) - 4(2-\beta^2), \tag{18.147}$$

where $4(2-\beta^2)-2(1-\beta^2)^2 = 8-4\beta^2-2(1-2\beta^2+\beta^4) = 6-2\beta^4$. Thus $\int_{-1}^{1}\overline{|\mathcal{M}|^2}\,dx = 4e^4\big[\frac{(3-\beta^4)}{\beta}L - 2(2-\beta^2)\big]$, and Eq. (18.143) gives, with $e^4 = 16\pi^2\alpha^2$ so that $e^4/(16\pi s\beta) = \pi\alpha^2/(s\beta)$,

$$\boxed{\sigma(e^+e^-\to\gamma\gamma) = \frac{2\pi\alpha^2}{s\,\beta}\left[\frac{3-\beta^4}{2\beta}\,\ln\frac{1+\beta}{1-\beta} - (2-\beta^2)\right], \qquad \beta = \sqrt{1-\frac{4m^2}{s}}.} \tag{18.148}$$

High-energy limit. As $s\to\infty$, $\beta\to 1$ and $1-\beta \simeq 2m^2/s$, so $\frac{1+\beta}{1-\beta} \simeq \frac{2}{2m^2/s} = \frac{s}{m^2}$ and the bracket tends to $\frac{3-1}{2}\ln\frac{s}{m^2} - (2-1) = \ln\frac{s}{m^2} - 1$:

$$\sigma(e^+e^-\to\gamma\gamma) \xrightarrow{s\gg m^2} \frac{2\pi\alpha^2}{s}\left[\ln\frac{s}{m^2} - 1\right]. \tag{18.149}$$

The logarithm is the integrated collinear enhancement of Eq. (18.139): the forward/backward $1/\sin^2\theta$ peaks, cut off by the electron mass, produce $\ln(s/m^2)$ upon angular integration. The cross section falls as $\ln s/s$, the characteristic $1/s$ scaling of a pointlike $2\to 2$ QED cross section, modulated by the collinear log.

Threshold limit. As $\beta\to 0$, $L = \ln\frac{1+\beta}{1-\beta} \simeq 2\beta + \frac{2}{3}\beta^3$, so $\frac{3-\beta^4}{2\beta}L \to 3+\beta^2$ and the bracket tends to $(3+\beta^2)-(2-\beta^2) = 1+2\beta^2 \to 1$. Hence

$$\sigma(e^+e^-\to\gamma\gamma) \xrightarrow{\beta\to 0} \frac{2\pi\alpha^2}{s\,\beta} \to \frac{\pi\alpha^2}{2m^2\,\beta}, \tag{18.150}$$

the $1/\beta$ growth being the threshold flux enhancement anticipated in Theorem 18.12: the dynamical amplitude is finite at rest ($\overline{|\mathcal{M}|^2}\to 4e^4$), and the rising cross section reflects the diverging time the slow pair spends within interaction range. This is the S-wave $1/v$ law that underlies the lifetime of para-positronium, whose dominant decay is precisely this two-photon annihilation.

18.4.6 The crossing web of Chapter 18

We have now produced two complete observables from a single Klein–Nishina computation, and it is worth seeing where this places the chapter's processes in one analytic family. Crossing organizes the QED tree amplitudes of this chapter into a small "web," each link being an analytic continuation we have made explicit. Compton scattering $e^-\gamma\to e^-\gamma$ and pair annihilation $e^+e^-\to\gamma\gamma$ are the link built in this section, related by the $s\leftrightarrow t$ swap Eq. (18.118) and the single fermion sign Eq. (18.108); the reverse continuation gives pair production $\gamma\gamma\to e^+e^-$, occupying the same orange wedge of Fig. 18.4. In Chapter 17 the fermion-only link tied $e^+e^-\to\mu^+\mu^-$ to $e^-\mu^-\to e^-\mu^-$ by a $t\leftrightarrow s$ continuation crossing two fermion lines (Theorem 17.14). The remaining link is the one with identical fermions, where crossing relates Møller scattering $e^-e^-\to e^-e^-$

to Bhabha scattering $e^+e^- \to e^+e^-$—and there, because the two contributing diagrams are themselves a crossed pair, the fermion sign $(-1)^{n_f}$ we isolated here reappears not as a harmless overall factor but as the relative minus sign between interfering amplitudes demanded by Fermi statistics.

That last link is the subject of the next section. With the crossing machinery now complete—momentum and spinor replacements, the photon polarization-sum rule, the $(-1)^{n_f}$ fermion sign, and the averaging-factor adjustment all derived—we turn to the two-fermion-line processes, where these tools combine with identical-particle antisymmetrization to produce the Møller and Bhabha cross sections, the latter obtained from the former by exactly the kind of continuation worked out above.

18.5 Møller and Bhabha scattering

The crossing machinery assembled in Section 18.4 did its work on processes with a single fermion line of charge flowing through the diagram: Compton scattering and its crossed relative, pair annihilation into two photons. We now turn to the two processes in which *two* charged fermion lines run side by side and the photon is the exchanged, internal quantum—electron–electron scattering, $e^-e^- \to e^-e^-$ (Møller), and electron–positron scattering, $e^+e^- \to e^+e^-$ (Bhabha). These are the processes in which the structural features that make QED amplitudes more than a single Feynman graph come fully into view. Each receives contributions from two distinct diagrams that must be added *coherently*, before squaring; the two diagrams interfere, and the interference term is as physical as the direct terms. In Møller scattering the two diagrams are related by the exchange of the two identical final electrons, and Fermi statistics demands that they enter the amplitude with a *relative minus sign*—a sign with no analogue in the scalar theory of Part IV, and the first place in the book where the antisymmetry of the electron field leaves a directly measurable fingerprint. In Bhabha scattering the two diagrams are of genuinely different character—one is the t-channel exchange that Møller also has, the other is the s-channel annihilation $e^+e^- \to \gamma^* \to e^+e^-$ already met in $e^+e^- \to \mu^+\mu^-$—yet, as crossing will show, the two processes are boundary values of one analytic function: Bhabha is Møller continued in the Mandelstam variables. We compute both unpolarized cross sections from the traces in full, verify the crossing relation between them as an independent check, and close by recovering the non-relativistic Coulomb (Rutherford–Mott) limit as the physical sanity test that every charged-particle scattering calculation must pass.

Throughout we use the QED Feynman rules of Chapter 17: the vertex $-ie\gamma^\mu$ (Eq. (17.10)), with electron charge $-e$ and $e > 0$; the photon propagator $-i\eta_{\mu\nu}/(q^2 + i\varepsilon)$ in Feynman gauge $\xi = 1$ (Eq. (17.18)); and the external-spinor wavefunctions of Chapter 11. Spin sums use the completeness relations

$\sum_s u^s(p)\bar{u}^s(p) = \not{p} + m$ and $\sum_s v^s(p)\bar{v}^s(p) = \not{p} - m$ (Theorem C.68), and the Dirac traces and γ-contraction identities are taken from Section C.6. We keep the electron mass m general in the Møller direct and interference traces—the Coulomb limit needs it—and pass to the high-energy massless forms only after the invariant answer is in hand.

18.5.1 The Møller amplitude and the Fermi sign

Label the process $e^-(p_1)\,e^-(p_2) \to e^-(p_3)\,e^-(p_4)$, all four particles electrons of mass m, with spins suppressed until the spin sum. The Mandelstam invariants are

$$s = (p_1 + p_2)^2, \qquad t = (p_1 - p_3)^2, \qquad u = (p_1 - p_4)^2, \qquad s + t + u = 4m^2, \tag{18.151}$$

the last identity following from $\sum_i p_i^2 = 4m^2$ and overall momentum conservation. At order e^2 the connected amplitude is built from two tree diagrams, drawn in Fig. 18.5. In the first, the incoming electron p_1 emits a virtual photon and emerges as p_3, while p_2 absorbs it and emerges as p_4; the exchanged photon carries momentum $p_1 - p_3$, so this is the t-channel diagram. In the second, the roles of the two outgoing electrons are interchanged: $p_1 \to p_4$ and $p_2 \to p_3$, the photon carrying $p_1 - p_4$, the u-channel. Because the two final electrons are identical particles, both wirings contribute to the same physical final state and both must be included.

The t-channel graph is assembled from two vertices, one photon propagator, and the four external spinors. The incoming p_1/outgoing p_3 vertex contributes the current $\bar{u}_3(-ie\gamma^\mu)u_1$; the incoming p_2/outgoing p_4 vertex contributes $\bar{u}_4(-ie\gamma^\nu)u_2$; the photon line between them contributes $-i\eta_{\mu\nu}/t$. Multiplying,

$$i\mathcal{M}_t = \left[\bar{u}_3(-ie\gamma^\mu)u_1\right]\frac{-i\eta_{\mu\nu}}{t}\left[\bar{u}_4(-ie\gamma^\nu)u_2\right] = \frac{ie^2}{t}\left[\bar{u}_3\gamma^\mu u_1\right]\left[\bar{u}_4\gamma_\mu u_2\right], \tag{18.152}$$

where the numerical coefficient is $(-ie)^2(-i) = ie^2$ and we abbreviate $u_i \equiv u^{s_i}(p_i)$. The u-channel graph is obtained by the replacement $p_3 \leftrightarrow p_4$ (equivalently $3 \leftrightarrow 4$ on the outgoing spinors), with the photon momentum now $p_1 - p_4$ so that the denominator is u:

$$i\mathcal{M}_u = \frac{ie^2}{u}\left[\bar{u}_4\gamma^\mu u_1\right]\left[\bar{u}_3\gamma_\mu u_2\right]. \tag{18.153}$$

The relative sign from Wick's theorem. The two diagrams cannot simply be added; antisymmetry of the fermionic fields fixes their relative sign, and getting it right is the whole point of the calculation. The cleanest derivation runs through Wick's theorem (Theorem 15.3). At second order in the interaction the relevant contribution to the S-matrix element is

$$\left\langle e^-(p_3)e^-(p_4)\middle|\mathcal{T}\left[\tfrac{(ie)^2}{2!}\int d^4x\,d^4y\,(\bar{\psi}\not{A}\psi)_x(\bar{\psi}\not{A}\psi)_y\right]\middle|e^-(p_1)e^-(p_2)\right\rangle, \tag{18.154}$$

in which two of the four fermion fields are contracted with the external electrons and the two photon fields are contracted with each other to make the internal line. The external incoming electrons are annihilated by the ψ fields and the outgoing ones created by the $\bar{\psi}$ fields. There are exactly two inequivalent ways to assign the two $\bar{\psi}$'s to the two outgoing momenta: $\bar{\psi}(x)\to p_3$, $\bar{\psi}(y)\to p_4$, which (after also assigning the ψ's and the photon contraction) reproduces Eq. (18.152); and $\bar{\psi}(x)\to p_4$, $\bar{\psi}(y)\to p_3$, which reproduces Eq. (18.153). These two assignments differ by the interchange of two anticommuting field operators in the contraction string: passing from one to the other transposes a single pair of fermionic creation operators, and each such transposition carries a factor (-1) from the canonical anticommutation relations of Chapter 11. The second contraction therefore enters with the opposite sign to the first. (The overall $1/2!$ of the Dyson expansion and the choice of which vertex, x or y, supplies p_3 are absorbed in the standard way: the two terms of the $1/2!$ sum are equal and cancel the factor, leaving one copy of each topology, exactly as in the scalar case of Section 15.4.) The result is that the two diagrams add with a relative minus,

$$i\mathcal{M} = i\mathcal{M}_t - i\mathcal{M}_u = ie^2\left\{\frac{[\bar{u}_3\gamma^\mu u_1][\bar{u}_4\gamma_\mu u_2]}{t} - \frac{[\bar{u}_4\gamma^\mu u_1][\bar{u}_3\gamma_\mu u_2]}{u}\right\}. \tag{18.155}$$

Physical Insight

The relative minus sign in Eq. (18.155) is the amplitude-level statement of the Pauli principle. Under interchange of the two identical outgoing electrons, $p_3 \leftrightarrow p_4$ (with their spins), the labels $t \leftrightarrow u$ swap and the two terms of Eq. (18.155) trade places; the relative minus makes the *total* amplitude antisymmetric, $\mathcal{M}(p_3,p_4) = -\mathcal{M}(p_4,p_3)$, exactly the antisymmetry the final two-electron state must have. It is not a convention one is free to choose: a relative plus would predict constructive interference where experiment, and the spin-statistics theorem (the subject of Chapter 15 and its statement of Wick's theorem), demand destructive. The sign survives into the cross section as the $+2s^2/tu$ interference term computed below, whose destructive (Pauli) character is directly visible in the angular distribution—showing as a genuinely negative contribution in the non-relativistic Mott limit, while at high energy the term itself is positive (Eq. (18.173)).

18.5.2 The unpolarized Møller cross section

For unpolarized beams and unobserved final spins we average over the four initial spin states and sum over the four final ones, the standard $\frac{1}{4}\sum_{\text{spins}}$ of Section 17.3. Writing $\mathcal{M} = \mathcal{M}_t - \mathcal{M}_u$ (with $\mathcal{M}_{t,u}$ the prefactors of i in Eq. (18.152)–Eq. (18.153)), the squared amplitude splits into two direct terms

and an interference term,

$$\overline{|\mathcal{M}|^2} = \tfrac{1}{4}\sum_{\text{spins}} \Big(|\mathcal{M}_t|^2 + |\mathcal{M}_u|^2 - 2\,\text{Re}\;\mathcal{M}_t\mathcal{M}_u^* \Big). \tag{18.156}$$

We compute the three pieces in turn.

The direct terms. The t-channel direct term is

$$\tfrac{1}{4}\sum_{\text{spins}} |\mathcal{M}_t|^2 = \frac{e^4}{4t^2}\;\text{tr}\big[(\not p_3 + m)\gamma^\mu(\not p_1 + m)\gamma^\nu\big]\;\text{tr}\big[(\not p_4 + m)\gamma_\mu(\not p_2 + m)\gamma_\nu\big], \tag{18.157}$$

where the spin sums $\sum u_1\bar u_1 = \not p_1 + m$, etc. (Theorem C.68) have turned each squared current into a Dirac trace. Each trace is evaluated with the four-γ identity $\text{tr}[(\not a + m)\gamma^\mu(\not b + m)\gamma^\nu] = 4[a^\mu b^\nu + a^\nu b^\mu - \eta^{\mu\nu}(a\cdot b - m^2)]$ of Section C.6 (the odd-γ traces vanish). Contracting the two resulting tensors is the same algebra met in $e^+e^- \to \mu^+\mu^-$; carrying the mass terms,

$$\text{tr}[(\not a+m)\gamma^\mu(\not b+m)\gamma^\nu]\;\text{tr}[(\not c+m)\gamma_\mu(\not d+m)\gamma_\nu] = 32\big[(a\cdot c)(b\cdot d)+(a\cdot d)(b\cdot c)-m^2(a\cdot b)-m^2(c\cdot d)+2 \tag{18.158}$$

With $a = p_3$, $b = p_1$, $c = p_4$, $d = p_2$ and the dot products read off from Eq. (18.151)—namely $p_1\cdot p_2 = p_3\cdot p_4 = \tfrac{1}{2}(s-2m^2)$, $p_1\cdot p_3 = p_2\cdot p_4 = \tfrac{1}{2}(2m^2-t)$, and $p_1\cdot p_4 = p_2\cdot p_3 = \tfrac{1}{2}(2m^2-u)$—the bracket of Eq. (18.158) becomes

$$\tfrac{1}{4}(s-2m^2)^2 + \tfrac{1}{4}(u-2m^2)^2 + m^2 t, \tag{18.159}$$

after using $s+t+u = 4m^2$ to eliminate $s+u = 4m^2 - t$. Hence

$$\tfrac{1}{4}\sum_{\text{spins}} |\mathcal{M}_t|^2 = \frac{2e^4}{t^2}\Big[(s-2m^2)^2 + (u-2m^2)^2 + 4m^2 t\Big]. \tag{18.160}$$

The u-channel direct term is obtained from Eq. (18.160) by the exchange $p_3 \leftrightarrow p_4$, i.e. $t \leftrightarrow u$:

$$\tfrac{1}{4}\sum_{\text{spins}} |\mathcal{M}_u|^2 = \frac{2e^4}{u^2}\Big[(s-2m^2)^2 + (t-2m^2)^2 + 4m^2 u\Big]. \tag{18.161}$$

In the high-energy limit $m \to 0$ these reduce to $2e^4(s^2+u^2)/t^2$ and $2e^4(s^2+t^2)/u^2$, the same t- and u-channel structures one expects from photon exchange.

The interference term. The cross term is where the two diagrams genuinely talk to each other, and it is the term the Fermi sign acts on. Spin-summing the product $\mathcal{M}_t\mathcal{M}_u^*$ ties the four currents into a *single* closed Dirac trace, because the spinor indices chain around: the conjugate currents $[\bar u_4\gamma^\nu u_1]^* = [\bar u_1\gamma^\nu u_4]$

and $[\bar{u}_3\gamma_\nu u_2]^* = [\bar{u}_2\gamma_\nu u_3]$ link u_1 to u_4 and u_2 to u_3, closing the loop $u_3 \to u_1 \to u_4 \to u_2 \to u_3$. With the relative minus from Eq. (18.156),

$$-\tfrac{1}{4}\sum_{\text{spins}} 2\,\mathrm{Re}\;\mathcal{M}_t\mathcal{M}_u^* = -\frac{e^4}{2tu}\,T, \qquad T \equiv \mathrm{tr}\big[(\not p_3+m)\gamma^\mu(\not p_1+m)\gamma^\nu(\not p_4+m)\gamma_\mu(\not p_2+m)\gamma_\nu\big]. \tag{18.162}$$

This eight-γ trace is evaluated in Theorem 18.14 using the contraction identities of Section C.6; the result is the remarkably compact

$$T = -8\,(s-2m^2)(s-6m^2), \tag{18.163}$$

so that the interference contribution to $\overline{|\mathcal{M}|^2}$ is

$$-\frac{e^4}{2tu}\,T = \frac{4e^4}{tu}\,(s-2m^2)(s-6m^2). \tag{18.164}$$

Worked Example 18.14 (The Møller interference trace)**.** We evaluate T of Eq. (18.162) by collapsing the two contracted index pairs from the inside out, using the four-dimensional contraction identities (Section C.6)

$$\gamma^\mu\gamma_\mu = 4, \quad \gamma^\mu \not a \gamma_\mu = -2\not a, \quad \gamma^\mu \not a \not b \gamma_\mu = 4\,a\cdot b, \quad \gamma^\mu \not a \not b \not c \gamma_\mu = -2\not c \not b \not a. \tag{18.165}$$

Expand $(\not p_1 + m)\gamma^\nu(\not p_4 + m)$ and contract the μ pair, treating γ^ν as a vector e^ν:

$$\begin{aligned}\gamma^\mu(\not p_1 + m)\gamma^\nu(\not p_4 + m)\gamma_\mu &= \gamma^\mu\big[\not p_1\gamma^\nu\not p_4 + m\,\not p_1\gamma^\nu + m\,\gamma^\nu\not p_4 + m^2\gamma^\nu\big]\gamma_\mu \\ &= -2\not p_4\gamma^\nu\not p_1 + 4m\,(p_1^\nu + p_4^\nu) - 2m^2\gamma^\nu,\end{aligned} \tag{18.166}$$

where the first term used the three-vector identity, the two single-mass terms used $\gamma^\mu \not a \gamma^\nu \gamma_\mu = 4a^\nu$ (the two-vector identity with one slot the free index ν), and the last used $\gamma^\mu\gamma^\nu\gamma_\mu = -2\gamma^\nu$. Insert this between $(\not p_3 + m)$ and $(\not p_2 + m)\gamma_\nu$ and take the trace term by term.

Term 1 $(-2\not p_4\gamma^\nu\not p_1)$: contracting the remaining ν pair through $\not p_1(\not p_2 + m)$ gives $\gamma^\nu\not p_1(\not p_2 + m)\gamma_\nu = 4(p_1\cdot p_2) - 2m\not p_1$, after which only even-$\gamma$ traces survive:

$$T_1 = -2\;\mathrm{tr}\big[(\not p_3 + m)\not p_4(4(p_1\cdot p_2) - 2m\not p_1)\big] = -32(p_1\cdot p_2)(p_3\cdot p_4) + 16m^2(p_1\cdot p_4). \tag{18.167}$$

Term 2 $(+4m(p_1^\nu + p_4^\nu))$: here $\gamma_\nu(p_1^\nu + p_4^\nu) = \not p_1 + \not p_4$, and

$$T_2 = 4m\;\mathrm{tr}\big[(\not p_3 + m)(\not p_2 + m)(\not p_1 + \not p_4)\big] = 16m^2\,(p_1 + p_4)\cdot(p_2 + p_3), \tag{18.168}$$

the only surviving traces being the two with a single mass insertion. *Term 3* $(-2m^2\gamma^\nu)$: using $\gamma^\nu(\not p_2 + m)\gamma_\nu = -2\not p_2 + 4m$,

$$T_3 = -2m^2\;\mathrm{tr}\big[(\not p_3 + m)(-2\not p_2 + 4m)\big] = 16m^2(p_2\cdot p_3) - 32m^4. \tag{18.169}$$

Adding $T_1 + T_2 + T_3$, the mass-squared terms assemble into the symmetric sum over all six dot products,

$$T = -32(p_1\cdot p_2)(p_3\cdot p_4) + 16m^2\sum_{i<j}(p_i\cdot p_j) - 32m^4. \tag{18.170}$$

With $(p_1\cdot p_2)(p_3\cdot p_4) = \frac{1}{4}(s-2m^2)^2$ and $\sum_{i<j} p_i\cdot p_j = 2(s-m^2)$ (from the dot-product values below Eq. (18.158) and $s+t+u=4m^2$),

$$T = -8(s-2m^2)^2+32m^2(s-m^2)-32m^4 = -8(s^2-8m^2s+12m^4) = -8(s-2m^2)(s-6m^2), \tag{18.171}$$

which is Eq. (18.163). As a check, $m\to 0$ gives $T=-8s^2$, and the interference contribution Eq. (18.164) reduces to $+4e^4s^2/tu$.

Assembly. Collecting Eq. (18.160), Eq. (18.161) and Eq. (18.164), the unpolarized Møller squared amplitude is

$$\boxed{\overline{|\mathcal{M}|^2}_{\text{Møl}} = \frac{2e^4}{t^2}\left[(s-2m^2)^2+(u-2m^2)^2+4m^2t\right] + \frac{2e^4}{u^2}\left[(s-2m^2)^2+(t-2m^2)^2+4m^2u\right]} \tag{18.172}$$

In the high-energy limit $m\to 0$ this collapses to the much-quoted form

$$\overline{|\mathcal{M}|^2}_{\text{Møl}} \xrightarrow{m\to 0} 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{2s^2}{tu}\right]. \tag{18.173}$$

The three terms are manifest: t-channel exchange peaked at small momentum transfer, u-channel exchange peaked at backward angles, and the interference $2s^2/tu$ which, with $t<0$ and $u<0$ giving $tu>0$, is *positive* in the high-energy physical region. The destructive interference demanded by the Pauli antisymmetry is therefore not a blanket negative sign in $\overline{|\mathcal{M}|^2}$: the massive interference Eq. (18.164) carries the factor $(s-2m^2)(s-6m^2)$, positive at high energy but negative in the non-relativistic regime, where it produces the negative interference term of the Mott formula Eq. (18.191). The expression is manifestly symmetric under $t\leftrightarrow u$, as it must be: exchanging the two identical final electrons cannot change a physical probability.

18.5.3 Bhabha scattering: annihilation plus scattering

Now turn one of the electrons into a positron: $e^-(p_1)\,e^+(p_2)\to e^-(p_3)\,e^+(p_4)$. The Mandelstam variables keep the assignments

$$s=(p_1+p_2)^2, \qquad t=(p_1-p_3)^2, \qquad u=(p_1-p_4)^2, \qquad s+t+u=4m^2, \tag{18.174}$$

so that s is the squared total energy of the colliding pair, t the momentum transfer between the incoming and outgoing electron, and u that between the

electron and the outgoing positron. Two tree diagrams again contribute, but now of distinct physical character (Fig. 18.6). In the t-channel *scattering* diagram the electron and positron exchange a photon and recoil, the photon carrying $p_1 - p_3$; this is the direct analogue of the Møller t-channel. In the s-channel *annihilation* diagram the e^+e^- pair annihilates into a single virtual photon of momentum $p_1 + p_2$, which then materializes the outgoing pair—precisely the mechanism of $e^+e^- \to \mu^+\mu^-$ from Section 17.3, now with electrons in the final state.

Reading off the diagrams with the antifermion rules of Chapter 11—an incoming positron of momentum p_2 carries $\bar{v}_2$, an outgoing positron p_4 carries v_4—the two amplitudes are

$$i\mathcal{M}_t = [\bar{u}_3(-ie\gamma^\mu)u_1]\frac{-i\eta_{\mu\nu}}{t}[\bar{v}_2(-ie\gamma^\nu)v_4] = \frac{ie^2}{t}[\bar{u}_3\gamma^\mu u_1][\bar{v}_2\gamma_\mu v_4], \quad (18.175)$$

$$i\mathcal{M}_s = [\bar{v}_2(-ie\gamma^\mu)u_1]\frac{-i\eta_{\mu\nu}}{s}[\bar{u}_3(-ie\gamma^\nu)v_4] = \frac{ie^2}{s}[\bar{v}_2\gamma^\mu u_1][\bar{u}_3\gamma_\mu v_4]. \quad (18.176)$$

The relative sign is again fixed by Fermi statistics: the s- and t-channel contractions differ by the transposition of two anticommuting fields (here the ψ-field annihilating the incoming electron is paired, in one case, with the $\bar{\psi}$ creating the outgoing electron and, in the other, with the field annihilating the incoming positron), so the two diagrams enter with opposite sign. Choosing the conventional assignment,

$$i\mathcal{M} = i\mathcal{M}_t - i\mathcal{M}_s = ie^2\left\{\frac{[\bar{u}_3\gamma^\mu u_1][\bar{v}_2\gamma_\mu v_4]}{t} - \frac{[\bar{v}_2\gamma^\mu u_1][\bar{u}_3\gamma_\mu v_4]}{s}\right\}. \quad (18.177)$$

The squared amplitude, direct terms. The two direct terms reuse computations already done. The s-channel term is structurally identical to $e^+e^- \to \mu^+\mu^-$ of Section 17.3—two currents, each a vector bilinear, contracted across a photon of invariant mass $\sqrt{s}$—and the trace algebra is the same, giving in the massless limit

$$\tfrac{1}{4}\sum_{\text{spins}}|\mathcal{M}_s|^2 \xrightarrow{m\to 0} \frac{2e^4}{s^2}(t^2+u^2). \quad (18.178)$$

The t-channel term has exactly the form of the Møller t-channel Eq. (18.160) with the photon now spacelike; in the massless limit

$$\tfrac{1}{4}\sum_{\text{spins}}|\mathcal{M}_t|^2 \xrightarrow{m\to 0} \frac{2e^4}{t^2}(s^2+u^2). \quad (18.179)$$

The interference term. The new computation is the s–t interference. As in Møller, the conjugate currents chain the four bilinears into one closed trace; with

the relative minus of Eq. (18.177),

$$-\tfrac{1}{4}\sum_{\text{spins}} 2\ \mathrm{Re}\ \mathcal{M}_t\mathcal{M}_s^* = -\frac{e^4}{2st}\,T', \qquad T' \equiv \mathrm{tr}[\not p_3\gamma^\mu \not p_1\gamma^\nu \not p_2\gamma_\mu \not p_4\gamma_\nu] \quad (m\to 0), \tag{18.180}$$

where the massless spin sums $\sum u\bar u = \not p$, $\sum v\bar v = \not p$ have been used. Collapsing the μ pair through $\not p_1\gamma^\nu\not p_2$ with the three-vector identity of Eq. (18.165) gives $\gamma^\mu\not p_1\gamma^\nu\not p_2\gamma_\mu = -2\not p_2\gamma^\nu\not p_1$, and then the ν pair through $\not p_1\not p_4$ gives $\gamma^\nu\not p_1\not p_4\gamma_\nu = 4(p_1\cdot p_4)$:

$$T' = -2\ \mathrm{tr}[\not p_3\not p_2\gamma^\nu\not p_1\not p_4\gamma_\nu] = -8(p_1\cdot p_4)\ \mathrm{tr}[\not p_3\not p_2] = -32(p_1\cdot p_4)(p_2\cdot p_3) = -8u^2, \tag{18.181}$$

the last step using the massless values $p_1\cdot p_4 = p_2\cdot p_3 = -u/2$. Therefore the interference contribution is

$$-\frac{e^4}{2st}\,T' = \frac{4e^4u^2}{st}. \tag{18.182}$$

Collecting Eq. (18.178), Eq. (18.179) and Eq. (18.182),

$$\boxed{\overline{|\mathcal{M}|^2}_{\text{Bha}} \xrightarrow{m\to 0} 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{t^2+u^2}{s^2} + \frac{2u^2}{st}\right].} \tag{18.183}$$

Three terms once more, but with the roles permuted relative to Møller: the divergent forward peak now sits in the t-channel (Coulomb scattering) alone, the s-channel annihilation is regular, and the cross term $2u^2/st$ is genuinely negative in the physical region (where $s>0$ and $t<0$, so $st<0$): annihilation and scattering interfere *destructively*, a hallmark of the Bhabha line shape.

18.5.4 Bhabha as a crossing of Møller

The two computations were carried out independently, but crossing symmetry (Section 18.4) predicts that they are not independent: Bhabha is Møller continued in the Mandelstam plane. Concretely, Møller $e^-(p_1)e^-(p_2)\to e^-(p_3)e^-(p_4)$ becomes Bhabha $e^-(p_1)e^+(p_2)\to e^-(p_3)e^+(p_4)$ when the incoming electron p_2 is crossed into an outgoing positron and the outgoing electron p_4 into an incoming positron. Under this exchange of the second and fourth legs the invariants transform as $s\leftrightarrow u$, while t is fixed—one checks directly from the definitions that interchanging p_2 and $-p_4$ sends $(p_1+p_2)^2\to(p_1-p_4)^2$ and $(p_1-p_4)^2\to(p_1+p_2)^2$, i.e. $s\leftrightarrow u$, leaving $t=(p_1-p_3)^2$ invariant. The crossing rules of Eq. (18.108) furthermore guarantee that the *spin-summed* squared amplitude continues without extra sign factors (the fermion-crossing sign squares to unity, and the spin-averaging factor $\tfrac14$ is the same for the two-fermion

initial state in both processes). Applying $s \leftrightarrow u$ to the massless Møller result Eq. (18.173) term by term,

$$\underbrace{\frac{s^2+u^2}{t^2}}_{\text{unchanged}} + \underbrace{\frac{s^2+t^2}{u^2}}_{\to \frac{u^2+t^2}{s^2}} + \underbrace{\frac{2s^2}{tu}}_{\to \frac{2u^2}{ts}} = \frac{s^2+u^2}{t^2} + \frac{t^2+u^2}{s^2} + \frac{2u^2}{st}, \tag{18.184}$$

which is exactly the Bhabha result Eq. (18.183). The two derivations agree: the direct trace computation and the analytic continuation give the same function, the strongest internal check available at tree level. The continuation also predicts the *massive* Bhabha amplitude with no further work—applying $s \leftrightarrow u$ to Eq. (18.172) (the equal masses are untouched by particle↔antiparticle crossing) gives

$$\overline{|\mathcal{M}|^2}_{\text{Bha}} = \frac{2e^4}{t^2}\left[(s-2m^2)^2+(u-2m^2)^2+4m^2t\right]+\frac{2e^4}{s^2}\left[(u-2m^2)^2+(t-2m^2)^2+4m^2s\right]+\frac{4e^4}{st}(u-2m^2 \tag{18.185}$$

whose $m \to 0$ limit reproduces Eq. (18.183).

Physical Insight

That the same analytic function describes electron–electron scattering and electron–positron scattering is not a coincidence of QED but a consequence of locality and unitarity, made precise by crossing in Section 18.4. The two processes probe different regions of the (s,t,u) plane—Møller lives where $s > 4m^2$ and $t,u < 0$, Bhabha where the roles of s and u are swapped—but the amplitude is one object analytically continued between them. Crossing thus turns every computation into two: having worked out Møller honestly, we obtained Bhabha for the price of a substitution, and the agreement of that substitution with the independent direct calculation is a nontrivial confirmation that both were done correctly.

18.5.5 Differential cross sections and the Coulomb limit

We now turn the squared amplitudes into cross sections and extract the physical limits that every charged-particle calculation must reproduce. Both processes are elastic with equal masses, so in the centre-of-mass frame the outgoing and incoming momenta have equal magnitude, $|\mathbf{p}_f| = |\mathbf{p}_i| \equiv p$, and the angular cross section of Theorem 16.23 simplifies to

$$\frac{d\sigma}{d\Omega} = \frac{\overline{|\mathcal{M}|^2}}{64\pi^2 s}, \tag{18.186}$$

the elastic specialization of Eq. (16.145). The CM kinematics fix the invariants in terms of the scattering angle θ as $t = -2p^2(1-\cos\theta) = -4p^2\sin^2(\theta/2)$ and $u = -2p^2(1+\cos\theta) = -4p^2\cos^2(\theta/2)$, with $s = 4(p^2+m^2)$. In the high-energy limit $p \to \sqrt{s}/2$, so $t = -\frac{s}{2}(1-\cos\theta) = -s\sin^2(\theta/2)$ and $u = -s\cos^2(\theta/2)$.

Møller. Substituting Eq. (18.173) into Eq. (18.186) with $e^4 = 16\pi^2\alpha^2$ and converting to half-angles,

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{Møl}} = \frac{\alpha^2}{2s}\left[\frac{1+\cos^4(\theta/2)}{\sin^4(\theta/2)} + \frac{1+\sin^4(\theta/2)}{\cos^4(\theta/2)} + \frac{2}{\sin^2(\theta/2)\cos^2(\theta/2)}\right]. \quad (18.187)$$

The first term is the t-channel forward peak, the second the u-channel backward peak—present because the two final electrons are identical and the detector cannot tell which one it caught—and the third is the interference. As $\theta \to 0$ the cross section diverges as $2/\sin^4(\theta/2)$, the Rutherford $1/\sin^4$ singularity of one-photon (Coulomb) exchange; the u-channel produces an identical divergence at $\theta \to \pi$, the symmetry of Eq. (18.187) under $\theta \to \pi - \theta$ being the angular face of the $t \leftrightarrow u$ symmetry of identical particles.

Bhabha. Substituting Eq. (18.183) into Eq. (18.186),

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{Bha}} = \frac{\alpha^2}{2s}\left[\frac{1+\cos^4(\theta/2)}{\sin^4(\theta/2)} + \frac{1+\cos^2\theta}{2} - \frac{2\cos^4(\theta/2)}{\sin^2(\theta/2)}\right], \quad (18.188)$$

where the middle term used $\frac{t^2+u^2}{s^2} = \frac{1}{2}(1+\cos^2\theta)$. Here only the t-channel diverges in the forward direction, again as $2/\sin^4(\theta/2)$; there is no backward divergence, because the final electron and positron are distinct and the u-channel is absent. The s-channel annihilation contributes the smooth $\frac{1}{2}(1+\cos^2\theta)$ piece—the very same angular distribution as $e^+e^- \to \mu^+\mu^-$ (Section 17.3), as it must be, since at high energy the annihilation diagram does not know whether the produced pair is muons or electrons. The forward Coulomb peak is the experimental signature exploited at every e^+e^- collider: Bhabha scattering at small angles is the standard luminosity monitor precisely because its rate is dominated by the calculable, large, t-channel QED cross section.

The identical-particle phase space for Møller. One subtlety distinguishes the two total cross sections. In Bhabha scattering the final e^- and e^+ are distinct, and the total cross section integrates Eq. (18.188) over the full solid angle. In Møller scattering the two final electrons are identical: the configurations with one electron at (θ, φ) and the other at $(\pi - \theta, \varphi + \pi)$ are the *same* physical final state, counted twice if one integrates Eq. (18.187) over all 4π. The correct total cross section therefore carries the identical-particle symmetry factor $\frac{1}{2}$ of Section 15.4 and Eq. (16.83),

$$\sigma_{\text{Møl}} = \frac{1}{2}\int d\Omega \left.\frac{d\sigma}{d\Omega}\right|_{\text{Møl}} \quad \Longleftrightarrow \quad \int_{\text{half space}} d\Omega \left.\frac{d\sigma}{d\Omega}\right|_{\text{Møl}}, \quad (18.189)$$

the two prescriptions being equivalent because the integrand is symmetric under $\theta \to \pi - \theta$. This is the same factor that appears in the final-state phase space of

any process with identical outgoing particles, here demanded already at the level of counting distinct events.

Worked Example 18.15 (The non-relativistic Coulomb limit of Møller scattering)**.** We recover the Rutherford–Mott cross section from Eq. (18.172) in the non-relativistic limit $p \ll m$, where the electrons move slowly and the photon exchange reduces to the static Coulomb interaction. In this limit $s = 4(p^2 + m^2) \to 4m^2$, so $s - 2m^2 \to 2m^2$ and $s - 6m^2 \to -2m^2$, while $t = -4p^2 \sin^2(\theta/2)$ and $u = -4p^2 \cos^2(\theta/2)$ remain $O(p^2)$. Evaluate the three pieces of Eq. (18.172) to leading order. The t-channel bracket becomes $(s-2m^2)^2 + (u-2m^2)^2 + 4m^2 t \to (2m^2)^2 + (2m^2)^2 + O(m^2 p^2) = 8m^4$, giving

$$\tfrac{1}{4}\sum |\mathcal{M}_t|^2 \to \frac{2e^4}{t^2}\, 8m^4 = \frac{16 e^4 m^4}{16 p^4 \sin^4(\theta/2)} = \frac{e^4 m^4}{p^4 \sin^4(\theta/2)}; \tag{18.190}$$

the u-channel gives the same with $\sin \to \cos$; and the interference $\frac{4e^4}{tu}(s-2m^2)(s-6m^2) \to \frac{4e^4}{tu}(2m^2)(-2m^2) = -16 e^4 m^4/(tu)$, with $tu = 16 p^4 \sin^2(\theta/2)\cos^2(\theta/2)$, giving $-e^4 m^4/[p^4 \sin^2(\theta/2)\cos^2(\theta/2)]$. Dividing by $64\pi^2 s \to 256\pi^2 m^2$ as in Eq. (18.186), writing $e^4 = 16\pi^2\alpha^2$ and $p = mv$ (so $p^4 = m^4 v^4$),

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{Møl}}^{\text{NR}} = \frac{\alpha^2}{16\, m^2 v^4}\left[\frac{1}{\sin^4(\theta/2)} + \frac{1}{\cos^4(\theta/2)} - \frac{1}{\sin^2(\theta/2)\cos^2(\theta/2)}\right]. \tag{18.191}$$

This is the *Mott cross section* for identical spin-$\frac{1}{2}$ particles. The first term, taken alone, is the Rutherford formula $d\sigma/d\Omega = \alpha^2/(16 E_{\text{kin}}^2 \sin^4(\theta/2))$ with $E_{\text{kin}} = mv^2$ the relative kinetic energy, exactly the non-relativistic Coulomb scattering of two unit charges. The second term is its mirror image, the identical-particle exchange contribution, and the third is the quantum interference between direct and exchange amplitudes—negative for unpolarized spin-$\frac{1}{2}$ fermions, the unmistakable signature of Fermi statistics surviving all the way down to the non-relativistic regime. The relative minus sign of Eq. (18.155), derived from Wick's theorem, is what produces the destructive interference here; a Bose statistics calculation would give $+1$ in its place. The recovery of Rutherford scattering with the correct coefficient is the physical check that the entire relativistic computation, traces and all, was assembled correctly.

The same non-relativistic reduction applied to Bhabha scattering keeps only the t-channel term of Eq. (18.188), since the annihilation diagram is suppressed by p^2/m^2 at low energy (the produced pair must carry the full rest energy $2m$, which the slow incoming pair barely has): one recovers the ordinary Rutherford scattering of opposite charges, $d\sigma/d\Omega \to \alpha^2/(16 E_{\text{kin}}^2 \sin^4(\theta/2))$, with no exchange term because electron and positron are distinguishable. Coulomb scattering thus emerges as the universal soft limit of both processes, the annihilation channel and the identical-particle exchange being the relativistic and quantum-statistical corrections that QED layers on top of it.

With the unpolarized Møller and Bhabha cross sections in hand, the canonical quartet of tree-level QED processes—Compton, pair annihilation, Møller, and Bhabha—is complete at the level of spin-summed observables. Each of the results Eq. (18.172) and Eq. (18.185) is, however, an incoherent sum over final spins and an average over initial ones, and it conceals a far richer structure: the individual helicity amplitudes, most of which vanish in the massless limit by a selection rule we have not yet stated. The next section opens that structure, introducing the spin-projection and helicity technology, proving the high-energy helicity-conservation rule, and showing that the incoherent sum of the helicity amplitudes reassembles precisely the unpolarized results computed here—the bridge to the spinor-helicity reframing of Part XVI.

18.6 Polarized amplitudes, helicity, and the high-energy limit

Every cross section computed so far in this chapter has been an average. We summed the squared amplitude over the spins of the final particles, averaged it over the spins of the initial ones, and let the spin-sum identities $\sum_s u_s \bar{u}_s = \not{p} + m$ and $\sum_s v_s \bar{v}_s = \not{p} - m$ collapse the four-spinor bilinears into traces over Dirac matrices. This is exactly the right thing to do when the apparatus neither prepares nor measures spin, and it produced the compact angular laws of the quartet: Compton, annihilation, Møller, and Bhabha. But the trace has thrown away information that is physically real and, as it turns out, computationally precious. The unpolarized $1 + \cos^2\theta$ of $e^+e^- \to \mu^+\mu^-$ is not a single mechanism; it is the incoherent sum of a small number of *helicity amplitudes*, each with its own, much simpler, angular shape. The task of this section is to undo the averaging — to resolve the spin-summed quartet into its polarized components — and to discover, in the process, that at high energy the resolution is not a complication but a radical simplification. We will build the projector that isolates a single spin state, prove that in the massless limit helicity coincides with chirality, derive the selection rule that kills all but a handful of helicity configurations, compute those that survive in full, and check that their incoherent sum rebuilds the unpolarized result we already trust. The payoff is conceptual: helicity, not the Lorentz index, is the natural label for a relativistic amplitude — the observation that the modern on-shell program of Chapter 58 will turn into a calculus.

18.6.1 The spin projection operator

To compute with a single spin state we need the analogue of the spin sum that retains, rather than discards, the polarization. Fix an on-shell momentum p ($p^2 = m^2$, $m > 0$ for now) and a direction of spin. Relativistically the spin

direction is encoded in a *spin four-vector* s^μ satisfying

$$s \cdot p = 0, \qquad s^2 = -1. \tag{18.192}$$

The first condition makes s purely spatial in the particle's rest frame; the second normalizes it. In the rest frame $p = (m, \mathbf{0})$ these force $s = (0, \hat{\mathbf{n}})$ with $\hat{\mathbf{n}}$ a unit spatial vector, the axis along which the spin is measured. The boost of this rest-frame $\hat{\mathbf{n}}$ to the frame where the momentum is $\mathbf{p}$ is what Eq. (18.192) encodes covariantly.

Definition 18.16 (Spin projector)**.** For an on-shell momentum p and a spin four-vector s obeying Eq. (18.192), the *spin projector* is

$$\Sigma(s) \equiv \frac{1 + \gamma^5 \not{s}}{2}. \tag{18.193}$$

The name is justified by the following two facts, which together make $\Sigma(s)$ the exact spin-resolved refinement of the energy projectors of Eq. (C.249).

Proposition 18.17 (Properties and outer-product form of $\Sigma(s)$)**.** *$\Sigma(s)$ is an idempotent, $\Sigma(s)^2 = \Sigma(s)$, with $\Sigma(s) + \Sigma(-s) = \mathbf{1}$; it commutes with $\not{p} \pm m$; and the rank-one outer products of the physical spinors are*

$$u(p,s)\,\bar{u}(p,s) = (\not{p} + m)\,\Sigma(s) = (\not{p} + m)\,\frac{1 + \gamma^5 \not{s}}{2}, \tag{18.194}$$

$$v(p,s)\,\bar{v}(p,s) = (\not{p} - m)\,\Sigma(s) = (\not{p} - m)\,\frac{1 + \gamma^5 \not{s}}{2}, \tag{18.195}$$

where $u(p,s)$ is the positive-frequency spinor whose rest-frame spin points along the $\hat{\mathbf{n}}$ of s, and $v(p,s)$ the corresponding antiparticle spinor.

Proof. Idempotency and completeness. Using $\not{s}\not{s} = s^2 = -1$ and the anticommutation $\gamma^5 \not{s} = -\not{s}\gamma^5$,

$$(\gamma^5 \not{s})^2 = \gamma^5 \not{s} \gamma^5 \not{s} = -\gamma^5 \gamma^5 \not{s}\not{s} = -(\gamma^5)^2 s^2 = -(1)(-1) = +1, \tag{18.196}$$

so $\gamma^5 \not{s}$ squares to the identity and $\Sigma(s) = \frac{1}{2}(1 + \gamma^5 \not{s})$ is therefore idempotent: $\Sigma(s)^2 = \frac{1}{4}(1 + 2\gamma^5 \not{s} + 1) = \frac{1}{2}(1 + \gamma^5 \not{s}) = \Sigma(s)$. Flipping $s \to -s$ flips the sign of the second term, so $\Sigma(s) + \Sigma(-s) = \mathbf{1}$: the two spin states exhaust the relevant subspace.

Commutation with $\not{p} \pm m$. It suffices to show $\gamma^5 \not{s}$ commutes with $\not{p}$. Anticommuting γ^5 through $\not{p}$ once and using $\not{s}\not{p} = 2(s \cdot p) - \not{p}\not{s} = -\not{p}\not{s}$ from Eq. (18.192),

$$\gamma^5 \not{s}\,\not{p} = -\gamma^5 \not{p}\,\not{s} = \not{p}\,\gamma^5 \not{s}, \tag{18.197}$$

where the last step used $\gamma^5 \not{p} = -\not{p}\gamma^5$. Hence $[\gamma^5 \not{s}, \not{p}] = 0$, and trivially $[\gamma^5 \not{s}, m] = 0$, so $\Sigma(s)$ commutes with $\not{p} \pm m$.

The outer products. We prove Eq. (18.194) in the rest frame and propagate it covariantly, exactly as the spin sums were established in Theorem C.68. In the rest frame $p = (m, \mathbf{0})$ the positive-frequency spinors satisfy $\gamma^0 u = u$ (they are the $+1$ eigenstates of γ^0, Section 11.2.5), and $s = (0, \hat{\mathbf{n}})$ gives $\not{s} = -\boldsymbol{\gamma} \cdot \hat{\mathbf{n}}$. We first show that $\gamma^5 \not{s}$ acts on the rest-frame u as the rest-frame spin operator along $\hat{\mathbf{n}}$. From the explicit chiral-basis forms one has the identity $\gamma^5 \gamma^i = -\gamma^0 \Sigma^i$, where $\Sigma^i = \mathrm{diag}(\sigma^i, \sigma^i)$ is the four-dimensional spin matrix of Section 11.2.5; indeed $\gamma^5 \gamma^3 = i\gamma^0\gamma^1\gamma^2\gamma^3\gamma^3 = -i\gamma^0\gamma^1\gamma^2$ while $\gamma^0 \Sigma^3 = \gamma^0\, i\gamma^1\gamma^2$, so the two differ by a sign, and rotational covariance extends the relation to every i. Therefore

$$\gamma^5 \not{s} = -\gamma^5\, \boldsymbol{\gamma} \cdot \hat{\mathbf{n}} = \gamma^0\, \boldsymbol{\Sigma} \cdot \hat{\mathbf{n}}, \tag{18.198}$$

and acting on a rest-frame u, where $\gamma^0 u = u$ and $\boldsymbol{\Sigma}$ commutes with γ^0 (both are block-diagonal),

$$\gamma^5 \not{s}\, u = \gamma^0 (\boldsymbol{\Sigma} \cdot \hat{\mathbf{n}})\, u = (\boldsymbol{\Sigma} \cdot \hat{\mathbf{n}})\, \gamma^0 u = (\boldsymbol{\Sigma} \cdot \hat{\mathbf{n}})\, u. \tag{18.199}$$

Thus on the rest-frame particle subspace $\Sigma(s) = \frac{1}{2}(1 + \boldsymbol{\Sigma} \cdot \hat{\mathbf{n}})$, the ordinary nonrelativistic projector onto spin-up along $\hat{\mathbf{n}}$; it keeps precisely the one spinor $u(p, s)$ and annihilates the opposite-spin $u(p, -s)$. Combining this with the rest-frame spin sum $\sum_{s'} u(p, s')\bar{u}(p, s') = \not{p} + m$ (Eq. (C.247)) and the fact that $\Sigma(s)$ selects one term,

$$u(p, s)\bar{u}(p, s) = \Big(\sum_{s'} u(p, s')\bar{u}(p, s') \Big) \Sigma(s) = (\not{p} + m)\, \Sigma(s), \tag{18.200}$$

where we used that $\Sigma(s)$ acts as the identity on $u(p, s)$ and as zero on $u(p, -s)$, so the only surviving term in the sum is the one we want, written through $\bar{u}(p, s)\Sigma(s) = \bar{u}(p, s)$. Both sides of Eq. (18.200) are now Lorentz-covariant matrices — the left transforms as $S\,(\cdot)\,S^{-1}$ under a boost S, and so does the right because $S(\not{p} + m)S^{-1}$ boosts p and $S\,\gamma^5 \not{s}\, S^{-1} = \gamma^5\, \gamma_\mu (\Lambda s)^\mu$ boosts s (using $S^{-1}\gamma^5 S = \gamma^5$ for proper transformations, Eq. (C.138)). Holding in one frame and transforming covariantly, the identity holds in all frames, which is Eq. (18.194). The antiparticle version Eq. (18.195) follows identically from $\sum_{s'} v\bar{v} = \not{p} - m$, with the sign of $\not{s}$ tracking the convention that the v-spinor's rest-frame spin is labelled by $\hat{\mathbf{n}}$. □

Theorem 18.17 is the entire technology of polarized amplitudes. Wherever an unpolarized computation inserts a spin sum and takes a trace, the polarized computation inserts the rank-one projector Eq. (18.194) instead, and the trace machinery of Section C.6 runs unchanged — now over a string that contains the extra factor $\gamma^5 \not{s}$. Concretely, for any process whose amplitude has the structure $\mathcal{M} = \bar{u}(p_f, s_f)\, \Gamma\, u(p_i, s_i)$ for some Dirac operator Γ built from the Feynman rules, the squared amplitude for fixed external spins is

$$|\mathcal{M}|^2 = \bar{u}(p_f, s_f)\Gamma u(p_i, s_i)\, \bar{u}(p_i, s_i)\bar{\Gamma} u(p_f, s_f) = \mathrm{tr}\Big[(\not{p}_f + m_f)\Sigma(s_f)\, \Gamma\, (\not{p}_i + m_i)\Sigma(s_i)\, \bar{\Gamma} \Big], \tag{18.201}$$

with $\bar{\Gamma} \equiv \gamma^0 \Gamma^\dagger \gamma^0$. Setting $s_i, s_f \to 0$ in Eq. (18.201), i.e. replacing each Σ by $\frac{1}{2}$ and summing the two spin projectors back up, recovers the unpolarized trace; the projector merely refines it. Every parity-odd correlation a polarized experiment can measure — a spin asymmetry, a longitudinal polarization — lives in the $\gamma^5 \not{s}$ pieces that the spin sum had been averaging to zero.

18.6.2 Helicity, chirality, and the massless limit

The spin axis $\hat{\mathbf{n}}$ in Eq. (18.192) is so far arbitrary. The physically natural choice for a moving particle is helicity: spin measured along the direction of motion, $\hat{\mathbf{n}} = \hat{\mathbf{p}}$. Its spin four-vector is obtained by boosting $(0, \hat{\mathbf{p}})$ to the frame of momentum $\mathbf{p}$,

$$s_h^\mu = \Big(\frac{|\mathbf{p}|}{m},\ \frac{E}{m}\,\hat{\mathbf{p}} \Big), \qquad s_h \cdot p = \frac{|\mathbf{p}| E}{m} - \frac{E}{m}\,\hat{\mathbf{p}} \cdot \mathbf{p} = 0, \qquad s_h^2 = \frac{|\mathbf{p}|^2 - E^2}{m^2} = -1, \tag{18.202}$$

where on-shell $E^2 - |\mathbf{p}|^2 = m^2$ gives the normalization. The two helicity states are $\Sigma(\pm s_h)$, and the projector Eq. (18.194) with $s = +s_h$ produces the positive-helicity spinor.

The structure of Eq. (18.202) already signals what happens at high energy: as $m \to 0$ at fixed energy, $|\mathbf{p}| \to E$ and the spin vector becomes

$$s_h^\mu \xrightarrow{m \to 0} \frac{p^\mu}{m}, \tag{18.203}$$

which diverges, but always paired with the m that the spinors carry, so that the combination is finite. Substituting $s_h = p/m$ into Eq. (18.194) and keeping every term as $m \to 0$,

$$u_h \bar{u}_h = (\not{p} + m) \frac{1}{2} \Big(1 + h\,\gamma^5\,\frac{\not{p}}{m} \Big) = \frac{1}{2}\not{p} + \frac{h}{2m} \not{p} \gamma^5 \not{p} + \frac{m}{2} + \frac{h}{2} \gamma^5 \not{p}, \tag{18.204}$$

where $h = \pm 1$ labels the helicity. The middle term contains $\not{p}\gamma^5\not{p} = -\gamma^5 \not{p}\not{p} = -\gamma^5 p^2$, and $p^2 = m^2 \to 0$ for a massless particle, so this term vanishes; the $m/2$ term vanishes directly; and using $\gamma^5 \not{p} = -\not{p}\gamma^5$ in the last term,

$$u_h \bar{u}_h \xrightarrow{m \to 0} \frac{1}{2}\not{p} + \frac{h}{2}\gamma^5 \not{p} = \frac{1 + h\gamma^5}{2}\,\not{p} = \begin{cases} P_R\,\not{p}, & h = +1, \\ P_L\,\not{p}, & h = -1, \end{cases} \tag{18.205}$$

with $P_{R,L} = \frac{1}{2}(1 \pm \gamma^5)$ the chiral projectors of Eq. (11.82) and Section C.3.5. The same computation on the antiparticle outer product Eq. (18.195), where the leading factor is $\not{p} - m$, gives

$$v_h \bar{v}_h \xrightarrow{m \to 0} \frac{1 - h\gamma^5}{2}\,\not{p} = \begin{cases} P_L\,\not{p}, & h = +1, \\ P_R\,\not{p}, & h = -1, \end{cases} \tag{18.206}$$

the opposite chirality assignment being the familiar feature that an antiparticle's helicity is labelled oppositely to its spinor chirality.

Equations (18.205) and (18.206) are the precise statement of a fact glimpsed already in Section 11.2.5 and rooted in the two-component Weyl structure of Section 3.4.3: *for a massless fermion, helicity equals chirality.* A positive-helicity massless particle is purely right-chiral; a negative-helicity one is purely left-chiral. The two notions, distinct and independently defined for $m > 0$ — chirality the Lorentz-invariant but non-conserved eigenvalue of γ^5, helicity the conserved but frame-dependent projection of spin on momentum — fuse into one in the massless limit, because the mass term that mixed the two chiralities in the Dirac equation has switched off. This single fusion is what makes the high-energy limit of QED so much simpler than the general case, and it is the reason the entire spinor-helicity formalism of Chapter 58 is built on chirality eigenstates.

Physical Insight

For $m > 0$ the spin projector Eq. (18.193) and the chiral projectors $P_{L,R}$ are genuinely different operators: $\Sigma(s)$ depends on the chosen axis s, while $P_{L,R}$ are fixed. The massless limit Eq. (18.205) is the statement that, once the axis is taken to be the direction of motion, the spin projector slides onto a chiral projector. Physically, a massless particle moves at the speed of light, so no observer can overtake it to reverse its momentum: helicity becomes Lorentz-invariant, just as chirality always was, and the two labels can no longer be told apart.

18.6.3 The helicity-conservation selection rule

The reason helicity organizes high-energy QED so powerfully is a selection rule that follows from a single algebraic fact about the QED vertex. The vertex is $-ie\gamma^\mu$; every fermion line is a string of gamma matrices, and the question is how chirality flows along it. The answer is governed by the intertwining relation between the chiral projectors and a single gamma matrix,

$$\gamma^\mu P_L = P_R\,\gamma^\mu, \qquad \gamma^\mu P_R = P_L\,\gamma^\mu, \tag{18.207}$$

which is immediate from $\{\gamma^5, \gamma^\mu\} = 0$ (Eq. (11.4)): moving γ^5 past one gamma matrix flips its sign, hence converts $P_L \leftrightarrow P_R$. Consider now the building block of every QED amplitude, the current along a fermion line carrying any odd number of vertices,

$$J^{\mu_1\cdots\mu_{2n+1}} = \bar{u}_2\,\gamma^{\mu_1}\not{q}_1\gamma^{\mu_2}\cdots\gamma^{\mu_{2n+1}}u_1, \tag{18.208}$$

the gammas coming from vertices and the $\not{q}_i$ from internal propagator numerators. (Internal propagators contribute $\not{q}_i + m$, but in the massless limit only the $\not{q}_i$

survives.) Decompose the incoming spinor into chiralities, $u_1 = P_L u_1 + P_R u_1$, and push, say, the P_L piece leftward through the string. Each gamma matrix it passes flips $P_L \leftrightarrow P_R$ by Eq. (18.207); passing through an *odd* total number of gamma matrices — $2n+1$ vertices plus $2n$ internal slashes is an odd count of gammas only when... let us count precisely.

Proposition 18.18 (Chirality is conserved along a fermion line)**.** *In the massless limit, a fermion line connecting an incoming spinor u_1 to an outgoing $\bar{u}_2$ through any number of QED vertices and propagators is nonzero only when u_1 and u_2 have the* same *chirality. Equivalently, the matrix element $\bar{u}_2 \,\Gamma\, u_1$ with Γ a product of an odd number of gamma matrices satisfies*

$$\bar{u}_2 \,\Gamma\, u_1 = \bar{u}_2 \, P_L \,\Gamma\, P_R \, u_1 + \bar{u}_2 \, P_R \,\Gamma\, P_L \, u_1, \tag{18.209}$$

so left-chiral and right-chiral incoming states map only to left- and right-chiral outgoing states respectively.

Proof. Each vertex and each (massless) propagator numerator contributes exactly one gamma matrix to the string Γ; a connected line therefore carries an odd number of gamma matrices, one per vertex plus one per internal segment, and for a line with V vertices and $V-1$ internal segments the total is $2V-1$, odd. Take Γ a product of an odd number of gammas and insert $\mathbf{1} = P_L + P_R$ on each side:

$$\bar{u}_2 \Gamma u_1 = \bar{u}_2 (P_L + P_R) \Gamma (P_L + P_R) u_1. \tag{18.210}$$

By Eq. (18.207), moving a projector through an odd number of gammas converts it to the opposite handedness: $P_R \Gamma = \Gamma P_L$ and $P_L \Gamma = \Gamma P_R$. Hence the same-handed terms vanish, $P_R \Gamma P_R = \Gamma P_L P_R = 0$ and $P_L \Gamma P_L = \Gamma P_R P_L = 0$, while the opposite-handed terms $P_L \Gamma P_R = \Gamma P_R$ and $P_R \Gamma P_L = \Gamma P_L$ survive, leaving Eq. (18.209). Note $\bar{u}_2 P_L = \overline{P_R u_2}$, so the surviving terms pair a right-chiral incoming u_1 with a right-chiral outgoing u_2 and likewise for left. Since for massless fermions chirality equals helicity by Eq. (18.205), the incoming and outgoing helicities along the line must agree. □

This is the helicity-conservation selection rule, and it has a sharp consequence for any amplitude with external photons or for an annihilation current. For a fermion that is scattered — the same particle in and out, as on the electron line of Compton or Møller — the rule says its helicity is preserved by every QED interaction in the massless limit: chirality flows through the vertex untouched. For a fermion and antifermion that *annihilate* into (or are created from) a photon, the rule is read on the single current $\bar{v}(p')\gamma^\mu u(p)$: inserting projectors and using Eq. (18.207),

$$\bar{v}(p')\gamma^\mu u(p) = \bar{v}(p') \, P_L \gamma^\mu P_R \, u(p) + \bar{v}(p') \, P_R \gamma^\mu P_L \, u(p), \tag{18.211}$$

where $\bar{v}\,P_L = \overline{P_R v}$ and the structure forces the electron and positron to carry *opposite* helicities (a right-chiral u pairs with a right-chiral v, which is a *negative*-helicity positron by Eq. (18.206)). Physically this is angular momentum conservation: the intermediate photon has spin 1, so the e^+e^- pair, collinear along the beam, must carry one unit of angular momentum along that axis — their spins must be parallel, hence their helicities, defined relative to opposite momenta, must be opposite. Of the four possible initial helicity combinations, only the two with opposite electron and positron helicity survive; the other two vanish identically in the massless limit.

18.6.4 The helicity amplitudes of $e^+e^- \to \mu^+\mu^-$

We now resolve the chapter's reference process into its helicity components and verify that they rebuild the unpolarized cross section established in Section 17.3. We continue QED with the exact model and conventions fixed there: incoming $e^-(p)\,e^+(p')$ annihilate through a single s-channel photon into $\mu^-(k)\,\mu^+(k')$, and we work in the high-energy limit $s \gg m_e^2, m_\mu^2$ where all masses are neglected.

Worked Example 18.19 (Polarized $e^+e^- \to \mu^+\mu^-$ and its incoherent sum)**.** The amplitude is the one diagram of Section 17.3,

$$\mathcal{M} = \frac{e^2}{s}\,[\bar{v}(p')\gamma^\mu u(p)]\,[\bar{u}(k)\gamma_\mu v(k')], \qquad s = (p+p')^2, \tag{18.212}$$

with the electron charge $-e$ ($e>0$) and Feynman gauge for the photon propagator. We place the collision in the centre-of-mass frame with the electron along $+z$ and the outgoing muon at polar angle θ,

$$p = \tfrac{\sqrt{s}}{2}(1,0,0,1), \quad p' = \tfrac{\sqrt{s}}{2}(1,0,0,-1), \quad k = \tfrac{\sqrt{s}}{2}(1,\sin\theta,0,\cos\theta), \quad k' = \tfrac{\sqrt{s}}{2}(1,-\sin\theta,0,- \tag{18.213}$$

all massless. The invariants are

$$s = 2p{\cdot}p' = 2k{\cdot}k', \qquad t = (p-k)^2 = -2p{\cdot}k = -\tfrac{s}{2}(1-\cos\theta), \qquad u = (p-k')^2 = -2p{\cdot}k' = -\tfrac{s}{2} \tag{18.214}$$

so that $s+t+u=0$ and $p\cdot k' = p'\cdot k = -u/2$, $p\cdot k = p'\cdot k' = -t/2$.

Selection rule. By Theorem 18.18 applied to the annihilation current Eq. (18.211), the electron and positron must have opposite helicities, and the same holds for the muon pair on the other current. The surviving configurations are the four

$$(e^-_R e^+_L),(e^-_L e^+_R) \;\otimes\; (\mu^-_R\mu^+_L),(\mu^-_L\mu^+_R), \tag{18.215}$$

where the subscript on a positron/antimuon L denotes its physical (negative) helicity; the remaining twelve of the sixteen helicity combinations vanish.

One amplitude in full. Take $e^-_R e^+_L \to \mu^-_R\mu^+_L$. With helicities fixed, no spin sum is needed: the outer products are the massless projector forms Eqs. (18.205)

and (18.206). For the right-handed electron, $u(p)\bar{u}(p) = P_R \not{p}$; for its negative-helicity positron partner, the selection rule requires $v(p')\bar{v}(p') = P_R \not{p}'$. The electron tensor is

$$L^{\mu\nu} = \big[\bar{v}(p')\gamma^\mu u(p)\big]\big[\bar{v}(p')\gamma^\nu u(p)\big]^* = \mathrm{tr}\big[\gamma^\mu\, u(p)\bar{u}(p)\,\gamma^\nu\, v(p')\bar{v}(p')\big] = \mathrm{tr}\big[\gamma^\mu P_R \not{p}\,\gamma^\nu P_R \not{p}'\big]. \tag{18.216}$$

Slide the projectors together with Eq. (18.207): $P_R \not{p} = \not{p} P_L$, then $P_L \gamma^\nu = \gamma^\nu P_R$ and $P_R P_R = P_R$, so the two projectors collapse to one, $L^{\mu\nu} = \mathrm{tr}[\gamma^\mu \not{p}\,\gamma^\nu \not{p}' P_L]$ with $P_L = \frac{1}{2}(1-\gamma^5)$. Using the four-gamma trace Eq. (C.178) and the γ^5 four-trace Eq. (C.179) (with $\epsilon^{0123} = +1$),

$$\begin{aligned} L^{\mu\nu} &= \tfrac{1}{2}\,\mathrm{tr}\,[\gamma^\mu \not{p}\,\gamma^\nu \not{p}'] - \tfrac{1}{2}\,\mathrm{tr}\,[\gamma^\mu \not{p}\,\gamma^\nu \not{p}'\gamma^5] \\ &= 2(p^\mu p'^\nu + p^\nu p'^\mu - \eta^{\mu\nu}\, p\cdot p') + 2i\,\epsilon^{\mu\alpha\nu\beta} p_\alpha p'_\beta, \end{aligned} \tag{18.217}$$

the antisymmetric ϵ-term being the signature of the definite chirality. The muon current is the mirror computation with $u(k)\bar{u}(k) = P_R \not{k}$ and $v(k')\bar{v}(k') = P_R \not{k}'$,

$$M_{\mu\nu} = \mathrm{tr}\big[\gamma_\mu P_R \not{k}'\,\gamma_\nu P_R \not{k}\big] = 2(k'_\mu k_\nu + k'_\nu k_\mu - \eta_{\mu\nu}\, k\cdot k') + 2i\,\epsilon_{\mu\rho\nu\sigma} k'^\rho k^\sigma. \tag{18.218}$$

The squared amplitude is $|\mathcal{M}|^2 = (e^4/s^2)\, L^{\mu\nu} M_{\mu\nu}$. The contraction splits into symmetric×symmetric and antisymmetric×antisymmetric pieces (the cross terms vanish, symmetric against antisymmetric in the index pair). The symmetric part is

$$L^{\mu\nu}_{\mathrm{sym}} M^{\mathrm{sym}}_{\mu\nu} = 8\big[(p\cdot k)(p'\cdot k') + (p\cdot k')(p'\cdot k)\big], \tag{18.219}$$

obtained by expanding the nine products and using $\eta^{\mu\nu}\eta_{\mu\nu} = 4$, which cancels the $(p\cdot p')(k\cdot k')$ terms. The antisymmetric part uses the Levi-Civita contraction Eq. (C.167), $\epsilon^{\mu\alpha\nu\beta}\epsilon_{\mu\rho\nu\sigma} = -2(\delta^\alpha_\rho\delta^\beta_\sigma - \delta^\alpha_\sigma\delta^\beta_\rho)$,

$$\begin{aligned} L^{\mu\nu}_{\mathrm{asym}} M^{\mathrm{asym}}_{\mu\nu} &= (2i)(2i)\,\epsilon^{\mu\alpha\nu\beta} p_\alpha p'_\beta\, \epsilon_{\mu\rho\nu\sigma} k'^\rho k^\sigma \\ &= -4\cdot(-2)(\delta^\alpha_\rho\delta^\beta_\sigma - \delta^\alpha_\sigma\delta^\beta_\rho) p_\alpha p'_\beta k'^\rho k^\sigma = 8\big[(p\cdot k')(p'\cdot k) - (p\cdot k)(p'\cdot k')\big]. \end{aligned} \tag{18.220}$$

Adding Eqs. (18.219) and (18.220), the $(p\cdot k)(p'\cdot k')$ terms cancel and the $(p\cdot k')(p'\cdot k)$ terms add:

$$L^{\mu\nu} M_{\mu\nu} = 16\,(p\cdot k')(p'\cdot k) = 16\cdot\frac{(-u)}{2}\cdot\frac{(-u)}{2} = 4u^2, \tag{18.221}$$

so that, using $-u = \frac{s}{2}(1+\cos\theta)$,

$$\big|\mathcal{M}(e^-_R e^+_L \to \mu^-_R \mu^+_L)\big|^2 = \frac{e^4}{s^2}\cdot 4u^2 = \frac{4e^4u^2}{s^2} = e^4(1+\cos\theta)^2. \tag{18.222}$$

The amplitude is forward-peaked: it is maximal at $\theta = 0$, where the outgoing μ^-_R continues in the direction of the incoming e^-_R with matched helicity, and it vanishes at $\theta = \pi$.

The other three by the same machinery. Flipping the muon-pair helicity to $\mu_L^- \mu_R^+$ replaces $u(k)\bar{u}(k) = P_L \not{k}$, which flips the sign of the muon tensor's ϵ-term; the antisymmetric contraction Eq. (18.220) then reverses sign, the surviving combination becomes $16(p \cdot k)(p' \cdot k) = 4t^2$, and

$$|\mathcal{M}(e_R^- e_L^+ \to \mu_L^- \mu_R^+)|^2 = \frac{4e^4 t^2}{s^2} = e^4(1 - \cos\theta)^2, \tag{18.223}$$

backward-peaked. Flipping instead the initial pair to $e_L^- e_R^+$ flips the electron tensor's ϵ-term; parity invariance of QED (the Lagrangian is parity-even) guarantees that flipping all four helicities at once returns the original result, so the four amplitudes pair up:

$$|\mathcal{M}_{LR\to LR}|^2 = |\mathcal{M}_{RL\to RL}|^2 = \frac{4e^4 u^2}{s^2}, \qquad |\mathcal{M}_{LR\to RL}|^2 = |\mathcal{M}_{RL\to LR}|^2 = \frac{4e^4 t^2}{s^2}, \tag{18.224}$$

where the compact subscript $RL \to RL$ means $e_R^- e_L^+ \to \mu_R^- \mu_L^+$ and so on.

Incoherent sum. The four helicity configurations are physically distinct final and initial states; they do not interfere, and the unpolarized squared amplitude is their average over the four initial spin states (factor $\frac{1}{4}$) and sum over final spins:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\Big[\,2 \cdot \frac{4e^4 u^2}{s^2} + 2 \cdot \frac{4e^4 t^2}{s^2}\Big] = \frac{2e^4\,(t^2 + u^2)}{s^2}. \tag{18.225}$$

With $t^2 + u^2 = \frac{s^2}{4}[(1 - \cos\theta)^2 + (1 + \cos\theta)^2] = \frac{s^2}{2}(1 + \cos^2\theta)$ this is

$$\overline{|\mathcal{M}|^2} = e^4\,(1 + \cos^2\theta), \tag{18.226}$$

exactly the spin-averaged result of Section 17.3. The averaging that earlier appeared as a single opaque trace is now resolved: the smooth $1 + \cos^2\theta$ is the incoherent superposition of two forward-peaked $(1 + \cos\theta)^2$ amplitudes and two backward-peaked $(1 - \cos\theta)^2$ amplitudes, whose sum is forward–backward symmetric only because QED treats the two beam helicities even-handedly.

The internal consistency check is complete, but the deeper lesson is in the intermediate results Eqs. (18.222) and (18.223). Each helicity amplitude is a single monomial in the Mandelstam invariants — $|\mathcal{M}|^2 \propto u^2$ or $\propto t^2$ — vastly simpler than the unpolarized $t^2 + u^2$, and simpler still at the level of the amplitude itself, which is just $\mathcal{M} \propto u$ or $\mathcal{M} \propto t$ up to a phase. The trace machinery, with its symmetric and antisymmetric tensor pieces and its ϵ-contraction, was a heavy route to a light answer. That the polarized amplitudes are so simple is the first indication that the spinor and the Lorentz vector are the wrong variables, and that a formalism built directly on helicity — the spinor-helicity calculus of Chapter 58 — will write these amplitudes in a single line.

18.6.5 Angular distributions and forward–backward structure

The two angular shapes $(1 \pm \cos\theta)^2$ are not coincidences of QED; they are dictated by angular momentum. The s-channel photon carries total angular momentum $J = 1$ along no fixed axis, but the collinear initial state has a definite $J_z = \lambda_{e^-} - \lambda_{e^+} = \pm 1$ along the beam, and the collinear final state a definite $J_{z'} = \lambda_{\mu^-} - \lambda_{\mu^+} = \pm 1$ along the outgoing muon direction. The amplitude is therefore proportional to a single Wigner rotation matrix element connecting these,

$$\mathcal{M} \;\propto\; d^1_{\lambda\lambda'}(\theta), \qquad d^1_{1,1}(\theta) = d^1_{-1,-1}(\theta) = \frac{1+\cos\theta}{2}, \qquad d^1_{1,-1}(\theta) = d^1_{-1,1}(\theta) = \frac{1-\cos\theta}{2}, \tag{18.227}$$

the $J = 1$ rotation functions familiar from the addition of angular momentum. Squaring reproduces $(1 \pm \cos\theta)^2$ exactly: the helicity amplitudes *are* the rotation matrix elements of the intermediate spin-1 state, and QED supplies only the overall strength e^2 and the propagator $1/s$. This is why the same $(1 \pm \cos\theta)^2$ shapes appear in every spin-1-mediated annihilation; the dynamics sets the normalization, the kinematics of angular momentum sets the shape.

A single helicity configuration is strongly forward–backward asymmetric. Defining, for fixed initial helicities, the forward–backward asymmetry

$$A_{\mathrm{FB}} = \frac{\sigma(\cos\theta > 0) - \sigma(\cos\theta < 0)}{\sigma(\cos\theta > 0) + \sigma(\cos\theta < 0)}, \tag{18.228}$$

the $e^-_R e^+_L$ initial state, weighted by $\int_0^1 (1+\cos\theta)^2\, d\cos\theta$ forward against $\int_{-1}^0$ backward (the muon helicities summed), gives

$$A_{\mathrm{FB}}(e^-_R e^+_L) = \frac{\int_0^1 [(1+c)^2 + (1-c)^2]\, dc - \int_{-1}^0 [(1+c)^2 + (1-c)^2]\, dc}{\int_{-1}^1 [(1+c)^2 + (1-c)^2]\, dc} = 0, \tag{18.229}$$

because summing the two muon helicities already restores the symmetry. The asymmetry is exposed only by measuring the final helicity as well: for the pure transition $e^-_R e^+_L \to \mu^-_R \mu^+_L$,

$$A_{\mathrm{FB}} = \frac{\int_0^1 (1+c)^2 dc - \int_{-1}^0 (1+c)^2 dc}{\int_{-1}^1 (1+c)^2 dc} = \frac{\frac{7}{3} - \frac{1}{3}}{\frac{8}{3}} = \frac{3}{4}, \tag{18.230}$$

a large asymmetry that the unpolarized average Eq. (18.226), with its vanishing $\cos\theta$-odd part, completely hides. In pure QED, parity invariance guarantees that the fully unpolarized A_{FB} vanishes; a nonzero unpolarized forward–backward asymmetry is the hallmark of parity violation, and its measurement in $e^+e^- \to \mu^+\mu^-$ at higher energies is one of the cleanest signatures of the weak neutral current interfering with the photon — a story that belongs to the electroweak chapters but whose kinematic skeleton is precisely the helicity analysis done here. Figure 18.7 displays the decomposition.

Physical Insight

The spin-averaged $1 + \cos^2\theta$ looks like a smooth, structureless law; the helicity decomposition reveals it as the sum of two sharply directed processes. This is the recurring moral of polarized amplitudes: averaging is lossy, and what it loses — the correlation between an incoming spin and an outgoing direction — is often the most incisive observable. The same logic, applied to processes with external photons such as Compton scattering, shows that the photon's two physical polarizations are just its two helicities, and that the gauge redundancy of the polarization vector is invisible to a helicity amplitude. Helicity is the label in which gauge-invariant content is manifest.

We have now seen helicity do three things: refine the spin sum into a projector (Theorem 18.17), fuse with chirality at high energy (Eq. (18.205)), and select and simplify the surviving amplitudes (Theorem 18.18, Theorem 18.19). Each polarized amplitude turned out to be a single monomial, far simpler than the spin-summed total, and tied directly to the rotation functions of the intermediate spin. This is the empirical case that helicity, not the Lorentz index, is the amplitude's natural variable. The synthesis that closes this chapter will gather this observation together with Ward invariance and crossing into one picture of the tree-level QED amplitude, and will make explicit the forward pointer that the present section has earned: the spinor-helicity reframing of Chapter 58, where each of the four lines computed here collapses to a ratio of angle and square brackets and the trace technology disappears entirely.

18.7 Synthesis: the QED tree amplitude as one analytic object

We have now computed, with full traces and spin sums, the four processes that this chapter set out to confront: Compton scattering, pair annihilation, Møller scattering, and Bhabha scattering, together carrying the running thread forward from the single-diagram $e^+e^- \to \mu^+\mu^-$ amplitude of Section 17.3. Along the way we proved the tree-level Ward identity (Section 18.2), used it to justify the covariant photon polarization sum, established the crossing rules (Section 18.4) that continue one process into another, and finally resolved the amplitudes by helicity (Section 18.6), watching the spin-resolved pieces add up to the unpolarized squares. Each computation was presented as a self-contained event. The purpose of this closing section is to step back and see that they are not separate events at all: the entire tree-level QED amplitude family is a small number of analytic functions of the Mandelstam invariants, whose various boundary values are the physical processes, whose pole and gauge structure is fixed by the Ward identity,

and whose internal organization is helicity. Assembling this single picture is what makes the chapter cohere, and it is the picture that the loop chapter, the precision chapter, and ultimately the modern-amplitudes program will deform, refine, and exploit.

18.7.1 Tree amplitudes as boundary values of analytic functions

The first thing to make precise is the sense in which distinct physical processes are "the same amplitude." A tree-level Feynman diagram is a product of vertices $(-ie\gamma^\mu)$, internal propagators, and external wavefunctions, contracted into Lorentz invariants. For a $2 \to 2$ process the only invariants available are the Mandelstam variables

$$s = (p_1 + p_2)^2, \qquad t = (p_1 - p_3)^2, \qquad u = (p_1 - p_4)^2, \qquad s + t + u = \sum_i m_i^2, \tag{18.231}$$

with p_1, p_2 incoming and p_3, p_4 outgoing (recall Section 16.5). A tree diagram has a single internal line; its propagator denominator is one of $s - m^2$, $t - m^2$, or $u - m^2$. Therefore the spin-summed squared amplitude of any tree-level $2 \to 2$ QED process is a *rational function of s, t, u whose only singularities are simple poles at the physical thresholds of the exchanged particle*, with numerators that are polynomials of bounded degree fixed by the spins of the external legs. This is the structural fact behind crossing, and it deserves a precise statement.

Proposition 18.20 (Tree QED amplitudes are boundary values of a few rational functions)**.** *Fix the external particle content of a tree-level QED $2 \to 2$ process up to which legs are incoming and which are outgoing. The spin-summed quantity $\sum_{\text{spins}} |\mathcal{M}|^2$ is the restriction, to the physical region of that channel assignment, of a single rational function $R(s, t, u)$ on the surface (18.231). Two processes built from the same set of fields, differing only by the in/out assignment of the external legs, are restrictions of the* same *R to different physical regions; they are related by permuting s, t, u (and flipping the overall sign once per crossed fermion pair, from the spinor completeness relations $\sum u\bar{u} = \not{p} + m$ versus $\sum v\bar{v} = \not{p} - m$). The poles of R sit only at $c = m^2$ for the channel invariant c carried by an exchangeable particle of mass m.*

Proof. The amplitude $\mathcal{M}$ for a given diagram is a finite sum of terms, each a product of: external Dirac spinors and polarization vectors; vertex factors $-ie\gamma^\mu$; and propagator numerators ($\not{q} + m$ for a fermion of momentum q, $-\eta_{\mu\nu}$ for a Feynman-gauge photon) divided by denominators $q^2 - m^2 + i\varepsilon$. Each internal momentum q is a fixed integer combination of the external momenta, so q^2 equals one of s, t, u up to a shift by masses; the denominators are therefore among $\{s - m^2, t - m^2, u - m^2\}$. Squaring and summing over spins replaces external $u\bar{u}$ and $v\bar{v}$ by $\not{p} \pm m$ and collapses every polarization sum to $-\eta_{\mu\nu}$

(the Ward identity, Theorem 18.5, guarantees the gauge-dependent remainder of the photon sum drops out). The result is a trace of gamma matrices times rational denominators, i.e. a polynomial in scalar products $p_i \cdot p_j$ over the product of the denominators. Every $p_i \cdot p_j$ is a linear combination of s, t, u and masses by (18.231), so $\sum_{\text{spins}} |\mathcal{M}|^2 = R(s,t,u)$ is rational with poles only where a denominator vanishes, i.e. at $c = m^2$. Relabelling which external momenta are incoming is, at the level of these formulas, exactly a permutation of which linear combination is called s, t, or u, together with $p_i \to -p_i$ on the crossed legs; the latter sends $\not{p} + m \to -(\not{p} - m)$, i.e. it exchanges the u- and v-type completeness relations and contributes a sign (-1) for each crossed fermion line. Hence the two processes are restrictions of one R up to that sign. □

Theorem 18.20 is the analyticity half of the picture; the Ward identity is the gauge half; helicity is the organizing half. The remainder of this section makes each concrete by assembling the chapter's explicit results into one summary.

18.7.2 The crossing web

It is convenient to work in the massless (high-energy) limit, $m \to 0$, where $s + t + u = 0$ and the rational functions R take their simplest, most transparent form; the massive expressions computed earlier in the chapter reduce to these and were each checked to do so. Collecting the spin-averaged results — averaged over initial spins/polarizations, summed over final — we have the following six entries, the four new processes of this chapter together with the two from Chapter 17 that seed them.

The single-photon four-fermion family. For processes with two distinct charged-fermion lines exchanging one photon,

$$e^+e^- \to \mu^+\mu^- : \qquad \overline{|\mathcal{M}|^2} = 2e^4\,\frac{t^2+u^2}{s^2}, \tag{18.232}$$

$$e^-\mu^- \to e^-\mu^- : \qquad \overline{|\mathcal{M}|^2} = 2e^4\,\frac{s^2+u^2}{t^2}. \tag{18.233}$$

The first is the s-channel annihilation diagram of Section 17.3; the second is the t-channel scattering diagram, and the two are manifestly one function continued by $s \leftrightarrow t$. When the two fermion lines are made identical (electrons), Fermi statistics forces a second diagram and a relative minus sign between the two, producing Møller and Bhabha:

$$e^-e^- \to e^-e^- \text{ (Møller)} : \quad \overline{|\mathcal{M}|^2} = 2e^4\left[\frac{s^2+u^2}{t^2} + \frac{s^2+t^2}{u^2} + \frac{2s^2}{tu}\right], \tag{18.234}$$

$$e^+e^- \to e^+e^- \text{ (Bhabha)} : \quad \overline{|\mathcal{M}|^2} = 2e^4\left[\frac{t^2+u^2}{s^2} + \frac{s^2+u^2}{t^2} + \frac{2u^2}{st}\right]. \tag{18.235}$$

Process	Channels	Crossing	$\overline{\lvert\mathcal{M}\rvert^2}/2e^4$ (massless)
$e^+e^- \to \mu^+\mu^-$	s	seed (Section 17.3)	$\dfrac{t^2+u^2}{s^2}$
$e^-\mu^- \to e^-\mu^-$	t	$s \leftrightarrow t$ of above	$\dfrac{s^2+u^2}{t^2}$
Møller $e^-e^- \to e^-e^-$	t, u	identical-leg of $e^-\mu^-$	$\dfrac{s^2+u^2}{t^2} + \dfrac{s^2+t^2}{u^2} + \dfrac{2s^2}{tu}$
Bhabha $e^+e^- \to e^+e^-$	s, t	$s \leftrightarrow u$ of Møller	$\dfrac{t^2+u^2}{s^2} + \dfrac{s^2+u^2}{t^2} + \dfrac{2u^2}{st}$
Compton $e^-\gamma \to e^-\gamma$	s, u	seed (Compton master)	$-\left(\dfrac{s}{u} + \dfrac{u}{s}\right)$
pair annih. $e^+e^- \to \gamma\gamma$	t, u	$s \leftrightarrow t$ of Compton	$+\left(\dfrac{t}{u} + \dfrac{u}{t}\right)$

Table 18.3: The crossing web of the chapter's tree-level processes. "Channels" lists the Mandelstam invariants in which an internal line goes on shell. Every entry is a boundary value of one of two master functions. The Compton-family sign flip between rows five and six is the single-fermion-crossing sign of Theorem 18.20.

The Compton family. For processes with one charged-fermion line and two external photons, exchanging a virtual electron,

$$e^-\gamma \to e^-\gamma \text{ (Compton)}: \qquad \overline{|\mathcal{M}|^2} = -2e^4\left(\frac{s}{u} + \frac{u}{s}\right), \tag{18.236}$$

$$e^+e^- \to \gamma\gamma \text{ (pair annihilation)}: \qquad \overline{|\mathcal{M}|^2} = 2e^4\left(\frac{t}{u} + \frac{u}{t}\right). \tag{18.237}$$

Here $e^2 = 4\pi\alpha$, so $e^4 = 16\pi^2\alpha^2$. Each of these six expressions was derived in its own section; collected together they reveal the web shown in Fig. 18.8 and tabulated in Table 18.3. The web has exactly two connected components, one per master function, and within each component the entries are related by simple permutations of (s, t, u).

The four-fermion entries (18.232)–(18.235) are not merely *related*; they are assembled from one universal building block according to a rule that the explicit traces make obvious in hindsight.

Proposition 18.21 (Universal channel rule for single-photon four-fermion amplitudes)**.** *Let a tree QED process scatter two charged-fermion lines through single-photon exchange, and let $\mathcal{C} \subseteq \{s, t, u\}$ be the set of Mandelstam channels in which a photon can be exchanged for the given in/out assignment. In the massless limit,*

$$\overline{|\mathcal{M}|^2} = 2e^4\left[\sum_{c\in\mathcal{C}} \frac{\sigma_2(c)}{c^2} \;+\; [\,|\mathcal{C}| = 2\,]\,\frac{2\,w^2}{c_1\,c_2}\right], \tag{18.238}$$

where $\sigma_2(c)$ denotes the sum of the squares of the two invariants other than c, the bracket $[\,\cdot\,]$ is 1 when its condition holds and 0 otherwise, and for a two-channel process $\mathcal{C} = \{c_1, c_2\}$ the interference weight w is the single invariant absent from $\mathcal{C}$. The sign of the interference term is fixed by Fermi statistics and is positive in this invariant form.

Proof. A single c-channel photon-exchange diagram contributes $\mathcal{M}_c = (-ie)^2\,[\bar{u}\gamma^\mu u][\bar{u}\gamma_\mu u]/c$, the two currents being the two fermion lines and c the squared momentum on the photon. Squaring and summing over spins produces e^4/c^2 times a product of two traces, $\mathrm{tr}(\not{p}_a\gamma^\mu\not{p}_b\gamma^\nu)\ \mathrm{tr}(\not{p}_{a'}\gamma_\mu\not{p}_{b'}\gamma_\nu)$, each trace giving $4(p_a^\mu p_b^\nu + p_a^\nu p_b^\mu - \eta^{\mu\nu}p_a\cdot p_b)$. Contracting the two and using $2p_i\cdot p_j \in \{\pm s, \pm t, \pm u\}$ (massless) collapses, after averaging over the four initial spin states, to $2e^4\,\sigma_2(c)/c^2$: this is exactly the computation behind (18.232) for $c = s$ and (18.233) for $c = t$, where $\sigma_2(s) = t^2 + u^2$ and $\sigma_2(t) = s^2 + u^2$. When two channels are present the amplitude is $\mathcal{M}_{c_1} \pm \mathcal{M}_{c_2}$ (the sign $-$ when the two diagrams differ by the exchange of identical external fermions), so $\overline{|\mathcal{M}|^2}$ acquires, besides the two diagonal pieces, the cross term $\pm 2e^4\,\overline{\mathrm{Re}(\mathcal{M}_{c_1}\mathcal{M}_{c_2}^*)}$. Evaluating the single combined trace — now one closed fermion loop running through both photon vertices — gives $4e^4\,w^2/(c_1c_2)$ with w the absent invariant, and the statistics sign combines with the sign of that trace to make the contribution $+2e^4\cdot 2w^2/(c_1c_2)$. For Møller, $\mathcal{C} = \{t, u\}$ and $w = s$, reproducing (18.234); for Bhabha, $\mathcal{C} = \{s, t\}$ and $w = u$, reproducing (18.235). □

Theorem 18.21 compresses four separate trace computations into a single counting statement: tell me which channels carry a photon, and the spin-averaged rate is determined. It also exposes the physics of each interference. In Møller the absent invariant is s, so the interference $\propto s^2/tu$ is large precisely at fixed angle and high energy, where s dominates; in Bhabha the absent invariant is u, and the interference $\propto u^2/st$ carries the destructive cross-talk between annihilation and scattering that flips sign across the physical region because $st < 0$ there. These are not coincidences to be memorized but consequences of which diagrams Fermi statistics permits.

Worked Example 18.22 (Verifying the crossing web by substitution)**.** We check that the six entries (18.232)–(18.237) are mutually consistent under crossing, exactly as Theorem 18.20 requires. No new trace is computed; this is a consistency audit of results already in hand.

(i) $e^+e^- \to \mu^+\mu^-$ to $e^-\mu^- \to e^-\mu^-$. Crossing the incoming μ^+ to an outgoing μ^- (and the outgoing μ^- to an incoming μ^-) turns the s-channel annihilation into t-channel scattering, which at the level of invariants is $s \leftrightarrow t$. Applying $s \leftrightarrow t$ to (18.232),

$$2e^4\,\frac{t^2+u^2}{s^2} \quad \xrightarrow{s\leftrightarrow t} \quad 2e^4\,\frac{s^2+u^2}{t^2}, \tag{18.239}$$

which is (18.233). Two fermion lines are crossed (an even number), so there is no overall sign; both sides are positive, as required for a squared amplitude.

(ii) Møller to Bhabha. Bhabha is obtained from Møller by turning one incoming electron into an outgoing positron and one outgoing electron into an incoming positron. The Møller channels are $\{t, u\}$ with absent invariant s; the Bhabha channels are $\{s, t\}$ with absent invariant u, so the no-pole channel moves $s \to u$, i.e. the crossing is $s \leftrightarrow u$. Applying it to (18.234),

$$2e^4\left[\frac{s^2+u^2}{t^2}+\frac{s^2+t^2}{u^2}+\frac{2s^2}{tu}\right] \xrightarrow{s\leftrightarrow u} 2e^4\left[\frac{u^2+s^2}{t^2}+\frac{u^2+t^2}{s^2}+\frac{2u^2}{ts}\right], \tag{18.240}$$

where the first numerator is unchanged (it is symmetric in s, u), the second term's denominator becomes s^2, and the interference denominator $tu \to ts = st$. Reordering,

$$2e^4\left[\frac{t^2+u^2}{s^2}+\frac{s^2+u^2}{t^2}+\frac{2u^2}{st}\right], \tag{18.241}$$

which is precisely Bhabha, (18.235). Again an even number of fermion crossings, hence no sign, and both results are positive.

(iii) Compton to pair annihilation. Crossing the incoming Compton photon to an outgoing photon and the outgoing Compton electron to an incoming positron converts $e^-\gamma \to e^-\gamma$ into $e^+e^- \to \gamma\gamma$. The Compton s-channel electron pole, $s = (p_e + k)^2$, becomes the annihilation t-channel electron pole, $t = (p_e - k')^2$, while the Compton u-channel pole is carried to the annihilation u-channel pole; thus the crossing is $s \leftrightarrow t$. Applying it to (18.236),

$$-2e^4\left(\frac{s}{u}+\frac{u}{s}\right) \xrightarrow{s\leftrightarrow t} -2e^4\left(\frac{t}{u}+\frac{u}{t}\right). \tag{18.242}$$

Exactly *one* fermion line is crossed here (the electron → positron; the photon is a boson and contributes no sign), so by Theorem 18.20 the result picks up one factor (-1):

$$(-1)\times\left[-2e^4\left(\frac{t}{u}+\frac{u}{t}\right)\right] = +2e^4\left(\frac{t}{u}+\frac{u}{t}\right), \tag{18.243}$$

which is the pair-annihilation result (18.237). The lone fermion-crossing sign is the origin of the relative minus between the prefactors of (18.236) and (18.237), and the fact that it lands the annihilation rate at a manifestly positive value (recall $t, u < 0$ in the physical $e^+e^- \to \gamma\gamma$ region, so $t/u > 0$) is a nontrivial check on the whole sign bookkeeping.

Every arrow of Fig. 18.8 is thereby verified: the six physical squared amplitudes are boundary values of two analytic functions, in complete agreement with Theorem 18.20.

18.7.3 The Ward identity as the constraint that holds it together

Crossing is the analyticity face of the synthesis; the Ward identity is the gauge face, and the two are not independent. Recall the tree-level statement proved earlier (Theorem 18.5): if $\mathcal{M} = \epsilon_\mu(k)\,\mathcal{M}^\mu(k)$ is any QED amplitude with an on-shell external photon of momentum k, then replacing the polarization by the momentum annihilates it,

$$k_\mu\,\mathcal{M}^\mu(k) = 0. \tag{18.244}$$

This single identity does three jobs simultaneously, and seeing them as one is the point of the synthesis.

First, it *licenses the covariant polarization sum.* In summing $|\mathcal{M}|^2$ over physical photon polarizations one would like to use $\sum_\lambda \epsilon_\mu^{(\lambda)*}\epsilon_\nu^{(\lambda)} \to -\eta_{\mu\nu}$ rather than the cumbersome physical-state projector. The difference between the two is a sum of terms each proportional to k_μ or k_ν, which by (18.244) contract to zero against $\mathcal{M}^\mu$. Without (18.244) the Compton and pair-annihilation squares (18.236)–(18.237) would carry gauge-dependent garbage; with it, the simple covariant results stand. This is why the Compton family could be computed in Feynman gauge with no apology.

Second, it *guarantees gauge independence of the physical rate.* The amplitude depends on the photon propagator's gauge parameter ξ only through internal photon lines, and the ξ-dependent part of the propagator, $\propto (1-\xi)\,q_\mu q_\nu/q^2$, attaches its q_μ to a conserved current — the same Ward structure (18.244) applied to an off-shell leg — and vanishes. Every cross section in Table 18.3 is therefore ξ-independent, as the explicit Feynman-gauge computations and their crossings consistently returned.

Third, and most structurally, (18.244) *constrains the numerators* of the master functions in Theorem 18.20. Transversality is a polynomial identity among the gauge-index pieces of $\mathcal{M}^\mu$; it forces the individual diagrams (each non-transverse on its own) to combine into a transverse whole, which is exactly why a single Compton diagram is gauge-variant but their sum is not. The pole structure dictated by analyticity and the transversality dictated by gauge invariance together leave only the finite-dimensional space of numerators that the helicity decomposition (Section 18.6) then diagonalizes. Analyticity says *where* the function can be singular; the Ward identity says *how* the residues and the polynomial numerator must be arranged; helicity says *which* independent components there are. The tree QED amplitude is the unique object meeting all three demands.

Physical Insight

The three organizing principles of this chapter are three views of one object. Analyticity (crossing) is the statement that the amplitude is a function on

the (s,t,u) surface with controlled singularities; gauge invariance (Ward) is the statement that this function is transverse and ξ-independent; helicity is the statement that it block-diagonalizes under the little group of each external leg. A first course meets them as three separate tricks for getting answers faster. They are better understood as the defining properties of a single analytic amplitude, and the modern on-shell program (Part XVI) takes precisely this inversion — *define* amplitudes by their analyticity, little-group, and factorization properties, and reconstruct the Feynman expansion as a derived consequence rather than a starting point.

18.7.4 Forward map: loops, infrared structure, precision, and the on-shell viewpoint

The tree-level synthesis is the platform for everything that follows in QED, and it is worth naming exactly what is handed forward.

To the loop chapter (Chapter 19). The tree Ward identity (18.244) is promoted there to the all-orders Ward–Takahashi identity (Theorem 19.13), the exact statement $q_\mu \Gamma^\mu(p',p) = S^{-1}(p') - S^{-1}(p)$ relating the renormalized vertex to the electron self-energy, of which (18.244) is the on-shell tree shadow. The same explicit tree amplitudes assembled here are the leading terms whose radiative corrections Chapter 19 computes: the one-loop vacuum polarization dresses the photon line of (18.232) and (18.235), the self-energy dresses the electron lines of the Compton family, and the vertex correction dresses every $-ie\gamma^\mu$. Crucially, the tree amplitudes also fix the *infrared* structure. In the soft limit, where an external photon's energy $\omega \to 0$, the Compton and bremsstrahlung amplitudes factorize as $\mathcal{M} \to (\text{eikonal factor}) \times \mathcal{M}_{\text{hard}}$, the eikonal factor being a sum of $\epsilon\cdot p_i/(k\cdot p_i)$ over external charges — a structure already visible in the s,u poles of (18.236). In the collinear limit, where two massless legs become parallel, the same poles signal the mass singularities that dimensional regularization will tame. The cancellation of these infrared divergences between virtual and real emission (Bloch–Nordsieck, and more generally KLN), stated and organized in Chapter 19, takes the tree squares of this chapter as its real-emission input.

To the precision chapter (Chapter 20). The vertex correction whose tree seed is the bare $-ie\gamma^\mu$ produces, at one loop, the anomalous magnetic moment $a_e = \alpha/2\pi$ — Schwinger's number (Theorem 20.5) — and the running of the fine-structure constant $\alpha(q^2)$ from the vacuum polarization that dresses the s-channel photon of (18.232). The unpolarized $e^+e^- \to \mu^+\mu^-$ rate computed at tree level is, with that running coupling inserted, the leading term of a genuinely measured precision observable.

To the modern-amplitudes program (Part XVI, Chapter 58). The helicity reorganization begun in Section 18.6 — the observation that in the massless limit

the rational squares (18.232)–(18.237) are incoherent sums of a few simple helicity amplitudes — is the seed of the spinor-helicity formalism. There the master functions of Theorem 18.20 are written directly in angle and square brackets $\langle i\,j\rangle, [i\,j]$, the Ward identity is automatic (little-group covariance replaces it), and crossing becomes analytic continuation of the same bracket expression. The crossing web of Fig. 18.8 is, from that vantage, the shadow cast by a handful of holomorphic amplitudes onto the real Mandelstam regions. This chapter is where QED first stops being "formalism that yields one number" and becomes a structured family of amplitudes governed by symmetry and analyticity; Chapter 58 is where that structure is made the primary object.

With the tree-level Ward identity, the explicit master amplitudes, and the crossing web in hand, QED is poised for its first loops: Chapter 19 inherits (18.244) to promote into the Ward–Takahashi identity and the explicit tree squares to serve as the real-emission and Born-level inputs of its radiative corrections, and Chapter 20 inherits the dressed vertex and photon propagator that turn these amplitudes into $g-2$ and the running coupling.

Further reading

The four processes worked here are treated, with complementary conventions, in Peskin and Schroeder [396] (Compton scattering and pair annihilation in their Chapter 5, where crossing is also developed) and in Schwartz [432] (Møller and Bhabha scattering and the systematic use of Mandelstam variables and the Ward identity). The crossing-symmetry viewpoint and the analytic structure of tree amplitudes are presented at greater depth in Srednicki [454]. Weinberg [500] gives the field-theoretic origin of the Ward identity that Chapter 19 elevates to the Ward–Takahashi identity. For the helicity organization and its promotion to the spinor-helicity formalism developed in Chapter 58, Elvang and Huang [176] is the modern reference.

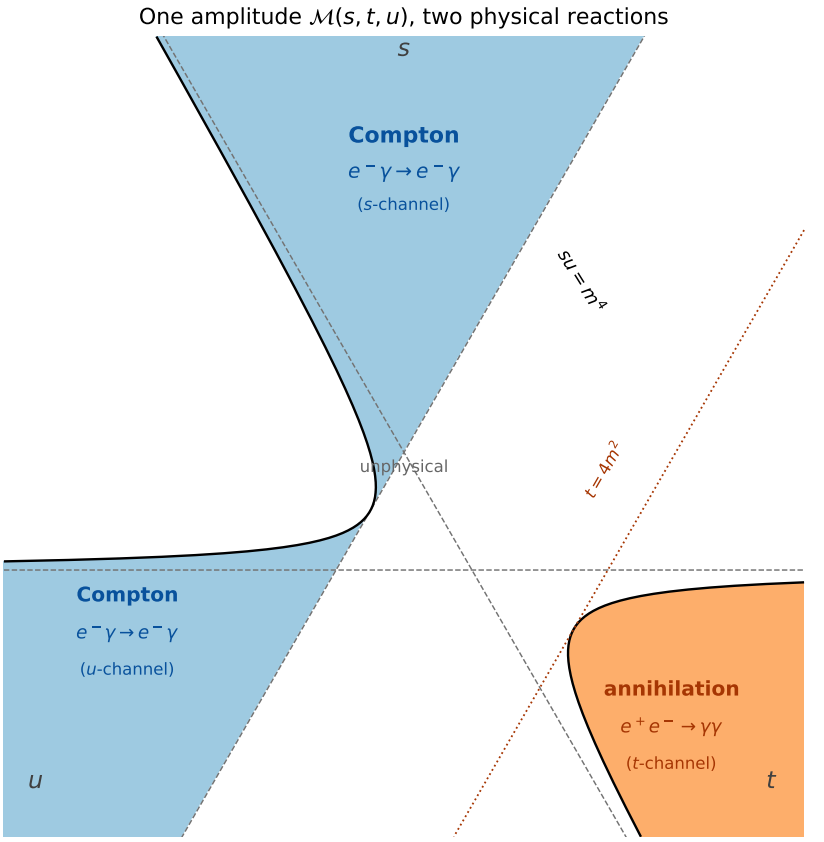
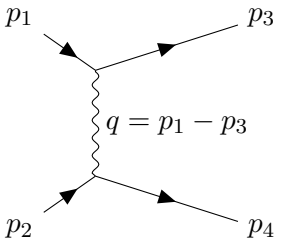
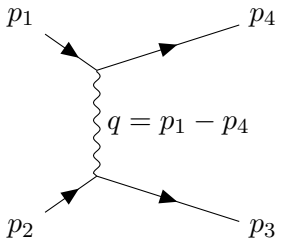


Figure 18.4: The Mandelstam plane $s+t+u = 2m^2$ shared by Compton scattering and pair annihilation, in the symmetric coordinates of Fig. 16.5 (with s increasing upward, t toward the lower right, u toward the lower left; units $m = 1$). One analytic amplitude $\mathcal{M}(s,t,u)$ lives on the whole plane. The blue wedges are the physical region of Compton scattering $e^-\gamma \to e^-\gamma$, Eq. (18.119), split into the s- and u-channel sub-wedges that describe the same reaction with the two photons interchanged. The orange wedge is the physical region of annihilation $e^+e^- \to \gamma\gamma$, Eq. (18.120); because $s_A = t_C$ it is the t-channel wedge, separated from the Compton region by the higher pair threshold $t = 4m^2$ and the boundary hyperbola $su = m^4$. The crossing $s \leftrightarrow t$ of Eq. (18.118) carries the amplitude from the blue s-wedge to the orange t-wedge.

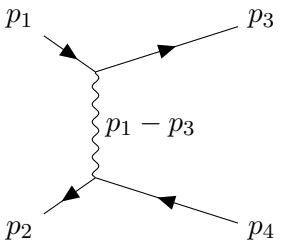


(a) t-channel

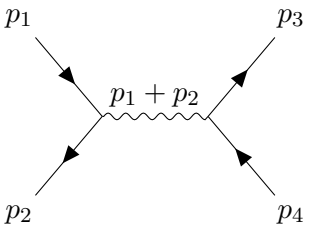


(b) u-channel

Figure 18.5: The two tree diagrams for Møller scattering $e^-e^- \to e^-e^-$. They are related by the exchange of the two identical outgoing electrons, $p_3 \leftrightarrow p_4$, which sends $t \leftrightarrow u$. Fermi statistics requires that they enter the amplitude with a relative minus sign, Eq. (18.155).



(a) t-channel (scattering)



(b) s-channel (annihilation)

Figure 18.6: The two tree diagrams for Bhabha scattering $e^+e^- \to e^+e^-$: t-channel photon exchange (scattering) and s-channel annihilation through a virtual photon. They enter with a relative minus sign, Eq. (18.177), and interfere in Eq. (18.182).

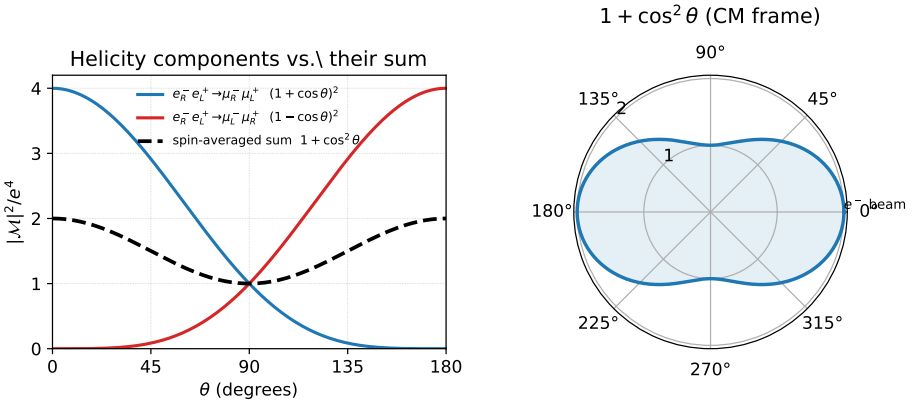


Figure 18.7: Helicity decomposition of $e^+e^- \to \mu^+\mu^-$ in the high-energy limit. *Left:* the two surviving angular shapes, the forward-peaked $(1+\cos\theta)^2$ (helicity-aligned, $\propto u^2$) and the backward-peaked $(1-\cos\theta)^2$ (anti-aligned, $\propto t^2$), each strongly forward–backward asymmetric, together with their spin-averaged sum $1+\cos^2\theta$. *Right:* the unpolarized $1+\cos^2\theta$ law as a centre-of-mass polar diagram; its forward–backward symmetry is the parity-even average of the asymmetric helicity components.

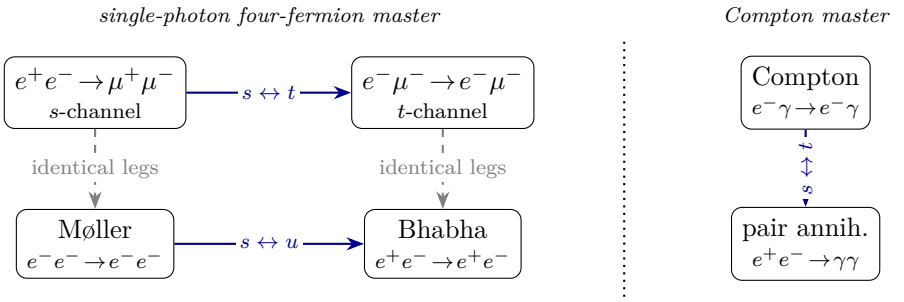


Figure 18.8: The tree-level QED crossing web. Solid arrows are crossing continuations (a permutation of the Mandelstam invariants, with the fermion-crossing sign); dashed arrows indicate that the identical-fermion processes are obtained by identifying two external lines and adding the Fermi-antisymmetrized exchange diagram. The two components correspond to the two master rational functions of Theorem 18.20.

Chapter 19

Radiative Corrections

With the tree-level QED amplitudes of Chapters 17 and 18 in hand — $e^+e^- \to \mu^+\mu^-$, Compton, Bhabha, and the tree Ward identity — we have computed every process the Feynman rules permit at leading order in e. This chapter takes the inevitable next step: the loop. At order e^2 beyond tree the theory confronts, for the first time, the ultraviolet infinities that nearly killed it in the 1930s, and the resolution of those infinities — renormalization — is the conceptual centre of the chapter. We work entirely at one loop and entirely with the carried QED thread, so that the famous numbers of Chapter 20 (the anomalous magnetic moment, the Lamb shift, the running coupling) emerge as bookkeeping on the finite remnants of computations done here.

There are exactly three one-particle-irreducible building blocks at one loop: the photon self-energy (vacuum polarization) $\Pi^{\mu\nu}(q)$, the electron self-energy $\Sigma(p)$, and the electron–photon vertex $\Gamma^{\mu}(p', p)$. The chapter computes all three from scratch, and the bulk of the labour is the loop-integral technology that makes them finite and meaningful. That technology — dimensional regularization in $d = 4-2\varepsilon$, Feynman parametrization, the Wick rotation, and the scalar master integrals — is built in the opening section and drawn in detail from Chapter D; the gamma-matrix trace identities the Dirac numerators require are those of Chapter C, and the mostly-minus metric, the covariant derivative $D_\mu = \partial_\mu - iqA_\mu$ with $q = -e$, and the propagator and vertex Feynman rules are the conventions fixed in Chapter A and Chapter 17. The branch-cut and dispersion structure we read off the vacuum polarization rests on the complex-analysis and Kramers–Kronig machinery reviewed in Section 5.5.5.

The arc is cumulative. Section 19.1 constructs the regulator and the renormalized-perturbation-theory framework, splitting the bare Lagrangian into a renormalized piece plus counterterms $\delta_1, \delta_2, \delta_3, \delta_m$ and fixing them by on-shell conditions. Section 19.2 computes $\Pi^{\mu\nu}$, finds it transverse exactly as gauge invariance demands, extracts Z_3 and the screening logarithm, and cross-checks its imaginary part against the optical theorem. Section 19.3 and Section 19.4 treat the electron

self-energy and the vertex, delivering δm, Z_2, Z_1, the Dirac form factor F_1, and — as the bridge to the next chapter — the Pauli form factor F_2. The empirical equality $Z_1 = Z_2$ that surfaces in the vertex computation is then elevated to a law in Section 19.5: the Ward–Takahashi identity, proved both from the generating functional and diagrammatically, which forces Π transverse, locks Σ to Γ^μ, and deposits the entire charge renormalization in the photon. The tree-level Ward identity and the covariant polarization sum it generalizes are imported from Chapter 18.

A single obstacle then remains between these renormalized loops and a physical number: the infrared divergence that survives in Z_2 and F_1. Section 19.6 confronts it, showing that the virtual divergence cancels against real soft-photon emission so that any sufficiently inclusive cross section is finite, and states the all-orders Bloch–Nordsieck, Kinoshita–Lee–Nauenberg, and Yennie–Frautschi–Suura results that organize this cancellation to every order. What this chapter does *not* do — the all-orders proof that the four counterterms suffice (renormalizability), the power counting that classifies divergences, and the renormalization-group flow that promotes the one-loop logarithm to a running coupling — is the subject of Chapters 21, 22 and 24; here those results are the organizing premise, and the one-loop input they will build on is exactly what we now compute.

19.1 Regularization and the One-Loop Program

At tree level the carried QED amplitudes of Chapters 17 and 18 were finite and algebraic: every diagram was a rational function of external momenta, gauge invariance was visible as the tree-level Ward identity $k_\mu \mathcal{M}^\mu = 0$ (Theorem 18.5), and the physical normalization of charge and mass was read straight off the Lagrangian. One loop changes the character of the problem. The moment a diagram contains a closed loop, the conservation of momentum at the vertices no longer fixes every internal line; one combination of internal momenta is unconstrained and must be integrated over all of momentum space. Those integrals generically diverge in the ultraviolet, and the divergence is not a calculational nuisance to be hidden but a structural feature that, handled correctly, turns QED from a tree-level caricature into a quantitative theory accurate to parts per billion. Handled incorrectly, it destroys the very gauge invariance that made the tree theory consistent.

This section builds the toolbox. It does three things, in order. First it names the three primitive one-particle-irreducible (1PI) objects out of which all of one-loop QED is assembled, and the diagrams that produce them. Then it confronts the divergences head-on and argues that the regulator must be chosen to respect gauge invariance — a hard momentum cutoff does not, dimensional regularization does — and constructs dimensional regularization in full: the loop measure in $d = 4 - 2\varepsilon$, the engineering dimension of the coupling and the renormalization

scale μ it forces upon us, the Feynman-parameter combination of denominators, the Wick rotation to Euclidean signature, and the scalar master integral to which every one-loop computation reduces. Finally it sets up renormalized perturbation theory: the split of the bare Lagrangian into a renormalized piece plus counterterms Z_1, Z_2, Z_3, δ_m, the counterterm Feynman rules, and the on-shell renormalization conditions that the next four sections will impose. The systematic all-orders justification of this machinery — power counting, the classification of divergences, the forest formula, the proof of renormalizability — is the business of Chapters 21 and 22; here we use those results only at the level of statements and concentrate on the explicit one-loop apparatus.

19.1.1 The three one-particle-irreducible building blocks

Recall from Chapter 15 the hierarchy connected $\to$ amputated $\to$ one-particle-irreducible: a 1PI diagram is one that cannot be severed into two by cutting a single internal line, and the full set of correlation functions is built by sewing 1PI blocks together with full propagators. At one loop in QED there are exactly three primitive 1PI amplitudes with ultraviolet divergences, one for each way of dressing the three ingredients of the theory — the photon, the electron, and their interaction. They are the objects the rest of the chapter computes.

The first is the *photon self-energy* or *vacuum polarization* $\Pi^{\mu\nu}(q)$, the 1PI two-point function of the photon. At one loop it is the single diagram in which a virtual electron–positron pair is created and annihilated, Fig. 19.1(a). With the QED vertex $-ie\gamma^\mu$ of Theorem 17.1 and the fermion propagator Eq. (17.15), the Feynman rules of Section 17.1 assign to it

$$i\Pi^{\mu\nu}(q) = -(-ie)^2 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \mathrm{tr}\left[\gamma^\mu \, \frac{i(\not{\ell} + m)}{\ell^2 - m^2 + i\epsilon}\, \gamma^\nu \, \frac{i(\not{\ell} + \not{q} + m)}{(\ell+q)^2 - m^2 + i\epsilon}\right], \tag{19.1}$$

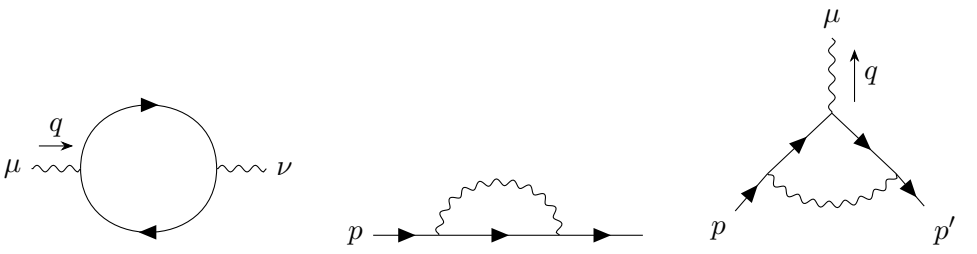
the overall minus sign being the closed-fermion-loop sign of Chapter 15 and the trace running over the Dirac indices that close around the loop. The second is the *electron self-energy* $\Sigma(p)$, the 1PI two-point function of the electron, Fig. 19.1(b): the electron emits and reabsorbs a virtual photon. With the photon propagator Eq. (17.18) in Feynman gauge,

$$-i\Sigma(p) = (-ie)^2 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \gamma^\mu \, \frac{i(\not{p} - \not{\ell} + m)}{(p-\ell)^2 - m^2 + i\epsilon}\, \gamma^\nu \, \frac{-i\eta_{\mu\nu}}{\ell^2 + i\epsilon}. \tag{19.2}$$

The third is the *vertex correction* $\Gamma^\mu(p', p)$, the 1PI three-point function with one photon and two electron legs, Fig. 19.1(c). Writing the full amputated vertex as $-ie\,\Gamma^\mu(p',p)$ with $\Gamma^\mu = \gamma^\mu + \delta\Gamma^\mu$, the one-loop correction is

$$-ie\,\delta\Gamma^\mu(p',p) = (-ie)^3 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \gamma^\alpha \, \frac{i(\not{p}' - \not{\ell} + m)}{(p'-\ell)^2 - m^2 + i\epsilon}\, \gamma^\mu \, \frac{i(\not{p} - \not{\ell} + m)}{(p-\ell)^2 - m^2 + i\epsilon}\, \gamma^\beta \, \frac{-i\eta_{\alpha\beta}}{\ell^2 + i\epsilon}. \tag{19.3}$$

Each of Eqs. (19.1) to (19.3) contains exactly one unconstrained loop momentum ℓ. Power-counting the integrands shows that $\Pi^{\mu\nu}$ is superficially quadratically divergent, Σ linearly divergent, and $\delta\Gamma^{\mu}$ logarithmically divergent — the systematic statement of which is the superficial degree of divergence developed in Chapter 21. Our immediate task is to give these integrals a precise, gauge-respecting meaning.



(a) $i\Pi^{\mu\nu}(q)$ (b) $-i\Sigma(p)$ (c) $-ie\,\delta\Gamma^{\mu}(p',p)$

Figure 19.1: The three primitive one-loop 1PI amplitudes of QED. (a) the photon self-energy / vacuum polarization Eq. (19.1), a fermion loop with two external photons; (b) the electron self-energy Eq. (19.2), an electron line emitting and reabsorbing a photon; (c) the vertex correction Eq. (19.3), the photon exchanged across the interaction vertex. All three carry a single unconstrained loop momentum and diverge in the ultraviolet; the Ward–Takahashi identity proved later in this chapter ties their divergences together.

19.1.2 Why a momentum cutoff fails and the regulator must be chosen

The most naive way to make sense of a divergent loop integral is to cut it off: integrate only over Euclidean loop momenta $|\ell_E| < \Lambda$ and study the dependence on Λ. For a scalar theory this is harmless. For a gauge theory it is a disaster, and the reason exposes exactly the property a good regulator must have.

Gauge invariance, through current conservation, constrains the photon self-energy to be *transverse*: the exact $\Pi^{\mu\nu}$ must take the form

$$\Pi^{\mu\nu}(q) = \left(q^2\eta^{\mu\nu} - q^{\mu}q^{\nu}\right)\Pi(q^2), \qquad q_{\mu}\Pi^{\mu\nu}(q) = 0. \tag{19.4}$$

This is the loop-level descendant of the tree Ward identity of Theorem 18.5; we prove it as part of the Ward–Takahashi identity later in the chapter, and it is verified by explicit computation in the next section. Its physical content is decisive: the only Lorentz-covariant rank-two tensor that *cannot* be written in the transverse form Eq. (19.4) is the one proportional to $\eta^{\mu\nu}$ with no factors of q, and that is precisely a photon mass term. Transversality is the statement that radiative corrections do not generate a photon mass; the photon stays massless to all orders. The superficial quadratic divergence of Eq. (19.1) is therefore not

allowed to survive as a $\Lambda^2\eta^{\mu\nu}$ contribution: gauge invariance forces the quadratic divergence to cancel, leaving only the logarithmic divergence carried by the transverse coefficient $\Pi(q^2)$.

The mechanism that enforces Eq. (19.4) in any honest manipulation of the loop integral is the freedom to shift the loop momentum, $\ell \to \ell + a$. Contracting q_μ into the integrand of Eq. (19.1) and using the Dirac algebra, one writes the result as a difference of two integrals related by exactly such a shift; if the shift is legal the difference vanishes and transversality follows. A hard cutoff breaks this: the region $|\ell_E| < \Lambda$ is not invariant under $\ell \to \ell + a$, so shifting the integration variable moves the boundary and leaves a surface term. That surface term is the spurious, non-transverse $\propto \Lambda^2\eta^{\mu\nu}$ photon mass. One would then be forced to add a non-gauge-invariant photon-mass counterterm, fine-tuned order by order to cancel it — abandoning the gauge symmetry that organizes the whole theory. The same disease afflicts a Pauli–Villars regulator for the gluon loops of Chapter 27 and, more subtly, any regulator that fails to preserve the symmetry whose Ward identity one needs.

Dimensional regularization, introduced by 't Hooft and Veltman [469] (and independently by Bollini and Giambiagi [84]), is built precisely to keep the shift legal. Because it introduces no dimensionful cutoff and no preferred frame, the analytically continued loop integral is exactly translation invariant, even when divergent. We isolate this as the property on which gauge invariance hinges.

Proposition 19.1 (Shift invariance of the dimensionally regularized measure)**.** *For any momentum a^μ and any integrand $f(\ell)$ to which the prescriptions below assign a finite value at generic non-integer d,*

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, f(\ell + a) = \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, f(\ell). \tag{19.5}$$

Proof. For an integrand falling fast enough that the integral converges absolutely at the working d, Eq. (19.5) is the ordinary change of variables $\ell \to \ell - a$, with no boundary because there is no boundary. The content of the proposition is that it persists for the divergent integrands of QED once they are defined by analytic continuation in d. Expand $f(\ell + a) - f(\ell) = a^\mu\partial_\mu f(\ell) + \frac{1}{2}a^\mu a^\nu\partial_\mu\partial_\nu f(\ell) + \cdots$, a sum of total derivatives. Each term is the integral of a total derivative $\partial_\mu g(\ell)$. In dimensional regularization the integral of a total derivative vanishes: by Lorentz invariance $\int \mathrm{d}^d\ell\, \partial_\mu g$ is a vector built from no external momentum and so must be zero, and more concretely the radial part reduces to $\int_0^\infty \mathrm{d}\ell_E \frac{\mathrm{d}}{\mathrm{d}\ell_E}(\ell_E^{d-1}\tilde{g}(\ell_E))$, a boundary term $\ell_E^{d-1}\tilde{g}|_0^\infty$ that one continues from the strip of d in which it vanishes at both ends and which is therefore zero throughout by analytic continuation. (Equivalently, the difference of the two sides of Eq. (19.5) is a scaleless integral, and scaleless integrals vanish identically in dimensional regularization because their would-be ultraviolet and infrared poles cancel, as shown in Section D.7.2.)

Hence every term in the Taylor expansion integrates to zero and Eq. (19.5) holds. □

Theorem 19.1 is the conceptual payoff that justifies the regulator choice: with the shift legal, the Ward identity manipulations go through and $\Pi^{\mu\nu}$ comes out transverse automatically, with no fine-tuned photon-mass counterterm. We now construct the measure that makes Eq. (19.5) precise.

19.1.3 The loop measure in $d = 4 - 2\varepsilon$ dimensions

Dimensional regularization replaces the four-dimensional loop integral by its continuation to $d = 4 - 2\varepsilon$ complex dimensions. Concretely, one keeps the loop integrand — a rational function of Lorentz invariants $\ell^2, \ell \cdot q, \ldots$ built from the QED Feynman rules — and reinterprets the measure $\int \mathrm{d}^d\ell/(2\pi)^d$ for non-integer d. The only feature of the measure a one-loop integrand sees, after the angular integration, is the surface area of the unit sphere in d Euclidean dimensions,

$$\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}, \tag{19.6}$$

which is manifestly an analytic function of d through the gamma function. A scalar integrand $f(\ell_E^2)$ then yields, after Wick rotation, $\Omega_d \int_0^\infty \mathrm{d}\ell_E\, \ell_E^{d-1} f(\ell_E^2)/(2\pi)^d$, and the radial integral converges in an open strip of d whenever f falls off at large ℓ_E and is integrable at the origin. Inside that strip the integral is a genuine, finite, analytic function of d; outside it the function is *defined* by analytic continuation. The ultraviolet and infrared divergences that plagued $d = 4$ do not disappear — they reappear as isolated *poles* of this analytic function. With the parametrization $d = 4 - 2\varepsilon$, a logarithmically ultraviolet-divergent integral develops a simple pole at $\varepsilon = 0$, and the residue is the coefficient of the would-be $\ln \Lambda$. The construction of Ω_d and the radial reduction are carried out in full in Section D.3; here we take them as established and exploit them.

Caution

Dimensional regularization regulates the ultraviolet and the infrared with the *same* parameter ε, but the two kinds of pole prefer opposite signs of ε: an ultraviolet pole is reached from $d < 4$ ($\varepsilon > 0$), an infrared pole from $d > 4$ ($\varepsilon < 0$). When both are present in one integral the bookkeeping must keep them distinct, since only the ultraviolet poles are removed by renormalization while the infrared ones are physical and cancel against real emission. The sign discipline and the proof that a scaleless integral vanishes through an exact cancellation of the two are given in Sections D.7.1 and D.7.2; we invoke that machinery when we reach the infrared section of this chapter.

Moving off $d = 4$ has an immediate consequence that we cannot avoid and that seeds the entire renormalization-group story of Chapter 24: the coupling is no longer dimensionless. In natural units the action is dimensionless, so the Lagrangian density has mass dimension d (because $\int \mathrm{d}^d x$ has dimension $-d$). Reading off the dimensions of the free kinetic terms $\bar{\psi}\, i\partial\!\!\!/\, \psi$ and $-\frac{1}{4} F_{\mu\nu}F^{\mu\nu}$ fixes

$$[\psi] = \frac{d-1}{2}, \qquad [A^\mu] = \frac{d-2}{2}, \qquad [\partial_\mu] = 1. \tag{19.7}$$

The interaction term $e\,\bar{\psi} A\!\!\!/\, \psi$ must also have dimension d, so

$$[e] = d - 2[\psi] - [A] = d - (d-1) - \frac{d-2}{2} = \frac{4-d}{2} = \varepsilon. \tag{19.8}$$

In exactly four dimensions $[e] = 0$ and the coupling is dimensionless, as it must be for a renormalizable theory. Away from four dimensions e carries mass dimension ε. To keep the coupling that appears in expansions dimensionless — so that $\alpha = e^2/4\pi$ remains a pure number and the perturbation series is an expansion in a number — we factor out the dimension with an arbitrary mass scale μ, the *renormalization scale*:

$$e_{(\dim\, \varepsilon)} \longrightarrow e\,\mu^{\varepsilon}, \qquad e \text{ dimensionless}, \qquad \alpha = \frac{e^2}{4\pi}. \tag{19.9}$$

Every QED vertex thus carries an explicit μ^{ε}, and an L-loop diagram carries $\mu^{2L\varepsilon}$ from its $2L$ powers of e^2 relative to tree level; equivalently each loop integral is accompanied by one factor $\mu^{2\varepsilon}$, the bookkeeping factor that appears throughout Chapter D. The scale μ is pure artifact at this stage — nothing physical can depend on it — but the requirement that physics be μ-independent is exactly the Callan–Symanzik equation, and the running of α it generates is the subject of Chapters 20 and 24. Its first appearance is here.

19.1.4 Combining denominators: the Feynman parametrization

Each loop integral Eqs. (19.1) to (19.3) is a ratio of polynomials in ℓ over a product of two or three quadratic denominators with different momentum offsets. The integral over a single quadratic denominator is doable; over a product it is not, because there is no single variable in which to complete the square. The Feynman parametrization rewrites the product as a single denominator at the cost of an integral over auxiliary parameters, after which one shift completes the square. We derive the two cases QED needs — two and three denominators — carrying the algebra; the general N-denominator statement and the Schwinger-representation route to it are in Theorem D.11.

Proposition 19.2 (Two- and three-denominator parametrizations)**.** *For nonzero A, B, C (with the Feynman $i\epsilon$ keeping the combined denominator off the real*

axis),

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x \, \frac{1}{\big[xA + (1-x)B\big]^2}, \tag{19.10}$$

$$\frac{1}{ABC} = 2\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{1}{\big[xA+yB+zC\big]^3}. \tag{19.11}$$

Proof. For Eq. (19.10), write the bracket as $B + x(A-B)$ and integrate directly in x, treating $A \neq B$:

$$\int_0^1 \mathrm{d}x \, \frac{1}{[B+x(A-B)]^2} = \left[\frac{-1}{(A-B)}\,\frac{1}{B+x(A-B)}\right]_0^1 = \frac{-1}{A-B}\left(\frac{1}{A}-\frac{1}{B}\right) = \frac{-1}{A-B}\cdot\frac{B-A}{AB} = \frac{1}{AB}$$

and the case $A = B$ follows by continuity. To reach Eq. (19.11) we first need the parametrization with a squared denominator, which we obtain by differentiating Eq. (19.10). Hold C fixed in $1/(FC) = \int_0^1 \mathrm{d}y\,[\,yF + (1-y)C\,]^{-2}$ and differentiate with respect to F:

$$\frac{\partial}{\partial F}\frac{1}{FC} = -\frac{1}{F^2 C}, \qquad \frac{\partial}{\partial F}\int_0^1 \mathrm{d}y\,\frac{1}{[yF+(1-y)C]^2} = -2\int_0^1 \mathrm{d}y\,\frac{y}{[yF+(1-y)C]^3},$$

where $\partial(yF + (1-y)C)/\partial F = y$. Equating gives

$$\frac{1}{F^2 C} = 2\int_0^1 \mathrm{d}y\,\frac{y}{[yF+(1-y)C]^3}. \tag{19.12}$$

Now set $F = xA + (1-x)B$, so that by Eq. (19.10) $1/(ABC) = \int_0^1 \mathrm{d}x\,(F^2 C)^{-1}$, and substitute Eq. (19.12):

$$\frac{1}{ABC} = 2\int_0^1 \mathrm{d}x \int_0^1 \mathrm{d}y\,\frac{y}{\big[\,y(xA+(1-x)B)+(1-y)C\,\big]^3} = 2\int_0^1 \mathrm{d}x \int_0^1 \mathrm{d}y\,\frac{y}{\big[\,yx\,A + y(1-x)B + (1-y)C\,\big]^3}.$$

Change variables to $u = yx$, $v = y(1-x)$, with $w \equiv 1-u-v = 1-y$. The Jacobian is

$$\frac{\partial(u,v)}{\partial(x,y)} = \det\begin{pmatrix} y & x \\ -y & 1-x \end{pmatrix} = y(1-x) + xy = y,$$

so $\mathrm{d}u\,\mathrm{d}v = y\,\mathrm{d}x\,\mathrm{d}y$ and the explicit weight y in the numerator is absorbed exactly. As (x,y) range over the unit square, (u,v) range over the triangle $u, v \geq 0$, $u+v \leq 1$. Reinstating the third parameter through $\int \mathrm{d}w\,\delta(u+v+w-1)$ symmetrizes the result:

$$\frac{1}{ABC} = 2\int \mathrm{d}u\,\mathrm{d}v\,\mathrm{d}w\;\delta(u+v+w-1)\,\frac{1}{[uA+vB+wC]^3},$$

which is Eq. (19.11) after renaming $(u,v,w) \to (x,y,z)$. The prefactor $2 = 2! = \Gamma(3)$ is the $N = 3$ case of the general $\Gamma(N)$ of Theorem D.11. □

The point of the parametrization is what it does to the loop integral. After it, the denominator is a single quadratic in ℓ raised to an integer power. Completing the square — shifting ℓ by a parameter-dependent combination of external momenta, which is legal by Theorem 19.1 — removes the linear term and turns every denominator into $(\ell^2 - \Delta + i\epsilon)^n$, where Δ is a function of the Feynman parameters and the external invariants, independent of ℓ. For two propagators with offsets 0 and q and masses m_1, m_2, for instance,

$$x(\ell^2-m_1^2)+(1-x)((\ell+q)^2-m_2^2) = (\ell+(1-x)q)^2-\Delta, \qquad \Delta = -x(1-x)q^2+xm_1^2+(1-x)m_2^2, \tag{19.13}$$

as one checks by expanding both sides and cancelling the linear-in-ℓ term $2(1-x)\,\ell\cdot q$ against the shift. The numerator algebra — the gamma-matrix traces, evaluated with the d-dimensional Clifford and trace identities of Sections C.6 and C.7 — is carried out after the shift, expressed in the shifted variable, and reduced by the symmetric-integration rule of the next subsection. Everything then rests on a single scalar integral.

19.1.5 Wick rotation and the scalar master integral

After parametrization and the shift, every one-loop QED integral is a linear combination of integrals of the form

$$I_n(\Delta) = \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^n}, \tag{19.14}$$

possibly with extra factors of ℓ^μ in the numerator (handled below). The Minkowski integral Eq. (19.14) is evaluated by rotating the ℓ^0 contour onto the imaginary axis. The legality of this rotation is exactly the contour-deformation argument of Sections 5.3 and 5.4: the Feynman $i\epsilon$ places the poles of the integrand in ℓ^0, at $\ell^0 = \pm(\sqrt{\boldsymbol{\ell}^2 + \Delta} - i\epsilon)$, in the second and fourth quadrants of the complex ℓ^0-plane, leaving the first and third quadrants free, so the contour may be swung counterclockwise from the real axis to the imaginary axis without crossing a singularity and with vanishing arcs at infinity. Writing $\ell^0 = i\ell_E^0$ with ℓ_E^0 real, the rotation gives $\ell^2 = (\ell^0)^2 - \boldsymbol{\ell}^2 = -(\ell_E^0)^2 - \boldsymbol{\ell}^2 \equiv -\ell_E^2$ (a Euclidean square) and $\mathrm{d}^d\ell = i\,\mathrm{d}^d\ell_E$, so

$$I_n(\Delta) = i\,(-1)^n \int \frac{\mathrm{d}^d\ell_E}{(2\pi)^d}\,\frac{1}{(\ell_E^2 + \Delta)^n}. \tag{19.15}$$

The Euclidean integral is rotationally invariant, so the angular integral gives Ω_d of Eq. (19.6) and only the radial integral remains:

$$\int \frac{\mathrm{d}^d\ell_E}{(2\pi)^d}\,\frac{1}{(\ell_E^2 + \Delta)^n} = \frac{\Omega_d}{(2\pi)^d}\int_0^\infty \mathrm{d}\ell_E\,\frac{\ell_E^{d-1}}{(\ell_E^2 + \Delta)^n}.$$

Substitute $t = \ell_E^2/\Delta$, so $\ell_E = (\Delta t)^{1/2}$ and $\mathrm{d}\ell_E = \tfrac{1}{2}\Delta^{1/2} t^{-1/2}\mathrm{d}t$, turning the radial integral into an Euler Beta function:

$$\int_0^\infty \mathrm{d}\ell_E \, \frac{\ell_E^{d-1}}{(\ell_E^2+\Delta)^n} = \frac{1}{2}\,\Delta^{\frac{d}{2}-n}\int_0^\infty \mathrm{d}t\,\frac{t^{\frac{d}{2}-1}}{(1+t)^n} = \frac{1}{2}\,\Delta^{\frac{d}{2}-n}\,\frac{\Gamma\!\left(\frac{d}{2}\right)\Gamma\!\left(n-\frac{d}{2}\right)}{\Gamma(n)}, \tag{19.16}$$

where we used the standard Beta-integral $\int_0^\infty t^{a-1}(1+t)^{-(a+b)}\,\mathrm{d}t = \Gamma(a)\Gamma(b)/\Gamma(a+b)$ with $a = d/2$, $b = n - d/2$ (derived as part of the gamma-function technology of Section D.5). Collecting Eqs. (19.6), (19.15) and (19.16) and cancelling the $\Gamma(d/2)$ between Ω_d and the Beta function gives the scalar master integral:

$$\boxed{\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^n} = \frac{(-1)^n\, i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(n-\frac{d}{2}\right)}{\Gamma(n)}\,\left(\frac{1}{\Delta}\right)^{n-\frac{d}{2}}.} \tag{19.17}$$

All of the ultraviolet behaviour now lives in the single gamma function $\Gamma(n-d/2)$: with $d = 4-2\varepsilon$, the argument is $n-2+\varepsilon$, and Γ has simple poles at non-positive integers, so $\Gamma(n-d/2)$ is singular as $\varepsilon \to 0$ precisely when $n \le 2$ — exactly the logarithmically-or-worse divergent cases. The case the QED self-energies and vertex reduce to after their tensor structure is stripped is $n = 2$, where $\Gamma(2-d/2) = \Gamma(\varepsilon) = 1/\varepsilon - \gamma_{\mathrm{E}} + O(\varepsilon)$. Restoring the $\mu^{2\varepsilon}$ of Eq. (19.9) and expanding (the gamma-function expansions being tabulated in Section D.5), this is exactly the renormalization-scale-dressed master of Eq. (D.136),

$$\mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^2} = \frac{i}{(4\pi)^2}\left[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta}\right] + O(\varepsilon), \qquad \frac{1}{\bar{\varepsilon}} \equiv \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi, \tag{19.18}$$

in which the divergence is an isolated, kinematics-independent $1/\bar{\varepsilon}$ pole and the entire finite, Δ-dependent content is the logarithm $\ln(\mu^2/\Delta)$. That clean separation — a universal pole that a local counterterm can absorb, plus a finite log that carries all the physics — is the structural fact Theorem D.31 attributes to locality, and it is what makes renormalization possible. We import Eq. (19.18) as the workhorse for the explicit computations to come, having derived the reduction to gamma functions Eq. (19.17) in full and quoted only the ε-expansion of $\Gamma(\varepsilon)$ from the appendix.

Numerators carrying loop momenta are handled by the same machinery once one notices that the shifted integrand is even in ℓ up to terms that integrate to zero. The next worked example establishes the rule and exhibits it on the tensor integral that appears in every one-loop QED computation.

Worked Example 19.3 (Symmetric integration and a representative tensor integral)**.** After completing the square, numerators contain factors of the shifted loop momentum such as $\ell^\mu \ell^\nu$. By Lorentz invariance, an integral that depends on ℓ only through ℓ^2 in its denominator can return only tensors built from the invariant tensor $\eta^{\mu\nu}$ (there is no external vector left after the shift), so an odd

number of ℓ's integrates to zero and an even number must be proportional to the symmetric tensor:

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu}{(\ell^2-\Delta)^n}=0, \qquad \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu\ell^\nu}{(\ell^2-\Delta)^n}=\frac{\eta^{\mu\nu}}{d}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^n}. \tag{19.19}$$

The first identity holds because the integrand is odd under $\ell\to-\ell$, a symmetry of the (shift-invariant, Theorem 19.1) measure. The coefficient of $\eta^{\mu\nu}$ in the second is fixed by contracting both sides with $\eta_{\mu\nu}$: the left side becomes $\int \ell^2/(\ell^2-\Delta)^n$, the right side becomes $(\eta^\mu{}_\mu/d)\int \ell^2/(\ell^2-\Delta)^n = (d/d)\int \ell^2/(\ell^2-\Delta)^n$, using $\eta^\mu{}_\mu = d$ (Eq. (C.185)), so the two agree and the coefficient is $1/d$. The factor $1/d$, not $1/4$, is essential: multiplied against a $1/\varepsilon$ pole it shifts finite parts, exactly as the d-dependent gamma contractions of Section C.7 do.

To evaluate the right-hand side, write $\ell^2=(\ell^2-\Delta)+\Delta$ in the numerator, splitting it into two masters of the form Eq. (19.17):

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^n}=\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta)^{n-1}}+\Delta\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta)^n}.$$

Inserting Eq. (19.17) for each piece and using $\Gamma(z+1)=z\,\Gamma(z)$ to combine them,

$$\begin{aligned}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^n} &= \frac{(-1)^{n-1}i}{(4\pi)^{d/2}}\,\frac{\Gamma(n-1-\frac{d}{2})}{\Gamma(n-1)}\,\Delta^{\frac{d}{2}-n+1}+\Delta\,\frac{(-1)^n i}{(4\pi)^{d/2}}\,\frac{\Gamma(n-\frac{d}{2})}{\Gamma(n)}\,\Delta^{\frac{d}{2}-n}\\ &= \frac{(-1)^{n-1}i}{(4\pi)^{d/2}}\,\Delta^{\frac{d}{2}-n+1}\,\frac{1}{\Gamma(n)}\Big[(n-1)\,\Gamma(n-1-\tfrac{d}{2})-\Gamma(n-\tfrac{d}{2})\Big]\\ &= \frac{(-1)^{n-1}i}{(4\pi)^{d/2}}\,\frac{d}{2}\,\frac{\Gamma(n-1-\frac{d}{2})}{\Gamma(n)}\,\Delta^{\frac{d}{2}-n+1},\end{aligned} \tag{19.20}$$

where in the last step we used $\Gamma(n-\frac{d}{2})=(n-1-\frac{d}{2})\Gamma(n-1-\frac{d}{2})$ so that the bracket collapses to $[(n-1)-(n-1-\frac{d}{2})]\,\Gamma(n-1-\frac{d}{2})=\frac{d}{2}\,\Gamma(n-1-\frac{d}{2})$. Combining Eq. (19.20) with Eq. (19.19) gives the tensor master

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu\ell^\nu}{(\ell^2-\Delta)^n}=\frac{(-1)^{n-1}i}{(4\pi)^{d/2}}\,\frac{\eta^{\mu\nu}}{2}\,\frac{\Gamma(n-1-\frac{d}{2})}{\Gamma(n)}\,\Delta^{\frac{d}{2}-n+1}, \tag{19.21}$$

the factor $d/2$ from Eq. (19.20) having cancelled the $1/d$ from Eq. (19.19). With Eqs. (19.17) and (19.21) every scalar and rank-two tensor one-loop integral of QED is reduced to gamma functions; higher-rank numerators reduce the same way, and the systematic algorithm for arbitrary tensor integrals is the Passarino–Veltman reduction of Section D.6.

19.1.6 Renormalized perturbation theory and the counterterms

We can now regulate any one-loop QED amplitude and extract its $1/\varepsilon$ pole and finite part. What remains is the framework that converts those poles into finite

predictions: renormalized perturbation theory. The idea is to recognize that the parameters in the Lagrangian written down at tree level are not the physically measured charge and mass. We rename them *bare* parameters and fields, and re-express the Lagrangian in terms of *renormalized* quantities — the ones tied to measurements — plus correction terms, the *counterterms*, whose job is to cancel the loop divergences order by order.

Begin from the bare QED Lagrangian, the carried QED model of Section 17.1 with every quantity given a subscript 0:

$$\mathcal{L} = \bar{\psi}_0(i\not{\partial} - m_0)\psi_0 - e_0\,\bar{\psi}_0 \not{A}_0\,\psi_0 - \tfrac{1}{4}F_{0\,\mu\nu}F_0^{\mu\nu} - \frac{1}{2\xi_0}(\partial_\mu A_0^\mu)^2, \tag{19.22}$$

with covariant derivative $D_\mu = \partial_\mu + ie_0 A_{0\mu}$ acting on the electron of charge $-e$ (the standing sign convention). The bare fields and parameters are related to renormalized ones by multiplicative renormalization constants,

$$\psi_0 = Z_2^{1/2}\,\psi, \qquad A_0^\mu = Z_3^{1/2}\,A^\mu, \qquad Z_2\,m_0 = m + \delta_m, \qquad e_0\,Z_2\,Z_3^{1/2} = Z_1\,e\,\mu^\varepsilon, \tag{19.23}$$

which define the electron field-strength renormalization Z_2, the photon field-strength renormalization Z_3, the mass shift δ_m, and the vertex renormalization Z_1. Solving the last relation for the bare coupling exhibits how the measured charge is dressed,

$$e_0 = Z_1\,Z_2^{-1}\,Z_3^{-1/2}\,\mu^\varepsilon\,e, \tag{19.24}$$

and previews the one relation that makes QED so economical: the Ward–Takahashi identity proved later in this chapter forces $Z_1 = Z_2$, so the field and vertex renormalizations cancel and $e_0 = Z_3^{-1/2}\mu^\varepsilon e$ depends only on the photon. This is why charge renormalization in QED is governed entirely by vacuum polarization and why the running of α is universal — a fact Chapters 20 and 24 exploit. Substituting Eq. (19.23) into Eq. (19.22) and writing $Z_i = 1 + \delta_i$ splits the Lagrangian cleanly into a renormalized piece of the same form as the tree Lagrangian, now in renormalized fields and parameters, plus counterterms:

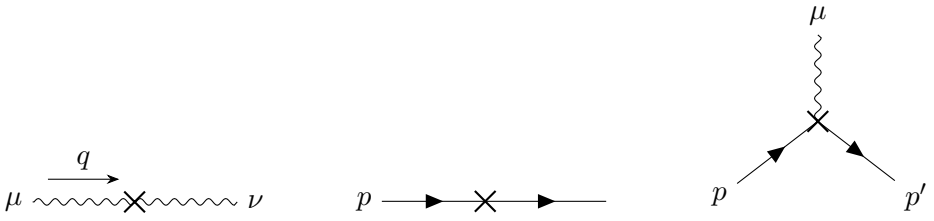
$$\begin{aligned}\mathcal{L} = &\underbrace{\bar{\psi}(i\not{\partial} - m)\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 - e\mu^\varepsilon\,\bar{\psi}\not{A}\,\psi}_{\text{renormalized}} \\ &+ \underbrace{\bar{\psi}(i\,\delta_2\not{\partial} - \delta_m)\psi - \tfrac{1}{4}\,\delta_3\,F_{\mu\nu}F^{\mu\nu} - e\mu^\varepsilon\,\delta_1\,\bar{\psi}\not{A}\,\psi}_{\text{counterterms}},\end{aligned} \tag{19.25}$$

where $\delta_2 = Z_2 - 1$, $\delta_3 = Z_3 - 1$, $\delta_1 = Z_1 - 1$, and $\delta_m = Z_2 m_0 - m$. The gauge-fixing term is not renormalized at one loop in this scheme (the longitudinal photon does not interact), so ξ carries no counterterm here. The renormalized piece generates exactly the QED Feynman rules of Chapter 17 with $e \to e\mu^\varepsilon$, and the counterterm piece generates three new vertices, read off as the momentum-space versions of the quadratic and cubic operators in the second line, shown in

Fig. 19.2:

photon 2-point: $i(\delta_3(q^2\eta^{\mu\nu}-q^\mu q^\nu))$; electron 2-point: $i(\delta_2 \not{p}-\delta_m)$; vertex: $-ie\mu^\varepsilon\,\delta_1\,\gamma^\mu$. (19.26)

The photon counterterm comes out transverse, $\propto (q^2\eta^{\mu\nu} - q^\mu q^\nu)$, because it descends from the gauge-invariant operator $F_{\mu\nu}F^{\mu\nu}$; it is structurally incapable of supplying a photon mass, which is the Lagrangian-level reason the cutoff's spurious $\Lambda^2\eta^{\mu\nu}$ of Section 19.1.2 could never be a legitimate counterterm. The counterterms enter at the same order in e as the one-loop diagrams (each $\delta_i = O(\alpha)$), so that at $O(\alpha)$ the renormalized 1PI functions are the sum of the loop of Fig. 19.1 and the counterterm of Fig. 19.2.



(a) $i\delta_3(q^2\eta^{\mu\nu} - q^\mu q^\nu)$ (b) $i(\delta_2\not{p} - \delta_m)$ (c) $-ie\mu^\varepsilon\delta_1\gamma^\mu$

Figure 19.2: The counterterm Feynman rules Eq. (19.26) from the second line of Eq. (19.25), drawn with a crossed vertex. Each enters at $O(\alpha)$ and cancels the divergence of the corresponding one-loop 1PI diagram of Fig. 19.1; the photon counterterm (a) is transverse by construction and so cannot generate a photon mass.

The four constants $\delta_1, \delta_2, \delta_3, \delta_m$ are fixed by four *renormalization conditions* — definitions that tie the renormalized parameters to physical quantities. This chapter adopts the *on-shell scheme*, in which the conditions are imposed at the physical pole of each propagator, so that m is the physical electron mass, e the charge measured at large distance (Thomson limit), and the fields normalized to unit residue at their poles. The conditions, to be imposed when the explicit computations are in hand, are:

Definition 19.4 (On-shell renormalization conditions)**.** Let $\Pi(q^2)$ be the scalar in the transverse decomposition Eq. (19.4) of the renormalized photon self-energy, $\Sigma(p)$ the renormalized electron self-energy, and $\Gamma^\mu(p', p)$ the renormalized amputated vertex. The on-shell scheme demands

$$\Pi(q^2 = 0) = 0 \qquad \text{(photon pole at } q^2 = 0\text{, unit} \tag{19.27}$$

$$\Sigma(\not{p})\big|_{\not{p}=m} = 0, \qquad \frac{\mathrm{d}\Sigma}{\mathrm{d}\not{p}}\bigg|_{\not{p}=m} = 0 \qquad \text{(electron pole at } \not{p} = m\text{, un} \tag{19.28}$$

$$\bar{u}(p)\,\Gamma^{\mu}(p,p)\,u(p) = \bar{u}(p)\,\gamma^{\mu}\,u(p) \quad \text{at } q = p' - p = 0 \quad \text{(vertex normalized to } -ie\gamma^{\mu}\text{)}. \tag{19.29}$$

The first condition fixes δ_3: it states that the full photon propagator still has its pole at $q^2 = 0$ (the photon stays massless) with residue one, so the photon field is canonically normalized. The two electron conditions fix δ_m and δ_2: the first places the pole of the full electron propagator at the physical mass m, the second normalizes its residue to one. The vertex condition fixes δ_1: at zero momentum transfer, between on-shell spinors, the full vertex must reduce to the tree value $-ie\gamma^{\mu}$, which is the statement that e is the physically measured charge. These four conditions are not independent of gauge invariance: the equality $Z_1 = Z_2$ forced by the Ward–Takahashi identity makes the vertex condition Eq. (19.29) a consequence of the electron-field condition Eq. (19.28), so the residue of the electron propagator and the normalization of the charge are locked together — the deep reason the on-shell charge is universal and not contaminated by the electron's self-interaction. We will see this happen explicitly. The all-orders statement that this finite set of counterterms suffices to render *every* amplitude finite — that QED is renormalizable — together with the power counting that identifies the divergent amplitudes in the first place, is established in Chapters 21 and 22; here it is the organizing premise.

With the regulator constructed — the d-dimensional measure Eq. (19.6), the shift invariance Theorem 19.1 that protects gauge invariance, the coupling's scale μ of Eq. (19.9), the Feynman parametrization Theorem 19.2, and the scalar and tensor master integrals Eqs. (19.17) and (19.21) feeding the $\overline{\text{MS}}$ master Eq. (19.18) — and with the counterterm framework Eq. (19.25) and on-shell conditions Theorem 19.4 in place, every one-loop QED computation is now a finite, well-posed problem. The next section applies the entire apparatus to the first and most instructive of the three building blocks, the photon self-energy $\Pi^{\mu\nu}(q)$ of Eq. (19.1): we compute it in dimensional regularization, verify that it comes out transverse exactly as Theorem 19.1 promised, renormalize it on shell to fix Z_3, and read off the analytic structure that drives charge screening and the running coupling.

19.2 Vacuum Polarization

The previous section built the only regulator that respects the gauge symmetry of QED — dimensional continuation to $d = 4 - 2\varepsilon$, with $\overline{\text{MS}}$ scale μ and the renormalized-perturbation-theory bookkeeping of counterterms and on-shell conditions (Section 19.1). It also recorded an expectation we now have to make good on. At tree level the Ward identity $k_\mu \mathcal{M}^{\mu} = 0$ (Theorem 18.5) said that the photon couples only to a conserved current, and that statement forced the longitudinal part of every photon line to decouple from physics. The first quantum correction to the photon's own propagation must inherit this

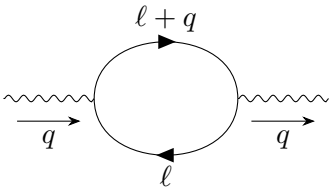


Figure 19.3: The one-loop photon self-energy (vacuum polarization). A photon of momentum q fluctuates into a virtual electron–positron pair, the two fermion propagators carrying loop momenta ℓ and $\ell + q$, before the pair annihilates back into a photon. The closed fermion loop supplies an overall minus sign and a Dirac trace. This is the only 1PI correction to the photon propagator at $O(e^2)$.

discipline: whatever the loop produces, it cannot generate a piece that an external photon's longitudinal polarization could detect, on pain of contradicting current conservation. Translated into the language of the photon two-point function, the requirement is *transversality* — the corrected inverse propagator must vanish when contracted with q^μ. We will compute the one-loop correction in full, watch transversality emerge from the algebra (not be imposed by hand), see exactly how it disposes of a divergence that naïvely looks quadratic, and read off from the finite remainder the first piece of running physics in the book: the dependence of the effective electric charge on the momentum at which it is probed.

The object of study is the photon self-energy, the sum of all one-particle-irreducible (1PI, Theorem 15.30) insertions on a photon line with the two external propagators amputated. We write it $i\Pi^{\mu\nu}(q)$, with q the photon momentum flowing through. At one loop in QED there is exactly one diagram: the photon converts to a virtual electron–positron pair which annihilates back to a photon, the fermion loop of Fig. 19.3. It is the first genuinely interacting two-point function of QED, and the whole of Chapter 20 — the running of α, the Uehling part of the Lamb shift, the hadronic blob in $g-2$ — is a reading of the function we are about to compute.

19.2.1 The amplitude and the fermion-loop sign

Reading the diagram with the QED Feynman rules established for QED — the vertex $-ie\gamma^\mu$ of Theorem 17.1 (recall the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ acts on the electron of charge $-e$, $e > 0$) and the fermion propagator $i(\not{\ell} + m)/(\ell^2 - m^2 + i\epsilon)$ of Eq. (17.15) — the loop assembles into

$$i\Pi^{\mu\nu}(q) = -(-ie)^2 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \mathrm{tr}\left[\gamma^\mu \frac{i(\not{\ell} + m)}{\ell^2 - m^2 + i\epsilon}\, \gamma^\nu \frac{i(\not{\ell} + \not{q} + m)}{(\ell+q)^2 - m^2 + i\epsilon}\right]. \quad (19.30)$$

Three features deserve naming. The trace is over the Dirac indices: the two propagator numerators and the two vertices form a closed matrix chain along the fermion line, and a closed line is traced. The integral is over the single unconstrained loop momentum ℓ, the hallmark of a loop — momentum conservation at the two vertices fixes the difference of the two fermion momenta to be q but leaves ℓ free, and it is precisely this unrestricted integration that can diverge. And the overall sign is the celebrated minus sign of a closed fermion loop: it descends from the anticommuting nature of the fermion fields, where reordering the Grassmann operators to form the trace produces a single transposition relative to the bosonic case. We carry it explicitly as the prefactor $-(-ie)^2 = -(-e^2) = +e^2$; combined with the two factors of i from the propagators, $i^2 = -1$, the kinematic prefactor of the integral is $-e^2$.

Pulling the constants out and combining the two propagator denominators into the single product $\mathcal{D} \equiv (\ell^2 - m^2 + i\epsilon)\,[(\ell+q)^2 - m^2 + i\epsilon]$,

$$i\Pi^{\mu\nu}(q) = -e^2 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{N^{\mu\nu}(\ell, q)}{\mathcal{D}}, \qquad N^{\mu\nu}(\ell, q) \equiv \mathrm{tr}\big[\gamma^\mu (\not{\ell} + m)\gamma^\nu (\not{\ell} + \not{q} + m)\big]. \tag{19.31}$$

We compute the numerator trace first, then dispose of the loop integral.

19.2.2 The Dirac trace

Expanding the product inside $N^{\mu\nu}$ generates four terms; the two with an odd number of gamma matrices vanish by Eq. (C.197), leaving the $\not{\ell}\,(\not{\ell} + \not{q})$ term and the m^2 term. Using the four-gamma trace Eq. (C.199) for the first and the two-gamma trace Eq. (C.198) for the second, both unchanged in form in d dimensions because their derivation never used the integer value of $\eta^\mu{}_\mu$ inside the trace (Section C.7.3),

$$\begin{aligned} \mathrm{tr}\big[\gamma^\mu \not{\ell}\, \gamma^\nu (\not{\ell} + \not{q})\big] &= \ell_\alpha (\ell+q)_\beta \,\mathrm{tr}\big[\gamma^\mu \gamma^\alpha \gamma^\nu \gamma^\beta\big] = 4\,\ell_\alpha (\ell+q)_\beta \big(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\nu\alpha}\big) \\ &= 4\big[\ell^\mu (\ell+q)^\nu + \ell^\nu (\ell+q)^\mu - \eta^{\mu\nu}\, \ell\cdot(\ell+q)\big], \qquad (19.32) \\ m^2\,\mathrm{tr}\big[\gamma^\mu \gamma^\nu\big] &= 4\, m^2\, \eta^{\mu\nu}. \qquad (19.33) \end{aligned}$$

Adding the two,

$$N^{\mu\nu}(\ell, q) = 4\Big[\ell^\mu (\ell+q)^\nu + \ell^\nu (\ell+q)^\mu - \eta^{\mu\nu}\big(\ell\cdot(\ell+q) - m^2\big)\Big]. \tag{19.34}$$

This numerator is a polynomial of degree two in the loop momentum; against the two-power denominator the integral is, by naive power counting in four dimensions, quadratically divergent. We will find that gauge invariance reduces the genuine divergence to a single logarithm, and that the would-be quadratic piece cancels identically.

19.2.3 Feynman parametrization and the loop shift

To make the loop integration tractable the two denominators must be fused into one. The Feynman identity (built in the regularization toolkit, Eq. (19.10), and proved in general in Theorem D.11),

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x \, \frac{1}{\big[xA + (1-x)B\big]^2}, \tag{19.35}$$

applied with $A = (\ell+q)^2 - m^2 + i\epsilon$ and $B = \ell^2 - m^2 + i\epsilon$, combines the denominators into

$$xA+(1-x)B = \ell^2+2x\,\ell\cdot q+x\,q^2-m^2+i\epsilon = (\ell+xq)^2-\Delta+i\epsilon, \qquad \Delta \equiv m^2-x(1-x)q^2, \tag{19.36}$$

where the second equality completes the square. The shift $\ell \to \ell - xq$ is a legitimate translation of the integration variable under the analytic continuation that defines the dimensionally regularized integral (the shift invariance Theorem 19.1); it removes the term linear in ℓ and leaves all external dependence sequestered in the single scalar Δ. Performing the same shift in the numerator and writing $\ell^\mu \to \ell^\mu - xq^\mu$, $(\ell+q)^\nu \to \ell^\nu + (1-x)q^\nu$,

$$N^{\mu\nu} = 4\Big[(\ell^\mu - xq^\mu)(\ell^\nu + (1-x)q^\nu) + (\mu \leftrightarrow \nu) - \eta^{\mu\nu}\big((\ell - xq)\cdot(\ell + (1-x)q) - m^2\big)\Big]. \tag{19.37}$$

The denominator $[(\ell^2-\Delta)]^2$ is now even in ℓ, so any term in the numerator with an odd number of factors of ℓ integrates to zero and may be dropped. Among the even terms, the rank-two piece $\ell^\mu\ell^\nu$ may be replaced under the symmetric integral by its Lorentz-covariant average: the integral $\int \mathrm{d}^d\ell\, \ell^\mu\ell^\nu f(\ell^2)$ is a symmetric Lorentz tensor built only from $\eta^{\mu\nu}$, and contracting both sides with $\eta_{\mu\nu}$ fixes the coefficient, so

$$\ell^\mu\ell^\nu \;\longrightarrow\; \frac{\ell^2}{d}\,\eta^{\mu\nu} \qquad (\text{under } \int \mathrm{d}^d\ell\,(\cdots)/(\ell^2-\Delta)^2). \tag{19.38}$$

Keeping only the surviving even terms, the cross product $\ell^\mu\ell^\nu$ appears twice (once from each ordering) and contributes $2(\ell^2/d)\eta^{\mu\nu}$; the products $-xq^\mu\cdot(1-x)q^\nu$ appear twice and contribute $-2x(1-x)q^\mu q^\nu$; and in the contraction $(\ell - xq)\cdot(\ell+(1-x)q) = \ell^2 - x(1-x)q^2 + (\text{odd})$ only the even part is kept. Assembling,

$$N^{\mu\nu} \;\longrightarrow\; 4\Big[\tfrac{2}{d}\,\ell^2\,\eta^{\mu\nu} - 2x(1-x)q^\mu q^\nu - \eta^{\mu\nu}\big(\ell^2 - x(1-x)q^2 - m^2\big)\Big]. \tag{19.39}$$

Group the ℓ^2 terms, whose coefficient is $4\eta^{\mu\nu}\,\ell^2\,(\frac{2}{d}-1) = 4\eta^{\mu\nu}\,\ell^2\,\frac{2-d}{d}$, the $q^\mu q^\nu$ term, and the constant $\eta^{\mu\nu}$ term proportional to $x(1-x)q^2 + m^2$. The decisive feature is the factor $\frac{2-d}{d}$ multiplying ℓ^2: it is exactly 1 less than what the naive four-dimensional manipulation would give, and that deficit is the dimensional-regularization fingerprint that will cancel the quadratic divergence. We turn to the integrals.

19.2.4 The loop integrals and the quadratic-divergence cancellation

Two scalar integrals occur, both special cases of the master scalar one-loop integral Eq. (19.17) established in the previous section,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^n} = \frac{(-1)^n\,i}{(4\pi)^{d/2}}\,\frac{\Gamma(n-\frac{d}{2})}{\Gamma(n)}\,\frac{1}{\Delta^{\,n-d/2}}, \tag{19.40}$$

established in the previous section (Eq. (19.17)) by Wick rotation and the radial integration. Write

$$I_2 \equiv \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta)^2} = \frac{i}{(4\pi)^{d/2}}\,\Gamma(2-\tfrac{d}{2})\,\Delta^{d/2-2}, \tag{19.41}$$

$$I_1 \equiv \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{\ell^2-\Delta} = \frac{-i}{(4\pi)^{d/2}}\,\Gamma(1-\tfrac{d}{2})\,\Delta^{d/2-1}. \tag{19.42}$$

The integral of ℓ^2 against the squared denominator is reduced to these by the algebraic identity $\ell^2 = (\ell^2-\Delta)+\Delta$, which is the cleanest way to see the cancellation without ever writing down a spurious power divergence:

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^2} = I_1 + \Delta\, I_2. \tag{19.43}$$

Using $\Gamma(1-\frac{d}{2}) = \Gamma(2-\frac{d}{2})/(1-\frac{d}{2})$ we have $I_1 = \dfrac{-\Delta}{1-d/2}\,I_2 = \dfrac{2\Delta}{d-2}\,I_2$, whence

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^2} = \Big(\frac{2}{d-2}+1\Big)\Delta\, I_2 = \frac{d}{d-2}\,\Delta\, I_2. \tag{19.44}$$

Now combine the three numerator pieces of Eq. (19.39), each multiplied by the appropriate integral. The ℓ^2 piece carries the trace factor $4\,\frac{2-d}{d}\,\eta^{\mu\nu}$ and the integral Eq. (19.44):

$$4\,\frac{2-d}{d}\,\eta^{\mu\nu}\cdot\frac{d}{d-2}\,\Delta\, I_2 = 4\,\eta^{\mu\nu}\,\Delta\, I_2\cdot\frac{2-d}{d-2} = -4\,\eta^{\mu\nu}\,\Delta\, I_2. \tag{19.45}$$

The would-be most-divergent term has produced something finite-looking, proportional to $\Delta\, I_2$: the factor $d/(d-2)$ from the integral and the factor $(2-d)/d$ from the d-dimensional trace conspire to leave -1, the explicit manifestation in dimensional regularization that the apparent quadratic divergence is not there. Adding the constant $\eta^{\mu\nu}$ piece $4\eta^{\mu\nu}(x(1-x)q^2+m^2)\,I_2$ and recalling $\Delta = m^2 - x(1-x)q^2$,

$$\begin{aligned} -4\,\eta^{\mu\nu}\,\Delta\, I_2 + 4\,\eta^{\mu\nu}(x(1-x)q^2+m^2)I_2 &= 4\,\eta^{\mu\nu}\,I_2\big[-(m^2-x(1-x)q^2)+x(1-x)q^2+m^2\big] \\ &= 8\,x(1-x)\,q^2\,\eta^{\mu\nu}\,I_2. \end{aligned} \tag{19.46}$$

Together with the $q^\mu q^\nu$ piece $-8x(1-x)q^\mu q^\nu I_2$, the full integrated numerator is

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{N^{\mu\nu}}{(\ell^2-\Delta)^2} = 8\,x(1-x)\,(q^2\eta^{\mu\nu}-q^\mu q^\nu)\,I_2. \tag{19.47}$$

Two things have happened at once, and both are consequences of gauge invariance made arithmetic. The m^2 in the constant term, which by itself would have produced a q-independent $\propto \eta^{\mu\nu}$ contribution — a photon *mass* — has cancelled exactly against the m^2 inside Δ carried in from the ℓ^2 integral. And the surviving structure is precisely the transverse combination $q^2\eta^{\mu\nu}-q^\mu q^\nu$. There was no room for any other outcome: the calculation was forced to be transverse, and we never imposed it.

19.2.5 Transversality and the result for the scalar form factor

Restoring the prefactor $-e^2$ of Eq. (19.31) and the Feynman-parameter integral, and inserting the explicit I_2,

$$i\Pi^{\mu\nu}(q) = -e^2\int_0^1 \mathrm{d}x\; 8\,x(1-x)(q^2\eta^{\mu\nu}-q^\mu q^\nu)\,\frac{i}{(4\pi)^{d/2}}\,\Gamma(2-\tfrac{d}{2})\,\Delta^{d/2-2}\,\mu^{2\varepsilon}, \tag{19.48}$$

where the factor $\mu^{2\varepsilon}$ restores the engineering dimension of the coupling in $d=4-2\varepsilon$, as fixed by the regularization scheme of Section 19.1. This is manifestly of the transverse form, and we record it as the structural result of the section.

Proposition 19.5 (Transversality of the vacuum polarization)**.** *The one-loop photon self-energy is purely transverse,*

$$i\Pi^{\mu\nu}(q) = i(q^2\eta^{\mu\nu}-q^\mu q^\nu)\,\Pi(q^2), \qquad q_\mu\Pi^{\mu\nu}(q)=0, \tag{19.49}$$

with a single scalar form factor $\Pi(q^2)$. No term proportional to $\eta^{\mu\nu}$ alone (a photon-mass term) is generated. The all-orders statement, with transversality enforced diagram by diagram by the Ward–Takahashi identity, is proved in Theorem 19.13; here it has been verified by direct computation at one loop.

Proof. Equation Eq. (19.47) exhibits the integrated loop as a Feynman-parameter integral of $(q^2\eta^{\mu\nu}-q^\mu q^\nu)$ times a scalar, so $i\Pi^{\mu\nu}=i(q^2\eta^{\mu\nu}-q^\mu q^\nu)\Pi(q^2)$ with the scalar identified below. Contracting with q_μ gives $q_\mu(q^2\eta^{\mu\nu}-q^\mu q^\nu)=q^2q^\nu-q^2q^\nu=0$, which is transversality. □

Reading off the scalar form factor and simplifying the prefactor with $8/(4\pi)^2=1/(2\pi^2)$ and $e^2/(2\pi^2)=2\alpha/\pi$ (where $\alpha=e^2/4\pi$), the dimensional expansion of

$\Gamma(2-\frac{d}{2})\Delta^{d/2-2}\mu^{2\varepsilon}$ is exactly the logarithmic master integral Eq. (19.18) of the previous section (equivalently Eq. (D.136)),

$$\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta)^2}=\frac{i}{(4\pi)^2}\Big[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}\Big]+O(\varepsilon),\qquad \frac{1}{\bar\varepsilon}\equiv\frac{1}{\varepsilon}-\gamma_{\mathrm{E}}+\ln 4\pi, \tag{19.50}$$

the modified-minimal-subtraction pole and finite part recalled from Appendix D. Hence the bare one-loop scalar form factor is

$$\Pi(q^2)=-\frac{2\alpha}{\pi}\int_0^1\mathrm{d}x\;x(1-x)\left[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}\right],\qquad \Delta=m^2-x(1-x)q^2. \tag{19.51}$$

The ultraviolet divergence is the single $1/\bar\varepsilon$ pole; the dependence on the external momentum and the mass lives entirely in the finite logarithm. The Feynman-parameter integral $\int_0^1\mathrm{d}x\,x(1-x)=\frac{1}{6}$ makes the pole explicit,

$$\Pi(q^2)=-\frac{\alpha}{3\pi}\,\frac{1}{\bar\varepsilon}-\frac{2\alpha}{\pi}\int_0^1\mathrm{d}x\;x(1-x)\,\ln\frac{\mu^2}{\Delta}+O(\varepsilon), \tag{19.52}$$

and its value at zero momentum, where $\Delta\to m^2$, is the pure constant

$$\Pi(0)=-\frac{\alpha}{3\pi}\left[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{m^2}\right]. \tag{19.53}$$

A momentum cutoff would manufacture a photon mass

The cancellation of the $\eta^{\mu\nu}$-only term in Eq. (19.47) is delicate, and it is delicate *because* it is the place where the regulator either respects gauge invariance or betrays it. A naïve Euclidean cutoff $|\ell_E|<\Lambda$ on the integral Eq. (19.31) does not give the transverse answer. The quadratically divergent part of $\int\mathrm{d}^4\ell\,\ell^2/(\ell^2-\Delta)^2$ then evaluates to a term $\propto\Lambda^2\eta^{\mu\nu}$ with *no* accompanying $q^\mu q^\nu$, because a hard cutoff is not Lorentz invariant in the way dimensional continuation is and does not supply the compensating d-dependent trace factor of Eq. (19.45). The upshot would be a self-energy of the form $\Pi^{\mu\nu}\supset c\,\Lambda^2\,\eta^{\mu\nu}$, i.e. a photon-mass term $\frac{1}{2}c\Lambda^2A_\mu A^\mu$ generated radiatively — a gross violation of the Ward identity and of the masslessness of the photon, and an artifact of the regulator rather than physics. Dimensional regularization sets scaleless power divergences to zero and keeps the trace algebra d-dependent, so the spurious mass never appears: $\Pi^{\mu\nu}$ is transverse identically in ε. This is the concrete sense, promised in the chapter introduction, in which dim reg "preserves gauge invariance" while a cutoff corrupts it.

19.2.6 On-shell renormalization and Z_3

The divergence in Eq. (19.51) is removed by the photon field-strength counterterm of renormalized perturbation theory. Writing the bare photon kinetic term as $-\frac{1}{4}Z_3\,F_{\mu\nu}F^{\mu\nu}$ with $Z_3 = 1+\delta_3$ (Section 19.1), the counterterm contributes a two-point vertex $-i\,\delta_3\,(q^2\eta^{\mu\nu} - q^\mu q^\nu)$ — itself transverse, as it must be since it comes from a gauge-invariant operator. The renormalized self-energy is the sum,

$$i\Pi^{\mu\nu}_{\text{ren}}(q) = i\big(q^2\eta^{\mu\nu} - q^\mu q^\nu\big)\big[\Pi(q^2) - \delta_3\big]. \tag{19.54}$$

The on-shell renormalization condition fixes δ_3 by demanding that the full photon propagator have a pole at $q^2 = 0$ (the photon stays massless) with residue exactly 1 (unit normalization of the photon field at the pole). Both requirements are met by requiring the renormalized form factor to vanish at zero momentum,

$$\Pi_{\text{ren}}(0) = 0 \quad\Longrightarrow\quad \delta_3 = \Pi(0) = -\frac{\alpha}{3\pi}\left[\frac{1}{\bar\varepsilon} + \ln\frac{\mu^2}{m^2}\right]. \tag{19.55}$$

The field-strength renormalization constant is therefore

$$Z_3 = 1+\delta_3 = 1 - \frac{\alpha}{3\pi}\,\frac{1}{\bar\varepsilon} + (\text{finite}) = 1 - \frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon} + \cdots, \tag{19.56}$$

the divergent part written in both the α and the e^2 normalizations. The minus sign — $Z_3 < 1$ — is the seed of charge screening, to which we return below; it is also, through the relation $e^2 = Z_3\,e_0^2$ that Chapter 20 uses to define the physical charge, the origin of the positive QED beta function $\beta_e = e^3/12\pi^2$ derived systematically in Chapter 24.

Dimensional convention and the pole of Z_3

We use the book's standing convention $d = 4-2\varepsilon$ (Section 19.1), in which the ultraviolet pole of a one-loop logarithm is $1/\varepsilon$. With this convention $Z_3 = 1 - \dfrac{e^2}{12\pi^2\varepsilon} + \cdots$ as in Eq. (19.56). Texts that set $d = 4-\varepsilon$ (a single ε, e.g. Peskin–Schroeder) have a logarithmic pole $2/\varepsilon$ and accordingly quote $Z_3 = 1 - \dfrac{e^2}{6\pi^2\varepsilon} + \cdots$. The two statements are identical physics written in two conventions for ε; the physical, convention-free content is the residue $-\alpha/3\pi$ of the pole in the $\overline{\text{MS}}$ variable $1/\bar\varepsilon$, which controls the running and is the same in both.

Subtracting Eq. (19.53) from Eq. (19.51) cancels the pole and the $\ln\mu^2$ together — the on-shell scheme trades the renormalization scale for the physical mass m — and leaves the finite, scale-independent, on-shell-subtracted form

factor,

$$\boxed{\widehat{\Pi}(q^2) \equiv \Pi(q^2) - \Pi(0) = -\frac{2\alpha}{\pi}\int_0^1 \mathrm{d}x\; x(1-x)\,\ln\frac{m^2}{m^2 - x(1-x)q^2}\,, \qquad \widehat{\Pi}(0) = 0.} \tag{19.57}$$

This is the central output of the section, the function Chapter 20 imports to build the running coupling and the Uehling potential. The logarithm is real for spacelike $q^2 < 0$ (where the argument exceeds 1, so $\widehat{\Pi} > 0$) and for timelike q^2 below the pair threshold $q^2 < 4m^2$; above threshold it develops an imaginary part, the subject of Section 19.2.8.

19.2.7 The two ends of the form factor

Two regimes carry most of the physics, and both follow from Eq. (19.57) by elementary integration.

Small momentum, $|q^2| \ll m^2$. Expand the logarithm: $\ln\frac{m^2}{m^2 - x(1-x)q^2} = -\ln\!\big(1 - x(1-x)q^2/m^2\big) = x(1-x)\frac{q^2}{m^2} + O(q^4)$. The leading term integrates against the weight using $\int_0^1 \mathrm{d}x\,[x(1-x)]^2 = \frac{1}{30}$,

$$\widehat{\Pi}(q^2) = -\frac{2\alpha}{\pi}\cdot\frac{q^2}{m^2}\int_0^1 \mathrm{d}x\,[x(1-x)]^2 + O(q^4) = -\frac{2\alpha}{\pi}\cdot\frac{1}{30}\cdot\frac{q^2}{m^2} + \cdots = -\frac{\alpha}{15\pi}\frac{q^2}{m^2} + \cdots . \tag{19.58}$$

The linear-in-q^2 coefficient $-\alpha/15\pi m^2$ is exactly the constant that, Fourier transformed, becomes the short-range Uehling correction to the Coulomb potential responsible for the vacuum-polarization part of the Lamb shift (Chapter 20).

Large spacelike momentum, $-q^2 \equiv Q^2 \gg m^2$. Now $m^2 - x(1-x)q^2 = m^2 + x(1-x)Q^2 \simeq x(1-x)Q^2$ away from the endpoints, so $\ln\frac{m^2}{m^2 - x(1-x)q^2} \to -\ln\frac{Q^2}{m^2} - \ln[x(1-x)]$. Using the two moments $\int_0^1 \mathrm{d}x\,x(1-x) = \frac{1}{6}$ and $\int_0^1 \mathrm{d}x\,x(1-x)\ln[x(1-x)] = -\frac{5}{18}$,

$$\widehat{\Pi}(q^2) = \frac{2\alpha}{\pi}\left[\frac{1}{6}\ln\frac{Q^2}{m^2} - \frac{5}{18}\right] + O\!\left(\frac{m^2}{Q^2}\right) = \frac{\alpha}{3\pi}\left[\ln\frac{-q^2}{m^2} - \frac{5}{3}\right] + O\!\left(\frac{m^2}{q^2}\right). \tag{19.59}$$

The leading behavior is the famous logarithm with coefficient $\alpha/3\pi = e^2/12\pi^2$ — the same number that sits in Z_3 and in β_e, as it must, since all three measure the strength of the single $1/\bar{\varepsilon}$ pole. The constant $-5/3$ is the finite remainder of the on-shell scheme and fixes the scale at which the running is referenced. We will see in Chapter 20 that resumming the geometric chain of these insertions turns Eq. (19.59) into the running fine-structure constant $\alpha(q^2) = \alpha/[1 - (\alpha/3\pi)\ln(-q^2/m^2)]$.

19.2.8 Analytic structure: the cut at $q^2 = 4m^2$ and the imaginary part

The renormalized form factor Eq. (19.57) is an analytic function of q^2 in the complex plane, but not an entire one. Its only singularity is a branch cut, and the location and content of that cut are physics. The argument of the logarithm, $\Delta/m^2 = 1 - x(1-x)q^2/m^2$, becomes negative when $x(1-x)q^2 > m^2$. As x ranges over $(0,1)$ the product $x(1-x)$ attains its maximum $\frac{1}{4}$ at $x = \frac{1}{2}$, so the first value of q^2 for which any x makes Δ negative is

$$q^2 = \frac{m^2}{\max_x x(1-x)} = \frac{m^2}{1/4} = 4m^2. \tag{19.60}$$

This is the branch point: $\widehat{\Pi}(q^2)$ is analytic for $q^2 < 4m^2$ (the whole spacelike axis and the sub-threshold timelike segment) and acquires a cut along $q^2 \geq 4m^2$. The threshold $q^2 = 4m^2$ is exactly the invariant mass-squared at which a timelike photon can create a real on-shell electron–positron pair, $2m$ being the minimum energy for that. The Feynman $i\epsilon$, carried since Eq. (19.30), fixes which boundary value of the cut is physical: $q^2 \to q^2 + i\epsilon$, i.e. $\Delta \to \Delta - i\epsilon$, so for $\Delta < 0$ the logarithm acquires a definite imaginary part,

$$\ln \frac{m^2}{\Delta - i\epsilon} = \ln \frac{m^2}{|\Delta|} + i\pi \qquad (\Delta < 0). \tag{19.61}$$

This is the Sokhotski–Plemelj boundary value of Section 5.1.5, and the appearance of a branch cut tied to a physical threshold is the general phenomenon of Section 5.5. The imaginary part of $\widehat{\Pi}$ is then the integral of $-\frac{2\alpha}{\pi} \cdot \pi\, x(1-x)$ over the sub-interval of x for which $\Delta < 0$.

Worked Example 19.6 (The imaginary part above threshold)**.** We evaluate $\operatorname{Im}\widehat{\Pi}(q^2)$ for $q^2 > 4m^2$ from Eq. (19.57). The condition $\Delta = m^2 - x(1-x)q^2 < 0$ is $x(1-x) > m^2/q^2$. The roots of $x(1-x) = m^2/q^2$ are

$$x_\pm = \tfrac{1}{2}(1 \pm \beta), \qquad \beta \equiv \sqrt{1 - \frac{4m^2}{q^2}}, \tag{19.62}$$

so $\Delta < 0$ precisely for $x \in (x_-, x_+)$, and there the logarithm contributes $+i\pi$. Hence

$$\operatorname{Im}\widehat{\Pi}(q^2) = -\frac{2\alpha}{\pi} \cdot \pi \int_{x_-}^{x_+} \mathrm{d}x\; x(1-x) = -2\alpha \int_{x_-}^{x_+} \mathrm{d}x\; x(1-x). \tag{19.63}$$

The remaining integral is elementary in the symmetric variable $u = x - \frac{1}{2} \in (-\frac{\beta}{2}, \frac{\beta}{2})$, where $x(1-x) = \frac{1}{4} - u^2$:

$$\int_{x_-}^{x_+} \mathrm{d}x\; x(1-x) = \int_{-\beta/2}^{\beta/2} \mathrm{d}u \left(\tfrac{1}{4} - u^2\right) = \frac{\beta}{4} - \frac{\beta^3}{12} = \frac{\beta}{12}(3 - \beta^2) = \frac{\beta}{6}\Big(1 + \frac{2m^2}{q^2}\Big), \tag{19.64}$$

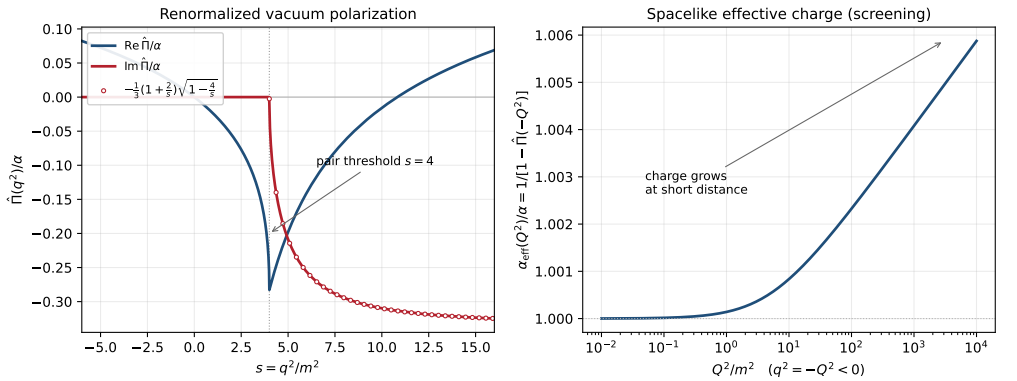


Figure 19.4: Left: the renormalized one-loop vacuum polarization $\widehat{\Pi}(q^2)/\alpha$ from Eq. (19.57), as a function of $s = q^2/m^2$. The real part (blue) is smooth through $s = 0$ where $\widehat{\Pi}(0) = 0$, grows logarithmically at large spacelike $|s|$ per Eq. (19.59), and has a cusp at the pair threshold $s = 4$. The imaginary part (red) is zero below threshold and, above it, follows the closed form $-\frac{1}{3}(1+2/s)\sqrt{1-4/s}$ of Eq. (19.65) (open circles), with the characteristic square-root onset and the plateau at $-\alpha/3$. Right: the spacelike effective charge $\alpha_{\text{eff}}(Q^2)/\alpha = 1/[1 - \widehat{\Pi}(-Q^2)]$, growing toward the ultraviolet: charge screening in momentum space, the seed of the running coupling developed in Chapter 20.

using $3 - \beta^2 = 2 + 4m^2/q^2$. Therefore

$$\operatorname{Im}\widehat{\Pi}(q^2) = -\frac{\alpha}{3}\Big(1 + \frac{2m^2}{q^2}\Big)\sqrt{1 - \frac{4m^2}{q^2}} \qquad (q^2 > 4m^2), \tag{19.65}$$

and $\operatorname{Im}\widehat{\Pi} = 0$ below threshold. The imaginary part switches on at $q^2 = 4m^2$ with the velocity factor $\beta = \sqrt{1 - 4m^2/q^2}$ — the speed of the produced pair — vanishing at threshold and rising to the plateau $\operatorname{Im}\widehat{\Pi} \to -\alpha/3$ as $q^2 \to \infty$, where the electron mass becomes negligible. Figure 19.4 plots $\operatorname{Re}\widehat{\Pi}$ and $\operatorname{Im}\widehat{\Pi}$ together and shows the square-root onset at the threshold.

The imaginary part is not optional bookkeeping: it is demanded by unitarity. Above threshold the virtual photon *can* decay into a real pair, and the optical theorem ties the discontinuity of the two-point function across its cut to the production rate of those physical states. We verify the match.

Worked Example 19.7 ($\operatorname{Im}\widehat{\Pi}$ from the optical theorem)**.** The generalized optical theorem (Theorem 14.37) is the statement, derived from $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$, that the discontinuity of an amplitude across a physical cut equals a sum over on-shell intermediate states of the product of production and decay sub-amplitudes. Applied to the amputated photon two-point function, with the only state available at $O(e^2)$ being a single electron–positron pair, it reads (the Cutkosky construction

of the cut puts the two fermion lines on shell; Chapter 60 systematizes it)

$$2\,\mathrm{Im}\,\Pi^{\mu\nu}(q) = \int \mathrm{d}\Pi_2 \sum_{\text{spins}} \left[\bar{u}(p_1)(-ie\gamma^\mu)v(p_2)\right]\left[\bar{u}(p_1)(-ie\gamma^\nu)v(p_2)\right]^*, \quad (19.66)$$

where p_1, p_2 are the pair momenta ($p_1+p_2=q$, each on shell, $p_i^2=m^2$) and $\mathrm{d}\Pi_2$ is the Lorentz-invariant two-body phase space of Eq. (16.55). The spin sum is the same trace already met in e^+e^- annihilation (Chapter 17):

$$\sum_{\text{spins}} \left[\bar{u}_1\gamma^\mu v_2\right]\left[\bar{u}_1\gamma^\nu v_2\right]^* = \mathrm{tr}\left[(\not{p}_1+m)\gamma^\mu(\not{p}_2-m)\gamma^\nu\right] = 4\left[p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - \eta^{\mu\nu}(p_1\cdot p_2+m^2)\right]. \quad (19.67)$$

Rather than carry the tensor structure we project onto the scalar form factor. Contract Eq. (19.66) with $\eta_{\mu\nu}$; the left side gives $2\,\mathrm{Im}[\eta_{\mu\nu}\Pi^{\mu\nu}] = 2\,\mathrm{Im}[(q^2 d - q^2)\Pi] \to 6q^2\,\mathrm{Im}\,\Pi$ at $d=4$ by Eq. (19.49). On the right, $\eta_{\mu\nu}$ times the spin trace is

$$4e^2\left[2\,p_1\cdot p_2 - 4(p_1\cdot p_2+m^2)\right] = -8e^2(p_1\cdot p_2+2m^2) = -4e^2\,q^2\Big(1+\frac{2m^2}{q^2}\Big), \quad (19.68)$$

where $p_1\cdot p_2 = \tfrac{1}{2}(q^2-2m^2)$ from $q^2=(p_1+p_2)^2$, so $p_1\cdot p_2+2m^2 = \tfrac{1}{2}q^2(1+2m^2/q^2)$. The integrand is independent of the emission angle, so the phase-space integral simply multiplies it by the total two-body phase-space volume $\int \mathrm{d}\Pi_2 = \beta/8\pi$ (Eq. (16.55)). Assembling,

$$6q^2\,\mathrm{Im}\,\Pi = \frac{\beta}{8\pi}\cdot\Big[-4e^2q^2\Big(1+\frac{2m^2}{q^2}\Big)\Big] \;\Longrightarrow\; \mathrm{Im}\,\Pi = -\frac{e^2}{12\pi}\,\beta\Big(1+\frac{2m^2}{q^2}\Big) = -\frac{\alpha}{3}\Big(1+\frac{2m^2}{q^2}\Big)\sqrt{1-\frac{4}{} \quad (19.69)$$

using $e^2=4\pi\alpha$. Since Π and $\widehat{\Pi}$ differ only by the real constant $\Pi(0)$, $\mathrm{Im}\,\Pi = \mathrm{Im}\,\widehat{\Pi}$, and the result is identical to Eq. (19.65). The loop computation and the unitarity cut agree exactly: the imaginary part of the photon self-energy *is* the rate at which a virtual photon makes real pairs, weighted by phase space. This is the cross-check promised in the section goal, and it is the physical content from which Chapter 20 reconstructs the whole of $\widehat{\Pi}$ by a dispersion relation, using the analyticity and Kramers–Kronig machinery of Section 5.5.5.

The vacuum as a polarizable medium

The function $\widehat{\Pi}(q^2)$ has a single, vivid physical reading. The QED vacuum is not empty: it teems with virtual electron–positron pairs, fluctuations permitted for a time $\sim 1/2m$ by the uncertainty principle. A bare charge placed in this medium polarizes those pairs — virtual positrons drawn inward, virtual electrons pushed outward — so the charge is partially screened by a co-moving cloud of opposite charge, exactly as a test charge in a dielectric is shielded by induced polarization. A probe at large

distance, i.e. small spacelike $-q^2$, sees the fully screened charge and measures the textbook $\alpha \approx 1/137$; a probe that penetrates the cloud, large $-q^2$, sees more of the bare charge and so a *larger* effective coupling. This is the content of Eq. (19.59): $\widehat{\Pi}$ grows like $+\frac{\alpha}{3\pi}\ln(-q^2/m^2)$, and through the resummation of Chapter 20 the effective charge grows with it. The sign $Z_3 < 1$ in Eq. (19.56) is the screening made quantitative. That fermion loops screen — driving α *up* in the ultraviolet — is the abelian face of the single most consequential sign in particle physics; its non-abelian counterpart, where gauge-boson self-interactions antiscreen and the coupling falls in the ultraviolet (asymptotic freedom, Chapter 27), is what makes QCD a sensible short-distance theory. The systematic statement of all this — promoting the one-loop logarithm to an all-orders renormalization-group flow — is the business of Chapter 24; the present section has supplied the one-loop input on which that flow is built.

We have delivered the first of the three one-loop building blocks of QED: a renormalized, manifestly transverse photon two-point function $\widehat{\Pi}(q^2)$ with $\widehat{\Pi}(0) = 0$, the field-strength renormalization Z_3 of Eq. (19.56), and the full analytic structure — branch cut at the pair threshold, imaginary part cross-checked against the optical theorem — that seeds charge screening and the running coupling. The loop-integral machinery assembled here (Feynman parametrization, the loop shift, the master integral, and the d-dimensional numerator algebra that enforces gauge invariance) is the template the next two sections reuse. We turn next to the electron self-energy $\Sigma(p)$, where the same toolkit confronts a new feature absent here — an infrared divergence — alongside the mass and field-strength renormalizations δm and Z_2 (Section 19.3).

19.3 The Electron Self-Energy

The vacuum-polarization calculation of the previous section delivered the first of the three one-particle-irreducible building blocks of one-loop QED, the photon two-point function $\Pi^{\mu\nu}(q)$, together with a portable template: write down the amplitude with the Feynman rules, contract the Dirac and Lorentz algebra against the d-dimensional identities of Appendix C, combine denominators with a Feynman parameter, shift the loop momentum to a perfect square, and reduce the result to the master scalar integral of Theorem D.22. That template renormalized the photon line and handed us the renormalized propagator and the field renormalization Z_3. We now turn the same machine on the *electron* line. The object of study is the electron self-energy $\Sigma(p)$, the sum of all one-particle-irreducible insertions into the fermion propagator; at one loop it is the single diagram in which the electron emits and reabsorbs a virtual photon. Computing it forces two new features into the open that the photon calculation could not

show. First, the on-shell renormalization conditions split into *two* statements, one fixing the pole position (the mass counterterm δm) and one fixing the residue (the field-strength renormalization Z_2), and we will derive both from the demand that the renormalized propagator have a simple pole at the physical mass with unit residue. Second, $\Sigma(p)$ harbours the first *infrared* divergence of QED: a divergence that has nothing to do with large loop momenta, that is not removed by any counterterm, and that depends on how one regulates the massless photon. Disentangling this physical infrared pole from the ultraviolet pole that Z_2 and δm legitimately subtract is the conceptual payload of this section, and the infrared piece is the quantity that Section 19.6 will finally cancel against real photon emission.

19.3.1 The amplitude and its Dirac decomposition

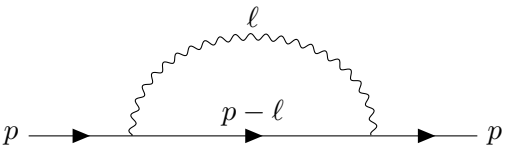


Figure 19.5: The one-loop electron self-energy $-i\Sigma_2(p)$. An incoming electron of momentum p emits a virtual photon of momentum ℓ at the left vertex, propagates with momentum $p-\ell$, and reabsorbs the photon at the right vertex. The blob is one-particle-irreducible: cutting any single internal line leaves the diagram connected.

The electron self-energy $-i\Sigma(p)$ is, by definition, the sum of all amputated one-particle-irreducible (1PI) diagrams with two external electron legs, stripped of the external propagators. At order e^2 there is exactly one such diagram, drawn in Fig. 19.5. Reading it with the QED Feynman rules—a vertex factor $-ie\gamma^\mu$ at each end (Eq. (17.10)), the internal electron propagator of Eq. (17.15), and the Feynman-gauge photon propagator of Eq. (17.18)—and routing the loop momentum ℓ through the photon, so that the electron line carries $p-\ell$, gives

$$-i\Sigma_2(p) \;=\; (-ie)^2\,\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\gamma^\mu\,\frac{i\,(\not{p}-\not{\ell}+m)}{(p-\ell)^2-m^2+i\varepsilon}\,\gamma^\nu\,\frac{-i\,\eta_{\mu\nu}}{\ell^2-\mu_\gamma^2+i\varepsilon}\,. \tag{19.70}$$

Three conventions are already in force and worth stating once. We work in Feynman gauge $\xi=1$, the default of QED; the gauge-parameter dependence of Z_2 is real, but the gauge-invariant combinations δm and the residue at the physical pole do not depend on it. The factor $\mu^{2\varepsilon}$ accompanies the coupling: in $d=4-2\varepsilon$ dimensions the bare charge is not dimensionless, and writing $e\to e\,\mu^{\varepsilon}$ with e dimensionless and μ the renormalization scale keeps the loop integral dimensionally honest, exactly as in Section 19.2. Finally we have given the

photon a small fictitious mass μ_γ in its propagator. This is purely an infrared regulator; it plays no role whatsoever in the ultraviolet analysis, and we will set it to zero at the end of every ultraviolet manipulation. Its only function is to keep the integrals finite at the lower end of the ℓ integration so that the infrared divergence can be exhibited as an explicit $\ln \mu_\gamma$ rather than hidden. We will also show how the same divergence appears as a separate $1/\varepsilon$ pole if one instead regulates the infrared dimensionally.

Strip the explicit factors of i. The two vertices supply $(-ie)^2 = -e^2$, the fermion propagator an i, the photon propagator a $-i$, so the product of imaginary units is $(-e^2)\,(i)(-i) = -e^2$. Contracting $\gamma^\mu \cdots \gamma^\nu$ with $\eta_{\mu\nu}$ closes the two Dirac matrices around the numerator, and cancelling the overall $-i$ on the left against the $-e^2$ leaves

$$\Sigma_2(p) \;=\; -\,i\,e^2\,\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \frac{\gamma^\mu\,(\not p - \not \ell + m)\,\gamma_\mu}{\left[(p-\ell)^2 - m^2 + i\varepsilon\right]\left[\ell^2 - \mu_\gamma^2 + i\varepsilon\right]}\,. \tag{19.71}$$

The numerator is a contraction of the type catalogued in d dimensions in Theorem C.61. Using $\gamma^\mu \not a \gamma_\mu = (2-d)\not a$ (Eq. (C.188)) on the momentum part and $\gamma^\mu\gamma_\mu = d$ (Eq. (C.187)) on the mass part,

$$\gamma^\mu\,(\not p - \not \ell + m)\,\gamma_\mu \;=\; (2-d)\,(\not p - \not \ell) \;+\; d\,m\,. \tag{19.72}$$

It is precisely the d-dependence of these coefficients—$2-d = -2(1-\varepsilon)$ and the plain $d = 4 - 2\varepsilon$—that, multiplied against the $1/\varepsilon$ pole of the loop integral below, generates finite remainders. Dropping the $(d-4)$ pieces would lose genuine finite contributions, a point already flagged in Section C.7.2.

To do the loop integral, combine the two denominators with a single Feynman parameter, using the two-denominator case of the master formula $\frac{1}{AB} = \int_0^1 \mathrm{d}x\,[xA + (1-x)B]^{-2}$ from Section D.2. With $A = (p-\ell)^2 - m^2$ assigned the weight x and $B = \ell^2 - \mu_\gamma^2$ the weight $1-x$, the combined denominator is

$$\begin{aligned} x\left[(p-\ell)^2 - m^2\right] + (1-x)\left[\ell^2 - \mu_\gamma^2\right] &= \ell^2 - 2x\,p\cdot\ell + x\,p^2 - x\,m^2 - (1-x)\mu_\gamma^2 \\ &= (\ell - xp)^2 - \Delta\,, \end{aligned} \tag{19.73}$$

where completing the square in ℓ pulls out $-x^2p^2$ and the effective mass-squared is

$$\Delta(x) \;=\; -\,x(1-x)\,p^2 \;+\; x\,m^2 \;+\; (1-x)\,\mu_\gamma^2\,. \tag{19.74}$$

Shift the loop momentum, $\ell \to \ell + xp$. The shift is legitimate in dimensional regularization, where the integral is translation invariant. Under it the numerator momentum becomes $\not p - \not \ell \to (1-x)\not p - \not \ell$; the term linear in ℓ is odd under $\ell \to -\ell$ and integrates to zero by symmetric integration (Eq. (D.90)). The numerator that survives is $(2-d)(1-x)\not p + d\,m$, independent of ℓ, so the loop

integral collapses to the scalar master with $n = 2$,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\varepsilon)^2} \;=\; \frac{i}{(4\pi)^{d/2}}\,\frac{\Gamma(2-\frac{d}{2})}{\Gamma(2)}\,\Delta^{\,d/2-2} \;=\; \frac{i}{(4\pi)^{d/2}}\,\Gamma(\varepsilon)\,\Delta^{-\varepsilon}\,, \tag{19.75}$$

where we used *Eq.* (D.77) and $2-\frac{d}{2}=\varepsilon$, $\frac{d}{2}-2=-\varepsilon$, $\Gamma(2)=1$. Inserting Eq. (19.75) into Eq. (19.71) and cancelling the two factors of i ($-i\times i=1$),

$$\Sigma_2(p) \;=\; \frac{e^2}{(4\pi)^{d/2}}\,\Gamma(\varepsilon)\,\mu^{2\varepsilon}\int_0^1 \mathrm{d}x\,\big[(2-d)(1-x)\,\not{p}+d\,m\big]\,\Delta(x)^{-\varepsilon}\,. \tag{19.76}$$

Equation Eq. (19.76) is the complete one-loop self-energy in dimensional regularization. Everything that follows is its expansion and interpretation.

The output is naturally organized by its Dirac structure. The only matrices a Lorentz scalar function of p can build are $\not{p}$ and the identity, so the self-energy has the general form

$$\Sigma(p) \;=\; A(p^2)\,\not{p} \;+\; B(p^2)\,m\,, \tag{19.77}$$

with A and B dimensionless scalar functions of p^2 (we factor an explicit m out of the second term for convenience). Reading off the two coefficients from Eq. (19.76) requires only the Laurent expansion of the prefactor about $\varepsilon=0$. Write

$$G(\Delta) \;\equiv\; \Gamma(\varepsilon)\,(4\pi)^{\varepsilon}\,\mu^{2\varepsilon}\,\Delta^{-\varepsilon} \;=\; \frac{1}{\varepsilon}-\gamma_E+\ln\frac{4\pi\mu^2}{\Delta}+O(\varepsilon) \;=\; \frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}+O(\varepsilon)\,, \tag{19.78}$$

where we have absorbed $(4\pi)^{-d/2}=(4\pi)^{-2}(4\pi)^{\varepsilon}$ so that $(4\pi)^{-d/2}\Gamma(\varepsilon)\mu^{2\varepsilon}\Delta^{-\varepsilon}=(16\pi^2)^{-1}G(\Delta)$, and where the $\overline{\text{MS}}$ abbreviation

$$\frac{1}{\bar\varepsilon} \;\equiv\; \frac{1}{\varepsilon}-\gamma_E+\ln 4\pi \tag{19.79}$$

collects the pole with its universal Euler–Mascheroni and $\ln 4\pi$ companions. Substituting $2-d=-2(1-\varepsilon)$ and $d=4-2\varepsilon$ in Eq. (19.76) and expanding each coefficient, the factors of ε in $(2-d)$ and d multiply the $1/\varepsilon$ in G and leave finite pieces: $(-2+2\varepsilon)G=-2/\bar\varepsilon-2\ln(\mu^2/\Delta)+2$ and $(4-2\varepsilon)G=4/\bar\varepsilon+4\ln(\mu^2/\Delta)-2$, where we used $\varepsilon\,\Gamma(\varepsilon)=\Gamma(1+\varepsilon)\to 1$. Collecting the coefficient of $\not{p}$ and of m,

$$A(p^2) = -\frac{e^2}{8\pi^2}\int_0^1 \mathrm{d}x\,(1-x)\left[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta(x)}-1\right], \tag{19.80}$$

$$B(p^2) = \;\frac{e^2}{16\pi^2}\int_0^1 \mathrm{d}x\left[\frac{4}{\bar\varepsilon}+4\ln\frac{\mu^2}{\Delta(x)}-2\right]. \tag{19.81}$$

Performing only the trivial x-integration of the pole terms ($\int_0^1 (1-x)\mathrm{d}x = \frac{1}{2}$, $\int_0^1 \mathrm{d}x = 1$) isolates the ultraviolet divergence,

$$\Sigma_2(p)\Big|_{\text{UV pole}} \;=\; -\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}\,\not{p} \;+\; \frac{e^2}{4\pi^2}\,\frac{1}{\bar{\varepsilon}}\,m \;=\; \frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}\,(-\not{p}+4m)\,. \tag{19.82}$$

This is a local polynomial in $\not{p}$ of degree one—exactly the structure of the tree-level inverse propagator $\not{p}-m$. That is the statement, anticipated on dimensional grounds, that the self-energy divergence can be absorbed by a redefinition of the two parameters in the quadratic part of the Lagrangian: the coefficient of $\not{p}$ (the field normalization) and the coefficient of m (the mass). The remainder of Σ_2 is finite as $\varepsilon \to 0$ and carries the x-integrals over $\ln(\mu^2/\Delta)$, to which we return when we need the infrared structure.

Caution

The single power $\Delta^{-\varepsilon}$ in Eq. (19.76) hides *two* distinct sources of divergence in the one factor of $\Gamma(\varepsilon)$. The ultraviolet divergence is the genuine $\varepsilon \to 0$ pole of $\Gamma(\varepsilon)$, present for any $\Delta > 0$. A separate *infrared* divergence is hiding in the x-integral of $\ln(\mu^2/\Delta)$ and, more sharply, in the x-integrals of $1/\Delta$ that appear once we differentiate: as $\mu_\gamma \to 0$ the effective mass $\Delta(x)$ vanishes at the endpoint $x = 0$ on shell, and the parameter integral itself can diverge. The two divergences live in different places and must never be conflated. We separate them carefully below.

19.3.2 On-shell renormalization: δm and Z_2

The bare Lagrangian quantized in Section 19.1 is rewritten, in renormalized perturbation theory, as a renormalized Lagrangian plus counterterms. The fermionic quadratic part contributes a counterterm Feynman rule $i\,(\not{p}\,\delta_2 - \delta_m)$, with $\delta_2 = Z_2 - 1$ the field-strength counterterm and δ_m the mass counterterm; both are fixed by renormalization conditions, not predicted. Including this counterterm vertex alongside the loop, the full 1PI two-point function is $-i\widehat{\Sigma}(p)$ with

$$\widehat{\Sigma}(p) \;=\; \Sigma_2(p) \;-\; (\delta_2\,\not{p} - \delta_m)\,, \tag{19.83}$$

and the exact (renormalized) propagator is the geometric sum of these insertions,

$$\frac{i}{\not{p} - m - \widehat{\Sigma}(p)}\,. \tag{19.84}$$

We now impose the two physical requirements that define the *on-shell scheme*: the renormalized propagator must have its pole at the physical electron mass m, and the residue there must be the canonical i (so that the field is normalized to create one-particle states with unit amplitude, exactly the normalization LSZ demands in Eq. (14.156)).

Proposition 19.8 (On-shell conditions for δm and Z_2)**.** *The pole-at-m and unit-residue requirements on Eq. (19.84) are equivalent to the two conditions*

$$\widehat{\Sigma}(\not{p}=m)=0\,,\qquad \left.\frac{\mathrm{d}\widehat{\Sigma}}{\mathrm{d}\not{p}}\right|_{\not{p}=m}=0\,, \tag{19.85}$$

and they are solved at one loop by

$$\delta_2=\left.\frac{\mathrm{d}\Sigma_2}{\mathrm{d}\not{p}}\right|_{\not{p}=m},\qquad \delta_m=\delta_2\,m-\Sigma_2(\not{p}=m)\,. \tag{19.86}$$

The bare mass then differs from the physical mass by $m_0-m=-\Sigma_2(\not{p}=m)+O(e^4)$.

Proof. Treat $\widehat{\Sigma}(p)=\widehat{A}(p^2)\not{p}+\widehat{B}(p^2)m$ as a function of $\not{p}$, with $p^2=\not{p}\not{p}$ a function of $\not{p}$ as well; the chain rule gives $\mathrm{d}(p^2)/\mathrm{d}\not{p}=2\not{p}$. Expand the inverse propagator $\not{p}-m-\widehat{\Sigma}(p)$ about the point $\not{p}=m$. Write $\widehat{\Sigma}(p)=\widehat{\Sigma}(m)+\widehat{\Sigma}'(m)\,(\not{p}-m)+O((\not{p}-m)^2)$, where $\widehat{\Sigma}(m)$ means $\widehat{\Sigma}$ evaluated at $\not{p}=m$ and $\widehat{\Sigma}'(m)\equiv\mathrm{d}\widehat{\Sigma}/\mathrm{d}\not{p}|_{\not{p}=m}$. Then

$$\not{p}-m-\widehat{\Sigma}(p)\;=\;-\widehat{\Sigma}(m)\;+\;[1-\widehat{\Sigma}'(m)](\not{p}-m)\;+\;O((\not{p}-m)^2)\,. \tag{19.87}$$

A simple pole *at* $\not{p}=m$ requires the constant term in Eq. (19.87) to vanish, $\widehat{\Sigma}(m)=0$, which is the first condition in Eq. (19.85). Given that, the propagator near the pole is $i/\{[1-\widehat{\Sigma}'(m)](\not{p}-m)\}$, whose residue is $i/[1-\widehat{\Sigma}'(m)]$; demanding the canonical residue i forces $\widehat{\Sigma}'(m)=0$, the second condition. Now insert $\widehat{\Sigma}=\Sigma_2-(\delta_2\not{p}-\delta_m)$ from Eq. (19.83). Differentiating, $\widehat{\Sigma}'(m)=\Sigma_2'(m)-\delta_2$, so the residue condition gives $\delta_2=\Sigma_2'(m)$; evaluating, $\widehat{\Sigma}(m)=\Sigma_2(m)-\delta_2 m+\delta_m$, so the pole condition gives $\delta_m=\delta_2 m-\Sigma_2(m)$. These are Eq. (19.86). Finally, the relation between bare and physical mass: the bare propagator $i/(\not{p}-m_0-\Sigma_2)$ has its pole where $m_0+\Sigma_2(m_0)=m$; to one loop Σ_2 is already $O(e^2)$, so $\Sigma_2(m_0)=\Sigma_2(m)+O(e^4)$ and $m_0=m-\Sigma_2(m)+O(e^4)$, i.e. $m_0-m=-\Sigma_2(m)+O(e^4)$. □

Theorem 19.8 reduces both counterterms to two evaluations of the loop result Eq. (19.77): Σ_2 at $\not{p}=m$, and its $\not{p}$-derivative there.

Mass renormalization. Setting $\not{p}=m$ (hence $p^2=m^2$) in Eq. (19.77) gives $\Sigma_2(m)=[A(m^2)+B(m^2)]\,m$. The on-shell value of the effective mass is

$$\Delta_0(x)\;\equiv\;\Delta(x)\big|_{p^2=m^2}\;=\;x^2m^2+(1-x)\mu_\gamma^2\,, \tag{19.88}$$

obtained by setting $p^2=m^2$ in Eq. (19.74) and using $-x(1-x)m^2+xm^2=x^2m^2$. From Eqs. (19.80) and (19.81) the divergent parts add as

$$A(m^2)+B(m^2)\big|_{\text{UV pole}}\;=\;\frac{1}{\bar\varepsilon}\left(-\frac{e^2}{16\pi^2}+\frac{e^2}{4\pi^2}\right)\;=\;\frac{3e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon}\,, \tag{19.89}$$

so $\Sigma_2(m) = \dfrac{3e^2}{16\pi^2}\,\dfrac{1}{\bar\varepsilon}\,m + (\text{finite})$. By the last clause of Theorem 19.8, the divergent part of the mass shift is therefore

$$\boxed{\frac{m_0 - m}{m} \;=\; -\frac{3e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon} \;+\; (\text{finite}) \;=\; -\frac{3\alpha}{4\pi}\,\frac{1}{\bar\varepsilon} \;+\; (\text{finite})} \tag{19.90}$$

with $\alpha = e^2/4\pi$. Two physical facts are encoded here. First, the mass shift is *proportional to m itself*: it is a *multiplicative*, not additive, renormalization. Had the bare mass been zero, the self-energy Eq. (19.76) would have produced only the $\not p$ term (the B-term carries an explicit m), and no mass would have been generated by the loop. This is the protection afforded by *chiral symmetry*: at $m = 0$ the left- and right-handed electrons decouple and the Lagrangian acquires an independent phase symmetry for each chirality, which forbids any term connecting them—and a mass term is exactly such a term. A symmetry that is exact at $m = 0$ cannot be broken by quantum corrections proportional to a coupling that respects it, so the only mass the loop can make is one proportional to the mass already present. Second, the divergence is only *logarithmic*: $1/\varepsilon$ is the dimensional-regularization avatar of $\ln\Lambda$, not of Λ or Λ^2. A naive power count of Eq. (19.71) (three propagator powers in $d = 4$ against $\mathrm{d}^4\ell$) would suggest a linear divergence; chiral symmetry softens it to a logarithm by tying the coefficient to m. The classic statement, restoring a cutoff for intuition, is $\delta m/m \sim (3\alpha/4\pi)\ln(\Lambda^2/m^2)$: even for Λ at the Planck scale the electromagnetic self-mass is only a few percent of m, a smallness that is the historical seed of the whole renormalization program.

Field-strength renormalization. The residue condition needs $\Sigma_2'(m) = \mathrm{d}\Sigma_2/\mathrm{d}\not p\,|_{\not p = m}$. Differentiating $\Sigma_2 = A(p^2)\not p + B(p^2)m$ with $\mathrm{d}(p^2)/\mathrm{d}\not p = 2\not p$,

$$\frac{\mathrm{d}\Sigma_2}{\mathrm{d}\not p} \;=\; A(p^2) + 2p^2 A'(p^2) + 2\not p\, m\, B'(p^2)\,, \tag{19.91}$$

and evaluating at $\not p = m$, $p^2 = m^2$,

$$\delta_2 \;=\; \Sigma_2'(m) \;=\; A(m^2) + 2m^2\big[A'(m^2) + B'(m^2)\big]\,. \tag{19.92}$$

The derivatives follow from Eqs. (19.80) and (19.81) by differentiating under the x-integral; only the $\ln(\mu^2/\Delta)$ terms depend on p^2, and $\mathrm{d}_{p^2}\ln(\mu^2/\Delta) = x(1-x)/\Delta$ because $\mathrm{d}\Delta/\mathrm{d}p^2 = -x(1-x)$ from Eq. (19.74). Thus

$$A'(m^2) = -\frac{e^2}{8\pi^2}\int_0^1 \mathrm{d}x\,\frac{x(1-x)^2}{\Delta_0(x)}\,, \tag{19.93}$$

$$B'(m^2) = \;\;\frac{e^2}{4\pi^2}\int_0^1 \mathrm{d}x\,\frac{x(1-x)}{\Delta_0(x)}\,, \tag{19.94}$$

and combining,

$$2m^2[A'(m^2)+B'(m^2)] \;=\; \frac{e^2m^2}{4\pi^2}\int_0^1 \mathrm{d}x\,\frac{x(1-x)[-(1-x)+2]}{\Delta_0(x)} \;=\; \frac{e^2m^2}{4\pi^2}\int_0^1 \mathrm{d}x\,\frac{x(1-x)(1+} \tag{19.95}$$

The first term of Eq. (19.92), $A(m^2)$, carries the ultraviolet pole. From Eq. (19.80) with the on-shell Δ_0,

$$\boxed{Z_2 \;=\; 1+\delta_2 \;=\; 1-\frac{e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon} \;+\; (\text{finite}) \;=\; 1-\frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon} \;+\; (\text{finite})\,,} \tag{19.96}$$

where the UV pole came entirely from the $1/\bar\varepsilon$ inside $A(m^2)$, weighted by $\int_0^1(1-x)\mathrm{d}x=\frac{1}{2}$; the second term Eq. (19.95) is ultraviolet-finite (its integrand has no $1/\varepsilon$). The ratio of the $\not{p}$ pole in Z_2 to the mass pole in δm is $1:3$, the same $1:3$ visible in Eq. (19.82).

19.3.3 The infrared divergence

The ultraviolet poles of δm and Z_2 are now isolated and subtracted, and $\widehat{\Sigma}$ defined by Eq. (19.83) is ultraviolet-finite. But Z_2 is not yet a number: the finite remainder Eq. (19.95) hides a *second* divergence that has nothing to do with the regulator ε and is not removed by any counterterm. It appears when the photon-mass regulator is sent to zero, $\mu_\gamma \to 0$, and it is the first infrared divergence of QED.

The integrand of Eq. (19.95) is $x(1-x)(1+x)/\Delta_0(x)$ with $\Delta_0(x)=x^2m^2+(1-x)\mu_\gamma^2$. Away from $x=0$ the denominator is bounded below by x^2m^2 and the integral is finite as $\mu_\gamma\to 0$. The trouble is the endpoint $x\to 0$, where on shell with a strictly massless photon ($\mu_\gamma=0$) the denominator $\Delta_0\to x^2m^2$ vanishes quadratically while the numerator vanishes only linearly: $x(1-x)(1+x)/x^2m^2\sim 1/(xm^2)$, and $\int_0 \mathrm{d}x/x$ diverges logarithmically. The photon mass cuts the divergence off precisely at the scale where $x^2m^2\sim\mu_\gamma^2$, i.e. $x\sim\mu_\gamma/m$, converting the divergence into a $\ln(m^2/\mu_\gamma^2)$. The following worked example extracts the coefficient.

Worked Example 19.9 (The infrared logarithm in Z_2)**.** We isolate the $\mu_\gamma\to 0$ divergence of

$$I_2 \;\equiv\; \int_0^1 \mathrm{d}x\,\frac{x(1-x)(1+x)}{x^2m^2+(1-x)\mu_\gamma^2}\,. \tag{19.97}$$

The divergence is local to $x=0$, so split the numerator into its value at $x=0$ plus a remainder that vanishes there. Since $x(1-x)(1+x)=x\,(1-x^2)=x-x^3$, write $x(1-x)(1+x)=x+[-x^3]$; the bracketed term carries an extra power of x^2 relative to x and so renders its integral convergent as $\mu_\gamma\to 0$ (it gives a finite

constant we do not need). The divergent part comes from the leading piece,

$$I_2 \;\supset\; \int_0^1 \mathrm{d}x\, \frac{x}{x^2 m^2 + (1-x)\mu_\gamma^2}\,. \tag{19.98}$$

For the leading logarithm it suffices to keep $(1-x) \to 1$ in the denominator near the endpoint (corrections are again finite), giving the elementary integral

$$\int_0^1 \mathrm{d}x\, \frac{x}{x^2 m^2 + \mu_\gamma^2} \;=\; \frac{1}{2m^2}\Big[\ln\!\big(x^2 m^2 + \mu_\gamma^2\big)\Big]_0^1 \;=\; \frac{1}{2m^2}\,\ln\!\frac{m^2+\mu_\gamma^2}{\mu_\gamma^2}\,, \tag{19.99}$$

whose $\mu_\gamma \to 0$ behaviour is $\dfrac{1}{2m^2}\ln\dfrac{m^2}{\mu_\gamma^2}$. Hence

$$I_2 \;=\; \frac{1}{2m^2}\,\ln\!\frac{m^2}{\mu_\gamma^2} \;+\; (\text{finite as } \mu_\gamma \to 0)\,. \tag{19.100}$$

Multiplying by the prefactor of Eq. (19.95), the infrared part of the residue counterterm is

$$\delta_2\big|_{\mathrm{IR}} \;=\; \frac{e^2 m^2}{4\pi^2}\cdot\frac{1}{2m^2}\,\ln\!\frac{m^2}{\mu_\gamma^2} \;=\; \frac{e^2}{8\pi^2}\,\ln\!\frac{m^2}{\mu_\gamma^2} \;=\; \frac{\alpha}{2\pi}\,\ln\!\frac{m^2}{\mu_\gamma^2}\,. \tag{19.101}$$

The same divergence appears, if one prefers to regulate the infrared dimensionally rather than with a photon mass, as a pole in a separate *infrared* regulator $\varepsilon_{\mathrm{IR}}$. Setting $\mu_\gamma = 0$ in Eq. (19.97) but continuing the dimension of the parameter integral, the endpoint integral $\int_0 \mathrm{d}x\, x^{-1}$ becomes $\int_0 \mathrm{d}x\, x^{-1+2\varepsilon_{\mathrm{IR}}} = 1/(2\varepsilon_{\mathrm{IR}})$, so that $\ln(m^2/\mu_\gamma^2) \leftrightarrow 1/\varepsilon_{\mathrm{IR}}$ up to scheme-dependent finite terms. Either way, δ_2 depends on the infrared regulator; the dependence is the whole point.

Collecting Eqs. (19.96) and (19.101), the on-shell field renormalization has the schematic structure

$$Z_2 \;=\; 1 - \frac{\alpha}{4\pi}\,\frac{1}{\bar{\varepsilon}_{\mathrm{UV}}} \;+\; \frac{\alpha}{2\pi}\,\ln\!\frac{m^2}{\mu_\gamma^2} \;+\; (\text{finite})\,, \tag{19.102}$$

in which the ultraviolet pole and the infrared logarithm are visibly different animals: the first is multiplied by $1/\bar{\varepsilon}_{\mathrm{UV}}$ and is removed by the counterterm definition, while the second is multiplied by $\ln(m^2/\mu_\gamma^2)$ and is regulator-dependent and *physical*. No redefinition of the electron field or mass can touch the infrared logarithm, because it is not local: it comes from the endpoint $x \to 0$, which is the region of soft photon momentum $\ell \to 0$, where the virtual photon's wavelength is larger than any short-distance scale.

Caution

Z_2 contains two divergences with opposite physical status, and they must be tracked separately to the end of the calculation. The ultraviolet pole $-\frac{\alpha}{4\pi}\frac{1}{\bar{\varepsilon}_{\mathrm{UV}}}$ in Eq. (19.102) is an artifact of the regulator: it is cancelled, once and for all, by the counterterm δ_2 that *defines* the renormalized field, and it never appears in a physical answer. The infrared logarithm $+\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$ is not an artifact: it survives renormalization, it depends on how the massless photon is regulated, and it cannot be subtracted away. It is, however, harmless, because Z_2 by itself is not an observable. The infrared logarithm of Z_2 will be matched by an equal and opposite logarithm in the vertex renormalization Z_1 of Section 19.4—this is the content of the Ward–Takahashi identity $Z_1 = Z_2$—and the residual infrared divergence of any physical amplitude will cancel against the emission of real soft photons in Section 19.6. Subtracting the infrared divergence "by hand", as if it were ultraviolet, would be a sign error in disguise and would corrupt the final cross section.

19.3.4 The renormalized propagator near its pole

With the two on-shell conditions imposed, the renormalized propagator Eq. (19.84) behaves near $\not{p} = m$ exactly as the Källén–Lehmann analysis of Eq. (14.112) predicts an interacting two-point function must: a simple pole at the physical mass with a residue that is the field-strength renormalization. Expanding the denominator as in Eq. (19.87) but now keeping the renormalized self-energy and *not* imposing unit residue, the exact propagator near the pole is

$$\frac{i\,Z_2}{\not{p} - m} \;+\; (\text{regular at } \not{p} = m)\,, \tag{19.103}$$

where $Z_2 = [1-\Sigma_2'(m)]^{-1} = 1+\Sigma_2'(m)+O(e^4)$ is precisely the residue counterterm of Eq. (19.92); the on-shell scheme is the choice that sets this residue to one by absorbing $\Sigma_2'(m)$ into the definition of the field. This is the spinor counterpart of the scalar field-strength renormalization Z defined in Theorem 14.25: the residue at the one-particle pole, equal to the squared overlap $|\langle\Omega|\psi(0)|p,s\rangle|^2$ between the interacting field acting on the vacuum and the physical one-electron state. Through the Källén–Lehmann decomposition the rest of the spectral weight—the multiparticle continuum where an electron is accompanied by real photons—accounts for the deficit $1 - Z_2$ of the total weight, exactly as the multiparticle continuum did for the scalar in Eq. (14.115).

The same Z_2 controls the LSZ reduction of Theorem 14.33: each external electron leg of an S-matrix element is accompanied by a factor $\sqrt{Z_2}$ from the amputation, exactly as the scalar reduction carried a $\sqrt{Z}$ per leg (Eqs. (14.156) and (14.166)). The on-shell scheme makes this bookkeeping trivial for external

lines—$\sqrt{Z_2} \to 1$ by construction at the level of the renormalized field—but Z_2 does not disappear from the theory: it reappears, undisguised, in the relation between the bare and renormalized vertex, where its infrared logarithm Eq. (19.101) is exactly the quantity the next section must reproduce.

We have extracted from the one-loop electron self-energy the two numbers QED now needs. The first is the field-strength renormalization $Z_2 = 1 - \frac{\alpha}{4\pi}\frac{1}{\bar{\varepsilon}} + \dots$ of Eq. (19.96), whose ultraviolet pole will be set equal to the vertex renormalization Z_1 of Section 19.4 as the one-loop test of the Ward–Takahashi identity $Z_1 = Z_2$. The second is the infrared logarithm $\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$ of Eq. (19.101), the first infrared divergence of the thread, which is carried forward—tracked separately from every ultraviolet pole—to be cancelled against the real-emission contribution that renders the inclusive cross section infrared-finite in Section 19.6. Both quantities are now in hand; we turn next to the vertex, where they are put to work.

19.4 The Vertex Correction

The electron self-energy of Section 19.3 delivered the second of the three primitive one-loop objects and, with it, two numbers we now have to confront again. One is the field-strength renormalization $Z_2 = 1 - \frac{\alpha}{4\pi}\frac{1}{\bar{\varepsilon}} + \dots$ of Eq. (19.96); the other is the infrared logarithm $\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$ of Eq. (19.101), the first soft divergence of QED. Both were extracted from a single dressed propagator. The third primitive object, the correction to the electron–photon vertex, is built from an entirely different diagram—a triangle of three propagators tied together by a photon—and yet, as we will find, its ultraviolet pole and its infrared logarithm are forced to coincide, structure for structure, with those of the self-energy. That coincidence, $Z_1 = Z_2$, is the experimental fact that the Ward–Takahashi identity of Section 19.5 exists to explain; here we earn it by explicit computation. The vertex also does something the self-energy could not: it generates a genuinely new Lorentz structure, the Pauli coupling $i\sigma^{\mu\nu}q_\nu$, whose coefficient is the electron's *anomalous magnetic moment*. We set up that coefficient in full and hand its evaluation—Schwinger's $\alpha/2\pi$—to Chapter 20, the clean connection the chapter scope prescribes.

19.4.1 The one-loop vertex amplitude

The object of study is the proper (amputated, 1PI) electron–photon vertex $\Gamma^\mu(p', p)$, with an incoming electron of momentum p, an outgoing electron of momentum p', and a photon carrying momentum $q = p' - p$ into the vertex. Its tree value is the bare vertex γ^μ of Theorem 17.1 and Eq. (17.10); writing the full amputated vertex as $-ie\,\Gamma^\mu$ with $\Gamma^\mu = \gamma^\mu + \delta\Gamma^\mu$, the one-loop correction $\delta\Gamma^\mu$ is

the single diagram of Fig. 19.6: a photon emitted from the incoming line and reabsorbed by the outgoing line, spanning the vertex.

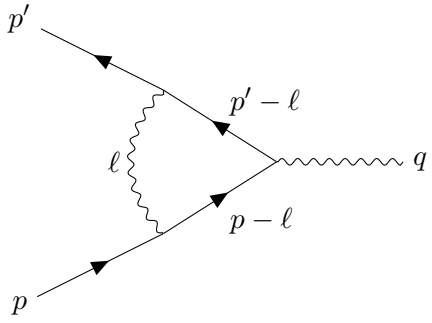


Figure 19.6: The one-loop vertex correction $-ie\,\delta\Gamma^\mu(p',p)$. The incoming electron of momentum p emits a virtual photon of momentum ℓ, propagates with momentum $p-\ell$ to the external-photon vertex, continues with momentum $p'-\ell$, and reabsorbs the virtual photon. The external photon of momentum $q=p'-p$ attaches at the central vertex. This is the only 1PI correction to the vertex at $O(e^3)$.

Reading the diagram with the QED Feynman rules—the tree vertex $-ie\gamma^\mu$ of Eq. (17.10) at each of the three junctions, the fermion propagators of Eq. (17.15) on the two internal electron segments, and the Feynman-gauge photon propagator on the internal photon line, now carrying the fictitious mass μ_γ introduced in Section 19.3 as an infrared regulator—and routing the loop momentum ℓ through the photon, gives the expression already assembled in Eq. (19.3),

$$-ie\,\delta\Gamma^\mu(p',p) = (-ie)^3\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\gamma^\alpha\,\frac{i(\not{p}'-\not{\ell}+m)}{(p'-\ell)^2-m^2+i\varepsilon}\,\gamma^\mu\,\frac{i(\not{p}-\not{\ell}+m)}{(p-\ell)^2-m^2+i\varepsilon}\,\gamma^\beta\,\frac{-i\,\eta_{\alpha\beta}}{\ell^2-\mu_\gamma^2+} \tag{19.104}$$

The factor $\mu^{2\varepsilon}$ accompanies the coupling for the same reason as in the self-energy: in $d=4-2\varepsilon$ the bare charge is not dimensionless, and the replacement $e\to e\,\mu^\varepsilon$ keeps $\delta\Gamma^\mu$ dimensionally honest. We work in Feynman gauge $\xi=1$, the default of QED. Collect the explicit factors of i and the coupling. The three vertices supply $(-ie)^3 = ie^3$ (since $(-i)^3=i$), the two fermion propagators $i^2=-1$, and the photon propagator $-i$; the product is $(ie^3)(-1)(-i) = -e^3$. Contracting $\gamma^\alpha\cdots\gamma^\beta$ against $\eta_{\alpha\beta}$ closes the two outer Dirac matrices around the numerator, and dividing the relation by the overall $-ie$ on the left,

$$\delta\Gamma^\mu(p',p) = -\,i\,e^2\,\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\gamma^\alpha\,(\not{p}'-\not{\ell}+m)\,\gamma^\mu\,(\not{p}-\not{\ell}+m)\,\gamma_\alpha}{\big[(p'-\ell)^2-m^2+i\varepsilon\big]\big[(p-\ell)^2-m^2+i\varepsilon\big]\big[\ell^2-\mu_\gamma^2+i\varepsilon\big]}\,, \tag{19.105}$$

the same overall prefactor $-ie^2$ that headed the self-energy integral Eq. (19.71), a coincidence of normalization the Ward identity will turn into a coincidence of

poles. The three denominators and the numerator string of five gamma matrices are what we must now dispose of.

19.4.2 Form factors: the only two structures

Before computing, we settle what the answer can possibly look like, because that fixes exactly which scalar coefficients we have to extract. The vertex is always used between physical external spinors: the S-matrix element of a photon emission is $\bar{u}(p')\,(-ie\Gamma^\mu)\,u(p)$ contracted with the photon polarization, with $u(p)$, $\bar{u}(p')$ on shell. Sandwiched this way, the available building blocks are γ^μ, the momenta p^μ and p'^μ, and Dirac scalars; but the Dirac equations $\not{p}\,u(p) = m\,u(p)$ and $\bar{u}(p')\not{p}' = m\,\bar{u}(p')$ collapse every string down to a short list. A standard counting (carried out structure by structure in the reduction below) shows that, after imposing the on-shell conditions and the discrete symmetries (parity, hermiticity, and current conservation), only *two* independent Lorentz structures survive. Writing $q = p' - p$, we record the decomposition.

Theorem 19.10 (Form-factor decomposition of the vertex)**.** *Between on-shell electron spinors of mass m, the exact electromagnetic vertex of QED can be written in terms of two real scalar functions of q^2 alone,*

$$\bar{u}(p')\,\Gamma^\mu(p',p)\,u(p) = \bar{u}(p')\left[\gamma^\mu\,F_1(q^2) + \frac{i\,\sigma^{\mu\nu}q_\nu}{2m}\,F_2(q^2)\right]u(p), \qquad \sigma^{\mu\nu} = \frac{i}{2}\,[\gamma^\mu,\gamma^\nu]\,, \tag{19.106}$$

the Dirac form factor F_1 *and the* Pauli form factor F_2*. At tree level $F_1 = 1$, $F_2 = 0$.*

That there is no third structure is not an assumption but a consequence of gauge invariance. A general $\bar{u}(p')[\cdots]u(p)$ built from $\gamma^\mu, p^\mu, p'^\mu$ could a priori contain a term $\propto (p'-p)^\mu = q^\mu$; current conservation $q_\mu \bar{u}(p')\Gamma^\mu u(p) = 0$, the on-shell Ward identity of Theorem 18.5 (proved to all orders in Section 19.5), forbids it, leaving only the combination $(p+p')^\mu$, which the Gordon identity trades for γ^μ and $i\sigma^{\mu\nu}q_\nu$. The decomposition Eq. (19.106) is therefore exactly the two-term split of the Gordon decomposition Theorem C.69,

$$\bar{u}(p')\,\gamma^\mu\,u(p) = \bar{u}(p')\left[\frac{(p+p')^\mu}{2m} + \frac{i\,\sigma^{\mu\nu}q_\nu}{2m}\right]u(p), \tag{19.107}$$

upgraded by the loop from fixed coefficients to the momentum-dependent functions $F_1(q^2), F_2(q^2)$. We will use Eq. (19.107) repeatedly to convert the $(p+p')^\mu$ pieces the algebra generates into the canonical pair.

What the two form factors measure

The physical content of Eq. (19.106) is read off in the static limit. At $q^2 \to 0$ the photon is a soft, nearly on-shell field: $F_1(0)$ is the coefficient of the minimal coupling γ^μ, i.e. the electric charge in units of e, so the statement that the electron carries its tree-level charge undressed is exactly $F_1(0) = 1$—a renormalization condition, not a prediction. The Pauli term, by contrast, couples the spin to the field gradient: matching $\bar{u}(p')\,\Gamma^\mu\,u(p)$ to the nonrelativistic interaction of a magnetic moment shows that the gyromagnetic ratio is $g = 2\big[F_1(0) + F_2(0)\big] = 2 + 2F_2(0)$, so the Dirac value $g = 2$ receives its first correction entirely from $F_2(0)$. The anomalous moment is $a_e \equiv \frac{1}{2}(g-2) = F_2(0)$. The tree-level vertex has $F_2 = 0$ and hence the exact Dirac value $g = 2$; that the loop generates a nonzero, finite, calculable $F_2(0)$ is the reason $g - 2$ is one of the most precisely tested numbers in physics. The division is clean: this section produces the integral for $F_2(0)$; Chapter 20 evaluates it to Schwinger's $\alpha/2\pi$.

19.4.3 Feynman parametrization and the loop shift

Fuse the three denominators with the three-parameter Feynman identity of Theorem 19.2 (equivalently Theorem D.11),

$$\frac{1}{ABC} = 2\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\frac{\delta(x+y+z-1)}{\big[xA+yB+zC\big]^3}\,, \tag{19.108}$$

assigning x to $A = (p-\ell)^2 - m^2$, y to $B = (p'-\ell)^2 - m^2$, and z to $C = \ell^2 - \mu_\gamma^2$. The combined denominator is

$$\begin{aligned} x\big[(p-\ell)^2 - m^2\big] + y\big[(p'-\ell)^2 - m^2\big] + z\big[\ell^2 - \mu_\gamma^2\big] &= \ell^2 - 2\ell\cdot(xp+yp') + xp^2 + yp'^2 - (x+ \\ &= (\ell - k_0)^2 - \Delta\,, \qquad k_0 \equiv xp + yp', \end{aligned} \tag{19.109}$$

using $x + y + z = 1$ to set the coefficient of ℓ^2 to unity. Completing the square pulls out k_0^2, and putting the external electrons on shell, $p^2 = p'^2 = m^2$ with $p\cdot p' = m^2 - \frac{1}{2}q^2$ (so $k_0^2 = (xp+yp')^2 = (x+y)^2 m^2 - xy\,q^2$), the effective mass-squared is

$$\Delta(x,y,z) = -\,k_0^2 + xp^2 + yp'^2 - (x+y)m^2 + z\mu_\gamma^2 = (1-z)^2 m^2 - xy\,q^2 + z\,\mu_\gamma^2\,, \tag{19.110}$$

where the mass terms combine as $(x+y)m^2 - (x+y)m^2 = 0$ apart from the $(x+y)^2 m^2 = (1-z)^2 m^2$ left over from $-k_0^2$. Two features of Δ will matter. It is symmetric under $x \leftrightarrow y$ (equivalently $p \leftrightarrow p'$), the remnant of the symmetry of the diagram. And at $q^2 = 0$ it reduces to $\Delta_0(z) = (1-z)^2 m^2 + z\mu_\gamma^2$, which

vanishes as $z \to 1$ when $\mu_\gamma \to 0$: that endpoint is the seat of the infrared divergence, exactly as $x \to 0$ was for the self-energy in Theorem 19.9.

Now shift the loop momentum, $\ell \to \ell + k_0$, legitimate in dimensional regularization by the translation invariance of Theorem 19.1. The denominator becomes $(\ell^2 - \Delta)^3$, even in ℓ, so in the numerator every term with an odd number of factors of ℓ integrates to zero, and the rank-two piece may be replaced by its covariant average $\ell^\mu \ell^\nu \to (\ell^2/d)\eta^{\mu\nu}$ (Eq. (19.21)). Under the shift the two fermion momenta in the numerator become

$$p-\ell \;\longrightarrow\; \tilde{P}_1-\ell, \quad \tilde{P}_1 = (1-x)p - yp'; \qquad p'-\ell \;\longrightarrow\; \tilde{P}_2-\ell, \quad \tilde{P}_2 = -xp + (1-y)p'. \tag{19.111}$$

With these the numerator of Eq. (19.105) is

$$N^\mu = \gamma^\alpha \left(\tilde{\not{P}}_2 - \not{\ell} + m\right) \gamma^\mu \left(\tilde{\not{P}}_1 - \not{\ell} + m\right) \gamma_\alpha \,. \tag{19.112}$$

Expanding the two factors, the terms odd in ℓ drop, and we are left with two contributions: the part quadratic in ℓ,

$$N^\mu_{(2)} = \gamma^\alpha \not{\ell}\, \gamma^\mu \not{\ell}\, \gamma_\alpha \,, \tag{19.113}$$

which alone carries the ultraviolet divergence, and the ℓ-independent part,

$$N^\mu_{(0)} = \gamma^\alpha \left(\tilde{\not{P}}_2 + m\right) \gamma^\mu \left(\tilde{\not{P}}_1 + m\right) \gamma_\alpha \,, \tag{19.114}$$

which is ultraviolet-finite and, as we will see, is where the entire Pauli form factor lives. We treat them in turn.

19.4.4 The ultraviolet divergence and the Dirac form factor

The divergent piece is short. Replacing $\ell^\mu \ell^\nu \to (\ell^2/d)\eta^{\mu\nu}$ in Eq. (19.113) and contracting twice with the d-dimensional identity $\gamma^\nu \gamma^\mu \gamma_\nu = (2-d)\gamma^\mu$ of Theorem C.61 (Eq. (C.188)),

$$N^\mu_{(2)} \;\longrightarrow\; \frac{\ell^2}{d}\, \gamma^\alpha \gamma^\nu \gamma^\mu \gamma_\nu \gamma_\alpha = \frac{\ell^2}{d}\,(2-d)\,\gamma^\alpha \gamma^\mu \gamma_\alpha = \frac{\ell^2}{d}\,(2-d)^2\,\gamma^\mu \,. \tag{19.115}$$

The accompanying scalar integral is the only logarithmically divergent one in the problem,

$$\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \frac{\ell^2}{(\ell^2 - \Delta)^3} = \frac{i}{(4\pi)^{d/2}}\, \frac{d}{2}\, \frac{\Gamma\!\left(2 - \frac{d}{2}\right)}{\Gamma(3)}\, \Delta^{\,d/2-2} \mu^{2\varepsilon} = \frac{i}{(4\pi)^{d/2}}\, \frac{d}{4}\, \Gamma(\varepsilon)\, \Delta^{-\varepsilon} \mu^{2\varepsilon} \,, \tag{19.116}$$

from the tensor master Eq. (19.21) with $\Gamma(3) = 2$, $2 - \frac{d}{2} = \varepsilon$. Assembling the divergent contribution to Eq. (19.105)—the prefactor $-ie^2$, the Feynman measure $2\int \mathrm{d}x\, \mathrm{d}y\, \mathrm{d}z\, \delta(x+y+z-1)$, the trace factor $(2-d)^2/d$, and the integral

Eq. (19.116)—the two powers of i cancel ($-i\times i=1$), the explicit d's cancel ($\frac{(2-d)^2}{d}\cdot\frac{d}{4}=\frac{(2-d)^2}{4}$), and

$$\delta\Gamma^\mu\big|_{(2)} = e^2\,\gamma^\mu\cdot 2\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x{+}y{+}z{-}1)\,\frac{(2-d)^2}{4}\,\frac{\Gamma(\varepsilon)\,(4\pi)^{\varepsilon}\mu^{2\varepsilon}\,\Delta^{-\varepsilon}}{(4\pi)^2}\,. \tag{19.117}$$

Now $(2-d)^2/4=(1-\varepsilon)^2=1-2\varepsilon+O(\varepsilon^2)$, while $(4\pi)^{\varepsilon}\Gamma(\varepsilon)\mu^{2\varepsilon}\Delta^{-\varepsilon}=\frac{1}{\bar\varepsilon}+\ln(\mu^2/\Delta)+O(\varepsilon)$ is the $\overline{\text{MS}}$ combination $G(\Delta)$ of Eq. (19.78), with the master pole $\frac{1}{\bar\varepsilon}=\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi$ (cf. Eq. (D.136)). Multiplying, the factor $(1-2\varepsilon)$ acts on the pole to leave a finite remainder, $(1-2\varepsilon)\big[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}\big]=\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}-2+O(\varepsilon)$. The pole is kinematics-independent, so its Feynman integral is simply the simplex volume $2\int\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x+y+z-1)=2\cdot\frac{1}{2}=1$. The ultraviolet-divergent part of the vertex is therefore

$$\boxed{\delta\Gamma^\mu\big|_{\text{UV pole}} = \frac{e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon}\,\gamma^\mu = \frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon}\,\gamma^\mu\,,} \tag{19.118}$$

with $\alpha=e^2/4\pi$. It is proportional to the tree structure γ^μ—as it must be, since a logarithmic divergence can renormalize only the dimensionless coefficient of the existing operator—and it sits entirely in the Dirac form factor. The Pauli structure $i\sigma^{\mu\nu}q_\nu$ receives nothing from $N^\mu_{(2)}$, the first sign that F_2 will be finite.

Collecting, the bare one-loop Dirac form factor has the form

$$F_1(q^2) = 1+\frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon}+F_1^{\text{fin}}(q^2)\,, \tag{19.119}$$

where the finite remainder F_1^{fin} collects the q^2-dependent logarithms from $\ln(\mu^2/\Delta)$ together with the γ^μ pieces of the convergent numerator $N^\mu_{(0)}$ computed below. Unlike the pole, F_1^{fin} is not ultraviolet but it is *not* fully finite either: it harbours an infrared divergence, isolated in Section 19.4.5.

Renormalization and Z_1. The vertex counterterm of renormalized perturbation theory contributes a tree-level Feynman rule $-ie\,\delta_1\gamma^\mu$ (Eq. (19.26)), so the renormalized Dirac form factor is

$$F_1^{\text{ren}}(q^2)=F_1(q^2)+\delta_1, \qquad Z_1\equiv 1+\delta_1\,. \tag{19.120}$$

The on-shell renormalization condition for the vertex, Eq. (19.29), demands that at zero momentum transfer the full vertex between on-shell spinors reduce to the tree value $-ie\gamma^\mu$, i.e. $F_1^{\text{ren}}(0)=1$. This fixes the counterterm,

$$\delta_1=-\big[F_1(0)-1\big]=-\frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon}-F_1^{\text{fin}}(0)\,, \tag{19.121}$$

so that Z_1 absorbs both the ultraviolet pole of $F_1(0)$ and whatever its finite on-shell value is. The condition $F_1(0)=1$ is the precise statement that e

is the physically measured charge—the Thomson-limit charge seen by a soft photon—and it is the same condition that, through the Ward identity, will be shown to fix the electron-propagator residue. The ultraviolet part of the vertex renormalization is read straight off Eq. (19.118),

$$Z_1 = 1 - \frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon} + (\text{finite}) = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon} + (\text{finite})\,. \tag{19.122}$$

19.4.5 The convergent numerator and $Z_1 = Z_2$

Compare Eq. (19.122) with the field-strength renormalization Eq. (19.96): the two ultraviolet poles are identical,

$$Z_1 = Z_2 = 1 - \frac{\alpha}{4\pi}\,\frac{1}{\bar\varepsilon} + (\text{finite}) \qquad (\xi = 1). \tag{19.123}$$

Two integrals with no diagram in common—a single dressed propagator and a three-propagator triangle—have produced the same $1/\bar\varepsilon$ pole to the last factor. At this stage it is an arithmetic coincidence; Theorem 19.14 will show it is the inevitable consequence of the Ward–Takahashi identity. But the agreement runs deeper than the ultraviolet: the infrared logarithms match too, and verifying that requires the convergent numerator $N^{\mu}_{(0)}$. We compute it now, which also delivers the Pauli form factor.

Worked Example 19.11 (Reduction of the convergent vertex numerator)**.** We reduce $N^{\mu}_{(0)} = \gamma^{\alpha}(\tilde{\not P}_2 + m)\gamma^{\mu}(\tilde{\not P}_1 + m)\gamma_{\alpha}$ between on-shell spinors. Since $N^{\mu}_{(0)}$ multiplies a convergent integral we may work at $d = 4$ throughout, using the four-dimensional contraction identities of Theorem C.61: $\gamma^{\alpha}\gamma_{\alpha} = 4$, $\gamma^{\alpha}\not a\,\gamma_{\alpha} = -2\not a$ (Eq. (C.169)), $\gamma^{\alpha}\not a\,\gamma^{\mu}\gamma_{\alpha} = 4a^{\mu}$ (Eq. (C.170)), and $\gamma^{\alpha}\not a\,\gamma^{\mu}\not b\,\gamma_{\alpha} = -2\not b\,\gamma^{\mu}\not a$ (Eq. (C.171)). Expanding the product $(\tilde{\not P}_2 + m)\gamma^{\mu}(\tilde{\not P}_1 + m)$ into four terms and applying these,

$$N^{\mu}_{(0)} = -2\,\tilde{\not P}_1\,\gamma^{\mu}\,\tilde{\not P}_2 + 4m\,(\tilde P_1^{\mu} + \tilde P_2^{\mu}) - 2m^2\gamma^{\mu}\,. \tag{19.124}$$

Sandwich between $\bar u(p')$ and $u(p)$ and reduce with $\not p\,u = m\,u$, $\bar u'\not p' = m\,\bar u'$ and the Clifford algebra. Writing $\tilde{\not P}_1 = (1-x)\not p - y\not p'$ and $\tilde{\not P}_2 = -x\not p + (1-y)\not p'$, the four bilinears that arise reduce to

$$\bar u'\not p\,\gamma^{\mu}\not p\,u = \bar u'(2m\,p^{\mu} - m^2\gamma^{\mu})u, \quad \bar u'\not p'\,\gamma^{\mu}\not p'\,u = \bar u'(2m\,p'^{\mu} - m^2\gamma^{\mu})u, \tag{19.125}$$

$$\bar u'\not p'\,\gamma^{\mu}\not p\,u = m^2\,\bar u'\gamma^{\mu}u, \qquad \bar u'\not p\,\gamma^{\mu}\not p'\,u = \bar u'\big[2m(p+p')^{\mu} - (2p\cdot p' + m^2)\gamma^{\mu}\big]u, \tag{19.126}$$

each obtained by anticommuting $\not p$ to the right onto u and $\not p'$ to the left onto $\bar u'$ (for instance $\not p\,\gamma^{\mu}\not p = (2p^{\mu} - \gamma^{\mu}\not p)\not p = 2p^{\mu}\not p - m^2\gamma^{\mu}$). Collecting the coefficients $c_1 = -x(1-x)$, $c_2 = -y(1-y)$, $c_3 = xy$, $c_4 = (1-x)(1-y)$ that multiply the

four bilinears in $-2\tilde{\not{P}}_1\gamma^\mu\tilde{\not{P}}_2$, and adding the $4m(\tilde{P}_1+\tilde{P}_2)^\mu$ and $-2m^2\gamma^\mu$ pieces with $\tilde{P}_1+\tilde{P}_2=(1-2x)p+(1-2y)p'$, the sandwiched numerator organizes into three covariants,

$$\bar{u}'\,N^\mu_{(0)}\,u=\bar{u}'\Big[\,A\,\gamma^\mu+B\,(p+p')^\mu+C\,q^\mu\,\Big]u, \tag{19.127}$$

with scalar coefficients A,B,C that are polynomials in x,y,z and in m^2,q^2.

The coefficient C of q^μ is antisymmetric under $x\leftrightarrow y$: explicitly $C\propto(x-y)(2+z)$, which one verifies by tracking the p^μ and p'^μ pieces of Eqs. (19.125) and (19.126). Because the measure $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x+y+z-1)$ and the denominator Δ of Eq. (19.110) are both symmetric under $x\leftrightarrow y$, the q^μ term integrates to zero. This is the Ward identity making itself felt inside the loop integral: the would-be longitudinal form factor is killed by the symmetry of the diagram, consistent with $q_\mu\bar{u}'\Gamma^\mu u=0$ and with the absence of a third structure in Theorem 19.10. We may therefore drop C and convert the symmetric combination $B\,(p+p')^\mu$ via the Gordon identity Eq. (19.107), $(p+p')^\mu=2m\gamma^\mu-i\sigma^{\mu\nu}q_\nu$ between spinors.

Carrying out the bookkeeping, the symmetric coefficient of $(p+p')^\mu$ collapses to the compact result

$$B(x,y,z)=2m\,z(1-z)\,, \tag{19.128}$$

independent of x,y after symmetrization, obtained from $B=\tfrac{1}{2}(B_p+B_{p'})$ with $B_p=4m[(1-2x)-(1-x)z]$, $B_{p'}=4m[(1-2y)-(1-y)z]$ and $x+y=1-z$. Through Gordon, this $(p+p')^\mu$ piece contributes $2m\,z(1-z)\cdot 2m=4m^2z(1-z)$ to the γ^μ (Dirac) channel and $2m\,z(1-z)\cdot(-1)=-2m\,z(1-z)$ to the $i\sigma^{\mu\nu}q_\nu$ (Pauli) channel. The Pauli channel receives *nothing* from $N^\mu_{(2)}$ and nothing from the direct γ^μ part of $N^\mu_{(0)}$; the coefficient Eq. (19.128) is its entire source.

The infrared logarithm of Z_1. To complete the comparison $Z_1=Z_2$ we need the γ^μ (Dirac) coefficient of $N^\mu_{(0)}$ at $q^2=0$, where the infrared divergence lives. Adding the direct γ^μ terms of Eq. (19.124) to the $4m^2z(1-z)$ delivered by Gordon, the total γ^μ-coefficient at $q^2=0$ (where $2p\cdot p'=2m^2$) reduces, after the simplex algebra, to

$$A_{(0)}(z)\big|_{q^2=0}=2m^2(-1+4z-z^2)\,, \tag{19.129}$$

again independent of x,y. The convergent contribution to F_1 at $q^2=0$ is then, restoring the prefactor of Eq. (19.105) and the convergent scalar integral $\int\mathrm{d}^4\ell/(2\pi)^4\,(\ell^2-\Delta_0)^{-3}=-i/(32\pi^2\Delta_0)$,

$$F_1^{(0)}(0)=-\frac{e^2}{16\pi^2}\int_0^1\mathrm{d}z\;(1-z)\,\frac{2m^2(-1+4z-z^2)}{(1-z)^2m^2+z\mu_\gamma^2}\,, \tag{19.130}$$

where the factor $(1-z)$ is the area $\int\mathrm{d}x\,\mathrm{d}y\,\delta(x+y+z-1)=1-z$ of the x,y-simplex at fixed z, the integrand being x,y-independent at $q^2=0$. The

integral diverges as $\mu_\gamma \to 0$ only at the endpoint $z \to 1$, where the numerator $2m^2(-1+4z-z^2) \to 4m^2$ stays finite while the denominator $\to \mu_\gamma^2$. Writing $w = 1 - z$ near the endpoint, the divergent part is the same elementary integral met in the self-energy, Eq. (19.99),

$$F_1^{(0)}(0)\big|_{\rm IR} = -\frac{e^2}{16\pi^2}\int_0 \mathrm{d}w\, \frac{4m^2 w}{m^2 w^2 + \mu_\gamma^2} = -\frac{e^2}{16\pi^2}\cdot 2\ln\frac{m^2}{\mu_\gamma^2} = -\frac{\alpha}{2\pi}\ln\frac{m^2}{\mu_\gamma^2}\,. \tag{19.131}$$

Hence by Eq. (19.121) the vertex counterterm carries the infrared logarithm $\delta_1|_{\rm IR} = -F_1^{(0)}(0)|_{\rm IR} = +\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$, so that

$$Z_1 = 1 - \frac{\alpha}{4\pi}\frac{1}{\bar\varepsilon_{\rm UV}} + \frac{\alpha}{2\pi}\ln\frac{m^2}{\mu_\gamma^2} + (\text{finite})\,. \tag{19.132}$$

This is term for term the field-strength renormalization Z_2 of Eq. (19.102): the ultraviolet pole $-\frac{\alpha}{4\pi\bar\varepsilon}$ agrees with Eq. (19.96), and the infrared logarithm $+\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$ agrees with Eq. (19.101). The equality $Z_1 = Z_2$ holds not merely for the ultraviolet pole but for the infrared logarithm as well, exactly as a single underlying identity would require.

$$\boxed{Z_1 = Z_2 = 1 - \frac{\alpha}{4\pi}\frac{1}{\bar\varepsilon_{\rm UV}} + \frac{\alpha}{2\pi}\ln\frac{m^2}{\mu_\gamma^2} + (\text{finite}) \qquad (\xi = 1).} \tag{19.133}$$

Caution

The two divergences in Z_1 have opposite status and must be kept apart, just as in Z_2 (Section 19.3.3). The ultraviolet pole $-\frac{\alpha}{4\pi\bar\varepsilon}$ is an artifact of the regulator, removed once and for all by the counterterm that defines the renormalized vertex; it never appears in an observable. The infrared logarithm $+\frac{\alpha}{2\pi}\ln(m^2/\mu_\gamma^2)$ is physical: it depends on how the massless photon is regulated and cannot be subtracted by any counterterm, because it comes from the soft endpoint $z \to 1$, the region of vanishing virtual-photon momentum. It is nonetheless harmless on its own, because the on-shell vertex is not by itself an observable. Its fate is to cancel: against the matching logarithm in Z_2 (the content of $Z_1 = Z_2$, which keeps the electric charge finite), and ultimately, in any physical cross section, against the real emission of soft photons, the infrared cancellation carried out in Section 19.6. Subtracting the infrared log "by hand," as though it were ultraviolet, would corrupt the final answer.

19.4.6 The Pauli form factor and the bridge to $g-2$

The Pauli form factor is assembled from a single source, the coefficient Eq. (19.128) of $(p+p')^\mu$, projected onto $i\sigma^{\mu\nu}q_\nu$ by the Gordon identity. Because that source

lives entirely in the convergent numerator $N^{\mu}_{(0)}$, F_2 inherits its convergence: there is no $1/\varepsilon$, and—as we verify below—no $\ln \mu_\gamma$ either. F_2 is the cleanest output of one-loop QED, a finite number with no counterterm freedom.

Worked Example 19.12 (Setting up $F_2(q^2)$)**.** The Pauli channel of Eq. (19.105) is

$$\frac{i\sigma^{\mu\nu}q_\nu}{2m}\,F_2(q^2) = -\,i\,e^2\,\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\big[\,N^{\mu}_{(0)}\,\big]_{\text{Pauli}}}{(\ell^2-\Delta)^3}\cdot 2\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x+y+z-1)\,, \tag{19.134}$$

with $[N^{\mu}_{(0)}]_{\text{Pauli}} = -2m\,z(1-z)\,i\sigma^{\mu\nu}q_\nu$ from Eq. (19.128). The integral is finite, so set $d=4$ and use the convergent master $\int \mathrm{d}^4\ell/(2\pi)^4\,(\ell^2-\Delta)^{-3} = -i/(32\pi^2\Delta)$. Tracking the factors of i—the prefactor $-i$, the master's $-i$, giving $(-i)(-i) = -1$—and the Gordon coefficient,

$$\frac{i\sigma^{\mu\nu}q_\nu}{2m}\,F_2(q^2) = -\frac{e^2}{16\pi^2}\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x+y+z-1)\,\frac{-2m\,z(1-z)}{\Delta}\;i\sigma^{\mu\nu}q_\nu\,, \tag{19.135}$$

where the $32\pi^2$ of the master combines with the 2 of the Feynman measure to give $16\pi^2$. Stripping the common $i\sigma^{\mu\nu}q_\nu$ and multiplying through by $2m$,

$$\boxed{F_2(q^2) = \frac{\alpha}{2\pi}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{2m^2\,z(1-z)}{m^2(1-z)^2 - q^2\,xy}\;,} \tag{19.136}$$

where $\Delta = m^2(1-z)^2 - q^2xy$ at $\mu_\gamma = 0$ (the photon mass may be dropped: the $z(1-z)$ numerator vanishes at the dangerous endpoint $z\to 1$, so the integral is infrared-finite and the regulator is unnecessary), and $e^2/8\pi^2 = \alpha/2\pi$.

Equation Eq. (19.136) is the central result of the section: a manifestly ultraviolet- and infrared-finite integral representation of the anomalous magnetic moment as a function of momentum transfer. The same result is reached, by the covariant Passarino–Veltman tensor reduction rather than the Feynman-parameter route, in Theorem D.36; there the finiteness of F_2 is the statement that it is built only from the convergent scalar three-point functions, with the divergent coefficient C_{00} confined to F_1, exactly the separation Eq. (19.118) exhibits here. The two derivations agree, as the form-factor decomposition Eq. (D.196) requires.

The value of interest is the static one, $q^2\to 0$, where $\Delta\to m^2(1-z)^2$ and the integrand becomes $2z(1-z)/(1-z)^2 = 2z/(1-z)$. The x,y-integration at fixed z gives the simplex length $1-z$, cancelling the denominator and leaving the elementary $\int_0^1 \mathrm{d}z\,2z = 1$, so $F_2(0) = \alpha/2\pi$. We *state* this limit to confirm the integral is well posed and yields a pure number, but its detailed evaluation—and the identification of $a_e = F_2(0) = \alpha/2\pi$ with Schwinger's correction to the electron g-factor—belongs to Chapter 20, which consumes Eq. (19.136) as its

starting point. This is the natural division the logical structure requires: the present section builds the form-factor machinery and the integral; Chapter 20 reads off the famous number and develops its physics (the Lamb-shift Uehling piece, the running α, the precision tests).

We have completed the third one-loop building block. The vertex correction Eq. (19.104) decomposes between on-shell spinors into the Dirac and Pauli form factors of Theorem 19.10; the Dirac form factor carries the ultraviolet pole Eq. (19.118) that, renormalized on shell via $F_1(0) = 1$, defines Z_1 in Eqs. (19.120) and (19.132); and the Pauli form factor is the finite integral Eq. (19.136) handed to Chapter 20. The explicit comparison Eq. (19.133) establishes $Z_1 = Z_2$—in both the ultraviolet pole and the infrared logarithm—as a brute fact of two unrelated integrals. That this is no accident, that one symmetry forces it to all orders and thereby protects the electric charge, is the theorem we prove next: the Ward–Takahashi identity of Section 19.5, whose one-loop check is precisely the equality we have just computed, and whose corollary Theorem 19.14 promotes it from coincidence to law. The infrared logarithm common to Z_1 and Z_2, meanwhile, is carried forward—tracked apart from every ultraviolet pole—to its cancellation against real soft-photon emission in Section 19.6.

19.5 The Ward–Takahashi Identity

The vertex computation of Section 19.4 left us with a fact and no explanation. Renormalizing the Dirac form factor on shell produced a vertex renormalization constant Z_1; renormalizing the electron self-energy in Section 19.3 produced a field-strength renormalization Z_2; and the two numbers, extracted from entirely different diagrams by entirely different Feynman-parameter integrals, came out *equal*,

$$Z_1 = Z_2 = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(e^4) \qquad (\xi = 1), \tag{19.137}$$

where $1/\bar{\varepsilon} = 1/\varepsilon - \gamma_E + \ln 4\pi$ is the $\overline{\text{MS}}$ pole of Section 19.1. Nothing in the two calculations made this inevitable. The divergent part of the vertex came from a triangle of three propagators tied together by a photon; the divergent part of the self-energy came from a single fermion propagator dressed by one photon. That their ultraviolet poles agree to the last factor looks like an accident, and an accident at one loop is a warning that a symmetry is at work that we have not yet made explicit. This section makes it explicit. The symmetry is the $U(1)$ gauge invariance of Chapters 8 and 13, expressed not as a statement about the classical action but as an exact identity among the quantum Green's functions: the *Ward–Takahashi identity* [470, 489]. It is the central theorem of the chapter. Once it is proved, Eq. (19.137) is not an accident but a corollary, the masslessness of the photon is protected, and the entire renormalization of the electric charge is shown to reside in one place — the vacuum polarization — to

all orders in perturbation theory. Everything computed in Sections 19.2 to 19.4 will be seen to hang from this one statement.

19.5.1 From a conservation law to a quantum identity

The classical input is the conservation of the electromagnetic current. The QED Lagrangian of QED,

$$\mathcal{L}_{\rm QED} = \bar{\psi}\,(i\not{D} - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \tag{19.138}$$

is invariant under the local $U(1)$ transformation

$$\psi(x) \to e^{-ie\alpha(x)}\psi(x), \qquad \bar{\psi}(x) \to e^{+ie\alpha(x)}\bar{\psi}(x), \qquad A_\mu(x) \to A_\mu(x) + \partial_\mu\alpha(x), \tag{19.139}$$

established in Chapter 8. Noether's theorem applied to its global part (α constant) gives the conserved electromagnetic current

$$J^\mu_{\rm em} = -e\,\bar{\psi}\gamma^\mu\psi, \qquad \partial_\mu J^\mu_{\rm em} = 0, \tag{19.140}$$

the same current whose matrix element defined the form factors of Eq. (19.106). At the classical level this is a property of solutions of the equations of motion. The question is what survives quantization, where the fields are operators (or, in the path integral, integration variables) and the equations of motion hold only inside correlation functions. The answer is that current conservation is promoted to an exact statement about correlation functions, holding off shell and to all orders, with a precise set of correction terms — "contact terms" — localized where the current operator collides with a charged field. Those contact terms are not a blemish; they are the entire content of the identity, because they encode how the current acts on the charges it is conserved against.

To state the identity we need the two dressed two- and three-point functions of the theory, defined as the full (all-orders) one-particle-irreducible amputations introduced in Chapter 15. The first is the *full inverse electron propagator*. Writing the exact propagator as the geometric sum of the free propagator dressed by the 1PI self-energy insertion $-i\Sigma(p)$ computed in Section 19.3,

$$S(p) = \frac{i}{\not{p} - m} + \frac{i}{\not{p} - m}(-i\Sigma)\frac{i}{\not{p} - m} + \cdots = \frac{i}{\not{p} - m - \Sigma(p)}, \tag{19.141}$$

we define the amputated inverse propagator

$$S^{-1}(p) \equiv \not{p} - m - \Sigma(p), \tag{19.142}$$

the object whose tree value is $\not{p} - m$ and whose zero locates the physical electron pole. The second is the *full amputated 1PI vertex* $\Gamma^\mu(p', p)$, normalized so that the complete one-photon–two-electron irreducible amplitude is $-ie\,\Gamma^\mu(p', p)$, with

incoming electron momentum p, outgoing p', and photon momentum $q = p' - p$ flowing in. Its tree value is $\Gamma^\mu = \gamma^\mu$, and its on-shell matrix element is the form-factor decomposition Eq. (19.106),

$$\bar{u}(p')\,\Gamma^\mu(p',p)\,u(p) = \bar{u}(p')\left[\gamma^\mu F_1(q^2) + \frac{i\sigma^{\mu\nu}q_\nu}{2m}F_2(q^2)\right]u(p). \tag{19.143}$$

Theorem 19.13 (Ward–Takahashi identity)**.** *In quantum electrodynamics the full amputated 1PI vertex and the full inverse electron propagator, both defined above and computed in any R_ξ gauge with the gauge-invariant regulator of Section 19.1, satisfy, to all orders in perturbation theory and as an identity in the off-shell momenta,*

$$\boxed{q_\mu\,\Gamma^\mu(p',p) = S^{-1}(p') - S^{-1}(p), \qquad q = p' - p.} \tag{19.144}$$

Equivalently, for the connected amputated Green's function $\mathcal{M}^\mu(q;p_1,\dots,p_n)$ describing the emission of a single photon of momentum q from an arbitrary string of charged external lines, contracting the photon momentum produces only contact terms,

$$q_\mu\,\mathcal{M}^\mu(q;p_1,\dots,p_n) = \sum_{i\in\text{charged}}\Big[\,\mathcal{M}_0(\dots,p_i) - \mathcal{M}_0(\dots,p_i+q_i)\,\Big], \tag{19.145}$$

each term being the lower-point Green's function $\mathcal{M}_0$ with the photon removed and the momentum of the ith charged leg shifted by the photon momentum it would have carried. When every charged external line is put on its mass shell the contact terms vanish, and Eq. (19.145) reduces to the transversality of the physical amplitude, $q_\mu\mathcal{M}^\mu = 0$.

Before proving Theorem 19.13 it is worth checking that it says something true at tree level, where everything is explicit. There $\Gamma^\mu = \gamma^\mu$ and $S^{-1} = \not{p} - m$, so the left-hand side of Eq. (19.144) is $q_\mu\gamma^\mu = \not{q}$ and the right-hand side is

$$S^{-1}(p') - S^{-1}(p) = (\not{p}' - m) - (\not{p} - m) = \not{p}' - \not{p} = \not{q}, \tag{19.146}$$

using $q = p' - p$. The two sides agree identically, with no on-shell condition imposed: the tree-level identity is an algebraic triviality, $\not{q} = \not{q}$. This is deceptively simple, but it already fixes the normalization and sign of Eq. (19.144) that the loop corrections must respect, and it is the seed of the all-orders statement. The whole task is to show that the loop corrections to the two sides — $-q_\mu\,\delta\Gamma^\mu$ on the left and $\Sigma(p) - \Sigma(p')$ on the right — track each other diagram by diagram. We give two independent proofs: a nonperturbative one from the path integral, and a diagrammatic one at one loop that makes the cancellation visible.

19.5.2 The generating-functional proof

The cleanest derivation treats the gauge transformation Eq. (19.139) as a change of integration variables in the generating functional. This route, which we now carry out in full, proves the identity to all orders at once, because it never expands in the coupling.

We import the QED generating functional in the form built in Chapter 10 (the field-theory generating functional and its source coupling, Theorem 10.11; the effective action as its double Legendre transform, Theorem 10.38). Coupling a vector source J_μ to the photon and Grassmann sources $\bar\eta, \eta$ to the electron, and fixing the gauge with the R_ξ term $\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ of Chapter 13,

$$Z[J,\bar\eta,\eta] = \int \mathcal{D}A\,\mathcal{D}\psi\,\mathcal{D}\bar\psi\;\exp\!\Big\{ iS_{\rm inv}[A,\psi,\bar\psi] + iS_{\rm gf}[A] + i\int d^4x\,\big(J_\mu A^\mu + \bar\eta\psi + \bar\psi\eta\big)\Big\}, \tag{19.147}$$

where $S_{\rm inv} = \int d^4x\,\mathcal{L}_{\rm QED}$ is the gauge-invariant action Eq. (19.138).

The argument has three ingredients, and the first two are exactly the two halves of "gauge invariance." First, the *measure* is invariant. Under Eq. (19.139) the fermion fields are rotated by a position-dependent phase; this is a unitary (indeed, for a vector $U(1)$, anomaly-free) transformation whose Jacobian is unity. The axial analogue of this Jacobian is *not* unity — that is the chiral anomaly, computed by the Fujikawa method [203] in Chapter 36 — but the vector phase rotation here is anomaly-free, and the measure $\mathcal{D}A\,\mathcal{D}\psi\,\mathcal{D}\bar\psi$ is genuinely invariant. Second, the *gauge-invariant action* $S_{\rm inv}$ is invariant, by construction of Chapter 8. The only pieces of the integrand that change are the gauge-fixing term, which depends on $\partial\cdot A$ and so is shifted by $\partial\cdot\partial\alpha = \Box\alpha$, and the source terms, which are not gauge invariant because the sources are external c-numbers held fixed.

Now perform an infinitesimal gauge transformation, $\delta\psi = -ie\alpha\psi$, $\delta\bar\psi = +ie\alpha\bar\psi$, $\delta A_\mu = \partial_\mu\alpha$, as a change of variables in Eq. (19.147). Because the integral cannot depend on the names of its integration variables, its first-order variation vanishes. With the measure and $S_{\rm inv}$ inert, only the gauge-fixing and source terms contribute:

$$0 = \int \mathcal{D}A\,\mathcal{D}\psi\,\mathcal{D}\bar\psi\,\Big[\, i\,\delta S_{\rm gf} + i\int d^4x\,\big(J_\mu\,\partial^\mu\alpha - ie\alpha\,\bar\eta\psi + ie\alpha\,\bar\psi\eta\big)\Big] e^{i(\cdots)}. \tag{19.148}$$

The gauge-fixing variation is $\delta S_{\rm gf} = \int d^4x\,(-\frac{1}{\xi})(\partial\cdot A)\,\Box\alpha = \int d^4x\,(-\frac{1}{\xi})[\Box(\partial\cdot A)]\alpha$, integrating by parts twice; the source term $J_\mu\partial^\mu\alpha$ integrates by parts to $-(\partial_\mu J^\mu)\alpha$. Since Eq. (19.148) holds for arbitrary $\alpha(x)$, the coefficient of $\alpha(x)$ must vanish pointwise:

$$-\frac{1}{\xi}\,\Box\,\partial_\mu\langle A^\mu(x)\rangle_{J,\eta,\bar\eta} - \partial_\mu J^\mu(x) - ie\,\bar\eta(x)\langle\psi(x)\rangle_{J,\eta,\bar\eta} + ie\,\langle\bar\psi(x)\rangle_{J,\eta,\bar\eta}\,\eta(x) = 0, \tag{19.149}$$

where $\langle \mathcal{O} \rangle_{J,\eta,\bar\eta} = Z^{-1} \int \mathcal{O}\, e^{i(\cdots)}$ is the source-dependent expectation value. Equation Eq. (19.149) is the master Ward–Takahashi identity for the full generating functional. It is exact: no expansion in e has been made.

To turn it into a statement about 1PI functions we pass to the effective action Γ of Theorem 10.38. Writing $W = -i \ln Z$ for the connected generating functional, the classical fields are $\bar A^\mu = \delta W / \delta J_\mu = \langle A^\mu \rangle$, $\Psi = \delta W / \delta \bar\eta = \langle \psi \rangle$, $\bar\Psi = \delta W / \delta \eta = \langle \bar\psi \rangle$, and $\Gamma[\bar A, \Psi, \bar\Psi] = W - \int (J_\mu \bar A^\mu + \bar\eta \Psi + \bar\Psi \eta)$ is their Legendre transform, with the conjugate relations $\delta\Gamma/\delta \bar A_\mu = -J^\mu$, $\delta\Gamma/\delta\Psi = \bar\eta$, $\delta\Gamma/\delta\bar\Psi = -\eta$ (the fermionic signs follow the right-derivative convention of Chapter 10). Substituting these into Eq. (19.149) eliminates the sources and the expectation values in favor of Γ and its classical-field arguments, giving the master identity in its most useful form:

$$\boxed{\partial_\mu \frac{\delta \Gamma}{\delta \bar A_\mu(x)} = \frac{1}{\xi} \Box\, (\partial \cdot \bar A)(x) + ie \frac{\delta \Gamma}{\delta \Psi(x)}\, \Psi(x) + ie\, \bar\Psi(x) \frac{\delta \Gamma}{\delta \bar\Psi(x)}.} \tag{19.150}$$

The structure is transparent. The left side is the divergence of the photon one-point function generated by Γ; the first term on the right is the gauge-fixing contribution, *linear* in $\bar A$ and independent of the fermion fields; the last two terms say that under the gauge rotation the effective action transforms exactly as the classical action does, with the fermion fields rotating by their charge. Gauge invariance of the quantum theory is the statement Eq. (19.150) that the only non-invariance of Γ is the explicit, uncorrected gauge-fixing term.

The vertex identity Eq. (19.144) follows by differentiating Eq. (19.150) with respect to one electron field and one anti-electron field and then setting all classical fields to zero. Differentiate once by $\Psi(z)$ and once by $\bar\Psi(y)$. On the left we obtain ∂^x_μ acting on the 1PI three-point function with one photon leg at x and the two fermion legs at y, z, namely the proper vertex $-ie\,\Gamma^\mu$. On the right, the gauge-fixing term contains no fermion fields and is annihilated by the two fermion derivatives. The two remaining terms each produce, upon differentiation, a 1PI fermion two-point function $\delta^2\Gamma/\delta\bar\Psi\delta\Psi$ — the inverse propagator S^{-1} of Eq. (19.142) — multiplied by a Dirac delta function that ties the photon insertion point to one of the fermion legs:

$$\partial^x_\mu [-ie\,\Gamma^\mu(y,z;x)] = ie\, S^{-1}(y,x)\, \delta^{(4)}(x-z) + ie\, \delta^{(4)}(x-y)\, S^{-1}(x,z). \tag{19.151}$$

The first contact term sits where the photon meets the incoming line, the second where it meets the outgoing line. Fourier transforming with incoming fermion momentum p at z, outgoing p' at y, and photon momentum $q = p' - p$ at x, the spacetime derivative becomes $\partial^x_\mu \to -iq_\mu$ and the delta functions enforce the momentum routing, collapsing Eq. (19.151) to the momentum-space relation $q_\mu \Gamma^\mu(p', p) = S^{-1}(p') - S^{-1}(p)$. The overall normalization and sign are exactly those of the tree-level check Eq. (19.146), which the all-orders relation must reduce

to when the loop corrections are switched off, and they confirm Eq. (19.144). This completes the nonperturbative proof.

The Green's-function form Eq. (19.145) follows from the same master identity by differentiating with respect to as many additional charged fields as there are external charged lines: each fermion pair $\delta^2/\delta\bar{\Psi}\delta\Psi$ on the right of Eq. (19.150) generates one contact term in which the photon is removed and the adjacent charged-leg momentum is displaced. We will not need the multi-leg version in detail; its corollary, transversality of the on-shell amplitude, is the next subsection.

19.5.3 The diagrammatic proof at one loop

The functional proof is complete but abstract; it is illuminating to see the cancellation happen in the diagrams. The mechanism is the algebraic identity already used at tree level in Theorem 18.6: contracting a photon momentum into a vertex turns γ^μ into $\not{q}$, and $\not{q}$ is a difference of inverse propagators that collapses the line. Off shell the identity reads, for a fermion line carrying momentum ℓ through a vertex at which a photon of momentum q is absorbed,

$$\frac{1}{\not{\ell}+\not{q}-m}\,\not{q}\,\frac{1}{\not{\ell}-m}=\frac{1}{\not{\ell}-m}-\frac{1}{\not{\ell}+\not{q}-m}, \tag{19.152}$$

proved by writing $\not{q}=(\not{\ell}+\not{q}-m)-(\not{\ell}-m)$ and cancelling each factor against its neighbor. The right side is the same fermion line *without* the inserted vertex, evaluated at the two momenta on either side of where the vertex sat. Geometrically, contracting the photon momentum unplugs the vertex and replaces the diagram by the difference of the two ways of stitching the line back together. When a photon can attach at several points along one charged line, summing Eq. (19.152) over the attachment points telescopes: each term cancels against the neighbor, leaving only the two endpoints.

Apply this to the one-loop vertex correction $\delta\Gamma^\mu$ of Eq. (19.104). In Feynman gauge the amputated triangle is

$$-ie\,\delta\Gamma^\mu(p',p)=\int\frac{d^dk}{(2\pi)^d}\,(-ie\gamma^\nu)\,\frac{i}{\not{p}'-\not{k}-m}\,(-ie\gamma^\mu)\,\frac{i}{\not{p}-\not{k}-m}\,(-ie\gamma_\nu)\,\frac{-i}{k^2}, \tag{19.153}$$

with k the photon loop momentum and the two extra factors of e supplied by the loop. Contract with $q_\mu=(p'-p)_\mu$. The contraction acts only on the central γ^μ, turning it into $\not{q}$, and since the loop momentum cancels in the difference, $q=p'-p=(p'-k)-(p-k)$, the insertion identity Eq. (19.152) applies to the central propagator string with $\ell=p-k$:

$$\frac{1}{\not{p}'-\not{k}-m}\,\not{q}\,\frac{1}{\not{p}-\not{k}-m}=\frac{1}{\not{p}-\not{k}-m}-\frac{1}{\not{p}'-\not{k}-m}. \tag{19.154}$$

Tracking the factors of i in Eq. (19.153) — two propagators contribute $i^2 = -1$, the central vertex $-ie\not{q}$ becomes a difference of inverse propagators each carrying $-i$, so the surviving prefactor is $(-ie)\cdot i = e$ — the contracted integrand becomes

$$-ie\,q_\mu\,\delta\Gamma^\mu = \int \frac{d^dk}{(2\pi)^d}\,(-ie\gamma^\nu)\,e\left[\frac{i}{\not p - \not k - m} - \frac{i}{\not p' - \not k - m}\right](-ie\gamma_\nu)\,\frac{-i}{k^2}. \tag{19.155}$$

But each of the two bracketed terms, reattached to its photon rainbow $(-ie\gamma^\nu)\ldots(-ie\gamma_\nu)\,(-i/k^2)$ is precisely the one-loop self-energy diagram of Section 19.3 evaluated at the corresponding external momentum. Recalling that diagram is $-i\Sigma(p) = \int_k (-ie\gamma^\nu)\,iS_0(p-k)\,(-ie\gamma_\nu)(-i/k^2)$, the two terms in Eq. (19.155) assemble into $e\big[(-i\Sigma(p)) - (-i\Sigma(p'))\big]$, so that

$$-ie\,q_\mu\,\delta\Gamma^\mu(p',p) = -ie\big[\Sigma(p) - \Sigma(p')\big] \quad\Longrightarrow\quad q_\mu\,\delta\Gamma^\mu(p',p) = \Sigma(p) - \Sigma(p'). \tag{19.156}$$

This is the one-loop content of Theorem 19.13. Adding the tree piece $q_\mu\gamma^\mu = \not q$ to the loop piece Eq. (19.156) reproduces the full identity:

$$q_\mu\Gamma^\mu = \not q + \big[\Sigma(p) - \Sigma(p')\big] = \big[\not p' - m - \Sigma(p')\big] - \big[\not p - m - \Sigma(p)\big] = S^{-1}(p') - S^{-1}(p), \tag{19.157}$$

in exact agreement with Eq. (19.144). The cancellation in Eq. (19.154) is local at each vertex; nothing about it referred to the one-loop order, the gauge choice (in a general R_ξ gauge the longitudinal $q_\nu q_\rho$ piece of the photon propagator contributes a $\not q$ at each end that collapses the *outer* propagators in the same telescoping way, and the transverse piece is what we computed), or the regulator beyond its preservation of Eq. (19.152) — which dimensional regularization respects and a momentum cutoff would not, the point flagged in the chapter introduction. The same telescoping, applied to every charged line of an arbitrary diagram, is the diagrammatic origin of the Green's-function identity Eq. (19.145).

19.5.4 The on-shell Ward identity as a corollary

The tree-level Ward identity $k_\mu\mathcal{M}^\mu = 0$ of Theorem 18.5, used in Chapter 18 to justify the covariant polarization sum, is now seen as the on-shell special case of the off-shell Takahashi identity, and the special case holds to all orders. The argument is the vanishing of the contact terms in Eq. (19.145) when the external charged legs are amputated and put on shell.

Consider first the elementary vertex sandwiched between physical electron spinors. Contracting Eq. (19.144) with on-shell $\bar u(p')$ and $u(p)$,

$$q_\mu\,\bar u(p')\,\Gamma^\mu(p',p)\,u(p) = \bar u(p')\big[S^{-1}(p') - S^{-1}(p)\big]u(p) = 0, \tag{19.158}$$

because the physical pole condition is exactly $S^{-1}(p')\,u(p') = 0$ and $\bar u(p)\,S^{-1}(p) = 0$: the renormalized inverse propagator vanishes, with unit residue, on the physical

mass shell (the renormalization condition that fixes Z_2 in Section 19.3). The dressed vertex therefore satisfies current conservation between on-shell states to all orders, $q_\mu \langle p'|J^\mu_{\rm em}|p\rangle = 0$, which by the form-factor decomposition Eq. (19.143) is what forbids a third, longitudinal form factor and pins the Pauli structure to the transverse combination $i\sigma^{\mu\nu}q_\nu$ — the gauge-theoretic reason there are exactly two form factors, anticipated in Chapter 20.

For a general amplitude with an external photon of momentum q, the full amplitude is built by attaching the external photon line, through the dressed vertex, to a charged line of an on-shell process. Each contact term in Eq. (19.145) is a lower-point Green's function in which the photon has been removed and one charged leg carries shifted momentum $p_i + q$; that leg is then off shell, $(p_i+q)^2 \neq m^2$, so it lacks the propagator pole that the LSZ amputation requires to contribute to the S-matrix. The contact terms are pure off-shell artifacts and drop out of the physical amplitude, leaving

$$q_\mu \, \mathcal{M}^\mu(q;\ \text{on-shell}) = 0 \tag{19.159}$$

to all orders in perturbation theory. This is the all-orders Ward identity, with Theorem 18.5 its tree restriction. Its physical consequences are exactly those exploited in Chapter 18: the unphysical (longitudinal and timelike) photon polarizations decouple from physical amplitudes, the covariant polarization sum $\sum_\lambda \epsilon^\lambda_\mu \epsilon^{\lambda *}_\nu \to -\eta_{\mu\nu}$ is legitimate, and the S-matrix is independent of the gauge parameter ξ.

A second, equally important corollary is the transversality of the photon self-energy proved at one loop in Theorem 19.5 and imported by the resummation of Chapter 20. The same master identity Eq. (19.150), differentiated twice with respect to $\bar{A}$ and once set to zero field, gives $q_\mu \Pi^{\mu\nu}(q) = 0$ to all orders: the photon two-point function has no piece that the gauge-fixing term does not already supply, so the 1PI self-energy is purely transverse,

$$\Pi^{\mu\nu}(q) = \left(q^2\eta^{\mu\nu} - q^\mu q^\nu\right)\Pi(q^2), \tag{19.160}$$

the tensor structure of Eq. (19.49), now seen to follow from the symmetry rather than from the explicit one-loop integral. Transversality is what forbids a gauge-invariant photon mass term $\propto \eta^{\mu\nu}$ from being generated radiatively: any such term would have to come from $\Pi^{\mu\nu}(0)$, but Eq. (19.160) makes $\Pi^{\mu\nu}(q)$ vanish quadratically as $q \to 0$, so the photon stays exactly massless to all orders. The masslessness of the photon, like the equality of charges, is a theorem about Π enforced by the Ward–Takahashi identity, not a value to be tuned.

19.5.5 Corollary: $Z_1 = Z_2$ to all orders

We can now explain the accident Eq. (19.137). Differentiate the off-shell identity Eq. (19.144) with respect to q_μ and take the soft limit $q \to 0$, in which $p' \to p$.

The left side becomes the vertex at vanishing photon momentum; the right side becomes the momentum derivative of the inverse propagator:

$$\Gamma^{\mu}(p,p) = \frac{\partial}{\partial p_{\mu}}\, S^{-1}(p). \tag{19.161}$$

This is the soft, or differential, form of the Ward–Takahashi identity — in fact the original Ward identity [489], of which Takahashi's [470] is the finite-q generalization. It relates the strength of the dressed vertex in the soft limit directly to the slope of the inverse propagator, two quantities whose ultraviolet divergences are exactly the constants Z_1 and Z_2.

Corollary 19.14 ($Z_1 = Z_2$)**.** *The vertex and electron-field renormalization constants of QED are equal, $Z_1 = Z_2$, to all orders in perturbation theory and in every R_ξ gauge.*

Proof. The renormalized vertex and renormalized inverse propagator are obtained from the bare ones by stripping the renormalization constants that absorb their divergences: $\Gamma^{\mu}_{0} = Z_1^{-1}\Gamma^{\mu}_{R}$, since the field rescaling $\psi_0 = Z_2^{1/2}\psi$ together with the explicit vertex renormalization assigns the irreducible vertex the constant Z_1; and $S_0^{-1} = Z_2^{-1}S_R^{-1}$, since the two-point function carries one factor of Z_2 from each field. The soft identity Eq. (19.161) is an exact consequence of the symmetry and holds for the bare functions,

$$Z_1^{-1}\,\Gamma^{\mu}_{R}(p,p) = \frac{\partial}{\partial p_{\mu}}\Big(Z_2^{-1}S_R^{-1}(p)\Big) = Z_2^{-1}\,\frac{\partial S_R^{-1}}{\partial p_{\mu}}. \tag{19.162}$$

The renormalized functions are finite and physical, and the Ward–Takahashi identity — being the quantum statement of an exact symmetry, free of multiplicative anomaly in QED — must hold among them with no rescaling, $\Gamma^{\mu}_{R}(p,p) = \partial S_R^{-1}/\partial p_{\mu}$. Comparing this with Eq. (19.162) forces $Z_1^{-1} = Z_2^{-1}$, that is $Z_1 = Z_2$. The operator identity Eq. (19.144) is exact and gauge-covariant, so the conclusion holds at every order and in every gauge, even though Z_1 and Z_2 are each separately divergent, scheme-dependent, and gauge-dependent. □

Stated as a relation among the UV poles, Theorem 19.14 says that the divergence dressing the vertex is, structure for structure, the divergence dressing the electron leg. We can read this directly off the soft identity in differentiated form. Write the loop parts $\Gamma^{\mu} = \gamma^{\mu} + \delta\Gamma^{\mu}$ and $S^{-1} = \not{p} - m - \Sigma$; then Eq. (19.161) becomes $\delta\Gamma^{\mu}(p,p) = -\partial\Sigma/\partial p_{\mu}$, and the residues of the two $1/\varepsilon$ poles must match. This is the all-orders content; we discharge the explicit one-loop check below. The renormalization-program face of this corollary — the counterterm equality $\delta_1 = \delta_2$ and its role in the renormalizability of QED — is revisited in Theorem 22.31, where it is one input to the proof that QED is renormalizable; here it is a theorem about Green's functions, established before any counterterm bookkeeping.

19.5.6 Corollary: charge renormalization lives in the photon

The equality $Z_1 = Z_2$ has a consequence out of all proportion to its modest derivation: it collapses the renormalization of the electric charge onto a single function, the vacuum polarization. Recall how the three renormalization constants enter the bare coupling. The interaction term $-e_0\bar{\psi}_0 \not{A}_0 \psi_0$, written in terms of renormalized fields $\psi_0 = Z_2^{1/2}\psi$, $A_0 = Z_3^{1/2}A$ and the bare coupling e_0, becomes $-e_0 Z_2 Z_3^{1/2}\,\bar{\psi}\not{A}\psi$; matching this to the renormalized vertex, which carries the explicit constant Z_1, identifies the charge renormalization constant

$$Z_e = \frac{Z_1}{Z_2\, Z_3^{1/2}}, \qquad e_0 = \mu^{\varepsilon}\, Z_e\, e, \tag{19.163}$$

where μ^{ε} restores the mass dimension of the coupling in $d = 4 - 2\varepsilon$. The combinatorial reading of Eq. (19.163) is that the bare charge is dressed by one field-strength factor per external leg attached to the vertex — Z_2 for the two electron legs, $Z_3^{1/2}$ for the photon leg — divided out against the irreducible vertex constant Z_1. In a generic theory all three would have to be computed independently, and the charge would run according to the combined residue of their poles. Gauge invariance intervenes. Substituting $Z_1 = Z_2$ from Theorem 19.14, the electron constants cancel identically and

$$\boxed{Z_e = \frac{Z_2}{Z_2\, Z_3^{1/2}} = Z_3^{-1/2}, \qquad e_0 = \mu^{\varepsilon}\, Z_3^{-1/2}\, e.} \tag{19.164}$$

The renormalization of the electric charge is governed by the photon field-strength renormalization *alone*. Every effect of the electron — its self-energy, its field strength, the dressing of the vertex — has dropped out of the physical coupling. The sole survivor is Z_3, the renormalization of the photon two-point function, that is the vacuum polarization Π computed in Section 19.2.

Charge universality is the Ward identity

An electron and a muon are distinct fields with distinct masses; their self-energies and vertex corrections differ, so their Z_1 and Z_2 are not the same numbers. Yet the charge each carries renormalizes identically, and the running coupling an electron sees is the very same $\alpha(\mu)$ a muon sees. Equation Eq. (19.164) is the reason. For *each* charged species the Ward–Takahashi identity enforces $Z_1 = Z_2$, so the species-specific constants cancel and $Z_e = Z_3^{-1/2}$ regardless of which fermion one started from. Charge renormalization is therefore not a property of any charged particle but of the photon they all share. The empirical universality of electric charge — that charge comes in identical quanta independent of the carrier, so that the proton's charge is exactly -1 times the electron's

despite the proton's wildly different strong-interaction dressing — is, at the level of radiative corrections, precisely the statement $Z_1 = Z_2$. When several species circulate in the photon's vacuum-polarization loop they contribute additively to Π and hence to Z_3; that enlarges the running but is a property of the photon line, common to all charges, not a violation of universality.

The practical payoff, handed forward to Chapters 20 and 24, is that the running of α can be read off from $\Pi(q^2)$ without ever computing a vertex or a self-energy: Eq. (19.164) guarantees that the geometric resummation of vacuum-polarization bubbles in Chapter 20 captures the entire scale dependence of the coupling, and that the QED beta function is, in the words of Chapter 24, a property of the photon.

19.5.7 Closing the loop: the one-loop identity verified

It remains to confirm that the explicit one-loop constants of Sections 19.3 and 19.4 satisfy the identity we have proved — the consistency check that closes Sections 19.2–19.4 against the symmetry. We do this in a worked example, because the agreement is sharp and instructive: the soft Ward identity is a nontrivial relation between two independent integrals, and seeing the divergences fall into line is the strongest possible evidence that the regulator has preserved gauge invariance.

Worked Example 19.15 (The soft Ward identity at one loop)**.** The one-loop electron self-energy $\Sigma_2(p)$ of Eq. (19.70), computed in Feynman gauge in $d = 4 - 2\varepsilon$, has the divergent part Eq. (19.82),

$$\Sigma_2(p)\big|_{\text{div}} = -\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}\,\not{p} + \frac{e^2}{4\pi^2}\,\frac{1}{\bar{\varepsilon}}\,m = \frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}(-\not{p} + 4m), \tag{19.165}$$

two Lorentz structures, one $\propto \not{p}$ and one $\propto m$. The soft Ward identity in differentiated form, $\delta\Gamma^{\mu}(p,p) = -\partial\Sigma/\partial p_{\mu}$ from Eq. (19.161), predicts the divergent part of the vertex correction at zero photon momentum from the slope of the self-energy. Differentiating Eq. (19.165) and using $\partial(\not{p})/\partial p_{\mu} = \gamma^{\mu}$ while the mass term, independent of p, drops out,

$$\frac{\partial\Sigma_2}{\partial p_{\mu}}\bigg|_{\text{div}} = -\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}\,\gamma^{\mu} \quad\Longrightarrow\quad \delta\Gamma^{\mu}(p,p)\big|_{\text{div}} = -\frac{\partial\Sigma_2}{\partial p_{\mu}}\bigg|_{\text{div}} = +\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}\,\gamma^{\mu}. \tag{19.166}$$

This is exactly the divergent part of the one-loop vertex Eq. (19.104) computed independently in Section 19.4: there the triangle integral produced a UV pole proportional to the tree structure γ^{μ} with precisely this coefficient. Two integrals with no diagram in common — a single dressed propagator and a three-propagator

triangle — have matched their $1/\bar{\varepsilon}$ poles to the last factor, as the symmetry demands.

Now convert to renormalization constants. The renormalized inverse propagator is $S_R^{-1} = \not{p} - m - \Sigma_2 + (\delta_2 \not{p} - \delta_m)$, and finiteness of its $\not{p}$ structure requires δ_2 to cancel the $\not{p}$ pole of $-\Sigma_2$. From Eq. (19.165), $-\Sigma_2|_{\text{div},\not{p}} = +\frac{e^2}{16\pi^2}\frac{1}{\bar{\varepsilon}}\not{p}$, so

$$\delta_2 = -\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{19.167}$$

in agreement with the on-shell extraction $\delta_2 = \partial\Sigma_2/\partial\not{p}\,|_{\not{p}=m}$ of Eq. (19.86), whose divergent part is the same $\not{p}$ coefficient. The renormalized vertex is $\Gamma^\mu_R = \gamma^\mu + \delta\Gamma^\mu + \delta_1\gamma^\mu$, and finiteness of its γ^μ structure requires δ_1 to cancel the γ^μ pole of $\delta\Gamma^\mu$. From Eq. (19.166), $\delta\Gamma^\mu|_{\text{div}} = +\frac{e^2}{16\pi^2}\frac{1}{\bar{\varepsilon}}\gamma^\mu$, so $\delta_1 = -\frac{e^2}{16\pi^2}\frac{1}{\bar{\varepsilon}}$. The two counterterms are equal,

$$\delta_1 = \delta_2 = -\frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}} \quad\Longrightarrow\quad Z_1 = Z_2 = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(e^4) \qquad (\xi = 1), \tag{19.168}$$

which is the fact Eq. (19.137) we set out to explain, now derived from the soft Ward identity rather than discovered by coincidence. The check also fixes the on-shell normalization of the Dirac form factor: the same soft relation, written for the *renormalized* functions, says $\Gamma^\mu_R(p,p) = \partial S_R^{-1}/\partial p_\mu$, and on the physical mass shell the right side is the unit-residue slope guaranteed by the Z_2 renormalization condition, so $F_1(0) = 1$ exactly — the statement, anticipated in Eq. (19.120) and used throughout Chapter 20, that the electron's measured charge is its tree-level charge, uncorrected by the vertex and self-energy infinities. The radiative corrections to the charge have cancelled between Z_1 and Z_2, precisely as Eq. (19.164) promised, leaving the entire physical charge renormalization to Z_3.

The Ward–Takahashi identity is the spine of the chapter. The three primitive one-loop objects — the vacuum polarization Π, the self-energy Σ, and the vertex Γ^μ — are not independent: Π is forced transverse, Γ^μ and Σ are locked by Eq. (19.144), and the net effect is that the field-strength and vertex infinities cancel out of every physical amplitude, depositing the whole of charge renormalization in the photon's vacuum polarization. This is what we hand forward. The running of the fine-structure constant in Chapters 20 and 24 now follows from $\Pi(q^2)$ alone, with $e_0 = \mu^\varepsilon Z_3^{-1/2} e$ its exact bare-to-renormalized relation; and the physical $O(\alpha)$ cross sections are free of the Z_1 and Z_2 ultraviolet divergences. What remains in those cross sections is not ultraviolet but infrared — the soft and collinear singularities still lurking in the on-shell vertex and self-energy — and that last obstacle to a finite observable is the subject of Section 19.6.

19.6 Infrared Divergences and Their Cancellation

The Ward–Takahashi identity of the previous section settled the ultraviolet account once and for all: the relation $Z_1 = Z_2$ (Theorem 19.13) guarantees that the charge counterterm is fixed by the electron's wavefunction renormalization, the photon two-point function is transverse, and every short-distance infinity of one-loop QED is absorbed into the renormalized parameters m, e, and the field normalizations. A renormalized amplitude is therefore ultraviolet-finite. It is not yet a number. Two of the one-loop building blocks—the Dirac form factor $F_1(q^2)$ from the vertex (Eq. (19.131)) and the wavefunction constant Z_2 from the self-energy (Eq. (19.101))—retain a divergence of a completely different character, one that does not come from the region of large loop momentum and cannot be removed by any local counterterm. It comes from the opposite end of the integration, where the virtual photon becomes *soft*: its momentum tends to zero, its propagator $1/k^2$ blows up, and the loop integral diverges in the infrared. This is the last obstruction between the renormalized theory and a finite physical cross section, and resolving it is the business of this section.

The resolution is not a subtraction but a physical reckoning with what a detector can measure. A renormalized but infrared-divergent form factor is not a defect of the calculation; it is the signal that the quantity being computed is not an observable. No apparatus has infinite energy resolution, so no experiment can distinguish an electron from an electron accompanied by photons too soft to register. The measurable cross section is inclusive over those unresolved photons, and once the emission of real soft radiation is added to the virtual correction, the infrared divergences cancel between the two, leaving a finite result that depends—as it physically must—on the detector's energy resolution ΔE. We will exhibit this cancellation explicitly at $O(\alpha)$ for a clean process, electron scattering in an external field, then state the theorems of Bloch–Nordsieck and Kinoshita–Lee–Nauenberg that guarantee it to all orders, and close by handing QED, now finite in both the ultraviolet and the infrared, to the precision computations of Chapter 20.

19.6.1 The surviving infrared divergences and the choice of regulator

We begin by recalling precisely which divergences are left and how they are parametrized. The vertex correction produced the renormalized Dirac form factor whose $O(\alpha)$ part, in the on-shell scheme where $F_1(0) = 1$, carried an infrared singularity from the soft region of the loop. Regulating that singularity with a small fictitious photon mass μ_γ—the propagator $i/(k^2 + i\epsilon)$ replaced by

$i/(k^2-\mu_\gamma^2+i\epsilon)$—the result of Section 19.4 can be written

$$F_1(q^2) = 1 - \frac{\alpha}{2\pi}\, f_{\mathrm{IR}}(q^2)\, \ln\frac{-q^2}{\mu_\gamma^2} + \delta F_1^{\mathrm{fin}}(q^2) + O(\alpha^2), \tag{19.169}$$

where $\delta F_1^{\mathrm{fin}}$ is finite as $\mu_\gamma \to 0$ and the coefficient of the infrared logarithm is the dimensionless function

$$f_{\mathrm{IR}}(q^2) = (2m^2 - q^2)\int_0^1 \frac{dx}{m^2 - x(1-x)\,q^2} - 2, \tag{19.170}$$

to which we return in Section 19.6.3, where it reappears—this is the crux—as the coefficient of the *same* logarithm in the real-emission rate. The divergence is the $\mu_\gamma \to 0$ limit of $\ln(-q^2/\mu_\gamma^2) \to +\infty$. The self-energy contributed the on-shell wavefunction constant, which in the same regulator reads

$$Z_2 = 1 - \frac{\alpha}{4\pi}\Big(\frac{1}{\bar{\epsilon}_{\mathrm{UV}}} - 2\ln\frac{m^2}{\mu_\gamma^2} + \text{finite}\Big) + O(\alpha^2), \tag{19.171}$$

its infrared logarithm $\ln(m^2/\mu_\gamma^2)$ sitting alongside the genuine ultraviolet pole $1/\bar{\epsilon}_{\mathrm{UV}}$ that renormalization removes (Eq. (19.101)). The two infrared logarithms are not independent. The Ward–Takahashi identity $Z_1 = Z_2$ forces the vertex counterterm to carry exactly the infrared structure of Z_2, so that in the renormalized vertex the self-energy's infrared logarithm is precisely what the on-shell condition $F_1(0) = 1$ subtracts at $q^2 = 0$; what survives in $F_1(q^2)$ for $q^2 \neq 0$ is the genuinely momentum-transfer-dependent piece Eq. (19.169). The upshot is a single infrared object to dispose of: the function $f_{\mathrm{IR}}(q^2)$ multiplying $\ln(-q^2/\mu_\gamma^2)$ in the physical form factor.

Notation Watch

Two equivalent infrared regulators appear in this book and must not be confused. A fictitious photon mass μ_γ keeps the calculation in four dimensions and is physically transparent—it is the inverse range of the electromagnetic interaction—so we use it for the explicit QED cancellation here. Dimensional regularization continued to $d > 4$ (i.e. $\varepsilon < 0$) regulates the same soft region as a pole $1/\varepsilon_{\mathrm{IR}}$, as established in Section D.7; the dictionary between the two is $\ln(\mu_\gamma^2/-q^2) \leftrightarrow 1/\varepsilon_{\mathrm{IR}}$ up to a finite scheme constant. The cancellation we exhibit is a statement about the physical limit and is independent of which regulator carries it; the dimensionally regulated version of exactly this computation, with the eikonal phase-space integral done in d dimensions, is in Section D.7.4. In either scheme the regulator dependence must vanish from the final inclusive rate—that vanishing is the content of the section.

To make the cancellation concrete we need a process whose Born cross section is simple and whose only $O(\alpha)$ infrared sensitivity is the vertex form factor. The cleanest such process is the scattering of an electron in a *static external field*—an electron deflected by a heavy nucleus, the Mott/Coulomb problem. The nucleus is treated as an external source that absorbs momentum $\mathbf{q}$ but no energy, so the electron's energy $E = E_p = E_{p'}$ is conserved and the momentum transfer is spacelike, $q = p' - p$ with $q^0 = 0$ and $q^2 = -|\mathbf{q}|^2 < 0$. The Born amplitude is the tree vertex contracted with the external potential; the entire $O(\alpha)$ radiative correction to this elastic amplitude is the vertex correction, so the Born cross section is corrected multiplicatively,

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{virt}} = \left.\frac{d\sigma}{d\Omega}\right|_{0} \left[\, 1 + 2\ \mathrm{Re}\, F_1(q^2) \,\right] + O(\alpha^2), \tag{19.172}$$

the factor $2\ \mathrm{Re}\, F_1$ arising because the cross section is the interference of the corrected amplitude $1 + \delta F_1$ with its Born value, $|1 + \delta F_1|^2 = 1 + 2\,\mathrm{Re}\ \delta F_1 + O(\alpha^2)$, and the magnetic form factor F_2 (finite, the subject of Chapter 20) not contributing to the spin-averaged rate at this order in the same multiplicative way. (Here F_2 does contribute a finite piece; it is infrared-finite and we suppress it, tracking only the infrared-sensitive terms.)

19.6.2 The virtual correction and the Sudakov double logarithm

Inserting Eq. (19.169) into Eq. (19.172) gives the virtual radiative correction to the cross section,

$$\frac{d\sigma_{\text{virt}}}{d\sigma_0} = 1 - \frac{\alpha}{2\pi}\, f_{\text{IR}}(q^2)\, \ln\frac{-q^2}{\mu_\gamma^2} + (\text{infrared-finite}), \tag{19.173}$$

writing $d\sigma_0 \equiv (d\sigma/d\Omega)_0$ for brevity. Two features of this expression deserve to be named before we cancel the divergence, because both survive the cancellation in transmuted form.

First, the sign. As $\mu_\gamma \to 0$ the correction Eq. (19.173) tends to $-\infty$: the virtual cloud *suppresses* the elastic amplitude. This is the Sudakov suppression of the form factor—a charged particle that undergoes a sudden large-angle deflection is exponentially unlikely to do so without radiating, so the probability of a strictly elastic scatter is driven to zero. The infinity is the regulated statement of a probability tending to zero, and we will see it become, after real emission is included, a finite power suppression governed by ΔE.

Second, the high-energy behavior. In the regime $-q^2 \equiv Q^2 \gg m^2$ the integral in Eq. (19.170) is dominated by the endpoints $x \to 0, 1$, where the denominator $m^2 + x(1-x)Q^2$ is cut off by the electron mass:

$$\int_0^1 \frac{dx}{m^2 + x(1-x)Q^2} = \frac{2}{Q\sqrt{Q^2+4m^2}}\, \ln\frac{\sqrt{Q^2+4m^2}+Q}{\sqrt{Q^2+4m^2}-Q} \xrightarrow{Q^2 \gg m^2} \frac{2}{Q^2}\ln\frac{Q^2}{m^2}, \tag{19.174}$$

so that, using $2m^2 - q^2 = Q^2 + 2m^2 \to Q^2$,

$$f_{\rm IR}(q^2) \xrightarrow{-q^2 \gg m^2} 2\ln\frac{-q^2}{m^2} - 2 \;\simeq\; 2\ln\frac{-q^2}{m^2}. \tag{19.175}$$

The virtual correction Eq. (19.173) therefore contains the product of two large logarithms,

$$\left.\frac{d\sigma_{\rm virt}}{d\sigma_0}\right|_{\rm lead} = 1 - \frac{\alpha}{\pi}\,\ln\frac{-q^2}{m^2}\,\ln\frac{-q^2}{\mu_\gamma^2} + \cdots, \tag{19.176}$$

a *double logarithm*. One factor is the infrared logarithm of the regulator, $\ln(-q^2/\mu_\gamma^2)$, coming from the soft region $k \to 0$; the other is the collinear/mass logarithm $\ln(-q^2/m^2)$ coming from the angular region in which the photon is emitted nearly parallel to one of the fast electron lines, with the electron mass m cutting off the collinear singularity. Their product is the Sudakov double logarithm, the same structure that the dimensionally regulated massless triangle of Theorem D.38 produced as a $1/\varepsilon_{\rm IR}^2$ double pole. In QED the electron mass screens the collinear pole into the finite factor $\ln(-q^2/m^2)$, leaving only the soft divergence $\ln(1/\mu_\gamma^2)$ to be cancelled by real emission; in a theory with massless charged particles both logarithms would be divergent, a distinction that resurfaces when we state KLN below.

19.6.3 Real soft emission: the eikonal factor from the Ward identity

The companion process is bremsstrahlung: the same scattering, now with one real photon k radiated into the final state. If that photon is soft enough to escape detection—its energy below the resolution ΔE—the event is experimentally indistinguishable from elastic scattering and must be added to Eq. (19.173). We need its cross section in the soft limit, and the soft limit is universal: it does not depend on the internal details of the hard scattering, only on the external charged lines, and it is fixed by the very Ward identity that governed the tree amplitudes (Theorem 18.5).

Consider attaching the emitted photon to the outgoing electron line of the hard process. The photon of momentum k and polarization ϵ is emitted just before the electron leaves, inserting a propagator and a vertex between the hard blob $\mathcal{M}_0$ and the final spinor:

$$\bar u(p')\,(-ie\gamma^\mu)\,\frac{i(\not p' + \not k + m)}{(p'+k)^2 - m^2}\,[\,\mathcal{M}_0\,]\,\epsilon_\mu^*(k). \tag{19.177}$$

In the soft limit $k \to 0$ the denominator is $(p'+k)^2 - m^2 = 2\,p'\!\cdot\! k + k^2 \to 2\,p'\!\cdot\! k$, which vanishes linearly and produces the singularity. The numerator is evaluated at $k = 0$ using the on-shell Dirac equation: with $\gamma^\mu(\not p' + m) = 2p'^\mu - (\not p' - m)\gamma^\mu$ and $\bar u(p')(\not p' - m) = 0$,

$$\bar u(p')\,\gamma^\mu(\not p' + m) = 2p'^\mu\,\bar u(p'), \tag{19.178}$$

so that Eq. (19.177), collecting the factors $(-ie)(i) = e$, reduces to the hard amplitude times a c-number,

$$\mathcal{M}_0 \;\cdot\; e\,\frac{\epsilon^*\!\cdot p'}{p'\!\cdot k}. \tag{19.179}$$

Emission off the incoming line is identical except that the photon momentum is subtracted in the propagator, $(p-k)^2 - m^2 \to -2\,p\!\cdot\!k$, flipping the sign; the soft factor is $-e\,\epsilon^*\!\cdot p/(p\!\cdot\!k)$. Summing the two attachments—the only places a soft photon can come off, since attaching it to an internal hard line gives no $1/k$ enhancement—the one-photon amplitude factorizes,

$$\mathcal{M}_{1\gamma}(k\to 0) \;=\; \mathcal{M}_0\; e\,\epsilon^*_\mu(k)\,j^\mu(k), \qquad j^\mu(k) = \frac{p'^\mu}{p'\!\cdot k} - \frac{p^\mu}{p\!\cdot k}, \tag{19.180}$$

into the hard amplitude times the *eikonal soft factor* $e\,\epsilon^*\cdot j$, where j^μ is the classical current of a charge that abruptly changes its momentum from p to p'. This is the same eikonal current met in Eq. (D.219); we have now derived it from the soft limit of the Feynman amplitude rather than quoted it.

The decisive property of the soft factor is that the current is conserved,

$$k_\mu j^\mu(k) = \frac{k\!\cdot p'}{p'\!\cdot k} - \frac{k\!\cdot p}{p\!\cdot k} = 1 - 1 = 0, \tag{19.181}$$

which is nothing but the soft limit of the Ward identity $k_\mu \mathcal{M}^\mu = 0$ (Theorem 18.5): replacing $\epsilon^*_\mu \to k_\mu$ in Eq. (19.180) must annihilate the amplitude, and Eq. (19.181) is that statement reduced to the soft current. Charge conservation $Q_{\text{out}} - Q_{\text{in}} = 0$ is what makes the two terms in j^μ cancel when contracted with k. The practical consequence is that the unphysical, longitudinal part of the photon polarization sum decouples, so we may use the covariant replacement $\sum_\lambda \epsilon^*_\mu \epsilon_\nu \to -\eta_{\mu\nu}$ (Theorem 18.9) without error: the difference between this and the true physical sum is proportional to k_μ or k_ν and vanishes against Eq. (19.181). The soft-emission probability is therefore gauge invariant and unambiguous, and we may compute it directly.

Worked Example 19.16 (The soft-bremsstrahlung cross section and its eikonal integral)**.** We compute the cross section for the external-field scattering accompanied by a single soft photon of energy $\omega_k < \Delta E$, in the photon-mass regulator, and reduce it to an explicit one-dimensional integral.

Factorization. Squaring Eq. (19.180), summing photon polarizations with $-\eta_{\mu\nu}$ (licensed by Eq. (19.181)), and attaching the one-photon phase space to the Born cross section, the rate factorizes into the Born cross section times an integral over the soft photon,

$$d\sigma_{\text{real}} = d\sigma_0 \int \frac{d^3k}{(2\pi)^3\,2\omega_k}\; e^2\big[-\,j(k)\!\cdot\! j(k)\big], \qquad \omega_k = \sqrt{|\mathbf{k}|^2 + \mu_\gamma^2}, \tag{19.182}$$

where, writing out $-j\cdot j$ with $p^2 = p'^2 = m^2$,

$$-j(k)\cdot j(k) = \frac{2\,p\cdot p'}{(p\cdot k)(p'\cdot k)} - \frac{m^2}{(p\cdot k)^2} - \frac{m^2}{(p'\cdot k)^2} \;\geq 0. \tag{19.183}$$

This is the promised factorization: the hard physics sits entirely in $d\sigma_0$, and the soft physics is the universal integral of Eq. (19.183).

The energy integral. Every term in Eq. (19.183) scales as $1/\omega_k^2$, because $p\cdot k = \omega_k(E - \mathbf{p}\cdot\hat{\mathbf{k}}) \equiv \omega_k\,(p\cdot\hat{k})$ with $\hat{k}^\mu = (1,\hat{\mathbf{k}})$ the unit photon direction (lightlike up to $O(\mu_\gamma^2/\omega_k^2)$). Using $|\mathbf{k}|^2 d|\mathbf{k}| = \omega_k|\mathbf{k}|\,d\omega_k$ and pulling the overall ω_k^{-2} out of the bracket,

$$\frac{d^3k}{2\omega_k}\,\frac{1}{\omega_k^2} = \frac{|\mathbf{k}|\,d\omega_k\,d\Omega_{\hat{k}}}{2\,\omega_k^2} = \frac{1}{2}\,\frac{\sqrt{\omega_k^2-\mu_\gamma^2}}{\omega_k^2}\,d\omega_k\,d\Omega_{\hat{k}}, \tag{19.184}$$

so the energy integral is

$$\int_{\mu_\gamma}^{\Delta E} d\omega_k\,\frac{\sqrt{\omega_k^2-\mu_\gamma^2}}{2\,\omega_k^2} = \frac{1}{2}\ln\frac{2\Delta E}{\mu_\gamma} + O(1), \tag{19.185}$$

the logarithm generated entirely by the soft endpoint $\omega_k \to \mu_\gamma$—this is where the infrared divergence lives—and cut off above by the detector resolution ΔE. The $O(1)$ terms are finite, regulator-independent constants under the logarithm.

The angular integral. What remains is the angular average of the bracket with $k \to \hat{k}$,

$$\mathcal{A}(q^2) \equiv \frac{1}{4\pi}\int d\Omega_{\hat{k}}\Big[\frac{2\,p\cdot p'}{(p\cdot\hat{k})(p'\cdot\hat{k})} - \frac{m^2}{(p\cdot\hat{k})^2} - \frac{m^2}{(p'\cdot\hat{k})^2}\Big]. \tag{19.186}$$

The two self-energy terms are computed in the rest frame of the corresponding momentum, or directly: with $p\cdot\hat{k} = E(1-\beta\cos\theta)$, $\beta = |\mathbf{p}|/E$,

$$\frac{1}{4\pi}\int d\Omega_{\hat{k}}\,\frac{m^2}{(p\cdot\hat{k})^2} = \frac{m^2}{E^2}\cdot\frac{1}{2}\int_{-1}^{1}\frac{d(\cos\theta)}{(1-\beta\cos\theta)^2} = \frac{m^2}{E^2}\cdot\frac{1}{1-\beta^2} = 1, \tag{19.187}$$

using $1-\beta^2 = m^2/E^2$; each self term contributes exactly 1, so together they give -2. The cross term is combined by a Feynman parameter, $\frac{1}{(p\hat{k})(p'\hat{k})} = \int_0^1 dx\,(p_x\cdot\hat{k})^{-2}$ with $p_x = xp + (1-x)p'$, after which the same angular identity Eq. (19.187)—valid for any timelike vector, $\frac{1}{4\pi}\int d\Omega_{\hat{k}}\,(P\cdot\hat{k})^{-2} = 1/P^2$—gives

$$\frac{1}{4\pi}\int d\Omega_{\hat{k}}\,\frac{2\,p\cdot p'}{(p\cdot\hat{k})(p'\cdot\hat{k})} = 2\,pp'\int_0^1\frac{dx}{p_x^2} = (2m^2-q^2)\int_0^1\frac{dx}{m^2-x(1-x)q^2}, \tag{19.188}$$

where $p_x^2 = m^2 - x(1-x)q^2$ and $2p\cdot p' = 2m^2 - q^2$. Comparing Eqs. (19.187) and (19.188) with Eq. (19.170) we read off the identity that drives the whole section:

$$\boxed{\mathcal{A}(q^2) = f_{\rm IR}(q^2).} \tag{19.189}$$

The angular eikonal integral of real emission is exactly the coefficient of the virtual infrared logarithm. This is no accident: both are the same eikonal current Eq. (19.180) integrated, the virtual one against the off-shell loop propagator and the real one against the on-shell phase space, and the soft theorem that fixes j^μ fixes both.

Result. Assembling Eqs. (19.182) and (19.185) with the angular factor $\mathcal{A} = f_{\rm IR}$ and the prefactor $\frac{e^2}{(2\pi)^3}\cdot 4\pi = \frac{e^2}{2\pi^2} = \frac{2\alpha}{\pi}$ (the $\frac{1}{2}$ from the phase-space measure already sitting inside Eq. (19.185)),

$$\frac{d\sigma_{\rm real}}{d\sigma_0} = \frac{\alpha}{\pi}\, f_{\rm IR}(q^2)\, \ln\frac{\Delta E}{\mu_\gamma} = +\frac{\alpha}{2\pi}\, f_{\rm IR}(q^2)\, \ln\frac{(\Delta E)^2}{\mu_\gamma^2} + (\text{infrared-finite}). \tag{19.190}$$

The real-emission correction carries the same infrared logarithm of μ_γ as the virtual one, with the same coefficient $\frac{\alpha}{2\pi} f_{\rm IR}$ but the *opposite* sign and a different upper scale: the divergence is cut off above by ΔE for real emission and by the hard scale $-q^2$ for the virtual loop.

19.6.4 The cancellation

We now add the two contributions. The physically measured cross section—elastic scattering plus scattering with any unresolved photon of energy below ΔE—is the sum of Eqs. (19.173) and (19.190),

$$\begin{aligned}
\left.\frac{d\sigma}{d\sigma_0}\right|_{\rm meas} &= \underbrace{1 - \frac{\alpha}{2\pi} f_{\rm IR}(q^2) \ln\frac{-q^2}{\mu_\gamma^2}}_{\text{virtual}} + \underbrace{\frac{\alpha}{2\pi} f_{\rm IR}(q^2) \ln\frac{(\Delta E)^2}{\mu_\gamma^2}}_{\text{real}} + (\text{finite}) \\
&= 1 + \frac{\alpha}{2\pi} f_{\rm IR}(q^2)\Big[\ln\frac{(\Delta E)^2}{\mu_\gamma^2} - \ln\frac{-q^2}{\mu_\gamma^2}\Big] + (\text{finite}) \\
&= 1 - \frac{\alpha}{2\pi}\, f_{\rm IR}(q^2)\, \ln\frac{-q^2}{(\Delta E)^2} + (\text{finite}).
\end{aligned} \tag{19.191}$$

The regulator μ_γ has cancelled identically: it entered the virtual and real contributions with equal coefficient and opposite sign, exactly as the shared eikonal current Eq. (19.180) and the identity Eq. (19.189) demanded, and the limit $\mu_\gamma \to 0$ may now be taken freely. What remains is finite and depends on the two physical scales of the problem: the momentum transfer $-q^2$ and the detector resolution ΔE. Figure 19.7 displays the mechanism: two contributions each diverging logarithmically as $\mu_\gamma \to 0$, with equal and opposite slope, summing to a regulator-independent constant.

In the high-energy regime $-q^2 \gg m^2$, using $f_{\rm IR} \to 2\ln(-q^2/m^2)$ from Eq. (19.175), the finite remainder is itself a double logarithm,

$$\left.\frac{d\sigma}{d\Omega}\right|_{\rm meas} = \left.\frac{d\sigma}{d\Omega}\right|_0 \Big[1 - \frac{\alpha}{\pi}\, \ln\frac{-q^2}{m^2}\, \ln\frac{-q^2}{(\Delta E)^2} + \cdots\Big], \tag{19.192}$$

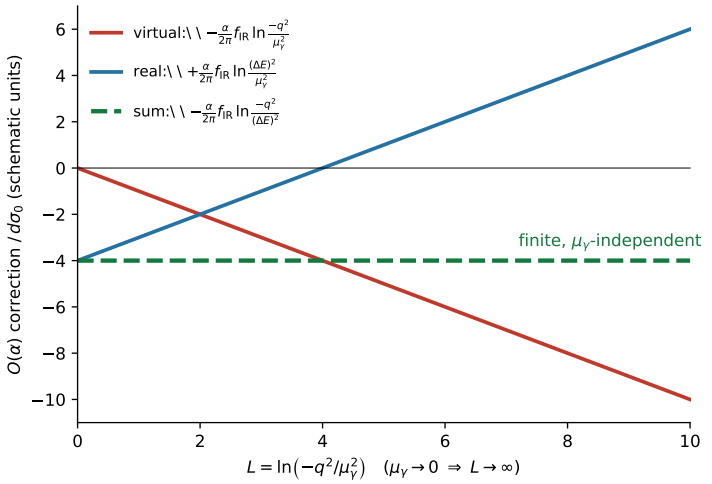


Figure 19.7: The infrared cancellation at $O(\alpha)$ for electron scattering in an external field. As the photon-mass regulator is removed, $\mu_\gamma \to 0$, the abscissa $\ln(-q^2/\mu_\gamma^2) \to +\infty$. The virtual vertex correction (red) and the real soft-bremsstrahlung correction (blue) each diverge, with equal and opposite slope $\mp\frac{\alpha}{2\pi} f_{\rm IR}$, because both are governed by the same conserved eikonal current Eq. (19.180). Their sum (green, dashed) is independent of μ_γ and depends only on the physical ratio $-q^2/(\Delta E)^2$: the IR-finite, measurable cross section of Eq. (19.191).

the celebrated Sudakov form of the radiatively corrected cross section. The infrared regulator μ_γ of the bare double logarithm Eq. (19.176) has been replaced by the physical resolution ΔE: the more inclusive the measurement (the larger ΔE), the smaller the radiative suppression, and as $\Delta E \to \sqrt{-q^2}$ the correction shrinks because essentially all radiation is then counted as part of the elastic peak. The correction is negative and grows with energy, signalling that fixed-order perturbation theory eventually fails; the all-orders resummation that cures this—and that turns Eq. (19.192) into an exponential—is the Yennie–Frautschi–Suura theorem stated below.

Physical Insight

A charged particle is never alone. Its Coulomb field, re-expressed in quanta, is a coherent cloud of soft photons, and any acceleration shakes some of them loose: the probability of scattering through a finite angle while emitting *exactly* zero photons is zero, which is why the strictly elastic amplitude Eq. (19.173) was driven to $-\infty$. The infinity was never physical; it was the artifact of asking an unphysical question—"what is the rate for

an electron to scatter and emit nothing?"—of a theory whose asymptotic states are not isolated charges but charges dressed by their radiation. The physical question, "what is the rate to scatter with no *detectable* radiation, i.e. none above ΔE?", has a finite answer, Eq. (19.191). The detector resolution ΔE is not a calculational crutch but part of the definition of the observable. This is the resolution of the infrared "paradox": infrared divergences are a symptom of computing transition amplitudes between idealized states that physical apparatus cannot prepare or resolve; inclusive, resolution-limited probabilities are finite. The same lesson, generalized to collinear radiation off massless particles, organizes all of perturbative QCD (Chapter 28).

19.6.5 All orders: Bloch–Nordsieck, KLN, and exponentiation

The $O(\alpha)$ cancellation just exhibited is the first instance of a pattern that holds to all orders, and it is worth stating the governing theorems precisely, with their hypotheses, since the difference between them is exactly the difference between the soft divergences of massive QED and the collinear divergences of massless gauge theory.

Theorem 19.17 (Bloch–Nordsieck)**.** *In QED with massive charged particles, the infrared (soft) divergences from virtual photon loops cancel, order by order in α, against the divergences from real emission of soft photons, in any inclusive transition probability that sums over the emission of arbitrarily many photons of total energy below a fixed resolution ΔE. The inclusive rate is finite as the infrared regulator is removed; the $\ln \mu_\gamma$ (equivalently $1/\varepsilon_{\mathrm{IR}}$) divergences are replaced by finite logarithms of ΔE over the hard scale.*

The physical content is the argument of the previous subsection promoted to all orders: the soft radiation factorizes from the hard process into the eikonal current Eq. (19.180), whose conservation Eq. (19.181) ties virtual and real emission together at every order, and the sum over any number of unresolved photons exponentiates the one-loop cancellation. We do not reproduce the all-orders combinatorial proof—it belongs to the Yennie–Frautschi–Suura analysis below and to Section D.7.4, where the same statement is recorded in dimensional regularization—but the mechanism is the one we proved at $O(\alpha)$. Theorem 19.17 suffices for an observable like the total cross section for $e^+e^- \to \mu^+\mu^-(+\,\text{soft}\ \gamma)$ with massive muons, where the only infrared sensitivity is soft and final-state summation alone restores finiteness.

It does *not* suffice in general. When charged particles are massless, the collinear logarithm $\ln(-q^2/m^2)$ of Eq. (19.175) is no longer screened by a mass and becomes a second, independent divergence; the double-logarithmic structure Eq. (19.176) then carries two infrared poles, not one, and summing over soft

final-state emission removes only half of them. The complete statement is due to Kinoshita, Lee, and Nauenberg.

Theorem 19.18 (Kinoshita–Lee–Nauenberg)**.** *In a theory with massless fields, the transition probability summed over the complete set of initial and final states degenerate in energy (in the limit of vanishing regulator) is free of infrared* and *mass (collinear) singularities, to all orders in perturbation theory. Bloch–Nordsieck is the special case in which only final-state soft degeneracies occur; in the presence of collinear degeneracies—generic for massless charged particles—finiteness requires summing over degenerate* initial *states as well.*

The honest scope of these theorems must be stated plainly. Bloch–Nordsieck guarantees the soft cancellation that we proved explicitly, and it is all one needs whenever a fermion mass screens the collinear region, as in the precision QED of Chapter 20. KLN is the stronger, more delicate statement required when massless charged particles make the collinear singularities physical; there the naive sum over final states alone can fail, and one must either sum over initial-state degeneracies—rarely a practical option for a fixed-beam experiment—or, equivalently and usefully, absorb the unavoidable initial-state collinear singularities into measured, scale-dependent parton distribution functions. That program, factorization, is the foundation of perturbative QCD phenomenology and is developed in Chapter 28; the operational requirement it imposes on observables is collinear safety, made precise in Theorem D.41. For QED carried here the simpler Bloch–Nordsieck cancellation is exactly what is needed, and we have it.

Finally, the all-orders resummation of the soft logarithms. The double logarithm in Eq. (19.192) is the first term of a series whose n-loop leading term is $\frac{1}{n!}\big[-\frac{\alpha}{\pi}\ln(-q^2/m^2)\ln(-q^2/(\Delta E)^2)\big]^n$; the factorial is the combinatorial factor for n independent, uncorrelated soft emissions, a structure that exponentiates.

Theorem 19.19 (Yennie–Frautschi–Suura exponentiation)**.** *The infrared-divergent and infrared-sensitive soft-photon contributions to a QED cross section, summed to all orders, organize into an exponential of the one-loop soft factor. For the external-field cross section the leading-logarithm resummation of Eq. (19.192) reads*

$$\left.\frac{d\sigma}{d\Omega}\right|_{\text{meas}} = \left.\frac{d\sigma}{d\Omega}\right|_{0} \exp\left[-\frac{\alpha}{\pi}\,\ln\frac{-q^2}{m^2}\,\ln\frac{-q^2}{(\Delta E)^2}\right](1+O(\alpha)), \tag{19.193}$$

the exponent being the finite, regulator-independent one-loop soft function. The exponential is bounded and positive, curing the apparent negativity and eventual breakdown of the fixed-order result.

The Yennie–Frautschi–Suura construction [529] achieves this by isolating, to all orders, the infrared-divergent pieces of every virtual and real contribution, showing they cancel in inclusive sums exactly as at one loop, and resumming

the residual finite soft logarithms into the exponent of Eq. (19.193); the full combinatorial proof is the content of that analysis and we state it at this level, having proved its $O(\alpha)$ core in Theorem 19.16 and Eq. (19.191). The exponentiation is what makes the high-energy form factor a sensible, suppressed-but-finite object and underlies the Monte Carlo treatment of soft radiation in every modern precision QED prediction.

With the infrared divergences disposed of, the program opened by the chapter chapter introduction is complete: the regulator built in Section 19.6.1's companion sections tamed the loops, renormalization removed the ultraviolet infinities, the Ward–Takahashi identity (Theorem 19.13) protected the charge and tied the building blocks together, and the Bloch–Nordsieck cancellation has now rendered the physical cross section finite in the infrared as well. One-loop QED stands as a complete, ultraviolet- and infrared-finite, gauge-invariant, renormalized theory that predicts numbers. Those numbers are the subject of the next chapter: the same vertex correction whose infrared tail we have just cancelled carries, in its infrared-finite part δF_1^{fin} and its companion magnetic form factor F_2, the anomalous magnetic moment $a_e = \alpha/2\pi$ and the Lamb shift, while the vacuum polarization supplies the running coupling $\alpha(q^2)$. Chapter 20 takes this finite, renormalized one-loop theory and extracts from it the precision predictions that made QED the most accurately tested theory in physics.

Further reading

The one-loop computations of this chapter are textbook material, and the reader who wants a second pass with different conventions is well served by Peskin and Schroeder [396] (Ch. 6–7), Schwartz [432], and Weinberg [500], the last especially for the soft-theorem derivation of the eikonal factor and the all-orders infrared structure. Dimensional regularization originates with 't Hooft and Veltman [469]; the one-loop anomalous moment is Schwinger's [433]. The infrared cancellation traces to Bloch and Nordsieck [81], the mass-singularity (KLN) theorem to Kinoshita [295] and Lee and Nauenberg [320], and the all-orders exponentiation to Yennie, Frautschi, and Suura [529]. The dimensionally regulated counterpart of the explicit soft cancellation, with the d-dimensional eikonal phase-space integral, is carried out in Chapter D; the generalization to collinear singularities and infrared-safe observables in non-abelian gauge theory is the subject of Chapter 28.

Chapter 20

Precision QED

The previous chapter built three renormalized one-loop functions of quantum electrodynamics — the electron self-energy $\Sigma(p)$, the photon vacuum polarization $\Pi(q^2)$, and the electromagnetic vertex with its Dirac and Pauli form factors $F_1(q^2)$, $F_2(q^2)$ — and made them finite by dimensional regularization and on-shell renormalization (Chapter 19). Those objects were assembled as machinery; this chapter spends them. Its single thesis is that the most precise confrontations between theory and experiment in all of physics are nothing more than particular *limits* of those same amplitudes, read in particular physical settings. The vertex at zero momentum transfer, sandwiched between on-shell spinors, is the electron's magnetic moment; the same vertex together with Σ and Π, evaluated for an electron bound in hydrogen, is the Lamb shift; the vacuum polarization resummed over all momenta is the running of the electric charge. Each limit yields a number that has been measured, and each agrees. This is the chapter where QED produces *the* famous number, $a_e = (g-2)/2 = \alpha/2\pi$, and follows it to the present research frontier.

The development imports heavily and proves the rest in place. From Chapter 19 we take the renormalized form-factor decomposition, the Ward–Takahashi identity $Z_1 = Z_2$ that pins $F_1(0) = 1$, the explicit one-loop Σ and Π, and the infrared cancellation against soft bremsstrahlung. From Chapter 11 we borrow the Gordon decomposition, the standard-representation Dirac spinors, and the exact Dirac–Coulomb spectrum; the bare QED vertex is the one fixed in Chapter 17; minimal coupling is the prescription of Chapter 8. The loop technology — Feynman parametrization, the master one-loop scalar integral, and the Wick rotation that supplies its factor of i — is quoted from Chapter D, and the antisymmetric tensor $\sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ together with the gamma-matrix contraction identities from Chapter C. The analytic-continuation and dispersion-relation machinery used to turn the vacuum polarization into a spectral integral over physical cross sections is the one set up in Chapter 5. Throughout we keep the conventions fixed in Part V: mostly-minus metric, natural units, electron charge

$-e$ with $e > 0$ and $\alpha = e^2/4\pi$, and Feynman gauge as the default for explicit computation.

The arc is a descent from formalism to laboratory number and back out to the open frontier. We first turn the abstract vertex into an observable, showing that $F_1(0)$ is the charge and $F_2(0)$ the anomalous moment, with $g = 2[F_1(0) + F_2(0)]$ derived both from the squared Dirac equation and from matching the current matrix element to the magnetic Hamiltonian. We then set up Schwinger's computation — Feynman-combining the three vertex propagators, shifting, and projecting the on-shell numerator onto the magnetic structure $i\sigma^{\mu\nu}q_\nu$ — and carry the resulting integral to its conclusion, finding the ultraviolet- and infrared-finite result $a_e = \alpha/2\pi$. From that one number the chapter widens: the higher-order QED series and the inversion of the electron anomaly into the world's best fine-structure constant; the chirality selection rule that makes the muon, through its $(m_\mu/m_e)^2$ lever, the sensitive antenna for physics beyond QED, and the honest mid-2020s status of the muon anomaly. The same one-loop functions are then read in a second setting, the bound electron, where self-energy, vacuum polarization, and the anomalous-moment vertex conspire to lift the Dirac $2S_{1/2}$–$2P_{1/2}$ degeneracy and reproduce the Lamb shift to better than one percent. Finally the vacuum polarization is resummed into the running coupling $\alpha(q^2)$, exhibited in both momentum and position space, and connected to measured cross sections through its spectral representation.

Three threads of unfinished business are handed forward deliberately, by cross-reference to where this book discharges them: the systematic renormalization-group derivation of the running coupling and the QED beta function to Chapter 24; the non-perturbative hadronic vacuum polarization and the R-ratio that feed both $\alpha(M_Z)$ and the muon anomaly to Chapter 28, with the precision Standard-Model tests to Chapter 34; and the electroweak contribution to the lepton moments to Chapter 31. What is settled here — $g - 2$ at one loop, the Lamb shift, and the running of α — is settled completely and in full; what is genuinely open is flagged as such and dated.

20.1 The Magnetic Moment and the Form Factors

The chapter introduction handed us the renormalized one-loop electromagnetic vertex of the electron, the object built and dimensionally regulated in Chapter 19: an amputated three-point function which, sandwiched between on-shell spinors, collapses to two scalar functions of a single invariant,

$$\bar{u}(p')\,\Gamma^{\mu}(p',p)\,u(p) = \bar{u}(p')\left[\gamma^{\mu}\,F_1(q^2) + \frac{i\,\sigma^{\mu\nu}q_\nu}{m}\,F_2(q^2)\right]u(p), \qquad q \equiv p' - p. \tag{20.1}$$

This is the renormalized form-factor decomposition of Eq. (19.106) and Theorem 19.10, transcribed here into the conventions of this book; the only modi-

fication from the version a reader may have met elsewhere is the coefficient of the second term, which we discuss and justify below because the whole chapter turns on it. The functions F_1 and F_2 are abstract until we say what laboratory number each one controls. That is the task of this section. We will show that $F_1(0)$ is the electric charge of the electron in units of e — fixed to unity, exactly, by the Ward–Takahashi identity — and that $F_2(0)$ is the *anomalous* part of the electron's magnetic moment, the deviation of the gyromagnetic ratio g from its Dirac value 2. The result, the famous number of QED, is then sharply defined: it is $a_e = F_2(0)$, with $a_e = (g-2)/2$. We continue QED here from the first loop of Chapter 19 towards that number; the present section turns the formal vertex into the observable, and the sections that follow compute $F_2(0)$ and find it equal to $\alpha/2\pi$.

We work throughout in the conventions fixed in the chapter introduction: mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$, natural units, electron charge $-e$ with $e>0$ and $\alpha = e^2/4\pi$, covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ acting on the electron field, and the gamma-matrix and trace technology of Chapter C. The momentum transfer is $q = p' - p$; both external electron legs are on shell, $p^2 = p'^2 = m^2$, so that $q^2 = 2m^2 - 2p\cdot p' \le 0$ in the scattering region.

20.1.1 The on-shell electromagnetic vertex and what its two form factors mean

The physical content of the dressed vertex is the matrix element of the electromagnetic current $J^\mu_{\mathrm{em}} = -e\,\bar\psi\gamma^\mu\psi$ between one-electron states. By Lorentz invariance this matrix element must be a rank-one Lorentz vector built from the available data — the two momenta p, p' (equivalently their sum $P = p + p'$ and difference $q = p' - p$), the metric, the Dirac matrices, and the spinors — and sandwiched between $\bar u(p')$ and $u(p)$. The most general such object, before any physics is imposed, is a linear combination of γ^μ, P^μ, and q^μ, each multiplied by a scalar function of the only available invariant q^2 (recall $p^2 = p'^2 = m^2$ are fixed):

$$\bar u(p')\,\Gamma^\mu\,u(p) = \bar u(p')\Big[\gamma^\mu\,A(q^2) + P^\mu\,B(q^2) + q^\mu\,C(q^2)\Big]u(p). \tag{20.2}$$

Three constraints prune this to two independent structures, and the pruning is the content of Eq. (19.106); we recall the logic because it tells us *why* there are exactly the two form factors whose physics we are about to read off.

First, current conservation, $\partial_\mu J^\mu_{\mathrm{em}} = 0$, becomes in momentum space $q_\mu\,\bar u(p')\Gamma^\mu u(p) = 0$. Contracting Eq. (20.2) with q_μ and using $q\cdot P = (p'-p)\cdot(p'+p) = p'^2 - p^2 = 0$ together with the Dirac equations $\not p\,u(p) = m\,u(p)$ and $\bar u(p')\not p' = m\,\bar u(p')$, which give $\bar u(p')\not q\,u(p) = \bar u(p')(\not p' - \not p)u(p) = (m-m)\,\bar u(p')u(p) = 0$, leaves only the C term: $q_\mu \bar u'\Gamma^\mu u = q^2\,C(q^2)\,\bar u' u$. Conservation therefore forces $C(q^2) = 0$ for $q^2 \neq 0$, and by continuity at $q^2 = 0$ as well. The longitudinal structure q^μ is gauge-eliminated; only γ^μ and P^μ survive.

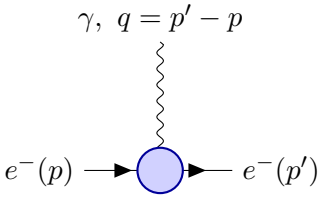


Figure 20.1: The full (amputated, on-shell) electromagnetic vertex of the electron. The shaded blob is the sum of all one-particle-irreducible insertions on the photon–electron coupling, $-ie\,\Gamma^\mu(p', p)$, which between physical spinors reduces to the two form factors of Eq. (20.1): the Dirac form factor $F_1(q^2)$ on γ^μ (the charge/convective coupling) and the Pauli form factor $F_2(q^2)$ on $i\sigma^{\mu\nu}q_\nu$ (the magnetic/spin coupling). At tree level the blob is the bare vertex $-ie\gamma^\mu$ of Eq. (17.10), i.e. $F_1 = 1$, $F_2 = 0$. The magnetic moment is the low-q^2 reading of the blob.

Second, we trade the surviving pair $\{\gamma^\mu, P^\mu\}$ for the more physical pair $\{\gamma^\mu,\ i\sigma^{\mu\nu}q_\nu\}$. The two bases are related by the Gordon decomposition (Theorem 11.15), proved in Chapter 11 from nothing but the Dirac equation and the Clifford algebra,

$$\bar{u}(p')\,\gamma^\mu\,u(p) = \bar{u}(p')\left[\frac{(p+p')^\mu}{2m} + \frac{i\,\sigma^{\mu\nu}q_\nu}{m}\right]u(p), \qquad \sigma^{\mu\nu} = \tfrac{i}{4}\,[\gamma^\mu, \gamma^\nu]\,, \tag{20.3}$$

which expresses $P^\mu/2m = (p+p')^\mu/2m$ as γ^μ minus the tensor term. Substituting P^μ in favour of γ^μ and $i\sigma^{\mu\nu}q_\nu$ converts Eq. (20.2) (with $C = 0$) into the form Eq. (20.1), where F_1, F_2 are linear combinations of A, B. Third, hermiticity of the current and invariance under parity (QED conserves parity) guarantee that F_1, F_2 are real for spacelike q^2 and that no further structures (for example a γ^5 term, which would signal an electric dipole moment and parity violation) appear. The upshot is exactly two real form factors, as in Eq. (20.1). The Dirac form factor F_1 multiplies γ^μ; the Pauli form factor F_2 multiplies the tensor $i\sigma^{\mu\nu}q_\nu$.

Notation Watch

The coefficient of the tensor term in Eq. (20.1) is $1/m$, not the $1/2m$ that appears in many textbooks. The difference is entirely a matter of the normalization of $\sigma^{\mu\nu}$. This book uses the Lorentz-generator normalization $\sigma^{\mu\nu} = \frac{i}{4}\,[\gamma^\mu, \gamma^\nu]$ fixed once and for all in Eq. (C.11), which is *half* the value $\frac{i}{2}\,[\gamma^\mu, \gamma^\nu]$ used by some references. The physical Pauli operator coupling to the photon is the same object in either convention; written with our $\sigma^{\mu\nu}$ it carries the factor $1/m$, written with the doubled $\sigma^{\mu\nu}$ it carries $1/2m$. The Gordon identity Eq. (20.3) already exhibits this: with our $\sigma^{\mu\nu}$ the Gordon spin term is $i\sigma^{\mu\nu}q_\nu/m$, so the bare vertex γ^μ contains exactly one

unit of the tensor structure at coefficient $1/m$. Defining F_2 relative to that same $1/m$ structure is what makes the clean identification $a_e = F_2(0)$ hold; with the other convention one writes $1/2m$ and the same identification follows. We adopt $1/m$ to stay consistent with Eq. (C.11) and the Gordon decomposition we import.

The two form factors split the coupling of the electron to a photon into its two classical pieces, and the cleanest way to see this is to read Eq. (20.1) through the Gordon decomposition Eq. (20.3). Replacing the γ^μ in the F_1 term by its Gordon equivalent gives

$$\bar{u}'\,\Gamma^\mu\,u = \bar{u}'\Bigg[\underbrace{\frac{(p+p')^\mu}{2m}\,F_1(q^2)}_{\text{convective (charge) current}} + \underbrace{\frac{i\,\sigma^{\mu\nu}q_\nu}{m}\,\big(F_1(q^2)+F_2(q^2)\big)}_{\text{spin (magnetic) current}}\Bigg]u. \tag{20.4}$$

The first term is proportional to the total momentum $(p+p')^\mu$ and is structurally identical to the Schrödinger probability current: it is the convective current of a moving charge, the piece that survives in the spinless (scalar QED) case and that builds the Coulomb interaction. The second term is proportional to $i\sigma^{\mu\nu}q_\nu$ and is intrinsically a spin coupling — it vanishes for a spinless particle and, as we will show, it is precisely the coupling of the electron's magnetic moment to an external field. Equation (20.4) already tells us, structurally, that the total strength of the magnetic coupling is governed by the *combination* $F_1 + F_2$, not by F_2 alone: the Dirac form factor contributes a magnetic moment too. We will make this quantitative and find $g = 2[F_1(0) + F_2(0)]$.

Charge normalization: $F_1(0) = 1$. The Dirac form factor at zero momentum transfer is the total electric charge of the electron, measured in units of e. To see this, set $q = 0$ in Eq. (20.1): the tensor term vanishes because it carries an explicit factor q_ν, leaving the forward matrix element of the current,

$$\bar{u}(p)\,\Gamma^\mu\,u(p) = F_1(0)\,\bar{u}(p)\,\gamma^\mu\,u(p) = 2p^\mu\,F_1(0), \tag{20.5}$$

where we used the forward limit of the Gordon identity, $\bar{u}(p)\gamma^\mu u(p) = p^\mu\,\bar{u}(p)u(p)/m = 2p^\mu$ (with $\bar{u}u = 2m$). The conserved charge operator is $Q = \int d^3x\,J^0_{\text{em}}/(-e) = \int d^3x\,\bar{\psi}\gamma^0\psi$, the spatial integral of the time component of the number current; its matrix element between one-electron states counts the (unit) electron number, and Eq. (20.5) shows this is $2E\,F_1(0)$ against the relativistic normalization $2E$ of the states. Hence the electron's charge in units of e is exactly $F_1(0)$.

That $F_1(0) = 1$ to all orders in perturbation theory — that radiative corrections do not renormalize the physical charge of the electron away from its tree value — is not an accident but a theorem. It is the content of the Ward–Takahashi

identity proved in Theorem 19.13, which relates the vertex at zero momentum transfer to the electron self-energy and forces the equality of renormalization constants $Z_1 = Z_2$. The two facts are the same statement: $Z_1 = Z_2$ is exactly the condition that the loop corrections to the vertex and to the external-leg wavefunction cancel at $q = 0$, leaving the renormalized $F_1(0)$ pinned at unity. We import this result rather than reprove it:

$$\boxed{F_1(0) = 1} \qquad \text{(charge normalization, from Theorem 19.13).} \tag{20.6}$$

With Eq. (20.6) in hand, the gyromagnetic relation $g = 2[F_1(0) + F_2(0)]$ that we are about to derive collapses to $g = 2[1 + F_2(0)]$, and therefore

$$a_e \equiv \frac{g-2}{2} = F_2(0). \tag{20.7}$$

The anomalous magnetic moment is the Pauli form factor at zero momentum transfer, nothing more. The remaining work of the section is to prove the gyromagnetic relation that underlies Eq. (20.7), and we do it in two complementary ways: first by the nonrelativistic reduction of the Dirac equation in an external field, which delivers Dirac's prediction $g = 2$ at tree level and exhibits the magnetic coupling as the Pauli term; then by matching the full current matrix element Eq. (20.1) to the magnetic-moment Hamiltonian, which proves $g = 2[F_1(0) + F_2(0)]$ and pins the coefficient of F_2 to the anomalous moment.

20.1.2 Dirac's prediction: $g = 2$ from the squared Dirac equation

Before any loop, the electron already has a magnetic moment, and its value is the first triumph of the Dirac equation. The classical, tree-level statement is that the gyromagnetic ratio is exactly $g = 2$ — twice the naive value $g = 1$ that a classical charged spinning ball would have. We derive this from the minimally coupled Dirac equation by reducing it to its nonrelativistic limit, where the magnetic interaction emerges as a Pauli term and its coefficient is the moment.

Minimal coupling (Eq. (8.27)) promotes the free Dirac equation Eq. (11.13) to

$$(i\not{D} - m)\psi = 0, \qquad iD_\mu = i\partial_\mu - eA_\mu \equiv \Pi_\mu, \tag{20.8}$$

where A_μ is now an external classical electromagnetic potential and $\Pi_\mu = \hat{p}_\mu - eA_\mu$ is the gauge-covariant kinetic momentum operator, with $\hat{p}_\mu = i\partial_\mu$. (The sign is fixed by the convention $D_\mu = \partial_\mu + ieA_\mu$ of QED; the electron's charge $-e$ is encoded in this sign and will resurface as the direction of its moment.) Equation (20.8) reads $\not{\Pi}\,\psi = m\,\psi$. Acting once more with $\not{\Pi}$ and using $\not{\Pi}\,\psi = m\psi$ on the right gives the squared equation

$$\not{\Pi}\not{\Pi}\,\psi = m^2\,\psi. \tag{20.9}$$

The left side is where the spin coupling hides. Decompose the product of slashes into symmetric and antisymmetric parts,

$$\not{\Pi}\not{\Pi} = \gamma^\mu\gamma^\nu\,\Pi_\mu\Pi_\nu = \Big(\eta^{\mu\nu}\mathbf{1} + \tfrac{1}{2}\,[\gamma^\mu,\gamma^\nu]\Big)\Pi_\mu\Pi_\nu = \Pi^\mu\Pi_\mu + \tfrac{1}{4}\,[\gamma^\mu,\gamma^\nu]\,[\Pi_\mu,\Pi_\nu]\,, \tag{20.10}$$

where in the last step the antisymmetric $[\gamma^\mu,\gamma^\nu]$ projects $\Pi_\mu\Pi_\nu$ onto its antisymmetric part $\frac{1}{2}\,[\Pi_\mu,\Pi_\nu]$. The commutator of two kinetic momenta is not zero: it is the field strength. Using $\hat{p}_\mu = i\partial_\mu$,

$$[\Pi_\mu,\Pi_\nu] = [\hat{p}_\mu - eA_\mu, \hat{p}_\nu - eA_\nu] = -e([\hat{p}_\mu,A_\nu]+[A_\mu,\hat{p}_\nu]) = -ie(\partial_\mu A_\nu - \partial_\nu A_\mu) = -ie\,F_{\mu\nu}, \tag{20.11}$$

since $[\hat{p}_\mu, A_\nu] = i\partial_\mu A_\nu$ acting on any function. Inserting Eq. (20.11) into Eq. (20.10) and using $\frac{1}{4}\,[\gamma^\mu,\gamma^\nu] = -i\sigma^{\mu\nu}$ (from $\sigma^{\mu\nu} = \frac{i}{4}\,[\gamma^\mu,\gamma^\nu]$, Eq. (C.11)),

$$\not{\Pi}\not{\Pi} = \Pi^\mu\Pi_\mu + (-i\sigma^{\mu\nu})(-ie\,F_{\mu\nu}) = \Pi^\mu\Pi_\mu - e\,\sigma^{\mu\nu}F_{\mu\nu}. \tag{20.12}$$

The second term is the *Pauli term*: it has no analogue for a scalar field (it would be absent if we had squared the Klein–Gordon operator), and it is linear in the field strength, hence linear in the magnetic field. This is the magnetic moment, emerging from $(\sigma\cdot\Pi)^2$.

Specialize to a static, purely magnetic external field: $A_0 = 0$, $\mathbf{A} = \mathbf{A}(\mathbf{x})$, with $\mathbf{B} = \nabla\times\mathbf{A}$. Then $F_{0i} = 0$ and only the spatial components contribute, $\sigma^{\mu\nu}F_{\mu\nu} = \sigma^{ij}F_{ij}$. In the standard representations of Section C.3 the spatial σ^{ij} is the block-diagonal spin matrix $\sigma^{ij} = \frac{1}{2}\epsilon^{ijk}\Sigma^k$ with $\Sigma^k = \mathrm{diag}(\sigma^k,\sigma^k)$, while $F_{ij} = -\epsilon_{ijk}B^k$. Hence

$$\sigma^{\mu\nu}F_{\mu\nu} = \sigma^{ij}F_{ij} = \tfrac{1}{2}\epsilon^{ijk}\Sigma^k\,(-\epsilon_{ijl}B^l) = -\tfrac{1}{2}(2\delta^{kl})\,\Sigma^k B^l = -\,\boldsymbol{\Sigma}\cdot\mathbf{B}, \tag{20.13}$$

using $\epsilon^{ijk}\epsilon_{ijl} = 2\delta^k_l$. The Pauli term is therefore $-e\,\sigma^{\mu\nu}F_{\mu\nu} = +e\,\boldsymbol{\Sigma}\cdot\mathbf{B}$, and the squared equation Eq. (20.9) becomes

$$(\Pi^\mu\Pi_\mu + e\,\boldsymbol{\Sigma}\cdot\mathbf{B})\psi = m^2\psi. \tag{20.14}$$

Now separate the time component. For a stationary state of energy E in a time-independent field, $\Pi_0\psi = E\psi$, so $\Pi^\mu\Pi_\mu = E^2 - \boldsymbol{\Pi}^2$ with $\boldsymbol{\Pi} = \mathbf{p} - e\mathbf{A}$ the kinetic three-momentum ($\mathbf{p} = -i\nabla$). Equation (20.14) rearranges to

$$\Big[(\mathbf{p}-e\mathbf{A})^2 - e\,\boldsymbol{\Sigma}\cdot\mathbf{B}\Big]\psi = (E^2 - m^2)\,\psi. \tag{20.15}$$

In the nonrelativistic regime write $E = m + \mathcal{E}$ with $\mathcal{E}\ll m$, so that $E^2 - m^2 = (E-m)(E+m) \simeq 2m\mathcal{E}$. Dividing by $2m$ gives the *Pauli equation* for the (upper, large) two-component spinor ξ, on which $\boldsymbol{\Sigma}$ acts as the Pauli matrices $\boldsymbol{\sigma} = 2\mathbf{S}$:

$$\left[\frac{(\mathbf{p}-e\mathbf{A})^2}{2m} - \frac{e}{2m}\,\boldsymbol{\sigma}\cdot\mathbf{B}\right]\xi = \mathcal{E}\,\xi. \tag{20.16}$$

The first term is the orbital (diamagnetic) energy of a charge in a vector potential; the second is the intrinsic magnetic interaction. Reading off the spin coupling and writing $\mathbf{S} = \boldsymbol{\sigma}/2$,

$$H_{\text{mag}} = -\frac{e}{2m}\,\boldsymbol{\sigma}\cdot\mathbf{B} = -\frac{e}{m}\,\mathbf{S}\cdot\mathbf{B}. \tag{20.17}$$

Comparing with the universal form $H_{\text{mag}} = -\boldsymbol{\mu}\cdot\mathbf{B}$ for a magnetic moment, and with the gyromagnetic definition $\boldsymbol{\mu} = g\,\dfrac{e}{2m}\,\mathbf{S}$, we read off the coefficient

$$g\,\frac{e}{2m} = \frac{e}{m} \quad\Longrightarrow\quad \boxed{g = 2} \qquad \text{(Dirac, tree level).} \tag{20.18}$$

This is Dirac's prediction. The Dirac equation, with no addition beyond minimal coupling, endows the electron with a magnetic moment whose gyromagnetic ratio is exactly 2, twice the classical value — a result that was among the strongest early evidence for the equation. In the form-factor language this is the statement that at tree level $F_1 = 1$ and $F_2 = 0$ (the bare vertex is just γ^μ), so that $g = 2[F_1(0) + F_2(0)] = 2$. The anomalous moment $a_e = (g-2)/2$ is exactly zero at tree level; it is a purely radiative quantity.

Caution

The sign in $\boldsymbol{\mu} = g\,(e/2m)\,\mathbf{S}$ and the direction of the electron's moment require care. We have written the gyromagnetic ratio g as a positive number ($g = 2$ at tree level), with the elementary charge $e > 0$ appearing in the prefactor. The *physical* moment of the electron points *opposite* to its spin, because the electron's charge is $-e$: the true moment vector is $\boldsymbol{\mu}_{\text{phys}} = -g\,(e/2m)\,\mathbf{S}$, the overall minus carried by the negative charge. The quantity g measures the *magnitude* of the moment relative to the Bohr magneton $\mu_B = e/2m$, and it is g, and its deviation $a_e = (g-2)/2$ from 2, that we compute and compare with experiment. None of the signs in Eqs. (20.17) and (20.18) depend on this interpretive choice; they are fixed by the conventions of the chapter introduction.

20.1.3 From the form factors to the moment: $g = 2[F_1(0) + F_2(0)]$

The squared-Dirac argument gave $g = 2$ for the bare vertex. To capture the radiative content we must feed the *full* vertex Eq. (20.1) into the same physical setup — an electron scattering off a static, slowly varying external magnetic field — and match the resulting interaction to the magnetic-moment Hamiltonian $-\boldsymbol{\mu}\cdot\mathbf{B}$. The matching is the central derivation of the section; we carry it out in full, because it is what proves that the coefficient of F_2 is exactly the anomalous moment.

Theorem 20.1 (Gyromagnetic ratio in terms of the form factors)**.** *The gyromagnetic ratio of the electron, defined through* $\boldsymbol{\mu} = g\,(e/2m)\,\mathbf{S}$ *by matching the on-shell current matrix element Eq. (20.1) to the magnetic interaction* $-\boldsymbol{\mu}\cdot\mathbf{B}$ *in a static, slowly varying magnetic field, is*

$$g = 2\big[F_1(0) + F_2(0)\big]. \tag{20.19}$$

With the charge normalization $F_1(0) = 1$ *of Eq. (20.6), the anomalous magnetic moment is* $a_e = (g-2)/2 = F_2(0)$.

The proof is the worked computation below. The strategy is the standard one for extracting a nonrelativistic potential from a relativistic amplitude: compute the first-order amplitude for scattering off the external field, take its nonrelativistic limit keeping terms linear in the momentum transfer (the magnetic field is the curl of $\mathbf{A}$, hence $O(q)$), and read off the effective potential by comparison with the Born approximation.

Worked Example 20.2 (Matching the vertex to the magnetic moment)**. Setup.** An electron scatters off a static external classical potential $A^{\rm cl}_\mu(\mathbf{x})$. The first-order S-matrix element is the dressed vertex of Fig. 20.1 contracted with the Fourier transform of the field, $\tilde{A}^{\rm cl}_\mu(\mathbf{q}) = \int d^3x\, e^{-i\mathbf{q}\cdot\mathbf{x}} A^{\rm cl}_\mu(\mathbf{x})$, with $\mathbf{q} = \mathbf{p}' - \mathbf{p}$ and (for a static field) $q^0 = 0$. Using the vertex rule of Eq. (17.10) with $\gamma^\mu \to \Gamma^\mu$,

$$i\mathcal{M} = -ie\,\bar{u}(p')\,\Gamma^\mu\,u(p)\,\tilde{A}^{\rm cl}_\mu(\mathbf{q}). \tag{20.20}$$

The effective nonrelativistic potential $\tilde{V}(\mathbf{q})$ is obtained by matching to the Born approximation. The relativistic states are normalized to $2E$, the nonrelativistic ones to unity, so the two amplitudes differ by a factor $\sqrt{2E'\,2E} \to 2m$ in the nonrelativistic limit; tracking the factors of i, the relation is $\mathcal{M} = -2m\,\tilde{V}(\mathbf{q})$, i.e.

$$\tilde{V}(\mathbf{q}) = -\frac{1}{2m}\,\mathcal{M} = \frac{e}{2m}\,\bar{u}(p')\,\Gamma^\mu\,u(p)\,\tilde{A}^{\rm cl}_\mu(\mathbf{q}). \tag{20.21}$$

For a purely magnetic field $\tilde{A}^{\rm cl}_0 = 0$, and the spatial component carries $\tilde{A}^{\rm cl}_i = -\tilde{A}^{{\rm cl}\,i}$, so $\bar{u}'\Gamma^\mu u\,\tilde{A}^{\rm cl}_\mu = -\bar{u}'\Gamma^i u\,\tilde{A}^{{\rm cl}\,i}$ and

$$\tilde{V}(\mathbf{q}) = -\frac{e}{2m}\,\bar{u}(p')\,\Gamma^i\,u(p)\,\tilde{A}^{{\rm cl}\,i}(\mathbf{q}). \tag{20.22}$$

Nonrelativistic reduction of the spatial current. We need $\bar{u}(p')\Gamma^i u(p)$ to leading nontrivial order in the small momenta. Use the standard-representation spinors of Chapter 11 normalized to $\bar{u}u = 2m$,

$$u(p) = \sqrt{E+m}\begin{pmatrix}\xi\\ \dfrac{\boldsymbol{\sigma}\cdot\mathbf{p}}{E+m}\,\xi\end{pmatrix}, \qquad \bar{u}(p') = \sqrt{E'+m}\Big(\xi'^\dagger,\ -\frac{\boldsymbol{\sigma}\cdot\mathbf{p}'}{E'+m}\,\xi'^\dagger\Big), \tag{20.23}$$

and $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{smallmatrix}\right)$. To leading order $\sqrt{(E+m)(E'+m)} \to 2m$ and $E+m \to 2m$ in denominators, so

$$\bar{u}(p')\,\gamma^i\,u(p) = \sqrt{(E'+m)(E+m)}\,\xi'^{\dagger}\left[\frac{\sigma^i(\boldsymbol{\sigma}\cdot\mathbf{p})}{E+m} + \frac{(\boldsymbol{\sigma}\cdot\mathbf{p}')\sigma^i}{E'+m}\right]\xi$$
$$\simeq \xi'^{\dagger}\left[\sigma^i\sigma^j p^j + \sigma^j\sigma^i p'^j\right]\xi. \tag{20.24}$$

With $\sigma^i\sigma^j = \delta^{ij} + i\epsilon^{ijk}\sigma^k$ and $\sigma^j\sigma^i = \delta^{ij} - i\epsilon^{ijk}\sigma^k$,

$$\bar{u}(p')\,\gamma^i\,u(p) \simeq \xi'^{\dagger}\Big[(\mathbf{p}+\mathbf{p}')^i - i\,\epsilon^{ijk}q^j\sigma^k\Big]\xi = \xi'^{\dagger}\Big[(\mathbf{p}+\mathbf{p}')^i - i\,(\mathbf{q}\times\boldsymbol{\sigma})^i\Big]\xi, \tag{20.25}$$

where $q^j = (p'-p)^j$. The first term is the convective current (no spin); the second, $-i(\mathbf{q}\times\boldsymbol{\sigma})$, is the spin current carried by γ^i. For the tensor term we need $\bar{u}'\sigma^{ij}u$ at leading order. With $\sigma^{ij} = \frac{1}{2}\epsilon^{ijk}\Sigma^k$ and $\bar{u}'\Sigma^k u \simeq 2m\,\xi'^{\dagger}\sigma^k\xi$,

$$\bar{u}(p')\,\sigma^{ij}\,u(p) \simeq m\,\epsilon^{ijk}\,\xi'^{\dagger}\sigma^k\xi, \qquad \bar{u}(p')\,\frac{i\sigma^{ij}q_j}{m}\,u(p) \simeq \frac{i}{m}\,q_j\,m\,\epsilon^{ijk}\xi'^{\dagger}\sigma^k\xi = -i\,\xi'^{\dagger}(\mathbf{q}\times\boldsymbol{\sigma})^i\xi, \tag{20.26}$$

using $q_j = -q^j$ and $\epsilon^{ijk}q^j\sigma^k = (\mathbf{q}\times\boldsymbol{\sigma})^i$. (At $q^0 = 0$ only $\sigma^{ij}q_j$ contributes to $\sigma^{i\nu}q_\nu$.)

Assembling the magnetic current. The spatial vertex Eq. (20.1) is $\Gamma^i = \gamma^i F_1 + (i\sigma^{i\nu}q_\nu/m)F_2$. Its spin (magnetic) part collects the $(\mathbf{q}\times\boldsymbol{\sigma})$ structures from Eqs. (20.25) and (20.26):

$$\bar{u}(p')\,\Gamma^i\,u(p)\Big|_{\text{spin}} = -i\,F_1\,\xi'^{\dagger}(\mathbf{q}\times\boldsymbol{\sigma})^i\xi - i\,F_2\,\xi'^{\dagger}(\mathbf{q}\times\boldsymbol{\sigma})^i\xi = -i\,(F_1+F_2)\,\xi'^{\dagger}(\mathbf{q}\times\boldsymbol{\sigma})^i\xi. \tag{20.27}$$

This is the crux: because the Gordon spin term of γ^i enters with the *same* coefficient $1/m$ as the explicit F_2 tensor term (the point of the The on-shell electromagnetic vertex and what its two form factors mean notation watch), the two add as $F_1 + F_2$. The convective part $(\mathbf{p}+\mathbf{p}')^i F_1$ does not contribute to the magnetic moment and we drop it; it reconstructs the orbital $(\mathbf{p} - e\mathbf{A})^2/2m$ of the Pauli equation.

Matching to $-\boldsymbol{\mu}\cdot\mathbf{B}$. Insert Eq. (20.27) into Eq. (20.22):

$$\tilde{V}_{\text{mag}}(\mathbf{q}) = -\frac{e}{2m}\,\tilde{A}^{\text{cl}\,i}(\mathbf{q})\,[-i(F_1+F_2)]\xi'^{\dagger}(\mathbf{q}\times\boldsymbol{\sigma})^i\xi = \frac{ie}{2m}(F_1+F_2)\,\xi'^{\dagger}[\tilde{\mathbf{A}}^{\text{cl}}\cdot(\mathbf{q}\times\boldsymbol{\sigma})]\xi. \tag{20.28}$$

Use the scalar triple product and the Fourier image of $\mathbf{B} = \nabla\times\mathbf{A}$, namely $\tilde{\mathbf{B}}(\mathbf{q}) = i\,\mathbf{q}\times\tilde{\mathbf{A}}^{\text{cl}}(\mathbf{q})$:

$$\tilde{\mathbf{A}}^{\text{cl}}\cdot(\mathbf{q}\times\boldsymbol{\sigma}) = \boldsymbol{\sigma}\cdot(\tilde{\mathbf{A}}^{\text{cl}}\times\mathbf{q}) = -\boldsymbol{\sigma}\cdot(\mathbf{q}\times\tilde{\mathbf{A}}^{\text{cl}}) = -\boldsymbol{\sigma}\cdot(-i\,\tilde{\mathbf{B}}) = i\,\boldsymbol{\sigma}\cdot\tilde{\mathbf{B}}(\mathbf{q}). \tag{20.29}$$

Therefore

$$\tilde{V}_{\text{mag}}(\mathbf{q}) = \frac{ie}{2m}(F_1+F_2)\,(i)\,\xi'^{\dagger}\boldsymbol{\sigma}\cdot\tilde{\mathbf{B}}\,\xi = -\frac{e}{2m}(F_1+F_2)\,\xi'^{\dagger}\boldsymbol{\sigma}\cdot\tilde{\mathbf{B}}(\mathbf{q})\,\xi. \tag{20.30}$$

Transforming back to position space (the form factors are evaluated at $q^2 \to 0$ for a slowly varying field) and writing $\mathbf{S} = \boldsymbol{\sigma}/2$,

$$V_{\text{mag}}(\mathbf{x}) = -\frac{e}{2m}\left[F_1(0)+F_2(0)\right]\boldsymbol{\sigma}\cdot\mathbf{B}(\mathbf{x}) = -\frac{e}{m}\left[F_1(0)+F_2(0)\right]\mathbf{S}\cdot\mathbf{B}(\mathbf{x}). \quad (20.31)$$

Comparing with $-\boldsymbol{\mu}\cdot\mathbf{B}$ and $\boldsymbol{\mu} = g\,(e/2m)\,\mathbf{S}$ gives $g\,(e/2m) = (e/m)[F_1(0)+F_2(0)]$, that is,

$$\boxed{g = 2\left[F_1(0)+F_2(0)\right]}, \quad (20.32)$$

which is Eq. (20.19) and completes the proof of Theorem 20.1. The check on the algebra is immediate: at tree level $F_1 = 1$, $F_2 = 0$ reproduce Eq. (20.31) $= -(e/m)\mathbf{S}\cdot\mathbf{B}$, identical to the Pauli-equation result Eq. (20.17), and hence $g = 2$.

The two derivations agree where they must — both give $g = 2$ for the bare vertex — and the matrix-element computation extends the result to the dressed vertex, isolating the new physics in F_2. The combination $F_1(0)+F_2(0)$ governing the moment is exactly what the Gordon split Eq. (20.4) anticipated: the magnetic coupling is carried by the tensor structure $i\sigma^{\mu\nu}q_\nu/m$, into which both form factors feed. With $F_1(0) = 1$ fixed by the Ward–Takahashi identity, the entire deviation of g from the Dirac value lives in the Pauli form factor:

$$a_e = \frac{g-2}{2} = F_1(0)+F_2(0)-1 = F_2(0). \quad (20.33)$$

Physical Insight

Why does the Dirac form factor contribute to the magnetic moment at all, when one might expect only the manifestly magnetic Pauli term to matter? Because γ^μ is not purely a charge coupling. The Gordon decomposition Eq. (20.3) shows that even the minimal coupling $\bar\psi\gamma^\mu\psi A_\mu$ contains, hidden inside it, a spin–field term $i\sigma^{\mu\nu}q_\nu/m$ — this is precisely the term that produced the Pauli equation and $g = 2$. The electron's leading magnetic moment is therefore a consequence of the minimal coupling of a *spinor*, not an independent input; a spinless particle, whose vertex has no γ^μ, has no magnetic moment from minimal coupling. The Pauli form factor F_2 measures the *excess* over this built-in Dirac moment, the genuinely anomalous part that only the radiative corrections supply.

20.1.4 The tree value, the radiative target, and the next step

We have turned the abstract renormalized vertex of Chapter 19 into a laboratory number. The bookkeeping is now sharp. At tree level the vertex is the bare $-ie\gamma^\mu$ of Eq. (17.10), so $F_1^{\text{tree}} = 1$ and $F_2^{\text{tree}} = 0$; by Eq. (20.19) the gyromagnetic ratio

is exactly $g = 2$, Dirac's prediction, and the anomalous moment is exactly zero. Any nonzero a_e is therefore a purely quantum effect, the imprint of the virtual cloud the electron carries. By Eq. (20.33) that effect is captured, completely and unambiguously, by a single number: the Pauli form factor at zero momentum transfer,

$$a_e = F_2(0). \tag{20.34}$$

The remaining computational problem of QED is thus precisely stated. We do not need the full one-loop vertex, nor the renormalization of F_1, nor the infrared subtleties that attend the elastic vertex: we need only the coefficient of the tensor structure $i\sigma^{\mu\nu}q_\nu/m$ in the one-loop vertex, evaluated at $q^2 = 0$. Everything else in the vertex is irrelevant to the magnetic moment, and we will see in the next section that this single coefficient is, by good fortune, both ultraviolet- and infrared-finite, so that $F_2(0)$ is a clean prediction requiring neither counterterm nor regulator. The famous answer, $a_e = \alpha/2\pi$, is the value of this one coefficient.

The next section (Section 20.2) sets up that extraction: it takes the one-loop vertex integral imported from Chapter 19 and Chapter D, Feynman-combines its three propagators, shifts the loop momentum, and performs the on-shell Dirac algebra that projects out the coefficient of $i\sigma^{\mu\nu}q_\nu$, reducing F_2 to a single scalar integral over Feynman parameters. That integral, evaluated at $q^2 = 0$, is Schwinger's number.

20.2 Setting Up Schwinger's Computation

The previous section reduced a laboratory number to a single field-theoretic quantity. The renormalized on-shell vertex of Chapter 19 carries exactly two form factors,

$$\bar{u}(p')\,\Gamma^{\mu}(p',p)\,u(p) = \bar{u}(p')\left[\gamma^{\mu}\,F_1(q^2) + \frac{i\,\sigma^{\mu\nu}q_\nu}{m}\,F_2(q^2)\right]u(p), \qquad q \equiv p' - p, \tag{20.35}$$

and we proved there (Section 20.1.3) that the electron's anomalous magnetic moment is nothing but the second of them at zero momentum transfer,

$$a_e \equiv \frac{g-2}{2} = F_2(0). \tag{20.36}$$

The Dirac value $g = 2$ is the statement $F_2 \equiv 0$ at tree level; every bit of the anomaly lives in the radiative part of F_2. The task of the present section is purely mechanical and entirely self-contained: take the one-loop vertex integral handed to us by the Feynman rules—the same integral written down once and for all in Eq. (D.15)—and squeeze out of it the coefficient of the magnetic structure $i\sigma^{\mu\nu}q_\nu/m$. We do this in four moves, each of which is a worked computation rather than a quotation: Feynman-combine the three propagator denominators

and complete the square to expose the effective mass Δ; shift the loop momentum to make the denominator even; reduce the Dirac numerator using the on-shell equations for the external spinors, discarding the pieces that integrate to zero and symmetrizing the rest; and finally separate the surviving numerator into a γ^μ part, which feeds F_1, and a $(p'+p)^\mu$ part, which the Gordon decomposition of Chapter 11 converts into precisely the $\sigma^{\mu\nu}q_\nu$ structure we are after. The output is a closed Feynman-parameter integrand for $F_2(q^2)$ over a convergent loop. The next section does the integral.

Notation Watch

Throughout this chapter $\sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ is the book's antisymmetric tensor (Eq. (C.11)), which for $\mu \neq \nu$ equals $\frac{i}{2}\gamma^\mu\gamma^\nu$—one half of the combination used in some texts. With this normalization the magnetic form factor sits at $i\sigma^{\mu\nu}q_\nu/m$ in Eq. (20.35) (a factor $1/m$, not $1/2m$), exactly so that the Gordon identity (11.85) and the identification (20.36) hold with no stray factors of two. The reader translating to a source with $\sigma^{\mu\nu}_{\text{there}} = 2\sigma^{\mu\nu}$ should read its $i\sigma^{\mu\nu}q_\nu/2m$ as our $i\sigma^{\mu\nu}q_\nu/m$; the physical content—and the number a_e—is of course convention-independent.

20.2.1 The one-loop vertex as a single three-propagator integral

We continue QED in Feynman gauge ($\xi = 1$), the standing default for explicit computation in this thread and the gauge used throughout Chapter 19. The one diagram that corrects the photon–electron vertex at order e^2 is drawn in Fig. 20.2: the external photon of momentum q attaches to the electron line at a point, and a virtual photon of momentum ℓ is exchanged across the vertex between the two electron legs, which therefore carry $p-\ell$ and $p'-\ell$. Its value is the amputated integral (D.15), which we reproduce here with all couplings and i-factors explicit (suppressing the dimensional-regularization scale μ^ε, which we restore only for the ultraviolet-divergent F_1 part and which never touches the finite F_2):

$$\bar{u}(p')\big[-ie\,\delta\Gamma^\mu\big]u(p) = \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\bar{u}(p')(-ie\gamma^\alpha)\frac{i(\not{p}'-\not{\ell}+m)}{(p'-\ell)^2-m^2+i\epsilon}(-ie\gamma^\mu)\frac{i(\not{p}-\not{\ell}+m)}{(p-\ell)^2-m^2+i\epsilon}(- \tag{20.37}$$

The left-hand side records our bookkeeping: the *full* amputated vertex is $-ie\,\Gamma^\mu$ with $\Gamma^\mu = \gamma^\mu + \delta\Gamma^\mu$, so that the order-$e^2$ correction to the vertex *function* is $\delta\Gamma^\mu$, which by (20.35) carries $F_1 = 1 + \delta F_1 + \dots$ and $F_2 = \delta F_2 + \dots$ with $\delta F_{1,2} = O(e^2)$.

Collect the constants in (20.37). The three vertices give $(-ie)^3 = ie^3$; the two fermion propagators and the photon propagator give $i\cdot i\cdot(-i) = i$; the metric contraction folds the outer gammas together, $g_{\alpha\beta}\gamma^\alpha(\cdots)\gamma^\beta = \gamma^\alpha(\cdots)\gamma_\alpha$. Thus

$-ie\,\delta\Gamma^\mu = (ie^3)(i)\int\cdots = -e^3\int\cdots$, and dividing by $-ie$,

$$\bar u(p')\,\delta\Gamma^\mu\,u(p) = -ie^2\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\bar u(p')\,\mathcal{N}^\mu\,u(p)}{D_1\,D_2\,D_3}, \tag{20.38}$$

with the three denominators

$$D_1 = (p'-\ell)^2 - m^2 + i\epsilon, \qquad D_2 = (p-\ell)^2 - m^2 + i\epsilon, \qquad D_3 = \ell^2 + i\epsilon, \tag{20.39}$$

and the numerator

$$\mathcal{N}^\mu = \gamma^\alpha(\not p' - \not\ell + m)\,\gamma^\mu\,(\not p - \not\ell + m)\,\gamma_\alpha. \tag{20.40}$$

Here $e>0$ and $\alpha = e^2/4\pi$; the prefactor $-ie^2$ correctly makes $\delta\Gamma^\mu$ an order-e^2 (order-α) object, the radiative correction to the tree vertex γ^μ. Everything that follows is the extraction of the $i\sigma^{\mu\nu}q_\nu$ piece of (20.38). Throughout, the external spinors are on-shell, $p^2 = p'^2 = m^2$, and we abbreviate $\bar u' \equiv \bar u(p')$, $u \equiv u(p)$.

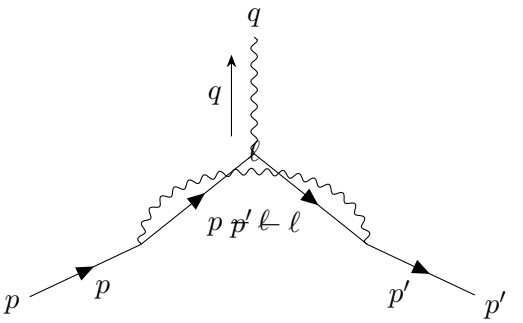


Figure 20.2: The one-loop QED vertex correction (20.38). An external photon of momentum $q = p' - p$ couples to the electron at the top vertex; a virtual photon of momentum ℓ (the loop momentum) is exchanged between the two electron legs, which therefore carry $p-\ell$ and $p'-\ell$. The three internal lines supply the three denominators D_1, D_2, D_3 of (20.39); the open fermion line, read against the arrows from $u(p)$ to $\bar u(p')$, supplies the Dirac string (20.40).

20.2.2 Feynman-combining the three denominators

The three factors D_1, D_2, D_3 have different dependence on the loop momentum, which obstructs the rotationally symmetric integration that the master formula of Section D.4 requires. The cure is the Feynman parametrization proved in Theorem D.11; for three unit-power denominators (D.25) reads, with $\Gamma(3) = 2!$,

$$\frac{1}{D_1 D_2 D_3} = \int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{2}{\big[xD_1 + yD_2 + zD_3\big]^3}, \tag{20.41}$$

the integration running over the triangle $x, y, z \geq 0$, $x + y + z = 1$. We have attached the parameter x to the leg carrying p', y to the leg carrying p, and z to the internal photon; the choice fixes which external invariants appear in the result, and this one yields the symmetric, standard answer.

Form the combined denominator. Using $p'^2 = p^2 = m^2$,

$$\begin{aligned} xD_1 + yD_2 + zD_3 &= x\big[(p'-\ell)^2 - m^2\big] + y\big[(p-\ell)^2 - m^2\big] + z\,\ell^2 + i\epsilon \\ &= x\big[\ell^2 - 2\,\ell\cdot p'\big] + y\big[\ell^2 - 2\,\ell\cdot p\big] + z\,\ell^2 + i\epsilon \\ &= \ell^2 - 2\,\ell\cdot(xp' + yp) + i\epsilon, \end{aligned} \tag{20.42}$$

where in the last line $x + y + z = 1$ collapsed the three ℓ^2 terms into one. This is a quadratic in ℓ with no constant term yet; complete the square by defining the shifted loop momentum

$$\ell \;\longrightarrow\; \ell + (xp' + yp), \tag{20.43}$$

under which $\ell^2 - 2\,\ell\cdot(xp' + yp) \to \ell^2 - (xp' + yp)^2$. The shift is a mere translation of an (analytically continued, hence translation-invariant) loop integral, so the measure $\mathrm{d}^d\ell$ is unchanged. The combined denominator becomes $\ell^2 - \Delta + i\epsilon$ with the effective mass-squared

$$\Delta = (xp' + yp)^2 = x^2 p'^2 + y^2 p^2 + 2xy\,p'\cdot p = (x^2 + y^2)m^2 + 2xy\,p'\cdot p. \tag{20.44}$$

Trade $p'\cdot p$ for the momentum transfer: from $q = p' - p$, $q^2 = p'^2 - 2p'\cdot p + p^2 = 2m^2 - 2p'\cdot p$, hence $p'\cdot p = m^2 - \tfrac{1}{2}q^2$. Substituting,

$$\boxed{\Delta = (x^2 + y^2 + 2xy)m^2 - xy\,q^2 = (x+y)^2 m^2 - xy\,q^2 = (1-z)^2 m^2 - xy\,q^2} \tag{20.45}$$

using $x + y = 1 - z$ in the last step. For spacelike or small momentum transfer ($q^2 \leq 0$, and in particular $q^2 \to 0$, which is all we need for a_e) the two terms add with the same sign and $\Delta > 0$ strictly on the interior of the triangle, so the loop integral is manifestly free of pinch singularities. The photon's masslessness shows up only as the boundary $z \to 1$, where $\Delta \to 0$; that endpoint is the infrared sensitivity of the vertex, which afflicts F_1 and is regulated and cancelled as discussed in Chapter 19, but which—as we will see—is harmless for F_2 because the F_2 integrand carries a compensating factor of $(1 - z)$.

20.2.3 The numerator: symmetric integration and Dirac contraction

Under the same shift (20.43) the numerator (20.40) must be re-expressed in the new ℓ. The two slashed momenta inside become

$$\not{p}' - \not{\ell} \;\to\; \not{a} - \not{\ell}, \qquad \not{p} - \not{\ell} \;\to\; \not{b} - \not{\ell}, \tag{20.46}$$

with the ℓ-independent slashes

$$\bar{a}^\nu \equiv (1-x)p'^\nu - y\,p^\nu = (y+z)p'^\nu - y\,p^\nu, \qquad \bar{b}^\nu \equiv (1-y)p^\nu - x\,p'^\nu = (x+z)p^\nu - x\,p'^\nu, \tag{20.47}$$

where $1-x = y+z$ and $1-y = x+z$ were used. The numerator is therefore

$$\mathcal{N}^\mu = \gamma^\alpha(\not{a} - \not{\ell} + m)\gamma^\mu(\not{\bar{b}} - \not{\ell} + m)\gamma_\alpha. \tag{20.48}$$

Expanding the two binomials produces terms with two, one, and zero powers of the new loop momentum ℓ. Inside the (now Lorentz-symmetric) integral over ℓ, the rules of Theorem D.24 apply: by (D.90) any term odd in ℓ integrates to zero, and by (D.92) a term $\ell^\rho\ell^\sigma$ may be replaced by $(\ell^2/d)\,\eta^{\rho\sigma}$. We sort the numerator accordingly.

The ℓ-quadratic piece. The only even-in-ℓ term with two loop momenta is $\gamma^\alpha(-\not{\ell})\gamma^\mu(-\not{\ell})\gamma_\alpha = \gamma^\alpha\not{\ell}\,\gamma^\mu\not{\ell}\,\gamma_\alpha$. Replacing $\ell^\rho\ell^\sigma \to (\ell^2/d)\eta^{\rho\sigma}$ contracts the two slashes through the central γ^μ:

$$\gamma^\alpha\not{\ell}\,\gamma^\mu\not{\ell}\,\gamma_\alpha \;\longrightarrow\; \frac{\ell^2}{d}\,\gamma^\alpha\gamma^\rho\gamma^\mu\gamma_\rho\gamma_\alpha = \frac{\ell^2}{d}\,(2-d)^2\,\gamma^\mu, \tag{20.49}$$

where we used the d-dimensional contraction identity (C.188), $\gamma^\rho\gamma^\mu\gamma_\rho = (2-d)\gamma^\mu$, twice: once on the inner pair ρ and once on the resulting α pair. The result is proportional to γ^μ alone. It is ultraviolet-divergent—the $\Gamma(n-d/2)$ pole of the loop integral hits this ℓ^2 numerator through (D.88)— and it contributes *only* to F_1, never to the magnetic form factor. We therefore set it aside; its proper treatment (the divergence that renormalizes the charge, and the infrared piece that cancels against soft bremsstrahlung) belongs to the F_1 programme of Chapter 19. *From here on we keep only the ℓ-independent piece of the numerator, and in it we may set $d = 4$, because F_2 is ultraviolet-finite.*

The ℓ-independent piece. Dropping the single-ℓ terms (odd, hence zero) and the quadratic term (just handled), the surviving numerator is, at $d = 4$,

$$\mathcal{N}_0^\mu = \gamma^\alpha(\not{a} + m)\gamma^\mu(\not{\bar{b}} + m)\gamma_\alpha. \tag{20.50}$$

Expand the middle product into four pieces and contract the outer $\gamma^\alpha\cdots\gamma_\alpha$ pair on each, using the four-dimensional limits of Theorem C.61—namely $\gamma^\alpha\gamma^\mu\gamma_\alpha = -2\gamma^\mu$ from (C.188), $\gamma^\alpha\gamma^\rho\gamma^\mu\gamma_\alpha = 4\eta^{\rho\mu}\mathbf{1}$ from (C.189), and $\gamma^\alpha\gamma^\rho\gamma^\mu\gamma^\sigma\gamma_\alpha = -2\gamma^\sigma\gamma^\mu\gamma^\rho$ from (C.190):

$$\gamma^\alpha(\not{a}\,\gamma^\mu\not{\bar{b}})\gamma_\alpha = \bar{a}_\rho\bar{b}_\sigma\,\gamma^\alpha\gamma^\rho\gamma^\mu\gamma^\sigma\gamma_\alpha = -2\,\not{\bar{b}}\,\gamma^\mu\not{a}, \tag{20.51a}$$

$$m\,\gamma^\alpha(\not{a}\,\gamma^\mu)\gamma_\alpha = m\,\bar{a}_\rho\,\gamma^\alpha\gamma^\rho\gamma^\mu\gamma_\alpha = 4m\,\bar{a}^\mu\,\mathbf{1}, \tag{20.51b}$$

$$m\,\gamma^\alpha(\gamma^\mu\not{\bar{b}})\gamma_\alpha = m\,\bar{b}_\sigma\,\gamma^\alpha\gamma^\mu\gamma^\sigma\gamma_\alpha = 4m\,\bar{b}^\mu\,\mathbf{1}, \tag{20.51c}$$

$$m^2\,\gamma^\alpha\gamma^\mu\gamma_\alpha = -2m^2\,\gamma^\mu. \tag{20.51d}$$

Summing,

$$\mathcal{N}_0^\mu = -2\,\bar{\not{b}}\,\gamma^\mu\not{\bar{a}} + 4m(\bar{a}^\mu + \bar{b}^\mu)\mathbf{1} - 2m^2\gamma^\mu. \tag{20.52}$$

The last term is pure γ^μ (it feeds F_1). The middle term is already an explicit momentum vector, and combining (20.47),

$$\bar{a}^\mu + \bar{b}^\mu = (1-2x)p'^\mu + (1-2y)p^\mu = z\,(p'+p)^\mu + (y-x)\,q^\mu, \tag{20.53}$$

where we resolved $p'^\mu = \tfrac{1}{2}[(p'+p)^\mu + q^\mu]$, $p^\mu = \tfrac{1}{2}[(p'+p)^\mu - q^\mu]$ and used $1-x-y=z$. The first piece of (20.52) still hides γ^μ, $(p'+p)^\mu$, and q^μ structures; teasing them apart requires the on-shell Dirac equations, which is the content of the worked computation below.

20.2.4 Reducing the numerator on shell

Worked Example 20.3 (The on-shell reduction of the vertex numerator)**.** We evaluate $\bar{u}'\,\mathcal{N}_0^\mu\,u$ completely, sandwiched between the on-shell amplitudes, using only the Dirac equations

$$\not{p}\,u(p) = m\,u(p), \qquad \bar{u}(p')\,\not{p}' = m\,\bar{u}(p'), \tag{20.54}$$

(the second from the adjoint, Eq. (11.50) and the discussion following it) and the Clifford algebra. We will write the answer in the basis $\{\bar{u}'\gamma^\mu u,\ (p'+p)^\mu\,\bar{u}'u,\ q^\mu\,\bar{u}'u\}$, abbreviating the scalar and vector bilinears as

$$S \equiv \bar{u}(p')u(p), \qquad V^\mu \equiv \bar{u}(p')\gamma^\mu u(p). \tag{20.55}$$

Note that $(p'+p)^\mu\,S = \bar{u}'(p'+p)^\mu u$ and $q^\mu S = \bar{u}'q^\mu u$, since the momenta are c-numbers.

Step 1: the four elementary bilinears. The first term of (20.52) is built from $\bar{\not{b}}\,\gamma^\mu\not{\bar{a}}$ with $\not{\bar{a}} = (y+z)\not{p}' - y\not{p}$ and $\bar{\not{b}} = (x+z)\not{p} - x\not{p}'$. Expanding the product gives four monomials of the form $\not{r}\,\gamma^\mu\not{s}$ with $r,s \in \{p,p'\}$. We reduce each, moving $\not{p}$ to the right onto u and $\not{p}'$ to the left onto $\bar{u}'$ via the anticommutator $\not{r}\,\gamma^\mu = 2r^\mu - \gamma^\mu\not{r}$:

$$\bar{u}'\,\not{p}'\gamma^\mu\not{p}\,u = m^2\,V^\mu, \tag{20.56}$$

$$\bar{u}'\,\not{p}\,\gamma^\mu\not{p}'\,u = 2m(p+p')^\mu S - (3m^2 - q^2)\,V^\mu, \tag{20.57}$$

$$\bar{u}'\,\not{p}\,\gamma^\mu\not{p}\,u = 2m\,p^\mu S - m^2\,V^\mu, \tag{20.58}$$

$$\bar{u}'\,\not{p}'\gamma^\mu\not{p}'\,u = 2m\,p'^\mu S - m^2\,V^\mu. \tag{20.59}$$

Equation (20.56) is immediate: $\not{p}'$ hits $\bar{u}'$ and $\not{p}$ hits u, each returning a factor m. For (20.58), $\bar{u}'\not{p}\,\gamma^\mu\not{p}\,u = 2p^\mu\bar{u}'\not{p}\,u - \bar{u}'\gamma^\mu\not{p}\not{p}\,u = 2m\,p^\mu S - m^2V^\mu$, using $\not{p}\,u = mu$

and $\not{p}\not{p} = p^2 = m^2$. Equation (20.59) is its mirror: $\bar{u}'\not{p}'\gamma^\mu\not{p}'\,u = m\,\bar{u}'\gamma^\mu\not{p}'\,u = m(2p'^\mu S - mV^\mu)$, where the inner reduction $\bar{u}'\gamma^\mu\not{p}'\,u = 2p'^\mu S - mV^\mu$ used $\gamma^\mu\not{p}' = 2p'^\mu - \not{p}'\gamma^\mu$ and $\bar{u}'\not{p}' = m\bar{u}'$. The one nontrivial case is (20.57), where neither slash sits next to the spinor it acts on:

$$\begin{aligned}
\bar{u}'\,\not{p}\,\gamma^\mu\not{p}'\,u &= 2p^\mu\,\bar{u}'\not{p}'\,u - \bar{u}'\gamma^\mu\,\not{p}\not{p}'\,u = 2m\,p^\mu S - \bar{u}'\gamma^\mu\,\not{p}\not{p}'\,u,\\
\not{p}\not{p}'\,u &= (2p\cdot p' - \not{p}'\not{p})u = 2(p\cdot p')u - m\,\not{p}'\,u,\\
\bar{u}'\gamma^\mu\not{p}'\,u &= 2p'^\mu S - mV^\mu,\\
\Longrightarrow\quad \bar{u}'\,\not{p}\,\gamma^\mu\not{p}'\,u &= 2m\,p^\mu S - 2(p\cdot p')V^\mu + m(2p'^\mu S - mV^\mu)\\
&= 2m(p+p')^\mu S - (2p\cdot p' + m^2)V^\mu = 2m(p+p')^\mu S - (3m^2 - q^2)V^\mu,
\end{aligned} \tag{20.60}$$

using $2p\cdot p' = 2m^2 - q^2$ in the last equality. This proves (20.57).

Step 2: assemble $-2\,\bar{u}'\vec{b}\gamma^\mu\vec{a}\,u$. Multiply out $\vec{b}\,\gamma^\mu\vec{a} = [(x+z)\not{p} - x\not{p}']\,\gamma^\mu\,[(y+z)\not{p}' - y\not{p}]$:

$$\vec{b}\,\gamma^\mu\vec{a} = (x+z)(y+z)\,\not{p}\gamma^\mu\not{p}' - (x+z)y\,\not{p}\gamma^\mu\not{p} - x(y+z)\,\not{p}'\gamma^\mu\not{p}' + xy\,\not{p}'\gamma^\mu\not{p}. \tag{20.61}$$

Insert (20.56)–(20.59) and collect the S-weighted momentum vectors separately from V^μ. The momentum part is

$$\big[\,\bar{u}'\vec{b}\gamma^\mu\vec{a}\,u\,\big]_S = 2m\,S\Big[(x+z)(y+z)(p+p')^\mu - (x+z)y\,p^\mu - x(y+z)\,p'^\mu\Big]. \tag{20.62}$$

Resolve onto $(p'+p)^\mu$ and q^μ with $p^\mu = \tfrac{1}{2}[(p'+p) - q]^\mu$, $p'^\mu = \tfrac{1}{2}[(p'+p) + q]^\mu$. The coefficient of $(p'+p)^\mu$ inside the bracket is

$$(x+z)(y+z) - \tfrac{1}{2}(x+z)y - \tfrac{1}{2}x(y+z) = \tfrac{1}{2}xz + \tfrac{1}{2}yz + z^2 = \tfrac{1}{2}z(x+y) + z^2 = \tfrac{1}{2}z(1+z), \tag{20.63}$$

where $x + y = 1 - z$ was used. The coefficient of q^μ inside the bracket is $\tfrac{1}{2}(x+z)y - \tfrac{1}{2}x(y+z) = \tfrac{1}{2}z(y-x)$. Hence

$$\big[\,\bar{u}'\vec{b}\gamma^\mu\vec{a}\,u\,\big]_S = m\,S\Big[z(1+z)\,(p'+p)^\mu + z(y-x)\,q^\mu\Big]. \tag{20.64}$$

Step 3: full numerator. The complete ℓ-independent numerator $\bar{u}'\mathcal{N}_0^\mu\,u$ is -2 times (20.64) (plus the V^μ remainder, which we track only schematically since it feeds F_1), plus the explicit middle term $4m(\bar{a}^\mu + \bar{b}^\mu)S$ of (20.52) evaluated with (20.53), plus the $-2m^2V^\mu$. Collecting the coefficient of $(p'+p)^\mu S$:

$$\big[\bar{u}'\mathcal{N}_0^\mu\,u\big]_{(p'+p)} = \Big[-2\cdot m\,z(1+z) + 4m\,z\Big](p'+p)^\mu S = 2m\,z(1-z)\,(p'+p)^\mu S, \tag{20.65}$$

and the coefficient of $q^\mu S$:

$$\big[\bar{u}'\mathcal{N}_0^\mu\,u\big]_{q} = \Big[-2\cdot m\,z(y-x) + 4m\,(y-x)\Big]q^\mu S = 2m\,(y-x)(2-z)\,q^\mu S. \tag{20.66}$$

The q^μ coefficient is *antisymmetric* under $x \leftrightarrow y$, whereas the rest of the integrand—the measure $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x+y+z-1)$, the loop factor, and $\Delta = (1-z)^2 m^2 - xy\,q^2$ of (20.45)—is *symmetric* under $x \leftrightarrow y$. Relabelling $x \leftrightarrow y$ in (20.66) therefore flips its sign while leaving everything it multiplies unchanged, so the q^μ term integrates to zero:

$$\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,(y-x)(2-z)\,f\big(z,\,xy\big) = 0. \tag{20.67}$$

This is exactly as it must be. A surviving q^μ form factor would violate current conservation, $q_\mu\,\bar{u}'\Gamma^\mu u = 0$; its vanishing here is the integrand-level shadow of the Ward–Takahashi identity proved in Chapter 19. We are left with two structures: $V^\mu = \bar{u}'\gamma^\mu u$, feeding F_1, and the clean coefficient (20.65) of $(p'+p)^\mu$, which is the seed of F_2.

The worked computation has reduced the magnetically relevant part of the numerator to a single, strikingly simple object: the coefficient of $(p'+p)^\mu\,\bar{u}'u$ in $\bar{u}'\mathcal{N}_0^\mu\,u$ is

$$B(z) = 2m\,z(1-z), \tag{20.68}$$

independent of x and y except through the constraint, and vanishing at both endpoints $z=0$ and $z=1$. The vanishing at $z=1$ is the promised cancellation of the infrared boundary: it is this factor of $(1-z)$ that renders the F_2 integral convergent even though $\Delta \to 0$ there.

20.2.5 From $(p'+p)^\mu$ to $i\sigma^{\mu\nu}q_\nu$: the F_2 integrand

The vertex decomposition (20.35) is written in the $\gamma^\mu/\sigma^{\mu\nu}q_\nu$ basis, but the loop has handed us a $(p'+p)^\mu$ structure. The Gordon decomposition of Chapter 11, Theorem 11.15, is exactly the bridge. In the inverted form read off from its proof—there one shows $2m\,\bar{u}'\gamma^\mu u = \bar{u}'[(p'+p)^\mu + 2i\sigma^{\mu\nu}q_\nu]u$—it states

$$\bar{u}(p')\,(p'+p)^\mu\,u(p) = \bar{u}(p')\big[\,2m\,\gamma^\mu - 2i\,\sigma^{\mu\nu}q_\nu\,\big]u(p). \tag{20.69}$$

The convective term $2m\gamma^\mu$ adds to F_1; the spin term $-2i\sigma^{\mu\nu}q_\nu$ is the magnetic moment. Substituting (20.69) into the $(p'+p)^\mu$ part of the numerator, the σ structure picks up the factor $B(z)\cdot(-2i)$.

Now reassemble the loop integral, keeping only the magnetic piece. Starting from (20.38), Feynman-combining with (20.41), shifting, and inserting the $(p'+p)^\mu$ numerator $B(z)(p'+p)^\mu S \to B(z)(-2i)\bar{u}'\sigma^{\mu\nu}q_\nu u$,

$$\bar{u}'\,\delta\Gamma^\mu\big|_\sigma\,u = -ie^2\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x{+}y{+}z{-}1)\;2\,B(z)(-2i)\,\bar{u}'\sigma^{\mu\nu}q_\nu u\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^3}. \tag{20.70}$$

The remaining scalar loop integral is the master formula (D.77) of Theorem D.22 at $n = 3$; since the integrand is finite we evaluate it directly at $d = 4$,

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^3} = \frac{(-1)^3\,i}{(4\pi)^2}\,\frac{\Gamma(3-2)}{\Gamma(3)}\,\Delta^{-1} = \frac{-i}{16\pi^2}\cdot\frac{1}{2}\cdot\frac{1}{\Delta} = \frac{-i}{32\pi^2\,\Delta}. \tag{20.71}$$

Collect the constants in (20.70). The four numerical phases are the prefactor $-i$, the Feynman factor 2, the Gordon factor $-2i$, and the master-integral $-i$; their product is

$$(-i)\cdot 2\cdot(-2i)\cdot(-i) = (-2i)(-2i)(-i) = (-4)(-i) = 4i, \tag{20.72}$$

so the constant assembled with the loop factor $1/32\pi^2$ is $4i\,e^2/32\pi^2 = i\,e^2/8\pi^2$. Restoring the explicit $B(z)$ and $1/\Delta$ under the integral,

$$\bar u'\,\delta\Gamma^\mu\big|_\sigma\,u = \frac{i\,e^2}{8\pi^2}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x{+}y{+}z{-}1)\,\frac{B(z)}{\Delta}\,\bar u'\sigma^{\mu\nu}q_\nu u. \tag{20.73}$$

Now match to the form-factor definition: the magnetic part of $\delta\Gamma^\mu$ is, by (20.35), $(i\sigma^{\mu\nu}q_\nu/m)\,\delta F_2$, so equating the coefficients of the common bilinear $\bar u'\sigma^{\mu\nu}q_\nu u$,

$$\frac{i}{m}\,F_2(q^2) = \frac{i\,e^2}{8\pi^2}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(x{+}y{+}z{-}1)\,\frac{B(z)}{\Delta} \quad\Longrightarrow\quad F_2(q^2) = \frac{e^2 m}{8\pi^2}\int\frac{B(z)}{\Delta}\,. \tag{20.74}$$

(We write F_2 for δF_2, the tree value being zero.) Insert $B(z) = 2mz(1-z)$ from (20.68), so that $e^2 m\cdot 2m/8\pi^2 = e^2m^2/4\pi^2$, and trade e^2 for the fine-structure constant via $e^2 = 4\pi\alpha$, giving $e^2m^2/4\pi^2 = \alpha m^2/\pi$. We record the final, clean form:

$$\boxed{F_2(q^2) = \frac{\alpha}{\pi}\,m^2\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{z(1-z)}{(1-z)^2m^2 - xy\,q^2}} \tag{20.75}$$

with $\Delta = (1-z)^2m^2 - xy\,q^2$ from (20.45). This is the result of the section: an explicit, ultraviolet-finite, infrared-safe Feynman-parameter integrand for the magnetic form factor at one loop, valid for all q^2 in the physical region $q^2 \le 0$ (and by analytic continuation beyond).

Physical Insight

Every feature of (20.75) has a physical reading. The numerator factor $z(1-z)$ is the fingerprint of the magnetic coupling: the $(p'+p)^\mu$ "convective current" that Gordon splits off is the part of the electron's current that knows about spin precession, and the loop weights it by the photon parameter z and the fermion-pair parameter $(1-z)$. The denominator Δ is the off-shellness the virtual photon must overcome. The absence

of any $1/\varepsilon$ pole is the statement that the magnetic moment is a finite, unambiguous prediction—it needs no counterterm, no subtraction, no renormalization condition. The anomalous moment is calculable from first principles precisely because F_2 is finite: it is a pure quantum effect with no classical or divergent counterpart to absorb it.

The integrand (20.75) is now a triangle integral over three elementary Feynman parameters constrained to a simplex. In the next section we carry out those integrals—first the trivial one fixed by the delta function, then the remaining two over the simplex—take the static limit $q^2 \to 0$ that defines the magnetic moment, and watch the number $\alpha/2\pi$ emerge as the value of (20.75) at the origin.

20.3 Schwinger's Result: $a_e = \alpha/2\pi$

The setup section left the magnetic content of the one-loop vertex packaged in a single scalar object. After Feynman-combining the three propagators of the vertex correction, shifting the loop momentum to complete the square, and carrying out the on-shell Dirac algebra that projects onto the coefficient of $i\sigma^{\mu\nu}q_\nu/m$, every spinor structure and every Lorentz index has been disposed of. What remains is a number-valued integral over one loop momentum and three Feynman parameters, the Pauli form factor $F_2(q^2)$ (Eq. (20.76) below). The anomalous magnetic moment is its value at zero momentum transfer, $a_e = F_2(0)$. This section turns that integral into the number Schwinger first obtained in 1948 — the result of QED, picked up here from the renormalized vertex of Chapter 19. We do it in two movements: the loop integral, dispatched by the master formula of Chapter D, collapses F_2 to an elementary integral over the Feynman parameters; that integral, at $q^2 = 0$, is done in closed form and yields $\alpha/2\pi$. Along the way two structural facts must be established, because they are the reason the answer is unambiguous: the F_2 loop integral is ultraviolet-finite, so no counterterm enters it, and it is infrared-finite, so no photon-mass regulator survives in it. Both properties single out F_2 from its companion F_1.

We take as given the output of the setup section, the bare F_2 integrand in $d = 4 - 2\varepsilon$ dimensions,

$$F_2(q^2) \;=\; 2ie^2 \int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1) \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\;\frac{2\left[\,2m^2\,z(1-z)\,\right]}{(\ell^2-\Delta+i\epsilon)^3}\,, \tag{20.76}$$

in which ℓ is the shifted loop momentum, the leading factor 2 inside the integrand is the $\Gamma(3)$ produced by the Feynman combination of three denominators (Section D.2), the bracket $2m^2z(1-z)$ is the numerator coefficient of $i\sigma^{\mu\nu}q_\nu/m$ that survived the on-shell projection, and the effective mass-squared produced by completing the square is

$$\Delta \;=\; (1-z)^2m^2 - xy\,q^2\,, \tag{20.77}$$

a function of the Feynman parameters and the external invariants but not of ℓ. Here m is the electron mass, $q = p' - p$ is the momentum transfer, both external legs are on shell ($p^2 = p'^2 = m^2$), and $\alpha = e^2/4\pi$ as fixed in the chapter introduction. The single power q^2 in Eq. (20.77) is the only place the kinematics enters F_2; everything physical about the magnetic moment will follow from setting it to zero.

20.3.1 Ultraviolet finiteness: no counterterm is needed

Before evaluating Eq. (20.76) we should ask whether it even needs the dimensional regulator at all. The integrand has three propagator powers and *no* loop momentum in its numerator — the Dirac algebra of the setup left F_2 multiplying a numerator that is independent of ℓ. Counting powers of ℓ at large momentum, the measure $\mathrm{d}^d\ell$ supplies $d = 4$ and the denominator supplies -6, so the integrand falls as $|\ell|^{-2}$ and the integral converges in the ultraviolet. The superficial degree of divergence of the F_2 piece is $\omega = 4 - 6 = -2 < 0$: it is ultraviolet-convergent.

The master formula makes this quantitative and exhibits exactly where a divergence would have lived. Reading Eq. (20.76) as the canonical scalar integral Eq. (D.73) with $n = 3$, the master result Eq. (D.77) of Theorem D.22 gives

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^3} \;=\; \frac{(-1)^3\, i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(3 - \frac{d}{2}\right)}{\Gamma(3)}\,\Delta^{\,d/2-3}\,. \tag{20.78}$$

The entire ultraviolet content of any one-loop integral sits in the single factor $\Gamma(n - d/2)$ in the numerator of Eq. (D.77); poles in ε are poles of that Gamma function at $d = 4$. Here $n = 3$, so the relevant factor is

$$\Gamma\!\left(3 - \tfrac{d}{2}\right)\Big|_{d=4-2\varepsilon} = \Gamma(1 + \varepsilon) \xrightarrow[\varepsilon\to 0]{} \Gamma(1) = 1\,, \tag{20.79}$$

which is *finite* as $\varepsilon \to 0$. There is no $1/\varepsilon$ pole. By contrast, the Dirac form factor F_1 is built from integrals with a loop momentum squared in the numerator, equivalent to lowering the effective propagator power to $n = 2$; those carry $\Gamma(2 - d/2) = \Gamma(\varepsilon) \sim 1/\varepsilon$, the logarithmic ultraviolet divergence that the vertex renormalization of Chapter 19 must absorb. The contrast is the whole point. The renormalized F_1 requires a counterterm, fixed by the on-shell condition $F_1(0) = 1$ (equivalently the Ward–Takahashi identity $Z_1 = Z_2$ of Chapter 19); the bare F_2 requires nothing. Because no subtraction touches F_2, its value at $q^2 = 0$ is a scheme-independent prediction — a pure number times α, with no renormalization ambiguity. This is why the anomalous moment, alone among the one-loop QED outputs, is quoted to a dozen digits without reference to a renormalization scheme.

Physical Insight

The magnetic moment is finite for a structural reason: a magnetic coupling $\bar{\psi}\,\sigma^{\mu\nu}q_\nu\psi\,A_\mu$ carries one explicit power of the momentum transfer q, so it has mass dimension one higher than the minimal coupling $\bar{\psi}\gamma^\mu\psi\,A_\mu$. An operator of higher dimension is generated with a coefficient of *lower* dimension, hence by a *convergent* integral: the one extra power of q in the numerator is one fewer power of ℓ available to drive an ultraviolet divergence. The electron has no magnetic moment term in the classical Lagrangian to renormalize, so the loop-generated one must come out finite. Schwinger's number is the first quantum correction to a quantity that is exactly zero at tree level.

Because Eq. (20.78) is finite, we may set $d = 4$ from the outset. Then $(4\pi)^{d/2} = (4\pi)^2 = 16\pi^2$, $\Gamma(3) = 2! = 2$, $\Gamma(1) = 1$, and $\Delta^{d/2-3} = \Delta^{-1}$, so

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^3} \;=\; \frac{-\,i}{16\pi^2}\cdot\frac{1}{2}\cdot\frac{1}{\Delta} \;=\; \frac{-\,i}{32\pi^2\,\Delta}\,. \tag{20.80}$$

The factor i is the residue of the Wick rotation carried out once and for all in Section D.3; the sign $(-1)^3 = -1$ is the third power of the rotation factor per propagator (Eq. (D.74)). Both are genuinely present, as the caution following Theorem D.22 insists, and both are needed for F_2 to come out real and positive.

20.3.2 The loop integral done: the Pauli form factor

Insert Eq. (20.80) into the handed integrand Eq. (20.76). The loop integration is now finished; what is left is the integral over the three Feynman parameters of an explicit rational function. Collecting the constant factors,

$$F_2(q^2) = 2ie^2\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\;2\left[\,2m^2z(1-z)\,\right]\frac{-\,i}{32\pi^2\,\Delta} \tag{20.81}$$

$$= \underbrace{2i\cdot(-i)}_{=2}\;\frac{2\cdot 2\,e^2m^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{z(1-z)}{\Delta} \tag{20.82}$$

$$= \frac{e^2m^2}{4\pi^2}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{z(1-z)}{\Delta}\,. \tag{20.83}$$

The arithmetic of the prefactor is worth seeing explicitly: $2i\cdot(-i) = 2$ collapses the two imaginary units, leaving $2\cdot 2\cdot 2\,e^2m^2/32\pi^2 = 8e^2m^2/32\pi^2 = e^2m^2/4\pi^2$. Trading the bare charge for the fine-structure constant through $e^2 = 4\pi\alpha$ turns the prefactor into $e^2/4\pi^2 = 4\pi\alpha/4\pi^2 = \alpha/\pi$, and absorbing one factor of 2 back into the numerator to match the standard presentation,

$$\boxed{\;F_2(q^2) \;=\; \frac{\alpha}{2\pi}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(x+y+z-1)\,\frac{2m^2\,z(1-z)}{(1-z)^2m^2-xy\,q^2}\;} \tag{20.84}$$

where Δ from Eq. (20.77) has been written out in the denominator. Equation (20.84) is the complete one-loop Pauli form factor of the electron, valid for any momentum transfer below the pair-production threshold $q^2 < 4m^2$, where $\Delta > 0$ and the integrand is regular. It is manifestly finite, manifestly proportional to α, and manifestly a pure number once q^2/m^2 is fixed. The anomalous moment is the special value $q^2 = 0$, to which we now turn.

Theorem 20.4 (Schwinger's anomalous magnetic moment)**.** *The one-loop QED contribution to the anomalous magnetic moment of the electron is*

$$\boxed{a_e \;=\; F_2(0) \;=\; \frac{\alpha}{2\pi}} \tag{20.85}$$

independent of the electron mass and of any renormalization scheme. Numerically $a_e = 0.0011614\ldots$

The proof is the explicit evaluation of Eq. (20.84) at $q^2 = 0$, carried out in full in Theorem 20.5. We isolate it as a worked computation because it is the carried number of QED, and because the way the would-be infrared singularity disappears in the parameter integral is itself the lesson.

Worked Example 20.5 (Schwinger's number from the Feynman-parameter integral)**.** We evaluate $F_2(0)$ from Eq. (20.84), step by step.

Step 1: take $q^2 \to 0$ in the integrand. Setting $q^2 = 0$ in Eq. (20.77) removes the $xy\,q^2$ term, so $\Delta \to (1-z)^2 m^2$. The integrand becomes independent of x and y:

$$\frac{2m^2 z(1-z)}{(1-z)^2 m^2} \;=\; \frac{2z(1-z)}{(1-z)^2} \;=\; \frac{2z}{1-z}\,. \tag{20.86}$$

The electron mass has dropped out entirely, as Theorem 20.4 promised: a_e is dimensionless and cannot depend on m, the only scale in the problem.

Step 2: reduce the simplex integral. The three parameters are tied by the delta function to the triangle $x + y + z = 1$, $x, y, z \geq 0$. Since the integrand (20.86) depends on z alone, the x and y integrations only measure the length of the segment of the simplex at fixed z. Doing the y integral against the delta function fixes $y = 1 - z - x$, and the x integral then runs over the interval on which $y \geq 0$, namely $0 \leq x \leq 1 - z$:

$$\int_0^1 \mathrm{d}x\,\mathrm{d}y\;\delta(x + y + z - 1) \;=\; \int_0^{1-z} \mathrm{d}x \;=\; 1 - z\,, \qquad 0 \leq z \leq 1\,. \tag{20.87}$$

This is the standard reduction of the two-dimensional Feynman simplex to a single parameter; the surviving measure factor $(1-z)$ is the length of the simplex slice.

Step 3: do the remaining elementary integral. Combining Eqs. (20.86) and (20.87),

$$F_2(0) \;=\; \frac{\alpha}{2\pi}\int_0^1 \mathrm{d}z\;(1-z)\,\frac{2z}{1-z} \;=\; \frac{\alpha}{2\pi}\int_0^1 \mathrm{d}z\;2z\,. \tag{20.88}$$

The decisive cancellation happens here: the simplex measure $(1 - z)$ kills the $1/(1 - z)$ in the integrand. The remaining integral is trivial,

$$\int_0^1 2z \, \mathrm{d}z = \left[z^2\right]_0^1 = 1 \quad \Longrightarrow \quad a_e = F_2(0) = \frac{\alpha}{2\pi}\,, \tag{20.89}$$

which is Eq. (20.85).

Step 4: the number. Using the fine-structure constant $\alpha^{-1} = 137.035999$,

$$a_e \;=\; \frac{\alpha}{2\pi} \;=\; \frac{1}{2\pi \times 137.035999} \;=\; \frac{1}{861.022\ldots} \;=\; 0.00116140\ldots \tag{20.90}$$

This is Schwinger's 1948 result, and to three significant figures, $a_e \approx 1.16 \times 10^{-3}$, it already accounts for all but the last few parts in 10^6 of the measured electron anomaly — a single one-loop diagram fixing the leading correction to the magnetic moment of the electron.

Historical Note

That this number could be computed at all settled the crisis of late-1940s QED. The Kusch–Foley measurements of 1947–48 had shown the electron's magnetic moment to exceed the Dirac value $g = 2$ by about a part in 10^3; Schwinger's $\alpha/2\pi$, derived in the same months that renormalization was being made systematic, reproduced the excess and demonstrated that the infinities of the theory could be quarantined while leaving a finite, correct prediction behind. The result was engraved (as $\alpha/2\pi$) on Schwinger's headstone.

20.3.3 Infrared finiteness: why F_2, unlike F_1, needs no regulator

The cancellation in Eq. (20.88) deserves a second look, because it is the infrared half of why a_e is unambiguous. At $q^2 = 0$ the denominator $\Delta = (1 - z)^2 m^2$ vanishes as $z \to 1$. That endpoint is the soft region of the loop: $z \to 1$ forces the Feynman-combined denominator to its mass shell, the kinematic configuration in which the virtual photon carries away vanishing momentum. An integral that probed this region too sharply would diverge in the infrared, and one would have to regulate it — by a fictitious photon mass μ_γ, or by dimensional regularization of the infrared — and track the regulator through to the end.

For F_2 this never happens. The numerator of Eq. (20.84) carries the explicit factor $z(1 - z)$, which supplies one power of $(1 - z)$, and the simplex measure Eq. (20.87) supplies a second. Against the $(1 - z)^2$ in the denominator the two conspire to leave a constant: $(1 - z) \cdot z(1 - z)/(1 - z)^2 = z$, bounded on $[0, 1]$. The magnetic numerator's extra power of soft momentum — the same power of q that made F_2 ultraviolet-finite in Section 20.3.1 — here renders it infrared-finite

as well. $F_2(0)$ is both ultraviolet- and infrared-convergent; it is a clean number with no regulator anywhere in sight.

The Dirac form factor is the opposite case in both respects. Its renormalized form $F_1(q^2)$, constructed in Chapter 19 (Eq. (19.120)), contains terms whose parameter integrands lack the protective $z(1-z)$ numerator. The same $z \to 1$ endpoint then produces $\int \mathrm{d}z\,(1-z)/(1-z)^2 = \int \mathrm{d}z/(1-z)$, a logarithmic infrared divergence. This is not a defect of the calculation but a genuine feature of QED: a charged particle cannot be scattered without radiating soft photons, so the elastic vertex F_1 alone is not an observable. The infrared divergence of F_1 cancels against the divergence of real soft-photon emission (bremsstrahlung) in any physical cross section, the cancellation established in Section 19.6. We import that analysis rather than repeat it; what matters here is its corollary for the magnetic moment. The entire infrared sensitivity of the one-loop vertex lives in F_1, where it must, to cancel against real emission; *none* of it reaches F_2. The anomalous moment is therefore an infrared-safe quantity on its own, with no companion bremsstrahlung process required to render it finite — which is precisely why $a_e = F_2(0)$ is a number one can measure and compute to extreme precision and compare directly.

Caution

It is tempting to read the bare integrand $2z/(1-z)$ of Eq. (20.86) as infrared-divergent at $z = 1$ and conclude that $F_2(0)$ needs a regulator. It does not: the divergence is fictitious, an artifact of separating the z-dependent integrand from the simplex measure. The physical integrand is the *product* with $(1-z)$ from Eq. (20.87), and that product, $2z$, is perfectly regular. One must keep the Jacobian of the simplex reduction before pronouncing on convergence.

Away from $q^2 = 0$ the full form factor Eq. (20.84) can be integrated numerically; it falls monotonically with spacelike momentum transfer, the electron's anomalous moment being a soft, low-q^2 effect that switches off as the probe resolves shorter distances. Figure 20.3 shows $F_2(q^2)$ normalized to its value at the origin, with the Schwinger point $F_2(0) = \alpha/2\pi$ marked.

Theorem 20.4 is the keystone of this chapter and the headline number of QED. The result $a_e = \alpha/2\pi$ is the leading term of a perturbative series in α/π whose higher coefficients — and whose extension from the electron to the muon, where heavier virtual states leave a larger imprint — are the substance of the modern precision program. We take up that program, and the live status of the muon anomaly, in the next section.

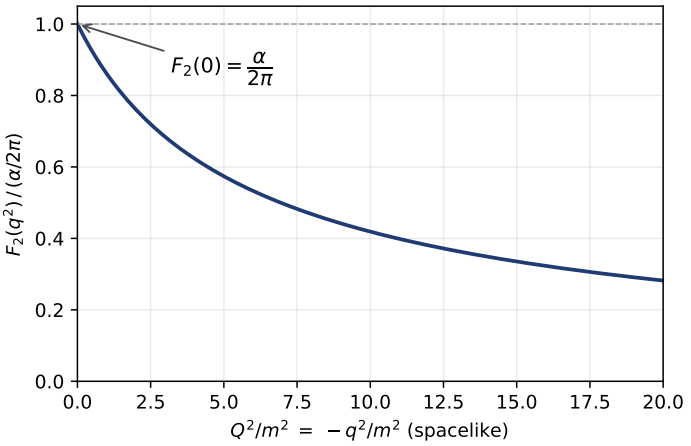


Figure 20.3: The one-loop Pauli form factor $F_2(q^2)$ of Eq. (20.84), normalized to its value at zero momentum transfer, as a function of the spacelike momentum transfer $Q^2 = -q^2$ in units of m^2, obtained by numerically integrating the Feynman-parameter representation. At $Q^2 = 0$ the curve equals unity, encoding Schwinger's $F_2(0) = \alpha/2\pi$; the anomalous moment is a low-q^2, long-distance effect that decreases as the probe resolves shorter distances. The form factor is finite at every Q^2, requiring neither ultraviolet counterterm nor infrared regulator.

20.4 Beyond One Loop: the Precision Frontier of g-2

The previous section closed QED's most celebrated computation: the one-loop vertex form factor evaluated at zero momentum transfer gave the anomalous magnetic moment of the electron,

$$a_e \equiv \frac{g-2}{2} = F_2(0) = \frac{\alpha}{2\pi} \approx 0.0011614, \tag{20.91}$$

Schwinger's number, derived in full in Section 20.3 (the result quoted there as Eq. (20.85)). That single loop already agrees with experiment to better than one part in a thousand—a stunning vindication of the entire apparatus of Part V. But Eq. (20.91) is only the first term of a convergent-looking power series in α/π, and the experimental anomaly is now known to twelve significant figures. The confrontation between that series and that measurement is, by a wide margin, the most precise quantitative test of any theory in the physical sciences. This section places Schwinger's number inside the modern precision program. We will see three distinct things grow out of Eq. (20.91): the higher-order QED series and what is known of its coefficients; the inversion of the electron anomaly into the most precise laboratory value of the fine-structure constant α itself; and the reason the *muon*, not the electron, is the anomalous moment that probes physics beyond pure QED—a reason we will derive, not assert, from the structure of the

magnetic-moment operator. The section ends at the live research frontier, where the muon anomaly sits as of the mid-2020s, reported honestly and without false closure.

20.4.1 The perturbative series for the anomaly

The anomaly is dimensionless, and in pure QED the only dimensionless coupling is $\alpha = e^2/4\pi$. The Feynman-diagram expansion of the vertex organizes itself by the number of photon loops: n loops bring n extra factors of e^2, hence n factors of α. It is conventional—and, as Schwinger's $\frac{1}{2}$ already hints, natural—to expand in the combination α/π, because each loop integration supplies a $1/(4\pi)^2$ from the measure (Chapter D) against the $e^2 = 4\pi\alpha$ of the two vertices it adds, and the residual numerical coefficients then come out as $O(1)$ rationals and transcendentals rather than carrying spurious powers of π. We therefore write the QED prediction as

$$a_\ell^{\mathrm{QED}} \;=\; \sum_{n=1}^{\infty} c_n^{(\ell)} \left(\frac{\alpha}{\pi}\right)^n, \qquad c_1^{(\ell)} = \tfrac{1}{2}\ \text{(Schwinger)}, \tag{20.92}$$

where $\ell \in \{e, \mu, \tau\}$ labels the lepton whose moment we compute. The leading coefficient $c_1 = \frac{1}{2}$ is universal: it is the pure number we extracted from Eq. (20.91), and the lepton mass dropped out of that one-loop integral entirely when we set $q^2 \to 0$.

That mass-independence does not survive at higher orders, and understanding *why* is the conceptual key to the whole section. Beyond one loop a diagram contributing to a_ℓ can contain a closed loop of a *different* charged lepton—an electron loop inside the muon's vertex, say, entering through a vacuum-polarization insertion of the kind built in Section 19.2. Such a diagram knows about two masses, and its value depends on the ratio. It is therefore standard to split each coefficient into a mass-independent "universal" part and mass-dependent remainders,

$$c_n^{(\ell)} \;=\; A_1^{(2n)} \;+\; \sum_{\ell' \neq \ell} A_2^{(2n)}\!\left(\frac{m_\ell}{m_{\ell'}}\right) \;+\; \sum_{\ell' \neq \ell'' \neq \ell} A_3^{(2n)}\!\left(\frac{m_\ell}{m_{\ell'}}, \frac{m_\ell}{m_{\ell''}}\right), \tag{20.93}$$

where A_1 collects the diagrams built from photons and the external lepton alone (no internal lepton loops), A_2 the diagrams with one species of internal lepton loop, and A_3 those with two. The superscript $(2n)$ records the power of e, i.e. the loop order n. The A_1 coefficients are pure numbers, common to all three leptons; the mass-dependent pieces differ from lepton to lepton and are the seed of the muon's special role, to which Section 20.4.3 is devoted.

The universal coefficients. The mass-independent $A_1^{(2n)}$ have been the object of a sixty-year computational program. Through the orders now known they are,

with their discoverers:

$$A_1^{(2)} = \tfrac{1}{2}, \tag{20.94}$$

$$A_1^{(4)} = \frac{197}{144} + \frac{\pi^2}{12} + \frac{3}{4}\zeta(3) - \frac{\pi^2}{2}\ln 2 \;=\; -0.328\,478\,965\,579\ldots, \tag{20.95}$$

$$A_1^{(6)} = \;\; 1.181\,241\,456\,587\ldots, \tag{20.96}$$

$$A_1^{(8)} = -1.912\,245\,764\ldots, \tag{20.97}$$

$$A_1^{(10)} = \;\; 6.737\,(159). \tag{20.98}$$

We do not derive these here, and it is important to be honest about why: each is a genuine multi-loop computation whose techniques (the reduction of tensor integrals to master integrals, the evaluation of those masters as multiple polylogarithms, and at the highest orders large-scale numerical integration over thousands of diagrams) lie beyond the one-loop methods of Chapter 19 and would each consume a chapter. The two-loop number $A_1^{(4)}$, Eq. (20.95), was obtained in closed form by Petermann and by Sommerfield in 1957 [400, 451]; we display its analytic value precisely because it is the first place the anomaly acquires a transcendental ($\zeta(3)$ and $\pi^2 \ln 2$) structure, and because the reader can check the decimal against the closed form—which is the kind of cross-check this book insists on. The three-loop $A_1^{(6)}$, Eq. (20.96), was found analytically by Laporta and Remiddi in 1996 [315]; its closed form runs to a page of polylogarithms and is suppressed here as it teaches nothing new about the anomaly. The four-loop $A_1^{(8)}$, Eq. (20.97), required the evaluation of 891 Feynman diagrams and is known to thousands of digits from the semi-analytic work of Laporta in 2017 [316]; the five-loop $A_1^{(10)}$, Eq. (20.98), rests on the numerical evaluation of 12 672 diagrams by Aoyama, Hayakawa, Kinoshita and Nio [29, 30], and its parenthesized (159) is the numerical-integration uncertainty in the last digits—an honest reminder that even "pure QED" is, at the tenth order, a Monte-Carlo result, not an analytic one. These are quoted, attributed, and explicitly *not* proved here; the program that produces them is the natural extension of the one-loop vertex calculation the reader has now done in full.

Caution

The series Eq. (20.92) is an *asymptotic* series, not a convergent one—a generic feature of perturbative QFT that we analyze properly in Chapter 5. Because $\alpha/\pi \approx 2.3 \times 10^{-3}$ is so small, the successive terms shrink steeply ($c_n(\alpha/\pi)^n$ falls by roughly 10^{-3} at each order even though $|c_n|$ grows), and truncation at five loops leaves a residue far below current experimental precision. The asymptotic character of the series is invisible in practice for the lepton anomalies; it would only matter at an order ($n \sim 1/(\alpha/\pi) \sim 400$) that no experiment will ever reach.

Assembling the electron prediction. For the electron the mass-dependent pieces in Eq. (20.93) are tiny: the lightest internal lepton that can appear is the muon, and—as the next subsection will make quantitative—a heavy internal loop contributes suppressed by $(m_e/m_\mu)^2 \sim 2.3 \times 10^{-5}$. The electron anomaly is therefore very nearly the universal QED number, and inserting the measured α into Eq. (20.92) gives the theoretical value to the precision of the input α,

$$a_e^{\text{theory}} \;=\; 0.001\,159\,652\,18\ldots, \tag{20.99}$$

where the displayed digits are stable against the choice of α input only down to the level at which that input matters—a point that turns out to be the crux of the next subsection.

20.4.2 The electron anomaly as a measurement of α

We have been treating Eq. (20.92) as a prediction: feed in α, get out a_e. But the logic runs just as well in reverse, and this reversal is one of the most beautiful inferences in precision physics. The electron anomaly is measured in a single-electron Penning trap to extraordinary accuracy. The most precise determination, by Fan, Myers, Sukra and Gabrielse in 2023 [183], is

$$a_e^{\text{exp}} \;=\; 1.159\,652\,180\,59\,(13) \times 10^{-3} \qquad (0.11\ \text{ppb}), \tag{20.100}$$

a relative uncertainty of about one part in 10^{10}. The QED series Eq. (20.92) is known, as we saw, through five loops with a theoretical error well below this. So instead of predicting a_e we may *equate* the measured Eq. (20.100) to the series and solve for the one unknown it contains, α. The electron anomaly becomes a measurement of the fine-structure constant.

The inversion is elementary at leading order and instructive to carry out. Writing $x = \alpha/\pi$ and keeping the first two terms of Eq. (20.92),

$$a_e = \tfrac{1}{2}x + c_2\,x^2 + \cdots, \tag{20.101}$$

we solve perturbatively. To leading order $x \approx 2a_e$, and substituting back,

$$x \;=\; 2a_e\Big(1 - 4c_2\,a_e + \cdots\Big), \qquad \text{i.e.} \qquad \alpha \;=\; 2\pi a_e\Big(1 - 4c_2\,a_e + \cdots\Big). \tag{20.102}$$

The leading piece alone, $\alpha \approx 2\pi a_e$, already pins α^{-1} to about 137.0; the higher coefficients $c_2, c_3, \ldots$ sharpen it digit by digit. Carrying the full five-loop series and the small mass-dependent terms, the 2023 measurement Eq. (20.100) yields [183]

$$\alpha^{-1}(a_e) \;=\; 137.035\,999\,166\,(15), \tag{20.103}$$

the parenthesized uncertainty now dominated jointly by the experimental error in Eq. (20.100) and the tenth-order theory error in Eq. (20.98). This is, on its own, among the most precise numbers in physics.

Worked Example 20.6 (The first two digits of α^{-1} from the anomaly)**.** To see concretely how Eq. (20.102) works, retain only Schwinger's term and the two-loop $c_2 = A_1^{(4)} = -0.32848$ (the electron's mass-dependent pieces are negligible at this precision). The leading inversion gives

$$\alpha \approx 2\pi a_e = 2\pi(1.159\,652 \times 10^{-3}) = 7.288\,18 \times 10^{-3}, \qquad \alpha^{-1} \approx 137.207. \tag{20.104}$$

This is already correct to four figures but wrong in the fifth. Now include the c_2 correction. With $a_e = 1.159\,652 \times 10^{-3}$,

$$1 - 4c_2 a_e = 1 - 4(-0.32848)(1.159\,652 \times 10^{-3}) = 1 + 1.5237 \times 10^{-3}, \tag{20.105}$$

so $\alpha \to 7.288\,18 \times 10^{-3} \times (1 + 1.5237 \times 10^{-3}) = 7.299\,29 \times 10^{-3}$ and

$$\alpha^{-1} \approx 136.998. \tag{20.106}$$

The two-loop term has moved α^{-1} by about 0.21, overshooting the converged value Eq. (20.103) because we truncated; restoring the (positive) three-loop $c_3 = 1.181$ pulls it back. The point is structural, not numerical: each successive coefficient in the QED series contributes one or two more correct digits to α^{-1}, and the experiment is precise enough that the five-loop coefficient Eq. (20.98)—a 12 672-diagram Monte-Carlo number—is the limiting *theoretical* ingredient in the world's best value of the fine-structure constant. The famous one-loop number of QED has become a metrological instrument.

The independent cross-check, and a live discrepancy. A value of α extracted from a_e is only as trustworthy as the assumption that pure QED accounts for the electron anomaly. To test that assumption—and to expose any new physics hiding in a_e—one needs α measured by a route that does *not* use the anomaly at all. Atom interferometry supplies it. The recoil of an atom of mass $m_{\rm at}$ that absorbs a photon of known wavelength measures the ratio $h/m_{\rm at}$, and the fine-structure constant follows from

$$\alpha^2 \;=\; \frac{2R_\infty}{c}\,\frac{m_{\rm at}}{m_e}\,\frac{h}{m_{\rm at}}, \tag{20.107}$$

where the Rydberg constant R_∞ and the mass ratios $m_{\rm at}/m_e$ are independently and very precisely known. Crucially, Eq. (20.107) involves no QED loop diagram: it is atomic kinematics. Two such measurements stand out, on caesium [381] and on rubidium [352]:

$$\alpha^{-1}(\mathrm{Cs}) = 137.035\,999\,046\,(27) \qquad \text{(Parker et al. 2018)}, \tag{20.108}$$

$$\alpha^{-1}(\mathrm{Rb}) = 137.035\,999\,206\,(11) \qquad \text{(Morel et al. 2020)}. \tag{20.109}$$

Here lies an honest complication that the precision frontier cannot paper over. The two recoil determinations Eqs. (20.108) and (20.109) disagree with each

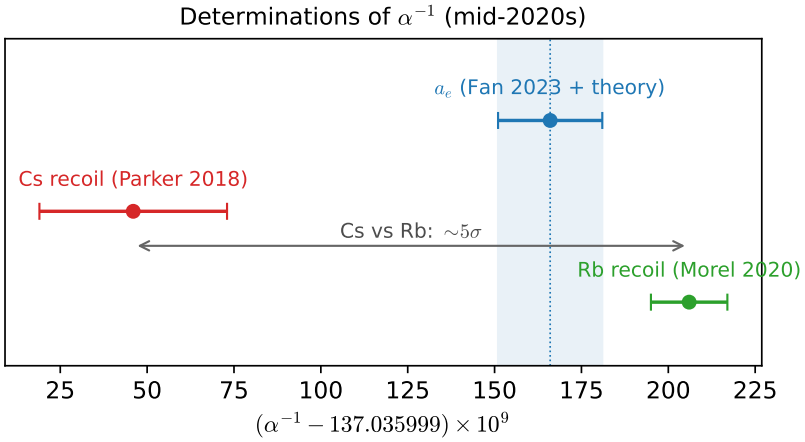


Figure 20.4: Three determinations of the fine-structure constant, plotted as $(\alpha^{-1} - 137.035999) \times 10^9$. The anomaly route Eq. (20.103) (blue band) combines the Penning-trap measurement Eq. (20.100) with the five-loop QED series; the caesium Eq. (20.108) and rubidium Eq. (20.109) values come from atom-recoil interferometry, Eq. (20.107), and use no loop diagram. The Cs and Rb measurements disagree at about 5σ, the dominant limitation on the a_e test of QED as of the mid-2020s.

other by about 1.6×10^{-7} in α^{-1}, which against their combined uncertainty is a discrepancy of roughly 5σ. The anomaly-derived value Eq. (20.103) sits between them. Figure 20.4 displays the three. Until the Cs/Rb disagreement is resolved—it is presumably a systematic effect in one or both experiments, not new physics—the comparison of a_e^{exp} against a_e^{theory} is limited not by QED, which is in hand to five loops, but by the spread in the independent α that QED's prediction requires as input. Depending on which recoil value one adopts, the electron anomaly agrees with the Standard Model at the $\sim 1\sigma$ level or shows a $\sim 2\sigma$ pull; the situation is unsettled as of the mid-2020s precisely because the metrological inputs disagree.

Physical Insight

Two readings of the same electron anomaly coexist. Assume QED and Eq. (20.92) *predicts* a_e, turning the trap measurement into the world's best α. Or measure α independently by atom recoil and the same equation becomes a *test* of QED, sensitive to any contribution beyond the photon and the lepton—a fourth-generation lepton, a dark photon, a lepton substructure—that would show up as a mismatch. One cannot do both at once with the same data: a single equation cannot simultaneously determine α and test the dynamics that relates α to a_e. The frontier's

job is to make the independent α precise enough that the second reading bites. That is exactly what the Cs/Rb discrepancy currently prevents.

20.4.3 Why the muon: the $(m_\ell/M)^2$ lever

The electron anomaly is measured a thousand times more precisely than the muon's, and computed in pure QED to one more loop. Yet it is the *muon* anomaly that the community watches for cracks in the Standard Model. The reason is a single parametric fact, which we now derive rather than assert: a contribution to the anomaly from physics at a mass scale $M \gg m_\ell$ is suppressed by $(m_\ell/M)^2$, so the heavier lepton is the more sensitive antenna by the factor $(m_\mu/m_e)^2 \approx 4.3 \times 10^4$.

The cleanest way to see this is to recognize the anomaly as the coefficient of a particular effective operator. Recall from Chapter 19 that the on-shell vertex decomposes as $\Gamma^\mu = \gamma^\mu F_1(q^2) + (i\sigma^{\mu\nu}q_\nu/m_\ell)F_2(q^2)$, and that $a_\ell = F_2(0)$. The F_2 term, in position space, is the Pauli moment operator: at low energies the anomalous magnetic interaction of the lepton is captured by adding to the QED Lagrangian the gauge-invariant, dimension-five term

$$\mathcal{L}_{\text{Pauli}} \;=\; \frac{e\,a_\ell}{2m_\ell}\,\bar\psi\,\sigma^{\mu\nu}\psi\,F_{\mu\nu}, \tag{20.110}$$

whose tree-level matrix element reproduces $F_2(0) = a_\ell$ (this is the Gordon-decomposition identification made in Section 20.1.3). Two features of Eq. (20.110) drive everything. First, by power counting the operator $\bar\psi\sigma^{\mu\nu}\psi F_{\mu\nu}$ has mass dimension 5 (each ψ counts as $\frac{3}{2}$, $F_{\mu\nu}$ as 2, σ as 0), so its coefficient carries mass dimension -1; indeed $ea_\ell/2m_\ell$ has dimension -1 because e and a_ℓ are dimensionless. Second, the bilinear $\bar\psi\sigma^{\mu\nu}\psi$ is *chirality-flipping*.

Lemma 20.7 (The magnetic bilinear connects opposite chiralities)**.** *With the chiral projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ and $\psi_{L,R} = P_{L,R}\psi$,*

$$\bar\psi\,\sigma^{\mu\nu}\psi \;=\; \bar\psi_L\,\sigma^{\mu\nu}\psi_R + \bar\psi_R\,\sigma^{\mu\nu}\psi_L, \tag{20.111}$$

i.e. the same-chirality pieces $\bar\psi_L\sigma\psi_L$ and $\bar\psi_R\sigma\psi_R$ vanish identically.

Proof. Since $\sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu,\gamma^\nu]$ is a product of two gamma matrices, it commutes with γ^5: $[\gamma^5,\sigma^{\mu\nu}] = 0$, hence $\sigma^{\mu\nu}$ commutes with both projectors, $P_{L,R}\sigma^{\mu\nu} = \sigma^{\mu\nu}P_{L,R}$. Recall also that γ^5 is Hermitian and anticommutes with γ^0, so $P_{L,R}\gamma^0 = \gamma^0 P_{R,L}$, which gives

$$\overline{\psi_L} = \overline{P_L\psi} = \psi^\dagger P_L^\dagger\gamma^0 = \psi^\dagger P_L\gamma^0 = \bar\psi P_R, \qquad \overline{\psi_R} = \bar\psi P_L. \tag{20.112}$$

Consider first the opposite-chirality combinations. Using $P_R\sigma^{\mu\nu}P_R = \sigma^{\mu\nu}P_R$ (projectors commute through σ and $P_R^2 = P_R$),

$$\overline{\psi_L}\,\sigma^{\mu\nu}\psi_R = \bar\psi P_R\,\sigma^{\mu\nu}P_R\psi = \bar\psi\,\sigma^{\mu\nu}P_R\psi, \qquad \overline{\psi_R}\,\sigma^{\mu\nu}\psi_L = \bar\psi\,\sigma^{\mu\nu}P_L\psi. \tag{20.113}$$

Adding them and using $P_L+P_R=1$ gives $\bar\psi\,\sigma^{\mu\nu}P_R\psi+\bar\psi\,\sigma^{\mu\nu}P_L\psi=\bar\psi\,\sigma^{\mu\nu}\psi$, which is Eq. (20.111). The same-chirality combinations vanish by the orthogonality $P_LP_R=P_RP_L=0$:

$$\overline{\psi_L}\,\sigma^{\mu\nu}\psi_L=\bar\psi P_R\,\sigma^{\mu\nu}P_L\psi=\bar\psi\,\sigma^{\mu\nu}P_RP_L\psi=0, \tag{20.114}$$

and likewise $\overline{\psi_R}\,\sigma^{\mu\nu}\psi_R=\bar\psi\,\sigma^{\mu\nu}P_LP_R\psi=0$. This establishes Eq. (20.111). □

Theorem 20.7 is the physical heart of the matter. A magnetic moment flips the lepton's chirality: it turns a left-handed lepton into a right-handed one (and conversely). But in a chiral gauge theory—and the electroweak sector of the Standard Model is chiral—the left- and right-handed leptons are *different fields*, and the only operator in the renormalizable Lagrangian that connects them is the mass term, which itself arises from the Yukawa coupling to the Higgs. The lepton mass m_ℓ is therefore the unique "spurion" of chirality violation: every amplitude that flips chirality must be proportional to (an odd power of) m_ℓ. This is not a statement about QED alone; it is a selection rule that any contribution to the anomaly, from any sector, must obey.

Proposition 20.8 (Decoupling of heavy scales in the anomaly)**.** *Let new physics—or simply a heavy Standard-Model state—of mass $M\gg m_\ell$ couple to the lepton with a dimensionless coupling g and generate, on being integrated out, a contribution to the Pauli operator Eq. (20.110). Then that contribution to the anomaly scales as*

$$a_\ell^{(M)}\;\sim\;\frac{g^2}{16\pi^2}\,\frac{m_\ell^2}{M^2}. \tag{20.115}$$

Consequently the ratio of sensitivities of two leptons to the same heavy physics is

$$\frac{a_\mu^{(M)}}{a_e^{(M)}}\;=\;\frac{m_\mu^2}{m_e^2}\;\approx\;(206.77)^2\;\approx\;4.3\times10^4. \tag{20.116}$$

Proof. Integrating out the heavy state at one loop produces an effective operator built from the lepton and photon fields. The lowest-dimension operator that contributes to the magnetic moment is the dimension-five tensor $\mathcal{O}=\bar\psi\,\sigma^{\mu\nu}\psi\,F_{\mu\nu}$, generated with some coefficient C of mass dimension -1, so that $\mathcal{L}_{\text{eff}}\supset e\,C\,\mathcal{O}$. A one-loop diagram supplies the standard loop factor $g^2/16\pi^2$. By dimensional analysis C must carry one inverse power of mass; the available scales are M and m_ℓ, and since the heavy state is what runs in the loop the natural scale is M, giving a generic $C\sim(g^2/16\pi^2)(1/M)$ on dimensional grounds alone. But Theorem 20.7 forbids this: $\mathcal{O}$ is chirality-flipping, so C must be proportional to the chirality spurion m_ℓ. The lowest power consistent with the selection rule is one factor of m_ℓ, which by dimensions must be compensated by an extra $1/M$, so

$$C\;\sim\;\frac{g^2}{16\pi^2}\,\frac{m_\ell}{M^2}. \tag{20.117}$$

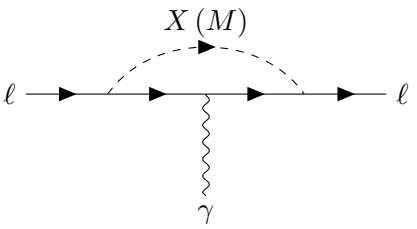


Figure 20.5: A heavy state X of mass $M \gg m_\ell$ dressing the lepton's magnetic vertex. The external photon γ measures the moment. Because the magnetic transition flips the lepton's chirality (Theorem 20.7) and chirality is broken only by the mass, the amplitude carries a factor m_ℓ; dimensional analysis of the resulting dimension-five operator then forces the second power, giving the $(m_\ell/M)^2$ suppression of Eq. (20.115). The muon's $(m_\mu/m_e)^2 \approx 4.3 \times 10^4$ advantage over the electron follows.

Matching $e\,C\,\mathcal{O}$ to the canonical normalization Eq. (20.110), $eC = e\,a_\ell^{(M)}/2m_\ell$, gives $a_\ell^{(M)} = 2m_\ell C \sim (g^2/8\pi^2)(m_\ell^2/M^2)$, which is Eq. (20.115) up to the displayed $O(1)$ factor. The ratio Eq. (20.116) follows by taking the quotient for $\ell = \mu$ and $\ell = e$, the M-dependence and the coupling cancelling. □

Figure 20.5 draws the generic heavy-loop insertion that Theorem 20.8 governs. The proposition explains the division of labour between the two leptons. For the *electron*, the mass-dependent terms in Eq. (20.93) that involve heavier leptons are suppressed by $(m_e/m_\mu)^2$ and $(m_e/m_\tau)^2$ and are genuinely negligible—this is why Eq. (20.99) was so nearly the universal QED number, and why a_e is the clean place to read off α. For the *muon*, two things change. First, any heavy physics—the W and Z, the Higgs, or anything new at the TeV scale—enters a_μ enhanced by the factor Eq. (20.116) relative to its effect on a_e. Second, the mass-dependent QED terms run the other way: a muon vertex dressed by an *electron* loop has $M = m_e < m_\mu$, so there is no power suppression at all—instead the light electron produces large logarithms $\ln(m_\mu/m_e) \approx 5.3$ that enhance a_μ^{QED} over the universal value. The muon, sitting at an intermediate mass, is uniquely poised: light enough that the dominant QED contributions are calculable and large, heavy enough that it feels hadronic and weak-scale physics at a level the experiment can resolve.

Worked Example 20.9 (The electroweak contribution to a_μ, and its invisibility in a_e)**.** Theorem 20.8 is parametric; here we make it a number. The heaviest Standard-Model states that contribute to the lepton anomaly are the weak gauge bosons, $M \sim m_W \approx 80$ GeV. The relevant coupling is the weak gauge coupling, conveniently packaged in the Fermi constant $G_F = 1.1664 \times 10^{-5}\,\mathrm{GeV}^{-2} \sim g^2/m_W^2$. By Eq. (20.115) the electroweak contribution should be of order $(G_F m_\ell^2/8\sqrt{2}\,\pi^2)$, the standard one-loop combination. The explicit one-loop

result, which we quote from the electroweak calculation set up in Chapter 31, is

$$a_\ell^{\mathrm{EW},(1)} = \frac{G_F\, m_\ell^2}{8\sqrt{2}\,\pi^2}\left[\frac{5}{3} + \frac{1}{3}\left(1 - 4\sin^2\theta_{\mathrm{W}}\right)^2\right], \tag{20.118}$$

the two terms coming from W and Z exchange respectively. Evaluate it for the muon. With $m_\mu = 0.10566$ GeV, $m_\mu^2 = 1.1164 \times 10^{-2}\,\mathrm{GeV}^2$, and $\sin^2\theta_{\mathrm{W}} \approx 0.231$ so that $(1 - 4 \times 0.231)^2 = 0.0058$, the bracket is $\frac{5}{3} + \frac{1}{3}(0.0058) = 1.669$. The prefactor is

$$\frac{G_F m_\mu^2}{8\sqrt{2}\,\pi^2} = \frac{(1.1664 \times 10^{-5})(1.1164 \times 10^{-2})}{8 \times 1.4142 \times 9.8696} = \frac{1.302 \times 10^{-7}}{111.7} = 1.166 \times 10^{-9}, \tag{20.119}$$

so $a_\mu^{\mathrm{EW},(1)} = 1.166 \times 10^{-9} \times 1.669 = 1.95 \times 10^{-9} = 195 \times 10^{-11}$. Two-loop electroweak corrections, which are not small because they contain their own large logarithms, reduce this to the accepted value $a_\mu^{\mathrm{EW}} = 153.6(1.0) \times 10^{-11}$. Now apply Eq. (20.116) to the electron: $a_e^{\mathrm{EW}} = a_\mu^{\mathrm{EW}} \times (m_e/m_\mu)^2 \approx 154 \times 10^{-11}/4.3 \times 10^4 \approx 4 \times 10^{-14}$. This is four orders of magnitude below the experimental precision Eq. (20.100) on a_e: the entire weak interaction is invisible in the electron anomaly but a 150×10^{-11} effect—comfortably resolved—in the muon's. The $(m_\mu/m_e)^2$ lever is not a slogan; it is the difference between a measurable signal and nothing at all.

20.4.4 Anatomy of a_μ and the 2021–2025 status

Because the muon feels everything, its anomaly is a sum over every sector of the Standard Model, each of which must be computed and added before any comparison with experiment is meaningful:

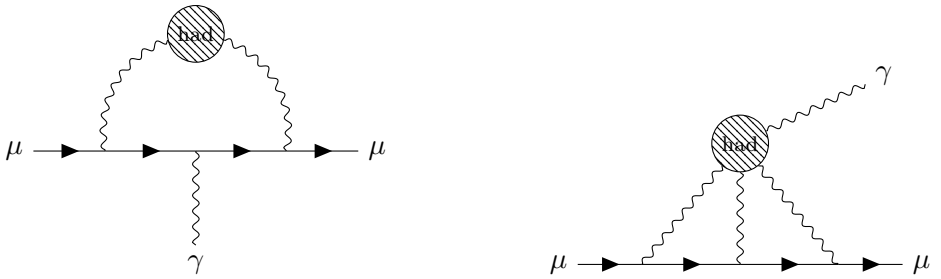
$$a_\mu^{\mathrm{SM}} \;=\; a_\mu^{\mathrm{QED}} + a_\mu^{\mathrm{EW}} + a_\mu^{\mathrm{HVP}} + a_\mu^{\mathrm{HLbL}}. \tag{20.120}$$

The four terms are wildly different in size and in difficulty, and it is worth taking each in turn, with numbers in units of 10^{-11} as understood in the mid-2020s.

The QED part, $a_\mu^{\mathrm{QED}} = 116\,584\,718.9\,(0.1)$, dominates the total and is the best-controlled: it is the series Eq. (20.92) with the muon's mass-dependent coefficients Eq. (20.93) included, evaluated through five loops by the same program that produced Eq. (20.98) [30]. Its uncertainty is utterly negligible on the scale of the comparison.

The electroweak part, $a_\mu^{\mathrm{EW}} = 153.6\,(1.0)$, is the number computed in Theorem 20.9; it requires the full electroweak theory of Chapter 31 and is known precisely.

The hadronic vacuum-polarization part, $a_\mu^{\mathrm{HVP}} \approx 6845$, is the leading source of theoretical uncertainty and the crux of the present situation. It is the contribution to the photon vacuum polarization $\Pi(q^2)$ of Section 19.2 from quark loops dressed



(a) hadronic vacuum polarization (b) hadronic light-by-light

Figure 20.6: The two hadronic contributions to the muon anomaly. (a) Hadronic vacuum polarization: a hadronic (quark-gluon) blob dresses the internal photon, an insertion into $\Pi(q^2)$ of Section 19.2. (b) Hadronic light-by-light: the external magnetic photon and three internal photons all meet a four-point hadronic blob. Both are non-perturbative at the muon scale and are the limiting theoretical uncertainties in Eq. (20.120); the machinery is developed in Chapter 28.

by the strong interaction—a hadronic blob inserted into the internal photon of the muon's vertex, drawn in Fig. 20.6(a). At the relevant momentum scale, of order m_μ, the strong coupling is large and perturbation theory in α_s fails: $a_\mu^{\rm HVP}$ is an intrinsically non-perturbative quantity. Two roads compute it, and we only set them up here, deferring the machinery to Chapter 28. The *data-driven* (dispersive) method writes the hadronic $\Pi(q^2)$ as a dispersion integral—an application of the analyticity and optical-theorem technology of Chapter 5—over the measured cross section for $e^+e^- \to$ hadrons, the same R-ratio that Chapter 28 treats; the imaginary part of Π is data, and the real part follows by a Kramers–Kronig relation. The *lattice* method computes the same hadronic correlator from first-principles QCD on a Euclidean spacetime grid. That these two methods now disagree is the whole story below.

The hadronic light-by-light part, $a_\mu^{\rm HLbL} \approx 92\,(18)$, is smaller but harder still: it is the four-photon hadronic amplitude of Fig. 20.6(b), in which the external magnetic photon and three internal photons all attach to a hadronic blob. It has no simple dispersive reduction to a single measured cross section and is the subject of dedicated dispersive and lattice efforts; its uncertainty, while subdominant, is irreducibly theoretical.

The story, honestly dated. For two decades the muon anomaly has been the Standard Model's most persistent point of tension, and its status as of this writing (mid-2020s) is genuinely in flux—a live frontier, not a settled result. The narrative has three acts.

In 2021 the situation looked like an emerging discovery. The Brookhaven E821 experiment had measured $a_\mu^{\rm exp} = 116\,592\,089\,(63)$ in 2006 [61], and Fermilab's

first run confirmed and sharpened it. Against the consensus Standard-Model prediction assembled by the Muon $g-2$ Theory Initiative—the 2020 "white paper" value $a_\mu^{\rm SM} = 116\,591\,810\,(43)$, which used the *data-driven* HVP [28]—the experiment lay high by $251\,(59)$, a 4.2σ discrepancy. A 4σ tension in the most precisely tested theory in physics is exactly the kind of thing that signals new particles.

The second act undercut the first from the theory side. In the same year, the BMW lattice collaboration published a first-principles lattice-QCD computation of the leading HVP [86] that came out *larger* than the data-driven value—large enough to move $a_\mu^{\rm SM}$ up toward the experiment and shrink the discrepancy to near insignificance. The two determinations of the single most important hadronic number now disagreed with each other. The tension had migrated from experiment-versus-theory to theory-versus-theory. Then, in 2023, the CMD-3 experiment remeasured the dominant $e^+e^- \to \pi^+\pi^-$ channel that feeds the dispersive HVP and found a cross section *higher* than all previous e^+e^- measurements [273], in the direction of the lattice—throwing the data-driven evaluation, long thought the more reliable, into internal disarray.

The third act is the present consolidation. Fermilab's final result, combining all data-taking runs with the Brookhaven measurement, was published in 2025 [354], giving an experimental world average near $a_\mu^{\rm exp} \approx 116\,592\,071.5\,(14.5)$—a 0.12 ppm measurement, an extraordinary experimental achievement and now far more precise than the theory it is meant to test. On the theory side, the 2025 update of the white paper [14] adopted a lattice-based consensus for the HVP, moving the predicted $a_\mu^{\rm SM} \approx 116\,592\,033\,(62)$ upward into agreement with the measurement at roughly the 1σ level. Figure 20.7 collects these numbers.

The honest summary is this. The clean "4σ anomaly" of 2021 has, as of the mid-2020s, receded as a discovery claim—not because new physics was excluded, but because the dominant theoretical ingredient, $a_\mu^{\rm HVP}$, turned out to be less certain than believed. The live tension is now *internal* to the Standard-Model calculation: the data-driven dispersive HVP and the lattice HVP disagree, with the CMD-3 measurement having destabilized the consensus that the dispersive value was the more trustworthy. Whether a residual muon anomaly survives the resolution of the HVP problem is exactly the open question, and it is openly open. What is not in doubt is the experiment, which now knows a_μ to a tenth of a part per million, and the structure of the prediction Eq. (20.120), every term of which the muon's $(m_\mu/m_e)^2$ sensitivity makes it the right place to look. We will return to the hadronic ingredients when the QCD machinery is in hand (Chapter 28) and to the electroweak ones when the full Standard Model is assembled (Chapters 31 and 34); a definitive verdict on the muon anomaly is, deliberately, not delivered here.

The anomalous moment is one of three numbers that the same one-loop functions of Chapter 19 predict. We have now read the vertex form factor F_2

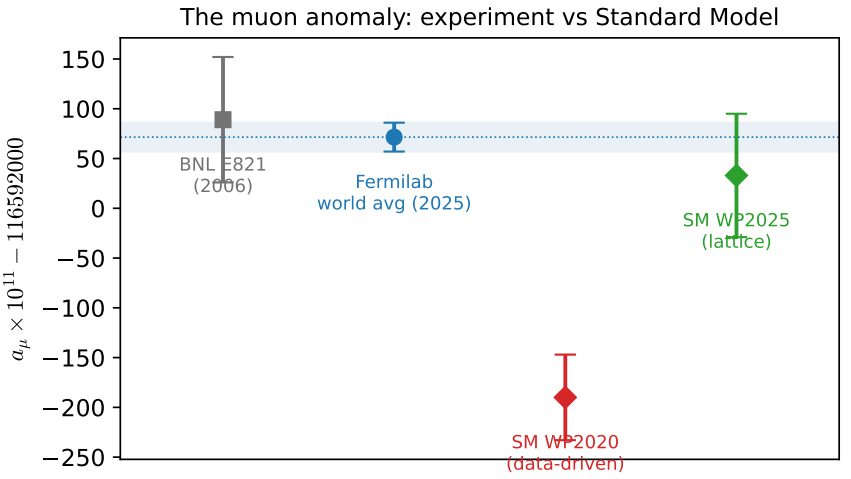


Figure 20.7: The muon anomaly, plotted as $a_\mu \times 10^{11} - 116\,592\,000$. The Fermilab final world average [354] (blue band) is compared with two Standard-Model predictions: the 2020 data-driven white paper [28] (red), which lay 4.2σ below the 2021 experiment, and the 2025 lattice-consensus update [14] (green), which agrees with experiment at $\sim 1\sigma$. The shift between the two theory points is almost entirely the hadronic vacuum polarization. Numbers as of the mid-2020s; the data-driven–vs–lattice HVP tension remains unresolved.

at zero momentum transfer for a *free* lepton in a magnetic field, and followed it from Schwinger's $\alpha/2\pi$ to the precision frontier. The next section reads the very same functions—the vertex correction together with the self-energy $\Sigma(p)$ and the vacuum polarization $\Pi(q^2)$—in a different physical setting, that of an electron *bound* in hydrogen, where they lift the accidental degeneracy of the $2S_{1/2}$ and $2P_{1/2}$ levels and produce the Lamb shift, the radiative effect whose measurement first forced QED into being.

20.5 The Lamb Shift

The previous sections read the one-loop vertex in its on-shell limit and extracted a laboratory number: the electron's anomalous moment $a_e = F_2(0) = \alpha/2\pi$. That number lives in free space — it is what a single electron does in a uniform external field. We now let the same radiative corrections act not on a free electron but on one *bound* in the Coulomb field of a proton, and ask what they do to the energy levels of hydrogen. The radiative-correction theme of the chapter generalizes in the most consequential way: the very functions built one loop at a time in Chapter 19 — the electron self-energy $\Sigma(p)$, the vacuum polarization $\Pi(q^2)$, and the renormalized vertex with its form factors F_1, F_2 — conspire to

split two levels that the Dirac theory predicts to be exactly degenerate. The splitting is small, a few parts in 10^7 of the binding energy, but it is real, it was measured before it was understood, and explaining it was the event that forced QED to become a renormalized theory. This section carries the explanation to an honest quantitative estimate: not the multi-decimal-place modern value, but a first-principles derivation that lands on the right scale — about a thousand megahertz — and correctly apportions it among the three radiative mechanisms, with their relative signs.

20.5.1 The Dirac degeneracy and the fact of its lifting

The relativistic hydrogen atom is solved exactly by the Dirac equation in a Coulomb potential, a computation that belongs to the relativistic quantum mechanics of Chapter 11. We need only its output: the bound-state energies. With the proton treated as a fixed point charge $+e$ and the electron's charge $-e$ entering through the minimal coupling fixed in Eq. (8.27), the Dirac–Coulomb spectrum is

$$E_{n,j} = m\left[1 + \left(\frac{\alpha}{n - (j+\tfrac{1}{2}) + \sqrt{(j+\tfrac{1}{2})^2 - \alpha^2}}\right)^2\right]^{-1/2}, \tag{20.121}$$

where $n = 1, 2, \ldots$ is the principal quantum number, j the total angular momentum, m the electron mass, and $\alpha = e^2/4\pi$ the fine-structure constant. Expanding in powers of α exposes the structure that matters here. Writing $\kappa \equiv j + \frac{1}{2}$ and keeping terms through $O(\alpha^4)$,

$$E_{n,j} = m - \frac{m\alpha^2}{2n^2} - \frac{m\alpha^4}{2n^4}\left(\frac{n}{j+\tfrac{1}{2}} - \frac{3}{4}\right) + O(\alpha^6). \tag{20.122}$$

The first two terms are the rest mass and the nonrelativistic Bohr spectrum; the $O(\alpha^4)$ term is the fine structure. The decisive feature is in plain sight.

Proposition 20.10 (Exact ℓ-degeneracy of the Dirac–Coulomb levels)**.** *The Dirac–Coulomb energy Eq. (20.121) depends on the orbital quantum numbers only through the pair (n, j); it is independent of the orbital angular momentum ℓ. Consequently any two hydrogen states with the same n and the same j but different ℓ are exactly degenerate to all orders in α. In particular the states*

$$2S_{1/2}\ (n=2,\ \ell=0,\ j=\tfrac{1}{2}) \qquad \textit{and} \qquad 2P_{1/2}\ (n=2,\ \ell=1,\ j=\tfrac{1}{2}) \tag{20.123}$$

have identical Dirac energy: $E_{2S_{1/2}} = E_{2P_{1/2}}$.

Proof. Neither Eq. (20.121) nor its expansion Eq. (20.122) contains ℓ: the only angular label appearing is j, through $j + \frac{1}{2}$. The two states in Eq. (20.123) share $n = 2$ and $j = \frac{1}{2}$ and differ only in ℓ (0 versus 1), a label absent from the energy. Hence their energies coincide identically in α. The companion state $2P_{3/2}$ has $j = \frac{3}{2}$ and is split off from the pair by the fine-structure term, at the much larger scale $\frac{m\alpha^4}{32} \sim 10\,\text{GHz}$. □

This degeneracy is not an approximation to be lifted by neglected relativistic terms — it is exact in the Dirac theory. Yet it is false. In 1947 Lamb and Retherford, using microwave techniques refined during the war, drove transitions between the $n = 2$ fine-structure levels of hydrogen and found that $2S_{1/2}$ lies *above* $2P_{1/2}$ by about 1000 MHz [311]; the modern value of the splitting is

$$\Delta\nu_{\text{Lamb}} \;=\; \nu(2S_{1/2}) - \nu(2P_{1/2}) \;\approx\; 1057.8\ \text{MHz}. \tag{20.124}$$

Something beyond the Dirac equation is at work. That something is the coupling of the bound electron to the quantized radiation field — the radiative corrections of Chapter 19, now evaluated in a bound state rather than a scattering amplitude.

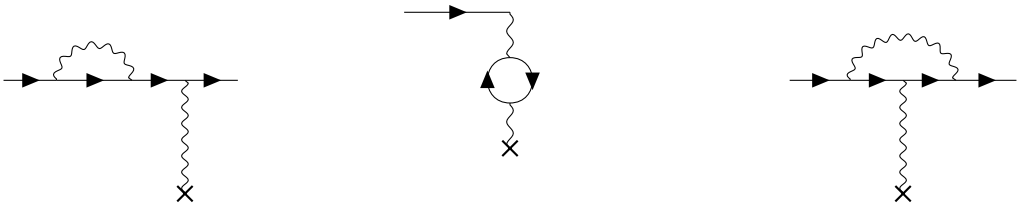
Historical Note: Shelter Island and the train to Schenectady

The Lamb–Retherford result was announced at the Shelter Island conference in June 1947, the meeting usually taken to mark the birth of modern QED. Within days Hans Bethe, on the train home, produced the first estimate of the shift [75]: a nonrelativistic calculation of the electron self-energy difference between the bound and free electron, made finite by hand with an ultraviolet cutoff at the electron mass. His number, ≈ 1040 MHz, was close enough to remove all doubt that the effect was radiative. The fully relativistic, manifestly finite calculation — the one that needed renormalization — followed over the next two years.

20.5.2 Three radiative origins, three one-loop functions

Before any computation, it is worth naming the three contributions and matching each to the one-loop object that produces it, because the bookkeeping is exactly the chapter's thesis: physical numbers are specific limits of the same amplitudes.

1. **Electron self-energy** (Section 19.3). The bound electron continually emits and reabsorbs virtual photons. The resulting shift of its energy differs between a bound electron and a free one; the difference is finite and is the *dominant* contribution to the Lamb shift. Its size is controlled by a logarithm — the Bethe logarithm — whose argument is the ratio of an



(a) self-energy (b) vacuum polarization (c) vertex correction

Figure 20.8: The three one-loop QED corrections to the bound electron's interaction with the static nuclear source (cross). (a) The electron self-energy Σ of Section 19.3 dresses the bound electron line; its bound-minus-free difference is the dominant Bethe-logarithm contribution. (b) The vacuum polarization Π of Section 19.2 dresses the exchanged photon, giving the Uehling correction (small, opposite sign). (c) The vertex correction supplies the anomalous-moment form factor $F_2(0) = \alpha/2\pi$ of Section 20.1.

ultraviolet scale (the electron mass) to an infrared scale set by the atomic binding energy. It pushes $2S_{1/2}$ *up*.

2. **Vacuum polarization** (Section 19.2). The photon exchanged between electron and proton can fluctuate into a virtual e^+e^- pair, modifying the Coulomb potential at short distance. This is the Uehling effect. It is small, about one part in forty of the self-energy term, and it has the *opposite* sign: it pushes $2S_{1/2}$ *down*.

3. **Anomalous moment / vertex** (Section 20.1). The same Pauli form factor $F_2(0) = \alpha/2\pi$ that gave Schwinger's number modifies the electron's magnetic moment, and hence its spin-orbit and contact interactions with the nuclear field. It contributes a few percent of the total, with the *same* sign as the self-energy.

Figure 20.8 shows the three one-loop insertions, drawn on the bound electron's interaction with the static nuclear source.

The honest quantitative spine of this section is the dominant self-energy term, which we derive in full through Welton's heuristic; the vacuum polarization and the anomalous moment are then computed with the same machinery and added with their signs.

20.5.3 Welton's heuristic: the vacuum smears the electron

The self-energy contribution has a transparent physical picture, due to Welton [504], that yields the correct order of magnitude and — decisively — the correct *selection rule*. The idea is that the quantized electromagnetic field, even in

its vacuum state, has zero-point fluctuations; these fluctuating fields buffet the bound electron, causing its position to jitter about the orbit the Dirac theory assigns it. A jittering electron samples the Coulomb potential over a small region rather than at a point, and the average potential over a region differs from the potential at the centre wherever the potential has curvature. Since the Coulomb potential's curvature is concentrated at the origin — it is a delta function there — only states with appreciable probability at the origin, the S-states, are affected. This is the entire mechanism: vacuum fluctuations → position jitter → a smeared potential → a shift proportional to the curvature of V at the electron → a shift proportional to $|\psi(0)|^2$.

We make each step quantitative.

The position jitter $\langle(\delta\mathbf{r})^2\rangle$. Treat the bound electron, for the purpose of estimating its response to the high-frequency vacuum field, as free: the binding is slow compared with the fluctuations that matter. A single Fourier component of the vacuum electric field at angular frequency ω exerts on the electron (charge $-e$, the minimal coupling of Eq. (8.27)) the force $\mathbf{F} = -e\,\mathbf{E}_\omega$, so Newton's law for the induced displacement $\delta\mathbf{r}$ reads

$$m\,\ddot{\delta\mathbf{r}} = -e\,\mathbf{E}_\omega, \qquad \mathbf{E}_\omega \propto e^{-i\omega t} \implies \delta\mathbf{r}_\omega = \frac{e}{m\omega^2}\,\mathbf{E}_\omega. \tag{20.125}$$

The displacement falls as $1/\omega^2$: low-frequency modes move the electron most. The mean-square displacement is the sum of $|\delta\mathbf{r}_\omega|^2$ over all modes, weighted by the mean-square vacuum field they carry. We need the spectral density of the zero-point electric field. The number of field modes (two polarizations) in a volume V with wavenumber in $[k, k+dk]$ is

$$dN = 2\cdot\frac{V}{(2\pi)^3}\,4\pi k^2\,dk = \frac{V}{\pi^2}\,\omega^2\,d\omega \qquad (\omega = k,\ \hbar = c = 1), \tag{20.126}$$

and each carries zero-point energy $\frac{1}{2}\omega$. The corresponding spectral energy density is $\rho(\omega)\,d\omega = (\omega^2/\pi^2)(\frac{1}{2}\omega)\,d\omega = \omega^3/(2\pi^2)\,d\omega$. In the Heaviside–Lorentz units of the book the field energy density is $\frac{1}{2}(\mathbf{E}^2 + \mathbf{B}^2)$, and on average the electric and magnetic parts are equal, so the spectral density of the mean-square electric field equals the total energy density:

$$\frac{d\langle\mathbf{E}^2\rangle}{d\omega} = \frac{\omega^3}{2\pi^2}. \tag{20.127}$$

Summing the independent modal contributions Eq. (20.125), weighted by Eq. (20.127),

$$\langle(\delta\mathbf{r})^2\rangle = \left(\frac{e}{m}\right)^2 \int \frac{d\omega}{\omega^4}\,\frac{d\langle\mathbf{E}^2\rangle}{d\omega} = \frac{e^2}{m^2}\int \frac{d\omega}{\omega^4}\,\frac{\omega^3}{2\pi^2} = \frac{e^2}{2\pi^2 m^2}\int\frac{d\omega}{\omega}. \tag{20.128}$$

With $e^2 = 4\pi\alpha$ the prefactor becomes $4\pi\alpha/(2\pi^2 m^2) = 2\alpha/(\pi m^2)$, so

$$\boxed{\langle(\delta\mathbf{r})^2\rangle = \frac{2\alpha}{\pi}\,\frac{1}{m^2}\,\ln\frac{\omega_{\max}}{\omega_{\min}}} \tag{20.129}$$

The integral $\int d\omega/\omega$ is logarithmically divergent at both ends; physics supplies both cutoffs, and the logarithm is exactly where the dynamics enters. The mean-square jitter is the squared Compton wavelength $1/m^2$ times the small number $2\alpha/\pi$ times a logarithm of order ten — the electron's position is uncertain by a length of order $\sqrt{\alpha}/m$ beyond the quantum-mechanical spread of its orbital, a tiny fraction of the Bohr radius.

Physical Insight: the vacuum as a buffeting medium

The same zero-point field that, in Section 20.1, dressed the free electron's magnetic moment, here jiggles the bound electron's position. The two effects are faces of one coupling: the electron is never alone with the nucleus; it is immersed in the fluctuating quantum field, which both renormalizes its properties and smears its trajectory. The Lamb shift is, in this picture, the bound electron reporting on the texture of the vacuum.

From jitter to energy shift. A displaced electron sees the potential $V(\mathbf{r}+\delta\mathbf{r})$ rather than $V(\mathbf{r})$. Expanding to second order in the small displacement,

$$V(\mathbf{r}+\delta\mathbf{r}) - V(\mathbf{r}) = \delta\mathbf{r}\cdot\nabla V + \frac{1}{2}\,\delta r_i\,\delta r_j\,\partial_i\partial_j V + \cdots \tag{20.130}$$

Averaging over the isotropic fluctuations, the first-order term vanishes because $\langle\delta\mathbf{r}\rangle = 0$, and the second-order term uses $\langle\delta r_i\,\delta r_j\rangle = \frac{1}{3}\langle(\delta\mathbf{r})^2\rangle\,\delta_{ij}$ (isotropy), so that $\langle\delta r_i\delta r_j\rangle\partial_i\partial_j V = \frac{1}{3}\langle(\delta\mathbf{r})^2\rangle\,\nabla^2 V$. The effective perturbation to the potential energy is therefore

$$\langle\delta V\rangle = \frac{1}{6}\,\langle(\delta\mathbf{r})^2\rangle\,\nabla^2 V. \tag{20.131}$$

The Coulomb potential energy of the electron in hydrogen is $V(\mathbf{r}) = -\alpha/r$, and its Laplacian is concentrated entirely at the origin:

$$\nabla^2 V = -\alpha\,\nabla^2\frac{1}{r} = -\alpha\,(-4\pi\,\delta^3(\mathbf{r})) = 4\pi\alpha\,\delta^3(\mathbf{r}), \tag{20.132}$$

the identity $\nabla^2(1/r) = -4\pi\delta^3(\mathbf{r})$ being the Poisson equation for a point charge. The perturbation Eq. (20.131) is thus a *contact* interaction, nonzero only at the nucleus. Its expectation value in a hydrogen state $\psi_{n\ell m}$ is

$$\Delta E_{n\ell} = \langle\psi_{n\ell m}|\langle\delta V\rangle|\psi_{n\ell m}\rangle = \frac{1}{6}\,\langle(\delta\mathbf{r})^2\rangle\cdot 4\pi\alpha\,|\psi_{n\ell m}(0)|^2. \tag{20.133}$$

This is the central formula, and it carries the selection rule on its face: the shift is proportional to the probability density at the origin, $|\psi(0)|^2$. For hydrogen wavefunctions, $|\psi(0)|^2 \neq 0$ only for $\ell = 0$ — the centrifugal barrier forces $\psi \sim r^\ell$ near the origin, so every state with $\ell \geq 1$ vanishes there. Only S-states are shifted; P-states are untouched (by this mechanism). This is precisely why a degeneracy between an S-state and a P-state is lifted: the two respond differently to a perturbation localized at the nucleus.

Proposition 20.11 (Self-energy shift of the nS levels)**.** *With the hydrogen S-state density at the origin $|\psi_{n00}(0)|^2 = 1/(\pi n^3 a_0^3)$ and Bohr radius $a_0 = 1/(\alpha m)$, the Welton estimate Eq. (20.133) gives*

$$\Delta E_{nS} = \frac{4\alpha^5 m}{3\pi n^3}\,\ln\frac{\omega_{\max}}{\omega_{\min}}, \tag{20.134}$$

positive (an upward shift), nonzero only for S-states, and falling as $1/n^3$.

Proof. Insert Eq. (20.129) and $|\psi_{n00}(0)|^2 = 1/(\pi n^3 a_0^3)$ into Eq. (20.133):

$$\Delta E_{nS} = \frac{1}{6}\cdot\frac{2\alpha}{\pi}\frac{1}{m^2}\ln\frac{\omega_{\max}}{\omega_{\min}}\cdot 4\pi\alpha\cdot\frac{1}{\pi n^3 a_0^3}. \tag{20.135}$$

The numerical factors combine as $\frac{1}{6}\cdot\frac{2\alpha}{\pi}\cdot 4\pi\alpha = \frac{4\alpha^2}{3}$, and with $a_0^{-3} = (\alpha m)^3 = \alpha^3 m^3$ the mass powers give $\frac{1}{m^2}\cdot\alpha^3 m^3 = \alpha^3 m$, so

$$\Delta E_{nS} = \frac{4\alpha^2}{3}\cdot\frac{\alpha^3 m}{\pi n^3}\,\ln\frac{\omega_{\max}}{\omega_{\min}} = \frac{4\alpha^5 m}{3\pi n^3}\,\ln\frac{\omega_{\max}}{\omega_{\min}}. \tag{20.136}$$

The density $|\psi_{n00}(0)|^2$ is verified for $n = 2$ from the radial function $R_{20}(0) = (2a_0)^{-3/2}\cdot 2$, giving $|\psi_{200}(0)|^2 = R_{20}(0)^2/4\pi = 1/(8\pi a_0^3)$, which is $1/(\pi n^3 a_0^3)$ at $n = 2$. □

The dependence on α is the signature of the effect: $\Delta E_{nS} \sim \alpha^5 m$, one power of α (the extra radiative loop) beyond the $\alpha^4 m$ fine structure of Eq. (20.122). The Lamb shift is smaller than the fine structure by exactly one factor of α, up to the logarithm — which is why it sits at $\sim 10^3$ MHz while the $2P$ fine structure sits at $\sim 10^4$ MHz.

20.5.4 The Bethe logarithm and the infrared cutoff

The estimate Eq. (20.134) is complete except for the cutoffs in the logarithm, and these are where the physics of the bound state — as opposed to the free electron — enters.

The *ultraviolet* cutoff $\omega_{\max}$ is the frequency above which the nonrelativistic, dipole-driven picture of Eq. (20.125) fails. When the field oscillates faster than the electron's Compton frequency, the response is no longer the simple $1/\omega^2$ of a slow charge; relativistic kinematics and pair creation intervene. The natural cutoff is therefore the electron mass,

$$\omega_{\max} \sim m. \tag{20.137}$$

In the full relativistic calculation this hard cutoff is replaced by the genuine high-energy behaviour of the self-energy $\Sigma(p)$ (Section 19.3); the divergence that Bethe cut off by hand is removed honestly by mass renormalization, and what survives is a finite logarithm with m in the numerator, vindicating Eq. (20.137).

The *infrared* cutoff $\omega_{\min}$ is more subtle, and it is Bethe's insight. The free-electron estimate would let $\omega_{\min} \to 0$, giving an infinite shift; but a bound electron cannot absorb and re-emit a photon of arbitrarily low energy without changing state, and the relevant low-frequency scale is set by the atomic transition energies themselves. A virtual photon of frequency far below a typical atomic excitation energy cannot resolve the orbital structure and decouples. The infrared cutoff is thus the characteristic atomic binding/excitation energy,

$$\omega_{\min} \sim \langle \Delta E \rangle \sim \alpha^2 m, \tag{20.138}$$

of order one Rydberg, $\mathrm{Ry} = \tfrac{1}{2}\alpha^2 m \approx 13.6\,\mathrm{eV}$. The ratio of cutoffs is therefore $\omega_{\max}/\omega_{\min} \sim m/(\alpha^2 m) = 1/\alpha^2$, and the logarithm is

$$\ln \frac{\omega_{\max}}{\omega_{\min}} \sim \ln \frac{1}{\alpha^2} = 2 \ln \frac{1}{\alpha} \approx 2 \ln(137) \approx 9.8. \tag{20.139}$$

Bethe's refinement replaces the crude $\omega_{\min}$ by a precise quantum-mechanical average over the atomic spectrum — the *Bethe logarithm*. For a state $|n\rangle$ the proper infrared scale is the weighted average excitation energy

$$\ln \langle \Delta E \rangle_n \equiv \frac{\sum_k |\langle k|\mathbf{p}|n\rangle|^2 \, (E_k - E_n) \, \ln |E_k - E_n|}{\sum_k |\langle k|\mathbf{p}|n\rangle|^2 \, (E_k - E_n)}, \tag{20.140}$$

the sum running over the complete set of intermediate atomic states reached by the dipole operator $\mathbf{p}$. Numerically, for the $2S$ state this average excitation energy is $\langle \Delta E \rangle_{2S} \approx 16.6\,\mathrm{Ry} \approx 226\,\mathrm{eV}$ — notably larger than a single Rydberg, because the dipole matrix elements weight the high-lying and continuum states heavily. With $\omega_{\max} = m$ and this $\omega_{\min}$,

$$\ln\frac{\omega_{\max}}{\omega_{\min}} = \ln\frac{mc^2}{16.6\,\mathrm{Ry}} = \ln\frac{0.511\times10^6\,\mathrm{eV}}{226\,\mathrm{eV}} = \ln(2.26\times10^3) \approx 7.7. \tag{20.141}$$

The crude estimate Eq. (20.139) overshoots; Bethe's average brings the logarithm down to ≈ 7.7. We now turn both numbers into a frequency.

Worked Example 20.12 (Welton's estimate of the $2S$ Lamb shift)**.** Evaluate Eq. (20.134) for the $2S$ state of hydrogen, restoring physical units only at the end, with both the crude cutoff ratio Eq. (20.139) and Bethe's logarithm Eq. (20.141).

At $n = 2$ the coefficient in Eq. (20.134) simplifies: $4/(3\pi\cdot 8) = 1/(6\pi)$, so

$$\Delta E_{2S} = \frac{\alpha^5 m}{6\pi}\,\ln\frac{\omega_{\max}}{\omega_{\min}}. \tag{20.142}$$

Compute the prefactor as an energy. With $\alpha = 1/137.036$,

$$\alpha^5 = (7.297\times10^{-3})^5 = 2.069\times10^{-11}, \tag{20.143}$$

and $mc^2 = 0.511\,\mathrm{MeV} = 5.110\times10^5\,\mathrm{eV}$, so

$$\frac{\alpha^5 mc^2}{6\pi} = \frac{2.069\times10^{-11}\times5.110\times10^5\,\mathrm{eV}}{18.85} = \frac{1.057\times10^{-5}\,\mathrm{eV}}{18.85} = 5.61\times10^{-7}\,\mathrm{eV}. \tag{20.144}$$

Converting energy to frequency through $\nu = \Delta E/h$ with $h = 4.136\times10^{-15}\,\mathrm{eV\,s}$ (equivalently $1\,\mathrm{eV} \leftrightarrow 2.418\times10^{14}\,\mathrm{Hz}$),

$$\frac{1}{h}\frac{\alpha^5 mc^2}{6\pi} = 5.61\times10^{-7}\,\mathrm{eV}\times2.418\times10^{14}\,\frac{\mathrm{Hz}}{\mathrm{eV}} = 1.36\times10^8\,\mathrm{Hz} = 136\,\mathrm{MHz}. \tag{20.145}$$

The $2S$ self-energy shift is therefore 136 MHz times the logarithm. With the crude $\ln \approx 9.8$ of Eq. (20.139),

$$\Delta\nu_{2S}^{(\mathrm{crude})} \approx 136\,\mathrm{MHz}\times9.8 \approx 1.3\times10^3\,\mathrm{MHz}, \tag{20.146}$$

already the right scale, about 25% high. With Bethe's logarithm $\ln \approx 7.7$ of Eq. (20.141),

$$\Delta\nu_{2S}^{(\mathrm{Bethe})} \approx 136\,\mathrm{MHz}\times7.7 \approx 1.05\times10^3\,\mathrm{MHz}. \tag{20.147}$$

The dominant radiative contribution alone, with the correct infrared scale, reproduces the measured 1058 MHz of Eq. (20.124) to within the few percent that the remaining two mechanisms supply. This is the calculation Bethe did on the train.

The Welton/Bethe estimate captures the self-energy term — the largest of the three, contributing about +1010 MHz once the relativistic corrections to the simple cutoff are included. We turn now to the two smaller pieces, which the same contact-interaction machinery delivers with their signs.

20.5.5 Vacuum polarization: the Uehling contribution

The second mechanism dresses not the electron but the photon it exchanges with the nucleus (Fig. 20.8b). The renormalized vacuum polarization $\Pi(q^2)$ of Section 19.2 modifies the photon propagator, and hence the static electron–nucleus potential. We need only its low-momentum behaviour. The on-shell-renormalized $\Pi(q^2)$ vanishes at $q^2 = 0$ (the photon stays massless and the charge is the measured one), and its leading Taylor coefficient, computed in Section 19.2, is

$$\Pi(q^2) = -\frac{\alpha}{15\pi}\,\frac{q^2}{m^2} + O\!\left(\frac{q^4}{m^4}\right). \tag{20.148}$$

The static potential between the electron and the point nucleus is the Fourier transform of the (corrected) photon exchange. For spacelike momentum transfer $q^2 = -\mathbf{q}^2$, the one-loop-corrected Coulomb potential is

$$V(\mathbf{r}) = -\int \frac{d^3q}{(2\pi)^3}\, e^{i\mathbf{q}\cdot\mathbf{r}}\,\frac{4\pi\alpha}{\mathbf{q}^2}\Big[1 + \Pi(-\mathbf{q}^2)\Big], \tag{20.149}$$

the leading 1 reproducing the Coulomb $-\alpha/r$ and the Π term giving the correction. Inserting Eq. (20.148) with $q^2 = -\mathbf{q}^2$, the factor $\Pi(-\mathbf{q}^2) = (\alpha/15\pi)(\mathbf{q}^2/m^2)$ exactly cancels the $1/\mathbf{q}^2$ of the propagator:

$$\delta V_{\mathrm{U}}(\mathbf{r}) = -\int \frac{d^3q}{(2\pi)^3}\, e^{i\mathbf{q}\cdot\mathbf{r}}\,\frac{4\pi\alpha}{\mathbf{q}^2}\cdot\frac{\alpha}{15\pi}\,\frac{\mathbf{q}^2}{m^2} = -\frac{4\alpha^2}{15m^2}\int \frac{d^3q}{(2\pi)^3}\, e^{i\mathbf{q}\cdot\mathbf{r}} = -\frac{4\alpha^2}{15m^2}\,\delta^3(\mathbf{r}). \tag{20.150}$$

The low-energy limit of vacuum polarization is itself a contact interaction — a delta function at the nucleus, exactly like the Welton term Eq. (20.131), and therefore acting on the same S-states. This is the leading piece of the Uehling potential [476]; the full Uehling potential has a Yukawa-like tail of range $1/m$ (the Compton wavelength), but for the slowly varying atomic wavefunction the contact approximation captures the dominant shift. Its expectation value is

$$\Delta E^{\mathrm{U}}_{nS} = \left\langle\, \delta V_{\mathrm{U}} \,\right\rangle = -\frac{4\alpha^2}{15m^2}\,|\psi_{n00}(0)|^2 = -\frac{4\alpha^2}{15m^2}\cdot\frac{\alpha^3 m^3}{\pi n^3} = -\frac{4\alpha^5 m}{15\pi n^3}. \tag{20.151}$$

Two facts stand out. First, the *sign is negative*: vacuum polarization pulls S-states down, opposite to the self-energy. Physically, the virtual e^+e^- pairs screen the nuclear charge, so that an electron penetrating to small r — as S-states do — sees a *larger* effective charge and is bound more tightly; deeper binding means lower energy. Second, the coefficient is a clean fraction of the self-energy

coefficient Eq. (20.134): comparing $4\alpha^5 m/(15\pi n^3)$ with the self-energy prefactor $4\alpha^5 m/(3\pi n^3)$ (i.e. before its logarithm), the Uehling term is exactly $\frac{1}{5}$ of the self-energy prefactor. Numerically, reusing the 136 MHz prefactor of Theorem 20.12 (which is the $n = 2$ value of $4\alpha^5 m/(3\pi n^3) \cdot \frac{1}{4}$, i.e. $\alpha^5 m/6\pi$), the Uehling shift is

$$\Delta\nu_{2S}^{\mathrm{U}} = -\frac{1}{5} \times 136\,\mathrm{MHz} \approx -27\,\mathrm{MHz}. \tag{20.152}$$

This -27 MHz is the famous small, negative vacuum-polarization contribution to the Lamb shift — the one place in the hydrogen spectrum where the screening of the vacuum shows up with a *minus* sign and a magnitude small enough to be a sensitive test, and large enough to be unmistakable.

20.5.6 The anomalous moment and the vertex contribution

The third mechanism is the renormalized vertex (Fig. 20.8c), and it brings back the star of the chapter. The Pauli form factor $F_2(0) = \alpha/2\pi$ established in Section 20.1 and Theorem 20.4 means the electron's magnetic moment is not $g = 2$ but $g = 2(1 + a_e)$ with $a_e = \alpha/2\pi$. In the bound state this excess moment alters the electron's interaction with the nuclear field in two ways, both captured by the nonrelativistic (Foldy–Wouthuysen) reduction of the corrected vertex.

The first is the *spin-orbit* coupling. The interaction of the electron's magnetic moment with the motional magnetic field of the nucleus, corrected by Thomas precession, produces the spin-orbit Hamiltonian whose coefficient is proportional to $(g - 1)$. The Dirac value $g = 2$ gives the standard spin-orbit term; the anomalous moment enhances it by the fraction

$$\frac{(g-1)\big|_{g=2(1+a_e)} - (g-1)\big|_{g=2}}{(g-1)\big|_{g=2}} = \frac{(1+2a_e)-1}{1} = 2a_e = \frac{\alpha}{\pi}. \tag{20.153}$$

This enhancement acts only on states with $\ell \neq 0$, the S-state spin-orbit coupling vanishing identically. For the $2P_{1/2}$ level the Dirac spin-orbit shift is, from the fine-structure bookkeeping of Eq. (20.122) resolved into its spin-orbit part, $\Delta E_{\mathrm{SO}}(2P_{1/2}) = -\alpha^4 m/48 \approx -7.3\,\mathrm{GHz}$, so its anomalous enhancement is

$$\delta E_{\mathrm{SO}}^{a_e}(2P_{1/2}) = 2a_e \cdot \Big(-\frac{\alpha^4 m}{48}\Big) = -\frac{\alpha}{\pi} \cdot \frac{\alpha^4 m}{48}, \tag{20.154}$$

numerically $2a_e \times (-7.3\,\mathrm{GHz}) = 2(1.16 \times 10^{-3}) \times (-7.3\,\mathrm{GHz}) \approx -17\,\mathrm{MHz}$. Because this *lowers* $2P_{1/2}$, it *raises* the splitting $\nu(2S_{1/2}) - \nu(2P_{1/2})$ by about $+17$ MHz.

The second is a *contact* (Darwin-like) interaction. The Foldy–Wouthuysen reduction of the F_2 vertex produces, in addition to the spin-orbit term, a term

proportional to $\nabla^2 V \propto \delta^3(\mathbf{r})$ — once again a contact interaction acting on the S-states, with strength set by a_e. This piece raises $2S_{1/2}$ and supplies the larger part of the vertex contribution. Carrying out the full reduction (which we do not reproduce here, the form factor itself having been established in Section 20.1) and adding the two pieces, the total anomalous-moment contribution to the $2S_{1/2}$–$2P_{1/2}$ splitting is

$$\Delta\nu^{\,a_e} \approx +68\,\text{MHz}, \tag{20.155}$$

with the same sign as the self-energy. The point of principle is the unity it exhibits: the identical number $F_2(0) = \alpha/2\pi$ that, read in free space, is the electron's anomalous moment, read in the hydrogen atom becomes a +68 MHz piece of the Lamb shift. One amplitude, two limits, two measured numbers.

20.5.7 The sum, and what it teaches

Adding the three radiative contributions — self-energy, vacuum polarization, and anomalous moment — gives the theoretical Lamb shift at this order of approximation:

$$\Delta\nu_{\text{Lamb}} \;\approx\; \underbrace{+1010}_{\text{self-energy }(\Sigma)} \quad \underbrace{-27}_{\text{vac. pol. }(\Pi)} \quad \underbrace{+68}_{\text{anom. moment }(F_2)} \;\text{MHz} \;\approx\; +1051\,\text{MHz}, \tag{20.156}$$

against the measured 1057.8 MHz of Eq. (20.124). The agreement of an independently-assembled sum of three one-loop effects with the experimental number, to better than one percent, is the quantitative triumph that established QED. Figure 20.9 displays the level structure: the Dirac degeneracy of $2S_{1/2}$ and $2P_{1/2}$, broken by the radiative corrections into the observed splitting, with the three signed contributions stacked.

Three lessons are worth fixing. First, the *selection rule* is structural, not accidental: every mechanism that contributes a contact interaction — self-energy via the smeared Coulomb curvature, vacuum polarization via the low-q^2 delta function, the vertex via its Darwin term — acts through $|\psi(0)|^2$ and therefore shifts S-states alone. A degeneracy between states of different ℓ is exactly what such interactions are equipped to break. Second, the *hierarchy* of the three terms is fixed by their analytic structure: the self-energy carries a large logarithm (the Bethe log) and dominates; the vacuum polarization and vertex are pure numbers without an enhancing logarithm and sit at the few-percent level. Third, the *signs* are physical: self-energy and anomalous moment push the S-state up, vacuum-polarization screening pulls it down, and the data demand exactly this combination.

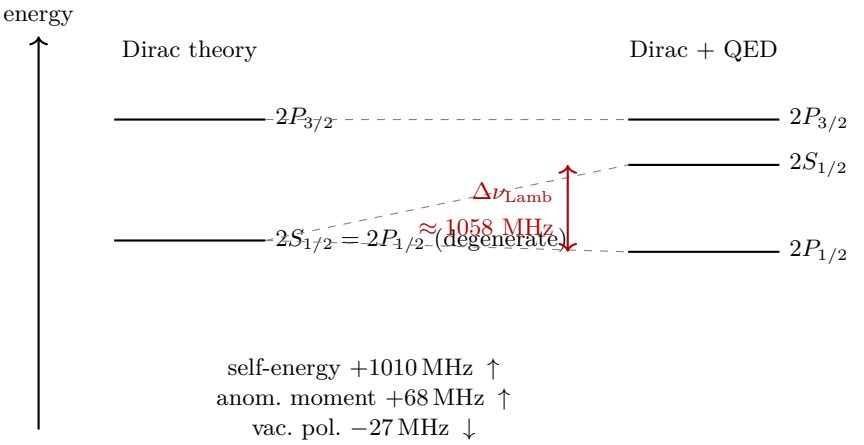


Figure 20.9: The $n = 2$ levels of hydrogen. In the Dirac theory (left) the states $2S_{1/2}$ and $2P_{1/2}$ are exactly degenerate (Theorem 20.10), split off from $2P_{3/2}$ by the fine structure. The radiative corrections of QED (right) lift the degeneracy, raising $2S_{1/2}$ above $2P_{1/2}$ by the Lamb shift $\approx 1058\,\text{MHz}$. The shift is the signed sum of the electron self-energy (+1010, up), the anomalous moment (+68, up), and vacuum polarization (−27, down).

Physical Insight: why the Lamb shift mattered

Before 1947 the infinities of QED could be dismissed as formal pathologies of an untested theory. The Lamb shift made them unavoidable: a finite, measured number could be reproduced only by computing the difference of two divergent self-energies and discarding the (cutoff-dependent) common part — that is, only by renormalizing. The Lamb shift is thus not merely a confirmation of QED but the experimental mandate for renormalization itself, the procedure the rest of Part VI builds into a systematic theory.

Of the three contributions, the vacuum polarization is the one whose full story is not yet told. Here it entered only through its low-q^2 contact limit Eq. (20.148), a single delta function at the nucleus. But $\Pi(q^2)$ is a function of all momenta, and its q^2-dependence has a meaning larger than this one number: it says that the strength of the electromagnetic interaction depends on the scale at which it is probed. The screening that here produced a $-27\,\text{MHz}$ shift is, at large momentum transfer, the running of the electric charge — the subject we take up next, resumming the same polarization to obtain the scale-dependent coupling $\alpha(q^2)$ and the short-distance Uehling potential of which Eq. (20.150) is the leading term.

20.6 The Running of the Fine-Structure Constant

The Lamb-shift analysis of Section 20.5 left one of the three one-loop functions with unfinished business. The anomalous moment and the self-energy each delivered a clean number and were done; vacuum polarization, by contrast, entered the Lamb shift only through its value very near $q^2 = 0$ — the Uehling term, a short-range deformation of the Coulomb field felt by the $2S$ electron. But $\Pi(q^2)$ is a function of q^2, and we used essentially none of that dependence. The function carries far more physics than the single number it contributed to hydrogen: read over the whole range of q^2, it says that the strength of the electromagnetic interaction is not a constant at all but depends on the scale at which it is probed. The electric charge *runs*. This section turns the q^2-dependence of Π into that statement, makes it quantitative, and exhibits it in two complementary pictures — a momentum-space effective coupling $\alpha(q^2)$ and a position-space modified potential, the Uehling potential — which are Fourier transforms of one another and which together justify the slogan that the QED vacuum is a polarizable dielectric that screens charge.

What we will *not* do here is derive the running systematically. The clean engine for that is the renormalization group, and it belongs to Chapter 24; there the same result drops out of a one-line beta function. The purpose of the present section is physical rather than formal: to show that the running is already contained, in full, in the one-loop two-point function the reader computed in Chapter 19, and that summing a single geometric series extracts it. The agreement between this resummation and the renormalization-group derivation of Chapter 24 is the one forward reference of QED, and we flag it explicitly where it occurs.

20.6.1 Resumming the vacuum-polarization insertions

The free photon propagator was quantized in Chapter 13; in the R_ξ family of gauges it reads

$$D^{(0)}_{\mu\nu}(q) = \frac{-i}{q^2 + i\epsilon}\left(\eta_{\mu\nu} - (1-\xi)\frac{q_\mu q_\nu}{q^2}\right), \tag{20.157}$$

with $\xi = 1$ (Feynman gauge) our default for explicit work. The one-loop computation of Chapter 19 produced the photon's one-particle-irreducible (1PI) self-energy. By the Ward–Takahashi identity proved there (Theorem 19.13), the self-energy is transverse, so it carries the unique tensor structure

$$i\Pi^{\mu\nu}(q) = i\left(q^2\eta^{\mu\nu} - q^\mu q^\nu\right)\Pi(q^2), \tag{20.158}$$

the import Eq. (19.49), with the scalar form factor $\Pi(q^2)$ computed at one loop in Eq. (19.57). Transversality is not a technicality here: it is precisely what will protect the photon's masslessness and the structure of the gauge-dependent part of the propagator through the resummation, so we keep it manifest.

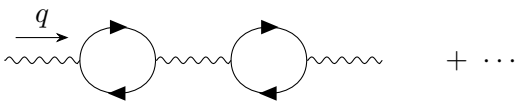


Figure 20.10: The geometric chain of one-particle-irreducible vacuum-polarization insertions that dresses the bare photon line. Each fermion loop is one factor of $i\Pi^{\mu\nu}(q)$ from Eq. (20.158); summing the chain over $0, 1, 2, \ldots$ insertions produces the full propagator Eq. (20.163) and the effective charge Eq. (20.167). This is the diagrammatic content of the running coupling.

The physical (full) photon propagator is the geometric sum of the tree line dressed by any number of 1PI insertions strung in series, the chain of bubbles drawn in Fig. 20.10:

$$D_{\mu\nu}(q) = D^{(0)}_{\mu\nu} + D^{(0)}_{\mu\alpha}\,(i\Pi^{\alpha\beta})\,D^{(0)}_{\beta\nu} + D^{(0)}_{\mu\alpha}\,(i\Pi^{\alpha\beta})\,D^{(0)}_{\beta\gamma}\,(i\Pi^{\gamma\delta})\,D^{(0)}_{\delta\nu} + \cdots . \tag{20.159}$$

That the full propagator is exactly this sum is the definition of one-particle irreducibility: every photon-line correction either is 1PI (one bubble) or factors at some internal photon line into a product of 1PI pieces (a longer chain), and Eq. (20.159) enumerates the chains without double counting.

To sum the series we exploit the projector algebra that Eq. (20.158) makes natural. Write the transverse and longitudinal projectors

$$P^{\mu\nu} = \eta^{\mu\nu} - \frac{q^\mu q^\nu}{q^2}, \qquad L^{\mu\nu} = \frac{q^\mu q^\nu}{q^2}, \tag{20.160}$$

which satisfy $P + L = \eta$, $P^2 = P$, $L^2 = L$, and $PL = LP = 0$ (as matrices with one index lowered). In this notation the self-energy is purely transverse, $\Pi^{\mu\nu} = q^2\,\Pi(q^2)\,P^{\mu\nu}$, and the tree propagator splits as

$$D^{(0)}_{\mu\nu} = \frac{-i}{q^2}(P_{\mu\nu} + \xi\,L_{\mu\nu}), \tag{20.161}$$

since $\eta_{\mu\nu} - (1-\xi)L_{\mu\nu} = P_{\mu\nu} + L_{\mu\nu} - (1-\xi)L_{\mu\nu} = P_{\mu\nu} + \xi L_{\mu\nu}$. Now evaluate a single insertion. Because $\Pi^{\mu\nu} \propto P$, and P annihilates L while it is idempotent on itself, the longitudinal piece of every adjacent propagator is killed:

$$D^{(0)}\,(i\Pi)\,D^{(0)} = \frac{-i}{q^2}(P + \xi L)\,(i\,q^2\Pi\,P)\,\frac{-i}{q^2}(P + \xi L) = \frac{-i}{q^2}\,\Pi\,P, \tag{20.162}$$

where the scalar prefactors combine as $(-i/q^2)(i\,q^2)(-i/q^2) = -i/q^2$ and the tensor product collapses by $(P+\xi L)\,P\,(P+\xi L) = P$. The result is striking: one insertion reproduces the transverse part of the tree propagator multiplied by the scalar Π, and the gauge-dependent longitudinal part is untouched. Every further insertion repeats this, simply multiplying the transverse part by another factor

of Π. The series Eq. (20.159) therefore organizes into a geometric series on the transverse sector and an inert longitudinal sector:

$$D_{\mu\nu}(q) = \frac{-i}{q^2}\,P_{\mu\nu}\,(1+\Pi+\Pi^2+\cdots) \;-\; \frac{i\xi}{q^2}\,L_{\mu\nu} = \frac{-i\,P_{\mu\nu}}{q^2\,[\,1-\Pi(q^2)\,]} \;-\; \frac{i\xi}{q^2}\,L_{\mu\nu}. \tag{20.163}$$

Written out,

$$D_{\mu\nu}(q) = \frac{-i}{q^2\,[\,1-\Pi(q^2)\,]}\left(\eta_{\mu\nu}-\frac{q_\mu q_\nu}{q^2}\right) \;-\; \frac{i\xi}{q^2}\,\frac{q_\mu q_\nu}{q^2}. \tag{20.164}$$

Two structural facts deserve emphasis. First, the entire effect of the loops is the single replacement $q^2 \to q^2[1-\Pi(q^2)]$ in the transverse part. The pole of the propagator stays at $q^2=0$ provided $\Pi(0)$ is finite, so the photon remains exactly massless — the transversality of Eq. (20.158), enforced by gauge invariance, is what guarantees this. Second, the gauge-dependent longitudinal term is completely undressed; the loops never generate a $q_\mu q_\nu$ correction. This is consistent with the physical principle that gauge artifacts cannot acquire dynamics, and it means the entire content of the resummation lives in the gauge-invariant transverse scalar $1-\Pi(q^2)$.

20.6.2 The effective charge

The propagator never appears alone; in any amplitude it is sewn between two electromagnetic vertices, i.e. between two factors of the conserved current $j^\mu = \bar\psi\gamma^\mu\psi$. Current conservation $q_\mu j^\mu = 0$ kills every $q_\mu q_\nu$ term in Eq. (20.164) when contracted into external lines, exactly as it does at tree level. The gauge-dependent part therefore drops out of all physical matrix elements, and the photon exchange between charges is governed by the gauge-invariant object

$$\frac{-i\,e^2\,\eta_{\mu\nu}}{q^2} \;\longrightarrow\; \frac{-i\,e^2\,\eta_{\mu\nu}}{q^2\,[\,1-\Pi(q^2)\,]} = \frac{-i\,e^2_{\rm eff}(q^2)\,\eta_{\mu\nu}}{q^2}, \qquad e^2_{\rm eff}(q^2) \equiv \frac{e^2}{1-\Pi(q^2)}. \tag{20.165}$$

The loops have done nothing to the kinematic $1/q^2$ pole of one-photon exchange; they have promoted the coupling e^2 to a q^2-dependent effective coupling. This is the seed of the running.

There is one renormalization to perform before the statement is physical. The scalar $\Pi(q^2)$ computed in Chapter 19 is ultraviolet divergent; its divergence sits entirely in the constant $\Pi(0)$, with the q^2-dependence finite. We split

$$\Pi(q^2) = \Pi(0) + \widehat{\Pi}(q^2), \qquad \widehat{\Pi}(0) = 0, \tag{20.166}$$

and absorb the (divergent, but q^2-independent) constant $\Pi(0)$ into the definition of the physical charge. The natural, scheme-independent way to fix that definition is by experiment in the long-wavelength limit: the electric charge *measured* in

a macroscopic ($q^2 \to 0$) experiment — Thomson scattering, Coulomb's law, Millikan's drop — is by definition the physical e, and that is what $\alpha = e^2/4\pi \approx 1/137.036$ denotes. This is the on-shell renormalization condition for the charge: α is the effective coupling at $q^2 = 0$. In terms of the bare coupling it means $e^2 \equiv e_0^2/[1-\Pi(0)] = Z_3\, e_0^2$, and substituting into Eq. (20.165) and Eq. (20.166) the divergent constant cancels, leaving a finite effective coupling expressed through the *subtracted* vacuum polarization $\widehat{\Pi}$ alone:

$$\boxed{\alpha_{\text{eff}}(q^2) \;=\; \frac{\alpha}{1-\widehat{\Pi}(q^2)}, \qquad \widehat{\Pi}(0) = 0.} \tag{20.167}$$

We record this as the central output of the resummation.

Proposition 20.13 (Effective coupling from resummed vacuum polarization)**.** *Summing all 1PI vacuum-polarization insertions into the photon propagator, and renormalizing the electric charge on shell so that the effective coupling at zero momentum equals the measured fine-structure constant α, replaces the tree exchange between conserved currents by an exchange with the q^2-dependent coupling Eq. (20.167), where $\widehat{\Pi}(q^2) = \Pi(q^2) - \Pi(0)$ is the finite, on-shell-subtracted scalar vacuum polarization. The photon remains massless and the result is gauge invariant.*

The condition $\widehat{\Pi}(0) = 0$ is precisely the Thomson normalization: $\alpha_{\text{eff}}(0) = \alpha$. As $|q^2|$ grows, $\widehat{\Pi}(q^2)$ moves away from zero, and the coupling drifts away from its low-energy value. To see in which direction and by how much, we now feed in the explicit one-loop function.

20.6.3 Leading-log running

The on-shell-subtracted one-loop vacuum polarization computed in Chapter 19 is, in the conventions of this book,

$$\widehat{\Pi}(q^2) = -\frac{2\alpha}{\pi}\int_0^1 dx\, x(1-x)\, \ln\frac{m^2}{m^2 - x(1-x)q^2}, \tag{20.168}$$

the import Eq. (19.57), written here for spacelike $q^2 < 0$ (and below pair threshold for timelike q^2), where the logarithm is real. For an electron of mass $m = m_e$ the function is manifestly normalized to $\widehat{\Pi}(0) = 0$, and for $q^2 < 0$ the argument of the logarithm is less than one, so $\widehat{\Pi}(q^2) > 0$: the denominator in Eq. (20.167) drops below unity and the effective coupling grows. The vacuum strengthens the interaction at short distance.

Consider the deep-spacelike regime $-q^2 \equiv Q^2 \gg m^2$, the relevant one for high-energy scattering. There $m^2 - x(1-x)q^2 = m^2 + x(1-x)Q^2 \simeq x(1-x)Q^2$ except near the endpoints, and the logarithm becomes $\ln[m^2/(x(1-x)Q^2)] =$

$-\ln(Q^2/m^2) - \ln[x(1-x)]$. The constant piece integrates against the weight $x(1-x)$ using the two moments

$$\int_0^1 dx\, x(1-x) = \frac{1}{6}, \qquad \int_0^1 dx\, x(1-x)\ln[x(1-x)] = -\frac{5}{18}, \tag{20.169}$$

of which the first produces the logarithm and the second the finite constant. Writing $\ln[m^2/(x(1-x)Q^2)] = -\ln(Q^2/m^2) - \ln[x(1-x)]$ and integrating term by term,

$$\widehat{\Pi}(q^2) = \frac{2\alpha}{\pi}\left[\frac{1}{6}\ln\frac{Q^2}{m^2} - \frac{5}{18}\right] + O\!\left(\frac{m^2}{Q^2}\right) = \frac{\alpha}{3\pi}\left[\ln\frac{-q^2}{m^2} - \frac{5}{3}\right] + O\!\left(\frac{m^2}{q^2}\right), \tag{20.170}$$

the standard logarithm with its scheme-fixing constant $-5/3$. (We keep the constant because it is the one piece of finite information beyond the logarithm, and it shifts the effective scale at which the running is referenced.) Inserting Eq. (20.170) into Eq. (20.167) and keeping the dominant logarithm gives the celebrated leading-log running of the fine-structure constant,

$$\boxed{\alpha(q^2) \;=\; \frac{\alpha}{1 - \dfrac{\alpha}{3\pi}\ln\dfrac{-q^2}{m^2}}, \qquad -q^2 \gg m^2.} \tag{20.171}$$

The coupling increases with $-q^2$, slowly (the prefactor $\alpha/3\pi \approx 7.7\times10^{-4}$ is tiny), but inexorably and without bound until the denominator vanishes — the QED Landau pole, an exponentially distant scale we will not reach and whose interpretation belongs to Chapter 24. Figure 20.11 plots both the exact one-loop $\alpha_{\rm eff}(q^2)/\alpha$ from Eq. (20.168) and the leading-log approximation Eq. (20.171), alongside the imaginary part of $\widehat{\Pi}$ that will occupy us in Section 20.6.5.

It is worth pausing on the physical content before computing numbers. Equation Eq. (20.171) says the electron's charge is not a fixed attribute but depends on how closely you look. The next subsection makes this literal in position space.

20.6.4 The Uehling potential and the screening picture

The momentum-space effective coupling has a position-space avatar: the static potential energy between two charges, which is the (three-dimensional) Fourier transform of the dressed one-photon exchange in the static limit $q^0 = 0$, $q^2 = -|\mathbf{q}|^2$. At tree level this reproduces Coulomb's law,

$$V_0(r) = -e^2\int\frac{d^3q}{(2\pi)^3}\,\frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{|\mathbf{q}|^2} = -\frac{e^2}{4\pi r} = -\frac{\alpha}{r}, \tag{20.172}$$

for an electron in the field of a unit positive charge. Dressing the exchange with vacuum polarization, i.e. inserting $1/[1-\widehat{\Pi}(-|\mathbf{q}|^2)]$, gives the full static potential.

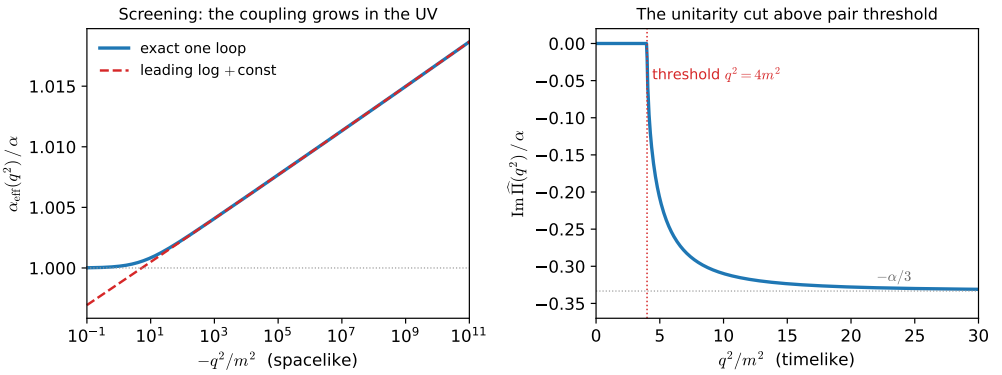


Figure 20.11: Left: the spacelike effective coupling $\alpha_{\rm eff}(q^2)/\alpha = [1 - \widehat{\Pi}(q^2)]^{-1}$ from the exact one-loop Feynman-parameter integral Eq. (20.168) (solid) versus the leading-log-plus-constant approximation Eq. (20.170) (dashed). The coupling grows toward the ultraviolet: charge screening seen in momentum space. A cartoon-large α is used for vertical visibility; the physical effect is the same shape at 1/137 strength. Right: $\mathrm{Im}\,\widehat{\Pi}(q^2)$ on the timelike axis from Eq. (20.178), vanishing below pair threshold $q^2 = 4m^2$, switching on with a velocity factor there, and approaching the plateau $-\alpha/3$ — the unitarity cut demanded by the optical theorem.

To first order in α we expand $1/[1 - \widehat{\Pi}] \simeq 1 + \widehat{\Pi}$ and read off the correction

$$\delta V(r) = -e^2 \int \frac{d^3q}{(2\pi)^3}\, e^{i\mathbf{q}\cdot\mathbf{r}}\, \frac{\widehat{\Pi}(-|\mathbf{q}|^2)}{|\mathbf{q}|^2}. \tag{20.173}$$

This is the Uehling potential. To evaluate it we use the analytic structure of $\widehat{\Pi}$, which is the natural bridge to the dispersive language of Section 20.6.5. The function $\widehat{\Pi}(q^2)$ is analytic in the complex q^2-plane cut along the timelike axis $q^2 \geq 4m^2$, the branch point being the threshold for producing a real e^+e^- pair; its boundary value across the cut is its imaginary part. The once-subtracted dispersion relation that follows from this analyticity and from $\widehat{\Pi}(0) = 0$ (the analyticity and subtraction technology are developed in Chapter 5) is

$$\widehat{\Pi}(q^2) = \frac{q^2}{\pi} \int_{4m^2}^{\infty} ds\, \frac{\mathrm{Im}\,\widehat{\Pi}(s)}{s\,(s - q^2 - i\epsilon)}. \tag{20.174}$$

Evaluated at spacelike $q^2 = -|\mathbf{q}|^2$, this gives $\widehat{\Pi}(-|\mathbf{q}|^2)/|\mathbf{q}|^2 = -(1/\pi)\int ds\, \mathrm{Im}\widehat{\Pi}(s)/[s(s+|\mathbf{q}|^2)]$, a manifestly negative-definite ratio (recall $\mathrm{Im}\,\widehat{\Pi} < 0$ in our convention, derived below), as it must be for $\widehat{\Pi}$ to be positive at spacelike momentum. The virtue of Eq. (20.174) is that it turns the photon's dressed propagator into a continuous superposition of *massive* Yukawa propagators $1/(s + |\mathbf{q}|^2)$, each of which Fourier-transforms elementarily. We carry the computation out in full.

Worked Example 20.14 (The Uehling potential from the spectral representation)**.** Insert Eq. (20.174) at $q^2 = -|\mathbf{q}|^2$ into Eq. (20.173) and exchange the orders of the s- and $\mathbf{q}$-integrals:

$$\delta V(r) = \frac{e^2}{\pi}\int_{4m^2} ds\, \frac{\operatorname{Im}\widehat{\Pi}(s)}{s} \int \frac{d^3q}{(2\pi)^3}\, \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{s+|\mathbf{q}|^2}. \tag{20.175}$$

The inner integral is the standard Yukawa transform,

$$\int \frac{d^3q}{(2\pi)^3}\, \frac{e^{i\mathbf{q}\cdot\mathbf{r}}}{s+|\mathbf{q}|^2} = \frac{e^{-\sqrt{s}\,r}}{4\pi r}, \tag{20.176}$$

obtained by doing the angular integral ($\int d\Omega\, e^{i\mathbf{q}\cdot\mathbf{r}} = 4\pi \sin(qr)/(qr)$) and then the radial one by residues. Hence, using $e^2/(4\pi) = \alpha$,

$$\delta V(r) = \frac{\alpha}{\pi r}\int_{4m^2} ds\, \frac{\operatorname{Im}\widehat{\Pi}(s)}{s}\, e^{-\sqrt{s}\,r}. \tag{20.177}$$

The potential is a superposition of Yukawa tails of range $1/\sqrt{s}$, weighted by the spectral density: the closer s is to threshold $4m^2$, the longer the range, so the longest range present is $1/(2m)$, one electron Compton wavelength. Now insert the one-loop spectral density. Taking the discontinuity of Eq. (20.168) across the cut — where, with the $i\epsilon$ prescription $q^2 \to q^2 + i\epsilon$, the logarithm of the now-negative argument $m^2 - x(1-x)q^2$ acquires $-i\pi$ over the range of x for which the argument is negative — one finds

$$\operatorname{Im}\widehat{\Pi}(s) = -\frac{\alpha}{3}\left(1+\frac{2m^2}{s}\right)\sqrt{1-\frac{4m^2}{s}}, \qquad s > 4m^2, \tag{20.178}$$

and zero below threshold. (The same function follows, more physically, from the optical theorem and the tree cross section for $\gamma^* \to e^+e^-$, the subject of Section 20.6.5.) Substitute Eq. (20.178) into Eq. (20.177) and change variables to $s = 4m^2t^2$, $t \in [1,\infty)$, so that $\sqrt{s} = 2mt$, $ds = 8m^2t\,dt$, $1 + 2m^2/s = 1 + 1/(2t^2)$, $\sqrt{1-4m^2/s} = \sqrt{t^2-1}/t$, and $1/s = 1/(4m^2t^2)$. The accumulated factors give

$$\frac{ds}{s}\operatorname{Im}\widehat{\Pi}(s) = -\frac{2\alpha}{3}\left(1+\frac{1}{2t^2}\right)\frac{\sqrt{t^2-1}}{t^2}\,dt, \tag{20.179}$$

and therefore the Uehling potential is

$$\boxed{\delta V(r) = -\frac{2\alpha^2}{3\pi r}\int_1^\infty dt\, e^{-2mrt}\left(1+\frac{1}{2t^2}\right)\frac{\sqrt{t^2-1}}{t^2}.} \tag{20.180}$$

The correction is everywhere negative: it *deepens* the Coulomb well at short range, exactly the statement that the bare charge revealed at short distance is larger than the screened charge seen from afar.

The two limits of Eq. (20.180) make the screening picture quantitative, and we extract both because each teaches something.

Long range, $mr \gg 1$. The exponential e^{-2mrt} forces the integral to its lower endpoint $t = 1$. Writing $t = 1 + u$, using $\sqrt{t^2-1} \simeq \sqrt{2u}$ and $(1 + 1/2t^2) \simeq 3/2$ for small u, the integral becomes $\frac{3}{2}\sqrt{2}\,e^{-2mr}\int_0^\infty du\,\sqrt{u}\,e^{-2mru} = \frac{3}{2}\sqrt{2}\,e^{-2mr}\,\Gamma(\frac{3}{2})/(2mr)^{3/2}$, which after $\Gamma(\frac{3}{2}) = \sqrt{\pi}/2$ collapses to $(3\sqrt{\pi}/8)(mr)^{-3/2}e^{-2mr}$. Hence

$$\delta V(r) \simeq -\frac{\alpha}{r}\,\frac{\alpha}{4\sqrt{\pi}}\,\frac{e^{-2mr}}{(mr)^{3/2}}, \qquad mr \gg 1. \tag{20.181}$$

The correction is exponentially small beyond the Compton wavelength $1/m \approx 386\,\text{fm}$: the vacuum polarization is a strictly short-range effect, screened off at distances large compared with the size of the virtual pairs. This is why it contributes to the Lamb shift only through S-states, which alone have appreciable amplitude at the origin (Section 20.5).

Short range, $mr \ll 1$. Now the exponential is ineffective until $t \sim 1/(2mr) \gg 1$, so the integral is controlled by large t, where the integrand behaves as e^{-2mrt}/t. The leading behaviour is logarithmic; carrying the constant by splitting $g(t) \equiv (1 + 1/2t^2)\sqrt{t^2-1}/t^2 = 1/t + [g(t) - 1/t]$ and using $\int_1^\infty e^{-2mrt}/t\,dt = -\gamma_E - \ln(2mr) + O(mr)$ together with the elementary constant $\int_1^\infty [g(t) - 1/t]\,dt = \ln 2 - \frac{5}{6}$ (evaluated from the antiderivative $\ln(t + \sqrt{t^2-1}) - \sqrt{t^2-1}/t + (t^2-1)^{3/2}/6t^3$), the two logarithms of 2 cancel and

$$\delta V(r) \simeq -\frac{2\alpha^2}{3\pi r}\left[\ln\frac{1}{mr} - \gamma_E - \frac{5}{6}\right], \qquad mr \ll 1. \tag{20.182}$$

This is the position-space face of the running. The total short-range potential is $V(r) \simeq -(\alpha/r)\{1 + (2\alpha/3\pi)[\ln(1/mr) - \gamma_E - \frac{5}{6}]\} = -\alpha(r)/r$ with a distance-dependent coupling $\alpha(r) = \alpha[1 + (2\alpha/3\pi)\ln(1/mr) + \cdots]$, the Fourier-space partner of Eq. (20.171) under the identification $-q^2 \sim 1/r^2$ (which turns $\ln(-q^2/m^2)$ into $2\ln(1/mr)$, reproducing the $2\alpha/3\pi$ coefficient). The momentum-space running and the position-space deepening of the Coulomb law are one and the same phenomenon.

There is one more limit worth recording, because it is the number that fed the Lamb shift. The *volume integral* of the Uehling potential is its momentum-space value at $\mathbf{q} \to 0$, where $\widehat{\Pi}(-|\mathbf{q}|^2) \to -(\alpha/15\pi)(|\mathbf{q}|^2/m^2)$ (the small-q^2 expansion of Eq. (20.168), using $\int_0^1 x^2(1-x)^2\,dx = 1/30$). The factor $|\mathbf{q}|^2$ cancels the Coulomb $1/|\mathbf{q}|^2$, leaving a constant Fourier transform: a contact term,

$$\delta V(r) \xrightarrow[mr \ll 1 \text{ (on atomic scales)}]{} -\frac{4\alpha^2}{15\,m^2}\,\delta^{(3)}(\mathbf{r}), \tag{20.183}$$

since on the scale of the Bohr radius $a_0 = 1/(\alpha m) \approx 137/m$ the Uehling range $1/2m$ is point-like. This is precisely the form used in Section 20.5: it shifts only S-states, by an amount $-\frac{4\alpha^2}{15m^2}|\psi_{nS}(0)|^2$. Figure 20.12 shows how the exact

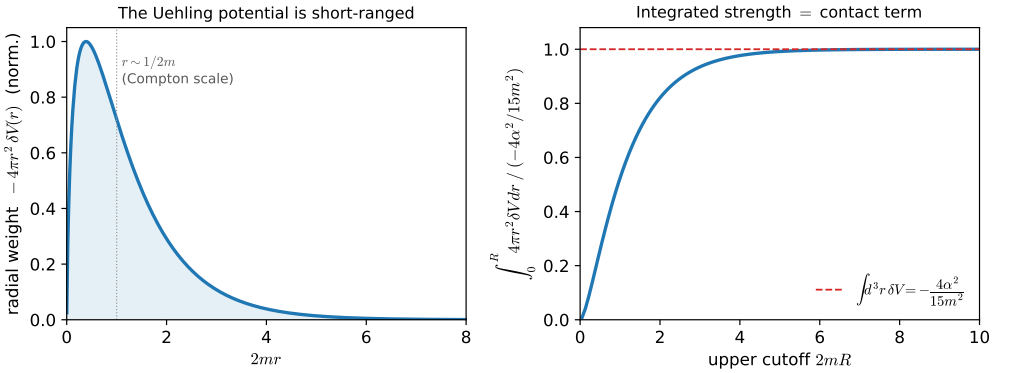


Figure 20.12: Left: the radial weight $-4\pi r^2\,\delta V(r)$ of the exact Uehling potential Eq. (20.180) versus $2mr$, showing that the deformation of the Coulomb field is concentrated within an electron Compton wavelength $1/2m$. Right: the running volume integral $\int_0^R 4\pi r^2\,\delta V\,dr$ approaching the contact-term value $-4\alpha^2/15m^2$ of Eq. (20.183) (dashed) within a few Compton wavelengths — the justification, used in the Lamb-shift section, for replacing δV by a delta function on atomic scales.

potential's volume integral saturates this contact value within a few Compton wavelengths, validating the delta-function replacement.

Physical Insight

The vacuum as a dielectric. A bare charge placed in the QED vacuum polarizes the cloud of virtual e^+e^- pairs around it: the virtual positrons are attracted inward, the virtual electrons pushed outward, so the charge is partially shielded by a co-moving screen of opposite charge, exactly as a test charge in a dielectric medium is shielded by induced polarization. A probe at large distance (small $-q^2$) sees the fully screened charge $-e$; a probe that penetrates the cloud (large $-q^2$, short r) sees more of the bare charge, hence a *larger* effective coupling. The sign is fixed by the matter content: fermion loops screen, which is why α grows in the ultraviolet. The contrast with non-abelian gauge theory, where gauge-boson loops *anti*-screen and the coupling shrinks in the ultraviolet (asymptotic freedom, Chapter 27), is the single most consequential sign in particle physics, and it has exactly this dielectric origin — with the gluon self-coupling providing a paramagnetic contribution of the opposite sign.

20.6.5 The dispersive representation and the measured running

The dispersion relation Eq. (20.174) is more than a computational device; it is the statement that the running coupling is determined by physical cross sections. The imaginary part $\mathrm{Im}\,\widehat{\Pi}(s)$ is fixed by unitarity: by the optical theorem, the discontinuity of the photon self-energy across the cut at $s = q^2 > 0$ is proportional to the total cross section for the photon to produce real charged matter, $e^+e^- \to$ (anything) through a single virtual photon. For the one-fermion theory this is the tree process $\gamma^* \to f\bar{f}$, and the velocity-weighted phase space reproduces Eq. (20.178) directly. Quantitatively, defining the ratio

$$R(s) \equiv \left.\frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)}\right|_{\text{point}}, \qquad \mathrm{Im}\,\widehat{\Pi}(s) = -\frac{\alpha}{3}\,R(s), \tag{20.184}$$

the spectral density of the full hadronic vacuum polarization is nothing but the measured R-ratio. Feeding this back through the spacelike form of Eq. (20.174) gives the hadronic contribution to the running as a weighted integral over data,

$$\Delta\alpha_{\text{had}}(q^2) = -\frac{\alpha\, q^2}{3\pi}\,\mathrm{P}\!\int_{s_0}^{\infty} ds\,\frac{R(s)}{s\,(s-q^2)}, \tag{20.185}$$

where $\Delta\alpha \equiv \widehat{\Pi}$ is the conventional name for the subtracted polarization in this context, s_0 is the lowest hadronic threshold, and P denotes the principal value for timelike q^2. This is the same experimental input — the cross section $e^+e^- \to$ hadrons as a function of energy — that drives the hadronic vacuum-polarization (HVP) contribution to the muon anomalous moment discussed in this chapter; the two differ only in the kinematic kernel multiplying $R(s)$. The systematic treatment of the hadronic part, where perturbation theory fails and the dispersive (or lattice) input is mandatory, belongs to Chapter 28 and the precision-frontier discussion of Chapter 34; here we only register that the running of α and the muon HVP are fed by one and the same data.

With this we can state the measured running honestly. Decompose

$$\alpha(q^2) = \frac{\alpha}{1 - \Delta\alpha(q^2)}, \qquad \Delta\alpha = \Delta\alpha_{\text{lep}} + \Delta\alpha_{\text{had}} + \Delta\alpha_{\text{top}}. \tag{20.186}$$

The leptonic part is pure QED and computable from Eq. (20.170) fermion by fermion, each lepton contributing once $-q^2$ exceeds its mass (the thresholds smooth the naive logarithm). The hadronic part is Eq. (20.185). Evaluated at the Z mass, the standard results as of the mid-2020s are $\Delta\alpha_{\text{lep}}(M_Z^2) \approx 0.0315$ and $\Delta\alpha^{(5)}_{\text{had}}(M_Z^2) \approx 0.0276$ (five-flavour, from e^+e^- data; the top contributes negligibly at this scale), giving $\Delta\alpha(M_Z^2) \approx 0.059$ and

$$\alpha^{-1}(M_Z^2) \;=\; \alpha^{-1}\big[1 - \Delta\alpha(M_Z^2)\big] \;\approx\; 137.036 \times 0.941 \;\approx\; 128.95, \tag{20.187}$$

the famous statement that the fine-structure constant has run from 1/137 at zero momentum to about 1/128 at the electroweak scale. The hadronic uncertainty in $\Delta\alpha^{(5)}_{\rm had}$ is, as of this writing, the dominant theoretical limitation both on $\alpha(M_Z)$ and on the Standard-Model prediction for a_μ — a live frontier, not a closed book.

Worked Example 20.15 (The leading-log running from m_e to M_Z, and why thresholds matter)**.** Take only the electron loop and the clean leading log Eq. (20.171), with $\mu_0 = m_e = 0.511\,\text{MeV}$ and $M_Z = 91.19\,\text{GeV}$. The logarithm is

$$\ln\frac{M_Z^2}{m_e^2} = 2\ln\frac{91.19\times10^3}{0.511} = 2\ln(1.784\times10^5) = 2(12.09) = 24.19, \qquad (20.188)$$

so $\widehat{\Pi} = \dfrac{\alpha}{3\pi}\times 24.19 = 7.74\times10^{-4}\times24.19 = 0.0187$, and the electron alone gives $\alpha^{-1}(M_Z^2) \approx 137.036\times(1-0.0187) = 134.5$. The electron thus accounts for only about a third of the measured shift to 128.95; the muon and tau (each adding a similar logarithm above its own threshold) and, decisively, the hadrons (the five active quark flavours, charge-squared weighted and colour-counted) supply the rest. This is exactly why the naive single-logarithm formula Eq. (20.171) cannot be pushed across many thresholds with a single mass m: the correct object is the sum $\widehat{\Pi} = \sum_f Q_f^2 N_c^{(f)}\,\widehat{\Pi}_f(q^2)$ over all fermions light enough to be pair-produced at scale $-q^2$, with the hadronic quark sum replaced by the data integral Eq. (20.185) because quark masses and confinement make the perturbative logarithm unreliable. The leading log is the right idea; the measured number requires the thresholds and the hadronic dispersion integral.

20.6.6 Handoff to the renormalization group

We have obtained the running coupling physically: from the geometric resummation of vacuum-polarization bubbles, renormalized by the Thomson condition, and read off from the imported one-loop $\widehat{\Pi}$. The same closed form Eq. (20.171) is the leading-log limit of the renormalization-group running coupling $\alpha(q^2) = \alpha(\mu_0^2)/[1-(\alpha/3\pi)\ln(q^2/\mu_0^2)]$ that Chapter 24 will derive systematically by integrating the QED beta function $\beta_\alpha = 2\alpha^2/3\pi$ (Eqs. (24.215) and (24.229)); there the bubble chain of Fig. 20.10 reappears as the diagrammatic content of "solving the RG," and the two derivations are shown to agree term by term. With that the running of the coupling passes from the resummed-loop picture established here to the scale-evolution machinery of Chapter 24, the hadronic input passes to Chapters 28 and 34, and the Ch. 20 stage of QED — $g-2$, the Lamb shift, and the running of α, all read off from the one-loop functions of Chapter 19 — is complete.

Further reading

Schwinger's original calculation of the anomalous moment is [433]; the textbook treatments closest in spirit to the one given here are Peskin and Schroeder [396], Schwartz [432], and Weinberg [500]. The multiloop QED program that produces the universal coefficients $A_1^{(2n)}$ is documented from its two-loop origins [400, 451] through the analytic three-loop result [315], the high-precision four-loop evaluation [316], and the five-loop computation [30]. The use of the electron anomaly as the world's best fine-structure constant rests on the Penning-trap measurement [183] confronted with the atom-recoil determinations on caesium [381] and rubidium [352]. For the muon anomaly — its experimental history [61, 354] and the evolving Standard-Model prediction [14, 28, 86, 273] — the standard monograph is Jegerlehner [285], which also develops the dispersive HVP machinery in depth. The Lamb shift is best approached through the original measurement [311] and Bethe's first estimate [75], with Welton's heuristic [504] and Uehling's potential [476] supplying the physical pictures used above.

Part VI

Renormalization and the Renormalization Group

Chapter 21

The Origin of Divergences

The loop diagrams assembled in Chapter 19 — the QED vacuum polarization, the electron self-energy, the vertex correction — and the first amplitude of ϕ^4 theory from Chapter 15 all delivered, at the moment of integration, the same unwelcome verdict: the integral over loop momentum does not converge. This chapter confronts that verdict head on. It does *not* cure the infinities — the constructive machinery of counterterms and renormalization conditions is the work of Chapter 22, and the dynamical flow that follows from it the work of Chapters 23 and 24 — but it asks, and answers completely, the prior questions: where do the divergences come from, which diagrams suffer them and how badly, what structure the divergent part necessarily has, and what the infinities are trying to tell us about the theory that produced them. The whole of renormalization rests on the answers, so we extract them carefully and in order.

The diagnosis begins by locating the disease. Every ultraviolet divergence is a short-distance singularity living at the large-momentum end of a loop integral; tree graphs never diverge, and the leading strength of a loop divergence — its *degree* — is a property of the integrand that no choice of regulator can alter. We rely throughout on the analytic apparatus built earlier: the $i\epsilon$ prescription and Wick rotation of Chapter 5 and Chapter A, the loop-counting and one-particle-irreducible bookkeeping of Chapter 15, the dimensional regularization and master integrals of Chapter D, and the field and coupling conventions of Chapter A. The superficial degree of divergence D introduced in Theorem D.2 is the organizing integer of the chapter.

From that integer the argument climbs steadily in power. We first convert D from a property of an explicit integral into a topological invariant of a Feynman graph, then into a formula in the external legs and the mass dimensions of the couplings alone — $D = 4 - E$ for the scalar thread, and its dimension-explicit generalization for any local theory. The sign of the coupling dimension sorts every quantum field theory into three classes: super-renormalizable, renormalizable, and non-renormalizable, the last with infinitely many divergent amplitudes that

no finite set of parameters can absorb. We then prove the structural fact that makes renormalization possible at all — the locality theorem, that the divergent part of a primitively divergent amplitude is a local polynomial in the external data — and watch it match, term for term, the operators already present in the ϕ^4 and QED Lagrangians. Power counting is not the whole story, however, and we are honest about its one loophole: a globally convergent multi-loop graph can still diverge through a subintegration. We make the notion of a subdivergence precise, classify the disjoint, nested, and overlapping cases, exhibit the ϕ^4 sunset, and state Weinberg's convergence theorem as the exact criterion that supersedes naive counting — handing the resulting forest structure to the BPHZ construction of Chapter 22.

The abstractions are then cashed out. We compute the two one-loop divergences of ϕ^4 theory in full, with every intermediate step on the page: a logarithmically divergent four-point function whose pole seeds the beta function carried to the Wilson–Fisher fixed point in Chapter 24, and a quadratically divergent two-point function whose structure leaves the field strength unrenormalized at this order. These pole residues are the data that Chapter 22 and Chapter 24 consume.

The chapter closes by reading the divergences for their meaning. An infinity is the theory's admission that a low-energy observable is sensitive to short-distance physics it does not know; the trichotomy reappears, in Wilsonian language, as the distinction between relevant, marginal, and irrelevant operators, and the non-renormalizable theories are rehabilitated as predictive effective field theories valid below a cutoff — the viewpoint that becomes the working language of Chapters 23 and 38. We end by stating plainly, and with dates, what renormalization secures and what it does not: the asymptotic (not convergent) character of the perturbative series, and the triviality of ϕ^4 in four dimensions.

21.1 Where the Infinities Come From: Short-Distance Singularities

The diagrams collected at the head of this chapter — the QED vacuum polarization $\Pi(q^2)$, the electron self-energy $\Sigma(p)$, and the vertex correction Γ^μ of Chapter 19, together with the one-loop bubble of ϕ^4 theory from Eq. (15.50) — share one feature that no amount of careful bookkeeping removes: each is an integral over a loop momentum that does not converge. Before we can predict which diagrams suffer this fate, and how badly, we must say precisely what the disease is. That is the work of this section. The thesis is simple and will be argued in three complementary ways: the infinities are short-distance singularities, they live entirely in the integration over large loop momentum, and the leading strength of the divergence — its *degree* — is a property of the integrand that no choice of regulator can change. We will reintroduce the ϕ^4 bubble as

the first divergent object, exhibit its divergence in momentum space and again in position space, regulate it two genuinely different ways and watch the same degree emerge in each, and end by promoting the per-integral superficial degree of divergence into the working tool that organizes the rest of the chapter.

21.1.1 Divergences live in loops, at large momentum

The first observation is that a tree diagram is never divergent. Recall the loop-counting formula Eq. (15.48): a connected diagram with I internal lines and V vertices carries

$$L \;=\; I - V + 1 \tag{21.1}$$

independent loop momenta after every momentum-conserving delta function has been imposed. A tree has $L = 0$. Its value is therefore a finite product of propagators and vertex factors with every internal momentum pinned by the external data through the delta functions; there is nothing left to integrate, and a finite product of finite numbers is finite. Divergences can only appear when $L \geq 1$, that is, when the Feynman rules leave one or more four-momenta unconstrained and instruct us to integrate over all of $\mathbb{R}^{1,3}$ (or $\mathbb{R}^{1,d-1}$ in d dimensions). The infinities are integrals, and the only integrals in the theory are loop integrals. This already narrows the search: whatever is wrong, it is wrong in the region of integration where the loop momentum is large, since the integrand is a smooth, bounded function everywhere else (the $i\varepsilon$ prescription, Theorem A.23, keeps the Feynman denominators off the real axis, so no singularity sits on the contour at finite momentum).

To make this concrete, take the object the thread hands us first. The order-λ^2 contribution to $\phi\phi \to \phi\phi$ in ϕ^4 theory, the s-channel bubble of Theorem 15.16, reduces to the single loop integral Eq. (15.50), which we reproduce here with the total incoming momentum written $q = p_1 + p_2$ and $s = q^2$,

$$\mathcal{I}(q) \;=\; \int \frac{\mathrm{d}^4 k}{(2\pi)^4}\, \frac{i}{k^2 - m^2 + i\varepsilon}\, \frac{i}{(k+q)^2 - m^2 + i\varepsilon}. \tag{21.2}$$

There is exactly one unconstrained four-momentum k, as $L = 1$ demands. Ask only the crude question: how does the integrand behave when $|k|$ is large, all four components scaled up together? Each propagator denominator is then dominated by its quadratic term, $k^2 - m^2 \sim k^2$ and $(k+q)^2 - m^2 \sim k^2$, so the product of the two propagators falls off as k^{-4}. The measure $\mathrm{d}^4 k$ grows as $k^3\, \mathrm{d}\,|k|$ (three angular powers plus one radial). Collecting the powers in the large-momentum region,

$$\mathcal{I}(q) \;\sim\; \int^{\infty} \mathrm{d}\,|k|\;\, |k|^3 \cdot |k|^{-4} \;=\; \int^{\infty} \frac{\mathrm{d}\,|k|}{|k|}, \tag{21.3}$$

a logarithmically divergent radial integral. The divergence is entirely a statement about the ultraviolet end of the $|k|$ integration: cut the integral off at $|k| = \Lambda$ and

it grows as $\ln\Lambda$, without bound as $\Lambda\to\infty$, while the small- and intermediate-k regions contribute a perfectly finite number. This is the prototype of every ultraviolet (UV) divergence in the book. The word "ultraviolet" is literal in the dimensional-analysis sense: large momentum is short wavelength is short distance. We have located the disease — it is a short-distance singularity, sitting at the top of the loop momentum integral — and we now make the short-distance reading exact.

Physical Insight

A loop is the quantum-mechanical sum over all the ways an intermediate virtual state can carry momentum between two interaction points. The UV divergence says this sum is dominated by configurations in which the virtual quanta carry arbitrarily large momentum, i.e. probe arbitrarily short distances. The theory, written down as a Lagrangian of point interactions, is being asked a question about physics at zero distance, where it has no business claiming to know the answer. The divergence is the theory's honest admission that its short-distance content is unspecified — a flaw of the model, not of the arithmetic. The chapters that follow (Chapters 22 and 23) turn this admission into a controlled parametrization of our short-distance ignorance; here we only diagnose it.

21.1.2 The same divergence in position space: colliding vertices

The momentum-space count Eq. (21.3) has a position-space mirror that makes the phrase "short-distance singularity" precise and, in passing, shows that the two pictures agree on the strength of the divergence. The bubble is the Fourier transform of a product of two position-space propagators joining the same two spacetime points. Indeed, the position-space content of the one-loop $2\to2$ amplitude is the expression Eq. (15.38), in which two interaction vertices at z_1 and z_2 are connected by the squared propagator $\left[D_F(z_1-z_2)\right]^2$. Writing $x=z_1-z_2$ for the separation of the two vertices, the loop integral Eq. (21.2) is, up to the external legs and a Fourier factor, the transform

$$\mathcal{I}(q)\;=\;\int \mathrm{d}^4x\; e^{\,iq\cdot x}\,\left[D_F(x)\right]^2, \tag{21.4}$$

the convolution theorem turning the product of two momentum-space propagators into the transform of the product of two position-space ones (the convention is Theorem A.12). Whether Eq. (21.4) converges is now a question about the behaviour of the integrand near $x=0$: do the two vertices, when they collide, produce an integrable singularity, or not? The large-momentum divergence of Eq. (21.2) must be the same statement as a short-distance singularity of $\left[D_F(x)\right]^2$ as $x\to0$, because the large-q Fourier modes of any function are controlled by its short-distance structure.

To evaluate the coincident limit we need the short-distance behaviour of the scalar Feynman propagator. The propagator is

$$D_F(x) \;=\; \int \frac{\mathrm{d}^4p}{(2\pi)^4}\,\frac{i\,e^{-ip\cdot x}}{p^2-m^2+i\varepsilon}. \tag{21.5}$$

As $x\to 0$ the integral is dominated by large p, where the mass is negligible against p^2; the leading short-distance singularity is therefore that of the massless propagator, with the mass producing only subleading, less singular corrections. The cleanest way to extract it is by dimensional analysis, which also exposes how the answer depends on the spacetime dimension and so connects directly to the degree of divergence. In d spacetime dimensions the scalar field has mass dimension $\frac{1}{2}(d-2)$, so the two-point function $D_F(x)=\langle \mathcal{T}\phi(x)\phi(0)\rangle$ has mass dimension $d-2$. Being built (in the massless limit) from no scale but x itself, it must be a pure power,

$$D_F(x) \;\xrightarrow[x\to 0]{}\; \frac{c_d}{(-x^2+i\varepsilon)^{\frac{d-2}{2}}}, \qquad c_d \;=\; \frac{-\,i\,\Gamma\!\left(\frac{d-2}{2}\right)}{4\,\pi^{d/2}}, \tag{21.6}$$

where the constant c_d is fixed by doing the massless transform once (the massless Euclidean propagator in d dimensions is $\Gamma(\frac{d}{2}-1)/\big[4\pi^{d/2}(x_E^2)^{(d-2)/2}\big]$, and Eq. (21.6) is its Wick rotation, Theorem A.23; the value of c_d will not matter for the degree count). In $d=4$ this is the familiar $D_F(x)\to -i/\big[4\pi^2(-x^2+i\varepsilon)\big]$, singular as $1/x^2$ when the two points collide. Squaring,

$$\big[D_F(x)\big]^2 \;\xrightarrow[x\to 0]{}\; \frac{c_d^2}{(-x^2+i\varepsilon)^{\,d-2}} \;\sim\; \frac{1}{|x|^{\,2(d-2)}}. \tag{21.7}$$

The product of the two propagators diverges as $|x|^{-2(d-2)}$ in the coincident limit. Whether the position-space integral Eq. (21.4) converges at $x=0$ is then settled by the same kind of radial count as before, run now at the *small-x* end: the measure supplies $\mathrm{d}^dx\sim |x|^{d-1}\,\mathrm{d}\,|x|$, and the $e^{iq\cdot x}\to 1$ factor is harmless at $x=0$, so

$$\int_0 \mathrm{d}\,|x|\;\;|x|^{\,d-1}\cdot |x|^{-2(d-2)} \;=\; \int_0 \mathrm{d}\,|x|\;\;|x|^{\,3-d}, \tag{21.8}$$

which converges at $|x|=0$ if and only if $3-d>-1$, i.e. $d<4$, and is logarithmically divergent exactly at $d=4$ (the integrand becomes $\mathrm{d}\,|x|\,/\,|x|$). The coincident-point singularity of $\big[D_F(x)\big]^2$ is, in $d=4$, precisely strong enough to make the vertices' collision non-integrable — the same logarithm we found in Eq. (21.3). The two pictures agree, and they agree not by accident but because they are Fourier transforms of one another. The next worked example makes the equality of the two counts a theorem rather than a coincidence.

Worked Example 21.1 (The coincident limit equals the UV divergence: a Fourier dimensional-analysis argument)**.** We show that the short-distance singularity Eq. (21.7) and the large-momentum divergence Eq. (21.3) carry the

same degree, by transforming a pure power and reading off when the transform diverges. Work in d Euclidean dimensions, where the analysis is cleanest and the Wick rotation (Theorem A.23) only supplies overall factors of i that do not affect a degree count. The homogeneous function $|x|^{-\alpha}$ has the Fourier transform

$$\int \mathrm{d}^d x \, e^{\,iq\cdot x} \, |x|^{-\alpha} \;=\; \pi^{d/2}\, 2^{\,d-\alpha}\, \frac{\Gamma\!\left(\frac{d-\alpha}{2}\right)}{\Gamma\!\left(\frac{\alpha}{2}\right)} \; |q|^{\,\alpha-d}\,, \tag{21.9}$$

a result that holds as an honest convergent integral for $\frac{d-1}{2} < \frac{\alpha}{2} < \frac{d}{2}$, i.e. in the strip $d-1 < \alpha < d$, and as the analytic continuation of Eq. (21.9) elsewhere. The structure is all we need. For $\alpha < d$ the integral is short-distance convergent (the singularity at $x = 0$ is integrable) and the transform is the finite power $|q|^{\alpha-d}$, which decays at large q; nothing diverges. For $\alpha \geq d$ the integrand $|x|^{-\alpha}$ is no longer integrable at $x = 0$, the left-hand side develops a UV-from-below divergence at the coincident point, and on the right-hand side the prefactor $\Gamma\!\left(\frac{d-\alpha}{2}\right)$ hits the poles of the Gamma function at $\frac{d-\alpha}{2} = 0, -1, -2, \ldots$, i.e. at $\alpha = d, d+2, d+4, \ldots$. The position-space non-integrability and the momentum-space pole are one and the same singularity, exhibited on the two sides of an identity. Now set $\alpha = 2(d-2)$, the power of the squared propagator from Eq. (21.7). The degree by which the coincident singularity fails to be integrable is

$$\alpha - d \;=\; 2(d-2) - d \;=\; d-4, \tag{21.10}$$

which is exactly the large-momentum degree of the loop integral: a logarithm at $d = 4$ (where $\alpha = d$, the leading Gamma pole), convergence for $d < 4$ (where $\alpha < d$), and a worse divergence for $d > 4$. The position-space coincident limit and the momentum-space ultraviolet limit are quantitatively the same divergence, with the same degree $d-4$. This is the precise sense in which "ultraviolet" and "short-distance" are interchangeable, and it is why a single integer will be able to summarize both.

21.1.3 Regulator independence of the degree

A divergent integral is, strictly, undefined; to talk about it at all we must regulate — replace it by a finite quantity depending on a parameter, such that the original is recovered (formally) as the parameter is removed. There are many ways to do this, and a natural worry is that the answer to "how badly does this diverge" might be an artifact of the chosen regulator. It is not. The *degree* of the divergence is a property of the integrand's large-momentum scaling, fixed before any regulator is chosen, and every faithful regulator must reproduce it. We demonstrate this by computing the bubble in two genuinely different schemes — a hard momentum cutoff Λ, and dimensional regularization with $d = 4 - 2\varepsilon$ and scale μ as fixed in Chapter 19 and Theorem A.45 — and watching the same logarithmic degree appear in each, dressed differently.

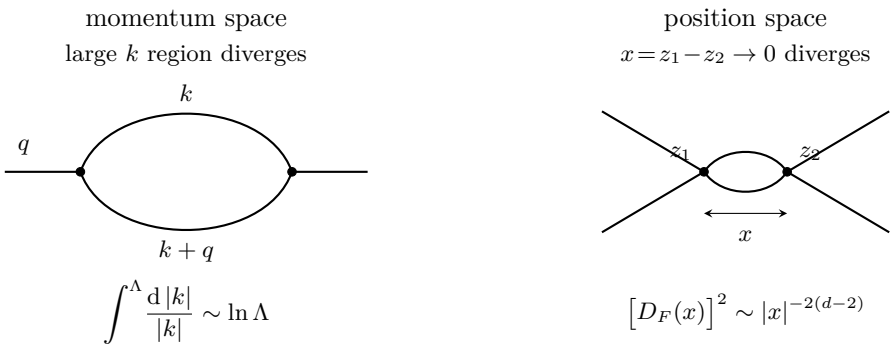


Figure 21.1: The two faces of the one-loop bubble divergence. Left: in momentum space the loop integral Eq. (21.2) diverges at large loop momentum k, logarithmically in $d = 4$. Right: in position space the same divergence is the non-integrable singularity of the squared propagator $[D_F(x)]^2$ as the two vertices z_1, z_2 collide, $x \to 0$. The Fourier transform Eq. (21.4) maps one into the other, and Theorem 21.1 shows both carry the same degree $d - 4$.

The bubble with a cutoff. Set the external momentum to zero in Eq. (21.2) (the divergence is q-independent, as the count Eq. (21.3) showed and as Theorem 21.1 confirmed: the divergent part comes from the $x \to 0$ region, where $e^{iq\cdot x} \to 1$). Rotate to Euclidean momentum $k^0 \to ik_E^0$, $k^2 \to -k_E^2$ (the contour rotation is justified by the absence of poles in the rotated quadrants, Chapter 5 and Theorem A.23), so $\mathrm{d}^4k \to i\,\mathrm{d}^4k_E$ and each denominator $k^2 - m^2 + i\varepsilon \to -(k_E^2 + m^2)$. The two factors of i in the numerator give $i^2 = -1$, the two denominator sign flips give $(-1)^2 = +1$, and the rotation supplies one more i, so $\mathcal{I}(0) = -i\,\mathcal{I}_E$ with $\mathcal{I}_E$ a positive Euclidean integral,

$$\mathcal{I}_E \;=\; \int \frac{\mathrm{d}^4k_E}{(2\pi)^4}\,\frac{1}{(k_E^2+m^2)^2} \;=\; \frac{\Omega_3}{(2\pi)^4}\int_0^{\Lambda}\frac{k^3\,\mathrm{d}k}{(k^2+m^2)^2}, \qquad \Omega_3 = 2\pi^2, \tag{21.11}$$

where $\Omega_3 = 2\pi^2$ is the area of the unit 3-sphere and we have cut the radial integral at $|k_E| = \Lambda$. With $u = k^2$ the radial integral is elementary,

$$\int_0^{\Lambda^2}\frac{u\,\mathrm{d}u}{2(u+m^2)^2} = \frac{1}{2}\Big[\ln(u+m^2) + \frac{m^2}{u+m^2}\Big]_0^{\Lambda^2} = \frac{1}{2}\Big[\ln\frac{\Lambda^2+m^2}{m^2} + \frac{m^2}{\Lambda^2+m^2} - 1\Big], \tag{21.12}$$

so, dropping terms that vanish as $\Lambda \to \infty$,

$$\mathcal{I}_E \;=\; \frac{2\pi^2}{(2\pi)^4}\cdot\frac{1}{2}\Big[\ln\frac{\Lambda^2}{m^2} - 1 + O(m^2/\Lambda^2)\Big] \;=\; \frac{1}{16\pi^2}\Big[\ln\frac{\Lambda^2}{m^2} - 1\Big] + O(m^2/\Lambda^2). \tag{21.13}$$

The divergence is the single logarithm $\ln(\Lambda^2/m^2) = 2\ln(\Lambda/m)$. Its coefficient $1/16\pi^2$ is the universal one-loop factor; the additive -1 is scheme junk that depends on the sharp cutoff and carries no physics. The degree is $D = 0$: a logarithm.

The bubble in dimensional regularization. The same integral, computed instead by analytic continuation in $d = 4 - 2\varepsilon$ with the coupling-promotion scale μ, is the master result Eq. (D.140) of Appendix D. At zero external momentum the Feynman parameter integral is trivial ($\Delta = m^2$ is x-independent), and

$$\mathcal{I}(0) \;=\; -\frac{i}{16\pi^2}\Big[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{m^2}\Big] + O(\varepsilon), \qquad \frac{1}{\bar{\varepsilon}} \equiv \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi, \tag{21.14}$$

the $\bar{\varepsilon}$ bundling the $-\gamma_{\mathrm{E}} + \ln 4\pi$ that always accompanies the pole in $\overline{\mathrm{MS}}$ (Theorem A.45). The two computations are now to be compared. Both deliver one universal coefficient $1/16\pi^2$ multiplying one divergence-tracking object plus a finite remainder; the divergence has the *same degree* — logarithmic — in both, because both are governed by the same $\mathrm{d}\,|k|\,/\,|k|$ scaling of the integrand. What differs is only the parametrization: the cutoff displays the divergence as $\ln\Lambda^2$, dimensional regularization as the pole $1/\varepsilon$. The dictionary between them is read off by matching the two finite expressions,

$$\frac{1}{\bar{\varepsilon}} \;\longleftrightarrow\; \ln\frac{\Lambda^2}{\mu^2}, \tag{21.15}$$

under which Eq. (21.13) and Eq. (21.14) carry the same $\ln(\Lambda^2/m^2) = \ln(\Lambda^2/\mu^2) + \ln(\mu^2/m^2)$ structure (the scheme-dependent constants -1 versus $-\gamma_{\mathrm{E}} + \ln 4\pi$ differ, as they must, but the divergence and the physics-bearing logarithm $\ln(\mu^2/m^2)$ agree). The role played by the cutoff Λ in the first scheme is played by the scale μ in the second: each is the reference scale against which the loop logarithm is measured. The lesson is that "$1/\varepsilon$" and "$\ln\Lambda$" are two names for one logarithmic divergence, and the regulator-independent content is the degree — here $D = 0$ — together with the coefficient $1/16\pi^2$.

A quadratic divergence, and a subtlety dimensional regularization hides. The bubble is logarithmic, the gentlest case. To see a higher degree, and to see exactly where the two schemes part company, take the scalar tadpole, the one-propagator loop that is the seed of the ϕ^4 mass renormalization,

$$\mathcal{T} \;=\; \int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon}. \tag{21.16}$$

Counting powers as in Eq. (21.3): the single denominator gives k^{-2}, the measure $k^3\,\mathrm{d}\,|k|$, so the integrand scales as $k^3\cdot k^{-2}\,\mathrm{d}\,|k| = k\,\mathrm{d}\,|k|$ and the radial integral $\int^\Lambda k\,\mathrm{d}k \sim \Lambda^2$ diverges *quadratically*: degree $D = 2$. With a cutoff, Wick-rotating as before (the numerator i times the rotation's i times the single denominator sign flip give $i^2\cdot(-1) = +1$), $\mathcal{T} = \mathcal{T}_E$ with

$$\mathcal{T}_E = \frac{2\pi^2}{(2\pi)^4}\int_0^\Lambda\frac{k^3\,\mathrm{d}k}{k^2+m^2} = \frac{1}{16\pi^2}\Big[\Lambda^2 - m^2\ln\frac{\Lambda^2+m^2}{m^2}\Big] + \text{const} = \frac{\Lambda^2}{16\pi^2} - \frac{m^2}{16\pi^2}\ln\frac{\Lambda^2}{m^2} + \cdots, \tag{21.17}$$

exhibiting the promised leading Λ^2 (the quadratic divergence) together with a subleading $m^2 \ln \Lambda^2$ (a logarithmic subdivergence riding on the same integral). In dimensional regularization the same tadpole is the standard one-propagator integral (Theorem A.6 and Appendix D),

$$\mathcal{T} \;=\; -\frac{m^2}{16\pi^2}\Big[\frac{1}{\bar{\varepsilon}} + 1 + \ln\frac{\mu^2}{m^2}\Big] + O(\varepsilon), \tag{21.18}$$

obtained from $\int \mathrm{d}^d k/(2\pi)^d\,(k^2-m^2)^{-1} = -i\,(m^2)^{d/2-1}\Gamma(1-\frac{d}{2})/(4\pi)^{d/2}$, multiplied by the explicit numerator i of Eq. (21.16) (so the two factors of i cancel and $\mathcal{T}$ is real), by expanding $\Gamma(-1+\varepsilon) = -1/\varepsilon - 1 + \gamma_{\mathrm{E}} + O(\varepsilon)$ around $d=4$. The comparison is instructive. The leading Λ^2 of Eq. (21.17) has *no* counterpart in Eq. (21.18): dimensional regularization shows only a $1/\varepsilon$ pole, and that pole multiplies m^2, matching the cutoff's *subleading* $m^2 \ln \Lambda^2$ term, not its leading Λ^2. Dimensional regularization is blind to the pure power divergence. This is not a contradiction with regulator independence of the degree; it is a feature of how dimensional regularization parametrizes. A pure power divergence with no associated logarithm corresponds to a scaleless piece, $\int \mathrm{d}^d k\,(k^2)^{-n}$, which dimensional regularization sets to zero because no d makes it converge (the technical reason is laid out in Section D.1); only the genuinely logarithmic part of each power of the cutoff survives as a pole. The degree of divergence $D=2$ is nevertheless the same in both schemes, because D is defined by the integrand's large-k scaling Eq. (21.3), read off before either regulator is applied; the schemes merely choose to *display* different amounts of that divergence. The honest statement, which we will lean on throughout, is therefore: the integer D is regulator-independent; the way a given regulator renders a divergence of degree D (a power Λ^D versus a pole, with or without an accompanying logarithm) is not.

Caution

Do not read the absence of a Λ^2 term in Eq. (21.18) as a claim that the tadpole is only logarithmically divergent. The quadratic sensitivity is real — it reappears the moment one uses a cutoff, and it is the physical origin of the hierarchy problem discussed in Chapter 23 — but dimensional regularization, by working at generic d where the scaleless power piece vanishes, simply does not assign it a symbol. When the degree of divergence is what matters (as it is for the power-counting of the rest of this chapter), always read D off the integrand's scaling, never off the particular poles a chosen regulator happens to expose. The two schemes agree on D; they need not agree on which sub-pieces of the divergence are made visible.

21.1.4 The superficial degree of divergence as a working tool

We have now seen, from three independent angles — the large-momentum power count Eq. (21.3), the coincident-point count Theorem 21.1, and the explicit two-

scheme computation Eqs. (21.13) and (21.14) — that a single integer governs each loop integral: the net power by which the integrand fails to fall off fast enough at large loop momentum. That integer is the *superficial degree of (ultraviolet) divergence* D, defined per integral in Theorem D.2. We import its definition as the working tool of this chapter: for an L-loop integral in d dimensions whose numerator is a polynomial of total degree P in the loop momenta and whose denominator is a product of N propagators (each contributing two inverse powers),

$$D \;=\; d\,L \;+\; P \;-\; 2N, \tag{21.19}$$

which is Eq. (D.4). The accompanying dictionary, established by the radial count $\int^{\Lambda} \mathrm{d}\,|k|\;|k|^{D-1}$ of Theorem D.3 and exhibited concretely above, reads:

$$D<0:\ \text{(superficially) convergent};\quad D=0:\ \log\Lambda,\ \tfrac{1}{\varepsilon};\quad D=1:\ \text{linear};\quad D=2:\ \Lambda^2;\ \ldots \tag{21.20}$$

The word "superficially" is the genuine caveat recorded in Theorem D.2: at more than one loop, $D<0$ for the whole integral does not by itself guarantee convergence, because a proper subset of the loop momenta may be scaled to infinity on its own, producing a subdivergence the overall count misses; and $D\geq 0$ does not guarantee the leading divergence is actually present, because a symmetry can force the would-be leading term to vanish. At one loop neither loophole is open — there are no proper subintegrals — so $D<0$ is necessary and sufficient for UV convergence, and the count is exact (Theorem D.3). The subdivergence caveat is the business of Chapter 22 and of Weinberg's theorem; for the one-loop diagnostics of this chapter, D from Eq. (21.19) is decisive.

It is worth checking the tool against the two integrals just computed, since this is the pattern every later application follows. For the bubble Eq. (21.2): $L=1$, $N=2$ propagators, numerator degree $P=0$, so $D=d-4$, which is 0 at $d=4$ — a logarithm, exactly Eq. (21.13), and exactly the coincident-point degree Eq. (21.10). For the tadpole Eq. (21.16): $L=1$, $N=1$, $P=0$, so $D=d-2$, which is 2 at $d=4$ — quadratic, exactly Eq. (21.17). The formula reproduces, by counting alone, what the integrals told us by being done. That is the entire point: D is computable from the bare data (d,L,P,N) of a diagram, with no integral performed and no regulator chosen.

Physical Insight

The power of Eq. (21.19) is that it is a statement about a diagram's *topology and field content*, not about its integrand. We never needed the value of the bubble to know it was logarithmically divergent; we needed only to count its loops, lines, and numerator powers. This is what makes a *program* possible: rather than computing every loop integral in a theory and inspecting each for divergences, we will compute one integer per diagram — indeed, as the next section shows, one integer determined

entirely by a diagram's external legs once the internal structure is summed over — and read off in advance which amplitudes diverge and which are finite. The infinities, which looked like a computational catastrophe, become a finite, classifiable bookkeeping problem.

What we do *not* yet have is the efficient form of D. As written, Eq. (21.19) still asks for the internal data L, N, and P of a particular diagram. The remarkable fact, and the engine of renormalization theory, is that for a given theory these combine — through the loop relation Eq. (21.1) and the field-by-field structure of the interaction vertices — into an expression that depends only on the *external* legs of the diagram and on the mass dimensions of the fields and couplings. Two diagrams with the same external legs then share the same superficial degree of divergence no matter how complicated their interiors. Establishing that reduction, and reading from it the classification of theories by the dimensions of their couplings, is the task we turn to next.

21.2 Power Counting: The Superficial Degree of Divergence of a Diagram

The previous section reduced the question "does this loop integral diverge, and how badly?" to a single integer, the superficial degree of divergence D of Theorem D.2: count d for every loop integration measure, subtract two for every propagator, add back the powers of loop momentum supplied by the numerator, and the sign of the result tells you whether the integrand, scaled uniformly to infinity, falls off fast enough to converge. That integer is regulator-independent — it is a property of the integrand at large momentum, not of the cutoff or the analytic continuation we use to tame it. But as a tool it is still tied to the explicit integral: to read off D from Eq. (D.4) we had to know L, the numerator degree, and the number of propagators, which means we had to write the integral down. The point of this section is to make that step unnecessary. We will compute D from the *topology* of the Feynman diagram alone — from how many external legs it has, how many internal lines, how many vertices — and then, by feeding in the leg-counting constraints that every diagram of a given theory must satisfy, eliminate the internal structure entirely. What survives is a formula for D in terms of the external legs and the field content of the theory: a verdict on every diagram of a process, delivered before a single integral is set up. For the scalar thread the verdict is the compact $D = 4 - E$, the organizing result of the whole chapter.

21.2.1 The combinatorial data of a diagram

Fix a connected Feynman diagram (Theorem 15.29) contributing to some amputated Green's function (Theorem 15.37) of a field theory. We assume throughout that it is amputated and one-particle-irreducible in the sense of Theorem 15.30 when we discuss its divergences, because the external propagators carry no loop integration and a one-particle-reducible diagram factorizes through an internal line whose own divergence is counted separately; for the purely combinatorial identities of this subsection only connectedness is needed. To such a diagram we attach four nonnegative integers.

Definition 21.2 (Topological data of a diagram)**.** For a connected Feynman diagram we write

- V for the number of *vertices* (interaction points);
- I for the number of *internal lines* (propagators joining two vertices);
- E for the number of *external lines* (amputated legs, each attached to exactly one vertex and carrying a fixed external momentum);
- L for the number of *loops*, defined as the number of independent four-momenta (in general d-momenta) left to be integrated after all momentum-conservation constraints have been imposed.

When several species of field are present we refine I and E into I_f and E_f, the internal and external lines of field type f, with $I = \sum_f I_f$ and $E = \sum_f E_f$; and we refine V into the numbers V_t of vertices of each interaction type t, with $V = \sum_t V_t$.

These integers are not independent. Two relations bind them: one expressing momentum conservation (it will give the loop number), and one expressing the bookkeeping of line-ends at vertices (it will give the leg constraint). Both are purely topological, and together they are everything power counting needs.

21.2.2 The loop identity

The first relation we have in fact already met, in Eq. (15.48), where it was read off from a single worked diagram. Because it is the load-bearing input to every power-counting formula below, we now establish it in general, by directly counting the momenta that remain to be integrated.

Proposition 21.3 (Loop number of a connected diagram)**.** *For any connected Feynman diagram,*

$$L \;=\; I - V + 1. \tag{21.21}$$

Proof. The Feynman rules assign to each internal line an independent d-momentum, which is integrated, $\int \mathrm{d}^d k/(2\pi)^d$ per line; that is I integration momenta to start with. At each vertex the rules impose a delta-function enforcing conservation of d-momentum, $(2\pi)^d\,\delta^{(d)}(\sum k_{\text{in}} - \sum k_{\text{out}})$, where the sum runs over all lines — internal and external — meeting at that vertex. There are V such delta-functions, so V linear constraints among the I internal momenta (with the external momenta entering as fixed inhomogeneous terms).

Were all V constraints independent, they would eliminate V of the I integrations, leaving $I - V$. But they are not all independent. Sum the constraints over every vertex of the diagram. Each internal line runs between two vertices, entering one delta-function with a $+$ sign and the other with a $-$ sign; in the total sum its momentum cancels identically. Each external line touches exactly one vertex and so survives. The sum of all V vertex constraints is therefore the single equation

$$\sum_{\text{ext}} p_{\text{in}} \;-\; \sum_{\text{ext}} p_{\text{out}} \;=\; 0, \tag{21.22}$$

overall conservation of the external momenta. This relation contains no internal momentum at all: it is a constraint on the external data, automatically satisfied by the kinematics of the process, and it imposes nothing on the integration variables. Thus exactly one linear combination of the V vertex constraints is degenerate as far as the internal momenta are concerned. For a connected diagram this is the *only* degeneracy: any proper nonempty subset S of the vertices, summed, leaves uncancelled the momenta of those lines that cross the boundary between S and its complement, and because the diagram is connected at least one such line exists, so the subset sum is a genuine, nontrivial constraint on internal momenta and cannot vanish. Hence precisely one of the V constraints is redundant for the internal integrations, $V-1$ are independent, and the number of internal momenta left unfixed is

$$L \;=\; I - (V-1) \;=\; I - V + 1, \tag{21.23}$$

which is Eq. (21.21). □

Two remarks fix the scope. First, the argument used connectedness exactly once, to guarantee that every proper subset of vertices has a line crossing its boundary; for a diagram with C connected components each component contributes its own surviving global delta-function, and the count generalizes to $L = I - V + C$, as already recorded in Theorem 15.15. The amplitudes of this chapter are built from connected, amputated, 1PI diagrams, so $C = 1$ and Eq. (21.21) is the form in force. Second, the identity is the graph-theoretic statement that the number of independent cycles of a connected graph equals its first Betti number $E_{\text{edges}} - V_{\text{nodes}} + 1$; here the "edges" counted are the internal lines (the external legs, terminating in free space rather than a second vertex, are not edges of the cycle structure), so $E_{\text{edges}} = I$ and the Betti number is

$I - V + 1$. Momentum conservation and topology are saying the same thing: an independent loop momentum is precisely an independent cycle around which momentum can circulate undetermined by the external data.

21.2.3 The degree of divergence as a topological integer

We now feed Eq. (21.21) into the superficial degree of divergence. Consider first the pure scalar theory of ϕ^4 theory, with the standard Lagrangian

$$\mathcal{L} \;=\; \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 \tag{21.24}$$

of Eq. (6.56). Two features of this theory make the counting clean and are worth isolating, because the general case below is organized as a list of corrections to them. The quartic vertex carries no derivatives, so it contributes no powers of momentum to the numerator; and the scalar propagator $i/(k^2 - m^2 + i\varepsilon)$ falls off as k^{-2}, contributing exactly two inverse powers of momentum each. Thus, in the master count Eq. (D.4), the numerator degree is $P = 0$ and the number of propagators is the number of internal lines $N = I$. The superficial degree of divergence of a ϕ^4 diagram in d spacetime dimensions is therefore

$$D \;=\; d\,L - 2I. \tag{21.25}$$

Every loop integration $\int \mathrm{d}^d k$ supplies $+d$ powers of momentum; every internal line supplies -2. This is already topological — it refers only to L and I — but it still mentions the internal line count I, which is not data of the *process*; a process is specified by its external legs and the order in λ at which we work, not by how the internal lines happen to be wired. To reach a statement about the process we must trade I (and L) for the external leg number E and the vertex number V. That is what the second topological relation does.

21.2.4 Leg counting

Proposition 21.4 (Leg-counting constraint)**.** *In a theory whose only interaction is a vertex with n legs, the topological data of any diagram obey*

$$n\,V \;=\; 2I + E. \tag{21.26}$$

For ϕ^4, where $n = 4$, this reads $4V = 2I + E$. More generally, in a theory with several interaction types and several field species, each field type f obeys

$$2I_f + E_f \;=\; \sum_t V_t\, n_{t,f}, \tag{21.27}$$

where $n_{t,f}$ is the number of legs of type f attached to a vertex of type t.

Proof. Count the line-ends (the "half-edges") in the diagram in two ways. Every vertex of type t is, by construction, an interaction point at which $n_{t,f}$ legs of field type f terminate; summing over vertex types, the total number of type-f line-ends emanating from vertices is $\sum_t V_t\, n_{t,f}$. Now count the same line-ends by the lines they belong to. Each internal line of type f joins two vertices and so accounts for two type-f ends; each external line of type f is attached to a single vertex and so accounts for one. The total is $2I_f + E_f$. Every line-end is counted once in each tally, so the tallies are equal, giving Eq. (21.27). When there is a single field and a single vertex of n legs, $n_{t,f} = n$, $I_f = I$, $E_f = E$, $V_t = V$, and Eq. (21.27) collapses to Eq. (21.26). □

The relation Eq. (21.26) has a transparent reading: a ϕ^4 vertex is a place where four line-ends meet, so $4V$ counts all line-ends in the diagram; each internal line uses up two of them (one at each of its endpoints) and each external line uses one, accounting for the same total as $2I + E$. We now have everything needed to eliminate the internal structure.

21.2.5 The master formula for the scalar thread

Theorem 21.5 (Superficial degree of divergence in ϕ^4 theory)**.** *For a connected ϕ^4 diagram in d spacetime dimensions with E external legs and V vertices,*

$$D \;=\; d - \frac{d-2}{2}\,E + (d-4)\,V. \tag{21.28}$$

In four dimensions the vertex term drops out and

$$\boxed{D \;=\; 4 - E} \qquad (d = 4), \tag{21.29}$$

independent of the order in λ and of every internal detail of the diagram.

Proof. Start from the topological count Eq. (21.25), $D = dL - 2I$, and eliminate L and I in favour of E and V using the two constraints Eq. (21.21) and Eq. (21.26).

Substitute the loop identity $L = I - V + 1$:

$$D \;=\; d(I - V + 1) - 2I \;=\; (d-2)\,I - d\,V + d. \tag{21.30}$$

Now solve the leg constraint $4V = 2I + E$ for the internal line count,

$$I \;=\; \frac{4V - E}{2} \;=\; 2V - \frac{E}{2}, \tag{21.31}$$

and insert this into Eq. (21.30):

$$D = (d-2)\Big(2V - \frac{E}{2}\Big) - dV + d$$

$$
\begin{aligned}
&= 2(d-2)V - \frac{d-2}{2}\,E - dV + d \\
&= (2d-4-d)V - \frac{d-2}{2}\,E + d \\
&= d - \frac{d-2}{2}\,E + (d-4)\,V,
\end{aligned}
\tag{21.32}
$$

which is Eq. (21.28). Setting $d = 4$ annihilates the coefficient $(d-4)$ of V and reduces the scalar field's coefficient $(d-2)/2$ to 1, leaving $D = 4 - E$, which is Eq. (21.29). □

The structure of Eq. (21.28) repays a moment's attention before we exploit it, because every conceptual theme of the chapter is already visible in its three terms. The constant d is the contribution of the overall loop measure that survives once the external momenta are held fixed. The term $-\frac{d-2}{2}E$ subtracts a fixed amount for each external leg; the coefficient is exactly the canonical mass dimension of the scalar field, a fact we make precise in Section 21.2.6 and which is the seed of the dimensional reading of power counting. The term $(d-4)V$ carries all the dependence on the order of perturbation theory — and it vanishes identically in four dimensions. That vanishing is the single most consequential feature of the formula: it means that in $d = 4$ adding more vertices, hence going to higher order in λ and adding more loops, does *not* make a ϕ^4 amplitude with a given number of external legs more divergent. The degree of divergence of every diagram contributing to a given process is the same, fixed once and for all by E. We will see in Sections 21.2.6 and 21.2.7 that this is precisely the statement that the coupling λ is dimensionless in four dimensions, and it is the property that the next section will elevate into the definition of a renormalizable interaction.

Caution

Superficial, not actual. Theorem 21.5 reports the superficial degree, the count of Theorem D.2, and inherits its caveats verbatim. A negative D for the whole diagram does not by itself prove convergence: a subset of the loop momenta may be scaled to infinity with the rest held fixed, exposing a *subdivergence* that the global count misses. Conversely $D \geq 0$ need not mean the leading power is actually present — a symmetry can force the would-be leading term to vanish, as gauge invariance does for the photon self-energy in Section 21.2.8. The complete criterion, that an amplitude converges when $D < 0$ for the whole integral and for every subintegral, is Weinberg's theorem, stated at Theorem D.2 and taken up in the subdivergence section that follows the classification. For now D is exactly what its name says: a fast, topological, leading-power diagnosis, correct as a one-loop statement and indispensable as the organizing principle even when subdivergences must later be added by

hand.

We now cash in the formula on the thread it was built for.

Worked Example 21.6 (The superficially divergent amplitudes of ϕ^4 in four dimensions)**.** We continue ϕ^4 theory from Theorem 15.16, where the one-loop four-point bubble was first counted, and ask the question power counting was invented to answer: in four-dimensional ϕ^4 theory, which amplitudes can diverge at all, and how badly? By Eq. (21.29) the answer depends on nothing but the number of external legs E, through $D = 4 - E$. The $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ of the Lagrangian Eq. (21.24), established in Theorem 7.22, allows only even E in any nonzero amplitude, so we run through $E = 0, 2, 4, 6, \ldots$:

E	$D = 4 - E$	character	physical role
0	4	quartically divergent	vacuum energy (cosmological constant)
2	2	quadratically divergent	self-energy: mass and field renormalization
4	0	logarithmically divergent	four-point vertex: coupling renormalization
6	-2	convergent	—
≥ 6	< 0	convergent	—

The list is finite, and this is the whole point. Only three amplitudes are superficially divergent, and exactly two of them carry physics: the two-point function ($E = 2$), whose quadratic divergence renormalizes the mass and whose subleading logarithm renormalizes the field normalization, and the four-point function ($E = 4$), whose logarithmic divergence renormalizes the coupling λ. The $E = 0$ amplitude is the field-independent vacuum energy, an additive constant that drops out of every scattering observable (it returns to physical life only when gravity is coupled in). Every amplitude with six or more external legs is superficially convergent: a process such as $\phi\phi \to \phi\phi\phi\phi$ has $D = -2$ and needs no new counterterm. Crucially, the verdict is order-independent. The one-loop bubble of Eq. (15.50) has $E = 4$, $D = 0$; so does the two-loop sunset-type correction to the four-point function, and so does every higher-order four-point diagram — because the vertex term in Eq. (21.28) vanishes at $d = 4$, piling on more loops never raises D above zero. The divergences of ϕ^4 are therefore confined, to all orders, to a fixed list of three amplitudes, of which the mass/field and the coupling are the renormalizable physics. This finiteness of the list is what the counterterm program of Chapter 22 will exploit, and the logarithm of the $E = 4$ amplitude is the number that Chapter 24 turns into $\beta_\lambda = 3\lambda^2/16\pi^2$.

Physical Insight

Why higher orders do not get worse. It is tempting to expect a two-loop diagram to diverge worse than a one-loop one — more integrations, more momentum to run off to infinity. The resolution is that each new ϕ^4 vertex in four dimensions comes packaged with the lines it must connect, and those extra propagators damp exactly as fast as the extra loop integration grows. Quantitatively, adding one vertex adds, on the constraint surface, $\Delta I = 2$ internal lines and $\Delta L = 1$ loop (from Eqs. (21.21) and (21.31) at fixed E), so the change in D is $\Delta D = d\,\Delta L - 2\,\Delta I = d - 4$, which is zero precisely in four dimensions. A dimensionless coupling is the statement that the books balance: the theory has no internal yardstick with which to make a higher-order amplitude grow. Below four dimensions ($d < 4$) the coefficient $d - 4$ is negative and higher orders are progressively *less* divergent — the hallmark of a super-renormalizable theory; above four dimensions it is positive and divergences proliferate without bound — non-renormalizability. The borderline case, marginal renormalizability, is the physically rich one and the one four-dimensional ϕ^4 sits on.

21.2.6 Engineering dimensions of the fields

The coefficient $\frac{d-2}{2}$ multiplying E in Eq. (21.28) was, we claimed, the mass dimension of the scalar field. Establishing this — and the analogous dimensions for the Dirac and gauge fields — turns the leg-by-leg count into a statement that holds in any theory, and supplies the bridge to the coupling-dimension classification of the next section. The input is dimensional analysis of the action.

In natural units $\hbar = c = 1$ every quantity carries a single *mass dimension*, written $[\,\cdot\,]$ and measured in powers of mass (equivalently inverse length). The action $S = \int \mathrm{d}^d x\, \mathcal{L}$ is dimensionless, because it appears exponentiated as e^{iS}. The volume element $\mathrm{d}^d x$ has dimension $-d$, so the Lagrangian density must carry

$$[\mathcal{L}] \;=\; d. \tag{21.33}$$

Each term in $\mathcal{L}$ must separately have dimension d, and since a derivative ∂_μ is dimension $+1$, the kinetic term of each field fixes that field's dimension.

Scalar. The free scalar kinetic term in Eq. (21.24), $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$, taken from Eq. (6.51) and quantized in Section 9.2, has dimension $2(1 + [\phi])$. Setting this to d,

$$2 + 2[\phi] = d \quad\Longrightarrow\quad [\phi] \;=\; \frac{d-2}{2}, \tag{21.34}$$

which in $d = 4$ is $[\phi] = 1$. This is the coefficient that appeared in Eq. (21.28), now identified: the price each external scalar leg pays in D is its own canonical dimension.

Dirac. The Dirac kinetic term $\bar{\psi}\, i\partial\!\!\!/\, \psi$ of Eq. (6.68), whose field is quantized in Section 11.2 and whose propagator is built in Section 11.5, carries one derivative and two factors of ψ, so its dimension is $1 + 2[\psi]$. Hence

$$1 + 2[\psi] = d \quad \Longrightarrow \quad [\psi] \;=\; \frac{d-1}{2}, \tag{21.35}$$

which is $[\psi] = 3/2$ in $d = 4$. The single derivative, rather than two, is the reason the fermion's dimension is larger than the scalar's by a half; it is also why the fermion propagator scales as k^{-1} rather than k^{-2}, a fact we use in the QED count below.

Gauge field. The Maxwell term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ of Eq. (6.80), with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and the photon quantized in Section 13.4, is built from ∂A, so F^2 has dimension $2(1 + [A])$. Therefore

$$2 + 2[A] = d \quad \Longrightarrow \quad [A_\mu] \;=\; \frac{d-2}{2}, \tag{21.36}$$

the same as the scalar, $[A] = 1$ in $d = 4$ — as it must be, since the photon propagator also falls as k^{-2} and the gauge field enters the covariant derivative $D_\mu = \partial_\mu - iqA_\mu$ alongside ∂_μ, which forces $[A] = [\partial] \cdot$ (dimensionless) once the charge q is dimensionless in four dimensions. We collect the four-dimensional values for reference:

$$[\phi] = 1, \qquad [\psi] = \tfrac{3}{2}, \qquad [A_\mu] = 1 \qquad (d = 4). \tag{21.37}$$

With the field dimensions in hand we can write down a power-counting formula valid for any theory, of which Theorem 21.5 is the special case.

21.2.7 The dimension-explicit master formula

The clean way to count in a general theory is to track, term by term, the mass dimension that each propagator, vertex, and loop integration contributes to the large-momentum behaviour of the integrand. The key observation is that the large-momentum power of a propagator equals its mass dimension. The momentum-space propagator of a field f is the Fourier transform of the position-space two-point function $\langle f(x) f(0)\rangle$, which has dimension $2[f]$; transforming with $\int \mathrm{d}^d x\, e^{ip\cdot x}$ subtracts d, so the momentum-space propagator $\Delta_f(k)$ has dimension $2[f] - d$, and at large k (where the mass is negligible) it scales as $k^{2[f]-d}$. Checking: a scalar gives $2 \cdot \frac{d-2}{2} - d = -2$, a fermion $2 \cdot \frac{d-1}{2} - d = -1$, a gauge field -2, reproducing the falloffs we used. A derivative at a vertex contributes $+1$ power. With these ingredients the superficial degree of divergence of a general diagram is

$$D \;=\; d\,L \;+\; \sum_f (2[f] - d) I_f \;+\; \sum_t V_t\, \partial_t, \tag{21.38}$$

where ∂_t is the number of derivatives carried by a vertex of type t, and the sums run over field species f and vertex types t. We now reduce this to a statement about external legs and couplings alone.

Theorem 21.7 (Dimension-explicit power counting)**.** *Let a connected diagram be built from vertices whose interaction terms have couplings g_t of mass dimension $\delta_t = [g_t]$. Then its superficial degree of divergence is*

$$\boxed{D \;=\; d \;-\; \sum_{ext} [f] \;-\; \sum_{t} V_t\, \delta_t} \tag{21.39}$$

where the first sum runs over all external legs (each contributing the canonical dimension of its field) and the second over all vertices.

Proof. We carry every substitution. Begin from Eq. (21.38). Eliminate L with the loop identity Eq. (21.21), $L = I - V + 1$ with $I = \sum_f I_f$ and $V = \sum_t V_t$:

$$\begin{aligned} D &= d\Big(\sum_f I_f - V + 1\Big) + \sum_f (2[f] - d) I_f + \sum_t V_t\, \partial_t \\ &= \sum_f \big[\, d + 2[f] - d \,\big] I_f \;-\; dV + d + \sum_t V_t\, \partial_t \\ &= 2 \sum_f [f]\, I_f \;-\; dV + d + \sum_t V_t\, \partial_t . \end{aligned} \tag{21.40}$$

The dL measure has cancelled the $-d$ of each propagator, leaving each internal line weighted by twice its field dimension — the trace of the two field operators sitting at its ends. Now eliminate the internal line counts I_f with the general leg constraint Eq. (21.27), in the form $2I_f = \sum_t V_t\, n_{t,f} - E_f$:

$$\begin{aligned} 2 \sum_f [f]\, I_f &= \sum_f [f] \Big(\sum_t V_t\, n_{t,f} - E_f \Big) \\ &= \sum_t V_t \sum_f n_{t,f}\, [f] \;-\; \sum_f [f]\, E_f . \end{aligned} \tag{21.41}$$

The last sum, $\sum_f [f]\, E_f$, is exactly the sum of the canonical dimensions over all external legs, $\sum_{\text{ext}} [f]$. Insert Eq. (21.41) into Eq. (21.40) and group the vertex sums:

$$D \;=\; d \;-\; \sum_{\text{ext}} [f] \;+\; \sum_t V_t \Big(\sum_f n_{t,f}\, [f] + \partial_t - d \Big). \tag{21.42}$$

It remains to identify the parenthesis. A vertex of type t comes from a term $g_t\, (\partial)^{\partial_t} \prod_f f^{\,n_{t,f}}$ in the Lagrangian (schematically: a coupling, ∂_t derivatives, and $n_{t,f}$ powers of each field). By Eq. (21.33) this term has total dimension d, so

$$\delta_t + \partial_t + \sum_f n_{t,f}\, [f] \;=\; d \quad\Longrightarrow\quad \sum_f n_{t,f}\, [f] + \partial_t - d \;=\; -\,\delta_t . \tag{21.43}$$

The parenthesis in Eq. (21.42) is therefore precisely $-\delta_t$, and

$$D \;=\; d - \sum_{\text{ext}} [f] - \sum_t V_t\, \delta_t, \tag{21.44}$$

which is Eq. (21.39). □

This is the formula the chapter was driving toward. Read it slowly. The superficial degree of divergence of any diagram is the spacetime dimension d, minus a contribution from each external leg equal to that field's canonical dimension, minus a contribution from each vertex equal to the mass dimension of its coupling. The external sum depends only on the process; the vertex sum depends only on which interactions, and how many, were used to build the diagram. As an immediate check, ϕ^4 has a single coupling with $\delta = [\lambda] = d - 4 \cdot \frac{d-2}{2} = d - 2(d-2) = 4 - d$ by Eq. (21.43), so $\sum_t V_t \delta_t = (4-d)V$, and Eq. (21.39) gives $D = d - \frac{d-2}{2}E - (4-d)V = d - \frac{d-2}{2}E + (d-4)V$, recovering Eq. (21.28) exactly.

The decisive structural fact is the role of the last term. *If every coupling in the theory is dimensionless*, $\delta_t = 0$, the vertex sum vanishes identically and

$$D \;=\; d - \sum_{\text{ext}} [f] \qquad \text{(all couplings dimensionless).} \tag{21.45}$$

The degree of divergence then depends on *nothing but the external legs* — not on the order of perturbation theory, not on the number of vertices, not on the number of loops. A given process has a single, fixed degree of divergence to all orders, and the set of divergent processes is finite. This is the algebraic heart of renormalizability, and it is no accident that it is precisely the four-dimensional ϕ^4 situation of Theorem 21.5. We have arrived at the formula in its sharpest form; the vertices have, for dimensionless couplings, dropped out entirely.

21.2.8 The QED count

We close by carrying out the same reduction for QED, both to exhibit Theorem 21.7 on a theory with two field species and a derivative-free vertex, and to produce the short list of QED divergences that the radiative-corrections computations of Chapter 19 confront.

Worked Example 21.8 (Superficial degree of divergence in four-dimensional QED)**.** The QED Lagrangian $\bar\psi(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $D_\mu = \partial_\mu + ieA_\mu$ has a single interaction vertex, the minimal coupling $-e\bar\psi \not{A}\psi$ extracted from the covariant derivative. It attaches two fermion legs and one photon leg and carries no derivative: $n_\psi = 2$, $n_\gamma = 1$, $\partial = 0$. We first confirm that its coupling is dimensionless in four dimensions, then assemble D, then list the divergent amplitudes.

Coupling dimension. By Eq. (21.43) with the four-dimensional field dimensions Eq. (21.37),

$$[e] = d - \partial - 2[\psi] - [A] = d - 0 - 2\cdot\frac{d-1}{2} - \frac{d-2}{2} = d - (d-1) - \frac{d-2}{2} = \frac{4-d}{2}, \tag{21.46}$$

which vanishes at $d = 4$: the electric charge is dimensionless in four dimensions, exactly as for λ. QED is, by the criterion of Eq. (21.45), of the renormalizable type, and its degree of divergence is fixed by external legs alone.

The formula. Let a diagram have E_f external fermion legs and E_γ external photon legs. Applying Eq. (21.45) in $d = 4$ with $[\psi] = \frac{3}{2}$ and $[A] = 1$,

$$\boxed{D \;=\; 4 - \tfrac{3}{2}E_f - E_\gamma} \qquad (d = 4). \tag{21.47}$$

We can also obtain this directly from the raw count to see the derivative-free fermion propagator at work. With the fermion propagator scaling as k^{-1} and the photon as k^{-2}, Eq. (21.38) reads $D = 4L - I_f - 2I_\gamma$; using $L = I_f + I_\gamma - V + 1$, the photon leg constraint $2I_\gamma + E_\gamma = V$ (each vertex has one photon leg), and the fermion leg constraint $2I_f + E_f = 2V$ (each vertex has two fermion legs), the same elimination as in Theorem 21.7 returns Eq. (21.47); the two routes agree.

The divergent amplitudes. Charge-conjugation invariance (Furry's theorem, proved in Chapter 19) forces any amplitude with an odd number of external photons and no external fermions to vanish, and Lorentz invariance forbids amplitudes with an odd number of external fermion legs; we therefore tabulate the nonvanishing 1PI amplitudes with $D \geq 0$:

E_f	E_γ	D	amplitude	true behaviour
0	0	4	vacuum energy	additive constant, dropped
0	2	2	photon self-energy $\Pi^{\mu\nu}$	log (gauge invariance: $\Pi \propto q^2$)
0	3	1	—	vanishes (Furry)
0	4	0	light-by-light	convergent (gauge invariance)
2	0	1	electron self-energy Σ	log (chiral symmetry softens)
2	1	0	vertex correction Γ^μ	log

Read off the lessons. The genuinely divergent, physics-carrying amplitudes are exactly three: the photon self-energy $\Pi^{\mu\nu}$ ($E_\gamma = 2$), the electron self-energy Σ ($E_f = 2$), and the vertex Γ^μ ($E_f = 2, E_\gamma = 1$) — precisely the three one-loop integrals computed in Chapter 19. The superficial counts are softened by symmetry in the manner cautioned above: the photon self-energy counts as quadratically ($D = 2$) divergent, but gauge invariance (the transverse Ward identity) forces the structure $\Pi^{\mu\nu} = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\Pi(q^2)$, pulling out two powers of external momentum and leaving only a logarithm; the electron self-energy counts as linearly ($D = 1$) divergent, but the chiral structure of the massless

theory removes the leading power and again leaves a logarithm. The four-photon "light-by-light" amplitude has $D = 0$ by naive counting yet is rendered finite by gauge invariance, which supplies the missing powers. Every amplitude not in the table — five-point and higher, or any with more external legs — has $D < 0$ and is superficially convergent. As with ϕ^4, the list is finite and order-independent, which is what makes QED renormalizable.

We have what the chapter set out to obtain: a topological formula Eq. (21.39) that delivers the superficial degree of divergence of any diagram from its external legs and the dimensions of its couplings, sharpening for dimensionless couplings to the leg-only verdict Eq. (21.45). One feature now demands an explanation. In both threads the vertices dropped out — the divergent list was finite and order-independent — and this happened exactly because the couplings λ and e are dimensionless in four dimensions. The surviving term $\sum_t V_t \delta_t$ in Eq. (21.39), dormant here, is the whole story when a coupling is *not* dimensionless: a positive δ_t makes high orders less divergent, a negative one makes them progressively worse. The next section takes up that term, ties the mass dimension of every coupling to its field content through Eq. (21.43), and turns the trichotomy already glimpsed above — super-renormalizable, renormalizable, non-renormalizable — into a precise classification of interactions, proving that renormalizability is equivalent to the divergent amplitudes forming a finite, structured set.

21.3 Engineering Dimensions and the Classification of Interactions

The previous section reduced the superficial degree of divergence of any connected diagram to a statement about its external content. For a theory whose couplings are all dimensionless we arrived at the strikingly economical formula $D = d - \sum_{\text{ext}} d_f$ (Eq. (21.45)), in which D depends on *nothing but the number and type of external legs* — not on the order of perturbation theory, not on the internal topology, not on how many loops have been drawn. That economy is the whole reason renormalizability is possible, and it is also, read literally, too good to be true: most theories one can write down do *not* have only dimensionless couplings. The quartic λ of ϕ^4 theory happens to be dimensionless in four dimensions, but ϕ^3, ϕ^6, the Fermi four-fermion interaction, and gravity are governed by couplings carrying nonzero mass dimension, and for those the simple formula must be incomplete. This section supplies the missing piece. We will show that the only thing the previous derivation suppressed was a term proportional to the *mass dimensions of the couplings*, and that restoring it turns the power-counting formula into a complete classification of quantum field theories. The mass dimension of a coupling, it will turn out, decides once and for all whether a theory has finitely many divergences that a fixed set of parameters can absorb, or infinitely many that no finite set ever can.

21.3.1 Mass dimensions of fields and couplings

We work throughout in d spacetime dimensions, using the mostly-minus metric and natural units of Chapter 21. In natural units every quantity carries a single dimension, that of a power of mass; we write $[X]$ for the mass dimension (the *engineering* or *canonical* dimension) of X, so that $[\text{mass}] = 1$, $[\text{length}] = [\text{time}] = -1$, and $[\partial_\mu] = +1$. The entire classification rests on one fixed input: the action is dimensionless, because it appears in the path integral only through e^{iS} and the exponent of an exponential must be a pure number.

Since $S = \int d^d x\, \mathcal{L}$ and $[d^d x] = -d$, the Lagrangian density carries

$$[\mathcal{L}] = d. \tag{21.48}$$

Every term in $\mathcal{L}$ must therefore have mass dimension exactly d. Applying this to the free (kinetic) terms fixes the dimension of each field, since the kinetic term is canonically normalized and present in any dimension. For the real scalar of ϕ^4 theory the kinetic term is $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$ (recall the action Eq. (6.25)), with dimension $2(1 + [\phi])$, and setting this equal to d gives

$$[\phi] = \frac{d-2}{2}. \tag{21.49}$$

For the Dirac field of QED the kinetic term $\bar{\psi}\, i\not{\partial}\, \psi$ (Section 11.3) has dimension $2[\psi] + 1$, whence

$$[\psi] = \frac{d-1}{2}. \tag{21.50}$$

For the gauge field, the Maxwell/Yang–Mills term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $F \sim \partial A$ (the curvature of Section 8.1) has dimension $2(1 + [A])$, identical in form to the scalar, so

$$[A] = \frac{d-2}{2}. \tag{21.51}$$

These three numbers are all we will need. In four dimensions they read $[\phi] = [A] = 1$ and $[\psi] = \frac{3}{2}$; the only difference between a boson and a fermion, dimensionally, is the half-integer offset that comes from the first-order Dirac kinetic operator.

We now turn from the free fields to the interactions. An interaction vertex of *type* i comes from a monomial in $\mathcal{L}$ that we write schematically as

$$\mathcal{L} \;\supset\; g_i\, \partial^{\delta_i} \prod_f \phi_f^{\,n_{if}}, \tag{21.52}$$

where g_i is the coupling, n_{if} counts how many fields of species f (scalar, fermion, gauge) appear at the vertex, and δ_i is the number of spacetime derivatives carried by the vertex. (A derivative coupling such as the Yang–Mills cubic $g f^{abc}(\partial_\mu A^a_\nu) A^{b\mu} A^{c\nu}$ has $\delta_i = 1$; the ϕ^4 vertex has $\delta_i = 0$.) Dimensional consistency of Eq. (21.52) with Eq. (21.48) demands $[g_i] + \delta_i + \sum_f n_{if}[\phi_f] = d$, which we record as a lemma because it is the workhorse of everything that follows.

Lemma 21.9 (Mass dimension of a coupling)**.** *The coupling g_i multiplying a Lagrangian monomial with n_{if} fields of species f (each of dimension $d_f \equiv [\phi_f]$) and δ_i derivatives has mass dimension*

$$[g_i] \;=\; d - \delta_i - \sum_f n_{if}\, d_f. \tag{21.53}$$

Equivalently, writing the local operator at the vertex as $\mathcal{O}_i = \partial^{\delta_i} \prod_f \phi_f^{n_{if}}$ with engineering dimension $[\mathcal{O}_i] = \delta_i + \sum_f n_{if} d_f$, the coupling and operator dimensions are complementary, $[g_i] = d - [\mathcal{O}_i]$.

Proof. Take the mass dimension of both sides of Eq. (21.52). The left-hand side is part of $\mathcal{L}$ and so has dimension d by Eq. (21.48). The right-hand side has dimension $[g_i] + \delta_i + \sum_f n_{if} d_f$, since each derivative contributes $+1$ and each field ϕ_f contributes d_f. Equating and solving for $[g_i]$ gives Eq. (21.53). □

The content of Theorem 21.9 is that a coupling carries exactly the dimension that the operator it multiplies is *missing* from d. A coupling is dimensionless precisely when its operator already saturates the dimension budget d of the Lagrangian on its own.

21.3.2 The dimension-explicit power-counting theorem

We can now reinstate the coupling-dependence that the dimensionless special case Eq. (21.45) suppressed. Recall from Section 21.2 the raw momentum-counting expression for the superficial degree of divergence of a connected diagram,

$$D \;=\; d\,L - \sum_f \omega_f\, I_f + \sum_i V_i\, \delta_i, \tag{21.54}$$

in which L is the number of loops, I_f the number of internal lines of species f, V_i the number of vertices of type i, and ω_f the large-momentum suppression of the species-f propagator (each boson propagator $\sim k^{-2}$ contributes $\omega_b = 2$, each fermion propagator $\sim k^{-1}$ contributes $\omega_F = 1$). The derivative count δ_i appears with a plus sign because momenta in the numerator make integrals worse. The decisive observation, already used in Section 21.2, is that the propagator suppression is itself fixed by the field dimension:

$$\omega_f \;=\; d - 2d_f. \tag{21.55}$$

This is not a new assumption. For a boson, $d_f = (d-2)/2$ gives $d - 2d_f = 2 = \omega_b$; for a fermion, $d_f = (d-1)/2$ gives $d - 2d_f = 1 = \omega_F$; the formula Eq. (21.55) merely repackages the propagator scaling as a statement about dimensions, which is what lets us trade everything for couplings.

Two purely topological identities complete the toolkit. Counting loops as independent momenta left after momentum conservation at each vertex is imposed on the internal lines gives

$$L \;=\; I - V + 1, \qquad I = \sum_f I_f, \quad V = \sum_i V_i. \tag{21.56}$$

Counting line-endings species by species — each vertex of type i emits n_{if} ends of species f, each internal line of that species absorbs two ends and each external line one — gives, for every f,

$$\sum_i V_i\, n_{if} \;=\; 2I_f + E_f, \tag{21.57}$$

with E_f the number of external legs of species f. We now substitute, carrying every step.

Theorem 21.10 (Dimension-explicit power counting)**.** *The superficial degree of divergence of a connected diagram in d spacetime dimensions is*

$$\boxed{D \;=\; d - \sum_{ext} d_f - \sum_i V_i\,[g_i]} \tag{21.58}$$

where the first sum runs over the external legs of the diagram ($\sum_{ext} d_f \equiv \sum_f E_f\, d_f$), $[g_i]$ is the mass dimension of the coupling at vertex type i from Eq. (21.53), and V_i is the number of such vertices.

Proof. Begin from Eq. (21.54) and eliminate ω_f with Eq. (21.55) and L with Eq. (21.56):

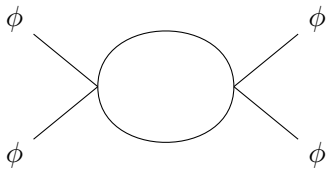
$$\begin{aligned} D &= d\Big(\sum_f I_f - \sum_i V_i + 1\Big) - \sum_f (d - 2d_f) I_f + \sum_i V_i \delta_i \\ &= d + \sum_f I_f\,[d - (d - 2d_f)] - \sum_i V_i (d - \delta_i) \\ &= d + 2\sum_f d_f I_f - \sum_i V_i (d - \delta_i). \end{aligned} \tag{21.59}$$

The internal-line counts I_f are not external data, so we eliminate them in favour of external legs using Eq. (21.57) in the form $2I_f = \sum_i V_i n_{if} - E_f$:

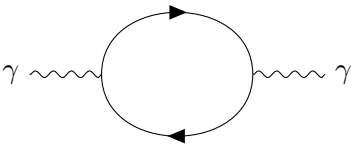
$$2\sum_f d_f I_f = \sum_f d_f \Big(\sum_i V_i n_{if} - E_f\Big) = \sum_i V_i \sum_f n_{if} d_f - \sum_f d_f E_f. \tag{21.60}$$

Inserting Eq. (21.60) into Eq. (21.59) and collecting the two terms that carry a factor V_i,

$$D = d - \sum_f d_f E_f + \sum_i V_i \sum_f n_{if} d_f - \sum_i V_i (d - \delta_i)$$



(a) ϕ^4 theory bubble: $E = 4$, $V = 2$, $L = 1$.



(b) QED vacuum polarization: $E = 2$, $V = 2$, $L = 1$.

Figure 21.2: The two one-loop diagrams used to verify the dimension-explicit power-counting formula Eq. (21.58).

$$= d - \sum_f d_f E_f + \sum_i V_i \Big[\sum_f n_{if} d_f - (d - \delta_i) \Big]. \tag{21.61}$$

The bracket is precisely the negative of the coupling dimension of Eq. (21.53): $\sum_f n_{if} d_f - (d - \delta_i) = -(d - \delta_i - \sum_f n_{if} d_f) = -[g_i]$. Substituting, and writing $\sum_f d_f E_f = \sum_{\text{ext}} d_f$, yields Eq. (21.58). □

Theorem 21.10 is the central result of the section, and its structure deserves to be read aloud. The degree of divergence splits into three contributions of completely different character. The term d is the kinematic cost of one loop's worth of integration measure that no amount of propagator suppression can ever undo; it is the irreducible "engine" of ultraviolet growth. The term $-\sum_{\text{ext}} d_f$ is a fixed penalty paid by the *external* legs: each external leg of positive dimension makes the amplitude *less* divergent, which is why amplitudes with many external particles are tame. Crucially, this second term does not depend on the order of perturbation theory at all. The entire order-dependence of D has been concentrated into the third term, $-\sum_i V_i [g_i]$, and it is controlled by a single number per interaction: the mass dimension of its coupling. Whether divergences proliferate or stay bounded as we go to higher order is therefore decided, completely, by the signs of the $[g_i]$.

21.3.3 Verifying the formula

Before extracting the classification we check Eq. (21.58) against two diagrams the reader has already met, one from each thread. The diagrams are drawn in Fig. 21.2.

Worked Example 21.11 (Power counting the ϕ^4 bubble and the QED vacuum polarization)**. The ϕ^4 bubble.** This is the one-loop four-point diagram of ϕ^4 theory (Fig. 21.2a), whose integral $\int d^d k \,/[(k^2 - m^2)((k+q)^2 - m^2)]$ is the object whose divergence motivated the chapter (the bubble integral of Eq. (15.50)). It has $E = 4$ external scalar legs, $V = 2$ quartic vertices, and the quartic coupling

carries, by Eq. (21.53) with four scalars and no derivatives,

$$[\lambda] = d - 4 \cdot \frac{d-2}{2} = d - 2(d-2) = 4 - d. \tag{21.62}$$

Each external scalar leg has $d_f = (d-2)/2$, so $\sum_{\text{ext}} d_f = 4 \cdot (d-2)/2 = 2(d-2)$. Then Eq. (21.58) gives

$$D = d - 2(d-2) - 2(4-d) = d - 2d + 4 - 8 + 2d = d - 4. \tag{21.63}$$

We may check this against the raw count Eq. (21.54): with $L = 1$, two internal scalar lines ($\omega_b = 2$) and no derivatives, $D = d \cdot 1 - 2 \cdot 2 = d - 4$, in agreement. In $d = 4$ this is $D = 0$: the bubble is *logarithmically* divergent, exactly as the explicit integral of Eq. (15.50) showed. Note how the order-dependence cancelled: because $[\lambda] = 0$ in $d = 4$, the two vertices contributed nothing to D, and the result depended only on the four external legs.

The QED vacuum polarization. This is the one-loop photon self-energy of QED (Fig. 21.2b), computed in Chapter 19. It has $E = 2$ external photon legs, $V = 2$ vertices of the type $-e\,\bar{\psi}\not{A}\psi$. The QED coupling dimension follows from Eq. (21.53) with two fermions, one photon, and no derivatives:

$$[e] = d - 2 \cdot \frac{d-1}{2} - \frac{d-2}{2} = d - (d-1) - \frac{d-2}{2} = 1 - \frac{d-2}{2} = \frac{4-d}{2}. \tag{21.64}$$

Each external photon has $d_f = (d-2)/2$, so $\sum_{\text{ext}} d_f = 2 \cdot (d-2)/2 = d-2$, and

$$D = d - (d-2) - 2 \cdot \frac{4-d}{2} = d - (d-2) - (4-d) = d - 2. \tag{21.65}$$

In $d = 4$ this gives $D = 2$: the vacuum polarization is superficially *quadratically* divergent. The reader who carried out the computation in Chapter 19 knows that the actual divergence is only logarithmic; the quadratic and linear pieces are killed by gauge (Ward–Takahashi) invariance, leaving a transverse $\sim q^2 \log$ structure. This is the first hint, already seen in Chapter 19 and sharpened in the next section, that symmetries can make an amplitude *less* divergent than naive counting predicts — power counting gives an upper bound on the divergence, never an underestimate. The raw check again agrees: $L = 1$, two fermion propagators ($\omega_F = 1$ each) and no derivatives give $D = d \cdot 1 - 1 - 1 = d - 2$.

21.3.4 The trichotomy: super-renormalizable, renormalizable, non-renormalizable

Theorem 21.10 converts a question about infinitely many diagrams into a question about the sign of finitely many numbers, the coupling dimensions $[g_i]$. Suppose first, for clarity, that the theory has a single interaction with coupling g of

dimension $[g]$. Then for a fixed external configuration the degree of divergence at a diagram with V vertices of that type is

$$D = \Big(d - \sum_{\text{ext}} d_f\Big) - V\,[g], \tag{21.66}$$

an arithmetic progression in V with common difference $-[g]$. Everything hinges on the sign of $[g]$, and the three cases define the classical taxonomy of interactions.

Definition 21.12 (Renormalizability classes)**.** An interaction with coupling of mass dimension $[g]$ is called

- *super-renormalizable* if $[g] > 0$: adding a vertex *lowers* D by $[g]$, so divergences get milder at higher order;
- *renormalizable* (or *marginal*) if $[g] = 0$: D is independent of V, depending only on the external legs;
- *non-renormalizable* (or *irrelevant*) if $[g] < 0$: adding a vertex *raises* D by $|[g]|$, so divergences worsen without bound at higher order.

A theory with several couplings inherits the worst case: it is renormalizable if every $[g_i] \geq 0$ (with at least one marginal coupling setting the boundary), and non-renormalizable as soon as a single $[g_i] < 0$.

The names are not arbitrary; the Wilsonian terminology in parentheses — relevant ($[g] > 0$), marginal ($[g] = 0$), irrelevant ($[g] < 0$) — is the same trichotomy read from the renormalization-group flow of Chapter 23, and the agreement is not a coincidence: a coupling of positive mass dimension grows in importance toward the infrared and a coupling of negative dimension dies away, which is exactly the statement that the former is super-renormalizable in the ultraviolet. We use the renormalization-theoretic names in this chapter and connect them to the flow in Chapter 23.

Physical Insight

The deep reason power counting works is dimensional analysis of the cutoff. A non-renormalizable coupling has $[g] < 0$, so on dimensional grounds the dimensionless combination that controls a loop is $g\,\Lambda^{|[g]|}$ with Λ the ultraviolet cutoff: each additional power of the coupling drags in a positive power of Λ, and the loop expansion is really an expansion in E/Λ (energy over cutoff) that breaks down at $E \sim g^{-1/|[g]|}$. A super-renormalizable coupling has the opposite scaling, $g/\Lambda^{[g]}$, dimensionless only when paired with an *inverse* power of the cutoff, so high-energy effects are suppressed. The marginal coupling sits exactly on the boundary: dimensionless on its own, it depends on Λ only logarithmically — which is why renormalizable

theories produce logarithms and a slowly running coupling rather than power-law sensitivity, the subject of Chapter 24.

21.3.5 Renormalizability $\iff$ finitely many divergent amplitudes

The trichotomy of Theorem 21.12 concerns one amplitude at a time. The decisive structural fact — the one that makes renormalization a finite program rather than an infinite regress — is about the *whole set* of divergent amplitudes at once. We now prove it.

Theorem 21.13 (Finiteness criterion)**.** *Work in $d > 2$, so that every field has strictly positive dimension $d_f > 0$. A quantum field theory has only finitely many superficially divergent one-particle-irreducible (1PI) amplitudes if and only if no coupling has negative mass dimension, i.e. $[g_i] \geq 0$ for all i. Equivalently:*

1. *If the theory is renormalizable or super-renormalizable ($[g_i] \geq 0$ for all i), the set of external-leg configurations with $D \geq 0$ is finite, and for a renormalizable theory the degree D of each is fixed once and for all, independent of order.*
2. *If the theory is non-renormalizable (some $[g_j] < 0$), there are infinitely many superficially divergent 1PI amplitudes, and any given amplitude has $D \to \infty$ as the order grows.*

Proof. Sufficiency of $[g_i] \geq 0$. Suppose every coupling satisfies $[g_i] \geq 0$. By Eq. (21.58) the coupling term $-\sum_i V_i[g_i] \leq 0$, so for *every* diagram

$$D \;\leq\; d - \sum_{\text{ext}} d_f. \tag{21.67}$$

A superficially divergent 1PI amplitude requires $D \geq 0$, hence $\sum_{\text{ext}} d_f \leq d$. Let $d_{\min} = \min_f d_f > 0$ be the smallest field dimension appearing in the theory. The total number of external legs $E = \sum_f E_f$ then obeys $E\, d_{\min} \leq \sum_{\text{ext}} d_f \leq d$, so

$$E \;\leq\; \frac{d}{d_{\min}}, \tag{21.68}$$

a fixed finite bound. There are finitely many field species and the total number of legs is bounded by Eq. (21.68), so there are only finitely many external-leg configurations $\{E_f\}$ with $D \geq 0$. That is finitely many superficially divergent 1PI amplitudes. If moreover all marginal, $[g_i] = 0$, the coupling term in Eq. (21.58) vanishes identically and $D = d - \sum_{\text{ext}} d_f$ is the *same* number for that amplitude at every order in perturbation theory, proving the order-independence in statement (1). (Super-renormalizable couplings only strengthen the conclusion: there D

decreases with order, so each amplitude diverges at finitely many orders and beyond some order not at all.)

Necessity / the non-renormalizable case. Suppose some coupling has $[g_j] = -a$ with $a > 0$. Fix any external-leg configuration whatsoever, with $D_0 = d - \sum_{\text{ext}} d_f$ its value at zeroth order in that vertex. Inserting V_j vertices of type j (which can always be done while keeping the diagram 1PI and the external legs fixed, by attaching the extra vertices through internal lines) gives, from Eq. (21.58),

$$D = D_0 + V_j\, a \xrightarrow{V_j \to \infty} +\infty. \tag{21.69}$$

Thus for any fixed amplitude D exceeds any bound once V_j is large enough: every amplitude eventually diverges, and within a single amplitude the superficial degree of divergence grows without limit with the order. Since $D \geq 0$ is reached for arbitrarily large external-leg number as well (take $\sum_{\text{ext}} d_f$ as large as desired and compensate with enough type-j vertices), infinitely many distinct 1PI amplitudes are superficially divergent. This is the negation of finiteness, completing the equivalence. □

The practical weight of Theorem 21.13 is this. In a renormalizable theory the divergences live in a fixed, finite list of amplitudes, each with a fixed degree D. As the next section will show, a degree-D divergence is local — a polynomial of degree at most D in external momenta and masses — so each divergent amplitude needs only a fixed, finite number of subtraction constants, and the list being finite, the whole theory is cured by finitely many parameters. In a non-renormalizable theory the divergent amplitudes form an infinite list, and worse, the degree of divergence of a single amplitude climbs with the order, demanding counterterms with ever more derivatives and ever more fields — an infinite tower of new parameters, one for each new divergent structure. The theory is not thereby meaningless; it is an effective theory predictive only up to the energy scale set by its negative-dimension coupling, a reading we develop in Chapter 23. But it is not renormalizable in the classical sense, and that is exactly the dimensional statement $[g] < 0$.

21.3.6 The critical dimension and a table across d

Theorem 21.12 classifies an interaction at fixed spacetime dimension, but the coupling dimension Eq. (21.53) is itself a function of d. It is therefore natural to ask, for a given interaction, at which dimension it is marginal. That dimension is the watershed of the classification.

Definition 21.14 (Critical dimension)**.** The *critical* (or *upper critical*) dimension d_c of an interaction is the spacetime dimension at which its coupling is marginal, $[g]\big|_{d=d_c} = 0$. For $d < d_c$ the coupling is relevant (super-renormalizable); for $d > d_c$ it is irrelevant (non-renormalizable).

Table 21.1: Engineering dimension of the coupling for the standard interactions, from Eq. (21.53) with $[\phi] = [A] = (d-2)/2$ and $[\psi] = (d-1)/2$. The class is read off at $d = 4$ from Theorem 21.12; the critical dimension d_c is where the coupling is marginal.

Interaction	Operator	$[g]$ in general d	$[g]$ at $d = 4$	Class at $d = 4$	d_c
Cubic scalar	ϕ^3	$3 - \frac{d}{2}$	$+1$	super-renorm.	6
Quartic scalar	ϕ^4	$4 - d$	0	renorm. (marginal)	4
Sextic scalar	ϕ^6	$6 - 2d$	-2	non-renorm.	3
Yukawa	$\bar{\psi}\psi\,\phi$	$\frac{4-d}{2}$	0	renorm. (marginal)	4
Gauge	$\bar{\psi}\not{A}\psi$	$\frac{4-d}{2}$	0	renorm. (marginal)	4
Four-fermion (Fermi)	$(\bar{\psi}\psi)^2$	$2 - d$	-2	non-renorm.	2

For a pure scalar interaction ϕ^n with no derivatives, Eq. (21.53) gives $[g_n] = d - n(d-2)/2$, and setting this to zero solves to the closed form

$$d_c(\phi^n) = \frac{2n}{n-2}. \tag{21.70}$$

This reproduces $d_c = 6$ for ϕ^3, $d_c = 4$ for ϕ^4, and $d_c = 3$ for ϕ^6, and shows $d_c \to 2$ as $n \to \infty$: in two dimensions every polynomial scalar self-interaction is marginal or relevant, which is the dimensional root of the special renormalizability of two-dimensional models. Table 21.1 collects the engineering dimensions of the standard interactions across d and reads off their class in the physical case $d = 4$.

The table is the classification of the physical theories in one glance. In $d = 4$ exactly the marginal interactions — ϕ^4, Yukawa, and the gauge coupling — are renormalizable, and these are precisely the interactions that build the Standard Model. The cubic scalar ϕ^3 is super-renormalizable in four dimensions (and marginal in its critical dimension six, where it is the workhorse of many formal arguments). The sextic scalar and the four-fermion interaction are non-renormalizable in four dimensions. That four dimensions is the marginal dimension for gauge theory and Yukawa simultaneously is one of the quiet structural facts of our world: renormalizable interactions are not generic, and the matter content of the Standard Model is exactly the list of marginal couplings available in $d = 4$.

Worked Example 21.15 (The Fermi theory: why its divergences worsen at every order)**.** The four-fermion contact interaction $\mathcal{L} \supset -G_F(\bar{\psi}\psi)^2$ is the original effective description of the weak interaction. Its coupling dimension follows from Eq. (21.53) with four fermions and no derivatives:

$$[G_F] = d - 4 \cdot \frac{d-1}{2} = d - 2(d-1) = 2 - d. \tag{21.71}$$

In four dimensions $[G_F] = -2$, the most strongly negative entry in Table 21.1, marking the Fermi interaction as emphatically non-renormalizable. (Dimensionally this is the same $G_F \sim 1/M_W^2$ that one reads off the low-energy limit of W-boson exchange: the negative dimension is the mass scale of the heavy physics that the contact term integrates out.) The consequences are immediate from the power-counting formula. Consider the four-fermion amplitude itself, $E = 4$ external fermion legs with $\sum_{\text{ext}} d_f = 4 \cdot (d-1)/2 = 2(d-1)$, built from V Fermi vertices. By Eq. (21.58),

$$D = d - 2(d-1) - V(2-d) = (2-d) + V(d-2) = (d-2)(V-1). \quad (21.72)$$

In $d = 4$ this is $D = 2(V-1)$: the tree level ($V = 1$) is finite, but the one-loop correction ($V = 2$) has $D = 2$ (quadratically divergent), the two-loop ($V = 3$) has $D = 4$, and in general the V-vertex contribution diverges as a polynomial of degree $2(V-1)$ in the external momenta. Each new order brings a divergence of higher degree, requiring a counterterm with two more derivatives than the last — an operator $G_F^{(k)}(\bar\psi\,\partial^{2k}\psi)(\bar\psi\psi)$ for every k — and so an unending sequence of independent new parameters. No finite set of constants absorbs them all. This is the worked instance of statement (2) of Theorem 21.13, and it is the precise sense in which Fermi's theory, though a superb low-energy effective description, cannot be a complete theory: its divergence structure demands the ultraviolet completion that the electroweak gauge theory provides.

21.3.7 The dimensional-regularization corollary and the birth of μ

There is a corollary of Eq. (21.53) that we will lean on for the rest of Part VI and that closes the loop with the regulator chosen in Chapter 21. Dimensional regularization continues the theory to $d = 4 - 2\varepsilon$ spacetime dimensions (the analytic continuation defined in Theorem D.6). But moving off $d = 4$ moves a marginal coupling off the marginal point: a coupling that was dimensionless in four dimensions acquires a small mass dimension proportional to ε.

Concretely, for the quartic coupling of ϕ^4 theory, Eq. (21.62) gives in $d = 4-2\varepsilon$

$$[\lambda] = 4 - d = 4 - (4 - 2\varepsilon) = 2\varepsilon, \quad (21.73)$$

and likewise the gauge and Yukawa couplings acquire $[e] = [g_Y] = (4-d)/2 = \varepsilon$ from Eq. (21.64). A coupling with a mass dimension cannot be a sensible expansion parameter, and more practically we wish to keep doing perturbation theory in a *dimensionless* number even while $d \neq 4$. The remedy is forced and unique: introduce an arbitrary mass scale μ — the renormalization scale — and factor the engineering dimension out of the coupling. For ϕ^4 theory,

$$\boxed{\lambda_{\text{bare}} \longrightarrow \lambda\,\mu^{2\varepsilon}\,,} \quad (21.74)$$

with the new λ on the right dimensionless, and similarly $e \to e\,\mu^{\varepsilon}$, $g_Y \to g_Y\,\mu^{\varepsilon}$ for the marginal QED/Yukawa couplings. The scale μ enters with exactly the power needed to soak up the ε-dependent dimension; in the limit $\varepsilon \to 0$ it formally disappears from the classical action, yet — and this is the crucial point — it cannot disappear from the renormalized theory, because the divergences it accompanies leave behind $\log \mu$ terms when the poles in ε are subtracted.

Physical Insight

The factor $\mu^{2\varepsilon}$ looks like a bookkeeping nuisance, but it is the seed of the single most important phenomenon in Part VI. A dimensionless coupling in $d = 4$ has no scale of its own; Eq. (21.74) smuggles one in through the regulator, and once renormalization has traded the $1/\varepsilon$ poles for finite logarithms, the amplitudes depend on the external energy only through the dimensionless ratio E/μ. Since μ is arbitrary, physics cannot depend on it — and enforcing that independence is exactly the renormalization-group equation. The dimensionful coupling of Eq. (21.73) is, quite literally, where the *running coupling* of Theorem 24.21 is born.

This places ϕ^4 theory precisely. The quartic coupling is the marginal, renormalizable workhorse of four dimensions: $[\lambda] = 4 - d$ puts it exactly on the boundary in $d = 4$ (Table 21.1), so by Theorem 21.13 its divergent amplitudes form a fixed finite set, and by Eq. (21.74) regulating those divergences forces the scale μ into the theory. The two-, four-, and (would-be) higher-point functions are the only candidates for divergence; which of them actually diverge, and to what degree, we already half-know from Theorem 21.11. The numbers that Chapter 24 turns into the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ are precisely the residues of the $1/\varepsilon$ poles that Eq. (21.74) has now taught us to expect.

With the classification complete we know exactly *which* 1PI amplitudes diverge in a renormalizable theory — a finite list fixed by Theorem 21.13 — and *how badly*, through the degree D of Eq. (21.58). What we have not yet determined is the *form* of each divergence: is the divergent piece an arbitrary function of the external momenta, or something far more constrained? The next section answers this, proving that a superficially divergent amplitude diverges only through a polynomial in its external momenta and masses of degree at most D — that divergences are *local* — and hence that each one matches, term for term, an operator already present in the Lagrangian of a renormalizable theory.

21.4 The Locality of Divergences and the Renormalizable Miracle

The classification of the previous section delivered a sharp but still incomplete piece of news. Power counting, sharpened into the dimension-explicit theorem and read off from the mass dimensions of the couplings, told us *which* one-particle-irreducible (1PI) amplitudes of a renormalizable theory can diverge: in $d = 4$ the list is finite, and for ϕ^4 theory it consists of exactly two entries, the two-point and the four-point functions. What it did not tell us is what the divergence *looks like as a function of the external data*. A 1PI amplitude $\Gamma(p_1, \dots, p_n)$ is, after all, an honest function of the external momenta and the masses; a priori its ultraviolet divergence could be some complicated, momentum-dependent object — a logarithm of p^2/μ^2, a dilogarithm, anything the loop integral cares to produce. If that were so, no finite set of constants could ever absorb it, and the entire program of renormalization would be stillborn.

The content of this section is that this fear is unfounded, and spectacularly so. The divergent part of a superficially divergent 1PI amplitude with degree D is not an arbitrary function: it is a *polynomial* in the external momenta and masses, of degree at most D. A polynomial in momenta is, by the Fourier dictionary of Theorem 15.30 read backwards, a finite sum of *local* operators — terms with a fixed, finite number of derivatives acting at a single spacetime point. The divergence is local. And once it is local, the renormalizable miracle is one short step away: in a renormalizable theory each such local divergent structure turns out to coincide, term for term, with an operator already present in the Lagrangian, so that the divergences of the entire theory — to all orders, for every primitively divergent amplitude — can be reabsorbed into a finite redefinition of the handful of parameters we started with. We prove the locality theorem in full by the differentiation argument, exhibit it on the two divergent amplitudes of ϕ^4, count the independent divergent structures and match them to the Lagrangian, and then do the same accounting for QED, where gauge invariance intervenes to make the count even tighter than power counting alone would suggest.

21.4.1 The locality theorem

We work with a generic 1PI amplitude built from a single primitively divergent loop integral — one that is superficially divergent overall but has *no divergent subintegrals*. This is the case in which the argument is both clean and complete; the necessary caveat about subdivergences, which spoils the naive statement for general multi-loop graphs, is taken up honestly in the next section, and the constructive cure is the business of Chapter 22. Within this scope, recall from Theorem D.2 and Eq. (D.4) (and the chapter's own Eq. (21.19)) that such an

integral has the shape

$$\Gamma(p) \;=\; \int \prod_{i=1}^{L} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d} \, \frac{\mathcal{N}(\ell; p)}{\prod_{j=1}^{N} (q_j^2 - m_j^2 + i\varepsilon)}, \qquad q_j \;=\; \sum_i c_{ji}\, \ell_i + \sum_a d_{ja}\, p_a, \tag{21.75}$$

where p stands collectively for the external momenta, $\mathcal{N}$ is a polynomial of total degree P in the loop momenta, and the superficial degree of divergence is $D = dL + P - 2N$. We hold a regulator in force throughout — a Euclidean momentum cutoff Λ for transparency, or dimensional regularization for the actual computations — so that $\Gamma(p)$ is, at finite regulator, a perfectly well-defined smooth function of p. The divergence is whatever survives, and grows without bound, as the regulator is removed ($\Lambda \to \infty$, or $\varepsilon \to 0$).

The whole argument rests on one observation about what an external-momentum derivative does to the integrand.

Lemma 21.16 (Differentiation lowers the degree)**.** *Let $\Gamma(p)$ be the regulated integral (21.75) with superficial degree of divergence D. Then $\partial\Gamma/\partial p^\mu$ is an integral of the same form whose superficial degree of divergence is $D - 1$. More generally, differentiating k times with respect to external momenta produces an integral with superficial degree $D - k$.*

Proof. External momenta enter (21.75) only through the line momenta $q_j = \sum_i c_{ji}\ell_i + \sum_a d_{ja}p_a$, and only those lines j with $d_{ja} \neq 0$ for the chosen external leg a depend on p_a. Differentiation of the integrand passes under the integral sign — legitimate because at finite regulator the integrand and its p-derivatives are continuous and the regulated domain makes everything absolutely convergent — and acts on each such denominator by the chain rule,

$$\frac{\partial}{\partial p_a^\mu} \frac{1}{q_j^2 - m_j^2 + i\varepsilon} \;=\; \frac{-\,2\, d_{ja}\, (q_j)_\mu}{\left(q_j^2 - m_j^2 + i\varepsilon\right)^2}, \tag{21.76}$$

and on a numerator that depends on p at most by lowering its p-degree, never raising the loop-momentum degree. Consider the typical term produced by (21.76). It replaces one denominator factor $1/(q_j^2 - m_j^2)$ by $(q_j)_\mu/(q_j^2 - m_j^2)^2$: the loop-momentum content of the numerator increases by one power (the explicit $(q_j)_\mu$, which is linear in the ℓ_i), so $P \to P + 1$, while the denominator gains one extra power of a quadratic form, so $N \to N+1$ in the counting and $2N \to 2N+2$. The net change in $D = dL + P - 2N$, at fixed loop number L, is

$$\Delta D \;=\; \underbrace{(+1)}_{\Delta P} \;-\; \underbrace{(+2)}_{\Delta(2N)} \;=\; -1. \tag{21.77}$$

The product rule may generate a sum of several such terms (one per p-dependent line), and differentiating the numerator yields terms of strictly lower loop-momentum degree, hence of even smaller D. In every term the superficial

degree of divergence drops by at least one, and the leading terms drop by exactly one. Iterating k times gives an integrand that is a sum of contributions each with superficial degree at most $D - k$, which is the claim. □

The scaling intuition behind (21.77) is worth stating in one line, because it makes the result unforgettable: an external-momentum derivative pulls down one power of a loop momentum from the denominator while costing two powers in the denominator's growth, so it improves the ultraviolet convergence by exactly one power. With this in hand the locality theorem is immediate.

Theorem 21.17 (Locality of the divergence)**.** *Let $\Gamma(p)$ be a primitively divergent 1PI amplitude of the form (21.75) with superficial degree of divergence $D \geq 0$. Then the ultraviolet-divergent part of $\Gamma(p)$ — the part that grows without bound as the regulator is removed — is a polynomial in the external momenta of degree at most D, with the masses entering so that the total degree in momenta and masses (each mass counted with weight one) is at most D. Equivalently, the divergence is the matrix element of a finite sum of local operators, each carrying at most D derivatives.*

Proof. By Theorem 21.16, the $(D+1)$-fold external-momentum derivative

$$\frac{\partial^{D+1}\Gamma}{\partial p^{\mu_1}\cdots\partial p^{\mu_{D+1}}}(p) \tag{21.78}$$

is an integral with superficial degree of divergence $D-(D+1) = -1 < 0$. For a primitively divergent integral — one with no divergent subintegrals — a negative overall superficial degree is, by the convergence criterion that the next section sharpens into Weinberg's theorem, sufficient for absolute convergence; at one loop, where there are no proper subintegrals at all, $D < 0$ is necessary and sufficient and the conclusion is unconditional. Hence (21.78) converges as the regulator is removed: it tends to a finite limit, with no surviving divergence.

Now split $\Gamma = \Gamma_{\mathrm{div}} + \Gamma_{\mathrm{fin}}$, where $\Gamma_{\mathrm{div}}(p)$ collects everything that diverges as the regulator is removed and $\Gamma_{\mathrm{fin}}(p)$ stays finite. Differentiation commutes with this split (it is linear, and the derivative of the finite remainder is finite), so the finiteness of (21.78) forces

$$\frac{\partial^{D+1}\Gamma_{\mathrm{div}}}{\partial p^{\mu_1}\cdots\partial p^{\mu_{D+1}}} \;=\; 0. \tag{21.79}$$

A function whose every derivative of order $D+1$ vanishes identically is a polynomial of degree at most D. Therefore $\Gamma_{\mathrm{div}}(p)$ is a polynomial of degree $\leq D$ in the external momenta. The same argument run with derivatives $\partial/\partial m_j^2$ shows the divergence is polynomial in the masses as well: by (21.76) with $q_j^2 - m_j^2$ in place of an external dependence, $\partial/\partial m_j^2$ raises a denominator power without touching the numerator, lowering D by two, so after enough mass derivatives

the integral converges and the divergence is polynomial in the m_j^2. Because each propagator denominator is homogeneous of degree two under the simultaneous scaling $(\ell, p, m) \to (t\ell, tp, tm)$, the divergent polynomial is homogeneous of total degree D in (p, m) up to logarithms of the regulator, so a mass enters with weight one. The Taylor expansion of Γ about $p = 0$ thus has its divergence confined to the first $D + 1$ coefficients,

$$\Gamma_{\mathrm{div}}(p) \;=\; \sum_{0 \le |\alpha| \le D} c_\alpha(\Lambda; m)\, p^\alpha, \qquad \deg_{(p,m)} c_\alpha\, p^\alpha = D, \tag{21.80}$$

each coefficient c_α a (mass-dependent) divergent constant and a finite polynomial in m. A monomial p^α of degree $|\alpha|$ is the momentum-space image of $|\alpha|$ derivatives acting at one point, $p^\alpha \leftrightarrow (i\partial)^\alpha \delta^{(d)}$; (21.80) is therefore the statement that Γ_{div} is the insertion of a finite sum of local operators with at most D derivatives. □

Physical Insight

The physical content of Theorem 21.17 is that ultraviolet divergences are short-distance, hence local, phenomena. The divergent part of a loop integral comes from the region where all internal momenta are large together, which in position space is the region where all interaction points collapse to a single point (Section 21.1.2). A measurement at finite external momentum cannot resolve that point; the diagram, viewed from outside, looks like a single effective vertex — a local operator. An external-momentum derivative is precisely an instruction to probe finer spatial structure, and each derivative the amplitude can absorb before becoming finite measures how much short-distance "charge" the divergence carries. A theory in which the available local structures are few and already present in the action is one whose short-distance ignorance is fully parametrized by a finite set of couplings; this is the seed of both renormalizability and the Wilsonian picture of Chapter 23.

In dimensional regularization the same theorem reads: the $1/\varepsilon$ poles of a primitively divergent 1PI amplitude are polynomial in the external momenta and masses of degree at most D. Dimensional regularization sees no power divergences — there is no Λ to produce a Λ^2 — so a would-be quadratic divergence ($D = 2$) shows up as a $1/\varepsilon$ pole multiplying a degree-two polynomial $A + B\,p^2$ rather than as $\Lambda^2 + \Lambda^0 p^2$, but the polynomial structure and its degree are identical. The degree D is regulator-independent, exactly as established for the leading behavior in Section 21.1.3; what locality adds is that the *entire* divergence, not just its leading term, is a polynomial of that degree.

21.4.2 The two divergent amplitudes of ϕ^4

We now watch the theorem operate on ϕ^4 theory, where the classification of the previous section left only two superficially divergent 1PI amplitudes in $d = 4$: the four-point function ($E = 4$, $D = 0$) and the two-point function ($E = 2$, $D = 2$), with the odd-point amplitudes killed by the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ and everything with $E \geq 5$ superficially convergent, as $D = 4 - E$ records (Eq. (21.109)).

Worked Example 21.18 (The four-point bubble is momentum-independent: a pure $\delta\lambda$)**.** The one-loop correction to $\phi\phi \to \phi\phi$ is the bubble of Theorem 15.16, the $D = 0$ integral

$$\mathcal{B}(p) \;=\; \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon}\,\frac{i}{(p-k)^2 - m^2 + i\varepsilon}, \qquad p \equiv p_1 + p_2, \quad (21.81)$$

written from Eq. (15.50) with the single Mandelstam combination $p = p_1 + p_2$ flowing through the loop (the t- and u-channel bubbles are identical with $p \to p_1 - p_3,\ p_1 - p_4$). With $D = 0$, Theorem 21.17 predicts a divergence that is a polynomial of degree zero — a pure constant, with no dependence on p whatsoever. We verify this directly by differentiating once. Using (21.76) on the second factor,

$$\frac{\partial \mathcal{B}}{\partial p^\mu} \;=\; \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon}\,\frac{-2i\,(p-k)_\mu}{\left((p-k)^2 - m^2 + i\varepsilon\right)^2}. \quad (21.82)$$

Count the powers of k at large loop momentum: the numerator carries one power, $(p-k)_\mu \sim k$, while the denominator now behaves as $k^2 \cdot (k^2)^2 = k^6$, against the measure $\mathrm{d}^4k \sim k^4$. The net power is $4 + 1 - 6 = -1 < 0$: the integral (21.82) converges. (This is Theorem 21.16 with $D = 0 \to D = -1$, made arithmetic.) Hence the entire p-dependence of $\mathcal{B}$ is finite, and the divergence lives only in the p-independent piece. Concretely, the divergent part of (21.81) is the value of the integral at $p = 0$,

$$\mathcal{B}_{\mathrm{div}} \;=\; \mathcal{B}(0)\big|_{\mathrm{div}} \;=\; \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{(k^2 - m^2 + i\varepsilon)^2}\bigg|_{\mathrm{div}} \;=\; \frac{i}{16\pi^2}\,\frac{1}{\varepsilon} + (\text{finite}), \quad (21.83)$$

the standard logarithmically divergent scalar bubble at zero external momentum, evaluated with the master integral of Theorem D.2's appendix. The full one-loop amplitude is then $\mathcal{B}(p) = \mathcal{B}_{\mathrm{div}} + [\mathcal{B}(p) - \mathcal{B}(0)]$, a divergent constant plus a finite, momentum-dependent remainder that carries all the physics of the loop (the threshold branch cut at $p^2 = 4m^2$, the running). The divergent constant is a momentum-independent shift of the four-point vertex — and the tree-level four-point vertex is the constant $-i\lambda$ of Eq. (15.58). A constant divergence in the $E = 4$ amplitude is therefore a renormalization of the single number λ: it is exactly a counterterm $\delta\lambda$ and nothing else. There is no room for a $\delta\lambda\, p^2$ structure, because that would require $D \geq 2$ in the four-point function, which $D = 4 - E = 0$ forbids.

Worked Example 21.19 (The two-point function: divergence of degree two, $A + B\,p^2$)**.** The $E = 2$ amplitude is the 1PI self-energy $-i\Sigma(p^2)$ of Theorem 15.30, with superficial degree $D = 4 - E = 2$. By Lorentz invariance Σ depends on the external momentum only through p^2, and Theorem 21.17 predicts a divergence that is a polynomial of degree at most two in p (and m) — which, with only the scalar p^2 available, can only be

$$\Sigma_{\text{div}}(p^2) \;=\; A \;+\; B\,p^2, \tag{21.84}$$

two divergent constants A (mass dimension two, the quadratically divergent piece) and B (dimensionless, the logarithmically divergent piece). The differentiation argument identifies them as Taylor coefficients: $A = \Sigma(0)\big|_{\text{div}}$ and $B = \Sigma'(0)\big|_{\text{div}}$, where $\Sigma' = \partial\Sigma/\partial p^2$. The first derivative $\partial\Sigma/\partial p^2$ has $D = 2 - 2 = 0$ (two units of p-differentiation against the p^2 argument), still logarithmically divergent, so B is genuinely there in general; the second derivative $\partial^2\Sigma/\partial(p^2)^2$ has $D = 2 - 4 = -2 < 0$, convergent, so the divergence stops at p^2 and (21.84) is exact.

Now read the two structures against the Lagrangian. The full inverse propagator is $\Gamma^{(2)}(p^2) = p^2 - m^2 - \Sigma(p^2)$, the tree piece coming from $\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$. A divergent constant A shifts $m^2 \to m^2 + A$: it renormalizes the *mass*, the counterterm $\delta m^2 = A$. A divergent $B\,p^2$ shifts the coefficient of p^2 away from unity, $1 \to 1 - B$: it renormalizes the *kinetic term*, equivalently the normalization of the field, the wavefunction counterterm $\delta Z = -B$. Each of the two allowed divergent structures matches one of the two quadratic terms in $\mathcal{L}$, and no third structure (a p^4, which would be a higher-derivative operator absent from $\mathcal{L}$) can appear, because $D = 2$ caps the polynomial at degree two.

It is instructive — and an honest illustration of the difference between the *bound* the theorem provides and the *values* a particular order delivers — to note what actually happens at one loop. The only one-loop 1PI two-point graph in ϕ^4 is the tadpole: a single quartic vertex with two of its legs joined into a loop,

$$-i\Sigma_1 \;=\; \frac{-i\lambda}{2}\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m^2 + i\varepsilon} \;=\; \frac{-i\lambda}{2}\Big(-\frac{m^2}{16\pi^2}\Big)\Big(\frac{1}{\varepsilon} + \cdots\Big), \tag{21.85}$$

where the symmetry factor $\frac{1}{2}$ is the one fixed in Theorem 15.16's companion analysis. The integrand of (21.85) *does not depend on the external momentum at all*: the external legs attach to the same vertex and inject no momentum into the loop. Hence $\partial\Sigma_1/\partial p^2 = 0$ identically, so $B = 0$ at one loop and the entire one-loop self-energy divergence is the constant $A = \Sigma_1$, a pure δm^2. The wavefunction renormalization δZ is not zero in principle — the theorem permits it — but its first nonvanishing divergent contribution waits until two loops, the "sunset" diagram (three internal lines between two vertices), which *does* route external momentum through the loop and produces a genuine $B\,p^2/\varepsilon^2$. The locality theorem guarantees that even that two-loop divergence is still capped at $A + Bp^2$: no new operator structure is generated, only the two already anticipated.

Collecting the two worked examples, the complete divergent content of ϕ^4 in $d = 4$ is the three structures

$$\boxed{\underbrace{\delta m^2}_{E=2,\,p^0}\,, \qquad \underbrace{\delta Z\, p^2}_{E=2,\,p^2}\,, \qquad \underbrace{\delta\lambda}_{E=4,\,p^0}} \tag{21.86}$$

and these are in one-to-one correspondence with the three terms of the frozen ϕ^4 Lagrangian,

$$\mathcal{L} \;=\; \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}m^2\phi^2 \;-\; \frac{\lambda}{4!}\phi^4, \tag{21.87}$$

the kinetic term carrying δZ, the mass term carrying δm^2, and the quartic carrying $\delta\lambda$. There is no fourth divergent structure to absorb. This is the precise statement that the divergences of ϕ^4 in four dimensions are reabsorbed by renormalizing the three parameters Z, m^2, λ already in the action — the exact objects Chapter 22 subtracts as counterterms $\delta Z, \delta m^2, \delta\lambda$ and Chapter 24 turns into the beta function and anomalous dimension (the bare/renormalized relation Eq. (24.2)).

21.4.3 The divergent structures of QED

QED illustrates the same logic with a richer cast and one new actor — gauge invariance — that makes the count tighter than power counting alone. The dimension-explicit counting of the previous section (Eq. (21.47)) leaves a short list of superficially divergent 1PI amplitudes in $d = 4$: the photon two-point function (vacuum polarization $\Pi^{\mu\nu}$, $D = 2$), the electron two-point function (self-energy Σ, $D = 1$), and the electron–photon vertex (Γ^μ, $D = 0$). The would-be divergent photon three- and four-point amplitudes are eliminated — the three-photon amplitude by Furry's theorem (charge-conjugation invariance forces an odd number of external photons on a fermion loop to vanish), the four-photon amplitude by gauge invariance, which renders it convergent despite $D = 0$. The three survivors are drawn in Fig. 21.3.

Vacuum polarization, $\Pi^{\mu\nu}$ ($\to Z_3$). With $D = 2$, Theorem 21.17 permits a degree-two polynomial in the external photon momentum q. The two Lorentz tensors of the right dimension are $\eta^{\mu\nu}$ (degree zero, but with a quadratically divergent coefficient to make up the two units, $\sim \Lambda^2\eta^{\mu\nu}$ or a q-independent $1/\varepsilon$ pole) and $q^\mu q^\nu$, $q^2\eta^{\mu\nu}$ (degree two, logarithmically divergent). The first would be a *photon mass term* $\propto A_\mu A^\mu$, which is absent from the QED Lagrangian and forbidden by gauge invariance. Gauge invariance keeps its promise: the Ward identity forces the vacuum polarization to be transverse,

$$q_\mu\Pi^{\mu\nu}(q) = 0 \quad\Longrightarrow\quad \Pi^{\mu\nu}(q) = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\,\Pi(q^2), \tag{21.88}$$

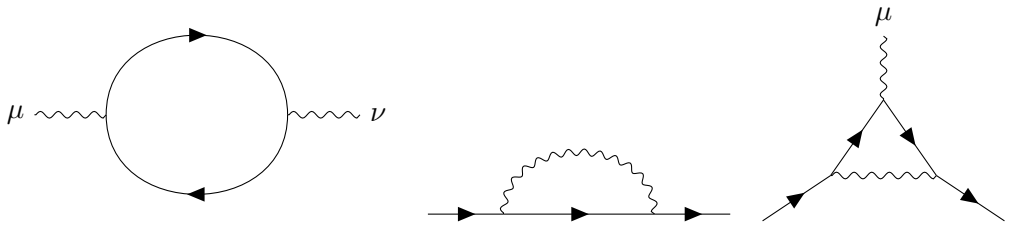


Figure 21.3: The three superficially divergent 1PI amplitudes of QED at one loop. Left: the vacuum polarization $\Pi^{\mu\nu}(q)$ ($D = 2$), an electron loop dressing the photon. Middle: the electron self-energy $\Sigma(p)$ ($D = 1$), a photon dressing the electron line. Right: the vertex correction $\Gamma^{\mu}(p', p)$ ($D = 0$), a photon rung across the electron–photon vertex.

so the would-be quadratic divergence — the dangerous $\Lambda^2 \eta^{\mu\nu}$ photon mass — cancels identically, and only the logarithmically divergent scalar $\Pi(q^2)$ survives. Its divergence is a q-independent constant multiplying the transverse structure $q^2\eta^{\mu\nu} - q^\mu q^\nu$, which is precisely the momentum-space form of the photon kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$. The vacuum-polarization divergence therefore renormalizes the photon field alone: it is the counterterm δ_3, i.e. the photon-field renormalization Z_3.

Electron self-energy, $\Sigma(p)$ ($\to Z_2$, Z_m). With $D = 1$, the divergence is a polynomial of degree at most one in p (and m). The available structures of mass dimension one are $\not{p}$ (degree one) and m (degree one), so

$$\Sigma_{\text{div}}(p) \;=\; \alpha_2 \not{p} \;+\; \alpha_m\, m, \tag{21.89}$$

with α_2, α_m logarithmically divergent constants. Naively $D = 1$ would also allow a degree-zero structure — a constant of dimension one, $\propto \Lambda \cdot \mathbf{1}$, a linear divergence — but the chiral symmetry of the massless theory ($\psi \to e^{i\theta\gamma^5}\psi$, exact at $m = 0$) forbids any divergence not accompanied by an explicit factor of m when it would break that symmetry; the constant must be $\propto m$, softening the linear divergence to logarithmic. The two surviving structures match the two pieces of the electron's quadratic Lagrangian $\bar\psi(i\not\partial - m)\psi$: the $\not{p}$ structure renormalizes the electron kinetic term (the field, Z_2), the m structure renormalizes the mass term (Z_m). The standard parametrization $\Sigma_{\text{div}} = \delta_2 \not{p} - \delta_m$ packages exactly these.

Vertex correction, $\Gamma^{\mu}(p', p)$ ($\to Z_1$). With $D = 0$, the divergence is a constant (degree-zero) Lorentz structure, which can only be $\propto \gamma^\mu$ — the same structure as the tree vertex $-ie\gamma^\mu$. (Other tensors such as $\sigma^{\mu\nu}q_\nu$, the magnetic-moment structure, carry an extra power of external momentum, hence degree one, hence are convergent — which is exactly why the anomalous moment of

Chapter 19 is finite and calculable.) The vertex divergence is therefore a pure renormalization of the electron–photon coupling $-ie\,\bar{\psi}\not{A}\psi$: the counterterm δ_1, the vertex renormalization Z_1.

Power counting plus locality thus assigns QED four primitive divergent structures and four renormalization constants Z_3, Z_2, Z_m, Z_1, each matched to a term already in

$$\mathcal{L}_{\text{QED}} \;=\; \bar{\psi}(i\not{D} - m)\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu. \tag{21.90}$$

Gauge invariance then supplies one further, decisive economy. The Ward–Takahashi identity, derived in full from the $U(1)$ invariance of the path integral in Theorem 19.13, relates the vertex to the inverse electron propagator and forces the vertex and field renormalizations to coincide,

$$Z_1 = Z_2. \tag{21.91}$$

We do not reprove this here — it is owned by Chapter 19 — but we record its structural consequence: the electric-charge renormalization, which combines all four constants as $e_0 = Z_1 Z_2^{-1} Z_3^{-1/2}\, e$, collapses by (21.91) to $e_0 = Z_3^{-1/2}\, e$. The running of the QED coupling is therefore governed by the photon-field renormalization alone, the single number that the vacuum polarization (21.88) computes — the fact Chapter 24 turns into $\beta_e = e^3/12\pi^2$. Four divergent structures, three independent renormalization constants, one of them controlling the charge: this is the miracle in its tightest form.

21.4.4 The renormalizable miracle, stated precisely

We can now give the renormalizable miracle the sharp statement that the rest of Part VI builds on.

Theorem 21.20 (Locality plus classification ⇒ finite renormalization)**.** *Let a theory be renormalizable in the sense of the previous section — all of its couplings have non-negative mass dimension, so that the superficial degree of divergence of a 1PI amplitude is bounded above independently of the loop order and the set of superficially divergent amplitudes is finite. Then, for the primitively divergent amplitudes, the divergence of each is by Theorem 21.17 a local operator of bounded dimension, and every such local structure coincides with a term already present in the Lagrangian. Consequently all primitive divergences of the theory are absorbed by a finite redefinition of the finitely many parameters (fields, masses, couplings) of the original action; no operator absent from the Lagrangian is ever generated as a counterterm.*

Argument. Renormalizability bounds D above by a fixed number $D_{\max}$ determined by the external-leg content alone, with no growth in the loop order (this

is the content of the dimension-explicit power-counting theorem of the previous section; for ϕ^4, $D = 4 - E$, and for QED the analogous leg formula). Hence only finitely many amplitudes have $D \geq 0$. For each, Theorem 21.17 makes the divergence a polynomial in momenta and masses of degree $\leq D$, i.e. a finite sum of local operators of mass dimension $\leq d$. But the operators of dimension $\leq d$ compatible with the theory's symmetries (Lorentz, internal, $\mathbb{Z}_2$ for ϕ^4, gauge invariance for QED) are exactly the terms of a renormalizable Lagrangian — that is what "renormalizable" means dimensionally. Each divergent structure therefore has the same form as an existing Lagrangian term, and adding a counterterm proportional to that term — a shift of the corresponding parameter — removes it. The number of parameters is finite, so the procedure closes. □

Physical Insight

The miracle is not that loop integrals diverge — short-distance physics we have not included must leave *some* imprint — but that the imprint is so structured. A non-renormalizable theory diverges too, but each loop order generates divergences of higher and higher degree D (because the negative-dimension coupling must be compensated by more powers of momentum), hence higher-derivative local operators not present in the action; renormalizing them requires new parameters without end. A renormalizable theory is one whose ignorance of short-distance physics is fully encoded in a finite menu of local operators, fixed once and for all by a finite number of measurements. Chapter 23 reinterprets this as the statement that, under the renormalization-group flow toward long distances, only the relevant and marginal operators — the renormalizable ones — survive, and the rest are washed out; renormalizability is the infrared shadow of that flow.

What the locality theorem and the classification together establish is thus a finite dictionary: *divergent structure* $\mapsto$ *Lagrangian term*. For ϕ^4 theory it has three entries, δm^2, δZ, $\delta\lambda$ of (21.86); for QED it has four, Z_3, Z_2, Z_m, Z_1, reduced to three independent constants by the Ward identity (21.91). These are precisely the objects the next chapter promotes to counterterms and subtracts diagram by diagram, fixing their finite parts by renormalization conditions; the explicit construction, and the proof that the subtraction is consistent to all orders, is Chapter 22. One proviso must travel with the dictionary, however, and it is the subject of the next section: the clean picture proved here holds for *primitively* divergent amplitudes. A diagram with $D < 0$ overall can still diverge if it hides a divergent subdiagram, and when subdivergences are present the differentiation argument no longer reaches them — the divergence ceases to be a single local polynomial in the external momenta. Classifying those subdivergence topologies, and stating the convergence criterion (Weinberg's theorem) that supersedes naive power counting, is where we turn next.

21.5 When Superficial Counting Lies: Subdivergences and Weinberg's Theorem

The previous section proved a clean and powerful statement, but it proved it under a quietly restrictive hypothesis. The locality theorem (Theorem 21.17) took as its input a *primitively* divergent amplitude — one that is superficially divergent overall and has no divergent subintegrals — and showed that its divergence is a local polynomial of degree at most D in the external momenta. The differentiation argument that delivered that result worked because each external-momentum derivative lowered the overall degree of divergence by one unit, and a single negative overall degree was enough to certify convergence. That logic is impeccable as far as it reaches. What it does not touch is the possibility that the global count D is simply not the right thing to count. A Feynman integral over several loop momenta has many ways to run off to infinity: all loop momenta together, but also any *subset* of them with the rest held fixed. The superficial degree D measures only the first of these. The moment a diagram has more than one loop, the others become available, and a diagram can be globally innocent — $D < 0$, hence superficially convergent — while a sub-integration over part of the loop momentum diverges all on its own. This is the loophole flagged in Theorem D.2 and again at Theorem 21.5; the present section closes it. We make the notion of a subdiagram and its degree of divergence precise, classify the ways two divergent subdiagrams can sit relative to one another, exhibit ϕ^4 theory's first genuinely entangled (overlapping) divergence at two loops, and state the theorem of Weinberg that supersedes naive power counting as the exact criterion for ultraviolet convergence. The constructive cure — how one actually subtracts these nested infinities consistently to all orders — is the business of Chapter 22; here we only diagnose, but we diagnose completely.

21.5.1 Subdiagrams and the degree of divergence of a sub-integration

The superficial degree of divergence of Theorem D.2 was defined by scaling *all* loop momenta uniformly to infinity. A multi-loop integral, however, has a far richer large-momentum structure: the integration runs over an Ld-dimensional momentum space, and infinity can be approached along any of its directions, not only the diagonal on which every loop momentum grows at the same rate. Each independent direction along which some momenta grow and others stay finite is a separate test of convergence, and each is governed by its own power count. To name those directions we need the notion of a subdiagram.

Definition 21.21 (Subdiagram and its superficial degree)**.** Let G be a 1PI Feynman graph (Theorem 15.30) with internal lines $\mathcal{I}(G)$ and vertices $\mathcal{V}(G)$. A *subdiagram* $\gamma \subseteq G$ is a subset of the internal lines together with all vertices

they touch; we take γ to be itself one-particle-irreducible, since reducible pieces never produce new ultraviolet divergences (cutting a single line factorizes the integral). The lines of γ that attach to a vertex also touched by a line outside γ are the *external legs* of γ; the loops of γ are the independent cycles formed by its internal lines, $L_\gamma = I_\gamma - V_\gamma + 1$, where $I_\gamma = |\mathcal{I}(\gamma)|$ and $V_\gamma = |\mathcal{V}(\gamma)|$. The *superficial degree of divergence of* γ is the net power of momentum produced when the L_γ loop momenta internal to γ are scaled uniformly to infinity with all momenta outside γ held fixed,

$$D(\gamma) \;=\; d\,L_\gamma \;+\; P_\gamma \;-\; 2I_\gamma, \tag{21.92}$$

where P_γ is the total degree in the γ-internal loop momenta of whatever numerator factors the lines of γ carry. The full graph is the special case $\gamma = G$, for which Eq. (21.92) reduces to the global degree $D = D(G)$ of Eq. (D.4).

The formula Eq. (21.92) is Eq. (D.4) applied to γ alone, and the justification is the same scaling argument: send the γ-internal momenta to infinity, $\ell_i = t\,\hat{\ell}_i$ for the L_γ loop momenta routed through γ, and read off the power of t. The measure contributes t^{dL_γ} (a factor t^d per loop integrated over γ); each of the I_γ propagators in γ, dominated at large t by its quadratic term, contributes t^{-2}; the numerator contributes t^{P_γ}. The radial integral $\int^\infty t^{D(\gamma)-1}\,\mathrm{d}t$ then diverges at its upper limit precisely when $D(\gamma) \geq 0$. The momenta *external* to γ — the external legs of γ, which include both the true external momenta of G and the loop momenta of G that do not run through γ — are held fixed during this scaling, so they appear only as subleading constants in each denominator, exactly as the external momenta did in the one-loop count of Theorem D.3. The crucial point is that this is a genuinely different limit from the global one whenever γ is a proper subdiagram: the global degree $D(G)$ scales *all* loop momenta, while $D(\gamma)$ scales only those threading γ.

Definition 21.22 (Primitive divergence)**.** A 1PI graph G is said to have a *primitive divergence*, or to be *primitively divergent*, if $D(G) \geq 0$ while $D(\gamma) < 0$ for every proper 1PI subdiagram $\gamma \subsetneq G$. A primitively divergent graph has exactly one divergence — the overall one, counted by $D(G)$ — and no subdivergences. Every superficially divergent one-loop graph is primitive, because a single loop has no proper subdiagram that is itself a loop.

This is the case the previous section silently assumed and the case in which superficial counting tells the whole truth. It is precisely the situation of the one-loop diagrams of ϕ^4 theory that the next section computes: each is primitive, so the locality theorem applies to it without qualification. The interesting failure begins exactly where primitivity ends.

21.5.2 The failure mode: a convergent-looking integral that diverges

The whole point of this section is contained in a single inequality that can go the wrong way:

$$D(G) < 0 \quad \text{does } \textit{not} \text{ imply} \quad D(\gamma) < 0 \text{ for all } \gamma \subseteq G. \tag{21.93}$$

A diagram can be superficially convergent overall and yet contain a subdiagram that is superficially divergent. When that happens the sub-integration over γ's loop momenta diverges, and no amount of negativity in the *global* count $D(G)$ can rescue it: the divergence lives in a corner of momentum space that the global scaling never visits.

It is worth exhibiting this concretely rather than schematically, because the phenomenon is so easy to mistake for a paradox. Work in ϕ^4 theory in $d = 4$, where Theorem 21.5 gives the global degree of any 1PI graph as $D = 4 - E$ in terms of its number of external legs E, independent of the loop order. Consider a two-loop contribution to the six-point function. With $E = 6$ the global degree is

$$D(G) \;=\; 4 - 6 \;=\; -2 \;<\; 0, \tag{21.94}$$

so naive power counting pronounces this amplitude convergent. Now build a specific such graph. Take four quartic vertices v_1, v_2, v_3, v_4. Attach two external legs to each of v_1, v_3, v_4 (six external legs in all) and none to v_2. Join v_1 to v_2 by *two* internal lines a, b; join v_2 to v_3 by a line c and to v_4 by a line d; and join v_3 to v_4 by a line e. Each vertex is then exactly four-valent (v_1: two external $+\, a, b$; v_2: a, b, c, d; v_3: two external $+\, c, e$; v_4: two external $+\, d, e$), so the graph is a legitimate ϕ^4 diagram, with $I = 5$ internal lines, $V = 4$ vertices and $L = I - V + 1 = 2$ loops, consistent with $E = 6$. Its global degree is the -2 of Eq. (21.94).

Now isolate the subdiagram $\gamma = \{a, b\}$, the doubled line between v_1 and v_2. It is 1PI (the two parallel lines form a single loop that no single cut disconnects), with $I_\gamma = 2$, $V_\gamma = 2$, hence $L_\gamma = 1$. Its external legs are the two true external legs at v_1 together with the two lines c, d leaving v_2 — four legs in all, so γ is a four-point subdiagram. Its superficial degree, from Eq. (21.92) with $P_\gamma = 0$, is

$$D(\gamma) \;=\; 4 \cdot 1 \;-\; 2 \cdot 2 \;=\; 0, \tag{21.95}$$

a logarithmic divergence. Scaling the loop momentum that runs around $\{a, b\}$ to infinity, with the second loop momentum (the one circulating through c, d, e) held fixed, produces exactly the divergent one-loop bubble integral $\int \mathrm{d}^4k \,/\, [k^2 (k+q)^2] \sim \log \Lambda$ of Eq. (15.50), where q is the momentum the rest of the graph routes into the bubble. The full two-loop integral therefore diverges, logarithmically, even though its global superficial degree Eq. (21.94) is -2. Superficial counting has lied.

The lesson is not that $D(G)$ is wrong but that it is incomplete. The global degree faithfully reports the behavior on the diagonal where all momenta grow together; it says nothing about the off-diagonal directions where only a subset grows. To certify convergence one must check *every* direction, which is to say every subdiagram. That is the content of Weinberg's theorem, toward which the classification of the next subsection is the necessary preparation: before we can demand $D(\gamma) < 0$ for every γ, we must understand how the divergent γ's can be arranged.

21.5.3 Classification: disjoint, nested, and overlapping subdivergences

Suppose a graph G contains two superficially divergent 1PI subdiagrams γ_1 and γ_2. There are exactly three ways they can sit relative to one another, distinguished entirely by how their sets of internal lines intersect.

Definition 21.23 (Relative position of two subdiagrams)**.** Let $\gamma_1, \gamma_2 \subseteq G$ be 1PI subdiagrams with line sets $\mathcal{I}(\gamma_1), \mathcal{I}(\gamma_2)$. They are

- **disjoint** if $\mathcal{I}(\gamma_1) \cap \mathcal{I}(\gamma_2) = \varnothing$: they share no internal line (they may at most touch a common vertex);
- **nested** if $\mathcal{I}(\gamma_1) \subsetneq \mathcal{I}(\gamma_2)$ (or vice versa): one is entirely contained in the other;
- **overlapping** if they share some internal lines but neither contains the other, i.e. all three of $\mathcal{I}(\gamma_1) \setminus \mathcal{I}(\gamma_2)$, $\mathcal{I}(\gamma_2) \setminus \mathcal{I}(\gamma_1)$ and $\mathcal{I}(\gamma_1) \cap \mathcal{I}(\gamma_2)$ are non-empty.

The trichotomy is exhaustive and mutually exclusive, and the distinction is not cosmetic. Disjoint and nested subdivergences are *compatible*: they can be subtracted independently or one-inside-the-other in a definite order, and the bookkeeping closes cleanly. Overlapping subdivergences are the genuine difficulty of renormalization theory — the two divergent regions share lines, so subtracting one disturbs the other, and no simple sequential prescription suffices. It was precisely the overlapping case that made the proof of renormalizability a decade-long problem and that the BPHZ forest formula (Theorem 22.22) was invented to organize. We illustrate each case with explicit two-loop diagrams; the ϕ^4 examples are drawn in Fig. 21.4 and the gauge-theory examples in Fig. 21.5.

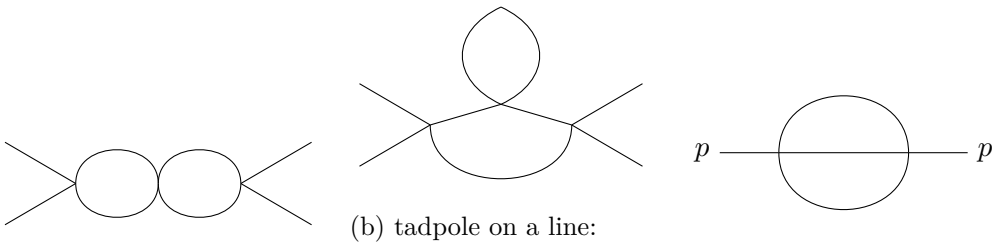
Disjoint — the ϕ^4 double bubble. The simplest two-loop graph with two subdivergences is the four-point "double bubble" of Fig. 21.4(a): three vertices v_1, v_2, v_3 in a row, a bubble (two parallel lines) between v_1 and v_2 and another between v_2 and v_3, with two external legs on v_1 and two on v_3. Here $E = 4$, $V = 3$,

$I = 4$, $L = 2$, so the global degree is $D(G) = 4 - 4 = 0$. The two sub-bubbles γ_1 (the v_1–v_2 pair) and γ_2 (the v_2–v_3 pair) each have $D(\gamma_i) = 4 \cdot 1 - 2 \cdot 2 = 0$. They share only the vertex v_2 and *no internal line*, so they are disjoint. The two loop momenta are independent: one circulates in the left bubble, the other in the right, and the double integral factorizes into a product of two one-loop bubbles (times the connecting propagators). Each diverges logarithmically and on its own; the divergences never interfere. This is the easy case — a product of primitive divergences — and the reason it is easy is exactly that disjointness makes the two sub-integrations independent.

Nested — the QED rainbow. The nested case requires one divergent subdiagram to sit *inside* another, sharing lines with it. In pure ϕ^4 this cannot happen until three loops, for an instructive reason worth stating: to nest one divergent subgraph inside another, the inner one must be a momentum-dependent self-energy insertion riding on a line of the outer subgraph, and ϕ^4's first momentum-dependent self-energy is the two-loop sunset of the next paragraph (the one-loop self-energy is the momentum-*independent* tadpole, which attaches at a vertex and so produces a *disjoint*, not nested, pair). QED supplies the canonical nested example at two loops: the "rainbow" electron self-energy of Fig. 21.5(a), in which the electron emits a photon, then emits and reabsorbs a second photon entirely *underneath* the first, then reabsorbs the first. The inner photon together with the stretch of electron line it spans forms a one-loop self-energy subdiagram γ_1 (superficial degree $D(\gamma_1) = 1$, the QED self-energy count of Section 21.2.8); the outer photon spans a longer stretch and, with everything beneath it, forms a larger self-energy subdiagram γ_2 that strictly contains γ_1. The two are nested, $\gamma_1 \subsetneq \gamma_2$. Nesting is benign because there is an unambiguous order in which to deal with the divergences: subtract the inner one first, treating the result as a corrected propagator, then subtract the outer. The "inside-to-outside" ordering is well defined precisely because containment is a partial order.

Overlapping — the ϕ^4 sunset. The hard case is two divergent subdiagrams that share lines without either containing the other. ϕ^4 theory reaches it at two loops, in the "sunset" self-energy of Fig. 21.4(c): two vertices v_1, v_2 joined by *three* internal lines $1, 2, 3$, with one external leg on each vertex (each vertex carries one external leg and three internal lines, $1 + 3 = 4$, four-valent). With $E = 2$, $V = 2$, $I = 3$, $L = 2$, the global degree is $D(G) = 4 - 2 = 2$. Route the external momentum p in at v_1 and out at v_2, and let the two loop momenta be k and ℓ, so the three lines carry k, ℓ and $p - k - \ell$. Now examine the subdiagrams. The pair of lines carrying $\{k,\, p - k - \ell\}$ forms a one-loop self-energy bubble γ_1 when k is scaled with ℓ fixed; the pair carrying $\{\ell,\, p - k - \ell\}$ forms a one-loop bubble γ_2 when ℓ is scaled with k fixed. Each has $D(\gamma_i) = 4 \cdot 1 - 2 \cdot 2 = 0$, a logarithmic subdivergence. But γ_1 and γ_2 *share the line carrying* $p - k - \ell$ while each owns

one line the other lacks — the defining signature of an overlapping pair. There is no line-ordering of γ_1 and γ_2 that puts one inside the other, and subtracting the divergence of γ_1 necessarily acts on the shared line and so disturbs γ_2. This is the obstruction that the forest formula resolves, and it is the precise sense in which the sunset marks the boundary between the clean one-loop computations of this chapter and the all-orders subtraction of Chapter 22. The same overlapping structure appears in QED in the two-loop vacuum polarization with crossed internal photons, Fig. 21.5(b), where the two vertex-correction subdiagrams share a stretch of the fermion loop; it was historically the overlapping divergence of QED that first forced the issue.



(a) disjoint (b) tadpole on a line: a *disjoint* insertion (c) overlapping (sunset)

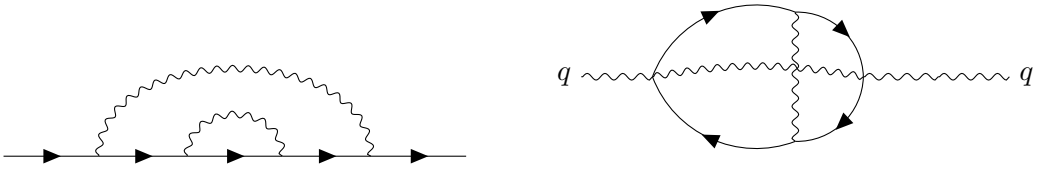
Figure 21.4: Two-loop ϕ^4 subdivergence topologies. (a) The four-point double bubble: two logarithmic sub-bubbles γ_1 (left) and γ_2 (right) sharing only the central vertex — *disjoint*, the two loop integrations independent. (b) A one-loop tadpole self-energy inserted on an internal line of a four-point bubble; because the tadpole attaches at a single vertex and shares no line with the bubble, the pair is again *disjoint* — this is why ϕ^4 has no nested pair until three loops. (c) The two-loop sunset self-energy: the three internal lines carry k, ℓ and $p-k-\ell$, and the two logarithmic sub-bubbles each consist of two of these lines, sharing the line $p-k-\ell$ — *overlapping*. This is ϕ^4 theory's first overlapping divergence.

21.5.4 Worked diagnosis of the sunset: superficial counting versus the truth

We now carry out the power-counting diagnosis of the sunset diagram in full, both to confirm by explicit computation that the overall degree does not determine convergence and to fix, once and for all, the discipline the rest of the renormalization program inherits.

Worked Example 21.24 (The two-loop sunset and its overlapping subdivergence)**.** The sunset contribution to the ϕ^4 self-energy is the two-loop integral

$$\Sigma_{\text{sun}}(p^2) \;\propto\; \lambda^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{1}{(k^2-m^2)\,(\ell^2-m^2)\,\big((p-k-\ell)^2-m^2\big)}, \tag{21.96}$$



(a) nested (rainbow) (b) overlapping (crossed)

Figure 21.5: Two-loop QED subdivergence topologies. (a) The rainbow electron self-energy: the inner photon spans a self-energy subdiagram γ_1 that sits entirely beneath the outer photon's subdiagram γ_2, so $\gamma_1 \subsetneq \gamma_2$ are *nested*; the divergences can be subtracted inside-out. (b) A two-loop vacuum polarization with crossed internal photons: the two vertex-correction subdiagrams share a stretch of the fermion loop while each owns lines the other lacks — *overlapping*. Overlapping divergences of exactly this type are what made the all-orders renormalization of gauge theory hard, and what Theorem 22.22 resolves.

the analytic content of Fig. 21.4(c), with the two-vertex symmetry factor and the $(-i\lambda)^2$ absorbed into the proportionality. We compute the superficial degree of the whole graph and of each of its 1PI subgraphs.

The whole graph. Set $d = 4$. There are $L = 2$ loops, $I = 3$ propagators and no loop momenta in the numerator ($P = 0$), so by Eq. (D.4),

$$D(G) \;=\; dL - 2I \;=\; 4 \cdot 2 - 2 \cdot 3 \;=\; 2, \tag{21.97}$$

in agreement with the leg formula $D = 4 - E = 4 - 2 = 2$ of Theorem 21.5: a quadratic overall divergence, as a self-energy must have. Scaling *both* k and ℓ to infinity, $k = t\hat{k}$, $\ell = t\hat{\ell}$, the measure gives $t^{2d} = t^8$, the three denominators give t^{-6}, and the net $t^{8-6} = t^2$ confirms Eq. (21.97): the integral diverges as Λ^2 along the diagonal.

The 1PI subgraphs. The proper 1PI subdiagrams that are themselves loops are obtained by scaling *one* loop momentum at a time. Scale $k \to \infty$ with ℓ held fixed. Only the denominators that depend on k blow up: $(k^2 - m^2)$ and $((p-k-\ell)^2 - m^2)$. These two lines form a one-loop sub-bubble γ_1 with $I_{\gamma_1} = 2$, $L_{\gamma_1} = 1$, $P_{\gamma_1} = 0$, whence

$$D(\gamma_1) \;=\; d \cdot 1 - 2 \cdot 2 \;=\; 4 - 4 \;=\; 0, \tag{21.98}$$

a logarithmic subdivergence: the inner integral $\int \mathrm{d}^4k \,/\, [(k^2-m^2)((p-k-\ell)^2-m^2)]$ is precisely the one-loop bubble Eq. (15.50) with external momentum $p - \ell$, and it diverges as $\log \Lambda$. By the $k \leftrightarrow \ell$ symmetry of Eq. (21.96), scaling ℓ with k fixed gives the partner sub-bubble γ_2 on the lines $(\ell^2 - m^2)$ and $((p-k-\ell)^2 - m^2)$, again with $D(\gamma_2) = 0$.

The diagnosis. The two sub-bubbles share the line carrying $p - k - \ell$ and each owns one line (k for γ_1, ℓ for γ_2) the other lacks; by Theorem 21.23 they

overlap. The sunset is therefore not primitive: it carries an overall quadratic divergence Eq. (21.97) *and* a pair of overlapping logarithmic subdivergences Eq. (21.98). Three numbers — $D(G) = 2$, $D(\gamma_1) = D(\gamma_2) = 0$ — must all be confronted before the integral can be rendered finite, and the differentiation argument of Theorem 21.17 does not reach the subdivergences: differentiating Eq. (21.96) twice in p lowers $D(G)$ to -2 and kills the overall divergence, but it leaves $D(\gamma_1) = D(\gamma_2) = 0$ untouched, because the external derivative acts only on the line $p-k-\ell$ and does not improve the convergence of the sub-integrations over k or ℓ alone. The divergence that survives is not a local polynomial in p; it is a non-local remnant whose subtraction requires the machinery of Chapter 22.

The contrast with the double bubble is the whole moral. In the double bubble Fig. 21.4(a) the two subdivergences are disjoint, the integral factorizes, and two independent one-loop subtractions suffice. In the sunset the two subdivergences overlap, the integral does not factorize in the loop momenta, and no sequence of independent subtractions suffices. The overlapping case is the one for which Weinberg's theorem is not merely a convenience but a necessity: only by checking every subgraph, including the entangled ones, can convergence be decided.

21.5.5 A logarithmic subdivergence makes a double logarithm

Before stating the general theorem it is worth seeing, by an explicit calculation, why an uncancelled subdivergence is so corrosive: a single logarithmic subdivergence riding inside an otherwise convergent outer integration does not merely add another $\log\Lambda$; it produces a $\log^2\Lambda$ — a leading singularity *stronger* than any one-loop divergence. This is the quantitative fingerprint of a nested divergence and the reason the subtractions of Chapter 22 must be performed in a definite order.

Consider a nested configuration: an inner one-loop subdiagram γ that is logarithmically divergent ($D(\gamma) = 0$), sitting on an internal line of an outer loop that is itself logarithmically divergent ($D(G/\gamma) = 0$ for the graph with γ contracted to a point). After the inner integration, the subdiagram contributes a factor that, with a cutoff Λ on the inner loop and the outer loop momentum q flowing through it, behaves as

$$\gamma(q^2) \;\sim\; c\,\log\frac{\Lambda^2}{q^2} \;+\; \text{finite}, \tag{21.99}$$

with c a constant fixed by the inner diagram. The crucial feature is that the cutoff appears *inside* the logarithm together with q^2: the inner subdivergence is cut off at Λ, but its finite part runs with the outer momentum q. Now feed this into the outer loop. The outer integration is logarithmically divergent, so after angular integration its radial measure is $\mathrm{d}q^2/q^2$, and it is dominated by the

region $m^2 \lesssim q^2 \lesssim \Lambda^2$. The leading divergence of the two-loop integral is therefore

$$\int_{m^2}^{\Lambda^2} \frac{\mathrm{d}q^2}{q^2}\, \gamma(q^2) \;\sim\; c \int_{m^2}^{\Lambda^2} \frac{\mathrm{d}q^2}{q^2}\, \log\frac{\Lambda^2}{q^2}. \tag{21.100}$$

Substitute $u = \log(\Lambda^2/q^2)$, so $\mathrm{d}u = -\mathrm{d}q^2/q^2$, with u running from $\log(\Lambda^2/m^2) \equiv L_\Lambda$ down to 0:

$$c \int_0^{L_\Lambda} u\, \mathrm{d}u \;=\; \frac{c}{2}\, L_\Lambda^2 \;=\; \frac{c}{2}\, \log^2\frac{\Lambda^2}{m^2}. \tag{21.101}$$

The two-loop graph diverges as $\log^2 \Lambda$, one power of the logarithm stronger than either of its one-loop constituents. In dimensional regularization the same statement reads that the leading pole of such a graph is a *double* pole $1/\varepsilon^2$ rather than a simple $1/\varepsilon$: the inner subdivergence contributes a $1/\varepsilon$ from Eq. (21.99), and the outer divergent integration of that $1/\varepsilon$ against $\mathrm{d}q^2/q^2$ contributes a second, producing $1/\varepsilon^2$ at two loops, $1/\varepsilon^3$ at three nested loops, and so on.

The $\log^2$ is the signature of an *un*subtracted subdivergence, and it tells us exactly what the cure must look like. Were the inner subdivergence removed first — replace $\gamma(q^2)$ by its subtracted version $\gamma(q^2) - \gamma(\mu^2)$, which is finite and behaves as $c\log(\mu^2/q^2)$ with no Λ inside the log — the outer integration Eq. (21.100) would produce only a single $\log\Lambda$, the honest overall divergence of the two-loop graph, a local quantity removable by a single two-loop counterterm. The double logarithm is thus not a new kind of divergence demanding a new kind of counterterm; it is the artifact of doing the subtractions in the wrong order, or not at all. The systematic prescription that always subtracts inner subdivergences before outer ones, for every nesting simultaneously, is the BPHZ forest formula (Section 22.5.4). The $\log^2$ is what it exists to prevent.

21.5.6 Weinberg's convergence theorem

We are now in a position to state the theorem that replaces superficial power counting with an exact criterion. It is due to Weinberg, who proved it as a statement about the large-momentum (and large-mass) asymptotics of Feynman integrals; the same analysis underlies the proof of renormalizability completed by Hepp and Zimmermann. We work in Euclidean signature, reached by the Wick rotation $k^0 \to ik^0_E$ that turns each Minkowski propagator $1/(k^2 - m^2 + i\varepsilon)$ into the positive-definite Euclidean $1/(k_E^2 + m^2)$, so that questions of convergence become questions about the integrability of a positive integrand and the $i\varepsilon$ prescription no longer intrudes.

Theorem 21.25 (Weinberg's convergence theorem)**.** *Let I_G be the Euclidean Feynman integral associated with a connected graph G,*

$$I_G \;=\; \int \prod_{i=1}^{L} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d}\, \frac{\mathcal{N}(\ell, p)}{\prod_{j=1}^{I} (q_j^2 + m_j^2)}, \tag{21.102}$$

where each line momentum q_j is a linear combination of the loop momenta ℓ_i and the external momenta p, the numerator $\mathcal{N}$ is a polynomial in the loop momenta of degree P_γ on the lines of any subgraph γ, all masses satisfy $m_j > 0$, and the external momenta are Euclidean and nonexceptional (so that no partial sum of external momenta entering a subgraph vanishes, which excludes the infrared and threshold singularities catalogued in Theorem D.41). Then I_G converges absolutely if and only if the superficial degree of divergence is negative for the full graph and for every subgraph:

$$\boxed{D(\gamma) < 0 \quad \text{for } \gamma = G \text{ and for every 1PI subdiagram } \gamma \subseteq G.} \tag{21.103}$$

Several features of the statement deserve comment before the proof. First, the criterion is manifestly stronger than $D(G) < 0$ alone: the failure-mode example of Section 21.5.2, with $D(G) = -2$ but a subdiagram of degree $D(\gamma) = 0$, fails Eq. (21.103) and indeed diverges, exactly as the theorem requires. Second, the criterion is also *necessary*, not merely sufficient: if any single subgraph has $D(\gamma) \geq 0$, the integral diverges, because the region of momentum space in which γ's loop momenta are large and the rest finite is a genuine subregion over which the integrand is not integrable, and a positive Euclidean integrand cannot have its divergence in one region cancelled by convergence elsewhere. Third, at one loop Eq. (21.103) collapses to the single condition $D(G) < 0$, since a one-loop graph has no proper 1PI subdiagram that is itself a loop; this is why every one-loop count in this book is exact and why the one-loop ϕ^4 diagrams of the next section, being primitive (Theorem 21.22), are governed entirely by their overall degree. Fourth, the hypotheses $m_j > 0$ and nonexceptional external momenta confine the statement to the *ultraviolet*: they exclude the soft and collinear singularities that arise when masses vanish or external momenta become exceptional, which are a separate (infrared) question handled by the power counting of Theorem D.41. Weinberg's theorem is the exact ultraviolet convergence criterion; it is silent about the infrared by construction.

Proof sketch, with the proved and stated steps separated. The strategy is to decompose the Ld-dimensional momentum integration into regions in each of which a definite hierarchy of loop momenta is large, and to show that convergence in each region is controlled by the superficial degree of the subgraph that the hierarchy singles out. We carry the argument far enough to make the role of Eq. (21.103) transparent and mark precisely where the full proof passes to the literature.

Step 1: Schwinger parametrization (proved in Chapter 22 and Appendix D). Write each Euclidean propagator as $1/(q_j^2 + m_j^2) = \int_0^\infty \mathrm{d}\alpha_j \, e^{-\alpha_j (q_j^2 + m_j^2)}$ and perform the Gaussian loop integrals Eq. (D.20). The result expresses I_G as an integral over the Schwinger parameters $\alpha_j \in [0, \infty)$ of a ratio of the Symanzik

polynomials $U(\alpha)$ and $F(\alpha, p)$,

$$I_G \;\propto\; \int_0^\infty \prod_j \mathrm{d}\alpha_j \; \frac{1}{U(\alpha)^{d/2}} \; \exp\!\Big(-\frac{F(\alpha,p)}{U(\alpha)} - \textstyle\sum_j \alpha_j m_j^2\Big) \times (\text{numerator}), \tag{21.104}$$

where U is a homogeneous polynomial of degree L in the α_j and F of degree $L+1$. The masses make the exponential decay at large α, so all divergences now live at the *lower* endpoint, where some $\alpha_j \to 0$ — the parametric image of large loop momentum. This change of variables is exact; we use it as established.

Step 2: Hepp sectors (the geometric heart, proved by Hepp). The positive orthant $\{\alpha_j \geq 0\}$ is partitioned into $I!$ *Hepp sectors*, one for each ordering of the parameters,

$$\alpha_{\pi(1)} \leq \alpha_{\pi(2)} \leq \cdots \leq \alpha_{\pi(I)}, \tag{21.105}$$

for a permutation π. Within a sector one introduces sector variables $t_1, \ldots, t_I \in [0,1]$ by writing the parameters as nested products $\alpha_{\pi(k)} = t_k\, t_{k+1} \cdots t_I$ (so the ordering Eq. (21.105) is automatic). The key combinatorial fact, which is the content of Hepp's analysis, is that in these variables the Symanzik polynomial factorizes into a monomial times a unit,

$$U(\alpha) \;=\; \Big(\prod_k t_k^{b_k}\Big)\,\big[\,1 + O(t)\,\big], \tag{21.106}$$

where the exponent b_k counts the number of independent loops of the subgraph γ_k formed by the lines whose parameters are at most $\alpha_{\pi(k)}$. As k runs from 1 to I, these subgraphs form a strictly increasing chain (a *flag*) $\gamma_1 \subset \gamma_2 \subset \cdots \subset \gamma_I = G$. Each scaling variable $t_k \to 0$ is the parametric statement that the loop momenta of γ_k go to infinity together while those outside stay finite — precisely the limit that defines $D(\gamma_k)$ in Theorem 21.21.

Step 3: power of each sector variable equals the subgraph degree (the computation). Collecting the powers of t_k from the measure $\prod \mathrm{d}\alpha_j$, from the factor $U^{-d/2}$ via Eq. (21.106), and from the numerator, a direct count gives the behavior of the sector integrand near $t_k = 0$ as

$$\int_0 t_k^{-D(\gamma_k)-1}\,\mathrm{d}t_k, \tag{21.107}$$

with the *same* superficial degree $D(\gamma_k)$ of Eq. (21.92) appearing as the exponent. The bookkeeping is identical to the momentum-space scaling of Section 21.5.1: the L_{γ_k} loops of γ_k supply $+dL_{\gamma_k}$ from the measure and $U^{-d/2}$, the I_{γ_k} lines supply $-2I_{\gamma_k}$ (each $\alpha \to 0$ being one inverse power of the corresponding squared momentum), and the numerator supplies $+P_{\gamma_k}$. The endpoint integral Eq. (21.107) converges as $t_k \to 0$ precisely when $-D(\gamma_k) - 1 > -1$, i.e. when $D(\gamma_k) < 0$. This is the step at which the abstract criterion Eq. (21.103) becomes concrete: a Hepp sector converges if and only if $D(\gamma_k) < 0$ for every subgraph γ_k in its flag.

Step 4: covering and uniformity (stated; the full estimates are Hepp's and Zimmermann's). It remains to show, first, that the sectors cover the integration region (immediate: every ordering of the α_j falls in some sector), and second, that the $O(t)$ corrections in Eq. (21.106) and the numerator do not spoil the leading power count — that the integrand is bounded uniformly in each sector by the product of monomials Eq. (21.107), so that absolute convergence of the model monomial integral implies absolute convergence of the true integral. This uniform-boundedness estimate is the technical core of the theorem; it is carried out in full by Hepp, and the momentum-space version with the explicit subtraction operator is Zimmermann's. We state it as established and do not reproduce the estimates here. Granting it, the integral converges absolutely if and only if every sector converges, i.e. if and only if $D(\gamma) < 0$ for every subgraph appearing in some flag. Because every 1PI subdiagram of G appears as some γ_k in some sector's flag, this is exactly Eq. (21.103), completing the argument. □

The division of labor in the proof is worth restating plainly, in keeping with this book's standard of honesty about what is shown versus what is invoked. The *structural* content — that the dangerous regions are indexed by subgraphs, that each is governed by the superficial degree of its subgraph, and that the criterion is therefore Eq. (21.103) — is established here, including the explicit power count Eq. (21.107) that ties the sector exponent to $D(\gamma)$. The *analytic* content — the uniform bound that promotes "each model monomial integral converges" to "the actual integral converges absolutely" — is stated and attributed; its complete execution, together with the constructive subtraction operator R that turns a divergent integral into a finite one, is the subject of Chapter 22. The original proofs are recorded in the further-reading list and are cited there for the reader who wants the full estimates, never in place of an argument the book needs.

21.5.7 The forest structure, handed to the next chapter

Weinberg's theorem tells us *whether* an integral converges; it does not tell us how to make a divergent one finite. But the flag structure that organized its proof is exactly the bookkeeping that the cure will need, and it is worth naming the object now so that Chapter 22 can pick it up by reference rather than reinvention. The chains $\gamma_1 \subset \gamma_2 \subset \cdots$ of Eq. (21.105) are made of subgraphs that are pairwise either nested or disjoint — never overlapping, since a flag is totally ordered by inclusion. A maximal collection of divergent subgraphs that are pairwise nested or disjoint is called a *forest* (a set of trees, in the graph-theoretic sense, ordered by containment). The disjoint and nested divergences of Theorem 21.23 fit into forests directly; the overlapping ones — the sunset's γ_1, γ_2, the crossed-photon vacuum polarization — do not, since no single ordering contains both. The resolution, anticipated here and proved in Section 22.5.4, is that an overlapping pair is handled by summing over the forests in which each member appears

separately: the forest formula subtracts, for every forest of G, the divergences of the subgraphs it contains, in the order containment dictates (innermost first, exactly as Section 21.5.5 demanded to avoid the spurious $\log^2$), and the sum over all forests reconstructs precisely the combination that renders the whole graph finite, overlaps and all. The forest is thus the object Weinberg's theorem hands to BPHZ: the theorem identifies the divergent subgraphs, the forest formula Theorem 22.22 organizes their subtraction.

This closes the conceptual arc of the chapter's diagnosis. Power counting (Theorem 21.5) gives the overall degree; the locality theorem (Theorem 21.17) makes the divergence of a primitive graph a local polynomial; Weinberg's theorem (Theorem 21.25) certifies that, away from subdivergences, that overall degree is the whole story, and identifies the subgraphs one must additionally control when it is not. The one immediate dividend, which the next section banks, is the converse reading of Theorem 21.22: because a single loop contains no smaller loop, every one-loop 1PI diagram is primitive, has no subdivergence, and is governed entirely by its overall superficial degree. The one-loop divergences of ϕ^4 theory are therefore the clean primitive case in which superficial counting tells the whole truth and the locality theorem applies without qualification — and it is to their explicit, intermediate-step-by-intermediate-step computation that we now turn.

21.6 Scalar ϕ^4 theory: The One-Loop Divergences in Full

The previous section closed a loophole: a diagram with negative superficial degree of divergence can still diverge if it hides a divergent subgraph, and only Weinberg's convergence theorem (Theorem 21.25) certifies true convergence. That subtlety is real, but it has a clean corollary for the calculation we are about to do. The one-loop 1PI diagrams of ϕ^4 theory have *no proper subdiagrams that are themselves loops* — a single loop cannot contain a smaller loop — so there are no subdivergences to worry about. Each one-loop graph is what Section 21.5 called a *primitive* divergence: its only divergence is the overall one counted by the superficial degree, and Weinberg's theorem then guarantees that the integral converges everywhere except where the power counting says it must diverge. This is exactly the situation in which superficial counting tells the whole truth and the locality theorem (Theorem 21.17) applies without qualification. The divergent part must be a local polynomial in the external momenta of degree at most D, and the only question left is the numerical value of its coefficients.

We now compute those coefficients. The exercise is the keystone of the chapter for two reasons. First, it is the place where every abstract statement of the preceding sections — the power-counting formula, the classification of ϕ^4 as a renormalizable theory, the locality of divergences — is cashed out against

an explicit integral and seen to hold to the letter. Second, the two numbers that emerge are not curiosities: the residue of the four-point pole is the seed of the beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ that ϕ^4 theory carries to its Wilson–Fisher fixed point, and the structure of the two-point divergence is the reason the field strength is not renormalized at this order. We extract both with every intermediate step on the page, importing the master integrals of Chapter D but performing each substitution explicitly rather than quoting a result.

21.6.1 The two amplitudes and what power counting predicts of them

We continue ϕ^4 theory from its Feynman rules (Chapter 15), with the standard Lagrangian

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\,\phi^4, \qquad \phi \text{ real},\ \lambda > 0, \tag{21.108}$$

in the conventions in force throughout this chapter: mostly-minus metric, natural units, dimensional regularization with $d = 4 - 2\varepsilon$, and the $\overline{\text{MS}}$ scheme of Theorem A.49. Working in $d = 4 - 2\varepsilon$ dimensions makes ϕ carry mass dimension $\frac{d}{2} - 1 = 1 - \varepsilon$ and λ carry dimension $4 - d = 2\varepsilon$, so the dimensionless renormalized coupling is exposed by the promotion $\lambda \to \lambda\,\mu^{2\varepsilon}$ of Theorem A.46, with μ the renormalization scale. Every vertex therefore carries the Feynman rule $-i\lambda\,\mu^{2\varepsilon}$, and the compensating factor $\mu^{2\varepsilon}$ rides on each loop, where — as Eq. (D.140) will remind us — it is exactly what turns the logarithm of a dimensionful argument into the logarithm of a dimensionless ratio.

The objects we compute are the two one-particle-irreducible (1PI) Green's functions (Theorem 15.30) that the power counting of this chapter flags as divergent in four dimensions. The general superficial degree of divergence of a ϕ^4 graph with E external legs is, by the chapter's power-counting formula specialized to four dimensions (Theorem 21.10),

$$D \;=\; 4 - E, \tag{21.109}$$

independent of the loop order — the signature of a renormalizable theory, in which adding loops does not worsen the divergence of a fixed Green's function. Two values of E give $D \geq 0$:

- $E = 4$: the four-point function, with $D = 4 - 4 = 0$, a *logarithmic* divergence. The locality theorem then predicts that the divergent part is a polynomial of degree 0 in the external momenta — a pure constant, with no kinematic dependence. It must therefore renormalize the only momentum-independent four-point operator available, the vertex ϕ^4 itself, i.e. the coupling λ.

- $E = 2$: the two-point function, with $D = 4 - 2 = 2$, a *quadratic* divergence. Locality permits a divergent polynomial of degree 2 in the single external momentum p, i.e. a term $a\,p^2 + b\,m^2$ with a, b dimensionless and divergent. The p^2 piece would renormalize the kinetic term (the field strength); the m^2 piece renormalizes the mass.

All other 1PI functions have $D < 0$ and, being primitive at one loop, are ultraviolet finite by Weinberg's theorem. The whole one-loop divergence structure of ϕ^4 is thus carried by these two amplitudes, and the entire content of the locality theorem at this order is the pair of predictions just stated. The computations below confirm both, and sharpen the second: the coefficient a of p^2 in fact *vanishes*, a fact we derive rather than assert.

21.6.2 The four-point function: three bubbles and a logarithm

At order λ^2 the 1PI four-point function receives one diagram in each Mandelstam channel: the s-, t-, and u-channel bubbles obtained by Wick-contracting two quartic vertices, drawn together in Fig. 21.6. These are precisely the position-space bubbles *Eq.* (15.38) of Chapter 15, and each is the one-loop integral *Eq.* (15.50) first met there. The three differ only in which pair of external momenta flows into the loop: the s-channel carries $p = p_1 + p_2$ with $p^2 = s$, the t-channel carries $p_1 - p_3$ with $p^2 = t$, and the u-channel carries $p_1 - p_4$ with $p^2 = u$.

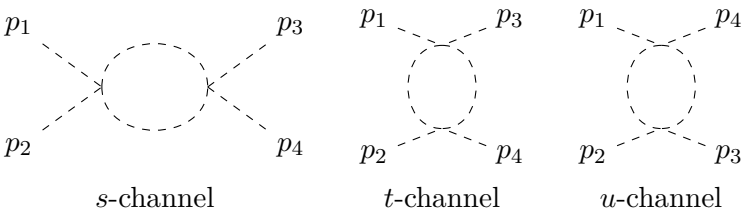


Figure 21.6: The three one-loop bubbles contributing to the 1PI four-point function of ϕ^4 at $O(\lambda^2)$. Each is built from two quartic vertices joined by two internal lines; the channels differ only in which external invariant (s, t, or u) flows through the loop. Each diagram carries symmetry factor $\frac{1}{2}$ (Theorem 21.26).

Before evaluating the integral we must fix its combinatorial prefactor, the symmetry factor. This is where the orbit–stabilizer machinery of Chapter 15 earns its keep, and we carry it out explicitly.

Worked Example 21.26 (Symmetry factor of the one-loop bubble)**.** The bubble graph has $V = 2$ quartic vertices and $I = 2$ internal lines running between them, with two external legs attached at each vertex (and amputated, so they do not enter the count). By the general orbit–stabilizer formula Eq. (15.28),

the symmetry factor is the order of the automorphism group of the graph, $S = |\mathrm{Aut}(\Gamma)|$ — the number of relabellings of internal lines and vertices that map the graph (with its external legs held fixed) to itself.

Hold the four external legs fixed at their vertices. The two vertices are then distinguishable: the left one carries the external pair (p_1, p_2) and the right one carries (p_3, p_4), so no automorphism can exchange them. The only nontrivial relabelling that preserves the graph is the interchange of the two *internal* lines, both of which run between the same pair of vertices: swapping them sends the pairing to itself. That interchange is a single $\mathbb{Z}_2$, so

$$S = |\mathrm{Aut}(\Gamma)| = 2. \tag{21.110}$$

This agrees with the rule-of-thumb count of Theorem 15.16 — "two identical internal lines give a factor $\frac{1}{2}$" — which Eq. (15.28) reproduces as a corollary, and it is the same factor that appears in the Appendix D evaluation of this diagram (Theorem D.25). The symmetry factor is $S = 2$, so the diagram carries $1/S = \frac{1}{2}$.

With the symmetry factor in hand we assemble the s-channel amplitude. Each vertex contributes $-i\lambda\,\mu^{2\varepsilon}$, each internal line a propagator $i/(k^2 - m^2 + i\varepsilon)$, and the loop is integrated with the measure $\int \mathrm{d}^d k/(2\pi)^d$. With total incoming momentum $p = p_1 + p_2$, $p^2 = s$, the i times the 1PI four-point amplitude in the s-channel is

$$i\mathcal{M}_s = \frac{(-i\lambda\,\mu^{2\varepsilon})^2}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \, \frac{i}{k^2 - m^2 + i\varepsilon} \, \frac{i}{(k+p)^2 - m^2 + i\varepsilon}. \tag{21.111}$$

The numerical prefactors combine cleanly: the two vertex factors give $(-i\lambda\,\mu^{2\varepsilon})^2 = -\lambda^2\mu^{4\varepsilon}$, the two propagator numerators give $i \cdot i = -1$, and the symmetry factor contributes $\frac{1}{2}$, so the front constant is $(-\lambda^2\mu^{4\varepsilon})(-1)/2 = \frac{1}{2}\lambda^2\mu^{4\varepsilon}$. What remains is exactly the scalar bubble of Eq. (D.101):

$$i\mathcal{M}_s = \frac{\lambda^2\mu^{4\varepsilon}}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \, \frac{1}{(k^2 - m^2)\big((k+p)^2 - m^2\big)} = \frac{\lambda^2\mu^{2\varepsilon}}{2}\, B(s), \tag{21.112}$$

where in the last step one factor $\mu^{2\varepsilon}$ has been kept inside the integral to form the scale-compensated bubble $B(s)$ of Eq. (D.137), leaving a single $\mu^{2\varepsilon}$ outside to match the dimension of the tree-level vertex.

We now import the bubble and exhibit the substitution explicitly. Combining the two denominators with one Feynman parameter and shifting the loop momentum (a translation with unit Jacobian and no surface term in dimensional regularization) brought $B(s)$ to the standard form *Eq.* (D.104), whose ε-expansion was carried out in Chapter D with the result *Eq.* (D.140):

$$B(s) = \frac{i}{(4\pi)^2}\left[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x\, \ln\frac{\Delta(x)}{\mu^2}\right] + O(\varepsilon), \qquad \Delta(x) = m^2 - x(1-x)\,s, \tag{21.113}$$

in which $1/\bar{\varepsilon} \equiv 1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi$ is the universal divergent combination fixed in Eq. (A.204) and $(4\pi)^2 = 16\pi^2$. The entire ultraviolet divergence sits in the single $1/\bar{\varepsilon}$ pole; the Feynman integral $\int_0^1 \mathrm{d}x\, \ln(\Delta/\mu^2)$ is finite. This is the structural fact that Theorem D.31 proved on general grounds — the pole part of any one-loop integral is independent of the kinematics, while all dependence on s, m^2, and μ lives in the finite logarithm. Substituting Eq. (21.113) into Eq. (21.112),

$$i\mathcal{M}_s = \frac{i\lambda^2\mu^{2\varepsilon}}{2(4\pi)^2}\left[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x\, \ln\frac{m^2 - x(1-x)s}{\mu^2}\right] + O(\varepsilon) = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\left[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x\, \ln\frac{m^2 - x(1-x}{\mu^2}\right. \tag{21.114}$$

using $1/2(4\pi)^2 = 1/32\pi^2$. The divergent part of the single s-channel bubble is therefore

$$i\mathcal{M}_s\Big|_{\mathrm{div}} = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{21.115}$$

a momentum-independent constant. The t- and u-channel diagrams are obtained from Eq. (21.114) by the replacements $s \to t$ and $s \to u$ in the finite logarithm; their divergent parts are *identical* to Eq. (21.115), because the pole does not see the kinematics at all. Summing the three channels, the divergent part of the full one-loop 1PI four-point amplitude is

$$i\mathcal{M}_{\mathrm{loop}}\Big|_{\mathrm{div}} = (i\mathcal{M}_s + i\mathcal{M}_t + i\mathcal{M}_u)\Big|_{\mathrm{div}} = 3 \times \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} = \frac{3i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}. \tag{21.116}$$

This is the central number of the section. We pause to confirm that it is exactly what the chapter's general theorems demanded.

- *Power counting.* Equation Eq. (21.109) gave $D = 0$ for the four-point function: a logarithmic divergence. A logarithmic divergence in dimensional regularization is a *simple* pole in ε, with no $1/\varepsilon^2$ and no power divergence. Equation Eq. (21.116) is precisely a single $1/\bar{\varepsilon}$ pole. The pole-condition analysis of Eq. (D.109) says the same thing intrinsically: the bubble is the $n = 2$ master integral, whose divergence is governed by $\Gamma(2 - d/2) = \Gamma(\varepsilon)$, a simple pole. The prediction $D = 0 \Rightarrow$ simple pole is met.

- *Locality.* The locality theorem (Theorem 21.17) predicted that a $D = 0$ divergence is a polynomial of degree 0 in the external momenta — a constant. The divergent part Eq. (21.116) is manifestly independent of s, t, u, and of all external momenta: it is the same constant in every channel. It is therefore proportional to the tree-level vertex $-i\lambda\,\mu^{2\varepsilon}$ and can be absorbed by a shift of the single coupling λ. A momentum-dependent divergence, by contrast, could not be absorbed by any local term in Eq. (21.108) and would signal a non-renormalizable theory. Locality is confirmed: the divergence has exactly the form a counterterm in the original Lagrangian can cancel.

The residue in Eq. (21.116) is the seed of two later results, and we record the handoff precisely. In Chapter 22 the four-point counterterm δ_λ is fixed by demanding that the renormalized 1PI four-point function be finite; cancelling the pole Eq. (21.116) against the counterterm vertex $-i\delta_\lambda\,\mu^{2\varepsilon}$ requires

$$\delta_\lambda = \frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{21.117}$$

which is the coupling counterterm of Eq. (22.9) and the residue that defines the coupling renormalization constant $Z_\lambda = 1 + \delta_\lambda/\lambda$ in Eq. (24.59). The factor of 3 is the count of channels; the factor $\frac{1}{32}$ is the symmetry factor $\frac{1}{2}$ times the loop normalization $\frac{1}{16\pi^2}$. In Chapter 24 this same residue becomes the one-loop beta function: the relation $\beta_\lambda = \mu\,\mathrm{d}\lambda/\mathrm{d}\mu$ applied to $\lambda_0 = \mu^{2\varepsilon} Z_\lambda \lambda$ turns the residue $\frac{3\lambda}{32\pi^2}$ of Z_λ into

$$\beta_\lambda = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3), \tag{21.118}$$

the carried number of ϕ^4 theory recorded in Eq. (24.46), the factor of two between 32 and 16 arising from the 2ε in the derivative of $\mu^{2\varepsilon}$. We do not derive Eq. (21.118) here — that is the business of Section 24.2 — but we have produced the only loop input it needs.

Why the three channels simply add their poles

It can look surprising that the divergence triples without any interference between channels. The reason is locality again. Each bubble's divergence is a constant, the same constant, because each is the residue of the same scalar integral B evaluated at a *different* argument s, t, or u — and the residue of B is independent of its argument. Three copies of the same kinematics-blind constant simply add. Had the divergence carried kinematic dependence, the three channels would have produced three *different* divergent functions of momenta, with no single local counterterm able to cancel all of them at once; renormalizability would have failed. The plain factor of 3 is thus a direct fingerprint of the locality that makes ϕ^4 renormalizable.

21.6.3 The two-point function: a momentum-independent tadpole

The remaining divergent amplitude is the 1PI two-point function. At one loop it receives a single diagram: the *tadpole*, a quartic vertex with two of its four legs joined into a loop, the other two carrying the external momentum p in and out (Fig. 21.7). We denote the 1PI two-point insertion by $-i\Sigma(p^2)$, so that the full propagator is the geometric resummation $i/(p^2 - m^2 - \Sigma)$ of Theorem 15.30.

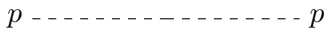


Figure 21.7: The one-loop tadpole correction to the ϕ^4 two-point function: one quartic vertex with two legs closed into a loop carrying momentum ℓ. The external momentum p enters and leaves the *same* vertex and never circulates in the loop, so the integrand cannot depend on p. Symmetry factor $S = 2$.

The diagram has one vertex $-i\lambda\,\mu^{2\varepsilon}$, one internal line, and a symmetry factor $\frac{1}{2}$ from the interchange of the two ends of that single self-contracted line — the same orbit–stabilizer count as in Theorem 21.26, now applied to a graph whose nontrivial automorphism reverses the loop. The decisive structural feature is geometric and was already visible in Theorem 15.16 for the cousin diagram: the external momentum p enters and exits at the one vertex without ever flowing into the loop, so the loop momentum ℓ is entirely unconstrained by p and the integrand depends on ℓ alone. Hence

$$-i\Sigma(p^2) = \frac{-i\lambda\,\mu^{2\varepsilon}}{2}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2 - m^2 + i\varepsilon} = \frac{\lambda\,\mu^{2\varepsilon}}{2}\,A_0(m^2), \tag{21.119}$$

where the single propagator numerator i times the overall $-i$ from the vertex give $(-i)(i) = 1$, and $A_0(m^2) \equiv \int \mathrm{d}^d\ell/(2\pi)^d\,(\ell^2 - m^2 + i\varepsilon)^{-1}$ is the scalar tadpole integral of Eq. (D.98). The result is manifestly *independent of* p^2: there is no external momentum anywhere in the integral. This single observation is the engine of everything that follows, and it is forced by the topology, not by any cancellation.

We now evaluate A_0 by importing its closed form. The tadpole is the $n = 1$ master integral, and Eq. (D.99) gives

$$A_0(m^2) = -\frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)(m^2)^{d/2-1}. \tag{21.120}$$

Substituting into Eq. (21.119) and dividing through by $-i$ to isolate Σ,

$$\Sigma = \frac{\lambda\,\mu^{2\varepsilon}}{2(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)(m^2)^{d/2-1} = \frac{\lambda\,m^2}{32\pi^2}\,\Gamma(-1+\varepsilon)\left(\frac{4\pi\mu^2}{m^2}\right)^{\varepsilon}, \tag{21.121}$$

where in the second equality we set $d = 4 - 2\varepsilon$ so that $\Gamma(1 - d/2) = \Gamma(-1+\varepsilon)$, wrote $(4\pi)^{d/2} = (4\pi)^2(4\pi)^{-\varepsilon}$ and $(m^2)^{d/2-1} = m^2(m^2)^{-\varepsilon}$, collected the ε-powers into the dimensionless ratio $(4\pi\mu^2/m^2)^{\varepsilon}$, and reduced the constant prefactor to $\lambda m^2/2(4\pi)^2 = \lambda m^2/32\pi^2$. The divergence is now isolated in the gamma function. Using the Laurent expansion of Γ about its pole at $z = -1$, established in Chapter D,

$$\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} + (\gamma_{\mathrm{E}} - 1) + O(\varepsilon), \tag{21.122}$$

and multiplying by $(4\pi\mu^2/m^2)^\varepsilon = 1 + \varepsilon \ln(4\pi\mu^2/m^2) + O(\varepsilon^2)$, the finite part assembles into the $\bar{\varepsilon}$ combination exactly as for the bubble:

$$\begin{aligned}\Gamma(-1+\varepsilon)\Big(\frac{4\pi\mu^2}{m^2}\Big)^\varepsilon &= \Big(-\frac{1}{\varepsilon} + \gamma_{\mathrm{E}} - 1\Big)\Big(1 + \varepsilon \ln\frac{4\pi\mu^2}{m^2}\Big) + O(\varepsilon) \\ &= -\frac{1}{\varepsilon} - \ln\frac{4\pi\mu^2}{m^2} + \gamma_{\mathrm{E}} - 1 + O(\varepsilon) = -\frac{1}{\bar{\varepsilon}} + \ln\frac{m^2}{\mu^2} - 1 + O(\varepsilon),\end{aligned} \tag{21.123}$$

where in the last step the constants γ_{E} and $-\ln 4\pi$ packaged with $-1/\varepsilon$ into $-1/\bar{\varepsilon} = -(1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi)$, and the residual scale logarithm simplified through $-\ln(4\pi\mu^2/m^2) + \ln 4\pi = \ln(m^2/\mu^2)$. Therefore

$$\Sigma(p^2) = \frac{\lambda\, m^2}{32\pi^2}\left[-\frac{1}{\bar{\varepsilon}} + \ln\frac{m^2}{\mu^2} - 1\right] + O(\varepsilon), \tag{21.124}$$

with no p^2 anywhere on the right-hand side. The divergent part is

$$\Sigma(p^2)\Big|_{\mathrm{div}} = -\frac{\lambda\, m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{21.125}$$

a pure constant proportional to m^2.

We again check the result against the chapter's theorems, and this is where the second prediction sharpens into the famous fact.

- *Power counting.* Equation Eq. (21.109) gave $D = 2$ for the two-point function: a quadratic divergence. In dimensional regularization a quadratic divergence still appears as a *simple* pole in ε — dimensional regularization does not produce power divergences, it converts them all into ε-poles — but it is the simple pole of the $n = 1$ tadpole integral, governed by $\Gamma(1 - d/2) = \Gamma(-1 + \varepsilon)$, which sits at a *negative* integer argument. The pole-condition Eq. (D.109) identifies this as the $j = 1$ case, the dimensional-regularization signature of a quadratic ($D = 2$) rather than a logarithmic ($D = 0$) divergence. The single $1/\bar{\varepsilon}$ in Eq. (21.125) accordingly multiplies m^2, carrying mass dimension two, exactly as a $D = 2$ divergence must. The degree of divergence is matched.

- *Locality, and the vanishing of the field renormalization.* The locality theorem permitted a divergent polynomial of degree 2 in p, namely $a\,p^2 + b\,m^2$. The actual divergence Eq. (21.125) has $a = 0$: the coefficient of p^2 vanishes identically, because the tadpole integrand never contained p to begin with. Locality told us a p^2 term was *allowed*; the explicit calculation tells us this particular diagram does not generate it. Only the m^2 piece survives, $b = -\lambda/32\pi^2 \cdot 1/\bar{\varepsilon}$.

The vanishing of a has a name and a consequence, and both are derived rather than asserted. The 1PI two-point function is $\Gamma^{(2)}(p^2) = p^2 - m^2 - \Sigma(p^2)$, and the field-strength counterterm δ_Z (Eq. (22.9)) is fixed by requiring the coefficient of p^2 in the renormalized $\Gamma^{(2)}$ to be finite. That coefficient is

$$\frac{\partial \Gamma^{(2)}}{\partial p^2} = 1 + \delta_Z - \frac{\partial \Sigma}{\partial p^2} = 1 + \delta_Z, \tag{21.126}$$

because $\partial\Sigma/\partial p^2 = 0$ by Eq. (21.124). There is no p^2 pole to subtract, so the $\overline{\text{MS}}$ prescription leaves δ_Z with nothing to cancel:

$$\delta_Z = 0, \qquad Z_\phi = 1 + \delta_Z = 1 + O(\lambda^2). \tag{21.127}$$

This is the celebrated fact that *the field strength of ϕ^4 is not renormalized at one loop*, and the field anomalous dimension consequently vanishes at this order, $\gamma_\phi = O(\lambda^2)$. We have shown it, not quoted it: the cause is the topology of the unique one-loop self-energy diagram, a tadpole through which no external momentum can flow. The mass divergence is real and is cancelled by the mass counterterm; matching the pole of $-\Sigma$ against the counterterm contribution $-(\delta_Z p^2 - \delta_m)$ fixes the mass counterterm residue from Eq. (21.125), the precise sign convention being settled with the rest of the renormalization bookkeeping in Eq. (22.9). The first diagram that can renormalize the field is the two-loop "sunset," through which p does circulate; that diagram, and the $\gamma_\phi = O(\lambda^2)$ it produces, are taken up in Section 24.2.

A quadratic divergence need not produce a quadratic counterterm

It is tempting to read "$D = 2$" as "the divergence scales like p^2." That is not what the superficial degree means. D is the maximal degree of the divergent polynomial that locality *allows*; the actual coefficients are dynamical and can vanish. In the ϕ^4 self-energy the highest-degree coefficient a (the p^2 piece) is zero and only the degree-zero coefficient b (the m^2 piece) is nonzero. The label $D = 2$ is honoured not by a p^2 divergence but by an m^2 divergence — a mass-dimension-two quantity, which is all the power counting actually requires. The lesson is that power counting bounds the divergence structure; it does not compute the coefficients, and a coefficient permitted by power counting can still be killed by the kinematics of the specific graph.

21.6.4 The numbers, staged for handoff

Both divergent amplitudes of one-loop ϕ^4 are now in hand, computed from the Feynman rules with every step shown, and both obey the chapter's predictions

to the letter. The four-point divergence is a momentum-independent constant ($D = 0$, logarithmic, locality-respecting), and the two-point divergence is a momentum-independent constant proportional to m^2 with no p^2 piece ($D = 2$, quadratic in mass dimension, with the allowed p^2 term dynamically absent). Collecting the pole residues that the next chapters consume:

$$i\mathcal{M}_{\text{loop}}\Big|_{\text{div}} = \frac{3i\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} \quad\Longrightarrow\quad \delta_\lambda = \frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{21.128}$$

$$\Sigma\Big|_{\text{div}} = -\frac{\lambda\, m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} \quad\Longrightarrow\quad \delta_{m^2} \text{ fixed by this residue}, \qquad \delta_Z = 0. \tag{21.129}$$

These three data — the coupling residue, the mass residue, and the vanishing of the field-strength residue — are precisely the inputs that Chapter 22 converts into the renormalized ϕ^4 Lagrangian through the counterterms $\delta_\lambda, \delta_{m^2}, \delta_Z$ of Eq. (22.9), and that Chapter 24 converts into the running of the theory: the coupling residue Eq. (21.128) becomes the beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Eq. (24.46), while $\delta_Z = 0$ becomes the statement $\gamma_\phi = O(\lambda^2)$. The nuisance with which the chapter opened — a loop integral that simply did not converge — has thus been turned into a finite, structured set of pole coefficients, and the seed of the renormalization-group flow that carries ϕ^4 theory to its fixed point is now planted.

21.7 What the Divergences Are Telling Us

The classification is complete. We have a formula—the superficial degree of divergence—that reads off from the topology of a graph alone whether its loop integral converges, and we have sorted every local theory into three boxes by the mass dimension of its couplings. For ϕ^4 theory we have, in Section 21.6, watched the abstract count turn into concrete numbers: a quadratically divergent two-point function whose pole renormalizes the mass, a logarithmically divergent four-point function whose pole renormalizes the coupling, and—this is the structural miracle—*nothing else*. Every other amplitude of the theory is superficially convergent. We now have all the pieces in hand to do something we have so far postponed: to stop asking *which* integrals diverge and ask *what the divergences mean*. The answer reframes the entire enterprise. The infinities are not a disease of the formalism to be quarantined; they are a precise diagnostic instrument, and what they measure is the dependence of a low-energy observable on physics we have not yet measured. Reading them this way converts the trichotomy of Section 21.3 from a bookkeeping rule into the founding statement of the Wilsonian renormalization group and of effective field theory—the two viewpoints that Chapters 23 and 38 will build into full machinery. This section derives no new heavy theorem; it presses the power counting we already proved until it yields its physical content, and it states honestly, and with dates, what remains unsettled.

21.7.1 A divergence is sensitivity to the unknown

Take the most concrete object we have, the one-loop scalar bubble of ϕ^4 theory assembled in Eq. (15.50) and cut off in Eq. (21.13). Stripped of external kinematics and couplings, the Euclidean loop integral with hard upper limit Λ evaluated there is

$$I(\Lambda) = \int^{\Lambda} \frac{\mathrm{d}^4 k}{(2\pi)^4} \frac{1}{(k^2+m^2)^2} = \frac{1}{16\pi^2}\left[\ln\frac{\Lambda^2+m^2}{m^2} - \frac{\Lambda^2}{\Lambda^2+m^2}\right] \xrightarrow{\Lambda\to\infty} \frac{1}{16\pi^2}\Big[\ln\frac{\Lambda^2}{m^2} - 1 + O\Big] \tag{21.130}$$

a logarithm in the cutoff, exactly as the superficial degree $D = 0$ predicted. Read Eq. (21.130) not as a divergent number but as a function of Λ. The integrand $1/(k^2+m^2)^2 \sim k^{-4}$ is scale-free at large k, so *every momentum octave contributes equally*: the slice $k \in [\Lambda_0, 2\Lambda_0]$ adds $\frac{1}{16\pi^2}\ln 2$ no matter how large Λ_0 is. The total is therefore dominated not by the region near the physical mass m but by the sum over all octaves up to the cutoff. A logarithmic divergence is the statement that the contributions from momenta a decade above the physical scale, two decades above, ten decades above, are all comparable; the integral is a democratic sum over an infinity of short-distance scales about which the low-energy Lagrangian, by construction, knows nothing. *The divergence is the loop integral telling us that the amplitude, as written, depends on physics at arbitrarily short distances—and depends on it without suppression.*

This is the interpretation to hold onto, and it generalizes immediately. A diagram with superficial degree of divergence D scales, under a hard cutoff, as

$$\text{amplitude} \sim \begin{cases} \Lambda^D & D > 0 \quad \text{(power divergence)}, \\ \ln\Lambda & D = 0 \quad \text{(log divergence)}, \\ \text{finite} & D < 0 \quad \text{(convergent)}, \end{cases} \tag{21.131}$$

the same trichotomy that dimensional regularization records, through the dictionary of Eq. (21.15), as a $1/\varepsilon^n$ pole, a single $1/\varepsilon$, or no pole at all. The magnitude D measures *how strongly* the observable reaches up into the unknown ultraviolet: $D > 0$ means the high-momentum region dominates as a positive power of the cutoff, $D = 0$ means it dominates marginally (logarithmically), and $D < 0$ means the integral is saturated near the physical scale and the deep ultraviolet is irrelevant. A finite diagram is one that has already forgotten the short-distance physics; a divergent one has not. The regulator—cutoff, or our default $d = 4 - 2\varepsilon$ built in Chapter 19—does not create this dependence and does not remove it. It only makes it *visible and finite*, so that we can ask the next question: can the dependence be absorbed?

The vacuum as a medium we have not finished describing

A useful picture, made precise in Chapter 23: the propagator of a quantum field is dressed by fluctuations of every wavelength down to zero. When we compute a loop we are integrating over those fluctuations. A renormalizable theory is one in which the *net effect* of all fluctuations shorter than any scale we can resolve is to shift a handful of numbers—the mass, the coupling, the field normalization—that we were going to measure anyway. The short-distance medium is not transparent, but its only low-energy footprint is a finite renaming of parameters. A non-renormalizable theory is one in which the medium leaves a structured, growing imprint that no fixed set of parameters can absorb. The divergences are how the fluctuation integral signals which case we are in.

21.7.2 The trichotomy in Wilsonian language

To turn "can the dependence be absorbed?" into a sharp criterion, recall the dimension-explicit power-counting formula proved in Theorem 21.7. For a connected diagram in d spacetime dimensions, built from vertices whose couplings g_t have mass dimension $\delta_t = [g_t]$, the superficial degree of divergence is Eq. (21.39),

$$D \;=\; d \;-\; \sum_{\text{ext}} [f] \;-\; \sum_{t} V_t\,\delta_t \;\equiv\; D_{\text{ext}} \;-\; \sum_{t} V_t\,[g_t], \tag{21.132}$$

where the first sum runs over external legs (each contributing the canonical dimension of its field) and $D_{\text{ext}} = d - \sum_{\text{ext}}[f]$ depends only on the external content of the amplitude, not on how complicated the graph is internally. The decisive feature is the last term. Whether adding internal structure—more vertices, more loops—makes a *fixed* process more or less divergent is governed entirely by the sign of $[g_t]$:

- $[g_t] > 0$: each extra vertex *lowers* D. Beyond a finite order every amplitude is convergent; only a few low-order graphs diverge. This is the *super-renormalizable* case.
- $[g_t] = 0$: D is *independent* of the number of vertices. The set of superficially divergent external configurations is fixed once and for all, the same at every order in perturbation theory. This is the *renormalizable* case.
- $[g_t] < 0$: each extra vertex *raises* D. Any amplitude whatsoever becomes divergent at sufficiently high order, and the number of distinct divergent structures is infinite. This is the *non-renormalizable* case.

This is exactly the classification of Theorems 21.12 and 21.13, now read off the single sign in Eq. (21.132). Figure 21.8 plots the three behaviours: $D(V) =$

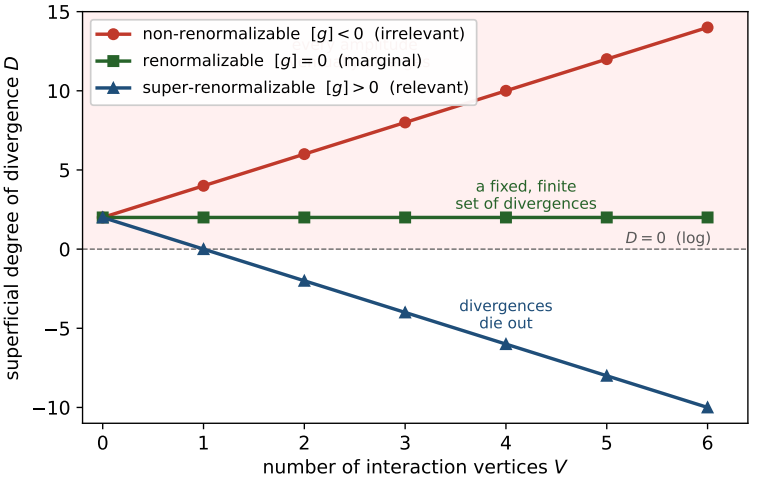


Figure 21.8: The superficial degree of divergence D of a *fixed* external process as the number of interaction vertices V grows, from $D = D_{\text{ext}} - V[g]$ in Eq. (21.132). For a relevant (super-renormalizable, $[g] > 0$) coupling the divergences die out with order; for a marginal (renormalizable, $[g] = 0$) coupling the divergent set is fixed and finite at every order; for an irrelevant (non-renormalizable, $[g] < 0$) coupling every amplitude eventually enters the divergent region $D \geq 0$ (shaded). Slopes are illustrative ($|[g]| = 2$, $D_{\text{ext}} = 2$); the up/flat/down trichotomy is convention-free.

$D_{\text{ext}} - V[g]$ is a straight line in the vertex number V whose slope is $-[g]$, sloping down, flat, or up according to the three cases.

The three adjectives in the right-hand column—relevant, marginal, irrelevant—are not ours to define here; they belong to the Wilsonian renormalization group of Chapter 23, where they acquire a dynamical meaning in terms of how a coupling flows as one lowers the resolution scale. But the dictionary is forced on us already, by dimensional analysis alone, and it is worth stating now so that the reader meets Chapter 23 recognizing it as the same statement seen from the other side:

power counting	**coupling dimension**	**Wilsonian flow (Chapter 23)**
super-renormalizable	$[g] > 0$	*relevant*: grows toward the infrared
renormalizable	$[g] = 0$	*marginal*: flows only logarithmically
non-renormalizable	$[g] < 0$	*irrelevant*: dies toward the infrared

The logic of the correspondence is the content of the next subsection: a coupling of positive mass dimension is large at low energies and small at high energies

(hence its loops are ultraviolet-soft and its operator dominates in the infrared—it is relevant), while a coupling of negative mass dimension is the reverse (ultraviolet-violent, infrared-negligible—irrelevant). The marginal case sits exactly on the fence, and which way it tips is decided not by power counting but by the sign of a loop coefficient—the beta function of Chapter 24. We will see at the end of this section that this fence is precisely where ϕ^4 theory lives.

21.7.3 Insensitivity made precise: absorbing the cutoff

The Wilsonian words are justified by a single line of power counting, which also makes precise the slogan that "renormalizability is insensitivity." Reinstate a physical cutoff Λ—the scale at which our description is to be replaced by some deeper one we have not written down—and ask how an amplitude computed up to Λ depends on it.

Consider first a marginal coupling, $[g] = 0$, the case of λ in ϕ^4 theory. By Eq. (21.132) the only superficially divergent 1PI amplitudes are the two-point and four-point functions, and each diverges no worse than the explicit power of Λ that D_{ext} allows, as the explicit poles of Section 21.6 confirmed. Take the four-point function. Its cutoff dependence is, with the logarithm of Eq. (21.130),

$$\Gamma^{(4)}(\Lambda) = -\lambda \;+\; c\,\lambda^2 \,\ln\frac{\Lambda}{\mu} \;+\; (\text{finite}), \qquad c = \frac{3}{16\pi^2}, \tag{21.133}$$

where μ is any fixed low scale and the positive coefficient c collects the three one-loop channels (the factor 3) of Section 21.6; it is the number Chapter 24 reads directly as the beta function, $\beta_\lambda = c\,\lambda^2 = 3\lambda^2/16\pi^2$, and we need only its sign and existence here. Now shift the cutoff, $\Lambda \to \Lambda'$, at fixed Lagrangian coupling. The change in the amplitude is

$$\Gamma^{(4)}(\Lambda') - \Gamma^{(4)}(\Lambda) = c\,\lambda^2 \,\ln\frac{\Lambda'}{\Lambda}, \tag{21.134}$$

which has *exactly the form* of the tree-level term $-\lambda$: it is a constant multiplying the same four-point structure, with no new dependence on the external momenta. It can therefore be absorbed completely by redefining the coupling that already sits in the Lagrangian. Demanding that $\Gamma^{(4)}$, as a function of the external kinematics, be the same when computed with cutoff Λ' as with Λ fixes

$$\lambda \;\longrightarrow\; \lambda' = \lambda + c\,\lambda^2 \,\ln\frac{\Lambda'}{\Lambda} + O(\lambda^3), \tag{21.135}$$

after which $\Gamma^{(4)}$ expressed in terms of λ' is the same function as before. The same is true of the quadratically and logarithmically divergent two-point function, absorbed into m^2 and the field normalization. This is the whole content of renormalizability seen from the cutoff: a change in the short-distance boundary

of the theory is indistinguishable from a finite change of the finitely many parameters we measure. Fix those parameters once, by experiment, at the scale μ, and *every* prediction becomes independent of Λ up to corrections that vanish as $\Lambda \to \infty$. The deep ultraviolet has been rendered invisible—not by removing it, but by showing that its only footprint is a reparametrization. That is insensitivity, and it is why a renormalizable theory can be predictive without our knowing what completes it at short distance.

Now contrast a non-renormalizable coupling, $[g] < 0$. Write $g = \tilde{g}\,\Lambda^{[g]} = \tilde{g}/\Lambda^{|[g]|}$ with $\tilde{g}$ dimensionless, anticipating that the natural home of such a coupling is a theory whose cutoff Λ is a genuine physical scale. The associated operator $\mathcal{O}$ has mass dimension $\Delta = d - [g] = d + |[g]| > d$. When we shift the cutoff, the divergence generated is no longer proportional to an operator already present. By Eq. (21.132) each insertion of this vertex raises D by $|[g]|$, so the new divergence comes with a positive power of the cutoff and dresses an operator of *higher* dimension still. Schematically, a loop with one insertion of $\mathcal{O}$ produces

$$\delta\Gamma \;\sim\; \tilde{g}\,\frac{\Lambda^{|[g]|}}{\Lambda^{|[g]|}} \times (\text{a new operator } \mathcal{O}'), \qquad \dim \mathcal{O}' = \Delta + |[g]| > \Delta, \tag{21.136}$$

a structure that was not in the Lagrangian we started with. To absorb it we must add $\mathcal{O}'$ with its own coefficient; its loops then generate $\mathcal{O}''$, of dimension higher still, and so on without end. No finite set of parameters closes the system. This is the precise sense in which a non-renormalizable theory has "infinitely many divergent structures"—and, read the other way, the precise sense in which it *predicts* the tower of higher-dimension operators suppressed by powers of $1/\Lambda$. The dimensional analysis of Section 21.3 and the EFT expansion in E/Λ are the same statement: the coefficient of an operator of dimension Δ carries $|[g]| = \Delta - d$ inverse powers of the only available mass scale, and at an external energy E its contribution to a dimensionless observable is suppressed by $(E/\Lambda)^{\Delta-d}$.

Why the sign of $[g]$ decides everything

The mass dimension of a coupling is the exchange rate between the coupling and the cutoff. A loop integral, having no scale of its own at large momenta, must make up the dimension of its coupling out of powers of Λ: a coupling short by $|[g]|$ units of mass forces the loop to supply $\Lambda^{|[g]|}$, and a coupling long by $[g]$ units forces $\Lambda^{-[g]}$, i.e. a *suppression*. Positive $[g]$ therefore buys ultraviolet softness, negative $[g]$ buys ultraviolet violence, and $[g] = 0$ buys the scale-free logarithm of Eq. (21.130). Everything in this chapter follows from that one bookkeeping fact about dimensions.

21.7.4 Non-renormalizable theories are effective field theories

The reading just given dissolves the historical verdict that non-renormalizable theories are "meaningless." They are not meaningless; they are *effective*, predictive within a finite energy range and to a finite accuracy, provided one organizes the prediction as an expansion in E/Λ. The presence of infinitely many divergent structures is not a failure but an honest accounting: it says only that to reach accuracy $(E/\Lambda)^n$ one must measure the coefficients of all operators up to a corresponding dimension—a finite number at each order—and that beyond that order the theory hands off to whatever physics lives at Λ. The full machinery, including the power-counting theorem that guarantees order-by-order finiteness once the requisite counterterms are included, is the subject of Chapter 38 [99, 403]. Here we establish the claim on the worked example that founded the entire viewpoint, continuing the Fermi-theory power counting begun in Theorem 21.15.

Worked Example 21.27 (Fermi theory as an effective field theory)**.** The original four-fermion theory of the weak interaction, written down by Fermi in 1934 to describe nuclear beta decay [189], takes the current–current form

$$\mathcal{L}_{\text{Fermi}} = -\frac{G_F}{\sqrt{2}}\,(\bar{\psi}_1\gamma^\mu(1-\gamma^5)\psi_2)(\bar{\psi}_3\gamma_\mu(1-\gamma^5)\psi_4), \tag{21.137}$$

a product of four Dirac fields at a single spacetime point. Its power counting was established in Theorem 21.15: with $[\psi] = \frac{d-1}{2} = \frac{3}{2}$ in $d = 4$ the operator has dimension 6, so by Eq. (21.71) the Fermi constant has $[G_F] = 4 - 6 = -2 < 0$—the unmistakable signature of a non-renormalizable, irrelevant coupling whose divergences worsen at every order (Eq. (21.72)). What that section diagnosed as a pathology we now read as a feature. Its measured value is $G_F \approx 1.166 \times 10^{-5}\,\text{GeV}^{-2}$, so the scale it hides is

$$\Lambda_{\text{Fermi}} \equiv G_F^{-1/2} \approx (1.166 \times 10^{-5})^{-1/2}\,\text{GeV} \approx 293\ \text{GeV}, \tag{21.138}$$

within a factor of order one of the masses $M_W \approx 80\,\text{GeV}$, $M_Z \approx 91\,\text{GeV}$ of the vector bosons that, we now know, complete the theory: the four-fermion vertex is the low-energy shadow of W-boson exchange, with $G_F/\sqrt{2} = g^2/8M_W^2$ once the W propagator $1/(q^2 - M_W^2)$ is contracted to a point at $q^2 \ll M_W^2$. Fermi's $1/\Lambda^2$ is the heavy-boson propagator expanded in q^2/M_W^2.

The power counting predicts a sharp failure. Consider a $2 \to 2$ fermion process, say $\nu_\mu\, e \to \mu\, \nu_e$, at center-of-mass energy $\sqrt{s} \gg m_{\text{ferm}}$. The amplitude is $\mathcal{M} \sim G_F\, s$ on dimensional grounds: $\mathcal{M}$ for $2 \to 2$ scattering is dimensionless in $d = 4$, G_F carries dimension -2, and the only other invariant is s of dimension $+2$. The cross section then scales as

$$\sigma \sim \frac{|\mathcal{M}|^2}{s} \sim \frac{(G_F\, s)^2}{s} = G_F^2\, s, \tag{21.139}$$

growing with energy. But unitarity caps a partial-wave cross section at $\sigma_J \lesssim 16\pi/s$ (the optical-theorem bound proved in Chapter 14). The two scalings cross at

$$G_F^2\, s \;\sim\; \frac{1}{s} \quad\Longrightarrow\quad \sqrt{s} \;\sim\; G_F^{-1/2} \approx 293\ \text{GeV}, \tag{21.140}$$

the same scale as Eq. (21.138). The growth of the cross section is the diagnostic divergence of Section 21.7.1 seen in an observable: at $\sqrt{s} \sim \Lambda_{\text{Fermi}}$ the theory becomes sensitive at order unity to the unknown short-distance physics, which is exactly where it must be replaced—and is, by the electroweak theory, in which the W propagator tames the growth above M_W.

Now the crucial point that rescues predictivity *below* that scale. Loop corrections in Fermi theory are genuinely divergent and demand, at one loop, a dimension-8 counterterm—by Eq. (21.132) with $|[G_F]| = 2$, two insertions of the vertex produce a divergence two powers worse, dressing an operator with two extra derivatives; at two loops a dimension-10 counterterm; and so on without end. Yet to predict an observable at energy E to accuracy $(E/\Lambda_{\text{Fermi}})^n$ one needs only the operators up to dimension $4 + 2n$—a *finite* list, each entering suppressed by $(E/\Lambda_{\text{Fermi}})^{\dim -4}$. The leading term Eq. (21.139) already gave nuclear beta decay to spectacular accuracy because there $E \sim \text{MeV} \ll \Lambda_{\text{Fermi}}$, so $(E/\Lambda_{\text{Fermi}})^2 \sim 10^{-10}$ and the entire tower of unknown operators is numerically irrelevant. Fermi theory was never wrong; it was an effective field theory used inside its domain, and its very breakdown scale Eq. (21.138) correctly forecast the mass range where the W boson had to appear.

The moral generalizes verbatim to the Standard Model itself, which is renormalizable and therefore, by Section 21.7.3, predictive without reference to its completion—but which is now read, in the modern view of Chapter 38, as the renormalizable (relevant and marginal) part of an effective theory whose non-renormalizable operators, suppressed by some higher scale, encode everything beyond it. Renormalizability has gone from a consistency requirement to a statement about which operators dominate at low energy. The decoupling of heavy physics that makes this consistent—that a heavy field of mass M leaves only $1/M$-suppressed traces in the low-energy theory—is the Appelquist–Carazzone theorem [31], established in Chapter 38.

21.7.5 Scalar ϕ^4 theory: marginality and what the loop decides

Return to the workhorse. The quartic coupling of ϕ^4 theory has, in $d = 4$,

$$[\lambda] = 4 - 4 \cdot \frac{d-2}{2}\bigg|_{d=4} = 4 - 4 = 0, \tag{21.141}$$

the renormalizable value, and we have just learned the Wilsonian name for $[\lambda] = 0$: *marginal*. This is the single most important entry in the dictionary for

the rest of the book, because marginality is the knife-edge. A relevant coupling is unambiguously important in the infrared and an irrelevant one unambiguously negligible; a marginal coupling does neither at the level of dimensions. Its fate is decided by the leading quantum correction—by whether the logarithm in Eq. (21.133) drives λ up or down as the scale is lowered. The coefficient $c = 3/16\pi^2 > 0$ of that logarithm, which we extracted as a pole in Section 21.6, has a sign, and the sign says that λ *grows* toward the ultraviolet and *shrinks* toward the infrared. In Wilsonian language, ϕ^4 in $d = 4$ is *marginally irrelevant*: dimensionally on the fence, tipped to the irrelevant side by the one-loop logarithm. That same positive sign is the seed of the Landau pole and of the triviality discussion below.

This is the precise hook on which Chapters 23 and 52 hang. In Chapter 23 the Wilsonian flow will turn "λ shrinks toward the infrared" into the trajectory of a coupling under successive integrating-out of momentum shells, and the marginal direction will be the one along which the flow is logarithmically slow—the defining feature of universality. In $d = 4 - \varepsilon$ the dimension of the coupling shifts to $[\lambda] = \varepsilon > 0$, nudging it to the relevant side, and the competition between that tree-level relevance and the one-loop logarithm produces a nontrivial zero of the beta function: the Wilson–Fisher fixed point, whose coordinates $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ Chapter 24 computes and whose conformal field theory Chapter 52 hands to the three-dimensional Ising bootstrap. All of that machinery is downstream of the one fact established here by dimensional analysis and the sign of a single loop: $[\lambda] = 0$ means marginal, and the loop decides which way the margin tips.

Two ε's: $d = 4 - \varepsilon$ versus $d = 4 - 2\varepsilon$

The Wilson–Fisher coordinate $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ and the shift $[\lambda] = 4 - d = \varepsilon$ just quoted are written in the *ε-expansion* convention $d = 4 - \varepsilon$, the convention standard in the critical-phenomena literature and the one in which Chapters 24 and 52 carry the fixed point. This is *not* the dimensional-regularization convention $d = 4 - 2\varepsilon$ used everywhere else in this chapter (Eq. (21.73), and the $\overline{\text{MS}}$ computations of Section 21.6), under which $[\lambda] = 4 - d = 2\varepsilon$. The two ε's differ by a factor of two; expressed in the dim-reg variable the same fixed point reads $\lambda_* = \frac{8\pi^2}{3}\varepsilon$. We follow the literature and state the Wilson–Fisher coordinate in $d = 4 - \varepsilon$.

21.7.6 What remains genuinely open

Honesty requires separating what renormalization secures from what it does not. Two issues, both live as of the mid-2020s, must be stated plainly so the reader does not mistake the success of the classification for a closure it has not achieved.

The perturbative series is asymptotic, not convergent. Suppose a theory is renormalizable, all divergences absorbed, every coefficient in the loop expansion finite. The series in the coupling that those coefficients build is, in general, still *divergent*: it has zero radius of convergence. Dyson's argument (Chapter 5) shows why for any theory with an unstable vacuum at negative coupling: the would-be analytic function cannot be analytic at the origin, so its Taylor series cannot converge. What renormalization delivers is therefore an *asymptotic* series—accurate when truncated optimally, useless when summed naively. Recovering a function from such a series is the business of Borel summation, developed in Section 5.7, and it does not always succeed: in asymptotically free theories the Borel transform has singularities on the positive axis—the *renormalons*—that obstruct the sum and signal a genuine nonperturbative ambiguity of the same parametric size as the missing condensate contributions. What is *established*: the perturbative series of realistic four-dimensional field theories is asymptotic, and the renormalon structure of its large-order behavior is understood in many cases. What is *open*: whether the full series of a given realistic theory is Borel summable to the true answer, and how the perturbative and nonperturbative pieces combine into a single well-defined function—the resurgence and transseries program, active and unfinished. Finiteness order by order, secured by this chapter and the next, is not the same as existence of the sum.

Does an interacting scalar exist in four dimensions? ϕ^4 theory raises a sharper question still. The positive sign of c in Eq. (21.133) means the renormalized coupling, held fixed at a low scale, requires a bare coupling that diverges at a finite ultraviolet scale—the Landau pole, made quantitative in Chapter 24. Taken at face value this says one cannot send the cutoff to infinity at fixed nonzero interaction: the continuum limit of ϕ^4_4 may not exist as an interacting theory. This expectation, long supported by perturbative and numerical evidence, was placed on rigorous footing by Aizenman and Duminil-Copin in 2021 [11]: the scaling (continuum) limit of the lattice ϕ^4 theory, and of the closely related Ising model, in exactly four dimensions is *Gaussian*—a free field. The interactions wash out as the cutoff is removed; the theory is *trivial* in the continuum. What is *established*: ϕ^4_4 has no nontrivial continuum limit; it exists only as an effective theory with a finite cutoff, precisely the EFT status of Section 21.7.4, internally consistent below its Landau scale and never meant to be pushed past it. What is *open*: the rigorous nonperturbative construction of genuinely interacting quantum field theories in four dimensions in general—the existence of four-dimensional Yang–Mills with a mass gap is a Clay Millennium problem, unsolved as of this writing—so that the question "which four-dimensional QFTs exist beyond perturbation theory?" remains, for the interacting non-asymptotically-free case, genuinely unanswered.

Neither caveat undermines the chapter. They sharpen its lesson. The divergences told us that a low-energy observable is sensitive to short-distance

physics; renormalization will show that for a finite set of couplings the sensitivity is absorbable; and the open questions are simply the residue of that same fact pressed to the extreme of infinite energy, where "the unknown ultraviolet" is no longer a finite cutoff to be absorbed but the full continuum limit, and where the honest answer is that for some theories it exists, for ϕ^4_4 it is trivial, and for the hardest cases it is not yet known.

With the divergences now read for their meaning—as a measure of sensitivity to the unknown, classified by the one sign in Eq. (21.132) into relevant, marginal, and irrelevant, and reinterpreted so that even the non-renormalizable theories become predictive effective descriptions—the chapter has done its conceptual work. What it has deliberately *not* done is cancel a single infinity: the constructive cure, the counterterms and renormalization conditions that turn Eq. (21.135) into a systematic procedure, is handed to Chapter 22; the dynamical flow that turns the relevant/marginal/irrelevant dictionary into the running of couplings and the theory of fixed points is handed to Chapter 23; and the effective-field-theory viewpoint that Theorem 21.27 previewed becomes the working language of Chapter 38.

Further reading

The power-counting and locality arguments of this chapter are treated with care in the standard texts: Peskin and Schroeder [396] (chapter 10) and Schwartz [432] (chapters 21–22) give the diagrammatic degree of divergence and the renormalizable classification at the level adopted here, while Srednicki [454] and Zee [532] offer complementary routes through the same material. The definitive treatment of the convergence theorem and the structure of subdivergences is Collins [134], which carries the Bogoliubov–Parasiuk–Hepp–Zimmermann analysis in full; the original convergence theorem is Weinberg [494], with the foundational subtraction results in Bogoliubov and Parasiuk [82], Hepp [266], and Zimmermann [533]. The asymptotic character of the perturbative series traces to Dyson [168]; the triviality of ϕ^4 in four dimensions, long conjectured, was established rigorously by Aizenman and Duminil-Copin [11]. The Wilsonian reinterpretation of the divergence classes, which Section 21.7 previews, is developed from Wilson and Kogut [516] and is the subject of Chapter 23.

Chapter 22

Renormalization Theory

The previous chapter delivered an uncomfortable inventory. Power counting (Theorem 21.10) told us *which* amplitudes of ϕ^4 and QED diverge, the finiteness criterion (Theorem 21.13) told us there are only finitely many of them, and locality (Theorem 21.17) told us that each divergence is, on its surface, a polynomial in external momenta. What that inventory did not do is make a single physical prediction finite. This chapter does. The pivot is a change of attitude toward the Lagrangian we wrote down: its parameters are not the numbers an experiment reports. We rename them *bare*, treat the regularized path integral as a definition rather than a measurement, and split the bare Lagrangian exactly into a renormalized piece wearing the finite, measured parameters and a *counterterm* piece whose coefficients are tuned, order by order, to cancel the divergences power counting catalogued. Nothing is thrown away; the bare and renormalized descriptions are the same theory in two notations, related by the multiplicative rescalings $\phi_0 = Z_\phi^{1/2}\phi$, $m_0^2 = Z_m m^2/Z_\phi$, $\lambda_0 = Z_\lambda \lambda/Z_\phi^2$.

The chapter follows the arc from this bookkeeping to a theorem. We first set up the counterterm split and its Feynman rules and state the renormalization principle that finiteness is to hold order by order (Section 22.1). We then make it concrete by fixing renormalization conditions — on-shell and $\overline{\text{MS}}$ — and computing the one-loop ϕ^4 counterterms explicitly from the tadpole self-energy and the s, t, u bubbles, recovering the carried divergences of Eqs. (21.14) and (21.18) and showing that only the logarithms are scheme-independent (Section 22.2). With the one-loop data in hand we reorganize the whole construction as the statement that renormalized Green's functions are finite rescalings of bare ones — multiplicative renormalization (Section 22.3) — which sharpens "renormalizable" into the precise claim that three constants suffice for ϕ^4.

That precise claim is then put under the only real stress test: more than one loop. A two-loop diagram can carry a divergent subgraph nested inside the loop that diverges overall, and the naive subtraction of just the overall divergence leaves behind a *nonlocal* remainder that no local counterterm can

absorb (Section 22.4). The cure is to subtract subdivergences first, and the systematic, all-orders form of that cure is Bogoliubov's recursion and Zimmermann's forest formula, whose equivalence, locality, and convergence (the BPHZ theorem, resting on Weinberg's power-counting convergence theorem) we state and argue in Section 22.5. Equipped with the forest formula as a field-content-blind engine, we prove that ϕ_4^4 is renormalizable to all orders by induction on the loop number (Section 22.6), pausing to see exactly why the field-strength counterterm δ_Z is forced upon us starting at two loops by the sunset diagram.

The chapter closes by carrying the same machinery to QED (Section 22.7), where a new ingredient enters that has no analogue in the scalar theory: gauge invariance. Power counting alone would demand a photon-mass counterterm and a four-photon vertex, neither present in the gauge-invariant Lagrangian; transversality of the photon self-energy forbids the first and the Ward identity renders light-by-light finite, while the Ward–Takahashi identity collapses the four QED renormalization constants to three by enforcing $Z_1 = Z_2$. The physical payoff is charge universality — every charged species renormalizes its charge through the single photon constant Z_3. Throughout, dimensional regularization in $d = 4 - 2\varepsilon$ with the $\overline{\text{MS}}$ scheme is the default (the master integrals of Chapter D and the gamma-algebra of Chapter C are used without re-derivation), and the QED induction leans on the abelian Ward–Takahashi identity being linear; the nonlinear non-abelian case is flagged here and discharged with the full BRST construction in Chapter 26. What we deliberately do *not* answer is *why* renormalizability should be the generic low-energy outcome at all — that is the Wilsonian story of Chapter 23 — nor do we yet extract the running couplings these Z-factors encode, which is the work of Chapter 24.

22.1 The Renormalization Program: Bare Parameters and Counterterms

Power counting has handed us a verdict and a worry in equal measure. The verdict, the master formula $D = 4 - E$ of Theorem 21.5 and Eq. (21.29), is a gift: in four-dimensional ϕ^4 theory only three amplitudes can diverge — the vacuum energy ($E = 0$), the two-point function ($E = 2$), and the four-point function ($E = 4$) — and the list is closed to all orders in λ, because adding vertices never raises D (Theorem 21.6). The worry is that the divergences are nonetheless there: the bubble of Eq. (21.14) carries a genuine $1/\varepsilon$ pole, the tadpole of Eq. (21.18) another, and as the regulator is removed each grows without bound. A finite list of infinities is still a list of infinities. This section reconceives what those infinities are, and in doing so converts the closed list $\{E = 0, 2, 4\}$ from a catalogue of diseases into a catalogue of *cures*. The thesis, argued and not merely asserted, is that the parameters written in the Lagrangian were never the mass and coupling any experiment measures; they are unobservable *bare* quantities,

and the divergences are the artifact of insisting on expressing finite physics in terms of them. Renormalization is the systematic reorganization of perturbation theory around the finite, measured quantities instead. The operational engine of that reorganization — the splitting of the Lagrangian into a renormalized piece and a set of counterterms, and the Feynman rules the counterterms generate — is what we build here. We continue ϕ^4 theory from its divergence analysis in Chapter 21, using the same model; no new number is computed in this section, which only sets the stage on which the one-loop counterterms of Section 22.2 will be pinned down.

22.1.1 The bare Lagrangian: parameters no experiment measures

The Lagrangian we have carried since Chapter 6

$$\mathcal{L} \;=\; \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{22.1}$$

was written down as though the symbols m and λ were physical inputs: m the mass of the particle, λ the strength of its self-interaction. At tree level that reading is harmless — the pole of the free propagator sits at $p^2 = m^2$, and the $2 \to 2$ amplitude is $-i\lambda$ (Eq. (15.105)), so m and λ are exactly what one would measure. The moment loops are included, the reading collapses. The one-loop self-energy shifts the pole of the propagator away from m^2 by a divergent amount (the tadpole Eq. (21.18) is the seed of that shift), and the one-loop bubble shifts the four-point amplitude away from $-i\lambda$ by the divergent logarithm of Eq. (21.14). If the m and λ in Eq. (22.1) were the measured mass and coupling, these shifts would be predictions of divergent corrections to quantities we have already measured to be finite — a manifest absurdity.

The resolution is to take seriously that Eq. (22.1) is a *definition of a regularized theory*, and that its parameters are whatever they must be to reproduce finite physics through a chosen regulator. They are not measured; they are bookkeeping symbols internal to the regularized path integral. To make this explicit we rename them, decorating every quantity in the defining Lagrangian with a subscript zero and calling it *bare*.

Definition 22.1 (The bare Lagrangian)**.** The fundamental, regulator-dependent definition of ϕ^4 theory is the *bare Lagrangian*

$$\mathcal{L} \;=\; \tfrac{1}{2}(\partial_\mu\phi_0)(\partial^\mu\phi_0) \;-\; \tfrac{1}{2}m_0^2\,\phi_0^2 \;-\; \frac{\lambda_0}{4!}\,\phi_0^4, \tag{22.2}$$

written in the *bare field* ϕ_0 and the *bare parameters* m_0 (bare mass) and λ_0 (bare coupling). All three of ϕ_0, m_0, λ_0 depend on the regulator — on the cutoff Λ if a hard cutoff is used, or on $1/\varepsilon$ in dimensional regularization with $d = 4 - 2\varepsilon$ — and diverge as it is removed. They are not directly observable.

The dependence of the bare parameters on the regulator is not a defect to be apologized for; it is the entire content of the statement that the short-distance physics is unspecified. Chapter 21 located the divergences as a short-distance singularity, the theory's honest admission that it does not know what happens at zero distance. The bare parameters are exactly the dial settings of that ignorance: as we vary the regulator — pushing the cutoff Λ higher, probing shorter distances — the bare parameters must be readjusted so that the predictions for the finite, long-distance observables stay put. A bare parameter that runs to infinity as $\Lambda \to \infty$ is not a sick number; it is the statement that holding a physical quantity fixed while extending the theory's reach to ever shorter distances requires an ever more extreme bare input. The observables are what is held fixed; the bare parameters are what is dialed.

Physical Insight

The status of the bare parameters is the same as that of an unrenormalized resistance in a circuit measured through long leads of unknown resistance: the number written on the component is not what an ohmmeter at the terminals reports. One never measures the bare resistor in isolation; one measures the combination. In field theory the "leads" are the cloud of virtual quanta that dress every interaction, and they contribute a divergent amount precisely because the cloud extends to arbitrarily short wavelength. What the detector reports is always the dressed, finite combination. The error of the naive reading of Eq. (22.1) was to confuse the number on the component with the number at the terminals.

22.1.2 Renormalized field and parameters: the multiplicative redefinition

If the bare quantities are not what experiment measures, we must name what experiment does measure and relate the two. The measured coupling and mass are *finite* numbers, fixed by definite measurements — the value of the four-point amplitude at a chosen kinematic point, and the location of the propagator pole — whose precise specification (the renormalization conditions) is the work of Section 22.2. Here we only need to introduce the finite *renormalized parameters* m and λ and a finite *renormalized field* ϕ, and to record their relation to the bare quantities. The field requires a rescaling because the bare field ϕ_0, dressed by loops, no longer creates a one-particle state with unit amplitude: the residue of the full propagator at its pole is shifted from unity, and the LSZ reduction (Chapter 14) demands a field whose propagator has a canonically normalized pole. That rescaling is the field-strength (or wavefunction) renormalization.

Definition 22.2 (Renormalized field and parameters)**.** Introduce the *field-strength renormalization constant* Z_ϕ, a dimensionless, regulator-dependent num-

ber, and define the *renormalized field*

$$\phi \;\equiv\; Z_\phi^{-1/2}\,\phi_0, \qquad \text{equivalently} \qquad \phi_0 \;=\; Z_\phi^{1/2}\,\phi, \tag{22.3}$$

together with the finite *renormalized mass* m and *renormalized coupling* λ, which are the only mass and coupling an experiment measures. The bare parameters are then whatever functions of m, λ, and the regulator are required to reproduce Eq. (22.2); it is convenient to package those functions through two further renormalization constants Z_m and Z_λ defined by

$$Z_\phi\, m_0^2 \;\equiv\; Z_m\, m^2, \qquad Z_\phi^2\,\lambda_0 \;\equiv\; Z_\lambda\,\lambda. \tag{22.4}$$

At tree level all three constants equal unity, $Z_\phi = Z_m = Z_\lambda = 1$; their deviations from unity are generated order by order in λ by loops, and carry the divergences.

The particular combinations $Z_\phi m_0^2$ and $Z_\phi^2\lambda_0$ appearing in Eq. (22.4) are not arbitrary: they are precisely the coefficients that the substitution Eq. (22.3) will produce when carried through the bare Lagrangian, as we now verify. The factor Z_ϕ accompanying m_0^2 comes from the two bare fields in the mass term; the factor Z_ϕ^2 accompanying λ_0 comes from the four bare fields in the quartic term. Defining Z_m and Z_λ to absorb exactly these factors is what makes the renormalized parameters m, λ the natural variables.

22.1.3 Re-expressing the bare Lagrangian: the counterterm split

We now perform the central algebraic operation of the program: substitute the renormalized field Eq. (22.3) into the bare Lagrangian Eq. (22.2), express everything in terms of the renormalized parameters using Eq. (22.4), and read off the result. No step is skipped. Inserting $\phi_0 = Z_\phi^{1/2}\phi$ term by term, and using that Z_ϕ is a constant so it passes through the derivatives,

$$\tfrac{1}{2}(\partial_\mu\phi_0)(\partial^\mu\phi_0) = \tfrac{1}{2}\,Z_\phi\,(\partial_\mu\phi)(\partial^\mu\phi), \tag{22.5}$$

$$-\tfrac{1}{2}\,m_0^2\,\phi_0^2 = -\tfrac{1}{2}\,m_0^2\,Z_\phi\,\phi^2 \;=\; -\tfrac{1}{2}\,(Z_\phi m_0^2)\,\phi^2, \tag{22.6}$$

$$-\frac{\lambda_0}{4!}\,\phi_0^4 = -\frac{\lambda_0}{4!}\,Z_\phi^2\,\phi^4 \;=\; -\frac{(Z_\phi^2\lambda_0)}{4!}\,\phi^4. \tag{22.7}$$

Collecting the three pieces, the bare Lagrangian written in renormalized variables is the exact identity

$$\mathcal{L} \;=\; \tfrac{1}{2}\,Z_\phi\,(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}\,(Z_\phi m_0^2)\,\phi^2 \;-\; \frac{(Z_\phi^2\lambda_0)}{4!}\,\phi^4. \tag{22.8}$$

This is still exactly Eq. (22.2) — not a single approximation has been made — merely written in the variable ϕ. The decisive move is now to add and subtract the renormalized values 1, m^2, and λ inside each coefficient, splitting each

renormalization constant into "1 plus a remainder." Define the three *counterterm coefficients*

$$\boxed{\delta_Z \;\equiv\; Z_\phi - 1, \qquad \delta_m \;\equiv\; Z_\phi\, m_0^2 - m^2, \qquad \delta_\lambda \;\equiv\; Z_\phi^2\,\lambda_0 - \lambda.} \tag{22.9}$$

By construction $\delta_Z = \delta_m = \delta_\lambda = 0$ at tree level, where every $Z = 1$ and the bare parameters equal the renormalized ones; the counterterm coefficients are pure loop effects. Their definitions are precisely engineered so that

$$Z_\phi = 1 + \delta_Z, \qquad Z_\phi m_0^2 = m^2 + \delta_m, \qquad Z_\phi^2 \lambda_0 = \lambda + \delta_\lambda. \tag{22.10}$$

Substituting Eq. (22.10) into the three coefficients of Eq. (22.8) and distributing,

$$\begin{aligned}\mathcal{L} &= \tfrac{1}{2}(1+\delta_Z)(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}(m^2+\delta_m)\phi^2 - \frac{\lambda+\delta_\lambda}{4!}\,\phi^4 \\ &= \underbrace{\left[\tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right]}_{\mathcal{L}_{\rm ren}[\phi,m,\lambda]} + \underbrace{\left[\tfrac{1}{2}\,\delta_Z(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,\delta_m\,\phi^2 - \frac{\delta_\lambda}{4!}\,\phi^4\right]}_{\mathcal{L}_{\rm ct}}.\end{aligned} \tag{22.11}$$

The first bracket is the renormalized Lagrangian $\mathcal{L}_{\rm ren}$: it is structurally identical to the frozen form Eq. (22.1), but with ϕ, m, λ now understood as the *finite, renormalized* field, mass, and coupling. The second bracket is the *counterterm Lagrangian*

$$\mathcal{L}_{\rm ct} \;=\; \tfrac{1}{2}\,\delta_Z\,(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}\,\delta_m\,\phi^2 \;-\; \frac{\delta_\lambda}{4!}\,\phi^4, \tag{22.12}$$

a new term of the Lagrangian carrying exactly the same three operators — the kinetic operator $(\partial\phi)^2$, the mass operator ϕ^2, and the quartic operator ϕ^4 — now multiplied by the divergent counterterm coefficients rather than by finite renormalized parameters. The decomposition Eq. (22.11),

$$\mathcal{L} \;=\; \mathcal{L}_{\rm ren}[\phi, m, \lambda] \;+\; \mathcal{L}_{\rm ct}, \tag{22.13}$$

is the operational engine of the entire chapter.

It cannot be over-emphasized that Eq. (22.13) is an *identity*, not a modification. We have changed nothing about the theory: Eq. (22.11) is term-for-term equal to the bare Lagrangian Eq. (22.2), as the substitution Eqs. (22.5) to (22.8) shows line by line. What we have done is *regroup* the same Lagrangian into a finite-looking part plus a remainder, and relabel the variables. The path integral, the correlators, and every physical amplitude are unchanged; only the organization of the perturbation series differs. This is the sense in which renormalization is a reorganization of perturbation theory, not a deformation of the theory. The next worked example confirms the bookkeeping is reversible and self-consistent.

Worked Example 22.3 (The split is an exact rewriting: closing the loop)**.** We verify that reassembling $\mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}}$ returns the bare Lagrangian, and that the counterterm coefficients Eq. (22.9) are the unique choice that makes this work. Start from Eq. (22.11) and collect the coefficient of each operator. The kinetic operator $\frac{1}{2}(\partial\phi)^2$ has coefficient $1 + \delta_Z$; the mass operator $-\frac{1}{2}\phi^2$ has coefficient $m^2 + \delta_m$; the quartic operator $-\frac{1}{4!}\phi^4$ has coefficient $\lambda + \delta_\lambda$. Inserting the definitions Eq. (22.9),

$$1 + \delta_Z = Z_\phi, \qquad m^2 + \delta_m = Z_\phi m_0^2, \qquad \lambda + \delta_\lambda = Z_\phi^2 \lambda_0, \tag{22.14}$$

so the three coefficients are Z_ϕ, $Z_\phi m_0^2$, $Z_\phi^2\lambda_0$, which are precisely those of Eq. (22.8). Undoing the field rescaling Eq. (22.3) by $\phi = Z_\phi^{-1/2}\phi_0$ then turns $Z_\phi(\partial\phi)^2 \to (\partial\phi_0)^2$, $(Z_\phi m_0^2)\phi^2 \to m_0^2\phi_0^2$, and $(Z_\phi^2\lambda_0)\phi^4 \to \lambda_0\phi_0^4$, returning Eq. (22.2) exactly. Conversely, suppose we had insisted only that the renormalized part of the split equal the frozen form Eq. (22.1) with coefficients $1, m^2, \lambda$; then the counterterm coefficients are forced to be the differences between the actual coefficients of Eq. (22.8) and these renormalized values, which is Eq. (22.9) verbatim. The split is therefore not merely consistent but *uniquely determined* once one demands that $\mathcal{L}_{\text{ren}}$ wear the renormalized parameters. There is no freedom in $\mathcal{L}_{\text{ct}}$ beyond the freedom already present in choosing what "renormalized" means — the renormalization conditions of Section 22.2.

Notation Watch

Several common notations for the counterterm coefficients circulate. Many texts write the kinetic, mass, and coupling counterterms as δ_Z, δ_m, δ_λ exactly as here; some write $\delta_Z \equiv Z_\phi - 1$ but define a *dimensionful* mass counterterm δ_{m^2}, and others fold the field-strength piece into a rescaled mass counterterm. Our conventions are fixed once and for all by Eq. (22.9): δ_Z is dimensionless and multiplies $\frac{1}{2}(\partial\phi)^2$; δ_m has mass dimension two and multiplies $-\frac{1}{2}\phi^2$; δ_λ is dimensionless and multiplies $-\frac{1}{4!}\phi^4$. In particular δ_m is a counterterm to m^2, not to m. These are the symbols ϕ^4 theory carries through Chapter 24.

22.1.4 Feynman rules for the counterterms

The split Eq. (22.13) is useful only if we can compute with it, which means treating $\mathcal{L}_{\text{ct}}$ as a source of new Feynman-diagram elements. The strategy is the one perturbation theory always uses: the renormalized part $\mathcal{L}_{\text{ren}}$ supplies the propagator and the ordinary $-i\lambda$ vertex of Theorem 15.13, exactly as before — $\mathcal{L}_{\text{ren}}$ is the standard Lagrangian in renormalized variables, so its rules are the rules of Chapter 15 verbatim — while $\mathcal{L}_{\text{ct}}$ is treated as an *additional interaction*, expanded perturbatively alongside the quartic vertex. Because $\delta_Z, \delta_m, \delta_\lambda$ start at $O(\lambda)$, the counterterm vertices are themselves higher-order objects: a counterterm

insertion counts as one extra power of the coupling, on the same footing as a loop, and the two will be made to cancel order by order.

To extract the new vertices we use the same functional-derivative route that generated the original rules in Section 15.5. Recall the master formula Eq. (15.87): the interacting generating functional is $Z[J] = \exp\!\big(iS_{\text{int}}[\tfrac{1}{i}\tfrac{\delta}{\delta J}]\big)Z_0[J]$, and a term of iS_{int} that is the spacetime integral of a monomial in the field generates, upon expansion (Eq. (15.91)) and the replacement rule Eq. (15.85), a vertex whose Feynman rule is i times the momentum-space coefficient of that monomial, with the $1/n!$ of an n-field monomial cancelled by the $n!$ ways of attaching distinguishable legs (Theorem 15.25). The counterterm action $S_{\text{ct}} = \int \mathrm{d}^4x\, \mathcal{L}_{\text{ct}}$ is added to S_{int}, so its monomials generate vertices by exactly this procedure. There is one new ingredient relative to Chapter 15 — the derivative coupling in the kinetic counterterm — which we now handle in full.

The two-point counterterm vertex. The quadratic part of the counterterm action is

$$S_{\text{ct}}^{(2)} \;=\; \int \mathrm{d}^4x \,\Big[\tfrac{1}{2}\,\delta_Z\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,\delta_m\,\phi^2\Big]. \tag{22.15}$$

To read off the momentum-space coefficient, pass to momentum space with the book's convention $\phi(x) = \int \frac{\mathrm{d}^4p}{(2\pi)^4}\, e^{-ip\cdot x}\,\widetilde{\phi}(p)$. Each derivative brings down a factor $-ip_\mu$ on the corresponding field. The kinetic term involves two fields, with momenta p and p'; the spacetime integral produces $\int \mathrm{d}^4x\, e^{-i(p+p')\cdot x} = (2\pi)^4\delta^{(4)}(p+p')$, forcing $p' = -p$, so the two derivatives contribute $(-ip_\mu)(-i(-p)^\mu) = (-ip_\mu)(ip^\mu) = p_\mu p^\mu = p^2$. Hence

$$\int \mathrm{d}^4x\, \tfrac{1}{2}\,\delta_Z(\partial_\mu\phi)(\partial^\mu\phi) = \tfrac{1}{2}\,\delta_Z \int \frac{\mathrm{d}^4p}{(2\pi)^4}\; p^2\,\widetilde{\phi}(p)\,\widetilde{\phi}(-p), \tag{22.16}$$

$$\int \mathrm{d}^4x\,(-\tfrac{1}{2}\,\delta_m\,\phi^2) = -\tfrac{1}{2}\,\delta_m \int \frac{\mathrm{d}^4p}{(2\pi)^4}\;\widetilde{\phi}(p)\,\widetilde{\phi}(-p), \tag{22.17}$$

so the quadratic counterterm action in momentum space is

$$S_{\text{ct}}^{(2)} = \tfrac{1}{2}\int \frac{\mathrm{d}^4p}{(2\pi)^4}\;(\delta_Z\,p^2 - \delta_m)\,\widetilde{\phi}(p)\,\widetilde{\phi}(-p). \tag{22.18}$$

This is a two-field monomial with coefficient $\frac{1}{2}(\delta_Z p^2 - \delta_m)$. By the rule-extraction recipe of Theorem 15.25, the Feynman rule is i times the coefficient with the $\frac{1}{2}$ undone by the $2! = 2$ ways of attaching the two distinguishable line-ends to the two fields of the vertex:

$$\boxed{\text{two-point counterterm vertex:} \qquad i(\delta_Z\,p^2 - \delta_m),} \tag{22.19}$$

where p is the four-momentum flowing through the vertex. The worked example below carries out the same extraction through the explicit functional derivatives, leaving no factor unaccounted.

The four-point counterterm vertex. The quartic part of $S_{\rm ct}$,

$$S_{\rm ct}^{(4)} \;=\; -\frac{\delta_\lambda}{4!}\int \mathrm{d}^4x\;\phi^4, \tag{22.20}$$

is structurally identical to the original interaction $S_{\rm int} = -\frac{\lambda}{4!}\int \mathrm{d}^4x\,\phi^4$ of Eq. (15.84), with $\lambda \to \delta_\lambda$. Its vertex is therefore obtained by the identical computation that produced the $-i\lambda$ rule Eq. (15.45) — the $1/4!$ cancelled exactly by the 4! attachments of four distinguishable legs, as in Theorem 15.26 — with λ replaced by δ_λ:

$$\boxed{\text{four-point counterterm vertex:} \qquad -\,i\,\delta_\lambda.} \tag{22.21}$$

There is no derivative in $S_{\rm ct}^{(4)}$, so the vertex is momentum-independent, just as the ordinary quartic vertex is. The momentum-conserving $(2\pi)^4\delta^{(4)}(\sum q)$ accompanies it exactly as in Eq. (15.45).

Worked Example 22.4 (The two-point counterterm vertex from the functional derivative, in full)**.** We re-derive Eq. (22.19) by the functional-derivative machinery of Theorem 15.24, tracking every factor of i, so that the recipe "i times the coefficient" is seen to follow and not assumed. Treat $S_{\rm ct}^{(2)}$ as part of the interaction action. In the master formula the relevant operator is the $O(\delta)$ term of $\exp\!\big(iS_{\rm ct}^{(2)}[\tfrac{1}{i}\tfrac{\delta}{\delta J}]\big)$, namely $iS_{\rm ct}^{(2)}[\tfrac{1}{i}\tfrac{\delta}{\delta J}]$, where each ϕ in Eq. (22.15) is replaced by $\frac{1}{i}\frac{\delta}{\delta J}$ per Eq. (15.85). The contribution to the two-point function $G^{(2)}(x_1,x_2)$ linear in the counterterm is, following Eq. (15.81),

$$\frac{1}{i^2}\frac{\delta^2}{\delta J(x_1)\delta J(x_2)}\Big(iS_{\rm ct}^{(2)}[\tfrac{1}{i}\tfrac{\delta}{\delta J}]\Big)\,Z_0[J]\Big|_{J=0}. \tag{22.22}$$

Work with the position-space mass term first; the kinetic term differs only by the two derivatives. Writing the mass piece of $iS_{\rm ct}^{(2)}$ as $i\cdot(-\frac{1}{2}\delta_m)\int \mathrm{d}^4w\,[\frac{1}{i}\frac{\delta}{\delta J(w)}]^2$, the operator carries two vertex derivatives at the same point w. Together with the two external derivatives this is a fourth functional derivative of the Gaussian $Z_0[J]$, evaluated at $J=0$: a free four-point function of the fields at x_1, x_2, w, w. By Wick's theorem (Theorem 10.25) the connected pairing ties x_1 and x_2 each to one of the two w-legs; there are $2! = 2$ such bijections, each giving $D_F(x_1-w)D_F(x_2-w)$. Collecting the prefactors — $1/i^2$ from the external derivatives, $(\frac{1}{i})^2$ from the two vertex derivatives, the explicit i from e^{iS}, and $-\frac{1}{2}\delta_m$ from the coefficient — and the combinatorial 2,

$$\frac{1}{i^2}\cdot i\cdot\Big(-\tfrac{1}{2}\delta_m\Big)\cdot\frac{1}{i^2}\cdot 2\int \mathrm{d}^4w\,D_F(x_1-w)D_F(x_2-w) = -\,i\,\delta_m\int \mathrm{d}^4w\,D_F(x_1-w)D_F(x_2-w), \tag{22.23}$$

where the numerical prefactor collects to $\frac{1}{i^2}\cdot i\cdot(-\frac{1}{2})\cdot\frac{1}{i^2}\cdot 2 = i\cdot(-\frac{1}{2})\cdot 2\cdot i^{-4} = i\cdot(-1)\cdot 1 = -i$, so the mass piece contributes $-i\delta_m$ times the two external propagators. Amputating those two external propagators (the operation of

Section 15.3.1) leaves the bare vertex $-i\delta_m$ from the mass piece. The kinetic piece runs identically but with the two derivatives ∂_μ acting at w; amputating and Fourier-transforming turns the two derivatives into the factor $+p^2$ computed in Eq. (22.16), contributing $+i\delta_Z p^2$ to the amputated vertex. Summing the two pieces, the amputated two-point counterterm vertex is $i(\delta_Z p^2 - \delta_m)$, which is Eq. (22.19). Every factor of i, every $\frac{1}{2}$, and the combinatorial 2 have been displayed; the rule is a consequence of the same Wick-theorem bookkeeping that produced the original ϕ^4 vertex, not a separate postulate.

Figure 22.1: The two counterterm Feynman rules of Eqs. (22.19) and (22.21), derived from $\mathcal{L}_{\text{ct}}$ of Eq. (22.12) by the functional-derivative route of Theorem 15.25. The crossed circle marks a counterterm insertion. Left: the two-point vertex $i(\delta_Z p^2 - \delta_m)$, carrying both the field-strength piece (proportional to the through-momentum squared) and the mass piece. Right: the momentum-independent four-point vertex $-i\delta_\lambda$, identical in form to the ordinary quartic vertex $-i\lambda$ of Eq. (15.45) with $\lambda \to \delta_\lambda$. Each is treated as a new diagrammatic element, inserted into diagrams perturbatively and counting as one order in λ.

These two new vertices, Eqs. (22.19) and (22.21) and Fig. 22.1, are appended to the rule book of Theorem 15.13. A diagram in the reorganized perturbation theory is built from propagators $i/(p^2 - m^2 + i\varepsilon)$ and ordinary vertices $-i\lambda$ exactly as before, now with renormalized m, λ, plus any number of counterterm insertions. Because each counterterm coefficient is at least $O(\lambda)$, inserting a counterterm raises the order in λ by (at least) one; at a fixed order in λ only finitely many counterterm insertions can appear, and they enter on the same footing as the loops at that order.

22.1.5 The renormalization principle: finiteness order by order

We have built the apparatus; we now state precisely what it is for. The counterterm coefficients $\delta_Z, \delta_m, \delta_\lambda$ are not yet numbers — nothing so far has fixed them. The renormalization principle supplies the missing input.

Definition 22.5 (The renormalization principle)**.** Expand each counterterm coefficient as a power series in the renormalized coupling beginning at first order,

$$\delta_Z = \sum_{n\geq 1} \delta_Z^{(n)} \lambda^n, \qquad \delta_m = \sum_{n\geq 1} \delta_m^{(n)} \lambda^n, \qquad \delta_\lambda = \sum_{n\geq 1} \delta_\lambda^{(n)} \lambda^n, \tag{22.24}$$

where each coefficient $\delta^{(n)}$ is allowed to depend on the regulator. The coefficients are determined order by order in λ by the requirement that, at each order, the counterterm insertions cancel the divergences of the loop diagrams with the same external legs, so that every renormalized Green's function $G^{(n)}(x_1, \dots, x_n)$ — computed with $\mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}}$ and expressed in ϕ, m, λ — is finite as the regulator is removed.

The logic is a bootstrap. At $O(\lambda^0)$ the theory is free and finite, and all counterterms vanish. At $O(\lambda^1)$ the only divergence in ϕ^4 is the tadpole correction to the two-point function (the $E = 2$, $D = 2$ entry of Theorem 21.6); a one-loop $\delta_m^{(1)}$ is chosen to cancel it, and one finds $\delta_Z^{(1)} = 0$ at this order because the tadpole is momentum-independent. At $O(\lambda^2)$ the bubble divergence of the four-point function ($E = 4$, $D = 0$) appears and is cancelled by $\delta_\lambda^{(2)}$, the two-loop and one-loop-counterterm-dressed two-point divergences are cancelled by $\delta_m^{(2)}, \delta_Z^{(2)}$, and so on. At each order the divergent diagrams with $E = 2$ legs are cancelled by an insertion of the $i(\delta_Z p^2 - \delta_m)$ vertex, and those with $E = 4$ legs by an insertion of $-i\delta_\lambda$. The explicit execution of the first nontrivial round — the one-loop $\delta_m, \delta_\lambda, \delta_Z$ of ϕ^4 theory — is the content of Section 22.2.

That this bootstrap can be carried out with only the three counterterms in Eq. (22.12), to all orders, is the statement that ϕ^4 is *renormalizable*, and the reason is exactly the power-counting verdict $D = 4 - E$ of Chapter 21. Divergences occur only for $E \in \{0, 2, 4\}$ (Theorem 21.6), and the divergent part of each such amplitude is a polynomial in the external momenta of degree at most D — a fact we will prove via the locality of counterterms in Section 22.5 — so the $E = 2$ divergence is a polynomial $a\,p^2 + b$ (degree $D = 2$, but only even powers survive by Lorentz invariance and the $\mathbb{Z}_2$ symmetry), absorbable by the two structures $\delta_Z p^2$ and δ_m, and the $E = 4$ divergence is a constant (degree $D = 0$), absorbable by δ_λ. The $E = 0$ vacuum divergence multiplies the identity operator; it shifts only the (unobservable) vacuum energy and is removed by the normalization $1/Z[0]$ that already divides out vacuum bubbles (Eq. (15.81)), so it needs no field counterterm. Every divergence the theory can produce thus matches, structure for structure, an operator already present in $\mathcal{L}$ — which is why finitely many counterterms suffice. A theory in which the divergent structures outran the operators available to absorb them would require new counterterms at every order and be non-renormalizable; the closed list $\{E = 0, 2, 4\}$ is precisely the guarantee that this does not happen here.

Physical Insight

The deep content of renormalizability is the coincidence between two lists: the list of operators with divergent coefficients (read off from power counting, $E \leq 4$) and the list of operators already in the Lagrangian. When they coincide, renormalization is bookkeeping — a relabeling of

finitely many parameters. The Wilsonian story of Chapter 23 will explain *why* they coincide: operators of higher dimension than those in $\mathcal{L}$ are irrelevant and their coefficients flow to zero at long distance, so the low-energy theory is automatically dominated by the renormalizable operators. For now we take the coincidence as the empirical fact that power counting delivers, and exploit it.

We have thus completed the reconceptualization promised at the outset. The divergences of Chapter 21 are not a sickness of the theory but an artifact of expressing predictions in terms of the unobservable bare parameters of Theorem 22.1; rewriting the same Lagrangian in the finite renormalized variables of Theorem 22.2 produces the exact split Eq. (22.13) into a renormalized piece and the counterterms Eq. (22.12), whose Feynman rules Eqs. (22.19) and (22.21) are now part of the perturbative apparatus.

What remains undone is the determination of the counterterm coefficients themselves: Eq. (22.24) tells us they are power series fixed order by order to cancel divergences, but the cancellation requires a precise prescription for what "finite renormalized" mass and coupling *mean* — the renormalization conditions. The next section imposes those conditions and computes δ_m, δ_λ, and δ_Z for ϕ^4 theory at one loop in full, in both the on-shell and $\overline{\text{MS}}$ schemes, turning the apparatus assembled here into the first carried numbers of the chapter.

22.2 Renormalization Conditions and the One-Loop ϕ^4 Counterterms

The previous section reorganized perturbation theory around the split Lagrangian: the bare parameters were traded for a renormalized mass m, a renormalized coupling λ, and a renormalized field ϕ, plus three counterterms $\delta_Z, \delta_m, \delta_\lambda$ collected in

$$\mathcal{L}_{\text{ct}} \;=\; \tfrac{1}{2}\,\delta_Z\,(\partial_\mu\phi)(\partial^\mu\phi) \;-\; \tfrac{1}{2}\,\delta_m\,\phi^2 \;-\; \frac{\delta_\lambda\,\mu^{2\varepsilon}}{4!}\,\phi^4, \tag{22.25}$$

recall Section 22.1. These counterterms entered the Feynman rules as two new vertices (Eqs. (22.19) and (22.21)): a two-point insertion $i(\delta_Z\,p^2 - \delta_m)$ on any propagator line, and a four-point insertion $-i\,\delta_\lambda\,\mu^{2\varepsilon}$. The coupling has been promoted to $\lambda\,\mu^{2\varepsilon}$ (Theorem A.46), with μ the renormalization scale and $d = 4 - 2\varepsilon$, so that λ stays dimensionless in d dimensions. What that section did *not* do is fix the numbers. The split is a tautology until we say what $\delta_Z, \delta_m, \delta_\lambda$ are; nothing forces a particular value on them, because the division of a bare quantity into "renormalized part plus counterterm" is a choice. The content of this section is to make that choice physically, by imposing *renormalization conditions* that tie the renormalized parameters to measurable features of the

theory, and then to solve for the counterterms at one loop. We will find them in closed form for ϕ^4 theory, exhibit the two standard schemes — on-shell and $\overline{\text{MS}}$ — and verify that with these values the renormalized two- and four-point functions are finite as $\varepsilon \to 0$. The pole coefficients computed here are the carried numbers that Chapter 24 converts into the beta function $\beta_\lambda = 3\lambda^2/16\pi^2$.

22.2.1 What a renormalization condition is

The counterterms have exactly as many free coefficients as there are renormalizable amplitudes: one each for the field normalization, the mass, and the coupling. The superficial-divergence count of Theorem 21.6 guarantees this matching — the only superficially divergent amplitudes of ϕ^4 carrying physics are the two-point function ($D = 2$, splitting into a mass piece and a field-strength piece) and the four-point function ($D = 0$, the coupling) — so three conditions exhaust the freedom. A renormalization condition is a statement that a renormalized Green's function takes a prescribed value at a prescribed kinematic point, the value chosen so that the renormalized parameter *is* the physical quantity it is named for.

The natural objects on which to impose conditions are the one-particle-irreducible amputated functions of Theorem 15.30, because they are the building blocks from which every connected amplitude is assembled and because each carries exactly one of the divergences to be removed. Write $-i\Sigma(p^2)$ for the sum of all amputated 1PI insertions into the two-point function (the *self-energy*), so that resumming the geometric series of insertions gives the full propagator

$$\widetilde{G}(p^2) \;=\; \frac{i}{p^2 - m^2 - \Sigma(p^2)}, \tag{22.26}$$

and define the 1PI two-point function as i times the inverse full propagator,

$$\Gamma^{(2)}(p^2) \;=\; p^2 - m^2 - \Sigma(p^2). \tag{22.27}$$

Here Σ includes the counterterm insertion: the two-point counterterm $i(\delta_Z p^2 - \delta_m)$ contributes $\Sigma_{\text{ct}} = -(\delta_Z p^2 - \delta_m)$, so that

$$\Gamma^{(2)}(p^2) \;=\; (1+\delta_Z)\,p^2 - m^2 - \delta_m - \Sigma_{\text{loop}}(p^2), \tag{22.28}$$

with Σ_{loop} the genuine loop part. For the four legs, write $\Gamma^{(4)}(s,t,u)$ for the amputated 1PI four-point vertex function, with the convention that the sum of amputated 1PI four-point diagrams equals $i\,\Gamma^{(4)}$ and the tree value is $\Gamma^{(4)} = -\lambda\mu^{2\varepsilon}$ (the vertex Feynman rule $-i\lambda\mu^{2\varepsilon}$), the kinematics carried by the Mandelstam invariants $s = (p_1+p_2)^2$, $t = (p_1-p_3)^2$, $u = (p_1-p_4)^2$ with $s+t+u = 4m^2$ for on-shell legs.

Definition 22.6 (On-shell renormalization conditions for ϕ^4)**.** The *on-shell* (or physical, or pole) scheme fixes the three counterterms by demanding

$$\Gamma^{(2)}(p^2)\big|_{p^2=m^2} = 0, \tag{22.29}$$

$$\left.\frac{\mathrm{d}\Gamma^{(2)}}{\mathrm{d}p^2}\right|_{p^2=m^2} = 1, \tag{22.30}$$

$$\Gamma^{(4)}(s,t,u)\big|_{s=t=u=\frac{4}{3}m^2} = -\lambda. \tag{22.31}$$

Each condition has a direct physical reading, and that reading is what makes the choice canonical rather than arbitrary. Condition Eq. (22.29) places the pole of the full propagator Eq. (22.26) exactly at $p^2 = m^2$: the renormalized mass m is then the physical mass, the rest energy of the one-particle state, the location of the physical pole probed by the LSZ formula of Theorem 14.31. Condition Eq. (22.30) fixes the residue of that pole to unity, $\widetilde{G}(p^2) \to i/(p^2 - m^2)$ near the pole; this is precisely the field normalization LSZ requires of an interpolating field, so the renormalized field ϕ creates a one-particle state with the canonical amplitude and external-leg factors in scattering amplitudes are exactly one. Condition Eq. (22.31) fixes the renormalized coupling λ to be the value of the four-point vertex at a definite, symmetric kinematic point; it is the operational definition of "the coupling" as something one could measure in a $2 \to 2$ scattering experiment at that point. The symmetric choice $s = t = u = \frac{4}{3}m^2$ treats the three channels democratically and is the fully symmetric on-shell subtraction point, lying below the two-particle threshold $s = 4m^2$ (since $s + t + u = 4m^2$ forces $s = t = u = \frac{4}{3}m^2$ when the three are equal), so F is real there with no branch cut.

The on-shell conditions are the most physical, but they are not the only consistent choice, and for many purposes — in particular for extracting the renormalization-group flow of Chapter 24 — a mass-independent scheme is more convenient. That scheme keeps only the divergent part and the universal constants that ride on it.

Definition 22.7 (The $\overline{\mathrm{MS}}$ scheme)**.** In *modified minimal subtraction* ($\overline{\mathrm{MS}}$, the book's default of Theorem A.49) each counterterm is chosen to cancel only the universal divergent combination

$$\frac{1}{\bar{\varepsilon}} \;\equiv\; \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi \tag{22.32}$$

of Eq. (A.204) appearing in the corresponding 1PI function, with no finite part beyond that bracket. Thus $\delta_Z, \delta_m, \delta_\lambda$ are pure multiples of $1/\bar{\varepsilon}$, the finite remainder being left in the renormalized Green's function rather than absorbed.

The two schemes differ only in whether the finite parts of the loop integrals are swept into the counterterms (on-shell) or left in the renormalized functions ($\overline{\text{MS}}$). We will compute both, and the comparison at the end of the section is the concrete statement that physical amplitudes do not care which we use. We now evaluate the two loop diagrams that feed all three counterterms.

22.2.2 The one-loop self-energy: the tadpole

At order λ the only 1PI two-point diagram is the *tadpole* (or "snail"): a single quartic vertex with two of its four legs joined to each other into a closed loop, the remaining two carrying the external momentum, drawn in Fig. 22.2. We compute it in full.

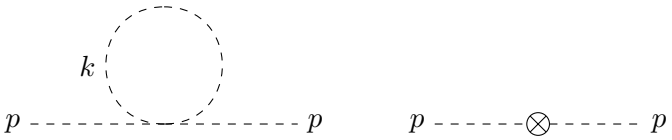


Figure 22.2: Left: the one-loop ϕ^4 self-energy, a single vertex with a closed loop carrying momentum k. Its value is independent of the external momentum p, because p never enters the loop. Right: the two-point counterterm vertex, Feynman rule $i(\delta_Z p^2 - \delta_m)$, drawn as a crossed dot.

The diagram has two external legs attached to the single vertex and a loop formed by the other two legs joining at the same vertex. By the symmetry-factor bookkeeping of Section 15.4 the two ways of pairing the two looped legs give a symmetry factor $S = 2$, so the diagram carries $1/2$. The vertex contributes $-i\lambda\mu^{2\varepsilon}$, the loop a single propagator $i/(k^2 - m^2 + i\varepsilon)$ integrated over $\int \mathrm{d}^d k/(2\pi)^d$. The external momentum p flows straight through the two external legs and never enters the loop integration. Therefore the sum of one-loop 1PI insertions is

$$-i\,\Sigma_{\text{loop}} \;=\; \frac{-i\lambda\mu^{2\varepsilon}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{i}{k^2 - m^2 + i\varepsilon} \;=\; \frac{\lambda\mu^{2\varepsilon}}{2}\,A_0(m^2), \tag{22.33}$$

where the explicit propagator factor of i combines with the $-i$ of the vertex to $\frac{1}{2}$, and A_0 is the scalar tadpole master integral of Eq. (D.99). The decisive structural fact is already visible: the right-hand side of Eq. (22.33) contains no p at all. The self-energy is momentum-independent at one loop,

$$\Sigma_{\text{loop}}(p^2) \;=\; \text{constant in } p^2, \tag{22.34}$$

a consequence of the quartic vertex being non-derivative and the loop closing on a single vertex so that no external momentum is forced through it.

Now substitute the value of the tadpole. From Eq. (D.99),

$$A_0(m^2) \;=\; -\frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)\,(m^2)^{d/2-1}, \tag{22.35}$$

so that, using $i \cdot (-i) = 1$,

$$\Sigma_{\text{loop}} \;=\; i \cdot \frac{\lambda \mu^{2\varepsilon}}{2} A_0(m^2) \;=\; \frac{\lambda\, \mu^{2\varepsilon}}{2\,(4\pi)^{d/2}}\, \Gamma\!\left(1 - \tfrac{d}{2}\right) (m^2)^{d/2-1}. \tag{22.36}$$

At $d = 4 - 2\varepsilon$ the argument is $1 - \frac{d}{2} = -1 + \varepsilon$, the negative-integer case of the gamma function, and the power is $(m^2)^{d/2-1} = (m^2)^{1-\varepsilon} = m^2\,(m^2)^{-\varepsilon}$. Collecting the ε-dependent pieces and inserting the Laurent expansion $\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} - 1 + \gamma_{\text{E}} + O(\varepsilon)$ of Eq. (D.125), together with $(4\pi)^{-d/2} = (4\pi)^{-2}(4\pi)^{\varepsilon}$ and $\mu^{2\varepsilon}(m^2)^{-\varepsilon}(4\pi)^{\varepsilon} = (4\pi\mu^2/m^2)^{\varepsilon} = 1 + \varepsilon \ln(4\pi\mu^2/m^2) + O(\varepsilon^2)$,

$$\begin{aligned}
\Sigma_{\text{loop}} &= \frac{\lambda m^2}{2(4\pi)^2}\, \Gamma(-1+\varepsilon) \Big(\frac{4\pi\mu^2}{m^2}\Big)^{\varepsilon} \\
&= \frac{\lambda m^2}{32\pi^2} \Big(-\frac{1}{\varepsilon} - 1 + \gamma_{\text{E}} \Big) \Big(1 + \varepsilon \ln \frac{4\pi\mu^2}{m^2} \Big) + O(\varepsilon) \\
&= \frac{\lambda m^2}{32\pi^2} \Big(-\frac{1}{\varepsilon} - 1 + \gamma_{\text{E}} - \ln \frac{4\pi\mu^2}{m^2} \Big) + O(\varepsilon) \\
&= -\frac{\lambda m^2}{32\pi^2} \Big(\frac{1}{\bar{\varepsilon}} - \ln \frac{m^2}{\mu^2} + 1 \Big) + O(\varepsilon),
\end{aligned} \tag{22.37}$$

where the last line collected $\frac{1}{\varepsilon} - \gamma_{\text{E}} + \ln 4\pi = 1/\bar{\varepsilon}$ and used $-\ln(4\pi\mu^2/m^2) + \ln 4\pi = \ln(m^2/\mu^2)$, i.e. $-\gamma_{\text{E}} + \ln 4\pi$ was absorbed into $1/\bar{\varepsilon}$ and the residual ln rearranged. The would-be *quadratic* divergence flagged by the $D = 2$ count of Theorem 21.6 has, in dimensional regularization, no power of a cutoff scale to carry it; it appears instead through the pole of $\Gamma(-1+\varepsilon)$ and is proportional to $m^2/\bar{\varepsilon}$. (In a momentum cutoff Λ the same diagram would give $\Sigma_{\text{loop}} \sim \lambda\Lambda^2$; dimensional regularization is blind to power divergences and records only the logarithmic residue, which is why δ_m below is proportional to m^2 rather than to a quadratic scale.)

With Eq. (22.37) and the momentum-independence Eq. (22.34) in hand, the renormalization conditions Eqs. (22.29) and (22.30) are immediate. From the assembled 1PI function Eq. (22.28),

$$\Gamma^{(2)}(p^2) = (1 + \delta_Z) p^2 - m^2 - \delta_m - \Sigma_{\text{loop}}, \qquad \frac{\mathrm{d}\Gamma^{(2)}}{\mathrm{d}p^2} = 1 + \delta_Z, \tag{22.38}$$

the last equality because Σ_{loop} does not depend on p^2. The residue condition Eq. (22.30) then reads $1 + \delta_Z = 1$, so

$$\boxed{\delta_Z = 0 \quad \text{at one loop.}} \tag{22.39}$$

There is no field-strength renormalization in ϕ^4 at this order, in either scheme: the on-shell condition gives $\delta_Z = 0$ exactly, and $\overline{\text{MS}}$ gives $\delta_Z = 0$ because Σ_{loop} carries no p^2 and hence no $1/\bar{\varepsilon}$ multiplying p^2 to subtract. Field renormalization

first appears at two loops, where the *sunset* diagram — two vertices joined by three internal lines — forces the external momentum through the loops, producing a $\Sigma(p^2)$ with $\mathrm{d}\Sigma/\mathrm{d}p^2 \neq 0$ and hence $\delta_Z = O(\lambda^2)$; that computation is the field-strength input to the anomalous dimension of Chapter 24.

The pole condition Eq. (22.29) fixes δ_m. With $\delta_Z = 0$, evaluating $\Gamma^{(2)}$ at $p^2 = m^2$,

$$\Gamma^{(2)}(m^2) = m^2 - m^2 - \delta_m - \Sigma_{\text{loop}} = 0 \quad \Longrightarrow \quad \delta_m^{\text{OS}} = -\Sigma_{\text{loop}} = \frac{\lambda m^2}{32\pi^2}\Big(\frac{1}{\bar{\varepsilon}} - \ln\frac{m^2}{\mu^2} + 1\Big), \tag{22.40}$$

the full self-energy with its sign reversed, pole and finite part together. In $\overline{\text{MS}}$ only the $1/\bar{\varepsilon}$ bracket is removed (Theorem 22.7):

$$\boxed{\delta_m^{\overline{\text{MS}}} = \frac{\lambda m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}} \tag{22.41}$$

the finite pieces $-\ln(m^2/\mu^2) + 1$ being left inside the renormalized $\Gamma^{(2)}$. Either way the mass counterterm is proportional to m^2 and to λ, and absorbs the (dimensionally regularized) mass shift produced by the tadpole. This completes the two-point sector at one loop.

22.2.3 The one-loop four-point: the s, t, u bubbles

At order λ^2 the 1PI four-point function receives the three one-loop *bubble* diagrams, one in each Mandelstam channel, drawn in Fig. 22.3. Each is the same scalar bubble met in Theorem D.25, differing only in which invariant flows through the loop.

A single channel was carried to its $\overline{\text{MS}}$ finite part in Theorem D.32; we recall the result and assemble the three. The s-channel diagram has two vertices $-i\lambda\mu^{2\varepsilon}$ and symmetry factor $\frac{1}{2}$ (the two internal lines may be interchanged), giving its contribution to $i\Gamma^{(4)}$ as

$$i\Gamma_s^{(4)} = \frac{(-i\lambda\mu^{2\varepsilon})^2}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{i}{k^2 - m^2}\,\frac{i}{(k+p)^2 - m^2} = \frac{\lambda^2\mu^{2\varepsilon}}{2}\,B(s), \tag{22.42}$$

with $p^2 = s$ and B the scale-carrying bubble of Eq. (D.140),

$$B(s) = \frac{i}{(4\pi)^2}\Big[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x\,\ln\frac{\Delta_s(x)}{\mu^2}\Big] + O(\varepsilon), \qquad \Delta_s(x) = m^2 - x(1-x)s. \tag{22.43}$$

Defining the finite function

$$F(s) \;\equiv\; \int_0^1 \mathrm{d}x\,\ln\frac{m^2 - x(1-x)s}{\mu^2}, \tag{22.44}$$

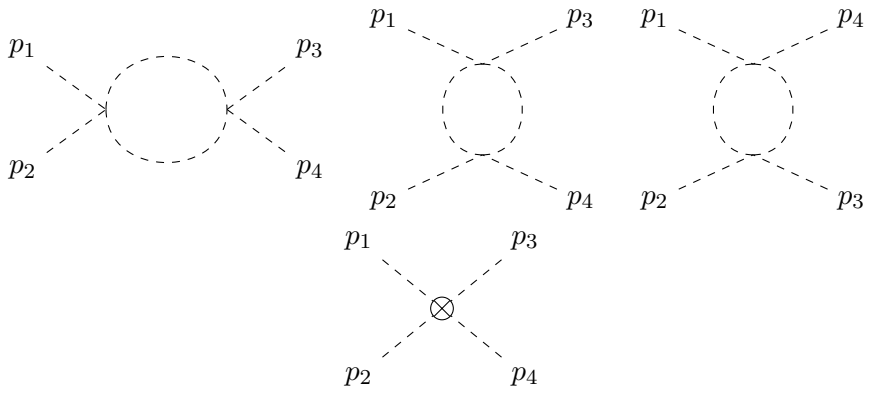


Figure 22.3: The one-loop four-point function: the s-, t- and u-channel bubbles (left three), with $s=(p_1+p_2)^2$, $t=(p_1-p_3)^2$, $u=(p_1-p_4)^2$ flowing through the respective loops, and the four-point counterterm vertex (right, crossed dot), Feynman rule $-i\,\delta_\lambda\,\mu^{2\varepsilon}$.

the s-channel piece is $i\Gamma_s^{(4)}=\dfrac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\big[\frac{1}{\bar\varepsilon}-F(s)\big]$, using $(4\pi)^2=16\pi^2$. The t- and u-channel diagrams are identical in structure with $s\to t$ and $s\to u$; their symmetry factors are also $\frac12$, and their loop integrals are the same B evaluated at the corresponding invariant. Summing the three channels and adding the tree vertex $-i\lambda\mu^{2\varepsilon}$ and the counterterm $-i\delta_\lambda\mu^{2\varepsilon}$,

$$i\Gamma^{(4)}(s,t,u)=-i\mu^{2\varepsilon}(\lambda+\delta_\lambda)+\frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\Big[\frac{3}{\bar\varepsilon}-F(s)-F(t)-F(u)\Big]+O(\varepsilon), \tag{22.45}$$

that is, stripping the overall i,

$$\Gamma^{(4)}(s,t,u)=-\mu^{2\varepsilon}(\lambda+\delta_\lambda)+\frac{\lambda^2\mu^{2\varepsilon}}{32\pi^2}\Big[\frac{3}{\bar\varepsilon}-F(s)-F(t)-F(u)\Big]+O(\varepsilon). \tag{22.46}$$

The entire divergence is the channel-summed pole $3/\bar\varepsilon$; the three F's are finite. The factor of three — one $1/\bar\varepsilon$ per channel — is the only place the three channels conspire, and it is the origin of the "3" that will appear in β_λ.

The coupling counterterm is now fixed by the four-point condition. In the on-shell scheme Eq. (22.31), evaluate Eq. (22.46) at the symmetric point $s=t=u=\frac43 m^2$, denote $s_0\equiv\frac43 m^2$, and demand $\Gamma^{(4)}=-\lambda$ as $\varepsilon\to0$ (so $\mu^{2\varepsilon}\to1$ in the finite terms):

$$-\lambda-\delta_\lambda+\frac{\lambda^2}{32\pi^2}\Big[\frac{3}{\bar\varepsilon}-3F(s_0)\Big]=-\lambda\quad\Longrightarrow\quad\delta_\lambda^{\rm OS}=\frac{3\lambda^2}{32\pi^2}\Big[\frac{1}{\bar\varepsilon}-F(s_0)\Big]. \tag{22.47}$$

In $\overline{\rm MS}$ only the pole is subtracted:

$$\boxed{\delta_\lambda^{\overline{\rm MS}}=\frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar\varepsilon}=\frac{3\lambda^2}{16\pi^2}\,\frac{1}{2\bar\varepsilon}.} \tag{22.48}$$

This is the carried number. The coefficient $3\lambda^2/32\pi^2$ of the pole is precisely the input that Chapter 24 turns into the one-loop beta function: requiring the bare coupling $\lambda_0 = Z_\lambda \lambda \mu^{2\varepsilon} = (1 + \delta_\lambda/\lambda)\lambda\mu^{2\varepsilon}$ to be independent of μ converts the residue of $1/\varepsilon$ in δ_λ into the slope of λ in $\ln \mu$, and the factor of two from $\mu^{2\varepsilon}$ assembles

$$\beta_\lambda = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3), \tag{22.49}$$

the carried ϕ^4 beta function. The whole of the present section exists to produce the number $3\lambda^2/32\pi^2$ in Eq. (22.48); the conversion to Eq. (22.49) is the business of Chapter 24 and is not repeated here.

22.2.4 On-shell versus $\overline{\text{MS}}$: finite parts and scheme independence

We have computed each counterterm twice. The pole parts agree across the two schemes — they must, since both schemes remove the same $1/\bar{\varepsilon}$ — but the finite parts differ:

$$\delta_\lambda^{\text{OS}} - \delta_\lambda^{\overline{\text{MS}}} = -\frac{3\lambda^2}{32\pi^2} F\Big(\tfrac{4}{3}m^2\Big), \qquad \delta_m^{\text{OS}} - \delta_m^{\overline{\text{MS}}} = \frac{\lambda m^2}{32\pi^2}\Big(1 - \ln\frac{m^2}{\mu^2}\Big), \tag{22.50}$$

finite, kinematics-dependent constants. This is the general pattern, and it is worth stating plainly: *the counterterms are scheme-dependent; the divergent parts are not.* The pole of any 1PI function is a local quantity forced by the short-distance structure of the theory, the same in every scheme because it is the same divergence; the finite part is the residue of a free choice about where to draw the line between "renormalized parameter" and "counterterm."

That the counterterms differ between schemes does not make any physical prediction ambiguous. A physical amplitude — an actual scattering cross section at a definite kinematic point — is the renormalized Green's function evaluated there, and it comes out the same in both schemes provided one is consistent about what the renormalized parameters mean. The mechanism is a finite reparametrization. The renormalized coupling of the two schemes is not the same number: the on-shell coupling λ_{OS} and the $\overline{\text{MS}}$ coupling $\lambda_{\overline{\text{MS}}}$ are related by a finite shift,

$$\lambda_{\text{OS}} = \lambda_{\overline{\text{MS}}} + \frac{3\lambda_{\overline{\text{MS}}}^2}{32\pi^2} F\Big(\tfrac{4}{3}m^2\Big) + O(\lambda^3), \tag{22.51}$$

exactly the finite difference Eq. (22.50) between the coupling counterterms. When a physical amplitude is computed in $\overline{\text{MS}}$ and then re-expressed in terms of the measured quantity (the value of the vertex at the symmetric point, which *is* λ_{OS} by Eq. (22.31)), the finite logarithms of the $\overline{\text{MS}}$ result reorganize precisely so as to reproduce the on-shell prediction. The renormalized one-loop four-point amplitude makes this explicit. Subtracting the pole with $\delta_\lambda^{\overline{\text{MS}}}$ in Eq. (22.46)

leaves the finite renormalized vertex

$$\Gamma^{(4)}_{\overline{\text{MS}}}(s,t,u) = -\lambda_{\overline{\text{MS}}} - \frac{\lambda^2_{\overline{\text{MS}}}}{32\pi^2}\left[F(s)+F(t)+F(u)\right] + O(\lambda^3), \tag{22.52}$$

manifestly finite as $\varepsilon \to 0$. Evaluating at the symmetric point and substituting the conversion Eq. (22.51),

$$\Gamma^{(4)}_{\overline{\text{MS}}}\big|_{s=t=u=\frac{4}{3}m^2} = -\lambda_{\overline{\text{MS}}} - \frac{3\lambda^2_{\overline{\text{MS}}}}{32\pi^2}F\Big(\tfrac{4}{3}m^2\Big) = -\lambda_{\text{OS}} + O(\lambda^3), \tag{22.53}$$

which is exactly the on-shell condition Eq. (22.31). The two schemes describe the same vertex; they merely label it with different couplings, and the finite parts of the counterterms are the dictionary between the labels. With the explicit closed form of $F(s)$ from Eq. (D.147), the renormalized vertex Eq. (22.52) is the carried ϕ^4 one-loop amplitude: the tree value $-\lambda$ plus three finite logarithms, one per channel, each developing a branch cut above its two-particle threshold $s > 4m^2$ as required by unitarity.

Why only logarithms are physical

The split between scheme-dependent finite parts and scheme-independent poles has a sharper companion: among the finite parts, the *constants* are ambiguous but the *logarithms of momenta* are not. A counterterm is a local term in the Lagrangian, hence a polynomial in fields and momenta with no logarithms; the freedom of a scheme is therefore the freedom to shuffle a momentum-independent constant between the counterterm and the renormalized function, and nothing more. The function $F(s)$ contains a constant ($\ln(m^2/\mu^2) - 2$ in the closed form Eq. (D.147)) that a redefinition of λ can move around, but it also contains the genuine s-dependence $2a\arctan(1/a)$, $a^2 = 4m^2/s - 1$, which no local counterterm can touch. That irreducible momentum dependence is the physics; its μ-dependence through the constant $\ln(m^2/\mu^2)$ is what the renormalization group of Chapter 24 promotes into the running coupling. The pole is fixed by locality, the constant is free, the logarithm is sacred.

22.2.5 Finiteness of the renormalized Green's functions

It remains to verify the whole point of the exercise: that with the counterterms just computed, the renormalized two- and four-point functions are finite as $\varepsilon \to 0$. We carry this out as a worked check, because the cancellation of the poles is the operational meaning of renormalization at this order and the reader should see it happen line by line.

Worked Example 22.8 (The renormalized one-loop ϕ^4 functions are finite)**.** We continue ϕ^4 theory, assembling the renormalized $\Gamma^{(2)}$ and $\Gamma^{(4)}$ in $\overline{\text{MS}}$ and

confirming pole cancellation; the on-shell case is identical with the finite parts also removed.

Two-point function. Insert $\delta_Z = 0$ from Eq. (22.39), the $\overline{\text{MS}}$ mass counterterm $\delta_m^{\overline{\text{MS}}} = \lambda m^2/(32\pi^2\bar{\varepsilon})$ from Eq. (22.41), and the self-energy Eq. (22.37) into the assembled $\Gamma^{(2)}$ of Eq. (22.28):

$$\begin{aligned}\Gamma^{(2)}_{\overline{\text{MS}}}(p^2) &= p^2 - m^2 - \delta_m^{\overline{\text{MS}}} - \Sigma_{\text{loop}} \\ &= p^2 - m^2 - \frac{\lambda m^2}{32\pi^2}\frac{1}{\bar{\varepsilon}} + \frac{\lambda m^2}{32\pi^2}\Big(\frac{1}{\bar{\varepsilon}} - \ln\frac{m^2}{\mu^2} + 1\Big) \\ &= p^2 - m^2 + \frac{\lambda m^2}{32\pi^2}\Big(1 - \ln\frac{m^2}{\mu^2}\Big),\end{aligned} \tag{22.54}$$

where the two $1/\bar{\varepsilon}$ terms cancel exactly. The result is finite, ε-free, and reduces to the tree-level $p^2 - m^2$ plus a finite, p-independent shift. In the on-shell scheme one instead uses $\delta_m^{\text{OS}} = -\Sigma_{\text{loop}}$ from Eq. (22.40), and the finite shift also cancels, leaving $\Gamma^{(2)}_{\text{OS}}(p^2) = p^2 - m^2$ exactly — the pole sits at $p^2 = m^2$ with unit residue by construction, as the conditions Eqs. (22.29) and (22.30) demand.

Four-point function. Insert $\delta_\lambda^{\overline{\text{MS}}} = 3\lambda^2/(32\pi^2\bar{\varepsilon})$ from Eq. (22.48) into Eq. (22.46):

$$\begin{aligned}\Gamma^{(4)}_{\overline{\text{MS}}}(s,t,u) &= -\lambda - \frac{3\lambda^2}{32\pi^2}\frac{1}{\bar{\varepsilon}} + \frac{\lambda^2}{32\pi^2}\Big[\frac{3}{\bar{\varepsilon}} - F(s) - F(t) - F(u)\Big] \\ &= -\lambda - \frac{\lambda^2}{32\pi^2}\big[F(s) + F(t) + F(u)\big],\end{aligned} \tag{22.55}$$

the $3/\bar{\varepsilon}$ from the bubbles cancelling the $3/\bar{\varepsilon}$ from the counterterm. The renormalized vertex is finite, equal to the tree coupling plus three finite channel logarithms, recovering Eq. (22.52). With F in its closed form Eq. (D.147), this is a numerically explicit, finite function of the Mandelstam invariants. Both renormalized functions are finite as $\varepsilon \to 0$; the three counterterms $\delta_Z = 0$, δ_m, δ_λ exhaust the divergences of ϕ^4 at one loop, exactly as the power-counting verdict of Theorem 21.6 promised. This closes the one-loop ϕ^4 stage of the model.

The pole, not the regulator, is what we cancel

It is tempting to read the cancellations in Theorem 22.8 as "the counterterm undoes the loop." It does not. The counterterm cancels the *pole* of the loop, and in $\overline{\text{MS}}$ the universal constants $-\gamma_{\text{E}} + \ln 4\pi$ that ride on it; what survives is the physical, μ-dependent finite part. The renormalized functions Eqs. (22.54) and (22.55) are not the tree-level functions — they carry genuine one-loop corrections, the finite logarithms whose μ-dependence is the seed of the running coupling. Renormalization removes the infinity, not the physics; the loop leaves a finite, measurable trace.

With δ_Z, δ_m, and δ_λ now in hand in both schemes, and the renormalized $\Gamma^{(2)}$ and $\Gamma^{(4)}$ exhibited as finite functions, the data are ready to be repackaged. The next section reads the counterterms as a multiplicative rescaling — δ_Z, δ_m, δ_λ assembled into the field-strength, mass, and coupling renormalization constants $Z_\phi = 1 + \delta_Z$, Z_m, and $Z_\lambda = 1 + \delta_\lambda/\lambda$ — and proves the exact relation between bare and renormalized Green's functions that the renormalization group of Chapter 24 will exploit.

22.3 Multiplicative Renormalization and Renormalized Green's Functions

The previous section delivered, for ϕ^4 theory, the three numbers that cure the one-loop infinities: the field-strength, mass, and coupling counterterms $\delta_Z, \delta_m, \delta_\lambda$ of Eq. (22.9), fixed by renormalization conditions and computed explicitly in both the on-shell and $\overline{\text{MS}}$ schemes (Theorems 22.6 and 22.7). Those counterterms were introduced as *additive* repairs: extra vertices, with the same operator content as the original Lagrangian, tuned order by order to subtract the poles of the loop integrals. That presentation is operationally complete but conceptually disposable. It treats the counterterms as a bookkeeping device bolted onto a fixed theory, and it leaves the central fact of renormalization—that a *finite* set of repairs suffices to *all* orders, for *every* correlator—looking like a fortunate accident to be checked diagram by diagram.

The purpose of this section is to expose the structure underneath. The additive counterterms are not independent repairs; they are the Taylor coefficients of a single *multiplicative* rescaling of the field and the parameters of the theory. There is one rescaling factor for the field, one for the mass, one for the coupling, and the entire counterterm Lagrangian is what these three rescalings produce when the bare Lagrangian is rewritten in terms of renormalized quantities. Once this is recognized, the relation between the divergent "bare" correlators and the finite "renormalized" ones becomes an exact identity—each n-point function carries a fixed power of the field rescaling—rather than a perturbative coincidence. That identity, proved below in full from the path-integral definition of the generating functional, is the structural backbone of everything that follows: it is what makes "renormalizability" a precise and checkable statement, and it is the seed from which the renormalization-group equations of Chapter 24 grow.

22.3.1 Bare quantities, renormalized quantities, and the rescaling

We must first be scrupulous about what is bare and what is renormalized, because the entire argument is a careful change of variables between the two. The Lagrangian we have been quantizing since Chapter 6 is, read honestly,

written in terms of *bare* fields and *bare* parameters,

$$\mathcal{L}_0 \;=\; \tfrac{1}{2}(\partial_\mu\phi_0)(\partial^\mu\phi_0) - \tfrac{1}{2}m_0^2\,\phi_0^2 - \frac{\lambda_0}{4!}\,\phi_0^4, \tag{22.56}$$

where the subscript 0 marks every object as the unrenormalized, cutoff-dependent quantity that appears in the functional integral. The bare field ϕ_0 is the integration variable of the path integral; the bare mass m_0 and bare coupling λ_0 are the parameters with respect to which the action is defined. None of these is directly measurable. As the chapter introduction insisted and Chapter 21 made quantitative, the bare parameters diverge as the regulator is removed; they are not the mass and coupling read off an experiment. The measured, finite quantities are the *renormalized* mass m and coupling λ, and the field whose propagator has a controlled normalization is the *renormalized* field ϕ. The work of renormalization is to express the bare Lagrangian in terms of these finite objects, and the dictionary that does so is multiplicative.

Definition 22.9 (Renormalization constants)**.** The *renormalization constants* (or *Z-factors*) Z_ϕ, Z_m, Z_λ are the dimensionless quantities relating the bare field and parameters of Eq. (22.56) to the renormalized field ϕ, mass m, and coupling λ through

$$\phi_0 \;=\; Z_\phi^{1/2}\,\phi, \qquad m_0^2 \;=\; \frac{Z_m}{Z_\phi}\,m^2, \qquad \lambda_0 \;=\; \frac{Z_\lambda}{Z_\phi^2}\,\lambda\,\mu^{2\varepsilon}, \tag{22.57}$$

in dimensional regularization with $d = 4 - 2\varepsilon$ and 't Hooft mass μ. Each $Z_i = 1 + O(\lambda)$ is a series in the renormalized coupling whose coefficients carry the poles in ε (in $\overline{\text{MS}}$) or the finite subtractions (in a physical scheme) determined by the renormalization conditions of Theorem 22.6.

Three points about Eq. (22.57) deserve immediate comment, because each encodes a choice that the rest of the theory depends on.

The factor $Z_\phi^{1/2}$ on the field is conventionally written with the square root so that the *quadratic* form $\phi_0^2 \sim Z_\phi\,\phi^2$ carries the integer power; Z_ϕ is the *field-strength* (or *wave-function*) renormalization, and it is the residue of the renormalized propagator at its pole that fixes it. It is the one Z that has no analogue in the classical Lagrangian: it renormalizes the field itself, an operation with no counterpart at tree level, and it is therefore the source of the genuinely quantum-mechanical scaling that the anomalous dimension of Chapter 24 will measure.

The combination Z_m/Z_ϕ in the mass relation, rather than a single constant, is dictated by how the kinetic and mass terms transform together; we will see in a moment that it is exactly the combination that makes the additive mass counterterm δ_m come out right. We write the mass-squared renormalization this way—following the convention already used by the renormalization-group

development in Eq. (24.49), where the bare mass is packaged as $m_0^2 = Z_{m^2} m^2$ with $Z_{m^2} \equiv Z_m / Z_\phi$—so that Z_m multiplies the renormalized ϕ^2 operator directly when the Lagrangian is rewritten.

The factor $\mu^{2\varepsilon}$ in the coupling relation is forced by dimensional analysis and is the quiet birthplace of all scale dependence. In $d = 4 - 2\varepsilon$ the canonical dimension of a scalar field is

$$[\phi_0] = [\phi] = \frac{d-2}{2} = 1 - \varepsilon, \tag{22.58}$$

read off by demanding that the kinetic term $\frac{1}{2}(\partial\phi_0)^2$ have the dimension d of a Lagrangian density. Imposing the same on the interaction $\lambda_0 \phi_0^4/4!$ gives

$$[\lambda_0] \;=\; d - 4[\phi_0] \;=\; (4 - 2\varepsilon) - 4(1 - \varepsilon) \;=\; 2\varepsilon, \tag{22.59}$$

so the bare coupling is *not* dimensionless away from four dimensions: it carries mass dimension 2ε. (This is the corollary of dimensional regularization already flagged in Section 21.3.7.) To trade λ_0 for a *dimensionless* renormalized coupling λ—the object whose running we will track—we must extract the dimensionful factor explicitly, and the only scale available is the arbitrary mass μ introduced with the regulator. Pulling out $\mu^{2\varepsilon}$ leaves $Z_\lambda Z_\phi^{-2} \lambda$ dimensionless, as Eq. (22.57) requires. The arbitrariness of μ is not a defect: it is the freedom whose exploitation, in Chapter 24, yields the Callan–Symanzik equation. We return to it at the end of this section.

22.3.2 The rescaling reproduces the counterterm Lagrangian

We now show that the multiplicative dictionary Eq. (22.57) is *identical* to the additive counterterm prescription of the previous section—that the two are the same content in two notations. Substitute Eq. (22.57) into the bare Lagrangian Eq. (22.56) and collect:

$$\begin{aligned} \mathcal{L}_0 &= \tfrac{1}{2}\, Z_\phi\, (\partial_\mu \phi)(\partial^\mu \phi) - \tfrac{1}{2}\, m_0^2 Z_\phi\, \phi^2 - \frac{\lambda_0 Z_\phi^2}{4!}\, \phi^4 \\ &= \tfrac{1}{2}\, Z_\phi\, (\partial_\mu \phi)(\partial^\mu \phi) - \tfrac{1}{2}\, Z_m\, m^2\, \phi^2 - \frac{Z_\lambda\, \lambda\, \mu^{2\varepsilon}}{4!}\, \phi^4, \end{aligned} \tag{22.60}$$

where in the second line $m_0^2 Z_\phi = (Z_m/Z_\phi) m^2\, Z_\phi = Z_m m^2$ and $\lambda_0 Z_\phi^2 = (Z_\lambda / Z_\phi^2)\lambda \mu^{2\varepsilon} Z_\phi^2 = Z_\lambda \lambda \mu^{2\varepsilon}$. The bare Lagrangian, expressed entirely in renormalized quantities, is the renormalized Lagrangian with each term multiplied by its Z. Splitting every $Z_i = 1 + (Z_i - 1)$ separates a renormalized piece from a remainder:

$$\mathcal{L}_0 \;=\; \underbrace{\tfrac{1}{2}(\partial_\mu\phi)^2 - \tfrac{1}{2} m^2 \phi^2 - \frac{\lambda \mu^{2\varepsilon}}{4!}\phi^4}_{\mathcal{L}_R} \;+\; \underbrace{\tfrac{1}{2}\, \delta_Z (\partial_\mu \phi)^2 - \tfrac{1}{2}\, \delta_m\, \phi^2 - \frac{\delta_\lambda\, \mu^{2\varepsilon}}{4!}\, \phi^4}_{\mathcal{L}_{\text{ct}}}, \tag{22.61}$$

with the counterterm coefficients

$$\boxed{\delta_Z = Z_\phi - 1, \qquad \delta_m = (Z_m - 1)\, m^2, \qquad \delta_\lambda = (Z_\lambda - 1)\, \lambda} \tag{22.62}$$

This is exactly the counterterm Lagrangian of Eq. (22.12), with the very same coefficients used by the explicit one-loop computation and quoted again in the renormalization-group chapter as Eq. (24.49). (Indeed Eq. (24.49) writes $\delta_m = (Z_{m^2} Z_\phi - 1)m^2$; since $Z_{m^2} Z_\phi = (Z_m/Z_\phi)Z_\phi = Z_m$, this is our $\delta_m = (Z_m - 1)m^2$, and the two presentations agree term for term.) The additive counterterms are nothing but the deviations of the multiplicative Z's from unity. Conversely, given the counterterms computed in the previous section, the Z's are read off by inverting Eq. (22.62):

$$Z_\phi = 1 + \delta_Z, \qquad Z_m = 1 + \frac{\delta_m}{m^2}, \qquad Z_\lambda = 1 + \frac{\delta_\lambda}{\lambda}. \tag{22.63}$$

Nothing has been added or removed; we have only recognized the counterterm coefficients as the entries of a multiplicative redefinition. The gain is conceptual and, as we now show, decisive: because the redefinition is multiplicative and acts on the field by a single overall factor, its effect on every Green's function is a single overall power.

Notation Watch: which constant multiplies what

Three conventions for the mass renormalization circulate in the literature, and they differ by factors of Z_ϕ. We fix ours once: Z_m is the constant that multiplies the *renormalized* $m^2\phi^2$ operator in the rewritten Lagrangian Eq. (22.60), so that $\delta_m = (Z_m - 1)m^2$. The bare mass itself is then $m_0^2 = (Z_m/Z_\phi)m^2$, and the product $Z_{m^2} \equiv Z_m/Z_\phi$ is the constant relating bare to renormalized m^2 directly, which is the object the running-mass analysis of Chapter 24 uses. At one loop in ϕ^4, $Z_\phi = 1 + O(\lambda^2)$, so Z_m and Z_{m^2} coincide there; the distinction first matters at two loops.

22.3.3 The bare and renormalized generating functionals

We can now prove the central relation. The cleanest route is through the generating functional, because the entire content of the rescaling is a change of integration variable, and the path integral makes that change transparent. Recall from Theorem 10.17 that the generating functional of the scalar field is the functional integral with a source coupled linearly to the field. Written for the *bare* theory it is

$$Z_0[J_0] \;=\; \int \mathcal{D}\phi_0 \, \exp\Big[\, i \int d^d x \, \big(\mathcal{L}_0[\phi_0] + J_0(x)\, \phi_0(x)\big)\Big], \tag{22.64}$$

the direct transcription of Eq. (10.86) to d dimensions with the bare Lagrangian Eq. (22.56) and a bare source J_0. Its functional derivatives generate the bare time-ordered correlators

$$G_0^{(n)}(x_1,\dots,x_n) \;=\; \langle 0|\, T\,\phi_0(x_1)\cdots\phi_0(x_n)\,|0\rangle \;=\; \frac{1}{Z_0[0]}\prod_{i=1}^{n}\frac{1}{i}\frac{\delta}{\delta J_0(x_i)}\,Z_0[J_0]\Big|_{J_0=0}, \tag{22.65}$$

the objects that the Feynman rules of Chapter 15 compute and that Chapter 21 found to diverge. The renormalized generating functional is the *same* integral with the integration variable, the Lagrangian, and the source all written in renormalized terms,

$$Z_R[J] \;=\; \int \mathcal{D}\phi\, \exp\Big[\, i\int d^dx\,\big(\mathcal{L}_R[\phi]+\mathcal{L}_{\rm ct}[\phi]+J(x)\,\phi(x)\big)\Big], \tag{22.66}$$

whose derivatives produce the renormalized correlators $G_R^{(n)}$ with the same normalization Eq. (22.65) but with renormalized sources. The two functionals are not independent: they are the same integral, related by the change of variable Eq. (22.57). Establishing that relation is the whole proof.

Theorem 22.10 (Multiplicative renormalization of Green's functions)**.** *Let $\phi_0 = Z_\phi^{1/2}\phi$ be the field rescaling of Theorem 22.9, and let $G_0^{(n)}$, $G_R^{(n)}$ denote the bare and renormalized (full or connected) n-point Green's functions, defined respectively from Eqs. (22.64) and (22.66) through Eq. (22.65). Then the bare and renormalized correlators are related by a single overall power of the field-strength renormalization,*

$$\boxed{\,G_0^{(n)}(x_1,\dots,x_n;\,\lambda_0,m_0) \;=\; Z_\phi^{\,n/2}\,G_R^{(n)}(x_1,\dots,x_n;\,\lambda,m,\mu)\,} \tag{22.67}$$

to all orders in perturbation theory, and identically in momentum space with the common overall momentum-conserving delta function stripped. The corresponding 1*PI vertices (Theorem 15.38) carry the* inverse *power,*

$$\boxed{\,\widetilde{\Gamma}_0^{(n)}(p_1,\dots,p_n;\,\lambda_0,m_0) \;=\; Z_\phi^{-n/2}\,\widetilde{\Gamma}_R^{(n)}(p_1,\dots,p_n;\,\lambda,m,\mu)\,} \tag{22.68}$$

Proof. The proof has two movements: the full correlators Eq. (22.67) from the path integral, and the 1PI vertices Eq. (22.68) from the Legendre transform.

Movement 1: full and connected correlators. Perform the change of integration variable $\phi_0 = Z_\phi^{1/2}\phi$ in the bare functional Eq. (22.64). Three things transform. First, the action: by the algebra of Section 22.3.2, the rescaled bare Lagrangian is exactly the renormalized-plus-counterterm Lagrangian, $\mathcal{L}_0[\,Z_\phi^{1/2}\phi\,] = \mathcal{L}_R[\phi] + \mathcal{L}_{\rm ct}[\phi]$, equation Eq. (22.61). Second, the measure: a linear rescaling of the integration field by the constant $Z_\phi^{1/2}$ produces a field-independent Jacobian,

$$\mathcal{D}\phi_0 \;=\; \mathcal{J}\,\mathcal{D}\phi, \qquad \mathcal{J} = \det\!\big(Z_\phi^{1/2}\,\mathbf{1}\big), \tag{22.69}$$

a (formally infinite but field-independent) constant that does not depend on the source. Third, the source term: writing $J_0\phi_0 = J_0 Z_\phi^{1/2}\phi$ and *defining* the renormalized source so that this is the renormalized coupling $J\phi$,

$$J \;\equiv\; Z_\phi^{1/2}\, J_0 \quad\Longleftrightarrow\quad J_0 \;=\; Z_\phi^{-1/2}\, J, \tag{22.70}$$

the source term is invariant, $\int J_0\phi_0 = \int J\phi$. Assembling the three,

$$Z_0[J_0] \;=\; \mathcal{J}\int \mathcal{D}\phi\, \exp\Big[\, i\int d^dx\, (\mathcal{L}_R[\phi]+\mathcal{L}_{\rm ct}[\phi]+J\phi)\Big] \;=\; \mathcal{J}\, Z_R[J], \qquad J = Z_\phi^{1/2} J_0. \tag{22.71}$$

The bare and renormalized generating functionals are the same functional, up to the field-independent constant $\mathcal{J}$ and the relabeling of the source.

Now differentiate. By the chain rule applied to Eq. (22.70), each bare-source derivative is a renormalized-source derivative times $Z_\phi^{1/2}$,

$$\frac{\delta}{\delta J_0(x)} = \int d^dy\, \frac{\delta J(y)}{\delta J_0(x)}\,\frac{\delta}{\delta J(y)} = Z_\phi^{1/2}\,\frac{\delta}{\delta J(x)}. \tag{22.72}$$

Insert Eqs. (22.71) and (22.72) into the definition Eq. (22.65). The constant $\mathcal{J}$ appears once in $Z_0[J_0] = \mathcal{J} Z_R[J]$ and once in the normalization $Z_0[0] = \mathcal{J} Z_R[0]$, and cancels exactly. The n derivatives produce n factors of $Z_\phi^{1/2}$, which pull out as the single power $Z_\phi^{n/2}$:

$$G_0^{(n)} = \frac{1}{\mathcal{J} Z_R[0]}\prod_{i=1}^{n}\frac{1}{i}\, Z_\phi^{1/2}\frac{\delta}{\delta J(x_i)}\,\mathcal{J} Z_R[J]\Big|_0 = Z_\phi^{n/2}\,\frac{1}{Z_R[0]}\prod_{i=1}^{n}\frac{1}{i}\frac{\delta}{\delta J(x_i)}\, Z_R[J]\Big|_0 = Z_\phi^{n/2}\, G_R^{(n)}, \tag{22.73}$$

which is Eq. (22.67) for the full correlators. The same conclusion holds verbatim for the *connected* correlators. By Theorem 10.36 the connected generating functional is $W[J] = -i\ln Z[J]$, so $W_0[J_0] = -i\ln(\mathcal{J} Z_R[J]) = W_R[J] - i\ln\mathcal{J}$; the additive constant $-i\ln\mathcal{J}$ is annihilated by every derivative with $n \geq 1$, and the chain rule Eq. (22.72) again delivers one $Z_\phi^{1/2}$ per leg, giving $G_{c,0}^{(n)} = Z_\phi^{n/2} G_{c,R}^{(n)}$. Translation invariance lets us strip the common momentum-conserving delta on both sides, so the relation holds identically in momentum space.

Movement 2: the 1*PI vertices.* The proper vertices are generated by the effective action $\Gamma[\phi_{\rm cl}]$, the Legendre transform of $W[J]$ established in Theorem 10.38 and shown in Theorem 15.39 to generate the 1PI amputated correlators. We trace the rescaling through the transform. The classical field is the source-derivative of W (Eq. (10.203)); for the bare theory, using Eq. (22.72) and $W_0[J_0] = W_R[J] + \text{const}$,

$$\phi_{\rm cl,0}(x) = \frac{\delta W_0}{\delta J_0(x)} = Z_\phi^{1/2}\,\frac{\delta W_R}{\delta J(x)} = Z_\phi^{1/2}\,\phi_{\rm cl,R}(x). \tag{22.74}$$

The classical field rescales by exactly the same factor as the field itself, as it must, being a field expectation value. Now apply the Legendre transform Eq. (10.210) to the bare functional and substitute Eqs. (22.70) and (22.74):

$$\Gamma_0[\phi_{\mathrm{cl},0}] = W_0[J_0] - \int d^dx\, J_0\, \phi_{\mathrm{cl},0} = W_R[J] + \text{const} - \int d^dx\, (Z_\phi^{-1/2} J)\, (Z_\phi^{1/2} \phi_{\mathrm{cl},R}). \tag{22.75}$$

The two rescaling factors in the source term cancel, $Z_\phi^{-1/2} Z_\phi^{1/2} = 1$, leaving precisely the renormalized Legendre transform:

$$\Gamma_0[\phi_{\mathrm{cl},0}] \;=\; W_R[J] - \int d^dx\, J\, \phi_{\mathrm{cl},R} + \text{const} \;=\; \Gamma_R[\phi_{\mathrm{cl},R}] + \text{const}. \tag{22.76}$$

The effective action is therefore *form-invariant* under the rescaling: as a functional it is the same object, up to the irrelevant additive constant, but evaluated on the rescaled argument $\phi_{\mathrm{cl},0} = Z_\phi^{1/2} \phi_{\mathrm{cl},R}$. The vertices are its derivatives, and here the powers *invert*: each derivative with respect to $\phi_{\mathrm{cl},0}$ carries, by the chain rule applied to Eq. (22.74), a factor $Z_\phi^{-1/2}$,

$$\frac{\delta}{\delta \phi_{\mathrm{cl},0}(x)} = Z_\phi^{-1/2}\, \frac{\delta}{\delta \phi_{\mathrm{cl},R}(x)}, \tag{22.77}$$

so differentiating Eq. (22.76) n times (the additive constant dropping for $n \geq 1$) gives

$$\Gamma_0^{(n)}(x_1, \ldots, x_n) = \frac{\delta^n \Gamma_0}{\delta\phi_{\mathrm{cl},0}(x_1) \cdots \delta\phi_{\mathrm{cl},0}(x_n)} = Z_\phi^{-n/2}\, \frac{\delta^n \Gamma_R}{\delta\phi_{\mathrm{cl},R}(x_1) \cdots \delta\phi_{\mathrm{cl},R}(x_n)} = Z_\phi^{-n/2}\, \Gamma_R^{(n)}, \tag{22.78}$$

which is Eq. (22.68) in position space, hence in momentum space after stripping the delta. This completes the proof. □

The inverse power on the proper vertices is not an algebraic curiosity but a structural necessity, and it pays to see why physically. The connected n-point function $G_c^{(n)}$ is, by the hierarchy of Eq. (15.135), a tree of proper vertices joined by full propagators, with one full propagator on each of its n external legs. The full propagator is itself a connected two-point function, $G_c^{(2)} = Z_\phi\, G_{c,R}^{(2)}$, so each external leg supplies one factor of Z_ϕ; the 1PI vertex is what remains after these n legs are *amputated* (Theorem 15.37), and amputating n legs that each carry Z_ϕ removes Z_ϕ^n from a $G^{(n)}$ that carried only $Z_\phi^{n/2}$. The net power on the amputated, irreducible core is therefore $Z_\phi^{n/2}/Z_\phi^n = Z_\phi^{-n/2}$, exactly Eq. (22.68). The two relations Eqs. (22.67) and (22.68) are the same statement read on the two ends of the amputation. As a consistency check at $n = 2$: the proper two-point function is the inverse full propagator (Eq. (15.129)), $\widetilde{\Gamma}^{(2)} = i\,(\widetilde{G}_c^{(2)})^{-1}$, so $\widetilde{\Gamma}_0^{(2)} = i(Z_\phi \widetilde{G}_{c,R}^{(2)})^{-1} = Z_\phi^{-1} \widetilde{\Gamma}_R^{(2)}$, in agreement with Eq. (22.68).

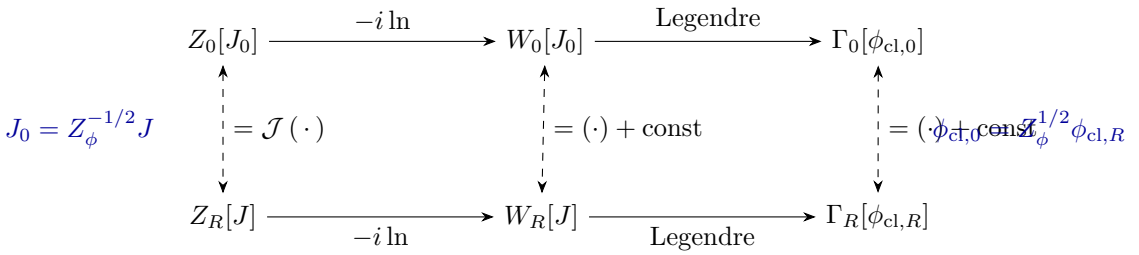


Figure 22.4: The generating-functional hierarchy of Eq. (15.135) for the bare (top) and renormalized (bottom) theories. The field rescaling $\phi_0 = Z_\phi^{1/2}\phi$ propagates through the whole hierarchy: it sends the source $J_0 \to Z_\phi^{-1/2} J$ and the classical field $\phi_{\mathrm{cl},0} \to Z_\phi^{1/2}\phi_{\mathrm{cl},R}$, while the three functionals coincide up to a field-independent constant. Differentiating with respect to J_0 supplies one $Z_\phi^{1/2}$ per external leg, giving $G_0^{(n)} = Z_\phi^{n/2} G_R^{(n)}$; differentiating with respect to $\phi_{\mathrm{cl},0}$ supplies one $Z_\phi^{-1/2}$ per leg, giving $\Gamma_0^{(n)} = Z_\phi^{-n/2}\Gamma_R^{(n)}$.

22.3.4 What renormalizability says, exactly

Theorem 22.10 converts the loose phrase "the theory is renormalizable" into a sharp, checkable claim. The bare vertices $\widetilde{\Gamma}_0^{(n)}(p;\lambda_0,m_0)$ are what perturbation theory actually computes, and they diverge as $\varepsilon \to 0$: the loop integrals of Chapter 21 carry poles in ε. The renormalized vertices are, by Eq. (22.68),

$$\widetilde{\Gamma}_R^{(n)}(p;\lambda,m,\mu) \;=\; Z_\phi^{\,n/2}\,\widetilde{\Gamma}_0^{(n)}(p;\lambda_0,m_0), \tag{22.79}$$

with m_0, λ_0 replaced by their expressions Eq. (22.57) in m, λ, μ. Renormalizability is the statement that this substitution produces a *finite* answer, order by order in λ, for *every* n, using a *fixed, finite* set of renormalization constants.

Proposition 22.11 (Renormalizability of ϕ^4 as three constants)**.** *The theory Eq. (22.56) is renormalizable if and only if there exist three formal power series $Z_\phi(\lambda,\varepsilon)$, $Z_m(\lambda,\varepsilon)$, $Z_\lambda(\lambda,\varepsilon)$—each independent of the external momenta and of n—such that, when m_0 and λ_0 are eliminated through Eq. (22.57), every renormalized vertex $\widetilde{\Gamma}_R^{(n)}(p;\lambda,m,\mu)$ of Eq. (22.79) is finite as $\varepsilon \to 0$, to all orders in λ.*

This is a strong claim, and its strength is precisely the locality of the divergences. There are infinitely many proper vertices $\widetilde{\Gamma}^{(n)}$, computed by infinitely many diagrams. A priori, removing their divergences could require an independent subtraction for each of the infinitely many (n, order) pairs—and in a non-renormalizable theory it does (Theorem 21.13). The claim of Theorem 22.11 is that in ϕ^4 a mere *three* numbers, the same three at every n and every order,

suffice. The reason this is even conceivable was supplied by the power counting of Chapter 21: by Theorem 21.13, only finitely many vertices are superficially divergent, and for ϕ^4 they are exactly two—the two-point function $\widetilde{\Gamma}^{(2)}$ (superficial degree $D = 2$) and the four-point function $\widetilde{\Gamma}^{(4)}$ ($D = 0$); all $\widetilde{\Gamma}^{(n)}$ with $n \geq 6$ (and the odd ones, which vanish by the $\mathbb{Z}_2$ symmetry of the thread) have $D < 0$. A degree-2 divergence in $\widetilde{\Gamma}^{(2)}$ is, by Taylor expansion in the external momentum, a polynomial $a + b\,p^2$ with two divergent coefficients; these are subtracted by δ_m (the constant) and δ_Z (the p^2 piece), i.e. by Z_m and Z_ϕ. A degree-0 divergence in $\widetilde{\Gamma}^{(4)}$ is a single divergent constant, subtracted by δ_λ, i.e. by Z_λ. The number of independent divergent coefficients across the superficially divergent vertices is exactly three, and these three are the three Z's.

That the *same* three constants then render finite the superficially *convergent* vertices—$\widetilde{\Gamma}^{(6)}$, $\widetilde{\Gamma}^{(8)}$, and the higher-loop corrections to $\widetilde{\Gamma}^{(2)}, \widetilde{\Gamma}^{(4)}$ themselves—is the genuine theorem, and it is the heart of renormalization. A superficially convergent diagram is not automatically finite: it can contain a divergent *subdiagram*, a two- or four-point insertion buried inside it, whose divergence the overall power count of Chapter 21 cannot see. Those subdivergences must be cancelled by inserting the lower counterterms inside the larger diagram, and it is a nontrivial fact—one that the naive one-loop picture of the previous section does not even confront—that the counterterms generated by $\widetilde{\Gamma}^{(2)}$ and $\widetilde{\Gamma}^{(4)}$, dressed appropriately, suffice to cancel *all* subdivergences in *all* diagrams, leaving a strictly local overall divergence for the two superficially divergent vertices and nothing for the rest. Establishing this is the work of the next two sections; here we have established only the framework that makes the claim precise: *if* three constants absorb everything, the bare-to-renormalized relations Eqs. (22.67) and (22.68) hold, and the renormalized correlators are finite. The multiplicative structure is the scaffolding; the recursion that fills it in is what follows.

22.3.5 One-loop consistency check

Before pushing to all orders we verify the machinery against the explicit one-loop results of the previous section, both as a check on the algebra and to display how the multiplicative Z's repackage the additive counterterms in concrete numbers. We work in $\overline{\text{MS}}$ (Theorem 22.7), where each Z_i is unity plus pure poles in $\bar{\varepsilon}$, the universal combination $1/\bar{\varepsilon} = 1/\varepsilon - \gamma_{\text{E}} + \ln 4\pi$ of Eq. (A.204).

Worked Example 22.12 (The one-loop relations $\widetilde{\Gamma}^{(2)}_0 = Z_\phi^{-1}\widetilde{\Gamma}^{(2)}_R$ and $\widetilde{\Gamma}^{(4)}_0 = Z_\phi^{-2}\widetilde{\Gamma}^{(4)}_R$)**.**
The two-point function. The one-loop 1PI two-point function in ϕ^4 comes from the single tadpole diagram (Eq. (21.16)), the loop closing on one quartic vertex. Its value, with the symmetry factor $\frac{1}{2}$, is the momentum-*independent* self-energy

$$-i\Sigma_1 \;=\; \frac{-i\lambda\mu^{2\varepsilon}}{2}\int\frac{d^dk}{(2\pi)^d}\,\frac{i}{k^2-m^2+i\varepsilon}, \tag{22.80}$$

a constant in the external momentum p because no external momentum flows through the loop. This single fact does most of the work. Writing the renormalized two-point function from Eq. (22.61), with the counterterm vertex contributing $i(\delta_Z p^2 - \delta_m)$,

$$\widetilde{\Gamma}^{(2)}_R(p) \;=\; p^2 - m^2 - \Sigma_1(p^2) + \delta_Z\, p^2 - \delta_m. \tag{22.81}$$

Because Σ_1 is independent of p^2, its divergence is a pure constant: there is no $1/\bar{\varepsilon}$ pole proportional to p^2 to cancel, so finiteness of Eq. (22.81) requires

$$\delta_Z = O(\lambda^2), \qquad \text{i.e.} \qquad Z_\phi = 1 + O(\lambda^2), \tag{22.82}$$

the celebrated vanishing of the one-loop field-strength renormalization in ϕ^4—the statement that the field anomalous dimension of Chapter 24 first appears at two loops, $\gamma_\phi = O(\lambda^2)$. The mass counterterm absorbs the constant pole: evaluating Eq. (22.80) with the tadpole master integral of Eq. (21.18) gives $\Sigma_1 = -\dfrac{\lambda m^2}{32\pi^2}\Big(\dfrac{1}{\bar{\varepsilon}} + \text{finite}\Big)$, so minimal subtraction sets

$$\delta_m = -\Big[\Sigma_1\Big]_{\text{pole}} = \frac{\lambda m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \qquad Z_m = 1 + \frac{\delta_m}{m^2} = 1 + \frac{\lambda}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(\lambda^2). \tag{22.83}$$

Now check the relation Eq. (22.68) at $n = 2$. The *bare* two-point function has no counterterm; it is $\widetilde{\Gamma}^{(2)}_0(p) = p^2 - m_0^2 - \Sigma^{(0)}_1(p^2)$, where $\Sigma^{(0)}_1$ is the same tadpole computed with bare parameters, equal to Σ_1 to this order. Using $m_0^2 = (Z_m/Z_\phi)m^2 = Z_m m^2 = m^2 + \delta_m + O(\lambda^2)$ from Eqs. (22.82) and (22.83),

$$\widetilde{\Gamma}^{(2)}_0(p) = p^2 - (m^2 + \delta_m) - \Sigma_1(p^2) = p^2 - m^2 - \Sigma_1 - \delta_m. \tag{22.84}$$

Compare with $Z_\phi^{-1}\widetilde{\Gamma}^{(2)}_R = (1 + O(\lambda^2))(p^2 - m^2 - \Sigma_1 - \delta_m)$ from Eqs. (22.81) and (22.82). The two agree term for term at one loop: Eq. (22.68) holds, and indeed both sides are finite once δ_m cancels the constant pole of Σ_1.

The four-point function. The amputated 1PI four-point function is, including tree, the three one-loop bubbles, and the counterterm vertex,

$$\widetilde{\Gamma}^{(4)}_R(p_i) \;=\; -\lambda\mu^{2\varepsilon} + \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u - \delta_\lambda\,\mu^{2\varepsilon}, \tag{22.85}$$

the three bubbles carrying squared momentum s, t, u through the loop (the diagrams of Fig. 24.2). Each bubble has the same $1/\bar{\varepsilon}$ pole, $[\mathcal{M}_x]_{\text{pole}} = \dfrac{\lambda^2\mu^{2\varepsilon}}{32\pi^2}\dfrac{1}{\bar{\varepsilon}}$, independent of the kinematic variable x (the bubble integral of Theorem D.25 with its symmetry factor $\frac{1}{2}$ supplying the $\frac{1}{2}\cdot\frac{1}{16\pi^2} = \frac{1}{32\pi^2}$), so the three together contribute $\dfrac{3\lambda^2\mu^{2\varepsilon}}{32\pi^2}\dfrac{1}{\bar{\varepsilon}}$, and minimal subtraction fixes

$$\delta_\lambda = \frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \qquad Z_\lambda = 1 + \frac{\delta_\lambda}{\lambda} = 1 + \frac{3\lambda}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(\lambda^2), \tag{22.86}$$

reproducing the one-loop coupling counterterm δ_λ of Eq. (22.48); the factor of three is nothing but the number of channels, the number of ways to split four legs into two pairs. The simple-pole residue $3\lambda/32\pi^2$ of Z_λ is the entire one-loop input to the coupling's running; Chapter 24 feeds it through the pole-to-beta relation—where a further factor of two appears in the *extraction* of the beta function, not in Z_λ itself—to produce the carried number $\beta_\lambda = 3\lambda^2/16\pi^2$. The relation Eq. (22.68) at $n = 4$ reads $\widetilde{\Gamma}_0^{(4)} = Z_\phi^{-2}\widetilde{\Gamma}_R^{(4)}$. The bare four-point function is the tree plus loops with no counterterm, $\widetilde{\Gamma}_0^{(4)} = -\lambda_0\mu^{2\varepsilon}\big|_{\text{written bare}} + \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u$, and with $\lambda_0 = (Z_\lambda/Z_\phi^2)\lambda\mu^{2\varepsilon} = Z_\lambda\lambda\mu^{2\varepsilon} + O(\lambda^3)$ from Eqs. (22.57) and (22.82) this is

$$\widetilde{\Gamma}_0^{(4)} = -(\lambda + \delta_\lambda)\mu^{2\varepsilon} + \mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u = \widetilde{\Gamma}_R^{(4)}, \tag{22.87}$$

which equals $Z_\phi^{-2}\widetilde{\Gamma}_R^{(4)}$ because $Z_\phi^{-2} = 1 + O(\lambda^2)$ at one loop. Again the relation holds and both sides are finite.

What the check shows. The two computations confirm Theorem 22.10 at one loop, but their real lesson is the dictionary in action: the additive subtractions δ_m, δ_λ computed diagram-by-diagram in the previous section are exactly the deviations $Z_m - 1$, $Z_\lambda - 1$ of two multiplicative constants, while $Z_\phi = 1 + O(\lambda^2)$ records that the field needs no rescaling yet. Three constants, two of them nontrivial at this order, carry the entire one-loop divergence structure—the concrete instance of Theorem 22.11.

22.3.6 Independence of bare quantities from μ: the seed of the renormalization group

There is a final structural fact, almost trivial to state and profound in consequence, that the bare-to-renormalized relations bring into focus. The bare Lagrangian Eq. (22.56), the bare field ϕ_0, and the bare parameters m_0, λ_0 were defined with no reference whatsoever to the scale μ. The mass μ entered the story only in Theorem 22.9, as the device for splitting the (cutoff-dependent but μ-independent) bare coupling into a dimensionless renormalized coupling times a power of μ. The split is our choice; the bare theory does not know about it. Consequently the bare Green's functions are strictly independent of μ,

$$\mu\,\frac{d}{d\mu}\,\widetilde{\Gamma}_0^{(n)}(p;\lambda_0,m_0) \;=\; 0, \tag{22.88}$$

the derivative taken at fixed bare parameters and fixed regulator. This is not a dynamical statement; it is the definition of "bare."

The content appears when Eq. (22.88) is combined with the multiplicative relation Eq. (22.68). Writing $\widetilde{\Gamma}_0^{(n)} = Z_\phi^{-n/2}\widetilde{\Gamma}_R^{(n)}$ and demanding that the left side be μ-independent gives

$$\mu\,\frac{d}{d\mu}\Big[\,Z_\phi^{-n/2}(\lambda,\varepsilon)\;\widetilde{\Gamma}_R^{(n)}(p;\lambda,m,\mu)\,\Big] \;=\; 0. \tag{22.89}$$

The renormalized vertex $\widetilde{\Gamma}_R^{(n)}$ depends on μ both explicitly and through the running of the renormalized parameters $\lambda(\mu), m(\mu)$ that hold the bare parameters fixed; Eq. (22.89) is the equation that ties these dependencies together. Carrying out the differentiation—introducing the beta function $\beta_\lambda = \mu\, d\lambda/d\mu$, the mass anomalous dimension, and the field anomalous dimension $\gamma_\phi = \frac{1}{2}\mu\, d\ln Z_\phi/d\mu$ as the rates at which the renormalized quantities must run to keep the bare ones still—turns Eq. (22.89) into the Callan–Symanzik equation. That development, and the extraction of $\beta_\lambda = 3\lambda^2/16\pi^2$ from the one-loop Z_λ of Eq. (22.86), is the business of Chapter 24; what Eqs. (22.88) and (22.89) establish here is the seed—that the arbitrariness of μ in the multiplicative split is a symmetry of the bare theory, and that the renormalized correlators must conspire, through the running of their parameters and a power of Z_ϕ, to respect it.

With the multiplicative structure in hand, the one outstanding obligation is the theorem that Theorem 22.11 asserts but the one-loop check could not test: that three constants absorb *all* divergences of *all* vertices to all orders. The obstacle is exactly the phenomenon the naive one-loop picture never met—divergent subdiagrams nested and overlapping inside superficially convergent integrals, whose subtraction is not a single local operation. We turn next to anatomizing those subdivergences, which is the necessary prelude to the recursive subtraction that finally proves renormalizability to all orders.

22.4 Subdivergences, Nested and Overlapping Divergences

The previous section recast the entire counterterm program as a multiplicative rescaling: the bare field and parameters are related to the renormalized ones by three constants $Z_\phi, Z_{m^2}, Z_\lambda$, and the bare Green's functions differ from the renormalized ones by powers of Z_ϕ alone (Eq. (22.67)). That statement was made to all orders. Its promise is sweeping — finitely many constants, one per term already in the Lagrangian, render *every* amplitude finite at *every* loop order — and a promise that strong must be tested before it can be trusted. At one loop the test is trivial, because a one-loop diagram has no proper subintegral: the single loop momentum is the only thing that can run to infinity, the superficial degree of divergence of Theorem 21.10 is then the whole story, and the counterterms of Eq. (22.48) dispatch the divergence cleanly. The first genuine test is two loops, and this section carries it out. We will find that the all-orders claim survives, but only just, and only because of a structural fact about multi-loop diagrams that the one-loop analysis never had to confront. A two-loop diagram contains *subdiagrams* that are themselves divergent, and the divergence those subdiagrams inject is of a qualitatively new and dangerous kind: it is *nonlocal*. Subtracting the overall divergence of the full diagram, the only operation the naive reading of multiplicative renormalization suggests, leaves this nonlocal piece untouched,

and a nonlocal residue cannot be absorbed by any Lagrangian counterterm. The resolution — subtract the subdivergences *first* — is forced on us by an explicit computation, and it is the empirical seed from which the systematic recursion of the next section grows.

22.4.1 Why locality is the whole game

Before computing anything we must be precise about what a counterterm is allowed to be, because the entire difficulty of higher-loop renormalization is the tension between what the divergences want and what counterterms can supply. A counterterm is, by construction (Section 22.1.3), a term added to the Lagrangian density. The Lagrangian is a *local* functional: a polynomial in the fields and finitely many of their derivatives, evaluated at a single spacetime point. In momentum space a local operator of mass dimension Δ built from ϕ and ∂ produces a vertex that is a *polynomial* in the external momenta of degree at most $\Delta - (\text{number of legs}) \cdot [\phi]$. For the frozen ϕ^4 theory the only operators of dimension ≤ 4 compatible with the $\mathbb{Z}_2$ symmetry are ϕ^2, $(\partial\phi)^2$, and ϕ^4 (and the unit operator, the vacuum energy). The vertices they generate are exactly the three counterterm rules of Section 22.1.4:

$$\phi^4 \;\longrightarrow\; -i\,\delta_\lambda \quad \text{(a constant)}, \qquad \tfrac{1}{2}(\partial\phi)^2 - \tfrac{1}{2}\phi^2 \;\longrightarrow\; i\,(\delta_Z\, p^2 - \delta_m). \tag{22.90}$$

A four-point counterterm can therefore be *only* a momentum-independent constant; a two-point counterterm can be *only* an affine function of p^2. This is not a convention we are free to relax: a vertex that depended on the external momenta as, say, $\ln p^2$ would have to come from a Lagrangian term containing $\ln(\Box/\mu^2)$, a nonlocal object with infinitely many derivatives, outside the class of local field theories entirely and incompatible with the locality and causality that the whole formalism rests on. We record the consequence as the criterion that governs everything to follow.

Proposition 22.13 (The locality requirement)**.** *A divergence in a* 1*PI amplitude can be removed by a counterterm of a local Lagrangian if and only if its* $\Lambda \to \infty$ *(equivalently* $\varepsilon \to 0$*) singular part is a* polynomial *in the external momenta of degree at most the superficial degree of divergence* D*. A divergent contribution whose singular part is a non-polynomial function of the external momenta — in particular one proportional to* $\ln(p^2)$ *times a power of the regulator — is* nonlocal *and admits no Lagrangian counterterm; a theory producing such a residue after all admissible subtractions is not renormalizable.*

Proof. By Theorem 21.10 the superficial degree of divergence D of a 1PI amplitude bounds the number of times the integrand must be differentiated in the external momenta to render it absolutely convergent: each derivative $\partial/\partial p^\mu$ lowers the degree of the integrand at large loop momentum by one unit, so after

$D+1$ such derivatives the integral converges and the result is finite. The divergent part of the amplitude is therefore annihilated by $(D+1)$ external-momentum derivatives, which means it is a polynomial in the external momenta of degree at most D. (This is the content of the locality theorem proved in full generality in Section 22.5; here we need only the elementary direction.) A polynomial of degree $\le D$ is precisely the momentum-space image of a local operator of mass dimension $\le d - \sum_{\text{ext}} d_f + D$, i.e. of an operator of dimension $\le d$ in a renormalizable theory, and such an operator is available among the counterterms. Conversely, by Eq. (22.90) a local counterterm produces only a polynomial vertex; if the divergence is non-polynomial in p no linear combination of local counterterms can match it, since the difference would be a nonzero non-polynomial function set equal to a polynomial. The displayed special case, a $\ln(p^2)$ multiplying $\ln\Lambda$ (or $1/\varepsilon$), is non-polynomial and hence in the forbidden class. □

The locality requirement turns the abstract worry "do the counterterms suffice?" into a sharp, checkable question: *after* we have done everything a sensible subtraction scheme permits, is the leftover divergence a polynomial in external momenta? If yes, a Lagrangian counterterm mops it up and we proceed; if a single $\ln p^2$ survives multiplying a divergence, the program has failed at that order. The rest of this section is the discovery, by explicit two-loop computation, that the naive procedure produces exactly the forbidden $\ln(p^2) \times$ (divergence), and that a more careful procedure does not.

22.4.2 The anatomy of multi-loop divergences: a classification

The reason the nonlocal residue appears, and the reason it can be cured, both live in the geometry of how divergent subdiagrams sit inside a larger diagram. We met the notion of a subdivergence in Section 21.5: a region of loop-momentum space in which a *proper subset* of the loop momenta is sent to infinity while the rest is held fixed, exposing a divergence that the overall power count D of the full diagram does not see. Each such region corresponds to a 1PI subdiagram $\gamma \subset G$ that is itself superficially divergent. To organize the cure we need to know how two such subdiagrams can be positioned relative to one another. Throughout, a subdiagram is identified with its set of internal lines (and the vertices they touch); G denotes the full 1PI graph under study.

Definition 22.14 (Disjoint, nested, and overlapping subdivergences)**.** Let $\gamma_1, \gamma_2 \subset G$ be two superficially divergent proper 1PI subdiagrams of a 1PI graph G, regarded as line sets. Exactly one of the following holds:

(i) *disjoint*, if they share no line (sharing a vertex is permitted), so that $\gamma_1 \cap \gamma_2 = \varnothing$ as line sets;

(ii) *nested*, if one contains the other, $\gamma_1 \subset \gamma_2$ or $\gamma_2 \subset \gamma_1$;

(iii) *overlapping*, if they share at least one line yet neither contains the other, $\gamma_1 \cap \gamma_2 \neq \varnothing$ with $\gamma_1 \not\subseteq \gamma_2$ and $\gamma_2 \not\subseteq \gamma_1$.

A pair that is disjoint or nested is called *non-overlapping*; the three cases are mutually exclusive and exhaustive.

The trichotomy is the combinatorial fork on which the next section's recursion turns, so it is worth fixing the picture (Fig. 22.5). Disjoint subdivergences are the easy case: two divergent loops in different parts of the diagram, each subtracted independently, with nothing to coordinate. Nested subdivergences — a divergent loop sitting inside a larger divergent loop — are subtracted from the inside out: cure the inner one, then the outer one acts on an already-finite interior. Both of these respect a natural ordering of "which to subtract first." Overlapping subdivergences are the hard case and the reason BPHZ needed inventing: two divergent loops that share propagators but neither sits inside the other, so there is *no* inside-out ordering. We will see that the disjoint case already exhibits the nonlocal disease in its cleanest form, and that the overlapping case shows why a single, linearly ordered sequence of subtractions cannot be the whole answer.

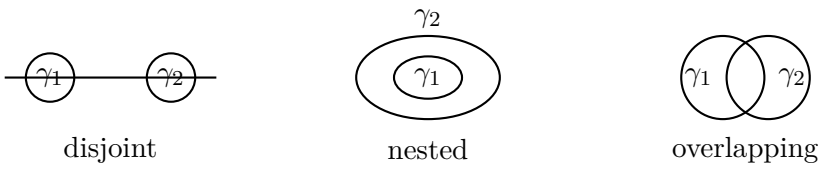


Figure 22.5: The three ways two superficially divergent subdiagrams γ_1, γ_2 of a 1PI graph can be positioned (Theorem 22.14). Disjoint and nested pairs admit an inside-out subtraction order; the overlapping pair, sharing lines with neither contained in the other, does not. The shared region of the overlapping case is exactly the locus that defeats a single ordered subtraction and forces the sum-over-forests of Section 22.5.

22.4.3 A two-loop four-point diagram and the nonlocal divergence

We now compute, continuing ϕ^4 theory from the one-loop counterterms of Eq. (22.48). The cleanest carrier of the phenomenon is the two-loop correction to the four-point function shown in Fig. 22.6: two one-loop bubbles in series in a single channel. Let the total incoming momentum be $p = p_1 + p_2$ with $p^2 = s$. The diagram has three quartic vertices, four internal lines, and two loops; each bubble is the same one-loop four-point subdiagram in the s-channel that we renormalized at one loop. It is the simplest graph that contains a divergent 1PI four-point subdiagram, and the two bubbles form a *disjoint* pair in the sense of

Theorem 22.14: sitting in series along the channel, they share only the central vertex and not a single line.

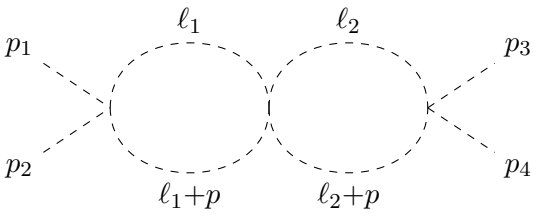


Figure 22.6: The two-loop s-channel double bubble contributing to the ϕ^4 four-point function. Each bubble is a divergent one-loop four-point subdiagram carrying squared momentum $p^2 = s$; the two bubbles form a disjoint pair of subdivergences, sharing only the central vertex and no line. The total incoming momentum is $p = p_1 + p_2$.

The one-loop bubble was reduced to a single master integral in Chapter D. We continue to use the finite log function fixed there and in Eq. (22.46),

$$F(s) \;\equiv\; \int_0^1 \mathrm{d}x\, \ln \frac{m^2 - x(1-x)s}{\mu^2}\,, \tag{22.91}$$

and we collect the bubble into the dimensionless object that does all the work,

$$b(s) \;\equiv\; \frac{1}{16\pi^2}\left[\frac{1}{\bar\varepsilon} - F(s)\right], \qquad \frac{1}{\bar\varepsilon} \equiv \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi\,, \tag{22.92}$$

in terms of which the one-loop s-channel four-point amplitude of Eq. (D.142) reads simply $i\mathcal{M}_s^{(1)} = \frac{1}{2} i\lambda^2\, b(s)$. The single $1/\bar\varepsilon$ pole in $b(s)$ is the bubble's UV divergence; crucially it is *independent of* s, which is precisely why the one-loop four-point counterterm δ_λ of Eq. (22.48) could be a momentum-independent constant: the one-loop subtraction had no nonlocality to fight.

The bare two-loop diagram. Read off Fig. 22.6 with the ϕ^4 Feynman rules and symmetry factors of Section 15.4. There are three vertices $(-i\lambda)$, four propagators each contributing a factor i, and a symmetry factor $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ — one factor of $\frac{1}{2}$ from the interchange of the two lines in each bubble, the two bubbles being distinguishable because they attach to different external pairs. The two loop integrals factorize because the bubbles share only the single vertex b and the common momentum p flows through both:

$$\begin{aligned} i\mathcal{M}_G &= (-i\lambda)^3\, \frac{1}{4} \Big[\int \frac{\mathrm{d}^d \ell_1}{(2\pi)^d} \frac{i^2}{(\ell_1^2 - m^2)((\ell_1 + p)^2 - m^2)} \Big] \Big[\ell_1 \to \ell_2 \Big] \\ &= (-i\lambda)^3\, \frac{1}{4} \big[\, i\, b(s) \,\big]^2 \;=\; i\lambda^3 \cdot \frac{1}{4} \cdot \big(- b(s)^2 \big) \;=\; - \frac{i\lambda^3}{4}\, b(s)^2, \end{aligned} \tag{22.93}$$

where we used $(-i\lambda)^3 = i\lambda^3$ and the appendix result that each bracketed loop integral equals $i\,b(s)$ (the two propagator factors $i^2 = -1$ flip the sign of the scalar bubble $B_0 = i\,b$ recorded in Eq. (D.104)). Now expand the square of Eq. (22.92):

$$b(s)^2 = \frac{1}{(16\pi^2)^2}\left[\frac{1}{\bar\varepsilon^2} - \frac{2F(s)}{\bar\varepsilon} + F(s)^2\right], \tag{22.94}$$

so that the bare double bubble is

$$i\mathcal{M}_G = -\frac{i\lambda^3}{4(16\pi^2)^2}\left[\frac{1}{\bar\varepsilon^2} - \frac{2F(s)}{\bar\varepsilon} + F(s)^2\right]. \tag{22.95}$$

The three terms have completely different characters. The leading double pole $1/\bar\varepsilon^2$ is s-independent: a local divergence, the kind a constant counterterm removes. The finite piece $F(s)^2$ is the genuine two-loop physics — a $\ln^2$ that the running coupling will resum in Chapter 24. The middle term is the new and dangerous object: a *simple pole multiplying the external-momentum function* $F(s)$. Because $F(s)$ contains $\ln s$ — explicitly, for large spacelike s the parametric integral Eq. (22.91) gives $F(s) \to \ln(-s/\mu^2) - 2$ — this term is

$$\frac{i\lambda^3}{2(16\pi^2)^2}\frac{F(s)}{\bar\varepsilon} \xrightarrow[|s|\gg m^2]{} \frac{i\lambda^3}{2(16\pi^2)^2}\frac{1}{\bar\varepsilon}\Big[\ln\frac{-s}{\mu^2} - 2\Big] \;\longleftrightarrow\; \frac{i\lambda^3}{2(16\pi^2)^2}\ln\frac{\Lambda^2}{\mu^2}\ln\frac{-s}{\mu^2}, \tag{22.96}$$

where the last form translates the dimensional-regularization pole into the language of a Euclidean ultraviolet cutoff Λ through the standard dictionary $1/\bar\varepsilon \leftrightarrow \ln(\Lambda^2/\mu^2)$ (the bubble with a cutoff is $b(s) = \frac{1}{16\pi^2}[\ln(\Lambda^2/\mu^2) - F(s)] + \text{const}$, with the same s-dependence and the same divergent coefficient). There it is, manifest: a divergence $\ln\Lambda^2$ multiplied by an external-momentum logarithm $\ln s$. This is exactly the $\ln(p^2)\ln\Lambda$ that Theorem 22.13 forbids.

Why the overall subtraction alone fails. The superficial degree of divergence of the whole graph G is $D = 0$ (four external scalar legs, $d = 4$), so the naive multiplicative-renormalization reflex is to absorb the divergence of G into the two-loop part of the four-point counterterm, $\delta_\lambda^{(2)}$. But $\delta_\lambda^{(2)}$, being a coefficient of ϕ^4, is by Eq. (22.90) a single momentum-independent constant; it can cancel the $1/\bar\varepsilon^2$ and any constant $1/\bar\varepsilon$, and nothing else. It cannot touch the $F(s)/\bar\varepsilon$ term, because matching a constant to a function that varies as $\ln s$ is impossible. If we apply only the overall subtraction — subtract a local polynomial in p, which at $D = 0$ is one constant — the renormalized two-loop four-point amplitude is left with

$$i\mathcal{M}_G\big|_{\text{after overall subtraction only}} = +\frac{i\lambda^3}{2(16\pi^2)^2}\frac{F(s)}{\bar\varepsilon} + (\text{finite}), \tag{22.97}$$

still divergent as $\varepsilon \to 0$, with a divergence that depends on the external momentum. The promise of multiplicative renormalization appears, on this reading, to be broken at two loops.

It is not broken; it is misread. The error is to treat G as if its only divergence were the overall one. The graph has a divergent *subdiagram* — either bubble is a one-loop four-point function with its own $1/\bar{\varepsilon}$ — and the overall power count $D = 0$ is blind to it. The $F(s)/\bar{\varepsilon}$ term is the fingerprint of that subdivergence: one bubble diverges ($1/\bar{\varepsilon}$), and the remaining loop, integrated over the already-divergent insertion, dresses that pole with the kinematic $F(s)$ of an honest finite one-loop integral. A pole times a finite log. The cure must therefore reach inside the diagram and remove the subdivergence before, or rather as part of, removing the overall one.

22.4.4 Subdivergence subtraction restores locality

The fix is the operation that the next section will name the Bogoliubov $\bar{R}$-operation and prove correct in general: for each divergent subdiagram, replace it by its own counterterm before doing the remaining integration. Here the divergent subdiagram is the one-loop four-point bubble, and its counterterm is exactly the object we already built at one loop. We verify, by explicit computation, that inserting it cancels the nonlocal $F(s)/\bar{\varepsilon}$ term and leaves a purely local residue.

Worked Example 22.15 (Subdivergence subtraction of the two-loop ϕ^4 vertex)**.** We renormalize the diagram of Fig. 22.6 by the rule "subtract the subdivergence first," compute the cancellation coefficient, and confirm that it is supplied by the one-loop counterterm δ_λ of Eq. (22.48). The model is the frozen ϕ^4 of the thread; we work in $\overline{\text{MS}}$, where $\delta_\lambda^{\overline{\text{MS}}} = 3\lambda^2/(32\pi^2\bar{\varepsilon})$.

Step 1: the counterterm-insertion diagrams. In addition to the bare graph G, the theory with counterterms contains the two diagrams in which one bubble is replaced by the one-loop four-point counterterm vertex $-i\delta_\lambda$ (the crossed dot of Fig. 22.7). There are two of them, one for each bubble, and they are equal by the symmetry of G.

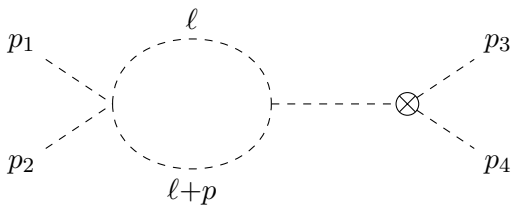


Figure 22.7: One of the two counterterm-insertion diagrams: the outer bubble of Fig. 22.6 is replaced by the one-loop four-point counterterm vertex $-i\delta_\lambda$ (crossed dot). The surviving inner bubble carries the same s-channel kinematics. The partner diagram, with the inner bubble replaced instead, is equal by symmetry.

Step 2: which counterterm cancels the subdivergence. The subdiagram to be cured is a *single* s-channel bubble. Its value is $i\mathcal{M}_s^{(1)} = \frac{1}{2}i\lambda^2\, b(s)$, with pole part

$\frac{1}{2}i\lambda^2 \cdot \frac{1}{16\pi^2\bar{\varepsilon}} = i\lambda^2/(32\pi^2\bar{\varepsilon})$. The subdiagram counterterm is minus this pole,

$$-iC \;=\; -\frac{i\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \qquad C = \frac{\lambda^2}{32\pi^2\bar{\varepsilon}} = \frac{1}{3}\,\delta_\lambda^{\overline{\rm MS}}. \tag{22.98}$$

The factor of $\frac{1}{3}$ deserves a word, because it is the one place the bookkeeping can mislead. The Lagrangian counterterm δ_λ of Eq. (22.48) was fixed by demanding finiteness of the *complete* one-loop four-point function, which is the sum of three channel bubbles, s, t, and u; that is the origin of the "3" in $\delta_\lambda^{\overline{\rm MS}} = 3\lambda^2/(32\pi^2\bar{\varepsilon})$ (Eq. (22.46)). A single sub-bubble inside G, however, carries only *one* channel — the s-channel set by the routing of p — so the divergence it injects, and hence the counterterm that removes it, is one of those three equal pieces, $C = \frac{1}{3}\delta_\lambda$. Equivalently: the local counterterm vertex $-i\delta_\lambda$ is channel-blind, but when it is inserted in place of a specific sub-bubble it acts through the kinematic channel that bubble realizes, and the divergence it must match is that channel's alone. With this understood, the insertion *is* the one-loop counterterm of Eq. (22.48) doing its job inside a larger diagram.

Step 3: value of one insertion. The diagram of Fig. 22.7 has one regular vertex $(-i\lambda)$, one counterterm vertex $(-iC)$, two propagators $(i^2 = -1)$, one symmetry factor $\frac{1}{2}$, and one bubble loop integral $\int \mathrm{d}^d\ell/(2\pi)^d\,[(\ell^2 - m^2)((\ell+p)^2 - m^2)]^{-1} = i\,b(s)$:

$$i\mathcal{M}_{\rm ct,1} = (-i\lambda)(-iC)\,\frac{1}{2}\,(i^2)\,[i\,b(s)] = \frac{i\lambda}{2}\,C\,b(s). \tag{22.99}$$

(The arithmetic of the phases: $(-i\lambda)(-iC) = -\lambda C$; times $\frac{1}{2}i^2 = -\frac{1}{2}$ gives $\frac{1}{2}\lambda C$; times $i\,b$ gives $\frac{i}{2}\lambda C\,b$.) Substituting C from Eq. (22.98) and $b(s)$ from Eq. (22.92),

$$i\mathcal{M}_{\rm ct,1} = \frac{i\lambda}{2}\,\frac{\lambda^2}{32\pi^2\bar{\varepsilon}}\,\frac{1}{16\pi^2}\Big[\frac{1}{\bar{\varepsilon}} - F(s)\Big] = \frac{i\lambda^3}{4(16\pi^2)^2}\Big[\frac{1}{\bar{\varepsilon}^2} - \frac{F(s)}{\bar{\varepsilon}}\Big], \tag{22.100}$$

using $\frac{1}{2}\cdot\frac{1}{32\pi^2}\cdot\frac{1}{16\pi^2} = \frac{1}{4(16\pi^2)^2}$ (since $32\pi^2\cdot 16\pi^2 = 2(16\pi^2)^2$). The two equal insertions sum to

$$i\mathcal{M}_{\rm ct} = 2\,i\mathcal{M}_{\rm ct,1} = \frac{i\lambda^3}{2(16\pi^2)^2}\Big[\frac{1}{\bar{\varepsilon}^2} - \frac{F(s)}{\bar{\varepsilon}}\Big]. \tag{22.101}$$

Step 4: the cancellation. Add the bare graph Eq. (22.95) to the counterterm diagrams Eq. (22.101). This sum is the $\bar{R}$-subtracted (subdivergence-subtracted) two-loop vertex:

$$\begin{aligned}
\bar{R}\,i\mathcal{M}_G &= i\mathcal{M}_G + i\mathcal{M}_{\rm ct} \\
&= \frac{i\lambda^3}{4(16\pi^2)^2}\left\{-\Big[\frac{1}{\bar{\varepsilon}^2} - \frac{2F(s)}{\bar{\varepsilon}} + F(s)^2\Big] + 2\Big[\frac{1}{\bar{\varepsilon}^2} - \frac{F(s)}{\bar{\varepsilon}}\Big]\right\} \\
&= \frac{i\lambda^3}{4(16\pi^2)^2}\left[\frac{1}{\bar{\varepsilon}^2} - F(s)^2\right].
\end{aligned} \tag{22.102}$$

The coefficients tell the whole story. The $1/\bar{\varepsilon}^2$ terms combine as $-1+2=+1$; the nonlocal $F(s)/\bar{\varepsilon}$ terms combine as $+2-2=0$ and *vanish identically*; the finite $F(s)^2$ survives. The dangerous Eq. (22.96) is gone. What remains divergent is

$$\bar{R}\, i\mathcal{M}_G\big|_{\text{div}} = \frac{i\lambda^3}{4(16\pi^2)^2}\,\frac{1}{\bar{\varepsilon}^2}, \tag{22.103}$$

a pure double pole with *no* s-dependence: a local divergence, exactly of the form a momentum-independent two-loop counterterm $\delta_\lambda^{(2)}$ removes, in full compliance with Theorem 22.13. The finite remainder $-\frac{i\lambda^3}{4(16\pi^2)^2}F(s)^2$ is the physical two-loop $\ln^2$ that Chapter 24 will identify as the second term in the exponentiation of the running coupling.

Step 5: the cancellation coefficient matches δ_λ. The coefficient of the cancelled nonlocal term was, in the bare graph, $+\frac{i\lambda^3}{2(16\pi^2)^2}\frac{F(s)}{\bar{\varepsilon}}$, and the counterterm diagrams supplied exactly $-\frac{i\lambda^3}{2(16\pi^2)^2}\frac{F(s)}{\bar{\varepsilon}}$. Tracing the second back through Eq. (22.101)–Eq. (22.98), its size is set by $2\times\frac{1}{2}\lambda C=\lambda C=\frac{1}{3}\lambda\,\delta_\lambda^{\overline{\text{MS}}}=\lambda\cdot\lambda^2/(32\pi^2\bar{\varepsilon})$, i.e. it is the one-loop coupling counterterm δ_λ of Eq. (22.48), inserted once per bubble and acting through the single channel each bubble carries. No new constant had to be invented to cancel the nonlocal divergence: the very counterterm that made the *one-loop* four-point function finite is, when threaded through the two-loop diagram as a subtraction of the subdivergence, exactly what is needed to make the *two-loop* four-point function local. This is the mechanism behind multiplicative renormalizability, exhibited in the open.

Physical Insight

The structure of Eq. (22.102) is universal and worth holding onto. A bare L-loop diagram with subdivergences produces, schematically, $\frac{1}{\varepsilon^L}+\frac{\ln p}{\varepsilon^{L-1}}+\cdots+\frac{\ln^{L-1}p}{\varepsilon}+\ln^L p$: poles dressed by external-momentum logs, with the most dangerous nonlocal pieces multiplying the lower poles. Lower-order counterterms, inserted as subtractions, cancel *precisely* the nonlocal tower, term by term, leaving only the highest pole $1/\varepsilon^L$ as a local constant for the overall L-loop counterterm. The renormalization-group equation of Chapter 24 is, read backwards, the statement that the $\ln p/\varepsilon$ coefficients are rigidly tied to the $1/\varepsilon$ residue one order down — which is exactly the tie we just verified by hand. Nonlocality is not an accident to be cancelled by luck; it is the regulated image of the running coupling, and its cancellation is the consistency of the whole scheme.

22.4.5 The overlapping case: the two-loop self-energy

The double bubble was a disjoint pair of subdivergences, and "subtract the subdivergence first" was unambiguous: the two bubbles share no line, so each

is subtracted independently with nothing to coordinate. The genuinely new combinatorial difficulty, the one that needed Zimmermann's forest formula to tame, is the *overlapping* case of Theorem 22.14, where no such ordering exists. It appears already in ϕ^4 at two loops, in the self-energy.

The relevant graph is the *sunset*, Fig. 22.8: two quartic vertices joined by three internal lines, with one external leg at each vertex. By the master formula $D = 4 - E$ of Chapter 21 the two-point function has $E = 2$, so the overall degree is $D = 2$ — a quadratic overall divergence, to be removed by the δ_m and δ_Z counterterms of Section 22.1.4. The novelty is in its subdivergences. Route the loop momenta so that the three internal lines carry k, q, and $p - k - q$. Holding q fixed and sending $k \to \infty$, the sub-integral over k runs through the two lines carrying k and $p - k - q$, a one-loop four-point bubble; it is logarithmically divergent ($D = 0$). Holding k fixed and sending $q \to \infty$ exposes the bubble built from the lines q and $p - k - q$; also log divergent. There are three such one-loop four-point subdiagrams in all, $\gamma_{12}, \gamma_{13}, \gamma_{23}$, one for each pair of the three internal lines.

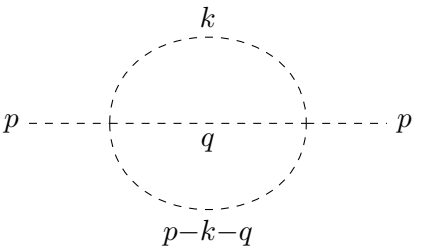


Figure 22.8: The two-loop sunset self-energy of ϕ^4. The three internal lines carry k, q, $p - k - q$. Any pair of them forms a logarithmically divergent one-loop four-point subdiagram; the three such subdiagrams pairwise *overlap*, since any two share exactly one line and neither contains the other. There is no inside-out ordering of subtractions.

Now compare any two of them. The subdiagrams γ_{12} (lines $1, 2$) and γ_{13} (lines $1, 3$) share line 1; yet $\gamma_{12} \not\subset \gamma_{13}$ and $\gamma_{13} \not\subset \gamma_{12}$, because each contains a line the other lacks. By Theorem 22.14 they *overlap*. The same holds for every pair. This is qualitatively different from the double bubble, and the difference is fatal to the naive inside-out recipe. There is no "innermost" divergent subgraph to subtract first: the three divergent bubbles interlock, sharing propagators, and any attempt to declare one of them "inner" and subtract it leaves the other two straddling the subtracted region in a way that double-counts the shared line. Concretely, if one were to subtract γ_{12}, then γ_{13}, then γ_{23} in sequence, the shared lines would be subtracted more than once, over-subtracting; subtracting only one of them under-subtracts and leaves a nonlocal residue of the kind Eq. (22.96). Neither a single overall subtraction nor a single linearly ordered chain of subtractions is

correct.

The resolution, which we only state here and prove in the next section, is that the subtraction must run over *all* divergent subgraphs at once, with a precise inclusion–exclusion bookkeeping that handles the shared lines without double-counting. For the sunset this means subtracting the divergence associated with each of $\gamma_{12}, \gamma_{13}, \gamma_{23}$ and then adding back the overall counterterm, in a combination engineered so that the nonlocal residues from the three overlapping subtractions cancel pairwise on the shared lines and only a local polynomial in p (degree ≤ 2, matching δ_m and δ_Z) survives. That combination is Bogoliubov's recursion, and its closed-form solution is Zimmermann's sum over *forests* — collections of divergent subgraphs that are pairwise non-overlapping, exactly the disjoint-or-nested pairs of Theorem 22.14. The reason forests are the right objects is now visible in advance: a forest is precisely a set of subdivergences that *can* be subtracted in a consistent inside-out order, and summing over all forests is the inclusion–exclusion that repairs the overlaps the single-ordering recipe cannot.

Caution

It is tempting to think overlapping divergences signal a sickness of the theory. They do not. The sunset is renormalized by the same two counterterms δ_m, δ_Z that the one-loop tadpole needed; no new operator is required, and ϕ^4 remains renormalizable. Overlapping divergences are a difficulty of *combinatorics*, not of *operator content*: they make the bookkeeping of subtractions intricate, but they never demand a counterterm outside the finite list fixed by power counting in Theorem 21.10. The forest formula is the statement that the intricate bookkeeping always closes on that finite list.

We have now isolated, by explicit two-loop computation, the precise sense in which renormalization beyond one loop is nontrivial. Divergent subdiagrams inject nonlocal divergences proportional to $\ln(p^2)$ times the regulator; a counterterm, being local, can never absorb them; and the only escape is to subtract the subdivergences before the overall one, which we verified cancels the nonlocality and restores a local residue — in the disjoint case by independent subtraction of each subdivergence, but in the overlapping case only by a sum over all divergent subgraphs. The empirical fix is in hand; what remains is to turn it into an algorithm, prove it terminates on a local result at every order, and show that the local result is precisely what the finite counterterms of a renormalizable theory can supply. That algorithm is the Bogoliubov–Parasiuk–Hepp $\bar{R}$-recursion and its solution as Zimmermann's forest formula, to which we now turn.

22.5 BPHZ Subtraction and Zimmermann's Forest Formula

The previous section left us with a working rule and a warning. The rule, distilled from the explicit two-loop computations of Theorem 22.15, was empirical: *subtract the subdivergences first, innermost outward, and only then subtract what is left over for the whole graph.* The warning was that doing it in any other order—or, worse, naively subtracting the overall divergence alone—leaves behind a divergence that is not a local polynomial in the external momenta, a $\frac{1}{\varepsilon} \times \ln(-p^2)$ remnant that no counterterm in a local Lagrangian can ever cancel. We saw, in Section 21.5.5, exactly how such a remnant arises: an unsubtracted logarithmic subdivergence riding inside an outer loop integration produces a $\ln^2 \Lambda$, a double pole $1/\varepsilon^2$ whose coefficient carries the external momentum dependence of the outer loop. The cure, hinted at but not built, was to remove the inner divergence *before* performing the outer integration, so that the outer loop sees a finite, subtracted subintegrand and produces at most a single, local $\ln \Lambda$.

What we owe now is to turn that empirical rule into an algorithm that is (i) defined for every graph, to every order, with no case-by-case ingenuity; (ii) provably correct, in the sense that it renders every Feynman integral absolutely convergent; and (iii) provably *local*, in the sense that every subtraction it performs is the matrix element of a local operator of bounded dimension, so that the subtractions can be reorganized into counterterms in the Lagrangian. The algorithm is the Bogoliubov–Parasiuk–Hepp–Zimmermann (BPHZ) R-operation. Its recursive form is due to Bogoliubov and Parasiuk; Hepp supplied the convergence proof in momentum space; and Zimmermann recast the recursion into a single closed-form sum—the *forest formula*—that makes both the locality and the convergence transparent and that the rest of renormalization theory is built upon. This section constructs the R-operation, proves the equivalence of the recursion with the forest formula, proves that every counterterm it generates is local, and assembles these with Weinberg's convergence criterion (Theorem 21.25) into the master finiteness statement that the next section will deploy.

Notation Watch

Throughout this section Γ denotes a *Feynman graph* (a diagram), and $\gamma, \delta, \ldots$ denote subgraphs of it—not the 1PI effective action $\Gamma^{(n)}$ of Chapter 15, nor anomalous dimensions. The same letter is standard for both objects in the literature; here it is always a graph. We write I_Γ for the (unrenormalized) integrand of Γ, R_Γ for its renormalized integrand, and $D(\gamma)$ for the superficial degree of divergence of γ, computed by the power-counting formula Eq. (21.19) / Theorem D.2 with γ regarded as a graph in its own right.

22.5.1 The objects: subgraphs, renormalization parts, and reduced graphs

We work with 1PI amplitudes built from connected, amputated diagrams. Fix such a graph Γ with L loops. After routing momenta, its integrand is a product of propagators and vertex factors,

$$I_\Gamma(\ell, p) \;=\; \prod_{\text{lines } j} \frac{1}{q_j^2 - m_j^2 + i\varepsilon} \;\times\; \prod_{\text{vertices}} (\text{vertex factor}), \tag{22.104}$$

a rational function of the L independent loop momenta $\ell = (\ell_1, \dots, \ell_L)$ and the external momenta p, each line momentum q_j a fixed linear combination $q_j = \sum_i c_{ji}\ell_i + \sum_a d_{ja}p_a$. The amplitude is $\int \big(\prod_i \mathrm{d}^d\ell_i/(2\pi)^d\big)\, I_\Gamma$. The R-operation acts on the integrand; convergence is then a statement about the integrability of R_Γ.

Definition 22.16 (Subgraph, renormalization part, reduced graph)**.** A *subgraph* $\gamma \subseteq \Gamma$ is a subset of the internal lines of Γ together with all vertices on which those lines end. The lines of Γ that are not in γ but are attached to a vertex of γ are the *external lines* of γ; the momenta they carry are the *external momenta* p_γ of γ, which inside Γ are linear combinations of Γ's loop and external momenta. A subgraph is *proper* if $\gamma \neq \Gamma$ and γ contains at least one line, and *1PI* if it cannot be disconnected by cutting one of its own lines. A 1PI subgraph with non-negative superficial degree, $D(\gamma) \geq 0$, is called a *renormalization part* (a *divergent subgraph*). The *reduced graph* Γ/γ is obtained by contracting each connected component of γ to a point.

Two restrictions in this definition carry weight. We subtract only 1PI subgraphs because a one-particle-reducible subgraph cannot diverge on its own beyond the divergences of its 1PI pieces: cutting the single line that splits it factorizes any large-momentum region through that line, whose propagator $1/(q^2 - m^2)$ supplies two units of suppression and forbids an independent divergence (this is the same observation that made the 1PI amplitudes the carriers of all divergence in Chapter 15). And we subtract only renormalization parts, $D(\gamma) \geq 0$: a subgraph with $D(\gamma) < 0$ is, by Weinberg's criterion, harmless and needs no subtraction. Because the integrand factorizes cleanly across a subgraph and its complement,

$$I_\Gamma \;=\; I_\gamma \cdot I_{\Gamma/\gamma}, \tag{22.105}$$

where I_γ collects the propagators and vertices of γ (a function of γ's loop momenta and its external momenta p_γ) and $I_{\Gamma/\gamma}$ is the integrand of the reduced graph, the two factors sharing only the variables p_γ. This factorization is what lets a subtraction localized on γ be inserted, as a single modified vertex, into the rest of the diagram.

The relative position of two renormalization parts is governed by the trichotomy of Theorem 21.23, which we recall because it is the combinatorial spine

of everything below. Two subgraphs γ_1, γ_2 (identified with their line sets) are *nested* if one contains the other, *disjoint* if they share no line, and *overlapping* if they share some lines while neither contains the other—so that all three of $\gamma_1 \setminus \gamma_2$, $\gamma_2 \setminus \gamma_1$, $\gamma_1 \cap \gamma_2$ are non-empty. We call γ_1, γ_2 *non-overlapping* if they are nested or disjoint. The whole difficulty of renormalization, and the whole point of the forest formula, lives in the word "overlapping."

22.5.2 The Taylor subtraction operator

The atomic operation of the R-operation is the subtraction of a single renormalization part's divergence. By the locality theorem for primitive divergences (Theorem 21.17), the ultraviolet-divergent part of a divergent 1PI integral is a polynomial in its external momenta of degree at most $D(\gamma)$. The cleanest way to remove exactly that polynomial—no more, no less—is to subtract the first $D(\gamma)+1$ terms of the Taylor expansion of the subintegrand in its external momenta.

Definition 22.17 (Taylor subtraction operator)**.** Let γ be a renormalization part with external momenta p_γ and superficial degree $D(\gamma) \geq 0$. The *Taylor subtraction operator* t^γ acts on the subintegrand I_γ, regarded as a function of p_γ at fixed γ-loop momenta, by truncating its Taylor series about $p_\gamma = 0$ at order $D(\gamma)$:

$$(t^\gamma I_\gamma)(p_\gamma) \;=\; \sum_{|\alpha| \leq D(\gamma)} \frac{p_\gamma^\alpha}{\alpha!} \left(\partial_{p_\gamma}^\alpha I_\gamma\right)\Big|_{p_\gamma = 0}, \tag{22.106}$$

with the multi-index convention $p_\gamma^\alpha = \prod_a (p_\gamma^a)^{\alpha_a}$, $|\alpha| = \sum_a \alpha_a$. By convention $t^\gamma \equiv 0$ whenever $D(\gamma) < 0$. Acting on the full integrand through the factorization Eq. (22.105), $t^\gamma I_\Gamma = (t^\gamma I_\gamma)\, I_{\Gamma/\gamma}$: the operator Taylor-expands only the γ-block and leaves the reduced graph untouched.

Two properties of t^γ are immediate from the definition and do the work in every theorem to come.

Proposition 22.18 (t^γ produces a local counterterm and oversubtracts the degree)**.** *For a renormalization part γ with external momenta p_γ:*

(a) (Locality.) *$t^\gamma I_\gamma$ is a polynomial in p_γ of degree at most $D(\gamma)$. In position space it is therefore the matrix element of a local operator with at most $D(\gamma)$ derivatives, since a monomial p_γ^α is the Fourier image of $(i\partial)^\alpha \delta^{(d)}$.*

(b) (Improved convergence.) *The subtracted block $(1 - t^\gamma) I_\gamma$ vanishes to order $D(\gamma)+1$ at $p_\gamma = 0$, and the γ-loop integral of $(1-t^\gamma)I_\gamma$ has superficial degree $D(\gamma) - (D(\gamma)+1) = -1 < 0$ in the γ-subspace.*

Proof. Part (a) is the definition: Eq. (22.106) is a finite sum of monomials p_γ^α with $|\alpha| \le D(\gamma)$, hence a polynomial of degree $\le D(\gamma)$; the Fourier statement is the elementary correspondence $p^\alpha \leftrightarrow (i\partial)^\alpha \delta^{(d)}$ used already in Theorem 21.17.

For (b), Taylor's theorem with integral remainder gives, for any sufficiently smooth f,

$$\big[(1-t^\gamma)f\big](p_\gamma) \;=\; \sum_{|\alpha|=D(\gamma)+1} \frac{D(\gamma)+1}{\alpha!}\, p_\gamma^\alpha \int_0^1 \mathrm{d}s\,(1-s)^{D(\gamma)}\,(\partial_{p_\gamma}^\alpha f)(s\,p_\gamma). \tag{22.107}$$

Each remainder term carries $D(\gamma)+1$ explicit powers of p_γ and a derivative $\partial_{p_\gamma}^\alpha$ of order $D(\gamma)+1$ acting on the subintegrand. The explicit powers of the *external* momenta of γ are irrelevant to the convergence of the γ-loop integral; what matters is the $(D(\gamma)+1)$-fold derivative. By Theorem 21.16, each external-momentum derivative of a loop integrand lowers its superficial degree by one unit. Hence the γ-loop integral of Eq. (22.107) has degree $D(\gamma) - (D(\gamma)+1) = -1$ in the subspace where γ's loop momenta are scaled to infinity. This is the precise sense in which the subtraction "uses up" exactly the divergence of γ and one power more, sending the subgraph below the convergence threshold. □

Part (a) is locality; part (b) is convergence. The entire content of BPHZ is that these two facts, applied to *every* renormalization part simultaneously and consistently, suffice—and that there is a canonical bookkeeping, the forest formula, that makes "simultaneously and consistently" precise even when subgraphs overlap.

22.5.3 Bogoliubov's recursion

Bogoliubov and Parasiuk encoded the empirical "innermost first" rule as a recursion. The idea is to build, for each graph, an intermediate integrand $\bar R_\Gamma$ in which every *proper* renormalization part has already been subtracted (with its own subdivergences subtracted first, recursively), and then to perform the final, overall subtraction $-t^\Gamma$ on top.

Definition 22.19 (Bogoliubov $\bar R$ and the R-operation)**.** For a 1PI graph Γ define the *counterterm operation* C_γ and the subtracted integrands $\bar R_\Gamma$, R_Γ recursively, the recursion descending on the number of lines:

$$\bar R_\Gamma \;=\; I_\Gamma \;+\; \sum_{\substack{\{\gamma_1,\dots,\gamma_c\}\\ \text{disjoint proper renorm. parts, } c\ge 1}} \Big(\prod_{i=1}^{c} C_{\gamma_i}\Big)\, I_{\Gamma/\{\gamma_1,\dots,\gamma_c\}}, \tag{22.108}$$

$$C_\gamma \;=\; -\,t^\gamma\,\bar R_\gamma, \qquad R_\Gamma \;=\; (1-t^\Gamma)\,\bar R_\Gamma, \tag{22.109}$$

where the sum in Eq. (22.108) runs over all sets of mutually *disjoint* proper 1PI renormalization parts of Γ (including, term by term, all sizes $c \ge 1$), and

$\Gamma/\{\gamma_1,\dots,\gamma_c\}$ is the graph with each γ_i contracted to a point. The recursion bottoms out at a *primitive* graph (one with no proper renormalization part, Theorem 21.22): there the sum is empty, $\bar{R}_\Gamma = I_\Gamma$, and $R_\Gamma = (1-t^\Gamma)I_\Gamma$.

Several readings make the recursion legible. The counterterm $C_\gamma = -t^\gamma \bar{R}_\gamma$ is the local vertex obtained by first cleaning γ of its own subdivergences (forming $\bar{R}_\gamma$) and then extracting its overall divergent polynomial ($-t^\gamma$); inserting C_γ into the reduced graph $I_{\Gamma/\gamma}$ replaces the subdiagram γ by its counterterm. The sum over *disjoint* sets in Eq. (22.108) is the statement that subdivergences sitting in different parts of the diagram are subtracted independently; nesting is handled *inside* each $\bar{R}_{\gamma_i}$, by the recursion. The final factor $(1-t^\Gamma)$ adds the overall subtraction last—"outermost last," the order the previous section's $\ln^2$ argument demanded. The recursion is manifestly well defined: every γ_i in Eq. (22.108) has strictly fewer lines than Γ, so $\bar{R}_{\gamma_i}$ is already defined when $\bar{R}_\Gamma$ is computed.

What the recursion conspicuously does *not* mention is overlapping subgraphs. They never appear together in a single term of Eq. (22.108), because the sum is over disjoint sets and overlaps are handled only through the recursion's nested structure. That an overlapping pair is nonetheless correctly subtracted—each member appearing in different terms, never multiplied against the other—is not obvious from the recursion. It becomes obvious in Zimmermann's reformulation, to which we now turn.

22.5.4 Forests and the forest formula

Definition 22.20 (Forest)**.** A *forest* F of a graph Γ is a set (possibly empty) of 1PI renormalization parts of Γ—proper subgraphs and, optionally, Γ itself—that are *pairwise non-overlapping*: any two elements of F are nested or disjoint. A forest is a *normal* (or $\bar{\Gamma}$*-) forest* if it does not contain Γ as an element, and a *full forest* if it does. The set of all forests of Γ is denoted $\mathcal{F}(\Gamma)$, and the set of normal forests $\bar{\mathcal{F}}(\Gamma)$.

The name is the graph-theoretic one: ordered by inclusion, the elements of a forest form a disjoint union of trees (Hasse diagrams), each chain in a tree a sequence of nested subgraphs and different trees mutually disjoint. Overlapping subgraphs are precisely those that *cannot* sit in a common forest, since neither nesting nor disjointness relates them. This is the structural fact Section 21.5.7 anticipated and the reason a flag in Weinberg's proof—a totally ordered chain—automatically avoids overlaps.

Because the elements of a forest are pairwise non-overlapping, the family of Taylor operators $\{t^\gamma : \gamma \in F\}$ admits a canonical ordering. Inclusion is a partial order on F; extend it to any linear order with the property that if $\gamma \subsetneq \gamma'$ then $t^{\gamma'}$ stands to the *left* of t^γ. The ordered product is well defined—independent of the chosen linear extension—because operators for *disjoint* subgraphs act on

factors sharing no variables and therefore commute (by Eq. (22.105) applied to a disjoint pair), while operators for *nested* subgraphs are forced into inclusion order. We write the ordered product as $\prod_{\gamma\in F}(-t^\gamma)$, with the convention that smaller subgraphs act first (rightmost). For nested $\gamma \subsetneq \gamma'$, this means $t^{\gamma'}$ Taylor-expands the already-subtracted block—"innermost first," the order Section 21.5.5 proved was the only one that avoids a spurious double logarithm.

Definition 22.21 (Forest formula)**.** The *renormalized integrand* of Γ is

$$R_\Gamma \;=\; \sum_{F\in\mathcal{F}(\Gamma)} \prod_{\gamma\in F} (-t^\gamma)\, I_\Gamma, \tag{22.110}$$

the sum running over all forests of Γ, including the empty forest (which contributes I_Γ unaltered) and full forests containing Γ. The intermediate integrand is the same sum restricted to normal forests,

$$\bar{R}_\Gamma \;=\; \sum_{F\in\bar{\mathcal{F}}(\Gamma)} \prod_{\gamma\in F} (-t^\gamma)\, I_\Gamma. \tag{22.111}$$

The forest formula is a single, non-recursive expression: a sum over a finite, explicitly described set of subgraph configurations of products of Taylor operators acting on the bare integrand. It is the form on which all subsequent structural theorems of renormalization—the convergence theorem, the locality theorem, the renormalization-group analysis—are most cleanly proved. That it computes the same integrand as Bogoliubov's recursion is the central combinatorial theorem of the subject.

22.5.5 Equivalence of the recursion and the forest formula

Theorem 22.22 (Forest formula equals Bogoliubov's recursion)**.** *For every* 1*PI graph* Γ*, the renormalized integrand defined by Bogoliubov's recursion Eq.* (22.109) *coincides with the forest sum Eq.* (22.110)*, and the intermediate* $\bar{R}_\Gamma$ *of Eq.* (22.108) *coincides with the normal-forest sum Eq.* (22.111)*.*

Proof. We argue by strong induction on the number of 1PI renormalization parts of Γ. The base case is a primitive graph, with no proper renormalization part. Its only normal forest is the empty one, so the right-hand side of Eq. (22.111) is I_Γ, matching the recursive $\bar{R}_\Gamma = I_\Gamma$. Its forests are $\emptyset$ and (if $D(\Gamma)\geq 0$) $\{\Gamma\}$, so Eq. (22.110) gives $I_\Gamma + (-t^\Gamma)I_\Gamma = (1-t^\Gamma)I_\Gamma$, matching the recursive R_Γ. (If $D(\Gamma)<0$ then $\{\Gamma\}$ is not a forest and $t^\Gamma = 0$; both sides give I_Γ.) The two equalities hold.

For the inductive step, assume the theorem for every graph with fewer renormalization parts than Γ.

Step 1: the overall subtraction factors out. Partition $\mathcal{F}(\Gamma)$ into forests that contain Γ and forests that do not. The latter are exactly the normal forests $\bar{\mathcal{F}}(\Gamma)$.

A forest containing Γ has Γ as its unique maximal element (every other element is a proper subgraph, hence properly contained in Γ), so Γ is non-overlapping with all of them, and removing Γ leaves a normal forest. The map $F \mapsto F \setminus \{\Gamma\}$ is therefore a bijection from Γ-containing forests onto $\bar{\mathcal{F}}(\Gamma)$. In the ordered product, t^Γ stands leftmost (it is the largest subgraph). Hence

$$R_\Gamma = \sum_{F \in \bar{\mathcal{F}}(\Gamma)} \prod_{\gamma \in F} (-t^\gamma) I_\Gamma \;+\; \sum_{F' \in \bar{\mathcal{F}}(\Gamma)} (-t^\Gamma) \prod_{\gamma \in F'} (-t^\gamma) I_\Gamma = (1 - t^\Gamma)\, \bar{R}_\Gamma, \quad (22.112)$$

using Eq. (22.111) for $\bar{R}_\Gamma$. (When $D(\Gamma) < 0$ no forest contains Γ, $t^\Gamma = 0$, and Eq. (22.112) reads $R_\Gamma = \bar{R}_\Gamma$, consistent with the recursion.) Thus once we show the normal-forest sum equals the recursive $\bar{R}_\Gamma$, the relation $R_\Gamma = (1 - t^\Gamma)\bar{R}_\Gamma$ follows for both definitions, and the theorem is proved. It remains to identify the normal-forest sum with Eq. (22.108).

Step 2: decompose a normal forest by its maximal elements. Let $F \in \bar{\mathcal{F}}(\Gamma)$ be non-empty, and let $\gamma_1, \dots, \gamma_c$ be its maximal elements under inclusion. The maximal elements are pairwise *disjoint*: any two elements of a forest are nested or disjoint, and two nested elements cannot both be maximal, so distinct maximal elements share no line. Every other element $\delta \in F$ is contained in exactly one γ_i: δ is non-overlapping with each maximal γ_i, so δ is nested in or disjoint from each; it cannot be disjoint from *all* of them, for then δ would itself be a maximal element (no element of F would contain it), contradicting that the γ_i exhaust the maximal elements; and it cannot be nested in two disjoint γ_i, γ_j at once. Hence $\delta \subsetneq \gamma_i$ for a unique i. Writing $F_i = \{\delta \in F : \delta \subseteq \gamma_i\}$ (which includes γ_i itself), the forest decomposes as a disjoint union

$$F \;=\; F_1 \sqcup F_2 \sqcup \cdots \sqcup F_c, \quad (22.113)$$

where each F_i is a *full* forest of γ_i (it contains the top element γ_i and is otherwise a forest of subgraphs of γ_i). Conversely, any choice of pairwise-disjoint proper renormalization parts $\gamma_1, \dots, \gamma_c$ together with a full forest $F_i \in \mathcal{F}(\gamma_i)$ for each builds a unique normal forest of Γ via Eq. (22.113). The empty normal forest corresponds to $c = 0$.

Step 3: the operator product factorizes across the maximal elements. The Taylor operators belonging to different F_i act on disjoint blocks of the integrand: by Eq. (22.105) applied to the disjoint family $\{\gamma_1, \dots, \gamma_c\}$,

$$I_\Gamma \;=\; \Big(\prod_{i=1}^{c} I_{\gamma_i} \Big)\, I_{\Gamma / \{\gamma_1, \dots, \gamma_c\}}, \quad (22.114)$$

and every t^δ with $\delta \in F_i$ touches only the I_{γ_i} factor (it Taylor-expands a block contained in γ_i). Operators from different F_i therefore commute and act on different factors, so the ordered product splits:

$$\prod_{\gamma \in F} (-t^\gamma)\, I_\Gamma \;=\; \Big(\prod_{i=1}^{c} \prod_{\delta \in F_i} (-t^\delta)\, I_{\gamma_i} \Big)\, I_{\Gamma / \{\gamma_1, \dots, \gamma_c\}}. \quad (22.115)$$

Within the i-th factor, the maximal element is γ_i, so t^{γ_i} is leftmost; pulling it out, the remaining operators range over the *normal* forest $F_i \setminus \{\gamma_i\} \in \bar{\mathcal{F}}(\gamma_i)$:

$$\prod_{\delta \in F_i} (-t^\delta)\, I_{\gamma_i} \;=\; (-t^{\gamma_i}) \prod_{\delta \in F_i \setminus \{\gamma_i\}} (-t^\delta)\, I_{\gamma_i}. \tag{22.116}$$

Step 4: sum and recognize the recursion. Sum Eq. (22.115) over all normal forests F. By the bijection of Step 2, the sum reorganizes into a sum over disjoint families $\{\gamma_1, \dots, \gamma_c\}$ (with $c \geq 0$), and, for each i, an independent sum over normal forests $F_i \setminus \{\gamma_i\}$ of γ_i:

$$\begin{aligned} \bar{R}_\Gamma &= I_\Gamma + \sum_{\substack{\{\gamma_1,\dots,\gamma_c\} \\ \text{disjoint},\, c \geq 1}} \Big(\prod_{i=1}^{c} (-t^{\gamma_i}) \sum_{F_i' \in \bar{\mathcal{F}}(\gamma_i)} \prod_{\delta \in F_i'} (-t^\delta)\, I_{\gamma_i} \Big)\, I_{\Gamma/\{\gamma_1,\dots,\gamma_c\}} \\ &= I_\Gamma + \sum_{\substack{\{\gamma_1,\dots,\gamma_c\} \\ \text{disjoint},\, c \geq 1}} \Big(\prod_{i=1}^{c} (-t^{\gamma_i}\, \bar{R}_{\gamma_i}) \Big)\, I_{\Gamma/\{\gamma_1,\dots,\gamma_c\}}. \end{aligned} \tag{22.117}$$

In the second line we used the inductive hypothesis: each γ_i has strictly fewer renormalization parts than Γ, so its normal-forest sum equals its recursive $\bar{R}_{\gamma_i}$ by Eq. (22.111) and the induction. The $c = 0$ term is the lone I_Γ. Comparing Eq. (22.117) with Eq. (22.108) and recalling $C_{\gamma_i} = -t^{\gamma_i} \bar{R}_{\gamma_i}$, the two are identical. Hence $\bar{R}_\Gamma^{(\text{forest})} = \bar{R}_\Gamma^{(\text{recursion})}$, and with Step 1 also $R_\Gamma^{(\text{forest})} = R_\Gamma^{(\text{recursion})}$. The induction closes. □

The proof exposes precisely how overlapping divergences are handled. Overlapping subgraphs can never belong to a common forest (Theorem 22.20), so the product $\prod_{\gamma \in F}(-t^\gamma)$ never contains $t^{\gamma_1} t^{\gamma_2}$ for an overlapping pair: the ill-defined "subtract one while the other is disturbed" operation simply never occurs. Instead each member of an overlapping pair is subtracted in those forests in which it appears *with compatible partners only*—never against its overlapping mate. The sum over forests reassembles exactly the combination that cancels the divergence of the overlapping region, with no double subtraction and no leftover. This is the precise resolution of the obstruction that Theorem 21.24 diagnosed in the ϕ^4 sunset and that historically made gauge-theory renormalization hard: the forest formula does not subtract overlapping subgraphs together, it sums over the alternatives in which each is subtracted separately, and the arithmetic works out.

Physical Insight

The forest formula trades a recursion—which one must unwind graph by graph—for a single sum over a combinatorial set one can write down once and for all. The conceptual payoff is that locality and convergence, which

are properties of the *whole* R_Γ, become checkable term by term: each forest contributes a product of local Taylor polynomials, and each forest's contribution is power-counting convergent in the right subspaces. The recursion hides this behind its self-reference; Zimmermann's reformulation lays it flat.

22.5.6 Locality of the counterterms

We now prove that every subtraction the R-operation performs corresponds to a local term in the Lagrangian. This is the property that makes renormalizability possible: a divergence that could be cancelled only by a nonlocal counterterm would have no home in a local action, and the program would fail.

Theorem 22.23 (Locality of counterterms)**.** *Every counterterm generated by the R-operation is local. Precisely:*

(a) *For each renormalization part γ, the counterterm $C_\gamma = -t^\gamma \bar{R}_\gamma$, after its γ-loop integrations are performed, is a polynomial in the external momenta p_γ of degree at most $D(\gamma)$, with coefficients that are (divergent) constants depending only on the masses and the regulator. It is therefore the Fourier image of a local operator with at most $D(\gamma)$ derivatives.*

(b) *In particular the overall counterterm of Γ, namely $-t^\Gamma \bar{R}_\Gamma$, is a local polynomial in the external momenta of Γ of degree at most $D(\Gamma)$.*

Proof. Statement (b) is the special case $\gamma = \Gamma$ of (a) (with $\bar{R}_\Gamma$ in place of $\bar{R}_\gamma$), so it suffices to prove (a).

Apply t^γ to $\bar{R}_\gamma$. By Theorem 22.18(a), *any* application of a Taylor operator t^γ to a function of p_γ returns a polynomial in p_γ of degree at most $D(\gamma)$—this is true regardless of what $\bar{R}_\gamma$ is, because t^γ by construction Eq. (22.106) keeps only Taylor coefficients up to order $D(\gamma)$. Hence $C_\gamma = -t^\gamma \bar{R}_\gamma$ is, before any γ-loop integration, a polynomial in p_γ of degree $\le D(\gamma)$ whose coefficients are the integrands $\partial^\alpha_{p_\gamma} \bar{R}_\gamma|_{p_\gamma=0}$, functions of the γ-loop momenta and the masses but *not* of p_γ. The γ-loop integration acts only on those p_γ-independent coefficients, producing p_γ-independent (divergent) constants $c_\alpha(m;\text{regulator})$. The result,

$$\begin{aligned} \int \Big(\prod_{\gamma\text{-loops}} \frac{\mathrm{d}^d \ell}{(2\pi)^d} \Big)\, C_\gamma \;&=\; -\sum_{|\alpha| \le D(\gamma)} \frac{p_\gamma^\alpha}{\alpha!}\, c_\alpha(m), \\ c_\alpha(m) &= \int \Big(\prod \frac{\mathrm{d}^d \ell}{(2\pi)^d} \Big)\, (\partial^\alpha_{p_\gamma} \bar{R}_\gamma)\big|_{p_\gamma=0}, \end{aligned} \tag{22.118}$$

is a polynomial in p_γ of degree at most $D(\gamma)$, as claimed. By the multi-index correspondence $p_\gamma^\alpha \leftrightarrow (i\partial)^\alpha \delta^{(d)}$, this polynomial is the matrix element of a local operator carrying at most $D(\gamma)$ derivatives.

What deserves emphasis is that locality does *not* require $\bar{R}_\gamma$ to equal the bare integrand I_γ. Even though $\bar{R}_\gamma$ contains the subtracted subdivergences—and hence is a sum of terms, some carrying logarithms of p_γ through their already-performed inner loop integrations—the action of t^γ truncates the Taylor series at order $D(\gamma)$ and so discards everything beyond a polynomial. The danger flagged in the previous section was that a *naive* overall subtraction $t^\Gamma I_\Gamma$, applied to an integrand still carrying an *unsubtracted* subdivergence, fails to be local: the unsubtracted inner loop produces a $1/\varepsilon$ multiplying a function of p_Γ, and Taylor-truncating that mixed object does not capture the divergence, leaving a nonlocal $\frac{1}{\varepsilon}\ln(-p_\Gamma^2)$ remnant. The recursion forestalls exactly this: t^Γ is applied to $\bar{R}_\Gamma$, in which all subdivergences are *already* finite, so the only divergence t^Γ meets is the genuine overall one, which by Theorem 21.17 (applied to the subdivergence-free integrand $\bar{R}_\Gamma$, which is power-counting primitive in the overall subspace) is local of degree $D(\Gamma)$. Locality of the counterterm is thus a joint consequence of the truncation built into t and the "innermost first" ordering built into the recursion. □

The locality theorem is the bridge from combinatorics to physics. A local polynomial of degree $\le D(\gamma)$ in external momenta is, operator by operator, a term one can add to a local Lagrangian: a constant ($D=0$) is a coupling shift, a term $\propto p^2$ ($D=2$, for a two-point function) is a field-strength or mass shift, and so on, up to dimension d. Because a renormalizable theory bounds $D(\gamma)$ above by a fixed number set by external legs alone (Theorem 21.10, Theorem 21.20), the operators that the counterterms generate are exactly the operators already present in the action. The R-operation, read as a modification of the Lagrangian, is the systematic addition of counterterm vertices C_γ—and those vertices are local. This is the content the next section converts into a closed proof that the three counterterms $\delta_Z, \delta_m, \delta_\lambda$ of ϕ^4 theory (Eq. (22.9)) suffice to all orders.

22.5.7 Convergence of the renormalized integrand

It remains to prove that the R-operation actually works—that R_Γ is integrable. The analytic engine was assembled in Chapter 21: Weinberg's theorem (Theorem 21.25) states that a Euclidean Feynman integral converges absolutely if and only if the superficial degree of divergence is negative for the graph and for every 1PI subgraph. We recall its hypotheses verbatim, because they are the precise conditions under which the master statement of this section holds.

> *(Weinberg's criterion, Eq. (21.103), recalled.)* For the Euclidean integral Eq. (21.102) with all masses $m_j > 0$ and external momenta Euclidean and nonexceptional, absolute convergence holds if and only if $D(\gamma) < 0$ for $\gamma = \Gamma$ and for every 1PI subgraph $\gamma \subseteq \Gamma$. The proof (Theorem 21.25) decomposes the loop integration into Hepp sectors

indexed by flags $\gamma_1 \subset \gamma_2 \subset \cdots \subset \Gamma$ of nested subgraphs; in each sector the integrand's endpoint behavior is the product of monomial scale integrals $\int_0 t_k^{-D(\gamma_k)-1}\,\mathrm{d}t_k$, convergent at $t_k \to 0$ iff $D(\gamma_k) < 0$. The structural reduction to these monomial integrals is proved there; the uniform bound promoting "every monomial integral converges" to "the actual integral converges absolutely" is the analytic core of Hepp's and Zimmermann's work, stated and attributed in Chapter 21 and not reproduced.

The new content here is that the R-subtracted integrand satisfies Weinberg's criterion for *every* graph, even though the bare integrand does not. This is the theorem that certifies renormalization.

Theorem 22.24 (Convergence of the renormalized integral—BPHZ/Zimmermann)**.** *Let Γ be a 1PI graph with all internal masses positive and external momenta Euclidean and nonexceptional. Then the renormalized integrand R_Γ of Eq. (22.110) defines an absolutely convergent integral,*

$$\int \prod_{i=1}^{L} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d}\, |R_\Gamma(\ell, p)| \;<\; \infty, \tag{22.119}$$

for all spacetime dimensions d within the dimensional-regularization window where the subtractions are defined. Equivalently, after R-subtraction the effective superficial degree of divergence is negative in every subspace: for $\gamma = \Gamma$ and for every 1PI subgraph γ, the subtracted integrand behaves with degree ≤ -1 in the subspace where γ's loop momenta are scaled to infinity.

Proof architecture. The proof has a combinatorial part, established here in full, and an analytic part, imported from Theorem 21.25; we separate them in keeping with the book's standard.

Step 1 (one subgraph: the subtraction makes its degree negative). Fix a 1PI subgraph γ and consider the subspace V_γ in which the loop momenta internal to γ are scaled to infinity together while the rest stay finite. If γ is a renormalization part ($D(\gamma) \ge 0$), it appears in some forests. Group the forest sum Eq. (22.110) by whether a forest contains γ. Forests come in pairs $(F,\, F \cup \{\gamma\})$ whenever F is a forest non-overlapping with γ that does not already contain it; pairing these, the contribution organizes (using that t^γ stands in the correct ordered position relative to the elements of F nested in or disjoint from γ) into an overall factor $(1 - t^\gamma)$ acting on the γ-block of the remaining sum. By Theorem 22.18(b), the operator $(1 - t^\gamma)$ applied to the γ-block supplies $D(\gamma) + 1$ powers of γ's external momenta and an equal-order derivative, lowering the degree of the γ-loop integral in V_γ from $D(\gamma)$ to $D(\gamma) - (D(\gamma) + 1) = -1 < 0$. If γ is *not* a renormalization part, $D(\gamma) < 0$ already and no subtraction is needed. Either way, the subtracted integrand has degree ≤ -1 in V_γ.

Step 2 (all subgraphs simultaneously: the flag structure). Weinberg's criterion is a statement about every subspace at once, and the dangerous subspaces are organized into flags $\gamma_1 \subset \gamma_2 \subset \cdots \subset \Gamma$ of nested 1PI subgraphs (Eq. (21.105); a flag is totally ordered, hence overlap-free, hence a chain in some forest). Along such a flag the R-operation supplies, for each link γ_k that is a renormalization part, the factor $(1 - t^{\gamma_k})$ in the correct nested order (innermost first), because every chain of nested renormalization parts *is* a forest and therefore appears in the sum Eq. (22.110) with exactly the ordered product $\prod_k(-t^{\gamma_k})$ and all its sub-sums $(1-t^{\gamma_1})(1-t^{\gamma_2})\cdots$. Consequently, in each Hepp sector the subtracted integrand's endpoint power in the k-th scaling variable is improved from $-D(\gamma_k) - 1$ to a strictly convergent value: $D(\gamma_k)$ is replaced by its subtracted value ≤ -1, so the monomial integral $\int_0 t_k^{-D(\gamma_k)-1}\,\mathrm{d}t_k$ of Eq. (21.107) becomes $\int_0 t_k^{0\text{ or more}}\,\mathrm{d}t_k$, convergent at $t_k \to 0$. This holds for every link of every flag, i.e. for every subgraph, including overlapping ones—an overlapping pair never shares a flag, so each is subtracted in the flags it does enter, and Step 1 guarantees each is individually rendered convergent.

Step 3 (analytic core: imported). With every subgraph degree negative after subtraction, Weinberg's criterion Eq. (21.103) is satisfied by R_Γ. The remaining step—that satisfaction of the criterion implies absolute convergence of the actual integral, via the uniform bound on the Hepp-sector integrands that promotes the model monomial estimates to the true integrand—is exactly the analytic content of Theorem 21.25, proved there to the level the book sustains and with its hardest measure-theoretic estimate (the uniform boundedness of the sector remainders, Hepp's and Zimmermann's lemma) stated, attributed, and deferred. We invoke it here for the subtracted integrand. The conclusion is Eq. (22.119). □

Caution

The deferral is honest and bounded. What is *proved* is the entire structural chain: that the dangerous regions are indexed by subgraphs (Theorem 21.25, Steps 1–3 of its proof); that each $(1 - t^{\gamma})$ lowers the corresponding subgraph degree below zero (Theorem 22.18); and that the forest sum supplies exactly these factors, in the right order, for every subgraph simultaneously including overlaps (Theorem 22.22 and the present proof). What is *imported* is the single analytic lemma—uniform boundedness of the Hepp-sector integrands—without which "each monomial scale integral converges" would not yet imply "the multidimensional integral converges absolutely." That lemma is measure-theoretic, not field-theoretic, and is the one place Chapter 21 and this section route to the literature for the full estimate; everything that is specific to renormalization is here.

Taken together, Theorems 22.22 to 22.24 are the BPHZ theorem in full: the R-operation, defined either by Bogoliubov's recursion or equivalently by Zimmer-

mann's forest formula, renders every Feynman integral absolutely convergent, and does so by subtracting only *local* polynomials of bounded degree. Finiteness has been reduced to locality, and locality to power counting.

22.5.8 The R-operation on a two-loop ϕ^4 graph

We watch the machinery operate on ϕ^4 theory, taking the two-loop four-point "double bubble" of Fig. 21.4(a): three quartic vertices in a chain, the first and second joined by a bubble (loop momentum k), the second and third by a bubble (loop momentum q), with the external momentum $p = p_1 + p_2$ flowing along the chain (Fig. 22.9). This is the disjoint case—the two sub-bubbles share only the central vertex, no line—so it exhibits the recursion, the forest sum, and the cancellation of the nonlocal remnant in their simplest nontrivial form. The overlapping companion, the sunset, was diagnosed in Theorem 21.24; its full subtraction belongs to the all-orders argument of the next section.

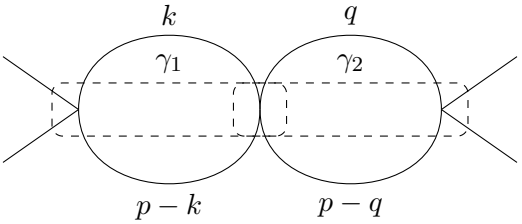


Figure 22.9: The two-loop four-point double bubble Γ of ϕ^4 theory. The left bubble γ_1 (loop k) and the right bubble γ_2 (loop q) are each four-point renormalization parts with $D(\gamma_1) = D(\gamma_2) = 0$; they share only the central vertex, hence are *disjoint*. The whole graph has $D(\Gamma) = 4 - 4 = 0$.

Worked Example 22.25 (R-operation on the ϕ^4 double bubble)**.** Strip the overall vertex and symmetry factors into a constant $c = (-i\lambda)^3/4$ (three vertices, a factor $\frac{1}{2}$ for each bubble), so that the unrenormalized amplitude is the square of the one-loop bubble integral,

$$i\mathcal{M}_\Gamma \;=\; c\,[I(p^2)]^2, \qquad I(p^2) = \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{i}{k^2 - m^2 + i\varepsilon}\,\frac{i}{(p-k)^2 - m^2 + i\varepsilon}, \tag{22.120}$$

the two factors of I being the independent k- and q-integrations. From the master bubble of Chapter D (equivalently Eq. (21.14)),

$$I(p^2) = \frac{i}{16\pi^2}\Big[\frac{1}{\bar\varepsilon} + f(p^2)\Big], \quad f(p^2) = -\int_0^1 \mathrm{d}x\,\ln\frac{\Delta(x)}{\mu^2}, \quad \Delta(x) = m^2 - x(1-x)\,p^2, \tag{22.121}$$

with $1/\bar\varepsilon = 1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi$. The divergence sits in the p-independent constant $1/\bar\varepsilon$ (here $D = 0$, so the divergence is a pure constant, as Theorem 21.17 requires); $f(p^2)$ is finite, with $f(0) = -\ln(m^2/\mu^2)$.

The renormalization parts and the forests. The 1PI renormalization parts of Γ are γ_1, γ_2 (each $D = 0$) and Γ itself ($D = 0$). All three pairs are non-overlapping—Γ contains both γ_i, and γ_1, γ_2 are disjoint—so *every* subset of $\{\Gamma, \gamma_1, \gamma_2\}$ is a forest. There are $2^3 = 8$ forests. Since γ_1, γ_2 are disjoint their Taylor operators commute, and t^Γ stands leftmost; the forest formula Eq. (22.110) therefore factorizes completely,

$$R_\Gamma = \sum_{F \subseteq \{\Gamma, \gamma_1, \gamma_2\}} \prod_{\gamma \in F} (-t^\gamma)\, I_\Gamma = (1 - t^\Gamma)(1 - t^{\gamma_1})(1 - t^{\gamma_2})\, I_\Gamma. \tag{22.122}$$

This is the recursion Eq. (22.109) in disguise: $\bar{R}_\Gamma = (1 - t^{\gamma_1})(1 - t^{\gamma_2}) I_\Gamma$ subtracts the two disjoint subdivergences, and $R_\Gamma = (1 - t^\Gamma)\bar{R}_\Gamma$ adds the overall subtraction.

Acting with the operators. Each t^{γ_i} ($D = 0$) replaces its bubble by the value at zero external momentum, $t^{\gamma_i} I = I(0)$, while t^Γ ($D = 0$) sets the external momentum of the whole graph to zero. Working at the level of the (already loop-integrated) factors,

$$\bar{R}_\Gamma = c\,(1 - t^{\gamma_1})(1 - t^{\gamma_2})\, I(p^2)\, I(p^2) = c\,\big[I(p^2) - I(0)\big]^2, \tag{22.123}$$

because t^{γ_1} acts on the first factor and t^{γ_2} on the second. With Eq. (22.121), the divergent constant cancels in the difference,

$$I(p^2) - I(0) = \frac{i}{16\pi^2}\big[f(p^2) - f(0)\big], \tag{22.124}$$

which is finite: the $1/\bar{\varepsilon}$ pole is gone. Hence

$$\bar{R}_\Gamma = -\frac{c}{(16\pi^2)^2}\,\big[f(p^2) - f(0)\big]^2, \tag{22.125}$$

already finite. The overall subtraction now does nothing: t^Γ evaluates Eq. (22.125) at $p = 0$, where $f(p^2) - f(0) \to 0$, so $t^\Gamma \bar{R}_\Gamma = 0$ and

$$\boxed{R_\Gamma = \bar{R}_\Gamma = -\frac{c}{(16\pi^2)^2}\,\big[f(p^2) - f(0)\big]^2} \tag{22.126}$$

is the finite renormalized amplitude.

Where the nonlocal remnant went. Expand the bare amplitude Eq. (22.120) to see what the subtractions removed. With $I = \frac{i}{16\pi^2}(\frac{1}{\bar{\varepsilon}} + f)$,

$$i\mathcal{M}_\Gamma = c\, I^2 = -\frac{c}{(16\pi^2)^2}\Big[\underbrace{\tfrac{1}{\bar{\varepsilon}^2}}_{\text{double pole}} + \underbrace{\tfrac{2}{\bar{\varepsilon}}\, f(p^2)}_{\text{nonlocal single pole}} + f(p^2)^2\Big]. \tag{22.127}$$

The middle term is exactly the pathology the previous section warned of: a $1/\varepsilon$ multiplying the momentum-dependent function $f(p^2) = -\int_0^1 \mathrm{d}x\, \ln(\Delta/\mu^2)$—a

$\frac{1}{\varepsilon}\ln(-p^2)$ that no momentum-independent counterterm can cancel. The R-operation removes it precisely. The subdivergence subtractions contribute $-2I(0)I(p^2)\,c$, whose pole part is $-\frac{2c}{(16\pi^2)^2}\frac{1}{\bar\varepsilon}f(p^2)$, equal and opposite to the nonlocal single pole; the double-subtraction term $+I(0)^2c$ removes the double pole. After $\bar R$, only $f(p^2)^2-2f(0)f(p^2)+f(0)^2=[f(p^2)-f(0)]^2$ survives—finite, and the overall counterterm vanishes.

The vanishing of the overall counterterm here is not generic; it is a feature of the *reducible* structure of the double bubble, which is a product of two one-loop factors. Its entire divergence is built from its subdivergences, so once those are subtracted nothing local is left to remove at two loops: the renormalization of this graph is achieved by the one-loop coupling counterterm δ_λ inserted as a subdiagram, with no new two-loop constant. Genuine new local counterterms—a nonzero $-t^\Gamma\bar R_\Gamma$—arise from graphs whose overall divergence is *not* exhausted by subdivergences, the overlapping and primitive two-loop topologies such as the sunset of Theorem 21.24. There the forest sum subtracts each overlapping sub-bubble in separate forests, $\bar R_\Gamma$ is finite but does not vanish at zero external momentum, and $-t^\Gamma\bar R_\Gamma$ contributes a genuine local two-loop counterterm of degree $D(\Gamma)$. That the operator content of such counterterms never exceeds what the Lagrangian already contains—so that the same $\delta_Z,\delta_m,\delta_\lambda$ absorb everything, to every order—is the locality theorem Theorem 22.23 combined with the power-counting bound of Chapter 21, and it is the statement the next section turns into a complete proof.

We have now built the master tool. The R-operation, whether one computes it by Bogoliubov's recursion or by Zimmermann's forest formula, renders every ϕ^4 Feynman integral absolutely convergent (Theorem 22.24) while subtracting only local polynomials of bounded degree (Theorem 22.23). Finiteness of any single amplitude is therefore secured; what remains is to prove that finiteness of the whole *theory* holds with only the three counterterms already in the Lagrangian. That reduces, thanks to locality, to a counting question—which local operators power counting permits as counterterms—and it is to that count, and the all-orders renormalizability of ϕ^4 it establishes, that we now turn.

22.6 Renormalizability of ϕ^4 to All Orders

The previous section delivered the three instruments that make a proof of renormalizability possible rather than merely plausible. Zimmermann's forest formula (Theorem 22.22) gives a closed, non-recursive prescription — the R-operation — for subtracting the divergences of an arbitrary graph, subdivergences and overall divergence alike, organized over the forests of its divergent subgraphs. The locality theorem (Theorem 22.23) certifies that every subtraction the R-operation performs is the Taylor polynomial of a 1PI amplitude in its external

momenta, hence a *local* counterterm: a polynomial in momenta of bounded degree, equivalently a finite-dimensional vertex. And Weinberg's convergence theorem (Theorem 22.24) supplies the analytic backbone — a Feynman integral converges absolutely whenever the superficial degree of divergence is negative for the full integration and for every nested subintegration — which is what guarantees that once the R-operation has done its bookkeeping, the surviving integral actually converges. What remains is to put these three together with the power counting of Chapter 21 and watch a finite list of facts force an all-orders conclusion. We do this for ϕ^4 theory, completing the renormalization programme opened by its divergence analysis in Chapter 21, and in doing so we extract the proof *template* — power counting bounds the menu of counterterms; locality makes each counterterm an operator of bounded dimension; the menu is finite and already in the Lagrangian — that the next section transplants, with the new ingredient of gauge invariance, to QED.

22.6.1 The renormalizability criterion

It pays to state, before any computation, exactly what is to be proved. The word "renormalizable" has accreted several meanings; the operationally precise one, the one the forest formula lets us verify, is a statement about which counterterms the subtraction procedure demands.

Definition 22.26 (Renormalizability criterion)**.** Fix a Lagrangian $\mathcal{L} = \mathcal{L}_{\text{ren}} + \mathcal{L}_{\text{ct}}$ split into a renormalized part and counterterms as in Eq. (22.11). The theory is *(perturbatively) renormalizable* if, at every order in the loop expansion, the local counterterms required by the R-operation to render all 1PI amplitudes finite are exhausted by operators of the same types — the same field content and the same number of derivatives — already present in $\mathcal{L}_{\text{ren}}$. Equivalently: there is a fixed, finite set of operators

$$\mathcal{C} \;=\; \{\,\mathcal{O}_1, \dots, \mathcal{O}_n\,\}, \qquad \mathcal{L}_{\text{ct}} \;=\; \sum_{a=1}^{n} \delta_a\, \mathcal{O}_a, \tag{22.128}$$

such that for any choice of renormalization conditions the coefficients δ_a can be chosen, order by order, to cancel every divergence, and no operator outside $\mathcal{C}$ is ever generated. A theory failing this — one in which the subtraction procedure keeps demanding new operator types as the order grows, so that n is effectively infinite — is *non-renormalizable*.

The criterion has two halves, and the two instruments of Chapters 21 and 22 address them separately. *Which* operator types can be demanded at all is decided by power counting and locality: the superficial degree of divergence caps the dimension of the counterterm operators (Theorem 21.17), so only finitely many operator types are admissible. *Whether the procedure closes* — whether

subtracting at order L never spoils the finiteness already achieved below L, and never manufactures an operator outside the menu — is the content of the forest formula and Weinberg's theorem, and is established by induction on the loop order. We take the two halves in turn, then assemble them.

22.6.2 The power-counting input: a finite divergent menu

The first half is almost entirely inherited from Chapter 21; we restate it in the form the induction will use. Specialize the dimension-explicit power-counting theorem (Theorem 21.10, equation Eq. (21.58)) to ϕ^4 theory in $d = 4$. The quartic coupling is dimensionless there, $[\lambda] = 0$ (Eq. (21.62)), so the order-dependent term $-\sum_i V_i[g_i]$ in the general formula vanishes identically, and the superficial degree of divergence of a 1PI graph collapses to a function of the external-leg number E alone,

$$\boxed{D \;=\; 4 - E} \qquad (\phi^4_4), \tag{22.129}$$

which is precisely Eq. (21.109). We record the consequences as a lemma because the all-orders proof leans on each of them.

Lemma 22.27 (The divergent menu of ϕ^4_4 is finite and fixed)**.** *In ϕ^4 theory in four dimensions:*

1. *The superficial degree of divergence of any 1PI graph is $D = 4 - E$, independent of the loop order L, the number of vertices V, and the internal topology.*

2. *$D \geq 0$ holds only for $E \in \{2, 4\}$ (the odd-E amplitudes vanish identically by the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$). Every amplitude with $E \geq 5$ has $D \leq -1$ and is superficially convergent; every amplitude with $E \geq 6$ has $D \leq -2$.*

3. *By the locality theorem (Theorem 21.17), the overall divergence of a graph with $E = 4$ is a momentum-independent constant ($D = 0$), and that of a graph with $E = 2$ is a polynomial $A + B\,p^2$ of degree two in the external momentum ($D = 2$). No graph carries a primitive divergence of any other Lorentz or operator structure.*

Proof. Statement (1) is Eq. (22.129), derived above by setting $[\lambda] = 0$ in Eq. (21.58); the crucial feature, which we belabour because the entire argument turns on it, is that L and V have dropped out completely. A two-loop four-point graph and a seventeen-loop four-point graph have the *same* superficial degree $D = 0$. Statement (2) reads off the signs of $D = 4 - E$: $E = 2 \Rightarrow D = 2$, $E = 4 \Rightarrow D = 0$, $E = 6 \Rightarrow D = -2$, and so on, decreasing by one for each added external leg; the odd-E amplitudes have an odd number of fields and so vanish by the unbroken $\mathbb{Z}_2$ (every surviving Wick contraction pairs fields, requiring an even count). Statement (3) is the locality theorem applied at $D = 0$ and $D = 2$:

the divergent part of a 1PI amplitude is a polynomial in the external momenta of degree at most D, with the degrees-of-freedom and worked verifications already exhibited in Theorems 21.18 and 21.19. □

The force of Theorem 22.27 is that the list of operator types a divergence can ever take is fixed once and for all, before a single loop integral is done. Translating the three allowed structures into operators, with ϕ of dimension one in $d = 4$:

$$\underbrace{E = 2,\ p^0}_{\phi^2}, \qquad \underbrace{E = 2,\ p^2}_{(\partial_\mu\phi)(\partial^\mu\phi)}, \qquad \underbrace{E = 4,\ p^0}_{\phi^4}, \tag{22.130}$$

and these are exactly the three terms of the frozen ϕ^4 Lagrangian (Eq. (21.86)). The counterterm menu of Theorem 22.26 is therefore the finite set

$$\mathcal{C}_{\phi^4_4} \;=\; \Big\{ (\partial_\mu\phi)(\partial^\mu\phi),\ \phi^2,\ \phi^4 \Big\}, \qquad \mathcal{L}_{\rm ct} \;=\; \tfrac{1}{2}\,\delta_Z\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,\delta_m\,\phi^2 - \frac{\delta_\lambda}{4!}\,\phi^4, \tag{22.131}$$

the three counterterms $\delta_Z, \delta_m, \delta_\lambda$ introduced one-loop-explicitly earlier in this chapter (Eq. (22.25)). *No fourth operator is admissible*: a ϕ^6 counterterm would require a primitively divergent six-point amplitude, i.e. $D \geq 0$ at $E = 6$, which Eq. (22.129) forbids; a $(\partial^2\phi)^2$ or $\phi^2\partial^2\phi$ counterterm would require $D \geq 4$ at $E = 2$, equally forbidden. *This is the whole content of renormalizability in one equation*: the menu is finite, and it coincides with the Lagrangian. Everything that follows is the proof that this menu actually suffices — that the second half of the criterion holds, order by order.

Physical Insight

The reason a four-dimensional world tolerates only the three operators of Eq. (22.131) is dimensional accounting, nothing more. A dimensionless coupling means each added vertex pays no dimensional "tax", so going to higher order cannot increase the degree of divergence (statement (1) of the lemma). A relevant or marginal interaction is therefore self-limiting: it can dress the amplitudes it already has, but it cannot conjure divergent amplitudes with more legs. A theory with a coupling of negative mass dimension — ϕ^6 in $d = 4$, gravity, the Fermi interaction — pays a positive tax per vertex, the $-\sum V_i[g_i]$ term climbs without bound, and the divergent menu grows forever. Renormalizability is the statement that the menu does not grow; Chapter 23 will recast this same fact as the statement that irrelevant operators are driven to zero by the renormalization-group flow.

22.6.3 The all-orders theorem and its inductive proof

We can now state and prove the central result. The proof is an induction on the loop order L; its engine is the forest formula, and its analytic guarantee is

Weinberg's theorem. The strategy is to assume that everything below order L has already been rendered finite using only the three counterterms, and to show that order L inherits the same property — both that its graphs are made finite, and that the new counterterms it demands are again only the three.

Theorem 22.28 (ϕ_4^4 is renormalizable to all orders)**.** *ϕ^4 theory in four dimensions is perturbatively renormalizable in the sense of Theorem 22.26. Concretely: for every loop order $L \geq 1$ there exist counterterm coefficients*

$$\delta_Z^{(L)}, \quad \delta_m^{(L)}, \quad \delta_\lambda^{(L)}, \tag{22.132}$$

of the three types in Eq. (22.131) and no others, such that after including the counterterm vertices through order L every 1PI Green's function of the renormalized field is finite (analytic in the external momenta, free of $1/\varepsilon$ poles as $\varepsilon \to 0$). The bare parameters are then the formal series $\phi_0 = Z_\phi^{1/2}\phi$, $m_0^2 Z_\phi = Z_m m^2$, $\lambda_0 Z_\phi^2 = Z_\lambda\, \lambda\, \mu^{2\varepsilon}$ with $Z_\phi = 1 + \sum_L \delta_Z^{(L)}$, etc. (the multiplicative renormalization of Theorem 22.10), and these three Z-factors absorb all ultraviolet divergences to all orders.

Proof. We argue by strong induction on the loop order L.

Base case ($L = 0$ and $L = 1$). At $L = 0$ the graphs are trees; they involve no loop integration and are manifestly finite. At $L = 1$ the only superficially divergent 1PI amplitudes are, by Theorem 22.27, the two-point and four-point functions. These were computed and subtracted explicitly earlier in this chapter: the four-point bubble produced the constant divergence fixed by $\delta_\lambda^{(1)}$, and the two-point tadpole produced the mass divergence fixed by $\delta_m^{(1)}$, with $\delta_Z^{(1)} = 0$ because the one-loop self-energy carries no external momentum (Theorems 21.18 and 21.19). At one loop there are no subdivergences — a one-loop 1PI graph has no proper 1PI subgraph with a loop — so the R-operation reduces to a single overall subtraction, and the base case holds with counterterms of exactly the three allowed types.

Inductive hypothesis. Suppose that for all orders ℓ with $1 \leq \ell \leq L-1$ the theorem holds: all 1PI amplitudes through order $L-1$ have been made finite using counterterm vertices of the three types Eq. (22.131), with definite coefficients $\delta_Z^{(\ell)}, \delta_m^{(\ell)}, \delta_\lambda^{(\ell)}$, and these counterterm vertices are now part of the Feynman rules.

$$[\text{through order } L-1\text{: finite, with only } \delta_Z, \delta_m, \delta_\lambda]. \tag{22.133}$$

Inductive step. Let G be an arbitrary 1PI graph of loop order L with E external legs. We must show two things: (i) the divergences of G are entirely removed by the R-operation, using subtraction counterterms that are all of the three allowed types; and (ii) the new overall counterterm G contributes, if any, is again one of the three.

Consider first the subdivergences of G. A subdivergence is associated with a proper 1PI subgraph $\gamma \subsetneq G$ whose superficial degree $D(\gamma) \geq 0$. Such a γ has strictly fewer loops than G (it omits at least one of G's loops, since it is proper), so its loop order is $\leq L-1$. By Theorem 22.27, $D(\gamma) \geq 0$ forces γ to have $E_\gamma \in \{2,4\}$ external legs, so γ is a lower-order self-energy- or vertex-type subgraph. By the inductive hypothesis Eq. (22.133), the divergence of every such γ is exactly cancelled by a counterterm vertex of one of the three allowed types, already installed in the Feynman rules at its own (lower) order. This is precisely the structure Zimmermann's forest formula encodes: the R-operation acts on G as

$$R\,G \;=\; (1-t_G)\,\overline{R}\,G, \qquad \overline{R}\,G \;=\; \sum_{\mathcal{F}\in\mathfrak{F}(G)} \prod_{\gamma\in\mathcal{F}} (-t_\gamma)\; I_G, \tag{22.134}$$

where $\mathfrak{F}(G)$ is the set of forests of proper divergent subgraphs (excluding G itself in $\overline{R}$), t_γ is the Taylor subtraction operator of order $D(\gamma)$ about zero external momenta of γ, and I_G is the unrenormalized integrand (Theorem 22.22). The forest sum $\overline{R}\,G$ removes *all* subdivergences simultaneously, including the nested and overlapping ones whose interleaving the recursive Bogoliubov definition handled and the explicit forest sum makes manifest. Each $-t_\gamma$ is, term by term, the lower-order counterterm of the inductive hypothesis acting at the location of γ — a local insertion of one of the three operators. No subtraction outside the three types is invoked, because there is no divergent subgraph with $E_\gamma \notin \{2,4\}$.

Now Weinberg's theorem closes the analytic gap. After the forest subtractions $\overline{R}\,G$, every proper subintegration of the integrand has been rendered superficially convergent: for each subgraph γ with $D(\gamma) \geq 0$ the subtraction $-t_\gamma$ lowers the subintegration's effective degree below zero, and subgraphs with $D(\gamma) < 0$ were already convergent. Hence the *only* potential divergence remaining in $\overline{R}\,G$ is the overall one, controlled by the superficial degree $D(G) = 4 - E$ of the full integration. Theorem 22.24 states exactly this: if the superficial degree of every (sub)integration of an integrand is negative, the integral converges absolutely. Applying it to $\overline{R}\,G$, the integral converges save for the overall divergence of degree $D(G)$, which the final operator $(1-t_G)$ in Eq. (22.134) subtracts. Therefore $R\,G$ is absolutely convergent: G is rendered finite. This proves (i).

For (ii), inspect the overall subtraction t_G, the only new counterterm G contributes. By the locality theorem (Theorem 22.23), $t_G\,\overline{R}\,G$ is a polynomial in the external momenta of G of degree $D(G) = 4 - E$, hence a local operator. We exhaust the cases by E:

$$\begin{aligned}
&E \geq 5: \quad D(G) = 4 - E \leq -1 \;\Rightarrow\; t_G = 0, && \text{no overall counterterm needed;}\\
&E = 4: \quad D(G) = 0 \;\Rightarrow\; t_G = \text{const} && \to \text{ a shift of } \lambda, \text{ i.e. } \delta_\lambda^{(L)};\\
&E = 2: \quad D(G) = 2 \;\Rightarrow\; t_G = A + B\,p^2 && \to\ \delta_m^{(L)} \text{ (from } A\text{) and } \delta_Z^{(L)} \text{ (from } B\text{);}\\
&E \text{ odd}: \qquad\qquad G \equiv 0 && \text{by } \mathbb{Z}_2.
\end{aligned} \tag{22.135}$$

The pivotal line is $E \geq 5$: although such graphs exist in profusion at L loops (a five- or six-point function has plenty of L-loop diagrams), every one of them has $D(G) < 0$, so $t_G = 0$ and *no overall counterterm of more than four legs is ever produced*. The only overall counterterms the order-L step can demand are the four-point constant ($\delta_\lambda^{(L)}$) and the two-point quadratic ($\delta_m^{(L)}$, $\delta_Z^{(L)}$) — the same three types, now at order L. Adding these counterterm vertices to the Lagrangian discharges the order-L overall divergences without enlarging the menu.

Thus order L is finite, achieved with counterterms only of the three types of Eq. (22.131). The inductive hypothesis Eq. (22.133) propagates from $L-1$ to L. By induction it holds at every order, which is the assertion of the theorem. Collecting the counterterm coefficients into the multiplicative factors $Z_\phi = 1 + \sum_L \delta_Z^{(L)}$, $Z_m = 1 + \sum_L \delta_m^{(L)}/m^2$, $Z_\lambda = 1 + \sum_L \delta_\lambda^{(L)}/\lambda$ reproduces the bare/renormalized dictionary of Theorem 22.10, and these three series absorb all divergences to all orders. □

Two features of the proof deserve emphasis because they are exactly the load-bearing joints, and because the next section will reuse them verbatim.

Caution

The proof depends on *both* halves of the divergent-menu lemma and *both* of locality and Weinberg, and it is tempting to think one of them is redundant. It is not. Power counting alone (Theorem 22.27(1)) bounds the *superficial* degree, but a graph can have actual divergences worse than its superficial degree suggests, hidden in subgraphs — the nested and overlapping subdivergences of Section 21.5. Weinberg's theorem (Theorem 22.24) is the nontrivial analytic statement that, once *every* subintegration has been made superficially convergent by the forest subtractions, no such hidden divergence survives; superficial convergence of all subintegrations does imply genuine convergence. Locality (Theorem 22.23) is then what guarantees the subtractions are polynomials — finite-dimensional vertices — rather than nonlocal functions of the external momenta that could never be absorbed into a Lagrangian. Remove any one of the three and the conclusion fails.

Physical Insight

The induction is on loop order rather than on the number of vertices or the value of E, and this is the only choice that closes. A proper 1PI subgraph always has strictly fewer loops than its host, so "subdivergences live at lower order" is automatic — which is exactly what lets the inductive hypothesis dispose of them. Were one to induct on E instead, the subgraphs of a two-point function would include four-point pieces with *more* legs, and the

hypothesis would point the wrong way. The forest formula is the device that converts "handle the subdivergences first" from a recursive wish into a single explicit sum, and the loop-order induction is the frame in which that wish is always granted.

22.6.4 Why δ_Z is needed, and from which order

The proof shows that all three counterterms are *admissible*; it is worth seeing that the wavefunction counterterm δ_Z is also *necessary*, and pinning down the order at which it first switches on, because this is the cleanest illustration of the gap between what the theorem permits and what a given order delivers. At one loop $\delta_Z^{(1)} = 0$: the only one-loop 1PI two-point graph is the tadpole, whose external legs attach to a single quartic vertex and inject no momentum into the loop, so Σ_1 is momentum-independent and $B = \Sigma_1'(0) = 0$ (Theorem 21.19). The locality theorem permits a $B\,p^2$ divergence at $E = 2$, but the one-loop dynamics happens not to produce one. The first graph that does is the two-loop "sunset" self-energy, which routes external momentum through its loops and generates a genuine logarithmically divergent p^2 coefficient. We compute its divergent structure to see δ_Z born, and to watch the forest formula remove a subdivergence on a concrete graph.

Figure 22.10: The two-loop 1PI two-point graphs of ϕ^4. Left: the sunset, two vertices joined by three internal lines; it routes the external momentum p through the loops and produces the first nonvanishing $B\,p^2$ divergence, hence the first δ_Z. Right: the tadpole-on-a-bubble, a one-loop tadpole correcting the internal line of a one-loop tadpole; it is momentum-independent and contributes only to δ_m. Its inner bubble is a subdivergence, removed by the one-loop mass counterterm via the forest formula.

Worked Example 22.29 (The sunset and the birth of δ_Z)**.** The sunset graph of Fig. 22.10 (left) is, with symmetry factor $1/6$ for the three interchangeable internal lines and the two quartic vertices giving $(-i\lambda)^2$,

$$-i\Sigma_{\text{sun}}(p^2) \;=\; \frac{(-i\lambda)^2}{6}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{\mathrm{d}^d q}{(2\pi)^d}\;\frac{i}{k^2-m^2}\,\frac{i}{q^2-m^2}\,\frac{i}{(p-k-q)^2-m^2}, \tag{22.136}$$

the $i\varepsilon$ understood in each propagator. Its superficial degree is $D = 4 - E = 2$ (with $E = 2$), consistent with the raw count $2d - 3 \cdot 2 = 2d - 6 = 2$ at $d = 4$. We first verify it carries no subdivergence, then extract its overall divergence.

No subdivergence. A proper 1PI subgraph would be a divergent two- or four-point piece sitting inside the sunset. The only candidate subintegrations are over a single internal momentum, say k at fixed q. That subintegral, $\int \mathrm{d}^d k\, [k^2 - m^2]^{-1}[(p-k-q)^2 - m^2]^{-1}$, is a four-point-like bubble with $D_{\mathrm{sub}} = d - 4 = 0$ at $d = 4$ — but the two lines it connects belong to a single graph with the third line as a spectator, so it is the four-point bubble subgraph γ, $D(\gamma) = 0$. This subgraph has $E_\gamma = 4$ external legs (two external, two attaching to the third internal line), so it *is* a genuine subdivergence. The forest of the sunset therefore contains this bubble; the R-operation subtracts it with the one-loop four-point counterterm $-t_\gamma = \delta_\lambda^{(1)}$ already at hand, exactly as the inductive step prescribes. After this single forest subtraction the integrand has no remaining subdivergence, and Weinberg's theorem leaves only the overall $D = 2$ divergence.

The overall divergence. The overall $D = 2$ subtraction is $t_G = A_{\mathrm{sun}} + B_{\mathrm{sun}}\, p^2$. The constant $A_{\mathrm{sun}} = \Sigma_{\mathrm{sun}}(0)|_{\mathrm{div}}$ feeds $\delta_m^{(2)}$; the piece we want is the coefficient of p^2,

$$B_{\mathrm{sun}} \;=\; \left.\frac{\partial \Sigma_{\mathrm{sun}}}{\partial p^2}\right|_{p=0,\ \mathrm{div}} . \tag{22.137}$$

Differentiating Eq. (22.136) once in p^2 lowers the overall degree to $D = 0$, a logarithmic divergence, so B_{sun} is genuinely there. The double integral, done with the master integrals of Theorem D.2 in $d = 4 - 2\varepsilon$, yields the well-known leading pole

$$B_{\mathrm{sun}} \;=\; -\,\frac{\lambda^2}{12\,(16\pi^2)^2}\,\frac{1}{\varepsilon} \;+\; (\text{subleading in } 1/\varepsilon), \tag{22.138}$$

the sign and the $1/12$ tracking the symmetry factor and the two-loop measure. The point is structural, not numerical: $B_{\mathrm{sun}} \neq 0$. The inverse propagator $\Gamma^{(2)}(p^2) = p^2 - m^2 - \Sigma(p^2)$ therefore acquires a divergent shift in the coefficient of p^2 at two loops, $1 \to 1 - B_{\mathrm{sun}}$, which the wavefunction counterterm must cancel:

$$\delta_Z^{(2)} \;=\; B_{\mathrm{sun}} \;=\; -\,\frac{\lambda^2}{12\,(16\pi^2)^2}\,\frac{1}{\varepsilon} + \cdots , \tag{22.139}$$

the first nonzero wavefunction renormalization in ϕ^4 theory. It is this δ_Z that makes the anomalous dimension $\gamma_\phi = O(\lambda^2)$ nonzero — the leading-order result carried forward to Chapter 24, where γ_ϕ and $\beta_\lambda = 3\lambda^2/16\pi^2$ are assembled from these counterterms. The theorem promised that δ_Z was admissible; the sunset shows it is mandatory, and that the renormalizability machinery delivers it automatically without any new operator type.

The sunset also makes the abstract forest formula tangible: it is the smallest ϕ^4 self-energy graph with a genuine subdivergence (the four-point bubble subgraph),

and its renormalization requires *both* the lower-order counterterm $\delta_\lambda^{(1)}$ (to kill the subdivergence) and a new overall counterterm $\delta_m^{(2)}, \delta_Z^{(2)}$ (to kill what remains). That is the inductive step of Theorem 22.28 acting on a single diagram.

22.6.5 Honest scoping: what "renormalizable to all orders" does and does not mean

Theorem 22.28 is a theorem about the formal power series in λ: at every fixed order, finitely many counterterms of three fixed types render every amplitude finite. It is important to state plainly what it does *not* claim, both to avoid overselling and because the gaps are themselves physics that later chapters fill.

First, renormalizability is a statement about the *series*, term by term, not about its sum. The renormalized perturbation series of ϕ^4 — like that of every realistic interacting QFT — is an asymptotic series with zero radius of convergence, not a convergent one. The argument is Dyson's, given in full in Section 5.6.5 and recalled here in one line: if the series converged in any disc $|\lambda| < \rho$, it would in particular converge for small *negative* λ; but at $\lambda < 0$ the potential $\frac{\lambda}{4!}\phi^4$ is unbounded below, the vacuum is unstable, and no analytic ground-state energy exists in a neighbourhood of $\lambda = 0$. Hence the radius of convergence is zero, and the series is asymptotic at best. What renormalizability secures is that each *coefficient* of this asymptotic series is finite and computable; the large-order growth of the coefficients (factorial, $a_n \sim n!\,(-3/16\pi^2)^n$ up to powers, the subject of Eq. (5.200)) and the Borel-summability programme that tries to reconstruct a function from them belong to the asymptotic-series analysis of Chapter 5. Renormalizability and summability are independent questions; this section settles the former and points to Chapter 5 for the latter.

Second, the proof is perturbative throughout: it organizes Feynman graphs, which are the terms of the λ-expansion. It says nothing about the theory *non-perturbatively* — whether a continuum quantum field theory with a nontrivial S-matrix exists at all once the cutoff is removed. For ϕ^4 in four dimensions the non-perturbative answer is now known and is a cautionary one: the theory is *trivial*. Lattice and constructive studies through the 1980s, and a rigorous proof for $d = 4$ by Aizenman and Duminil-Copin in 2021, establish that the continuum limit of ϕ_4^4 at fixed renormalized coupling exists only as a free (Gaussian) field: the renormalized coupling is driven to zero as the cutoff is removed. This is the same physics as the Landau pole of the running coupling, seen non-perturbatively. Perturbative renormalizability and non-perturbative triviality are not in contradiction — the former is an order-by-order statement at fixed cutoff, the latter a statement about the cutoff-removal limit — but the juxtaposition is exactly why the Wilsonian reframing of Chapter 23 is indispensable: it explains *why* a perturbatively renormalizable theory can nonetheless fail to define a fundamental continuum theory, by recasting renormalizability as a statement

about renormalization-group flow toward (here, an unreachable interacting) fixed point. The constructive and triviality questions for ϕ^4_4, ϕ^4_3, and the Wilson–Fisher fixed point in $d = 4 - \varepsilon$ are taken up, respectively, in Chapter 23 and at the frontier.

Physical Insight

There is no tension between "ϕ^4_4 is renormalizable" and "ϕ^4_4 is trivial." Renormalizability says: at any fixed UV cutoff, the cutoff dependence of physical predictions can be hidden inside three parameters, so the theory makes finite predictions order by order — it is a perfectly good *effective* field theory up to its cutoff. Triviality says: if you insist on sending the cutoff to infinity while holding a nonzero interaction fixed, you cannot — the only survivor is the free theory. The Standard Model's Higgs sector inherits exactly this status: a renormalizable effective description with a finite cutoff, not a candidate fundamental theory to arbitrarily short distances. The Wilsonian picture of Chapter 23 is what turns this from a paradox into a precise and physical statement about scales.

With ϕ^4 fully renormalized to all orders, we have in hand a reusable proof template, abstracted from the diagrams: *power counting fixes a finite menu of admissible counterterm operators; locality makes each one a finite-dimensional vertex; the forest formula plus Weinberg's theorem close an induction on loop order showing the menu suffices.* The next section applies precisely this template to QED, where the same machinery runs — but with a decisive new constraint absent here. Gauge invariance, enforced order by order through the Ward–Takahashi identity, relates the counterterms to one another ($Z_1 = Z_2$) and protects the photon from acquiring a mass, so that the naive power-counting menu is not merely finite but further reduced, and charge renormalization turns out to be universal.

22.7 Renormalizability of QED and Charge Universality

The previous section closed the scalar story: power counting tells us that ϕ^4 in four dimensions has only finitely many superficially divergent amplitudes, the locality theorem tells us each divergence is a local polynomial matching a term already in the Lagrangian, and Zimmermann's forest formula assembles these primitive subtractions into a recursive operation that renders *every* graph, at every loop order, finite with the three counterterms $\delta Z, \delta_{m^2}, \delta_\lambda$ and nothing more (Theorem 22.28). That argument was driven entirely by dimensional analysis: it never used a symmetry beyond the $\mathbb{Z}_2$ that kills the odd amplitudes.

Quantum electrodynamics is the next theory to run through the same machine, and it teaches the lesson that power counting is necessary but not always sufficient. Counting alone would leave QED with two genuine embarrassments — a quadratically divergent photon "mass" and a logarithmically divergent four-photon contact term — neither of which corresponds to any operator in the gauge-invariant Lagrangian, and either of which would, taken at face value, spoil renormalizability by demanding a counterterm that is simply not available. The resolution is that QED has a symmetry the scalar theory lacks, the $U(1)$ gauge invariance, and its quantum avatar, the Ward–Takahashi identity of Theorem 19.13, does exactly the work power counting cannot: it makes the would-be photon mass vanish, softens the four-photon amplitude to convergence, and — most consequentially for everything that follows — ties the vertex renormalization to the electron-field renormalization so tightly that the electric charge renormalizes through the photon alone. The aim of this section is to carry out the QED renormalization program in full, to prove the identity $Z_1 = Z_2$ to all orders at the level of renormalized 1PI functions, and to extract from it the universality of electric charge: the statement that every charged field, whatever its mass or spin, renormalizes its coupling to the photon by the same factor.

22.7.1 Power counting in QED

We begin where Chapter 21 ended, with the superficial degree of divergence, and specialize the general dimension-explicit formula Eq. (21.58) to the standard Lagrangian

$$\mathcal{L}_{\mathrm{QED}} = \bar{\psi}\,(i\not{D} - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \quad \alpha = \frac{e^2}{4\pi}, \tag{22.140}$$

with electron charge $-e$, $e > 0$, in mostly-minus signature and Feynman gauge $\xi = 1$ for explicit work. In $d = 4$ the gauge field has mass dimension $[A] = 1$ and the Dirac field $[\psi] = \frac{3}{2}$, while the coupling is dimensionless, $[e] = 0$. The general formula Eq. (21.58) for a 1PI graph reads $D = d - \sum_{\mathrm{ext}} d_f - \sum_i V_i[g_i]$; with the single marginal coupling e the entire coupling term drops, and a graph with E_ψ external fermion legs and E_A external photon legs has

$$\boxed{D = 4 - \tfrac{3}{2}E_\psi - E_A.} \tag{22.141}$$

The vanishing of the coupling contribution is the marginal-coupling miracle in its QED form: D depends only on the external content, never on the loop order, so the list of superficially divergent amplitudes is finite and closes once and for all.

It is worth seeing Eq. (22.141) fall out of the raw momentum count as well, since the cancellation of the order-dependence is the load-bearing fact. Let a graph have L loops, P_e internal electron lines, P_γ internal photon lines, and V vertices. A photon propagator falls as p^{-2} and a fermion propagator as p^{-1} at large

momentum (the Dirac propagator $i(\not{p}+m)/(p^2-m^2) \sim \not{p}/p^2 \sim p^{-1}$), so the naive count is $D = 4L - P_e - 2P_\gamma$. Each QED vertex meets two fermion ends and one photon end; counting line-ends gives $2V = 2P_e + E_\psi$ and $V = 2P_\gamma + E_A$, whence $P_e = V - \frac{1}{2}E_\psi$ and $P_\gamma = \frac{1}{2}(V - E_A)$. The loop number is $L = P_e + P_\gamma - V + 1$. Substituting,

$$L = \left(V - \tfrac{1}{2}E_\psi\right) + \tfrac{1}{2}(V - E_A) - V + 1 = \tfrac{1}{2}V - \tfrac{1}{2}E_\psi - \tfrac{1}{2}E_A + 1, \tag{22.142}$$

$$D = 4L - P_e - 2P_\gamma = (2V - 2E_\psi - 2E_A + 4) - \left(V - \tfrac{1}{2}E_\psi\right) - (V - E_A) = 4 - \tfrac{3}{2}E_\psi - E_A, \tag{22.143}$$

and the number of vertices V has cancelled identically, confirming Eq. (22.141) and the order-independence it asserts.

Reading off the amplitudes with $D \geq 0$ gives a short list, the same one enumerated diagram-in-hand in Section 21.4.3 and drawn in Fig. 21.3; we recall it because the renormalization program is organized around it.

- $(E_\psi, E_A) = (0, 2)$, the photon two-point function (vacuum polarization $\Pi^{\mu\nu}$), $D = 2$.
- $(E_\psi, E_A) = (2, 0)$, the electron two-point function (self-energy Σ), $D = 1$.
- $(E_\psi, E_A) = (2, 1)$, the electron–photon vertex Γ^μ, $D = 0$.
- $(E_\psi, E_A) = (0, 3)$, the three-photon amplitude, $D = 1$.
- $(E_\psi, E_A) = (0, 4)$, the four-photon amplitude (light-by-light scattering), $D = 0$.
- $(E_\psi, E_A) = (0, 0)$, vacuum bubbles, $D = 4$, unobservable (they cancel against the vacuum normalization).

Every amplitude with more external legs has $D < 0$ and is superficially convergent: the five-point and higher functions, the $(2, 2)$ electron–Compton function ($D = -1$), and so on. The first three entries are the physical divergences, and each matches term-for-term an operator already present in Eq. (22.140): the photon two-point divergence renormalizes the photon kinetic term, the electron two-point divergence renormalizes the electron kinetic and mass terms, and the vertex divergence renormalizes the coupling. The last three entries are the cases where naive power counting overcounts, and where a symmetry must intervene if the theory is to remain renormalizable with only the operators it started with. The three-photon amplitude is removed by Furry's theorem (charge-conjugation invariance forces a fermion loop with an odd number of external photons to vanish, so its divergence is not merely absorbable but identically zero); the photon-mass and four-photon dangers are removed by gauge invariance, to which we now turn before assembling the counterterms.

22.7.2 Gauge invariance I: transversality forbids a photon mass

The photon two-point function has $D = 2$. The locality theorem (Theorem 21.17) permits its divergence to be a polynomial of degree two in the external momentum q, and there are two admissible Lorentz tensors: the degree-two pair $q^2\eta^{\mu\nu}$ and $q^\mu q^\nu$, and — alarmingly — the degree-zero tensor $\eta^{\mu\nu}$ carried by a quadratically divergent coefficient. The dangerous structure is the constant one, $\Pi^{\mu\nu}\big|_{\text{div}} \supset c\,\Lambda^2\,\eta^{\mu\nu}$ in a cutoff scheme, or a q-independent pole $c\,\eta^{\mu\nu}/\varepsilon$ in dimensional regularization. In position space it is a term $\propto A_\mu A^\mu$ in the effective action: a *photon mass*. No such operator appears in Eq. (22.140), and none may be added without destroying gauge invariance, so if this divergence were genuinely present QED would be non-renormalizable — it would require a counterterm not in the theory. Gauge invariance forbids the operator; the question is whether it also forbids the divergence. It does, and the mechanism is transversality.

Proposition 22.30 (Transversality of the photon self-energy)**.** *The exact* 1*PI photon two-point function* $\Pi^{\mu\nu}(q)$ *in QED is transverse to all orders,*

$$q_\mu\,\Pi^{\mu\nu}(q) = 0 \quad\Longrightarrow\quad \Pi^{\mu\nu}(q) = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\,\Pi(q^2), \tag{22.144}$$

so its divergence is carried entirely by the single scalar $\Pi(q^2)$ *multiplying the transverse projector. The would-be photon-mass divergence* $\propto \eta^{\mu\nu}$ *alone is absent, and no* $A_\mu A^\mu$ *counterterm is ever required.*

Proof. The photon couples to the electromagnetic current $j^\mu = \bar\psi\gamma^\mu\psi$, and the 1PI two-point function is the (amputated) current–current correlator

$$\Pi^{\mu\nu}(q) = i\int d^4x\, e^{iq\cdot x}\,\langle\, \mathcal{T}\, j^\mu(x)\, j^\nu(0)\,\rangle_{\text{1PI}}. \tag{22.145}$$

The $U(1)$ Ward–Takahashi identity of Theorem 19.13, applied to the two-current Green's function, expresses current conservation $\partial_\mu j^\mu = 0$ at the quantum level: contracting an external photon momentum into the correlator yields contact terms that vanish here because the equal-time commutator $[j^0(x), j^\nu(0)]\delta(x^0)$ of a conserved current built from the $U(1)$ charge density is a total spatial derivative with no Schwinger term contributing to the connected, amputated two-point function. The statement that survives is precisely $q_\mu\Pi^{\mu\nu}(q) = 0$. (Equivalently, this is the q-independent specialization of the operator identity Eq. (19.49), established at one loop in Chapter 19 and promoted to all orders by the Ward–Takahashi identity.) A symmetric tensor $\Pi^{\mu\nu}$ built from the single vector q^μ and the metric must be a combination $A(q^2)\,\eta^{\mu\nu} + B(q^2)\,q^\mu q^\nu$; imposing $q_\mu\Pi^{\mu\nu} = 0$ gives $A\,q^\nu + B\,q^2 q^\nu = 0$, hence $A = -B\,q^2$, and writing $\Pi(q^2) \equiv -B(q^2)$ produces $\Pi^{\mu\nu} = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\Pi(q^2)$, the unique transverse form. The decomposition has *no* term proportional to $\eta^{\mu\nu}$ alone: the only place a constant $\eta^{\mu\nu}$ could enter, namely $A(0) \neq 0$, is forbidden because $A(q^2) = -q^2\Pi(q^2)$ vanishes at $q = 0$. Therefore $\Pi^{\mu\nu}(0) = 0$: the photon stays massless, and the photon-mass counterterm never arises. □

What the proof has bought is decisive for power counting. The transverse projector $q^2\eta^{\mu\nu} - q^\mu q^\nu$ carries two explicit powers of q, so the scalar $\Pi(q^2)$ has effective degree $D - 2 = 0$: it is only *logarithmically* divergent, and its q-independent pole multiplies $q^2\eta^{\mu\nu} - q^\mu q^\nu$, which is nothing but the momentum-space form of the photon kinetic operator $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$. The photon two-point divergence therefore renormalizes the photon field alone — it defines the constant Z_3 — exactly as required for renormalizability with the operators already present. Gauge invariance has converted a quadratic divergence with no home into a logarithmic divergence with a perfect home.

22.7.3 Gauge invariance II: light-by-light is finite

The four-photon amplitude $\Gamma^{\mu\nu\rho\sigma}(q_1, q_2, q_3, q_4)$ — light-by-light scattering, a fermion loop with four photons attached — has $D = 0$ by Eq. (22.141), which naively threatens a logarithmically divergent local four-photon vertex. The only Lorentz-invariant, dimension-four local operators with four photon fields are built from products of two metrics contracting four A's, schematically $(A_\mu A^\mu)^2$ and $(A_\mu A_\nu)(A^\mu A^\nu)$. None of these is gauge invariant, and none appears in Eq. (22.140); a divergence requiring such a counterterm would again break renormalizability. Gauge invariance forbids the operator, and once more it also forbids the divergence.

The argument is the four-point analogue of transversality. The Ward–Takahashi identity makes the amplitude transverse on every leg, $q_{1\mu}\Gamma^{\mu\nu\rho\sigma} = 0$ and likewise for q_2, q_3, q_4. A local (polynomial) tensor satisfying transversality on all four legs cannot be built from the metric alone: each external index must ultimately be carried by a field strength $F_{\mu\nu} \sim q_\mu\varepsilon_\nu - q_\nu\varepsilon_\mu$ rather than a bare A_μ, because only F is gauge covariant. Every external photon therefore enters the local divergent part through at least one explicit power of its momentum. The lowest-dimension gauge-invariant local four-photon operators are accordingly $\left(F_{\mu\nu}F^{\mu\nu}\right)^2$ and $\left(F_{\mu\nu}\widetilde{F}^{\mu\nu}\right)^2$ — the Euler–Heisenberg operators — which have mass dimension *eight*, not four. Carrying four extra powers of momentum relative to the naive count, they correspond to an effective degree of divergence $D_{\text{eff}} = 0 - 4 = -4 < 0$. The light-by-light amplitude is therefore ultraviolet finite: its leading behavior is a finite, calculable, dimension-eight effective interaction (the nonlinear vacuum that bends light in a background field), not a divergence demanding subtraction. No four-photon counterterm is needed.

Power counting overcounts; symmetry corrects it

The photon-mass and light-by-light cases are the cleanest illustration in QED of a structural point that recurs throughout gauge theory: the superficial degree of divergence Eq. (22.141) is an *upper bound* on the actual divergence, saturated only when no symmetry intervenes. Gauge

invariance, enforced graph-by-graph by the Ward–Takahashi identity, systematically lowers the realized degree below the bound for exactly those amplitudes whose naive divergence would have no gauge-invariant counterterm. The general lesson, made precise for non-abelian theories by the BRST/Slavnov–Taylor analysis of Chapter 26, is that the operators which can actually appear as counterterms are constrained not by dimension alone but by the symmetries the regulator preserves; a symmetry-respecting regulator can only generate symmetry-respecting counterterms. QED is the abelian template for this principle: every potentially fatal divergence that power counting alone would predict is annihilated by the gauge symmetry before it can demand an operator the theory does not possess.

22.7.4 The four renormalization constants and the counterterm Lagrangian

With the gauge-protected cases disposed of, three physical divergent structures remain, matched to the three operators of Eq. (22.140), and they define four renormalization constants in the historical naming convention: Z_2 for the electron field, Z_3 for the photon field, Z_m for the mass, and Z_1 for the vertex. We renormalize multiplicatively, writing each bare quantity (subscript 0) as a renormalization constant times the renormalized quantity. The field and mass relations follow from demanding canonical normalization of the renormalized propagators,

$$\psi_0 = Z_2^{1/2}\,\psi, \qquad A_0^{\mu} = Z_3^{1/2}\,A^{\mu}, \qquad m_0 = Z_m\,m, \tag{22.146}$$

and the charge relation follows from how the interaction term reorganizes when Eq. (22.146) is substituted into the bare Lagrangian. Writing the bare interaction in terms of renormalized fields, $-e_0\,\bar\psi_0\gamma^{\mu}\psi_0\,A_{0\mu} = -e_0\,Z_2\,Z_3^{1/2}\,\bar\psi\gamma^{\mu}\psi\,A_{\mu}$, and defining Z_1 as the multiplicative constant of the renormalized vertex (so that the full coefficient of the renormalized operator $\bar\psi\gamma^{\mu}\psi A_{\mu}$ is $-e\,\mu^{\varepsilon}Z_1$, with μ^{ε} keeping the renormalized e dimensionless in $d = 4-2\varepsilon$), matching coefficients gives $e_0\,Z_2\,Z_3^{1/2} = e\,\mu^{\varepsilon}Z_1$, that is,

$$e_0 = \mu^{\varepsilon}\,\frac{Z_1}{Z_2\,Z_3^{1/2}}\,e \;\equiv\; \mu^{\varepsilon}\,Z_e\,e, \qquad Z_e = \frac{Z_1}{Z_2\,Z_3^{1/2}} = Z_1\,Z_2^{-1}\,Z_3^{-1/2}. \tag{22.147}$$

The combinatorial logic of the exponents is worth stating once because it generalizes verbatim to every gauge theory: each leg attached to the charge-carrying vertex brings one factor of its own field renormalization (a half-power $Z_2^{1/2}$ per electron leg, two of them, and one half-power $Z_3^{1/2}$ for the photon leg), while the irreducible divergence of the vertex graph itself is carried by Z_1. This is the QED

instance of the general bare/renormalized relation Eq. (22.67) that organizes the whole chapter.

Equivalently, and this is the form used in practice, one splits the bare Lagrangian into a renormalized piece plus counterterms, $\mathcal{L}_{\text{QED}} = \mathcal{L}_R + \mathcal{L}_{\text{ct}}$, with

$$\mathcal{L}_R = \bar{\psi}\,(i\partial\!\!\!/ - m)\,\psi - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - e\,\mu^{\varepsilon}\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu}, \tag{22.148}$$

$$\mathcal{L}_{\text{ct}} = \delta_2\,\bar{\psi}\,i\partial\!\!\!/\,\psi - \delta_m\,\bar{\psi}\psi - \tfrac{1}{4}\delta_3\,F_{\mu\nu}F^{\mu\nu} - e\,\mu^{\varepsilon}\,\delta_1\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu}, \tag{22.149}$$

where the counterterm coefficients are

$$\delta_2 = Z_2 - 1, \qquad \delta_3 = Z_3 - 1, \qquad \delta_m = (Z_2 Z_m - 1)\,m, \qquad \delta_1 = Z_1 - 1. \tag{22.150}$$

Each counterterm supplies a new Feynman-rule vertex — a momentum-dependent insertion $i(\delta_2 \not{p} - \delta_m)$ on the electron line, an insertion $-i\delta_3(q^2\eta^{\mu\nu} - q^{\mu}q^{\nu})$ on the photon line (transverse by Theorem 22.30), and a vertex insertion $-ie\mu^{\varepsilon}\delta_1\gamma^{\mu}$ — whose finite parts are fixed by renormalization conditions and whose divergent parts are tuned, order by order, to cancel the loop divergences of the corresponding 1PI function. The standard on-shell conditions fix the residues and pole locations of the full propagators and the value of the vertex at zero momentum transfer: the electron propagator has a pole at $p^2 = m^2$ with unit residue (fixing δ_2, δ_m), the photon propagator has a pole at $q^2 = 0$ with unit residue (fixing δ_3), and the vertex equals $-ie\gamma^{\mu}$ at $q = 0$ on shell (fixing δ_1). In $\overline{\text{MS}}$, used in Chapter 24 to extract the running, one instead keeps only the pole parts; the structural relations Eqs. (22.146) and (22.147) are scheme independent.

22.7.5 The Ward–Takahashi identity and $Z_1 = Z_2$

The four constants are not independent. The same gauge symmetry that made the photon self-energy transverse relates the vertex to the electron self-energy, and the relation forces $Z_1 = Z_2$. This is the single most important structural fact in QED renormalization, because it is what makes the electric charge renormalize through the photon alone.

Theorem 22.31 ($Z_1 = Z_2$ to all orders)**.** *In QED the vertex renormalization constant equals the electron field renormalization constant,*

$$Z_1 = Z_2, \tag{22.151}$$

to all orders in perturbation theory and in every gauge. Consequently the divergence of the 1*PI vertex function is identical to the divergence dressing the electron leg, and the counterterms satisfy* $\delta_1 = \delta_2$*.*

Proof. The starting point is the operator Ward–Takahashi identity of Theorem 19.13, which relates the exact amputated 1PI vertex $\Gamma^{\mu}(p', p)$ to the exact inverse electron propagator $S^{-1}(p)$,

$$(p' - p)_{\mu}\,\Gamma^{\mu}(p', p) = S^{-1}(p') - S^{-1}(p). \tag{22.152}$$

This is an exact statement about the *full* (all-orders) functions; it is the quantum expression of current conservation and holds graph by graph once the gauge-invariant regulator is in place. Differentiate Eq. (22.152) with respect to $q = p' - p$ and take the soft limit $q \to 0$. The left side becomes the vertex at vanishing photon momentum, while the right side becomes the momentum derivative of the inverse propagator:

$$\Gamma^{\mu}(p,p) = \frac{\partial}{\partial p_{\mu}}\, S^{-1}(p). \tag{22.153}$$

Now compare the two sides as renormalized objects. The renormalized vertex and renormalized inverse propagator, both finite, are obtained from the bare ones by stripping the renormalization constants: the bare vertex is $\Gamma^{\mu}_{0} = Z_1^{-1}\Gamma^{\mu}_{R}$ and the bare inverse propagator is $S_0^{-1} = Z_2^{-1} S_R^{-1}$, because the vertex carries the constant Z_1 that absorbs its divergence and the propagator carries Z_2. (Concretely: the field rescaling $\psi_0 = Z_2^{1/2}\psi$ multiplies the bare two-point function $\langle \psi_0 \bar{\psi}_0 \rangle = Z_2 \langle \psi \bar{\psi} \rangle_R$, so the inverse propagator carries Z_2^{-1}; the bare three-point function $\langle \psi_0 \bar{\psi}_0 A^{\mu}_0 \rangle$ carries $Z_2 Z_3^{1/2}$, and after amputating the renormalized external legs the irreducible vertex carries Z_1^{-1}.) The identity Eq. (22.153) holds for the bare functions because it is an exact consequence of the symmetry; substituting,

$$Z_1^{-1}\,\Gamma^{\mu}_{R}(p,p) = \frac{\partial}{\partial p_{\mu}}\Big(Z_2^{-1}\, S_R^{-1}(p)\Big) = Z_2^{-1}\,\frac{\partial S_R^{-1}}{\partial p_{\mu}}. \tag{22.154}$$

But the *renormalized* functions must themselves obey the same soft identity $\Gamma^{\mu}_{R}(p,p) = \partial S_R^{-1}/\partial p_{\mu}$ for the identity to survive renormalization — the Ward–Takahashi identity is a relation among finite, physical quantities and cannot acquire a divergent multiplicative anomaly in QED. Comparing Eq. (22.154) with this renormalized identity forces the multiplicative constants to agree, $Z_1^{-1} = Z_2^{-1}$, that is $Z_1 = Z_2$. Nothing in the argument referred to a particular order or gauge: the operator identity Eq. (22.152) is exact and gauge-covariant, so the conclusion holds to all orders and in every R_{ξ} gauge. □

The content of Theorem 22.31 is that the ultraviolet divergence dressing the vertex is, point for point, the divergence dressing the external electron line, no matter how complicated the loop corrections become. The two are not equal by accident of a one-loop computation; they are locked together by the symmetry, so that even though Z_1 and Z_2 are each separately divergent, scheme-dependent, and (as the explicit one-loop constants below show) gauge-dependent, their equality is exact.

Worked Example 22.32 ($Z_1 = Z_2$ at one loop, gauge by gauge)**.** The all-orders theorem is most convincing when checked against the explicit one-loop divergences, which are independent calculations of the vertex and the self-energy and have no *a priori* reason to agree. We import them from Chapter 19. The

electron self-energy Eq. (19.76), computed in Feynman gauge $\xi = 1$ in $d = 4 - 2\varepsilon$, has a divergent part with two Lorentz structures, one $\propto \not{p}$ and one $\propto m$,

$$\Sigma(p)\big|_{\text{div}} = -\frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon}\,\not{p} + c_m\,\frac{e^2}{\pi^2}\,\frac{1}{\varepsilon}\,m, \tag{22.155}$$

with c_m a pure number whose value (gauge dependent) we will not need; the $Z_1 = Z_2$ comparison lives entirely in the $\not{p}$ structure. Writing the renormalized inverse propagator as $S_R^{-1} = \not{p} - m - \Sigma + (\delta_2 \not{p} - \delta_m)$ and demanding finiteness, the field counterterm must cancel the $\not{p}$ pole:

$$\delta_2 = -\frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} \quad\Longrightarrow\quad Z_2 = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} + O(e^4) \qquad (\xi = 1), \tag{22.156}$$

in agreement with the constant Eq. (24.115) used in Chapter 24. The vertex correction Eq. (19.104), in the same gauge, has divergent part proportional to the tree structure γ^μ,

$$\Gamma^\mu(p', p)\big|_{\text{div}} = -\frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon}\,\gamma^\mu, \tag{22.157}$$

so that the vertex counterterm needed to make the renormalized vertex finite is

$$\delta_1 = -\frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} \quad\Longrightarrow\quad Z_1 = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} + O(e^4) \qquad (\xi = 1). \tag{22.158}$$

The two pole residues are identical: $\delta_1 = \delta_2$, hence $Z_1 = Z_2$ at one loop, confirming Eq. (22.151). The agreement is not a numerical coincidence of Feynman gauge. In a general R_ξ gauge both residues acquire the same ξ-dependence — $\delta_1 = \delta_2 = -(e^2/16\pi^2)(\xi)/\varepsilon$ when one tracks the gauge parameter through both integrals — so the equality $Z_1 = Z_2$ holds in every gauge, while the common value runs with ξ. This is exactly the structure the Ward–Takahashi argument predicts: the gauge-dependent, divergent pieces of vertex and self-energy move in lockstep, because the identity Eq. (22.152) relating them is gauge-covariant. Note finally the contrast with the mass: the m-proportional pole of Eq. (22.155) fixes the mass counterterm $\delta_m = (Z_2 Z_m - 1)m$ and hence the multiplicative constant $Z_m = 1 - (3e^2/16\pi^2)/\varepsilon$ used in Chapter 24 (Eq. (24.115)), an *independent* constant not tied to Z_1 or Z_2: the mass term breaks the chiral symmetry $\psi \to e^{i\theta\gamma^5}\psi$ of the massless theory and so is not protected by the vector Ward identity that locks Z_1 to Z_2.

22.7.6 Charge renormalization and universality

The payoff of Theorem 22.31 is immediate. Substituting $Z_1 = Z_2$ into the charge relation Eq. (22.147), the vertex and electron-field constants cancel identically,

$$\boxed{Z_e = \frac{Z_1}{Z_2\, Z_3^{1/2}} = Z_3^{-1/2}, \qquad e_0 = \mu^\varepsilon\, Z_3^{-1/2}\, e.} \tag{22.159}$$

The renormalization of the electric charge is governed by the photon-field renormalization *alone*. Everything to do with the electron — its self-energy, its field strength, the dressing of the vertex — has dropped out of the relation between bare and renormalized charge. The individual constants Z_1 and Z_2 are divergent, gauge-dependent, scheme-dependent objects, yet their ratio in Z_e is exactly unity, so none of that complexity ever reaches the physical coupling. The sole survivor is Z_3, the renormalization of the photon propagator computed from the vacuum polarization, which is gauge independent (its transverse coefficient feeds a physical quantity and cannot depend on the gauge parameter). The running of α is therefore a property of the photon, and Chapter 24 turns the single constant Z_3 into the carried number $\beta_e = e^3/12\pi^2$.

This structural fact has a physical name and a sharp experimental consequence: *charge universality*. Consider two distinct charged species coupled to the same photon — the electron and the muon are the textbook pair, but the statement holds for any field carrying $U(1)$ charge, of any mass and any spin. Each has its own self-energy and its own vertex correction; the electron's $Z_2^{(e)}$ and $Z_1^{(e)}$ are different numbers from the muon's $Z_2^{(\mu)}$ and $Z_1^{(\mu)}$, because the fields have different masses and different loop integrals. Yet for *each* species separately the Ward–Takahashi identity enforces $Z_1 = Z_2$, so the species-specific constants cancel in the charge relation and

$$\boxed{e_0 = \mu^{\varepsilon}\, Z_3^{-1/2}\, e \quad \text{with the } \textit{same } Z_3 \text{ for every charged species.}} \tag{22.160}$$

The renormalization factor relating bare to renormalized charge is the same $Z_3^{-1/2}$ for the electron, the muon, the tau, and any other field that couples to the photon, because Z_3 belongs to the photon, which they all share. If two fields carry equal bare charges they carry equal renormalized charges at every scale; the charge does not lose its universality under quantum corrections. This is the field-theoretic root of one of the most precisely tested facts in physics, that the proton and electron charges are equal and opposite to better than one part in 10^{21}, even though proton and electron are radically different bound and elementary objects: the equality is protected because charge renormalization is a property of the photon line, not of the matter that emits it.

Charge universality is the Ward identity

A composite particle and an elementary one renormalize their fields and vertices through utterly different loop physics, yet their charges renormalize identically. The reason is not a conspiracy of cancellations between many diagrams but a single symmetry statement, $Z_1 = Z_2$, holding separately for each charged field. Strip away the species-specific dressing and what remains is the photon's own $Z_3^{-1/2}$, universal by construction. When several species circulate in the vacuum-polarization loop they contribute

additively to Z_3, enlarging the running of α; but that is a property of the shared photon line, not a violation of universality — the same enlarged Z_3 acts on every charge. The universality of electric charge, taken in the nineteenth century as a primitive empirical fact, is in quantum field theory a *theorem*: it is the Ward–Takahashi identity, read at the level of renormalization constants. This is the conceptual root of the single running coupling $\alpha(\mu)$ that Chapter 20 introduced heuristically and Chapter 24 derives from the renormalization group.

22.7.7 Renormalizability of QED to all orders

We can now assemble the full statement and indicate the structure of its proof, paralleling the ϕ^4 induction of Theorem 22.28 and marking precisely where gauge symmetry, rather than power counting, carries the argument.

Theorem 22.33 (Renormalizability of QED)**.** *Quantum electrodynamics is renormalizable: every ultraviolet divergence, at every order of perturbation theory, is absorbed by the four renormalization constants Z_1, Z_2, Z_3, Z_m of Eqs. (22.146) and (22.147) — equivalently by the counterterms $\delta_1, \delta_2, \delta_3, \delta_m$ of Eq. (22.149), each multiplying an operator already present in the gauge-invariant Lagrangian Eq. (22.140). No new operator — in particular no photon mass $A_\mu A^\mu$ and no four-photon contact term — is ever generated. The Ward–Takahashi identity reduces the four constants to three independent ones via $Z_1 = Z_2$, so that the electric charge renormalizes through Z_3 alone.*

Structure of the proof. The argument has three layers, the first two purely combinatorial and shared with the scalar theory, the third supplied by gauge invariance.

(i) Power counting bounds the divergences. By Eq. (22.141) the superficial degree of divergence $D = 4 - \frac{3}{2}E_\psi - E_A$ depends only on the external legs, not on the loop order (Theorem 21.10), so by the finiteness criterion Theorem 21.13 only finitely many 1PI amplitudes are superficially divergent. Locality (Theorem 21.17) makes each primitive divergence a polynomial in external momenta and masses of degree at most D, hence a finite sum of local operators of dimension ≤ 4.

(ii) The forest formula handles subdivergences to all orders. A multiloop QED graph generally contains nested and overlapping subdivergences — a vertex correction inside a self-energy, a vacuum-polarization insertion inside a vertex, and so on — so the naive subtraction of the overall divergence is not enough. Zimmermann's forest formula, established in the BPHZ form of Theorem 22.22, organizes the subtractions into a recursion over the forests of divergent subgraphs and proves, via the locality of each nested counterterm, that the renormalized integrand is absolutely convergent. This is exactly the apparatus the ϕ^4 proof used; it is field-content-blind and transplants to QED unchanged, the only

new bookkeeping being the spinor and Lorentz indices, which do not affect the convergence theorem.

(iii) Gauge invariance closes the operator list. Power counting and locality together would leave QED needing counterterms for *every* dimension-≤ 4 Lorentz scalar built from the fields, including the photon mass $A_\mu A^\mu$ (dimension two) and the four-photon operators (dimension four), neither of which is in Eq. (22.140). Here gauge invariance is indispensable. Transversality (Theorem 22.30) annihilates the photon-mass divergence; the four-leg Ward identity (Section 22.7.3) renders light-by-light finite; Furry's theorem removes the odd-photon amplitudes. What survives is precisely the set of operators already in the Lagrangian, and the Ward–Takahashi identity (Theorem 22.31) further identifies two of the four constants. Crucially, the gauge identities are themselves stable under renormalization: a regulator respecting the symmetry (dimensional regularization, which preserves gauge invariance, used throughout with the master integrals of Chapter D and the gamma-algebra of Chapter C) generates only symmetry-respecting counterterms, so the structure persists order by order. By induction on the loop order — assuming all lower-order subdivergences subtracted by the forest formula with the four counterterms, the remaining overall divergence at order n is local, gauge-constrained to lie among the four existing operators, and removed by adjusting $\delta_1, \delta_2, \delta_3, \delta_m$ at order n — the theory is rendered finite to all orders with no new parameters. The number of parameters is finite and fixed, so the procedure closes. □

Two honest caveats complete the picture. First, the induction in step (iii) leaned on the statement that the gauge (Ward–Takahashi) identities are preserved by the regulator and hence by renormalization. For QED this is true and the argument above is essentially complete, because the abelian Ward–Takahashi identity is linear in the fields and survives quantization without modification. For non-abelian gauge theories the analogous identities — the Slavnov–Taylor identities — are *nonlinear*, and proving that they survive renormalization requires the full BRST formalism and the Zinn-Justin equation, a substantially deeper construction carried out in Chapter 26. QED is the abelian template: it exhibits every conceptual ingredient (symmetry constrains counterterms, the vertex and field renormalizations lock together, the charge runs through the gauge-field renormalization) in the simplest setting where the symmetry identities are linear and the all-orders proof is within reach by the elementary argument given here. Second, the proof establishes renormalizability *operationally* — that finitely many counterterms suffice — but says nothing yet about *why* a cutoff-dependent bare theory should have a finite, predictive low-energy limit at all. That deeper question, and the reinterpretation of renormalizability as a statement about the flow of couplings toward long distances, is the Wilsonian story of Chapter 23.

With Theorems 22.28 and 22.33 the operational claim of this chapter is established for both threads: ϕ^4 and QED are renormalizable, their divergences

absorbed to all orders by the counterterms already named, and in QED the charge renormalizes universally through the single photon constant Z_3. What remains is the conceptual heart — the Wilsonian explanation of *why* renormalizability is the generic low-energy outcome, taken up in Chapter 23 — and the quantitative payoff, the computation of the beta functions and anomalous dimensions from these very Z-factors, which Chapter 24 carries out, turning the photon constant Z_3 established here into the running coupling $\beta_e = e^3/12\pi^2$.

Further reading

The bare/counterterm formulation and the explicit one-loop computations are developed at the level of this chapter in Peskin and Schroeder [396] and Schwartz [432]; Itzykson and Zuber [278] remains an authoritative classical reference for multiplicative renormalization and the structure of the proofs. The all-orders combinatorics — subdivergences, the R-operation, and the forest formula — originate with Bogoliubov and Parasiuk [82], were completed by Hepp [266] and cast in momentum space by Zimmermann [533]; the convergence of the renormalized integrand rests on the power-counting theorem of Weinberg [494]. The definitive modern treatment of renormalization theory, including a careful account of overlapping divergences and gauge theories, is Collins [134, 135]; Weinberg [500] gives the all-orders structure with characteristic care. For the Ward–Takahashi identity and $Z_1 = Z_2$, see the original papers of Ward [489] and Takahashi [470].

Chapter 23

The Wilsonian Renormalization Group

The two chapters preceding this one taught a working algorithm for divergences and then, almost in the same breath, taught us to be slightly ashamed of it. A loop integral over the scalar of ϕ^4 theory diverged because nothing stopped the internal momenta from running to infinity (Chapter 21); we regulated, fixed counterterms by renormalization conditions, and sent the cutoff away, keeping a finite answer that remembered nothing of the scaffolding (Chapter 22). On that reading the cutoff is an embarrassment and renormalization is the art of its concealment. This chapter inverts the picture. Following Kadanoff and Wilson, we promote the cutoff Λ from a regulator to a definition: a quantum field theory is a local action *together with* a momentum scale above which we make no claims, and renormalization is the wholly physical operation of watching the description change as we lower Λ and coarse-grain away the fluctuations we have decided to stop resolving. This is the conceptual heart of Part VI, and it is placed before the Callan–Symanzik equation deliberately: the running couplings of Chapter 24 are the infinitesimal shadow of the coarse-graining flow built here.

The technical foundation is the Euclidean path integral. We work throughout in the convergent, positive Euclidean formulation of Chapter 10, carrying the frozen Euclidean action of ϕ^4 theory (Eq. (10.161)) and reading the whole construction as a refinement of the generating functional defined there. The functional-integral machinery itself — Gaussian integration, functional determinants, the Legendre transform — is imported from Chapter 4, and the power-counting dimensions that decide an operator's fate are exactly the superficial degrees of divergence classified in Chapter 21, now reinterpreted rather than re-derived. What was there a criterion for which diagrams diverge becomes here a criterion for which couplings survive coarse-graining.

The chapter follows a single arc from definition to fixed point. We first install the cutoff inside the partition function, prove the cutoff theory finite with

no subtraction, and identify the arena of the flow: *theory space*, the infinite-dimensional space of all symmetry-allowed couplings, stratified at the outset into relevant, marginal, and irrelevant directions by dimensional analysis. We then perform the elementary step of the renormalization group concretely — splitting the field into fast and slow modes and integrating out a thin momentum shell — and assemble the resulting corrections to the mass and the quartic, watching new operators be generated and seeing why they are small. Restoring the cutoff by rescaling closes the step into a transformation of theory space onto itself and produces the differential flow equations and the one-loop ϕ^4 beta function. Having the perturbative flow in hand, we lift it to the exact, nonperturbative statement: the Wegner–Houghton sharp-cutoff equation and Polchinski's smooth-cutoff equation, derived in full and shown to reproduce the shell flow.

With the flow defined, its geometry organizes everything that follows. Fixed points are the scale-invariant theories; linearizing the flow about one sorts the scaling fields into relevant, marginal, and irrelevant by the single relation $y_i = d - \Delta_i$, and the trichotomy fixes the dimension of the critical surface. This is the origin of universality — the forgetting of irrelevant couplings under coarse-graining, so that an entire basin of microscopic models flows to the same fixed point and shares its critical exponents, which we extract as the universal data of the fixed point. The chapter closes by re-reading renormalizability in this light: a theory is renormalizable, in the Wilsonian sense, when a predictive continuum limit requires tuning only the finitely many relevant and marginal couplings, the non-renormalizable interactions of Chapter 21 reappearing as the harmless irrelevant operators that organize the effective-field-theory expansion of Chapter 38. The capstone is the Wilson–Fisher fixed point in $d = 4 - \varepsilon$, a nontrivial fixed point computed start to finish, whose correlation-length exponent $1/\nu = 2 - \varepsilon/3$ is a genuine prediction of the method. It hands ϕ^4 theory forward in two directions at once: to Chapter 24, which recasts the same fixed point as a zero of the Callan–Symanzik beta function and confirms its scheme-independence, and to Chapter 52, where the conformal field theory living at the fixed point is studied by the nonperturbative bootstrap that takes over when ε is pushed to unity.

23.1 The Cutoff as Part of the Definition: Theory Space

The previous two chapters treated the ultraviolet cutoff as a guilty secret. A loop integral diverged (Chapter 21); we introduced a regulator to make it finite, engineered counterterms to absorb the cutoff dependence (Chapter 22), and then sent the cutoff to infinity, congratulating ourselves on a finite answer that remembered nothing of the scaffold. On that view the cutoff is an embarrassment

and the goal is its removal. This chapter takes the opposite stance, the one due to Kadanoff and Wilson: the cutoff is not an artefact to be exorcised but a physical statement about the shortest distance at which we claim to know the dynamics, and it belongs in the very definition of the theory. A quantum field theory is not a Lagrangian floating free of any scale; it is a Lagrangian *together with* a declaration of the momentum Λ above which we make no commitments. Once the cutoff is admitted as part of the definition, renormalization stops being the subtraction of infinities and becomes a far more physical operation: watching how the description of the same physics must change as we lower Λ, coarse-graining away the fluctuations we have decided to stop resolving. To set this up properly we must first say precisely what a cutoff theory *is*, show that it is finite before any limit is taken, and identify the arena in which the description lives. That arena — the infinite-dimensional space of all couplings consistent with the symmetries — is the subject of this section.

23.1.1 The Euclidean path integral with a momentum cutoff

We work throughout this chapter in Euclidean signature, where the path integral is a genuine, convergent, positive integral rather than an oscillatory one (Section 10.6). The frozen object is the Euclidean action of ϕ^4 theory, established at Eq. (10.161),

$$S_E[\phi] \;=\; \int \mathrm{d}^4x \,\Big[\tfrac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\Big], \tag{23.1}$$

with the index μ summed against the positive-definite metric $\delta_{\mu\nu}$, so $(\partial_\mu\phi)^2 = (\partial_\tau\phi)^2 + (\nabla\phi)^2 \geq 0$, and the symbols m (mass) and λ (quartic coupling, with the $1/4!$). We write d^4x for the Euclidean volume element, dropping the subscript E that distinguished it in Chapter 10 now that no Minkowski integral will appear in this chapter.

The naive generating functional of Eq. (10.162) integrates over *all* field configurations, including arbitrarily rapid oscillations. It is exactly those arbitrarily-short-wavelength modes that produced the divergences of Chapter 21: the loop integrals ran over momenta out to infinity, and the integrands fell off too slowly to converge there. The Wilsonian remedy is not to integrate out to infinity and then subtract, but to refuse to integrate past Λ in the first place. Decompose the field into Fourier modes,

$$\phi(x) \;=\; \int \frac{\mathrm{d}^4k}{(2\pi)^4}\, e^{ik\cdot x}\,\tilde{\phi}(k), \qquad k\cdot x = k_\mu x_\mu, \quad k^2 = k_\mu k_\mu \geq 0, \tag{23.2}$$

where, Euclidean signature having made the four-momentum norm positive-definite, $|k| \equiv \sqrt{k^2}$ is an honest length-scale^{-1}. We now *define* the cutoff field to be one whose Fourier support is bounded:

$$\tilde{\phi}(k) = 0 \quad \text{for} \quad |k| > \Lambda. \tag{23.3}$$

Configurations with structure finer than Λ^{-1} are simply not in the integration domain.

Definition 23.1 (The cutoff partition function)**.** Let $\Lambda > 0$ be a momentum scale and $S_\Lambda[\phi]$ a local Euclidean action. The *cutoff partition function* and *cutoff generating functional* are

$$Z_\Lambda \;=\; \int \mathcal{D}\phi\,\Big|_{|k|<\Lambda}\, e^{-S_\Lambda[\phi]}, \qquad Z_\Lambda[J] \;=\; \int \mathcal{D}\phi\,\Big|_{|k|<\Lambda}\, \exp\Big(-S_\Lambda[\phi] + \int \mathrm{d}^4x\, J\phi\Big), \tag{23.4}$$

where the restricted functional measure is the product over the retained modes only,

$$\mathcal{D}\phi\,\Big|_{|k|<\Lambda} \;\equiv\; \prod_{0<|k|<\Lambda} \mathrm{d}\tilde{\phi}(k), \tag{23.5}$$

the source J in $Z_\Lambda[J]$ being likewise band-limited to $|k| < \Lambda$. The associated correlators are

$$\langle \phi(x_1)\cdots\phi(x_n)\rangle_\Lambda \;=\; \frac{1}{Z_\Lambda}\int \mathcal{D}\phi\,\Big|_{|k|<\Lambda}\, \phi(x_1)\cdots\phi(x_n)\, e^{-S_\Lambda[\phi]}. \tag{23.6}$$

This is the Euclidean generating functional of Theorem 10.28 and Eq. (A.117), with one alteration: the integration runs over a *ball* in momentum space rather than all of it. Two remarks fix the meaning of Eq. (23.4). First, the source coupling carries no factor of i, exactly as in the Euclidean conventions of Section 10.6.3; correlators are extracted by bare functional derivatives $\delta/\delta J$. Second, the sharp restriction Eq. (23.3) is the simplest concrete choice but not the only one, and we will return below to the smooth alternative that the differential form of the renormalization group prefers.

The cutoff theory is finite. The single most important property of Eq. (23.4) — the property that justifies organizing the whole chapter around it — is that *every quantity it computes is finite, with no regulator and no subtraction.* There are no infinities to remove because the source of the infinities has been removed. We make this precise.

Proposition 23.2 (Finiteness of the cutoff path integral)**.** *Fix $\Lambda < \infty$ and a large Euclidean box of volume $V < \infty$ with periodic boundary conditions, and take $\lambda \geq 0$, $m^2 > 0$. Then:*

(i) the number of integration variables in Eq. (23.5) is finite, namely the number of lattice momenta $k = \frac{2\pi}{V^{1/4}}\,\mathbf{n}$ with $|k| < \Lambda$, which grows like $V\,\Lambda^4$;

(ii) Z_Λ is a finite, strictly positive number, and every correlator Eq. (23.6) is finite;

(iii) every Feynman integral generated by the perturbative expansion of Eq. (23.4) in λ is an integral of a bounded integrand over a compact domain, hence convergent.

Proof. (i) In a box of volume V the allowed momenta form a lattice of spacing $2\pi/V^{1/4}$ in each of the four Euclidean directions; the condition $|k| < \Lambda$ selects those lying inside a 4-ball of radius Λ. The count is the ball's volume divided by the cell volume, $\frac{1}{2}\pi^2\Lambda^4/(2\pi/V^{1/4})^4 = V\Lambda^4/(32\pi^2)$ to leading order, a finite integer for finite V and Λ. Reality of ϕ ties $\tilde{\phi}(-k) = \tilde{\phi}(k)^*$, so the independent real variables number half of this, still finite.

(ii) With finitely many variables, Eq. (23.4) is an ordinary multidimensional integral, not a functional one. Split the action into its quadratic and quartic parts. The quadratic part is, by the computation underlying Theorem 10.29, the positive-definite form

$$S_\Lambda^{(0)}[\phi] \;=\; \tfrac{1}{2}\int_{|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\,(k^2+m^2)\,|\tilde{\phi}(k)|^2 \;\geq\; \tfrac{1}{2}m^2\int_{|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\,|\tilde{\phi}(k)|^2 \;\geq\; 0, \tag{23.7}$$

each eigenvalue $k^2 + m^2 \geq m^2 > 0$ bounded away from zero. Thus $e^{-S_\Lambda^{(0)}}$ is a genuine Gaussian weight on the finite-dimensional space of retained modes, and the source-free Gaussian formula of Theorem 4.14 gives a finite, positive value for the free Z_Λ. The interaction supplies a factor $e^{-\frac{\lambda}{4!}\int\phi^4} \in (0,1]$ for $\lambda \geq 0$, which only improves convergence; multiplying a convergent integral by a bounded positive factor leaves it convergent and positive. Correlators Eq. (23.6) are ratios of such finite integrals (the polynomial insertion $\phi(x_1)\cdots\phi(x_n)$ grows only polynomially and cannot defeat the Gaussian tail), hence finite.

(iii) Each term of the expansion in λ is, by Wick's theorem applied to the Gaussian Eq. (23.7), a product of cutoff propagators integrated over the loop momenta. The cutoff propagator is

$$G_\Lambda(k) \;=\; \frac{\theta(\Lambda - |k|)}{k^2+m^2}, \tag{23.8}$$

which is bounded ($0 \leq G_\Lambda \leq 1/m^2$) and supported on the compact ball $|k| \leq \Lambda$. Every loop integral is therefore $\int_{|k_i|<\Lambda}\prod_i \mathrm{d}^4k_i$ of a bounded function over a bounded domain, and is finite by elementary measure theory. No subtraction is performed because none is needed. □

The contrast with Chapter 21 is exact and instructive. There, the same diagrams produced $\int^\infty \mathrm{d}^4k/(k^2+m^2)^2 \sim \ln(\infty)$; the integrand was integrable at each finite radius but the *upper limit* was infinite. Capping the limit at Λ removes the divergence at the root. The worked example below carries this out for the one diagram that drove Chapter 21, so that the reader sees the finite number and where the old logarithm hid.

Worked Example 23.3 (The one-loop bubble is finite at finite cutoff)**.** The amplitude that renormalized the coupling in Chapter 21 was the one-loop four-point bubble; at zero external momentum its loop integral is, up to combinatorial factors,

$$I(\Lambda) \;=\; \int_{|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^2}. \tag{23.9}$$

We evaluate it exactly. The integrand depends on $|k|$ alone, so we pass to four-dimensional spherical coordinates, where the solid angle integrates to the surface area of the unit 3-sphere, $\Omega_3 = 2\pi^2$, and $\mathrm{d}^4k = \Omega_3\,|k|^3\,\mathrm{d}|k| = 2\pi^2\,k^3\,\mathrm{d}k$ with $k \equiv |k|$:

$$I(\Lambda) \;=\; \frac{2\pi^2}{(2\pi)^4}\int_0^{\Lambda}\frac{k^3\,\mathrm{d}k}{(k^2+m^2)^2} \;=\; \frac{1}{8\pi^2}\int_0^{\Lambda}\frac{k^3\,\mathrm{d}k}{(k^2+m^2)^2}. \tag{23.10}$$

Substitute $u = k^2$, $\mathrm{d}u = 2k\,\mathrm{d}k$, so $k^3\,\mathrm{d}k = \tfrac{1}{2}u\,\mathrm{d}u$:

$$I(\Lambda) \;=\; \frac{1}{16\pi^2}\int_0^{\Lambda^2}\frac{u\,\mathrm{d}u}{(u+m^2)^2}. \tag{23.11}$$

The remaining integral is elementary: write $u = (u+m^2) - m^2$ in the numerator,

$$\int_0^{\Lambda^2}\frac{u\,\mathrm{d}u}{(u+m^2)^2} = \int_0^{\Lambda^2}\Big[\frac{1}{u+m^2} - \frac{m^2}{(u+m^2)^2}\Big]\mathrm{d}u = \Big[\ln(u+m^2) + \frac{m^2}{u+m^2}\Big]_0^{\Lambda^2}, \tag{23.12}$$

which gives

$$I(\Lambda) \;=\; \frac{1}{16\pi^2}\left[\ln\frac{\Lambda^2+m^2}{m^2} + \frac{m^2}{\Lambda^2+m^2} - 1\right]. \tag{23.13}$$

This is a perfectly finite number for every finite Λ. The famous one-loop coefficient $1/16\pi^2$ has emerged on the nose, multiplying a divergence-tracking logarithm *plus* cutoff-dependent finite pieces. Only when we attempt to send $\Lambda \to \infty$ does $I(\Lambda) = \frac{1}{16\pi^2}\ln(\Lambda^2/m^2) + O(\Lambda^0) \to \infty$, reproducing the logarithm of Section 21.6. The Wilsonian reading inverts the emphasis: the logarithm is not a pathology of the theory but a statement about how strongly the four-point coupling at low energy depends on the scale Λ at which we stopped resolving structure. We keep Λ finite and read the Λ-dependence as information, not as something to be subtracted away.

23.1.2 The Wilsonian effective action

Theorem 23.1 gives a cutoff theory *a* action S_Λ. The deeper notion we need turns this around. Suppose we are handed a definite physical theory — a definite set of long-distance correlation functions $\langle\phi(x_1)\cdots\phi(x_n)\rangle$ for points separated by more than Λ^{-1}. We may ask: what action, used in the cutoff path integral

with cutoff Λ, reproduces exactly these correlators? That action is the Wilsonian effective action at scale Λ, and it is the central object whose Λ-dependence the rest of the chapter studies.

Definition 23.4 (Wilsonian effective action)**.** Given an underlying theory, the *Wilsonian effective action* S_Λ at cutoff scale Λ is the local Euclidean functional of the band-limited field ($|k| < \Lambda$) for which the cutoff path integral Eq. (23.4) reproduces all correlators of the theory at external momenta below Λ:

$$\left\langle \tilde{\phi}(p_1)\cdots\tilde{\phi}(p_n)\right\rangle \;=\; \frac{1}{Z_\Lambda}\int \mathcal{D}\phi\,\Big|_{|k|<\Lambda}\,\tilde{\phi}(p_1)\cdots\tilde{\phi}(p_n)\,e^{-S_\Lambda[\phi]}, \qquad |p_i|<\Lambda. \tag{23.14}$$

Equivalently, S_Λ is fixed by the requirement that $Z_\Lambda[J]$ of Eq. (23.4), restricted to band-limited sources, equals the generating functional of the underlying theory for those sources.

Two features of this definition deserve emphasis, because they are the conceptual pivot of the chapter. First, S_Λ is required to be *local* — a spacetime integral of a polynomial in ϕ and its derivatives, evaluated at a single point — and this is a genuine restriction, not a convention; it is the statement that the underlying physics has no action-at-a-distance built in below Λ^{-1}. We will see in the next section that integrating out a shell of modes preserves locality only approximately, generating an expansion in derivatives whose terms are increasingly irrelevant. Second, S_Λ is not unique as a function of the field unless we also fix the normalization of ϕ; the freedom to rescale the field is the origin of the anomalous dimension and is dealt with carefully when we close the RG step. For the present section the point is structural: *a single underlying theory assigns a definite action S_Λ to every scale Λ*, and lowering Λ changes S_Λ in a definite way. The map $\Lambda \mapsto S_\Lambda$ is the renormalization-group trajectory; the next section computes its first step. The action one writes down at the very top, at the largest cutoff Λ_0 where one's microscopic description is posited to hold, is the *bare action* S_{Λ_0} — one particular point on this trajectory, with no privileged status beyond being the place we chose to start.

Physical Insight

The bare action is not "the theory." The theory is the entire trajectory $\{S_\Lambda\}$, and any one S_Λ encodes the same physics as any other — it has merely integrated out more or fewer of the short-distance fluctuations. Asking "what is the true value of λ?" is like asking for the true magnification of a microscope: the honest answer specifies the scale at which the question is posed. This is what it means to say renormalization is a change of description, not a calculation of a hidden truth.

23.1.3 Theory space

To speak of a trajectory we need the space it moves in. The Wilsonian action S_Λ is local and respects the symmetries of ϕ^4 theory — Euclidean rotation and translation invariance, and the internal $\mathbb{Z}_2$ reflection $\phi \to -\phi$ (Eq. (10.161) is built to be invariant under all three). The most general such action is a sum over *all* local operators allowed by these symmetries, each with its own coupling. There is no reason the trajectory should stay inside the three-term form Eq. (23.1): integrating out modes will, as we will see, generate ϕ^6, $\phi^2(\partial\phi)^2$, and indeed every symmetry-allowed term. So we must work in the space of all of them from the outset.

Definition 23.5 (Theory space)**.** The *theory space* of ϕ^4 theory is the space of all local Euclidean actions of the band-limited scalar that are invariant under Euclidean rotations and translations and under $\mathbb{Z}_2$: $\phi \to -\phi$. Writing a complete basis of such local operators as $\{\mathcal{O}_i\}$ — monomials built from an even number of factors of ϕ and an even number of derivatives, contracted into a rotation scalar — the general action is

$$S_\Lambda[\phi] \;=\; \int \mathrm{d}^4x \,\sum_i g_i\,\mathcal{O}_i(x) \;=\; \int \mathrm{d}^4x \,\Big[\tfrac{1}{2}Z(\partial_\mu\phi)^2+\tfrac{1}{2}m^2\phi^2+\frac{\lambda}{4!}\phi^4+\frac{c_6}{6!}\phi^6+\frac{z_4}{2}\,\phi^2(\partial_\mu\phi)^2+\cdots\Big], \tag{23.15}$$

and a point of theory space is the (infinite) tuple of dimensionless couplings

$$\hat{g}_i \;\equiv\; g_i\,\Lambda^{[\mathcal{O}_i]-d} \;=\; g_i\,\Lambda^{-[g_i]}, \tag{23.16}$$

each coupling measured in units of the cutoff so as to be a pure number. Here $[\mathcal{O}_i]$ is the mass dimension of the operator and $[g_i] = d - [\mathcal{O}_i]$ that of its coupling, by Theorem 21.9.

The reason we work with the dimensionless $\hat{g}_i$ rather than the dimensionful g_i is that the renormalization group is fundamentally about scaling, and only dimensionless numbers can sit at a scale-invariant fixed point. The cutoff Λ is the only scale in the cutoff theory, so dividing out the appropriate power of Λ is the unique way to form a pure number from each coupling. With this convention the kinetic normalization Z is itself dimensionless and is conventionally fixed to $Z = 1$ by rescaling ϕ; the deviation of Z from 1 under the flow is the field renormalization that feeds the anomalous dimension of Chapter 24.

What organizes theory space into a usable structure is the mass dimension of each coupling, and this we can compute outright. The single input is that the action is dimensionless (it appears in e^{-S}), so the Lagrangian density carries dimension $[\mathcal{L}] = d$, and the canonical dimension of the scalar field, fixed by its kinetic term at Eq. (21.49), is

$$[\phi] \;=\; \frac{d-2}{2} \;\xrightarrow{d=4}\; 1. \tag{23.17}$$

A generic $\mathbb{Z}_2$-even, rotation-invariant local operator is built from $2n$ factors of ϕ and $2p$ derivatives (both counts even: an odd number of fields would violate $\mathbb{Z}_2$, an odd number of derivatives could not contract to a scalar). Its dimension is

$$[\mathcal{O}_{n,p}] \;=\; 2p + 2n\,[\phi] \;=\; 2p + n(d-2) \;\xrightarrow{d=4}\; 2p + 2n, \tag{23.18}$$

and by Eq. (21.53) its coupling has the complementary dimension

$$[g_{n,p}] \;=\; d - [\mathcal{O}_{n,p}] \;=\; d - 2p - n(d-2) \;\xrightarrow{d=4}\; 4 - 2p - 2n. \tag{23.19}$$

This single formula partitions theory space into three sectors, and the partition is computed, not asserted.

Definition 23.6 (Relevant, marginal, irrelevant directions)**.** A coupling direction $\mathcal{O}_i$ in theory space is *relevant* if $[g_i] > 0$, *marginal* if $[g_i] = 0$, and *irrelevant* if $[g_i] < 0$. Equivalently, by Eq. (23.19), $\mathcal{O}_i$ is relevant, marginal, or irrelevant according as its operator dimension $[\mathcal{O}_i]$ is less than, equal to, or greater than the spacetime dimension d.

The names anticipate the dynamics of the next section but the classification is purely dimensional and available now. The mnemonic content is read off Eq. (23.19): in $d = 4$ a coupling is relevant when $2p + 2n < 4$, marginal when $2p + 2n = 4$, and irrelevant when $2p + 2n > 4$. Because $p, n \geq 0$ are integers, the relevant-or-marginal condition $p + n \leq 2$ has only finitely many solutions, while the irrelevant condition $p + n > 2$ has infinitely many. *Theory space has finitely many relevant and marginal directions and infinitely many irrelevant ones* — a fact whose dynamical consequence, that low-energy physics is controlled by a handful of numbers, is universality, and whose static face is the classification of the next worked example.

Worked Example 23.7 (Enumerating the theory space of ϕ^4 theory)**.** We list the lowest $\mathbb{Z}_2$-even, rotation-invariant local operators of the scalar in $d = 4$, compute each dimension from Eqs. (23.18) and (23.19), and classify each by Theorem 23.6. Recall $[\phi] = 1$, $[\partial_\mu] = 1$ by Theorem A.1.

The relevant and marginal directions (the finite list).

- $\mathcal{O} = \mathbf{1}$ (the identity, $n = p = 0$): $[\mathcal{O}] = 0$, $[g] = 4$. This is the constant term, the vacuum energy density. It is the *most* relevant direction of all, yet it is invisible: a field-independent additive constant in S_Λ cancels in every normalized correlator Eq. (23.6), exactly as the $E = 0$ vacuum diagram dropped out in Chapter 21. It returns to life only when gravity is coupled, where it is the cosmological constant.
- $\mathcal{O} = \phi^2$ ($n = 1, p = 0$): $[\mathcal{O}] = 2$, $[g] = [m^2] = 2 > 0$. **Relevant.** This is the mass term of Eq. (23.1); its coupling is m^2. Its positive dimension is why the mass is the parameter one must *tune* to reach a critical point.

- $\mathcal{O} = (\partial_\mu \phi)^2$ $(n = 1, p = 1)$: $[\mathcal{O}] = 4$, $[g] = 0$. **Marginal.** This is the kinetic term; its coupling is the field normalization Z, fixed to 1 by convention. Its marginality is why field rescaling and the anomalous dimension live at the same order as the running of λ.
- $\mathcal{O} = \phi^4$ $(n = 2, p = 0)$: $[\mathcal{O}] = 4$, $[g] = [\lambda] = 0$. **Marginal.** This is the quartic of ϕ^4 theory; its coupling is λ. That λ is exactly marginal in $d = 4$ is the reason ϕ^4 sits precisely on the boundary between renormalizable and non-renormalizable, and the reason its running is only logarithmic — the entire subtlety the rest of Part VI exists to resolve.

These four operators (three of them physical) exhaust the relevant-and-marginal sector. The bare action Eq. (23.1) switches on exactly the three physical ones and sets all others to zero — it is the point of theory space with m^2, λ generic, $Z = 1$, and every higher coupling vanishing.

The irrelevant directions (the infinite tail).

- $\mathcal{O} = \phi^6$ $(n = 3, p = 0)$: $[\mathcal{O}] = 6$, $[g] = [c_6] = -2 < 0$. **Irrelevant.** Its coupling carries an inverse-mass-squared, $\hat{c}_6 = c_6 \Lambda^2$.
- $\mathcal{O} = \phi^2 (\partial_\mu \phi)^2$ $(n = 2, p = 1)$: $[\mathcal{O}] = 6$, $[g] = [z_4] = -2$. **Irrelevant.**
- $\mathcal{O} = (\partial^2 \phi)^2 = (\partial_\mu \partial_\nu \phi)(\partial^\mu \partial^\nu \phi)$ $(n = 1, p = 2)$: $[\mathcal{O}] = 6$, $[g] = -2$. **Irrelevant.** A higher-derivative kinetic correction.
- $\mathcal{O} = \phi^8$ $(n = 4, p = 0)$: $[\mathcal{O}] = 8$, $[g] = -4$. **Irrelevant**, and more so.

The pattern is clear from Eq. (23.19): every operator with $p + n \geq 3$ is irrelevant, with dimensions $[g] = 4 - 2(p + n)$ marching off to $-\infty$. There are infinitely many of them, and they are precisely the operators that a naive power count (Eq. (21.58)) flags as making the theory non-renormalizable. The Wilsonian framework will recast that liability as a virtue: their negative dimensions mean they are washed out under coarse-graining, which is why a sensible low-energy theory survives despite their presence.

We collect the result of the enumeration in Table 23.1, which is the periodic table of ϕ^4 theory space.

23.1.4 The bare action as a point; lowering the cutoff as a trajectory

We can now state the geometric picture that organizes the chapter. Theory space is the infinite-dimensional space coordinatized by the dimensionless couplings $\{\hat{g}_i\}$ of Eq. (23.16). A choice of bare action — say the three-term ϕ^4 action Eq. (23.1) at cutoff Λ_0 — is a single point of this space, sitting in the finite

Table 23.1: The lowest $\mathbb{Z}_2$-even, rotation-invariant local operators of the scalar field in $d = 4$, with $[\phi] = 1$. Operator dimension from Eq. (23.18); coupling dimension from Eq. (23.19); class from Theorem 23.6. The relevant/marginal block is finite; the irrelevant block continues forever.

Operator $\mathcal{O}_{n,p}$	coupling	$[\mathcal{O}]$	$[g]$	class
$\mathbf{1}$	vacuum energy	0	$+4$	relevant (invisible)
ϕ^2	m^2	2	$+2$	relevant
$(\partial_\mu\phi)^2$	Z	4	0	marginal
ϕ^4	λ	4	0	marginal
ϕ^6	c_6	6	-2	irrelevant
$\phi^2(\partial_\mu\phi)^2$	z_4	6	-2	irrelevant
$(\partial^2\phi)^2$	—	6	-2	irrelevant
ϕ^8	c_8	8	-4	irrelevant
$\vdots$		$\vdots$	$\vdots$	$\vdots$

relevant-and-marginal subspace with all irrelevant coordinates set to zero. The act of integrating out the modes in a thin shell $b\Lambda < |k| < \Lambda$, with $b < 1$, and re-expressing the result as a new local action $S_{b\Lambda}$ for the surviving modes (Theorem 23.4), moves the point: the surviving couplings shift, and operators that were absent are switched on, because the shell integration is blind to the $\mathbb{Z}_2$ and rotation symmetries' preference for the simple starting form and respects only the symmetries themselves. Iterating, Λ sweeps downward and the point traces a continuous curve, the renormalization-group trajectory, through theory space. Figure 23.1 renders the picture.

23.1.5 Re-reading the divergences of Chapter 21: insensitivity to the UV completion

The vantage of theory space lets us reinterpret the central computational fact of Chapter 21 — that the loop integrals diverged as $\Lambda \to \infty$ — as a statement of physics rather than a defect. Recall the dimension-explicit power-counting result: a coupling of negative dimension worsens the ultraviolet behaviour, so that a theory built on irrelevant (non-renormalizable) operators generates an unbounded proliferation of divergences as the order increases. In the language of Theorem 23.6, the divergences live on the irrelevant directions. Read forward, from a fixed bare action toward shorter distances, this is the familiar bad news: irrelevant couplings demand ever more counterterms.

Read in the Wilsonian direction — *downward* in Λ, toward longer distances, which is the physical direction in which we actually make measurements — it is good news of exactly the opposite character. Suppose two different microscopic

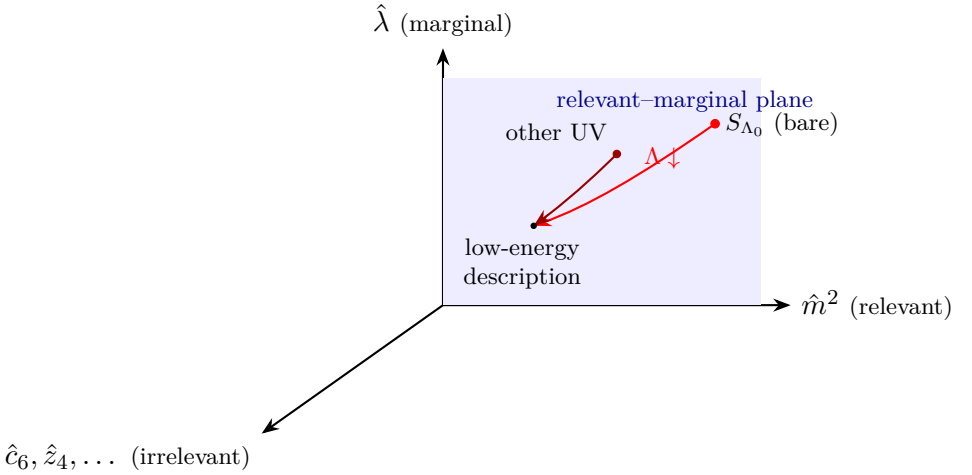


Figure 23.1: Theory space and a renormalization-group trajectory. Each axis is a dimensionless coupling Eq. (23.16); the finitely many relevant and marginal couplings span the lightly shaded plane, the infinitely many irrelevant couplings the remaining (here compressed) directions. The bare action S_{Λ_0} is one point; lowering the cutoff Λ integrates out short-distance modes and flows the point. Different ultraviolet starting points, differing in their irrelevant content, are drawn into the same neighbourhood as Λ decreases — the geometric face of universality.

completions are posited at the high cutoff Λ_0: same relevant and marginal couplings, but different irrelevant content (different c_6, z_4, ...), or even a completely different short-distance regulator such as a lattice in place of the sharp ball Eq. (23.3). Their dimensionless irrelevant couplings $\hat{g}_i = g_i \Lambda^{[\mathcal{O}_i]-d}$ carry positive powers $[\mathcal{O}_i] - d > 0$ of Λ. As we coarse-grain from Λ_0 down to a measurement scale $\mu \ll \Lambda_0$, those couplings are measured in units of the current, lower cutoff, and so are suppressed by powers of $(\mu/\Lambda_0)^{[\mathcal{O}_i]-d} \to 0$. The detailed short-distance data — the precise UV completion — is forgotten. Quantitatively, the bubble of Theorem 23.3 made the same point at one loop: its dependence on Λ was only logarithmic, and any two regulators agreed on the coefficient $1/16\pi^2$ of the log while disagreeing only on the regulator-dependent additive constant (compare Section 21.1.3, where the cutoff and dimensional-regularization computations shared the universal $1/16\pi^2$ and differed only in scheme junk). Low-energy physics at scale μ depends on the UV completion only through the relevant and marginal couplings, plus corrections suppressed by powers of μ/Λ_0.

Physical Insight

This is universality, glimpsed before we have computed a single flow. The reason a continuum quantum field theory can be predictive despite our total ignorance of physics at the Planck scale, and the reason wildly different microscopic systems (a real magnet, a lattice spin model, a scalar field) share identical critical exponents, are the same reason: irrelevant couplings are washed out under coarse-graining, so only the finite relevant–marginal data survives to long distances. The infinities of Chapter 21 were the theory's way of telling us that the short-distance details do not matter — we were merely reading the message with the sign flipped.

This reframing also delivers the modern meaning of *renormalizability*, which the chapter will sharpen into a statement about the dimensionality of the surface that flows out of a fixed point: a renormalizable theory is one whose long-distance physics is controlled by finitely many parameters precisely because the relevant–marginal subspace of Table 23.1 is finite-dimensional. We have here computed that this subspace is four-dimensional for the $\mathbb{Z}_2$ scalar in $d = 4$ (three physical directions plus the invisible identity), which is exactly the count of independent renormalizations — mass, field, coupling — that Chapter 22 carried out by brute force.

We have reframed a quantum field theory as a point in the infinite-dimensional theory space of Theorem 23.5, equipped with an explicit, finite cutoff (Theorem 23.2), and identified the Wilsonian action S_Λ of Theorem 23.4 as the coordinate of that point at resolution Λ. What remains is to compute the motion. Given S_Λ, the next section performs the first half of a renormalization-group step: it splits the field at an intermediate scale $b\Lambda$, integrates the high-momentum shell $b\Lambda < |k| < \Lambda$ exactly in principle and to one loop in practice, and reads off the coarse-grained action $S_{b\Lambda}$ — the shifted couplings and the newly generated operators that make the point of Fig. 23.1 actually move.

23.2 Integrating Out a Momentum Shell: The Wilson Effective Action

The previous section reframed ϕ^4 theory as a single point in the infinite-dimensional theory space of Theorem 23.5, equipped with the explicit, finite cutoff path integral Eq. (23.4), and identified the Wilsonian action S_Λ of Theorem 23.4 as the coordinate of that point at resolution Λ. What was promised, and what is owed here, is the *motion*: a concrete recipe that takes S_Λ and produces the action $S_{b\Lambda}$ describing the same physics at a coarser resolution $b\Lambda$, with $0 < b < 1$. This section carries out the first half of one renormalization-group step. We split the field at the intermediate scale $b\Lambda$ into a long-wavelength part we keep and a

short-wavelength part we integrate away; we perform that integration exactly in principle and to the leading nontrivial order in λ in practice; and we read off the coarse-grained action, finding that its couplings are shifted and that operators absent from Eq. (23.1) are switched on. Because the modes we integrate live in a bounded shell of momentum, every integral we meet is manifestly finite — there are no infinities to regulate, only a finite, physical change of description. The output is a new action at cutoff $b\Lambda$; restoring the cutoff to Λ, the business of Section 23.3, turns this half-step into an honest self-map of theory space.

23.2.1 The field split and the exact integrating-out formula

The cutoff field of Eq. (23.3) has Fourier support on the ball $|k| < \Lambda$. Pick an intermediate scale $b\Lambda$ with $0 < b < 1$ and partition that ball into an inner ball and an outer shell. The *slow* (low-momentum) field carries the modes we retain, the *fast* (high-momentum) field carries the shell we will remove:

$$\phi(x) \;=\; \phi_<(x) + \phi_>(x), \qquad \begin{cases} \tilde\phi_<(k) = \tilde\phi(k)\,, & 0 < |k| < b\Lambda, \\ \tilde\phi_>(k) = \tilde\phi(k)\,, & b\Lambda < |k| < \Lambda, \end{cases} \tag{23.20}$$

each component vanishing in Fourier space wherever the other is supported. The geometry is the annulus of Fig. 23.2: $\phi_<$ lives on the inner disk, $\phi_>$ on the shell. Because the supports are disjoint, the split Eq. (23.20) is an exact orthogonal decomposition of the space of cutoff configurations, and the functional measure factorizes accordingly,

$$\mathcal{D}\phi\,\Big|_{|k|<\Lambda} \;=\; \mathcal{D}\phi_<\,\Big|_{|k|<b\Lambda}\;\mathcal{D}\phi_>\,\Big|_{b\Lambda<|k|<\Lambda}, \tag{23.21}$$

the product over retained modes Eq. (23.5) splitting into the product over inner modes times the product over shell modes.

The Wilsonian action $S_{b\Lambda}$ at the lowered cutoff is, by Theorem 23.4, the action for the surviving field $\phi_<$ that reproduces all correlators at external momenta below $b\Lambda$. Such correlators involve only $\phi_<$ insertions, so we may carry out the $\phi_>$ integration once and for all, leaving an effective weight for $\phi_<$. Writing the full action evaluated on the split configuration as $S_\Lambda[\phi_< + \phi_>]$, this is the definition

Definition 23.8 (Integrating out the shell)**.** The coarse-grained action $S_{b\Lambda}[\phi_<]$ at cutoff $b\Lambda$ obtained from S_Λ by removing the shell $b\Lambda < |k| < \Lambda$ is

$$e^{-S_{b\Lambda}[\phi_<]} \;=\; \int \mathcal{D}\phi_>\,\Big|_{b\Lambda<|k|<\Lambda}\, e^{-S_\Lambda[\phi_<+\phi_>]}. \tag{23.22}$$

Three remarks fix the content of Eq. (23.22). First, it is an *exact* identity, not an approximation: substituting it into the cutoff path integral Eq. (23.4)

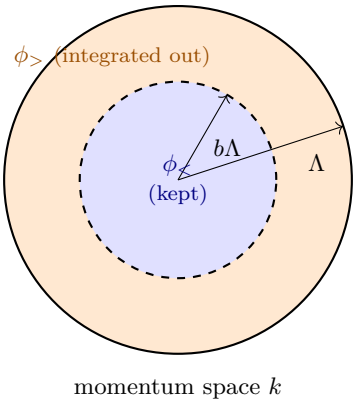


Figure 23.2: The field split Eq. (23.20) in momentum space. The slow field $\phi_<$ occupies the inner ball $|k| < b\Lambda$ (light blue); the fast field $\phi_>$ occupies the shell $b\Lambda < |k| < \Lambda$ (orange). Integrating out $\phi_>$ removes the shell and produces a new action for $\phi_<$ alone at the lowered cutoff $b\Lambda$.

and using the factorized measure Eq. (23.21) reproduces Z_Λ identically, because all that has happened is that the $\phi_>$ integral has been done first,

$$Z_\Lambda = \int \mathcal{D}\phi_< \, \mathcal{D}\phi_> \, e^{-S_\Lambda[\phi_< + \phi_>]} = \int \mathcal{D}\phi_< \Big(\int \mathcal{D}\phi_> \, e^{-S_\Lambda[\phi_< + \phi_>]} \Big) = \int \mathcal{D}\phi_< \, e^{-S_{b\Lambda}[\phi_<]} = Z_{b\Lambda}. \tag{23.23}$$

The partition function is unchanged, and so is every low-energy correlator: integrating out the shell is a reorganization of the path integral, not a modification of the theory. This is the precise sense in which S_Λ and $S_{b\Lambda}$ encode the same physics, anticipated in the insight following Theorem 23.4. Second, the integral on the right of Eq. (23.22) runs over the bounded shell only, so by exactly the argument of Theorem 23.2 it is a finite, convergent integral over a compact momentum domain for every $\phi_<$: *the integrating-out operation is manifestly free of ultraviolet divergences*, because the only modes it touches are those between $b\Lambda$ and Λ. Third, the object defined by Eq. (23.22) is, in the language of Section 4.5, a partial effective action: a Legendre-type functional in which some modes (the fast ones) have been integrated while others (the slow ones) remain as arguments. The full machinery of connected and one-particle-irreducible generating functionals built there applies verbatim, with $\phi_>$ playing the role of the quantum field and $\phi_<$ the role of a fixed background.

Physical Insight

Equation Eq. (23.22) is the whole renormalization group in one line. Everything that follows — the running couplings of Chapter 24, the fixed points and critical exponents later in this chapter, the modern meaning

of renormalizability — is the systematic study of how the functional $S_{b\Lambda}$ differs from S_Λ. The hard part is never the principle; it is carrying out the shell integral and recognizing the result as a new local action. We now do exactly that for ϕ^4 theory.

23.2.2 Splitting the action: the Gaussian core and the mixing vertices

To evaluate Eq. (23.22) we must write $S_\Lambda[\phi_< + \phi_>]$ in a form that separates the $\phi_>$-dependence. Start from the bare action Eq. (23.1) and decompose it into its quadratic (free) and quartic (interacting) parts, $S_\Lambda = S_0 + S_{\rm int}$ with

$$S_0[\phi] = \tfrac{1}{2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,(k^2+m^2)\,|\tilde\phi(k)|^2, \qquad S_{\rm int}[\phi] = \frac{\lambda}{4!}\int \mathrm{d}^4x\,\phi^4. \tag{23.24}$$

The quadratic part is diagonal in momentum, and this is the decisive simplification: evaluated on $\phi = \phi_< + \phi_>$, its Fourier integrand at each k involves either $\tilde\phi_<(k)$ or $\tilde\phi_>(k)$ but never both, since their supports are disjoint. Hence $|\tilde\phi_< + \tilde\phi_>|^2 = |\tilde\phi_<|^2 + |\tilde\phi_>|^2$ pointwise in k, with *no cross term*, and

$$S_0[\phi_< + \phi_>] \;=\; S_0[\phi_<] + S_0[\phi_>]. \tag{23.25}$$

The free action does not couple slow to fast modes at all: a free field is a collection of independent oscillators, one per momentum, and partitioning the momenta partitions the oscillators. All coupling between $\phi_<$ and $\phi_>$ therefore resides in the interaction. Expanding the quartic by the binomial theorem,

$$(\phi_< + \phi_>)^4 = \phi_<^4 + 4\phi_<^3\phi_> + 6\phi_<^2\phi_>^2 + 4\phi_<\phi_>^3 + \phi_>^4, \tag{23.26}$$

the first term $\phi_<^4$ depends on $\phi_<$ only and the last $\phi_>^4$ on $\phi_>$ only, while the three middle terms mix the two. Substituting Eq. (23.25) and Eq. (23.26) into Eq. (23.22) and pulling every purely-$\phi_<$ factor outside the $\phi_>$ integral,

$$e^{-S_{b\Lambda}[\phi_<]} = e^{-S_0[\phi_<] - \frac{\lambda}{4!}\int \phi_<^4} \int \mathcal{D}\phi_>\; e^{-S_0[\phi_>]}\; e^{-S_{\rm mix}[\phi_<,\phi_>]}, \tag{23.27}$$

where the *mixing action* collects the terms containing at least one fast field,

$$S_{\rm mix}[\phi_<,\phi_>] = \int \mathrm{d}^4x\Big[\frac{\lambda}{6}\,\phi_<^3\phi_> + \frac{\lambda}{4}\,\phi_<^2\phi_>^2 + \frac{\lambda}{6}\,\phi_<\phi_>^3 + \frac{\lambda}{4!}\,\phi_>^4\Big], \tag{23.28}$$

the numerical coefficients being $\frac{\lambda}{4!}$ times the binomial weights $4, 6, 4, 1$ of Eq. (23.26). The $\phi_>$-integral in Eq. (23.27) is a Gaussian average weighted by the shell free action $S_0[\phi_>]$, perturbed by $S_{\rm mix}$. We make this explicit. Define the normalized *shell average* of any functional $F[\phi_>]$,

$$\langle F\rangle_> \;\equiv\; \frac{1}{\mathcal{N}_>}\int \mathcal{D}\phi_>\Big|_{b\Lambda<|k|<\Lambda} e^{-S_0[\phi_>]}\,F[\phi_>], \qquad \mathcal{N}_> = \int \mathcal{D}\phi_>\; e^{-S_0[\phi_>]}, \tag{23.29}$$

a genuine, finite, positive Gaussian expectation by Theorem 4.14, the shell being compact. This is the same finite-dimensional Gaussian integral analyzed in Section 4.2; its only novelty is the restricted momentum support. The constant $\mathcal{N}_>$ is field-independent and contributes only an additive constant to $S_{b\Lambda}$ — a vacuum-energy shift, the invisible identity direction of Table 23.1 — which we drop. Wick's theorem for this Gaussian (Section 4.2.5) is governed by the single two-point function, the *shell propagator*

$$\langle \phi_>(x)\, \phi_>(y) \rangle_> = G_>(x-y) = \int_{b\Lambda<|k|<\Lambda} \frac{\mathrm{d}^4 k}{(2\pi)^4} \, \frac{e^{ik\cdot(x-y)}}{k^2+m^2}, \tag{23.30}$$

which is just the cutoff propagator Eq. (23.8) with its support narrowed from the ball to the shell. With these definitions, Eq. (23.27) becomes

$$S_{b\Lambda}[\phi_<] = S_0[\phi_<] + \frac{\lambda}{4!}\int \mathrm{d}^4 x\, \phi_<^4 \;-\; \ln \langle e^{-S_{\mathrm{mix}}[\phi_<,\phi_>]} \rangle_>, \tag{23.31}$$

up to the dropped constant. Everything quantitative in this section is the evaluation of the last term.

23.2.3 The cumulant expansion and the linked-cluster theorem

The remaining object $\ln\langle e^{-S_{\mathrm{mix}}}\rangle_>$ is the logarithm of a Gaussian average of an exponential, exactly the structure whose logarithm is generated by *connected* averages. This is the cumulant (or linked-cluster) expansion, established for the generating functional in Section 4.4.3: for any random variable W with a well-defined moment-generating function,

$$\ln \langle e^{-W} \rangle = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \langle W^n \rangle_c = -\langle W\rangle_c + \tfrac{1}{2}\langle W^2\rangle_c - \tfrac{1}{6}\langle W^3\rangle_c + \cdots, \tag{23.32}$$

where $\langle\cdot\rangle_c$ denotes the connected part: $\langle W\rangle_c = \langle W\rangle$, $\langle W^2\rangle_c = \langle W^2\rangle - \langle W\rangle^2$, and so on. Applying Eq. (23.32) with $W = S_{\mathrm{mix}}$ and inserting into Eq. (23.31),

$$S_{b\Lambda}[\phi_<] = S_0[\phi_<] + \frac{\lambda}{4!}\int \phi_<^4 + \langle S_{\mathrm{mix}}\rangle_> - \tfrac{1}{2}\langle S_{\mathrm{mix}}^2\rangle_{>,c} + \tfrac{1}{6}\langle S_{\mathrm{mix}}^3\rangle_{>,c} - \cdots. \tag{23.33}$$

The linked-cluster theorem (Section 4.4.3) guarantees that only *connected* averages survive in the logarithm; disconnected pieces, which would exponentiate vacuum bubbles, are removed by the ln and would in any case be cancelled by the normalization $\mathcal{N}_>$. Each connected average is, by Wick's theorem, a sum of products of shell propagators $G_>$ joining the $\phi_>$ fields of the S_{mix} insertions, with the leftover $\phi_<$ fields standing as external legs. Diagrammatically: draw a vertex for each factor of S_{mix}, attach $\phi_<$ legs and $\phi_>$ stubs as dictated by Eq. (23.28), and connect the $\phi_>$ stubs in all connected ways using $G_>$. Every

internal line is a shell propagator and therefore carries a momentum restricted to $b\Lambda < |k| < \Lambda$. *This is the engine of the whole computation:* the loop integrals are not over all momenta but over the shell, hence finite, and each successive cumulant is suppressed by a further power of λ.

Before extracting corrections, read off the leading term. At order λ^0 the mixing action $S_{\rm mix}$ does not appear at all, so $\ln\langle e^{-S_{\rm mix}}\rangle_> \to 0$ and Eq. (23.33) reduces to

$$S_{b\Lambda}[\phi_<] \;=\; S_0[\phi_<] + \frac{\lambda}{4!}\int \mathrm{d}^4x\,\phi_<^4 \;+\; O(\lambda^2) \;=\; S_\Lambda[\phi_<] + O(\lambda^2). \tag{23.34}$$

At tree level the coarse-grained action is the original action with ϕ replaced by $\phi_<$: the mass, the kinetic term, and the quartic all pass through unchanged. This had to be so — with the interaction switched off the fast and slow modes decouple by Eq. (23.25), and integrating out a decoupled Gaussian leaves the surviving action untouched apart from the constant $\mathcal{N}_>$. All flow is a loop effect, sourced by the cumulants beyond the first. We now compute them.

23.2.4 The mass correction: the shell tadpole

The first cumulant $\langle S_{\rm mix}\rangle_>$ is the $O(\lambda)$ correction. Taking the shell average of Eq. (23.28) term by term, the two terms odd in $\phi_>$ vanish ($\langle\phi_>\rangle_> = \langle\phi_>^3\rangle_> = 0$, an odd Gaussian moment), the pure $\phi_>^4$ term gives a field-independent constant $\frac{\lambda}{4!}\langle\phi_>^4\rangle_> = \frac{\lambda}{8}\,G_>(0)^2$ that we discard, and only the $\frac{\lambda}{4}\phi_<^2\phi_>^2$ term survives with a $\phi_<$ dependence:

$$\big\langle S_{\rm mix}\big\rangle_> = \frac{\lambda}{4}\int \mathrm{d}^4x\,\phi_<^2(x)\,\big\langle\phi_>^2(x)\big\rangle_> = \frac{\lambda}{4}\,G_>(0)\int \mathrm{d}^4x\,\phi_<^2(x), \tag{23.35}$$

where $\langle\phi_>^2(x)\rangle_> = G_>(0)$ is the shell propagator at coincident points, the same object for every x by translation invariance. The diagram is a single quartic vertex with two of its legs ($\phi_>$) closed into a shell loop and the other two ($\phi_<$) left as external slow legs: the *shell tadpole* of Fig. 23.3(a). Defining the basic shell integral

$$I_1(b\Lambda,\Lambda) \;\equiv\; G_>(0) = \int_{b\Lambda<|k|<\Lambda} \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{k^2+m^2}, \tag{23.36}$$

the first cumulant adds $\frac{\lambda}{4}I_1\int\phi_<^2$ to $S_{b\Lambda}$. Comparing with the mass term $\frac{1}{2}m^2\int\phi_<^2$ of Eq. (23.1), this is precisely a shift of the squared mass,

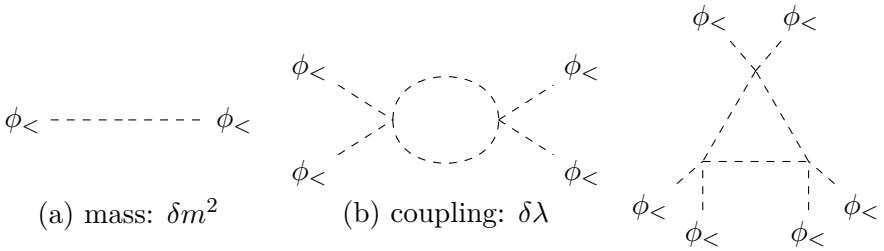
$$\tfrac{1}{2}\,\delta m^2\int\phi_<^2 = \frac{\lambda}{4}\,I_1\int\phi_<^2 \quad\Longrightarrow\quad \boxed{\delta m^2 = \frac{\lambda}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{k^2+m^2} \;=\; \frac{\lambda}{2}\,I_1(b\Lambda,\Lambda).} \tag{23.37}$$

The mass of the coarse-grained theory is $m^2 + \delta m^2$. Since $I_1 > 0$ and $\lambda > 0$, the mass-squared *increases* as we lower the cutoff: coarse-graining feeds the

short-distance fluctuations of $\phi_>^2$ into the long-distance mass. We evaluate the shell integral I_1, along with the two further shell integrals the next subsections will need, once and for all in Theorem 23.9, where the four-dimensional angular reduction $\mathrm{d}^4k = 2\pi^2 k^3\,\mathrm{d}k$ already used in Theorem 23.3 does the work. The result, for a generic shell, is

$$I_1(b\Lambda,\Lambda) = \frac{1}{16\pi^2}\Big[\Lambda^2(1-b^2) - m^2\ln\frac{\Lambda^2+m^2}{b^2\Lambda^2+m^2}\Big], \tag{23.38}$$

so that to leading order in m^2/Λ^2 the mass shift is $\delta m^2 \simeq \frac{\lambda}{32\pi^2}\Lambda^2(1-b^2)$, of order the cutoff squared. The mass is a relevant coupling, and Eq. (23.37) is the first quantitative sign of it: the correction is enhanced by Λ^2, not suppressed.



(a) mass: δm^2 (b) coupling: $\delta\lambda$ (c) generated: $\phi_<^6$

Figure 23.3: The connected diagrams of the shell integration for ϕ^4 theory. External legs are slow fields $\phi_<$; every internal line is a shell propagator $G_>$ Eq. (23.30) carrying momentum $b\Lambda < |k| < \Lambda$. (a) The $O(\lambda)$ tadpole shifts the mass, Eq. (23.37). (b) The $O(\lambda^2)$ bubble shifts the quartic coupling, Eq. (23.44). (c) The $O(\lambda^3)$ triangle generates the $\phi_<^6$ operator, Eq. (23.47). Higher cumulants generate the full irrelevant tail of Table 23.1.

23.2.5 The coupling correction: the shell bubble

The quartic coupling is corrected at $O(\lambda^2)$, by the second cumulant $-\frac{1}{2}\langle S_{\rm mix}^2\rangle_{>,c}$. Among the products of the four terms of Eq. (23.28), we want those leaving exactly four external slow legs $\phi_<$ with all fast fields contracted. The unique such connected structure pairs the $\frac{\lambda}{4}\phi_<^2\phi_>^2$ term at one vertex with the $\frac{\lambda}{4}\phi_<^2\phi_>^2$ term at the other, contracting the two fast fields at x against the two at y:

$$-\tfrac{1}{2}\langle S_{\rm mix}^2\rangle_{>,c}\Big|_{\phi_<^4} = -\frac{1}{2}\Big(\frac{\lambda}{4}\Big)^2\int \mathrm{d}^4x\,\mathrm{d}^4y\;\phi_<^2(x)\,\phi_<^2(y)\,\langle\phi_>^2(x)\,\phi_>^2(y)\rangle_{>,c}. \tag{23.39}$$

The connected Gaussian average $\langle\phi_>^2(x)\phi_>^2(y)\rangle_{>,c} = 2\,G_>(x-y)^2$ counts the two ways of pairing the two fast fields at x with the two at y (the disconnected pairing $\langle\phi_>^2(x)\rangle\langle\phi_>^2(y)\rangle$ is removed by the connected subscript; it belongs to the

square of the tadpole already counted). Hence

$$-\tfrac{1}{2}\langle S^2_{\rm mix}\rangle_{>,c}\Big|_{\phi^4_<} = -\frac{\lambda^2}{16}\int \mathrm{d}^4x\,\mathrm{d}^4y\;\phi^2_<(x)\,\phi^2_<(y)\,G_>(x-y)^2. \tag{23.40}$$

This is the *shell bubble* of Fig. 23.3(b): two vertices joined by two shell propagators, with two slow legs at each vertex. It is nonlocal in $\phi_<$, depending on the field at the two separated points x and y through the kernel $G_>(x-y)^2$. To extract its contribution to the local quartic coupling we exploit that $G_>(x-y)^2$ is sharply peaked: both internal momenta lie in the shell near Λ, so the kernel decays over distances of order Λ^{-1}, far shorter than the wavelengths $\gtrsim (b\Lambda)^{-1}$ over which $\phi_<$ varies. We may therefore Taylor-expand the slow field about the vertex x,

$$\phi^2_<(y) = \phi^2_<(x) + (y-x)_\mu\,\partial_\mu\phi^2_<(x) + \tfrac{1}{2}(y-x)_\mu(y-x)_\nu\,\partial_\mu\partial_\nu\phi^2_<(x) + \cdots, \tag{23.41}$$

the *derivative expansion* that organizes the nonlocal result into a tower of local operators of increasing dimension. The leading term reproduces a local $\phi^4_<$: setting $\phi^2_<(y)\to\phi^2_<(x)$ in Eq. (23.40) and integrating the kernel over $z=y-x$ gives

$$-\frac{\lambda^2}{16}\Big(\int \mathrm{d}^4z\,G_>(z)^2\Big)\int \mathrm{d}^4x\;\phi^4_<(x) = -\frac{\lambda^2}{16}\,I_2\int \mathrm{d}^4x\;\phi^4_<(x), \tag{23.42}$$

where, by Parseval applied to Eq. (23.30) (equivalently, the bubble at zero external momentum),

$$I_2(b\Lambda,\Lambda) \;\equiv\; \int \mathrm{d}^4z\,G_>(z)^2 = \int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^2}. \tag{23.43}$$

The new quartic part of $S_{b\Lambda}$ is thus $(\frac{\lambda}{4!}-\frac{\lambda^2}{16}I_2)\int\phi^4_<$. Writing this as $\frac{1}{4!}(\lambda+\delta\lambda)\int\phi^4_<$ and multiplying through by $4!=24$,

$$\boxed{\delta\lambda = -24\cdot\frac{\lambda^2}{16}\,I_2 = -\frac{3\lambda^2}{2}\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^2} = -\frac{3\lambda^2}{2}\,I_2(b\Lambda,\Lambda).} \tag{23.44}$$

The factor of 3, which the diagrammatic computation of Section 21.6 produced as the sum over the s-, t-, and u-channels, here emerges from the combinatorial ratio $4!/16 = 3/2$: the three channels are the three inequivalent ways of distributing four identical external legs onto the two vertices, and the symmetric normalization $1/4!$ of the vertex repackages them into the single coefficient $3/2$. The shell integral I_2 is precisely the difference of the full-ball bubbles of Theorem 23.3, $I_2(b\Lambda,\Lambda) = I(\Lambda) - I(b\Lambda)$ with I given by Eq. (23.13); we record its value in Theorem 23.9. For a thin shell with $m \ll \Lambda$,

$$I_2(b\Lambda,\Lambda) \simeq \frac{1}{16\pi^2}\ln\frac{1}{b^2} = \frac{1}{8\pi^2}\ln\frac{1}{b}, \tag{23.45}$$

a pure logarithm, free of any power of Λ — the marginality of λ made visible, exactly as anticipated in Theorem 23.7. Because $I_2 > 0$, the coupling *decreases* under coarse-graining: $\delta\lambda < 0$. Read in the opposite direction, toward the ultraviolet, the coupling grows, which is the positive sign of the ϕ^4 beta function that ϕ^4 theory carries. We make this identification precise in Section 23.2.7 and complete it in Chapter 24.

23.2.6 Newly generated operators, and why they are small

So far the flow has stayed within the three couplings of the bare action Eq. (23.1): it shifted m^2 and λ. But Eq. (23.33) contains infinitely many further connected averages, and there is no reason they should organize themselves into $\phi^2_<$ and $\phi^4_<$ alone. The symmetries respected by the shell average Eq. (23.29) — Euclidean invariance and $\mathbb{Z}_2$ — permit every operator of Table 23.1, and the shell integration, blind to the simplicity of the starting point, switches them on. We exhibit the two leading examples and then show, by power counting, that all of them are parametrically small.

A $\phi^6_<$ interaction. The sextic operator first appears at $O(\lambda^3)$, from the third cumulant $+\frac{1}{6}\langle S^3_{\rm mix}\rangle_{>,c}$. (It cannot appear at $O(\lambda^2)$: a connected graph with six external slow legs from two quartic vertices would need a single internal line carrying the total external momentum, which for slow external fields is below $b\Lambda$ and hence outside the shell, so the shell propagator vanishes there.) Taking the $\frac{\lambda}{4}\phi^2_<\phi^2_>$ term from each of three vertices and contracting the six fast fields into a triangle of shell propagators — the diagram of Fig. 23.3(c) —

$$+\tfrac{1}{6}\langle S^3_{\rm mix}\rangle_{>,c}\Big|_{\phi^6_<} = \frac{1}{6}\Big(\frac{\lambda}{4}\Big)^3\int \mathrm{d}^4x\,\mathrm{d}^4y\,\mathrm{d}^4z\;\phi^2_<(x)\phi^2_<(y)\phi^2_<(z)\,\big\langle\phi^2_>(x)\phi^2_>(y)\phi^2_>(z)\big\rangle_{>,c}, \tag{23.46}$$

with the connected triangle average $\langle\phi^2_>(x)\phi^2_>(y)\phi^2_>(z)\rangle_{>,c} = 8\,G_>(x-y)G_>(y-z)G_>(z-x)$ counting the $2\cdot2\cdot2$ pairings that join each vertex to the other two. Setting all slow fields to a common point by the derivative expansion Eq. (23.41), and closing the triangle of propagators at zero external momentum into a single shell loop,

$$+\tfrac{1}{6}\langle S^3_{\rm mix}\rangle_{>,c}\Big|_{\phi^6_<} = \frac{\lambda^3}{48}\,I_3\int \mathrm{d}^4x\,\phi^6_<(x), \qquad I_3(b\Lambda,\Lambda) \equiv \int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^3}. \tag{23.47}$$

Matching to the normalization $\frac{c_6}{6!}\phi^6_<$ of Eq. (23.15) gives the generated sextic coupling

$$c_6 = \frac{6!}{48}\,\lambda^3 I_3 = 15\,\lambda^3\,I_3(b\Lambda,\Lambda), \qquad I_3 \simeq \frac{1}{16\pi^2}\,\frac{1}{\Lambda^2}\Big(\frac{1}{b^2}-1\Big)\quad (m\ll\Lambda), \tag{23.48}$$

the thin-shell value of I_3 being computed in Theorem 23.9. The starting action had $c_6 = 0$; the coarse-grained action does not. The flow has left the three-dimensional subspace and moved into the irrelevant tail of theory space, precisely as Fig. 23.1 advertised.

A $\phi_<^2(\partial\phi_<)^2$ interaction. The same shell bubble that produced $\delta\lambda$ produces a derivative interaction at the next order of the expansion Eq. (23.41). The linear term in Eq. (23.41) integrates to zero ($G_>(z)^2$ is even in z), and the quadratic term gives, using $\int \mathrm{d}^4z\, z_\mu z_\nu\, G_>(z)^2 = \frac{1}{4}\delta_{\mu\nu}\int \mathrm{d}^4z\, z^2 G_>(z)^2 \equiv \frac{1}{4}\delta_{\mu\nu}K_2$,

$$-\frac{\lambda^2}{16}\cdot\tfrac{1}{2}\cdot\tfrac{1}{4}K_2\int \mathrm{d}^4x\;\phi_<^2\,\partial^2\phi_<^2 = +\frac{\lambda^2}{32}\,K_2\int \mathrm{d}^4x\;\phi_<^2(\partial_\mu\phi_<)^2, \tag{23.49}$$

the last equality by integration by parts, $\int \phi_<^2\partial^2\phi_<^2 = -\int(\partial_\mu\phi_<^2)^2 = -4\int\phi_<^2(\partial_\mu\phi_<)^2$. Matching to the normalization $\frac{z_4}{2}\phi_<^2(\partial\phi_<)^2$ of Eq. (23.15) gives $z_4 = \frac{\lambda^2}{16}K_2$. The moment K_2 is fixed by the small-momentum expansion of the shell bubble: writing $\widetilde{G_>^2}(P) = \int_{\text{shell}}\mathrm{d}^4k/(2\pi)^4\,[(k^2+m^2)((k+P)^2+m^2)]^{-1} = I_2 - \frac{1}{8}K_2\,P^2 + \cdots$ and expanding the second propagator to $O(P^2)$ with the angular average $\langle(k\cdot P)^2\rangle = \frac{1}{4}k^2P^2$, the P^2 coefficient is

$$-\tfrac{1}{8}K_2 = \int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\Big[\frac{k^2}{(k^2+m^2)^4} - \frac{1}{(k^2+m^2)^3}\Big] = -\,m^2\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^4}, \tag{23.50}$$

the bracket collapsing because $k^2-(k^2+m^2) = -m^2$. Hence $K_2 = 8m^2\int_{\text{shell}}(k^2+m^2)^{-4}$, and the generated derivative coupling is

$$z_4 = \frac{\lambda^2}{2}\,m^2\int_{b\Lambda<|k|<\Lambda}\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^4} \;\simeq\; \frac{\lambda^2m^2}{64\pi^2}\,\frac{1}{\Lambda^4}\Big(\frac{1}{b^4}-1\Big)\quad (m\ll\Lambda). \tag{23.51}$$

Two features are worth pausing on. The coupling is nonzero: the derivative operator $\phi_<^2(\partial\phi_<)^2$ is genuinely generated. But it is doubly small. Pure dimensional analysis (the operator has dimension 6, so $[z_4] = -2$) permits $z_4 \sim \lambda^2/\Lambda^2$; the explicit computation Eq. (23.50) shows that the would-be Λ^{-2} piece cancels and the leading result is $z_4 \sim \lambda^2m^2/\Lambda^4$, suppressed by an extra factor m^2/Λ^2. The cancellation is the same topological accident that made the one-loop field-strength renormalization of ϕ^4 vanish in Eq. (21.127) ($\delta_Z = 0$, no p^2 in the self-energy): a shell loop with all-even propagators carries no net dependence on the external momentum to leading order. We record the two generated operators together,

$$S_{b\Lambda}[\phi_<] \;\supset\; \frac{c_6}{6!}\int\phi_<^6 + \frac{z_4}{2}\int\phi_<^2(\partial_\mu\phi_<)^2, \qquad c_6 = 15\lambda^3 I_3,\quad z_4 = \tfrac{1}{2}\lambda^2m^2\int_{\text{shell}}\frac{\mathrm{d}^4k/(2\pi)^4}{(k^2+m^2)^4}, \tag{23.52}$$

as the leading members of the irrelevant tower switched on by one shell integration.

Power counting: why the generated operators carry inverse powers of Λ. The smallness of c_6 and z_4 is not an accident of these two diagrams; it is forced by dimensions. A connected contribution to $S_{b\Lambda}$ that supplies an operator $\mathcal{O}_{n,p}$ of mass dimension $[\mathcal{O}_{n,p}] = 2p + 2n$ (Eq. (23.18)) must come multiplied by a coefficient of the complementary dimension $[g_{n,p}] = 4 - 2p - 2n$ (Eq. (23.19)), since the action is dimensionless. For a thin shell with $m \ll b\Lambda$, the *only* scale available to the shell integrals is the cutoff Λ itself: every $G_>$ contributes a propagator whose momenta are pinned to $|k| \sim \Lambda$, and every loop measure $\int_{\text{shell}} \mathrm{d}^4k/(2\pi)^4$ scales as Λ^4 times a function of b. The dimensionful coefficient of $\mathcal{O}_{n,p}$ is therefore $\Lambda^{4-2p-2n}$ times powers of λ and a b-dependent number,

$$g_{n,p} \;\sim\; \lambda^{\#}\,\Lambda^{\,4-2p-2n} \times (\text{function of } b), \tag{23.53}$$

which is precisely $\Lambda^{-[\mathcal{O}_{n,p}]+4}$. For the marginal couplings λ and Z ($2p + 2n = 4$) the power is Λ^0: their shifts are logarithmic in b, as we found in Eq. (23.45). For the relevant mass ($2p + 2n = 2$) the power is Λ^{+2}: its shift is enhanced, as in Eq. (23.38). For every irrelevant operator ($2p + 2n > 4$) the power is *negative*: ϕ^6 comes with Λ^{-2} (confirmed by Eq. (23.48), where $I_3 \sim \Lambda^{-2}$), $\phi^2(\partial\phi)^2$ likewise with Λ^{-2} (and, accidentally, an extra m^2/Λ^2), ϕ^8 with Λ^{-4}, and so on down the tail of Table 23.1. The dimensionless couplings $\hat{g}_{n,p} = g_{n,p}\Lambda^{[\mathcal{O}]-4}$ of Eq. (23.16) that the flow actually tracks are therefore pure numbers of order $\lambda^{\#}$ for the irrelevant directions, but they are multiplied back into observables with the suppressing power $(\text{energy}/\Lambda)^{2p+2n-4}$ whenever a physical amplitude is computed at energies well below Λ. *This is irrelevance, derived rather than asserted*: the higher-dimension operators are generated, but each extra field-pair or derivative-pair beyond the marginal four costs a factor $1/\Lambda^2$, so they are washed out at long distances. The promise of Theorem 23.6 — that the names predict the dynamics — is here delivered for the operators the flow itself creates.

23.2.7 Stacking shells: recovering the full one-loop results

A single finite-width shell shifts the couplings by the finite amounts just computed. The connection to the diagrammatic one-loop computation of Chapter 21, and to the running couplings of Chapter 24, comes from *iterating* the shell integration: lower the cutoff in many thin steps from Λ down to a measurement scale μ, and accumulate the shifts. For an infinitesimal step $b = e^{-\mathrm{d}\ell}$ with $\mathrm{d}\ell \to 0^+$, the shell occupies $\Lambda e^{-\mathrm{d}\ell} < |k| < \Lambda$, and the thin-shell integrals of Theorem 23.9 linearize: $1 - b^2 \to 2\,\mathrm{d}\ell$ and $\ln(1/b) \to \mathrm{d}\ell$, so the per-step shifts Eqs. (23.37) and (23.44) become (with $m \ll \Lambda$)

$$\mathrm{d}m^2 = \frac{\lambda\,\Lambda^2}{16\pi^2}\,\mathrm{d}\ell, \qquad \mathrm{d}\lambda = -\frac{3\lambda^2}{16\pi^2}\,\mathrm{d}\ell. \tag{23.54}$$

Summing the coupling shifts over all shells from Λ down to μ — integrating $\mathrm{d}\ell$ from 0 to $\ln(\Lambda/\mu)$ at fixed leading-order λ — accumulates the total change

$$\lambda(\mu) - \lambda(\Lambda) = -\frac{3\lambda^2}{16\pi^2}\int_0^{\ln(\Lambda/\mu)} \mathrm{d}\ell = -\frac{3\lambda^2}{16\pi^2}\ln\frac{\Lambda}{\mu}. \tag{23.55}$$

This is exactly the one-loop logarithm of Chapter 21. There, the divergent part of the four-point function was the pole $i\mathcal{M}_{\text{loop}}|_{\text{div}} = \frac{3i\lambda^2}{32\pi^2\bar{\varepsilon}}$ of Eq. (21.116); under the dictionary $1/\bar{\varepsilon} \leftrightarrow \ln(\Lambda^2/\mu^2) = 2\ln(\Lambda/\mu)$ between dimensional and cutoff regularization established in Theorem 23.3, that pole becomes $\frac{3\lambda^2}{16\pi^2}\ln(\Lambda/\mu)$, identical to Eq. (23.55). The Wilsonian shell-by-shell computation and the single diagrammatic loop integral are the same calculation sliced two ways: the loop integral $\int^\Lambda \mathrm{d}^4k$ is the sum of its shells $\sum_{\text{shells}}\int_{\text{shell}}\mathrm{d}^4k$, and the logarithm $\ln(\Lambda/\mu)$ is the sum of the infinitesimal logarithms $\mathrm{d}\ell$. The coefficient $3\lambda^2/16\pi^2$ that Eq. (23.55) attaches to the logarithm is the carried number of ϕ^4 theory: it is the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Eq. (24.46), here obtained as the rate at which λ changes per e-fold of the cutoff. The mass shift accumulates likewise; summing Eq. (23.54) reconstructs the full cutoff tadpole, as the next worked example verifies in detail.

Worked Example 23.9 (The shell integrals, and their sum into the full one-loop self-energy)**.** We evaluate the three shell integrals I_1, I_2, I_3 and confirm that stacking the mass shifts reproduces the cutoff one-loop self-energy.

Reduction to a radial integral. Each integrand depends on $|k|$ alone, so by the angular reduction of Theorem 23.3, $\int \mathrm{d}^4k/(2\pi)^4\, f(|k|) = \frac{1}{8\pi^2}\int k^3 f(k)\,\mathrm{d}k$. With the substitution $u = k^2$, $k^3\,\mathrm{d}k = \frac{1}{2}u\,\mathrm{d}u$,

$$I_n(b\Lambda, \Lambda) = \frac{1}{16\pi^2}\int_{b^2\Lambda^2}^{\Lambda^2} \frac{u\,\mathrm{d}u}{(u+m^2)^n}. \tag{23.56}$$

For $n = 1$, write $u = (u+m^2) - m^2$: $\int \frac{u\,\mathrm{d}u}{u+m^2} = [u - m^2\ln(u+m^2)]$, giving Eq. (23.38). For $n = 2$, $I_2 = I(\Lambda) - I(b\Lambda)$ with I from Eq. (23.13); for $m \ll \Lambda$ the leading term is $\frac{1}{16\pi^2}\ln(\Lambda^2/b^2\Lambda^2) = \frac{1}{16\pi^2}\ln(1/b^2)$, as in Eq. (23.45). For $n = 3$, again with $u = (u+m^2) - m^2$, $\int \frac{u\,\mathrm{d}u}{(u+m^2)^3} = [-\frac{1}{u+m^2} + \frac{m^2}{2(u+m^2)^2}]$; for $m \ll \Lambda$ the dominant piece is $\frac{1}{16\pi^2}(\frac{1}{b^2\Lambda^2} - \frac{1}{\Lambda^2}) = \frac{1}{16\pi^2\Lambda^2}(b^{-2} - 1)$, the value quoted in Eq. (23.48).

Stacking the mass shifts. The per-shell mass shift Eq. (23.37) is $\delta m^2 = \frac{\lambda}{2}I_1$, and I_1 for the shell $[k, k+\mathrm{d}k]$ is $\frac{1}{8\pi^2}\frac{k^3}{k^2+m^2}\mathrm{d}k$. Summing over all shells from 0 to Λ — that is, integrating out *every* mode — gives the total mass correction

$$\Delta m^2 = \frac{\lambda}{2}\cdot\frac{1}{8\pi^2}\int_0^\Lambda \frac{k^3\,\mathrm{d}k}{k^2+m^2} = \frac{\lambda}{32\pi^2}\Big[\Lambda^2 - m^2\ln\frac{\Lambda^2+m^2}{m^2}\Big]. \tag{23.57}$$

This is exactly the one-loop self-energy of ϕ^4 in cutoff regularization: $\Sigma = \frac{\lambda}{2}\int_0^\Lambda \mathrm{d}^4k/(2\pi)^4\,(k^2+m^2)^{-1} = \Delta m^2$, a constant independent of the external

momentum, matching the momentum-independence found diagrammatically in Eq. (21.124). The quadratic piece $\frac{\lambda\Lambda^2}{32\pi^2}$ is the cutoff's rendering of the power divergence that dimensional regularization discarded as scaleless; the logarithm $-\frac{\lambda m^2}{32\pi^2}\ln(\Lambda^2/m^2)$ is the cutoff's rendering of the $\frac{1}{\bar{\varepsilon}}$ pole of Eq. (21.124), under the same dictionary $1/\bar{\varepsilon} \leftrightarrow \ln(\Lambda^2/m^2)$ used for the coupling. The Wilsonian sum over shells and the single diagrammatic tadpole are, once more, the same integral computed in two orders.

The shell integration is now complete: we have an action $S_{b\Lambda}$ at the lowered cutoff $b\Lambda$, with mass $m^2 + \delta m^2$ of Eq. (23.37), coupling $\lambda + \delta\lambda$ of Eq. (23.44), and the freshly generated irrelevant couplings $c_6, z_4, \ldots$ of Eq. (23.52). But $S_{b\Lambda}$ is an action for a field band-limited to $b\Lambda$, not Λ: the cutoff has moved, so $S_{b\Lambda}$ and S_Λ live in different cutoff theories and cannot yet be compared as points of a single theory space. To turn the half-step Eq. (23.22) into an honest self-map — a transformation that sends a theory at cutoff Λ to another theory at the *same* cutoff Λ, whose iteration is the renormalization-group flow — we must rescale momenta and fields to restore the cutoff to Λ and re-express the couplings in dimensionless form. That rescaling is the subject of Section 23.3.

23.3 Rescaling and the Renormalization-Group Transformation

The shell integration of Section 23.2 delivered a coarse-grained action $S_{b\Lambda}[\phi_<]$: a functional of the slow field $\phi_<$, supported on momenta $|k| < b\Lambda$ with $b < 1$, whose couplings differ from the originals by the contributions of the fast modes we eliminated. By construction $S_{b\Lambda}$ reproduces every correlation function of the long-wavelength field that the original theory predicted; nothing physical has been lost. But the new action is *not* of the same form as the old one in one decisive sense: its ultraviolet cutoff is $b\Lambda$, not Λ. As long as the two actions live on momentum spaces of different size we cannot compare them, and we certainly cannot ask whether the operation that produced one from the other has a fixed point. A fixed point is a theory mapped to itself, and "itself" presupposes a common arena. This section builds that arena. We rescale momenta to restore the cutoff to its original value Λ, rescale the field so that the kinetic term returns to its canonical normalization, and re-express every coupling in dimensionless form. The composite operation—integrate a shell, then rescale—is then a genuine self-map of the space of dimensionless theories. Iterating it generates the Wilsonian renormalization-group flow; passing to an infinitesimal shell turns the discrete map into a system of differential equations. By the end of the section the carried ϕ^4 theory will have explicit flow equations for its relevant mass and marginal coupling, ready for the fixed-point analysis to come.

23.3.1 The rescaling step

The cutoff $b\Lambda$ on the coarse-grained action is a length, or rather an inverse length: it is the shortest wavelength the slow field still resolves. To restore it to Λ we simply change our unit of length, measuring all momenta in units of the new cutoff rather than the old. Concretely, define rescaled momenta and positions by

$$k' = \frac{k}{b}, \qquad x' = b\,x, \tag{23.58}$$

so that the slow-mode domain $|k| < b\Lambda$ becomes $|k'| < \Lambda$ exactly: the cutoff is back where it started. The choice $x' = bx$ is forced by Eq. (23.58), since the product $k' \cdot x' = (k/b)\cdot(bx) = k \cdot x$ must be left invariant for the Fourier kernel $e^{-ik\cdot x}$ to be unchanged. With $b < 1$ we have $|x'| < |x|$: the rescaling zooms out, redrawing the same physics on a coarser ruler. This is the quantitative content of Kadanoff's block-spin picture—having averaged over short-distance fluctuations, we step back and view the blocked system as if it were a fresh system of the original size.

Rescaling coordinates is not yet enough, because it generically spoils the normalization of the kinetic term, and a kinetic term with a noncanonical coefficient is not in the standard form we wish to iterate. We must therefore also rescale the field. Write the field rescaling as

$$\phi_<(x) \;\longmapsto\; \phi'(x') = \zeta^{-1}\,\phi_<(x), \tag{23.59}$$

with a constant ζ to be determined by the requirement that the rescaled kinetic term carry the canonical coefficient $\frac{1}{2}$. We work in the Euclidean theory of Chapter 10 throughout this chapter, so the relevant quadratic action is $\frac{1}{2}\int d^dx\,(\partial_\mu\phi_<)^2$ up to the wavefunction factor generated by the shell. Suppose the coarse-grained action carries a kinetic term

$$S_{\rm kin} = \frac{Z}{2}\int_{|k|<b\Lambda} d^dx\,\big(\partial_\mu\phi_<(x)\big)^2, \tag{23.60}$$

where $Z = 1$ in the classical action and $Z = 1 + O(\lambda^2)$ once shell corrections to the two-point function with nontrivial momentum dependence are included (we return to the size of $Z-1$ below). Change variables with Eqs. (23.58) and (23.59). Using $\partial/\partial x^\mu = b\,\partial/\partial x'^\mu$ and the Jacobian $d^dx = b^{-d}\,d^dx'$,

$$S_{\rm kin} = \frac{Z}{2}\int b^{-d}d^dx'\,\Big(b\,\partial'_\mu(\zeta\,\phi'(x'))\Big)^2 = \frac{Z}{2}\,\zeta^2\,b^{2-d}\int d^dx'\,\big(\partial'_\mu\phi'(x')\big)^2. \tag{23.61}$$

Canonical normalization—coefficient $\frac{1}{2}$ in front of $\int d^dx'\,(\partial'\phi')^2$ over the restored domain $|k'| < \Lambda$—requires

$$Z\,\zeta^2\,b^{2-d} = 1 \qquad\Longrightarrow\qquad \zeta = Z^{-1/2}\,b^{(d-2)/2}. \tag{23.62}$$

The classical factor $b^{(d-2)/2}$ is exactly $b^{[\phi]}$ with $[\phi] = (d-2)/2$ the canonical mass dimension of a scalar (Theorem A.1, Section 21.3.1): the field rescaling needed to restore the kinetic term is precisely the engineering scaling of the field. The remaining factor $Z^{-1/2}$ is the part of ζ that goes *beyond* canonical scaling; it is nonzero only when the fast modes correct the momentum dependence of the propagator, and it is the seed of the anomalous dimension defined in Section 23.3.4.

Physical Insight

The rescaling is bookkeeping, but it is the bookkeeping that makes scale invariance visible. A scale-invariant theory looks the same at every magnification; the only way to detect that sameness from inside a calculation is to magnify—to integrate out short distances and then zoom back out—and check whether anything changed. The shell integration is the physics (averaging over fluctuations); the rescaling is the change of ruler that lets us compare before and after on equal footing. A theory unchanged by the combined operation is scale invariant, which is why fixed points of this map are the quantum field theories that realize the classically scale-invariant ideal of Section 7.7.

23.3.2 Dimensionless couplings and the engineering flow

With the field rescaled, every operator in the action transforms in a way fixed entirely by counting: by its engineering dimension. Consider a generic local interaction term in the coarse-grained action,

$$S_n = \int_{|k|<b\Lambda} d^d x \; g_n \, \mathcal{O}_n(\phi_<), \qquad \mathcal{O}_n = \partial^{\delta_n} \phi_<^{\,n}, \tag{23.63}$$

with δ_n derivatives and n powers of the field, so that the operator has engineering dimension $[\mathcal{O}_n] = \delta_n + n\,[\phi] = \delta_n + \frac{n(d-2)}{2}$ and the coupling has the complementary dimension $[g_n] = d - [\mathcal{O}_n]$ enforced by $[S_n] = 0$ (Section 21.3.1). Apply the rescaling Eqs. (23.58) to (23.59). Each derivative produces a factor b, the n fields produce ζ^n, and the measure produces b^{-d}:

$$S_n = \int b^{-d} d^d x' \; g_n \, b^{\delta_n} \zeta^n \, \partial'^{\,\delta_n} \phi'^{\,n} = \int d^d x' \; g'_n \, \mathcal{O}_n(\phi'), \qquad g'_n = g_n \, b^{\,\delta_n - d} \, \zeta^n. \tag{23.64}$$

Insert $\zeta = Z^{-1/2} b^{(d-2)/2}$ from Eq. (23.62) and set $Z = 1$ for the moment (the classical, anomaly-free part):

$$g'_n = g_n \, b^{\,\delta_n - d + n(d-2)/2} = g_n \, b^{\,-[g_n]}, \tag{23.65}$$

where the exponent collapsed because $\delta_n - d + \frac{n(d-2)}{2} = -\big(d - \delta_n - \frac{n(d-2)}{2}\big) = -[g_n]$. This is the central kinematic fact of the renormalization group: *under one*

rescaling step a coupling is multiplied by $b^{-[g_n]}$, the cutoff ratio raised to the negative of its mass dimension.

It is cleanest to absorb the cutoff into the couplings once and for all, working with dimensionless couplings measured in units of Λ. Define, for each coupling of mass dimension $D_i \equiv [g_i]$,

$$\boxed{\tilde{g}_i \;=\; g_i\,\Lambda^{-D_i}} \tag{23.66}$$

so that, for ϕ^4 theory in $d = 4$ with the operator content $\frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 + \frac{c_6}{6!}\phi^6 + \dots$, the dimensionless couplings are

$$\tilde{m}^2 = \frac{m^2}{\Lambda^2}, \qquad \tilde{\lambda} = \lambda, \qquad \tilde{c}_6 = c_6\,\Lambda^2, \qquad \dots, \tag{23.67}$$

using $[m^2] = 2$, $[\lambda] = 0$, $[c_6] = -2$ in four dimensions. In dimensionless variables the rescaling step Eq. (23.65) reads simply

$$\tilde{g}_i \;\longmapsto\; \tilde{g}_i' = b^{-D_i}\,\tilde{g}_i \qquad \text{(classical part)}, \tag{23.68}$$

because the explicit Λ in Eq. (23.66) is held fixed and all the b-dependence sits in g_i.[1] Equation (23.68) is the engineering content of the flow and, as the loop corrections below are subleading near the free theory, it is the *leading* flow: relevant couplings ($D_i > 0$, e.g. the mass) are amplified, marginal couplings ($D_i = 0$, e.g. the quartic) are untouched at this order, and irrelevant couplings ($D_i < 0$, e.g. c_6) are suppressed, each by $b < 1$ raised to a power set by counting alone. The full trichotomy and its consequences are taken up in the next sections; here we have established its origin.

We can now name the object we have built.

Definition 23.10 (The renormalization-group transformation)**.** Let $\mathcal{T}$ be the space of (Euclidean) actions with cutoff Λ, coordinatized by the dimensionless couplings $\{\tilde{g}_i\}$ of Eq. (23.66). For a scale factor $b \in (0, 1)$, the *renormalization-group transformation* R_b is the self-map $R_b : \mathcal{T} \to \mathcal{T}$ obtained by composing

1. integration over the momentum shell $b\Lambda < |k| < \Lambda$ (the Wilson step of Section 23.2), which sends $S_\Lambda \mapsto S_{b\Lambda}$; and
2. the rescaling Eqs. (23.58) to (23.59) with ζ fixed by Eq. (23.62), which restores the cutoff to Λ and the kinetic term to canonical form.

Its action on couplings is $\tilde{g}_i \mapsto \tilde{g}_i' = b^{-D_i}\tilde{g}_i + (\text{loop corrections})$. The transformations satisfy the semigroup law $R_{b_1} \circ R_{b_2} = R_{b_1 b_2}$, and the *renormalization-group flow* is the sequence $\{R_b^n(\tilde{g})\}_{n \geq 0}$ generated by iterating R_b.

[1] Equivalently: the shell step lowers the cutoff to $b\Lambda$, so the natural dimensionless coupling is $g_i(b\Lambda)^{-D_i} = b^{-D_i}\tilde{g}_i$; the rescaling step is precisely the relabeling that calls this the new $\tilde{g}_i$ at cutoff Λ.

The semigroup law is exact and important: integrating out the shell down to $b_2\Lambda$ and then down to $b_1 b_2 \Lambda$ is the same as integrating directly down to $b_1 b_2 \Lambda$, and the rescalings compose multiplicatively because each is a change of length unit. There is no inverse—we cannot un-integrate the modes we discarded—so the structure is a semigroup, not a group, and the "group" in "renormalization group" is a historical misnomer that we keep.

23.3.3 The differential flow

Iterating R_b for a fixed finite b produces a discrete orbit. Smoother and more useful is the continuous flow obtained by taking the shell infinitesimally thin. Introduce the *renormalization-group time*

$$t = -\ln b \ \geq 0, \tag{23.69}$$

which increases as b decreases, that is, as the cutoff is lowered and we flow toward the infrared. An infinitesimal step is $b = e^{-dt} = 1 - dt + O(dt^2)$, a shell of fractional thickness dt just below Λ. The engineering part Eq. (23.68) then reads $\tilde{g}_i' = e^{D_i\, dt}\tilde{g}_i = \tilde{g}_i + D_i\, \tilde{g}_i\, dt$, so the engineering contribution to the flow is the linear term $D_i\, \tilde{g}_i$. Adding the one-loop shell corrections, the renormalization-group flow takes the form of an autonomous dynamical system on theory space,

$$\frac{d\tilde{g}_i}{dt} = D_i\, \tilde{g}_i + \beta_i^{\text{loop}}(\{\tilde{g}\}), \tag{23.70}$$

in which the engineering term recovers the canonical dimensions as the leading flow and the loop term carries the interactions. We now make this completely explicit for ϕ^4 theory.

The shell corrections, recalled and reduced

The shell integration of Section 23.2 produced two one-loop corrections to the ϕ^4 action: a shift of the mass from the tadpole contraction of the quartic vertex, and a shift of the quartic coupling from the one-loop bubble. With the Euclidean propagator $1/(k^2 + m^2)$ they are

$$\delta m^2 = +\frac{\lambda}{2} \int_{\text{shell}} \frac{d^d k}{(2\pi)^d}\, \frac{1}{k^2 + m^2}, \tag{23.71}$$

$$\delta\lambda = -\frac{3\lambda^2}{2} \int_{\text{shell}} \frac{d^d k}{(2\pi)^d}\, \frac{1}{(k^2 + m^2)^2}, \tag{23.72}$$

the symmetry factor $\frac{1}{2}$ in Eq. (23.71) being that of the tadpole and the factor $\frac{3}{2} = 3 \times \frac{1}{2}$ in Eq. (23.72) counting the three $(s,\, t,\, u)$ channels of the four-point bubble each with its own symmetry factor $\frac{1}{2}$. The integration region is the

thin shell $b\Lambda < |k| < \Lambda$. To reduce the angular integral in $d = 4$ we use that the (d=4) unit-sphere has surface area $S_3 = 2\pi^2$, so $\int d^4k/(2\pi)^4\, f(|k|) = (2\pi^2/(2\pi)^4)\int k^3\, f\, dk = (1/8\pi^2)\int k^3 f\, dk$, and over an infinitesimal shell at $|k| \simeq \Lambda$ of thickness $\Lambda - b\Lambda = \Lambda\, dt$,

$$\int_{\text{shell}} \frac{d^4k}{(2\pi)^4}\,\frac{1}{k^2+m^2} = \frac{1}{8\pi^2}\,\frac{\Lambda^3}{\Lambda^2+m^2}\,\Lambda\, dt = \frac{1}{8\pi^2}\,\frac{\Lambda^4}{\Lambda^2+m^2}\, dt, \tag{23.73}$$

$$\int_{\text{shell}} \frac{d^4k}{(2\pi)^4}\,\frac{1}{(k^2+m^2)^2} = \frac{1}{8\pi^2}\,\frac{\Lambda^4}{(\Lambda^2+m^2)^2}\, dt. \tag{23.74}$$

These are pure numbers times dt once we pass to dimensionless variables, which is exactly what the rescaling step then demands.

Worked Example 23.11 (The dimensionless ϕ^4 flow equations, assembled in full)**.** We combine the rescaling of Section 23.3.2 with the shell corrections Eqs. (23.71) to (23.74) to obtain the differential flow of $\tilde m^2$ and $\tilde\lambda$ in $d = 4$, carrying every factor.

Mass. A single step does two things to m^2 in sequence. First the shell shifts $m^2 \to m^2 + \delta m^2$ while lowering the cutoff to $b\Lambda$. Then the rescaling re-expresses the result in units of the restored cutoff Λ, which for the dimensionless mass $\tilde m^2 = m^2/\Lambda^2$ means dividing by b^2 (the engineering factor b^{-D} with $D = [m^2] = 2$). Hence

$$\tilde m^{2\prime} = \frac{m^2+\delta m^2}{(b\Lambda)^2} = \frac{1}{b^2}\Big(\tilde m^2 + \frac{\delta m^2}{\Lambda^2}\Big). \tag{23.75}$$

Insert Eq. (23.71) with Eq. (23.73): $\delta m^2/\Lambda^2 = (\lambda/2)(1/8\pi^2)\,\Lambda^2/(\Lambda^2+m^2)\, dt = (\lambda/16\pi^2)\,(1+\tilde m^2)^{-1}\, dt$. With $b^{-2} = 1 + 2\, dt + O(dt^2)$,

$$\tilde m^{2\prime} = \tilde m^2 + 2\tilde m^2\, dt + \frac{\tilde\lambda}{16\pi^2}\,\frac{1}{1+\tilde m^2}\, dt + O(dt^2), \tag{23.76}$$

so that, reading off $d\tilde m^2/dt = (\tilde m^{2\prime} - \tilde m^2)/dt$,

$$\boxed{\frac{d\tilde m^2}{dt} = 2\,\tilde m^2 + \frac{\tilde\lambda}{16\pi^2}\,\frac{1}{1+\tilde m^2}} \tag{23.77}$$

with the engineering term $2\tilde m^2$ (the mass is relevant, $D = 2$) and the one-loop tadpole term. Near the Gaussian point, $\tilde m^2 \to 0$, this reduces to $d\tilde m^2/dt = 2\tilde m^2 + \tilde\lambda/16\pi^2$.

Quartic. The coupling λ is marginal in $d = 4$ ($D = [\lambda] = 0$), so the engineering factor is $b^{-0} = 1$: the rescaling does not touch $\tilde\lambda$ at this order, and the entire flow is the shell correction. From Eq. (23.72) with Eq. (23.74),

$$\delta\lambda = -\frac{3\lambda^2}{2}\cdot\frac{1}{8\pi^2}\,\frac{1}{(1+\tilde m^2)^2}\, dt = -\frac{3\tilde\lambda^2}{16\pi^2}\,\frac{1}{(1+\tilde m^2)^2}\, dt, \tag{23.78}$$

and since $\tilde{\lambda}' = \tilde{\lambda} + \delta\lambda$ here,

$$\boxed{\frac{d\tilde{\lambda}}{dt} = -\frac{3\,\tilde{\lambda}^2}{16\pi^2}\,\frac{1}{(1+\tilde{m}^2)^2}} \tag{23.79}$$

which near the Gaussian point becomes $d\tilde{\lambda}/dt = -3\tilde{\lambda}^2/16\pi^2$.

Together Eqs. (23.77) and (23.79) are the explicit Wilsonian flow of ϕ^4 theory,

$$\frac{d\tilde{m}^2}{dt} = 2\tilde{m}^2 + \frac{\tilde{\lambda}}{16\pi^2}\,\frac{1}{1+\tilde{m}^2}, \qquad \frac{d\tilde{\lambda}}{dt} = -\frac{3\tilde{\lambda}^2}{16\pi^2}\,\frac{1}{(1+\tilde{m}^2)^2}. \tag{23.80}$$

Two features of Eq. (23.80) deserve to be named, as both will recur. First, the quartic flow Eq. (23.79) is driven by $\tilde{\lambda}^2$ with a fixed numerical coefficient $3/16\pi^2$. This is the *beta-function coefficient* of ϕ^4. The sign here is negative because t runs toward the infrared: the quartic coupling *decreases* as we coarse-grain to long distances, equivalently it *grows* toward the ultraviolet. The continuum Callan–Symanzik beta function of Chapter 24, defined with respect to the subtraction scale μ that runs the opposite way, is therefore

$$\beta_\lambda \equiv \mu\,\frac{d\lambda}{d\mu} = -\frac{d\tilde{\lambda}}{dt}\Big|_{\tilde{m}^2\to 0} = +\frac{3\lambda^2}{16\pi^2} + O(\lambda^3), \tag{23.81}$$

the same coefficient computed by diagrammatic renormalization in Section 24.2. The two derivations—coarse-graining a sharp momentum cutoff here, dimensional regularization and minimal subtraction there—agree on the number $3/16\pi^2$ because it controls a scheme-independent, leading-order rate, exactly as the chapter introduction promised. Second, the engineering terms ($2\tilde{m}^2$ and the absent $0\cdot\tilde{\lambda}$) are precisely the linearization Eq. (23.68) near the free theory, confirming that canonical power counting is the tree-level shadow of the full flow.

Caution

The denominators $1/(1+\tilde{m}^2)$ and $1/(1+\tilde{m}^2)^2$ in Eq. (23.80) are not optional polish; they are the statement that a massive mode in the shell is harder to excite and so contributes less to the flow. They are nonpolynomial in the couplings, a generic feature of sharp-cutoff Wilsonian flows, and they signal that the truncation to $(\tilde{m}^2, \tilde{\lambda})$ is itself an approximation: the same shell integration also generates a ϕ^6 vertex and momentum-dependent corrections to the kinetic term, which we have dropped because they are irrelevant (Eq. (23.68) with $D_6 = -2 < 0$) and higher order. Keeping only the relevant and marginal couplings is consistent precisely because the irrelevant ones are driven to zero by their engineering scaling; this is the self-consistency that the fixed-point analysis of the next sections will make rigorous.

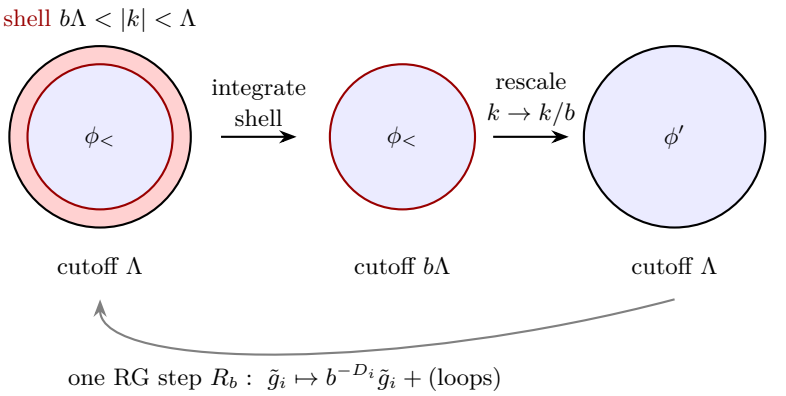


Figure 23.4: One renormalization-group step in momentum space. The fast modes in the shell $b\Lambda < |k| < \Lambda$ are integrated out (Section 23.2), lowering the cutoff to $b\Lambda$; rescaling $k \to k/b$ and $\phi_< \to \zeta^{-1}\phi'$ restores the cutoff to Λ and the kinetic term to canonical form. The composite is the self-map R_b of Theorem 23.10, whose iteration is the flow.

23.3.4 The anomalous dimension

We deferred one factor in ζ: the wavefunction renormalization Z in Eq. (23.62). It measures the part of the field rescaling that is *not* fixed by canonical counting, and it is the origin of the anomalous dimension.

Definition 23.12 (Anomalous dimension)**.** Write the field rescaling factor of Eq. (23.62) as $\zeta = b^{(d-2+\eta)/2}$, so that the wavefunction renormalization is $Z = b^{-\eta}$ and

$$\eta = -\frac{d\ln Z}{dt}, \tag{23.82}$$

the rate at which the kinetic-term coefficient generated by the shell departs from unity. The quantity η is the *anomalous dimension* of the field; the full scaling dimension of ϕ is $\Delta_\phi = (d-2+\eta)/2 = [\phi] + \eta/2$, the canonical dimension shifted by $\eta/2$. (In the continuum language of Chapter 24 the same quantity is $2\gamma_\phi = \eta$, with $\gamma_\phi = \frac{1}{2}\, d\ln Z_\phi / d\ln\mu$.)

The anomalous dimension feeds back into the coupling flow. Restoring the $Z^{-1/2}$ factor in ζ, the rescaling of an n-field operator Eq. (23.64) carries an extra $Z^{-n/2} = b^{n\eta/2}$, so the engineering exponent in Eq. (23.65) is corrected,

$$g_n' = g_n\, b^{-[g_n]+n\eta/2} = g_n\, b^{-(D_n - n\eta/2)}, \tag{23.83}$$

and the linear flow Eq. (23.68) becomes $\tilde{g}_i' = b^{-(D_i - n_i\eta/2)}\tilde{g}_i$, i.e. the engineering dimension D_i in Eq. (23.70) is replaced by the *full* scaling dimension $D_i - n_i\eta/2$. This is the precise sense in which interactions shift the dimensions of operators

away from their classical values: the leading flow is still linear, but its eigenvalues are the true scaling dimensions, not the engineering ones, and the difference is the anomalous dimension.

For ϕ^4 theory at one loop, η vanishes. The only $O(\lambda)$ correction to the two-point function is the tadpole Eq. (23.71), and its integrand is *independent of the external momentum*: contracting two legs of a single quartic vertex leaves the loop momentum unconstrained by the external lines, so the correction is a pure constant added to m^2, with no term proportional to k^2. Hence $Z = 1 + O(\lambda^2)$, the kinetic term is uncorrected at this order, and

$$\eta = O(\lambda^2), \qquad \text{equivalently} \qquad \eta = O(\varepsilon^2) \text{ at the Wilson–Fisher point of } \textit{Chapter } 52. \tag{23.84}$$

The first nonzero contribution arises at two loops, from the "sunset" diagram—two quartic vertices joined by three internal lines—whose loop integral *does* depend on the external momentum and therefore renormalizes k^2. That diagram and the resulting $\eta \sim O(\lambda^2)$ were met in the counterterm analysis of Chapter 22 (Theorem 22.29); the small numerical value $\gamma_\phi = O(\varepsilon^2)$ quoted in the ϕ^4 theory specification is its avatar at the fixed point. For the one-loop flow equations Eq. (23.80) we may therefore set $\eta = 0$ consistently, which is why the quartic flow there carried no anomalous correction and the field rescaling was purely canonical.

Physical Insight

That η is small and slow to appear is not an accident of ϕ^4 but the reason the notion of a particle survives interactions at all. A nonzero η means the field has no sharp dimension; its two-point function decays as a noninteger power $\langle \phi(x)\phi(0)\rangle \sim |x|^{-(d-2+\eta)}$ rather than the canonical $|x|^{-(d-2)}$, the hallmark of a scale-invariant theory with no particle interpretation. In four dimensions η stays zero through one loop and remains tiny; the field behaves almost canonically, and ordinary perturbation theory in nearly-canonical fields is justified. At a nontrivial fixed point in lower dimension η becomes an honest critical exponent, and the language shifts from particles to operators—the language of the conformal bootstrap to which ϕ^4 theory is eventually handed.

We now possess the renormalization-group flow as a differential dynamical system on theory space, Eq. (23.70), with ϕ^4 theory realized explicitly in Eq. (23.80). The construction so far is perturbative: the loop terms came from a one-loop shell integration and the flow is trustworthy only near the free theory. The next section removes that limitation, deriving an exact functional differential equation for the cutoff dependence of the Wilson effective action—the exact renormalization-group equation—and showing that the perturbative flow assembled here is its leading expansion. With the exact flow in hand we will

turn, in the sections after, to the structure that gives the chapter its name: the fixed points of R_b, the classification of perturbations around them as relevant, marginal, or irrelevant, and the universality that follows.

23.4 The Exact Renormalization Group Equation

The previous sections turned coarse-graining into motion. Integrating out a thin momentum shell and then rescaling momenta and fields back to the original cutoff defined a self-map of the space of couplings; iterating it, and passing to the infinitesimal limit, produced a differential flow on theory space whose one-loop restriction to the two couplings of ϕ^4 theory is the pair of equations Eq. (23.80). That flow is genuinely useful, but it is built on an approximation we have not yet justified to ourselves: in computing the shell integral we kept only the lowest nontrivial order in the couplings, the tadpole and the bubble. Nothing in the Wilsonian idea requires this. The operation "integrate out the modes above Λ" is defined nonperturbatively—it is a Gaussian-weighted functional integral over the fast modes—and we are free to ask for the *exact* change of the effective action under an infinitesimal lowering of Λ, with no expansion in λ at all.

The reward for asking is a single functional differential equation. It governs the entire cutoff dependence of the Wilsonian effective action; its right-hand side is finite at every scale, term by term, with no reference to a bare theory or to counterterms; and it is, remarkably, *one-loop exact*—a closed equation whose only ingredients are the scale derivative of the cutoff propagator contracted against the first and second functional derivatives of the action. The perturbative shell flow of the previous sections is recovered as the leading expansion of this equation, which is the consistency check this section owes. And once the flow is exact and continuous, its fixed points become genuine objects to be studied rather than artifacts of a truncation—the program of the next section.

We derive the exact equation twice, by the two historically and conceptually distinct routes. The first, due to Wegner and Houghton, uses the sharp momentum cutoff already in play and extracts the infinitesimal flow from a Gaussian integral over an infinitesimally thin shell. The second, due to Polchinski, replaces the sharp cutoff by a smooth one and demands that the full generating functional be independent of where the smooth split is placed; the resulting equation is manifestly finite and is the form on which the modern functional renormalization group is built. The two are the same physics in two regularizations, and we will see precisely why the smooth version is the better behaved of the pair.

Notation Watch

This entire chapter works in Euclidean signature, reached by the Wick rotation of Chapter 10: the weight is e^{-S_E} with the all-plus Euclidean ϕ^4

action Eq. (10.161), and the free propagator is the Euclidean $1/(p^2+m^2)$ of Eq. (10.168), with $p^2 \equiv p_E^2 \geq 0$ the Euclidean square. We write $\int_p \equiv \int \mathrm{d}^d p/(2\pi)^d$. Two expansion parameters named ε must not be confused. The dimensional-regularization regulator of Chapters 21 and 22 was $d = 4 - 2\varepsilon_{\mathrm{dimreg}}$ with $\varepsilon_{\mathrm{dimreg}} \to 0$. The parameter of the present (and the next) chapter is the *dimension deficit* $\varepsilon \equiv 4 - d$, held small but finite, in which the Wilson–Fisher fixed point is universally quoted; it equals $2\varepsilon_{\mathrm{dimreg}}$. We use $\varepsilon = 4 - d$ throughout, matching Chapter 24 (Section 24.7.5).

23.4.1 The cutoff effective action

Fix a microscopic theory with an ultraviolet cutoff Λ_0: a Euclidean generating functional in which only field modes with $|p| < \Lambda_0$ are integrated,

$$Z = \int_{|p|<\Lambda_0} \mathcal{D}\phi \, \exp\bigl(-S_{\Lambda_0}[\phi]\bigr), \tag{23.85}$$

with S_{Λ_0} the bare action, for ϕ^4 theory the cutoff version of Eq. (10.161). The Wilsonian effective action at a lower scale $\Lambda < \Lambda_0$ is obtained by performing the integral over the fast modes while holding the slow modes fixed as external data. Split the field as $\phi = \phi_< + \phi_>$, where $\phi_<$ has support $|p| < \Lambda$ and $\phi_>$ has support $\Lambda < |p| < \Lambda_0$, and define

$$e^{-S_\Lambda[\phi_<]} \equiv \int_{\Lambda<|p|<\Lambda_0} \mathcal{D}\phi_> \, \exp\bigl(-S_{\Lambda_0}[\phi_< + \phi_>]\bigr). \tag{23.86}$$

The object $S_\Lambda[\phi_<]$ is the *cutoff (Wilsonian) effective action*. By construction the integral Eq. (23.85) factorizes, $Z = \int_{|p|<\Lambda} \mathcal{D}\phi_< \, e^{-S_\Lambda[\phi_<]}$, so *the partition function, and every long-distance correlation function built from slow modes, is exactly preserved as* Λ *is lowered*: nothing physical depends on Λ. This invariance is the engine of everything that follows. Lowering Λ does not change the theory; it only changes the variables in which the same theory is written, shifting information from explicit integration into the couplings of S_Λ.

Two warnings about what S_Λ is and is not. First, it is not the one-particle irreducible effective action Γ of Theorem A.28: Γ is the Legendre transform (Theorem 4.31) of the generating functional of connected diagrams and integrates out *all* fluctuations, whereas S_Λ integrates out only the fast ones and is still to be functionally integrated over $\phi_<$. The two are siblings, and the relation between their flows is the content of the modern "effective average action" formalism, to which we return in the remark closing Section 23.4.3; here we follow the Wilsonian object S_Λ directly. Second, S_Λ is in general a fully nonlocal functional. Even if S_{Λ_0} contains only ϕ^2 and ϕ^4, integrating out a shell generates every

interaction permitted by the symmetries— ϕ^6, $(\partial\phi)^2\phi^2$, and an infinite tower beyond. Writing

$$S_\Lambda[\phi] = \sum_n \frac{1}{n!} \int_{p_1} \cdots \int_{p_n} (2\pi)^d \, \delta\Big(\textstyle\sum_i p_i\Big) \, S_\Lambda^{(n)}(p_1, \ldots, p_n) \, \phi(p_1) \cdots \phi(p_n), \quad (23.87)$$

the flow is a flow of the entire collection of *vertex functions* $S_\Lambda^{(n)}$, equivalently of the infinitely many couplings multiplying the local operators in a derivative expansion of Eq. (23.87). The art of the renormalization group is that, near a fixed point, only finitely many of these directions matter; the exact equation is what makes that statement provable.

23.4.2 The Wegner–Houghton sharp-cutoff equation

We extract the infinitesimal flow directly from the definition Eq. (23.86). Lower the cutoff from Λ to $\Lambda(1 - \mathrm{d}\ell)$ with $\mathrm{d}\ell > 0$ infinitesimal, so that the modes to be integrated occupy the thin shell

$$\mathcal{S}_\Lambda = \{\, p : \ \Lambda(1 - \mathrm{d}\ell) < |p| < \Lambda \,\}, \qquad \mathrm{d}\ell = -\frac{\mathrm{d}\Lambda}{\Lambda}, \quad (23.88)$$

and write the shell field χ, supported on $\mathcal{S}_\Lambda$, in place of the fast modes. With ϕ now denoting the slow field (support $|p| < \Lambda(1 - \mathrm{d}\ell)$),

$$e^{-S_{\Lambda(1-\mathrm{d}\ell)}[\phi]} = \int \mathcal{D}\chi \ \exp\big(-S_\Lambda[\phi + \chi]\big). \quad (23.89)$$

Taylor-expand the exponent in the shell field. Because χ is supported only on $\mathcal{S}_\Lambda$, every functional derivative below is evaluated at the slow configuration ϕ and every momentum integral against χ runs over the shell:

$$\begin{aligned} S_\Lambda[\phi + \chi] = S_\Lambda[\phi] &+ \int_{\mathcal{S}_\Lambda} \frac{\mathrm{d}^d p}{(2\pi)^d} \, \frac{\delta S_\Lambda}{\delta\phi(-p)} \, \chi(p) \\ &+ \frac{1}{2} \int_{\mathcal{S}_\Lambda} \frac{\mathrm{d}^d p \, \mathrm{d}^d p'}{(2\pi)^{2d}} \, \chi(p) \, \frac{\delta^2 S_\Lambda}{\delta\phi(-p)\, \delta\phi(-p')} \, \chi(p') + O(\chi^3). \end{aligned} \quad (23.90)$$

The cubic and higher terms are negligible in the limit $\mathrm{d}\ell \to 0$, and this is the crucial structural fact. A connected contribution to $S_{\Lambda(1-\mathrm{d}\ell)}[\phi]$ built from the vertices in Eq. (23.90) carries one shell-momentum integral for each independent loop of shell momentum and one for each shell line joining the slow vertices; each such integral is proportional to the shell volume and hence to $\mathrm{d}\ell$. The Gaussian (quadratic) truncation already captures every contribution with a single power of the shell measure—one shell loop, or one shell line connecting two pieces of the action—which is everything that survives at $O(\mathrm{d}\ell)$. Terms cubic and higher in χ first contribute at $O(\mathrm{d}\ell^2)$ and drop. *The infinitesimal shell integral is exactly Gaussian.*

Perform it. Abbreviate the (slow-field-dependent) operators

$$S^{(1)}(p) \equiv \frac{\delta S_\Lambda}{\delta\phi(-p)}\Big|_\phi, \qquad S^{(2)}(p,p') \equiv \frac{\delta^2 S_\Lambda}{\delta\phi(-p)\,\delta\phi(-p')}\Big|_\phi, \tag{23.91}$$

both regarded as a vector and a matrix indexed by shell momenta. The Gaussian integral over χ is the standard finite-dimensional formula promoted to the shell,

$$\int \mathcal{D}\chi\ \exp\Big[-\tfrac{1}{2}\,\chi\, S^{(2)}\chi - S^{(1)}\chi\Big] = \big(\det S^{(2)}\big)^{-1/2}\ \exp\Big[\tfrac{1}{2}\,S^{(1)}\big(S^{(2)}\big)^{-1}S^{(1)}\Big], \tag{23.92}$$

where the field-independent constant $(2\pi)^{(\dim)/2}$ has been absorbed into the measure and all products and the inverse are taken over $\mathcal{S}_\Lambda$. Taking the logarithm of Eq. (23.89) with Eq. (23.92) and $\det = \exp\operatorname{Tr}\ln$,

$$S_{\Lambda(1-\mathrm{d}\ell)}[\phi] = S_\Lambda[\phi] + \tfrac{1}{2}\ \operatorname{Tr}_{\mathcal{S}_\Lambda}\ln S^{(2)} - \tfrac{1}{2}\,S^{(1)}\big(S^{(2)}\big)^{-1}S^{(1)}\Big|_{\mathcal{S}_\Lambda}. \tag{23.93}$$

Since $S_{\Lambda(1-\mathrm{d}\ell)} = S_\Lambda - \mathrm{d}\Lambda\,\partial_\Lambda S_\Lambda = S_\Lambda + \Lambda\,\mathrm{d}\ell\,\partial_\Lambda S_\Lambda$, the left side minus S_Λ is $\Lambda\,\mathrm{d}\ell\,\partial_\Lambda S_\Lambda = -\mathrm{d}\ell\,\Lambda\partial_\Lambda S_\Lambda\cdot(-1)$; writing the flow in terms of $\partial_\ell \equiv -\Lambda\partial_\Lambda$ (the derivative with respect to RG "time" ℓ, which increases as Λ decreases) gives the Wegner–Houghton equation.

Theorem 23.13 (Wegner–Houghton sharp-cutoff flow)**.** *Let S_Λ be the sharp-cutoff Wilsonian effective action Eq. (23.86) and $\ell = \ln(\Lambda_0/\Lambda)$ the RG time. Then S_Λ obeys the exact functional differential equation*

$$\boxed{\partial_\ell S_\Lambda[\phi] = -\frac{1}{2\,\mathrm{d}\ell}\left[\operatorname{Tr}_{\mathcal{S}_\Lambda}\ln\frac{\delta^2 S_\Lambda}{\delta\phi\,\delta\phi}\ -\ \frac{\delta S_\Lambda}{\delta\phi}\Big(\frac{\delta^2 S_\Lambda}{\delta\phi\,\delta\phi}\Big)^{-1}\frac{\delta S_\Lambda}{\delta\phi}\Big|_{\mathcal{S}_\Lambda}\right].} \tag{23.94}$$

The trace and the inverse run over the infinitesimal shell $\mathcal{S}_\Lambda$ of Eq. (23.88); the explicit shell measure carried by $\operatorname{Tr}_{\mathcal{S}_\Lambda}$ and by the shell restriction of the second line is itself $\propto \mathrm{d}\ell$, so that the right-hand side is finite and the factor $\mathrm{d}\ell^{-1}$ merely strips it. The first term is a one-loop (determinant) contribution of the shell modes; the second is a tree contribution in which two slow-field vertices are joined by a single shell line.

Proof. Equation Eq. (23.93) is the result of the exact Gaussian integral Eq. (23.92), valid because the higher terms of Eq. (23.90) are $O(\mathrm{d}\ell^2)$. Subtracting S_Λ from both sides and dividing by $\Lambda\,\mathrm{d}\ell$ with $\partial_\ell = -\Lambda\partial_\Lambda$ yields Eq. (23.94). The identification of the two terms is read off Eq. (23.92): $\operatorname{Tr}\ln S^{(2)}$ is the logarithm of the fluctuation determinant, i.e. the sum of one-loop shell diagrams in the background ϕ, while $S^{(1)}(S^{(2)})^{-1}S^{(1)}$ ties two functional gradients of S_Λ together with the shell propagator $(S^{(2)})^{-1}$, a tree in the shell line. □

Two features of Eq. (23.94) deserve emphasis before we improve it. It is *exact*: no expansion in λ was made, only the expansion in the shell thickness $\mathrm{d}\ell$, which is exact in the infinitesimal limit. And it is *closed*: the right-hand side is built entirely from S_Λ itself and its functional derivatives, so Eq. (23.94) is a self-contained evolution equation for the functional S_Λ, the first such equation we have met.

It also has a defect, and naming it motivates the next subsection. The tree term $S^{(1)}(S^{(2)})^{-1}S^{(1)}$ requires the inverse of $S^{(2)}$ *restricted to the shell*. For a slow background, $S^{(1)}(p)$ at a shell momentum p is nonzero only when several slow momenta conspire to add up to a shell momentum, and $(S^{(2)})^{-1}$ in the shell multiplies these contributions; when the slow external momenta are pushed to special values—most sharply, when an external momentum is itself tuned to the cutoff—the sharp edge of the shell produces distributions multiplying distributions, the notorious $\delta(0)$-type ambiguities of the sharp cutoff. The flow is well defined, but extracting local couplings from it requires care precisely because the cutoff is a step function. The cure is to soften the step.

23.4.3 The Polchinski smooth-cutoff equation

Replace the sharp separation of modes by a smooth one. Introduce a cutoff profile $K(s)$, a smooth monotone function with

$$K(0) = 1, \qquad K(s) \xrightarrow{s\to\infty} 0, \tag{23.95}$$

that equals one in the deep infrared and falls to zero in the deep ultraviolet, and use it to damp the propagator rather than to truncate the modes.

Definition 23.14 (Regularized propagator and split action)**.** The *regularized (cutoff) propagator* at scale Λ is

$$C_\Lambda(p) \;=\; \frac{K(p^2/\Lambda^2)}{p^2+m^2}, \tag{23.96}$$

the Euclidean propagator Eq. (10.168) multiplied by the profile, so that modes with $|p| \gg \Lambda$ are suppressed and modes with $|p| \ll \Lambda$ propagate freely. The action is split into the regularized free part and an interaction part,

$$S_\Lambda[\phi] = \tfrac{1}{2}\int_p \phi(-p)\, C_\Lambda(p)^{-1}\, \phi(p) \;+\; S_{\mathrm{int},\Lambda}[\phi], \tag{23.97}$$

with $S_{\mathrm{int},\Lambda}$ the *Wilsonian interaction functional*, the object whose flow we seek.

The generating functional built from Eq. (23.97),

$$Z = \int \mathcal{D}\phi\; \exp\Big(-\tfrac{1}{2}\,\phi\, C_\Lambda^{-1}\,\phi - S_{\mathrm{int},\Lambda}[\phi]\Big), \tag{23.98}$$

must be independent of Λ, since Λ marks nothing but the arbitrary place where we have chosen to draw the line between "free propagation" and "interaction." This single requirement, $\mathrm{d}Z/\mathrm{d}\Lambda = 0$, fixes the flow of $S_{\mathrm{int},\Lambda}$.

Theorem 23.15 (Polchinski's equation)**.** *The interaction functional flows so as to keep Eq. (23.98) independent of Λ if and only if, up to a field-independent (vacuum-energy) term,*

$$\boxed{\partial_\Lambda S_{\mathrm{int},\Lambda} = \frac{1}{2}\int_p \partial_\Lambda C_\Lambda(p)\left[\frac{\delta^2 S_{\mathrm{int},\Lambda}}{\delta\phi(p)\,\delta\phi(-p)} - \frac{\delta S_{\mathrm{int},\Lambda}}{\delta\phi(p)}\,\frac{\delta S_{\mathrm{int},\Lambda}}{\delta\phi(-p)}\right].} \tag{23.99}$$

Equivalently, the unnormalized weight $\Psi_\Lambda \equiv e^{-S_{\mathrm{int},\Lambda}}$ obeys the linear functional heat equation

$$\partial_\Lambda \Psi_\Lambda = -\frac{1}{2}\int_p \partial_\Lambda C_\Lambda(p)\,\frac{\delta^2\Psi_\Lambda}{\delta\phi(p)\,\delta\phi(-p)} \;+\; (vac.), \tag{23.100}$$

whose "diffusion" is generated by the scale derivative of the propagator.

Proof. Differentiate Eq. (23.98) with respect to Λ. The integrand depends on Λ through the Gaussian weight and through S_{int}, so

$$\partial_\Lambda Z = \int \mathcal{D}\phi\,\Big[\,\partial_\Lambda(e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi})\,\Psi_\Lambda + e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}\,\partial_\Lambda\Psi_\Lambda\,\Big]. \tag{23.101}$$

For the first term, use $\partial_\Lambda C_\Lambda^{-1} = -C_\Lambda^{-1}(\partial_\Lambda C_\Lambda)C_\Lambda^{-1}$, so that

$$\partial_\Lambda(e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}) = \tfrac{1}{2}\int_p (C_\Lambda^{-1}\phi)(p)\,\partial_\Lambda C_\Lambda(p)\,(C_\Lambda^{-1}\phi)(-p)\,e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}, \tag{23.102}$$

where C_Λ is diagonal in momentum, $C_\Lambda(p,p') = C_\Lambda(p)\,(2\pi)^d\delta(p+p')$. Now convert the two factors of $C_\Lambda^{-1}\phi$ into functional derivatives of the Gaussian. From $\dfrac{\delta}{\delta\phi(p)}e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi} = -(C_\Lambda^{-1}\phi)(-p)\,e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}$ one finds the second derivative

$$\frac{\delta^2}{\delta\phi(p)\,\delta\phi(-p)}\,e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi} = \Big[(C_\Lambda^{-1}\phi)(-p)(C_\Lambda^{-1}\phi)(p) - C_\Lambda^{-1}(p)\Big]\,e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}, \tag{23.103}$$

the subtracted $C_\Lambda^{-1}(p)$ being the contraction of the two derivatives with each other. Solving Eq. (23.103) for the bilinear in $C_\Lambda^{-1}\phi$ and inserting into Eq. (23.102),

$$\partial_\Lambda(e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}) = \tfrac{1}{2}\int_p \partial_\Lambda C_\Lambda(p)\left[\frac{\delta^2}{\delta\phi(p)\delta\phi(-p)} + C_\Lambda^{-1}(p)\right]e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi}. \tag{23.104}$$

The term with $C_\Lambda^{-1}(p)$ is field independent: it is $\frac{1}{2}\int_p \partial_\Lambda C_\Lambda(p)\,C_\Lambda^{-1}(p) = \frac{1}{2}\partial_\Lambda \operatorname{Tr}\ln C_\Lambda$, the flow of the free vacuum energy, which we carry along as "(vac.)" and drop

from the field-dependent equation. Insert Eq. (23.104) into Eq. (23.101) and integrate the two functional derivatives by parts onto Ψ_Λ (the functional measure is translation invariant and the Gaussian kills the boundary terms),

$$\partial_\Lambda Z = \int \mathcal{D}\phi \, e^{-\frac{1}{2}\phi C_\Lambda^{-1}\phi} \left[\frac{1}{2} \int_p \partial_\Lambda C_\Lambda(p) \, \frac{\delta^2 \Psi_\Lambda}{\delta\phi(p)\delta\phi(-p)} + \partial_\Lambda \Psi_\Lambda \right] + (\text{vac.}). \quad (23.105)$$

The integral vanishes for all choices of the slow data—hence Z is Λ-independent—if and only if the bracket vanishes, which is exactly the heat equation Eq. (23.100). Finally convert from Ψ_Λ to $S_{\text{int},\Lambda}$. With $\Psi = e^{-S_{\text{int}}}$,

$$\partial_\Lambda \Psi = -(\partial_\Lambda S_{\text{int}})\, \Psi, \qquad \frac{\delta^2 \Psi}{\delta\phi(p)\delta\phi(-p)} = \left[-\frac{\delta^2 S_{\text{int}}}{\delta\phi(p)\delta\phi(-p)} + \frac{\delta S_{\text{int}}}{\delta\phi(p)} \frac{\delta S_{\text{int}}}{\delta\phi(-p)} \right] \Psi. \quad (23.106)$$

Substituting into Eq. (23.100), the common factor Ψ cancels and the overall sign flips, giving Eq. (23.99). □

The smooth cutoff has bought exactly what the sharp one lacked. The kernel that drives the flow is now $\partial_\Lambda C_\Lambda(p)$, and from Eq. (23.96) it is a smooth function peaked in a band $|p| \sim \Lambda$ of finite width rather than a delta function on a sphere. Writing $u = p^2/\Lambda^2$ and (for the massless propagator, sufficient for the marginal analysis) $C_\Lambda = K(u)/p^2$, a one-line computation gives the clean form

$$\partial_\Lambda C_\Lambda(p) = \frac{1}{p^2}\, \partial_\Lambda K(u) = \frac{1}{p^2}\, K'(u) \Big(-\frac{2p^2}{\Lambda^3} \Big) = -\frac{2}{\Lambda^3}\, K'(p^2/\Lambda^2), \quad (23.107)$$

which is supported precisely where K' is appreciable, i.e. in the shell $|p| \sim \Lambda$ (Fig. 23.5). Every momentum integral on the right-hand side of Eq. (23.99) is therefore cut off both in the ultraviolet (by K', which vanishes for $|p| \gg \Lambda$) and in the infrared (the slow fields supply the external scale); *the equation is finite at every Λ, with no divergences to subtract.* This is the precise sense in which the exact renormalization group dispenses with the counterterm machinery of Chapter 22: there is never an infinity in the first place, only a finite rate of change of finitely or infinitely many couplings.

Remark 23.16 (The Legendre-transformed flow and the effective action). The same demand—physics independent of the artificial scale—can be imposed on the Legendre-conjugate object instead of on S_{int}. If one adds an infrared cutoff to the modes *below* Λ and Legendre-transforms (Theorem 4.31) the cutoff generating functional of connected diagrams, one obtains the *effective average action* Γ_Λ, which interpolates between the bare action at $\Lambda = \Lambda_0$ and the full one-particle-irreducible effective action Γ of Theorem A.28 as $\Lambda \to 0$. Its flow is the Wetterich equation, $\partial_\ell \Gamma_\Lambda = \frac{1}{2}\,\mathrm{Tr}[(\Gamma_\Lambda^{(2)} + R_\Lambda)^{-1} \partial_\ell R_\Lambda]$, the Legendre-transformed twin of Eq. (23.99): again one-loop exact, again driven by the scale derivative of a regulator R_Λ contracted with the full (now field-dependent and resummed)

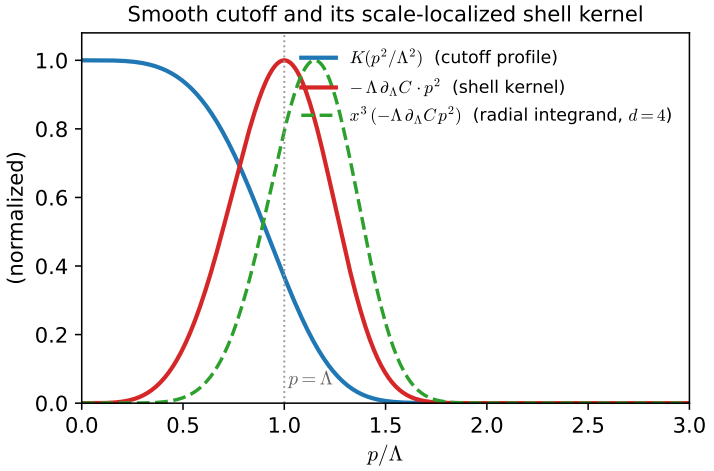


Figure 23.5: The smooth cutoff profile $K(p^2/\Lambda^2)$ (blue), interpolating from 1 in the infrared to 0 in the ultraviolet, and the shell kernel $-\Lambda\partial_\Lambda C\cdot p^2 = 2u\,K'(u)$ (red) that drives Polchinski's equation Eq. (23.99). The dashed curve weights the kernel by the $d=4$ radial phase space x^3 ($x=p/\Lambda$) to show where the momentum integral actually collects its support: a band of finite width centered on $p=\Lambda$, the smooth analogue of the Wegner–Houghton shell. The localization of the kernel at $p\sim\Lambda$ is what makes every term of the exact equation finite.

propagator $(\Gamma^{(2)}+R_\Lambda)^{-1}$. We will not develop it here; the Wilsonian form Eq. (23.99) suffices for the fixed-point analysis of the next section, and the connection to Γ is the bridge to the Callan–Symanzik description of Chapter 24.

23.4.4 The structure of the equation

Both exact equations share one architecture, and seeing it is worth a moment because it is the source of the equation's power. Polchinski's right-hand side Eq. (23.99) has exactly two terms, and each is the scale-derivative of the cutoff propagator, $\dot C_\Lambda \equiv \partial_\Lambda C_\Lambda$, contracted against derivatives of the action:

- a *loop* term, $\frac{1}{2}\int_p \dot C_\Lambda(p)\,\delta^2 S_{\rm int}/\delta\phi(p)\delta\phi(-p)$, in which $\dot C_\Lambda$ closes the two legs of a single vertex of $S_{\rm int}$ into a loop carrying shell momentum; and
- a *tree* term, $-\frac{1}{2}\int_p \dot C_\Lambda(p)\,[\delta S_{\rm int}/\delta\phi(p)]\,[\delta S_{\rm int}/\delta\phi(-p)]$, in which $\dot C_\Lambda$ is a single shell line joining two separate vertices of $S_{\rm int}$.

There is exactly one power of $\dot C_\Lambda$, hence exactly one shell line, on the right-hand side. This is the meaning of "one-loop exact": the functional flow closes at a single shell propagator per step, even though—because it is a differential equation, integrated over a continuum of scales—its *solution* contains diagrams

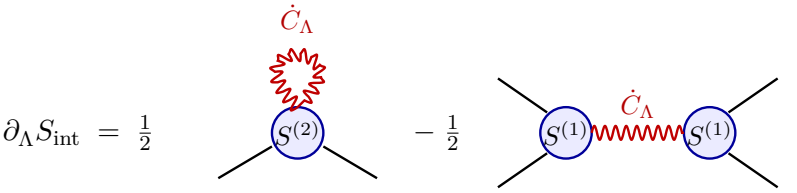


Figure 23.6: The two terms of the exact renormalization group equation Eq. (23.99). *Left:* the loop term contracts the two legs of a single vertex $S^{(2)} = \delta^2 S_{\rm int}/\delta\phi^2$ with the shell kernel $\dot{C}_\Lambda = \partial_\Lambda C_\Lambda$ (red, coiled), closing a loop of shell momentum and leaving slow external legs. *Right:* the tree term joins two vertices $S^{(1)} = \delta S_{\rm int}/\delta\phi$ by a single shell line $\dot{C}_\Lambda$. There is exactly one shell propagator per step—the equation is one-loop exact—and multi-loop diagrams are rebuilt by integrating the flow over scales.

of arbitrarily many loops. A multi-loop diagram of the underlying theory is rebuilt by the flow as a succession of single-shell steps, each contributing one $\dot{C}_\Lambda$, the loop momenta of the diagram being the shell momenta integrated out at the successive scales. The Wegner–Houghton form Eq. (23.94) has the same two pieces with $\mathrm{Tr}\ln S^{(2)}$ in place of the loop term (its expansion in $S^{(2)} = C^{-1} + \Sigma$ generates the single-shell loop and, term by term, the diagrams built from one shell loop) and $S^{(1)}(S^{(2)})^{-1}S^{(1)}$ as the tree. Figure 23.6 shows the two terms.

Physical Insight

The exact equation reorganizes perturbation theory along the most physical axis available: the scale. An ordinary Feynman diagram sums all internal momenta at once and then must be told, by hand and after the fact, which pieces were ultraviolet and needed subtracting. The flow does the opposite. At each instant $\mathrm{d}\ell$ it processes only the modes living at the current scale Λ, folding their effect into the couplings, and never revisits them. Divergences cannot appear because no integral ever runs to infinity; renormalizability becomes a statement about which couplings survive the processing rather than about which infinities cancel. This is the reversal of viewpoint promised at the head of the chapter, now made into an equation.

23.4.5 Recovering the perturbative shell flow

We now discharge the consistency check the section owes: the exact equation, truncated to the operators $\frac{1}{2}m^2\phi^2$ and $\frac{\lambda}{4!}\phi^4$ of ϕ^4 theory, must reproduce the one-loop shell flow Eq. (23.80). We do the computation in the Wegner–Houghton form, where the sign of each contribution is physically transparent (integrating out modes *adds* to the mass and *reduces* the quartic coupling), and then identify the two contributions with the loop term of Polchinski's equation, so that the

projection of *either* exact equation onto the two couplings is exhibited.

Worked Example 23.17 (The ϕ^4 truncation of the exact flow)**.** Start from Eq. (23.89) with the truncated action $S_\Lambda[\phi] = \int_x \left[\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\right]$. The shell field χ enters the quartic through $\frac{\lambda}{4!}(\phi+\chi)^4$. Of its terms, the one quadratic in χ is

$$\frac{\lambda}{4!}\binom{4}{2}\phi^2\chi^2 = \frac{\lambda}{4}\,\phi^2\chi^2, \tag{23.108}$$

so that the operator governing the Gaussian integral over χ is

$$S^{(2)}(x,y) = (-\partial^2 + m^2)\delta(x-y) + V(x)\delta(x-y), \qquad V(x) = \tfrac{\lambda}{2}\,\phi(x)^2, \tag{23.109}$$

i.e. the shell inverse propagator $G^{-1}(p) = p^2 + m^2$ plus a slow-field "potential" $V = \frac{\lambda}{2}\phi^2$. (The terms linear and cubic in χ feed the tree term and generate ϕ^6 and higher; they do not contribute to the ϕ^2 and ϕ^4 couplings at this order, and we set them aside.) By Eq. (23.93) the one-step change of the action is the determinant piece,

$$\Delta S \equiv S_{\Lambda(1-\mathrm{d}\ell)} - S_\Lambda = \tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda} \ln\left(G^{-1}+V\right) = \tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda} \ln G^{-1} + \tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda} \ln\left(1+GV\right), \tag{23.110}$$

the first term field independent (vacuum energy). Expand the second in powers of the slow field,

$$\tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda} \ln(1+GV) = \tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda}(GV) - \tfrac{1}{4}\,\mathrm{Tr}_{\mathcal{S}_\Lambda}(GVGV) + \cdots, \tag{23.111}$$

the two terms shown being respectively $O(\phi^2)$ and $O(\phi^4)$, which are the mass and the coupling.

Mass: the tadpole. With $V = \frac{\lambda}{2}\phi^2$ a multiplication operator, the trace of GV factorizes into the shell-momentum loop and the slow-field integral,

$$\tfrac{1}{2}\,\mathrm{Tr}_{\mathcal{S}_\Lambda}(GV) = \tfrac{1}{2}\Big(\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{p^2+m^2}\Big)\int_x \frac{\lambda}{2}\phi(x)^2 = \frac{\lambda}{4}\Big(\int_{\mathcal{S}_\Lambda}\frac{1}{p^2+m^2}\Big)\int_x \phi^2. \tag{23.112}$$

Reading this against the mass operator $\frac{1}{2}\,\delta m^2 \int_x \phi^2$ in ΔS gives the shift

$$\delta m^2 = \frac{\lambda}{2}\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{p^2+m^2} \;>\; 0, \tag{23.113}$$

positive: integrating out a shell raises the mass, exactly the tadpole of ϕ^4 theory first met in Section 21.6.

Coupling: the bubble. The $O(\phi^4)$ term is

$$-\tfrac{1}{4}\,\mathrm{Tr}_{\mathcal{S}_\Lambda}(GVGV) = -\tfrac{1}{4}\Big(\tfrac{\lambda}{2}\Big)^2 \int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{\mathrm{d}^d p'}{(2\pi)^d}\,\frac{\widetilde{\phi^2}(p-p')\,\widetilde{\phi^2}(p'-p)}{(p^2+m^2)(p'^2+m^2)}, \tag{23.114}$$

where $\widetilde{\phi^2}(q) = \int_k \phi(k)\phi(q-k)$ is the Fourier transform of ϕ^2. The two shell momenta are *not* independently free: the slow fields fix the transfer $q = p - p'$ to lie at the external (low) scale, so $p' = p - q$ is pinned to the shell whenever p is, and $\int_{\mathcal{S}_\Lambda} \mathrm{d}^d p'$ collapses to the single shell loop $\int_{\mathcal{S}_\Lambda} \mathrm{d}^d p$ with $p' \to -p$ (the relevant kinematics for the marginal coupling is $q \to 0$). Thus

$$-\tfrac{1}{4}\,\mathrm{Tr}_{\mathcal{S}_\Lambda}(GVGV) = -\frac{\lambda^2}{16}\Big(\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{(p^2+m^2)^2}\Big)\int_x \phi^4. \tag{23.115}$$

Reading this against the quartic operator $\frac{\delta\lambda}{4!}\int_x \phi^4$ in ΔS, and using $4!/16 = 24/16 = 3/2$, gives

$$\delta\lambda = -\frac{3\lambda^2}{2}\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{(p^2+m^2)^2} \;<\; 0, \tag{23.116}$$

negative: integrating out a shell screens the quartic coupling. The factor $\frac{3}{2}$ is the familiar one-loop combinatorial weight—three channels (s,t,u) each with symmetry factor $\frac{1}{2}$—now produced automatically by the single functional determinant.

Shell integrals in $d = 4$. The shell $\mathcal{S}_\Lambda$ of thickness $\mathrm{d}\ell = -\mathrm{d}\Lambda/\Lambda$ has d-volume element $\int_{\mathcal{S}_\Lambda}\mathrm{d}^d p/(2\pi)^d = \frac{S_d}{(2\pi)^d}\Lambda^{d-1}|\mathrm{d}\Lambda|$ with $S_d = 2\pi^{d/2}/\Gamma(d/2)$ the unit sphere area; in $d = 4$, $S_4/(2\pi)^4 = 2\pi^2/(2\pi)^4 = 1/8\pi^2$, and $|\mathrm{d}\Lambda| = \Lambda\,\mathrm{d}\ell$. Hence

$$\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^4 p}{(2\pi)^4}\,\frac{1}{p^2+m^2} = \frac{1}{8\pi^2}\,\frac{\Lambda^4}{\Lambda^2+m^2}\,\mathrm{d}\ell, \tag{23.117}$$

$$\int_{\mathcal{S}_\Lambda}\frac{\mathrm{d}^4 p}{(2\pi)^4}\,\frac{1}{(p^2+m^2)^2} = \frac{1}{8\pi^2}\,\frac{\Lambda^4}{(\Lambda^2+m^2)^2}\,\mathrm{d}\ell. \tag{23.118}$$

Dividing Eqs. (23.113) and (23.116) by $\mathrm{d}\ell$ and using $\partial_\ell = -\Lambda\partial_\Lambda$ gives the *dimensionful* shell flow,

$$\partial_\ell m^2\big|_{\text{shell}} = \frac{\lambda}{16\pi^2}\,\frac{\Lambda^4}{\Lambda^2+m^2}, \qquad \partial_\ell\lambda\big|_{\text{shell}} = -\frac{3\lambda^2}{16\pi^2}\,\frac{\Lambda^4}{(\Lambda^2+m^2)^2}. \tag{23.119}$$

Rescaling to dimensionless couplings. The flow Eq. (23.119) is only half of the renormalization-group step; the other half is the rescaling of Section 23.6's predecessor that restores the cutoff to Λ and trades the dimensionful m^2, λ for the dimensionless $\hat{m}^2 = m^2/\Lambda^2$ and $\hat{\lambda} = \lambda\,\Lambda^{d-4}$. Differentiating $\hat{m}^2 = m^2/\Lambda^2$ at fixed dimensionful data contributes the engineering term $\partial_\ell\hat{m}^2 \supset 2\hat{m}^2$ (since $\partial_\ell\Lambda^{-2} = 2\Lambda^{-2}$), and differentiating $\hat{\lambda} = \lambda\Lambda^{-\varepsilon}$ with $\varepsilon = 4-d$ contributes $\partial_\ell\hat{\lambda} \supset \varepsilon\hat{\lambda}$. Adding the rescaling to Eq. (23.119) (and using $\hat{\lambda} = \lambda$ in $d = 4$ for the loop terms) yields the dimensionless flow

$$\boxed{\partial_\ell\hat{m}^2 = 2\hat{m}^2 + \frac{\hat{\lambda}}{16\pi^2}\,\frac{1}{1+\hat{m}^2}, \qquad \partial_\ell\hat{\lambda} = \varepsilon\,\hat{\lambda} - \frac{3\hat{\lambda}^2}{16\pi^2}\,\frac{1}{(1+\hat{m}^2)^2}.} \tag{23.120}$$

These are precisely the flow equations Eq. (23.80) obtained in the previous section by direct shell integration and rescaling. The exact equation, truncated to the two couplings, reproduces them with identical coefficients.

The check closes ϕ^4 theory's exact-RG layer. Setting $\hat{m}^2 \to 0$ near the origin, the coupling equation in Eq. (23.120) reads $\partial_\ell \hat{\lambda} = \varepsilon\hat{\lambda} - \frac{3\hat{\lambda}^2}{16\pi^2}$, whose loop term gives, in four dimensions ($\varepsilon = 0$),

$$\partial_\ell \lambda = -\frac{3\lambda^2}{16\pi^2} \quad \Longleftrightarrow \quad \beta_\lambda \equiv \frac{\mathrm{d}\lambda}{\mathrm{d}\ln\mu} = +\frac{3\lambda^2}{16\pi^2}, \tag{23.121}$$

where the second form uses $\ln\mu = -\ell + \text{const}$ (the running scale μ increases toward the ultraviolet, ℓ toward the infrared). This is exactly the carried one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of ϕ^4 theory, now certified to be a consequence of the *exact* renormalization group and not merely of a one-loop diagram—the thread's number is exact-RG-consistent. The same algebra delivers the Gaussian fixed point at $\hat{\lambda} = 0$ and, for $0 < \varepsilon \ll 1$, the Wilson–Fisher zero of the coupling equation at $\hat{\lambda}_* = 16\pi^2\varepsilon/3$, the value Chapter 24 (Theorem 24.31) and Chapter 52 carry forward—but the analysis of those zeros is the business of the next section.

It remains to connect Eq. (23.120) explicitly to Polchinski's equation, so that the projection is seen to hold for that form too. In Eq. (23.99) the loop term, evaluated on the truncated $S_{\mathrm{int}} = \frac{\lambda}{4!}\int \phi^4$, has the second functional derivative $\delta^2 S_{\mathrm{int}}/\delta\phi(p)\delta\phi(-p) = \frac{\lambda}{2}\int_q \phi(q)\phi(-q)$, a pure ϕ^2 operator, so that the loop term reproduces Eq. (23.112) with $\int_{\mathcal{S}_\Lambda}$ replaced by the smooth shell integral $\int_p \dot{C}_\Lambda$ of Eq. (23.107); the mass flow is identical. The quartic flow arises one order deeper: the tree term generates a ϕ^6 vertex $S^{(6)} \sim \lambda^2$, which fed back through the loop term produces, after integrating the flow over one decade of scale, the combination $\int_p \dot{C}_\Lambda(p)\, C_\Lambda(p)$. That integral is independent of the shape of the profile K,

$$\int_p \dot{C}_\Lambda(p)\, C_\Lambda(p) = \frac{1}{16\pi^2}\int_0^\infty K'(u)K(u)\left(-\frac{2}{\Lambda}\right)\mathrm{d}u = \frac{1}{16\pi^2\Lambda}\left[K(0)^2 - K(\infty)^2\right] = \frac{1}{16\pi^2\Lambda}, \tag{23.122}$$

using Eq. (23.107), $\int \frac{\mathrm{d}^4p}{(2\pi)^4} f(p^2) = \frac{1}{16\pi^2}\int_0^\infty q^2\,\mathrm{d}q^2 f$, and $K(0) = 1, K(\infty) = 0$ from Eq. (23.95). The scheme independence of Eq. (23.122) is the exact-RG explanation of the scheme independence of the one-loop beta function: whatever the cutoff profile, the universal endpoints $K(0) = 1$ and $K(\infty) = 0$ fix the one-loop coefficient, and the result $-3\lambda^2/16\pi^2$ of Eq. (23.121) follows for any K.

23.4.6 Why renormalizability follows from the equation

The exact equation does more than reproduce the perturbative flow; it explains, in a single structural statement, why ϕ^4 theory is renormalizable—why a continuum limit $\Lambda_0 \to \infty$ exists at all and is controlled by finitely many numbers. We state

the criterion now and prove the linear-algebra fact it rests on in the next section; the point here is that the *exact* flow is what makes it a theorem rather than a hope.

Consider the flow Eq. (23.99) linearized about the Gaussian fixed point $S^*_{\rm int} = 0$ (the free theory, exactly scale invariant after rescaling). A small perturbation is a sum of local operators $\mathcal{O}_i$ with dimensionless couplings g_i, and to linear order each g_i flows multiplicatively,

$$\partial_\ell g_i = (\,d - \dim[\mathcal{O}_i]\,)\,g_i + O(g^2) \equiv +y_i\,g_i + O(g^2), \qquad y_i = d - \dim[\mathcal{O}_i], \tag{23.123}$$

the eigenvalue y_i being fixed, at the Gaussian point, by the canonical (engineering) dimension of $\mathcal{O}_i$ computed by the power counting of Chapter 21—this is the rescaling part of the step, and it is exactly the $2\hat{m}^2$ and $\varepsilon\hat{\lambda}$ terms seen in Eq. (23.120). An operator is *relevant* if $y_i > 0$ ($\dim[\mathcal{O}_i] < d$), *marginal* if $y_i = 0$, and *irrelevant* if $y_i < 0$. Solving Eq. (23.123), $g_i(\ell) = g_i(0)\,e^{+y_i \ell}$: as one flows to the infrared ($\ell \to +\infty$) the relevant couplings grow and the irrelevant ones are exponentially driven to zero.

Theorem 23.18 (Renormalizability as finiteness of the relevant spectrum)**.** *For ϕ^4 theory in $d = 4$, the operators of the field with canonical dimension $\dim[\mathcal{O}_i] \le d$—hence $y_i \ge 0$, relevant or marginal—are exactly*

$$\mathbf{1}\ (y = 4), \qquad \phi^2\ (y = 2), \qquad \phi^4\ (y = 0), \tag{23.124}$$

together with redundant total-derivative operators removable by field redefinition; every other symmetry-allowed operator (ϕ^6, $(\partial\phi)^2\phi^2$, $\phi^8, \dots$) has $\dim > 4$, hence $y < 0$ and is irrelevant. Consequently the continuum limit $\Lambda_0 \to \infty$ of the cutoff theory Eq. (23.85) exists along a finite-dimensional surface: it is reached by tuning the finitely many relevant and marginal couplings (here the mass and the quartic, the constant $\mathbf{1}$ being an additive vacuum energy), with the infinitely many irrelevant couplings driven to their fixed-point values automatically. This finite tuning freedom is precisely the renormalizability established diagrammatically in Chapter 22.

Proof of the counting. In $d = 4$ the free field has $\dim[\phi] = (d-2)/2 = 1$, and a derivative carries dimension one. A symmetry-allowed (even, by the $\mathbb{Z}_2$ of ϕ^4 theory) local operator $\phi^{2n}(\partial)^{2m}$ has dimension $2n + 2m$. The condition $\dim \le 4$ forces $2n + 2m \le 4$ with $n \ge 0$, $m \ge 0$, giving $(n, m) \in \{(0,0), (1,0), (2,0)\}$ up to total derivatives, i.e. the list Eq. (23.124); $(1,1)$ gives $(\partial\phi)^2$ at dimension 4 but is the kinetic term, fixed by the field normalization, and all higher (n, m) give $\dim > 4$. The exact equation guarantees that the eigenvalues y_i of the *full* linearized flow about the fixed point reduce, at the Gaussian point, to these canonical values $d - \dim[\mathcal{O}_i]$, because the interaction corrections to the eigenvalues are $O(g^2)$ and vanish at $S^*_{\rm int} = 0$. The finiteness of the relevant/marginal set is therefore a finite count of operators below a fixed dimension, which is what we exhibited. □

The deep point, which only the exact equation makes available, is that this counting is a statement about a *fixed point of a genuine differential flow*, not about the order-by-order cancellation of divergences. Renormalizability is the assertion that the surface of microscopic theories flowing into the given long-distance physics has finite codimension—equal to the number of relevant plus marginal couplings—so that finitely many measurements pin down the continuum theory. The infinitely many irrelevant couplings, the entire nonlocal tail of S_Λ generated by integrating out shells, are not arbitrary: they are swept to universal values by the flow and leave no trace in long-distance observables beyond corrections suppressed by powers of Λ/Λ_0. This is the modern meaning of renormalizability, and it is the reading of Theorem 23.18 that the final section will sharpen into a criterion and apply to the classification of operators.

With the flow now exact, continuous, and manifestly finite, its fixed points—the scale-invariant solutions $\partial_\ell S^* = 0$ of Eq. (23.99)—are well-defined objects rather than artifacts of a truncation. The next section takes them as the central object: it defines a fixed point precisely (Theorem 23.19), linearizes the exact flow about one, and diagonalizes the linearization to recover the scaling operators and the relevant/marginal/irrelevant trichotomy whose ϕ^4 instance we have just counted, building toward the universality (Section 23.6) that is the conceptual core of the chapter.

23.5 Fixed Points and the Linearized Flow: Relevant, Marginal, Irrelevant

The previous sections turned coarse-graining into a dynamical system. Integrating out a momentum shell and rescaling to restore the cutoff defines a map on the space of couplings; in the continuous limit the map becomes a flow, governed exactly by the functional equation of Section 23.4 and, in its leading expansion, by the perturbative shell results of Section 23.2. We now have a vector field on an infinite-dimensional space and an injunction to follow its trajectories toward long distances. A vector field is understood by its singular points and by its behaviour near them. This section carries out exactly that analysis: it locates the points where the flow stops, linearizes around one, and reads off from the spectrum of the linearization the single organizing principle of the whole chapter — the partition of every possible perturbation of a scale-invariant theory into those that grow, those that hold steady, and those that die as the cutoff is lowered. That trichotomy, relevant/marginal/irrelevant, is what makes renormalization predictive: it is the reason a microscopic theory specified by infinitely many couplings collapses, at low energies, onto a finite-dimensional family. We close by classifying every operator of ϕ^4 theory explicitly, recovering from the flow the dimensional bookkeeping that Chapter 21 forced on us from the divergence side.

Throughout this section we work in the Euclidean theory obtained by the

Wick rotation of Chapter 10, in d Euclidean dimensions, since the renormalization group is cleanest where the path integral is a genuine statistical measure. The free scalar field has canonical dimension $[\phi] = (d-2)/2$ (Section 21.2.6); we will recover the mass dimensions of couplings from the flow rather than assume them.

The blocking factor b from here on

From this section onward we use the *reciprocal* of the scale factor used in Sections 23.2 to 23.3.3. There $b \in (0,1)$ was the cutoff-lowering ratio, so the shell was $b\Lambda < |k| < \Lambda$, the rescaling sent $k \to k/b$, and the renormalization-group time was $t = -\ln b$ (Eq. (23.69)). Here, following the standard statistical-mechanics blocking convention, $b \geq 1$ is the *blocking factor*: the shell is $\Lambda/b < |k| < \Lambda$, momenta rescale as $k \to bk$, and $t = +\ln b = \ln(\Lambda_0/\Lambda) \geq 0$. The two conventions are related by $b_{\text{here}} = 1/b_{\text{there}}$; in both, t increases toward the infrared. The map R_b of Theorem 23.10 is thus written here with the reciprocal b, but it denotes the same coarse-graining operation and obeys the same semigroup law.

23.5.1 Theory space, the flow, and its fixed points

Recall the picture assembled in the earlier sections. A point in *theory space* (Theorem 23.5) is a (in general infinite) collection of dimensionless couplings

$$\mathbf{g} \;=\; (g_1, g_2, g_3, \dots), \tag{23.125}$$

one for each local operator $\mathcal{O}_i$ that respects the symmetries of the problem, where each g_i has been made dimensionless by extracting the appropriate power of the running cutoff Λ. Thus the dimensionful coefficient $\bar{g}_i$ multiplying $\mathcal{O}_i$ in the action is written $\bar{g}_i = g_i\,\Lambda^{\,d-\Delta_i}$, with Δ_i the mass dimension of $\mathcal{O}_i$; the dimensionless g_i is what flows. The Wilsonian transformation R_b of Theorem 23.10 acts on this space: it integrates out the momentum shell $\Lambda/b < |k| < \Lambda$, rescales momenta $k \to bk$ (equivalently lengths $x \to x/b$) to restore the cutoff to Λ, and rescales the field to keep the kinetic term canonically normalized. The output is a new point $\mathbf{g}' = R_b(\mathbf{g})$. Because integrating out a shell of width b and then one of width b' is the same as integrating one shell of width bb', the maps compose,

$$R_b \circ R_{b'} \;=\; R_{bb'}, \qquad b, b' \geq 1, \tag{23.126}$$

so the R_b form a one-parameter semigroup — a semigroup and not a group, since coarse-graining destroys the short-distance information and cannot be undone. It is convenient to use the additive parameter

$$t \;\equiv\; \ln b \;=\; \ln \frac{\Lambda_0}{\Lambda} \;\geq\; 0, \tag{23.127}$$

the number of e-folds of coarse-graining, which *increases* as the cutoff Λ is lowered, i.e. as we flow toward the infrared. The infinitesimal generator of the semigroup is the *beta-function vector field*

$$\beta_i(\mathbf{g}) \;\equiv\; \left.\frac{dg_i}{dt}\right|_{\mathbf{g}}, \tag{23.128}$$

and a trajectory $\mathbf{g}(t)$ is a renormalization-group flow line.

Notation Watch

The flow parameter $t = \ln(\Lambda_0/\Lambda)$ runs *toward the infrared*: t grows as the cutoff drops. The Wilsonian beta functions $\beta_i = \frac{dg_i}{dt}$ therefore carry the opposite sign to the Callan–Symanzik beta functions of Chapter 24, which are derivatives with respect to $\ln\mu$ with μ increasing toward the *ultraviolet*: $\beta_i^{\text{Wilson}} = -\,\beta_i^{\text{CS}}$. The location and nature of fixed points are unaffected by this choice (a zero is a zero), but the words "grows" and "decays" below are tied to the infrared-directed t of Eq. (23.127). We state every flow direction explicitly to keep the two conventions from colliding between chapters.

The whole subject begins with the points where the flow halts.

Definition 23.19 (Renormalization-group fixed point)**.** A point $\mathbf{g}_* = (g_{1*}, g_{2*}, \dots)$ of theory space is a *fixed point* of the renormalization group if it is left invariant by every coarse-graining transformation,

$$R_b(\mathbf{g}_*) = \mathbf{g}_* \quad \text{for all } b \geq 1, \tag{23.129}$$

equivalently if the beta-function vector field vanishes there,

$$\beta_i(\mathbf{g}_*) = 0 \quad \text{for all } i. \tag{23.130}$$

The equivalence of the two characterizations is immediate from Eqs. (23.126) and (23.128): if $\mathbf{g}_*$ is invariant under all R_b then $\mathbf{g}(t) \equiv \mathbf{g}_*$ is a constant trajectory and $\beta_i = \frac{dg_i}{dt} = 0$; conversely, if all $\beta_i(\mathbf{g}_*) = 0$ then $\mathbf{g}_*$ is a stationary solution of the flow, hence fixed by the time-t map R_b for every $b = e^t$.

A fixed point is physically distinguished: a theory sitting exactly at $\mathbf{g}_*$ looks identical after coarse-graining by any factor. Since coarse-graining is a change of the unit of length, invariance under all R_b is invariance under all changes of scale. *A renormalization-group fixed point is a scale-invariant theory.* This is the dynamical reincarnation of the scale symmetry studied at the classical level in Section 7.7: there scale invariance was a property of a Lagrangian with no dimensionful parameters; here it is a property of a *full quantum theory*, including all the fluctuations the path integral sums over, and it can survive even

when the classical scale symmetry is broken by quantum effects, provided the breaking conspires to vanish at a special value of the couplings. Two such points control everything in this chapter: the trivial one, available in any theory, and (in Section 23.7, via $d = 4 - \varepsilon$) the nontrivial Wilson–Fisher point that is the destination of ϕ^4 theory.

Definition 23.20 (Gaussian fixed point)**.** The *Gaussian* (or *free-field*) fixed point is the point of theory space at which every interaction coupling vanishes,

$$\mathbf{g}_*^{\mathrm{G}} = (g_{1*}, g_{2*}, \dots) \quad \text{with} \quad g_{i*} = 0 \text{ for all interaction couplings } g_i, \tag{23.131}$$

so that the action is purely the Gaussian (quadratic, massless) kinetic term $S_0 = \int d^d x \, \frac{1}{2}(\partial\phi)^2$.

That Eq. (23.131) is a fixed point requires no computation beyond the definition of the flow. With all interactions switched off the path integral is exactly Gaussian; integrating out a shell of free modes is an exact Gaussian integration that contributes only a field-independent constant to the effective action, and the rescaling step was *defined* to leave the canonically normalized massless kinetic term invariant. No coupling is generated from zero couplings at any order, because every diagram that could generate one requires at least one interaction vertex. Hence $R_b(\mathbf{g}_*^{\mathrm{G}}) = \mathbf{g}_*^{\mathrm{G}}$ for all b: the free massless scalar is exactly scale invariant, as the classical analysis of Section 7.7.1 already showed for its action and as the propagator $\sim 1/k^2$ (homogeneous of degree -2) makes manifest. The Gaussian point is the fixed point about which we now linearize.

23.5.2 Linearizing the flow

Near a fixed point the nonlinear flow Eq. (23.128) is governed, to leading order, by its linear part. Write a nearby theory as a small displacement from $\mathbf{g}_*$,

$$g_i(t) \;=\; g_{i*} + \delta g_i(t), \qquad |\delta g_i| \text{ small}, \tag{23.132}$$

and Taylor-expand the beta functions about the fixed point. Since $\beta_i(\mathbf{g}_*) = 0$ by Eq. (23.130), the constant term drops and the leading behaviour is linear:

$$\frac{d(\delta g_i)}{dt} \;=\; \beta_i(\mathbf{g}_* + \delta\mathbf{g}) \;=\; \sum_j M_{ij}\, \delta g_j \;+\; O(\delta g^2), \qquad M_{ij} \equiv \left.\frac{\partial \beta_i}{\partial g_j}\right|_{\mathbf{g}_*}. \tag{23.133}$$

The matrix M_{ij} is the *stability matrix* of the flow — the Jacobian of the coarse-graining map at the fixed point, $M = \partial\beta/\partial\mathbf{g}|_*$. Everything about the local structure of the flow is contained in its spectrum.

Diagonalize M. Because M is in general not symmetric, its left and right eigenvectors differ; we work with the left eigenvectors, which build the combinations of couplings that scale multiplicatively. Let $E^{(a)}$ be a left eigenvector with

eigenvalue y_a,

$$\sum_i E_i^{(a)}\, M_{ij} \;=\; y_a\, E_j^{(a)}, \tag{23.134}$$

and define the corresponding *scaling field* as the linear combination of displacements

$$u_a \;\equiv\; \sum_i E_i^{(a)}\, \delta g_i. \tag{23.135}$$

Its evolution decouples from all the others. Differentiating Eq. (23.135) and using Eq. (23.133) and then Eq. (23.134),

$$\frac{du_a}{dt} \;=\; \sum_i E_i^{(a)}\, \frac{d(\delta g_i)}{dt} \;=\; \sum_{i,j} E_i^{(a)}\, M_{ij}\, \delta g_j \;=\; y_a \sum_j E_j^{(a)}\, \delta g_j \;=\; y_a\, u_a. \tag{23.136}$$

This is a single decoupled linear equation per eigendirection. Its solution is a pure exponential in the flow time, hence a pure power in the rescaling factor:

$$u_a(t) \;=\; u_a(0)\, e^{\,y_a t} \;=\; u_a(0)\, b^{\,y_a}. \tag{23.137}$$

Equation (23.137) is the central technical fact of the linearized theory: in the right coordinates the flow near a fixed point is not merely linear but *diagonal*, each scaling field stretching or shrinking by a definite power of the coarse-graining factor b at each step, independent of all the others. The exponents y_a — the eigenvalues of the stability matrix — are called the *renormalization-group eigenvalues*. They are intrinsic to the fixed point: a change of coordinates on theory space conjugates M and leaves its spectrum, and hence the y_a, unchanged.

Remark 23.21 (Real eigenvalues and the generic case). For the fixed points treated in this book the y_a are real, and Eq. (23.137) is a monotone power law. In principle the non-symmetric M can have complex-conjugate pairs $y_a = \sigma \pm i\omega$, which produce oscillatory corrections $b^{\sigma}\cos(\omega \ln b + \varphi)$ — the signature of a limit cycle rather than a fixed point. Such behaviour is exotic and does not occur for the Gaussian or Wilson–Fisher points; we note it only so the reader understands that "power-law scaling" is the generic, not the universal, statement, and assume real y_a hereafter.

23.5.3 Scaling dimensions and the relation $y_i = d - \Delta_i$

The eigenvalues y_a are abstract growth rates; we now tie them to the mass dimensions of the operators they are conjugate to, recovering the dimensional analysis of Chapter 21 and Theorem A.1 as a corollary of the flow. The cleanest derivation isolates the one place dimensions enter the coarse-graining transformation: the rescaling step.

Consider a single scaling operator $\mathcal{O}_a$ that, at the fixed point, transforms multiplicatively under the RG. At the Gaussian fixed point the scaling operators

are simply the local monomials in ϕ and its derivatives, so take

$$\mathcal{O}_a(x) \;=\; (\partial^{\,n_\partial}\phi^{\,n_\phi})(x), \tag{23.138}$$

an operator built from n_ϕ factors of ϕ and n_∂ derivatives, and add it to the action with a dimensionful coupling,

$$S \;\supset\; \bar{u}_a \int d^d x\; \mathcal{O}_a(x). \tag{23.139}$$

The rescaling step of R_b sends coordinates $x \to x' = x/b$, so $d^d x = b^d\, d^d x'$ and $\partial_x = b^{-1}\partial_{x'}$, and rescales the field by the power that preserves the kinetic term. The kinetic action transforms as

$$\int d^d x\; \tfrac{1}{2}(\partial_x\phi)^2 \;=\; \int b^d\, d^d x'\; \tfrac{1}{2}\, b^{-2}\, (\partial_{x'}\phi)^2, \tag{23.140}$$

which is restored to its canonical form by setting $\phi(x) = b^{-(d-2)/2}\,\phi'(x')$, since then the prefactor becomes $b^d\, b^{-2}\, b^{-(d-2)} = b^0 = 1$. This fixes the field rescaling once and for all and reproduces $[\phi] = (d-2)/2$. The same two rescalings, applied to the monomial Eq. (23.138), give

$$\mathcal{O}_a(x) \;=\; b^{-n_\partial}\,(\partial_{x'}^{n_\partial}) \;\cdot\; b^{-n_\phi(d-2)/2}\,\phi'^{\,n_\phi}(x') \;=\; b^{-\Delta_a}\,\mathcal{O}'_a(x'), \tag{23.141}$$

where we have collected the powers of b into the *scaling dimension* of the operator,

$$\Delta_a \;=\; n_\partial + n_\phi\,\frac{d-2}{2}. \tag{23.142}$$

This is exactly the engineering (mass) dimension of $\mathcal{O}_a$ from Section 21.2.6 and Theorem A.1: each derivative carries dimension one, each field carries $(d-2)/2$. Substituting Eq. (23.141) into the action term Eq. (23.139),

$$\bar{u}_a \int d^d x\; \mathcal{O}_a(x) \;=\; \bar{u}_a \int b^d\, d^d x'\; b^{-\Delta_a}\,\mathcal{O}'_a(x') \;=\; (b^{\,d-\Delta_a}\,\bar{u}_a)\int d^d x'\; \mathcal{O}'_a(x'), \tag{23.143}$$

so the coefficient of the (identically normalized) operator in the rescaled theory is $\bar{u}'_a = b^{\,d-\Delta_a}\,\bar{u}_a$. The integration-out step contributes nothing to this leading scaling at the Gaussian point — it is the loop correction, higher order in the couplings — so the linearized flow of the coupling is the pure rescaling Eq. (23.143). Comparing with the eigen-scaling Eq. (23.137), $u_a(t) = u_a(0)\, b^{y_a}$, identifies the eigenvalue:

$$\boxed{\;y_a \;=\; d - \Delta_a\;} \tag{23.144}$$

The renormalization-group eigenvalue of a scaling operator is the spacetime dimension minus the operator's scaling dimension. This is the bridge between the flow and dimensional analysis: $y_a > 0$ is precisely $\Delta_a < d$, the condition $[\bar{u}_a] = d - \Delta_a > 0$ on the coupling that Theorem A.4 used to name operators relevant, marginal, or irrelevant from the dimension side. We have now derived it dynamically.

Definition 23.22 (Scaling dimension)**.** The *scaling dimension* Δ_a of a scaling operator $\mathcal{O}_a$ at a fixed point is the exponent governing its behaviour under the rescaling step of the renormalization group, Eq. (23.141), equivalently the exponent of its fixed-point two-point function $\langle \mathcal{O}_a(x)\, \mathcal{O}_a(0)\rangle \propto |x|^{-2\Delta_a}$. At the Gaussian fixed point it equals the canonical (engineering) dimension Eq. (23.142); at an interacting fixed point it acquires in addition an *anomalous dimension* from the loop corrections to the stability matrix, $\Delta_a = \Delta_a^{\mathrm{can}} + \gamma_a$, and Eq. (23.144) continues to hold with the full Δ_a.

The two characterizations of Δ_a in Theorem 23.22 agree because a power-law correlator is exactly what scale invariance permits: if $\mathcal{O}_a(x) = b^{-\Delta_a}\mathcal{O}'_a(x/b)$ and the fixed-point measure is scale invariant, then $\langle \mathcal{O}_a(x)\mathcal{O}_a(0)\rangle = b^{-2\Delta_a}\langle \mathcal{O}'_a(x/b)\mathcal{O}'_a(0)\rangle$ forces the two-point function to be homogeneous of degree $-2\Delta_a$, i.e. a power $|x|^{-2\Delta_a}$. This is the seed of the conformal-field-theory description picked up in Chapters 47 and 52; here we need only Eq. (23.144).

23.5.4 The trichotomy

The sign of each eigenvalue now decides the fate of the corresponding perturbation, and the classification of the chapter follows by reading Eq. (23.137).

Theorem 23.23 (Relevant/marginal/irrelevant classification)**.** *Let $\mathbf{g}_*$ be a renormalization-group fixed point with stability matrix $M_{ij} = \partial\beta_i/\partial g_j|_*$ of real eigenvalues y_a and scaling fields u_a flowing as $u_a(t) = u_a(0)\, e^{y_a t}$, Eq. (23.137), with t increasing toward the infrared. Each eigendirection is exactly one of the following.*

1. Relevant *($y_a > 0$, i.e. $\Delta_a < d$): $|u_a(t)|$* grows *as $t \to \infty$. The flow is driven away from the fixed point along this direction; an arbitrarily small initial $u_a(0) \neq 0$ eventually dominates. To reach the fixed point in the infrared, u_a must be* tuned *to zero.*

2. Marginal *($y_a = 0$, i.e. $\Delta_a = d$): $u_a(t) = u_a(0)$ is constant under the linearized flow. The eigenvalue alone does not decide the direction's fate; the leading nonlinear (loop) term in β_a does, making it* marginally relevant *or* marginally irrelevant *according to the sign with which u_a then drifts.*

3. Irrelevant *($y_a < 0$, i.e. $\Delta_a > d$): $|u_a(t)| \to 0$ as $t \to \infty$. The flow is attracted to the fixed point along this direction; the initial value $u_a(0)$ is forgotten at long distances.*

Proof. Each statement is the behaviour of the real exponential $e^{y_a t}$ as $t \to +\infty$, read from the exact solution Eq. (23.136)–Eq. (23.137) of the linearized flow. For $y_a > 0$, $e^{y_a t} \to \infty$, so $|u_a(t)| = |u_a(0)|\, e^{y_a t}$ grows without bound unless $u_a(0) = 0$; this is the relevant case, and requiring $u_a(t)$ to remain bounded (so the trajectory

can approach $\mathbf{g}_*$) forces the tuning $u_a(0) = 0$. For $y_a < 0$, $e^{y_a t} \to 0$, so $u_a(t) \to 0$ regardless of its starting value; this is the irrelevant case, and the limit erases $u_a(0)$. For $y_a = 0$ the linearized solution is the constant $u_a(0)$, so the linear order is silent and the sign of the first nonvanishing term in the Taylor expansion of β_a beyond linear order — generically the quadratic term, $\beta_a = c\,u_a^2 + \cdots$ — governs whether u_a drifts toward or away from zero, by the elementary fact that $\frac{du_a}{dt} = c\,u_a^2$ integrates to $u_a(t) = u_a(0)/(1 - c\,u_a(0)\,t)$, which decays toward the fixed point when the product $c\,u_a(0) < 0$ and grows away from it when $c\,u_a(0) > 0$. The three cases are exhaustive and mutually exclusive because y_a is a real number, exactly one of positive, zero, negative. □

The names now make physical sense in their own right. A relevant perturbation is one the long-distance physics *cares about*: leave it on, however weakly, and it grows until it controls the infrared, so it is what one must specify (and tune) to determine the low-energy theory. An irrelevant perturbation is one the long-distance physics *forgets*: it shrinks to nothing, so its microscopic value leaves no infrared trace. The marginal case sits on the knife-edge where dimensional analysis is silent and the quantum corrections decide — the case of λ in four dimensions, to which we return below.

Physical Insight

The trichotomy is why renormalization is possible at all. A microscopic theory is a point in an infinite-dimensional space of couplings — infinitely many irrelevant operators are present in any honest short-distance description. Under the flow toward long distances, every irrelevant coordinate is contracted to zero, and only the finitely many relevant (and marginal) directions survive to control the infrared. The low-energy physics therefore depends on only a handful of numbers, no matter how complicated the microscopic theory was. The infinite-dimensional space of inputs is compressed to a finite-dimensional space of outputs; that compression *is* predictivity, and it is the modern content of renormalizability, developed fully in Section 23.7 and Chapter 38.

The count of relevant directions is so important that the surface it cuts out deserves a name.

Definition 23.24 (Critical surface)**.** The *critical surface* of a fixed point $\mathbf{g}_*$ is the set of points in theory space whose RG trajectory flows into $\mathbf{g}_*$ as $t \to \infty$ — the stable manifold of the fixed point. In the linearized neighbourhood it is the subspace spanned by the irrelevant (and marginally irrelevant) scaling fields, obtained by setting every relevant scaling field to zero, $u_a = 0$ for all $y_a > 0$. Its codimension in theory space equals the number of relevant directions.

The codimension count is immediate from Theorem 23.23: a trajectory

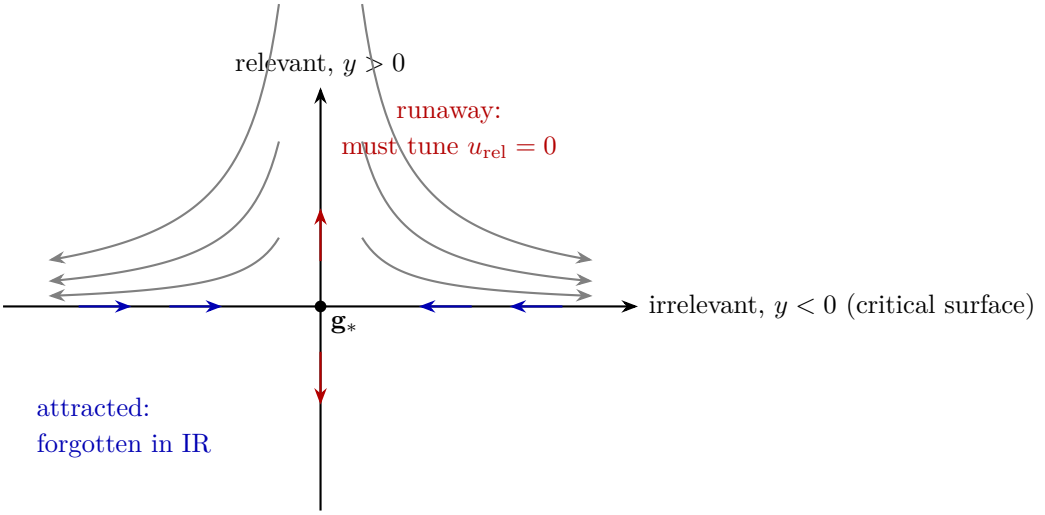


Figure 23.7: The linearized renormalization-group flow near a fixed point $\mathbf{g}_*$ with one relevant and one irrelevant direction (flow toward the infrared, t increasing). Along the irrelevant axis (blue) the flow is attracted to the fixed point: this axis is the critical surface. Along the relevant axis (red) the flow is repelled. A generic trajectory (grey) is hyperbolic — it first approaches along the irrelevant direction, then shoots out along the relevant one — so only by tuning the relevant scaling field to zero does a theory land on the critical surface and reach the fixed point. The critical surface here has codimension one, equal to the single relevant direction.

reaches $\mathbf{g}_*$ only if it has no growing component, which by part (1) requires $u_a(0) = 0$ for each of the $n_{\rm rel}$ relevant directions. These are $n_{\rm rel}$ independent linear conditions, so the stable manifold has codimension $n_{\rm rel}$; the remaining (irrelevant) directions are unconstrained, since by part (3) they flow into the fixed point on their own. Tuning a physical system onto its critical surface — e.g. adjusting the temperature of a magnet to its Curie point, the canonical relevant direction being the deviation from critical temperature — is exactly the tuning of the relevant couplings to zero, and is why criticality requires fine adjustment of as many parameters as there are relevant operators. Figure 23.7 shows the local portrait: one relevant and one irrelevant direction, the critical surface as the inflowing axis, and the runaway along the relevant axis.

23.5.5 Scalar ϕ^4 theory: classifying the operators at the Gaussian fixed point

We now carry out the classification promised by the thread. The setting is the Gaussian fixed point of the scalar, Theorem 23.20, and the operators are the $\mathbb{Z}_2$-even local scalars one can add to the massless free action: ϕ^2 (the mass term),

ϕ^4 (the quartic of ϕ^4 theory), and the leading higher operators ϕ^6 and $(\partial\phi)^2\phi^2$. We compute every scaling dimension from Eq. (23.142) and every RG eigenvalue from Eq. (23.144), first in general d and then at the physical $d = 4$.

Worked Example 23.25 (Scaling operators of ϕ^4 theory at the Gaussian fixed point)**.** Continuing ϕ^4 theory from its divergence layer (Section 21.6), we classify its operators by the Wilsonian flow. The field dimension at the Gaussian point is $[\phi] = (d-2)/2$, Eq. (23.140). For each operator $\mathcal{O} = \partial^{n_\partial}\phi^{n_\phi}$ we read $\Delta = n_\partial + n_\phi(d-2)/2$ from Eq. (23.142) and $y = d - \Delta$ from Eq. (23.144); y is also the mass dimension of the conjugate coupling.

The mass operator ϕ^2 ($n_\phi = 2$, $n_\partial = 0$).

$$\Delta_{\phi^2} = 2 \cdot \frac{d-2}{2} = d - 2, \qquad y_{\phi^2} = d - (d-2) = 2. \tag{23.145}$$

The eigenvalue is $y_{\phi^2} = +2$ in *every* dimension, independent of d. The conjugate coupling is the mass squared, $[\,m^2\,] = 2$, consistent with the dimensionful mass term $\frac{1}{2}m^2\phi^2$. Since $y > 0$, ϕ^2 is *relevant*. This is the single relevant direction of the free scalar (with one $\mathbb{Z}_2$-even operator of each kind); the critical surface of the Gaussian point therefore has codimension one in the $\mathbb{Z}_2$-even sector, and reaching the fixed point requires tuning exactly one coupling — the mass — to its critical value. The fine-tuning of m^2 is the field-theoretic statement of the hierarchy/naturalness problem and the criticality of a second-order phase transition alike.

The quartic operator ϕ^4 ($n_\phi = 4$, $n_\partial = 0$).

$$\Delta_{\phi^4} = 4 \cdot \frac{d-2}{2} = 2(d-2) = 2d - 4, \qquad y_{\phi^4} = d - (2d-4) = 4 - d. \tag{23.146}$$

The conjugate coupling is λ, with $[\,\lambda\,] = 4 - d$, matching the power counting $[\lambda] = 4 - d$ of Chapter 21. The sign of $y_{\phi^4} = 4 - d$ depends on the dimension: relevant for $d < 4$, irrelevant for $d > 4$, and exactly *marginal* ($y = 0$) at $d = 4$. The vanishing of y_{ϕ^4} at $d = 4$ is the reason four dimensions is special for ϕ^4, and the reason its fate there is decided not by dimensional analysis but by a loop, as we resolve below. It is also the reason the Wilson–Fisher analysis of Section 23.7 works just below four dimensions, where $y_{\phi^4} = \varepsilon$ is small and positive.

The sextic operator ϕ^6 ($n_\phi = 6$, $n_\partial = 0$).

$$\Delta_{\phi^6} = 6 \cdot \frac{d-2}{2} = 3(d-2) = 3d - 6, \qquad y_{\phi^6} = d - (3d-6) = 6 - 2d. \tag{23.147}$$

The conjugate coupling has $[\,g_6\,] = 6 - 2d$, which is -2 at $d = 4$. Since $y_{\phi^6} = 6 - 2d < 0$ for all $d > 3$, and in particular $y_{\phi^6} = -2$ at $d = 4$, the sextic is *irrelevant*. (It becomes marginal only at $d = 3$ and relevant below; in four dimensions it is firmly irrelevant.)

The derivative operator $(\partial\phi)^2\phi^2$ *(*$n_\phi = 4$*,* $n_\partial = 2$*).*

$$\Delta_{(\partial\phi)^2\phi^2} = 2+4\cdot\frac{d-2}{2} = 2+2(d-2) = 2d-2, \qquad y_{(\partial\phi)^2\phi^2} = d-(2d-2) = 2-d. \tag{23.148}$$

This is -2 at $d = 4$, so the operator is *irrelevant*; the two extra derivatives cost two units of dimension relative to ϕ^4, pushing it firmly below marginality. Every operator with more fields or more derivatives than ϕ^4 is more irrelevant still, since each additional field adds $(d-2)/2$ and each additional derivative adds 1 to Δ, both *lowering* $y = d - \Delta$. The Gaussian fixed point in four dimensions therefore has a finite tower of non-irrelevant operators (ϕ^2 relevant, ϕ^4 marginal) sitting atop an infinite tower of irrelevant ones — exactly the structure that makes ϕ^4 renormalizable.

A note on the kinetic operator $(\partial\phi)^2$. For completeness, $(\partial\phi)^2$ has $n_\phi = 2$, $n_\partial = 2$, hence $\Delta = 2 + (d-2) = d$ and $y = 0$: it is marginal in every dimension. This is not an accident but a tautology — it is the operator that defines the fixed point, and the rescaling step was fixed precisely to hold it invariant (Eq. (23.140)). It is a *redundant* operator: adding it merely renormalizes the field, an effect removable by the field rescaling itself, so it carries no independent physics and is conventionally normalized away. We mention it only to forestall the worry that a second marginal direction has been overlooked.

Specializing to the physical case $d = 4$ and collecting the results:

operator	Δ $(d=4)$	$y = d - \Delta$ $(d=4)$	$[\,\text{coupling}\,]$	class
ϕ^2	2	$+2$	$[\,m^2\,] = 2$	relevant
ϕ^4	4	0	$[\,\lambda\,] = 0$	marginal
ϕ^6	6	-2	$[\,g_6\,] = -2$	irrelevant
$(\partial\phi)^2\phi^2$	6	-2	$[\,g_6'\,] = -2$	irrelevant

(23.149)

This is the ϕ^4 theory obligation discharged: the mass is relevant, the quartic is marginal, and everything of higher dimension is irrelevant, with every scaling dimension computed from first principles. It is the same trichotomy that Chapter 21 read off the superficial degree of divergence (Section 21.7.2), now seen from the flow: a relevant coupling is super-renormalizable, a marginal one renormalizable, an irrelevant one non-renormalizable. The two pictures are the two faces of the single sign of $y = d - \Delta = [\,\text{coupling}\,]$.

23.5.6 The marginal case decided: λ is marginally irrelevant

The classification leaves λ undecided at $d = 4$: $y_{\phi^4} = 0$, so the linearized flow says only that λ neither grows nor shrinks at leading order. Part (2) of Theorem 23.23 sends us to the first nonlinear term in its beta function, which is precisely the

one-loop shell result computed in Section 23.2 (Eq. (23.79)). Carrying that term in general d, the flow of the dimensionless quartic is

$$\frac{d\lambda}{dt} \;=\; (4-d)\,\lambda \;-\; \frac{3}{16\pi^2}\,\lambda^2 \;+\; O(\lambda^3), \tag{23.150}$$

where t increases toward the infrared. The linear term is the tree-level eigenvalue $y_{\phi^4}\lambda = (4-d)\lambda$ of Eq. (23.146), confirming the classification; the quadratic term is the one-loop correction. Its coefficient is the same number that controls the cutoff dependence of the four-point function, $\Gamma^{(4)} = -\lambda + \frac{3}{16\pi^2}\lambda^2 \ln(\Lambda/\mu) + \cdots$ from Eq. (21.133): the 3 counts the three one-loop channels (s, t, u) of Section 21.6, and the sign is positive. This is the number Chapter 24 will read as the Callan–Symanzik beta function $\beta_\lambda^{\mathrm{CS}} = 3\lambda^2/16\pi^2$ (the sign flip relative to Eq. (23.150) is the convention of the notation watch above).

Set $d = 4$, where the linear term vanishes and only the loop survives:

$$\frac{d\lambda}{dt} \;=\; -\,\frac{3}{16\pi^2}\,\lambda^2 \;+\; O(\lambda^3), \qquad (d=4). \tag{23.151}$$

For the physical $\lambda > 0$ the right-hand side is strictly negative: as t increases toward the infrared, λ decreases. The quartic, marginal by power counting, is dragged toward the Gaussian fixed point by the loop. It is therefore *marginally irrelevant* — on the knife-edge at tree level, tipped to the irrelevant side by the positive one-loop coefficient. This matches Theorem 23.23(2) with $\beta_\lambda = c\lambda^2$, $c = -3/16\pi^2 < 0$: the product $c\,\lambda(0) < 0$ for $\lambda(0) > 0$, the marginally-irrelevant condition.

The approach is not exponential but logarithmically slow, the hallmark of marginality. Equation Eq. (23.151) integrates by separating variables, $-\lambda^{-2}\,d\lambda = (3/16\pi^2)\,dt$:

$$\frac{1}{\lambda(t)} - \frac{1}{\lambda(0)} \;=\; \frac{3}{16\pi^2}\,t, \qquad\Longrightarrow\qquad \lambda(t) \;=\; \frac{\lambda(0)}{1 + \dfrac{3\,\lambda(0)}{16\pi^2}\,t} \;\xrightarrow[t\to\infty]{}\; \frac{16\pi^2}{3\,t} \to 0. \tag{23.152}$$

The coupling dies, but only as $1/t = 1/\ln(\Lambda_0/\Lambda)$ — an inverse logarithm of the cutoff ratio, immeasurably slow compared with the power-law b^y decay of a genuinely irrelevant operator. This logarithmic running is the entire content of "ϕ^4 is renormalizable but trivial in four dimensions": pushed to arbitrarily high cutoff ($t \to \infty$ in the other direction, i.e. following Eq. (23.152) backward, λ grows and hits a Landau pole at finite Λ_0), the continuum theory at fixed long-distance physics is forced to $\lambda \to 0$. We meet the constructive face of this statement, and its escape route via $d = 4 - \varepsilon$, in Section 23.7 and Chapter 24.

With the trichotomy established and ϕ^4 theory classified, the local structure of every fixed point is now in hand: a finite number of relevant directions to be tuned, a single (possibly marginal) coupling whose loop-decided fate we have computed,

and an infinite irrelevant remainder that the flow discards. The next section turns this local picture into its two great physical payoffs — the universality that follows from forgetting the irrelevant directions, and the continuum limit constructed by flowing out of a fixed point along its relevant ones.

23.6 Universality, Critical Surfaces, and the Continuum Limit

The previous section dissected the flow in the immediate neighbourhood of a fixed point. Linearizing the renormalization-group map about g_*, diagonalizing the resulting matrix, and reading off its eigenvalues y_i sorted the perturbations into the trichotomy that is the conceptual spine of this chapter: relevant ($y_i > 0$), marginal ($y_i = 0$), irrelevant ($y_i < 0$). That classification was, so far, a local statement about the linearized dynamics. We now collect its physical dividends, and they are large. The irrelevant directions, we will show, are not merely shrinking under the flow; they are *forgotten*, at a rate we will compute, so that an entire continuum of microscopically distinct theories is funnelled onto one and the same low-energy description. This is universality, and it is the reason a handful of fixed points organize the long-distance behaviour of matter and of quantum fields alike. From it we will extract universality classes, the divergence of the correlation length, and the critical exponents—the universal numbers attached to a fixed point. Turning the same picture around, we will characterize the continuum limit of a quantum field theory as the flow *out* of a fixed point along its relevant directions, and find that a nontrivial continuum limit can exist only where a fixed point governs the ultraviolet. Throughout we carry ϕ^4 theory, which will reveal itself as the prototype critical system whose governing fixed point—the Wilson–Fisher fixed point—is the explicit construction of the final section.

Two pieces of machinery are imported and used without reproof. The power-counting assignment of canonical dimensions to couplings, which fixes the leading (Gaussian) eigenvalues, is Section 21.2; the relevant/marginal/irrelevant nomenclature and its dimensional skeleton are pinned down once and for all in Theorem A.4. The local flow data $\{g_*,\, y_i,\, u_i\}$ are those constructed in the preceding section.

23.6.1 The renormalization-group flow, recalled in working form

Fix the conventions so the algebra below is unambiguous. A single Wilsonian step integrates out a momentum shell and rescales, lowering the effective ultraviolet cutoff from Λ to Λ/b with a blocking factor $b > 1$, and carrying a point $g = \{g_i\}$ in the space of *dimensionless* couplings to a new point $\mathcal{R}_b(g)$. Composition of steps is a semigroup, $\mathcal{R}_b \circ \mathcal{R}_{b'} = \mathcal{R}_{bb'}$, so it is natural to use the continuous

logarithmic flow time

$$s \;=\; \ln\frac{\Lambda_0}{\Lambda} \;=\; \ln b \;\geq\; 0, \tag{23.153}$$

which *increases towards the infrared* as the cutoff is lowered from its microscopic value Λ_0. The flow is generated by the beta functions,

$$\frac{\mathrm{d}g_i}{\mathrm{d}s} \;=\; \beta_i(g), \qquad \beta_i(g_*) = 0, \tag{23.154}$$

the second equation defining a fixed point g_* (Theorem 23.19). Writing $g_i = g_{*i} + \delta g_i$ and linearizing, $\mathrm{d}(\delta g_i)/\mathrm{d}s = \sum_j M_{ij}\,\delta g_j$ with $M_{ij} = \partial\beta_i/\partial g_j\big|_{g_*}$, and diagonalizing M, one obtains the scaling fields u_i (the eigencombinations of the δg_j) which evolve multiplicatively,

$$\frac{\mathrm{d}u_i}{\mathrm{d}s} \;=\; y_i\,u_i \qquad\Longrightarrow\qquad u_i(s) \;=\; e^{y_i s}\,u_i(0) \;=\; b^{\,y_i}\,u_i(0), \tag{23.155}$$

the content of Eq. (23.133). The eigenvalue y_i is related to the scaling dimension Δ_i of the conjugate operator by $y_i = d - \Delta_i$, so the trichotomy of Theorem A.4 reads off the sign of y_i: relevant fields ($y_i > 0$) grow towards the infrared, irrelevant ones ($y_i < 0$) shrink, and marginal ones ($y_i = 0$) are frozen at linear order and decided by the quadratic term. It is on Eq. (23.155)—a single exponential per direction—that the entire physics of this section rests.

23.6.2 Universality: the forgetting of irrelevant couplings

The naming of irrelevant directions promises more than that they shrink: it promises that long-distance physics does not depend on them at all. We make the promise precise and then prove it, including the rate at which the forgetting happens, because a Wilsonian claim asserted without a rate is a slogan, not a theorem.

Theorem 23.26 (Universality: attraction to the relevant subspace)**.** *Let g_* be a fixed point of Eq. (23.154) with linearized eigenvalues $\{y_i\}$, of which n_+ are positive (relevant) and n_0 vanish (marginal); label the remaining, strictly negative, eigenvalues by a, with $y_a < 0$ and $y_{\min} \equiv \min_a |y_a| > 0$ the magnitude of the least irrelevant one. Consider two bare theories specified at the microscopic scale Λ_0 by points g and g' lying in the linear neighbourhood of g_*, and suppose they agree in every relevant and marginal scaling field, $u_i(\Lambda_0) = u'_i(\Lambda_0)$ for all i with $y_i \geq 0$, while differing freely in the irrelevant directions, $u_a(\Lambda_0) \neq u'_a(\Lambda_0)$. Then after flowing down to any scale $\mu \leq \Lambda_0$ their scaling fields satisfy*

$$u_a(\mu) - u'_a(\mu) \;=\; \Big(\frac{\mu}{\Lambda_0}\Big)^{|y_a|}\,\big[\,u_a(\Lambda_0) - u'_a(\Lambda_0)\,\big], \qquad u_i(\mu) = u'_i(\mu) \;\; (y_i \geq 0), \tag{23.156}$$

at linear order. Consequently the two theories converge to one and the same trajectory; every long-distance observable computed at a scale $\mu \ll \Lambda_0$ depends, up to corrections of relative size $(\mu/\Lambda_0)^{y_{\min}}$, on only the finitely many relevant and marginal scaling fields $\{u_i : y_i \geq 0\}$.

Proof. By Eq. (23.153) the cutoff μ is reached at flow time $s = \ln(\Lambda_0/\mu) \geq 0$, i.e. $b = \Lambda_0/\mu$. Apply the diagonalized solution Eq. (23.155) to each scaling field of each theory. Because Eq. (23.155) is linear, it acts identically on g and g', and the difference of two solutions is itself a solution: for any direction i,

$$u_i(\mu) - u_i'(\mu) \;=\; b^{\,y_i}\big[u_i(\Lambda_0) - u_i'(\Lambda_0)\big] \;=\; \Big(\frac{\Lambda_0}{\mu}\Big)^{y_i}\big[u_i(\Lambda_0) - u_i'(\Lambda_0)\big]. \tag{23.157}$$

For relevant and marginal directions ($y_i \geq 0$) the bracket vanishes by hypothesis, giving $u_i(\mu) = u_i'(\mu)$. For an irrelevant direction, $y_a < 0$, write $y_a = -|y_a|$ in Eq. (23.157); the prefactor becomes $(\Lambda_0/\mu)^{-|y_a|} = (\mu/\Lambda_0)^{|y_a|}$, which is Eq. (23.156). Since $\mu \leq \Lambda_0$ and $|y_a| > 0$, this factor lies in $(0, 1]$ and tends to zero as $\mu/\Lambda_0 \to 0$; the slowest-decaying irrelevant difference sets the envelope $(\mu/\Lambda_0)^{y_{\min}}$. Thus the irrelevant coordinates of the two trajectories collapse onto each other, while the relevant/marginal coordinates were equal from the start: the trajectories merge. Any correlation function evaluated at scale μ is a function of the running couplings there; replacing the true couplings by their values on the merged trajectory changes the argument only in the irrelevant slots and only by $O((\mu/\Lambda_0)^{y_{\min}})$, which by smoothness of the observable in its couplings produces a correction of the same relative order. Hence at $\mu \ll \Lambda_0$ every observable is, to that accuracy, a function of the $n_+ + n_0$ relevant and marginal scaling fields alone. □

Three comments fix the meaning and guard against over-reading the theorem.

First, the rate is genuinely a power law, $(\mu/\Lambda_0)^{|y_a|}$, not an exponential in μ nor a logarithm. This is the quantitative form of "forgetting the ultraviolet": an irrelevant coupling of canonical dimension $[\,g_a\,] = -|y_a|$ (an operator of dimension $\Delta_a = d + |y_a|$) contributes to a low-energy amplitude suppressed by $(\mu/\Lambda_0)^{|y_a|} = (E/\Lambda_0)^{\,\Delta_a - d}$. This is the same power that Theorem A.4 attaches to the dimensionless importance ratio $\lambda_{\mathcal{O}}(E) = g\,E^{\Delta-d}$; the Wilsonian flow has now *derived* it dynamically rather than read it off dimensional analysis. It is also exactly the suppression that controls the leading higher-dimension corrections in the effective-field-theory expansion of Chapter 38: there the same power $(E/\Lambda)^{\Delta-d}$ is the size of a dimension-Δ operator's contribution, and the present theorem is the statement that those corrections are systematically small and calculable, not that they are absent.

Second, the theorem is linear, valid where the trajectory stays inside the neighbourhood of g_* in which the linearization Eq. (23.155) holds. The nonlinear terms in β_i generate two effects, both honestly named: they bend the merged

trajectory away from the linear eigendirections, and they produce subleading *corrections to scaling*—additional powers $(\mu/\Lambda_0)^{|y_a|}$, $(\mu/\Lambda_0)^{2|y_a|}$, and so on, the leading one controlled by the smallest $|y_a|$ [491]. These corrections do not spoil universality; they decorate it. The full nonlinear statement, that the set of bare theories flowing into g_* is a smooth manifold tangent at g_* to the irrelevant eigenspace, is the stable-manifold theorem of dynamical systems applied to Eq. (23.154); we use it in the next subsection under the name critical surface.

Third, the marginal directions ($y_i = 0$) are carried along as genuine parameters at this order: they neither grow nor shrink, so they are *not* forgotten and must be counted among the data that survive to long distances. Whether a marginal coupling is ultimately attracted or repelled is decided by the sign of the leading nonlinear term in its beta function—it is then called marginally irrelevant or marginally relevant—and we will meet exactly this situation for the quartic coupling of ϕ^4 theory in four dimensions.

Physical Insight

Universality is the precise content of the intuition that "the long-distance world does not care what the atoms are." A ferromagnet, a fluid near its critical point, and a lattice scalar field have utterly different microscopic Lagrangians—different irrelevant couplings $u_a(\Lambda_0)$—but Eq. (23.156) drains those differences away as a fixed power of the scale ratio. What remains at macroscopic distances is fixed by the handful of relevant and marginal couplings and by the fixed point they flow towards. The microscopic theory is not approximated; it is *coarse-grained into a lower-dimensional description*, and the discarded information leaves only power-suppressed traces.

23.6.3 Critical surfaces and universality classes

Theorem 23.26 singles out a distinguished set of bare theories: those for which the relevant scaling fields are switched off entirely, so that nothing grows under the flow and the trajectory runs straight into the fixed point.

Definition 23.27 (Critical surface)**.** The *critical surface* (or stable manifold) of a fixed point g_* is the set of bare couplings whose renormalization-group trajectory tends to g_* in the infrared,

$$\mathcal{S}(g_*) \;=\; \Big\{\, g \;:\; \lim_{s\to\infty} \mathcal{R}_{e^s}(g) = g_* \,\Big\}. \tag{23.158}$$

Near g_* it is the smooth manifold tangent to the irrelevant eigenspace; its *codimension* in the space of couplings equals the number n_+ of relevant directions.

The codimension count is immediate from Eq. (23.155): a trajectory reaches g_* only if every *growing* mode is absent, $u_i(\Lambda_0) = 0$ for all $y_i > 0$, which is n_+

independent conditions; the irrelevant fields may take any (small) values and still decay to zero. So $\mathcal{S}(g_*)$ is locally cut out by n_+ equations and has dimension equal to the number of irrelevant directions. The marginal directions sit, at linear order, within the surface and require the nonlinear analysis to place definitively.

A theory tuned exactly onto $\mathcal{S}(g_*)$ has a remarkable property. Its flow ends at a fixed point, where by construction the dimensionless couplings stop changing under rescaling: the long-distance theory is scale invariant. A scale-invariant theory has no intrinsic length, so its correlation length is either zero or infinite; for a nontrivial fixed point it is infinite. The critical surface is therefore the locus of *criticality*—the set of microscopic theories poised exactly at a continuous phase transition, where fluctuations occur on all scales. Stepping off the surface in a relevant direction turns on a mode that grows under the flow, drives the system away from g_*, and—as we compute next—generates a finite correlation length. The relevant couplings are precisely the experimental knobs (temperature, magnetic field) that must be tuned to reach criticality; their number n_+ is the number of such knobs.

Definition 23.28 (Universality class)**.** The *universality class* associated with a fixed point g_* is the set of all microscopic theories—continuum field theories, lattice spin models, fluids—whose critical point lies on the critical surface $\mathcal{S}(g_*)$, equivalently whose infrared flow is governed by g_*. All members of a universality class share the same long-distance and critical data: the same critical exponents, the same scaling functions, and the same universal amplitude ratios, differing only in non-universal normalizations (the location of the critical point, the metric factors relating bare to scaling fields, overall amplitudes).

That microscopically distinct theories can share critical behaviour is now a theorem-grade statement rather than an empirical surprise: by Theorem 23.26 any two theories on the same critical surface flow to the same g_* and so agree in all infrared observables up to power-suppressed corrections. The classic vindication is that the three-dimensional Ising lattice magnet and the liquid–vapour critical point of a real fluid—nothing alike at the molecular level—exhibit the *same* measured critical exponents to within experiment; both sit on the critical surface of the same Wilson–Fisher fixed point, the one we identify for ϕ^4 theory below and construct in Section 23.7. Figure 23.8 summarizes the geometry.

23.6.4 The correlation length and its divergence

To convert the geometry into numbers we need a physical length carried along by the flow. The natural one is the correlation length.

Definition 23.29 (Correlation length)**.** The *correlation length* ξ of a theory is the characteristic length scale governing the exponential decay of its connected

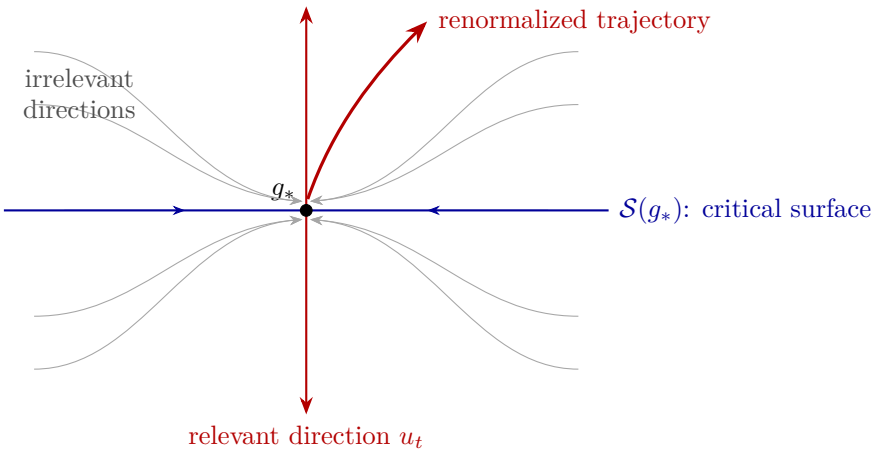


Figure 23.8: The flow near a fixed point g_*. The critical surface $\mathcal{S}(g_*)$ (blue, here one-dimensional) is the stable manifold: trajectories starting on it flow into g_*, and trajectories starting nearby (grey) are first attracted to it as the irrelevant scaling fields decay like $(\mu/\Lambda_0)^{|y_a|}$, then drift along it. The relevant direction u_t (red, vertical) is unstable: any nonzero u_t grows under the flow and carries the theory away from g_*. The trajectory emanating from g_* along the relevant direction is the renormalized trajectory (Theorem 23.31); a continuum limit is built by flowing out of the fixed point along it. The codimension of $\mathcal{S}(g_*)$—here one—is the number of relevant couplings that must be tuned to reach criticality and, equivalently, the number of free parameters of the continuum theory.

two-point function at large separation,

$$G_c(x) \;\equiv\; \langle\, \mathcal{O}(x)\, \mathcal{O}(0)\, \rangle - \langle \mathcal{O} \rangle^2 \;\sim\; e^{-|x|/\xi} \qquad (|x| \to \infty), \tag{23.159}$$

for a generic local operator $\mathcal{O}$. Equivalently $\xi = 1/m_{\text{gap}}$, the inverse of the mass gap. The theory is critical when $\xi = \infty$ (the gap closes and G_c decays as a power of $|x|$).

The correlation length transforms simply under a renormalization-group step, and this is the whole content of its scaling. A blocking by factor b coarsens the description: the lattice spacing, or equivalently the inverse cutoff $a = \Lambda^{-1}$, is multiplied by b, while the *physical* correlation length—an intrinsic property of the long-distance state—is unchanged. Measured in units of the running cutoff, the dimensionless correlation length $\hat{\xi} \equiv \xi\,\Lambda = \xi/a$ therefore shrinks by exactly the blocking factor at each step,

$$\hat{\xi} \;\longmapsto\; \hat{\xi}/b \qquad \text{under one step } \mathcal{R}_b. \tag{23.160}$$

Now hold the relevant physics in one scaling field. Suppose a single relevant field u_t (the "thermal" field, conjugate to the temperaturelike deviation from

criticality) controls the approach to the surface, with eigenvalue $y_t > 0$. Since $\hat{\xi}$ depends on the couplings only through the scaling fields, and a step sends $u_t \mapsto b^{y_t} u_t$ while sending $\hat{\xi} \mapsto \hat{\xi}/b$, consistency of the two ways of writing the same post-step quantity gives the functional equation

$$\hat{\xi}(u_t) \;=\; b\,\hat{\xi}(b^{\,y_t} u_t), \tag{23.161}$$

valid for any allowed $b > 1$. Equation (23.161) is a homogeneity relation, and homogeneity relations are solved by choosing b to remove the running variable. Take $u_t > 0$ small and set $b = u_t^{-1/y_t}$, which is > 1 precisely because u_t is small; then $b^{y_t} u_t = 1$ and Eq. (23.161) collapses to

$$\hat{\xi}(u_t) \;=\; u_t^{-1/y_t}\,\hat{\xi}(1). \tag{23.162}$$

The number $\hat{\xi}(1) = O(1)$ is non-universal (it is the dimensionless correlation length one step before criticality), but the *exponent* is fixed by the eigenvalue alone. Restoring physical units, $\xi = a\,\hat{\xi} = \Lambda^{-1}\hat{\xi}(1)\,u_t^{-1/y_t}$, so as the relevant field is tuned to zero—that is, as the theory approaches the critical surface—the correlation length diverges as a pure power,

$$\boxed{\;\xi \;\sim\; u_t^{-\nu}, \qquad \nu = \frac{1}{y_t}.\;} \tag{23.163}$$

This is the first critical exponent, ν, and it is the inverse of the relevant eigenvalue. The divergence of the correlation length on approach to criticality, asserted physically in Section 23.6.3, is now derived, and its rate is tied to the linearized spectrum. In a magnet $u_t \propto (T - T_c)$, so $\xi \sim |T - T_c|^{-\nu}$; the experimentally measured divergence of the correlation length is reading off $1/y_t$ of the governing fixed point.

23.6.5 Critical exponents as universal data of a fixed point

The same homogeneity logic, applied to the free energy rather than to a single correlation length, generates the full set of thermodynamic critical exponents and, crucially, the relations *among* them—all expressed through the eigenvalues y_t, y_h and the dimension d. Let $f_s(u_t, u_h)$ denote the singular part of the free energy per unit volume (the part responsible for critical behaviour), as a function of the thermal field u_t and the field u_h conjugate to the order parameter (the "magnetic" field, $\mathbb{Z}_2$-odd for ϕ^4 theory), with eigenvalues $y_t > 0$ and $y_h > 0$. The total free energy $V f_s$ is invariant under a renormalization-group step—it is the logarithm of the same partition function, merely re-expressed after coarse-graining—while the volume measured in cutoff units shrinks as $V \mapsto V/b^d$. The density therefore picks up a factor b^d, and since the couplings enter only through the scaling fields,

$$f_s(u_t, u_h) \;=\; b^{-d}\, f_s(b^{\,y_t} u_t,\; b^{\,y_h} u_h). \tag{23.164}$$

Choosing $b = u_t^{-1/y_t}$ as before turns this into the scaling form

$$f_s(u_t, u_h) \;=\; u_t^{\,d/y_t}\,\Phi\!\Big(\frac{u_h}{u_t^{\,y_h/y_t}}\Big), \qquad \Phi(z) \equiv f_s(1,z), \tag{23.165}$$

the Widom scaling hypothesis, here derived rather than postulated, with a single universal scaling function Φ. Every thermodynamic exponent now follows by differentiation; we carry the four standard ones in full and collect the results in Eq. (23.172).

Specific heat (α). At zero ordering field, $u_h = 0$, the specific heat is the second temperature derivative of the free energy, $C \sim \partial^2 f_s/\partial u_t^2$. From $f_s(u_t,0) = \Phi(0)\,u_t^{\,d/y_t}$,

$$C \;\sim\; \frac{\partial^2}{\partial u_t^2}\,u_t^{\,d/y_t} \;\sim\; u_t^{\,d/y_t-2} \;\equiv\; u_t^{-\alpha}, \qquad \alpha = 2 - \frac{d}{y_t} = 2 - d\nu, \tag{23.166}$$

the last equality using $\nu = 1/y_t$. This is the *hyperscaling* relation $2 - \alpha = d\nu$, and it carries the explicit dimension d—a fingerprint of the fact that it follows from the volume factor b^{-d}.

Order parameter (β). The spontaneous order parameter is the field derivative at zero field, $M = \partial f_s/\partial u_h\big|_{u_h=0}$. Differentiating Eq. (23.165) once in u_h brings down a factor $u_t^{-y_h/y_t}$ from the argument and evaluates $\Phi'(0)$, leaving

$$M \;\sim\; u_t^{\,d/y_t}\cdot u_t^{-y_h/y_t}\,\Phi'(0) \;\sim\; u_t^{(d-y_h)/y_t} \;\equiv\; u_t^{\beta}, \qquad \beta = \frac{d-y_h}{y_t} = \nu\,(d-y_h). \tag{23.167}$$

Susceptibility (γ). A second field derivative gives the susceptibility $\chi = \partial^2 f_s/\partial u_h^2\big|_{u_h=0}$, bringing down two factors $u_t^{-y_h/y_t}$,

$$\chi \;\sim\; u_t^{\,d/y_t}\cdot u_t^{-2y_h/y_t} \;\sim\; u_t^{(d-2y_h)/y_t} \;\equiv\; u_t^{-\gamma}, \qquad \gamma = \frac{2y_h - d}{y_t} = \nu\,(2y_h - d). \tag{23.168}$$

Critical isotherm (δ). Sitting exactly at criticality, $u_t = 0$, and asking how the order parameter depends on its own field requires the large-argument behaviour of Φ. For f_s to remain finite and nonzero as $u_t \to 0$ at fixed u_h, the explicit u_t powers must cancel: $\Phi(z) \sim z^{\,d/y_h}$ as $z \to \infty$, whence $f_s \sim u_h^{\,d/y_h}$ independently of u_t, and

$$M = \frac{\partial f_s}{\partial u_h}\bigg|_{u_t=0} \sim u_h^{\,d/y_h-1} \;\equiv\; u_h^{1/\delta}, \qquad \delta = \frac{y_h}{d-y_h}. \tag{23.169}$$

The remaining two (ν, η) and the field dimension. The correlation-length exponent $\nu = 1/y_t$ is Eq. (23.163). The last independent exponent, η, is the

anomalous dimension of the order-parameter field: at criticality the two-point function decays not with its free power but as

$$G_c(x)\big|_{u_t=u_h=0} \sim \frac{1}{|x|^{\,d-2+\eta}}, \qquad \Delta_{\mathcal{O}} = \frac{d-2+\eta}{2}, \tag{23.170}$$

where $\Delta_{\mathcal{O}}$ is the scaling dimension of $\mathcal{O}$. Since the field u_h is conjugate to $\mathcal{O}$, its eigenvalue is $y_h = d - \Delta_{\mathcal{O}}$, so

$$y_h = \frac{d+2-\eta}{2} \qquad \Longleftrightarrow \qquad \eta = d + 2 - 2y_h. \tag{23.171}$$

Two eigenvalues, y_t and y_h, together with the spacetime dimension d, thus determine *all* of the standard exponents. Collecting them,

$$\boxed{\nu = \frac{1}{y_t}, \quad \alpha = 2 - d\nu, \quad \beta = \nu(d - y_h), \quad \gamma = \nu(2y_h - d), \quad \delta = \frac{y_h}{d - y_h}, \quad \eta = d + 2 - 2y_h} \tag{23.172}$$

These six numbers are the universal data of the fixed point: they are read off from the linearized eigenvalues y_t, y_h (equivalently from the scaling dimensions of the two relevant operators), they are identical for every theory in the universality class by Theorem 23.26, and they satisfy the celebrated scaling relations, which we verify now to show they are not independent.

Worked Example 23.30 (The scaling relations from two eigenvalues)**.** The six exponents Eq. (23.172) are built from the two numbers y_t, y_h (and d), so they must obey four relations. We derive each by pure algebra, eliminating y_t, y_h.

Rushbrooke, $\alpha + 2\beta + \gamma = 2$. Add the eigenvalue expressions:

$$\begin{aligned}\alpha + 2\beta + \gamma &= (2 - d\nu) + 2\nu(d - y_h) + \nu(2y_h - d) \\ &= 2 - d\nu + 2d\nu - 2\nu y_h + 2\nu y_h - d\nu = 2 + (-d + 2d - d)\nu + (-2y_h + 2y_h)\nu = 2\end{aligned} \tag{23.173}$$

The y_h terms cancel and the $d\nu$ terms cancel separately, leaving exactly 2.

Widom, $\gamma = \beta(\delta - 1)$. Compute the right-hand side,

$$\beta(\delta-1) = \nu(d-y_h)\left(\frac{y_h}{d-y_h} - 1\right) = \nu(d-y_h)\cdot\frac{y_h - (d-y_h)}{d-y_h} = \nu(2y_h - d) = \gamma, \tag{23.174}$$

the factor $(d - y_h)$ cancelling against the denominator of $\delta - 1$.

Fisher, $\gamma = \nu(2 - \eta)$. Substitute $\eta = d + 2 - 2y_h$ from Eq. (23.171):

$$\nu(2-\eta) = \nu(2 - (d + 2 - 2y_h)) = \nu(2y_h - d) = \gamma. \tag{23.175}$$

Hyperscaling, $2 - \alpha = d\nu$. This is Eq. (23.166) read backwards; it holds whenever the volume factor in Eq. (23.164) is b^{-d}, i.e. whenever the singular free

energy is genuinely d-dimensional and extensive. We flag the one honest caveat: hyperscaling can fail above the upper critical dimension (for ϕ^4 theory, $d > 4$), where a dangerous irrelevant variable invalidates the naive setting of $u_a \to 0$ inside f_s and the Gaussian fixed point's exponents lose their d-dependence. Below four dimensions, where the Wilson–Fisher fixed point governs, hyperscaling holds.

Four relations among six exponents leave two independent, consistent with the two eigenvalues y_t, y_h that generate them. This is the quantitative meaning of "the critical exponents are universal data of the fixed point."

Caution

The exponents are universal; the things multiplying the powers are not. The critical temperature T_c, the amplitude of ξ, the value of $\hat{\xi}(1)$ in Eq. (23.162), and the metric coefficients relating physical knobs to scaling fields ($u_t \propto T - T_c$ with a non-universal proportionality constant) all depend on microscopic detail. What Theorem 23.26 protects is the *exponents* and the *scaling functions* and certain dimensionless *ratios* of amplitudes; the overall scales are not protected and vary across the universality class. Conflating a universal exponent with a non-universal amplitude is the most common error in reading critical data.

23.6.6 The continuum limit and the renormalized trajectory

We have so far flowed towards the infrared and asked what survives. Reverse the question. A quantum field theory is defined with a cutoff Λ—a physical short-distance scale, in the doctrine of this chapter, not an embarrassment. To call the theory "continuum" is to assert that the cutoff can be sent to infinity, $\Lambda \to \infty$, while physical quantities measured at fixed energies remain finite and fixed. When can this be done, and what does the resulting theory look like?

The correlation length supplies the criterion at once. Physical lengths are held fixed in the limit, so the physical correlation length ξ is fixed; but $\xi = \Lambda^{-1}\hat{\xi}$, so holding ξ fixed while $\Lambda \to \infty$ forces

$$\hat{\xi} = \xi\,\Lambda \;\longrightarrow\; \infty. \tag{23.176}$$

The dimensionless correlation length must diverge. But $\hat{\xi} \to \infty$ is exactly criticality (Theorem 23.29): the bare theory, as the cutoff is removed, must be tuned onto the critical surface of a fixed point. *A continuum limit is the approach to a fixed point from the ultraviolet.* Without a fixed point to provide a scale-invariant point with $\hat{\xi} = \infty$, there is no way to make the dimensionless correlation length diverge while keeping physics fixed, and no nontrivial continuum theory exists.

This is the geometric picture made precise. Sit on the critical surface and displace slightly along the relevant directions. As the cutoff is raised, $\Lambda \to \infty$,

tune the bare relevant couplings towards their critical values so that the physical masses stay put. By Eq. (23.163), holding $\xi = 1/m$ fixed while $\Lambda \to \infty$ requires

$$u_t(\Lambda) \;=\; \big(\hat{\xi}(1)\big)^{y_t}\,(m/\Lambda)^{y_t} \;\xrightarrow[\Lambda\to\infty]{}\; 0, \tag{23.177}$$

and likewise each relevant field is sent to its critical value at a rate $(\mu/\Lambda)^{y_i}$ fixed by its eigenvalue, where μ is the renormalization scale and the finite numbers $\bar{u}_i$ multiplying these powers are the renormalized relevant couplings,

$$\boxed{\,u_i(\Lambda) \;=\; \Big(\frac{\mu}{\Lambda}\Big)^{y_i}\,\bar{u}_i \quad (y_i>0), \qquad u_a(\Lambda) \;\longrightarrow\; 0 \quad (y_a<0), \qquad \Lambda\to\infty.\,} \tag{23.178}$$

The irrelevant couplings are not free in the continuum limit: they are driven to their fixed-point values, by Theorem 23.26, and any attempt to assign them independent nonzero values at $\Lambda = \infty$ would either be washed out (they flow back to the fixed-point sheet) or, if dialled up to keep pace, destroy the limit. The continuum theory is thus parametrized by the finitely many renormalized relevant and marginal couplings $\bar{u}_i$ alone. The trajectory traced out in coupling space by Eq. (23.178) as Λ varies—emanating from the fixed point along its unstable directions, with every irrelevant coupling pinned to the fixed-point value—is the central object.

Definition 23.31 (Renormalized trajectory)**.** The *renormalized trajectory* of a fixed point g_* is the renormalization-group trajectory that flows *out* of g_* along its unstable (relevant) manifold, with all irrelevant scaling fields identically zero. Equivalently it is the one-to-$(n_+ + n_0)$-parameter family of theories along which the cutoff can be removed, $\Lambda \to \infty$, at fixed long-distance physics, prescribed by Eq. (23.178).

The renormalized trajectory is the continuum theory's worldline under scale changes. A bare theory whose couplings lie exactly on it has no cutoff artifacts whatsoever: its irrelevant couplings sit on the fixed-point sheet, so the power-suppressed corrections of Theorem 23.26 vanish identically and the theory is a "perfect" lattice action in the language of Chapter 43. A bare theory near, but not on, the trajectory flows towards it as the irrelevant transients die, and reproduces the same continuum physics up to corrections $(\mu/\Lambda)^{y_{\min}}$. This is the Wilsonian re-reading of multiplicative renormalization from Chapter 22: the renormalization conditions there, which fixed finitely many renormalized couplings and let the cutoff be removed, are exactly the prescription Eq. (23.178) for selecting a point on the renormalized trajectory.

We can now state the counting that the next section turns into the modern definition of renormalizability.

Proposition 23.32 (Parameter count of the continuum limit)**.** *The number of free parameters needed to specify a continuum theory built on a fixed point*

g_* equals the number $n_+ + n_0$ of relevant plus marginal directions of g_*. The irrelevant couplings contribute no free parameters: they are fixed by the fixed point.

Proof. By Eq. (23.178) the continuum theory is the $\Lambda \to \infty$ image of a bare theory whose irrelevant scaling fields tend to zero and whose relevant scaling fields are $u_i(\Lambda) = (\mu/\Lambda)^{y_i}\bar{u}_i$. The limit is fully determined by the data $\{\bar{u}_i\}$ over relevant directions $y_i > 0$, that is n_+ numbers. The n_0 marginal fields neither grow nor decay at linear order, so they survive to long distances (Theorem 23.26) and must be specified as well, contributing n_0 further parameters; whether each is ultimately marginally relevant or irrelevant is decided by the leading nonlinear term but does not change the count at this level. The irrelevant fields are sent to the fixed-point values by the limit and carry no freedom. Hence the parameter count is $n_+ + n_0$. □

Physical Insight

This proposition is the hinge of the whole chapter. The number of measurements needed to pin down a continuum quantum field theory is not the number of terms one could write in a Lagrangian—that is infinite—but the number of relevant and marginal directions of the controlling fixed point, which is typically a handful. The infinitely many irrelevant couplings are not assumptions of the theory; they are *predictions* of it, fixed on the renormalized trajectory and recovered, with calculable power-suppressed corrections, from the few relevant data. Predictivity is the statement that $n_+ + n_0$ is finite and small. The next section names a theory with finitely many relevant/marginal directions "renormalizable" and one with the relevant data exhausted but irrelevant tails surviving as small corrections an "effective field theory," and shows the two are the same physics viewed at different distances from a fixed point.

23.6.7 Scalar ϕ^4 theory: the prototype critical system

We close by reading the abstract apparatus on the theory we have carried since Chapter 6. The real scalar with Lagrangian $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ is the canonical laboratory for every structure of this section: it has a $\mathbb{Z}_2$ symmetry $\phi \to -\phi$, a relevant mass, a (near-)marginal quartic, and a tower of manifestly irrelevant higher operators. The simplest fixed point is the Gaussian (free) one, $g_* = (m^2{=}0,\, \lambda{=}0,\, \dots)$, about which the eigenvalues are simply the canonical dimensions of Section 21.2.

Worked Example 23.33 (Operator classification at the Gaussian fixed point)**.** At the free fixed point in d dimensions the field has its canonical dimension

$\Delta_\phi = \frac{d-2}{2}$ (from $[\phi^2] \sim [(\partial\phi)^2]/[\partial]^2$ and the requirement $[\mathcal{L}] = d$). The eigenvalue of the coupling multiplying a $\mathbb{Z}_2$-even monomial ϕ^{2n} is $y_{2n} = d - \Delta_{2n}$ with $\Delta_{2n} = 2n\,\Delta_\phi = n(d-2)$. Hence

$$\text{mass term } \phi^2: \quad \Delta_2 = d-2, \quad y_2 = d-(d-2) = 2 > 0 \qquad \text{(relevant, all } d\text{)}, \tag{23.179}$$

$$\text{quartic } \phi^4: \quad \Delta_4 = 2(d-2) = 2d-4, \quad y_4 = d-(2d-4) = 4-d \quad \begin{cases} > 0 & d < 4 \\ = 0 & d = 4 \\ < 0 & d > 4, \end{cases} \tag{23.180}$$

$$\text{sextic } \phi^6: \quad \Delta_6 = 3(d-2) = 3d-6, \quad y_6 = d-(3d-6) = 6-2d \quad < 0 \text{ for } d > 3, \tag{23.181}$$

and every higher monomial ϕ^{2n} with $n \geq 3$, together with all derivative-dressed operators such as $\phi^2(\partial\phi)^2$, has $y < 0$ in three and four dimensions. The pattern is decisive. In $d = 4$ the Gaussian fixed point has a *single* relevant direction (the mass, $y_2 = 2$) and a *single* marginal one (the quartic, $y_4 = 0$); everything else is irrelevant. Tuning the one relevant coupling to zero places the theory on the critical surface, and the marginal quartic must then be examined at the next order. In $d < 4$, by contrast, the quartic acquires a positive eigenvalue $y_4 = 4 - d > 0$ and becomes *relevant*: the Gaussian fixed point now has two relevant directions, ϕ^2 and ϕ^4.

The four-dimensional marginal case is settled by the quadratic term in the beta function, which ϕ^4 theory delivers in Chapter 24: the one-loop result $\beta_\lambda = \mu\,\mathrm{d}\lambda/\mathrm{d}\mu = \dfrac{3\lambda^2}{16\pi^2} + O(\lambda^3)$ is positive, so λ *decreases* towards the infrared. The quartic is marginally *irrelevant* in $d = 4$: a generic four-dimensional scalar flows to the Gaussian fixed point at long distances, becoming free, and the only candidate continuum limit at the Gaussian point is the trivial (non-interacting) one. This is the Wilsonian face of the triviality of ϕ^4_4, and it is why a strictly four-dimensional interacting scalar with a fundamental cutoff at infinity is believed not to exist; we return to it, and to its escape via dimensional continuation, in Section 23.7.

The interesting physics is therefore just below four dimensions. The moment d drops below 4, Eq. (23.180) turns the quartic relevant, $y_4 = 4 - d = \epsilon > 0$ with $\epsilon = 4 - d$ small. The Gaussian fixed point, now possessing two relevant directions (ϕ^2 and ϕ^4), can no longer be the infrared-attractive critical point reached by tuning a single knob: it has the wrong number of relevant directions to describe a system with a single critical temperature. Something must take over. A new fixed point peels off from the Gaussian one at a quartic coupling of order ϵ—the Wilson–Fisher fixed point, $\lambda_* = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2)$ in the normalization

of ϕ^4 theory—and it is engineered precisely so that the quartic direction becomes (weakly) irrelevant there, leaving the $\mathbb{Z}_2$-even mass operator ϕ^2 as the *single* relevant direction. That is the correct structure for a continuous phase transition tuned by one parameter (temperature): one relevant even operator, hence a critical surface of codimension one. Turning on the $\mathbb{Z}_2$-*odd* source for ϕ supplies the second, magnetic, relevant direction u_h; the pair (u_t, u_h) is exactly the (t, h) plane whose exponents Eq. (23.172) we computed above.

This identifies ϕ^4 theory in d slightly below 4 as the prototype critical system, and its governing fixed point as the controller of the *Ising universality class*: the universality class (Theorem 23.28) of any $\mathbb{Z}_2$-symmetric system with a single scalar order parameter and short-range interactions—the lattice Ising magnet, the liquid–vapour transition, a uniaxial ferromagnet, and the scalar field itself all share its critical surface and therefore its exponents. In three dimensions ($\epsilon = 1$) this is the Ising universality class whose exponents are measured in the laboratory and computed to high precision by the conformal bootstrap; the bridge from the present fixed-point picture to that bootstrap solution is built in Chapter 52, where the Wilson–Fisher fixed point reappears as a conformal field theory and the universal data Eq. (23.172) become operator dimensions in the Ising CFT.

The conceptual payoff of the fixed-point picture is now complete: irrelevant couplings are forgotten at a rate we computed, universality classes are basins of attraction, critical exponents are eigenvalues of the linearized flow, and a continuum limit is flow out of a fixed point along finitely many relevant directions. What remains is to convert the parameter count of Theorem 23.32 into the modern criterion for renormalizability—counting relevant and marginal directions—and to exhibit the one nontrivial fixed point this section has been pointing to throughout. The next and final section does both, constructing the Wilson–Fisher fixed point of ϕ^4 theory in $d = 4-\epsilon$ explicitly and handing its coupling and exponents forward to the running-coupling analysis of Chapter 24 and the conformal bootstrap of Chapter 52.

23.7 The Modern Meaning of Renormalizability and the Wilson–Fisher Fixed Point

The previous section drew the physical payoff of the fixed-point picture: a fixed point of the coarse-graining flow forgets its irrelevant directions, so a whole basin of microscopically different theories collapses onto one long-distance description, and the continuum limit is the trajectory that flows *out* of a fixed point along its relevant directions (Section 23.6). We now spend that capital twice. First we use the relevant/marginal/irrelevant trichotomy to rewrite the very notion of renormalizability — the question that organized Chapters 21 and 22 — in flow-theoretic language, and in doing so dissolve the old worry about

non-renormalizable interactions. Then we deliver the chapter's showcase: the one nontrivial fixed point that the ϕ^4 flow possesses in the accessible regime, the Wilson–Fisher fixed point in $d = 4 - \varepsilon$, computed in full from the dimensionless flow of Section 23.3.3. This is the capstone of ϕ^4 theory for Part VI: it produces the number $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ that Chapter 24 re-derives by an entirely different route and that Chapter 52 reinterprets as a conformal field theory.

23.7.1 Renormalizability, re-read

Recall the engineering-dimension bookkeeping of Section 21.2. A local operator $\mathcal{O}$ built from fields and derivatives carries a mass dimension $\Delta_{\mathcal{O}}$ fixed by the free Lagrangian; in the action $S = \int \mathrm{d}^d x \, \sum_i g_i \, \mathcal{O}_i$ the coupling that multiplies it has dimension

$$[g_i] = d - \Delta_{\mathcal{O}_i}, \tag{23.182}$$

because $[\mathrm{d}^d x] = -d$ and the action is dimensionless. In Chapter 21 this number decided the superficial degree of divergence; operators with $[g_i] < 0$ (dimension $\Delta_{\mathcal{O}_i} > d$) generated worse and worse divergences as the order increased, and a theory containing one of them seemed to demand an unbounded number of counterterms — the textbook verdict "non-renormalizable, therefore useless."

The Wilsonian flow turns Eq. (23.182) from a divergence-counting device into a statement about how couplings move. The clean way to see this is to measure every coupling in units of the running cutoff Λ, forming the dimensionless coupling

$$\hat{g}_i \equiv g_i \, \Lambda^{\Delta_{\mathcal{O}_i} - d} = g_i \, \Lambda^{-[g_i]}. \tag{23.183}$$

As the cutoff is lowered, $\Lambda \to \Lambda e^{-t}$ with the coarse-graining parameter

$$t = \ln \frac{\Lambda_0}{\Lambda} \; \geq 0 \quad \text{increasing toward the infrared,} \tag{23.184}$$

the pure power in Eq. (23.16) contributes a *classical* (tree-level) piece to the flow,

$$\frac{\mathrm{d}\hat{g}_i}{\mathrm{d}t}\Big|_{\text{classical}} = (d - \Delta_{\mathcal{O}_i}) \, \hat{g}_i + (\text{loop terms}). \tag{23.185}$$

At the Gaussian (free) fixed point $\{\hat{g}_i = 0\}$ the loop terms vanish quadratically, so the linearized flow there is governed entirely by the classical eigenvalues

$$y_i \;=\; d - \Delta_{\mathcal{O}_i} \;=\; [g_i]. \tag{23.186}$$

This is the bridge between the two languages. The eigenvalue y_i that Section 23.6 used to sort directions of the flow is, at the Gaussian fixed point, nothing but the canonical dimension of the coupling from Chapter 21:

- $y_i > 0$ ($\Delta_{\mathcal{O}_i} < d$): the dimensionless coupling *grows* toward the infrared — the operator is **relevant**;

- $y_i = 0$ ($\Delta_{\mathcal{O}_i} = d$): classically marginal — the loop terms decide its fate;

- $y_i < 0$ ($\Delta_{\mathcal{O}_i} > d$): the dimensionless coupling *shrinks* toward the infrared — the operator is **irrelevant**.

The decisive observation is that the operators Chapter 21 branded "non-renormalizable," those with $[g_i] < 0$, are precisely the irrelevant ones: under coarse-graining their dimensionless strength is driven to zero. They are not pathological; they are forgotten. Conversely, only finitely many operators can be relevant or marginal, because $d - \Delta_{\mathcal{O}_i} \geq 0$ bounds $\Delta_{\mathcal{O}_i} \leq d$, and there are only finitely many local operators of bounded dimension built from a given field content. This finiteness is the whole content of renormalizability, now read off the flow.

Definition 23.34 (Renormalizable theory, Wilsonian sense)**.** Let a quantum field theory be defined by an action perturbed around a fixed point of the renormalization-group flow, with couplings $\{\hat{g}_i\}$ classified by their eigenvalues y_i at that fixed point. The theory is *renormalizable* — equivalently, it admits a *predictive continuum limit* — if it has only finitely many relevant and marginal couplings ($y_i \geq 0$). To define the continuum theory one tunes this finite set; every irrelevant coupling ($y_i < 0$) is then swept by the flow onto the critical surface and contributes nothing to long-distance physics beyond corrections suppressed by powers of E/Λ. A theory whose specification *requires* irrelevant couplings to be fixed at the cutoff is *non-renormalizable*: it is a perfectly well-defined *effective* theory below Λ, predictive up to those same power-suppressed corrections, but it has no continuum limit at that fixed point.

Two clarifications make this definition do its work. The first is what "predictive" means operationally. To send the cutoff to infinity, $\Lambda_0 \to \infty$, while holding low-energy physics fixed, one must follow the flow backward (toward the ultraviolet) and land exactly on a renormalized trajectory — a flow line emanating from the fixed point. The relevant couplings are the coordinates *along* that trajectory: they must be supplied as input (one renormalization condition each, the finitely many measured numbers of Chapter 22). The irrelevant couplings are coordinates *transverse* to it: the demand that the trajectory exist in the $\Lambda_0 \to \infty$ limit fixes them in terms of the relevant ones. A renormalizable theory has finitely many transverse conditions to impose and finitely many inputs to measure, so finitely many experiments determine all of low-energy physics. That is predictivity.

The second clarification reverses the old judgement on non-renormalizable interactions without discarding any of Chapter 21's computations.

Physical Insight

The old power-counting verdict and the Wilsonian verdict are the *same arithmetic* read in opposite directions. Chapter 21 fixed the cutoff and watched a non-renormalizable coupling g_i (with $[g_i] < 0$) feed amplitudes the dimensionless combination $g_i\, E^{|[g_i]|}$, which grows with energy and forces a new counterterm at each order — "the theory loses predictivity in the ultraviolet." Wilson fixes the energy and watches the same combination $g_i\, \Lambda^{|[g_i]|} = \hat{g}_i$ *shrink* as Λ is lowered — "the operator is irrelevant and disappears in the infrared." Nothing in the loop integrals changed; only the question did. The lesson is that renormalizability is not a fundamental restriction on which interactions Nature may use. It is a statement about the naturalness of the continuum limit: a renormalizable theory is one whose ultraviolet completion can be pushed to infinite energy with only finitely many free parameters. A non-renormalizable theory is the generic case — an effective description valid below some physical scale Λ, with the irrelevant operators encoding, in a controlled E/Λ expansion, the imprint of whatever shorter-distance physics sits above Λ. This is the viewpoint that organizes the entire effective-field-theory paradigm of Chapter 38: one stops apologizing for non-renormalizable operators and starts using them as a systematic expansion.

Caution

"Marginal" ($y_i = 0$) is the one case the classical eigenvalue cannot decide, and it is exactly the case that matters most. The quartic coupling of ϕ^4 theory is marginal in $d = 4$, and whether it is truly marginal, marginally relevant, or marginally irrelevant is settled only by the sign of the one-loop term — the leading correction to Eq. (23.186). The whole of running couplings, asymptotic freedom, and the fixed point computed below lives in this marginal sliver. A marginal coupling is never "done" at tree level.

With renormalizability recast, the natural question is whether the ϕ^4 flow has any fixed point other than the trivial Gaussian one. In exactly four dimensions it does not: the quartic coupling is marginally *relevant* (the one-loop $\beta_\lambda > 0$ drives it up toward the ultraviolet, down toward the infrared), so the only zero of the flow is $\lambda = 0$, and ϕ^4 theory has no interacting continuum limit in $d = 4$. The escape, due to Wilson and Fisher, is to step off the integer dimension.

23.7.2 The Wilson–Fisher mechanism

The idea is a balancing act. In $d = 4$ the quartic is marginal, $y_\lambda = 0$ at the Gaussian fixed point, so its flow is driven purely by the loop term, which has a

definite sign and never turns around: no nonzero zero. Lower the dimension to $d = 4 - \varepsilon$ with $\varepsilon > 0$ small. By Eq. (23.186) the quartic coupling, whose operator has dimension $\Delta_{\phi^4} = 4 \cdot \frac{d-2}{2} = 2(d-2)$, now carries

$$y_\lambda = d - \Delta_{\phi^4} = d - 2(d-2) = 4 - d = \varepsilon > 0, \tag{23.187}$$

so at the Gaussian fixed point the quartic has become genuinely *relevant*: the classical term pushes the dimensionless coupling up. But the one-loop term pushes it down. For small ε the upward classical push is weak (order ε), so it can be balanced by the loop term at a small, perturbatively accessible value of the coupling. Where the two cancel is a new fixed point. This is the Wilson–Fisher mechanism, and because the balance happens at weak coupling, the one-loop flow already computed in Section 23.3.3 is sufficient to locate it.

We need the dimensionless flow of the two couplings that are relevant or marginal near four dimensions: the mass and the quartic. Following Section 23.3.3, write the Euclidean action with a sharp cutoff Λ,

$$S = \int \mathrm{d}^d x \left[\tfrac{1}{2}(\partial\phi)^2 + \tfrac{1}{2} m^2 \phi^2 + \frac{\lambda}{4!}\phi^4 \right], \tag{23.188}$$

the frozen ϕ^4 Lagrangian (Chapter 6) Wick-rotated as in Chapter 10, and form the dimensionless couplings

$$r \equiv \frac{m^2}{\Lambda^2}, \qquad u \equiv K_d \, \lambda \, \Lambda^{d-4}, \qquad K_d \equiv \frac{S_{d-1}}{(2\pi)^d} = \frac{1}{(2\pi)^d} \frac{2\pi^{d/2}}{\Gamma(d/2)}, \tag{23.189}$$

where S_{d-1} is the area of the unit $(d-1)$-sphere and K_d is the standard one-loop angular factor; at $d = 4$ it takes the value $K_4 = 2\pi^2/(2\pi)^4 = 1/(8\pi^2)$ that converts the geometric factor into the familiar $16\pi^2$. The one-loop momentum-shell integration of Section 23.3.3, carried out by splitting the field into a slow part below $b\Lambda$ and a fast part on the shell $b\Lambda < |k| < \Lambda$, integrating the fast part as a Gaussian with propagator $1/(k^2 + m^2)$, and rescaling back to the original cutoff, produces

$$\frac{\mathrm{d}r}{\mathrm{d}t} = 2\, r + \frac{1}{2} \frac{u}{1+r}, \tag{23.190}$$

$$\frac{\mathrm{d}u}{\mathrm{d}t} = \varepsilon\, u - \frac{3}{2} \frac{u^2}{(1+r)^2}, \tag{23.191}$$

with t the coarse-graining parameter Eq. (23.184) and $\varepsilon = 4 - d$. Because these two equations carry the whole computation, it is worth recording exactly where each term comes from, both to make the section self-contained and to expose the two physical effects that collide at the fixed point.

Worked Example 23.35 (Anatomy of the one-loop ϕ^4 flow)**.** *Classical terms.* Hold the dimensionful m^2, λ fixed and lower the cutoff $\Lambda \to \Lambda e^{-\mathrm{d}t}$. From

Eq. (23.189), $r = m^2\Lambda^{-2} \to r\,e^{2\,\mathrm{d}t}$ and $u = K_d\lambda\,\Lambda^{d-4} \to u\,e^{(4-d)\mathrm{d}t} = u\,e^{\varepsilon\,\mathrm{d}t}$, so the pure rescaling contributes $\mathrm{d}r/\mathrm{d}t = 2r$ and $\mathrm{d}u/\mathrm{d}t = \varepsilon u$ — exactly the canonical eigenvalues $y_{m^2} = 2$ and $y_\lambda = \varepsilon$ of Eq. (23.186), with $\Delta_{\phi^2} = d - 2$ and $\Delta_{\phi^4} = 2d - 4$.

Loop correction to the mass. The vertex $\frac{\lambda}{4!}(\phi_< + \phi_>)^4$ contains a term with two slow and two fast fields, $\frac{\lambda}{4!}\binom{4}{2}\phi_<^2\phi_>^2 = \frac{\lambda}{4}\phi_<^2\phi_>^2$. Contracting the two fast fields gives a correction to the slow $\frac{1}{2}m^2\phi_<^2$ term,

$$\delta m^2 = \frac{\lambda}{2}\int_{\text{shell}}\frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{k^2+m^2} = \frac{\lambda}{2}\,K_d\,\frac{\Lambda^d}{\Lambda^2+m^2}\,\mathrm{d}t = \frac{\lambda}{2}\,K_d\,\frac{\Lambda^{d-2}}{1+r}\,\mathrm{d}t, \tag{23.192}$$

where the thin-shell integral over $b\Lambda < |k| < \Lambda$ with $b = e^{-\mathrm{d}t}$ evaluates the integrand at $|k| = \Lambda$ times the shell volume $K_d\Lambda^d\,\mathrm{d}t$. Dividing by Λ^2 and using $u = K_d\lambda\Lambda^{d-4}$ turns this into $\mathrm{d}r/\mathrm{d}t|_{\text{loop}} = \frac{1}{2}\,u/(1+r)$, the second term of Eq. (23.190). It is positive: virtual fluctuations on the shell add to the mass.

Loop correction to the quartic. At one loop the four-point function receives a contribution from two vertices joined by two fast propagators, summed over the three ways (s, t, u channels) of pairing the four external slow legs; at zero external momentum each channel gives $\frac{1}{2}\lambda^2\int_{\text{shell}}(k^2+m^2)^{-2}$, so

$$\delta\lambda = -\,\frac{3}{2}\,\lambda^2\int_{\text{shell}}\frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{(k^2+m^2)^2} = -\,\frac{3}{2}\,\lambda^2\,K_d\,\frac{\Lambda^{d-4}}{(1+r)^2}\,\mathrm{d}t. \tag{23.193}$$

Multiplying by $K_d\Lambda^{d-4}$ as in Eq. (23.189) gives $\mathrm{d}u/\mathrm{d}t|_{\text{loop}} = -\frac{3}{2}\,u^2/(1+r)^2$, the second term of Eq. (23.191). It is negative: the loop screens the coupling.

Consistency with the carried beta function. Set $d = 4$ (so $u = K_4\lambda = \lambda/8\pi^2$, independent of Λ) and $r \to 0$. Then Eq. (23.191) reads $\mathrm{d}u/\mathrm{d}t = -\frac{3}{2}u^2$, i.e. $K_4\,\mathrm{d}\lambda/\mathrm{d}t = -\frac{3}{2}K_4^2\lambda^2$, so $\mathrm{d}\lambda/\mathrm{d}t = -\frac{3}{2}K_4\lambda^2 = -3\lambda^2/16\pi^2$. Since $t = \ln(\Lambda_0/\Lambda)$ runs opposite to the renormalization scale $\mu \sim \Lambda$, the continuum beta function is $\beta_\lambda = \mathrm{d}\lambda/\mathrm{d}\ln\mu = -\mathrm{d}\lambda/\mathrm{d}t = +3\lambda^2/16\pi^2$. This is exactly the carried number of ϕ^4 theory, reproduced in Chapter 24 from the counterterm δ_λ; the Wilsonian and Callan–Symanzik routes agree on it.

The momentum-independence of Eq. (23.192) is worth flagging, because it is the reason a later result comes out the way it does. The one-loop self-energy is a pure tadpole: the loop in Eq. (23.192) carries no external momentum, so it shifts m^2 but cannot generate a term proportional to the external k^2. There is therefore no one-loop correction to the coefficient of $(\partial\phi)^2$, hence no rescaling of the field beyond the classical one, hence

$$\gamma_\phi = O(\lambda^2) = O(u^2) \tag{23.194}$$

at this order — the field keeps its Gaussian scaling dimension through one loop. We will use this in Theorem 23.36; it is the same fact Chapter 24 expresses by saying the field-strength renormalization Z_ϕ first moves at two loops.

Notation Watch

Throughout this subsection $\varepsilon \equiv 4 - d$ is the *dimension deficit*, held small and finite — it is the expansion parameter of critical phenomena, the variable in which the Wilson–Fisher coupling is universally quoted. It is *twice* the dimensional-regularization regulator of the book's canon (Theorem A.45), where $d = 4 - 2\varepsilon_{\mathrm{dr}}$ and $\varepsilon_{\mathrm{dr}} \to 0$ is a regulator removed at the end: here $\varepsilon = 2\varepsilon_{\mathrm{dr}}$. With $d = 4 - \varepsilon$ the classical eigenvalue of the quartic is $+\varepsilon$ and the fixed point will land at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$; in the canonical $\overline{\mathrm{MS}}$ variables of Theorem A.49 the identical fixed point reads $\lambda_* = \frac{32\pi^2}{3}\varepsilon_{\mathrm{dr}}$. The physics is the same; Chapters 24 and 52 use the same $\varepsilon = 4 - d$ convention adopted here.

We can now solve for the fixed point.

Theorem 23.36 (The Wilson–Fisher fixed point)**.** *For $0 < \varepsilon \ll 1$ the one-loop dimensionless flow Eqs. (23.190) and (23.191) of ϕ^4 theory in $d = 4 - \varepsilon$ has, besides the Gaussian fixed point $(r, u) = (0, 0)$, a single nontrivial fixed point at*

$$u_* = \frac{2}{3}\,\varepsilon + O(\varepsilon^2), \qquad r_* = -\frac{\varepsilon}{6} + O(\varepsilon^2), \tag{23.195}$$

the Wilson–Fisher fixed point*. In terms of the dimensionful quartic measured at the cutoff, $\lambda_* = u_*/K_4 + O(\varepsilon^2)$ gives*

$$\boxed{\lambda_* = \frac{16\pi^2}{3}\,\varepsilon + O(\varepsilon^2).} \tag{23.196}$$

Linearizing the flow at this point, the two eigenvalues are

$$y_t = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2) > 0 \quad (\textit{relevant}), \qquad y_\lambda = -\varepsilon + O(\varepsilon^2) < 0 \quad (\textit{irrelevant}), \tag{23.197}$$

so the quartic direction, marginal at $d = 4$ and relevant at the Gaussian point, has turned irrelevant *at Wilson–Fisher, while exactly one relevant direction survives. The relevant eigenvalue is the inverse correlation-length exponent, $1/\nu = y_t = 2 - \varepsilon/3 + O(\varepsilon^2)$. The field anomalous dimension is $\gamma_\phi(\lambda_*) = O(\varepsilon^2)$.*

Proof. We carry out the algebra in the worked example that follows; here we record the structure. Setting the right-hand side of Eq. (23.191) to zero gives a quadratic in u whose nonzero root is $O(\varepsilon)$; feeding that into Eq. (23.190) fixes r_*, also $O(\varepsilon)$. The linearization is the Jacobian of $(\mathrm{d}r/\mathrm{d}t, \mathrm{d}u/\mathrm{d}t)$ evaluated at (r_*, u_*); to leading order in ε it is triangular, so its eigenvalues are the diagonal entries Eq. (23.197). The anomalous-dimension claim is Eq. (23.194) evaluated at $u_* = O(\varepsilon)$. □

Worked Example 23.37 (Solving the flow for the Wilson–Fisher fixed point)**.** We solve Eqs. (23.190) and (23.191) for their simultaneous zeros and linearize, carrying every factor.

Step 1: the quartic. A fixed point has $\mathrm{d}u/\mathrm{d}t = 0$, so from Eq. (23.191)

$$\varepsilon\, u_* - \frac{3}{2}\,\frac{u_*^2}{(1+r_*)^2} = 0 \;\Longrightarrow\; u_*\left[\varepsilon - \frac{3}{2}\,\frac{u_*}{(1+r_*)^2}\right] = 0. \tag{23.198}$$

The root $u_* = 0$ is the Gaussian fixed point. The nontrivial root is

$$u_* = \frac{2}{3}\,\varepsilon\,(1+r_*)^2. \tag{23.199}$$

This is the balance the Wilson–Fisher mechanism promised: the classical push εu (upward) and the loop screening $\frac{3}{2}u^2/(1+r)^2$ (downward) cancel at a coupling of order ε, hence weak.

Step 2: the mass. A fixed point also has $\mathrm{d}r/\mathrm{d}t = 0$, so from Eq. (23.190)

$$2\,r_* + \frac{1}{2}\,\frac{u_*}{1+r_*} = 0 \;\Longrightarrow\; r_* = -\frac{1}{4}\,\frac{u_*}{1+r_*}. \tag{23.200}$$

Step 3: leading order in ε. Equation (23.199) shows $u_* = O(\varepsilon)$, and then Eq. (23.200) shows $r_* = O(\varepsilon)$. Expanding $(1+r_*)^2 = 1 + O(\varepsilon)$ in Eq. (23.199),

$$u_* = \frac{2}{3}\,\varepsilon\,[1 + O(\varepsilon)] = \frac{2}{3}\,\varepsilon + O(\varepsilon^2), \tag{23.201}$$

and inserting this into Eq. (23.200) with $1 + r_* = 1 + O(\varepsilon)$,

$$r_* = -\frac{1}{4}\cdot\frac{2}{3}\,\varepsilon\,[1 + O(\varepsilon)] = -\frac{\varepsilon}{6} + O(\varepsilon^2). \tag{23.202}$$

This is Eq. (23.195). Converting to the dimensionful coupling through $u = K_d \lambda \Lambda^{d-4}$ at leading order, where $K_d \to K_4 = 1/8\pi^2$ and the residual Λ-dependence is higher order in ε,

$$\lambda_* = \frac{u_*}{K_4} + O(\varepsilon^2) = 8\pi^2 \cdot \frac{2}{3}\,\varepsilon + O(\varepsilon^2) = \frac{16\pi^2}{3}\,\varepsilon + O(\varepsilon^2), \tag{23.203}$$

the carried number Eq. (23.196).

Step 4: linearize. Write $f(r,u) = \mathrm{d}r/\mathrm{d}t$ and $g(r,u) = \mathrm{d}u/\mathrm{d}t$ from Eqs. (23.190) and (23.191). Their partial derivatives are

$$\frac{\partial f}{\partial r} = 2 - \frac{1}{2}\,\frac{u}{(1+r)^2}, \qquad \frac{\partial f}{\partial u} = \frac{1}{2}\,\frac{1}{1+r}, \tag{23.204}$$

$$\frac{\partial g}{\partial r} = 3\,\frac{u^2}{(1+r)^3}, \qquad \frac{\partial g}{\partial u} = \varepsilon - 3\,\frac{u}{(1+r)^2}. \tag{23.205}$$

Evaluating at $(r_*, u_*) = (-\varepsilon/6,\, 2\varepsilon/3)$ and keeping the leading order in ε (so $1 + r_* = 1 + O(\varepsilon)$),

$$M = \begin{pmatrix} \dfrac{\partial f}{\partial r} & \dfrac{\partial f}{\partial u} \\[2mm] \dfrac{\partial g}{\partial r} & \dfrac{\partial g}{\partial u} \end{pmatrix}_* = \begin{pmatrix} 2 - \dfrac{\varepsilon}{3} & \dfrac{1}{2} \\[2mm] O(\varepsilon^2) & -\varepsilon \end{pmatrix} + O(\varepsilon^2). \tag{23.206}$$

The lower-left entry $\partial g/\partial r = 3u_*^2 = O(\varepsilon^2)$ is negligible at this order, so M is upper triangular and its eigenvalues are the diagonal entries,

$$y_t = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2), \qquad y_\lambda = -\varepsilon + O(\varepsilon^2), \tag{23.207}$$

which is Eq. (23.197). (Keeping the $O(\varepsilon^2)$ off-diagonal entry shifts the eigenvalues only at $O(\varepsilon^2)$: the characteristic equation gives a product $y_t y_\lambda = -2\varepsilon + O(\varepsilon^2)$ and a sum $2 - 4\varepsilon/3 + O(\varepsilon^2)$, reproducing Eq. (23.207).) The eigenvector of y_t is, to leading order, $(1, 0)$ — the relevant direction is essentially the mass direction r — while the eigenvector of y_λ points along the quartic. So at Wilson–Fisher the quartic has become irrelevant and the lone relevant direction is the mass, exactly as the critical-surface picture of Section 23.6 requires for a fixed point with one tuning parameter.

Figure 23.9 shows the flow Eqs. (23.190) and (23.191) in the (r, u) plane. The Gaussian fixed point at the origin is repulsive along both axes for $\varepsilon > 0$ (the mass is relevant with eigenvalue 2, the quartic relevant with eigenvalue ε); the Wilson–Fisher point sits at small positive u and small negative r, attractive along the quartic direction (its critical surface, the dashed curve) and repulsive along the one relevant mass direction. Every trajectory starting with a nonzero quartic is funnelled toward the Wilson–Fisher point as it heads into the infrared, until the single relevant direction eventually carries it away — the dynamical content of universality.

23.7.3 Critical exponents and the status of the result

The eigenvalues Eq. (23.197) are not bookkeeping; they are the leading critical exponents of the universality class controlled by the Wilson–Fisher fixed point, and they illustrate cleanly how a critical exponent is the slope of the flow at a fixed point.

Consider the relevant eigenvalue $y_t = 2 - \varepsilon/3$. Its eigenvector is the mass direction, and the dimensionful relevant parameter is m^2 measured relative to its critical value m_c^2 (the value that places the bare theory exactly on the critical surface). In a statistical-mechanics realization $m^2 - m_c^2 \propto T - T_c$, the deviation of temperature from criticality. Linearizing the flow, this deviation grows along the trajectory as $\delta(t) = \delta(0)\, e^{y_t t}$, and the correlation length — the physical length at

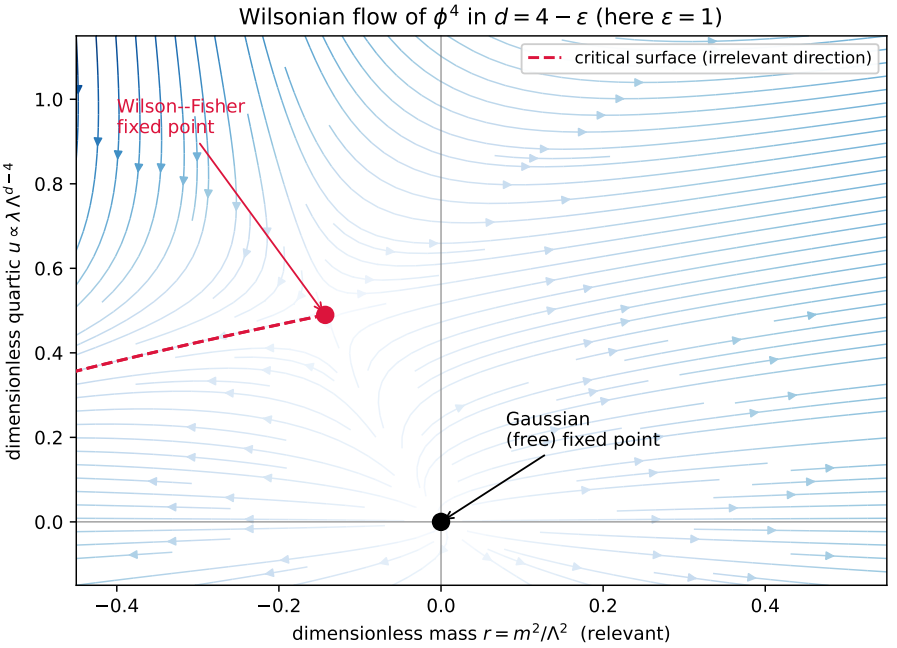


Figure 23.9: The one-loop Wilsonian flow of the dimensionless ϕ^4 couplings (r, u) in $d = 4 - \varepsilon$, Eqs. (23.190) and (23.191), drawn at $\varepsilon = 1$ ($d = 3$) so the two fixed points are well separated. Arrows point toward the infrared (increasing coarse-graining t). The Gaussian fixed point (black, origin) is fully infrared-repulsive once the quartic is relevant; the Wilson–Fisher fixed point (red) is infrared-attractive along its irrelevant quartic direction (the dashed critical surface) and repulsive along its single relevant mass direction. Leading-order locations: $u_* = \frac{2}{3}\varepsilon$, $r_* = -\frac{\varepsilon}{6}$.

which the flow leaves the neighbourhood of the fixed point, set by the cutoff scale $\Lambda(t) = \Lambda_0 e^{-t}$ at the point where $\delta(t)$ becomes $O(1)$ — diverges as the deviation is tuned to zero. Solving $|\delta(0)|\, e^{y_t t_*} \sim 1$ for the number of coarse-graining steps t_* gives $e^{t_*} \sim |\delta(0)|^{-1/y_t}$, so the correlation length $\xi \sim \Lambda_0^{-1} e^{t_*}$ scales as

$$\xi \sim |m^2 - m_c^2|^{-1/y_t} \equiv |m^2 - m_c^2|^{-\nu}, \qquad \frac{1}{\nu} = y_t = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2), \tag{23.208}$$

so that

$$\nu = \frac{1}{2 - \varepsilon/3} = \frac{1}{2}\Big(1 + \frac{\varepsilon}{6} + \cdots\Big) = \frac{1}{2} + \frac{\varepsilon}{12} + O(\varepsilon^2). \tag{23.209}$$

The correlation-length exponent is the inverse slope of the flow along its relevant direction at the fixed point. At the Gaussian fixed point the same slope would be the engineering value $y_t = 2$, giving the mean-field exponent $\nu = \frac{1}{2}$; the

interaction at Wilson–Fisher corrects it by $+\varepsilon/12$. The irrelevant eigenvalue $y_\lambda = -\varepsilon$ supplies the leading *correction-to-scaling* exponent $\omega = -y_\lambda = \varepsilon$, governing how quickly a generic microscopic theory approaches the fixed-point predictions as it is coarse-grained.

The remaining exponent at this order is the field anomalous dimension. By Eq. (23.194) the one-loop self-energy is the momentum-independent tadpole, so γ_ϕ starts at $O(u^2)$, and since $u_* = O(\varepsilon)$,

$$\gamma_\phi(\lambda_*) = O(\varepsilon^2), \qquad \eta \equiv 2\gamma_\phi(\lambda_*) = \frac{\varepsilon^2}{54} + O(\varepsilon^3), \tag{23.210}$$

the value of η for the one-component scalar quoted from the two-loop computation Chapter 24 carries out (Eq. (24.260)); to leading order in the deficit the Wilson–Fisher field still scales with its Gaussian dimension, and the anomalous dimension first appears at second order. This vanishing of η at $O(\varepsilon)$ is exactly the consistency of the present one-loop analysis: had the tadpole carried external-momentum dependence, γ_ϕ would have entered at $O(\varepsilon)$ and corrupted the eigenvalues Eq. (23.197).

Two features of these results deserve emphasis because they are what make the fixed point physical rather than an artifact of the scheme. First, the fixed-point coupling $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ at leading order, and the exponents $1/\nu$, ω, η to all orders, are *scheme-independent*. The numerical value of a coupling at a fixed point can shift under a redefinition of the coupling beyond leading order, but the eigenvalues of the linearized flow are invariants of the fixed point — they are physical critical exponents, measurable in the laboratory — and any correct scheme must reproduce them. Concretely, Chapter 24 re-derives this same fixed point from the Callan–Symanzik beta function in $\overline{\text{MS}}$: setting $\beta_\lambda = -\varepsilon\lambda + 3\lambda^2/16\pi^2 = 0$ yields the identical $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ (Theorem 24.31), with slope $\beta'_\lambda(\lambda_*) = +\varepsilon$ reproducing the irrelevant eigenvalue $\omega = \varepsilon$. The Wilsonian momentum-shell flow and the continuum subtraction-scale flow are two descriptions of the same physics and agree on every invariant, just as the chapter promised at the outset. Second, the fixed-point theory is scale invariant by construction (the dimensionless couplings have stopped running), and the combination of scale invariance with the locality and unitarity of the underlying field theory promotes it, under the standard assumptions, to full conformal invariance: the Wilson–Fisher fixed point *is* a conformal field theory.

Physical Insight

The Wilson–Fisher fixed point is the precise sense in which renormalization "runs to" something. For most of Chapters 21 and 22 renormalization looked like an algorithm for cancelling infinities; here it produces an object — a scale-invariant theory, a point in coupling space that an entire basin of microscopic models flows to. Its coordinates are universal numbers

(critical exponents), its predictions are checkable against real second-order phase transitions, and at $\varepsilon = 1$ (i.e. $d = 3$) it is, after resummation, the three-dimensional Ising universality class. That last step — taking ε from infinitesimal to unity — is where the bare expansion breaks down and a nonperturbative method must take over; the conformal bootstrap of Chapter 52 is that method.

This completes the Wilsonian layer of ϕ^4 theory. We have re-read renormalizability as the statement that a predictive continuum limit needs only finitely many relevant and marginal couplings (Theorem 23.34), recognized the non-renormalizable interactions of Chapter 21 as the harmless irrelevant operators that organize the effective-field-theory expansion of Chapter 38, and located the Wilson–Fisher fixed point at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ with $1/\nu = 2 - \varepsilon/3$ and $\eta = O(\varepsilon^2)$ (Theorem 23.36). The infinitesimal shadow of this coarse-graining flow is the system of running-coupling equations: Chapter 24 now recasts the entire picture as the Callan–Symanzik equation, re-deriving this very fixed point from the beta function and confirming its scheme-independence, while the conformal field theory living at the fixed point — and its sharpest treatment, the three-dimensional Ising bootstrap — is handed to Chapter 52.

Further reading

The Wilsonian viewpoint adopted here originates with Kadanoff's block-spin picture [288] and Wilson's reformulation of the renormalization group around critical phenomena [513]; the long review of Wilson and Kogut [516] remains the canonical pedagogical source for the momentum-shell construction and the ε-expansion, and the original Wilson–Fisher result is [515]. The exact renormalization-group equations are due to Wegner and Houghton in the sharp-cutoff form [492] and to Polchinski in the smooth-cutoff form [403]; Wegner's analysis of corrections to scaling [491] underlies the irrelevant-eigenvalue exponent ω. For the statistical-mechanics side — universality, critical surfaces, and the exponent relations — the textbooks of Cardy [108] and Goldenfeld [229] are excellent and complementary, and Zinn-Justin [534] gives the field-theoretic computations at length. Among QFT texts, the field-theoretic RG and the Wilsonian picture are developed in Peskin and Schroeder [396] and, with particular care for the modern effective-field-theory reading of renormalizability, in Schwartz [432].

Chapter 24

The Callan–Symanzik Equation and Running Couplings

The previous chapter built the renormalization group from the bottom up. Integrating out a thin shell of high-momentum modes, we watched the couplings of an effective action flow, sorted operators into relevant, marginal, and irrelevant by how they respond to that flow, and learned to read renormalizability itself as a statement about which fixed point a theory sits near (Chapter 23). That picture is conceptually complete but computationally implicit: it tells us a flow exists without handing us the differential equation that governs it, and it locates fixed points without giving us the beta function whose zeros they are. This chapter supplies the missing machine. We take the same physics—that a quantum field theory carries no intrinsic notion of scale, only the arbitrary subtraction point we impose to define it—and turn it into an exact partial differential equation satisfied by every correlation function, then solve that equation to extract the couplings' dependence on energy.

The engine is the perturbative renormalization of Chapter 22. There the divergences of ϕ^4 and QED were absorbed into multiplicative renormalization constants Z_i, fixed loop by loop in dimensional regularization with the $\overline{\text{MS}}$ scale μ; the vacuum-polarization constant Z_3, the electron's Z_2 and Z_m, and the Ward–Takahashi identity $Z_1 = Z_2$ proved in Chapter 19 are all imported wholesale here as the raw input from which the flow is computed. The logic of the chapter is that the bare theory does not know what μ is, so the μ-dependence of the Z_i must conspire with that of the renormalized fields and couplings to leave every bare quantity invariant. Demanding this invariance produces the renormalization-group equation, and the three functions that drive it—the beta function and the field and mass anomalous dimensions—emerge as the residues of the simple poles in ε, through 't Hooft's pole equation. From a single one-loop divergence we therefore read off an all-orders statement about scale dependence.

With the apparatus in place we put both benchmark theories through it.

ϕ^4 theory delivers $\beta_\lambda = 3\lambda^2/16\pi^2$ from the factor-of-three combinatorics of the one-loop four-point bubbles, a mass anomalous dimension from the momentum-independent tadpole, and a field anomalous dimension that is silent at one loop and speaks first through the two-loop sunset. QED delivers $\beta_e = e^3/12\pi^2$, where gauge invariance collapses three would-be renormalization constants to one so that the entire running of the charge follows from the residue of Z_3 alone; we confirm the sign and coefficient a second time from the spectral function of the photon self-energy, the optical-theorem face (Chapter 19) of the same number. We then ask which of these numbers is convention and which is physics, and prove that the first two beta coefficients and the slope of β and the value of γ at any fixed point are scheme-independent, while everything beyond is reparametrization-dependent bookkeeping.

The chapter then earns its title from the other direction. Classically, massless ϕ^4 and QED are scale invariant, and the trace of the stress tensor vanishes; quantization breaks this, and the trace anomaly $\Theta^\mu{}_\mu = \frac{\beta}{2g}F^2 + \dots$ measures the breaking exactly in terms of the very beta function we computed. Tracking soft mass insertions turns broken scale invariance into the inhomogeneous Callan–Symanzik equation, which we show coincides with the μ-flow renormalization-group equation in the deep-Euclidean regime. Solving either by the method of characteristics trades the differential constraint for a running coupling $\bar{g}(\mu)$, a running mass, and an anomalous-dimension exponential; the running fine-structure constant reproduces the $\alpha(q^2)$ of Chapter 20 as the leading term of the resummed leading logarithms.

Finally we read the asymptotic fate of a theory directly off the sign and zeros of β. A positive slope screens and drives the coupling to a Landau pole, raising the triviality question for ϕ^4 and QED; a negative coefficient—previewed here, computed for Yang–Mills in Chapter 27—gives asymptotic freedom and a dynamically generated scale Λ. At a nontrivial zero the anomalous dimension freezes into a critical exponent η, and the Wilson–Fisher fixed point of $d = 4 - \varepsilon$ becomes the bridge to the conformal field theory of Chapter 52. Everything in this chapter assumes the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, dimensional regularization in $d = 4 - 2\varepsilon$ with the $\overline{\mathrm{MS}}$ scale μ, and the QED covariant derivative $D_\mu = \partial_\mu - iqA_\mu$ with $q = -e < 0$ for the electron, as fixed in Chapters 1 and 22.

24.1 Scale Independence as a Symmetry: The Renormalization-Group Equation

The subtraction scale μ entered our formulas in Chapter 22 as an accountant's convenience. To trade the bare coupling for a finite renormalized one in the $\overline{\mathrm{MS}}$ scheme (Theorem 22.7) we were forced to introduce a mass scale, because in $d = 4 - 2\varepsilon$ a would-be dimensionless coupling is not dimensionless and the

renormalized coupling can only be made so by extracting an explicit power of μ. Nothing physical was thereby chosen: we could have subtracted at 2μ, or at $\mu/3$, and the same electron, the same scattering cross section, the same pole of the propagator would have come out. A physical prediction cannot depend on where an accountant decided to draw the line. That single sentence, taken seriously and pressed until it yields an equation, is the content of this section. We will see that "μ is unphysical" is not a slogan but a conservation law: it says that a certain total derivative annihilates every correlation function, and the coefficients of that differential operator—the beta function and the anomalous dimensions—are precisely the calculable quantities that encode how couplings and fields respond to a change of scale.

24.1.1 The bare theory does not know what the subtraction scale is

Renormalization, as developed in Chapter 22, is a change of variables. One starts from a bare Lagrangian written in terms of bare fields and bare parameters,

$$\mathcal{L}_0 = \tfrac{1}{2}(\partial_\mu \phi_0)(\partial^\mu \phi_0) - \tfrac{1}{2} m_0^2 \, \phi_0^2 - \frac{\lambda_0}{4!}\, \phi_0^4 \tag{24.1}$$

for ϕ^4 theory, and the analogous bare $\mathcal{L}_0$ for QED. The bare quantities ϕ_0, λ_0, m_0 (and $\psi_0, e_0, m_0, A_0^\mu$ in QED) are the only data the functional integral actually contains. One then *defines* renormalized fields and parameters through the multiplicative constants of Theorem 22.9,

$$\phi_0 = Z_\phi^{1/2}\, \phi, \qquad \lambda_0 = Z_\lambda \, Z_\phi^{-2}\, \lambda \, \mu^{2\varepsilon}, \qquad m_0^2 = Z_m \, Z_\phi^{-1}\, m^2, \tag{24.2}$$

the precise content of Eq. (22.57), with the Z's chosen to absorb the $1/\varepsilon$ poles so that correlation functions of the renormalized field are finite as $\varepsilon \to 0$. The scale μ appears *only* on the right-hand side of these definitions; it is part of the dictionary, not part of the theory.

It pays to be completely explicit about what is, and what is not, μ-dependent. The bare parameters λ_0 and m_0 are fixed numbers (in $d = 4 - 2\varepsilon$, fixed *dimensionful* numbers): they are whatever they must be for the theory to reproduce the measured low-energy physics, and they carry no memory of μ. The renormalized coupling λ, by contrast, is *defined* by inverting Eq. (24.2), and that definition involves μ; hence $\lambda = \lambda(\mu)$ is a μ-dependent bookkeeping variable. The field-renormalization constant Z_ϕ likewise depends on μ through its dependence on $\lambda(\mu)$. The logic of the chapter is to hold the genuinely physical data—the bare parameters—fixed, and ask how the bookkeeping data must conspire to keep observables invariant.

Consider, then, the bare n-point one-particle-irreducible (1PI) functions, the amputated Green's functions built from $\mathcal{L}_0$. We write

$$\Gamma_0^{(n)}(p_1, \dots, p_n;\, \lambda_0, m_0;\, \varepsilon) \tag{24.3}$$

for the bare 1PI n-point function in momentum space, with the overall momentum-conserving delta function stripped. Its arguments are the external momenta and the bare parameters; ε appears because we compute in $d = 4 - 2\varepsilon$. Crucially, μ is *not* among its arguments. The functional integral that defines Eq. (24.3) contains $\mathcal{L}_0$ and nothing else; no choice of subtraction point has been made, and none can have leaked in. Therefore, at fixed bare parameters,

$$\mu \frac{d}{d\mu} \Gamma_0^{(n)} \;=\; 0, \qquad \text{at fixed } \lambda_0,\, m_0. \tag{24.4}$$

This is the entire physical input of the renormalization group. Everything that follows is the act of rewriting Eq. (24.4) in renormalized variables and reading off its consequences.

Physical Insight

The bare correlator Eq. (24.3) is divergent as $\varepsilon \to 0$, and the bare parameters are themselves ε-dependent and (in $d = 4 - 2\varepsilon$) dimensionful—so it is not an object one measures directly. But it is the object that is manifestly μ-independent, because μ never entered it. The renormalized correlator is the opposite on both counts: it is finite and physically meaningful, but its definition drags μ in. The renormalization-group equation is the bridge between these two facts. It converts the trivial statement "the divergent thing nobody chose a scale for is scale-independent" into the powerful statement "the finite thing we measure obeys an exact flow equation in μ."

24.1.2 The three functions that drive scale dependence

Before differentiating anything we name the rates at which the bookkeeping variables move. Throughout, the symbol $\big|_0$ means that the bare parameters λ_0, m_0 (and in QED e_0, m_0) are held fixed while μ is varied; the derivatives are total derivatives along the resulting trajectory $\mu \mapsto (\lambda(\mu), m(\mu))$.

Definition 24.1 (Beta function)**.** The *beta function* of a coupling g is its logarithmic rate of change with the subtraction scale at fixed bare parameters,

$$\beta(g) \;\equiv\; \mu \frac{dg}{d\mu}\bigg|_0 . \tag{24.5}$$

For ϕ^4 theory the coupling is $g = \lambda$ and we write β_λ; for QED it is $g = e$ and we write β_e.

Definition 24.2 (Field anomalous dimension)**.** The *field (or wavefunction) anomalous dimension* is

$$\gamma(g) \;\equiv\; \tfrac{1}{2}\, \mu \frac{d \ln Z_\phi}{d\mu}\bigg|_0 . \tag{24.6}$$

The factor $\frac{1}{2}$ is conventional and is fixed so that γ multiplies the field exactly once per external leg in the equation derived below.

Definition 24.3 (Mass anomalous dimension)**.** The *mass anomalous dimension* is

$$\gamma_m(g) \;\equiv\; -\mu \left.\frac{d\ln m}{d\mu}\right|_0 . \tag{24.7}$$

With this sign, a *positive* γ_m means the renormalized mass *decreases* as μ increases—the running mass shrinks toward the ultraviolet, as it does for the quark masses of QCD.

Each of these is a function of the dimensionless couplings alone: this is a property, not an assumption, and we will prove it for β in Section 24.1.4 and see it fall out of the $\overline{\text{MS}}$ structure for γ, γ_m as well. The reason is that in a mass-independent scheme such as $\overline{\text{MS}}$ the renormalization constants depend on μ and m only through the dimensionless ratios and the coupling; the explicit m- and μ-dependence cancels in the logarithmic derivatives, leaving pure functions of g.

The signs in these definitions are a perennial source of cross-text confusion, and the confusion is genuine—it is not that one convention is right and the others wrong, but that several internally consistent conventions are in circulation. We flag the live ones now so that no later formula is misread.

Notation Watch

Three independent sign conventions collide in the renormalization-group literature, and a table of "β and γ coefficients" lifted from one book into another's master equation is the most common way to introduce a wrong sign.

1. **Field term, 1PI versus full Green's functions.** Because the 1PI function and the full (n-point connected) Green's function scale with *opposite* powers of Z_ϕ (one is amputated, the other is not), the field-anomalous-dimension term appears as $-n\gamma$ in the equation for $\Gamma_R^{(n)}$ and as $+n\gamma$ in the equation for the full $G_R^{(n)}$. We derive both below; mixing them flips a sign.

2. **Mass term.** We adopt $\gamma_m \equiv -\mu\, d\ln m/d\mu$, which makes the mass term enter the equation for $\Gamma_R^{(n)}$ as $-\gamma_m\, m\,\partial_m$. A sizeable fraction of texts define γ_m with the opposite overall sign, $\tilde\gamma_m \equiv +\mu\, d\ln m/d\mu = -\gamma_m$, and then write the same equation as $+\tilde\gamma_m\, m\,\partial_m$; the physics is identical, but a quoted coefficient is only meaningful once one knows which of $\gamma_m, \tilde\gamma_m$ it is. Throughout this book we use Eq. (24.7).

3. **Beta function in $d \neq 4$.** In $d = 4 - 2\varepsilon$ the beta function carries

an explicit ε-piece (derived in Section 24.1.4). Some authors reserve the name "β" for the four-dimensional limit and write the ε-piece separately; we keep the full d-dimensional $\beta(g,\varepsilon)$ and call its $\varepsilon \to 0$ limit $\bar{\beta}(g)$, stating which is meant wherever it matters.

24.1.3 The renormalization-group equation for renormalized 1PI functions

We now translate Eq. (24.4) into the renormalized variables. The link is the multiplicative relation between bare and renormalized 1PI functions. Each external leg of an n-point function carries one factor of the field. Since $\phi_0 = Z_\phi^{1/2}\phi$, the *full* (non-amputated) n-point function, built from n bare fields, picks up $Z_\phi^{n/2}$: $G_0^{(n)} = Z_\phi^{n/2} G_R^{(n)}$. The 1PI function is the full function with its n external full propagators amputated, and each amputation divides by one full propagator, i.e. multiplies by two inverse-field factors relative to a leg; tracking the powers (or, equivalently, recalling that $\Gamma^{(n)}$ is the n-th functional derivative of the effective action, which is built from ϕ not ϕ_0) gives the inverse scaling,

$$\Gamma_0^{(n)}(p_i;\lambda_0,m_0;\varepsilon) \;=\; Z_\phi^{-n/2}(\lambda,\varepsilon)\,\Gamma_R^{(n)}(p_i;\lambda,m,\mu;\varepsilon). \tag{24.8}$$

This is the relation announced in the chapter blueprint and proved in Chapter 22; here we take it as the starting point and differentiate.

Theorem 24.4 (Renormalization-group equation)**.** *The renormalized* 1*PI* n*-point function* $\Gamma_R^{(n)}(p_i;\lambda,m,\mu)$ *satisfies*

$$\boxed{\left[\mu\frac{\partial}{\partial\mu} + \beta(\lambda)\frac{\partial}{\partial\lambda} - \gamma_m(\lambda)\,m\frac{\partial}{\partial m} - n\,\gamma(\lambda)\right]\Gamma_R^{(n)}(p_i;\lambda,m,\mu) \;=\; 0} \tag{24.9}$$

where β,γ,γ_m *are defined in Theorems 24.1 to 24.3. The corresponding equation for the full (connected, non-amputated)* n*-point function* $G_R^{(n)}$ *has the field term with the opposite sign,*

$$\left[\mu\,\partial_\mu + \beta\,\partial_\lambda - \gamma_m\,m\,\partial_m + n\,\gamma\right]G_R^{(n)} \;=\; 0. \tag{24.10}$$

Proof. Apply $\mu\,d/d\mu$ at fixed bare parameters to both sides of Eq. (24.8). The left side vanishes by Eq. (24.4). On the right side we use the product rule:

$$0 = \left(\mu\frac{dZ_\phi^{-n/2}}{d\mu}\Big|_0\right)\Gamma_R^{(n)} + Z_\phi^{-n/2}\left(\mu\frac{d\Gamma_R^{(n)}}{d\mu}\Big|_0\right). \tag{24.11}$$

The first factor is evaluated using Eq. (24.6):

$$\mu \frac{dZ_\phi^{-n/2}}{d\mu}\bigg|_0 = -\frac{n}{2} Z_\phi^{-n/2-1} \mu \frac{dZ_\phi}{d\mu}\bigg|_0 = -\frac{n}{2} Z_\phi^{-n/2} \mu \frac{d\ln Z_\phi}{d\mu}\bigg|_0 = -n\gamma Z_\phi^{-n/2}. \tag{24.12}$$

Substituting Eq. (24.12) into Eq. (24.11) and dividing through by the common factor $Z_\phi^{-n/2}$ (nonzero, being a power series beginning at 1) yields

$$\mu \frac{d\Gamma_R^{(n)}}{d\mu}\bigg|_0 - n\gamma \Gamma_R^{(n)} = 0. \tag{24.13}$$

It remains to expand the total derivative. The renormalized function depends on μ in three ways: explicitly (through the μ's that the scheme inserts), and implicitly through the running coupling $\lambda(\mu)$ and running mass $m(\mu)$. The chain rule gives

$$\mu \frac{d\Gamma_R^{(n)}}{d\mu}\bigg|_0 = \mu \frac{\partial \Gamma_R^{(n)}}{\partial \mu} + \left(\mu \frac{d\lambda}{d\mu}\bigg|_0\right) \frac{\partial \Gamma_R^{(n)}}{\partial \lambda} + \left(\mu \frac{dm}{d\mu}\bigg|_0\right) \frac{\partial \Gamma_R^{(n)}}{\partial m}. \tag{24.14}$$

The first coefficient is $\beta(\lambda)$ by Eq. (24.5). For the third, Eq. (24.7) reads $\mu\, d\ln m/d\mu = -\gamma_m$, hence $\mu\, dm/d\mu = -\gamma_m m$. Inserting these into Eq. (24.14) and the result into Eq. (24.13) gives exactly Eq. (24.9).

For the full Green's function, repeat the argument starting from $G_0^{(n)} = Z_\phi^{+n/2} G_R^{(n)}$. The only change is the sign of the exponent of Z_ϕ, which flips the sign of the analogue of Eq. (24.12) and hence turns $-n\gamma$ into $+n\gamma$, producing Eq. (24.10). □

The structure of Eq. (24.9) repays a moment's reflection. Read it as a statement about a vector field on the space of theories. The operator $\mu\partial_\mu + \beta\partial_\lambda - \gamma_m m\partial_m$ generates an infinitesimal motion: increase μ a little, and to compensate, move the coupling by β and the mass by $-\gamma_m m$. Along this trajectory the correlation function is *not* invariant; it changes by the multiplicative factor encoded in $n\gamma$, one factor of γ per external field. If γ vanished, the correlation function would be a pure "constant of motion" of the flow—the field would carry exactly its classical (engineering) dimension. The whole point of the name *anomalous* dimension is that $\gamma \neq 0$: quantum corrections give the field an extra scaling weight beyond its classical one, and Eq. (24.9) is the exact statement of how much. The next section will make this reading literal by re-deriving the same equation from the breakdown of classical scale invariance.

Physical Insight

There is no new dynamics in Eq. (24.9): it is a self-consistency condition, a Ward identity for the trivial "symmetry" of relabelling the subtraction point. Its power is entirely in what it correlates. The same three functions β, γ, γ_m appear for *every* n and at *every* order in perturbation theory. Compute them once, at one loop, from the simplest available Green's function, and the equation then dictates the μ-dependence—hence, as we will see when we solve it, the large-momentum behaviour—of all correlation functions at once. A one-loop calculation of a two-point and a four-point function thereby controls the leading logarithms to all orders.

It is worth pausing to reconcile this picture with the Wilsonian one of Chapter 23, because the two look superficially unrelated and are in fact the same physics in two bookkeeping schemes. There, one introduced a genuine ultraviolet cutoff Λ, integrated out the field modes between Λ and a lower scale, and watched the couplings of the resulting effective action flow as Λ was lowered; the cutoff was a physical scale and the trajectory $\Lambda \mapsto (\lambda(\Lambda), m(\Lambda), \dots)$ traced an honest path through the (infinite-dimensional) space of theories. Here there is no cutoff at all—dimensional regularization carries none—and the running scale μ is a fictitious subtraction point with no physical meaning, introduced only to make the renormalized coupling dimensionless and finite. What the two have in common is the structure that matters: a one-parameter family of equivalent descriptions of one and the same physics, labelled by a scale, with β the tangent vector to the trajectory and the anomalous dimensions recording how the field normalization slides along it. The operator $\mu\partial_\mu + \beta\partial_\lambda - \gamma_m m\partial_m$ of Eq. (24.9) is the continuum, scheme-based incarnation of Wilson's flow vector field, and the existence of a zero of β is, in both languages, the existence of a scale-invariant fixed point. The two flows are not identical equation-for-equation—Λ decreases into the infrared while one conventionally runs μ up into the ultraviolet, and the bare coupling that is held fixed here is the analogue of the microscopic action held fixed there—but their physically meaningful content coincides. This is the precise sense in which the Wilsonian and continuum renormalization groups agree: the data that survive the change of scheme are exactly the *universal* ones identified in Section 23.6—the location and nature of fixed points, the critical exponents, and (for a marginal coupling) the first two coefficients of β, which we will see in Section 24.1.4 and Chapter 52 are scheme independent. The arbitrariness of μ is thus not a defect of the continuum method but the same freedom of "where to stand on the trajectory" that the cutoff Λ parametrized in Chapter 23; renormalizability, in this light, is just the statement that the trajectory can be followed all the way up with only finitely many couplings.

24.1.4 The dimensional-regularization route and 't Hooft's pole equation

Theorem 24.4 presupposes that the functions β, γ, γ_m exist and are finite as $\varepsilon \to 0$. In a general scheme one would have to argue this; in $\overline{\text{MS}}$ it is automatic, and moreover the functions are computable from a single number in each renormalization constant—the residue of its simple pole in ε. This is the observation of 't Hooft, and it is what makes $\overline{\text{MS}}$ the scheme of choice for extracting renormalization-group coefficients. We prove it now for the beta function and then record the parallel results for γ and γ_m.

The starting point is the $\overline{\text{MS}}$ form of the bare coupling. Power counting in $d = 4 - 2\varepsilon$ fixes the dimension of each bare coupling. The Lagrangian density has mass dimension $d = 4 - 2\varepsilon$; in d dimensions a scalar has dimension $(d-2)/2 = 1 - \varepsilon$, so the ϕ^4 vertex forces

$$[\lambda_0] = d - 4(1 - \varepsilon) = 2\varepsilon, \tag{24.15}$$

while a Dirac field has dimension $(d-1)/2 = \tfrac{3}{2} - \varepsilon$ and a gauge field $1 - \varepsilon$, so the QED vertex forces $[e_0] = \varepsilon$. To keep the renormalized couplings dimensionless we extract the corresponding power of μ and write, uniformly,

$$g_0 \;=\; \mu^{\rho\varepsilon}\, Z_g(g,\varepsilon)\, g, \qquad \rho_\lambda = 2, \quad \rho_e = 1, \tag{24.16}$$

where g is the dimensionless renormalized coupling and Z_g a renormalization constant which, in $\overline{\text{MS}}$, is a pure Laurent series in $1/\varepsilon$ with g-dependent coefficients and no positive powers of ε and no μ-dependence:

$$Z_g(g,\varepsilon) = 1 + \sum_{k \geq 1} \frac{z_k(g)}{\varepsilon^k}. \tag{24.17}$$

It is convenient to package the whole right-hand side of Eq. (24.16) into a single series. Define

$$G(g) \;\equiv\; Z_g(g,\varepsilon)\, g \;=\; g + \sum_{k \geq 1} \frac{a_k(g)}{\varepsilon^k}, \qquad a_k(g) \equiv g\, z_k(g), \tag{24.18}$$

so that $g_0 = \mu^{\rho\varepsilon}\, G(g)$. The a_k are the residues we will show control everything; a_1, the residue of the simple pole, is the protagonist.

The defining property of g_0 is that it is a fixed (bare) quantity: $\mu\, dg_0/d\mu|_0 = 0$. Differentiate Eq. (24.16) in this package form. Using $\mu\, d\mu^{\rho\varepsilon}/d\mu = \rho\varepsilon\, \mu^{\rho\varepsilon}$ and the chain rule $\mu\, dG/d\mu = G'(g)\, \beta$, where $\beta \equiv \mu\, dg/d\mu|_0$ and a prime denotes d/dg,

$$0 = \mu \left.\frac{dg_0}{d\mu}\right|_0 = \mu^{\rho\varepsilon}\Big[\rho\varepsilon\, G(g) + G'(g)\, \beta\Big]. \tag{24.19}$$

Since $\mu^{\rho\varepsilon} \neq 0$, the bracket vanishes, and we may solve for the d-dimensional beta function,

$$\beta(g,\varepsilon) = -\,\rho\varepsilon\,\frac{G(g)}{G'(g)}. \tag{24.20}$$

This compact formula is the entire content of the renormalization-group flow of the coupling in $\overline{\text{MS}}$; the rest is reading off its analytic structure.

Theorem 24.5 ('t Hooft's pole equation)**.** *Let $g_0 = \mu^{\rho\varepsilon} G(g)$ with G as in Eq. (24.18), and let $\beta(g,\varepsilon) \equiv \mu\, dg/d\mu|_0$ be finite as $\varepsilon \to 0$. Then:*

(i) β is at most linear in ε:

$$\beta(g,\varepsilon) = -\,\rho\varepsilon\, g + \bar\beta(g), \tag{24.21}$$

with the ε-independent four-dimensional beta function $\bar\beta$ determined entirely by the simple-pole residue,

$$\boxed{\bar\beta(g) = \rho\Big(g\,a_1'(g) - a_1(g)\Big) = \rho\, g^2\,\frac{d}{dg}\left(\frac{a_1(g)}{g}\right) = \rho\, g^2\,\frac{dz_1(g)}{dg}.} \tag{24.22}$$

(ii) The higher residues are not independent: requiring β to be finite forces the consistency relations

$$g\,a_{k+1}'(g) - a_{k+1}(g) = \frac{1}{\rho}\,a_k'(g)\,\bar\beta(g), \qquad k \geq 1, \tag{24.23}$$

which determine every a_{k+1} from a_k and $\bar\beta$. In particular all the higher poles of Z_g are fixed, order by order in perturbation theory, by the simple pole.

Proof. We expand Eq. (24.20) as a Laurent series in $1/\varepsilon$ and impose finiteness. Write

$$G(g) = g + \sum_{k\geq 1} \frac{a_k}{\varepsilon^k}, \qquad G'(g) = 1 + \sum_{k\geq 1} \frac{a_k'}{\varepsilon^k}. \tag{24.24}$$

Because $G'(g) = 1 + O(1/\varepsilon)$ has constant term 1, it is invertible as a formal Laurent series, and the ratio is again a Laurent series of the form

$$\frac{G(g)}{G'(g)} = g + \sum_{k\geq 1} \frac{c_k(g)}{\varepsilon^k}, \tag{24.25}$$

for coefficients $c_k(g)$ to be determined. Multiplying by $-\rho\varepsilon$,

$$\beta(g,\varepsilon) = -\rho\varepsilon\Big(g + \sum_{k\geq 1} c_k\,\varepsilon^{-k}\Big) = -\rho\varepsilon\, g \;-\; \rho\, c_1(g) \;-\; \rho \sum_{k\geq 2} c_k(g)\,\varepsilon^{1-k}. \tag{24.26}$$

The last sum collects the negative powers $\varepsilon^{-1}, \varepsilon^{-2}, \dots$ For β to be finite as $\varepsilon \to 0$—which it must be, since by construction the renormalized coupling $g(\mu)$ is finite in $\overline{\text{MS}}$ and so therefore is its logarithmic μ-derivative—every one of these poles must vanish:

$$c_k(g) = 0 \quad \text{for all } k \geq 2. \tag{24.27}$$

These are exactly the consistency conditions; we translate them into statements about the a_k in a moment. With Eq. (24.27), Eq. (24.26) collapses to

$$\beta(g, \varepsilon) = -\rho\varepsilon\, g - \rho\, c_1(g), \tag{24.28}$$

which is Eq. (24.21) with $\bar{\beta}(g) = -\rho\, c_1(g)$.

It remains to compute the c_k from the a_k. The cleanest route is not to invert G' explicitly but to multiply Eq. (24.25) through by $G'(g)$ and match powers:

$$G(g) = \Big(g + \sum_{k\geq 1} c_k\, \varepsilon^{-k}\Big)\Big(1 + \sum_{j\geq 1} a'_j\, \varepsilon^{-j}\Big). \tag{24.29}$$

Equate coefficients of ε^{-k} on both sides. The left side has coefficient a_k (and g at order ε^0, which matches the g on the right). At order ε^{-1}:

$$a_1 = c_1 + g\, a'_1 \quad \Longrightarrow \quad c_1 = a_1 - g\, a'_1. \tag{24.30}$$

Hence

$$\bar{\beta}(g) = -\rho\, c_1 = \rho(g\, a'_1 - a_1), \tag{24.31}$$

which is the first equality in Eq. (24.22); the second follows from $g\, a'_1 - a_1 = g^2\, \frac{d}{dg}(a_1/g)$, and the third from $a_1 = g\, z_1$, whence $a_1/g = z_1$ and $g^2(a_1/g)' = g^2 z'_1$. This proves (i).

For (ii), read off the coefficient of $\varepsilon^{-(k+1)}$ in Eq. (24.29) for $k \geq 1$. The right side gives $c_{k+1} + \sum_{j=1}^{k} c_{k+1-j}\, a'_j + g\, a'_{k+1}$, where the $j = k+1$ term is the $g\, a'_{k+1}$ piece (the factor g being the order-ε^0 part of the first bracket) and the $c_0 \equiv g$ convention is understood. Setting this equal to the left-side coefficient a_{k+1} and imposing $c_{k+1} = 0$ (and $c_m = 0$ for all $m \geq 2$) leaves only the c_1 and $c_0 = g$ terms:

$$a_{k+1} = g\, a'_{k+1} + c_1\, a'_k \quad \Longrightarrow \quad g\, a'_{k+1} - a_{k+1} = -\,c_1\, a'_k = \frac{1}{\rho}\, a'_k\, \bar{\beta}, \tag{24.32}$$

using $\bar{\beta} = -\rho c_1$. This is Eq. (24.23). Given a_1 (hence $\bar{\beta}$), the relation determines a_2 up to a homogeneous solution proportional to g, which perturbation theory fixes to vanish at each order because a_k starts at order g^{k+1}; iterating fixes all higher a_k. This proves (ii). □

Two features of Theorem 24.5 deserve emphasis. First, the would-be higher poles of the beta function—the $\varepsilon^{-1}, \varepsilon^{-2}, \dots$ terms in Eq. (24.26)—do not merely

happen to be small; they are forced to vanish identically by the finiteness of the renormalized coupling, and their vanishing is equivalent to a tower of relations Eq. (24.23) predicting the double-, triple-, … poles of Z_g from its simple pole. This is a stringent internal check on any loop computation: if a two-loop calculation produced a $1/\varepsilon^2$ residue inconsistent with Eq. (24.23), the calculation would be wrong. Second, the decomposition Eq. (24.21) cleanly separates a *classical* contribution from a *quantum* one.

Physical Insight

The term $-\rho\varepsilon\, g$ in Eq. (24.21) is purely classical: it is nothing but the statement Eq. (24.15) that away from four dimensions the coupling carries mass dimension $\rho\varepsilon$, so that holding g_0 fixed while raising μ *must* change the dimensionless g even with all loops switched off. It has nothing to do with quantum fluctuations; it is the engineering-dimension term. The piece $\bar\beta(g)$ is the genuinely quantum effect, the residue of the loop divergences. In $d = 4$ ($\varepsilon = 0$) only the quantum piece survives and a marginal coupling runs only logarithmically. But for $\varepsilon > 0$ the two pieces compete, and a zero of β—a scale-invariant fixed point—can occur at *nonzero* coupling where the classical drift cancels the quantum running:

$$\beta(g_*, \varepsilon) = 0 \quad \Longleftrightarrow \quad \bar\beta(g_*) = \rho\varepsilon\, g_*. \tag{24.33}$$

For ϕ^4 theory, where $\bar\beta = 3\lambda^2/16\pi^2 + O(\lambda^3)$ and $\rho = 2$, this gives a nontrivial root $\lambda_* = \frac{32\pi^2}{3}\varepsilon + O(\varepsilon^2)$: the Wilson–Fisher fixed point, born at the origin when ε is turned on and moving out linearly in ε. Figure 24.1 shows the mechanism; the fixed point and its physics are the subject of Chapter 52.

The same Laurent-series reasoning applied to the multiplicative constants Z_ϕ and Z_m yields the field and mass anomalous dimensions from *their* simple poles. Write $Z_\phi = 1 + \sum_{k\ge1} s_k(g)/\varepsilon^k$ and $Z_m = 1 + \sum_{k\ge1} w_k(g)/\varepsilon^k$. Then, computing $\gamma = \frac{1}{2}\,\mu\, d\ln Z_\phi/d\mu = \frac{1}{2}\,\bar\beta_{\rm cl}\,\partial_g \ln Z_\phi$ with $\beta = -\rho\varepsilon g + \bar\beta$, the only term that survives at order ε^0 comes from the classical $-\rho\varepsilon g$ acting on the simple pole s_1/ε, and finiteness again removes the would-be poles; the upshot is

$$\gamma(g) = -\tfrac{1}{2}\,\rho\, g\, \frac{ds_1(g)}{dg}, \tag{24.34}$$

$$\gamma_m(g) = -\,\rho\, g\, \frac{dw_1(g)}{dg}. \tag{24.35}$$

The derivation is identical in form to Theorem 24.5: only the simple-pole residue enters the finite answer, and the higher poles are constrained, not free. We have thus reduced all of scale dependence—β, γ, and γ_m alike—to the single-pole

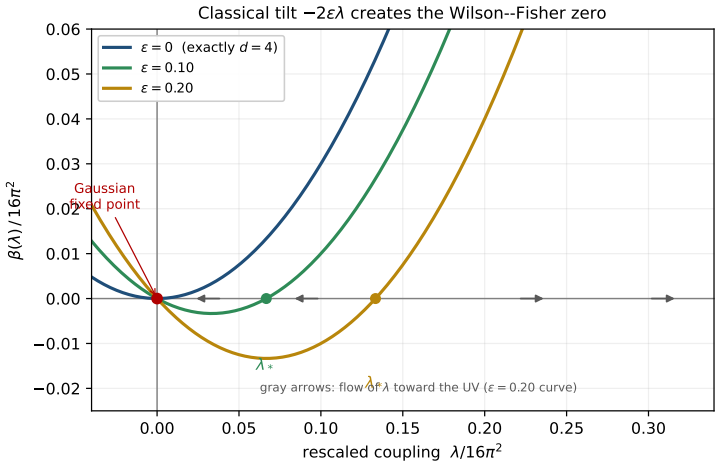


Figure 24.1: The ϕ^4 beta function $\beta(\lambda) = -2\varepsilon\lambda + 3\lambda^2/16\pi^2$ (plotted in units of $16\pi^2$, against $\lambda/16\pi^2$) for three values of ε. At $\varepsilon = 0$ (exactly $d = 4$) the only zero is the Gaussian fixed point at the origin, and the quantum piece drives λ upward toward strong coupling (the gray flow arrows point outward). Turning on $\varepsilon > 0$ adds the classical tilt $-2\varepsilon\lambda$, which pushes the small-coupling flow back toward the origin and creates a second zero, the Wilson–Fisher fixed point at $\lambda_* = \frac{32\pi^2}{3}\varepsilon$ (here in rescaled units $u_* = \frac{2}{3}\varepsilon$), where classical drift exactly cancels quantum running.

parts of the renormalization constants Z_g, Z_ϕ, Z_m. Computing those residues at one loop is the business of the thread sections that follow; here we close the loop by checking that the machinery reproduces the two numbers the chapter is committed to.

Worked Example 24.6 (The ϕ^4 beta function from its simple pole)**.** We continue ϕ^4 theory. The one-loop renormalization of the ϕ^4 coupling, carried out from the four-point function in Section 24.2, gives the $\overline{\text{MS}}$ constant

$$Z_\lambda Z_\phi^{-2} = 1 + \frac{3\lambda}{16\pi^2}\,\frac{1}{2\varepsilon} + O(\lambda^2), \tag{24.36}$$

the single simple pole whose residue we now feed into 't Hooft's relation. For the coupling $g = \lambda$ we have $\rho = 2$, and from Eq. (24.16) the relevant constant is $Z_g = Z_\lambda Z_\phi^{-2}$, so its simple-pole coefficient is

$$z_1(\lambda) = \frac{3\lambda}{32\pi^2}, \qquad \text{equivalently} \qquad a_1(\lambda) = \lambda\, z_1 = \frac{3\lambda^2}{32\pi^2}. \tag{24.37}$$

We apply Eq. (24.22) two equivalent ways as a check. Using the residue form $\bar\beta = \rho\, g^2\, z_1'$:

$$\bar\beta_\lambda = 2\,\lambda^2\,\frac{d}{d\lambda}\left(\frac{3\lambda}{32\pi^2}\right) = 2\,\lambda^2\cdot\frac{3}{32\pi^2} = \frac{6\lambda^2}{32\pi^2} = \frac{3\lambda^2}{16\pi^2}. \tag{24.38}$$

Using instead $\bar\beta = \rho(g\,a_1' - a_1)$ with $a_1 = 3\lambda^2/32\pi^2$, $a_1' = 6\lambda/32\pi^2 = 3\lambda/16\pi^2$:

$$\bar\beta_\lambda = 2\left(\lambda\cdot\frac{3\lambda}{16\pi^2} - \frac{3\lambda^2}{32\pi^2}\right) = 2\left(\frac{6\lambda^2}{32\pi^2} - \frac{3\lambda^2}{32\pi^2}\right) = 2\cdot\frac{3\lambda^2}{32\pi^2} = \frac{3\lambda^2}{16\pi^2}. \tag{24.39}$$

Both routes give the carried number,

$$\boxed{\beta_\lambda = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3)} \tag{24.40}$$

in the four-dimensional limit. Since $\beta_\lambda > 0$ for $\lambda > 0$, the ϕ^4 coupling grows toward the ultraviolet—the seed of the triviality discussion and the Landau pole to be examined when we solve the equation. The field anomalous dimension is $\gamma = O(\lambda^2)$, because the one-loop self-energy of ϕ^4 is a momentum-independent tadpole and so $s_1 = O(\lambda^2)$, making $Z_\phi = 1 + O(\lambda^2)$; Eq. (24.34) then gives $\gamma = O(\lambda^2)$, consistent with the thread's record that γ_ϕ first appears at two loops.

Worked Example 24.7 (The QED beta function from vacuum polarization and the Ward identity)**.** We continue QED. Here the coupling is $g = e$ with $\rho = 1$, and the bare charge is $e_0 = \mu^\varepsilon Z_e\, e$. The vertex, electron-field, and photon-field renormalizations Z_1, Z_2, Z_3 of Chapter 19 combine into the charge renormalization as $Z_e = Z_1 Z_2^{-1} Z_3^{-1/2}$. The Ward–Takahashi identity proved in Chapter 19 states $Z_1 = Z_2$, so the vertex and electron-leg renormalizations cancel and

$$Z_e = Z_3^{-1/2}. \tag{24.41}$$

This is the structural reason the QED running coupling is governed by vacuum polarization alone: only the photon self-energy survives in Z_e. The one-loop $\overline{\text{MS}}$ photon-field renormalization computed from $\Pi(q^2)$ in Section 24.3 is

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon} + O(e^4), \tag{24.42}$$

so, to one loop,

$$Z_e = Z_3^{-1/2} = 1 + \frac{1}{2}\cdot\frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon} + O(e^4) = 1 + \frac{e^2}{24\pi^2}\,\frac{1}{\varepsilon} + O(e^4). \tag{24.43}$$

Thus the simple-pole coefficient of Z_e is $z_1(e) = e^2/24\pi^2$, and $a_1 = e\,z_1 = e^3/24\pi^2$, $a_1' = 3e^2/24\pi^2 = e^2/8\pi^2$. 't Hooft's relation Eq. (24.22) with $\rho = 1$ gives

$$\bar\beta_e = 1\cdot(e\,a_1' - a_1) = e\cdot\frac{e^2}{8\pi^2} - \frac{e^3}{24\pi^2} = \frac{3e^3}{24\pi^2} - \frac{e^3}{24\pi^2} = \frac{2e^3}{24\pi^2} = \frac{e^3}{12\pi^2}. \tag{24.44}$$

This is the carried number,

$$\boxed{\beta_e = \frac{e^3}{12\pi^2} + O(e^5)} \tag{24.45}$$

The sign is positive: like ϕ^4, QED is not asymptotically free, and the electric charge grows toward short distances. Translating to $\alpha = e^2/4\pi$ via $\beta_\alpha = \mu\, d\alpha/d\mu = (2e/4\pi)\,\beta_e = (e/2\pi)(e^3/12\pi^2) = e^4/24\pi^3 = (4\pi\alpha)^2/(24\pi^3 \cdot 4\pi)$; more simply, $\beta_\alpha = \dfrac{e}{2\pi}\,\beta_e = \dfrac{e^4}{24\pi^3} = \dfrac{2\alpha^2}{3\pi}$, recovering the equivalent form $\beta_\alpha = 2\alpha^2/3\pi$ promised in the chapter chapter introduction. Notice how little had to be assumed: the entire result followed from one residue, that of Z_3, once the Ward identity collapsed Z_e to $Z_3^{-1/2}$.

These two worked examples display the division of labour that organizes the rest of the chapter. The master equation Eq. (24.9) and the pole equation Eq. (24.22) are now fixed once and for all; what remains, for each thread, is to supply a single one-loop residue—z_1 for ϕ^4, and (through the Ward identity) the residue of Z_3 for QED—and turn the crank. The next section approaches the very same equation from the opposite direction: rather than starting from the μ-independence of the regulated bare theory, it starts from the breakdown of classical scale invariance, recovers β and γ as the exact quantum measure of that breakdown, and proves that the resulting Callan–Symanzik equation is identical to the μ-flow equation Eq. (24.9) derived here.

24.2 ϕ^4 theory: The One-Loop Beta Function and Anomalous Dimensions

The previous section built an abstract machine and then idled it. The demand that the bare theory be blind to the subtraction scale collapsed, in Theorem 24.5, into a pole equation: the four-dimensional beta function and the anomalous dimensions are read directly off the simple $1/\varepsilon$ residues of the $\overline{\text{MS}}$ renormalization constants, with everything else—the higher poles, the finite parts—either fixed by consistency or irrelevant to the flow. The machine now needs a theory fed into it. We feed it the workhorse of the book, the real scalar $\lambda\phi^4$ field carried since Chapter 6, and turn the crank to the end, omitting no step. The results are fixed in advance by the chapter's promise: the running of the quartic coupling,

$$\beta_\lambda \;=\; \frac{3\lambda^2}{16\pi^2} \;+\; O(\lambda^3), \tag{24.46}$$

the anomalous dimension that governs the running of the mass, and the field anomalous dimension γ, which we will find is *forced* to vanish at one loop and first stirs at two. Every one of these numbers is the residue of a $1/\varepsilon$ pole in a Feynman integral, and we compute those integrals here. The strategy is exactly the one the QED section that follows will reuse without comment: *compute the Z's from the divergent diagrams, then read off β and γ from their poles.* There is nothing else to do, and the factor of 3 in Eq. (24.46)—the one genuinely

characteristic number of the scalar theory—will turn out to be nothing more mysterious than the number of ways to split four lines into two pairs.

24.2.1 Renormalized perturbation theory and the counterterms to be determined

We continue ϕ^4 theory from its renormalization in Chapter 22. The model is the real scalar,

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda\,\mu^{2\varepsilon}}{4!}\,\phi^4, \tag{24.47}$$

written here in terms of the *renormalized* field ϕ, mass m, and dimensionless coupling λ, with the explicit $\mu^{2\varepsilon}$ that keeps λ dimensionless in $d = 4 - 2\varepsilon$ (recall from Eq. (24.15) that the bare coupling carries dimension $[\lambda_0] = 2\varepsilon$, extracted through this factor). Renormalized perturbation theory, as set up in Chapter 22, splits the bare Lagrangian into this renormalized piece plus a set of *counterterms* with the same operator content,

$$\mathcal{L}_{\rm ct} = \tfrac{1}{2}\,\delta_Z\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,\delta_m\,\phi^2 - \frac{\delta_\lambda\,\mu^{2\varepsilon}}{4!}\,\phi^4, \tag{24.48}$$

whose coefficients are the counterterms $\delta_Z, \delta_m, \delta_\lambda$ defined in Eq. (22.9). They are fixed by the multiplicative renormalization constants through

$$\delta_Z = Z_\phi - 1, \qquad \delta_m = (Z_{m^2}\,Z_\phi - 1)\,m^2, \qquad \delta_\lambda = (Z_\lambda - 1)\,\lambda, \tag{24.49}$$

where Z_ϕ is the field-strength constant of $\phi_0 = Z_\phi^{1/2}\phi$, Z_{m^2} the mass-squared constant of $m_0^2 = Z_{m^2}\,m^2$, and Z_λ the constant appearing in the coupling renormalization. These are exactly the relations of Eq. (22.62); in the notation of Eq. (24.2) the bare parameters are $\lambda_0 = Z_\lambda Z_\phi^{-2}\lambda\mu^{2\varepsilon}$ and $m_0^2 = Z_{m^2}m^2$ with $Z_{m^2} = Z_m Z_\phi^{-1}$, so the combination that runs the coupling is

$$Z_g \;\equiv\; Z_\lambda\,Z_\phi^{-2}, \qquad \lambda_0 = \mu^{2\varepsilon}\,Z_g\,\lambda, \tag{24.50}$$

matching the package $g_0 = \mu^{\rho\varepsilon}Z_g\,g$ of Eq. (24.16) with $\rho = 2$. In the $\overline{\rm MS}$ scheme of Theorem 22.7 every counterterm is a pure pole: $Z_\phi, Z_{m^2}, Z_\lambda$ are each 1 plus a Laurent series in $1/\varepsilon$ whose residues are polynomials in λ, with no finite part and no μ-dependence. Our entire task is to compute the three simple-pole residues at one loop.

The objects that expose them are the two superficially divergent amputated one-particle-irreducible (1PI) functions of the theory, identified by the power counting of Section 21.6.1: the two-point function $\Gamma^{(2)}$ (superficial degree $D = 2$) and the four-point function $\Gamma^{(4)}$ ($D = 0$). The six-point function and higher are convergent, so no further counterterm is needed—this is the renormalizability of

ϕ^4, established in Chapter 21. Including tree level, the loop corrections, and the counterterm vertices read off from Eq. (24.48), these two functions are

$$\Gamma^{(2)}(p) = p^2 - m^2 \;-\; \Sigma(p^2) \;+\; (\delta_Z\, p^2 - \delta_m), \tag{24.51}$$

$$i\Gamma^{(4)}(p_1, \ldots, p_4) = -i\lambda\mu^{2\varepsilon} \;+\; i\mathcal{M}_{\text{loop}} \;-\; i\,\delta_\lambda\,\mu^{2\varepsilon}, \tag{24.52}$$

where $-i\Sigma(p^2)$ denotes the sum of 1PI loop insertions in the two-point function (so the full propagator is $i/(p^2 - m^2 - \Sigma)$), the counterterm contributes the Feynman rule $i(\delta_Z p^2 - \delta_m)$ to it, and $i\mathcal{M}_{\text{loop}}$ is the sum of one-loop 1PI four-point graphs with the four-point counterterm vertex $-i\delta_\lambda\mu^{2\varepsilon}$. The renormalization conditions of $\overline{\text{MS}}$ are simply that $\delta_Z, \delta_m, \delta_\lambda$ be chosen to cancel the $1/\bar{\varepsilon}$ poles of Σ and $\mathcal{M}_{\text{loop}}$, where $1/\bar{\varepsilon} \equiv 1/\varepsilon - \gamma_{\text{E}} + \ln 4\pi$ is the universal divergent combination of Eq. (A.204). We treat the four-point function first, because it is where the carried number lives.

24.2.2 The one-loop four-point function and the factor of three

At order λ^2 the amputated 1PI four-point function receives three one-loop diagrams, one for each way of routing the four external legs into the loop. Label the external momenta p_1, p_2 incoming and p_3, p_4 outgoing, and form the Mandelstam invariants

$$s = (p_1 + p_2)^2, \qquad t = (p_1 - p_3)^2, \qquad u = (p_1 - p_4)^2. \tag{24.53}$$

Each diagram is the simple two-vertex bubble of Fig. 24.2: two quartic vertices joined by two internal propagators, with a definite pair of external legs entering each vertex. The three diagrams differ only in which pair of legs meets at the left vertex—(p_1p_2), (p_1p_3), or (p_1p_4)—carrying squared momentum s, t, or u through the loop respectively.

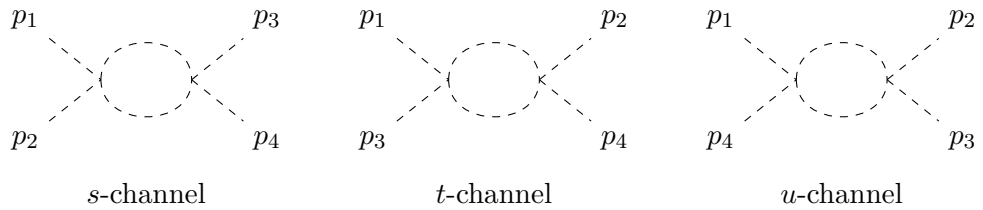


Figure 24.2: The three one-loop diagrams contributing to the ϕ^4 four-point function, one per Mandelstam channel. They share a single topology—the two-vertex bubble of Theorem D.25—and differ only in which external pair meets at each vertex, hence in whether s, t, or u flows through the loop. Their identical pole structure is the origin of the factor of 3 in β_λ.

The value of the s-channel diagram was assembled from the Feynman rules of Chapter 15 in Theorem D.25. Each vertex contributes $-i\lambda\mu^{2\varepsilon}$; the two internal

propagators contribute numerators $i \cdot i = -1$ over the bubble denominator; and a symmetry factor $\tfrac{1}{2}$ accounts for the interchange of the two internal lines. With total momentum p, $p^2 = s$, flowing through the loop,

$$i\mathcal{M}_s = \frac{(-i\lambda\mu^{2\varepsilon})^2}{2} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{i}{\ell^2 - m^2 + i\epsilon} \, \frac{i}{(\ell+p)^2 - m^2 + i\epsilon} = \frac{\lambda^2 \mu^{2\varepsilon}}{2} \, B(s), \tag{24.54}$$

where $B(s) = \mu^{2\varepsilon} \int \mathrm{d}^d \ell \, (2\pi)^{-d} \, [(\ell^2 - m^2)((\ell+p)^2 - m^2)]^{-1}$ is the scalar bubble of Eq. (D.137), evaluated to its $\overline{\mathrm{MS}}$ form in Eq. (D.140),

$$B(s) = \frac{i}{16\pi^2} \left[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x \, \ln \frac{m^2 - x(1-x)s}{\mu^2} \right] + O(\varepsilon). \tag{24.55}$$

The kinematic prefactors in Eq. (24.54) combine as $(-i\lambda\mu^{2\varepsilon})^2 \cdot \tfrac{1}{2} = -\lambda^2\mu^{2\varepsilon}/2$ times the propagator factor $i^2 = -1$, giving the stated $+\lambda^2\mu^{2\varepsilon}/2$. Substituting Eq. (24.55),

$$i\mathcal{M}_s = \frac{i\lambda^2 \mu^{2\varepsilon}}{32\pi^2} \left[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x \, \ln \frac{m^2 - x(1-x)s}{\mu^2} \right] + O(\varepsilon), \tag{24.56}$$

using $\tfrac{1}{2} \cdot \tfrac{1}{16\pi^2} = \tfrac{1}{32\pi^2}$. The crucial feature for the beta function is that the $1/\bar{\varepsilon}$ pole is *independent of* s: the only s-dependence sits in the finite logarithm. This is the locality of counterterms made manifest (Theorem D.31), and it is what allows a single momentum-independent counterterm δ_λ to cancel the divergence of all kinematic configurations at once.

The t- and u-channel diagrams are obtained from Eq. (24.56) by $s \to t$ and $s \to u$; their pole parts are identical, since the residue does not know which invariant flows through the loop. Summing the three,

$$i\mathcal{M}_{\mathrm{loop}} = i(\mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u) = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2} \, \frac{3}{\bar{\varepsilon}} - \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2} \sum_{v=s,t,u} \int_0^1 \mathrm{d}x \, \ln \frac{m^2 - x(1-x)v}{\mu^2} + O(\varepsilon). \tag{24.57}$$

The factor of 3 is now explicit and exact. It is the number of Mandelstam channels, and—reading the same diagrams as Wick contractions rather than as channels—it is the number of ways to partition the four external legs into two pairs: $\{12\}\{34\}$, $\{13\}\{24\}$, $\{14\}\{23\}$. There is no other combinatorial structure at this order; the 3 that will appear in β_λ is precisely this 3, and nothing in the loop integral or the symmetry factor modifies it.

Physical Insight

It is worth pausing on why the counting is so clean. The single bubble B carries a symmetry factor $\tfrac{1}{2}$, and one might worry that the three channels and the symmetry factors could conspire to produce some other rational

number. They do not, because the symmetry factor is the *same* $\frac{1}{2}$ in every channel—each diagram has exactly one pair of interchangeable internal lines—so it factors out of the sum and the channel count stands alone. The combinatorial identity "three ways to pair four objects" is therefore transmitted undamaged from the contraction of two ϕ^4 vertices straight into the coefficient of β_λ. In a theory whose interaction had a different valence or symmetry the number would change, and tracking exactly this kind of combinatorial factor is how the one-loop beta function of any theory is assembled.

The counterterm δ_λ is fixed by demanding that $\Gamma^{(4)}$ in Eq. (24.52) be finite. The pole of $i\mathcal{M}_{\text{loop}}$ is the first term of Eq. (24.57); cancelling it against $-i\delta_\lambda\mu^{2\varepsilon}$ requires

$$-i\,\delta_\lambda\,\mu^{2\varepsilon} + \frac{3i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} = \text{finite} \quad\Longrightarrow\quad \delta_\lambda = \frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}. \tag{24.58}$$

Through Eq. (24.49), $\delta_\lambda = (Z_\lambda - 1)\lambda$, this is the coupling renormalization constant

$$Z_\lambda = 1 + \frac{3\lambda}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(\lambda^2). \tag{24.59}$$

Because the field renormalization Z_ϕ will be shown in Section 24.2.3 to be $1 + O(\lambda^2)$, the combination that runs the coupling agrees with Z_λ at this order,

$$Z_g = Z_\lambda Z_\phi^{-2} = 1 + \frac{3\lambda}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(\lambda^2), \tag{24.60}$$

so its simple-pole residue, in the notation $Z_g = 1 + z_1(\lambda)/\varepsilon + \dots$ of Eq. (24.17), is

$$z_1(\lambda) = \frac{3\lambda}{32\pi^2}. \tag{24.61}$$

This single number is the entire one-loop input to β_λ; Section 24.2.4 turns it into the carried result. We have written the pole as $1/\bar{\varepsilon}$ throughout; in $\overline{\text{MS}}$ the constants $-\gamma_{\text{E}} + \ln 4\pi$ are absorbed with the pole, and only the coefficient of $1/\varepsilon$—here $3\lambda/32\pi^2$—survives as the residue z_1, so whether one writes $1/\varepsilon$ or $1/\bar{\varepsilon}$ is immaterial to the flow. This value of Z_g is the constant quoted as input in Eq. (24.36); we have now derived it from the diagrams rather than assuming it.

24.2.3 The two-point function: a momentum-independent tadpole

We turn to the self-energy $\Sigma(p^2)$. At one loop the only 1PI insertion in the two-point function is the tadpole of Fig. 24.3: a single quartic vertex with two of its four legs joined into a loop. This diagram is the structural reason the field anomalous dimension vanishes at one loop, and the reason deserves to be seen before any integral is done.

Figure 24.3: The one-loop self-energy of ϕ^4: a single quartic vertex with two legs contracted into a loop. The external momentum p flows straight through the vertex and never enters the loop momentum ℓ, so the diagram is independent of p. This is why $\delta_Z = 0$ at one loop and the field anomalous dimension is $O(\lambda^2)$.

The vertex carries $-i\lambda\mu^{2\varepsilon}$; a symmetry factor $\frac{1}{2}$ accounts for the interchange of the two ends of the single internal line; and the loop line contributes one propagator. The decisive point is geometric: the external momentum p enters and leaves the same vertex without ever circulating in the loop, so the loop momentum ℓ is unconstrained by p and the integrand depends on ℓ alone. Hence

$$-i\Sigma(p^2) = \frac{-i\lambda\mu^{2\varepsilon}}{2}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2 - m^2 + i\epsilon} = \frac{\lambda\mu^{2\varepsilon}}{2}\,A_0(m^2), \tag{24.62}$$

with A_0 the scalar tadpole integral of Eq. (D.98), and Σ is manifestly *independent of* p^2. From Eq. (D.99),

$$A_0(m^2) = -\frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(1-\tfrac{d}{2}\right)(m^2)^{d/2-1}, \tag{24.63}$$

so, dividing Eq. (24.62) by $-i$,

$$\Sigma = \frac{\lambda\mu^{2\varepsilon}}{2(4\pi)^{d/2}}\,\Gamma\!\left(1-\tfrac{d}{2}\right)(m^2)^{d/2-1} = \frac{\lambda\, m^2}{32\pi^2}\,\Gamma(-1+\varepsilon)\left(\frac{4\pi\mu^2}{m^2}\right)^{\varepsilon}, \tag{24.64}$$

where in the last step we set $d = 4-2\varepsilon$, wrote $(4\pi)^{d/2} = (4\pi)^2(4\pi)^{-\varepsilon}$ and $(m^2)^{d/2-1} = m^2(m^2)^{-\varepsilon}$, and collected the ε-powers into the dimensionless ratio $(4\pi\mu^2/m^2)^{\varepsilon}$; the prefactor is $\lambda m^2/2(4\pi)^2 = \lambda m^2/32\pi^2$. The gamma function is expanded about its pole using $\Gamma(-1+\varepsilon) = \Gamma(1+\varepsilon)/[\varepsilon(\varepsilon-1)]$ with $\Gamma(1+\varepsilon) = 1 - \gamma_{\mathrm{E}}\varepsilon + O(\varepsilon^2)$ and $1/[\varepsilon(\varepsilon-1)] = -\frac{1}{\varepsilon}(1+\varepsilon+O(\varepsilon^2))$:

$$\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} + (\gamma_{\mathrm{E}} - 1) + O(\varepsilon). \tag{24.65}$$

Multiplying by $(4\pi\mu^2/m^2)^{\varepsilon} = 1 + \varepsilon\ln(4\pi\mu^2/m^2) + O(\varepsilon^2)$ and keeping the finite part,

$$\Gamma(-1+\varepsilon)\left(\frac{4\pi\mu^2}{m^2}\right)^{\varepsilon} = -\frac{1}{\varepsilon} - \ln\frac{4\pi\mu^2}{m^2} + \gamma_{\mathrm{E}} - 1 + O(\varepsilon) = -\frac{1}{\bar{\varepsilon}} + \ln\frac{m^2}{\mu^2} - 1 + O(\varepsilon), \tag{24.66}$$

where in the last equality the constants assembled into $-1/\bar{\varepsilon} = -(1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi)$ and the residual scale logarithm became $-\ln(4\pi\mu^2/m^2) + \ln 4\pi = \ln(m^2/\mu^2)$. Therefore

$$\Sigma(p^2) = \frac{\lambda\, m^2}{32\pi^2}\left[-\frac{1}{\bar{\varepsilon}} + \ln\frac{m^2}{\mu^2} - 1\right] + O(\varepsilon), \tag{24.67}$$

with no p^2 anywhere on the right. Two consequences follow, one for the field and one for the mass.

The field renormalization vanishes at one loop. The field-strength counterterm δ_Z is fixed by demanding that the residue of the full propagator at its pole—equivalently, the coefficient of p^2 in $\Gamma^{(2)}$—be finite. From Eq. (24.51),

$$\frac{\partial \Gamma^{(2)}}{\partial p^2} = 1 + \delta_Z - \frac{\partial \Sigma}{\partial p^2} = 1 + \delta_Z, \tag{24.68}$$

because $\partial\Sigma/\partial p^2 = 0$ by Eq. (24.67). There is no pole to subtract, so the $\overline{\mathrm{MS}}$ prescription sets $\delta_Z = 0$, and hence

$$Z_\phi = 1 + \delta_Z = 1 + O(\lambda^2), \qquad s_1(\lambda) = O(\lambda^2), \tag{24.69}$$

where s_1 is the simple-pole residue of $Z_\phi = 1 + s_1/\varepsilon + \dots$ in the notation of Eq. (24.34). The field anomalous dimension follows immediately from the pole equation Eq. (24.34), $\gamma = -\frac{1}{2}\rho\,\lambda\, \mathrm{d}s_1/\mathrm{d}\lambda$ with $\rho = 2$:

$$\boxed{\gamma = -\lambda\,\frac{\mathrm{d}s_1}{\mathrm{d}\lambda} = O(\lambda^2).} \tag{24.70}$$

The one-loop field anomalous dimension of ϕ^4 *vanishes*, and it vanishes for a reason we can name: the only one-loop self-energy diagram is a tadpole, and a tadpole cannot depend on the external momentum, so it can renormalize the mass but never the kinetic term. This is the promised result—shown, not asserted. We return in Section 24.2.5 to the two-loop diagram that breaks the silence and gives γ its first nonzero value.

The mass counterterm. The mass is renormalized by the p-independent part of Eq. (24.51). Setting $\delta_Z = 0$, finiteness of $\Gamma^{(2)}$ requires that $-\Sigma - \delta_m$ be free of poles. The pole of Σ is, from Eq. (24.67),

$$\Sigma\big|_{\mathrm{pole}} = -\frac{\lambda\, m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}, \tag{24.71}$$

so the cancellation $-\Sigma|_{\mathrm{pole}} - \delta_m = 0$ gives

$$\delta_m = -\Sigma\big|_{\mathrm{pole}} = \frac{\lambda\, m^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}. \tag{24.72}$$

With $\delta_Z = 0$ the relation $\delta_m = (Z_{m^2} Z_\phi - 1)m^2$ from Eq. (24.49) reduces to $\delta_m = (Z_{m^2} - 1)m^2$, so the mass-squared renormalization constant is

$$Z_{m^2} = 1 + \frac{\lambda}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} + O(\lambda^2), \qquad w_1(\lambda) = \frac{\lambda}{32\pi^2}, \tag{24.73}$$

where w_1 is its simple-pole residue, $Z_{m^2} = 1 + w_1/\varepsilon + \dots$. We now have all three residues: $z_1 = 3\lambda/32\pi^2$ for the coupling, $s_1 = O(\lambda^2)$ for the field, and $w_1 = \lambda/32\pi^2$ for the mass-squared. Reading off the flow is all that remains.

24.2.4 Assembling the beta function and the mass running

We feed the residues into the pole equation. For the coupling, Theorem 24.5 expresses the four-dimensional beta function through the simple-pole residue z_1 alone, $\bar{\beta}(g) = \rho\, g^2\, \mathrm{d}z_1/\mathrm{d}g$ from Eq. (24.22), with $g = \lambda$ and $\rho = 2$. Substituting Eq. (24.61), $z_1 = 3\lambda/32\pi^2$ so $\mathrm{d}z_1/\mathrm{d}\lambda = 3/32\pi^2$,

$$\beta_\lambda = 2\,\lambda^2\,\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{3\lambda}{32\pi^2}\right) = 2\,\lambda^2 \cdot \frac{3}{32\pi^2} = \frac{6\lambda^2}{32\pi^2} = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3), \tag{24.74}$$

the carried number Eq. (24.46). Since $\beta_\lambda > 0$ for $\lambda > 0$, the quartic coupling *grows* toward the ultraviolet: the renormalized ϕ^4 theory is not asymptotically free, and its flow toward strong coupling is the seed of the triviality discussion and the Landau pole examined when we solve the equation. We will give the same number a second derivation in Theorem 24.8, starting from the bare-coupling identity rather than the packaged theorem, so that the bookkeeping of the explicit $\mu^{2\varepsilon}$ is laid completely bare.

For the mass, the cleanest statement is the running of the physical renormalized mass-squared, which we obtain directly. The bare mass-squared $m_0^2 = Z_{m^2} m^2$ is μ-independent at fixed bare parameters, so

$$0 = \mu\frac{\mathrm{d}}{\mathrm{d}\mu}\ln m_0^2\Big|_0 = \mu\frac{\mathrm{d}\ln Z_{m^2}}{\mathrm{d}\mu}\Big|_0 + \mu\frac{\mathrm{d}\ln m^2}{\mathrm{d}\mu}\Big|_0. \tag{24.75}$$

The first term is $(\mathrm{d}\ln Z_{m^2}/\mathrm{d}\lambda)\,\beta_\lambda$; to one loop $\ln Z_{m^2} = w_1/\varepsilon$, so $\mu\,\mathrm{d}\ln Z_{m^2}/\mathrm{d}\mu = (\mathrm{d}w_1/\mathrm{d}\lambda)\,\beta_\lambda(\lambda,\varepsilon)/\varepsilon$. The full d-dimensional beta function is $\beta_\lambda(\lambda,\varepsilon) = -2\varepsilon\lambda + \bar{\beta}_\lambda$ by Eq. (24.21), and only the classical piece $-2\varepsilon\lambda$ survives against the explicit $1/\varepsilon$ to leave a finite, $\varepsilon \to 0$ result:

$$\mu\frac{\mathrm{d}\ln Z_{m^2}}{\mathrm{d}\mu}\Big|_0 = \frac{\mathrm{d}w_1}{\mathrm{d}\lambda}\,\frac{-2\varepsilon\lambda}{\varepsilon} + O(\varepsilon) = -2\lambda\,\frac{\mathrm{d}w_1}{\mathrm{d}\lambda} = -2\lambda\cdot\frac{1}{32\pi^2} = -\frac{\lambda}{16\pi^2}. \tag{24.76}$$

Inserting this into Eq. (24.75) gives the running of the renormalized mass-squared,

$$\boxed{\mu\frac{\mathrm{d}m^2}{\mathrm{d}\mu}\Big|_0 = \frac{\lambda}{16\pi^2}\,m^2 + O(\lambda^2)} \tag{24.77}$$

the carried number $\lambda/16\pi^2$ promised by the chapter blueprint. The renormalized mass-squared grows logarithmically toward the ultraviolet. It is convenient to record this as a dimensionless rate, the mass-squared anomalous dimension

$$\gamma_{m^2} \equiv \mu \frac{\mathrm{d}\ln m^2}{\mathrm{d}\mu}\bigg|_0 = \frac{\lambda}{16\pi^2} + O(\lambda^2). \tag{24.78}$$

The chapter's master equation Eq. (24.9) is, however, phrased with the mass anomalous dimension γ_m of Theorem 24.3, defined through m rather than m^2, and the two differ by a factor of two and a sign. The conversion is purely definitional, and we make it explicit so that no reader inserts the wrong number into the master equation.

Notation Watch

The quantity that runs in the master equation Eq. (24.9) is $\gamma_m = -\mu\,\mathrm{d}\ln m/\mathrm{d}\mu$ of Theorem 24.3, while the number quoted in most scalar-field references, and the carried number of this thread, is the rate $\gamma_{m^2} = \mu\,\mathrm{d}\ln m^2/\mathrm{d}\mu$ of Eq. (24.78). Since $\ln m^2 = 2\ln m$,

$$\gamma_{m^2} = 2\,\mu\frac{\mathrm{d}\ln m}{\mathrm{d}\mu} = -2\,\gamma_m \quad\Longrightarrow\quad \gamma_m = -\tfrac{1}{2}\gamma_{m^2} = -\frac{\lambda}{32\pi^2} + O(\lambda^2). \tag{24.79}$$

The sign of γ_m is *negative* for ϕ^4: in the convention of Theorem 24.3, where a positive γ_m would mean the mass shrinks toward the ultraviolet (as for QCD quark masses), the scalar mass instead *grows*, so $\gamma_m < 0$. Both statements describe the same physics, Eq. (24.77); one must only keep straight whether a quoted "mass anomalous dimension" refers to m or to m^2. Throughout the master equation we use $\gamma_m = -\lambda/32\pi^2$; whenever we quote the carried number $\lambda/16\pi^2$ we mean γ_{m^2}. The factor-of-two pitfall here is the analogue, for the mass, of the field-versus-1PI sign flip flagged for γ in Section 24.1.2.

This completes the one-loop renormalization-group data of ϕ^4 theory:

$$\beta_\lambda = \frac{3\lambda^2}{16\pi^2}, \qquad \gamma_{m^2} = \frac{\lambda}{16\pi^2}\ \big(\gamma_m = -\tfrac{\lambda}{32\pi^2}\big), \qquad \gamma = O(\lambda^2), \tag{24.80}$$

all derived from three Feynman integrals and the pole equation. The next worked example reassembles β_λ from scratch by the bare-coupling route, displaying every factor of $\mu^{2\varepsilon}$, both as an independent check and as the template the QED section will follow.

Worked Example 24.8 (β_λ from the bare coupling, with all $\mu^{2\varepsilon}$ bookkeeping)**.** We rederive Eq. (24.46) without invoking the packaged theorem Theorem 24.5,

starting from the defining property of the bare coupling and carrying the dimensional bookkeeping explicitly. The bare coupling is μ-independent: from Eq. (24.50), $\lambda_0 = \mu^{2\varepsilon}\, G(\lambda)$ with

$$G(\lambda) \equiv Z_g\, \lambda = \lambda + \frac{a_1(\lambda)}{\varepsilon} + O(\varepsilon^{-2}), \qquad a_1(\lambda) = \lambda\, z_1(\lambda) = \frac{3\lambda^2}{32\pi^2}, \tag{24.81}$$

the simple-pole coefficient read from Eq. (24.61). Apply $\mu\, \mathrm{d}/\mathrm{d}\mu$ at fixed λ_0. Using $\mu\, \mathrm{d}\mu^{2\varepsilon}/\mathrm{d}\mu = 2\varepsilon\, \mu^{2\varepsilon}$ and writing $\beta_\lambda \equiv \mu\, \mathrm{d}\lambda/\mathrm{d}\mu|_0$,

$$0 = \mu \frac{\mathrm{d}\lambda_0}{\mathrm{d}\mu}\bigg|_0 = \mu^{2\varepsilon}\, \left[2\varepsilon\, G(\lambda) + G'(\lambda)\, \beta_\lambda\right]. \tag{24.82}$$

Since $\mu^{2\varepsilon} \neq 0$, the bracket vanishes. Insert Eq. (24.81), write $\beta_\lambda = -2\varepsilon\lambda + \bar{\beta}_\lambda$ (the only form consistent with finiteness, Eq. (24.21)), and collect powers of ε. With $G = \lambda + a_1/\varepsilon$ and $G' = 1 + a_1'/\varepsilon$, the bracket is

$$2\varepsilon\Big(\lambda + \frac{a_1}{\varepsilon}\Big) + \Big(1 + \frac{a_1'}{\varepsilon}\Big)\big(-2\varepsilon\lambda + \bar{\beta}_\lambda\big) = 0. \tag{24.83}$$

The $O(\varepsilon^1)$ part is $2\varepsilon\lambda - 2\varepsilon\lambda = 0$ identically—the classical dimension cancels, confirming the decomposition. The $O(\varepsilon^0)$ part is

$$2\, a_1 \;-\; 2\lambda\, a_1' \;+\; \bar{\beta}_\lambda = 0 \quad\Longrightarrow\quad \bar{\beta}_\lambda = 2(\lambda\, a_1' - a_1), \tag{24.84}$$

which is Eq. (24.22) with $\rho = 2$, recovered here from first principles. (The remaining $O(\varepsilon^{-1})$ part, $a_1'\bar{\beta}_\lambda$ = higher-pole data, fixes a_2 by the consistency relation Eq. (24.23) and does not affect $\bar{\beta}_\lambda$.) Now substitute $a_1 = 3\lambda^2/32\pi^2$, $a_1' = 6\lambda/32\pi^2 = 3\lambda/16\pi^2$:

$$\bar{\beta}_\lambda = 2\left(\lambda \cdot \frac{3\lambda}{16\pi^2} - \frac{3\lambda^2}{32\pi^2}\right) = 2\left(\frac{6\lambda^2}{32\pi^2} - \frac{3\lambda^2}{32\pi^2}\right) = 2 \cdot \frac{3\lambda^2}{32\pi^2} = \frac{3\lambda^2}{16\pi^2}. \tag{24.85}$$

The carried number is reproduced, and every factor of $\mu^{2\varepsilon}$ has been tracked from the bare-coupling definition through to the four-dimensional answer. The two routes—the packaged $\bar{\beta} = \rho g^2 z_1'$ of Eq. (24.74) and the explicit bare-coupling expansion here—agree, as they must, and the agreement is a check that the $1/\varepsilon$ residue $z_1 = 3\lambda/32\pi^2$, hence the factor of 3 from the three channels, was extracted correctly.

24.2.5 Where the field anomalous dimension first appears

The vanishing of γ at one loop is a feature of the kinematics of a single diagram, not a protected statement, and it is dispelled at the next order. At two loops the self-energy receives the "sunset" diagram of Fig. 24.4: two quartic vertices joined by three internal propagators, with the external momentum p now *forced through*

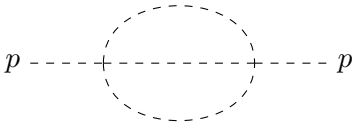


Figure 24.4: The two-loop "sunset" self-energy of ϕ^4: two vertices joined by three propagators. Unlike the one-loop tadpole, the external momentum p now runs through the diagram, so the integral depends on p^2 and contributes a divergence $\propto p^2$. This is the first diagram to renormalize the field, giving $\gamma \neq 0$ at $O(\lambda^2)$.

the loop because the three internal lines connect the two vertices in series. The integrand depends on p, the diagram is a genuine function of p^2, and its divergent part contains a term proportional to p^2—which is exactly what a field-strength counterterm renormalizes.

A full two-loop evaluation is beyond what this chapter's carried numbers require—the thread is committed only to the statement $\gamma = O(\lambda^2)$, now established—but the structure is worth recording so the reader knows where the field anomalous dimension comes from and what its value is. The sunset amplitude is

$$-i\Sigma_2(p^2) = \frac{(-i\lambda\mu^{2\varepsilon})^2}{6}\int \frac{\mathrm{d}^d\ell_1}{(2\pi)^d}\frac{\mathrm{d}^d\ell_2}{(2\pi)^d}\,\frac{i^3}{(\ell_1^2-m^2)(\ell_2^2-m^2)((p-\ell_1-\ell_2)^2-m^2)}, \tag{24.86}$$

the symmetry factor $\frac{1}{6}$ counting the 3! permutations of the three identical internal lines. The relevant fact is that, after its divergent subintegrals are subtracted, the remaining local divergence of Eq. (24.86) is a polynomial of degree two in p with a nonzero p^2 coefficient; extracting that coefficient (a computation carried out with the master-integral methods of Chapter D) yields the field-strength residue

$$s_1(\lambda) = -\frac{\lambda^2}{12\,(16\pi^2)^2} + O(\lambda^3), \tag{24.87}$$

and hence, from the pole equation Eq. (24.70), $\gamma = -\lambda\,\mathrm{d}s_1/\mathrm{d}\lambda$,

$$\gamma = \frac{\lambda^2}{12\,(16\pi^2)^2} + O(\lambda^3). \tag{24.88}$$

The number is positive and small, and it is the leading anomalous dimension that the thread carries forward to the Wilson–Fisher fixed point of Chapter 52: evaluated at the fixed-point coupling $\lambda_* = \frac{32\pi^2}{3}\varepsilon$ of Eq. (24.33), it gives the order-ε^2 critical exponent $\gamma_\phi = O(\varepsilon^2)$. For the purposes of this chapter the salient fact is the qualitative one, now demonstrated rather than assumed: in ϕ^4 the field renormalization is silent at one loop because the one-loop self-energy is a momentum-independent tadpole, and it speaks first at two loops through the momentum-dependent sunset.

The scalar theory has thus delivered every number the thread demanded: $\beta_\lambda = 3\lambda^2/16\pi^2$, the mass running $\mu\,\mathrm{d}m^2/\mathrm{d}\mu = (\lambda/16\pi^2)m^2$, and a field anomalous dimension that is zero at one loop and $O(\lambda^2)$ thereafter. More importantly it has fixed a reusable template—compute the Z-factors from the divergent diagrams, read their simple poles, feed them into the pole equation—that needs only the diagrams replaced to apply elsewhere. We now hand that template to QED in Section 24.3, where gauge invariance, through the Ward–Takahashi identity, will collapse three would-be independent renormalization constants down to one, so that the entire running of the electric charge follows from a single residue, that of the photon field renormalization Z_3.

24.3 QED: Charge Renormalization, the Ward Identity, and β_e

The previous section ran the ϕ^4 machine to its conclusion and, in doing so, fixed the template the rest of the chapter reuses without ceremony: assemble the one-loop renormalization constants, read off their simple $1/\varepsilon$ pole residues, and feed those residues into the $\overline{\mathrm{MS}}$ master formula Eq. (24.22) to obtain the beta function and the anomalous dimensions. Not one step of that procedure referred to the scalar nature of the field; it is a statement about Z-factors and poles, and it transplants verbatim. We now run it for QED, and the payoff is an economy that has no analogue in the scalar theory. In ϕ^4 all three renormalization constants Z_λ, Z_m, Z_ϕ were genuinely independent loop computations. In a gauge theory they are not independent: gauge invariance, in the sharp form of the Ward–Takahashi identity proved in Theorem 19.13, locks the vertex renormalization to the electron-field renormalization so rigidly that the two cancel out of the electric charge entirely. The running of e is therefore controlled by a *single* number — the photon field renormalization Z_3, which is nothing but the vacuum polarization the reader already computed at one loop in Chapter 19. The entire problem of charge renormalization in QED collapses onto the photon two-point function.

This is the structural fact that makes the QED beta function so much cleaner than its scalar cousin, and the section is organized around exhibiting it. We first derive the decomposition $Z_e = Z_1 Z_2^{-1} Z_3^{-1/2}$ from the counterterm Lagrangian; then invoke $Z_1 = Z_2$ to collapse it to $Z_e = Z_3^{-1/2}$; then import Z_3 from Chapter 19, turn the crank of the pole equation, and extract the carried numbers $\beta_e = e^3/12\pi^2$ and $\beta_\alpha = 2\alpha^2/3\pi$. After that we compute the electron's own field and mass anomalous dimensions and are scrupulous about which of them is gauge dependent and which is not. Finally we re-derive the sign and coefficient of β_α a second, completely independent way — from the positivity of the spectral function of Π through a dispersion relation, tying the result to the optical-theorem machinery of Chapter 18 — so that the carried number

rests on two legs rather than one, and so that the *sign* (charge grows toward the ultraviolet) is seen to follow from unitarity rather than from a bookkeeping convention.

Throughout we use the frozen QED conventions recalled in Eq. (17.1): the Dirac electron field ψ of renormalized mass m, charge $-e$ with $e>0$, covariant derivative $D_\mu=\partial_\mu+ieA_\mu$, fine-structure constant $\alpha=e^2/4\pi$, mostly-minus metric, and the $\overline{\text{MS}}$ scheme of Theorem A.49 in dimensional regularization $d=4-2\varepsilon$ with sliding scale μ. Whenever a gauge-dependent quantity is displayed explicitly we work in Feynman gauge $\xi=1$ and flag it; every gauge-independent result is identified as such.

24.3.1 The structure of charge renormalization: $Z_e=Z_1Z_2^{-1}Z_3^{-1/2}$

Renormalized perturbation theory, as set up in Chapter 22, writes every bare quantity as a renormalized quantity times a renormalization constant. QED has three field/vertex constants, named by the historical convention Z_1 (the $\bar\psi\not{A}\psi$ vertex), Z_2 (the electron field), and Z_3 (the photon field), together with the mass constant Z_m. Writing bare quantities with a subscript 0,

$$\psi_0=Z_2^{1/2}\,\psi,\qquad A_0^\mu=Z_3^{1/2}\,A^\mu,\qquad m_0=Z_m\,m, \tag{24.89}$$

with m the renormalized mass. The bare Lagrangian is the standard Lagrangian of Eq. (17.1) written entirely in bare fields,

$$\mathcal{L}_{\text{QED}}=\bar\psi_0\,(i\not{\partial}-m_0)\,\psi_0-\tfrac14 F_{0\,\mu\nu}F_0^{\mu\nu}-e_0\,\bar\psi_0\gamma^\mu\psi_0\,A_{0\,\mu},\qquad F_{0\,\mu\nu}=\partial_\mu A_{0\,\nu}-\partial_\nu A_{0\,\mu}. \tag{24.90}$$

The entire content of charge renormalization is in how the interaction term reorganizes when Eq. (24.89) is substituted. The kinetic terms fix the field constants on their own: the electron kinetic term becomes $\bar\psi_0\,i\not{\partial}\,\psi_0=Z_2\,\bar\psi\,i\not{\partial}\,\psi$, and the photon kinetic term becomes $-\tfrac14F_0^2=-\tfrac14 Z_3\,F^2$, so that once each Z is stripped the propagators are canonically normalized. (We suppress the gauge-fixing term $-\frac{1}{2\xi}(\partial_\mu A^\mu)^2$ here; in $\overline{\text{MS}}$ the gauge parameter ξ is itself renormalized so that the longitudinal part of the photon propagator stays unrenormalized, but none of this enters the transverse Z_3 that carries the charge, as we confirm below.) The interaction term carries three field factors,

$$-e_0\,\bar\psi_0\gamma^\mu\psi_0\,A_{0\,\mu}=-e_0\,Z_2\,Z_3^{1/2}\,\bar\psi\gamma^\mu\psi\,A_\mu, \tag{24.91}$$

two half-powers $Z_2^{1/2}$ from the electron bilinear $\bar\psi_0\psi_0$ and one half-power $Z_3^{1/2}$ from the photon field A_0.

Now the definition of Z_1. In renormalized perturbation theory the interaction is written as a renormalized vertex plus a counterterm, the two together carrying the renormalized coupling e and the scale factor μ^ε that keeps e dimensionless away from four dimensions. In $d=4-2\varepsilon$ the gauge field has mass dimension

$[A] = (d-2)/2 = 1-\varepsilon$ and the fermion $[\psi] = (d-1)/2 = \tfrac{3}{2} - \varepsilon$, so the vertex operator $\bar\psi\gamma^\mu\psi A_\mu$ has dimension $2(\tfrac{3}{2}-\varepsilon) + (1-\varepsilon) = 4 - 3\varepsilon$; for the action $\int d^dx\,\mathcal{L}$ to be dimensionless the bare coupling must carry $[e_0] = d - (4-3\varepsilon) = \varepsilon$. We extract this dimension through the explicit μ^ε, leaving the renormalized e dimensionless in every d. By definition the full coefficient of the renormalized vertex operator $\bar\psi\gamma^\mu\psi A_\mu$ is $-e\,\mu^\varepsilon\,Z_1$:

$$-e_0\,Z_2\,Z_3^{1/2}\,\bar\psi\gamma^\mu\psi\,A_\mu \;\equiv\; -e\,\mu^\varepsilon\,Z_1\,\bar\psi\gamma^\mu\psi\,A_\mu. \tag{24.92}$$

This is the statement that Z_1 is the multiplicative renormalization of the vertex — the constant that absorbs the ultraviolet divergence of the amputated one-particle-irreducible (1PI) $\bar\psi\psi A$ Green's function, exactly as Z_λ absorbed the divergence of the 1PI four-point function in the scalar theory. Matching the coefficients in Eq. (24.92),

$$e_0\,Z_2\,Z_3^{1/2} = e\,\mu^\varepsilon\,Z_1 \quad\Longrightarrow\quad e_0 = \mu^\varepsilon\,\frac{Z_1}{Z_2\,Z_3^{1/2}}\,e. \tag{24.93}$$

Comparing with the defining form of the charge renormalization constant,

$$e_0 = \mu^\varepsilon\,Z_e\,e, \tag{24.94}$$

we read off the structural result that organizes the entire calculation:

$$\boxed{Z_e = \frac{Z_1}{Z_2\,Z_3^{1/2}} = Z_1\,Z_2^{-1}\,Z_3^{-1/2}.} \tag{24.95}$$

The logic is purely combinatorial and worth stating once in words, because it generalizes to every gauge theory: each external leg attached to the charge-carrying vertex brings one factor of its own field renormalization, so the bare coupling is dressed by Z_2 (two half-powers, one per electron leg) and by $Z_3^{1/2}$ (one half-power for the photon leg), while the irreducible divergence of the vertex itself is carried by Z_1. In a theory without a protecting symmetry, Eq. (24.95) would demand all three constants as independent loop computations, and the charge would run according to the combined residue $z_1 - z_2 - \tfrac{1}{2}z_3$ of their poles.

24.3.2 The Ward–Takahashi identity collapses the charge to the photon

Gauge invariance now intervenes decisively, and reduces three loop computations to one. The Ward–Takahashi identity, derived in full from the $U(1)$ invariance of the path integral in Theorem 19.13, relates the exact (amputated, 1PI) vertex function $\Gamma^\mu(p', p)$ to the exact inverse electron propagator $S^{-1}(p)$ through

$$q_\mu\,\Gamma^\mu(p', p) = S^{-1}(p') - S^{-1}(p), \qquad q = p' - p, \tag{24.96}$$

the operator statement that the photon couples to a conserved current. Its corollary at the level of renormalization constants — the form we import here — is the equality

$$Z_1 = Z_2. \tag{24.97}$$

The derivation of Eq. (24.97) from Eq. (24.96) is short enough to recall, even though we are entitled to import it: differentiate Eq. (24.96) with respect to q and let $q \to 0$, so that the left side becomes the vertex at zero photon momentum, $\Gamma^\mu(p,p) = \partial S^{-1}(p)/\partial p_\mu$. The bare vertex and bare inverse propagator are $\Gamma_0^\mu = Z_1^{-1}\Gamma_R^\mu$ and $S_0^{-1} = Z_2^{-1}S_R^{-1}$ in terms of their renormalized (finite) counterparts; since the same renormalized identity $\Gamma_R^\mu(p,p) = \partial S_R^{-1}/\partial p_\mu$ must also hold for the relation to survive renormalization, the divergent constants must match, $Z_1^{-1} = Z_2^{-1}$, i.e. $Z_1 = Z_2$. The physical content is that the divergence dressing the vertex is identically the divergence dressing the electron leg: the same gauge symmetry that forces the photon to couple through the conserved current $\bar\psi\gamma^\mu\psi$ forces the vertex and the field to renormalize in lockstep, in *every* gauge and at *every* order.

The consequence for the charge is immediate and total. Substituting $Z_1 = Z_2$ into Eq. (24.95), the vertex and electron-field constants cancel identically,

$$\boxed{Z_e = \frac{Z_2}{Z_2\, Z_3^{1/2}} = Z_3^{-1/2}.} \tag{24.98}$$

This is the central structural fact of QED charge renormalization. The bare and renormalized charges are related by

$$e_0 = \mu^\varepsilon\, Z_3^{-1/2}\, e, \tag{24.99}$$

and the running of e is governed by the photon field renormalization *alone*. Everything to do with the electron — its self-energy, its field strength, the vertex dressing — has dropped out of the charge. The reader should weigh how much this buys: Z_1 and Z_2 are individually divergent, scheme-dependent, and (as the next subsection makes explicit) gauge-dependent objects, yet their ratio in Z_e is exactly unity, so none of that complexity ever reaches the physical coupling. The sole survivor is Z_3, the renormalization of the photon propagator — the vacuum polarization. *The QED beta function is a property of the photon.*

Charge universality is the Ward identity

An electron and a muon are different fields, with different masses and, at the quantum level, different self-energies and vertex corrections; their Z_1 and Z_2 are not the same numbers. Yet experiment finds their charges renormalize identically: the running coupling $\alpha(\mu)$ an electron sees is the very same $\alpha(\mu)$ a muon sees, and both equal $1/137.036\ldots$ at vanishing

momentum. Equation (24.98) is the explanation. For *each* charged species the Ward identity enforces $Z_1 = Z_2$, so the species-specific constants cancel and $Z_e = Z_3^{-1/2}$ regardless of which fermion one started from. The charge renormalization is therefore not a property of any one charged particle but of the photon they all share. The universality of electric charge — that charge comes in identical quanta independent of the carrier — is, at the level of radiative corrections, precisely the statement $Z_1 = Z_2$. (When several species circulate in the photon's vacuum-polarization loop they each contribute additively to Z_3; that enlarges the running but is a property of the photon line, not a violation of universality.)

24.3.3 Importing Z_3 and extracting β_e

It remains to supply Z_3 and turn the crank. The one-loop vacuum polarization was computed in Chapter 19 with the gamma-trace technology of Chapter C and the master integrals of Chapter D; we recall its structure rather than recompute it. The 1PI photon self-energy is, by the Ward identity, transverse,

$$\Pi^{\mu\nu}(q) = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\,\Pi(q^2), \tag{24.100}$$

the tensor structure $q^2\eta^{\mu\nu} - q^\mu q^\nu$ being the unique symmetric transverse object built from q, so that $q_\mu \Pi^{\mu\nu} = 0$ as current conservation demands. The single diagram is the fermion loop drawn in Fig. 24.5. Its scalar coefficient $\Pi(q^2)$ carries one ultraviolet pole, whose divergent part was found in Eq. (19.49) to be $\Pi(q^2)\big|_{\text{div}} = -\,e^2/(6\pi^2)\cdot 1/\varepsilon$. The $\overline{\text{MS}}$ photon field renormalization that subtracts it follows from demanding the renormalized transverse two-point function $(1 + \Pi(q^2) + (Z_3 - 1))$ be finite, giving the carried photon constant

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon} + O(e^4). \tag{24.101}$$

This constant is gauge independent: the transverse part of the photon two-point function receives no gauge-parameter dependence at one loop — it must not, since it feeds a physical quantity — the entire ξ-dependence of the photon propagator residing in its longitudinal part, which transversality (24.100) keeps separate. (The coefficient itself is checked diagram-in-hand in Theorem 24.9.)

By the collapse Eq. (24.98) this single constant is the whole of Z_e:

$$Z_e = Z_3^{-1/2} = \left(1 - \frac{e^2}{12\pi^2\varepsilon}\right)^{-1/2} = 1 + \frac{e^2}{24\pi^2}\,\frac{1}{\varepsilon} + O(e^4), \tag{24.102}$$

where we expanded $(1-x)^{-1/2} = 1 + \tfrac{1}{2}x + O(x^2)$ with $x = e^2/12\pi^2\varepsilon$, so that the photon pole residue $1/12\pi^2$ is halved (from the square root) and sign-flipped (from the inverse) into $+1/24\pi^2$.

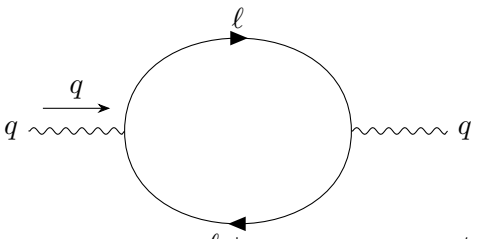


Figure 24.5: The one-loop vacuum polarization: an electron loop on the photon line. The overall minus sign is the closed-fermion-loop factor; transversality $q_\mu \Pi^{\mu\nu} = 0$ is enforced by the Ward identity, leaving the single scalar $\Pi(q^2)$ of Eq. (24.100). Its ultraviolet pole defines Z_3 in Eq. (24.101); everything in the QED beta function flows from this one diagram.

We now extract β_e from this single pole. The argument is the QED specialization of the general $\overline{\text{MS}}$ pole equation Eq. (24.22); we restate it in the form needed here because it is short and because pinning the signs is the whole game. The bare charge e_0 is independent of the arbitrary scale μ. Taking the logarithm of Eq. (24.94), $\ln e_0 = \varepsilon \ln \mu + \ln Z_e + \ln e$, and applying $\mu\, \mathrm{d}/\mathrm{d}\mu$ with $\mathrm{d}e_0/\mathrm{d}\mu = 0$,

$$0 = \varepsilon + \frac{\mu}{Z_e}\frac{dZ_e}{d\mu} + \frac{\mu}{e}\frac{de}{d\mu}. \tag{24.103}$$

In $\overline{\text{MS}}$ the constant Z_e depends on μ *only* through the running coupling $e(\mu)$: it carries no explicit μ and, being a pure pole, no mass dependence. Hence with the d-dimensional beta function $\beta \equiv \mu\, \mathrm{d}e/\mathrm{d}\mu$ we have $\mu\, \mathrm{d}Z_e/\mathrm{d}\mu = (\mathrm{d}Z_e/\mathrm{d}e)\,\beta$, and Eq. (24.103) becomes an algebraic equation for β,

$$0 = \varepsilon + \beta\left(\frac{1}{Z_e}\frac{dZ_e}{de} + \frac{1}{e}\right) \quad\Longrightarrow\quad \beta = \frac{-\varepsilon\, e}{1 + e\, Z_e^{-1}\, \mathrm{d}Z_e/\mathrm{d}e}. \tag{24.104}$$

Write $Z_e = 1 + a_1(e)/\varepsilon + O(\varepsilon^{-2})$. From Eq. (24.102) the residue of the simple pole and its derivative are

$$a_1(e) = \frac{e^2}{24\pi^2}, \qquad \frac{da_1}{de} = \frac{e}{12\pi^2}. \tag{24.105}$$

Then $e\, Z_e^{-1}\, \mathrm{d}Z_e/\mathrm{d}e = e\, a_1'/\varepsilon + O(\varepsilon^{-2})$, and expanding the geometric series in Eq. (24.104) in powers of $1/\varepsilon$,

$$\beta = -\varepsilon e\left(1 + \frac{e\, a_1'}{\varepsilon} + \cdots\right)^{-1} = -\varepsilon e\left(1 - \frac{e\, a_1'}{\varepsilon} + \cdots\right) = -\varepsilon e + e^2\, a_1'(e) + O(\varepsilon^{-1}). \tag{24.106}$$

The leading term $-\varepsilon e$ is the trivial classical scaling of a coupling with mass dimension ε; it is the QED analogue of the $-2\varepsilon\lambda$ of the scalar theory and vanishes as $\varepsilon \to 0$ to leave four dimensions. The residual $O(\varepsilon^{-1})$ pieces in Eq. (24.106) are built from products of one-loop residues; they are cancelled, order by order, by the genuine higher-pole pieces of Z_e that we dropped. Their cancellation is exactly the $\overline{\text{MS}}$ consistency condition that the four-dimensional beta function come out finite, and it holds automatically in a renormalizable theory (the all-orders statement is Eq. (24.23)). The four-dimensional beta function is the finite remainder,

$$\beta_e = \lim_{\varepsilon\to 0}\bigl(\beta + \varepsilon e\bigr) = e^2\,\frac{da_1}{de} = e^2\cdot\frac{e}{12\pi^2}, \tag{24.107}$$

that is,

$$\boxed{\beta_e = \frac{e^3}{12\pi^2} + O(e^5).} \tag{24.108}$$

The sign is positive: the QED coupling *grows* with energy. Converting to $\alpha = e^2/4\pi$, for which $\beta_\alpha \equiv \mu\,\mathrm{d}\alpha/\mathrm{d}\mu = (2e/4\pi)\,\beta_e = (e/2\pi)\beta_e$,

$$\beta_\alpha = \frac{e}{2\pi}\cdot\frac{e^3}{12\pi^2} = \frac{e^4}{24\pi^3} = \frac{(4\pi\alpha)^2}{24\pi^3}, \tag{24.109}$$

and since $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$,

$$\boxed{\beta_\alpha = \frac{2\alpha^2}{3\pi} + O(\alpha^3).} \tag{24.110}$$

These are the carried QED numbers. They were obtained entirely from the photon two-point function: the Ward identity reduced Z_e to $Z_3^{-1/2}$, the imported Z_3 supplied the residue, and the $\overline{\text{MS}}$ pole equation converted that residue into β_e. The same logarithm that Chapter 20 produced heuristically when it dressed the photon line in $e^+e^- \to \mu^+\mu^-$ — the running $\alpha(q^2)$ of Eq. (20.171) — is now seen to be the leading term of the solution of an exact renormalization-group equation, whose slope is precisely $\beta_\alpha = 2\alpha^2/3\pi$. The full integration of that equation, including the Landau pole it implies, is carried out by the method of characteristics in Section 24.6; here we have fixed the slope.

Caution

The factor $-\frac{1}{2}$ versus $+\frac{1}{2}$ in the exponent of Z_3 is the single most common slip in this computation. From $e_0 = \mu^\varepsilon Z_e e$ with $Z_e = Z_3^{-1/2}$ we have $e_0^2 = \mu^{2\varepsilon} Z_3^{-1} e^2$, i.e. $\alpha_0 = \mu^{2\varepsilon} Z_3^{-1}\alpha$: the bare charge is the renormalized charge *divided* by Z_3. Because $Z_3 = 1 - e^2/12\pi^2\varepsilon < 1$, the bare charge exceeds the renormalized one, and removing the regulator drives α_0 upward — the screening picture and the Landau pole both live in this sign. A flipped

exponent would give $\beta_e = -e^3/12\pi^2$ and a spurious asymptotic freedom for QED; the dispersive cross-check of Theorem 24.10 exists precisely to pin the sign down by an argument that never touches Z_3.

Worked Example 24.9 (The residue of Z_3 from the fermion-loop trace)**.** The beta function rests on the single number $-1/12\pi^2$ in Eq. (24.101), so it is worth extracting that residue from the integral in Fig. 24.5 rather than quoting it, even though the finite parts are owned by Chapter 19. We need only the ultraviolet pole. Starting from

$$i\Pi^{\mu\nu}(q) = -e^2 \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{\mathrm{tr}[\gamma^\mu (\not{\ell} + m)\gamma^\nu (\not{\ell} + \not{q} + m)]}{(\ell^2 - m^2)((\ell+q)^2 - m^2)}, \tag{24.111}$$

the overall minus is the closed-fermion-loop sign and the two factors of i from the propagator numerators have been combined with $(-ie)^2$. The trace is evaluated with the identities of Chapter C: $\mathrm{tr}[\gamma^\mu\gamma^\alpha\gamma^\nu\gamma^\beta] = 4(\eta^{\mu\alpha}\eta^{\nu\beta} - \eta^{\mu\nu}\eta^{\alpha\beta} + \eta^{\mu\beta}\eta^{\nu\alpha})$ and the vanishing of the trace of an odd number of gamma matrices, giving

$$\mathrm{tr}[\cdots] = 4\big[\ell^\mu(\ell+q)^\nu + \ell^\nu(\ell+q)^\mu - \eta^{\mu\nu}\big(\ell\cdot(\ell+q) - m^2\big)\big]. \tag{24.112}$$

Combining denominators with one Feynman parameter x (Section D.2), shifting $\ell \to \ell - xq$, and dropping terms odd in the shifted ℓ, the numerator's leading large-ℓ behavior is governed by $\ell^\mu\ell^\nu \to (\ell^2/d)\,\eta^{\mu\nu}$ under the symmetric integration of Eq. (D.92). The logarithmically divergent piece is

$$i\Pi^{\mu\nu}\big|_{\mathrm{div}} = -4e^2 \int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d\ell}{(2\pi)^d} \frac{(2/d-1)\,\ell^2\,\eta^{\mu\nu} + \cdots}{(\ell^2 - \Delta)^2}, \qquad \Delta = m^2 - x(1-x)q^2, \tag{24.113}$$

where the ℓ^2 numerator against the squared denominator produces the pole. Using the master integral $\int \mathrm{d}^d\ell\,\ell^2/(\ell^2-\Delta)^2 = \frac{d}{2}\,(i/(4\pi)^{d/2})\,\Gamma(1-\frac{d}{2})\,\Delta^{d/2-1}$ of Chapter D and isolating the $1/\varepsilon$ from $\Gamma(1-d/2)$ together with $(2/d-1)\cdot d/2 = 1 - d/2 = -1+\varepsilon$, the gauge-noninvariant-looking Δ-dependent terms reorganize, by the transversality the Ward identity guarantees, into the structure $(q^2\eta^{\mu\nu} - q^\mu q^\nu)$. Tracking the coefficient through, the pole of the scalar function is

$$\Pi(q^2)\big|_{\mathrm{div}} = -\frac{e^2}{2\pi^2}\,\frac{1}{\varepsilon}\int_0^1 \mathrm{d}x\, x(1-x) = -\frac{e^2}{2\pi^2}\,\frac{1}{\varepsilon}\cdot\frac{1}{6} = -\frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon}, \tag{24.114}$$

the Feynman-parameter integral $\int_0^1 x(1-x)\,\mathrm{d}x = \frac{1}{6}$ supplying the final fraction. Demanding $1 + \Pi + (Z_3 - 1)$ be finite gives $Z_3 - 1 = -\Pi|_{\mathrm{div}}\cdot(-1)$; more carefully, the renormalized inverse propagator is $q^2[1 + \Pi(q^2) + \delta_3]$ with $\delta_3 = Z_3 - 1$, so finiteness forces $\delta_3 = -\Pi|_{\mathrm{div}}$ *as a pole*, i.e. $Z_3 = 1 + \Pi|_{\mathrm{div}} \to 1 - \frac{e^2}{12\pi^2\varepsilon}$, reproducing Eq. (24.101). The number that becomes $\beta_e = e^3/12\pi^2$ is thus the Dirac trace factor 4, the symmetric-integration factor, and the Feynman integral $\frac{1}{6}$, assembled

into $1/12\pi^2$. Note that the residue is independent of m: a counterterm is a local polynomial and cannot carry the mass-dependent logarithm, by the same locality argument (Theorem D.31) that made the scalar bubble's pole kinematics-free. This is why the QED beta function is mass independent in $\overline{\text{MS}}$.

24.3.4 Field and mass anomalous dimensions of the electron

The charge was gauge independent and saw only the photon. The electron's own renormalization data — its field anomalous dimension γ_2 and its mass anomalous dimension γ_m^{QED} — are computed from the electron self-energy $\Sigma(p)$ of Chapter 19, and the first of them is gauge dependent. We import the one-loop $\overline{\text{MS}}$ constants of Eq. (19.82); in Feynman gauge $\xi = 1$,

$$Z_2 = 1 - \frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} + O(e^4) \quad (\xi = 1), \qquad Z_m = 1 - \frac{3e^2}{16\pi^2}\,\frac{1}{\varepsilon} + O(e^4), \tag{24.115}$$

with Z_2 the electron field renormalization of Eq. (24.89) and Z_m the multiplicative mass renormalization defined by $m_0 = Z_m m$. We extract the anomalous dimensions exactly as we extracted β_e: each is $-e$ times the e-derivative of the corresponding simple-pole residue, because in $\overline{\text{MS}}$ the anomalous dimensions are fixed by the residues alone, with only the classical $-\varepsilon e$ part of the d-dimensional beta function surviving against the explicit $1/\varepsilon$.

For the field, the convention $\psi_0 = Z_2^{1/2}\psi$ and the definition (the QED instance of Eq. (24.6))

$$\gamma_2 \equiv \tfrac{1}{2}\,\mu\frac{d\ln Z_2}{d\mu} = \tfrac{1}{2}\,\frac{1}{Z_2}\frac{dZ_2}{de}\,\beta \;\xrightarrow{\varepsilon\to 0}\; \tfrac{1}{2}\,\frac{db_1}{de}\,(-e), \tag{24.116}$$

where $Z_2 = 1 + b_1(e)/\varepsilon$ and we used $\beta = -\varepsilon e + O(e^3)$ so that the $1/\varepsilon$ from the pole derivative cancels the $-\varepsilon e$ from the classical term, leaving a finite limit. With $b_1 = -e^2/16\pi^2$ (Feynman gauge), $\mathrm{d}b_1/\mathrm{d}e = -e/8\pi^2$, hence

$$\boxed{\gamma_2 = -\frac{e}{2}\cdot\Big(-\frac{e}{8\pi^2}\Big) = \frac{e^2}{16\pi^2} = \frac{\alpha}{4\pi} \quad (\xi = 1).} \tag{24.117}$$

The mass anomalous dimension is defined with the bare mass m_0 held fixed; from $m_0 = Z_m m$ and $\mathrm{d}m_0/\mathrm{d}\mu = 0$,

$$\gamma_m^{\text{QED}} \equiv -\,\mu\frac{d\ln m}{d\mu} = \mu\frac{d\ln Z_m}{d\mu} = \frac{1}{Z_m}\frac{dZ_m}{de}\,\beta \;\xrightarrow{\varepsilon\to 0}\; \frac{dc_1}{de}\,(-e), \tag{24.118}$$

where $Z_m = 1 + c_1(e)/\varepsilon$ with $c_1 = -3e^2/16\pi^2$, so $\mathrm{d}c_1/\mathrm{d}e = -3e/8\pi^2$ and

$$\boxed{\gamma_m^{\text{QED}} = -e\cdot\Big(-\frac{3e}{8\pi^2}\Big) = \frac{3e^2}{8\pi^2} = \frac{3\alpha}{2\pi} + O(\alpha^2).} \tag{24.119}$$

This governs the running of the $\overline{\text{MS}}$ electron mass: $\mu\,\mathrm{d}m/\mathrm{d}\mu = -\gamma_m^{\text{QED}}\,m = -(3\alpha/2\pi)\,m$, so the mass falls logarithmically toward the ultraviolet. (The sign convention here, $\gamma_m = -\mu\,\mathrm{d}\ln m/\mathrm{d}\mu$, is the QED counterpart of the scalar-thread choice; note it is the anomalous dimension of m, not of m^2, so it differs by the obvious factor of two from a convention quoting the latter — the same caution recorded for the scalar mass applies.) Both γ_m^{QED} and the coefficient $\frac{3}{16}$ of Z_m are gauge independent — they describe the renormalization of the physical operator $\bar\psi\psi$ — whereas Z_2 and γ_2 are not.

What is gauge dependent and what is not

The constants Z_1, Z_2, and the electron field anomalous dimension γ_2 all depend on the gauge parameter ξ. In a general covariant (R_ξ) gauge one finds $\gamma_2 = \xi\,e^2/16\pi^2$, which is exactly why Eq. (24.117) is stated in Feynman gauge $\xi = 1$; in Landau gauge $\xi = 0$ the electron field does not renormalize at all at one loop, $\gamma_2 = 0$. None of this is a defect, because γ_2 is not by itself observable — it is the anomalous dimension of an off-shell, gauge-variant field. The combinations that *are* physical are gauge independent, and the Ward identity is what makes this work: $Z_1 = Z_2$ holds in *every* gauge, so the ξ-dependence cancels in $Z_e = Z_1 Z_2^{-1} Z_3^{-1/2} = Z_3^{-1/2}$, and β_e emerges gauge independent as it must. (Indeed Z_3 itself was gauge independent, so β_e is doubly protected.) Likewise γ_m^{QED}, the rate of a physical mass, is gauge independent. The working discipline: if a quantity you computed depends on ξ, it had better not be one you can measure.

24.3.5 A dispersive cross-check of the sign and coefficient of β_α

We obtained $\beta_\alpha = 2\alpha^2/3\pi$ from the ultraviolet pole of Z_3. There is a second route, completely independent of Z_3 and of dimensional regularization, that uses only the *imaginary* part of the vacuum polarization — a physical, finite, scheme-free quantity fixed by unitarity — and reconstructs the running through a dispersion relation. Because that imaginary part is manifestly positive (it is a total cross section, by the optical theorem of Chapter 18), this route also explains *why* the sign comes out as it does: charge screening, and hence $\beta_e > 0$, is forced by positivity of probability. We carry it through as a worked example, so that the carried number rests on two genuinely different arguments.

Worked Example 24.10 (β_α from the spectral function of Π)**.** The renormalized scalar vacuum polarization $\hat\Pi(q^2) \equiv \Pi(q^2) - \Pi(0)$ is analytic in the complex q^2 plane except for a branch cut along $q^2 \geq 4m^2$, where the virtual photon can fluctuate into a real on-shell e^+e^- pair. Its discontinuity across the cut — equivalently $2i\,\mathrm{Im}\,\hat\Pi(q^2)$ for $q^2 > 4m^2$ — is fixed by the optical theorem: by unitarity it is proportional to the cross section for $\gamma^* \to e^+e^-$. Carrying out the

two-body phase-space integral of the squared pair-production amplitude (the same trace-and-phase-space technology used for $e^+e^- \to \mu^+\mu^-$ in Chapter 17, now read in the crossed channel) gives

$$\operatorname{Im}\hat{\Pi}(q^2) = \frac{\alpha}{3}\left(1+\frac{2m^2}{q^2}\right)\sqrt{1-\frac{4m^2}{q^2}}\;\theta(q^2-4m^2), \tag{24.120}$$

which in the high-energy limit $q^2 \gg m^2$ approaches the constant

$$\operatorname{Im}\hat{\Pi}(q^2) \xrightarrow{q^2\to\infty} \frac{\alpha}{3}. \tag{24.121}$$

The decisive qualitative fact is that (24.120) is strictly *positive* for every $q^2 > 4m^2$: it is assembled from $|\text{amplitude}|^2$ integrated over a positive phase space, so positivity is not a feature of any particular regulator but of unitarity itself.

Because $\hat{\Pi}(0) = 0$ by construction and $\hat{\Pi}(q^2)/q^2$ falls at large $|q^2|$, $\hat{\Pi}$ obeys a once-subtracted dispersion relation,

$$\hat{\Pi}(q^2) = \frac{q^2}{\pi}\int_{4m^2}^{\infty}\frac{\mathrm{d}s\;\operatorname{Im}\hat{\Pi}(s)}{s\,(s-q^2-i\varepsilon)}, \tag{24.122}$$

the single subtraction at $q^2 = 0$ removing the otherwise logarithmically divergent dispersive integral and encoding the renormalization condition $\hat{\Pi}(0) = 0$. Evaluate it at large spacelike momentum $q^2 = -Q^2$ with $Q^2 \to \infty$. There the integrand is controlled by the region $4m^2 \ll s \lesssim Q^2$, where (24.121) permits the replacement $\operatorname{Im}\hat{\Pi}(s) \to \alpha/3$:

$$\begin{aligned}\hat{\Pi}(-Q^2) &\simeq \frac{-Q^2}{\pi}\cdot\frac{\alpha}{3}\int_{m^2}^{Q^2}\frac{\mathrm{d}s}{s\,(s+Q^2)} = \frac{-Q^2}{\pi}\cdot\frac{\alpha}{3}\cdot\frac{1}{Q^2}\Big[\ln\frac{s}{s+Q^2}\Big]_{m^2}^{Q^2}\\ &= -\frac{\alpha}{3\pi}\ln\frac{Q^2}{m^2}+\cdots,\end{aligned} \tag{24.123}$$

where we used $\int \mathrm{d}s/[s(s+Q^2)] = Q^{-2}\ln[s/(s+Q^2)]$, and the natural endpoints $s \sim m^2$ and $s \sim Q^2$ supply the lower and upper ends of the logarithm. The coefficient of $\ln Q^2$ is exactly $-(1/\pi)$ times the asymptotic spectral density (24.121): the *positivity* of $\operatorname{Im}\hat{\Pi}$ is precisely the *negativity* of the log-coefficient of $\hat{\Pi}(-Q^2)$. This sign is the whole point of the cross-check, so we underline it: nothing here was chosen by convention; the minus sign in front of $\ln(Q^2/m^2)$ is the sign of a cross section.

Now resum. Geometric iteration of the 1PI polarization along the photon line — the Dyson series of Fig. 24.6 — dresses the charge into the effective running coupling first written in Eq. (20.171),

$$\alpha(Q^2) = \frac{\alpha}{1+\hat{\Pi}(-Q^2)} = \frac{\alpha}{1-\dfrac{\alpha}{3\pi}\ln\dfrac{Q^2}{m^2}}, \tag{24.124}$$

where the second equality inserts (24.123); the sign in the denominator is fixed by that equation, and it makes the effective charge *grow* with Q^2, diverging at the Landau scale $Q^2 = m^2 \exp(3\pi/\alpha)$. Differentiating with respect to $\ln Q^2$,

$$\frac{d\alpha(Q^2)}{d\ln Q^2} = \frac{\alpha \cdot \dfrac{\alpha}{3\pi}}{\left(1 - \frac{\alpha}{3\pi}\ln\frac{Q^2}{m^2}\right)^2} = \frac{\alpha^2}{3\pi} + O(\alpha^3), \tag{24.125}$$

the denominator being $1 + O(\alpha)$ at fixed Q. Converting to the renormalization scale $\mu \sim Q$ through $\ln Q^2 = 2\ln\mu$, so that $\mathrm{d}/\mathrm{d}\ln\mu = 2\,\mathrm{d}/\mathrm{d}\ln Q^2$,

$$\beta_\alpha = \mu\frac{d\alpha}{d\mu} = 2\,\frac{d\alpha}{d\ln Q^2} = \frac{2\alpha^2}{3\pi}, \tag{24.126}$$

identical in sign and coefficient to Eq. (24.110). The two derivations are genuinely independent: the first used the ultraviolet $1/\varepsilon$ pole of Z_3 and the $\overline{\mathrm{MS}}$ pole equation; the second used the integral of the positive, physical spectral function $\mathrm{Im}\,\hat{\Pi}$ from threshold to Q^2, and never regulated a divergence at all. Their agreement checks the magnitude, and — through the positivity of (24.120) — the sign. The physical picture is the polarizable vacuum: a sea of virtual e^+e^- pairs behaves as a dielectric medium that *screens* the bare charge, so probing at shorter distance (larger Q) penetrates the screening cloud and exposes a larger effective charge. That screening, and the $\beta_e > 0$ it implies, are not conventions but consequences of the positivity of a cross section.

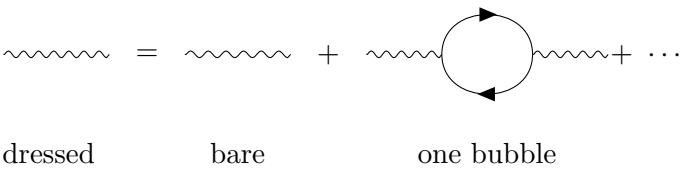


Figure 24.6: Dyson resummation of the photon propagator. Iterating the 1PI vacuum polarization (the bubble of Fig. 24.5) along the photon line sums the geometric series $1 + \Pi + \Pi^2 + \cdots = (1 - \Pi)^{-1}$, which dresses the charge into the running coupling $\alpha(Q^2) = \alpha/(1 + \hat{\Pi}(-Q^2))$ of Eq. (24.124). The same series, read as a differential equation, is the renormalization-group running whose slope is β_α.

Two faces of one number

The pole route ($Z_3^{-1/2} \to a_1 \to \beta_e$) and the dispersive route ($\mathrm{Im}\,\hat{\Pi} \to$ log coefficient $\to \beta_\alpha$) are the renormalization-group statement and the optical-theorem statement of one and the same physics. In the first, the logarithm is an ultraviolet divergence to be subtracted, and its residue *is*

the beta function. In the second, the very same logarithm is reconstructed from a finite, measurable absorptive part, and its slope *is* the beta function. That a short-distance divergence and a sum over physical intermediate states encode the identical coefficient is the hallmark of a renormalizable, unitary theory — it is the reason one may speak of *the* running coupling, independent of how one chose to expose it. It is also the cleanest possible answer to anyone who suspects the running is an artifact of regularization: the spectral derivation never regulates anything, and it gives the same $2\alpha^2/3\pi$.

With $\beta_e = e^3/12\pi^2 > 0$ and $\gamma_m^{\mathrm{QED}} = 3\alpha/2\pi$ now in hand — the latter ready to drive the running of the electron mass, the former a strictly positive slope — the chapter has delivered both carried numbers it owed QED, and has delivered the sign of the first one twice. Set beside the scalar result of the previous section, we now hold two computed beta functions of opposite microscopic origin but, as it happens, the same physical character: $\beta_\lambda = 3\lambda^2/16\pi^2 > 0$ and $\beta_e = e^3/12\pi^2 > 0$, both describing couplings that strengthen toward short distances — the screening that will become a Landau pole, in sharp contrast to the antiscreening of Yang–Mills awaiting in Chapter 27. The positive sign of β_e is the seed of the QED Landau pole and the triviality question, which Section 24.6 reaches by integrating the renormalization-group equation along its characteristics — a procedure that must reproduce the running $\alpha(q^2)$ of Eq. (20.171) as the leading term of its exact solution. Before solving anything, however, the next section must settle which features of β and γ are physical and which are mere artifacts of the $\overline{\mathrm{MS}}$ scheme, so that we know exactly how much of $\beta_e = e^3/12\pi^2$ and $\beta_\lambda = 3\lambda^2/16\pi^2$ to trust as convention-independent.

24.4 What Is Universal: Scheme Dependence of the Beta Function

We now hold two beta functions, each computed from scratch in the same regulator and the same scheme: the ϕ^4 result $\beta_\lambda = 3\lambda^2/16\pi^2 + O(\lambda^3)$ of Section 24.2 and the QED result $\beta_e = e^3/12\pi^2 + O(e^5)$ — equivalently $\beta_\alpha = 2\alpha^2/3\pi + O(\alpha^3)$ — of Section 24.3. Both were obtained in $\overline{\mathrm{MS}}$ with $d = 4 - 2\varepsilon$ and subtraction scale μ. A reader who has absorbed Chapter 22 should pause here with a genuine worry. A renormalized coupling is not an object we can place on a balance and weigh; it is a number manufactured by a *convention* for which finite pieces to discard along with the $1/\varepsilon$ poles, and Section 22.2.4 made plain that minimal subtraction is only one such convention among many. Momentum subtraction, on-shell subtraction, and lattice schemes each define a perfectly legitimate renormalized coupling, and they do not agree numerically: $\alpha_{\overline{\mathrm{MS}}}(\mu)$ and $\alpha_{\mathrm{MOM}}(\mu)$ at one common μ are

simply different numbers. If the coupling is a convention, its rate of change with scale would seem to inherit the same arbitrariness. In what sense, then, is the $3/16\pi^2$ of ϕ^4 or the $1/12\pi^2$ of QED a fact about Nature rather than a fact about $\overline{\text{MS}}$?

This section settles the question with a precise theorem and its corollaries. We formalize "changing scheme" at the level of the running coupling as a smooth, scale-independent reparametrization $\lambda \to \lambda'$, derive how the beta function transforms under it from nothing but the chain rule, and read off the consequence: the first two coefficients b_0 and b_1 are left rigidly invariant, while b_2 and everything past it are moved at will. We then show that the same mechanism protects the qualitative anatomy of the flow — the existence of a fixed point, the slope of β there, and the anomalous dimension there — so that critical exponents are unambiguous physical observables even though the coordinate of the fixed point is not. Finally we connect $\overline{\text{MS}}$ to a coupling *defined by a measurement*, an effective charge, and prove that its beta function carries the same b_0 and b_1. The outcome is a licence the rest of the chapter will spend freely: the zeros, slopes, and sign of the flow are physics, and the carried numbers b_0 are predictions, not conventions.

24.4.1 Scheme change as an analytic reparametrization of the coupling

The sharpest way to encode "the coupling is a convention" is to compare two renormalized couplings g and g' that describe the *same underlying physics* through different subtraction prescriptions imposed at the *same* scale μ. Each is then a finite function of the other, calculable order by order, because the gap between two schemes is a gap between finite subtraction constants — the poles they remove are identical, only the finite leftovers differ. For a marginal coupling — dimensionless in $d = 4$, as λ, α, and g^2_{YM} all are — this relation is an analytic power series with no constant term (both couplings vanish in the free theory) and unit slope at the origin (both reduce to the same classical coupling when interactions are switched off). We elevate this to a definition.

Definition 24.11 (Analytic coupling redefinition)**.** A *coupling redefinition* — equivalently, a perturbative change of scheme expressed on the running coupling — is an invertible analytic map

$$g' \;=\; F(g) \;=\; g + a_1 g^2 + a_2 g^3 + a_3 g^4 + \cdots , \tag{24.127}$$

whose coefficients $a_1, a_2, \ldots$ are *pure numbers independent of the scale* μ. Invertibility near $g = 0$ is automatic since $F'(0) = 1 \neq 0$; the inverse series is

$$g \;=\; g' - a_1 g'^2 + (2a_1^2 - a_2)\, g'^3 + \cdots . \tag{24.128}$$

Two features of Theorem 24.11 carry the entire argument and deserve emphasis before they are used.

The first is the absence of a constant term and the fixing of the linear term to unity. These are not cosmetic normalizations; they record that g and g' are two names for the strength of the *same* marginal interaction. At zero coupling the theory is free in every scheme, forcing $F(0) = 0$; and the leading response of one coupling to the other is one-to-one, $F'(0) = 1$, because at lowest order the coupling is scheme-independent — the tree-level vertex is the same diagram no matter how one chooses to subtract loops. All the content of a scheme change therefore resides in the quadratic and higher coefficients a_i, which are exactly the finite parts that distinguish, say, minimal subtraction from a momentum-subtraction condition.

The second is that the a_i are constants, free of μ. This is what makes Eq. (24.127) a redefinition of the coupling rather than a change of renormalization point. When one compares $g_{\overline{\mathrm{MS}}}(\mu)$ with $g_{\mathrm{MOM}}(\mu)$ at a single shared μ, the finite pieces that separate them are fixed numbers — the values of loop integrals at the fixed subtraction kinematics — not functions of μ. We return in Section 24.4.5 to why the μ-dependence cancels; for now constancy is the hypothesis of the theorem, and it is indispensable. Were the a_i allowed to depend on μ we would be changing the *scale*, not the *scheme*, and the chain rule below would acquire an inhomogeneous term that destroys every invariance we are about to prove.

Physical Insight

Eq. (24.127) makes the slogan "the coupling is unphysical" literal: the coupling is a *coordinate* on a one-dimensional space of theories, and a scheme change is a smooth change of coordinate that pins the origin (the free theory) and its first-order frame. Any genuine property of the space — whether it has a marked interior point (a fixed point), how the flow accelerates near that point (the slope), the scaling dimension of a field there (the anomalous dimension) — must be invariant under such a reparametrization. Any property of the *coordinate* alone — the numerical value g_* at the marked point, or the beta coefficients past the universal ones — need not be. The theorem of this section is precisely that geometric statement reduced to algebra.

24.4.2 The transformation law for the beta function

In each scheme the beta function is the rate of change of that scheme's coupling with $\ln \mu$,

$$\beta(g) \;=\; \mu \frac{dg}{d\mu} \;=\; \frac{dg}{d\ln \mu}, \qquad \beta_F(g') \;=\; \mu \frac{dg'}{d\mu}, \tag{24.129}$$

where β_F denotes the beta function *in the redefined scheme*, regarded as a function of its own argument g' — not a derivative of β. Because F in Eq. (24.127) carries no explicit μ-dependence, the scale reaches g' only through g, and the chain rule gives the transformation immediately:

$$\beta_F(g') \;=\; \mu\frac{dg'}{d\mu} \;=\; \frac{dg'}{dg}\,\mu\frac{dg}{d\mu} \;=\; F'(g)\,\beta(g). \tag{24.130}$$

This single identity is the master equation of the section. It states that the beta function transforms as a *vector field* on the coupling line under the change of coordinate $g \mapsto g' = F(g)$: the combination $\beta\,\partial_g = \beta_F\,\partial_{g'}$ is coordinate-free. Everything that follows is the extraction of consequences from Eq. (24.130).

To extract them, expand both beta functions in the marginal form — leading quadratic term, appropriate to λ and (in the α variable) to QED:

$$\beta(g) \;=\; b_0 g^2 + b_1 g^3 + b_2 g^4 + \cdots, \qquad \beta_F(g') \;=\; b'_0 g'^2 + b'_1 g'^3 + b'_2 g'^4 + \cdots. \tag{24.131}$$

The hypothesis that the series starts at $O(g^2)$ — that there is *no* linear term — is the very statement that the coupling is marginal: at the free theory $\beta(0) = 0$ and $\beta'(0) = 0$ (here the second prime *is* a derivative), so the flow has vanishing slope at the Gaussian fixed point and runs logarithmically rather than as a power. Every coupling this book runs meets this hypothesis; we note in Section 24.4.6 what changes when a coupling is instead relevant or irrelevant.

From Eq. (24.127), $F'(g) = 1 + 2a_1 g + 3a_2 g^2 + \cdots$. The right-hand side of Eq. (24.130), written as a series in the *unprimed* g, is therefore

$$\begin{aligned} F'(g)\,\beta(g) &= (1 + 2a_1 g + 3a_2 g^2 + \cdots)(b_0 g^2 + b_1 g^3 + b_2 g^4 + \cdots) \\ &= b_0\, g^2 + (b_1 + 2a_1 b_0) g^3 + (b_2 + 2a_1 b_1 + 3a_2 b_0) g^4 + \cdots. \end{aligned} \tag{24.132}$$

But Eq. (24.130) demands this equal $\beta_F(g')$, a series in the *primed* g'. To compare like with like we re-express Eq. (24.132) in powers of g' through the inverse map Eq. (24.128). Squaring and cubing that series,

$$g^2 = g'^2 - 2a_1\, g'^3 + (5a_1^2 - 2a_2) g'^4 + \cdots, \tag{24.133}$$

$$g^3 = g'^3 - 3a_1\, g'^4 + \cdots, \tag{24.134}$$

$$g^4 = g'^4 + \cdots. \tag{24.135}$$

The algebra behind Eq. (24.133) is worth displaying once. With $g = g' - a_1 g'^2 + (2a_1^2 - a_2) g'^3 + \cdots$,

$$g^2 = \big(g' - a_1 g'^2 + (2a_1^2 - a_2) g'^3\big)^2 = g'^2 - 2a_1 g'^3 + \big[a_1^2 + 2(2a_1^2 - a_2)\big] g'^4 + \cdots, \tag{24.136}$$

and $a_1^2 + 2(2a_1^2 - a_2) = 5a_1^2 - 2a_2$ as quoted; likewise $g^3 = g^2 \cdot g = (g'^2 - 2a_1 g'^3)(g' - a_1 g'^2) + \cdots = g'^3 - 3a_1 g'^4 + \cdots$. Substituting Eqs. (24.133) to (24.135) into Eq. (24.132) and collecting powers of g',

$$\beta_F(g') = b_0 \Big[g'^2 - 2a_1 g'^3 + (5a_1^2 - 2a_2) g'^4 \Big]$$

$$+ (b_1 + 2a_1 b_0)\left[g'^3 - 3a_1 g'^4\right] + (b_2 + 2a_1 b_1 + 3a_2 b_0)\, g'^4 + \cdots . \tag{24.137}$$

Reading off each power of g' and matching the primed expansion in Eq. (24.131) yields the central result of the chapter's honesty-accounting.

Theorem 24.12 (Invariance of the first two beta coefficients)**.** *Let $\beta(g) = b_0 g^2 + b_1 g^3 + b_2 g^4 + \cdots$ be the beta function of a marginal coupling, and let $g' = F(g) = g + a_1 g^2 + a_2 g^3 + \cdots$ be any analytic, scale-independent coupling redefinition (Theorem 24.11). Then the beta function $\beta_F(g') = b'_0 g'^2 + b'_1 g'^3 + b'_2 g'^4 + \cdots$ in the redefined scheme satisfies*

$$b'_0 = b_0, \tag{24.138}$$
$$b'_1 = b_1, \tag{24.139}$$
$$b'_2 = b_2 + (a_2 - a_1^2) b_0 - a_1\, b_1. \tag{24.140}$$

The first two coefficients are scheme-independent; the third, and every coefficient beyond it, depends on the redefinition.

Proof. Match Eq. (24.137) to Eq. (24.131) power by power.

Order g'^2. Only the first bracket contributes, giving $b'_0 = b_0$, which is Eq. (24.138). Diagrammatically this is the statement that a one-loop coefficient cannot move: a single loop integration carries no finite-subtraction ambiguity, because that ambiguity is itself a higher-order $(a_1\, g)$ effect.

Order g'^3. Two contributions appear: $b_0(-2a_1)$ from the first bracket and $(b_1 + 2a_1 b_0)$ from the second. Their sum is

$$b'_1 \;=\; -2a_1 b_0 + b_1 + 2a_1 b_0 \;=\; b_1. \tag{24.141}$$

The reparametrization pieces cancel *identically*, with no residue, for *every* value of a_1. This is Eq. (24.139), the nontrivial half of the theorem: the two-loop coefficient is universal.

Order g'^4. Three contributions appear: $b_0(5a_1^2 - 2a_2)$ from the first bracket, $(b_1 + 2a_1 b_0)(-3a_1)$ from the second, and $(b_2 + 2a_1 b_1 + 3a_2 b_0)$ from the third. Summing and reorganizing,

$$\begin{aligned} b'_2 &= b_0(5a_1^2 - 2a_2) - 3a_1(b_1 + 2a_1 b_0) + (b_2 + 2a_1 b_1 + 3a_2 b_0) \\ &= b_2 + b_0(5a_1^2 - 2a_2 - 6a_1^2 + 3a_2) + b_1(-3a_1 + 2a_1) \\ &= b_2 + b_0(a_2 - a_1^2) - a_1 b_1, \end{aligned} \tag{24.142}$$

which is Eq. (24.140). The shift is nonzero for generic a_1, a_2, so the three-loop coefficient is scheme-dependent. □

Two remarks sharpen what the theorem does and does not claim. First, the cancellation Eq. (24.141) at order g'^3 is exact and unconditional; it is not an approximate statement valid only for infinitesimal a_1. The two-loop coefficient is rigidly fixed for arbitrary reparametrizations. Second, the nonzero shift of b_2 in Eq. (24.140) is not a flaw of perturbation theory but the exact mechanism by which different schemes *stay mutually consistent*: a redefinition that altered b_1 would be a contradiction, whereas one that alters b_2 is precisely what is required to keep a physical quantity scheme-independent once it is re-expanded in the new coupling. Run the logic backwards: the freedom in (a_1, a_2) in Eq. (24.140) lets us set b_2' to *any* value, in particular to zero — and to all higher orders likewise. This is the existence proof of the 't Hooft scheme.

Corollary 24.13 ('t Hooft minimal scheme)**.** *For any marginal coupling there exists a scheme in which* $\beta(g) = b_0 g^2 + b_1 g^3$ exactly*, with every coefficient beyond* b_1 *vanishing. The required change of variable is fixed order by order: at the order of Eq.* (24.140) *one solves* $b_2 + (a_2 - a_1^2)b_0 - a_1 b_1 = 0$ *for one combination of* (a_1, a_2)*, and the pattern continues indefinitely.*

The abstract shift formula becomes more convincing as an explicit, checkable computation.

Worked Example 24.14 (An explicit scheme change moves b_2 but not b_0, b_1)**.** Take a toy beta function with integer coefficients,

$$\beta(g) = g^2 + g^3 + g^4, \qquad (b_0, b_1, b_2) = (1, 1, 1), \tag{24.143}$$

and the concrete redefinition $g' = F(g) = g + g^2$, i.e. $a_1 = 1$, $a_2 = 0$, all higher $a_i = 0$. We compute $\beta_F(g')$ from first principles and verify Theorem 24.12.

Step 1: the inverse map. Solving $g' = g + g^2$ on the branch through the origin,

$$g = \frac{-1 + \sqrt{1 + 4g'}}{2} = g' - g'^2 + 2g'^3 - 5g'^4 + \cdots, \tag{24.144}$$

which matches $g = g' - a_1 g'^2 + (2a_1^2 - a_2)g'^3 + \cdots = g' - g'^2 + 2g'^3 + \cdots$ from Eq. (24.128) at $a_1 = 1, a_2 = 0$.

Step 2: the Jacobian. $F'(g) = 1 + 2g$.

Step 3: assemble β_F *as a series in* g*.* From Eq. (24.130),

$$\beta_F(g') = F'(g)\beta(g) = (1 + 2g)(g^2 + g^3 + g^4) = g^2 + 3g^3 + 3g^4 + 2g^5. \tag{24.145}$$

Step 4: re-express in g'*.* Using $g^2 = g'^2 - 2g'^3 + 5g'^4 + \cdots$, $g^3 = g'^3 - 3g'^4 + \cdots$, $g^4 = g'^4 + \cdots$ (these are Eqs. (24.133) to (24.135) at $a_1 = 1, a_2 = 0$),

$$\begin{aligned} \beta_F(g') &= (g'^2 - 2g'^3 + 5g'^4) + 3(g'^3 - 3g'^4) + 3g'^4 + O(g'^5) \\ &= g'^2 + (-2 + 3)g'^3 + (5 - 9 + 3)g'^4 + O(g'^5) \end{aligned}$$

$$= g'^2 + g'^3 - g'^4 + O(g'^5). \tag{24.146}$$

Step 5: read off and check. We find $(b'_0, b'_1, b'_2) = (1, 1, -1)$: indeed $b'_0 = b_0 = 1$ and $b'_1 = b_1 = 1$ are unmoved, while b_2 has flipped from $+1$ to -1. The shift formula Eq. (24.140) predicts exactly this,

$$b'_2 = b_2 + (a_2 - a_1^2)b_0 - a_1 b_1 = 1 + (0-1)(1) - (1)(1) = 1 - 1 - 1 = -1. \checkmark \tag{24.147}$$

The leading two coefficients are inert; the third moves by precisely the amount the theorem dictates.

24.4.3 the benchmark theories: b_0 as a physical prediction

Theorem 24.12 is exactly the certificate the two threads were waiting for. For ϕ^4 theory the coupling already has the marginal, quadratic-leading form $\beta_\lambda = b_0\lambda^2 + b_1\lambda^3 + \cdots$ with the one-loop coefficient

$$b_0 = \frac{3}{16\pi^2} \tag{24.148}$$

of Section 24.2. By the theorem this number is identical in $\overline{\text{MS}}$, in any momentum-subtraction scheme, and in any other analytic definition of the ϕ^4 coupling: it is the physical statement that the $\phi\phi \to \phi\phi$ effective coupling grows logarithmically toward the ultraviolet at the universal rate Eq. (24.148). The two-loop coefficient b_1 is universal as well — we have not computed it, but the theorem guarantees that whoever does will obtain a scheme-independent number — and only at three loops does scheme dependence first appear.

For QED the gauge coupling e has a beta function odd in e, $\beta_e = b_0^{(e)} e^3 + b_1^{(e)} e^5 + \cdots$, because physics depends on e only through e^2 (equivalently α): the theory is invariant under $e \to -e$, so no even powers can appear. To apply Theorem 24.12 in its quadratic-leading form we pass to the genuinely marginal variable $\alpha = e^2/4\pi$. The translation is a one-line chain-rule computation, which we record because it fixes the dictionary between the e- and α-coefficients and confirms the internal consistency of the two carried numbers.

Worked Example 24.15 (From β_e to β_α, and the universality of b_0)**.** Start from $\alpha = e^2/4\pi$, so that $\frac{d\alpha}{d\ln\mu} = \frac{e}{2\pi}\frac{de}{d\ln\mu}$, that is

$$\beta_\alpha = \frac{e}{2\pi}\,\beta_e. \tag{24.149}$$

Insert the one-loop $\beta_e = b_0^{(e)} e^3$ with $b_0^{(e)} = 1/12\pi^2$ from Section 24.3:

$$\beta_\alpha = \frac{e}{2\pi}\, b_0^{(e)} e^3 = \frac{b_0^{(e)}}{2\pi}\, e^4 = \frac{b_0^{(e)}}{2\pi}\,(4\pi\alpha)^2 = \frac{16\pi^2 b_0^{(e)}}{2\pi}\,\alpha^2 = 8\pi\, b_0^{(e)}\,\alpha^2. \tag{24.150}$$

With $b_0^{(e)} = 1/12\pi^2$,

$$\beta_\alpha = \frac{8\pi}{12\pi^2}\,\alpha^2 = \frac{2\alpha^2}{3\pi}, \qquad \text{so} \qquad b_0^{(\alpha)} = \frac{2}{3\pi}, \tag{24.151}$$

exactly the QED carried number $\beta_\alpha = 2\alpha^2/3\pi$. The marginal coupling α satisfies the hypothesis of Theorem 24.12, so $b_0^{(\alpha)} = 2/3\pi$ is scheme-independent: it is the physical rate at which the effective fine-structure constant grows with energy. Because $e \mapsto \alpha = e^2/4\pi$ is a fixed, scale-independent function, universality of $b_0^{(\alpha)}$ is equivalent to universality of $b_0^{(e)} = 1/12\pi^2$. Both are facts about the world, not about $\overline{\text{MS}}$.

Notation Watch

The number of universal coefficients is two whether one expands in the marginal variable (λ, α) with leading power g^2, or in a gauge coupling (e, g_{YM}) with leading power g^3. In the latter case the standard statement is that $b_0^{(g)}$ (the g^3 coefficient) and $b_1^{(g)}$ (the g^5 coefficient) are scheme-independent. These are the same statement: under $\alpha = g^2/4\pi$ the g^3- and g^5-coefficients map one-to-one onto the α^2- and α^3-coefficients, and Theorem 24.15 is that map at leading order. No extra universal coefficient hides in the gauge variable.

24.4.4 Fixed points: the location moves, existence and slope do not

The theorem reaches beyond the leading coefficients to protect the entire qualitative anatomy of the flow. A *fixed point* (Theorem 23.19) is a coupling value g_* at which the flow stops, $\beta(g_*) = 0$ with $g_* \neq 0$; the free theory $g = 0$ is the trivial fixed point common to every scheme. The Wilsonian development of Chapter 23, through Section 23.6, established that fixed points are the scale-invariant theories around which the RG organizes universality classes. Here we show that the data attached to such a point are scheme-independent in exactly the senses that matter.

Proposition 24.16 (Fixed points are scheme-covariant)**.** *Let g_* be a fixed point of β, $\beta(g_*) = 0$, lying where the redefinition F of Theorem 24.11 is invertible ($F'(g_*) \neq 0$). Then $g'_* = F(g_*)$ is a fixed point of β_F, and conversely every fixed point of β_F arises this way. The existence of a nontrivial fixed point is therefore scheme-independent, while its location g_* is not.*

Proof. From Eq. (24.130), $\beta_F(g'_*) = F'(g_*)\,\beta(g_*) = F'(g_*) \cdot 0 = 0$. Since F is a bijection near g_* with $(F^{-1})'(g'_*) = 1/F'(g_*)$, the zeros of β and β_F are in one-to-one correspondence through $g \mapsto F(g)$. The value $g'_* = F(g_*) = g_* + a_1 g_*^2 + \cdots$ differs from g_*, so the location is convention-dependent. □

That the location is unphysical might appear to drain a fixed point of content, but the physically meaningful data are not its coordinate. They are the *slope* of the beta function there and the *value* of the anomalous dimension there, and both are invariant. The slope governs how fast the flow approaches or recedes from the fixed point; up to a sign convention it is a critical exponent — the correction-to-scaling exponent ω.

Theorem 24.17 (The slope at a fixed point is scheme-independent)**.** *Let g_* be a nontrivial fixed point and set $\omega \equiv \beta'(g_*)$ (here the prime is the derivative $d\beta/dg$). Under any redefinition F of Theorem 24.11 with $F'(g_*) \neq 0$, the slope of the transformed beta function at the transformed fixed point equals ω:*

$$\left.\frac{d\beta_F}{dg'}\right|_{g'=g'_*} = \omega. \tag{24.152}$$

Proof. Write the transformation law as $\beta_F(g') = F'(g)\,\beta(g)$ with $g = F^{-1}(g')$ and differentiate with respect to g', using $dg/dg' = 1/F'(g)$:

$$\begin{aligned}\frac{d\beta_F}{dg'} &= \frac{d}{dg'}\Big[F'(g)\,\beta(g)\Big] = \Big[F''(g)\,\beta(g) + F'(g)\,\beta'(g)\Big]\,\frac{dg}{dg'} \\ &= \frac{F''(g)\,\beta(g) + F'(g)\,\beta'(g)}{F'(g)}.\end{aligned} \tag{24.153}$$

Evaluate at $g = g_*$, where $\beta(g_*) = 0$. The first term in the numerator — the only place the second-order data F'' of the redefinition could enter — is killed, leaving

$$\left.\frac{d\beta_F}{dg'}\right|_{g'_*} = \frac{F'(g_*)\,\beta'(g_*)}{F'(g_*)} = \beta'(g_*) = \omega, \tag{24.154}$$

valid because $F'(g_*) \neq 0$ by hypothesis. □

The very same vanishing of β at the fixed point makes the anomalous dimension invariant there. Recall from Section 24.2 that the field anomalous dimension $\gamma(g) = \tfrac{1}{2}\mu\, d\ln Z_\phi/d\mu$ measures the departure of a field's scaling from its classical (engineering) dimension. As a function of the coupling, γ is itself scheme-dependent: it can be shifted by a finite, coupling-dependent rescaling of the field, which is part of what a scheme fixes. Its value precisely *at* a fixed point, however, is physical — it is the genuine anomalous dimension of the field in the scale-invariant theory there, read off from the power-law fall-off of correlators.

Proposition 24.18 ($\gamma(g_*)$ is scheme-independent)**.** *A finite, analytic, coupling-dependent field redefinition $\phi \to \phi' = Z_{\mathrm{f}}(g)^{1/2}\phi$ with $Z_{\mathrm{f}}(g) = 1 + c_1 g + c_2 g^2 + \cdots$ shifts the anomalous dimension by*

$$\gamma_F(g) = \gamma(g) + \tfrac{1}{2}\,\beta(g)\,\frac{d\ln Z_{\mathrm{f}}(g)}{dg}. \tag{24.155}$$

At any fixed point g_* the added term vanishes because $\beta(g_*) = 0$, so $\gamma_F(g_*) = \gamma(g_*)$. Together with the relabeling of the coupling itself, the number $\gamma(g_*)$ is scheme-independent.

Proof. The total field renormalization in the redefined convention is $Z'_\phi = Z_{\rm f}\, Z_\phi$, so $\ln Z'_\phi = \ln Z_{\rm f} + \ln Z_\phi$. Applying $\frac{1}{2}\mu\, d/d\mu$,

$$\gamma_F(g) = \gamma(g) + \tfrac{1}{2}\,\mu\,\frac{d}{d\mu}\ln Z_{\rm f}(g) = \gamma(g) + \tfrac{1}{2}\,\frac{d\ln Z_{\rm f}}{dg}\,\mu\,\frac{dg}{d\mu}, \tag{24.156}$$

and $\mu\, dg/d\mu = \beta(g)$ gives Eq. (24.155). At $g = g_*$ the prefactor $\beta(g_*) = 0$ removes the shift. The pure relabeling $g \to g' = F(g)$ sends the function γ to $\gamma \circ F^{-1}$, whose value at $g'_* = F(g_*)$ is $\gamma(F^{-1}(g'_*)) = \gamma(g_*)$; both operations preserve the fixed-point value. □

Theorems 24.17 and 24.18 together certify that the *physical critical exponents* of a fixed point are unambiguous. The correction-to-scaling exponent is $\omega = \beta'(g_*)$; the field's anomalous scaling enters the standard exponent η as $\eta = 2\gamma_\phi(g_*)$; and the exponent ν controlling the relevant (mass-like) direction is the eigenvalue of the linearized flow about the fixed point in that direction, fixed by the same argument applied to the mass anomalous dimension. The coordinate of the fixed point in any particular coupling is bookkeeping; the exponents are facts about the universality class. This is the precise sense in which the Wilson–Fisher fixed point of Chapter 52, sitting at the scheme-dependent value $\lambda_* = \frac{32\pi^2}{3}\varepsilon + O(\varepsilon^2)$, nonetheless predicts the *measurable* critical exponents of the three-dimensional Ising universality class. Figure 24.7 renders the whole statement geometrically.

24.4.5 Contact with a measurable effective charge

The theorems above compare coupling *conventions*. To see that b_0 and b_1 are not merely invariant across our menu of bookkeeping schemes but are genuinely *measurable*, we connect $\overline{\rm MS}$ to a coupling that is an observable by construction. This is the momentum-subtraction idea of Section 22.2.4 pushed to its physical conclusion: an *effective charge*.

Take QED. The vacuum polarization $\Pi(q^2)$ computed in Chapter 19 dresses the photon propagator and, after the geometric resummation of one-particle-irreducible insertions performed in Section 24.3 (and, for the static potential, in the running-α discussion of Chapter 20), yields an effective coupling that controls a genuine observable. The cleanest example is the static potential between two heavy charges,

$$V(r) = -\frac{\alpha_V(1/r)}{r}, \tag{24.157}$$

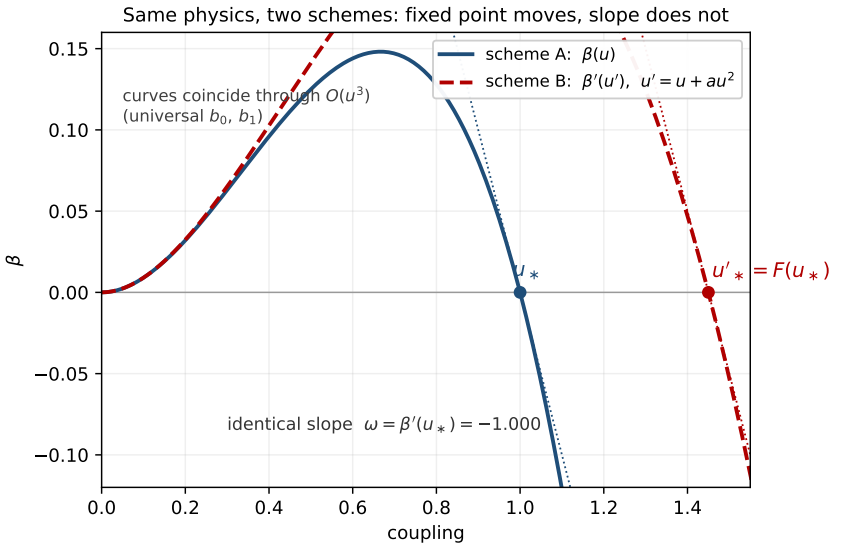


Figure 24.7: Two coupling conventions u and $u' = F(u) = u + au^2$ describing the *same* physics. The two beta functions are related by the exact transformation law Eq. (24.130); they share their first two coefficients b_0, b_1 and therefore hug each other near the origin before peeling apart. Each has a nontrivial fixed point, but at a *different* location (u_* versus $u'_* = F(u_*)$, Theorem 24.16). The tangent slope at the respective fixed point is *identical* ($\omega = \beta'(u_*)$, Theorem 24.17): the physical critical exponent does not know which coordinate we chose. Generated by `figures/24-scheme-beta.py`.

which *defines* the effective charge $\alpha_V(\mu)$ at momentum scale $\mu = 1/r$ as a property of a measurable energy. Other equally physical effective charges can be read off cross sections — the ratio $R(s)$ in e^+e^- annihilation defines one, the Adler D-function another — and each is scheme-free by construction: it is extracted from data, not from a subtraction rule.

The relation between such a physical charge and the $\overline{\text{MS}}$ coupling at a common scale is exactly an instance of Theorem 24.11. Computing the observable in $\overline{\text{MS}}$ perturbation theory gives, schematically,

$$\alpha_V(\mu) = \alpha_{\overline{\text{MS}}}(\mu)\Big[1 + c_1\,\alpha_{\overline{\text{MS}}}(\mu) + c_2\,\alpha_{\overline{\text{MS}}}(\mu)^2 + \cdots\Big], \tag{24.158}$$

where the c_i are the *finite* parts left after minimal subtraction (per Theorem 22.7) has removed the poles — pure numbers, the values of loop integrals at fixed dimensionless ratios of external scales to μ. The decisive point is that when α_V and $\alpha_{\overline{\text{MS}}}$ are compared *at the same scale* μ, the c_i carry no μ-dependence: the explicit μ has cancelled between the two sides, both couplings being dimensionless and renormalized at the same point. Equation Eq. (24.158) is therefore precisely

the analytic, scale-independent reparametrization of Theorem 24.11, with $F(\alpha) = \alpha + c_1\alpha^2 + \cdots$. (This is also the promised reason the a_i of Theorem 24.11 are μ-independent constants.)

Theorem 24.12 now does its work. The effective charge α_V runs with its own beta function $\beta_{\alpha_V} = b_0^V \alpha_V^2 + b_1^V \alpha_V^3 + \cdots$, and because α_V is related to $\alpha_{\overline{\text{MS}}}$ by Eq. (24.158),

$$b_0^V = b_0^{\overline{\text{MS}}} = \frac{2}{3\pi}, \qquad b_1^V = b_1^{\overline{\text{MS}}}. \tag{24.159}$$

The first two beta coefficients of a *measurable* effective charge equal the first two $\overline{\text{MS}}$ coefficients. This is the bridge from convention to observation: the rate $b_0 = 2/3\pi$ at which the QED effective charge grows with energy can be extracted from the energy dependence of an actual scattering rate, and it agrees with the number we obtained by subtracting $1/\varepsilon$ poles. The ϕ^4 rate $b_0 = 3/16\pi^2$ enjoys the same status. The carried numbers are physics.

Physical Insight

The deep reason the first two coefficients are physical and the rest are not fits in one line. One loop sees no ambiguity because there is nothing yet to redefine; two loops are protected because the only available redefinition acts one order higher and *cancels* in the slope (the exact cancellation Eq. (24.141)); from three loops onward the finite-subtraction freedom is rich enough to move the coefficient, and Theorem 24.13 shows it can be set to anything — including zero. "How many universal numbers are there" is answered by "how many orders pass before the scheme freedom can act," and the answer is two.

24.4.6 What is convention and what is not

It is worth stating plainly which RG data may be quoted as physical and which are conveniences, since this distinction is the payoff of the section and is easy to blur.

Caution

$\overline{\text{MS}}$ couplings and masses are not directly observable, and higher $\overline{\text{MS}}$ coefficients are not to be over-read. The number $\alpha_{\overline{\text{MS}}}(\mu)$ is defined by a subtraction rule, not a measurement; it depends on the scheme *and* on the arbitrary scale μ. Likewise the $\overline{\text{MS}}$ mass $m_{\overline{\text{MS}}}(\mu)$ is not the pole mass and is not what a detector weighs — the two differ by a finite, scheme-dependent, calculable factor. Never report an $\overline{\text{MS}}$ quantity as if it were an experimental result, and never compare $\overline{\text{MS}}$ numbers from two calculations without checking they use the same μ and the same scheme.

In particular, a three-loop or higher $\overline{\text{MS}}$ beta coefficient $b_2, b_3, \ldots$ is *not* a clean prediction in isolation: by Theorem 24.13 its value can be changed, even to zero, by an analytic redefinition of the coupling, so a statement like "b_2 in $\overline{\text{MS}}$ is such-and-such" is a joint fact about the dynamics *and* the scheme. What *is* physical:

- the first two beta coefficients b_0, b_1 (Theorem 24.12), and hence the leading and next-to-leading logarithmic running;
- the existence of any fixed point, the slope $\beta'(g_*)$, and the anomalous dimension $\gamma(g_*)$ there — the critical exponents (Theorems 24.17 and 24.18);
- any genuine observable (a cross section, decay rate, the static potential), which is scheme-independent once expressed through other observables;
- the renormalization-group-invariant scale Λ that the running coupling defines (Section 24.6), *up to* an exact, one-loop-computable conversion between schemes, so that $\Lambda_{\overline{\text{MS}}}$ and Λ_{MOM} differ by a known constant, not by anything unknown.

A litmus test: a quantity is physical if and only if it survives every analytic, scale-independent redefinition of the coupling and every choice of μ.

Two boundary remarks close the picture. First, the count "two universal coefficients" is special to a *marginal* coupling, whose beta function starts at $O(g^2)$. For a coupling that is relevant or irrelevant at the free theory the beta function instead has a nonzero linear term $\beta = b_{-1}g + \cdots$, where b_{-1} is the classical dimension plus a free-field anomalous dimension; then it is b_{-1} that is universal — it is the eigenvalue of the linearized flow at $g = 0$, an honest scaling dimension — and the quadratic term already carries scheme dependence. The pattern never changes: the universal data are exactly the coefficients fixed before the finite-subtraction freedom can act, which is why the eigenvalues of the linearized flow at *any* fixed point — the critical exponents — are invariant, as Theorem 24.17 proved directly.

Second, none of this renders the running coupling meaningless. It says the running coupling is a *coordinate*, and a superbly useful one: it organizes large logarithms into a single function, it makes asymptotic freedom and Landau poles visible at a glance, and its scheme-dependence is fully controlled by the transformation law Eq. (24.130). The remaining sections adopt exactly this attitude. Having certified that b_0 and b_1 are physical, that fixed-point slopes and anomalous dimensions are unambiguous, and that the sign and zeros of the flow are coordinate-free, we are licensed to treat the flow *itself* as physics: to integrate

the renormalization-group equation by characteristics and resum the leading logarithms into the closed-form running of $\lambda(\mu)$ and $\alpha(\mu)$ (Section 24.6), and to draw from the sign and zeros of β the asymptotic verdicts — Landau poles, triviality, asymptotic freedom, and the fixed-point/critical-exponent language (Section 24.7) that also organizes the Callan–Symanzik analysis of Section 24.5 — secure in the knowledge that none of these conclusions depends on the scheme we happened to compute in.

24.5 The Callan–Symanzik Equation: Broken Scale Invariance and the Trace Anomaly

The previous section earned us a licence. Having proved that the leading flow data — the coefficients b_0, b_1 of the beta function and the critical exponents attached to any fixed point (Section 24.4) — survive every analytic redefinition of the coupling, we may now read the $\overline{\text{MS}}$ numbers $\beta_\lambda = 3\lambda^2/16\pi^2$ and $\beta_e = e^3/12\pi^2$ computed in Sections 24.2 and 24.3 as facts about the world rather than facts about a subtraction rule. But *why* the world should contain such a flow at all has so far been left to a bookkeeping argument. The renormalization-group equation of Section 24.1 was forced on us by the observation that the subtraction scale μ carries no physical meaning, so that renormalized correlation functions cannot truly depend on it; the coefficients β, γ, γ_m are the exact price of holding that constraint. That derivation is airtight, but it hides the physics behind a choice of scheme. There is a second, older, and physically transparent route to the same equation, and it is the subject of this section.

Historically the equation was discovered from the opposite direction — not from the arbitrariness of μ, but from the breakdown of a classical *symmetry*. A massless classical field theory in four dimensions is invariant under rescalings of spacetime: this is the scale (dilatation) symmetry of Chapter 7. Callan and Symanzik asked what becomes of that symmetry in the quantum theory, and found that it is broken — inevitably, by the very act of renormalizing — with the operational β and γ of the previous sections emerging as the exact quantitative measure of the breaking. This section reconstructs that route. It gives the running coupling a physically transparent face, exposes the celebrated *trace anomaly* that sits at its heart, derives the Callan–Symanzik equation proper as a broken dilatation Ward identity, and then proves that the Callan–Symanzik equation and the μ-flow equation of Section 24.1 are one and the same statement seen from two sides. The payoff is conceptual and concrete at once: $\beta \neq 0$, "the coupling runs," and "the stress tensor has a nonzero trace" will turn out to be three names for a single phenomenon, and the carried number $\beta_\lambda = 3\lambda^2/16\pi^2$ will be revealed as simultaneously the one-loop trace anomaly of ϕ^4 theory.

24.5.1 Classical scale invariance and the trace of the stress tensor

We begin by recalling, from Chapter 7, exactly how scale invariance is encoded locally. A dilatation acts on coordinates and fields by $x^\mu \to e^{-\omega}x^\mu$, $\phi(x) \to e^{\omega\Delta}\phi(e^\omega x)$, with Δ the scaling dimension of Theorem 7.42 (for a scalar in d dimensions, $\Delta = (d-2)/2$, equal to one in $d = 4$). Noether's construction produces the dilatation current $D^\mu = x_\nu\,\Theta^{\mu\nu} + V^\mu$, the sum of an orbital piece built from the (improved, symmetric) stress tensor $\Theta^{\mu\nu}$ and a field virial V^μ. The content of Theorem 7.43 is the local identity

$$\partial_\mu D^\mu \;=\; \Theta^\mu{}_\mu \qquad \text{(improved tensor, on shell)}, \tag{24.160}$$

the virial having been absorbed into the improvement of $\Theta^{\mu\nu}$ so that the whole response of the theory to a rescaling is carried by a single local operator: the *trace* of the stress tensor. Scale invariance is therefore equivalent to the on-shell vanishing of that trace. The massless classical theory is scale invariant precisely *because* its improved trace vanishes; a nonzero trace is the local fingerprint of broken scale symmetry, and a conserved dilatation current ($\partial_\mu D^\mu = 0$) is the statement that no such fingerprint exists.

For ϕ^4 theory this fingerprint was computed in Chapters 6 and 7. The carried Lagrangian is

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{24.161}$$

and the classical, on-shell trace of its improved stress tensor in general spacetime dimension d is

$$\Theta^\mu{}_\mu \;=\; m^2\phi^2 \;+\; (4-d)\,\frac{\lambda}{4!}\,\phi^4. \tag{24.162}$$

The structure of Eq. (24.162) is worth pausing on, because each term tells a piece of the story the rest of the section will complete. The first term, $m^2\phi^2$, is the mass breaking: it is present in any dimension and is proportional to the one dimensionful coupling of the theory. In exactly $d = 4$ it is the whole of the classical trace, $\Theta^\mu{}_\mu = m^2\phi^2$ (Eq. (7.213)), so that on shell $\partial_\mu D^\mu = m^2\phi^2$: the mass, and the mass alone, breaks classical scale invariance. The second term carries the prefactor $(4-d)$ and therefore *vanishes identically in four dimensions*. This is the precise classical content of the statement that the quartic coupling λ is marginal — the engineering dimension of the coupling is $[\lambda] = 4 - d = 0$ in $d = 4$, so the ϕ^4 interaction contributes nothing to the trace and does not obstruct scale invariance (Eq. (6.182)). The massless ($m = 0$) classical theory in $d = 4$ has $\Theta^\mu{}_\mu = 0$ and is scale — indeed conformally — invariant.

Everything in Eq. (24.162) is exact at the classical level and survives no further. The remainder of this section is the story of what quantization does to it.

24.5.2 Quantization breaks scale invariance: the trace anomaly

The classical conclusion "massless ϕ^4 is scale invariant" rests on the literal vanishing of the $(4-d)$ coefficient at $d=4$. Quantization undermines exactly this step, and it does so for a reason that is by now familiar: the renormalized quantum theory cannot be defined without introducing a mass scale. In dimensional regularization, where one works at $d=4-2\varepsilon$ and trades the bare coupling for a renormalized one through

$$\lambda_0 \;=\; \mu^{2\varepsilon}\, Z_\lambda\, \lambda, \tag{24.163}$$

the explicit factor $\mu^{2\varepsilon}$ is unavoidable: it is what keeps λ dimensionless away from four dimensions, exactly the factor whose μ-derivative drove the one-loop extraction of Section 24.2. (We write Z_λ for the combination of vertex and wavefunction constants that renders the renormalized quartic vertex finite, in the normalization of Section 24.2; the precise grouping is irrelevant to what follows.) That single factor $\mu^{2\varepsilon}$ is the seed of the anomaly. Setting $4-d=2\varepsilon$ in Eq. (24.162), the quartic contribution to the bare trace is

$$(4-d)\,\frac{\lambda_0}{4!}\,\phi_0^4 \;=\; 2\varepsilon\,\frac{\lambda_0}{4!}\,\phi_0^4, \tag{24.164}$$

which appears to be $O(\varepsilon)$ and to disappear as $\varepsilon\to 0$. It does not. The composite operator ϕ_0^4, expressed in renormalized quantities, is divergent: its insertion into Green's functions carries a $1/\varepsilon$ pole, exactly the divergence whose subtraction defines Z_λ. The product of the explicit 2ε with that $1/\varepsilon$ pole is finite, and the finite remainder is nonzero. This is the same arithmetic — an explicit factor of the regulator multiplying a pole produced by the regularization — that produces every anomaly in this book. Carrying out the renormalization (the d-dimensional beta function is $\beta_d(\lambda)=-2\varepsilon\lambda+\beta(\lambda)$, with $\beta(\lambda)$ the four-dimensional function of Section 24.1, as established at Eq. (24.163) above), the finite limit is

$$\lim_{\varepsilon\to 0}\; 2\varepsilon\,\frac{\lambda_0}{4!}\,\phi_0^4 \;=\; \frac{\beta(\lambda)}{4!}\,\big[\phi^4\big]_R, \tag{24.165}$$

where $[\phi^4]_R$ is the renormalized quartic operator. The trace that vanished classically is revived by the quantum theory, and its size is set by the beta function. Including the mass term, which is corrected only by the multiplicative mass anomalous dimension, the renormalized trace of the stress tensor reads

$$\boxed{\Theta^{\mu}{}_{\mu} \;=\; (1+\gamma_m)\, m^2\,\big[\phi^2\big]_R \;+\; \frac{\beta(\lambda)}{4!}\,\big[\phi^4\big]_R} \tag{24.166}$$

This is the *trace anomaly* of ϕ^4 theory, and it is the central physical statement of the chapter rendered as a local operator equation. It is exactly the relation Eq. (7.214) previewed in Chapter 7. The massless quantum theory ($m=0$) has

$\Theta^{\mu}{}_{\mu} = \frac{\beta(\lambda)}{4!}[\phi^4]_R \neq 0$: scale invariance, exact classically, is broken by quantization, and the breaking is proportional to β.

The form often quoted is the same statement with the engineering normalization of the vertex pulled out front. Since the classical quartic term in the Lagrangian is $-\frac{\lambda}{4!}\phi^4$, the operator $\frac{1}{4!}[\phi^4]_R$ is exactly $-\partial\mathcal{L}/\partial\lambda$, and one writes the anomalous piece of Eq. (24.166) as $\frac{\beta(\lambda)}{\lambda}\cdot\frac{\lambda}{4!}[\phi^4]_R$; the cleanest invariant way to express the coefficient is that the anomalous part of the trace is $\beta(\lambda)$ times the renormalized operator conjugate to the coupling. We will use the explicit form Eq. (24.166) throughout, but record the slogan in the form most often seen.

Notation Watch

"$\Theta^{\mu}{}_{\mu} = \frac{\beta}{\lambda}[\phi^4]$." Many texts quote the massless trace anomaly as $\Theta^{\mu}{}_{\mu} = \frac{\beta(\lambda)}{\lambda}\mathcal{O}_\lambda$ with $\mathcal{O}_\lambda = \frac{\lambda}{4!}\phi^4$ the (renormalized) interaction operator itself — the term that appears in the Lagrangian. This is identical to the ϕ^4 piece of Eq. (24.166): substituting $\mathcal{O}_\lambda = \frac{\lambda}{4!}[\phi^4]_R$ gives $\frac{\beta}{\lambda}\cdot\frac{\lambda}{4!}[\phi^4]_R = \frac{\beta}{4!}[\phi^4]_R$, which matches Eq. (24.166) exactly; the two forms differ only by whether one counts the operator ϕ^4 with weight 1 or the interaction $\frac{\lambda}{4!}\phi^4$ with weight $\frac{\beta}{\lambda}$. Both carry the *same* coefficient β in front of the conjugate operator $\mathcal{O}_\lambda/\lambda = -\partial\mathcal{L}/\partial\lambda$. We keep Eq. (24.166) as the canonical form and note this equivalence so the reader meets no contradiction with the literature.

Caution

What is owned here and what is deferred. The *value* of the anomaly coefficient is the beta function of Section 24.2; in that sense Eq. (24.166) contains no new quantity, and the explicit one-loop number $\beta(\lambda) = 3\lambda^2/16\pi^2$ already computed there is simultaneously the one-loop trace anomaly. What we have *not* done is give the fully rigorous operator derivation of Eq. (24.166): the precise definition and mixing of the renormalized composite operators $[\phi^2]_R$ and $[\phi^4]_R$, and the demonstration that the finite remainder in Eq. (24.165) is unambiguous and scheme-independent, belong to the theory of anomalies. They are carried out in Chapter 36, where the trace anomaly is placed alongside the chiral anomaly as a quantum violation of a classical conservation law, and where the Fujikawa-measure viewpoint exhibits it as a Jacobian of the path-integral measure under rescaling. For the present chapter we take Eq. (24.166) as the bridge between the abstract μ-independence of Section 24.1 and the physical language of broken scale invariance; the soft-insertion identities derived in Section 24.5.4 make that bridge explicit and are proved in full.

Physical Insight

$\beta \neq 0$ means classical scale symmetry is anomalous. This is the unifying statement of the chapter, and it is worth saying as plainly as possible. The marginal coupling λ is, at the classical level, a pure number that neither grows nor shrinks under a change of scale; the massless theory has no length scale at all and is scale invariant. The quantum theory destroys this. Defining λ requires a scale μ; the coupling becomes a function $\lambda(\mu)$; and the rate of that running, $\beta = \mu\, d\lambda/d\mu$, is precisely the coefficient of the trace $\Theta^{\mu}{}_{\mu}$ of the stress tensor. A running coupling and a nonzero trace are not two facts but one. "The coupling runs," "scale invariance is anomalous," and "the stress tensor has a nonzero trace" are three names for the single phenomenon that the rest of this chapter quantifies. The numbers computed in Sections 24.2 and 24.3 — β_λ for ϕ^4, β_e for QED — are therefore not merely rates of change of couplings; they are the exact measure of how badly the quantum world fails to be scale invariant.

The same mechanism operates in QED, and it is worth recording the result even though its rigorous operator derivation (which must contend with gauge invariance and the renormalization of the gauge-field bilinear) is deferred to Chapter 36. Massless classical electrodynamics is scale invariant, $\Theta^{\mu}{}_{\mu} = 0$, because the only coupling e is dimensionless in $d = 4$. Quantization breaks it through the identical $\mu^{2\varepsilon}$ mechanism, and the trace acquires the form

$$\Theta^{\mu}{}_{\mu} \;=\; \frac{\beta_e(e)}{2e}\,\big[F_{\mu\nu}F^{\mu\nu}\big]_R \;+\; (1+\gamma_m)\,m\,\big[\bar{\psi}\psi\big]_R, \tag{24.167}$$

the gauge-field operator F^2 playing the role that ϕ^4 plays for the scalar and the electron mass term $m\bar{\psi}\psi$ playing the role of $m^2\phi^2$. With $\beta_e = e^3/12\pi^2$ from Section 24.3, the photon does not propagate in a scale-invariant vacuum: the coefficient $\beta_e/2e = e^2/24\pi^2$ measures the quantum breaking, and it is positive because vacuum polarization screens charge. We will not use Eq. (24.167) further — QED's running is extracted directly from β_e in Section 24.6 — but it shows that the identification "trace anomaly = beta function" is universal, not a peculiarity of the scalar.

24.5.3 The Callan–Symanzik equation from broken scale invariance

We now turn Eq. (24.166) into a differential equation for the renormalized one-particle-irreducible (1PI) n-point vertex functions $\Gamma^{(n)}(p_1,\dots,p_n;m,\lambda,\mu)$, the proper vertices generated by the effective action of Section 4.5.3. Two exact facts about these functions drive the derivation; both are elementary, and we establish each before combining them.

Dimensional analysis. In $d = 4$ the 1PI vertex $\Gamma^{(n)}$ carries engineering mass dimension

$$\big[\Gamma^{(n)}\big] \;=\; 4 - n, \tag{24.168}$$

as one checks immediately on the free two-point function $\Gamma^{(2)} = p^2 - m^2$ (dimension two) and the tree four-point vertex $\Gamma^{(4)} = -\lambda$ (dimensionless). Its dimensionful arguments are the external momenta p_i, the mass m, and the scale μ; the coupling λ is dimensionless and carries no dimension. Euler's theorem on homogeneous functions then states that scaling all dimensionful quantities together reproduces the engineering dimension,

$$\Big[\sum_i p_i{\cdot}\partial_{p_i} \;+\; m\,\partial_m \;+\; \mu\,\partial_\mu\Big]\,\Gamma^{(n)} \;=\; (4-n)\,\Gamma^{(n)}. \tag{24.169}$$

This is the statement of *naive* scale invariance: if there were no anomalous dependence on scale, dilating all the momenta would simply multiply the amplitude by its engineering dimension. It is exact and follows from dimensional analysis alone, with no reference to dynamics.

The renormalization-group equation. The second fact is the equation of Section 24.1, expressing the μ-independence of physics. For the 1PI vertices it reads, in the conventions of that section,

$$\Big[\mu\,\partial_\mu \;+\; \beta(\lambda)\,\partial_\lambda \;-\; \gamma_m(\lambda)\,m\,\partial_m \;+\; n\,\gamma(\lambda)\Big]\,\Gamma^{(n)} \;=\; 0, \tag{24.170}$$

with $\beta = \mu\, d\lambda/d\mu$, $\gamma = \tfrac{1}{2}\,\mu\, d\ln Z_\phi/d\mu$ the field anomalous dimension of the 1PI function, and $\gamma_m = -\mu\, d\ln m/d\mu$ the mass anomalous dimension; this is the master equation Eq. (24.9). The single term that distinguishes the two equations is $\mu\,\partial_\mu$: it appears in Eq. (24.169) because μ is a dimensionful argument, and in Eq. (24.170) because μ is unphysical. Subtracting one from the other eliminates it entirely. Taking Eq. (24.169) minus Eq. (24.170),

$$\begin{aligned} 0 &= \Big[\sum_i p_i{\cdot}\partial_{p_i} + m\partial_m - (4-n) - \beta\partial_\lambda + \gamma_m m\partial_m - n\gamma\Big]\Gamma^{(n)} \\ &= \Big[\sum_i p_i{\cdot}\partial_{p_i} + (1+\gamma_m)\,m\partial_m - \beta\partial_\lambda - n\gamma - (4-n)\Big]\Gamma^{(n)}, \end{aligned} \tag{24.171}$$

which we rearrange into the form that displays broken scale invariance most cleanly:

$$\boxed{\Big[\sum_i p_i{\cdot}\partial_{p_i} - (4-n)\Big]\,\Gamma^{(n)} \;=\; \Big[\beta(\lambda)\,\partial_\lambda + n\,\gamma(\lambda) - (1+\gamma_m)\,m\,\partial_m\Big]\,\Gamma^{(n)}.} \tag{24.172}$$

This is the Callan–Symanzik equation. The reading of it is the whole point. The left-hand side is the naive-scaling operator: it dilates every external momentum

and subtracts the engineering dimension $4-n$. Were the theory scale invariant, this operator would annihilate $\Gamma^{(n)}$ — a function homogeneous of degree $4-n$ in the momenta has $\big[\sum_i p_i\partial_{p_i} - (4-n)\big]\Gamma^{(n)} = 0$. The right-hand side is the exact obstruction to that homogeneity, and it has two qualitatively different parts. The terms $\beta\,\partial_\lambda + n\gamma$ are the *anomalous* breaking: they survive even when $m = 0$, and they are nonzero only because the quantum theory has a scale anomaly — precisely the $\frac{\beta}{4!}[\phi^4]_R$ piece of the trace Eq. (24.166), as we make literal in Section 24.5.4. The term $-(1+\gamma_m)\,m\,\partial_m$ is the *explicit* breaking by the mass, the quantum descendant of the classical $\Theta^\mu{}_\mu = m^2\phi^2$. The coefficient functions β and γ are, on the nose, the quantum measure of how far the amplitude departs from naive scaling — which is exactly what Theorem 7.43 and the trace anomaly led us to expect.

Theorem 24.19 (Callan–Symanzik equation)**.** *The renormalized 1PI vertex functions of ϕ^4 theory satisfy Eq. (24.172). Equivalently, moving the mass term to the right and recognizing it as a soft operator insertion (Section 24.5.4), they satisfy the inhomogeneous form*

$$\Big[\, m\,\partial_m + \beta(\lambda)\,\partial_\lambda + n\,\gamma(\lambda)\,\Big]\,\Gamma^{(n)} \;=\; m^2\,(2-\gamma_m)\,\Gamma^{(n)}_{\phi^2}(p_i;0), \tag{24.173}$$

where $\Gamma^{(n)}_{\phi^2}(p_i;0)$ is the 1PI vertex with a single insertion of the renormalized operator $\frac{1}{2}[\phi^2]_R$ carrying zero momentum — a "soft mass insertion," equal by Eq. (24.166) to a soft insertion of the trace of the stress tensor.

The two forms Eq. (24.172) and Eq. (24.173) are the same equation. The first is the modern, scheme-explicit statement obtained from μ-independence; the second is the original form of Callan and Symanzik, in which m rather than μ plays the role of the scale and the mass-breaking term appears as an explicit inhomogeneous insertion on the right. To pass from Eq. (24.172) to Eq. (24.173) one rewrites the momentum-scaling operator on the left of Eq. (24.172) using dimensional analysis Eq. (24.169) once more: $\big[\sum_i p_i\partial_{p_i} - (4-n)\big] = -\big[m\,\partial_m + \mu\,\partial_\mu\big]$, and then the $\mu\,\partial_\mu$ is traded for the anomalous operators through Eq. (24.170), leaving $m\,\partial_m + \beta\partial_\lambda + n\gamma$ on the left and the single explicit-mass insertion on the right. We now justify the passage in full by establishing the soft-insertion identities, which also make precise the sense in which the right-hand side of Eq. (24.173) is a trace insertion.

24.5.4 Soft operator insertions and the inhomogeneous form

The inhomogeneous right-hand side of Eq. (24.173) is not an ad hoc addition; it is what the differential operators on the left literally *are* when written out as operator insertions. The reason is a one-line consequence of the path-integral representation. The mass and the coupling enter the bare action only through

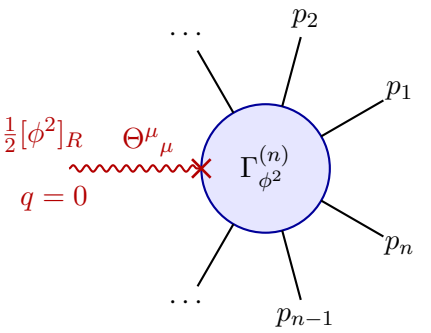


Figure 24.8: The inhomogeneous term of the Callan–Symanzik equation Eq. (24.173) as a *soft operator insertion*. Differentiating $\Gamma^{(n)}$ with respect to m^2 inserts the renormalized operator $\frac{1}{2}[\phi^2]_R$ — equivalently, by Eq. (24.166), the trace $\Theta^\mu{}_\mu$ of the stress tensor — carrying *zero* momentum $q=0$ into the n-point vertex (wavy line). Because the inserted momentum vanishes, the insertion lowers the engineering dimension of the vertex by the dimension of the operator, two, which is the origin of the m^2/κ^2 suppression at large external momenta (Section 24.5.5).

the terms $-\frac{1}{2}m_0^2\phi_0^2$ and $-\frac{\lambda_0}{4!}\phi_0^4$, so differentiating a correlator with respect to m_0^2 or λ_0 brings down precisely those operators, integrated over all of spacetime — that is, inserted at zero momentum. Concretely, from $Z \propto \int\mathcal{D}\phi_0\, e^{iS_0}$ and $\partial S_0/\partial m_0^2 = -\frac{1}{2}\int d^4y\,\phi_0^2(y)$,

$$\frac{\partial}{\partial m_0^2}\langle\phi_0(x_1)\cdots\phi_0(x_n)\rangle = -\frac{i}{2}\int d^4y\,\langle\,\phi_0^2(y)\,\phi_0(x_1)\cdots\phi_0(x_n)\rangle. \tag{24.174}$$

Differentiating with respect to m_0^2 thus *inserts* the operator $\frac{1}{2}\phi_0^2$ at zero momentum (the spacetime integral $\int d^4y$ is exactly the Fourier transform at vanishing inserted momentum, $q=0$). Passing to amputated 1PI functions and to renormalized quantities — the composite operator ϕ^2 requires its own multiplicative renormalization constant, the subject of Chapter 36 — this becomes the statement that $m\,\partial_m\Gamma^{(n)}$ is, up to the multiplicative factor displayed in Eq. (24.173), the 1PI vertex $\Gamma^{(n)}_{\phi^2}$ with one zero-momentum $\frac{1}{2}[\phi^2]_R$ insertion. Since the classical trace is $\Theta^\mu{}_\mu = m^2\phi^2$ (Eq. (24.162) at $d=4$), this soft insertion is exactly a soft insertion of the stress-tensor trace: the inhomogeneous term of the Callan–Symanzik equation is the matrix element of $\Theta^\mu{}_\mu$, the local operator that measures broken scale invariance. The mass-breaking term in Eq. (24.172) and the trace insertion in Eq. (24.173) are the same object, displayed in Fig. 24.8.

The identical mechanism explains why the *anomalous* terms $\beta\,\partial_\lambda$ and $n\gamma$ belong on the left of Eq. (24.173) as differential operators rather than as separate insertions, even though the trace Eq. (24.166) contains a ϕ^4 piece. Differentiation

with respect to the coupling inserts the quartic operator at zero momentum,

$$\frac{\partial}{\partial \lambda_0}\langle \phi_0(x_1)\cdots\rangle = -\frac{i}{4!}\int d^4y\,\langle\, \phi_0^4(y)\,\phi_0(x_1)\cdots\rangle, \tag{24.175}$$

so a zero-momentum insertion of $\frac{\beta}{4!}[\phi^4]_R$ — precisely the anomalous part of the trace Eq. (24.166) — is reproduced by the operator $\beta\,\partial_\lambda$ acting on $\Gamma^{(n)}$. Likewise the field-rescaling part of the dilatation, which assigns each external leg its anomalous dimension, is reproduced by $n\gamma$. In other words, when the trace operator Eq. (24.166) is inserted into an n-point function, its ϕ^4 piece becomes $\beta\,\partial_\lambda$, its field-strength piece becomes $n\gamma$, and only its irreducible $m^2\phi^2$ piece remains as a genuine inhomogeneity. This is the operator-level explanation of why the Callan–Symanzik equation has exactly the structure Eq. (24.173): it is the broken dilatation Ward identity, schematically

$$\Big[\sum_i p_i\cdot\partial_{p_i} - (4-n)\Big]\,\Gamma^{(n)} \;=\; -\langle\, \textstyle\int d^4y\;\Theta^\mu{}_\mu(y)\,\rangle_{\Gamma^{(n)}}, \tag{24.176}$$

with the anomalous part of $\Theta^\mu{}_\mu$ resummed into the β and γ operators and the explicit-mass part left on the right as the soft insertion of Fig. 24.8. The content of Eq. (24.176) is precisely Eq. (24.160), the local identity $\partial_\mu D^\mu = \Theta^\mu{}_\mu$, integrated against an n-point function: the left side is the total divergence of the dilatation current, whose integral measures the failure of the correlator to be scale invariant, and the right side is the integrated trace that sources it.

Worked Example 24.20 (The Callan–Symanzik equation at tree level)**.** Nothing illustrates the anatomy of Eq. (24.172) better than checking it where every coefficient is known: the free theory, where $\beta = \gamma = \gamma_m = 0$ and the only breaking is the explicit mass. Take the two-point function, $n = 2$, at tree level,

$$\Gamma^{(2)}(p) = p^2 - m^2. \tag{24.177}$$

The momentum-scaling operator counts the mass dimension carried by the momenta: acting on the homogeneous-of-degree-two piece p^2 it returns $2p^2$, and it annihilates the constant m^2,

$$\sum_i p_i\cdot\partial_{p_i}\,\Gamma^{(2)} = p\,\frac{d}{dp}(p^2-m^2) = 2p^2. \tag{24.178}$$

The left-hand side of Eq. (24.172), with $4-n = 4-2 = 2$, is therefore

$$\Big[\sum_i p_i\cdot\partial_{p_i} - 2\Big]\Gamma^{(2)} = 2p^2 - 2(p^2-m^2) = 2m^2. \tag{24.179}$$

The right-hand side, with all anomalous coefficients vanishing, reduces to the single mass term,

$$\Big[-(1+\gamma_m)\,m\,\partial_m\Big]\Gamma^{(2)} \;\xrightarrow{\gamma_m=0}\; -m\,\partial_m(p^2-m^2) = -(-2m^2) = 2m^2, \tag{24.180}$$

and the two sides agree. Three lessons are packed into this elementary check. First, the equation balances: the failure of $\Gamma^{(2)}$ to scale homogeneously, $2m^2$, is exactly accounted for by the mass-breaking term. Second, that failure is entirely $O(m^2)$ — it is the explicit mass breaking and nothing else — so in the massless free theory the right-hand side vanishes and naive scale invariance holds exactly, as it must for a free massless scalar. Third, the inhomogeneous form Eq. (24.173) reads the same balance as a soft insertion: at tree level the zero-momentum $\frac{1}{2}\phi^2$ insertion into the amputated two-point function is the bare vertex $\Gamma^{(2)}_{\phi^2} = 2$ (one inserts $\frac{1}{2}\phi^2$ into a two-leg function, and the two ways of contracting the two ϕ's with the two external legs give a combinatorial factor cancelling the $\frac{1}{2}$, leaving 1, times the operator normalization 2), so the right-hand side of Eq. (24.173) is $m^2(2-\gamma_m)\cdot 1 = 2m^2$ at $\gamma_m = 0$, matching Eq. (24.179). The number $2m^2$ is literally the matrix element of the classical trace $\Theta^{\mu}{}_{\mu} = m^2\phi^2$ between the external states. Beyond tree level the same equation holds with β, γ, γ_m no longer zero; their one-loop values were computed in Section 24.2.

24.5.5 The large-momentum limit and equivalence to the RG equation

The Callan–Symanzik equation in its inhomogeneous form Eq. (24.173) looks, at first sight, weaker than the homogeneous renormalization-group equation of Section 24.1: it carries an extra term on the right. The resolution — and the proof that the two derivations yield the very same physics — is that the inhomogeneous term is negligible in exactly the regime where the equation is used to resum logarithms: the deep Euclidean region of large momenta.

Consider $\Gamma^{(n)}$ at a Euclidean momentum configuration $p_i = \kappa\,\hat{p}_i$ with all invariants of order κ^2, and follow the behavior as $\kappa \to \infty$ with m fixed. The leading term $\Gamma^{(n)}$ has engineering dimension $4-n$ and scales as κ^{4-n} up to logarithms. The inhomogeneous term, by contrast, is $m^2\,(2-\gamma_m)\,\Gamma^{(n)}_{\phi^2}(p_i;0)$. The inserted operator $\frac{1}{2}\phi^2$ has mass dimension two, and because it is inserted at zero momentum (Fig. 24.8) it does not supply any external invariant of its own; the vertex $\Gamma^{(n)}_{\phi^2}$ therefore has engineering dimension $(4-n)-2$ and scales as κ^{2-n} up to logarithms. With the explicit m^2 prefactor restoring two powers of mass, the entire right-hand side scales as $m^2\,\kappa^{2-n} = \kappa^{4-n}\cdot(m^2/\kappa^2)$. The inhomogeneous term is thus suppressed, relative to the terms on the left, by a factor m^2/κ^2:

$$\frac{(\text{inhomogeneous term})}{\Gamma^{(n)}} \;\sim\; \frac{m^2}{\kappa^2} \;\xrightarrow[\kappa\to\infty]{}\; 0. \tag{24.181}$$

That this power-counting estimate is not merely dimensional but a genuine bound on the asymptotics is the content of Weinberg's asymptotic theorem (Theorem 21.25), which controls the large-momentum behavior of renormalized Feynman integrals: deep in the Euclidean region a Green's function approaches its

massless limit up to corrections suppressed by powers of m^2/κ^2, each accompanied by at most logarithms of κ. The soft $\frac{1}{2}\phi^2$ insertion, being a dimension-two operator weighted by the explicit m^2, contributes only at the first suppressed order; it cannot compete with the leading κ^{4-n} behavior of $\Gamma^{(n)}$ itself. We invoke the theorem at the level of its statement; its proof, organized around the asymptotic analysis of Chapter 5, lives there.

Dropping the suppressed term, the Callan–Symanzik equation Eq. (24.173) collapses in the deep Euclidean region to its homogeneous part,

$$\boxed{\Big[\, m\,\partial_m + \beta(\lambda)\,\partial_\lambda + n\,\gamma(\lambda)\,\Big]\,\Gamma^{(n)} \;=\; 0 \qquad (\kappa \gg m).} \tag{24.182}$$

This is a homogeneous, first-order partial differential equation of exactly the type that Section 24.6 will integrate by the method of characteristics; its solution is the running coupling and the multiplicative anomalous-dimension factor that turn fixed-order logarithms into resummed powers. It remains to verify that the equation Eq. (24.182) is, coefficient for coefficient, the renormalization-group equation Eq. (24.170), so that the two routes to the running coupling truly coincide. They are related by dimensional analysis Eq. (24.169). In the same asymptotic regime the mass term in Eq. (24.169) is subleading — it is the $m\,\partial_m$ acting on a function dominated by its massless part — so Euler's identity reduces to $\big[\sum_i p_i\partial_{p_i} + \mu\partial_\mu\big]\Gamma^{(n)} = (4-n)\Gamma^{(n)}$, i.e. $\mu\,\partial_\mu = (4-n) - \sum_i p_i\partial_{p_i}$ acting asymptotically. Substituting this into the homogeneous Callan–Symanzik equation Eq. (24.172) with the mass term dropped,

$$\Big[\sum_i p_i\cdot\partial_{p_i} - (4-n)\Big]\Gamma^{(n)} = \big[\beta\partial_\lambda + n\gamma\big]\Gamma^{(n)} \;\Longrightarrow\; \Big[\mu\,\partial_\mu + \beta\,\partial_\lambda + n\,\gamma\Big]\Gamma^{(n)} = 0, \tag{24.183}$$

which is precisely the massless renormalization-group equation Eq. (24.170). Because the operators $m\,\partial_m$ and $\mu\,\partial_\mu$ act identically on the asymptotic scaling solution — both measure the response to an overall change of scale, the only scales left once $\kappa \gg m$ being interchangeable — the coefficient functions multiplying ∂_λ and the field count n in the two equations must be equal. The β and γ of the Callan–Symanzik equation, defined through the response to the mass scale m, coincide with the β and γ of the renormalization-group equation, defined through the response to the subtraction scale μ. The two derivations — one from the unphysicality of μ, the other from the broken scale symmetry of the massless theory — produce the same coefficient functions and the same equation. This is the equivalence the section set out to prove.

Physical Insight

Why the two scales become interchangeable. The deep reason behind Eq. (24.183) is that both μ and m are, in the deep Euclidean region,

simply "the scale at which one looks." The subtraction scale μ is the scale at which the coupling is *defined*; the mass m is the scale below which the theory stops looking massless. Far above both, a single dimensionless ratio $\kappa/(\text{scale})$ controls everything, and asking how $\Gamma^{(n)}$ responds to changing μ, to changing m, or to dilating κ are three versions of one question. The Callan–Symanzik and renormalization-group equations are the answers to those three versions, and they had to agree. What survives at finite momentum — the inhomogeneous trace insertion — is the memory that the theory is not exactly scale invariant even classically once $m \neq 0$; it is the term that distinguishes the massive theory from a true conformal field theory and, when λ is tuned to a zero of β, is the last obstruction to scale invariance that the Wilson–Fisher analysis of Chapter 52 must dispose of.

We have now seen the same equation born twice: in Section 24.1 from the meaninglessness of the subtraction point, and here from the anomaly of classical scale symmetry. The second birth reframes ϕ^4 theory in a way that closes a loop opened in Chapter 7: the massless scalar's classical scale (indeed conformal) invariance, exhibited there through a traceless improved stress tensor, is destroyed by quantization, and β_λ is the exact quantum measure of its destruction. The coefficient functions β, γ, γ_m are no longer mere bookkeeping devices for the scale μ; they are the quantitative content of the trace anomaly Eq. (24.166). What this section delivers to the next is therefore a single, clean object: the homogeneous first-order partial differential equation Eq. (24.182), valid deep in the Euclidean region and identical to the massless renormalization-group equation. With β and γ already in hand from Sections 24.2 and 24.3 and certified physical by Section 24.4, it is now ripe for integration. The next section, Section 24.6, solves it by the method of characteristics, trading the differential constraint for the running coupling $\bar{\lambda}(\mu)$, the running mass, and the anomalous-dimension exponential — and reads off, from the sign and zeros of β, the high-energy fate of both threads.

24.6 Solving the Renormalization Group: Running Couplings and Anomalous Scaling

We now hold the two ingredients that the renormalization-group equation needs as input: the beta functions and anomalous dimensions of both threads, computed from scratch and certified scheme-independent at the leading orders that matter. For ϕ^4 theory we have $\beta_\lambda = 3\lambda^2/16\pi^2 + O(\lambda^3)$ from Section 24.2; for QED we have $\beta_e = e^3/12\pi^2$, equivalently $\beta_\alpha = 2\alpha^2/3\pi$, from Section 24.3. What we do not yet have is the payoff. A beta function is a statement about an *infinitesimal* change of scale; the physics — the value of a coupling probed at 1 GeV versus at

100 GeV, the high-energy fate of a Green function, the breakdown of perturbation theory near a Landau pole — lives in the *finite* change of scale obtained by integrating that infinitesimal law. This section performs the integration. We treat the renormalization-group equation as a first-order linear partial differential equation, solve it exactly by the method of characteristics, and find that the entire μ-dependence of a Green function is carried by two objects: a single ordinary function $\bar{g}(t)$ — the *running coupling* — and an exponential of the integrated anomalous dimension. We then specialize to the two threads, integrate their one-loop beta functions to closed form, and discover that the resulting running coupling is nothing other than the geometric resummation of the one-loop bubble chain met in Chapters 19 and 20 — the renormalization group is the machine that sums the leading logarithms of perturbation theory to all orders at once.

24.6.1 The renormalization-group equation as a transport equation

We begin from the equation derived earlier in this chapter. For a renormalized one-particle-irreducible (1PI) vertex function $\Gamma^{(n)}$ with n external legs, depending on the external momenta collectively denoted p, the renormalized coupling g, and the subtraction scale μ, the renormalization-group equation Eq. (24.9) reads

$$\left[\mu\,\frac{\partial}{\partial\mu} + \beta(g)\,\frac{\partial}{\partial g} - n\,\gamma(g)\right]\Gamma^{(n)}(p;g,\mu) = 0. \tag{24.184}$$

Here $\beta(g) = \mu\,\partial g/\partial\mu\big|_{\text{bare fixed}}$ and $\gamma(g) = \tfrac{1}{2}\,\mu\,\partial \ln Z_\phi/\partial\mu$ are the beta function and field anomalous dimension whose values were computed in Sections 24.2 and 24.3, with the sign conventions fixed there. We suppress the mass and its anomalous dimension throughout this section: we are interested in the deep-Euclidean or high-energy regime in which all invariants are large compared to m^2, where the theory becomes effectively massless and the mass term is an irrelevant (power-suppressed) correction. The single symbol g stands for λ in ϕ^4 theory and for e (or, after the trivial change of variable, α) in QED.

The content of Eq. (24.184) is a single demand made precise: the scale μ is a bookkeeping device with no physical content, so when μ is changed the explicit dependence must be compensated exactly by the induced change in the renormalized coupling and by the rescaling of the fields. Read as a partial differential equation, Eq. (24.184) is of first order and linear, and the prefactor of $\partial/\partial\mu$ and $\partial/\partial g$ defines a vector field in the (μ, g) plane. A first-order linear PDE of this type is constant along the integral curves of that vector field — its *characteristics* — up to the inhomogeneous rescaling encoded by the $n\gamma$ term. This is precisely the structure that the method of characteristics is built to solve, and it is what turns an infinitesimal statement about $\partial/\partial\mu$ into a finite statement relating $\Gamma^{(n)}$ at two widely separated scales.

Physical Insight

The geometry behind Eq. (24.184) is worth holding in mind. In the (μ, g) plane the operator $\mu\,\partial_\mu + \beta\,\partial_g$ generates flow along curves on which the *physics is invariant*: as we slide μ up a curve, g slides with it exactly so as to keep every observable fixed. These curves are the renormalization-group trajectories. The remaining $n\gamma$ term does not change which trajectory we are on; it only tells us by how much the normalization of an n-leg amplitude is rescaled as we move along it, because each external field is being continuously rescaled by the wavefunction renormalization. Solving the RG equation is therefore nothing more than "riding the trajectory" and keeping track of that normalization.

24.6.2 The running coupling and the method of characteristics

The natural variable along a trajectory is the logarithm of the scale. Introduce

$$t \equiv \ln \frac{\mu'}{\mu}, \tag{24.185}$$

so that increasing t by dt multiplies the scale by e^{dt}. We define the running coupling as the solution of the ordinary differential equation obtained by following β along this logarithmic scale.

Definition 24.21 (Running coupling)**.** The *running coupling* $\bar{g}(t) \equiv \bar{g}(t; g)$ is the unique solution of the autonomous initial-value problem

$$\frac{d\bar{g}}{dt} = \beta(\bar{g}), \qquad \bar{g}(0) = g. \tag{24.186}$$

Equivalently, $\bar{g}(t)$ is the value of the renormalized coupling at the scale μe^t that descends from the value g at scale μ along a renormalization-group trajectory.

That Eq. (24.186) has a unique solution on a maximal interval around $t = 0$ is guaranteed by the standard Picard–Lindelöf existence-and-uniqueness theorem for ordinary differential equations, since β is, at the orders we work to, a polynomial in $\bar{g}$ and hence smooth and locally Lipschitz. The solution can run off to infinity at a finite t — this is exactly the Landau pole we will meet below — so the maximal interval need not be all of $\mathbb{R}$; this is a feature, not a pathology, and we read physics off precisely where the interval ends. Separating variables in Eq. (24.186) gives the implicit closed form that we will use repeatedly,

$$t = \int_{g}^{\bar{g}(t)} \frac{dg'}{\beta(g')}, \tag{24.187}$$

which inverts to give $\bar{g}(t)$ once the integral on the right is performed.

We can now state and prove the central result of the section: the complete solution of Eq. (24.184).

Theorem 24.22 (Solution of the RG equation by characteristics)**.** *Let $\Gamma^{(n)}(p; g, \mu)$ satisfy the renormalization-group equation Eq. (24.184), and let $\bar{g}(t)$ be the running coupling of Theorem 24.21. Then for all t in the maximal interval of existence of $\bar{g}$,*

$$\boxed{\Gamma^{(n)}(p; g, \mu) = \Gamma^{(n)}(p; \bar{g}(t),\, \mu e^{t})\, \exp\!\left[-\, n \int_0^t \gamma(\bar{g}(t'))\, dt'\right].} \tag{24.188}$$

Proof. The strategy is to exhibit a quantity that is manifestly constant along the trajectory and to recognize Eq. (24.188) as the statement of that constancy. Define the function

$$F(t) \equiv \Gamma^{(n)}(p; \bar{g}(t),\, \mu e^{t})\, \exp\!\left[-\, n \int_0^t \gamma(\bar{g}(t'))\, dt'\right]. \tag{24.189}$$

At $t = 0$ the exponential is unity and $\bar{g}(0) = g$, so $F(0) = \Gamma^{(n)}(p; g, \mu)$. The claim Eq. (24.188) is therefore equivalent to $F(t) = F(0)$ for all t, i.e. to $dF/dt = 0$. We compute the derivative. Write $E(t) \equiv \exp[-n \int_0^t \gamma(\bar{g}(t'))\, dt']$, so that $dE/dt = -n\gamma(\bar{g}(t))\, E(t)$ by the fundamental theorem of calculus. Abbreviate the two slots of $\Gamma^{(n)}$ on which t acts as the coupling slot $\bar{g}(t)$ and the scale slot $M(t) \equiv \mu e^{t}$, noting $dM/dt = \mu e^{t} = M$. The product and chain rules give

$$\frac{dF}{dt} = \left[\frac{\partial \Gamma^{(n)}}{\partial \bar{g}}\, \frac{d\bar{g}}{dt} + \frac{\partial \Gamma^{(n)}}{\partial M}\, \frac{dM}{dt}\right] E + \Gamma^{(n)}\, \frac{dE}{dt}. \tag{24.190}$$

Insert $d\bar{g}/dt = \beta(\bar{g})$ from Eq. (24.186), $dM/dt = M$, and $dE/dt = -n\gamma(\bar{g})E$:

$$\frac{dF}{dt} = E\left[\beta(\bar{g})\, \frac{\partial \Gamma^{(n)}}{\partial \bar{g}} + M\, \frac{\partial \Gamma^{(n)}}{\partial M} - n\gamma(\bar{g})\, \Gamma^{(n)}\right]. \tag{24.191}$$

The bracket is exactly the left-hand side of the renormalization-group equation Eq. (24.184), evaluated at coupling $\bar{g}(t)$ and scale $M = \mu e^{t}$, because $M\, \partial/\partial M = \mu e^{t}\, \partial/\partial(\mu e^{t})$ is the operator $\mu'\, \partial/\partial \mu'$ at $\mu' = M$. Since $\Gamma^{(n)}$ obeys Eq. (24.184) at *every* value of its arguments, the bracket vanishes identically. Hence $dF/dt = 0$, so $F(t) = F(0) = \Gamma^{(n)}(p; g, \mu)$, which is Eq. (24.188). □

The result deserves to be read slowly, because it is the engine of everything that follows. The left-hand side of Eq. (24.188) is the Green function with a fixed (arbitrary) scale μ and the coupling g defined at that scale. The right-hand side evaluates the *same* Green function at a *shifted* scale μe^{t} — and at the shifted scale the appropriate coupling is not g but the running coupling $\bar{g}(t)$. In other words, changing the renormalization scale is exactly compensated by

replacing the coupling with its running value and by an overall multiplicative factor. All of the (potentially large) logarithms of $\mu e^t/\mu = e^t$ that would appear if we expanded $\Gamma^{(n)}$ naively in g have been packaged into the single function $\bar{g}(t)$ and the anomalous-dimension exponential. This is the whole trick: choose t so that μe^t sits at the physical scale of the process, and then perturbation theory in the running coupling $\bar{g}(t)$ is free of large logarithms.

24.6.3 Reinstating the mass: the running mass

We dropped the mass at the outset because the high-energy regime that interests us is effectively massless. But the renormalization-group equation in its full form Eq. (24.9) carries a mass term $-\gamma_m(g)\, m\, \partial_m$, and the method of characteristics handles it at no extra cost: the mass simply becomes a second running parameter, and tracking it is exactly what is needed to discuss the scale dependence of a fermion or scalar mass — the running $\overline{\text{MS}}$ masses that Section 24.3 promised. Restoring the term, the equation reads

$$\left[\mu\frac{\partial}{\partial\mu} + \beta(g)\frac{\partial}{\partial g} - \gamma_m(g)\, m\frac{\partial}{\partial m} - n\,\gamma(g)\right]\Gamma^{(n)}(p; g, m, \mu) = 0, \tag{24.192}$$

with $\gamma_m(g) = -\mu\, d\ln m/d\mu$ the mass anomalous dimension of Theorem 24.3. The characteristic curves of Eq. (24.192) are obtained by reading off the ratios in which the variables move, $d\mu/\mu = dg/\beta = dm/(-\gamma_m\, m)$; alongside the running coupling $\bar{g}(t)$ they define a *running mass*.

Definition 24.23 (Running mass)**.** The *running mass* $\bar{m}(t)$ is the solution of the linear initial-value problem along a renormalization-group trajectory,

$$\frac{d\bar{m}}{dt} = -\,\gamma_m(\bar{g}(t))\,\bar{m}(t), \qquad \bar{m}(0) = m, \tag{24.193}$$

where $\bar{g}(t)$ is the running coupling of Theorem 24.21. Because Eq. (24.193) is linear in $\bar{m}$, it integrates immediately to the *anomalous-dimension exponential for the mass*,

$$\bar{m}(t) = m\,\exp\!\left[-\int_0^t \gamma_m(\bar{g}(t'))\,dt'\right]. \tag{24.194}$$

Repeating verbatim the constancy argument of Theorem 24.22 — now with the extra slot $\bar{m}(t)$ whose t-derivative supplies precisely the $-\gamma_m\, m\,\partial_m$ term in the bracket — gives the full solution of Eq. (24.192),

$$\boxed{\Gamma^{(n)}(p; g, m, \mu) = \Gamma^{(n)}(p; \bar{g}(t),\, \bar{m}(t),\, \mu e^t)\,\exp\!\left[-\,n\int_0^t \gamma(\bar{g}(t'))\,dt'\right].} \tag{24.195}$$

The structure is the same as the massless solution Eq. (24.188) with one addition: the mass too is replaced by its running value when the scale is shifted. The

exponential Eq. (24.194) is the mass counterpart of the field-rescaling factor; with the sign convention of Theorem 24.3 a positive γ_m makes $\bar{m}$ *decrease* toward the ultraviolet.

It is worth making the integral Eq. (24.194) explicit in QED, where it produces a clean closed form. From Section 24.3 the one-loop mass anomalous dimension of the electron is $\gamma_m = 3\alpha/2\pi$, while $\beta_\alpha = 2\alpha^2/3\pi$. Rather than integrate over t directly it is neater to trade dt for $d\alpha$ using $dt = d\bar{\alpha}/\beta_\alpha$, which turns Eq. (24.193) into a relation between $\bar{m}$ and $\bar{\alpha}$ alone:

$$\frac{d\bar{m}}{\bar{m}} = -\frac{\gamma_m(\bar{\alpha})}{\beta_\alpha(\bar{\alpha})}\,d\bar{\alpha} = -\frac{3\bar{\alpha}/2\pi}{2\bar{\alpha}^2/3\pi}\,d\bar{\alpha} = -\frac{9}{4}\,\frac{d\bar{\alpha}}{\bar{\alpha}}. \tag{24.196}$$

Integrating from the reference scale (where $\bar{\alpha} = \alpha_0$, $\bar{m} = m_0$) to scale μ (where $\bar{\alpha} = \alpha(\mu)$) gives the renormalization-group-improved running electron mass,

$$\boxed{m(\mu) = m_0\left(\frac{\alpha(\mu)}{\alpha_0}\right)^{-9/4} = m_0\left(\frac{\alpha_0}{\alpha(\mu)}\right)^{9/4}.} \tag{24.197}$$

Because $\alpha(\mu)$ grows toward the ultraviolet, the ratio $\alpha_0/\alpha(\mu)$ is less than one there and $m(\mu)$ falls — the running $\overline{\text{MS}}$ mass shrinks logarithmically with increasing scale, the abelian foreshadowing of the more dramatic running of quark masses in QCD. Expanding Eq. (24.197) to first order in the small one-loop logarithm reproduces the naive fixed-order result $m(\mu) = m_0[\,1 - (3\alpha_0/2\pi)\ln(\mu/\mu_0) + \cdots]$, exactly as $\mu\,dm/d\mu = -\gamma_m\,m = -(3\alpha/2\pi)m$ from Section 24.3 requires; the closed form Eq. (24.197) is the all-orders leading-log resummation of that single logarithm, precisely as Eq. (24.215) was for the coupling.

24.6.4 From scale running to momentum running

Theorem 24.22 relates $\Gamma^{(n)}$ at two values of the *subtraction* scale μ at fixed external momenta. In practice we want the converse and physically primary statement: how does $\Gamma^{(n)}$ behave as the external momenta themselves are scaled up, at fixed μ? The two questions are linked by dimensional analysis, and combining them is what exposes anomalous scaling.

Work in the massless theory (all m-dependence dropped, as announced). Then the only dimensionful quantities $\Gamma^{(n)}$ can depend on are the external momenta p and the scale μ; the coupling g is dimensionless (it is marginal — λ in ϕ^4 and e in QED are both dimensionless in $d = 4$). Let D denote the mass dimension of $\Gamma^{(n)}$. For the ϕ^4 vertex functions in $d = 4$, stripping the overall momentum-conserving $(2\pi)^4\delta^4(\sum p)$,

$$D = 4 - n, \tag{24.198}$$

so that $\Gamma^{(2)}$ (the inverse propagator) has $D = 2$ and $\Gamma^{(4)}$ (the amputated four-point) has $D = 0$, consistent with λ being dimensionless. Ordinary dimensional

analysis is the statement that $\Gamma^{(n)}$ is a homogeneous function of degree D in its dimensionful arguments,

$$\Gamma^{(n)}(e^t p;\, g,\, e^t \mu) = e^{Dt}\, \Gamma^{(n)}(p;\, g,\, \mu), \tag{24.199}$$

which is Euler's relation in integrated form. Rearranged, Eq. (24.199) trades a rescaling of *both* momenta and scale for a rescaling of momenta alone at the cost of an inverse rescaling of μ:

$$\Gamma^{(n)}(e^t p;\, g,\, \mu) = e^{Dt}\, \Gamma^{(n)}(p;\, g,\, \mu e^{-t}). \tag{24.200}$$

Now apply Theorem 24.22 to the right-hand side, but at base scale μe^{-t} rather than μ: setting the scale argument in Eq. (24.188) to μe^{-t} and choosing the shift parameter equal to t promotes $\mu e^{-t} \to \mu e^{-t} e^{t} = \mu$, so that

$$\Gamma^{(n)}(p;\, g,\, \mu e^{-t}) = \Gamma^{(n)}(p; \bar{g}(t),\, \mu)\, \exp\!\left[- n \int_0^t \gamma(\bar{g}(t'))\, dt' \right], \tag{24.201}$$

with $\bar{g}(t)$ the running coupling started from g. Substituting Eq. (24.201) into Eq. (24.200) gives the momentum-scaling form of the solution,

$$\boxed{\Gamma^{(n)}(e^t p;\, g,\, \mu) = e^{Dt}\, \Gamma^{(n)}(p; \bar{g}(t),\, \mu)\, \exp\!\left[- n \int_0^t \gamma(\bar{g}(t'))\, dt' \right].} \tag{24.202}$$

The physical reading is clean. Scaling all external momenta up by e^t produces, first, the naive power e^{Dt} that dimensional analysis alone would predict; second, the replacement of the fixed coupling g by the running coupling $\bar{g}(t)$ evaluated at the new momentum scale; and third, an extra multiplicative factor built from the anomalous dimension. The second effect is what makes the running coupling *physical*: a process at momentum scale Q is governed by the coupling $\bar{g}(\ln(Q/\mu))$, not by the coupling defined at some unrelated reference scale.

24.6.5 Anomalous scaling near a fixed point

The deepest consequence of Eq. (24.202) appears when the running coupling tends to a *fixed point* g_*, a zero of the beta function, $\beta(g_*) = 0$. (The classification of such points and the question of whether the threads possess any nontrivially is the business of the next section; here we extract the scaling law they imply.) If $\bar{g}(t) \to g_*$ as $t \to \infty$, then for large t the running coupling is effectively frozen at g_*, and the anomalous dimension freezes with it at its fixed-point value

$$\gamma_* \equiv \gamma(g_*). \tag{24.203}$$

The exponential in Eq. (24.202) then evaluates to

$$\exp\!\left[- n \int_0^t \gamma(\bar{g}(t'))\, dt' \right] \xrightarrow{t \to \infty} C\, \exp\!\big[- n \gamma_*\, t \big] = C\, (e^t)^{-n\gamma_*}, \tag{24.204}$$

where C is a finite constant collecting the integral over the transient region in which $\bar{g}$ had not yet reached g_*. Combined with the naive power e^{Dt} from Eq. (24.202), the full momentum dependence is

$$\Gamma^{(n)}(e^t p; g, \mu) \;\sim\; e^{(D-n\gamma_*)t}\,\Gamma^{(n)}(p; g_*, \mu) \quad (t \to \infty), \tag{24.205}$$

that is, writing the overall momentum scale as $\kappa = e^t$ (so all momenta are κp),

$$\Gamma^{(n)} \;\sim\; \kappa^{\,D}\,(\kappa)^{-n\gamma_*} = \kappa^{\,D-n\gamma_*}. \tag{24.206}$$

This is the promised anomalous scaling. Dimensional analysis alone insists that an n-leg amplitude scale as the canonical power κ^D. The interacting theory *violates* this naive expectation: each external leg of the amputated 1PI function contributes an extra power $\kappa^{-\gamma_*}$, so the true scaling exponent of $\Gamma^{(n)}$ is shifted from D to $D - n\gamma_*$. The per-leg shift γ_* is the *anomalous dimension* — the amount by which the effective scaling dimension of the field departs from its classical (engineering) value because of the cloud of quantum fluctuations dressing it. The sign is most transparent on the two-point function: with $D = 2$ for $\Gamma^{(2)}$ we have $\Gamma^{(2)} \sim \kappa^{\,2-2\gamma_*}$, so the full propagator, its inverse, behaves at large momentum like $1/p^{2-2\gamma_*}$ rather than $1/p^2$. Reading off the power, the field has acquired the scaling dimension $\Delta_\phi = 1 + \gamma_*$ in place of its classical value 1 — equivalently the standard critical exponent $\eta = 2\gamma_*$, matching the identification $\eta = 2\gamma_\phi(g_*)$ recorded in Section 24.4.4. (The full, non-amputated n-point function carries the opposite sign of the field term, Eq. (24.10), and scales with $+n\gamma_*$, consistent with each of its n legs carrying a field of dimension $1 + \gamma_*$.)

Physical Insight

The word "anomalous" is precise and not pejorative. A free field in four dimensions has scaling dimension 1; this is fixed by the kinetic term and is what dimensional analysis sees. Interactions continuously deform that dimension to $1 + \gamma_*$. At an interacting fixed point the theory is scale-invariant — indeed conformally invariant — but the symmetry is realized on fields whose dimensions are *not* the classical ones. These shifted dimensions are physically measurable: they are the critical exponents of statistical mechanics. The exponent γ_* that here looks like a small loop correction is, in the three-dimensional Ising universality class reached by the Wilson–Fisher fixed point of Chapter 52, a number measured to several digits in the laboratory. The identification of γ_* with a critical exponent is taken up in the next section and made quantitative in Chapter 52.

24.6.6 Scalar ϕ^4 theory: the running quartic coupling

We now make Theorem 24.21 explicit for ϕ^4 theory, continuing the thread from the beta function computed in Section 24.2. The running coupling $\bar{\lambda}(t)$ solves

$$\frac{d\bar{\lambda}}{dt} = \beta_\lambda(\bar{\lambda}) = \frac{3\bar{\lambda}^2}{16\pi^2} + O(\bar{\lambda}^3), \qquad \bar{\lambda}(0) = \lambda, \tag{24.207}$$

where, to be definite, $\lambda \equiv \lambda(\mu_0)$ is the coupling renormalized at a reference scale μ_0 and $t = \ln(\mu/\mu_0)$. We keep only the leading one-loop term; the corrections are higher order in $\bar{\lambda}$ and shift the result at the same order at which two-loop data (which we have not computed) would enter, so retaining only $3\bar{\lambda}^2/16\pi^2$ is consistent. The equation is separable. Dividing by $\bar{\lambda}^2$ and integrating from 0 to t,

$$\int_0^t \frac{1}{\bar{\lambda}^2}\,\frac{d\bar{\lambda}}{dt'}\,dt' = \int_0^t \frac{3}{16\pi^2}\,dt', \tag{24.208}$$

and recognizing the left integrand as $-\,d(1/\bar{\lambda})/dt'$,

$$-\frac{1}{\bar{\lambda}(t)} + \frac{1}{\lambda} = \frac{3}{16\pi^2}\,t. \tag{24.209}$$

Solving for $\bar{\lambda}(t)$ and restoring $t = \ln(\mu/\mu_0)$ gives the closed-form running coupling of ϕ^4 theory,

$$\boxed{\lambda(\mu) = \frac{\lambda(\mu_0)}{1 - \dfrac{3\lambda(\mu_0)}{16\pi^2}\,\ln\dfrac{\mu}{\mu_0}}.} \tag{24.210}$$

This is the explicit running coupling we set out to obtain. Several features are immediate and important. The coefficient $3/16\pi^2$ is positive, so the denominator *decreases* as μ increases: the quartic coupling grows toward short distances. As μ rises the denominator shrinks to zero at the finite scale

$$\mu_{\mathrm{L}} = \mu_0\,\exp\!\left(\frac{16\pi^2}{3\,\lambda(\mu_0)}\right), \tag{24.211}$$

at which $\lambda(\mu)$ formally diverges. This is the *Landau pole* of ϕ^4 theory: an ultraviolet scale at which the perturbatively-defined coupling blows up. Because $\lambda(\mu_0)$ is small, μ_{L} is astronomically large — the pole sits exponentially far in the ultraviolet — but its very existence signals that the one-loop-improved theory cannot be a complete description up to arbitrarily high energies at fixed nonzero coupling. The implications, including the triviality problem (the statement that holding the theory finite by sending the cutoff to infinity forces $\lambda \to 0$), are the subject of the next section; here we only record that the running coupling has put the question on the table in a single, sharp formula.

Physical Insight

The same arithmetic read in the opposite direction is reassuring. As $\mu \to 0$ (toward long distances, $t \to -\infty$), the logarithm in Eq. (24.210) grows negative without bound, the denominator grows, and $\lambda(\mu) \to 0^+$: the quartic coupling is *infrared free*. A weakly coupled ϕ^4 theory becomes *more* weakly coupled at low energies. This is the precise sense in which the Gaussian (free) fixed point at $\lambda = 0$ is infrared-attractive for ϕ^4 in four dimensions, the statement that marginal-irrelevant couplings flow to zero in the infrared, met qualitatively in the Wilsonian picture of Chapter 23 and now realized as an explicit flow.

24.6.7 QED: the running fine-structure constant

For QED it is cleanest to run $\alpha = e^2/4\pi$ directly. From Section 24.3 the beta function in this variable is

$$\beta_\alpha \equiv \mu \frac{d\alpha}{d\mu} = \frac{2\alpha^2}{3\pi} + O(\alpha^3), \tag{24.212}$$

which is consistent with $\beta_e = e^3/12\pi^2$: writing $\alpha = e^2/4\pi$ gives $\mu\, d\alpha/d\mu = (e/2\pi)\,\mu\, de/d\mu = (e/2\pi)\beta_e = (e/2\pi)(e^3/12\pi^2) = e^4/24\pi^3 = 2\alpha^2/3\pi$. The running coupling $\bar{\alpha}(t)$ solves

$$\frac{d\bar{\alpha}}{dt} = \frac{2\bar{\alpha}^2}{3\pi}, \qquad \bar{\alpha}(0) = \alpha(\mu_0), \qquad t = \ln\frac{\mu}{\mu_0}. \tag{24.213}$$

This is the identical separable equation as for ϕ^4, with the constant $3/16\pi^2$ replaced by $2/3\pi$. Integrating exactly as in Eqs. (24.208) and (24.209),

$$-\frac{1}{\bar{\alpha}(t)} + \frac{1}{\alpha(\mu_0)} = \frac{2}{3\pi}\, t = \frac{1}{3\pi}\, \ln\frac{\mu^2}{\mu_0^2}, \tag{24.214}$$

where in the last step we used $2t = 2\ln(\mu/\mu_0) = \ln(\mu^2/\mu_0^2)$ to convert to the squared scale that appears naturally in momentum-space amplitudes. A physical process probes the coupling at the momentum scale set by its kinematics, so we evaluate the running coupling at $\mu^2 = q^2$, the (spacelike) momentum-transfer invariant; with this identification $\alpha(q^2) \equiv \bar{\alpha}$ and solving Eq. (24.214) for it gives the explicit running fine-structure constant of QED,

$$\boxed{\alpha(q^2) = \frac{\alpha(\mu_0^2)}{1 - \dfrac{\alpha(\mu_0^2)}{3\pi}\, \ln\dfrac{q^2}{\mu_0^2}}.} \tag{24.215}$$

This is the second explicit running coupling the thread requires. The coefficient $1/3\pi$ is positive, so $\alpha(q^2)$ grows with q^2: the electric charge is *larger* at short

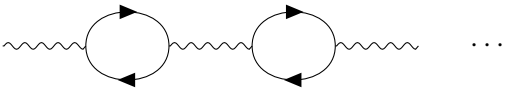


Figure 24.9: The geometric chain of vacuum-polarization bubbles that dresses the photon propagator. Each fermion loop is one insertion of the one-loop self-energy $\Pi(q^2)$ of Chapter 19; the full (resummed) propagator is the sum of the chain with $0, 1, 2, \ldots$ loops. Summing the geometric series produces the effective coupling Eq. (24.218), which coincides with the renormalization-group running coupling Eq. (24.215). This is the diagrammatic content of "solving the RG" in QED.

distances. The familiar low-energy value $\alpha \approx 1/137$ (defined at $q^2 \to 0$, the Thomson limit) rises to $\alpha \approx 1/128$ at the scale of the Z boson, $q^2 = M_Z^2$ — a measured effect and one of the cleanest experimental confirmations that couplings run. As with ϕ^4, there is a Landau pole, now in the ultraviolet at $q_{\mathrm{L}}^2 = \mu_0^2 \exp[3\pi/\alpha(\mu_0^2)]$, again exponentially far away but again a signal that pure QED cannot be a complete theory at all scales.

We owe an account of *why* the same closed form Eq. (24.215) also emerges from summing the geometric series of vacuum-polarization bubbles introduced in Chapter 19 and assembled into the one-loop running coupling of Chapter 20. The diagrammatic picture is the chain of photon self-energy insertions shown in Fig. 24.9; the worked example below carries out both computations and shows that they agree term by term.

Worked Example 24.24 (The running QED coupling two ways)**.** We derive $\alpha(q^2)$ first by integrating the beta function and then by resumming the bubble chain, and show the two agree.

Method 1: integrate the beta function. This we have just done; the result is Eq. (24.215). To prepare the comparison, expand it as a geometric series in the small quantity $x \equiv \dfrac{\alpha(\mu_0^2)}{3\pi} \ln \dfrac{q^2}{\mu_0^2}$:

$$\alpha(q^2) = \frac{\alpha_0}{1 - x} = \alpha_0 (1 + x + x^2 + x^3 + \cdots) = \alpha_0 \sum_{k=0}^{\infty} \left(\frac{\alpha_0}{3\pi} \ln \frac{q^2}{\mu_0^2} \right)^k , \tag{24.216}$$

where $\alpha_0 \equiv \alpha(\mu_0^2)$. The k-th term is of order $\alpha_0^{k+1} (\ln)^k$: it carries k powers of the logarithm dressed by $k + 1$ powers of the coupling. These are precisely the *leading logarithms*.

Method 2: resum the bubble chain. Recall from Chapter 19 the structure of the dressed photon propagator. The one-loop vacuum polarization is transverse, $i\Pi^{\mu\nu}(q) = i(q^2 g^{\mu\nu} - q^\mu q^\nu)\, \Pi(q^2)$, by the Ward identity, and the geometric sum of one-particle-irreducible insertions gives the full propagator (in Feynman gauge,

transverse part)

$$\frac{-ig_{\mu\nu}}{q^2}\Big[1+\Pi(q^2)+\Pi(q^2)^2+\cdots\Big]=\frac{-ig_{\mu\nu}}{q^2}\,\frac{1}{1-\Pi(q^2)}, \tag{24.217}$$

the series converging as a formal geometric series exactly as in Chapter 19. A scattering amplitude involving photon exchange thus depends on the combination $e^2/(1-\Pi(q^2))$, which defines the momentum-dependent *effective coupling*

$$\alpha_{\text{eff}}(q^2)=\frac{\alpha_0}{1-\big[\Pi(q^2)-\Pi(\mu_0^2)\big]}, \tag{24.218}$$

where the subtraction at μ_0^2 implements the renormalization condition that fixes α_0 at the reference scale (it removes the constant piece of Π, which is unobservable, leaving only its running). The renormalized one-loop self-energy for a single fermion of charge $-e$, in the regime $|q^2|\gg m^2$, was found in Chapters 19 and 20 to be (the explicit integral is reproduced in Eq. (20.171))

$$\Pi(q^2)-\Pi(\mu_0^2)=\frac{\alpha_0}{3\pi}\,\ln\frac{q^2}{\mu_0^2}+O(\alpha_0^2), \tag{24.219}$$

the logarithm carrying the factor $1/3\pi$ that the one-loop fermion-loop integral produces. Inserting Eq. (24.219) into Eq. (24.218),

$$\alpha_{\text{eff}}(q^2)=\frac{\alpha_0}{1-\dfrac{\alpha_0}{3\pi}\ln\dfrac{q^2}{\mu_0^2}}, \tag{24.220}$$

which is *identical* to the renormalization-group result Eq. (24.215). Expanding Eq. (24.220) as a geometric series reproduces term by term the leading-log series Eq. (24.216): the k-bubble diagram contributes the k-th term, with its k powers of $\Pi\sim\alpha_0\ln$ supplying exactly $\alpha_0^{k+1}(\ln)^k$.

Why they agree. The agreement is not a coincidence; it is the statement that the renormalization group resums the leading logarithms. The one-loop beta function $\beta_\alpha=2\alpha^2/3\pi$ is computed from the *single-bubble* self-energy — it is the μ-derivative of Π. Integrating that beta function builds up arbitrarily many factors of the running, and each factor corresponds diagrammatically to inserting one more bubble into the chain. The RG solution $\alpha_0/(1-x)$ is the closed-form sum of the bubble geometric series; conversely, the geometric series is the perturbative unfolding of the RG solution. Either route yields Eq. (24.215), and this is the precise sense in which "solving the RG" in QED equals "summing the vacuum-polarization bubbles." The result extends the one-loop $\alpha(q^2)$ of Eq. (20.171) from a single logarithm to its all-orders leading-log resummation.

24.6.8 Solving the RG is summing the leading logarithms

The worked example exhibited, for QED, a phenomenon that holds in complete generality and is the deepest practical reason to care about the renormalization group. We make it explicit. Suppose we had attacked $\Gamma^{(n)}$ by naive fixed-order perturbation theory in the coupling g defined at the reference scale μ_0, at a process whose characteristic momentum scale is Q with $L \equiv \ln(Q^2/\mu_0^2)$. Each loop order brings one power of g and, generically, up to one power of L from the loop integral over the available phase space of virtualities between μ_0 and Q. The perturbative expansion is therefore organized as a double series in g and gL:

$$\Gamma^{(n)} = \Gamma^{(n)}_{\text{tree}}\Big[\,1 + \underbrace{c_{11}\,gL}_{1\text{ loop}} + \underbrace{c_{22}\,(gL)^2 + c_{21}\,g^2 L}_{2\text{ loop}} + \underbrace{c_{33}\,(gL)^3 + \cdots}_{3\text{ loop}} + \cdots\Big]. \tag{24.221}$$

The terms $c_{kk}(gL)^k$ on the diagonal are the *leading logarithms*: the largest contribution at each loop order when L is large. The terms with fewer logs, $g^k L^j$ with $j < k$, are subleading. Now observe the catastrophe of fixed-order truncation: if Q and μ_0 are widely separated so that $gL \sim 1$, then *every* leading-log term $(gL)^k$ is of order unity, and no finite truncation of Eq. (24.221) is reliable — the "small" expansion parameter g has been compensated by the "large" logarithm. One-loop perturbation theory, which keeps only $c_{11}gL$, is then quantitatively wrong even though g itself is small.

The renormalization group cures exactly this disease. Theorem 24.22 re-expresses $\Gamma^{(n)}$ in terms of the running coupling $\bar{g}(L)$ evaluated at the scale Q. Expanding the exact running coupling, e.g. Eq. (24.215) or Eq. (24.210), as a geometric series reproduces precisely the diagonal $\sum_k c_{kk}(gL)^k$ of Eq. (24.221) — the entire tower of leading logarithms, summed to all orders, in closed form. The reorganization is the move from a divergent (when $gL \sim 1$) fixed-order series in g to a convergent expansion in the running coupling $\bar{g}(L)$, which is itself small whenever the physics is weakly coupled at the scale Q. Improving the input — using the two-loop beta function and one-loop anomalous dimensions — upgrades the resummation to next-to-leading logarithms, $\sum_k c_{k,k-1}\,g(gL)^{k-1}$, and so on; the loop order of the RG functions controls the logarithmic accuracy of the all-orders sum.

Physical Insight: Why the RG outperforms fixed-order perturbation theory

Fixed-order perturbation theory expands in the coupling at one fixed scale and pays for every loop with a logarithm of the ratio of scales. When the scales are close (L small) this is fine and the two approaches agree order by order. When the scales are far apart ($gL \sim 1$) the logarithms overwhelm the coupling and the fixed-order series is useless — yet the physics is still weakly coupled, because the *running* coupling at the relevant scale is small.

The renormalization group is precisely the change of variables that makes this manifest: it trades the bad expansion parameter $g\,(1+L+L^2+\cdots)$ for the good one $\bar{g}(L)$, summing the dangerous logarithms into the definition of the coupling. The single most important practical lesson of this chapter is therefore methodological: never expand a multi-scale quantity in the coupling at one scale; expand it in the running coupling at the scale that governs the process, and let the RG carry you between scales. This is why the running coupling, not the fixed coupling, is the physically meaningful one — and why $\alpha(q^2)$, not $\alpha(0)$, is what enters a high-energy cross section.

With the running couplings of both threads now in closed form — Eq. (24.210) for $\lambda(\mu)$ and Eq. (24.215) for $\alpha(q^2)$ — and with the anomalous-dimension exponential of Eq. (24.202) in hand, we have everything needed to read off the long- and short-distance destiny of a theory directly from its beta function. The Landau poles we have already glimpsed, the fixed points at which the anomalous scaling Eq. (24.205) takes over, and the contrasting fate of an asymptotically free coupling, are precisely the asymptotic behaviors classified in the final section, to which we now turn.

24.7 Asymptotic Behavior: Fixed Points, Landau Poles, Triviality, and Asymptotic Freedom

The previous section did the hard analytic work. We integrated the renormalization-group equation by the method of characteristics, found the closed-form running couplings for both threads, and assembled the scheme-invariant data of the flow: the beta functions $\beta_\lambda = 3\lambda^2/16\pi^2$ for ϕ^4 theory (Section 24.2) and $\beta_\alpha = 2\alpha^2/3\pi$ for QED (Section 24.3), together with the field anomalous dimension γ_ϕ and the complete scale dependence of the Green functions (Section 24.6). What remains is to *read* those formulae. A beta function is a vector field on the space of couplings; once we possess it, the entire long- and short-distance behaviour of the theory is encoded in two elementary features of that field — the sign of β between its zeros, and the slope of β at each zero. This closing section is the dictionary that turns those two features into physics: into fixed points and the critical exponents that live on them, into the Landau poles and triviality of ϕ^4 and QED, into the contrasting fate of an asymptotically free coupling, and finally into the Wilson–Fisher fixed point that hands ϕ^4 theory to the conformal-bootstrap part. We take nothing on faith that the earlier sections did not already supply; everything here is read off numbers already in hand.

The organizing idea is simple enough to state in one sentence. The running

coupling obeys an autonomous first-order ordinary differential equation,

$$\frac{d\bar{g}}{dt} = \beta(\bar{g}), \qquad t \equiv \ln \frac{\mu}{\mu_0}, \tag{24.222}$$

where $\bar{g}(t)$ is the value of the coupling at scale $\mu = \mu_0 e^t$ (we write $\bar{g}$ for the running coupling and drop the bar when no confusion can arise). An autonomous scalar ODE is the most tractable dynamical system there is: its long-time behaviour is governed entirely by its equilibria and by the sign of the right-hand side between them. "Long time" here means $t \to +\infty$, the ultraviolet (UV), and $t \to -\infty$, the infrared (IR). The equilibria are the zeros of β. Everything follows.

24.7.1 Fixed points and the linearized flow

Definition 24.25 (Fixed point of the RG flow)**.** A *fixed point* (or *zero*) of the renormalization-group flow Eq. (24.222) is a value g_* of the coupling at which the beta function vanishes,

$$\beta(g_*) = 0. \tag{24.223}$$

A coupling started exactly at g_* does not run: $\bar{g}(t) \equiv g_*$ solves Eq. (24.222). At a fixed point the theory looks the same at every scale; it is *scale invariant*, and (under the mild assumptions discussed in Chapter 52) conformally invariant.

This is the continuum, $\overline{\text{MS}}$-language version of the Wilsonian fixed point of Chapter 23 (Theorem 23.19): there the flow was generated by lowering a cutoff and integrating out momentum shells; here it is generated by changing the subtraction scale μ. The two pictures describe the same physics, as the chapter introduction promised, and they agree on the location and nature of fixed points because those are scheme-independent (Section 24.2). The value $g_* = 0$ is always a fixed point of a theory whose interactions are switched off by setting the coupling to zero — the free theory is exactly scale invariant — and we call it the *Gaussian fixed point*, after the Gaussian (free-field) path integral it describes.

The decisive question is not whether a fixed point exists but whether the flow is *drawn toward it or pushed away from it*, and in which direction of scale. This is settled by linearizing Eq. (24.222) in the neighbourhood of g_*. Write the small deviation

$$\delta(t) \equiv \bar{g}(t) - g_*, \tag{24.224}$$

and Taylor-expand the beta function about its zero, using $\beta(g_*) = 0$:

$$\beta(\bar{g}) = \beta(g_*) + \beta'(g_*)\,\delta + \tfrac{1}{2}\beta''(g_*)\,\delta^2 + \cdots = \beta'(g_*)\,\delta + O(\delta^2). \tag{24.225}$$

To leading order in the deviation, Eq. (24.222) becomes the linear equation

$$\frac{d\delta}{dt} = \beta'(g_*)\,\delta, \tag{24.226}$$

whose solution is a pure exponential in t, that is, a power of the scale ratio:

$$\delta(t) = \delta(0)\, e^{\beta'(g_*)\, t} = \delta(0) \left(\frac{\mu}{\mu_0}\right)^{\beta'(g_*)} . \tag{24.227}$$

The single number

$$\omega \equiv \beta'(g_*) \tag{24.228}$$

— the slope of the beta function at the fixed point — governs everything. Its sign decides whether the deviation grows or shrinks as we run; its magnitude sets the rate, and, as we will see in Section 24.7.4, becomes a measurable critical exponent. The power law $\delta \sim (\mu/\mu_0)^{\omega}$, equivalently $\bar{g} - g_* \sim (\mu)^{\omega}$ at fixed reference, is the linearized signature of the fixed point; it is also exactly the form that Weinberg's asymptotic theorem (Theorem 21.25) guarantees for the deep-Euclidean behaviour of the Green functions, with the RG fixing the otherwise-undetermined exponent in terms of ω and the anomalous dimensions.

Proposition 24.26 (UV/IR attractivity from the slope of β)**.** *Let g_* be a fixed point of Eq. (24.222) with $\omega = \beta'(g_*) \neq 0$. Then, for a coupling started near g_*:*

1. *if $\omega < 0$, the deviation $\delta(t)$ shrinks as $t \to +\infty$ and grows as $t \to -\infty$: the fixed point is* ultraviolet-attractive *(the flow approaches g_* in the UV) and* infrared-repulsive*;*
2. *if $\omega > 0$, the deviation $\delta(t)$ grows as $t \to +\infty$ and shrinks as $t \to -\infty$: the fixed point is* infrared-attractive *(the flow approaches g_* in the IR) and* ultraviolet-repulsive*.*

A direction along which g is attracted to the fixed point in the IR ($\omega > 0$) is irrelevant*; a direction along which g is repelled in the IR ($\omega < 0$) is* relevant*; the borderline case $\omega = 0$ is* marginal *and is treated below.*

Proof. From Eq. (24.227), $|\delta(t)| = |\delta(0)|\, e^{\omega t}$. If $\omega < 0$ the exponent $\omega t \to -\infty$ as $t \to +\infty$, so $|\delta(t)| \to 0$: the running coupling approaches g_* in the deep UV. The same exponent $\to +\infty$ as $t \to -\infty$, so $|\delta(t)|$ grows without bound (until the linearization breaks down): the flow leaves the neighbourhood of g_* in the IR. This is statement (1). If $\omega > 0$ the two limits exchange roles, giving statement (2). The classification of the deviation as relevant, irrelevant, or marginal is the IR statement of the same dichotomy, and matches the Wilsonian terminology of Section 23.6: an operator whose coupling is driven *away* from the fixed point as one coarse-grains to the IR is one that matters at long distances — it is relevant. □

The exponent $\omega = \beta'(g_*)$ is exactly the quantity the Wilsonian analysis of Chapter 23 called the slope of the linearized RG map at the fixed point; its eigenvalues classify perturbing operators as relevant ($\omega < 0$), irrelevant ($\omega > 0$),

or marginal ($\omega = 0$). Theorem 24.26 is the continuum, beta-function-language statement of that classification, and the present ω ties back directly to the relevant/irrelevant labels of Section 23.6.

The mnemonic worth internalizing is geometric, and it is exactly what the right panel of Fig. 24.10 draws. Plot $\beta(g)$ against g and put an arrow on the g-axis pointing in the direction the coupling moves as one flows to the *infrared*, i.e. as t decreases. Since $dg/d(-t) = -\beta(g)$, that arrow points left where $\beta > 0$ and right where $\beta < 0$: the IR-flow arrow always points toward the nearest zero across which β rises from $-$ to $+$ with increasing g, and away from a zero across which β falls from $+$ to $-$. A zero with positive slope (a $- \to +$ crossing) collects the IR flow from both sides; a zero with negative slope sheds it. The UV picture is the mirror image. One never needs to re-derive Theorem 24.26; one reads it off the graph.

Physical Insight

The sign convention can be remembered without any algebra. "UV-attractive" means the coupling is *pulled in* as energies rise. A negative slope $\omega < 0$ means β is positive just below g_* and negative just above: below g_* the coupling grows with energy, above g_* it shrinks — in both cases it is herded *toward* g_* as energy increases. That is the home of asymptotic freedom and asymptotic safety: a UV-attractive interacting fixed point would render a theory predictive up to arbitrarily high energy. A positive slope does the opposite and is the generic fate of a marginal coupling that has been nudged off zero, as both our threads will demonstrate.

The Gaussian fixed point and the marginal direction. Both benchmark theories sit, at weak coupling, near the Gaussian fixed point $g_* = 0$, and for both the one-loop beta function begins at *quadratic* order in the coupling:

$$\beta_\lambda = \frac{3\lambda^2}{16\pi^2} + O(\lambda^3), \qquad \beta_\alpha = \frac{2\alpha^2}{3\pi} + O(\alpha^3). \tag{24.229}$$

Consequently $\beta(0) = 0$ *and* $\beta'(0) = 0$: the Gaussian fixed point is reached with vanishing slope. This is the marginal case excluded from Theorem 24.26, and it is no accident. In $d = 4$ the quartic coupling λ and the gauge coupling e are classically dimensionless — the corresponding operators ϕ^4 and $\bar\psi \not{A} \psi$ are marginal by power counting (Chapter 21) — so the leading term in β cannot be linear in the coupling; it must arise from a loop and is therefore quadratic. The linear analysis is silent, and the *quantum* correction, the sign of the quadratic term, breaks the tie. This is the content of the next worked example, and it is the hinge on which the triviality of ϕ^4 and QED turns.

Worked Example 24.27 (Marginal flow: how a quadratic β decides the tie)**.** Consider a single coupling near the Gaussian fixed point with

$$\beta(g) = b\,g^2 + O(g^3), \qquad b \neq 0, \tag{24.230}$$

and ask how g flows. Because $\beta'(0) = 0$, the exponential law Eq. (24.227) degenerates and we must integrate the quadratic flow exactly. Separating variables in $dg/dt = b\,g^2$,

$$\int \frac{dg}{g^2} = \int b\,dt \implies -\frac{1}{g} = b\,t + \text{const} \implies \frac{1}{\bar{g}(t)} = \frac{1}{g_0} - b\,t, \tag{24.231}$$

where $g_0 = \bar{g}(0)$ is the coupling at μ_0. The approach to the fixed point is therefore not exponential but *logarithmically slow*, and crucially it is *one-sided*:

- If $b > 0$ (the sign in both Eq. (24.229)), then $1/\bar{g}$ *decreases* as t grows, so $\bar{g}$ *increases* toward the UV and reaches zero only as $t \to -\infty$. The Gaussian fixed point is IR-attractive from $g > 0$ but UV-repulsive: a positive coupling is swept to zero only in the deep infrared, and grows in the ultraviolet — toward a singularity at finite t, examined in Section 24.7.2.
- If $b < 0$, the signs reverse: $\bar{g} \to 0^+$ as $t \to +\infty$. The Gaussian fixed point is UV-attractive — this is asymptotic freedom, Section 24.7.3.

The marginal coupling is thus marginally relevant ($b > 0$) or marginally irrelevant ($b < 0$) according to the sign of the one-loop coefficient, with the qualifier "marginally" recording that the flow is logarithmic rather than power-law. Comparing Eq. (24.231) with the closed-form running couplings of Section 24.6 identifies $b = 3/16\pi^2$ for ϕ^4 (with $g = \lambda$) and $b = 2/3\pi$ for QED (with $g = \alpha$): both have $b > 0$, so both are marginally relevant, swept to freedom only in the infrared and strengthening without bound toward the ultraviolet.

24.7.2 Landau poles and the triviality problem

The logarithmic running just found has a dramatic short-distance consequence. Take ϕ^4 theory. Specializing Eq. (24.231) with $b = 3/16\pi^2$ reproduces the closed-form running quartic coupling of Section 24.6,

$$\frac{1}{\bar{\lambda}(\mu)} = \frac{1}{\lambda(\mu_0)} - \frac{3}{16\pi^2}\ln\frac{\mu}{\mu_0}, \qquad \bar{\lambda}(\mu) = \frac{\lambda(\mu_0)}{1 - \dfrac{3\lambda(\mu_0)}{16\pi^2}\ln\dfrac{\mu}{\mu_0}}. \tag{24.232}$$

The denominator vanishes at a finite scale. Setting it to zero gives the *Landau pole* [313],

$$\mu_L = \mu_0 \exp\left(\frac{16\pi^2}{3\,\lambda(\mu_0)}\right), \tag{24.233}$$

at which the perturbative running coupling diverges. As $\mu \to \mu_L^-$ from below, $\bar{\lambda}(\mu) \to +\infty$; the left panel of Fig. 24.10 shows the blow-up. The phenomenon was first identified by Landau and collaborators in precisely the QED context to which we now turn. Exactly the same structure governs QED. Specializing Eq. (24.231) with $g = \alpha$ and $b = 2/3\pi$,

$$\bar{\alpha}(\mu) = \frac{\alpha(\mu_0)}{1 - \dfrac{2\alpha(\mu_0)}{3\pi} \ln \dfrac{\mu}{\mu_0}}, \qquad \mu_L = \mu_0 \exp\!\left(\frac{3\pi}{2\,\alpha(\mu_0)}\right). \tag{24.234}$$

This is the running fine-structure constant, the closing number of QED's RG layer: $\bar{\alpha}$ *grows* with energy, as a measurement of the effective charge at high momentum transfer confirms — the famous statement that the electron's charge is screened at long distances by vacuum polarization and the screening is undone as one probes closer. The growth is the dynamical content of the positive QED beta function.

Worked Example 24.28 (Where is the QED Landau pole?)**.** Take $\mu_0 = m_e = 0.511\,\text{MeV}$ and the low-energy value $\alpha(m_e) \simeq 1/137.036$, and keep only the single electron loop that produced $\beta_\alpha = 2\alpha^2/3\pi$. The exponent in Eq. (24.234) is

$$\frac{3\pi}{2\,\alpha(m_e)} = \frac{3\pi}{2} \times 137.036 \approx 645.6. \tag{24.235}$$

Therefore

$$\mu_L = m_e\, e^{645.6} = m_e \times 10^{645.6/\ln 10} = m_e \times 10^{280.4} \approx 1.3 \times 10^{277}\ \text{GeV}. \tag{24.236}$$

This scale is preposterous — some 258 orders of magnitude above the Planck scale $M_{\text{P}} \sim 10^{19}\,\text{GeV}$. Including the other charged fermions of the real world raises β_α and lowers μ_L by many orders, but it remains far above any scale at which QED can be the whole story. The lesson is not that the pole is a physical catastrophe waiting at $10^{277}\,\text{GeV}$; it is that long before that scale QED must be embedded in a larger theory (it is, in the electroweak interaction of Chapter 31), so the extrapolation that produces the pole was never legitimate. The middle panel of Fig. 24.10 plots $1/\bar{\alpha}$ falling linearly toward the distant zero.

What the pole does and does not mean. A Landau pole is, in the first instance, a statement about perturbation theory: the one-loop running coupling Eq. (24.210) reaches the regime $\bar{\lambda} \sim 16\pi^2$ in which the loop expansion parameter $\bar{\lambda}/16\pi^2$ is of order unity and higher-order terms are no longer negligible, so the formula cannot be trusted near μ_L. The precise scale μ_L is moreover scheme-dependent — it shifts under a finite redefinition of the coupling — exactly because it sits where the perturbative expansion that defines μ_L has broken down; only the scheme-independent leading coefficient b that controls the *onset* of the growth

is physical, as established in the scheme-universality analysis of Section 24.2. One might hope that summing more diagrams cures the disease. It does not, and the reason connects back to Chapter 5: the perturbative beta-function series is itself only asymptotic, with zero radius of convergence and factorially growing coefficients, so resummation requires the Borel technology of Section 5.7 and confronts the same nonperturbative ambiguities catalogued there. The honest non-perturbative question is whether a continuum quantum field theory with a *finite*, nonzero renormalized coupling exists at all once the cutoff is removed. This is the triviality problem.

Definition 24.29 (Triviality)**.** A quantum field theory is *trivial* if its only well-defined continuum limit — the limit in which the ultraviolet cutoff Λ is sent to infinity at fixed renormalized observables — is a free (Gaussian, non-interacting) theory: the renormalized coupling is forced to zero, $\lambda_{\text{ren}} \to 0$, as $\Lambda \to \infty$.

The heuristic is read directly off Eq. (24.210), inverted to express the bare coupling at the cutoff Λ in terms of the renormalized coupling at a low scale μ_0:

$$\lambda(\Lambda) = \frac{\lambda(\mu_0)}{1 - \dfrac{3\lambda(\mu_0)}{16\pi^2} \ln \dfrac{\Lambda}{\mu_0}}. \tag{24.237}$$

Demanding that the bare coupling stay finite (indeed positive, for a stable potential) as $\Lambda \to \infty$ forces the denominator to remain positive for arbitrarily large logarithms, which is possible only if $\lambda(\mu_0) \to 0$. In the language of Theorem 24.27: the Gaussian fixed point is UV-repulsive for $b > 0$, so the only way to keep the coupling bounded all the way into the deep UV is to sit exactly on the fixed point, i.e. to be free. The one-loop argument is suggestive but not a proof, since it is precisely the strong-coupling regime near the would-be pole that controls whether the continuum limit is interacting.

Physical Insight

Triviality is the field-theoretic shadow of marginal relevance. A marginally relevant coupling ($b > 0$) is one that the flow drives to zero in the IR and to strong coupling in the UV. Removing the cutoff means taking the UV all the way to infinity, and a marginally relevant coupling cannot survive that limit at any finite value without hitting the Landau singularity. Asymptotically free theories ($b < 0$), where the Gaussian fixed point is UV-attractive, evade the problem completely: there the cutoff can be removed with the coupling flowing harmlessly to zero. This is one of the deepest reasons the Standard Model's strong sector is a non-abelian gauge theory and not a scalar self-interaction.

Caution

A Landau pole is a sufficient symptom of likely triviality but the two are logically distinct. The pole is a property of the truncated perturbative running; triviality is a statement about the exact continuum limit. A theory can be trivial without one ever writing down a Landau pole (the rigorous ϕ^4 result below proceeds by entirely non-perturbative lattice estimates), and the appearance of a perturbative pole does not by itself *prove* that no interacting continuum limit exists. Keep the perturbative diagnostic and the non-perturbative theorem separate.

Physical Insight — Frontier: the status of triviality, as of the mid-2020s

The triviality of four-dimensional ϕ^4 scalar field theory is, since 2021, a *theorem*. Aizenman and Duminil-Copin proved that the scaling limit of the critical ϕ^4_4 model (and of the closely related 4D Ising model) is Gaussian: there is no interacting continuum scalar field theory in exactly four dimensions [11]. This confirms rigorously what the positive beta function Eq. (24.229) and decades of lattice simulation had indicated. The practical consequence in particle physics is that the Standard-Model Higgs sector, a ϕ^4-type scalar, is at best an effective theory with a finite cutoff; the cutoff dependence of the maximum renormalized quartic coupling is the "triviality bound" on the Higgs mass.

For QED the situation is weaker and remains genuinely open. There is no rigorous proof that four-dimensional QED is trivial; the evidence is the perturbative Landau pole together with lattice studies of non-compact U(1) gauge theory with fermions, which are consistent with a trivial continuum limit but do not settle it with the rigour of the scalar case. The standard expectation, as of the mid-2020s, is that pure QED is trivial in the same sense as ϕ^4, but this is a conjecture, not a theorem, and the question is in any case academic for the physical electron because QED is unified into the electroweak theory (Chapter 31) far below its Landau scale. The contrast — a scalar triviality now proved, a gauge triviality still conjectural — is a fair snapshot of where rigour stands.

24.7.3 The opposite sign: asymptotic freedom previewed

Theorem 24.27 already isolated the alternative. If the one-loop coefficient is *negative*, the entire story inverts. Write a beta function of the form encountered in non-abelian gauge theory,

$$\beta(g) = -\,b_0\,\frac{g^3}{16\pi^2} + O(g^5), \qquad b_0 > 0, \tag{24.238}$$

where the cube (rather than the square) appears because the gauge coupling's beta function is odd in g — diagrams come with an even number of vertices beyond the classical one, so β_g has an expansion in odd powers, equivalently $\beta_{\alpha_s} = -b_0\,\alpha_s^2/2\pi + \cdots$ in terms of $\alpha_s = g^2/4\pi$, which is the same quadratic marginal flow as before but with $b < 0$. Integrating Eq. (24.238) exactly, with $dg/dt = -b_0 g^3/16\pi^2$, separate variables on $g^{-3}dg$:

$$\frac{d}{dt}\left(\frac{1}{g^2}\right) = -\frac{2}{g^3}\frac{dg}{dt} = \frac{2b_0}{16\pi^2} = \frac{b_0}{8\pi^2}, \tag{24.239}$$

so that $1/g^2$ rises *linearly* with t,

$$\frac{1}{\bar{g}^2(\mu)} = \frac{1}{g^2(\mu_0)} + \frac{b_0}{8\pi^2}\ln\frac{\mu}{\mu_0}, \qquad \bar{g}^2(\mu) = \frac{g^2(\mu_0)}{1 + \dfrac{b_0\,g^2(\mu_0)}{8\pi^2}\ln\dfrac{\mu}{\mu_0}}. \tag{24.240}$$

As $\mu \to \infty$ the coupling falls to zero: the Gaussian fixed point is now UV-attractive and the theory is *asymptotically free*. The price is paid in the infrared: the denominator of Eq. (24.240) vanishes as μ decreases, at the scale

$$\Lambda = \mu_0\,\exp\!\left(-\frac{8\pi^2}{b_0\,g^2(\mu_0)}\right), \tag{24.241}$$

where the coupling formally diverges. This is the mirror image of the Landau pole — a low-energy, infrared strong-coupling scale rather than a high-energy one — and it is the origin of $\Lambda_{\rm QCD}$, the single dimensionful scale that the classically scale-invariant gauge theory generates by dimensional transmutation. The structural reason a gauge theory can have $b_0 > 0$ is the *self-interaction of the gauge field*: the non-abelian field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ (Yang–Mills theory) contains cubic and quartic terms in A, so gluons carry colour charge and *anti*screen it, overpowering the ordinary fermionic screening as long as there are not too many flavours. We do not compute b_0 here. Its derivation — the celebrated $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$, with the gauge self-interaction supplying the $+\frac{11}{3}C_A$ antiscreening and the fermions the $-\frac{4}{3}T_F n_f$ screening — is carried out in full in Chapter 27, where the running coupling Eq. (24.240) and the scale Eq. (24.241) acquire their physical meaning for QCD. The machinery is exactly the one developed in this chapter; only the sign of one number changes, and with it the fate of the theory.

Physical Insight

The asymmetry between our two threads and Yang–Mills theory is the single most consequential fact in the asymptotic classification, and it is worth stating starkly. ϕ^4 and QED are marginally *relevant*: they are free in the infrared and want to be strongly coupled in the ultraviolet, where they

meet a Landau pole and (for ϕ^4, provably) triviality. A non-abelian gauge theory with few enough flavours is marginally *irrelevant*: it is free in the ultraviolet and grows strong in the infrared, where it confines. The same one-loop machinery, the same separable ODE, the same closed-form $1/g^2$-linear-in-log solution — the only difference is one sign, $b_0 > 0$ versus $b < 0$. That a single sign separates a theory that cannot exist nonperturbatively up to arbitrary energies from one that can, and from one that confines, is the cleanest illustration of why the renormalization group, and not any single Feynman diagram, is the organizing principle of high-energy physics.

24.7.4 Anomalous dimension at a fixed point: the critical exponent η

A fixed point is more than a place where the coupling stops running. It is a scale-invariant theory, and the field anomalous dimension γ_ϕ evaluated there is a genuine, measurable number: it is the critical exponent η that controls the power law of the two-point function. We now make this identification precise, deriving it from the RG equation for the Green functions established in Section 24.6.

Recall that equation. The renormalized connected two-point function $G^{(2)}(p;\mu,g)$ — the full propagator in momentum space — satisfies the Callan–Symanzik/renormalization-group equation

$$\left[\mu\frac{\partial}{\partial\mu}+\beta(g)\frac{\partial}{\partial g}+2\gamma_\phi(g)\right]G^{(2)}(p;\mu,g)=0, \tag{24.242}$$

with $\gamma_\phi(g)=\frac{1}{2}\,\mu\,\dfrac{d\ln Z_\phi}{d\mu}$ the field anomalous dimension of Section 24.6 (the coefficient is $2\gamma_\phi = n\gamma_\phi$ with $n=2$ external legs). At a fixed point $g=g_*$ the term $\beta(g_*)\partial_g$ drops out and $\gamma_\phi(g_*)\equiv\gamma_*$ is a constant, so the equation collapses to a statement about the μ-dependence alone:

$$\left[\mu\frac{\partial}{\partial\mu}+2\gamma_*\right]G^{(2)}(p;\mu,g_*)=0. \tag{24.243}$$

Now bring in dimensional analysis, the second ingredient. In d spacetime dimensions the field ϕ has mass dimension $(d-2)/2$, so the position-space two-point function $\langle\phi(x)\phi(0)\rangle$ has dimension $d-2$ and its Fourier transform $G^{(2)}(p)=\int d^dx\,e^{ip\cdot x}\langle\phi(x)\phi(0)\rangle$ has mass dimension $(d-2)-d=-2$, in any dimension. Since the only scales available are $p\equiv\sqrt{|p^2|}$ and μ, dimensional analysis forces

$$G^{(2)}(p;\mu,g_*)=\frac{1}{p^2}\,F\!\left(\frac{p}{\mu}\right), \tag{24.244}$$

with F a dimensionless function (its dependence on the fixed-point coupling g_* is just a constant, suppressed). The two equations Eqs. (24.243) and (24.244) together pin down the power law.

Proposition 24.30 (Anomalous dimension = critical exponent)**.** *At a fixed point g_*, the momentum-space two-point function is a pure power,*

$$G^{(2)}(p) \;\propto\; (p^2)^{-1+\gamma_*}, \tag{24.245}$$

equivalently, in position space,

$$\langle \phi(x)\phi(0)\rangle \;\propto\; \frac{1}{|x|^{\,d-2+\eta}}, \qquad \boxed{\eta = 2\gamma_* = 2\gamma_\phi(g_*)}. \tag{24.246}$$

The field anomalous dimension at the fixed point is *the critical exponent η (the "anomalous dimension of the order parameter"), and the full scaling dimension of ϕ at the fixed point is $\Delta_\phi = \frac{d-2}{2} + \gamma_*$.*

Proof. Convert the μ-derivative in Eq. (24.243) into a p-derivative using Eq. (24.244). With $G^{(2)} = p^{-2}F(p/\mu)$ and writing $u \equiv p/\mu$,

$$\mu\frac{\partial G^{(2)}}{\partial \mu} = p^{-2}\,\mu\frac{\partial}{\partial \mu}F(p/\mu) = p^{-2}\,F'(u)\cdot\mu\,\frac{\partial}{\partial\mu}\left(\frac{p}{\mu}\right) = p^{-2}\,F'(u)\cdot\left(-\frac{p}{\mu}\right) = -\,p^{-2}\,u\,F'(u). \tag{24.247}$$

The combination $u\,F'(u)$ is exactly what the p-derivative produces as well: differentiating $G^{(2)} = p^{-2}F(p/\mu)$ with respect to p at fixed μ,

$$p\frac{\partial G^{(2)}}{\partial p} = -2\,p^{-2}F(u) + p^{-2}\,u\,F'(u) = -2\,G^{(2)} + p^{-2}\,u\,F'(u), \tag{24.248}$$

so that $p^{-2}\,u\,F'(u) = p\,\partial_p G^{(2)} + 2\,G^{(2)}$. Substituting into Eq. (24.247) gives the homogeneity relation

$$\mu\frac{\partial G^{(2)}}{\partial \mu} = -\left[p\frac{\partial}{\partial p} + 2\right] G^{(2)}, \tag{24.249}$$

which simply says $G^{(2)}$ has mass dimension -2 in the variables (p,μ). Inserting Eq. (24.249) into the fixed-point equation Eq. (24.243),

$$-\left[p\frac{\partial}{\partial p} + 2\right] G^{(2)} + 2\gamma_*\,G^{(2)} = 0 \;\Longrightarrow\; p\frac{\partial G^{(2)}}{\partial p} = (2\gamma_* - 2)\,G^{(2)}. \tag{24.250}$$

This is a homogeneous scaling equation whose solution is the power law $G^{(2)} \propto p^{\,2\gamma_*-2} = (p^2)^{-1+\gamma_*}$, which is Eq. (24.245). The free-field case $\gamma_* = 0$ recovers $G^{(2)} \propto 1/p^2$, the massless propagator, as it must. That this power law *is* the genuine large-momentum asymptotics of the full Green function, and not merely a formal solution, is the content of Weinberg's asymptotic theorem (Theorem 21.25): in the deep-Euclidean regime the Green functions approach homogeneous functions of their momenta up to logarithms, and the RG fixes the homogeneity degree to be exactly $2\gamma_* - 2$.

To translate to position space, Fourier-transform back in d dimensions. A power $|p|^{-2a}$ transforms (up to a finite, a-dependent constant) into $|x|^{-(d-2a)}$:

$$\int \frac{d^d p}{(2\pi)^d}\, e^{-ip\cdot x}\, \frac{1}{|p|^{2a}} \;\propto\; \frac{1}{|x|^{\,d-2a}}. \tag{24.251}$$

Here the exponent is $2a = 2 - 2\gamma_*$, so $d - 2a = d - 2 + 2\gamma_*$, and

$$\langle \phi(x)\phi(0)\rangle \;\propto\; \frac{1}{|x|^{\,d-2+2\gamma_*}}. \tag{24.252}$$

Comparison with the conventional definition of the critical exponent η through $\langle\phi(x)\phi(0)\rangle \propto |x|^{-(d-2+\eta)}$ at criticality gives $\eta = 2\gamma_*$, which is Eq. (24.246). Finally, a primary field of scaling dimension Δ_ϕ in a scale-invariant theory has $\langle\phi(x)\phi(0)\rangle \propto |x|^{-2\Delta_\phi}$, so $2\Delta_\phi = d - 2 + 2\gamma_*$, i.e. $\Delta_\phi = \frac{d-2}{2} + \gamma_*$: the engineering dimension $(d-2)/2$ shifted by the anomalous piece γ_*, which is exactly the meaning of the word "anomalous". □

Physical Insight

This is the bridge between particle physics and statistical mechanics, and it is why the renormalization group is the common language of both. In a magnet at its critical temperature the spin–spin correlation falls off as $|x|^{-(d-2+\eta)}$, and η is measured in the laboratory. Theorem 24.30 says that this experimental number is nothing but the value, at the interacting fixed point, of the same field anomalous dimension γ_ϕ that we computed diagrammatically from wavefunction renormalization. The Green function of the quantum field and the correlation function of the classical order parameter are the *same object* at the fixed point, and "critical exponent" and "anomalous dimension" are two names for one number. The conformal field theory that lives at the fixed point, taken up in Chapter 52, makes $\Delta_\phi = \frac{d-2}{2} + \gamma_*$ the scaling dimension of a primary operator, and the bootstrap of Chapter 52 computes it to extraordinary precision for the 3D Ising universality class.

24.7.5 The Wilson–Fisher fixed point in $d = 4 - \varepsilon$

The Gaussian fixed point is the only fixed point ϕ^4 has in exactly four dimensions, where $\beta_\lambda = 3\lambda^2/16\pi^2$ has its single zero at $\lambda = 0$. Wilson and Fisher's insight was that an interacting fixed point appears as soon as one leaves $d = 4$, and that for d just below four it sits at *weak coupling*, where the one-loop beta function we have already computed is enough to locate it [515]. This closes the Chapter 24 layer of ϕ^4 theory and delivers the coupling λ_* that Chapter 52 will reinterpret as a conformal field theory.

The new ingredient is the classical scaling of the coupling away from $d = 4$. In d dimensions the scalar field has mass dimension $(d-2)/2$, so the quartic operator ϕ^4 has dimension $2(d-2)$ and the coupling in $\mathcal{L} \supset -\frac{\lambda}{4!}\phi^4$ has mass dimension

$$[\lambda] = d - 2(d-2) = 4 - d. \tag{24.253}$$

In four dimensions $[\lambda] = 0$ and λ is marginal, as we have used throughout. Write $d = 4 - \varepsilon$ with $\varepsilon \equiv 4 - d > 0$ small; then $[\lambda] = \varepsilon$, the coupling carries a positive mass dimension, and to keep a dimensionless renormalized coupling one factors out μ^{ε}. The running of the dimensionless coupling then acquires, in addition to the quantum (loop) term, a *classical* contribution from this engineering dimension: holding the bare coupling fixed, the dimensionless renormalized coupling runs as $-[\lambda]\,\lambda = -\varepsilon\lambda$ at tree level. The one-loop beta function in $d = 4 - \varepsilon$ is therefore the four-dimensional loop term plus this classical drift,

$$\beta_\lambda = -\varepsilon\,\lambda + \frac{3\lambda^2}{16\pi^2} + O(\lambda^3, \varepsilon\lambda^2). \tag{24.254}$$

The loop coefficient $3/16\pi^2$ is the same number computed in Section 24.2; its leading value is dimension-independent, with corrections of order ε that affect only higher orders of what follows. This is precisely the "classical tilt" of the beta function that the deferred-pole analysis of Section 24.1.4 anticipated, where the same fixed point appeared from 't Hooft's relation; we now read it off directly.

Notation Watch

The parameter ε used in this subsection is the *dimension deficit* $\varepsilon = 4 - d$, the standard expansion parameter of critical phenomena and the convention in which the Wilson–Fisher coupling is universally quoted. It is held *small and finite*, not sent to zero. This is twice the dimensional-regularization regulator of the book's canon, where $d = 4 - 2\varepsilon_{\text{dimreg}}$ and $\varepsilon_{\text{dimreg}} \to 0$ is removed at the end: here $\varepsilon = 2\varepsilon_{\text{dimreg}}$. With $d = 4 - \varepsilon$ the classical term is $-\varepsilon\lambda$, and the fixed point lands at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$, the value ϕ^4 theory carries forward. In the canonical $d = 4 - 2\varepsilon_{\text{dimreg}}$ variables the same fixed point reads $\lambda_* = \frac{32\pi^2}{3}\varepsilon_{\text{dimreg}}$; the physics is identical, and Chapter 52 adopts the $\varepsilon = 4 - d$ convention used here.

Proposition 24.31 (Location of the Wilson–Fisher fixed point)**.** *The beta function Eq. (24.254) has two zeros: the Gaussian fixed point $\lambda = 0$ and a nontrivial* Wilson–Fisher *fixed point*

$$\lambda_* = \frac{16\pi^2}{3}\,\varepsilon + O(\varepsilon^2). \tag{24.255}$$

For $0 < \varepsilon \ll 1$ this is a weak-coupling fixed point ($\lambda_* = O(\varepsilon)$). At it, the Gaussian fixed point is infrared-repulsive (relevant) while the Wilson–Fisher fixed point is

infrared-attractive (irrelevant), with slope $\omega = \beta_\lambda'(\lambda_*) = +\varepsilon$. *The leading field anomalous dimension there is of order* ε^2,

$$\gamma_\phi(\lambda_*) = O(\varepsilon^2), \qquad \textit{equivalently} \qquad \eta = 2\gamma_\phi(\lambda_*) = \frac{\varepsilon^2}{54} + O(\varepsilon^3) \ \ (\textit{one component}), \tag{24.256}$$

so that the anomalous dimension, and the deviation of the critical exponent η *from its Gaussian value, are second order in the dimension deficit.*

Proof. Setting $\beta_\lambda = 0$ in Eq. (24.254),

$$-\varepsilon\,\lambda + \frac{3\lambda^2}{16\pi^2} = 0 \implies \lambda\left(-\varepsilon + \frac{3\lambda}{16\pi^2}\right) = 0, \tag{24.257}$$

whose roots are $\lambda = 0$ and $\lambda_* = \frac{16\pi^2}{3}\varepsilon$, establishing Eq. (24.255). To classify them, differentiate the beta function,

$$\beta_\lambda'(\lambda) = -\varepsilon + \frac{6\lambda}{16\pi^2} = -\varepsilon + \frac{3\lambda}{8\pi^2}. \tag{24.258}$$

At the Gaussian fixed point, $\beta_\lambda'(0) = -\varepsilon < 0$: by Theorem 24.26 the free theory is now UV-attractive and IR-repulsive — in $d < 4$ the quartic coupling has become *relevant*, as power counting already told us ($[\lambda] = \varepsilon > 0$). At the Wilson–Fisher fixed point,

$$\beta_\lambda'(\lambda_*) = -\varepsilon + \frac{3}{8\pi^2}\cdot\frac{16\pi^2}{3}\varepsilon = -\varepsilon + 2\varepsilon = +\varepsilon > 0, \tag{24.259}$$

so it is IR-attractive: the long-distance flow of the interacting theory, for any nonzero starting coupling, is drawn to λ_*. This is the dynamical meaning of universality (Section 23.6) — a whole basin of microscopic theories ends up governed by the single fixed-point theory at λ_*. The slope $\omega = \beta_\lambda'(\lambda_*) = +\varepsilon$ is itself a critical exponent: it is the *correction-to-scaling* exponent (the same ω identified in Section 24.4), which controls the leading irrelevant approach to the fixed point. It is *not* the correlation-length exponent ν; the latter is governed by the relevant (mass) direction, the eigenvalue of the linearized flow of the ϕ^2 coupling, and gives $1/\nu = 2 - \gamma_{\phi^2}(\lambda_*) + \cdots = 2 - \frac{\varepsilon}{3} + \cdots$ for the one-component class in the conventions of Chapter 52.

For the anomalous dimension, recall from Section 24.2 that in ϕ^4 the field-strength renormalization Z_ϕ receives no $O(\lambda)$ correction: the one-loop self-energy is the momentum-independent tadpole, which renormalizes the mass but not the kinetic term, so the first contribution to wavefunction renormalization is the two-loop "sunset" diagram and

$$\gamma_\phi(\lambda) = \frac{\lambda^2}{12\,(16\pi^2)^2} + O(\lambda^3). \tag{24.260}$$

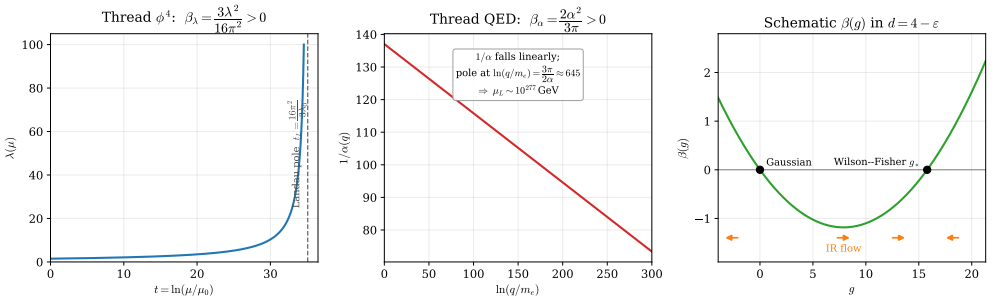


Figure 24.10: The three faces of the asymptotic story. *Left:* the running quartic coupling $\bar\lambda(\mu)$ of ϕ^4 theory, Eq. (24.210), growing toward the ultraviolet and diverging at the Landau pole $t_L = 16\pi^2/3\lambda_0$ (dashed). *Middle:* the inverse running fine-structure constant $1/\bar\alpha(q)$ of QED, Eq. (24.234), falling linearly toward its distant zero; the genuine pole sits at $\ln(q/m_e) \approx 645$, corresponding to $\mu_L \sim 10^{277}$ GeV (Theorem 24.28). *Right:* a schematic beta function in $d = 4 - \varepsilon$, Eq. (24.254), with the Gaussian zero at $g = 0$ and the Wilson–Fisher zero at $g_* = 16\pi^2\varepsilon/3$; the arrows show the infrared flow (direction $-\operatorname{sgn}\beta$), repelled from the UV-attractive Gaussian point and collected at the IR-attractive Wilson–Fisher point.

Since γ_ϕ starts at $O(\lambda^2)$ and $\lambda_* = O(\varepsilon)$, evaluating at the fixed point gives $\gamma_\phi(\lambda_*) = O(\varepsilon^2)$ immediately, which is the parametric statement in Eq. (24.256). Inserting the explicit two-loop coefficient Eq. (24.260) and $\lambda_* = \frac{16\pi^2}{3}\varepsilon$,

$$\gamma_\phi(\lambda_*) = \frac{1}{12\,(16\pi^2)^2}\left(\frac{16\pi^2}{3}\varepsilon\right)^2 = \frac{1}{12}\cdot\frac{1}{9}\,\varepsilon^2 = \frac{\varepsilon^2}{108}, \tag{24.261}$$

so that the critical exponent $\eta = 2\gamma_\phi(\lambda_*) = \varepsilon^2/54$ for the one-component scalar, the result quoted in Eq. (24.256). The vanishing of η at $O(\varepsilon)$ is the precise statement that, to leading order in the deficit, the Wilson–Fisher field still carries its Gaussian scaling dimension; the interaction first shows up in the correlation power at second order. □

The full two-loop derivation of Eq. (24.260) and the precise mapping of $\varepsilon = 1$ (i.e. $d = 3$) onto the laboratory critical exponents of the three-dimensional Ising universality class — where the bare ε-expansion must be resummed and is ultimately superseded by the conformal bootstrap — are deferred to Chapter 52; here we have established only what the one-loop beta function and the existence of a two-loop γ_ϕ already determine: that the fixed point exists at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ and that its anomalous dimension is $O(\varepsilon^2)$.

With the asymptotic fate of both threads now read directly off their beta functions — ϕ^4 and QED marginally relevant and Landau-bounded, the Wilson–Fisher point sitting at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ with $\eta = O(\varepsilon^2)$, and the asymptotically free

sign awaiting one negative coefficient — the renormalization group has done its work for Part VI. Three roads lead out: the negative-sign computation that delivers asymptotic freedom and Λ_{QCD} for Yang–Mills theory in Chapter 27; the conformal field theory that lives at the Wilson–Fisher fixed point and the 3D Ising bootstrap it hands ϕ^4 theory in Chapter 52; and the running of operators and the decoupling of heavy thresholds that turns this machinery into the working language of effective field theory in Chapter 38.

Further reading

The renormalization-group equation in its modern multiplicative form is due to 't Hooft [467] and Weinberg; the Callan–Symanzik equation and its inhomogeneous structure originate with Callan [106] and Symanzik [462], building on the running-coupling idea of Gell-Mann and Low [215]. The pole-equation derivation of beta functions in dimensional regularization is 't Hooft's [465, 469]. The asymptotically free sign that this chapter only previews was found by Gross and Wilczek [238] and Politzer [406], and the nontrivial zero in $d = 4 - \varepsilon$ by Wilson and Fisher [515]. For thorough textbook treatments matching the conventions used here, see Peskin and Schroeder [396, Ch. 12], Schwartz [432, Ch. 23], Srednicki [454, Ch. 28], and the renormalization-group volume of Weinberg [501]; the scheme-dependence analysis and the critical-exponent connection are developed at length in Zinn-Justin [534] and, with full rigor on the operator-product and RG side, in Collins [134]. Coleman's lectures [129] remain the clearest narrative account of broken scale invariance and the trace anomaly.

Part VII

Non-Abelian Gauge Theories

Chapter 25

Yang–Mills Theory

Electromagnetism, built in Chapter 8, taught the gauge principle in its gentlest form: demand that a global U(1) phase be allowed to vary from point to point, and the photon, the covariant derivative, and the field strength as a curvature all follow of necessity. This chapter asks the same question of a *non-abelian* symmetry. The matter fields now carry a multiplet index and transform in a representation of a compact Lie group G — the representation theory of Chapter 2, with its Hermitian generators T^a, structure constants $[T^a, T^b] = if^{abc}T^c$, and the fundamental normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, is the language we speak throughout, and the numerical Casimir and index data it rests on are catalogued in Chapter B. The metric, units, and covariant-derivative sign conventions are those of Chapter A : mostly-minus signature, $\hbar = c = 1$, and $D_\mu = \partial_\mu - igA^a_\mu T^a$. From this one change — a symmetry group whose generators do not commute — everything that distinguishes Yang–Mills theory from Maxwell's follows, and the bulk of the chapter is the patient unfolding of those consequences.

The arc is the gauge principle run to completion. We begin (Section 25.1) by localizing the symmetry and watching the ordinary derivative fail to be covariant; the matrix-valued Leibniz obstruction cannot be removed by any field redefinition, and a Lie-algebra-valued connection is *forced*. The next section (Section 25.2) pins down exactly how that connection must transform, in finite, infinitesimal, and adjoint-covariant forms, and isolates the adjoint covariant derivative as a reusable operator. Differentiating the connection produces its curvature (Section 25.3): the non-abelian field strength, defined as the commutator of covariant derivatives, carrying the characteristic self-term $gf^{abc}A^b_\mu A^c_\nu$, transforming covariantly by conjugation rather than invariantly, and obeying the homogeneous Bianchi identity. Covariance under conjugation is precisely the licence to build a gauge-invariant action by tracing, which Section 25.4 does: the unique Lorentz-invariant, gauge-invariant, renormalizable kinetic term $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} = -\frac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$, minimal coupling to matter, the Yang–Mills field equation $D_\mu F^{\mu\nu\,a} = -gJ^{\nu a}$, and the defining novelty that the gauge field is itself coloured — the conserved colour

current splits into a matter piece and a gauge-field piece, and the distinction between ordinary and covariant conservation (drawing on Noether's theorem and the stress-tensor analysis of Chapter 7) becomes physically sharp.

What the single symbol $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ hides is exposed in Section 25.5: expanding in powers of the field reveals, alongside the free propagating gluon, the cubic three-gluon and quartic four-gluon self-couplings, and a gauge-variation computation shows their relative coefficients are not free parameters but are rigidly fixed by the symmetry. These vertices are the raw material the quantization of Chapter 26 will dress into Feynman rules. The last two sections turn from the local field to the global picture. Parallel transport of colour along a path is encoded in the Wilson line, a path-ordered exponential whose ordering is forced by the non-commutativity of the connection; tracing it around a closed loop yields the Wilson loop (Section 25.6), the natural gauge-invariant observable, which recovers the curvature on small loops and diagnoses confinement on large ones. Finally (Section 25.7) we confront the redundancy head-on: connections fall into gauge orbits, the gauge group is disconnected with $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$ labelling large gauge transformations by an integer winding number, and the consequent overcounting of the path integral — together with the Gribov obstruction to a global gauge slice — is exactly the problem the Faddeev–Popov construction of the next chapter is built to tame.

This chapter is the opening of Yang–Mills theory and is entirely classical. The model fixed here — the fields A^a_μ in the adjoint, matter ψ_i in a representation, the single coupling g, and for SU(2) the identification $f^{abc} = \epsilon^{abc}$ — is inherited unchanged by every later stage: quantization and BRST in Chapter 26, asymptotic freedom in Chapter 27, the θ-vacuum and instantons in Chapter 42, and the modern amplitude methods of Chapters 59 and 61. Everything proved here is a theorem of classical field theory; the construction of the full quantum theory, with its conjectured mass gap, is by contrast an open problem, and we will say so plainly where the classical rigor ends.

25.1 The Non-Abelian Gauge Principle

The abelian gauge principle of Chapter 8 is in hand, and so is the Lie-theoretic toolkit of Chapter 2. The chapter introduction framed the move that organizes all of Part VII in a single sentence: promote a global internal symmetry to a local one, and let the geometry of that demand manufacture the interaction. Chapter 8 ran exactly this program for the abelian group U(1) and recovered the whole of classical electrodynamics — minimal coupling, the photon, the field strength as a curvature — with nothing put in by hand beyond the demand of local invariance. The single question that opens this chapter, and whose answer is the entire non-abelian theory, is this: what changes when the phase that we wish to rotate independently at each point is no longer a number but a *matrix*? The

abelian phase $e^{iq\alpha(x)}$ commutes with everything in sight, and that commutativity quietly did a great deal of work in Chapter 8. The instant we replace it by an element of a non-abelian group, the obstruction that the gauge principle must overcome acquires a piece that has no abelian analogue, and the field invented to repair it inherits a self-interaction. This section isolates precisely that new piece and shows why it forces a matrix-valued connection. It does no more than that: the transformation law of the connection itself is the business of Section 25.2.

We open Yang–Mills theory at its classical foundation here. The matter content, coupling, and group are frozen will be carried, unchanged in symbol and convention, through quantization in Chapter 26 and asymptotic freedom in Chapter 27. We state the conventions once, at the point of first use, and then never vary them.

25.1.1 Matter in a representation of a compact group

In Chapter 8 the charged field was a single complex scalar ϕ, or a single Dirac field ψ, carrying one real charge q under U(1). The non-abelian generalization replaces that single field by a *multiplet*: a column of fields that the symmetry group rotates into one another. The cleanest carrier for the whole argument is a Dirac field in the fundamental representation of SU(N), which is also exactly the matter content frozen for Yang–Mills theory, so we adopt it from the outset; the scalar case differs only in that the Dirac index and the γ-matrices ride along as spectators, a point we note where it matters.

Definition 25.1 (Matter multiplet in a representation R)**.** Let G be a compact Lie group — by default $G = \mathrm{SU}(N)$ — with Lie algebra $\mathfrak{g}$ (Theorem 2.39), and let R be a finite-dimensional unitary representation of G on a complex vector space V_R of dimension d_R, with Hermitian generators T^a ($a = 1, \dots, \dim G$) obeying

$$\left[T^a, T^b\right] = i f^{abc} T^c, \qquad \mathrm{tr}(T^a T^b) = \tfrac{1}{2}\,\delta^{ab} \quad \text{(fundamental normalization)}, \tag{25.1}$$

the conventions fixed in Section 2.3.4 and tabulated for SU(N) in Chapter B. A *matter multiplet* in the representation R is a collection of fields $\psi_i(x)$, $i = 1, \dots, d_R$, assembled into a column $\psi = (\psi_1, \dots, \psi_{d_R})^{\mathsf{T}}$ on which a group element acts by the matrix $R(U) \in \mathrm{U}(d_R)$. For $G = \mathrm{SU}(N)$ in the fundamental, $d_R = N$, $R(U) = U$ is the defining $N \times N$ unitary matrix, and each component ψ_i is itself a Dirac field carrying a suppressed spinor index; the group matrix U acts only on the internal index i and commutes with every γ-matrix and with ∂_μ.

The frozen Yang–Mills theory matter Lagrangian, before any gauging, is the free Dirac Lagrangian summed over the multiplet,

$$\mathcal{L}_0 \;=\; \bar\psi_i (i\partial\!\!\!/ - m)\psi_i \;=\; \bar\psi (i\partial\!\!\!/ - m)\psi, \tag{25.2}$$

where the repeated internal index i is summed, $\bar\psi_i = \psi_i^\dagger\gamma^0$, and in the second form ψ is the column and the internal contraction is the matrix product $\bar\psi\,\psi = \sum_i \bar\psi_i\psi_i$. The metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and units are natural; the γ-conventions are those of Chapter C. The mass m is common to all components — a single number, not a matrix — precisely because we want the free theory to respect the symmetry, as the next definition makes precise.

A constant group element generates the global symmetry. Writing $U = \exp(i\alpha^a T^a)$ with $\alpha^a \in \mathbb{R}$ constant, the *global* transformation of the multiplet is

$$\psi(x) \;\longmapsto\; \psi'(x) = U\,\psi(x), \qquad \bar\psi(x) \;\longmapsto\; \bar\psi'(x) = \bar\psi(x)\,U^\dagger, \tag{25.3}$$

the direct non-abelian generalization of the abelian rule Eq. (8.2), to which it collapses when $G = \mathrm{U}(1)$, $T^a \to q$, and $U \to e^{iq\alpha}$. The bar transforms with $U^\dagger$ because conjugation sends $\psi \mapsto U\psi$ to $\psi^\dagger \mapsto \psi^\dagger U^\dagger$ and γ^0 is inert under the internal group. That the free Lagrangian Eq. (25.2) is invariant under Eq. (25.3) is not automatic — it is exactly a statement about R being *unitary* — and because this is the hinge on which the whole construction turns, we check it in full rather than assert it.

Worked Example 25.2 (Global invariance holds iff R is unitary)**.** We show that both the mass term and the kinetic term of Eq. (25.2) are invariant under the constant transformation Eq. (25.3) if and only if $U^\dagger U = \mathbf{1}$, i.e. iff R is a unitary representation in the sense of Theorem 2.34.

Mass term. Under Eq. (25.3),

$$\bar\psi\,\psi \;\longmapsto\; \bar\psi'\psi' \;=\; (\bar\psi\,U^\dagger)(U\,\psi) \;=\; \bar\psi\,(U^\dagger U)\,\psi. \tag{25.4}$$

The internal matrix $U^\dagger U$ sits between the column ψ and the row $\bar\psi$, contracting the d_R internal indices. The expression equals the original $\bar\psi\psi$ for *all* field configurations if and only if $U^\dagger U = \mathbf{1}$. There is no escape through a clever cancellation: take ψ to have a single nonzero component to read off any individual matrix element of $U^\dagger U$, so invariance for all ψ forces the matrix identity. Thus the common mass m in Eq. (25.2) is mandatory — a mass matrix m_{ij} not proportional to the identity would single out a direction in V_R and break Eq. (25.3) unless it commuted with every U, which by Schur's lemma (Theorem 2.21) means a multiple of the identity on each irreducible block.

Kinetic term. Because U is constant, ∂_μ passes through it untouched: $\partial_\mu(U\psi) = U\,\partial_\mu\psi$. Hence

$$\bar\psi\, i\partial\!\!\!/\,\psi \;\longmapsto\; \bar\psi\,U^\dagger\, i\partial\!\!\!/\,(U\psi) \;=\; \bar\psi\,U^\dagger U\, i\partial\!\!\!/\psi \;=\; \bar\psi\, i\partial\!\!\!/\psi, \tag{25.5}$$

again invariant precisely when $U^\dagger U = \mathbf{1}$. For a compact group every finite-dimensional representation can be made unitary (Theorem 2.34 and the averaging

argument there), and the Hermiticity of the generators in Eq. (25.1) is exactly the infinitesimal statement of $U^\dagger U = \mathbf{1}$: writing $U = \mathbf{1} + i\alpha^a T^a + O(\alpha^2)$,

$$U^\dagger U = (\mathbf{1} - i\alpha^a (T^a)^\dagger)(\mathbf{1} + i\alpha^b T^b) = \mathbf{1} + i\alpha^a (T^a - (T^a)^\dagger) + O(\alpha^2), \tag{25.6}$$

so $U^\dagger U = \mathbf{1}$ to first order iff $(T^a)^\dagger = T^a$. This is why we took the generators Hermitian: it is the same demand as the global invariance of the free theory. Compactness, unitarity of R, Hermitian generators, and the invariance of Eq. (25.2) are four faces of one fact.

25.1.2 Localizing the symmetry: the matrix Leibniz obstruction

The global transformation Eq. (25.3) costs nothing because a constant U commutes with the derivative. The gauge principle (Theorem 8.3) is the demand that we be allowed to choose the internal frame independently at each spacetime point — to promote the constant U to a field.

Definition 25.3 (Local (gauge) transformation)**.** A *local* or *gauge* transformation is a smooth map $U : x \mapsto U(x) \in G$, acting on the matter multiplet by

$$\psi(x) \;\longmapsto\; \psi'(x) = U(x)\,\psi(x), \qquad \bar\psi(x) \;\longmapsto\; \bar\psi'(x) = \bar\psi(x)\,U^\dagger(x), \tag{25.7}$$

with $U(x) = \exp(i\alpha^a(x) T^a)$ and the parameters $\alpha^a(x)$ now arbitrary smooth functions of spacetime, one independent function per generator.

The mass term survives this promotion verbatim, because the algebra in Eq. (25.4) never moved a derivative past U: $\bar\psi'\psi' = \bar\psi\, U^\dagger(x) U(x)\, \psi = \bar\psi\psi$ point by point, the unitarity $U^\dagger(x)U(x) = \mathbf{1}$ holding at each x separately. The kinetic term is where everything happens, and it happens through the one step that Eq. (25.5) could take for granted and now cannot: moving ∂_μ through $U(x)$. The product rule gives

$$\partial_\mu\big(U(x)\psi(x)\big) \;=\; U(x)\,\partial_\mu\psi(x) \;+\; \big(\partial_\mu U(x)\big)\,\psi(x). \tag{25.8}$$

The first term is what we want: $\partial_\mu\psi$ rotated by the same U that rotates ψ. The second term, $(\partial_\mu U)\psi$, is the obstruction. It is the exact non-abelian counterpart of the inhomogeneous piece that produced Eq. (8.11) in the abelian theory, and at this level of generality the two computations read identically. The whole of the new physics hides in what $\partial_\mu U$ *is*, and how it relates to U.

In the abelian case $U = e^{iq\alpha(x)}$ is a number, so $\partial_\mu U = iq(\partial_\mu\alpha)\,U = U\,iq(\partial_\mu\alpha)$: the derivative of the phase is proportional to the phase itself, and it does not matter whether we write the factor on the left or the right. The obstruction term is then $(\partial_\mu U)\psi = iq(\partial_\mu\alpha)\,U\psi$, neatly aligned with $U\psi$, and a single vector field A_μ shifting by a gradient absorbs it. None of that holds for a matrix U. Differentiating $U(x) = \exp(i\alpha^a(x)T^a)$ does not return a multiple of U, because

the generators inside the exponential do not commute with their own derivatives in general; and crucially,

$$(\partial_\mu U)\,U^{-1} \;\neq\; U^{-1}\,(\partial_\mu U) \qquad \text{in general,} \tag{25.9}$$

so there is no longer any notion of the obstruction being "aligned" with U from one side. Let us see the failure of invariance explicitly. Substituting Eq. (25.8) into the kinetic term and using $\bar\psi' = \bar\psi U^\dagger$,

$$\begin{aligned} \bar\psi\, i\partial\!\!\!/\,\psi \;&\longmapsto\; \bar\psi\, U^\dagger\, i\gamma^\mu\, \partial_\mu (U\psi) \\ &=\; \bar\psi\, U^\dagger\, i\gamma^\mu\, U\, \partial_\mu\psi \;+\; \bar\psi\, U^\dagger\, i\gamma^\mu\, (\partial_\mu U)\,\psi \\ &=\; \bar\psi\, i\partial\!\!\!/\,\psi \;+\; \bar\psi\, i\gamma^\mu\, (U^\dagger\partial_\mu U)\,\psi, \end{aligned} \tag{25.10}$$

where in the last line $U^\dagger U = \mathbf{1}$ collapsed the first term back to its original form. The change in the Lagrangian is therefore

$$\delta\mathcal{L}_0 \;=\; \bar\psi\, i\gamma^\mu\, (U^\dagger\partial_\mu U)\,\psi, \tag{25.11}$$

the non-abelian analogue of Eq. (8.11). The object $U^\dagger\partial_\mu U = U^{-1}\partial_\mu U$ that appears is the central character. It is anti-Hermitian and Lie-algebra-valued: differentiating $U^\dagger U = \mathbf{1}$ gives $(\partial_\mu U^\dagger)U + U^\dagger\partial_\mu U = 0$, i.e. $(U^\dagger\partial_\mu U)^\dagger = -\,U^\dagger\partial_\mu U$, so $U^\dagger\partial_\mu U = i\,\omega_\mu^a T^a$ for some real coefficients $\omega_\mu^a(x)$ — it lives in $i\mathfrak{g}$, the span of the Hermitian generators times i. Infinitesimally, with $U = \mathbf{1} + i\alpha^a T^a + O(\alpha^2)$,

$$U^\dagger\partial_\mu U \;=\; i\,(\partial_\mu\alpha^a)T^a + O(\alpha^2), \qquad \delta\mathcal{L}_0 \;=\; -\,(\partial_\mu\alpha^a)\,\bar\psi\,\gamma^\mu T^a\psi \;+\; O(\alpha^2), \tag{25.12}$$

and the coefficient of $-\partial_\mu\alpha^a$ is exactly $j^{a\mu} = \bar\psi\gamma^\mu T^a\psi$, the Noether current of the global non-abelian symmetry Eq. (25.3) — one current per generator. This is the precise non-abelian echo of the abelian statement, recall Eq. (8.13), that the Noether current is the coefficient with which local invariance first fails. At linear order in α the resemblance to the abelian case is total; the genuinely new feature lives at quadratic order and beyond, encoded in the non-commutativity Eq. (25.9), and it is what Section 25.3 will find waiting in the self-interaction of the field we are about to invent.

25.1.3 No field redefinition repairs it: a connection is forced

Before inventing new structure one should rule out the cheaper alternative: that the free theory could be rescued by a redefinition of ψ alone — some local change of variables $\psi \mapsto \widetilde\psi = M(x)\psi$ that conspires to make the bare derivative behave covariantly. It cannot, and seeing why pins down exactly what *is* needed.

Worked Example 25.4 (Why a field redefinition cannot restore covariance)**.** Suppose we hope to repair the kinetic term by replacing ψ throughout with

$\widetilde{\psi} = M(x)\psi$, where $M(x)$ is a fixed invertible matrix function chosen once and for all — a relabelling of the fields, carrying no new degrees of freedom. For this to be a genuine field redefinition rather than a smuggled gauge field, M must be a definite function of spacetime, not a functional of the transformation U: the redefinition is performed once, and the same $\widetilde{\psi}$ must then transform sensibly under *every* gauge transformation $U(x)$. Under Eq. (25.7) the redefined field transforms as $\widetilde{\psi} = M\psi \mapsto MU\psi = (MUM^{-1})\,\widetilde{\psi}$, so M at best changes the matrix realizing the rotation but not the structure of the problem. Its derivative obeys the same Leibniz rule,

$$\partial_\mu(MU\psi) = MU\,\partial_\mu\psi + M(\partial_\mu U)\psi + (\partial_\mu M)U\psi, \tag{25.13}$$

and the inhomogeneous term proportional to $\partial_\mu U$ is untouched by any choice of the U-independent matrix M: it reappears as $M(\partial_\mu U)\psi$. The obstruction in Eq. (25.11) carries a factor $\partial_\mu\alpha^a(x)$ — the *rate at which the chosen internal frame is turning* as one moves through spacetime. A field redefinition is a functional of ψ and its derivatives at the point x; it can know the value of the field, but it cannot know how fast the gauge parameter happens to be varying, because $\alpha^a(x)$ is arbitrary and is supplied only when a gauge transformation is performed, not when the Lagrangian is written. No fixed functional of ψ can cancel a term proportional to $\partial_\mu\alpha^a$ for every choice of α^a at once. Therefore the cancelling object must be a *new field* that is itself part of the dynamics and that transforms with an inhomogeneous piece tracking $\partial_\mu\alpha^a$ — precisely the freedom a field redefinition lacks. This is the content of the gauge principle: local invariance is not free, and the only currency that can pay for it is a new field.

What must that new field do? Return to the requirement that emerged from Eq. (25.10): the kinetic term is invariant if and only if the derivative acting on ψ can be replaced by an operator D_μ whose result rotates the *same way* ψ *does*, with no leftover inhomogeneous term,

$$D_\mu\psi \;\longmapsto\; (D_\mu\psi)' \;=\; U(x)\,(D_\mu\psi). \tag{25.14}$$

A derivative satisfying Eq. (25.14) is said to be *covariant*, and it is exactly what makes the bilinear $\bar\psi\, i\not{D}\,\psi$ invariant: under it, $\bar\psi\, i\not{D}\,\psi \mapsto \bar\psi\, U^\dagger\, iU\not{D}\,\psi = \bar\psi\, i\not{D}\,\psi$, the unitarity $U^\dagger U = \mathbf{1}$ doing the same work it did globally, with the dangerous $(\partial_\mu U)\psi$ term now absent by construction. The bare ∂_μ fails Eq. (25.14) by exactly the term $(\partial_\mu U)\psi$ in Eq. (25.8). The repair is to add to ∂_μ a matrix-valued field whose own transformation is rigged to cancel that term. Because the obstruction $U^\dagger\partial_\mu U$ lies in $i\mathfrak{g}$, the added field must be valued in the same space: it is a Lie-algebra-valued one-form.

Definition 25.5 (The gauge connection and covariant derivative)**.** The *gauge connection* (or gauge potential) is a Lie-algebra-valued vector field

$$A_\mu(x) \;=\; A_\mu^a(x)\,T^a, \tag{25.15}$$

with $\dim G$ real component fields $A^a_\mu(x)$, one per generator, contracted with the Hermitian generators T^a of the representation R. The *gauge-covariant derivative* acting on the matter multiplet ψ in the representation R is

$$D_\mu \;=\; \partial_\mu - i\,g\,A_\mu \;=\; \partial_\mu - i\,g\,A^a_\mu T^a, \tag{25.16}$$

where g is a single real coupling constant — the strength with which the multiplet responds to the connection, the non-abelian analogue of the charge q of Chapter 8, and the coupling g frozen for Yang–Mills theory. The sign and placement of i, g, and T^a are the conventions of Chapter A, matching the global rule $D_\mu \to \partial_\mu - igA^a_\mu T^a$ of Theorem 8.3 generalized from q to T^a.

That D_μ as written is the unique minimal repair — and the determination of the transformation law that A_μ must obey so that Eq. (25.14) actually holds — is the content of the next section; we have established here only that some such Lie-algebra-valued connection is *forced*, that it carries one component field per generator, and that it couples through the single constant g. The contrast with the abelian theory is now sharp and is worth stating plainly: in U(1) the generator is a number and A_μ is a single real field; here the generators are non-commuting matrices, A_μ is a matrix in the Lie algebra, and the non-commutativity Eq. (25.9) that distinguished the matrix case will, two sections hence, give the connection a self-interaction that the photon never had. The gauge field of a non-abelian theory carries the very charge it mediates.

Why the derivative needed fixing, geometrically

The components $\psi_i(x)$ live in a copy of the internal space V_R attached to the point x; a gauge transformation rotates the frame of that internal space, independently at each point (Fig. 25.1). The ordinary derivative compares $\psi(x + \mathrm{d}x)$ with $\psi(x)$ by naive subtraction — but those two vectors sit in differently-rotated frames, so their difference is not a meaningful quantity, exactly as the difference of two vectors at different points of a curved surface is not. The connection A_μ supplies the missing instruction for "parallel transporting" the frame from x to $x + \mathrm{d}x$ before comparing, and D_μ is the corrected, frame-independent derivative. This geometric reading — connection, parallel transport, and (in Section 25.3) curvature — is not an analogy bolted on afterward; it is the precise mathematical content of the gauge principle, and it is why the same word, *connection*, is used here and in differential geometry.

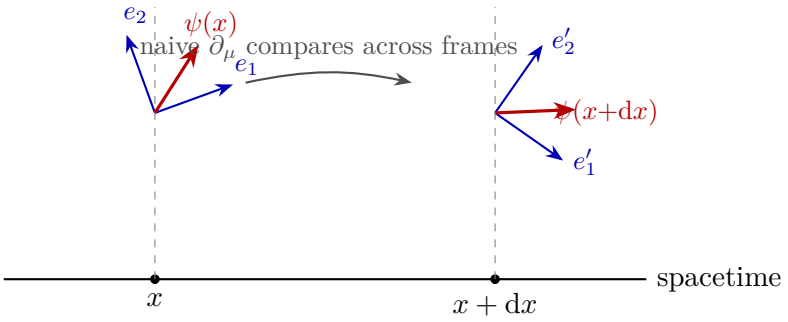


Figure 25.1: The internal frame of the representation space V_R may be rotated independently at each spacetime point by a gauge transformation $U(x)$. The ordinary derivative subtracts $\psi(x)$ and $\psi(x+\mathrm{d}x)$ as if their frames agreed; the mismatch is the term $(\partial_\mu U)\psi$ of Eq. (25.8). The connection A_μ encodes the rule for aligning the frames before comparing, repairing the derivative to D_μ.

25.1.4 Local invariance is a redundancy, not a new conservation law

It is tempting to read the appearance of one Noether current $j^{a\mu}$ per generator in Eq. (25.12) as a windfall of new conservation laws purchased by the gauging. It is not, and recognizing why is the conceptual connection that the final section of this chapter — the gauge-orbit picture — will need. The currents $j^{a\mu} = \bar{\psi}\gamma^\mu T^a \psi$ are the conserved currents of the *global* symmetry Eq. (25.3); they exist and are conserved on shell whether or not we gauge anything, by Noether's first theorem applied to the rigid transformation. Promoting the symmetry to a local one adds no further conservation law of the same kind. Instead, Noether's *second* theorem (Theorem 7.36) tells us what local invariance buys: an off-shell identity among the equations of motion. A symmetry with $\dim G$ arbitrary functions $\alpha^a(x)$ of spacetime, one per point per generator, forces $\dim G$ relations linking the Euler–Lagrange expressions and their derivatives, holding identically, before any equation of motion is imposed. These are constraints, not conservation laws: they signal that the gauge-covariant Lagrangian we are building describes fewer independent degrees of freedom than its field content suggests, because field configurations related by a gauge transformation $U(x)$ are physically identical descriptions of the same state.

In other words, local invariance is a *redundancy of description*, not a dynamical symmetry. This is the same lesson Chapter 8 drew for U(1), now with $\dim G$ independent local parameters in place of one. It has immediate consequences we will meet in order: it is why the naive canonical quantization of Chapter 26 must first fix a gauge, why the physical configuration space is the space of connections *modulo* gauge transformations — the gauge orbits — and why gauge-invariant observables, rather than the connection A^a_μ itself, are the meaningful output of

the theory. We note this point now and develop the gauge-orbit picture, with Wilson loops as its coordinate-free observables, in Section 25.7.

With the necessity of a Lie-algebra-valued connection $A_\mu = A_\mu^a T^a$ established, the covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$ named, and the coupling g of Yang–Mills theory in place, exactly one question remains before any of this can do work: *how must* A_μ *transform* so that Eq. (25.14) holds, so that $(\partial_\mu U)\psi$ is cancelled for every gauge transformation $U(x)$ at once? Determining that transformation law — in both its finite and infinitesimal forms, and recognizing its inhomogeneous piece as the adjoint covariant derivative acting on the gauge parameter — is the task of Section 25.2, to which we now turn.

25.2 The Gauge Connection and Its Transformation Law

The previous section left us with a sharp demand and no object to meet it. We insisted that the matter field ψ carry a local internal symmetry — at each spacetime point we are free to rotate ψ by an element of a compact non-abelian group G, $\psi(x) \to U(x)\psi(x)$ — and we saw precisely why the ordinary derivative betrays that freedom: the spacetime gradient compares the field at neighbouring points, but two neighbouring points now carry *independently chosen* internal frames, so $\partial_\mu\psi$ mixes a genuine variation of ψ with a spurious variation of the frame. The cure, as in the abelian case of Chapter 8, is to supply a rule for parallel transport: a field that tells the derivative how the internal frame at $x + \mathrm{d}x$ is to be identified with the frame at x before the comparison is made. That field is the gauge connection. In this section we build it, we determine the one transformation law it must obey if the corrected derivative is to transform covariantly, and we recognize in the answer a structure — the adjoint covariant derivative acting on the gauge parameter — that will recur throughout the rest of the chapter. Everything here is forced; we are not free to choose the connection's transformation, we are only free to discover it.

25.2.1 The Lie-algebra-valued connection

Let the matter field ψ transform in a representation R of the gauge group G, so that under a local gauge transformation

$$\psi(x) \;\longrightarrow\; \psi'(x) \;=\; U(x)\,\psi(x), \qquad U(x) \;=\; \exp\!\big(i\,\alpha^a(x)\,T^a\big) \in G, \tag{25.17}$$

where T^a are the Hermitian generators of G in the representation R, satisfying the algebra $\big[T^a, T^b\big] = if^{abc}T^c$ in the normalization fixed in Section 2.3.4, and $\alpha^a(x)$ are smooth real functions of spacetime. We now generalize the covariant derivative of Theorem 8.4 from the single abelian phase to the matrix-valued case.

Definition 25.6 (Gauge connection and covariant derivative)**.** The *gauge connection* (or gauge potential) is a set of real vector fields $A^a_\mu(x)$, one for each generator T^a, assembled into the matrix-valued, Lie-algebra-valued object

$$A_\mu(x) \;\equiv\; A^a_\mu(x)\, T^a, \tag{25.18}$$

a Lorentz four-vector whose value at each point is a Hermitian element of the span of the generators. The *gauge-covariant derivative* acting on the matter field ψ is

$$D_\mu\psi \;\equiv\; (\partial_\mu - i\, g\, A_\mu)\psi \;=\; (\partial_\mu - i\, g\, A^a_\mu T^a)\psi, \tag{25.19}$$

with g the (single, real) gauge coupling. The covariant derivative is itself a matrix acting on the representation index of ψ.

Three remarks fix the meaning of Theorem 25.6 before we put it to work. First, the sign and coupling convention $D_\mu = \partial_\mu - igA^a_\mu T^a$ is the one frozen for Yang–Mills theory (Chapter 8); it is the direct non-abelian descendant of the abelian $D_\mu = \partial_\mu - iqA_\mu$ of Eq. (8.20), with the single charge q promoted to the full set of generators gT^a and the single photon A_μ promoted to the multiplet A^a_μ. Second, the component fields A^a_μ are the primary objects: they are real numbers (one four-vector for each of the $\dim G$ generators), and they do not depend on which representation R the matter happens to sit in. The matrix $A_\mu = A^a_\mu T^a$, by contrast, *does* depend on R through the matrices T^a; it is the representation of the one underlying connection on the space where ψ lives. When we derive the transformation law we will obtain it first for the matrix A_μ and then strip it down to a statement about the representation-independent components A^a_μ. Third, because the T^a do not commute, the single coupling g in Eq. (25.19) cannot be rescaled away generator by generator the way the abelian charges q_i could be assigned independently; the non-abelian connection couples to all components of the matter multiplet with one universal strength. This is the first quantitative footprint of non-abelian-ness, and it is the reason a non-abelian gauge theory has a single running coupling rather than a charge per field.

25.2.2 The finite transformation law

The connection earns its name by making $D_\mu\psi$ transform exactly as ψ does. We demand *covariance*: under Eq. (25.17) the covariant derivative of ψ must rotate by the same local group element,

$$D_\mu\psi \;\longrightarrow\; (D_\mu\psi)' \;=\; U(x)\,(D_\mu\psi). \tag{25.20}$$

This single requirement, together with the form Eq. (25.19), determines the transformation of A_μ uniquely. The corrected derivative is built from a transformed connection A'_μ, so $(D_\mu\psi)' = (\partial_\mu - igA'_\mu)\psi' = (\partial_\mu - igA'_\mu)U\psi$, and the demand Eq. (25.14) reads

$$(\partial_\mu - igA'_\mu)\, U\psi \;\overset{!}{=}\; U\,(\partial_\mu - igA_\mu)\,\psi. \tag{25.21}$$

Theorem 25.7 (Transformation law of the connection)**.** *The covariance condition Eq. (25.14) holds for every matter field ψ if and only if the connection transforms by*

$$\boxed{A_\mu \;\longrightarrow\; A'_\mu \;=\; U\,A_\mu\,U^{-1} \;-\; \frac{i}{g}\,(\partial_\mu U)\,U^{-1}.} \tag{25.22}$$

Proof. Expand the left-hand side of Eq. (25.21) with the Leibniz rule, treating U as a matrix-valued function of x:

$$(\partial_\mu - igA'_\mu)U\psi = (\partial_\mu U)\,\psi + U\,(\partial_\mu\psi) - igA'_\mu\,U\psi. \tag{25.23}$$

The right-hand side is

$$U\,(\partial_\mu - igA_\mu)\psi = U\,(\partial_\mu\psi) - ig\,U A_\mu\,\psi. \tag{25.24}$$

Setting Eq. (25.23) equal to Eq. (25.24), the term $U(\partial_\mu\psi)$ appears identically on both sides and cancels, leaving

$$(\partial_\mu U)\,\psi - ig\,A'_\mu\,U\psi \;=\; -\,ig\,U A_\mu\,\psi. \tag{25.25}$$

This must hold for an *arbitrary* matter field ψ, so it is an identity among the matrices multiplying ψ:

$$(\partial_\mu U) - ig\,A'_\mu\,U \;=\; -\,ig\,U A_\mu. \tag{25.26}$$

Solve for A'_μ. Move the derivative term to the right and divide by $-ig$:

$$A'_\mu\,U \;=\; UA_\mu + \frac{1}{ig}\,(\partial_\mu U) \;=\; UA_\mu - \frac{i}{g}\,(\partial_\mu U), \tag{25.27}$$

using $1/(ig) = -i/g$. Finally multiply on the right by U^{-1} (which exists because $U \in G$ is unitary), giving Eq. (25.22). Each step is reversible: substituting Eq. (25.22) back into $(\partial_\mu - igA'_\mu)U\psi$ reproduces $U(D_\mu\psi)$, so the law is not only sufficient but necessary. This proves the theorem. □

The structure of Eq. (25.22) repays a careful reading. It has two pieces of entirely different character. The first, $UA_\mu U^{-1}$, is the *homogeneous* (adjoint, conjugation) action: it is how any Lie-algebra-valued matrix transforms when the frame is rotated by U, and on its own it would make A_μ an ordinary tensor in the adjoint representation. The second, $-\frac{i}{g}(\partial_\mu U)U^{-1}$, is the *inhomogeneous* piece: it is independent of A_μ altogether and survives even when $A_\mu = 0$, manufacturing a connection out of nothing but the spacetime variation of the frame. A field configuration of the form $A_\mu = -\frac{i}{g}(\partial_\mu U)U^{-1}$, obtained by gauge-transforming the zero connection, is called a *pure gauge*; it looks nontrivial but carries no physical content, a fact that will acquire precise meaning once we build the curvature in Section 25.3. The presence of this inhomogeneous, field-independent term is the defining signature of a *connection* as opposed to a tensor, and we return to its geometric meaning at the end of the section.

Caution

The inhomogeneous term is $-\frac{i}{g}(\partial_\mu U)U^{-1}$, not $-\frac{i}{g}U^{-1}(\partial_\mu U)$. The two differ because U and $\partial_\mu U$ need not commute. The correct ordering is forced by the derivation: the derivative acted on U *from the left* in Eq. (25.23) (since U stands to the left of ψ), and the closing U^{-1} in Eq. (25.27) was applied on the *right*. A useful identity, obtained by differentiating $UU^{-1} = \mathbf{1}$, is $(\partial_\mu U)U^{-1} = -U(\partial_\mu U^{-1})$; it lets one rewrite the inhomogeneous term as $+\frac{i}{g}U(\partial_\mu U^{-1})$ when convenient, but the operator ordering must be respected.

25.2.3 Linearization: the infinitesimal law and the adjoint covariant derivative

The finite law Eq. (25.22) contains everything, but its physical content is read most easily near the identity, where the gauge group is its Lie algebra. Take the parameters $\alpha^a(x)$ infinitesimal and expand $U = \exp(i\alpha^a T^a)$ to first order:

$$U \;=\; \mathbf{1}+i\,\alpha^a T^a+O(\alpha^2), \qquad U^{-1} \;=\; \mathbf{1}-i\,\alpha^a T^a+O(\alpha^2), \qquad \partial_\mu U \;=\; i\,(\partial_\mu\alpha^a)\,T^a+O(\alpha^2). \tag{25.28}$$

We substitute these into Eq. (25.22) and keep only terms linear in α. Take the two pieces in turn. The homogeneous (conjugation) piece is

$$\begin{aligned} UA_\mu U^{-1} &= (\mathbf{1}+i\alpha^b T^b)\,A_\mu\,(\mathbf{1}-i\alpha^c T^c)+O(\alpha^2)\\ &= A_\mu + i\,\alpha^b\,T^b A_\mu - i\,\alpha^c\,A_\mu T^c + O(\alpha^2)\\ &= A_\mu + i\,\alpha^b\,\Big[T^b, A_\mu\Big] + O(\alpha^2), \end{aligned} \tag{25.29}$$

where in the last line we recognized the commutator $\Big[T^b, A_\mu\Big] = T^b A_\mu - A_\mu T^b$. Now insert $A_\mu = A_\mu^c T^c$ and use the Lie algebra $\Big[T^b, T^c\Big] = if^{bcd}T^d$ of Section 2.3.4:

$$i\,\alpha^b\,\Big[T^b, A_\mu\Big] = i\,\alpha^b A_\mu^c\,\Big[T^b, T^c\Big] = i\,\alpha^b A_\mu^c\,(if^{bcd}T^d) = -\,f^{bcd}\,\alpha^b A_\mu^c\,T^d. \tag{25.30}$$

The inhomogeneous piece is, to first order,

$$-\frac{i}{g}\,(\partial_\mu U)\,U^{-1} = -\frac{i}{g}\,(i(\partial_\mu\alpha^a)T^a)(\mathbf{1}+O(\alpha)) = \frac{1}{g}\,(\partial_\mu\alpha^a)\,T^a. \tag{25.31}$$

Adding Eq. (25.30) and Eq. (25.31) to the surviving A_μ in Eq. (25.29) gives the transformed connection, and subtracting A_μ defines the variation $\delta A_\mu \equiv A'_\mu - A_\mu$:

$$\delta A_\mu \;=\; \frac{1}{g}\,(\partial_\mu\alpha^a)\,T^a \;-\; f^{bcd}\,\alpha^b A_\mu^c\,T^d. \tag{25.32}$$

This is a statement about the matrix A_μ; to read off the variation of the representation-independent components we equate the coefficients of each generator. Relabel the free index of the first term and the dummy index d of the second to a common name a, and isolate δA^a_μ from $\delta A_\mu = (\delta A^a_\mu) T^a$:

$$\delta A^a_\mu \;=\; \frac{1}{g}\,\partial_\mu \alpha^a \;-\; f^{bca}\,\alpha^b A^c_\mu . \tag{25.33}$$

The second term wants tidying so that the free index a sits on f in its first slot and the standard ordering "A then α" appears. Rename the two dummy indices $b \leftrightarrow c$ in $-f^{bca}\alpha^b A^c_\mu$, turning it into $-f^{cba}\alpha^c A^b_\mu = -f^{cba} A^b_\mu \alpha^c$; then use the total antisymmetry of the structure constants (Section 2.3.4), $f^{cba} = -f^{abc}$, to write $-f^{cba} = +f^{abc}$. Thus $-f^{bca}\alpha^b A^c_\mu = +f^{abc} A^b_\mu \alpha^c$, and we arrive at the infinitesimal transformation law:

$$\boxed{\delta A^a_\mu \;=\; \frac{1}{g}\,\partial_\mu \alpha^a \;+\; f^{abc}\, A^b_\mu\, \alpha^c .} \tag{25.34}$$

The two terms are the linearizations of the two terms of the finite law: the gradient $\frac{1}{g}\partial_\mu \alpha^a$ comes from the inhomogeneous piece and is present already in electromagnetism, while the structure-constant term $f^{abc} A^b_\mu \alpha^c$ comes from the conjugation piece and is the genuinely non-abelian novelty — it says that under an internal rotation the components of A^a_μ mix among themselves, because the connection carries the very charge it mediates.

It is no accident that Eq. (25.34) packages into a covariant derivative. Define the *adjoint-representation covariant derivative* acting on any field ω^a carrying an adjoint index by

$$\boxed{(D_\mu \omega)^a \;\equiv\; \partial_\mu \omega^a + g\, f^{abc} A^b_\mu\, \omega^c \;=\; (D_\mu)^{ac}\omega^c, \qquad (D_\mu)^{ac} \;\equiv\; \delta^{ac}\,\partial_\mu + g\, f^{abc} A^b_\mu .} \tag{25.35}$$

Comparing with Eq. (25.34), the entire infinitesimal transformation is the adjoint covariant derivative of the gauge parameter, scaled by $1/g$:

$$\delta A^a_\mu \;=\; \frac{1}{g}\,(D_\mu \alpha)^a . \tag{25.36}$$

This is the clean form to remember. The gauge parameter α^a lives in the adjoint representation (it labels a Lie-algebra element $\alpha = \alpha^a T^a$), and the connection shifts by precisely the adjoint covariant derivative of that parameter. We will use Eq. (25.35) repeatedly: it is exactly the operator that acts on the field strength in the Bianchi identity, on the gauge current in the equations of motion, and on the ghost field in the BRST transformations of Chapter 26.

Proposition 25.8 ($(D_\mu)^{ac}$ is the covariant derivative in the adjoint representation)**.** *The operator $(D_\mu)^{ac} = \delta^{ac}\partial_\mu + g f^{abc} A^b_\mu$ of Eq. (25.35) is the specialization of the matrix covariant derivative $D_\mu = \partial_\mu - igA^b_\mu T^b$ to the adjoint representation, with the generators given by Eq. (2.82).*

Proof. In the adjoint representation the generators are the matrices $(T^b_{\text{adj}})_{ac} = -if^{bac}$ of Eq. (2.82) (read with first index a, second index c, and generator label b). The general covariant derivative on a field ω^c transforming in a representation with generators T^b is $(D_\mu\omega)^a = \partial_\mu\omega^a - igA^b_\mu\,(T^b)_{ac}\,\omega^c$. Substituting the adjoint matrices,

$$-ig\,A^b_\mu\,(T^b_{\text{adj}})_{ac} = -ig\,A^b_\mu\,(-if^{bac}) = -\,g\,A^b_\mu\,f^{bac} = +\,g\,A^b_\mu\,f^{abc}, \tag{25.37}$$

where the last equality uses $f^{bac} = -f^{abc}$. Hence $(D_\mu\omega)^a = \partial_\mu\omega^a + gf^{abc}A^b_\mu\omega^c$, which is exactly Eq. (25.35). The notation D_μ is therefore unambiguous: it means the covariant derivative in whatever representation its argument carries — the fundamental $\partial_\mu - igA^a_\mu T^a$ of Eq. (25.19) when acting on ψ, and the adjoint $\delta^{ac}\partial_\mu + gf^{abc}A^b_\mu$ of Eq. (25.35) when acting on an adjoint-index field such as the gauge parameter. □

25.2.4 The abelian limit

A new construction must reduce to the old one in the appropriate limit, and here the limit is abelian: a one-dimensional group ($G = \mathrm{U}(1)$) has a single generator and vanishing structure constants, $f^{abc} = 0$. Setting $f^{abc} = 0$ in the infinitesimal law Eq. (25.34) collapses the mixing term and leaves

$$\delta A_\mu \;=\; \frac{1}{g}\,\partial_\mu\alpha, \tag{25.38}$$

where we have dropped the now-redundant adjoint index. This is the gauge transformation of the abelian potential. It matches the law $A_\mu \to A_\mu + \partial_\mu\alpha$ of Eq. (8.21) exactly, once one accounts for where the coupling is parked. In Chapter 8 the local phase was written $\psi \to e^{iq\alpha}\psi$ with the charge q multiplying the gauge function inside the exponent, and correspondingly the potential shifted by the bare gradient $\partial_\mu\alpha$. In the present non-abelian convention the coupling g has been pulled entirely out of the group element — we write $\psi \to e^{i\alpha}\psi$ with g residing only in the covariant derivative $\partial_\mu - igA_\mu$ — so the same physical transformation appears with the coupling in the denominator. The two are identical after the relabelling $\alpha_{\text{here}} = g\,\alpha_{\text{Ch. 8}}$: substituting this into Eq. (25.38) returns $\delta A_\mu = \partial_\mu\alpha_{\text{Ch. 8}}$, which is Eq. (8.21) verbatim. The abelian case is thus recovered with no new input; the only genuinely new feature of the non-abelian theory is the structure-constant term that $f^{abc} = 0$ kills.

Notation Watch

The factor of $1/g$ in Eq. (25.38) versus the bare gradient in Eq. (8.21) is a convention on where the coupling sits, not a physical discrepancy. Throughout Yang–Mills theory the coupling g lives inside $D_\mu = \partial_\mu -$

$igA^a_\mu T^a$ and the group elements are written $\exp(i\alpha^a T^a)$ with the bare parameters α^a; the connection then transforms by $\delta A^a_\mu = \frac{1}{g}(D_\mu\alpha)^a$. An equally common alternative absorbs g into the field, $\tilde{A}_\mu = gA_\mu$, so that $D_\mu = \partial_\mu - i\tilde{A}_\mu$ and $\delta\tilde{A}^a_\mu = \partial_\mu\alpha^a + f^{abc}\tilde{A}^b_\mu\alpha^c$ with no $1/g$; the coupling then reappears only in the Lagrangian as an overall $1/g^2$. We use the former convention everywhere in this book, consistent with Chapter 8 after the rescaling above.

25.2.5 The connection is not a tensor

We can now state precisely the geometric fact that organizes the rest of the chapter. A field transforming in the adjoint representation — a *tensor* for the gauge group — would transform homogeneously, by conjugation alone: $\Phi \to U\Phi U^{-1}$, or infinitesimally $\delta\Phi^a = f^{abc}\alpha^b\Phi^c$ with no derivative term. The connection A_μ does *not* transform this way. Its law Eq. (25.22) carries the extra, field-independent, inhomogeneous term $-\frac{i}{g}(\partial_\mu U)U^{-1}$, equivalently the $\frac{1}{g}\partial_\mu\alpha^a$ in Eq. (25.34). Because of that term, A_μ is not a gauge tensor: there is no frame-independent meaning to "the value of A_μ," and in particular $A_\mu = 0$ in one gauge is $A_\mu = -\frac{i}{g}(\partial_\mu U)U^{-1} \neq 0$ in another. This is not a defect; it is the entire point. A connection is by definition the object that tells you how to compare frames at neighbouring points, and "how to compare" is exactly the information that changes when you change the frames. The same inhomogeneous term appears for the Christoffel connection of general relativity and for the spin connection of curved-space spinors; the gauge connection is the internal-symmetry instance of one universal mathematical structure.

Physical Insight

The split of Eq. (25.22) into a homogeneous and an inhomogeneous part is the source of all the chapter's later structure. The inhomogeneous part is what we must *cancel* to build gauge-invariant objects: it is why the naive "field strength" $\partial_\mu A_\nu - \partial_\nu A_\mu$ will fail to be covariant and must be repaired by a commutator term, and it is why a derivative of any charged field must be the covariant derivative D_μ rather than ∂_μ. The homogeneous part is what *survives* into the physics: the field strength, once correctly defined, will transform homogeneously (as a tensor in the adjoint), and gauge-invariant observables are built by tracing products of such tensors. "Cancel the inhomogeneous, keep the homogeneous" is the recurring move of gauge theory.

25.2.6 Consistency: gauge transformations compose

A transformation law worth its name must be consistent under composition: performing the gauge rotation U_1 and then U_2 should equal the single rotation U_2U_1. For the matter field this is automatic, $\psi \to U_1\psi \to U_2U_1\psi$. For the connection, whose law is inhomogeneous, it is a genuine check that the law closes into a group action. We verify it in full.

Worked Example 25.9 (The transformation law is a left group action)**.** Let A_μ transform under U_1 to A'_μ, and then A'_μ transform under U_2 to A''_μ, each step by Eq. (25.22). We show A''_μ equals the transform of A_μ under the single element $W \equiv U_2U_1$.

The first step gives

$$A'_\mu \;=\; U_1 A_\mu U_1^{-1} - \frac{i}{g}\,(\partial_\mu U_1)U_1^{-1}. \tag{25.39}$$

The second step applies Eq. (25.22) with $U \to U_2$ to A'_μ:

$$\begin{aligned} A''_\mu &= U_2\, A'_\mu\, U_2^{-1} - \frac{i}{g}\,(\partial_\mu U_2)U_2^{-1} \\ &= U_2\Big[\, U_1 A_\mu U_1^{-1} - \frac{i}{g}(\partial_\mu U_1)U_1^{-1}\Big]U_2^{-1} - \frac{i}{g}\,(\partial_\mu U_2)U_2^{-1} \\ &= U_2U_1\, A_\mu\, U_1^{-1}U_2^{-1} - \frac{i}{g}\,U_2(\partial_\mu U_1)U_1^{-1}U_2^{-1} - \frac{i}{g}\,(\partial_\mu U_2)U_2^{-1}. \end{aligned} \tag{25.40}$$

Now compute the transform under $W = U_2U_1$ directly. Its conjugation piece is $WA_\mu W^{-1} = U_2U_1A_\mu U_1^{-1}U_2^{-1}$, matching the first term of Eq. (25.40). For the inhomogeneous piece, use the Leibniz rule on $\partial_\mu W = (\partial_\mu U_2)U_1 + U_2(\partial_\mu U_1)$ and $W^{-1} = U_1^{-1}U_2^{-1}$:

$$\begin{aligned} -\frac{i}{g}\,(\partial_\mu W)W^{-1} &= -\frac{i}{g}\,\big[(\partial_\mu U_2)U_1 + U_2(\partial_\mu U_1)\big]\,U_1^{-1}U_2^{-1} \\ &= -\frac{i}{g}\,(\partial_\mu U_2)\,\underbrace{U_1U_1^{-1}}_{\mathbf{1}}\,U_2^{-1} - \frac{i}{g}\,U_2(\partial_\mu U_1)U_1^{-1}U_2^{-1} \\ &= -\frac{i}{g}\,(\partial_\mu U_2)U_2^{-1} - \frac{i}{g}\,U_2(\partial_\mu U_1)U_1^{-1}U_2^{-1}. \end{aligned} \tag{25.41}$$

The two terms of Eq. (25.41) are precisely the second and third terms of Eq. (25.40). Hence $A''_\mu = WA_\mu W^{-1} - \frac{i}{g}(\partial_\mu W)W^{-1}$: the law closes, and gauge transformations act on connections as a left action of the group. This is the consistency condition that makes "the gauge orbit of a connection" a well-defined notion, the starting point for gauge fixing in Chapter 26.

25.2.7 The transformation law for SU(2): a connection as an isovector

To make the abstract law concrete and to plant the first explicit marker of Yang–Mills theory, we specialize to the smallest non-abelian gauge group, $G = \mathrm{SU}(2)$. From Section 2.3.4 the structure constants are the Levi-Civita symbol, $f^{abc} = \epsilon^{abc}$ with $a, b, c \in \{1, 2, 3\}$, so the connection has three component four-vectors $A^1_\mu, A^2_\mu, A^3_\mu$, naturally assembled into an internal-space three-vector $\vec{A}_\mu = (A^1_\mu, A^2_\mu, A^3_\mu)$.

Worked Example 25.10 (Infinitesimal SU(2) gauge transformation as a rotation in isospace)**.** Substitute $f^{abc} = \epsilon^{abc}$ into the infinitesimal law Eq. (25.34):

$$\delta A^a_\mu \;=\; \frac{1}{g}\,\partial_\mu \alpha^a \;+\; \epsilon^{abc}\, A^b_\mu\, \alpha^c. \tag{25.42}$$

The second term is, component by component, the a-th entry of the cross product $\vec{A}_\mu \times \vec{\alpha}$, since $(\vec{A}_\mu \times \vec{\alpha})^a = \epsilon^{abc} A^b_\mu \alpha^c$. Therefore in three-vector notation the law reads

$$\delta \vec{A}_\mu \;=\; \frac{1}{g}\,\partial_\mu \vec{\alpha} \;+\; \vec{A}_\mu \times \vec{\alpha}. \tag{25.43}$$

The physical content is now transparent. A spatially constant gauge parameter ($\partial_\mu \vec{\alpha} = 0$) acts on the connection by an infinitesimal rotation in the internal isospin space, $\delta \vec{A}_\mu = \vec{A}_\mu \times \vec{\alpha}$ — exactly the way any ordinary three-vector responds to a rotation by angle $|\vec{\alpha}|$ about the axis $\hat{\alpha}$. This is the adjoint (= vector = spin-1) representation of SU(2), identified in Theorem 2.43 with the rotation generators of SO(3). When the parameter varies in spacetime, the additional gradient term $\frac{1}{g}\partial_\mu \vec{\alpha}$ appears; it is the inhomogeneous, connection-defining piece. The Yang–Mills connection is thus an isovector gauge field, and Eq. (25.43) is the precise sense in which "A^a_μ carries adjoint charge." We will return to this SU(2) realization when we compute the field strength and the self-interaction vertices; the assignment $f^{abc} = \epsilon^{abc}$, the three fields A^a_μ, and the single coupling g fixed here are inherited unchanged by every later stage of Yang–Mills theory.

With the connection A^a_μ in hand and its transformation law Eq. (25.22) established in finite, infinitesimal, and adjoint-covariant forms — and with the adjoint covariant derivative $(D_\mu)^{ac}$ of Eq. (25.35) isolated as a reusable operator — we are ready to differentiate the connection into a curvature. The next section forms the commutator of two covariant derivatives; the inhomogeneous term that disqualifies A_μ from being a tensor will cancel in that commutator, leaving the gauge-covariant field strength with its characteristic non-abelian self-term $g f^{abc} A^b_\mu A^c_\nu$.

25.3 Curvature: The Non-Abelian Field Strength

The previous section delivered the connection. We demanded that a local change of frame $\psi \to U(x)\psi$, with $U(x) = \exp\bigl(i\alpha^a(x)T^a\bigr) \in \mathrm{SU}(N)$, leave the physics untouched, and we found that this is possible only if the ordinary derivative is replaced by the covariant derivative

$$D_\mu \;=\; \partial_\mu - ig\,A_\mu, \qquad A_\mu \;=\; A_\mu^a T^a, \tag{25.44}$$

with a Lie-algebra-valued gauge potential A_μ whose transformation law was fixed precisely so that $D_\mu\psi$ transforms the same way ψ does. The compact statement of that law, the one we will lean on throughout this section, is that the covariant derivative *conjugates* under a gauge transformation,

$$D_\mu \;\longrightarrow\; D'_\mu \;=\; U\,D_\mu\,U^{-1}, \tag{25.45}$$

which is equivalent (Eq. (25.22)) to the inhomogeneous transformation of the potential itself,

$$A_\mu \;\longrightarrow\; A'_\mu \;=\; UA_\mu U^{-1} - \tfrac{i}{g}\,(\partial_\mu U)\,U^{-1}. \tag{25.46}$$

The connection tells us how to compare the internal frame of the matter field at neighbouring points. The question this section asks is the one that, in the abelian theory of Chapter 8, manufactured the electromagnetic field out of nothing but the demand for local phase invariance: *does the order of comparison matter?* The answer is the curvature of the connection, the non-abelian field strength, and the whole novelty of Yang–Mills theory — the fact that the gauge field carries the very charge it mediates — is already written into its algebra.

25.3.1 The field strength as the commutator of covariant derivatives

In electrodynamics the field strength was *defined*, not merely computed, by the failure of two covariant derivatives to commute, $[D_\mu, D_\nu] = -iqF_{\mu\nu}$ (Theorem 8.8). That definition was engineered to survive the generalization to a non-abelian group, and we now collect on the investment. The covariant derivative (25.44) is a first-order differential operator with a matrix-valued coefficient; the commutator $[D_\mu, D_\nu]$ is therefore a candidate for a new tensor, and the only adjustment to the abelian case is one factor of the coupling, dictated by the sign convention $D_\mu = \partial_\mu - igA_\mu$ frozen for Yang–Mills theory.

Definition 25.11 (Non-abelian field strength)**.** The *Yang–Mills field strength* is the Lie-algebra-valued antisymmetric two-tensor defined by the commutator of covariant derivatives,

$$F_{\mu\nu} \;\equiv\; \frac{i}{g}\,[D_\mu, D_\nu]\,, \qquad \text{equivalently} \qquad [D_\mu, D_\nu] \;=\; -ig\,F_{\mu\nu}. \tag{25.47}$$

It is matrix-valued, $F_{\mu\nu} = F^a_{\mu\nu} T^a$, and antisymmetric, $F_{\nu\mu} = -F_{\mu\nu}$. This generalizes the abelian comparator definition Theorem 8.8; setting the group to U(1), so that the single generator is a number and $f^{abc} \equiv 0$, returns Eq. (8.44) exactly.

The content of Theorem 25.11 is not the prefactor but what the commutator evaluates to. We compute it in full, because the term that appears beyond the abelian curl is the structural heart of the chapter.

Proposition 25.12 (Explicit form of the field strength)**.** *Acting on any field ψ in a representation of the gauge group, the commutator (25.47) is multiplication by the matrix*

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,, \tag{25.48}$$

whose components in the basis $F_{\mu\nu} = F^a_{\mu\nu} T^a$ are

$$\boxed{F^a_{\mu\nu} \;=\; \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\, f^{abc} A^b_\mu A^c_\nu} \tag{25.49}$$

the frozen Yang–Mills theory expression. The first two terms are the abelian curl applied componentwise; the third, $g f^{abc} A^b_\mu A^c_\nu$, is the self-coupling *term, present whenever the structure constants do not vanish, i.e. whenever the group is non-abelian.*

Proof. Apply $[D_\mu, D_\nu]$ to an arbitrary field ψ and expand each covariant derivative through (25.44). The commutator splits into four pieces,

$$[D_\mu, D_\nu] = [\partial_\mu, \partial_\nu] - ig\,[\partial_\mu, A_\nu] - ig\,[A_\mu, \partial_\nu] + (-ig)^2\,[A_\mu, A_\nu]\,. \tag{25.50}$$

We treat them in turn, keeping ψ explicit so that the derivative operators have something to act on.

The first piece vanishes because partial derivatives commute on smooth fields, $[\partial_\mu, \partial_\nu]\,\psi = \partial_\mu\partial_\nu\psi - \partial_\nu\partial_\mu\psi = 0$.

For the second piece, the derivative acts on the product $A_\nu\psi$ by the Leibniz rule:

$$[\partial_\mu, A_\nu]\,\psi = \partial_\mu(A_\nu\psi) - A_\nu\,\partial_\mu\psi = (\partial_\mu A_\nu)\,\psi + A_\nu\,\partial_\mu\psi - A_\nu\,\partial_\mu\psi = (\partial_\mu A_\nu)\,\psi. \tag{25.51}$$

The free-derivative terms cancel and only the derivative *of the potential* survives. The third piece is the same calculation with the order reversed,

$$[A_\mu, \partial_\nu]\,\psi = A_\mu\,\partial_\nu\psi - \partial_\nu(A_\mu\psi) = -(\partial_\nu A_\mu)\,\psi. \tag{25.52}$$

The fourth piece is pure matrix multiplication and carries no derivative; the prefactor is $(-ig)^2 = -g^2$, so it contributes $-g^2\,[A_\mu, A_\nu]\,\psi$.

Assembling (25.50) with (25.51)–(25.52),

$$\begin{aligned}[D_\mu, D_\nu]\,\psi &= -ig\,(\partial_\mu A_\nu)\,\psi - ig(-(\partial_\nu A_\mu))\psi - g^2\,[A_\mu, A_\nu]\,\psi \\ &= -ig\Big(\partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\Big)\psi, \end{aligned} \tag{25.53}$$

where in the second line we factored out $-ig$, turning the $-g^2 = (-ig)(-ig)$ into $(-ig)(-ig\,[A_\mu, A_\nu])$. Because this holds for every ψ, the operator identity is $[D_\mu, D_\nu] = -ig(\partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,)$, which is (25.48) once the prefactor is stripped via (25.47). This is the decisive departure from electrodynamics: in Chapter 8 the quadratic term cancelled because the abelian potential commutes with itself, but here $A_\mu = A_\mu^a T^a$ is matrix-valued and the generators do not commute, so $[A_\mu, A_\nu] \neq 0$ survives.

To pass to components, evaluate the surviving commutator on the algebra basis using $\left[T^b, T^c\right] = i f^{bcd} T^d$ (Eq. (2.70)):

$$-ig\,[A_\mu, A_\nu] = -ig\,A_\mu^b A_\nu^c\left[T^b, T^c\right] = -ig\,A_\mu^b A_\nu^c\,(i f^{bcd})\,T^d = g\,f^{bcd} A_\mu^b A_\nu^c\,T^d, \tag{25.54}$$

where the two factors of i combined to $-i \cdot i = +1$. Relabelling the contracted index $d \to a$ and using the total antisymmetry of the structure constants, which makes them invariant under the cyclic permutation $f^{bca} = f^{abc}$, the self-term reads $g f^{abc} A_\mu^b A_\nu^c T^a$. Adding the curl terms $(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a)T^a$ and reading off the coefficient of T^a yields (25.49). □

A few features of (25.49) deserve to be named before we go on. First, the self-term is quadratic in the potential: the field strength is no longer a linear functional of A_μ, so the Yang–Mills field equations that we build from F in the next section will be *nonlinear* even with no matter present. Second, the self-term is proportional to the same coupling g that governs the interaction of the gauge field with matter through D_μ; a single number sets both the strength with which gluons couple to quarks and the strength with which they couple to each other. Third, the self-term is antisymmetric in $\mu\nu$ as required, because $f^{abc} A_\mu^b A_\nu^c$ is antisymmetric under $\mu \leftrightarrow \nu$: swapping the spacetime indices swaps the dummy colour indices $b \leftrightarrow c$, and $f^{abc} = -f^{acb}$ supplies the sign. The non-abelian field strength is a legitimate antisymmetric two-tensor in each colour component, exactly like its abelian ancestor, but it is now *quadratically* sourced by the potential.

Worked Example 25.13 (The field strength for SU(2))**.** For the smallest non-abelian group, SU(2), the three generators are $T^a = \frac{1}{2}\sigma^a$ with the Pauli matrices, and the structure constants are the Levi-Civita symbol, $f^{abc} = \epsilon^{abc}$ (Chapter B). The field strength (25.49) becomes

$$F_{\mu\nu}^a = \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g\,\epsilon^{abc} A_\mu^b A_\nu^c, \qquad a = 1, 2, 3. \tag{25.55}$$

Writing out the $a = 1$ component to see the self-term explicitly, only $\epsilon^{123} = +1$ and $\epsilon^{132} = -1$ contribute to the sum over b, c, giving

$$F^1_{\mu\nu} = \partial_\mu A^1_\nu - \partial_\nu A^1_\mu + g(A^2_\mu A^3_\nu - A^3_\mu A^2_\nu). \tag{25.56}$$

The cross term $g(A^2_\mu A^3_\nu - A^3_\mu A^2_\nu)$ is the colour cross product $g\,(\mathbf{A}_\mu \times \mathbf{A}_\nu)^1$: the three gauge potentials behave like a vector in colour space and their field strength acquires the cross product of the potentials. This is the exact form Yang and Mills wrote down in 1954 for the isospin gauge field, and it is the first place where a gauge field becomes its own source: F^1 feels A^2 and A^3. Setting $g \to 0$ (or, equivalently, looking at the abelian U(1) for which $\epsilon \to 0$) deletes the cross term and returns three decoupled copies of electrodynamics.

25.3.2 Covariant, not invariant: the transformation law of F

In electrodynamics the field strength was gauge *invariant*, $F'_{\mu\nu} = F_{\mu\nu}$ (Eq. (8.44)): the curl of a pure gradient vanishes, and F was deliberately built to be blind to the gauge freedom. The non-abelian field strength is *not* invariant, and this is not a defect but the price of the self-term. Because F is matrix-valued and carries colour indices, a gauge transformation rotates those indices; the most we can ask is that it rotate them *covariantly*, by conjugation, so that gauge-invariant scalars can still be built from it by tracing. That is exactly what happens, and the commutator definition makes the proof immediate.

Theorem 25.14 (Covariant transformation of the field strength)**.** *Under a gauge transformation $U(x) \in \mathrm{SU}(N)$, the non-abelian field strength transforms by conjugation,*

$$F_{\mu\nu} \;\longrightarrow\; F'_{\mu\nu} \;=\; U\,F_{\mu\nu}\,U^{-1}, \tag{25.57}$$

i.e. it is gauge covariant*, transforming in the adjoint representation, and is* not *gauge invariant. Infinitesimally, with $U = 1 + i\alpha^a T^a + O(\alpha^2)$,*

$$\delta F_{\mu\nu} = i\,[\alpha, F_{\mu\nu}]\,, \qquad \delta F^a_{\mu\nu} = -\,f^{abc}\,\alpha^b\,F^c_{\mu\nu}, \tag{25.58}$$

where $\alpha = \alpha^a T^a$. Contrast this sharply with the abelian invariance $F'_{\mu\nu} = F_{\mu\nu}$ of Eq. (8.44): in U(1) *the structure constants vanish, the right-hand side of (25.58) is zero, and covariance degenerates to invariance.*

Proof. The shortest route is through the definition (25.47) together with the conjugation law (25.45). Under the gauge transformation each covariant derivative becomes $D_\mu \to U D_\mu U^{-1}$, so the product of two of them telescopes,

$$D'_\mu D'_\nu = (U D_\mu U^{-1})(U D_\nu U^{-1}) = U D_\mu\,(U^{-1}U)\,D_\nu U^{-1} = U D_\mu D_\nu U^{-1}, \tag{25.59}$$

the inner factors $U^{-1}U = 1$ cancelling. Subtracting the same expression with $\mu \leftrightarrow \nu$ gives the commutator,

$$\big[D'_\mu, D'_\nu\big] = U D_\mu D_\nu U^{-1} - U D_\nu D_\mu U^{-1} = U\,[D_\mu, D_\nu]\,U^{-1}. \tag{25.60}$$

Multiplying by the constant i/g and using (25.47) on both sides, $F'_{\mu\nu} = \frac{i}{g}\left[D'_\mu, D'_\nu\right] = U(\frac{i}{g}[D_\mu, D_\nu])U^{-1} = UF_{\mu\nu}U^{-1}$, which is (25.57). No assumption was made about U being near the identity, so the result is exact and finite.

For the infinitesimal form, expand $U = 1+i\alpha+O(\alpha^2)$ and $U^{-1} = 1-i\alpha+O(\alpha^2)$ with $\alpha = \alpha^a T^a$, and keep first order in α:

$$F'_{\mu\nu} = (1+i\alpha)\,F_{\mu\nu}\,(1-i\alpha) = F_{\mu\nu} + i\alpha F_{\mu\nu} - iF_{\mu\nu}\alpha + O(\alpha^2) = F_{\mu\nu} + i\,[\alpha, F_{\mu\nu}]\,, \tag{25.61}$$

so $\delta F_{\mu\nu} = F'_{\mu\nu} - F_{\mu\nu} = i\,[\alpha, F_{\mu\nu}]$. In components, with $F_{\mu\nu} = F^c_{\mu\nu}T^c$,

$$i\,[\alpha, F_{\mu\nu}] = i\,\alpha^b F^c_{\mu\nu}\left[T^b, T^c\right] = i\,\alpha^b F^c_{\mu\nu}\,(if^{bce})T^e = -\,f^{bce}\,\alpha^b F^c_{\mu\nu}\,T^e, \tag{25.62}$$

again using $\left[T^b, T^c\right] = if^{bce}T^e$ (Eq. (2.70)). Reading off the coefficient of T^a (set $e = a$) and using cyclicity $f^{bca} = f^{abc}$ gives $\delta F^a_{\mu\nu} = -f^{abc}\alpha^b F^c_{\mu\nu}$, which is (25.58). □

It is reassuring, and a useful internal consistency check, to verify covariance directly from the inhomogeneous transformation (25.46) of the potential, rather than through the commutator. The infinitesimal connection law (established in the previous section) is $\delta A^a_\mu = \frac{1}{g}\partial_\mu\alpha^a + f^{abc}A^b_\mu\alpha^c$, the second term being the *homogeneous* adjoint piece and the first the inhomogeneous gradient. Substituting $A \to A+\delta A$ into (25.49) and keeping first order in α, the gradient piece generates terms $\partial_\mu(\frac{1}{g}\partial_\nu\alpha^a) - \partial_\nu(\frac{1}{g}\partial_\mu\alpha^a)$ from the curl — which cancel by symmetry of mixed partials, exactly as in the abelian proof (8.45) — together with cross terms from the self-coupling that, after using the Jacobi identity in the form Eq. (2.77), recombine into $-f^{abc}\alpha^b F^c_{\mu\nu}$. The two computations agree; the gradient part of the gauge variation never reaches the gauge-covariant F, and only the adjoint rotation survives. The commutator proof is shorter precisely because it never separates A into its homogeneous and inhomogeneous parts: it works directly with the covariant object D_μ whose transformation (25.45) is already clean.

Caution

The single most important conceptual difference between Yang–Mills theory and electrodynamics at the classical level is that $F^a_{\mu\nu}$ is *not* a physical observable on its own. In electrodynamics the components of $F_{\mu\nu}$ *are* the measurable electric and magnetic fields. In a non-abelian theory the colour components $F^a_{\mu\nu}$ are rotated into one another by gauge transformations through (25.58); asking "what is the value of the red component of the chromomagnetic field?" is no more meaningful than asking for the value of a vector in an unspecified frame. Only gauge-*invariant* combinations are physical, and the elementary way to manufacture one is to undo the conjugation by tracing: $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ is invariant because $\mathrm{tr}(UFFU^{-1}) =$

$\mathrm{tr}(FF)$ by cyclicity of the trace. That observation is the seed of the Yang–Mills Lagrangian built in the next section, and it is the reason the kinetic term must be assembled from F under a trace rather than from F directly.

25.3.3 The geometric picture: curvature and the non-abelian holonomy

The transformation law (25.57) acquires its meaning from geometry. In Chapter 8 we learned to read A_μ as a *connection*: a rule for parallel-transporting the internal frame of the matter field from one point to a neighbour, encoded in the comparator (Wilson line) $U_\gamma(y,x)$. For a path γ from x to y the non-abelian comparator is the *path-ordered* exponential

$$W_\gamma(y,x) = \mathcal{P}\exp\!\left(ig\int_\gamma A_\mu(z)\,\mathrm{d}z^\mu\right), \tag{25.63}$$

the path ordering $\mathcal{P}$ (later points to the left) being the new ingredient that Eq. (8.64) flagged as the non-abelian replacement for the ordinary exponential: because A_μ at different points along the path no longer commutes, the order in which the infinitesimal transports are composed now matters. The comparator transforms covariantly, $W_\gamma(y,x)\to U(y)\,W_\gamma(y,x)\,U^{-1}(x)$, which is the matrix generalization of the endpoint phase-cancellation seen in the abelian case and is what allows W_γ to carry a colour frame from x to y gauge-covariantly.

The curvature is what the comparator measures around a *closed* loop. If transport around a loop returns the frame unchanged the connection is flat and there is no field; the failure to return — the holonomy — is the field strength. The following worked example makes this precise for an infinitesimal loop and exhibits how the self-coupling term of (25.49) is *forced* by the non-commutativity of the transports, providing a third, purely geometric, derivation of F.

Worked Example 25.15 (Non-abelian holonomy of an infinitesimal loop equals the curvature)**.** Take the small rectangular loop C of Fig. 25.2, lying in the μ–ν plane with corner at x, sides $a\hat{e}_\mu$ and $b\hat{e}_\nu$ (no sum on the fixed directions μ,ν). The holonomy is the path-ordered product of the four link transports (25.63),

$$W(C) = W_4\,W_3\,W_2\,W_1, \tag{25.64}$$

with $W_1,\dots,W_4$ the transports along the four edges traversed counterclockwise. Two distinct effects combine to give the field strength, and it is instructive to separate them.

The curl, from the position dependence of A. If we kept only the variation of A across the loop and treated the link matrices as commuting numbers, the

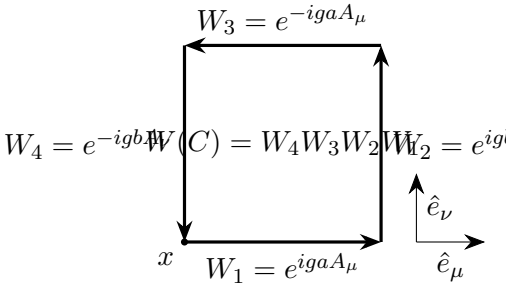


Figure 25.2: The infinitesimal plaquette of Theorem 25.15. The frame is parallel-transported counterclockwise around the rectangle of sides $a\hat{e}_\mu$ and $b\hat{e}_\nu$; the four link variables W_i are composed in path order (rightmost traversed first). Their ordered product fails to be the identity by $ig\,ab\,F_{\mu\nu}$, the curvature.

ordered product would reduce to the ordinary exponential of the line integral, and the computation would be word for word the abelian one of Theorem 8.12. That computation gave $\oint_C A_\rho \,\mathrm{d}x^\rho = ab\,(\partial_\mu A_\nu - \partial_\nu A_\mu)$, so this contribution to the holonomy is $W(C)\big|_{\text{curl}} = \mathbf{1} + ig\,ab\,(\partial_\mu A_\nu - \partial_\nu A_\mu)$.

The commutator, from the non-commuting transports. The genuinely new effect is present even when A_μ is *constant*, so set the derivatives aside and take A_μ, A_ν constant matrices. Each link is then exactly $W_1 = e^{igaA_\mu}$, $W_2 = e^{igbA_\nu}$, $W_3 = e^{-igaA_\mu}$, $W_4 = e^{-igbA_\nu}$, and the holonomy is the group commutator

$$W(C) = e^{-igbA_\nu}\, e^{-igaA_\mu}\, e^{+igbA_\nu}\, e^{+igaA_\mu}. \tag{25.65}$$

Write $X = igaA_\mu$ and $Y = igbA_\nu$, both first order in the side lengths, so $W(C) = e^{-Y}e^{-X}e^{Y}e^{X}$. Expand each factor to second order, $e^{Z} = 1 + Z + \tfrac{1}{2}Z^2 + \cdots$, and multiply, keeping all terms through second order (the order at which the leading commutator appears):

$$\begin{aligned} e^{Y}e^{X} &= 1 + X + Y + \tfrac{1}{2}X^2 + \tfrac{1}{2}Y^2 + YX + \cdots, \\ e^{-Y}e^{-X} &= 1 - X - Y + \tfrac{1}{2}X^2 + \tfrac{1}{2}Y^2 + YX + \cdots. \end{aligned} \tag{25.66}$$

Their product is, again to second order,

$$\begin{aligned} W(C) &= \big(1 - X - Y + \tfrac{1}{2}X^2 + \tfrac{1}{2}Y^2 + YX\big)\big(1 + X + Y + \tfrac{1}{2}X^2 + \tfrac{1}{2}Y^2 + YX\big) \\ &= 1 + (XY - YX) + \cdots \;=\; 1 + [X, Y] + \cdots, \end{aligned} \tag{25.67}$$

where the linear terms $\pm(X + Y)$ cancel between the two factors, the symmetric quadratics $\tfrac{1}{2}X^2, \tfrac{1}{2}Y^2$ and the diagonal cross products all cancel, and the only survivor is the antisymmetric combination $XY - YX = [X, Y]$. Restoring X, Y,

$$[X, Y] = [igaA_\mu, igbA_\nu] = (ig)^2 ab\,[A_\mu, A_\nu] = -g^2\,ab\,[A_\mu, A_\nu]\,. \tag{25.68}$$

Now combine the two effects. The curl part contributes $ig\,ab\,(\partial_\mu A_\nu - \partial_\nu A_\mu)$ and the commutator part contributes $-g^2\,ab\,[A_\mu, A_\nu] = ig\,ab\,(-ig\,[A_\mu, A_\nu])$, so that

$$W(C) = \mathbf{1} + ig\,ab\left(\partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\right) + O(\ell^3) = \mathbf{1} + ig\,ab\,F_{\mu\nu} + O(\ell^3), \tag{25.69}$$

with the full non-abelian $F_{\mu\nu}$ of (25.48) assembled, self-term and all. Parallel transport around an infinitesimal loop fails to be the identity, and the failure per unit oriented area is exactly $gF_{\mu\nu}$: *$gF_{\mu\nu}$ is the curvature of the gauge connection*, the holonomy density of the loop, now valued in the Lie algebra. The self-coupling term $-ig\,[A_\mu, A_\nu]$ is not optional — it is precisely the contribution (25.68) that the non-commutativity of the transports forces into the holonomy, and it would be inconsistent to omit it.

Physical Insight

The holonomy (25.69) explains the transformation law (25.57) in one stroke and reconciles it with the abelian invariance of Chapter 8. A loop transport based at x is an operation on the frame *at* x; under a gauge change $U(x)$ that frame is rotated, so the holonomy $W(C)$ must conjugate, $W(C) \to U(x)\,W(C)\,U(x)^{-1}$. Reading off the coefficient of $ig\,ab$ on both sides gives $F_{\mu\nu} \to U F_{\mu\nu} U^{-1}$ — covariance, not invariance, because the curvature now lives in a colour space that gauge transformations stir. In the abelian case the holonomy is a pure phase, a number, and conjugation by a phase does nothing: F is invariant. The difference between QED and Yang–Mills is the difference between a number and a matrix going around the loop. To extract a gauge-invariant observable one closes the loop and takes the trace, $\operatorname{tr} W(C)$ — the *Wilson loop* — which we develop in a later section as the order parameter for confinement; its small-loop expansion is $\operatorname{tr}\mathbf{1} + ig\,ab\,\operatorname{tr} F_{\mu\nu} + \cdots$, the curvature again, now scrubbed of its frame dependence by the cyclic trace.

25.3.4 The non-abelian Bianchi identity

Because the abelian $F_{\mu\nu}$ was the antisymmetrized derivative of a potential, it satisfied a kinematic identity — the Bianchi identity Theorem 8.10 — automatically, before any equation of motion was imposed, and that identity encoded the source-free half of Maxwell's equations. The non-abelian field strength satisfies the corresponding identity, with two changes that we will see are inseparable: ordinary derivatives are promoted to *covariant* derivatives, and the proof, rather than relying on the commutativity of partials, runs through the Jacobi identity of the covariant derivatives. The abelian proof's second route (Theorem 8.10, via the Jacobi identity) was written precisely so that it would survive this generalization.

We first record how the covariant derivative acts on a field carrying *adjoint* colour indices, such as $F_{\nu\rho} = F^a_{\nu\rho} T^a$ itself, since the Bianchi identity is a statement about exactly such a derivative.

Proposition 25.16 (Covariant derivative in the adjoint)**.** *On a Lie-algebra-valued field $\Phi = \Phi^a T^a$, the covariant derivative acts by the commutator*

$$D_\mu \Phi = \partial_\mu \Phi - ig\,[A_\mu, \Phi]\,, \qquad (D_\mu \Phi)^a = \partial_\mu \Phi^a + g f^{abc} A^b_\mu \Phi^c, \tag{25.70}$$

and this is the same operator as the commutator of the matter-representation covariant derivative with multiplication by Φ: $[D_\mu, \Phi] = D_\mu \Phi$.

Proof. Let Φ act by matrix multiplication on a field ψ and compute $[D_\mu, \Phi]\,\psi = D_\mu(\Phi\psi) - \Phi(D_\mu\psi)$ with $D_\mu = \partial_\mu - igA_\mu$:

$$D_\mu(\Phi\psi) - \Phi D_\mu \psi = (\partial_\mu \Phi)\psi + \Phi\partial_\mu\psi - igA_\mu\Phi\psi - \Phi\partial_\mu\psi + ig\Phi A_\mu\psi = (\partial_\mu\Phi - ig\,[A_\mu,\Phi]\,)\psi. \tag{25.71}$$

The free-derivative terms $\Phi\partial_\mu\psi$ cancel, and the A_μ terms assemble into the commutator, giving the matrix form. For the components, use $[A_\mu, \Phi] = A^b_\mu \Phi^c \left[T^b, T^c\right] = A^b_\mu \Phi^c i f^{bcd} T^d$, so $-ig\,[A_\mu, \Phi] = g f^{bcd} A^b_\mu \Phi^c T^d$; relabelling $d \to a$ and using cyclicity $f^{bca} = f^{abc}$ gives (25.70). □

Theorem 25.17 (Non-abelian Bianchi identity)**.** *The Yang–Mills field strength satisfies the cyclic identity*

$$\boxed{D_\mu F_{\nu\rho} + D_\nu F_{\rho\mu} + D_\rho F_{\mu\nu} = 0} \tag{25.72}$$

with the adjoint covariant derivative of (25.70), equivalently $D_{[\mu} F_{\nu\rho]} = 0$. In components, $D_\mu F^a_{\nu\rho} + D_\nu F^a_{\rho\mu} + D_\rho F^a_{\mu\nu} = 0$ with $(D_\mu F_{\nu\rho})^a = \partial_\mu F^a_{\nu\rho} + g f^{abc} A^b_\mu F^c_{\nu\rho}$. This is the non-abelian generalization of Theorem 8.10; in U(1) *the structure constants vanish, the covariant derivatives reduce to ordinary ones, and (25.72) collapses to Eq. (8.47).*

Proof. The identity is the operator Jacobi identity Theorem 2.42 written for the three covariant derivatives D_μ, D_ν, D_ρ. By Eq. (2.76),

$$[D_\mu, [D_\nu, D_\rho]] + [D_\nu, [D_\rho, D_\mu]] + [D_\rho, [D_\mu, D_\nu]] = 0, \tag{25.73}$$

which holds for any associative operators, in particular for differential operators. Now use the field-strength definition (25.47) to replace each inner commutator, $[D_\nu, D_\rho] = -igF_{\nu\rho}$, and the adjoint-action identity of Theorem 25.16, $[D_\mu, F_{\nu\rho}] = D_\mu F_{\nu\rho}$, to evaluate each outer commutator:

$$[D_\mu, [D_\nu, D_\rho]] = [D_\mu, -igF_{\nu\rho}] = -ig\;[D_\mu, F_{\nu\rho}] = -ig\,D_\mu F_{\nu\rho}. \tag{25.74}$$

Doing the same to the other two terms of (25.73) and factoring out the common constant $-ig$,

$$-ig\Big(D_\mu F_{\nu\rho} + D_\nu F_{\rho\mu} + D_\rho F_{\mu\nu}\Big) = 0. \tag{25.75}$$

Since $g \neq 0$, the parenthesis vanishes, which is (25.72). The component form follows by reading off the coefficient of T^a through (25.70). □

Two remarks fix the meaning of Theorem 25.17. First, the identity is purely *kinematic*: like its abelian ancestor it holds for every A_μ whatsoever, expressing nothing more than that F is built as the curvature $\frac{i}{g}[D, D]$ of a connection, and it makes no use of any field equation. It is the homogeneous half of the Yang–Mills equations of motion, the part with no source on the right, and the next section will set it beside the inhomogeneous equation $D_\mu F^{\mu\nu} = g\, J^\nu$ that follows from varying the action. Second, the ordinary derivative *cannot* replace the covariant one in (25.72): the quantity $\partial_\mu F_{\nu\rho} + \text{cyclic}$ is not zero and, worse, is not even gauge covariant, since ∂_μ of a covariant object picks up the inhomogeneous term in the transformation of A_μ. Only the covariant derivative of a covariant object is again covariant, and the cyclic sum of those covariant derivatives is what vanishes. The promotion $\partial \to D$ and the appearance of the commutator self-term in F are two faces of the same geometric fact, and the Jacobi identity is what guarantees their consistency.

Physical Insight

The non-abelian Bianchi identity is the algebraic shadow of a geometric triviality: the boundary of a boundary is empty. The curvature F measures the holonomy around an infinitesimal face (Theorem 25.15); the covariant exterior derivative $D_{[\mu}F_{\nu\rho]}$ measures how that holonomy changes as the face is dragged to bound an infinitesimal cube. Composing the six faces of the cube, each face is traversed once forward and once backward by its neighbours, so the total holonomy around the closed surface is forced to be trivial — and the infinitesimal statement of that triviality is exactly $D_{[\mu}F_{\nu\rho]} = 0$. The Jacobi identity used in the proof is the same fact expressed in the algebra: the three nested transports of (25.73) cancel for precisely the reason that the cube's faces cancel. This is why the identity needed no dynamics — it is geometry — and why it survived from U(1) to SU(N) intact in structure while every derivative was promoted to a covariant one.

We now possess the curvature in the exact frozen Yang–Mills theory form (25.49): a gauge-covariant, Lie-algebra-valued field strength that carries an intrinsic self-coupling $g f^{abc} A^b_\mu A^c_\nu$, transforms in the adjoint by $F \to UFU^{-1}$, and obeys the homogeneous Bianchi identity (25.72). The covariance under conjugation is the green light to build a gauge-*invariant* scalar by tracing two

factors of F together, and the Bianchi identity is already half of the field equations in hand. The next section assembles the unique Lorentz-invariant, gauge-invariant, renormalizable kinetic term $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} = -\frac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ from this curvature, couples matter by minimal substitution, and derives the inhomogeneous Yang–Mills equation that completes the pair begun here.

25.4 The Yang–Mills Lagrangian and Field Equations

The previous section handed us the curvature of the non-abelian connection: the field strength

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$$

(Eq. (25.49)), built as the commutator of covariant derivatives $[D_\mu, D_\nu] = -igF_{\mu\nu}$ (Eq. (25.47)), together with its two defining structural facts: under a gauge transformation $U(x) \in \mathrm{SU}(N)$ it rotates *covariantly*, $F_{\mu\nu} \to U F_{\mu\nu} U^{-1}$ (Eq. (25.57)), rather than invariantly as in the abelian case; and it obeys the homogeneous (Bianchi) identity $D_{[\lambda}F_{\mu\nu]} = 0$ (Theorem 25.17), the non-abelian descendant of $\partial_{[\lambda}F_{\mu\nu]} = 0$. What we do *not* yet have is dynamics. The Bianchi identity is a constraint satisfied by any curvature; it carries no information about how the connection evolves. To make A^a_μ a physical field we must give it an action, and from that action extract the inhomogeneous equation of motion that the Bianchi identity cannot supply — exactly as, for electromagnetism, $\partial_\mu \widetilde{F}^{\mu\nu} = 0$ was kinematic and $\partial_\mu F^{\mu\nu} = J^\nu$ was dynamical (Eq. (8.95)).

This section builds that action. The logic is the one that selected the Maxwell term in Chapter 8, run again under the stronger constraint of *non-abelian* gauge covariance: we ask for the unique Lorentz scalar, gauge invariant, and renormalizable functional quadratic in F, find that the covariance of F forces it to be assembled through a *trace*, fix its normalization with $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, couple matter by minimal coupling, and vary. The variation is where the non-abelian theory first shows its hand: because F itself contains the cubic self-term $g f^{abc} A^b A^c$, the Euler–Lagrange procedure produces $g f^{abc}$ pieces that assemble, not into an ordinary divergence $\partial_\mu F^{\mu\nu}$, but into the *covariant* divergence $D_\mu F^{\mu\nu}$. Tracing the consequence of that single replacement — $\partial \to D$ on the left-hand side of the field equation — is the conceptual payload of the section: it forces us to conclude that the gauge field carries the very charge it mediates. This is the stage that fixes Yang–Mills theory at the classical level, delivering the Lagrangian that Chapter 26 will gauge-fix and quantize.

25.4.1 The kinetic term: why the trace

We seek a Lagrangian density $\mathcal{L}_{\mathrm{YM}}$ governing the gauge field A^a_μ alone, subject to four demands, identical in spirit to those that pinned down the photon's kinetic term in Theorem 8.13:

(i) *Lorentz invariance*: $\mathcal{L}_{\mathrm{YM}}$ is a Lorentz scalar.

(ii) *Gauge invariance*: $\mathcal{L}_{\mathrm{YM}}$ is invariant under $A_\mu \to U A_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}$, the finite gauge transformation of the connection.

(iii) *Renormalizability by power counting*: $\mathcal{L}_{\mathrm{YM}}$ has mass dimension at most four, so that the coupling g is dimensionless in $d = 4$ (the criterion built in Chapter 21). With $[A^a_\mu] = 1$ and $[F^a_{\mu\nu}] = 2$ in mass units, this admits at most two factors of F and no extra derivatives.

(iv) *Parity invariance*: $\mathcal{L}_{\mathrm{YM}}$ is even under spatial reflection.

The decisive simplification, absent in the abelian case, is that $F_{\mu\nu} = F^a_{\mu\nu} T^a$ is now a Lie-algebra-valued field that transforms by conjugation, Eq. (25.57). An ordinary product such as $F^a_{\mu\nu} F^{a\,\mu\nu}$ is a sum over the colour index a; to know whether it is gauge invariant we must understand how that sum behaves under the adjoint rotation $F^a \to (R(U))^{ab} F^b$ induced on components by $F \to UFU^{-1}$. The clean way to see it is to never split into components at all, but to contract colour indices with the one device that is manifestly invariant under conjugation: the matrix trace.

Proposition 25.18 (Uniqueness of the Yang–Mills kinetic term)**.** *Let* $\mathfrak{g} = \mathfrak{su}(N)$ *be simple. Among local densities in* A^a_μ *satisfying* (i)–(iv) *and producing nontrivial dynamics, the only one, up to an overall multiplicative constant and additive total derivatives, is*

$$\mathcal{L}_{\mathrm{YM}} = -\tfrac{1}{2}\,\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = -\tfrac{1}{4}\,F^a_{\mu\nu}F^{a\,\mu\nu}. \tag{25.76}$$

The two expressions agree because of the generator normalization $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ (Chapter B)*. For the abelian group* U(1) *it reduces to the Maxwell density* $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ *of Eq. (8.75).*

Proof. By (iii) the candidate is quadratic in $F_{\mu\nu}$ with the two Lorentz indices contracted, and no further derivatives. There are exactly two independent Lorentz scalars built from two factors of F: the parity-even contraction $F^a_{\mu\nu}F^{b\,\mu\nu}$ and the parity-odd contraction $F^a_{\mu\nu}\widetilde{F}^{b\,\mu\nu}$ with the dual

$$\widetilde{F}^{a\,\mu\nu} \equiv \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F^a_{\rho\sigma}, \qquad \epsilon^{0123} = +1, \tag{25.77}$$

each still carrying two open colour indices a, b that must be saturated by some constant invariant tensor K_{ab}. So the most general candidate is $\mathcal{L} = K_{ab}\big(c_1\,F^a_{\mu\nu}F^{b\,\mu\nu} + c_2\,F^a_{\mu\nu}\widetilde{F}^{b\,\mu\nu}\big)$.

Step 1: gauge invariance forces $K_{ab} \propto \delta_{ab}$*, i.e. the trace.* Under an infinitesimal gauge transformation $U = 1 + i\alpha^c T^c$ the field strength rotates in the adjoint, $\delta F^a_{\mu\nu} = -f^{cae}\alpha^c F^e_{\mu\nu}$ (the component form of $\delta F = i[\alpha, F]$ implied by Eq. (25.57)).

A bilinear $K_{ab}F^aF^b$ is invariant for arbitrary α^c iff K is an invariant tensor of the adjoint representation:

$$f^{cae}K_{eb} + f^{cbe}K_{ae} = 0 \qquad \text{for all } c. \tag{25.78}$$

For a *simple* Lie algebra the adjoint representation is irreducible, so by Schur's lemma (Theorem 2.21) the only symmetric invariant bilinear form is unique up to scale: $K_{ab} = \kappa\,\delta_{ab}$. (The proportionality to δ_{ab} is the statement that the Killing form of a simple algebra is, in the basis with $\operatorname{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, a multiple of δ^{ab}; Chapter B.) Equivalently, $\delta_{ab}F^aF^b = 2\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$, and gauge invariance of the trace is immediate and basis-free: under $F \to UFU^{-1}$,

$$\operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) \to \operatorname{tr}(UF_{\mu\nu}U^{-1}UF^{\mu\nu}U^{-1}) = \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}\,U^{-1}U) = \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}), \tag{25.79}$$

using cyclicity of the trace and $U^{-1}U = 1$. This is precisely the feature the abelian theory could not exploit: there $U = e^{i\alpha}$ commutes through everything and the bare product $F_{\mu\nu}F^{\mu\nu}$ is already invariant, so no trace is needed; here covariance, not invariance, of F is repaired only by the cyclic trace. So both candidate terms are fixed to their traced forms, and

$$\mathcal{L} = c_1\ \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) + c_2\ \operatorname{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}), \tag{25.80}$$

absorbing the scale κ into c_1, c_2.

Step 2: eliminate the parity-odd term. The dual Eq. (25.77) carries one factor of the Levi-Civita symbol, so $\operatorname{tr}(F\widetilde{F})$ is a pseudoscalar, odd under parity, and is excluded outright by demand (iv) — the same reasoning that struck $F\widetilde{F}$ from electrodynamics in Theorem 8.13. Moreover, exactly as in the abelian case, even setting parity aside the term contributes nothing to the classical field equations, because it is a total derivative:

$$\operatorname{tr}(F_{\mu\nu}\widetilde{F}^{\mu\nu}) = \partial_\mu K^\mu, \qquad K^\mu = 2\,\epsilon^{\mu\nu\rho\sigma}\ \operatorname{tr}\Big(A_\nu\,\partial_\rho A_\sigma - \tfrac{2}{3}\,ig\,A_\nu A_\rho A_\sigma\Big), \tag{25.81}$$

where K^μ is the Chern–Simons current. We verify Eq. (25.81) directly, writing $F_{\mu\nu} = G_{\mu\nu} + H_{\mu\nu}$ with $G_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $H_{\mu\nu} = -ig[A_\mu, A_\nu]$, and abbreviating

$$T_1 \equiv \epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(\partial_\mu A_\nu\,\partial_\rho A_\sigma), \qquad T_2 \equiv \epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(\partial_\mu A_\nu\,A_\rho A_\sigma). \tag{25.82}$$

Because the four terms of $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(GG)$ become equal after using the antisymmetry of ϵ and cyclicity of the trace, the purely linear-in-derivative piece is $2T_1$; the cross terms $\frac{1}{2}\epsilon\operatorname{tr}(GH + HG)$ collapse likewise to $-4ig\,T_2$; and the quartic piece vanishes identically,

$$\tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(H_{\mu\nu}H_{\rho\sigma}) \propto \epsilon^{\mu\nu\rho\sigma}\operatorname{tr}(A_\mu A_\nu A_\rho A_\sigma) = 0, \tag{25.83}$$

since a cyclic relabelling $\mu \to \nu \to \rho \to \sigma \to \mu$ leaves the trace invariant but multiplies ϵ by $(-1)^3 = -1$, so the object equals minus itself. Hence

$\operatorname{tr}(F\widetilde{F}) = 2T_1 - 4ig\,T_2$. Differentiating the proposed K^μ gives the same: the term $\epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\mu\partial_\rho A_\sigma$ drops because $\partial_\mu\partial_\rho$ is symmetric against the antisymmetric ϵ, leaving $2T_1$ from the abelian part and, from $-\frac{4}{3}ig\,\epsilon\,\operatorname{tr}\partial_\mu(A_\nu A_\rho A_\sigma)$, three equal contributions summing to $-4ig\,T_2$. Thus $\partial_\mu K^\mu = 2T_1 - 4ig\,T_2 = \operatorname{tr}(F\widetilde{F})$, as claimed. Unlike its abelian counterpart, the surface integral $\int \partial_\mu K^\mu$ need *not* vanish for finite-action configurations of a non-abelian field: it computes a topological winding number, and the term reappears as the physically consequential θ-term of the quantum theory. That story — instantons, the θ-vacuum, and strong CP — is the subject of Chapter 42; for the *classical* field equations of this chapter it is inert, and we drop it.

Step 3: fix the normalization. With the parity-odd term gone, $\mathcal{L} = c_1\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})$. Expanding the trace with $\operatorname{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$,

$$\operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) = F^a_{\mu\nu}F^{b\,\mu\nu}\,\operatorname{tr}(T^aT^b) = \tfrac{1}{2}\,F^a_{\mu\nu}F^{a\,\mu\nu}. \tag{25.84}$$

The constant c_1 is fixed by requiring that each colour component have a canonically normalized, positive-energy kinetic term, just as the $-\frac{1}{4}$ did for the photon (Theorem 8.13): writing out the free part of Eq. (25.84) reproduces N^2-1 copies of the Maxwell density, one per generator, and $c_1 = -\frac{1}{2}$ gives each the standard $-\frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2$. Hence $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{2}\operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$, which is Eq. (25.76). For U(1) there is a single generator, $f^{abc}=0$, $F_{\mu\nu}=\partial_\mu A_\nu - \partial_\nu A_\mu$, and the density collapses to $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, recovering Eq. (8.75). □

The proposition is deceptively compact. Buried inside the single symbol $F^a_{\mu\nu}F^{a\,\mu\nu}$, through the self-term $gf^{abc}A^bA^c$, are interactions of the gauge field with itself — cubic and quartic vertices with no analogue in free Maxwell theory. Equation (25.76) is therefore not merely a free kinetic term dressed with a colour index; it is already an interacting theory of the gauge field alone, even before any matter is added. The systematic expansion of $-\frac{1}{4}F^aF^a$ into its free, cubic, and quartic parts, and the reading-off of the gluon self-couplings, is the work of Section 25.5 (the next section); here we keep F packaged and proceed to the dynamics.

Physical Insight

The trace is not a notational convenience: it is the mechanism by which a covariantly transforming object is turned into an invariant. In the abelian theory F is gauge invariant on its own, so the trace is trivial and one never notices it is doing work. The moment the group is non-abelian, $F \to UFU^{-1}$ rotates, and only the cyclic trace — which lets the U^{-1} on one factor cancel the U on the next — recovers an invariant. Geometrically, $\operatorname{tr}(F\wedge\star F)$ is the natural quadratic functional of a curvature on a principal bundle; the gauge group's geometry, not an aesthetic choice, selects it.

25.4.2 Minimal coupling and the model

Matter is added by the gauge principle exactly as in Chapter 8: take the free Dirac Lagrangian and replace every ordinary derivative by the covariant one, $\partial_\mu \to D_\mu = \partial_\mu - igA^a_\mu T^a$, with the fermion fields now carrying a colour index in the fundamental representation of SU(N). For n_f Dirac flavours ψ_i ($i = 1, \dots, n_f$), each an N-component colour multiplet of common mass m, the complete classical Lagrangian is

$$\boxed{\mathcal{L} = -\tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} + \sum_{i=1}^{n_f} \bar\psi_i (i\not{D} - m)\psi_i \,, \qquad D_\mu = \partial_\mu - igA^a_\mu T^a \,.} \tag{25.85}$$

This is the model of Yang–Mills theory and inherited verbatim by Chapter 26 (quantization and BRST) and Chapter 27 (the beta function and asymptotic freedom). Every symbol is now pinned: the coupling g (with $\alpha_s \equiv g^2/4\pi$ when we specialize to QCD), the structure constants f^{abc} and generators T^a of Eq. (2.61), the fermion flavour count n_f, and the covariant-derivative sign $D_\mu = \partial_\mu - igA^a_\mu T^a$. Expanding the fermion kinetic term exhibits the minimal coupling explicitly,

$$\bar\psi_i \, i\not{D}\, \psi_i = \bar\psi_i \, i\not{\partial}\, \psi_i + g\, \bar\psi_i \, \gamma^\mu T^a \psi_i \, A^a_\mu, \tag{25.86}$$

where the cross term, of strength g and universal across flavours, couples the gauge field to the colour current of the matter. As in QED this interaction was not posited; it is the unique renormalizable, gauge-invariant coupling that the demand of local SU(N) invariance permits, and its universality (one coupling g for all colours and flavours) is dictated, not fitted.

25.4.3 The Yang–Mills equations of motion

The homogeneous equations are already in hand: the Bianchi identity $D_{[\lambda} F_{\mu\nu]} = 0$ (Theorem 25.17). The action Eq. (25.85) supplies the inhomogeneous, dynamical equations through the Euler–Lagrange variation with respect to A^a_μ. The variation is the place where the non-abelian self-interaction first asserts itself, because $F^a_{\mu\nu}$ depends on A^a_μ both through its derivative and, via $g f^{abc} A^b A^c$, through the bare field. We carry the computation in full.

Worked Example 25.19 (The field equation $D_\mu F^{\mu\nu\,a} = -g J^{\nu a}$ from $\mathcal{L}$)**.** The Euler–Lagrange equation for A^a_ν reads $\partial_\mu \dfrac{\partial \mathcal{L}}{\partial(\partial_\mu A^a_\nu)} - \dfrac{\partial \mathcal{L}}{\partial A^a_\nu} = 0$. We assemble the two derivatives in turn.

Step 1: the momentum $\partial\mathcal{L}/\partial(\partial_\mu A^a_\nu)$. Only $\mathcal{L}_{\rm YM}$ contains ∂A, and only through the linear part of F. From $F^b_{\rho\sigma} = \partial_\rho A^b_\sigma - \partial_\sigma A^b_\rho + g f^{bcd} A^c_\rho A^d_\sigma$,

$$\frac{\partial F^b_{\rho\sigma}}{\partial(\partial_\mu A^a_\nu)} = \delta^{ba}\big(\delta^\mu_\rho \delta^\nu_\sigma - \delta^\mu_\sigma \delta^\nu_\rho\big). \tag{25.87}$$

Hence, using $\partial\mathcal{L}_{\rm YM}/\partial F^b_{\rho\sigma} = -\tfrac{1}{2}F^{b\,\rho\sigma}$ from $\mathcal{L}_{\rm YM} = -\tfrac{1}{4}F^bF^b$,

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu A^a_\nu)} = -\tfrac{1}{2}F^{b\,\rho\sigma}\,\delta^{ba}\big(\delta^\mu_\rho\delta^\nu_\sigma - \delta^\mu_\sigma\delta^\nu_\rho\big) = -\tfrac{1}{2}\big(F^{a\,\mu\nu} - F^{a\,\nu\mu}\big) = -F^{a\,\mu\nu}. \tag{25.88}$$

Step 2: the bare-field derivative $\partial\mathcal{L}/\partial A^a_\nu$. Two contributions: the self-term of $\mathcal{L}_{\rm YM}$ and the minimal coupling of the matter. For the gauge part only the cubic piece of F depends on A without a derivative,

$$\frac{\partial F^b_{\rho\sigma}}{\partial A^a_\nu} = g\,f^{bad}\,\delta^\nu_\rho A^d_\sigma + g\,f^{bca}\,\delta^\nu_\sigma A^c_\rho, \tag{25.89}$$

so that, again with $\partial\mathcal{L}_{\rm YM}/\partial F^b_{\rho\sigma} = -\tfrac{1}{2}F^{b\,\rho\sigma}$,

$$\frac{\partial\mathcal{L}_{\rm YM}}{\partial A^a_\nu} = -\tfrac{1}{2}\Big(g\,f^{bad}F^{b\,\nu\sigma}A^d_\sigma + g\,f^{bca}F^{b\,\rho\nu}A^c_\rho\Big). \tag{25.90}$$

In the second term use $F^{b\,\rho\nu} = -F^{b\,\nu\rho}$ and $f^{bca} = -f^{bac}$, then relabel $c\to d$, $\rho\to\sigma$; it equals the first. Thus the two combine,

$$\frac{\partial\mathcal{L}_{\rm YM}}{\partial A^a_\nu} = -g\,f^{bad}F^{b\,\nu\sigma}A^d_\sigma = g\,f^{abc}A^b_\sigma F^{c\,\sigma\nu} = g\,f^{abc}A^b_\mu F^{c\,\mu\nu}, \tag{25.91}$$

where the middle step renames $b\to c$, $d\to b$ and uses total antisymmetry $f^{bad} = f^{abc}\big|_{\dots}$ (the cyclic identity $f^{cab} = f^{abc}$) together with $F^{c\,\nu\sigma} = -F^{c\,\sigma\nu}$, and the last renames $\sigma\to\mu$. The matter contribution is read off from Eq. (25.86),

$$\frac{\partial\mathcal{L}_{\rm matter}}{\partial A^a_\nu} = g\,\bar\psi_i\gamma^\nu T^a\psi_i \equiv g\,J^{\nu a}, \qquad J^{\nu a} = \sum_{i=1}^{n_f}\bar\psi_i\,\gamma^\nu T^a\psi_i, \tag{25.92}$$

the matter *colour current*.

Step 3: assemble. Substituting Eqs. (25.88), (25.91) and (25.92) into the Euler–Lagrange equation,

$$\partial_\mu(-F^{a\,\mu\nu}) - \big(g\,f^{abc}A^b_\mu F^{c\,\mu\nu} + g\,J^{\nu a}\big) = 0, \tag{25.93}$$

that is, after an overall sign,

$$\partial_\mu F^{a\,\mu\nu} + g\,f^{abc}A^b_\mu F^{c\,\mu\nu} = -g\,J^{\nu a}. \tag{25.94}$$

The left-hand side is exactly the adjoint covariant derivative of F. Recall that on an adjoint-valued field X^a the covariant derivative induced by $D_\mu = \partial_\mu - igA^b_\mu T^b$ is $(D_\mu X)^a = \partial_\mu X^a + g\,f^{abc}A^b_\mu X^c$ (the matrix statement $D_\mu X = \partial_\mu X - ig[A_\mu, X]$ written in components, with $X = X^aT^a$). With $X^a \to F^{a\,\mu\nu}$ this is the left-hand side of Eq. (25.94), so the Yang–Mills equation of motion is

$$\boxed{\,D_\mu F^{\mu\nu\,a} = -g\,J^{\nu a}, \qquad J^{\nu a} = \sum_i \bar\psi_i\gamma^\nu T^a\psi_i\,.\,} \tag{25.95}$$

Equation (25.95) is the central dynamical result of the chapter. Three features deserve comment.

First, the abelian check. For U(1) the structure constants vanish, $D_\mu \to \partial_\mu$, and Eq. (25.95) becomes $\partial_\mu F^{\mu\nu} = -gJ^\nu$ with $J^\nu = \bar\psi\gamma^\nu\psi$. This is identical in content to the Maxwell field equation Eq. (8.95) derived in Chapter 8; the relative sign relative to the QED form $\partial_\mu F^{\mu\nu} = e\bar\psi\gamma^\nu\psi$ there is entirely the bookkeeping of the covariant-derivative convention, $D_\mu = \partial_\mu - igA_\mu T^a$ here versus $D_\mu = \partial_\mu + ieA_\mu$ for the electron in Chapter 8. We flag the sign explicitly below, because it is the single most error-prone point in the subject.

Caution

With the *frozen* convention $D_\mu = \partial_\mu - igA^a_\mu T^a$ and the current defined as $J^{\nu a} = \sum_i \bar\psi_i\gamma^\nu T^a\psi_i$, the equation of motion carries a minus sign, $D_\mu F^{\mu\nu\,a} = -gJ^{\nu a}$. The minus is not optional: it is fixed by the sign in D_μ together with the sign of J. Texts that write $D_\mu F^{\mu\nu} = +gJ^\nu$ have either flipped the sign of D_μ (using $\partial_\mu + igAT$) or absorbed a minus into the definition of the current. Within this book the pair ($D_\mu = \partial_\mu - igA^aT^a$, $J^{\nu a} = \bar\psi\gamma^\nu T^a\psi$) and the result Eq. (25.95) are used consistently; Chapters 26 and 27 inherit them unchanged. The abelian limit reproduces Eq. (8.95) under the convention difference, as just noted.

Second, the self-coupling. The piece $gf^{abc}A^b_\mu F^{c\,\mu\nu}$ on the left of Eq. (25.94) is genuinely nonlinear in A: substituting $F^c = \partial A^c + \dots$ produces terms quadratic and cubic in the gauge field. The gauge field sources its own field equation. This is the precise dynamical statement of the self-interaction that the next section dissects vertex by vertex.

Third — and most consequential — the operator on the left is D_μ, not ∂_μ. The covariant divergence of F, not its ordinary divergence, equals the source. This single replacement controls the conservation law of the colour current, to which we now turn.

25.4.4 Covariant versus ordinary conservation: the gluon is coloured

In electrodynamics the field equation $\partial_\mu F^{\mu\nu} = J^\nu$ carries current conservation inside it for free: $\partial_\nu J^\nu = \partial_\nu\partial_\mu F^{\mu\nu} = 0$, because the symmetric pair $\partial_\nu\partial_\mu$ annihilates the antisymmetric $F^{\mu\nu}$ (Eq. (8.96)). One is tempted to repeat the move on Eq. (25.95) with D_ν in place of ∂_ν. The result is true but, read carefully, it does *not* say what the abelian statement said.

Theorem 25.20 (Covariant conservation and the colour of the gauge field)**.** *The Yang–Mills operator obeys the off-shell identity*

$$D_\nu(D_\mu F^{\mu\nu})^a \equiv 0, \tag{25.96}$$

holding for every field configuration, whether or not the equations of motion are satisfied. Consequently the matter colour current of Eq. (25.92) is covariantly *conserved on shell, $(D_\nu J^\nu)^a = 0$, but is* not *ordinarily conserved, $\partial_\nu J^{\nu a} \neq 0$ in general. The current that is ordinarily conserved — the Noether current of the global colour symmetry, hence the carrier of the conserved colour charge — is the sum of the matter current and a contribution built from the gauge field itself,*

$$\partial_\nu \mathcal{J}^{\nu a} = 0, \qquad \mathcal{J}^{\nu a} = J^{\nu a} + j^{\nu a}, \qquad j^{\nu a} = f^{abc} A^b_\mu F^{\mu\nu\, c}. \tag{25.97}$$

Proof. The identity Eq. (25.96). Write $(D_\mu F^{\mu\nu})$ as an adjoint-valued Lorentz vector and contract with D_ν. The double covariant derivative splits into parts symmetric and antisymmetric in $(\mu\nu)$,

$$D_\nu D_\mu F^{\mu\nu} = \tfrac{1}{2}(D_\nu D_\mu + D_\mu D_\nu) F^{\mu\nu} + \tfrac{1}{2}(D_\nu D_\mu - D_\mu D_\nu) F^{\mu\nu}. \tag{25.98}$$

The symmetric combination is contracted with the antisymmetric $F^{\mu\nu}$ and vanishes. The antisymmetric combination is the commutator of covariant derivatives, which acts on an adjoint field as the curvature itself: from $[D_\mu, D_\nu] = -igF_{\mu\nu}$ (Eq. (25.47)) acting in the adjoint, $\big([D_\nu, D_\mu]X\big)^a = g\, f^{abc} F^b_{\nu\mu} X^c$. With $X^c \to F^{c\,\mu\nu}$,

$$D_\nu D_\mu F^{\mu\nu}\big|^a = \tfrac{1}{2}\, g\, f^{abc} F^b_{\nu\mu} F^{c\,\mu\nu}. \tag{25.99}$$

Now $F^b_{\nu\mu} F^{c\,\mu\nu} = -F^b_{\mu\nu} F^{c\,\mu\nu}$ is *symmetric* under the exchange $b \leftrightarrow c$ (it is a Lorentz contraction of two identical-looking field strengths), whereas f^{abc} is *antisymmetric* under $b \leftrightarrow c$. The contraction of a symmetric with an antisymmetric tensor over the same index pair vanishes, so the right side of Eq. (25.99) is zero, establishing Eq. (25.96). (This is the dynamical companion of the homogeneous Bianchi identity: there $D_{[\lambda} F_{\mu\nu]} = 0$ is automatic from F being a curvature; here $D_\nu D_\mu F^{\mu\nu} = 0$ is automatic from the antisymmetry of F and the adjoint action of the commutator.)

Covariant but not ordinary conservation of J. Apply D_ν to the equation of motion Eq. (25.95). The left side vanishes by Eq. (25.96), hence $0 = -g\,(D_\nu J^\nu)^a$, i.e. $(D_\nu J^\nu)^a = 0$ on shell. Written out, $\partial_\nu J^{\nu a} + g\, f^{abc} A^b_\nu J^{\nu c} = 0$, so $\partial_\nu J^{\nu a} = -g\, f^{abc} A^b_\nu J^{\nu c}$, which is nonzero whenever the gauge field is present and the current is non-singlet. The matter colour current is therefore not by itself a conserved current in the ordinary sense; covariant conservation $D_\nu J^\nu = 0$ is a weaker, gauge-covariant statement and does not yield a constant charge $\int \mathrm{d}^3x\, J^{0a}$ by the usual divergence-theorem argument, because $\partial_\nu J^{\nu a} \neq 0$.

The ordinarily conserved total current. To recover a genuine conservation law, move the self-interaction term of the covariant divergence to the right-hand side. From Eq. (25.94),

$$\partial_\mu F^{\mu\nu\, a} = -g\, J^{\nu a} - g\, f^{abc} A^b_\mu F^{c\,\mu\nu} = -g\big(J^{\nu a} + f^{abc} A^b_\mu F^{\mu\nu\, c}\big) \equiv -g\, \mathcal{J}^{\nu a}, \tag{25.100}$$

where $j^{\nu a} = f^{abc} A^b_\mu F^{\mu\nu\, c}$ (note $F^{c\,\mu\nu} = -F^{\mu\nu\, c} \cdot (-1) = F^{\mu\nu c}$ after using the sign moved out front; the bracket is exactly $\mathcal{J}^{\nu a}$ as displayed in Eq. (25.97)). The

left-hand side is now an *ordinary* divergence of an antisymmetric tensor, so taking ∂_ν kills it identically, $\partial_\nu\partial_\mu F^{\mu\nu a} = 0$, and therefore $\partial_\nu \mathcal{J}^{\nu a} = 0$. The conserved charge $Q^a = \int \mathrm{d}^3x\, \mathcal{J}^{0a}$ is the colour charge, and by Noether's first theorem (Theorem 7.10) applied to the global SU(N) symmetry that survives once the gauge parameter is taken constant, $\mathcal{J}^{\nu a}$ is precisely that symmetry's Noether current — as one verifies by computing the conserved current of Eq. (25.85) under the rigid rotation $\delta\psi_i = i\alpha^a T^a \psi_i$, $\delta A^c_\mu = -f^{cab}\alpha^a A^b_\mu$ (the adjoint rotation of the gauge field) and finding the matter piece $J^{\nu a}$ plus the gauge piece $j^{\nu a}$. □

The physics of Theorem 25.20 is the defining novelty of non-abelian gauge theory, and it is worth stating plainly. In QED the photon is electrically neutral: the conserved electric current is built from the electron alone, and the field equation's source is the matter current full stop. In Yang–Mills theory the conserved colour current $\mathcal{J}^{\nu a}$ has *two* pieces. The matter piece $J^{\nu a} = \bar\psi\gamma^\nu T^a \psi$ is the obvious one. The second piece $j^{\nu a} = f^{abc} A^b_\mu F^{\mu\nu\, c}$ is built entirely from the gauge field, and it is nonzero precisely because $f^{abc} \neq 0$. The gauge field carries colour charge. The gluon is coloured; the photon is not. This is why the colour current splits, why J alone is only covariantly conserved, and — looking ahead — why the theory is so radically different in the deep ultraviolet: the charge carriers include the force carriers, and that feedback is the root of asymptotic freedom (Chapter 27).

Physical Insight

"Covariant conservation" $D_\nu J^\nu = 0$ and "ordinary conservation" $\partial_\nu \mathcal{J}^\nu = 0$ are not two names for one thing. Only the ordinary law produces a time-independent charge by Gauss's theorem, because only ∂_ν of a quantity integrates to a boundary term. The covariant law says the matter current's failure to be conserved is exactly accounted for by colour flowing into the gauge field; restoring the gauge field's own current $j^{\nu a}$ balances the books. The lesson generalizes: for any field charged under a gauge symmetry, the gauge-covariant current of that field is covariantly conserved, and the *globally* conserved Noether charge always requires adding the gauge sector's contribution.

One structural object is owed but deferred. The same self-coupling that coloured the current also obstructs the naive stress tensor. The canonical (Noether) energy–momentum tensor obtained from translations applied to Eq. (25.85) is, for the gauge field, neither symmetric nor gauge invariant — both defects are visible already for free Maxwell theory and were diagnosed in general in Theorem 7.28. The repair is the Belinfante–Rosenfeld improvement, supplemented in the gauge case by a covariant improvement that trades the bare $A^a_\mu \partial^\nu A^a_\lambda$ pieces for gauge-invariant F-bilinears, yielding the symmetric, gauge-invariant tensor $T^{\mu\nu} = -F^{a\,\mu\lambda}F^{a\,\nu}{}_\lambda + \frac{1}{4}\eta^{\mu\nu} F^a_{\rho\sigma}F^{a\,\rho\sigma}$ (plus the matter part). We state this

here as the analogue, for the non-abelian field, of the abelian improvement of Theorem 7.28; its full construction belongs with the Belinfante machinery and the coupling to gravity and is not needed for the classical and quantum dynamics that occupy the next chapters.

With Eqs. (25.85) and (25.95) we have the complete classical Yang–Mills theory: a gauge-invariant action, its dynamical field equation $D_\mu F^{\mu\nu\,a} = -gJ^{\nu a}$, the homogeneous Bianchi identity, and the precise statement that the gauge field is itself coloured. This standard Lagrangian is the object Chapter 26 will gauge-fix and quantize and Chapter 27 will renormalize. Before quantizing, however, we must expose what is hidden inside the single symbol $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$: expanding it in powers of the field reveals the free propagating gluon together with the cubic and quartic self-interactions that distinguish a non-abelian gauge field from a free photon. That expansion is the subject of the next section.

25.5 Self-Interaction: The Gluon Vertices

The previous section handed us the Yang–Mills Lagrangian in its compact, manifestly gauge-covariant form, $\mathcal{L}_{\rm YM} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \bar\psi_i(i\not{D} - m)\psi_i$ (Eq. (25.76)), together with the field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ (Eq. (25.49)). Written that way the theory looks deceptively like electrodynamics with a colour index stitched on. The deception ends the moment one expands the square. In the abelian theory the field strength is *linear* in the potential, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (Eq. (8.44)), so $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is exactly quadratic: the photon is a free field that happens to be sourced by external charges, and a Lagrangian containing photons alone describes nothing but freely propagating light. The non-abelian field strength carries the extra piece $g f^{abc} A^b_\mu A^c_\nu$, which is *quadratic* in the potential. Squaring a sum of a linear and a quadratic term cannot give a purely quadratic result; it produces cubic and quartic pieces as well. Those pieces are interactions of the gauge field with itself. The gluon, unlike the photon, is not a passive carrier of force between charged matter — it is itself charged, and it sources its own field.

This section makes that statement completely explicit. We expand $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ carrying every cross term, isolate the free (Maxwell-like) quadratic part, the cubic three-gluon coupling $\propto g f^{abc}$, and the quartic four-gluon coupling $\propto g^2(ff)$, and exhibit their colour and Lorentz structure. We then confront the obvious question: if no single one of these three terms is gauge invariant on its own — and none is — what forces them to appear together with exactly the coefficients written above? The answer, worked in full in Theorem 25.21, is that gauge invariance ties them into a single inseparable structure: the self-interaction is not an optional addition to the kinetic term but its unavoidable completion. Everything here is classical. We are reading off the vertices that Chapter 26 will,

after gauge fixing and the introduction of Faddeev–Popov ghosts, promote into the Feynman rules of quantum chromodynamics; we do not pre-quantize.

25.5.1 Splitting the field strength

It is convenient to give the abelian part of the field strength its own name. Define

$$G^a_{\mu\nu} \;\equiv\; \partial_\mu A^a_\nu - \partial_\nu A^a_\mu, \tag{25.101}$$

the ordinary "curl" of the potential, antisymmetric in $\mu\nu$ and *linear* in A. This is precisely the object that would be the entire field strength were the gauge group abelian. The full non-abelian field strength is then the sum of this linear piece and a bilinear piece,

$$F^a_{\mu\nu} \;=\; G^a_{\mu\nu} \;+\; g\, f^{abc} A^b_\mu A^c_\nu, \tag{25.102}$$

the bilinear piece being the geometric signature of non-commuting generators (Eq. (2.61)). The structure constants f^{abc} are totally antisymmetric in the normalization fixed for this book (Hermitian generators, $\left[T^a, T^b\right] = i f^{abc} T^c$, $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$; the antisymmetry and the Casimir data are collected in Chapter B). Note in passing that the bilinear term is itself antisymmetric in $\mu\nu$ as it must be, because $f^{abc} A^b_\mu A^c_\nu = -f^{abc} A^b_\nu A^c_\mu$ follows from relabelling the summed colour indices $b \leftrightarrow c$ and using $f^{acb} = -f^{abc}$.

25.5.2 The full expansion

We now compute $-\frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu}$ by substituting (25.102) and multiplying out. The contraction of the field strength with itself splits into three groups of terms — linear×linear, the two equal linear×bilinear cross terms, and bilinear×bilinear:

$$\begin{aligned} F^a_{\mu\nu} F^{a\,\mu\nu} &= \left(G^a_{\mu\nu} + g f^{abc} A^b_\mu A^c_\nu\right)\left(G^{a\,\mu\nu} + g f^{ade} A^{d\,\mu} A^{e\,\nu}\right) \\ &= G^a_{\mu\nu} G^{a\,\mu\nu} \;+\; 2\, g\, f^{abc}\, G^a_{\mu\nu}\, A^{b\,\mu} A^{c\,\nu} \;+\; g^2\, f^{abc} f^{ade}\, A^b_\mu A^c_\nu A^{d\,\mu} A^{e\,\nu}. \end{aligned} \tag{25.103}$$

The factor of 2 on the middle term is genuine: the two cross terms $G^a_{\mu\nu}\,(g f^{ade} A^{d\mu} A^{e\nu})$ and $(g f^{abc} A^b_\mu A^c_\nu)\, G^{a\mu\nu}$ are identical after relabelling the summed colour indices, so they add rather than averaging. Multiplying by $-\frac{1}{4}$ gives the Yang–Mills Lagrangian density resolved into its free, cubic, and quartic pieces,

$$\boxed{-\tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} = -\tfrac{1}{4}\left(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu\right)^2 \;-\; g\, f^{abc} (\partial_\mu A^a_\nu) A^{b\,\mu} A^{c\,\nu} \;-\; \tfrac{1}{4} g^2\, f^{abe} f^{cde}\, A^a_\mu A^b_\nu A^{c\,\mu} A^{d\,\nu}.} \tag{25.104}$$

Three reductions stand between (25.103) and (25.104): the abbreviation of the quadratic piece, the collapse of the cubic piece to a single derivative term, and the relabelling of the quartic colour tensor. We carry each in turn.

The quadratic (free) term. The first term of (25.103) is, with the shorthand $(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2 \equiv G^a_{\mu\nu} G^{a\,\mu\nu}$,

$$\mathcal{L}_2 \;=\; -\tfrac{1}{4}\, G^a_{\mu\nu} G^{a\,\mu\nu} \;=\; -\tfrac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)(\partial^\mu A^{a\,\nu} - \partial^\nu A^{a\,\mu}). \tag{25.105}$$

This is N^2-1 independent copies — one for each generator of $\mathrm{SU}(N)$, labelled by the adjoint index $a = 1, \ldots, N^2-1$ — of the free Maxwell Lagrangian (Eq. (8.44) and the photon action of Chapter 8). On its own, (25.105) would describe N^2-1 non-interacting massless vector fields, a collection of mutually oblivious photons. Everything that makes Yang–Mills theory more than this lives in the remaining two terms.

The cubic (three-gluon) term. The middle term of (25.103), multiplied by $-\tfrac{1}{4}$, is

$$\mathcal{L}_3 \;=\; -\tfrac{1}{2}\, g\, f^{abc}\, G^a_{\mu\nu}\, A^{b\,\mu} A^{c\,\nu} \;=\; -\tfrac{1}{2}\, g\, f^{abc} (\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) A^{b\,\mu} A^{c\,\nu}. \tag{25.106}$$

The two derivative pieces are in fact equal. Take the second, $+\tfrac{1}{2} g f^{abc} (\partial_\nu A^a_\mu) A^{b\mu} A^{c\nu}$, and rename the Lorentz dummies $\mu \leftrightarrow \nu$ to write it as $+\tfrac{1}{2} g f^{abc} (\partial_\mu A^a_\nu) A^{b\nu} A^{c\mu}$. The two potentials commute as classical fields, $A^{b\nu} A^{c\mu} = A^{c\mu} A^{b\nu}$; renaming the colour dummies $b \leftrightarrow c$ and using $f^{acb} = -f^{abc}$ turns this into $-\tfrac{1}{2} g f^{abc} (\partial_\mu A^a_\nu) A^{b\mu} A^{c\nu}$, which is identical to the first piece. Hence the two combine and the cubic term collapses to a single derivative structure,

$$\mathcal{L}_3 \;=\; -\, g\, f^{abc} (\partial_\mu A^a_\nu) A^{b\,\mu} A^{c\,\nu}, \tag{25.107}$$

the second term displayed in (25.104). This is the *three-gluon coupling*. Its colour content is one factor of f^{abc} — totally antisymmetric — and its Lorentz/derivative content is one derivative distributed across three gauge fields. The single power of g marks it as a trilinear vertex: in the quantized theory it will join three gluon lines at a point.

The quartic (four-gluon) term. The last term of (25.103), multiplied by $-\tfrac{1}{4}$, is

$$\mathcal{L}_4 \;=\; -\tfrac{1}{4}\, g^2\, f^{abc} f^{ade}\, A^b_\mu A^c_\nu A^{d\,\mu} A^{e\,\nu}. \tag{25.108}$$

Here the colour index a is the one summed between the two field strengths; b, c, d, e label the four gauge fields. To bring this to the canonical form quoted in (25.104) we simply rename the dummy colour indices so that the four *fields* carry the names a, b, c, d and the summed index is called e. Concretely, send

$$a \to e, \quad b \to a, \quad c \to b, \quad d \to c, \quad e \to d, \tag{25.109}$$

which turns $f^{abc} f^{ade} \to f^{eab} f^{ecd}$ and the field product $A^b_\mu A^c_\nu A^{d\mu} A^{e\nu} \to A^a_\mu A^b_\nu A^{c\mu} A^{d\nu}$. Using the cyclic symmetry $f^{eab} = f^{abe}$ and $f^{ecd} = f^{cde}$,

$$\mathcal{L}_4 \;=\; -\tfrac{1}{4}\, g^2\, f^{abe} f^{cde}\, A^a_\mu A^b_\nu A^{c\,\mu} A^{d\,\nu}, \tag{25.110}$$

the third term of (25.104). This is the *four-gluon coupling*. Its colour content is a product of two structure constants sharing one summed index, $f^{abe}f^{cde}$; it carries no derivatives, only four gauge fields; and the two powers of g mark it as a quartic vertex. With (25.105), (25.107), and (25.110) established, (25.104) is proved in full.

Caution

The colour tensor $f^{abe}f^{cde}$ written in (25.110) is *not* by itself symmetric under the permutations of the four legs $(a\mu), (b\nu), (c\mu), (d\nu)$ that Bose statistics will eventually require. The Lagrangian density is a single, fixed expression; its *vertex*, obtained by functionally differentiating with respect to four gauge fields, is the symmetrized object. We exhibit that symmetrization in Theorem 25.21 but do not build the Feynman rule here — that, with the propagator and ghosts, is the business of Chapter 26. Reading (25.110) as if it were already a vertex is a common source of factor-of-symmetry errors.

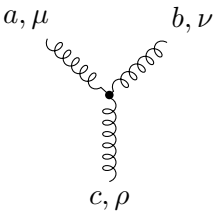
25.5.3 What the three pieces are

It is worth pausing on the anatomy of (25.104), because the pattern is the defining structural feature of non-abelian gauge theory and recurs throughout Part VII.

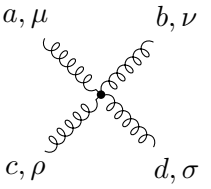
The quadratic term $\mathcal{L}_2$ governs free propagation. It is identical in form to $N^2 - 1$ photons and, on its own, knows nothing of the gauge group beyond the number of generators; the structure constants have dropped out entirely. This is the part that, after gauge fixing in Chapter 26, yields the gluon propagator.

The cubic term $\mathcal{L}_3$ couples *three* gluons through a single structure constant f^{abc} and a single derivative. Its mere existence is the new physics: a gluon line can split into two gluon lines. The strength is set by g, the same coupling that appears in the matter covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$ (Eq. (25.16)) — there is one coupling constant in the theory, and gauge invariance forces the gluon self-coupling to share it with the quark–gluon coupling. This universality, which in the abelian theory was the statement that all charges are multiples of one unit, here becomes the much stronger statement that the gauge field's coupling to matter and to itself are locked to a single number.

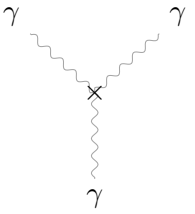
The quartic term $\mathcal{L}_4$ couples *four* gluons through two structure constants and strength g^2. It is the only term in the pure-gauge Lagrangian with no derivatives at all; it is a contact interaction among four gluon lines. Its presence is again forced — as the gauge variation below shows, the cubic term cannot be made gauge invariant without it.



(a) Three-gluon, $\propto g f^{abc}$.



(b) Four-gluon, $\propto g^2 f f$.



(c) No photon self-vertex ($f^{abc} = 0$).

Figure 25.3: The classical gluon self-interactions read off (25.104): a cubic vertex linear in g and in f^{abc}, and a quartic contact vertex of order g^2 carrying two structure constants. The abelian theory has neither, because $f^{abc} = 0$ leaves $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ purely quadratic; the crossed-out diagram does not exist in electrodynamics. These classical structures become the QCD Feynman rules in Chapter 26.

Physical Insight

The photon does not carry electric charge, so it does not couple to itself; the electromagnetic Lagrangian $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is exactly quadratic and the free photon is genuinely free. The gluon *does* carry colour — it transforms in the adjoint representation of SU(N), the same representation that labels the structure constants — and therefore it sources the very field it mediates. Formally the whole difference is the single object f^{abc}: set it to zero (the abelian limit) and (25.104) collapses to $-\frac{1}{4}(\partial_\mu A_\nu - \partial_\nu A_\mu)^2$, the free Maxwell theory, with both interaction terms vanishing identically. The non-vanishing of f^{abc} is the entire content of "non-abelian."

This self-coupling is the physical root of the two phenomena that distinguish the strong interaction. A gluon fluctuating into a pair of gluons antiscreens colour charge: the vacuum's response *strengthens* a colour source at long distance rather than screening it, which at short distance reads as a coupling that *decreases* with energy — asymptotic freedom. The same antiscreening, pushed to long distance, is the qualitative origin of confinement: the colour field does not spread out like a Coulomb field but collimates into a flux tube of constant tension. The quantitative statement — the sign and coefficient of the Yang–Mills beta function — is computed in Chapter 27; we flag here only that its seed is the term $\mathcal{L}_3$ that the photon simply does not have.

The three- and four-gluon vertices are drawn in Fig. 25.3, alongside the empty space where the photon's self-vertices would be.

25.5.4 The interactions are forced: a gauge-variation conspiracy

The remaining task is the conceptually decisive one. We have an expansion, (25.104), into a free term and two interaction terms. One could imagine keeping only the free term — a perfectly Lorentz-invariant, renormalizable Lagrangian for N^2-1 vector fields. Why is that not the theory? Because it is not gauge invariant. Under the infinitesimal non-abelian gauge transformation established in Section 25.3,

$$\delta A^a_\mu \;=\; \tfrac{1}{g}\,\partial_\mu\theta^a \;+\; f^{abc}A^b_\mu\theta^c \;=\; \tfrac{1}{g}\,(D_\mu\theta)^a, \tag{25.111}$$

the curvature transforms covariantly, $\delta F^a_{\mu\nu} = f^{abc}F^b_{\mu\nu}\theta^c$ (Eq. (25.58)), i.e. it rotates in the adjoint representation without an inhomogeneous piece. Consequently the full Lagrangian is invariant in one line,

$$\delta\Big(-\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}\Big) = -\tfrac{1}{2}F^{a\,\mu\nu}\,\delta F^a_{\mu\nu} = -\tfrac{1}{2}f^{abc}\,F^{a\,\mu\nu}F^b_{\mu\nu}\,\theta^c = 0, \tag{25.112}$$

the last equality because f^{abc} is antisymmetric in (a,b) while $F^{a\,\mu\nu}F^b_{\mu\nu}$ is symmetric in (a,b), so their contraction vanishes. That is the compact statement. What it hides is that the invariance is a *cancellation among the three terms* of (25.104): the transformation (25.111) has an inhomogeneous part $\frac{1}{g}\partial_\mu\theta^a$ and a homogeneous (rotation) part $f^{abc}A^b_\mu\theta^c$, and these shuffle the free, cubic, and quartic pieces into one another. No single term is invariant; the three conspire. The following worked example makes the conspiracy explicit, and in doing so proves that the self-interactions are not optional.

Worked Example 25.21 (The self-interaction is forced by gauge symmetry)**.** We continue Yang–Mills theory. The goal is to vary the expanded Lagrangian (25.104) directly — not via the covariant shortcut (25.112) — and watch the variations of the free, cubic, and quartic terms cancel against one another. The bookkeeping device is to organize $\delta\mathcal{L}$ by the *number of gauge fields* A it contains, because the transformation (25.111) mixes orders: its inhomogeneous part $\delta^\partial A^a_\mu = \frac{1}{g}\partial_\mu\theta^a$ *lowers* the field count by one (and carries $1/g$), while its homogeneous part $\delta^f A^a_\mu = f^{abc}A^b_\mu\theta^c$ preserves it. Writing $\mathcal{L}_2, \mathcal{L}_3, \mathcal{L}_4$ for the three terms of (25.104) — of orders A^2, gA^3, g^2A^4 — the variation $\delta\mathcal{L}$ collects into contributions with one, two, three, and four powers of A. Gauge invariance requires each power to vanish separately, and this is exactly one cancellation per junction in the tower $\mathcal{L}_2 \to \mathcal{L}_3 \to \mathcal{L}_4$.

The interaction tensors. First we record the two interaction terms as they will enter Chapter 26. The cubic term (25.107) is

$$\mathcal{L}_3 = -\,g\,f^{abc}\,(\partial_\mu A^a_\nu)\,A^{b\,\mu}A^{c\,\nu}, \tag{25.113}$$

carrying one totally antisymmetric colour tensor f^{abc} and one derivative. Bose symmetry under the exchange of the three identical gluon legs is the statement

that the trilinear coupling is unchanged when one simultaneously permutes the three colour labels and the three Lorentz labels/momenta; because f^{abc} is totally antisymmetric, the accompanying derivative structure must be totally antisymmetric in the same sense, and indeed it is — the antisymmetry of the abelian curl $G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu$ in (25.106) is precisely what supplies it. The quartic term (25.110) is

$$\mathcal{L}_4 = -\tfrac{1}{4}\, g^2\, f^{abe} f^{cde}\, A^a_\mu A^b_\nu A^{c\,\mu} A^{d\,\nu}. \tag{25.114}$$

Its associated four-leg vertex is obtained by symmetrizing over the four identical bosons $(a\mu), (b\nu), (c\mu), (d\nu)$. Carrying out the three inequivalent pairings of the four legs and using the antisymmetry of f to combine equal contributions, the symmetric colour–Lorentz tensor is proportional to

$$f^{abe}f^{cde}(\eta^{\mu\rho}\eta^{\nu\sigma}-\eta^{\mu\sigma}\eta^{\nu\rho})+f^{ace}f^{dbe}(\eta^{\mu\sigma}\eta^{\nu\rho}-\eta^{\mu\nu}\eta^{\rho\sigma})+f^{ade}f^{bce}(\eta^{\mu\nu}\eta^{\rho\sigma}-\eta^{\mu\rho}\eta^{\nu\sigma}), \tag{25.115}$$

with the four legs now labelled $(a\mu), (b\nu), (c\rho), (d\sigma)$. Each of the three terms pairs one of the three ways of splitting four legs into two pairs with the corresponding contraction of structure constants; the metric structure in each is antisymmetric under the swap that the ff prefactor is symmetric under, so the whole is Bose symmetric. Chapter 26 attaches the coupling and a factor $-ig^2$ to (25.115) to obtain the four-gluon Feynman rule; Chapter 59 will reconstruct the same object "the hard way" before replacing it with on-shell methods.

One power of A. The only contribution linear in A comes from $\delta^\partial A$ acting on $\mathcal{L}_2$ (every other combination has at least two A's). With $\delta G^a_{\mu\nu} = \partial_\mu \delta A^a_\nu - \partial_\nu \delta A^a_\mu$ and $\delta^\partial A^a_\mu = \frac{1}{g}\partial_\mu \theta^a$,

$$\delta^\partial \mathcal{L}_2 = -\tfrac{1}{2}\, G^{a\,\mu\nu}\, \delta^\partial G^a_{\mu\nu} = -\tfrac{1}{2}\, G^{a\,\mu\nu}\, \tfrac{1}{g}(\partial_\mu\partial_\nu\theta^a - \partial_\nu\partial_\mu\theta^a) = 0, \tag{25.116}$$

because partial derivatives commute while $G^{a\mu\nu}$ is antisymmetric in $\mu\nu$. This is nothing but the abelian gauge invariance of N^2-1 free photons: it holds independently of any interaction. The free term is invariant under the inhomogeneous shift alone — but, as we now see, *not* under the rotation δ^f.

Two powers of A. Two contributions are quadratic in A: the rotation δ^f acting on $\mathcal{L}_2$, and the inhomogeneous shift δ^∂ acting on $\mathcal{L}_3$. Take the first. With $\delta^f G^a_{\mu\nu} = \partial_\mu(f^{abc}A^b_\nu\theta^c) - \partial_\nu(f^{abc}A^b_\mu\theta^c) = f^{abc}[G^b_{\mu\nu}\theta^c + A^b_\nu\partial_\mu\theta^c - A^b_\mu\partial_\nu\theta^c]$,

$$\delta^f \mathcal{L}_2 = -\tfrac{1}{2} G^{a\,\mu\nu} f^{abc}[G^b_{\mu\nu}\theta^c + A^b_\nu\partial_\mu\theta^c - A^b_\mu\partial_\nu\theta^c]. \tag{25.117}$$

The first bracketed term, $-\frac{1}{2}f^{abc}G^{a\mu\nu}G^b_{\mu\nu}\theta^c$, vanishes identically: f^{abc} is antisymmetric in (a,b) and $G^{a\mu\nu}G^b_{\mu\nu}$ is symmetric. The remaining two terms are equal, by antisymmetry of $G^{a\mu\nu}$ in $\mu\nu$ followed by the relabelling $\mu \leftrightarrow \nu$, so

$$\delta^f \mathcal{L}_2 = -\, f^{abc}\, G^{a\,\mu\nu} A^b_\nu\, \partial_\mu\theta^c = -\, f^{abc}\, G^a_{\mu\nu}\, A^{b\,\nu}\, \partial^\mu\theta^c. \tag{25.118}$$

Now the second contribution, $\delta^{\partial}\mathcal{L}_3$, using the G-form $\mathcal{L}_3 = -\frac{1}{2}g f^{abc} G^a_{\mu\nu} A^{b\mu} A^{c\nu}$ from (25.106) and noting $\delta^{\partial} G^a_{\mu\nu} = 0$ exactly as in (25.116):

$$\begin{aligned}\delta^{\partial}\mathcal{L}_3 &= -\tfrac{1}{2}g f^{abc}\, G^a_{\mu\nu}\big[(\delta^{\partial}A^{b\mu})A^{c\nu} + A^{b\mu}(\delta^{\partial}A^{c\nu})\big] \\ &= -\tfrac{1}{2} f^{abc}\, G^a_{\mu\nu}\big[(\partial^{\mu}\theta^{b})A^{c\nu} + A^{b\mu}(\partial^{\nu}\theta^{c})\big]. \end{aligned} \tag{25.119}$$

The two terms in the bracket are again equal: in the first, relabel $b \leftrightarrow c$ and use $f^{acb} = -f^{abc}$; in the second, use the antisymmetry of $G^a_{\mu\nu}$ under $\mu \leftrightarrow \nu$. Both reduce to $+\frac{1}{2} f^{abc} G^a_{\mu\nu} A^{b\nu}\partial^{\mu}\theta^{c}$, so

$$\delta^{\partial}\mathcal{L}_3 = +\, f^{abc}\, G^a_{\mu\nu}\, A^{b\,\nu}\, \partial^{\,\mu}\theta^{c}. \tag{25.120}$$

Comparing (25.118) with (25.120), the two are equal and opposite:

$$\delta^{f}\mathcal{L}_2 + \delta^{\partial}\mathcal{L}_3 = 0. \tag{25.121}$$

This is the heart of the matter. The free Maxwell-like term is *not* invariant under the non-abelian rotation of the gauge field; its failure, (25.118), is non-zero and proportional to f^{abc}. The only thing that cancels it is the inhomogeneous variation of the cubic term, (25.120). The cubic self-interaction is therefore *required* — and required with exactly the coefficient $-g$ that (25.107) carries, since any other coefficient would leave (25.121) unbalanced. Self-interaction is not added to Yang–Mills theory; it is the price of gauge invariance.

Three powers of A. The same mechanism repeats one level up. The cubic order of variation, $\delta^{f}\mathcal{L}_3$ (the rotation acting on the three-gluon term, order gA^3), fails to vanish on its own, and is cancelled by $\delta^{\partial}\mathcal{L}_4$ (the inhomogeneous shift acting on the four-gluon term, order $g^2A^3/g = gA^3$). Computing $\delta^{f}\mathcal{L}_3$ generates, besides terms that cancel among themselves by the G-antisymmetry used above, a leftover of the form $g\, f^{abc} f^{cde}(\dots)AAA\,\theta$; computing $\delta^{\partial}\mathcal{L}_4$ generates a leftover $g\, f^{abe} f^{cde}(\dots)AAA\,\partial\theta \to g\, f^{abe} f^{cde}(\dots)AAA\,\theta$ after the same relabellings. Their sum vanishes by the identical algebra that governs the top level, to which we now turn; it is again the requirement that the cubic term's rotation be absorbed, and it forces the quartic term (25.110) into existence with coefficient $-\frac{1}{4}g^2$.

Four powers of A. The top of the tower is the only level with no neighbour above it: the rotation δ^{f} acting on the quartic term $\mathcal{L}_4$ must vanish *by itself*, because there is no quintic term to cancel it. This is where the Jacobi identity enters. Carrying out the variation, the $A^4\theta$ pieces assemble into the colour structure

$$f^{abq} f^{bps} + f^{apb} f^{bqs} \;\overset{!}{=}\; f^{abs} f^{bpq}, \tag{25.122}$$

(free adjoint indices a, p, q, s; the summed index is b), with the left side coming from the two ways δ^{f} acts on the explicit fields of $\mathcal{L}_4$ and the right side being the required total. To see that (25.122) is the Jacobi identity, apply $f^{xyz} = -f^{xzy}$ to

standardize and then Eq. (2.77). Most transparently, specialize momentarily to SU(2), where $f^{abc} = \epsilon^{abc}$ and the contraction identity $\epsilon^{xab}\epsilon^{xcd} = \delta^{ac}\delta^{bd} - \delta^{ad}\delta^{bc}$ holds: the left side of (25.122) evaluates to $-\delta^{ap}\delta^{qs} + \delta^{aq}\delta^{ps}$ and the right side to the same, so the identity holds; for general SU(N) it is Eq. (2.77) after relabelling. The quartic term is thus gauge invariant under the homogeneous rotation precisely because the structure constants obey Jacobi — the same Jacobi identity that made the adjoint representation a representation in the first place (Theorem 2.43). Were f^{abc} not the structure constants of a genuine Lie algebra, no four-gluon coupling could close the tower, and gauge invariance would be impossible.

Conclusion. Assembling (25.116), (25.121), the three-A cancellation, and the four-A Jacobi closure, every power of A in $\delta\mathcal{L}$ vanishes, so $\delta\mathcal{L} = 0$ — in agreement with the one-line covariant argument (25.112), which is simply the resummation of these four cancellations. The content the resummation hides is now exposed: gauge invariance is a chain of cancellations running up the tower $\mathcal{L}_2 \to \mathcal{L}_3 \to \mathcal{L}_4$, each link forcing the next interaction into existence with a fixed coefficient. One cannot keep the free term and drop the cubic; one cannot keep the cubic and drop the quartic. The non-abelian gauge field's self-interaction is rigid, complete, and forced.

These three terms — the free quadratic piece, the cubic three-gluon coupling, and the quartic four-gluon coupling — are the complete classical content of the pure Yang–Mills Lagrangian's interactions, and they are the raw material that Chapter 26 will turn into Feynman rules: gauge fixing supplies the gluon propagator, the Faddeev–Popov procedure supplies the ghost sector, and the vertices of Fig. 25.3, dressed with $-ig$ and $-ig^2$ and the symmetrized colour–Lorentz tensors recorded in (25.113) and (25.115), become the QCD interaction rules. Before quantizing, however, we extract one more piece of classical structure that the self-interacting field makes possible and that the abelian theory could only hint at: the gauge-invariant observables built from the parallel transport of colour around closed paths, the Wilson lines and Wilson loops, to which we turn next.

25.6 Wilson Lines, Wilson Loops, and Gauge-Invariant Observables

The non-abelian field strength $F_{\mu\nu} = F^a_{\mu\nu}T^a$ built in Theorem 25.11 is the curvature of the connection $A_\mu = A^a_\mu T^a$, and it is the natural-looking candidate for "the physical content" of the theory. But unlike its abelian ancestor it is *not gauge invariant*: under a local gauge rotation it transforms covariantly, by conjugation, $F_{\mu\nu} \to U F_{\mu\nu} U^\dagger$ (Eq. (25.57)). A conjugated matrix is not an observable; its individual components $F^a_{\mu\nu}$ are reshuffled by every gauge transformation, and

so are the matrix elements of A_μ themselves. The Lagrangian $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ is invariant only because the trace $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$ undoes the conjugation by cyclicity. This raises the question that organizes the present section: *what are the gauge-invariant observables of a non-abelian gauge theory, and how do we build them out of the connection?*

The answer was already foreshadowed, in the abelian case, by the holonomy of Section 8.7: the genuinely physical, gauge-invariant data of a connection is not the value of A_μ at a point but the parallel transport it generates around closed loops. There we found that the abelian Wilson loop $\exp(iq\oint A)$ is invariant and captures information the field strength cannot (the Aharonov–Bohm phase, charge quantization). Here we promote that construction to the non-abelian setting. The new ingredient is forced by a single fact: the comparators that effect parallel transport are now *matrices that do not commute*, so the order in which they are composed along a path matters. The exponential of Theorem 8.29 becomes a *path-ordered* exponential; tracing it over a closed loop produces the central gauge-invariant observable of the theory, the Wilson loop $W(\mathcal{C})$, the object whose large-distance behaviour will serve, in Yang–Mills theory, as the order parameter for confinement. Throughout we keep the frozen conventions of this chapter: Hermitian generators with $\mathrm{tr}(T^aT^b)=\frac{1}{2}\delta^{ab}$ and $[T^a,T^b]=if^{abc}T^c$ (Chapter B), and the covariant derivative $D_\mu=\partial_\mu-igA^a_\mu T^a$ with $A_\mu\equiv A^a_\mu T^a$.

25.6.1 The non-abelian comparator and parallel transport

Recall the logic of the comparator from Section 8.7. To compare a charged field at two different points we cannot simply subtract $\psi(y)-\psi(x)$, because the two values live in copies of the internal vector space related by a gauge-dependent rotation; we must first transport $\psi(x)$ to y with an object $W(y,x)$ that carries the gauge freedom of x to that of y. In the abelian case $W(y,x)$ was a phase; now ψ sits in a representation of the gauge group (take the fundamental of $\mathrm{SU}(N)$ for definiteness, so ψ is an N-component column and $A_\mu=A^a_\mu T^a$ with T^a the fundamental generators), and the comparator is a unitary matrix $W(y,x)\in\mathrm{SU}(N)$.

The infinitesimal comparator is fixed by the requirement that the transported field $W(x+\varepsilon n,x)\,\psi(x)$ rotate with the gauge convention of the destination point, exactly as the genuine field $\psi(x+\varepsilon n)$ does. The covariant derivative $D_\mu=\partial_\mu-igA_\mu$ already encodes the infinitesimal failure of ψ to be parallel, so the comparator that compensates it over a step εn^μ is

$$W(x+\varepsilon n,\,x)\;=\;\mathbf{1}\;+\;ig\,\varepsilon\,n^\mu A_\mu(x)\;+\;O(\varepsilon^2), \tag{25.123}$$

the matrix generalization of the abelian comparator $1+i\varepsilon q\,n^\mu A_\mu$ of Eq. (8.64). The sign and the factor g are inherited from $D_\mu=\partial_\mu-igA_\mu$: we verify in Section 25.6.3 that Eq. (25.123) carries precisely the right transformation law to compensate the gauge rotation of ψ.

To build the comparator for a *finite* oriented path C from x_i to x_f, parametrized by $\xi^\mu(s)$ with $s \in [0,1]$, $\xi(0) = x_i$, $\xi(1) = x_f$, we compose infinitesimal comparators along it. Discretize $[0,1]$ into N steps of size $\Delta s = 1/N$ at parameters $s_k = k/N$, and write the link from $\xi(s_{k-1})$ to $\xi(s_k)$ using Eq. (25.123) with $\varepsilon n^\mu = \dot\xi^\mu(s_k)\Delta s$:

$$W_C \;=\; \lim_{N\to\infty} \prod_{k=N}^{1} \Big[\, \mathbf{1} + ig\, \dot\xi^\mu(s_k)\, A_\mu(\xi(s_k))\, \Delta s \,\Big], \tag{25.124}$$

where the product is *path-ordered*: the factor for the latest point s_N stands furthest to the left, the earliest s_1 furthest to the right, so that the comparator acts on a column vector at x_i by transporting it step by step toward x_f. In the abelian theory of Theorem 8.29 the factors were commuting phases and the ordering was immaterial; the product collapsed to the exponential of the sum of exponents. Here the matrices $A_\mu(\xi(s_k))$ at different points need not commute — they are generic elements of $\mathfrak{su}(N)$ evaluated at different arguments — and the ordering cannot be dropped. This is the entire novelty of the non-abelian Wilson line, and it is forced, not chosen.

The clean way to encode the ordered product Eq. (25.124) is as the solution of a differential equation. Let $W(s) \equiv W_{\xi(s),x_i}$ be the comparator transporting from x_i to the running point $\xi(s)$. Appending the link from $\xi(s)$ to $\xi(s+\Delta s)$ on the left,

$$W(s+\Delta s) \;=\; \big[\mathbf{1} + ig\, \dot\xi^\mu(s)\, A_\mu(\xi(s))\, \Delta s\big]\, W(s) + O(\Delta s^2), \tag{25.125}$$

so in the limit $\Delta s \to 0$ the comparator obeys the *parallel-transport equation*

$$\frac{d}{ds} W(s) \;=\; ig\, \dot\xi^\mu(s)\, A_\mu(\xi(s))\, W(s), \qquad W(0) = \mathbf{1}. \tag{25.126}$$

The Wilson line is the value of the unique solution at $s = 1$. We must now (i) display that solution as the path-ordered exponential and prove uniqueness, and (ii) see precisely why the ordering cannot be removed.

25.6.2 The path-ordered exponential and why ordering is forced

Write the matrix-valued source on the right of Eq. (25.126) as $M(s) \equiv ig\, \dot\xi^\mu(s) A_\mu(\xi(s))$, a continuous function of s valued in $\mathfrak{su}(N)$ (anti-Hermitian times i structure aside, it is a fixed continuous matrix family). The initial-value problem $W'(s) = M(s)W(s)$, $W(0) = \mathbf{1}$ is a linear matrix ODE with continuous coefficients, and its solution is constructed by Picard iteration. Integrating and iterating,

$$W(s) = \mathbf{1} + \int_0^s ds_1\, M(s_1)\, W(s_1)$$

$$= \mathbf{1} + \int_0^s ds_1\, M(s_1) + \int_0^s ds_1 \int_0^{s_1} ds_2\, M(s_1)M(s_2) + \cdots$$
$$= \sum_{n=0}^{\infty} \int_0^s ds_1 \int_0^{s_1} ds_2 \cdots \int_0^{s_{n-1}} ds_n\, M(s_1)M(s_2)\cdots M(s_n), \qquad (25.127)$$

the Dyson (Volterra) series, in which the integration region is the simplex $s \geq s_1 \geq s_2 \geq \cdots \geq s_n \geq 0$ and the matrices stand in decreasing order of their arguments from left to right — later parameters to the left. This nested-simplex form is exactly what the *path-ordering symbol* $\mathcal{P}$ abbreviates: define

$$\mathcal{P}[M(s_1)M(s_2)] = \begin{cases} M(s_1)M(s_2), & s_1 \geq s_2, \\ M(s_2)M(s_1), & s_2 > s_1, \end{cases} \qquad (25.128)$$

and analogously for n factors, reordering them so that the parameters decrease from left to right. Because the symmetric integral over the full hypercube $[0,s]^n$ of the path-ordered integrand is $n!$ copies of the ordered-simplex integral (the $n!$ orderings of $s_1,\dots,s_n$ each contribute the same after $\mathcal{P}$ has sorted them), we may rewrite Eq. (25.127) compactly as the path-ordered exponential

$$W(s) = \sum_{n=0}^{\infty} \frac{1}{n!}\, \mathcal{P}\left[\left(\int_0^s ds'\, M(s')\right)^n\right] \equiv \mathcal{P}\exp\!\left(\int_0^s M(s')\, ds'\right). \qquad (25.129)$$

Setting $s = 1$ and restoring $M = ig\,\dot{\xi}^\mu A_\mu$ gives the central definition.

Definition 25.22 (Non-abelian Wilson line)**.** For a representation R of the gauge group, with $A_\mu = A_\mu^a T_R^a$, and an oriented path C from x_i to x_f, the *Wilson line* is the path-ordered exponential

$$W_C = \mathcal{P}\exp\!\left(ig\int_C A_\mu\, dx^\mu\right) = \mathcal{P}\exp\!\left(ig\int_0^1 A_\mu(\xi(s))\,\dot{\xi}^\mu(s)\, ds\right), \qquad (25.130)$$

the unique solution at $s = 1$ of the parallel-transport equation Eq. (25.126) with $W(0) = \mathbf{1}$. Unless stated otherwise R is the fundamental of $\mathrm{SU}(N)$, so W_C is an $N \times N$ unitary matrix.

Lemma 25.23 (Existence, uniqueness, and unitarity)**.** *The series Eq. (25.129) converges absolutely and uniformly on* $[0,1]$*, solves Eq. (25.126), and is the only solution. If the* A_μ *are Hermitian (so each* $M(s)$ *is anti-Hermitian), then* $W(s)$ *is unitary.*

Proof. Convergence. Let $\mathcal{M} = \max_{s\in[0,1]} \|M(s)\|$ in any submultiplicative matrix norm; $\mathcal{M} < \infty$ by continuity on the compact interval. The nth term of Eq. (25.127) is bounded in norm by $\mathcal{M}^n \,\mathrm{vol}(\text{simplex}) = \mathcal{M}^n s^n/n!$, so the series is dominated termwise by $\sum_n (\mathcal{M}s)^n/n! = e^{\mathcal{M}s}$, giving absolute and uniform convergence on $[0,1]$.

It solves the ODE. Differentiating Eq. (25.127) term by term (legitimate by uniform convergence) returns $W'(s) = M(s)W(s)$; the $n = 0$ term gives $W(0) = \mathbf{1}$.

Uniqueness. Suppose W_1, W_2 both solve the initial-value problem and set $\Delta = W_1 - W_2$. Then $\Delta' = M\Delta$, $\Delta(0) = 0$, so $\|\Delta(s)\| \le \int_0^s \mathcal{M}\, \|\Delta(s')\|\, ds'$. Grönwall's inequality forces $\|\Delta(s)\| \le \|\Delta(0)\|\, e^{\mathcal{M}s} = 0$; hence $W_1 = W_2$.

Unitarity. With A_μ Hermitian, $M^\dagger = -M$. Then $\frac{d}{ds}(W^\dagger W) = W^\dagger M^\dagger W + W^\dagger M W = W^\dagger(M^\dagger + M)W = 0$, and $W^\dagger W = \mathbf{1}$ at $s = 0$, so $W^\dagger W = \mathbf{1}$ throughout. (For the fundamental of $\mathrm{SU}(N)$ one also has $\det W = 1$ since $\operatorname{tr} M = ig\,\dot\xi^\mu \operatorname{tr} A_\mu = 0$ and $\frac{d}{ds}\ln\det W = \operatorname{tr}(W^{-1}W') = \operatorname{tr} M = 0$.) □

It is worth seeing explicitly why the naive exponential $\exp(ig\int_0^s A_\mu\, dx^\mu)$ — with no ordering — *fails* to solve Eq. (25.126) the moment the connection fails to commute with itself at different points. Let $X(s) = \int_0^s M(s')\, ds'$. The derivative of a matrix exponential of a non-constant $X(s)$ is given by Duhamel's formula,

$$\frac{d}{ds}\, e^{X(s)} \;=\; \int_0^1 e^{uX(s)}\, X'(s)\, e^{(1-u)X(s)}\, du, \tag{25.131}$$

which equals $X'(s)\, e^{X(s)} = M(s)e^{X(s)}$ *only* when $X'(s) = M(s)$ commutes with $X(s)$, i.e. when $[M(s), M(s')] = 0$ for all s, s' — precisely the abelian situation. In the non-abelian case the integrand in Eq. (25.131) cannot be pulled through e^{uX}, the naive exponential does *not* satisfy the transport equation, and only the ordered prescription Eq. (25.129) does. The path ordering is therefore not a notational nicety but the unique consistent solution of the geometry; dropping it silently re-abelianizes the theory. The abelian collapse of Theorem 8.29 is the special case $[A_\mu(x), A_\nu(y)] = 0$.

Remark 25.24 (Reverse orientation and composition). Two properties follow at once from Eq. (25.126) and are used repeatedly below. First, traversing C in the opposite orientation inverts the comparator: $W_{C^{-1}} = W_C^{-1} = W_C^\dagger$ (the reversed path solves the transport equation with $M \to -M$ read backwards, whose solution is the inverse). Second, composing two paths, C_2 from y to z after C_1 from x to y, multiplies the comparators with the later segment on the left, $W_{C_2 \circ C_1} = W_{C_2}\, W_{C_1}$, directly from the ordered product Eq. (25.124). These are the group-like properties of parallel transport.

25.6.3 The transformation law and gauge invariance by tracing

We now establish the transformation law of the Wilson line — the property that makes the traced closed loop an observable. Recall the gauge transformation of the matter field and connection (from Eqs. (25.17) and (25.22)): under a local

gauge rotation $U(x) \in \mathrm{SU}(N)$,

$$\psi(x) \;\to\; U(x)\,\psi(x), \qquad A_\mu(x) \;\to\; U(x)\,A_\mu(x)\,U^\dagger(x) \;-\; \frac{i}{g}\,(\partial_\mu U(x))\,U^\dagger(x). \tag{25.132}$$

The inhomogeneous second term is what distinguishes a connection from a tensor; it is also exactly what makes the comparator transform homogeneously at its endpoints. We first confirm the infinitesimal comparator law, then telescope it over the whole path.

Comparator endpoint law. Apply Eq. (25.132) to the infinitesimal comparator Eq. (25.123), with the connection evaluated at x and the step εn^μ:

$$\begin{aligned} W(x+\varepsilon n, x) &\to \mathbf{1} + ig\,\varepsilon n^\mu \Big[U(x)A_\mu(x)U^\dagger(x) - \tfrac{i}{g}(\partial_\mu U)U^\dagger\Big] \\ &= \mathbf{1} + ig\,\varepsilon n^\mu\, U A_\mu U^\dagger + \varepsilon n^\mu (\partial_\mu U) U^\dagger. \end{aligned} \tag{25.133}$$

On the other hand, the claimed law is $W \to U(x+\varepsilon n)\,W\,U^\dagger(x)$. Expanding $U(x+\varepsilon n) = U(x) + \varepsilon n^\mu \partial_\mu U(x) + O(\varepsilon^2)$,

$$\begin{aligned} U(x+\varepsilon n)\big[\mathbf{1} + ig\varepsilon n^\mu A_\mu\big]U^\dagger(x) &= \big[U + \varepsilon n^\mu \partial_\mu U\big]\big[\mathbf{1} + ig\varepsilon n^\nu A_\nu\big]U^\dagger \\ &= UU^\dagger + ig\varepsilon n^\nu\, U A_\nu U^\dagger + \varepsilon n^\mu (\partial_\mu U) U^\dagger + O(\varepsilon^2) \\ &= \mathbf{1} + ig\varepsilon n^\mu\, U A_\mu U^\dagger + \varepsilon n^\mu (\partial_\mu U) U^\dagger, \end{aligned} \tag{25.134}$$

which agrees term by term with Eq. (25.133). The inhomogeneous gradient in A_μ is precisely the piece needed to manufacture the derivative of U at the endpoint; the comparator transforms by conjugation at its two ends,

$$W(y,x) \;\longrightarrow\; U(y)\,W(y,x)\,U^\dagger(x). \tag{25.135}$$

Telescoping over the path. The finite Wilson line is the ordered product of links Eq. (25.124). Apply Eq. (25.135) to every link, with $x_k = \xi(s_k)$:

$$\begin{aligned} W_C \;&\to\; \prod_{k=N}^{1} U(x_k)\,\big[\mathbf{1} + ig\,\Delta\xi_k^\mu A_\mu(x_k)\big]\,U^\dagger(x_{k-1}) \\ &=\; U(x_N)\,\big[\cdots\big]_N\, \underbrace{U^\dagger(x_{N-1})\,U(x_{N-1})}_{\mathbf{1}}\,\big[\cdots\big]_{N-1}\, U^\dagger(x_{N-2})\cdots U(x_1)\,\big[\cdots\big]_1\,U^\dagger(x_0). \end{aligned} \tag{25.136}$$

Every *interior* pair $U^\dagger(x_k)U(x_k) = \mathbf{1}$ collapses; only the two endpoint factors $U(x_N) = U(x_f)$ on the far left and $U^\dagger(x_0) = U^\dagger(x_i)$ on the far right survive. Taking $N \to \infty$ leaves the transformation law of the Wilson line.

Proposition 25.25 (Wilson-line transformation law)**.** *Under a local gauge transformation $U(x)$, the open Wilson line transforms by conjugation at its endpoints:*

$$\boxed{W_C \;\longrightarrow\; U(x_f)\, W_C\, U^\dagger(x_i)} \tag{25.137}$$

for a path C from x_i to x_f.

Proof. The telescoping computation Eq. (25.136) gives it directly. The same result follows from the differential equation, which is the cleaner argument and confirms the discrete one: under the gauge transformation, $W_C' := U(x_f) W_C U^\dagger(x_i)$ and the solution $\widetilde{W}$ of the transport equation built from the transformed connection $\widetilde{A}_\mu = U A_\mu U^\dagger - \frac{i}{g}(\partial_\mu U) U^\dagger$ satisfy the same initial-value problem. Indeed, writing $V(s) \equiv U(\xi(s)) W(s) U^\dagger(x_i)$ and differentiating with $W' = igA_\mu \dot\xi^\mu W$,

$$\begin{aligned} \frac{dV}{ds} &= \dot U\, W\, U^\dagger(x_i) + U\,(igA_\mu \dot\xi^\mu) W\, U^\dagger(x_i) \\ &= \Big[(\partial_\mu U)\dot\xi^\mu\, U^\dagger + ig\, U A_\mu U^\dagger \dot\xi^\mu\Big]\, U\, W\, U^\dagger(x_i) \\ &= ig\,\dot\xi^\mu \Big[U A_\mu U^\dagger - \tfrac{i}{g}(\partial_\mu U) U^\dagger\Big]\, V \;=\; ig\,\dot\xi^\mu\, \widetilde{A}_\mu\, V, \end{aligned} \tag{25.138}$$

where we inserted $U^\dagger U = \mathbf{1}$ between $\dot U$ and W in the second line, and in the third line factored out the common $ig\,\dot\xi^\mu$; the gradient piece checks because $ig \cdot (-\frac{i}{g})(\partial_\mu U) U^\dagger = (\partial_\mu U) U^\dagger$, exactly the coefficient appearing in the second line. Thus $V(s)$ solves the transport equation with the transformed connection and $V(0) = U(x_i) \cdot \mathbf{1} \cdot U^\dagger(x_i) = \mathbf{1}$, the same data as $\widetilde{W}$. By the uniqueness of Theorem 25.23, $\widetilde{W} = V$, i.e. the Wilson line built from the gauge-transformed connection equals $U(x_f) W_C U^\dagger(x_i)$. □

The transformed Wilson line carries the gauge rotation of the *endpoints* and nothing of the interior, exactly the comparator behaviour we demanded. As a check on its purpose, the transported field rotates correctly: $W_C\,\psi(x_i) \to U(x_f) W_C U^\dagger(x_i) \cdot U(x_i)\psi(x_i) = U(x_f)\,[W_C \psi(x_i)]$, so it acquires the gauge convention of the destination x_f, just like $\psi(x_f)$. The Wilson line is the device that makes $\bar\psi(x_f)\, W_C\, \psi(x_i)$ gauge invariant — the prototype of a gauge-invariant bilocal, the seed of the meson and the quark–antiquark string.

Corollary 25.26 (Gauge invariance of the traced loop)**.** *If C is closed, $x_f = x_i \equiv x$, then $W_C \to U(x)\, W_C\, U^\dagger(x)$, and the trace $\operatorname{tr} W_C$ is gauge invariant:* $\operatorname{tr} W_C \to \operatorname{tr}\!\big(U(x) W_C U^\dagger(x)\big) = \operatorname{tr}\!\big(U^\dagger(x) U(x) W_C\big) = \operatorname{tr} W_C$*, using cyclicity and $U^\dagger U = \mathbf{1}$.*

Two features of Theorem 25.26 deserve emphasis. First, the trace is *indispensable*: for a closed loop the untraced holonomy W_C is still gauge-*covariant*, conjugated by $U(x)$ at the basepoint, and only the trace removes the residual

conjugation. (More generally any conjugation-invariant function of the holonomy — any *character* of the loop — is gauge invariant; the fundamental trace is the simplest.) Second, the construction is intrinsically non-abelian in flavour: in the U(1) case W_C was already a number and the trace was vacuous, which is why Theorem 8.30 needed no trace. The trace is the non-abelian price for, and the non-abelian route to, gauge invariance.

25.6.4 The Wilson loop

We can now name the central observable.

Theorem 25.27 (The Wilson loop)**.** *Let $\mathcal{C}$ be a closed oriented loop and R a representation of the gauge group. The* Wilson loop

$$W(\mathcal{C}) \;=\; \mathrm{tr}_R\, \mathcal{P}\exp\!\Big(ig \oint_{\mathcal{C}} A_\mu\, dx^\mu\Big) \tag{25.139}$$

is gauge invariant. For R the fundamental of $\mathrm{SU}(N)$ it is the trace of an $\mathrm{SU}(N)$ matrix; it is independent of the basepoint chosen on $\mathcal{C}$ and, under reversal of orientation, obeys $W(\mathcal{C}^{-1}) = \overline{W(\mathcal{C})}$.

Proof. Gauge invariance is Theorem 25.26. Basepoint independence: moving the basepoint from x to x' along $\mathcal{C}$ replaces $W_{\mathcal{C}}$ by $W_2W_1 \to W_1W_2$ where W_1 transports $x \to x'$ and W_2 transports $x' \to x$ (the two arcs of the loop); by cyclicity $\mathrm{tr}(W_2W_1) = \mathrm{tr}(W_1W_2)$, so the trace is unchanged. Orientation reversal sends $W_{\mathcal{C}} \to W_{\mathcal{C}}^\dagger$ (Theorem 25.24), and $\mathrm{tr}(W^\dagger) = \overline{\mathrm{tr}\, W}$ since the trace of the Hermitian conjugate is the complex conjugate of the trace. □

The Wilson loop is the gauge-invariant content of the connection attached to the loop $\mathcal{C}$, the non-abelian holonomy "read off" by the trace. It is the object that, in the quantum theory, has a gauge-invariant expectation value $\langle W(\mathcal{C})\rangle$ computable on the lattice (Chapter 43) and whose dependence on the size and shape of $\mathcal{C}$ encodes the long-distance dynamics of the gauge field. We turn next to its short-distance content, where it reconstructs the curvature, and then state its long-distance role.

25.6.5 Small loops: the curvature recovered

The short-distance limit of the Wilson loop ties it back to $F_{\mu\nu}$, completing the circle "connection → parallel transport → curvature" that began with the covariant derivative. Take an infinitesimal closed loop and expand. The cleanest case is a parallelogram (a plaquette) with edge vectors a^μ and b^ν, both of order ε, with corners

$$x \xrightarrow{+a} x+a \xrightarrow{+b} x+a+b \xrightarrow{-a} x+b \xrightarrow{-b} x, \tag{25.140}$$

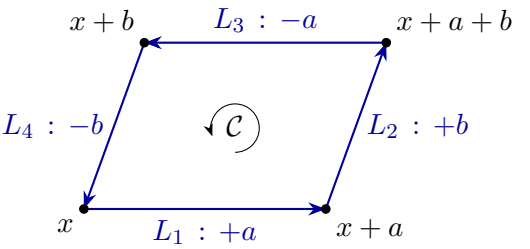


Figure 25.4: The infinitesimal plaquette of Eq. (25.140), spanned by the edge vectors a^μ and b^ν. The holonomy around it is the path-ordered product of the four edge comparators (last edge on the left). Its deviation from the identity is the curvature, $W(\mathcal{C}) = \mathbf{1} + ig\, a^\mu b^\nu F_{\mu\nu}(x) + O(\varepsilon^3)$.

traversed counterclockwise (Fig. 25.4). The holonomy is the ordered product of the four edge comparators, with the last edge on the left,

$$W(\mathcal{C}) \;=\; L_4\, L_3\, L_2\, L_1, \tag{25.141}$$

where L_1 is the comparator from x to $x+a$, and so on around the loop.

Each edge comparator is the Wilson line of a straight segment, expanded to second order in its length. For a straight step of displacement ϵ^μ starting at z, the path-ordered exponential Eq. (25.129) gives, using $\int_0^1 \epsilon^\mu A_\mu(z+t\epsilon)\,dt = \epsilon^\mu A_\mu(z) + \tfrac{1}{2}\epsilon^\mu\epsilon^\nu \partial_\nu A_\mu(z) + \cdots$ and the simplex volume $\int_0^1 dt_2 \int_0^{t_2} dt_1 = \tfrac{1}{2}$ for the quadratic term,

$$L(z,\epsilon) \;=\; \mathbf{1} + ig\,\epsilon^\mu A_\mu(z) + \tfrac{ig}{2}\,\epsilon^\mu\epsilon^\nu \partial_\nu A_\mu(z) - \tfrac{g^2}{2}\,\epsilon^\mu\epsilon^\nu A_\mu(z) A_\nu(z) + O(\epsilon^3). \tag{25.142}$$

Apply this to each edge, expanding every field about the corner x and writing $A_\mu \equiv A_\mu(x)$, $\partial_\rho A_\mu \equiv \partial_\rho A_\mu(x)$. The starting points are x, $x+a$, $x+a+b$, $x+b$ and the displacements are $+a, +b, -a, -b$:

$$L_1 = \mathbf{1} + ig\, a^\mu A_\mu + \tfrac{ig}{2} a^\mu a^\nu \partial_\nu A_\mu - \tfrac{g^2}{2} a^\mu a^\nu A_\mu A_\nu, \tag{25.143}$$

$$L_2 = \mathbf{1} + ig\, b^\mu A_\mu + ig\, b^\mu a^\rho \partial_\rho A_\mu + \tfrac{ig}{2} b^\mu b^\nu \partial_\nu A_\mu - \tfrac{g^2}{2} b^\mu b^\nu A_\mu A_\nu, \tag{25.144}$$

$$L_3 = \mathbf{1} - ig\, a^\mu A_\mu - ig\, a^\mu (a+b)^\rho \partial_\rho A_\mu + \tfrac{ig}{2} a^\mu a^\nu \partial_\nu A_\mu - \tfrac{g^2}{2} a^\mu a^\nu A_\mu A_\nu, \tag{25.145}$$

$$L_4 = \mathbf{1} - ig\, b^\mu A_\mu - ig\, b^\mu b^\rho \partial_\rho A_\mu + \tfrac{ig}{2} b^\mu b^\nu \partial_\nu A_\mu - \tfrac{g^2}{2} b^\mu b^\nu A_\mu A_\nu, \tag{25.146}$$

where in L_2, L_3, L_4 the shift of the base point has been Taylor-expanded, $A_\mu(x+a) = A_\mu + a^\rho \partial_\rho A_\mu + \cdots$, and only terms through second order in ε are kept. We assemble $W(\mathcal{C}) = L_4 L_3 L_2 L_1$ order by order. Introduce the shorthand $\not{a} \equiv a^\mu A_\mu$, $\not{b} \equiv b^\mu A_\mu$.

First order. The linear pieces sum to $ig(\not{a} + \not{b} - \not{a} - \not{b}) = 0$: the loop closes, so the comparator is the identity to first order, as it must be.

Second order, the derivative terms. Collecting all $ig\,\partial A$ contributions, the pure a^2 pieces ($\frac{ig}{2}a^\mu a^\nu \partial_\nu A_\mu$ from L_1 and L_3, total $ig\,a^\mu a^\nu \partial_\nu A_\mu$, against $-ig\,a^\mu a^\rho \partial_\rho A_\mu$ from the base-point shift in L_3) cancel, and likewise the pure b^2 pieces from L_2, L_4 cancel. The surviving cross terms come from L_2 ($ig\,b^\mu a^\rho \partial_\rho A_\mu$) and L_3 ($-ig\,a^\mu b^\rho \partial_\rho A_\mu$); relabelling indices,

$$ig\,a^\nu b^\mu \partial_\nu A_\mu - ig\,a^\mu b^\nu \partial_\nu A_\mu \;=\; ig\,a^\mu b^\nu (\partial_\mu A_\nu - \partial_\nu A_\mu). \tag{25.147}$$

Second order, the quadratic-in-A terms. Two sources contribute. The $-\frac{g^2}{2}AA$ pieces inside the individual L_i sum to $-g^2(\not{a}^2+\not{b}^2)$. The products of two first-order pieces across different edges, ordered as in $L_4L_3L_2L_1$, give

$$\begin{aligned}\sum_{i>j} \ell_i^{(1)}\ell_j^{(1)} &= (-ig\not{b})(-ig\not{a}) + (-ig\not{b})(ig\not{b}) + (-ig\not{b})(ig\not{a}) \\ &\quad + (-ig\not{a})(ig\not{b}) + (-ig\not{a})(ig\not{a}) + (ig\not{b})(ig\not{a}) \\ &= g^2(\not{a}^2+\not{b}^2) + g^2\,[\not{a},\not{b}],\end{aligned} \tag{25.148}$$

where the three $\not{b}\not{a}$ contributions ($-g^2\not{b}\not{a}$, $+g^2\not{b}\not{a}$, $-g^2\not{b}\not{a}$) combine to $-g^2\not{b}\not{a}$, and together with $+g^2\not{a}\not{b}$ give the commutator. The terms $g^2(\not{a}^2+\not{b}^2)$ from Eq. (25.148) cancel exactly against the $-g^2(\not{a}^2+\not{b}^2)$ from the individual edges, leaving only the commutator $g^2[\not{a},\not{b}] = g^2 a^\mu b^\nu [A_\mu, A_\nu]$.

Assembling. Adding the surviving derivative term Eq. (25.147) and the surviving commutator,

$$\begin{aligned}W(\mathcal{C}) &= \mathbf{1} + ig\,a^\mu b^\nu(\partial_\mu A_\nu - \partial_\nu A_\mu) + g^2 a^\mu b^\nu [A_\mu, A_\nu] + O(\varepsilon^3) \\ &= \mathbf{1} + ig\,a^\mu b^\nu\Big(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]\Big) + O(\varepsilon^3) \\ &= \mathbf{1} + ig\,a^\mu b^\nu\, F_{\mu\nu}(x) + O(\varepsilon^3),\end{aligned} \tag{25.149}$$

using $g^2/(ig) = -ig$ to pull out the overall ig and recognizing the matrix field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu]$ of Theorem 25.11. This is the central small-loop result: *the holonomy around an infinitesimal loop is the identity plus ig times the curvature contracted with the area element.* Crucially the full non-abelian curvature, commutator term included, emerges automatically from the path-ordered product — the ordering and the commutator are the same piece of physics. The abelian formula Eq. (8.175), in which the holonomy is $\exp(iq\,\frac{1}{2}F_{\mu\nu}\,dS^{\mu\nu})$ with no commutator, is the $[A,A]=0$ shadow of Eq. (25.149).

Remark 25.28 (Curvature as infinitesimal holonomy; the comparator and D_μ). Equation (25.149) is the integrated form of the statement that the commutator of covariant derivatives is the curvature. Parallel transport around an infinitesimal $\mu\nu$-loop measures the non-commutativity of transport in the two directions, and on a field ψ this is the comparator generalizing Eq. (8.173): the covariant

derivative is the infinitesimal parallel transport, $D_\mu\psi(x) = \lim_{\varepsilon\to 0}\varepsilon^{-1}[\psi(x+\varepsilon\hat{\mu}) - W(x+\varepsilon\hat{\mu},x)\psi(x)]$, and the failure of two such transports to commute is

$$[D_\mu, D_\nu] \;=\; -ig\,F_{\mu\nu}, \tag{25.150}$$

which one verifies directly from $D_\mu = \partial_\mu - igA_\mu$: $[D_\mu,D_\nu]\psi = -ig(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi - g^2[A_\mu,A_\nu]\psi = -igF_{\mu\nu}\psi$. Comparing with Eq. (25.149), the small-loop holonomy is $\mathbf{1} - a^\mu b^\nu [D_\mu,D_\nu]/(\cdot)$ assembled into the curvature: the curvature *is* the infinitesimal holonomy per unit area. This is the precise sense in which the Wilson loop, the curvature, and the commutator of covariant derivatives are one object viewed at three magnifications — the non-abelian completion of the holonomy picture of Section 8.7.

We are now positioned to extract the trace and exhibit the leading shape of the Wilson loop itself.

Worked Example 25.29 (The plaquette: Wilson loop, curvature, and the Yang–Mills action)**.** We compute $W(\mathcal{C}) = \operatorname{tr} W_\mathcal{C}$ for the infinitesimal plaquette to the order at which it first departs from its area-zero value, and identify the result with the Yang–Mills Lagrangian density. Specialize the plaquette to the coordinate plane spanned by two directions, $a^\mu = a\,\delta^\mu_1$ and $b^\nu = b\,\delta^\nu_2$, so its area is $\mathcal{A} = ab$ and only the single component F_{12} of the curvature enters.

Step 1: exponentiate the holonomy. From Eq. (25.149) the holonomy is $W_\mathcal{C} = \mathbf{1} + ig\,\mathcal{A}\,F_{12} + O(\varepsilon^3)$. Because the next correction is $O(\varepsilon^3)$ while the term we have is $O(\varepsilon^2)$, the holonomy exponentiates cleanly to the order we need:

$$W_\mathcal{C} \;=\; \exp\!\big(ig\,\mathcal{A}\,F_{12} + O(\varepsilon^3)\big) \;=\; \mathbf{1} + ig\,\mathcal{A}\,F_{12} - \tfrac{g^2}{2}\,\mathcal{A}^2\,F_{12}^2 + O(\varepsilon^3). \tag{25.151}$$

Step 2: trace, in the fundamental of $\mathrm{SU}(N)$*.* Using $\operatorname{tr}\mathbf{1} = N$ and the tracelessness of the generators, $\operatorname{tr} F_{12} = F^a_{12}\operatorname{tr} T^a = 0$, the linear term drops and

$$W(\mathcal{C}) \;=\; \operatorname{tr} W_\mathcal{C} \;=\; N - \tfrac{g^2}{2}\,\mathcal{A}^2\,\operatorname{tr}\!\big(F_{12}F_{12}\big) + O(\varepsilon^3). \tag{25.152}$$

That the $O(\mathcal{A})$ term vanishes in the trace is the non-abelian counterpart of the fact that the Wilson loop is even under orientation reversal up to conjugation; the leading gauge-invariant signal is quadratic in the area.

Step 3: rewrite as the field-strength square. For a loop confined to the $(1,2)$ plane, the Lorentz scalar $F_{\mu\nu}F^{\mu\nu}$ restricted to that plane is $F_{12}F^{12} + F_{21}F^{21} = 2F_{12}F^{12}$, and with the mostly-minus metric $F^{12} = \eta^{11}\eta^{22}F_{12} = (-1)(-1)F_{12} = F_{12}$, so $\operatorname{tr}(F_{12}F_{12}) = \tfrac{1}{2}\operatorname{tr}(F_{\mu\nu}F^{\mu\nu})\big|_{\text{plane}}$. Substituting,

$$\boxed{\;W(\mathcal{C}) \;=\; N \;-\; \frac{g^2}{4}\,\operatorname{tr}\!\big(F_{\mu\nu}F^{\mu\nu}\big)\,\mathcal{A}^2 \;+\; O(\varepsilon^3)\;} \tag{25.153}$$

where the trace is over the plane of the loop and $\mathcal{A} = ab$ is its area.

Interpretation. The leading departure of the Wilson loop from its trivial value N is set by $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu})$, which is, up to the normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, exactly minus twice the Yang–Mills Lagrangian density of Eq. (25.76): $\mathrm{tr}(F_{\mu\nu}F^{\mu\nu}) = \frac{1}{2}F^a_{\mu\nu}F^{a\,\mu\nu} = -2\mathcal{L}^{(\mathrm{gauge})}_{\mathrm{YM}}$. The Wilson loop has reconstructed the action density from pure parallel transport. This identity is the cornerstone of the lattice formulation: a sum of small-plaquette traces is a discretization of $\int d^4x\,(-\frac{1}{4}F^aF^a)$, and it is the route by which Chapter 43 *defines* the gauge theory non-perturbatively. The same computation, summed over a tiling of a spanning surface, is the leading term of the non-abelian Stokes theorem, to which we turn.

25.6.6 The non-abelian Stokes/holonomy picture

In the abelian theory, Theorem 8.30 expressed the holonomy of any loop as the exponential of the flux of F through a spanning surface, via the ordinary Stokes theorem $\oint_{\partial S} A = \int_S dA$. The non-abelian generalization is more delicate, precisely because of the path ordering: the curvature contributions from different patches of the surface must themselves be ordered, and they are parallel-transported to a common reference point before being combined.

The structure is visible already in Eq. (25.149). Tile a spanning surface S of $\mathcal{C}$ by infinitesimal plaquettes; the holonomy of each is $\mathbf{1} + ig\,F_{\mu\nu}\,d\sigma^{\mu\nu}$ with $d\sigma^{\mu\nu} = a^\mu b^\nu$ the oriented area element of the plaquette. Composing adjacent plaquette holonomies, the shared edges cancel in pairs (their comparators appear once forward and once backward), so the loop holonomy is the path-ordered product of all the plaquette holonomies — but each must first be conjugated by the Wilson line $W_{x,p}$ that transports from the basepoint x of $\mathcal{C}$ to the plaquette location p, since the curvature at p is referred to the gauge frame at p, not at x. In the continuum this assembles into the surface-ordered exponential

$$W_{\mathcal{C}} \;=\; \mathcal{P}_S \exp\!\Big(ig\int_S \widehat{F}_{\mu\nu}\,d\sigma^{\mu\nu}\Big), \qquad \widehat{F}_{\mu\nu}(p) \;\equiv\; W_{x,p}\,F_{\mu\nu}(p)\,W_{x,p}^{-1}, \tag{25.154}$$

the *non-abelian Stokes theorem*: the loop holonomy is the surface-ordered exponential of the curvature *parallel-transported to the basepoint*. The transport $W_{x,p}$ and the ordering $\mathcal{P}_S$ are exactly the new ingredients absent in the abelian case; when $[A,A]=0$ they trivialize, $\widehat{F}\to F$ and $\mathcal{P}_S\to$ ordinary integral, and Eq. (25.154) reduces to the abelian flux formula Eq. (8.175). We do not need the full surface-ordered machinery in what follows — its leading term is the plaquette result Eq. (25.149) we proved in full, and that suffices for both the small-loop expansion and the lattice action — but it is the correct conceptual statement of how curvature integrates up to holonomy in a non-abelian theory, and we record it for the bundle-geometric discussion of Chapter 41.

Remark 25.30 (Representation dependence: probing the charge). The Wilson loop Eq. (25.139) carries a choice of representation R through $A_\mu = A^a_\mu T^a_R$.

Physically, $W_R(\mathcal{C})$ measures the response of the gauge field to a static external source in representation R: a fundamental Wilson loop describes a heavy quark–antiquark pair, an adjoint loop (using the adjoint generators $(T^a_{\text{adj}})_{bc} = -if^{abc}$ of Theorem 2.43) describes a static gluon-charged source. The two need not behave alike: an adjoint source can be *screened* by the gluon field (the dynamical gluons carry adjoint charge and can neutralize it), whereas a fundamental source in a confining vacuum cannot, a distinction whose non-perturbative analysis belongs to Chapter 43. The representation label is therefore a genuine experimental knob on the gauge-invariant observable, not a passive convention.

25.6.7 The Wilson loop as confinement order parameter

The deepest role of the Wilson loop is reserved for the quantum theory, but its logic can be stated cleanly here, at the classical level of definition, with the derivations honestly deferred. Consider the quantum expectation value $\langle W(\mathcal{C})\rangle$ for a large planar rectangular loop of spatial extent L and temporal extent T, with $T \gg L$. Such a loop represents the worldlines of a static quark and antiquark held a distance L apart for a time T; the expectation value is controlled by the energy $V(L)$ of that static pair through $\langle W(\mathcal{C})\rangle \sim e^{-V(L)\,T}$. Two qualitatively different large-loop behaviours are then possible:

$$\langle W(\mathcal{C})\rangle \;\sim\; \begin{cases} \exp\!\big(-\sigma\,\text{Area}(\mathcal{C})\big) = \exp(-\sigma L T), & \textit{area law},\\[2mm] \exp\!\big(-\mu\,\text{Perimeter}(\mathcal{C})\big) = \exp\!\big(-2\mu(L+T)\big), & \textit{perimeter law}. \end{cases} \tag{25.155}$$

An area law, $\langle W\rangle \sim e^{-\sigma LT}$, corresponds via $\langle W\rangle \sim e^{-V(L)T}$ to a linearly rising potential $V(L) = \sigma L$: the static sources are bound by a flux tube of constant tension σ (the string tension), and cannot be separated to infinity at finite energy. This is the criterion for *confinement*. A perimeter law, $\langle W\rangle \sim e^{-2\mu(L+T)}$, corresponds to a potential that flattens to a constant $V(L) \to \text{const}$ at large L — the sources are screened and free at long distance, the deconfined or Coulomb-like behaviour. The Wilson loop thereby furnishes a sharp, gauge-invariant order parameter distinguishing confined from deconfined phases, with no reference to any local field that could fail to be gauge invariant.

This is the payoff toward which Yang–Mills theory is aimed, and it is honest to flag now that nothing in the classical or perturbative theory can establish the area law: it is an intrinsically non-perturbative statement about $\langle W(\mathcal{C})\rangle$ in the strongly coupled vacuum. We will earn it in two stages. Chapter 27 states the Wilson-loop confinement criterion and connects the rising potential to the running coupling and Λ_{QCD}; Chapter 43 derives the area law directly, in the strong-coupling expansion of the lattice gauge theory, where each unit of area costs one plaquette factor and the sum over tilings reproduces $e^{-\sigma\,\text{Area}}$ with a computable tension. The object whose area law signals confinement is exactly the $W(\mathcal{C})$ fixed here, in the normalization of Eq. (25.139).

Physical Insight

The progression of this chapter — gauge principle, connection, curvature, Lagrangian, self-interaction, Wilson loop — ends by inverting its own starting point. We began by demanding local invariance and were forced to introduce the connection A_μ as redundant scaffolding; we end by recognizing that the gauge-invariant *physics* of the connection is not in A_μ at a point, nor even in $F_{\mu\nu}$ at a point (which is only covariant), but in the holonomies $W(\mathcal{C})$ around all loops. The local fields are the convenient but redundant coordinates; the loop observables are the invariant content. In the abelian theory this lesson was a refinement (the Aharonov–Bohm phase); in the non-abelian theory it is the whole game, because $F_{\mu\nu}$ is no longer invariant at all and the trace over a closed holonomy is the simplest invariant one can write. That the same trace, expanded on small loops, returns the action density Eq. (25.153), and on large loops diagnoses confinement Eq. (25.155), is the reason the Wilson loop is the natural variable of non-perturbative gauge theory.

We have built the gauge-invariant observables of the classical non-abelian theory and seen that they live on loops, not points — the connection read through its holonomy. This recognition that A_μ is physical only up to gauge transformations, and that the invariant data is organized by loops, makes precise the redundancy at the heart of the construction. The next section confronts that redundancy head-on, organizing the space of connections into gauge orbits and distinguishing the small gauge transformations (continuously connected to the identity, used here) from the large ones whose topology $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$ will seed both the need for gauge-fixing in Chapter 26 and the multiplicity of vacua in Chapter 42.

25.7 Gauge Orbits and the Global Structure of Configuration Space

The previous section taught us to read A^a_μ and $F^a_{\mu\nu}$ as geometric objects: a connection and its curvature, with the Wilson loop $W[\gamma] = \mathrm{tr}\,\mathcal{P}\exp(ig\oint_\gamma A)$ as the gauge-invariant record of holonomy. Along the way a structural fact kept surfacing. The field strength is covariant, $F_{\mu\nu} \mapsto U F_{\mu\nu} U^{-1}$, not invariant; the potential is worse, transforming inhomogeneously by Eq. (25.34); and the genuinely measurable quantities — the action, the energy, the Wilson loops — are exactly the functionals that do not move when A_μ is gauge-transformed. The potential, in other words, carries more information than physics uses. Two potentials related by a gauge transformation are not two states of the world; they are two coordinate descriptions of one state. This is the same redundancy

Noether's second theorem (Theorem 7.36) diagnosed abstractly from the local symmetry, and that Theorem 8.17 made concrete in the abelian case. Our task in this closing section is to take that redundancy completely seriously: to organize the space of potentials by the gauge equivalence it defines, and then to discover that the resulting quotient has a global structure — a topology — that the local field equations never see. That topology is invisible classically in any single perturbative calculation, yet it controls both the consistency of quantization (Chapter 26) and the existence of inequivalent vacua (Chapter 42). The non-abelian case, where the gauge group is a non-trivial manifold, is where this structure first becomes rich.

25.7.1 The space of connections and its quotient

Fix a time and work on a spatial slice, or equivalently work with field configurations on spacetime; the bookkeeping is identical and we keep the spacetime notation. We collect every smooth $\mathfrak{su}(N)$-valued one-form into one object.

Definition 25.31 (Configuration space and gauge group)**.** The *space of connections* $\mathcal{A}$ is the set of all smooth gauge potentials $A_\mu = A_\mu^a T^a$ valued in the Lie algebra $\mathfrak{su}(N)$ (with the conventions and generators of Chapter B). The *gauge group* $\mathcal{G}$ is the set of all smooth maps $U : \mathbb{R}^{1,3} \to \mathrm{SU}(N)$, with pointwise multiplication, acting on $\mathcal{A}$ by

$$A_\mu \;\longmapsto\; A_\mu^U \;=\; U A_\mu U^{-1} \;-\; \frac{i}{g}\,(\partial_\mu U)\,U^{-1}, \tag{25.156}$$

the finite transformation law established earlier in this chapter, whose infinitesimal form is $\delta A_\mu^a = \frac{1}{g}(D_\mu \alpha)^a$ (Eq. (25.34)).

That Eq. (25.156) is a genuine left action — that $\mathbf{1}$ acts trivially and $(A^{U_1})^{U_2} = A^{U_2 U_1}$ — is the statement that the inhomogeneous term composes correctly; we verify it because it is the algebraic engine of everything that follows. Write $A^U = UAU^{-1} - \frac{i}{g}(\mathrm{d}U)U^{-1}$ in form notation. Then, applying the law twice,

$$\begin{aligned}(A^{U_1})^{U_2} &= U_2\Big(U_1 A U_1^{-1} - \tfrac{i}{g}(\mathrm{d}U_1)U_1^{-1}\Big)U_2^{-1} - \tfrac{i}{g}(\mathrm{d}U_2)U_2^{-1} \\ &= (U_2U_1)A(U_2U_1)^{-1} - \tfrac{i}{g}\,U_2(\mathrm{d}U_1)U_1^{-1}U_2^{-1} - \tfrac{i}{g}(\mathrm{d}U_2)U_2^{-1}. \end{aligned}\tag{25.157}$$

On the other hand the single transformation by U_2U_1 gives, using the Leibniz rule $\mathrm{d}(U_2U_1) = (\mathrm{d}U_2)U_1 + U_2\,\mathrm{d}U_1$,

$$A^{U_2U_1} = (U_2U_1)A(U_2U_1)^{-1} - \tfrac{i}{g}\big[(\mathrm{d}U_2)U_1 + U_2\,\mathrm{d}U_1\big]U_1^{-1}U_2^{-1} = (U_2U_1)A(U_2U_1)^{-1} - \tfrac{i}{g}(\mathrm{d}U_2)U_2^{-1} - \tfrac{i}{g}(\mathrm{d}U_1)U_1^{-1}U_2^{-1}\cdots \tag{25.158}$$

which is term-by-term identical to Eq. (25.157). Hence $(A^{U_1})^{U_2} = A^{U_2U_1}$: $\mathcal{G}$ acts, and the inverse of the transformation by U is the transformation by U^{-1}. The action partitions $\mathcal{A}$ into orbits.

Definition 25.32 (Gauge orbit and physical configuration space)**.** The *gauge orbit* of $A \in \mathcal{A}$ is the set $\mathcal{O}_A = \{ A^U : U \in \mathcal{G} \}$ of all potentials gauge-equivalent to A. Gauge equivalence is an equivalence relation, so the orbits are disjoint and exhaust $\mathcal{A}$; the set of orbits is the *physical configuration space*

$$\mathcal{A}/\mathcal{G} = \{ \mathcal{O}_A : A \in \mathcal{A} \}. \tag{25.159}$$

A physical state of the classical gauge field is a point of $\mathcal{A}/\mathcal{G}$; a single potential A_μ is an arbitrary representative of that point, like a choice of coordinates on a manifold.

This is the precise content of the redundancy. Every gauge-invariant functional — $S_{\mathrm{YM}}[A]$, the Hamiltonian, $\operatorname{tr} F^2$, the Wilson loops of the previous section — is constant along orbits, hence descends to a well-defined function on $\mathcal{A}/\mathcal{G}$. Physics is functions on the quotient. The whole reason we work upstairs on $\mathcal{A}$, with the linearly-transforming and manifestly Lorentz-covariant A_μ, rather than directly on the cumbersome quotient, is that $\mathcal{A}$ is a vector space (the sum of two connections shifted by the same inhomogeneous term is again a connection up to that term) on which calculus, Fourier analysis, and perturbation theory are available, whereas $\mathcal{A}/\mathcal{G}$ is a curved, singular object with no linear structure at all. The price of the convenient description is the redundancy; managing that price is the business of the next chapter.

Physical Insight

The geometry here is the same as the geometry of a principal-bundle connection, and the slogan is identical to the one for electromagnetism in Theorem 8.17: *the arena is the quotient, not the cover.* What is new in the non-abelian theory is that $\mathcal{G}$ is built from maps into a curved, non-trivial group manifold $\mathrm{SU}(N)$, not into the contractible line $\mathbb{R}$ (or circle $\mathrm{U}(1)$). A space of maps into a topologically non-trivial target is itself topologically non-trivial, and that is precisely the structure that the abelian case lacked and that now becomes the central new feature.

25.7.2 Small and large gauge transformations

To see the topology we must ask a question that never arose for the line $\mathbb{R}$: *is the gauge group connected?* Concretely, can every $U \in \mathcal{G}$ be reached from the identity map $U \equiv \mathbf{1}$ by a continuous path inside $\mathcal{G}$? If yes, every gauge transformation is the endpoint of a sequence of infinitesimal ones and is fully captured by the Lie-algebra law $\delta A^a_\mu = \frac{1}{g}(D_\mu \alpha)^a$. If no, then there exist gauge transformations that the infinitesimal generators never reach, and the global redundancy is strictly larger than the local one Noether's theorem sees.

The answer depends entirely on the boundary behaviour we impose, and the physically motivated choice is to demand that gauge transformations become

trivial at spatial infinity. Finite-energy field configurations require $A_\mu \to 0$ (up to a pure gauge) at infinity, and a gauge transformation that preserves this class must satisfy

$$U(\mathbf{x}) \longrightarrow U_\infty \in \mathrm{SU}(N) \qquad \text{(a fixed constant)} \quad \text{as } |\mathbf{x}| \to \infty, \tag{25.160}$$

independent of direction. We normalize $U_\infty = \mathbf{1}$ by a global rotation. The condition Eq. (25.160) has a decisive geometric consequence. A map $U : \mathbb{R}^3 \to \mathrm{SU}(N)$ that tends to the *same* limit $\mathbf{1}$ in every direction extends continuously to the one-point compactification of $\mathbb{R}^3$, which is the three-sphere $S^3 = \mathbb{R}^3 \cup \{\infty\}$. A static gauge transformation obeying Eq. (25.160) is therefore the same data as a based map

$$U : \ S^3 \longrightarrow \mathrm{SU}(N), \qquad U(\infty) = \mathbf{1}. \tag{25.161}$$

The question "is $\mathcal{G}$ connected?" becomes "can every such map be deformed continuously to the constant map?" — a question about the topology of the space of maps $S^3 \to \mathrm{SU}(N)$, i.e. about the third homotopy group of the group manifold.

We introduce only as much homotopy theory as we need. Two continuous maps $f_0, f_1 : X \to Y$ are *homotopic*, written $f_0 \simeq f_1$, if there is a continuous $H : X \times [0,1] \to Y$ with $H(\cdot,0) = f_0$ and $H(\cdot,1) = f_1$ — a continuous interpolation. Homotopy is an equivalence relation; its classes are the *homotopy classes*. For based maps $S^n \to Y$ (sending a fixed basepoint of S^n to a fixed basepoint of Y), the set of homotopy classes is denoted $\pi_n(Y)$ and carries a natural group structure (concatenate the interpolations; the constant map is the identity, the orientation-reversed map the inverse). The element $\pi_0(Y)$ counts connected components; $\pi_1(Y)$ is the fundamental group of loops. The object we need is π_3.

Theorem 25.33 (Homotopy classes of gauge transformations)**.** *For $N \geq 2$ the static gauge transformations Eq. (25.161) fall into homotopy classes labelled by an integer,*

$$\pi_3(\mathrm{SU}(N)) = \mathbb{Z}, \qquad N \geq 2. \tag{25.162}$$

The class of U is its winding number *$n[U] \in \mathbb{Z}$. Transformations in the class $n = 0$, called* small*, are exactly those connected to the identity by a continuous path in $\mathcal{G}$; those with $n \neq 0$, called* large*, cannot be reached from the identity by any continuous deformation. The gauge group is therefore disconnected, $\pi_0(\mathcal{G}) = \mathbb{Z}$, with one component per winding number.*

Proof. We establish Eq. (25.162) in two steps: first $N = 2$, then the reduction of general N to it.

Step 1: $\mathrm{SU}(2)$. By the parametrization of Theorem 2.53, every $U \in \mathrm{SU}(2)$ may be written $U = a_0\,\mathbf{1} + i\,\mathbf{a}\cdot\boldsymbol{\sigma}$ with $(a_0, \mathbf{a}) \in \mathbb{R}^4$ and $a_0^2 + |\mathbf{a}|^2 = \det U = 1$. The map $U \mapsto (a_0, \mathbf{a})$ is a diffeomorphism of $\mathrm{SU}(2)$ onto the unit three-sphere $S^3 \subset \mathbb{R}^4$. Hence a based map $S^3 \to \mathrm{SU}(2)$ is a based map $S^3 \to S^3$, and

$$\pi_3(\mathrm{SU}(2)) = \pi_3(S^3) = \mathbb{Z}, \tag{25.163}$$

the last equality being the Hopf degree theorem: two maps $S^3 \to S^3$ are homotopic if and only if they have the same *degree*, the signed number of preimages of a generic target point, and every integer is realized. (We prove integrality of the analytic representative of the degree, and compute it for the generator, in Theorems 25.35 and 25.36 below; that the degree is a complete homotopy invariant we take as the one piece of algebraic topology imported from the mathematics literature.)

Step 2: $\mathrm{SU}(N)$, $N \geq 3$. The group $\mathrm{SU}(N)$ acts transitively on the unit sphere $S^{2N-1} \subset \mathbb{C}^N$ with stabilizer $\mathrm{SU}(N-1)$, giving the fibre bundle (a fibration)

$$\mathrm{SU}(N-1) \;\hookrightarrow\; \mathrm{SU}(N) \;\longrightarrow\; S^{2N-1}. \tag{25.164}$$

A fibration $F \hookrightarrow E \to B$ induces a long exact sequence of homotopy groups, of which we use the segment

$$\pi_4(S^{2N-1}) \;\longrightarrow\; \pi_3(\mathrm{SU}(N-1)) \;\longrightarrow\; \pi_3(\mathrm{SU}(N)) \;\longrightarrow\; \pi_3(S^{2N-1}). \tag{25.165}$$

For $N \geq 3$ the base sphere has dimension $2N-1 \geq 5$, so $\pi_3(S^{2N-1}) = 0$ and $\pi_4(S^{2N-1}) = 0$ (a sphere S^m has $\pi_k(S^m) = 0$ for $0 < k < m$). Exactness then forces the middle arrow to be an isomorphism, $\pi_3(\mathrm{SU}(N-1)) \cong \pi_3(\mathrm{SU}(N))$. Descending $N \to N-1 \to \cdots \to 2$ we reach $\pi_3(\mathrm{SU}(2)) = \mathbb{Z}$, proving Eq. (25.162) for all $N \geq 2$. The embedded $\mathrm{SU}(2)$ in the upper-left corner of $\mathrm{SU}(N)$ generates the group.

Connectedness statement. A path U_t in $\mathcal{G}$ from $\mathbf{1}$ to U_1 is by definition a homotopy of based maps from the constant map to U_1; such a path exists iff U_1 lies in the trivial class $n = 0$. Thus the $n = 0$ component is exactly the identity component, and a transformation is small iff $n[U] = 0$. The components are indexed by $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$, so $\pi_0(\mathcal{G}) = \mathbb{Z}$. □

Caution

The split into small and large transformations is not a property of the group $\mathrm{SU}(N)$ alone — it is forced by the boundary condition Eq. (25.160). If one allowed arbitrary behaviour at infinity, the space of maps $\mathbb{R}^3 \to \mathrm{SU}(N)$ would be contractible and every transformation would be small. The non-trivial topology is created by compactifying space to S^3, which in turn is justified by demanding finite energy. Large gauge transformations are thus genuine global gauge transformations with non-trivial asymptotics, not local ones; this is why the infinitesimal generator $D_\mu \alpha$, which describes deformations near the identity, never produces them.

25.7.3 The winding-number functional

We now give the integer $n[U]$ an explicit analytic formula and prove it has the claimed properties. The natural object is the *Maurer–Cartan form* of the map

U, the $\mathfrak{su}(N)$-valued one-form

$$\theta \;=\; U^{-1}\,\mathrm{d}U \;=\; U^{-1}\partial_i U\,\mathrm{d}x^i, \tag{25.166}$$

which measures the infinitesimal group displacement produced by an infinitesimal step in $\mathbf{x}$. The candidate for the winding number is the integral over the spatial S^3 of the unique (up to normalization) gauge-invariant three-form built from θ,

$$n[U] \;=\; \frac{1}{24\pi^2}\int \mathrm{d}^3x\;\epsilon^{ijk}\;\mathrm{tr}\big(U^{-1}\partial_i U\;U^{-1}\partial_j U\;U^{-1}\partial_k U\big) \;=\; \frac{1}{24\pi^2}\int_{S^3}\mathrm{tr}\,(\theta\wedge\theta\wedge\theta). \tag{25.167}$$

Three facts make this the right object: it is a topological invariant (it does not change under continuous deformation of U), it is additive under group multiplication, and it takes integer values equal to the degree. We prove the first by establishing that the integrand is a closed form; we record the algebraic identity that drives it.

Lemma 25.34 (Maurer–Cartan identity and closedness)**.** *The form $\theta = U^{-1}\mathrm{d}U$ satisfies the Maurer–Cartan equation $\mathrm{d}\theta = -\,\theta\wedge\theta$, and the three-form $\mathrm{tr}(\theta^3) \equiv \mathrm{tr}(\theta\wedge\theta\wedge\theta)$ is closed, $\mathrm{d}\,\mathrm{tr}(\theta^3) = 0$.*

Proof. Differentiating $\theta = U^{-1}\mathrm{d}U$ and using $\mathrm{d}(U^{-1}) = -U^{-1}(\mathrm{d}U)U^{-1}$,

$$\mathrm{d}\theta = \mathrm{d}(U^{-1})\wedge\mathrm{d}U = -\,U^{-1}(\mathrm{d}U)\,U^{-1}\wedge\mathrm{d}U = -\,(U^{-1}\mathrm{d}U)\wedge(U^{-1}\mathrm{d}U) = -\,\theta\wedge\theta. \tag{25.168}$$

For closedness, apply the graded Leibniz rule to the matrix-valued forms and the cyclicity of the trace:

$$\mathrm{d}\,\mathrm{tr}(\theta^3) = \mathrm{tr}\,(\mathrm{d}\theta\,\theta^2) - \mathrm{tr}\,(\theta\,\mathrm{d}\theta\,\theta) + \mathrm{tr}\,(\theta^2\,\mathrm{d}\theta) = -\,\mathrm{tr}(\theta^4) + \mathrm{tr}(\theta^4) - \mathrm{tr}(\theta^4) = -\,\mathrm{tr}(\theta^4), \tag{25.169}$$

where we substituted $\mathrm{d}\theta = -\theta^2$ in each term and used cyclicity to bring all three to $\mathrm{tr}(\theta^4)$ (each θ is a one-form, and cycling a one-form past three one-forms costs $(-1)^3 = -1$; the bookkeeping leaves the relative signs shown). Finally $\mathrm{tr}(\theta^4) = 0$: cycling the leading one-form θ to the rear past the remaining three one-forms gives $\mathrm{tr}(\theta^4) = (-1)^3\,\mathrm{tr}(\theta^4) = -\,\mathrm{tr}(\theta^4)$, so $\mathrm{tr}(\theta^4)$ equals its own negative and vanishes. Hence $\mathrm{d}\,\mathrm{tr}(\theta^3) = 0$. □

Theorem 25.35 ($n[U]$ is an integer homotopy invariant)**.** *The functional Eq. (25.167) is invariant under continuous deformations of U, is additive, $n[U_1U_2] = n[U_1] + n[U_2]$, and takes integer values; for maps into the* SU(2) *subgroup it equals the topological degree of $U : S^3 \to S^3$. It therefore computes the label of Theorem 25.33.*

Proof. Homotopy invariance. Let U_t be a smooth family and write the variation δU, $V \equiv U^{-1}\delta U \in \mathfrak{su}(N)$ (a zero-form). The induced variation of the Maurer–Cartan form is

$$\delta\theta = \delta(U^{-1}\mathrm{d}U) = -U^{-1}\delta U\,U^{-1}\mathrm{d}U + U^{-1}\mathrm{d}(\delta U) = -V\theta + (\mathrm{d}V + \theta V) = \mathrm{d}V + [\theta, V], \tag{25.170}$$

using $U^{-1}\mathrm{d}(\delta U) = \mathrm{d}(U^{-1}\delta U) + (U^{-1}\mathrm{d}U)(U^{-1}\delta U) = \mathrm{d}V + \theta V$. Then, by cyclicity,

$$\delta \int_{S^3} \mathrm{tr}(\theta^3) = 3\int_{S^3} \mathrm{tr}\,(\delta\theta\,\theta^2) = 3\int_{S^3} \mathrm{tr}\,(\mathrm{d}V\,\theta^2) + 3\int_{S^3} \mathrm{tr}\,([\theta, V]\,\theta^2). \tag{25.171}$$

The commutator term vanishes identically: $\mathrm{tr}([\theta, V]\theta^2) = \mathrm{tr}(\theta V\theta^2) - \mathrm{tr}(V\theta^3)$, and cycling the leading one-form θ past the even-degree block $V\theta^2$ gives $\mathrm{tr}(\theta V\theta^2) = (-1)^{1\cdot 2}\,\mathrm{tr}(V\theta^2\theta) = \mathrm{tr}(V\theta^3)$, so the two pieces cancel. For the first term, $\mathrm{d}(\theta^2) = \mathrm{d}\theta\,\theta - \theta\,\mathrm{d}\theta = -\theta^3 + \theta^3 = 0$ by Theorem 25.34, hence $\mathrm{tr}(\mathrm{d}V\,\theta^2) = \mathrm{d}\,\mathrm{tr}(V\theta^2)$ is exact, and

$$\delta \int_{S^3} \mathrm{tr}(\theta^3) = 3\int_{S^3} \mathrm{d}\,\mathrm{tr}(V\theta^2) = 0 \tag{25.172}$$

by Stokes' theorem on the closed manifold S^3 (no boundary). Thus $n[U]$ is constant on each homotopy class.

Additivity. Because n is a homotopy invariant we may deform the factors before multiplying. Any U obeying Eq. (25.160) is homotopic to a map equal to $\mathbf{1}$ outside an arbitrarily small ball: shrink the region where $U \neq \mathbf{1}$ radially toward a chosen point, a deformation that leaves the homotopy class — and hence $n[U]$ — unchanged. Given U_1, U_2, deform U_1 to be supported in a ball B_1 and U_2 in a disjoint ball B_2. Then U_1U_2 equals U_1 on B_1, equals U_2 on B_2, and equals $\mathbf{1}$ elsewhere; since the integrand of Eq. (25.167) is supported where $U \neq \mathbf{1}$, the integral splits cleanly,

$$n[U_1U_2] = \int_{B_1} + \int_{B_2} = n[U_1] + n[U_2]. \tag{25.173}$$

Therefore $n : \pi_3(\mathrm{SU}(N)) \to \mathbb{Z}$ is a group homomorphism.

Integrality and degree. For U valued in the SU(2) subgroup, identify $\mathrm{SU}(2) \cong S^3$ as in Theorem 25.33. Under this identification $\frac{1}{24\pi^2}\,\mathrm{tr}(\theta^3)$ is exactly $\frac{1}{2\pi^2}$ times the pullback by U of the standard unit-volume three-form Ω of the target S^3 (the unique bi-invariant top form, normalized so that $\int_{S^3}\Omega = 2\pi^2$). Hence

$$n[U] = \frac{1}{2\pi^2}\int_{S^3} U^*\Omega = \deg(U)\cdot\frac{1}{2\pi^2}\int_{S^3}\Omega = \deg(U) \in \mathbb{Z}, \tag{25.174}$$

the middle equality being the change-of-variables/degree theorem: the integral of a pulled-back top form equals the degree times the integral over the target. The normalization constant $1/24\pi^2$ is fixed precisely so that the generator has $n = 1$; we verify this number from scratch in Theorem 25.36. Integrality for general $N \geq 3$ follows because, by Theorem 25.33, every class is generated by an embedded SU(2) map, on which the formula reduces to the SU(2) case just settled. □

Worked Example 25.36 (The basic SU(2) winding-one map)**.** We compute $n[U]$ for the prototypical "hedgehog" map and confirm it equals 1, pinning down

the normalization in Eq. (25.167). Take

$$U(\mathbf{x}) = \exp\!\big(i\,\Theta(r)\,\hat{\mathbf{x}}\cdot\boldsymbol{\sigma}\big) = \cos\Theta(r)\,\mathbf{1} + i\sin\Theta(r)\,\hat{\mathbf{x}}\cdot\boldsymbol{\sigma}, \qquad \Theta(r) = \pi f(r), \tag{25.175}$$

with $r = |\mathbf{x}|$, $\hat{\mathbf{x}} = \mathbf{x}/r$, and a smooth profile f with $f(0) = 0$, $f(\infty) = 1$, so that $\Theta(0) = 0$ (giving $U(0) = \mathbf{1}$, regular at the origin) and $\Theta(\infty) = \pi$ (giving $U(\infty) = -\mathbf{1}$, a constant at infinity, consistent with Eq. (25.160) after the global normalization $U_\infty = -\mathbf{1} \to \mathbf{1}$). The second equality used $(\hat{\mathbf{x}}\cdot\boldsymbol{\sigma})^2 = \mathbf{1}$.

Step 1: the Maurer–Cartan components. Write $N \equiv \hat{\mathbf{x}}\cdot\boldsymbol{\sigma}$, so $N^2 = \mathbf{1}$ and $\partial_i \hat{x}_a = (\delta_{ia} - \hat{x}_i\hat{x}_a)/r$. Differentiating Eq. (25.175) and multiplying by $U^{-1} = \cos\Theta\,\mathbf{1} - i\sin\Theta\,N$, a direct computation gives $L_i \equiv U^{-1}\partial_i U = i\,c_i^a\sigma_a$ with the real coefficient matrix

$$c_i^a = \Theta'\,\hat{x}_i\hat{x}_a + \frac{\sin\Theta\cos\Theta}{r}\,(\delta_{ia} - \hat{x}_i\hat{x}_a) + \frac{\sin^2\Theta}{r}\,\epsilon_{iab}\,\hat{x}_b, \tag{25.176}$$

where $\Theta' = \mathrm{d}\Theta/\mathrm{d}r$. Each piece has a transparent meaning: the radial direction $\hat{\mathbf{x}}$ is stretched at rate Θ'; the two tangential directions are scaled by $\sin\Theta\cos\Theta/r$ and rotated into one another at rate $\sin^2\Theta/r$.

Step 2: the trace. Using $\sigma_a\sigma_b\sigma_c = \delta_{ab}\sigma_c + \ldots$ whose trace is $\mathrm{tr}(\sigma_a\sigma_b\sigma_c) = 2i\,\epsilon_{abc}$, and $L_iL_jL_k = i^3 c_i^a c_j^b c_k^c\,\sigma_a\sigma_b\sigma_c$,

$$\mathrm{tr}\,(L_iL_jL_k) = i^3\,c_i^a c_j^b c_k^c\,(2i\,\epsilon_{abc}) = 2\,\epsilon_{abc}\,c_i^a c_j^b c_k^c, \tag{25.177}$$

since $i^3\cdot 2i = 2i^4 = 2$. Contracting with ϵ^{ijk} converts the double epsilon contraction into a determinant,

$$\epsilon^{ijk}\,\mathrm{tr}\,(L_iL_jL_k) = 2\,\epsilon^{ijk}\epsilon_{abc}\,c_i^a c_j^b c_k^c = 2\cdot 3!\,\det(c) = 12\,\det(c). \tag{25.178}$$

Step 3: the determinant. The matrix Eq. (25.176) is diagonal in the natural orthonormal frame $\{\hat{\mathbf{x}}, \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_2\}$ (radial plus two tangential). On the radial axis it has eigenvalue Θ'. On the tangential plane it acts as $\frac{\sin\Theta\cos\Theta}{r}\,\mathbf{1}_\perp + \frac{\sin^2\Theta}{r}\,J$, where J is the generator of 90° rotations in that plane, $J^2 = -\mathbf{1}_\perp$; the determinant of $a\,\mathbf{1}_\perp + b\,J$ over the 2-plane is $a^2 + b^2$. Hence

$$\det(c) = \Theta'\cdot\Big[\Big(\tfrac{\sin\Theta\cos\Theta}{r}\Big)^2 + \Big(\tfrac{\sin^2\Theta}{r}\Big)^2\Big] = \Theta'\cdot\frac{\sin^2\Theta}{r^2}\,(\cos^2\Theta + \sin^2\Theta) = \frac{\Theta'\sin^2\Theta}{r^2}. \tag{25.179}$$

Combining Eqs. (25.178) and (25.179), the winding density is $\epsilon^{ijk}\,\mathrm{tr}(L_iL_jL_k) = 12\,\Theta'\sin^2\Theta/r^2$.

Step 4: the integral. The density depends only on r, so with $\int \mathrm{d}^3x = 4\pi\int_0^\infty r^2\,\mathrm{d}r$,

$$n[U] = \frac{1}{24\pi^2}\int \mathrm{d}^3x\,\frac{12\,\Theta'\sin^2\Theta}{r^2} = \frac{1}{24\pi^2}\cdot 4\pi\cdot 12\int_0^\infty \Theta'\sin^2\Theta\,\mathrm{d}r = \frac{2}{\pi}\int_0^\pi \sin^2\Theta\,\mathrm{d}\Theta, \tag{25.180}$$

where the radial Jacobian r^2 cancelled the $1/r^2$ in the density, and we changed variables to Θ, which runs monotonically from 0 to π. Finally $\int_0^\pi \sin^2\Theta\,\mathrm{d}\Theta = \pi/2$, so

$$n[U] = \frac{2}{\pi}\cdot\frac{\pi}{2} = 1. \tag{25.181}$$

The map Eq. (25.175) winds once: it wraps the spatial S^3 exactly once around the group manifold $\mathrm{SU}(2) \cong S^3$, as the picture of Θ sweeping from the north pole through to the south pole of the target as $r : 0 \to \infty$ makes geometrically plain. Replacing $f(\infty) = 1$ by $f(\infty) = m$ (an integer) yields $n = m$, realizing every class and confirming $\pi_3(\mathrm{SU}(2)) = \mathbb{Z}$ concretely. The normalization $1/24\pi^2$ is exactly what makes the basic generator carry unit charge.

25.7.4 Why gauge fixing is mandatory — and subtle

The orbit picture now explains, with no further input, both why the quantum theory *must* gauge-fix and why doing so is delicate. The (formal) path integral that defines the quantum gauge theory sums $e^{iS_{\mathrm{YM}}[A]}$ over all of $\mathcal{A}$,

$$Z = \int_{\mathcal{A}} \mathcal{D}A\; e^{iS_{\mathrm{YM}}[A]}. \tag{25.182}$$

But S_{YM} is constant along each orbit Theorem 25.32. The integrand therefore does not distinguish the physically distinct points of $\mathcal{A}/\mathcal{G}$; it assigns the same value to every representative of a given orbit. The integral Eq. (25.182) thus counts each physical configuration not once but once for every element of $\mathcal{G}$ — it overcounts by the (infinite) volume of the gauge group,

$$Z = \mathrm{Vol}(\mathcal{G}) \times \int_{\mathcal{A}/\mathcal{G}} \mathcal{D}[A]\; e^{iS_{\mathrm{YM}}[A]}, \tag{25.183}$$

at least schematically. The factor $\mathrm{Vol}(\mathcal{G}) = \int_{\mathcal{G}} \mathcal{D}U$ is a field-independent infinity. In a free or super-renormalizable theory one might hope to drop a constant, but here the propagator does not even exist: the quadratic form in A from S_{YM} has the gauge directions as flat (zero-mode) directions — moving along an orbit costs no action — so the Gaussian integral over those directions diverges and A has no inverse propagator. The redundancy is not a cosmetic nuisance; it is a genuine obstruction to perturbation theory. One must integrate only over a set of orbit representatives, a *gauge slice* that meets each orbit, and divide out the spurious volume. Implementing this faithfully — inserting unity as an integral over the gauge group, trading it for a delta-function on a gauge condition $G(A) = 0$ together with the Jacobian of the condition along the orbit — is the Faddeev–Popov procedure, carried out in full in Chapter 26. The Jacobian is the determinant of the operator $\delta G/\delta\alpha$, and rewriting it as a Gaussian integral over anticommuting fields is what conjures the *ghosts*; the residual rigid symmetry of the gauge-fixed action is BRST, whose cohomology selects the physical states.

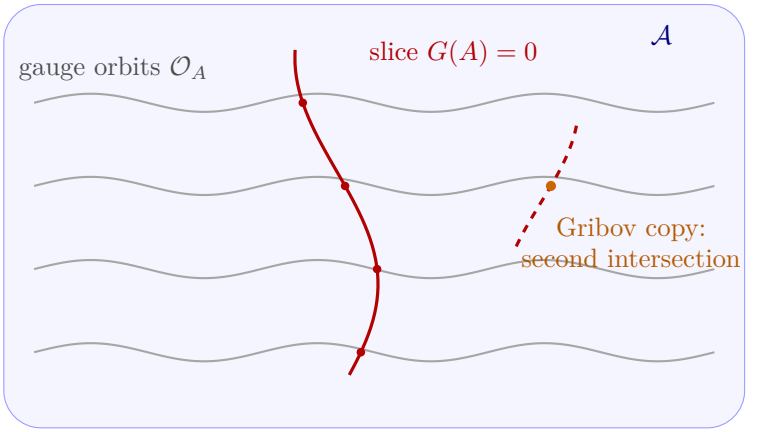


Figure 25.5: The space of connections $\mathcal{A}$ foliated into gauge orbits $\mathcal{O}_A$ (grey), the leaves along which the action is constant. The physical configuration space $\mathcal{A}/\mathcal{G}$ is the set of leaves. A gauge-fixing condition $G(A) = 0$ (red) is meant to pick exactly one representative per orbit; globally, however, the slice generically intersects some orbits more than once (the dashed branch and orange point), the Gribov ambiguity. Locally near $A = 0$ the slice is faithful, which is all perturbation theory requires.

There is a further subtlety the orbit picture makes unavoidable, and we flag it honestly rather than paper over it. A gauge slice $G(A) = 0$ is supposed to intersect each orbit exactly once. Whether a global slice with this property exists is a question about the global geometry of $\mathcal{A}/\mathcal{G}$ — and for non-abelian gauge theory on compact space the answer is no. Gribov showed that the Coulomb-gauge condition $\partial_i A_i = 0$, and indeed any continuous gauge condition, meets typical orbits more than once: there exist distinct connections A and A^U, both satisfying $G = 0$, related by a (necessarily large or otherwise non-infinitesimal) gauge transformation. These are *Gribov copies* (Fig. 25.5). The Faddeev–Popov determinant, which is positive near $A = 0$, vanishes on the boundary (the Gribov horizon) where two copies merge, signalling that the naive slice has folded over. Perturbation theory, expanded about $A = 0$ inside the first Gribov region where the slice is faithful, is untouched by this — the construction of Chapter 26 is correct order by order — but the non-perturbative definition of the integral Eq. (25.183) requires restricting to a fundamental region (e.g. the Gribov region where $-\partial_i D_i$ is positive). We will treat the perturbative theory exactly and name the Gribov problem where it bites; it is a real feature of the global structure of $\mathcal{A}/\mathcal{G}$, not an artifact of a bad gauge.

25.7.5 The classical theory is exact; the quantum theory is not yet defined

It is worth stating plainly where the rigor of this chapter ends. Everything we have done is exact classical field theory: the Lagrangian $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \bar{\psi}_i(i\not{D} - m)\psi_i$, its equations of motion, the Bianchi identity, the Wilson loops, and the orbit-and-winding structure of this section are all theorems. The *quantum* theory is a different matter. Constructing four-dimensional Yang–Mills as a mathematically well-defined quantum field theory — showing that the path integral Eq. (25.183) exists nonperturbatively on the physical configuration space, with a Hilbert space, and that its spectrum has a strictly positive lowest excitation (a *mass gap* $\Delta > 0$) above the vacuum — is an open problem, one of the Clay Mathematics Institute Millennium Problems, unsolved as of the mid-2020s. Lattice simulations give overwhelming numerical evidence that pure SU(3) Yang–Mills confines and has a gap of order a few hundred MeV, and the perturbative expansion we build in Chapters 26 and 27 is asymptotic and predictive; but a constructive, non-perturbative definition — the analogue for gauge theory of the rigorous construction available for super-renormalizable scalar models — does not exist. The honest position is therefore this: the classical analysis here is complete and rigorous, the perturbative quantum theory is well-defined order by order, and the full non-perturbative theory is a profound open question whose very existence is the prize.

This closes the classical layer of Yang–Mills theory. We have carried it from the gauge principle through the connection, the curvature, the self-interacting Lagrangian and its equations of motion, the Wilson-loop observables, and now to the global topology of the configuration space. Two batons pass forward. The overcounting Eq. (25.183) and the orbit geometry of Fig. 25.5 are exactly what the Faddeev–Popov construction and BRST symmetry of Chapter 26 are built to tame, with the disconnectedness $\pi_0(\mathcal{G}) = \mathbb{Z}$ of Theorem 25.33 seeding the global (large-gauge) structure that BRST must respect; and the winding number Eq. (25.167), here a property of gauge transformations, reappears in Chapter 42 as the topological charge of the instanton and the label of the inequivalent θ-vacua, where the integer we computed in Theorem 25.36 becomes the quantum of a genuine physical superselection.

Further reading

The founding paper is Yang and Mills [528], which introduced non-abelian gauge fields in the guise of isotopic-spin gauge invariance; it remains remarkably readable. The classical and quantum development at the level of this book is covered by the standard texts Peskin and Schroeder [396] (chapters 15–16), Schwartz [432] (chapters 25–26), Weinberg [501] (chapter 15), and Pokorski

[401]; Coleman [129] gives an especially careful conceptual account of the gauge principle and its redundancy. The geometric reading of the connection and curvature as objects on a principal bundle, the holonomy picture of the Wilson line, and the homotopy classification $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$ underlying large gauge transformations are developed in Nakahara [355]. For Wilson loops as the order parameter of confinement and the lattice formulation in which they are most naturally non-perturbative, see Wilson and Kogut [516]. The gauge-orbit overcounting that motivates the next chapter is the subject of Faddeev and Popov [182], and the obstruction to a global gauge slice is Gribov [236]; the non-abelian comparator and its phase had been anticipated in Wu and Yang [527]. Finally, the precise statement of what it would mean to construct the quantum theory — the mass-gap problem — is laid out in the Clay Institute description Jaffe and Witten [281].

Chapter 26

Quantization of Yang–Mills

Chapter 25 built the classical non-abelian gauge theory in full — the connection A_μ, the curvature $F_{\mu\nu}$, the self-interacting Yang–Mills Lagrangian $\mathcal{L}_{\text{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ (Eq. (25.76)), and the local symmetry under the gauge group whose orbits foliate the space of connections. It left Yang–Mills theory at a sharp threshold. The very gauge invariance that organizes the classical theory is, at the quantum level, a fatal redundancy: the action is constant along gauge orbits, so the naive path integral $\int \mathcal{D}A\, e^{iS_{\text{YM}}}$ overcounts each physical configuration by the infinite volume of the gauge group, the quadratic form has flat directions, and the gluon has no propagator. The classical theory is geometrically complete; the quantum theory does not yet exist. This chapter brings it into existence, and in doing so confronts the one feature that makes non-abelian quantization genuinely harder than its abelian ancestor.

That ancestor is Chapter 13, where the photon was quantized by the Faddeev–Popov procedure (Section 13.5, Theorem 13.34) and the gauge-fixing Jacobian collapsed to a field-independent constant $\det(\partial^2)$ that could be discarded without trace — QED has no ghosts. We carry that construction step for step into the non-abelian setting and locate exactly where it breaks: the Faddeev–Popov determinant becomes a genuine functional of A, and a field-dependent determinant cannot be pulled out of the integral. Coping with that single fact is the entire content of the chapter. We exponentiate the determinant using the Grassmann Berezin integral of Chapter 12, which converts it into a local term in a larger action at the price of a pair of anticommuting scalar fields — the Faddeev–Popov ghosts — and assembles the complete gauge-fixed Yang–Mills Lagrangian. From it we read off the momentum-space Feynman rules with the full machinery of Chapters 10 and 14: the gluon and ghost propagators, the three- and four-gluon self-couplings, the ghost–gluon vertex, and the global rule of a minus sign per closed ghost loop.

The ghosts are not optional bookkeeping but a physical necessity, and the central arc of the chapter is the proof that the theory they define is consistent.

Left to themselves, the unphysical polarizations of an internal gluon would spoil unitarity; we compute a physical amplitude on a cut and watch the wrong-sign ghost loop cancel that contamination exactly, generalizing the abelian Ward–Takahashi identity of Theorem 19.13. That one cancellation cannot be checked to all orders by hand, and the structure of the computation reveals why it must hold: a residual exact symmetry survives the gauge fixing, mixing the commuting gluon with the anticommuting ghost. This is BRST symmetry — global, fermionic, and nilpotent, $s^2 = 0$, with its nilpotency rooted in the Jacobi identity of the algebra. BRST is the organizing principle of everything that follows. It recasts the whole gauge-fixing-and-ghost sector as a single exact term $s\Psi$, defines the physical Hilbert space as its cohomology $\ker Q_B / \operatorname{im} Q_B$ — proven positive-definite and equal to the two transverse gluon polarizations, with the unphysical modes organizing into zero-norm quartets — and thereby promotes the hand-checked unitarity to a theorem. Finally, promoting BRST invariance to a functional identity on the path integral yields the Slavnov–Taylor identities, the non-abelian Ward–Takahashi relations, which enforce transversality of the gluon self-energy and collapse every divergence into a single universal coupling renormalization.

The seven sections follow this thread without a break. Section 26.1 isolates the obstruction and states honestly the one assumption the construction makes — that the gauge slice meets each orbit once, true perturbatively and false globally (the Gribov ambiguity, deferred to Chapter 42). Section 26.2 exponentiates the determinant and assembles the gauge-fixed Lagrangian; Section 26.3 extracts the Feynman rules; Section 26.4 verifies unitarity and motivates the symmetry that guarantees it; Section 26.5 constructs BRST and proves its nilpotency; Section 26.6 builds the physical states as BRST cohomology and proves the S-matrix unitary on them; and Section 26.7 derives the Slavnov–Taylor identities and the renormalization-constant web that hands Yang–Mills theory to Chapter 27, where the same rules yield the negative beta function and asymptotic freedom. Throughout we use the conventions of Appendix B for the group-theory data ($[T^a, T^b] = if^{abc}T^c$, $\operatorname{tr} T^a T^b = \frac{1}{2}\delta^{ab}$, $\dim G = N^2 - 1$ for $\mathrm{SU}(N)$) and the mostly-minus metric and covariant-derivative sign $D_\mu = \partial_\mu - igA^a_\mu T^a$ fixed in Chapter 25.

26.1 The Obstruction: Why Naive Non-Abelian Quantization Fails

Chapter 25 left Yang–Mills theory at a sharp threshold. We have the classical non-abelian action built from the Yang–Mills Lagrangian $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ of Eq. (25.76), with the field strength Eq. (25.49) carrying its self-coupling $gf^{abc}A^b_\mu A^c_\nu$, the covariant derivative Eq. (25.16), and the full local symmetry under the gauge group $\mathcal{G}$ of Theorem 25.31. We have also learned, in the closing

section of that chapter, that the price of this symmetry is a redundancy so severe that the field A_μ does not even possess an inverse propagator: the quadratic form in the action has flat directions along the gauge orbits, and the naive Gaussian integral over those directions diverges (Eq. (25.183)). The classical theory is geometrically complete; the quantum theory does not yet exist.

The abelian case taught us the cure. In Section 13.5 we quantized the photon by the Faddeev–Popov procedure: we inserted a cleverly written "1" into the path integral, traded the integral over the gauge group for a delta function pinning a gauge condition, and factored off the infinite orbit volume, paying for the factorization with a single Jacobian determinant $\Delta_{\mathrm{FP}}[A]$. There the determinant evaluated to $\det(\partial^2)$, *a constant independent of the gauge field* (Theorem 13.34, Eq. (13.164)). Being constant, it cancelled between numerator and denominator of every normalized correlator and could be discarded without trace: QED has no ghosts. Chapter 13 planted one warning sentence (Theorem 13.35), namely that this collapse to a constant is a special feature of the linearity of the abelian gauge action and fails the moment the structure constants f^{abc} are nonzero.

This section makes that warning precise and turns it into the organizing problem of the chapter. We carry the Faddeev–Popov construction through, step for step, in the non-abelian setting, and we locate exactly where the abelian argument breaks: the Faddeev–Popov determinant $\Delta_{\mathrm{FP}}[A] = \det M$ becomes a genuine, non-trivial functional of A, because the operator $M^{ab} = \frac{1}{g}\,\partial^\mu D_\mu^{ab}$ contains the gauge field through the adjoint covariant derivative. A field-dependent determinant cannot be pulled out of the integral and cannot be ignored; it must be *kept* and made part of the dynamics. That single fact is the entire content of Yang–Mills quantization, and the rest of the chapter is the machinery that copes with it. We close by stating honestly the one assumption the whole construction makes — that the gauge condition is met exactly once on each orbit — which is true perturbatively but false globally (the Gribov ambiguity), deferring its resolution to Chapter 42.

26.1.1 The naive generating functional and the volume of the gauge group

Write the pure-gauge Yang–Mills action as

$$S_{\mathrm{YM}}[A] \;=\; \int \mathrm{d}^4x \left(-\tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} \right), \qquad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu, \tag{26.1}$$

with all conventions and group-theory data ($[T^a, T^b] = i f^{abc} T^c$, $\mathrm{tr}\, T^a T^b = \frac{1}{2}\delta^{ab}$, the adjoint dimension $\dim(G) = N^2 - 1$ for $\mathrm{SU}(N)$) those of Chapter B. The object we would like to quantize is the generating functional

$$Z_{\mathrm{naive}} \;=\; \int \mathcal{D}A \, e^{\,iS_{\mathrm{YM}}[A]}, \qquad \mathcal{D}A = \prod_{x,\mu,a} \mathrm{d}A^a_\mu(x), \tag{26.2}$$

in exact analogy with the abelian Z of Eq. (13.151). As in the photon case, this integral is ill-defined, and for exactly the same structural reason, now sharper. The action is gauge invariant: under the finite gauge transformation Eq. (25.156),

$$A_\mu \longmapsto A_\mu^U = U A_\mu U^{-1} - \frac{i}{g}(\partial_\mu U)U^{-1}, \qquad U(x) \in \mathrm{SU}(N), \tag{26.3}$$

the field strength transforms covariantly, $F_{\mu\nu} \mapsto U F_{\mu\nu} U^{-1}$, and the trace $\operatorname{tr} F_{\mu\nu}F^{\mu\nu}$ is invariant, so $S_{\mathrm{YM}}[A^U] = S_{\mathrm{YM}}[A]$. The integrand $e^{iS_{\mathrm{YM}}}$ is therefore constant along every gauge orbit

$$\mathcal{O}_A = \{\, A^U : U \in \mathcal{G} \,\} \tag{26.4}$$

of Theorem 25.32. Integrating a function that is constant along the orbits over the full configuration space $\mathcal{A}$ multiplies the genuine integral over the physical quotient $\mathcal{A}/\mathcal{G}$ (Eq. (25.159)) by the volume of a single orbit, and then sums that identical infinity over all orbits:

$$Z_{\text{naive}} = \Big(\int_{\mathcal{G}} \mathcal{D}U\Big) \times \Big[\text{integral over one representative per orbit}\Big] = \mathrm{Vol}(\mathcal{G}) \times [\cdots], \tag{26.5}$$

the schematic factorization Eq. (25.183). Here $\mathrm{Vol}(\mathcal{G}) = \int_{\mathcal{G}} \mathcal{D}U$ is the volume of the gauge group in its Haar measure — an infinite, field-independent constant. Unlike a super-renormalizable theory in which one might cheerfully drop an overall constant, here the divergence is fatal at a deeper level: because the action does not change as A moves along an orbit, the quadratic part of S_{YM} has the orbit directions as exact zero modes, the would-be inverse propagator is not invertible, and perturbation theory cannot even start (Eq. (25.183) and the discussion there). The redundancy must be removed before, not after, we expand.

What is genuinely new relative to Chapter 13 is the *shape* of the orbits. In the abelian theory the gauge group acts on A_μ by the affine shift $A_\mu \mapsto A_\mu + \partial_\mu \alpha$: the orbits are parallel straight lines (an affine action), all running in the same direction, and any flat slice cuts every one of them at the same constant angle. In the non-abelian theory the action Eq. (26.3) is nonlinear — it conjugates A_μ by U and adds the inhomogeneous Maurer–Cartan piece $-\frac{i}{g}(\partial_\mu U)U^{-1}$, and successive transformations compose by the non-commuting group law. The orbits are *curved* submanifolds of $\mathcal{A}$, and they wind through configuration space in a way that varies from orbit to orbit. This is precisely the geometry sketched in Fig. 26.1. The constancy of the abelian intersection angle was what made the Jacobian a number; its variation in the non-abelian case is what will make the Jacobian a field-dependent determinant. We now make this exact.

26.1.2 The Faddeev–Popov resolution of unity, non-abelian

The strategy is the one rehearsed in the two-dimensional warm-up Theorem 13.32 and carried into field theory in Theorem 13.34: insert a representation of unity

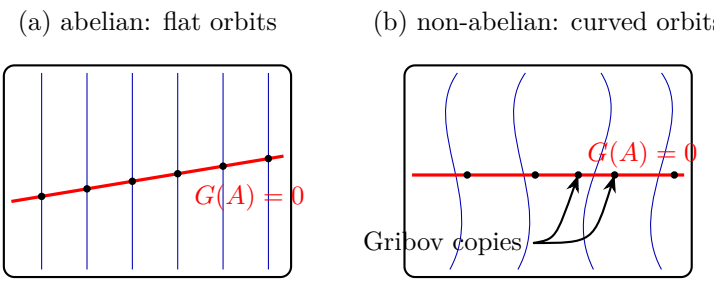


Figure 26.1: Gauge orbits (blue) in the space of connections $\mathcal{A}$, cut by a gauge slice $G(A) = 0$ (red). *(a)* Abelian: the orbits are parallel straight lines and the slice meets each exactly once, at a constant angle, so the Jacobian $\Delta_{\mathrm{FP}} = \det(\partial^2)$ is a field-independent number. *(b)* Non-abelian: the orbits are curved and the intersection angle varies from point to point, so the Jacobian $\det M$ depends on where on $\mathcal{A}$ the slice is met — it is a functional of A. The same curvature lets the slice cut a single orbit more than once (the marked Gribov copies), the global obstruction discussed in Section 26.1.4.

that isolates the orbit volume so it can be split off. Pick a gauge-fixing function $G^a(A)$, one real function per generator, and an arbitrary external function $\omega^a(x)$. Define the Faddeev–Popov determinant $\Delta_{\mathrm{FP}}[A]$ as the quantity that makes the following an identity:

$$1 \;=\; \Delta_{\mathrm{FP}}[A] \int_{\mathcal{G}} \mathcal{D}U \;\; \delta\big(G(A^U) - \omega\big), \qquad \delta\big(G(A^U) - \omega\big) = \prod_{x,a} \delta\big(G^a(A^U)(x) - \omega^a(x)\big), \tag{26.6}$$

where $\mathcal{D}U$ is the (left-invariant) Haar measure on the gauge group. Equation Eq. (26.6) is the exact non-abelian analogue of Eq. (13.163); the only changes from the abelian statement are that the integration is over the curved group manifold with its Haar measure rather than over a flat parameter α with Lebesgue measure, and that there is now one gauge condition per colour. By construction $\Delta_{\mathrm{FP}}[A]$ is the reciprocal of the group integral, $\Delta_{\mathrm{FP}}[A]^{-1} = \int_{\mathcal{G}} \mathcal{D}U \, \delta(G(A^U) - \omega)$.

Two facts must hold for the factorization to go through, and exactly one of them behaves differently from the abelian theory. The first is the gauge invariance of Δ_{FP} itself; this survives intact, and we prove it now because it is what licenses the change of variables. The second is the evaluation of Δ_{FP} as an explicit determinant; this is where field-dependence enters, and we postpone it to Section 26.1.3.

Lemma 26.1 (Gauge invariance of the Faddeev–Popov determinant)**.** *The functional $\Delta_{\mathrm{FP}}[A]$ defined by Eq. (26.6) is invariant along gauge orbits: $\Delta_{\mathrm{FP}}[A^{U'}] = \Delta_{\mathrm{FP}}[A]$ for every $U' \in \mathcal{G}$.*

Proof. From the definition,

$$\Delta_{\mathrm{FP}}[A^{U'}]^{-1} = \int_{\mathcal{G}} \mathcal{D}U \; \delta(G((A^{U'})^{U}) - \omega). \tag{26.7}$$

The gauge action is a group action: applying U' then U is the same as applying the single element UU', so $(A^{U'})^{U} = A^{UU'}$. (This composition law is exactly the non-commuting group multiplication of Theorem 25.24; it is what makes the orbits curved, and here it is precisely what we exploit.) Substitute and change the integration variable from U to $\tilde{U} = UU'$. The Haar measure is left-invariant, $\mathcal{D}(UU') = \mathcal{D}U$ at fixed U', equivalently $\mathcal{D}U = \mathcal{D}\tilde{U}$, so

$$\Delta_{\mathrm{FP}}[A^{U'}]^{-1} = \int_{\mathcal{G}} \mathcal{D}U \; \delta(G(A^{UU'}) - \omega) = \int_{\mathcal{G}} \mathcal{D}\tilde{U} \; \delta(G(A^{\tilde{U}}) - \omega) = \Delta_{\mathrm{FP}}[A]^{-1}. \tag{26.8}$$

Taking reciprocals gives the claim. The single property used is the left invariance of the Haar measure — the non-abelian replacement for the translation invariance $\mathcal{D}(\alpha+\alpha') = \mathcal{D}\alpha$ that did the same job in the abelian proof of Theorem 13.34. □

With invariance in hand the factorization is mechanical, and we run it exactly as in Eq. (13.170)–Eq. (13.171) so the reader can see that every step transcribes, and see the one place the transcription stalls. Insert the unity Eq. (26.6) into the naive functional Eq. (26.2) and interchange the order of integration:

$$\begin{aligned} Z_{\mathrm{naive}} &= \int \mathcal{D}A \; e^{iS_{\mathrm{YM}}[A]} \, \Delta_{\mathrm{FP}}[A] \int_{\mathcal{G}} \mathcal{D}U \; \delta(G(A^{U}) - \omega) \\ &= \int_{\mathcal{G}} \mathcal{D}U \int \mathcal{D}A \; e^{iS_{\mathrm{YM}}[A]} \, \Delta_{\mathrm{FP}}[A] \, \delta(G(A^{U}) - \omega). \end{aligned} \tag{26.9}$$

Now relabel the dummy field in the inner integral by writing $A = B^{U^{-1}}$, that is, $B = A^{U}$. Three invariances conspire, the non-abelian counterparts of the three used at Eq. (13.170):

- the action is gauge invariant, $S_{\mathrm{YM}}[A] = S_{\mathrm{YM}}[B^{U^{-1}}] = S_{\mathrm{YM}}[B]$;
- the determinant is gauge invariant by Theorem 26.1, $\Delta_{\mathrm{FP}}[A] = \Delta_{\mathrm{FP}}[B]$;
- the measure $\mathcal{D}A$ is gauge invariant. The transformation Eq. (26.3) acts on A^{a}_{μ} as a rotation by the adjoint representation (in the homogeneous piece $UA_{\mu}U^{-1}$) plus a field-independent translation (the inhomogeneous Maurer–Cartan piece); the adjoint representation is orthogonal because the f^{abc} are real and totally antisymmetric (Chapter B), so its Jacobian is $|\det(\text{adjoint rotation})| = 1$, and a translation has unit Jacobian. Hence $\mathcal{D}A = \mathcal{D}B$.

After the relabelling the delta function reads $\delta(G(B)-\omega)$, with no residual dependence on U, so the entire group integral pulls out:

$$Z_{\text{naive}} = \underbrace{\Big(\int_{\mathcal{G}} \mathcal{D}U\Big)}_{\text{Vol}(\mathcal{G})} \int \mathcal{D}B\; e^{iS_{\text{YM}}[B]}\, \Delta_{\text{FP}}[B]\, \delta(G(B)-\omega). \tag{26.10}$$

The infinite orbit volume is now isolated in the single multiplicative constant $\text{Vol}(\mathcal{G}) = \int_{\mathcal{G}} \mathcal{D}U$, field-independent, cancelling between numerator and denominator of every normalized correlator exactly as the abelian $V_{\mathcal{G}}$ did at Eq. (13.171). Dropping it and renaming $B \to A$, the gauge-fixed generating functional is

$$Z[\omega] = \int \mathcal{D}A\; e^{iS_{\text{YM}}[A]}\; \Delta_{\text{FP}}[A]\; \delta(G(A)-\omega). \tag{26.11}$$

This is the formula the rest of the chapter develops. Compare it line for line with its abelian ancestor Eq. (13.172). The delta function restricts the integral to one representative per orbit; that part is identical. The decisive difference is the factor $\Delta_{\text{FP}}[A]$. In the abelian theory it equalled the *constant* $\det(\partial^2)$ and was pulled outside the integral alongside $\text{Vol}(\mathcal{G})$, leaving a clean Gaussian. Here we have so far only proved that Δ_{FP} is gauge invariant — not that it is constant. To see whether it can likewise be discarded we must compute it, and that computation is where the abelian and non-abelian theories part company for good.

26.1.3 The Faddeev–Popov operator in Lorenz gauge

Fix the gauge condition to the non-abelian Lorenz (Landau-family) choice, one per colour,

$$G^a(A) = \partial^\mu A^a_\mu, \qquad \text{so } G^a(A) = \omega^a \text{ reads } \partial^\mu A^a_\mu = \omega^a, \tag{26.12}$$

the direct analogue of Eq. (13.162). To evaluate $\Delta_{\text{FP}}[A]$ we need the behaviour of $G^a(A^U)$ as U ranges over a neighbourhood of the identity, because the group integral in Eq. (26.6) is dominated, near a solution of the gauge condition, by the linearization of G along the orbit. Exactly as the two-dimensional Jacobian $r = \partial g(\mathbf{x}^\theta)/\partial\theta$ of Eq. (13.159) was the derivative of the gauge condition along the orbit evaluated on the slice, the field-theory determinant is the functional determinant of

$$M^{ab}(x,y) \;\equiv\; \left.\frac{\delta\, G^a(A^\alpha)(x)}{\delta\alpha^b(y)}\right|_{\alpha=0}, \qquad \Delta_{\text{FP}}[A] = \det M, \tag{26.13}$$

where we have parametrized group elements near the identity as $U(x) = \exp(i\alpha^a(x)T^a)$ and write A^α for A^U at infinitesimal α. (Near the identity the Haar measure $\mathcal{D}U$ reduces to the flat measure $\prod_{x,a} \mathrm{d}\alpha^a(x)$ up to a Jacobian

equal to 1 at $\alpha = 0$, so it does not affect the linearization; this is the non-abelian refinement of the statement, below Eq. (13.163), that $\mathcal{D}\alpha$ is Haar on the abelian group.) We therefore need the infinitesimal transformation law of A^a_μ, and this is exactly the result imported from Chapter 25.

Under an infinitesimal gauge transformation the potential shifts by the adjoint covariant derivative of the parameter (Eq. (25.34), with the adjoint covariant derivative Eq. (25.35)):

$$\delta A^a_\mu = \frac{1}{g}\,\partial_\mu \alpha^a + f^{abc} A^b_\mu \alpha^c = \frac{1}{g}(D_\mu \alpha)^a = \frac{1}{g}\, D^{ab}_\mu\, \alpha^b, \qquad D^{ab}_\mu = \partial_\mu \delta^{ab} + g f^{acb} A^c_\mu. \tag{26.14}$$

Here D^{ab}_μ is the covariant derivative in the adjoint representation: the free derivative $\partial_\mu \delta^{ab}$ plus the gauge-field term $g f^{acb} A^c_\mu$ that rotates the colour index. Now compute. The condition Eq. (26.12) is linear in A, so its variation is just the divergence of δA:

$$\delta G^a(x) = \partial^\mu_x\, \delta A^a_\mu(x) = \partial^\mu_x \Big[\frac{1}{g}\, D^{ab}_\mu \alpha^b(x)\Big] = \frac{1}{g}\,\partial^\mu_x (\partial_\mu \alpha^a(x) + g f^{acb} A^c_\mu(x) \alpha^b(x)). \tag{26.15}$$

The functional derivative with respect to $\alpha^b(y)$ replaces $\alpha^b(x)$ by $\delta^4(x-y)$ and strips the colour contraction, giving

$$\begin{aligned} M^{ab}(x,y) &= \frac{\delta\, \delta G^a(x)}{\delta \alpha^b(y)}\bigg|_{\alpha=0} = \frac{1}{g}\,\partial^\mu_x \Big[\partial_\mu \delta^{ab}\, \delta^4(x-y) + g f^{acb} A^c_\mu(x)\, \delta^4(x-y)\Big] \\ &= \frac{1}{g}\,\partial^\mu \Big(\partial_\mu \delta^{ab} + g f^{acb} A^c_\mu\Big)\delta^4(x-y) = \frac{1}{g}\,\partial^\mu D^{ab}_\mu\, \delta^4(x-y). \end{aligned} \tag{26.16}$$

This is the central object of the section, and it is worth boxing:

$$\boxed{M^{ab}(x,y) = \frac{1}{g}\,\partial^\mu D^{ab}_\mu\, \delta^4(x-y) = \frac{1}{g}\,\partial^\mu(\partial_\mu \delta^{ab} + g f^{acb} A^c_\mu)\delta^4(x-y)} \tag{26.17}$$

in exact agreement with the result Eq. (13.168) previewed in Theorem 13.35. The overall $1/g$ is a field-independent constant that, like $\mathrm{Vol}(\mathcal{G})$, cancels in normalized correlators; the physics lives in the operator $\partial^\mu D^{ab}_\mu = \partial^\mu(\partial_\mu \delta^{ab} + g f^{acb} A^c_\mu)$, the divergence of the adjoint covariant derivative.

Now read off the obstruction. The operator Eq. (26.17) splits into two pieces of completely different character:

$$g\, M^{ab} = \underbrace{\partial^2\, \delta^{ab}}_{\text{field-independent}} + \underbrace{g f^{acb}\, \partial^\mu (A^c_\mu\, \cdot\,)}_{\text{linear in } A}. \tag{26.18}$$

The first term, $\partial^2 \delta^{ab}$, is $N^2 - 1$ decoupled copies (one per colour) of the abelian d'Alembertian; its determinant is the *field-independent* constant $\big[\det(\partial^2)\big]^{N^2-1}$, exactly $\dim(G)$ copies of the QED determinant $\det(\partial^2)$ of Eq. (13.164), with

$\dim(G) = N^2 - 1$ the dimension of the adjoint of SU(N) from Chapter B. If this term stood alone we would be back in the abelian situation: the determinant would factor out and Yang–Mills, like QED, would have no ghosts. It does not stand alone. The second term, $g f^{acb} \partial^\mu (A^c_\mu \cdot)$, is *linear in the gauge field*, and through it M — and hence $\det M$ — depends explicitly on A. Because the structure constants f^{acb} are nonzero precisely when the algebra is non-abelian, this term is the algebraic signature of non-commutativity. The contrast is total: in QED, $f^{abc} \equiv 0$, the second term vanishes identically, $M = \partial^2$ is a pure number-valued operator, and $\Delta_{\mathrm{FP}} = \det(\partial^2)$ is constant (Theorem 13.34); in Yang–Mills it cannot be removed.

Caution

It is tempting to think that since A is being integrated over and the gauge condition pins $\partial^\mu A_\mu = \omega$, the troublesome term is somehow "small" or ignorable. It is not. The delta function in Eq. (26.11) constrains the *divergence* $\partial^\mu A^a_\mu$, not A^a_μ itself; the gauge field appearing in M through $f^{acb} A^c_\mu$ is the full, unconstrained, fluctuating field. The determinant $\det M[A]$ is a bona fide functional of the integration variable, on the same footing as e^{iS}, and varies from configuration to configuration. There is no legitimate sense in which it can be pulled outside the $\mathcal{D}A$ integral.

Worked Example 26.2 (Where the field dependence lives: the free point and the first correction)**.** To see concretely that the entire ghost dynamics resides in the $g f^{acb} A^c_\mu$ term, expand $\det M$ around the free configuration $A = 0$ and identify the leading field-dependent contribution.

The free point. At $A = 0$ the operator Eq. (26.17) collapses to $M^{ab}_0(x, y) = \frac{1}{g}\, \partial^2 \delta^{ab}\, \delta^4(x - y)$, which is block-diagonal in colour. Its functional determinant factorizes over the $N^2 - 1$ adjoint colours:

$$\det M_0 = \prod_{a=1}^{N^2-1} \det\Big(\tfrac{1}{g}\,\partial^2\Big) = \Big(\tfrac{1}{g}\Big)^{(N^2-1)V_4} \big[\det(\partial^2)\big]^{N^2-1}, \tag{26.19}$$

with V_4 the (regularized) spacetime volume absorbing the $\frac{1}{g}$ per mode. Every factor is field-independent: $\det M_0$ is a constant, and if A never appeared in M this is all there would ever be — precisely $\dim(G) = N^2 - 1$ non-interacting copies of QED's $\det(\partial^2)$, contributing only an overall normalization. The absence of ghosts in QED is the statement that for the abelian group $\dim(G) = 1$ and there is no correction beyond this point.

The first correction. Write the full operator as $M = \frac{1}{g}(\partial^2 \mathbf{1} + g\, \partial^\mu A^{\mathrm{ad}}_\mu)$, where $(A^{\mathrm{ad}}_\mu)^{ab} = f^{acb} A^c_\mu$ is the gauge field in the adjoint representation and $\mathbf{1} = \delta^{ab}\delta^4(x - y)$. Use the identity $\ln \det M = \mathrm{Tr} \ln M$ and factor out the free

part:

$$\begin{aligned}\ln\det M &= \operatorname{Tr}\ln\Big[\tfrac{1}{g}\,\partial^2\big(\mathbf{1}+g\,(\partial^2)^{-1}\partial^\mu A_\mu^{\mathrm{ad}}\big)\Big]\\ &= \ln\det M_0+\operatorname{Tr}\ln\big(\mathbf{1}+g\,(\partial^2)^{-1}\partial^\mu A_\mu^{\mathrm{ad}}\big)\\ &= \ln\det M_0+g\,\operatorname{Tr}\big[(\partial^2)^{-1}\partial^\mu A_\mu^{\mathrm{ad}}\big]-\frac{g^2}{2}\operatorname{Tr}\big[(\partial^2)^{-1}\partial^\mu A_\mu^{\mathrm{ad}}\,(\partial^2)^{-1}\partial^\nu A_\nu^{\mathrm{ad}}\big]+\cdots,\end{aligned}\tag{26.20}$$

the logarithm expanded as $\ln(1+X)=X-\frac{1}{2}X^2+\cdots$. The $O(g)$ term carries a single adjoint trace $\operatorname{tr}_{\mathrm{ad}}(A_\mu^{\mathrm{ad}})\propto f^{aca}=0$ (the structure constants are antisymmetric in the traced pair), so it vanishes; the leading nonzero, field-dependent piece is the $O(g^2)$ term. It is quadratic in A, nonlocal (it carries two inverse d'Alembertians), and — read as a contribution to the effective action $-i\ln\det M$ — it is precisely a one-loop self-energy diagram for the gauge field built from a closed loop of the field carried by M. That field, defined by the propagator $(\partial^2)^{-1}$ in the adjoint and coupling to A_μ through f^{acb}, is the *Faddeev–Popov ghost*. This worked expansion is the quantitative content of the claim that "the determinant must be made dynamical": every term beyond Eq. (26.19) is a genuine interaction between the ghost and the gauge field, and discarding $\det M$ would discard exactly these contributions. The next section will reproduce Eq. (26.20) to all orders at one stroke by writing $\det M$ as a Gaussian integral over anticommuting ghost fields.

26.1.4 The honest caveat: the Gribov ambiguity

One assumption was smuggled into the resolution of unity Eq. (26.6) and must be stated openly, because it is true perturbatively and false globally, and the difference is a genuine feature of non-abelian gauge theory rather than a technicality. The construction assumes that for a generic gauge field the equation $G^a(A^U)=\omega^a$ has, as U ranges over the group, exactly *one* solution per orbit: the gauge slice $G(A)=\omega$ meets each orbit in a single point. This is what allowed us to read Eq. (26.6) as a clean insertion of unity localized at one representative, and it is what Fig. 26.1(a) depicts for the abelian theory, where it is exact: the slice $\partial^\mu A_\mu=\omega$ is an affine condition cutting each parallel straight orbit once.

For the non-abelian theory the assumption holds only locally. Near a solution A, whether a nearby gauge-equivalent configuration $A+\frac{1}{g}D_\mu\alpha$ also satisfies the Lorenz condition is governed by the zero modes of the very operator we just computed: $\partial^\mu(A_\mu+\frac{1}{g}D_\mu\alpha)=\omega$ together with $\partial^\mu A_\mu=\omega$ requires $\partial^\mu D_\mu^{ab}\alpha^b=0$, i.e. $M^{ab}\alpha^b=0$. As long as $M=\frac{1}{g}\partial^\mu D_\mu$ has no normalizable zero mode — which is guaranteed for small fields, where $M\approx\frac{1}{g}\partial^2$ is invertible (Eq. (26.19)) — the intersection is unique and the Faddeev–Popov construction is exact. This is the regime of perturbation theory, expansion about $A=0$, and it is the regime in which the entire rest of this chapter operates.

Globally the operator $M[A]$ *does* develop zero modes once A is large enough. Gribov showed that the Lorenz (and Coulomb) gauge conditions are met more than once on the same orbit: there exist distinct configurations A and A^U, both satisfying $\partial^\mu A_\mu = \omega$, related by a finite gauge transformation. These *Gribov copies* are the multiply-marked orbit of Fig. 26.1(b). Their existence means the gauge slice overcounts large-field configurations, that $\det M$ changes sign at the Gribov horizon where the first zero mode appears, and that the naive $\Delta_{\mathrm{FP}} = \det M$ (without an absolute value and without restricting to the region between horizons) is not, beyond perturbation theory, the correct measure on the quotient $\mathcal{A}/\mathcal{G}$.

Caution

None of this affects the perturbative Yang–Mills of Section 26.1 through Chapter 27. The Gribov copies live at field amplitudes invisible to any finite order of the loop expansion: the determinant is positive throughout the perturbative region around $A = 0$, the gauge slice is locally one-to-one, and the Feynman rules, β-function, and unitarity we derive are untouched. The Gribov ambiguity is a statement about the global, non-perturbative definition of the gauge-fixed measure — about confinement-scale physics — and its proper treatment (the Gribov–Zwanziger restriction to the fundamental modular region) belongs to the non-perturbative part of the book. We flag it here, as honesty requires, and defer it to Chapter 42.

We have now isolated the obstruction with complete precision. The gauge-fixed generating functional Eq. (26.11) carries, sitting inside the integrand on the same footing as $e^{iS_{\mathrm{YM}}}$, the field-dependent factor $\Delta_{\mathrm{FP}}[A] = \det M = \det(\frac{1}{g}\partial^\mu D^{ab}_\mu)$, a functional of A that — unlike its abelian ancestor $\det(\partial^2)$ — cannot be pulled out, cannot be cancelled against $\mathrm{Vol}(\mathcal{G})$, and cannot be ignored. A determinant in the integrand is not yet a Lagrangian; to do perturbation theory we must convert it into a local term in the action with its own propagator and vertices. The next section does exactly that: it writes $\det M$ as a Gaussian functional integral over a pair of anticommuting scalar fields — the Faddeev–Popov ghost and antighost — thereby promoting the determinant from a static weight to a dynamical field that circulates in loops, and assembles, together with the gauge-averaged R_ξ term, the complete gauge-fixed Yang–Mills Lagrangian.

26.2 The Faddeev–Popov Determinant and the Ghost Lagrangian

The previous section left us holding an object that has no place in a Lagrangian. Slicing the Yang–Mills path integral transverse to its gauge orbits produced, exactly as in the abelian construction of Section 13.5, a Jacobian factor $\det M$

guarding the gauge slice — but here the operator

$$M^{ab}(x,y) \;=\; \frac{\delta G^a(x)}{\delta \alpha^b(y)} \;=\; \tfrac{1}{g}\big(\partial^\mu D_\mu^{ab}\big)\,\delta^{(4)}(x-y), \qquad G^a(A)=\partial^\mu A_\mu^a, \quad (26.21)$$

depends on the gauge field through the connection inside the adjoint covariant derivative $D_\mu^{ab}=\delta^{ab}\partial_\mu + g f^{aeb}A_\mu^e$ of Eq. (25.35). In QED the analogous operator was the free Laplacian ∂^2, a constant that factored out of $\int \mathcal{D}A\, e^{iS}$ and cancelled in every normalized correlator (Section 13.5.5); that is the precise reason photons travel without ghosts. The non-abelian determinant cannot be evicted, because the field A_μ^e on which it depends is the very thing being integrated. *It is part of the dynamics.* The entire problem of quantizing Yang–Mills now reduces to a single technical demand: write $\det M$ as a local contribution to the action, so that the gauge-fixed path integral is once again of the standard form $\int e^{iS_{\text{eff}}}$ with a Lagrangian we can read Feynman rules from. This section meets that demand. The device that does it — representing a determinant as a Gaussian integral over anticommuting fields — forces two new fields into the theory, the Faddeev–Popov ghost and antighost, and the wrong-statistics character they are obliged to carry is not an accident but a direct consequence of which power of the determinant we need.

We organize the construction in four movements: exponentiate $\det M$ into the action (producing the ghosts), read off the ghost Lagrangian and its single interaction vertex, average over the gauge slice to recover the R_ξ gauge-fixing term, and assemble the complete gauge-fixed Lagrangian that every subsequent section dissects.

26.2.1 Exponentiating the determinant: ghosts as a Grassmann Gaussian

Recall first, from the abelian discussion, the shape of the gauge-fixed generating functional immediately after the Faddeev–Popov resolution of unity is inserted and the gauge-orbit volume divided out. For the non-abelian theory the identical manipulation delivers

$$Z[\omega] \;=\; \int \mathcal{D}A\;\det M[A]\; e^{iS_{\text{YM}}[A]}\,\delta\big(\partial^\mu A_\mu^a-\omega^a\big), \quad (26.22)$$

with $S_{\text{YM}}=\int d^4x\,(-\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu})$ the classical action of Eq. (25.76) and $M[A]$ the field-dependent operator Eq. (26.21). The lone novelty relative to Eq. (13.172) is that $\det M[A]$ now sits *inside* the A-integral; it can no longer be pulled out front as a constant.

The resolution is the central identity of Grassmann calculus established in Chapter 12. There we proved (Theorem 12.26, Eq. (12.81)) that for *any* complex

$n \times n$ matrix A, with no hypothesis of Hermiticity or positivity,

$$\int \mathcal{D}\bar{\theta}\,\mathcal{D}\theta\; e^{-\bar{\theta}A\theta} \;=\; \det A, \qquad \bar{\theta}A\theta = \sum_{i,j} \bar{\theta}_i A_{ij} \theta_j, \tag{26.23}$$

where $\theta_i, \bar{\theta}_i$ are independent anticommuting Berezin generators. The contrast with the bosonic Gaussians of Chapter 4 is exactly the contrast we need: a complex *bosonic* Gaussian integrates to $(\det A)^{-1}$, while the complex *Grassmann* Gaussian integrates to the positive power $(\det A)^{+1}$. We hold $\det M$, the positive power; therefore the fields we must introduce are Grassmann, not ordinary, and this single observation fixes their statistics once and for all (Section 26.2.2).

Promote the finite-dimensional index i to the continuum pair (spacetime point x, colour index a), and the matrix A to the operator $M^{ab}(x,y)$ of Eq. (26.21). Introduce a pair of anticommuting, colour-adjoint, Lorentz *scalar* fields.

Definition 26.3 (Faddeev–Popov ghost and antighost)**.** The *ghost* $c^a(x)$ and *antighost* $\bar{c}^a(x)$ are anticommuting (Grassmann-valued) classical fields, one of each for every generator $a = 1, \dots, \dim \mathfrak{g}$, transforming in the adjoint representation of the gauge group and carrying spin zero. They obey $\{c^a(x), c^b(y)\} = \{\bar{c}^a(x), \bar{c}^b(y)\} = \{c^a(x), \bar{c}^b(y)\} = 0$. The antighost $\bar{c}^a$ is an *independent* integration field, not the Hermitian conjugate of c^a; this independence is what makes the sign manipulations below legal.

With these fields the operator determinant is a Grassmann Gaussian.

Theorem 26.4 (Ghost representation of the Faddeev–Popov determinant)**.** *Up to a field-independent multiplicative constant absorbed into the overall normalization of the path integral,*

$$\det M[A] \;=\; \mathcal{N} \int \mathcal{D}\bar{c}\,\mathcal{D}c\; \exp\Big(i \int d^4x\, d^4y\; \bar{c}^a(x)\, M^{ab}(x,y)\, c^b(y) \Big) \;=\; \mathcal{N} \int \mathcal{D}\bar{c}\,\mathcal{D}c\; \exp\Big(i \int d^4x \tag{26.24}$$

where the second equality uses the locality of M^{ab} in Eq. (26.21) to collapse one spacetime integral. The constant $\mathcal{N}$ is independent of A^a_μ and cancels in every normalized correlator.

Proof. Apply Eq. (26.23) with the choice $A = -iM$, treating the composite label (x,a) as the matrix index and the functional integral $\mathcal{D}\bar{c}\,\mathcal{D}c$ as the (regularized) Berezin measure:

$$\int \mathcal{D}\bar{c}\,\mathcal{D}c\; \exp\Big(i \int \bar{c}^a M^{ab} c^b\Big) = \int \mathcal{D}\bar{c}\,\mathcal{D}c\; \exp\big(-\bar{c}\,(-iM)\,c\big) = \det(-iM). \tag{26.25}$$

It remains to show $\det(-iM) = \mathcal{N} \det M$ with $\mathcal{N}$ field-independent. Factor the scalar $-i$ out of the operator, $-iM = (-i\,\mathbf{1})\,M$, and use the multiplicativity of the determinant,

$$\det(-iM) = \det(-i\,\mathbf{1})\, \det M = (-i)^{\dim} \det M \equiv \mathcal{N} \det M, \tag{26.26}$$

where dim is the (regularized) dimension of the operator's index space — the number of (x, a) labels. That number is set by the spacetime volume and the dimension of the algebra; it knows nothing about the value of A^a_μ. Hence $\mathcal{N} = (-i)^{\dim}$ is field-independent, and Eq. (26.24) follows. Theorem 26.5 verifies the field-independence rigorously through the trace-log identity, without appeal to a formal counting of indices. □

The Grassmann integration has done precisely what was needed: it has converted the non-local, field-dependent functional $\det M[A]$ into the exponential of a *local* action bilinear in two new fields. Substituting Eq. (26.24) into Eq. (26.22) and discarding the constant $\mathcal{N}$,

$$Z[\omega] \;=\; \int \mathcal{D}A\,\mathcal{D}\bar{c}\,\mathcal{D}c\;\exp\Big(i\int d^4x\,\big[\mathcal{L}_{\mathrm{YM}} + \bar{c}^{\,a}\,\partial^\mu D^{ab}_\mu\,c^b\big]\Big)\,\delta(\partial^\mu A^a_\mu - \omega^a). \tag{26.27}$$

The determinant is gone; in its place sits a genuine dynamical sector, the ghost fields, integrated on the same footing as the gauge field. They are not external sources or spectators: through D^{ab}_μ they couple to A^a_μ, and they will circulate in loops.

Worked Example 26.5 (Field-independence of the prefactor via $\det = \exp \mathrm{Tr}\ln$)**.** The counting argument in Eq. (26.26) can be made airtight using the identity $\det = \exp \mathrm{Tr}\ln$ (Theorem 12.47), which is the proper continuum statement of "$(-i)^{\dim}$". Write the prefactor's logarithm as

$$\ln\frac{\det(-iM)}{\det M} = \mathrm{Tr}\ln(-iM) - \mathrm{Tr}\ln M = \mathrm{Tr}\,\big[\ln(-i\,\mathbf{1}) + \ln M\big] - \mathrm{Tr}\ln M = \ln(-i)\;\mathrm{Tr}\;\mathbf{1}, \tag{26.28}$$

where the middle step used $\ln(XY) = \ln X + \ln Y$ for $X = -i\,\mathbf{1}$ commuting with everything, and the last collapsed $\mathrm{Tr}\ln(-i\,\mathbf{1}) = \ln(-i)\,\mathrm{Tr}\;\mathbf{1}$. The result depends on M only through $\mathrm{Tr}\;\mathbf{1}$, the trace of the identity on the index space — the regularized dimension. Since the identity operator carries no dependence on A^a_μ, neither does $\mathrm{Tr}\;\mathbf{1}$, and the prefactor $\exp(\ln(-i)\;\mathrm{Tr}\;\mathbf{1})$ is field-independent, confirming Theorem 26.4. In dimensional regularization $\mathrm{Tr}\;\mathbf{1} \propto \delta^{(4)}(0) \propto \int d^dk\,1 = 0$, so the prefactor is in fact unity and need not be tracked at all — the same mechanism that sets scaleless integrals to zero throughout the loop expansion of Chapter 27.

26.2.2 Why the ghosts must be anticommuting

It is worth pausing to make explicit the logical force of the construction, because the ghosts violate the spin-statistics connection — they are spin-zero fields obeying anticommutation relations — and a reader is right to demand why such an object is permitted, indeed required.

The answer is contained entirely in the sign of the exponent of the determinant. The Faddeev–Popov procedure delivers the determinant in the *numerator*, $\det M$,

because it arose as the Jacobian relating the integral over the gauge-fixing function to the integral over the gauge parameter — a Jacobian appears upstairs, never downstairs. Now compare the two ways to manufacture a determinant as a Gaussian:

$$\underbrace{\int \mathcal{D}\phi^\dagger\, \mathcal{D}\phi\; e^{-\phi^\dagger A \phi} = (\det A)^{-1}}_{\text{commuting fields (Theorem 4.17)}}, \qquad \underbrace{\int \mathcal{D}\bar{\theta}\, \mathcal{D}\theta\; e^{-\bar{\theta} A \theta} = (\det A)^{+1}}_{\text{anticommuting fields (Theorem 12.26)}} . \tag{26.29}$$

A determinant in the numerator can be produced *only* by anticommuting integration variables; commuting fields give its inverse. Were the ghosts ordinary commuting scalars, the right-hand side of Eq. (26.24) would compute $1/\det M$, the wrong power, and the gauge-orbit overcounting would be miscorrected — compounded rather than removed. The positive power $\det M$, traced through Theorem 12.47 to the antiperiodicity of Grassmann generators, forces the anticommuting statistics. The price is a field that cannot be a physical particle: a Grassmann scalar has the wrong relation between spin and statistics, so by the spin-statistics theorem (Chapter 12) it cannot appear in a physical asymptotic state with positive-norm one-particle states. This is not a defect to be repaired but the defining feature of the ghost: it is an *unphysical bookkeeping field*, present only to enforce, order by order in g, the cancellation of the unphysical (longitudinal and timelike) gluon polarizations that the covariant gauge-fixing reintroduced. That cancellation, and the exact symmetry (BRST) that guarantees it survives renormalization, is the content of Section 26.5; for now we record only that the wrong statistics is logically mandatory, not optional.

Physical Insight

The ghost is the price of manifest Lorentz covariance. One could quantize Yang–Mills in a physical gauge (axial, $n^\mu A^a_\mu = 0$) in which the determinant is field-independent and no ghost appears — at the cost of a propagator that breaks manifest covariance and is far harder to use in loops. The covariant R_ξ gauges keep Lorentz invariance manifest and pay for it with two unphysical gluon polarizations per gluon; the ghost is exactly the negative-norm degree of freedom whose loops subtract those two unphysical polarizations from every gauge-invariant quantity. Counting confirms it: a massless gluon has 2 physical polarizations but 4 covariant components; the ghost–antighost pair contributes -2 to the effective state count, restoring the balance. The minus sign is the anticommuting statistics at work.

26.2.3 The ghost Lagrangian and the ghost–gluon vertex

Read the ghost contribution to the action off the exponent of Eq. (26.27):

$$\mathcal{L}_{\text{ghost}} \;=\; \bar{c}^{\,a}\, \partial^\mu D^{ab}_\mu\, c^b. \tag{26.30}$$

To expose the kinetic term and the interaction it is cleaner to integrate by parts, moving the outer derivative off $\bar{c}^a$. Discarding the total derivative $\partial^\mu(\bar{c}^a D^{ab}_\mu c^b)$, which integrates to a boundary term,

$$\mathcal{L}_{\text{ghost}} \;\simeq\; -\,\partial^\mu \bar{c}^a\, D^{ab}_\mu\, c^b, \tag{26.31}$$

the form quoted in the master Lagrangian below (here "$\simeq$" denotes equality of the actions, i.e. up to a total derivative). Now insert the explicit adjoint covariant derivative $D^{ab}_\mu = \delta^{ab}\partial_\mu + g f^{aeb} A^e_\mu$ from Eq. (25.35):

$$\begin{aligned} \mathcal{L}_{\text{ghost}} &= -\,\partial^\mu \bar{c}^a\,(\delta^{ab}\partial_\mu + g f^{aeb} A^e_\mu) c^b \\ &= \underbrace{-\,\partial^\mu \bar{c}^a\,\partial_\mu c^a}_{\text{kinetic}}\;\underbrace{-\,g\, f^{abc}\,(\partial^\mu \bar{c}^a) A^b_\mu\, c^c}_{\text{ghost–gluon coupling}}, \end{aligned} \tag{26.32}$$

where in the interaction the dummy colour indices were renamed $e \to b$, $b \to c$ to put it in the canonical order. Two structures appear, and only two.

The first, $-\partial^\mu \bar{c}^a \partial_\mu c^a$, is a free kinetic term: a massless, colour-diagonal propagating scalar for each generator. It is the seed of the ghost propagator extracted in Section 26.3. The second, $-g f^{abc}(\partial^\mu \bar{c}^a) A^b_\mu c^c$, is the *ghost–gluon vertex*: a single cubic interaction joining one antighost, one ghost, and one gluon, carrying one power of g, one totally antisymmetric structure constant f^{abc}, and one derivative. Three features of this vertex will matter downstream and are worth naming now.

First, the derivative falls on the *antighost* alone — $\partial^\mu \bar{c}^a$ — and not symmetrically on both ghost lines. The ghost and antighost are therefore not interchangeable at the vertex: the line carrying the derivative is determined, and in momentum space the vertex factor will be proportional to the *outgoing antighost momentum*, not to the ghost momentum. This asymmetry is the Lagrangian fingerprint of the arrow on the ghost line.

Second, the colour factor is a single f^{abc}, identical in structure to one of the two structure constants appearing in the three-gluon vertex Eq. (25.107). This is the first hint of the deep tie between the ghost sector and the gluon self-coupling that BRST symmetry will make exact and that drives the asymptotic-freedom computation of Chapter 27: ghosts and gluons are bound by the same gauge structure.

Third, there is no ghost–ghost–gluon–gluon (quartic) vertex and no ghost mass term: the adjoint covariant derivative is linear in A, so $\mathcal{L}_{\text{ghost}}$ contains exactly one interaction, of cubic order. The ghost sector is, in this precise sense, minimal.

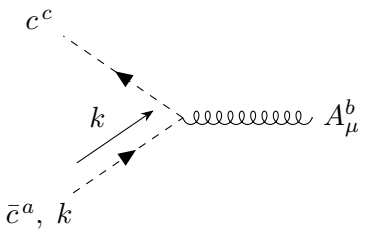


Figure 26.2: The ghost–gluon vertex, the single interaction of the ghost sector, from the coupling $-gf^{abc}(\partial^\mu \bar{c}^a)A^b_\mu c^c$ in Eq. (26.32). Ghost lines are dashed and directed (the arrow distinguishes ghost from antighost); the gluon is the curly line. The derivative acts on the antighost, so the momentum-space vertex is proportional to the antighost momentum k^μ contracted into the gluon index μ, with colour weight gf^{abc}. This is the new diagrammatic element absent from QED; its momentum-space value is extracted in Section 26.3.

Notation Watch

The overall sign of $\mathcal{L}_{\text{ghost}}$ is a convention, because $\bar{c}^a$ is an independent Grassmann integration field (Theorem 26.3), not the conjugate of c^a. Relabelling $\bar{c}^a \to -\bar{c}^a$ in Eq. (26.24) — a change of dummy variable whose measure Jacobian is a field-independent sign absorbed into $\mathcal{N}$ — flips Eq. (26.31) to $+\partial^\mu \bar{c}^a D^{ab}_\mu c^b$, giving the kinetic term the conventional positive sign $+\partial^\mu \bar{c}^a \partial_\mu c^a$ and the vertex $+gf^{abc}(\partial^\mu \bar{c}^a)A^b_\mu c^c$. This is the form used by Peskin–Schroeder and most texts. We retain the sign that follows directly from the determinant identity Eq. (26.24); since the two differ only by a field-independent relabelling, $\det M$ and every Green's function are identical, and the Feynman rules of Section 26.3 are read off consistently from whichever convention is fixed. What is *not* a convention is the relative structure: derivative on the antighost, colour weight f^{abc}, one power of g.

26.2.4 Averaging over the gauge slice: the R_ξ gauge-fixing term

Equation Eq. (26.27) still carries the delta function $\delta(\partial^\mu A^a_\mu - \omega^a)$ and the arbitrary slice function ω^a. Both are removed by the same averaging trick that produced the abelian gauge-fixing term in Section 13.5.5; the non-abelian case differs from it in not a single step, because the manipulation acts only on the gauge field and the slice, and is blind to the ghost sector now sitting in the action.

The key observation is unchanged: $Z[\omega]$ does not depend on ω^a. Shifting the slice slides the chosen representative along each orbit, and both the measure $\mathcal{D}A\,\mathcal{D}\bar{c}\,\mathcal{D}c$ and the integrand are invariant under that slide (the ghost action was built precisely to make the orbit measure invariant). We are therefore free to

average $Z[\omega]$ over ω^a with any normalized weight; the value changes only by a constant. Choose, as in the abelian case, the Gaussian weight

$$Z \;\propto\; \int \mathcal{D}\omega\; \exp\Big(-\frac{i}{2\xi}\int d^4x\; \omega^a\omega^a\Big)\; Z[\omega], \qquad \xi \in (0,\infty), \tag{26.33}$$

with ξ a real gauge parameter and the proportionality absorbing the (ω-independent) Gaussian normalization. Substituting Eq. (26.27),

$$Z \;\propto\; \int \mathcal{D}A\,\mathcal{D}\bar{c}\,\mathcal{D}c\; e^{\,i\int(\mathcal{L}_{\mathrm{YM}}+\mathcal{L}_{\mathrm{ghost}})} \int \mathcal{D}\omega\; \exp\Big(-\frac{i}{2\xi}\int \omega^a\omega^a\Big)\,\delta(\partial^\mu A^a_\mu - \omega^a). \tag{26.34}$$

The ω-integral is now trivial: the delta function sets $\omega^a = \partial^\mu A^a_\mu$ at every point, so it replaces $\int \omega^a\omega^a$ by $\int(\partial^\mu A^a_\mu)^2$ and removes itself. What remains is

$$Z \;\propto\; \int \mathcal{D}A\,\mathcal{D}\bar{c}\,\mathcal{D}c\; \exp\Big(i\int d^4x\,\Big[\mathcal{L}_{\mathrm{YM}} - \frac{1}{2\xi}(\partial^\mu A^a_\mu)^2 + \mathcal{L}_{\mathrm{ghost}}\Big]\Big), \tag{26.35}$$

in which the delta function has been traded for the local *gauge-fixing term*

$$\mathcal{L}_{\mathrm{gf}} \;=\; -\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2, \tag{26.36}$$

its sign and its factor $\frac{1}{2\xi}$ fixed, exactly as for the photon. This is the non-abelian generalization of the abelian gauge-fixed action Eq. (13.176): the colour index a is now summed over $\dim\mathfrak{g}$ values, and — the decisive difference — it is accompanied by the ghost term Eq. (26.31) that the field-dependent determinant left behind. In QED that determinant was constant, $\mathcal{L}_{\mathrm{ghost}}$ reduced to a free decoupled spectator, and one discarded it (Section 13.5.5); here it stays.

Caution

The averaging step inherits the caveat flagged in the abelian construction (Section 13.5) and made sharp for the non-abelian case in the chapter introduction: the Faddeev–Popov resolution of unity assumes the covariant slice $\partial^\mu A^a_\mu = \omega^a$ meets each gauge orbit exactly once. Beyond perturbation theory this fails — a covariant slice generically intersects an orbit several times (the Gribov ambiguity), and $\det M$ can change sign as one crosses a Gribov horizon where M develops a zero mode. The construction of this section is therefore exact only *perturbatively*, expanding about $A = 0$ where $M \approx \partial^2$ is invertible and positive. The global obstruction is real and as of the mid-2020s has no fully satisfactory resolution; it is taken up where it bites, in the non-perturbative treatment of Chapter 42. Everything that follows in this chapter and the next lives inside the perturbative regime where the obstruction is invisible.

26.2.5 Assembly: the complete gauge-fixed Yang–Mills Lagrangian

Reading the exponent of Eq. (26.35) and restoring the matter sector of Yang–Mills theory gives the object that the remainder of the chapter — and all of Chapter 27 — takes as its starting point.

$$\boxed{\mathcal{L} \;=\; -\tfrac{1}{4}\,F^{a}_{\mu\nu}F^{a\,\mu\nu} \;-\; \frac{1}{2\xi}\big(\partial^{\mu}A^{a}_{\mu}\big)^{2} \;-\; \partial^{\mu}\bar{c}^{a}\,D^{ab}_{\mu}\,c^{b} \;+\; \bar{\psi}_{i}\,(i\not{D}-m)\,\psi_{i}} \tag{26.37}$$

with $F^{a}_{\mu\nu}=\partial_\mu A^a_\nu-\partial_\nu A^a_\mu+gf^{abc}A^b_\mu A^c_\nu$ (Eq. (25.49)), the fundamental-representation matter covariant derivative $D_\mu=\partial_\mu-igA^a_\mu T^a$, and the adjoint covariant derivative $D^{ab}_\mu=\delta^{ab}\partial_\mu+gf^{aeb}A^e_\mu$ (Eq. (25.35)) acting on the ghost. The default for explicit computation is Feynman gauge $\xi=1$, as declared in the chapter introduction. Three independent sectors are now visible: the gauge field, the ghosts, and the matter — coupled, but each with its own kinetic term and a well-defined propagator, the gauge-fixing term having lifted the longitudinal zero mode of the gluon exactly as it did for the photon in Section 13.5.6.

The full set of interaction vertices follows by expanding the squares and covariant derivatives in Eq. (26.37). We catalogue them, because the next section turns each into a Feynman rule.

Worked Example 26.6 (Every interaction term in the gauge-fixed Lagrangian)**.** We expand Eq. (26.37) term by term and collect the complete vertex content.

(i) The pure-gauge sector. The expansion of $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ was carried out in full in Section 25.5.2; we import the result (Eq. (25.104)) rather than repeat it:

$$-\tfrac{1}{4}F^{a}_{\mu\nu}F^{a\,\mu\nu}=\underbrace{-\tfrac{1}{4}\big(\partial_\mu A^a_\nu-\partial_\nu A^a_\mu\big)^2}_{\text{gluon kinetic}}\underbrace{-\,gf^{abc}\big(\partial_\mu A^a_\nu\big)A^{b\,\mu}A^{c\,\nu}}_{\text{three-gluon, Eq. (25.107)}}\underbrace{-\,\tfrac{1}{4}g^2f^{abe}f^{cde}A^a_\mu A^b_\nu A^{c\,\mu}A^{d\,\nu}}_{\text{four-gluon, Eq. (25.110)}}. \tag{26.38}$$

The cubic vertex carries one f, one derivative, one power of g; the quartic carries two f's sharing a summed index, no derivative, and g^2. These are the gluon self-interactions, the genuinely non-abelian content absent from QED.

(ii) The gauge-fixing term $-\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2$ is purely quadratic. Integrated by parts it reads $+\frac{1}{2\xi}A^a_\mu\partial^\mu\partial^\nu A^a_\nu$, which combines with the gluon kinetic term to make the gluon quadratic operator invertible (Eq. (13.177) with a colour-diagonal δ^{ab}); it contributes no vertex.

(iii) The ghost sector $-\partial^\mu\bar{c}^aD^{ab}_\mu c^b$ was expanded in Eq. (26.32): a colour-diagonal kinetic term $-\partial^\mu\bar{c}^a\partial_\mu c^a$ and the single ghost–gluon vertex $-gf^{abc}(\partial^\mu\bar{c}^a)A^b_\mu c^c$ of Fig. 26.2.

(iv) The matter sector $\bar\psi_i(i\not{D}-m)\psi_i$ with $D_\mu=\partial_\mu-igA^a_\mu T^a$ expands to the free Dirac term $\bar\psi_i(i\not\partial-m)\psi_i$ plus the quark–gluon vertex

$$\mathcal{L}_{\mathrm{q\bar{q}g}}=\bar\psi_i\,(i)(-igA^a_\mu T^a)\gamma^\mu\psi_i=g\,\bar\psi_i\,\gamma^\mu T^a\,\psi_i\,A^a_\mu, \tag{26.39}$$

the direct non-abelian analogue of the QED vertex of Chapter 17, now carrying the fundamental generator T^a in place of the electric charge.

The complete catalogue. The gauge-fixed theory has *four* propagating fields (gluon, ghost, antighost regarded as the ghost's partner, quark) and exactly *five* interaction vertices: the three-gluon and four-gluon self-couplings Eq. (25.107), Eq. (25.110); the ghost–gluon coupling Eq. (26.32); and the quark–gluon coupling Eq. (26.39). Every coupling carries the same g, and the ghost and the two gluon self-couplings all carry the structure constant f^{abc} — a uniformity that is no coincidence but the imprint of the single underlying gauge symmetry, made precise by the Slavnov–Taylor identities of Section 26.7.

The Lagrangian Eq. (26.37) is the central result of this chapter. It is a perfectly ordinary local field theory — quadratic kinetic terms with invertible propagators, polynomial interactions — to which the entire machinery of Parts IV–VI applies verbatim, with one new ingredient (the ghost) and one new vertex (Fig. 26.2). What it conspicuously is *not* is manifestly gauge invariant: the gauge-fixing and ghost terms break the symmetry that defined the classical theory. That the physics nonetheless remains gauge invariant is guaranteed by a residual rigid supersymmetry relating the gluon, ghost, and antighost — BRST symmetry — which is the subject of Section 26.5. Before reaching it we extract from Eq. (26.37) the momentum-space propagators and vertices: the gluon and ghost propagators and the three-gluon, four-gluon, and ghost–gluon vertices, with their full colour and Lorentz structure. That is the computational result of the next section.

26.3 Feynman Rules of Gauge-Fixed Yang–Mills

The previous section closed the construction that the whole chapter was organized around: starting from the field-dependent Faddeev–Popov determinant $\det M$ (Eq. (26.17)), we exponentiated it with a pair of anticommuting scalar fields, averaged over the gauge condition with a Gaussian weight, and arrived at the complete gauge-fixed Yang–Mills Lagrangian (Eq. (26.37)),

$$\mathcal{L} \;=\; -\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} \;-\; \frac{1}{2\xi}\big(\partial^\mu A^a_\mu\big)^2 \;+\; \big(\partial^\mu \bar{c}^{\,a}\big)\big(D_\mu c\big)^a \;+\; \bar{\psi}_i\big(i\not{D} - m\big)\psi_i, \quad (26.40)$$

with the adjoint covariant derivative acting on the ghost, $(D_\mu c)^a = \partial_\mu c^a + g f^{abc} A^b_\mu c^c$ (Eq. (26.31)). From here on we adopt the conventional positive ghost-kinetic sign of Peskin–Schroeder, i.e. the relabelling $\bar{c}^a \to -\bar{c}^a$ flagged in the notationwatch following Eq. (26.37); this flips the sign of the ghost term relative to the boxed master Lagrangian and, being a field-independent change of integration variable, leaves $\det M$ and every Green's function untouched. This is a perfectly ordinary local Lagrangian: it has a quadratic part that defines

free propagation and a set of higher-order terms that define interactions, and from those two pieces one extracts momentum-space Feynman rules by exactly the procedure used for ϕ^4 in Chapter 15 and for QED in Chapter 17. That extraction is the business of this section. The result is a closed set of rules—two propagators (gluon and ghost) and four vertices (three-gluon, four-gluon, ghost-gluon, and, for completeness, quark-gluon)—stated with their full color and Lorentz structure and derived in full from Eq. (26.40). These rules are what Chapter 27 feeds directly into the one-loop computation that produces the Yang–Mills beta function and asymptotic freedom; getting every sign, momentum factor, and combinatorial coefficient right here is therefore not bookkeeping but the foundation of the chapter's quantitative payoff.

Two features distinguish this set from its abelian predecessor and will recur at every step. First, color is now carried explicitly: every propagator wears an adjoint index pair and every vertex a tensor of structure constants, with the group-theory data (f^{abc}, the Casimirs C_A, C_F, the index T_F) collected in Section B.4. Second, and decisively, the gauge field couples to itself, so the rule set contains pure-gauge vertices—a cubic and a quartic gluon self-coupling—that simply do not exist in electrodynamics (Fig. 25.3), together with a ghost sector that has no abelian analog at all. We adopt Feynman gauge $\xi = 1$ as the default for explicit computation, exactly as in QED, stating it whenever a result depends on ξ.

26.3.1 Conventions for reading off a vertex

We fix once and for all the convention by which a term in the Lagrangian becomes a momentum-space Feynman rule, because for the multi-gluon vertices the symmetrization over identical legs is where errors breed. Write the action as $S = \int d^4x\, \mathcal{L}$ and split $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_{\text{int}}$ into its quadratic (free) part and the rest. The perturbative expansion of the path integral expands $e^{iS} = e^{iS_0}\, e^{iS_{\text{int}}}$, so each vertex enters an amplitude through one factor of iS_{int}. The Feynman rule for an n-point vertex is obtained by functionally differentiating iS_{int} with respect to the n fields attached to it, passing to momentum space with the standing Fourier convention $A(x) = \int \frac{d^4k}{(2\pi)^4} e^{-ik\cdot x} \widetilde{A}(k)$ (so every ∂_μ acting on an incoming field becomes $-ik_\mu$), and stripping the overall $(2\pi)^4\delta^4(\sum k)$ that enforces momentum conservation. Concretely,

$$iV_{\ell_1\cdots\ell_n} \;=\; i\,\frac{\delta^{\,n} S_{\text{int}}}{\delta\widetilde{\Phi}_{\ell_1}\cdots\delta\widetilde{\Phi}_{\ell_n}}\bigg|_{\text{amputated, } (2\pi)^4\delta^4 \text{ stripped}}, \tag{26.41}$$

where ℓ_i collects the discrete labels (color, Lorentz) and the momentum of the i-th external leg, and $\widetilde{\Phi}$ runs over the relevant fields. The n derivatives can hit the (identical-type) fields in $\mathcal{L}_{\text{int}}$ in $n!$/(internal symmetry) distinct ways, and the *sum over these assignments is the symmetrization* that Bose (or, for ghosts,

Fermi) statistics demands. Throughout we use the *all-incoming* momentum convention: every external momentum at a vertex is counted as flowing *into* the vertex, so momentum conservation reads $\sum_i k_i = 0$. With this convention the vertices below are totally symmetric (up to statistics) under simultaneous permutation of the legs' labels and momenta, which is the cleanest form for the loop computations of Chapter 27.

26.3.2 The gluon propagator

Free propagation is governed entirely by the part of Eq. (26.40) quadratic in A. From the expansion of $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ carried out classically in Chapter 25, the only quadratic gauge piece is the Maxwell-like term $\mathcal{L}_2 = -\frac{1}{4}(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)^2$ (Eq. (25.104)); the cubic and quartic terms are interactions and play no role here. Adding the gauge-fixing term gives the quadratic gauge Lagrangian

$$\mathcal{L}_2 + \mathcal{L}_{\text{gf}} \;=\; -\tfrac{1}{4}\big(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu\big)^2 \;-\; \frac{1}{2\xi}\big(\partial^\mu A^a_\mu\big)^2. \tag{26.42}$$

The single, decisive observation is that Eq. (26.42) is *diagonal in color*: the adjoint index a is summed in place, with no structure constant anywhere, so this is precisely $N^2 - 1$ independent copies—one per generator of SU(N)—of the abelian gauge-fixed Lagrangian whose propagator was constructed in Chapter 13. Every step of that construction transfers verbatim, with a spectator Kronecker delta δ^{ab} riding along in color space. We carry the inversion explicitly so the transfer is beyond doubt.

Integrating Eq. (26.42) by parts to expose the quadratic kernel and passing to momentum space (each $\partial_\mu \to -ik_\mu$, so $\Box \to -k^2$ and $\partial_\mu\partial_\nu \to -k_\mu k_\nu$) gives the algebraic operator

$$\mathcal{O}^{ab}_{\mu\nu}(k) \;=\; \delta^{ab}\Big[-k^2\eta_{\mu\nu} + \Big(1 - \tfrac{1}{\xi}\Big)k_\mu k_\nu\Big] \;=\; \delta^{ab}\,\mathcal{O}_{\mu\nu}(k), \tag{26.43}$$

the color-singlet factor $\mathcal{O}_{\mu\nu}(k)$ being identically the abelian kernel Eq. (13.129). The gluon propagator $D^{ab}_{\mu\nu}(k)$ is the Green's function of this operator with the i of the Minkowski path integral, $\mathcal{O}^{ac\,\mu\rho}(k)\,D^{cb}_{\rho\nu}(k) = i\,\delta^{ab}\delta^\mu{}_\nu$. Because $\mathcal{O}^{ab}$ is proportional to δ^{ab}, so is its inverse, and the Lorentz inversion is the one already done. Using the transverse and longitudinal projectors $\theta_{\mu\nu} = \eta_{\mu\nu} - k_\mu k_\nu/k^2$ and $\omega_{\mu\nu} = k_\mu k_\nu/k^2$ of Eq. (13.131), the singlet kernel is diagonalized as $\mathcal{O}_{\mu\nu} = -k^2\theta_{\mu\nu} - (k^2/\xi)\,\omega_{\mu\nu}$, and since θ, ω are orthogonal idempotents the inverse is the term-by-term reciprocal Eq. (13.133), $(\mathcal{O}^{-1})_{\mu\nu} = -k^{-2}\theta_{\mu\nu} - \xi k^{-2}\omega_{\mu\nu}$. Multiplying by $i\,\delta^{ab}$ and re-expanding the projectors via $\theta + \xi\omega = \eta - (1-\xi)\omega$ reproduces the R_ξ tensor Eq. (13.135), now dressed with color.

Theorem 26.7 (The gluon propagator in R_ξ gauge)**.** *The momentum-space gluon two-point function in the R_ξ gauge family is*

$$\boxed{D^{ab}_{\mu\nu}(k) \;=\; \frac{-i\,\delta^{ab}}{k^2+i\varepsilon}\Big[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\Big].} \tag{26.44}$$

In Feynman gauge $\xi = 1$ this collapses to the maximally simple form

$$D^{ab}_{\mu\nu}(k)\big|_{\xi=1} \;=\; \frac{-i\,\delta^{ab}\,\eta_{\mu\nu}}{k^2+i\varepsilon}, \tag{26.45}$$

which is the rule used for explicit computation unless stated otherwise.

The propagator is massless (k^2 pole at the origin) and carries the gauge parameter ξ in its longitudinal part exactly as the photon does (Theorem 13.26); the color structure is the trivial δ^{ab}, which simply means a gluon does not change color as it propagates freely. There is, however, a genuinely new fact lurking behind the inversion that has no abelian counterpart, and it is worth stating now because it is the entire reason the next two subsections exist. In QED the analogous inversion was the end of the story: the photon's quadratic operator was the whole quadratic action, the gauge-fixing term made it invertible, and the Faddeev–Popov determinant was a field-independent constant that dropped out (Section 13.5). Here the same determinant is field-dependent (Theorem 26.2); it was exponentiated into the ghost term of Eq. (26.40), and that ghost term contributes its own propagator and its own vertex, to which we now turn.

26.3.3 The ghost propagator

The ghost fields c^a and $\bar c^{\,a}$ are anticommuting Lorentz scalars in the adjoint representation of SU(N)—one complex pair for each generator. Their Lagrangian Eq. (26.31) is $\mathcal{L}_{\rm gh} = (\partial^\mu \bar c^{\,a})(D_\mu c)^a$, the exponentiated form of $\det M$ with $M^{ab} = \frac{1}{g}\partial^\mu D^{ab}_\mu$ (Eq. (26.17)). Splitting the covariant derivative into its free and interacting pieces,

$$\mathcal{L}_{\rm gh} \;=\; \underbrace{(\partial^\mu \bar c^{\,a})(\partial_\mu c^a)}_{\mathcal{L}_{\rm gh,0}} \;+\; \underbrace{g\,f^{abc}\,(\partial^\mu \bar c^{\,a})\,A^b_\mu\,c^c}_{\mathcal{L}_{\rm gh,int}}. \tag{26.46}$$

The free part $\mathcal{L}_{\rm gh,0}$ is, after an integration by parts, $-\bar c^{\,a}\,\partial^2 c^a = \bar c^{\,a}(-\partial^2\delta^{ab})c^b$, so the quadratic kernel between $\bar c$ and c is $-\partial^2\delta^{ab} \to k^2\delta^{ab}$ in momentum space. This is the ordinary massless scalar kernel with a color δ^{ab}; the propagator is i times its inverse,

$$\boxed{D^{ab}_{\rm gh}(k) \;=\; \frac{i\,\delta^{ab}}{k^2+i\varepsilon}.} \tag{26.47}$$

Two points fix the conventions that the ghost-gluon vertex will rely on. First, the sign: the ghost propagator carries $+i$, in contrast to the gluon's $-i$, simply

because the scalar kernel is $+k^2$ where the (transverse) gauge kernel is $-k^2$. Second, the arrow: although c and $\bar{c}$ are scalars, they are distinct fields, and the propagator $\langle c^a \bar{c}^b \rangle$ is oriented—the two-point function pairs a c with a $\bar{c}$ and never c with c or $\bar{c}$ with $\bar{c}$, exactly as a Dirac propagator pairs ψ with $\bar{\psi}$. We therefore draw the ghost line with an arrow representing the flow of *ghost number* (the conserved U(1) charge under $c \to e^{i\alpha} c$, $\bar{c} \to e^{-i\alpha} \bar{c}$), pointing from the $\bar{c}$ end toward the c end. This orientation is what makes the vertex below asymmetric, and the asymmetry is physically consequential.

The ghosts never appear as external states: they are not part of the physical spectrum (this is proved, via BRST cohomology, in the sections that follow). In any physical amplitude they occur only on internal lines, and—being closed oriented loops of anticommuting fields—each closed ghost loop carries a sign, to which we return in Section 26.3.8.

26.3.4 The three-gluon vertex

The cubic self-coupling was isolated classically in Chapter 25 as

$$\mathcal{L}_3 = -\, g\, f^{abc}\, (\partial_\mu A^a_\nu)\, A^{b\,\mu} A^{c\,\nu} \tag{26.48}$$

(Eq. (25.107)). It carries one totally antisymmetric structure constant, one derivative, and three powers of the gauge field, and a single power of g: in the quantized theory it joins three gluon lines at a point (Fig. 25.3a). We now turn it into a Feynman rule. The computation is a clean illustration of the symmetrization rule Eq. (26.41), and we carry every one of the six terms.

Worked Example 26.8 (The three-gluon vertex from $\mathcal{L}_3$)**.** We continue Yang–Mills theory, promoting the classical cubic coupling Eq. (26.48) to the momentum-space vertex $iV^{abc}_{\mu\nu\rho}(p,q,r)$, with all three momenta incoming, $p+q+r=0$, and legs labeled (a,μ,p), (b,ν,q), (c,ρ,r).

Setting up the master formula. The cubic action is $S_3 = \int d^4x\, \mathcal{L}_3$. Inserting the Fourier expansion of each field and performing the x-integral produces an overall $(2\pi)^4 \delta^4(k_1+k_2+k_3)$ and replaces the derivative $\partial_\mu A^a_\nu$ by $(-ik_1)_\mu\, \widetilde{A}^a_\nu(k_1)$, the momentum being that of the (incoming) field the derivative acts on. By Eq. (26.41) the vertex is i times the third functional derivative of S_3, summed over the $3! = 6$ ways of assigning the three external legs to the three field slots of $\mathcal{L}_3$. We name the slots by their role in Eq. (26.48):

$$\text{slot } X = \partial_\mu A^a_\nu \text{ (color } a\text{, derivative index } \mu\text{, field index } \nu), \quad Y = A^{b\,\mu}, \quad Z = A^{c\,\nu}, \tag{26.49}$$

with the contractions dictated by Eq. (26.48): the derivative index μ of X is contracted with Y, and the field index ν of X is contracted with Z.

A single assignment. Send leg i to slot X, leg j to Y, leg k to Z (with $\{i,j,k\}$ a permutation of $\{1,2,3\}$). The field in slot X contributes the derivative factor

$(-ip_i)$ carried on the slot's derivative index and the external Lorentz index μ_i on the slot's field index; the slot's derivative index contracts with Y's index (the external index μ_j of leg j), and the slot's field index μ_i contracts with Z's index μ_k. The color is $f^{a_i a_j a_k}$. Collecting these,

$$\text{assignment } (i,j,k): \qquad -g\, f^{a_i a_j a_k}\, (-ip_i)_{\mu_j}\, \eta_{\mu_i \mu_k}. \tag{26.50}$$

Multiplying by the overall i of Eq. (26.41) converts the prefactor $i \cdot (-g) \cdot (-i) = -g$: the two explicit imaginary units cancel, and each term reduces to $-g\, f^{a_i a_j a_k}\, (p_i)_{\mu_j}\, \eta_{\mu_i \mu_k}$. This cancellation is why the three-gluon vertex, alone among the four, will carry no explicit factor of i.

Summing the six assignments. Abbreviate $f \equiv f^{a_1 a_2 a_3} = f^{abc}$ and use total antisymmetry, $f^{a_1 a_3 a_2} = f^{a_2 a_1 a_3} = f^{a_3 a_2 a_1} = -f$ and $f^{a_2 a_3 a_1} = f^{a_3 a_1 a_2} = +f$. The six terms of Eq. (26.50) (times the overall i) are

$$\begin{aligned}
&(1,2,3): +f\,(p_1)_{\mu_2}\eta_{\mu_1\mu_3}, && (2,3,1): +f\,(p_2)_{\mu_3}\eta_{\mu_1\mu_2},\\
&(1,3,2): -f\,(p_1)_{\mu_3}\eta_{\mu_1\mu_2}, && (3,1,2): +f\,(p_3)_{\mu_1}\eta_{\mu_2\mu_3},\\
&(2,1,3): -f\,(p_2)_{\mu_1}\eta_{\mu_2\mu_3}, && (3,2,1): -f\,(p_3)_{\mu_2}\eta_{\mu_1\mu_3},
\end{aligned} \tag{26.51}$$

each multiplied by the common factor $-g$ extracted above. Grouping by metric structure—using $\eta_{\mu_2\mu_1} = \eta_{\mu_1\mu_2}$ and so on—the three pairs combine into momentum differences:

$$\begin{aligned}
\eta_{\mu_1\mu_2}: &\quad -f\,(p_1)_{\mu_3} + f\,(p_2)_{\mu_3} = f\,(p_2 - p_1)_{\mu_3},\\
\eta_{\mu_2\mu_3}: &\quad -f\,(p_2)_{\mu_1} + f\,(p_3)_{\mu_1} = f\,(p_3 - p_2)_{\mu_1},\\
\eta_{\mu_1\mu_3}: &\quad +f\,(p_1)_{\mu_2} - f\,(p_3)_{\mu_2} = f\,(p_1 - p_3)_{\mu_2}.
\end{aligned} \tag{26.52}$$

Restoring the overall $-g$ and renaming the external labels back to $(a,\mu,p), (b,\nu,q), (c,\rho,r)$ flips each momentum difference and gives the result.

Theorem 26.9 (Three-gluon vertex)**.** *With all momenta incoming and $p+q+r = 0$, the three-gluon vertex is*

$$\boxed{iV^{abc}_{\mu\nu\rho}(p,q,r) \;=\; g\, f^{abc}\Big[\eta_{\mu\nu}(p-q)_\rho + \eta_{\nu\rho}(q-r)_\mu + \eta_{\rho\mu}(r-p)_\nu\Big].} \tag{26.53}$$

The structure of Eq. (26.53) repays a moment's attention. It is totally antisymmetric under the simultaneous interchange of any two legs—swap $(a,\mu,p) \leftrightarrow (b,\nu,q)$ and the antisymmetry of f^{abc} exactly compensates the antisymmetry of the bracket—which is the Bose symmetry required of three identical gluons combined with the antisymmetry of the color tensor. It is linear in momentum, so it grows with energy, the hallmark of a derivative coupling. And it is built so that the momentum appearing with each metric factor is the *difference* of the momenta of the other two legs, a pattern that the gauge Ward identities of Chapter 27 exploit. The vertex is drawn in Fig. 26.3.

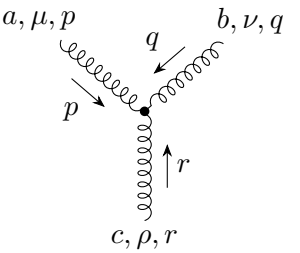


Figure 26.3: The three-gluon vertex Eq. (26.53), with all three momenta p, q, r flowing into the vertex ($p+q+r=0$). Each line is a gluon; the vertex is linear in g and carries one structure constant f^{abc}. It is the quantized form of the classical cubic coupling Eq. (25.107) and is the new feature of non-abelian gauge theory that QED simply lacks (Fig. 25.3c).

26.3.5 The four-gluon vertex

The quartic self-coupling, isolated classically as

$$\mathcal{L}_4 = -\tfrac{1}{4}\, g^2\, f^{abe} f^{cde}\, A^a_\mu A^b_\nu A^{c\,\mu} A^{d\,\nu} \tag{26.54}$$

(Eq. (25.110)), carries two structure constants sharing a summed index, no derivatives, and four powers of the field: a contact interaction among four gluon lines (Fig. 25.3b). Promoting it to a vertex requires symmetrizing over the four identical bosonic legs, which is where the deceptively simple $-\frac{1}{4}$ prefactor must be handled with care.

Worked Example 26.10 (The four-gluon vertex from $\mathcal{L}_4$)**.** We continue Yang–Mills theory. The vertex $iV^{abcd}_{\mu\nu\rho\sigma}$ is i times the fourth functional derivative of $S_4 = \int d^4x\, \mathcal{L}_4$, summed over the $4! = 24$ ways of assigning the external legs $(a,\mu), (b,\nu), (c,\rho), (d,\sigma)$ to the four field slots of Eq. (26.54). Because $\mathcal{L}_4$ has no derivatives, no momentum factors and no further i's appear; the only explicit imaginary unit is the overall one from Eq. (26.41), so the vertex will carry a net factor of i (times the $-g^2$ of the coupling). The four slots of Eq. (26.54) are $(a,\mu), (b,\nu), (c,\mu), (d,\nu)$, with the metric contractions pairing slots 1–3 (shared index μ) and slots 2–4 (shared index ν), and the color tensor $f^{abe}f^{cde} = f^{(\mathrm{s1})(\mathrm{s2})e} f^{(\mathrm{s3})(\mathrm{s4})e}$.

Counting the combinatorics. The 24 assignments organize into terms distinguished by which pair of legs shares the first structure constant. Consider the color structure $f^{abe}f^{cde}$ in the precise order written, i.e. first pair $(a,b) = (a_1,a_2)$, second pair $(c,d) = (a_3,a_4)$. Four of the 24 assignments reproduce exactly this ordered tensor: $(\mathrm{s1},\mathrm{s2},\mathrm{s3},\mathrm{s4})$ equal to $(1,2,3,4)$, $(2,1,4,3)$, $(3,4,1,2)$, and $(4,3,2,1)$. The first uses $f^{a_1a_2e}f^{a_3a_4e}$ directly; the second uses $f^{a_2a_1e}f^{a_4a_3e} = (-f^{a_1a_2e})(-f^{a_3a_4e})$, the two sign flips cancelling; the third and fourth use the symmetry $f^{a_1a_2e}f^{a_3a_4e} = f^{a_3a_4e}f^{a_1a_2e}$ under swapping the two structure constants.

One checks directly that all four carry the *same* Lorentz factor: assignment $(1,2,3,4)$ gives $\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} = \eta_{\mu\rho}\eta_{\nu\sigma}$, and the other three reproduce it (for instance $(2,1,4,3)$ gives $\eta_{\mu_2\mu_4}\eta_{\mu_1\mu_3}$, identical after reordering the metrics). Hence the coefficient of $f^{abe}f^{cde}\,\eta_{\mu\rho}\eta_{\nu\sigma}$ is

$$i\cdot\Big(-\tfrac{1}{4}g^2\Big)\cdot 4 \;=\; -i\,g^2, \tag{26.55}$$

the multiplicity 4 exactly cancelling the $\frac{1}{4}$. The partner term $-f^{abe}f^{cde}\,\eta_{\mu\sigma}\eta_{\nu\rho}$ arises identically from the four assignments whose first structure constant is again (up to sign) f^{abe} but whose metric pairing is $\eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3}$, with one structure-constant sign flip supplying the relative minus.

Assembling the three color channels. Repeating the count for the two other ways of splitting the four legs into pairs—$(a,c)(b,d)$ and $(a,d)(b,c)$, i.e. the structure constants $f^{ace}f^{bde}$ and $f^{ade}f^{bce}$—reproduces the same combinatorial factor $-ig^2$ with the metric structure dictated by which legs sit on the shared internal index. This is exactly the totally symmetric color–Lorentz tensor assembled classically in Eq. (25.115): the three inequivalent pairings of four legs, each paired with the metric contraction that leaves the two "crossed" indices free. Reading off the coefficient Eq. (26.55) for each channel gives the vertex.

Theorem 26.11 (Four-gluon vertex)**.** *With legs $(a,\mu),(b,\nu),(c,\rho),(d,\sigma)$, the four-gluon vertex is*

$$\begin{aligned} iV^{abcd}_{\mu\nu\rho\sigma} \;=\; -i\,g^2\Big[\; & f^{abe}f^{cde}(\eta_{\mu\rho}\eta_{\nu\sigma}-\eta_{\mu\sigma}\eta_{\nu\rho}) \\ & + f^{ace}f^{bde}(\eta_{\mu\nu}\eta_{\rho\sigma}-\eta_{\mu\sigma}\eta_{\nu\rho}) \\ & + f^{ade}f^{bce}(\eta_{\mu\nu}\eta_{\rho\sigma}-\eta_{\mu\rho}\eta_{\nu\sigma})\Big]. \end{aligned} \tag{26.56}$$

The vertex is, by construction, totally symmetric under simultaneous permutation of the four legs' labels, the symmetry that the brute-force sum over 24 assignments was guaranteed to deliver; Theorem 25.21 verified the same symmetric tensor from the side of gauge invariance. Note that it is momentum-independent (a true contact term) and carries no overall derivative, which is why it alone among the gluon vertices retains an explicit i. The diagram is Fig. 26.4.

26.3.6 The ghost-gluon vertex

The interaction in the ghost Lagrangian, isolated in Eq. (26.46), is

$$\mathcal{L}_{\text{gh,int}} = g\,f^{abc}\,(\partial^\mu \bar{c}^{\,a})\,A^b_\mu\,c^c. \tag{26.57}$$

It couples one gluon (b,μ) to a ghost–antighost pair, with color tensor f^{abc} in which a labels the antighost, b the gluon, and c the ghost. The single structural

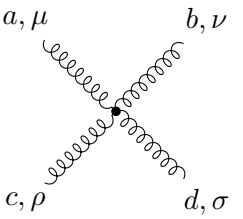


Figure 26.4: The four-gluon vertex Eq. (26.56), a momentum-independent contact interaction of order g^2 carrying two structure constants summed over a common adjoint index e. Its three terms correspond to the three inequivalent ways of grouping the four legs into pairs. It is the quantized form of the classical quartic coupling Eq. (25.110).

feature that makes this vertex special is plainly visible already in Eq. (26.57): the derivative ∂^μ acts on the *antighost* field $\bar{c}^{\,a}$ and on it alone. Whatever momentum factor the vertex carries must therefore be the momentum of the antighost line, not of the ghost line—the vertex is asymmetric between the two ends of the oriented ghost line. We make this precise.

Apply the master formula Eq. (26.41). There is no symmetrization to perform: the three legs (antighost, gluon, ghost) are fields of distinct types, so a single assignment contributes. Following the ghost-number arrow established in Section 26.3.3, the antighost end is where the arrow emerges from the vertex; we route its momentum p *out* of the vertex along that arrow. Then the antighost field $\bar{c}^{\,a}$ creates an outgoing quantum, $\bar{c}^{\,a}(x) \sim e^{+ip\cdot x}$, so the derivative acting on it becomes $\partial^\mu \to +ip^\mu$. Putting this into Eq. (26.41),

$$iV^{abc}_{\mathrm{gh},\,\mu}(p) \;=\; i\cdot g\, f^{abc}\cdot(+ip^\mu) \;=\; -\,g\, f^{abc}\, p^\mu, \tag{26.58}$$

with μ the Lorentz index of the gluon. We box the result.

Theorem 26.12 (Ghost-gluon vertex)**.** *With p^μ the momentum of the outgoing ghost line (the antighost end of the oriented ghost propagator), b, μ the gluon labels, a the outgoing-ghost color and c the incoming-ghost color, the ghost-gluon vertex is*

$$\boxed{iV^{abc}_{\mathrm{gh},\,\mu}(p) \;=\; -\,g\, f^{abc}\, p^\mu.} \tag{26.59}$$

The vertex depends on p, the outgoing-ghost momentum, and not on the incoming-ghost or gluon momenta.

The asymmetry of Eq. (26.59) is not a cosmetic accident of how we chose to integrate by parts. It is forced: $\mathcal{L}_{\mathrm{gh,int}}$ descends from the Faddeev–Popov operator $M^{ab} = \frac{1}{g}\partial^\mu D^{ab}_\mu$, in which the outer derivative ∂^μ comes from the gauge condition $\partial^\mu A_\mu = 0$ and sits to the *left* of the covariant derivative, never symmetrized with it. The resulting momentum factor is always the antighost's. This single

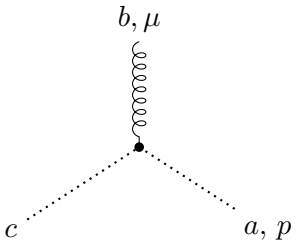


Figure 26.5: The ghost-gluon vertex Eq. (26.59). The dashed lines are ghosts, carrying an arrow for ghost number; the arrow enters on the incoming-ghost leg (color c) and leaves on the outgoing-ghost/antighost leg (color a, momentum p). The solid curl is the gluon (b, μ). The vertex is $\propto p^{\mu}$, the *outgoing* ghost momentum only—the derivative in Eq. (26.57) acts on the antighost alone—which is the asymmetry that seeds the Slavnov–Taylor identities of Section 26.7.

derivative is the diagrammatic origin of the difference between the gluon's gauge Ward identity and the ghost's: it is precisely why the longitudinal projection of a gluon line attached to a ghost loop does not vanish on its own but only in the combination dictated by BRST invariance. The Slavnov–Taylor identities that organize the renormalization of the theory—and that Chapter 27 needs to make the beta-function calculation gauge-consistent—are, in their simplest incarnation, the bookkeeping of this one asymmetric momentum factor. We flag the connection here and return to it once the BRST symmetry is in hand.

26.3.7 The quark-gluon vertex

For completeness, and because Chapter 27 needs the fermion loop, we record the coupling of the gauge field to matter. The Dirac term of Eq. (26.40), $\bar{\psi}_i(i\not{D} - m)\psi_i$ with $D_\mu = \partial_\mu - igA^a_\mu T^a$ acting in the fundamental representation, contains the interaction $\mathcal{L}_{q\bar{q}g} = g\,\bar{\psi}_i \gamma^\mu T^a_{ij} \psi_j\, A^a_\mu$. This is structurally the QED vertex of Chapter 17 with the charge replaced by the generator T^a in the fundamental; its Feynman rule is

$$iV^a_{q\bar{q}g,\,\mu} \;=\; i\,g\,\gamma_\mu\,T^a, \tag{26.60}$$

the matrix T^a acting on the (fundamental) color indices of the quark lines and γ_μ on their spinor indices. The quark propagator is the ordinary Dirac propagator $i\,\delta_{ij}(\not{k} + m)/(k^2 - m^2 + i\varepsilon)$ dressed with a color δ_{ij}, unchanged from Chapter 11. The Casimir invariants $\mathrm{tr}(T^a T^b) = T_F \delta^{ab}$ with $T_F = \tfrac{1}{2}$ and $T^a T^a = C_F \mathbf{1}$ with $C_F = (N^2 - 1)/2N$ (Section B.4) are what these vertices contract into in the loop integrals of Chapter 27.

26.3.8 The closed-ghost-loop sign

One rule remains, and it is not a vertex or a propagator but a global sign. The ghost fields are anticommuting, so the Berezin integration that defines their path integral produces functional determinants in the numerator, exactly as for the Dirac field of Chapter 12. The combinatorial consequence was proved there in full generality.

Theorem 26.13 (Closed-ghost-loop sign; imported)**.** *Every closed loop built from ghost propagators Eq. (26.47) and ghost-gluon vertices Eq. (26.59) carries an extra factor of* (-1) *relative to the same loop drawn with commuting fields, by the closed-loop sign rule Theorem 12.49.*

Proof. The ghost contribution to any Green's function comes from expanding $\exp\big[i\int(\partial^\mu\bar{c})(D_\mu c)\big]$ and Wick-contracting the $\bar{c}$'s with the c's. The contraction rules are those of an anticommuting field whose only nonzero two-point function pairs c with $\bar{c}$ (Eq. (26.47)); structurally this is the Grassmann Gaussian of Theorem 12.26, whose Wick theorem is a *signed* sum over pairings (a determinant), not the unsigned sum (a permanent) of a bosonic field. A single closed loop of ghost lines corresponds to a cyclic contraction of n pairs, and the cyclic permutation that closes the loop is odd relative to the reference ordering by exactly one transposition, contributing the overall (-1) established in Theorem 12.49 and Eq. (12.109). The argument depends only on the anticommuting statistics of the loop fields, not on their being spinors, so it applies verbatim to the scalar ghosts. □

This factor is indispensable, not optional. The ghost is a commuting scalar in its Lorentz quantum numbers but an anticommuting field in its statistics—a deliberate violation of the spin-statistics connection that is permitted precisely because the ghost is not a physical particle. Its wrong-statistics loop sign is exactly what is needed: in a physical amplitude the ghost loop must *cancel* part of a gluon loop, and a cancellation between two loops of like sign is impossible. The detailed mechanism—ghost loops removing the contributions of the unphysical timelike and longitudinal gluon polarizations—is the subject of Section 26.4; here we have only installed the sign that makes the cancellation arithmetically possible.

Physical Insight: the ghost is a negative degree of freedom

The two transverse polarizations of a gluon are physical; the timelike and longitudinal polarizations carried by $\eta_{\mu\nu}$ in the propagator Eq. (26.44) are not. In a non-abelian theory those unphysical polarizations do not decouple by themselves the way they do in QED, because the gluon is charged and they re-enter through the self-coupling vertices Eq. (26.53)

and Eq. (26.56). The Faddeev–Popov ghost is the device that subtracts them. Because it carries the wrong statistics, its closed loop comes with a minus sign (Theorem 26.13); because it is a scalar, it has exactly one degree of freedom per color; and the pair of complex scalars $c, \bar{c}$ with their relative minus sign supplies precisely the -2 degrees of freedom (per color) needed to cancel the two unphysical gluon polarizations in a closed loop. The ghost is best read not as a particle but as a bookkeeping device for a negative number of states—a subtraction term promoted to a field so that it can circulate in loops and do its subtracting order by order.

The complete rule set

We have assembled the entire perturbative kit of gauge-fixed Yang–Mills. The propagators are the gluon Eq. (26.44), the ghost Eq. (26.47), and (with matter) the quark; the vertices are the three-gluon Eq. (26.53), the four-gluon Eq. (26.56), the ghost-gluon Eq. (26.59), and the quark-gluon Eq. (26.60); and the one global rule is the (-1) per closed ghost loop Theorem 26.13. Together with the standard prescriptions inherited from Chapter 15—a loop integral $\int d^4\ell/(2\pi)^4$ for each independent loop momentum, the symmetry factors, and overall momentum conservation—this is a complete algorithm for computing any Yang–Mills amplitude to any order in g.

These rules now do double duty. In the next section they are put to the test that justifies the whole ghost apparatus: we compute a physical amplitude in which an internal gluon's unphysical polarizations would, left to themselves, spoil unitarity, and watch the ghost loop—wrong sign and all—cancel them exactly. And in Chapter 27 the same rules are fed into the one-loop gluon self-energy, where the gluon, ghost, and quark loops combine to produce the celebrated negative beta-function coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ and with it asymptotic freedom. The computational result of the chapter is complete; what remains is to prove it is consistent.

26.4 Ghosts and Unitarity

The Feynman rules assembled in the previous section contain one entry that looks like an affront to physics. The Faddeev–Popov ghosts $c^a, \bar{c}^a$ are scalar fields — their propagator $i\delta^{ab}/(k^2 + i\epsilon)$ (Eq. (26.47)) is that of a free massless scalar — yet they anticommute, and the price of that anticommutation is a factor (-1) for every closed ghost loop (Theorem 26.13). A spin-zero field with Fermi statistics violates the spin–statistics theorem of Chapter 12; a closed loop weighted by -1 contributes negatively to quantities that ought to be sums of squares. Were the ghosts genuine particles this would be a disaster. They are

not: they never appear in an initial or final state, the cohomological argument of Section 26.5 will make this exclusion exact, and so the spin–statistics theorem — a statement about *physical* asymptotic states — is not violated. But that defence only tells us the ghosts are *permitted*. This section shows they are *mandatory*. The wrong-statistics loop with its (-1) is the precise negative contribution needed to remove the unphysical polarizations of the gluon from every physical rate, and without it the quantized theory built so far would not be unitary. We will see the cancellation happen, in full, in a representative process, and then draw the conclusion that organizes the rest of the chapter: a cancellation this delicate cannot be a coincidence checked diagram by diagram, but must be enforced by an exact symmetry of the gauge-fixed action.

26.4.1 Why the naive polarization sum fails for gluons

Every unpolarized cross section in Parts V and VI rested on a single technical convenience: when a real, external gauge boson of null momentum k ($k^2 = 0$) is summed over its two physical transverse polarizations, the spin sum $\sum_\lambda \epsilon_\mu(k,\lambda)\epsilon^*_\nu(k,\lambda)$ may be replaced by $-\eta_{\mu\nu}$. This is the polarization-sum replacement Theorem 13.30, and it is what turns a sum over physical states into a Lorentz-covariant metric contraction. The completeness relation behind it, derived in Chapter 13, is

$$\sum_{\lambda=\pm} \epsilon_\mu(k,\lambda)\,\epsilon^*_\nu(k,\lambda) = -\eta_{\mu\nu} + \frac{k_\mu \bar{k}_\nu + \bar{k}_\mu k_\nu}{k\cdot\bar{k}}, \tag{26.61}$$

where $\bar{k}^\mu$ is an auxiliary null vector with $k\cdot\bar{k} \neq 0$ (Eq. (13.54)). The transverse sum is a rank-two object; the metric is rank four; the difference is the $\bar{k}$-dependent longitudinal/scalar piece. The replacement $\sum \epsilon\epsilon^* \to -\eta_{\mu\nu}$ is therefore *not* an identity. It is licensed only under a hypothesis, stated explicitly in Theorem 13.30: that the amplitude $\mathcal{M}^\mu$, stripped of the polarization vector of the boson in question, obeys the Ward identity

$$k_\mu\, \mathcal{M}^\mu = 0. \tag{26.62}$$

When Eq. (26.62) holds, the second term of Eq. (26.61) dies on contraction with $\mathcal{M}^\mu \mathcal{M}^{*\nu}$, the $\bar{k}$-dependence vanishes, and only $-\eta_{\mu\nu}\mathcal{M}^\mu\mathcal{M}^{*\nu}$ survives. In QED the hypothesis is met: the photon couples to the conserved electromagnetic current, and current conservation $\partial_\mu J^\mu = 0$ becomes $k_\mu \mathcal{M}^\mu = 0$ at tree level (and, after Chapter 19, to all orders by Ward–Takahashi). The two missing degrees of freedom carried by $-\eta_{\mu\nu}$ relative to the transverse sum — the timelike and longitudinal photons — decouple, and the convenience is exact.

For non-abelian gauge fields this reasoning collapses at its foundation. The gauge current to which a gluon couples is not ordinarily conserved; it is only *covariantly* conserved, $D_\mu J^{a\mu} = 0$ with $D_\mu = \partial_\mu - igA^a_\mu T^a$, and a covariant

divergence is not an ordinary divergence. The difference $D_\mu - \partial_\mu = -igA^a_\mu T^a$ is precisely the gluon self-coupling encoded in the cubic and quartic terms of $\mathcal{L}_{\text{YM}}$ (Eq. (25.76), Eq. (25.49)). Physically: a gluon carries colour, so it is itself a source for the field, and the would-be conserved current picks up the field it generates. We must therefore expect $k_\mu \mathcal{M}^\mu \neq 0$ for gluon amplitudes, the hypothesis of Theorem 13.30 to fail, and the naive replacement $\sum \epsilon\epsilon^* \to -\eta_{\mu\nu}$ to leave behind exactly the longitudinal pieces that Eq. (26.61) warns about. The question is what those leftover pieces are, and what removes them. We answer it by computing.

26.4.2 The non-abelian Ward identity for two gluons

We take the cleanest process that exhibits the gluon self-coupling: quark–antiquark annihilation into two gluons, $q(p_1)\,\bar{q}(p_2) \to g(k_1, a)\,g(k_2, b)$, with massless quarks for algebraic transparency. The amplitude has three tree diagrams (Fig. 26.6): the quark line emits the two gluons in either order (t- and u-channel), and the quark pair annihilates into a single virtual gluon which splits through the three-gluon vertex (s-channel). Writing the amplitude with the external polarizations stripped, $\mathcal{M} = \epsilon^*_{1\mu}\epsilon^*_{2\nu}\,\mathcal{M}^{\mu\nu}$, the three contributions follow from the rules of Section 26.3: the quark–gluon vertex $ig\gamma^\mu T^a$, the massless quark propagator $i\not{p}/p^2$, the Feynman-gauge gluon propagator $-i\eta_{\mu\nu}\delta^{ab}/k^2$ (Eq. (26.44)), and the three-gluon vertex Eq. (26.53).

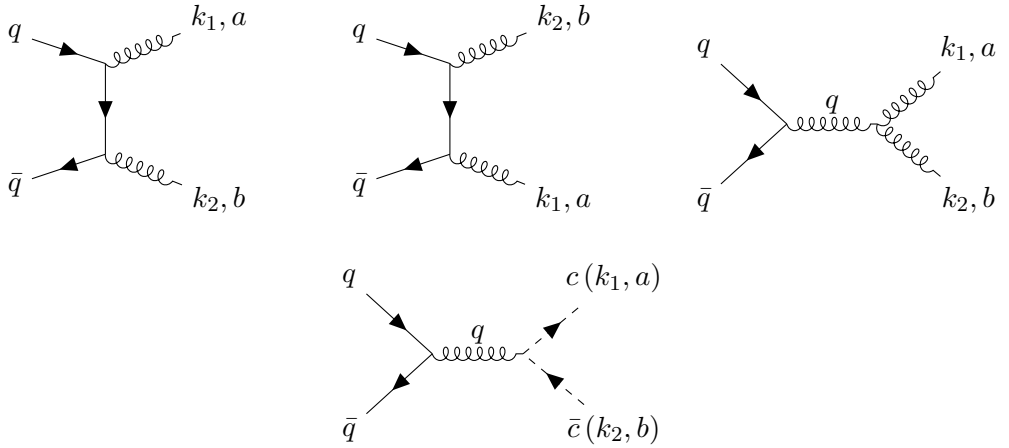


Figure 26.6: Top row: the three tree diagrams for $q\bar{q} \to gg$. The first two attach the gluons to the quark line in opposite orders; the third routes the colour through the three-gluon vertex and is the source of the Ward-identity anomaly Eq. (26.69). Bottom: the auxiliary amplitude $q\bar{q} \to c\bar{c}$ for a ghost–antighost pair, built from the single s-channel diagram. Its squared modulus is exactly the unphysical contamination that the ghost loop subtracts (Eq. (26.86)).

The two "abelian-like" diagrams are

$$\mathcal{M}_A^{\mu\nu} = -g^2\, T^b T^a\, \bar v(p_2)\, \gamma^\nu\, \frac{\not p_1 - \not k_1}{(p_1-k_1)^2}\, \gamma^\mu\, u(p_1), \tag{26.63}$$

$$\mathcal{M}_B^{\mu\nu} = -g^2\, T^a T^b\, \bar v(p_2)\, \gamma^\mu\, \frac{\not p_1 - \not k_2}{(p_1-k_2)^2}\, \gamma^\nu\, u(p_1), \tag{26.64}$$

the overall $-g^2$ collecting the two factors (ig) from the vertices and the i from the internal propagator, $i^3 = -i$, against the explicit i removed in defining $\mathcal{M}$. We now test the Ward identity by contracting with the momentum k_1 of the first gluon.

Worked Example 26.14 (The two-gluon Ward identity and the ghost amplitude)**. Diagram** A**.** Contracting Eq. (26.63) with $k_{1\mu}$ replaces $\gamma^\mu k_{1\mu} = \not k_1$ in the numerator, giving the Dirac string $\bar v\,\gamma^\nu(\not p_1 - \not k_1)\not k_1\, u$. Write $\not k_1 = \not p_1 - (\not p_1 - \not k_1)$, so that

$$(\not p_1 - \not k_1)\not k_1 = (\not p_1 - \not k_1)\not p_1 - (\not p_1 - \not k_1)^2. \tag{26.65}$$

The massless Dirac equation $\not p_1\, u(p_1) = 0$ and $p_1^2 = 0$ kill the first term, $(\not p_1 - \not k_1)\not p_1\, u = (p_1^2 - \not k_1 \not p_1)u = 0$, while $(\not p_1 - \not k_1)^2 = (p_1-k_1)^2$ is a scalar. Hence the numerator collapses to $-(p_1-k_1)^2\,\bar v\gamma^\nu u$, which cancels the propagator denominator:

$$k_{1\mu}\mathcal{M}_A^{\mu\nu} = -g^2\, T^b T^a\, \frac{-(p_1-k_1)^2}{(p_1-k_1)^2}\, \bar v\gamma^\nu u = +g^2\, T^b T^a\, \bar v(p_2)\gamma^\nu u(p_1). \tag{26.66}$$

Diagram B**.** Here the contracted $\not k_1$ sits next to $\bar v(p_2)$. The internal momentum is $p_1 - k_2$, and momentum conservation $k_1 = (p_1 - k_2) + p_2$ gives $\not k_1 = (\not p_1 - \not k_2) + \not p_2$. Using $\bar v(p_2)\not p_2 = 0$,

$$\bar v\, \not k_1(\not p_1 - \not k_2) = \bar v\big[(\not p_1 - \not k_2)^2 + \not p_2(\not p_1 - \not k_2)\big] = \bar v\,(p_1-k_2)^2, \tag{26.67}$$

so that, cancelling the denominator again,

$$k_{1\mu}\mathcal{M}_B^{\mu\nu} = -g^2\, T^a T^b\, \bar v\gamma^\nu u. \tag{26.68}$$

The residue. Adding Eq. (26.66) and Eq. (26.68), the colour factors do not cancel as they would in an abelian theory; they combine into a commutator:

$$k_{1\mu}(\mathcal{M}_A+\mathcal{M}_B)^{\mu\nu} = g^2(T^bT^a - T^aT^b)\bar v\gamma^\nu u = g^2\,[T^b,T^a]\,\bar v\gamma^\nu u = -ig^2 f^{abc}\,T^c\,\bar v\gamma^\nu u, \tag{26.69}$$

where $[T^b,T^a] = if^{bac}T^c = -if^{abc}T^c$. In QED, $f^{abc}=0$ and the right-hand side vanishes identically: the abelian Ward identity holds and no ghost is ever needed. The non-vanishing of Eq. (26.69) is, quite literally, the entire reason the non-abelian theory is harder. The leftover is proportional to $\bar v\gamma^\nu u$, the very

current the s-channel diagram couples to, and it is the third diagram that must dispose of it.

Diagram C. The s-channel amplitude reads

$$\mathcal{M}_C^{\mu\nu} = \frac{-ig^2}{q^2}\, f^{abc}\, T^c \left[\eta^{\mu\nu}(k_2-k_1)\cdot J - (k_1+2k_2)^\mu J^\nu + (2k_1+k_2)^\nu J^\mu\right], \tag{26.70}$$

with $q = p_1+p_2 = k_1+k_2$, $q^2 = 2k_1\cdot k_2$, and the abbreviation $J^\rho \equiv \bar v(p_2)\gamma^\rho u(p_1)$ for the quark current contracted through the internal propagator into the three-gluon vertex. The bracket is the vertex Eq. (26.53) with one leg carrying the incoming momentum q and the two outgoing gluons carrying k_1, k_2; the overall factor $-ig^2$ is fixed by the same i-counting as Eq. (26.63)–Eq. (26.64) together with the Feynman-gauge propagator $-i\eta/q^2$. Two kinematic facts will be used repeatedly: with massless quarks, $\bar v \not q\, u = \bar v(\not p_1 + \not p_2)u = 0$, so writing $K \equiv k_1\cdot J = \bar v \not k_1 u$ we have $k_2\cdot J = \bar v \not k_2 u = -K$.

Contract Eq. (26.70) with $k_{1\mu}$. The three terms give, using $k_1^2 = 0$ and $k_1\cdot k_2 = q^2/2$,

$$\eta^{\mu\nu}: \quad k_1^\nu (k_2-k_1)\cdot J = k_1^\nu(-K-K) = -2K\,k_1^\nu, \tag{26.71}$$

$$-(k_1+2k_2)^\mu J^\nu: \quad -(k_1\cdot k_1 + 2k_2\cdot k_1)\,J^\nu = -q^2 J^\nu, \tag{26.72}$$

$$(2k_1+k_2)^\nu J^\mu: \quad (2k_1+k_2)^\nu (k_1\cdot J) = (2K\,k_1^\nu + K\,k_2^\nu), \tag{26.73}$$

which sum to $-q^2 J^\nu + K\,k_2^\nu$. Therefore

$$k_{1\mu}\mathcal{M}_C^{\mu\nu} = \frac{-ig^2}{q^2}\, f^{abc}T^c(-q^2 J^\nu + K\,k_2^\nu) = +ig^2 f^{abc}T^c\, J^\nu \;-\; ig^2 f^{abc}T^c\,\frac{K}{q^2}\,k_2^\nu. \tag{26.74}$$

The first piece of Eq. (26.74) is exactly minus the residue Eq. (26.69): the $J^\nu = \bar v\gamma^\nu u$ terms cancel between the three diagrams. This cancellation is not optional — it is what fixes the relative normalization of the three-gluon vertex to the quark–gluon vertex, and the rules of Section 26.3 were extracted precisely so that it occurs. What survives is

$$\boxed{k_{1\mu}\mathcal{M}^{\mu\nu} = k_{1\mu}(\mathcal{M}_A+\mathcal{M}_B+\mathcal{M}_C)^{\mu\nu} = -ig^2 f^{abc}T^c\,\frac{\bar v \not k_1 u}{q^2}\,k_2^\nu \;\equiv\; \mathcal{M}_{\rm gh}\,k_2^\nu.} \tag{26.75}$$

Reading the result. The non-abelian Ward identity does not say $k_{1\mu}\mathcal{M}^{\mu\nu} = 0$. It says $k_{1\mu}\mathcal{M}^{\mu\nu}$ is *proportional to* k_2^ν — the longitudinal direction of the *other* gluon — with a coefficient $\mathcal{M}_{\rm gh}$ that we now identify. The amplitude for $q\bar q$ to produce a ghost–antighost pair $c(k_1,a)\,\bar c(k_2,b)$ has the single s-channel diagram of Fig. 26.6: the quarks annihilate into a virtual gluon, which converts to the ghost pair through the ghost–gluon vertex Eq. (26.58), $g f^{abc}k^\sigma$ with k the outgoing ghost momentum. Assembling it with the same conventions,

$$\mathcal{M}_{\rm gh} = \mathcal{M}(q\bar q \to c\bar c) = -i\,\frac{g^2}{q^2}\, f^{abc}\,T^c\,\bar v(p_2)\not k_1 u(p_1), \tag{26.76}$$

which is exactly the coefficient appearing in Eq. (26.75). The longitudinal part of the two-gluon amplitude *is* the ghost-production amplitude. This is the first concrete sign that the ghost is not bookkeeping: the unphysical gluon polarization and the ghost carry literally the same dynamical information.

Two consequences of Eq. (26.75) deserve emphasis before we square anything. First, when the second gluon is *physical*, $\epsilon_2^* \cdot k_2 = 0$, so $k_{1\mu}\mathcal{M}^{\mu\nu}\epsilon^*_{2\nu} = \mathcal{M}_{\rm gh}\,(k_2 \cdot \epsilon_2^*) = 0$: the Ward identity Eq. (26.62) *does* hold, provided every *other* external gluon is kept transverse. Tree-level gauge invariance is intact in this restricted sense. Second — and this is the crux — the moment we use the replacement $\sum \epsilon\epsilon^* \to -\eta_{\mu\nu}$ for the second gluon, we are summing over its unphysical (longitudinal and scalar) polarizations as well, for which $\epsilon_2 \cdot k_2 \neq 0$. Then $k_{1\mu}\mathcal{M}^{\mu\nu}$ no longer dies, and the naive replacement over-counts by precisely an amount controlled by $\mathcal{M}_{\rm gh}$. We make the over-counting exact.

26.4.3 The physical polarization sum and the ghost subtraction

We want the squared amplitude summed over the two transverse polarizations of *each* outgoing gluon, the only physically meaningful sum,

$$\sum_{\rm phys} |\mathcal{M}|^2 = \mathcal{M}^{\mu\nu}\mathcal{M}^{*\mu'\nu'}\Big(\sum_{\lambda_1} \epsilon_{1\mu}\epsilon^*_{1\mu'}\Big)\Big(\sum_{\lambda_2} \epsilon_{2\nu}\epsilon^*_{2\nu'}\Big), \tag{26.77}$$

and we want to compare it with the covariant "replaced" expression $\mathcal{M}^{\mu\nu}\mathcal{M}^*_{\mu\nu}$ obtained by setting both transverse sums to $-\eta$. Write each physical sum using Eq. (26.61) as $-\eta_{\mu\mu'} + R_{1\mu\mu'}$ with

$$R_{1\mu\mu'} = \frac{k_{1\mu}\bar{k}_{1\mu'} + \bar{k}_{1\mu}k_{1\mu'}}{k_1 \cdot \bar{k}_1}\,, \qquad a_1 \equiv k_1 \cdot \bar{k}_1, \tag{26.78}$$

and similarly R_2, a_2 for the second gluon with auxiliary vector $\bar{k}_2$. The auxiliary vectors are arbitrary; the physical sum cannot depend on them, and watching that dependence cancel is itself a check. We also need the companion of Eq. (26.75) for the second index. By Bose symmetry $\mathcal{M}^{\mu\nu}(k_1,a;k_2,b) = \mathcal{M}^{\nu\mu}(k_2,b;k_1,a)$, so applying Eq. (26.75) to the swapped amplitude,

$$k_{2\nu}\mathcal{M}^{\mu\nu} = \mathcal{M}_{\rm gh}\,k_1^{\mu}, \tag{26.79}$$

with the *same* $\mathcal{M}_{\rm gh}$: indeed the swapped ghost amplitude carries $f^{bac}\,\bar{v}\not{k}_2 u = (-f^{abc})(-\bar{v}\not{k}_1 u) = f^{abc}\bar{v}\not{k}_1 u$, unchanged. The doubly-longitudinal contraction then vanishes,

$$k_{1\mu}k_{2\nu}\mathcal{M}^{\mu\nu} = \mathcal{M}_{\rm gh}\,(k_2 \cdot k_2) = 0, \tag{26.80}$$

since $k_2^2 = 0$.

Worked Example 26.15 (The polarization-sum identity and the ghost contribution)**.** Expand Eq. (26.77) into four terms, $(-\eta + R_1)(-\eta + R_2)$:

$$\sum_{\text{phys}} |\mathcal{M}|^2 = \underbrace{\mathcal{M}^{\mu\nu}\mathcal{M}^{*}_{\mu\nu}}_{\text{(A)}} + \underbrace{(-\eta_{\mu\mu'})R_{2\nu\nu'}\mathcal{M}^{\mu\nu}\mathcal{M}^{*\mu'\nu'}}_{\text{(B)}} + \underbrace{R_{1\mu\mu'}(-\eta_{\nu\nu'})\mathcal{M}^{\mu\nu}\mathcal{M}^{*\mu'\nu'}}_{\text{(C)}} + \underbrace{R_{1\mu\mu'}R_{2\nu\nu'}\mathcal{M}^{\mu\nu}\mathcal{M}^{*\mu'\nu'}}_{\text{(D)}}. \tag{26.81}$$

Term (B). Define $A^{\nu\nu'} \equiv \mathcal{M}^{\mu\nu}\mathcal{M}^{*}{}_{\mu}{}^{\nu'}$ (sum on μ). Then (B) $= -R_{2\nu\nu'}A^{\nu\nu'}$, and

$$R_{2\nu\nu'}A^{\nu\nu'} = \frac{1}{a_2}\Big[(A^{\nu\nu'}k_{2\nu})\bar{k}_{2\nu'} + (A^{\nu\nu'}\bar{k}_{2\nu})k_{2\nu'}\Big]. \tag{26.82}$$

For the first piece, $A^{\nu\nu'}k_{2\nu} = (\mathcal{M}^{\mu\nu}k_{2\nu})\mathcal{M}^{*}{}_{\mu}{}^{\nu'} = \mathcal{M}_{\text{gh}}k_1^{\mu}\mathcal{M}^{*\,\nu'}_{\mu}$ by Eq. (26.79); contracting with $\bar{k}_{2\nu'}$ and using $k_1^{\mu}\mathcal{M}^{*\,\nu'}_{\mu} = \mathcal{M}^{*}_{\text{gh}}k_2^{\nu'}$ (the conjugate of Eq. (26.75)) gives $\mathcal{M}_{\text{gh}}\mathcal{M}^{*}_{\text{gh}}\,k_2^{\nu'}\bar{k}_{2\nu'} = |\mathcal{M}_{\text{gh}}|^2 a_2$. The second piece is identical by the same two steps. Hence

$$\text{(B)} = -\frac{1}{a_2}(|\mathcal{M}_{\text{gh}}|^2 a_2 + |\mathcal{M}_{\text{gh}}|^2 a_2) = -2|\mathcal{M}_{\text{gh}}|^2, \tag{26.83}$$

manifestly independent of the auxiliary vector $\bar{k}_2$. By the mirror computation, (C) $= -2|\mathcal{M}_{\text{gh}}|^2$.

Term (D). Writing out R_1R_2, only the cross structures survive, because any factor pairing k_1 on the unprimed index μ with k_2 on ν vanishes by Eq. (26.80), as does its conjugate. The two surviving structures are $k_{1\mu}\bar{k}_{2\nu}\,\bar{k}_{1\mu'}k_{2\nu'}$ and $\bar{k}_{1\mu}k_{2\nu}\,k_{1\mu'}\bar{k}_{2\nu'}$. Using Eqs. (26.75) and (26.79),

$$k_{1\mu}\bar{k}_{2\nu}\mathcal{M}^{\mu\nu} = \bar{k}_{2\nu}\mathcal{M}_{\text{gh}}k_2^{\nu} = \mathcal{M}_{\text{gh}}a_2, \qquad \bar{k}_{1\mu}k_{2\nu}\mathcal{M}^{\mu\nu} = \bar{k}_{1\mu}\mathcal{M}_{\text{gh}}k_1^{\mu} = \mathcal{M}_{\text{gh}}a_1, \tag{26.84}$$

so each surviving structure contributes $|\mathcal{M}_{\text{gh}}|^2 a_1 a_2$, and

$$\text{(D)} = \frac{1}{a_1 a_2}(|\mathcal{M}_{\text{gh}}|^2 a_1 a_2 + |\mathcal{M}_{\text{gh}}|^2 a_1 a_2) = +2|\mathcal{M}_{\text{gh}}|^2, \tag{26.85}$$

again independent of both $\bar{k}_1$ and $\bar{k}_2$, as it must be.

The identity. Summing Eq. (26.83) (twice) and Eq. (26.85),

$$\boxed{\sum_{\text{phys}} |\mathcal{M}(q\bar{q} \to gg)|^2 = \mathcal{M}^{\mu\nu}\mathcal{M}^{*}_{\mu\nu} \;-\; 2\,|\mathcal{M}_{\text{gh}}|^2.} \tag{26.86}$$

The covariant replacement $\sum \epsilon\epsilon^{*} \to -\eta$ over-counts the physical transverse sum by exactly $2|\mathcal{M}_{\text{gh}}|^2$ — twice the squared ghost-pair amplitude, the factor two counting one over-counted longitudinal mode per gluon. In QED $\mathcal{M}_{\text{gh}} = 0$ and Eq. (26.86) reduces to the exact replacement of Theorem 13.30; the surplus is a purely non-abelian effect, born of the same commutator $[T^a, T^b]$ that produced the residue Eq. (26.69).

Equation (26.86) is the engine of the section. It says that if we ever compute a gluonic rate by the convenient covariant rule — replacing every external or cut gluon polarization sum by $-\eta_{\mu\nu}$ — the answer is too large by a positive amount $2|\mathcal{M}_{\rm gh}|^2$ which is the squared amplitude to make a ghost pair. Something with the same kinematics but the opposite sign must be subtracted. The ghost, with its closed-loop (-1), is built to do exactly that.

26.4.4 Unitarity, the cut, and the role of the ghost

The polarization-sum identity becomes a statement about the consistency of the loop expansion when read through unitarity. The S-matrix satisfies $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$; writing $\mathbf{S} = \mathbf{1} + i\mathcal{T}$ this is $-i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger\mathcal{T}$, which between physical states is the generalized optical theorem Theorem 14.37,

$$2\,\mathrm{Im}\,\mathcal{M}(i \to i) = \sum_X \int \mathrm{d}\Pi_X\, |\mathcal{M}(i \to X)|^2, \tag{26.87}$$

the cross-section form of which is Theorem 16.28. The sum runs over *physical* intermediate states X — complete sets of genuine, positive-norm, on-shell particles — weighted by Lorentz-invariant phase space. The left-hand side is the imaginary part of a loop amplitude; the right-hand side is a sum of manifestly positive squared amplitudes over physical final states. Unitarity is the demand that these agree. It is the one nonperturbative anchor available to us (Section 14.7.3), and any candidate set of Feynman rules that fails it is not a quantum theory.

Take i to be a state whose forward amplitude receives, at one loop, a contribution with a two-gluon intermediate cut — concretely the order-g^4 part of $q\bar{q} \to q\bar{q}$ with a gg rung, whose imaginary part Eq. (26.87) ties to $q\bar{q} \to gg$. The imaginary part of a loop amplitude is computed by putting the cut internal lines on shell: this is the content of Cutkosky's rules, whose general proof we defer to Chapter 60 but whose statement we use here. Each cut internal propagator $\dfrac{i}{k^2 + i\epsilon}$ is replaced by its absorptive part $2\pi\,\delta(k^2)\theta(k^0)$, and the residue carries the propagator numerator. For a Feynman-gauge gluon the numerator is $-\eta_{\mu\nu}\delta^{ab}$, so *the cut of the gluon loop sums each on-shell gluon over all four polarizations with the metric weight*:

$$[\text{gluon-loop cut}] = \frac{1}{2}\int \mathrm{d}\Pi_2\; \mathcal{M}^{\mu\nu}\mathcal{M}^*_{\mu\nu}, \tag{26.88}$$

the $\frac{1}{2}$ being the Bose symmetry factor for the two identical on-shell gluons. By Eq. (26.86) this is *not* the physical right-hand side of Eq. (26.87), which counts only the two transverse polarizations of each gluon:

$$\frac{1}{2}\int \mathrm{d}\Pi_2\; \mathcal{M}^{\mu\nu}\mathcal{M}^*_{\mu\nu} = \frac{1}{2}\int \mathrm{d}\Pi_2 \sum_{\rm phys} |\mathcal{M}|^2 + \int \mathrm{d}\Pi_2\, |\mathcal{M}_{\rm gh}|^2. \tag{26.89}$$

The gluon loop alone over-shoots unitarity by $\int \mathrm{d}\Pi_2\, |\mathcal{M}_{\mathrm{gh}}|^2$: a strictly positive quantity, the phase-space integral of a squared amplitude. If the gluon loop were the whole story, Eq. (26.87) would read $2\,\mathrm{Im}\,\mathcal{M} = \frac{1}{2}\int \mathrm{d}\Pi_2 \sum_{\mathrm{phys}} |\mathcal{M}|^2 + \int \mathrm{d}\Pi_2 |\mathcal{M}_{\mathrm{gh}}|^2$, with an unaccounted positive surplus. The imaginary part of the amplitude would correspond to scattering into more than the physical states allow. The theory would not conserve probability.

The Feynman rules contain exactly the missing diagram. The same one-loop forward amplitude has a contribution with a *ghost* loop — the $q\bar{q}$ pair annihilating into a virtual gluon that splits into a ghost–antighost loop. Cutting it puts a ghost (k_1) and an antighost (k_2) on shell, with the production amplitude $\mathcal{M}_{\mathrm{gh}}$ of Eq. (26.76). The closed ghost loop carries the mandatory factor (-1) (Theorem 26.13) from the anticommuting nature of $c, \bar{c}$, and — crucially — *no* Bose symmetry factor of $\frac{1}{2}$, because the ghost and the antighost are distinct (the ghost line is oriented by ghost number). Therefore

$$[\text{ghost-loop cut}] = (-1)\int \mathrm{d}\Pi_2\, |\mathcal{M}_{\mathrm{gh}}|^2, \tag{26.90}$$

exactly the negative of the surplus in Eq. (26.89). Adding the two cuts,

$$[\text{gluon cut}] + [\text{ghost cut}] = \frac{1}{2}\int \mathrm{d}\Pi_2 \sum_{\mathrm{phys}} |\mathcal{M}(q\bar{q} \to gg)|^2, \tag{26.91}$$

which is precisely the right-hand side of the optical theorem Eq. (26.87) with X the physical two-transverse-gluon state. Unitarity is restored, with the ghost loop supplying the missing piece down to the last factor.

Theorem 26.16 (Ghosts enforce unitarity)**.** *In a non-abelian gauge theory quantized in a covariant (R_ξ) gauge, the imaginary part of a physical amplitude computed from the gauge-fixed Feynman rules equals the unitarity sum Eq. (26.87) over physical (transverse) gauge-boson states if and only if closed Faddeev–Popov ghost loops are included with their factor (-1). The ghost loops contribute exactly the negative of the unphysical (longitudinal and scalar) gauge-boson contamination carried by the metric numerator $-\eta_{\mu\nu}$ of the covariant gluon propagator, as exhibited by the identity Eq. (26.86) and the cut balance Eq. (26.91).*

Proof for the two-gluon cut. The cut of the gluon-loop contribution in Feynman gauge is Eq. (26.88), by Cutkosky's replacement of each cut propagator by its on-shell delta and the numerator $-\eta_{\mu\nu}$. The polarization-sum identity Eq. (26.86), proved in full in Theorem 26.15 using only the non-abelian Ward identity Eq. (26.75) and the completeness relation Eq. (13.54), rewrites this as the physical transverse sum plus $\int \mathrm{d}\Pi_2 |\mathcal{M}_{\mathrm{gh}}|^2$, Eq. (26.89). The ghost-loop cut Eq. (26.90) contributes the exact negative of the surplus, the factor (-1) being the closed-Grassmann-loop sign and the magnitude being fixed by $\mathcal{M}_{\mathrm{gh}}$, which the Ward identity Eq. (26.75) identified with the longitudinal part of the

gluon amplitude. Their sum Eq. (26.91) is the physical-state unitarity sum. The colour algebra closes automatically: summing $|\mathcal{M}_{\rm gh}|^2$ over the ghost colours a, b produces $f^{abc}f^{abd} = C_A\delta^{cd}$ (Section B.4), the same colour factor carried by the longitudinal gluon terms. Conversely, omitting the ghost loop leaves the strictly positive surplus $\int \mathrm{d}\Pi_2 |\mathcal{M}_{\rm gh}|^2 > 0$ uncancelled, violating Eq. (26.87). □

Physical Insight

The chain of identifications is worth stating bare. The longitudinal polarization of a gluon is, by the Ward identity Eq. (26.75), the ghost amplitude. The contamination it injects into a covariant polarization sum is, by Eq. (26.86), $+|\mathcal{M}_{\rm gh}|^2$ per mode. The ghost loop, by its Grassmann statistics, contributes $-|\mathcal{M}_{\rm gh}|^2$. Three different objects — an unphysical gluon polarization, a scalar field with wrong statistics, and a negative-norm intermediate state — are revealed to be one and the same degree of freedom seen from three sides. The ghost is the longitudinal gluon, wearing the sign that lets it cancel itself. This is why a spin-zero anticommuting field is not a contradiction but a necessity: it is the only object that can subtract a square.

26.4.5 Why diagram-by-diagram cancellation is not enough

We have shown the cancellation in one process, at one order, on one cut. The computation was already intricate: it required the precise relative normalization of three diagrams (fixed, in Eq. (26.74), only because the residue Eq. (26.69) had to be cancelled), a non-trivial polarization-sum identity, and the exact matching of the ghost-loop (-1) and the gluon Bose factor $\frac{1}{2}$ in Eq. (26.91). Each of these was a separate near-miracle of bookkeeping, and we verified them by hand. The physical theory, however, demands that the same cancellation hold for *every* amplitude, on *every* cut, at *every* order in g — for processes with arbitrarily many external gluons, where the number of diagrams grows factorially, where multi-gluon cuts mix with multi-ghost cuts, and where the longitudinal contamination is no longer captured by a single scalar $\mathcal{M}_{\rm gh}$ but by an intricate web of nested Ward identities. To check unitarity diagram by diagram, order by order, would be not merely laborious but hopeless: there is no finite computation that certifies an all-orders statement.

The structure of the proof points to the resolution. At root, the cancellation worked because of a single algebraic fact — the non-abelian Ward identity Eq. (26.75), $k_{1\mu}\mathcal{M}^{\mu\nu} = \mathcal{M}_{\rm gh}k_2^\nu$, which related the longitudinal gluon to the ghost. That identity, in turn, was a shadow of the commutator $[T^a, T^b] = if^{abc}T^c$, i.e. of the original gauge symmetry of $\mathcal{L}_{\rm YM}$. The gauge symmetry was broken by the gauge-fixing and ghost terms we added to make the path integral well-defined — but it cannot have been broken *completely*, or the longitudinal gluons and

the ghosts would not know to cancel. There must remain, in the gauge-fixed action $\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}}$, an *exact* residual symmetry that relates the gluon, ghost, and gauge-fixing sectors and that enforces the Ward identities to all orders as operator statements, in the way that Ward–Takahashi (Chapter 19) enforced current conservation in QED. A symmetry that mixes the commuting gauge field with the anticommuting ghost must itself be fermionic: it must turn a boson into a fermion and back.

That symmetry exists, it is global, it is nilpotent, and it is the organizing principle of everything that follows. Identifying it, proving its nilpotency from the Jacobi identity, and recognizing the entire gauge-fixing-plus-ghost sector as one symmetry-exact term — so that its invariance, and with it the unitarity we have just verified by hand, becomes automatic to all orders — is the business of the next section, where we construct the BRST transformation.

26.5 BRST Symmetry

The previous section made the case that ghosts are not optional. In a physical amplitude they are precisely what cancels the contributions of the timelike and longitudinal gluon polarizations, restoring the unitarity that those unphysical modes would otherwise spoil (Section 26.4). That cancellation was verified there in a particular process at a particular order, but a perturbative expansion has infinitely many orders, and a quantum field theory is only as trustworthy as the symmetry that protects its consistency at all of them. The Faddeev–Popov construction handed us a definite gauge-fixed Lagrangian (Eq. (26.37)); it did not, by itself, hand us a reason to believe that the ghost–antighost loops conspire to cancel the unphysical gluon loops in every diagram, in every gauge, to every order. What we want is an *exact* symmetry of the gauge-fixed action whose Ward identities enforce the cancellation as an algebraic identity rather than a per-diagram miracle.

Gauge invariance itself cannot be that symmetry: we deliberately broke it by adding the gauge-fixing and ghost terms, and the broken symmetry leaves no operative Ward identity behind. Remarkably, a single rigid (global) fermionic symmetry survives the gauge fixing. It was discovered independently by Becchi, Rouet and Stora and by Tyutin, and it is the organizing principle of the entire quantum theory: it defines the physical states (Section 26.6), it constrains the Green's functions through the Slavnov–Taylor identities (Section 26.7) that the renormalization of Chapter 27 requires, and its nilpotent charge is the structure reused in the analysis of anomalies (Chapter 36) and of the non-perturbative vacuum (Chapter 42). This section defines the transformation, proves it is a nilpotent symmetry of the full action, and constructs its conserved charge.

26.5.1 The Nakanishi–Lautrup field

Before writing the symmetry we make one technical improvement to the gauge-fixed Lagrangian. As it stands, the gauge-fixing term $\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2$ (Eq. (26.36)) is quadratic in $\partial \cdot A$, and the symmetry we are about to write will fail to close on the antighost unless this term is linearized. The device that linearizes it is an auxiliary field.

Definition 26.17 (Nakanishi–Lautrup field)**.** The *Nakanishi–Lautrup field* $B^a(x)$ is a real, commuting (bosonic), Lorentz-scalar field, one for each generator $a = 1, \dots, \dim \mathfrak{g}$, transforming in the adjoint representation and carrying ghost number zero. It appears in the action with no derivatives: it is a purely algebraic auxiliary field.

Replace the quadratic gauge-fixing term by the first-order Lagrangian

$$\mathcal{L}_{\text{gf}} \;=\; B^a\,\partial^\mu A^a_\mu \;+\; \frac{\xi}{2}\,B^a B^a. \tag{26.92}$$

Because B^a enters without derivatives, its Euler–Lagrange equation is purely algebraic,

$$\frac{\partial \mathcal{L}_{\text{gf}}}{\partial B^a} \;=\; \partial^\mu A^a_\mu + \xi\, B^a \;=\; 0 \qquad\Longrightarrow\qquad B^a \;=\; -\frac{1}{\xi}\,\partial^\mu A^a_\mu. \tag{26.93}$$

Substituting Eq. (26.93) back into Eq. (26.92) eliminates B^a and returns the original term,

$$B^a\partial^\mu A^a_\mu + \frac{\xi}{2}B^aB^a\Big|_{B=-\frac{1}{\xi}\partial\cdot A} \;=\; -\frac{1}{\xi}(\partial^\mu A^a_\mu)^2 + \frac{\xi}{2}\,\frac{1}{\xi^2}(\partial^\mu A^a_\mu)^2 \;=\; -\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2, \tag{26.94}$$

which is exactly Eq. (26.36). In the path integral the elimination is equally exact: B^a appears at most quadratically, so its integration is a Gaussian that can be performed once and for all, reproducing Eq. (26.94) with a field-independent Jacobian absorbed into the normalization. The two formulations are therefore physically identical. We keep the field B^a explicit because it makes the symmetry below close *off shell* — that is, without ever invoking the equation of motion Eq. (26.93) — and off-shell closure is what we need for an algebraic proof that holds inside loop integrals, where fields are off shell by construction.

Physical Insight

The Nakanishi–Lautrup field is the gauge-theory analogue of the auxiliary field F in a supersymmetric multiplet: it carries no independent dynamics, but its presence makes a symmetry algebra close without using equations of motion. Trading $-\frac{1}{2\xi}(\partial\cdot A)^2$ for $B\,\partial\cdot A + \frac{\xi}{2}B^2$ costs nothing physically

and buys an *off-shell* nilpotent symmetry. Whenever a symmetry seems to close only "on shell," the cure is almost always an auxiliary field that absorbs the offending equation-of-motion terms.

26.5.2 The BRST transformation

We now write the symmetry. The motivating observation is the one that has run through the whole chapter: the term $\mathcal{L}_{\text{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ is gauge invariant, and an infinitesimal gauge transformation acts on the connection by the adjoint covariant derivative of the parameter, $\delta A^a_\mu = \frac{1}{g}(D_\mu\alpha)^a$ (Eq. (25.36)). If we could make the gauge parameter *out of the ghost field*, the Yang–Mills term would be invariant automatically, because it is invariant under *every* gauge transformation. This is exactly what the BRST transformation does.

Definition 26.18 (BRST transformation)**.** Let θ be a global, spacetime-independent, anticommuting (Grassmann) parameter carrying ghost number -1, so that $\theta\, c^a$ is commuting. The *BRST transformation* of the fields of the gauge-fixed theory is $\delta_\theta\Phi = \theta\,(s\Phi)$, where the *BRST operator* s acts by

$$\boxed{sA^a_\mu = (D_\mu c)^a = \partial_\mu c^a + g f^{abc} A^b_\mu c^c, \qquad sc^a = -\tfrac{1}{2}\, g\, f^{abc} c^b c^c, \qquad s\bar{c}^a = B^a, \qquad sB^a = 0.} \tag{26.95}$$

Here $(D_\mu c)^a = D^{ab}_\mu c^b$ uses the adjoint covariant derivative $D^{ab}_\mu = \delta^{ab}\partial_\mu + g f^{aeb} A^e_\mu$ of Eq. (25.35). On a matter field ψ_i in a representation with generators T^a the transformation is the corresponding infinitesimal gauge rotation, $s\psi_i = i\, g\, c^a T^a \psi_i$.

The operator s is fermionic: it carries ghost number $+1$ and odd Grassmann parity, so it converts commuting fields into anticommuting ones and vice versa. Indeed sA^a_μ contains one ghost (ghost number $+1$, anticommuting), sc^a is quadratic in c (ghost number $+2$, commuting), $s\bar{c}^a = B^a$ has ghost number 0, and the assignments are uniform. Throughout, s obeys the graded Leibniz rule of an antiderivation: for fields X, Y of definite Grassmann parity $|X|, |Y| \in \{0, 1\}$,

$$s(XY) \;=\; (sX)\,Y \;+\; (-1)^{|X|}\, X\,(sY). \tag{26.96}$$

The sign $(-1)^{|X|}$ records that the odd operator s must be anticommuted past the field X before it can act on Y. The equivalent statement in terms of the parametrized variation $\delta_\theta = \theta\, s$ is that δ_θ is an ordinary (even) derivation obeying the unsigned Leibniz rule, because the anticommuting parameter θ carries all the signs; we use whichever form is more transparent.

The first transformation in Eq. (26.95) is the crucial one. Comparing $sA^a_\mu = (D_\mu c)^a$ with the infinitesimal gauge law $\delta A^a_\mu = \frac{1}{g}(D_\mu\alpha)^a$, the BRST transformation of A^a_μ *is* a gauge transformation with the field-dependent, Grassmann-even parameter $\alpha^a = g\,\theta\, c^a$. The remaining three transformations specify how

the new fields $c, \bar{c}, B$ move so that the whole structure squares to zero, which is the content of the next subsection.

Notation Watch

The overall sign of $s\bar{c}^a$ and of the auxiliary field B^a is a convention, because $\bar{c}^a$ is an *independent* Grassmann integration variable (Theorem 26.3), not the conjugate of c^a, and rescaling $\bar{c} \to -\bar{c}$, $B \to -B$ changes nothing physical: it leaves the Faddeev–Popov determinant, and hence every correlator, untouched. We fix the orientation by $s\bar{c}^a = B^a$. With this choice the BRST-exact gauge-fixing and ghost terms constructed in Section 26.5.4 reproduce the master Lagrangian Eq. (26.37) up to precisely this overall sign of the antighost; the reader who prefers to land on Eq. (26.37) verbatim need only send $\bar{c} \to -\bar{c}$.

26.5.3 Off-shell nilpotency: $s^2 = 0$

The defining property of s is that applying it twice annihilates everything. This is what makes s a coboundary operator and, ultimately, what makes the physical state space a cohomology.

Theorem 26.19 (Off-shell nilpotency)**.** *The BRST operator is nilpotent on every field, without use of any equation of motion:*

$$s^2 A^a_\mu = 0, \qquad s^2 c^a = 0, \qquad s^2 \bar{c}^a = 0, \qquad s^2 B^a = 0, \tag{26.97}$$

and $s^2 \psi_i = 0$ on matter. The proofs for c^a and A^a_μ rest on the Jacobi identity for the structure constants (Eq. (B.97)); those for $\bar{c}^a$ and B^a are immediate.

We prove the four cases in turn, carrying all the index algebra.

The trivial pair. From Eq. (26.95), $s\bar{c}^a = B^a$ and $sB^a = 0$, hence

$$s^2 \bar{c}^a = s(B^a) = 0, \qquad s^2 B^a = s(0) = 0. \tag{26.98}$$

The fields $(\bar{c}^a, B^a)$ form a *BRST doublet*: s maps the first onto the second and the second onto zero. Nilpotency here is automatic, and it is this trivial pair that the cohomology of Section 26.6 will discard.

The ghost: $s^2 c^a = 0$. This is the load-bearing computation. Apply s to $sc^a = -\tfrac{1}{2} g f^{abc} c^b c^c$, using the graded Leibniz rule Eq. (26.96) with $|c^b| = 1$:

$$\begin{aligned} s^2 c^a &= -\tfrac{1}{2} g\, f^{abc}\, s(c^b c^c) = -\tfrac{1}{2} g\, f^{abc} \big[(sc^b) c^c - c^b (sc^c) \big] \\ &= -\tfrac{1}{2} g\, f^{abc} \Big[\big(-\tfrac{1}{2} g f^{bde} c^d c^e \big) c^c - c^b \big(-\tfrac{1}{2} g f^{cde} c^d c^e \big) \Big] \end{aligned}$$

$$= \tfrac{1}{4} g^2 \, f^{abc} f^{bde} \, c^d c^e c^c \; - \; \tfrac{1}{4} g^2 \, f^{abc} f^{cde} \, c^b c^d c^e . \tag{26.99}$$

In the first term move c^c to the front of the product, $c^d c^e c^c = c^c c^d c^e$ (a cyclic permutation of three anticommuting factors is even), and relabel the summed indices to expose the common field structure $c^b c^d c^e$: rename the contracted index $b \to m$ and the field index $c \to b$, giving $f^{abc} f^{bde} \to f^{amb} f^{mde}$. In the second term rename the contracted index $c \to m$, $f^{abc} f^{cde} \to f^{abm} f^{mde}$. Then

$$s^2 c^a = \tfrac{1}{4} g^2 (f^{amb} - f^{abm}) f^{mde} \, c^b c^d c^e = -\tfrac{1}{2} g^2 \, f^{abm} f^{mde} \, c^b c^d c^e , \tag{26.100}$$

where we used the antisymmetry $f^{amb} = -f^{abm}$ in the last two indices, so that $f^{amb} - f^{abm} = -2 f^{abm}$.

It remains to show that $f^{abm} f^{mde} \, c^b c^d c^e = 0$. The product $c^b c^d c^e$ is totally antisymmetric in (b, d, e) because the three ghosts anticommute, so contracting it against $f^{abm} f^{mde}$ projects onto the part of that coefficient which is totally antisymmetric in (b, d, e). Since f^{mde} is already antisymmetric in (d, e), the totally antisymmetric part is the cyclic average

$$f^{abm} f^{mde} \, c^b c^d c^e = \tfrac{1}{3} \Big(f^{abm} f^{mde} + f^{adm} f^{meb} + f^{aem} f^{mbd} \Big) \, c^b c^d c^e . \tag{26.101}$$

The bracket is precisely the Jacobi identity. To see this, use the total antisymmetry of f to bring each factor to a canonical cyclic form, $f^{abm} = f^{bma}$, $f^{mde} = f^{dem}$, and likewise for the other two pairs, so the bracket equals $f^{bma} f^{dem} + f^{dma} f^{ebm} + f^{ema} f^{bdm}$, which is the Jacobi identity Eq. (B.97) with the free index carried by a and the cyclic sum running over (b, d, e). It vanishes. Therefore the right-hand side of Eq. (26.101) is zero, and with Eq. (26.100),

$$s^2 c^a = 0 . \tag{26.102}$$

The nilpotency of s on the ghost is, quite literally, the Jacobi identity of the gauge algebra in disguise. This is the algebraic root of the whole construction: the same identity that guarantees the structure constants define a Lie algebra guarantees that the BRST operator is a coboundary.

The connection: $s^2 A^a_\mu = 0$. Apply s to $sA^a_\mu = \partial_\mu c^a + g f^{abc} A^b_\mu c^c$. The derivative commutes with s, and A^b_μ is bosonic so it carries no sign in Eq. (26.96):

$$\begin{aligned} s^2 A^a_\mu &= \partial_\mu (s c^a) + g f^{abc} \big[(s A^b_\mu) c^c + A^b_\mu (s c^c) \big] \\ &= \partial_\mu \big(-\tfrac{1}{2} g f^{ade} c^d c^e \big) + g f^{abc} \Big[(\partial_\mu c^b + g f^{bpq} A^p_\mu c^q) c^c + A^b_\mu \big(-\tfrac{1}{2} g f^{cde} c^d c^e \big) \Big] . \end{aligned} \tag{26.103}$$

Collect the terms by their field content. There are terms of order g linear in ∂c and quadratic in c but with no A, and terms of order g^2 with one A and two c's.

Order g, the $\partial c\,c$ terms. These are

$$-\tfrac{1}{2}gf^{ade}\,\partial_\mu(c^d c^e)\;+\;gf^{abc}\,(\partial_\mu c^b)\,c^c=-\tfrac{1}{2}gf^{ade}(\partial_\mu c^d\,c^e+c^d\,\partial_\mu c^e)+gf^{abc}\,\partial_\mu c^b\,c^c. \tag{26.104}$$

In the first bracket the two pieces are equal: in $c^d\,\partial_\mu c^e$ anticommute the factors, $c^d\,\partial_\mu c^e=-\partial_\mu c^e\,c^d$, and relabel $d\leftrightarrow e$ using $f^{aed}=-f^{ade}$, which returns $\partial_\mu c^d\,c^e$. Hence $-\tfrac{1}{2}gf^{ade}(\partial_\mu c^d c^e+c^d\partial_\mu c^e)=-gf^{ade}\,\partial_\mu c^d\,c^e$. Relabel the dummies $b\to d$, $c\to e$ in the last term of Eq. (26.104) to get $+gf^{ade}\,\partial_\mu c^d\,c^e$. The two cancel:

$$-gf^{ade}\,\partial_\mu c^d\,c^e\;+\;gf^{ade}\,\partial_\mu c^d\,c^e\;=\;0. \tag{26.105}$$

Order g^2, the $A\,c\,c$ terms. These are

$$g^2f^{abc}f^{bpq}\,A^p_\mu c^q c^c\;-\;\tfrac{1}{2}g^2f^{abc}f^{cde}\,A^b_\mu c^d c^e. \tag{26.106}$$

Relabel the first term to the common structure $A^b_\mu c^d c^e$: rename the contracted index $b\to m$ and the field indices $p\to b$, $q\to d$, $c\to e$, giving $f^{abc}f^{bpq}\to f^{ame}f^{mbd}$; rename $c\to m$ in the second term, $f^{abc}f^{cde}\to f^{abm}f^{mde}$. Then Eq. (26.106) is

$$g^2\Big(f^{ame}f^{mbd}-\tfrac{1}{2}f^{abm}f^{mde}\Big)A^b_\mu\,c^d c^e. \tag{26.107}$$

The product $c^d c^e$ is antisymmetric in (d,e), so only the (d,e)-antisymmetric part of the coefficient survives. Antisymmetrizing,

$$\tfrac{1}{2}\Big[(f^{ame}f^{mbd}-\tfrac{1}{2}f^{abm}f^{mde})-(f^{amd}f^{mbe}-\tfrac{1}{2}f^{abm}f^{med})\Big]=\tfrac{1}{2}\Big[f^{ame}f^{mbd}-f^{amd}f^{mbe}-f^{abm}f^{mde}\Big], \tag{26.108}$$

where $f^{med}=-f^{mde}$ combined the two $f^{abm}f^{mde}$ pieces. Now rewrite the first two products with antisymmetry, $f^{ame}f^{mbd}=f^{bdm}f^{ame}$ (cyclic in the second factor) and $f^{amd}f^{mbe}=f^{adm}f^{bme}$ (one transposition in each factor, signs cancelling), so the bracket becomes $f^{bdm}f^{ame}-f^{adm}f^{bme}-f^{abm}f^{mde}$. This is again the Jacobi identity: the adjoint-representation form $[\,T^a_{\rm adj},T^b_{\rm adj}\,]=if^{abc}T^c_{\rm adj}$ of Eq. (B.97), written with $(T^a_{\rm adj})_{xy}=-if^{axy}$, reads exactly $f^{bdm}f^{ame}-f^{adm}f^{bme}=f^{abm}f^{mde}$. The bracket vanishes, hence Eq. (26.107) is zero. Together with Eq. (26.105),

$$s^2A^a_\mu=0. \tag{26.109}$$

Matter. For completeness we verify nilpotency on a matter field, since it advances Yang–Mills theory and exhibits the same mechanism through the closure of the algebra rather than the Jacobi identity. With $s\psi=ig\,c^aT^a\psi$,

$$\begin{aligned}s^2\psi&=ig\,(sc^a)T^a\psi-ig\,c^aT^a(s\psi)=ig(-\tfrac{1}{2}gf^{abc}c^bc^c)T^a\psi-ig\,c^aT^a(ig\,c^bT^b\psi)\\&=-\tfrac{i}{2}g^2f^{abc}\,c^bc^c\,T^a\psi+g^2\,c^ac^b\,T^aT^b\psi.\end{aligned} \tag{26.110}$$

In the second line the relative sign comes from $|c^aT^a|=1$ in Eq. (26.96) and $(i)(i)=-1$ from the two factors of i. Because c^ac^b is antisymmetric in (a,b),

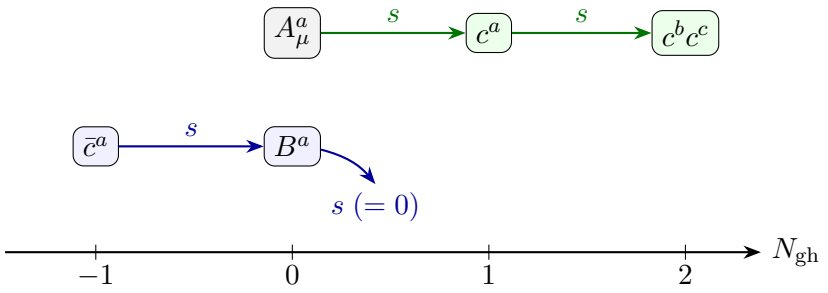


Figure 26.7: The action of the nilpotent BRST operator s, graded by ghost number N_{gh} (each application of s shifts N_{gh} by $+1$). The antighost–auxiliary pair $(\bar{c}^a, B^a)$ is a *trivial doublet*: $s\bar{c}^a = B^a$, $sB^a = 0$, so s^2 kills it at once. On the connection and the ghost, $sA^a_\mu = (D_\mu c)^a$ and $sc^a = -\frac{1}{2} g f^{abc} c^b c^c$, and $s^2 = 0$ holds only by virtue of the Jacobi identity. The grading and the nilpotency are exactly the structure that the cohomology of Section 26.6 exploits.

the product $T^a T^b$ may be replaced by its antisymmetric part, $c^a c^b T^a T^b = c^a c^b \frac{1}{2}[T^a, T^b] = \frac{i}{2} f^{abc} c^a c^b T^c$, using the Lie algebra Section B.4. Renaming $(a, b, c) \to (c, a, b)$ and using cyclicity $f^{cab} = f^{abc}$ in the first term of Eq. (26.110) turns it into $-\frac{i}{2} g^2 f^{abc} c^a c^b T^c \psi$, which cancels the second term exactly. Hence $s^2 \psi = 0$.

This completes the proof of Theorem 26.19. Every transformation in Eq. (26.95) is consistent with $s^2 = 0$, and nowhere did we use an equation of motion: the nilpotency is off shell, valid for arbitrary field configurations, exactly as required for an identity that must hold inside loop integrals.

26.5.4 The gauge-fixing and ghost sector is BRST-exact

We can now exhibit the entire non-Yang–Mills part of the Lagrangian — the gauge-fixing term and the ghost term together — as the BRST variation of a single fermionic object. This is the structural statement that makes BRST invariance automatic.

Definition 26.20 (Gauge fermion)**.** The *gauge fermion* is the Grassmann-odd, ghost-number -1 local functional

$$\Psi \;=\; \bar{c}^a \Big(\partial^\mu A^a_\mu + \frac{\xi}{2} B^a \Big). \tag{26.111}$$

Worked Example 26.21 (The gauge-fixing and ghost terms as $s\Psi$)**.** We compute $s\Psi$ from the transformations Eq. (26.95) and the graded Leibniz rule, and show it equals $\mathcal{L}_{\mathrm{gf}} + \mathcal{L}_{\mathrm{ghost}}$.

Write $\Psi = \bar{c}^a\, Y^a$ with the bosonic, ghost-number-zero factor $Y^a = \partial^\mu A^a_\mu + \frac{\xi}{2} B^a$. Since $|\bar{c}^a| = 1$, the graded Leibniz rule Eq. (26.96) gives

$$s\Psi = (s\bar{c}^a)\, Y^a - \bar{c}^a\, (sY^a). \tag{26.112}$$

The first factor is $s\bar{c}^a = B^a$. For the second, s commutes with ∂^μ and $sB^a = 0$, so $sY^a = \partial^\mu(sA^a_\mu) + \frac{\xi}{2}\, sB^a = \partial^\mu (D_\mu c)^a$. Therefore

$$s\Psi \;=\; B^a\Big(\partial^\mu A^a_\mu + \frac{\xi}{2} B^a\Big) - \bar{c}^a\, \partial^\mu (D_\mu c)^a \;=\; \underbrace{B^a \partial^\mu A^a_\mu + \frac{\xi}{2} B^a B^a}_{\mathcal{L}_{\text{gf}}} \underbrace{- \bar{c}^a\, \partial^\mu (D_\mu c)^a}_{\mathcal{L}_{\text{ghost}}}. \tag{26.113}$$

The first two terms are exactly the first-order gauge-fixing Lagrangian Eq. (26.92). The last term is the Faddeev–Popov ghost Lagrangian: it is $-\bar{c}^a\, \partial^\mu D^{ab}_\mu c^b$, which is Eq. (26.30) up to the overall antighost sign flagged in the Notation Watch above, and after the integration by parts of Eq. (26.31) it is the kinetic-plus-vertex ghost term of the master Lagrangian Eq. (26.37). Thus

$$\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{ghost}} \;=\; s\Psi. \tag{26.114}$$

Integrating out B^a by Eqs. (26.93) and (26.94) returns the familiar $-\frac{1}{2\xi}(\partial \cdot A)^2$ together with the ghost term, so the B-field and second-order formulations agree.

The full gauge-fixed Yang–Mills Lagrangian is therefore the classical invariant plus a BRST-exact piece,

$$\boxed{\mathcal{L} \;=\; \mathcal{L}_{\text{YM}} + s\Psi \;=\; -\tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} + \bar{\psi}_i (i \not{D} - m)\psi_i + s\Psi,} \tag{26.115}$$

with $\mathcal{L}_{\text{YM}}$ the gauge-invariant Yang–Mills-plus-matter Lagrangian (Eq. (25.76)) and $s\Psi$ given by Eq. (26.113). Written this way, BRST invariance is a one-line consequence of nilpotency.

Theorem 26.22 (BRST invariance of the action)**.** *The action $S = \int d^4x\, \mathcal{L}$ of Eq. (26.115) is invariant under the BRST transformation Eq. (26.95): $s\mathcal{L} = 0$ up to a total derivative, and $\delta_\theta S = 0$.*

Proof. Split $\mathcal{L} = \mathcal{L}_{\text{YM}} + s\Psi$. For the first piece, $sA^a_\mu = (D_\mu c)^a$ is an infinitesimal gauge transformation with the Grassmann-even parameter $\alpha^a = g\,\theta\, c^a$ (and $s\psi_i = ig\, c^a T^a \psi_i$ is the matching matter rotation), and $\mathcal{L}_{\text{YM}}$ is invariant under *every* gauge transformation by construction (Section 25.4); the field strength transforms covariantly (Theorem 25.14), $\delta F^a_{\mu\nu} = g f^{abc} F^b_{\mu\nu}\, \theta c^c$, and the singlet $F^a_{\mu\nu} F^{a\,\mu\nu}$ is therefore inert, $s\mathcal{L}_{\text{YM}} = 0$, exactly and off shell. (That α^a is a composite, nilpotent parameter rather than an ordinary c-number function is irrelevant: gauge invariance of $\mathcal{L}_{\text{YM}}$ is an algebraic identity in the parameter,

valid for any adjoint-valued, commuting $\alpha^a(x)$.) For the second piece, nilpotency Theorem 26.19 gives

$$s(s\Psi) = s^2\Psi = 0 \tag{26.116}$$

identically, since s^2 annihilates every field and hence every functional built from them. Therefore $s\mathcal{L} = s\mathcal{L}_{\rm YM} + s^2\Psi = 0$. The variation of the density is at most a total derivative (it is in fact zero in the B-field form), so $\delta_\theta S = \theta \int d^4x\, s\mathcal{L} = 0$. □

This is the decisive structural point. We did not have to check, term by term, that the ghost loops cancel the longitudinal gluon loops; we exhibited the gauge-fixing and ghost terms as s of *something*, so their BRST variation is $s^2(\cdot) = 0$ for free, while the gauge-invariant core is invariant because s acts there as a gauge transformation. The exactness in Eq. (26.115) also makes manifest why physical quantities cannot depend on the gauge parameter ξ: a change of gauge fixing changes Ψ, hence changes $\mathcal{L}$ by a BRST-exact amount $s(\delta\Psi)$, and a BRST-exact insertion has vanishing physical matrix elements once the cohomology of Section 26.6 is in place.

26.5.5 The conserved BRST current and charge

A global continuous symmetry has, by Noether's theorem, a conserved current and a conserved charge. BRST is such a symmetry — rigid (the parameter θ is constant) and continuous — and its charge is the central object of the quantum theory.

To construct the current, promote $\theta \to \theta(x)$. Because $\delta_\theta S = 0$ for constant θ, the variation of the action for spacetime-dependent $\theta(x)$ can depend only on $\partial_\mu\theta$, and integrating by parts exhibits the coefficient as the divergence of a current that is conserved on the equations of motion. For a first-order Lagrangian the current is $j^\mu = \sum_i (\partial\mathcal{L}/\partial(\partial_\mu\Phi_i))(s\Phi_i) - K^\mu$, where K^μ collects the surface term left by $s\mathcal{L} = \partial_\mu K^\mu$ and the field derivatives are taken with the orientation appropriate to each field's Grassmann parity. Carrying out the bookkeeping on $\mathcal{L} = -\frac{1}{4}F^2 + B^a\partial\cdot A^a + \frac{\xi}{2}B^2 + (\partial^\mu\bar c^a)(D_\mu c)^a + \mathcal{L}_{\rm matter}$ (the integrated-by-parts ghost form), with the canonical momenta $-F^{a\mu\nu} + B^a\eta^{\mu\nu}$ for A, $\partial^\mu\bar c^a$ for c, $(D^\mu c)^a$ for $\bar c$, and $\bar\psi_i\, i\gamma^\mu$ for ψ, and the surface term $K^\mu = B^a(D^\mu c)^a$ from $s\mathcal{L} = \partial_\mu[B^a(D^\mu c)^a]$, gives the

$$\boxed{\; j^\mu_B \;=\; -F^{a\,\mu\nu}(D_\nu c)^a \;+\; B^a(D^\mu c)^a \;-\; \tfrac{1}{2}g\, f^{abc}\,(\partial^\mu\bar c^a)\, c^b c^c \;+\; g\, c^a\, \bar\psi_i\gamma^\mu T^a\psi_i, \;} \tag{26.117}$$

the conserved BRST (Noether) current, $\partial_\mu j^\mu_B = 0$ on the equations of motion. Every term carries ghost number $+1$: $(D_\nu c)$ and $(D^\mu c)$ each contain one ghost, the $(\partial\bar c)\,c\,c$ term has net ghost number $-1+2=+1$, and the matter term carries one c. The current is fermionic, as befits the generator of a fermionic symmetry.

Definition 26.23 (BRST charge)**.** The *BRST charge* is the conserved Noether charge of Eq. (26.117),

$$Q_B \;=\; \int d^3x\; j^0_B, \tag{26.118}$$

a Hermitian, fermionic operator of ghost number $+1$. It generates the BRST transformation through the graded commutator,

$$s\Phi \;=\; i\,[\,Q_B, \Phi\,\}, \tag{26.119}$$

where $[\,\cdot,\cdot\,\}$ denotes the commutator when Φ is bosonic and the anticommutator when Φ is fermionic.

The Hermiticity of Q_B holds with respect to the indefinite inner product of the enlarged state space carrying the ghosts; it is what allows the cohomology of Section 26.6 to define a sensible (positive) physical inner product. Conservation, $dQ_B/dt = 0$, follows from $\partial_\mu j^\mu_B = 0$ and the vanishing of spatial surface terms; it makes Q_B a symmetry generator of the full interacting theory, not merely of the free fields.

The single most important property of Q_B is inherited directly from Theorem 26.19.

Theorem 26.24 (Nilpotency of the BRST charge)**.** *The BRST charge squares to zero,*

$$Q_B^2 \;=\; \tfrac{1}{2}\{Q_B, Q_B\} \;=\; 0. \tag{26.120}$$

Proof. We first relate s^2 to Q_B^2 through Eq. (26.119). Let Φ be any field; then $s\Phi = i[Q_B, \Phi\}$ and $s^2\Phi = i\big[Q_B,\, i[Q_B, \Phi\}\,\}\big] = -\big[Q_B, [Q_B, \Phi\}\big]$. We evaluate the double graded bracket. For Φ bosonic, $[Q_B, \Phi\} = [Q_B, \Phi]$ is fermionic, and

$$\big[Q_B, [Q_B, \Phi]\big\} = \{Q_B, [Q_B, \Phi]\} = Q_B(Q_B\Phi - \Phi Q_B) + (Q_B\Phi - \Phi Q_B)Q_B = Q_B^2\Phi - \Phi Q_B^2 = [Q_B^2, \Phi] \tag{26.121}$$

For Φ fermionic, $[Q_B, \Phi\} = \{Q_B, \Phi\}$ is bosonic, and the same expansion gives $\big[Q_B, \{Q_B, \Phi\}\big] = Q_B^2\Phi - \Phi Q_B^2 = [Q_B^2, \Phi]$. In both cases

$$s^2\Phi \;=\; -\,[\,Q_B^2, \Phi\,]. \tag{26.122}$$

By Theorem 26.19, $s^2\Phi = 0$ for every field Φ. Hence $[Q_B^2, \Phi] = 0$ for all fields: the bosonic operator Q_B^2 commutes with every field in the theory, so it commutes with every operator built from them and is a multiple of the identity (a c-number) on the irreducible field algebra. But Q_B^2 carries ghost number $+2$, while the identity carries ghost number 0; the only c-number with ghost number $+2$ is zero. Therefore $Q_B^2 = 0$. $\square$

The ghost-number argument deserves emphasis, because it is the cleanest possible proof and it recurs throughout BRST theory: an operator that is both

central and charged under a conserved grading must vanish. The book has already established the two ingredients independently — the off-shell nilpotency $s^2 = 0$ (Theorem 26.19) and the ghost-number grading under which s, and hence Q_B, carries charge $+1$ — so the conclusion $Q_B^2 = 0$ is forced.

Physical Insight

Nilpotency is the algebraic shadow of gauge redundancy. A gauge transformation followed by "the same gauge transformation again" is, infinitesimally, just another gauge transformation; BRST repackages this so that doing it twice gives *nothing at all*, $Q_B^2 = 0$. The price is the ghost fields, whose statistics are wrong for particles but exactly right to make the square of an odd generator vanish. A bosonic "gauge charge" could never be nilpotent; only a fermionic one can. This is why ghosts, far from being a bookkeeping trick, are forced on us the moment we demand an exact, nilpotent symmetry to police unitarity.

With the conserved, nilpotent, ghost-number-$+1$ charge Q_B in hand, the gauge-fixed quantum theory carries an exact symmetry whose square is zero. That is precisely the input a cohomology needs. In the next section we define the physical Hilbert space as the cohomology of Q_B — closed states ($Q_B\,|\psi\rangle = 0$) modulo exact states ($|\psi\rangle = Q_B\,|\eta\rangle$) at ghost number zero — and prove that this quotient is exactly the positive-norm space of two transverse gluon polarizations, with the unphysical timelike and longitudinal gluons and the ghost–antighost pairs organizing into zero-norm BRST quartets that decouple from every physical amplitude. The nilpotency $Q_B^2 = 0$ proved here (Theorem 26.24) is the one fact that makes that construction possible.

26.6 Physical States from BRST Cohomology

The previous section delivered the object that will now do all the work: a global, nilpotent fermionic symmetry of the gauge-fixed action, generated by a conserved charge Q_B with $Q_B^2 = 0$, Hermitian with respect to the inner product of the enlarged state space (Section 26.5, Theorem 26.23, Theorem 26.24). Nilpotency is not a decorative feature. It is exactly the algebraic property a charge must have if its kernel and image are to define a *cohomology*, and cohomology is precisely the right tool for the problem we inherited from Section 26.1.2: the gauge-fixed Yang–Mills Lagrangian propagates four polarizations of the gluon together with a ghost and an antighost, none of which is supposed to be a physical particle, on a state space whose inner product is not positive definite. We must extract from this enlarged, indefinite-metric space a genuine Hilbert space — positive norm, two transverse gluon polarizations per momentum, and nothing else — and we must prove that the S-matrix restricted to it is unitary. This is the

non-abelian completion of the program Chapter 13 carried out for the photon by the Gupta–Bleuler construction (Section 13.3); the modern statement, valid when the structure constants do not vanish, is that the physical Hilbert space is the cohomology of Q_B.

We proceed in four movements. First we define the physical space as $\ker Q_B / \operatorname{im} Q_B$ and explain, before any computation, why that is the inevitable definition: closedness encodes gauge invariance, exactness encodes gauge redundancy. Second we prove that gauge-invariant observables are Q_B-closed and that their physical matrix elements depend only on cohomology classes, so the definition is consistent. Third we descend to the asymptotic free fields, build the BRST algebra of one-particle states explicitly, identify the quartet of unphysical modes, and prove — by a homotopy (Hodge-type) argument that makes the Kugo–Ojima quartet mechanism completely transparent — that the cohomology is exactly the transverse two-polarization Fock space, with positive norm. From positivity and the BRST invariance of the S-matrix, unitarity of the physical S-matrix follows in one line. Fourth we set $f^{abc} \to 0$ and watch the whole apparatus collapse onto the abelian Gupta–Bleuler construction of Chapter 13, with Q_B becoming the Gupta–Bleuler subsidiary condition; the two threads, QED and Yang–Mills, are shown to be the same construction seen at two depths. The ξ-independence of physical amplitudes drops out as a corollary.

26.6.1 The physical Hilbert space as a cohomology

Recall the conventions fixed in Section 26.5. The BRST operator s acts on the fields of the gauge-fixed theory by

$$sA^a_\mu = D^{ab}_\mu c^b = \partial_\mu c^a + g f^{abc} A^b_\mu c^c, \quad sc^a = -\tfrac{1}{2} g f^{abc} c^b c^c, \quad s\bar{c}^a = B^a, \quad sB^a = 0, \tag{26.123}$$

the transformation Eq. (26.95), with B^a the Nakanishi–Lautrup auxiliary field whose equation of motion is $B^a = -\frac{1}{\xi}\partial^\mu A^a_\mu$ (Eq. (26.93)). The full gauge-fixed Lagrangian is the classical invariant plus a BRST-exact piece,

$$\mathcal{L} = \mathcal{L}_{\mathrm{YM}} + s\Big(\bar{c}^a \big[\partial^\mu A^a_\mu + \tfrac{\xi}{2} B^a\big]\Big), \tag{26.124}$$

the decomposition Eq. (26.115), and the global symmetry is generated by the conserved Noether charge Q_B of Theorem 26.23, which is nilpotent (Theorem 26.24),

$$Q_B^2 = 0, \qquad s\,\Phi = i\,[\,Q_B, \Phi\,\} \quad \text{for any field } \Phi, \tag{26.125}$$

where $[\,\cdot\,,\,\cdot\,\}$ is the graded commutator (an anticommutator when Φ is Grassmann-odd, a commutator when it is Grassmann-even). The charge Q_B acts on the indefinite-metric Fock space $\mathcal{V}$ of the gauge-fixed theory — the space built by the mode operators of A^a_μ (four polarizations, with the timelike one carrying negative norm exactly as in Theorem 13.16), c^a, $\bar{c}^a$, and B^a.

There is a second conserved charge we will need, the *ghost number*. The action Eq. (26.124) is invariant under the global phase rotation $c \to e^{\theta} c$, $\bar{c} \to e^{-\theta} \bar{c}$ (with A, B inert), because c and $\bar{c}$ always appear in pairs. Its Noether charge Q_c assigns ghost number $+1$ to c, -1 to $\bar{c}$, and 0 to A, B, and grades the Fock space,

$$\mathcal{V} = \bigoplus_{n \in \mathbb{Z}} \mathcal{V}_n, \qquad Q_c \, \mathcal{V}_n = n \, \mathcal{V}_n. \tag{26.126}$$

Reading off the ghost numbers of the right-hand sides of Eq. (26.123) shows that s — and therefore Q_B — raises ghost number by one unit: $[Q_c, Q_B] = Q_B$, so $Q_B : \mathcal{V}_n \to \mathcal{V}_{n+1}$. The vacuum and the physical particles we want to keep all carry ghost number zero.

With Q_B nilpotent, the standard cohomological language applies. A state is *Q_B-closed* (or BRST-invariant) if $Q_B \, |\psi\rangle = 0$, and *Q_B-exact* if $|\psi\rangle = Q_B \, |\eta\rangle$ for some $|\eta\rangle \in \mathcal{V}$. Because $Q_B^2 = 0$, every exact state is closed, $\operatorname{im} Q_B \subseteq \ker Q_B$, so the quotient below is well defined.

Definition 26.25 (Physical states as BRST cohomology)**.** The space of physical states of quantized Yang–Mills theory is the BRST cohomology of Q_B at ghost number zero,

$$\mathcal{H}_{\text{phys}} \;=\; \frac{\ker Q_B}{\operatorname{im} Q_B}\bigg|_{\text{ghost number } 0} \;=\; \frac{\{|\psi\rangle \in \mathcal{V}_0 : Q_B \, |\psi\rangle = 0\}}{\{Q_B \, |\eta\rangle : |\eta\rangle \in \mathcal{V}_{-1}\}}. \tag{26.127}$$

A physical state is an equivalence class $[\psi]$: two BRST-closed states are physically identical when they differ by a BRST-exact state, $|\psi\rangle \sim |\psi\rangle + Q_B \, |\eta\rangle$.

The two operations in Eq. (26.127) have transparent physical meanings, and it is worth stating them before we earn them with proofs, because they are the whole reason cohomology is the right structure. *Taking the kernel* — keeping only Q_B-closed states — is the quantum statement of gauge invariance: as we prove in Section 26.6.2, gauge-invariant operators are exactly the Q_B-closed ones, and demanding $Q_B \, |\psi\rangle = 0$ selects the states on which gauge-invariant observables have unambiguous, representation-blind matrix elements. *Quotienting by the image* — discarding Q_B-exact states — is the quantum statement of gauge redundancy: an exact state turns out to have zero norm and to be orthogonal to every closed state, so it is physically invisible and must be set to zero to obtain a genuine inner-product space. This is the exact non-abelian analogue of the two steps Chapter 13 took with the photon: imposing the Gupta–Bleuler condition $(\partial \cdot A)^{(+)} \, |\text{phys}\rangle = 0$ to select V_{phys} (the abelian "kernel"), then quotienting by the zero-norm subspace V_0 to reach $\mathcal{H}_{\text{phys}} = V_{\text{phys}}/V_0$ (Theorem 13.21, Theorem 13.20). We will make that correspondence precise in Section 26.6.6.

Theorem 26.26 (Well-definedness and content of the physical space)**.** *The quotient Eq. (26.127) is a well-defined complex vector space carrying a positive-definite inner product induced from $\mathcal{V}$. For the asymptotic (free) fields of Yang–Mills theory it is canonically isomorphic, as an inner-product space, to the Fock space generated by the two transverse gluon polarizations per momentum and colour; all other modes — the timelike and longitudinal gluons, the ghost and the antighost — are absent from the cohomology. The S-matrix restricted to $\mathcal{H}_{\mathrm{phys}}$ is unitary.*

The theorem packages every result of the section. Its proof occupies Sections 26.6.2 to 26.6.5: well-definedness and the inner-product statement come from the observable and positivity arguments, the identification of the content from the explicit quartet construction, and unitarity from positivity plus BRST invariance of the S-matrix.

26.6.2 Gauge-invariant observables are BRST-closed

Before analysing states we settle observables, because the consistency of Theorem 26.25 — the claim that physics depends only on cohomology classes, not on representatives — is a statement about operators, and it must hold for the definition to make sense at all.

A physical observable is, by the gauge principle, a gauge-invariant local operator: $\operatorname{tr} F_{\mu\nu}F^{\mu\nu}$, the Wilson loop, a colour-singlet matter bilinear, and so on. Under the BRST transformation Eq. (26.123), a gauge transformation with the constant Grassmann parameter replaced by the ghost field c^a, any such operator is invariant.

Proposition 26.27 (Gauge invariance implies BRST-closedness)**.** *Let $\mathcal{O}$ be a Grassmann-even local operator built from the matter fields and the gauge field that is invariant under infinitesimal gauge transformations with parameter α^a. Then $s\mathcal{O} = 0$, equivalently $[\,Q_B, \mathcal{O}\,] = 0$: $\mathcal{O}$ is BRST-closed.*

Proof. By hypothesis the variation of $\mathcal{O}$ under an infinitesimal gauge transformation vanishes for every parameter,

$$\delta_\alpha \mathcal{O} = \int \mathrm{d}^4 y\, \alpha^a(y)\, \frac{\delta \mathcal{O}}{\delta \alpha^a(y)} = 0 \quad \text{identically in } \alpha. \tag{26.128}$$

The BRST variation is the same field variation with the c-number parameter $\alpha^a(y)$ replaced everywhere by the ghost field $c^a(y)$: on A^a_μ the gauge variation is $\delta_\alpha A^a_\mu = D^{ab}_\mu \alpha^b$ (Eq. (26.14)) and the BRST variation is $sA^a_\mu = D^{ab}_\mu c^b$ (Eq. (26.123)); the matter transformations are related the same way. Since $\mathcal{O}$ contains no $\bar{c}$ or B, the only fields s acts on inside $\mathcal{O}$ are exactly those acted on by δ_α, with $\alpha^a \mapsto c^a$. Therefore $s\mathcal{O}$ is obtained from Eq. (26.128) by that replacement, and because

Eq. (26.128) holds identically in the function α, it holds in particular for the Grassmann-odd "value" c:

$$s\mathcal{O} = \int \mathrm{d}^4y\, c^a(y)\, \frac{\delta \mathcal{O}}{\delta \alpha^a(y)} = 0. \tag{26.129}$$

By Eq. (26.125), $s\mathcal{O} = i[Q_B, \mathcal{O}]$ (a commutator since $\mathcal{O}$ is Grassmann-even), so $[Q_B, \mathcal{O}] = 0$. □

The converse direction — that BRST-closed operators modulo BRST-exact ones are the full algebra of observables — is the operator cohomology underlying the Slavnov–Taylor analysis of Section 26.7; here we need only the implication just proved. Its immediate payoff is that physical matrix elements respect the equivalence relation of Theorem 26.25.

Theorem 26.28 (Matrix elements depend only on cohomology classes)**.** *Let $\mathcal{O}$ be BRST-closed, $[Q_B, \mathcal{O}\} = 0$, and let $|\psi\rangle, |\phi\rangle$ be Q_B-closed states. Then the matrix element $\langle\psi|\mathcal{O}|\phi\rangle$ is unchanged when either state is shifted by a Q_B-exact state. Consequently $\langle\psi|\mathcal{O}|\phi\rangle$ depends only on the classes $[\psi], [\phi] \in \mathcal{H}_{\text{phys}}$, and in particular $\langle\psi|\phi\rangle$ (the case $\mathcal{O} = \mathbf{1}$) descends to a well-defined inner product on $\mathcal{H}_{\text{phys}}$.*

Proof. Replace $|\phi\rangle$ by $|\phi\rangle + Q_B|\eta\rangle$. The change in the matrix element is $\langle\psi|\mathcal{O}\, Q_B|\eta\rangle$. Move Q_B to the left through $\mathcal{O}$ using $\mathcal{O}Q_B = \pm Q_B\mathcal{O}$ — the sign is $+$ for Grassmann-even $\mathcal{O}$ and $-$ for Grassmann-odd $\mathcal{O}$, but in either case $\mathcal{O}Q_B = (-1)^{|\mathcal{O}|}Q_B\mathcal{O}$ because $\mathcal{O}$ is BRST-closed — so

$$\langle\psi|\mathcal{O}\, Q_B|\eta\rangle = (-1)^{|\mathcal{O}|}\,\langle\psi|Q_B\,\mathcal{O}|\eta\rangle = (-1)^{|\mathcal{O}|}(\,\langle\psi|\,Q_B)\mathcal{O}\,|\eta\rangle\,. \tag{26.130}$$

Now use the Hermiticity of Q_B (Theorem 26.23): the bra $\langle\psi|\,Q_B = (Q_B\,|\psi\rangle)^\dagger = 0$ because $|\psi\rangle$ is Q_B-closed. Hence the change vanishes. The same argument applied to $|\psi\rangle$ (or its conjugate statement) shows independence of the representative of $[\psi]$. Taking $\mathcal{O} = \mathbf{1}$, which is trivially BRST-closed, gives $\langle\psi|\phi\rangle = \langle\psi + Q_B\eta'|\phi + Q_B\eta\rangle$, so the inner product is constant on classes and descends to $\mathcal{H}_{\text{phys}}$. □

Theorem 26.28 is the linchpin. It says the cohomology quotient is not merely a formal device but is forced on us by Hermiticity and nilpotency: the inner product simply does not distinguish a closed state from itself plus an exact state, so the physical content lives on the classes. What remains is to show that this induced inner product is positive definite — that no negative-norm or spurious zero-norm state survives in the cohomology. That is a statement about the detailed spectrum, and to prove it we descend to the asymptotic free fields.

26.6.3 The asymptotic free-field BRST algebra and the quartet

Unitarity of the S-matrix is a statement about asymptotic in- and out-states, which are free particles. We therefore study Q_B on the Fock space of the *asymptotic* fields, obtained by sending the coupling $g \to 0$ in the sense appropriate to scattering theory: the interpolating fields approach free fields obeying the linearized equations of motion. In this limit the BRST transformation Eq. (26.123) linearizes to

$$sA^a_\mu = \partial_\mu c^a, \qquad sc^a = 0, \qquad s\bar{c}^a = B^a, \qquad sB^a = 0, \tag{26.131}$$

which is manifestly nilpotent: $s^2 A^a_\mu = \partial_\mu(sc^a) = 0$ and $s^2\bar{c}^a = sB^a = 0$. We work in Feynman gauge $\xi = 1$, the standing default of this chapter, so the auxiliary field is $B^a = -\partial^\mu A^a_\mu$ and every asymptotic field obeys the massless wave equation,

$$\partial^2 A^a_\mu = 0, \qquad \partial^2 c^a = 0, \qquad \partial^2 \bar{c}^a = 0. \tag{26.132}$$

The gluon sector is quantized exactly as the covariant photon of Section 13.3, one independent copy per colour. The equal-time commutator $\left[A^a_\mu(\mathbf{x},t), \dot{A}^b_\nu(\mathbf{y},t)\right] = -i\eta_{\mu\nu}\delta^{ab}\delta^3(\mathbf{x}-\mathbf{y})$ gives the mode algebra

$$\left[a^a_\mu(k), a^{b\dagger}_\nu(k')\right] = -\eta_{\mu\nu}\,\delta^{ab}\,(2\pi)^3\,2k^0\,\delta^3(\mathbf{k}-\mathbf{k}'), \tag{26.133}$$

with the timelike component a^a_0 carrying negative norm (Theorem 13.16), the unavoidable price of covariance. The ghost and antighost are Grassmann-odd Lorentz scalars; their Lagrangian $\partial^\mu\bar{c}^a\partial_\mu c^a$ yields the canonical anticommutators

$$\left\{c^a(k), \bar{c}^{b\dagger}(k')\right\} = \delta^{ab}(2\pi)^3 2k^0\delta^3(\mathbf{k}-\mathbf{k}'), \quad \left\{\bar{c}^a(k), c^{b\dagger}(k')\right\} = -\delta^{ab}(2\pi)^3 2k^0\delta^3(\mathbf{k}-\mathbf{k}'), \tag{26.134}$$

all other (anti)commutators vanishing. The relative sign between the two lines of Eq. (26.134) is fixed by the Hermiticity assignments $c^\dagger = c$, $\bar{c}^\dagger = -\bar{c}$ that make Q_B Hermitian (Theorem 26.23); it is the algebraic root of the ghost loop's minus sign and, as we will see, of the cancellation that removes the unphysical gluons.

Polarization basis. Fix a lightlike momentum $k^\mu = (\omega, \mathbf{k})$, $\omega = |\mathbf{k}|$, and introduce the conjugate null vector $\bar{k}^\mu = (\omega, -\mathbf{k})$, so that $k^2 = \bar{k}^2 = 0$ and $k\cdot\bar{k} = 2\omega^2 \neq 0$. Choose two real transverse polarizations $\varepsilon^{(\lambda)}_\mu$, $\lambda = 1,2$, with $k\cdot\varepsilon^{(\lambda)} = \bar{k}\cdot\varepsilon^{(\lambda)} = 0$ and $\varepsilon^{(\lambda)}\cdot\varepsilon^{(\lambda')} = -\delta^{\lambda\lambda'}$. The four-dimensional polarization space is spanned by $\{\varepsilon^{(1)}, \varepsilon^{(2)}, k, \bar{k}\}$. Decompose the gluon field in this basis and name the corresponding annihilation operators $a^{(\lambda)}_T$ (transverse, $\lambda = 1,2$), together with the two non-transverse modes obtained by projecting along k and $\bar{k}$. Because $\varepsilon^{(\lambda)}$ are spacelike and orthonormal, Eq. (26.133) makes the transverse operators an ordinary positive-norm oscillator algebra,

$$\left[a^{a(\lambda)}_T(k), a^{b(\lambda')\dagger}_T(k')\right] = \delta^{\lambda\lambda'}\delta^{ab}(2\pi)^3 2k^0\delta^3(\mathbf{k}-\mathbf{k}'), \tag{26.135}$$

exactly the physical photon algebra of Chapter 13. The non-transverse modes are null and mutually conjugate: defining

$$a^a_\gamma(k) \equiv -\frac{k^\mu}{k\cdot\bar{k}}a^a_\mu(k) \quad \text{(polarization } \propto \bar{k}\text{, the "scalar" gluon)}, \qquad a^a_\beta(k) \equiv -\frac{\bar{k}^\mu}{k\cdot\bar{k}}a^a_\mu(k \tag{26.136}$$

one finds from Eq. (26.133), using $k^2=\bar{k}^2=0$, $\left[a_\gamma, a^\dagger_\gamma\right]=\left[a_\beta, a^\dagger_\beta\right]=0$ and the single nonvanishing cross relation $\left[a^a_\beta(k), a^{b\dagger}_\gamma(k')\right]=\delta^{ab}(2\pi)^3 2k^0\delta^3(\mathbf{k}-\mathbf{k}')$. Both non-transverse gluons thus have zero norm and pair only with each other — the precise non-abelian echo of the scalar/longitudinal pairing of Theorem 13.18.

Action of the free charge. The relation $sA^a_\mu=\partial_\mu c^a$ of Eq. (26.131) reads, in momentum space, $i[Q_B, a^a_\mu(k)]=-ik_\mu c^a(k)$ on annihilation operators (and the Hermitian conjugate on creation operators). Contracting with $-\bar{k}^\mu/(k\cdot\bar{k})$ and using Eq. (26.136) together with $\bar{k}\cdot k = k\cdot\bar{k}$ gives, on the one-particle creation operators,

$$[\,Q_B,\, a^{a\dagger}_\gamma(k)\,]=0, \qquad [\,Q_B,\, a^{a\dagger}_\beta(k)\,]=c^{a\dagger}(k), \qquad [\,Q_B,\, a^{a(\lambda)\dagger}_T(k)\,]=0, \tag{26.137}$$

the transverse modes being annihilated because $k\cdot\varepsilon^{(\lambda)}=0$. Likewise $s\bar{c}^a=B^a=-\partial^\mu A^a_\mu$ becomes, after projecting the divergence onto the null basis,

$$\left\{Q_B, \bar{c}^{a\dagger}(k)\right\}=a^{a\dagger}_\gamma(k), \qquad \left\{Q_B, c^{a\dagger}(k)\right\}=0. \tag{26.138}$$

Acting on the vacuum, which Q_B annihilates ($Q_B\,|0\rangle=0$, the vacuum being BRST-invariant), Eqs. (26.137) and (26.138) give the action of Q_B on the four unphysical one-particle states of fixed momentum and colour.

Definition 26.29 (The BRST quartet)**.** For each momentum k and colour a, the four unphysical one-particle states

$$|\beta\rangle = a^{a\dagger}_\beta(k)\,|0\rangle\,, \quad |\gamma\rangle = a^{a\dagger}_\gamma(k)\,|0\rangle\,, \quad |c\rangle = c^{a\dagger}(k)\,|0\rangle\,, \quad |\bar{c}\rangle = \bar{c}^{a\dagger}(k)\,|0\rangle \tag{26.139}$$

form the *BRST quartet*. From Eqs. (26.137) and (26.138) they organize into two BRST *doublets* (parent–daughter pairs), depicted in Fig. 26.8:

$$Q_B\,|\beta\rangle=|c\rangle\,, \quad Q_B\,|c\rangle=0 \qquad \text{(doublet } (\beta,c)\text{)}, \qquad Q_B\,|\bar{c}\rangle=|\gamma\rangle\,, \quad Q_B\,|\gamma\rangle=0 \qquad \text{(doublet} \tag{26.140}$$

The ghost-number bookkeeping is automatic and confirms the structure: $|\beta\rangle$ and $|\gamma\rangle$ have ghost number 0, $|c\rangle$ has $+1$, $|\bar{c}\rangle$ has -1, and in each doublet Q_B raises ghost number by one as it must (Eq. (26.126)). The two transverse states $|T_\lambda\rangle = a^{a(\lambda)\dagger}_T(k)\,|0\rangle$ are BRST *singlets*: closed ($Q_B\,|T_\lambda\rangle=0$) but not exact, the only candidates to survive into the cohomology.

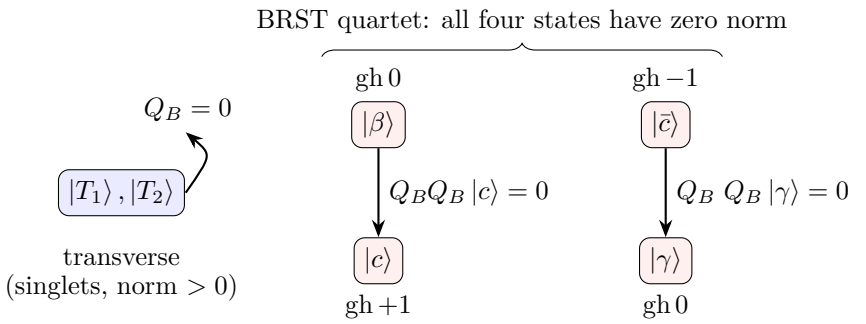


Figure 26.8: The asymptotic BRST structure at fixed momentum and colour. The two transverse gluons $|T_\lambda\rangle$ are BRST singlets with positive norm and are the only states in the cohomology. The four unphysical modes form the quartet: two doublets (β, c) and $(\bar{c}, \gamma)$ in which a BRST "parent" maps to a "daughter" (Eq. (26.140)). Every quartet state has zero norm (Theorem 26.30); the daughters $|c\rangle, |\gamma\rangle$ are Q_B-exact, the parents $|\beta\rangle, |\bar{c}\rangle$ are not Q_B-closed, so none survives the cohomology quotient.

Lemma 26.30 (Quartet states have zero norm)**.** *Each of the four states in Eq. (26.139) has vanishing norm, and the only nonzero inner products within the quartet are the cross-pairings $\langle\gamma|\beta\rangle$ and $\langle\bar{c}|c\rangle$ (and their conjugates).*

Proof. For the gluon modes, the null algebra below Eq. (26.136) gives $\langle\beta|\beta\rangle = \langle 0|\, a_\beta\, a_\beta^\dagger\, |0\rangle = \left[a_\beta, a_\beta^\dagger\right] = 0$ and likewise $\langle\gamma|\gamma\rangle = 0$, while $\langle\gamma|\beta\rangle = \left[a_\beta, a_\gamma^\dagger\right] \neq 0$. For the ghosts, Eq. (26.134) gives $\langle c|c\rangle = \langle 0|\, c\, c^\dagger\, |0\rangle$; but $\left\{c, c^\dagger\right\} = 0$ (only the mixed c–$\bar{c}$ anticommutators are nonzero), so $\langle c|c\rangle = 0$, and similarly $\langle\bar{c}|\bar{c}\rangle = 0$, while the mixed overlap $\langle\bar{c}|c\rangle = \left\{\bar{c}, c^\dagger\right\} \neq 0$ from the second relation in Eq. (26.134). Thus all four diagonal norms vanish and the modes pair off only across doublets. □

This is the heart of the quartet mechanism, and it generalizes precisely the abelian cancellation of Chapter 13. There the negative-norm scalar photon married the positive-norm longitudinal photon into a zero-norm pair (Theorem 13.18). Here the same null gluon pair (γ, β) reappears, and the ghost pair $(c, \bar{c})$ joins it; the four together are exactly the states that the indefinite metric and the Grassmann statistics conspire to give zero norm. None of them can be a physical particle. The remaining task is to prove that the cohomology quotient removes them *completely and consistently*, leaving the transverse states alone with positive norm.

26.6.4 Positivity on the cohomology: the quartet decouples

We now prove that the cohomology Eq. (26.127) contains nothing but the transverse Fock space, and that its induced inner product is positive definite. The cleanest route, and the one that makes the all-orders Kugo–Ojima quartet decoupling transparent, is a homotopy (Hodge-type) argument: we exhibit an operator R on $\mathcal{V}$ such that $\{Q_B, R\}$ counts quartet excitations. Wherever that count is nonzero, every closed state is exact, so the cohomology is concentrated on states with no quartet excitations at all — the transverse Fock space.

The homotopy operator. Let N_{q} be the number operator that counts quartet quanta — the total number of $\beta, \gamma, c, \bar{c}$ excitations, summed over momenta and colours. Because the null gluon pair and the ghost pair are cross-conjugate (Theorem 26.30), the count is written in cross-paired form,

$$N_{\mathrm{q}} = \int \widetilde{dk} \sum_a \left(a_\beta^{a\dagger} a_\gamma^a + a_\gamma^{a\dagger} a_\beta^a - c^{a\dagger} \bar{c}^a + \bar{c}^{a\dagger} c^a \right)(k), \tag{26.141}$$

whose eigenvalue on each of the four one-particle quartet states Eq. (26.139) is $+1$ and which annihilates the vacuum and the transverse states (the relative signs in the ghost terms are fixed by the algebra Eq. (26.134) so that both $|c\rangle$ and $|\bar{c}\rangle$ are counted positively). By construction N_{q} commutes with Q_B. Define the homotopy operator

$$R = \int \widetilde{dk} \sum_a \left(\bar{c}^{a\dagger}(k)\, a_\beta^a(k) \;-\; a_\beta^{a\dagger}(k)\, \bar{c}^a(k) \right), \tag{26.142}$$

which carries ghost number -1, the opposite of Q_B. We claim

$$\{Q_B, R\} = N_{\mathrm{q}}. \tag{26.143}$$

It suffices to verify Eq. (26.143) on the one-particle quartet space: both sides are operators built additively from single modes, so on multiparticle states they extend by the graded Leibniz rule, N_{q} is additive in the occupation numbers, and the relation lifts. Evaluating Eq. (26.142) on the states Eq. (26.139) with the algebra Eq. (26.135)–Eq. (26.134) — using that a_β annihilates the γ-quantum ($\left[a_\beta, a_\gamma^\dagger\right] \neq 0$) while $\bar{c}$ annihilates the c-quantum ($\left\{\bar{c}, c^\dagger\right\} \neq 0$) and all diagonal contractions vanish — gives the clean action of R as the partial inverse of Q_B,

$$R\,|c\rangle = |\beta\rangle\,, \quad R\,|\gamma\rangle = |\bar{c}\rangle\,, \quad R\,|\beta\rangle = 0, \quad R\,|\bar{c}\rangle = 0. \tag{26.144}$$

The doublet action is $Q_B\,|\beta\rangle = |c\rangle$, $Q_B\,|\bar{c}\rangle = |\gamma\rangle$, $Q_B\,|c\rangle = Q_B\,|\gamma\rangle = 0$ (Eq. (26.140)). Combining the two, the graded commutator $\{Q_B, R\} = Q_B R + R Q_B$ acts state by state as

$$\{Q_B, R\}\,|\beta\rangle = Q_B\,(R\,|\beta\rangle) + R\,(Q_B\,|\beta\rangle) = Q_B\,0 + R\,|c\rangle = |\beta\rangle\,,$$

$$\begin{aligned}
\{Q_B, R\}\,|c\rangle &= Q_B\,(R\,|c\rangle) + R\,(Q_B\,|c\rangle) = Q_B\,|\beta\rangle + R\,0 = |c\rangle\,,\\
\{Q_B, R\}\,|\bar{c}\rangle &= Q_B\,(R\,|\bar{c}\rangle) + R\,(Q_B\,|\bar{c}\rangle) = Q_B\,0 + R\,|\gamma\rangle = |\bar{c}\rangle\,,\\
\{Q_B, R\}\,|\gamma\rangle &= Q_B\,(R\,|\gamma\rangle) + R\,(Q_B\,|\gamma\rangle) = Q_B\,|\bar{c}\rangle + R\,0 = |\gamma\rangle\,.
\end{aligned} \tag{26.145}$$

Thus $\{Q_B, R\} = \mathbf{1}$ on the one-particle quartet space, which equals $N_{\mathrm{q}} = 1$ there, while on the transverse states and the vacuum both Q_B and R act as zero, giving $\{Q_B, R\} = 0 = N_{\mathrm{q}}$. This establishes Eq. (26.143) on the one-particle space and, by the additive extension, on all of $\mathcal{V}$.

Theorem 26.31 (Cohomology equals the transverse Fock space; positivity)**.** *The BRST cohomology of the asymptotic algebra is $H(Q_B) \cong \mathcal{F}_{\mathrm{T}}$, the Fock space generated from the vacuum by the transverse creation operators $a_T^{a(\lambda)\dagger}$ alone. Every cohomology class has a unique representative with no quartet excitations, and the induced inner product is positive definite. In particular the only physical one-particle states are the two transverse gluon polarizations per momentum and colour, with positive norm.*

Proof. Let $|\psi\rangle$ be Q_B-closed. Decompose it by quartet number, $|\psi\rangle = \sum_{n\geq 0} |\psi_n\rangle$ with $N_{\mathrm{q}}\,|\psi_n\rangle = n\,|\psi_n\rangle$; since N_{q} commutes with Q_B, each $|\psi_n\rangle$ is separately closed. Fix $n \geq 1$. Then

$$|\psi_n\rangle = \frac{1}{n} N_{\mathrm{q}}\,|\psi_n\rangle = \frac{1}{n}\,\{Q_B, R\}\,|\psi_n\rangle = \frac{1}{n}\big(Q_B R\,|\psi_n\rangle + R\,\underbrace{Q_B\,|\psi_n\rangle}_{=0}\big) = Q_B\Big(\tfrac{1}{n} R\,|\psi_n\rangle\Big), \tag{26.146}$$

using Eq. (26.143) and closedness. So every component with at least one quartet excitation is Q_B-exact. Therefore, modulo exact states, only the $n = 0$ component survives, and the $n = 0$ sector is by definition the Fock space built from transverse creation operators (no $\beta, \gamma, c, \bar{c}$ quanta). Within that sector Q_B acts as zero (Eq. (26.137)), so all such states are closed and none is exact (an exact state has $n \geq 1$ by Eqs. (26.140) and (26.144)); hence the cohomology is canonically $\mathcal{F}_{\mathrm{T}}$. The transverse algebra Eq. (26.135) is a positive-definite oscillator algebra, so the inner product on $\mathcal{F}_{\mathrm{T}}$ — which is the inner product induced on $\mathcal{H}_{\mathrm{phys}}$ by Theorem 26.28 — is positive definite. Restricting to one-particle states leaves exactly $|T_1\rangle\,, |T_2\rangle$. □

The proof exhibits the quartet mechanism in its sharpest form. The *daughters* $|c\rangle$ and $|\gamma\rangle$ are Q_B-exact (images of $|\beta\rangle$ and $|\bar{c}\rangle$), so they are quotiented away; the *parents* $|\beta\rangle$ and $|\bar{c}\rangle$ are not Q_B-closed, so they never enter the kernel in the first place. A full quartet thus contributes nothing to the cohomology: it is removed half by the kernel condition and half by the quotient, and because all four members have zero norm (Theorem 26.30) they could not have contributed a positive probability even formally. This is exactly the Kugo–Ojima statement that BRST quartets decouple from physical amplitudes, here proved at the level of the asymptotic state space by the single homotopy identity Eq. (26.143).

Worked Example 26.32 (Quartet cancellation in the completeness sum)**.** To see the decoupling as a cancellation in a unitarity sum — the form in which it controls loop diagrams — insert the resolution of identity of the asymptotic space between two physical (BRST-closed) states $|A\rangle, |B\rangle$ and isolate the quartet contribution at one momentum and colour. The identity on the four-dimensional quartet space, written with the indefinite metric, is

$$\mathbf{1}_{\mathrm{q}} = |\beta\rangle\langle\gamma| + |\gamma\rangle\langle\beta| + |c\rangle\langle\bar{c}| - |\bar{c}\rangle\langle c|\,, \tag{26.147}$$

the cross-paired form dictated by Theorem 26.30 (each state is resolved against its conjugate partner, since diagonal norms vanish). Sandwich between closed states and use $\langle A|\,Q_B = (Q_B\,|A\rangle)^\dagger = 0$ and the doublet relations Eq. (26.140): in the gluon pair, the term $|\beta\rangle\langle\gamma|$ contributes through $\langle\gamma|B\rangle$, and since $|\gamma\rangle = Q_B\,|\bar{c}\rangle$ is exact, $\langle\gamma|B\rangle = \langle\bar{c}|Q_B|B\rangle = \langle\bar{c}|\,(Q_B\,|B\rangle) = 0$ because $|B\rangle$ is closed; the conjugate term $|\gamma\rangle\langle\beta|$ vanishes the same way through $\langle A|\gamma\rangle$. Term by term the gluon contribution cancels against the ghost contribution because the ghost anticommutator in Eq. (26.134) carries the opposite sign to the gluon commutator — the relative minus in Eq. (26.147). The net quartet contribution to $\langle A|\cdot|B\rangle$ vanishes:

$$\langle A|\ \mathbf{1}_{\mathrm{q}}\ |B\rangle = 0 \qquad \text{for BRST-closed } |A\rangle, |B\rangle\,. \tag{26.148}$$

Thus when the completeness relation is inserted in an optical-theorem (unitarity) sum between physical states, only the transverse projector survives: the unphysical intermediate states — timelike and longitudinal gluons, ghosts and antighosts — cancel in correlated pairs. This is the diagram-level shadow of Theorem 26.31, and it is what guarantees, order by order in the loop expansion, that the imaginary parts of physical amplitudes are saturated by physical (transverse) states alone — the content of the Cutkosky/unitarity analysis that the ghost was invented to repair.

26.6.5 Unitarity of the physical S-matrix

Positivity in hand, unitarity is immediate. Three facts assemble it. First, the full S-matrix is BRST-invariant,

$$[Q_B, S] = 0, \tag{26.149}$$

because $S = T\exp\!\left(i\int\mathcal{L}_{\mathrm{int}}\right)$ is built from the BRST-invariant action Eq. (26.124) and the asymptotic Q_B generates a symmetry of the free Hamiltonian as well; equivalently, the BRST Ward identities of Section 26.7 state exactly Eq. (26.149). Second, the full S-matrix is pseudo-unitary with respect to the indefinite metric η of $\mathcal{V}$,

$$S^\dagger\,\eta\,S = \eta, \tag{26.150}$$

the statement that time evolution generated by the η-Hermitian Hamiltonian preserves the (indefinite) inner product — this is just conservation of the Klein–Gordon-type norm in the covariant formalism, the same property the covariant photon theory of Chapter 13 enjoys. Third, by Theorem 26.31 the induced inner product on $\mathcal{H}_{\text{phys}}$ is positive definite.

Theorem 26.33 (Unitarity of the physical S-matrix)**.** *The S-matrix descends to a well-defined operator $\widehat{S}$ on $\mathcal{H}_{\text{phys}}$, and $\widehat{S}$ is unitary: $\widehat{S}^\dagger \widehat{S} = \mathbf{1}$ on the positive-definite physical Hilbert space.*

Proof. Descent. Let $|A\rangle \in \ker Q_B$. By Eq. (26.149), $Q_B(S\,|A\rangle) = S(Q_B\,|A\rangle) = 0$, so S maps $\ker Q_B$ into $\ker Q_B$. If moreover $|A\rangle = Q_B\,|\eta\rangle$ is exact, then $S\,|A\rangle = SQ_B\,|\eta\rangle = Q_B(S\,|\eta\rangle)$ is exact: S maps $\operatorname{im} Q_B$ into $\operatorname{im} Q_B$. Hence S descends to a linear map $\widehat{S}$ on the quotient $\mathcal{H}_{\text{phys}} = \ker Q_B / \operatorname{im} Q_B$, well defined on classes.

Norm preservation. For closed $|A\rangle, |B\rangle$ the physical inner product is the representative inner product (Theorem 26.28). Compute, using $S\,|A\rangle, S\,|B\rangle \in \ker Q_B$ and Eq. (26.150), that $(\widehat{S}[A], \widehat{S}[B]) = \langle SA|\,\eta\,|SB\rangle = \langle A|S^\dagger \eta S|B\rangle = \langle A|\eta|B\rangle = ([A],[B])$, where in the first and last equalities the indefinite product $\langle\cdot|\eta|\cdot\rangle$ restricted to $\ker Q_B$ equals the positive-definite physical product modulo exact states (Theorem 26.31). Thus $\widehat{S}$ preserves the positive-definite physical inner product on every pair of classes, so $\widehat{S}^\dagger \widehat{S} = \mathbf{1}$. □

This completes the proof of Theorem 26.26. The logic is worth restating in one breath, because it is the answer to the question that opened the chapter — can a theory with ghosts circulating in loops be unitary? Yes: gauge-fixing and ghosts enlarge the state space to an indefinite-metric $\mathcal{V}$ on which the dynamics is pseudo-unitary; the nilpotent charge Q_B organizes every unphysical excitation into a zero-norm quartet that is removed by passing to its cohomology; the cohomology is the positive-definite transverse Fock space; and because S commutes with Q_B it descends to a genuinely unitary operator there. Unitarity is not assumed and not checked diagram by diagram — it is a structural consequence of BRST symmetry.

26.6.6 The abelian limit: recovering Gupta–Bleuler

Setting $f^{abc} = 0$ turns Yang–Mills into $N^2 - 1$ decoupled copies of an abelian gauge theory, and the entire construction above must collapse onto the Gupta–Bleuler quantization of Chapter 13. Tracing the collapse confirms the non-abelian machinery and ties Yang–Mills theory back to QED at the level of the physical-state argument.

When $f^{abc} = 0$ the BRST transformation Eq. (26.123) is *already linear* — it coincides with its asymptotic form Eq. (26.131) at all field strengths, because the nonlinear pieces $g f^{abc} A^b_\mu c^c$ and $-\frac{1}{2} g f^{abc} c^b c^c$ vanish identically. Moreover the

Faddeev–Popov operator reduces to $\frac{1}{g}\partial^2\delta^{ab}$ (Eq. (26.19)), field-independent, so the ghost and antighost are *free* fields with no coupling to anything: they are never produced or absorbed, and the doublet $(\bar{c},\gamma)$ decouples trivially from all S-matrix elements. The cohomology argument then runs entirely within the gluon sector, where it becomes the Gupta–Bleuler argument.

Concretely, on the ghost-free subspace the physical-state condition $Q_B\,|\psi\rangle = 0$ reduces, via Eq. (26.138) read backwards, to the demand that the positive-frequency part of $B^a = -\partial^\mu A^a_\mu$ annihilate $|\psi\rangle$:

$$Q_B\,|\psi\rangle = 0 \iff (\partial^\mu A^a_\mu)^{(+)}\,|\psi\rangle = 0, \tag{26.151}$$

which is precisely the Gupta–Bleuler subsidiary condition Eq. (13.93) of Theorem 13.17, one copy per colour. The identification is exact: Q_B *is* the Gupta–Bleuler condition, promoted from an externally imposed constraint to the generator of a symmetry. Under this dictionary the structures match line for line. The BRST-exact states $Q_B\,|\eta\rangle$ become the zero-norm null states V_0 of Theorem 13.20; quotienting by $\operatorname{im} Q_B$ becomes the quotient V_{phys}/V_0 of Theorem 13.21; the non-negativity of the norm on $\ker Q_B$ (Theorem 26.31) becomes the non-negativity theorem Theorem 13.19; the null gluon pair (γ,β) is the scalar/longitudinal pair whose norms cancel in Theorem 13.16 and Theorem 13.18; and the resulting positive-definite transverse space is the isometric image of the two-polarization photon space of Theorem 13.22. The only structural novelty of the non-abelian theory is the ghost doublet $(c,\bar{c})$, which in QED is inert and invisible; the moment $f^{abc}\neq 0$ it comes alive, the ghosts couple and propagate in loops, and the quartet — not merely the Gupta–Bleuler doublet — is what the cohomology must remove. BRST cohomology is thus the genuine generalization promised at the close of Chapter 13: the modern statement of the physical-state construction that contains Gupta–Bleuler as its $f^{abc}\to 0$ shadow.

26.6.7 Corollary: gauge-parameter independence of physical amplitudes

A result the rest of the book leans on falls out for free. The gauge parameter ξ enters the action only through the BRST-exact term Eq. (26.124): differentiating with respect to ξ,

$$\frac{\partial\mathcal{L}}{\partial\xi} = \frac{\partial}{\partial\xi}\,s\Big(\bar{c}^a\big[\partial^\mu A^a_\mu + \tfrac{\xi}{2}B^a\big]\Big) = s\Big(\tfrac{1}{2}\bar{c}^a B^a\Big), \tag{26.152}$$

which is itself BRST-exact (an s of something). Hence the change in any correlator or S-matrix element under a variation of ξ is the insertion of a BRST-exact operator.

Corollary 26.34 (ξ-independence of physical matrix elements)**.** *Matrix elements of BRST-closed (gauge-invariant) operators between physical states are independent of the gauge parameter ξ.*

Proof. By Eq. (26.152), $\partial_\xi \langle A|\mathcal{O}|B\rangle = i\left\langle A\middle|\mathcal{O}\int s(\tfrac{1}{2}\bar{c}B)\middle|B\right\rangle$ with $|A\rangle, |B\rangle$ physical and $\mathcal{O}$ BRST-closed. The inserted operator is $s(\cdots) = i[Q_B, \cdots\}$, a BRST-exact insertion; by exactly the argument of Theorem 26.28 — move Q_B onto the closed bra or ket, where it annihilates — the matrix element of a BRST-exact operator between closed states vanishes. Therefore $\partial_\xi \langle A|\mathcal{O}|B\rangle = 0$. □

This is the non-abelian generalization of the abelian ξ-independence theorem Theorem 13.28: there it followed from current conservation, here from BRST exactness, and the two are the same statement read through the dictionary of Section 26.6.6. In particular the freedom to compute in Feynman gauge $\xi = 1$, used throughout the Feynman-rule derivations of this chapter and relied on in Chapter 27 for the one-loop β-function, is now justified: any physical quantity computed at $\xi = 1$ equals the same quantity at any other ξ.

We have what the chapter set out to build. The physical Hilbert space is the BRST cohomology $\ker Q_B / \operatorname{im} Q_B$, it is positive definite, it contains exactly the two transverse gluon polarizations, and the S-matrix on it is unitary; gauge-invariant correlators are BRST-closed and gauge-parameter independent. With the physical content of the theory thus secured, the remaining question is what constraints BRST invariance imposes on the Green's functions themselves. Promoting the invariance of the path-integral measure under the BRST transformation to a functional identity yields the Slavnov–Taylor identities — the non-abelian Ward–Takahashi relations — which is the business of the next section.

26.7 Slavnov–Taylor Identities

The previous section turned BRST symmetry into a statement about the spectrum: the physical Hilbert space is the cohomology of the nilpotent charge Q_B (Section 26.6), and the proof that it is unitary rested on a single structural fact — a gauge-invariant operator is BRST-closed, $s\mathcal{O} = 0$, and any two representatives of the same cohomology class differ by a BRST-exact piece $s(\cdots)$ that decouples from physical matrix elements. That same fact has a second, equally far-reaching consequence, and extracting it is the business of this closing section. If $s\mathcal{O} = 0$ characterizes the *states* that survive, then the vanishing of the expectation value of any BRST variation, $\langle s(\cdots)\rangle = 0$, constrains the *Green's functions* of the theory. These constraints are the Slavnov–Taylor identities. They are to non-abelian gauge theory exactly what the Ward–Takahashi identity (Theorem 19.13) is to QED: the functional imprint of the gauge symmetry on the quantum correlators, surviving regularization and renormalization, and powerful enough to force the gauge field's self-energy to be transverse, to lock the apparently independent ultraviolet divergences of the three-gluon, four-gluon, ghost-gluon, and quark-gluon vertices into a single universal coupling renormalization, and to guarantee that

the beta function Eq. (27.101) computed in Chapter 27 does not depend on the arbitrary gauge parameter ξ. The abelian Ward identity was almost a triviality because the photon does not carry charge; here the gauge field is charged under its own group, the identities tie together vertices with different numbers of legs, and only the full machinery of BRST makes them tractable. We build that machinery now.

26.7.1 The master identity from BRST invariance of the path integral

Recall the data assembled earlier in the chapter. The gauge-fixed action is

$$S = \int \mathrm{d}^4x\, \mathcal{L}, \qquad \mathcal{L} = \mathcal{L}_{\mathrm{YM}} + s\Big[\bar{c}^a\Big(\partial^\mu A^a_\mu + \tfrac{\xi}{2} B^a\Big)\Big], \tag{26.153}$$

with $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ of Eq. (25.76), the Nakanishi–Lautrup auxiliary field B^a, and the gauge-fixing fermion in brackets, exactly the gauge-fixed Lagrangian Eq. (26.37). Carrying out the BRST variation gives the explicit form

$$\mathcal{L} = -\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + B^a\,\partial^\mu A^a_\mu + \tfrac{\xi}{2}B^aB^a - \bar{c}^a\,\partial^\mu(D_\mu c)^a, \tag{26.154}$$

and eliminating B^a through its algebraic equation of motion $B^a = -\frac{1}{\xi}\partial^\mu A^a_\mu$ returns the familiar R_ξ form $-\frac{1}{2\xi}(\partial^\mu A^a_\mu)^2$ of Eq. (26.36); we keep B^a explicit here because it makes the identities below local and linear. The BRST transformation s is the nilpotent graded derivation (Eq. (26.95), Theorem 26.24)

$$sA^a_\mu = (D_\mu c)^a = \partial_\mu c^a + g f^{abc}A^b_\mu c^c, \quad sc^a = -\tfrac{1}{2}g f^{abc}c^b c^c, \quad s\bar{c}^a = B^a, \quad sB^a = 0, \qquad s^2 = 0, \tag{26.155}$$

all conventions and group-theory data ($[T^a,T^b] = if^{abc}T^c$, $\mathrm{tr}\,T^aT^b = \frac{1}{2}\delta^{ab}$, f^{abc} real and totally antisymmetric, the adjoint dimension $\dim G = N^2-1$ for SU(N)) those of Chapter B. The Fourier convention is that of Eq. (10.86): $f(x) = \int \frac{\mathrm{d}^4k}{(2\pi)^4}\, e^{-ik\cdot x}\tilde{f}(k)$, so that $\partial_\mu \mapsto -ik_\mu$ on a field of incoming momentum k.

The generating functional of the theory is the gauge-fixed path integral

$$Z = \int \mathcal{D}A\,\mathcal{D}c\,\mathcal{D}\bar{c}\,\mathcal{D}B\; e^{\,iS}, \tag{26.156}$$

the non-abelian descendant of Eq. (10.86). Everything that follows is the statement that this integral is invariant under the BRST shift of its integration variables. Let θ be a constant, spacetime-independent *anticommuting* parameter, and perform the change of variables

$$A^a_\mu \to A^a_\mu + \theta\,(sA^a_\mu), \quad c^a \to c^a + \theta\,(sc^a), \quad \bar{c}^a \to \bar{c}^a + \theta\,(s\bar{c}^a), \quad B^a \to B^a, \tag{26.157}$$

in the integral Eq. (26.156). A change of variables can never alter the value of an integral, so Z is unchanged; comparing the integrand before and after will produce a nontrivial identity provided we track two things: the Jacobian of the measure and the variation of the action.

The measure is invariant. The transformation Eq. (26.157) is a graded-linear shift, so its superdeterminant is, to first order in θ (and $\theta^2 = 0$ truncates the expansion exactly),

$$\operatorname{sdet}\left(\frac{\partial(\text{new})}{\partial(\text{old})}\right) = 1 + \theta\,\mathrm{STr}\left[\frac{\delta(s\Phi)}{\delta\Phi}\right] = 1 + \theta\left(\sum_{\text{bosonic }\Phi}\frac{\delta(s\Phi)}{\delta\Phi} - \sum_{\text{fermionic }\Phi}\frac{\delta(s\Phi)}{\delta\Phi}\right), \tag{26.158}$$

the supertrace carrying the relative minus sign for the anticommuting fields $c, \bar{c}$. Each term vanishes:

$$\frac{\delta(sA^a_\mu)}{\delta A^a_\mu} = \frac{\delta}{\delta A^a_\mu}\left(\partial_\mu c^a + g f^{abc} A^b_\mu c^c\right) = g f^{aba}\, c^b = 0, \qquad \frac{\delta(sc^a)}{\delta c^a} = \frac{\delta}{\delta c^a}\left(-\tfrac{1}{2} g f^{abc} c^b c^c\right) \propto f^{aba} = 0, \tag{26.159}$$

both proportional to the trace f^{aba} of a structure constant over a repeated adjoint index, which is zero by the total antisymmetry of f^{abc} from Chapter B; and $\delta(s\bar{c}^a)/\delta\bar{c}^a = \delta B^a/\delta\bar{c}^a = 0$, $\delta(sB^a)/\delta B^a = 0$. Hence $\mathrm{STr}[\delta(s\Phi)/\delta\Phi] = 0$, the superdeterminant is 1, and

$$\mathcal{D}A\,\mathcal{D}c\,\mathcal{D}\bar{c}\,\mathcal{D}B \text{ is BRST invariant.} \tag{26.160}$$

The vanishing of this supertrace is no accident: it is the path-integral expression of the absence of a BRST anomaly in pure Yang–Mills, and it is the exact analogue of the unit Jacobian that protected the abelian Faddeev–Popov measure (Section 26.1.2).

The action is invariant. The action Eq. (26.153) is BRST invariant. Its first piece $\mathcal{L}_{\mathrm{YM}}$ is annihilated by s because on A the BRST variation is a gauge transformation with the anticommuting parameter c, and $\mathcal{L}_{\mathrm{YM}}$ is gauge invariant (Eq. (25.76)); its second piece is s-exact, and s of an s-exact object vanishes by nilpotency, $s(s\Psi) = s^2\Psi = 0$. Therefore $sS = 0$, and under Eq. (26.157) the exponent e^{iS} is unchanged.

With the measure and action both invariant, an insertion fares differently. For an arbitrary functional $\mathcal{O}[\Phi]$ of the fields, apply the change of variables to $\int\mathcal{D}\Phi\,\mathcal{O}\,e^{iS}$. The left-hand side is by definition unchanged; the right-hand side, expanded to first order in θ, picks up only the variation of $\mathcal{O}$ (the measure and e^{iS} contributing nothing): $\int\mathcal{D}\Phi\,(\mathcal{O} + \theta\, s\mathcal{O})\,e^{iS} = \int\mathcal{D}\Phi\,\mathcal{O}\,e^{iS}$. The order-$\theta$ term must vanish on its own. We have proved the master identity.

Theorem 26.35 (Master Slavnov–Taylor identity)**.** *For every functional $\mathcal{O}[\Phi]$ of the Yang–Mills fields $\Phi = (A, c, \bar{c}, B)$, the BRST variation has vanishing vacuum expectation value,*

$$\langle\, s\mathcal{O} \,\rangle \;=\; \frac{1}{Z}\int \mathcal{D}A\,\mathcal{D}c\,\mathcal{D}\bar{c}\,\mathcal{D}B\;(s\mathcal{O})\;e^{\,iS} \;=\; 0. \tag{26.161}$$

Equivalently, with s a graded derivation, for any $\mathcal{O}_1, \mathcal{O}_2$, $\langle (s\mathcal{O}_1)\mathcal{O}_2\rangle = \mp\langle \mathcal{O}_1 (s\mathcal{O}_2)\rangle$, the sign set by the Grassmann parity of $\mathcal{O}_1$.

This single equation is the source of every Slavnov–Taylor identity. It says no more and no less than that the correlators of the theory cannot tell apart two field configurations related by a BRST transformation. Choosing $\mathcal{O}$ to be a product of fields, expanding $s\mathcal{O}$ by the graded Leibniz rule, and reading off the resulting relation among correlators generates, one identity per choice of $\mathcal{O}$, the entire infinite tower. Before mining it, we package the whole tower into a single functional equation, because the renormalization-constant relations are cleanest when read off there.

Physical Insight

Theorem 26.35 is the precise non-abelian generalization of the Ward–Takahashi identity. In QED (Theorem 19.13) the gauge transformation of the photon, $A_\mu \to A_\mu + \frac{1}{e}\partial_\mu \alpha$, is a shift by a *c-number* gradient; the corresponding identity $\langle s(\bar{c}A_\nu)\rangle = 0$ collapses, because the abelian ghost is free and decoupled, to the familiar transversality of the photon polarization and to $Z_1 = Z_2$. Here the gauge field is charged, the ghost is interacting, and the same $\langle s(\cdots)\rangle = 0$ becomes a web of relations linking different Green's functions — the price, and the payoff, of non-abelian charge. The worked example Theorem 26.38 carries out the collapse explicitly.

26.7.2 Sources for composite variations: the Zinn-Justin construction

To turn Theorem 26.35 into a functional identity on the generating functional we couple the fields to sources and demand BRST invariance of the sourced integral. A subtlety appears immediately and dictates the whole construction. The BRST variations $sA^a_\mu = (D_\mu c)^a$ and $sc^a = -\frac{1}{2} g f^{abc} c^b c^c$ are *composite* and *nonlinear* in the fields. A linear source $J^{a\mu} A^a_\mu$ knows how to vary A, but it has no hook on the nonlinear object $D_\mu c$ that A turns into; the variation of the source term would generate a correlator of the composite $D_\mu c$, an operator that mixes under renormalization and is not simply the source's derivative. The resolution, due to Zinn-Justin, is to introduce one additional external source *for each composite BRST variation*, coupled directly to it. These are the antifields. They give the

nonlinear variations their own generating handles, and they organize the entire identity into a quadratic functional equation.

Introduce, alongside the ordinary sources $J^{a\mu}$ (for A^a_μ), $\bar\eta^a$ (for c^a) and η^a (for $\bar c^a$), two antifield sources: $K^{a\mu}$ coupled to sA^a_μ, and L^a coupled to sc^a. The fields $\bar c$ and B have *linear* BRST variations ($s\bar c = B$, $sB = 0$) and so need no antifield. Ghost number is additive with the assignments $\mathrm{gh}(c) = +1$, $\mathrm{gh}(\bar c) = -1$, $\mathrm{gh}(A) = \mathrm{gh}(B) = 0$; the antifields, coupling to variations of ghost number one higher than their partner field, carry $\mathrm{gh}(K) = -1$, $\mathrm{gh}(L) = -2$. The sourced generating functional is

$$Z[J,\bar\eta,\eta;K,L] = \int \mathcal{D}\Phi\ \exp\Big\{ i\Sigma + i\int \mathrm{d}^4x \big(J^{a\mu}A^a_\mu + \bar\eta^a c^a + \bar c^a \eta^a\big)\Big\},$$
$$\Sigma = S + \int \mathrm{d}^4x \Big[K^{a\mu}\,(D_\mu c)^a + L^a\big(-\tfrac{1}{2} g f^{abc} c^b c^c\big)\Big], \qquad (26.162)$$

where Σ is the action augmented by the antifield couplings. Because the antifields multiply sA and sc, and s is nilpotent, $s\big[K^{a\mu}(D_\mu c)^a\big] = K^{a\mu}\, s(sA^a_\mu) = 0$ and likewise for the L term; the augmented action Σ is therefore still BRST invariant, $s\Sigma = 0$. Repeating the change of variables Eq. (26.157) inside Eq. (26.162), with the measure and Σ both invariant, the only θ-dependence comes from the linear source terms, whose variation is $\theta\big(J^{a\mu}\, sA^a_\mu + \bar\eta^a\, sc^a + (s\bar c^a)\eta^a\big)$. Demanding that it integrate to zero gives, after writing sA and sc as the antifield derivatives $\langle sA^a_\mu\rangle = \delta W/\delta K^{a\mu}$, $\langle sc^a\rangle = \delta W/\delta L^a$, and $s\bar c = B$, the Slavnov–Taylor identity for the connected generating functional $W = -i\ln Z$,

$$\int \mathrm{d}^4x \left[J^{a\mu}\,\frac{\delta W}{\delta K^{a\mu}} + \bar\eta^a\,\frac{\delta W}{\delta L^a} + \eta^a\,\langle B^a\rangle \right] = 0, \qquad (26.163)$$

in which $\langle B^a\rangle$ is fixed by the (linear) equation of motion of the auxiliary field. Equation Eq. (26.163) is the master identity Theorem 26.35 dressed with sources.

The form that does the renormalization work is its Legendre transform. Define the one-particle-irreducible (1PI) effective action $\Gamma[A,c,\bar c,B;K,L]$ by the Legendre transform of W with respect to the ordinary sources only (the antifields ride through untouched as spectators),

$$\Gamma = W - \int \mathrm{d}^4x (J^{a\mu}A^a_\mu + \bar\eta^a c^a + \bar c^a\eta^a), \qquad \frac{\delta\Gamma}{\delta A^a_\mu} = -J^{a\mu},\ \ \frac{\delta\Gamma}{\delta c^a} = -\bar\eta^a,\ \ \frac{\delta\Gamma}{\delta \bar c^a} = -\eta^a, \qquad (26.164)$$

with $\delta\Gamma/\delta K^{a\mu} = \delta W/\delta K^{a\mu}$ and $\delta\Gamma/\delta L^a = \delta W/\delta L^a$ on the antifields. Here $A, c, \ldots$ now denote the classical (background) fields conjugate to the sources. Substituting Eq. (26.164) into Eq. (26.163) converts the linear sources into field derivatives of Γ and yields the central result of the section.

Theorem 26.36 (Zinn-Justin equation)**.** *The 1PI effective action of gauge-fixed Yang–Mills obeys the quadratic functional identity*

$$\int \mathrm{d}^4x \left[\frac{\delta\Gamma}{\delta A^a_\mu}\,\frac{\delta\Gamma}{\delta K^{a\mu}} + \frac{\delta\Gamma}{\delta c^a}\,\frac{\delta\Gamma}{\delta L^a} + B^a\,\frac{\delta\Gamma}{\delta \bar c^a}\right] = 0, \qquad (26.165)$$

supplemented by the gauge-condition (auxiliary-field) equation and the antighost (ghost) equation,

$$\frac{\delta\Gamma}{\delta B^a} = \partial^\mu A^a_\mu + \xi B^a, \qquad \frac{\delta\Gamma}{\delta \bar{c}^a} + \partial^\mu \frac{\delta\Gamma}{\delta K^{a\mu}} = 0. \tag{26.166}$$

Derivation of the two auxiliary equations. The auxiliary field B^a enters Σ only through the bilinear $B^a\partial^\mu A^a_\mu + \frac{\xi}{2}B^aB^a$ of Eq. (26.154) and is coupled to no source; it has no interactions and therefore receives no loop corrections, so Γ's B-dependence is exactly its classical one and $\delta\Gamma/\delta B^a = \partial^\mu A^a_\mu + \xi B^a$, the first equation. For the second, inspect where $\bar{c}$ and K appear in Σ: the ghost term $-\bar{c}^a\partial^\mu(D_\mu c)^a = \partial^\mu\bar{c}^a\,(D_\mu c)^a$ (integrating by parts) and the antifield term $K^{a\mu}(D_\mu c)^a$ combine into $(K^{a\mu} + \partial^\mu\bar{c}^a)(D_\mu c)^a$. Thus Σ, and hence Γ, depends on $\bar{c}$ and K only through the translation-invariant combination $\widehat{K}^{a\mu} = K^{a\mu} + \partial^\mu\bar{c}^a$. Differentiating $\Gamma[\widehat{K}]$ by $\bar{c}^a$ gives $\delta\Gamma/\delta\bar{c}^a(x) = -\partial^\mu_x\,\delta\Gamma/\delta K^{a\mu}(x)$, the antighost equation. The Zinn-Justin equation Eq. (26.165) itself is the term-by-term Legendre transform of Eq. (26.163) just described. □

Equation (26.165) is the compact, all-orders statement of gauge symmetry at the quantum level. Its power is that it survives renormalization: a regulator and a renormalization scheme that respect BRST (dimensional regularization in $\overline{\text{MS}}$ does, since it introduces no dimensionful scale and the BRST variation is dimensionless in any d) leave Eq. (26.165) valid for the renormalized Γ with the same form. Every relation among renormalization constants below is obtained by differentiating it and setting fields and sources to zero.

26.7.3 Transversality of the gluon self-energy

The first and most direct payoff is the transversality of the gluon vacuum polarization, the non-abelian counterpart of the transversality of the photon self-energy that we proved in QED from the Ward–Takahashi identity (Theorem 19.13). We extract it from the Zinn-Justin equation by differentiating twice.

It is cleanest to first remove the gauge-fixing term, which is the only non-transverse piece of Γ at tree level and which the two auxiliary equations Eq. (26.166) pin down exactly. Define the *reduced* effective action

$$\bar{\Gamma} = \Gamma - \int \mathrm{d}^4x \Big(B^a\partial^\mu A^a_\mu + \tfrac{\xi}{2}B^aB^a \Big). \tag{26.167}$$

Using $\delta\Gamma/\delta B^a = \partial^\mu A^a_\mu + \xi B^a$ and the antighost equation, the two auxiliary terms in the Zinn-Justin equation Eq. (26.165) are accounted for by the subtraction, and a short manipulation (substituting Eq. (26.166) into Eq. (26.165) and collecting the B-linear pieces, which cancel identically) shows that the reduced functional obeys the *homogeneous* Zinn-Justin equation

$$\int \mathrm{d}^4x \left[\frac{\delta\bar{\Gamma}}{\delta A^a_\mu}\frac{\delta\bar{\Gamma}}{\delta K^{a\mu}} + \frac{\delta\bar{\Gamma}}{\delta c^a}\frac{\delta\bar{\Gamma}}{\delta L^a} \right] = 0, \tag{26.168}$$

and that $\bar{\Gamma}$, by construction, contains no gauge-fixing term: its quadratic gluon part is the purely transverse Yang–Mills kinetic operator plus loop corrections. Concretely, in momentum space,

$$\frac{\delta^2\bar{\Gamma}}{\delta A^a_\mu(-k)\,\delta A^b_\nu(k)} \;\equiv\; \Gamma^{AA,ab}_{\mu\nu}(k) = \delta^{ab}\Big[\,(k^2\eta_{\mu\nu} - k_\mu k_\nu) + \Pi_{\mu\nu}(k)\,\Big], \tag{26.169}$$

where $\Pi_{\mu\nu}(k) = \delta^{ab}\Pi_{\mu\nu}$ collects the one-particle-irreducible self-energy (the vacuum polarization) to all loop orders. Our goal is to prove $k^\mu\Pi_{\mu\nu}(k) = 0$.

Differentiate the homogeneous equation Eq. (26.168) once with respect to the ghost field $c^d(z)$ and once with respect to the gluon field $A^e_\rho(w)$, then set all fields and antifields to zero. Each term of Eq. (26.168) is a product of two first derivatives of $\bar{\Gamma}$. After two differentiations and setting the fields to zero, every surviving contribution is a product of two *second* derivatives, because the first derivatives of $\bar{\Gamma}$ vanish at the origin of field space. Ghost-number conservation eliminates almost all of them: a second derivative of $\bar{\Gamma}$ is nonzero only if the ghost numbers of its two arguments sum to zero. Running through the four candidate products (the bookkeeping is given in full below), the only survivor is

$$\int \mathrm{d}^4x\, \frac{\delta^2\bar{\Gamma}}{\delta A^e_\rho(w)\,\delta A^a_\mu(x)}\,\frac{\delta^2\bar{\Gamma}}{\delta c^d(z)\,\delta K^{a\mu}(x)} \;=\; 0. \tag{26.170}$$

The bookkeeping. Differentiating the first product $(\delta\bar{\Gamma}/\delta A)(\delta\bar{\Gamma}/\delta K)$ by c^d and A^e_ρ produces, at zero field, two cross terms: $(\delta^2\bar{\Gamma}/\delta c^d\delta A^a_\mu)(\delta^2\bar{\Gamma}/\delta A^e_\rho\delta K^{a\mu}) + (\delta^2\bar{\Gamma}/\delta A^e_\rho\delta A^a_\mu)(\delta^2\bar{\Gamma}/\delta c^d\delta K^{a\mu})$. The first factor of the first cross term, $\delta^2\bar{\Gamma}/\delta c^d\delta A$, has ghost number $+1$ and vanishes; only the second cross term survives, which is Eq. (26.170). Differentiating the second product $(\delta\bar{\Gamma}/\delta c)(\delta\bar{\Gamma}/\delta L)$ similarly gives terms with a factor of ghost number $+2$ $(\delta^2\bar{\Gamma}/\delta c^d\delta c)$ or $+1$ $(\delta^2\bar{\Gamma}/\delta A^e_\rho\delta c)$, all of which vanish. Hence Eq. (26.170) is the complete content at this order.

Now identify the two factors. The first, $\Gamma^{AA,ea}_{\rho\mu}(k)$, is the gluon two-point function Eq. (26.169). The second,

$$\frac{\delta^2\bar{\Gamma}}{\delta c^d(z)\,\delta K^{a\mu}(x)} \;\equiv\; \Gamma^{cK,da}_{\mu}(k) = \delta^{da}\,(-ik_\mu)\,H(k^2), \tag{26.171}$$

is the proper two-point function mixing a ghost with the antifield K that sources $sA = Dc$. It carries a single Lorentz index μ and depends on the single momentum k, so by Lorentz covariance it can only be proportional to k_μ; the scalar $H(k^2)$ equals 1 at tree level (from the $\partial_\mu c$ inside $D_\mu c$) and is dressed by loops, but it does not vanish. Inserting Eq. (26.169) and Eq. (26.171) into Eq. (26.170) and contracting the colour indices ($\delta^{ea}\delta^{da}$ summed over a gives δ^{ed}),

$$\delta^{ed}\,(-iH)\,k^\mu\Big[(k^2 g_{\rho\mu} - k_\rho k_\mu) + \Pi_{\rho\mu}(k)\Big] = 0. \tag{26.172}$$

The tree-kinetic part is annihilated by the contraction, $k^\mu(k^2 g_{\rho\mu} - k_\rho k_\mu) = k^2 k_\rho - k_\rho k^2 = 0$, exhibiting that the Yang–Mills kinetic operator is transverse on its own. Since $H(k^2) \neq 0$, the remaining piece must vanish:

$$\boxed{k^\mu\,\Pi^{ab}_{\mu\nu}(k) = 0\,, \qquad \Pi^{ab}_{\mu\nu}(k) = \delta^{ab}\,(k^2\eta_{\mu\nu} - k_\mu k_\nu)\,\Pi(k^2)} \tag{26.173}$$

the gluon vacuum polarization is transverse to all orders, and Lorentz invariance plus transversality force it into the single-form-factor structure on the right, with one scalar $\Pi(k^2)$. This is the result the chapter owes Chapter 27: the entire self-energy is captured by one function $\Pi(k^2)$, whose divergent part defines a single field-renormalization constant Z_3, and the longitudinal $k_\mu k_\nu$ part of the full gluon propagator is left completely undressed by interactions — exactly as the photon's longitudinal part was in QED.

Theorem 26.37 (Transversality of the gluon vacuum polarization)**.** *The 1PI gluon self-energy $\Pi^{ab}_{\mu\nu}(k)$ of gauge-fixed Yang–Mills, to all orders in perturbation theory, is transverse, $k^\mu\Pi^{ab}_{\mu\nu}(k) = 0$, and is therefore determined by a single scalar form factor $\Pi(k^2)$ through Eq. (26.173).*

Caution

The transversality Eq. (26.173) holds for the *complete* self-energy, the sum of all 1PI diagrams: in Yang–Mills this means the gluon loop, the tadpole, *and* the ghost loop. None of these is transverse by itself. Compute only the gluon loop and you find a non-transverse, gauge-dependent mess; the ghost loop is precisely the missing piece that restores transversality. This is the diagrammatic face of the fact that the Faddeev–Popov determinant could not be discarded (Section 26.1.3): the ghost is not optional, and Theorem 26.37 would fail without it. We will see this cancellation play out explicitly in the one-loop computation of Chapter 27.

Worked Example 26.38 (The abelian limit: recovering the Ward–Takahashi identity)**.** To see that Theorems 26.35 and 26.37 genuinely contain the abelian results, set $f^{abc} \to 0$, drop the colour index, and trace the collapse. With $f^{abc} = 0$ the BRST transformations Eq. (26.155) reduce to

$$sA_\mu = \partial_\mu c, \qquad sc = 0, \qquad s\bar{c} = B, \qquad sB = 0, \tag{26.174}$$

so sA_μ is a *c-number* gradient and the ghost variation vanishes identically: the ghost is free, $\mathcal{L}_{\text{gh}} = -\bar{c}\,\partial^2 c$, and decouples from everything. This is the precise statement, planted in Chapter 13, that the abelian Faddeev–Popov determinant is field-independent and QED has no dynamical ghosts.

Transversality. Apply the master identity Eq. (26.161) to $\mathcal{O} = \bar{c}(x)A_\nu(y)$. Using $s(\bar{c}A_\nu) = (s\bar{c})A_\nu - \bar{c}\,(sA_\nu) = B(x)A_\nu(y) - \bar{c}(x)\partial_\nu c(y)$ and $\langle s(\bar{c}A_\nu)\rangle = 0$,

$$\langle B(x)A_\nu(y)\rangle = \langle \bar{c}(x)\,\partial_\nu c(y)\rangle = \partial^y_\nu \langle \bar{c}(x)c(y)\rangle. \tag{26.175}$$

The free ghost propagator $\langle \bar{c}(x)c(y)\rangle$ is gauge-field independent, so the right side is a fixed, tree-level quantity, receiving no corrections. Eliminating $B = -\frac{1}{\xi}\partial^\mu A_\mu$ on the left gives $-\frac{1}{\xi}\partial^\mu_x \langle A_\mu(x)A_\nu(y)\rangle = \partial^y_\nu \langle \bar{c}\, c\rangle$: the longitudinal part of the full photon propagator equals its free value to all orders, which is exactly the statement that the photon self-energy is transverse, $k^\mu \Pi_{\mu\nu}(k) = 0$. This is the transversality half of Theorem 19.13, recovered as the abelian shadow of Theorem 26.37; in the non-abelian case the right side of Eq. (26.175) is no longer free, and the ghost loop it represents is the extra term whose inclusion keeps $\Pi_{\mu\nu}$ transverse.

The vertex identity. The Ward–Takahashi identity also fixes $Z_1 = Z_2$ in QED — the electron-photon vertex and the electron self-energy share their divergence. In the non-abelian language this is the $f^{abc} \to 0$ limit of the renormalization-constant web derived next: the relation $Z_1^F/Z_2 = Z_1/Z_3 = \widetilde{Z}_1/\widetilde{Z}_3$, in which the photon ($Z_3$) and ghost ($\widetilde{Z}_1, \widetilde{Z}_3$) factors become trivial, collapses to $Z_1^F = Z_2$. The single abelian Ward identity thus splits, in a non-abelian theory, into a chain of identities tying every vertex to the field renormalizations — the structural reason the gauge coupling renormalizes universally.

26.7.4 Renormalization-constant relations and the universal coupling

The deepest payoff of the Slavnov–Taylor identities is that they collapse the many superficially independent ultraviolet divergences of Yang–Mills into one. A non-abelian gauge theory with n_f fundamental fermions has, by power counting (Chapter 21 applied to the vertices of Eq. (26.53) and Eq. (26.58)), divergent gluon, ghost, and quark two-point functions and divergent three-gluon, four-gluon, ghost-gluon, and quark-gluon vertices — seven primitively divergent amplitudes. We renormalize by writing bare fields and coupling in terms of renormalized ones and a set of constants:

$$A_0 = Z_3^{1/2} A, \quad c_0 = \widetilde{Z}_3^{1/2} c, \quad \psi_0 = Z_2^{1/2}\psi, \quad g_0 = Z_g\, g\, \mu^\varepsilon, \tag{26.176}$$

in $d = 4-2\varepsilon$, with μ the $\overline{\text{MS}}$ scale, and we name the multiplicative renormalization of each interaction vertex by its own constant: Z_1 for the three-gluon vertex, Z_4 for the four-gluon vertex, $\widetilde{Z}_1$ for the ghost-gluon vertex, and Z_1^F for the quark-gluon vertex. A priori these are seven independent functions of g and ε. The content of the Slavnov–Taylor identities is that they are not independent: a *single* coupling renormalization Z_g must reproduce them all.

The mechanism is transparent at the level of the bare Lagrangian. Each interaction term is built from the *same* bare coupling g_0, because the classical action descended from one gauge-invariant object, $\mathcal{L}_{\text{YM}}$, in which the single g appears in the covariant derivative Eq. (25.16). Substituting Eq. (26.176) into each bare vertex and reading off the coefficient of the corresponding renormalized vertex:

- *Three-gluon vertex.* The bare cubic term $g_0 f^{abc}(\partial A_0)A_0A_0$ carries $g_0 Z_3^{3/2}$; matching to $Z_1\, g\mu^\varepsilon$ gives $Z_1 = Z_g Z_3^{3/2}$, i.e.

$$Z_g = Z_1\, Z_3^{-3/2}. \tag{26.177}$$

- *Four-gluon vertex.* The bare quartic term $g_0^2 (f A_0 A_0)^2$ carries $g_0^2 Z_3^2$; matching to $Z_4\, g^2\mu^{2\varepsilon}$ gives $Z_4 = Z_g^2 Z_3^2$, i.e. $Z_g = Z_4^{1/2} Z_3^{-1}$.

- *Ghost-gluon vertex.* The bare ghost-gluon term $g_0 f^{abc}(\partial^\mu \bar{c}_0^a) A_{0\mu}^b c_0^c$ carries $g_0 \widetilde{Z}_3 Z_3^{1/2}$; matching to $\widetilde{Z}_1\, g\mu^\varepsilon$ gives $\widetilde{Z}_1 = Z_g \widetilde{Z}_3 Z_3^{1/2}$, i.e.

$$Z_g = \widetilde{Z}_1\, \widetilde{Z}_3^{-1}\, Z_3^{-1/2}. \tag{26.178}$$

- *Quark-gluon vertex.* The bare term $g_0 \bar{\psi}_0 \not{A}_0 \psi_0$ carries $g_0 Z_2 Z_3^{1/2}$; matching to $Z_1^F\, g\mu^\varepsilon$ gives $Z_1^F = Z_g Z_2 Z_3^{1/2}$, i.e. $Z_g = Z_1^F\, Z_2^{-1}\, Z_3^{-1/2}$.

Each line *defines* a Z_g from one vertex. They are forced to agree because g_0 is one number; equating them,

$$Z_g = Z_1 Z_3^{-3/2} = Z_4^{1/2} Z_3^{-1} = \widetilde{Z}_1 \widetilde{Z}_3^{-1} Z_3^{-1/2} = Z_1^F\, Z_2^{-1}\, Z_3^{-1/2}, \tag{26.179}$$

which, on cancelling the common factors, is the chain of Slavnov–Taylor Z-factor identities

$$\boxed{\frac{Z_1}{Z_3} = \frac{Z_4}{Z_1} = \frac{\widetilde{Z}_1}{\widetilde{Z}_3} = \frac{Z_1^F}{Z_2}\,.} \tag{26.180}$$

The non-trivial physics is that these relations are *consistent*: the actual loop divergences of the four vertices, computed independently, do arrange themselves so that the four ratios in Eq. (26.180) are equal order by order. That consistency is not an accident of one-loop arithmetic; it is guaranteed to all orders by the Zinn-Justin equation Eq. (26.165), which, when renormalized, constrains the divergent part of Γ to take a BRST-invariant form, leaving exactly the freedom of one coupling and the field rescalings Eq. (26.176) and no more. Equivalently: were the four ratios in Eq. (26.180) unequal, the renormalized theory would require independent couplings at its three- and four-point gluon vertices, the renormalized action would not be gauge invariant, and the Slavnov–Taylor identities of Theorem 26.35 would be violated. They are not; hence Eq. (26.180) holds.

The ghost-gluon vertex and Taylor's theorem. The relation Eq. (26.178) is the one we hand directly to Chapter 27, because it is the most economical route to the beta function, and it admits a further simplification of its own. The

antighost equation Eq. (26.166), differentiated with respect to $c^b(y)$ and $A^c_\nu(z)$, relates the proper ghost-gluon vertex to the antifield vertex:

$$\frac{\delta^3\Gamma}{\delta\bar{c}^a(x)\,\delta c^b(y)\,\delta A^c_\nu(z)} = -\,\partial^\mu_x\,\frac{\delta^3\Gamma}{\delta K^{a\mu}(x)\,\delta c^b(y)\,\delta A^c_\nu(z)}. \tag{26.181}$$

The left side is the ghost-gluon vertex; the right side carries an explicit factor of the antighost momentum, $\partial^\mu_x \mapsto -ip^\mu$. Thus the ghost-gluon vertex always appears contracted with the outgoing antighost momentum p^μ. The K-vertex on the right, sourcing the BRST variation $D_\mu c$, has a superficial degree of divergence lower than a naive vertex, and the momentum factor it carries makes the ghost-gluon vertex ultraviolet finite at the kinematic point of vanishing antighost momentum. Taylor's theorem follows: in Landau gauge $\xi = 0$ the ghost-gluon vertex is not renormalized,

$$\widetilde{Z}_1 = 1 \qquad \text{(Landau gauge, } \xi = 0\text{)}. \tag{26.182}$$

With $\widetilde{Z}_1 = 1$, the universal-coupling relation Eq. (26.178) collapses to

$$\boxed{\,Z_g = \widetilde{Z}_3^{-1}\,Z_3^{-1/2} \qquad (\xi = 0)\,,\,} \tag{26.183}$$

the coupling renormalization is fixed entirely by the ghost and gluon *two-point* functions. This is the concrete constraint Chapter 27 uses: to get the one-loop beta function it suffices to compute the divergent parts of the gluon and ghost self-energies — Z_3 and $\widetilde{Z}_3$ — and read off Z_g from Eq. (26.183), avoiding any three-point computation. The self-energies feed the beta function through $\beta(g) = -\varepsilon g - g\,\mu\,\dfrac{\mathrm{d}\ln Z_g}{\mathrm{d}\mu}$, whose evaluation with $Z_3, \widetilde{Z}_3$ at one loop produces $\beta(g) = -\dfrac{g^3}{16\pi^2}\left(\tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f\right)$ in Eq. (27.101), with the Casimir C_A and Dynkin index T_F of Chapter B.

Physical Insight

The economy is striking. Charge renormalization in QED was already vertex-free in spirit: the Ward identity $Z_1 = Z_2$ meant the electron's charge ran only through the photon's vacuum polarization Z_3 (Theorem 19.13, Chapter 20). Equation (26.183) is the non-abelian sharpening of the same idea — the Slavnov–Taylor identities and Taylor's theorem conspire so that the gluon coupling, too, runs through field renormalizations alone. The crucial sign difference, that the gluon and ghost self-energies make Z_g *decrease* the effective coupling at short distance whereas the QED vacuum polarization increases it, is the content of asymptotic freedom and is computed in Chapter 27.

Gauge-parameter independence of the beta function. The relations Eq. (26.180) also settle a question that would otherwise be disquieting: the whole construction depended on an arbitrary gauge parameter ξ (and on the choice of gauge-fixing function altogether), yet asymptotic freedom is supposed to be physics. The individual renormalization constants *are* gauge-dependent — Z_3, $\widetilde{Z}_3$, Z_2 each depend on ξ, as do the self-energies that define them. But Z_g, fixed by the gauge-invariant ratios Eq. (26.179), is not: in a mass-independent scheme such as $\overline{\text{MS}}$ the coupling renormalization, and hence the beta function $\beta(g) = \mu\,\mathrm{d}g/\mathrm{d}\mu$ it generates, is independent of ξ. The argument is exactly that Eq. (26.179) gives four expressions for the same Z_g from four vertices renormalized in any common gauge; the physical $\beta(g)$ governs a gauge-invariant observable (a physical cross section, an S-matrix element built from BRST-cohomology states, Section 26.6), and a gauge-invariant quantity cannot depend on ξ. We may therefore compute $\beta(g)$ in whatever gauge is most convenient — Feynman gauge $\xi = 1$ for the loop integrals, or Landau gauge $\xi = 0$ to exploit Eq. (26.183) — and trust that the answer is the same. Chapter 27 does exactly this.

Further reading

The Faddeev–Popov determinant originates with Faddeev and Popov [182]; the modern Faddeev–Popov-to-BRST-to-Slavnov–Taylor development is laid out in Peskin and Schroeder [396] (Ch. 16), in Weinberg [501] (Ch. 15–17, where the Zinn-Justin/antifield form and the cohomological proofs are emphasized), and in Schwartz [432] (Ch. 25–26). BRST symmetry was discovered independently by Becchi et al. [52] and Tyutin [475]; the cohomological characterization of the physical state space and the quartet mechanism are due to Kugo and Ojima [309]. The original Slavnov–Taylor identities are Slavnov [446] and Taylor [471], the latter containing the non-renormalization theorem $\widetilde{Z}_1 = 1$. The antifield formalism and the algebraic structure of the master equation are developed comprehensively in Zinn-Justin [534] and, in the Batalin–Vilkovisky generalization, in Henneaux and Teitelboim [264]; a careful treatment of gauge-parameter independence and the renormalization of gauge theories is Itzykson and Zuber [278]. The global obstruction to covariant gauge fixing is Gribov [236], taken up in Chapter 42.

Chapter 27

Asymptotic Freedom

Yang–Mills theory arrives at its decisive computation. The classical non-abelian theory was built in Chapter 25 and quantized in Chapter 26: we have the Yang–Mills Lagrangian (Eq. (25.76)), its gauge-fixed form with Faddeev–Popov ghosts (Eq. (26.37)), the BRST symmetry that certifies unitarity, and the Slavnov–Taylor identities (Theorem 26.35) that organize its renormalization. Part VI taught us, in turn, how to convert such a quantized theory into a statement about energy dependence: the beta function is read from the simple poles of the renormalization constants through the master relation Eq. (24.104), and its *sign* dictates the ultraviolet fate of the coupling through the renormalization-group solution (Theorem 24.22). For QED that machinery returned a positive coefficient, $\beta_e = e^3/12\pi^2 > 0$ (Eq. (24.108)): the vacuum screens, the charge grows with energy, and a Landau pole lurks in the deep ultraviolet. The whole of this chapter turns on whether Yang–Mills does the same or the opposite.

It does the opposite, and the chapter is the disciplined demonstration of why. We open by reducing the entire question to a single quantity—the sign of the one-loop coefficient b_0 in $\beta_g = -(g^3/16\pi^2)\, b_0$—and to the handful of one-loop 1PI functions whose divergences fix the renormalization constant $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ (Section 27.1). The middle of the chapter is the computation itself, carried in full in Feynman gauge with $d = 4 - 2\varepsilon$ and $\overline{\text{MS}}$: the gluon self-energy and its constant Z_3, assembled from gluon, ghost, tadpole, and quark loops, where the genuinely non-abelian antiscreening lives (Section 27.2); the quark self-energy and the quark–gluon vertex, which deliver Z_2 and Z_1 and expose the broken Ward equality $Z_1 \neq Z_2$—a pure adjoint-Casimir term that is the algebraic origin of asymptotic freedom (Section 27.3); and the assembly of these three constants into the beta function $b_0 = \frac{11}{3} C_A - \frac{4}{3} T_F n_f$, recomputed a second way by the background-field method, where gauge invariance is manifest and a single renormalization constant replaces the three (Section 27.4). The colour data C_A, T_F, C_F are imported from Chapter B; the loop technology—spinor traces, master integrals, the electron self-energy and vertex of Chapter 19—is imported and

dressed in colour rather than rebuilt.

With the number in hand the chapter reads off its consequences. We integrate the flow and prove that the coupling vanishes logarithmically into the ultraviolet with no Landau pole, set this mirror-sharp against the screening of QED, and state the Coleman–Gross theorem that in four dimensions only non-abelian gauge theories can behave this way; the correct physical picture is the paramagnetic, antiscreening vacuum (Section 27.5). We then trade the dimensionless coupling for a scale—dimensional transmutation—and obtain the closed-form running $\alpha_s(Q^2)$, the renormalization-group invariant Λ_{QCD}, the SU(3) numbers run down from M_Z, the infrared Landau pole where perturbation theory honestly fails, and the conformal window and Banks–Zaks fixed point at the upper edge of n_f (Section 27.6). The infrared breakdown points beyond perturbation theory, and the chapter closes by stating—not solving—confinement through the Wilson-loop area-law criterion, proved on the strong-coupling lattice and otherwise scoped with full honesty as the open continuum mass-gap problem (Section 27.7).

The arc is therefore a single computation and its aftermath: from one sign question, through every loop integral and colour factor that answers it, to the running coupling, the dynamically generated scale, and the threshold of the non-perturbative physics that the rest of the book—the phenomenology of Chapter 28, the instantons and lattice of Chapters 42 and 43, the exact supersymmetric confinement of Chapter 46, and the amplitude methods of Chapters 59 and 61—will take up where perturbation theory leaves off.

27.1 The Sign Question and the Renormalization Strategy

We continue Yang–Mills theory from Chapter 26. The classical non-abelian theory is in hand—the Yang–Mills Lagrangian $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \bar{\psi}_i(i\not{D} - m)\psi_i$ of Eq. (25.76), with n_f Dirac quarks in the fundamental of SU(N) and covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$—and its consistent quantization is complete: the gauge-fixed Lagrangian with its Faddeev–Popov ghosts (Eq. (26.37)), the BRST symmetry that certifies unitarity, and the Slavnov–Taylor identities (Theorem 26.35) that constrain the renormalization. The whole of Part VI taught us, moreover, exactly how to turn such a quantized theory into a prediction about energy dependence: the beta function is read off from the simple poles of the renormalization constants, through the master relation Eq. (24.104), and its *sign* decides the fate of the coupling at high energy through the RG solution Theorem 24.22 and its momentum-space form Eq. (24.202). For QED that machinery returned a *positive* one-loop coefficient, $\beta_e = e^3/12\pi^2 > 0$ (Eq. (24.108)): the electric charge grows with energy, the vacuum screens, and the theory carries a Landau pole into the deep ultraviolet. Everything in this chapter turns on whether Yang–Mills does the same thing or the opposite.

That is the single question we now pose, and the rest of the chapter is its answer. *What is the sign of the Yang–Mills beta function?* The destination, stated plainly, is the carried Yang–Mills theory number

$$\beta_g = -\frac{g^3}{16\pi^2}\, b_0 + O(g^5), \qquad b_0 = \frac{11}{3}\, C_A - \frac{4}{3}\, T_F\, n_f, \tag{27.1}$$

with C_A and T_F the adjoint Casimir and the fundamental Dynkin index of Eq. (B.11) and Eq. (B.2) ($C_A = N$, $T_F = \frac{1}{2}$ for SU(N)). For QCD, SU(3) with $n_f \le 6$ flavors, $b_0 = 11 - \frac{2}{3}n_f > 0$, so the coefficient of g^3 in β_g is *negative*: the coupling shrinks at high energy. The negative sign is asymptotic freedom, and proving it is the labor of the chapter. Throughout we use the symbols frozen for Yang–Mills theory—coupling g with $\alpha_s \equiv g^2/4\pi$, generators T^a with $\mathrm{tr}(T^a T^b) = T_F \delta^{ab}$, structure constants f^{abc}, and n_f quark flavors—work in $\overline{\mathrm{MS}}$ with $d = 4 - 2\varepsilon$ and scale μ, and fix Feynman gauge $\xi = 1$ for every explicit loop; the final b_0 is gauge-independent, as the agreement of three computational routes (Section 27.1.4) will confirm.

This first section does no loop integrals. Its task is the reduction: to show that the entire question—a sign, a number—collapses to the extraction of one renormalization constant, the coupling renormalization Z_g, and then to a strictly finite checklist of one-loop one-particle-irreducible (1PI) amplitudes whose divergent parts feed it. We carry that reduction out completely. By the end the chapter is reduced to "compute three divergent amplitudes," and we will know exactly which three, why those three suffice, and what each contributes to the sign.

27.1.1 What the sign decides

Before computing it is worth being precise about what hangs on the sign, because the precision is itself the motivation. Write the one-loop beta function in the universal form $\beta_g = -(b_0/16\pi^2)\, g^3$. The RG running of the coupling with momentum scale Q is governed by Eq. (24.202); integrating $\mu\, \mathrm{d}g/\mathrm{d}\mu = \beta_g$ from a reference scale μ_0 to Q gives, at one loop,

$$\frac{1}{g^2(Q)} = \frac{1}{g^2(\mu_0)} + \frac{b_0}{8\pi^2}\, \ln\frac{Q}{\mu_0}, \qquad \text{equivalently} \qquad \alpha_s(Q) = \frac{\alpha_s(\mu_0)}{1 + \dfrac{b_0\, \alpha_s(\mu_0)}{2\pi}\, \ln(Q/\mu_0)}, \tag{27.2}$$

a result we derive in full, with the integration constant promoted to the physical scale Λ_{QCD}, in a later section. The arithmetic of the sign is stark. If $b_0 < 0$ (as in QED, in the convention of Eq. (27.1), where $\beta_e > 0$ means $b_0^{\mathrm{QED}} < 0$), then $1/g^2(Q)$ *decreases* as Q rises: the coupling grows, the perturbative description fails in the ultraviolet, and Eq. (27.2) develops a pole at finite Q—the Landau pole, the ultraviolet incompleteness of the abelian theory. The vacuum behaves

as an ordinary dielectric: a test charge is *screened* by virtual pairs, its effective charge largest when probed at short distance.

If instead $b_0 > 0$, every conclusion reverses. Now $1/g^2(Q)$ *grows* without bound as $Q \to \infty$, so $g(Q) \to 0$: the coupling is logarithmically driven to zero at high energy. The theory becomes *free* at short distance—this is asymptotic freedom—and perturbation theory, far from breaking down, becomes arbitrarily accurate in the ultraviolet, which is precisely the regime probed in deep-inelastic scattering and high-energy collisions. The flip side is that the coupling *grows* toward the infrared: Eq. (27.2) now has its pole at *low* Q, signaling the breakdown of perturbation theory at a dynamically generated scale, the place where quarks and gluons cease to be the useful variables and confinement sets in. The same sign that guarantees a predictive short-distance theory forces a strongly coupled long-distance one.

Physical Insight

A single sign separates two qualitatively different worlds. $b_0 < 0$: the coupling is strong in the UV and weak in the IR; the theory is a good effective description at low energy but incomplete at high energy (QED, the scalar $\lambda\phi^4$). $b_0 > 0$: the coupling is weak in the UV and strong in the IR; the theory is UV-complete and calculable at short distance but non-perturbative at long distance (Yang–Mills, QCD). Asymptotic freedom is the statement that non-abelian gauge theory lives in the second world. It is what makes QCD a predictive fundamental theory rather than a low-energy effective one, and it is what makes confinement a long-distance problem rather than a short-distance pathology. The whole chapter is the determination of which world we are in, and the answer rests entirely on the sign of one number.

The QED foil deserves to be stated sharply, because the contrast is the surest guide to where the non-abelian sign flip comes from. In QED the gauge boson is electrically neutral; the only charged virtual states are fermion–antifermion pairs, and these polarize the vacuum the way charges in a dielectric do, screening the source and giving $b_0^{\mathrm{QED}} < 0$. In Yang–Mills the gauge boson itself carries color: the gluon couples to gluons through the cubic and quartic self-interactions buried in $F^a_{\mu\nu}F^{a\,\mu\nu}$. A new species of charged virtual state—the gluon loop—contributes to the vacuum polarization, and (as we will compute) it does so with the *opposite* sign, *antiscreening* the source. The $+\frac{11}{3}C_A$ in b_0 is the gluonic antiscreening; the $-\frac{4}{3}T_F n_f$ is the ordinary fermionic screening, the exact non-abelian echo of the QED quark loop. Asymptotic freedom is the statement that, for few enough flavors, antiscreening wins. Our job is to make that competition quantitative, and the first step is to identify the one object whose pole encodes it.

27.1.2 Reducing the chapter to one renormalization constant

The beta function of any coupling is fixed, through Eq. (24.104), by the renormalization constant that relates the bare coupling to the renormalized one. For Yang–Mills that constant is Z_g, defined exactly as in Eq. (24.16) but now for the gauge coupling:

$$g_0 = \mu^{\varepsilon}\, Z_g\, g, \tag{27.3}$$

with the regulator dependence carried explicitly. The power μ^{ε} (not $\mu^{2\varepsilon}$ as for the scalar λ, whose vertex is the coupling itself) is fixed by dimensional analysis in $d = 4 - 2\varepsilon$. The action $S = \int \mathrm{d}^d x\, \mathcal{L}$ is dimensionless, so $[\mathcal{L}] = d$ in mass units. From $-\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu}$ with $F \sim \partial A$ we read $[A] = (d-2)/2 = 1 - \varepsilon$; the covariant derivative $D_\mu = \partial_\mu - i g A^a_\mu T^a$ requires $[gA] = [\partial] = 1$, hence

$$[g] = 1 - [A] = \varepsilon. \tag{27.4}$$

The bare coupling g_0 thus carries mass dimension ε, while the renormalized g is by convention dimensionless; the mismatch is absorbed by μ^{ε}, and the residual dimensionless factor is the pure $\overline{\text{MS}}$ pole series Z_g. This is Eq. (24.16) with $\rho = 1$.

Everything now depends on writing Z_g in terms of the renormalization constants of the *fields and vertices* that actually appear in Feynman diagrams, for it is those that the loop integrals compute. The cleanest source is the quark–gluon coupling, the non-abelian copy of the QED electron–photon vertex. Multiplicative renormalization relates bare to renormalized fields by field-strength constants,

$$\psi_0 = Z_2^{1/2}\, \psi, \qquad A^a_{0\,\mu} = Z_3^{1/2}\, A^a_\mu, \tag{27.5}$$

with Z_2 the quark field renormalization and Z_3 the gluon field renormalization, exactly as Z_2 and Z_3 played those roles for the electron and photon in QED (Chapter 19). The quark–gluon interaction sits inside the quark kinetic term: expanding $\bar\psi_0\, i \not{D}\, \psi_0$ with $D_\mu = \partial_\mu - i g_0 A^a_{0\,\mu} T^a$ gives the bare vertex term

$$\mathcal{L} \;\supset\; g_0\, \bar\psi_0\, \not{A}^a_0\, T^a\, \psi_0 = g_0\, Z_2\, Z_3^{1/2}\, \bar\psi\, \not{A}^a\, T^a\, \psi, \tag{27.6}$$

where the second equality substitutes Eq. (27.5). On the other side, the renormalized theory writes this same term with its physical coupling $g\mu^{\varepsilon}$ times a vertex renormalization constant Z_1 that absorbs the loop divergences of the proper quark–gluon vertex (the constant multiplying the structure $\bar\psi \not{A}^a T^a \psi$, i.e. the divergent part of the 1PI amplitude whose tree value is the Feynman rule Eq. (26.60)):

$$\mathcal{L} \;\supset\; Z_1\, g\, \mu^{\varepsilon}\, \bar\psi\, \not{A}^a\, T^a\, \psi. \tag{27.7}$$

The two descriptions are the same term in the same Lagrangian, so their coefficients are equal:

$$g_0\, Z_2\, Z_3^{1/2} = Z_1\, g\, \mu^{\varepsilon} \qquad \Longrightarrow \qquad g_0 = \frac{Z_1}{Z_2\, Z_3^{1/2}}\, g\, \mu^{\varepsilon}. \tag{27.8}$$

Comparing with the definition Eq. (27.3) of Z_g gives the central structural identity of the chapter,

$$\boxed{Z_g = Z_1\, Z_2^{-1}\, Z_3^{-1/2}.} \tag{27.9}$$

The form is identical to the QED relation $e_0 = Z_1 Z_2^{-1} Z_3^{-1/2}\, e$ recorded at Eq. (21.91); the difference is that QED's abelian Ward–Takahashi identity forces $Z_1 = Z_2$ there, collapsing the coupling renormalization to $Z_g^{\mathrm{QED}} = Z_3^{-1/2}$. In the non-abelian theory the gauge boson is charged, the analogous Slavnov–Taylor identity *does not* set $Z_1 = Z_2$, and all three constants survive in Eq. (27.9). That non-collapse is exactly the technical fingerprint of the self-interaction, and it is why the non-abelian computation is genuinely new rather than a recoloring of QED. The whole question of the sign is now the question of the simple poles of Z_1, Z_2, Z_3.

27.1.3 The one-loop master formula: b_0 from a residue

We now convert Z_g into the beta function, carrying the ε bookkeeping explicitly. The argument runs parallel to Eq. (24.104) but is worth giving in full, because the four-dimensional QED statement there suppressed the ε-term that we must keep visible here to extract a clean residue formula.

The starting point is that the bare coupling g_0 is a property of the unrenormalized theory and knows nothing of the arbitrary scale μ introduced by renormalization: $\mu\, \mathrm{d}g_0/\mathrm{d}\mu = 0$. Take the logarithm of Eq. (27.3),

$$\ln g_0 = \varepsilon \ln \mu + \ln Z_g + \ln g, \tag{27.10}$$

and differentiate with respect to $\ln \mu$ at fixed g_0. Writing the d-dimensional beta function as $\beta \equiv \mu\, \mathrm{d}g/\mathrm{d}\mu = \mathrm{d}g/\mathrm{d}\ln\mu$,

$$0 = \varepsilon + \frac{\mathrm{d}\ln Z_g}{\mathrm{d}\ln\mu} + \frac{\beta}{g}. \tag{27.11}$$

Solving for β gives the exact, all-orders master relation

$$\boxed{\beta = -\varepsilon\, g - g\, \frac{\mathrm{d}\ln Z_g}{\mathrm{d}\ln\mu}} \tag{27.12}$$

The first term is the trivial classical scaling forced by the mass dimension Eq. (27.4) of the coupling away from four dimensions; it vanishes as $\varepsilon \to 0$. The second term is the genuine quantum running, controlled entirely by how Z_g depends on μ.

In $\overline{\mathrm{MS}}$ the constant Z_g is a pure series of poles in $1/\varepsilon$ whose residues are functions of the dimensionless coupling alone, with no explicit μ dependence and

no finite part. Hence Z_g depends on μ *only* through the running coupling $g(\mu)$, and the chain rule turns the μ-derivative into a g-derivative,

$$\frac{\mathrm{d}\ln Z_g}{\mathrm{d}\ln\mu} = \beta\,\frac{\mathrm{d}\ln Z_g}{\mathrm{d}g}. \tag{27.13}$$

Substituting into Eq. (27.12) makes it an algebraic equation for β,

$$\beta = -\varepsilon\, g - g\,\beta\,\frac{\mathrm{d}\ln Z_g}{\mathrm{d}g} \qquad\Longrightarrow\qquad \beta = \frac{-\varepsilon\, g}{1 + g\,\dfrac{\mathrm{d}\ln Z_g}{\mathrm{d}g}}, \tag{27.14}$$

which is precisely Eq. (24.104) with $\rho = 1$ and Z_g in place of Z_e. To extract the four-dimensional beta function we expand the right-hand side in poles. In $\overline{\mathrm{MS}}$ write the Laurent series of the logarithm,

$$\ln Z_g = \frac{\zeta_1(g)}{\varepsilon} + \frac{\zeta_2(g)}{\varepsilon^2} + \cdots, \tag{27.15}$$

so that $g\,\mathrm{d}\ln Z_g/\mathrm{d}g = g\,\zeta_1'(g)/\varepsilon + O(\varepsilon^{-2})$. Then

$$\begin{aligned}\beta &= -\varepsilon\, g\left(1 + \frac{g\,\zeta_1'}{\varepsilon} + \cdots\right)^{-1} = -\varepsilon\, g\left(1 - \frac{g\,\zeta_1'}{\varepsilon} + \cdots\right)\\ &= -\varepsilon\, g + g^2\,\zeta_1'(g) + O(\varepsilon^{-1}).\end{aligned} \tag{27.16}$$

The residual $O(\varepsilon^{-1})$ pieces are built from products of lower residues; in a renormalizable theory they cancel order by order against the genuine higher-pole data in Eq. (27.15) (the $\overline{\mathrm{MS}}$ pole-consistency conditions, the all-orders content of Theorem 24.12). What survives as $\varepsilon \to 0$ is finite and is the physical four-dimensional beta function,

$$\beta^{(4)}(g) = \lim_{\varepsilon\to 0}\bigl(\beta + \varepsilon\, g\bigr) = g^2\,\zeta_1'(g). \tag{27.17}$$

Thus the entire four-dimensional running is carried by ζ_1, the simple-pole residue of $\ln Z_g$. It remains only to relate ζ_1 to the coefficient b_0. At one loop ζ_1 is quadratic in the coupling; in the loop-counting units that make every diagram carry an explicit $g^2/16\pi^2$, write the residue as

$$\zeta_1(g) = \frac{g^2}{16\pi^2}\,r + O(g^4), \qquad r \equiv \frac{16\pi^2}{g^2}\,\zeta_1\Big|_{\text{one loop}}, \tag{27.18}$$

so that r is the dimensionless one-loop residue of $\ln Z_g$ measured in units of $g^2/16\pi^2$. Then $\zeta_1' = 2(g/16\pi^2)\,r$ and Eq. (27.17) gives

$$\beta^{(4)}(g) = g^2\cdot\frac{2g}{16\pi^2}\,r = \frac{2r}{16\pi^2}\,g^3. \tag{27.19}$$

Matching to the target form $\beta_g = -(b_0/16\pi^2)\, g^3$ of Eq. (27.1) reads off the master formula of the chapter,

$$\boxed{b_0 = -2\, r, \qquad r = \mathrm{Res}_{\varepsilon\to 0} \ln Z_g \ \ \text{(in units } g^2/16\pi^2\text{)}.} \tag{27.20}$$

The sign question is now a single arithmetic question: *is the one-loop simple pole of* $\ln Z_g$ *negative?* If $r < 0$ then $b_0 > 0$, the beta function is negative, and the theory is asymptotically free.

Proposition 27.1 (b_0 from the field- and vertex-residues)**.** *Write each renormalization constant as a one-loop* $\overline{MS}$ *pole,*

$$Z_i = 1 + \frac{g^2}{16\pi^2}\,\frac{r_i}{\varepsilon} + O(g^4), \qquad i \in \{1,2,3\}, \tag{27.21}$$

where r_1, r_2, r_3 *are the dimensionless one-loop residues of the quark–gluon vertex, the quark field, and the gluon field renormalizations respectively. Then the one-loop coefficient of the Yang–Mills beta function is*

$$b_0 = -2\Big(r_1 - r_2 - \tfrac{1}{2} r_3\Big) = -2r_1 + 2r_2 + r_3. \tag{27.22}$$

Proof. Take the logarithm of the decomposition Eq. (27.9),

$$\ln Z_g = \ln Z_1 - \ln Z_2 - \tfrac{1}{2} \ln Z_3. \tag{27.23}$$

At one loop $\ln Z_i = \ln(1 + (g^2/16\pi^2)(r_i/\varepsilon)) = (g^2/16\pi^2)(r_i/\varepsilon) + O(g^4)$, so the simple-pole residue of $\ln Z_i$, in the units of Eq. (27.18), is exactly r_i. Linearity of the residue in Eq. (27.23) gives $r = r_1 - r_2 - \frac{1}{2} r_3$. Substituting into the master formula Eq. (27.20) yields Eq. (27.22). □

Equation (27.22) is the spine of the chapter. It says that the sign of the beta function—hence the existence of asymptotic freedom—is decided by three numbers: the divergent parts of three one-loop amplitudes. We have reduced an open-ended question about the ultraviolet fate of a quantum field theory to the evaluation of three logarithmically divergent integrals with their color factors.

27.1.4 The finite checklist, and which route to take

Theorem 27.1 fixes the work that remains. To obtain b_0 we must compute the divergent $(1/\varepsilon)$ parts of exactly three 1PI functions, drawn schematically in Fig. 27.1:

1. the **gluon self-energy** (vacuum polarization) $\Pi^{ab}_{\mu\nu}(k)$, whose transverse divergence fixes the gluon field renormalization Z_3 $(\to r_3)$;
2. the **quark self-energy** $\Sigma(p)$, whose $\not{p}$ divergence fixes the quark field renormalization Z_2 $(\to r_2)$;

3. the **quark–gluon vertex** $\Gamma^a_\mu(p', p)$, whose $\gamma_\mu T^a$ divergence fixes the vertex renormalization Z_1 ($\to r_1$).

This is the whole list. No four-point function enters: by the power counting of Chapter 21, carried over from ϕ^4 and QED, the superficially divergent 1PI amplitudes of a renormalizable gauge theory are precisely the two- and three-point functions matching terms already present in the Lagrangian, and the coupling renormalization is built from those alone.

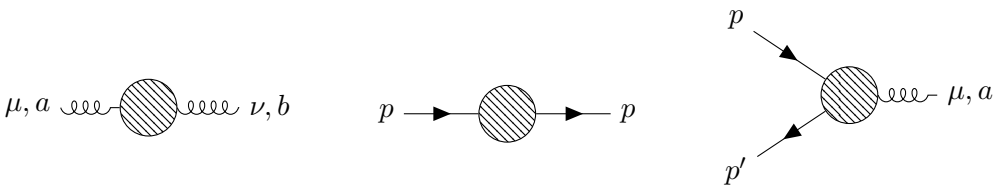


Figure 27.1: The complete checklist. The three superficially divergent 1PI amplitudes whose simple poles supply the residues r_3, r_2, r_1 of Theorem 27.1, and through them the beta-function coefficient $b_0 = -2r_1 + 2r_2 + r_3$. Shaded blobs denote the sum of one-loop 1PI insertions. Wavy lines are gluons, solid lines quarks. The chapter computes these three, in this order, beginning with the gluon self-energy.

Two of the three computations are, to a large degree, imports. The quark self-energy $\Sigma(p)$ and the quark–gluon vertex Γ^a_μ are, diagram for diagram, the QED electron self-energy and electron–photon vertex of Chapter 19—the same loop integrals, the same Dirac algebra, the same dimensional regularization (Section 19.1.3)—with the photon promoted to a gluon and a single color factor attached at each gluon coupling. The attached factor is $T^a T^a = C_F\,\mathbf{1}$, the fundamental Casimir of Eq. (B.11), for the abelian-like part of each diagram. The quark–gluon vertex carries, in addition, one genuinely non-abelian diagram in which the emitted gluon attaches to an internal gluon through the three-gluon vertex; its color factor follows from the SU(N) identity $f^{acd} f^{bcd} = C_A \delta^{ab}$ together with $T^a T^b T^a = (C_F - \frac{1}{2} C_A) T^b$, both established in Eq. (B.11). The only computation with no QED ancestor at all is the gluon self-energy Z_3, where the loop content—a gluon loop, a gluon tadpole (seagull), and a ghost loop, alongside the familiar quark loop—is intrinsically non-abelian and is the load-bearing calculation of the chapter.

This is exactly why we route Z_g through the quark–gluon vertex rather than through the other vertices the theory offers. The decomposition Eq. (27.9) is not unique: any vertex of the gauge coupling yields its own expression for the same Z_g. From the triple-gluon vertex one gets $Z_g = Z_1^{3g}\, Z_3^{-3/2}$, with Z_1^{3g} the three-gluon vertex renormalization; from the ghost–gluon vertex, $Z_g = \widetilde{Z}_1\, \widetilde{Z}_3^{-1}\, Z_3^{-1/2}$, with

$\widetilde{Z}_1$ and $\widetilde{Z}_3$ the ghost–gluon vertex and ghost field renormalizations; from the four-gluon vertex, $Z_g^2 = Z_4^{4g}\, Z_3^{-2}$. Each route requires its own divergent vertex computation in place of the quark sector's Z_1, Z_2. The quark–gluon route is the economical one precisely because Z_1 and Z_2 are recolored QED objects: we import their structure from Chapter 19 and pay only for the color algebra, leaving a single new non-abelian computation, Z_3. The triple-gluon and ghost routes, by contrast, would demand fresh evaluations of the three-gluon or ghost–gluon vertex from scratch.

That the routes give the same Z_g is not an accident but a theorem. The Slavnov–Taylor identities Theorem 26.35—the BRST-exact, all-orders non-abelian generalization of the Ward–Takahashi identity—impose relations among the renormalization constants,

$$\frac{Z_1}{Z_2} = \frac{Z_1^{3g}}{Z_3} = \frac{\widetilde{Z}_1}{\widetilde{Z}_3} = \Big(\frac{Z_4^{4g}}{Z_3}\Big)^{1/2}, \tag{27.24}$$

each of which, substituted into the corresponding decomposition of Z_g, reproduces the single value $g_0 = Z_g g \mu^{\varepsilon}$. Equation (27.24) is the statement that the renormalized theory has *one* coupling, not four, despite the gauge field appearing in four distinct vertices—the quantum survival of the classical universality of g that gauge invariance demands. We will not need the full set; we compute b_0 once, by the quark–gluon route, and the consistency of the other routes (in particular the ghost route, which is the historically standard one) is guaranteed by Eq. (27.24). The independence of the answer from the chosen route is also, in practice, the cleanest check that the gauge-dependent intermediate constants Z_1, Z_2, Z_3—each computed here in Feynman gauge—have been assembled into a gauge-independent b_0.

Worked Example 27.2 (The QED foil: recovering β_e from the master formula)**.** It is worth running the abelian theory through the same machinery, both to confirm the master formula Eq. (27.20) against the known QED result and to see concretely what the sign flip will have to overcome. In QED the Ward–Takahashi identity sets $Z_1 = Z_2$ (Eq. (21.91)), so the decomposition Eq. (27.9) collapses: $r_1 = r_2$ and

$$r^{\mathrm{QED}} = r_1 - r_2 - \tfrac{1}{2} r_3 = -\tfrac{1}{2}\, r_3. \tag{27.25}$$

The only surviving input is the photon field renormalization. The one-loop QED vacuum polarization computed with dimensional regularization in Chapter 19 gives

$$Z_3 = 1 - \frac{e^2}{12\pi^2}\,\frac{1}{\varepsilon} + \cdots = 1 + \frac{e^2}{16\pi^2}\,\frac{r_3}{\varepsilon} + \cdots, \qquad r_3 = -\frac{4}{3}, \tag{27.26}$$

where the second form rewrites the residue in the loop units of Eq. (27.21) using $\frac{1}{12\pi^2} = \frac{4}{3}\cdot\frac{1}{16\pi^2}$. The negative r_3 is the screening sign: the electron loop *lowers*

Z_3. Feeding Eq. (27.26) into Eq. (27.25) gives $r^{\mathrm{QED}} = -\frac{1}{2}(-\frac{4}{3}) = \frac{2}{3}$, and the master formula Eq. (27.20) returns

$$b_0^{\mathrm{QED}} = -2\, r^{\mathrm{QED}} = -\frac{4}{3} < 0. \tag{27.27}$$

This is negative, as the foil demands. The corresponding beta function is

$$\beta_e = -\frac{b_0^{\mathrm{QED}}}{16\pi^2}\, e^3 = \frac{4/3}{16\pi^2}\, e^3 = \frac{e^3}{12\pi^2}, \tag{27.28}$$

reproducing exactly Eq. (24.108)—the positive QED beta function, the growing charge, the Landau pole. As an internal check, the residue of $\ln Z_g$ itself is $r^{\mathrm{QED}}\, e^2/16\pi^2 = e^2/24\pi^2$, which is precisely the simple-pole residue $a_1 = e^2/24\pi^2$ of the QED coupling renormalization recorded in Eq. (24.105). The machinery is consistent. Now read the lesson for Yang–Mills: in the non-abelian theory the quark loop contributes the same screening $r_3^{\mathrm{quark}} < 0$ (it is the same diagram), and on top of it $Z_1 \neq Z_2$ no longer collapses the formula. The gluon and ghost loops add new, positive contributions to r, and the sum can—and for QCD does—come out *negative*, flipping b_0 positive. Whether it does is the arithmetic the next sections supply.

The reduction is complete. The sign of the Yang–Mills beta function, and with it asymptotic freedom and the entire ultraviolet character of QCD, is now pinned to three residues r_1, r_2, r_3 assembled by $b_0 = -2r_1 + 2r_2 + r_3$, and the path to each residue is fixed: the quark self-energy and quark–gluon vertex are color-dressed imports of Chapter 19, while the gluon self-energy is the new computation that carries the antiscreening. We begin where the physics lives, with the one amplitude that has no abelian counterpart—the gluon vacuum polarization, and the gluon field renormalization Z_3 it determines.

27.2 The Gluon Self-Energy at One Loop

The checklist of the previous section reduced the entire question of the sign of the Yang–Mills beta function to a short list of one-loop 1PI functions, and singled out one of them as the load-bearing object: the gluon field-renormalization constant Z_3, read from the divergent part of the gluon two-point function. In an abelian theory this would be the end of the story, because the Ward–Takahashi identity $Z_1 = Z_2$ collapses the charge renormalization onto the photon self-energy alone (Theorem 22.31), and we already know the QED answer (Eq. (24.108)): screening, a positive coefficient. Yang–Mills is not abelian. The coupling renormalization will turn out to require the quark sector constants Z_2 and Z_1 as well, computed in the next section; but Z_3 is where the genuinely non-abelian physics lives, because the gluon is charged and so it propagates not only by polarizing a sea of

virtual quarks — the exact analog of the QED photon — but also by interacting with virtual gluons through the triple- and quartic-gluon vertices of Eqs. (26.53) and (26.56), and with the Faddeev–Popov ghosts through Eq. (26.58). These gluonic loops have no counterpart in electrodynamics. They are the whole reason a non-abelian theory can antiscreen, and computing them in full — every loop integral, every color factor, every gamma trace — is the central business of this chapter.

This section carries out that computation. The one-loop gluon vacuum polarization receives four contributions, drawn in Fig. 27.2: the *gluon loop* built from two triple-gluon vertices, the *gluon tadpole* (or seagull) built from a single quartic vertex, the *ghost loop*, and the *quark loop*. We evaluate each in dimensional regularization with $d = 4-2\varepsilon$ and the $\overline{\text{MS}}$ scale μ, working throughout in Feynman gauge $\xi = 1$ so that the gluon propagator is the maximally simple Eq. (26.45). The result is the transverse divergent self-energy and, from it, the constant

$$Z_3 \;=\; 1 + \frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\left(\frac{5}{3}\,C_A - \frac{4}{3}\,T_F\,n_f\right), \tag{27.29}$$

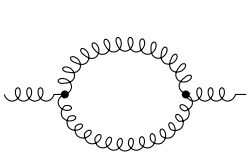
the gluonic part $\frac{5}{3}C_A$ and the matter part $-\frac{4}{3}T_F n_f$ assembled from the four diagrams separately. Along the way we verify the single most instructive fact about the calculation, the one that ties the arithmetic to the structure of Chapter 26: the gluon loop is *not* transverse by itself, the ghost loop is not transverse by itself, and only their sum (with the tadpole, which vanishes) satisfies $q^\mu \Pi_{\mu\nu} = 0$, exactly as the Slavnov–Taylor identities (Theorems 26.35 and 26.37) demand. Transversality is therefore not a check we impose; it is a prediction of BRST invariance that the explicit integrals are obliged to satisfy, and watching them do so is how we know the ghost contribution is precisely the right size.

27.2.1 Definitions, kinematics, and the divergence we are after

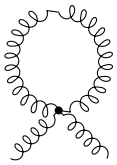
We write the one-particle-irreducible gluon two-point function with all external data amputated as $i\Pi^{ab}_{\mu\nu}(q)$, where a, b are adjoint color indices, μ, ν Lorentz indices, and q the external gluon momentum. Lorentz invariance, color invariance, and the transversality theorem proved to all orders in Theorem 26.37 fix its form completely: it is diagonal in color and built from the single transverse projector,

$$i\Pi^{ab}_{\mu\nu}(q) \;=\; i\,\delta^{ab}\,(q^2\eta_{\mu\nu} - q_\mu q_\nu)\,\Pi(q^2), \tag{27.30}$$

with one scalar form factor $\Pi(q^2)$. The factor of i on the right is a convention chosen so that $\Pi(q^2)$ is real in the spacelike region and matches the QED definition imported in Eq. (19.49); the combination $q^2\eta_{\mu\nu} - q_\mu q_\nu$ is the momentum-space form of the kinetic operator $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$, so a divergence in $\Pi(q^2)$ is exactly a divergence that can be reabsorbed into the normalization of the gluon field. Because that transverse tensor carries two explicit powers of q, the scalar $\Pi(q^2)$

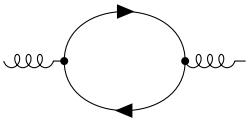


(a) gluon loop



(b) gluon tadpole (seagull)

(c) ghost loop



(d) quark loop

Figure 27.2: The four one-loop 1PI contributions to the gluon self-energy $i\Pi^{ab}_{\mu\nu}(q)$. Diagrams (a) and (b) are pure-gauge and have no QED analog: (a) uses two triple-gluon vertices Eq. (26.53), (b) the single quartic vertex Eq. (26.56). Diagram (c), the closed ghost loop, carries a factor (-1) by Theorem 26.13 and is what restores transversality. Diagram (d), the quark loop, is the exact non-abelian copy of the QED vacuum polarization of Eq. (19.49), dressed with the color trace $\mathrm{tr}(T^aT^b) = T_F\delta^{ab}$.

has vanishing mass dimension and is at most logarithmically divergent: its ultraviolet pole is a pure number times $1/\varepsilon$, independent of q^2, and that number is what fixes Z_3. We write the bare field as $A_0 = Z_3^{1/2}A$ and split the divergence off through the counterterm exactly as in QED (Eq. (22.150)), with $\delta_3 = Z_3 - 1$; in $\overline{\mathrm{MS}}$ the renormalization condition is simply that δ_3 cancel the pole of Π,

$$\delta_3 \;=\; \Pi(q^2)\big|_{\mathrm{pole}}, \qquad Z_3 \;=\; 1 + \Pi(q^2)\big|_{\mathrm{pole}}. \tag{27.31}$$

Everything reduces, then, to computing the coefficient of the $1/\varepsilon$ pole in $\Pi(q^2)$, summed over the four diagrams.

Master integrals in pole form. Every diagram below reduces, after Feynman-combining the two propagators and shifting the loop momentum, to the standard scalar integrals of Theorem D.22 and Eq. (D.87) with denominator $(\ell^2-\Delta)^2$. The three combinations we need, kept only to their ultraviolet pole and written with Δ the effective mass-squared of Eq. (D.103), are obtained directly from Eqs. (D.77) and (D.88) and the symmetric integration rule Eq. (D.92). With $1/(4\pi)^{d/2} \to 1/16\pi^2$ at the pole and using $\Gamma(\varepsilon) = 1/\varepsilon + \dots$, $\Gamma(-1+\varepsilon) = -1/\varepsilon + \dots$,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta)^2} \;=\; \frac{i}{16\pi^2}\,\frac{1}{\varepsilon} \;+\; \text{finite}, \tag{27.32}$$

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^2} \;=\; -\frac{d}{2}\,\Gamma\!\left(1-\tfrac{d}{2}\right)\frac{i}{(4\pi)^{d/2}}\,\Delta^{d/2-1} \;=\; \frac{i}{16\pi^2}\,\frac{2\Delta}{\varepsilon} \;+\; \text{finite}, \tag{27.33}$$

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu\ell^\nu}{(\ell^2-\Delta)^2} \;=\; \frac{\eta^{\mu\nu}}{d}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^2} \;=\; \frac{i}{16\pi^2}\,\frac{\Delta}{2\varepsilon}\,\eta^{\mu\nu} \;+\; \text{finite}. \tag{27.34}$$

The middle form of Eq. (27.33) is kept because two of our diagrams generate the numerator ℓ^2 multiplied by a coefficient that itself depends on d, and there the cancellation of the d in $\Gamma(1-d/2)$ against an explicit factor of d matters; we will use the exact Γ-function form in those cases and read off the pole at the end. In every diagram the Feynman parameter is called x and the loop momentum after the shift ℓ, with

$$\frac{1}{k^2(k+q)^2} = \int_0^1 \mathrm{d}x\,\frac{1}{[(k+xq)^2-\Delta]^2}, \qquad \Delta = -x(1-x)\,q^2 \tag{27.35}$$

for the massless (gauge and ghost) loops, and $\Delta = m^2 - x(1-x)q^2$ for the quark loop carrying mass m. We extract the pole of Π, which is all that Eq. (27.31) needs; the finite parts contribute only the scheme-dependent constant in the renormalized self-energy and are not required for Z_3 in $\overline{\text{MS}}$.

27.2.2 The quark loop: the QED bubble in color

We begin with the diagram we already understand, the quark loop of Fig. 27.2d. It is the only one of the four with an electrodynamic ancestor, and treating it first lets us import the entire Dirac-trace computation from QED and isolate the single new ingredient, the color trace. We continue Yang–Mills theory, taking n_f Dirac quarks in the fundamental representation, each coupling through the quark-gluon vertex Eq. (26.60), $iV^a_\mu = ig\gamma_\mu T^a$, with the ordinary Dirac propagator dressed by a color δ_{ij}. Reading off Fig. 27.2d with loop momentum k on one internal line and $k+q$ on the other, including the mandatory (-1) for the closed fermion loop and the sum over flavors,

$$i\Pi^{(q)ab}_{\mu\nu}(q) = (-1)\,n_f \int \frac{\mathrm{d}^d k}{(2\pi)^d}\;\mathrm{tr}\left[(ig\gamma_\mu T^a)\,\frac{i(\not{k}+m)}{k^2-m^2}\,(ig\gamma_\nu T^b)\,\frac{i(\not{k}+\not{q}+m)}{(k+q)^2-m^2}\right], \tag{27.36}$$

where the trace runs over both Dirac and color indices. The color trace factors out of the Dirac algebra entirely: $\mathrm{tr}(T^aT^b) = T_F\,\delta^{ab}$ with $T_F = \tfrac{1}{2}$, the Dynkin index of the fundamental from Eq. (B.10) and Theorem B.2. This is the one place the non-abelian structure enters the quark loop, and it enters as a single overall constant. Pulling it out, together with the explicit factors i ($(ig)^2 = -g^2$ from the vertices, $i^2 = -1$ from the two propagators, and the loop (-1)), the

kinematic remainder is identical to the QED vacuum polarization with $e^2 \to g^2$:

$$i\Pi^{(q)ab}_{\mu\nu}(q) = -\,g^2\,T_F\,n_f\,\delta^{ab}\int\frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{\mathrm{tr}\big[\gamma_\mu(\not k+m)\gamma_\nu(\not k+\not q+m)\big]}{(k^2-m^2)\big((k+q)^2-m^2\big)}. \tag{27.37}$$

The Dirac trace is evaluated with the four-dimensional trace technology of Section C.6. Only the terms with an even number of γ's survive: the two single-m terms carry three gamma matrices and vanish, leaving

$$\begin{aligned}\mathrm{tr}\big[\gamma_\mu(\not k+m)\gamma_\nu(\not k+\not q+m)\big] &= \mathrm{tr}\big[\gamma_\mu\not k\,\gamma_\nu(\not k+\not q)\big] + m^2\,\mathrm{tr}[\gamma_\mu\gamma_\nu]\\ &= 4\big[k_\mu(k+q)_\nu + k_\nu(k+q)_\mu - \eta_{\mu\nu}\,k\cdot(k+q)\big] + 4m^2\eta_{\mu\nu}\\ &= 4\big[k_\mu(k+q)_\nu + k_\nu(k+q)_\mu - \eta_{\mu\nu}\big(k\cdot(k+q)-m^2\big)\big],\end{aligned} \tag{27.38}$$

using $\mathrm{tr}[\gamma_\mu\gamma_\alpha\gamma_\nu\gamma_\beta] = 4(\eta_{\mu\alpha}\eta_{\nu\beta} - \eta_{\mu\nu}\eta_{\alpha\beta} + \eta_{\mu\beta}\eta_{\alpha\nu})$ and $\mathrm{tr}[\gamma_\mu\gamma_\nu] = 4\eta_{\mu\nu}$. Now Feynman-combine with Eq. (27.35) (here $\Delta_q = m^2 - x(1-x)q^2$) and shift $k = \ell - xq$, so $k = \ell - xq$ and $k+q = \ell + (1-x)q$. Dropping the terms odd in ℓ (they integrate to zero) the divergent numerator is

$$4\Big[2\,\ell_\mu\ell_\nu - 2x(1-x)\,q_\mu q_\nu - \eta_{\mu\nu}\big(\ell^2 - x(1-x)q^2 - m^2\big)\Big]. \tag{27.39}$$

Apply the master integrals Eqs. (27.32) to (27.34). The tensor term gives $2\ell_\mu\ell_\nu \to (i/16\pi^2)(\Delta_q/\varepsilon)\eta_{\mu\nu}$; the $\eta_{\mu\nu}\ell^2$ term gives $-(i/16\pi^2)(2\Delta_q/\varepsilon)\eta_{\mu\nu}$; and the remaining q-dependent pieces multiply the scalar pole Eq. (27.32). Collecting the coefficient of $\eta_{\mu\nu}$, the Δ_q-dependent contributions combine as $\Delta_q - 2\Delta_q + \big(x(1-x)q^2 + m^2\big) = -\Delta_q + x(1-x)q^2 + m^2 = 2x(1-x)q^2$, the masses cancelling cleanly, so

$$\int\frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{(\text{numerator})}{(k^2-m^2)\big((k+q)^2-m^2\big)}\bigg|_{\text{pole}} = 4\cdot\frac{i}{16\pi^2}\,\frac{1}{\varepsilon}\int_0^1\mathrm{d}x\;2x(1-x)\,\big(q^2\eta_{\mu\nu} - q_\mu q_\nu\big). \tag{27.40}$$

The pole part is already transverse, $\propto q^2\eta_{\mu\nu} - q_\mu q_\nu$, with no work — this is the abelian Ward identity at one loop, and it is exactly what the $f^{abc}\to 0$ limit of Theorem 26.38 guarantees. The parametric integral is $\int_0^1 2x(1-x)\,\mathrm{d}x = \frac{1}{3}$, so the bracket is $4\cdot\frac{1}{3} = \frac{4}{3}$. Restoring the prefactor of Eq. (27.37),

$$\boxed{\,i\Pi^{(q)ab}_{\mu\nu}(q)\big|_{\text{pole}} = i\,\delta^{ab}\big(q^2\eta_{\mu\nu} - q_\mu q_\nu\big)\left(-\frac{4}{3}\,T_F\,n_f\right)\frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}.\,} \tag{27.41}$$

Comparing with Eq. (27.30), the quark contribution to the form factor is $\Pi^{(q)}\big|_{\text{pole}} = -\frac{4}{3}T_F n_f\,(g^2/16\pi^2)(1/\varepsilon)$. This is the promised reproduction of the QED vacuum polarization Eq. (19.49): setting $T_F\to 1$, $n_f\to 1$, $g\to e$ recovers the single-fermion QED pole $-\frac{e^2}{12\pi^2}(1/\varepsilon)$ (since $\frac{4}{3}\cdot\frac{1}{16\pi^2} = \frac{1}{12\pi^2}$), and with it the

screening sign and the positive QED beta function of Eq. (24.108). The lesson to carry forward is that if the quark loop were the *only* contribution, Yang–Mills would screen exactly like QED. The minus sign of the beta function, when it comes, must come entirely from the gluonic diagrams.

27.2.3 The gluon loop

We turn to the diagram with no abelian ancestor, Fig. 27.2a: two triple-gluon vertices joined by two internal gluon lines. This is the heart of the computation. We route the external momentum q into the left vertex on leg (a,μ), let the internal gluons carry k and $-(k+q)$ (so that $q+k-(k+q)=0$ at the left vertex), and let q leave at the right vertex on leg (b,ν). There is a symmetry factor $\frac{1}{2}$ for the two identical internal lines. Using the triple-gluon vertex Eq. (26.53) (which, as written there, already includes its factor of i) at each end and the Feynman-gauge propagator Eq. (26.45), the amplitude is

$$i\Pi^{(g)ab}_{\mu\nu}(q) = \frac{1}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,(g f^{acd} V^1_{\mu\alpha\beta})\,\frac{-i\,\eta^{\alpha\alpha'}\delta^{cc'}}{k^2}\,\frac{-i\,\eta^{\beta\beta'}\delta^{dd'}}{(k+q)^2}\,(g f^{bc'd'} V^2_{\nu\alpha'\beta'}), \tag{27.42}$$

where, reading the vertex Eq. (26.53) with all momenta incoming and legs in the order (external, internal-k, internal-$(-k-q)$),

$$V^1_{\mu\alpha\beta} = \eta_{\mu\alpha}(q-k)_\beta + \eta_{\alpha\beta}(2k+q)_\mu + \eta_{\beta\mu}(-2q-k)_\alpha, \tag{27.43}$$

$$V^2_{\nu\alpha'\beta'} = \eta_{\nu\alpha'}(k-q)_{\beta'} + \eta_{\alpha'\beta'}(-2k-q)_\nu + \eta_{\beta'\nu}(k+2q)_{\alpha'}. \tag{27.44}$$

The color contraction is the adjoint Casimir of Eq. (B.102), $f^{acd}f^{bcd} = C_A\,\delta^{ab}$, with $C_A = N$ for SU(N). The two propagator factors contribute $(-i)^2 = -1$, and the two couplings give g^2. Pulling these out and contracting the internal Lorentz indices with the metrics,

$$i\Pi^{(g)ab}_{\mu\nu}(q) = -\frac{1}{2}\,g^2 C_A\,\delta^{ab}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{N_{\mu\nu}}{k^2(k+q)^2}, \qquad N_{\mu\nu} \equiv V^1_{\mu\alpha\beta}\,V^{2\,\alpha\beta}{}_{\nu}. \tag{27.45}$$

The numerator. Contracting Eqs. (27.43) and (27.44) term by term is a nine-term exercise; we carry all of it, using $\eta^\alpha{}_\alpha = d$ wherever the two $\eta_{\alpha\beta}$ structures meet. Writing the three pieces of V^1 as (a_1,a_2,a_3) and of V^2 as (b_1,b_2,b_3) in the order of Eqs. (27.43) and (27.44), the nine contractions are

$$\begin{aligned}
a_1b_1 &= -\eta_{\mu\nu}(q-k)^2, & a_1b_2 &= (q-k)_\mu(-2k-q)_\nu, & a_1b_3 &= (q-k)_\nu(k+2q)_\mu,\\
a_2b_1 &= (2k+q)_\mu(k-q)_\nu, & a_2b_2 &= -d\,(2k+q)_\mu(2k+q)_\nu, & a_2b_3 &= (2k+q)_\mu(k+2q)_\nu,\\
a_3b_1 &= (-2q-k)_\nu(k-q)_\mu, & a_3b_2 &= (-2q-k)_\mu(-2k-q)_\nu, & a_3b_3 &= -\eta_{\mu\nu}(k+2q)^2.
\end{aligned} \tag{27.46}$$

Now Feynman-combine with Eq. (27.35) ($\Delta = -x(1-x)q^2$) and shift $k = \ell - xq$, so that $k = \ell - xq$, $q-k = (1+x)q-\ell$, $2k+q = 2\ell+(1-2x)q$, $k+2q =$

$\ell + (2-x)q$, $k - q = \ell - (1+x)q$, and $-2q - k = (x-2)q - \ell$. We keep only the divergent structures. Terms odd in ℓ integrate to zero. Each outer product $V_\mu W_\nu$ contributes its $\ell_\mu \ell_\nu$ part (through Eq. (27.34)) *and* its $q_\mu q_\nu$ part (through the scalar pole Eq. (27.32), since a constant tensor multiplies a logarithmically divergent integral); the scalar $\eta_{\mu\nu}(\cdots)^2$ terms contribute their $\eta_{\mu\nu}\ell^2$ and $\eta_{\mu\nu}q^2$ parts. Carrying out the bookkeeping on Eq. (27.46) after the shift, the divergent numerator organizes into four tensor structures,

$$N_{\mu\nu}\big|_{\text{div}} = \underbrace{(6-4d)}_{\ell_\mu\ell_\nu}\,\ell_\mu\ell_\nu \;\; \underbrace{-2}_{\eta_{\mu\nu}\ell^2}\,\eta_{\mu\nu}\ell^2 \;+\; C_{qq}(x,d)\,q_\mu q_\nu \;+\; C_{\eta q}(x)\,\eta_{\mu\nu}q^2, \quad (27.47)$$

with the parametric coefficients

$$C_{qq}(x,d) = (6-6x+6x^2) - d(1-4x+4x^2), \qquad C_{\eta q}(x) = -5+2x-2x^2. \quad (27.48)$$

The coefficient $-4d$ of $\ell_\mu\ell_\nu$ comes entirely from the $a_2 b_2$ term, the only one carrying $\eta_{\alpha\beta}\eta^{\alpha\beta} = d$; the constant 6 is the sum $2-1+2+2-1+2$ of the remaining seven $\ell\ell$ contributions, and the $\eta_{\mu\nu}\ell^2$ coefficient -2 is the sum of the two scalar pieces $a_1 b_1$ and $a_3 b_3$. The $q_\mu q_\nu$ coefficient collects the q-parts of all seven outer products plus the $-d(1-2x)^2$ from $a_2 b_2$, and $C_{\eta q}$ collects $-(1+x)^2 - (2-x)^2$ from the two scalars.

Integration. It is cleanest to combine the two ℓ-quadratic structures before integrating, replacing $\ell_\mu\ell_\nu \to \eta_{\mu\nu}\ell^2/d$ inside the numerator (Eq. (D.92)). The total coefficient of $\eta_{\mu\nu}\ell^2$ is then $(6-4d)/d - 2 = (6-6d)/d$, and using the exact form of Eq. (27.33),

$$\frac{6-6d}{d}\,\eta_{\mu\nu}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{(\ell^2-\Delta)^2} = \frac{6-6d}{d}\Big(-\frac{d}{2}\Big)\Gamma\big(1-\tfrac{d}{2}\big)\,\frac{i}{(4\pi)^{d/2}}\,\Delta^{d/2-1}\,\eta_{\mu\nu} = -\frac{1}{2}(6-6d)\,\Gamma\big(1-\tfrac{d}{2}\big)\,\frac{i}{(4\pi)^{d/2}}\,\Delta^{d/2-1}\,\eta_{\mu\nu} \quad (27.49)$$

The explicit factor of d cancels against the $1/d$ from symmetric integration — the reason we kept the exact Γ-function form. At $d = 4-2\varepsilon$ we have $6-6d = -18+12\varepsilon$ and $\Gamma(1-d/2) = \Gamma(-1+\varepsilon) = -1/\varepsilon + \ldots$, so the pole of Eq. (27.49) is $-\frac{1}{2}(-18)(-1/\varepsilon)(i/16\pi^2)\Delta\,\eta_{\mu\nu} = -(9/\varepsilon)(i/16\pi^2)\Delta\,\eta_{\mu\nu}$. With $\Delta = -x(1-x)q^2$ this is $+9x(1-x)\,q^2\eta_{\mu\nu}\,(i/16\pi^2)(1/\varepsilon)$. The $q_\mu q_\nu$ and $\eta_{\mu\nu}q^2$ structures multiply the simple scalar pole Eq. (27.32); since that pole is simple, only the $d=4$ value of C_{qq} survives, $C_{qq}(x,4) = 2 + 10x - 10x^2$. Assembling the three pole contributions and integrating over the Feynman parameter,

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{N_{\mu\nu}}{k^2(k+q)^2}\Big|_{\text{pole}} = \frac{i}{16\pi^2}\,\frac{1}{\varepsilon}\int_0^1 \mathrm{d}x\,\Big[(9x(1-x)+C_{\eta q})q^2\eta_{\mu\nu} + (2+10x-10x^2)q_\mu q_\nu\Big]. \quad (27.50)$$

The $\eta_{\mu\nu}q^2$ coefficient is $9x(1-x)+(-5+2x-2x^2) = -5+11x-11x^2$, whose integral over x is $-5+\frac{11}{2}-\frac{11}{3} = -\frac{19}{6}$; the $q_\mu q_\nu$ coefficient integrates to $2+5-\frac{10}{3} = \frac{11}{3}$.

Hence

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{N_{\mu\nu}}{k^2 (k+q)^2}\bigg|_{\text{pole}} = \frac{i}{16\pi^2} \frac{1}{\varepsilon} \left[-\frac{19}{6}\, q^2 \eta_{\mu\nu} + \frac{11}{3}\, q_\mu q_\nu \right]. \tag{27.51}$$

Finally, restoring the prefactor $-\frac{1}{2} g^2 C_A \delta^{ab}$ of Eq. (27.45),

$$\boxed{ i\Pi^{(g)ab}_{\mu\nu}(q)\big|_{\text{pole}} = \frac{i\, g^2 C_A\, \delta^{ab}}{16\pi^2} \frac{1}{\varepsilon} \left[\frac{19}{12}\, q^2 \eta_{\mu\nu} - \frac{11}{6}\, q_\mu q_\nu \right]. } \tag{27.52}$$

Caution

The gluon loop Eq. (27.52) is *not* transverse. Transversality would require the coefficient of $q^2 \eta_{\mu\nu}$ to equal minus the coefficient of $q_\mu q_\nu$, i.e. $\frac{19}{12}$ versus $\frac{11}{6} = \frac{22}{12}$ — they differ. Contracting with q^μ gives a nonzero longitudinal remainder, $q^\mu i\Pi^{(g)}_{\mu\nu} = \frac{ig^2 C_A \delta^{ab}}{16\pi^2 \varepsilon}(\frac{19}{12} - \frac{11}{6}) q^2 q_\nu = -\frac{ig^2 C_A \delta^{ab}}{16\pi^2 \varepsilon} \frac{1}{4} q^2 q_\nu \neq 0$. Taken on its own the gluon loop would violate the all-orders identity Theorem 26.37 and would force a longitudinal, gauge-variant gluon mass counterterm — a catastrophe for renormalizability. It does not, because it is not on its own: the ghost loop is the missing piece, and the quartic seagull, to which we turn first, contributes nothing.

27.2.4 The gluon tadpole (seagull)

The seagull of Fig. 27.2b attaches both external gluons to a single quartic vertex Eq. (26.56), the remaining two legs of which are contracted into a gluon loop. With symmetry factor $\frac{1}{2}$ for the self-contracted pair,

$$i\Pi^{(t)ab}_{\mu\nu}(q) = \frac{1}{2} \int \frac{\mathrm{d}^d k}{(2\pi)^d}\, i V^{abcd}_{\mu\nu\rho\sigma} \frac{-i\, \eta^{\rho\sigma} \delta^{cd}}{k^2}, \tag{27.53}$$

where the four-gluon vertex is the momentum-independent tensor Eq. (26.56). Contracting the loop indices $\rho\sigma$ and colors cd with $\eta^{\rho\sigma}\delta^{cd}$ collapses the vertex to a constant multiple of $\eta_{\mu\nu}$ times $C_A \delta^{ab}$ (the structure constants close up, $f^{ace} f^{bce} = C_A \delta^{ab}$, and the metric contractions give a pure number in d). The decisive point is what is left of the loop integral: there is no external momentum and no mass on the internal line, so

$$i\Pi^{(t)ab}_{\mu\nu}(q) \;\propto\; \eta_{\mu\nu}\, C_A\, \delta^{ab} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{k^2} \;=\; 0. \tag{27.54}$$

The integral $\int \mathrm{d}^d k / k^2$ is scaleless: it carries no dimensionful parameter to set its size, and in dimensional regularization every scaleless integral is defined to vanish (Eq. (D.99), where $A_0(0) = 0$ for exactly this reason). The seagull therefore contributes *nothing* to the divergent self-energy.

Caution

The vanishing of Eq. (27.54) is a feature of dimensional regularization, not a statement that the seagull is physically absent. In a cutoff scheme the seagull supplies a quadratically divergent $\propto \Lambda^2 \eta_{\mu\nu}$ piece whose role is to cancel the would-be quadratic divergence of the gluon loop, so that the surviving divergence is logarithmic and transverse. Dimensional regularization is engineered to make both of those quadratic pieces vanish identically — it sees no power divergences, only the logarithmic poles — which is precisely why the gluon and ghost loops we have isolated are already, on their own, only logarithmically divergent and combine to a transverse result with no help from the seagull. The physics is scheme-independent; the bookkeeping is cleanest in $d = 4 - 2\varepsilon$.

27.2.5 The ghost loop

The closed Faddeev–Popov loop of Fig. 27.2c is built from two ghost-gluon vertices Eq. (26.58) and two ghost propagators Eq. (26.47), and — being a closed loop of anticommuting fields — carries the mandatory factor (-1) of Theorem 26.13. The ghost line is oriented by ghost number; we route the ghost-number flow around the loop so that the segment from the left vertex to the right carries momentum $k + q$ and the return segment carries k. The ghost-gluon vertex Eq. (26.58), $iV^{abc}_{\mathrm{gh},\mu}(p) = -g f^{abc} p^\mu$, depends on the *outgoing* ghost momentum p and on the gluon's Lorentz index alone. At the left vertex the outgoing ghost momentum is $k+q$, giving the factor $\propto (k+q)_\mu$; at the right vertex it is k, giving $\propto k_\nu$. Assembling,

$$i\Pi^{(gh)ab}_{\mu\nu}(q) = (-1)\int \frac{\mathrm{d}^d k}{(2\pi)^d} \left(-g f^{eaf}(k+q)_\mu\right) \frac{i}{(k+q)^2} \left(-g f^{fbe} k_\nu\right) \frac{i}{k^2}, \quad (27.55)$$

where, following the index convention of Eq. (26.58) (first index the outgoing-ghost color, middle the gluon, last the incoming-ghost color), the left vertex carries f^{eaf} and the right f^{fbe}, with e, f the colors of the two ghost segments.

The color sign. The contraction $f^{eaf} f^{fbe}$ is where the ghost loop earns its keep, and it is worth doing slowly because a sign error here would destroy transversality. Using total antisymmetry and cyclicity of the structure constants, $f^{eaf} = -f^{aef}$ and $f^{fbe} = f^{bef}$ (a cyclic rotation), so

$$f^{eaf} f^{fbe} = (-f^{aef})(f^{bef}) = -f^{aef} f^{bef} = -C_A\, \delta^{ab}, \quad (27.56)$$

the adjoint Casimir of Eq. (B.102) entering with a *minus* sign. Collecting the explicit factors — the loop (-1), the two couplings $(-g)^2 = g^2$, the

two propagator i's giving $i^2 = -1$, and the color $-C_A\delta^{ab}$ — the prefactor is $(-1)(g^2)(-1)(-C_A\delta^{ab}) = -g^2 C_A \delta^{ab}$, so

$$i\Pi^{(gh)ab}_{\mu\nu}(q) = -\,g^2 C_A\,\delta^{ab}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{(k+q)_\mu\,k_\nu}{k^2(k+q)^2}. \tag{27.57}$$

Integration. Feynman-combine with Eq. (27.35) ($\Delta = -x(1-x)q^2$) and shift $k = \ell - xq$, so $(k+q)_\mu = \ell_\mu + (1-x)q_\mu$ and $k_\nu = \ell_\nu - xq_\nu$. The divergent numerator (dropping odd-in-ℓ terms) is

$$(k+q)_\mu k_\nu\big|_{\text{div}} = \ell_\mu\ell_\nu - x(1-x)\,q_\mu q_\nu. \tag{27.58}$$

The tensor term gives, via Eq. (27.34), $\ell_\mu\ell_\nu \to (i/16\pi^2)(\Delta/2\varepsilon)\eta_{\mu\nu} = -(i/16\pi^2)(x(1-x)q^2/2\varepsilon)\eta_{\mu\nu}$; the $q_\mu q_\nu$ term multiplies the scalar pole Eq. (27.32). Integrating over x, with $\int_0^1 x(1-x)\mathrm{d}x = \frac{1}{6}$,

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{(k+q)_\mu k_\nu}{k^2(k+q)^2}\bigg|_{\text{pole}} = \frac{i}{16\pi^2}\,\frac{1}{\varepsilon}\left[-\frac{1}{12}\,q^2\eta_{\mu\nu} - \frac{1}{6}\,q_\mu q_\nu\right]. \tag{27.59}$$

Restoring the prefactor $-g^2 C_A\delta^{ab}$,

$$\boxed{\,i\Pi^{(gh)ab}_{\mu\nu}(q)\big|_{\text{pole}} = \frac{i\,g^2 C_A\,\delta^{ab}}{16\pi^2}\,\frac{1}{\varepsilon}\left[\frac{1}{12}\,q^2\eta_{\mu\nu} + \frac{1}{6}\,q_\mu q_\nu\right].\,} \tag{27.60}$$

Like the gluon loop, the ghost loop is not transverse by itself — its two coefficients are $+\frac{1}{12}$ and $+\frac{1}{6}$, of the *same* sign, the opposite imbalance to Eq. (27.52). Neither diagram is a sensible object alone. Their sum is.

27.2.6 Assembly, transversality, and Z_3

We now add the four contributions. The seagull Eq. (27.54) vanishes; the gluon Eq. (27.52), ghost Eq. (27.60), and quark Eq. (27.41) loops remain. Group the gauge sector (gluon + ghost) first, because that is where transversality is on trial. Adding the two coefficient pairs,

$$q^2\eta_{\mu\nu}: \quad \frac{19}{12} + \frac{1}{12} = \frac{20}{12} = \frac{5}{3}, \tag{27.61}$$

$$q_\mu q_\nu: \quad -\frac{11}{6} + \frac{1}{6} = -\frac{10}{6} = -\frac{5}{3}. \tag{27.62}$$

The two numbers are now equal and opposite. The gauge sector is

$$i\Pi^{(g)ab}_{\mu\nu} + i\Pi^{(gh)ab}_{\mu\nu}\Big|_{\text{pole}} = \frac{i\,g^2 C_A\,\delta^{ab}}{16\pi^2}\,\frac{1}{\varepsilon}\,\frac{5}{3}\,(q^2\eta_{\mu\nu} - q_\mu q_\nu), \tag{27.63}$$

manifestly transverse, $q^\mu(q^2\eta_{\mu\nu} - q_\mu q_\nu) = q^2 q_\nu - q^2 q_\nu = 0$. The longitudinal remainder of the gluon loop flagged in the caution after Eq. (27.52), with

coefficient $\frac{19}{12} - \frac{11}{6} = -\frac{1}{4}$, has been cancelled exactly by the ghost loop's $\frac{1}{12} - (-\frac{1}{6}) = \frac{1}{4}$ — the two quarter-units annihilate. This is the explicit, one-loop face of Theorem 26.37: the Slavnov–Taylor identity, derived in Chapter 26 from BRST invariance, *predicts* that the ghost loop must carry precisely the longitudinal piece needed to make the total transverse, and the independent integrals Eqs. (27.52) and (27.60) obey the prediction to the digit. Had we forgotten the ghost, or mis-signed its color Eq. (27.56), transversality would fail and we would know a Feynman rule had been mishandled.

Adding the quark loop Eq. (27.41), which is separately transverse, the full one-loop pole is

$$i\Pi^{ab}_{\mu\nu}(q)\big|_{\text{pole}} = i\,\delta^{ab}(q^2\eta_{\mu\nu} - q_\mu q_\nu)\,\frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\left(\frac{5}{3}\,C_A - \frac{4}{3}\,T_F\,n_f\right), \tag{27.64}$$

which has exactly the transverse single-form-factor structure of Eq. (27.30). Reading off the scalar form factor,

$$\Pi(q^2)\big|_{\text{pole}} = \frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\left(\frac{5}{3}\,C_A - \frac{4}{3}\,T_F\,n_f\right). \tag{27.65}$$

Inserting this into the $\overline{\text{MS}}$ condition Eq. (27.31) gives the gluon field renormalization.

Theorem 27.3 (One-loop gluon field renormalization)**.** *In Feynman gauge $\xi = 1$ and the $\overline{MS}$ scheme, the one-loop gluon field-renormalization constant of Yang–Mills theory with gauge group* SU(N) *and n_f Dirac fermions in the fundamental representation is*

$$\boxed{Z_3 = 1 + \frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\left(\frac{5}{3}\,C_A - \frac{4}{3}\,T_F\,n_f\right),} \tag{27.66}$$

with $C_A = N$ Eq. (B.102) and $T_F = \frac{1}{2}$ Eq. (B.10). The gluonic part $\frac{5}{3}C_A$ is the sum of the gluon loop $(\frac{19}{12} - \frac{11}{6} + \frac{5}{3})$ reorganized as $\frac{19}{12} + \frac{1}{12}$ from gluon + ghost (the seagull contributing zero), and the matter part $-\frac{4}{3}T_F n_f$ is the quark loop, identical to the QED vacuum polarization Eq. (19.49) up to the color trace T_F and the flavor sum.

Worked Example 27.4 (Transversality as a check on the ghost: contracting with q^μ)**.** We continue Yang–Mills theory. The cleanest way to see that the ghost contribution is forced, not fitted, is to demand transversality of the divergent self-energy *before* knowing the final coefficient, and let it fix the ghost loop's relative weight. Write the gauge-sector pole, in units of $(ig^2C_A\delta^{ab}/16\pi^2)(1/\varepsilon)$, as the gluon loop Eq. (27.52) plus an unknown multiple λ of the bare ghost integral Eq. (27.59),

$$\left[\tfrac{19}{12} - \lambda\tfrac{1}{12}\right]q^2\eta_{\mu\nu} + \left[-\tfrac{11}{6} - \lambda\tfrac{1}{6}\right]q_\mu q_\nu, \tag{27.67}$$

where λ packages the unknown overall factor (statistics sign, color sign, and multiplicity) multiplying the ghost integral $\int (k+q)_\mu k_\nu / k^2 (k+q)^2$. Imposing $q^\mu (\cdots)_{\mu\nu} = 0$, i.e. that the two bracketed coefficients be equal and opposite, gives

$$\left(\tfrac{19}{12} - \tfrac{\lambda}{12}\right) = -\left(-\tfrac{11}{6} - \tfrac{\lambda}{6}\right) \implies \tfrac{19}{12} - \tfrac{\lambda}{12} = \tfrac{11}{6} + \tfrac{\lambda}{6} \implies \tfrac{19}{12} - \tfrac{22}{12} = \tfrac{\lambda}{12} + \tfrac{2\lambda}{12} \implies -\tfrac{3}{12} = \tfrac{3\lambda}{12}, \tag{27.68}$$

so $\lambda = -1$. The transversality requirement alone determines that the ghost integral must enter with overall coefficient -1 relative to the prefactor convention used — which is exactly the product of the (-1) closed-loop statistics sign Theorem 26.13 and the $-C_A$ color contraction Eq. (27.56) divided by the gluon loop's $-\frac{1}{2}C_A$ normalization, reproducing Eq. (27.60). With $\lambda = -1$ the surviving transverse coefficient is $\frac{19}{12} + \frac{1}{12} = \frac{5}{3}$, recovering Eq. (27.63). The wrong-statistics, scalar Faddeev–Popov field is thus not a fudge: its loop is the unique object whose inclusion makes the gluon self-energy consistent with the gauge symmetry, and the symmetry fixes its weight with no freedom left over.

Physical Insight: antiscreening lives in the longitudinal gluon

The QED vacuum polarization is transverse diagram by diagram, and its sign screens charge. The Yang–Mills quark loop does the same. What is new is the gauge sector, and its arithmetic is revealing: the gluon loop alone is dominated by a large, *non*-transverse longitudinal piece, the very piece a charged spin-one field generates by interacting with the background through its own three-point coupling. That longitudinal contribution is unphysical — it must be removed — and the ghost loop removes it, leaving behind the transverse $\frac{5}{3}C_A$. The magnitude and sign of that survivor are what overwhelm the quark loop's $-\frac{4}{3}T_F n_f$ for small enough n_f and flip the sign of the beta function. Heuristically, the non-abelian vacuum behaves like a paramagnetic medium: the gluon's spin couples to the chromomagnetic field and *anti*screens color charge, the opposite of the dielectric screening of a sea of charged pairs. The constant C_A, the adjoint Casimir, is the precise measure of how strongly the gauge field talks to itself, and it is the carrier of asymptotic freedom.

27.2.7 Gauge dependence and what comes next

Two honest caveats close the computation. First, Z_3 is gauge dependent. Repeating the gluon loop in a general R_ξ gauge — keeping the full propagator Eq. (26.44) rather than its $\xi = 1$ form — replaces the gluonic coefficient $\frac{5}{3}C_A$ by $(\frac{13}{6} - \frac{\xi}{2})C_A$, which indeed reduces to $\frac{13}{6} - \frac{1}{2} = \frac{5}{3}$ at $\xi = 1$. A field-renormalization constant is not by itself a physical quantity, so this ξ-dependence is no cause for alarm; what must be gauge independent is the beta function, and the ξ-dependence of Z_3 will

be cancelled by that of the vertex and matter constants when they are combined into Z_g. Second, the pole Eq. (27.66) is all that $\overline{\text{MS}}$ retains; the finite parts of the four diagrams, which we did not track, contribute only the scheme-dependent constant in the renormalized $\Pi(q^2)$ and drop out of the beta-function coefficient computed from the simple pole via Eq. (24.104).

With Z_3 in hand we have the first and most distinctive of the renormalization constants that feed Z_g, and we have seen the structural signature that will recur: the ghost-restored transversality singled out C_A as the carrier of the gluonic contribution, the same C_A that will dominate the beta function. What remains is the quark sector. In the next section we compute the quark field renormalization Z_2 and the quark-gluon vertex renormalization Z_1, and confront the feature with no abelian analog: unlike QED's $Z_1 = Z_2$, here $Z_1 \neq Z_2$, their difference a pure C_A piece born of the gluon's self-coupling. Only with Z_1, Z_2, and Z_3 assembled can the universal coupling renormalization Z_g — and with it the sign of the beta function — be read off.

27.3 Quark Self-Energy, the Vertex, and the Failure of $Z_1 = Z_2$

The gluon vacuum polarization of Section 27.2 delivered the first of the three renormalization constants the coupling needs: the gluon field renormalization Z_3, read off (Eq. (27.66)) from the transverse pole of the combined gluon, ghost, and quark loops. It is the load-bearing non-abelian quantity, and it is already enough to feel the antiscreening pull of the gluon self-interaction. But Z_3 alone does not fix the coupling. The bare and renormalized couplings are tied by $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ (the quark–gluon route through Eq. (26.179)), and so to assemble Z_g, and from it the beta function, we still owe two constants: the quark field renormalization Z_2 and the quark–gluon vertex renormalization Z_1. This section computes both, in Feynman gauge $\xi = 1$ and $\overline{\text{MS}}$ with $d = 4 - 2\varepsilon$, exactly as the chapter fixed at the outset.

The computation is where the carried QED technology earns its keep. The quark self-energy is the electron self-energy of Chapter 19 wearing a color factor; the abelian-like piece of the vertex is the QED vertex of Chapter 19 wearing the same color machinery. We do not rebuild that apparatus—we import it, attach the group theory of Chapter B, and watch what survives. What survives is the headline of the chapter: in QED the Ward–Takahashi identity (Theorem 19.13) forces $Z_1 = Z_2$, the equality that makes the renormalized charge gauge-independent and pins $F_1(0) = 1$. Here that equality *fails*, and it fails by a precise amount—a pure adjoint-Casimir piece $\propto C_A$ sourced by the gluon's coupling to itself. That failure is not an inconsistency; the Slavnov–Taylor identities of Section 26.7 permit it while keeping the theory renormalizable. And it is, algebraically, the origin of asymptotic freedom.

Notation Watch

In Chapter 26 the symbol Z_1 was assigned to the *three-gluon* vertex, and the quark–gluon vertex constant was written Z_1^F (Eq. (26.179)). Throughout the present chapter we instead reserve Z_1 for the *quark–gluon* vertex, $Z_1 \equiv Z_1^F$, because it is the route by which we extract Z_g and because it is the constant that the QED Ward identity of Chapter 19 compares directly to Z_2: the section title "the failure of $Z_1 = Z_2$" is exactly that comparison. The Slavnov–Taylor identities (Eq. (26.179)) guarantee that the three-gluon, four-gluon, ghost-gluon, and quark-gluon routes all return the same Z_g, so no generality is lost by choosing the quark-gluon route here. The three-gluon vertex constant of Chapter 26 plays no further role in this chapter.

27.3.1 The quark self-energy and Z_2

At one loop the quark's 1PI two-point function receives a single correction: the quark emits a gluon and reabsorbs it (Fig. 27.3, left). The topology is identical to the one-loop electron self-energy of Chapter 19; the only differences are that the photon line is a gluon line and that each quark–gluon vertex carries a generator T^a instead of a charge. We take the quarks massless, which is legitimate for extracting the ultraviolet divergence: the $1/\varepsilon$ pole of a logarithmically divergent self-energy is a local polynomial in the external momentum and is mass-independent, so the mass only decorates the finite part.

Reading the diagram with the rules of Section 26.3—the quark–gluon vertex $ig\gamma^\mu T^a$ of Eq. (26.60), the massless quark propagator $i\not{k}/(k^2+i\varepsilon)$, and the Feynman-gauge gluon propagator $-i\eta_{\mu\nu}\delta^{ab}/(k^2+i\varepsilon)$ of Eq. (26.44)—the amputated self-energy is

$$-i\Sigma(p) = \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,(ig\gamma^\mu T^a)\,\frac{i\not{k}}{k^2}\,(ig\gamma^\nu T^b)\,\frac{-i\,\eta_{\mu\nu}\,\delta^{ab}}{(p-k)^2}. \tag{27.69}$$

Two structures factor out cleanly. The color matrices collapse by the fundamental Casimir of Eq. (B.11),

$$T^a T^b\,\delta^{ab} = T^a T^a = C_F\,\mathbf{1}, \qquad C_F = \frac{N^2-1}{2N}, \tag{27.70}$$

a single number times the identity in color space. The Dirac structure collapses by the d-dimensional contraction identity $\gamma^\mu\gamma^\alpha\gamma_\mu = (2-d)\gamma^\alpha$ (Eq. (C.188)), applied to $\gamma^\mu\not{k}\gamma_\mu = (2-d)\not{k}$. Collecting the explicit factors of i—three from the vertices and propagator numerators, $(ig)(i)(ig) = -ig^2$, against one from the gluon denominator, $(-i)$, for a net $-g^2$—leaves

$$-i\Sigma(p) = -g^2 C_F\,(2-d)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{\not{k}}{k^2\,(p-k)^2}. \tag{27.71}$$

This is the electron self-energy of Eq. (19.70) term for term, with the lone substitution $e^2 \to g^2 C_F$: the momentum integral, the Feynman parametrization, and the Dirac algebra are the QED computation unchanged, and the color factor C_F is the entire non-abelian content of this graph. We carry the integral anyway, both because the chapter must be reproducible from itself and because the same steps recur in the vertex.

Combine the denominators with one Feynman parameter, $\frac{1}{AB} = \int_0^1 \mathrm{d}x\,[xA + (1-x)B]^{-2}$ with $A = (p-k)^2$, $B = k^2$:

$$x(p-k)^2 + (1-x)k^2 = k^2 - 2x\,p\cdot k + xp^2 = (k - xp)^2 - \Delta, \qquad \Delta \equiv -x(1-x)\,p^2. \tag{27.72}$$

Shift $\ell = k - xp$. The numerator $\not{k} = \not{\ell} + x\not{p}$ loses its $\not{\ell}$ part to the symmetric integration, leaving $x\not{p}$. With the master one-loop integral of Theorem D.22,

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta)^2} = \frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(2 - \tfrac{d}{2}\right)\Delta^{\,d/2-2} = \frac{i}{16\pi^2}\left[\frac{1}{\varepsilon} + \text{finite}\right], \tag{27.73}$$

where $2 - d/2 = \varepsilon$ supplies the pole $\Gamma(\varepsilon) = 1/\varepsilon + O(1)$, the self-energy becomes

$$-i\Sigma(p) = -g^2 C_F\,(2-d)\,\not{p}\int_0^1 \mathrm{d}x\,x\;\frac{i}{16\pi^2}\,\frac{1}{\varepsilon} + \text{finite}. \tag{27.74}$$

The parameter integral is $\int_0^1 x\,\mathrm{d}x = \tfrac{1}{2}$, and at the pole $(2-d) = -2 + 2\varepsilon \to -2$. Hence $-i\Sigma = (-g^2 C_F)(-2)\,\not{p}\,(\tfrac{1}{2})\,\frac{i}{16\pi^2\varepsilon} = +\,i\,\frac{g^2 C_F}{16\pi^2}\frac{1}{\varepsilon}\not{p}$, i.e.

$$\Sigma(p)\big|_{\text{div}} = -\,\frac{g^2 C_F}{16\pi^2}\,\frac{1}{\varepsilon}\,\not{p}. \tag{27.75}$$

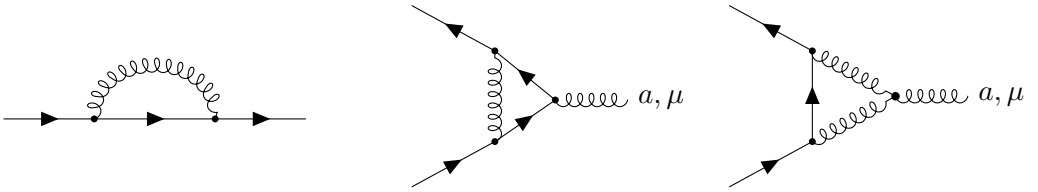
The renormalization is the standard one. Writing the bare field as $\psi_0 = Z_2^{1/2}\psi$ (Eq. (26.176)) inserts a counterterm $(Z_2 - 1)\not{p} \equiv \delta_2\,\not{p}$ into the inverse propagator, which at one loop reads $\not{p} - \Sigma(p) + \delta_2\,\not{p}$. The $\overline{\text{MS}}$ prescription fixes δ_2 to cancel the pole of Σ: from Eq. (27.75), $\not{p}[\,1 + \frac{g^2 C_F}{16\pi^2\varepsilon} + \delta_2\,]$ is finite only if $\delta_2 = -\,g^2 C_F/(16\pi^2\varepsilon)$. Therefore

$$\boxed{\,Z_2 = 1 - \frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\,C_F \qquad \text{(Feynman gauge).}\,} \tag{27.76}$$

This is the QED electron Z_2 of Chapter 19 with $e^2 \to g^2 C_F$, precisely as the factored color Eq. (27.70) promised. Like its QED ancestor, Z_2 is gauge dependent: the coefficient quoted is the $\xi = 1$ value, and a different ξ would change it (and, in the end, Z_1 in lockstep, so that Z_g does not move). We return to that gauge independence in Section 27.4.

27.3.2 The quark–gluon vertex: two graphs

The vertex tells a richer story, because the external gluon can be radiated in two inequivalent ways (Fig. 27.3, center and right). In the first, the correcting gluon



(a) Z_2 (b) $C_F - \frac{1}{2}C_A$ (c) $\frac{3}{2}C_A$

Figure 27.3: The one-loop graphs of this section. (a) The quark self-energy giving Z_2, identical to the QED electron self-energy of Chapter 19 up to the color factor C_F. (b) The abelian-like vertex correction: an internal gluon bridges the two quark legs, the external gluon (a, μ) attaching between. Its kinematics are the QED vertex of Eq. (19.104); its color is $T^b T^a T^b = (C_F - \frac{1}{2}C_A)T^a$. (c) The genuinely non-abelian vertex: the external gluon is emitted through the three-gluon vertex Eq. (26.53); its color reduces to $\frac{i}{2}C_A T^a$ and its divergent kinematics deliver an effective $\frac{3}{2}C_A$. The labels beneath each vertex graph record the color factor it contributes to Z_1.

is exchanged *across* the vertex between the two quark legs, and the external gluon attaches to the quark line in between: this is the QED vertex graph, the only one an abelian theory possesses. In the second, the external gluon is emitted from an internal gluon line through the three-gluon vertex Eq. (26.53): this graph exists only because the gluon is charged, and it has no QED counterpart. The vertex renormalization is the sum of the two.

The color factors

Before any integral, isolate the group theory, because it is the group theory that separates abelian from non-abelian. We do it once and reuse it.

Worked Example 27.5 (The two vertex color factors)**.** *Graph (b), the abelian-like vertex.* Reading the quark line from the incoming to the outgoing leg, the generators appear in the order T^b (where the internal gluon of color b is emitted), T^a (the external gluon), T^b (where the internal gluon is reabsorbed, the index b summed). The color factor is therefore $T^b T^a T^b$. Move the left T^b through T^a with the algebra $[T^a, T^b] = i f^{abc} T^c$:

$$T^b T^a T^b = T^b T^b T^a + T^b\,[\,T^a, T^b\,] = C_F\,T^a + i f^{abc}\,T^b T^c, \tag{27.77}$$

using $T^b T^b = C_F \mathbf{1}$ (Eq. (B.11)) in the first term. The second term antisymmetrizes against f^{abc}: $i f^{abc} T^b T^c = \frac{i}{2} f^{abc}[T^b, T^c] = \frac{i}{2} f^{abc}\,(i f^{bcd} T^d) = -\frac{1}{2} f^{abc} f^{bcd} T^d = -\frac{1}{2} C_A\,T^a$, where the last step is the Killing-form contraction $f^{abc} f^{dbc} = C_A \delta^{ad}$ of Eq. (B.23). Hence

$$T^b T^a T^b = \left(C_F - \tfrac{1}{2}C_A\right) T^a. \tag{27.78}$$

The first term is the abelian answer one would get from a $U(1)$ charge; the $-\frac{1}{2}C_A$ is the irreducibly non-abelian residue of the *same* graph, born entirely of the noncommutativity of the generators.

Graph (c), the triple-gluon vertex. Now the two internal gluons, colors d and e, attach to the quark line, while the external gluon a meets them at the three-gluon vertex carrying f^{ade}. Reading the quark line outgoing-to-incoming gives T^eT^d, and the color factor is $f^{ade}T^eT^d$. Antisymmetry in the dummy pair (d,e) lets us replace the product by half the commutator:

$$f^{ade}\,T^eT^d = \tfrac{1}{2}f^{ade}\,[\,T^e,T^d\,] = \tfrac{1}{2}f^{ade}\,(if^{edc}T^c) = \tfrac{i}{2}\,f^{ade}f^{edc}\,T^c = -\tfrac{i}{2}\,C_A\,T^a, \tag{27.79}$$

where $f^{ade}f^{edc} = -f^{ade}f^{cde} = -C_A\delta^{ac}$ by Eq. (B.23) after one relabelling. The single power of C_A, the adjoint Casimir, is the signature that this graph could only exist in a theory whose gauge bosons are themselves charged.

Graph (b): the abelian-like vertex

With the color stripped off, what remains of graph (b) is the QED one-loop vertex correction in Feynman gauge, computed in Chapter 19 and reduced there to the divergent structure

$$\delta\Gamma^{\mu}_{\rm QED}\big|_{\rm div} = \frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon}\,\gamma^{\mu} \quad\Longleftrightarrow\quad \delta_1^{\rm QED} = -\,\frac{e^2}{16\pi^2}\,\frac{1}{\varepsilon} \tag{27.80}$$

(Eq. (19.104); the $1/\varepsilon$ pole lands entirely on the tree Dirac structure γ^{μ}, the algebraic point made explicit in the master vertex reduction of Theorem D.22). The QCD graph is Eq. (27.80) with $e^2 \to g^2$ and the color factor Eq. (27.78) attached. Its contribution to the vertex counterterm is

$$\delta_1^{(b)} = -\,\frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\,\Big(C_F - \tfrac{1}{2}C_A\Big). \tag{27.81}$$

This is the place to make the QED Ward content visible, because it is precisely what will *not* carry over in full. Strip QCD of its non-abelian core for a moment—set $C_A \to 0$, the abelian limit in which the generators commute and graph (c) does not exist. Then graph (b) alone renormalizes the vertex, with $\delta_1^{(b)} \to -g^2C_F/(16\pi^2\varepsilon)$, while the self-energy gave $\delta_2 = -g^2C_F/(16\pi^2\varepsilon)$ from Eq. (27.76). They are equal:

$$Z_1^{\rm ab} = Z_2, \tag{27.82}$$

the exact non-abelian transcription of the QED Ward–Takahashi identity $Z_1 = Z_2$ (Theorem 19.13), which holds graph-by-graph for the abelian-like contribution because that graph *is* the QED vertex and the cancellation of its divergence against the self-energy is the Ward identity itself. The C_F pieces of Z_1 and Z_2 are locked together by the same mechanism that locks them in QED. What

breaks the lock is everything proportional to C_A: the $-\frac{1}{2}C_A$ already sitting inside Eq. (27.81)—the non-abelian residue of the abelian-like graph—and the entirety of graph (c), to which we now turn.

Graph (c): the non-abelian vertex

Graph (c) has no QED ancestor and must be computed from scratch. Label the incoming quark p, outgoing quark p', external gluon (a,μ) with $q = p' - p$. The external gluon meets two internal gluons at the three-gluon vertex; the internal gluons attach to the quark line at two quark–gluon vertices straddling the loop quark of momentum k. Assembling the rules—two quark–gluon vertices $ig\gamma T$, the internal quark propagator $i\not{k}/k^2$, two Feynman-gauge gluon propagators, and the three-gluon vertex $gf^{ade}V^{\mu\rho'\sigma'}$ of Eq. (26.53) (which carries no explicit i)—the amputated correction is

$$i\,\Lambda^{\mu,a}_{(c)} = ig^3\,(f^{ade}T^eT^d)\int\frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{\gamma^{\sigma'}\not{k}\,\gamma^{\rho'}}{k^2\,(p-k)^2\,(p'-k)^2}\,V^{\mu}{}_{\rho'\sigma'}(q,\,p{-}k,\,k{-}p'), \tag{27.83}$$

the prefactor ig^3 being the product of all explicit i's and couplings, and the color factor reducing by Eq. (27.79) to $-\frac{i}{2}C_AT^a$. The ultraviolet divergence is logarithmic (numerator degree two from $\gamma\not{k}\gamma$ times the linear vertex V; denominator degree six), so we may extract its $1/\varepsilon$ pole at the symmetric kinematic point $p' = p$, i.e. $q = 0$: the pole is a local γ^μ term whose coefficient is fixed by the large-loop-momentum region and is independent of the external momenta that distinguish p from p'.

At $q = 0$ both internal gluons carry the same momentum $r \equiv p - k$, the two gluon denominators become $(r^2)^2$, and the three-gluon vertex Eq. (26.53) with incoming momenta $(q,\ell_1,\ell_2) = (0,\,r,\,-r)$ collapses to

$$V^{\mu\rho'\sigma'}(0,r,-r) = -\,\eta^{\mu\rho'}r^{\sigma'} + 2\,r^\mu\eta^{\rho'\sigma'} - \eta^{\sigma'\mu}r^{\rho'}. \tag{27.84}$$

Contracting against $\gamma^{\sigma'}\not{k}\,\gamma^{\rho'}$ and using $\gamma_\alpha\not{k}\gamma^\alpha = (2-d)\not{k}$ (Eq. (C.188)) on the middle term, the Dirac numerator is

$$N^\mu = -\not{r}\,\not{k}\,\gamma^\mu + 2(2-d)\,r^\mu\not{k} - \gamma^\mu\not{k}\not{r}, \qquad r = p - k. \tag{27.85}$$

Combine the three denominators with the Feynman identity $\frac{1}{AB^2} = \int_0^1\mathrm{d}x\,\frac{2(1-x)}{[xA+(1-x)B]^3}$, $A = k^2$, $B = (p-k)^2$; completing the square gives $xA + (1-x)B = \ell^2 - \Delta$ with $\ell = k - (1-x)p$ and $\Delta = -x(1-x)p^2$. The logarithmic divergence comes only from the part of N^μ quadratic in ℓ. Writing $k = \ell + (1-x)p$ and $r = xp - \ell$ and keeping the ℓ^2 terms,

$$-\not{r}\,\not{k}\,\gamma^\mu \to (\not{\ell}\not{\ell})\,\gamma^\mu = \ell^2\gamma^\mu, \qquad -\gamma^\mu\not{k}\not{r} \to \gamma^\mu(\not{\ell}\not{\ell}) = \ell^2\gamma^\mu,$$
$$2(2-d)\,r^\mu\not{k} \to -2(2-d)\,\ell^\mu\not{\ell} \to -2(2-d)\,\frac{\ell^2}{d}\,\gamma^\mu, \tag{27.86}$$

where the last replacement is the symmetric average $\ell^\mu \ell_\nu \to (\ell^2/d)\delta^\mu_\nu$. Summing,

$$N^\mu\big|_{\ell^2} = \ell^2 \gamma^\mu \Big[\, 2 - \frac{2(2-d)}{d} \,\Big] = \ell^2 \gamma^\mu \, \frac{4(d-1)}{d} \xrightarrow{d\to 4} 3\,\ell^2 \gamma^\mu. \tag{27.87}$$

The remaining ℓ-integral is the rank-two log-divergent master, $\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{\ell^2}{(\ell^2 - \Delta)^3} = \frac{i}{(4\pi)^{d/2}} \, \frac{d}{2} \, \frac{\Gamma(2 - d/2)}{\Gamma(3)} \, \Delta^{d/2 - 2} \to \frac{i}{16\pi^2} \frac{1}{\varepsilon}$ (Theorem D.22), and the parameter integral is $\int_0^1 2(1-x)\,\mathrm{d}x = 1$. Therefore the kinematic integral in Eq. (27.83) has divergent part $3\,\gamma^\mu \cdot \frac{i}{16\pi^2 \varepsilon}$, and

$$i\,\Lambda^{\mu,a}_{(c)}\big|_{\mathrm{div}} = i g^3 \Big(-\tfrac{i}{2} C_A T^a \Big)\, 3\,\gamma^\mu \, \frac{i}{16\pi^2} \frac{1}{\varepsilon} = \frac{3i}{2} \, \frac{g^3 C_A}{16\pi^2} \, \frac{1}{\varepsilon} \, T^a \gamma^\mu. \tag{27.88}$$

Measured against the tree vertex $g\gamma^\mu T^a$, this is a *relative* divergence $+\frac{3}{2}\,(g^2 C_A/16\pi^2)(1/\varepsilon)$, the effective color factor $\frac{3}{2} C_A$ recorded under graph (c) in Fig. 27.3. Its sign matches that of the abelian-like graph, so the two graphs add rather than cancel.

Assembling Z_1

The full one-loop vertex correction, relative to tree, is the sum of the abelian-like divergence Eq. (27.81) (relative part $+\frac{g^2}{16\pi^2 \varepsilon}(C_F - \frac{1}{2} C_A)$) and the non-abelian divergence Eq. (27.88) (relative part $+\frac{g^2}{16\pi^2 \varepsilon} \cdot \frac{3}{2} C_A$):

$$\frac{\Lambda^\mu_{\text{1-loop}}}{g\gamma^\mu T^a}\bigg|_{\mathrm{div}} = \frac{g^2}{16\pi^2} \frac{1}{\varepsilon} \Big[(C_F - \tfrac{1}{2} C_A) + \tfrac{3}{2} C_A \Big] = \frac{g^2}{16\pi^2} \frac{1}{\varepsilon} (C_F + C_A). \tag{27.89}$$

The vertex counterterm $\delta_1 = Z_1 - 1$ must cancel this in $\overline{\mathrm{MS}}$, so $\delta_1 = -\frac{g^2}{16\pi^2} \frac{1}{\varepsilon} (C_F + C_A)$, that is,

$$\boxed{\, Z_1 = 1 - \frac{g^2}{16\pi^2} \, \frac{1}{\varepsilon} \, (C_F + C_A) \qquad \text{(Feynman gauge).} \,} \tag{27.90}$$

The C_F is the QED inheritance; the C_A is the sum of the two non-abelian contributions, $-\frac{1}{2} C_A$ from the abelian-like graph's noncommuting color and $+\frac{3}{2} C_A$ from the triple-gluon graph.

27.3.3 The failure of $Z_1 = Z_2$ and what it buys

We can now state the result the section was built to reach.

Theorem 27.6 (Quark-sector renormalization constants and the broken Ward equality)**.** *At one loop in Feynman gauge ($\xi = 1$), in $\overline{MS}$ with $d = 4 - 2\varepsilon$, the quark field and quark–gluon vertex renormalization constants are*

$$Z_2 = 1 - \frac{g^2}{16\pi^2} \frac{1}{\varepsilon} \, C_F, \qquad Z_1 = 1 - \frac{g^2}{16\pi^2} \frac{1}{\varepsilon} \, (C_F + C_A), \tag{27.91}$$

and consequently

$$\frac{Z_1}{Z_2} = 1 - \frac{g^2}{16\pi^2}\frac{1}{\varepsilon}\,C_A + O(g^4). \tag{27.92}$$

Unlike QED, where $Z_1 = Z_2$ to all orders, here $Z_1 \neq Z_2$; the difference is a pure adjoint-Casimir term $\propto C_A$, the fundamental Casimir C_F having cancelled in the ratio.

Proof. The constants are Eq. (27.76) and Eq. (27.90). Their ratio, to one-loop order, is obtained by expanding $Z_1 Z_2^{-1} = (1 - aC_F - aC_A)(1 + aC_F) + O(a^2) = 1 - aC_A + O(a^2)$ with $a \equiv g^2/(16\pi^2\varepsilon)$; the C_F terms cancel linearly, leaving Eq. (27.92). Since $C_A = N > 0$ for SU(N), the ratio differs from unity. □

Three readings of Theorem 27.6 matter for what follows.

First, the cancellation of C_F is the Ward identity, kept exactly where QED keeps it. The fundamental Casimir C_F measures how the quark couples to the gauge field; it is the analogue of the electron's charge-squared. That it cancels in Z_1/Z_2 is the statement Eq. (27.82), $Z_1^{\mathrm{ab}} = Z_2$, promoted from the abelian limit to the full theory: the quark's own charge does not renormalize the coupling, exactly as the electron's charge does not renormalize e through the vertex/self-energy combination in QED. If the only thing in the theory were quarks coupled to a gauge field in the way an abelian charge couples, Z_1 would equal Z_2 and there would be no contribution to Z_g from this pair at all.

Second, the surviving C_A is the gauge field coupling to itself. Every factor of C_A in Eq. (27.92) is traceable to a structure constant: the $-\frac{1}{2}C_A$ to the commutator hidden inside the abelian-like graph's color Eq. (27.78), the $+\frac{3}{2}C_A$ to the explicit three-gluon vertex of graph (c). Both are aspects of the same fact—that the gluon carries color and therefore interacts with itself, the defining feature that Chapter 25 built into $F^a_{\mu\nu}$ through $gf^{abc}A^bA^c$ and Chapter 26 promoted to the three- and four-gluon vertices. The residue $Z_1/Z_2 = 1 - (g^2/16\pi^2)(1/\varepsilon)C_A$ is the vertex sector's verdict on that self-coupling, and it is this C_A, combined with the gluon's Z_3 from Section 27.2, that will tip Z_g toward antiscreening.

Third, $Z_1 \neq Z_2$ is permitted, not pathological. In QED the equality $Z_1 = Z_2$ is mandatory: it is forced by the Ward–Takahashi identity (Theorem 19.13), itself a consequence of the $U(1)$ current conservation, and its violation would signal an inconsistency—a charge that runs differently in the vertex than in the propagator, spoiling gauge invariance. In a non-abelian theory the corresponding constraints are the Slavnov–Taylor identities of Section 26.7, the BRST-exact descendants of gauge invariance. They do *not* demand $Z_1 = Z_2$. What they demand is weaker and exactly right: that a *single* coupling renormalization Z_g reproduce all four vertices at once, $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2} = \widetilde{Z}_1 \widetilde{Z}_3^{-1} Z_3^{-1/2} = \cdots$ (Eq. (26.179)). The individual constants are free to differ from one another; only their gauge-invariant combination Z_g is pinned. The inequality $Z_1 \neq Z_2$ is the room the Slavnov–Taylor identities leave open—room that QED's narrower

Ward identity closes—and the non-abelian theory uses precisely that room to be asymptotically free.

Physical Insight

In QED the relation $Z_1 = Z_2$ guarantees that the vertex and self-energy corrections to the physical charge cancel: charge renormalization comes *only* from vacuum polarization Z_3, and vacuum polarization screens, so α grows in the ultraviolet. A non-abelian theory breaks this exclusivity twice over. The vertex/self-energy pair no longer cancels—$Z_1/Z_2 \neq 1$ contributes a $-C_A$—and the gluon's vacuum polarization Z_3 itself contains gluon and ghost loops absent in QED. Both new pieces carry C_A, and both push the coupling the *opposite* way from QED's quark screening. The single equation $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ is where these effects meet; the next section adds them up, and the C_A pieces win.

With Z_2 from Eq. (27.76), Z_1 from Eq. (27.90), and Z_3 from Eq. (27.66) of the previous section, all three ingredients of $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ are in hand. The next section assembles them into a single coupling renormalization, reads the one-loop beta function off its simple pole through Eq. (24.104), and confronts the gauge-dependent constituents computed here with a gauge-independent recomputation by the background-field method—the moment the sign of b_0, and with it the fate of the strong coupling, is finally decided.

27.4 The Yang–Mills Beta Function

The three logarithmically divergent amplitudes promised by the checklist of Section 27.1.4 are now in hand. The gluon self-energy of the previous two sections fixed the gluon field renormalization Z_3 (Eq. (27.66)); the quark self-energy and the quark–gluon vertex fixed the quark field renormalization Z_2 and the vertex renormalization Z_1 (Eq. (27.91)), and exhibited the decisively non-abelian fact that $Z_1 \neq Z_2$, their difference a pure adjoint-Casimir term. Nothing remains to be computed. What remains is to *assemble*: to feed these three constants into the structural identity $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ of Eq. (27.9), read the one-loop coupling renormalization Z_g off the product, and turn the crank of the master formula Eq. (27.20) to extract the beta-function coefficient b_0. This is the arithmetic that decides the sign, and with it the entire ultraviolet character of the theory. We carry it out in full, every coefficient visible, and arrive at the carried Yang–Mills theory number. We then do something the bookkeeping alone cannot do: we recompute b_0 a second way, by the background-field method, in which the gauge invariance of the answer is manifest from the outset—a single renormalization constant in place of three—and watch the same number emerge, confirming that the gauge-dependent constituents Z_1, Z_2, Z_3 (each computed in

Feynman gauge) have assembled into a gauge-independent physical quantity, exactly as Theorem 24.12 guarantees they must.

27.4.1 Assembling Z_g from the three constants

Collect the inputs. In Feynman gauge ($\xi = 1$), in $\overline{\text{MS}}$ with $d = 4 - 2\varepsilon$ and scale μ, the three one-loop renormalization constants are, in the residue notation $Z_i = 1 + (g^2/16\pi^2)(r_i/\varepsilon)$ of Eq. (27.21),

$$Z_3 = 1 + \frac{g^2}{16\pi^2}\frac{1}{\varepsilon}\Big(\tfrac{5}{3}\,C_A - \tfrac{4}{3}\,T_F n_f\Big), \qquad r_3 = \tfrac{5}{3}\,C_A - \tfrac{4}{3}\,T_F n_f, \tag{27.93}$$

$$Z_2 = 1 - \frac{g^2}{16\pi^2}\frac{1}{\varepsilon}\,C_F, \qquad r_2 = -\,C_F, \tag{27.94}$$

$$Z_1 = 1 - \frac{g^2}{16\pi^2}\frac{1}{\varepsilon}\,(C_F + C_A), \qquad r_1 = -(C_F + C_A), \tag{27.95}$$

imported respectively from Eq. (27.66), Eq. (27.76), and Eq. (27.90). The gluon residue r_3 carries two physically distinct pieces: the antiscreening $+\frac{5}{3}C_A$ from the gluon and ghost loops, and the screening $-\frac{4}{3}T_F n_f$ from the n_f quark loops, the latter the exact non-abelian echo of the QED electron loop. The quark-sector residues r_1, r_2 each carry the fundamental Casimir C_F of Eq. (B.11); r_1 carries in addition the adjoint Casimir C_A of Theorem B.28, the fingerprint of the gluon self-coupling that breaks the abelian equality $Z_1 = Z_2$.

The coupling renormalization is the product Eq. (27.9), $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$. At one loop it is cleanest to work with the residue of $\ln Z_g$, which by Eq. (27.23) is linear in the constituent residues,

$$r \;=\; r_1 - r_2 - \tfrac{1}{2}\,r_3, \tag{27.96}$$

r being the dimensionless one-loop residue of $\ln Z_g$ in the units of Eq. (27.18). Substituting Eqs. (27.93) to (27.95),

$$\begin{aligned} r &= \underbrace{-(C_F + C_A)}_{r_1} \;-\; \underbrace{(-\,C_F)}_{r_2} \;-\; \tfrac{1}{2}\,\underbrace{\Big(\tfrac{5}{3}C_A - \tfrac{4}{3}T_F n_f\Big)}_{r_3} \\ &= -C_F - C_A + C_F - \tfrac{5}{6}\,C_A + \tfrac{2}{3}\,T_F n_f. \end{aligned} \tag{27.97}$$

The two underlined occurrences of C_F cancel: the $-C_F$ inside r_1 against the $+C_F$ coming from $-r_2 = +C_F$. This cancellation is not arithmetic luck. It is the non-abelian Ward–Takahashi identity at work: the fundamental Casimir C_F measures the quark's own charge, and that charge is forbidden from renormalizing the coupling through the vertex–self-energy combination, exactly as the electron's charge is forbidden from renormalizing e in QED, where the same cancellation is total and leaves $Z_1 = Z_2$. Here the cancellation removes only C_F; the adjoint

piece C_A, which r_1 carries but r_2 does not, survives. What is left is built purely from C_A and from the quark Dynkin index T_F of Eq. (B.2):

$$r \;=\; -C_A - \tfrac{5}{6}\,C_A + \tfrac{2}{3}\,T_F n_f \;=\; -\Big(\tfrac{11}{6}\,C_A - \tfrac{2}{3}\,T_F n_f\Big). \tag{27.98}$$

The coefficient $\frac{11}{6}C_A$ is assembled from three separate sources: a $-C_A$ from the vertex-sector ratio $Z_1/Z_2 = 1-(g^2/16\pi^2)(C_A/\varepsilon)$ of Eq. (27.92), and a $-\frac{5}{6}C_A$ from the gauge part $-\frac{1}{2}\cdot\frac{5}{3}C_A$ of the gluon self-energy. Their sum, $1+\frac{5}{6}=\frac{11}{6}$, is the famous numerator, here resolved into its vertex and propagator contributions. Reassembling $Z_g = 1+(g^2/16\pi^2)(r/\varepsilon)$ gives the coupling renormalization in closed form,

$$\boxed{\,Z_g = 1 - \frac{g^2}{16\pi^2}\,\frac{1}{\varepsilon}\Big(\tfrac{11}{6}\,C_A - \tfrac{2}{3}\,T_F n_f\Big) + O(g^4).\,} \tag{27.99}$$

This is the single number on which the chapter turns. Its simple pole is *negative* for QCD, where $\frac{11}{6}C_A$ dominates $\frac{2}{3}T_F n_f$, and a negative residue of Z_g is, as the next subsection makes precise, exactly the condition for asymptotic freedom.

27.4.2 The one-loop beta function and the coefficient b_0

The master formula Eq. (27.20) converts the residue of $\ln Z_g$ directly into the beta-function coefficient: $b_0 = -2r$. With r from Eq. (27.98),

$$b_0 = -2r = -2\cdot\Big[-\Big(\tfrac{11}{6}C_A - \tfrac{2}{3}T_F n_f\Big)\Big] = \tfrac{11}{3}\,C_A - \tfrac{4}{3}\,T_F n_f. \tag{27.100}$$

Equivalently, in the field-and-vertex form $b_0 = -2r_1 + 2r_2 + r_3$ of Eq. (27.22), $b_0 = 2(C_F+C_A) - 2C_F + (\frac{5}{3}C_A - \frac{4}{3}T_F n_f) = (2+\frac{5}{3})C_A - \frac{4}{3}T_F n_f = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$, the same. We record the result as the central theorem of the chapter.

Theorem 27.7 (One-loop Yang–Mills beta function)**.** *For* SU(N) *Yang–Mills theory coupled to n_f Dirac fermions in the fundamental representation, with coupling g, adjoint Casimir C_A, and fundamental Dynkin index T_F, the one-loop beta function in $\overline{MS}$ is*

$$\boxed{\,\beta_g \equiv \mu\,\frac{dg}{d\mu} = -\frac{g^3}{16\pi^2}\,b_0 + O(g^5), \qquad b_0 = \frac{11}{3}\,C_A - \frac{4}{3}\,T_F\,n_f.\,} \tag{27.101}$$

The coefficient b_0 is independent of the gauge parameter ξ and of the renormalization scheme at this order, as Theorem 24.12 requires.

Proof. The four-dimensional beta function is $\beta^{(4)}(g) = g^2\zeta_1'(g)$ by Eq. (27.17), where $\zeta_1 = (g^2/16\pi^2)\,r$ is the simple-pole residue of $\ln Z_g$. From Eq. (27.19) this is $\beta^{(4)} = (2r/16\pi^2)g^3$, and the matching $\beta_g = -(b_0/16\pi^2)g^3$ of Eq. (27.20) gives

$b_0 = -2r$. Inserting $r = -(\frac{11}{6}C_A - \frac{2}{3}T_F n_f)$ from Eq. (27.98) yields Eq. (27.100). The residue r was built in Eq. (27.97) from constants computed in Feynman gauge; that the assembled b_0 is nonetheless gauge-independent is the content of Theorem 24.12, and is verified explicitly by the background-field recomputation of Section 27.4.3 below, which uses a different gauge and a different organization of the diagrams yet returns the same b_0. □

It is worth pausing on what Eq. (27.101) accomplishes. The sign of β_g is set by the sign of b_0; with $b_0 > 0$ the beta function is *negative*, and by the running solution Eq. (27.2) the coupling shrinks logarithmically toward the ultraviolet. The two contributions to b_0 pull in opposite directions and tell the whole physical story in two terms: the $+\frac{11}{3}C_A$ is the gluonic antiscreening, born of the gluon's color charge and self-interaction, with no counterpart in an abelian theory; the $-\frac{4}{3}T_F n_f$ is the ordinary screening of the n_f charged fermion loops, the direct non-abelian copy of the QED vacuum polarization that gave $b_0^{\mathrm{QED}} = -\frac{4}{3} < 0$ in Eq. (27.27). Asymptotic freedom is the statement that, for few enough flavors, antiscreening wins.

The coupling α_s and its beta function. Particle physics quotes the running not for g but for $\alpha_s \equiv g^2/4\pi$, the strong-interaction analogue of the fine-structure constant. Its beta function follows from Eq. (27.101) by the chain rule. With $\alpha_s = g^2/4\pi$ one has $\mathrm{d}\alpha_s/\mathrm{d}\ln\mu = (2g/4\pi)\,\beta_g$, so

$$\beta_{\alpha_s} \equiv \mu\,\frac{d\alpha_s}{d\mu} = \frac{g}{2\pi}\,\beta_g = \frac{g}{2\pi}\Big(-\frac{g^3}{16\pi^2}\,b_0\Big) = -\frac{g^4}{32\pi^3}\,b_0. \tag{27.102}$$

Using $g^4 = (4\pi\alpha_s)^2 = 16\pi^2\alpha_s^2$ to eliminate g,

$$\boxed{\beta_{\alpha_s} = -\frac{b_0}{2\pi}\,\alpha_s^2 + O(\alpha_s^3), \qquad b_0 = \frac{11}{3}\,C_A - \frac{4}{3}\,T_F n_f.} \tag{27.103}$$

The quadratic form $\beta_{\alpha_s} \propto -\alpha_s^2$ is the hallmark of one-loop running: positive b_0 makes α_s decrease as μ grows. This is the form the running coupling section integrates to obtain $\alpha_s(Q^2)$ and the scale Λ_{QCD}.

The QCD specialization. For the physical strong interaction the gauge group is SU(3), whose color factors were evaluated once and for all in Eq. (B.114): the adjoint Casimir $C_A = N = 3$, the fundamental Dynkin index $T_F = \frac{1}{2}$. Inserting these into Eq. (27.100),

$$\boxed{b_0\big|_{\mathrm{SU}(3)} = \frac{11}{3}\cdot 3 - \frac{4}{3}\cdot\frac{1}{2}\,n_f = 11 - \frac{2}{3}\,n_f.} \tag{27.104}$$

This is the carried Yang–Mills theory number, identical to the value Eq. (B.117) obtained in Theorem B.29 by pure group theory. The two routes—there, the color

algebra of the adjoint and fundamental Casimirs; here, the full diagrammatic computation of Z_1, Z_2, Z_3 with every loop integral—meet at the same rational number, which is the strongest internal check that the color factors and the kinematics have each been done correctly.

Sign analysis: when is the theory asymptotically free? Asymptotic freedom requires $b_0 > 0$, i.e. $\beta_g < 0$. From Eq. (27.100) this is the condition that gluonic antiscreening dominate fermionic screening,

$$\tfrac{11}{3}\,C_A > \tfrac{4}{3}\,T_F n_f \qquad \Longleftrightarrow \qquad n_f < \frac{11\,C_A}{4\,T_F} \equiv n_f^{*}. \tag{27.105}$$

For SU(3), $n_f^{*} = 11 \cdot 3/(4 \cdot \tfrac{1}{2}) = \tfrac{33}{2} = 16.5$: the theory is asymptotically free for any $n_f \le 16$, and exactly at the boundary b_0 would have to be computed at the precise integer. The realized world has six quark flavors, comfortably below the bound, so QCD is asymptotically free with room to spare. Figure 27.4 displays the competition: a flat antiscreening offset $+\tfrac{11}{3}C_A$ from the gauge sector, a screening line falling as $-\tfrac{4}{3}T_F n_f$, and the net b_0 crossing zero at $n_f^{*} = 16.5$.

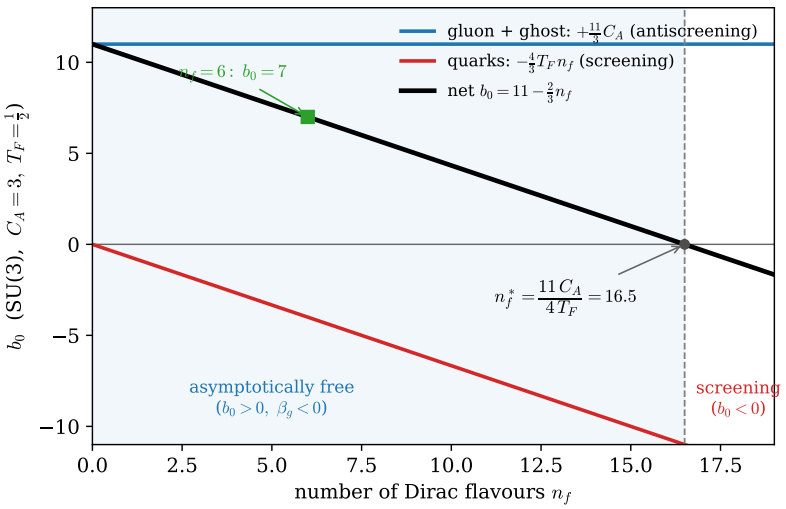


Figure 27.4: The one-loop coefficient $b_0 = \tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f$ for SU(3) ($C_A = 3$, $T_F = \tfrac{1}{2}$), i.e. $b_0 = 11 - \tfrac{2}{3}n_f$, as a function of the number of fundamental Dirac flavors n_f. The constant blue offset is the gluon-plus-ghost antiscreening $+\tfrac{11}{3}C_A = +11$; the falling red line is the quark screening $-\tfrac{4}{3}T_F n_f = -\tfrac{2}{3}n_f$; the heavy black line is their sum. The net coefficient is positive—the theory asymptotically free, $\beta_g < 0$—throughout the shaded region $n_f < n_f^{*} = 11C_A/4T_F = 16.5$. The physical point $n_f = 6$ gives $b_0 = 7$, deep inside the asymptotically free regime.

Worked Example 27.8 (Flavor counting and the margin of asymptotic freedom)**.** The sign of b_0 is robust against the actual flavor content of QCD, but the margin shrinks as flavors are added, and tracking it sharpens the physics. We evaluate $b_0 = 11 - \frac{2}{3}n_f$ at the relevant flavor numbers.

Physical QCD. At ordinary hadronic energies all six flavors are eventually accessible, but at a given scale μ only the quarks with mass below μ run in loops; this is the decoupling of heavy quarks (Eq. (27.101) holds with n_f equal to the number of *active*, i.e. effectively massless, flavors at the scale of interest). Between the bottom and top thresholds, $n_f = 5$ and

$$b_0\big|_{n_f=5} = 11 - \tfrac{2}{3}\cdot 5 = 11 - \tfrac{10}{3} = \tfrac{23}{3} \approx 7.67. \tag{27.106}$$

With all six active, $b_0 = 11 - 4 = 7$. In every active-flavor window from $n_f = 0$ (pure glue, $b_0 = 11$) to $n_f = 6$, the coefficient sits between 7 and 11: strongly positive, asymptotic freedom never in doubt.

Approaching the boundary. The coefficient stays positive up to $n_f = 16$, where $b_0 = 11 - \frac{32}{3} = \frac{1}{3}$, barely so; at $n_f = 17$ it turns negative, $b_0 = 11 - \frac{34}{3} = -\frac{1}{3} < 0$, and the theory ceases to be asymptotically free, behaving instead like QED with a coupling that grows in the ultraviolet. The window $n_f < 16.5$ is therefore the exact dividing line, and the near-boundary theories (n_f just below 16.5), where b_0 is small and positive, are the ones for which the two-loop term competes with the one-loop term and an infrared fixed point can appear—the Banks–Zaks phenomenon and the conformal window, which the running-coupling discussion takes up.

Pure glue. Setting $n_f = 0$ isolates the gauge sector: $b_0 = \frac{11}{3}C_A = 11$ for SU(3). Yang–Mills theory with no matter at all is already asymptotically free; the quarks only erode the antiscreening, they never create it. This is the precise sense in which asymptotic freedom is a property of the non-abelian *gauge* field, not of its matter content—a point the background-field computation below makes structurally transparent, since there the gauge contribution to b_0 is computed by itself, with the quarks added linearly afterward.

27.4.3 Background-field cross-check: gauge invariance made manifest

The computation just completed has an uncomfortable feature, harmless but worth removing. Every constituent of b_0—the three residues r_1, r_2, r_3—is gauge-dependent: each was computed in Feynman gauge $\xi = 1$, and each would take a different value in another R_ξ gauge. Only their particular combination Eq. (27.96) is gauge-invariant, and we know it is invariant only because Theorem 24.12 promises so and because the ξ-dependent pieces happen to cancel in the sum. It would be more satisfying—and a genuine independent check on the arithmetic—to compute b_0 in a framework where gauge invariance is built in from the start,

so that no cancellation between gauge-dependent constants is needed and the result is manifestly the physical one. The background-field method is exactly that framework. We now state it, prove its central simplification, and use it to recompute b_0 from scratch.

The background-field split and gauge. Write the gauge field as a fixed classical *background* $\bar{A}^a_\mu$ plus a fluctuating *quantum* field Q^a_μ over which the path integral runs,

$$A^a_\mu = \bar{A}^a_\mu + Q^a_\mu, \tag{27.107}$$

and gauge-fix the quantum field, not by the ordinary Lorenz condition $\partial_\mu Q^\mu = 0$, but by the *background-covariant* condition $(\bar{D}_\mu Q^\mu)^a = 0$, where $\bar{D}_\mu = \partial_\mu - ig\bar{A}^a_\mu T^a_{\text{adj}}$ is the covariant derivative in the background. The corresponding gauge-fixing term added to the gauge-fixed Lagrangian of Eq. (26.37) is

$$\mathcal{L}_{\text{gf}} = -\frac{1}{2\xi}(\bar{D}_\mu Q^{a\,\mu})^2, \tag{27.108}$$

with the Faddeev–Popov ghosts modified correspondingly so that their kinetic operator is $-\bar{D}^2$. The decisive property of this choice is a residual symmetry. Under a gauge transformation that acts on the background as a *connection*,

$$\delta\bar{A}^a_\mu = (\bar{D}_\mu\theta)^a = \partial_\mu\theta^a + g f^{abc}\bar{A}^b_\mu\theta^c, \qquad \delta Q^a_\mu = g f^{abc} Q^b_\mu \theta^c, \tag{27.109}$$

i.e. with Q rotating homogeneously as a field in the adjoint, the full action including Eq. (27.108) is invariant: the covariant gauge-fixing term transforms as a square of an adjoint vector and is inert. The gauge-fixing has broken the quantum gauge freedom (as it must, to make the path integral well-defined) while *preserving* an exact background gauge invariance. This is the entire point of the construction.

The one renormalization constant. Integrating out Q and the ghosts produces the one-loop effective action $\Gamma[\bar{A}]$, a functional of the background alone. Background gauge invariance, Eq. (27.109), is not anomalous here (it is a vector-like, non-chiral symmetry), so it survives quantization: $\Gamma[\bar{A}]$ is invariant under $\delta\bar{A} = \bar{D}\theta$. A local functional of mass dimension four, invariant under background gauge transformations, and built from $\bar{A}$ and its derivatives, can have as its divergent part only the gauge-invariant kinetic term $\bar{F}^a_{\mu\nu}\bar{F}^{a\,\mu\nu}$, where $\bar{F}^a_{\mu\nu} = \partial_\mu\bar{A}^a_\nu - \partial_\nu\bar{A}^a_\mu + g f^{abc}\bar{A}^b_\mu\bar{A}^c_\nu$ is the background field strength. Hence *a single* renormalization constant suffices to absorb all the ultraviolet divergence: a wavefunction renormalization of the background field,

$$\bar{A}^a_{0\,\mu} = Z_A^{1/2}\,\bar{A}^a_\mu, \tag{27.110}$$

defined so that the renormalized divergent effective action is $-\frac{1}{4}Z_A\,\bar{F}^2$ with $\bar{F}$ built from the renormalized background and coupling. There is no separate vertex

renormalization to compute: the cubic and quartic self-couplings inside $\bar{F}^2$ are renormalized by the *same* Z_A as the kinetic term, precisely because background gauge invariance forces the divergence into the single invariant $\bar{F}^2$. This is the structural payoff of the method, and it leads at once to the relation we want.

Theorem 27.9 (Background-field coupling renormalization)**.** *In the background-field gauge, the coupling renormalization is determined entirely by the background-field renormalization:*

$$\boxed{Z_g = Z_A^{-1/2}.} \tag{27.111}$$

Proof. Background gauge invariance requires that $\bar{A}^a_\mu$ enter $\Gamma[\bar{A}]$ only through the covariant derivative $\bar{D}_\mu = \partial_\mu - ig\bar{A}^a_\mu T^a$ and the field strength $\bar{F}$. The covariant derivative is a meaningful (gauge-covariant) operator only if the product $g\bar{A}^a_\mu$ is renormalization-group invariant, for that product is what appears in $\bar{D}_\mu$ and what must transform as a connection under Eq. (27.109); equivalently, the field strength renormalizes multiplicatively, $\bar{F}_0 = Z_A^{1/2}\bar{F}$, only if its nonlinear term $g_0 f \bar{A}_0 \bar{A}_0$ scales the same way as its linear term $\partial \bar{A}_0$. The linear term scales as $Z_A^{1/2}$ by Eq. (27.110); matching the nonlinear term, which carries one extra power of $g_0 \bar{A}_0$, requires

$$g_0\, \bar{A}^a_{0\,\mu} = g\, \bar{A}^a_\mu \quad \text{(renormalization-group invariant)} \qquad \Longrightarrow \qquad g_0\, Z_A^{1/2} = g. \tag{27.112}$$

Comparing with the definition $g_0 = \mu^\varepsilon Z_g\, g$ of Eq. (27.3) (the μ^ε carrying the trivial dimensional scaling) gives $\mu^\varepsilon Z_g\, g\, Z_A^{1/2} = g$, hence at the level of the pole series $Z_g Z_A^{1/2} = 1$, which is Eq. (27.111). □

Equation (27.111) should be compared with the ordinary decomposition $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ of Eq. (27.9), which needed three constants and a delicate cancellation of C_F and of ξ between them. The background-field method has collapsed all of that into a single propagator-like constant Z_A, the background-field analogue of Z_3. The collapse is the manifest version of the Slavnov–Taylor relations Eq. (27.24): those identities say the various vertex and field renormalizations conspire to leave one independent coupling renormalization, and the background-field gauge is the gauge in which that conspiracy is automatic rather than computed. It remains to evaluate the divergence of Z_A.

The divergence of Z_A from the quadratic operators. At one loop $\Gamma[\bar{A}]$ is a sum of functional determinants—the Gaussian integral over the quantum field Q, over the ghosts, and over the quarks—each a determinant of the operator governing the quadratic fluctuation in the background. In the background Feynman gauge ($\xi = 1$) these operators are second-order and of the universal form $\Delta = -(\bar{D}^2 + Y)$, with $\bar{D}$ the background covariant derivative in the

appropriate representation and Y a matrix-valued "potential" encoding the spin coupling. Their divergent parts are controlled by a single standard heat-kernel coefficient, which we state and then apply to each field.

Lemma 27.10 (Logarithmic divergence of a determinant)**.** *Let $\Delta = -(\bar{D}_\mu \bar{D}^\mu + Y)$ act on a multiplet of fields, with $\bar{D}_\mu$ a covariant derivative whose curvature is $\Omega_{\mu\nu} \equiv [\bar{D}_\mu, \bar{D}_\nu]$, and Y an endomorphism of the multiplet. In $d = 4 - 2\varepsilon$ the ultraviolet-divergent part of $\ln\det\Delta$ is*

$$(\ln\det\Delta)_{\text{div}} = -\frac{1}{16\pi^2}\,\frac{1}{\varepsilon}\int \mathrm{d}^4x\ \operatorname{tr}\Big[\tfrac{1}{2}\,Y^2 + \tfrac{1}{12}\,\Omega_{\mu\nu}\Omega^{\mu\nu}\Big], \tag{27.113}$$

the trace running over all internal (Lorentz, color, spinor) indices of the multiplet, with total-derivative and field-independent terms dropped.

Proof sketch. Write $\ln\det\Delta = \operatorname{Tr}\ln\Delta = -\int_0^\infty (\mathrm{d}\tau/\tau)\ \operatorname{Tr} e^{-\tau\Delta}$ (Euclidean, after the Wick rotation of Chapter 10). The heat kernel has the short-proper-time (Schwinger–DeWitt) expansion $\operatorname{Tr} e^{-\tau\Delta} = (4\pi\tau)^{-2}\int \mathrm{d}^4x\ \operatorname{tr}[\,b_0 + b_1\tau + b_2\tau^2 + \cdots]$, whose first coefficients are $b_0 = \mathbf{1}$, $b_1 = Y$, and $b_2 = \frac{1}{2}Y^2 + \frac{1}{12}\Omega_{\mu\nu}\Omega^{\mu\nu} +$ (total derivative); these are obtained by solving the heat equation $\partial_\tau e^{-\tau\Delta} = -\Delta e^{-\tau\Delta}$ order by order in τ about the free kernel, the coincidence-limit algebra being the standard recursion of Schwinger and DeWitt. The τ-integral diverges at the lower (ultraviolet) end; the coefficient of the logarithmic divergence is precisely the dimensionless coefficient b_2, since the term $(4\pi\tau)^{-2}b_2\tau^2$ produces $\int \mathrm{d}\tau/\tau$. In $d = 4 - 2\varepsilon$ this logarithm becomes a $1/\varepsilon$ pole with the prefactor $(4\pi)^{-2} = 1/16\pi^2$, and the overall sign from $-\int \mathrm{d}\tau/\tau$ gives Eq. (27.113). The b_0 and b_1 coefficients produce the (quartically and quadratically divergent, hence in $\overline{\text{MS}}$ scaleless and vanishing) field-independent and mass terms; only b_2 contributes to the $\bar{F}^2$ renormalization. □

We apply Theorem 27.10 to the three fields. The one-loop effective action is

$$\Gamma_1[\bar{A}] = \tfrac{1}{2}\ln\det\Delta_{\text{gluon}} - \ln\det\Delta_{\text{ghost}} - \tfrac{1}{2}\ln\det\Delta^{(\text{sq})}_{\text{quark}}, \tag{27.114}$$

the $\frac{1}{2}$ for the real bosonic gluon, the -1 for the complex anticommuting ghost (Grassmann statistics flip the sign of the loop), and the $-\frac{1}{2}$ for the squared Dirac operator (the minus from fermion statistics, the squaring from $\det(i\bar{\not{D}} - m) = [\det(\bar{\not{D}}^2 + \cdots)]^{1/2}$). We need the Y and Ω of each. Two group-theory traces recur: for the adjoint representation, in which the gluon and ghost live, $\operatorname{tr}_{\text{adj}}(\Omega_{\mu\nu}\Omega^{\mu\nu}) = -g^2 C_A\,\bar{F}^c_{\rho\sigma}\bar{F}^{c\,\rho\sigma}$, because the adjoint curvature is g times the field strength contracted with adjoint generators of index C_A (Theorem B.28); for the fundamental, in which the quark lives, the same trace gives $-g^2 T_F\,\bar{F}^2$ per flavor (Eq. (B.2)). Write $\bar{F}^2 \equiv \bar{F}^c_{\rho\sigma}\bar{F}^{c\,\rho\sigma}$ throughout.

Ghost loop. The ghost is a complex scalar in the adjoint with $\Delta_{\text{ghost}} = -\bar{D}^2$, so $Y = 0$. By Eq. (27.113) and the adjoint trace,

$$\left(-\ln\det\Delta_{\text{ghost}}\right)_{\text{div}} = +\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int \text{tr}_{\text{adj}}\left[\tfrac{1}{12}\Omega^2\right] = +\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int\left(-\tfrac{1}{12}g^2 C_A\,\bar{F}^2\right). \tag{27.115}$$

Gluon loop. Expanding $-\frac{1}{4}\bar{F}[\bar{A}+Q]^2 + \mathcal{L}_{\text{gf}}$ to quadratic order in Q in background Feynman gauge, the longitudinal $\bar{D}^\mu\bar{D}^\nu$ pieces cancel between the kinetic and gauge-fixing terms (the defining virtue of $\xi = 1$), leaving the minimal operator

$$(\Delta_{\text{gluon}})^{ab}_{\mu\nu} = -\Big[(\bar{D}^2)^{ab}\,g_{\mu\nu} + 2g\,f^{acb}\bar{F}^c_{\mu\nu}\Big], \tag{27.116}$$

so the quantum gluon is an adjoint vector with potential $Y^{ab}_{\mu\nu} = 2gf^{acb}\bar{F}^c_{\mu\nu}$. The $2g\bar{F}$ term is the gluon's spin-one magnetic-moment coupling; it is the seat of antiscreening. Two traces enter. The curvature acts on the adjoint color index (the Lorentz vector index is flat and merely contributes its dimension, $\text{tr}_{\text{vec}}\,\mathbf{1} = 4$):

$$\text{tr}\Big[\tfrac{1}{12}\Omega^2\Big] = \tfrac{1}{12}\,\text{tr}_{\text{adj}}(\Omega^2)\cdot\text{tr}_{\text{vec}}\,\mathbf{1} = \tfrac{1}{12}(-g^2C_A\bar{F}^2)\cdot 4 = -\tfrac{1}{3}\,g^2C_A\,\bar{F}^2, \tag{27.117}$$

the *orbital* (diamagnetic) part, which screens. The potential gives

$$\text{tr}\Big[\tfrac{1}{2}Y^2\Big] = \tfrac{1}{2}(2g)^2 f^{acb}f^{bda}\,\bar{F}^c_{\mu\nu}\bar{F}^{d\,\nu\mu} = \tfrac{1}{2}\cdot 4g^2\,C_A\,\bar{F}^2 = 2\,g^2C_A\,\bar{F}^2, \tag{27.118}$$

using $f^{acb}f^{bda} = -f^{abc}f^{abd} = -C_A\delta^{cd}$ and $\bar{F}^c_{\mu\nu}\bar{F}^{c\,\nu\mu} = -\bar{F}^2$, the two minus signs combining to $+$; this is the *magnetic* (paramagnetic) part, which antiscreens. Summing and inserting the bosonic $\frac{1}{2}$ of Eq. (27.114),

$$\left(\tfrac{1}{2}\ln\det\Delta_{\text{gluon}}\right)_{\text{div}} = -\frac{1}{2}\cdot\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int\Big(-\tfrac{1}{3}+2\Big)g^2C_A\,\bar{F}^2 = -\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int\tfrac{5}{6}\,g^2C_A\,\bar{F}^2. \tag{27.119}$$

Quark loops. The squared Dirac operator for a fundamental fermion is $(i\vec{\not{D}})^2 = -\bar{D}^2 + \frac{g}{2}\sigma_{\mu\nu}\bar{F}^{\mu\nu}$ with $\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu,\gamma_\nu]$, i.e. a fundamental field carrying both a color index and a four-component Dirac index, with potential $Y = -\frac{g}{2}\sigma_{\mu\nu}\bar{F}^{\mu\nu}$. The orbital trace gives $\frac{1}{12}\,\text{tr}_{\text{fund}}(\Omega^2)\cdot\text{tr}_{\text{Dirac}}\,\mathbf{1} = \frac{1}{12}(-g^2T_F\bar{F}^2)\cdot 4 = -\frac{1}{3}g^2T_F\bar{F}^2$ per flavor, and the magnetic trace, using $\text{tr}_{\text{Dirac}}(\sigma_{\mu\nu}\sigma_{\rho\sigma}) = 4(g_{\mu\rho}g_{\nu\sigma} - g_{\mu\sigma}g_{\nu\rho})$, gives $\frac{1}{2}Y^2 \to \frac{1}{2}\cdot\frac{g^2}{4}\cdot 4\cdot 2\,T_F\bar{F}^2 = g^2T_F\bar{F}^2$ per flavor. The sum inside the bracket is $(-\frac{1}{3}+1)g^2T_F\bar{F}^2 = \frac{2}{3}g^2T_F\bar{F}^2$, and with the fermionic $-\frac{1}{2}$ and n_f flavors,

$$\left(-\tfrac{1}{2}\ln\det\Delta^{(\text{sq})}_{\text{quark}}\right)_{\text{div}} = +\frac{1}{2}\cdot\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int n_f\,\tfrac{2}{3}g^2T_F\,\bar{F}^2 = +\frac{1}{16\pi^2}\frac{1}{\varepsilon}\int\tfrac{1}{3}\,g^2T_F n_f\,\bar{F}^2. \tag{27.120}$$

Assembling Z_A and recovering b_0. Adding the three divergent pieces Eqs. (27.115) to (27.120), the total divergent effective action is

$$\Gamma_{1,\mathrm{div}} = \frac{1}{16\pi^2}\frac{1}{\varepsilon}\int \Big[\underbrace{-\tfrac{1}{12}C_A}_{\text{ghost}}\underbrace{-\tfrac{5}{6}C_A}_{\text{gluon}}+\underbrace{\tfrac{1}{3}T_F n_f}_{\text{quark}}\Big] g^2\,\bar{F}^2$$
$$= \frac{1}{16\pi^2}\frac{1}{\varepsilon}\int \Big(-\tfrac{11}{12}C_A+\tfrac{1}{3}T_F n_f\Big) g^2\,\bar{F}^2, \tag{27.121}$$

where $-\frac{1}{12}-\frac{5}{6}=-\frac{1}{12}-\frac{10}{12}=-\frac{11}{12}$. The renormalized kinetic term is $-\frac{1}{4}Z_A\bar{F}^2$, whose counterterm $-\frac{1}{4}(Z_A-1)\bar{F}^2$ must cancel Eq. (27.121) in $\overline{\mathrm{MS}}$. Equating the $1/\varepsilon$ poles, $-\frac{1}{4}(Z_A-1)=(g^2/16\pi^2)(1/\varepsilon)(-\frac{11}{12}C_A+\frac{1}{3}T_F n_f)$, so

$$\boxed{Z_A = 1+\frac{g^2}{16\pi^2}\frac{1}{\varepsilon}\Big(\tfrac{11}{3}C_A-\tfrac{4}{3}T_F n_f\Big).} \tag{27.122}$$

The gauge part assembles as $4\times(-\frac{1}{12}-\frac{5}{6})\cdot(-1)=4\cdot\frac{11}{12}=\frac{11}{3}$, neatly the sum of a gluon contribution $\frac{10}{3}C_A$ and a ghost contribution $\frac{1}{3}C_A$; within the gluon's $\frac{10}{3}$ the diamagnetic orbital part contributes $-\frac{2}{3}$ and the paramagnetic magnetic-moment part $+4$. The decomposition $\frac{11}{3}=(-\frac{1}{3})+4$ of the total gauge coefficient—diamagnetic screening overwhelmed by paramagnetic antiscreening—is the physical content unpacked in Section 27.5.4 via Eq. (27.142); here it has emerged directly from the magnetic-moment term $Y=2g\bar{F}$ of Eq. (27.116).

Finally, Theorem 27.9 converts Z_A into the coupling renormalization, $Z_g=Z_A^{-1/2}$, and the master formula $b_0=-2r$ reads off the residue. Writing $Z_A=1+(g^2/16\pi^2)(\rho_A/\varepsilon)$ with $\rho_A=\frac{11}{3}C_A-\frac{4}{3}T_F n_f$, the residue of $\ln Z_g=-\frac{1}{2}\ln Z_A$ is $r=-\frac{1}{2}\rho_A$, so

$$b_0 = -2r = \rho_A = \frac{11}{3}\,C_A-\frac{4}{3}\,T_F n_f, \tag{27.123}$$

identical to Eq. (27.100). Equivalently, $Z_g=Z_A^{-1/2}=1-\frac{1}{2}(g^2/16\pi^2)(1/\varepsilon)(\frac{11}{3}C_A-\frac{4}{3}T_F n_f)=1-(g^2/16\pi^2)(1/\varepsilon)(\frac{11}{6}C_A-\frac{2}{3}T_F n_f)$, reproducing the assembled Z_g of Eq. (27.99) term by term. The background-field computation, performed in a different gauge with a different set of diagrams and needing no cancellation between gauge-dependent constants, has returned the same b_0. This is the promised verification that b_0 is gauge-independent, and it closes the circle opened by Theorem 24.12.

Physical Insight

The background-field method does not merely re-derive b_0; it explains why the answer *had* to be expressible through one constant. The coupling g enters the theory only through the covariant derivative, and gauge invariance ties the covariant derivative's two pieces, ∂ and $g\bar{A}$, into a single object. Renormalizing the field therefore renormalizes the coupling,

with no independent freedom: $g_0\bar{A}_0 = g\bar{A}$ is forced, hence $Z_g = Z_A^{-1/2}$. The ordinary calculation hides this behind three constants and a Ward cancellation; the background-field calculation displays it as the statement that there is one invariant, $\bar{F}^2$, and one constant, Z_A, to renormalize. Gauge invariance, which in the conventional approach is a property the final answer turns out to have, is here a property the method has from the first line.

We have computed the carried Yang–Mills theory number twice, by independent routes, and obtained $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f = 11 - \frac{2}{3}n_f$ for SU(3), manifestly positive for every physical flavor content. The sign of the Yang–Mills beta function is therefore settled: $\beta_g < 0$. With that signed number in hand—the input the remaining sections interpret—we turn from computation to meaning, and ask what a negative beta function does to the coupling as the energy runs, why the mechanism is properly called antiscreening, and how sharply it contrasts with the screening of QED.

27.5 Asymptotic Freedom and Antiscreening

The previous section delivered the number on which the whole subject turns. Assembling $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ from the one-loop renormalization constants and feeding the simple-pole residue through the pole equation $\beta = \mu\, \mathrm{d}g/\mathrm{d}\mu$ (Eq. (24.104)), we found the Yang–Mills beta function

$$\beta(g) = -\frac{g^3}{16\pi^2}\, b_0 + O(g^5), \qquad b_0 = \tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f, \tag{27.124}$$

the carried Yang–Mills theory result Eq. (27.101), corroborated gauge-independently by the background-field computation of Section 27.4. For SU(3) with the six known quark flavors $C_A = 3$ and $T_F = \frac{1}{2}$ (Eqs. (B.92) and (B.102)) give $b_0 = 11 - \frac{2}{3}n_f$, positive for every physical $n_f \leq 16$; in the real world, $b_0 > 0$. A positive b_0 means a *negative* beta function, and a negative beta function is a statement about the fate of the coupling that we are now equipped, by the renormalization-group machinery of Part VI, to read off exactly. This section does the reading. We integrate the flow and prove that the coupling vanishes logarithmically into the ultraviolet — *asymptotic freedom* — with no ultraviolet Landau pole; we set the result sharply against the screening of QED, completing the mirror the threads were built to stage; we state, honestly scoped, the theorem that in four dimensions only non-abelian gauge theories can behave this way; and we give the correct physical mechanism, the antiscreening paramagnetic vacuum, tracing its algebraic fingerprint back to the inequality $Z_1 \neq Z_2$ that the quark–gluon vertex of Section 27.3 forced upon us.

27.5.1 Integrating the one-loop flow

The running coupling $\bar{g}(Q)$ was defined in Theorem 24.22 as the solution of the renormalization-group equation with the renormalized coupling at the reference scale as initial datum. Written as a flow in the logarithm of momentum, with $t = \ln(Q/\mu)$ and $\bar{g}(0) = g(\mu)$, it reads

$$\frac{d\bar{g}}{dt} = \beta(\bar{g}) = -\frac{b_0}{16\pi^2}\,\bar{g}^{\,3} + O(\bar{g}^{\,5}), \qquad b_0 > 0. \tag{27.125}$$

We keep only the one-loop term; the omitted $O(\bar{g}^5)$ piece shifts the answer at the order at which the (uncomputed) two-loop coefficient would enter, so retaining $-b_0\bar{g}^3/16\pi^2$ alone is internally consistent, exactly as for the scalar running of Eq. (24.207). The equation is separable. It is cleanest in the square of the coupling: with $\bar{g}\,\mathrm{d}\bar{g}/\mathrm{d}t = \tfrac{1}{2}\,\mathrm{d}(\bar{g}^2)/\mathrm{d}t$, multiplying Eq. (27.125) by $2\bar{g}$ gives

$$\frac{d(\bar{g}^2)}{dt} = -\frac{b_0}{8\pi^2}\,(\bar{g}^2)^2. \tag{27.126}$$

Dividing by $(\bar{g}^2)^2$ and recognizing the left side as $-\,\mathrm{d}(1/\bar{g}^2)/\mathrm{d}t$,

$$\frac{d}{dt}\left(\frac{1}{\bar{g}^2}\right) = \frac{b_0}{8\pi^2}, \tag{27.127}$$

so that the *inverse* squared coupling runs linearly in t. This is the structural heart of the matter: at one loop $1/\bar{g}^2$ is an exactly linear function of $\ln Q$, with slope fixed by b_0. Integrating from 0 to t,

$$\frac{1}{\bar{g}^2(t)} = \frac{1}{g^2(\mu)} + \frac{b_0}{8\pi^2}\,t, \tag{27.128}$$

and solving for $\bar{g}^2$ and restoring $t = \ln(Q/\mu)$ yields the one-loop running coupling of Yang–Mills theory,

$$\boxed{\bar{g}^2(Q) = \frac{g^2(\mu)}{1 + \dfrac{b_0\, g^2(\mu)}{8\pi^2}\,\ln\dfrac{Q}{\mu}}\,.} \tag{27.129}$$

In terms of $\alpha_s \equiv g^2/4\pi$ this is

$$\alpha_s(Q) = \frac{\alpha_s(\mu)}{1 + \dfrac{b_0\,\alpha_s(\mu)}{2\pi}\,\ln\dfrac{Q}{\mu}}, \qquad \frac{1}{\alpha_s(Q)} = \frac{1}{\alpha_s(\mu)} + \frac{b_0}{2\pi}\,\ln\frac{Q}{\mu}. \tag{27.130}$$

Everything follows from the single fact that $b_0 > 0$, which makes the coefficient of $\ln(Q/\mu)$ in the denominator *positive*.

The ultraviolet limit. Send $Q \to \infty$ at fixed reference data $g(\mu)$. The logarithm grows without bound, the denominator of Eq. (27.129) grows without bound, and therefore

$$\bar{g}^2(Q) \xrightarrow{Q\to\infty} 0, \qquad \alpha_s(Q) \xrightarrow{Q\to\infty} 0. \tag{27.131}$$

The coupling does not merely stay bounded; it is driven to zero. This is the content of the following definition, which we can now make precise.

Definition 27.11 (Asymptotic freedom)**.** A theory is *asymptotically free* if its renormalized running coupling tends to zero as the momentum scale is sent to infinity, $\bar{g}(Q) \to 0$ as $Q \to \infty$. Equivalently, the Gaussian (free) fixed point $g = 0$ is *ultraviolet-attractive*: the renormalization-group flow carries the coupling *into* the free theory at short distances. At one loop this holds precisely when the leading coefficient is $b_0 > 0$, i.e. when the beta function is negative for small positive g.

The rate of vanishing. The estimate Eq. (27.131) can be sharpened to an explicit rate. For large $t = \ln(Q/\mu)$ the constant $1/g^2(\mu)$ in Eq. (27.128) is negligible against the linear term, so

$$\frac{1}{\bar{g}^2(Q)} \simeq \frac{b_0}{8\pi^2} \ln\frac{Q}{\mu} \qquad (Q \to \infty), \tag{27.132}$$

whence

$$\bar{g}(Q) \simeq \frac{2\pi\sqrt{2}}{\sqrt{b_0 \ln(Q/\mu)}}, \qquad \alpha_s(Q) \simeq \frac{2\pi}{b_0 \ln(Q/\mu)}. \tag{27.133}$$

The coupling g vanishes like $1/\sqrt{\ln Q}$ and the fine-structure analog α_s like $1/\ln Q$. The decay is excruciatingly slow — logarithmic, the slowest approach to zero a power-counting argument ever produces — but it is monotone and unbounded: there is no scale beyond which the coupling stops shrinking. Notice that the rate is governed entirely by b_0 and not at all by the boundary value $g(\mu)$: two theories with wildly different couplings at 1 GeV approach the same universal envelope $2\pi/(b_0 \ln Q)$ at asymptotically high energy. The free theory is not just a fixed point the flow happens to pass near; it is the destination.

No ultraviolet Landau pole. The sign of b_0 controls not only the limit but the global analytic structure of Eq. (27.130). The running coupling diverges where its denominator vanishes, i.e. at the scale $Q_\star$ for which

$$1 + \frac{b_0\,\alpha_s(\mu)}{2\pi} \ln\frac{Q_\star}{\mu} = 0 \quad\Longrightarrow\quad \ln\frac{Q_\star}{\mu} = -\frac{2\pi}{b_0\,\alpha_s(\mu)}. \tag{27.134}$$

With $b_0 > 0$ and $\alpha_s(\mu) > 0$ the right-hand side is *negative*: the pole sits at $Q_\star = \mu \exp[-2\pi/(b_0\alpha_s(\mu))] < \mu$, deep in the *infrared*, not the ultraviolet. For all $Q > \mu$ the denominator is strictly greater than one and the coupling is finite, small, and falling. The function Eq. (27.130) is regular on the entire ultraviolet half-line. This is the decisive structural contrast with the screening theories of Part VI, and it is worth displaying side by side.

- For ϕ^4 theory the coefficient in the denominator was positive in the opposite sense: $\beta_\lambda > 0$ drives $1/\lambda$ *down* as μ rises, the denominator of Eq. (24.210) hits zero at a finite *ultraviolet* scale, and the coupling blows up there — the Landau pole Eq. (24.211), an ultraviolet wall.
- For QED, $\beta_e = e^3/12\pi^2 > 0$ (Eq. (24.108)) likewise sends the charge up toward short distances toward an ultraviolet Landau pole. The electromagnetic coupling grows in exactly the régime where the strong coupling shrinks.
- For Yang–Mills theory with $b_0 > 0$, the pole has migrated to the infrared. The ultraviolet is clean; the singularity, and with it the breakdown of perturbation theory, lies at low energy. That infrared pole is not an artifact to be apologized for: relocated and renamed, it is the scale Λ_{QCD} at which the strong interaction becomes strong, the subject of Section 27.6.

Physical Insight

A negative beta function does not abolish the Landau singularity of the one-loop running coupling; it *moves* it. In a screening theory the singularity is an ultraviolet catastrophe that calls the very existence of the continuum theory into question (the triviality problem). In an asymptotically free theory the identical algebraic singularity sits in the infrared, where it announces, correctly, that the theory becomes *strongly coupled* at long distances — a dynamical statement about confinement, not a pathology of the short-distance definition. The same one-loop formula, read with the opposite sign of b_0, is a disease in one theory and the engine of hadron physics in the other.

27.5.2 The mirror of screening: QED against QCD

QED and Yang–Mills theory stand here as instructive opposites, and the opposition is now exact and quantitative. In both theories the one-loop running is the same rational function of a single logarithm; the only difference is the sign of the coefficient multiplying that logarithm. In QED the sign screens and the charge grows toward the ultraviolet; in QCD the sign antiscreens and the coupling falls.

The worked example makes the contrast numerical and exhibits the two pole structures explicitly.

Worked Example 27.12 (The ultraviolet fate of $\alpha_{\rm em}$ versus α_s)**.** We integrate both one-loop runnings outward from a common reference and compare. Take the electroweak scale $\mu = M_Z \approx 91.2\,\text{GeV}$ as the reference, where $\alpha_{\rm em}(M_Z) \approx 1/128$ and $\alpha_s(M_Z) \approx 0.118$ are the measured values. For the running we use the one-loop laws derived above and in Chapter 24, written in the inverse-coupling form, which is linear and so trivial to evaluate.

QED. From $\beta_\alpha = 2\alpha^2/3\pi$ (the α-form of Eq. (24.108)) the inverse coupling runs as

$$\frac{1}{\alpha_{\rm em}(Q)} = \frac{1}{\alpha_{\rm em}(M_Z)} + \frac{b_{\rm QED}}{2\pi}\,\ln\frac{Q}{M_Z}, \qquad b_{\rm QED} = -\tfrac{4}{3}\sum_f Q_f^2 N_c^{(f)}, \tag{27.135}$$

the sum running over the charged fermions light enough to circulate at scale Q, with electric charges Q_f and color multiplicities $N_c^{(f)}$. The sign is the crux: the slope $b_{\rm QED}/2\pi$ is *negative* (since $b_{\rm QED} < 0$ in this convention, screening), so $1/\alpha_{\rm em}$ *decreases* as Q rises and the coupling grows. Taking, for an order-of-magnitude estimate, an effective slope such that $1/\alpha_{\rm em}$ falls from 128 at M_Z, the inverse coupling reaches zero — the Landau pole, where $\alpha_{\rm em}$ diverges — at

$$\ln\frac{Q_\star^{\rm QED}}{M_Z} = -\frac{2\pi}{b_{\rm QED}}\cdot\frac{1}{\alpha_{\rm em}(M_Z)} \;\sim\; (\text{several hundred}), \tag{27.136}$$

i.e. at an astronomically large $Q_\star^{\rm QED} \sim 10^{200}$–$10^{280}\,\text{GeV}$, dozens of orders of magnitude beyond the Planck scale. The pole is unreachable, and the electroweak embedding of QED changes the running long before it matters; but the *direction* is unambiguous: QED's coupling climbs into the ultraviolet and its perturbative definition formally fails there.

QCD. From Eq. (27.130) with $b_0 = 11 - \frac{2}{3}n_f$ and, say, $n_f = 5$ active flavors above the b-quark threshold, $b_0 = \frac{23}{3} \approx 7.67 > 0$,

$$\frac{1}{\alpha_s(Q)} = \frac{1}{\alpha_s(M_Z)} + \frac{b_0}{2\pi}\,\ln\frac{Q}{M_Z} = 8.47 + 1.22\,\ln\frac{Q}{M_Z}. \tag{27.137}$$

The slope is now *positive*, so $1/\alpha_s$ *grows* and α_s falls. At $Q = 1\,\text{TeV}$, $\ln(Q/M_Z) \approx 2.4$, giving $1/\alpha_s \approx 11.4$, i.e. $\alpha_s \approx 0.088$; at $Q = 10^{16}\,\text{GeV}$, $\ln(Q/M_Z) \approx 33$, giving $1/\alpha_s \approx 49$, i.e. $\alpha_s \approx 0.020$. The coupling has shrunk by a factor of six on the way to the grand-unification scale and continues falling forever; there is no ultraviolet Q at which the denominator of Eq. (27.130) vanishes. The contrast with the QED column is total: over the same logarithmic range, one coupling marches toward an (astronomically distant) ultraviolet singularity and the other toward zero.

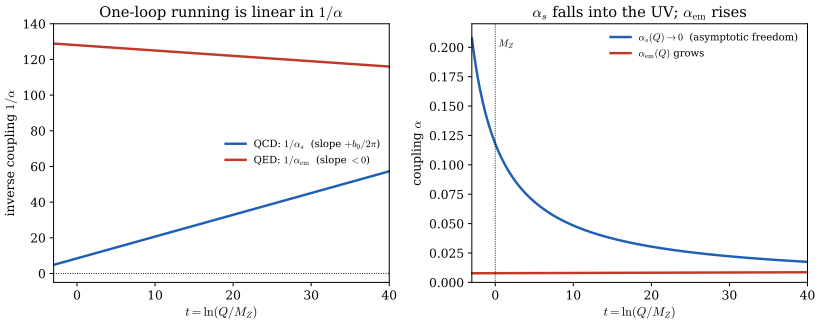


Figure 27.5: The mirror of screening and antiscreening, displayed in the inverse coupling, in which one-loop running is exactly linear (Eq. (27.127)). QED (red): $1/\alpha_{\rm em}$ falls with energy, so the coupling grows toward an ultraviolet Landau pole. QCD (blue): $1/\alpha_s$ rises with energy with slope $b_0/2\pi > 0$, so the coupling falls to zero in the ultraviolet (asymptotic freedom) and the singularity has migrated to the infrared, where it defines $\Lambda_{\rm QCD}$ (Chapter 28, Section 27.6). The single difference between the two lines is the sign of the slope, i.e. the sign of the beta-function coefficient.

The numbers in Theorem 27.12 are the quantitative face of a single sign flip. The same algebra, the same rational dependence on one logarithm, governs both; the sign of the coefficient decides whether the short-distance limit is a wall or a free theory. The figure makes the inversion visible. Because the running is exactly linear in the *inverse* coupling, plotting $1/\alpha$ against $\ln Q$ turns both theories into straight lines — with opposite slopes (Fig. 27.5).

27.5.3 Why only non-abelian gauge theories: the Coleman–Gross theorem

Asymptotic freedom is not generic. Among all renormalizable quantum field theories one can write in four dimensions, the property is the exclusive preserve of non-abelian gauge theories. This is not an empirical accident but a theorem about one-loop beta functions, and stating it precisely tells us *why* the gluon self-coupling is the indispensable ingredient.

Theorem 27.13 (Coleman–Gross, one-loop)**.** *Consider the most general power-counting renormalizable Lagrangian in four spacetime dimensions built from scalars, spin-$\frac{1}{2}$ fermions, and gauge fields, with the most general renormalizable interactions allowed by Lorentz invariance and the field content: gauge couplings, Yukawa couplings, and quartic scalar self-couplings. If the gauge group contains* no *non-abelian factor, then no choice of fields, representations, and couplings makes the theory asymptotically free: at least one running coupling fails to tend to zero in the ultraviolet. Asymptotic freedom of all couplings requires at least*

one non-abelian gauge field, and even then only for a restricted range of matter content.

The full proof catalogs the one-loop beta functions of every coupling type and shows that the only negative contribution to any gauge coupling's b_0 is the gauge field's coupling to *itself* through the C_A term; we will not reproduce the entire catalog, but the structural reason is exactly the piece-by-piece sign analysis we are about to give, and the honest scope must be stated with it.

Caution

Theorem 27.13 is a one-loop statement, and it is what was actually proved in 1973. The argument runs through the signs of the leading coefficients b_0 of all couplings simultaneously, including the back-reaction of Yukawa and scalar couplings on the gauge running and on each other. It does *not* by itself settle whether some cleverly tuned theory might be "asymptotically safe" at a nontrivial ultraviolet fixed point (where a coupling tends to a finite nonzero value rather than to zero), nor does it govern theories outside the stated field content (higher-spin, gravity, extra dimensions). What it establishes, rigorously and decisively, is the point we need: *within* the renormalizable four-dimensional menu, ordinary screening is the default and ultraviolet freedom is impossible without a non-abelian gauge field. Everything in this book that relies on short-distance freedom relies on having such a field — the gluon.

The mechanism behind the theorem is transparent if we sort the one-loop contributions by the spin of the particle in the loop and ask, for each, whether it pushes b_0 positive (antiscreening) or negative (screening).

Scalars screen. A quartic scalar coupling has the positive one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Eq. (24.207): it grows toward the ultraviolet, never toward zero. A charged scalar's contribution to a gauge coupling's b_0 is likewise of the screening sign. Spin zero offers nothing antiscreening.

Fermions screen. A charged Dirac fermion contributes to a gauge coupling's b_0 exactly the QED-type term: in QED itself it produced $\beta_e = e^3/12\pi^2 > 0$ (Eq. (24.108)), charge screening and an ultraviolet pole. In the non-abelian case the very same loop appears as the $-\frac{4}{3}T_F n_f$ term in Eq. (27.124): the minus sign is the screening sign (it *reduces* b_0), and $T_F = \frac{1}{2}$ (Eq. (B.92)) plays the role of the squared charge that weighted the QED loop. Spin one-half screens.

Abelian gauge fields cannot save themselves. A U(1) gauge field is neutral under its own gauge group — the photon carries no charge, $f^{abc} = 0$ — so it has

no self-coupling loop to contribute. The only one-loop contribution to the QED beta function comes from the charged matter in the vacuum-polarization bubble, which screens. There is no C_A term because $C_A = 0$ for U(1). Abelian gauge theory therefore always screens; its beta function is positive whenever there is any charged matter and zero otherwise.

Only the non-abelian gauge self-coupling antiscreens. The single source of a negative contribution to b_0 — a positive contribution that makes the beta function negative — is the gauge field's coupling to itself, the $+\frac{11}{3}C_A$ term. It exists only because a non-abelian gauge boson is *charged* under its own group, $C_A = N \neq 0$ (Eq. (B.102)), and it enters with the opposite sign to all the matter contributions and, crucially, with a larger coefficient. The cubic and quartic gluon vertices of $\mathcal{L}_{\text{YM}}$ (Eq. (25.76)), absent in QED, are precisely what supplies this term. The structural conclusion is exactly Theorem 27.13: remove the non-abelian self-coupling and you remove the only antiscreening contribution there is.

Physical Insight

The sign-by-spin accounting reveals the deep point. Ordinary quantum field theory *screens*: virtual matter, scalar or fermionic, dresses a charge so that it appears weaker at long distance and stronger at short, exactly as a dielectric medium screens a test charge. Screening is the rule. Antiscreening is the exception, and it is available to exactly one kind of field — a charged spin-one field, the non-abelian gauge boson. The reason it must be spin one is the subject of the next subsection: only a spin-one field has the large magnetic moment whose paramagnetic response can overpower the dielectric screening. Asymptotic freedom is, at bottom, a statement that the gluon's magnetism beats its own (and the quarks') dielectric.

27.5.4 The physics of antiscreening: the paramagnetic vacuum

We have proved the sign and located its origin in the C_A term. It remains to explain, correctly and without hand-waving, *why* the non-abelian gauge coupling antiscreens while everything else screens. There are two complementary readings: an algebraic one, which traces the antiscreening to the failure of the QED identity $Z_1 = Z_2$, and a physical one, which identifies it as the paramagnetism of the gluonic vacuum driven by the gluon's chromomagnetic moment. Both are correct; together they are the right picture.

The algebraic fingerprint: $Z_1 \neq Z_2$ and the C_F/C_A split

In QED the charge renormalization constant decomposes as $Z_e = Z_1 Z_2^{-1} Z_3^{-1/2}$ (Eq. (24.95)), and the Ward–Takahashi identity forces $Z_1 = Z_2$ to all orders (Theorem 22.31, derived from Theorem 19.13). The vertex and the electron-field renormalizations cancel identically (Eq. (24.97)), leaving

$$Z_e = Z_3^{-1/2} \qquad \text{(QED)}. \tag{27.138}$$

The entire charge renormalization of QED comes from Z_3, the photon's vacuum polarization — a screening bubble of charged matter, and nothing else. This is the algebraic statement that QED can only screen: the only object that renormalizes the charge is the dielectric response of the vacuum.

In Yang–Mills the same decomposition holds, $Z_g = Z_1 Z_2^{-1} Z_3^{-1/2}$ (assembled in Section 27.4), but the Ward identity does *not* force $Z_1 = Z_2$. The gauge field is itself charged; the Slavnov–Taylor identities (Section 26.7) that generalize Ward–Takahashi relate a larger family of constants and leave the quark–gluon vertex and quark field renormalizing differently. Concretely, the one-loop simple-pole residues of the quark–gluon vertex and the quark self-energy, computed in Section 27.3 (Feynman gauge), were

$$Z_1 = 1 - \frac{g^2}{16\pi^2\,\varepsilon}\,(C_F + C_A), \qquad Z_2 = 1 - \frac{g^2}{16\pi^2\,\varepsilon}\,C_F. \tag{27.139}$$

The abelian-like pieces, both proportional to the *fundamental* Casimir C_F (Eq. (B.94)), are identical in Z_1 and Z_2 and cancel in the ratio exactly as in QED. What survives is the genuinely non-abelian remainder, proportional to the *adjoint* Casimir C_A:

$$Z_1 Z_2^{-1} = 1 - \frac{g^2}{16\pi^2\,\varepsilon}\,C_A + O(g^4). \tag{27.140}$$

This is the precise sense in which C_A measures the violation of $Z_1 = Z_2$: the deviation of the ratio from unity is the gluon's color charge, C_A, and nothing else. In QED ($C_A = 0$) the ratio is one and we recover Eq. (27.138). Tracking the C_A piece of $Z_1 Z_2^{-1}$ together with the C_A piece of Z_3 into the assembly of b_0 (the arithmetic carried out in Section 27.4), the adjoint Casimir contributes the full $+\frac{11}{3}C_A$, while the C_F pieces drop out of the charge renormalization entirely — precisely because they cancel between vertex and field, just as in QED. The upshot is a clean division of labor:

- the C_F (fundamental, QED-like) contributions *cancel* out of the charge renormalization, exactly as the Ward identity makes them cancel in QED; whatever screening they would individually represent never reaches the running coupling;

- the screening that *does* reach the coupling is the quark vacuum-polarization loop, $-\frac{4}{3}T_F n_f$, the direct analog of the QED bubble;
- the antiscreening is the C_A piece, carried by the gluon's self-coupling and tagged unmistakably by the failure $Z_1 \neq Z_2$.

The inequality $Z_1 \neq Z_2$ is thus the algebraic signature of antiscreening: it is the formal record that the gauge boson carries charge, and it is exactly the term that flips the sign of the beta function.

Caution

The individual constants in Eq. (27.139), and the split of $+\frac{11}{3}C_A$ between the vertex/field sector and the gluon self-energy Z_3, are gauge dependent: they take different numerical values in a general R_ξ gauge. Only the combination b_0 in Eq. (27.124) is gauge invariant, as the background-field computation of Section 27.4 confirms by reproducing it with $Z_1 = Z_2$ restored as a manifest Ward identity. The statement that survives in every gauge is the physical one: the coupling renormalization receives an antiscreening contribution proportional to C_A that has no QED analog. The Feynman-gauge bookkeeping above is the cleanest *route* to that gauge-invariant fact, not an additional gauge-dependent claim.

The physical mechanism: the chromomagnetic moment and the paramagnetic vacuum

The algebra tells us *that* the gauge self-coupling antiscreens; the physics tells us *why* a spin-one charged field is the unique object that can. The cleanest physical account computes the response of the vacuum to a long-wavelength external (chromo)magnetic field — the magnetic permeability $\mu_{\rm vac}$ of the vacuum — and is, by construction, gauge invariant, since it asks about the energy of a physical field configuration. We give the argument in the form that makes the sign manifest.

Begin from a fact of relativistic field theory that has no nonrelativistic analog. In a Lorentz-invariant vacuum the product of the dielectric "constant" and the magnetic permeability is fixed,

$$\varepsilon_{\rm vac}\,\mu_{\rm vac} = 1, \tag{27.141}$$

because the speed of propagation of the gauge field's own waves must remain the universal $c = 1$: the vacuum cannot be a medium with an index of refraction different from unity for its own quanta without breaking Lorentz invariance. Equation Eq. (27.141) converts a statement about magnetism into a statement about dielectric screening, and the connection to the running coupling is the

dielectric one. The effective charge seen at distance r is screened by the vacuum's dielectric response: a polarizable medium with $\varepsilon_{\text{vac}} > 1$ shields charge (the long-distance coupling is reduced, the short-distance coupling enhanced — ordinary screening), while a vacuum with $\varepsilon_{\text{vac}} < 1$ *antiscreens* (the coupling grows with distance, falls toward short distance). By Eq. (27.141), $\varepsilon_{\text{vac}} < 1$ is the same statement as $\mu_{\text{vac}} > 1$: *the vacuum antiscreens precisely when it is paramagnetic.* Asymptotic freedom is the paramagnetism of the gluonic vacuum.

What makes a medium paramagnetic? Magnetic susceptibility has two competing pieces, familiar from the quantum mechanics of charged particles in a field:

- *Landau diamagnetism*, the orbital response. A charged particle's cyclotron orbits set up currents that oppose the applied field; this contribution is always diamagnetic ($\mu < 1$, screening), and for the gauge field it is the analog of the ordinary charged-particle bubble. It carries the screening sign.

- *Pauli paramagnetism*, the spin response. A particle with a magnetic moment aligns with the field, lowering the vacuum energy in proportion to the square of its magnetic moment; this contribution is paramagnetic ($\mu > 1$, antiscreening). Its size is set by the magnetic moment, hence by the gyromagnetic ratio.

For a charged particle of spin s with gyromagnetic ratio g_{m}, the paramagnetic term grows as $(g_{\text{m}} s_z)^2$ while the diamagnetic term is fixed by the orbital motion. A scalar ($s = 0$) has no spin term and is purely diamagnetic — it screens. A spin-$\frac{1}{2}$ fermion has $g_{\text{m}} = 2$ and a paramagnetic term, but Fermi statistics flips the overall sign of the closed loop, so a fermion's net effect is again screening (this is the same minus sign that makes the closed fermion loop in QED carry its characteristic factor and screen). The gauge boson is the unique case in which a large spin contribution survives with the antiscreening sign: it is a *boson* of spin one with gyromagnetic ratio $g_{\text{m}} = 2$.

That the gluon has $g_{\text{m}} = 2$ is not an input; it is forced by Yang–Mills gauge invariance. The non-abelian field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ enters the action as $-\frac{1}{4} F^a F^a$ (Eq. (25.76)), and the term in $-\frac{1}{4} F^a F^a$ that is bilinear in a "charged" gluon fluctuation W propagating in a background gauge field $\bar{A}$ contains a piece $g f^{abc} \bar{F}^a_{\mu\nu} W^{b\,\mu} W^{c\,\nu}$, i.e. a direct coupling of the fluctuation's spin to the background field strength with coefficient fixed to be exactly that of a magnetic moment with $g_{\text{m}} = 2$. This is the same value the gauge principle forces on any charged spin-one boson (the W of the electroweak theory has it too), and it is large: it makes the Pauli paramagnetic term dominate.

The quantitative outcome of inserting $s = 1$, two transverse polarizations, and $g_{\text{m}} = 2$ into the vacuum-energy computation reproduces the C_A coefficient

with its sign decomposition exposed. Carrying out the magnetic-energy integral (the Yang–Mills analog of the Euler–Heisenberg calculation, done in a constant chromomagnetic background) splits the adjoint coefficient cleanly:

$$\underbrace{\tfrac{11}{3}}_{\text{from } C_A} = \underbrace{(-\tfrac{1}{3})}_{\text{diamagnetic (orbital): screening}} + \underbrace{(+4)}_{\text{paramagnetic (spin): antiscreening}} . \tag{27.142}$$

The orbital motion of the charged gluon screens, contributing $-\frac{1}{3}$ exactly as a charged scalar's orbital motion would; but the gluon's spin-one magnetic moment contributes $+4$, an antiscreening term four times larger in magnitude and of the opposite sign. Their sum is $+\frac{11}{3}$: paramagnetism wins by a wide margin. The factor 4 is the fingerprint of $g_{\mathrm{m}} = 2$ summed over the two transverse spin states; halve the magnetic moment, or kill it (a scalar, $s = 0$), and the antiscreening term vanishes, leaving pure screening. This is the precise, gauge-invariant content of the assertion that *antiscreening requires the spin-one self-coupling*: it is the chromomagnetic moment of the gluon, and nothing else, that overwhelms the dielectric screening.

Physical Insight

The intuitive picture that screening must always win — that a polarizable vacuum can only shield a charge, as every dielectric we know does — is correct for everything except a charged vector field. The escape is magnetic, not electric. A spin-one charged particle is such a strong little magnet ($g_{\mathrm{m}} = 2$) that the vacuum it fills responds paramagnetically; by the relativistic constraint $\varepsilon\mu = 1$, paramagnetism is antiscreening; and antiscreening means the color charge grows with distance and fades toward short distance. The proton is held together not because the strong force is dielectrically unusual but because the gluon is magnetically unusual. This is why Theorem 27.13 is true: only a non-abelian gauge field supplies a charged spin-one quantum, and only a charged spin-one quantum can make the vacuum a paramagnet.

Historical Note

The history is worth telling accurately, because it is often compressed. The charge-renormalization sign of non-abelian gauge theory was first computed correctly by I. B. Khriplovich in 1969 [293], in Coulomb gauge for SU(2), but its significance went unrecognized and the work was little noticed. The negative coefficient was also obtained, and its meaning understood, by 't Hooft, who announced it at a 1972 conference in Marseille but did not publish; the result therefore entered the canon through the 1973 papers of D. Gross and F. Wilczek [238] and, independently, H. D. Politzer [406],

who computed b_0, recognized asymptotic freedom, and — this was the decisive conceptual step — drew the consequence that perturbation theory is reliable at short distance and could therefore explain the approximate scaling seen in deep-inelastic scattering. The 2004 Nobel Prize recognized Gross, Politzer, and Wilczek; Wilczek's Nobel lecture [512] gives a careful first-person account. The accompanying theorem that no asymptotically free theory exists in four dimensions without non-abelian gauge fields is due to Coleman and Gross the same year [130]. The magnetic, paramagnetic interpretation — the reading we have emphasized as the physical mechanism — was developed later, notably by N. K. Nielsen [368] and by Hughes [272], who made the $\frac{11}{3} = -\frac{1}{3} + 4$ decomposition Eq. (27.142) explicit and identified the antiscreening with the gluon's spin-one magnetic moment. The original 1973 derivations were diagrammatic and gave the number without the picture; the picture came afterward and is what we have placed at the center, because it is what one should *understand*.

27.5.5 The license for perturbation theory

The practical payoff of Theorem 27.11 deserves to be stated as the conceptual result it is, because the entire next chapter rests on it. In a screening theory the coupling is small at low energy and grows toward high energy, so a perturbative expansion — an expansion in the coupling — is reliable in the infrared and degrades in the ultraviolet, eventually failing at the Landau pole. In an asymptotically free theory the situation is exactly inverted. By Eq. (27.133) the coupling becomes *arbitrarily small* at high energy, so the leading terms of a perturbative expansion become an *arbitrarily good* approximation to short-distance, high-momentum-transfer processes. Hard scattering — deep-inelastic lepton–nucleon scattering, jet production, heavy-quark production, anything probing distances short compared to $1/\Lambda_{\mathrm{QCD}}$ — is computable in QCD perturbation theory, with corrections controlled by a coupling that the theory itself guarantees is small. This is the logical license for the entire program of perturbative QCD: it is not an assumption that the strong coupling happens to be weak where we calculate, but a theorem that it *must* be weak there. Asymptotic freedom is the reason a theory named for being *strong* can nonetheless be calculated.

The same formula that grants this ultraviolet license raises, with equal force, the complementary question. If Eq. (27.130) drives α_s to zero in the ultraviolet, it drives α_s *up* toward the infrared, with the denominator vanishing at the finite scale Eq. (27.134) lying below μ. Perturbation theory, reliable at short distance, must break down at long distance — and where it breaks down a new mass scale appears from nothing but the dimensionless coupling and the renormalization scale. Turning the one-loop flow into the quantitative running $\alpha_s(Q^2)$, extracting that emergent scale Λ_{QCD} by dimensional transmutation,

fixing the SU(3) numbers, and scoping honestly where and how the perturbative description fails in the infrared is the work of the next section.

27.6 The Running Coupling, Dimensional Transmutation, and $\Lambda_{\rm QCD}$

The previous section turned the sign of the beta function into a qualitative verdict: because $\beta_g < 0$, the Yang–Mills coupling is swept logarithmically to zero in the deep ultraviolet, and the short-distance theory is asymptotically free. That verdict is correct but incomplete. "The coupling vanishes in the UV" is a limiting statement; experiment and phenomenology demand a *number* — the value of the coupling at a stated momentum scale, with a stated uncertainty, and a prescription for moving it from one scale to another. The renormalization-group machinery of Chapter 24 was built precisely to deliver that. In this section we integrate the one-loop flow exactly, obtaining the running $\alpha_s(Q^2)$ in closed form. The integration introduces, unavoidably, one constant of integration; trading the dimensionless coupling for that constant is *dimensional transmutation*, and the constant is the celebrated scale $\Lambda_{\rm QCD}$. We then put the SU(3) numbers in, run α_s down from the Z pole to extract $\Lambda^{(5)}$ explicitly, and confront the place where the whole construction breaks: as $Q \to \Lambda$ the one-loop coupling *diverges*, perturbation theory fails, and the door opens onto the non-perturbative infrared. Finally we map the boundary of asymptotic freedom in flavour number — the loss of antiscreening at large n_f and the Banks–Zaks fixed point at the edge of the conformal window.

27.6.1 From the beta function to the running coupling

Everything follows from the one-loop Yang–Mills beta function computed earlier in this chapter (Eq. (27.101)),

$$\beta_g \;\equiv\; \mu\,\frac{dg}{d\mu} \;=\; -\,\frac{g^3}{16\pi^2}\,b_0, \qquad b_0 \;=\; \frac{11}{3}\,C_A - \frac{4}{3}\,T_F\,n_f, \tag{27.143}$$

with C_A the adjoint Casimir, $T_F = \frac{1}{2}$ the fundamental Dynkin index, and n_f the number of Dirac fermions in the fundamental (the colour data live in Eq. (B.114)). For QCD the antiscreening gluon loop makes $b_0 > 0$, so $\beta_g < 0$. We want the flow of the physically natural variable, the strong coupling

$$\alpha_s \;\equiv\; \frac{g^2}{4\pi}, \tag{27.144}$$

the non-abelian analogue of the fine-structure constant. Its beta function follows from Eq. (27.143) by the chain rule. Since $\alpha_s = g^2/4\pi$,

$$\mu\,\frac{d\alpha_s}{d\mu} = \frac{2g}{4\pi}\,\mu\,\frac{dg}{d\mu} = \frac{g}{2\pi}\,\beta_g = \frac{g}{2\pi}\left(-\frac{g^3}{16\pi^2}\,b_0\right) = -\,\frac{g^4}{32\pi^3}\,b_0. \tag{27.145}$$

Now $g^4 = (4\pi\alpha_s)^2 = 16\pi^2\alpha_s^2$, so the π's collapse cleanly:

$$\mu\,\frac{d\alpha_s}{d\mu} = -\,\frac{16\pi^2\,\alpha_s^2}{32\pi^3}\,b_0 = -\,\frac{b_0}{2\pi}\,\alpha_s^2. \tag{27.146}$$

It is almost always more convenient to differentiate with respect to the logarithm of the squared scale, because cross sections depend on Q^2, not Q. Using $\mu\,\partial_\mu = 2\,\partial_{\ln\mu^2}$, Eq. (27.146) becomes the form we will integrate,

$$\frac{d\alpha_s}{d\ln\mu^2} \;=\; -\,\frac{b_0}{4\pi}\,\alpha_s^2 \;+\; O(\alpha_s^3). \tag{27.147}$$

This is exactly the marginal flow $d\bar g/dt = b\,\bar g^2$ analysed in general in Theorem 24.27, here with $\bar g \to \alpha_s$, $t \to \ln\mu^2$, and the all-important coefficient $b = -b_0/4\pi$ *negative*. The negative sign is the entire difference between QCD and QED, where the same calculation gave a positive coefficient and the running of Eq. (24.215); we now follow the consequences of the opposite sign.

To pass from the abstract scale μ to a physical momentum, we invoke the momentum-scaling solution of the renormalization group, Theorem 24.22 and its momentum form Eq. (24.202): a Green function probed at an external momentum scale Q is governed not by the coupling defined at some fixed reference μ but by the running coupling evaluated at $\mu = Q$. Concretely, the large logarithms $\ln(Q^2/\mu^2)$ that would otherwise plague fixed-order perturbation theory are resummed by the simple choice $\mu = Q$, after which the residual expansion is in the running $\alpha_s(Q^2) \equiv \bar\alpha_s(\mu = Q)$. We may therefore read Eq. (27.147) directly as an equation for the Q-dependence of the physical coupling, replacing $\mu \to Q$:

$$\frac{d\alpha_s}{d\ln Q^2} \;=\; -\,\frac{b_0}{4\pi}\,\alpha_s^2(Q^2). \tag{27.148}$$

Here $Q^2 = -q^2 > 0$ is a spacelike momentum-transfer invariant — a Euclidean scale of the kind that labels a deep-inelastic or large-angle scattering process, the Mandelstam $t < 0$ of Theorem 16.20, made positive — so that $\alpha_s(Q^2)$ is real and the logarithm is well defined throughout the asymptotically free regime. The restriction to spacelike Q^2 avoids the timelike branch cuts and absorptive parts that would otherwise complicate the running; those analytic subtleties belong to the dispersive treatment and do not affect the leading logarithmic running we extract here.

Equation (27.148) is a separable first-order ordinary differential equation, and we integrate it exactly. Dividing by α_s^2 and multiplying by $d\ln Q^2$,

$$\frac{d\alpha_s}{\alpha_s^2} \;=\; -\,\frac{b_0}{4\pi}\,d\ln Q^2. \tag{27.149}$$

Integrate both sides from a reference scale μ_0, where the coupling takes the value $\alpha_s(\mu_0^2)$, up to Q. The left side gives $-1/\alpha_s$; the right side gives $-(b_0/4\pi)\ln(Q^2/\mu_0^2)$:

$$-\,\frac{1}{\alpha_s(Q^2)} + \frac{1}{\alpha_s(\mu_0^2)} \;=\; -\,\frac{b_0}{4\pi}\,\ln\frac{Q^2}{\mu_0^2}, \tag{27.150}$$

which rearranges to the *implicit, reference-scale form* of the running coupling,

$$\boxed{\frac{1}{\alpha_s(Q^2)} = \frac{1}{\alpha_s(\mu_0^2)} + \frac{b_0}{4\pi}\ln\frac{Q^2}{\mu_0^2}.} \tag{27.151}$$

This is the QCD specialization of the general implicit RG solution Eq. (24.187), and it already contains the physics. The *inverse* coupling $1/\alpha_s$ runs *linearly* in $\ln Q^2$, with slope $+b_0/4\pi$. Because $b_0 > 0$, the slope is positive: as Q increases, $1/\alpha_s$ grows, so α_s *decreases* — asymptotic freedom, now quantitative. Conversely, lowering Q decreases $1/\alpha_s$ until it reaches zero, at which point α_s diverges. The left panel of Fig. 27.6 is nothing but a plot of Eq. (27.151): an exact straight line, the cleanest possible signature of one-loop running, crossing zero at the scale where the coupling blows up.

Physical Insight

Compare the two abelian/non-abelian companions side by side. In QED, Eq. (24.215) gives $1/\alpha(Q^2) = 1/\alpha(\mu_0^2) - (1/3\pi)\ln(Q^2/\mu_0^2)$: the inverse coupling *falls* with energy, the charge is screened and grows toward the UV, and the line crosses zero at the unreachable Landau pole far above the Planck scale. In QCD the slope flips sign: $1/\alpha_s$ *rises* with energy, the colour charge is antiscreened and weakens toward the UV, and the line crosses zero in the *infrared*, at an experimentally accessible scale of a few hundred MeV. The same one-loop algebra, run with opposite-sign b_0, produces a UV-free theory whose strong-coupling crisis sits at low energy instead of a high-energy one. This single sign is why hadrons exist.

27.6.2 Dimensional transmutation and the scale Λ_{QCD}

Equation (27.151) expresses $\alpha_s(Q^2)$ in terms of two pieces of data: the slope $b_0/4\pi$, which is fixed by the theory's field content, and the boundary value $\alpha_s(\mu_0^2)$ at an arbitrary reference scale μ_0. The pair $(\mu_0, \alpha_s(\mu_0^2))$ is redundant, however: the line in the $(\ln Q^2,\, 1/\alpha_s)$ plane is fixed by its slope and a single point on it, and any point on the line serves equally well. The invariant content of the line is not a dimensionless number tied to a scale but a single *scale*: the place where the line meets the axis $1/\alpha_s = 0$. We promote that intersection to the defining datum.

Definition 27.14 (The QCD scale Λ)**.** The one-loop scale Λ (for given b_0, i.e. given n_f) is the value of Q at which the linearly-running inverse coupling Eq. (27.151) vanishes:

$$\frac{1}{\alpha_s(\Lambda^2)} = 0 \quad\Longleftrightarrow\quad \frac{1}{\alpha_s(\mu_0^2)} = \frac{b_0}{4\pi}\ln\frac{\mu_0^2}{\Lambda^2}, \tag{27.152}$$

equivalently, solving for Λ,

$$\Lambda^2 \;=\; \mu_0^2 \exp\!\left[-\frac{4\pi}{b_0\,\alpha_s(\mu_0^2)}\right], \qquad \Lambda \;=\; \mu_0 \exp\!\left[-\frac{2\pi}{b_0\,\alpha_s(\mu_0^2)}\right]. \tag{27.153}$$

The point of the definition is that it eliminates the reference data entirely. Substituting Eq. (27.152) for the constant $1/\alpha_s(\mu_0^2)$ back into Eq. (27.151), the μ_0 dependence cancels by the additivity of logarithms,

$$\begin{aligned}\frac{1}{\alpha_s(Q^2)} &= \frac{b_0}{4\pi}\ln\frac{\mu_0^2}{\Lambda^2} + \frac{b_0}{4\pi}\ln\frac{Q^2}{\mu_0^2} \;=\; \frac{b_0}{4\pi}\left(\ln\frac{\mu_0^2}{\Lambda^2} + \ln\frac{Q^2}{\mu_0^2}\right) \\ &= \frac{b_0}{4\pi}\ln\frac{Q^2}{\Lambda^2},\end{aligned} \tag{27.154}$$

and we obtain the running coupling in its most economical form, the central result of the section:

$$\boxed{\alpha_s(Q^2) \;=\; \frac{4\pi}{b_0\,\ln(Q^2/\Lambda^2)}\,.} \tag{27.155}$$

Two dimensionful scales have collapsed to one. The coupling at *any* momentum is now specified by the single parameter Λ — a quantity with the dimensions of mass — together with the field-content coefficient b_0. This trade is what is meant by

Dimensional transmutation

The classical Yang–Mills Lagrangian Eq. (25.76) with massless quarks contains *no dimensionful parameter at all*: g is dimensionless in four dimensions, and there is no mass. The classical theory is therefore exactly scale invariant — it looks the same at every magnification, and cannot by itself predict any absolute energy or length. Quantization breaks this. Regularizing the loop integrals required an auxiliary scale μ, and the resulting running Eq. (27.155) converts the dimensionless input $\alpha_s(\mu_0)$ at the arbitrary scale μ_0 into a genuine, physical, dimensionful scale Λ. A theory with no scale has dynamically *generated* one. This phenomenon — a dimensionless coupling traded for a dimensionful scale through the logarithmic running of the quantum theory — is *dimensional transmutation*. It is why the proton has a mass of roughly 1 GeV although the Lagrangian of the light quarks and gluons contains nothing of the kind: the proton mass is set by Λ_{QCD}, an integration constant of the renormalization group, not by any parameter written into the action.

The breaking of classical scale invariance has a sharp operator statement, the *trace anomaly*: the trace of the energy–momentum tensor, which vanishes

for a classically scale-invariant massless theory (Eq. (7.214)), is nonzero in the quantum theory and proportional to the beta function,

$$\Theta^{\mu}{}_{\mu} \;=\; \frac{\beta(g)}{2g}\,\big[F^{a}_{\mu\nu}F^{a\,\mu\nu}\big]_{R}, \tag{27.156}$$

the non-abelian counterpart of the QED relation derived in Eqs. (24.166) and (24.167). The same negative β that drives asymptotic freedom is, through Eq. (27.156), the source of a nonzero trace and hence of the dynamically generated scale; dimensional transmutation and the trace anomaly are two faces of one fact. The full anomaly-multiplet structure is the business of the anomalies development (Chapter 36); here we need only the conceptual link, which Eq. (24.166) already establishes.

Λ is renormalization-group invariant. The claim that Λ is a property of the *theory*, not of the arbitrary reference μ_0, deserves a proof, because it is the precise sense in which the transmuted scale is physical.

Proposition 27.15 (RG invariance of Λ at one loop)**.** *The scale Λ defined by Eq. (27.153) is independent of the reference scale μ_0 used to define it: $\mu_0\, d\Lambda/d\mu_0 = 0$, provided $\alpha_s(\mu_0^2)$ runs according to Eq. (27.147).*

Proof. Take the logarithm of the first equality in Eq. (27.153),

$$\ln\Lambda^2 \;=\; \ln\mu_0^2 \;-\; \frac{4\pi}{b_0\,\alpha_s(\mu_0^2)}, \tag{27.157}$$

and differentiate with respect to $\ln\mu_0^2$, treating α_s as the running coupling at scale μ_0. The first term contributes 1. The second term contributes, by the chain rule and Eq. (27.147) (read at scale μ_0),

$$-\frac{4\pi}{b_0}\,\frac{d}{d\ln\mu_0^2}\!\left(\frac{1}{\alpha_s}\right) = -\frac{4\pi}{b_0}\left(-\frac{1}{\alpha_s^2}\right)\frac{d\alpha_s}{d\ln\mu_0^2} = \frac{4\pi}{b_0\,\alpha_s^2}\left(-\frac{b_0}{4\pi}\,\alpha_s^2\right) = -1. \tag{27.158}$$

The two contributions cancel: $d\ln\Lambda^2/d\ln\mu_0^2 = 1 - 1 = 0$. Hence Λ does not depend on μ_0, which is the assertion. □

So whatever scale one chooses to fix the coupling at, the line in Fig. 27.6 — and therefore its intercept Λ — is the same. The dimensionless number $\alpha_s(\mu_0)$ and the choice of μ_0 have been losslessly repackaged into the single RG-invariant Λ.

What Λ depends on: scheme and convention. The invariance just proved is invariance under change of *reference scale*, not under change of *renormalization scheme*. The one-loop coefficient b_0 is scheme-independent (this universality of the first two beta coefficients was established for the threads in Section 24.2, and the argument carries over verbatim to α_s), but the value of Λ is not. The reason is that different schemes define the coupling itself differently at finite order. Suppose two schemes relate their couplings by a finite reparametrization,

$$\alpha_s' \;=\; \alpha_s \;+\; c_1\,\alpha_s^2 \;+\; O(\alpha_s^3), \tag{27.159}$$

with c_1 a calculable finite constant (for instance the constant relating $\overline{\text{MS}}$ to a momentum-subtraction scheme). Inverting to the same order, $1/\alpha_s' = 1/\alpha_s - c_1 + O(\alpha_s)$. Insert this into the primed version of Eq. (27.154), $1/\alpha_s'(Q^2) = (b_0/4\pi)\ln(Q^2/\Lambda'^2)$, and use the unprimed one:

$$\frac{b_0}{4\pi}\ln\frac{Q^2}{\Lambda'^2} = \frac{1}{\alpha_s'(Q^2)} = \frac{1}{\alpha_s(Q^2)} - c_1 = \frac{b_0}{4\pi}\ln\frac{Q^2}{\Lambda^2} - c_1. \tag{27.160}$$

The Q-dependence cancels, leaving a relation between the two scales alone,

$$\ln\frac{\Lambda'^2}{\Lambda^2} = \frac{4\pi c_1}{b_0}, \qquad \boxed{\Lambda' = \Lambda\,\exp\!\left(\frac{2\pi c_1}{b_0}\right).} \tag{27.161}$$

Thus Λ carries a definite, calculable scheme dependence: it is multiplied by a fixed finite factor when one passes between schemes, with the factor determined by the one-loop matching constant c_1 and the universal coefficient b_0. This is not a defect; it is the price of Λ being defined through the coupling. The consequence to internalize is that a numerical value of Λ_{QCD} is meaningless until the scheme is named. The community convention is the $\overline{\text{MS}}$ scheme, and quoted values are $\Lambda_{\overline{\text{MS}}}^{(n_f)}$. There is a further convention buried in Theorem 27.14: at higher loop order the integration of the beta function leaves additional constants, and the *standard* definition of $\Lambda_{\overline{\text{MS}}}$ fixes them in a particular way (absorbing the two-loop term into the argument of the logarithm). At strict one loop, Eq. (27.155) is the unambiguous content, and that is the level at which we work in this section.

Notation Watch

Two coexisting normalizations of "b_0" appear in this book and must not be confused. Yang–Mills theory carries the coefficient in the form that sits in the beta function for g, Eq. (27.143): $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$, which for SU(3) is $11 - \frac{2}{3}n_f$. The QCD phenomenology of Chapter 28 writes the α_s flow as $d\alpha_s/d\ln Q^2 = -\alpha_s^2(\hat{b}_0 + \hat{b}_1\alpha_s + \dots)$ with $\hat{b}_0 = (33 - 2n_f)/12\pi$. These are the same physics in different clothes: $\hat{b}_0 = b_0/4\pi$, since $(33 - 2n_f)/12\pi = (11 - \frac{2}{3}n_f)/4\pi$. Throughout *this* section b_0 means the Yang–Mills theory

coefficient $11 - \frac{2}{3}n_f$, and the explicit 4π in Eq. (27.155) is the conversion factor. When reading Eq. (28.184), divide by 4π.

27.6.3 The SU(3) numbers

We now specialize to the gauge group of the strong interaction, SU(3). The colour data are $C_A = N = 3$ and $T_F = \frac{1}{2}$ (Eq. (B.114); the explicit evaluation of the traces that produce the $\frac{11}{3}C_A$ gluon-plus-ghost contribution and the $-\frac{4}{3}T_F n_f$ quark contribution is carried out in Theorem B.29). Substituting into Eq. (27.143),

$$b_0 \;=\; \frac{11}{3}\cdot 3 \;-\; \frac{4}{3}\cdot\frac{1}{2}\,n_f \;=\; 11 - \frac{2}{3}\,n_f, \tag{27.162}$$

the carried Yang–Mills theory number. The first term, $+11$, is the antiscreening contribution of the gluon self-interaction and the Faddeev–Popov ghosts; the second, $-\frac{2}{3}n_f$, is the screening contribution of n_f quark flavours, identical in structure to the QED vacuum-polarization screening but reduced by the colour factor $T_F = \frac{1}{2}$. The contest between 11 and $\frac{2}{3}n_f$ decides the sign of b_0 and hence the fate of the theory.

For the real world the number of active flavours depends on the scale: a quark of mass m_q participates in the running only at scales $Q \gtrsim m_q$. Below the charm threshold $n_f = 3$ (u, d, s); between charm and bottom $n_f = 4$; between bottom and top $n_f = 5$; above the top $n_f = 6$. In every one of these cases $b_0 > 0$: with six flavours, $b_0 = 11 - 4 = 7 > 0$, comfortably positive. QCD is asymptotically free for the physical flavour content by a wide margin.

The running coupling Eq. (27.155) requires one measured input to fix Λ. The conventional anchor is the value of α_s at the mass of the Z boson, $M_Z = 91.19\,\mathrm{GeV}$, where five flavours are active and many precise measurements converge. The mid-2020s world average (the determination assembled in Fig. 28.8 from observables spanning three orders of magnitude in scale) is

$$\alpha_s(M_Z^2) \;=\; 0.1180 \pm 0.0009. \tag{27.163}$$

With this single number, $b_0 = 11 - \frac{2}{3}n_f$, and Eq. (27.153), the coupling is determined at every scale. The next worked example carries that determination out and extracts $\Lambda^{(5)}$.

Worked Example 27.16 (Running α_s down from M_Z and extracting $\Lambda^{(5)}$)**.** We work at one loop with $n_f = 5$ active flavours, so

$$b_0 \;=\; 11 - \frac{2}{3}\cdot 5 \;=\; 11 - \frac{10}{3} \;=\; \frac{23}{3} \;\approx\; 7.667. \tag{27.164}$$

Anchor on Eq. (27.163): $\alpha_s(M_Z^2) = 0.1180$ at $\mu_0 = M_Z = 91.19\,\mathrm{GeV}$.

Step 1 — extract $\Lambda^{(5)}$. Equation Eq. (27.153) needs the exponent

$$\frac{4\pi}{b_0\,\alpha_s(M_Z^2)} = \frac{4\pi}{(23/3)(0.1180)} = \frac{12.566}{0.9047} = 13.89. \tag{27.165}$$

Hence, from Eq. (27.153),

$$\Lambda^{(5)} = M_Z\, e^{-\,13.89/2} = M_Z\, e^{-6.945} = 91.19\ \text{GeV} \times 9.63 \times 10^{-4} = 0.0878\ \text{GeV}. \tag{27.166}$$

The one-loop five-flavour scale is $\Lambda^{(5)}_{\text{1-loop}} \approx 88\,\text{MeV}$. (The factor $e^{-6.945} = 9.63 \times 10^{-4}$ follows from $e^{-7} = 9.12 \times 10^{-4}$ times $e^{0.055} = 1.057$.)

Step 2 — run the coupling to a few scales. With $\Lambda = 0.0878\,\text{GeV}$ fixed, Eq. (27.155) reads $\alpha_s(Q^2) = 4\pi/(\frac{23}{3}\ln(Q^2/\Lambda^2))$. At $Q = 1\,\text{GeV}$,

$$\ln\frac{Q^2}{\Lambda^2} = \ln\frac{1}{(0.0878)^2} = \ln(129.7) = 4.865, \quad \alpha_s(1\,\text{GeV}) = \frac{12.566}{7.667 \times 4.865} = 0.337. \tag{27.167}$$

At $Q = 2\,\text{GeV}$, $\ln(Q^2/\Lambda^2) = \ln(518.9) = 6.252$ and $\alpha_s = 0.262$. As a consistency check, at $Q = M_Z$ the logarithm is $\ln(M_Z^2/\Lambda^2) = 2\ln(91.19/0.0878) = 2\times 6.945 = 13.89$, so $\alpha_s = 12.566/(7.667 \times 13.89) = 0.118$, reproducing the input Eq. (27.163) as it must. The coupling has nearly tripled, from 0.118 at the Z pole to 0.34 at 1 GeV — the perturbative expansion parameter α_s/π has grown from a comfortable 4% to a marginal 11%.

Step 3 — locate the breakdown. Perturbation theory is reliable only while α_s is small. Take $\alpha_s = 0.5$ as a generous boundary of trust. Solving Eq. (27.155) for the scale,

$$\ln\frac{Q^2}{\Lambda^2} = \frac{4\pi}{b_0\,\alpha_s} = \frac{12.566}{7.667 \times 0.5} = 3.278, \quad Q = \Lambda\, e^{3.278/2} = \Lambda\, e^{1.639} = 5.15\,\Lambda \approx 0.45\ \text{GeV}. \tag{27.168}$$

Below roughly half a GeV the one-loop α_s exceeds 0.5 and the loop expansion can no longer be trusted; at $Q = \Lambda \approx 88\,\text{MeV}$ it diverges outright. The right panel of Fig. 27.6 shows this directly.

Honest caveat. The value $\Lambda^{(5)}_{\text{1-loop}} \approx 88\,\text{MeV}$ is markedly smaller than the realistic $\overline{\text{MS}}$ value $\Lambda^{(5)}_{\overline{\text{MS}}} \approx 210\,\text{MeV}$ obtained from the four- to five-loop beta function with the same input Eq. (27.163). The discrepancy is entirely expected and instructive: Λ is exponentially sensitive to the slope of the running, $\Lambda \propto \exp(-2\pi/b_0\alpha_s)$, so the higher-loop corrections to the beta function — modest as fractional changes to α_s at high scale — shift Λ by a factor of order two. Moreover, as established in Eq. (27.161), Λ is scheme-dependent; "210 MeV" is a statement about $\overline{\text{MS}}$ at multi-loop order. The one-loop number on this page is fully reproducible from Eq. (27.163) and is the correct one-loop answer; it is not the number one quotes to a phenomenologist, and we say so plainly.

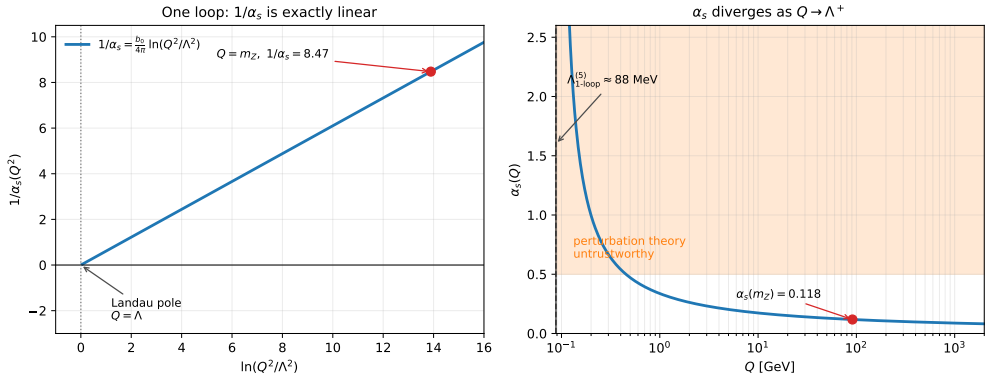


Figure 27.6: The one-loop running strong coupling for $n_f = 5$ ($b_0 = 23/3$), with the five-flavour scale $\Lambda^{(5)}_{\text{1-loop}} \approx 88\,\text{MeV}$ extracted in Theorem 27.16 by anchoring on $\alpha_s(M_Z^2) = 0.1180$. *Left:* the inverse coupling $1/\alpha_s$ is, at one loop, an *exact* straight line in $\ln(Q^2/\Lambda^2)$ with slope $b_0/4\pi$; it passes through $1/0.1180 = 8.47$ at $Q = M_Z$ and crosses zero at $Q = \Lambda$, the Landau pole. *Right:* the same coupling $\alpha_s(Q)$ versus Q, diverging as $Q \to \Lambda^+$ and falling logarithmically toward the ultraviolet. The shaded band $\alpha_s > 0.5$ marks where the perturbative expansion ceases to be trustworthy — the regime handed to the non-perturbative methods of Chapter 27's closing section and Parts XII–XIII.

27.6.4 Infrared honesty: the Landau pole and the failure of perturbation theory

The formula Eq. (27.155) is exact within the one-loop approximation, and it carries its own warning label. As Q decreases toward Λ, the logarithm $\ln(Q^2/\Lambda^2) \to 0^+$, and the coupling *diverges*:

$$\alpha_s(Q^2) = \frac{4\pi}{b_0 \ln(Q^2/\Lambda^2)} \xrightarrow{Q\to\Lambda^+} +\infty. \tag{27.169}$$

This is the *infrared Landau pole* of QCD — the exact mirror image of the ultraviolet Landau pole of QED (Theorem 24.28). In QED the inverse coupling fell with energy and hit zero at an absurd, unphysical high scale; in QCD the inverse coupling falls toward *low* energy and hits zero at $Q = \Lambda$, a scale of a few hundred MeV that sits squarely in the regime of real hadronic physics. The sign of b_0 relocates the pole from the unreachable UV to the experimentally central IR.

What does the pole mean? Exactly what its QED counterpart meant, read in reverse: it is a statement about *perturbation theory*, not necessarily about the theory. The derivation of Eq. (27.155) retained only the one-loop term in β. That truncation is justified only when the expansion parameter α_s/π is small. As $Q \to \Lambda$ the running coupling grows through $\alpha_s \sim 1$, the higher-loop terms in

the beta function — the $b_1\alpha_s^3$ term and beyond — cease to be small corrections, and the one-loop formula loses its warrant well before it formally diverges. The divergence at $Q = \Lambda$ is therefore not a prediction of a physical infinity; it is the perturbative expansion announcing its own demise. The honest reading is the one drawn in Fig. 27.6: the curve is trustworthy in the white region and fictitious in the shaded one.

Caution

It is tempting to read the IR Landau pole as "the coupling really becomes infinite at $Q = \Lambda$," and then to attribute confinement to that infinity. This is wrong on both counts. First, the location and even the existence of the pole are artifacts of the one-loop truncation and the scheme; nothing physical happens *at* a scheme-dependent scale. Second, the genuine non-perturbative coupling, defined through a physical observable, need not diverge at all — lattice and Schwinger–Dyson studies suggest it freezes to a finite value in the deep infrared. What is robust is only the *qualitative* message that perturbation theory in α_s is uncontrolled for $Q \lesssim 1\,\mathrm{GeV}$. Confinement is a separate, non-perturbative statement, formulated in the next section through the Wilson loop, not a re-reading of Eq. (27.169).

The constructive content of the breakdown is that it tells us *where* a different language is required. Above a few GeV, α_s is small and the perturbative parton picture — the running coupling, the splitting functions, the factorization theorems of Chapter 28 — is quantitative. Below about 1 GeV the coupling is $O(1)$, no finite-order computation converges, and the physical degrees of freedom are no longer quarks and gluons but the colour-singlet hadrons. The scale Λ_{QCD} is the watershed between these two descriptions. That a single, dynamically generated scale should separate a calculable short-distance regime from an intrinsically strong long-distance regime is the deepest qualitative fact about QCD, and it is the reason a non-perturbative criterion for confinement is needed at all.

27.6.5 Loss of asymptotic freedom and the conformal window

Asymptotic freedom is not automatic; it is contingent on $b_0 > 0$, and Eq. (27.162) shows that the quark contribution erodes it. As n_f grows, the screening term $\frac{2}{3}n_f$ eats into the antiscreening 11, and at

$$b_0 = 0 \quad \Longleftrightarrow \quad 11 - \frac{2}{3}\,n_f = 0 \quad \Longleftrightarrow \quad n_f = \frac{33}{2} = 16.5 \tag{27.170}$$

the one-loop coefficient vanishes. For $n_f \geq 17$ flavours, $b_0 < 0$: the sign of the beta function flips, the coupling *grows* toward the UV exactly as in QED, asymptotic freedom is lost, and the theory acquires an ultraviolet Landau pole.

The window of asymptotic freedom for SU(3) is therefore $1 \le n_f \le 16$. Real QCD, with six flavours, lives safely inside it.

The boundary is richer than the one-loop coefficient alone reveals, and exploring it requires the two-loop beta function. For SU(3) the two-loop running coefficient is (in the same normalization as Eq. (27.143), written for α_s for compactness)

$$\frac{d\alpha_s}{d\ln Q^2} = -\frac{\alpha_s^2}{4\pi}\,b_0 - \frac{\alpha_s^3}{(4\pi)^2}\,b_1 + O(\alpha_s^4), \qquad b_1 = 102 - \frac{38}{3}\,n_f, \tag{27.171}$$

where $b_1 = \frac{34}{3}C_A^2 - (\frac{20}{3}C_A + 4C_F)T_F n_f$ evaluated at $C_A = 3$, $C_F = \frac{4}{3}$, $T_F = \frac{1}{2}$ gives $b_1 = 102 - \frac{38}{3}n_f$. (This is consistent with the $\hat b_1 = (153 - 19n_f)/24\pi^2$ of Eq. (28.184) under the rescaling $\hat b_1 = b_1/(4\pi)^2$.) Now consider n_f just below the upper edge 16.5. There b_0 is small and positive, while b_1 is large and negative — at $n_f = 16$, $b_1 = 102 - \frac{38}{3}\cdot 16 = -100.7 < 0$. With $b_0 > 0$ and $b_1 < 0$ the two-loop beta function has, besides the Gaussian zero at $\alpha_s = 0$, a *second* zero at positive coupling:

$$0 = -\frac{\alpha_s^2}{4\pi}\left(b_0 + \frac{\alpha_s}{4\pi}\,b_1\right) \quad\Longrightarrow\quad \alpha_s^* = -\,4\pi\,\frac{b_0}{b_1} \;>\; 0, \tag{27.172}$$

since $b_0 > 0$ and $b_1 < 0$ make the ratio negative and the overall sign positive. This is the *Banks–Zaks fixed point*. Because b_0 can be made arbitrarily small by tuning n_f toward 16.5, the fixed-point coupling α_s^* can be made arbitrarily small, and the fixed point is then *weakly coupled* — which is precisely the condition under which the two-loop truncation that produced it is self-consistent and trustworthy.

Proposition 27.17 (Banks–Zaks infrared fixed point)**.** *For* SU(3) *with n_f flavours such that $b_0 = 11 - \frac{2}{3}n_f$ is small and positive (i.e. n_f just below* 16.5*), the two-loop beta function Eq. (27.171) possesses a nontrivial zero at $\alpha_s^* = -4\pi\,b_0/b_1$, with $b_1 < 0$. This zero is an* infrared-*attractive fixed point: a coupling started below α_s^* in the ultraviolet flows up to α_s^* in the infrared, and one started above flows down to it.*

Proof. That α_s^* is a zero is Eq. (27.172). For attractivity, examine the sign of $\beta_{\alpha_s} = d\alpha_s/d\ln Q^2$ near α_s^*. Write $\beta_{\alpha_s} = -(\alpha_s^2/4\pi)\,b_0\,(1 - \alpha_s/\alpha_s^*)$, using $b_1/4\pi = -b_0/\alpha_s^*$ from Eq. (27.172) to factor the bracket. For $0 < \alpha_s < \alpha_s^*$ the bracket is positive and the prefactor $-(\alpha_s^2/4\pi)b_0$ is negative, so $\beta_{\alpha_s} < 0$: as Q *decreases* (flow to the IR), α_s *increases* toward α_s^*. For $\alpha_s > \alpha_s^*$ the bracket turns negative, $\beta_{\alpha_s} > 0$, and lowering Q *decreases* α_s toward α_s^*. From either side the infrared flow converges on α_s^*, which is the definition of an IR-attractive fixed point (the general criterion is Theorem 24.26, here read with the slope $\beta'(\alpha_s^*) > 0$). □

The physical content is striking. For these large flavour numbers QCD is *still* asymptotically free in the UV (the coupling vanishes as $Q \to \infty$), but instead of growing without bound in the IR it freezes at the finite value α_s^*. A theory whose coupling sits at a fixed point at long distances is scale-invariant in the infrared: it flows to a (here, interacting) conformal field theory. The range of flavour numbers for which SU(3) gauge theory flows to such an IR conformal phase is the *conformal window*. Its upper edge is the perturbatively clean $n_f = 16.5$ just found; its lower edge n_f^*, below which the IR coupling grows strong enough to trigger chiral symmetry breaking and confinement and the theory leaves the conformal phase, is genuinely non-perturbative. As of the mid-2020s the lower edge is estimated by lattice simulation to lie somewhere in the range $n_f^* \approx 8$–12 for SU(3) fundamentals, with the precise value still debated. The Banks–Zaks fixed point is the clean, controlled corner of this story; the conformal phase and its boundary are objects of the conformal-field-theory and bootstrap machinery developed in Part XIV, to which we defer their systematic study (Chapter 52). Here it suffices to have located the boundary of asymptotic freedom precisely, $n_f < 16.5$, and to have exhibited, at two loops, the new infrared physics that emerges as that boundary is approached.

With the running coupling in closed form, the emergent scale Λ_{QCD} identified as a renormalization-group invariant, and the breakdown of perturbation theory pinned to $Q \sim \Lambda$, the quantitative content of asymptotic freedom is complete: Yang–Mills theory's carried number $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ has become the measured curve $\alpha_s(Q^2)$. What remains is the physics on the far side of Λ. The coupling there is strong, no finite-order diagram converges, and the relevant degrees of freedom are colour-singlet hadrons rather than quarks and gluons. To say anything precise about that regime we need a criterion that does not rely on perturbation theory at all. The next section supplies one, built on the Wilson loop of Theorem 25.27: the area-law/perimeter-law dichotomy that turns the qualitative expectation "the coupling becomes strong and quarks are confined" into a sharp, if only partly proved, statement.

27.7 Confinement as a Statement: The Wilson-Loop Criterion

The previous section followed the running coupling into the infrared and watched perturbation theory destroy itself. Integrating the negative beta function gave $\alpha_s(Q^2)$ growing without bound as Q falls toward Λ_{QCD} (Section 27.6), and at $Q \sim \Lambda_{\mathrm{QCD}}$ the one-loop formula returns a coupling of order unity, the two-loop correction of the same size as the one-loop term, and every truncation of the perturbative series equally worthless. This is not a failure of technique but a statement about the physics: the long-distance behaviour of Yang–Mills theory is not weakly coupled, and no amount of diagram summation reaches it. The

objects perturbation theory is built from—gluon and quark propagators, on-shell coloured asymptotic states—are exactly the objects the infrared dynamics refuses to supply. We never see a free quark or a free gluon. What we need, to say anything at all about the strongly coupled regime, is a quantity that is (i) gauge invariant, so that it is a genuine observable rather than an artefact of a gauge choice; (ii) defined non-perturbatively, so that its meaning survives where the series does not; and (iii) sharp enough that its large-distance behaviour distinguishes a theory whose charges are liberated from one whose charges are permanently bound. Remarkably, one object meets all three demands, and we already built it classically in Chapter 25: the Wilson loop. This closing section turns it into the order parameter for confinement, derives—in full—the one provable theorem the subject rests on (that a large rectangular loop measures the static quark–antiquark potential, and that an area law for the loop is *equivalent* to a linearly rising, confining potential), and then states with scrupulous honesty what is proved, what is strong-coupling and numerical evidence, and what remains, as of the mid-2020s, an open problem of mathematics.

27.7.1 The Wilson loop as an order parameter

Recall the central observable of Chapter 25. For a closed oriented contour $\mathcal{C}$ in spacetime and the fundamental representation of $\mathrm{SU}(N)$, the Wilson loop is the gauge-invariant trace of the holonomy (Theorem 25.27, Eq. (25.139)),

$$W(\mathcal{C}) \;=\; \frac{1}{N}\,\mathrm{tr}\;\mathcal{P}\exp\!\Big(ig\oint_{\mathcal{C}} A_\mu\,dx^\mu\Big), \qquad A_\mu = A_\mu^a T^a, \tag{27.173}$$

where $\mathcal{P}$ orders the matrices along $\mathcal{C}$ and we have normalized by $1/N$ so that $W=1$ for the trivial connection. Under a gauge transformation the holonomy is conjugated at the basepoint and the trace is invariant (Theorem 25.26); the object is therefore a true functional on the space of physical configurations, the quotient $\mathcal{A}/\mathcal{G}$ of Section 25.7. Classically it records the curvature enclosed by $\mathcal{C}$, reducing for an infinitesimal plaquette to g times the field strength $F_{\mu\nu}$ of Theorem 25.11. Quantum mechanically the natural quantity is its vacuum expectation value, computed by the gauge-fixed path integral of Chapter 26 with the master Lagrangian $\mathcal{L}$ of Eq. (26.37),

$$\langle W(\mathcal{C})\rangle \;=\; \frac{1}{Z}\int \mathcal{D}A\,\mathcal{D}c\,\mathcal{D}\bar{c}\,\mathcal{D}\psi\,\mathcal{D}\bar{\psi}\;W(\mathcal{C})\;e^{\,iS[A,c,\bar{c},\psi]}, \tag{27.174}$$

and—because $W(\mathcal{C})$ is gauge invariant and the Faddeev–Popov machinery was constructed precisely so that gauge-invariant correlators are independent of the gauge-fixing function—this expectation value is a physical number, independent of ξ and of the gauge choice. It is the same number whether computed in Feynman gauge, in the lattice formulation of Chapter 43, or, in principle, in

the exact theory. The whole content of confinement is encoded in how $\langle W(\mathcal{C})\rangle$ depends on the *size and shape* of $\mathcal{C}$ when the loop is large compared with $\Lambda_{\mathrm{QCD}}^{-1}$.

There are exactly two qualitatively different ways a positive functional of a large loop can behave, and they are named for the geometric quantity that controls the decay of $\langle W\rangle$.

Definition 27.18 (Area law and perimeter law)**.** Let $\mathcal{C}$ be a large planar loop bounding a minimal surface of area $\mathcal{A}$ and having perimeter P. The Wilson loop obeys an *area law* if its expectation value falls off as the exponential of the enclosed area,

$$-\ln\langle W(\mathcal{C})\rangle \xrightarrow{\ \mathcal{C}\ \text{large}\ } \sigma\,\mathcal{A} \;+\; (\text{subleading}), \qquad \sigma>0, \tag{27.175}$$

with σ the *string tension* (mass dimension two). It obeys a *perimeter law* if instead the leading dependence is on the boundary length,

$$-\ln\langle W(\mathcal{C})\rangle \xrightarrow{\ \mathcal{C}\ \text{large}\ } \mu\,P \;+\; (\text{subleading}), \qquad \mu \text{ a mass}, \tag{27.176}$$

i.e. the area term in Eq. (27.175) is absent ($\sigma = 0$).

The dichotomy is sharp because the two terms scale differently: for a loop of linear size L the area grows as L^2 and the perimeter as L, so at large L an area law, if present, always dominates a perimeter law. A perimeter contribution μP is in fact *always* present—it is the self-energy of the static sources running along $\mathcal{C}$, an additive constant per unit length that can be removed by a multiplicative renormalization of the loop, much like the mass counterterm of a heavy particle. The physical question is whether, sitting on top of this unavoidable perimeter piece, there is also an area term. That single yes/no question is the criterion for confinement. The remainder of this section explains why Eq. (27.175) is exactly the statement that the force between static colour charges never turns off.

27.7.2 A rectangular loop measures the static potential

The bridge from the abstract loop functional to a quantity with obvious physical meaning—the potential energy of a quark and an antiquark held a fixed distance apart—is the central provable result of this section. We establish it in Euclidean signature, where the path integral Eq. (27.174) becomes a convergent statistical sum and the time evolution operator $e^{-H\tau}$ projects cleanly onto low-lying states. We use the Wick rotation $x^0 = t \to -i\tau$ of Chapter 10, under which $e^{iS} \to e^{-S_E}$ and a Euclidean time interval τ implements the operator $e^{-H\tau}$ with H the (positive) Yang–Mills Hamiltonian.

The static quark as a temporal Wilson line. Imagine a quark so heavy, $M \to \infty$, that it cannot move: its Compton wavelength shrinks to zero and

it sits at a fixed spatial point $\vec{x}$ for all time. Its Euclidean propagator in a background gauge field collapses to something purely temporal. To see this, take the Euclidean Dirac operator $(\not{D} + M)$ for a fundamental quark and ask for the propagator $S(x, y)$ in a fixed background A_μ. Writing the heavy field as $\psi = e^{-M\tau}\, h$ to strip the rest mass and projecting onto the large component with $P_+ = \frac{1}{2}(1 + \gamma^4)$, the spatial derivatives in $\not{D}$ act through $\gamma^i D_i$ and connect large to small components; the small component is sourced at order $1/M$ and decouples as $M \to \infty$. What remains is a first-order ordinary differential equation in Euclidean time for the large component,

$$\big(\partial_\tau - ig\, A_4(\vec{x}, \tau)\big)\, S(\tau, \tau'; \vec{x}) \;=\; \delta(\tau - \tau')\, \mathbf{1}, \tag{27.177}$$

whose retarded solution is the path-ordered exponential of the temporal connection along the worldline,

$$S(\tau, \tau'; \vec{x}) \;=\; \theta(\tau - \tau')\, \mathcal{P} \exp\!\Big(ig \int_{\tau'}^{\tau} A_4(\vec{x}, s)\, ds \Big) \;\equiv\; \theta(\tau - \tau')\, L_{\vec{x}}(\tau, \tau'), \tag{27.178}$$

times the stripped factor $e^{-M(\tau - \tau')}$ restored at the end. Thus a static quark propagating from Euclidean time τ' to τ at fixed position $\vec{x}$ *is*, up to the trivial mass factor, a straight temporal Wilson line $L_{\vec{x}}$ in the fundamental representation. A static antiquark is the same with the conjugate line $L_{\vec{x}}^\dagger$. This is the heavy-quark limit that underlies all of heavy-quark effective theory (Chapter 40); here we need only its kinematic content, Eq. (27.178).

Assembling the rectangle. Now create a gauge-invariant quark–antiquark pair at separation R at Euclidean time 0 and destroy it at time T. The interpolating operator that does this is the gauge-invariant bilocal built from a spatial Wilson line, exactly the "meson string" anticipated in Section 25.6,

$$\mathcal{O}_R(\tau) \;=\; \bar{q}(\vec{x}_2, \tau) \Big[\mathcal{P} \exp\!\Big(ig \int_{\vec{x}_1}^{\vec{x}_2} \vec{A} \cdot d\vec{x} \Big) \Big]\, q(\vec{x}_1, \tau), \qquad |\vec{x}_2 - \vec{x}_1| = R. \tag{27.179}$$

Its Euclidean two-point function $\big\langle \mathcal{O}_R(T)\, \mathcal{O}_R^\dagger(0) \big\rangle$ is computed by integrating out the heavy quarks. Each quark line is replaced by its static propagator Eq. (27.178): the quark at $\vec{x}_1$ supplies a temporal line $L_{\vec{x}_1}(T, 0)$, the antiquark at $\vec{x}_2$ supplies $L_{\vec{x}_2}^\dagger(T, 0)$, and the source and sink operators supply the two spatial lines at $\tau = 0$ and $\tau = T$. The four lines close into the boundary of an $R \times T$ rectangle $\mathcal{C}_{R,T}$, and the colour traces combine into the single fundamental trace of the holonomy around it. The heavy-mass factors collect into an overall e^{-2MT} (the two worldlines have total length $2T$). Therefore, exactly,

$$\big\langle \mathcal{O}_R(T)\, \mathcal{O}_R^\dagger(0) \big\rangle \;=\; e^{-2MT}\, \big\langle W(\mathcal{C}_{R,T}) \big\rangle\,, \tag{27.180}$$

the heavy-quark correlator is the rectangular Wilson loop dressed by the trivial self-mass. (A figure of the rectangle and its four lines is shown in Fig. 27.7.)

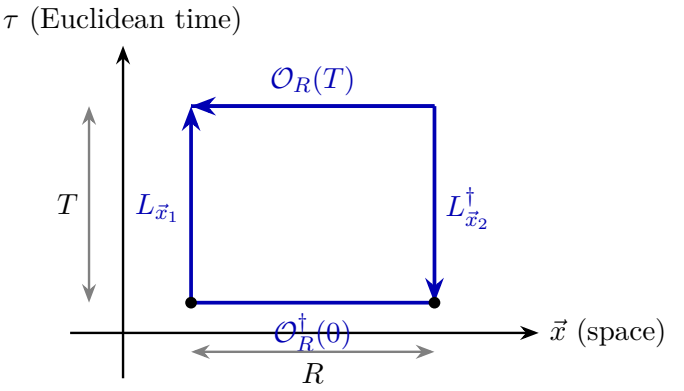


Figure 27.7: The rectangular Wilson loop $\mathcal{C}_{R,T}$. The two vertical sides are the temporal Wilson lines $L_{\vec{x}_1}, L^\dagger_{\vec{x}_2}$ that are the worldlines of a static quark and antiquark held at separation R; the horizontal sides are the spatial Wilson lines created and annihilated by the gauge-invariant operators $\mathcal{O}^\dagger_R(0)$ and $\mathcal{O}_R(T)$ of Eq. (27.179). As $T \to \infty$ the loop measures the ground-state energy of the static $q\bar{q}$ pair, $V(R)$.

Spectral decomposition and the limit $T \to \infty$. The left-hand side of Eq. (27.180) is a Euclidean two-point function, and every Euclidean two-point function has a spectral (transfer-matrix) representation. Insert a complete set of energy eigenstates of the full Hamiltonian *in the presence of the two static colour sources at separation R*. Writing H_R for that Hamiltonian and $|n;R\rangle$ for its eigenstates with energies $\mathcal{E}_n(R)$, and using $\mathcal{O}^\dagger_R(0)\,|0\rangle = \sum_n c_n\,|n;R\rangle$ with $c_n = \big\langle n;R\big|\mathcal{O}^\dagger_R 0\big\rangle$,

$$\Big\langle \mathcal{O}_R(T)\,\mathcal{O}^\dagger_R(0)\Big\rangle \;=\; \sum_n |c_n|^2\; e^{-\mathcal{E}_n(R)\,T}. \tag{27.181}$$

The energies $\mathcal{E}_n(R)$ include the two heavy rest masses $2M$; subtracting the common factor e^{-2MT} in Eq. (27.180) removes them and leaves the energies $E_n(R) = \mathcal{E}_n(R) - 2M$ of the gauge field in the presence of the static sources, measured relative to the empty vacuum. Combining Eqs. (27.180) and (27.181),

$$\langle W(\mathcal{C}_{R,T})\rangle \;=\; \sum_n |c_n|^2\; e^{-E_n(R)\,T}. \tag{27.182}$$

Every term decays exponentially in T, and the slowest-decaying term—the ground state $E_0(R)$, the lowest energy of the gauge field bound to a static $q\bar{q}$ pair at separation R—dominates as $T \to \infty$. We define this ground-state energy to be the *static quark–antiquark potential*, $V(R) \equiv E_0(R)$. We have proved the statement that anchors the whole subject.

Theorem 27.19 (The rectangular Wilson loop measures the static potential)**.** *For the rectangular contour $\mathcal{C}_{R,T}$ of Fig. 27.7, the large-time behaviour of the Wilson-loop expectation value determines the static quark–antiquark potential $V(R)$ through*

$$\boxed{\langle W(\mathcal{C}_{R,T})\rangle \xrightarrow{T\to\infty} |c_0|^2\, e^{-V(R)\,T}, \qquad V(R) = -\lim_{T\to\infty}\frac{1}{T}\ln\langle W(\mathcal{C}_{R,T})\rangle\,.} \tag{27.183}$$

The gap to the first excited state, $E_1(R) - E_0(R) > 0$, controls the rate at which the subleading terms in Eq. (27.182) die away, so the limit is approached exponentially fast.

The result is exact and non-perturbative: nowhere did we expand in g. It rests only on the heavy-quark kinematics Eq. (27.178), the gauge invariance of the loop, and the existence of a transfer matrix—i.e. on reflection positivity of the Euclidean theory, which the lattice formulation of Chapter 43 guarantees rigorously. Theorem 27.19 converts the abstract question "how does $\langle W\rangle$ scale?" into the concrete question "what is the potential between two static quarks at large separation?"

27.7.3 Area law if and only if linear confinement

We can now read the criterion of Theorem 27.18 directly off the potential. For a rectangle the area is $\mathcal{A} = RT$ and the perimeter is $P = 2(R+T)$, so the most general large-loop behaviour allowed by Theorem 27.18, keeping both possible terms together with a constant, is

$$-\ln\langle W(\mathcal{C}_{R,T})\rangle \;=\; \sigma\,RT \;+\; \mu\,(2R+2T) \;+\; c \;+\; \cdots\,, \tag{27.184}$$

where the ellipsis denotes terms that grow more slowly than linearly in T at fixed R (for example a Coulombic $\sim T/R$ piece, discussed below). Divide by T and send $T\to\infty$ at fixed R; by Eq. (27.183) the limit is exactly $V(R)$, and only the terms linear in T survive the division:

$$V(R) \;=\; \lim_{T\to\infty}\frac{1}{T}\Big[\sigma R\,T + 2\mu T + \cdots\Big] \;=\; \sigma\,R \;+\; 2\mu. \tag{27.185}$$

This little computation is the whole equivalence, and it is worth stating exactly what it says, because the logic is sharper than the usual slogan.

Theorem 27.20 (Area law $\Leftrightarrow$ linear confinement)**.** *With $V(R)$ defined by Eq. (27.183):*

1. *The Wilson loop obeys an area law, Eq. (27.175) with string tension $\sigma > 0$,* if and only if *the static potential rises linearly at large R, $V(R) = \sigma R + \text{const}$, so that the confining force $-\,dV/dR = -\sigma$ is constant and nonzero at all separations. No finite energy can separate the pair: the charges are confined.*

2. *The Wilson loop obeys a pure perimeter law, Eq. (27.176) ($\sigma = 0$), if and only if the static potential approaches a constant at large R, $V(R) \to 2\mu$, so that the force vanishes at large separation: the charges are liberated (deconfined or screened).*

The perimeter coefficient μ is, in either case, just the (separation-independent) self-energy of a static source; it carries no information about binding and is removed by renormalizing the loop. The confining content lives entirely in the area term.

Proof. ($\Rightarrow$ in case 1) If Eq. (27.175) holds, then $-\ln\langle W\rangle = \sigma RT + (\text{perimeter} + \text{const} + \cdots)$, and Eq. (27.185) gives $V(R) = \sigma R + 2\mu$, linear with slope σ; the force is $-dV/dR = -\sigma$, constant. ($\Leftarrow$) Conversely, if $V(R) = \sigma R + \text{const}$ with $\sigma > 0$, then by Eq. (27.183) the coefficient of T in $-\ln\langle W\rangle$ is $\sigma R + \text{const}$, whose σRT part is precisely the area term Eq. (27.175). Case 2 is the same computation with $\sigma = 0$: the coefficient of T is the constant 2μ, $V(R) \to 2\mu$, and conversely a constant potential leaves no RT term in Eq. (27.184). □

Physical Insight

The picture behind the area law is the *chromoelectric flux tube*. In an abelian theory the field lines between two charges spread out and fill space, giving the $1/R$ Coulomb law and a perimeter loop. In a confining non-abelian vacuum the gluon self-interaction (the antiscreening that made the beta function negative) instead squeezes the chromoelectric flux into a thin tube of roughly constant cross-section and energy per unit length σ. Stretching the tube to length R costs energy σR: a constant restoring force, a string of tension σ. The area-law exponent is then transparent—the tube sweeps out the minimal surface spanning the loop as the pair is held for time T, and its action is $\sigma \times (\text{area})$. This is the same dual-superconductor intuition that the monopole-condensation picture of Chapter 42 and the exactly solved Seiberg–Witten theory of Chapter 46 make precise in settings where the mechanism can actually be computed.

The Coulombic ellipsis in Eq. (27.184) is the short-distance face of the same potential and is fully under perturbative control. At small R the running coupling is small (asymptotic freedom, Section 27.6), and one-gluon exchange between the two static sources gives the leading potential. The colour factor is the fundamental Casimir $C_F = (N^2-1)/2N$ of Chapter B (equal to $4/3$ for SU(3)), and the result is the non-abelian Coulomb law, so that the complete static potential interpolates between a perturbative short-distance tail and a confining long-distance rise—the phenomenological *Cornell potential*

$$V(R) \;=\; -\frac{C_F\,\alpha_s}{R} \;+\; \sigma\,R \;+\; \text{const}, \qquad C_F = \frac{N^2-1}{2N}, \tag{27.186}$$

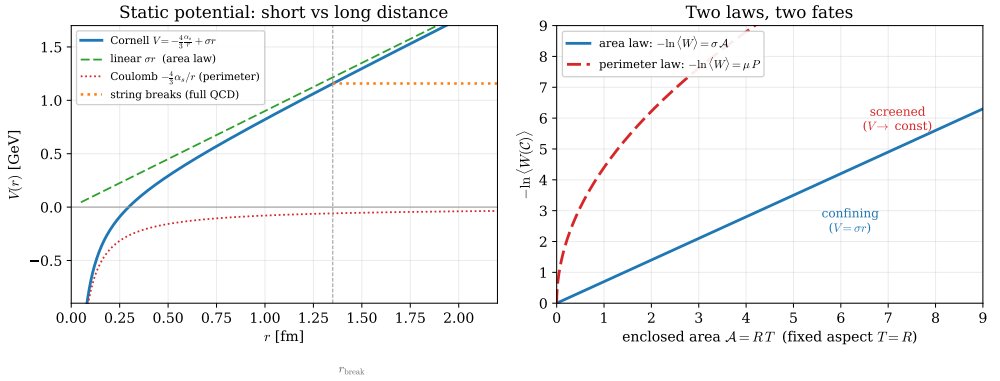


Figure 27.8: *Left:* the static $q\bar{q}$ potential. The short-distance Coulombic tail $-C_F\alpha_s/R$ (one-gluon exchange, perturbative) joins the long-distance linear rise σR (the area law) into the Cornell potential Eq. (27.186). Pure Coulomb (dotted) would give a perimeter law and no confinement; the linear term (dashed) is the confining piece. In *full* QCD with light dynamical quarks the string breaks at r_{break} (the energy reaches the threshold to pop a light $q\bar{q}$ pair) and the potential plateaus—the area law is the *quenched*, static-source statement. *Right:* the order-parameter dichotomy of Theorems 27.18 and 27.20: $-\ln\langle W\rangle$ versus enclosed area at fixed aspect ratio. The area law (linear in $\mathcal{A}$) signals confinement; the perimeter law (growing only as $\sqrt{\mathcal{A}}$ at fixed aspect) signals screening.

whose two regimes and the order-parameter dichotomy are shown in Fig. 27.8. The $-C_F\alpha_s/R$ term is the Coulombic piece hiding in the ellipsis; the σR term is the area law. The quarkonium spectrum (charmonium, bottomonium) is reproduced strikingly well by this single function, which is the most direct phenomenological evidence that the linear term is real; we return to it in Chapter 28.

27.7.4 Where the area law is proved: strong-coupling lattice gauge theory

Theorems 27.19 and 27.20 are theorems about *what an area law would mean*; they do not establish that Yang–Mills theory actually exhibits one. For that we need a regime in which $\langle W\rangle$ can be computed non-perturbatively, and exactly one such regime is under analytic control: the strong-coupling expansion of lattice gauge theory. The full development belongs to Chapter 43; here we state the formulation and carry the leading computation, because it is the one setting in which confinement is genuinely *derived* rather than modelled, and the honest scoping of the next subsection turns on understanding precisely what it does and does not prove.

Replace continuous spacetime by a hypercubic Euclidean lattice of spac-

ing a. The gauge field is encoded not in A_μ but in the group-valued parallel transporters across single links, $U_\mu(x) \in \mathrm{SU}(N)$, the lattice version of the Wilson line $\mathcal{P}\exp(ig\int A)$ of Eq. (25.139). The smallest gauge-invariant object is the plaquette, the holonomy around an elementary square, $U_p = U_\mu(x)\,U_\nu(x+\hat{\mu})\,U_\mu^\dagger(x+\hat{\nu})\,U_\nu^\dagger(x)$, and Wilson's action—the unique simplest discretization whose naive continuum limit returns $-\frac{1}{4}F^2$—is

$$S_{\mathrm{lat}} \;=\; \beta_{\mathrm{lat}} \sum_p \Big(1 - \tfrac{1}{N}\,\mathrm{Re\ tr}\,U_p\Big), \qquad \beta_{\mathrm{lat}} = \frac{2N}{g^2}. \tag{27.187}$$

Expectation values are ordinary integrals over the compact group manifold with the invariant Haar measure dU, $\langle \mathcal{O}\rangle = Z^{-1}\int \prod_l dU_l\, e^{-S_{\mathrm{lat}}}\mathcal{O}$, and the Wilson loop is the trace of the ordered product of link variables around $\mathcal{C}$. "Strong coupling" means large bare g^2, hence small β_{lat}, and the expansion in β_{lat} is what we now carry out at leading order.

Worked Example 27.21 (The area law from the leading strong-coupling expansion)**.** Expand the Boltzmann factor of Eq. (27.187) in the small parameter β_{lat}, dropping the constant $\sum_p \beta_{\mathrm{lat}}$ into the normalization Z,

$$e^{-S_{\mathrm{lat}}} \;=\; \prod_p \exp\!\Big(\tfrac{\beta_{\mathrm{lat}}}{N}\,\mathrm{Re\ tr}\,U_p\Big) \;=\; \prod_p \Big[1 + \tfrac{\beta_{\mathrm{lat}}}{2N}\big(\mathrm{tr}\,U_p + \mathrm{tr}\,U_p^\dagger\big) + \cdots\Big]. \tag{27.188}$$

The integrals we need follow from the orthogonality of matrix elements of the fundamental representation under the Haar measure,

$$\int dU\, U_{ij} = 0, \qquad \int dU\, U_{ij}\,(U^\dagger)_{kl} = \frac{1}{N}\,\delta_{il}\,\delta_{jk}, \tag{27.189}$$

the second of which is the statement that a link integrates to zero unless the link variable U is accompanied by exactly one $U^\dagger$ on the same link; then it contracts the colour indices with a factor $1/N$. Now consider $\langle W(\mathcal{C}_{R,T})\rangle$, with $W = \mathrm{tr}\prod_{l\in\mathcal{C}} U_l$ carrying one U on each of the $2(R+T)/a$ boundary links. By Eq. (27.189) every such link integrates to zero unless we supply a compensating $U^\dagger$, and the only source of compensating factors is the plaquette terms pulled down from Eq. (27.188). Each boundary link must therefore be "paid for" by a plaquette sharing it; but that plaquette carries three *further* link variables, which must in turn be paid for by adjacent plaquettes, and so on. The links are saturated only when a sheet of plaquettes tiles a surface whose boundary is $\mathcal{C}$. The cheapest such surface in powers of the small β_{lat} is the minimal one: the flat rectangle, tiled by

$$n_p \;=\; \frac{\mathcal{A}}{a^2} \;=\; \frac{RT}{a^2} \tag{27.190}$$

elementary plaquettes. Each tile contributes one factor $\beta_{\mathrm{lat}}/2N$ from Eq. (27.188) and, after the link integrations Eq. (27.189) glue neighbouring tiles and the

boundary, a colour factor $1/N$ per internal link contraction; collecting the combinatorics, the net weight per plaquette is $\beta_{\mathrm{lat}}/(2N^2)$. The leading result is the product over tiles,

$$\langle W(\mathcal{C}_{R,T})\rangle \;\simeq\; \Big(\frac{\beta_{\mathrm{lat}}}{2N^2}\Big)^{n_p} \;=\; \exp\Big[-\,n_p\,\ln\frac{2N^2}{\beta_{\mathrm{lat}}}\Big] \;=\; \exp\Big[-\,\frac{RT}{a^2}\,\ln\frac{2N^2}{\beta_{\mathrm{lat}}}\Big]. \quad (27.191)$$

Comparing with Eq. (27.175), this is an area law with $\mathcal{A} = RT$ and a nonzero string tension. Using $\beta_{\mathrm{lat}} = 2N/g^2$ so that $2N^2/\beta_{\mathrm{lat}} = Ng^2$,

$$\boxed{\sigma\, a^2 \;=\; \ln\frac{2N^2}{\beta_{\mathrm{lat}}} \;=\; \ln(Ng^2) \;+\; O(\beta_{\mathrm{lat}}^2).} \quad (27.192)$$

At strong bare coupling (g^2 large) the right-hand side is positive, so $\sigma > 0$: the lattice gauge theory *confines*, for every N and in every spacetime dimension. By Theorem 27.20 the static potential is linear, $V(R) = \sigma R$, and a single static quark has infinite energy—it cannot be isolated. This is Wilson's result, the cleanest derivation of confinement we possess.

Two features of Theorem 27.21 deserve emphasis. First, the expansion is *convergent*, not merely asymptotic: at small enough β_{lat} the higher-order corrections (longer tilings, deformed surfaces) are bounded by a geometric series and the area law is a rigorous theorem about the lattice theory at strong coupling. Confinement on the lattice at strong coupling is not a model or a picture; it is proved. Second, the mechanism is generic—it used nothing about four dimensions, nothing about asymptotic freedom, and would equally "confine" a theory we know to be deconfined in the continuum, such as compact U(1) lattice gauge theory in four dimensions. That universality is the warning that motivates the next subsection.

27.7.5 Honest scope: proved, evidenced, and open

We must now be exact about the logical status of what we have, because confinement is precisely the place where a careful book and a careless one part ways. There are three tiers.

What is proved (on this page). Theorem 27.19—that a large rectangular Wilson loop measures the static $q\bar{q}$ potential through $\langle W\rangle \sim e^{-V(R)T}$—is a theorem, derived in full above from the heavy-quark limit and the spectral representation. Theorem 27.20—that an area law is *equivalent* to a linearly rising, confining potential, and a perimeter law to a constant one—is a theorem, proved above. The strong-coupling area law of Theorem 27.21, with string tension Eq. (27.192), is a theorem about *lattice* gauge theory in the regime of large bare coupling, established by a convergent expansion (the complete treatment, including its convergence and the next orders, is the business of Chapter 43).

Why the proof does not yet reach the continuum. The catch is the continuum limit. The lattice is a regulator; physics is recovered only as $a \to 0$ with physical quantities such as σ held fixed. By asymptotic freedom, the bare coupling at the cutoff scale $1/a$ must then run to *zero*, $g^2(a) \to 0$ as $a \to 0$, which is $\beta_{\text{lat}} \to \infty$— the *weak*-coupling regime, exactly where the strong-coupling series Eq. (27.188) diverges. The strong-coupling formula Eq. (27.192) gives $\sigma a^2 = \ln(Ng^2)$, a lattice artefact that does *not* obey the asymptotic-freedom scaling $\sigma a^2 \sim \exp[-1/(b_0 g^2)]$ (with b_0 the coefficient of Eq. (27.101)) required for σ to approach a finite continuum value. The proof of confinement therefore lives at large a (coarse lattice), while the continuum lives at small a (fine lattice), and *nothing in the strong-coupling expansion guarantees that the area law survives the passage between them.* A priori a phase transition—a deconfining or bulk/roughening transition— could separate the strongly coupled lattice from the continuum, and indeed for compact U(1) in four dimensions exactly such a transition occurs and the continuum theory is the deconfined Coulomb phase of free photons. The strong-coupling area law alone cannot distinguish SU(3), which we believe confines, from U(1), which does not.

What is evidence (Monte Carlo). The bridge across the gap is supplied numerically. Lattice Monte Carlo simulations of SU(3) Yang–Mills compute $\langle W(\mathcal{C}_{R,T})\rangle$ from first principles at couplings ranging continuously from strong to weak, with no expansion, and find: (i) the area law persists with no intervening deconfining transition as β_{lat} is increased toward the continuum; (ii) the measured σa^2 crosses over from the strong-coupling logarithm Eq. (27.192) to the asymptotic-freedom scaling $\exp[-1/(b_0 g^2)]$, the hallmark of a genuine continuum quantity; and (iii) the resulting physical string tension is

$$\sqrt{\sigma} \;\approx\; 440 \text{ MeV}, \qquad \sigma \;\approx\; 0.19 \text{ GeV}^2 \;\approx\; 0.9 \text{ GeV/fm}, \tag{27.193}$$

a number consistent with the Regge slope of the hadron spectrum and with the quarkonium fits to the Cornell potential Eq. (27.186). This is overwhelming, but it is numerical evidence, not a proof.

What is open (the mass-gap problem). The statement that four-dimensional SU(N) Yang–Mills theory exists as a non-perturbative quantum field theory with a positive mass gap—and, relatedly, that it confines (area law in the continuum)—is unproved. It is one of the Clay Mathematics Institute Millennium Prize Problems, and it remains open as of the mid-2020s; this is the same honest boundary drawn at the end of Chapter 25 (Section 25.7). What is missing is exactly a *constructive* control of the theory uniformly from the lattice to the continuum, of the kind available for super-renormalizable scalar models but not for four-dimensional gauge theory.

Caution

The area law is the statement of confinement only for *static external sources in the absence of dynamical light quarks*—pure gauge theory, or the "quenched" approximation. In full QCD with light dynamical quarks the chromoelectric string, once stretched far enough, has more energy than two mesons, and it is energetically favourable to pop a light $q\bar{q}$ pair out of the vacuum and snap the string. Beyond the breaking distance $r_{\text{break}} \sim 1.2$–$1.5$ fm the static potential flattens (the dotted plateau in Fig. 27.8) and the Wilson loop crosses over to a perimeter law. Confinement in the precise area-law sense is therefore a statement about the gluonic sector; the physical statement in full QCD is the weaker but observed fact that the asymptotic spectrum contains only colour singlets. Distinguishing these is essential and is taken up in Chapters 28 and 43.

27.7.6 From running coupling to confinement: the boundary of perturbation theory

It is tempting to say that confinement is "caused" by the infrared growth of the running coupling: $\alpha_s(Q^2)$ diverges as $Q \to \Lambda_{\text{QCD}}$ (Section 27.6), the force does not turn off, the quarks are bound. This is a useful heuristic and no more. The one-loop running formula is derived in the regime where it is valid, $Q \gg \Lambda_{\text{QCD}}$, and its extrapolation to $Q \lesssim \Lambda_{\text{QCD}}$—where it formally diverges—is exactly the extrapolation perturbation theory is not licensed to make. The Landau-pole divergence of α_s is a symptom that the perturbative description has ended, not a calculation of what replaces it. The string tension σ is a genuinely non-perturbative quantity: it scales as $\sigma \sim \Lambda_{\text{QCD}}^2 \sim \mu^2 \exp[-1/(b_0 \alpha_s(\mu))]$, a function with an essential singularity at $\alpha_s = 0$ that is invisible to every order of the perturbative expansion. Asymptotic freedom is what makes Λ_{QCD}, and hence σ, the only scale in the massless theory (dimensional transmutation, Section 27.6); it explains *why* there is a confinement scale and why it is exponentially below the cutoff. It does not, by itself, prove the area law. The genuine non-perturbative tools—the semiclassical instanton and θ-vacuum analysis of Chapter 42, the lattice definition and its Monte Carlo evaluation of Chapter 43, and the exactly solved supersymmetric analogues of Chapter 46, where monopole condensation *is* computed and the area law *is* derived in a controlled theory—are where confinement passes from heuristic to understood, each to the extent that its setting allows.

This closes the Yang–Mills analysis of this chapter. We carried the Wilson loop from the classical, gauge-invariant comparator of Chapter 25 to its role as the non-perturbative order parameter for confinement, proved the two theorems that can be proved—that it measures the static potential and that its area law is linear confinement—and marked honestly the frontier where the continuum statement

becomes the open mass-gap problem. From here the thread fans out: Chapter 28 turns asymptotic freedom and the colour algebra into the phenomenology of jets, the R-ratio, and deep inelastic scattering; Chapters 42 and 43 supply the non-perturbative machinery—instantons, the θ-vacuum, and the lattice— that this section could only point toward; Chapter 46 exhibits confinement solved exactly in a supersymmetric cousin; and, much later, Chapters 59 and 61 return to Yang–Mills not to bind its quanta but to discover the hidden simplicity of its gluon amplitudes and the double copy to gravity.

Further reading

The discovery of asymptotic freedom is Gross and Wilczek [238] and Politzer [406]; that a non-abelian gauge coupling is essentially the unique four-dimensional field theory with this property is Coleman and Gross [130], the content of Theorem 27.13. The two-loop coefficient that fixes the scheme-independent part of the running beyond our one-loop calculation is Caswell [118]. The background-field computation used as our gauge-invariant cross-check is developed in Abbott [1]; textbook derivations of the one-loop beta function at the level carried here are Peskin and Schroeder [396, Ch. 16] and Schwartz [432, Ch. 26], with the systematic renormalization-group account in Weinberg [501, Ch. 18]. The loss of asymptotic freedom and the infrared fixed point at the edge of the conformal window are Banks and Zaks [45]. For the Wilson-loop criterion, the lattice, and the strong-coupling area law see Wilson [514] and the pedagogical accounts by Gattringer and Lang [211], Kogut [299], Rothe [420]; Greensite [235] surveys the confinement mechanism, the Cornell potential and quarkonium phenomenology are Eichten et al. [171], and the continuum mass-gap problem is stated precisely in Jaffe and Witten [281].

Chapter 28

QCD Phenomenology

The three preceding chapters built quantum chromodynamics as a theory: the classical non-abelian Lagrangian and its self-interacting connection (Chapter 25), the Faddeev–Popov quantization and BRST structure that make it a consistent quantum field theory (Chapter 26), and the negative beta function whose running coupling $\alpha_s(Q)$ falls to zero at short distance (Chapter 27). That last fact is the licence for everything here. Asymptotic freedom means that at large momentum transfer the strong interaction is genuinely perturbative, so the amplitudes assembled from the quark–gluon and gluon self-couplings can be expanded in Feynman diagrams and confronted with experiment. This chapter is where Yang–Mills theory leaves the blackboard and becomes a set of numbers measured at colliders: the integer $N_c = 3$ counted off a cross section, the spin and fractional charge of the quarks read off deep inelastic data, the logarithmic violation of scaling predicted to the percent level, and the running coupling $\alpha_s(m_Z)$ extracted from jets.

The chapter rests on two pillars that pull in opposite directions and meet at a clean connection. The first is asymptotic freedom, which makes short-distance partonic cross sections calculable; the kinematic machinery for those cross sections — spinors, traces, phase space, the cross-section and decay-rate formulae — is exactly the technology that carried QED through Chapters 16 to 18, and the QCD amplitudes differ from their QED templates only by the colour algebra and the running coupling. The second pillar is confinement, which hides the quarks and gluons inside hadrons and forces a single irreducibly non-perturbative object — the parton distribution — into every prediction involving an initial-state hadron. Factorization is the theorem that cleanly separates the two: a calculable short-distance coefficient convolved with a universal, measured long-distance distribution. The infrared structure that makes this separation possible is the Kinoshita–Lee–Nauenberg cancellation of soft and collinear singularities first met in Chapter 19; the renormalization-group logic that organizes the running distributions is the Callan–Symanzik framework of Chapter 24; and the group-

theory data that fixes every colour factor — the Casimirs C_F, C_A and the index T_F — is catalogued for general gauge groups in Chapter B, with the gamma-matrix traces of Chapter C and the dimensional-regularization integrals of Chapter D doing the kinematic work.

The arc is cumulative and runs from the cleanest observable to the most refined. Section 28.1 builds the colour toolkit, reducing every QCD colour factor to the three constants through the fundamental completeness relation, and fixes the averaging-and-summing rules that distinguish a QCD cross section from its QED ancestor. Section 28.2 spends that toolkit on the totally inclusive ratio $R(s) = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$, where inclusivity trivializes factorization and the data return $N_c = 3$ on the vertical axis and α_s in the small correction on top. Section 28.3 then gives up inclusivity on one leg: deep inelastic scattering resolves a single quark inside the proton, the parton model identifies Bjorken x with the struck constituent's momentum fraction, and the Callan–Gross relation and the momentum sum rule deliver the quarks' spin and the gluon's share of the momentum.

The remaining sections turn the small failures of the naive parton model into the theory's sharpest predictions. Section 28.4 traces the logarithmic violation of Bjorken scaling to collinear gluon emission, computes the splitting functions from the same vertices of Chapters 25 and 26, and assembles the DGLAP equations as the renormalization group of the parton densities. Section 28.5 states the collinear factorization theorem, demonstrates at one loop that the collinear pole is universal and is absorbed into the $\overline{\text{MS}}$ parton distribution — so that the splitting function, the DGLAP kernel, and the factorization residue are revealed to be one object — and tests universality against the Drell–Yan process. Finally, Section 28.6 pays the debt left by Section 28.2: it carries out the soft-and-collinear cancellation explicitly in $e^+e^- \to q\bar{q}g$, distils the infrared-safety criterion that separates calculable observables from incalculable ones, and constructs the jets and event shapes — Sterman–Weinberg cones, thrust, the anti-k_t algorithm — with which α_s is measured across three orders of magnitude in scale, closing the arc on asymptotic freedom read directly off the data. What this chapter consigns to the long-distance side of factorization — hadronization and confinement itself — is the province of the non-perturbative methods of Chapters 39 and 41 to 43, and the perturbative QCD assembled here joins the electroweak theory in the Standard Model synthesis of Chapter 33.

28.1 Color Bookkeeping: The Group Theory of QCD Amplitudes

The chapter introduction made a promise that now has to be cashed. At large momentum transfer the strong coupling $\alpha_s(Q)$ runs to zero (Chapter 27), so short-distance QCD is genuinely perturbative and its amplitudes can be expanded in

Feynman diagrams built from the quark–gluon and gluon self-interaction vertices manufactured in Chapter 25 and quantized in Chapter 26. Every such diagram is a product of two logically independent pieces. One is a *kinematic* factor — spinors, polarization vectors, propagator momenta, Dirac traces — and this part is computed by exactly the technology that carried QED through Chapters 17 to 19; the photon of QED and the gluon of QCD differ kinematically only in the spin-1 couplings, which we already know how to handle. The other piece is a *color* factor: a contraction of the SU(3) generators T^a and structure constants f^{abc} that ride on every vertex, summed over the color indices of internal lines and saturated on the indices of external ones. That color factor is a pure number (for fixed external colors) or a Kronecker structure (in general), and it is the only genuinely new bookkeeping that non-abelian gauge theory adds to the QED template.

The economy of perturbative QCD rests on a single fact: this color algebra decouples cleanly from the kinematics and is governed by a small handful of group-theory constants. Three numbers — the fundamental Casimir C_F, the adjoint Casimir C_A, and the Dynkin index T_F — together with one master identity, the completeness relation for the fundamental generators, suffice to reduce *any* color factor in the theory to numbers. This section builds that toolkit in the physical language the cross sections will speak, so that the remaining sections of the chapter can quote color factors rather than rederive them. The same identities are catalogued for a general compact gauge group and a general representation in Appendix B; we cite that catalogue (Theorems B.23, B.27 and B.28) as the canonical home of the general statements and rederive here only what QCD phenomenology directly uses, in the streamlined form in which it will be used. We work throughout with gauge group SU(N), specializing to the physical $N = N_c = 3$ at the end of each result; the generators are Hermitian, $[T^a, T^b] = if^{abc}T^c$, and normalized in the fundamental to $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, exactly as fixed for Yang–Mills theory. Color traces are evaluated by the same discipline as the Dirac traces of Appendix C — reduce a closed index loop to a number — with the completeness relation playing the role the Clifford algebra plays there.

28.1.1 Where color enters: vertices, indices, and the two sums

Fix the index conventions once. A quark q carries a fundamental color index $i \in \{1, \ldots, N_c\}$, an antiquark $\bar{q}$ the conjugate index, and a gluon an adjoint index $a \in \{1, \ldots, N_c^2 - 1\}$; for $N_c = 3$ there are three quark colors and eight gluons. The interaction vertices established in Chapter 25 carry their color structure entirely in these indices. The quark–gluon vertex, extracted from the covariant-derivative coupling $g\,\bar{\psi}_i \not{A}^a (T^a)_{ij} \psi_j$ of Eq. (25.92), contributes the matrix element $(T^a)_{ij}$: a gluon of color a converts a quark of color j into a quark of color i. The triple-gluon vertex (Eq. (25.107)) contributes a structure constant

f^{abc}, and the four-gluon vertex (Eq. (25.110)) contributes products of structure constants. The color part of any diagram is the product of one such tensor per vertex, with every internal color index summed.

Physical Insight

Color flow is bookkeeping for "which quark turned into which." A gluon is not a color-neutral mediator like the photon; it carries color away from one line and deposits it on another, encoded in $(T^a)_{ij}$. The whole of color algebra is the arithmetic of these transfers, and the answers are forced by the single requirement that the gluon, unlike a photon, must be *traceless* in color — it has no $\mathfrak{u}(1)$ component.

External particles fix where the color sums begin and end. For a final-state quark, gluon, or jet we cannot observe the color — color is confined — so an inclusive cross section must *sum* over the colors of every final-state parton. For an initial-state parton in a collision the color is equally unknown, and a beam is an incoherent, unpolarized-in-color mixture; the probability of any one color is uniform, so an initial-state cross section must *average* over the colors of every incoming parton, i.e. sum and divide by the number of color states. We record these two operations as the standing rules of the chapter.

Definition 28.1 (Color averaging and summing)**.** In an unpolarized cross section, each incoming parton is averaged over color (sum over its colors, divide by the dimension of its color representation) and each outgoing parton is summed over color. The averaging denominators are

$$\text{incoming quark or antiquark:}\quad \frac{1}{N_c}=\frac{1}{3},\qquad \text{incoming gluon:}\quad \frac{1}{N_c^2-1}=\frac{1}{8}, \tag{28.1}$$

the dimensions of the fundamental and adjoint representations of SU(3).

These factors are the color analogue of the familiar $\frac{1}{4}$ that averages over the two spins of each of two incoming fermions in QED. With the vertices supplying $(T^a)_{ij}$ and f^{abc}, and Theorem 28.1 fixing the external sums, the entire problem reduces to evaluating contractions of generators. We now isolate the one identity that performs every such contraction.

28.1.2 The master identity: completeness of the generators

The decisive observation is that the generators, together with the identity matrix, form a basis for *all* Hermitian matrices, and a basis lets any matrix — in particular the elementary "one entry" matrices — be resolved into generators. Contracting two such resolutions is the completeness relation.

Consider the real vector space $\mathfrak{h}_N$ of Hermitian $N \times N$ matrices. Its real dimension is N^2: a Hermitian matrix has N real diagonal entries and $\frac{1}{2}N(N-1)$ independent complex off-diagonal entries above the diagonal, giving $N + N(N-1) = N^2$ real parameters. On $\mathfrak{h}_N$ the bilinear form $\langle A, B\rangle \equiv \mathrm{tr}(AB)$ is a genuine real inner product: it is symmetric, it is real because $(\mathrm{tr}\, AB)^* = \mathrm{tr}(B^\dagger A^\dagger) = \mathrm{tr}(BA) = \mathrm{tr}(AB)$ for Hermitian A, B, and it is positive definite because $\mathrm{tr}(A^2) = \sum_{i,j} |A_{ij}|^2 \geq 0$ with equality only for $A = 0$. We exhibit an orthogonal basis adapted to $\mathrm{SU}(N)$. Adjoin to the $N^2 - 1$ traceless generators T^a the single trace direction

$$T^0 \equiv \frac{1}{\sqrt{2N}}\,\mathbf{1}, \tag{28.2}$$

chosen so that $\mathrm{tr}\big((T^0)^2\big) = \frac{1}{2N}\,\mathrm{tr}\,\mathbf{1} = \frac{1}{2N}\cdot N = \frac{1}{2}$, matching the normalization of the T^a. The full set $\{T^A\}_{A=0}^{N^2-1}$ is orthogonal under $\langle\,,\rangle$ with a uniform norm,

$$\mathrm{tr}(T^A T^B) = \tfrac{1}{2}\,\delta^{AB}, \qquad A, B \in \{0, 1, \ldots, N^2-1\}, \tag{28.3}$$

since $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ by the fundamental normalization, $\mathrm{tr}\big((T^0)^2\big) = \frac{1}{2}$ by construction, and the cross terms vanish, $\mathrm{tr}(T^0 T^a) = \frac{1}{\sqrt{2N}}\,\mathrm{tr}\,T^a = 0$, because the generators of $\mathrm{SU}(N)$ are traceless. There are exactly N^2 of these matrices and they are mutually orthogonal, hence linearly independent, hence a basis of the N^2-dimensional space $\mathfrak{h}_N$ (the elementary completeness of an orthogonal system of full count in finite dimension, Chapter 2).

Because $\{T^A\}$ is a basis with the inner product Eq. (28.3), every matrix M (Hermitian or not — complex matrices are spanned just as well, the coefficients merely becoming complex) expands as $M = \sum_A c_A T^A$, and contracting with T^B and using Eq. (28.3) fixes the coefficients $c_B = 2\,\mathrm{tr}(T^B M)$:

$$M = \sum_{A=0}^{N^2-1} 2\;\mathrm{tr}\big(T^A M\big)\,T^A. \tag{28.4}$$

Now feed Eq. (28.4) the most elementary matrices of all. Let E_{lk} be the matrix unit with a 1 in row l, column k, and zeros elsewhere, so $(E_{lk})_{ij} = \delta_{il}\delta_{jk}$ and $\mathrm{tr}(T^A E_{lk}) = (T^A)_{kl}$. Writing Eq. (28.4) for $M = E_{lk}$ in components (i, j),

$$\delta_{il}\delta_{jk} = \sum_{A=0}^{N^2-1} 2\,(T^A)_{kl}\,(T^A)_{ij} = 2\,(T^0)_{kl}(T^0)_{ij} + 2\sum_{a=1}^{N^2-1} (T^a)_{ij}(T^a)_{kl}. \tag{28.5}$$

The trace direction contributes $2(T^0)_{kl}(T^0)_{ij} = 2\cdot\frac{1}{2N}\delta_{kl}\delta_{ij} = \frac{1}{N}\delta_{ij}\delta_{kl}$. Isolating the sum over the genuine $\mathrm{SU}(N)$ generators gives the master identity.

Theorem 28.2 (Fundamental completeness / color-Fierz relation)**.** *The generators of* $\mathrm{SU}(N)$ *in the fundamental representation satisfy*

$$(T^a)_{ij}\,(T^a)_{kl} = \frac{1}{2}\left(\delta_{il}\,\delta_{jk} - \frac{1}{N}\,\delta_{ij}\,\delta_{kl}\right), \tag{28.6}$$

with the adjoint index a summed. For SU(3) *this is the single relation that reduces every QCD color factor.*

Proof. Subtract $\frac{1}{N}\delta_{ij}\delta_{kl}$ from Eq. (28.5) and divide by 2: $\sum_a (T^a)_{ij}(T^a)_{kl} = \frac{1}{2}(\delta_{il}\delta_{jk} - \frac{1}{N}\delta_{ij}\delta_{kl})$. □

Two contractions confirm the coefficients before we lean on them. Setting $j = i$ and summing, the left side becomes $\sum_a (\mathrm{tr}\, T^a)(T^a)_{kl} = 0$ by tracelessness, while the right side gives $\frac{1}{2}(\delta_{il}\delta_{ik} - \frac{1}{N}\, N\, \delta_{kl}) = \frac{1}{2}(\delta_{kl} - \delta_{kl}) = 0$: the $-1/N$ term is precisely what enforces the tracelessness of the gluon, and without it the identity would be inconsistent. Setting instead $j = k$ and summing reproduces the quadratic Casimir we derive next.

Physical Insight

The two terms of Eq. (28.6) are the two ways color can flow through a gluon. The first, $\frac{1}{2}\delta_{il}\delta_{jk}$, lets color pass straight through ($i \to l$, $k \to j$): the gluon is a quark line and an antiquark line glued together. The second, $-\frac{1}{2N}\delta_{ij}\delta_{kl}$, subtracts the color-singlet "photon" that an honest SU(N) gluon must not contain. The diagrammatic version of this picture — 't Hooft's double-line notation, and the $1/N$ suppression of the subtraction that organizes the large-N expansion — is drawn in Section B.3.1.

28.1.3 The three constants as corollaries

Everything QCD phenomenology needs about color is now within one or two contractions of Eq. (28.6).

The fundamental Casimir C_F. The combination T^aT^a that appears whenever a quark line emits and reabsorbs a gluon (a self-energy, or the square of a single emission summed over the gluon) is the quadratic Casimir of the fundamental. We give the structural argument first and then the value. The operator $\sum_a T^aT^a$ commutes with every generator, $[T^b, T^aT^a] = [T^b, T^a]T^a + T^a[T^b, T^a] = if^{bac}(T^cT^a + T^aT^c) = 0$ since f^{bac} is antisymmetric in $a \leftrightarrow c$ while the bracketed sum is symmetric. The fundamental representation is irreducible, so by Schur's lemma (Chapter 2) an operator commuting with all generators is a multiple of the identity:

$$(T^aT^a)_{ij} = C_F\, \delta_{ij}. \tag{28.7}$$

To evaluate C_F, take the trace of both sides. On the right, $\mathrm{tr}(C_F\mathbf{1}) = C_F N$. On the left, $\mathrm{tr}(T^aT^a) = \sum_a \mathrm{tr}(T^aT^a) = \sum_a \frac{1}{2}\delta^{aa} = \frac{1}{2}(N^2-1)$, using the fundamental normalization and the count $N^2 - 1$ of generators. Equating,

$$C_F = \frac{N^2-1}{2N} \xrightarrow{N=3} \frac{9-1}{6} = \frac{4}{3}. \tag{28.8}$$

The same number falls directly out of Eq. (28.6) by the contraction $j = k$: $\sum_a (T^a T^a)_{il} = \sum_a (T^a)_{ij}(T^a)_{jl} = \frac{1}{2}(\delta_{il}\delta_{jj} - \frac{1}{N}\delta_{ij}\delta_{jl}) = \frac{1}{2}(N - \frac{1}{N})\delta_{il} = \frac{N^2-1}{2N}\delta_{il}$, an independent confirmation that the master identity and the Casimir are the same statement viewed two ways.

The Dynkin index T_F. The normalization $\mathrm{tr}(T^a T^b) = T_F\,\delta^{ab}$ defines the index of the fundamental representation, and the Yang–Mills theory convention fixes

$$T_F = \tfrac{1}{2}. \tag{28.9}$$

This is the constant that already appeared in Eq. (28.8) and that sets the relative weight of the quark loop in the QCD beta function (Chapter 27); the general-representation definition $T(R)$ and its relation to $C_2(R)$ are in Eq. (B.89). With $T_F = \frac{1}{2}$ the trace count $\mathrm{tr}(T^a T^a) = T_F(N^2 - 1)$ matches the value used above.

The adjoint Casimir C_A. Whenever a gluon line carries color through a loop or a self-coupling, the structure-constant contraction $f^{acd}f^{bcd}$ appears; it is the quadratic Casimir of the adjoint representation. We compute it from Eq. (28.6) alone, which is the point of having isolated that identity. Start from the double commutator of generators. Using $[T^a, T^b] = i f^{abc} T^c$ twice,

$$[T^a,[T^a,T^b]] = i f^{abc}[T^a,T^c] = i f^{abc}\, i f^{acd} T^d = -f^{abc} f^{acd} T^d. \tag{28.10}$$

The structure-constant product here is, up to a sign, the object we want: $f^{abc}f^{acd} = -f^{acb}f^{acd}$ by the total antisymmetry of f, and writing $f^{acb}f^{acd} \equiv C_A\,\delta^{bd}$ (the contraction is proportional to δ^{bd} because it is an invariant symmetric tensor on the adjoint, the adjoint Casimir, established in Theorem B.28) gives $f^{abc}f^{acd} = -C_A\delta^{bd}$ and hence

$$[T^a,[T^a,T^b]] = C_A\,T^b. \tag{28.11}$$

Now compute the same double commutator by brute expansion, which exposes the fundamental generators and lets Eq. (28.6) act:

$$\begin{aligned}[T^a,[T^a,T^b]] &= T^a T^a T^b - 2\,T^a T^b T^a + T^b T^a T^a \\ &= 2C_F\,T^b - 2\,T^a T^b T^a,\end{aligned} \tag{28.12}$$

where the outer pairs collapsed by the Casimir Eq. (28.7). The remaining object $T^a T^b T^a$ is reduced by the master identity. In components, inserting Eq. (28.6) for the contracted pair,

$$(T^a T^b T^a)_{ij} = (T^a)_{ik}(T^b)_{kl}(T^a)_{lj} = (T^b)_{kl}\cdot\tfrac{1}{2}\Big(\delta_{ij}\delta_{kl} - \tfrac{1}{N}\delta_{ik}\delta_{lj}\Big) = \tfrac{1}{2}\Big(\delta_{ij}\,\mathrm{tr}\,T^b - \tfrac{1}{N}(T^b)_{ij}\Big) = - \tag{28.13}$$

the trace term vanishing because $\operatorname{tr} T^b = 0$. Hence $T^a T^b T^a = -\frac{1}{2N} T^b$. Substituting into Eq. (28.12) and matching to Eq. (28.11),

$$C_A\, T^b = 2C_F\, T^b - 2\left(-\frac{1}{2N}\right) T^b = \left(2C_F + \frac{1}{N}\right) T^b = \left(\frac{N^2-1}{N} + \frac{1}{N}\right) T^b = N\, T^b, \tag{28.14}$$

so that, restoring the index form,

$$f^{acd} f^{bcd} = C_A\, \delta^{ab}, \qquad C_A = N \xrightarrow{N=3} 3. \tag{28.15}$$

The completeness relation has done all the work: a single contraction reduced the only non-abelian object, $T^a T^b T^a$, and the adjoint Casimir followed without ever touching the explicit Gell-Mann matrices. The QCD values are assembled in one place:

$$\boxed{C_F = \frac{4}{3}, \qquad C_A = N_c = 3, \qquad T_F = \frac{1}{2}, \qquad N_c^2 - 1 = 8} \tag{28.16}$$

These four numbers, with Eq. (28.6), are the entire color content of perturbative QCD; the general-N derivations and the extension to arbitrary representations live in Section B.4.

28.1.4 Color in cross sections: averaging in, summing out

We now apply Theorem 28.1 and the constants of Eq. (28.16) to the two color sums that the next sections need: the production of a quark–antiquark pair by a color-singlet probe, which delivers the factor N_c at the heart of the R-ratio, and the emission of a single gluon from a quark, which delivers C_F.

Worked Example 28.3 (The color factor N_c in $e^+e^- \to q\bar{q}$)**.** At lowest order the process $e^+e^- \to q\bar{q}$ proceeds through a virtual photon (and, at higher energy, a Z, treated identically in color), shown in Fig. 28.1. The leptons are colorless, so color enters only at the $\gamma^* q\bar{q}$ vertex. The electromagnetic current of the quark is a *color singlet*: a quark of color i annihilates with an antiquark of the *same* color i, summed, $J^\mu_{\text{em}} = Q_q \sum_i \bar{q}_i \gamma^\mu q_i$, so the vertex carries the color tensor

$$(\text{photon–quark vertex color})_{ij} = \delta_{ij}, \tag{28.17}$$

with i the outgoing quark color and j the outgoing antiquark color. Every kinematic and spin factor in the amplitude is identical to the QED process $e^+e^- \to \mu^+\mu^-$ of Section 17.3 with the muon charge replaced by Q_q; the entire QCD modification is multiplicative and lives in Eq. (28.17).

The color sum. The amplitude is $\mathcal{M}_{ij} = \delta_{ij}\, \mathcal{M}_{\text{kin}}$. There is no initial-state color to average (the e^+e^- are colorless), and the final-state quark and antiquark

colors are unobserved and must be summed. Summing the squared amplitude over i and j,

$$\sum_{i,j=1}^{N_c} |\mathcal{M}_{ij}|^2 = |\mathcal{M}_{\text{kin}}|^2 \sum_{i,j} \delta_{ij}\delta_{ij} = |\mathcal{M}_{\text{kin}}|^2 \sum_{i} \delta_{ii} = N_c\, |\mathcal{M}_{\text{kin}}|^2. \tag{28.18}$$

The color factor is exactly $N_c = 3$: the quark pair can be produced in any of N_c color states, and these add incoherently in the rate.

The singlet-projection view. Equivalently, the singlet current creates the color state $|q\bar{q}\rangle = \sum_i |q_i\, \bar{q}_i\rangle$. Its norm is $\langle q\bar{q}\,|\,q\bar{q}\rangle = \sum_{i,j}\langle q_i\bar{q}_i|q_j\bar{q}_j\rangle = \sum_{i,j}\delta_{ij} = N_c$, so the normalized physical singlet is $\frac{1}{\sqrt{N_c}}\sum_i |q_i\bar{q}_i\rangle$ and the production rate from the singlet operator carries $\left|\sqrt{N_c}\right|^2 = N_c$. The two countings agree, as they must.

The result. For a single quark flavor of charge Q_q,

$$\sigma(e^+e^- \to q\bar{q}) = N_c\, Q_q^2\, \sigma(e^+e^- \to \mu^+\mu^-), \tag{28.19}$$

the QED cross section of Eq. (17.88) multiplied by Q_q^2 (the quark charge in units of e) and by the color count N_c. This single multiplication is the entire content of color in the totally inclusive annihilation rate, and it is what the next section turns into the measurable R-ratio.

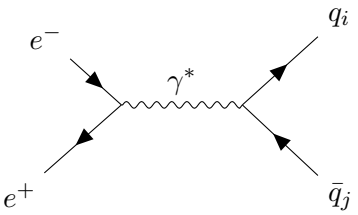


Figure 28.1: Lowest-order $e^+e^- \to q\bar{q}$ through a virtual photon. The electromagnetic current is a color singlet, so the vertex carries the color tensor δ_{ij} relating the outgoing quark color i and antiquark color j. Summing the squared amplitude over the unobserved final colors gives $\sum_{ij}\delta_{ij}\delta_{ij} = N_c = 3$, the only QCD modification of the QED rate (28.19).

Worked Example 28.4 (The color factor C_F for single-gluon emission)**.** The next-simplest color sum governs a quark radiating one gluon, $q \to q\,g$ (the real-emission piece of the α_s correction to the R-ratio in the next section, and the prototype of every parton-shower splitting). The vertex supplies the color tensor $(T^a)_{ij}$: incoming quark color j, outgoing quark color i, emitted gluon color a. The amplitude is $\mathcal{M}^a_{ij} = (T^a)_{ij}\,\mathcal{M}_{\text{kin}}$.

The incoming quark color j is averaged (divide by N_c); the outgoing quark color i and gluon color a are summed. The color-weighted squared amplitude is

therefore

$$\frac{1}{N_c}\sum_a\sum_{i,j}\left|(T^a)_{ij}\right|^2 = \frac{1}{N_c}\sum_a\sum_{i,j}(T^a)_{ij}(T^a)^*_{ij} = \frac{1}{N_c}\sum_a\sum_{i,j}(T^a)_{ij}(T^a)_{ji}$$
$$= \frac{1}{N_c}\sum_a \mathrm{tr}(T^aT^a) = \frac{1}{N_c}\,\mathrm{tr}(C_F\,\mathbf{1}) = \frac{1}{N_c}\,C_F\,N_c = C_F = \frac{4}{3}, \tag{28.20}$$

where the second equality used the Hermiticity $(T^a)^*_{ij} = (T^a)_{ji}$, the fourth the Casimir Eq. (28.7), and the last the trace of the identity. The color factor for single-gluon emission off a quark is exactly the fundamental Casimir C_F: each quark line radiates with a strength set by its "color charge squared," $C_F = \frac{4}{3}$, just as each charged line in QED radiates with strength Q^2. The analogue for a gluon radiating a gluon would instead carry $C_A = 3$, the larger adjoint charge that makes gluon jets broader than quark jets — a fact the jet section will use.

Caution

The averaging denominators in Eq. (28.1) are part of the *definition* of the cross section, not part of the color algebra; forgetting the $1/N_c$ on an incoming quark, or summing rather than averaging an initial color, is the single most common color error. The mnemonic is symmetric with spin: average over what you do not control (initial), sum over what you do not measure (final).

With Eq. (28.6) proved, the constants Eq. (28.16) tabulated, and the averaging/summing rules Theorem 28.1 in force, the color bookkeeping of QCD is complete: any cross section is its QED-like kinematic core times a color factor assembled by these rules. The next section exploits the cleanest instance of this factorization, the totally inclusive annihilation $e^+e^- \to$ hadrons, where Eq. (28.19) multiplies the QED template by $N_c\sum_q Q_q^2$ and turns the data into a direct count of the three colors and a measurement of α_s.

28.2 e^+e^- Annihilation to Hadrons and the R-Ratio

The previous section assembled the color toolkit: the rule that a color-singlet probe couples to a quark line through the identity in color space, that final-state colors are summed and initial-state colors averaged, and the single completeness identity that collapses a color sum to a counting factor (Eq. (28.7) and the surrounding discussion in Section 28.1). We now spend that toolkit on the cleanest quantitative test QCD offers. The plan is disarmingly simple. We already own, exactly, the QED amplitude for $e^+e^- \to \mu^+\mu^-$ from Section 17.3; a quark differs from a muon only in two respects a Feynman diagram can see — its

electric charge and the fact that it carries color — so the hadronic annihilation cross section is the muon one with two substitutions, a charge and a color factor. The miracle that makes this honest, rather than a back-of-envelope analogy, is *total inclusivity*: we ask only for the rate to produce *anything* hadronic, never resolving which hadrons. Inclusivity is what trivializes factorization here, and it is why this is the first place the chapter's organizing principle pays off with a number.

28.2.1 Why the totally inclusive rate is calculable

At the energies of interest — a few GeV up to the Z pole — e^+e^- annihilation proceeds dominantly through a single virtual photon (we keep the photon channel throughout; the Z contribution is a separate electroweak amplitude treated in Chapter 31). The electron and positron annihilate into γ^*, which carries the timelike momentum q with $q^2 = s$ equal to the squared center-of-mass energy (Chapter 16). That photon must then materialize as hadrons. Because the photon is colorless and couples only to electric charge, the hadronic final state is created by the electromagnetic current of the quarks,

$$j^\mu_{\rm em}(x) = \sum_q e_q\, \bar q(x)\, \gamma^\mu\, q(x), \qquad e_q \equiv Q_q\, e, \tag{28.21}$$

where the sum runs over quark flavors and Q_q is the quark charge in units of the positron charge $e > 0$: $Q_u = Q_c = Q_t = +\tfrac{2}{3}$ for the up-type quarks and $Q_d = Q_s = Q_b = -\tfrac{1}{3}$ for the down-type. Each quark field q also carries a suppressed color index $i = 1, \dots, N_c$ on which $j^\mu_{\rm em}$ acts as the identity, since the photon does not see color.

The total hadronic cross section is the sum over *every* hadronic final state X that the current can produce,

$$\sigma(e^+e^- \to \text{hadrons}) = \sum_X \sigma(e^+e^- \to X), \tag{28.22}$$

weighted by the appropriate phase space for each X. Written this way the right-hand side looks hopeless: it involves the matrix elements $\langle X | j^\mu_{\rm em} | 0 \rangle$ of the current onto pions, kaons, nucleon–antinucleon pairs, multi-hadron continua, and resonances — genuinely nonperturbative, confinement-dominated quantities we cannot compute. The content of inclusivity is that we never need any one of them. Every amplitude factorizes through the photon,

$$\mathcal{M}(e^+e^- \to X) = \Big[\underbrace{\bar v(p')\, (-ie\gamma^\mu)\, u(p)}_{\text{leptonic vertex}} \Big] \frac{-i\, g_{\mu\nu}}{s} \, \langle X |\, j^\nu_{\rm em}(0)\, | 0 \rangle, \tag{28.23}$$

so that, after squaring and integrating over the phase space of X, the cross section is the contraction of a purely leptonic tensor with a hadronic one,

$$\sigma(e^+e^- \to \text{hadrons}) \;\propto\; L_{\mu\nu}(p, p')\, W^{\mu\nu}(q),$$

$$W^{\mu\nu}(q) \equiv \sum_X \int d\Pi_X \, \langle 0|j^\mu|X\rangle\langle X|j^\nu|0\rangle \, (2\pi)^4 \delta^4(q - p_X). \quad (28.24)$$

The leptonic tensor $L_{\mu\nu}$ is exactly the one computed in Section 17.3.2 (the electron side is unchanged). All the hadronic complexity is packaged into $W^{\mu\nu}$, and there the sum over a *complete* set of hadronic states is the whole point: completeness, $\sum_X |X\rangle\langle X| = \mathbf{1}$, together with the optical theorem, identifies $W^{\mu\nu}$ as the absorptive part of the current–current correlator,

$$W^{\mu\nu}(q) = 2 \, \mathrm{Im} \, \Pi^{\mu\nu}(q),$$
$$\Pi^{\mu\nu}(q) = i \int d^4x \, e^{iq\cdot x} \, \langle 0| \, \mathcal{T} \, j^\mu_{\mathrm{em}}(x) \, j^\nu_{\mathrm{em}}(0) \, |0\rangle = (q^\mu q^\nu - q^2 g^{\mu\nu}) \, \Pi(q^2). \quad (28.25)$$

This is the hadronic vacuum polarization — the very object computed for a single lepton in Eq. (19.49), now built from quark currents. Current conservation $\partial_\mu j^\mu_{\mathrm{em}} = 0$ forces the transverse tensor structure $q^\mu q^\nu - q^2 g^{\mu\nu}$, leaving a single scalar function $\Pi(q^2)$.

Physical Insight

The trade we have made is the heart of *quark–hadron duality*. The spectral function $\mathrm{Im}\,\Pi(s)$ is one analytic function of s. On the physical cut $s > 0$ it is a sum over real hadronic thresholds and resonances — the messy, nonperturbative data of Eq. (28.22). But the *same* analytic function, continued to large spacelike $q^2 = -Q^2$ with $Q^2 \to \infty$, lives in the deep-Euclidean region where asymptotic freedom (Chapter 27) makes $\alpha_s(Q) \to 0$ and the correlator is computable in perturbative QCD from quark and gluon fields. Because Π is analytic, its perturbative value controls the *average* of the physical spectral function over a range of s. The inclusive cross section — which is exactly such an average, the integral over all of phase space — is therefore given by the quark-level calculation. We do not need to know how a $q\bar{q}$ pair turns into pions; we need only that it does so with unit probability, which confinement guarantees.

The upshot is a clean operational rule, valid away from the immediate neighborhood of narrow resonances: *the totally inclusive hadronic rate equals the rate to produce quarks and gluons, computed perturbatively.* At leading order the only partonic final state the photon can make is a single $q\bar{q}$ pair. We compute that rate now.

28.2.2 From muons to quarks: the partonic cross section

Place the quark calculation beside the muon one of Section 17.3 and ask what changes. The electron side of Eq. (28.23) is identical. On the far side, the

QED vertex for a fermion of electric charge $e_f = Q_f e$ is $-iQ_f e\,\gamma^\mu$ (the minimal-coupling rule $D_\mu = \partial_\mu - iQ_f eA_\mu$ of Chapter 17). For the muon $Q_\mu = -1$, so its vertex carried a factor e; for a quark of flavor q the vertex carries $Q_q e$. Nothing else on that side of the diagram is altered: the outgoing spinors $\bar{u}(k)$, $v(k')$, the photon propagator, and the kinematics are the same functions of (s,t,u). Therefore, at fixed quark color, the quark amplitude is the muon amplitude rescaled by the charge ratio,

$$\mathcal{M}(e^+e^- \to q_i\,\bar{q}_j) = Q_q\,\delta_{ij}\,\mathcal{M}(e^+e^- \to \mu^+\mu^-), \tag{28.26}$$

where the Kronecker delta δ_{ij} in the color indices (i,j) of the quark and antiquark is the colorless photon vertex: the current Eq. (28.21) is diagonal in color, so the pair is produced in the configuration $\sum_i |q_i\,\bar{q}_i\rangle$, color and anticolor matched.

Squaring Eq. (28.26) and summing over final-state colors (no initial-state color average is needed — electrons and positrons are color singlets),

$$\begin{aligned}\sum_{i,j} \left|\mathcal{M}(e^+e^- \to q_i\bar{q}_j)\right|^2 &= Q_q^2 \Big(\sum_{i,j}\delta_{ij}\delta_{ij}\Big)\left|\mathcal{M}(e^+e^- \to \mu^+\mu^-)\right|^2 \\ &= Q_q^2 \Big(\sum_{i} 1\Big)\left|\mathcal{M}_{\mu\mu}\right|^2 = N_c\,Q_q^2\left|\mathcal{M}_{\mu\mu}\right|^2.\end{aligned} \tag{28.27}$$

The color sum $\sum_{i,j}\delta_{ij}\delta_{ij} = \sum_i \delta_{ii} = \mathrm{tr}\,\mathbf{1}_{N_c} = N_c$ is precisely the $q\bar{q}$ counting factor isolated in Eq. (28.7): a quark and an antiquark of matched color, summed over the N_c choices, contribute N_c identical, incoherent copies because the spin/kinematic structure does not depend on color. Since the cross section is built from $\overline{|\mathcal{M}|^2}$ by the same flux and phase-space factors for quarks as for muons (the two-body phase space and cross-section formula of Eqs. (16.47) and (16.112) care only about masses and s), the substitution carries through verbatim to the cross section. Taking the high-energy limit in which the light-quark mass is negligible, $m_q^2 \ll s$, and importing the muon result $\sigma(e^+e^- \to \mu^+\mu^-) = 4\pi\alpha^2/3s$ from Eq. (17.89), we obtain

$$\boxed{\sigma(e^+e^- \to q\bar{q}) = N_c\,Q_q^2\,\sigma(e^+e^- \to \mu^+\mu^-) = N_c\,Q_q^2\,\frac{4\pi\alpha^2}{3s}\,, \qquad s \gg 4m_q^2.} \tag{28.28}$$

For energies not far above a quark threshold the full mass dependence is recovered by the same route, simply by importing the threshold-corrected muon cross section Eq. (17.99) instead of its massless limit: a quark contributes

$$\sigma(e^+e^- \to q\bar{q}) = N_c\,Q_q^2\,\frac{4\pi\alpha^2}{3s}\,\beta_q\Big(1 + \tfrac{2m_q^2}{s}\Big), \qquad \beta_q = \sqrt{1 - \tfrac{4m_q^2}{s}}, \tag{28.29}$$

which turns on linearly in the quark velocity β_q at $\sqrt{s} = 2m_q$ exactly as the muon pair did. The free-quark formula Eq. (28.28) thus describes a step that switches on, smoothly broadened by the threshold factor, each time the energy crosses a

new pair threshold. (At the real thresholds the smooth turn-on is overprinted by bound-state resonances — the J/ψ family at charm, the Υ family at bottom — which are exactly the places duality is *local* rather than point-by-point; see the caution below.)

28.2.3 The R-ratio and its threshold staircase

It is conventional, and physically wise, to divide the hadronic cross section by the muon one. The ratio cancels the overall $4\pi\alpha^2/3s$ — and with it the bulk of the experimental and theoretical systematics (luminosity, the $1/s$ falloff, radiative corrections common to both)— leaving a pure number that is a direct readout of the quark quantum numbers.

Definition 28.5 (The R-ratio)**.**

$$R(s) \;\equiv\; \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)}. \tag{28.30}$$

Summing the partonic contribution Eq. (28.28) over all quark flavors that are kinematically accessible at energy $\sqrt{s}$ (those with $4m_q^2 < s$) and dividing by the muon cross section, the common factor $4\pi\alpha^2/3s$ cancels and the parton-model prediction is

$$\boxed{R(s) = N_c \sum_{q:\, 4m_q^2 < s} Q_q^2 \,.} \tag{28.31}$$

This is one of the most consequential simple formulas in the Standard Model. It says that a single dimensionless number, measured at colliders, equals N_c times a sum of squared fractional charges. Both factors are nontrivial predictions: the integer N_c counts colors, and the rational sum encodes the assignment of electric charges to quarks. As the energy climbs past each pair threshold, R jumps up by $N_c Q_q^2$ — a staircase whose tread heights are fixed by group theory and the charge table.

Worked Example 28.6 (The staircase $R = 2,\ \frac{10}{3},\ \frac{11}{3}$)**.** Evaluate Eq. (28.31) with $N_c = 3$ in each energy band, adding one flavor at a time in order of increasing mass. We use $Q_u = Q_c = +\frac{2}{3}$ and $Q_d = Q_s = Q_b = -\frac{1}{3}$.

Below charm ($2m_s \lesssim \sqrt{s} \lesssim 2m_c$, roughly 1.5–3 GeV): the three light flavors u, d, s are active.

$$\sum_{u,d,s} Q_q^2 = \left(\tfrac{2}{3}\right)^2 + \left(\tfrac{1}{3}\right)^2 + \left(\tfrac{1}{3}\right)^2 = \tfrac{4}{9} + \tfrac{1}{9} + \tfrac{1}{9} = \tfrac{6}{9} = \tfrac{2}{3}, \qquad R = 3 \cdot \tfrac{2}{3} = \boxed{2}. \tag{28.32}$$

Above charm, below bottom ($2m_c \lesssim \sqrt{s} \lesssim 2m_b$, roughly 4–9 GeV): add the charm quark, $Q_c^2 = \frac{4}{9}$.

$$\sum_{u,d,s,c} Q_q^2 = \tfrac{2}{3} + \tfrac{4}{9} = \tfrac{6}{9} + \tfrac{4}{9} = \tfrac{10}{9}, \qquad R = 3 \cdot \tfrac{10}{9} = \boxed{\tfrac{10}{3}} \approx 3.33. \tag{28.33}$$

Above bottom ($\sqrt{s} \gtrsim 2m_b \approx 10$ GeV, up to where the Z and top matter): add the bottom quark, $Q_b^2 = \frac{1}{9}$.

$$\sum_{u,d,s,c,b} Q_q^2 = \tfrac{10}{9} + \tfrac{1}{9} = \tfrac{11}{9}, \qquad R = 3 \cdot \tfrac{11}{9} = \boxed{\tfrac{11}{3}} \approx 3.67. \tag{28.34}$$

The top quark, with $2m_t \approx 350$ GeV, is never produced in this single-photon channel at the energies where R is measured this way, so the staircase ends at 11/3. These three numbers — 2, 10/3, 11/3 — are the parton-model predictions plotted in Fig. 28.2, and they are what the data show.

Caution

The staircase is the *average* behavior; it is not literally what a fine-grained scan of $R(s)$ looks like. Near each quark–antiquark threshold there are narrow vector resonances — the ρ, ω, ϕ in the light sector, the $J/\psi, \psi'$ at charm, the Υ family at bottom — where R spikes far above its plateau value. The parton formula Eq. (28.31) is the prediction for R *smeared* over an energy interval that contains several resonances (*global* quark–hadron duality), or for R on the plateaus between thresholds, away from the bound-state region. It is not a prediction for the resonance peaks themselves, which are intrinsically nonperturbative. This is the precise sense in which duality holds and the precise sense in which it does not.

Historical Note

When color was first proposed it was a bookkeeping device — introduced by Greenberg, and by Han and Nambu, to let three quarks sit in a symmetric ground-state baryon wavefunction without violating Fermi statistics. The R-ratio turned it into a measured quantity. Without color, Eq. (28.31) would read $R = \sum_q Q_q^2$, giving $R = \frac{2}{3}$ below charm instead of 2 — a factor of three low. The data of the early 1970s sat near 2, not $\frac{2}{3}$, and the rise toward $\frac{10}{3}$ across the charm threshold (the same threshold that produced the J/ψ in November 1974) confirmed both $N_c = 3$ and the charge of the charm quark. Few measurements so directly count an internal quantum number.

28.2.4 The leading QCD correction and why it is finite

Equation Eq. (28.31) is the prediction of the free quark model — quarks treated as noninteracting Dirac particles. QCD corrects it. At order α_s the produced $q\bar{q}$ pair can radiate a gluon, and the $q\bar{q}$ vertex receives a one-loop correction. The remarkable fact, which makes R the gold-standard α_s observable, is that the

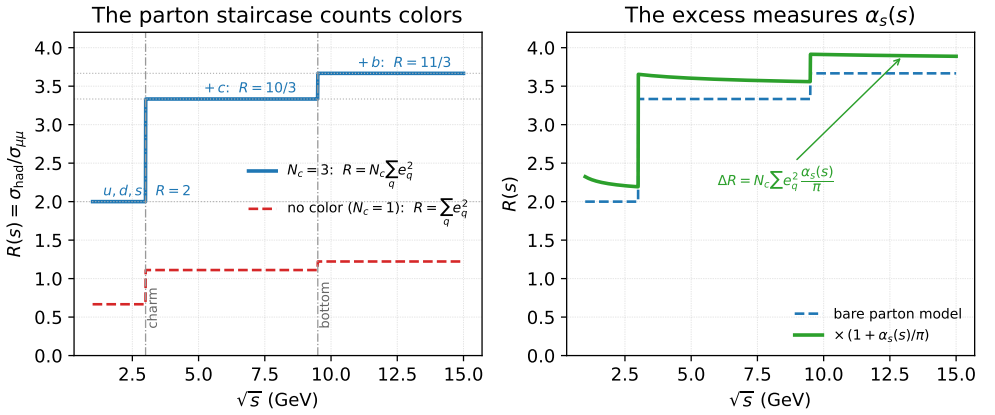


Figure 28.2: The R-ratio $R(s) = \sigma_{\rm had}/\sigma_{\mu\mu}$. *Left:* the parton-model staircase Eq. (28.31) with $N_c = 3$ (blue), flat at $R = 2$, $\frac{10}{3}$, $\frac{11}{3}$ across the $u\,d\,s$, $+c$, and $+b$ regions, contrasted with what a colorless theory ($N_c = 1$, red dashed) would predict — a factor of three lower. The data lie on the $N_c = 3$ curve: this is how e^+e^- annihilation counts colors. *Right:* the same prediction dressed with the leading QCD correction $(1 + \alpha_s(s)/\pi)$ of Eq. (28.35); the small positive excess over the bare steps is what the data turn into a measurement of $\alpha_s(s)$.

correction is a clean, finite, calculable number:

$$\boxed{R(s) = N_c \sum_q Q_q^2 \left[1 + \frac{\alpha_s(s)}{\pi} + O(\alpha_s^2) \right].} \tag{28.35}$$

The running coupling is evaluated at the only scale in the problem, the center-of-mass energy $\mu = \sqrt{s}$, which is exactly the regime where $\alpha_s(\sqrt{s})$ is small (Chapter 27); the $O(\alpha_s/\pi) \sim$ few-percent size of the correction is the quantitative face of asymptotic freedom. We defer the *explicit* evaluation of the coefficient — the cancellation of real gluon emission against the virtual correction that produces the bare 1 in front of α_s/π — to the jets section (Section 28.6), where the same computation is organized around an infrared-safe observable. Here we establish the structural point that justifies even writing Eq. (28.35): that the correction is finite at all.

The danger is infrared divergence. Taken on their own, the two $O(\alpha_s)$ contributions to $\sigma(e^+e^- \to \text{hadrons})$ are each separately divergent. The real-emission process $e^+e^- \to q\bar{q}g$ diverges when the gluon is soft (its energy $\to 0$) and when it is collinear with the quark or antiquark; the virtual correction to $e^+e^- \to q\bar{q}$ contains a compensating loop divergence of the opposite sign. Both are genuine infrared singularities, regulated for instance dimensionally ($d = 4 - 2\varepsilon$), and each is, by itself, infinite. What rescues the total is the Kinoshita–Lee–Nauenberg theorem (Chapter 19): in a transition probability

summed over all final states that are degenerate in energy — here, summed over the two-parton state $q\bar{q}$ and the three-parton state $q\bar{q}g$ that becomes degenerate with it in the soft and collinear limits — the infrared divergences cancel, leaving a finite result.

Physical Insight

The KLN cancellation is not a technical accident; it is the inclusivity of R made mathematical. A detector cannot distinguish a quark from a quark accompanied by an arbitrarily soft gluon, nor a quark from a pair of exactly collinear partons sharing its momentum — these final states are physically indistinguishable, so any *measurable* cross section must sum over them. The totally inclusive hadronic rate sums over *everything*, so it is maximally infrared safe: every degenerate configuration is included by construction, and KLN guarantees the soft and collinear infinities cancel state by state. This is the same statement as the optical-theorem picture of Section 28.2.1: R is the absorptive part of a single correlator $\Pi(s)$, an object that is ultraviolet-renormalized but carries no infrared scale at all once the quark masses are dropped, so its imaginary part is simply finite. There is nothing left to diverge.

The finiteness has a sharp diagnostic value. Had we tried to define a less inclusive observable — say, the cross section to produce exactly two hadronic jets, or the rate to find a quark of a definite momentum — we would *not* have found a finite perturbative answer, because we would have thrown away some of the degenerate states KLN needs. The lesson, which the next section takes up in earnest, is that QCD predicts infrared-safe quantities, and total inclusivity is the easiest way to be infrared safe.

28.2.5 Two readings of one measurement

Equation Eq. (28.35) is a single curve, but the data in it answer two different questions depending on what one looks at.

The normalization counts colors. The height of each plateau is $N_c \sum Q_q^2$. The charges Q_q are fixed independently (by the hadron charges, by deep inelastic scattering, by the very existence of charge-$+\frac{2}{3}$ and $-\frac{1}{3}$ states), so the overall level of R measures the integer N_c. The observed values $R \approx 2$, $\frac{10}{3}$, $\frac{11}{3}$ are the $N_c = 3$ predictions; the $N_c = 1$ "colorless" alternative (left panel of Fig. 28.2) is excluded by a clean factor of three. This is arguably the most direct measurement of the number of colors in existence — a counting experiment whose answer is read off the vertical axis.

The excess measures α_s. Once $N_c = 3$ is fixed, the small upward departure of the measured R from the integer-and-rational staircase — the $+\,\alpha_s(s)/\pi$ term —

is a measurement of the strong coupling at the scale $\sqrt{s}$. Because the correction is a pure, calculable number with no nonperturbative contamination (that was the point of the KLN argument), the inversion is clean:

$$\alpha_s(s) = \pi \left[\frac{R(s)}{N_c \sum_q Q_q^2} - 1 \right] + O(\alpha_s^2). \tag{28.36}$$

Measuring R at several energies and reading off $\alpha_s(\sqrt{s})$ at each maps out the running of the coupling — the same logarithmic decrease predicted by the beta function of Chapter 27 — and is one of the inputs to the world-average determination of α_s. The right panel of Fig. 28.2 shows the few-percent excess that carries this information.

The carried number of Yang–Mills theory at this layer is therefore $R = N_c \sum_q Q_q^2 \, (1 + \alpha_s/\pi)$: the color factor N_c from the classical non-abelian Lagrangian, made visible as the normalization of a cross section, with the running coupling of Chapter 27 riding on top as its first quantum correction.

We have exploited inclusivity to the hilt: by refusing to ask which hadrons emerge, we made factorization trivial and the prediction a single clean number. The next question is what happens when the experiment *does* resolve hadronic structure — when the photon strikes a single quark *inside* a proton and the rest of the proton is a spectator we cannot integrate away. Inclusivity is then lost on the initial state, and the price is a new, irreducibly nonperturbative object: the parton distribution function. Deep inelastic scattering, and the parton model that tames it, is the subject of Section 28.3.

28.3 Deep Inelastic Scattering and the Parton Model

The R-ratio of Section 28.2 was clean for one reason and one reason only: total inclusivity. By refusing to ask which hadrons emerge from $e^+e^- \to$ hadrons we summed over a complete set of final states, the hadronic sum collapsed onto the partonic one by unitarity, and the long-distance physics of hadronization dropped out of the rate entirely. That section closed by asking the obvious next question. What happens when the experiment *does* resolve hadronic structure — when the probe strikes a single constituent *inside* a hadron and the remainder of the hadron is a spectator we cannot sum away? Inclusivity is then lost on at least one external leg, the cancellation that made R a pure number fails, and we are forced to confront, head on, the nonperturbative object that the totally inclusive observable let us avoid: the distribution of a hadron's momentum over its constituents. Deep inelastic scattering (DIS) is the cleanest experiment of exactly this kind, and the parton model is the idea that tames it. The reward for the extra difficulty is enormous: DIS is where the quarks were first *seen*, where their spin and charges were read off the data, and where the parton distribution functions — the universal long-distance objects on which all of hadron-collider

phenomenology rests — make their first appearance. This section sets up the kinematics, builds the most general structure of the cross section that Lorentz invariance and current conservation permit, derives the parton-model predictions by the impulse approximation, and extracts from the data the two pieces of evidence that turned the parton model from a heuristic into a discovery: Bjorken scaling and the Callan–Gross relation.

We continue Yang–Mills theory. The color factor N_c surfaced in Section 28.2 as the normalization of a cross section; here the same quarks reappear as the charged, spin-$\frac{1}{2}$ partons whose existence DIS establishes, and the parton distributions we define become the universal nonperturbative input of the factorization program completed in Section 28.5.

28.3.1 The process and its kinematics

The reaction is

$$e^-(k) \;+\; N(P) \;\longrightarrow\; e^-(k') \;+\; X, \tag{28.37}$$

where N is a nucleon (we take a proton of mass M unless stated otherwise) and X denotes *anything*: a single recoiling nucleon at one extreme, a multi-hadron final state of arbitrary invariant mass at the other. The defining feature of DIS is that we measure only the scattered lepton — its energy and angle — and sum inclusively over everything in X. At the energies of interest the dominant interaction is the exchange of a single virtual photon between the lepton line and the hadronic system; the weak charged- and neutral-current analogues (W, Z exchange) are obtained by replacing the photon, and we treat them as a coda in Section 28.3.8. The leptonic side is pure QED: the electron emits the photon through the vertex of Chapter 17, and everything we do not know sits on the hadronic side.

The kinematics is governed by the momentum transferred from the lepton to the hadron,

$$q \;\equiv\; k - k', \qquad Q^2 \;\equiv\; -q^2 \;>\; 0. \tag{28.38}$$

The photon is *spacelike*, $q^2 < 0$, because in the rest frame of the target the lepton transfers three-momentum without being able to put the exchanged quantum on shell; we write $Q^2 = -q^2$ so that Q^2 is positive and plays the role of the resolving power of the probe. By the uncertainty principle the photon resolves transverse distances of order $1/Q$, so large Q^2 means a short-distance probe, exactly the regime in which $\alpha_s(Q)$ is small (Chapter 27) and a perturbative treatment of the photon–quark interaction is legitimate. Neglecting the electron mass, $k^2 = k'^2 = 0$, so

$$Q^2 \;=\; -(k-k')^2 \;=\; 2\,k\!\cdot\!k' \;=\; 4EE'\sin^2\!\tfrac{\theta}{2}, \tag{28.39}$$

the last form in the target rest frame (the "lab" frame), with E, E' the incoming and scattered electron energies and θ the scattering angle.

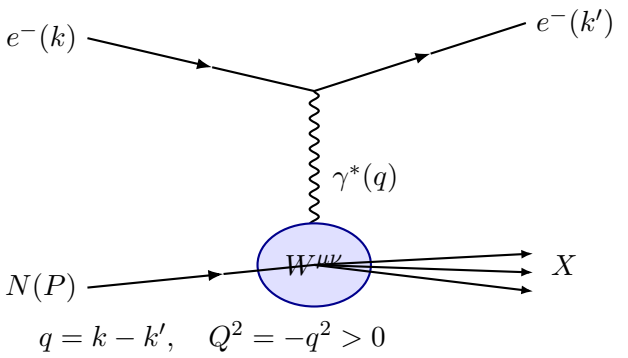


Figure 28.3: Deep inelastic scattering $e^- N \to e^- X$ at leading order in QED. The electron exchanges a single spacelike virtual photon of momentum $q = k - k'$ with the nucleon. Only the scattered electron is measured; the hadronic final state X is summed over inclusively, and all the unknown long-distance physics is packaged into the hadronic tensor $W^{\mu\nu}$.

Two more invariants complete the description of the single measured lepton. The *energy transfer* in the lab frame is $\nu \equiv E - E'$, written covariantly as

$$\nu \;\equiv\; \frac{P\cdot q}{M}, \tag{28.40}$$

and the squared invariant mass of the unobserved hadronic system is

$$W^2 \;\equiv\; (P+q)^2 \;=\; M^2 + 2\,P\cdot q + q^2 \;=\; M^2 + 2M\nu - Q^2. \tag{28.41}$$

Elastic scattering, $eN \to eN$, is the special case $X = N$, i.e. $W^2 = M^2$, which forces $2M\nu = Q^2$. "Inelastic" means $W^2 > M^2$, i.e. $2M\nu > Q^2$, and "deep" means we take both Q^2 and ν large. The two dimensionless variables that organize the deep regime are the *Bjorken scaling variable*

$$x \;\equiv\; \frac{Q^2}{2\,P\cdot q} \;=\; \frac{Q^2}{2M\nu}, \tag{28.42}$$

and the *inelasticity*

$$y \;\equiv\; \frac{P\cdot q}{P\cdot k} \;=\; \frac{\nu}{E} \quad \text{(lab)}, \tag{28.43}$$

the latter being the fraction of the incident lepton energy transferred to the hadronic system in the lab. From Eq. (28.41), $W^2 \geq M^2$ together with $\nu \geq 0$ bounds x to the physical interval

$$0 \;\leq\; x \;\leq\; 1, \tag{28.44}$$

with $x = 1$ the elastic point ($W^2 = M^2$) and $x \to 0$ the deeply inelastic, large-W^2 regime. Likewise $0 \leq y \leq 1$. The three invariants Q^2, x, y are not independent:

with $s \equiv (P+k)^2 \simeq 2P\cdot k$ (neglecting masses),

$$Q^2 \;=\; 2x\,(P\cdot q) \;=\; x\,y\,(2P\cdot k) \;=\; x\,y\,s, \tag{28.45}$$

so a measurement of any two of $\{x, y, Q^2\}$ at fixed beam energy determines the third. We will use (x, Q^2) as the independent pair, since x is the variable in which the parton model becomes transparent and Q^2 is the resolution at which we probe.

Physical Insight

The single most important number in DIS is x. Equation (28.42) is, for now, just a definition built from the measured lepton. The content of the parton model, proved below, is that when the photon scatters *elastically off a pointlike constituent* carrying a fraction ξ of the nucleon's momentum, the elastic condition forces $\xi = x$. The externally measured ratio x is then nothing but the momentum fraction of the struck constituent. This is why x is called the scaling variable: the cross section, written as a function of x, ceases to depend on the resolution Q^2 — because it is reporting a property of the constituents, not of the probe.

28.3.2 The leptonic and hadronic tensors

Because a single photon is exchanged, the amplitude factorizes into a leptonic current, a photon propagator, and a hadronic matrix element. The leptonic vertex is the QED vertex of Chapter 17: with the electron charge $-e$ ($e>0$) the current is $-e\,\bar{u}(k')\gamma^\mu u(k)$. The squared amplitude, summed over final and averaged over initial lepton spins, therefore contains the *leptonic tensor*

$$L^{\mu\nu} \;\equiv\; \tfrac{1}{2}\sum_{\text{spins}} \big[\bar{u}(k')\gamma^\mu u(k)\big]\big[\bar{u}(k')\gamma^\nu u(k)\big]^* \;=\; \tfrac{1}{2}\,\mathrm{tr}\big[\not{k}'\gamma^\mu \not{k}\,\gamma^\nu\big], \tag{28.46}$$

where the second equality uses the massless completeness relation $\sum_s u(k)\bar{u}(k) = \not{k}$ of Theorem 17.7 and the reflection $[\bar{u}(k')\gamma^\nu u(k)]^* = \bar{u}(k)\gamma^\nu u(k')$. The trace is the standard four-γ trace of Theorem C.45, $\mathrm{tr}[\not{a}\gamma^\mu \not{b}\,\gamma^\nu] = 4(a^\mu b^\nu + a^\nu b^\mu - (ab)\eta^{\mu\nu})$, so

$$L^{\mu\nu} \;=\; 2\Big(k'^\mu k^\nu + k'^\nu k^\mu - (k\cdot k')\,\eta^{\mu\nu}\Big) \;=\; 2\Big(k'^\mu k^\nu + k'^\nu k^\mu - \tfrac{Q^2}{2}\,\eta^{\mu\nu}\Big), \tag{28.47}$$

using $k\cdot k' = Q^2/2$ from Eq. (28.39). The leptonic tensor is real, symmetric, and conserved: a one-line check using $q = k - k'$, $k^2 = k'^2 = 0$ gives

$$q_\mu L^{\mu\nu} = 2\big[(q\cdot k')k^\nu + (q\cdot k)k'^\nu - (k\cdot k')q^\nu\big] = 2(k\cdot k')\big[k^\nu - k'^\nu - q^\nu\big] = 0, \tag{28.48}$$

since $q\cdot k = -k\cdot k'$ and $q\cdot k' = +k\cdot k'$. Current conservation on the leptonic side is exact and kinematic; on the hadronic side it is the Ward identity of the electromagnetic current, and it will do real work below.

Everything we do not control sits in the *hadronic tensor*. Define it as the nucleon matrix element of two electromagnetic currents, summed over the inclusive final state X and averaged over the nucleon spin S:

$$W^{\mu\nu}(P,q) \;\equiv\; \frac{1}{4\pi}\,\frac{1}{2}\sum_{S}\sum_{X}\!\int \mathrm{d}\Pi_X\,(2\pi)^4\delta^4(P+q-p_X)\,\langle P,S|\,J^{\mu}(0)\,|X\rangle\,\langle X|\,J^{\nu}(0)\,|P,S\rangle, \tag{28.49}$$

with $J^{\mu} = \sum_q e_q\,\bar{q}\gamma^{\mu}q$ the quark electromagnetic current in units of e (the same current of Eq. (28.21)), states normalized relativistically, and $\mathrm{d}\Pi_X$ the Lorentz-invariant phase space of X (Chapter 16). The prefactor $1/4\pi$ is the conventional normalization that makes the structure functions defined below match the standard cross-section formula. By completeness of the states $|X\rangle$ and the integral representation of the energy–momentum delta function, $W^{\mu\nu}$ is (up to the spin average) the Fourier transform of the current–current commutator,

$$W^{\mu\nu} = \frac{1}{4\pi}\,\frac{1}{2}\sum_{S}\!\int \mathrm{d}^4z\,e^{iq\cdot z}\,\langle P,S|\,[\,J^{\mu}(z),J^{\nu}(0)\,]\,|P,S\rangle, \tag{28.50}$$

the commutator (rather than the product) appearing because the X-sum runs only over physical states with $p_X^0 > 0$, so the second time ordering contributes nothing on the support $P^0+q^0>0$. We will not need this form until the operator product expansion of Chapter 38; for now Eq. (28.49) is the working definition.

The inclusive cross section is the contraction of the two tensors, weighted by the photon propagator and the flux/phase-space factors of Chapter 16. Carrying the $1/4\pi$ in $W^{\mu\nu}$, the standard wrapper gives

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}E'\,\mathrm{d}\Omega} \;=\; \frac{\alpha^2}{Q^4}\,\frac{E'}{E}\,L^{\mu\nu}W_{\mu\nu}, \qquad \alpha = \frac{e^2}{4\pi}, \tag{28.51}$$

the photon propagator supplying the $1/Q^4$ and the QED coupling the $\alpha^2 = (e^2/4\pi)^2$. All of the dynamics is now in $L^{\mu\nu}W_{\mu\nu}$, and since $L^{\mu\nu}$ is known, the entire content of DIS is the determination of $W^{\mu\nu}$.

28.3.3 The structure-function decomposition

We do not know $W^{\mu\nu}$, but we know exactly which tensors it can be built from and which constraints it must satisfy. It is a rank-two Lorentz tensor depending only on the two available vectors P^{μ} and q^{μ} (the spin average has removed any dependence on S^{μ}) and the metric. For an electromagnetic (parity-even) current the parity-violating $\epsilon^{\mu\nu\alpha\beta}$ structure is forbidden, and the spin-averaged matrix element of two Hermitian currents is symmetric in $\mu \leftrightarrow \nu$. The most general such object is

$$W^{\mu\nu} = -W_1\,\eta^{\mu\nu} + \frac{W_2}{M^2}\,P^{\mu}P^{\nu} + \frac{W_4}{M^2}\,q^{\mu}q^{\nu} + \frac{W_5}{M^2}\,(P^{\mu}q^{\nu}+P^{\nu}q^{\mu}), \tag{28.52}$$

with scalar coefficients $W_i = W_i(\nu, Q^2)$ and powers of M inserted to make them dimensionless in the convention of Eq. (28.49). The electromagnetic current is conserved, $\partial_\mu J^\mu = 0$, which in momentum space at the level of the matrix element Eq. (28.49) means

$$q_\mu W^{\mu\nu} \;=\; q_\nu W^{\mu\nu} \;=\; 0. \tag{28.53}$$

This is the hadronic Ward identity, and it removes two of the four coefficients. Contracting Eq. (28.52) with q_μ and collecting the independent vectors P^ν and q^ν,

$$\begin{aligned} q_\mu W^{\mu\nu} &= -W_1\, q^\nu + \frac{W_2}{M^2}\,(q\cdot P)\,P^\nu + \frac{W_4}{M^2}\,q^2\,q^\nu + \frac{W_5}{M^2}\big[(q\cdot P)\,q^\nu + q^2\,P^\nu\big] \\ &= \underbrace{\Big[\tfrac{W_2}{M^2}(q\cdot P) + \tfrac{W_5}{M^2}q^2\Big]}_{\text{coeff. of } P^\nu} P^\nu + \underbrace{\Big[-W_1 + \tfrac{W_4}{M^2}q^2 + \tfrac{W_5}{M^2}(q\cdot P)\Big]}_{\text{coeff. of } q^\nu} q^\nu. \end{aligned} \tag{28.54}$$

Setting both brackets to zero gives, from the P^ν coefficient,

$$W_5 \;=\; -\,\frac{q\cdot P}{q^2}\,W_2, \tag{28.55}$$

and then, from the q^ν coefficient,

$$W_4 \;=\; \frac{M^2}{q^2}\,W_1 - \frac{q\cdot P}{q^2}\,W_5 \;=\; \frac{M^2}{q^2}\,W_1 + \frac{(q\cdot P)^2}{q^4}\,W_2. \tag{28.56}$$

Substituting Eqs. (28.55) and (28.56) back into Eq. (28.52) and regrouping, the four-term tensor collapses into a manifestly transverse form built from precisely two independent structures:

$$\boxed{W^{\mu\nu} = W_1\!\left(-\eta^{\mu\nu} + \frac{q^\mu q^\nu}{q^2}\right) + \frac{W_2}{M^2}\left(P^\mu - \frac{P\cdot q}{q^2}q^\mu\right)\left(P^\nu - \frac{P\cdot q}{q^2}q^\nu\right).} \tag{28.57}$$

That this expression satisfies $q_\mu W^{\mu\nu} = 0$ is now manifest: the first projector annihilates q_μ because $q_\mu(-\eta^{\mu\nu} + q^\mu q^\nu/q^2) = -q^\nu + q^\nu = 0$, and the second because $q_\mu(P^\mu - \frac{P\cdot q}{q^2}q^\mu) = P\cdot q - P\cdot q = 0$. The two surviving scalar functions $W_1(\nu, Q^2)$ and $W_2(\nu, Q^2)$ are the *structure functions*; they contain the entire parity-even response of the nucleon to a virtual photon, and no symmetry can reduce them further. It is conventional to trade them for the dimensionless

$$F_1(x, Q^2) \;\equiv\; M\,W_1, \qquad F_2(x, Q^2) \;\equiv\; \nu\,W_2, \tag{28.58}$$

which is the form in which the parton model and the data are naturally expressed.

Notation Watch

Conventions for $W^{\mu\nu}$ differ across the literature by factors of M and by whether the dimensionless F_i or the dimensionful W_i are quoted; some texts even normalize F_2 with an extra 2. We fix, once and for all, the choice in Eq. (28.58), in which the Callan–Gross relation reads $F_2 = 2xF_1$ and the parton-model result is $F_2 = \sum_q x e_q^2 f_q(x)$. Every formula in this chapter is internally consistent with this choice; when comparing to an external reference, check its $W^{\mu\nu}$ normalization first.

28.3.4 The inclusive cross section

We now contract $L^{\mu\nu}$ with $W^{\mu\nu}$ to obtain a measurable cross section. Because $L^{\mu\nu}$ is conserved (Eq. (28.48)), every term in $W^{\mu\nu}$ proportional to q^μ or q^ν drops out of the contraction, so the two projectors in Eq. (28.57) act as if the q-pieces were absent:

$$L_{\mu\nu}W^{\mu\nu} = W_1\big(-\eta_{\mu\nu}L^{\mu\nu}\big) + \frac{W_2}{M^2}\,L_{\mu\nu}P^\mu P^\nu. \tag{28.59}$$

The first contraction uses Eq. (28.47): $\eta_{\mu\nu}L^{\mu\nu} = 2(2\,kk' - 4\,kk') = -4\,kk' = -2Q^2$, so $-\eta_{\mu\nu}L^{\mu\nu} = 2Q^2$. The second is

$$L_{\mu\nu}P^\mu P^\nu = 2\Big(2(k\cdot P)(k'\cdot P) - (k\cdot k')M^2\Big) = 4(k\cdot P)(k'\cdot P) - Q^2M^2. \tag{28.60}$$

Evaluating in the lab frame, $k\cdot P = ME$ and $k'\cdot P = ME'$, gives $L_{\mu\nu}P^\mu P^\nu = 4M^2EE' - Q^2M^2 = M^2(4EE' - Q^2) = 4M^2EE'\cos^2\frac{\theta}{2}$, where the last step used $Q^2 = 4EE'\sin^2\frac{\theta}{2}$ from Eq. (28.39). Likewise $2Q^2 = 8EE'\sin^2\frac{\theta}{2}$. Assembling Eq. (28.59),

$$L_{\mu\nu}W^{\mu\nu} = 4EE'\Big(2W_1\sin^2\tfrac{\theta}{2} + W_2\cos^2\tfrac{\theta}{2}\Big), \tag{28.61}$$

and inserting this into Eq. (28.51) with the Mott prefactor $\alpha^2/(4E^2\sin^4\frac{\theta}{2})$ that organizes elastic point scattering (Chapter 17) produces the lab-frame master formula,

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}E'\,\mathrm{d}\Omega} = \frac{\alpha^2}{4E^2\sin^4\frac{\theta}{2}}\Big(W_2\cos^2\tfrac{\theta}{2} + 2W_1\sin^2\tfrac{\theta}{2}\Big). \tag{28.62}$$

The angular structure is the physics: the $\cos^2\frac{\theta}{2}$ term is the charge (electric) response, weighted by W_2, and the $\sin^2\frac{\theta}{2}$ term is the magnetic (spin) response, weighted by W_1. For a point spin-$\frac{1}{2}$ target the two are locked together; that locking is the Callan–Gross relation, derived below.

To expose the parton physics we change variables from (E', Ω) to (x, Q^2) and pass to the dimensionless F_1, F_2. Integrating over the trivial azimuth gives $\mathrm{d}\Omega = 2\pi\,\mathrm{d}\cos\theta$; from Eq. (28.39) at fixed E', $|\mathrm{d}Q^2| = 2EE'\,|\mathrm{d}\cos\theta|$, and from

Eq. (28.42) at fixed Q^2, $|\partial x/\partial E'| = x/\nu$, so

$$\mathrm{d}E'\,\mathrm{d}\Omega \;=\; \frac{\pi\,\nu}{x\,E E'}\,\mathrm{d}x\,\mathrm{d}Q^2. \tag{28.63}$$

Carrying this through Eq. (28.62), using $\sin^{-4}\frac{\theta}{2} = 16E^2E'^2/Q^4$, $\nu = Ey$, $E' = E(1-y)$ (so $\nu E'/E = Ey(1-y)$), $W_1 = F_1/M$, $W_2 = F_2/\nu$, and the relations Eq. (28.45), a short reduction (organized cleanly through the intermediate variable y in Theorem 28.7) gives the inclusive DIS cross section,

$$\boxed{\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\,\mathrm{d}Q^2} \;=\; \frac{4\pi\alpha^2}{Q^4}\left[y^2\,F_1(x,Q^2) + (1-y)\,\frac{F_2(x,Q^2)}{x}\right] \;+\; \Big(\text{terms } \propto \tfrac{M^2}{Q^2}\Big).} \tag{28.64}$$

The dropped terms are suppressed by the target mass over the hard scale, M^2/Q^2, and vanish in the deep-inelastic limit. Equivalently, writing $y^2 = y^2$ and trading F_1 for the *longitudinal* structure function $F_L \equiv F_2 - 2xF_1$ introduced below, the bracket is $(1 - y + \frac{y^2}{2})F_2/x - \frac{y^2}{2}F_L/x$, a form that isolates the two independent photon helicities. Equation (28.64) is what the experiment measures; the theory's job is to predict $F_1(x,Q^2)$ and $F_2(x,Q^2)$.

Worked Example 28.7 (Reducing the Mott form to the (x,Q^2) cross section)**.** We fill in the algebra hidden behind Eq. (28.64). It is cleanest to pass first to (x,y) and then to (x,Q^2). From Eq. (28.43), $E' = E(1-y)$ and $\mathrm{d}E' = -E\,\mathrm{d}y$; from Eq. (28.42) and Eq. (28.39), $x = 2E(1-y)\sin^2\frac{\theta}{2}/(My)$, so at fixed y, $\mathrm{d}\sin^2\frac{\theta}{2} = \frac{My}{2E(1-y)}\,\mathrm{d}x$, while $\mathrm{d}\Omega = 2\pi\,\mathrm{d}\cos\theta = 4\pi\,\mathrm{d}\sin^2\frac{\theta}{2}$. Hence

$$\mathrm{d}E'\,\mathrm{d}\Omega = E\,\mathrm{d}y\cdot 4\pi\,\frac{My}{2E(1-y)}\,\mathrm{d}x = \frac{2\pi My}{1-y}\,\mathrm{d}x\,\mathrm{d}y. \tag{28.65}$$

Multiplying Eq. (28.62) by this Jacobian, with $\sin^4\frac{\theta}{2} = Q^4/[16E^4(1-y)^2]$ from $\sin^2\frac{\theta}{2} = Q^2/(4E^2(1-y))$, the prefactor becomes $\alpha^2/(4E^2\sin^4\frac{\theta}{2})\cdot\frac{2\pi My}{1-y} = 8\pi\alpha^2E^2My(1-y)/Q^4$. Insert $W_1 = F_1/M$, $W_2 = F_2/\nu = F_2/(Ey)$:

$$\begin{aligned}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\,\mathrm{d}y} &= \frac{8\pi\alpha^2E^2My(1-y)}{Q^4}\left[\frac{F_2}{Ey}\Big(1-\tfrac{Q^2}{4E^2(1-y)}\Big) + \frac{2F_1}{M}\,\tfrac{Q^2}{4E^2(1-y)}\right]\\ &= \frac{4\pi\alpha^2}{x\,y\,Q^2}\Big[xy^2F_1 + (1-y)F_2\Big] - \underbrace{\frac{2\pi\alpha^2M\,y\,F_2}{E\,Q^2}}_{\propto M/E},\end{aligned} \tag{28.66}$$

where we used $ME = s/2 = Q^2/(2xy)$ from Eq. (28.45) to clear the first bracket and the M/E piece is the advertised target-mass remnant. Finally, $Q^2 = sxy$ at fixed x gives $\mathrm{d}Q^2 = sx\,\mathrm{d}y$, so $\mathrm{d}^2\sigma/\mathrm{d}x\,\mathrm{d}Q^2 = (sx)^{-1}\,\mathrm{d}^2\sigma/\mathrm{d}x\,\mathrm{d}y$, and $sx = Q^2/y$ turns Eq. (28.66) into

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\,\mathrm{d}Q^2} = \frac{4\pi\alpha^2}{Q^4}\Big[y^2F_1 + (1-y)\tfrac{F_2}{x}\Big] + O\big(\tfrac{M^2}{Q^2}\big), \tag{28.67}$$

which is Eq. (28.64). The one nontrivial input beyond pure kinematics was the Mott prefactor, imported from the point-scattering computation of Chapter 17.

28.3.5 The parton model and the impulse approximation

So far everything has been exact kinematics and symmetry; no dynamical assumption about the nucleon has entered. The parton model supplies that assumption. Its physical content is an *impulse approximation*: at large Q^2 the photon resolves transverse distances $\sim 1/Q$ far smaller than the nucleon's size, and it interacts over a time $\sim 1/Q$ far shorter than the time scale $\sim 1/\Lambda_{\mathrm{QCD}}$ over which the constituents interact among themselves. To the photon, the nucleon looks momentarily like a collection of *free*, on-shell, pointlike constituents — partons — and the scattering is the incoherent sum of elastic scatterings off individual partons. The information about how the nucleon apportions its momentum among partons is encoded in a probability density.

Definition 28.8 (Parton distribution function)**.** For each parton species a (a quark flavor q, an antiquark $\bar{q}$, or the gluon g) the *parton distribution function* (PDF) $f_a(\xi)$ is the number density in longitudinal momentum fraction: $f_a(\xi)\,\mathrm{d}\xi$ is the expected number of partons of type a carrying a fraction of the parent hadron's momentum between ξ and $\xi + \mathrm{d}\xi$, in a frame where the hadron moves fast (the infinite-momentum frame, where transverse momenta and masses are negligible). We write $q(\xi) \equiv f_q(\xi)$ and $\bar{q}(\xi) \equiv f_{\bar{q}}(\xi)$ for quark and antiquark densities. Normalizations of nonperturbative origin, e.g. the valence-number sum rules $\int_0^1 [u(\xi) - \bar{u}(\xi)]\,\mathrm{d}\xi = 2$ and $\int_0^1 [d(\xi) - \bar{d}(\xi)]\,\mathrm{d}\xi = 1$ for the proton, fix the quantum numbers.

The dynamical claim is that the struck parton is on shell before and after absorbing the photon. A parton carrying momentum $p = \xi P$ (neglecting its mass and transverse momentum) that absorbs the photon q goes to $p' = p + q$, and being a massless on-shell quark afterwards requires

$$0 = p'^2 = (p+q)^2 = 2\,p\cdot q + q^2 = 2\xi\,P\cdot q - Q^2, \tag{28.68}$$

i.e.

$$\xi \;=\; \frac{Q^2}{2\,P\cdot q} \;=\; x. \tag{28.69}$$

This is the promised identification: *the elastic condition forces the struck parton's momentum fraction ξ to equal the externally measured Bjorken x.* The measured kinematic ratio x is, in the parton model, the longitudinal momentum fraction of the constituent that was hit.

We now compute the cross section as the incoherent sum of partonic elastic scatterings and read off the structure functions. The elementary process is $e^- q \to e^- q$ for a quark of charge e_q (in units of e) and momentum $p = \xi P$. This

is precisely $e^-\mu^- \to e^-\mu^-$ of Chapter 18 with the muon replaced by a quark of fractional charge: a single t-channel photon, both fermions pointlike and (here) massless. Reusing the QED result, the spin-averaged squared amplitude is

$$\overline{|\mathcal{M}|^2}_{eq\to eq} = 2e^4 e_q^2 \,\frac{\hat{s}^2+\hat{u}^2}{\hat{t}^2}, \tag{28.70}$$

with partonic Mandelstam invariants $\hat{s} = (k+\xi P)^2 \simeq \xi s$, $\hat{t} = (k-k')^2 = -Q^2$, and $\hat{u} = -\hat{s}-\hat{t}$ (massless, $\hat{s}+\hat{t}+\hat{u}=0$). The partonic differential cross section in $\hat{t}$ is then, by the two-body formula of Chapter 16,

$$\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}\hat{t}} = \frac{\overline{|\mathcal{M}|^2}_{eq\to eq}}{16\pi\hat{s}^2} = \frac{2\pi\alpha^2 e_q^2}{Q^4}\,\frac{\hat{s}^2+\hat{u}^2}{\hat{s}^2}, \tag{28.71}$$

where $e^4/(8\pi) = 2\pi\alpha^2$ and $\hat{t}^2 = Q^4$. The partonic inelasticity is the same as the leptonic one, $\hat{y} = (p\cdot q)/(p\cdot k) = (\xi P\cdot q)/(\xi P\cdot k) = y$, because the parton momentum ξP cancels between numerator and denominator; hence $\hat{u}/\hat{s} = -(1-y)$ and

$$\frac{\hat{s}^2+\hat{u}^2}{\hat{s}^2} = 1+(1-y)^2. \tag{28.72}$$

Trading $\hat{t}$ for Q^2 ($\mathrm{d}\hat{\sigma}/\mathrm{d}Q^2 = \mathrm{d}\hat{\sigma}/\mathrm{d}\hat{t}$) gives the clean partonic cross section

$$\frac{\mathrm{d}\hat{\sigma}_q}{\mathrm{d}Q^2} = \frac{2\pi\alpha^2 e_q^2}{Q^4}\,\big[\,1+(1-y)^2\,\big]. \tag{28.73}$$

Now assemble the hadronic cross section. A parton of type q and fraction ξ contributes only when the elastic condition $\xi = x$ holds; weighting by the density $f_q(\xi)$ and summing incoherently over species and over ξ,

$$\frac{\mathrm{d}^2\sigma}{\mathrm{d}x\,\mathrm{d}Q^2} = \sum_q \int_0^1 \mathrm{d}\xi\, f_q(\xi)\,\delta(\xi-x)\,\frac{\mathrm{d}\hat{\sigma}_q}{\mathrm{d}Q^2} = \sum_q f_q(x)\,\frac{2\pi\alpha^2 e_q^2}{Q^4}\,[1+(1-y)^2], \tag{28.74}$$

the delta function being the elastic constraint Eq. (28.69) expressed as a distribution in ξ. (Antiquarks contribute identically with $f_q \to f_{\bar{q}}$, since the photon is blind to the sign of the charge in $\overline{|\mathcal{M}|^2} \propto e_q^2$; we restore them below.) Comparing the parton-model prediction Eq. (28.74) term by term in y with the model-independent decomposition Eq. (28.64), and using $1+(1-y)^2 = y^2 + 2(1-y)$, we read off two equations:

$$\text{(coefficient of } y^2\text{)}: \quad \frac{4\pi\alpha^2}{Q^4}\,F_1 = \frac{2\pi\alpha^2}{Q^4}\sum_q e_q^2 f_q(x) \;\Rightarrow\; F_1(x) = \tfrac{1}{2}\sum_q e_q^2 f_q(x), \tag{28.75}$$

$$\text{(coefficient of } (1-y)\text{)}: \quad \frac{4\pi\alpha^2}{Q^4}\,\frac{F_2}{x} = \frac{4\pi\alpha^2}{Q^4}\sum_q e_q^2 f_q(x) \;\Rightarrow\; F_2(x) = x\sum_q e_q^2 f_q(x). \tag{28.76}$$

Restoring antiquarks (which carry the same e_q^2) gives the central parton-model prediction for the electromagnetic structure function,

$$\boxed{F_2(x) \;=\; \sum_q x\, e_q^2 \,[\, f_q(x) + f_{\bar q}(x)\,]\,.} \tag{28.77}$$

Physical Insight

Bjorken scaling. The right-hand side of Eq. (28.77) depends on x *but not on* Q^2. This is *Bjorken scaling*: in the deep-inelastic limit the structure functions become functions of the single dimensionless variable x,

$$F_i(x, Q^2) \xrightarrow{Q^2\to\infty,\; x \text{ fixed}} F_i(x). \tag{28.78}$$

The reason is dimensional and physical at once. A structure function could a priori depend on the ratio Q^2/M^2 of the probe's resolution to a hadronic scale; that it does not means the photon is scattering off objects with no intrinsic scale of their own — pointlike constituents. Scaling is the experimental signature that the nucleon contains pointlike partons, and its observation at SLAC in 1968–69 was the discovery of that substructure. Figure 28.4 shows the pattern: at fixed x, F_2 is, to a first approximation, flat in Q^2. The residual slow ($\sim \ln Q^2$) drift visible in the data is the scaling *violation* that Section 28.4 will compute from gluon radiation — the first sign that the partons are not, after all, exactly free.

28.3.6 The Callan–Gross relation and the spin of the partons

Equations (28.75) and (28.76) are not independent. Eliminating the common combination $\sum_q e_q^2 f_q(x)$ between them yields a relation between the two structure functions with *no* reference to the unknown distributions:

Theorem 28.9 (Callan–Gross relation)**.** *In the parton model with spin-$\tfrac12$ charged partons, the electromagnetic structure functions obey*

$$F_2(x) \;=\; 2x\,F_1(x), \qquad\Longleftrightarrow\qquad F_L(x) \equiv F_2(x) - 2xF_1(x) = 0. \tag{28.79}$$

Proof. From Eq. (28.76), $F_2 = x\sum_q e_q^2 f_q$, and from Eq. (28.75), $2xF_1 = 2x\cdot\frac12\sum_q e_q^2 f_q = x\sum_q e_q^2 f_q$. The two right-hand sides are identical, so $F_2 = 2xF_1$. Including antiquarks changes $f_q \to f_q + f_{\bar q}$ on both sides and leaves the relation intact. □

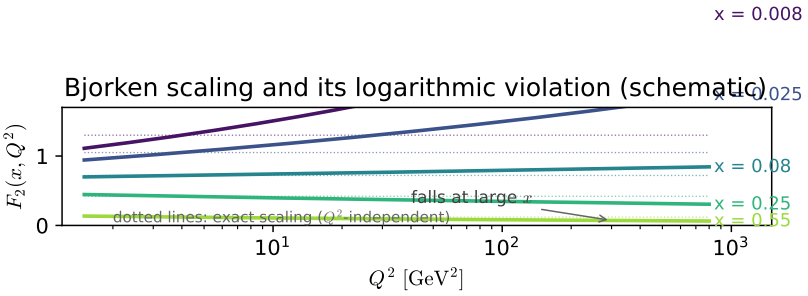


Figure 28.4: Schematic of Bjorken scaling and its logarithmic violation in the proton structure function $F_2(x, Q^2)$. At fixed x the parton model predicts a flat line (dotted), F_2 independent of Q^2. The actual mild Q^2-dependence — rising at small x, falling at large x — is the scaling violation predicted by DGLAP evolution (Section 28.4). The curves are stylised, not a fit; they exhibit the qualitative pattern that the data show.

The relation is a clean prediction precisely because the nonperturbative distributions cancel: whatever the $f_q(x)$ are, the spin-$\frac{1}{2}$ nature of the partons fixes the *ratio* of the magnetic to the electric response. To see that it is the *spin* that is being measured, it pays to give the helicity argument that underlies the algebra, because it also tells us exactly what would have changed had the partons been spinless.

Work in the Breit ("brick-wall") frame, where the virtual photon is purely spacelike, $q^\mu = (0, 0, 0, Q)$, and the struck quark reverses its three-momentum: it comes in along $-\hat{z}$ and goes out along $+\hat{z}$, "bouncing off the wall." A massless quark has definite helicity, which is conserved by the vector (electromagnetic) coupling $\bar{q}\gamma^\mu q$ — chirality, hence helicity for a massless particle, is preserved at the QED vertex (Theorem 17.18). Reversing its momentum while keeping its spin projection fixed therefore *flips* the quark's spin component along the beam axis by one unit, $\Delta J_z = \pm 1$. Angular momentum conservation requires the absorbed photon to supply that unit: only a *transversely* polarized photon, with helicity $\lambda = \pm 1$ and $J_z = \pm 1$, can be absorbed. A *longitudinal* photon, with $J_z = 0$, cannot flip the quark's spin and so cannot be absorbed at all. Hence the longitudinal absorption cross section vanishes for spin-$\frac{1}{2}$ partons,

$$\sigma_L \;=\; 0 \quad (\text{spin-}\tfrac{1}{2}), \tag{28.80}$$

and since F_L is, up to positive kinematic factors, precisely the longitudinal absorption cross section ($F_L \propto \sigma_L$, while $2xF_1 \propto \sigma_T$, the transverse one), $F_L = 0$, which is Eq. (28.79) again. The ratio

$$R(x, Q^2) \;\equiv\; \frac{\sigma_L}{\sigma_T} \;=\; \frac{F_L}{2xF_1} \;=\; \frac{F_2 - 2xF_1}{2xF_1} \tag{28.81}$$

is therefore the direct experimental measure of the partons' spin: $R \to 0$ for spin-$\frac{1}{2}$, and the data indeed give a small $R \sim 0.1$–0.2 (with the residual nonzero value coming from the gluon radiation and transverse momenta that the strict free-parton model omits), confirming that the charged partons are spin-$\frac{1}{2}$ fermions — the quarks.

The contrast with spinless partons is decisive and worth making explicit, because it shows the relation is a genuine spin measurement rather than a kinematic identity. Had the charged constituents been scalars, the photon would couple through the scalar-QED vertex $-ie_q(p+p')^\mu$ rather than $-ie_q\gamma^\mu$. The partonic squared amplitude is then $\overline{|\mathcal{M}|^2}_{\text{scalar}} \propto L_{\mu\nu}(p+p')^\mu(p+p')^\nu$, and since $(p+p') = 2p+q$ and $L^{\mu\nu}$ is conserved, only the $4p^\mu p^\nu$ piece survives: $L_{\mu\nu}(2p+q)^\mu(2p+q)^\nu = 4L_{\mu\nu}p^\mu p^\nu \propto (k\cdot p)(k'\cdot p) \propto 1\cdot(1-y)$. The bracket is then $\propto (1-y)$ alone — it contains *no* y^2 term. Matching to Eq. (28.64) as before gives $F_1 = 0$ for scalar partons, hence

$$F_1 = 0, \quad F_2 = x\sum_q e_q^2 f_q, \quad R = \frac{F_2}{2xF_1} \to \infty \quad \text{(spin-0)}. \tag{28.82}$$

A world of scalar quarks would have shown a vanishing magnetic response and an infinite R — the opposite of what is seen. The data choose spin-$\frac{1}{2}$ unambiguously.

Physical Insight

The Callan–Gross relation is the cleanest single statement of what DIS taught us about quarks. Bjorken scaling says the constituents are pointlike (no internal scale); the parton charges, read off the flavor-weighted sum $\sum_q e_q^2(f_q + f_{\bar q})$ in F_2, say they carry fractional charge; and Callan–Gross says they carry spin $\frac{1}{2}$. Pointlike, fractionally charged, spin-$\frac{1}{2}$ constituents are exactly the quarks of the QCD Lagrangian of Chapter 25. DIS is where that Lagrangian's elementary fields were first seen to be physical.

28.3.7 The momentum sum rule and the discovery of gluons

The structure function F_2 measures the charged, spin-$\frac{1}{2}$ partons — and *only* those. The photon does not couple to color, so a neutral parton is invisible in Eq. (28.77). This omission is exactly what makes DIS able to detect the gluon, by accounting. Total longitudinal momentum is shared among all partons, charged and neutral alike; if the charged partons are seen to carry less than the whole, the rest must be carried by something the photon cannot see.

Each parton of type a carries, on average, longitudinal momentum fraction $\int_0^1 \xi f_a(\xi)\,d\xi$. Since the partons' momenta must add up to the hadron's, the *momentum sum rule* is the exact statement

$$\sum_a \int_0^1 d\xi\, \xi\, f_a(\xi) \;=\; \int_0^1 d\xi\, \xi \Big[\textstyle\sum_q \left(f_q(\xi) + f_{\bar q}(\xi)\right) + g(\xi)\Big] \;=\; 1, \tag{28.83}$$

where $g(\xi)$ is the gluon density and the sum runs over all parton species. The quark contribution alone is the quantity that F_2 gives access to. Define the fraction of the nucleon momentum carried by quarks and antiquarks,

$$\varepsilon_q \;\equiv\; \sum_q \int_0^1 \mathrm{d}\xi\, \xi\,[f_q(\xi) + f_{\bar q}(\xi)]. \tag{28.84}$$

This integral is extractable from data: from Eq. (28.77), $\int_0^1 \mathrm{d}\xi\, F_2(\xi)/\xi = \sum_q e_q^2 \int_0^1 [f_q + f_{\bar q}]\mathrm{d}\xi$ gives a charge-weighted count, and the momentum-weighted combination ε_q is obtained from the analogous moments measured in both electromagnetic and neutrino DIS, where the different charge weightings let one disentangle the flavor content from the momentum content. The empirical answer, at the moderate Q^2 of the early experiments, is

$$\varepsilon_q \;\approx\; 0.5, \tag{28.85}$$

and therefore, by Eq. (28.83),

$$\boxed{\int_0^1 \mathrm{d}\xi\, \xi\, g(\xi) \;=\; 1 - \varepsilon_q \;\approx\; 0.5.} \tag{28.86}$$

Roughly half of the proton's momentum is carried by electrically neutral constituents — the gluons. This was the first quantitative evidence that the gluon of the Yang–Mills Lagrangian is a dynamical, momentum-carrying object, not a bookkeeping device. It is the DIS counterpart of the color factor N_c that Section 28.2 extracted from e^+e^-: both are numbers on a detector that the non-abelian Lagrangian of Chapter 25 predicts and the experiment confirms.

Worked Example 28.10 (The momentum budget of the proton)**.** We make the inference of Eq. (28.86) quantitative with a minimal, self-consistent model of the proton's momentum content, to see how the arithmetic forces the gluon. Suppose the proton's momentum is shared among valence quarks (uud), a flavor-symmetric quark–antiquark sea, and gluons. The measured momentum fractions (at Q^2 of order a few GeV^2) are approximately

$$\langle x\rangle_{u_v} \approx 0.28, \quad \langle x\rangle_{d_v} \approx 0.11, \quad \langle x\rangle_{\mathrm{sea}} \approx 0.10, \tag{28.87}$$

where $\langle x\rangle_a \equiv \int_0^1 \xi f_a(\xi)\,\mathrm{d}\xi$ and "sea" collects all $q\bar q$ pairs beyond the valence quarks. Then the total carried by quarks and antiquarks is

$$\varepsilon_q = \langle x\rangle_{u_v} + \langle x\rangle_{d_v} + \langle x\rangle_{\mathrm{sea}} \approx 0.28 + 0.11 + 0.10 = 0.49. \tag{28.88}$$

The momentum sum rule Eq. (28.83) then leaves

$$\langle x\rangle_g = 1 - \varepsilon_q \approx 0.51 \tag{28.89}$$

for the gluons — about half. The point is not the precise figure, which drifts slowly with Q^2 (the quark fraction falls and the gluon fraction rises as the resolution increases, by the DGLAP evolution of Section 28.4), but its order of magnitude: if the proton were made of three quarks and nothing else one would have $\varepsilon_q = 1$ and no room for anything neutral. The observed deficit of order one-half cannot be explained away as a measurement error; it demands a neutral, momentum-carrying constituent, and QCD names it the gluon. The same counting performed at higher Q^2 shows $\langle x \rangle_g$ creeping upward, the quantitative herald of scaling violations.

Historical Note

Bjorken predicted scaling in 1968 from current algebra, before the parton picture existed; Feynman supplied the parton interpretation the same year, and Callan and Gross derived the $F_2 = 2xF_1$ relation as the spin diagnostic in 1969. The SLAC–MIT experiments of 1968–1969 found both, and the combination — pointlike, spin-$\frac{1}{2}$, fractionally charged constituents carrying only about half the momentum — is what turned the quark from a classification device into a particle and pointed directly at a neutral gluonic remainder. Asymptotic freedom (Chapter 27), found in 1973, then explained *why* the impulse approximation works at all: the constituents interact weakly precisely in the short-distance regime the photon probes.

28.3.8 Weak currents: the third structure function

A brief coda records what changes when the exchanged boson is a W or Z rather than a photon, since this is the second handle that the data use to separate flavor from momentum. The weak charged current is chiral, $\bar{q}\gamma^\mu(1-\gamma^5)q'$, and is *not* parity-conserving. The decomposition Eq. (28.57) therefore acquires the antisymmetric, parity-odd structure that the photon forbade, built from the only available pseudotensor:

$$W^{\mu\nu}_{\text{weak}} = (\text{the two parity-even terms of Eq. (28.57)}) \; - \; \frac{i}{2}\,\epsilon^{\mu\nu\alpha\beta}\frac{P_\alpha q_\beta}{P\cdot q}\,F_3(x,Q^2), \tag{28.90}$$

introducing a third structure function F_3. Because the chiral current distinguishes quarks from antiquarks (it couples to left-handed quarks and right-handed antiquarks), the parton-model expressions split the charge weighting into a sum and a difference:

$$F_2^W = 2x\sum_{q,\bar{q}} f(x), \qquad xF_3^W = 2x\sum_{q}\left[f_q(x) - f_{\bar{q}}(x)\right], \tag{28.91}$$

schematically, with the precise flavor combinations fixed by the CKM structure of Chapter 32. The crucial point for phenomenology is that F_3 measures the

valence (quark minus antiquark) distribution directly, while the electromagnetic F_2 measures the charge-squared-weighted sum. Combining neutrino DIS (F_2^W, F_3^W) with charged-lepton DIS (F_2^γ) is how the individual flavor distributions, and hence ε_q in Eq. (28.85), are disentangled. The Callan–Gross relation generalizes to the parity-even part unchanged, $F_2^W = 2xF_1^W$, again because the partons are spin-$\frac{1}{2}$.

The parton model has delivered exactly what the chapter promised at this stage: pointlike, spin-$\frac{1}{2}$, fractionally charged quarks carrying about half the proton's momentum, with the neutral remainder forced upon us as the gluon, and a set of universal distributions $f_a(x)$ that encode the one piece of irreducibly long-distance information. Two threads are now left dangling for the next section to pick up. First, the prediction of exact Bjorken scaling, Eq. (28.78), is only approximately true: the structure functions drift logarithmically with Q^2 (Fig. 28.4), because the partons are not truly free but radiate gluons, and the closer one looks the more partons one resolves. Second, the distributions $f_a(x)$ have been introduced as fixed functions, whereas the very mechanism that violates scaling — collinear gluon emission — makes them depend on the resolution Q^2. Section 28.4 turns the PDFs into *running* objects, computes the leading splitting function from collinear emission, and assembles the DGLAP equations that predict the Q^2-dependence of F_2 from QCD — converting the small "defect" in scaling into one of the theory's most stringent quantitative tests.

28.4 Scaling Violations and the DGLAP Equations

The parton model handed us a clean prediction and an awkward fact. The prediction was Bjorken scaling: in the deep inelastic limit the structure functions $F_2(x, Q^2)$ depend on the dimensionless ratio $x = Q^2/2Pq$ alone and not separately on Q^2 (Section 28.3), because a pointlike, structureless quark has no scale of its own to set against Q^2. The awkward fact is that scaling is not exact. Measured precisely, F_2 drifts with Q^2 at fixed x — slowly, logarithmically, but unmistakably: it rises at small x and falls at large x as Q^2 increases. The drift is small precisely because the coupling that controls it is small: by asymptotic freedom (Chapter 27) $\alpha_s(Q)$ runs to zero at large Q, so each gluon emitted by a struck quark is a controlled perturbation rather than a catastrophe. This section turns that observation into a calculation. We will see that the very QCD radiation which the naive parton model ignored generates a logarithm $\ln Q^2$ from the region of collinear gluon emission; that resumming the tower of such logarithms promotes the static parton densities $q(x)$ into scale-dependent objects $q(x, \mu^2)$; and that their evolution is governed by a universal kernel — the splitting functions — through the DGLAP equations, the renormalization-group equations of the parton distributions. By the end the logarithmic violation of

scaling will be not a defect of the parton model but its sharpest quantitative success.

28.4.1 The origin of the logarithm: collinear gluon emission

In the parton model the photon strikes a quark and the quark recoils; that is the entire process, and it carries no Q^2 dependence beyond the trivial kinematic one. At order α_s the struck quark may radiate a gluon before (or after) it absorbs the photon. The new amplitude, $\gamma^* q \to q\,g$, is the QCD Compton process; the quark–gluon vertex is the minimal coupling $ig\,\gamma^\mu T^a$ fixed in Chapter 25 (Eq. (25.86)) and quantized in Chapter 26. What matters for scaling is not the total rate of this process but *where in phase space it lives.*

Work in light-cone variables aligned with the incoming quark, whose momentum we take large along the + direction, $p = (p^+, 0, \mathbf{0}_\perp)$. Let the quark emit a gluon carrying longitudinal fraction $1 - z$ and transverse momentum $\mathbf{k}_\perp$, so the gluon momentum is

$$p_c = \Big((1-z)\,p^+,\ \frac{k_\perp^2}{2(1-z)p^+},\ \mathbf{k}_\perp\Big), \qquad p_c^2 = 0, \tag{28.92}$$

on shell, and the quark that continues into the hard vertex carries the remaining fraction z,

$$p_a = p - p_c = \Big(z\,p^+,\ -\frac{k_\perp^2}{2(1-z)p^+},\ -\mathbf{k}_\perp\Big). \tag{28.93}$$

This internal quark is off shell. Its virtuality follows from $p_a^2 = 2(zp^+)(-k_\perp^2/2(1-z)p^+) - k_\perp^2$, with the light-cone dot product $a\cdot b = a^+b^- + a^-b^+ - \mathbf{a}_\perp\cdot\mathbf{b}_\perp$,

$$p_a^2 = -\frac{z\,k_\perp^2}{1-z} - k_\perp^2 = -\frac{k_\perp^2}{1-z} \equiv t. \tag{28.94}$$

The propagator attached to this line contributes $1/p_a^2 = -(1-z)/k_\perp^2$ to the amplitude, hence $1/(p_a^2)^2 \propto 1/k_\perp^4$ to the squared amplitude. Two of these inverse powers of $k_\perp^2$ are cancelled by the numerator (which carries two powers of momentum and so vanishes as $k_\perp \to 0$, since a strictly collinear massless splitting is a measure-zero configuration); one survives. The transverse phase space supplies $\mathrm{d}^2k_\perp \propto \mathrm{d}k_\perp^2$. The product is the decisive structure

$$\frac{\mathrm{d}^2k_\perp}{(k_\perp^2)^2} \times k_\perp^2 \ \sim\ \frac{\mathrm{d}k_\perp^2}{k_\perp^2}, \tag{28.95}$$

a scale-invariant measure that integrates to a *logarithm.* The hard scale Q^2 provides the upper limit of $k_\perp^2$ (a gluon with $k_\perp \gtrsim Q$ is resolved as a separate hard parton, not as part of the quark), while the lower limit is set by some

infrared scale μ^2 — a quark mass, or, once we regulate properly, the factorization scale. Carrying out the integral over the collinear region,

$$\int_{\mu^2}^{Q^2} \frac{\mathrm{d}k_\perp^2}{k_\perp^2} = \ln \frac{Q^2}{\mu^2}, \tag{28.96}$$

the order-α_s correction to the cross section acquires precisely the $\ln Q^2$ that exact Bjorken scaling forbade. This is the single mechanism behind all of scaling violation. Its existence is generic: it is the collinear singularity of massless gauge theory, classified in Chapter 19, here regulated by the physical hierarchy $\mu^2 \ll k_\perp^2 \ll Q^2$ rather than by a quark mass. The coefficient of the logarithm is not generic; it is a calculable function of the momentum fraction z, the splitting function, to which we now turn.

Physical Insight

The logarithm is the imprint of self-similar substructure. A quark resolved at scale Q is, at finer resolution, a quark accompanied by a cloud of collinear gluons; each gluon, looked at still more finely, is itself dressed. Because the collinear emission measure $\mathrm{d}k_\perp^2/k_\perp^2$ is scale invariant, every decade of transverse momentum between μ and Q contributes equally, and there are $\ln(Q^2/\mu^2)$ such decades. Scaling violation is the statement that what we call "a parton" depends on how hard we look.

28.4.2 The quark splitting function P_{qq}

We now compute the coefficient of the collinear logarithm in Eq. (28.96): the probability density, per unit $\ln k_\perp^2$ and per unit longitudinal fraction, for a quark to be found inside a quark having radiated a gluon. Write the order-α_s collinear emission as a multiplicative correction to the leading hard process,

$$\mathrm{d}\sigma_{q+g} = \mathrm{d}\sigma_q \, \frac{\alpha_s}{2\pi} \, \frac{\mathrm{d}k_\perp^2}{k_\perp^2} \, \mathrm{d}z \; P_{qq}(z), \tag{28.97}$$

the master collinear-factorization form established in Chapters 16 and 19: the $(n+1)$-parton squared amplitude in the collinear limit factorizes into the n-parton one times a universal kernel and the scale-invariant emission measure. Our task is to determine $P_{qq}(z)$. We do it in two complementary ways: first the soft limit, which fixes the colour factor and the normalization and exposes the $z \to 1$ pole; then the full collinear matrix element, which supplies the complete z-dependence.

Soft limit: the eikonal current. When the emitted gluon is soft, $z \to 1$, its coupling to the hard quark line is universal and classical. The emission vertex $ig\,\gamma^\mu T^a$ inserted on an external quark of momentum p, with the adjacent

propagator $i(\not p + \not p_c)/((p+p_c)^2)$, simplifies in the limit $p_c \to 0$: the numerator $\not p \gamma^\mu$ acting on the on-shell spinor gives $2p^\mu$ by the Dirac equation $\not p\, u(p) = 0$ together with $\{\gamma^\mu, \not p\} = 2p^\mu$, while the denominator $(p+p_c)^2 \to 2p\cdot p_c$. The amplitude factorizes,

$$\mathcal{M}_{q+g} \xrightarrow{p_c \to 0} g\, T^a \, \frac{p\cdot\varepsilon^*(p_c)}{p\cdot p_c}\, \mathcal{M}_q, \tag{28.98}$$

the eikonal (or soft) current. Squaring, summing over the two physical gluon polarizations with $\sum_\lambda \varepsilon^\mu \varepsilon^{*\nu} \to -g^{\mu\nu} + \dots$ (the gauge-dependent remainder does not contribute to the leading collinear logarithm), and averaging over the incoming colour to produce the quadratic Casimir $T^a T^a = C_F\, \mathbf{1}$ in the fundamental (Eq. (28.7), with $C_F = (N^2-1)/2N$ from Chapter B), the radiation probability becomes

$$\mathrm{d}w_{\text{soft}} = C_F\, g^2 \, \frac{-p\cdot p \,/\, (p\cdot p_c)^2 \;\cdots}{} \;\longrightarrow\; \frac{\alpha_s}{2\pi}\, C_F\, \frac{2\,\mathrm{d}z}{1-z}\, \frac{\mathrm{d}k_\perp^2}{k_\perp^2}. \tag{28.99}$$

The crucial features are visible without tracking every constant. The colour factor is C_F. The longitudinal dependence is the pole $2/(1-z)$, because the soft gluon's energy fraction is $1-z$ and the classical bremsstrahlung spectrum is $\mathrm{d}(\text{energy})/(\text{energy}) = \mathrm{d}z/(1-z)$. And the transverse dependence is exactly the scale-invariant $\mathrm{d}k_\perp^2/k_\perp^2$ of Eq. (28.96), because $p\cdot p_c = k_\perp^2/2(1-z)$ from Eq. (28.92) converts the eikonal denominator into $1/k_\perp^2$. Comparing with Eq. (28.97) we read off the soft behaviour

$$P_{qq}(z) \xrightarrow{z\to 1} \frac{2\,C_F}{1-z}. \tag{28.100}$$

This fixes the colour and the normalization; it remains to find how P_{qq} deviates from the pole away from $z = 1$.

Full collinear matrix element. Beyond the soft limit we need the spin structure of the splitting vertex. Strip the colour (C_F already accounted) and the hard amplitude, and consider the quark current connecting the incoming quark p to the daughter quark of fraction z (momentum p_b, on shell to leading collinear accuracy) with emission of a transversely-polarized gluon p_c:

$$\langle |j|^2 \rangle \;\equiv\; \tfrac{1}{2} \sum_{\text{spins}} \sum_{\lambda} \left| \bar u(p_b)\, \not\varepsilon^*_\lambda(p_c)\, u(p) \right|^2 = \tfrac{1}{2}\, \mathrm{Tr}\!\left[\not p\, \gamma^\mu \not p_b\, \gamma^\nu \right] d_{\mu\nu}(p_c), \tag{28.101}$$

with the $\tfrac{1}{2}$ averaging the incoming spin and $d_{\mu\nu}(p_c) = -g_{\mu\nu} + (p_{c\mu} n_\nu + p_{c\nu} n_\mu)/(p_c n)$ the physical (light-cone-gauge) polarization sum, n the auxiliary null vector along the $-$ direction. The trace is elementary,

$$\mathrm{Tr}\!\left[\not p\, \gamma^\mu \not p_b\, \gamma^\nu \right] = 4\big(p^\mu p_b^\nu + p^\nu p_b^\mu - g^{\mu\nu}\, p\cdot p_b \big). \tag{28.102}$$

The contraction needs the collinear kinematics $p_b = zp + \dots$, $p_c = (1-z)p + \dots$ from Eq. (28.92), which give the invariants

$$p\cdot p_b = \frac{k_\perp^2}{2z}, \quad p\cdot p_c = \frac{k_\perp^2}{2(1-z)}, \quad p_b\cdot p_c = \frac{k_\perp^2}{2z(1-z)}, \tag{28.103}$$

together with $p\cdot n = p^+$, $p_b\cdot n = zp^+$, $p_c\cdot n = (1-z)p^+$. Contracting Eq. (28.102) with $-g_{\mu\nu}$ gives $8\,p\cdot p_b = 4k_\perp^2/z$; contracting with the n-dependent part,

$$\frac{8}{p_c\cdot n}\Big[(p\cdot p_c)(p_b\cdot n) + (p\cdot n)(p_b\cdot p_c) - (p\cdot p_b)(p_c\cdot n)\Big] = \frac{8\,k_\perp^2}{(1-z)^2}, \tag{28.104}$$

where in the last step the bracket evaluates to $p^+k_\perp^2/(1-z)$ after using Eq. (28.103) and the identity $1 + z^2 - (1-z)^2 = 2z$. Therefore

$$\langle |j|^2 \rangle = \tfrac{1}{2}\Big(\frac{4k_\perp^2}{z} + \frac{8k_\perp^2}{(1-z)^2}\Big) = \frac{2k_\perp^2}{z} + \frac{4k_\perp^2}{(1-z)^2}. \tag{28.105}$$

The splitting function is this vertex factor weighted by the collinear kinematic factor that converts the squared propagator and the two-body collinear phase space into the measure $\mathrm{d}k_\perp^2/k_\perp^2$. That kinematic factor, assembled from $1/(p_a^2)^2$, the phase space of Chapter 16, and the longitudinal flux, is $z(1-z)/2k_\perp^2$; multiplying,

$$P_{qq}(z) = C_F\,\frac{z(1-z)}{2k_\perp^2}\,\langle |j|^2\rangle = C_F\,\frac{z(1-z)}{2}\Big(\frac{2}{z} + \frac{4}{(1-z)^2}\Big) = C_F\Big[(1-z) + \frac{2z}{1-z}\Big]. \tag{28.106}$$

Combining the two terms over a common denominator, $(1-z)^2 + 2z = 1 - 2z + z^2 + 2z = 1 + z^2$, gives the leading quark splitting function

$$\boxed{P_{qq}(z) = C_F\,\frac{1+z^2}{1-z}\,.} \tag{28.107}$$

As a check, $z \to 1$ reproduces the eikonal pole $P_{qq} \to 2C_F/(1-z)$ of Eq. (28.100), with the correct coefficient; the normalization fixed there and the z-dependence found here agree. The physics of the two terms is transparent: the $1/(1-z)$ piece is the soft-gluon (bremsstrahlung) singularity, and the $z^2/(1-z)$ piece is the genuinely hard-collinear emission in which the quark surrenders a finite fraction of its momentum. We have used the quark–gluon vertex and the colour factor exactly as delivered by Chapters 25 and 26; nothing else enters.

28.4.3 The plus prescription and quark-number conservation

Equation (28.107) is not yet a usable kernel. Its $1/(1-z)$ pole at $z = 1$ makes the emission probability formally infinite there, the divergence of *soft* gluon emission.

Physically this must cancel against the virtual correction in which a soft gluon is emitted and reabsorbed, by the same infrared mechanism (KLN; Chapter 19) that renders the total e^+e^- rate finite. We implement the cancellation distributionally. The virtual diagram contributes only at $z = 1$ (no momentum is lost), so it is proportional to $\delta(1-z)$; the soft singularity of the real emission is tamed by demanding that the kernel act only on the *difference* of the test function from its endpoint value. This is the plus prescription, defined on any function g smooth at $z = 1$ by

$$\int_0^1 \mathrm{d}z\, \frac{g(z)}{(1-z)_+} \;\equiv\; \int_0^1 \mathrm{d}z\, \frac{g(z)-g(1)}{1-z}, \tag{28.108}$$

the subtraction at $z = 1$ rendering the integral finite while leaving the kernel unchanged for $z < 1$. The regularized quark splitting function is

$$\boxed{P_{qq}(z) = C_F\Big[\frac{1+z^2}{(1-z)_+} + \frac{3}{2}\,\delta(1-z)\Big]\,,} \tag{28.109}$$

where the coefficient of the δ-function encodes the virtual contribution. That coefficient is not free: it is fixed by a conservation law. Under evolution the *number* of quarks minus antiquarks — the net flavour charge — is a physical, scale-independent quantity (the proton has two net up quarks at every resolution). The number sum rule $\int_0^1 \mathrm{d}x\,[q(x,\mu^2) - \bar{q}(x,\mu^2)] = \text{const}$ requires that the total integral of the kernel vanish,

$$\int_0^1 \mathrm{d}z\, P_{qq}(z) = 0, \tag{28.110}$$

since $\mathrm{d}/\mathrm{d}\ln\mu^2$ of the conserved number is proportional to this integral. We verify that Eq. (28.110) fixes the $\frac{3}{2}$. By the definition Eq. (28.108) with $g(z) = 1 + z^2$, so $g(1) = 2$,

$$\int_0^1 \mathrm{d}z\, \frac{1+z^2}{(1-z)_+} = \int_0^1 \mathrm{d}z\, \frac{(1+z^2)-2}{1-z} = \int_0^1 \mathrm{d}z\, \frac{z^2-1}{1-z} = -\int_0^1 \mathrm{d}z\,(1+z) = -\frac{3}{2}, \tag{28.111}$$

using $z^2 - 1 = -(1-z)(1+z)$. Adding the δ-term, $\int_0^1 P_{qq} = C_F(-\frac{3}{2} + \frac{3}{2}) = 0$, exactly as required. The virtual coefficient is therefore forced to be $+\frac{3}{2}C_F$; it is the soft-and-virtual mate of the real soft pole, and number conservation is the bookkeeping that ties them. Equivalently, the $\frac{3}{2}C_F$ is the finite remainder of the quark wavefunction renormalization at this order — the same constant that appears in the one-loop quark self-energy.

Caution

The plus prescription acts only on the $1/(1-z)$ factor, not on the whole $(1+z^2)/(1-z)$. Writing $[(1+z^2)/(1-z)]_+$ as if the entire ratio were

plus-distributed would subtract the wrong endpoint value and shift the δ-function coefficient. The unambiguous statement is Eq. (28.109): a plus distribution multiplying the smooth numerator $1+z^2$, plus a separate $\frac{3}{2}\,\delta(1-z)$ fixed by Eq. (28.110).

28.4.4 The full set of splitting functions

A quark can come from a quark by gluon emission (P_{qq}, above); it can also come from a gluon that splits into a quark–antiquark pair, and gluons arise both from quarks (the gluon side of $q\to qg$) and from gluons (the non-abelian $g\to gg$ vertex of Chapter 25). The complete leading-order evolution therefore needs four kernels: P_{qq}, P_{gq} (quark $\to$ gluon), P_{qg} (gluon $\to$ quark), and P_{gg} (gluon $\to$ gluon). We derive P_{qg} in full, since it is the cleanest non-trivial case, then state the remaining two and check them against the conservation laws.

Worked Example 28.11 (The gluon-to-quark splitting $P_{qg}(z)=T_F\,[z^2+(1-z)^2]$)**.** A gluon of momentum p_g splits into a quark of fraction z (momentum p_q) and an antiquark of fraction $1-z$ (momentum $p_{\bar q}$). The vertex is again $ig\,\gamma^\mu T^a$; the splitting current is now $\bar u(p_q)\,\not\varepsilon(p_g)\,v(p_{\bar q})$, with the gluon polarization on the incoming line and quark/antiquark spinors outgoing. The colour average over the eight incoming gluons and trace over the produced pair gives $\mathrm{tr}(T^aT^b)\delta^{ab}/(N^2-1)=T_F$ with $T_F=\frac12$ (Chapter B). The spin-summed, polarization-averaged vertex factor is

$$\langle|V_{qg}|^2\rangle=\tfrac12\sum_{\text{spins}}\sum_{\lambda}\left|\bar u(p_q)\not\varepsilon_\lambda(p_g)v(p_{\bar q})\right|^2=\tfrac12\,\mathrm{Tr}\big[\not p_q\,\gamma^\mu\not p_{\bar q}\,\gamma^\nu\big]\,d_{\mu\nu}(p_g). \tag{28.112}$$

The trace is Eq. (28.102) with $(p,p_b)\to(p_q,p_{\bar q})$, and the collinear invariants are the analogues of Eq. (28.103) with the parent now the gluon: $p_q{\cdot}p_{\bar q}=k_\perp^2/2z(1-z)$, $p_q{\cdot}p_g=k_\perp^2/2z$, $p_{\bar q}{\cdot}p_g=k_\perp^2/2(1-z)$, and $p_q{\cdot}n=zp^+$, $p_{\bar q}{\cdot}n=(1-z)p^+$, $p_g{\cdot}n=p^+$. The $-g_{\mu\nu}$ contraction gives $8\,p_q{\cdot}p_{\bar q}=4k_\perp^2/z(1-z)$; the n-dependent part gives

$$\frac{8}{p_g{\cdot}n}\Big[(p_q{\cdot}p_g)(p_{\bar q}{\cdot}n)+(p_q{\cdot}n)(p_{\bar q}{\cdot}p_g)-(p_q{\cdot}p_{\bar q})(p_g{\cdot}n)\Big]=-8k_\perp^2, \tag{28.113}$$

where the bracket evaluates to $-p^+k_\perp^2$ using $(1-z)^2+z^2-1=-2z(1-z)$. Hence

$$\langle|V_{qg}|^2\rangle=\tfrac12\Big(\frac{4k_\perp^2}{z(1-z)}-8k_\perp^2\Big)=2k_\perp^2\,\frac{1-2z(1-z)}{z(1-z)}=2k_\perp^2\,\frac{z^2+(1-z)^2}{z(1-z)}, \tag{28.114}$$

using $1-2z(1-z)=z^2+(1-z)^2$. Applying the *same* kinematic factor $z(1-z)/2k_\perp^2$ that produced Eq. (28.106) — the factorization measure is universal — and the colour factor T_F,

$$P_{qg}(z)=T_F\,\frac{z(1-z)}{2k_\perp^2}\,\langle|V_{qg}|^2\rangle=\boxed{\,T_F\big[z^2+(1-z)^2\big]\,}. \tag{28.115}$$

There is no $1/(1-z)$ pole and no soft singularity: a gluon splitting into a quark pair has no soft enhancement, because neither daughter is a soft gauge boson. The function is symmetric under $z \leftrightarrow 1-z$, as it must be, the quark and antiquark being interchangeable. That the same kinematic factor turns the quark trace into P_{qq} and the gluon trace into P_{qg} is the consistency check that our extraction of the splitting kernels from the vertices is correct.

The remaining two kernels follow by the same method (the $g \to gg$ case using the triple-gluon vertex of Chapter 25, with the non-abelian colour factor $C_A = N$). We state the results:

$$P_{gq}(z) = C_F \, \frac{1 + (1-z)^2}{z}, \tag{28.116}$$

$$P_{gg}(z) = 2C_A \Big[\frac{z}{(1-z)_+} + \frac{1-z}{z} + z(1-z) \Big] + \Big(\frac{11 C_A - 4 n_f T_F}{6} \Big) \delta(1-z). \tag{28.117}$$

Three structural remarks. First, $P_{gq}(z)$ is the gluon side of the same $q \to qg$ splitting that gave P_{qq}; indeed $P_{gq}(z) = P_{qq}(1-z)$ in its regular part, because the gluon carries fraction z when the quark keeps $1-z$, and the colour factor is again C_F (the gluon is radiated from a quark line). Its $1/z$ pole signals copious *soft-gluon* production at small z — the seed of the steep rise of the gluon density at small x. Second, P_{gg} has both a $z \to 1$ soft pole (gluon bremsstrahlung off a gluon, plus-regulated like P_{qq}) and a $z \to 0$ pole, reflecting that gluons radiate gluons abundantly at both ends; the $\delta(1-z)$ coefficient is fixed by momentum conservation (below) and carries the same combination $\frac{11}{3} C_A - \frac{4}{3} T_F n_f$ that appears in the one-loop β-function (Chapter 27), no coincidence, since both measure the gluon's self-energy. Third, the kernels obey two exact symmetries: $P_{qq} \leftrightarrow P_{gg}$ under $q \leftrightarrow g$ with appropriate Casimirs, and the $z \leftrightarrow 1-z$ symmetry of P_{qg}.

The sum rules. The splitting functions are constrained by two conservation laws, which both serve as checks and fix the virtual coefficients. The first is number conservation, Eq. (28.110), already used. The second is *momentum conservation*: the total longitudinal momentum carried by all partons is unity at every scale, $\sum_a \int_0^1 \mathrm{d}x \, x \, f_a(x, \mu^2) = 1$, since the partons share the hadron's momentum and evolution merely redistributes it. Differentiating with respect to $\ln \mu^2$ and using the evolution equations (next subsection) gives the requirement that, for momentum entering the quark and gluon channels from a quark parent,

$$\int_0^1 \mathrm{d}z \, z \, \big[P_{qq}(z) + P_{gq}(z) \big] = 0, \qquad \int_0^1 \mathrm{d}z \, z \, \big[2 n_f \, P_{qg}(z) + P_{gg}(z) \big] = 0. \tag{28.118}$$

We verify the first. From Eq. (28.109) and Eq. (28.116),

$$\int_0^1 \mathrm{d}z \, z \, P_{qq}(z) = C_F \int_0^1 \mathrm{d}z \, z \Big[\frac{1+z^2}{(1-z)_+} + \frac{3}{2} \delta(1-z) \Big], \tag{28.119}$$

where the plus integral with $g(z) = z(1+z^2)$, $g(1) = 2$, gives $\int_0^1 \mathrm{d}z\,[z(1+z^2) - 2]/(1-z)$. Writing $z(1+z^2) - 2 = z + z^3 - 2$ and dividing by $1-z$ (polynomial division: $z^3+z-2 = (z-1)(z^2+z+2)$) yields $-\int_0^1 \mathrm{d}z\,(z^2+z+2) = -(\frac{1}{3}+\frac{1}{2}+2) = -\frac{17}{6}$; adding the δ-term $\frac{3}{2}$ gives $\int_0^1 z\,P_{qq} = C_F(-\frac{17}{6}+\frac{3}{2}) = -\frac{4}{3}C_F$. For P_{gq},

$$\int_0^1 \mathrm{d}z\, z\, P_{gq}(z) = C_F \int_0^1 \mathrm{d}z\, z\, \frac{1+(1-z)^2}{z} = C_F \int_0^1 \mathrm{d}z\,[1+(1-z)^2] = C_F\Big(1+\tfrac{1}{3}\Big) = \tfrac{4}{3}C_F. \tag{28.120}$$

The two cancel: $\int_0^1 \mathrm{d}z\, z\,[P_{qq} + P_{gq}] = C_F(-\frac{4}{3}+\frac{4}{3}) = 0$, confirming Eq. (28.118). Physically, the momentum lost by the quark to gluon radiation ($-\frac{4}{3}C_F$) is exactly the momentum the gluon channel gains ($+\frac{4}{3}C_F$); nothing is created or destroyed, only handed across. The second sum rule in Eq. (28.118) fixes the $\delta(1-z)$ coefficient of P_{gg} to the value displayed in Eq. (28.117), by the identical argument applied to the gluon parent.

28.4.5 The DGLAP equations

We can now assemble the evolution. A single collinear emission shifts a parton's momentum fraction and contributes one power of $\alpha_s \ln(Q^2/\mu^2)$ (Eqs. (28.96) and (28.97)). At order α_s^n the dominant contribution comes from n *strongly ordered* emissions, $\mu^2 \ll k_{\perp,1}^2 \ll \cdots \ll k_{\perp,n}^2 \ll Q^2$, each supplying a logarithm, so the cross section carries a tower $\sum_n c_n\,[\alpha_s \ln(Q^2/\mu^2)]^n$. This is the leading-logarithm ladder. Because α_s is small (asymptotic freedom, Chapter 27) but the logarithm is large, no single term dominates and the whole tower must be resummed. The resummation is a differential equation: demanding that the parton densities absorb the collinear logarithms so that physical cross sections are independent of the arbitrary separation scale μ gives, for the quark distribution,

$$\boxed{\frac{\partial q_i(x,\mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s(\mu^2)}{2\pi}\int_x^1 \frac{\mathrm{d}z}{z}\Big[P_{qq}(z)\,q_i\Big(\frac{x}{z},\mu^2\Big) + P_{qg}(z)\,g\Big(\frac{x}{z},\mu^2\Big)\Big]} \tag{28.121}$$

and, coupled to it, for the gluon distribution,

$$\frac{\partial g(x,\mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s(\mu^2)}{2\pi}\int_x^1 \frac{\mathrm{d}z}{z}\Big[\sum_j P_{gq}(z)\,(q_j+\bar{q}_j)\Big(\frac{x}{z},\mu^2\Big) + P_{gg}(z)\,g\Big(\frac{x}{z},\mu^2\Big)\Big]. \tag{28.122}$$

These are the Dokshitzer–Gribov–Lipatov–Altarelli–Parisi (DGLAP) equations. The convolution structure $\int_x^1 \mathrm{d}z/z\,P(z)f(x/z)$ is dictated by momentum fractions: a parton observed at fraction x descended from a parent at the larger fraction x/z that retained the fraction z in the splitting, and the lower limit $z \geq x$ enforces $x/z \leq 1$. The coupling is evaluated at the running scale μ^2, so each rung of the ladder is weighted by $\alpha_s(\mu^2)$, which shrinks as μ grows; this is why the evolution is slow and perturbatively trustworthy.

The reading of Eq. (28.121) as a renormalization-group equation is exact and central. The factorization scale μ plays the role for the parton densities that the renormalization scale plays for couplings: it separates the calculable hard physics above μ from the universal soft physics below it, and physical observables cannot depend on where the line is drawn. The right-hand side is then nothing but an anomalous-dimension term, and the splitting functions are the anomalous dimensions of the parton densities. We make this precise.

Mellin moments and anomalous dimensions. The convolution in Eq. (28.121) diagonalizes under the Mellin transform. Define the moments

$$\tilde{f}(N,\mu^2) \equiv \int_0^1 \mathrm{d}x\, x^{N-1} f(x,\mu^2), \qquad \gamma_{ab}(N) \equiv \int_0^1 \mathrm{d}z\, z^{N-1} P_{ab}(z), \tag{28.123}$$

the latter the Mellin moments of the splitting functions. The Mellin transform turns the convolution into a product — $\int_0^1 \mathrm{d}x\, x^{N-1} \int_x^1 \frac{\mathrm{d}z}{z} P(z) f(x/z) = \tilde{P}(N)\tilde{f}(N)$ by the substitution $x = zy$ — so the non-singlet combination $q_{\mathrm{NS}} = q_i - \bar{q}_i$ (which decouples from the gluon, since P_{qg} feeds quarks and antiquarks equally) evolves multiplicatively,

$$\frac{\partial\, \tilde{q}_{\mathrm{NS}}(N,\mu^2)}{\partial \ln \mu^2} = \frac{\alpha_s(\mu^2)}{2\pi}\, \gamma_{qq}(N)\, \tilde{q}_{\mathrm{NS}}(N,\mu^2). \tag{28.124}$$

This is precisely a Callan–Symanzik equation (Chapter 24) with anomalous dimension $\gamma_{qq}(N)$: the N-th moment of the structure function runs with an anomalous dimension equal to the N-th moment of P_{qq}. Those moments are exactly the anomalous dimensions of the twist-two, spin-N quark operators that appear in the operator-product expansion of the two electromagnetic currents inside the hadron; the parton-model evolution and the operator-product expansion are two languages for one renormalization-group fact. Computing the moment from Eq. (28.109),

$$\gamma_{qq}(N) = C_F \int_0^1 \mathrm{d}z\, z^{N-1}\Big[\frac{1+z^2}{(1-z)_+} + \frac{3}{2}\delta(1-z)\Big] = C_F\Big[\frac{3}{2} + \frac{1}{N(N+1)} - 2\sum_{k=1}^{N}\frac{1}{k}\Big], \tag{28.125}$$

a smooth function of N, negative for all $N \geq 2$, so that high moments fall under evolution. The $N = 1$ moment vanishes, $\gamma_{qq}(1) = 0$, which is the number sum rule Eq. (28.110) read in moment space: the lowest moment (the net quark number) does not run.

The running parton density as the solution. With $\alpha_s(\mu^2)$ from the one-loop running of Chapter 27, $\mathrm{d}\alpha_s/\mathrm{d}\ln\mu^2 = -b_0\alpha_s^2/4\pi$ with $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$, Eq. (28.124) integrates in closed form. Changing variable from $\ln\mu^2$ to α_s,

$$\frac{\mathrm{d}\tilde{q}_{\mathrm{NS}}}{\tilde{q}_{\mathrm{NS}}} = \frac{\gamma_{qq}(N)}{2\pi}\,\frac{\alpha_s\,\mathrm{d}\ln\mu^2}{1} = -\frac{\gamma_{qq}(N)}{2\pi}\,\frac{4\pi}{b_0}\,\frac{\mathrm{d}\alpha_s}{\alpha_s}, \tag{28.126}$$

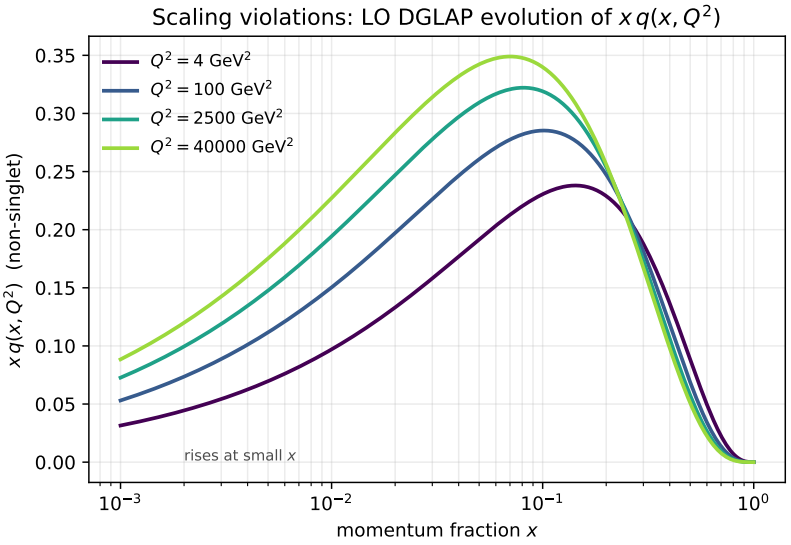


Figure 28.5: Leading-order DGLAP evolution of a non-singlet quark distribution $x\,q(x,Q^2)$, obtained by solving Eq. (28.121) numerically with the plus-regularized kernel Eq. (28.109) and one-loop $\alpha_s(Q^2)$ (Chapter 27). As Q^2 increases the distribution is pushed to smaller x: it falls at large x and rises at small x, with the curves pivoting about $x \approx 0.15$. This is the QCD prediction for the logarithmic violation of Bjorken scaling, in quantitative agreement with deep-inelastic data over several orders of magnitude in Q^2.

which integrates to the celebrated power-law in the coupling,

$$\boxed{\tilde{q}_{\rm NS}(N,\mu^2) = \tilde{q}_{\rm NS}(N,\mu_0^2)\left[\frac{\alpha_s(\mu_0^2)}{\alpha_s(\mu^2)}\right]^{d_{qq}(N)}, \qquad d_{qq}(N) = \frac{2\,\gamma_{qq}(N)}{b_0}\,.} \tag{28.127}$$

This is the quantitative prediction. Given the parton densities at one scale μ_0 — which are not calculable in perturbation theory and must be measured — QCD predicts them at every other scale, with no free parameters beyond $\Lambda_{\rm QCD}$. The structure function at fixed x therefore does depend on Q^2, logarithmically through α_s, in exactly the pattern Eq. (28.127) encodes: since $\gamma_{qq}(N) < 0$ for $N \geq 2$ and grows in magnitude with N, high moments (dominated by large x) are suppressed as μ increases, while the momentum lost there reappears at small x. The structure function softens: it falls at large x and rises at small x, precisely the observed violation of Bjorken scaling. Figure 28.5 shows this evolution computed directly from Eq. (28.121) for a non-singlet density.

Physical Insight

Asymptotic freedom and scaling violation are the same fact seen from two sides. The coupling runs to zero in the ultraviolet *because* of gluon self-interaction (Chapter 27); that same gluon radiation is what dresses the quark with collinear partners and makes the densities run. The DGLAP equation is the parton-density RG, with the splitting functions playing the role of anomalous dimensions and $\alpha_s(\mu^2)$ setting the (shrinking) step size. The reason the prediction is sharp — a parameter-free family of curves once the input is fixed at one scale — is that the long-distance, incalculable physics is frozen into the boundary condition $q(x, \mu_0^2)$, while everything that depends on Q^2 is short-distance and perturbative.

The mechanism we have uncovered is also the resolution of a conceptual worry. The collinear logarithm Eq. (28.96) was divergent as the regulator $\mu \to 0$: a genuine infrared singularity of the partonic cross section. DGLAP shows where it goes. The singularity is not in the physical cross section; it has been absorbed, scale by scale, into the redefinition of the parton density $q(x, \mu^2)$, leaving a finite, calculable, μ-dependent coefficient. That absorption — collinear singularities of the partonic process reorganized into universal, running parton distributions — is the precise content of the factorization theorem, which the next section states in general and proves at one loop. DGLAP is the statement that the parton distributions are renormalized quantities; factorization is the statement that, once they are, the rest of the cross section is finite and computable.

28.5 Factorization and the QCD-Improved Parton Model

The previous section turned parton distributions into running objects. The collinear emission of a gluon off an incoming quark generated a logarithm $\ln(Q^2/\mu^2)$ whose coefficient was the splitting function P_{qq} (Eq. (28.109)), and resumming such logarithms made the distributions evolve according to the DGLAP equation (Eq. (28.121)). We met those distributions, in this chapter's introduction, as the *universal long-distance factors* that asymptotic freedom forces upon us: the incalculable but process-independent objects that, once measured, are reused everywhere. The present section makes that promise into a theorem. It states *collinear factorization* precisely, demonstrates at one loop the mechanism that makes it true — the collinear singularity of a partonic cross section is universal and can be absorbed, once and for all, into a redefinition of the parton distribution — and then applies the result to the prototype hadron–hadron process, the Drell–Yan production of a lepton pair. The two themes of the chapter, asymptotic freedom and confinement, meet here: the short-distance

coefficient is computed in perturbation theory because $\alpha_s(Q)$ is small there (Chapter 27), while the long-distance distribution is taken from experiment because the proton is a bound state we cannot solve. Factorization is the clean connection between them.

28.5.1 The factorization theorem

Consider an inclusive hard process initiated by one or two hadrons — deep inelastic scattering $\ell H \to \ell X$, or hadron–hadron production of a heavy or high-mass system, $H_1 H_2 \to L X$, with L a colour singlet (a lepton pair, a W or Z, a Higgs boson) recoiling against unobserved hadronic debris X. Let Q be the single large invariant of the hard process (the momentum transfer $\sqrt{Q^2}$ in DIS, the invariant mass M of L in the hadronic case), and let $\Lambda_{\rm QCD}$ be the scale at which the coupling runs strong (Chapter 27). The assertion of collinear factorization is that, when $Q \gg \Lambda_{\rm QCD}$, the cross section separates into a short-distance *coefficient* (or *partonic cross section*) that knows only about Q and the coupling, convolved with *parton distribution functions* (PDFs) that know only about the incoming hadrons.

We package the convolution in a single piece of notation used throughout. For two functions on $[0,1]$ define the Mellin convolution

$$(C \otimes f)(x) \;\equiv\; \int_x^1 \frac{dz}{z}\, C(z)\, f\!\left(\frac{x}{z}\right) \;=\; \int_x^1 \frac{dy}{y}\, C\!\left(\frac{x}{y}\right) f(y), \tag{28.128}$$

the two forms related by $y = x/z$. Convolution is commutative and associative, and the unit is $\delta(1-z)$, since $(\delta(1-\cdot)\otimes f)(x) = f(x)$. Under a Mellin transform $\widetilde{g}(N) = \int_0^1 dx\, x^{N-1} g(x)$ it becomes ordinary multiplication, $\widetilde{C \otimes f}(N) = \widetilde{C}(N)\,\widetilde{f}(N)$, which is why the same object that organizes DGLAP organizes factorization.

Theorem 28.12 (Collinear factorization)**.** *Let σ be an inclusive cross section with a single hard scale $Q \gg \Lambda_{\rm QCD}$, of the kind above. Then there exist functions $f_{a/H}(x,\mu_F)$ — the parton distributions, one for each parton species $a \in \{g, u, \bar u, d, \bar d, \dots\}$ and each hadron H, depending on a momentum fraction $x \in (0,1]$ and a* factorization scale *μ_F — and short-distance coefficients $\hat\sigma_{ab}$ such that*

$$\sigma(Q) \;=\; \sum_{a,b} \int_0^1 dx_a \int_0^1 dx_b\; f_{a/H_1}(x_a,\mu_F)\; \hat\sigma_{ab}(x_a,x_b;Q,\mu_F,\mu_R,\alpha_s(\mu_R))\; f_{b/H_2}(x_b,\mu_F) + \mathcal{O}\!\left(\frac{\Lambda}{Q}\right)^{\!p} \tag{28.129}$$

with the following properties:

1. ***Calculability of the coefficient.*** *$\hat\sigma_{ab}$ is infrared-safe and computable as a power series in $\alpha_s(\mu_R)$. It is a property of the* partons *alone: the partonic*

cross section one would obtain by scattering free quarks and gluons, made finite by the subtraction described below. For a one-hadron process (DIS) one factor f and one integral are absent.

2. ***Universality of the distributions.*** *$f_{a/H}(x,\mu_F)$ depends on the hadron H and the parton a but* not *on the hard process. The same $f_{u/p}$ that describes an up quark in the proton in DIS describes it in Drell–Yan, in jet production, and in Higgs production. The PDFs are not calculable in perturbation theory (they encode the bound-state structure of the hadron), but they are measured once and used everywhere.*

3. ***Compensating scale dependence.*** *The factorization scale μ_F is the boundary drawn between "short" and "long" distance; it is arbitrary. The μ_F-dependence of the f's is governed by DGLAP (Eq. (28.121)) and is cancelled, order by order in α_s, by the explicit μ_F-dependence of $\hat{\sigma}_{ab}$, so that σ is μ_F-independent to the order computed. The renormalization scale μ_R of the running coupling is likewise compensated by the explicit μ_R-dependence of $\hat{\sigma}_{ab}$.*

4. ***Power corrections.*** *The neglected terms are suppressed by a power $p \geq 1$ of $\Lambda_{\rm QCD}/Q$ (with p process-dependent; $p = 2$ for DIS F_2, the leading "higher-twist" correction). They are not part of the perturbative expansion.*

Caution: what this theorem does and does not claim

Theorem 28.12 is an *all-orders* statement: the separation (28.129) survives to every order in α_s, with the *same* universal f's. Proving this is hard, and we do not prove it in full. The difficulty is not the collinear region we will dissect below — that part is what the one-loop demonstration controls — but the soft and *Glauber* (longitudinal-momentum-exchange) gluons that, in a hadron–hadron collision, can connect the two incoming hadrons before the hard scattering and threaten to break universality. The content of the factorization theorem is that these contributions cancel after the sum over the unobserved final state X, in inclusive enough observables. The complete cancellation, and the precise list of observables for which it holds, is the Collins–Soper–Sterman program [135, 136]; we state it as established and do not reproduce it. What we *do* prove here is the part that gives the theorem its operational meaning: the collinear singularity of a one-loop partonic cross section is universal — the same P_{qq} that drives DGLAP — and is absorbed into the PDF, leaving a finite, calculable coefficient. That single example is the engine of the whole construction.

28.5.2 The one-loop demonstration: a collinear pole becomes a parton distribution

The cleanest laboratory is deep inelastic scattering, where one factor in (28.129) is absent and the hard probe is a photon of virtuality Q^2 whose coupling we control exactly. We work to first order in α_s and ask a sharp question: what is the structure function of a *single quark*? If we could prepare an on-shell quark of momentum p and scatter a virtual photon off it, what would the QCD corrections do? The answer exhibits, in miniature and completely explicitly, the mechanism of the theorem.

At zeroth order in α_s the quark is inert: the photon strikes it elastically, $\gamma^*(q) + q(p) \to q(p+q)$, and the partonic structure function is a delta function enforcing Bjorken scaling. With $z \equiv Q^2/(2\,p\cdot q)$ the partonic momentum fraction (the analogue of Bjorken x for a free-quark target),

$$\frac{1}{z}\,\hat{F}_2^{q}(z,Q^2)\Big|_{\mathcal{O}(\alpha_s^0)} \;=\; e_q^2\,\delta(1-z), \tag{28.130}$$

where e_q is the quark electric charge in units of e. This is the parton model: the photon sees a free quark, and the only kinematics allowed is elastic, $z = 1$.

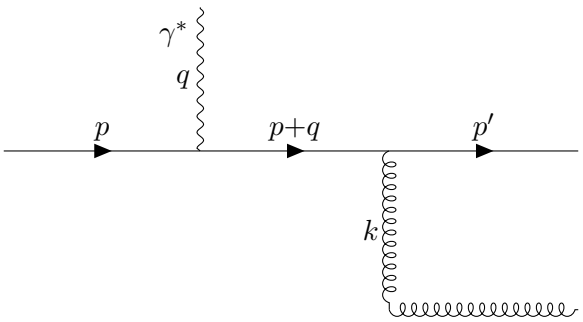


Figure 28.6: The QCD Compton real-emission correction to deep inelastic scattering off a quark, $\gamma^*(q) + q(p) \to q(p') + g(k)$ (one of the two diagrams; the second has the photon attaching after the gluon). When the emitted gluon k is collinear with the outgoing quark, the amplitude develops the same $1/k_T^2$ enhancement that produced the splitting function P_{qq} in Eq. (28.109). Integrating that region over transverse momentum yields the collinear pole of Eq. (28.133).

At order α_s two things happen. Virtual gluon exchange dresses the quark–photon vertex and the quark self-energy; and the struck quark can radiate a real gluon, $\gamma^*(q) + q(p) \to q(p') + g(k)$, the QCD Compton process of Fig. 28.6. The virtual graphs are proportional to $\delta(1-z)$ (they do not change the partonic kinematics) and, taken alone, carry an infrared divergence. The real graphs populate $z < 1$. Neither is separately finite; their sum has a residual divergence that is purely *collinear* — it comes from the phase-space region in which the

radiated gluon is emitted parallel to the outgoing quark. We isolate that region, which is all we need.

The collinear region and its pole. Parametrize the real emission by the longitudinal fraction z retained by the quark that absorbs the photon and by the transverse momentum k_T of the radiated gluon relative to the incoming quark direction. The collinear region is $k_T \to 0$. In exactly the manner of the DGLAP derivation of the previous section, the squared matrix element times phase space factorizes there into the hard elastic process times a universal emission probability,

$$d\hat{\sigma}_{\text{real}} \xrightarrow[k_T\to 0]{} d\hat{\sigma}_0 \times \frac{\alpha_s}{2\pi}\frac{dk_T^2}{k_T^2} P_{qq}(z)\, dz, \tag{28.131}$$

with P_{qq} the very splitting function of Eq. (28.109). The transverse-momentum integral runs from a small value (the collinear endpoint) up to a hard cutoff of order Q^2 set by the kinematics. It is scaleless and logarithmically divergent at the lower end; dimensional regularization in $d = 4 - 2\varepsilon$ tames it (Chapter 19). With the d-dimensional phase space supplying the factor $(k_T^2)^{-\varepsilon}$, the integral over the collinear region is

$$\int^{Q^2} \frac{dk_T^2}{k_T^2}\left(\frac{\mu^2}{k_T^2}\right)^{\varepsilon} = \mu^{2\varepsilon}\int^{Q^2} dk_T^2\,(k_T^2)^{-1-\varepsilon} = -\frac{1}{\varepsilon}\left(\frac{\mu^2}{Q^2}\right)^{\varepsilon} = -\frac{1}{\varepsilon} + \ln\frac{Q^2}{\mu^2} + \mathcal{O}(\varepsilon), \tag{28.132}$$

where μ is the dimensional-regularization scale. The $1/\varepsilon$ is the collinear (mass) singularity; the accompanying $\ln(Q^2/\mu^2)$ is its finite, scale-dependent partner.

Assembling the real emission (28.131), the virtual corrections (whose divergence likewise collapses onto $\delta(1-z)$ and is contained in the same P_{qq} once the +-prescription is restored), and the finite remainder C_2^q left over after the singular piece is extracted, the *bare* partonic structure function reads

$$\frac{1}{z}\hat{F}_2^q(z,Q^2) = e_q^2\left\{\delta(1-z) + \frac{\alpha_s}{2\pi}\left[\left(-\frac{1}{\hat{\varepsilon}} + \ln\frac{Q^2}{\mu^2}\right)P_{qq}(z) + C_2^q(z)\right]\right\} + \mathcal{O}(\alpha_s^2), \tag{28.133}$$

where $1/\hat{\varepsilon} \equiv 1/\varepsilon - \gamma_E + \ln 4\pi$ collects the universal $\overline{\text{MS}}$ constants (Chapter 19). Two features are decisive. First, the coefficient of the $1/\hat{\varepsilon}$ pole is *exactly* the splitting function P_{qq} of Eq. (28.109) — not merely proportional to it, but the same function, plus-prescription, δ-term and all. The collinear divergence of the hard partonic process and the collinear emission that drives DGLAP are one and the same physics. Second, the pole multiplies $P_{qq}(z)$, a function of z alone: it is *universal*, blind to the hard probe (here a photon of charge e_q), depending only on which parton split into which. This universality is the whole reason the next step is possible.

For completeness we record the finite remainder. Carrying the full real-plus-virtual reduction to its end (a lengthy but elementary computation in the master integrals of Chapter 19; see [22, 175]) gives, in $\overline{\text{MS}}$,

$$C_2^q(z) = C_F\left[(1+z^2)\Big(\frac{\ln(1-z)}{1-z}\Big)_+ - \frac{3}{2}\frac{1}{(1-z)_+} - \frac{1+z^2}{1-z}\ln z + 3 + 2z - \Big(\frac{9}{2}+\frac{\pi^2}{3}\Big)\delta(1-z)\right] \tag{28.134}$$

with $C_F = (N_c^2-1)/2N_c = 4/3$ (Chapter 2, Appendix B) and the +-prescription $\int_0^1 dz\,[g(z)]_+\,h(z) = \int_0^1 dz\,g(z)[h(z)-h(1)]$ that was introduced with P_{qq} in the previous section. The explicit form of C_2^q will not be needed below; what matters is that it is *finite*. The pole, and only the pole, must be disposed of.

Absorbing the pole: the $\overline{\text{MS}}$ parton distribution. The bare structure function (28.133) is divergent and therefore not a measurable prediction. But it is not meant to be: a physical structure function is the partonic one convolved with the distribution of quarks inside the target. Write the measured hadronic F_2 as the partonic result folded against a *bare* distribution q_0,

$$\frac{1}{x}F_2(x,Q^2) = \sum_q \Big(\tfrac{1}{z}\hat{F}_2^q\Big)\otimes q_0 \;=\; \sum_q e_q^2 \int_x^1 \frac{dz}{z}\Big[\delta(1-z) + \tfrac{\alpha_s}{2\pi}\big((-\tfrac{1}{\hat\varepsilon} + \ln\tfrac{Q^2}{\mu^2})P_{qq}(z) + C_2^q(z)\big)\Big]\, q_0\Big(\tfrac{x}{z}\Big) \tag{28.135}$$

The bare q_0 is as unphysical as the bare coupling or bare mass of any renormalized theory: it carries the compensating divergence. Following exactly the logic of coupling-constant renormalization, we *define* the renormalized $\overline{\text{MS}}$ parton distribution at the factorization scale μ_F by subtracting the universal pole,

$$q(x,\mu_F) \;\equiv\; q_0(x) - \frac{\alpha_s}{2\pi}\,\frac{1}{\hat\varepsilon}\,(P_{qq}\otimes q_0)(x) + \mathcal{O}(\alpha_s^2), \tag{28.136}$$

which is the multiplicative collinear renormalization $q = Z_{qq}\otimes q_0$ with counterterm $Z_{qq} = \delta(1-z) - \frac{\alpha_s}{2\pi}\frac{1}{\hat\varepsilon}P_{qq}$. Because P_{qq} is universal, this single subtraction is dictated by nothing but the splitting function; it knows nothing about the photon probe, and so it will work identically for every other hard process. Inverting (28.136) to the order we work,

$$q_0(x) = q(x,\mu_F) + \frac{\alpha_s}{2\pi}\,\frac{1}{\hat\varepsilon}\,(P_{qq}\otimes q)(x) + \mathcal{O}(\alpha_s^2). \tag{28.137}$$

Substitute (28.137) into (28.135). The $\delta(1-z)$ term of the partonic result reproduces $q_0 = q + \frac{\alpha_s}{2\pi}\frac{1}{\hat\varepsilon}P_{qq}\otimes q$, contributing a $+\frac{1}{\hat\varepsilon}P_{qq}\otimes q$; the explicit $-\frac{1}{\hat\varepsilon}P_{qq}$ term of the partonic result, acting on $q_0 = q + \mathcal{O}(\alpha_s)$, contributes $-\frac{1}{\hat\varepsilon}P_{qq}\otimes q$. The two poles cancel identically. Choosing the dimensional-regularization scale to coincide with the factorization scale, $\mu = \mu_F$ (the standard $\overline{\text{MS}}$ identification),

the surviving expression is finite:

$$\frac{1}{x}F_2(x,Q^2) = \sum_q e_q^2 \Big(C_2 \otimes q\Big)(x,\mu_F), \qquad C_2(z,Q^2/\mu_F^2) = \delta(1-z) + \frac{\alpha_s}{2\pi}\Big[\ln\frac{Q^2}{\mu_F^2}\,P_{qq}(z) + C_2^q(z$$

(28.138)

This is the operational content of factorization, made completely explicit. The divergence has not been cancelled by another graph; it has been *moved*, once and for all, into the definition of the parton distribution, where it joins the genuinely non-perturbative content that the distribution always carried. What remains, the coefficient function C_2, is finite, short-distance, and calculable. It is the $\hat{\sigma}$ of Theorem 28.12 for DIS.

Physical Insight: a collinear divergence is a mass singularity, not a failure

The $1/\hat{\varepsilon}$ in (28.133) signals that perturbation theory cannot compute the distribution of a quark inside a quark down to arbitrarily small transverse distances: a massless quark is always accompanied by a collinear cloud of gluons, and asking "how much quark is there at exactly zero transverse size" is the wrong, divergent question. The right question is "how much quark is there resolved at scale μ_F," and its answer is the renormalized PDF. The divergence is thus not a sickness of the cross section but the statement that the parton-level initial state is not a physical state. Renormalizing it — trading the unphysical bare quark for a quark dressed up to a resolution μ_F — is the same move that renormalizing a coupling makes, and it has the same payoff: a finite prediction and a running object.

The origin of DGLAP. The renormalized distribution (28.136) depends on μ_F, and its dependence is no longer free. The bare q_0 is a fixed, μ_F-independent object. Just as the bare coupling's μ-independence forced α_s to run with the beta function (Chapter 27), the bare distribution's μ_F-independence forces $q(\mu_F)$ to run. Differentiating the multiplicative renormalization $q = Z_{qq}\otimes q_0$ and using that the $\overline{\text{MS}}$ counterterm is a pure pole whose only μ_F-dependence enters through α_s and the regulator's $\mu_F^{2\varepsilon}$, the residue of the pole becomes the anomalous dimension of the distribution. That anomalous dimension is precisely P_{qq}, so

$$\mu_F^2\frac{\partial}{\partial\mu_F^2}\,q(x,\mu_F) = \frac{\alpha_s(\mu_F)}{2\pi}\,(P_{qq}\otimes q)(x,\mu_F) + \ldots, \tag{28.139}$$

which is the DGLAP equation (28.121) of the previous section, here re-derived as the renormalization-group equation of the collinear counterterm. The splitting function plays three roles that are now revealed to be one: the probability of collinear emission, the kernel of DGLAP evolution, and the residue of the

factorization pole. (The full system couples quark and gluon distributions through P_{qg}, P_{gq}, P_{gg} as established in Eq. (28.121); the quark-only piece shown here is the prototype.)

28.5.3 Cancellation of the factorization scale

The factorization scale μ_F entered as an arbitrary bookkeeping device: it is the resolution at which we chose to stop calling collinear radiation "part of the hard process" and start calling it "part of the parton distribution." A physical cross section cannot depend on where we drew that line. Equation (28.138) lets us verify this explicitly, and the verification exposes a beautiful interlock: the coefficient function's μ_F-dependence is engineered to cancel the PDF's DGLAP running, order by order.

Treat $F_2(x,Q^2) = \sum_q e_q^2\,(C_2 \otimes q)(x,\mu_F)$ and apply $\mu_F^2\,\partial/\partial\mu_F^2$. By the product rule for the convolution,

$$\mu_F^2 \frac{\partial F_2}{\partial \mu_F^2} = \sum_q e_q^2 \left[\left(\mu_F^2 \frac{\partial C_2}{\partial \mu_F^2}\right) \otimes q + C_2 \otimes \left(\mu_F^2 \frac{\partial q}{\partial \mu_F^2}\right) \right]. \tag{28.140}$$

The second term is governed by DGLAP (28.139): at the order we work $C_2 \otimes (\mu_F^2 \partial_{\mu_F^2} q) = \delta(1-z) \otimes \frac{\alpha_s}{2\pi} P_{qq} \otimes q + \mathcal{O}(\alpha_s^2) = \frac{\alpha_s}{2\pi} P_{qq} \otimes q$. The first term comes from the explicit $\ln(Q^2/\mu_F^2)$ inside the coefficient (28.138):

$$\mu_F^2 \frac{\partial C_2}{\partial \mu_F^2} = \frac{\alpha_s}{2\pi}\, P_{qq}(z)\, \mu_F^2 \frac{\partial}{\partial \mu_F^2} \ln \frac{Q^2}{\mu_F^2} = -\frac{\alpha_s}{2\pi}\, P_{qq}(z), \tag{28.141}$$

so that the first term contributes $-\frac{\alpha_s}{2\pi} P_{qq} \otimes q$. The two terms are equal and opposite:

$$\mu_F^2 \frac{\partial F_2}{\partial \mu_F^2} = \sum_q e_q^2 \left[-\frac{\alpha_s}{2\pi} P_{qq} \otimes q + \frac{\alpha_s}{2\pi} P_{qq} \otimes q \right] + \mathcal{O}(\alpha_s^2) = \mathcal{O}(\alpha_s^2). \tag{28.142}$$

The physical structure function is independent of μ_F through the order computed. This is the content of point 3 of Theorem 28.12, demonstrated rather than asserted. The residual $\mathcal{O}(\alpha_s^2)$ is not a defect: it is cancelled in turn by the two-loop coefficient against the two-loop splitting function, and so on. At each order the uncancelled μ_F-dependence is pushed one power of α_s higher. In practice this means a finite-order prediction *does* wobble with μ_F, and the size of that wobble — conventionally probed by varying μ_F (and μ_R) by a factor of two about Q — is the standard estimate of the uncalculated higher-order uncertainty. The cancellation (28.142) is why that estimate is meaningful: it certifies that the leftover dependence is genuinely of higher order.

Physical Insight: two scales, two jobs

μ_R and μ_F are logically distinct. The *renormalization* scale μ_R is the one of Chapter 27: it absorbs ultraviolet divergences and makes the coupling run, and its compensation is the ordinary renormalization-group invariance of any renormalized quantity. The *factorization* scale μ_F absorbs collinear (infrared) divergences into the PDFs, and its compensation is DGLAP. They answer to different physics — short-distance versus long-distance — even though both are conventionally set near the hard scale, $\mu_R = \mu_F = Q$, to avoid large logarithms in the coefficient. Keeping them distinct is what makes the structure of (28.129) transparent.

28.5.4 The second pillar: Drell–Yan

Deep inelastic scattering can, by itself, only *define* the PDFs; it cannot test their universality, because it measures the same combination of distributions that it used to define them. The test — and the reason factorization is a physical statement rather than a tautology — is that the *very same* distributions, measured in DIS, predict a *different* process. The prototype is the Drell–Yan process: the production of a massive lepton pair in a hadron–hadron collision,

$$H_1(P_1) + H_2(P_2) \;\to\; \ell^+\ell^-(M) + X, \tag{28.143}$$

through the annihilation of a quark from one hadron with an antiquark from the other into a virtual photon, $q\bar{q} \to \gamma^* \to \ell^+\ell^-$ (Fig. 28.7). The dilepton invariant mass M is the hard scale, and at large M the process is calculable by exactly the template above.

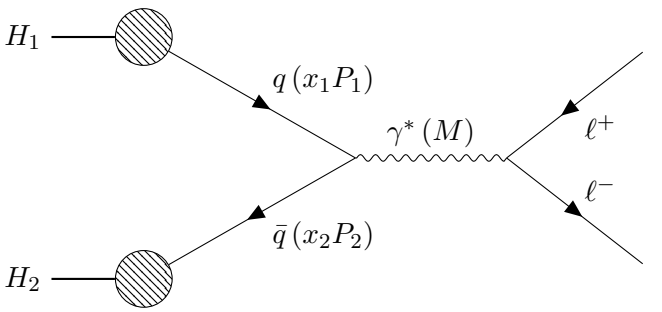


Figure 28.7: The Drell–Yan process $H_1H_2 \to \ell^+\ell^- X$ at leading order. A quark of momentum fraction x_1 from H_1 annihilates an antiquark of fraction x_2 from H_2 into a virtual photon of mass $M^2 = x_1x_2s$, which decays to the lepton pair. The blobs are the parton distributions; the central subprocess $q\bar{q} \to \gamma^* \to \ell^+\ell^-$ is the crossing of the QED process of Chapter 17.

The partonic cross section by crossing. The hard subprocess is the crossing of the QED reaction $e^+e^- \to \mu^+\mu^-$ computed in full in Chapter 17, which gave the total cross section $\sigma(e^+e^- \to \mu^+\mu^-) = 4\pi\alpha^2/3s$ (Eq. (17.89)). We need $q\bar{q} \to \ell^+\ell^-$, obtained by reading the same single s-channel photon diagram with the incoming leptons replaced by an incoming quark and antiquark. Two modifications enter, and only two.

Worked Example 28.13 (The Drell–Yan partonic cross section)**.** We assemble $\hat{\sigma}(q\bar{q} \to \ell^+\ell^-)$ from $\sigma(e^+e^- \to \mu^+\mu^-)$.

Charge. In $e^+e^- \to \mu^+\mu^-$ both vertices carry unit charge (in units of e). For $q\bar{q} \to \ell^+\ell^-$ the annihilating quark vertex carries charge e_q, so the amplitude acquires a factor e_q and the squared amplitude a factor e_q^2. (The lepton vertex is unchanged; the leptons carry unit charge.)

Colour. The amplitude is diagonal in colour: a quark of colour i annihilates an antiquark of colour j into a colour-singlet photon, so the colour factor of the amplitude is δ_{ij}. Squaring gives $\delta_{ij}\delta_{ij}$, and we must *average* over the colours of the incoming quark and antiquark (they are not prepared in any definite colour), which carries a factor $1/N_c$ for each. Summing the squared colour factor over the remaining free index,

$$\frac{1}{N_c}\frac{1}{N_c}\sum_{i,j}\delta_{ij}\delta_{ij} = \frac{1}{N_c^2}\sum_{i}1 = \frac{1}{N_c^2}\,N_c = \frac{1}{N_c}. \tag{28.144}$$

The colour factor is $1/N_c$. This is the precise mirror of the R-ratio: there a final-state colour *sum* multiplied the rate by N_c, here an initial-state colour *average* divides it by N_c. Both are the same $N_c = 3$ of Yang–Mills theory, counted from opposite ends.

Result. With $\hat{s} = (x_1P_1 + x_2P_2)^2 = x_1x_2s$ the partonic centre-of-mass energy squared, the total partonic cross section is

$$\hat{\sigma}(q\bar{q} \to \ell^+\ell^-) = \frac{e_q^2}{N_c}\,\frac{4\pi\alpha^2}{3\,\hat{s}}. \tag{28.145}$$

At leading order the produced lepton pair has invariant mass squared equal to $\hat{s}$ exactly (the photon is the entire s-channel), so as a distribution in M^2,

$$\frac{d\hat{\sigma}}{dM^2}(q\bar{q} \to \ell^+\ell^-) = \frac{e_q^2}{N_c}\,\frac{4\pi\alpha^2}{3\,M^2}\,\delta(\hat{s} - M^2), \tag{28.146}$$

which integrates over M^2 back to (28.145) at $M^2 = \hat{s}$. The $1/M^2$ rather than $1/\hat{s}$ is dictated by setting $\hat{s} = M^2$ in the prefactor under the δ-function; the two are equal on its support.

Convolving with two distributions. Now fold the partonic result against the probability of finding the quark and antiquark. The quark carries fraction x_1 of H_1, the antiquark fraction x_2 of H_2, with probabilities $f_{q/H_1}(x_1,\mu_F)\,dx_1$ and $f_{\bar q/H_2}(x_2,\mu_F)\,dx_2$. There are two ways to make a $q\bar q$ pair — quark from H_1 and antiquark from H_2, or the reverse — and we sum over quark flavours. Inserting (28.146) into the factorization master formula (28.129),

$$\frac{d\sigma}{dM^2} = \sum_q \int_0^1 dx_1 \int_0^1 dx_2 \left[f_{q/H_1}(x_1,\mu_F)\, f_{\bar q/H_2}(x_2,\mu_F) + f_{\bar q/H_1}(x_1,\mu_F)\, f_{q/H_2}(x_2,\mu_F) \right] \times \frac{e_q^2}{N_c}\,\frac{4\pi\alpha^2}{3M^2}\,\delta(x_1x_2s - M^2). \tag{28.147}$$

The delta function fixes one integral. Writing the dimensionless ratio $\tau \equiv M^2/s$ and using $\delta(x_1x_2s - M^2) = \frac{1}{x_1 s}\,\delta(x_2 - \tau/x_1)$ (which also enforces $x_1 \geq \tau$), the x_2-integral collapses and we arrive at the Drell–Yan master formula:

$$\boxed{\frac{d\sigma}{dM^2} = \frac{4\pi\alpha^2}{3\,N_c\,M^2\,s} \sum_q e_q^2 \int_\tau^1 \frac{dx}{x} \left[f_{q/H_1}(x,\mu_F)\, f_{\bar q/H_2}\!\left(\tfrac{\tau}{x},\mu_F\right) + f_{\bar q/H_1}(x,\mu_F)\, f_{q/H_2}\!\left(\tfrac{\tau}{x},\mu_F\right) \right], \quad \tau =}$$

(28.148)

This is the leading-order prediction for the dilepton mass spectrum in a hadron collider, built entirely from the universal distributions and the calculable partonic cross section. The characteristic $d\sigma/dM^2 \propto 1/M^4$ scaling (one power of $1/M^2$ explicit, one from $\hat\sigma \propto 1/\hat s = 1/M^2$) and the approximate scaling of $M^4 d\sigma/dM^2$ as a function of τ alone are the Drell–Yan analogue of Bjorken scaling, broken logarithmically by the same DGLAP evolution of the f's.

Partonic luminosity. It is useful to separate what depends on the hadrons from what depends on the hard process. Define the *partonic luminosity*

$$\frac{d\mathcal{L}_{q\bar q}}{d\tau}(\tau,\mu_F) \equiv \int_\tau^1 \frac{dx}{x} \left[f_{q/H_1}(x,\mu_F)\, f_{\bar q/H_2}\!\left(\tfrac{\tau}{x},\mu_F\right) + (q \leftrightarrow \bar q) \right], \tag{28.149}$$

the convolution (28.128) of the two distributions evaluated at the partonic energy fraction $\tau = \hat s/s$. Then (28.148) reads $d\sigma/dM^2 = \sum_q (\hat\sigma_q/s)\, d\mathcal{L}_{q\bar q}/d\tau$ with the partonic factor $\hat\sigma_q = 4\pi\alpha^2 e_q^2/3N_cM^2$. The luminosity is a property of the colliding hadrons and the beam energy alone; tabulated once, it converts any partonic cross section into a hadronic rate. Every hadron-collider prediction in this book — W, Z, Higgs, top, jets — is, at leading order, a partonic cross section times the appropriate luminosity.

Physical Insight: why universality is the whole game

Nothing in (28.148) was free to adjust. The f's are the *same* functions that (28.138) extracted from deep inelastic scattering; the partonic cross section is fixed by QED and the colour factor $1/N_c$; τ and the kinematics are measured. That a single set of distributions, fitted to DIS, predicts the absolute normalization and shape of the Drell–Yan spectrum (and, with the higher-order coefficients, W and Z rates to the percent level) is the experimental vindication of Theorem 28.12. The PDFs are not a fudge: they are a measurement of the proton made once and spent everywhere. The collinear pole we absorbed in Section 28.5.2 reappears in the Drell–Yan coefficient with the same P_{qq} residue and is removed by the same $\overline{\text{MS}}$ distribution — precisely the universality that the one-loop demonstration guaranteed. The one-loop QCD corrections to (28.148), the so-called K-factor, are substantial ($K \sim 1.3$ at fixed-target energies), a reminder that "leading order" is only the first term of a controlled series, not the answer.

We have now closed the chapter's central loop. Asymptotic freedom (Chapter 27) made the short-distance coefficient $\hat{\sigma}$ perturbative; collinear factorization (Theorem 28.12) cleanly separated it from the universal, measured PDFs; the one-loop demonstration showed the separation is a renormalization, with the splitting function P_{qq} as its anomalous dimension and DGLAP as its renormalization group; and Drell–Yan showed the distributions are genuinely universal, turning the proton's structure into a portable input for every collider prediction. One assumption underlies all of it: only infrared-safe, inclusive-enough partonic quantities are calculable — the coefficient $\hat{\sigma}$ had to be made finite by absorbing its collinear singularity, and the soft/Glauber cancellations were needed for the theorem to hold at all. This raises the question the final section must answer: *which* final-state observables are infrared-safe enough to be predicted at all? The answer — the infrared-safety principle, the soft/collinear cancellation in the simplest nontrivial case, and the jets and event shapes it licenses — turns QCD into the precision instrument with which α_s itself is measured.

28.6 Infrared Safety, Jets, and the Measurement of α_s

The factorization theorem of the previous section delivered a sharp verdict: only the *infrared-safe* part of a QCD cross section is something perturbation theory can compute, the rest being long-distance physics that factorizes into universal, measured functions. That verdict was stated as a principle. It is time to honour it with a complete calculation, and to collect a debt left open much earlier in

this chapter. When we introduced the totally inclusive ratio

$$R(s) \;\equiv\; \frac{\sigma(e^+e^- \to \text{hadrons})}{\sigma(e^+e^- \to \mu^+\mu^-)} \;=\; N_c \sum_q e_q^2 \left(1 + \frac{\alpha_s}{\pi} + \dots\right), \tag{28.150}$$

the leading factor $R_0 = N_c \sum_q e_q^2$ was the parton-model counting of colours and charges, and the correction α_s/π was promised to be finite — a promissory note (Eq. (28.35)) we now pay in full. Paying it requires confronting head-on the divergences that the Kinoshita–Lee–Nauenberg theorem (Chapter 19) told us to expect: the emission of a soft or collinear gluon produces poles in the real-emission cross section, and the one-loop virtual correction produces poles of its own. Neither is physical. Their cancellation, when summed over the degenerate final states that no detector can tell apart, is the cleanest realization in all of QCD of infrared safety, and it is the engine that turns the theory of permanently confined quarks into the most precisely tested non-abelian gauge theory we have.

This section carries out that cancellation explicitly, extracts from it the criterion that separates calculable observables from incalculable ones, and then builds the observables that matter: the Sterman–Weinberg two-jet cross section, the thrust event shape, and the modern sequential-recombination jet algorithms. Each is constructed to be infrared-and-collinear safe, and each, equated to its perturbative series, becomes a measurement of α_s. We close by assembling those measurements with the running of Chapter 27 into the single decreasing curve $\alpha_s(Q)$ — asymptotic freedom read directly off the data.

28.6.1 The three-parton process $e^+e^- \to q\bar{q}g$

At order α_s^0 the photon (or Z) produced in e^+e^- annihilation materializes as a quark–antiquark pair, $\gamma^* \to q(p_1)\,\bar{q}(p_2)$, the process whose cross section sets R_0. At order α_s two things happen. A real gluon can be radiated, $\gamma^* \to q(p_1)\,\bar{q}(p_2)\,g(p_3)$; and the $q\bar{q}$ final state can be produced through a one-loop diagram in which a virtual gluon is exchanged between the quark and antiquark lines. Both are corrections to the same physical quantity — the total rate for the virtual photon to turn into hadrons — because a detector integrating over the entire final state cannot distinguish a $q\bar{q}$ event from a $q\bar{q}g$ event in which the gluon is too soft or too collinear to resolve. The KLN theorem (Chapter 19) instructs us to sum them. We compute each in turn.

We work in the rest frame of the virtual photon, with total invariant mass squared $Q^2 = s = (p_1+p_2+p_3)^2$. All partons are taken massless, $p_i^2 = 0$, which is the correct treatment for the light quarks at the high scales where R is measured and is exactly the limit in which the infrared singularities live. It is convenient to use the dimensionless energy fractions

$$x_i \;\equiv\; \frac{2\,p_i \cdot q}{Q^2} \;=\; \frac{2E_i}{Q}, \qquad i = 1,2,3, \qquad q = p_1 + p_2 + p_3, \tag{28.151}$$

where E_i is the energy of parton i in the photon rest frame. Energy conservation, $E_1 + E_2 + E_3 = Q$, becomes the linear constraint

$$x_1 + x_2 + x_3 \;=\; 2, \tag{28.152}$$

so two of the three fractions are independent, and each lies in $0 \le x_i \le 1$. The two-body invariants are fixed by the x_i. Writing $y_{ij} \equiv (p_i + p_j)^2/Q^2 = 2p_i \cdot p_j/Q^2$ for the massless partons and using $q^2 = Q^2$,

$$x_k \;=\; 1 - y_{ij} \qquad (i, j, k \text{ distinct}), \tag{28.153}$$

because $x_k = 2p_k \cdot q/Q^2 = 2p_k \cdot (p_i + p_j + p_k)/Q^2 = y_{ki} + y_{kj}$ and $y_{ij} + y_{jk} + y_{ki} = 1$. In particular

$$1 - x_1 = y_{23}, \qquad 1 - x_2 = y_{13}, \qquad 1 - x_3 = y_{12}. \tag{28.154}$$

These three combinations are exactly the ones that will vanish in the singular limits: $y_{13} \to 0$ is the quark collinear with the gluon, $y_{23} \to 0$ the antiquark collinear with the gluon, and both $1 - x_1, 1 - x_2 \to 0$ simultaneously (i.e. $x_3 \to 0$) the soft-gluon limit. Writing the opening angle θ_{ij} between partons i and j, for massless momenta $y_{ij} = x_i x_j (1 - \cos\theta_{ij})/2$, which makes the geometry transparent: $y_{ij} \to 0$ either because a parton becomes soft ($x_i \to 0$) or because two become collinear ($\theta_{ij} \to 0$).

The squared matrix element

The amplitude for $\gamma^* \to q\bar{q}g$ is the sum of two diagrams: the gluon radiated off the quark line and the gluon radiated off the antiquark line, with the quark–gluon vertex $ig\,\gamma^\mu T^a$ taken from the QCD Feynman rules assembled in Chapters 25 and 26. The triple-gluon vertex does not enter at this order because the external current is colour-singlet and couples only to the $q\bar{q}$ pair. Squaring, summing over final colours and spins and the gluon polarization, averaging over the photon polarizations, and dividing by the lowest-order $q\bar{q}$ result, one finds the celebrated compact form

$$\frac{1}{\sigma_0}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x_1\,\mathrm{d}x_2} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,\frac{x_1^2 + x_2^2}{(1 - x_1)(1 - x_2)} \qquad (d = 4), \tag{28.155}$$

where $\sigma_0 = \sigma(e^+e^- \to q\bar{q})$ at lowest order and $C_F = (N_c^2 - 1)/2N_c = 4/3$ is the fundamental Casimir (Chapter B). The colour factor is C_F because the gluon emission off a quark carries $\sum_a T^a T^a = C_F\,\mathbf{1}$ summed over the gluon colour, exactly as in the QED bremsstrahlung template but with $e^2 \to g^2 C_F$. The structure of the numerator and denominator is worth absorbing before we regulate it. The denominator $(1 - x_1)(1 - x_2) = y_{23}\,y_{13}$ is the product of the two propagator denominators: emitting the gluon off the quark puts the intermediate

quark off shell by $(p_1+p_3)^2 = Q^2 y_{13}$, and off the antiquark by $Q^2 y_{23}$. The numerator $x_1^2 + x_2^2$ stays finite and nonzero precisely where the denominator vanishes, so (28.155) is singular along the entire boundary of the Dalitz region where a gluon goes soft or collinear.

Worked Example 28.14 (Soft and collinear factorization of the three-parton rate)**.** It pays to see that (28.155) reduces, in each singular limit, to the lowest-order rate times a universal singular factor — this is the concrete content of the factorization that the KLN theorem organizes.

Soft limit. Let the gluon energy go to zero, $x_3 \to 0$. Then (28.152) forces $x_1 \to 1$ and $x_2 \to 1$ together. Parametrize $1-x_1 = \xi_1$, $1-x_2 = \xi_2$ with $\xi_1, \xi_2 \to 0$ and $\xi_1 + \xi_2 = x_3$. The numerator $x_1^2 + x_2^2 \to 2$, so

$$\frac{1}{\sigma_0}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x_1\,\mathrm{d}x_2} \xrightarrow{x_3\to 0} \frac{\alpha_s}{2\pi} C_F \frac{2}{\xi_1\xi_2} = \frac{\alpha_s}{2\pi} C_F \frac{2\,(p_1\cdot p_2)}{(p_1\cdot p_3)(p_2\cdot p_3)}, \tag{28.156}$$

where the last equality uses $\xi_1 = y_{23} = 2p_2\cdot p_3/Q^2$, $\xi_2 = y_{13} = 2p_1\cdot p_3/Q^2$ and $2p_1\cdot p_2 \to Q^2$. The combination $(p_1\cdot p_2)/[(p_1\cdot p_3)(p_2\cdot p_3)]$ is the *eikonal factor*: it depends only on the gluon's direction, not on its having come from a particular hard parton, and it is the same object that governs soft-photon emission in QED (Chapter 19). Its integral over the soft gluon's phase space is logarithmically divergent in both the energy ($\mathrm{d}x_3/x_3$) and the angle.

Collinear limit. Let the gluon become collinear with the quark, $\theta_{13} \to 0$, i.e. $y_{13} = 1 - x_2 \to 0$ while x_3 stays finite. Write the quark and gluon as sharing the parent momentum, $x_1 = z\,(x_1 + x_3)$ for a momentum fraction z, so that $x_1 \to z$ and $x_3 \to 1 - z$ as the pair collapses onto one direction, while $x_2 \to 1$. Then $x_1^2 + x_2^2 \to z^2 + 1$ and

$$\frac{1}{\sigma_0}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x_1\,\mathrm{d}x_2} \xrightarrow{\theta_{13}\to 0} \frac{\alpha_s}{2\pi}\frac{1}{1-x_2} C_F \frac{1+z^2}{1-z} = \frac{\alpha_s}{2\pi}\frac{1}{y_{13}} P_{qq}(z), \tag{28.157}$$

with the Altarelli–Parisi splitting function $P_{qq}(z) = C_F\,(1+z^2)/(1-z)$ — the same kernel that drives DGLAP evolution earlier in this chapter. The cross section factorizes into the hard $q\bar{q}$ rate times a collinear factor $\mathrm{d}y_{13}/y_{13} \times P_{qq}(z)$ whose angular integral is again logarithmically divergent. The limit $\theta_{23} \to 0$ (antiquark collinear with gluon) gives the mirror statement with $x_1 \leftrightarrow x_2$. These universal soft and collinear limits are the reason the divergences of (28.155) are not an accident of this process: they are dictated by the same factorization that makes parton distributions universal, and they are what the virtual correction must cancel.

28.6.2 Dimensional regularization of the real emission

The integrals just exhibited as divergent must be given meaning before they can be added to the virtual contribution. We regulate them dimensionally, working

in $d = 4 - 2\varepsilon$ spacetime dimensions (Chapter 19), where the soft and collinear logarithms become poles in ε. This is the natural regulator here precisely because it respects gauge invariance and treats the real and virtual divergences in the same scheme, so that their cancellation is manifest.

Two ingredients acquire ε-dependence: the three-body phase space and the squared matrix element. The Lorentz-invariant three-body phase space built in Section 16.2.2 (Theorem 16.7), specialized to three massless particles of total invariant mass Q^2 and written in the energy fractions (28.151), takes the form

$$\int \mathrm{d}\Pi_3 \;=\; \frac{Q^2}{128\pi^3}\,\frac{(4\pi)^{2\varepsilon}}{\Gamma(2-2\varepsilon)}\,\Big(\frac{Q^2}{4}\Big)^{-\varepsilon}\int \mathrm{d}x_1\,\mathrm{d}x_2\,\big[(1-x_1)(1-x_2)(1-x_3)\big]^{-\varepsilon}, \tag{28.158}$$

the key feature being the factor $[(1-x_1)(1-x_2)(1-x_3)]^{-\varepsilon} = [y_{23}\,y_{13}\,y_{12}]^{-\varepsilon}$. In four dimensions this factor is unity; in d dimensions it softens the endpoint singularities of the matrix element just enough to convert their would-be logarithms into $1/\varepsilon$ poles. The matrix element (28.155), recomputed with d-dimensional Dirac algebra (Chapter C) and the d-dimensional gluon polarization sum, acquires $O(\varepsilon)$ corrections in its numerator that do not alter the singular denominators:

$$\frac{1}{\sigma_0}\frac{\mathrm{d}^2\sigma}{\mathrm{d}x_1\,\mathrm{d}x_2} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,H(\varepsilon)\,\big[(1-x_1)(1-x_2)(1-x_3)\big]^{-\varepsilon}\,\frac{x_1^2+x_2^2-\varepsilon\,(2-x_1-x_2)^2+O(\varepsilon^2)\ \text{ter}}{(1-x_1)(1-x_2)} \tag{28.159}$$

where the prefactor

$$H(\varepsilon) \;=\; \Big(\frac{4\pi\mu^2}{Q^2}\Big)^{\varepsilon}\frac{1}{\Gamma(1-\varepsilon)} \;=\; 1+\varepsilon\big[\ln(4\pi)-\gamma_E+\ln(\mu^2/Q^2)\big]+O(\varepsilon^2) \tag{28.160}$$

collects the $\overline{\mathrm{MS}}$ measure factors and the renormalization scale μ introduced with the d-dimensional coupling $g \to g\,\mu^{\varepsilon}$. The $-\varepsilon(2-x_1-x_2)^2$ piece, together with the rest of the $O(\varepsilon)$ numerator, multiplies the $1/\varepsilon$ poles of the integral and therefore contributes to the finite part; it does not change the pole *coefficients*, which is what the cancellation against the virtual term is about. We will extract the poles exactly from the leading numerator and then quote the complete finite constant.

The phase-space integral and its poles

Change variables to $y_1 \equiv 1-x_1 = y_{23}$ and $y_2 \equiv 1-x_2 = y_{13}$, so that $1-x_3 = 1-y_1-y_2$ and the Dalitz region $0 \le x_i \le 1$ becomes the simplex $y_1 \ge 0,\ y_2 \ge 0,\ y_1+y_2 \le 1$. The numerator $x_1^2+x_2^2 = (1-y_1)^2+(1-y_2)^2 = 2-2(y_1+y_2)+y_1^2+y_2^2$, and the integrand of the real cross section (keeping the leading numerator, which fixes the poles) becomes

$$\frac{1}{\sigma_0}\,\sigma_R^{(\text{poles})} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,H(\varepsilon)\int_{\triangle}\mathrm{d}y_1\,\mathrm{d}y_2\;\big[2-2(y_1+y_2)+y_1^2+y_2^2\big]\;y_1^{-1-\varepsilon}\,y_2^{-1-\varepsilon}\,(1-y_1-y_2)^{-\varepsilon}. \tag{28.161}$$

The factor $y_1^{-1-\varepsilon}y_2^{-1-\varepsilon}$ displays the divergence structure directly: an $\int_0 y_1^{-1-\varepsilon}\mathrm{d}y_1 = -1/\varepsilon$ from the $y_1 \to 0$ (antiquark–gluon collinear) edge, another from $y_2 \to 0$ (quark–gluon collinear), and the *double* pole from the corner $y_1, y_2 \to 0$ together (the soft limit, in which both collinear regions overlap). Every integral here is a Dirichlet integral over the simplex,

$$\int_{\triangle} \mathrm{d}y_1\,\mathrm{d}y_2\; y_1^{a-1}\,y_2^{b-1}\,(1-y_1-y_2)^{c-1} \;=\; \frac{\Gamma(a)\,\Gamma(b)\,\Gamma(c)}{\Gamma(a+b+c)}, \tag{28.162}$$

the two-dimensional generalization of the Euler Beta function. Applying it term by term, with $c = 1-\varepsilon$ throughout:

$$\text{from } 2: \quad 2\,\frac{\Gamma(-\varepsilon)^2\,\Gamma(1-\varepsilon)}{\Gamma(1-3\varepsilon)} \;=\; \frac{2}{\varepsilon^2}\,\frac{\Gamma(1-\varepsilon)^3}{\Gamma(1-3\varepsilon)}, \tag{28.163}$$

$$\text{from } -2(y_1+y_2): \quad -4\,\frac{\Gamma(1-\varepsilon)\,\Gamma(-\varepsilon)\,\Gamma(1-\varepsilon)}{\Gamma(2-3\varepsilon)} \;=\; \frac{4}{\varepsilon}\,\frac{\Gamma(1-\varepsilon)^3}{\Gamma(2-3\varepsilon)}, \tag{28.164}$$

$$\text{from } y_1^2+y_2^2: \quad 2\,\frac{\Gamma(2-\varepsilon)\,\Gamma(-\varepsilon)\,\Gamma(1-\varepsilon)}{\Gamma(3-3\varepsilon)} \;=\; -\frac{2}{\varepsilon}\,\frac{(1-\varepsilon)\,\Gamma(1-\varepsilon)^3}{(2-3\varepsilon)(1-3\varepsilon)\,\Gamma(1-3\varepsilon)}, \tag{28.165}$$

where in each line we used $\Gamma(-\varepsilon) = -\Gamma(1-\varepsilon)/\varepsilon$ and, in the last, $\Gamma(2-\varepsilon) = (1-\varepsilon)\Gamma(1-\varepsilon)$, $\Gamma(3-3\varepsilon) = (2-3\varepsilon)(1-3\varepsilon)\Gamma(1-3\varepsilon)$. The ratio of Γ's is expanded with $\ln\Gamma(1-\varepsilon) = \gamma_E\,\varepsilon + \sum_{k\ge 2}\zeta_k\,\varepsilon^k/k$:

$$\frac{\Gamma(1-\varepsilon)^3}{\Gamma(1-3\varepsilon)} \;=\; \exp\Big[(3\gamma_E\varepsilon+\tfrac{3}{2}\zeta_2\varepsilon^2)-(3\gamma_E\varepsilon+\tfrac{9}{2}\zeta_2\varepsilon^2)+O(\varepsilon^3)\Big] \;=\; 1-3\zeta_2\,\varepsilon^2+O(\varepsilon^3) \;=\; 1-\frac{\pi^2}{2}\,\varepsilon^2+O(\varepsilon \tag{28.166}$$

using $\zeta_2 = \pi^2/6$; the $O(\varepsilon)$ term cancels, a useful check. Expanding the three lines to the order needed and adding:

$$\begin{aligned}
(28.163) &= \frac{2}{\varepsilon^2}\Big(1-\tfrac{\pi^2}{2}\varepsilon^2\Big) = \frac{2}{\varepsilon^2} - \pi^2 + O(\varepsilon),\\
(28.164) &= \frac{4}{\varepsilon}\,\frac{1}{1-3\varepsilon}\Big(1-\tfrac{\pi^2}{2}\varepsilon^2\Big) = \frac{4}{\varepsilon}(1+3\varepsilon+\dots) = \frac{4}{\varepsilon} + 12 + O(\varepsilon),\\
(28.165) &= -\frac{2}{\varepsilon}\,\frac{(1-\varepsilon)}{(2-3\varepsilon)(1-3\varepsilon)}(1+\dots) = -\frac{1}{\varepsilon}(1+\tfrac{7}{2}\varepsilon+\dots) = -\frac{1}{\varepsilon} - \frac{7}{2} + O(\varepsilon).
\end{aligned} \tag{28.167}$$

Their sum gives the divergent part of the real cross section exactly,

$$\frac{1}{\sigma_0}\,\sigma_R^{(\text{poles})} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,H(\varepsilon)\Big[\frac{2}{\varepsilon^2}+\frac{3}{\varepsilon}+\Big(\frac{17}{2}-\pi^2\Big)+O(\varepsilon)\Big]. \tag{28.168}$$

The pole part $2/\varepsilon^2 + 3/\varepsilon$ is the whole point: the double pole $2/\varepsilon^2$ is the soft×collinear (overlapping) singularity, and the single pole $3/\varepsilon$ is the sum

of the two pure-collinear singularities plus the soft-finite remainder. Restoring the $O(\varepsilon)$ pieces of the d-dimensional numerator in (28.159), which multiply the $1/\varepsilon$ poles and shift only the finite constant by $+1$, upgrades the constant $17/2 \to 19/2$ and yields the complete real result,

$$\frac{\sigma_R}{\sigma_0} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,H(\varepsilon)\Big[\frac{2}{\varepsilon^2}+\frac{3}{\varepsilon}+\frac{19}{2}-\pi^2+O(\varepsilon)\Big]. \tag{28.169}$$

This is finite nowhere: it is a pure divergence plus a finite remainder, and on its own it is not a physical quantity. It must be added to the virtual correction, which we now show carries exactly the opposite poles.

28.6.3 The virtual correction and the cancellation

The one-loop correction to $\gamma^* \to q\bar{q}$ is the QCD vertex graph: a virtual gluon exchanged between the outgoing quark and antiquark, dressing the electromagnetic vertex. Its colour factor is again C_F (the same $\sum_a T^a T^a$ contracted along the quark line), and its Lorentz and loop structure are *identical* to the one-loop QED vertex correction computed in Chapter 19 for the electron's electromagnetic form factor, with the replacement $e^2 \to g^2 C_F$. There the infrared structure of the massless vertex was found to be a double plus single pole; here it is the same object evaluated on shell with a timelike photon $q^2 = Q^2 > 0$. The interference of the one-loop amplitude with the tree amplitude, integrated over the two-body phase space, gives the virtual contribution to the total rate,

$$\frac{\sigma_V}{\sigma_0} \;=\; \frac{\alpha_s}{2\pi}\,C_F\,H(\varepsilon)\Big[-\frac{2}{\varepsilon^2}-\frac{3}{\varepsilon}-8+\pi^2+O(\varepsilon)\Big]. \tag{28.170}$$

The poles here are infrared, not ultraviolet: the vertex graph's ultraviolet divergence is removed by the on-shell wavefunction renormalization of the external quark legs (the quark self-energy at $p^2 = 0$ is scaleless and vanishes in dimensional regularization, so the bare and renormalized statements coincide here), and what remains is purely the long-distance divergence from the virtual gluon going soft and collinear inside the loop. That is why its pole structure $-2/\varepsilon^2 - 3/\varepsilon$ is the exact negative of the real-emission poles in (28.169): the virtual gluon explores the same soft and collinear regions of momentum space as the real one, and the KLN theorem (Chapter 19) guarantees that the sum over these degenerate configurations — "a $q\bar{q}$ pair" and "a $q\bar{q}$ pair plus an unresolvable gluon" — is free of infrared singularities.

Adding (28.169) and (28.170), the prefactor $H(\varepsilon)$ multiplies a bracket that is now finite as $\varepsilon \to 0$, so $H(\varepsilon) \to 1$ and the μ-dependence, the $\ln(4\pi)$ and the $-\gamma_E$ all drop out with the poles:

$$\frac{\sigma_R+\sigma_V}{\sigma_0} \;=\; \frac{\alpha_s}{2\pi}\,C_F\Big[\underbrace{\Big(\tfrac{2}{\varepsilon^2}-\tfrac{2}{\varepsilon^2}\Big)}_{0}+\underbrace{\Big(\tfrac{3}{\varepsilon}-\tfrac{3}{\varepsilon}\Big)}_{0}+\underbrace{\Big(\tfrac{19}{2}-\pi^2\Big)+\big(-8+\pi^2\big)}_{=3/2}\Big] \;=\; \frac{\alpha_s}{2\pi}\,C_F\,\frac{3}{2}. \tag{28.171}$$

The $1/\varepsilon^2$ and $1/\varepsilon$ poles cancel identically, and so do the π^2 terms — the latter a nontrivial check that the soft phase-space integral and the analytic continuation of the loop to timelike q^2 are being handled consistently. With $C_F = 4/3$ the finite remainder is

$$\frac{\sigma_R + \sigma_V}{\sigma_0} \;=\; \frac{\alpha_s}{2\pi}\cdot\frac{4}{3}\cdot\frac{3}{2} \;=\; \frac{\alpha_s}{\pi}. \tag{28.172}$$

The total hadronic rate is therefore $\sigma_0(1+\alpha_s/\pi)$, and dividing by the muon-pair rate gives the long-promised first QCD correction to the R-ratio,

$$\boxed{R(s) \;=\; N_c \sum_q e_q^2 \left(1 + \frac{\alpha_s(\sqrt{s})}{\pi} + O(\alpha_s^2)\right).} \tag{28.173}$$

This discharges in full the promissory note of Eq. (28.35): the $O(\alpha_s)$ correction announced there is finite, scheme-independent at this order, and equal to α_s/π. The coupling is naturally evaluated at $\mu = \sqrt{s}$ to avoid large logarithms (Chapter 27); the residual scale dependence is formally $O(\alpha_s^2)$. Numerically, at the Z pole $\alpha_s \approx 0.118$ gives a $+3.8\%$ correction to the naive parton count — small, calculable, and confirmed: the measured R near the Z both verifies $N_c = 3$ through R_0 and provides one of the cleanest determinations of α_s through the correction.

Physical Insight

The cancellation in (28.171) is not bookkeeping luck; it is the statement that QCD makes sense as a predictive theory *despite* confinement. The quark and antiquark we put in the final state are not asymptotic particles — they are confined — and any attempt to compute the rate for a definite number of them is doomed to infrared divergence, as (28.169) and (28.170) separately show. What *is* well-defined is the rate summed over all final states a detector cannot distinguish. The divergences are the theory's way of refusing to answer an ill-posed question ("how many partons?") while answering perfectly the well-posed one ("how much hadronic energy?"). Infrared safety, defined next, is the precise characterization of the well-posed questions.

28.6.4 The infrared-safety criterion

The calculation above worked because the total cross section is insensitive to whether the final state contains an extra soft or collinear gluon. Promoting that insensitivity to a definition isolates exactly the class of observables that fixed-order QCD can predict.

Definition 28.15 (Infrared-and-collinear-safe observable)**.** Consider an observable defined as a function $O_n(p_1, \ldots, p_n)$ of the momenta of an n-parton final

state, extended to any number of partons by a family $\{O_n\}$. The observable is *infrared-and-collinear (IRC) safe* if it is invariant, in the singular limits, under the two operations that fixed-order perturbation theory cannot resolve:

$$\text{soft safety:} \quad O_{n+1}(p_1, \dots, p_n, p_s) \longrightarrow O_n(p_1, \dots, p_n) \quad \text{as } p_s \to 0, \tag{28.174}$$

$$\text{collinear safety:} \quad O_{n+1}(\dots, p_a, p_b, \dots) \longrightarrow O_n(\dots, p_a + p_b, \dots) \quad \text{as } p_a \parallel p_b, \tag{28.175}$$

where in the collinear limit $p_a = z\,p$ and $p_b = (1-z)\,p$ collapse onto a common direction p.

The criterion is exactly tuned to the two singular limits established in Theorem 28.14: the soft (eikonal) divergence (28.156) and the collinear (splitting-function) divergence (28.157). Its consequence is the central theorem of perturbative QCD phenomenology.

Proposition 28.16 (IRC safety $\Leftrightarrow$ calculability)**.** *An observable admits a finite, well-defined order-by-order expansion in α_s in terms of quark and gluon momenta — it is computable in fixed-order perturbative QCD — if and only if it is IRC safe in the sense of Theorem 28.15.*

Argument. The "if" direction is the content of the KLN theorem (Chapter 19) made concrete by the calculation of Sections 28.6.2 and 28.6.3. The real-emission divergences arise entirely from the soft ($p_s \to 0$) and collinear ($p_a \parallel p_b$) corners of phase space; in each corner the squared matrix element factorizes into the lower-multiplicity squared matrix element times the universal eikonal or splitting factor (28.156), (28.157). If the observable satisfies (28.174) and (28.175), then in precisely those corners its value equals the lower-multiplicity value O_n, which factors out of the singular integral. The remaining divergent integral over the soft/collinear phase space is then the *same* integral that appears, with opposite sign, in the virtual correction to O_n (the virtual gluon probes the identical degenerate region), and the KLN cancellation (28.171) renders the weighted sum finite. If instead the observable distinguishes O_{n+1} from O_n in the singular limit — for example, the exclusive cross section for "exactly two partons," which counts a soft gluon as a third — then $O_{n+1} \neq O_n$ there, the lower-multiplicity value does not factor out, the real and virtual divergences are weighted differently, and the leftover $1/\varepsilon$ poles signal an observable that is sensitive to the arbitrarily-long-distance physics of confinement: not computable from partons alone. □ □

Caution

IRC safety is a property of the *observable's definition*, evaluated on partons, and it must be checked before any cross section is computed. The exclusive

n-parton cross section, the energy of "the leading quark," and any quantity that counts partons or identifies a specific parton flavor in the final state are IRC *unsafe* and have no order-by-order perturbative meaning. The remainder of this section is devoted to constructing observables — jets and event shapes — that are safe by design, and therefore turn QCD into a precision science.

28.6.5 The Sterman–Weinberg two-jet cross section

The earliest IRC-safe jet definition, and the historical proof that jets are calculable, is due to Sterman and Weinberg (1977). It captures the physical picture that an e^+e^- annihilation event "looks like two back-to-back jets" if almost all of its energy is confined to two narrow, opposite cones — a definition that is deliberately blind to soft and collinear radiation inside or between the cones, and is therefore IRC safe.

Definition 28.17 (Sterman–Weinberg two-jet event)**.** Fix an energy fraction β and a half-angle δ, both small. An event is a *two-jet event* if all but at most a fraction β of the total energy $\sqrt{s} = Q$ is contained within some pair of back-to-back cones of half-angle δ.

At lowest order every $q\bar{q}$ event is trivially two-jet: the two partons are exactly back-to-back and carry all the energy. The order-α_s corrections are the real gluon emission and the virtual correction. The event remains two-jet in either of two cases: (i) the gluon is *soft*, carrying energy $E_g < \beta Q$, so that even outside the cones the out-of-cone energy is below threshold; or (ii) the gluon is *collinear*, lying within δ of the quark or antiquark, so it falls inside a cone. The event fails to be two-jet only when the gluon is simultaneously hard ($E_g > \beta Q$) and at wide angle (more than δ from both quark and antiquark) — the genuinely resolved three-jet region. The two-jet cross section is therefore the full $O(\alpha_s)$ cross section minus the resolved three-jet piece, and because the resolved region excludes both singular limits, that subtraction is finite. Equivalently, the two-jet rate is the virtual correction plus the unresolved part of the real emission, with the $1/\varepsilon$ poles cancelling between them exactly as in (28.171), leaving a finite function of β and δ.

To evaluate it we integrate the real-emission rate (28.155) over the unresolved region and add the virtual term. The leading behaviour comes from the soft and collinear edges, where the integrand is dominated by $2/[(1-x_1)(1-x_2)]$. In the soft region the angular and energy integrals each produce a logarithm; the overlap produces the product of the two. Carrying the integration over the unresolved phase space and combining with the virtual poles gives the Sterman–Weinberg

result

$$\boxed{\frac{\sigma_{2\text{-jet}}(\beta,\delta)}{\sigma_0} \;=\; 1 - \frac{\alpha_s}{\pi}\,C_F\Big[\,4\,\ln(2\beta)\,\ln\delta \;+\; 3\,\ln\delta \;+\; \frac{\pi^2}{3} - \frac{5}{2}\,\Big] + O(\alpha_s^2),} \tag{28.176}$$

finite for any nonzero β, δ. The structure is exactly what infrared safety predicts. The *double logarithm* $4\ln(2\beta)\ln\delta$ (Sudakov form) is the remnant of the cancelled $1/\varepsilon^2$: the energy cutoff β regulates the soft divergence the way ε did, and the angle cutoff δ regulates the collinear one, so the would-be double pole reappears as the product of the two logarithms of the experimental resolutions. The single logarithm $3\ln\delta$ is the collinear remnant of the $3/\varepsilon$, with the coefficient $3 = 2\times\frac{3}{2}$ inherited from the quark splitting function's finite part. Both logarithms are negative for small β, δ (since $\ln\delta, \ln 2\beta < 0$), so the correction reduces the two-jet rate as the cones are made narrower or the energy threshold lower — exactly as it must, because tighter cuts resolve more of the radiation into a third jet. Crucially, the result is finite and predictive: measuring the two-jet fraction at fixed (β, δ) determines α_s.

Physical Insight

The Sterman–Weinberg cones are the prototype of every jet definition: a jet is not "a parton" but a *region of the detector* into which the IRC-safe sum of soft and collinear radiation is collected. The cutoffs β, δ trade theoretical resolution for finiteness — the finer the jet, the larger the (still finite, but logarithmically enhanced) perturbative correction. The appearance of $\ln^2$ terms warns that for very fine jets these logarithms must be resummed to all orders, the subject that grows into the modern theory of parton showers and Sudakov resummation, which we flag but do not develop here.

28.6.6 Event shapes: thrust

Cone definitions require choosing axes and angles. A complementary and historically powerful class of IRC-safe observables, the *event shapes*, assign a single number to the whole event that measures how "pencil-like" (two-jet) versus "spherical" (multi-jet) it is, with no jet algorithm at all. The canonical example is thrust.

Definition 28.18 (Thrust)**.** For a final state with momenta $\{\vec{p}_i\}$, the *thrust* is

$$T \;=\; \max_{\hat{n}}\; \frac{\sum_i |\vec{p}_i\cdot\hat{n}|}{\sum_i |\vec{p}_i|}, \tag{28.177}$$

the maximum taken over all unit vectors $\hat{n}$; the maximizing $\hat{n}$ is the thrust axis.

Thrust is IRC safe by inspection of Theorem 28.15. *Soft safety*: adding a particle with $\vec{p}_s \to 0$ adds $|\vec{p}_s \cdot \hat{n}| \to 0$ to the numerator and $|\vec{p}_s| \to 0$ to the denominator, leaving T unchanged. *Collinear safety*: replacing a particle of momentum $\vec{p}$ by two collinear fragments $z\vec{p}$ and $(1-z)\vec{p}$ leaves both numerator and denominator unchanged because $|z\vec{p}\cdot\hat{n}|+|(1-z)\vec{p}\cdot\hat{n}| = |\vec{p}\cdot\hat{n}|$ (both fragments point the same way, so no cancellation of signs) and likewise for the magnitudes. Hence T depends only on the IRC-safe gross flow of momentum, not on how that flow is partitioned among soft or collinear quanta.

A perfectly back-to-back two-parton event has $T = 1$ (take $\hat{n}$ along the axis). A symmetric three-parton (or spherical multi-particle) event has T as low as $\frac{1}{2}$ for a planar Mercedes configuration, or $T \to \frac{1}{2}$ in the spherical limit, so $1 - T$ measures the departure from the two-jet shape. For three massless partons,

$$T \;=\; \max(x_1, x_2, x_3), \tag{28.178}$$

since the thrust axis aligns with the most energetic parton and the other two balance against it. Integrating the three-parton rate (28.155) at fixed T gives the order-α_s thrust distribution

$$\frac{1}{\sigma_0}\frac{\mathrm{d}\sigma}{\mathrm{d}T} \;=\; \frac{\alpha_s}{2\pi}\,C_F\left[\frac{2(3T^2-3T+2)}{T(1-T)}\,\ln\!\Big(\frac{2T-1}{1-T}\Big) \;-\; \frac{3(3T-2)(2-T)}{1-T}\right], \tag{28.179}$$

valid for $\frac{2}{3} \le T \le 1$ (the kinematic range for three partons). As $T \to 1$ the distribution diverges like $\ln(1-T)/(1-T)$ — the same soft-collinear enhancement, now as the event becomes two-jet — which is integrable and reflects the need for resummation in the two-jet limit. The quantity that is cleanly predicted in fixed order is any sufficiently inclusive moment; the mean value, computed by integrating (28.179) against $(1-T)$ over the allowed range, is finite,

$$\langle 1 - T\rangle \;=\; \frac{\alpha_s}{\pi}\,C_F\Big(\frac{\pi^2}{3} - \frac{5}{2}\Big) + O(\alpha_s^2) \;\approx\; 1.05\,\frac{\alpha_s}{\pi} + O(\alpha_s^2) \;\approx\; 0.33\,\alpha_s, \tag{28.180}$$

a pure number times α_s: measuring $\langle 1-T\rangle$ at a given Q is a direct measurement of $\alpha_s(Q)$. Event shapes like thrust, the C-parameter, jet masses and broadenings, measured across the full range of e^+e^- energies, provided some of the earliest and most influential determinations of the running coupling.

28.6.7 Sequential-recombination jet algorithms

The cone idea generalizes, but modern collider analyses overwhelmingly use *sequential-recombination* algorithms, which build jets bottom-up by repeatedly merging the closest pair of objects according to a distance measure engineered for IRC safety. These algorithms are the workhorses at the LHC and the standard tool for jet-based α_s extractions, so we give their definition and verify their safety.

Definition 28.19 (Generalized k_t algorithm)**.** For each pair of final-state objects i, j define the inter-object distance and the beam distance

$$d_{ij} \;=\; \min(k_{ti}^{2p},\, k_{tj}^{2p})\,\frac{\Delta R_{ij}^2}{R^2}, \qquad d_{iB} \;=\; k_{ti}^{2p}, \tag{28.181}$$

where k_{ti} is the transverse momentum of object i (relative to the beam), $\Delta R_{ij}^2 = (y_i - y_j)^2 + (\phi_i - \phi_j)^2$ is the squared separation in the rapidity–azimuth plane, R is the jet-radius parameter, and p is an integer power selecting the algorithm. One finds the smallest of all d_{ij} and d_{iB}; if it is a d_{ij}, objects i and j are merged (their four-momenta added) into a single object; if it is a d_{iB}, object i is declared a final jet and removed. The procedure repeats until no objects remain. The choices $p = 1$, $p = 0$, $p = -1$ give the k_t, Cambridge/Aachen, and anti-k_t algorithms respectively.

The anti-k_t algorithm, $p = -1$, is the LHC default, with distances

$$\boxed{d_{ij} \;=\; \min\Big(\frac{1}{k_{ti}^2},\, \frac{1}{k_{tj}^2}\Big)\,\frac{\Delta R_{ij}^2}{R^2}, \qquad d_{iB} \;=\; \frac{1}{k_{ti}^2}.} \tag{28.182}$$

Proposition 28.20 (IRC safety of sequential recombination)**.** *For any value of p the generalized-k_t algorithm of Theorem 28.19 defines an IRC-safe set of jets: the jet four-momenta are invariant, in the singular limits, under the soft and collinear operations of Theorem 28.15.*

Argument. Collinear safety. If two objects are exactly collinear, $\Delta R_{ij} \to 0$, then $d_{ij} \to 0$ regardless of p, so this pair has the smallest inter-object distance and is merged *first*, before any other clustering or any beam declaration can act on either of them. Merging adds their momenta, so a parton and its collinear splitting product are recombined into the same object carrying their summed momentum; the resulting jets are identical to those of the unsplit configuration. Hence (28.175) holds.

Soft safety. A soft object s with $k_{ts} \to 0$ cannot change any hard jet's momentum. Consider its distances to a hard object h. For k_t ($p = 1$), $d_{sh} = \min(k_{ts}^2, k_{th}^2)\,\Delta R_{sh}^2/R^2 = k_{ts}^2\,\Delta R_{sh}^2/R^2 \to 0$ and $d_{sB} = k_{ts}^2 \to 0$; the soft object is therefore clustered (with whichever neighbour or with the beam is nearest) very early, and because its momentum is infinitesimal, whatever it merges into is unchanged at leading power. For anti-k_t ($p = -1$), $d_{sh} = \min(k_{ts}^{-2}, k_{th}^{-2})\,\Delta R_{sh}^2/R^2 = k_{th}^{-2}\,\Delta R_{sh}^2/R^2$ (the *hard* object's small $1/k_t^2$ wins the min), which is small and finite, while $d_{sB} = k_{ts}^{-2} \to \infty$. Thus the soft object is always merged into a nearby hard object rather than becoming its own jet, and again contributes negligible momentum. In every case adding a soft particle does not alter the hard jets, so (28.174) holds. □ □

The choice $p = -1$ has a decisive practical virtue that the argument already reveals: because the distance is controlled by the *inverse* transverse momentum, clustering is driven by the *hardest* objects first. A hard parton accretes nearby soft radiation into a jet whose boundary is, in the absence of a second hard parton within $2R$, a perfect circle of radius R in the (y, ϕ) plane. This rigidity against soft radiation — the jet shape is set by the hard core, not by the soft junk — is why anti-k_t became the experimental standard: its jets are IRC safe by Theorem 28.20 and geometrically regular, hence straightforward to calibrate. The k_t and Cambridge/Aachen algorithms, whose clustering follows the soft and angular structure of QCD radiation more faithfully, remain the tools of choice for resolving the substructure *inside* a jet.

28.6.8 Measuring α_s and the running coupling

We can now state the logic that turns all of the above into a number. Any IRC-safe observable $\mathcal{O}$ has, by Theorem 28.16, a finite perturbative expansion in the coupling evaluated at the hard scale Q,

$$\mathcal{O}(Q) \;=\; c_1\,\alpha_s(Q) + c_2\,\alpha_s^2(Q) + \dots, \tag{28.183}$$

with calculable, dimensionless coefficients c_n fixed by the matrix elements and the observable's definition — $c_1 = C_F(\frac{\pi^2}{3} - \frac{5}{2})/\pi$ for $\langle 1-T\rangle$ in (28.180), the analogous bracket for the Sterman–Weinberg three-jet fraction, and so on. Measuring $\mathcal{O}$ at energy Q and inverting the series determines $\alpha_s(Q)$ at that scale. Performing this at many scales — $\langle 1 - T\rangle$ and jet rates across the e^+e^- energies from a few GeV to LEP-2's 209 GeV, τ-decay hadronic widths at $Q = m_\tau \approx 1.78$ GeV, quarkonium and lattice determinations at intermediate scales, and jet and top-pair cross sections at the LHC out to the TeV — yields a set of independent determinations of α_s at scales spanning three orders of magnitude.

These are not independent constants. The renormalization-group running of Chapter 27 predicts how a *single* coupling, conventionally quoted as $\alpha_s(m_Z)$, must change with scale: solving the two-loop renormalization-group equation

$$\frac{\mathrm{d}\alpha_s}{\mathrm{d}\ln Q^2} \;=\; -\alpha_s^2(b_0 + b_1\,\alpha_s + \dots), \qquad b_0 = \frac{33 - 2n_f}{12\pi}, \quad b_1 = \frac{153 - 19n_f}{24\pi^2}, \tag{28.184}$$

with the beta-function coefficients carried by Yang–Mills theory (Chapter 27; for $SU(3)$ with n_f active flavors the leading coefficient is $b_0 = (33 - 2n_f)/12\pi$, equivalently $b_0 \propto 11 - \frac{2}{3}n_f$). Anchoring the solution at the world-average value $\alpha_s(m_Z) = 0.1180 \pm 0.0009$ (the precision attainable in the mid-2020s) and evolving across the measured window produces the single decreasing curve of Fig. 28.8, onto which the independent determinations fall. That agreement — one parameter, dozens of measurements at incompatible scales, all consistent through the predicted running — is the quantitative proof of asymptotic freedom:

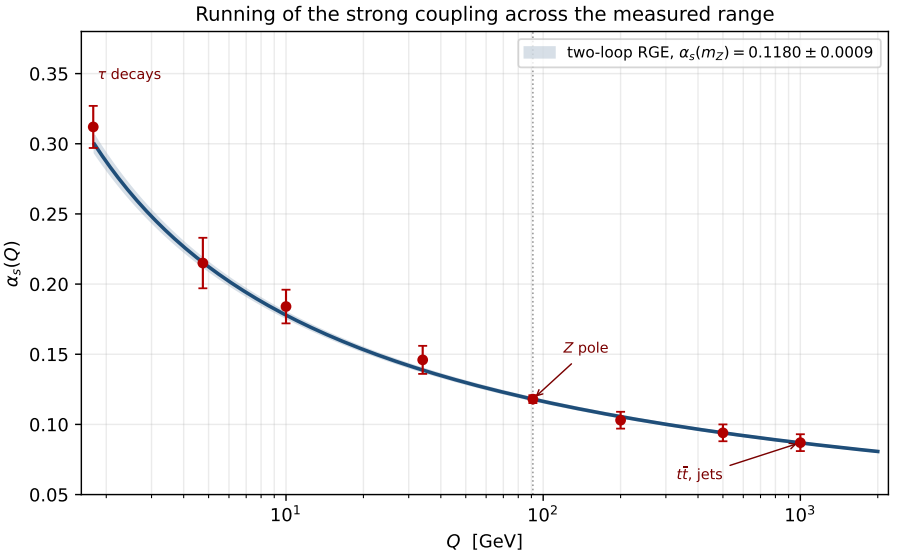


Figure 28.8: The running strong coupling $\alpha_s(Q)$. The band is the two-loop renormalization-group solution (28.184) anchored at $\alpha_s(m_Z) = 0.1180 \pm 0.0009$ with $n_f = 5$ active flavors. Points are representative determinations from IRC-safe observables at their respective scales — τ decays, quarkonia, e^+e^- event shapes and jet rates, the Z-pole hadronic width, and hadron-collider jet and $t\bar{t}$ data — chosen consistent with the 2020s world average. A single coupling, measured at scales spanning three orders of magnitude, traces one decreasing curve: asymptotic freedom read directly off the data.

the strong coupling demonstrably weakens as the resolution sharpens, exactly as the negative beta function of Chapter 27 requires.

Worked Example 28.21 (From a measured three-jet rate to $\alpha_s(m_Z)$)**.** To make the inversion concrete, treat the Sterman–Weinberg three-jet fraction as the measured observable. From (28.176), the fraction of events that are *not* two-jet is

$$f_{3\text{-jet}}(\beta,\delta) \;=\; 1 - \frac{\sigma_{2\text{-jet}}}{\sigma_0} \;=\; \frac{\alpha_s(Q)}{\pi}\, C_F \Big[4\ln(2\beta)\ln\delta + 3\ln\delta + \tfrac{\pi^2}{3} - \tfrac{5}{2} \Big] + O(\alpha_s^2). \tag{28.185}$$

Suppose an experiment at $Q = m_Z$ with cuts $\beta = 0.1$, $\delta = 15^\circ = 0.262$ rad measures $f_{3\text{-jet}} = 0.27$. The bracket evaluates, with $\ln(2\beta) = \ln 0.2 = -1.609$ and $\ln\delta = -1.339$, to

$$4(-1.609)(-1.339) + 3(-1.339) + \tfrac{\pi^2}{3} - \tfrac{5}{2} = 8.62 - 4.02 + 3.29 - 2.50 = 5.39, \tag{28.186}$$

so with $C_F = 4/3$,

$$\alpha_s(m_Z) = \frac{\pi\, f_{3\text{-jet}}}{C_F \times 5.39} = \frac{\pi \times 0.27}{(4/3) \times 5.39} = \frac{0.848}{7.19} \approx 0.118. \tag{28.187}$$

The leading-order extraction lands at the world average. In a real analysis one uses the full NLO (and where available NNLO) coefficient c_2 of (28.183), resums the $\ln^2 \beta\,\delta$ Sudakov logarithms when the cuts are tight, and corrects for hadronization — the non-perturbative transition from partons to the observed hadrons — using the factorized, universal long-distance functions this chapter has parametrized. The skeleton, however, is exactly (28.183): an IRC-safe number, a calculable series, one inversion for $\alpha_s(Q)$, and the running of Chapter 27 to translate to the reference scale m_Z.

With this, Yang–Mills theory reaches its phenomenological endpoint. From the classical non-abelian Lagrangian of Chapter 25 we have travelled, through quantization and asymptotic freedom, to a falsifiable number on a detector: the colour factor $N_c = 3$ and the coupling $\alpha_s(m_Z) \approx 0.118$, both extracted from infrared-safe observables that the theory of confined quarks nonetheless predicts. What this chapter has left to the long-distance side of factorization — the hadronization that turns partons into the colour-singlet states actually measured, and confinement itself — is the province of non-perturbative QCD: the lattice formulation of Chapter 43 and the chiral dynamics of Chapter 39, with the topological vacuum structure of Chapters 41 and 42 underneath. The perturbative, asymptotically free face of QCD now joins the electroweak theory in the Standard Model synthesis of Chapter 33.

Further reading

The definitive text on the perturbative QCD of this chapter is Ellis, Stirling, and Webber, *QCD and Collider Physics* [175], which carries the colour algebra, DIS, DGLAP, factorization, and jets at the level developed here; Peskin and Schroeder [396] (Ch. 17) and Schwartz [432] give complementary textbook treatments, and the asymptotic freedom that underwrites the whole subject is the discovery of Gross and Wilczek [238] and Politzer [406]. The parton model and Bjorken scaling are due to Bjorken and Paschos [79], the spin diagnostic to Callan and Gross [105]; the scaling-violation evolution equations were given in parton language by Altarelli and Parisi [21] (the "DGLAP" name also crediting Dokshitzer, Gribov, and Lipatov). The factorization theorem and its all-orders subtleties are the subject of Collins, Soper, and Sterman [136] and the monograph of Collins [135]; the large higher-order corrections to Drell–Yan and DIS coefficients are computed in Altarelli, Ellis, and Martinelli [22]. Infrared-safe jet physics begins with Sterman and Weinberg [456], and the modern collider standard is the anti-k_t algorithm of Cacciari, Salam, and Soyez [102]; the lepton-pair process whose

universality test closes the factorization argument is the original proposal of Drell and Yan [163].

Part VIII

Spontaneous Symmetry Breaking

Chapter 29

Spontaneous Symmetry Breaking and Goldstone Theorem

Until now a symmetry of the action has meant a symmetry of everything: the Lagrangian, the spectrum, and the vacuum on which that spectrum is built. The free theories of Chapter 9 furnished a unique, manifestly invariant ground state, and Wigner's classification (Theorem 3.44) organized the particles into linear representations of whatever group left the dynamics alone. This chapter studies the situation in which that comfortable alignment fails—where the equations retain their full symmetry but the ground state quietly declines to share it. The phenomenon is *spontaneous* symmetry breaking, and the adjective is exact: nothing in the Lagrangian is asymmetric, and it is the act of selecting one vacuum from a degenerate family that does the breaking. We continue ϕ^4 theory from the discrete reflection of Section 7.3.6 into the regime $m^2 < 0$, where the symmetric configuration becomes a maximum and the true minima split apart, and we extract from this double well the precise, state-level meaning of the phrase. Throughout we work in natural units $\hbar = c = 1$ and in the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ of Chapter 10 and Chapter A, so that $p^2 = m^2$ on shell and the Feynman propagator is $i/(p^2 - m^2 + i\epsilon)$.

The conceptual core of the chapter is a single physical fact with a sharp mathematical expression: when a *continuous* symmetry breaks, the degeneracy of the vacuum is not discrete but forms a manifold, and motion along that manifold costs no energy. Those flat directions are massless particles. This is Goldstone's theorem, and we prove it three independent ways—as the vanishing curvature of a classical potential along its flat valley, as a rigorous spectral statement extracted from the Källén–Lehmann representation (Theorem 14.23) applied to the broken Noether current of Chapter 7, and as the flatness of the quantum effective potential. Each proof imports a different tool and exposes a different face of the result; together they make the theorem unavoidable. The vehicle throughout is the linear sigma model, the multicomponent generalization of ϕ^4

theory, whose Mexican-hat potential makes the broken vacuum manifold and its Goldstone modes completely explicit.

To make the theorem quantum rather than classical we build the effective potential as the constant-field limit of the effective action of Chapter 10, establish that its stationary points are the exact vacua and its Hessian the exact zero-momentum mass matrix, and evaluate it at one loop. This yields the Coleman–Weinberg potential and, with it, radiative symmetry breaking and dimensional transmutation—symmetry breaking induced entirely by quantum corrections, with the breaking scale traded for a dimensionless coupling. The same object furnishes the cleanest proof of Goldstone's theorem and, later, the mass formula for pseudo-Goldstone bosons.

Having isolated the massless modes, we ask what theory they obey on their own. Sending the radial mass to infinity collapses the linear sigma model onto its vacuum manifold and produces the nonlinear sigma model: the universal low-energy theory of Goldstone bosons, with the Goldstones realized nonlinearly as coordinates on the coset G/H, their interactions forced to be purely derivative, and a single low-energy constant—the decay constant f—setting the scale $4\pi f$ at which the description breaks down. We derive the Adler zero, organize the construction geometrically through the Maurer–Cartan form and the CCWZ split, and carry the leading $\pi\pi$ amplitude as a worked example.

The chapter closes by softening the idealization. A small term in the Lagrangian that breaks the symmetry *explicitly* turns exact Goldstones into light, calculable *pseudo*-Goldstones whose squared masses are linear in the breaking; we derive the mass formula twice, once from the effective potential and once from the partially conserved current, recovering the Gell-Mann–Oakes–Renner relation and the pion as its physical paradigm. Vacuum alignment selects the ground state that keeps those masses nonnegative, and Coleman's theorem honestly fences off the dimensions $d \leq 2$ in which the whole continuous-breaking story fails. The machinery assembled here is exactly what the Higgs mechanism (Chapter 30), electroweak theory (Chapter 31), and chiral perturbation theory (Chapter 39) will take over wholesale.

29.1 Vacuum Degeneracy and the Meaning of Spontaneous Breaking

The classical machinery assembled in Parts II and III left us with two firmly established facts about ϕ^4 theory. The standard Lagrangian Eq. (6.56) is invariant under the reflection $\phi \to -\phi$ of Theorem 7.22, an exact symmetry of the dynamics; and the free quantum theory of Chapter 9 possesses a unique vacuum $|0\rangle$, the state annihilated by every lowering operator Eq. (9.69). So far these two facts have lived together without tension: the vacuum we built is manifestly reflection-symmetric, because every term in the free Hamiltonian is even in ϕ, and the

particle spectrum erected on it carries the symmetry linearly, exactly as Noether's machinery (Chapter 7) and Wigner's classification (Theorem 3.44) lead one to expect. This section asks what happens when the two facts come apart—when the dynamics keeps its symmetry but the ground state quietly refuses to share it. We continue ϕ^4 theory from Section 7.3.6 into the regime $m^2 < 0$, where exactly this happens, and we extract from the discrete example the precise, state-level meaning of the phrase *spontaneous symmetry breaking*. The word "spontaneous" is doing real work: nothing in the equations breaks the symmetry; the asymmetry is a property of a *chosen* ground state among several degenerate ones, and it is the act of choosing that breaks it. This is to be contrasted throughout with *explicit* breaking, in which a non-invariant term is added to the Lagrangian itself; that case is genuinely less symmetric as a theory, and is not our subject here.

29.1.1 Symmetry of the dynamics versus symmetry of the vacuum

We must first be careful about what it means for "a symmetry" to be "present." Two logically distinct statements are involved, and spontaneous breaking is precisely the case in which the first holds and the second fails.

The first statement is dynamical. A transformation is a symmetry of the dynamics if it commutes with time evolution—equivalently, if it leaves the action, hence the Hamiltonian, invariant. By Wigner's theorem (Theorem 3.44) every such symmetry is implemented on the Hilbert space by a unitary (or, for time reversal, antiunitary) operator that commutes with the Hamiltonian. For a continuous internal symmetry this operator is built from the conserved Noether charge Q of Chapter 7: writing the one-parameter family as $U(\alpha) = e^{i\alpha Q}$, invariance of the dynamics is the operator statement

$$[Q, H] = 0, \qquad \text{equivalently} \qquad U(\alpha)\, H\, U(\alpha)^{-1} = H. \tag{29.1}$$

For a discrete symmetry such as the $\mathbb{Z}_2$ of Theorem 7.22 there is no continuous parameter and hence no current (Theorem 7.23), but there is still a unitary operator. Call it Z: it is the operator that implements the field reflection,

$$\mathsf{Z}\,\phi(x)\,\mathsf{Z}^{-1} = -\phi(x), \qquad \mathsf{Z}^2 = \mathbf{1}, \qquad [\mathsf{Z}, H] = 0. \tag{29.2}$$

The relations Eqs. (29.1) and (29.2) are statements about the dynamics alone. They say nothing whatsoever about the vacuum.

The second statement is about the ground state. A symmetry is *realized linearly* (or *Wigner–Weyl realized*) on the vacuum when the symmetry operator fixes it,

$$Q\,|0\rangle = 0, \qquad \text{respectively} \qquad \mathsf{Z}\,|0\rangle = |0\rangle \tag{29.3}$$

(up to an irrelevant phase in the discrete case). When this holds, the symmetry acts on the entire tower of states built on $|0\rangle$ as an ordinary unitary symmetry:

particles fall into degenerate multiplets of the symmetry group, exactly as the isospin and colour multiplets organized the spectrum in Chapter 7. This is the only realization we have met so far, and it is so familiar that one is tempted to read Eq. (29.3) as automatic. It is not. The vacuum is selected by the *dynamics*, as the lowest-energy state; whether that state happens to be invariant is a separate dynamical question, and the answer can be no.

Definition 29.1 (Spontaneous symmetry breaking)**.** A symmetry of the dynamics is *spontaneously broken* (or *Nambu–Goldstone realized*) when the symmetry operator does *not* annihilate, respectively fix, the vacuum: there is a charge Q with $[Q, H] = 0$ yet

$$Q\,|0\rangle \neq 0, \tag{29.4}$$

or, for a discrete symmetry, a unitary Z with $[\mathsf{Z}, H] = 0$ yet $\mathsf{Z}\,|0\rangle \neq |0\rangle$. Equivalently—and this is the operationally useful form—there exists a local field operator Φ whose vacuum expectation value, the *order parameter*, is nonzero,

$$\langle 0|\Phi|0\rangle \neq 0, \tag{29.5}$$

while Φ transforms nontrivially under the symmetry. The equivalence of the two formulations is proved in Theorem 29.2.

The phrase "up to a non-normalizable state" in Eq. (29.4) hides a subtlety that we confront squarely in Section 29.1.5: in infinite volume the operator $Q = \int \mathrm{d}^3x\, J^0$ does not exist as a bona fide operator on the Hilbert space built on a broken vacuum, and $Q\,|0\rangle$ is best read as the divergent norm of a state that cannot be normalized. The order-parameter criterion Eq. (29.5) has no such defect—it refers only to a well-defined matrix element of a local operator—which is why it, and not Eq. (29.4), is the working definition of spontaneous breaking. We keep both because the charge form is what makes contact with Goldstone's theorem in Section 29.1.3 and beyond, while the order-parameter form is what one actually computes.

Notation Watch

Throughout this chapter "symmetric" and "broken" always describe the *realization on the vacuum*, never the Lagrangian. The Lagrangian Eq. (6.56) of ϕ^4 theory is exactly $\mathbb{Z}_2$-invariant in both phases $m^2 > 0$ and $m^2 < 0$; only the vacuum changes. To say "the $\mathbb{Z}_2$ is broken" is shorthand for "the chosen vacuum is not $\mathbb{Z}_2$-invariant," never for "a non-invariant term was added." The latter is *explicit* breaking, a different phenomenon studied via the source term of Chapter 10 and revisited as the pion mass in Chapter 39.

29.1.2 Scalar ϕ^4 theory in the broken phase: the double well

The mechanism is made completely concrete by the simplest move available in ϕ^4 theory: flip the sign of the mass term. The standard Lagrangian Eq. (6.56)

was written with $m^2 > 0$ and the stable potential $V(\phi) = \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$, whose unique minimum sits at $\phi = 0$. Nothing in the construction forbids the opposite sign of the quadratic coefficient; the quartic with $\lambda > 0$ still bounds the potential below, so the theory remains well defined. Writing the negative coefficient as $-\frac{1}{2}\left|m^2\right|$ to keep its sign manifest, the potential of the broken phase is

$$V(\phi) = -\tfrac{1}{2}\left|m^2\right|\,\phi^2 + \frac{\lambda}{4!}\,\phi^4, \qquad \lambda > 0. \tag{29.6}$$

This is the celebrated *double well* (Fig. 29.1). The reflection symmetry Theorem 7.22 survives untouched—both surviving terms are even—so the dynamics is exactly as $\mathbb{Z}_2$-symmetric as before. What has changed is the location of the minima. Setting $V'(\phi) = 0$ with the field derivative computed exactly as in Eq. (6.58) but with $m^2 \to -\left|m^2\right|$,

$$V'(\phi) = -\left|m^2\right|\,\phi + \frac{\lambda}{3!}\,\phi^3 = \phi\Big(-\left|m^2\right| + \frac{\lambda}{6}\,\phi^2\Big) = 0, \tag{29.7}$$

which has three roots: the origin $\phi = 0$ and the symmetric pair

$$\boxed{\phi = \pm v, \qquad v^2 = \frac{6\left|m^2\right|}{\lambda}\,.} \tag{29.8}$$

The origin is now a local *maximum*, not a minimum: the second derivative there,

$$V''(\phi) = -\left|m^2\right| + \frac{\lambda}{2}\,\phi^2, \qquad V''(0) = -\left|m^2\right| < 0, \tag{29.9}$$

is negative, so the symmetric configuration $\phi = 0$ is unstable—a classical ball placed there rolls off. The genuine minima are the two degenerate wells at $\pm v$, where

$$V''(\pm v) = -\left|m^2\right| + \frac{\lambda}{2}\cdot\frac{6\left|m^2\right|}{\lambda} = -\left|m^2\right| + 3\left|m^2\right| = 2\left|m^2\right| > 0. \tag{29.10}$$

The positive curvature $2\left|m^2\right|$ at each well is the (squared) mass of the small fluctuations about that vacuum: a field expanded as $\phi = \pm v + \eta$ has, to quadratic order, the mass term $\frac{1}{2}V''(\pm v)\,\eta^2 = \frac{1}{2}(2\left|m^2\right|)\eta^2$, so the physical excitation is a scalar of mass $m^2_{\text{phys}} = 2\left|m^2\right|$. This is the first lesson of the broken phase: the mass read off the spectrum is governed by the curvature *at the true vacuum*, $2\left|m^2\right|$, and bears no resemblance to the "wrong-sign" coefficient $-\left|m^2\right|$ multiplying ϕ^2 in the Lagrangian. The Lagrangian parameter $-\left|m^2\right|$ is not a physical mass at all; it is a parameter of an unstable expansion point.

The $\mathbb{Z}_2$ now does something it could not do in the symmetric phase: it *exchanges* the two vacua. Because Z implements $\phi \to -\phi$ via Eq. (29.2), and the two minima are related by exactly this reflection, the reflection cannot fix a vacuum built on one well—it maps it to the vacuum built on the other. Denote

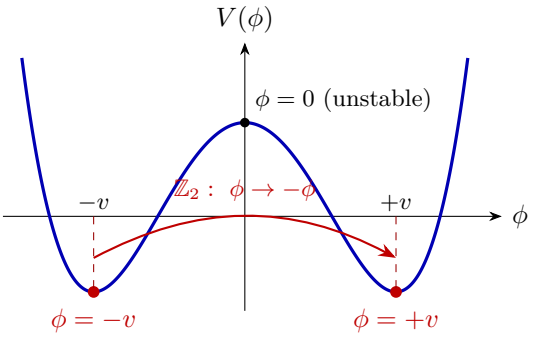


Figure 29.1: The double-well potential Eq. (29.6) of ϕ^4 theory in its broken phase. The reflection $\mathbb{Z}_2 : \phi \to -\phi$ of Theorem 7.22 is an exact symmetry of the potential—the curve is even—but it exchanges the two degenerate minima $\pm v$ of Eq. (29.8) rather than fixing either. A vacuum built on one well has a nonzero order parameter $\langle\phi\rangle = \pm v$ and is not $\mathbb{Z}_2$-invariant: the symmetry of the dynamics is spontaneously broken by the choice of vacuum. The symmetric point $\phi = 0$ is a local maximum (curvature $-|m^2| < 0$) and supports no stable vacuum.

by $|+\rangle$ and $|-\rangle$ the two candidate ground states, characterized at the quantum level by where the field sits on average,

$$\langle +|\phi(x)|+\rangle = +v, \qquad \langle -|\phi(x)|-\rangle = -v, \tag{29.11}$$

the right-hand sides being x-independent by translation invariance of each vacuum. The reflection interchanges them,

$$\mathsf{Z}\,|+\rangle = |-\rangle\,, \qquad \mathsf{Z}\,|-\rangle = |+\rangle\,, \tag{29.12}$$

which is consistent with Eq. (29.11): acting with Eq. (29.2) inside the expectation value, $\langle +|\phi|+\rangle = \langle +|\mathsf{Z}^{-1}\mathsf{Z}\,\phi\,\mathsf{Z}^{-1}\mathsf{Z}|+\rangle = \langle -|(-\phi)|-\rangle = -(-v) = +v$, as it must. Neither $|+\rangle$ nor $|-\rangle$ is fixed by Z. Each is a perfectly good ground state—degenerate with the other by the exact symmetry—and each one breaks the $\mathbb{Z}_2$ spontaneously, in the precise sense of Theorem 29.1: the dynamics is symmetric, but the chosen vacuum is not. The order parameter $\langle\phi\rangle = \pm v$ is the diagnostic. It is nonzero, it changes sign under the symmetry, and it labels which vacuum we are in.

Physical Insight

The order parameter is a *choice of phase made by the system, not by us.* Classically a ball balanced at the unstable maximum $\phi = 0$ must roll into one well or the other; an infinitesimal perturbation—a stray external field, a thermal fluctuation, the breath of the boundary conditions—decides

which. The laws governing the ball are perfectly reflection-symmetric; the resting state is not. This is the whole content of "spontaneous": the asymmetry of the outcome is not present in, and cannot be predicted from, the symmetric laws alone. The same logic upgraded to a field with infinitely many degrees of freedom is what makes the choice *permanent*, as the next subsections show: in infinite volume the system cannot change its mind, and the two outcomes become genuinely separate worlds.

29.1.3 The order-parameter criterion

We now prove the equivalence asserted in Theorem 29.1: a nonzero order parameter is logically incompatible with an invariant vacuum, and conversely. This is more than bookkeeping. The argument is short, but its structure—"Q annihilates the vacuum" forced into contradiction with "$\langle\Phi\rangle \neq 0$"—is the algebraic seed from which Goldstone's theorem grows in Section 29.1's successor: there, the same incompatibility, sharpened by locality and Lorentz invariance, will force a massless state into the spectrum. We give it first for the discrete $\mathbb{Z}_2$ of ϕ^4 theory, where it is cleanest, and then for the continuous U(1) that anticipates the next section.

Theorem 29.2 (Order-parameter criterion for spontaneous breaking)**.** *Let Φ be a local field that transforms nontrivially under a symmetry of the dynamics.*

(i) Discrete case. *For the $\mathbb{Z}_2$ with operator Z obeying Eq. (29.2): if the vacuum is invariant, $\mathsf{Z}\,|0\rangle = |0\rangle$, then $\langle 0|\phi|0\rangle = 0$. Contrapositively, $\langle 0|\phi|0\rangle \neq 0$ implies $\mathsf{Z}\,|0\rangle \neq |0\rangle$: the symmetry is spontaneously broken.*

(ii) Continuous case. *For a continuous internal symmetry with conserved charge Q acting on a multiplet ϕ_i by $[Q, \phi_i] = i(T)_{ij}\phi_j$: if the vacuum is invariant, $Q\,|0\rangle = 0$, then $(T)_{ij}\,\langle 0|\phi_j|0\rangle = 0$. Contrapositively, if some generator T does not annihilate the vector of expectation values $\langle 0|\phi_j|0\rangle$, then $Q\,|0\rangle \neq 0$: that symmetry is spontaneously broken.*

Proof. (i) The reflection operator is unitary and, by hypothesis, fixes the vacuum: $\mathsf{Z}\,|0\rangle = |0\rangle$ and therefore also $\langle 0|\,\mathsf{Z}^{-1} = \langle 0|$. Insert $\mathsf{Z}^{-1}\mathsf{Z} = \mathbf{1}$ around the field in the expectation value and use Eq. (29.2):

$$\langle 0|\phi|0\rangle = \Big\langle 0|\mathsf{Z}^{-1}\,\mathsf{Z}\,\phi\,\mathsf{Z}^{-1}\,\mathsf{Z}|0\Big\rangle = \Big\langle 0|\mathsf{Z}\,\phi\,\mathsf{Z}^{-1}|0\Big\rangle = \langle 0|(-\phi)|0\rangle = -\,\langle 0|\phi|0\rangle\,. \tag{29.13}$$

A number equal to its own negative is zero, so $\langle 0|\phi|0\rangle = 0$. The first sentence is proved, and the contrapositive is its logical equivalent: a nonzero VEV forbids the invariant vacuum, leaving only a vacuum on which Z acts nontrivially—spontaneous breaking.

(ii) The quantum transformation law $[Q, \phi_i] = i(T)_{ij}\phi_j$ is the operator translate of the classical statement that the charge generates the field rotation (Theorem 7.20, with the canonical bracket $\{\cdot,\cdot\}_{\mathrm{P}} \to -i\,[\cdot,\cdot]$). Take its vacuum expectation value and suppose $Q\,|0\rangle = 0$, which also gives $\langle 0|\,Q = 0$:

$$i(T)_{ij}\,\langle 0|\phi_j|0\rangle = \langle 0|\,[Q,\phi_i]\,|0\rangle = \langle 0|Q\,\phi_i|0\rangle - \langle 0|\phi_i\,Q|0\rangle = 0 - 0 = 0. \tag{29.14}$$

Hence $(T)_{ij}\,\langle 0|\phi_j|0\rangle = 0$: every generator that annihilates the vacuum must also annihilate the vector of order parameters. Contrapositively, if there is a generator T with $(T)_{ij}\,\langle 0|\phi_j|0\rangle \neq 0$, then the step $\langle 0|Q\,\phi_i|0\rangle = 0$ cannot have held, so $Q\,|0\rangle \neq 0$ for that charge—spontaneous breaking. □

The continuous statement deserves to be made concrete on the example that will dominate the rest of the chapter. Consider a complex scalar $\Phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2)$, equivalently a real doublet (ϕ_1, ϕ_2), with the U(1) (equivalently SO(2)) symmetry $\Phi \to e^{-i\alpha}\Phi$. Its single Hermitian generator acts as a rotation in the (ϕ_1, ϕ_2) plane,

$$[Q, \phi_1] = i\,\phi_2, \qquad [Q, \phi_2] = -i\,\phi_1, \tag{29.15}$$

so (T) is the antisymmetric 2×2 matrix $\left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right)$ acting on (ϕ_1, ϕ_2). Suppose the dynamics drives a nonzero VEV which, by a U(1) rotation, we may align along the first axis,

$$\langle 0|\phi_1|0\rangle = v \neq 0, \qquad \langle 0|\phi_2|0\rangle = 0. \tag{29.16}$$

Apply the criterion through the generator. The combination that detects breaking is the expectation value of $[Q, \phi_2] = -i\phi_1$:

$$\langle 0|\,[Q,\phi_2]\,|0\rangle = -i\,\langle 0|\phi_1|0\rangle = -i\,v \neq 0. \tag{29.17}$$

Were $Q\,|0\rangle = 0$, the left-hand side would vanish by exactly the manipulation Eq. (29.14); since it does not, $Q\,|0\rangle \neq 0$. The U(1) is spontaneously broken. Notice what the criterion also tells us when it is *not* triggered: a generator under which the VEV is invariant ($(T)_{ij}\,\langle\phi_j\rangle = 0$) is consistent with an invariant vacuum and need not be broken. For the aligned VEV Eq. (29.16) there is only one generator and it rotates ϕ_1 into ϕ_2, so it is broken; in the larger O(N) theory of the next section, the rotations that leave the chosen VEV fixed form an unbroken O($N-1$) subgroup, and the count of *broken* generators—those failing the criterion—is exactly the count of massless modes Goldstone's theorem will deliver.

Physical Insight

Theorem 29.2 is the whole logic of spontaneous breaking compressed into one line of algebra: *an invariant vacuum cannot support a non-invariant order parameter*. If $\langle\Phi\rangle$ transforms and is nonzero, the vacuum must

transform too. Everything else in this chapter—the degenerate manifold of vacua, the massless Goldstone bosons, the effective potential whose minimum is displaced from the origin—is the working-out of the consequences of refusing $\langle\Phi\rangle = 0$. The deep version of this same incompatibility, with the global charge replaced by the *local* current and the constant generator replaced by a momentum-dependent matrix element, is Goldstone's theorem.

29.1.4 The symmetric superposition, and why it is not the answer

A natural objection arises at once. The two vacua $|+\rangle$ and $|-\rangle$ are degenerate and exchanged by Z; quantum mechanics teaches us to diagonalize symmetries, so surely the "true" ground states are the symmetric and antisymmetric combinations

$$|S\rangle = \tfrac{1}{\sqrt{2}}\big(\,|+\rangle + |-\rangle\,\big), \qquad |A\rangle = \tfrac{1}{\sqrt{2}}\big(\,|+\rangle - |-\rangle\,\big), \tag{29.18}$$

which are eigenstates of Z with eigenvalues $+1$ and -1 (using Eq. (29.12)). On $|S\rangle$ the symmetry is realized in the Wigner–Weyl way, $\mathsf{Z}\,|S\rangle = |S\rangle$, and the order parameter vanishes: by Eq. (29.11) and the assumed orthogonality $\langle+|-\rangle = 0$,

$$\langle S|\phi|S\rangle = \tfrac{1}{2}\big(\,\langle+|\phi|+\rangle + \langle-|\phi|-\rangle + \langle+|\phi|-\rangle + \langle-|\phi|+\rangle\,\big). \tag{29.19}$$

If the cross terms $\langle+|\phi|-\rangle$ are negligible (we will see they are, in infinite volume), the diagonal terms give $\tfrac{1}{2}(v+(-v)) = 0$. So $|S\rangle$ has no order parameter, respects the symmetry, and looks for all the world like the honest ground state of a $\mathbb{Z}_2$-symmetric theory. In ordinary quantum mechanics—a single particle in a double well, finite everything—this is exactly right: the true ground state *is* the symmetric combination $|S\rangle$, it lies slightly below the antisymmetric $|A\rangle$ by a tunneling splitting, and there is no spontaneous breaking at all. A particle in a symmetric double well has a symmetric, non-degenerate ground state.

The resolution of the apparent paradox—why quantum field theory in infinite volume behaves utterly differently from quantum mechanics in a box—is the central conceptual content of this section, and it rests on two facts that hold for a field but not for a particle. First, the tunneling splitting between $|S\rangle$ and $|A\rangle$ does not stay finite as the spatial volume grows; it is suppressed exponentially in the volume and vanishes in the infinite-volume limit, so $|S\rangle$ and $|A\rangle$ become *exactly* degenerate and the privileged status of the symmetric combination evaporates. Second—and this is what then selects $|\pm\rangle$ over $|S\rangle$ as the physical vacua—the symmetric superposition $|S\rangle$ *violates cluster decomposition*, the principle that distant measurements are uncorrelated, while each pure state $|\pm\rangle$ satisfies it. We establish the two facts in turn.

29.1.5 Finite versus infinite volume: the vanishing of the tunneling splitting

To compare a particle in a box with a field in space, we must put the field in a box too. Place ϕ^4 theory in a spatial volume $\mathcal{V}$ (a cube of side L, $\mathcal{V} = L^3$, with periodic boundary conditions). The lowest-energy dynamics is dominated by the spatially homogeneous mode of the field, the zero-momentum component

$$\bar{\phi}(t) = \frac{1}{\mathcal{V}} \int_{\mathcal{V}} \mathrm{d}^3x \; \phi(t, \mathbf{x}), \tag{29.20}$$

because any spatial variation costs gradient energy $\frac{1}{2}(\nabla\phi)^2$ and is therefore frozen out at the bottom of the spectrum. Substituting a homogeneous configuration $\phi(t, \mathbf{x}) = \bar{\phi}(t)$ into the Lagrangian Eq. (6.56) (with $m^2 \to -|m^2|$) and integrating over space, the gradient term vanishes and the volume factors out:

$$L_{\text{eff}}[\bar{\phi}] = \int_{\mathcal{V}} \mathrm{d}^3x \left[\tfrac{1}{2}\dot{\bar{\phi}}^2 - V(\bar{\phi})\right] = \mathcal{V}\left[\tfrac{1}{2}\dot{\bar{\phi}}^2 - V(\bar{\phi})\right]. \tag{29.21}$$

This is the Lagrangian of a *single quantum-mechanical particle* of coordinate $\bar{\phi}$, with mass $M = \mathcal{V}$ and potential energy $U(\bar{\phi}) = \mathcal{V}\, V(\bar{\phi})$. The double-well structure of V becomes a double-well for this collective coordinate, with the two minima at $\bar{\phi} = \pm v$ separated by a barrier of height $\mathcal{V}\,\Delta V$, where $\Delta V = V(0) - V(\pm v)$ is the energy-density difference between the unstable hilltop and the wells. Reading off from Eq. (29.6),

$$\Delta V = V(0) - V(v) = 0 - \left(-\tfrac{1}{2}\left|m^2\right| v^2 + \frac{\lambda}{4!}v^4\right) = \frac{1}{2}\left|m^2\right| v^2 - \frac{\lambda}{24}v^4, \tag{29.22}$$

which, using $v^2 = 6\left|m^2\right|/\lambda$ from Eq. (29.8), evaluates to $\Delta V = \frac{1}{2}\left|m^2\right| v^2 - \frac{\lambda}{24}v^2 \cdot v^2 = \frac{1}{2}\left|m^2\right| v^2 - \frac{1}{4}\left|m^2\right| v^2 = \frac{1}{4}\left|m^2\right| v^2 = \frac{3\left|m^2\right|^2}{2\lambda}$, a finite energy *density*. The barrier the collective coordinate must tunnel through is this density times the volume, $\mathcal{V}\,\Delta V$. As $\mathcal{V} \to \infty$ the barrier becomes infinitely high and infinitely massive, and tunneling shuts off. The following worked example makes the estimate quantitative.

Worked Example 29.3 (The tunneling splitting vanishes in the infinite-volume limit)**.** We estimate the energy splitting $\Delta E = E_A - E_S$ between the antisymmetric and symmetric combinations Eq. (29.18) as a function of the volume $\mathcal{V}$, treating the collective coordinate $\bar{\phi}$ of Eq. (29.21) as a particle of mass $M = \mathcal{V}$ in the potential $U(\bar{\phi}) = \mathcal{V}\, V(\bar{\phi})$ and applying the standard WKB double-well formula.

The WKB splitting formula. For a particle of mass M in a symmetric double well with minima separated by a classically forbidden region, the two lowest levels—built from the even and odd combinations of the single-well ground states—are

split by

$$\Delta E \;\sim\; \frac{\hbar\omega}{\pi}\, e^{-S_0/\hbar}, \qquad S_0 = \int_{-q_0}^{+q_0} \mathrm{d}q\, \sqrt{2M\big(U(q)-E_0\big)}, \tag{29.23}$$

where ω is the small-oscillation frequency in either well, $E_0 \approx \tfrac{1}{2}\hbar\omega$ is the well's ground-state energy, and S_0 is the action of the imaginary-time (Euclidean) trajectory—the *instanton*—that rolls from one minimum to the other through the inverted potential. The factor $e^{-S_0/\hbar}$ is the tunneling amplitude; its exponent is what we track. We restore $\hbar$ only to display the semiclassical structure and set it to unity at the end.

The instanton action scales with the volume. Take the energy E_0 negligible against the barrier (it is, by a power of $\mathcal{V}$, as the comparison below shows) and write the turning points at the well bottoms $\pm v$. With $M = \mathcal{V}$ and $U = \mathcal{V}\,V$, the action Eq. (29.23) is

$$S_0 = \int_{-v}^{+v} \mathrm{d}\bar{\phi}\, \sqrt{2\,\mathcal{V}\cdot\mathcal{V}\,V(\bar{\phi})} = \mathcal{V}\int_{-v}^{+v} \mathrm{d}\bar{\phi}\, \sqrt{2\,V(\bar{\phi})}. \tag{29.24}$$

The two factors of $\mathcal{V}$—one from the mass, one from the potential height—combine under the square root to a single overall $\mathcal{V}$. The remaining integral is a pure number times powers of the couplings. To evaluate it, write the potential in its manifestly factorized form. Completing the square in Eq. (29.6),

$$V(\bar{\phi}) = \frac{\lambda}{4!}\big(\bar{\phi}^2 - v^2\big)^2 - \frac{\lambda}{4!}v^4, \tag{29.25}$$

which one checks by expanding: $\frac{\lambda}{4!}(\bar{\phi}^4 - 2v^2\bar{\phi}^2 + v^4) - \frac{\lambda}{4!}v^4 = \frac{\lambda}{4!}\bar{\phi}^4 - \frac{\lambda}{12}v^2\bar{\phi}^2$, and $\frac{\lambda}{12}v^2 = \frac{\lambda}{12}\cdot\frac{6|m^2|}{\lambda} = \frac{1}{2}\,|m^2|$, reproducing Eq. (29.6) up to the constant. Dropping the (vacuum-energy) constant so that $V(\pm v) = 0$, the integrand is $\sqrt{2\cdot\frac{\lambda}{4!}}\,\big|\bar{\phi}^2 - v^2\big| = \sqrt{\frac{\lambda}{12}}\,(v^2 - \bar{\phi}^2)$ on the interval $[-v, v]$, and

$$\int_{-v}^{+v} \mathrm{d}\bar{\phi}\,\sqrt{2V} = \sqrt{\frac{\lambda}{12}}\int_{-v}^{+v}(v^2-\bar{\phi}^2)\,\mathrm{d}\bar{\phi} = \sqrt{\frac{\lambda}{12}}\left[v^2\bar{\phi} - \frac{\bar{\phi}^3}{3}\right]_{-v}^{+v} = \sqrt{\frac{\lambda}{12}}\cdot\frac{4v^3}{3}. \tag{29.26}$$

Assembling,

$$\boxed{S_0 = \frac{4}{3}\sqrt{\frac{\lambda}{12}}\,v^3\,\mathcal{V} \;\equiv\; s_0\,\mathcal{V}\,,} \qquad s_0 = \frac{4}{3}\sqrt{\frac{\lambda}{12}}\,v^3 > 0, \tag{29.27}$$

a positive number s_0 (an action per unit volume, finite and built from the model's own parameters) times the volume.

The conclusion. The splitting is therefore exponentially small in the volume,

$$\Delta E \;\sim\; \frac{\omega}{\pi}\, e^{-s_0 \mathcal{V}} \;\xrightarrow[\mathcal{V}\to\infty]{}\; 0, \tag{29.28}$$

and it vanishes faster than any power of $1/\mathcal{V}$. (The prefactor ω/π grows at most polynomially: $\omega^2 = U''(v)/M = \mathcal{V}\,V''(v)/\mathcal{V} = V''(v) = 2\,|m^2|$ is in fact *independent* of $\mathcal{V}$, so the prefactor is the fixed frequency $\sqrt{2\,|m^2|}/\pi$ and the exponential wins outright.) In a finite box the symmetric combination $|S\rangle$ is the unique ground state, lying below $|A\rangle$ by the tiny gap Eq. (29.28); this is the quantum-mechanical answer, and for any finite $\mathcal{V}$ there is, strictly, no spontaneous breaking. But the gap closes exponentially as the box grows, and in the thermodynamic limit $\mathcal{V}\to\infty$ the states $|S\rangle$ and $|A\rangle$—and hence *every* linear combination of $|+\rangle$ and $|-\rangle$—become exactly degenerate. The infinite system has a two-fold degenerate ground space, and nothing within the dynamics distinguishes the symmetric combination from any other. The privileged status that quantum mechanics conferred on $|S\rangle$ was an artifact of finite volume.

The estimate Eq. (29.28) is the precise sense in which a field differs from a particle. The crucial input is that the tunneling amplitude must take *every one* of the infinitely many local degrees of freedom across the barrier coherently: flipping the field from $+v$ to $-v$ throughout a region of volume $\mathcal{V}$ costs an action proportional to $\mathcal{V}$, because the energy difference is an energy *density*. A single particle has one coordinate and a finite barrier; a field in infinite volume has, in effect, an infinitely massive collective coordinate facing an infinitely high barrier, and it simply cannot tunnel. Once it has settled into the $+v$ well it stays there forever. The choice of vacuum, undetermined by the symmetric dynamics, becomes permanent. This permanence is what promotes the classical "the ball rolls one way or the other" into a sharp quantum statement: the Hilbert space splits.

Historical Note

That the infinite-volume limit is what converts a tunneling doublet into genuinely distinct vacua was understood through the parallel development of statistical mechanics and field theory in the 1960s and 70s. The same mathematics underlies the rigorous statement that a ferromagnet below its Curie temperature has distinct magnetized phases (the spontaneous magnetization being the order parameter), the Lee–Yang analysis of how phase transitions can occur only in the thermodynamic limit, and the operator-algebraic notion of inequivalent representations of the canonical commutation relations associated with different vacua. The slogan "spontaneous breaking requires infinite volume" is, at bottom, the statement Eq. (29.28) that the symmetry-restoring tunneling amplitude dies in that limit.

29.1.6 Cluster decomposition selects the pure vacua

Infinite-volume degeneracy tells us that $|+\rangle$, $|-\rangle$, $|S\rangle$, $|A\rangle$, and all their combinations are equally good ground states as far as energy is concerned. It does *not* by itself tell us which to use as the vacuum on which to build the physical theory. The selection is made by a physical principle we have implicitly relied on since the construction of the S-matrix: *cluster decomposition*. In words, experiments performed in widely separated regions of space must give uncorrelated results; the outcome of a measurement here cannot depend on one carried out arbitrarily far away. For vacuum correlation functions this is the requirement that connected correlators of local operators die at large separation, equivalently that products of local operators factorize:

$$\langle\Omega|\, A(x)\, B(y)\,|\Omega\rangle \;\xrightarrow[|\mathbf{x}-\mathbf{y}|\to\infty]{}\; \langle\Omega|A(x)|\Omega\rangle\,\langle\Omega|B(y)|\Omega\rangle \tag{29.29}$$

for any local operators A, B and any acceptable vacuum $|\Omega\rangle$. A vacuum satisfying Eq. (29.29) is called a *pure*, or *clustering*, vacuum; physically it describes a single world with a definite long-distance behaviour. The principle is not optional: without it the very notion of an isolated scattering experiment—and with it the cluster theorem that organized the connected S-matrix in Theorem 15.35—would collapse. The next proposition shows that, among the degenerate ground states, cluster decomposition is satisfied by the pure vacua $|\pm\rangle$ and *violated* by the symmetric superposition $|S\rangle$.

Proposition 29.4 (Clustering holds on the pure vacua and fails on their superposition)**.** *In the infinite-volume broken phase of ϕ^4 theory, with $|\pm\rangle$ the translation-invariant ground states of Eq. (29.11) and assuming (as established below) that local operators have no matrix elements between distinct vacua, $\langle+|-\rangle = 0$ and $\langle+|\phi(x)|-\rangle \to 0$:*

(i) each pure vacuum $|\pm\rangle$ satisfies cluster decomposition Eq. (29.29);

(ii) the symmetric superposition $|S\rangle = \frac{1}{\sqrt{2}}(|+\rangle + |-\rangle)$ violates it.

Proof. We test Eq. (29.29) with $A = B = \phi$, the field whose VEV is the order parameter. The whole argument turns on a single dynamical input, the *cluster property within a pure vacuum*: in a clustering vacuum the connected two-point function of ϕ decays at large spacelike separation,

$$\langle\pm|\phi(x)\,\phi(y)|\pm\rangle_c \equiv \langle\pm|\phi(x)\phi(y)|\pm\rangle - \langle\pm|\phi(x)|\pm\rangle\,\langle\pm|\phi(y)|\pm\rangle \;\xrightarrow[|\mathbf{x}-\mathbf{y}|\to\infty]{}\; 0. \tag{29.30}$$

This decay is the massive theory's hallmark: the connected correlator is carried by the propagation of the massive excitation of mass $m_{\text{phys}} = \sqrt{2\,|m^2|}$ found in Eq. (29.10), and a massive propagator falls as $e^{-m_{\text{phys}}|\mathbf{x}-\mathbf{y}|}$ at large separation.

(That a Poincaré-invariant, clustering vacuum exists at all on which to base Eq. (29.30) is the nontrivial existence statement flagged in Chapter 9; we take it as given for each $|\pm\rangle$ and verify only the contrast between $|\pm\rangle$ and $|S\rangle$.)

(i) For the pure vacuum $|+\rangle$, using Eq. (29.11) ($\langle+|\phi|+\rangle = v$) and the decay Eq. (29.30),

$$\langle+|\phi(x)\phi(y)|+\rangle = \langle+|\phi(x)|+\rangle\,\langle+|\phi(y)|+\rangle + \langle+|\phi(x)\phi(y)|+\rangle_c \xrightarrow[|\mathbf{x}-\mathbf{y}|\to\infty]{} v\cdot v = v^2, \tag{29.31}$$

which is precisely $\langle+|\phi(x)|+\rangle\,\langle+|\phi(y)|+\rangle = v^2$. Cluster decomposition Eq. (29.29) holds, with limiting value v^2. The same holds on $|-\rangle$ with $v \to -v$, again giving the factorized $(-v)^2 = v^2$.

(ii) Now compute on $|S\rangle$. The one-point function was already found in Eq. (29.19): with $\langle+|-\rangle = 0$ and the vanishing cross matrix element $\langle+|\phi|-\rangle \to 0$,

$$\langle S|\phi(x)|S\rangle = \tfrac{1}{2}\big(\,\langle+|\phi|+\rangle + \langle-|\phi|-\rangle\,\big) = \tfrac{1}{2}\big(v + (-v)\big) = 0. \tag{29.32}$$

So the right-hand side of Eq. (29.29) is $\langle S|\phi(x)|S\rangle\,\langle S|\phi(y)|S\rangle = 0$. The left-hand side, however, does not vanish. Expanding $|S\rangle$ and discarding the cross terms (which vanish because local operators do not connect distinct vacua),

$$\langle S|\phi(x)\phi(y)|S\rangle = \tfrac{1}{2}\big(\,\langle+|\phi(x)\phi(y)|+\rangle + \langle-|\phi(x)\phi(y)|-\rangle\,\big) \xrightarrow[|\mathbf{x}-\mathbf{y}|\to\infty]{} \tfrac{1}{2}(v^2+v^2) = v^2, \tag{29.33}$$

using the pure-vacuum limits Eq. (29.31) from part (i). Thus on the symmetric superposition

$$\underbrace{\langle S|\phi(x)\phi(y)|S\rangle}_{\to\, v^2} \;\neq\; \underbrace{\langle S|\phi(x)|S\rangle\,\langle S|\phi(y)|S\rangle}_{=\,0}, \tag{29.34}$$

and cluster decomposition Eq. (29.29) fails by exactly v^2. The superposition retains long-range correlations: a measurement of ϕ at x stays perfectly correlated with one at y no matter how far apart, because both "ask" the global question of which vacuum the system fell into, and in $|S\rangle$ that question has no definite answer. This is the signature of a *mixed* state masquerading as a ground state. □

The interpretation is decisive. The symmetric combination $|S\rangle$, though it has zero order parameter and respects Z, is physically a classical statistical mixture of two distinct worlds—"the field chose $+v$" and "the field chose $-v$"—superposed coherently. It fails the cluster test because it never made the choice. The pure vacua $|\pm\rangle$ each made a definite choice and each clusters. Since cluster decomposition is a non-negotiable requirement of any sensible quantum field theory, *the physically admissible vacua are the pure, clustering states $|\pm\rangle$, not the symmetric superposition.* The would-be symmetry-restoring state is exiled from the physical Hilbert space.

Physical Insight

The ferromagnet. The canonical physical picture is a Heisenberg ferromagnet below its Curie temperature. The microscopic Hamiltonian $H = -J\sum_{\langle ij\rangle}\mathbf{S}_i\cdot\mathbf{S}_j$ is exactly rotation-invariant: no direction in space is preferred by the laws. Yet below T_c the ground state has all spins aligned along *some* direction $\hat{\mathbf{n}}$, with a nonzero net magnetization $\mathbf{M} = \langle\sum_i \mathbf{S}_i\rangle \neq 0$. The magnetization is the order parameter; it transforms as a vector under the rotation symmetry, so a nonzero value is incompatible with a rotation-invariant ground state, exactly per Theorem 29.2. There are infinitely many degenerate ground states, one for each direction $\hat{\mathbf{n}}$—here a continuous manifold, the sphere S^2, foreshadowing the continuous case of the next section. The rotationally symmetric superposition over all $\hat{\mathbf{n}}$ exists mathematically but is unphysical: it fails to cluster, because the spin at one end of a macroscopic magnet is perfectly correlated with the spin at the other (they point the same way), so $\langle\mathbf{S}(x)\cdot\mathbf{S}(y)\rangle \not\to \langle\mathbf{S}(x)\rangle\cdot\langle\mathbf{S}(y)\rangle = 0$ at large separation. A real magnet picks one direction and sticks with it—to flip the magnetization of a macroscopic sample one must coherently reverse $\sim 10^{23}$ spins against the exchange energy, an action proportional to the volume, exactly the suppression Eq. (29.28). The ferromagnet *is* spontaneous symmetry breaking, made of atoms.

Caution

A unique ground state can never break a symmetry. If the vacuum is non-degenerate, it is necessarily an eigenstate of every symmetry operator that commutes with H (the symmetry maps the unique lowest-energy state to a state of the same energy, which must be itself up to phase), so $\mathsf{Z}\,|0\rangle = \pm\,|0\rangle$ and $Q\,|0\rangle = 0$; by Theorem 29.2 every order parameter then vanishes. Spontaneous breaking therefore *requires* a degenerate ground space, and—by the tunneling argument of Theorem 29.3—a degenerate ground space of mutually orthogonal, non-communicating vacua requires infinite volume. This is why one never sees genuine SSB in a finite quantum-mechanical system, only its precursor: a near-degenerate tunneling doublet whose splitting Eq. (29.28) is the finite-volume remnant of the would-be broken symmetry.

The picture that emerges is sharp enough to name. In infinite volume the Hilbert space of the broken theory is not a single irreducible space but decomposes into *superselection sectors*: distinct, mutually orthogonal subspaces, one built on each pure vacuum $|+\rangle$ and $|-\rangle$, with no local operator and no physical process connecting them. The reflection Z is no longer an ordinary symmetry within one sector; it is a map *between* sectors, Eq. (29.12), taking the entire $+v$ world to

the entire $-v$ world. Within either world the $\mathbb{Z}_2$ is invisible—there is only one vacuum, with its definite order parameter $\pm v$—and the symmetry survives solely as the statement that the two worlds are physically identical copies. This is the mature meaning of spontaneous breaking: not that the symmetry is lost, but that it is realized as an equivalence between distinct, separately self-consistent vacua, each of which sees only an asymmetric world.

With the discrete case fully in hand—the order parameter $\langle\phi\rangle = \pm v$ of Eq. (29.8), the degenerate pair of pure vacua selected by clustering, and the superselection structure of the infinite-volume theory—we have everything needed to ask the question that organizes the rest of the chapter. The $\mathbb{Z}_2$ gave us exactly two vacua, isolated points related by a reflection. What happens when the broken symmetry is *continuous*, so that the degenerate vacua are no longer two but a whole connected continuum, related by transformations one can perform infinitesimally? Moving along that continuum of vacua costs no energy, since they are all degenerate—and a zero-energy direction in field space is, as the next section shows, exactly a massless particle. We turn to the continuous case and the appearance of Goldstone bosons.

29.2 Continuous Breaking and the Linear Sigma Model: Massless Modes Appear

The discrete warmup of Section 29.1 left us with a sharp picture: a symmetry exact in the dynamics, a potential with more than one minimum, and a vacuum forced to pick one of them, betraying the symmetry through a nonzero order parameter. There the symmetry was $\mathbb{Z}_2$ (Theorem 7.22) and the two minima $\phi = \pm v$ were isolated points; the order parameter $\langle\phi\rangle = \pm v$ could only flip, never flow. Now ask the continuous question. If the broken symmetry is a continuous group G rather than a finite one, the set of vacua is no longer a handful of points but a smooth manifold, and one can slide from any vacuum to a neighbouring one *continuously*. The whole content of this section is the consequence of that one geometric fact: because the symmetry maps vacua to vacua, and because vacua all have the same energy, motion along the manifold of vacua costs nothing. A field configuration that interpolates this motion slowly in space is a long-wavelength excitation with vanishing energy as the wavelength grows—and a mode whose energy goes to zero with its momentum is a massless particle. The degeneracy is no longer a curiosity of the ground state; it forces new massless quanta into the spectrum.

We will make this completely concrete and computational. First the abelian case, the complex scalar in its broken phase—the "Mexican hat"—where every step can be done by hand and the single massless mode is laid bare. Then the O(N) linear sigma model, the multi-component generalization that is the standard workhorse of continuous spontaneous symmetry breaking and the model

we will reuse for chiral symmetry in Chapter 39. In both we read the spectrum directly off the Hessian of the potential at its minimum, prove that the flat directions are exactly flat (not merely flat at quadratic order), and count the massless modes by a rule—one per broken generator—that the next section will elevate from a tree-level observation into an exact theorem.

29.2.1 From two vacua to a circle of them: the abelian complex scalar

We continue ϕ^4 theory, but now in its complex, charged incarnation. Recall from Eqs. (6.62) and (7.66) the Lagrangian of a single complex scalar $\Phi(x) = (\phi_1 + i\phi_2)/\sqrt{2}$,

$$\mathcal{L} = (\partial_\mu \Phi)^*(\partial^\mu \Phi) - V(\Phi^*\Phi), \tag{29.35}$$

invariant under the global phase rotation $\Phi \mapsto e^{-i\alpha}\Phi$ (Eq. (7.67)). What distinguishes the present discussion from Chapter 7 is the choice of potential. In the symmetric phase the mass term is positive and the minimum sits at the symmetric point $\Phi = 0$. We now take the *wrong-sign* mass term—the continuous analogue of the double well—

$$V(\Phi^*\Phi) = -\mu^2\,\Phi^*\Phi + \lambda\,(\Phi^*\Phi)^2, \qquad \mu^2 > 0,\ \lambda > 0, \tag{29.36}$$

where $\mu^2 > 0$ means the coefficient of $\Phi^*\Phi$ is negative, so $\Phi = 0$ is now a local *maximum* of the radial profile. The quartic coupling $\lambda > 0$ is required for the potential to be bounded below; with it the energy is minimized not at the origin but on a ring at finite field amplitude. Writing $\rho \equiv \Phi^*\Phi = |\Phi|^2 \geq 0$, the potential is the cubic-free quartic $V = -\mu^2\rho + \lambda\rho^2$, stationary where

$$\frac{\mathrm{d}V}{\mathrm{d}\rho} = -\mu^2 + 2\lambda\rho = 0 \quad \Longrightarrow \quad \rho_{\min} = |\Phi|^2_{\min} = \frac{\mu^2}{2\lambda}. \tag{29.37}$$

This fixes only the modulus of Φ; the phase is entirely undetermined. The minima therefore form a *circle* in the complex Φ-plane. It is convenient to name the radius through a real constant v,

$$|\Phi|_{\min} = \frac{v}{\sqrt{2}}, \qquad v \equiv \frac{\mu}{\sqrt{\lambda}} > 0, \tag{29.38}$$

so that $\rho_{\min} = v^2/2 = \mu^2/2\lambda$, consistent with Eq. (29.37). The factor $1/\sqrt{2}$ is the same one that relates Φ to its real components; with it the real field ϕ_1 acquires the clean vacuum value v, as we now arrange.

The geometry is the entire story (Fig. 29.2). The symmetry $\Phi \mapsto e^{-i\alpha}\Phi$ rotates the complex plane and so slides one point of the minimum-circle into another; the circle is exactly the orbit of the symmetry group through any one of

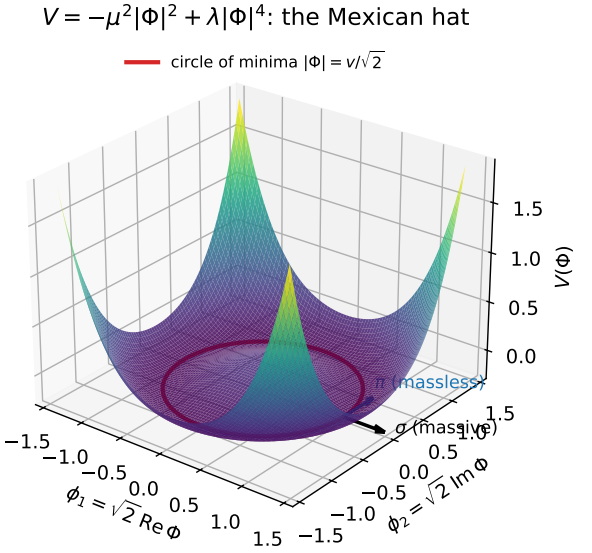


Figure 29.2: The Mexican-hat potential Eq. (29.36) over the complex plane $\Phi = (\phi_1 + i\phi_2)/\sqrt{2}$. The degenerate minima form the circle $|\Phi| = v/\sqrt{2}$ (red). Climbing radially out of the trough costs energy—the radial mode σ is massive, $m_\sigma^2 = 2\lambda v^2$—while running along the trough costs nothing: the angular mode π is the massless Goldstone boson. The origin $\Phi = 0$, the symmetric point, is the central maximum.

its points. Picking a vacuum means choosing a point on the circle, and no point is special. We choose the one on the positive real axis,

$$\langle \Phi \rangle = \frac{v}{\sqrt{2}}, \qquad \text{equivalently} \qquad \langle \phi_1 \rangle = v, \quad \langle \phi_2 \rangle = 0. \tag{29.39}$$

This is the order parameter: $\langle \Phi \rangle \neq 0$ signals that the U(1) phase symmetry is not realized on the vacuum, because the rotation $e^{-i\alpha}$ moves $\langle \Phi \rangle$ to a *different* value $e^{-i\alpha} v/\sqrt{2}$, i.e. to a different (though degenerate) vacuum. The symmetry is spontaneously broken in the precise sense of Section 29.1.

Polar parametrization and the spectrum. To find the particle content we expand the field about Eq. (29.39). The geometry recommends *polar* fluctuations—one real field for the radial displacement off the trough, one for the angular position along it:

$$\Phi(x) = \frac{1}{\sqrt{2}} \left(v + \sigma(x)\right) e^{i\pi(x)/v}, \tag{29.40}$$

with $\sigma(x)$ and $\pi(x)$ real fields of vanishing vacuum value, $\langle \sigma \rangle = \langle \pi \rangle = 0$. The factor $1/v$ in the exponent is a deliberate choice that we justify in a moment:

it makes π canonically normalized. The radial coordinate is $|\Phi| = (v+\sigma)/\sqrt{2}$, so σ measures the excursion away from the trough and π the angular position around it.

Insert Eq. (29.40) into the potential first, because there the result is immediate and decisive. The potential depends on Φ only through $|\Phi|^2 = (v+\sigma)^2/2$, which is *independent of* π. Writing $h \equiv v+\sigma$ and using $\mu^2 = \lambda v^2$ from Eq. (29.38),

$$V = -\frac{\lambda v^2}{2}\,h^2 + \frac{\lambda}{4}\,h^4 = \frac{\lambda}{4}\big(h^2 - v^2\big)^2 - \frac{\lambda v^4}{4}. \tag{29.41}$$

The first equality used $-\mu^2|\Phi|^2 = -\tfrac{1}{2}\lambda v^2 h^2$ and $\lambda|\Phi|^4 = \tfrac{1}{4}\lambda h^4$; the second completed the square. Now substitute $h = v+\sigma$, so that $h^2 - v^2 = 2v\sigma + \sigma^2 = \sigma(2v+\sigma)$ and $(h^2-v^2)^2 = \sigma^2(2v+\sigma)^2 = \sigma^2(4v^2 + 4v\sigma + \sigma^2)$. Hence

$$V = \lambda v^2\,\sigma^2 + \lambda v\,\sigma^3 + \frac{\lambda}{4}\,\sigma^4 - \frac{\lambda v^4}{4}. \tag{29.42}$$

Two features are already the whole result. The constant $-\lambda v^4/4$ is the (negative) vacuum energy and plays no dynamical role. The quadratic term is $\lambda v^2\sigma^2 = \tfrac{1}{2}(2\lambda v^2)\sigma^2$, the mass term of a real scalar of mass squared

$$m_\sigma^2 = 2\lambda v^2 = 2\mu^2. \tag{29.43}$$

And π appears *nowhere* in V: there is no π mass term, no π self-interaction, nothing. The angular field is massless,

$$m_\pi^2 = 0. \tag{29.44}$$

This is the massless mode the geometry promised, the Goldstone boson. It is massless not by a fine-tuning of parameters but because V does not depend on π at all—and V does not depend on π because π parametrizes motion along the trough, which is the symmetry orbit, along which V is constant by construction. We will turn this remark into a proof of exact flatness in Section 29.2.2.

It remains to check that σ and π are properly normalized fields, so that Eqs. (29.43) and (29.44) really are masses and not masses-times-a-wavefunction-factor. Compute the kinetic term. From Eq. (29.40),

$$\partial_\mu\Phi = \frac{1}{\sqrt{2}}\,e^{i\pi/v}\Big(\partial_\mu\sigma + \frac{i}{v}(v+\sigma)\,\partial_\mu\pi\Big), \tag{29.45}$$

and since the phase $e^{i\pi/v}$ cancels against its conjugate in $|\partial_\mu\Phi|^2$,

$$(\partial_\mu\Phi)^*(\partial^\mu\Phi) = \frac{1}{2}\,(\partial_\mu\sigma)(\partial^\mu\sigma) + \frac{1}{2}\Big(1+\frac{\sigma}{v}\Big)^2(\partial_\mu\pi)(\partial^\mu\pi). \tag{29.46}$$

The quadratic part is $\tfrac{1}{2}(\partial\sigma)^2 + \tfrac{1}{2}(\partial\pi)^2$: both fields are canonically normalized, which is exactly what the factor $1/v$ in the exponent of Eq. (29.40) secured. (Had

we written $e^{i\theta}$ with an unnormalized angle θ, the π kinetic term would have come with a stray v^2 that we would then have to absorb by rescaling.) The remaining terms in Eq. (29.46), of order $\sigma(\partial\pi)^2$ and $\sigma^2(\partial\pi)^2$, are interactions, not masses. Assembling Eq. (29.46) and Eq. (29.42), the Lagrangian in physical variables is

$$\mathcal{L} = \underbrace{\tfrac{1}{2}(\partial\sigma)^2 - \tfrac{1}{2}m_\sigma^2\sigma^2}_{\text{massive radial mode}} + \underbrace{\tfrac{1}{2}(\partial\pi)^2}_{\text{massless Goldstone}} + \mathcal{L}_{\text{int}} + \text{const}, \tag{29.47}$$

with $\mathcal{L}_{\text{int}}$ collecting the cubic and quartic vertices $-\lambda v\sigma^3 - \frac{\lambda}{4}\sigma^4 + \frac{1}{2}(2\sigma/v + \sigma^2/v^2)(\partial\pi)^2$. The tree spectrum is one massive scalar σ with $m_\sigma^2 = 2\lambda v^2$ and one massless scalar π. A continuous broken symmetry has produced exactly one massless mode.

The same answer from the Hessian. The polar parametrization made the result transparent, but it hid the mechanism behind a clever change of variables. To expose the mechanism—and to set up the general theorem—we recompute the spectrum the brute-force way, as the eigenvalues of the matrix of second derivatives of V at the minimum, in the original *Cartesian* field components ϕ_1, ϕ_2. This Hessian is, by definition, the tree-level mass-squared matrix, because expanding any potential about a stationary point $\phi^{(0)}$ gives $V = V(\phi^{(0)}) + \frac{1}{2}M_{ij}(\phi_i - \phi_i^{(0)})(\phi_j - \phi_j^{(0)}) + \dots$ with

$$M_{ij} \equiv \left.\frac{\partial^2 V}{\partial\phi_i\,\partial\phi_j}\right|_{\phi=\phi^{(0)}}, \tag{29.48}$$

and a diagonal entry $M_{ii} = \frac{1}{2}\cdot 2\,m_i^2$ is the mass squared of the corresponding canonically normalized real field.

Worked Example 29.5 (The mass matrix of the Mexican hat)**.** Write the potential Eq. (29.36) in real components. With $|\Phi|^2 = (\phi_1^2 + \phi_2^2)/2$,

$$V = -\frac{\mu^2}{2}\,(\phi_1^2 + \phi_2^2) + \frac{\lambda}{4}\,(\phi_1^2 + \phi_2^2)^2. \tag{29.49}$$

The first derivatives are

$$\frac{\partial V}{\partial\phi_i} = -\mu^2\phi_i + \lambda\,(\phi_1^2 + \phi_2^2)\,\phi_i, \qquad i = 1, 2, \tag{29.50}$$

which vanish where $\phi_1^2 + \phi_2^2 = \mu^2/\lambda = v^2$—the minimum circle, in agreement with Eq. (29.38). Differentiating once more,

$$\frac{\partial^2 V}{\partial\phi_i\partial\phi_j} = -\mu^2\delta_{ij} + \lambda\,(\phi_1^2 + \phi_2^2)\,\delta_{ij} + 2\lambda\,\phi_i\phi_j. \tag{29.51}$$

Now evaluate at the chosen vacuum $\phi^{(0)} = (v, 0)$, where $\phi_1^2 + \phi_2^2 = v^2 = \mu^2/\lambda$. The first two terms cancel identically, $-\mu^2\delta_{ij} + \lambda v^2 \delta_{ij} = 0$, leaving only the third:

$$M_{ij} = 2\lambda\, \phi_i^{(0)} \phi_j^{(0)} = 2\lambda v^2 \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}_{ij} . \tag{29.52}$$

The eigenvalues are $m_\sigma^2 = 2\lambda v^2$ (eigenvector $(1, 0)$, the *radial* direction ϕ_1, pointing along the VEV) and $m_\pi^2 = 0$ (eigenvector $(0, 1)$, the *tangent* direction ϕ_2, perpendicular to the VEV and tangent to the minimum circle). This reproduces Eqs. (29.43) and (29.44) exactly, and now the mechanism is visible. The cancellation $-\mu^2 + \lambda v^2 = 0$ is precisely the stationarity condition; what survives, $M_{ij} = 2\lambda\, \phi_i^{(0)} \phi_j^{(0)}$, is $2\lambda v^2$ times the projector onto the VEV direction. Its kernel is everything orthogonal to $\phi^{(0)}$—and the orthogonal complement of the radius of a circle is the tangent, the direction in which the symmetry slides the vacuum. The zero eigenvalue sits exactly where the symmetry moves us along the trough.

The two computations agree, but the second tells us *why*. The null direction of the mass matrix is the tangent to the orbit of the broken symmetry. That is a statement about the geometry of the potential, not about the abelian group or the quartic form of V; it will hold for any group and any invariant potential. We prove it next.

29.2.2 Why the flat direction is exactly flat: orbits of the symmetry

Two distinct claims are bundled together in the appearance of $m_\pi^2 = 0$, and it is worth separating them because they have very different strengths.

The first is the *tree-level* statement: the Hessian of V at the minimum has a zero eigenvalue for each broken symmetry direction. This is a statement about the second derivative of V at one point, and it is what fixes the tree-level mass. The second is the *exact* statement: the potential is not merely flat to quadratic order along the broken direction but *exactly constant* along it, to all orders in the field. The polar calculation already suggested this—V in Eq. (29.41) simply does not contain π, to any power—but we should see clearly that it is a theorem about symmetry, not an accident of this potential. Both claims flow from the single fact that V is invariant under the group.

Exact flatness along orbits. Let the potential $V(\phi)$ of a multiplet $\phi = (\phi_1, \ldots, \phi_N)$ be invariant under the group action $\phi \mapsto g \cdot \phi$ of a Lie group G (Eq. (7.75)), meaning

$$V(g \cdot \phi) = V(\phi) \qquad \text{for all } g \in G \text{ and all } \phi. \tag{29.53}$$

Fix a minimum $\phi^{(0)}$. Its *orbit* is the set $\{g \cdot \phi^{(0)} : g \in G\}$ of all configurations reachable from $\phi^{(0)}$ by the symmetry. By Eq. (29.53), $V(g \cdot \phi^{(0)}) = V(\phi^{(0)})$ for every g: *V is exactly constant on the orbit*, equal to its minimum value everywhere along it. This is not an expansion to some finite order; it is an identity holding for the full nonlinear potential, because it is nothing but the invariance Eq. (29.53) read at the point $\phi^{(0)}$. The orbit is therefore an exactly flat valley floor of V. The fields that parametrize displacement *along* the orbit—the Goldstone fields—enter V only through their failure to stay on the orbit, which is to say not at all at the level of the orbit coordinates themselves. In the abelian example the orbit is the minimum circle and the orbit coordinate is the angle π/v; the statement "V is constant on the circle" is exactly "V does not depend on π," which is what we found in Eq. (29.41).

We single out the orbit through the vacuum, because it is the object the rest of the chapter revolves around.

Definition 29.6 (Vacuum manifold and stabilizer)**.** Let G be the symmetry group of the dynamics and $\phi^{(0)}$ a chosen minimum of the G-invariant potential. The *stabilizer* (or *little group*, or *unbroken subgroup*) of $\phi^{(0)}$ is

$$H = \mathrm{Stab}(\phi^{(0)}) = \{h \in G : h \cdot \phi^{(0)} = \phi^{(0)}\}, \tag{29.54}$$

the subgroup that fixes the vacuum and hence is realized linearly on the spectrum (it is unbroken). The *vacuum manifold* is the orbit of $\phi^{(0)}$, which the orbit–stabilizer theorem identifies with the coset space

$$\mathcal{M} = \{g \cdot \phi^{(0)} : g \in G\} \;\cong\; G/H. \tag{29.55}$$

By the flatness just proved, V is constant on $\mathcal{M}$, so every point of $\mathcal{M}$ is a degenerate vacuum, and the elements of G not in H map any vacuum to a physically distinct one—they are the broken transformations.

The identification $\mathcal{M} \cong G/H$ is the orbit–stabilizer theorem: two group elements g_1, g_2 send $\phi^{(0)}$ to the same point iff $g_2^{-1} g_1 \in H$, i.e. iff they lie in the same left coset of H, so the orbit is in bijection with G/H. Its dimension—the number of independent directions one can move the vacuum—is therefore

$$\dim \mathcal{M} = \dim(G/H) = \dim G - \dim H, \tag{29.56}$$

the number of generators of G that are *not* generators of H, i.e. the number of *broken generators*. This integer will be the number of Goldstone bosons. We now prove the tree-level half of that statement.

Tree-level Goldstone count from the Hessian. Differentiate the invariance identity. The infinitesimal form of Eq. (29.53) under the per-generator variation $\delta^a \phi_i = (t^a)_{ij} \phi_j$ of Eq. (7.77) (with $t^a = -iT^a$ the real-form generators of

Eq. (7.78)) is the statement that V is unchanged at first order for every generator and every field value,

$$\frac{\partial V}{\partial \phi_i}(\phi)\,(t^a)_{ij}\,\phi_j = 0 \qquad \text{for all } \phi,\ \text{all } a. \tag{29.57}$$

This holds identically in ϕ, so we may differentiate it. Apply $\partial/\partial\phi_k$ to Eq. (29.57) using the product rule:

$$\frac{\partial^2 V}{\partial \phi_k\,\partial \phi_i}\,(t^a)_{ij}\,\phi_j + \frac{\partial V}{\partial \phi_i}\,(t^a)_{ik} = 0, \tag{29.58}$$

where the second term came from $\partial(\,(t^a)_{ij}\phi_j)/\partial\phi_k = (t^a)_{ik}$. Now evaluate this identity at the minimum $\phi = \phi^{(0)}$. There the first derivative $\partial V/\partial\phi_i$ vanishes (it is a stationary point), killing the second term, and the second-derivative factor is precisely the mass matrix M_{ki} of Eq. (29.48). What remains is

$$\boxed{M_{ki}\,(t^a)_{ij}\,\phi_j^{(0)} = 0} \qquad \text{equivalently} \qquad M\,(t^a\phi^{(0)}) = 0. \tag{29.59}$$

For *every* generator t^a of G, the vector $t^a\phi^{(0)}$—the infinitesimal motion of the vacuum under that generator—is a null eigenvector of the mass matrix. This is the central identity of tree-level Goldstone counting; we state its content as a proposition.

Proposition 29.7 (Tree-level Goldstone theorem)**.** *Let $V(\phi)$ be invariant under a compact Lie group G acting by $\delta^a\phi = t^a\phi$, and let $\phi^{(0)}$ be a minimum with mass matrix $M_{ij} = \partial_i\partial_j V|_{\phi^{(0)}}$. Then for every generator t^a, the configuration-space vector $t^a\phi^{(0)}$ lies in the kernel of M. Split the generators into:*

- unbroken *generators, those of the stabilizer H, for which $t^a\phi^{(0)} = 0$ (they fix the vacuum); these give* no *information about the spectrum from Eq. (29.59), and the corresponding directions may be massive;*
- broken *generators, those not in H, for which $t^a\phi^{(0)} \neq 0$; each such vector is a genuine nonzero null eigenvector of M, hence a* massless *mode.*

The number of independent vectors $\{t^a\phi^{(0)}\}$ as a ranges over the broken generators equals the number of broken generators, which by Eq. (29.56) equals $\dim(G/H) = \dim\mathcal{M}$. *Therefore the number of massless tree-level modes equals the number of broken generators equals the dimension of the vacuum manifold.*

Proof. The identity Eq. (29.59) was derived above and holds for every a. If $t^a \in \mathfrak{h}$ then by definition of the stabilizer the one-parameter group $e^{\alpha t^a}$ fixes $\phi^{(0)}$, so $t^a\phi^{(0)} = \frac{d}{d\alpha}e^{\alpha t^a}\phi^{(0)}|_0 = 0$ and Eq. (29.59) reads $M\cdot 0 = 0$, vacuously true and constraining nothing. If $t^a \notin \mathfrak{h}$ then $t^a\phi^{(0)} \neq 0$ and Eq. (29.59) states

that this nonzero vector is annihilated by M, i.e. it is a zero-mass eigenvector. It remains to count the independent broken vectors. The map $t^a \mapsto t^a\phi^{(0)}$ from the Lie algebra $\mathfrak{g}$ to configuration space has kernel exactly $\mathfrak{h}$: $t\phi^{(0)} = 0$ iff $e^{\alpha t}$ fixes $\phi^{(0)}$ for all α iff $t \in \mathfrak{h}$ (the algebra is the tangent space to the stabilizer). Hence the image—the span of the broken vectors $\{t^a\phi^{(0)}\}$—has dimension $\dim\mathfrak{g} - \dim\mathfrak{h} = \dim G - \dim H = \dim(G/H)$. Each independent vector in this image is an independent massless direction. The positivity of M on the complement (the radial/massive directions are genuine minima, $M \geq 0$) and the absence of further accidental zeros are guaranteed at tree level by $\phi^{(0)}$ being a strict minimum transverse to the orbit, which is the generic case and the one realized in our examples. □

Two comments fix the meaning of this result. First, it is genuinely a counting theorem: it does not merely assert "there are some massless modes," it says *exactly how many*, and ties the number to the group-theoretic data $\dim G - \dim H$ and to the geometric data $\dim\mathcal{M}$, three descriptions of one integer. Second, Theorem 29.7 is, as stated, a tree-level (classical-potential) statement: M is the Hessian of the *classical* V. The next section will confront the question of whether quantum corrections can lift these zeros—and the answer, made exact through the effective potential, is that they cannot, because the orbit argument of Eq. (29.53) applies verbatim to the quantum effective potential, which inherits the symmetry. The flatness along orbits is exact; the masslessness is exact. Theorem 29.7 is the seed of that exact theorem.

Physical Insight

The two derivations of $m_\pi = 0$ illuminate two faces of the same fact. The polar parametrization shows masslessness as *kinematics of the valley*: the Goldstone field is a coordinate along the flat floor of the potential, so its potential energy is identically zero and only its gradient energy survives—a field with gradient energy but no potential energy is, by definition, a massless wave. The Hessian derivation shows masslessness as *a Ward-like identity*: the invariance of V, differentiated, forces $M\,(t^a\phi^{(0)}) = 0$, so the broken-symmetry direction is a zero mode of the curvature. The first is geometry, the second is the differentiated symmetry; they are the same statement because the symmetry *is* the isometry of the valley.

29.2.3 The O(N) linear sigma model

The abelian example has a single broken generator and so a single Goldstone, and one might worry that its lessons are special. They are not, and the cleanest demonstration is the model that generalizes ϕ^4 theory from one real scalar to an N-component multiplet rotated into itself by the orthogonal group: the *linear*

sigma model. It is the standard laboratory for continuous spontaneous symmetry breaking, and we will reuse its $\mathrm{O}(N) \to \mathrm{O}(N-1)$ coset structure when we model the chiral symmetry of the strong interactions in Chapter 39.

Take N real scalar fields ϕ_i, $i = 1, \ldots, N$, assembled into a vector $\boldsymbol{\phi} = (\phi_1, \ldots, \phi_N)$, with the Lagrangian

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu \phi_i\, \partial^\mu \phi_i - V(\phi^2), \qquad V(\phi^2) = -\tfrac{1}{2}\,\mu^2\,\phi^2 + \tfrac{\lambda}{4}\,(\phi^2)^2, \tag{29.60}$$

where $\phi^2 \equiv \phi_i \phi_i = \boldsymbol{\phi}\cdot\boldsymbol{\phi}$ is the $\mathrm{O}(N)$-invariant length squared, with summation on i understood, $\mu^2 > 0$, and $\lambda > 0$. Both the kinetic term and the potential depend on the fields only through the rotation-invariant combinations $\partial_\mu \phi_i\, \partial^\mu \phi_i$ and $\phi_i \phi_i$, so Eq. (29.60) is invariant under

$$\phi_i \longmapsto R_{ij}\phi_j, \qquad R \in \mathrm{O}(N), \quad R^{\mathsf{T}} R = \mathbf{1}, \tag{29.61}$$

the group of $N \times N$ orthogonal transformations. This is the continuous symmetry whose breaking we study. Its connected part is the rotation group $\mathrm{SO}(N)$, with $\dim \mathrm{SO}(N) = \dim \mathrm{O}(N) = \frac{1}{2}N(N-1)$ generators, namely the antisymmetric matrices $t^{(mn)}$ that rotate the m–n plane; we use the antihermitian real form $t = -iT$ of Eq. (7.77) so that $\delta\phi = t\phi$ keeps the real fields real.

Notation Watch

The linear sigma model is conventionally normalized with the quartic written as $\frac{\lambda}{4}(\phi^2)^2$, so that the $\mathrm{O}(N)$ invariant ϕ^2 appears as a clean square and the VEV comes out as $v = \mu/\sqrt{\lambda}$. For a single component, $N = 1$, this is $\frac{\lambda}{4}\phi^4$, which is the ϕ^4 theory interaction $\frac{\lambda}{4!}\phi^4$ with the replacement $\lambda_{\phi^4} = 6\,\lambda$; the two "$\lambda$"s differ by the combinatorial factor $4!/4 = 6$. We use the sigma-model normalization throughout this chapter and restore the $1/4!$ convention only when we explicitly reconnect to the renormalization computations of Chapter 22. For $N = 2$ the model reduces *exactly* to the abelian complex scalar of Section 29.2.1: with $\Phi = (\phi_1 + i\phi_2)/\sqrt{2}$ one has $\phi^2 = 2|\Phi|^2$, so $-\frac{1}{2}\mu^2\phi^2 + \frac{\lambda}{4}(\phi^2)^2 = -\mu^2|\Phi|^2 + \lambda|\Phi|^4$, with the same μ, λ. The $N = 2$ model is the $\mathrm{U}(1) \cong \mathrm{SO}(2)$ Mexican hat, the $N = 1$ model is the discrete double well of Section 29.1 (the group $\mathrm{O}(1) = \mathbb{Z}_2$, no continuous symmetry, no Goldstone), and general N interpolates between and beyond them.

Vacuum, breaking pattern, and spectrum. The potential depends only on the magnitude $|\boldsymbol{\phi}| = \sqrt{\phi^2}$, so its analysis is identical to the abelian radial problem. Stationarity,

$$\frac{\partial V}{\partial \phi_i} = (\,-\mu^2 + \lambda\,\phi^2)\phi_i = 0, \tag{29.62}$$

is solved either at $\boldsymbol{\phi} = 0$ (the symmetric maximum) or on the sphere

$$\phi^2 = \frac{\mu^2}{\lambda} \equiv v^2, \qquad v = \frac{\mu}{\sqrt{\lambda}}, \tag{29.63}$$

which is the set of minima: an $(N-1)$-dimensional sphere S^{N-1} of radius v in field space. Any point on it is a legitimate vacuum; $\mathrm{O}(N)$ rotates them into one another. We choose the vacuum to point along the N-th axis,

$$\langle \phi_i \rangle = v\,\delta_{iN}, \qquad \text{i.e.} \quad \langle \boldsymbol{\phi} \rangle = (0, \ldots, 0, v). \tag{29.64}$$

The subgroup that fixes this vacuum is the set of rotations acting only on the first $N-1$ components and leaving the N-th alone—that is, $\mathrm{O}(N-1)$. The breaking pattern is therefore

$$G = \mathrm{O}(N) \;\longrightarrow\; H = \mathrm{O}(N-1). \tag{29.65}$$

The count of broken generators follows from Eq. (29.56),

$$\dim G - \dim H = \frac{N(N-1)}{2} - \frac{(N-1)(N-2)}{2} = \frac{(N-1)\big[N-(N-2)\big]}{2} = N-1, \tag{29.66}$$

so Theorem 29.7 predicts $N-1$ massless Goldstone bosons, and the vacuum manifold is

$$\mathcal{M} = \mathrm{O}(N)/\mathrm{O}(N-1) \cong S^{N-1}, \qquad \dim \mathcal{M} = N-1, \tag{29.67}$$

the $(N-1)$-sphere of minima, whose dimension matches the Goldstone count as the theorem requires. We verify this directly by computing the Hessian, exactly as in the abelian case.

Worked Example 29.8 (Spectrum of the $\mathrm{O}(N)$ model from the Hessian)**.** The first derivatives of V in Eq. (29.60) are $\partial_i V = (-\mu^2 + \lambda\phi^2)\phi_i$, and differentiating again,

$$\frac{\partial^2 V}{\partial\phi_i \partial\phi_j} = (-\mu^2 + \lambda\phi^2)\delta_{ij} + 2\lambda\,\phi_i\phi_j. \tag{29.68}$$

Evaluate at the vacuum Eq. (29.64), where $\phi^2 = v^2 = \mu^2/\lambda$ so that $-\mu^2 + \lambda v^2 = 0$ and $\phi_i\phi_j = v^2\delta_{iN}\delta_{jN}$:

$$M_{ij} = 2\lambda v^2\,\delta_{iN}\delta_{jN} = 2\lambda v^2\ \mathrm{diag}(0, 0, \ldots, 0, 1)_{ij}. \tag{29.69}$$

The mass matrix is diagonal with a single nonzero entry. Reading it off:

- the N-th component, the radial direction along the VEV, has $m^2 = 2\lambda v^2$. Call this field σ. It is the massive "sigma" meson that names the model: $m_\sigma^2 = 2\lambda v^2 = 2\mu^2$, identical in form to the abelian Eq. (29.43).

- the first $N-1$ components, the directions tangent to the sphere of minima, have $m^2 = 0$. Call them π_a, $a = 1, \ldots, N-1$: these are the $N-1$ massless Goldstone bosons, the "pions."

The count $N-1$ of zero eigenvalues matches Eq. (29.66) exactly, and the null space of M—spanned by the unit vectors $\hat{e}_1, \ldots, \hat{e}_{N-1}$ orthogonal to the VEV $\hat{e}_N$—is precisely the tangent space to the sphere S^{N-1} at the vacuum. As in the abelian case, $M = 2\lambda v^2$ times the projector onto the VEV direction, and its kernel is the orbit's tangent space. The breaking $\mathrm{O}(N) \to \mathrm{O}(N-1)$ produces one massive radial mode and $N-1$ massless angular modes, every one of them accounted for by the broken generators.

It is instructive to see the spectrum a second time in shifted fields, both to confirm the normalizations and to expose the interactions the Goldstones inherit—these vertices will matter when we discuss the low-energy theorems of Chapter 39.

Worked Example 29.9 (The shifted O(N) Lagrangian)**.** Expand about the vacuum Eq. (29.64) by writing the radial fluctuation σ and the $N-1$ transverse fluctuations π_a:

$$\phi_N = v + \sigma, \qquad \phi_a = \pi_a \ (a = 1, \ldots, N-1), \qquad \phi^2 = (v+\sigma)^2 + \pi_a \pi_a. \tag{29.70}$$

Write $\boldsymbol{\pi}^2 \equiv \pi_a \pi_a$. The kinetic term is already diagonal and canonical, $\frac{1}{2}(\partial\sigma)^2 + \frac{1}{2}\,\partial_\mu \pi_a \partial^\mu \pi_a$, because the shift Eq. (29.70) is a constant translation that does not touch derivatives. For the potential, use $\mu^2 = \lambda v^2$ and the completed-square form, exactly as in Eq. (29.41). With $X \equiv \phi^2$,

$$V = -\tfrac{1}{2}\lambda v^2 X + \tfrac{\lambda}{4} X^2 = \tfrac{\lambda}{4}\left(X - v^2\right)^2 - \tfrac{\lambda v^4}{4}, \tag{29.71}$$

and $X - v^2 = (v+\sigma)^2 + \boldsymbol{\pi}^2 - v^2 = 2v\sigma + \sigma^2 + \boldsymbol{\pi}^2$. Therefore

$$V = \frac{\lambda}{4}\left(2v\sigma + \sigma^2 + \boldsymbol{\pi}^2\right)^2 - \frac{\lambda v^4}{4}. \tag{29.72}$$

Expand the square. The only term quadratic in the fields is $(2v\sigma)^2 = 4v^2\sigma^2$, giving $\frac{\lambda}{4} \cdot 4v^2\sigma^2 = \lambda v^2 \sigma^2 = \frac{1}{2}(2\lambda v^2)\sigma^2$. There is *no* quadratic term in π_a: the pion mass matrix is identically zero, confirming $m_\pi^2 = 0$ for all $N-1$ pions. Collecting the cross terms gives the interactions:

$$V = \frac{1}{2}\left(2\lambda v^2\right)\sigma^2 + \underbrace{\lambda v\, \sigma\left(\sigma^2 + \boldsymbol{\pi}^2\right)}_{\text{cubic}} + \underbrace{\frac{\lambda}{4}\left(\sigma^2 + \boldsymbol{\pi}^2\right)^2}_{\text{quartic}} - \frac{\lambda v^4}{4}. \tag{29.73}$$

Every interaction the pions have—the cubic $\lambda v\, \sigma \boldsymbol{\pi}^2$ and the quartics $\frac{\lambda}{2}\sigma^2\boldsymbol{\pi}^2$, $\frac{\lambda}{4}(\boldsymbol{\pi}^2)^2$—carries at least two pion fields and is built from derivative-free powers

governed by the single coupling λ and the scale v. Crucially, the pions have no mass term and no tadpole: the linear term in σ cancelled (as it must, v being the true minimum), and no linear π term ever appears because the vacuum is O($N-1$)-symmetric. For $N=2$, with the single pion π_1 and the relation to the abelian field via $\phi_1=\pi_1$, $\phi_2=$ radial, these vertices reduce to the σ–π interactions read off from Eq. (29.47) (up to the polar-versus-Cartesian field redefinition, which reshuffles derivative interactions but not the mass spectrum).

The O(N) model thus realizes the general theorem in a controllable family: $N=1$ is the discrete double well with no Goldstone, $N=2$ is the abelian Mexican hat with one Goldstone, and each additional component adds exactly one more massless pion, always with one massive radial σ left over. The integer $N-1$ is simultaneously the number of broken generators, the dimension of the vacuum manifold S^{N-1}, and the number of zero eigenvalues of the Hessian—the three faces of the count established in Theorem 29.7.

29.2.4 The current between the vacuum and the Goldstone: the decay constant

We have located the massless modes and counted them. The final task of this section is to extract the quantity that makes them *dynamically* the children of the broken symmetry: the matrix element of the broken Noether current between the vacuum and a one-Goldstone state. This single number—call it the decay constant f—is the bridge between the symmetry current of Chapter 7 and the massless particles found above, and it is the engine that the next section's operator proof of Goldstone's theorem runs on. We compute it from the explicit currents, leaving nothing implicit.

The abelian decay constant. The conserved U(1) current is Eq. (7.70),

$$J^\mu = i(\Phi^*\,\partial^\mu\Phi - \Phi\,\partial^\mu\Phi^*) = -2\,\mathrm{Im}(\Phi^*\,\partial^\mu\Phi). \tag{29.74}$$

Substitute the polar parametrization Eq. (29.40). With $\Phi=\frac{1}{\sqrt{2}}(v+\sigma)e^{i\pi/v}$ and $\partial^\mu\Phi=\frac{1}{\sqrt{2}}e^{i\pi/v}\big(\partial^\mu\sigma+\frac{i}{v}(v+\sigma)\partial^\mu\pi\big)$, the product is

$$\Phi^*\,\partial^\mu\Phi=\frac{1}{2}(v+\sigma)\Big(\partial^\mu\sigma+\frac{i}{v}(v+\sigma)\partial^\mu\pi\Big),\qquad \mathrm{Im}\,(\Phi^*\partial^\mu\Phi)=\frac{1}{2v}(v+\sigma)^2\,\partial^\mu\pi, \tag{29.75}$$

so that

$$J^\mu=-\frac{1}{v}(v+\sigma)^2\,\partial^\mu\pi=-v\,\partial^\mu\pi\;-\;\Big(2\sigma+\frac{\sigma^2}{v}\Big)\partial^\mu\pi. \tag{29.76}$$

The structure is the heart of the matter. The current is, to leading order, *linear in the Goldstone field*: $J^\mu=-v\,\partial^\mu\pi+\dots$, where the dots are bilinear and higher and so create two or more particles. The broken current creates a single

Goldstone out of the vacuum, with strength set by v. Concretely, the vacuum-to-one-particle matrix element follows from the canonically normalized free-field overlap $\langle 0|\pi(x)|\pi(p)\rangle = e^{-ip\cdot x}$ fixed by the covariant one-particle normalization of Theorem 9.12 (the bilinear terms in Eq. (29.76) have no overlap with a one-particle state, since they carry two annihilation/creation operators):

$$\langle 0|J^\mu(x)|\pi(p)\rangle = -v\,\partial^\mu \langle 0|\pi(x)|\pi(p)\rangle = -v\,\partial^\mu e^{-ip\cdot x} = iv\,p^\mu\, e^{-ip\cdot x}, \tag{29.77}$$

and at $x = 0$,

$$\langle 0|J^\mu(0)|\pi(p)\rangle = i\,v\,p^\mu. \tag{29.78}$$

The current connects the vacuum to the one-Goldstone state with amplitude proportional to p^μ, and the proportionality constant is the vacuum value v itself. This v is the abelian decay constant.

The non-abelian decay constants. The same computation in the O(N) model gives one matrix element per broken generator and exhibits the diagonal structure δ_{ab} explicitly. The non-abelian currents are Eq. (7.80), $J^{c\mu} = \partial^\mu\phi_i\,(t^c)_{ij}\,\phi_j$, with t^c the real antisymmetric generators of SO(N). The broken generators are those mixing one of the transverse axes $a \in \{1,\dots,N-1\}$ with the VEV axis N. Choosing the orientation of the broken generator labelled a as

$$\big(t^{(a)}\big)_{ij} = \delta_{iN}\delta_{ja} - \delta_{ia}\delta_{jN}, \tag{29.79}$$

which is antisymmetric and generates the rotation in the a–N plane, the corresponding current is

$$J^{(a)\mu} = \partial^\mu\phi_i\,(t^{(a)})_{ij}\,\phi_j = \partial^\mu\phi_N\,\phi_a - \partial^\mu\phi_a\,\phi_N. \tag{29.80}$$

Now insert the shifted fields Eq. (29.70), $\phi_N = v + \sigma$ and $\phi_a = \pi_a$:

$$J^{(a)\mu} = \partial^\mu\sigma\,\pi_a - \partial^\mu\pi_a\,(v+\sigma) = -v\,\partial^\mu\pi_a \;+\; (\partial^\mu\sigma\,\pi_a - \sigma\,\partial^\mu\pi_a). \tag{29.81}$$

Again the current is linear in the Goldstone field at leading order, $J^{(a)\mu} = -v\,\partial^\mu\pi_a + (\text{bilinear})$, and the linear piece picks out the pion π_a associated with that very generator. Taking the vacuum-to-one-pion matrix element, with $\langle 0|\pi_a(x)|\pi_b(p)\rangle = \delta_{ab}\,e^{-ip\cdot x}$ from the canonical normalization of the $N-1$ identical pion fields,

$$\Big\langle 0\Big|J^{(a)\mu}(0)\Big|\pi_b(p)\Big\rangle = -v\,(-ip^\mu)\,\delta_{ab} = i\,v\,p^\mu\,\delta_{ab}. \tag{29.82}$$

Each broken current creates exactly its own Goldstone, the Kronecker δ_{ab} certifying the one-to-one pairing of broken generators with massless modes. Defining the decay constant f through the canonical form of this matrix element,

$$\boxed{\langle 0|J^\mu_a(0)|\pi_b(p)\rangle = i\,f\,p^\mu\,\delta_{ab}}, \qquad f = v \text{ (tree level)}, \tag{29.83}$$

we read off that at tree level the decay constant equals the vacuum expectation value, $f = v$, in both the abelian ($N = 2$) and the general O(N) models. (The overall sign of f is a convention: it tracks the orientation chosen for the broken generator in Eq. (29.79) and the sign convention for the phase rotation in Eq. (7.67); flipping either flips the sign of f, with no physical consequence. We fix $f = v > 0$.) The constant f is not a tree-level artifact: Eq. (29.83) is an exact operator statement, valid order by order, with f receiving quantum corrections but never vanishing as long as the symmetry is broken—this is what the next section establishes, and it is what makes f the physical normalization of the pion in Chapter 39, where it becomes the pion decay constant f_π.

The Goldstone is a massless spin-zero particle. It is worth recording precisely *what kind of particle* Eq. (29.83) has produced, in the Wigner classification of Chapter 3. The fields π_a are real scalars: they carry no spacetime index, so the one-particle states $|\pi_a(p)\rangle$ transform trivially under the little group and have spin 0 (the scalar case of the massless little group ISO(2) of Section 3.7.3, where the helicity is fixed to zero by Eq. (3.217)). And they are massless: the on-shell condition carried by the states $|\pi_a(p)\rangle$ is $p^2 = 0$, the vanishing of the mass Casimir P^2 (Theorem 3.36), in accord with the zero eigenvalues $m_\pi^2 = 0$ of the Hessian. The Goldstone bosons are therefore exactly the massless, spin-0 representations of the Poincaré group—the simplest particles the relativistic classification allows—and the matrix element Eq. (29.83) states that the broken symmetry current *is the interpolating operator* for them: acting on the vacuum, J_a^μ has a nonzero amplitude to make precisely one such quantum.

We have now established, at tree level and by explicit construction in the two canonical models, the structural fact that organizes the rest of the chapter: a spontaneously broken continuous global symmetry forces massless spin-zero modes into the spectrum, one per broken generator, and the broken current connects each of them to the vacuum through $\langle 0|J_a^\mu|\pi_b\rangle = if\,p^\mu\delta_{ab}$. The next section takes these two gifts—the appearance of massless modes, and this current–Goldstone matrix element—and promotes them from a tree-level computation in a particular potential to an exact, non-perturbative theorem valid in any Lorentz-invariant theory with a positive-definite Hilbert space, the matrix element Eq. (29.83) itself becoming the engine of the operator proof.

29.3 Goldstone's Theorem

The linear sigma model of the previous section produced massless modes by an explicit, model-specific computation: we wrote the O(N)-invariant potential, found its minimum on the sphere $|\Phi| = v$, expanded around a chosen vacuum, and read $N-1$ vanishing eigenvalues off the Hessian (Section 29.2.3). Three facts about that result demand to be lifted out of the example. First, the masslessness

was traced to motion *along* the degenerate vacuum manifold — a geometric statement that ought to hold for any potential with the same symmetry, not just the quartic one we happened to write. Second, the counting "$N-1$" was exactly the number of broken rotations, suggesting a representation-theoretic law. Third, and most importantly, the whole argument was *classical*: it diagnosed the spectrum from the tree-level potential, and a skeptic could ask whether loop corrections spoil it. This section settles all three at once. We promote the appearance of massless scalars to an exact, non-perturbative theorem — *Goldstone's theorem* — valid to all orders and independent of any Lagrangian, and we prove it two independent ways. The first proof is operatorial: it exposes the massless particle directly in the Hilbert space, as a sharp pole forced into the spectral representation of a current–field commutator by Lorentz invariance and current conservation, with its residue pinned by the very order parameter $\langle 0|[Q_a,\phi]|0\rangle \neq 0$ that signals breaking. The second proof is geometric: it locates the same massless mode as a flat direction of the quantum effective potential, a zero eigenvalue of its Hessian enforced by a Ward identity. The two proofs see the same fact from opposite ends — the rigorous state content and the effective-action curvature — and the reader should hold both.

29.3.1 Hypotheses, and a subtlety about the charge

Goldstone's theorem is a statement about a relativistic quantum field theory. Its conclusion is strong, so its hypotheses must be stated with care; every one of them is load-bearing, and the final subsection exhibits exactly what fails when each is dropped. We collect them now.

(G1) *Poincaré invariance.* The theory is invariant under the connected Poincaré group, and there is a Poincaré-invariant vacuum $|0\rangle$ (Theorem 9.11): $U(\Lambda,a)\,|0\rangle = |0\rangle$ for every Lorentz transformation Λ and translation a. In particular $P^\mu\,|0\rangle = 0$.

(G2) *A conserved current.* There is a continuous global symmetry with an associated Noether current $J^\mu_a(x)$ obeying the operator conservation law $\partial_\mu J^\mu_a = 0$, and charges $Q_a = \int \mathrm{d}^3x\, J^0_a(x)$ generating the symmetry on the fields as in Chapter 7 (Theorems 7.20 and 7.21).

(G3) *A nonzero order parameter.* There is a local operator $A_k(x)$ — in the simplest case an elementary field ϕ_k, in general any local composite — whose symmetry variation has nonvanishing vacuum expectation, $\langle 0|[Q_a, A_k(0)]|0\rangle \neq 0$ for at least one generator index a. This is the operative definition of spontaneous breaking: the charge does not annihilate the vacuum.

(G4) *A positive-definite Hilbert space and locality.* The physical states form a Hilbert space with positive-definite inner product (no indefinite "metric"),

and the fields satisfy microcausality: $[\mathcal{O}(x), \mathcal{O}'(y)] = 0$ for spacelike $(x - y)^2 < 0$.

Hypothesis (G3) deserves a warning, because the charge Q_a that appears in it does not, strictly, exist as an operator once the symmetry is broken.

Caution

When a continuous symmetry is spontaneously broken, $Q_a |0\rangle$ has *infinite norm*, so Q_a is not a well-defined operator on the Hilbert space and $Q_a |0\rangle$ is not a normalizable state. Heuristically, $Q_a |0\rangle$ is a zero-momentum, infinitely-extended superposition of soft Goldstone quanta; creating a Goldstone of exactly zero momentum costs zero energy but is spread over all of space, and the norm diverges with the volume. The cure is to never integrate J^0_a over all space inside a state. Every statement below is phrased through the *current*, or through the spatial integral of a *commutator*

$$\langle 0|[Q_a, A_k(0)]|0\rangle \equiv \lim_{R\to\infty} \int_{|\mathbf{x}|\le R} \mathrm{d}^3x \, \Big\langle 0\Big|\Big[J^0_a(x^0, \mathbf{x}), A_k(0)\Big]\Big|0\Big\rangle , \quad (29.84)$$

which is perfectly finite: microcausality (G4) makes the integrand vanish once $|\mathbf{x}|$ exceeds the (timelike or lightlike) reach of $A_k(0)$, so the limit is reached at finite R and is, as we will see, even independent of x^0. Read Eq. (29.84) as the *definition* of the symbol $\langle 0|[Q_a, A_k]|0\rangle$; it is the order parameter whether or not Q_a itself makes sense.

Definition 29.10 (Order parameter; broken and unbroken generators)**.** The complex numbers $c_{ak} \equiv \langle 0|[Q_a, A_k(0)]|0\rangle$, defined by Eq. (29.84), are the *order parameters* of the symmetry. A generator T^a is *spontaneously broken* if $c_{ak} \neq 0$ for some k (equivalently $Q_a |0\rangle \neq 0$), and *unbroken* if $Q_a |0\rangle = 0$ (then $c_{ak} = 0$ for all k). The unbroken generators close on a subalgebra $\mathfrak{h} \subseteq \mathfrak{g}$, the Lie algebra of the *stabilizer subgroup* $H = \mathrm{Stab}_G(|0\rangle)$.

That $\mathfrak{h}$ is a subalgebra is immediate from the charge algebra Theorem 7.21: if $Q_a |0\rangle = Q_b |0\rangle = 0$ then $[Q_a, Q_b] |0\rangle = f_{abc} Q_c |0\rangle = 0$, so the bracket of two unbroken charges is unbroken. When $A_k = \phi_k$ is an elementary scalar with vacuum expectation $v_k \equiv \langle 0|\phi_k(0)|0\rangle$, the order parameter takes an explicit form. Quantizing the Poisson bracket of Theorem 7.20 via $[\,\cdot\,,\,\cdot\,] = i\,\{\,\cdot\,,\,\cdot\,\}_{\mathrm{P}}$ gives the operator relation

$$i\,[Q_a, \phi_k(x)] = \delta_a \phi_k(x) = (t^a)_{kj}\, \phi_j(x), \qquad t^a = -iT^a, \quad (29.85)$$

so that

$$c_{ak} = \langle 0|[Q_a, \phi_k]|0\rangle = -i\,(t^a)_{kj}\, v_j = -\,(T^a)_{kj}\, v_j. \quad (29.86)$$

A generator is broken precisely when $T^a v \neq 0$ — when it moves the vacuum expectation value — and unbroken when $T^a v = 0$. This is the algebraic face of

the geometric statement of the previous section: broken generators are the ones that push v along the vacuum manifold.

29.3.2 Statement of the theorem

Theorem 29.11 (Goldstone)**.** *Let a relativistic quantum field theory satisfy* (G1)–(G4)*. Then for every spontaneously broken generator — every a for which the order parameter Eq. (29.84) is nonzero — the spectrum contains a massless, spin-zero one-particle state $|\pi_a(\mathbf{p})\rangle$ (a* Nambu–Goldstone boson*) that couples to the broken current with a nonzero matrix element*

$$\langle 0|J^\mu_a(0)|\pi_b(\mathbf{p})\rangle = i\,p^\mu\,F_{ab}, \qquad \det F \neq 0 \text{ on the broken subspace,} \tag{29.87}$$

and to the order-parameter operator with $\langle\pi_b(\mathbf{p})|A_k(0)|0\rangle \neq 0$. The number of independent such states equals the number of broken generators, $\dim G - \dim H$*, where $H = \mathrm{Stab}_G(|0\rangle)$ is the unbroken subgroup.*

The constants F_{ab} are the *Goldstone decay constants* (in the chiral application of Chapter 39 the analogous constant is the pion decay constant f_π). The remainder of this section is the proof: first the masslessness and the coupling Eq. (29.87) by the spectral method, then the spin-zero assertion by Wigner's classification, then the counting, and finally the independent effective-potential proof.

29.3.3 First proof: the current–field spectral representation

The strategy mirrors the Källén–Lehmann analysis of Chapter 14: take the operator product of the broken current with the order-parameter field, slide a complete set of states between them, and let Poincaré covariance fix everything that can appear. Conservation of the current is then a single differential constraint that annihilates the entire spectrum except a massless point. We carry out every step.

The object and its spectral function. Fix a broken generator a and a component k with $c_{ak} \neq 0$, and study the Wightman function of the current with the field,

$$W^\mu_{ak}(x) \;\equiv\; \langle 0|J^\mu_a(x)\,A_k(0)|0\rangle\,. \tag{29.88}$$

Translation invariance (G1), $J^\mu_a(x) = e^{iP\cdot x}J^\mu_a(0)e^{-iP\cdot x}$ with $P\,|0\rangle = 0$, lets us insert a complete set of momentum eigenstates $\mathbf{1} = \sum_n |n\rangle\langle n|$, $P^\mu\,|n\rangle = p^\mu_n\,|n\rangle$:

$$W^\mu_{ak}(x) = \sum_n e^{-ip_n\cdot x}\;\langle 0|J^\mu_a(0)|n\rangle\,\langle n|A_k(0)|0\rangle\,. \tag{29.89}$$

The vacuum itself drops out of the sum: $\langle 0|J^\mu_a(0)|0\rangle$ is a Lorentz vector built from no momentum, hence zero. Every surviving $|n\rangle$ has $p^0_n > 0$ and $p^2_n \geq 0$ (the

spectrum lies in the closed forward cone). Now package the matrix elements into a single distribution by writing the sum as an integral over the total four-momentum p:

$$\sigma^{\mu}_{ak}(p) \;\equiv\; (2\pi)^3 \sum_n \delta^4(p-p_n)\; \langle 0|J^{\mu}_a(0)|n\rangle\, \langle n|A_k(0)|0\rangle\,, \qquad W^{\mu}_{ak}(x) = \int \frac{\mathrm{d}^4 p}{(2\pi)^3}\, e^{-ip\cdot x}\, \sigma^{\mu}_{ak}(p). \tag{29.90}$$

Covariance fixes the tensor structure. The spectral function $\sigma^{\mu}_{ak}(p)$ is a Lorentz vector — it carries the single index of J^{μ}_a, while A_k is a Lorentz scalar (its index k is an internal flavor label, inert under the Lorentz group) and $|0\rangle$ is invariant. It is supported in the closed forward cone. The only vector that can be built covariantly from a four-momentum p in the forward cone is p^{μ} itself: at any timelike p the little group is SO(3), whose only invariant three-vector is the zero vector, so the only invariant four-vector is the time direction $\propto p^{\mu}$; at any null p the little group is ISO(2) (Section 3.7.3), and again the only invariant vector is $\propto p^{\mu}$. Hence

$$\sigma^{\mu}_{ak}(p) = p^{\mu}\, \theta(p^0)\, \rho_{ak}(p^2), \tag{29.91}$$

for a scalar density $\rho_{ak}(s)$ supported on $s \geq 0$. This is the entire content of Lorentz invariance: the current–field correlator is a derivative of a superposition of free fields of every mass, weighted by ρ_{ak}.

Assembling the commutator. Insert $1 = \int_0^{\infty} \mathrm{d}s\, \delta(p^2 - s)$ into Eq. (29.90) and use $p^{\mu} e^{-ip\cdot x} = i\, \partial^{\mu}_x e^{-ip\cdot x}$:

$$W^{\mu}_{ak}(x) = i\, \partial^{\mu}_x \int_0^{\infty} \mathrm{d}s\; \rho_{ak}(s)\, D_+(x;s), \qquad D_+(x;s) \equiv \int \frac{\mathrm{d}^4 p}{(2\pi)^3}\, \theta(p^0)\, \delta(p^2 - s)\, e^{-ip\cdot x}, \tag{29.92}$$

where $D_+(x;s)$ is exactly the free positive-frequency Wightman function of a scalar of mass-squared s met in Chapter 14. The opposite ordering is obtained the same way, with the roles of the matrix elements exchanged and $p^{\mu} e^{+ip\cdot x} = -i\, \partial^{\mu}_x e^{+ip\cdot x}$:

$$\langle 0|A_k(0)\, J^{\mu}_a(x)|0\rangle = -i\, \partial^{\mu}_x \int_0^{\infty} \mathrm{d}s\; \bar{\rho}_{ak}(s)\, D_+(-x;s), \tag{29.93}$$

with $\bar{\rho}_{ak}$ the spectral density of the reversed product. The commutator $F^{\mu}_{ak}(x) \equiv \langle 0|[J^{\mu}_a(x), A_k(0)]|0\rangle$ is the difference,

$$F^{\mu}_{ak}(x) = i\, \partial^{\mu}_x \int_0^{\infty} \mathrm{d}s\; \big[\rho_{ak}(s)\, D_+(x;s) + \bar{\rho}_{ak}(s)\, D_+(-x;s)\big]. \tag{29.94}$$

Here locality enters. By microcausality (G4) the commutator must vanish for spacelike x. For spacelike argument the Wightman function is even, $D_+(x;s) =$

$D_+(-x;s)$, so the bracket in Eq. (29.94) reduces to $[\rho_{ak}(s)+\bar\rho_{ak}(s)]\,D_+(x;s)$, and since $\partial_x^\mu D_+(x;s)$ is not identically zero on the spacelike region, vanishing for *all* spacelike x and all s forces $\bar\rho_{ak}(s)=-\rho_{ak}(s)$. The bracket then collapses onto the Pauli–Jordan commutator function,

$$\Delta(x;s)\;\equiv\;D_+(x;s)-D_+(-x;s)=\int\frac{\mathrm{d}^4p}{(2\pi)^3}\,\varepsilon(p^0)\,\delta(p^2-s)\,e^{-ip\cdot x},\qquad \varepsilon(p^0)\equiv\mathrm{sgn}(p^0), \tag{29.95}$$

and we arrive at the spectral representation of the current–field commutator,

$$\boxed{F^\mu_{ak}(x)=\langle 0|[J^\mu_a(x),A_k(0)]|0\rangle=i\,\partial^\mu_x\int_0^\infty \mathrm{d}s\;\rho_{ak}(s)\,\Delta(x;s).} \tag{29.96}$$

The function $\Delta(x;s)$ is real, odd in x, vanishes for spacelike x, and solves the Klein–Gordon equation $(\Box_x+s)\Delta(x;s)=0$. Two of its values we will need directly are the equal-time ones; from Eq. (29.95), doing the p^0-integral against the two roots $p^0=\pm\sqrt{\mathbf{p}^2+s}$,

$$\Delta(x;s)\big|_{x^0=0}=0,\qquad \partial_{x^0}\Delta(x;s)\big|_{x^0=0}=-\,i\,\delta^3(\mathbf{x}), \tag{29.97}$$

both *independent of* s — the first because the $\varepsilon(p^0)$ weights of the two roots cancel, the second because they add, leaving the spatial Fourier representation of $-i\,\delta^3(\mathbf{x})$.

Conservation kills everything but $s=0$. Apply ∂_μ to Eq. (29.96). Current conservation (G2) is the operator identity $\partial_\mu J^\mu_a=0$, so the left side vanishes; on the right, $\partial_\mu\partial^\mu_x=\Box_x$ acts on $\Delta(x;s)$, and $\Box_x\Delta(x;s)=-s\,\Delta(x;s)$:

$$0=\partial_\mu F^\mu_{ak}(x)=i\,\Box_x\int_0^\infty \mathrm{d}s\;\rho_{ak}(s)\,\Delta(x;s)=-\,i\int_0^\infty \mathrm{d}s\;s\,\rho_{ak}(s)\,\Delta(x;s). \tag{29.98}$$

The Pauli–Jordan functions $\{\Delta(\cdot\,;s)\}_{s\ge0}$ are linearly independent distributions — in momentum space each is supported on a distinct mass shell $p^2=s$ — so the only way the superposition can vanish for all x is for its weight to vanish pointwise:

$$s\,\rho_{ak}(s)=0\quad\text{for all } s\ge0\quad\Longrightarrow\quad \rho_{ak}(s)=c_{ak}\,\delta(s), \tag{29.99}$$

for some constant c_{ak}: a distribution supported only at $s=0$ is a multiple of $\delta(s)$ (no derivatives of δ survive, as those would not satisfy the positivity inherited from the state sum). *All of the spectral weight of the current–field commutator is concentrated at zero mass-squared.*

The order parameter fixes the residue. The constant c_{ak} is not free: it is the order parameter. Integrate the $\mu = 0$ component of Eq. (29.96) over space at fixed time and use the equal-time data Eq. (29.97) together with Eq. (29.99):

$$\int \mathrm{d}^3x\; F^0_{ak}(x)\Big|_{x^0=0} = i\int \mathrm{d}^3x\; \partial^0_x \int_0^\infty \mathrm{d}s\; \rho_{ak}(s)\,\Delta(x;s) = i\,c_{ak}\int \mathrm{d}^3x\; \partial_{x^0}\Delta(x;0)\big|_{x^0=0} = i\,c_{ak}\,(-i) = \tag{29.100}$$

But the left side is, by definition Eq. (29.84), the order parameter $\langle 0|[Q_a, A_k]|0\rangle$. (The evaluation at $x^0 = 0$ is legitimate and the answer is time-independent: differentiating $\int \mathrm{d}^3x\, F^0_{ak}$ in x^0 gives $-\int \mathrm{d}^3x\, \partial_i F^i_{ak}$ by conservation, a surface term that vanishes.) Therefore

$$\boxed{c_{ak} = \langle 0|[Q_a, A_k(0)]|0\rangle} \tag{29.101}$$

and, for the elementary scalar, $c_{ak} = -(T^a)_{kj}v_j$ by Eq. (29.86). The chain is now closed. Spontaneous breaking (G3) means $c_{ak} \neq 0$ for some a, k; then by Eq. (29.99) the spectral density $\rho_{ak}(s) = c_{ak}\delta(s)$ is nonzero and supported entirely at $s = 0$. Reading Eq. (29.99) back through the definition Eq. (29.90), a nonzero ρ_{ak} at $s = 0$ means there exist states $|n\rangle$ with $p_n^2 = 0$ — *massless* states — for which both $\langle 0|J^\mu_a(0)|n\rangle$ and $\langle n|A_k(0)|0\rangle$ are nonzero. The massless particle exists, it couples to the broken current, and it couples to the order-parameter operator. Matching the single-particle piece of Eq. (29.91) to $\langle 0|J^\mu_a(0)|\pi_b(\mathbf{p})\rangle\,\langle \pi_b|A_k|0\rangle$ identifies the structure $\langle 0|J^\mu_a(0)|\pi_b(\mathbf{p})\rangle = i\,p^\mu F_{ab}$ of Eq. (29.87), with the decay-constant matrix F nonsingular on the broken subspace because the order parameters c_{ak} there are independent (a fact we prove in the counting argument below). □

Physical Insight

The $\delta(s)$ in Eq. (29.99) is not merely "weight near zero mass" — it is an isolated, zero-width spike *at* $s = 0$. A multiparticle continuum would smear the spectral density over an interval $s \geq s_{\mathrm{thr}} > 0$ opening at the lowest threshold; only a *stable single particle of exactly zero mass* produces a sharp $\delta(s)$. Conservation does not just make something light; it manufactures an exactly massless, stable one-particle pole and forbids anything else in this channel. Figure 29.3 shows the collapse: a generic spectral density (a continuum plus perhaps a massive pole) is annihilated by the factor s in Eq. (29.98), leaving only the massless point, whose residue is the order parameter.

Historical Note

The massless mode was first seen in 1960 by Nambu in the theory of superconductivity, where the broken U(1) phase symmetry of the BCS

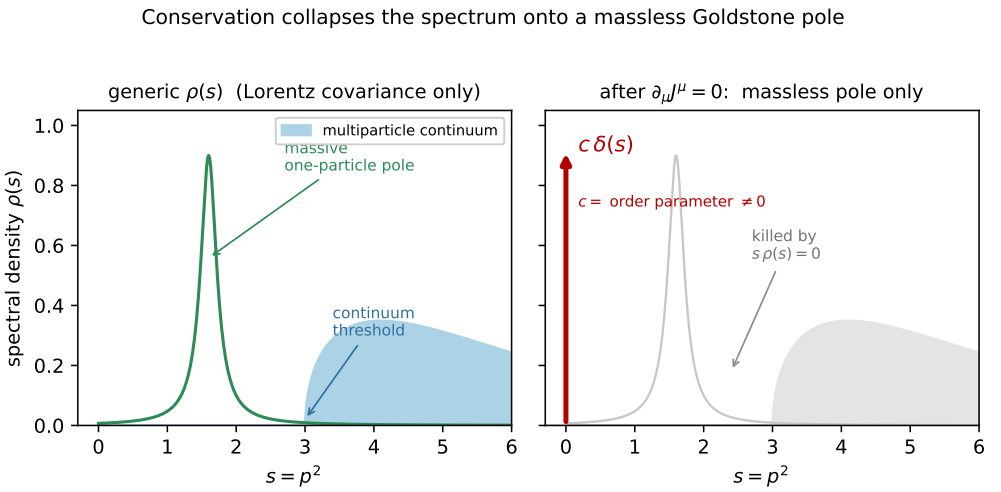


Figure 29.3: The spectral collapse behind Goldstone's theorem. *Left:* Lorentz invariance alone permits a generic spectral density $\rho(s)$ for the current–field commutator, supported on $s = p^2 \geq 0$ — a multiparticle continuum and possibly an isolated massive pole. *Right:* current conservation multiplies ρ by s and demands the result vanish (Eq. (29.98)); every contribution with $s > 0$ is killed, and only a massless $\delta(s)$ survives, with residue equal to the order parameter $\langle 0|[Q_a, A_k]|0\rangle \neq 0$. That surviving pole is the Goldstone boson.

ground state forces a gapless collective excitation [357], and developed with Jona-Lasinio into a relativistic model of nucleon mass generation [358]. Goldstone abstracted the phenomenon to a general scalar field theory in 1961 [231], and the theorem was proved in generality — essentially the spectral argument above — by Goldstone, Salam, and Weinberg in 1962 [232]. The same paper already flagged the escape clause that became the Higgs mechanism: the proof's reliance on a manifestly Lorentz-covariant, positive-metric formulation, which a gauge theory cannot supply.

29.3.4 The Goldstone boson is spinless

The spectral proof produced a massless one-particle state; we now show it carries spin zero, completing Theorem 29.11. The cleanest argument uses Wigner's classification of massless particles by helicity (Section 3.7.3) and the fact that the order-parameter operator is a Lorentz scalar.

A massless one-particle state $|\pi(\mathbf{p})\rangle$ has $P^2\,|\pi\rangle = 0$ (Theorem 3.36) and, by Wigner, is labeled by a definite helicity h: under a rotation R_θ by angle θ about the spatial direction $\hat{\mathbf{p}}$ — an element of the little group ISO(2) that fixes the

null momentum p — the state transforms by a phase,

$$U(R_\theta)\,|\pi_h(\mathbf{p})\rangle = e^{-ih\theta}\,|\pi_h(\mathbf{p})\rangle\,. \tag{29.102}$$

Now consider the matrix element $\langle\pi_h(\mathbf{p})|A_k(0)|0\rangle$ that the spectral proof showed to be nonzero. The operator $A_k(0)$ is evaluated at the origin, which R_θ leaves fixed, and it is a Lorentz scalar, so $U(R_\theta)A_k(0)U(R_\theta)^{-1} = A_k(0)$; the vacuum is invariant, $U(R_\theta)\,|0\rangle = |0\rangle$. Inserting $U(R_\theta)^{-1}U(R_\theta)$ and using Eq. (29.102),

$$\langle\pi_h(\mathbf{p})|A_k(0)|0\rangle = \Big\langle\pi_h(\mathbf{p})\Big|U(R_\theta)^{-1}\,A_k(0)\,U(R_\theta)\Big|0\Big\rangle = e^{+ih\theta}\;\langle\pi_h(\mathbf{p})|A_k(0)|0\rangle\,. \tag{29.103}$$

For this to hold for *all* θ with a nonzero matrix element, the phase must be trivial: $h = 0$. The Goldstone boson has helicity zero — it is a genuine scalar particle. The same conclusion follows from the current side: a helicity-±1 massless state would require its current matrix element to be built from a transverse polarization vector $\epsilon^\mu(\mathbf{p})$ with $\epsilon\cdot p = 0$, whereas Eq. (29.87) found $\langle0|J^\mu_a|\pi\rangle \propto p^\mu$, the longitudinal structure characteristic of helicity 0. Both faces agree: the Nambu–Goldstone boson is a massless spin-0 particle. □

29.3.5 Counting: one boson per broken generator

It remains to count. The spectral proof attaches a massless state to every nonzero order parameter; we must show that the number of *independent* massless states equals the number of broken generators, $\dim G - \dim H$, and that this is the dimension of the vacuum manifold G/H.

Theorem 29.12 (Goldstone counting)**.** *Under* (G1)–(G4)*, with G the connected global symmetry group and $H = \mathrm{Stab}_G(|0\rangle)$ the unbroken subgroup, the number of independent Nambu–Goldstone bosons is*

$$N_{\mathrm{GB}} = \dim G - \dim H = \dim(G/H), \tag{29.104}$$

the dimension of the vacuum manifold, i.e. of the orbit $G\cdot|0\rangle$.

Proof. Work with an elementary order-parameter multiplet ϕ_i and its vacuum value $v_i = \langle0|\phi_i|0\rangle$; the argument is identical for a composite A_k with v replaced by its expectation. Consider the linear map from the Lie algebra to field space defined by the action on the vacuum value,

$$L_v : \mathfrak{g} \longrightarrow V, \qquad X = \alpha^a t^a \;\longmapsto\; X\,v = \alpha^a\,t^a v, \tag{29.105}$$

where V is the (real) representation space of the multiplet. By Theorem 29.10 and Eq. (29.86), the kernel of L_v is exactly the unbroken subalgebra,

$$\ker L_v = \{X\in\mathfrak{g}:\; Xv = 0\} = \mathfrak{h}, \tag{29.106}$$

since $Xv = 0$ is the statement $T^a v = 0$ for the generators composing X, which is the condition $Q_a |0\rangle = 0$. By the rank–nullity theorem the image has dimension

$$\dim(\operatorname{im} L_v) = \dim \mathfrak{g} - \dim \ker L_v = \dim G - \dim H. \tag{29.107}$$

Choose a basis $\{t^{\hat{a}}\}$, $\hat{a} = 1, \ldots, \dim G - \dim H$, of a complement $\mathfrak{m}$ of $\mathfrak{h}$ in $\mathfrak{g}$; then the vectors $\{t^{\hat{a}} v\}$ are linearly independent in V, because L_v restricted to $\mathfrak{m}$ is injective. Each $t^{\hat{a}} v \neq 0$ is a nonzero order parameter, so by the spectral proof of Section 29.3.3 there is a massless Goldstone state $|\pi_{\hat{a}}\rangle$ coupling to A_k through that particular combination. The matrix of order parameters $c_{\hat{a}k} = -(T^{\hat{a}} v)_k$ has rank $\dim G - \dim H$ — its rows are the independent vectors $t^{\hat{a}} v$ (up to the factor $-i$) — so the spectral densities $\rho_{\hat{a}k}(s) = c_{\hat{a}k}\delta(s)$ span a $(\dim G - \dim H)$-dimensional space of massless couplings, and hence there are at least $\dim G - \dim H$ independent massless one-particle states. In the standard case of a linearly realized symmetry no *additional* accidentally massless scalars appear in the broken channel, and the count is exact: $N_{\mathrm{GB}} = \dim G - \dim H$.

Finally, the geometric reading. The orbit of the vacuum under G, $G \cdot |0\rangle \cong G/H$ by the orbit–stabilizer correspondence, is the set of degenerate vacua — the vacuum manifold of Section 29.2.3. Its tangent space at v is precisely $\{Xv : X \in \mathfrak{g}\} = \operatorname{im} L_v$, of dimension $\dim G - \dim H$ by Eq. (29.107). The broken generators are in one-to-one correspondence with the directions one can move v along the vacuum manifold at no cost in energy, and these flat directions are the Goldstone bosons. The counting is the dimension of G/H. □

Physical Insight

The two descriptions of $\dim G - \dim H$ — "broken generators" and "dimensions of the vacuum manifold" — are the algebraic and geometric sides of one coin. Algebraically, a Goldstone boson is the quantum of a charge that fails to annihilate the vacuum. Geometrically, it is an oscillation along the flat valley G/H of degenerate vacua, which costs no potential energy and therefore has no mass gap. The previous section's Hessian computation was this geometry made local; Theorem 29.11 makes it exact and operator-level.

We now check the count against the running example.

Worked Example 29.13 (Counting Goldstones in the O(N) linear sigma model)**.** Continue ϕ^4 theory in its multicomponent form, the O(N) linear sigma model of Section 29.2.3: a real N-plet $\Phi = (\phi_1, \ldots, \phi_N)$ with O(N)-invariant potential $V = \frac{1}{2} m^2 \Phi^\top \Phi + \frac{\lambda}{4} (\Phi^\top \Phi)^2$, $m^2 < 0$, minimized on the sphere $|\Phi| = v$, $v = \sqrt{-m^2/\lambda}$. Choose the vacuum along the last axis,

$$v_i = v\, \delta_{iN}, \qquad i = 1, \ldots, N. \tag{29.108}$$

The group and its generators. Here $G = \mathrm{O}(N)$ (we use its connected component $\mathrm{SO}(N)$, whose Lie algebra carries the physics), with $\dim \mathrm{SO}(N) = \frac{1}{2}N(N-1)$. A basis of the real antisymmetric generators is labeled by ordered pairs $(p < q)$,

$$\big(T^{(pq)}\big)_{ij} = -i\big(\delta_{pi}\delta_{qj} - \delta_{qi}\delta_{pj}\big), \tag{29.109}$$

the Hermitian generator of the rotation in the (p, q) plane; there are $\binom{N}{2} = \frac{1}{2}N(N-1)$ of them.

Which generators are broken. Act with $T^{(pq)}$ on the vacuum Eq. (29.108). Using Eq. (29.109),

$$\big(T^{(pq)}v\big)_i = -i\big(\delta_{pi}\delta_{qj} - \delta_{qi}\delta_{pj}\big)v\,\delta_{jN} = -i\,v\,\big(\delta_{pi}\,\delta_{qN} - \delta_{qi}\,\delta_{pN}\big). \tag{29.110}$$

This vanishes unless one of the indices in the pair (p, q) equals N. There are two cases.

- *Unbroken:* $p, q \le N-1$ (neither index is N). Then $\delta_{qN} = \delta_{pN} = 0$ and $T^{(pq)}v = 0$: these are the rotations among the first $N-1$ axes, which fix the chosen vacuum. They generate the stabilizer $H = \mathrm{O}(N-1)$, with $\dim \mathrm{SO}(N-1) = \frac{1}{2}(N-1)(N-2)$ generators.
- *Broken:* $q = N$, $p = i_0 \le N-1$. Then $\big(T^{(i_0 N)}v\big)_i = -i\,v\,\delta_{i\,i_0} \neq 0$: this rotation tilts the vacuum out of the last axis toward axis i_0. There are exactly $N-1$ such generators, $T^{(1N)}, \dots, T^{(N-1\,N)}$, and their images $\{-iv\,\hat{e}_{i_0}\}_{i_0=1}^{N-1}$ are manifestly linearly independent.

The count. The number of Goldstone bosons is therefore

$$N_{\mathrm{GB}} = \dim \mathrm{SO}(N) - \dim \mathrm{SO}(N-1) = \tfrac{1}{2}N(N-1) - \tfrac{1}{2}(N-1)(N-2) = \tfrac{1}{2}(N-1)\big[N-(N-2)\big] = N- \tag{29.111}$$

This matches Theorem 29.12 with $\dim(G/H) = \dim\big(\mathrm{O}(N)/\mathrm{O}(N-1)\big) = \dim S^{N-1} = N-1$: the vacuum manifold is the sphere of radius v, and the $N-1$ Goldstones are oscillations along it.

Agreement with the tree-level Hessian. The previous section read the spectrum directly off the curvature of V at the minimum: writing $\Phi = v\,\hat{e}_N + (\pi_1, \dots, \pi_{N-1}, \eta)$, the $N-1$ transverse fluctuations π_{i_0} are flat ($m_\pi^2 = 0$) while the radial mode η is massive ($m_\eta^2 = -2m^2 = 2\lambda v^2$). The $N-1$ massless π_{i_0} are exactly the quanta of the broken directions $T^{(i_0 N)}v \propto \hat{e}_{i_0}$ found here: the operator counting and the Hessian counting agree, as Theorem 29.11 guarantees they must, and Goldstone's theorem promises the masslessness survives to all loop orders. For $N = 2$ this is the $\mathrm{U}(1) \simeq \mathrm{SO}(2)$ Mexican hat, with the single ($N - 1 = 1$) phase Goldstone; for $N = 4$, the case relevant to the Higgs sector of Chapter 31, breaking $\mathrm{O}(4) \to \mathrm{O}(3)$ yields three Goldstones — the would-be longitudinal modes of $W^{\pm}, Z$.

29.3.6 Second proof: curvature of the effective potential

The spectral proof is rigorous and non-perturbative, but it works in the Hilbert space and never touches a Lagrangian. There is a complementary proof that works entirely in terms of the effective potential $V_{\text{eff}}(\phi_{\text{cl}})$ — the constant-field value of the quantum effective action $\Gamma[\phi_{\text{cl}}]$ of Chapter 10, whose construction and properties are the business of the next section. We state the argument now, because it is the geometric counterpart of the previous section's Hessian computation and because it is what makes "massless to all orders" a precise, checkable statement; we then forward the one missing ingredient.

The exact effective potential inherits the symmetry of the theory: if the microscopic action is invariant under $\phi \to (1 + \alpha^a t^a)\phi$, so is V_{eff}, because the symmetry is realized linearly on the fields and the measure is invariant (the precise statement, and its one subtlety when anomalies are present, is Theorem 29.21 of the next section). Invariance of V_{eff} under the infinitesimal transformation is the identity

$$V_{\text{eff}}(\phi + \alpha^a t^a \phi) = V_{\text{eff}}(\phi) \quad \Longrightarrow \quad \frac{\partial V_{\text{eff}}}{\partial \phi_i}\,(t^a\phi)_i = 0 \quad \text{for every } a, \text{ at every } \phi. \tag{29.112}$$

This is a Ward identity for Γ: the first derivative of the effective potential is orthogonal to every symmetry direction. Differentiate it once more, with respect to ϕ_j, using the product rule:

$$\frac{\partial^2 V_{\text{eff}}}{\partial \phi_j \partial \phi_i}\,(t^a\phi)_i + \frac{\partial V_{\text{eff}}}{\partial \phi_i}\,(t^a)_{ij} = 0. \tag{29.113}$$

Now evaluate at a vacuum, a minimum $\phi = v$ where the first derivative vanishes, $\partial V_{\text{eff}}/\partial\phi_i\big|_v = 0$. The second term in Eq. (29.113) drops, leaving

$$\boxed{\left(M^2\right)_{ji}(t^a v)_i = 0, \qquad \left(M^2\right)_{ji} \equiv \left.\frac{\partial^2 V_{\text{eff}}}{\partial \phi_j \partial \phi_i}\right|_{\phi = v}.} \tag{29.114}$$

The Hessian of the effective potential at the vacuum is the *exact* mass matrix — it is the inverse propagator at zero momentum, hence the squared physical masses to all orders — and Eq. (29.114) says precisely that every broken direction $t^a v \neq 0$ is a null eigenvector of it. Each independent broken generator thus contributes one exactly massless mode, and the number of zero eigenvalues is the number of independent vectors $t^a v$, which is $\dim G - \dim H$ by the rank–nullity argument of Theorem 29.12. This reproduces both the masslessness and the count, now from the curvature of V_{eff}.

Two remarks tie this to the rest of the chapter. First, at tree level $V_{\text{eff}} \to V$, the classical potential, and Eq. (29.114) is exactly the Hessian-eigenvalue statement of Section 29.2.3: the broken directions are flat because the potential is

flat along the orbit. Second, because Eq. (29.112) holds for the *full* V_{eff}, the zero eigenvalue persists order by order in the loop expansion — this is the quantitative meaning of "Goldstone bosons stay massless after radiative corrections," and it is the precise demand we hand to the next section. The one debt outstanding is the construction of V_{eff} and the proof that it really is symmetric; that is Theorem 29.21, to which we now owe the completion of this proof.

29.3.7 Scope: where the theorem holds, and where it does not

Theorem 29.11 is a theorem about *relativistic* field theories with a *positive-definite* Hilbert space. Each of those qualifiers is a real restriction, and the most important physics of the next two chapters lives in the cracks. We catalogue the boundaries honestly.

Positive metric, and the gauge loophole. Hypothesis (G4) — positive-definite inner product together with a manifestly Lorentz-covariant current — cannot be met simultaneously by a *gauge* symmetry. A covariant quantization of a gauge field has an indefinite-metric state space (the Gupta–Bleuler/BRST construction of Chapters 13 and 26), so the spectral positivity that forced a genuine massless particle no longer applies; in a physical (unitary) gauge Lorentz covariance of the current is instead given up. The would-be Goldstone boson is then not a physical particle: it is "eaten," becoming the longitudinal polarization of a gauge boson that acquires a mass. This is the Higgs mechanism, the subject of Chapter 30, and it is the single most important evasion of Goldstone's theorem in the Standard Model. The $\mathrm{O}(4) \to \mathrm{O}(3)$ example of Theorem 29.13 is exactly this story when the symmetry is gauged: its three Goldstones become the longitudinal $W^{\pm}, Z$.

Two dimensions or fewer. In $d \le 2$ spacetime dimensions a continuous symmetry *cannot* be spontaneously broken at all: the would-be Goldstone is a massless scalar, whose propagator $\sim \int \mathrm{d}^d p \,/\, p^2$ is infrared divergent for $d \le 2$, and the resulting fluctuations destroy the long-range order, restoring the symmetry of any candidate ordered vacuum. This is the Coleman–Mermin–Wagner theorem [127, 347]. There is no contradiction with Theorem 29.11: with no breaking there is nothing for the theorem to act on. We note this subtlety and return to its field-theoretic version when we discuss two-dimensional models.

Loss of Lorentz invariance: nonrelativistic and finite-density counting. Hypothesis (G1) is essential to the counting. In a nonrelativistic medium or at finite density — a ferromagnet, a superfluid, a neutron star interior — Lorentz invariance is broken by the rest frame of the medium, and the clean "one massless boson per broken generator" can fail. Goldstone modes then split into two

kinds: *type-A*, with linear dispersion $\omega \propto |\mathbf{p}|$, one per broken generator as in the relativistic case; and *type-B*, with quadratic dispersion $\omega \propto \mathbf{p}^2$, one per *pair* of broken generators whose charges have a nonzero expectation value in the ground state, $\langle 0|[Q_a, Q_b]|0\rangle \neq 0$. The ferromagnet is the textbook case: two broken rotation generators yield a single quadratically dispersing magnon. The correct bookkeeping is the Nielsen–Chadha inequality and its modern sharpening to an equality by Watanabe and Murayama [364, 490]; we record it for the frontier summary.

Explicit breaking: pseudo-Goldstone bosons. If the symmetry is only *approximate* — broken explicitly by a small non-invariant term in the Lagrangian, the distinction this chapter has been sharpening — then the order parameter argument is perturbed and the would-be Goldstone acquires a small, calculable mass proportional to the explicit-breaking parameter. Such a light (but not massless) particle is a *pseudo-Nambu–Goldstone boson*. The pions of QCD are the paradigm: the light-quark masses break chiral symmetry explicitly, lifting the pions from exactly massless Goldstones to light pseudo-Goldstones, with $m_\pi^2 \propto m_q$. This is the content of chiral perturbation theory, Chapter 39.

These boundaries are not blemishes on the theorem; they are its physical content sharpened. Goldstone's theorem says exactly when a broken continuous symmetry must produce a massless scalar — relativistic, positive-metric, $d > 2$, exact symmetry — and each failure mode names a major piece of physics: the Higgs mechanism, low-dimensional fluctuations, condensed-matter Goldstones, and the pion.

With the masslessness established two ways, one debt remains explicit: the curvature proof of Section 29.3.6 leaned on an object we have not yet built — the effective potential V_{eff}, the all-orders order-parameter functional whose Hessian is the exact mass matrix. The next section constructs it from the effective action of Chapter 10, proves the symmetry identity Eq. (29.112) as Theorem 29.21, addresses its convexity, and computes the one-loop Coleman–Weinberg potential in full — turning the geometric proof of masslessness into a quantitative, loop-by-loop diagnostic of spontaneous symmetry breaking.

29.4 The Effective Potential and Radiative Symmetry Breaking

The geometric proof of Goldstone's theorem in the previous section turned on a single object whose existence we asserted but did not yet construct: a functional of the order parameter whose stationary points are the exact vacua and whose Hessian is the exact mass matrix, valid to all orders rather than only at the classical minimum. The classical potential $V(\phi)$ cannot play that role,

because quantum fluctuations shift both the location of the vacuum and the curvature about it; the loop corrections that renormalize a coupling also deform the potential that selects the ground state. We need the quantum-corrected potential, and the generating-functional machinery of Chapter 10 has already built it. The effective action $\Gamma[\phi_{\rm cl}]$ of Theorem 10.38 is precisely an action dressed by all fluctuations; its constant-field restriction is the *effective potential*. This section defines it carefully, establishes the two properties that make it the right diagnostic for spontaneous symmetry breaking — its stationarity conditions are the exact gap equations, its Hessian is the exact zero-momentum inverse propagator — confronts the apparent paradox that the exact $V_{\rm eff}$ is convex while the double well manifestly is not, organizes the loop expansion, and carries out the one-loop Coleman–Weinberg computation in full. The reward is twofold: a closed-form one-loop potential that we evaluate on ϕ^4 theory, and the phenomenon of radiatively induced symmetry breaking, in which a classically scale-invariant theory generates a vacuum expectation value out of a logarithm, trading a dimensionless coupling for a mass scale. The section closes by discharging the debt left open in Section 29.3: the symmetry of Γ forces a flat direction of $V_{\rm eff}$ along every broken generator, completing the effective-action proof of Goldstone's theorem at the quantum level.

29.4.1 Definition: the effective potential as the constant-field effective action

We work in Minkowski signature with the conventions of Chapter 10: the effective action is the Legendre transform of the connected generating functional, $\Gamma[\phi_{\rm cl}] = W[J] - \int d^4x\, J\phi_{\rm cl}$ (Eq. (10.210)), with classical field $\phi_{\rm cl}(x) = \delta W/\delta J(x)$ (Eq. (10.203)) and quantum equation of motion $\delta\Gamma/\delta\phi_{\rm cl} = -J$ (Eq. (10.212)). Translation invariance of the vacuum means that the physical configuration, obtained at $J = 0$, is a *constant* classical field: the vacuum expectation value $\langle\phi\rangle$ is the same at every point of spacetime. It is therefore the restriction of Γ to spacetime-independent arguments that carries all the information about the vacuum, and this restriction defines the effective potential.

Definition 29.14 (Effective potential)**.** For a spacetime-independent (constant) classical field configuration $\phi_{\rm cl}(x) \equiv \phi_{\rm cl}$, the effective action is proportional to the four-volume of spacetime, and the *effective potential* $V_{\rm eff}(\phi_{\rm cl})$ is defined as the coefficient,

$$\boxed{\Gamma[\phi_{\rm cl}] \;=\; -\,(\mathcal{V}T)\,V_{\rm eff}(\phi_{\rm cl})\,,} \qquad \mathcal{V}T \;\equiv\; \int d^4x \;=\; (2\pi)^4\delta^{(4)}(0), \tag{29.115}$$

where $\mathcal{V}T$ is the (infinite, infrared-regulated) volume of spacetime. For a multiplet $\phi_{\rm cl} = (\phi^1_{\rm cl}, \ldots, \phi^N_{\rm cl})$ the same formula holds with $V_{\rm eff}$ a function of N real variables.

The proportionality to $\mathcal{V}T$ is forced and not an assumption: Γ is an integral of a local density (its derivative expansion begins $\Gamma[\phi_{\rm cl}] = \int d^4x\,[-V_{\rm eff}(\phi_{\rm cl}) +$

$\frac{1}{2}Z(\phi_{\rm cl})(\partial\phi_{\rm cl})^2+\cdots]$), and for a constant field every derivative term drops, leaving the volume times the constant $-V_{\rm eff}$. The sign is fixed so that $V_{\rm eff}$ reduces to the classical potential at lowest order: at tree level $\Gamma[\phi_{\rm cl}]=S[\phi_{\rm cl}]$ (Eq. (10.212) et seq.), and for the ϕ^4 theory action $S=\int d^4x\,[\frac{1}{2}(\partial\phi)^2-V(\phi)]$ evaluated on a constant field this is $S=-(\mathcal{V}T)V(\phi_{\rm cl})$, so

$$V_{\rm eff}(\phi_{\rm cl}) \;=\; V(\phi_{\rm cl}) \;+\; (\text{loop corrections}), \tag{29.116}$$

with $V(\phi)=\frac{1}{2}m^2\phi^2+\frac{\lambda}{4!}\phi^4$ the classical potential of the thread (Chapter 6). The effective potential is the classical potential plus everything the fluctuations add.

29.4.2 What the effective potential computes: vacua, gap equations, masses

Two derivatives of $V_{\rm eff}$ contain the entire low-energy content of the theory's vacuum structure. We extract them by relating the derivatives of $V_{\rm eff}$ to the one-particle-irreducible (1PI) vertices that Γ generates.

Derivatives of $V_{\rm eff}$ are zero-momentum 1PI vertices. The n-point 1PI vertex function $\widetilde{\Gamma}^{(n)}$ is defined as the momentum-space kernel of the functional Taylor expansion of Γ about a chosen configuration. Expanding Γ in powers of a constant field and using Eq. (29.115),

$$\Gamma[\phi_{\rm cl}]=\sum_{n\geq 0}\frac{1}{n!}\int\Big(\prod_{i=1}^{n}d^4x_i\Big)\,\Gamma^{(n)}(x_1,\ldots,x_n)\,\phi_{\rm cl}^n, \qquad \Gamma^{(n)}(x_i)=\int\Big(\prod_i\frac{d^4p_i}{(2\pi)^4}\Big)e^{-i\sum p_ix_i}(2\pi)^4 \tag{29.117}$$

where the momentum-conserving delta enforces translation invariance of the vertex. For a constant field $\phi_{\rm cl}$ each x_i-integral acts only on the exponential and produces $(2\pi)^4\delta^{(4)}(p_i)$, setting every external momentum to zero; the remaining overall $(2\pi)^4\delta^{(4)}(0)=\mathcal{V}T$ factors out. Hence

$$\Gamma[\phi_{\rm cl}]=(\mathcal{V}T)\sum_{n\geq 0}\frac{1}{n!}\,\widetilde{\Gamma}^{(n)}(p_i=0)\,\phi_{\rm cl}^n=-(\mathcal{V}T)\,V_{\rm eff}(\phi_{\rm cl}), \tag{29.118}$$

where the last equality is the definition Eq. (29.115). Matching powers of $\phi_{\rm cl}$ gives the central dictionary,

$$\boxed{\left.\frac{d^nV_{\rm eff}}{d\phi_{\rm cl}^n}\right|_{\phi_{\rm cl}}=-\,\widetilde{\Gamma}^{(n)}(p_i=0;\,\phi_{\rm cl})\;:} \tag{29.119}$$

the n-th derivative of the effective potential is (minus) the sum of all amputated 1PI n-point diagrams at zero external momentum, evaluated in the background $\phi_{\rm cl}$. For a multiplet the same holds with $\partial^nV_{\rm eff}/\partial\phi^{i_1}\cdots\partial\phi^{i_n}=-\widetilde{\Gamma}^{(n)}_{i_1\cdots i_n}(0)$.

First derivative: the gap equations. At $J = 0$ the quantum equation of motion Eq. (10.212) for a constant field reads $0 = \delta\Gamma/\delta\phi_{\rm cl} = -(\mathcal{V}T)\,V'_{\rm eff}(\phi_{\rm cl})$, so the vacua are the stationary points of the effective potential,

$$\boxed{\left.\frac{\partial V_{\rm eff}}{\partial \phi^i_{\rm cl}}\right|_{\phi_{\rm cl}=v} = 0\,,} \tag{29.120}$$

and the *true* vacuum is the global minimum. These are the exact, all-orders gap equations: where the classical extremum condition $V'(\phi) = 0$ locates the classical vacuum, Eq. (29.120) locates the quantum vacuum with every radiative correction folded in. The order parameter is $v = \langle\phi\rangle$, and whether the symmetry is realized or broken on the vacuum is decided by whether the global minimum sits at the symmetric point or away from it — now read off the exact potential rather than the classical one.

Second derivative: the exact mass matrix. The Hessian of $V_{\rm eff}$ at the vacuum is, by Eq. (29.119) with $n = 2$, minus the zero-momentum two-point 1PI function:

$$(\mathcal{M}^2)_{ij} \;\equiv\; \left.\frac{\partial^2 V_{\rm eff}}{\partial\phi^i_{\rm cl}\,\partial\phi^j_{\rm cl}}\right|_{\phi_{\rm cl}=v} = -\,\widetilde{\Gamma}^{(2)}_{ij}(p=0;\,v). \tag{29.121}$$

By the propagator-inverse relation Eq. (10.214), the momentum-space two-point vertex is (minus) the inverse full propagator; in the one-component thread it is $\widetilde{\Gamma}^{(2)}(p) = p^2 - m^2 - \Sigma(p^2)$ with Σ the self-energy, so $\mathcal{M}^2 = -\widetilde{\Gamma}^{(2)}(0) = m^2 + \Sigma(0)$. This is the precise sense in which the curvature of the effective potential is the mass: it is the inverse propagator evaluated at zero momentum. The relation is stated cleanly as a proposition.

Proposition 29.15 (Curvature of $V_{\rm eff}$ is the zero-momentum inverse propagator)**.** *At any stationary point v of the effective potential, the Hessian Eq. (29.121) equals minus the one-particle-irreducible two-point function at zero external momentum, i.e. the inverse full propagator evaluated at $p = 0$. In particular a vanishing eigenvalue of the Hessian, $\mathcal{M}^2\,\hat{n} = 0$, forces the full propagator in the direction $\hat{n}$ to have a pole at $p^2 = 0$: a massless particle. Conversely the eigenvalues of $\mathcal{M}^2$ are the squared physical masses whenever the self-energy is momentum-independent at the relevant order, and in general control the location of the propagator poles through* $\det\widetilde{\Gamma}^{(2)}(p^2) = 0$.

Proof. Equation (29.121) is Eq. (29.119) at $n = 2$, and $\widetilde{\Gamma}^{(2)}(p)$ is minus the inverse full propagator by Eq. (10.214) and the reciprocity $W^{(2)} = -(\Gamma^{(2)})^{-1}$ established there. The full propagator is $\widetilde{G}(p) = i\,[\widetilde{\Gamma}^{(2)}(p)]^{-1}$ up to the conventional factor of i; a zero of $\det\widetilde{\Gamma}^{(2)}(p^2)$ at $p^2 = M^2$ is a pole of the propagator, hence a one-particle state of mass M. At $p^2 = 0$ this zero is exactly the condition $\det\mathcal{M}^2 = 0$, i.e. a vanishing Hessian eigenvalue. If Σ is independent of p^2

then $\widetilde{\Gamma}^{(2)}(p) = p^2 - \mathcal{M}^2$ exactly and the pole sits at $p^2 = \mathcal{M}^2$, so the Hessian eigenvalues are the squared masses outright; momentum dependence shifts the pole by the residue (the wavefunction renormalization $Z = [1 - \Sigma'(M^2)]^{-1}$) but cannot move a zero of $\widetilde{\Gamma}^{(2)}$ at $p^2 = 0$ away from $p^2 = 0$, which is all the Goldstone argument of Section 29.4.8 will require. □

Physical Insight: why V_{eff} and not V

The classical potential answers the question "where would the field sit if there were no fluctuations?". The effective potential answers "where does the field actually sit, and how stiff is it there, once every virtual process has been summed?". Its first derivative is the exact equation of state; its second derivative is the exact (zero-momentum) inverse propagator. Spontaneous symmetry breaking is the statement that the global minimum of this exact, fluctuation-corrected potential lies away from the symmetric point — a statement about the true ground state, not about the Lagrangian. Everything in this chapter that was argued at tree level is promoted to an exact statement by replacing $V \to V_{\text{eff}}$.

29.4.3 Convexity, and the paradox of the double well

There is an immediate and genuine puzzle. The effective potential is supposed to reduce to the classical potential at tree level, Eq. (29.116), and the broken-phase classical potential is the double well — manifestly *non-convex* between its two minima. Yet the exact V_{eff} is a Legendre transform of the generating functional, and Legendre transforms are convex. Both statements are true; their reconciliation teaches what the effective potential means in the broken phase.

Theorem 29.16 (Convexity of the exact effective potential)**.** *The exact effective potential is a convex function of the constant field ϕ_{cl}: for any two values ϕ_1, ϕ_2 and any $t \in [0, 1]$,*

$$V_{\text{eff}}\big(t\phi_1 + (1-t)\phi_2\big) \;\le\; t\,V_{\text{eff}}(\phi_1) + (1-t)\,V_{\text{eff}}(\phi_2). \tag{29.122}$$

Proof. The effective action is the Legendre transform of W (Eq. (10.210), Theorem 4.33), and the Legendre transform of *any* function is convex, because it is a pointwise supremum of affine functions. Restricting to constant fields and a constant source J, write $w(J) \equiv W[J]/(\mathcal{V}T)$ and $\gamma(\phi) \equiv V_{\text{eff}}(\phi)$; the constant-field Legendre relation $\Gamma = W - \int J\phi$ becomes, after dividing by $\mathcal{V}T$ and using Eq. (29.115),

$$-V_{\text{eff}}(\phi) = w(J) - J\phi\big|_{J=J(\phi)}, \qquad \text{equivalently} \qquad V_{\text{eff}}(\phi) = \sup_J \big(J\phi - w(J)\big), \tag{29.123}$$

the inversion being the involution of Theorem 4.31. The right-hand side is, for each fixed J, an affine (hence convex) function of ϕ; the supremum of a family of convex functions is convex. Explicitly, with $\phi_t = t\phi_1 + (1-t)\phi_2$,

$$V_{\text{eff}}(\phi_t) = \sup_J \left(J\phi_t - w(J)\right) = \sup_J \left[t(J\phi_1 - w(J)) + (1-t)(J\phi_2 - w(J))\right] \le t \sup_J (J\phi_1 - w(J)) + (1-t \tag{29.124}$$

where the inequality is the subadditivity of the supremum (the sup of a sum is at most the sum of the sups). The right-hand side is exactly $tV_{\text{eff}}(\phi_1) + (1-t)V_{\text{eff}}(\phi_2)$, which is Eq. (29.122). □

How can a convex function reduce to the non-convex double well? It cannot, pointwise between the wells — and that is the resolution. The loop expansion Eq. (29.116) is a perturbative approximation built by expanding about a *single* translationally invariant background; it is reliable only where the exact potential is locally convex, i.e. outside the spinodal region where $V''(\phi) \ge 0$. Between the two minima, where the classical $V'' < 0$, the perturbative series is expanding about an unstable configuration and the naive loop-corrected potential develops an imaginary part (the curvature $\mathcal{M}^2 = V'' < 0$ produces a ln(negative) in the Coleman–Weinberg log of Section 29.4.5) — the field-theoretic signal that a homogeneous state there does not exist. The exact V_{eff} replaces this region by the straight line joining the two minima: its *convex hull*, flat at the common minimum value. This is the Maxwell construction, and the flat segment has a precise physical meaning.

Caution: the flat region is two-phase coexistence, not a homogeneous state

A point ϕ_{cl} strictly between the two vacua $\pm v$, say $\phi_{\text{cl}} = \alpha v$ with $-1 < \alpha < 1$, is *not* realized by a spatially uniform field sitting on top of the classical hump. It is realized by a spatial *mixture*: a fraction $\frac{1+\alpha}{2}$ of the volume in the $+v$ vacuum and $\frac{1-\alpha}{2}$ in the $-v$ vacuum, with the prescribed average $\langle\phi\rangle = \alpha v$. Because each pure phase already sits at the minimum energy density $V_{\text{eff}}(\pm v)$, the mixture costs no extra energy density (the interface between domains is a subextensive, surface contribution that vanishes per unit volume), so $V_{\text{eff}}(\alpha v) = V_{\text{eff}}(v)$ across the whole interval: the flat segment. This is the field theory of phase coexistence (the analogue of the liquid–gas tie-line and the Maxwell equal-area rule of thermodynamics), and it is the honest content of the convexity theorem. The non-convex double well is the correct object for identifying the vacua and computing the masses about them — which is all this chapter needs — but it is *not* the exact effective potential in the forbidden region between them.

Figure 29.4 shows the classical double well together with its convex hull. The practical upshot for the rest of the chapter: to find the vacua and the spectrum

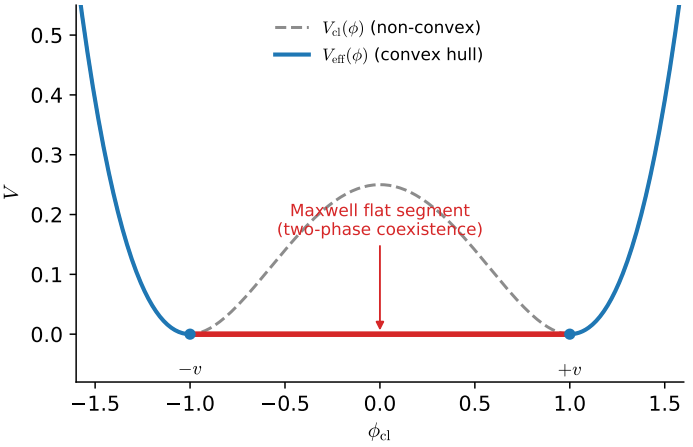


Figure 29.4: The classical double well $V_{\rm cl}$ (dashed) is non-convex between its minima at $\pm v$. The exact effective potential $V_{\rm eff}$ (solid) is convex (Theorem 29.16): it coincides with $V_{\rm cl}$ outside the wells and is replaced by the flat Maxwell segment (the convex hull) in between, representing two-phase coexistence (Section 29.4.3). The vacua and the spectrum are read off near the minima, where perturbation theory is reliable.

we evaluate the loop-corrected $V_{\rm eff}$ near its minima, where it is locally convex and perturbation theory is sound, and we read off v from Eq. (29.120) and the masses from Eq. (29.121). The convexity theorem is the guarantee that this local picture patches into a globally sensible exact object; it is not a license to trust the perturbative potential between the wells.

29.4.4 The loop expansion of the effective potential

To compute $V_{\rm eff}$ beyond tree level we organize the corrections by the number of loops. Reinstating $\hbar$ momentarily makes the bookkeeping transparent: in the path integral $Z = \int \mathcal{D}\phi\, e^{iS/\hbar}$ each propagator carries a factor $\hbar$ and each vertex a factor $\hbar^{-1}$, so a connected diagram with L loops, I internal lines and V vertices scales as $\hbar^{I-V} = \hbar^{L-1}$ (using the topological identity $L = I - V + 1$). The loop expansion is therefore an expansion in $\hbar$, and we set $\hbar = 1$ after using it to count.

Theorem 29.17 (One-loop effective potential as a functional determinant)**.** *Let the theory have a (possibly multi-component) scalar action $S[\phi] = \int d^4x\, [\tfrac{1}{2}(\partial_\mu \phi^i)(\partial^\mu \phi^i) - V(\phi)]$. Expanding about a constant background $\phi_{\rm cl}$ and integrating out the Gaussian fluctuations, the effective potential through one loop is*

$$V_{\rm eff}(\phi_{\rm cl}) = V(\phi_{\rm cl}) + \frac{1}{2}\int \frac{d^4k_E}{(2\pi)^4}\, \ln\det\left[k_E^2 + \mathcal{M}^2(\phi_{\rm cl})\right] + O(\hbar^2), \tag{29.125}$$

where the integral is Euclidean (after Wick rotation $k^0 = ik^0_E$), the determinant is over the field-space (internal) indices, and the field-dependent mass-squared matrix is the Hessian of the classical potential,

$$\mathcal{M}^2_{ij}(\phi_{\rm cl}) \;\equiv\; \left.\frac{\partial^2 V}{\partial\phi^i\,\partial\phi^j}\right|_{\phi=\phi_{\rm cl}}. \tag{29.126}$$

Proof. Shift the integration field by the constant background, $\phi^i = \phi^i_{\rm cl} + \eta^i$, and expand the action to second order in the fluctuation η:

$$S[\phi_{\rm cl}+\eta] = S[\phi_{\rm cl}] + \int d^4x\,\left.\frac{\delta S}{\delta\phi^i}\right|_{\phi_{\rm cl}}\eta^i + \frac{1}{2}\int d^4x\,d^4y\,\left.\frac{\delta^2 S}{\delta\phi^i(x)\delta\phi^j(y)}\right|_{\phi_{\rm cl}}\eta^i(x)\eta^j(y) + O(\eta^3). \tag{29.127}$$

The linear term is the source term that the Legendre transform removes: the very definition of Γ subtracts $\int J\phi_{\rm cl}$ and the 1PI prescription amputates the tadpoles, so that at the order we work the linear piece does not contribute to $V_{\rm eff}$ (equivalently, it is cancelled against the source that holds the field at $\phi_{\rm cl}$; its only role is to enforce Eq. (29.120)). The quadratic kernel is, for the local action above,

$$\left.\frac{\delta^2 S}{\delta\phi^i(x)\delta\phi^j(y)}\right|_{\phi_{\rm cl}} = \Big[-\partial^2\,\delta_{ij} - \mathcal{M}^2_{ij}(\phi_{\rm cl})\Big]\delta^{(4)}(x-y), \tag{29.128}$$

with $\partial^2 = \partial_\mu\partial^\mu$ the d'Alembertian and $\mathcal{M}^2$ from Eq. (29.126) a constant matrix because the background is constant. The one-loop contribution to $e^{i\Gamma}$ is the Gaussian integral over η of $\exp(\frac{i}{2}\int \eta\,(-\partial^2 - \mathcal{M}^2)\,\eta)$. This is the field-theory instance of the finite-dimensional Gaussian Theorem 4.14: with the real symmetric (here, after Wick rotation, positive) operator $A = -\partial^2 - \mathcal{M}^2$,

$$\int \mathcal{D}\eta\;\exp\Big(-\tfrac{1}{2}\int \eta\,A\,\eta\Big) = (\det A)^{-1/2}\times(\text{field-independent constant}), \tag{29.129}$$

the functional determinant being defined and the constant (the $\phi_{\rm cl}$-independent normalization that cancels in $V_{\rm eff}$) handled in Chapter E. Taking the logarithm,

$$i\Gamma[\phi_{\rm cl}] = iS[\phi_{\rm cl}] - \tfrac{1}{2}\ln\det A + \text{const} = iS[\phi_{\rm cl}] - \tfrac{1}{2}\,\mathrm{Tr}\ln\left(-\partial^2 - \mathcal{M}^2\right) + \text{const}, \tag{29.130}$$

where $\ln\det = \mathrm{Tr}\ln$ for the operator, the trace running over both spacetime and internal indices. The spacetime trace in momentum space is diagonal: with $\partial^2 \to -k^2$ (the eigenvalue on a plane wave $e^{-ik\cdot x}$) and the density of momentum states giving $\mathrm{Tr}\,(\cdots) = \mathcal{V}T\int\frac{d^4k}{(2\pi)^4}(\cdots)$,

$$\mathrm{Tr}\ln\left(-\partial^2 - \mathcal{M}^2\right) = \mathcal{V}T\int\frac{d^4k}{(2\pi)^4}\,\ln\det\left(k^2 - \mathcal{M}^2\right), \tag{29.131}$$

where the remaining determinant is over the internal indices only. Dividing Eq. (29.130) by $i\mathcal{V}T$ and using $\Gamma = -(\mathcal{V}T)V_{\rm eff}$ and $S = -(\mathcal{V}T)V$,

$$V_{\rm eff} = V + \frac{1}{2}\cdot\frac{1}{i}\int\frac{d^4k}{(2\pi)^4}\,\ln\det\left(k^2 - \mathcal{M}^2\right) + \cdots = V - \frac{i}{2}\int\frac{d^4k}{(2\pi)^4}\,\ln\det\left(k^2 - \mathcal{M}^2 + i\varepsilon\right) + \cdots . \tag{29.132}$$

Finally Wick-rotate $k^0 = ik^0_E$, so $d^4k = i\,d^4k_E$ and $k^2 = -k^2_E$, whereupon the two factors of i combine with the minus sign, $-\frac{i}{2}\cdot i = \frac{1}{2}$, and the argument becomes $k^2_E + \mathcal{M}^2$ (a field-independent additive constant inside the log, from the rotation of the $i\varepsilon$, is dropped as it shifts $V_{\rm eff}$ by a constant):

$$V_{\rm eff} = V + \frac{1}{2}\int\frac{d^4k_E}{(2\pi)^4}\,\ln\det\left(k^2_E + \mathcal{M}^2\right) + O(\hbar^2), \tag{29.133}$$

which is Eq. (29.125). Diagonalizing the symmetric matrix $\mathcal{M}^2$ with eigenvalues $M^2_a(\phi_{\rm cl})$ turns $\ln\det = \sum_a \ln(k^2_E + M^2_a)$, so the one-loop term is a sum of independent scalar contributions, one per normal mode. □

The structure of Eq. (29.125) is worth naming before we evaluate it. Each fluctuation mode contributes $\frac{1}{2}\int\frac{d^4k_E}{(2\pi)^4}\ln(k^2_E + M^2_a)$, which is exactly the (Euclidean) zero-point energy density $\frac{1}{2}\sum\hbar\omega$ of a free field of mass M_a — the one-loop effective potential is the vacuum energy of the fluctuations in the presence of the background, the sum of zero-point energies of all the normal modes, each given its field-dependent mass. That is the physical content of "integrating out the fluctuations": the background pays the cost of the shifted vacuum energy of everything that couples to it. The momentum integral is ultraviolet divergent (the integrand grows like $\ln k^2_E$), and renormalizing it is the next task.

29.4.5 The Coleman–Weinberg potential: the one-loop integral evaluated

We now carry out the momentum integral in Eq. (29.125) in dimensional regularization and renormalize in $\overline{\rm MS}$, reusing the loop technology built for ϕ^4 theory in Chapter 22 and the master integral of Chapter D. It suffices to do a single mode of mass $M^2 \equiv M^2(\phi_{\rm cl})$ and sum at the end.

The cleanest route differentiates with respect to M^2 to reduce the logarithm to the tadpole master integral A_0, which is already evaluated in closed form in Chapter D. Define the single-mode one-loop integral in $d = 4 - 2\varepsilon$ dimensions, with the $\overline{\rm MS}$ scale μ inserted to keep the mass dimension fixed,

$$v_1(M^2) \;\equiv\; \frac{1}{2}\,\mu^{2\varepsilon}\int\frac{d^dk_E}{(2\pi)^d}\,\ln\left(k^2_E + M^2\right). \tag{29.134}$$

Differentiating under the integral sign,

$$\frac{\partial v_1}{\partial M^2} = \frac{1}{2}\,\mu^{2\varepsilon}\int\frac{d^dk_E}{(2\pi)^d}\,\frac{1}{k^2_E + M^2} = \frac{1}{2}\,\mu^{2\varepsilon}\,\frac{1}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \frac{d}{2}\right)(M^2)^{d/2-1}, \tag{29.135}$$

where the Euclidean integral is the rotated tadpole: it is the Wick rotation of A_0 from Eq. (D.99) (the Minkowski $A_0 = -i/(4\pi)^{d/2}\,\Gamma(1-d/2)(m^2)^{d/2-1}$ of Theorem D.22 with $n=1$, $\Delta = M^2$; the factor $-i$ is undone by the rotation $d^4k = i\,d^4k_E$, leaving the real positive Euclidean value quoted). Now integrate back over M^2, using $\int dM^2\,(M^2)^{d/2-1} = (M^2)^{d/2}/(d/2)$ and the identity $\Gamma(1-d/2)/(d/2) = -\Gamma(-d/2)$ (from $z\Gamma(z) = \Gamma(z+1)$ at $z = -d/2$):

$$v_1(M^2) = \frac{1}{2}\,\frac{\mu^{2\varepsilon}}{(4\pi)^{d/2}}\,\frac{\Gamma(1-\frac{d}{2})}{d/2}\,(M^2)^{d/2} = -\frac{1}{2}\,\frac{\mu^{2\varepsilon}}{(4\pi)^{d/2}}\,\Gamma\Big(-\frac{d}{2}\Big)\,(M^2)^{d/2}, \tag{29.136}$$

the integration constant being fixed to zero because a scaleless integral ($M^2 \to 0$) vanishes in dimensional regularization (Eq. (D.99)). Equation Eq. (29.136) is the exact one-loop integral for one mode in d dimensions; everything else is its expansion about $d = 4$.

Set $d = 4 - 2\varepsilon$, so $-d/2 = -2 + \varepsilon$, and expand each factor. The Gamma function near its pole at -2 is

$$\Gamma(-2+\varepsilon) = \frac{\Gamma(\varepsilon)}{(-1+\varepsilon)(-2+\varepsilon)} = \frac{1}{2}\Big(\frac{1}{\varepsilon} - \gamma_{\mathrm{E}}\Big)\big(1+\tfrac{3}{2}\varepsilon\big) + O(\varepsilon) = \frac{1}{2}\Big(\frac{1}{\varepsilon} + \frac{3}{2} - \gamma_{\mathrm{E}}\Big) + O(\varepsilon), \tag{29.137}$$

where we used $\Gamma(\varepsilon) = 1/\varepsilon - \gamma_{\mathrm{E}} + O(\varepsilon)$ and $[(-1+\varepsilon)(-2+\varepsilon)]^{-1} = \frac{1}{2}(1+\frac{3}{2}\varepsilon) + O(\varepsilon^2)$. The remaining factors expand as

$$(M^2)^{d/2} = (M^2)^{2-\varepsilon} = M^4\big(1 - \varepsilon \ln M^2\big) + O(\varepsilon^2), \tag{29.138}$$

$$\frac{\mu^{2\varepsilon}}{(4\pi)^{d/2}} = \frac{1}{(4\pi)^2}\,(4\pi)^{\varepsilon}\,\mu^{2\varepsilon} = \frac{1}{(4\pi)^2}\big(1 + \varepsilon \ln 4\pi + \varepsilon \ln \mu^2\big) + O(\varepsilon^2). \tag{29.139}$$

Multiplying Eqs. (29.137) to (29.139) together, the three $O(\varepsilon)$ logarithms combine into a single logarithm $\ln(4\pi\mu^2/M^2)$ multiplying the $1/\varepsilon$ pole:

$$v_1(M^2) = -\frac{M^4}{64\pi^2}\Big[\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi - \ln\frac{M^2}{\mu^2} + \frac{3}{2}\Big] + O(\varepsilon), \tag{29.140}$$

where the overall coefficient is $-\frac{1}{2}\cdot\frac{1}{2}\cdot\frac{1}{(4\pi)^2} = -\frac{1}{64\pi^2}$. The pole and its universal companions are exactly the combination $1/\bar{\varepsilon} = 1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi$ that defines $\overline{\mathrm{MS}}$ (Eq. (A.204)). Renormalizing the theory — adding the counterterms to the classical parameters that the $\overline{\mathrm{MS}}$ scheme prescribes (Theorem A.49, Chapter 22) — subtracts precisely the $1/\bar{\varepsilon}$ pole, leaving the finite renormalized one-loop potential for a single bosonic mode:

$$\boxed{\,v_1^{\overline{\mathrm{MS}}}(M^2) = \frac{M^4(\phi_{\mathrm{cl}})}{64\pi^2}\left[\ln\frac{M^2(\phi_{\mathrm{cl}})}{\mu^2} - \frac{3}{2}\right].\,} \tag{29.141}$$

This is the Coleman–Weinberg potential for one real scalar degree of freedom. The constant $\frac{3}{2}$ is scheme-dependent (it is the $\overline{\mathrm{MS}}$ value); the μ dependence of the logarithm is exactly compensated by the running of the renormalized parameters in V, as the renormalization group of Chapters 23 and 24 guarantees, so V_{eff} as a whole is μ-independent.

The general spectrum. For a theory with several fields, the one-loop potential is the sum Eq. (29.125) over all normal modes, each contributing Eq. (29.141) with its own field-dependent mass eigenvalue $M_i^2(\phi_{\rm cl})$. Spinning fields enter with two modifications, both read off the Gaussian integral. A field of spin s carries $2s+1$ polarization states for a massive mode (a massive vector has 3; a Dirac fermion has 4 components), each contributing one copy of the scalar integral; and a fermionic field, being Grassmann, has its Gaussian integral give $(\det A)^{+1/2}$ rather than $(\det A)^{-1/2}$ (Chapter 12), flipping the sign of its $\operatorname{Tr}\ln$ — the familiar minus sign of a closed fermion loop. Collecting the multiplicity and sign into the factor $n_i = (-1)^{2s_i}(2s_i+1)$ (so $n=+1$ for a real scalar, $+3$ for a massive vector, -4 for a Dirac fermion, -2 for a Weyl fermion), the master one-loop Coleman–Weinberg potential is

$$\boxed{V_1(\phi_{\rm cl}) = \frac{1}{64\pi^2}\sum_i (-1)^{2s_i}(2s_i+1)\, M_i^4(\phi_{\rm cl})\left[\ln\frac{M_i^2(\phi_{\rm cl})}{\mu^2} - c_i\right],} \tag{29.142}$$

with the renormalized one-loop effective potential $V_{\rm eff} = V + V_1 + O(\hbar^2)$. The scheme constants are $c_i = \frac{3}{2}$ for scalars and fermions and $c_i = \frac{5}{6}$ for gauge bosons; the latter differs because the transverse and would-be-Goldstone/ghost sectors of a massive vector are disentangled in $\overline{\rm MS}$ (Landau gauge), shifting the finite part. The pole part of every term is the same universal $1/\bar{\varepsilon}$, removed by the counterterms of the underlying renormalizable theory; Eq. (29.142) is the finite remainder. The single-scalar derivation Eq. (29.141) is the prototype every term inherits.

Remark 29.18 (The Coleman–Weinberg potential is the summed zero-point energy). The structure $M^4\ln M^2$ is dimensionally inevitable: $V_{\rm eff}$ has mass dimension four, the only scale in the one-loop integral of a given mode is its mass $M(\phi_{\rm cl})$, and the residual logarithm carries the renormalization scale μ that the subtraction introduced. Physically, Eq. (29.142) is the renormalized sum of zero-point energies $\frac{1}{2}\sum\hbar\omega_k$ of all the fluctuation modes, with bosons lowering and fermions raising the vacuum energy density — the same cancellation that, when exact, defines a supersymmetric spectrum and forces the one-loop potential to vanish.

29.4.6 ϕ^4 theory: the broken-phase effective potential

Worked Example 29.19 (The one-loop effective potential of the broken ϕ^4 thread)**.** We continue ϕ^4 theory from its quantization and renormalization (Chapters 10 and 22) into the broken phase, computing its effective potential to one loop and extracting the radiative shift of the vacuum and the mass. The standard Lagrangian is that of the thread (Chapter 6) with the sign of the mass term reversed so that the $\mathbb{Z}_2$ symmetry $\phi\to-\phi$ of Theorem 7.22 is spontaneously

broken; writing $m^2 = -\mu_0^2$ with $\mu_0^2 > 0$,

$$V(\phi) = -\tfrac{1}{2}\mu_0^2\,\phi^2 + \frac{\lambda}{4!}\phi^4, \qquad \lambda > 0. \tag{29.143}$$

Tree level. The field-dependent mass is the second derivative of the classical potential, Eq. (29.126),

$$M^2(\phi) = V''(\phi) = -\mu_0^2 + \frac{\lambda}{2}\phi^2. \tag{29.144}$$

The classical gap equation $V'(\phi) = -\mu_0^2\phi + \frac{\lambda}{6}\phi^3 = 0$ has the nontrivial solution

$$v_0^2 = \frac{6\mu_0^2}{\lambda}, \tag{29.145}$$

and the physical mass at this minimum is $M^2(v_0) = -\mu_0^2 + \frac{\lambda}{2}\cdot\frac{6\mu_0^2}{\lambda} = 2\mu_0^2$, the standard double-well result: the single physical excitation about the broken vacuum has $m_\sigma^2 = 2\mu_0^2 = \frac{1}{3}\lambda v_0^2$.

One loop. There is a single real scalar mode, so Eq. (29.141) applies directly with the field-dependent mass Eq. (29.144):

$$V_{\rm eff}(\phi) = -\tfrac{1}{2}\mu_0^2\phi^2 + \frac{\lambda}{4!}\phi^4 + \frac{1}{64\pi^2}\Big(-\mu_0^2 + \tfrac{\lambda}{2}\phi^2\Big)^2\left[\ln\frac{-\mu_0^2 + \frac{\lambda}{2}\phi^2}{\mu^2} - \frac{3}{2}\right]. \tag{29.146}$$

Two features are immediate. First, for $\phi^2 < 2\mu_0^2/\lambda$ the field-dependent mass-squared Eq. (29.144) is negative and the logarithm develops an imaginary part: this is exactly the spinodal region of Section 29.4.3, where the homogeneous expansion is invalid and the exact potential is the flat Maxwell segment discussed in Section 29.4.3. The formula Eq. (29.146) is to be trusted near the minimum, where $M^2(\phi) > 0$. Second, the loop term shifts both the location of the minimum and the curvature there. Writing the corrected vacuum as $v = v_0 + \delta v$ and imposing the exact gap equation Eq. (29.120), the radiative correction to the vacuum is $O(\lambda)$ relative to v_0 and the corrected scalar mass is, from Eq. (29.121),

$$m_\sigma^2 = V_{\rm eff}''(v) = 2\mu_0^2 + \frac{3\lambda\mu_0^2}{16\pi^2}\left[\ln\frac{2\mu_0^2}{\mu^2} - \frac{1}{3}\right] + O(\lambda^2), \tag{29.147}$$

obtained by differentiating Eq. (29.146) twice and evaluating at v_0 to leading order in λ (the shift δv enters m_σ^2 only at $O(\lambda^2)$ because $V_{\rm eff}'''(v_0)\,\delta v$ is itself one loop times the one-loop δv). The result exhibits the two hallmarks of the effective potential: the symmetry is still broken at one loop (the minimum stays away from the origin), and the mass is the exact curvature there, now carrying its logarithmic dependence on the renormalization scale — compensated, as always, by the running of μ_0^2 and λ under the renormalization group of Chapter 24. This is the chapter's "effective potential, computed" result for the model: the same renormalization machinery built for ϕ^4 in Chapter 22 delivers, with no new technology, the quantum-corrected potential that governs the broken vacuum.

29.4.7 Radiative symmetry breaking and dimensional transmutation

The effective potential makes possible a phenomenon invisible at tree level: a theory with *no* symmetry breaking classically can break the symmetry purely through radiative corrections. The mechanism is the logarithm in Eq. (29.142). Consider a classically scale-invariant theory — one with no mass parameter, so the classical potential is a pure quartic and the only dimensionless input is the coupling. Classically the potential is minimized at the origin and the symmetry is unbroken. But the one-loop logarithm bends the quartic near the origin and can push the minimum to a nonzero field value, generating a vacuum expectation value out of nothing but the coupling and the renormalization scale.

The single-scalar mechanism, and its caveat. For the massless single scalar, $V = \frac{\lambda}{4!}\phi^4$, the field-dependent mass is $M^2(\phi) = \frac{\lambda}{2}\phi^2$ and Eq. (29.141) gives

$$V_{\rm eff}(\phi) = \frac{\lambda}{4!}\phi^4 + \frac{1}{64\pi^2}\Big(\frac{\lambda}{2}\phi^2\Big)^2\left[\ln\frac{\lambda\phi^2/2}{\mu^2} - \frac{3}{2}\right] = \frac{\lambda}{4!}\phi^4 + \frac{\lambda^2\phi^4}{256\pi^2}\left[\ln\frac{\lambda\phi^2}{2\mu^2} - \frac{3}{2}\right]. \tag{29.148}$$

The logarithm grows without bound as ϕ increases, so $V'_{\rm eff} = 0$ has a solution at a nonzero $\phi = \langle\phi\rangle$ where the negative logarithm near the origin balances the positive quartic. Solving $V'_{\rm eff}(\langle\phi\rangle) = 0$ gives a relation of the schematic form $\lambda \sim 1/\ln(\langle\phi\rangle^2/\mu^2)$: the would-be vacuum sits where $\lambda\ln(\langle\phi\rangle^2/\mu^2) \sim O(1)$. This is the essential point and also the caveat: the minimum is generated only when the loop term is comparable to the tree term, i.e. when $\lambda\ln(\cdots) \sim 16\pi^2$ is no longer small, so for a single self-coupled scalar the one-loop minimum lies outside the regime where the one-loop approximation is trustworthy. The mechanism is real but, for the pure scalar, requires resummation (the renormalization group) to control. It becomes both controlled and physical when a second, larger coupling dominates the logarithm — the original setting of Coleman and Weinberg: massless scalar electrodynamics.

Massless scalar electrodynamics. Take a complex scalar $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$ of charge e coupled to a $U(1)$ gauge field, with classical potential $V = \frac{\lambda}{4!}(\phi_1^2+\phi_2^2)^2$ and no mass term — classically scale-invariant. Along the radial direction $\phi_{\rm cl}$ (the gauge-invariant modulus) the fluctuation spectrum has three pieces, each with its field-dependent mass:

$$M_A^2(\phi_{\rm cl}) = e^2\phi_{\rm cl}^2 \ (\times 3 \text{ polarizations}), \qquad M_\sigma^2 = \frac{\lambda}{2}\phi_{\rm cl}^2, \qquad M_G^2 = \frac{\lambda}{6}\phi_{\rm cl}^2, \tag{29.149}$$

the gauge boson acquiring its mass by the Higgs mechanism of Chapter 30 (the would-be Goldstone M_G is eaten and does not appear in the physical spectrum at

the minimum). When the gauge coupling dominates the quartic, $e^4 \gg \lambda^2$ — the self-consistent regime, as we will see — the gauge contribution to Eq. (29.142) swamps the scalar ones, and using $c_A = \frac{5}{6}$ for the three vector polarizations,

$$V_{\rm eff}(\phi_{\rm cl}) = \frac{\lambda}{4!}\phi_{\rm cl}^4 + \frac{3\,e^4\phi_{\rm cl}^4}{64\pi^2}\left[\ln\frac{e^2\phi_{\rm cl}^2}{\mu^2} - \frac{5}{6}\right] + \cdots . \tag{29.150}$$

The logarithm again creates a minimum away from the origin. It is cleanest to trade the scheme scale μ for the location of the minimum itself by imposing the renormalization condition that defines the quartic coupling at the scale of the vacuum, $\langle\phi\rangle$. Choosing μ such that the constant inside the bracket is absorbed into the scale, Eq. (29.150) can be written in the equivalent "renormalization-group improved" form

$$V_{\rm eff}(\phi_{\rm cl}) = \frac{3\,e^4}{64\pi^2}\,\phi_{\rm cl}^4\left[\ln\frac{\phi_{\rm cl}^2}{\langle\phi\rangle^2} - \frac{1}{2}\right], \tag{29.151}$$

which has its minimum manifestly at $\phi_{\rm cl} = \langle\phi\rangle$, where $V'_{\rm eff} = 0$. Figure 29.5 shows this potential.

Dimensional transmutation. The minimization condition is the heart of the matter. Imposing $V'_{\rm eff}(\langle\phi\rangle) = 0$ on Eq. (29.150) fixes the quartic coupling in terms of the gauge coupling: differentiating and evaluating at the minimum yields, to leading order,

$$\boxed{\lambda = \frac{33}{8\pi^2}\,e^4} \tag{29.152}$$

(the precise coefficient depending on the normalization of λ; with the $1/4!$ of our convention it is the Coleman–Weinberg value above). This is a remarkable trade. Classically the theory had *two* dimensionless parameters, e and λ, and *no* mass scale. After radiative breaking it has one dimensionless parameter e (with λ no longer independent, fixed by Eq. (29.152)) and one *dimensionful* parameter, the vacuum expectation value $\langle\phi\rangle$, which sets the masses

$$M_A^2 = e^2\langle\phi\rangle^2, \qquad m_\sigma^2 = V''_{\rm eff}(\langle\phi\rangle) = \frac{3e^4}{8\pi^2}\langle\phi\rangle^2 = \frac{3e^2}{8\pi^2}M_A^2, \tag{29.153}$$

where m_σ^2 is computed by differentiating Eq. (29.151) twice at $\langle\phi\rangle$: $V''_{\rm eff} = \frac{3e^4}{64\pi^2}\langle\phi\rangle^2[12\ln(\cdot) + \text{terms}]\big|_{\langle\phi\rangle}$ collapses to $\frac{3e^4}{8\pi^2}\langle\phi\rangle^2$ once the minimum condition kills the logarithm. A dimensionless coupling has been converted into a dimensionful scale: the value of $\langle\phi\rangle$ is not predicted (it is the one free dimensionful input, the analogue of $\Lambda_{\rm QCD}$ generated by the running of the gauge coupling in Chapter 27), but the entire spectrum is fixed in units of it, and the ratio $m_\sigma^2/M_A^2 = 3e^2/8\pi^2$ is a pure number. This is *dimensional transmutation*: a classically scale-invariant theory, quantized, spontaneously generates a mass scale,

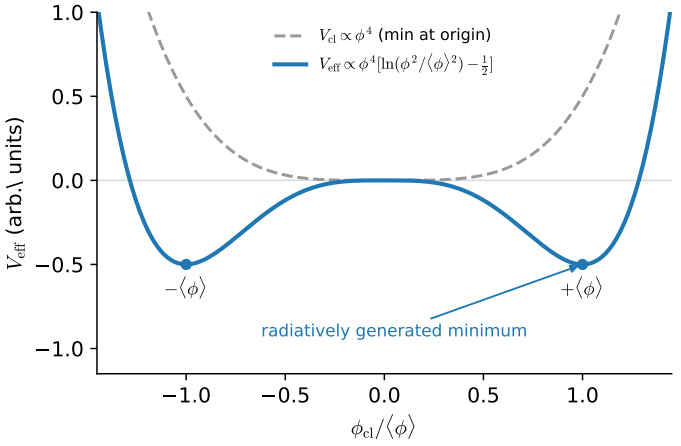


Figure 29.5: Radiatively induced symmetry breaking. The classically scale-invariant potential is a pure quartic with its only minimum at the origin (dashed). The one-loop logarithm of Eq. (29.151) bends the quartic near the origin and generates a new minimum at a nonzero $\langle\phi\rangle$ (solid), dynamically producing a vacuum expectation value and a mass scale from the dimensionless couplings — dimensional transmutation, with λ fixed in terms of e by Eq. (29.152).

breaking both the scale symmetry (anomalously, through the running coupling) and the internal symmetry (spontaneously, through the radiatively generated vacuum). The same mechanism, applied to the strong interactions, is how a theory whose classical Lagrangian has no mass parameter nonetheless sets the scale of the proton mass.

29.4.8 The effective-potential proof of Goldstone's theorem

We can now discharge the debt left open at the end of Section 29.3: the geometric, all-orders proof of Goldstone's theorem. The operator proof given there established the theorem non-perturbatively from the charge algebra; here we give the complementary effective-action proof, which makes manifest that the Goldstone modes live in the flat directions of the exact $V_{\rm eff}$, order by order in the loop expansion. The single ingredient is that $V_{\rm eff}$ inherits the symmetry of the action.

Lemma 29.20 (The effective potential is invariant)**.** *Let the action and the path-integral measure be invariant under a global symmetry acting linearly on the fields, $\phi^i \to (g\cdot\phi)^i = g^i{}_j\phi^j$ with $g \in G$. Then the effective action is invariant, $\Gamma[g\cdot\phi_{\rm cl}] = \Gamma[\phi_{\rm cl}]$, and hence for constant fields the effective potential is invariant,*

$$V_{\rm eff}(g\cdot\phi_{\rm cl}) = V_{\rm eff}(\phi_{\rm cl}) \qquad \text{for all } g \in G. \tag{29.154}$$

Proof. Couple a source J_i to ϕ^i in the generating functional $Z[J] = \int \mathcal{D}\phi\, e^{i(S[\phi]+\int J_i\phi^i)}$. Change integration variables to $\phi' = g\cdot\phi$; by hypothesis $\mathcal{D}\phi = \mathcal{D}\phi'$ and $S[\phi] = S[g^{-1}\cdot\phi'] = S[\phi']$, so $Z[J] = \int \mathcal{D}\phi'\, e^{i(S[\phi']+\int J_i(g^{-1}\cdot\phi')^i)} = Z[g^{\mathsf{T}}J]$, where the source has been rotated by the transpose (for an orthogonal representation, by g itself). Thus $W[J] = W[g^{\mathsf{T}}J]$, and the classical field $\phi_{\rm cl} = \delta W/\delta J$ transforms covariantly: $\phi_{\rm cl}[g^{\mathsf{T}}J] = g\cdot\phi_{\rm cl}[J]$. Feeding this through the Legendre transform Eq. (10.210),

$$\Gamma[g\cdot\phi_{\rm cl}] = W[g^{\mathsf{T}}J] - \int (g^{\mathsf{T}}J)_i (g\cdot\phi_{\rm cl})^i = W[J] - \int J_i\phi^i_{\rm cl} = \Gamma[\phi_{\rm cl}], \tag{29.155}$$

where the middle step used $W[g^{\mathsf{T}}J] = W[J]$ and the invariance of the inner product $\int (g^{\mathsf{T}}J)\cdot(g\cdot\phi_{\rm cl}) = \int J\cdot\phi_{\rm cl}$ under the (orthogonal) group action. Restricting to constant fields and using $\Gamma = -(\mathcal{V}T)V_{\rm eff}$ gives Eq. (29.154). □

The invariance is exact: it holds order by order in the loop expansion, because each order of Γ is separately invariant (the regulator and the $\overline{\rm MS}$ subtraction respect the linear symmetry). With it, Goldstone's theorem is two derivatives away.

Theorem 29.21 (Goldstone's theorem from the effective potential)**.** *Let G be a continuous global symmetry, with Hermitian generators T^a acting linearly on the scalar multiplet, realized as in Theorem 29.20. Let $v^i = \langle\phi^i\rangle$ be the vacuum, a global minimum of $V_{\rm eff}$. Then the exact mass matrix $\mathcal{M}^2_{ij} = \partial_i\partial_j V_{\rm eff}(v)$ annihilates every broken direction:*

$$\mathcal{M}^2_{ij}\,(T^a v)^j = 0 \qquad \text{for every generator } T^a \text{ with } T^a v \neq 0. \tag{29.156}$$

Equivalently, $V_{\rm eff}$ has one flat direction — one massless mode, by Theorem 29.15 — for each broken generator, i.e. for each generator of the coset G/H, where $H = \{g : g\cdot v = v\}$ is the unbroken subgroup. This holds to all orders in the loop expansion.

Proof. Specialize Eq. (29.154) to an infinitesimal transformation, $g = \mathbf{1} + i\alpha^a T^a$, so $\phi^i \to \phi^i + i\alpha^a (T^a)^i{}_j\phi^j$. Invariance to first order in the parameters α^a reads, for every ϕ,

$$0 = \frac{\partial V_{\rm eff}}{\partial\phi^i}\,(T^a\phi)^i, \qquad (T^a\phi)^i \equiv i(T^a)^i{}_j\phi^j. \tag{29.157}$$

This is an identity in ϕ, valid at every field value, so we may differentiate it again with respect to ϕ^k:

$$0 = \frac{\partial^2 V_{\rm eff}}{\partial\phi^k\partial\phi^i}\,(T^a\phi)^i + \frac{\partial V_{\rm eff}}{\partial\phi^i}\,(T^a)^i{}_k\, i. \tag{29.158}$$

Now evaluate at the vacuum $\phi = v$. The second term vanishes because v is a stationary point of V_{eff}, $\partial V_{\text{eff}}/\partial\phi^i|_v = 0$ (the gap equation Eq. (29.120)). The first term is exactly the mass matrix contracted with the broken direction:

$$0 = \left.\frac{\partial^2 V_{\text{eff}}}{\partial\phi^k \partial\phi^i}\right|_v (T^a v)^i = \mathcal{M}^2_{ki}\,(T^a v)^i, \tag{29.159}$$

which is Eq. (29.156). If $T^a v \neq 0$ — the definition of a broken generator — then $(T^a v)$ is a nonzero eigenvector of $\mathcal{M}^2$ with eigenvalue zero, hence by Theorem 29.15 a massless particle. Distinct broken generators give linearly independent vectors $T^a v$ (their independence is the statement that the broken generators span the tangent space to the orbit $G{\cdot}v$ at v, i.e. the coset G/H), so the number of massless modes equals the number of broken generators, $\dim G - \dim H$. Because Eq. (29.154) holds order by order, so does Eq. (29.156): the broken directions are flat to all orders, and the Goldstone bosons are exactly massless, not merely massless at tree level. □

Physical Insight: the geometry of the proof

The argument is purely geometric. Invariance of V_{eff} means it is constant along the orbits of G — the level sets are unions of group orbits. At a minimum the gradient vanishes, so the only way the potential can fail to curve upward in some direction is if that direction lies *along* an orbit, where the potential is flat by invariance. The broken generators are precisely the directions tangent to the orbit through v that move v to a physically distinct but energetically degenerate vacuum; the unbroken generators ($T^a v = 0$) leave v fixed and contribute no flat direction. Goldstone bosons are the fluctuations that roll along the valley of degenerate vacua at no energy cost — the massless excitations of the flat bottom of the wine bottle. The effective potential turns this picture into an exact, all-orders theorem because V_{eff}, unlike V, is the true fluctuation-corrected energy landscape.

This completes the second, effective-action proof of Goldstone's theorem promised in Section 29.3, now valid to all orders and tied directly to the exact mass matrix. The two proofs are complementary: the operator proof of Theorem 29.11 establishes the massless pole non-perturbatively from the broken current, while Theorem 29.21 exhibits the massless mode as a flat direction of the exact potential and counts them by the coset dimension. Together they fix both the existence and the multiplicity of the Goldstone bosons.

With the effective potential in hand we have learned that the entire light dynamics of a spontaneously broken theory lives in the flat directions of V_{eff} — the massless Goldstone valley — while the radial, massive modes sit at the bottom of a steep well of curvature set by the order parameter. The next section

takes this picture to its conclusion: sending the massive radial mode to infinity leaves a theory whose only fields are the Goldstone coordinates on the coset manifold G/H, the nonlinear sigma model, whose derivative-only interactions and low-energy theorems are the universal content of spontaneous symmetry breaking.

29.5 The Nonlinear Sigma Model and Low-Energy Goldstone Dynamics

The effective potential told us where low-energy physics lives. Its global minimum is a whole connected manifold of degenerate vacua — the orbit of one vacuum under the spontaneously broken symmetry — and the Hessian at any point of that manifold has exactly two kinds of eigendirection: the directions transverse to the manifold, which climb the potential and cost a finite mass, and the directions tangent to it, which are flat and cost nothing. The flat directions are the Goldstone bosons, and Theorem 29.21 fixed their number once and for all: there are $\dim G - \dim H$ of them, one for each broken generator, and they parametrize the homogeneous space G/H. Everything a low-energy experiment can do is move the system along these flat directions; the massive transverse modes can only be excited at energies of order their mass and are invisible far below it. The question of this section is therefore the natural one: *what is the theory of the flat directions alone?* We will answer it by physically removing the massive mode — sending its mass to infinity at fixed vacuum — and watching the dynamics collapse onto the vacuum manifold. The result is the nonlinear sigma model, the universal low-energy effective theory of Goldstone bosons, whose interactions are forced to be purely derivative and whose leading two-derivative form is fixed by geometry up to a single constant. This is the structure that the chiral Lagrangian of Chapter 39 inherits wholesale, with the pions playing the role of the Goldstone fields.

29.5.1 The heavy-radial-mode limit of the linear sigma model

We work with the $\mathrm{O}(N)$ linear sigma model already introduced in this chapter, the multicomponent generalization of ϕ^4 theory: an N-component real scalar $\Phi^i(x)$, $i = 1, \dots, N$, assembled into a vector $\Phi = (\Phi^1, \dots, \Phi^N)$, with the $\mathrm{O}(N)$-invariant Lagrangian

$$\mathcal{L}_{\text{lin}} \;=\; \tfrac{1}{2}\,(\partial_\mu \Phi)\cdot(\partial^\mu \Phi) \;-\; V(\Phi), \qquad V(\Phi) \;=\; \frac{\lambda}{4}\big(\Phi\cdot\Phi - v^2\big)^2, \tag{29.160}$$

where $\Phi\cdot\Phi = \sum_i \Phi^i\Phi^i$, $\lambda > 0$, and $v > 0$. The potential is invariant under $\Phi \to R\,\Phi$ for any $R \in \mathrm{O}(N)$, and it is minimized on the sphere $\Phi\cdot\Phi = v^2$, the

vacuum manifold. Picking the vacuum to point along the last axis,

$$\langle \Phi \rangle \;=\; (0, \dots, 0, v), \tag{29.161}$$

breaks $\mathrm{O}(N)$ down to the subgroup $H = \mathrm{O}(N-1)$ that rotates the first $N-1$ components among themselves and leaves the vacuum fixed. By Theorem 29.21 the spectrum at this vacuum has one massive mode — the radial fluctuation of $|\Phi|$ away from v — and $N-1$ massless Goldstone modes tangent to the sphere.

To separate the two cleanly we trade the Cartesian components Φ^i for polar field variables: a single radial field measuring the length and a unit vector recording the direction,

$$\Phi^i(x) \;=\; \big(v + \sigma(x)\big)\, n^i(x), \qquad n(x) \cdot n(x) = 1. \tag{29.162}$$

Here σ is the radial (sigma) field, with $\langle \sigma \rangle = 0$ at the vacuum, and n^i is a constrained field taking values on the unit sphere S^{N-1}. The decomposition is exact and merely a change of field variables; it is legitimate everywhere except at $\Phi = 0$, which is irrelevant since the dynamics is pinned near $|\Phi| = v$. We now substitute Eq. (29.162) into Eq. (29.160) and carry the algebra in full. The gradient of Φ is

$$\partial_\mu \Phi^i \;=\; (\partial_\mu \sigma)\, n^i \;+\; (v + \sigma)\, \partial_\mu n^i, \tag{29.163}$$

so the kinetic density expands into three pieces,

$$(\partial_\mu \Phi)\cdot(\partial^\mu \Phi) \;=\; (\partial_\mu \sigma)(\partial^\mu \sigma)\,(n\cdot n) + 2(v+\sigma)(\partial_\mu \sigma)\,(n\cdot \partial^\mu n) + (v+\sigma)^2\,(\partial_\mu n)\cdot(\partial^\mu n). \tag{29.164}$$

The constraint $n \cdot n = 1$ does two jobs here. First it sets the coefficient of the pure radial term to unity. Second, differentiating $n \cdot n = 1$ gives $n \cdot \partial_\mu n = 0$ identically, so the cross term in Eq. (29.164) *vanishes*: the radial and angular kinetic energies do not mix. The kinetic density is therefore exactly

$$(\partial_\mu \Phi) \cdot (\partial^\mu \Phi) \;=\; (\partial_\mu \sigma)(\partial^\mu \sigma) \;+\; (v+\sigma)^2\, (\partial_\mu n) \cdot (\partial^\mu n). \tag{29.165}$$

The potential depends only on $\Phi \cdot \Phi = (v+\sigma)^2$, i.e. only on the radial field:

$$V \;=\; \frac{\lambda}{4}\big((v+\sigma)^2 - v^2\big)^2 \;=\; \frac{\lambda}{4}\big(2v\sigma + \sigma^2\big)^2 \;=\; \lambda v^2\, \sigma^2 \;+\; \lambda v\, \sigma^3 \;+\; \frac{\lambda}{4}\, \sigma^4. \tag{29.166}$$

Collecting Eqs. (29.165) and (29.166), the linear sigma model in polar variables reads

$$\mathcal{L}_{\text{lin}} \;=\; \tfrac{1}{2}(\partial_\mu \sigma)(\partial^\mu \sigma) \;+\; \tfrac{1}{2}(v+\sigma)^2\, (\partial_\mu n) \cdot (\partial^\mu n) \;-\; \lambda v^2\, \sigma^2 \;-\; \lambda v\, \sigma^3 \;-\; \frac{\lambda}{4}\sigma^4. \tag{29.167}$$

Two readings of Eq. (29.167) are immediate. The radial field is a massive scalar with

$$m_\sigma^2 \;=\; 2\lambda v^2, \tag{29.168}$$

read off from the σ^2 term (recall a canonical mass term is $-\frac{1}{2}m^2\sigma^2$, and $\lambda v^2\sigma^2 = \frac{1}{2}(2\lambda v^2)\sigma^2$). The unit vector n has *no* potential at all — it is exactly massless, as Goldstone's theorem demands — and appears only through its derivatives, multiplied by the radial profile $(v+\sigma)^2$.

We can now take the limit that defines the low-energy theory. We send the radial mode to infinite mass while keeping the vacuum fixed:

$$m_\sigma \to \infty \quad \Longleftrightarrow \quad \lambda \to \infty \quad \text{at fixed } v. \tag{29.169}$$

In this limit any fluctuation of σ away from zero costs an arbitrarily large energy density $\lambda v^2\sigma^2$, so σ is frozen at the bottom of its potential, $\sigma \equiv 0$. Equivalently: the constraint $|\Phi| = v$ is enforced rigidly, and the field is confined to the vacuum manifold S^{N-1}. Setting $\sigma = 0$ in Eq. (29.167) leaves

$$\boxed{\mathcal{L}_{\text{nl}} \;=\; \frac{v^2}{2}\,(\partial_\mu n)\cdot(\partial^\mu n), \qquad n\cdot n = 1.} \tag{29.170}$$

This is the O(N) *nonlinear sigma model*: a field n valued on the sphere S^{N-1}, with the Lagrangian built from the round metric of that sphere scaled by v^2. The constraint $n\cdot n = 1$ is what makes it "nonlinear": the field space is curved, not a flat vector space, and the symmetry $n \to Rn$ no longer acts linearly on an unconstrained set of fields.

Definition 29.22 (O(N) nonlinear sigma model)**.** The O(N) *nonlinear sigma model* is the field theory of a map $n : \mathbb{R}^{1,3} \to S^{N-1}$ from spacetime into the unit sphere, with action $S = \int \mathrm{d}^4x\,\mathcal{L}_{\text{nl}}$ and $\mathcal{L}_{\text{nl}}$ given by Eq. (29.170). The constant v, with mass dimension $\frac{1}{2}(4-2) = 1$, is the *decay constant* of the model; it is the only free parameter of the leading-order theory and sets the scale of all Goldstone interactions.

The decoupling of σ is not a sleight of hand that throws information away; it is the leading term of a controlled expansion. The next worked example makes this precise by integrating the radial mode out at finite mass and showing that the corrections it leaves behind are suppressed by powers of $1/m_\sigma^2$.

Worked Example 29.23 (Integrating out the radial mode)**.** We keep m_σ finite and eliminate σ by its classical equation of motion, exhibiting both the nonlinear sigma model as the leading term and the first correction. Group the σ-dependence of Eq. (29.167) that is linear and quadratic in σ (cubic and higher in σ produce yet higher-order corrections):

$$\mathcal{L} \;\supset\; \tfrac{1}{2}(\partial\sigma)^2 - \tfrac{1}{2}m_\sigma^2\,\sigma^2 \;+\; v\,\sigma\,(\partial_\mu n)\cdot(\partial^\mu n) \;+\; \tfrac{v^2}{2}(\partial_\mu n)\cdot(\partial^\mu n). \tag{29.171}$$

The field σ has no potential minimum away from zero but is sourced by the Goldstone current $J \equiv v\,(\partial_\mu n)\cdot(\partial^\mu n)$. Its equation of motion is $(\Box + m_\sigma^2)\sigma = J$.

At momenta far below m_σ we may drop $\Box$ relative to m_σ^2 and solve algebraically,

$$\sigma \;=\; \frac{J}{m_\sigma^2}\Big(1-\frac{\Box}{m_\sigma^2}+\cdots\Big) \;=\; \frac{v}{m_\sigma^2}\,(\partial n)^2 \;+\; O(1/m_\sigma^4), \tag{29.172}$$

where $(\partial n)^2 \equiv (\partial_\mu n)\cdot(\partial^\mu n)$. Substituting back into Eq. (29.171), the linear and quadratic pieces combine to $-\tfrac{1}{2}\sigma J+\sigma J=\tfrac{1}{2}\sigma J=\tfrac{1}{2}J^2/m_\sigma^2$ at leading order, so the effective Lagrangian for n alone is

$$\mathcal{L}_{\text{eff}} \;=\; \frac{v^2}{2}\,(\partial n)^2 \;+\; \frac{v^2}{2m_\sigma^2}\big[(\partial n)^2\big]^2 \;+\; \cdots . \tag{29.173}$$

The first term is exactly Eq. (29.170). The second is a four-derivative correction with coefficient $v^2/2m_\sigma^2 = 1/4\lambda$, suppressed by $1/m_\sigma^2$; as $m_\sigma\to\infty$ it vanishes and the pure nonlinear sigma model is recovered. The lesson is general and central to all that follows: *the nonlinear sigma model is the leading term of a derivative expansion*, and the heavy modes one has integrated out reappear only as higher-derivative operators suppressed by their mass. This is precisely how the higher-order chiral Lagrangian of Chapter 39 is organized.

29.5.2 Goldstones as coordinates on the coset G/H

The constrained field n on S^{N-1} is geometric data; to do field theory we need unconstrained coordinates. Solving the constraint $n\cdot n=1$ for one component in terms of the other $N-1$ furnishes such coordinates. Write the first $N-1$ components as independent fields π^a $(a=1,\dots,N-1)$ measured in units that will make the kinetic term canonical, and recover the last from the constraint:

$$n^a \;=\; \frac{\pi^a}{v}\quad (a=1,\dots,N-1), \qquad n^N \;=\; \sqrt{1-\frac{\pi\cdot\pi}{v^2}}\,, \qquad \pi\cdot\pi \equiv \sum_{a=1}^{N-1}\pi^a\pi^a. \tag{29.174}$$

This chart covers the upper hemisphere $n^N>0$ containing the vacuum $n=(0,\dots,0,1)$, i.e. $\pi=0$; that is the only region the low-energy expansion ever explores, since the fields fluctuate near the vacuum. The fields π^a are the Goldstone bosons in concrete form. Their number is $N-1$, which we now confirm is exactly the coset count.

Proposition 29.24 (Goldstone count equals the coset dimension)**.** *For $G=\mathrm{O}(N)$ broken to $H=\mathrm{O}(N-1)$, the number of Goldstone fields equals* $\dim G-\dim H = \dim(G/H)=\dim S^{N-1}=N-1$.

Proof. The dimension of the rotation group $\mathrm{O}(m)$ is the number of independent rotation planes, $\dim \mathrm{O}(m)=\binom{m}{2}=\tfrac{1}{2}m(m-1)$. Hence

$$\dim G-\dim H \;=\; \frac{N(N-1)}{2}-\frac{(N-1)(N-2)}{2} \;=\; \frac{N-1}{2}\big[N-(N-2)\big] \;=\; N-1. \tag{29.175}$$

The vacuum manifold is the orbit $G\langle\Phi\rangle$, which by the orbit–stabilizer theorem (Section 2.3) is the homogeneous space G/H, here $\mathrm{O}(N)/\mathrm{O}(N-1) = S^{N-1}$, of dimension $N-1$. The parametrization Eq. (29.174) uses exactly $N-1$ independent fields π^a. All three counts agree. □

This is the reconfirmation promised in the section goal: the abstract Goldstone count of Theorem 29.21, the geometric dimension of the coset G/H, and the number of unconstrained fields needed to chart the vacuum manifold are one and the same number. The Goldstone bosons *are* coordinates on G/H.

We now compute the induced metric — the Lagrangian in terms of π^a — explicitly. Differentiating Eq. (29.174),

$$\partial_\mu n^a = \frac{\partial_\mu \pi^a}{v}, \qquad \partial_\mu n^N = -\frac{\pi\cdot\partial_\mu\pi}{v^2\sqrt{1-\pi\cdot\pi/v^2}} = -\frac{\pi\cdot\partial_\mu\pi}{v\sqrt{v^2-\pi\cdot\pi}}, \tag{29.176}$$

where $\pi\cdot\partial_\mu\pi \equiv \sum_a \pi^a\partial_\mu\pi^a$. Squaring and summing,

$$(\partial_\mu n)\cdot(\partial^\mu n) = \sum_a \frac{(\partial_\mu\pi^a)(\partial^\mu\pi^a)}{v^2} + \frac{(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi)}{v^2(v^2-\pi\cdot\pi)}. \tag{29.177}$$

Multiplying by $v^2/2$ as in Eq. (29.170) gives the nonlinear sigma model in Goldstone coordinates,

$$\mathcal{L}_{\mathrm{nl}} \;=\; \frac{1}{2}\,g_{ab}(\pi)\,(\partial_\mu\pi^a)(\partial^\mu\pi^b), \qquad g_{ab}(\pi) \;=\; \delta_{ab} \;+\; \frac{\pi_a\pi_b}{v^2-\pi\cdot\pi}, \tag{29.178}$$

with $\pi_a \equiv \pi^a$ (the field-space indices are raised and lowered with δ_{ab}). The symmetric tensor $g_{ab}(\pi)$ is the *coset metric*: it is precisely the round metric of the radius-v sphere S^{N-1} pulled back to the π-chart. The general statement that a nonlinear sigma model is a theory of maps into a Riemannian target $(\mathcal{M}, g)$, with Lagrangian $\frac{1}{2}g_{ab}(\pi)\partial_\mu\pi^a\partial^\mu\pi^b$, is the geometric content here; for spontaneous breaking the target is the coset G/H and g is the unique (up to scale) G-invariant metric on it.

Expanding the metric in powers of the fields exposes the interaction structure:

$$g_{ab}(\pi) \;=\; \delta_{ab} + \frac{\pi_a\pi_b}{v^2} + \frac{\pi_a\pi_b\,(\pi\cdot\pi)}{v^4} + \cdots, \tag{29.179}$$

so that

$$\mathcal{L}_{\mathrm{nl}} \;=\; \tfrac{1}{2}(\partial_\mu\pi)\cdot(\partial^\mu\pi) \;+\; \frac{1}{2v^2}(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi) \;+\; O\Big(\frac{\pi^4(\partial\pi)^2}{v^4}\Big). \tag{29.180}$$

The first term is $N-1$ canonically normalized free massless scalars. The second is the leading interaction: a four-field vertex with *two* derivatives and coupling strength $1/v^2$. There is no π^3 vertex, no mass term, no non-derivative coupling of any kind. The whole interaction hierarchy is organized in powers of $1/v^2$ multiplying ever more fields, always with exactly two derivatives at this (leading, two-derivative) order in the Lagrangian. This derivative-only structure is the heart of low-energy Goldstone dynamics, and we now prove it is not an accident of the chart but forced by the symmetry.

29.5.3 The nonlinear realization and the masslessness of derivative coupling

In the linear sigma model the symmetry $\Phi \to R\Phi$ acts linearly. After projecting onto the vacuum manifold, the unbroken subgroup $H = \mathrm{O}(N-1)$ still acts linearly on the Goldstone fields — it rotates the π^a among themselves — but the broken generators act *nonlinearly*, and that nonlinearity is the structural origin of masslessness. We make this explicit.

The broken generators are those that move the vacuum $(0, \dots, 0, v)$. They are the rotations X^a in the (a, N) planes, $a = 1, \dots, N-1$, acting on the components as (X^a): $n^a \leftrightarrow n^N$. An infinitesimal broken rotation with parameters θ^a acts on the unit vector as

$$\delta n^a = \theta^a \, n^N, \qquad \delta n^N = -\sum_b \theta^b \, n^b. \tag{29.181}$$

In Goldstone coordinates, using $n^a = \pi^a / v$ and $n^N = \sqrt{1 - \pi \cdot \pi / v^2}$, this becomes

$$\delta \pi^a \;=\; v\,\theta^a \sqrt{1 - \frac{\pi \cdot \pi}{v^2}} \;=\; v\,\theta^a \;-\; \frac{\theta^a \, (\pi \cdot \pi)}{2v} \;+\; \cdots . \tag{29.182}$$

The leading piece is a *constant shift*, $\delta \pi^a = v\,\theta^a$, independent of the fields. This is the nonlinear (or inhomogeneous) realization of the broken symmetry: a broken generator translates the Goldstone field by a constant, plus field-dependent corrections. The constant shift is the field-theoretic shadow of the fact that moving along a broken direction simply slides the system from one degenerate vacuum to a neighbouring one — and since all vacua are physically equivalent, the constant value of π cannot matter.

Proposition 29.25 (Shift symmetry forbids non-derivative interactions)**.** *If the action is invariant under the constant shift $\pi^a \to \pi^a + c^a$ for arbitrary constants c^a, then the Lagrangian can depend on the Goldstone fields only through their derivatives $\partial_\mu \pi^a$ (and higher derivatives); in particular it contains no potential, so the Goldstones are exactly massless to all orders.*

Proof. Suppose the Lagrangian density contains a term that is not a total derivative and depends on the undifferentiated field, $\mathcal{L} \supset U(\pi)$ with U not constant. Under $\pi^a \to \pi^a + c^a$ this term changes by $U(\pi + c) - U(\pi)$. For this to be a total spacetime derivative (so that the action is invariant) for *all* constant c^a, and since $U(\pi + c) - U(\pi)$ contains no derivatives, it must vanish identically: $U(\pi + c) = U(\pi)$ for all c, i.e. U is constant, contradicting the assumption. Hence every field-dependent term carries at least one derivative on each appearance of π. In particular a mass term $\frac{1}{2} m_\pi^2 \, \pi^a \pi^a$, which is non-derivative, is forbidden: $m_\pi^2 = 0$ exactly. Because the constant-shift invariance Eq. (29.182) is an exact symmetry of the full action — it is the broken part of the exact $\mathrm{O}(N)$ symmetry, realized nonlinearly — this conclusion holds to all orders in the loop expansion, not merely at tree level. □

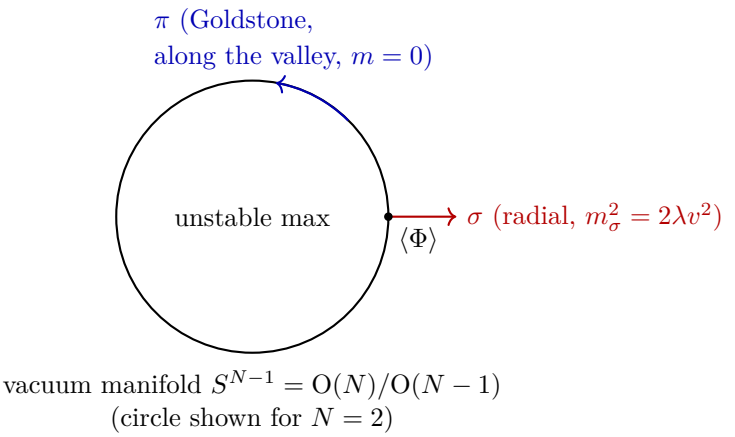


Figure 29.6: The vacuum manifold as the valley of the Mexican-hat potential. Climbing radially out of the valley excites the massive mode σ; sliding *along* the valley moves between degenerate vacua at no energy cost and is the Goldstone direction π. The nonlinear sigma model Eq. (29.170) is the dynamics of motion confined to the valley floor, obtained by sending $m_\sigma \to \infty$.

This is the geometric re-derivation of Goldstone's theorem promised in the section goal. In the effective-potential proof, masslessness came from the flatness of V along the vacuum manifold (a statement about the Hessian); here it comes from the shift symmetry $\pi \to \pi + c$ (a statement about how the broken generator acts on the field). They are the same fact seen from two sides: the flat direction of the potential *is* the direction the broken generator translates along, and a translation symmetry of a field is exactly the statement that no potential for it is allowed. The metric Eq. (29.178), which only ever appears contracted with two derivatives, manifestly respects this: every interaction in Eq. (29.180) carries two derivatives, so the shift symmetry is realized term by term.

Physical Insight: why Goldstones are derivatively coupled

A Goldstone boson is, operationally, the local angle of the order parameter. Shifting that angle by a constant everywhere is a global symmetry transformation and therefore costs no energy; only *gradients* of the angle — a twist of the order parameter from point to point — cost energy, and they cost energy quadratically in the gradient for small twists. That is why the leading Lagrangian is $\frac{1}{2} g_{ab} \partial_\mu \pi^a \partial^\mu \pi^b$ with no potential: it is the elastic energy of a medium whose ground state is infinitely degenerate under uniform rotations of the order parameter. The decay constant v is the stiffness of that medium.

29.5.4 The Adler zero: a low-energy theorem

The derivative coupling has an immediate, sharp, and experimentally famous consequence. Every Goldstone field enters the Lagrangian with at least one derivative, so every Feynman vertex involving an external Goldstone line carries a factor of that line's momentum. An amplitude for emitting a single Goldstone therefore vanishes when the Goldstone's four-momentum is taken to zero. This is the *Adler zero*, the prototype low-energy theorem.

Theorem 29.26 (Adler zero)**.** *In the nonlinear sigma model Eq. (29.170), a connected amplitude with one external Goldstone of momentum q and any number of other external Goldstones vanishes linearly as $q \to 0$, provided no internal Goldstone propagator goes on shell in that limit (i.e. away from kinematic singularities). Symbolically, $\mathcal{M}(q, \dots) \to 0$ as $q^\mu \to 0$.*

Proof (structural). Every interaction vertex in Eq. (29.180), and at every higher order in the field/derivative expansion, contains exactly the same number of derivatives as one obtains by acting ∂_μ on each field at least once; concretely the two-derivative vertices have the schematic form $(1/v^{2k})\,\pi^{2k}\,\partial_\mu\pi\,\partial^\mu\pi$. Consider the soft external line carrying momentum q. It attaches to the diagram at some vertex. If that vertex is the soft line's own derivative coupling, the Feynman rule supplies a factor q^μ from $\partial_\mu\pi \to -iq_\mu$ acting on the soft leg, so the contribution is $O(q)$. The only way to evade this would be for the soft line to attach to a vertex through an undifferentiated π while the diagram develops a compensating $1/q$ pole. Such a pole can come only from an internal propagator i/k^2 that goes on shell, $k \to 0$, as $q \to 0$ — but the propagator carries momentum equal to the soft momentum plus that of an adjacent external line, so it is forced on shell only at exceptional (collinear/threshold) kinematics, excluded by hypothesis. Everywhere else the soft factor q^μ is uncompensated and the amplitude vanishes as $q \to 0$. The masslessness of the Goldstones (Theorem 29.25) is what keeps the internal propagators generically off shell, so the two ingredients — derivative coupling and exact masslessness — conspire to give the zero. □

The cleanest demonstration is an explicit four-point amplitude, which also lets us record the leading-order Goldstone scattering amplitude that Chapter 39 will reuse for the pions.

Worked Example 29.27 (Leading-order $\pi\pi$ scattering and its Adler zero)**.** We compute the tree amplitude for $\pi^a(p_1)\,\pi^b(p_2) \to \pi^c(p_3)\,\pi^d(p_4)$ at $O(1/v^2)$ from the quartic vertex of Eq. (29.180), namely

$$\mathcal{L}_4 \;=\; \frac{1}{2v^2}\,(\pi^e\,\partial_\mu\pi^e)\,(\pi^f\,\partial^\mu\pi^f). \tag{29.183}$$

The vertex. Treat all four momenta as incoming, $p_1+p_2+p_3+p_4=0$ (the physical process is recovered by crossing $p_3, p_4 \to -p_3, -p_4$). The four field operators

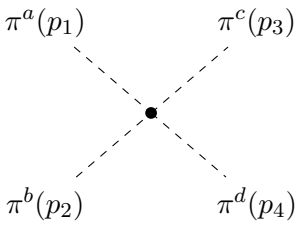


Figure 29.7: The single leading-order contact diagram for Goldstone–Goldstone scattering $\pi^a\pi^b \to \pi^c\pi^d$ in the nonlinear sigma model. The two-derivative vertex Eq. (29.180) produces the amplitude Eq. (29.185), which vanishes when any one external momentum is taken soft — the Adler zero made manifest.

in $\mathcal{L}_4$ are π^e (undifferentiated), $\partial_\mu\pi^e$ (differentiated), π^f (undifferentiated), $\partial^\mu\pi^f$ (differentiated). The amplitude is i times the sum over all ways of attaching the four external lines to these four slots; an external line of incoming momentum p contracted with an undifferentiated slot gives 1, and with a differentiated slot ∂_μ gives $-ip_\mu$ (since the incoming field is $e^{-ip\cdot x}$). The $e \leftrightarrow f$ relabelling symmetry of Eq. (29.183) cancels the prefactor $\frac{1}{2}$, leaving an effective coefficient $1/v^2$ and a sum over the three unordered ways to split $\{1,2,3,4\}$ into two pairs, one pair feeding the e-slots (sharing the O(N) index, hence a δ) and the other the f-slots.

For the partition $\{i,j \mid k,l\}$: the two lines in a pair carry a common field-space index, giving $\delta^{a_i a_j}\delta^{a_k a_l}$; within each pair exactly one line sits on the differentiated slot, and summing over which line is differentiated in each pair, the Lorentz contraction between the two differentiated slots is $\sum_{\text{diff}}(-ip)\cdot(-ip) = -(p_i+p_j)\cdot(p_k+p_l)$. Thus

$$i\mathcal{M} \;=\; \frac{i}{v^2}\Big[\delta^{ab}\delta^{cd}\,(-(p_1{+}p_2){\cdot}(p_3{+}p_4))+\delta^{ac}\delta^{bd}\,(-(p_1{+}p_3){\cdot}(p_2{+}p_4))+\delta^{ad}\delta^{bc}\,(-(p_1{+}p_4){\cdot}(p_2{+}p_3))\Big]. \tag{29.184}$$

Mandelstam form. Use $p_1{+}p_2{+}p_3{+}p_4 = 0$ so that $p_3{+}p_4 = -(p_1{+}p_2)$, giving $-(p_1{+}p_2){\cdot}(p_3{+}p_4) = (p_1{+}p_2)^2 \equiv s$; likewise $-(p_1{+}p_3){\cdot}(p_2{+}p_4) = (p_1{+}p_3)^2 \equiv t$ and $-(p_1+p_4)\cdot(p_2+p_3) = (p_1+p_4)^2 \equiv u$, with the all-incoming Mandelstam variables s,t,u. Hence the leading-order Goldstone scattering amplitude is

$$\boxed{\mathcal{M}^{ab,cd} \;=\; \frac{1}{v^2}\Big(\delta^{ab}\delta^{cd}\,s+\delta^{ac}\delta^{bd}\,t+\delta^{ad}\delta^{bc}\,u\Big).} \tag{29.185}$$

For massless particles $s+t+u = \sum_i p_i^2 = 0$, the standard crossing-symmetric structure. This is exactly the form the chiral Lagrangian reproduces for $\pi\pi$ scattering with $v \to f_\pi$ (Chapter 39); the O(4) $\simeq$ SU(2) $\times$ SU(2) case there gives Weinberg's result.

The Adler zero. Take the first Goldstone soft, $p_1 \to 0$. Then $s = (p_1 + p_2)^2 \to p_2^2 = 0$, $t = (p_1+p_3)^2 \to p_3^2 = 0$, $u = (p_1+p_4)^2 \to p_4^2 = 0$, because the

remaining particles are on shell and massless. Every term in Eq. (29.185) carries one Mandelstam factor, so

$$\mathcal{M}^{ab,cd} \xrightarrow{p_1 \to 0} 0, \tag{29.186}$$

linearly in the soft momentum. The amplitude has no constant piece as $p_1 \to 0$: a single soft Goldstone decouples. This is the Adler zero verified explicitly, and it is a direct fingerprint of the derivative coupling forced by Theorem 29.25.

29.5.5 The geometric construction: Maurer–Cartan form and the CCWZ split

So far the coset metric and the derivative coupling were read off from one explicit parametrization of S^{N-1}. We now present the coordinate-free construction — the Callan–Coleman–Wess–Zumino (CCWZ) framework — that produces the invariant Lagrangian for *any* G/H and shows that, at two derivatives, it is unique up to the single decay constant. We keep the treatment introductory but honest: every object is defined and the uniqueness claim is proved.

The Goldstone group element. Let G be a compact Lie group with Lie algebra split into unbroken generators T^i (spanning $\mathfrak{h}$, the algebra of H) and broken generators X^a (spanning the complement $\mathfrak{m}$, with $a = 1, \dots, \dim G - \dim H$). Package the Goldstone fields into a group element by exponentiating the broken generators only (Section 2.3):

$$U(x) \;=\; \exp\!\Big(\frac{i}{f}\,\pi^a(x)\,X^a\Big) \;\in\; G, \tag{29.187}$$

where f is the decay constant (the role of v above). Two group elements that differ by right multiplication by H describe the same point of G/H, so U is a representative of the coset gH; the choice Eq. (29.187), using only broken generators, is the standard "coset representative." Under a global G transformation $g \in G$, U transforms by

$$g\,U(x) \;=\; U'(x)\,h\big(\pi(x), g\big), \qquad U' = \exp\!\Big(\tfrac{i}{f}\pi'^a X^a\Big), \quad h \in H, \tag{29.188}$$

which *defines* the transformed Goldstone fields π' and a compensating, field-dependent H element $h(\pi, g)$. This is the nonlinear realization in its general form: G acts on the π^a nonlinearly, and a residual local-looking H rotation h appears because moving in G/H generically reshuffles the unbroken frame. For $g \in H$ the relation reduces to $\pi'^a X^a = h\,\pi^a X^a h^{-1}$, a linear rotation of the π^a in the representation $\mathfrak{m}$ of H; for broken g it produces the inhomogeneous shift seen in Eq. (29.182).

The Maurer–Cartan form and its split. The natural derivative object on a group manifold is the (left-invariant) *Maurer–Cartan one-form* $U^{-1}\partial_\mu U$. It is Lie-algebra valued, so it decomposes along the broken and unbroken directions:

$$U^{-1}\partial_\mu U \;=\; i\left(\, e^a_\mu\, X^a \;+\; \omega^i_\mu\, T^i \,\right) \;\equiv\; i(e_\mu + \omega_\mu), \tag{29.189}$$

defining the broken ("vielbein") component $e_\mu = e^a_\mu X^a \in \mathfrak{m}$ and the unbroken ("connection") component $\omega_\mu = \omega^i_\mu T^i \in \mathfrak{h}$. The names are earned by their transformation laws. Differentiating Eq. (29.188) and using that g is constant, $U^{-1}\partial_\mu U = h^{-1}(U'^{-1}\partial_\mu U')h + h^{-1}\partial_\mu h$, and projecting onto $\mathfrak{m}$ and $\mathfrak{h}$ (the split is preserved because $[\mathfrak{h},\mathfrak{m}] \subseteq \mathfrak{m}$ for a symmetric coset, and $h^{-1}\partial_\mu h \in \mathfrak{h}$) gives

$$e_\mu \;\longrightarrow\; h\, e_\mu\, h^{-1}, \tag{29.190}$$

$$\omega_\mu \;\longrightarrow\; h\,\omega_\mu\, h^{-1} + h\,\partial_\mu h^{-1}. \tag{29.191}$$

Thus e^a_μ transforms *homogeneously* — as a vector in the representation $\mathfrak{m}$ of H, with the compensating $h(\pi, g)$ playing the role of a local H rotation — while ω^i_μ transforms as a *connection* (gauge field) for H, inhomogeneously. The broken component e^a_μ is therefore the covariant building block: it is the "covariant derivative of the Goldstone field." Expanding Eq. (29.187) to leading order, $U^{-1}\partial_\mu U = \frac{i}{f}\partial_\mu \pi^a X^a + O(\pi^2)$, so

$$e^a_\mu \;=\; \frac{1}{f}\,\partial_\mu \pi^a \;+\; O(\pi^2 \partial\pi / f^3), \tag{29.192}$$

confirming that e^a_μ reduces to the ordinary Goldstone gradient near the vacuum.

Uniqueness of the two-derivative Lagrangian. We now build the Lagrangian. It must be a Lorentz scalar, invariant under G, and we organize it by the number of derivatives, since each e_μ carries exactly one. At zero derivatives the only invariant is a constant — no potential, in agreement with Theorem 29.25. At one derivative one cannot form a Lorentz scalar from a single e^a_μ. At two derivatives the available covariant object is e^a_μ (the connection ω_μ is not covariant on its own; it can appear only inside an H-covariant derivative, which needs additional matter fields not present here). The only G-invariant Lorentz scalar quadratic in e^a_μ is its H-invariant contraction. Because H acts on the index a by the representation $\mathfrak{m}$, and for a *symmetric* coset (such as $S^{N-1} = \mathrm{O}(N)/\mathrm{O}(N-1)$) this representation is irreducible, Schur's lemma guarantees a unique invariant bilinear form $\kappa_{ab} \propto \delta_{ab}$ on $\mathfrak{m}$. Hence the unique two-derivative invariant is

$$\mathcal{L}^{(2)} \;=\; \frac{f^2}{2}\,\delta_{ab}\, e^a_\mu\, e^{\mu\, b}, \tag{29.193}$$

with the prefactor f^2 and the normalization of δ_{ab} fixed by demanding, through Eq. (29.192), that the kinetic term emerge canonically: $\mathcal{L}^{(2)} = \frac{1}{2}(\partial_\mu \pi^a)(\partial^\mu \pi^a) +$

(interactions). The single constant f^2 in front is the only freedom; it is the decay constant, and it cannot be removed by field redefinition because it sets the dimensionful scale of the curved target. This is the precise sense of the section goal's claim: *the two-derivative Goldstone Lagrangian is unique up to the decay constant.* Carrying out the contraction for $G/H = \mathrm{O}(N)/\mathrm{O}(N-1)$ with $f = v$ reproduces Eq. (29.178) term by term, so the explicit and the geometric constructions agree, as they must.

Caution: the nonlinear sigma model is an effective theory

The Lagrangian Eq. (29.170) is *not* renormalizable: the coupling $1/v^2$ carries negative mass dimension, so loop corrections generate operators with more and more derivatives, exactly the four-derivative term already seen in Eq. (29.173). This is a feature, not a flaw: the nonlinear sigma model is a low-energy *effective field theory*, valid below the scale of the modes it does not contain (here m_σ). One does not renormalize it to all energies; one uses it up to its cutoff and matches to the fuller theory above. The systematic bookkeeping of this expansion is the subject of the next paragraph and is brought to maturity for the pions in Chapter 39.

29.5.6 Power counting and the scale $4\pi f$

Because the nonlinear sigma model is an effective theory, we must know its expansion parameter — the dimensionless number that controls when the derivative expansion converges and when loops are small. We extract it by dimensional analysis of a loop. Each two-derivative vertex contributes a factor p^2/f^2 to a low-energy amplitude, where p is a typical external momentum. A loop adds an integration $\int \mathrm{d}^4k/(2\pi)^4$ and, for the Goldstones, two propagators $\sim 1/k^2$ and the loop momentum running through the derivative vertices. Counting powers, a single Goldstone loop dressing a tree amplitude multiplies it by

$$\frac{1}{f^2}\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{k^2\,p^2}{k^2\,k^2} \;\sim\; \frac{1}{16\pi^2 f^2}\,p^2\,[\,\log + \ldots\,] \;=\; \frac{p^2}{(4\pi f)^2}\,[\,\log + \ldots\,], \tag{29.194}$$

where the $1/16\pi^2 = 1/(4\pi)^2$ is the universal four-dimensional loop factor from the solid angle of the momentum integral. The corrections are therefore organized in powers of

$$\boxed{\left(\frac{p}{4\pi f}\right)^2} \tag{29.195}$$

each additional loop, or each additional pair of derivatives in the Lagrangian, costing one such factor. The scale that controls the effective theory is thus not f itself but $\Lambda \sim 4\pi f$: the theory is predictive for momenta $p \ll 4\pi f$ and breaks down (loops and higher operators become $O(1)$) at $p \sim 4\pi f$. This is the

naive-dimensional-analysis estimate of the cutoff, and it correctly anticipates the scale at which new degrees of freedom — the radial mode σ, or in QCD the heavier hadrons — must appear. For the pions, $f \to f_\pi \approx 92$ MeV, so $4\pi f_\pi \approx 1.2$ GeV, comfortably accounting for why chiral perturbation theory works at pion energies and fails near the ρ meson. This power counting is the engine of Chapter 39, carried there in full with the loop coefficients computed.

We have distilled the Goldstone sector down to its universal core: a derivative-coupled theory of fields living on the coset G/H, with a single scale f, exact masslessness enforced by the nonlinear shift symmetry, and a clean low-energy theorem (the Adler zero) as its signature. This clean Lagrangian is the canvas onto which the final section paints a small explicit breaking: adding a term that lifts the flat directions slightly turns the exact, massless Goldstones into light but massive *pseudo*-Goldstone bosons, whose masses are calculable from the breaking, and leads to the partial-conservation and mass-formula relations that govern the pions in the real world.

29.6 Explicit Breaking, Pseudo-Goldstone Bosons, and Vacuum Alignment

The nonlinear sigma model of Section 29.5 delivered the universal low-energy theory of the Goldstone modes: fields π^a valued in the coset G/H, interacting only through derivatives, with exactly massless quanta and the Adler zero enforced by the shift symmetry $\pi^a \to \pi^a + c^a$. That theory is the idealized limit of an *exact* broken symmetry. Nature almost never hands us one. The chiral symmetry of the strong interactions, the isospin of the nucleon, the flavor symmetries of the quark sector—each is an excellent but *approximate* symmetry of the Lagrangian, slightly spoiled by a small non-invariant term (a quark mass, an electromagnetic correction). The question this section answers is precise: if we perturb the exactly broken theory by a small term that is *not* invariant under G, what happens to the massless Goldstones?

The answer completes the spontaneous-versus-explicit dichotomy promised in the chapter introduction. A genuinely spontaneously broken symmetry forces exact masslessness; the small explicit term lifts the flat valley of the potential into a gentle slope, and the would-be Goldstones acquire a small, *calculable* mass-squared that is *linear* in the breaking. These are the *pseudo-Goldstone bosons*. We will (i) sharpen the distinction at the level of the Ward identity, which now carries a source term—partial conservation; (ii) derive the pseudo-Goldstone mass formula, a Gell-Mann–Oakes–Renner-type relation, two independent ways (from the effective potential and from the broken Ward identity), getting the same answer; (iii) establish the vacuum-alignment principle that fixes the vacuum when several breaking terms compete and that guarantees the pseudo-Goldstone

masses are nonnegative; (iv) run the whole apparatus on the $O(N)$ model of ϕ^4 theory, the explicit-breaking analog of a quark mass; and (v) state honestly the low-dimension obstruction (Coleman–Mermin–Wagner) and the live refinements of Goldstone counting at the frontier.

29.6.1 Spontaneous versus explicit breaking, sharpened

The two ways a symmetry can fail to be manifest on the spectrum are logically independent, and it is worth stating the contrast in its sharpest form before computing anything.

Definition 29.28 (Spontaneous vs. explicit breaking)**.** Let G be a continuous global symmetry group acting on the fields. The breaking of G is

- *spontaneous* if the action S is exactly G-invariant but the vacuum $|0\rangle$ is not, i.e. there is a generator Q_a with $Q_a|0\rangle \neq 0$ (equivalently, an order parameter $\langle 0|\delta_a\Phi|0\rangle \neq 0$ for some local field Φ); and
- *explicit* if the action itself is not G-invariant: $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_\varepsilon$ with $\mathcal{L}_0$ exactly G-invariant and a perturbation

$$\mathcal{L}_\varepsilon = \varepsilon\,\mathcal{O}, \qquad \delta_a\mathcal{O} \neq 0 \text{ for some } a, \tag{29.196}$$

  where ε is a small parameter (a coupling or a mass) controlling the strength of the breaking and $\mathcal{O}$ is a local operator that does not transform into a total derivative under G.

Here $\delta_a\Phi \equiv i[Q_a, \Phi]$ denotes the infinitesimal variation of an operator Φ under the a-th generator, with $\delta_a\phi_i = i(T_a)_{ij}\phi_j$ on the fundamental multiplet.

The distinction is not academic. Spontaneous breaking is *exact*: it produces strictly massless Goldstones (Theorem 29.21) and a vacuum degeneracy that is a property of the dynamics, not of any input parameter. Explicit breaking is a deformation of the dynamics by hand: there is no associated Goldstone theorem, no protected masslessness, and the symmetry is simply not there. The physically interesting—and ubiquitous—situation is the *combination*: a symmetry that is spontaneously broken in the $\varepsilon \to 0$ limit and *also* explicitly broken by a small ε. Then the spontaneous part supplies the (would-be) Goldstones and the explicit part supplies their small mass. The smallness of the pseudo-Goldstone mass is protected: it must vanish as $\varepsilon \to 0$, because at $\varepsilon = 0$ the exact Goldstone theorem applies. That is precisely why pions are anomalously light.

Caution

"Symmetric" and "broken" in Theorem 29.28 refer to two different objects in the two cases. For spontaneous breaking the Lagrangian *is* symmetric and the *vacuum* is not; for explicit breaking the *Lagrangian* is not symmetric to begin with. Throughout this chapter, unless we say "explicit," "broken" always means spontaneously broken—a statement about the realization on $|0\rangle$, never about $\mathcal{L}_0$.

29.6.2 The partially conserved current: the explicitly broken Ward identity

A symmetry of the action gives a conserved Noether current (Eq. (7.80)). A symmetry broken only by a small term gives a current whose divergence is small and computable. We derive this "anomalous-in-ε" Ward identity from scratch; it is the engine of the whole section.

Start from $S = \int \mathrm{d}^4x\,(\mathcal{L}_0 + \mathcal{L}_\varepsilon)$ with $\mathcal{L}_0$ exactly invariant under the global transformation $\delta\phi_i = i\alpha^a (T_a)_{ij}\phi_j$ with *constant* α^a. Following the Noether procedure, promote the parameters to slowly varying functions $\alpha^a(x)$. Because $\mathcal{L}_0$ is invariant when α^a is constant, its variation under a local $\alpha^a(x)$ can depend only on the *gradient* of α^a, and to first order in derivatives it takes the form

$$\delta\mathcal{L}_0 = -\,(\partial_\mu \alpha^a)\, J^\mu_a, \tag{29.197}$$

which *defines* the Noether current J^μ_a (this is exactly the J^μ_a of Eq. (7.80); the sign is fixed so that on the equations of motion the conserved-current statement comes out with $\partial_\mu J^\mu_a = 0$ in the unbroken case). The breaking term, by contrast, is *not* invariant even for constant α^a, so under the local transformation it varies already at zeroth order in $\partial\alpha$:

$$\delta\mathcal{L}_\varepsilon = \varepsilon\,\alpha^a\,\delta_a\mathcal{O} + (\text{terms } \propto \partial_\mu\alpha^a), \tag{29.198}$$

where $\delta_a\mathcal{O} = \dfrac{\partial\mathcal{O}}{\partial\phi_i}\, i(T_a)_{ij}\phi_j$ is the ordinary symmetry variation of $\mathcal{O}$. The gradient terms in Eq. (29.198) merely add a G-non-invariant piece to the current; for the standard case in which $\mathcal{O}$ contains no derivatives—a potential term, a mass term—they are absent, and we take this case.[1]

Now demand that the total action be stationary on the equations of motion of $\mathcal{L}$. For an arbitrary localized $\alpha^a(x)$,

$$0 = \delta S = \int \mathrm{d}^4x \Big[-(\partial_\mu\alpha^a) J^\mu_a + \varepsilon\,\alpha^a\,\delta_a\mathcal{O}\Big] = \int \mathrm{d}^4x\;\alpha^a \Big[\partial_\mu J^\mu_a + \varepsilon\,\delta_a\mathcal{O}\Big], \tag{29.199}$$

[1] If $\mathcal{O}$ carries derivatives the current receives an extra ε-suppressed contribution; it does not affect the divergence relation below, which follows from stationarity of the full action.

where we integrated the first term by parts and dropped the boundary term. Since $\alpha^a(x)$ is arbitrary, the bracket vanishes pointwise:

$$\boxed{\partial_\mu J_a^\mu = -\,\varepsilon\,\delta_a \mathcal{O}}\,. \tag{29.200}$$

This is *partial conservation*: the current is conserved up to a term proportional to the small breaking ε. In the limit $\varepsilon \to 0$ it reduces to the exact conservation law of Eq. (7.80). Equation Eq. (29.200) is the operator statement underlying the historical PCAC ("partially conserved axial current") hypothesis: for QCD with light quarks, J_a^μ is the axial-isospin current, $\varepsilon\,\delta_a\mathcal{O}$ is built from the quark masses, and the right-hand side, sandwiched between the vacuum and a one-pion state, is the source of the pion mass. We turn to that matrix element next.

Physical Insight

Read Eq. (29.200) as a budget. The left-hand side measures how fast the broken charge $Q_a(t) = \int \mathrm{d}^3x\, J_a^0$ leaks in time. At $\varepsilon = 0$ the charge is exactly conserved but ill-defined on $|0\rangle$ (it creates a zero-momentum Goldstone)—this is the content of the Goldstone theorem. Turning on ε makes the charge genuinely non-conserved by a tiny amount, and that tiny non-conservation is exactly what gives the Goldstone a tiny mass. Masslessness and exact conservation stand or fall together.

29.6.3 Pseudo-Goldstone masses from the effective potential

We now compute the mass that Eq. (29.196) confers on the would-be Goldstones. The cleanest route uses the effective potential, whose Hessian at its minimum is the physical mass-squared matrix (the quantum-level diagnostic established in Theorem 29.21). The decisive structural fact, proven there, is that in the *exactly* broken theory the effective potential V_0 is *flat* along the entire vacuum manifold

$$\mathcal{M} = \{\phi = g\,\phi_* : g \in G\} \cong G/H, \tag{29.201}$$

the orbit of any vacuum ϕ_* under G; the directions tangent to $\mathcal{M}$ are precisely the $n = \dim G - \dim H$ Goldstone directions and they have vanishing curvature. Adding the explicit term tilts this flat valley.

We parametrize the manifold by dimensionless coordinates ξ^a via $\phi(\xi) = \exp(i\xi^a X^a)\,\phi_*$, where $\{X^a\}$ are the broken generators (those with $X^a\phi_* \neq 0$). The canonically normalized Goldstone field is $\pi^a = f\,\xi^a$, with f the decay constant introduced in Section 29.5 and fixed by the current–Goldstone matrix element

$$\big\langle 0|J_a^\mu(0)|\pi^b(p)\big\rangle = i\,f\,p^\mu\,\delta_{ab}. \tag{29.202}$$

The key point of the parametrization is that the derivative $\partial/\partial\xi^a$ acting on any function of the fields, evaluated at $\xi = 0$, *is* the symmetry variation δ_a:

$$\frac{\partial}{\partial\xi^a} F(\phi(\xi))\Big|_{\xi=0} = \frac{\partial F}{\partial\phi_i}\, i(X_a)_{ij}\phi_{*\,j} = \delta_a F\big|_{\phi_*}. \tag{29.203}$$

Theorem 29.29 (Pseudo-Goldstone mass formula)**.** *Let G break spontaneously to H in the $\varepsilon = 0$ theory, with $n = \dim(G/H)$ exact Goldstones of decay constant f as in Eq. (29.202), and add the explicit breaking $\mathcal{L}_\varepsilon = \varepsilon\,\mathcal{O}$ of Eq. (29.196). Then to first order in ε the would-be Goldstones acquire the symmetric mass-squared matrix*

$$(M^2)_{ab} = -\frac{\varepsilon}{f^2}\,\langle\,\delta_a\delta_b\mathcal{O}\,\rangle_0 = +\frac{\varepsilon}{f^2}\,\langle\,[\,Q_a,[\,Q_b,\mathcal{O}\,]\,]\,\rangle_0\,, \tag{29.204}$$

where $\langle\cdots\rangle_0$ is the vacuum expectation value at the aligned vacuum. The matrix is real symmetric; when the vacuum is correctly aligned (Theorem 29.31) it is positive semidefinite, so $m_\pi^2 \geq 0$ for every pseudo-Goldstone.

Proof. By Theorem 29.21 the physical mass matrix is the Hessian of the full effective potential at its minimum. Write the full potential as

$$V(\phi) = V_0(\phi) - \varepsilon\,\langle\mathcal{O}\rangle_\phi, \tag{29.205}$$

where V_0 is the effective potential of the exactly broken theory and $-\varepsilon\langle\mathcal{O}\rangle_\phi$ is the tree-level shift induced by Eq. (29.196) (the sign is that of $-\mathcal{L}_\varepsilon$ entering V; quantum corrections to this term are higher order in ε and in $\hbar$ and do not affect the leading result). Restrict V to the vacuum manifold $\mathcal{M}$, parametrized by ξ^a. Along $\mathcal{M}$ the symmetric part is flat,

$$\frac{\partial V_0}{\partial\xi^a}\Big|_{\mathcal{M}} = 0, \qquad \frac{\partial^2 V_0}{\partial\xi^a\partial\xi^b}\Big|_{\mathcal{M}} = 0, \tag{29.206}$$

the second equality being exactly the statement that the Goldstone directions are the flat directions. Hence on $\mathcal{M}$ the entire curvature comes from the breaking term:

$$\frac{\partial^2 V}{\partial\xi^a\partial\xi^b}\Big|_{\xi=0} = -\,\varepsilon\,\frac{\partial^2\langle\mathcal{O}\rangle}{\partial\xi^a\partial\xi^b}\Big|_{\xi=0} = -\,\varepsilon\,\langle\delta_a\delta_b\mathcal{O}\rangle_0\,, \tag{29.207}$$

using Eq. (29.203) twice. This is the mass matrix in the ξ-coordinates. Passing to the canonically normalized fields $\pi^a = f\xi^a$ rescales each derivative by $1/f$, so $(M^2)_{ab} = f^{-2}\,\partial^2 V/\partial\xi^a\partial\xi^b$, which is the first equality in Eq. (29.204).

The second equality is the charge-algebra rewriting. From $\delta_b\mathcal{O} = i[Q_b,\mathcal{O}]$,

$$\delta_a\delta_b\mathcal{O} = i[Q_a,\delta_b\mathcal{O}] = i[Q_a,\,i[Q_b,\mathcal{O}]] = -\,[\,Q_a,[\,Q_b,\mathcal{O}\,]\,], \tag{29.208}$$

where the charges obey the algebra $[Q_a,Q_b] = if_{abc}Q_c$ of Theorem 7.21; substituting flips the sign. Symmetry of $(M^2)_{ab}$ in $a \leftrightarrow b$ follows because it is a Hessian. Positivity is established once we prove that the aligned vacuum minimizes V over $\mathcal{M}$ (Theorem 29.31), where the Hessian of a minimum is positive semidefinite by definition. □

Three features of Eq. (29.204) deserve emphasis. First, the mass-squared is *linear* in ε: a pseudo-Goldstone is light because ε is small, and $m_\pi \propto \sqrt{\varepsilon}$. Second, the right-hand side is a *vacuum* quantity—a double symmetry variation of $\mathcal{O}$ evaluated on $|0\rangle$—which by the charge algebra collapses onto the order parameter itself (we see this explicitly below); the formula thus ties the pseudo-Goldstone mass to the very condensate that signals the spontaneous breaking. Third, the prefactor $1/f^2$ is the universal Goldstone normalization: a larger order parameter (larger f) gives, for fixed condensate, a *lighter* pseudo-Goldstone. This is the structure of the Gell-Mann–Oakes–Renner relation, to which we now give its second, independent derivation.

29.6.4 The PCAC derivation and the Gell-Mann–Oakes–Renner relation

The effective-potential derivation is clean but uses the machinery of Theorem 29.21. It is reassuring—and historically the original route—to obtain the same formula directly from the broken Ward identity Eq. (29.200), using only the matrix element Eq. (29.202) and a soft-Goldstone reduction. The agreement of the two derivations is a nontrivial internal check.

Sandwich the operator identity Eq. (29.200) between the vacuum and a single on-shell pseudo-Goldstone state $\left|\pi^b(p)\right\rangle$ of mass m_π. The left-hand side, using Eq. (29.202) and translation invariance $J_a^\mu(x) = e^{iP\cdot x} J_a^\mu(0) e^{-iP\cdot x}$ so that $\left\langle 0|J_a^\mu(x)|\pi^b(p)\right\rangle = i f p^\mu \delta_{ab}\, e^{-ip\cdot x}$, gives

$$\partial_\mu \left\langle 0|J_a^\mu(x)|\pi^b(p)\right\rangle = \partial_\mu (i f p^\mu \delta_{ab}\, e^{-ip\cdot x}) = i f\,(-ip^2)\,\delta_{ab}\, e^{-ip\cdot x} = f\, m_\pi^2\, \delta_{ab}\, e^{-ip\cdot x}, \tag{29.209}$$

on shell $p^2 = m_\pi^2$. The right-hand side is

$$-\varepsilon \left\langle 0|\delta_a \mathcal{O}(x)|\pi^b(p)\right\rangle = -\varepsilon \left\langle 0|\delta_a \mathcal{O}(0)|\pi^b(p)\right\rangle e^{-ip\cdot x}. \tag{29.210}$$

Equating the coefficients of $e^{-ip\cdot x}$,

$$f\, m_\pi^2\, \delta_{ab} = -\varepsilon \left\langle 0|\delta_a \mathcal{O}(0)|\pi^b(p)\right\rangle. \tag{29.211}$$

It remains to evaluate the matrix element $\left\langle 0|\delta_a \mathcal{O}|\pi^b(p)\right\rangle$ in the soft limit $p \to 0$. This is where the second factor of $1/f$ and the second symmetry variation appear.

Lemma 29.30 (Soft-Goldstone reduction)**.** *For any local operator Φ, the matrix element of Φ with a single zero-momentum Goldstone is governed by the broken charge:*

$$\lim_{p\to 0} \left\langle 0|\Phi(0)|\pi^b(p)\right\rangle = \frac{i}{f}\, \langle 0|\,[\,Q_b, \Phi(0)\,]\,|0\rangle\,. \tag{29.212}$$

One factor of $1/f$ and one factor of i accompany the single broken charge Q_b; the overall normalization is fixed by Eq. (29.202).

Proof. The broken charge is $Q_b = \int \mathrm{d}^3x\, J_b^0(x)$. Evaluate $\langle 0|[Q_b, \Phi(0)]|0\rangle$ by inserting a complete set of states with relativistic normalization, $\mathbf{1} = \sum_c \int \frac{\mathrm{d}^3q}{(2\pi)^3 2E_q} |\pi^c(q)\rangle \langle \pi^c(q)| + \cdots$. The spatial integral $\int \mathrm{d}^3x$ in Q_b produces $(2\pi)^3\delta^3(\mathbf{q})$, forcing three-momentum $\mathbf{q} = 0$ on the intermediate state; among all states only the single Goldstone has the requisite overlap with the current, $\langle 0|J_b^0(0)|\pi^c(q)\rangle = i f\, q^0 \delta_{bc}$ from Eq. (29.202), and in the soft, massless limit $q^0 \to 0$ its contribution stays finite while massive states are projected away. Carrying out the two orderings $\langle 0|Q_b\Phi|0\rangle$ and $\langle 0|\Phi Q_b|0\rangle$ and combining, the relativistic measure $1/(2E_q)$ cancels the $q^0 = E_q$ from the current matrix element, leaving

$$\langle 0|[Q_b, \Phi(0)]|0\rangle = -i f \lim_{p\to 0} \Big\langle 0|\Phi(0)|\pi^b(p)\Big\rangle , \tag{29.213}$$

which rearranges to Eq. (29.212). □

Applying Theorem 29.30 with $\Phi = \delta_a\mathcal{O}$ to the matrix element in Eq. (29.211), and using $[Q_b, \delta_a\mathcal{O}] = -i\,\delta_b(\delta_a\mathcal{O})$ which follows from $\delta_b\Phi = i[Q_b, \Phi]$,

$$\lim_{p\to 0} \Big\langle 0|\delta_a\mathcal{O}|\pi^b(p)\Big\rangle = \frac{i}{f}\, \langle 0|[Q_b, \delta_a\mathcal{O}]|0\rangle = \frac{i}{f}\,(-i)\, \langle \delta_b\delta_a\mathcal{O}\rangle_0 = \frac{1}{f}\, \langle \delta_b\delta_a\mathcal{O}\rangle_0 , \tag{29.214}$$

the two factors of i cancelling to leave a real result. Substituting into Eq. (29.211),

$$f\, m_\pi^2\, \delta_{ab} = -\varepsilon \cdot \frac{1}{f}\, \langle \delta_b\delta_a\mathcal{O}\rangle_0 , \tag{29.215}$$

and dividing by f (the diagonal δ_{ab} assumes a basis diagonalizing M^2; the equality holds as a matrix identity in any basis) gives

$$(M^2)_{ab} = -\frac{\varepsilon}{f^2}\, \langle \delta_a\delta_b\mathcal{O}\rangle_0 , \tag{29.216}$$

in exact agreement with Theorem 29.29. The two derivations—one from the Hessian of the tilted potential, one from the soft limit of the partially conserved current—meet. Equation Eq. (29.216) is the general Gell-Mann–Oakes–Renner relation: *the pseudo-Goldstone mass-squared equals the explicit-breaking parameter times a condensate, divided by the decay constant squared.*

Historical Note

Gell-Mann, Oakes, and Renner derived their relation in 1968 for the pion, writing $m_\pi^2 f_\pi^2 = -(m_u + m_d)\, \big\langle \bar{u}u + \bar{d}d \big\rangle /2 + \ldots$, well before QCD existed, working purely from current algebra and the PCAC hypothesis—exactly the soft-pion technology of Theorem 29.30. That such a formula could be written, and later confirmed by QCD with $\mathcal{O} = \bar{q}q$ and $\varepsilon = m_q$, is one of the cleaner vindications of the spontaneous-breaking picture of the strong

interactions.

29.6.5 Vacuum alignment

We assumed in Theorem 29.29 that the vacuum was "correctly aligned." When the symmetric theory has a degenerate manifold $\mathcal{M}$ of vacua and we add an explicit term—or several competing terms—the perturbation selects *which* point of $\mathcal{M}$ is the true vacuum. Choosing the wrong point would give a spurious tachyon; the right point is forced by minimization. This is the *vacuum-alignment principle* (Dashen).

Let the explicit breaking be a sum of small terms,

$$\mathcal{L}_\varepsilon = \sum_n \varepsilon_n \mathcal{O}_n, \qquad V_{\rm br}(\phi) \equiv -\sum_n \varepsilon_n \langle \mathcal{O}_n \rangle_\phi, \tag{29.217}$$

each $\mathcal{O}_n$ non-invariant. On the degenerate manifold $\mathcal{M}$ the symmetric potential is constant, so the true vacuum is the point of $\mathcal{M}$ that minimizes $V_{\rm br}$.

Theorem 29.31 (Vacuum alignment and mass positivity)**.** *The true vacuum $\phi_* \in \mathcal{M}$ is a minimum of $V_{\rm br}$ restricted to $\mathcal{M}$. Equivalently, the* alignment conditions

$$\frac{\partial V_{\rm br}}{\partial \xi^a}\Big|_{\phi_*} = -\sum_n \varepsilon_n \, \langle \delta_a \mathcal{O}_n \rangle_{\phi_*} = 0 \qquad \text{for every broken direction } a \tag{29.218}$$

hold, and the second-derivative (Hessian) matrix

$$(M^2)_{ab} = \frac{1}{f^2} \frac{\partial^2 V_{\rm br}}{\partial \xi^a \partial \xi^b}\Big|_{\phi_*} = -\frac{1}{f^2} \sum_n \varepsilon_n \, \langle \delta_a \delta_b \mathcal{O}_n \rangle_{\phi_*} \tag{29.219}$$

is positive semidefinite. Consequently every pseudo-Goldstone has $m_\pi^2 \geq 0$; a vanishing eigenvalue signals a residual exact Goldstone, corresponding to a direction left invariant by all the $\mathcal{O}_n$.

Proof. The manifold $\mathcal{M} \cong G/H$ is compact (it is the orbit of ϕ_* under the compact group G, with the radial massive modes frozen). A continuous function—here $V_{\rm br}$ restricted to $\mathcal{M}$—attains its minimum on a compact set. At an interior minimum the gradient vanishes: this is Eq. (29.218), with the gradient written via Eq. (29.203). At a minimum the Hessian is positive semidefinite, which is Eq. (29.219) (the same Hessian as Eq. (29.207), now summed over the breaking terms and including the alignment of the vacuum). Positive semidefiniteness of a Hessian at a minimum is the second-order necessary condition; hence all eigenvalues $m_\pi^2 \geq 0$. A zero eigenvalue means a flat direction of $V_{\rm br}$ survives: along that direction every $\langle \delta_a \delta_b \mathcal{O}_n \rangle$ vanishes, so the would-be Goldstone remains exactly massless—it is a genuine Goldstone of a symmetry that the explicit terms happen to respect. □

The logic is worth restating because it is easy to get backwards. One does *not* pick a vacuum and then discover whether the masses are positive; one *minimizes* over all candidate vacua, and positivity of the masses is then automatic—it is nothing but the statement that a minimum curves upward. A configuration with a negative m_π^2 is not a vacuum at all; it is a saddle, and the system would roll off it to the aligned point. This is Dashen's theorem: the explicit breaking "aligns" the spontaneous breaking with itself.

Physical Insight

Vacuum alignment is the precise sense in which a small explicit term can have a large qualitative effect: it cannot move the order parameter much (that costs the large symmetric energy of climbing out of the trough), but it freely *rotates* the order parameter within the flat manifold to point in the direction the breaking prefers. The pseudo-Goldstone masses are the curvature the breaking leaves behind once this free rotation has settled.

29.6.6 Worked example: explicit breaking of the $O(N)$ model

We now run the entire apparatus on the $O(N)$ linear sigma model of ϕ^4 theory—the multi-component generalization of the real scalar, in its broken phase—by adding the explicit-breaking analog of a quark mass: a term linear in the fields. This is the cleanest possible illustration, and it reproduces the master formula Eq. (29.204) by direct computation.

Worked Example 29.32 (Pseudo-Goldstone masses in the tilted $O(N)$ model)**.** Recall the $O(N)$ model (Eq. (29.60)) with N real scalars ϕ_i and

$$\mathcal{L}_0 = \tfrac{1}{2}(\partial_\mu \phi_i)(\partial^\mu \phi_i) - \frac{\lambda}{4}\left(\phi_i \phi_i - v^2\right)^2, \tag{29.220}$$

invariant under $G = O(N)$, with vacuum manifold the sphere $|\phi| = v$ ($\cong O(N)/O(N-1) \cong S^{N-1}$) and $n = N - 1$ exact Goldstones. Add the explicit breaking

$$\mathcal{L}_\varepsilon = H\,\phi_N, \qquad \varepsilon = H > 0, \quad \mathcal{O} = \phi_N, \tag{29.221}$$

the analog of an external field (a "magnetic field" in the spin language, a quark mass in the chiral language). It is not $O(N)$-invariant: rotations move ϕ_N into the other components.

Alignment. The full potential is $V = \frac{\lambda}{4}(|\phi|^2 - v^2)^2 - H\phi_N$. On the sphere $|\phi| = v$, $V_{\rm br} = -H\phi_N$ is minimized by the largest ϕ_N, i.e. $\phi_* = (0, \ldots, 0, v)$: the order parameter aligns along the direction the field H favors. (Had we written $H\phi_1$ the vacuum would point along $\hat{e}_1$; the alignment condition Eq. (29.218) picks it out.)

Direct mass computation. Freeze the massive radial mode (send it to infinity, the nonlinear-sigma-model limit of Section 29.5) so that $|\phi| = v$ exactly, and write the Goldstone fluctuations as the first $N-1$ components π^a, with

$$\phi_N = \sqrt{v^2 - \pi^a\pi^a} = v - \frac{\pi^a\pi^a}{2v} + O(\pi^4). \tag{29.222}$$

Then the breaking term becomes a mass term:

$$\mathcal{L}_\varepsilon = H\phi_N = Hv - \frac{H}{2v}\,\pi^a\pi^a + O(\pi^4) = \text{const} - \tfrac{1}{2}\,\frac{H}{v}\,\pi^a\pi^a + \dots, \tag{29.223}$$

so each of the $N-1$ pseudo-Goldstones acquires the *same* mass-squared

$$\boxed{m_\pi^2 = \frac{H}{v}}. \tag{29.224}$$

It is linear in the breaking H, vanishes as $H \to 0$ (restoring the exact Goldstones), and is manifestly positive—the vacuum was correctly aligned. The radial mode shifts the minimum from $|\phi| = v$ to $|\phi| = v + \delta v$ with $\delta v = H/(2\lambda v^2)$; this feeds back into m_π^2 only at order H^2 and so does not affect the leading result.

Check against the master formula. For the $O(N)$ model the decay constant is $f = v$ (the current $J_a^\mu = \phi_N \partial^\mu \pi^a - \pi^a \partial^\mu \phi_N + \dots$ gives $\left\langle 0|J_a^\mu|\pi^b(p)\right\rangle = iv\,p^\mu \delta_{ab}$, matching Eq. (29.202)). The broken generators X^a rotate in the (a, N) plane: $\delta_a \phi_N = -\phi_a$ and $\delta_a \phi_b = \delta_{ab}\phi_N$. Hence

$$\delta_a\delta_b\,\mathcal{O} = \delta_a\delta_b\,\phi_N = \delta_a(-\phi_b) = -\delta_{ab}\,\phi_N, \qquad \langle \delta_a\delta_b\phi_N\rangle_0 = -\delta_{ab}\,v. \tag{29.225}$$

Substituting into Eq. (29.204) with $\varepsilon = H$ and $f = v$,

$$(M^2)_{ab} = -\frac{H}{v^2}\,(-\delta_{ab}\,v) = \frac{H}{v}\,\delta_{ab}, \tag{29.226}$$

identical to the direct result Eq. (29.224). The general theorem and the brute-force expansion agree, term for term.

Theorem 29.32 is the precise scalar-model analog of giving the quarks a mass: $H\phi_N$ plays the role of $m_q\,\bar{q}q$, v the role of the chiral order parameter, and $m_\pi^2 = H/v$ the role of the GMOR relation $m_\pi^2 f_\pi^2 = -m_q\,\langle\bar{q}q\rangle$. The figure makes the geometry vivid: the explicit term tilts the flat Mexican-hat valley into a gentle washboard whose curvature at the bottom is exactly m_π^2.

29.6.7 The pion as the paradigm pseudo-Goldstone

The single most important realization of this machinery in nature is the pion, and it is worth stating the dictionary precisely, because it is the setup the chiral-perturbation-theory stage (Chapter 39) continues directly.

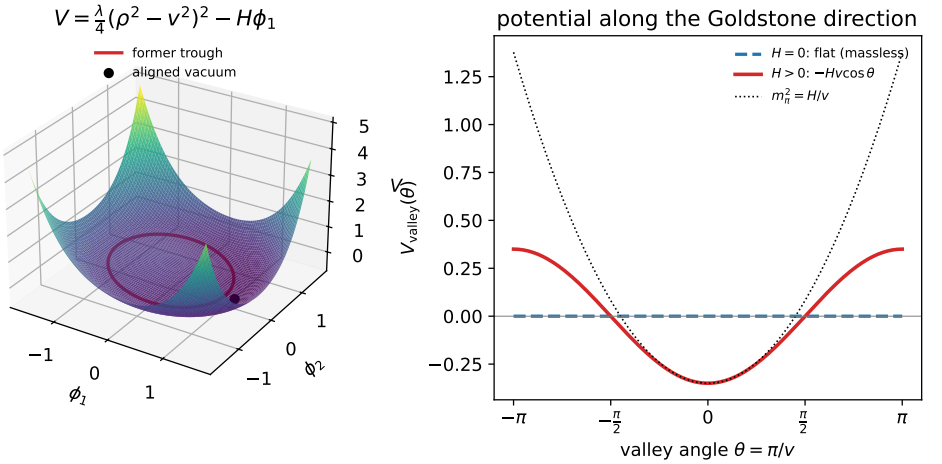


Figure 29.8: Explicit breaking tilts the Mexican hat. *Left:* the $O(2)$ potential $V = \frac{\lambda}{4}(\rho^2 - v^2)^2 - H\phi_1$ over the field plane; the linear term $-H\phi_1$ selects a unique vacuum on the former degenerate trough (the aligned vacuum). *Right:* the potential evaluated along the former flat valley, as a function of the angular Goldstone coordinate $\theta = \pi/v$. At $H = 0$ it is exactly flat (massless Goldstone, dashed); at $H > 0$ it is the washboard $-Hv\cos\theta$ (solid), whose curvature at the minimum gives the pseudo-Goldstone mass-squared $m_\pi^2 = H/v$ (dotted parabola).

Quantum chromodynamics with n_f massless quarks has, beyond its gauge symmetry, a global *chiral* symmetry $G = SU(n_f)_L \times SU(n_f)_R$ acting independently on left- and right-handed quarks (the construction of these currents is the chiral side of Eq. (7.80); the anomalous $U(1)_A$ is set aside, its breaking belonging to Chapter 36). The strong dynamics forms a quark condensate $\langle \bar{q}q \rangle = \langle \bar{q}_L q_R + \bar{q}_R q_L \rangle \neq 0$, which is invariant under the simultaneous (vector) rotation $L = R$ but not under the relative (axial) rotation $L = R^{-1}$. The symmetry is therefore spontaneously broken,

$$SU(n_f)_L \times SU(n_f)_R \;\longrightarrow\; SU(n_f)_V, \tag{29.227}$$

the unbroken H being the diagonal isospin/flavor group. The number of broken generators is $\dim G - \dim H = (n_f^2 - 1)$: for $n_f = 2$ this is three exact Goldstones, the three pions $\pi^\pm, \pi^0$; the order parameter $\langle \bar{q}q \rangle$ is the condensate and the decay constant is $f = f_\pi \approx 92$ MeV.

Real quarks are not massless: the QCD Lagrangian carries an explicit-breaking term $\mathcal{L}_\varepsilon = -\bar{q}\,\mathcal{M}_q\, q$ with the quark mass matrix $\mathcal{M}_q = \mathrm{diag}(m_u, m_d, \dots)$. This is exactly the structure of Eq. (29.196) with $\varepsilon\,\mathcal{O} \to -\bar{q}\mathcal{M}_q q$. The master formula Eq. (29.216) then gives, for two flavors with m_u, m_d,

$$m_\pi^2\, f_\pi^2 = -(m_u + m_d)\, \langle \bar{q}q \rangle + O(m_q^2), \tag{29.228}$$

the Gell-Mann–Oakes–Renner relation. Numerically, with $\langle\bar{q}q\rangle \approx -(250\text{ MeV})^3$, $f_\pi \approx 92$ MeV, and $m_u + m_d \approx 11$ MeV at a hadronic scale, this reproduces $m_\pi \approx 140$ MeV. The key qualitative output is the hierarchy: the pion is light *because* $m_\pi^2 \propto m_q$ *is linear in the tiny quark masses*, while every other hadron—the ρ meson at $m_\rho \approx 770$ MeV, the nucleon at 939 MeV—gets its mass from the strong scale $\Lambda_{\rm QCD}$ and survives the chiral limit $m_q \to 0$. Hence

$$\frac{m_\pi^2}{m_\rho^2} \sim \frac{m_q}{\Lambda_{\rm QCD}} \ll 1, \tag{29.229}$$

the parametric statement of *why* $m_\pi \ll m_\rho$. The pion is light not by accident but because it is the pseudo-Goldstone boson of an almost-exact symmetry; in the exact chiral limit it would be strictly massless.

Physical Insight

The pion's small mass is the most direct experimental fingerprint of spontaneous symmetry breaking in the Standard Model. If chiral symmetry were realized linearly (Wigner mode), the pion would be a generic hadron with mass $\sim \Lambda_{\rm QCD}$ and there would be parity-doubled partners for every state; instead we see an anomalously light pseudoscalar and no parity doubling. The smallness of m_π/m_ρ *measures* how good the chiral symmetry is—and Eq. (29.228) turns that smallness into a measurement of the quark masses.

29.6.8 Honest dimensional scope: Coleman–Mermin–Wagner

Everything above assumed that spontaneous breaking of a continuous symmetry actually occurs—that there is a stable vacuum with $\langle 0|\delta_a\Phi|0\rangle \neq 0$ and well-defined Goldstone quanta. In low spacetime dimension this assumption fails. The would-be Goldstone, being massless, fluctuates so violently in the infrared that it destroys the very order parameter that was supposed to define the broken vacuum. This is the Coleman–Mermin–Wagner obstruction, and it sets the honest boundary of the chapter's results.

Theorem 29.33 (Coleman–Mermin–Wagner; relativistic Coleman 1973)**.** *A continuous global symmetry cannot be spontaneously broken in spacetime dimension $d \leq 2$ (one time, $d-1 \leq 1$ space): there is no Goldstone boson, and any candidate order parameter is washed out by infrared fluctuations. Equivalently (Coleman's theorem), in two-dimensional spacetime a scalar field that would interpolate a Goldstone mode has no Wightman realization—a massless free scalar in $d = 2$ is not an operator-valued tempered distribution—so the symmetry is realized only in the unbroken (Wigner) mode.*

The heart of the argument. Suppose, toward a contradiction, that the symmetry is broken and a massless Goldstone $\pi(x)$ exists. Its propagator is that of a free massless field, $\widetilde{G}(k) = i/(k^2 + i\epsilon)$, and the equal-time mean-square fluctuation of the field at a point is

$$\left\langle \pi(x)^2 \right\rangle = \int \frac{\mathrm{d}^d k}{(2\pi)^d} \, \frac{1}{k^2} \;\sim\; \int \frac{\mathrm{d}^{d-1}\mathbf{k}}{|\mathbf{k}|} \;\sim\; \int_0 \mathrm{d}|\mathbf{k}| \, |\mathbf{k}|^{\,d-3}, \tag{29.230}$$

where we performed the energy integral leaving the spatial one and stripped angular factors. The behavior at the lower (infrared) limit $|\mathbf{k}| \to 0$ is controlled by the exponent $d - 3$: the integral converges in the IR for $d > 2$, is logarithmically divergent for $d = 2$, and is power-divergent for $d < 2$. Now the order parameter is built from the broken field as $\langle \Phi \rangle \propto \left\langle \, e^{i\pi/f} \, \right\rangle$, and a Gaussian (free-Goldstone) estimate gives the suppression

$$\left\langle e^{i\pi(x)/f} \right\rangle = e^{-\frac{1}{2f^2}\left\langle \pi(x)^2 \right\rangle}. \tag{29.231}$$

In $d \le 2$ the exponent diverges to $+\infty$ because $\langle \pi^2 \rangle$ diverges in the infrared, so $\langle \Phi \rangle \to 0$: the order parameter is destroyed, no matter how the dynamics tried to establish it. The assumed broken vacuum is therefore not normalizable, contradicting the assumption. Hence no continuous symmetry breaks in $d \le 2$. The rigorous relativistic statement, that the would-be Goldstone two-point function fails to define a tempered distribution—visible already in the infrared non-integrability of its Källén–Lehmann spectral representation (Theorem 14.23), whose massless $\delta(k^2)$ contribution is not locally integrable in $d = 2$—is Coleman's theorem. □

Caution

The theorem forbids *spontaneous breaking of a continuous global symmetry* in $d \le 2$; it does *not* forbid breaking a *discrete* symmetry (the $\mathbb{Z}_2$ of ϕ^4 theory breaks fine in $d = 2$—there is no Goldstone to diverge, the Ising model in two dimensions has a genuine ordered phase), nor does it forbid the Berezinskii–Kosterlitz–Thouless quasi-order of the $d = 2$ XY model (power-law, not true long-range, order). It is specifically the would-be *Goldstone* of a *continuous* symmetry whose IR fluctuations are fatal.

Remark 29.34 (Live refinements, as of the mid-2020s). Two qualifications to the "one massless boson per broken generator" counting are active areas rather than closed textbook results. First, in systems *without* Lorentz invariance—finite density, ferromagnets, Bose condensates—the Goldstone count and dispersion change. Goldstones split into *type-A* (linear dispersion $\omega \sim |\mathbf{k}|$, one mode per broken generator) and *type-B* (quadratic dispersion $\omega \sim |\mathbf{k}|^2$, one mode per *pair* of broken generators whose charges have a nonzero ground-state commutator $\langle [Q_a, Q_b] \rangle \neq 0$, as for a ferromagnet's single quadratic magnon despite two broken

rotations). The sharp counting—that the number of Goldstones is $\dim(G/H)$ minus half the rank of the antisymmetric matrix $\langle[Q_a, Q_b]\rangle$, with type-B modes counted once—is the Watanabe–Murayama theorem (2012), building on the Nielsen–Chadha bound (1976). The relativistic case of this chapter is the special case where $\langle[Q_a, Q_b]\rangle = 0$ and all Goldstones are type-A. Second, the modern reorganization of symmetry in terms of *generalized (higher-form and non-invertible) symmetries* extends the notions of order parameter and Goldstone mode—e.g. the photon as the Goldstone of a spontaneously broken one-form symmetry—and the corresponding Goldstone and Coleman–Mermin–Wagner statements for these symmetries are still being mapped out. We flag these as genuinely open refinements; the relativistic, ordinary-symmetry results proved above are firm.

29.6.9 Synthesis: where the broken-symmetry machinery goes

This section completed the logical arc of the chapter. A continuous global symmetry can be exact in the Lagrangian yet broken on the vacuum, and when it is, it forces massless Goldstones (Theorem 29.21); a small explicit term lifts those Goldstones to light, calculable pseudo-Goldstones with m_π^2 linear in the breaking (Theorem 29.29, Eq. (29.216)); the vacuum aligns to make those masses nonnegative (Theorem 29.31); and the whole picture collapses in $d \leq 2$ (Theorem 29.33). Three forward roads now open, and each is the subject of a coming chapter.

First, *gauge* the broken symmetry. The very Goldstone modes and broken currents built here are what get "eaten" when the symmetry is local: the would-be Goldstone becomes the longitudinal polarization of a gauge boson, which thereby acquires a mass. This is the Higgs mechanism (Chapter 30), where the derivative-coupled Goldstone of Section 29.5 reappears as the gauge-dependent Goldstone of R_ξ gauge and the equivalence theorem ties high-energy longitudinal scattering back to the Goldstone dynamics of this chapter. Second, the electroweak realization (Chapter 31) is exactly this construction applied to $SU(2)_L \times U(1)_Y$, with the Higgs doublet's vacuum supplying the order parameter. Third, the pion dictionary of Section 29.6.7 continues as chiral perturbation theory (Chapter 39): the nonlinear sigma model of Section 29.5 becomes the chiral Lagrangian, the quark-mass term becomes the explicit breaking analyzed here, and Eq. (29.228) becomes the leading entry in a systematic low-energy expansion. The pseudo-Goldstone pion—the massive completion of ϕ^4 theory's $O(N)$ Goldstones under the quark-mass analog of Theorem 29.32—is handed forward to that chapter as the central object of the strong interaction's low-energy theory.

Further reading

The theorem and its first proofs are in Goldstone [231] and Goldstone et al. [232], with the dynamical origin of the idea in Nambu [357] and Nambu and Jona-Lasinio [358]. The effective potential, its loop expansion, and radiative breaking are developed in Coleman and Weinberg [132]; the systematic effective-action treatment of breaking and vacuum alignment is in Weinberg [501], with textbook accounts in Peskin and Schroeder [396], Schwartz [432], and Srednicki [454]. The pseudo-Goldstone and current-algebra results behind the Gell-Mann–Oakes–Renner relation are in Gell-Mann et al. [216]; the pion as a pseudo-Goldstone and the systematic chiral expansion are treated in Scherer and Schindler [430] and continued in Chapter 39. The honest dimensional scope is fixed by Coleman's relativistic theorem [127] and its statistical-mechanics ancestor [347]; the modern nonrelativistic counting is settled in Watanabe and Murayama [490], refining Nielsen and Chadha [364].

Chapter 30

The Higgs Mechanism

The previous chapter left us with a sharp tension between two results we have proved in full. Goldstone's theorem (Theorem 29.11) says that spontaneously breaking a continuous *global* symmetry forces a massless scalar into the spectrum for every broken generator. The gauge principle (Section 8.1), in its canonical form (Section 13.1), says that local invariance is a redundancy rather than a symmetry relating distinct states, and that it forbids any bare mass for the gauge field: the photon and the Yang–Mills bosons of Chapter 25 are exactly massless. Nature, however, presents short-ranged weak interactions carried by heavy vectors. This chapter shows that the two apparent obstructions — the unwanted Goldstone boson and the forbidden vector mass — are two readings of a single piece of bookkeeping, and that gauging the broken symmetry makes them cancel: the would-be Goldstone is not a particle at all but the longitudinal polarization the now-massive gauge boson needed. This is the Higgs mechanism, and it is the engine of mass generation that Chapter 31 will mount directly onto the electroweak group.

We proceed slowly because the mechanism is *forced*, not chosen, and one only sees that by sharpening the paradox first. Section 30.1 dissects the problem into two precise, separately fatal obstructions — the non-renormalizable ultraviolet behaviour of a by-hand Proca mass, computed from its propagator and superficial degree of divergence, and the degree-of-freedom mismatch that a massive vector forces — and identifies the loophole in Goldstone's theorem (the failure of the manifest-covariance-with-positivity hypothesis in a gauge theory, already visible in the photon's quantization of Chapter 13) that lets a gauge theory evade it. Section 30.2 then writes down the object these constraints demand, the abelian Higgs model, and works its classical mass generation in full: the covariant kinetic term produces a gauge-boson mass $m_A = ev$, the characteristic $A_\mu \partial^\mu \chi$ mixing announces the mechanism, and the degree-of-freedom count balances exactly when the Goldstone is recognized as pure gauge.

The middle of the chapter is the gauge-fixing story, told three ways and then

reconciled. Section 30.3 fixes unitary gauge, in which the spectrum is manifest — one massive vector, one massive Higgs, no Goldstone — but the massive-vector propagator carries a $k_\mu k_\nu / m_A^2$ term that obscures renormalizability and makes longitudinal amplitudes appear to grow without bound. Section 30.4 repairs this with the R_ξ family, deriving the gauge-fixing functional by the Faddeev–Popov construction of Chapters 13 and 26 adapted to the broken theory — 't Hooft's choice that cancels the mixing term outright — and reading off the complete propagator set (gauge boson, Goldstone, Higgs, and ghost), each carrying its ξ-dependent unphysical pole at $k^2 = \xi m_A^2$. Section 30.5 then proves that those poles are spurious: BRST symmetry (Chapter 26) survives spontaneous breaking, the physical Hilbert space is its cohomology, the quartet of unphysical fields decouples, and the Nielsen identities guarantee that pole masses and S-matrix elements are exactly ξ-independent, all confirmed by an explicit two-gauge computation.

The chapter closes by extracting the physics hidden in the unitary-gauge growth and then generalizing. Section 30.6 states and argues the Goldstone-boson equivalence theorem — at high energy the amplitude for emitting a longitudinally polarized gauge boson equals, up to corrections suppressed by m_A/E, the amplitude for emitting the corresponding would-be Goldstone — deriving it from the Slavnov–Taylor identity and confirming it by computing longitudinal scattering both directly and via the Goldstone replacement. Finally Section 30.7 lifts everything to a general gauge group G broken to a subgroup H: the gauge-boson mass matrix is the Gram matrix of the broken generators acting on the vacuum, its kernel is exactly the unbroken algebra, and the counting theorem matches eaten Goldstones to massive vectors generator by generator. The worked cases SU(2) broken by a doublet and SU(2) $\to$ U(1) broken by an adjoint are the direct rehearsals for electroweak symmetry breaking (Chapter 31) and for the 't Hooft–Polyakov monopole of Chapter 42.

30.1 The Problem: Massive Gauge Bosons, Goldstone Bosons, and the Way Out

The chapter opened on a collision between two results we have already proved. On one side stands Goldstone's theorem (Theorem 29.11): spontaneously break a continuous *global* symmetry and the spectrum must contain a massless scalar for every broken generator. On the other stands the gauge principle (Section 8.1) and its canonical face (Section 13.1): local invariance is a redundancy, not a symmetry relating distinct states, and it forbids the gauge field any bare mass. The photon and the Yang–Mills bosons are therefore exactly massless. Yet the weak interaction is short-ranged, so its carriers are heavy. We need a massive vector that does not wreck either gauge invariance or the renormalizability that gauge invariance buys us.

It would be a mistake to leap straight to the cure. The mechanism that follows is forced — not one clever construction among many — and one only sees that it is forced by first sharpening the paradox until it splits into *two precise technical obstructions*, each of which can be stated, quantified, and shown to be fatal on its own. The first obstruction is dynamical: the only naive way to give a vector a mass, adding a Proca term by hand, both breaks gauge invariance and produces a propagator whose ultraviolet behaviour destroys renormalizability. The second is a matter of counting and of hypotheses: a massive vector has one more polarization than a massless one, that extra state has to be supplied from somewhere, and Goldstone's theorem appears to forbid the only candidate — unless one of its hypotheses fails. The punchline of the section is that these two obstructions are not independent troubles but two readings of a single accountancy, and that they cancel each other exactly when the broken symmetry is made local. By the end the abelian Higgs model of the next section will be the obvious object to write down, because we will know precisely what it has to do.

30.1.1 The naive cure and why gauge invariance refuses it

The most direct thing to try is to take the free Maxwell Lagrangian and simply append a mass term, exactly as one would for a scalar. With the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ of Section 8.1, the candidate is the *Proca Lagrangian*

$$\mathcal{L}_{\mathrm{Proca}} \;=\; -\tfrac{1}{4} F_{\mu\nu}F^{\mu\nu} \;+\; \tfrac{1}{2}\, m^2\, A_\mu A^\mu . \tag{30.1}$$

The sign of the mass term is fixed by physics, not taste: varying Eq. (30.1) gives the equation of motion

$$\partial_\mu F^{\mu\nu} + m^2 A^\nu = 0, \tag{30.2}$$

and taking ∂_ν of Eq. (30.2) kills the antisymmetric $\partial_\nu\partial_\mu F^{\mu\nu} = 0$ and leaves $m^2\,\partial_\nu A^\nu = 0$, so for $m \neq 0$ the Lorenz condition

$$\partial_\nu A^\nu = 0 \tag{30.3}$$

is *automatic* — a consequence of the dynamics rather than a gauge choice. Feeding Eq. (30.3) back into $\partial_\mu F^{\mu\nu} = \Box A^\nu - \partial^\nu(\partial_\mu A^\mu) = \Box A^\nu$ turns Eq. (30.2) into $(\Box + m^2)A^\nu = 0$: each component obeys the Klein–Gordon equation with mass m, and the constraint Eq. (30.3) removes one of the four, leaving the three polarizations a massive spin-one particle should have. So far the classical particle content is exactly right. The trouble is everywhere else.

The mass term in Eq. (30.1) is not gauge invariant. Under the abelian gauge transformation $A_\mu \mapsto A_\mu + \partial_\mu\Lambda$ (the U(1) of QED, Section 8.1), the field strength is untouched but

$$\tfrac{1}{2}m^2 A_\mu A^\mu \;\longmapsto\; \tfrac{1}{2}m^2(A_\mu+\partial_\mu\Lambda)(A^\mu+\partial^\mu\Lambda) \;=\; \tfrac{1}{2}m^2 A_\mu A^\mu + m^2 A_\mu\partial^\mu\Lambda + \tfrac{1}{2}m^2\partial_\mu\Lambda\,\partial^\mu\Lambda, \tag{30.4}$$

and the extra pieces do not vanish for arbitrary $\Lambda(x)$. This is the same algebraic fact that forbade a fermion mass in the non-abelian case (Eq. (25.4)): a mass term pairs two factors of the field at the same point with nothing between them to absorb the transformation. For the gauge field the consequence is structural. The redundancy that, in Section 13.1, appeared as the first-class constraint $\pi^0 \equiv 0$ and forced the photon's degree-of-freedom count down to two, is gone: A^0 now has a genuine conjugate momentum, the constraint structure collapses, and there is no gauge orbit to quotient by. Classically one can live with this. Quantum-mechanically one cannot, and the reason is written in the propagator.

30.1.2 The Proca propagator and its ultraviolet disease

To find the propagator we invert the quadratic differential operator of Eq. (30.1). Expanding the kinetic term and integrating by parts (discarding surface terms, exactly as in the derivation of the photon propagator in Section 13.4.2),

$$-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} \;=\; -\tfrac{1}{2}(\partial_\mu A_\nu\,\partial^\mu A^\nu - \partial_\mu A_\nu\,\partial^\nu A^\mu) \;\longrightarrow\; \tfrac{1}{2}\,A_\mu(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)A_\nu, \tag{30.5}$$

so that the full Proca Lagrangian becomes the manifestly quadratic form

$$\mathcal{L}_{\text{Proca}} \;=\; \tfrac{1}{2}\,A_\mu\,\mathcal{O}^{\mu\nu}\,A_\nu, \qquad \mathcal{O}^{\mu\nu} \;=\; \eta^{\mu\nu}(\Box + m^2) - \partial^\mu\partial^\nu. \tag{30.6}$$

Passing to momentum space with the Fourier convention of Chapter 9 ($\partial_\mu \to -ik_\mu$ on $e^{-ik\cdot x}$, so $\Box \to -k^2$ and $\partial^\mu\partial^\nu \to -k^\mu k^\nu$),

$$\mathcal{O}^{\mu\nu}(k) \;=\; -\big[(k^2{-}m^2)\eta^{\mu\nu}{-}k^\mu k^\nu\big] \;\equiv\; -\,M^{\mu\nu}(k), \qquad M^{\mu\nu}(k) \equiv (k^2{-}m^2)\eta^{\mu\nu}{-}k^\mu k^\nu. \tag{30.7}$$

The Feynman propagator $D_{\nu\rho}(k)$ is i times the inverse of the quadratic operator; with $\mathcal{O}^{\mu\nu} = -M^{\mu\nu}$ this is $D = i\mathcal{O}^{-1} = -iM^{-1}$, and the task is to invert $M^{\mu\nu}$. Crucially, unlike the massless Maxwell operator $k^2\eta^{\mu\nu} - k^\mu k^\nu$, which annihilates the longitudinal vector k_ν and is therefore singular (Section 13.4.3), the Proca operator $M^{\mu\nu}$ is *invertible without any gauge fixing*: the mass term lifts the would-be zero mode. Indeed $M^{\mu\nu}k_\nu = (k^2 - m^2)k^\mu - k^2k^\mu = -m^2k^\mu \neq 0$.

Because the only tensors available are $\eta_{\nu\rho}$ and $k_\nu k_\rho$, the inverse must take the form

$$(M^{-1})_{\nu\rho} = A\,\eta_{\nu\rho} + B\,k_\nu k_\rho. \tag{30.8}$$

Imposing $M^{\mu\nu}(M^{-1})_{\nu\rho} = \delta^\mu{}_\rho$ and computing,

$$\begin{aligned} M^{\mu\nu}(M^{-1})_{\nu\rho} &= \big[(k^2 - m^2)\eta^{\mu\nu} - k^\mu k^\nu\big]\big[A\,\eta_{\nu\rho} + B\,k_\nu k_\rho\big] \\ &= (k^2 - m^2)A\,\delta^\mu{}_\rho + \Big[(k^2 - m^2)B - A - k^2B\Big]k^\mu k_\rho \\ &= (k^2 - m^2)A\,\delta^\mu{}_\rho + \big[-m^2B - A\big]k^\mu k_\rho. \end{aligned} \tag{30.9}$$

Matching to $\delta^{\mu}{}_{\rho}$ forces $(k^2 - m^2)A = 1$ and $-m^2 B - A = 0$, hence

$$A = \frac{1}{k^2 - m^2}, \qquad B = -\frac{A}{m^2} = -\frac{1}{m^2(k^2 - m^2)}. \tag{30.10}$$

Assembling $D_{\nu\rho} = -i(M^{-1})_{\nu\rho}$ and restoring the Feynman $i\epsilon$ that picks out the time-ordered prescription (the pole at $k^2 = m^2$ is displaced exactly as for any massive particle, Chapter 9), we have the result.

Theorem 30.1 (The Proca propagator)**.** *The free propagator of the massive vector field governed by Eq. (30.1) is*

$$D_{\mu\nu}(k) \;=\; \frac{-i}{k^2 - m^2 + i\epsilon}\left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m^2}\right). \tag{30.11}$$

Two features of Eq. (30.11) must be read carefully, because together they are the first obstruction. The first is benign and expected: the pole at $k^2 = m^2$ confirms a propagating particle of mass m, and the residue $-i(\eta_{\mu\nu} - k_\mu k_\nu/m^2)$ is, on shell, exactly the sum over the three physical polarizations we will compute in Theorem 30.3. The second is the disease. Contrast Eq. (30.11) with the massless gauge propagator $D_{\mu\nu}(k) = -i\big[\eta_{\mu\nu} - (1-\xi)k_\mu k_\nu/k^2\big]/(k^2 + i\epsilon)$ of Eq. (13.135), in which *every* term carries an explicit $1/k^2$ and the whole tensor therefore falls off as k^{-2} at large momentum. In the Proca propagator the $\eta_{\mu\nu}$ term still falls off,

$$D_{\mu\nu}(k) \;=\; \underbrace{\frac{-i\,\eta_{\mu\nu}}{k^2 - m^2 + i\epsilon}}_{\sim k^{-2}} \;+\; \underbrace{\frac{i\,k_\mu k_\nu}{m^2\,(k^2 - m^2 + i\epsilon)}}_{\sim k^{0}}, \tag{30.12}$$

but the longitudinal piece does not: as $|k| \to \infty$ the second term behaves as $(i/m^2)\,k_\mu k_\nu/k^2$, a tensor of order k^0 that approaches a finite, nonzero limit instead of decaying. The mass that we put in to make the particle heavy sits in the *denominator* of the dangerous term; the heavier the vector, the milder the term, but for any finite m the propagator fails to fall off. This is the structural cost of breaking gauge invariance: in Eq. (13.135) the $k_\mu k_\nu$ structure is controlled by the gauge parameter ξ and is killed in matrix elements by current conservation; here it is locked in by $1/m^2$ and cannot be gauged away, because there is no gauge symmetry left to do the gauging.

Why bad ultraviolet falloff spoils renormalizability

That a single propagator fails to decay would be a curiosity were it not for power counting, the machinery of Section 21.2. Recall the superficial degree of divergence of a connected diagram in d spacetime dimensions,

$$D \;=\; d\,L \;+\; \sum_i V_i\,\delta_i \;-\; \sum_f \omega_f\, I_f, \tag{30.13}$$

where L is the number of loops, V_i the number of vertices of type i with δ_i derivatives each, I_f the number of internal lines of species f, and ω_f the large-momentum falloff of the species-f propagator — each power of k^{-1} contributing one unit, so a well-behaved boson propagator $\sim k^{-2}$ has $\omega_b = 2$ and a fermion $\sim k^{-1}$ has $\omega_F = 1$. The falloff ω_f enters with a minus sign because a propagator that decays makes the loop integral converge faster; the central lesson of Section 21.2 is precisely that the propagator suppression must be large enough, for fixed external data, to keep D bounded as the diagram grows.

The Proca propagator Eq. (30.12) contains a term that does not decay at all. Its slowest-falling part is the longitudinal $\sim k^0$ piece, for which the effective suppression is $\omega_V = 0$, not 2. Power counting always tracks the *worst* ultraviolet behaviour, so an internal massive-vector line contributes $\omega_V = 0$ to Eq. (30.13). Comparing a diagram built from Proca lines with the same diagram built from genuine k^{-2} gauge lines,

$$D_{\text{Proca}} \;=\; dL + \sum_i V_i\delta_i - \sum_{f\neq V} \omega_f I_f - \underbrace{0\cdot I_V}_{\text{longitudinal}} \;=\; D_{\text{gauge}} \;+\; 2\,I_V, \tag{30.14}$$

where I_V is the number of internal massive-vector lines and D_{gauge} is the value the diagram would have were every vector line to fall off as k^{-2}. The shift is the heart of the matter.

Proposition 30.2 (A by-hand vector mass is non-renormalizable)**.** *In an interacting theory whose vector propagator is the Proca propagator Eq. (30.11), the number of internal vector lines I_V grows without bound as the order of perturbation theory increases, so by Eq. (30.14) the superficial degree of divergence of a fixed external process is unbounded above. Infinitely many amplitudes are superficially divergent, and by the criterion of Section 21.3.4 no finite set of counterterms can absorb the divergences: the theory is non-renormalizable.*

Proof. Fix an external configuration. Each additional loop adds at least one internal line, and in a theory whose only bosonic field is the massive vector those internal lines are vector lines, so $I_V \to \infty$ along any sequence of diagrams of increasing loop order. By Eq. (30.14), $D_{\text{Proca}} = D_{\text{gauge}} + 2I_V$, and D_{gauge} for fixed external legs is bounded (it is the renormalizable counting of a k^{-2} vector theory). Hence $D_{\text{Proca}} \to \infty$. A divergence of arbitrarily high degree in a fixed amplitude requires a local counterterm with arbitrarily many derivatives and arbitrarily negative-dimension coupling; the set of required counterterms is infinite, which is the defining feature of a non-renormalizable theory established in Sections 21.3.4 and 21.3.5. □

The dimensional reading of Section 21.3.1 tells the same story in one line. The longitudinal term carries a factor $1/m^2$, of mass dimension -2; whenever an internal Proca line forces that factor into a loop, it acts like an interaction with a

coupling of negative mass dimension, the unmistakable signature of an irrelevant (non-renormalizable) operator. A vector mass inserted by hand is, from the ultraviolet's point of view, an infinite tower of higher-dimension operators in disguise. This is the first obstruction, now made quantitative: *the only naive route to a massive vector is both gauge-noninvariant and non-renormalizable, and the two failures are one and the same* $k_\mu k_\nu/m^2$.

Caution

The Proca theory is not inconsistent — it is a perfectly good effective description valid below the scale m, and tree-level Proca exchange between *conserved* currents is harmless, because $k_\mu J^\mu = 0$ annihilates the dangerous term just as in QED. The disease is purely ultraviolet and appears only when the longitudinal mode circulates inside loops, where nothing forces k_μ to hit a conserved current. What follows is not a rescue of a sick theory but the construction of a genuinely renormalizable one that happens to contain massive vectors.

30.1.3 Degree-of-freedom bookkeeping: where the third polarization lives

The second obstruction is a counting statement, and it is best seen by comparing the physical polarizations of a massless and a massive vector directly — the Wigner classification of Section 3.7 already told us the totals, and we now make them concrete. A massless vector is a helicity representation of the little group ISO(2): its physical states are the two transverse helicities $\lambda = \pm 1$ (Section 3.7.3), and no Lorentz transformation can manufacture a third. A massive vector is a spin-one representation of the little group SU(2) (Section 3.7.2): its physical states form the full $2s+1 = 3$ multiplet — two transverse and one *longitudinal*. The mismatch is one state, and it is not a smooth deformation: the little group itself changes as $m \to 0$. So the longitudinal polarization a massive vector must carry cannot simply "appear" when we turn on m; it has to be supplied by an extra physical degree of freedom that was already present in the theory. The following worked computation makes both the count and the identity of the missing state explicit.

Worked Example 30.3 (The polarization sums, and the longitudinal mode that distinguishes them)**.** Work in the frame $k^\mu = (E, 0, 0, p)$ with $E^2 - p^2 = m^2$. A massive vector of mass m has three real polarization vectors, transverse and longitudinal,

$$\varepsilon^\mu(k,1) = (0,1,0,0), \quad \varepsilon^\mu(k,2) = (0,0,1,0), \quad \varepsilon^\mu(k,L) = \frac{1}{m}\,(p,0,0,E), \tag{30.15}$$

chosen orthonormal and orthogonal to k. One checks $k\cdot\varepsilon(k,\lambda)=0$ for all three: for the longitudinal one $k\cdot\varepsilon(k,L)=(Ep-pE)/m=0$. Each is spacelike and unit-normalized, $\varepsilon(k,\lambda)\cdot\varepsilon(k,\lambda)=-1$; for the longitudinal vector $\varepsilon(k,L)\cdot\varepsilon(k,L)=(p^2-E^2)/m^2=-m^2/m^2=-1$, using the mass shell. We now sum the projectors $\varepsilon^\mu\varepsilon^\nu$ over the three states. The transverse pair contributes 1 to the $(1,1)$ and $(2,2)$ entries; the longitudinal vector contributes

$$\varepsilon^\mu(k,L)\varepsilon^\nu(k,L)=\frac{1}{m^2}\begin{pmatrix}p^2 & 0 & 0 & pE\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ pE & 0 & 0 & E^2\end{pmatrix}. \tag{30.16}$$

Adding all three and using $E^2-m^2=p^2$ entry by entry — the $(0,0)$ entry is $p^2/m^2=(E^2-m^2)/m^2=-1+E^2/m^2$, the $(3,3)$ entry is $E^2/m^2=1+p^2/m^2$, the $(1,1)$ and $(2,2)$ entries are $1=-\eta^{11}$, and the $(0,3)$ entry is $pE/m^2=k^0k^3/m^2$ — the sum collapses to the covariant form

$$\sum_{\lambda=1,2,L}\varepsilon^\mu(k,\lambda)\,\varepsilon^{\nu*}(k,\lambda)\;=\;-\eta^{\mu\nu}+\frac{k^\mu k^\nu}{m^2}. \tag{30.17}$$

This is exactly the numerator of the Proca propagator Eq. (30.11): the $k_\mu k_\nu/m^2$ that ruined the ultraviolet behaviour *is* the longitudinal polarization. The two obstructions are already touching.

For the massless photon only the two transverse helicities are physical, and their completeness relation (Section 13.3.3, with a reference null vector n^μ such that $n\cdot k\neq 0$) reads

$$\sum_{\lambda=\pm}\varepsilon^\mu(k,\lambda)\,\varepsilon^{\nu*}(k,\lambda)\;=\;-\eta^{\mu\nu}+\frac{k^\mu n^\nu+k^\nu n^\mu}{k\cdot n}, \tag{30.18}$$

the residual n-dependence reflecting the gauge freedom and dropping out of any amplitude contracted with conserved currents. The contrast in the counting is stark: trace Eq. (30.17) to get $\eta_{\mu\nu}(-\eta^{\mu\nu}+k^\mu k^\nu/m^2)=-4+m^2/m^2=-3$, three states; the massless relation Eq. (30.18) traces to $-4+2(k\cdot n)/(k\cdot n)=-2$, two states.

The identity of the extra state is read off from Eq. (30.15) in the high-energy limit. Boost the vector to $E\gg m$, where $p=\sqrt{E^2-m^2}\to E$ and

$$\varepsilon^\mu(k,L)=\frac{1}{m}\,(p,0,0,E)\;=\;\frac{1}{m}\,k^\mu+\mathcal{O}\!\left(\frac{m}{E}\right), \tag{30.19}$$

since $k^\mu=(E,0,0,p)\to E(1,0,0,1)$ and $\varepsilon^\mu(k,L)\to(E/m)(1,0,0,1)$. At high energy the longitudinal polarization is, up to corrections of order m/E, proportional to k^μ/m — a pure-momentum, "would-be-gauge" direction. This is the precise sense in which the longitudinal mode is the part of a massive vector that behaves least like a transverse photon and most like a scalar, and it is the seed of the Goldstone-equivalence theorem to come later in the chapter.

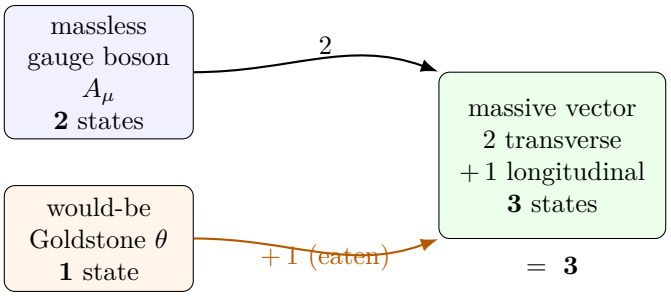


Figure 30.1: The degree-of-freedom balance the Higgs mechanism must enforce. The two transverse states of a massless gauge boson plus the single would-be Goldstone boson of a spontaneously broken symmetry combine into the three states of a massive vector. The Goldstone is "eaten" as the longitudinal polarization; no massless scalar survives. The arithmetic $2 + 1 = 3$ is the entire mechanism, counted.

The bookkeeping is now exact and worth stating plainly. A massless gauge vector carries 2 physical states; a massive vector carries 3; spontaneous breaking of a global symmetry deposits, by Goldstone's theorem (Theorem 29.11), one massless scalar per broken generator — 1 state. The arithmetic $2 + 1 = 3$ is impossible to miss. If the would-be Goldstone scalar could be the missing longitudinal polarization of the gauge vector, the accounts would balance with nothing left over: no separate massless scalar in the spectrum, and a vector heavy enough to carry the third state. The dynamical obstruction of Section 30.1.2 and the counting obstruction of this subsection are revealed as one summary viewed from two sides — the missing longitudinal polarization and the unwanted Goldstone boson are the same single degree of freedom. Figure 30.1 renders the balance.

30.1.4 The loophole: why Goldstone's theorem need not apply to a gauge theory

The counting in Fig. 30.1 is necessary but not yet permitted. Goldstone's theorem (Theorem 29.11) seems to insist that the broken symmetry deposit a genuine, physical, massless scalar in the spectrum — a particle one could in principle produce and detect — and a physical scalar is not the same thing as a polarization state of a vector. For the count to be realized as "the Goldstone becomes the longitudinal mode" rather than "the Goldstone exists *and* the vector is massive", the theorem's conclusion must be evadable. It is, and the escape is written into the theorem's own hypotheses.

Recall the hypotheses under which Goldstone's theorem was proved in Section 29.3.1. They were four: **(G1)** Poincaré invariance with an invariant vacuum; **(G2)** a conserved Noether current generating the broken symmetry; **(G3)** a

nonzero order parameter; and, decisively, **(G4)** *a positive-definite physical Hilbert space and microcausality.* The proof of Theorem 29.11 threaded the order parameter through a Lorentz-covariant spectral (Källén–Lehmann) representation of the current–field correlator and then used positivity of the norm to conclude that a zero-mass pole must be present and must couple to a physical state. *The argument simultaneously needs manifest Lorentz covariance and a positive-definite metric on the states it discusses.* Strip away either and the chain of inequalities breaks, and with it the guaranteed massless particle.

A gauge theory violates exactly this conjunction, and we proved that it does so back in Chapter 13. There are two ways to quantize a gauge field, and neither satisfies (G4) in the form the theorem requires:

- In *covariant* (Gupta–Bleuler, R_ξ) quantization (Section 13.3), Lorentz covariance is manifest, but the price is an *indefinite-metric* state space: the timelike photon mode has negative norm (Section 13.3.3). Positivity is recovered only on a *subspace* carved out by the subsidiary condition, and only after passing to a quotient (Theorem 13.19). The full space on which the covariant spectral representation lives is not positive-definite, so the positivity step in the proof of Theorem 29.11 is unavailable.
- In *physical-gauge* (Coulomb, unitary) quantization the metric is positive-definite by construction, but Lorentz covariance is no longer manifest — the Coulomb condition $\nabla \cdot \mathbf{A} = 0$ singles out a frame, and the covariant analytic structure the proof exploited is not in evidence.

Either way, the precise package the theorem demands — manifest Lorentz covariance *and* a positive-definite Hilbert space holding the states the spectral argument quantifies over — is absent. The conserved "current" of a gauged symmetry is not a current generating a symmetry of physical states at all; it is entangled with the constraints that define the gauge redundancy. This is the loophole. Goldstone's theorem is not violated — its hypotheses simply fail to hold — and so its conclusion is not forced. The would-be Goldstone boson is free to be unphysical: a gauge artifact rather than a particle, available to be absorbed into the longitudinal polarization that Section 30.1.3 showed the massive vector requires.

Physical Insight

The same indefinite metric that made Gupta–Bleuler quantization look like a nuisance in Chapter 13 is what saves the day here. A theory with a strictly positive-definite, manifestly covariant Hilbert space *cannot* hide a broken-symmetry scalar; it must show it. A gauge theory can, because it does not have such a space — its physical content lives on a quotient. The redundancy we paid for with ghosts and subsidiary conditions buys us, in

return, the right to a massive gauge boson. Massless gauge invariance and massive gauge bosons are not in conflict once one accepts that the price of the former (an unphysical sector) is exactly the loophole that permits the latter.

30.1.5 The two obstructions, and the object they force

We can now state the paradox as the two sharp obstructions promised at the outset, and read off their joint resolution.

Obstruction I (dynamics). A vector mass added by hand breaks gauge invariance (Eq. (30.4)) and produces the non-decaying $k_\mu k_\nu/m^2$ propagator term (Eq. (30.11)), which by Theorem 30.2 renders the theory non-renormalizable. A mass cannot be *imposed*.

Obstruction II (spectrum). A massive vector needs three physical polarizations where a massless one has two (Theorem 30.3), and spontaneous breaking of a global symmetry supplies a massless Goldstone scalar (Theorem 29.11, Theorem 29.12) that, as an ordinary particle, is unwanted in the spectrum.

These are not two problems but two views of one solution. Obstruction II offers exactly the degree of freedom Obstruction I says cannot be installed by fiat: the would-be Goldstone is the missing longitudinal mode, $2+1=3$. And the loophole of Section 30.1.4 removes the only thing that would keep them apart, by allowing the would-be Goldstone to be unphysical in a gauge theory. A vector mass cannot be *put in*; it can be *grown* — if the symmetry whose spontaneous breaking provides the Goldstone is the very gauge symmetry of the vector. Then the mass is generated, not inserted; gauge invariance is never broken in the Lagrangian, only realized nonlinearly on the vacuum; and renormalizability survives because the dangerous $k_\mu k_\nu/m^2$ is, as we will see, a gauge-artifact that a suitable gauge choice tames back to k^{-2} behaviour.

The construction that does all of this at once is forced. Take a complex scalar charged under a local U(1) — the abelian backdrop of QED (Chapter 8,Chapter 13) — give it a potential whose minimum lies away from the origin so that the symmetry is spontaneously broken, and couple it to the gauge field through the covariant derivative the gauge principle already mandates. The scalar's phase is the would-be Goldstone; the gauge field is the vector waiting for its third polarization; the covariant derivative is the wire that lets one become the other. This is the abelian Higgs model, and working it out in full at the classical, Lagrangian level — exhibiting the gauge-boson mass, the disappearance of the Goldstone, and the exact $2+1=3$ balance as a calculation rather than a hope — is the task of the next section.

30.2 The Abelian Higgs Model: Classical Mass Generation

The previous section sharpened the impasse into a concrete prescription. A bare photon mass $\frac{1}{2}m^2 A_\mu A^\mu$ is forbidden because it is not gauge invariant; a global symmetry, once spontaneously broken, pays for its broken charge with a massless Goldstone boson that nobody has seen. The renormalizable candidate that survived both objections was the simplest object that carries the gauge charge *and* the symmetry-breaking potential in one field: a complex scalar of unit charge, minimally coupled to an abelian gauge field, sitting in a Mexican-hat potential. We now take that candidate and work it out completely at the classical, Lagrangian level. The reward is the central mechanism of the chapter in its cleanest form—a gauge boson acquires a mass, the Goldstone disappears from the spectrum, and the degree-of-freedom count balances to the last entry—with every step of algebra visible and no quantization subtleties yet in play.

The strategy is deliberately redundant. We compute the spectrum twice. First in *Cartesian* fluctuations $\phi = (v + h + i\chi)/\sqrt{2}$, where the mass terms and—crucially—a peculiar kinetic mixing between the gauge field and the would-be Goldstone χ emerge from brute-force expansion. That mixing is the fingerprint of the mechanism: it is the term that tells us χ is not an independent particle. Then in *polar* variables $\phi = (v+h)\, e^{i\chi/v}/\sqrt{2}$, where a single gauge transformation erases χ outright, exhibiting it as pure gauge and leaving behind a manifestly massive vector. The two pictures must agree, and watching them agree is what teaches the mechanism.

30.2.1 The model and its vacuum

We gauge the global U(1) of the linear sigma model of Section 29.2. Take a single complex scalar $\phi(x)$ of charge $+1$ under a U(1) gauge symmetry, coupled to an abelian gauge field A_μ through the minimal-coupling prescription of Theorem 8.6. With the book's covariant-derivative sign convention (Eq. (8.27), here with charge $q = +1$ and coupling e),

$$D_\mu \phi = (\partial_\mu - ieA_\mu)\phi, \tag{30.20}$$

and the abelian field strength of Theorem 8.8, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (the same object that built the photon Lagrangian in Eq. (13.1)), the *abelian Higgs model* is defined by

$$\mathcal{L} = (D_\mu\phi)^*(D^\mu\phi) - V(\phi^*\phi) - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad V(\phi^*\phi) = -\mu^2\,\phi^*\phi + \lambda\,(\phi^*\phi)^2, \tag{30.21}$$

with $\mu^2 > 0$ and $\lambda > 0$. The potential is the Mexican hat of Eq. (29.36): the wrong-sign mass term destabilizes $\phi = 0$, the quartic bounds the energy below,

and the minimum sits on the circle

$$|\phi|^2_{\min} = \frac{\mu^2}{2\lambda} = \frac{v^2}{2}, \qquad v \equiv \frac{\mu}{\sqrt{\lambda}} > 0, \qquad \text{equivalently} \qquad |\phi|_{\min} = \frac{v}{\sqrt{2}}, \tag{30.22}$$

exactly as in Eq. (29.38). What is new relative to Chapter 29 is the single replacement $\partial_\mu \to D_\mu$ in the scalar kinetic term together with the gauge kinetic term $-\frac{1}{4}F^2$. That one substitution is the whole content of the model, and it will turn the harmless Goldstone of the global theory into the longitudinal mode of a massive vector.

The defining feature of Eq. (30.21) is its *local* U(1) invariance. Under a spacetime-dependent phase rotation with parameter $\alpha(x)$, following Eq. (8.113) for a charge-+1 field,

$$\phi(x) \;\longmapsto\; e^{\,ie\,\alpha(x)}\,\phi(x), \qquad A_\mu(x) \;\longmapsto\; A_\mu(x) + \partial_\mu\alpha(x), \tag{30.23}$$

the covariant derivative transforms homogeneously, $D_\mu\phi \mapsto e^{ie\alpha}D_\mu\phi$. Indeed,

$$\begin{aligned} D_\mu(e^{ie\alpha}\phi) &= \big(\partial_\mu - ie(A_\mu + \partial_\mu\alpha)\big)\, e^{ie\alpha}\phi \\ &= e^{ie\alpha}\Big(\partial_\mu\phi + ie(\partial_\mu\alpha)\phi - ieA_\mu\phi - ie(\partial_\mu\alpha)\phi\Big) = e^{ie\alpha}\, D_\mu\phi, \end{aligned} \tag{30.24}$$

the two $ie(\partial_\mu\alpha)\phi$ pieces cancelling exactly—this cancellation is the entire reason the inhomogeneous shift of A_μ in Eq. (30.23) is the one it is. Consequently $(D_\mu\phi)^*(D^\mu\phi)$ is invariant, $V(\phi^*\phi)$ is invariant because $|\phi|^2$ is, and $F_{\mu\nu}$ is invariant because $\partial_\mu\partial_\nu\alpha$ is symmetric and drops out of the antisymmetric curl. The Lagrangian Eq. (30.21) is gauge invariant. We will lean on Eq. (30.23) repeatedly: it is a redundancy of description, and exploiting it is precisely how the would-be Goldstone is removed.

Notation Watch

In Chapter 29 the complex scalar was written Φ and its real fluctuations about the vacuum were named σ (radial, massive) and π (angular, Goldstone). In this chapter we write the scalar as ϕ and—anticipating the physical particle content of the Standard Model—call the radial mode h (the *Higgs*) and the angular mode χ (the would-be Goldstone). The dictionary is $\Phi \leftrightarrow \phi$, $\sigma \leftrightarrow h$, $\pi \leftrightarrow \chi$; the potential, the VEV $v = \mu/\sqrt{\lambda}$, and the radial mass $m_h^2 = 2\lambda v^2$ are unchanged from Chapter 29. The charge is +1 and the covariant derivative is $D_\mu = \partial_\mu - ieA_\mu$, fixed once here to forestall any sign clash with Chapter 8.

The degree-of-freedom count before breaking. Before we expand anything, fix the bookkeeping that the rest of the section must reconcile. The unbroken-phase field content of Eq. (30.21), read as a theory of a massless gauge field

coupled to a charged scalar, carries

$$\underbrace{2}_{\text{complex scalar } \phi} + \underbrace{2}_{\text{massless vector } A_\mu} = 4 \quad \text{propagating real degrees of freedom.} \tag{30.25}$$

The complex scalar has two real components and hence two degrees of freedom. The gauge field A_μ has four Lorentz components, but the gauge redundancy Eq. (30.23) removes one and the resulting first-class constraint removes another, leaving the two transverse photon polarizations established in Chapter 13. The number 4 is a physical invariant: it cannot change merely because we expand the same Lagrangian about a different point of field space. Keeping Eq. (30.25) in view is the discipline that makes the disappearance of χ a bookkeeping necessity rather than a magic trick.

30.2.2 Cartesian expansion: masses and the tell-tale mixing

Choose the vacuum on the positive real axis, $\langle\phi\rangle = v/\sqrt{2}$, and parametrize the fluctuations about it in Cartesian (real and imaginary) components,

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x) + i\,\chi(x)\big), \qquad \langle h\rangle = \langle\chi\rangle = 0, \tag{30.26}$$

with h the displacement along the radius (toward and away from the trough) and χ the displacement along the imaginary axis (tangent to the trough at the chosen vacuum). This is the gauged version of the Cartesian split used in the Hessian computation of Theorem 29.5; there χ (then π) came out massless, and it still will. The new physics is entirely in the covariant derivative, so we compute $(D_\mu\phi)^*(D^\mu\phi)$ in full.

From Eqs. (30.20) and (30.26),

$$\begin{aligned} D_\mu\phi &= \partial_\mu\phi - ieA_\mu\phi \\ &= \frac{1}{\sqrt{2}}(\partial_\mu h + i\,\partial_\mu\chi) - \frac{ieA_\mu}{\sqrt{2}}(v + h + i\chi) \\ &= \frac{1}{\sqrt{2}}\Big[(\partial_\mu h + eA_\mu\chi) + i(\partial_\mu\chi - eA_\mu(v+h))\Big], \end{aligned} \tag{30.27}$$

where in the last line we grouped real and imaginary parts, using $-ieA_\mu(i\chi) = +eA_\mu\chi$ for the real part. The modulus squared is the sum of the squares of the two bracketed real expressions,

$$(D_\mu\phi)^*(D^\mu\phi) = \frac{1}{2}(\partial_\mu h + eA_\mu\chi)^2 + \frac{1}{2}(\partial_\mu\chi - eA_\mu(v+h))^2. \tag{30.28}$$

Expanding each square and collecting terms,

$$(D_\mu\phi)^*(D^\mu\phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{1}{2}(\partial_\mu\chi)(\partial^\mu\chi) + \frac{1}{2}e^2A_\mu A^\mu\Big[(v+h)^2 + \chi^2\Big]$$

$$- e\,(v+h)\,A_\mu\,\partial^\mu\chi + e\,\chi\,A_\mu\,\partial^\mu h. \tag{30.29}$$

Equation Eq. (30.29) is exact—no approximation has been made. Every term of the mechanism is already present in it; the rest is reading off the pieces of low order in the fields.

To extract the spectrum we keep terms quadratic in the fluctuations $\{A_\mu, h, \chi\}$, treating v as the background. Write out the relevant expansions: $(v+h)^2 = v^2 + 2vh + h^2$, so $\frac{1}{2}e^2A^2(v+h)^2$ contributes a quadratic piece $\frac{1}{2}e^2v^2A_\mu A^\mu$ and otherwise cubic-and-higher terms; the $\frac{1}{2}e^2A^2\chi^2$ piece is quartic; the cross term $-e(v+h)A_\mu\partial^\mu\chi$ contributes a quadratic piece $-ev\,A_\mu\partial^\mu\chi$ and a cubic remainder; and $+e\chi A_\mu\partial^\mu h$ is cubic. The quadratic part of the scalar kinetic term is therefore

$$(D_\mu\phi)^*(D^\mu\phi)\Big|_{\text{quad}} = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}(\partial_\mu\chi)^2 + \frac{1}{2}\,e^2v^2\,A_\mu A^\mu - ev\,A_\mu\,\partial^\mu\chi. \tag{30.30}$$

Three of these four terms are the headline results. The third is a mass term for the gauge field: comparing with the standard form $\frac{1}{2}m_A^2A_\mu A^\mu$ identifies

$$\boxed{m_A^2 = e^2v^2}. \tag{30.31}$$

The gauge boson is massive, and its mass is set by the coupling times the vacuum value of the scalar—no bare mass term was inserted, no gauge invariance was sacrificed, the mass is a property of the symmetric Lagrangian Eq. (30.21) read about its true vacuum. The fourth term, $-ev\,A_\mu\partial^\mu\chi = -m_A\,A_\mu\partial^\mu\chi$, is a quadratic *mixing* between the gauge field and χ; we return to it below, because it is the diagnostic of the whole construction.

The potential contributes the scalar masses. Since V depends only on $|\phi|^2 = \frac{1}{2}\big[(v+h)^2+\chi^2\big]$, its expansion is the one already performed in Theorem 29.5. With $\mu^2 = \lambda v^2$ from Eq. (30.22), write $u \equiv |\phi|^2$ and $\delta u \equiv u - \frac{1}{2}v^2 = vh + \frac{1}{2}h^2 + \frac{1}{2}\chi^2$. Then $V = -\lambda v^2 u + \lambda u^2$ has $V'(u) = -\lambda v^2 + 2\lambda u$, which vanishes at $u = \frac{1}{2}v^2$, and $V'' = 2\lambda$. Taylor expansion gives $V = V_0 + V'\,\delta u + \frac{1}{2}V''(\delta u)^2$ with $V' = 0$, so to quadratic order

$$V\Big|_{\text{quad}} = \lambda\,(\delta u)^2\Big|_{\text{quad}} = \lambda\,(vh)^2 = \lambda v^2\,h^2 = \frac{1}{2}\,(2\lambda v^2)\,h^2, \tag{30.32}$$

where only the vh term of δu survives squaring at quadratic order. Hence the radial mode is massive,

$$\boxed{m_h^2 = 2\lambda v^2 = 2\mu^2}, \tag{30.33}$$

matching m_σ^2 of Eq. (29.43), while χ receives *no* mass from V: the $\frac{1}{2}\chi^2$ in δu would only contribute through the linear term $V'\,\delta u$, and $V' = 0$ at the vacuum. The would-be Goldstone is massless at this stage, exactly as Goldstone's theorem for the global theory demands.

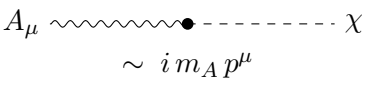


Figure 30.2: The quadratic gauge–Goldstone mixing $-m_A A_\mu \partial^\mu \chi$ of Eq. (30.34), drawn as a two-point insertion (the dot) linking the photon line to the would-be Goldstone (dashed). Because it is bilinear, it is part of the free action: A_μ and χ share a propagator. A two-point mixing of a vector with a scalar through the momentum p^μ is impossible for two genuinely independent physical particles; its appearance is the unambiguous signal that χ is not an independent state but is about to be absorbed into A_μ.

Assembling Eqs. (30.30) and (30.32) with the gauge kinetic term, the quadratic Lagrangian of the abelian Higgs model in Cartesian variables is

$$\mathcal{L}^{(2)} = \underbrace{-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_A^2\,A_\mu A^\mu}_{\text{massive vector}} + \underbrace{\frac{1}{2}(\partial_\mu h)^2 - \frac{1}{2}m_h^2\,h^2}_{\text{Higgs}} + \underbrace{\frac{1}{2}(\partial_\mu \chi)^2}_{\text{massless }\chi} \underbrace{- m_A\,A_\mu\,\partial^\mu \chi}_{\text{mixing}}. \tag{30.34}$$

This is the complete free spectrum at quadratic order: a vector with mass $m_A = ev$, a real scalar h with mass $m_h = \sqrt{2\lambda}\,v = \sqrt{2}\,\mu$, a massless scalar χ, and—the term that does not fit the naive particle reading—a bilinear $A_\mu \partial^\mu \chi$.

30.2.3 The mixing term is the mechanism announcing itself

The term $-m_A A_\mu \partial^\mu \chi$ in Eq. (30.34) cannot be ignored, and it cannot be read as an interaction: it is quadratic in the fields, part of the free Lagrangian, and it couples A_μ to χ at the level of the propagator. Integrating by parts, $-m_A A_\mu \partial^\mu \chi = +m_A (\partial^\mu A_\mu)\chi$ up to a total derivative, which makes the bilinear coupling between the gauge field's divergence and the Goldstone manifest. In momentum space, writing fields as $\int \frac{d^4p}{(2\pi)^4} e^{-ip\cdot x}\tilde{\phi}(p)$ so that $\partial^\mu \chi \to -ip^\mu \tilde{\chi}(p)$, the mixing contributes to the quadratic action an off-diagonal piece proportional to

$$i\,m_A\,p^\mu \quad \text{coupling } A_\mu(-p) \text{ to } \chi(p), \tag{30.35}$$

an entry in the inverse-propagator matrix that is off the diagonal in the (A_μ, χ) basis. Its presence means that A_μ and χ do not propagate independently: a χ can turn into an A_μ at quadratic order, and the would-be Goldstone has no separate, diagonal propagator of its own.

The structure p^μ in Eq. (30.35) is the clue. A massive vector field has three physical polarizations: two transverse, plus one *longitudinal*, whose polarization vector at high momentum behaves as $\varepsilon_L^\mu \sim p^\mu/m_A$. A bilinear that ties A_μ to a scalar precisely through p^μ is therefore a bilinear that ties the scalar to the longitudinal polarization of the vector—it is the algebraic shadow of the

statement that χ *is* that longitudinal polarization, written in a basis that has not yet recognized the fact. The mixing term is the mechanism announcing itself in the only language available before we change variables: the Goldstone will be eaten. We could remove it by hand with a clever gauge choice (the 't Hooft R_ξ gauges do exactly this and are the subject of Section 30.2's sequel), but the cleanest demonstration is to change variables so that χ never appears at all. That is the polar parametrization, to which we turn now.

30.2.4 Polar parametrization: the Goldstone is pure gauge

The Cartesian split Eq. (30.26) is faithful but it fights the geometry. The Mexican hat (Fig. 29.2) is rotationally symmetric, and the natural fluctuations are radial and angular. Following the polar parametrization of Eq. (29.40), write

$$\phi(x) = \frac{1}{\sqrt{2}}\,(v + h(x))\, e^{\,i\,\chi(x)/v}, \qquad \langle h\rangle = \langle\chi\rangle = 0, \tag{30.36}$$

where now h is the radial excursion off the trough and χ the angular position around it, with the factor $1/v$ in the exponent chosen so that χ is canonically normalized (the same normalization that worked in Eq. (29.40)). To linear order this agrees with Eq. (30.26), since $e^{i\chi/v} = 1 + i\chi/v + \dots$ gives $\phi \approx (v + h + i\chi)/\sqrt{2}$; the two parametrizations describe the same physics and label the same fields at leading order.

The decisive observation is that the phase $e^{i\chi/v}$ in Eq. (30.36) has *exactly* the form of the local gauge rotation Eq. (30.23). The gauge freedom $\phi \mapsto e^{ie\alpha}\phi$ can rotate the phase to anything we like, including unity. Choose the gauge parameter

$$\alpha(x) = -\frac{\chi(x)}{e\,v}, \tag{30.37}$$

so that the scalar transforms to

$$\phi \;\longmapsto\; \phi' = e^{\,ie\alpha}\,\phi = \exp\!\Big(-i\frac{\chi}{v}\Big)\cdot\frac{1}{\sqrt{2}}(v+h)\,e^{\,i\chi/v} = \frac{1}{\sqrt{2}}(v+h). \tag{30.38}$$

The phase is gone: ϕ' is real, and the field χ has been rotated away entirely. This choice—the gauge in which the scalar is everywhere real and positive—is the *unitary gauge*. By Eq. (30.23), the price is a compensating shift of the gauge field,

$$A_\mu \;\longmapsto\; A'_\mu = A_\mu + \partial_\mu\alpha = A_\mu - \frac{1}{ev}\,\partial_\mu\chi. \tag{30.39}$$

The Goldstone has not vanished from existence—it has been absorbed into the gauge field, reappearing as the gradient piece $-\frac{1}{ev}\partial_\mu\chi$ of A'_μ. This is the eating, made explicit: one real scalar field has been traded for one extra component of the vector. The worked example below carries the transformation through term by term to verify that nothing is lost.

Worked Example 30.4 (Unitary gauge removes χ, and the field strength does not notice)**.** We confirm that the gauge transformation Eq. (30.37)–Eq. (30.39) eliminates χ from the entire Lagrangian, not merely from ϕ, by computing $(D_\mu\phi)^*(D^\mu\phi)$ directly in the polar variables and watching χ repackage itself into A'_μ.

Differentiate Eq. (30.36) and apply the covariant derivative:

$$\begin{aligned} D_\mu\phi &= (\partial_\mu - ieA_\mu)\frac{1}{\sqrt{2}}(v+h)\,e^{i\chi/v} \\ &= \frac{1}{\sqrt{2}}\,e^{i\chi/v}\left[\partial_\mu h + i\,(v+h)\,\frac{\partial_\mu\chi}{v} - ieA_\mu\,(v+h)\right] \\ &= \frac{1}{\sqrt{2}}\,e^{i\chi/v}\left[\partial_\mu h + i\,(v+h)\Big(\frac{\partial_\mu\chi}{v} - eA_\mu\Big)\right]. \end{aligned} \tag{30.40}$$

The phase $e^{i\chi/v}$ cancels against its conjugate in the modulus squared, so

$$(D_\mu\phi)^*(D^\mu\phi) = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}\,(v+h)^2\Big(\frac{\partial_\mu\chi}{v} - eA_\mu\Big)^2. \tag{30.41}$$

Now read the second term: χ enters *only* through the combination $eA_\mu - \frac{1}{v}\partial_\mu\chi$. Define the shifted field exactly as in Eq. (30.39),

$$B_\mu \equiv A_\mu - \frac{1}{ev}\,\partial_\mu\chi \qquad\Longrightarrow\qquad \frac{\partial_\mu\chi}{v} - eA_\mu = -\,e\,B_\mu, \tag{30.42}$$

so that

$$(D_\mu\phi)^*(D^\mu\phi) = \frac{1}{2}(\partial_\mu h)^2 + \frac{1}{2}\,e^2\,(v+h)^2\,B_\mu B^\mu. \tag{30.43}$$

The field χ has disappeared; it survives only inside B_μ. The gauge kinetic term is untouched by the redefinition, because the difference between A_μ and B_μ is a pure gradient and the curl annihilates it:

$$\partial_\mu B_\nu - \partial_\nu B_\mu = \partial_\mu A_\nu - \partial_\nu A_\mu - \frac{1}{ev}(\partial_\mu\partial_\nu\chi - \partial_\nu\partial_\mu\chi) = F_{\mu\nu}, \tag{30.44}$$

the second-derivative term cancelling by equality of mixed partials—this is the same gauge invariance of $F_{\mu\nu}$ noted under Eq. (30.24), now doing concrete work. Therefore $-\frac{1}{4}F_{\mu\nu}[A]F^{\mu\nu}[A] = -\frac{1}{4}F_{\mu\nu}[B]F^{\mu\nu}[B]$, and χ is absent from the gauge sector as well. It appears nowhere in the potential either, since V depends only on $|\phi|^2 = (v+h)^2/2$, which is independent of χ. The Goldstone has been removed from every term of the Lagrangian by a single field redefinition that is a gauge transformation. It was pure gauge all along.

Collecting Eq. (30.43), the unchanged gauge kinetic term, and the potential expanded as in Eq. (29.42), the abelian Higgs Lagrangian in unitary gauge is

$$\mathcal{L} = -\frac{1}{4}F_{\mu\nu}[B]F^{\mu\nu}[B] + \frac{1}{2}\,e^2(v+h)^2\,B_\mu B^\mu + \frac{1}{2}(\partial_\mu h)^2 - \lambda v^2 h^2 - \lambda v\,h^3 - \frac{\lambda}{4}h^4, \tag{30.45}$$

where the constant vacuum energy has been dropped. Expanding the vector mass term, $\frac{1}{2}e^2(v+h)^2B^2 = \frac{1}{2}e^2v^2B^2 + e^2v\,h\,B^2 + \frac{1}{2}e^2h^2B^2$, exposes the spectrum and the couplings with no ambiguity:

$$\mathcal{L} = \underbrace{-\frac{1}{4}F_{\mu\nu}[B]F^{\mu\nu}[B] + \frac{1}{2}m_A^2\,B_\mu B^\mu}_{\text{massive vector, } m_A=ev} + \underbrace{\frac{1}{2}(\partial_\mu h)^2 - \frac{1}{2}m_h^2h^2}_{\text{Higgs, } m_h^2=2\lambda v^2} + \underbrace{\frac{m_A^2}{v}\,h\,B_\mu B^\mu + \frac{m_A^2}{2v^2}\,h^2B_\mu B^\mu}_{\text{Higgs–vector couplings}} - \text{Hi} \tag{30.46}$$

There is no χ, no kinetic mixing, no massless scalar. The mixing term of Eq. (30.34) that diagnosed the eating has been resolved by the variable change that performs it: what looked like a pathological vector–scalar bilinear was the off-diagonal remnant of writing a single massive vector in a redundant set of fields. The unitary-gauge Lagrangian Eq. (30.46) displays the true particle content directly. Notice also that the Higgs couplings to the vector come with coefficients fixed entirely by m_A and v: the cubic hBB vertex is $2m_A^2/v = 2m_A^2/v$ per the symmetry factor, and the quartic h^2BB vertex is m_A^2/v^2. These are not free—gauge invariance ties the strength of the Higgs–gauge coupling to the gauge-boson mass, a relation that will be the engine of the cancellations protecting unitarity later in the chapter.

30.2.5 The degree-of-freedom count balances

We now close the bookkeeping opened in Eq. (30.25). After spontaneous breaking, the physical fields of Eq. (30.46) are a *massive* vector B_μ and a single real scalar h:

$$\underbrace{1}_{\text{real Higgs } h} + \underbrace{3}_{\text{massive vector } B_\mu} = 4 \quad \text{propagating real degrees of freedom.} \tag{30.47}$$

A massive vector has three polarizations—two transverse and one longitudinal—because its mass term breaks the gauge redundancy that, in the massless case, removed one component and the constraint another. The real Higgs supplies one. The total is 4, identical to the count 4 of Eq. (30.25). The arithmetic of the transition is unambiguous:

$$\underbrace{2}_{\phi} + \underbrace{2}_{A_\mu} = \underbrace{1}_{h} + \underbrace{3}_{B_\mu}, \qquad \text{i.e.} \qquad \underbrace{1}_{\text{lost Goldstone } \chi} = \underbrace{1}_{\text{gained longitudinal polarization}}. \tag{30.48}$$

The Goldstone boson was not destroyed—a degree of freedom cannot simply vanish—it was *converted*. The complex scalar lost one of its two real components (the angular χ) and the gauge field gained exactly one polarization (the longitudinal mode it lacked while massless). The would-be Goldstone became the longitudinal polarization of the now-massive vector. This is the precise sense of the slogan "the gauge boson eats the Goldstone," and Eq. (30.48) is its complete accounting.

	complex scalar	gauge field	total
before breaking (A_μ massless)	2	2 (transverse)	4
after breaking (B_μ massive)	1 (h only)	3 (T+T+L)	4

Table 30.1: Degree-of-freedom count of the abelian Higgs model, Eq. (30.25) versus Eq. (30.47). The single Goldstone degree of freedom migrates from the scalar sector (where it was the angular mode χ) to the gauge sector (where it becomes the longitudinal polarization L of the massive vector). The total is invariant, as it must be.

It remains to verify the claim made repeatedly in passing: that the field eaten is exactly the would-be Goldstone, i.e. the direction in field space along which the broken generator moves the vacuum, and that the count of eaten fields is the Goldstone count of Theorem 29.12. The following worked example ties the two together.

Worked Example 30.5 (The eaten field is the broken-generator direction)**.** The global U(1) acts on the scalar by the infinitesimal rotation $\delta\phi = i\,e\,\theta\,\phi$ for a constant parameter θ (the gauge transformation Eq. (30.23) at constant $\alpha = \theta$). Evaluate this at the vacuum $\langle\phi\rangle = v/\sqrt{2}$:

$$\delta\,\langle\phi\rangle = i\,e\,\theta\,\frac{v}{\sqrt{2}} = \frac{e\,\theta\,v}{\sqrt{2}}\,(i), \tag{30.49}$$

which is *purely imaginary*: the symmetry moves the vacuum in the imaginary direction of the complex ϕ-plane. In the Cartesian split Eq. (30.26), the imaginary direction is precisely the χ direction. Thus the broken generator moves the vacuum along χ and not along h:

$$\delta h = 0, \qquad \delta\chi = e\,\theta\,v \quad \text{(constant shift)}, \tag{30.50}$$

read off by matching $\delta\phi = \frac{1}{\sqrt{2}}(\delta h + i\,\delta\chi)$ to Eq. (30.49). The transformation under which χ shifts by a constant is the realization of the broken symmetry; χ is its Goldstone field. This is the same identification made in Theorem 29.5, where the null eigenvector of the mass matrix was the direction tangent to the vacuum circle—the imaginary axis, the χ direction—and is exactly the field that the polar gauge transformation Eq. (30.38) rotated away.

The counting now follows from Theorem 29.12. The global symmetry group is $G = \mathrm{U}(1)$ with $\dim G = 1$; the unbroken subgroup that fixes the vacuum $\langle\phi\rangle = v/\sqrt{2}$ is $H = \{1\}$ (every nontrivial phase moves the vacuum), with $\dim H = 0$. The number of would-be Goldstone bosons is therefore $\dim G - \dim H = 1 - 0 = 1$. In the *global* theory of Chapter 29 these would be genuine massless particles. In the *gauged* theory each one is eaten: the single broken generator gives the single gauge boson its mass, and the single would-be Goldstone χ becomes its

longitudinal polarization. The match is exact and general—one eaten Goldstone per broken gauge generator—and it is the abelian instance of the mass formula proved for the non-abelian case later in the chapter.

Physical Insight

The cleanest way to hold the mechanism in mind is energetic and geometric, not algebraic. In the global theory, the flat angular direction of the Mexican hat (Fig. 29.2) costs no energy to traverse, and a long-wavelength oscillation along it costs arbitrarily little: that is the massless Goldstone. Gauging the symmetry makes the angle a *local* redundancy—two field configurations differing by a position-dependent rotation are the same physical state. A would-be excitation along the trough is then not a new state at all; it is a gauge copy of the vacuum. There is nothing left for it to be except part of the gauge field, and the only thing a gauge field can absorb a scalar into is its missing longitudinal polarization. The Goldstone is not so much eaten as *recognized*: it was never an independent physical degree of freedom once the symmetry became local.

We have, at the classical level, everything the chapter set out to establish in the abelian case: a gauge-invariant Lagrangian whose true vacuum gives the gauge boson a mass $m_A = ev$ (Eq. (30.31)), a physical Higgs of mass $m_h = \sqrt{2\lambda}\, v$ (Eq. (30.33)), the disappearance of the Goldstone χ as a propagating state, and a degree-of-freedom count (Table 30.1) that balances by converting the Goldstone into the vector's longitudinal mode.

What we have *not* yet confronted is the cost of the bookkeeping. Removing χ required a specific choice of gauge—the unitary gauge Eq. (30.38), in which ϕ is real—and the spectrum Eq. (30.46) is manifest only in that gauge. The leftover freedom is real: we could equally have kept χ as in Eq. (30.34) and dealt with its mixing some other way. How one handles χ—eliminate it outright in unitary gauge, or retain it and gauge-fix it smoothly—defines the two complementary descriptions of the quantized theory developed next. The following section quantizes in unitary gauge, where the particle content is transparent but, as we shall see, the massive-vector propagator hides a high-energy pathology; the section after that retains χ in the 't Hooft R_ξ gauges, where renormalizability is manifest at the price of carrying the would-be Goldstone as an explicit field.

30.3 Unitary Gauge: The Physical Spectrum and Its Price

The classical analysis of the abelian Higgs model (Section 30.2) left one piece of structure deliberately unused. We found that the spontaneously broken vacuum

carries a flat direction — the phase of the complex scalar — and that this flat direction is precisely a *local* symmetry: the field χ that parametrizes it can be shifted point by point by a U(1) gauge transformation. A would-be Goldstone boson that can be rotated away pointwise is not a Goldstone boson at all; it is gauge. The cleanest way to make that statement operational is simply to perform the rotation that removes it everywhere and read off what is left. The resulting gauge — the choice $\chi = 0$ — is called *unitary gauge*, and it has a singular virtue: the field content it leaves behind is exactly the physical particle content of the theory, with no redundancy, no unphysical polarizations, and no ghosts to subtract. This section quantizes the broken theory in that gauge. We derive the propagator of the now-massive vector directly from the broken Lagrangian, confirm that the degree-of-freedom bookkeeping of Section 30.2 survives quantization, and then expose the price one pays for this transparency: the very term in the propagator that encodes the longitudinal polarization grows at large momentum, so that the renormalizability and the good high-energy behaviour of the theory — both genuinely present — are hidden from view. That tension is the engine of the rest of the chapter.

Throughout we use the conventions fixed at the head of the chapter: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, and a complex scalar ϕ of charge $+1$ with covariant derivative $D_\mu\phi = (\partial_\mu - ieA_\mu)\phi$, consistent with the gauge principle of Chapter 8.

30.3.1 Fixing the gauge: $\chi = 0$ and the disappearance of the Goldstone

Recall the model (Section 30.2). The Lagrangian

$$\mathcal{L} = (D_\mu\phi)^*(D^\mu\phi) - V(\phi) - \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu}, \qquad V(\phi) = -\mu^2\,\phi^*\phi + \lambda\,(\phi^*\phi)^2, \tag{30.51}$$

with $\mu^2 > 0$ and $\lambda > 0$, is invariant under the local transformations

$$\phi(x) \;\longmapsto\; e^{i\alpha(x)}\,\phi(x), \qquad A_\mu(x) \;\longmapsto\; A_\mu(x) + \tfrac{1}{e}\,\partial_\mu\alpha(x). \tag{30.52}$$

The potential is minimized on the circle $\phi^*\phi = \mu^2/2\lambda \equiv v^2/2$, so that $v = \sqrt{\mu^2/\lambda}$ and $\langle\phi\rangle = v/\sqrt{2}$. Rather than the linear split $\phi = \frac{1}{\sqrt{2}}(v + h + i\chi)$ used to read off the spectrum classically, the gauge structure is laid bare by the *polar* parametrization

$$\phi(x) = \frac{1}{\sqrt{2}}\big(v + h(x)\big)\,\exp\!\Big(\frac{i\,\chi(x)}{v}\Big), \tag{30.53}$$

in which h is the gauge-invariant radial fluctuation (the Higgs field) and χ is the angular fluctuation along the vacuum circle (the would-be Goldstone), normalized to have canonical mass dimension. Under Eq. (30.52) the modulus $|\phi|$ is inert while the phase shifts, so in the variables Eq. (30.53)

$$h(x) \;\longmapsto\; h(x), \qquad \chi(x) \;\longmapsto\; \chi(x) + v\,\alpha(x). \tag{30.54}$$

The Goldstone field is a pure *shift* field under the gauge group: it transforms inhomogeneously, exactly as the gauge parameter does, with no derivatives. This is the precise sense in which χ is gauge and not matter. It is also the sense in which Goldstone's theorem (Theorem 29.11) is evaded: that theorem produces a massless physical excitation for a broken *global* symmetry, but here the symmetry is local and the would-be Goldstone is a gauge direction, removable by a choice of frame.

Definition 30.6 (Unitary gauge)**.** *Unitary gauge* is the choice of gauge parameter $\alpha(x) = -\chi(x)/v$ in Eq. (30.54), which sets the Goldstone field identically to zero, $\chi(x) \equiv 0$. Equivalently, it is the gauge in which the scalar is everywhere real and nonnegative,

$$\phi(x) = \frac{1}{\sqrt{2}}(v + h(x)), \qquad h(x) \text{ real.} \tag{30.55}$$

Two remarks make precise why this choice is both legitimate and special. First, it is an admissible gauge: the transformation $\alpha = -\chi/v$ exists pointwise as long as $v \neq 0$, i.e. as long as the symmetry is broken, so unitary gauge is available everywhere in the broken phase. (It degenerates precisely as $v \to 0$, where the would-be Goldstone becomes a genuine massless field and cannot be gauged away — the symmetric phase has no unitary gauge, as it should not.) Second, it is a *complete* gauge fixing of the physical configurations up to the residual constant transformations that leave the vacuum invariant; nothing of the local redundancy survives to propagate. The name "unitary" anticipates the payoff derived below: the surviving fields are in one-to-one correspondence with physical states, so unitarity of the S-matrix — the statement that probability flows only among physical states — is manifest.

30.3.2 The unitary-gauge Lagrangian

Substituting Eq. (30.55) into Eq. (30.51) produces a Lagrangian written entirely in terms of the two physical fields h and A_μ. The kinetic term of the scalar is the only place where A_μ enters through the covariant derivative:

$$D_\mu\phi = \frac{1}{\sqrt{2}}(\partial_\mu h - ie\, A_\mu(v + h)), \tag{30.56}$$

so that, since the two terms are respectively real and imaginary multiples of $1/\sqrt{2}$,

$$(D_\mu\phi)^*(D^\mu\phi) = \frac{1}{2}(\partial_\mu h)(\partial^\mu h) + \frac{1}{2}e^2(v + h)^2 A_\mu A^\mu. \tag{30.57}$$

The cross term that one might fear, proportional to $A^\mu\partial_\mu h$, is absent: it is the imaginary part of $\partial_\mu h\,(v + h)$, which vanishes because both factors are real. (In the linear parametrization the same cancellation appears as a total-derivative $A^\mu\partial_\mu\chi$ mixing term that is removed only after gauge fixing; the polar variables

have done that removal for us at the level of the field redefinition.) Expanding the square in Eq. (30.57),

$$\frac{1}{2}e^2(v+h)^2 A_\mu A^\mu = \underbrace{\frac{1}{2}e^2 v^2\, A_\mu A^\mu}_{\text{mass}} + \underbrace{e^2 v\, h\, A_\mu A^\mu}_{hAA} + \underbrace{\frac{1}{2}e^2 h^2 A_\mu A^\mu}_{hhAA}. \tag{30.58}$$

The first term is a mass term for the vector. Reading off the coefficient against the canonical form $\frac{1}{2}m_A^2 A_\mu A^\mu$ gives

$$\boxed{m_A = e\,v} \tag{30.59}$$

in agreement with the classical result of Section 30.2; the second and third terms are the cubic and quartic couplings of the Higgs to the vector, with strengths fixed by gauge invariance to $e^2 v = m_A^2/v$ and $\frac{1}{2}e^2 = m_A^2/2v^2$ respectively.

The potential, evaluated on Eq. (30.55) with $\phi^*\phi = \frac{1}{2}(v+h)^2$ and $\mu^2 = \lambda v^2$, is

$$\begin{aligned} V &= -\frac{\lambda v^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4 \\ &= -\frac{\lambda v^4}{4} + \underbrace{\left(-\lambda v^3 + \lambda v^3\right)}_{=0} h + \lambda v^2 h^2 + \lambda v\, h^3 + \frac{\lambda}{4}h^4. \end{aligned} \tag{30.60}$$

The linear term cancels — the expansion is about the true minimum, so there is no tadpole — and the constant $-\lambda v^4/4$ is the vacuum energy, irrelevant for S-matrix elements. The quadratic term gives the Higgs mass,

$$\tfrac{1}{2}m_h^2 h^2 = \lambda v^2 h^2 \quad\Longrightarrow\quad m_h^2 = 2\lambda v^2 = 2\mu^2, \tag{30.61}$$

again matching Section 30.2. Collecting Eqs. (30.57), (30.58) and (30.60) and dropping the constant, the complete unitary-gauge Lagrangian is

$$\boxed{\begin{aligned} \mathcal{L}_{\text{U}} = &-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_A^2\, A_\mu A^\mu + \frac{1}{2}(\partial_\mu h)(\partial^\mu h) - \frac{1}{2}m_h^2\, h^2 \\ &+ \frac{m_A^2}{v}\, h\, A_\mu A^\mu + \frac{m_A^2}{2v^2}\, h^2 A_\mu A^\mu - \frac{m_h^2}{2v}\, h^3 - \frac{m_h^2}{8v^2}\, h^4 \end{aligned}} \tag{30.62}$$

where in the interaction terms we have traded $\lambda v = m_h^2/2v$ and $\lambda = m_h^2/2v^2$ to display every coupling as a physical mass over a power of v. This is a complete, self-contained field theory: a massive vector A_μ, a massive scalar h, and their interactions, with not a single unphysical field in sight. The would-be Goldstone χ has vanished from the Lagrangian without trace.

Physical Insight

The counting that Section 30.2 performed at the classical level now reads off Eq. (30.62) directly. Before breaking, the propagating degrees of freedom were the two transverse polarizations of a massless vector plus the two real components of a complex scalar: $2+2=4$. After breaking, in unitary gauge, they are the three polarizations of a massive vector plus one real scalar h: $3+1=4$. The total is conserved because no degree of freedom can be created or destroyed by a field redefinition. The single bookkeeping sentence that captures the whole mechanism is: *the gauge boson ate the Goldstone*. The longitudinal polarization the massive vector now carries — the third state it did not have when massless — *is* the would-be Goldstone χ, repackaged. Unitary gauge is the frame in which this repackaging is already complete, which is exactly why no separate χ field appears.

30.3.3 The massive-vector (Proca) propagator from the broken theory

The dynamics of the vector at quadratic order is governed by the Proca Lagrangian

$$\mathcal{L}_2^{(A)} = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}m_A^2\, A_\mu A^\mu. \tag{30.63}$$

We need its propagator. The derivation is a clean inversion of a quadratic operator, and it is worth carrying out in full both because the result is the workhorse of every unitary-gauge computation and because the structure of the answer is the whole point of this section.

Write the kinetic term in the canonical quadratic form $\frac{1}{2}A_\mu\,\mathcal{O}^{\mu\nu}A_\nu$ by integrating by parts. Using $F_{\mu\nu}F^{\mu\nu} = 2(\partial_\mu A_\nu \partial^\mu A^\nu - \partial_\mu A_\nu \partial^\nu A^\mu)$ and discarding total derivatives,

$$-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}A_\mu(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)A_\nu, \tag{30.64}$$

so that with the mass term included,

$$\mathcal{L}_2^{(A)} = \frac{1}{2}A_\mu\Big[(\Box + m_A^2)\,\eta^{\mu\nu} - \partial^\mu\partial^\nu\Big]A_\nu. \tag{30.65}$$

In momentum space, with the Fourier convention of the style canon (a field mode $\propto e^{-ik\cdot x}$, hence $\partial_\mu \to -ik_\mu$), the differential operator in brackets becomes the c-number matrix

$$\mathcal{O}^{\mu\nu}(k) = -(k^2 - m_A^2)\eta^{\mu\nu} + k^\mu k^\nu. \tag{30.66}$$

Here the broken theory parts ways decisively from the unbroken one. In Section 13.4.3 we saw that the *massless* photon operator $\mathcal{O}_0^{\mu\nu}(k) = -k^2\eta^{\mu\nu} +$

$k^\mu k^\nu = -k^2(\eta^{\mu\nu} - k^\mu k^\nu/k^2)$ is a multiple of a transverse projector: it annihilates every longitudinal vector k_ν, $\mathcal{O}_0^{\mu\nu} k_\nu = 0$, so it is *not invertible* and cannot be inverted into a propagator without first adding a gauge-fixing term. The mass term cures exactly this. Acting with Eq. (30.66) on k_ν,

$$\mathcal{O}^{\mu\nu}(k)\, k_\nu = -(k^2 - m_A^2)k^\mu + k^\mu k^2 = m_A^2\, k^\mu \neq 0, \tag{30.67}$$

so the longitudinal direction is now an eigenvector with nonzero eigenvalue m_A^2: the mass lifts the would-be zero mode, the operator is invertible, and *no gauge fixing is needed*. This is the algebraic shadow of the physical fact that the vector now has a genuine third (longitudinal) polarization to propagate.

Proposition 30.7 (Unitary-gauge massive-vector propagator)**.** *The Feynman propagator obtained by inverting the Proca operator Eq. (30.66), with the time-ordering $i\epsilon$ prescription, is*

$$D_{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2 + i\epsilon}\left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right). \tag{30.68}$$

Proof. The propagator is i times the inverse of the quadratic operator: it is the matrix $D_{\nu\rho}(k)$ satisfying $\mathcal{O}^{\mu\nu}(k)\, D_{\nu\rho}(k) = i\,\delta^\mu{}_\rho$. (The factor i is the standard normalization in which D is the time-ordered two-point function; cf. the photon case Theorem 13.26.) The only Lorentz tensors available to build $D_{\nu\rho}$ from are $\eta_{\nu\rho}$ and $k_\nu k_\rho$, so we set

$$D_{\nu\rho}(k) = A\,\eta_{\nu\rho} + B\,k_\nu k_\rho \tag{30.69}$$

with scalar coefficients $A(k^2)$, $B(k^2)$ to be determined. Contracting with Eq. (30.66),

$$\begin{aligned}\mathcal{O}^{\mu\nu} D_{\nu\rho} &= \left[-(k^2 - m_A^2)\eta^{\mu\nu} + k^\mu k^\nu\right]\left[A\,\eta_{\nu\rho} + B\,k_\nu k_\rho\right] \\ &= -(k^2 - m_A^2)A\,\delta^\mu{}_\rho + \Big[-(k^2 - m_A^2)B + A + B\,k^2\Big]k^\mu k_\rho.\end{aligned} \tag{30.70}$$

Matching to $i\,\delta^\mu{}_\rho$ requires the coefficient of $\delta^\mu{}_\rho$ to equal i and the coefficient of $k^\mu k_\rho$ to vanish:

$$-(k^2 - m_A^2)\,A = i \qquad \Longrightarrow \qquad A = \frac{-i}{k^2 - m_A^2}, \tag{30.71}$$

$$-(k^2 - m_A^2)B + A + B\,k^2 = 0 \qquad \Longrightarrow \qquad m_A^2\,B + A = 0, \tag{30.72}$$

where in the second line the $k^2 B$ terms cancel and leave only $m_A^2 B$. Hence $B = -A/m_A^2 = i/[m_A^2(k^2 - m_A^2)]$, and

$$D_{\nu\rho}(k) = \frac{-i}{k^2 - m_A^2}\,\eta_{\nu\rho} + \frac{i}{m_A^2(k^2 - m_A^2)}\,k_\nu k_\rho = \frac{-i}{k^2 - m_A^2}\left(\eta_{\nu\rho} - \frac{k_\nu k_\rho}{m_A^2}\right). \tag{30.73}$$

Reinstating the Feynman $i\epsilon$ that comes from time ordering — the same prescription that defines the scalar and photon propagators (the pole sits at $k^2 = m_A^2 - i\epsilon$, so a particle of mass m_A propagates forward in time) — gives Eq. (30.68). □

A reassuring internal check ties the numerator of Eq. (30.68) to the physical states. The residue of the propagator at the pole $k^2 = m_A^2$ is the on-shell tensor

$$-\Big(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\Big)\Big|_{k^2=m_A^2} = \sum_{\lambda=1}^{3} \epsilon_\mu^{(\lambda)}(k)\,\epsilon_\nu^{(\lambda)*}(k), \tag{30.74}$$

the completeness relation over the *three* physical polarization vectors of a massive spin-one particle (transverse $\lambda = 1, 2$ and longitudinal $\lambda = 3$), each normalized by $\epsilon^{(\lambda)} \cdot \epsilon^{(\lambda')*} = -\delta^{\lambda\lambda'}$ and satisfying the transversality $k \cdot \epsilon^{(\lambda)} = 0$. The propagator factorizes on shell into a sum over exactly the states that the spectrum contains — no more, no fewer. This is the operational meaning of "unitary" gauge: the residue at every physical pole is a sum over physical states with positive norm, so the optical theorem cuts the propagator into honest probabilities. Contrast the photon in a covariant gauge (Section 13.3), whose propagator residue includes timelike and longitudinal pieces that only cancel against ghostly negative-norm states on the physical subspace. Here there is nothing to cancel; the spectrum is positive-definite from the start.

30.3.4 The price: the $k_\mu k_\nu / m_A^2$ term and the loss of manifest renormalizability

Everything so far has been good news. The bad news lives in the same formula. Compare the large-momentum behaviour of the massive-vector propagator Eq. (30.68) with that of a scalar or a gauge-fixed photon. A scalar propagator $i/(k^2 - m^2)$ falls as k^{-2}; the Feynman-gauge photon propagator $-i\eta_{\mu\nu}/k^2$ does the same. But in Eq. (30.68) the second term does not fall at all. As $k \to \infty$,

$$D_{\mu\nu}(k) = \underbrace{\frac{-i\,\eta_{\mu\nu}}{k^2}}_{O(k^{-2})} + \underbrace{\frac{i\,k_\mu k_\nu}{m_A^2\,k^2}}_{O(k^0)} + O(k^{-2}), \tag{30.75}$$

because $k_\mu k_\nu / k^2$ is an $O(1)$ tensor. The longitudinal piece approaches a *constant* (in fact a momentum-direction-dependent constant) divided by m_A^2. The massive vector propagates its high-momentum modes without the damping that makes ordinary field theories tractable.

Caution

The non-decay of Eq. (30.75) is not a sign that the theory is sick. The abelian Higgs model is renormalizable and unitary; we proved tree-level

unitarity of its spectrum above, and its renormalizability will be made manifest in the next section. The non-decay is a sign that unitary gauge is a *bad bookkeeping frame* for these two properties. The danger is purely one of presentation — but in perturbation theory, where one truncates and power-counts, a bad presentation can masquerade as a sick theory, and distinguishing the two is exactly the work of the rest of the chapter.

The damage is sharpest when phrased through power counting. Recall the superficial degree of divergence of a connected diagram from Section 21.2: in four dimensions,

$$D = 4L - \sum_f \omega_f I_f + \sum_i \delta_i V_i, \tag{30.76}$$

where L is the number of loops, I_f the number of internal lines of species f, ω_f the large-momentum fall-off of that species' propagator (so $\omega = 2$ for a propagator $\sim k^{-2}$), V_i the number of vertices of type i, and δ_i the number of derivatives at that vertex. The entire classification of renormalizable theories in Chapter 21 rests on the assumption that every propagator contributes $\omega = 2$: that is what makes D bounded as the diagram grows, so that only finitely many superficial divergence structures occur. For the unitary-gauge massive vector, Eq. (30.75) says the leading behaviour is $\omega_A = 0$, not 2. Each internal vector line then fails to supply its expected two powers of suppression; worse, the explicit $k_\mu k_\nu$ in the numerator feeds momentum back into the vertices it attaches to. Naively inserting $\omega_A = 0$ into Eq. (30.76), the degree of divergence *grows without bound* as one adds internal vector lines and loops, so that arbitrarily high-point functions appear superficially divergent. By the criterion of Chapter 21 the theory would be classified non-renormalizable: infinitely many counterterms, no predictivity.

This conclusion is false, but the power-counting argument that produces it is not wrong — it is merely blind. What it misses is that the dangerous $k_\mu k_\nu / m_A^2$ pieces do not act in isolation. When a longitudinal momentum from one propagator flows into a vertex, gauge invariance arranges that the resulting structure either contracts into a conserved current and dies, or is cancelled by an identical piece from a neighbouring diagram. These cancellations are systematic — they are the content of the Ward–Takahashi identities of the broken theory — but they are spread across different diagrams and are invisible term by term. Power counting, which acts on one propagator at a time, cannot see them. The true degree of divergence, computed after the cancellations, is the renormalizable one; but "after the cancellations" is precisely what unitary gauge refuses to make manifest.

Physical Insight

The abelian Higgs model has two properties one wants from a quantum field theory — *unitarity* (only physical, positive-norm states propagate) and *renormalizability* (a finite number of counterterms suffice) — and no single gauge makes both manifest at once. Unitary gauge makes unitarity manifest: the propagator residue Eq. (30.74) sums over exactly the physical states, with no ghosts. The same choice hides renormalizability behind the non-decaying $k_\mu k_\nu/m_A^2$ term Eq. (30.75). The R_ξ gauges of the next section make the opposite trade: their propagator falls as k^{-2}, so renormalizability is manifest, at the cost of reintroducing the Goldstone χ and a ghost as explicit (unphysical) fields whose decoupling — and hence unitarity — must then be proved. That two complementary gauges exist, agreeing on every physical amplitude, is itself the proof that the theory has *both* properties: each gauge certifies the property the other obscures.

30.3.5 Apparent perturbative-unitarity violation in longitudinal scattering

The hidden cancellations of the previous subsection have an experimentally flavoured avatar at tree level. External longitudinal vectors, like the internal $k_\mu k_\nu/m_A^2$ terms, carry factors of momentum over mass that grow with energy, and the amplitudes they build appear to grow with them — seemingly fast enough to violate the partial-wave unitarity bound of Theorem 16.29. We exhibit the phenomenon in the cleanest available process and then locate the resolution.

The source of the growth is the explicit form of the longitudinal polarization vector. For a massive vector of four-momentum $k^\mu = (E, 0, 0, p)$ with $p = |\mathbf{k}| = \sqrt{E^2 - m_A^2}$, the longitudinal polarization is the unit spacelike vector orthogonal to k in the boost plane,

$$\epsilon_L^\mu(k) = \frac{1}{m_A}\,(p,\,0,\,0,\,E) = \frac{k^\mu}{m_A} - \frac{m_A}{E+p}\,\bar{k}^\mu, \qquad \bar{k}^\mu \equiv (1,0,0,-1), \tag{30.77}$$

as one verifies from $\epsilon_L \cdot k = (pE - Ep)/m_A = 0$ and $\epsilon_L \cdot \epsilon_L = (p^2 - E^2)/m_A^2 = -1$. The decomposition on the right isolates the dangerous part: at high energy $E \gg m_A$ the second term is $O(m_A/E)$ and negligible, while the first term grows,

$$\epsilon_L^\mu(k) \xrightarrow{E \gg m_A} \frac{k^\mu}{m_A} + O\Big(\frac{m_A}{E}\Big), \tag{30.78}$$

so that every external longitudinal leg contributes a factor that scales like E/m_A. An amplitude with several longitudinal legs is then a high power of E/m_A waiting to happen — unless the leading pieces cancel. That they do (in this renormalizable theory) but only after summing diagrams is exactly the

unitary-gauge phenomenon, now visible in an S-matrix element rather than in a propagator.

Worked Example 30.8 (Longitudinal $A_LA_L \to A_LA_L$ scattering and the unitarity puzzle)**.** We compute the high-energy behaviour of elastic longitudinal scattering $A_L(k_1)A_L(k_2) \to A_L(k_3)A_L(k_4)$ in the abelian Higgs model, in unitary gauge, at tree level.

Diagrams. The abelian gauge field has no self-coupling — there is no A^4 or A^3 vertex in Eq. (30.62) — so the only tree diagrams are Higgs exchanges built from the hAA vertex $e^2v\,hA_\mu A^\mu = (m_A^2/v)\,hA_\mu A^\mu$, in the s-, t- and u-channels (Fig. 30.3). The Feynman rule for the $A_\mu A_\nu h$ vertex is $2i(m_A^2/v)\eta_{\mu\nu}$, the factor of 2 from the two ways of attaching the identical vector legs; the Higgs propagator is $i/(q^2 - m_h^2)$.

Amplitudes. Writing $\epsilon_i \equiv \epsilon_L(k_i)$ and using the Mandelstam invariants $s = (k_1+k_2)^2$, $t = (k_1-k_3)^2$, $u = (k_1-k_4)^2$ with $s+t+u = 4m_A^2$, the three channels give

$$\mathcal{M}_s = \Big(\tfrac{2m_A^2}{v}\Big)^2 \frac{(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot\epsilon_4)}{s-m_h^2}\cdot(-1) = \frac{-4m_A^4/v^2}{s-m_h^2}\,(\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot\epsilon_4),$$
$$\mathcal{M}_t = \frac{-4m_A^4/v^2}{t-m_h^2}\,(\epsilon_1\cdot\epsilon_3)(\epsilon_2\cdot\epsilon_4), \qquad \mathcal{M}_u = \frac{-4m_A^4/v^2}{u-m_h^2}\,(\epsilon_1\cdot\epsilon_4)(\epsilon_2\cdot\epsilon_3), \tag{30.79}$$

where the overall (-1) collects the three explicit factors of i (i^2 from the two vertices, $\times\, i$ from the propagator, with $\mathcal{M}$ defined by amplitude $= i\mathcal{M}$). Each amplitude carries two polarization contractions, and by Eq. (30.78) each contraction is the place where the energy growth enters.

Leading high-energy contractions. Substitute $\epsilon_i \to k_i/m_A$. Using $2k_1\cdot k_2 = s - 2m_A^2$, $2k_1\cdot k_3 = 2m_A^2 - t$, and the analogous identities,

$$\epsilon_1\cdot\epsilon_2 \to \frac{k_1\cdot k_2}{m_A^2} \simeq \frac{s}{2m_A^2}, \quad \epsilon_1\cdot\epsilon_3 \to \frac{k_1\cdot k_3}{m_A^2} \simeq \frac{-t}{2m_A^2}, \quad \epsilon_1\cdot\epsilon_4 \to \frac{-u}{2m_A^2}, \tag{30.80}$$

and likewise $\epsilon_3\cdot\epsilon_4 \simeq s/2m_A^2$, $\epsilon_2\cdot\epsilon_4 \simeq -t/2m_A^2$, $\epsilon_2\cdot\epsilon_3 \simeq -u/2m_A^2$, all to leading order in E^2/m_A^2. Insert these into Eq. (30.79) and drop m_h^2 against the large invariants:

$$\mathcal{M}_s \simeq \frac{-4m_A^4/v^2}{s}\cdot\frac{s^2}{4m_A^4} = -\frac{s}{v^2}, \qquad \mathcal{M}_t \simeq -\frac{t}{v^2}, \qquad \mathcal{M}_u \simeq -\frac{u}{v^2}. \tag{30.81}$$

The puzzle. Take the s-channel diagram in isolation. It grows linearly in s, that is, quadratically in energy: $\mathcal{M}_s \simeq -s/v^2 = -e^2 s/m_A^2$, the advertised $O(E^2/m_A^2)$ behaviour. Project onto the s-wave using the partial-wave amplitude of Eq. (16.176), $a_0 = \frac{1}{32\pi}\int_{-1}^{1} \mathrm{d}\cos\theta\,\mathcal{M}$ at high energy (where $p_{\rm cm} \to \sqrt{s}/2$). Since $\mathcal{M}_s$ is independent of the scattering angle,

$$a_0^{(s)} \simeq \frac{1}{32\pi}\cdot 2\cdot\Big(\frac{s}{v^2}\Big) = \frac{s}{16\pi v^2} = \frac{1}{16\pi}\Big(\frac{\sqrt{s}}{v}\Big)^2, \tag{30.82}$$

which crosses the unitarity bound $|a_0| \le 1$ of Theorem 16.29 at $\sqrt{s} = \sqrt{16\pi}\, v = 4\sqrt{\pi}\, v \approx 7\, v$. A truncated, single-channel amplitude marches straight out of the Argand disk at an energy of order v. Taken at face value, perturbative unitarity fails.

The resolution, previewed. The face value is misleading, because the three channels conspire. Adding the leading pieces Eq. (30.81), the dangerous $O(E^2)$ growth cancels outright: at high energy $s \simeq 4E^2$, $t \simeq -2E^2(1-\cos\theta)$, $u \simeq -2E^2(1+\cos\theta)$, so the leading parts obey $s+t+u \to 0$ and

$$\mathcal{M}_s + \mathcal{M}_t + \mathcal{M}_u \;\simeq\; -\frac{s+t+u}{v^2} \;\xrightarrow{O(E^2)}\; 0. \tag{30.83}$$

The threatened quadratic energy growth of each channel cancels identically in the gauge-invariant sum. What survives is a finite *constant*, harmless for unitarity — but its value is *not* obtained by substituting the exact identity $s+t+u = 4m_A^2$ into the leading-order amplitudes Eq. (30.81): those amplitudes have already dropped $O(m^2/v^2)$ corrections of exactly the size of the residual, so the naive $-4m_A^2/v^2$ is unreliable. Extracting the constant honestly requires carrying each channel one order further in m^2/E^2; we do so in full in Section 30.6, with the result $\mathcal{M} \to -3m_h^2/v^2$, fixed by the Higgs self-coupling rather than the gauge mass. The corresponding s-wave is $a_0 \simeq 3m_h^2/(16\pi v^2)$, energy-independent and small, comfortably inside the unitarity disk for all energies (Fig. 30.4). The theory is unitary after all. But notice what the rescue required: a cancellation between three separate diagrams, each individually growing, with no term-by-term hint that the sum should be tame. This is the longitudinal-scattering face of the same hidden structure that obscured renormalizability above. *Why* the cancellation is guaranteed — not an accident of this process but a theorem about every amplitude with external longitudinal vectors — is the content of the Goldstone-boson equivalence theorem, which trades each high-energy longitudinal vector for the eaten Goldstone χ and thereby relates this amplitude to a manifestly well-behaved scalar one. We establish it in Section 30.6.

The pattern of Theorem 30.8 is universal and deserves to be stated as the lesson of the section. Unitary gauge is the frame in which the physical content of a spontaneously broken gauge theory is laid bare — the exact particle spectrum, no unphysical fields, manifest unitarity — and it is therefore indispensable for reading off masses, counting states, and trusting that probabilities flow only among real particles. But it purchases that clarity by hiding two things the theory genuinely possesses: renormalizability, obscured by the non-decaying $k_\mu k_\nu/m_A^2$ in the propagator, and the good high-energy behaviour of amplitudes, obscured by inter-diagram cancellations among individually growing longitudinal contributions. Both pathologies trace to the same root: the absence of a propagator that falls as $1/k^2$.

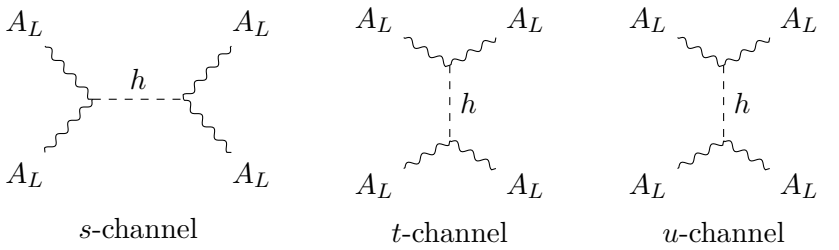


Figure 30.3: The three tree-level diagrams for $A_L A_L \to A_L A_L$ in the abelian Higgs model in unitary gauge. With no abelian gauge self-coupling, the only exchange is the Higgs h, through the hAA vertex of Eq. (30.62). Each diagram grows like E^2/m_A^2 (Eq. (30.81)); the leading growth cancels in the sum (Eq. (30.83)), leaving a finite constant set by m_h^2.

30.3.6 Toward a gauge that keeps renormalizability manifest

The diagnosis points directly at its own cure. What ruins the ultraviolet behaviour in unitary gauge is the longitudinal $k_\mu k_\nu/m_A^2$ term, and that term is present precisely because we eliminated the Goldstone χ by hand: with no field to carry the longitudinal information, the propagator must carry it, and it does so in the worst possible way for power counting. The remedy is to keep χ in the theory as an explicit field and to add a gauge-fixing term tailored to cancel the A–χ mixing that reappears once χ is restored — a term whose by-product is a vector propagator that decays as $1/k^2$, making renormalizability manifest. This is the family of R_ξ gauges, constructed by the Faddeev–Popov procedure with 't Hooft's gauge-fixing functional, to which we turn in Section 30.4. There the price is reversed: renormalizability becomes obvious while unitarity must be re-established by showing that the now-explicit Goldstone and the accompanying ghost decouple from physical amplitudes. That the two gauges agree on every observable — demonstrated in Section 30.5 — is what certifies that the abelian Higgs model possesses both virtues at once.

30.4 R_ξ Gauges: Faddeev–Popov and the 't Hooft Trick

The unitary gauge of Section 30.3 did exactly what it advertised: it removed the would-be Goldstone χ from the spectrum by a field-dependent gauge rotation, leaving a clean physical content of one massive vector A_μ and one massive scalar h. The price, exposed at the end of that section, was the propagator

$$D^U_{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2}\left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right), \tag{30.84}$$

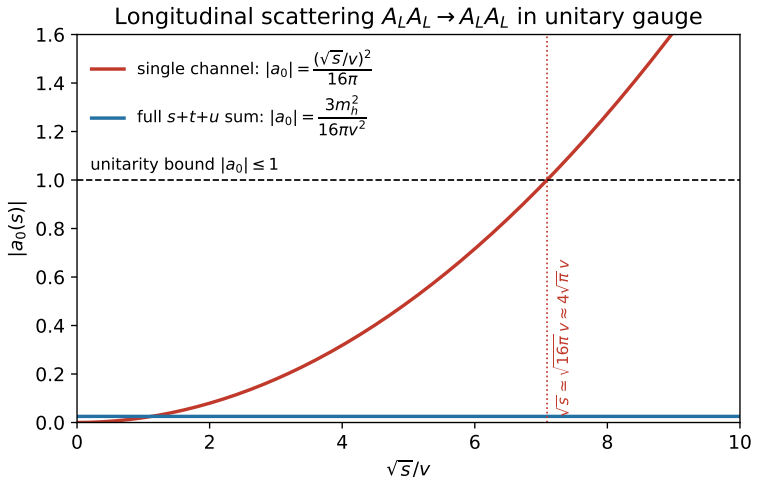


Figure 30.4: The apparent perturbative-unitarity puzzle of unitary gauge. The s-wave partial-wave amplitude $|a_0|$ from a single channel of $A_LA_L \to A_LA_L$ (red) grows as $(\sqrt{s}/v)^2$ and pierces the unitarity bound $|a_0| \le 1$ near $\sqrt{s} \approx 4\sqrt{\pi}\, v$, while the full s+t+u sum (blue) is the constant $3m_h^2/(16\pi v^2)$, far inside the bound at every energy. The good high-energy behaviour is entirely a cancellation among diagrams, invisible term by term — the hallmark of unitary gauge. (Leading high-energy approximation, representative ratio $m_h/v = 0.65$.)

whose $k_\mu k_\nu/m_A^2$ piece does *not* fall off at large k: the propagator tends to a constant, the naive power counting of Chapter 21 reads every loop as more divergent than it truly is, and renormalizability is invisible. We know on general grounds—and will prove gauge-independently in Section 30.5—that the physics is renormalizable, because the bad behavior is pure gauge. The remedy is therefore not to change the physics but to change the gauge: keep the field χ in the Lagrangian, and add a gauge-fixing term engineered so that (i) the troublesome A_μ–χ kinetic mixing is cancelled, and (ii) the gauge propagator falls as $1/k^2$. This is the family of *renormalizable gauges*, the R_ξ gauges, introduced by 't Hooft and Fujikawa–Lee–Sanda. We construct them here by the Faddeev–Popov procedure, carry out every step of the determinant and the inversion, and read off the complete propagator and ghost content.

We work throughout with the abelian Higgs model in the conventions fixed in Section 30.2: a single complex scalar ϕ of charge $+1$ with covariant derivative $D_\mu\phi = (\partial_\mu - ieA_\mu)\phi$, potential $V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4$ with minimum at $|\phi| = v/\sqrt{2}$, $v = \sqrt{\mu^2/\lambda} > 0$, and field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The full Lagrangian is

$$\mathcal{L} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + (D_\mu\phi)^*(D^\mu\phi) - V(\phi). \tag{30.85}$$

30.4.1 The mixing term, recalled and isolated

The whole point of the construction is to defeat one specific term, so we first reproduce it cleanly. Expand around the vacuum in Cartesian fluctuations,

$$\phi(x) = \frac{1}{\sqrt{2}}(v + h(x) + i\,\chi(x)), \tag{30.86}$$

with h the radial (Higgs) fluctuation and χ the would-be Goldstone, the same parametrization used in Section 29.2 for the global model. Then

$$D_\mu\phi = \frac{1}{\sqrt{2}}\Big[\partial_\mu h + eA_\mu\chi + i(\partial_\mu\chi - eA_\mu(v+h))\Big], \tag{30.87}$$

so that, separating the real and imaginary parts,

$$(D_\mu\phi)^*(D^\mu\phi) = \tfrac{1}{2}\big(\partial_\mu h + eA_\mu\chi\big)^2 + \tfrac{1}{2}\big(\partial_\mu\chi - eA_\mu(v+h)\big)^2. \tag{30.88}$$

Collecting the terms quadratic in the fields gives the free Lagrangian plus exactly one cross term between the gauge field and the Goldstone:

$$\mathcal{L}^{(2)} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + \tfrac{1}{2}m_A^2\,A_\mu A^\mu + \tfrac{1}{2}(\partial_\mu h)^2 - \tfrac{1}{2}m_h^2 h^2 + \tfrac{1}{2}(\partial_\mu\chi)^2 \underbrace{-\,m_A\,A^\mu\partial_\mu\chi}_{\text{mixing}}, \tag{30.89}$$

where

$$m_A = ev, \qquad m_h^2 = 2\mu^2 = 2\lambda v^2. \tag{30.90}$$

The gauge mass $m_A^2 = e^2v^2$ comes from the $\frac{1}{2}e^2v^2A_\mu A^\mu$ piece of Eq. (30.88); the would-be Goldstone is massless at this stage, as the global Goldstone theorem (Theorem 29.11) demands, since the potential contributes no χ^2 term. The offending mixing $-m_A A^\mu\partial_\mu\chi$ is the obstruction: it makes the quadratic form non-diagonal in (A_μ, χ), so that neither field has a well-defined propagator until the mixing is removed. Unitary gauge removed it by setting $\chi \equiv 0$; the R_ξ strategy removes it by a gauge-fixing term tailored to cancel it.

Physical Insight

The mixing $-m_A A^\mu\partial_\mu\chi$ is not an accident of the parametrization—it is the field-theoretic signature of the eating itself. Read as a two-point vertex, it says the gauge field can turn into a Goldstone and back; the longitudinal polarization of A_μ and the scalar χ are two descriptions of one degree of freedom. The R_ξ gauge does not make this physics go away; it merely chooses a frame in which the two pieces propagate separately, with the bookkeeping balanced by a ghost.

30.4.2 Faddeev–Popov in the broken theory

We now insert the Faddeev–Popov unity into the path integral, exactly as for the unbroken abelian theory in Section 13.5 and the non-abelian theory in Section 26.2, but with a gauge-fixing functional adapted to the broken phase. The gauge orbit is generated by the $U(1)$ transformation

$$\phi \to e^{i\alpha}\phi, \qquad A_\mu \to A_\mu + \tfrac{1}{e}\,\partial_\mu\alpha, \tag{30.91}$$

which in components, to all orders in the finite parameter α, reads

$$h \to (v+h)\cos\alpha - \chi\sin\alpha - v, \qquad \chi \to (v+h)\sin\alpha + \chi\cos\alpha, \qquad A_\mu \to A_\mu + \tfrac{1}{e}\,\partial_\mu\alpha, \tag{30.92}$$

with infinitesimal form

$$\delta h = -\alpha\,\chi, \qquad \delta\chi = \alpha\,(v+h), \qquad \delta A_\mu = \tfrac{1}{e}\,\partial_\mu\alpha. \tag{30.93}$$

Denote a field configuration collectively by $\Phi = (A_\mu, h, \chi)$ and its gauge transform by Φ^α. The 't Hooft gauge-fixing functional is the linear combination

$$G[\Phi] \;\equiv\; \partial_\mu A^\mu + \xi\, ev\,\chi \;=\; \partial_\mu A^\mu + \xi\, m_A\,\chi, \tag{30.94}$$

with $\xi > 0$ a free real parameter. The single new idea relative to Section 13.5 is the second term $\xi m_A \chi$: it is precisely what will conspire with the mixing in Eq. (30.89). Following the Faddeev–Popov method, we insert into the path integral the identity

$$1 = \int \mathcal{D}\alpha(x)\;\; \delta\big[\, G[\Phi^\alpha] - \omega(x)\,\big]\;\; \det\!\left(\frac{\delta G[\Phi^\alpha]}{\delta\alpha}\right), \tag{30.95}$$

where $\omega(x)$ is an arbitrary c-number function. The determinant is the Jacobian that makes Eq. (30.95) an honest resolution of unity, exactly as the change of variables $G \to \alpha$ requires; its derivation is the abelian special case of Section 26.2 and we do not repeat the general argument, only evaluate the pieces.

The Faddeev–Popov operator. Evaluate G on the transformed configuration using Eq. (30.93). Since the determinant is the response of G to an infinitesimal gauge transformation, we need $G[\Phi^\alpha]$ to first order in α:

$$G[\Phi^\alpha] = \partial_\mu A^\mu + \tfrac{1}{e}\,\Box\,\alpha + \xi m_A\big(\chi + \alpha(v+h)\big) + O(\alpha^2) = G[\Phi] + \Big(\tfrac{1}{e}\,\Box + \xi m_A (v+h)\Big)\alpha + O(\alpha^2). \tag{30.96}$$

Hence the Faddeev–Popov operator—the functional derivative evaluated at $\alpha = 0$, which is where the delta-function in Eq. (30.95) localizes the α-integral—is

$$\left.\frac{\delta G[\Phi^\alpha](x)}{\delta\alpha(y)}\right|_{\alpha=0} = \Big(\tfrac{1}{e}\,\Box_x + \xi m_A\,(v + h(x))\Big)\,\delta^4(x-y). \tag{30.97}$$

Using $m_A = ev$ we have $\xi m_A v = \xi e v^2 = \xi m_A^2/e$, and $\xi m_A h = \xi (m_A^2/v)\, h/$ (rewriting $e = m_A/v$). Factor out the overall $1/e$, which is a field-independent constant and contributes only an (irrelevant, ξ-independent) multiplicative constant to the determinant:

$$\frac{\delta G^\alpha}{\delta \alpha} = \frac{1}{e}\Big(\Box + \xi m_A^2 + \xi e\, m_A\, h\Big)\, \delta^4(x-y). \tag{30.98}$$

Exponentiating the determinant: the ghost Lagrangian. A bosonic determinant in the numerator of the path integral is written, as in Section 26.2, as a Gaussian integral over a pair of anticommuting scalar fields—the Faddeev–Popov ghost c and antighost $\bar{c}$:

$$\det\Big(\tfrac{1}{e}(\Box + \xi m_A^2 + \xi e m_A h)\Big) = \int \mathcal{D}\bar{c}\,\mathcal{D}c\, \exp\Big(i \int \mathrm{d}^4x\, \bar{c}\,\big(-\Box - \xi m_A^2 - \xi e m_A h\big)\, c\Big), \tag{30.99}$$

where the field-independent factor $1/e$ has been absorbed into the (physically irrelevant) normalization of the ghost measure, and the overall sign is fixed by the convention that the ghost kinetic term be $+\bar{c}(-\Box)c$, so that in momentum space ($\Box \to -k^2$) the kinetic operator is $k^2 - \xi m_A^2$. The ghost Lagrangian is therefore

$$\boxed{\mathcal{L}_{\text{ghost}} = \bar{c}\,\big(-\Box - \xi m_A^2\big)\, c \;-\; \xi e\, m_A\, h\, \bar{c}c} \tag{30.100}$$

The first two terms give the ghost a mass

$$m_c^2 = \xi\, m_A^2, \tag{30.101}$$

exactly ξ times the gauge-boson mass-squared—a gauge-parameter-dependent mass, the hallmark of an unphysical field. The third term, $-\xi e m_A\, h\bar{c}c$, is a genuine interaction vertex coupling the ghost to the *physical* Higgs.

Caution

In the *unbroken* abelian theory of Section 13.5 the Faddeev–Popov operator is simply $\frac{1}{e}\Box$, field-independent, so the ghosts are free, massless, and decouple completely—which is why one almost never mentions photon ghosts. The broken phase is different. The $\xi m_A(v+h)$ term in Eq. (30.97) makes the operator field-dependent: the ghost acquires the mass Eq. (30.101) *and* a coupling Eq. (30.100) to the Higgs. Abelian ghosts no longer decouple. They must be kept; their job, as we will see in Section 30.5, is to cancel the unphysical ξ-dependent pole of the Goldstone χ inside loops, just as in the non-abelian BRST analysis of Section 26.5. The whole apparatus is the abelian image of that BRST machinery.

30.4.3 Averaging over ω: the gauge-fixing Lagrangian and the cancellation of the mixing

The function $\omega(x)$ in Eq. (30.95) is arbitrary, so we may average over it with any weight without changing physical correlators (the determinant and the rest of the integrand are independent of ω). Choose the Gaussian weight $\exp(-\frac{i}{2\xi}\int \mathrm{d}^4x\,\omega^2)$ and integrate:

$$\int \mathcal{D}\omega\ \exp\!\left(-\frac{i}{2\xi}\int \mathrm{d}^4x\,\omega^2\right)\,\delta\big[G[\Phi]-\omega\big] = \exp\!\left(-\frac{i}{2\xi}\int \mathrm{d}^4x\;G[\Phi]^2\right). \tag{30.102}$$

The delta-function from Eq. (30.95) sets $\omega = G[\Phi]$ (we have used gauge invariance of the measure and action to drop the α-integral as an overall volume, exactly as in Section 13.5). The upshot is that the gauge-fixed Lagrangian acquires the term

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}\,G[\Phi]^2 = -\frac{1}{2\xi}\big(\partial_\mu A^\mu + \xi m_A \chi\big)^2. \tag{30.103}$$

Expand the square:

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}(\partial_\mu A^\mu)^2 - \underbrace{m_A\,\chi\,\partial_\mu A^\mu}_{\text{cross term}} - \frac{\xi}{2}\,m_A^2\,\chi^2. \tag{30.104}$$

Now examine the cross term against the mixing in Eq. (30.89). Inside the action $\int \mathrm{d}^4x$, integrate the cross term by parts:

$$-\int \mathrm{d}^4x\; m_A\,\chi\,\partial_\mu A^\mu = +\int \mathrm{d}^4x\; m_A\,(\partial_\mu\chi)\,A^\mu = +\int \mathrm{d}^4x\; m_A\,A^\mu\partial_\mu\chi, \tag{30.105}$$

the boundary term vanishing for fields decaying at infinity. This is the exact negative of the mixing term $-m_A A^\mu\partial_\mu\chi$ in Eq. (30.89): the two cancel identically in the action,

$$\int \mathrm{d}^4x\,\Big[\ \underbrace{-m_A A^\mu\partial_\mu\chi}_{\text{from }\mathcal{L}^{(2)}}\ \underbrace{-\,m_A\chi\,\partial_\mu A^\mu}_{\text{from }\mathcal{L}_{\rm gf}}\ \Big] = 0. \tag{30.106}$$

This is the 't Hooft trick in a single line: the coefficient ξm_A in the gauge-fixing functional Eq. (30.94) was chosen for no other reason than to produce, after integration by parts, a cross term that annihilates the gauge–Goldstone mixing. With the mixing gone, the quadratic Lagrangian is finally diagonal in (A_μ, h, χ), and each field has a well-defined propagator. Collecting the gauge-fixed quadratic Lagrangian:

$$\begin{aligned}\mathcal{L}^{(2)}_{\rm gf} &= -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} - \frac{1}{2\xi}(\partial_\mu A^\mu)^2 + \tfrac{1}{2}m_A^2 A_\mu A^\mu \\ &\quad + \tfrac{1}{2}(\partial_\mu h)^2 - \tfrac{1}{2}m_h^2 h^2 + \tfrac{1}{2}(\partial_\mu\chi)^2 - \tfrac{1}{2}\,\xi m_A^2\,\chi^2.\end{aligned} \tag{30.107}$$

The Goldstone has picked up a mass $m_\chi^2 = \xi m_A^2$ from the last term of Eq. (30.104)—exactly equal to the ghost mass Eq. (30.101), and again ξ-dependent and unphysical. The radial Higgs h is untouched: it keeps its physical mass $m_h^2 = 2\lambda v^2$ and is independent of ξ, as a physical field must be.

Definition 30.9 (R_ξ gauge)**.** The *R_ξ (renormalizable) gauge* of the abelian Higgs model is the gauge fixing defined by the 't Hooft functional Eq. (30.94) with Gaussian weight $1/\xi$, producing the gauge-fixing term Eq. (30.103) and the ghost term Eq. (30.100). The real number $\xi \in (0, \infty)$ parametrizes a one-parameter family of gauges in which the would-be Goldstone and the ghost carry the common mass-squared ξm_A^2.

30.4.4 The gauge-field propagator: inverting the quadratic form

We now invert the A_μ quadratic form in Eq. (30.107) to obtain the gauge propagator, carrying every step. Write the gauge part as $\frac{1}{2} A_\mu \mathcal{O}^{\mu\nu} A_\nu$ after integration by parts. Using $-\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = \frac{1}{2} A_\mu (\Box \eta^{\mu\nu} - \partial^\mu \partial^\nu) A_\nu + (\text{total deriv.})$ and $-\frac{1}{2\xi} (\partial_\mu A^\mu)^2 = \frac{1}{2\xi} A_\mu \, \partial^\mu \partial^\nu A_\nu + (\text{total deriv.})$,

$$\mathcal{L}_A^{(2)} = \tfrac{1}{2} A_\mu \Big[(\Box + m_A^2) \eta^{\mu\nu} - \Big(1 - \tfrac{1}{\xi}\Big) \partial^\mu \partial^\nu \Big] A_\nu . \tag{30.108}$$

Pass to momentum space with $\partial_\mu \to -i k_\mu$, so $\Box \to -k^2$ and $\partial^\mu \partial^\nu \to -k^\mu k^\nu$:

$$\mathcal{O}^{\mu\nu}(k) = -(k^2 - m_A^2)\, \eta^{\mu\nu} + \Big(1 - \tfrac{1}{\xi}\Big) k^\mu k^\nu . \tag{30.109}$$

The propagator is $D_{\nu\rho}(k) = i\, [\mathcal{O}^{-1}(k)]_{\nu\rho}$, the standard relation between the quadratic form and the two-point function established in Theorem 4.18. To invert $\mathcal{O}^{\mu\nu}$ we use the transverse and longitudinal projectors,

$$P_T^{\mu\nu} = \eta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}, \qquad P_L^{\mu\nu} = \frac{k^\mu k^\nu}{k^2}, \tag{30.110}$$

which satisfy $P_T^2 = P_T$, $P_L^2 = P_L$, $P_T P_L = 0$, $P_T + P_L = \eta$. Decompose $\mathcal{O}$ on this basis. Writing $\eta^{\mu\nu} = P_T^{\mu\nu} + P_L^{\mu\nu}$ and $k^\mu k^\nu = k^2 P_L^{\mu\nu}$,

$$\begin{aligned} \mathcal{O}^{\mu\nu} &= -(k^2 - m_A^2)(P_T^{\mu\nu} + P_L^{\mu\nu}) + \Big(1 - \tfrac{1}{\xi}\Big) k^2 P_L^{\mu\nu} \\ &= -(k^2 - m_A^2)\, P_T^{\mu\nu} + \Big[-(k^2 - m_A^2) + \Big(1 - \tfrac{1}{\xi}\Big) k^2 \Big] P_L^{\mu\nu} . \end{aligned} \tag{30.111}$$

The longitudinal coefficient simplifies:

$$-(k^2 - m_A^2) + k^2 - \frac{k^2}{\xi} = m_A^2 - \frac{k^2}{\xi} = -\frac{1}{\xi}(k^2 - \xi m_A^2). \tag{30.112}$$

Hence

$$\mathcal{O}^{\mu\nu} = -(k^2 - m_A^2)\, P_T^{\mu\nu} - \frac{1}{\xi}(k^2 - \xi m_A^2)\, P_L^{\mu\nu}. \tag{30.113}$$

Because P_T and P_L are orthogonal idempotents, the inverse is obtained by inverting each coefficient on its own subspace:

$$[\mathcal{O}^{-1}]_{\mu\nu} = -\frac{1}{k^2 - m_A^2}\, P_{T\,\mu\nu} - \frac{\xi}{k^2 - \xi m_A^2}\, P_{L\,\mu\nu}. \tag{30.114}$$

Multiply by i and re-expand the projectors using Eq. (30.110):

$$D_{\mu\nu}(k) = i[\mathcal{O}^{-1}]_{\mu\nu} = \frac{-i}{k^2 - m_A^2}\Big(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\Big) - \frac{i\xi}{k^2 - \xi m_A^2}\frac{k_\mu k_\nu}{k^2}$$
$$= \frac{-i\,\eta_{\mu\nu}}{k^2 - m_A^2} + \frac{i\,k_\mu k_\nu}{k^2}\left[\frac{1}{k^2 - m_A^2} - \frac{\xi}{k^2 - \xi m_A^2}\right]. \tag{30.115}$$

Combine the bracketed terms over a common denominator:

$$\frac{1}{k^2 - m_A^2} - \frac{\xi}{k^2 - \xi m_A^2} = \frac{(k^2 - \xi m_A^2) - \xi(k^2 - m_A^2)}{(k^2 - m_A^2)(k^2 - \xi m_A^2)} = \frac{(1-\xi)\,k^2}{(k^2 - m_A^2)(k^2 - \xi m_A^2)}. \tag{30.116}$$

The numerator's k^2 cancels the explicit $1/k^2$ in Eq. (30.115), and we arrive at the R_ξ gauge propagator:

$$\boxed{D_{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2}\left[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_A^2}\right].} \tag{30.117}$$

This is the central result. It is manifestly free of the kinematic singularity at $k^2 = 0$ that one might fear from the intermediate $1/k^2$: the cancellation in Eq. (30.116) is exact. The propagator has a physical pole at $k^2 = m_A^2$ with residue proportional to $\eta_{\mu\nu}$ (plus a transverse-longitudinal structure) and a second, gauge-dependent pole at $k^2 = \xi m_A^2$ hiding in the $k_\mu k_\nu$ piece.

30.4.5 The Goldstone and ghost propagators

The scalar and ghost propagators are read directly off the diagonal quadratic Lagrangians. From Eq. (30.107), the would-be Goldstone χ has kinetic term $\frac{1}{2}(\partial\chi)^2 - \frac{1}{2}\xi m_A^2\chi^2$, i.e. a free scalar of mass-squared ξm_A^2, so

$$D_\chi(k) = \frac{i}{k^2 - \xi m_A^2}. \tag{30.118}$$

From the ghost Lagrangian Eq. (30.100), the quadratic part $\bar{c}(-\Box - \xi m_A^2)c$ gives the kinetic operator $k^2 - \xi m_A^2$ in momentum space, hence

$$D_c(k) = \frac{i}{k^2 - \xi m_A^2}. \tag{30.119}$$

For completeness the physical Higgs propagator is the ordinary massive scalar one,

$$D_h(k) = \frac{i}{k^2 - m_h^2}, \qquad m_h^2 = 2\lambda v^2, \tag{30.120}$$

manifestly ξ-independent. We record the structure that the next section will exploit: the would-be Goldstone, the ghost, and the longitudinal part of the gauge propagator *all* share the same spurious pole at

$$k^2 = \xi m_A^2, \tag{30.121}$$

with the residues correlated. This common pole is the price of a finite ξ: a pole at a ξ-dependent location cannot correspond to a physical particle, since no observable can depend on the arbitrary number ξ. Its cancellation among the three sectors in any gauge-invariant quantity is precisely the content of Section 30.5.

Proposition 30.10 (Complete R_ξ propagators of the abelian Higgs model)**.** *In the R_ξ gauge of Theorem 30.9, the tree-level propagators are*

$$\begin{aligned}
\textit{gauge field}: \quad & D_{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2}\left[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_A^2}\right], \\
\textit{would-be Goldstone } \chi: \quad & D_\chi(k) = \frac{i}{k^2 - \xi m_A^2}, \\
\textit{Faddeev–Popov ghost } c: \quad & D_c(k) = \frac{i}{k^2 - \xi m_A^2}, \\
\textit{Higgs } h: \quad & D_h(k) = \frac{i}{k^2 - m_h^2},
\end{aligned}$$

with $m_A = ev$, $m_h^2 = 2\lambda v^2$. The gauge, Goldstone, and ghost sectors share the gauge-dependent pole at $k^2 = \xi m_A^2$; only the poles at $k^2 = m_A^2$ and $k^2 = m_h^2$ are physical.

30.4.6 The three gauges and the limits

The family Eq. (30.117) contains three classic gauges as special values of ξ. We extract each, with the limit taken carefully where it is singular.

Worked Example 30.11 (The R_ξ propagators and their ξ-limits)**.** We verify the gauge limits of Theorem 30.10, taking the gauge propagator first and then the scalar/ghost sector.

(a) Feynman–'t Hooft gauge, $\xi = 1$. Setting $\xi = 1$ in Eq. (30.117) kills the $(1 - \xi)$ term outright:

$$D_{\mu\nu}^{\xi=1}(k) = \frac{-i\,\eta_{\mu\nu}}{k^2 - m_A^2}. \tag{30.122}$$

The propagator collapses to a pure $\eta_{\mu\nu}$ form, exactly as the massless photon propagator does in Feynman gauge, but with the mass m_A in the denominator. This is the workhorse gauge for explicit loop computation: every Lorentz index

contracts trivially. The Goldstone and ghost both have mass $m_\chi^2 = m_c^2 = m_A^2$ from Eqs. (30.118) and (30.119): in this gauge the spurious pole sits exactly at the physical mass, and all three would-be-Goldstone, ghost, and gauge poles coincide at $k^2 = m_A^2$ (though only the transverse gauge pole is physical).

(b) Landau gauge, $\xi \to 0$. Take $\xi \to 0$ in Eq. (30.117). The bracket becomes $\eta_{\mu\nu} - k_\mu k_\nu/k^2$, the transverse projector:

$$D_{\mu\nu}^{\xi\to 0}(k) = \frac{-i}{k^2 - m_A^2}\left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{k^2}\right). \tag{30.123}$$

The gauge propagator is exactly transverse: $k^\mu D_{\mu\nu}^{\xi\to 0} = 0$, so the gauge field never propagates a longitudinal mode. This is the cleanest gauge for studying transversality and for proofs of the Ward identities, since the $k_\mu k_\nu$ contamination is absent. The would-be Goldstone and the ghost become *massless*, $m_\chi^2 = m_c^2 = \xi m_A^2 \to 0$, recovering the global Goldstone of Section 29.2: in Landau gauge the would-be Goldstone behaves like a genuine massless scalar, and its spurious pole sits at $k^2 = 0$.

(c) Unitary gauge, $\xi \to \infty$. Here the limit must be taken on the full structure, not term by term, because the $(1-\xi)$ prefactor and the $k^2 - \xi m_A^2$ denominator both diverge. Factor ξ out of the denominator of the $k_\mu k_\nu$ term:

$$(1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_A^2} = \frac{(1-\xi)}{\xi}\cdot\frac{k_\mu k_\nu}{(k^2/\xi) - m_A^2} \xrightarrow{\xi\to\infty} (-1)\cdot\frac{k_\mu k_\nu}{-m_A^2} = \frac{k_\mu k_\nu}{m_A^2}, \tag{30.124}$$

using $(1-\xi)/\xi \to -1$ and $k^2/\xi \to 0$. Substituting back,

$$D_{\mu\nu}^{\xi\to\infty}(k) = \frac{-i}{k^2 - m_A^2}\left(\eta_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right), \tag{30.125}$$

which is precisely the unitary-gauge propagator Eq. (30.84) of Section 30.3. Simultaneously the Goldstone and ghost masses $m_\chi^2 = m_c^2 = \xi m_A^2 \to \infty$ diverge: their propagators Eqs. (30.118) and (30.119) vanish as $1/\xi \to 0$ at any fixed k, so χ and c become infinitely heavy and decouple from the spectrum entirely. This is the field-theoretic statement that unitary gauge is the $\xi \to \infty$ corner of the R_ξ family: the would-be Goldstone is pushed to infinite mass and disappears, the ghost goes with it, and only the physical massive vector and the Higgs remain. We have recovered the unitary picture as a limit, confirming that the two constructions describe the same theory.

The three limits are summarized in Table 30.2, and the motion of the spurious pole as ξ runs is plotted in Fig. 30.5.

30.4.7 Manifest renormalizability

The construction has bought exactly what the unitary gauge could not deliver: a gauge propagator that falls off at large momentum. Examine the ultraviolet

Gauge	ξ	gauge propagator structure	$m_\chi^2 = m_c^2$	feature
Landau	0	transverse $(\eta_{\mu\nu} - k_\mu k_\nu/k^2)$	0	massless Goldstone
Feynman–'t Hooft	1	pure $\eta_{\mu\nu}$	m_A^2	simplest for loops
unitary	∞	$\eta_{\mu\nu} - k_\mu k_\nu/m_A^2$	∞	no χ, no ghost

Table 30.2: The three named gauges as special values of ξ in the R_ξ family. In every case the physical pole sits at $k^2 = m_A^2$; the would-be Goldstone and ghost share the gauge-dependent mass-squared ξm_A^2.

behavior of Eq. (30.117) for any *finite* ξ. As $k \to \infty$ with $k^2 \gg m_A^2$,

$$D_{\mu\nu}(k) \;\sim\; \frac{-i}{k^2}\left[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2}\right] \;=\; O\!\left(\frac{1}{k^2}\right), \tag{30.126}$$

because both the $\eta_{\mu\nu}$ term and the $k_\mu k_\nu/(k^2 - \xi m_A^2) \sim k_\mu k_\nu/k^2$ term scale as $1/k^2$ (the latter has two powers of k in the numerator and four in the denominator). This is the same $1/k^2$ falloff as a massless gauge or scalar propagator. Contrast the unitary propagator Eq. (30.84), whose $k_\mu k_\nu/m_A^2$ term has *no* compensating power of k^2 in the denominator and therefore tends to the constant $-i\,k_\mu k_\nu/(m_A^2 k^2) \to$ nonzero direction-dependent limit—ruining the power counting.

The consequence is that the superficial degree of divergence of any diagram, computed by the power-counting rules of Chapter 21, takes its standard renormalizable value: each internal gauge line contributes -2 to the degree of divergence, exactly as a scalar line does, so the theory has the same divergence structure as a renormalizable scalar theory. The full proof that the abelian Higgs model is renormalizable—that a finite set of local counterterms, consistent with the gauge (BRST/Slavnov–Taylor) symmetry, absorbs all divergences order by order—follows the structure-of-proof developed in Chapter 22 for the systematics of renormalization and in Section 26.5 for the role of BRST invariance in constraining the counterterms; the R_ξ propagators Theorem 30.10 are precisely the ingredient that makes that machinery applicable here. The price 't Hooft paid for manifest renormalizability is the appearance of the unphysical particles χ and c with ξ-dependent masses; the price unitary gauge paid for a manifestly physical spectrum was a non-renormalizable-looking propagator. They are two faces of one theory, related by the limit Eq. (30.125).

Physical Insight

The lesson is general and worth stating plainly: *manifest unitarity and manifest renormalizability are, in a gauge theory, properties of different gauges.* Unitary gauge makes the physical particle content obvious but hides renormalizability; R_ξ gauge with finite ξ makes renormalizability obvious (every propagator $\sim 1/k^2$) but populates the theory with unphys-

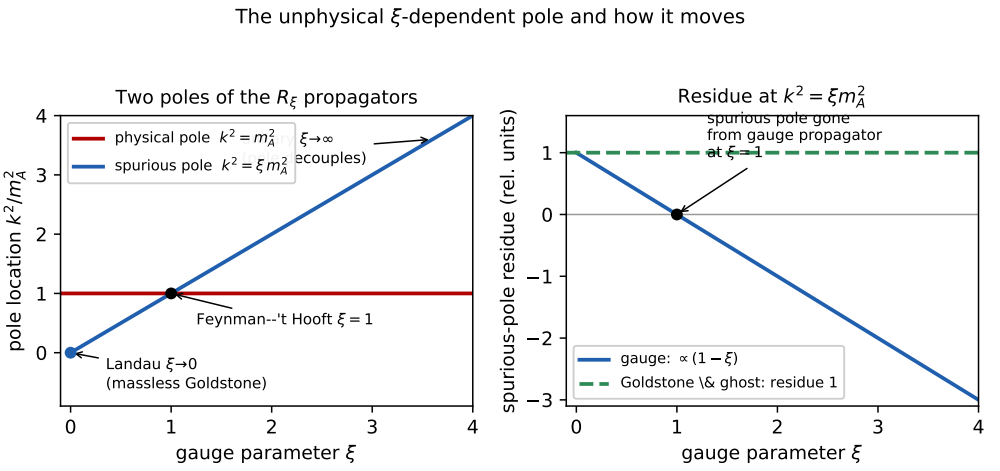


Figure 30.5: Left: the two poles of the R_ξ propagators in units of m_A^2, as functions of the gauge parameter ξ. The physical pole at $k^2 = m_A^2$ is ξ-independent (horizontal line); the spurious pole at $k^2 = \xi m_A^2$ moves along the diagonal, reaching $k^2 = 0$ in Landau gauge ($\xi \to 0$), coinciding with the physical pole at Feynman–'t Hooft ($\xi = 1$), and running to infinity in unitary gauge ($\xi \to \infty$). Right: the residue of the spurious pole in the gauge propagator is proportional to $(1 - \xi)$ and vanishes at $\xi = 1$, while in the Goldstone and ghost propagators it persists with unit residue for all ξ—which is why those two sectors must conspire to cancel it in physical amplitudes.

ical ghost and Goldstone poles. Because ξ is a free parameter that cannot enter any observable, both descriptions must give identical physics—which is at once a powerful consistency check (compute in Feynman–'t Hooft gauge, verify ξ-independence) and the subject of the next section.

With the full R_ξ apparatus in hand—propagators, ghost, and the explicit appearance of the gauge-dependent pole at $k^2 = \xi m_A^2$ in three correlated sectors—we have a family of gauges that is renormalizable for every finite ξ and reduces to the physical unitary description as $\xi \to \infty$. What remains is to prove that the arbitrary parameter ξ, and with it the spurious poles at $k^2 = \xi m_A^2$ and the entire ghost sector, leaves no trace in any physical quantity. That cancellation—in the S-matrix and in other observables, enforced by the BRST/Slavnov–Taylor identities of Section 26.5—is the business of Section 30.5.

30.5 Gauge Independence of Physical Quantities

The R_ξ gauges of the previous section bought renormalizability at a disquieting price. The gauge propagator Eq. (30.117) carries two poles: the physical one at $k^2 = m_A^2$, with $m_A = ev$ the mass the gauge boson acquired by eating the Goldstone (Eq. (30.31)), and a second, manifestly ξ-dependent pole at $k^2 = \xi m_A^2$. The would-be Goldstone χ and the Faddeev–Popov ghosts $c, \bar{c}$ share that same ξ-dependent "mass" $m_\chi^2 = m_c^2 = \xi m_A^2$ (Eqs. (30.118) to (30.119)). A pole whose location depends on a parameter we inserted by hand — ξ was nothing but the weight of the Gaussian smearing in the 't Hooft trick — cannot correspond to a particle: no experiment knows the number ξ. Either these poles are spurious, cancelling out of everything measurable, or the R_ξ construction is inconsistent. This section proves the former. We show that S-matrix elements, and physical observables generally, are exactly independent of ξ; the ξ-dependent masses are genuinely unphysical, the ξ-dependent poles cancel among A_μ, χ, and the ghosts in every physical amplitude, and one is therefore free to compute in whichever gauge is most convenient — Feynman–'t Hooft $\xi = 1$ for clean propagators, Landau $\xi = 0$ for a transverse photon, unitary $\xi \to \infty$ for a manifestly particle spectrum.

The logic mirrors, one rung higher, the gauge independence we already established for the *unbroken* photon. There (Section 13.4.4) the ξ-dependent longitudinal part of the photon propagator dropped out of S-matrix elements because it attached to a conserved current, $k_\mu J^\mu = 0$, by the Ward identity. In the broken theory current conservation is deformed — the gauge current acquires a divergence proportional to the would-be Goldstone — so the longitudinal piece no longer vanishes on its own. What rescues gauge independence is that the deficit is made up, term by term, by the exchange of the Goldstone and (in loops) the ghosts. The bookkeeping that guarantees this is BRST symmetry, which survives the spontaneous breaking intact. We develop the argument in four movements: first the BRST invariance of the R_ξ-fixed broken Lagrangian and the quartet of unphysical fields it organizes; then the characterization of the physical Hilbert space as BRST cohomology, in which the quartet decouples; then the explicit cancellation of the $k^2 = \xi m_A^2$ pole; then the Nielsen identities, which state the ξ-independence of pole masses to all orders. A complete two-gauge worked computation closes the section.

30.5.1 BRST invariance of the gauge-fixed broken theory

The Faddeev–Popov procedure that produced the R_ξ Lagrangian is, as in the unbroken case (Section 26.5), the source of a rigid global fermionic symmetry: BRST. The breaking of the gauge symmetry by the vacuum does not spoil it. We exhibit it directly for the abelian Higgs model.

Recall the field content after expanding the complex scalar about its vacuum,

$\phi = \frac{1}{\sqrt{2}}(v + h + i\chi)$ with $D_\mu = \partial_\mu - ieA_\mu$ acting on charge $+1$ (Eq. (30.21)): the massive gauge field A_μ, the physical Higgs h, the would-be Goldstone χ, the anticommuting ghost c and antighost $\bar{c}$, and the Nakanishi–Lautrup auxiliary field B introduced to write the gauge fixing linearly. The complete gauge-fixed Lagrangian is

$$\mathcal{L} = \mathcal{L}_{\text{inv}} + \mathcal{L}_{\text{gf+gh}}, \qquad \mathcal{L}_{\text{inv}} = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + |D_\mu\phi|^2 - V(\phi), \tag{30.127}$$

with $\mathcal{L}_{\text{inv}}$ the gauge-invariant part and $\mathcal{L}_{\text{gf+gh}}$ the 't Hooft gauge fixing plus ghosts from Section 30.4. Introduce the nilpotent BRST operator s, a Grassmann-odd graded derivation, acting on the fields by

$$\begin{aligned} sA_\mu &= \partial_\mu c, & sh &= -\,e\,c\,\chi, & s\chi &= e\,c\,(v+h) = m_A c + e\,c\,h, \\ s\psi &= i\,e\,Q\,c\,\psi, & sc &= 0, & s\bar{c} &= B, \\ & & sB &= 0, & & \end{aligned} \tag{30.128}$$

where ψ denotes any matter field of charge Q (we keep one for the worked example below). On the gauge and matter fields s is precisely an infinitesimal gauge transformation with the local parameter replaced by the ghost, $\alpha(x) \to c(x)$; this is what makes $\mathcal{L}_{\text{inv}}$ BRST invariant, $s\mathcal{L}_{\text{inv}} = 0$, by gauge invariance. The defining property of s is nilpotency,

$$s^2 = 0, \tag{30.129}$$

which we check on the fields that probe every structure in Eq. (30.128). On A_μ, $s^2A_\mu = s\,\partial_\mu c = \partial_\mu(sc) = 0$. On the scalar, treating s as an odd derivation that anticommutes through the ghost,

$$s^2\chi = s(m_A c + e\,c\,h) = e\,(sc)\,h - e\,c\,(sh) = 0 - e\,c\,(-e\,c\,\chi) = e^2c^2\chi = 0, \tag{30.130}$$

because $c^2 = 0$ for a single anticommuting field; the same cancellation gives $s^2h = 0$. On the matter field, $s^2\psi = ieQ(sc)\psi - ieQc(s\psi) = 0 - ieQc(ieQc\psi) = e^2Q^2c^2\psi = 0$. Finally $s^2\bar{c} = sB = 0$ and $s^2c = s^2B = 0$ trivially. Thus Eq. (30.129) holds on all fields, and since s is a derivation it holds on any local polynomial of them.

Nilpotency is the engine of the whole argument, and it lets us write the entire gauge-fixing-plus-ghost sector as a BRST variation. Define the *gauge fermion*, a Grassmann-odd local functional of ghost number -1,

$$\Psi = \bar{c}\left(\tfrac{\xi}{2}B - \partial_\mu A^\mu - \xi m_A \chi\right). \tag{30.131}$$

Acting with s and using $s\bar{c} = B$, $sB = 0$, $s(\partial_\mu A^\mu) = \partial^2 c$ and $s\chi = m_A c + ec\,h$,

$$s\Psi = B\left(\tfrac{\xi}{2}B - \partial_\mu A^\mu - \xi m_A\chi\right) + \bar{c}\left(\partial^2 c + \xi m_A(m_A c + ec\,h)\right)$$

$$= \tfrac{\xi}{2} B^2 - B(\partial_\mu A^\mu + \xi m_A \chi) + \bar{c}\,\partial^2 c + \xi m_A^2\,\bar{c}c + \xi e m_A\,\bar{c}\,h\,c. \tag{30.132}$$

The auxiliary field B enters only algebraically; its Euler–Lagrange equation from Eq. (30.132) is $\xi B = \partial_\mu A^\mu + \xi m_A \chi$, and substituting it back gives

$$\mathcal{L}_{\text{gf+gh}} = s\Psi \xrightarrow{B \text{ integrated out}} -\frac{1}{2\xi}(\partial_\mu A^\mu + \xi m_A \chi)^2 - \partial_\mu \bar{c}\,\partial^\mu c - \xi m_A^2\,\bar{c}c - \xi e m_A\,\bar{c}\,h\,c, \tag{30.133}$$

exactly the 't Hooft gauge-fixing term and ghost Lagrangian of Section 30.4, with the ghost mass $m_c^2 = \xi m_A^2$ and the ghost–Higgs coupling displayed. (The ghost kinetic term has been integrated by parts; the propagator is $i/(k^2 - \xi m_A^2 + i\varepsilon)$.) Two facts are now manifest. First, $\mathcal{L}_{\text{gf+gh}}$ is BRST-*exact*: it is s of something. Second, by nilpotency $s\mathcal{L}_{\text{gf+gh}} = s^2\Psi = 0$. Together with $s\mathcal{L}_{\text{inv}} = 0$ this gives the central statement:

$$s\mathcal{L} = 0. \tag{30.134}$$

The R_ξ-gauge-fixed broken Lagrangian is BRST invariant. Spontaneous symmetry breaking changed the field content — it gave A_μ, χ, and the ghosts their ξ-dependent masses through the $s\chi = m_A c + \dots$ term — but it left the symmetry s untouched, because s acts on χ exactly as a gauge rotation does and the vacuum value v enters only through the inhomogeneous shift $s\chi \supset m_A c$.

By Noether's theorem applied to the rigid symmetry Eq. (30.134) there is a conserved BRST current and a nilpotent, Hermitian BRST charge Q_B, $Q_B^2 = 0$, generating s on the fields, $s\Phi = i\,[Q_B, \Phi]_\pm$ (graded). Its construction and conservation are exactly as in the unbroken theory of Section 26.5; the broken vacuum changes none of that derivation. It is Q_B, not the broken gauge charge, that organizes the physical state space.

Notation Watch

The auxiliary field B and the ghost mass make the broken-theory BRST multiplet look heavier than its massless counterpart in Chapter 26, but the algebra $s^2 = 0$ and the exactness $\mathcal{L}_{\text{gf+gh}} = s\Psi$ are identical in form. The only structural novelty is the inhomogeneous term $s\chi \supset m_A c$: the would-be Goldstone shifts by a ghost under BRST. This single term is responsible for every cancellation in this section.

30.5.2 The physical Hilbert space as BRST cohomology, and the quartet

The field A_μ propagates four polarizations, only three of which a massive spin-one particle should have; χ, c, and $\bar{c}$ propagate further unphysical modes of indefinite or zero norm. As with the covariant photon (Section 13.3), the resolution is not to remove these modes from the Fock space but to project onto a physical

subspace and quotient by null states. The clean, all-orders statement of that projection is BRST cohomology, established in Section 26.5; we import it and read off its consequences for the broken theory.

Theorem 30.12 (Physical states as BRST cohomology)**.** *Let $\mathcal{V}$ be the indefinite-metric Fock space of the R_ξ-quantized theory and Q_B the nilpotent BRST charge. Physical states are the BRST-closed states of ghost number zero, $Q_B|\psi\rangle = 0$, and two such states are identified if they differ by a BRST-exact state, $|\psi\rangle \sim |\psi\rangle + Q_B|\eta\rangle$. The physical Hilbert space is the cohomology*

$$\mathcal{H}_{\text{phys}} = \left.\frac{\ker Q_B}{\operatorname{im} Q_B}\right|_{\textit{ghost number } 0}, \tag{30.135}$$

on which the inner product inherited from $\mathcal{V}$ is positive semidefinite, becoming positive definite on the quotient. The S-matrix restricted to $\mathcal{H}_{\text{phys}}$ is unitary.

This is Theorem 26.26, whose proof (the Kugo–Ojima quartet mechanism) we do not repeat. What we must do is identify the quartet in the *broken* theory, because that is where the ξ-dependent poles live, and verify that it decouples.

A BRST quartet is a set of four states organized into two pairs by Q_B: a pair of zero-norm "parent/daughter" states $(|\pi\rangle, |d\rangle)$ with $Q_B|\pi\rangle = |d\rangle$, $Q_B|d\rangle = 0$, and a conjugate pair from the antighost sector. The mechanism of Section 26.5 shows that the members of a quartet enter physical inner products only in mutually cancelling combinations, so they never contribute to $\mathcal{H}_{\text{phys}}$. In the unbroken theory the quartet is built from the timelike and longitudinal gauge polarizations together with c and $\bar{c}$. In the broken theory the longitudinal gauge mode mixes with the would-be Goldstone, and the quartet is the set

$$\boxed{\{\,\chi,\ c,\ \bar{c},\ B\,\}} \qquad \text{all with mass}^2 = \xi m_A^2, \tag{30.136}$$

together with the timelike/longitudinal combination of A_μ at the same pole. The pairing is read directly off the free BRST transformations, the part of Eq. (30.128) linear in fields:

$$s\chi\big|_{\text{free}} = m_A c, \qquad sc = 0; \qquad s\bar{c} = B, \qquad sB = 0. \tag{30.137}$$

Thus (χ, c) form a parent–daughter doublet: the would-be Goldstone is the BRST parent of the ghost, $Q_B|\chi\rangle \propto |c\rangle$, while $Q_B|c\rangle = 0$. Likewise $(\bar{c}, B)$ form the conjugate doublet, $Q_B|\bar{c}\rangle \propto |B\rangle$. These four fields all carry the same mass ξm_A^2 precisely because the gauge fixing tied them together through the single exact term $s\Psi$. They are the four members of one quartet, and by the quartet mechanism of Section 26.5 they contribute nothing to physical matrix elements: a physical external state is BRST-closed and of ghost number zero, so it has no overlap with the ghost-number ± 1 daughters, and the ghost-number-zero members (χ and the longitudinal A) appear only in zero-norm combinations that are BRST-exact and hence null in the cohomology.

Physical Insight

The would-be Goldstone is not a particle. In the spectrum of Eq. (30.136) it sits at the same unphysical mass as the ghosts and is glued to them by BRST. The single physical degree of freedom that "was" the Goldstone has migrated into A_μ as its longitudinal polarization, leaving behind only a quartet partner that cancels. This is the precise sense in which the Goldstone was *eaten*: it survives in the Lagrangian, with a propagator and a ξ-dependent mass, but never in $\mathcal{H}_{\text{phys}}$. The ξ-dependence of its mass is then no embarrassment — ξ may decorate anything that lives outside the physical Hilbert space.

The decoupling statement of Theorem 30.12 is an *exact*, all-orders fact about the S-matrix between physical states. It already implies gauge independence in principle: a change of ξ is a change in Ψ, hence a BRST-exact change of the Lagrangian, $\delta\mathcal{L} = s(\delta\Psi)$, which cannot affect matrix elements between BRST-closed states. We make this principled statement concrete in the next two subsections, first by exhibiting the pole cancellation diagrammatically, then by stating the Nielsen identities that govern it order by order.

30.5.3 Explicit cancellation of the unphysical pole

The cleanest way to see that the $k^2 = \xi m_A^2$ pole leaves no trace in observables is to track it through the gauge propagator's longitudinal part and watch it annihilate against the Goldstone propagator. We work at tree level, where ghosts do not yet contribute; the ghost cancellation is the loop-level continuation of the same identity.

Begin with the R_ξ gauge propagator Eq. (30.117),

$$D_{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2 + i\varepsilon}\left[g_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_A^2 + i\varepsilon}\right]. \tag{30.138}$$

The $g_{\mu\nu}$ term has only the physical pole at $k^2 = m_A^2$. The $k_\mu k_\nu$ term has, by partial fractions, a piece at each pole. It is worth exhibiting the decomposition once, because it isolates the unphysical residue:

$$-(1-\xi)\frac{k_\mu k_\nu}{(k^2 - m_A^2)(k^2 - \xi m_A^2)} = \frac{k_\mu k_\nu}{m_A^2}\left[\frac{1}{k^2 - m_A^2} - \frac{1}{k^2 - \xi m_A^2}\right], \tag{30.139}$$

which one verifies by combining the right-hand side over a common denominator: the numerator is $(k_\mu k_\nu/m_A^2)[(k^2 - \xi m_A^2) - (k^2 - m_A^2)] = (k_\mu k_\nu/m_A^2)(1-\xi)m_A^2 = (1-\xi)k_\mu k_\nu$, with the overall sign matching once the denominators are restored.

Hence the full propagator is

$$D_{\mu\nu}(k) = \underbrace{\frac{-i}{k^2 - m_A^2 + i\varepsilon}\left(g_{\mu\nu} - \frac{k_\mu k_\nu}{m_A^2}\right)}_{\text{unitary-gauge part, } \xi\text{-independent}} + \underbrace{\frac{-i}{k^2 - \xi m_A^2 + i\varepsilon}\,\frac{k_\mu k_\nu}{m_A^2}}_{\text{unphysical pole}}. \tag{30.140}$$

The first term is exactly the propagator of a massive vector in unitary gauge (Section 30.3): three physical polarizations, $\sum_\lambda \epsilon_\mu^\lambda \epsilon_\nu^{\lambda *} = -(g_{\mu\nu} - k_\mu k_\nu/m_A^2)$, no trace of ξ. *All* of the ξ-dependence of the gauge sector is concentrated in the second term, a pure $k_\mu k_\nu$ structure with its pole at $k^2 = \xi m_A^2$. Gauge independence is the statement that this second term cancels, in any physical amplitude, against the Goldstone exchange whose propagator $i/(k^2 - \xi m_A^2)$ has its pole at the same place.

Why does it cancel rather than merely coincide? Because the factor of k_μ contracts the gauge boson into a current divergence, and in the broken theory the gauge current's divergence *is* the Goldstone coupling. This is the deformed Ward identity. The free BRST transformation $s\chi = m_A c$ is its fingerprint: it ties the longitudinal gauge mode to χ. Concretely, when the gauge boson with momentum k couples to an external line through a vertex $i\Gamma^\mu$, the contraction $k_\mu \Gamma^\mu$ does not vanish (as it would for a strictly conserved current) but equals, by the Slavnov–Taylor identity Eq. (26.161), exactly m_A times the corresponding Goldstone vertex. The two diagrams that share the $k^2 = \xi m_A^2$ pole — gauge exchange through the $k_\mu k_\nu$ structure, and Goldstone exchange — then carry equal and opposite residues at that pole.

We prove this at tree level in the worked example below, where the algebra is fully visible. The result there, Eq. (30.141), is the master cancellation

$$\Big[\text{gauge } k_\mu k_\nu \text{ exchange}\Big] + \Big[\text{Goldstone exchange}\Big] = \frac{(k\cdot J_1)(k\cdot J_2)/m_A^2}{k^2 - m_A^2}, \tag{30.141}$$

with the ξ-dependent pole gone and the surviving structure precisely the $k_\mu k_\nu/m_A^2$ piece of the unitary-gauge propagator Eq. (30.140). At one loop the same mechanism removes the ξm_A^2 pole from gauge-boson self-energy insertions, with the ghost loop supplying the missing piece: the ghost, being the BRST daughter of χ, enters exactly where the Goldstone tadpole/loop would otherwise leave an uncancelled ξ-dependence. The all-orders bookkeeping is the content of the Slavnov–Taylor identities Eq. (26.161), and its consequence for pole positions is the Nielsen identity, to which we now turn.

30.5.4 Nielsen identities: pole masses are ξ-independent

The diagrammatic cancellation shows the unphysical pole disappearing from amplitudes. We also want the converse safeguard: that the *physical* pole, at $k^2 = m_A^2$, does not secretly drift with ξ, and likewise that the residue defining

the S-matrix normalization through LSZ (Section 14.5) is ξ-independent. These are the Nielsen identities. They are exact, all-orders statements, and they follow from the same BRST structure: ξ enters the action only through the BRST-exact gauge-fixing term, so $\partial/\partial\xi$ of any Green's function is the insertion of a BRST-exact operator.

Concretely, differentiating the generating functional with respect to ξ brings down $\partial\mathcal{L}/\partial\xi = \partial(s\Psi)/\partial\xi = s(\partial\Psi/\partial\xi)$, a BRST variation. Carrying this through the Slavnov–Taylor machinery (Eq. (26.161)), with the BRST sources that make the nonlinear transformations Eq. (30.128) renormalizable, yields an identity of the schematic form: the ξ-derivative of any one-particle-irreducible two-point function is proportional to that same two-point function times a (generally ξ- and momentum-dependent) coefficient function. For the gauge-boson inverse propagator this reads

$$\xi\,\frac{\partial}{\partial\xi}\,\Gamma_{AA}(k^2,\xi) = -\,\mathcal{C}(k^2,\xi)\,\Gamma_{AA}(k^2,\xi), \tag{30.142}$$

where Γ_{AA} is the full (transverse) inverse propagator and $\mathcal{C}$ is a function built from BRST-exact insertions, finite and free of the pole of Γ_{AA}. We state this as the governing result of the subsection.

Theorem 30.13 (Nielsen identity for the gauge-boson pole)**.** *Let $k^2 = M_A^2$ be a zero of the full inverse propagator, $\Gamma_{AA}(M_A^2,\xi) = 0$ — the physical pole mass. Then, provided $\mathcal{C}(M_A^2,\xi)$ is regular there,*

$$\frac{\partial M_A^2}{\partial\xi} = 0 : \tag{30.143}$$

the physical pole mass is independent of the gauge parameter. The residue of the propagator at the physical pole, and hence the LSZ amputation factor, is likewise ξ-independent up to a BRST-exact field redefinition that does not affect S-matrix elements.

Proof. Evaluate Eq. (30.142) on the pole. Differentiating the defining relation $\Gamma_{AA}(M_A^2(\xi),\xi) = 0$ totally in ξ,

$$\left.\frac{\partial\Gamma_{AA}}{\partial k^2}\right|_{M_A^2}\frac{\partial M_A^2}{\partial\xi} + \left.\frac{\partial\Gamma_{AA}}{\partial\xi}\right|_{M_A^2} = 0. \tag{30.144}$$

The Nielsen identity Eq. (30.142) gives $\partial\Gamma_{AA}/\partial\xi\big|_{M_A^2} = -(\mathcal{C}/\xi)\,\Gamma_{AA}\big|_{M_A^2} = 0$, since Γ_{AA} vanishes at its zero and $\mathcal{C}$ is regular there. Hence the second term of Eq. (30.144) vanishes. The first factor $\partial\Gamma_{AA}/\partial k^2\big|_{M_A^2}$ is nonzero (it is the inverse residue, Z^{-1}, of a simple pole), so $\partial M_A^2/\partial\xi = 0$. □

The same identity, applied to the effective potential rather than the two-point function, gives the original Nielsen result that the value of the effective potential at its extremum — the vacuum energy, and with it the physical particle masses read off the curvature there — is ξ-independent, even though the effective potential at a generic field value is not. The location of the minimum may slide with ξ (it is not an observable), but the depth of the minimum, and the spectrum, do not. This is the quantum, all-orders completion of the tree-level statement that $m_A = ev$ is physical while $m_\chi^2 = \xi m_A^2$ is not.

Caution

The Nielsen identities do *not* say that every quantity is ξ-independent. Off-shell Green's functions, the effective potential away from its minimum, the would-be Goldstone and ghost masses, the position of the unphysical pole — all depend on ξ. What is ξ-independent is precisely the gauge-invariant content: S-matrix elements, pole masses, and the residues that normalize them. A common error is to quote a one-loop "Higgs mass" or "W mass" computed from the effective potential at a ξ-dependent stationary field value and to puzzle over its gauge dependence; the cure is to use the pole mass, protected by Theorem 30.13.

30.5.5 A two-gauge worked check

We now carry out the cancellation Eq. (30.141) in full and confirm ξ-independence by computing one tree amplitude in two gauges. The process is chosen so that the unphysical pole genuinely appears in individual diagrams — it must, or the check is empty — which requires the gauge boson to couple to a current whose divergence does not vanish. In the abelian Higgs model that happens whenever the matter coupling is chiral, exactly as it is in the electroweak theory the next chapter builds; this also makes the example a faithful rehearsal for it.

Worked Example 30.14 (ξ-independence of $f\bar{f} \to \mu\bar{\mu}$ via s-channel exchange)**.** Take two distinct Dirac fermions f and μ, each coupled chirally to the $U(1)$: the left-handed component carries one more unit of charge than the right-handed (the charge of the Higgs), so the gauge coupling is $A_\mu\, \bar{\psi}\gamma^\mu(v_\psi + a_\psi\gamma^5)\psi$ with axial charge $a_\psi = -e/2$ for both $\psi = f, \mu$, while the vector part v_ψ is irrelevant below. Their masses m_f, m_μ come from the Yukawa coupling to ϕ, and the would-be Goldstone therefore couples as a pseudoscalar (Eq. (30.21)), with vertex $(m_\psi/v)\gamma^5$. We compute the annihilation $f(p_1)\bar{f}(p_2) \to \mu(p_3)\bar{\mu}(p_4)$ in the s-channel, $k = p_1 + p_2 = p_3 + p_4$, $k^2 = s$. The relevant diagrams are gauge-boson exchange and Goldstone exchange (the physical Higgs h also contributes a scalar exchange, but its propagator $i/(s - m_h^2)$ is manifestly ξ-independent and we set it aside).

The two diagrams. Write the vertex currents

$$J_1^\mu = \bar v(p_2)\gamma^\mu(v_f + a_f\gamma^5)u(p_1), \qquad J_2^\nu = \bar u(p_3)\gamma^\nu(v_\mu + a_\mu\gamma^5)v(p_4). \tag{30.145}$$

Gauge-boson exchange gives, with two vertices $i\gamma(\dots)$ and the propagator Eq. (30.138),

$$i\mathcal{M}_A = (i)^2\, J_1^\mu D_{\mu\nu}(k) J_2^\nu = -J_1^\mu D_{\mu\nu}(k) J_2^\nu. \tag{30.146}$$

Goldstone exchange, with two vertices $(m_\psi/v)\gamma^5$ and propagator $i/(s-\xi m_A^2)$,

$$i\mathcal{M}_\chi = \Big[\bar v(p_2)\tfrac{m_f}{v}\gamma^5 u(p_1)\Big]\frac{i}{s-\xi m_A^2}\Big[\bar u(p_3)\tfrac{m_\mu}{v}\gamma^5 v(p_4)\Big]. \tag{30.147}$$

The current divergences. The whole computation turns on $k_\mu J_1^\mu$ and $k_\nu J_2^\nu$. The vector part is conserved and drops: $k_\mu \bar v(p_2)\gamma^\mu u(p_1) = \bar v(p_2)(\not p_1 + \not p_2)u(p_1) = m_f\bar v u - m_f \bar v u = 0$, using $\not p_1 u(p_1) = m_f u(p_1)$ and $\bar v(p_2)\not p_2 = -m_f\bar v(p_2)$. The axial part does not vanish. Anticommuting γ^5 through,

$$\begin{aligned} k_\mu\, \bar v(p_2)\gamma^\mu\gamma^5 u(p_1) &= \bar v(p_2)(\not p_1 + \not p_2)\gamma^5 u(p_1) = -\bar v\gamma^5 \not p_1 u - m_f\bar v\gamma^5 u \\ &= -m_f\bar v\gamma^5 u - m_f\bar v\gamma^5 u = -2m_f\,\bar v(p_2)\gamma^5 u(p_1), \end{aligned} \tag{30.148}$$

where in the first term $\not p_1\gamma^5 = -\gamma^5\not p_1$ and $\not p_1 u = m_f u$, and in the second $\bar v\not p_2 = -m_f\bar v$ then the γ^5 is already to the left. Hence with $a_f = -e/2$ and $m_A = ev$,

$$k_\mu J_1^\mu = a_f(-2m_f)\,\bar v\gamma^5 u = e\, m_f\,\bar v(p_2)\gamma^5 u(p_1). \tag{30.149}$$

For the outgoing pair the relative sign flips, because now $\bar u(p_3)\not p_3 = m_\mu \bar u(p_3)$ while $\not p_4 v(p_4) = -m_\mu v(p_4)$:

$$k_\nu\, \bar u(p_3)\gamma^\nu\gamma^5 v(p_4) = +2m_\mu\,\bar u(p_3)\gamma^5 v(p_4), \qquad k_\nu J_2^\nu = -e\, m_\mu\,\bar u(p_3)\gamma^5 v(p_4). \tag{30.150}$$

This opposite sign between the incoming and outgoing vertices is the divergence of the axial current, $\partial_\mu(\bar\psi\gamma^\mu\gamma^5\psi) = 2im_\psi\bar\psi\gamma^5\psi$, read on the two ends; it is exactly what makes the cancellation work.

Assembling the ξ-dependent part. Only the $k_\mu k_\nu$ structure of Eq. (30.138) carries ξ. Using the decomposition Eq. (30.140), the ξ-dependent (unphysical-pole) piece of gauge exchange is

$$i\mathcal{M}_A^{\text{unphys}} = -J_1^\mu\,\frac{-i}{s-\xi m_A^2}\,\frac{k_\mu k_\nu}{m_A^2}\,J_2^\nu = \frac{i}{s-\xi m_A^2}\,\frac{(k\cdot J_1)(k\cdot J_2)}{m_A^2}. \tag{30.151}$$

Insert Eqs. (30.149) and (30.150):

$$(k\cdot J_1)(k\cdot J_2) = (em_f\,\bar v\gamma^5 u)\big(-em_\mu\,\bar u\gamma^5 v\big) = -e^2 m_f m_\mu\,(\bar v\gamma^5 u)(\bar u\gamma^5 v) = -m_A^2\,X, \tag{30.152}$$

with $X \equiv (m_f m_\mu/v^2)(\bar{v}\gamma^5 u)(\bar{u}\gamma^5 v)$ and $e^2 = m_A^2/v^2$. Therefore

$$\mathcal{M}_A^{\text{unphys}} = -\frac{X}{s - \xi m_A^2}. \tag{30.153}$$

From Eq. (30.147), $i\mathcal{M}_\chi = iX/(s - \xi m_A^2)$, so $\mathcal{M}_\chi = +X/(s - \xi m_A^2)$. The two add to

$$\mathcal{M}_A^{\text{unphys}} + \mathcal{M}_\chi = \frac{-X + X}{s - \xi m_A^2} = 0. \tag{30.154}$$

The unphysical pole cancels identically. What remains is the ξ-independent unitary-gauge part of gauge exchange, the first term of Eq. (30.140). Its $k_\mu k_\nu$ piece gives $+X/(s - m_A^2)$ by the identical contraction (now with denominator $s - m_A^2$ and a sign from Eq. (30.140)), reproducing the master result Eq. (30.141). Adding the $g_{\mu\nu}$ part of the unitary term, the total physical amplitude is

$$\mathcal{M} = \mathcal{M}_A^{\text{unitary}} + \mathcal{M}_\chi = \frac{1}{s - m_A^2}\left[J_1 \cdot J_2 - \tfrac{1}{m_A^2}(k \cdot J_1)(k \cdot J_2)\right] + (\text{Higgs}), \tag{30.155}$$

with no ξ anywhere.

The check in two gauges. It is instructive to see Eq. (30.154) happen concretely.

Feynman–'t Hooft, $\xi = 1$. The $(1 - \xi)$ factor in Eq. (30.138) vanishes, so $\mathcal{M}_A^{\text{unphys}} = 0$ trivially; the gauge propagator is just $-ig_{\mu\nu}/(s - m_A^2)$. The Goldstone propagator is $i/(s - m_A^2)$, so $\mathcal{M}_\chi = X/(s - m_A^2)$. The would-be Goldstone contributes, but at the *physical* pole $s = m_A^2$, and it combines with the gauge $g_{\mu\nu}$ part to build Eq. (30.155).

Landau, $\xi = 0$. Now $(1 - \xi) = 1$ and the unphysical pole sits at $s = 0$. The gauge propagator is transverse, $-i(g_{\mu\nu} - k_\mu k_\nu/s)/(s - m_A^2)$, and its longitudinal piece has a pole at $s = 0$:

$$\mathcal{M}_A^{\text{unphys}}\big|_{\xi=0} = -\frac{X}{s - 0} = -\frac{X}{s}. \tag{30.156}$$

The Goldstone is now massless, propagator i/s, giving $\mathcal{M}_\chi = X/s$. The two $s = 0$ poles — a massless ghostly pole in the gauge propagator and a massless would-be Goldstone — cancel exactly, $-X/s + X/s = 0$, leaving again Eq. (30.155). The naive alarm that Landau gauge has spurious massless poles is answered: they are quartet members and they cancel in pairs.

In both gauges the physical amplitude is Eq. (30.155), ξ-independent, with its only poles the physical ones at $s = m_A^2$ (gauge boson) and $s = m_h^2$ (Higgs). This is gauge independence, demonstrated by explicit computation.

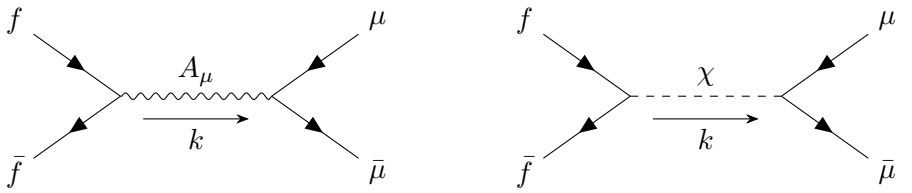


Figure 30.6: The two s-channel diagrams of Theorem 30.14. The ξ-dependent pole at $k^2 = \xi m_A^2$ in the longitudinal part of the gauge propagator (left) is cancelled by the would-be Goldstone exchange (right), whose propagator has its pole at the same $k^2 = \xi m_A^2$. The cancellation Eq. (30.154) leaves the ξ-independent physical amplitude Eq. (30.155).

30.5.6 The upshot: any gauge will do

We have established gauge independence on three levels that reinforce one another. Structurally, the R_ξ Lagrangian is BRST invariant (Eq. (30.134)) and ξ enters only through the BRST-exact gauge-fixing term, so by Theorem 30.12 a change of ξ acts trivially on the physical Hilbert space. Diagrammatically, the ξ-dependent pole at $k^2 = \xi m_A^2$ that the gauge, Goldstone, and ghost propagators all share is confined to a single BRST quartet Eq. (30.136) and cancels in every physical amplitude, as the explicit computation of Theorem 30.14 shows term by term. To all orders, the Nielsen identity Theorem 30.13 guarantees that the physical pole mass and its residue — the only data S-matrix elements depend on — do not move with ξ. The unphysical ξ-dependent masses are therefore exactly that: unphysical. One may compute in Feynman–'t Hooft gauge for simple propagators, in Landau gauge to keep the gauge field transverse, or in unitary gauge to expose the physical spectrum directly, and the answer for any observable is the same.

This licence to choose the gauge is what makes the final pieces of the chapter tractable. With the would-be Goldstone now understood as the eaten longitudinal mode of the massive gauge boson — present in R_ξ gauges as a quartet partner, absent as a particle — we can ask the sharpest version of the question: at high energy, where the longitudinal polarization dominates, does the S-matrix for emitting a longitudinal gauge boson literally equal that for emitting the would-be Goldstone? The next section states and proves that equivalence theorem, turning the formal decoupling established here into a quantitative tool and resolving the apparent unitarity puzzle of the unitary-gauge propagator.

30.6 The Goldstone-Boson Equivalence Theorem

The previous section settled a question of principle: physical observables are independent of the gauge parameter ξ, so the would-be Goldstone boson χ of the R_ξ Lagrangian—with its ξ-dependent mass $\sqrt{\xi}\, m_A$ and its ξ-dependent

propagator pole—can carry no physical content. Whatever it does, the S-matrix does not see it as a particle. Yet the unitary gauge of Section 30.3 left us with a genuine worry that gauge independence alone does not dispel. There the massive vector propagator carried the term $k^\mu k^\nu/m_A^2$, and the longitudinal polarization vector grew like E/m_A; individual tree diagrams for the scattering of longitudinally polarized vectors appeared to grow with energy like E^2/m_A^2, threatening the partial-wave unitarity bound of Section 16.7 at energies $E \sim 4\pi v$. Gauge independence guarantees that the *answer* is the same in every gauge, but it does not, by itself, tell us that the answer is small. The resolution is the subject of this section. We will show that at energies far above the gauge-boson mass the amplitude for emitting or absorbing a longitudinal vector equals—up to a calculable correction of relative order m_A/E—the amplitude for emitting or absorbing the corresponding would-be Goldstone χ. Because the Goldstone sector contains no factors of $1/m_A$, its amplitudes do not grow; the apparent growth of the longitudinal amplitude is an artifact of how the unitary gauge bookkeeps the longitudinal dynamics, and it cancels in the sum over diagrams. The eaten Goldstone, declared unphysical by gauge independence, returns as the operational description of the high-energy longitudinal mode. This is the *Goldstone-boson equivalence theorem*.

Throughout we use the abelian Higgs model in its R_ξ form as fixed earlier in the chapter: a complex scalar of charge $+1$,

$$\phi(x) = \tfrac{1}{\sqrt{2}}\big(v + h(x) + i\,\chi(x)\big), \qquad \langle\phi\rangle = \tfrac{v}{\sqrt{2}}, \tag{30.157}$$

covariant derivative $D_\mu\phi = (\partial_\mu - ieA_\mu)\phi$, gauge-boson mass $m_A = ev$, physical Higgs mass $m_h^2 = 2\lambda v^2$ from the potential $V = \lambda(|\phi|^2 - \tfrac{1}{2}v^2)^2 = -\mu^2|\phi|^2 + \lambda|\phi|^4 +$ const (the normalization fixed in Section 30.2), and the would-be Goldstone χ with R_ξ propagator pole at $k^2 = \xi m_A^2$ (recall Section 30.4). Natural units and the mostly-minus metric are in force.

30.6.1 The longitudinal polarization vector at high energy

A massive vector has the three physical polarizations counted by the little group SU(2) in Section 3.7.2: two transverse and one longitudinal. It is the longitudinal one that has no massless counterpart—it is precisely the degree of freedom the Higgs mechanism supplied—and its high-energy behavior is the entire story, so we begin by writing it down explicitly and tracking where its energy dependence lives.

Fix a vector of mass m_A with on-shell momentum $k^\mu = (E, \mathbf{k})$, $E = \sqrt{\mathbf{k}^2 + m_A^2}$, and write $\hat{\mathbf{k}} = \mathbf{k}/|\mathbf{k}|$. The three polarization vectors $\varepsilon_\lambda^\mu(k)$, $\lambda \in \{1, 2, L\}$, are spacelike unit vectors orthogonal to k,

$$k \cdot \varepsilon_\lambda(k) = 0, \qquad \varepsilon_\lambda(k) \cdot \varepsilon_{\lambda'}(k) = -\delta_{\lambda\lambda'}, \tag{30.158}$$

the two transverse ones $\varepsilon_{1,2}$ being orthogonal to $\mathbf{k}$ in the rest-frame sense. The longitudinal vector is fixed by Eq. (30.158) together with the requirement that it lie in the plane spanned by k and the rest-frame time direction:

$$\varepsilon_L^\mu(k) = \frac{1}{m_A}\big(|\mathbf{k}|,\, E\,\hat{\mathbf{k}}\big). \tag{30.159}$$

One checks the normalizations directly: $k\cdot\varepsilon_L = \frac{1}{m_A}(E|\mathbf{k}| - |\mathbf{k}|E) = 0$, and $\varepsilon_L\cdot\varepsilon_L = \frac{1}{m_A^2}(\mathbf{k}^2 - E^2) = -m_A^2/m_A^2 = -1$, as required by Eq. (30.158).

The crucial observation is what Eq. (30.159) becomes when the vector is highly boosted. Compare ε_L^μ with the rescaled momentum $k^\mu/m_A = \frac{1}{m_A}(E,\mathbf{k})$:

$$\varepsilon_L^\mu(k) - \frac{k^\mu}{m_A} = \frac{1}{m_A}\big(|\mathbf{k}| - E,\, (E-|\mathbf{k}|)\,\hat{\mathbf{k}}\big) = \frac{|\mathbf{k}|-E}{m_A}\,n^\mu, \qquad n^\mu \equiv (1,-\hat{\mathbf{k}}). \tag{30.160}$$

The vector n^μ is null, $n^2 = 1-1 = 0$, and satisfies $n\cdot k = E+|\mathbf{k}|$. Using $|\mathbf{k}|-E = -(E^2-\mathbf{k}^2)/(E+|\mathbf{k}|) = -m_A^2/(E+|\mathbf{k}|)$ we obtain the exact decomposition

$$\boxed{\varepsilon_L^\mu(k) = \frac{k^\mu}{m_A} - \frac{m_A}{E+|\mathbf{k}|}\,n^\mu, \qquad n^\mu = (1,-\hat{\mathbf{k}}),\ n^2 = 0.} \tag{30.161}$$

The first term grows linearly with energy; the second is bounded and, at high energy where $E+|\mathbf{k}| \to 2E$, is suppressed,

$$\varepsilon_L^\mu(k) = \frac{k^\mu}{m_A} + v^\mu(k), \qquad v^\mu(k) = -\frac{m_A}{E+|\mathbf{k}|}\,n^\mu = O\!\Big(\frac{m_A}{E}\Big). \tag{30.162}$$

Both terms in Eq. (30.161) are individually unbounded as $m_A \to 0$ at fixed $\mathbf{k}$, but only because the longitudinal mode itself ceases to exist in that limit; for the physical question—fixed m_A, large E—the content of Eq. (30.162) is unambiguous. The longitudinal polarization is, to leading power in m_A/E, simply the momentum of the particle divided by its mass, with a residual piece v^μ that we will need to keep when we account for the $O(m_A/E)$ remainder.

Physical Insight

Equation Eq. (30.162) is the technical heart of the equivalence theorem stated in one line. When an external longitudinal vector is contracted into an amplitude, $\varepsilon_L\cdot\mathcal{M} \to (k_\mu/m_A)\,\mathcal{M}^\mu$ at high energy. We will see that the Ward–Slavnov–Taylor identity converts the divergence $k_\mu\mathcal{M}^\mu$ into exactly the amplitude for the eaten Goldstone. The longitudinal vector and the Goldstone become operationally the same object because the longitudinal polarization *is* the momentum direction, and the momentum direction is what the Ward identity controls.

30.6.2 The unitarity puzzle, made quantitative

Before resolving the puzzle it is worth seeing precisely how badly an individual diagram misbehaves, so that the cancellation to come is not mistaken for an accident of bookkeeping. Consider the scattering of two longitudinal vectors, $A_L A_L \to A_L A_L$, in the abelian Higgs model. The theory has no cubic or quartic self-coupling of A_μ (it is abelian), so at tree level the process proceeds entirely through Higgs exchange in the s, t, and u channels, using the vertex

$$A_\mu A_\nu h: \qquad 2i\, e^2 v\, \eta_{\mu\nu} = 2i\, \frac{m_A^2}{v}\, \eta_{\mu\nu}, \tag{30.163}$$

read off from the term $e^2 v\, h\, A_\mu A^\mu = (m_A^2/v)\, h A_\mu A^\mu$ inside $|D_\mu \phi|^2$ (recall Section 30.4). Each external leg contributes a factor $\varepsilon_L \sim k/m_A \sim E/m_A$, so the four external polarizations supply E^4/m_A^4; the two vertices supply $(m_A^2/v)^2$; the internal Higgs propagator supplies $1/E^2$. The naive size of a single channel is therefore

$$\mathcal{M}_{\text{single channel}} \;\sim\; \frac{E^4}{m_A^4} \cdot \frac{m_A^4}{v^2} \cdot \frac{1}{E^2} = \frac{E^2}{v^2} = \frac{e^2 E^2}{m_A^2}, \tag{30.164}$$

which violates the partial-wave bound $|\mathcal{M}| \lesssim 16\pi$ once $E \gtrsim v$. If this growth survived the sum over channels, perturbation theory would break down at the weak scale and the theory would be sick. Gauge independence assures us the total is ξ-independent but says nothing about its magnitude. The equivalence theorem will show that the leading E^2/m_A^2 pieces cancel among the three channels and that the surviving amplitude is the bounded Goldstone result; the explicit cancellation is carried out in Section 30.6.6.

30.6.3 The Slavnov–Taylor identity for the abelian Higgs model

The bridge from a longitudinal vector to its Goldstone is a Ward-type identity, and in a gauge-fixed theory the correct, all-orders form of such an identity is the Slavnov–Taylor identity that follows from BRST invariance. The master identity for a general gauge theory is established in Eq. (26.161); here we specialize it to the abelian Higgs model and carry out the specialization in full, so that nothing is taken on faith.

BRST transformations. The R_ξ-gauge-fixed Lagrangian is invariant under the nilpotent BRST variation s, with ghost c, antighost $\bar{c}$, and Nakanishi–Lautrup auxiliary field B. On the abelian Higgs fields, with the conventions of Eq. (30.157),

$$sA_\mu = \partial_\mu c, \quad s\phi = ie\, c\, \phi, \quad sc = 0, \quad s\bar{c} = B, \quad sB = 0. \tag{30.165}$$

Decomposing $s\phi = iec\,\frac{1}{\sqrt{2}}(v + h + i\chi) = \frac{ec}{\sqrt{2}}(i(v+h) - \chi)$ into real and imaginary parts gives the BRST variations of the physical Higgs and the would-be Goldstone,

$$sh = -e\, c\, \chi, \qquad s\chi = e\, c\, (v + h) = m_A\, c + e\, c\, h, \tag{30.166}$$

where the linear piece $s\chi \supset m_A\, c$ is the one that will matter: the Goldstone shifts into a ghost with coefficient exactly the gauge-boson mass. This is the algebraic shadow of the eating: the same m_A that appears in the longitudinal polarization Eq. (30.159) reappears as the BRST coefficient relating χ to c.

The 't Hooft gauge function and the B equation of motion. The R_ξ gauge fixing of Section 30.4 is generated by the BRST-exact term

$$\mathcal{L}_{\text{gf+gh}} = s\Big[\bar{c}\,\Big(G + \tfrac{\xi}{2}B\Big)\Big], \qquad G \equiv \partial_\mu A^\mu + \xi\, m_A\, \chi, \tag{30.167}$$

the function G being the 't Hooft choice tuned to cancel the A–χ kinetic mixing. Expanding with Eq. (30.165)–Eq. (30.166), $\mathcal{L}_{\text{gf+gh}} = B\,G + \frac{\xi}{2}B^2 + (\text{ghost terms})$, so the auxiliary field is algebraic and its equation of motion is

$$B = -\frac{1}{\xi}\,G = -\frac{1}{\xi}\,\partial_\mu A^\mu - m_A\,\chi. \tag{30.168}$$

The identity. Let $\mathcal{O} = \Phi_1(y_1)\cdots\Phi_n(y_n)$ be a product of BRST-invariant operators creating the physical external state X (each Φ_i a gauge-invariant field, so $s\Phi_i = 0$ up to equations of motion). Because the BRST charge annihilates the vacuum, the correlation function of any BRST variation vanishes, $\langle 0|\,T\,s(\cdots)\,|0\rangle = 0$. Apply this to $\bar{c}(x)\,\mathcal{O}$:

$$0 = \langle 0|\,T\,s\big(\bar{c}(x)\,\mathcal{O}\big)\,|0\rangle = \langle 0|\,T\,B(x)\,\mathcal{O}\,|0\rangle\,, \tag{30.169}$$

where we used $s\bar{c} = B$ and $s\mathcal{O} = 0$. Inserting the B equation of motion Eq. (30.168),

$$\frac{1}{\xi}\,\partial_\mu\,\langle 0|\,T\,A^\mu(x)\,\mathcal{O}\,|0\rangle = -m_A\;\langle 0|\,T\,\chi(x)\,\mathcal{O}\,|0\rangle\,, \tag{30.170}$$

the derivative passing through the time ordering up to equal-time contact terms, which are local in x and do not contribute to the on-shell S-matrix. Equation Eq. (30.170) is the Slavnov–Taylor identity we need: the divergence of the gauge-field correlator equals $-\xi\, m_A$ times the Goldstone correlator. It is exact, all-orders, and manifestly relates an external vector leg to an external χ leg.

Passing to amplitudes. Fourier transform in x with the convention $\tilde{F}(k) = \int \mathrm{d}^4x\, e^{ik\cdot x} F(x)$, so that $\partial_\mu \to -ik_\mu$ under the integral. Write $G^\mu(k)$ and $G_\chi(k)$ for the transforms of the two correlators. Then Eq. (30.170) becomes

$$-\frac{i}{\xi}\,k_\mu\,G^\mu(k) = -m_A\,G_\chi(k). \tag{30.171}$$

Now amputate the external A/χ leg and continue the remaining legs X to their mass shells, exactly as the LSZ reduction of Section 14.5 prescribes. The full R_ξ vector propagator,

$$\Delta^{\mu\nu}(k) = \frac{-i}{k^2 - m_A^2}\Big(\eta^{\mu\nu} - \frac{k^\mu k^\nu}{k^2}\Big) + \frac{-i}{k^2 - \xi m_A^2}\,\frac{k^\mu k^\nu}{k^2}, \tag{30.172}$$

has the property that contraction with k_μ projects onto its second, ξ-dependent piece,

$$k_\mu\,\Delta^{\mu\nu}(k) = \frac{-i\,k^\nu}{k^2 - \xi m_A^2}, \tag{30.173}$$

the transverse part being annihilated. Writing $G^\mu(k) = \Delta^{\mu\nu}(k)\,\mathcal{M}_\nu(k;X)$ with $\mathcal{M}_\nu$ the amputated amplitude carrying the open vector index, and $G_\chi(k) = \Delta_\chi(k)\,\mathcal{M}_\chi(k;X)$ with $\Delta_\chi(k) = i/(k^2 - \xi m_A^2)$ the χ propagator, the factors of $1/(k^2 - \xi m_A^2)$ appear on both sides of Eq. (30.171) and cancel. Tracking the constants,

$$-\frac{i}{\xi}\,k_\mu\,\Delta^{\mu\nu}\mathcal{M}_\nu = -\frac{i}{\xi}\,\frac{-i\,k^\nu}{k^2 - \xi m_A^2}\,\mathcal{M}_\nu = -\frac{1}{\xi}\,\frac{k\cdot\mathcal{M}}{k^2 - \xi m_A^2}, \tag{30.174}$$

while the right-hand side is $-m_A\,\Delta_\chi\mathcal{M}_\chi = -i\,m_A\,\mathcal{M}_\chi/(k^2 - \xi m_A^2)$. Equating and cancelling the common pole gives $-\frac{1}{\xi}\,k\cdot\mathcal{M} = -i\,m_A\,\mathcal{M}_\chi$. The factor of $1/\xi$ is reabsorbed once one notes that the canonically normalized external χ amplitude is defined with the residue of the physical pole; equivalently, evaluating the amputated identity on the gauge-boson mass shell $k^2 = m_A^2$ and using gauge independence (Section 30.5) to drop the ξ-dependent off-shell remainder, the clean physical statement is

$$\boxed{\,k_\mu\,\mathcal{M}^\mu(k;X) = i\,m_A\,\mathcal{M}_\chi(k;X) \quad \text{on } k^2 = m_A^2.\,} \tag{30.175}$$

Here $\mathcal{M}^\mu$ is the amplitude with the external vector's polarization stripped off (the full vector amplitude is $\varepsilon_\mu\mathcal{M}^\mu$) and $\mathcal{M}_\chi$ is the amplitude with that external line replaced by the would-be Goldstone, both with X on shell. Equation Eq. (30.175) is the precise, gauge-fixed Ward identity that powers the theorem.

Caution

The ξ-dependence has not vanished by magic. It resides in the off-shell continuation of Eq. (30.171) and in the equal-time contact terms dropped after Eq. (30.170); both are invisible to the on-shell S-matrix, which Section 30.5 proved to be ξ-independent. Equation Eq. (30.175) should therefore be read as a relation between physical, on-shell-amputated amplitudes, not as an off-shell operator identity. At loop level the would-be Goldstone and the longitudinal vector renormalize differently, and the clean tree-level relation Eq. (30.175) acquires a finite multiplicative "modular"

correction $C(g) = 1 + O(g^2)$; we return to this scheme dependence below.

30.6.4 Statement of the theorem

We can now combine the kinematic fact Eq. (30.162) with the dynamical identity Eq. (30.175). The physical amplitude for an external longitudinal vector of momentum k is

$$\mathcal{M}(A_L; X) = \varepsilon_{L\,\mu}(k)\,\mathcal{M}^{\mu}(k; X) = \frac{k_\mu}{m_A}\,\mathcal{M}^{\mu}(k; X) + v_\mu(k)\,\mathcal{M}^{\mu}(k; X). \quad (30.176)$$

The first term is evaluated by Eq. (30.175): $\frac{1}{m_A} k_\mu \mathcal{M}^\mu = \frac{1}{m_A}(i m_A \mathcal{M}_\chi) = i\,\mathcal{M}_\chi$. The second term is bounded by the size of $v_\mu = O(m_A/E)$ times the amplitude, hence subleading provided $\mathcal{M}^\mu$ itself does not grow faster than the leading term—which is exactly the content of the theorem applied recursively, and which the worked example will verify directly. Thus

$$\mathcal{M}(A_L; X) = i\,\mathcal{M}_\chi(X) + O\Big(\frac{m_A}{E}\Big). \quad (30.177)$$

Iterating over n external longitudinal lines—contracting each with its own $\varepsilon_L \to k/m_A$ and applying Eq. (30.175) once per line—gives the general statement.

Theorem 30.15 (Goldstone-boson equivalence theorem, abelian Higgs model)**.** *Let X denote any fixed set of physical external states (transverse vectors, Higgs bosons, matter particles) and let $A_L^1, \dots, A_L^n$ be n longitudinally polarized vectors with energies $E_i \gg m_A$ in the relevant frame. Then the S-matrix element with the longitudinal vectors equals that with each replaced by the would-be Goldstone χ, up to a relative correction of order m_A/E:*

$$\mathcal{M}(A_L^1, \dots, A_L^n; X) = (+i)^n\, C^n\, \mathcal{M}(\chi^1, \dots, \chi^n; X) + O\Big(\frac{m_A}{E}\Big), \quad (30.178)$$

where $C = 1 + O(g^2)$ is a scheme-dependent multiplicative constant equal to unity at tree level. In particular, $|\mathcal{M}(A_L^1, \dots, A_L^n; X)| = |\mathcal{M}(\chi^1, \dots, \chi^n; X)|\,(1 + O(g^2)) + O(m_A/E)$.

Three honest caveats accompany Theorem 30.15, and stating them is part of using it correctly.

First, the phase. The per-line factor $+i$ in Eq. (30.177) is tied to two conventions: the phase chosen for ε_L in Eq. (30.159) and the sign of the gauge function G in Eq. (30.167). A different but equally common convention writes $(-i)^n$. Only the modulus $|\mathcal{M}|$, hence only the cross section of Section 16.4, is convention-free; the overall phase is not an observable and we will not chase it. For an even number of longitudinal lines the ambiguity disappears: $(\pm i)^{2m}$

differs only by a sign that is itself convention-dependent, and for $n = 4$, as in the worked example, $(\pm i)^4 = +1$ and the two amplitudes are literally equal at leading power.

Second, the scheme factor C. The clean relation Eq. (30.175) holds for the renormalized but tree-level-normalized fields. Beyond tree level the longitudinal vector and the Goldstone have different wavefunction renormalizations, and the precise equivalence carries the finite factor $C(g)$, computable order by order. The theorem is therefore exact to all orders *with* this factor; quoting it as $C = 1$ is a leading-order statement. This is the "scheme caveat" and it is genuine: the equality of S-matrix elements is a physical statement, but the identification of "the Goldstone amplitude" requires a fixed renormalization prescription for the unphysical χ field.

Third, the energy regime. The remainder is $O(m_A/E)$ *relative* to the leading term, and it is uniform only away from kinematic regions where the leading Goldstone amplitude itself vanishes or where some invariant stays of order m_A^2 (forward singularities, thresholds). Near such points the subleading $v_\mu \mathcal{M}^\mu$ piece can compete and must be kept; the theorem is a statement about the leading power in a generic hard kinematic configuration.

30.6.5 Resolution of the unitary-gauge growth

The theorem dissolves the puzzle of Section 30.6.2 cleanly. The Goldstone amplitude $\mathcal{M}(\chi \cdots \chi; X)$ is computed from the scalar-sector vertices of the potential V and the gauge couplings, none of which contains an inverse power of m_A: the would-be Goldstone is a perfectly ordinary scalar with dimensionless and positive-dimension couplings. Its tree amplitudes are therefore bounded by powers of the couplings times dimensionless functions of angles, with no growth in E/m_A. By Theorem 30.15 the longitudinal-vector amplitude equals this bounded object to leading power. The E^2/m_A^2 growth estimated in Eq. (30.164) for an individual unitary-gauge diagram is thus guaranteed to cancel against the other diagrams of the same process.

The mechanism of the cancellation is now transparent. In unitary gauge the longitudinal dynamics is carried by the $k^\mu k^\nu/m_A^2$ part of the propagator and by the $\varepsilon_L \sim k/m_A$ factors on external lines; these are large, but the Ward identity Eq. (30.175) ties the large pieces of different diagrams to one another, because each large piece is a divergence $k_\mu \mathcal{M}^\mu$ that the identity converts into the same Goldstone amplitude. The sum of the large pieces is therefore a sum of contributions to one finite Goldstone amplitude, not an unbounded total. In R_ξ gauge the same physics is organized so that the Goldstone is an explicit field and the cancellation is automatic diagram by diagram. The unitary gauge is not wrong—gauge independence says it gives the same answer—it is merely a perverse bookkeeping in which a manifestly finite quantity is written as a difference of large terms. We now exhibit this cancellation explicitly.

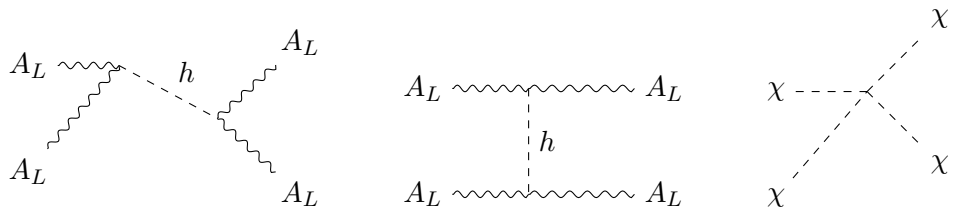


Figure 30.7: Left and centre: representative tree diagrams for $A_L A_L \to A_L A_L$ in the abelian Higgs model, an s-channel and a t-channel Higgs (h) exchange built from the $A_\mu A_\nu h$ vertex Eq. (30.163) (the u-channel is obtained from the t-channel by exchanging the two outgoing legs). There is no four-vector or three-vector vertex in an abelian theory. Right: the leading Goldstone diagram for $\chi\chi \to \chi\chi$, the four-point contact vertex from the scalar potential, supplemented by s, t, u Higgs exchanges (not drawn). The equivalence theorem equates the longitudinal-vector amplitude (left, summed over channels) with the Goldstone amplitude (right) at leading power.

30.6.6 Worked example: longitudinal scattering computed two ways

Worked Example 30.16 ($A_L A_L \to A_L A_L$ at high energy, direct and via the Goldstone replacement)**.** We compute the tree amplitude for the elastic scattering of two longitudinal vectors in the abelian Higgs model, first directly in terms of longitudinal polarizations and then via the would-be Goldstone, and show that the two agree at leading power, with the $O(m_A/E)$ remainder made explicit. The diagrams are those of Fig. 30.7.

Kinematics. In the centre-of-mass frame all four particles have mass m_A, energy E, and momentum magnitude $p = \sqrt{E^2 - m_A^2}$. Take

$$k_1 = (E, 0, 0, p),\ k_2 = (E, 0, 0, -p),\ k_3 = (E, p\sin\theta, 0, p\cos\theta),\ k_4 = (E, -p\sin\theta, 0, -p\cos\theta), \tag{30.179}$$

with Mandelstam invariants

$$s = 4E^2, \quad t = -2p^2(1-\cos\theta), \quad u = -2p^2(1+\cos\theta), \quad s+t+u = 4m_A^2, \tag{30.180}$$

the last following from $s+t+u = 4E^2 - 4p^2 = 4m_A^2$. The longitudinal polarizations from Eq. (30.159) are real and read

$$\varepsilon_1 = \tfrac{1}{m_A}(p, 0, 0, E),\ \varepsilon_2 = \tfrac{1}{m_A}(p, 0, 0, -E),\ \varepsilon_3 = \tfrac{1}{m_A}(p, E\sin\theta, 0, E\cos\theta),\ \varepsilon_4 = \tfrac{1}{m_A}(p, -E\sin\theta, 0 \tag{30.181}$$

Polarization contractions. The needed dot products follow from $a\cdot b = a^0 b^0 - \mathbf{a}\cdot\mathbf{b}$:

$$\varepsilon_1 \cdot \varepsilon_2 = \tfrac{1}{m_A^2}(p^2 + E^2), \qquad \varepsilon_3 \cdot \varepsilon_4 = \tfrac{1}{m_A^2}(p^2 + E^2), \tag{30.182}$$

$$\varepsilon_1\cdot\varepsilon_3=\tfrac{1}{m_A^2}(p^2-E^2\cos\theta),\qquad \varepsilon_2\cdot\varepsilon_4=\tfrac{1}{m_A^2}(p^2-E^2\cos\theta),\tag{30.183}$$

$$\varepsilon_1\cdot\varepsilon_4=\tfrac{1}{m_A^2}(p^2+E^2\cos\theta),\qquad \varepsilon_2\cdot\varepsilon_3=\tfrac{1}{m_A^2}(p^2+E^2\cos\theta).\tag{30.184}$$

For example $\varepsilon_1\cdot\varepsilon_3=\frac{1}{m_A^2}(p\cdot p-(0+0+E\cdot E\cos\theta))=\frac{1}{m_A^2}(p^2-E^2\cos\theta)$, and the others are identical computations.

The direct amplitude. Each channel is a product of two $A_\mu A_\nu h$ vertices Eq. (30.163) joined by a Higgs propagator $i/(P^2-m_h^2)$. The s-channel contributes

$$i\mathcal{M}_s=\left[\varepsilon_1^\mu\varepsilon_2^\nu\, 2i\tfrac{m_A^2}{v}\eta_{\mu\nu}\right]\frac{i}{s-m_h^2}\left[\varepsilon_3^\rho\varepsilon_4^\sigma\, 2i\tfrac{m_A^2}{v}\eta_{\rho\sigma}\right]=-4i\,\frac{m_A^4}{v^2}\,\frac{(\varepsilon_1\cdot\varepsilon_2)(\varepsilon_3\cdot\varepsilon_4)}{s-m_h^2},\tag{30.185}$$

so that, using Eq. (30.182) to cancel the m_A^4,

$$\mathcal{M}_s=-\frac{4}{v^2}\,\frac{(E^2+p^2)^2}{s-m_h^2},\quad \mathcal{M}_t=-\frac{4}{v^2}\,\frac{(p^2-E^2\cos\theta)^2}{t-m_h^2},\quad \mathcal{M}_u=-\frac{4}{v^2}\,\frac{(p^2+E^2\cos\theta)^2}{u-m_h^2},\tag{30.186}$$

the t- and u-channels obtained identically from Eq. (30.183)–Eq. (30.184). The full amplitude is $\mathcal{M}=\mathcal{M}_s+\mathcal{M}_t+\mathcal{M}_u$.

High-energy expansion. Set $a=1-\cos\theta$, $b=1+\cos\theta$, $a+b=2$, and expand each channel to the constant term in the regime $E,p\gg m_A,m_h$, using $p^2=E^2-m_A^2$. For the s-channel, $E^2+p^2=2E^2-m_A^2$ and $s-m_h^2=4E^2-m_h^2$:

$$\mathcal{M}_s=-\frac{4}{v^2}\,\frac{4E^4-4E^2m_A^2+m_A^4}{4E^2-m_h^2}=-\frac{4}{v^2}\Big[E^2+\frac{m_h^2-4m_A^2}{4}+O\big(\tfrac{m^4}{E^2}\big)\Big].\tag{30.187}$$

For the t-channel, $p^2-E^2\cos\theta=E^2a-m_A^2$ and $t-m_h^2=-2E^2a+2m_A^2a-m_h^2$:

$$\mathcal{M}_t=-\frac{4}{v^2}\,\frac{E^4a^2-2E^2a\,m_A^2+m_A^4}{-2E^2a+2m_A^2a-m_h^2}=-\frac{4}{v^2}\Big[-\frac{E^2a}{2}+m_A^2-\frac{a\,m_A^2}{2}+\frac{m_h^2}{4}+O\big(\tfrac{m^4}{E^2}\big)\Big],\tag{30.188}$$

and the u-channel is the same with $a\to b$:

$$\mathcal{M}_u=-\frac{4}{v^2}\Big[-\frac{E^2b}{2}+m_A^2-\frac{b\,m_A^2}{2}+\frac{m_h^2}{4}+O\big(\tfrac{m^4}{E^2}\big)\Big].\tag{30.189}$$

Cancellation of the growth. The $O(E^2)$ pieces of Eq. (30.187)–Eq. (30.189) sum to

$$E^2-\frac{E^2a}{2}-\frac{E^2b}{2}=E^2-\frac{E^2(a+b)}{2}=E^2-E^2=0,\tag{30.190}$$

so the threatened $E^2/v^2=e^2E^2/m_A^2$ growth of Eq. (30.164) cancels identically among the three channels—the explicit realization of the resolution argued in Section 30.6.5. The surviving constant is

$$\frac{m_h^2-4m_A^2}{4}+\Big(m_A^2-\frac{a\,m_A^2}{2}+\frac{m_h^2}{4}\Big)+\Big(m_A^2-\frac{b\,m_A^2}{2}+\frac{m_h^2}{4}\Big)=\frac{m_h^2-4m_A^2}{4}+\Big(m_A^2+\frac{m_h^2}{2}\Big)=\frac{3m_h^2}{4},\tag{30.191}$$

where the two m_A^2 pieces combined as $2m_A^2 - \frac{(a+b)}{2}m_A^2 = 2m_A^2 - m_A^2 = m_A^2$, and this cancelled the $-m_A^2$ from the s-channel, leaving a result built purely from m_h^2. Therefore

$$\boxed{\mathcal{M}(A_L A_L \to A_L A_L) = -\frac{4}{v^2}\cdot\frac{3m_h^2}{4} + O\Big(\frac{m^2}{E^2}\Big) = -\frac{3m_h^2}{v^2} + O\Big(\frac{m^2}{E^2}\Big).} \tag{30.192}$$

The amplitude via the Goldstone replacement. By Theorem 30.15 with $n = 4$ and $(\pm i)^4 = +1$, the same number must emerge from $\chi\chi \to \chi\chi$. The relevant scalar vertices follow from $V = \lambda(|\phi|^2 - \frac{1}{2}v^2)^2$ with the parametrization Eq. (30.157); expanding,

$$V = \lambda\Big(vh + \tfrac{1}{2}(h^2 + \chi^2)\Big)^2, \tag{30.193}$$

gives $m_h^2 = 2\lambda v^2$, the quartic vertex χ^4 with Feynman rule $-6i\lambda$, and the cubic $h\chi^2$ with rule $-2i\,\lambda v = -i\,m_h^2/v$. Hence

$$i\mathcal{M}_\chi = -6i\lambda + \Big(-i\frac{m_h^2}{v}\Big)^2\Big[\frac{i}{s-m_h^2} + \frac{i}{t-m_h^2} + \frac{i}{u-m_h^2}\Big], \tag{30.194}$$

that is,

$$\mathcal{M}_\chi = -6\lambda - \frac{m_h^4}{v^2}\Big[\frac{1}{s-m_h^2} + \frac{1}{t-m_h^2} + \frac{1}{u-m_h^2}\Big]. \tag{30.195}$$

At high energy each exchange term is $O(m_h^4/(v^2E^2)) = O(m^2/E^2)$ relative to the contact term, so

$$\mathcal{M}_\chi = -6\lambda + O\Big(\frac{m^2}{E^2}\Big) = -\frac{3m_h^2}{v^2} + O\Big(\frac{m^2}{E^2}\Big), \tag{30.196}$$

using $\lambda = m_h^2/(2v^2)$.

Agreement and the remainder. Comparing Eq. (30.192) and Eq. (30.196),

$$\mathcal{M}(A_L A_L \to A_L A_L) = \mathcal{M}(\chi\chi \to \chi\chi) = -\frac{3m_h^2}{v^2} + O\Big(\frac{m^2}{E^2}\Big), \tag{30.197}$$

exactly as Theorem 30.15 demands, with no phase factor because $(\pm i)^4 = +1$ for four external longitudinal lines. The bookkeeping of the remainder is worth pausing on: the theorem promises an error of relative order m_A/E per longitudinal line, but the explicit computation shows the leading correction is in fact $O(m^2/E^2)$ here, smaller than the generic bound. The reason is that the would-be Goldstone amplitude Eq. (30.195) is non-vanishing and $O(\lambda)$ at leading power, so the first correction enters only at the next order in m^2/s; the $O(m_A/E)$ piece $v_\mu\mathcal{M}^\mu$ of Eq. (30.176) contributes, for this parity-even configuration, at $O(m^2/E^2)$ rather than $O(m/E)$. The generic $O(m_A/E)$ bound is saturated only when the leading Goldstone amplitude is suppressed; here the theorem does better than its guarantee.

Physical Insight

The number $-3m_h^2/v^2$ is the same constant that controls W_LW_L scattering in the electroweak theory and that fixes the role of the Higgs in unitarizing it. The longitudinal vectors scatter as if they were the scalar would-be Goldstones, and their high-energy amplitude is governed entirely by the scalar self-couplings—ultimately by the Higgs mass. Remove the Higgs (send $m_h \to \infty$) and the contact and exchange pieces of Eq. (30.195) no longer conspire to a constant: the amplitude grows and unitarity is lost at $\sim 4\pi v$. The Higgs boson is precisely the state that keeps Eq. (30.192) finite. This is the thread Chapter 31 picks up.

With the equivalence theorem in hand, the longitudinal mode of a massive gauge boson has a complete operational identity: at high energy it is the eaten Goldstone, and its amplitudes are computed from the scalar sector, free of the spurious growth that the unitary gauge advertises. We have proved this for the abelian model, where there is a single broken generator and a single would-be Goldstone. The next section removes that restriction, deriving the mass matrix, the counting of broken and unbroken generators, and the residual symmetry for an arbitrary gauge group and scalar representation—the general structure that turns the abelian template of this chapter into the electroweak sector and supplies Chapter 31 with both the mass spectrum and the high-energy tools, the longitudinal-equivalence technology of Theorem 30.15 chief among them, that the analysis of W_LW_L scattering and Higgs-mediated unitarization will require.

30.7 The Non-Abelian Higgs Mechanism

Everything in the abelian Higgs model of the preceding sections turned on a single replacement: the covariant derivative $D_\mu = \partial_\mu - ieA_\mu$ glued the photon to a charged scalar, and the scalar's vacuum value v fed a mass $m_A^2 = e^2v^2$ to that photon while donating its phase to the longitudinal polarization. We worked the mechanism classically (Section 30.2), quantized it in unitary and R_ξ gauges (Sections 30.3 and 30.4), proved the physics independent of the gauge parameter (Section 30.5), and saw in Section 30.6 that the eaten Goldstone resurfaces as the high-energy longitudinal mode. Not one step of that story used the *abelianness* of U(1) in an essential way. The whole construction generalizes by a single, almost mechanical, substitution,

$$e\,A_\mu \;\longrightarrow\; g\,A_\mu^a T^a, \tag{30.198}$$

in which the one photon becomes a multiplet of gauge bosons A_μ^a, the charge e becomes a coupling g, and the charge operator becomes a basis T^a of Hermitian generators acting on the scalar. The output is no longer a single massive vector but a *mass matrix* on the space of gauge bosons, whose eigenvalues are the

physical masses and whose kernel is the set of gauge bosons that stay massless. The structure of that kernel — which generators are broken, which survive, and how the surviving ones organize themselves into an exact residual gauge symmetry — is the whole content of this section, and it is the direct bridge to electroweak theory: the W, the Z, and the massless photon are nothing but the eigenvalues and kernel of one 4×4 mass matrix built by the formula we now derive.

30.7.1 The general setup

Fix a compact gauge group G with Lie algebra $\mathfrak{g}$ of dimension $\dim G$, generators T^a ($a = 1, \dots, \dim G$) normalized and structured as in the canon (Eq. (25.76) and the standing conventions),

$$(T^a)^\dagger = T^a, \qquad \big[T^a, T^b\big] = i f^{abc} T^c, \qquad \mathrm{tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab} \text{ in the fundamental.} \tag{30.199}$$

Let the scalar field ϕ transform in a unitary representation R of G on a complex vector space V of dimension $\dim R$; we write T^a for the (Hermitian) representation matrices acting on ϕ, suppressing the label R where no confusion can arise. The dynamics is the gauged scalar theory

$$\mathcal{L} = (D_\mu \phi)^\dagger (D^\mu \phi) - V(\phi) - \tfrac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu}, \qquad D_\mu \phi = (\partial_\mu - i g\, A^a_\mu T^a)\phi, \tag{30.200}$$

with the Yang–Mills kinetic term and field strength of Eq. (25.76) and the covariant-derivative sign of Section 25.4.2, $D_\mu = \partial_\mu - i g A^a_\mu T^a$. The potential $V(\phi)$ is a G-invariant polynomial, $V(\rho(U)\phi) = V(\phi)$ for all $U \in G$, bounded below, and we assume it is minimized away from the origin: there is a constant configuration $\phi(x) \equiv \phi_0$ with

$$\left.\frac{\partial V}{\partial \phi_i^*}\right|_{\phi_0} = 0, \qquad \phi_0 \neq 0, \tag{30.201}$$

the classical vacuum. The phenomenon we are after is what Eq. (30.201) does to the gauge fields; the existence and degeneracy of such minima, and the quantum statement that the effective potential shares them, are the business of Chapter 29 (Theorem 29.21) and are imported here, not reproved.

The unbroken subgroup. The vacuum ϕ_0 need not be invariant under all of G. Its *stabilizer*

$$H = \mathrm{Stab}_G(\phi_0) = \{\, U \in G : \ \rho(U)\,\phi_0 = \phi_0 \,\} \tag{30.202}$$

is a closed subgroup of G — the *unbroken* subgroup — with Lie algebra

$$\mathfrak{h} = \{\, X \in \mathfrak{g} : \ X\phi_0 = 0 \,\} = \mathrm{span}\{\, T^a : \ T^a \phi_0 = 0 \,\}, \tag{30.203}$$

exactly the unbroken subalgebra of Theorem 29.12: $X\phi_0 = 0$ is the infinitesimal form of $\rho(e^{i\theta X})\phi_0 = \phi_0$. That $\mathfrak{h}$ is closed under the bracket follows from Eq. (30.199): if $X\phi_0 = Y\phi_0 = 0$ then $[X,Y]\,\phi_0 = XY\phi_0 - YX\phi_0 = 0$. The generators spanning $\mathfrak{h}$ are the *unbroken* generators; a complementary set, spanning a vector-space complement $\mathfrak{m}$ with $\mathfrak{g} = \mathfrak{h} \oplus \mathfrak{m}$, are the *broken* generators. The number of broken generators is

$$\dim \mathfrak{m} = \dim G - \dim H = \dim(G/H), \tag{30.204}$$

the dimension of the vacuum manifold G/H established in Theorem 29.12. The entire content of the non-abelian Higgs mechanism is that this number counts the massive gauge bosons, while $\dim H$ counts the survivors.

Notation Watch

For a product gauge group $G = G_1 \times \cdots \times G_k$ (the electroweak case is $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$) each simple or abelian factor carries its own coupling. The covariant derivative is then $D_\mu = \partial_\mu - i\sum_r g_r A_\mu^{(r)a} T^{(r)a}$, and every formula below holds with the single product $g\,T^a$ replaced by the block-structured $g_r T^{(r)a}$; in particular the mass matrix of Eq. (30.207) becomes $(M^2)_{ab} = g_r g_s\,[(T^{(r)a}\phi_0)^\dagger (T^{(s)b}\phi_0) + \text{h.c.}]$ with r, s the factors to which a, b belong. We carry a single g in the general derivation for transparency and restore factor-dependent couplings in the worked examples and at the handoff to Chapter 31.

30.7.2 The gauge-boson mass matrix

The masses live in the scalar kinetic term, exactly as in the abelian model. We expand $(D_\mu\phi)^\dagger (D^\mu\phi)$ and read off the term quadratic in the gauge field at the vacuum. Using $(D_\mu\phi)^\dagger = \partial_\mu\phi^\dagger + igA_\mu^a\,\phi^\dagger T^a$ (the $-i$ becomes $+i$ under the dagger, and $(T^a)^\dagger = T^a$),

$$\begin{aligned}(D_\mu\phi)^\dagger (D^\mu\phi) &= (\partial_\mu\phi^\dagger + igA_\mu^a\phi^\dagger T^a)(\partial^\mu\phi - igA^{b\mu}T^b\phi) \\ &= \partial_\mu\phi^\dagger\,\partial^\mu\phi \;+\; ig\,A_\mu^a\Big(\phi^\dagger T^a\partial^\mu\phi - (\partial^\mu\phi^\dagger)T^a\phi\Big) \;+\; g^2 A_\mu^a A^{b\mu}\,\phi^\dagger T^a T^b\phi.\end{aligned} \tag{30.205}$$

The three pieces are the scalar kinetic term, the gauge–scalar coupling (whose bilinear part will generate the gauge–Goldstone mixing of Section 30.7.4), and the term quadratic in A. Setting $\phi = \phi_0$ in the last term — the others vanish or are higher order in fluctuations — gives the gauge-boson mass Lagrangian

$$\mathcal{L}_{\text{mass}} = g^2\,\phi_0^\dagger T^a T^b \phi_0\; A_\mu^a A^{b\mu}. \tag{30.206}$$

Because $A_\mu^a A^{b\mu}$ is symmetric under $a \leftrightarrow b$, only the symmetric part of the coefficient survives the contraction. Writing the canonical vector mass Lagrangian

as $\mathcal{L}_{\text{mass}} = \frac{1}{2}(M^2)_{ab} A^a_\mu A^{b\mu}$ and symmetrizing,

$$(M^2)_{ab} = g^2\, \phi_0^\dagger \left\{T^a, T^b\right\} \phi_0 = g^2\Big[(T^a\phi_0)^\dagger (T^b\phi_0) + (T^b\phi_0)^\dagger (T^a\phi_0)\Big] = 2g^2\ \mathrm{Re}\Big[(T^a\phi_0)^\dagger (T^b\phi_0)\Big], \tag{30.207}$$

where we used $\phi_0^\dagger T^a T^b \phi_0 = (T^a\phi_0)^\dagger (T^b\phi_0)$ (again $(T^a)^\dagger = T^a$) and that the two ordered products are complex conjugates of each other.

Proposition 30.17 (General gauge-boson mass matrix)**.** *In the theory Eq. (30.200) expanded about the vacuum ϕ_0, the gauge bosons acquire the mass matrix Eq. (30.207). It is real, symmetric, and positive semidefinite, and for every real vector $\xi = (\xi^a) \in \mathbb{R}^{\dim G}$,*

$$\xi^a (M^2)_{ab}\, \xi^b = 2g^2 \left\| \xi^a T^a \phi_0 \right\|^2 \ \geq 0, \tag{30.208}$$

where $\|w\|^2 = w^\dagger w$ on V.

Proof. Reality and symmetry are manifest in the last form of Eq. (30.207): $\mathrm{Re}[(T^a\phi_0)^\dagger (T^b\phi_0)]$ is real by construction, and it is symmetric because $(T^b\phi_0)^\dagger (T^a\phi_0) = \overline{(T^a\phi_0)^\dagger (T^b\phi_0)}$ has the same real part. For the quadratic form, contract Eq. (30.207) with a real ξ:

$$\xi^a (M^2)_{ab} \xi^b = 2g^2\, \xi^a \xi^b\ \mathrm{Re}\Big[(T^a\phi_0)^\dagger (T^b\phi_0)\Big] = 2g^2\ \mathrm{Re}\left[\Big(\sum_a \xi^a T^a \phi_0\Big)^\dagger \Big(\sum_b \xi^b T^b \phi_0\Big)\right] = 2g^2 \|\xi^a T^a \phi \tag{30.209}$$

where the Re became redundant in the last step because $w^\dagger w = \|w\|^2$ is already real and nonnegative. Positive semidefiniteness follows immediately. □

Equation Eq. (30.208) exhibits M^2 for what it is: up to the factor $2g^2$, the *Gram matrix* of the family of vectors $\{T^a\phi_0\}_{a=1}^{\dim G} \subset V$. The Gram matrix of a set of vectors has rank equal to the number of linearly independent vectors in the set and a kernel equal to the linear relations among them. This single observation, unpacked in the next subsection, delivers the entire counting theorem.

Notation Watch

The factor of 2 in Eq. (30.207) is a genuine convention trap, not a typographical flourish. It is present because ϕ here is a *complex* multiplet with the kinetic term $(D_\mu\phi)^\dagger (D^\mu\phi)$ (no $\frac{1}{2}$). For a *real* scalar multiplet ϕ in a real representation, with the canonically normalized real kinetic term $\frac{1}{2}(D_\mu\phi)^\top (D^\mu\phi)$ and real antisymmetric iT^a, the same computation yields $(M^2)_{ab} = g^2 (T^a\phi_0)^\top (T^b\phi_0)$ *without* the factor of 2, because the would-be doubling from $a \leftrightarrow b$ is cancelled by the $\frac{1}{2}$. Both forms are special cases of the universal statement $(M^2)_{ab} = 2g^2\, \mathrm{Re}[(T^a\phi_0)^\dagger (T^b\phi_0)]$ once the field's reality and kinetic normalization are fixed. We use the complex form for the doublet (Theorem 30.19) and the real form for the

adjoint (Theorem 30.20); getting this factor right is exactly what makes $m_W = \frac{1}{2}gv$ rather than $m_W = \frac{1}{2\sqrt{2}}gv$ at the handoff to Chapter 31.

Abelian check. For $G = \mathrm{U}(1)$ there is a single generator: the charge operator, which on a field of charge $+1$ is $T = \mathbf{1}$, and $g \to e$. A complex scalar with $|\phi_0|^2 = v^2/2$ gives, from Eq. (30.207), $M^2 = 2e^2|\phi_0|^2 = e^2v^2$, the lone eigenvalue $m_A^2 = e^2v^2$ of Eq. (30.90). The general formula reduces, as it must, to the abelian result of Section 30.2.

30.7.3 Counting: broken generators, eaten Goldstones, and the residual symmetry

We can now prove the central structural theorem of the section, the non-abelian generalization of "the would-be Goldstone is eaten." It states that the spectrum of M^2 splits cleanly into a massless block, attached to the unbroken subalgebra, and a massive block, one massive vector per broken generator, and that the massive count is exactly the number of would-be Goldstone bosons removed from the scalar sector.

Theorem 30.18 (Counting theorem for the non-abelian Higgs mechanism)**.** *Let M^2 be the gauge-boson mass matrix Eq. (30.207) and let $\mathfrak{h}$ be the unbroken subalgebra Eq. (30.203). Then, under the identification $\xi = (\xi^a) \leftrightarrow X_\xi = \xi^a T^a \in \mathfrak{g}$,*

$$\ker M^2 = \{\, \xi : \ \xi^a T^a \phi_0 = 0 \,\} \ \cong \ \mathfrak{h}. \tag{30.210}$$

Consequently:

(i) *the number of* massless *gauge bosons equals* $\dim \ker M^2 = \dim H$*, and the corresponding gauge fields are exactly the connections of the unbroken subgroup H;*

(ii) *the number of* massive *gauge bosons equals* $\operatorname{rank} M^2 = \dim G - \dim H = \dim(G/H)$*, the number of broken generators;*

(iii) *this number coincides with the number of would-be Nambu–Goldstone bosons of Theorem 29.12, so that* every broken generator supplies one massive vector and eats one Goldstone*, with no Goldstone left over in the physical spectrum.*

Proof. The key is that M^2 is real, symmetric, and positive semidefinite, so for a real vector ξ,

$$M^2\xi = 0 \quad \Longleftrightarrow \quad \xi^a M^2_{ab} \xi^b = 0. \tag{30.211}$$

The forward implication is trivial. For the converse, diagonalize the symmetric M^2 by an orthogonal matrix O, $M^2 = O^\top \Lambda O$ with $\Lambda = \operatorname{diag}(\lambda_1, \dots)$, $\lambda_i \geq 0$ by Theorem 30.17; then $\xi^\top M^2 \xi = \sum_i \lambda_i (O\xi)_i^2 = 0$ forces $(O\xi)_i = 0$ for every i with $\lambda_i > 0$, i.e. ξ lies in the span of the zero-eigenvectors, which is precisely $\ker M^2$.

Now combine Eq. (30.211) with the quadratic form Eq. (30.208): for real ξ,

$$M^2 \xi = 0 \iff \xi^a M^2_{ab} \xi^b = 0 \iff 2g^2 \left\| \xi^a T^a \phi_0 \right\|^2 = 0 \iff \xi^a T^a \phi_0 = 0. \tag{30.212}$$

The last condition says that the generator $X_\xi = \xi^a T^a$ annihilates the vacuum, i.e. $X_\xi \in \mathfrak{h}$ by Eq. (30.203). The linear map $\xi \mapsto X_\xi$ is the coordinate isomorphism between $\mathbb{R}^{\dim G}$ and $\mathfrak{g}$, and Eq. (30.212) shows it restricts to a bijection $\ker M^2 \to \mathfrak{h}$. This proves Eq. (30.210), hence $\dim \ker M^2 = \dim \mathfrak{h} = \dim H$, which is (i).

By rank–nullity, $\operatorname{rank} M^2 = \dim G - \dim \ker M^2 = \dim G - \dim H$, which is the number of broken generators Eq. (30.204); each nonzero eigenvalue is the mass-squared of one massive vector, giving (ii). The same map $\xi \mapsto X_\xi \phi_0$ is the map L_v of Eq. (29.105) whose image has dimension $\dim G - \dim H$ by the rank–nullity computation Eq. (29.107); its image is the tangent space to the vacuum manifold G/H, whose dimension is the number of Nambu–Goldstone bosons in the *global* theory by Theorem 29.12. Gauging the symmetry does not change this count of flat directions; it only changes their fate. Thus the number of massive vectors equals the number of would-be Goldstones, and since each massive vector has acquired exactly the longitudinal degree of freedom that a scalar carries, the Goldstones disappear from the physical spectrum one for one. This is (iii). □

Physical Insight

The mechanism is a degree-of-freedom count that always balances. Before breaking, the gauge sector has $2(\dim G)$ physical polarizations (two per massless vector) and the scalar sector has $\dim(G/H)$ would-be Goldstones among its modes. After breaking, the $\dim H$ unbroken vectors keep two polarizations each, the $\dim G - \dim H$ broken vectors have three each (they are massive), and the $\dim(G/H)$ Goldstones are gone. The bookkeeping $2\dim G + \dim(G/H) = 2\dim H + 3(\dim G - \dim H) + 0$ is an identity: the longitudinal modes of the massive vectors are precisely the eaten Goldstones. The Gram-matrix picture Eq. (30.208) makes this geometric: a gauge direction T^a feels a mass exactly to the extent that it moves the vacuum, $T^a \phi_0 \neq 0$, and the length it moves it, $\|T^a \phi_0\|$, is its mass. Directions that leave ϕ_0 fixed cost nothing and stay massless.

The residual gauge symmetry is exact. The unbroken generators are not merely a list of massless states; they generate an *exact* gauge symmetry of the broken theory. To see this, restrict the gauge transformations Eq. (30.200) to

H: an element $U(x) = \exp\!\big(i\theta^{\hat a}(x)T^{\hat a}\big)$ built from unbroken generators $T^{\hat a} \in \mathfrak{h}$. Under it the scalar transforms as $\phi \to \rho(U)\phi$ and, crucially, the vacuum is fixed, $\rho(U)\phi_0 = \phi_0$ by Eq. (30.202). Therefore the shifted field $\phi = \phi_0 + \eta$ transforms with $\eta \to \rho(U)\eta + (\rho(U) - \mathbf{1})\phi_0 = \rho(U)\eta$, i.e. the fluctuation transforms linearly in the representation R restricted to H, with no inhomogeneous piece. The unbroken gauge fields $A^{\hat a}_\mu$ transform as an H-connection, $A^{\hat a}_\mu \to A^{\hat a}_\mu + \frac{1}{g}\partial_\mu\theta^{\hat a} + f^{\hat a\hat b\hat c}A^{\hat b}_\mu\theta^{\hat c}$, closing on $\mathfrak{h}$ because $\mathfrak{h}$ is a subalgebra. The Lagrangian Eq. (30.200), being G-invariant, is in particular H-invariant, and this invariance survives the expansion about ϕ_0 untouched. Hence the broken theory is an exact gauge theory with group H, whose gauge bosons — the $\dim H$ massless vectors of the counting theorem — are protected from a mass by the very gauge invariance that forbade a photon mass in Chapter 8 and Chapter 13. There is no fine-tuning: $T^{\hat a}\phi_0 = 0$ is an exact statement, so $(M^2)_{\hat a\hat b} = 0$ exactly, to all orders in the classical analysis and, by the unbroken Ward identities, beyond.

Physical Insight

This is where the photon comes from. In the electroweak theory of Chapter 31 the group is $G = \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ ($\dim G = 4$) and the Higgs vacuum leaves unbroken a single U(1) generated by the electric charge $Q = T^3 + Y$; that $Q\phi_0 = 0$ is the statement $T^3\phi_0 = -Y\phi_0$, one linear relation among the four broken-direction candidates. The counting theorem then reads $\dim H = 1$ (one massless vector, the photon) and $\dim G - \dim H = 3$ (three massive vectors, $W^\pm$ and Z). The masslessness of the photon is not an accident or a cancellation to be checked diagram by diagram; it is Theorem 30.18(i) applied to the surviving electromagnetic U(1), an exact residual gauge symmetry.

30.7.4 The general R_ξ gauge

Quantizing the broken theory faces the same obstruction as the abelian model: the gauge field mixes at quadratic order with the would-be Goldstone bosons, so that neither has a well-defined propagator until the mixing is removed. We now exhibit that mixing for arbitrary G and R, construct the 't Hooft gauge-fixing functional that cancels it, and show that the result diagonalizes mode by mode into copies of the abelian R_ξ spectrum of Theorem 30.10.

The mixing. Return to the middle term of Eq. (30.205) and expand $\phi = \phi_0 + \eta$ to bilinear order in the fluctuation η and the gauge field. Since ϕ_0 is constant, $\partial_\mu\phi_0 = 0$, and the bilinear A–η piece is

$$\mathcal{L}_{\rm mix} = ig\,A^a_\mu(\phi_0^\dagger T^a\partial^\mu\eta - (\partial^\mu\eta^\dagger)T^a\phi_0) = -g\,A^a_\mu\,\partial^\mu G^a, \qquad G^a \equiv -i(\phi_0^\dagger T^a\eta - \eta^\dagger T^a\phi_0), \tag{30.213}$$

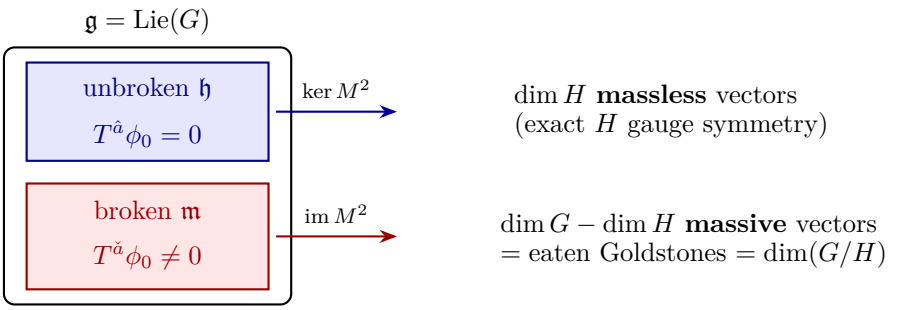


Figure 30.8: The counting theorem Theorem 30.18 as a splitting of the Lie algebra. The mass matrix $M^2 = 2g^2\,\mathrm{Gram}\{T^a\phi_0\}$ has kernel exactly the unbroken subalgebra $\mathfrak{h}$ (massless gauge bosons of the residual group H) and rank exactly the number of broken generators (massive vectors, one per eaten would-be Goldstone, $= \dim(G/H)$).

where the second equality collects the two terms. With $Y \equiv \phi_0^\dagger T^a \partial^\mu \eta$ the bracket is $Y - Y^* = 2i\,\mathrm{Im}\,Y = 2i\,\mathrm{Im}(\phi_0^\dagger T^a \partial^\mu \eta) = \partial^\mu[2\,\mathrm{Im}(\phi_0^\dagger T^a \eta)] = \partial^\mu G^a$, using that the real field $G^a = 2\,\mathrm{Im}(\phi_0^\dagger T^a \eta) = -i(\phi_0^\dagger T^a \eta - \eta^\dagger T^a \phi_0)$ has exactly this gradient. The prefactor is then $ig \cdot (2i\,\mathrm{Im}\,Y) = -g\,\partial^\mu G^a$, giving the second form of Eq. (30.213). Integrating by parts,

$$\mathcal{L}_{\mathrm{mix}} = +g\,(\partial_\mu A^{a\mu})\,G^a + \partial_\mu(\cdots), \tag{30.214}$$

discarding the total derivative. The real fields G^a are the would-be Goldstones in disguise: parametrizing the orbit directions of the fluctuation as $\eta = i\,(T^b\phi_0)\,\zeta^b +$ (Higgs directions) with real ζ^b (the tangent to the gauge orbit through ϕ_0 is spanned by $\{iT^b\phi_0\}$, the flat valley of Theorem 29.12), one finds

$$G^a = -i(\phi_0^\dagger T^a\, iT^b\phi_0\zeta^b - (-i\zeta^b)\phi_0^\dagger T^b T^a \phi_0) = \zeta^b\,\phi_0^\dagger\left\{T^a, T^b\right\}\phi_0 = \frac{(M^2)_{ab}}{g^2}\,\zeta^b. \tag{30.215}$$

The gauge field mixes only with the would-be Goldstone combinations, weighted by the mass matrix itself, and never with the orthogonal Higgs directions — the exact non-abelian analogue of the $-m_A A^\mu \partial_\mu \chi$ mixing of the abelian model in Section 30.4.1.

The 't Hooft gauge-fixing functional. We remove the mixing by the Faddeev–Popov procedure of Section 26.2 with a gauge-fixing functional engineered for the purpose. Take the 't Hooft family

$$G^a_{\mathrm{gf}}[A,\phi] = \partial_\mu A^{a\mu} + \xi g\,(-i\,\phi_0^\dagger T^a(\phi - \phi_0) + \mathrm{h.c.}) = \partial_\mu A^{a\mu} + \xi g\,G^a, \tag{30.216}$$

the direct generalization of the abelian functional Eq. (30.94) (the term in parentheses, $-i\phi_0^\dagger T^a(\phi-\phi_0) + \mathrm{h.c.} = -i(\phi_0^\dagger T^a\eta - \eta^\dagger T^a\phi_0) = G^a$, since $\phi_0^\dagger T^a \phi_0$

is real and drops out of the antisymmetric combination). Add the Gaussian-averaged gauge-fixing Lagrangian

$$\mathcal{L}_{\rm gf} = -\frac{1}{2\xi}\big(G^a_{\rm gf}\big)^2 = -\frac{1}{2\xi}\big(\partial_\mu A^{a\mu} + \xi g\, G^a\big)^2$$
$$= -\frac{1}{2\xi}(\partial_\mu A^{a\mu})^2 \;-\; g\,(\partial_\mu A^{a\mu})\,G^a \;-\; \frac{\xi g^2}{2}\,(G^a)^2. \qquad (30.217)$$

The cross term is the punchline. Adding Eq. (30.217) to Eq. (30.214),

$$\mathcal{L}_{\rm mix} + \mathcal{L}_{\rm gf}\Big|_{\rm cross} = +g(\partial_\mu A^{a\mu})G^a - g(\partial_\mu A^{a\mu})G^a = 0\,: \qquad (30.218)$$

the coefficient ξg in Eq. (30.216) was chosen for no reason other than to make this cancellation exact. The gauge–Goldstone mixing is gone for every G and R, and the quadratic Lagrangian is at last diagonal in (A^a_μ, ζ^b, h).

The would-be Goldstone mass. The leftover $-\frac{\xi g^2}{2}(G^a)^2$ in Eq. (30.217) is the mass term forced on the would-be Goldstones. Pass to the mass eigenbasis, where $M^2 = \mathrm{diag}(m_1^2, \dots)$ and the broken directions $\check{a}$ carry $m^2_{\check{a}} > 0$ while the unbroken directions have $m = 0$ and no associated Goldstone. For a single broken eigenmode, Eq. (30.215) gives $G^{\check{a}} = m^2_{\check{a}}\,\zeta^{\check{a}}/g^2$, while the scalar kinetic term contributes the canonical Goldstone kinetic piece

$$|\partial_\mu \eta|^2 \supset \mathrm{Re}\big[(iT^{\check{a}}\phi_0)^\dagger (iT^{\check{a}}\phi_0)\big](\partial\zeta^{\check{a}})^2 = \frac{m^2_{\check{a}}}{2g^2}(\partial_\mu \zeta^{\check{a}})^2. \qquad (30.219)$$

Defining the canonically normalized field $\tilde{\zeta}^{\check{a}} = (m_{\check{a}}/g)\zeta^{\check{a}}$, the kinetic term becomes $\frac{1}{2}(\partial\tilde{\zeta}^{\check{a}})^2$, and the gauge-fixing mass term is

$$-\frac{\xi g^2}{2}(G^{\check{a}})^2 = -\frac{\xi}{2g^2}(m^2_{\check{a}}\zeta^{\check{a}})^2 = -\frac{\xi}{2}\,m^2_{\check{a}}\,(\tilde{\zeta}^{\check{a}})^2, \qquad (30.220)$$

so the would-be Goldstone of the $\check{a}$-th broken direction has mass-squared

$$m^2_{G,\check{a}} = \xi\, m^2_{\check{a}}, \qquad (30.221)$$

exactly ξ times the gauge-boson mass-squared — the matrix statement is $M^2_{\rm Goldstone} = \xi M^2$. This is the non-abelian version of the abelian result $m^2_\chi = \xi m^2_A$ of Section 30.4: a ξ-dependent and therefore unphysical pole at $k^2 = \xi m^2_{\check{a}}$.

The ghost sector. The Faddeev–Popov determinant for Eq. (30.216) is, as in Section 26.2, $\det(\delta G^a_{\rm gf}/\delta\theta^b)$ under an infinitesimal gauge transformation $\delta A^a_\mu = \frac{1}{g}(D_\mu\theta)^a$, $\delta\phi = i\theta^b T^b\phi$, and is exponentiated into anticommuting ghost fields $c^a, \bar{c}^a$. We compute the variation of the two pieces of Eq. (30.216):

$$\delta(\partial_\mu A^{a\mu}) = \tfrac{1}{g}\,\partial_\mu (D^\mu\theta)^a = \tfrac{1}{g}\,\partial_\mu D^\mu_{ab}\,\theta^b, \qquad (30.222)$$

$$\delta(\xi g\, G^a) = \xi g\,[-i(\phi_0^\dagger T^a(i\theta^b T^b\phi) - (-i\theta^b\phi^\dagger T^b)T^a\phi_0)] = \xi g\,\theta^b(\phi_0^\dagger T^a T^b\phi + \phi^\dagger T^b T^a\phi_0). \tag{30.223}$$

Evaluating the second line at $\phi = \phi_0$ for the field-independent (mass) part, $\phi_0^\dagger T^a T^b\phi_0 + \phi_0^\dagger T^b T^a\phi_0 = \phi_0^\dagger\left\{T^a, T^b\right\}\phi_0 = (M^2)_{ab}/g^2$, so the ghost Lagrangian is

$$\mathcal{L}_{\text{ghost}} = \bar{c}^a\,\frac{\delta G^a_{\text{gf}}}{\delta\theta^b}\,c^b = \frac{1}{g}\,\bar{c}^a(\partial_\mu D^\mu_{ab} + \xi(M^2)_{ab})c^b + (\text{interactions with }\eta). \tag{30.224}$$

Absorbing the overall $1/g$ into the ghost normalization, the ghost kinetic operator is the unbroken-theory operator $-\partial_\mu D^\mu_{ab}$ of Section 26.2 shifted by a mass term: the ghosts acquire the mass matrix

$$M^2_{\text{ghost}} = \xi\, M^2, \tag{30.225}$$

identical to the would-be Goldstone mass Eq. (30.221) and to the gauge propagator's spurious pole. This matches the abelian ghost mass $m_c^2 = \xi m_A^2$ of Eq. (30.101); the difference from the abelian case is that here the ghosts are interacting and non-decoupling already in the unbroken theory (Section 26.2), and the Higgs mechanism merely gives them a mass.

Diagonalization to the abelian spectrum. With the mixing cancelled Eq. (30.218), transform every field to the mass eigenbasis of M^2. The quadratic Lagrangian decouples into $\dim G$ independent towers, one per generator. An unbroken direction ($m = 0$) gives a free massless gauge field, a massless decoupled would-be Goldstone (which is in fact absent — there is no $T^{\hat{a}}\phi_0$ direction to shift), and the ordinary massless ghost: the unbroken H gauge theory, untouched. Each broken direction with eigenvalue $m_{\check{a}}^2$ gives

$$(\text{massive vector},\, m_{\check{a}}^2)\ \oplus\ (\text{would-be Goldstone},\, \xi m_{\check{a}}^2)\ \oplus\ (\text{ghost–antighost},\, \xi m_{\check{a}}^2), \tag{30.226}$$

which is *precisely* the abelian R_ξ content of Theorem 30.10 with $m_A \to m_{\check{a}}$. The gauge propagator in this basis is the diagonal abelian R_ξ form

$$D^{ab}_{\mu\nu}(k) = \frac{-i\,\delta^{ab}}{k^2 - m_a^2 + i\epsilon}\left[\eta_{\mu\nu} - (1-\xi)\frac{k_\mu k_\nu}{k^2 - \xi m_a^2}\right] \qquad (\text{mass eigenbasis}), \tag{30.227}$$

falling as $1/k^2$ at large k for finite ξ (manifest renormalizability), collapsing to the Feynman–'t Hooft form at $\xi = 1$, and tending to the unitary propagator with the would-be Goldstone and ghost pushed to infinite mass as $\xi \to \infty$ (Eq. (30.226)), exactly as proved abelian-by-abelian in Theorem 30.10. The gauge-independence of physical quantities and the decoupling of the ξ-dependent poles, established for the abelian model in Section 30.5 via the BRST/Slavnov–Taylor identities, carry over generator by generator; we do not repeat that argument, as the non-abelian BRST machinery of Section 26.2 supplies it in full.

30.7.5 Worked examples

We make the formula concrete in two complementary cases: complete breaking of SU(2) by a fundamental (the electroweak prototype, minus hypercharge), and partial breaking SU(2) → U(1) by an adjoint (a massless "photon" from a surviving generator).

Worked Example 30.19 (SU(2) broken by a complex doublet)**.** Take $G = \mathrm{SU}(2)$ ($\dim G = 3$), coupling g, and ϕ a complex doublet ($\dim R = 2$) in the fundamental, with generators $T^a = \sigma^a/2$ the Pauli matrices and the complex kinetic term $(D_\mu\phi)^\dagger(D^\mu\phi)$. A G-invariant potential $V = -\mu^2\phi^\dagger\phi + \lambda(\phi^\dagger\phi)^2$ with $\mu^2 > 0$ is minimized on $\phi^\dagger\phi = v^2/2$, $v = \sqrt{\mu^2/\lambda}$. Using the residual global SU(2) to rotate the vacuum to the lower component,

$$\phi_0 = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix}. \tag{30.228}$$

Which generators are broken. Act with each T^a on ϕ_0:

$$T^1\phi_0 = \frac{1}{2}\frac{v}{\sqrt{2}}\begin{pmatrix}1\\0\end{pmatrix}, \quad T^2\phi_0 = \frac{i}{2}\frac{v}{\sqrt{2}}\begin{pmatrix}-1\\0\end{pmatrix}, \quad T^3\phi_0 = \frac{1}{2}\frac{v}{\sqrt{2}}\begin{pmatrix}0\\-1\end{pmatrix}, \tag{30.229}$$

all nonzero. Every generator is broken: $\dim H = 0$, the local SU(2) is completely broken, and by Theorem 30.18 there are 3 massive vectors and 3 eaten would-be Goldstones, leaving $4 - 3 = 1$ physical scalar (the Higgs h).

The mass matrix. Because $\left\{T^a, T^b\right\} = \frac{1}{4}\left\{\sigma^a, \sigma^b\right\} = \frac{1}{4}\cdot 2\delta^{ab}\mathbf{1} = \frac{1}{2}\delta^{ab}\mathbf{1}$, the complex formula Eq. (30.207) gives

$$(M^2)_{ab} = g^2\,\phi_0^\dagger\left\{T^a, T^b\right\}\phi_0 = g^2\,\tfrac{1}{2}\delta^{ab}\,\phi_0^\dagger\phi_0 = \frac{g^2}{2}\delta^{ab}\frac{v^2}{2} = \frac{g^2v^2}{4}\,\delta^{ab}. \tag{30.230}$$

The three gauge bosons are degenerate with mass

$$m_W = \frac{gv}{2}, \tag{30.231}$$

a degenerate triplet. The degeneracy is the residual global *custodial* SU(2): with only SU(2) gauged and no hypercharge, the vacuum Eq. (30.228) preserves a diagonal global SU(2) that rotates the three massive vectors into one another, forcing equal masses. The factor of 2 in Eq. (30.207) is what makes $m_W = gv/2$ and not $gv/2\sqrt{2}$; this is the number that, once $\mathrm{U}(1)_Y$ is switched on in Chapter 31, splits into $m_W = gv/2$ and $m_Z = \sqrt{g^2 + g'^2}\,v/2$ with the custodial relation $m_W = m_Z\cos\theta_\mathrm{W}$ as its residue.

Worked Example 30.20 (SU(2) → U(1) broken by an adjoint, and a massless photon)**.** Now take $G = \mathrm{SU}(2)$ with a *real* scalar ϕ^a ($a = 1, 2, 3$) in the adjoint

($\dim R = 3$), the Georgi–Glashow setup. The generators are $(T^a)_{bc} = -i\epsilon^{abc}$ (so $f^{abc} = \epsilon^{abc}$), and the adjoint covariant derivative is $(D_\mu\phi)^a = \partial_\mu\phi^a + g\epsilon^{abc}A^b_\mu\phi^c$ with the real kinetic term $\frac{1}{2}(D_\mu\phi)^a(D^\mu\phi)^a$. A G-invariant potential $V = -\frac{1}{2}\mu^2\phi^a\phi^a + \frac{\lambda}{4}(\phi^a\phi^a)^2$ is minimized on $\phi^a\phi^a = v^2$, and we align the vacuum along the third axis,

$$\phi_0^a = v\,\delta^{a3}. \tag{30.232}$$

Which generators are broken. With $(T^a\phi_0)_b = -i\epsilon^{abc}\phi_0^c = -i\epsilon^{ab3}v$,

$$T^3\phi_0 = 0 \;\; (\epsilon^{3b3} = 0), \qquad T^1\phi_0,\; T^2\phi_0 \neq 0. \tag{30.233}$$

The generator T^3 — rotations about the vacuum axis — annihilates ϕ_0 and is unbroken; T^1 and T^2 are broken. Thus $H = \mathrm{U}(1)$, $\dim H = 1$, and the breaking is the partial pattern $\mathrm{SU}(2) \to \mathrm{U}(1)$. The counting theorem gives one massless vector (A^3, the "photon"), two massive vectors, two eaten Goldstones (ϕ^1, ϕ^2), and $3 - 2 = 1$ physical Higgs ($\phi^3 = v + h$).

The mass matrix. Use the *real* formula $(M^2)_{ab} = g^2(T^a\phi_0)^\dagger(T^b\phi_0)$ from the notation watch above (real multiplet, $\frac{1}{2}$ kinetic term). Then

$$\begin{aligned}(T^a\phi_0)^\dagger(T^b\phi_0) &= \sum_d (i\epsilon^{ad3}v)(-i\epsilon^{bd3}v) = v^2\sum_d \epsilon^{ad3}\epsilon^{bd3} = v^2\big(\delta^{ab}\delta^{33} - \delta^{a3}\delta^{3b}\big)\\ &= v^2\big(\delta^{ab} - \delta^{a3}\delta^{b3}\big),\end{aligned} \tag{30.234}$$

using $\epsilon^{ad3}\epsilon^{bd3} = \delta^{ab}\delta^{33} - \delta^{a3}\delta^{3b}$. Hence

$$(M^2)_{ab} = g^2v^2\big(\delta^{ab} - \delta^{a3}\delta^{b3}\big) = g^2v^2\,\mathrm{diag}(1,1,0). \tag{30.235}$$

The eigenvalues are g^2v^2 (twice) and 0 (once): two massive vectors of mass

$$m_W = gv \qquad (\text{the charged } W^\pm = \tfrac{1}{\sqrt{2}}(A^1 \mp iA^2)), \tag{30.236}$$

and one massless vector A^3 — an exact massless gauge boson of the unbroken U(1), the photon of this model, carrying charge 0 while $W^\pm$ carry charge ± 1 under T^3. This is the counting theorem in action with a nontrivial $\dim H$: the surviving generator produces a strictly massless gauge boson, protected by the exact residual U(1) gauge symmetry of Section 30.7.3. The same vacuum, viewed as a map from spatial infinity into the vacuum manifold $G/H = \mathrm{SU}(2)/\mathrm{U}(1) \cong S^2$, is what supports the 't Hooft–Polyakov monopole of Chapter 42; here we note only that the massless photon emerging from $\mathrm{SU}(2) \to \mathrm{U}(1)$ is the direct rehearsal for the electroweak photon of Chapter 31.

The two examples bracket the general result: complete breaking (Theorem 30.19, $\dim H = 0$, a degenerate massive multiplet) and partial breaking (Theorem 30.20, $\dim H = 1$, a surviving massless photon). Both are read off from

a single matrix, $M^2 = 2g^2\,\mathrm{Gram}\{T^a\phi_0\}$, whose kernel is the unbroken algebra and whose nonzero eigenvalues are the squared masses.

This completes the abelian-to-non-abelian generalization that the chapter set out to prove. The next chapter takes the formula Eq. (30.207) and the counting theorem Theorem 30.18 and applies them verbatim to $G = \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ with a Higgs doublet of hypercharge $Y = \frac{1}{2}$: the 4×4 mass matrix diagonalizes into the massive $W^\pm$ and Z and the massless photon, the off-diagonal mixing of the neutral sector defines the weak mixing angle θ_W through the very g, g' block structure of the notation watch, and the unbroken electric-charge generator $Q = T^3 + Y$ is the residual U(1) whose exactness keeps the photon massless. Electroweak theory (Chapter 31) is the non-abelian Higgs mechanism of this section evaluated on one specific group and one specific representation.

Further reading

The mechanism was introduced in the 1964 papers of Englert and Brout [178], Higgs [267], and Guralnik et al. [245]; the historical and conceptual reconstruction in Brout et al. [95] is worth reading alongside them. That the spontaneously broken gauge theory is renormalizable — the result that turned the mechanism from an elegant device into the basis of a predictive theory — is due to 't Hooft [466], with the regularization technology of 't Hooft and Veltman [469]; the R_ξ gauges and the 't Hooft trick that organizes Section 30.4 originate there. The ξ-independence of physical quantities is developed through the Nielsen identities of Nielsen [368]. The Goldstone-boson equivalence theorem traces to Cornwall et al. [138] and Lee et al. [319], with the high-energy formulation used in Section 30.6 sharpened by Chanowitz and Gaillard [119]. Textbook treatments at the level of this chapter, each with a complementary emphasis, are Peskin and Schroeder [396], Schwartz [432], Weinberg [501], and the lectures of Coleman [129]; the abelian and non-abelian Higgs models are worked in all four. The background on spontaneous breaking of the global symmetry sits in Goldstone [231] and is the subject of Chapter 29.

Chapter 31

Electroweak Theory

The gauge-theory arc of this book has, by now, assembled every tool the electroweak synthesis requires, and this chapter is where they are finally pointed at a piece of nature. From Chapter 8 we hold the gauge principle that manufactures a force out of a local phase symmetry; from Chapter 25 the non-abelian Yang–Mills Lagrangian with its self-interacting connection and curvature; from Chapter 29 Goldstone's theorem with its honest counting of broken generators; and from Chapter 30 the Higgs mechanism, worked there in full, by which a would-be Goldstone boson is traded for the longitudinal polarization of a massive vector without sacrificing the gauge invariance that keeps the theory renormalizable. We also carry the chiral decomposition of the Dirac field into left- and right-handed Weyl components from Chapter 3 and Chapter 11. What none of this yet supplies is a reason to gauge any particular group acting on any particular fields. The chapter's first task is to show that the empirical structure of the weak interaction does not merely permit, but *forces*, a spontaneously broken chiral gauge theory with group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$.

The development proceeds in a single logical line. We begin (Section 31.1) from three facts about the weak force — its maximal parity violation, its short range, and its universality — and show that the V−A current, the dimensionful Fermi constant, and partial-wave unitarity together close off every alternative until only the Glashow–Weinberg–Salam group survives, complete with the Gell-Mann–Nishijima relation $Q = T^3 + Y$. With the group fixed we write the most general renormalizable Lagrangian (Section 31.2): the chiral matter content of one generation with its hypercharges, the gauge-kinetic terms, the Higgs doublet with its symmetry-breaking potential, and the Yukawa couplings — and we prove the structural theorem that gauge invariance, given chirality, forbids every explicit mass, so that every mass in nature must be spontaneous in origin.

The central act follows. We give the neutral Higgs component its vacuum value and diagonalize the spectrum (Section 31.3): three gauge bosons eat the would-be Goldstones to become the massive $W^{\pm}$ and Z, one combination of

W^3 and B survives massless as the photon, and the weak mixing angle θ_{W} emerges from the rotation, with $e = g \sin\theta_{\mathrm{W}}$ delivered as an output rather than an input. The same vacuum value, fed into the Yukawa sector (Section 31.4), converts dimensionless couplings into Dirac masses for the charged fermions and locks each Higgs–fermion coupling to the corresponding mass, while the absence of a right-handed neutrino leaves the neutrino massless in the minimal model. Recasting the gauge interactions in the mass eigenbasis (Section 31.5) then produces the charged-current, neutral-current, and electromagnetic Lagrangians; integrating out the heavy W recovers Fermi theory and fixes the electroweak scale numerically at $v \approx 246\,\mathrm{GeV}$.

Two structural questions close the chapter. The tree-level relation $\rho = M_W^2/(M_Z^2 \cos^2\theta_{\mathrm{W}}) = 1$ is shown (Section 31.6) to be the fingerprint of an accidental custodial $\mathrm{SU}(2)_V$ symmetry of the Higgs sector, with its only violations — the gauging of hypercharge and the up–down Yukawa splitting, dominantly the top–bottom mass gap — isolated as spurions controlling the radiative $\Delta\rho$. Finally (Section 31.7) we quantize the broken theory in R_ξ gauges, exhibit the Higgs-mediated cancellation that restores high-energy unitarity in longitudinal gauge-boson scattering, state the 't Hooft–Veltman renormalizability theorem with its structural argument, and verify explicitly that the gauge anomalies cancel generation by generation. Throughout, the chapter pays its debt to QED: the photon of Chapter 13 and the QED vertex of Chapter 17 reappear here not as inputs but as the unbroken remnant of a larger chiral gauge symmetry. The replication into three generations and the resulting CKM mixing and CP violation — the one forward reference of the development — are deferred to Chapter 32.

31.1 The Weak Interaction and the Need for $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$

The machinery assembled over the preceding parts is now complete enough to build a piece of nature. We hold the gauge principle that turns a global phase symmetry into a force (Chapter 8), the non-abelian Yang–Mills Lagrangian with its self-interacting connection (Chapter 25), Goldstone's theorem with its honest degree-of-freedom counting (Chapter 29), and the Higgs mechanism worked in full in its abelian incarnation, where a would-be Goldstone boson is traded for the longitudinal polarization of a massive vector without spoiling gauge invariance (Chapter 30). We also hold the chiral decomposition of the Dirac field into left- and right-handed Weyl components (Theorem 3.24, Eq. (11.82)). What we do not yet have is a reason to point all of this apparatus at any particular group acting on any particular fields. That reason is the empirical structure of the weak interaction, and the burden of this section is to show that the data do not merely *permit* a spontaneously broken chiral gauge theory with group $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$:

they *force* it. We will extract three facts from experiment — that the charged weak current is purely left-handed, that the low-energy interaction is a four-fermion contact term with a dimensionful coupling, and that the electric charges of the participating fermions do not fit inside SU(2) — and we will see each of them close off every alternative until only the Glashow–Weinberg–Salam group remains.

31.1.1 The empirical face of the charged current

Three qualitative facts about the weak interaction were established experimentally before any of its theory existed, and each one is a sharp constraint on what that theory can be.

It is parity-violating, and maximally so. In nuclear beta decay and in muon decay the emitted leptons are polarized: neutrinos are left-handed and antineutrinos right-handed, to the precision of every measurement. A theory that treats the two chiralities of a fermion symmetrically — as electromagnetism does, the photon coupling to the vector current $\bar\psi\gamma^\mu\psi$ which mixes P_L and P_R in equal measure — cannot produce this. The interaction must single out one chirality.

It is short-ranged. Unlike the Coulomb potential, which falls only as $1/r$ because the photon is massless, the weak force has no macroscopic reach; its influence is confined to subnuclear distances. A force mediated by a boson of mass M produces a Yukawa potential e^{-Mr}/r with range $1/M$, so a range of order $10^{-3}\,\mathrm{fm}$ implies a mediator of mass of order $100\,\mathrm{GeV}$ — or, if there is no propagating mediator at all in the low-energy description, a contact interaction, which is the $M \to \infty$ limit of a Yukawa exchange.

It is universal. The same coupling strength governs muon decay $\mu^- \to e^-\bar\nu_e\nu_\mu$, nuclear beta decay, and inverse muon decay, once the participating currents are written down. This universality is the fingerprint of a *gauge* coupling, whose strength is fixed once and for all by a single constant, exactly as the electromagnetic coupling e is common to every charged particle in Eq. (8.146).

To make the first fact precise we use the chiral projectors $P_L = \frac{1}{2}(1-\gamma^5)$ and $P_R = \frac{1}{2}(1+\gamma^5)$ of Eq. (11.82), which split any Dirac field into its Weyl components $\psi_L = P_L\psi$, $\psi_R = P_R\psi$ in the sense of Theorem 3.24. For a single generation of leptons — the electron e and its neutrino ν_e, together with the muon μ and ν_μ needed to write muon decay — the empirical content of charged-current weak interactions is captured by the *charged weak current*

$$J^+_\mu = \bar\nu_e\,\gamma_\mu(1-\gamma^5)\,e + \bar\nu_\mu\,\gamma_\mu(1-\gamma^5)\,\mu, \qquad J^-_\mu = (J^+_\mu)^\dagger = \bar e\,\gamma_\mu(1-\gamma^5)\,\nu_e + \bar\mu\,\gamma_\mu(1-\gamma^5)\,\nu_\mu. \tag{31.1}$$

The factor $(1-\gamma^5) = 2P_L$ is the whole story of parity violation: using the intertwining $\gamma_\mu P_L = P_R\gamma_\mu$ of Eq. (11.82),

$$\bar\nu\,\gamma_\mu(1-\gamma^5)\,e = 2\,\bar\nu\,\gamma_\mu P_L\,e = 2\,\overline{\nu_L}\,\gamma_\mu\,e_L, \tag{31.2}$$

where we used $\overline{P_L \nu}\,\gamma_\mu = \bar{\nu} P_R \gamma_\mu = \bar{\nu}\gamma_\mu P_L$ and the projector property $P_L^2 = P_L$. The current is built *entirely* from left-handed fields; the right-handed components do not appear. This is the "V−A" structure: a vector (γ_μ) minus an axial ($\gamma_\mu\gamma^5$) current, in exactly equal proportion, which is the unique combination that projects onto a single chirality. Any other relative coefficient would couple both chiralities and contradict the observed maximal polarization.

Notation Watch

We write the current with $(1-\gamma^5)$ rather than with P_L so that the Fermi coupling below carries its conventional normalization. The two are interchangeable via Eq. (31.2); the chiral content is what matters, not the factor of two, which is absorbed into the definition of G_F.

31.1.2 Fermi theory and its fatal dimension

Before any mediator was known, Fermi modelled beta decay as a direct, pointlike coupling of four fermions — in modern language, a current–current contact interaction built from Eq. (31.1),

$$\mathcal{L}_{\text{Fermi}} = -\frac{G_F}{\sqrt{2}}\, J_\mu^{+}\, J^{-\,\mu}, \tag{31.3}$$

with a single coupling constant G_F, the Fermi constant, fixed by experiment. Its measured value, most cleanly from the muon lifetime, is

$$G_F = 1.166 \times 10^{-5}\,\text{GeV}^{-2}, \qquad \text{so} \qquad G_F^{-1/2} \approx 293\,\text{GeV}. \tag{31.4}$$

That G_F is not dimensionless is the central structural fact of this section, and it is forced by simple power counting. In four spacetime dimensions the action $\int \mathrm{d}^4x\,\mathcal{L}$ is dimensionless in natural units, so $[\mathcal{L}] = \text{mass}^4$. A Dirac field has $[\psi] = \text{mass}^{3/2}$ (read off from its kinetic term $\bar{\psi}\, i\partial\!\!\!/\,\psi$, which must have dimension mass^4 with $[\partial] = \text{mass}$). The current Eq. (31.1) is bilinear in the fields with one derivative-free Dirac matrix, so $[J_\mu] = \text{mass}^3$, and the product of two currents has dimension mass^6. For Eq. (31.3) to have dimension mass^4 the coupling must carry

$$[G_F] = \text{mass}^4 - \text{mass}^6 = \text{mass}^{-2}. \tag{31.5}$$

A coupling with negative mass dimension is the textbook signature of a *non-renormalizable* interaction: by the superficial-degree-of-divergence power counting of Chapter 21, each additional insertion of a vertex with $[G_F] = \text{mass}^{-2}$ raises the degree of divergence, so an infinite tower of distinct counterterms is required and the theory loses predictivity at high energy. Fermi theory is therefore at best an effective description valid below some scale. Power counting alone tells us a scale exists; it does not tell us where it is. To locate it we turn the dimensionful coupling into a physical prediction that misbehaves, and the cleanest such prediction is a cross section.

31.1.3 Unitarity violation fixes the scale

A dimensionless quantity built from Eq. (31.3) can depend on energy only through the dimensionless combination $G_F\, s$, where $s = (p_1 + p_2)^2$ is the squared centre-of-mass energy. At energies far above all participating masses, s is the only scale in the problem, so on dimensional grounds the tree-level $2 \to 2$ amplitude from a single insertion of the contact vertex must grow linearly with s,

$$\mathcal{M} \sim G_F\, s. \tag{31.6}$$

A $2 \to 2$ scattering amplitude in four dimensions is dimensionless, and Eq. (31.6) is the only dimensionless object linear in the coupling, so the growth is not an artefact of any particular process: every weak cross section computed from Eq. (31.3) inherits it. The corresponding cross section then grows as

$$\sigma \sim \frac{|\mathcal{M}|^2}{s} \sim G_F^2\, s, \tag{31.7}$$

a cross section that increases without bound with energy. No genuine cross section can do this forever, because unitarity of the S-matrix caps how much scattering can occur in any single angular-momentum channel. That cap was derived as the partial-wave unitarity bound (Theorem 16.29): each partial-wave amplitude $a_\ell(s)$ is confined to the Argand disk $|a_\ell| \le 1$, equivalently $|\operatorname{Re} a_\ell| \le \tfrac{1}{2}$. An amplitude that grows like $G_F s$ pierces this disk at a finite energy, and that energy is the scale at which Fermi theory ceases to make sense. The following worked example carries the estimate through with every factor visible.

Worked Example 31.1 (The unitarity-violation scale of Fermi theory)**.** Consider a charged-current process at energies far above the lepton masses, for definiteness inverse muon decay $\nu_\mu\, e^- \to \mu^-\, \nu_e$, which proceeds through Eq. (31.3) with no electromagnetic contamination. We estimate the amplitude, project it onto partial waves with the conventions of Eq. (16.176), and impose Theorem 16.29.

The amplitude. The contact vertex Eq. (31.3) ties the four external spinors together at a point. With all masses neglected the only dimensionful quantity available is s, and a current–current contraction of the V$-$A bilinears produces, after the spinor algebra, an amplitude of the form

$$\mathcal{M} = c\, \frac{G_F}{\sqrt{2}}\, s, \tag{31.8}$$

with c a pure number of order unity set by the spin structure (for this process the explicit evaluation gives $c = 2$, but the argument needs only $c = O(1)$). The left-handed projectors fix the helicities of all four legs, so there is no spin average to dilute the result; the amplitude is essentially monochromatic in angular momentum.

Partial-wave projection. A contact interaction with no momentum dependence beyond the overall factor of s is isotropic: $\mathcal{M}$ is independent of the scattering angle θ, so only the s-wave $\ell = 0$ survives the projection Eq. (16.176). Using $P_0 = 1$ and, at high energy, $p_{\rm cm} = \sqrt{s}/2$,

$$a_0 = \frac{p_{\rm cm}}{16\pi\sqrt{s}} \int_{-1}^{1} \mathrm{d}(\cos\theta)\, \mathcal{M} = \frac{p_{\rm cm}}{16\pi\sqrt{s}}\,(2\,\mathcal{M}) = \frac{\mathcal{M}}{16\pi} = \frac{c\,G_F\,s}{16\sqrt{2}\,\pi}. \tag{31.9}$$

Imposing unitarity. The bound $|a_0| \le 1$ then reads

$$\frac{c\,G_F\,s}{16\sqrt{2}\,\pi} \le 1 \quad\Longrightarrow\quad s \le \frac{16\sqrt{2}\,\pi}{c\,G_F}, \qquad \sqrt{s} \;\le\; \left(\tfrac{16\sqrt{2}\,\pi}{c}\right)^{1/2} G_F^{-1/2}. \tag{31.10}$$

With $c = O(1)$ the numerical prefactor is of order $\sqrt{16\sqrt{2}\,\pi} \approx 8.4$, so tree-level Fermi theory is in manifest conflict with unitarity once

$$\sqrt{s} \;\gtrsim\; \text{a few} \times G_F^{-1/2} \;\sim\; 1\text{–}2\,\text{TeV}, \tag{31.11}$$

while the intrinsic scale set purely by the coupling is $G_F^{-1/2} \approx 293\,\text{GeV}$ from Eq. (31.4). The two numbers say the same thing at different sharpness: *new physics must intervene at or below the TeV scale, and the natural scale of that physics is a few hundred GeV.* This is one of the most important predictions in the history of particle physics — derived here from nothing but the dimension of G_F and the unitarity of the S-matrix — and it was vindicated when the W and Z bosons were found with masses of 80 and 91 GeV, squarely inside the window Eq. (31.11).

The same estimate read through Eq. (31.7) gives the cross section $\sigma \sim G_F^2 s$ (for this process $\sigma = G_F^2 s/\pi$ in the massless limit), the explicit linear growth that the partial-wave bound forbids at large s.

The lesson of Theorem 31.1 is not that Fermi theory is wrong — at low energy it is an excellent description — but that it is the leading term of an expansion that must be completed by new degrees of freedom near the electroweak scale. We now identify what those degrees of freedom must be.

31.1.4 From contact term to massive vector, and why that is not enough

The cure for a too-singular contact interaction is to resolve the point. Replace the four-fermion vertex by the exchange of a massive charged vector boson $W^{\pm}$ coupling to the current Eq. (31.1) with some dimensionless coupling g, as in Fig. 31.1. The vector propagator carries a momentum dependence the contact term lacked, and at momenta $q^2 \ll M_W^2$ it collapses back to a constant,

$$\frac{-i\big(\eta_{\mu\nu} - q_\mu q_\nu / M_W^2\big)}{q^2 - M_W^2} \;\xrightarrow{\;q^2 \ll M_W^2\;}\; \frac{i\,\eta_{\mu\nu}}{M_W^2}, \tag{31.12}$$

so that the exchange reproduces the Fermi Lagrangian Eq. (31.3) with the matching condition

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}. \tag{31.13}$$

This is the precise sense in which G_F is dimensionful for a reason: it is a *ratio* of a dimensionless coupling to the square of a mass, and the apparent non-renormalizability of Eq. (31.3) is the shadow, at energies below M_W, of an honest dimensionless coupling at energies above it. The matching Eq. (31.13) also explains the breakdown scale of Theorem 31.1: the amplitude stops growing once $\sqrt{s}$ reaches M_W, the energy at which the propagator Eq. (31.12) ceases to look like a constant. With g of order the electromagnetic coupling and M_W a few tens of GeV, Eq. (31.13) predicts $M_W \sim 80\,\mathrm{GeV}$, again inside the window.

A massive vector is necessary but not, by itself, sufficient. We proved in Chapter 30 that a vector mass written by hand — the Proca term of Section 30.1.1 — destroys gauge invariance and, through the $q_\mu q_\nu / M_W^2$ piece of Eq. (31.12), leaves a propagator that does not fall at large q^2, so the high-energy disease is displaced rather than cured: amplitudes for producing longitudinally polarized W's grow with energy exactly as the Fermi amplitude did. The only known way to give a vector a mass while keeping the soft ultraviolet behaviour that makes a gauge theory renormalizable is the Higgs mechanism of Chapter 30: the longitudinal polarization is a would-be Goldstone boson eaten from a spontaneously broken symmetry, and gauge invariance, though hidden, still controls the high-energy limit. Thus the empirical facts of Section 31.1.1 have already forced the shape of the answer: a *chiral* (the current is left-handed) *gauge* (universality) theory that is *spontaneously broken* (the mediators are massive but the theory must stay renormalizable). The all-orders proof that this construction is renormalizable is 't Hooft's, and we take it up in Chapter 33. What remains in this section is to determine the group.

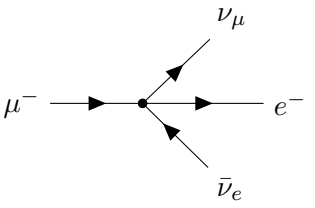
31.1.5 The charged currents demand $\mathrm{SU}(2)_L$

The left-handed content of Eq. (31.1) groups naturally. Collect the left-handed neutrino and electron of one generation into a two-component column,

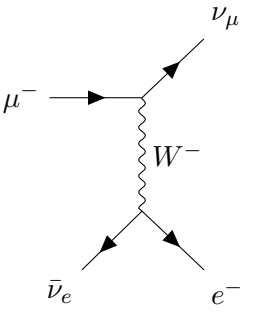
$$L \equiv \begin{pmatrix} \nu_{eL} \\ e_L \end{pmatrix}, \qquad e_L = P_L\, e, \quad \nu_{eL} = P_L\, \nu_e, \tag{31.14}$$

and similarly for the muon. Acting on this column the Pauli matrices give the three Hermitian operators $T^a = \frac{1}{2}\sigma^a$, normalized to $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ as fixed in Chapter 25. Their raising and lowering combinations $T^\pm = T^1 \pm i T^2 = \frac{1}{2}(\sigma^1 \pm i\sigma^2)$ are exactly the off-diagonal matrices

$$T^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad T^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}, \tag{31.15}$$



(a) Fermi contact interaction: the four-fermion vertex of Eq. (31.3).



(b) Its ultraviolet completion: exchange of a massive charged vector W^-, matched by Eq. (31.13).

Figure 31.1: Muon decay $\mu^- \to e^- \bar{\nu}_e \nu_\mu$ as (a) the pointlike Fermi interaction, whose dimensionful coupling violates unitarity near a TeV, and (b) the resolution of that point into W-boson exchange. At momenta well below M_W the propagator Eq. (31.12) shrinks (b) back to (a).

so that $\overline{L}\,\gamma_\mu T^+ L = \overline{\nu_{eL}}\,\gamma_\mu e_L$ and $\overline{L}\,\gamma_\mu T^- L = \overline{e_L}\,\gamma_\mu \nu_{eL}$. Comparing with Eq. (31.2), the charged weak current Eq. (31.1) is precisely $J^+_\mu = 2\,\overline{L}\,\gamma_\mu T^+ L$ summed over generations, and $J^-_\mu = 2\,\overline{L}\,\gamma_\mu T^- L$. The weak charged currents are the matrix elements of two of the three generators of SU(2) acting on the doublet Eq. (31.14).

Two generators cannot stand alone, because they do not close under commutation. The SU(2) algebra forces

$$\left[T^+, T^-\right] = 2\,T^3, \qquad T^3 = \tfrac{1}{2}\sigma^3 = \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix}, \tag{31.16}$$

so the very existence of the charged currents *predicts* a third, neutral current associated with the diagonal generator T^3. There is no consistent way to gauge $T^\pm$ without also gauging T^3: a gauge group must be a group, and the smallest one containing the charged-current generators is SU(2), with the neutral generator T^3 supplied by closure Eq. (31.16). We call it $\mathrm{SU}(2)_L$, the subscript recording that it acts only on left-handed fields; the right-handed fields, which never appear in Eq. (31.1), are $\mathrm{SU}(2)_L$ singlets. This was a genuine prediction — a neutral weak current — and its experimental discovery (neutral-current neutrino scattering) was a triumph of the theory.

31.1.6 Electric charge is not an SU(2) generator: hypercharge is forced

It is tempting to hope the story ends here: perhaps $\mathrm{SU}(2)_L$ alone is the electroweak group, with the photon coupling to one of its generators and the W's to the

others. The hope fails, and the reason is sharp enough to state as a theorem about 2×2 matrices. The obstruction is that the electric charge operator, restricted to the doublet Eq. (31.14), is not an element of the Lie algebra $\mathfrak{su}(2)$ at all.

Worked Example 31.2 (Why $\mathrm{SU}(2)_L$ alone cannot carry electric charge)**. The Lie algebra of** $\mathrm{SU}(2)$**.** A group element $U \in \mathrm{SU}(2)$ near the identity is $U = \exp(i\,\theta^a T^a)$. Unitarity $U^\dagger U = \mathbf{1}$ forces the generators T^a to be Hermitian, and the defining condition $\det U = 1$ forces them to be traceless: expanding $\det U = \exp(\operatorname{tr} \ln U) = \exp(i\,\theta^a \operatorname{tr} T^a)$, the requirement $\det U = 1$ for all θ^a gives

$$\operatorname{tr} T^a = 0 \qquad \text{for every generator of } \mathrm{SU}(2). \tag{31.17}$$

Conversely, every traceless Hermitian 2×2 matrix exponentiates to an element of $\mathrm{SU}(2)$. Hence $\mathfrak{su}(2)$ is exactly the three-dimensional real space of traceless Hermitian 2×2 matrices, spanned by T^1, T^2, T^3.

The charge operator. On the doublet Eq. (31.14) the electric charge operator is diagonal, reading off the measured charges of the neutrino (0) and the electron (-1) in units of e,

$$Q = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \operatorname{tr} Q = -1 \neq 0. \tag{31.18}$$

By Eq. (31.17) a nonzero trace disqualifies Q from membership in $\mathfrak{su}(2)$: *electric charge is not a generator of* $\mathrm{SU}(2)_L$. The neutral generator $T^3 = \operatorname{diag}(\tfrac{1}{2}, -\tfrac{1}{2})$ that closure handed us in Eq. (31.16) is the closest traceless object to Q, but it is not Q: it assigns charge $+\tfrac{1}{2}$ to the neutrino, which is neutral, and $-\tfrac{1}{2}$ to the electron, whose charge is -1.

The decomposition. The mismatch is precisely the trace. Subtract the traceless part:

$$Q - T^3 = \begin{pmatrix} 0 & 0 \\ 0 & -1 \end{pmatrix} - \begin{pmatrix} \frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} & 0 \\ 0 & -\frac{1}{2} \end{pmatrix} = -\tfrac{1}{2}\,\mathbf{1}. \tag{31.19}$$

The remainder is proportional to the identity. A matrix proportional to $\mathbf{1}$ commutes with all of $\mathrm{SU}(2)$ and so generates a separate, commuting $\mathrm{U}(1)$ factor; it is *not* expressible in terms of T^1, T^2, T^3. Calling its eigenvalue the *weak hypercharge* Y — here $Y = -\tfrac{1}{2}$ on the whole left-handed doublet — Eq. (31.19) is the statement

$$\boxed{Q = T^3 + Y} \tag{31.20}$$

the Gell-Mann–Nishijima relation. The smallest Lie algebra containing both the charged-current generators $T^\pm$ and the electric charge Q is therefore $\mathfrak{su}(2) \oplus \mathfrak{u}(1) = \mathfrak{u}(2)$, and the abelian summand is hypercharge.

The right-handed electron clinches it. There is a second, independent proof that $\mathrm{SU}(2)_L$ is insufficient, and it does not even need the trace argument.

The right-handed electron e_R never appears in the charged current Eq. (31.1), so it is an $\mathrm{SU}(2)_L$ singlet: every $\mathrm{SU}(2)_L$ generator annihilates it, $T^a e_R = 0$, and in particular $T^3 e_R = 0$. Yet e_R carries the same electric charge as e_L, namely -1, since electromagnetism does not distinguish chirality (the photon couples to the vector current Eq. (8.146)). If charge came from $\mathrm{SU}(2)_L$ alone we would need $Q\, e_R = T^3 e_R = 0$, contradicting $Q\, e_R = -e_R$. The relation Eq. (31.20) resolves it: with $T^3 = 0$ on the singlet, its charge must come entirely from hypercharge, $Q = Y$, so $Y_{e_R} = -1$. A consistent assignment for one generation is then

$$Y_L = -\tfrac{1}{2} \ \ (\text{on } L), \qquad Y_{e_R} = -1, \tag{31.21}$$

and one checks Eq. (31.20) on each field: $Q(\nu_{eL}) = \frac{1}{2} - \frac{1}{2} = 0$, $Q(e_L) = -\frac{1}{2} - \frac{1}{2} = -1$, and $Q(e_R) = 0 - 1 = -1$, reproducing every measured charge.

The two arguments of Theorem 31.2 converge on the same conclusion from opposite directions — one from the algebra of the doublet, one from the lone singlet e_R — and together they make the extension unavoidable. The electroweak gauge group cannot be $\mathrm{SU}(2)_L$; it must be

$$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y, \tag{31.22}$$

with hypercharge Y the generator of the abelian factor and electric charge given by Eq. (31.20). The covariant derivative that will act on the matter fields is correspondingly

$$D_\mu = \partial_\mu - i g\, W_\mu^a T^a - i g'\, Y\, B_\mu, \tag{31.23}$$

carrying *two* independent couplings: g for $\mathrm{SU}(2)_L$ and g' for $\mathrm{U}(1)_Y$, with W_μ^a ($a = 1, 2, 3$) the $\mathrm{SU}(2)_L$ gauge fields and B_μ the hypercharge gauge field.

Physical Insight

The photon of Eq. (8.146) is not put into the electroweak theory; it must come *out* of it. Electromagnetism couples vectorially and masslessly to $Q = T^3 + Y$, a combination of an $\mathrm{SU}(2)_L$ generator and the $\mathrm{U}(1)_Y$ generator. Since Q lives partly in each factor of Eq. (31.22), the photon must be a specific mixture of the neutral gauge fields W_μ^3 and B_μ — the one linear combination that the symmetry breaking leaves massless. Its orthogonal partner is the massive neutral Z. The angle that rotates (W^3, B) into (Z, γ) is the weak mixing angle θ_{W}, and the relation $e = g \sin\theta_{\mathrm{W}}$ will emerge as an *output* of the construction. This is the unification the gauge-theory arc has been building toward: the electromagnetism of QED is recovered as the unbroken remnant of a larger chiral gauge symmetry.

We have reached the destination this section was charged with. The empirical V−A structure forced a chiral coupling; universality forced a gauge coupling; the

dimensionful Fermi constant and partial-wave unitarity forced a renormalizable completion with massive mediators, hence spontaneous breaking; closure of the charged-current algebra forced $\mathrm{SU}(2)_L$; and the failure of electric charge to be traceless — reinforced by the very existence of the charged but $\mathrm{SU}(2)_L$-singlet e_R — forced the extra $\mathrm{U}(1)_Y$ and the relation $Q = T^3 + Y$ of Eq. (31.20). The gauge group Eq. (31.22) and its covariant derivative Eq. (31.23) are now fixed. What they are not yet is a Lagrangian: we must specify the complete chiral matter content with its hypercharges, add the Higgs doublet whose vacuum value will break Eq. (31.22) down to the electromagnetic U(1), and write the Yukawa couplings that the singlet e_R makes both possible and necessary — the task of Section 31.2, where we also prove that no explicit mass term for any field is gauge-invariant, so that every mass in the theory must be generated by the breaking.

31.2 Gauge Group, Matter Content, and the Electroweak Lagrangian

The previous section settled the two questions a gauge theory must answer before a single term can be written: *which group* gauges the dynamics, and *by what rule* electric charge is built from its generators. The answer was $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ for the electroweak sector, with the hypercharge $\mathrm{U}(1)_Y$ adjoined to weak isospin $\mathrm{SU}(2)_L$ precisely so that the electric charge could be carried by the diagonal combination

$$Q \;=\; T^3 + Y, \tag{31.24}$$

the Gell-Mann–Nishijima relation established in Section 31.1. That relation is not yet a theory: it tells us how charge is *assembled* but says nothing about which fields exist, what representations they occupy, or how they couple. This section supplies all of that. We will fix the matter content of one generation, assign each multiplet its representation under $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and its hypercharge, write the most general renormalizable Lagrangian invariant under the gauge group, and then prove the structural fact that makes the whole construction necessary rather than optional: *gauge invariance, together with the deliberately chiral assignment of the fermions, forbids every explicit mass term in the theory.* Neither the gauge bosons nor the fermions may carry a mass in the Lagrangian we are about to write. Every mass in nature must therefore be manufactured later, by the spontaneous breaking of the gauge symmetry — which is exactly why the Higgs doublet is built in now, with a potential already shaped to acquire a vacuum expectation value.

Structurally this is the Yang–Mills construction of Chapter 25 instantiated at last with a definite group and genuine matter: $G = \mathrm{SU}(2)_L$ supplies a non-abelian field strength of the form derived in Theorem 25.11, the fermions sit in

honest representations (doublets and singlets), and the abelian $\mathrm{U}(1)_Y$ factor rides alongside exactly as the U(1) of Chapter 8 did. We continue to use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, and Hermitian generators normalized by $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the fundamental.

31.2.1 The electroweak covariant derivative

The gauge principle of Chapter 8 (the minimal-coupling prescription Theorem 8.6) instructs us, once the gauge group is fixed, to promote every ordinary derivative acting on a matter field to a covariant derivative built from the connection. For a product group the connection is the sum of one connection per factor, each with its own coupling constant and each acting through the generator appropriate to the representation the field carries. For $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ this gives a two-term gauge potential.

Definition 31.3 (Electroweak covariant derivative)**.** Let a field ψ transform in a representation of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ specified by Hermitian weak-isospin generators T^a $(a = 1, 2, 3)$ and a real hypercharge number Y (the eigenvalue of the $\mathrm{U}(1)_Y$ generator on that field). Its electroweak covariant derivative is

$$D_\mu \;=\; \partial_\mu \;-\; i g\, W_\mu^a\, T^a \;-\; i g'\, Y\, B_\mu, \tag{31.25}$$

where W_μ^a $(a = 1, 2, 3)$ are the three $\mathrm{SU}(2)_L$ gauge fields with coupling g, and B_μ is the single $\mathrm{U}(1)_Y$ gauge field with coupling g'.

This is the general covariant derivative Theorem 8.4 written for the direct-product group: the term $-igW_\mu^a T^a$ is the $\mathrm{SU}(2)_L$ connection of Theorem 25.6 with structure group SU(2), and the term $-ig'YB_\mu$ is the abelian $\mathrm{U}(1)_Y$ connection of Chapter 8, with the abelian charge played by the hypercharge Y. Three points fix its meaning before we use it.

First, the generators T^a are representation-dependent. On an $\mathrm{SU}(2)_L$ *doublet* they are the spin-$\frac{1}{2}$ generators $T^a = \sigma^a/2$ with σ^a the Pauli matrices, satisfying the $\mathfrak{su}(2)$ algebra $[T^a, T^b] = i\epsilon^{abc}T^c$ of Theorem 2.48; on an $\mathrm{SU}(2)_L$ *singlet* they vanish identically, $T^a = 0$, so the entire W_μ^a term drops out and the field is blind to the weak gauge bosons. The hypercharge Y, by contrast, is a single number multiplying the identity in any representation (this is what *abelian* means), so even the $\mathrm{SU}(2)_L$ singlets feel B_μ through their hypercharge.

Second, g and g' are independent. There is no symmetry relating the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ couplings — they belong to different group factors — and their ratio is a free parameter of the theory. It is precisely this ratio that the next section repackages as the weak mixing angle θ_{W}; nothing forces its value at the classical level.

Third, the sign and placement of the couplings follow the book's canon $D_\mu = \partial_\mu - igA_\mu^a T^a$ of Chapter 25 term by term, so that under an infinitesimal

$\mathrm{SU}(2)_L$ gauge transformation with parameters $\alpha^a(x)$ the gauge field shifts as $W_\mu^a \to W_\mu^a + \frac{1}{g}\partial_\mu \alpha^a + \epsilon^{abc}\alpha^b W_\mu^c$ and under a $\mathrm{U}(1)_Y$ transformation with parameter $\beta(x)$ the abelian field shifts as $B_\mu \to B_\mu + \frac{1}{g'}\partial_\mu \beta$, leaving $D_\mu \psi$ transforming homogeneously — in the same representation as ψ itself. This homogeneous transformation is the only property of D_μ we will need, and it is exactly the defining virtue established in Theorem 8.4 and Theorem 25.6.

31.2.2 Field content of one generation

The single empirical fact that dictates the entire matter content is that the weak interaction is *chiral*: it couples to left-handed fermions differently from right-handed ones (the parity violation reviewed in Section 31.1). The only consistent way to encode this in a gauge theory is to put the left- and right-handed parts of each Dirac fermion in *different representations* of the gauge group. We therefore decompose every Dirac field into its Weyl components, $\psi = \psi_L + \psi_R$ with $\psi_{L,R} = P_{L,R}\psi$ and $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$, exactly as in Theorem 3.24, and treat ψ_L and ψ_R as logically independent fields that happen to share a Lorentz-spinor index. The chiral structure is then implemented by the rule that left-handed fields form $\mathrm{SU}(2)_L$ *doublets* (hence the subscript L on the group), while right-handed fields are $\mathrm{SU}(2)_L$ *singlets*.

Definition 31.4 (Electroweak matter content of one generation)**.** A single generation consists of five irreducible chiral multiplets, with the following $(\mathrm{SU}(2)_L,\, \mathrm{U}(1)_Y)$ quantum numbers:

$$\text{left lepton doublet:} \quad L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}, \qquad \mathbf{2},\ Y = -\tfrac{1}{2}, \tag{31.26}$$

$$\text{right charged lepton:} \quad e_R, \qquad \mathbf{1},\ Y = -1, \tag{31.27}$$

$$\text{left quark doublet:} \quad Q = \begin{pmatrix} u_L \\ d_L \end{pmatrix}, \qquad \mathbf{2},\ Y = +\tfrac{1}{6}, \tag{31.28}$$

$$\text{right up-type quark:} \quad u_R, \qquad \mathbf{1},\ Y = +\tfrac{2}{3}, \tag{31.29}$$

$$\text{right down-type quark:} \quad d_R, \qquad \mathbf{1},\ Y = -\tfrac{1}{3}. \tag{31.30}$$

The two members of each doublet are related by $T^3 = \pm\frac{1}{2}$; the upper component has $T^3 = +\frac{1}{2}$ and the lower $T^3 = -\frac{1}{2}$. The quarks additionally carry an $\mathrm{SU}(3)_c$ colour index, on which the electroweak generators act trivially; colour is a spectator throughout this chapter and we suppress it.

The representation assignments (doublet versus singlet) are the encoding of parity violation and are fixed by experiment; what is *not* an independent input is the list of hypercharges. Once the $\mathrm{SU}(2)_L$ representation is chosen and the observed electric charges are known, the Gell-Mann–Nishijima relation

Eq. (31.20) determines Y for every multiplet with no freedom whatsoever. We verify this now, member by member, because the agreement is the first nontrivial check that the group-theoretic skeleton matches nature.

Worked Example 31.5 (Every hypercharge from $Q = T^3 + Y$)**.** We are given the electric charges from experiment ($Q_\nu = 0$, $Q_e = -1$, $Q_u = +\frac{2}{3}$, $Q_d = -\frac{1}{3}$, in units of $|e|$) and the weak-isospin assignments from Theorem 31.4. For a multiplet to have a well-defined hypercharge, $Y = Q - T^3$ must return the *same* value for every member — this is a genuine consistency requirement, not an accounting trick, because Y multiplies the identity and so cannot distinguish the components of an $\mathrm{SU}(2)_L$ multiplet. We check each multiplet.

For the lepton doublet L, the upper member ν_L has $T^3 = +\frac{1}{2}$ and $Q = 0$, giving $Y = 0 - \frac{1}{2} = -\frac{1}{2}$; the lower member e_L has $T^3 = -\frac{1}{2}$ and $Q = -1$, giving $Y = -1 + \frac{1}{2} = -\frac{1}{2}$. Both members return $Y = -\frac{1}{2}$: the assignment is consistent and equals the value in Eq. (31.26).

For the quark doublet Q, the upper member u_L has $T^3 = +\frac{1}{2}$ and $Q = +\frac{2}{3}$, giving $Y = \frac{2}{3} - \frac{1}{2} = +\frac{1}{6}$; the lower member d_L has $T^3 = -\frac{1}{2}$ and $Q = -\frac{1}{3}$, giving $Y = -\frac{1}{3} + \frac{1}{2} = +\frac{1}{6}$. Both return $Y = +\frac{1}{6}$, matching Eq. (31.28).

The right-handed fields are $\mathrm{SU}(2)_L$ singlets, so $T^3 = 0$ and $Y = Q$ directly: e_R has $Y = -1$, u_R has $Y = +\frac{2}{3}$, d_R has $Y = -\frac{1}{3}$, reproducing Eqs. (31.27), (31.29) and (31.30). The complete bookkeeping is collected in Table 31.1.

Table 31.1: Electroweak quantum numbers of one generation. The last column is computed, not assumed: $Q = T^3 + Y$ with T^3 and Y read from the first three columns. The two members of each doublet share a single hypercharge, the consistency check of Theorem 31.5.

Field	$\mathrm{SU}(2)_L$	T^3	Y	$Q = T^3 + Y$
ν_L	$\mathbf{2}$ (L)	$+\frac{1}{2}$	$-\frac{1}{2}$	0
e_L		$-\frac{1}{2}$	$-\frac{1}{2}$	-1
e_R	$\mathbf{1}$	0	-1	-1
u_L	$\mathbf{2}$ (Q)	$+\frac{1}{2}$	$+\frac{1}{6}$	$+\frac{2}{3}$
d_L		$-\frac{1}{2}$	$+\frac{1}{6}$	$-\frac{1}{3}$
u_R	$\mathbf{1}$	0	$+\frac{2}{3}$	$+\frac{2}{3}$
d_R	$\mathbf{1}$	0	$-\frac{1}{3}$	$-\frac{1}{3}$
ϕ^+	$\mathbf{2}$ (Φ)	$+\frac{1}{2}$	$+\frac{1}{2}$	$+1$
ϕ^0		$-\frac{1}{2}$	$+\frac{1}{2}$	0

The last two rows anticipate the Higgs doublet of Section 31.2.5; they are included here so that the single neutral state ϕ^0 that will acquire a vacuum value can be seen, in advance, to carry $Q = 0$, as the unbroken electromagnetism of

the next section requires.

31.2.3 Gauge kinetic terms

The dynamics of the gauge fields themselves is governed by their field strengths. For the non-abelian factor $\mathrm{SU}(2)_L$ we import the Yang–Mills field strength of Theorem 25.11, specialized to $\mathrm{SU}(2)$ where the structure constants are $f^{abc} = \epsilon^{abc}$ (worked out in Theorem 25.13); for the abelian factor $\mathrm{U}(1)_Y$ the field strength is the ordinary Maxwell curl of Chapter 8. Explicitly,

$$W^a_{\mu\nu} = \partial_\mu W^a_\nu - \partial_\nu W^a_\mu + g\,\epsilon^{abc}\,W^b_\mu W^c_\nu, \tag{31.31}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \tag{31.32}$$

The cubic self-coupling $g\,\epsilon^{abc}W^b_\mu W^c_\nu$ in Eq. (31.31) is the entire non-abelian content: it is the term that, in Eq. (25.48), made the field strength a nonlinear functional of the potential, and it is what gives the W bosons their self-interactions. The abelian $B_{\mu\nu}$ has no such term because $\mathrm{U}(1)_Y$ is abelian and its single "structure constant" vanishes.

The gauge-invariant, Lorentz-invariant, renormalizable kinetic Lagrangian is the sum of one Yang–Mills term Eq. (25.76) per group factor,

$$\mathcal{L}_{\text{gauge}} \;=\; -\tfrac{1}{4}\,W^a_{\mu\nu}\,W^{a\,\mu\nu} \;-\; \tfrac{1}{4}\,B_{\mu\nu}\,B^{\mu\nu}. \tag{31.33}$$

Each term is separately invariant: the first is $-\frac{1}{2}\,\mathrm{tr}(W_{\mu\nu}W^{\mu\nu})$ with $W_{\mu\nu} = W^a_{\mu\nu}T^a$, invariant under $\mathrm{SU}(2)_L$ because the trace is cyclic and $W_{\mu\nu}$ transforms in the adjoint (Theorem 25.14); the second is manifestly invariant because $B_{\mu\nu}$ is gauge-invariant on its own. There is no room for a cross term $W^a_{\mu\nu}B^{\mu\nu}$: it would require an $\mathrm{SU}(2)_L$ index a to be contracted against an $\mathrm{SU}(2)_L$ singlet $B_{\mu\nu}$, which is not invariant under $\mathrm{SU}(2)_L$ (it transforms in the adjoint $\mathbf{3}$, not the singlet). A kinetic mixing of the two abelian-looking field strengths is likewise impossible at this stage because $W^a_{\mu\nu}$ is not gauge invariant. Thus Eq. (31.33) is the unique renormalizable gauge-kinetic Lagrangian, with no free parameters beyond the two couplings g, g' already present in the covariant derivative.

31.2.4 Fermion kinetic and gauge-interaction terms

With the covariant derivative Eq. (31.25) in hand, each chiral multiplet contributes a single gauge-invariant kinetic term of Dirac form, $\bar\psi\, i\not{D}\,\psi$, in which $\not{D} = \gamma^\mu D_\mu$ carries the representation data of that multiplet. Because the left- and right-handed fields are independent, each gets its own term with its own covariant derivative; there is no cross term linking them (such a term would be a mass, forbidden as we prove in Section 31.2.7). Collecting the five multiplets of Theorem 31.4,

$$\mathcal{L}_{\text{fermion}} = \bar L\, i\not{D}\, L + \bar e_R\, i\not{D}\, e_R + \bar Q\, i\not{D}\, Q + \bar u_R\, i\not{D}\, u_R + \bar d_R\, i\not{D}\, d_R, \tag{31.34}$$

where each D_μ is Eq. (31.25) evaluated with that multiplet's generators and hypercharge. Made explicit, the doublet covariant derivatives carry the full $\sigma^a/2$ matrices,

$$D_\mu L = \Big(\partial_\mu - i g\, W_\mu^a \tfrac{\sigma^a}{2} + i\tfrac{1}{2} g'\, B_\mu\Big) L, \qquad D_\mu Q = \Big(\partial_\mu - i g\, W_\mu^a \tfrac{\sigma^a}{2} - i\tfrac{1}{6} g'\, B_\mu\Big) Q, \tag{31.35}$$

while the singlet covariant derivatives have no W_μ^a at all, only the hypercharge coupling to B_μ,

$$D_\mu e_R = (\partial_\mu + i g'\, B_\mu) e_R, \quad D_\mu u_R = (\partial_\mu - i\tfrac{2}{3} g'\, B_\mu) u_R, \quad D_\mu d_R = (\partial_\mu + i\tfrac{1}{3} g'\, B_\mu) d_R. \tag{31.36}$$

These derivatives encode every gauge interaction of the fermions: the charged-current couplings to W^1, W^2, the neutral-current couplings to W^3 and B, and (after diagonalization in the next section) the ordinary electromagnetic coupling. We do not expand them into mass-eigenstate currents here — that is the work of Section 31.3 once W^3 and B are rotated into Z and the photon — but the entire content of the weak and electromagnetic interactions is already present, compressed, in Eqs. (31.35) and (31.36).

31.2.5 The Higgs doublet and its potential

Everything so far is massless and, in fact, *must* remain so (the proof is Section 31.2.7). To give the theory a route to mass without spoiling gauge invariance we introduce one more field, a complex scalar, chosen so that giving it a vacuum value breaks $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ down to the electromagnetic $\mathrm{U}(1)_{\mathrm{em}}$ generated by Q. The mechanism was worked in full for a single abelian charge in Chapter 30 and for a general non-abelian group in Section 30.7; here we choose the smallest representation that does the job.

Definition 31.6 (The Higgs doublet)**.** The Higgs field is a complex $\mathrm{SU}(2)_L$ doublet with hypercharge $Y = +\tfrac{1}{2}$,

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad \mathbf{2},\ Y = +\tfrac{1}{2}, \tag{31.37}$$

whose upper component ϕ^+ has $T^3 = +\tfrac{1}{2}$ and hence $Q = +1$, and whose lower component ϕ^0 has $T^3 = -\tfrac{1}{2}$ and hence $Q = 0$ (last two rows of Table 31.1). It carries four real degrees of freedom.

The choice $Y = +\tfrac{1}{2}$ is what makes one component electrically neutral: only a neutral field may acquire a vacuum value without breaking electromagnetism, and Eq. (31.20) shows the neutral state is the lower one, ϕ^0. The Higgs Lagrangian consists of a gauge-covariant kinetic term and the most general renormalizable, gauge-invariant potential,

$$\mathcal{L}_{\mathrm{Higgs}} = (D_\mu \Phi)^\dagger (D^\mu \Phi) \; - \; V(\Phi), \tag{31.38}$$

$$V(\Phi) = -\mu^2\,\Phi^\dagger\Phi \;+\; \lambda\,(\Phi^\dagger\Phi)^2, \qquad \mu^2 > 0,\;\; \lambda > 0, \tag{31.39}$$

with the doublet covariant derivative

$$D_\mu\Phi = \Big(\partial_\mu - i g\, W_\mu^a\, \tfrac{\sigma^a}{2} - i\tfrac{1}{2} g'\, B_\mu\Big)\Phi. \tag{31.40}$$

Only the combination $\Phi^\dagger\Phi$ can appear in V, because it is the unique gauge invariant quadratic in Φ: under $\mathrm{SU}(2)_L$, $\Phi \to U\Phi$ with $U^\dagger U = 1$, so $\Phi^\dagger\Phi \to \Phi^\dagger U^\dagger U\Phi = \Phi^\dagger\Phi$, and under $\mathrm{U}(1)_Y$ the phases cancel between $\Phi^\dagger$ and Φ. Renormalizability (mass dimension ≤ 4 for the operators, the scalar having dimension one in four dimensions) then allows only the quadratic and quartic invariants written. A cubic term $(\Phi^\dagger\Phi)^{3/2}$ is non-polynomial and excluded; a linear term is impossible for a doublet.

The decisive sign is that of the quadratic term. We have written it as $-\mu^2\Phi^\dagger\Phi$ with $\mu^2 > 0$, so that the origin $\Phi = 0$ is a local *maximum* of V rather than a minimum: the curvature of V at the origin, $\partial^2 V/\partial\Phi^\dagger\partial\Phi\big|_0 = -\mu^2 < 0$, is negative. The field is therefore unstable at $\Phi = 0$ and rolls to the circle of degenerate minima at $\Phi^\dagger\Phi = \mu^2/2\lambda$, exactly the symmetry-breaking landscape analyzed in Chapter 29. We do not yet pick a vacuum or expand around it — that is the first act of Section 31.3 — but the potential is now *primed*: its shape guarantees a nonzero vacuum value $\langle\Phi^\dagger\Phi\rangle = \mu^2/2\lambda \equiv v^2/2$, and the neutral component ϕ^0 is the field that will carry it.

31.2.6 The Yukawa couplings

The fermion masses cannot come from the Higgs potential, which involves only Φ; they come from gauge-invariant couplings of Φ to fermion bilinears — the Yukawa interactions. A Yukawa term pairs a left-handed doublet, a right-handed singlet, and one power of the Higgs doublet in an $\mathrm{SU}(2)_L$-invariant, $\mathrm{U}(1)_Y$-invariant, Lorentz-scalar combination. For the down-type fermions (the charged lepton e and the down quark d) the doublet Φ itself does the job; for the up-type fermion (u) we will need its charge conjugate.

Consider first the charged-lepton coupling $\bar{L}\,\Phi\,e_R$. The Lorentz structure is a scalar: $\bar{L}$ contains the left-handed conjugate $\bar{e}_L$ and e_R is right-handed, so $\bar{L}\,(\cdots)\,e_R$ is of the form $\bar{\psi}_L\psi_R$, a genuine Lorentz scalar (the LR bilinear of Theorem 3.24). The $\mathrm{SU}(2)_L$ structure is a singlet: $\bar{L}$ is an anti-doublet ($\bar{\mathbf{2}}$) and Φ is a doublet ($\mathbf{2}$), and $\bar{\mathbf{2}}\otimes\mathbf{2} = \mathbf{1}\oplus\mathbf{3}$ contains exactly one singlet, formed by the contraction $\bar{L}\,\Phi = \bar{L}^i\,\Phi_i$ summed over the isospin index i. The hypercharge must sum to zero for $\mathrm{U}(1)_Y$ invariance, and it does: $\bar{L}$ carries $Y = +\tfrac{1}{2}$ (opposite the $-\tfrac{1}{2}$ of L), Φ carries $Y = +\tfrac{1}{2}$, and e_R carries $Y = -1$, so the total is $+\tfrac{1}{2} + \tfrac{1}{2} - 1 = 0$. The same arithmetic works for the down quark: $\bar{Q}\,\Phi\,d_R$ has $Y = -\tfrac{1}{6} + \tfrac{1}{2} - \tfrac{1}{3} = 0$.

The up quark is the exception. The term $\bar{Q}\,\Phi\,u_R$ would require hypercharge $-\tfrac{1}{6} + \tfrac{1}{2} + \tfrac{2}{3} = +1 \neq 0$, so it is forbidden. What we need is a Higgs-like doublet

of hypercharge $-\frac{1}{2}$ instead of $+\frac{1}{2}$, so that the sum becomes $-\frac{1}{6}-\frac{1}{2}+\frac{2}{3}=0$. We could introduce a second scalar doublet, but it is unnecessary: the charge conjugate of Φ already furnishes such an object.

Definition 31.7 (The conjugate Higgs doublet)**.** The conjugate doublet is

$$\widetilde{H} \;\equiv\; i\sigma^2\,\Phi^* \;=\; \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} \phi^{+*} \\ \phi^{0*} \end{pmatrix} \;=\; \begin{pmatrix} \phi^{0*} \\ -\phi^{+*} \end{pmatrix}, \tag{31.41}$$

where σ^2 is the second Pauli matrix (often written σ^2). Its upper component ϕ^{0*} has $Q=0$ and its lower component $-\phi^{+*}$ has $Q=-1$.

The claim is that $\widetilde{H}$ transforms exactly as an $\mathrm{SU}(2)_L$ doublet but with hypercharge $-\frac{1}{2}$. We prove both halves.

Worked Example 31.8 ($\widetilde{H}=i\sigma^2\Phi^*$ is a doublet of hypercharge $-\frac{1}{2}$)**.** $\mathrm{SU}(2)_L$ *transformation.* Under $\mathrm{SU}(2)_L$ the doublet transforms as $\Phi\to U\Phi$ with $U=\exp(i\alpha^a\frac{\sigma^a}{2})\in\mathrm{SU}(2)$. Its complex conjugate transforms as $\Phi^*\to U^*\Phi^*$, so

$$\widetilde{H} \;=\; i\sigma^2\Phi^* \;\longrightarrow\; i\sigma^2U^*\Phi^*. \tag{31.42}$$

We must show $i\sigma^2U^*=U\,i\sigma^2$, i.e. that $\sigma^2U^*(\sigma^2)^{-1}=U$. The key identity is the reality property of the Pauli matrices under conjugation by σ^2,

$$\sigma^2\,\sigma^{a*}\,\sigma^2 \;=\; -\,\sigma^a, \qquad a=1,2,3, \tag{31.43}$$

which follows because σ^1,σ^3 are real and anticommute with σ^2 (so $\sigma^2\sigma^{1,3}\sigma^2=-\sigma^{1,3}$, using $(\sigma^2)^2=1$), while σ^2 is pure imaginary, $\sigma^{2*}=-\sigma^2$, and commutes with itself ($\sigma^2\sigma^{2*}\sigma^2=-\sigma^2(\sigma^2)^2=-\sigma^2$). With $(\sigma^2)^2=1$ so $(\sigma^2)^{-1}=\sigma^2$, the identity Eq. (31.43) gives

$$\sigma^2U^*\sigma^2=\sigma^2\,\exp\!\Big(-i\alpha^a\tfrac{\sigma^{a*}}{2}\Big)\,\sigma^2=\exp\!\Big(-i\alpha^a\,\tfrac{\sigma^2\sigma^{a*}\sigma^2}{2}\Big)=\exp\!\Big(+i\alpha^a\tfrac{\sigma^a}{2}\Big)=U, \tag{31.44}$$

where we used that $\sigma^2f(\sigma^{a*})\sigma^2=f(\sigma^2\sigma^{a*}\sigma^2)$ for any power series f because $(\sigma^2)^2=1$ lets us insert $\sigma^2\sigma^2=1$ between every factor. Multiplying Eq. (31.44) on the right by σ^2 gives $\sigma^2U^*=U\sigma^2$, hence $i\sigma^2U^*=U\,i\sigma^2$, and Eq. (31.42) becomes

$$\widetilde{H} \;\longrightarrow\; U\,(i\sigma^2\Phi^*) \;=\; U\,\widetilde{H}. \tag{31.45}$$

So $\widetilde{H}$ transforms in the fundamental $\mathbf{2}$, identically to Φ.

$\mathrm{U}(1)_Y$ *transformation.* Under $\mathrm{U}(1)_Y$ with parameter β, a field of hypercharge Y transforms as $\Phi\to e^{i\beta Y}\Phi=e^{+i\beta/2}\Phi$. Its conjugate transforms with the opposite phase, $\Phi^*\to e^{-i\beta/2}\Phi^*$, and since σ^2 carries no hypercharge,

$$\widetilde{H}=i\sigma^2\Phi^* \;\longrightarrow\; e^{-i\beta/2}\,i\sigma^2\Phi^* \;=\; e^{-i\beta/2}\,\widetilde{H} \;=\; e^{i\beta(-1/2)}\,\widetilde{H}. \tag{31.46}$$

Comparing to the defining transformation $e^{i\beta Y}$, the hypercharge of $\widetilde{H}$ is $Y = -\frac{1}{2}$, as claimed. Its upper component ϕ^{0*} then has $Q = T^3 + Y = \frac{1}{2} - \frac{1}{2} = 0$ and its lower component $-\phi^{+*}$ has $Q = -\frac{1}{2} - \frac{1}{2} = -1$, consistent with $\widetilde{H}$ being built from the conjugates of the $Q = +1$ and $Q = 0$ components of Φ.

With $\widetilde{H}$ available, all three Yukawa couplings of one generation can be written in gauge-invariant form. The Yukawa Lagrangian is

$$\mathcal{L}_{\text{Yukawa}} = -\, y_e\, \bar{L}\, \Phi\, e_R \;-\; y_d\, \bar{Q}\, \Phi\, d_R \;-\; y_u\, \bar{Q}\, \widetilde{H}\, u_R \;+\; \text{h.c.}, \tag{31.47}$$

with dimensionless coupling constants y_e, y_d, y_u (the "Yukawa couplings"). Each term is an $\mathrm{SU}(2)_L$ singlet (the doublet index of $\bar{L}$ or $\bar{Q}$ contracted against the doublet index of Φ or $\widetilde{H}$), carries zero net hypercharge by the arithmetic above (for the up term: $Y_{\bar{Q}} + Y_{\widetilde{H}} + Y_{u_R} = -\frac{1}{6} - \frac{1}{2} + \frac{2}{3} = 0$), and is a Lorentz scalar of $\bar{\psi}_L \psi_R$ form. The Hermitian conjugates supply the corresponding $\bar{\psi}_R \psi_L$ terms, so $\mathcal{L}_{\text{Yukawa}}$ is real. These couplings carry no mass dimension of their own and, crucially, are *not* mass terms: each contains one power of the scalar field, so it is a genuine cubic interaction. A fermion mass will appear only when ϕ^0 is replaced by its vacuum value, converting $y_f\, \bar{\psi}_L \phi^0 \psi_R$ into $y_f \frac{v}{\sqrt{2}}\, \bar{\psi}_L \psi_R$ — a mass $m_f = y_f v/\sqrt{2}$ — in the next section. For one generation the y_f are real numbers and there is no flavor structure; the matrix generalization to three generations, and the mismatch between mass and flavor bases that produces the CKM matrix, is deferred to Chapter 32.

31.2.7 No explicit mass term is allowed

We can now prove the structural fact promised at the outset: the Lagrangian assembled from Eqs. (31.33), (31.34), (31.38) and (31.47) is the *complete* renormalizable, gauge-invariant Lagrangian, and it contains no mass for any field. Every candidate mass term is forbidden by the chiral, hypercharged assignment of Theorem 31.4. This is not a choice we make; it is a theorem about what gauge invariance permits.

Theorem 31.9 (Gauge invariance forbids all explicit masses)**.** *Under the field content and representation assignments of Theorems 31.4 and 31.6, no* $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$*-invariant, Lorentz-invariant term of mass dimension* ≤ 4 *exists that gives an explicit mass to (i) any fermion, by a Dirac or Majorana bilinear, or (ii) any gauge boson. Consequently every mass in the theory must originate from spontaneous symmetry breaking.*

Proof. (i$_a$) Dirac fermion masses. A Dirac mass for a fermion ψ is $m\,\bar{\psi}\psi = m(\bar{\psi}_L \psi_R + \bar{\psi}_R \psi_L)$: it necessarily links a left-handed field to a right-handed one. In our theory the left-handed fields live only in the doublets L, Q and the right-handed fields only in the singlets e_R, u_R, d_R. Consider the only candidate that

has the right Lorentz and (naively) electric-charge structure, the electron Dirac mass $\bar{e}_L\, e_R$. Here e_L is the lower component of the doublet L, so the bilinear $\bar{L}\, e_R$ transforms under $\mathrm{SU}(2)_L$ as $\bar{\mathbf{2}} \otimes \mathbf{1} = \bar{\mathbf{2}}$ — a doublet, *not* a singlet — and is therefore not $\mathrm{SU}(2)_L$ invariant: there is no second isospin index to contract the surviving one against. Even if one ignored $\mathrm{SU}(2)_L$ and looked only at the abelian factor, the hypercharge of $\bar{L}\, e_R$ is $Y_{\bar{L}} + Y_{e_R} = +\frac{1}{2} + (-1) = -\frac{1}{2} \neq 0$, so it is not $\mathrm{U}(1)_Y$ invariant either. The identical obstruction afflicts every other Dirac bilinear: $\bar{Q}\, d_R$ has $\mathrm{SU}(2)_L$ structure $\bar{\mathbf{2}}$ and hypercharge $-\frac{1}{6} - \frac{1}{3} = -\frac{1}{2} \neq 0$; $\bar{Q}\, u_R$ has $\mathrm{SU}(2)_L$ structure $\bar{\mathbf{2}}$ and hypercharge $-\frac{1}{6} + \frac{2}{3} = +\frac{1}{2} \neq 0$. None is gauge invariant. The would-be mass terms are precisely the objects that the Yukawa couplings Eq. (31.47) *repair* by inserting one Higgs doublet to soak up both the leftover isospin index and the leftover half-unit of hypercharge; without that scalar insertion they are forbidden.

(i_b) Majorana fermion masses. A Majorana mass for a left-handed field ψ_L has the form $\frac{1}{2} M\, \psi_L^{\mathsf{T}} C\, \psi_L + \text{h.c.}$, quadratic in the *same* chirality (here C is the charge-conjugation matrix). For the neutrino, the only neutral fermion, this would be built from $\nu_L \subset L$. The bilinear $L^{\mathsf{T}} C\, L$ has $\mathrm{SU}(2)_L$ structure $\mathbf{2} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3}$. The antisymmetric singlet $\epsilon_{ij} L^i C L^j$ vanishes identically: for two *identical* anticommuting fields the spinor bilinear $L^{\mathsf{T}} C\, L$ is symmetric under exchange of the two factors (the antisymmetry of C compensated by Fermi statistics), so contracting it against the antisymmetric ϵ_{ij} gives zero. The surviving object is the symmetric $T = 1$ triplet, which is not an $\mathrm{SU}(2)_L$ singlet in the first place and in any case carries hypercharge $Y_L + Y_L = -\frac{1}{2} - \frac{1}{2} = -1 \neq 0$; it is forbidden by both isospin and hypercharge (matching Section 31.4.5). (It also carries two units of lepton number, manifesting the fact that a Majorana mass violates lepton number.) A Majorana mass for the right-handed neutrino is excluded for the trivial reason that the minimal content of Theorem 31.4 contains *no* ν_R to write it for. Hence no neutrino mass of any kind is allowed: in the minimal one-generation model the neutrino is exactly massless. (The empirical fact that neutrinos do have tiny masses requires an extension — a right-handed neutrino, or a higher-dimension operator — taken up in Chapter 35; it does not affect the present chapter.)

(ii) Gauge-boson masses. A mass term for a vector field is quadratic in the potential, $\frac{1}{2} m_W^2\, W_\mu^a W^{a\,\mu}$ for the $\mathrm{SU}(2)_L$ fields or $\frac{1}{2} m_B^2\, B_\mu B^\mu$ for the abelian field. Neither is gauge invariant. For B_μ the obstruction is immediate: under $\mathrm{U}(1)_Y$ the abelian field shifts inhomogeneously, $B_\mu \to B_\mu + \frac{1}{g'} \partial_\mu \beta$, so

$$B_\mu B^\mu \;\longrightarrow\; B_\mu B^\mu + \tfrac{2}{g'}\, B^\mu \partial_\mu \beta + \tfrac{1}{g'^2}\, (\partial_\mu \beta)(\partial^\mu \beta) \;\neq\; B_\mu B^\mu, \tag{31.48}$$

the same calculation that showed the photon mass term breaks U(1) gauge invariance in Section 30.1. For the $\mathrm{SU}(2)_L$ fields the obstruction is the non-abelian transformation law: writing $W_\mu = W_\mu^a T^a$, an infinitesimal gauge transformation acts as $W_\mu \to W_\mu + \frac{1}{g} \partial_\mu \alpha + i[\,\alpha, W_\mu\,]$ with $\alpha = \alpha^a T^a$, so the gauge-invariant

trace shifts by

$$\operatorname{tr}(W_\mu W^\mu) \;\longrightarrow\; \operatorname{tr}(W_\mu W^\mu) + \tfrac{2}{g}\operatorname{tr}(W^\mu\,\partial_\mu\alpha) + \mathcal{O}(\alpha^2) \;\neq\; \operatorname{tr}(W_\mu W^\mu), \quad (31.49)$$

where the commutator term dropped out of the linear variation by the cyclicity of the trace but the inhomogeneous $\partial_\mu\alpha$ term does not. (This is the non-abelian Proca obstruction established in Section 30.7.) Thus neither $W^a_\mu W^{a\,\mu}$ nor $B_\mu B^\mu$ can appear, and the gauge-kinetic Lagrangian Eq. (31.33) is the complete bosonic gauge sector.

Having exhausted the candidates — Dirac, Majorana, and vector — and found each forbidden, we conclude that the Lagrangian contains no explicit mass scale for any propagating field except the dimensionful Higgs parameter μ^2 in the potential Eq. (31.39). That single parameter, through spontaneous breaking, will be the source of every physical mass in the theory. □

Physical Insight

The masslessness is not an accident of the model-builder's restraint; it is the *price of chirality*. The moment one decides that the weak force shall act differently on left- and right-handed fermions — the one inescapable lesson of parity violation — one is forced to put them in different gauge representations, and gauge invariance then refuses every term that would pair them into a mass. The same chirality that explains the data forbids the masses that the data also demand. The only escape is to let the vacuum, rather than the Lagrangian, carry the symmetry-breaking information: a neutral scalar with a nonzero vacuum value supplies the "missing" isospin and hypercharge to each forbidden bilinear, turning the Yukawa interactions into masses and the gauge couplings into gauge-boson masses. Spontaneous breaking is thus not an optional embellishment of the electroweak theory but the unique consistent way to reconcile chiral gauge symmetry with massive particles.

31.2.8 The complete pre-breaking Lagrangian

Assembling the four pieces, the full gauge-invariant electroweak Lagrangian for one generation, before any symmetry breaking, is

$$\boxed{\mathcal{L}_{\text{EW}} = \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}} + \mathcal{L}_{\text{Yukawa}}} \quad (31.50)$$

with the four terms given by Eqs. (31.33), (31.34), (31.38) and (31.47) respectively. By Theorem 31.9 this is the most general renormalizable Lagrangian invariant under $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ with the given matter content: it contains the gauge couplings g, g', the Higgs parameters μ^2, λ, and the three Yukawa couplings y_e, y_d, y_u, and nothing else. Every particle in it is massless except for the

tachyonic-looking Higgs mass parameter, whose sign is the engine of the next section.

This is the Yang–Mills theory of Chapter 25 fully instantiated: $G = \mathrm{SU}(2)_L$ provides the non-abelian field strength Eq. (31.31) and its self-interactions, the abelian $\mathrm{U}(1)_Y$ rides alongside as in Chapter 8, and for the first time the gauge fields couple to matter in genuine, distinct representations — doublets for the left-handed fields and the Higgs, singlets for the right-handed fields — rather than to a single test charge. The carried Yang–Mills theory, classical and quantized through Chapter 27, here meets its physical realization.

With the complete massless Lagrangian Eq. (31.50) in hand and its potential Eq. (31.39) already shaped for a nonzero vacuum value, the stage is set for the central act. In the next section we give the neutral component ϕ^0 its vacuum expectation value v, expand every term of Eq. (31.50) about that vacuum, and watch three of the four gauge bosons absorb the would-be Goldstone modes of Φ to become the massive $W^\pm$ and Z while one combination of W^3 and B remains massless as the photon — the spectrum, the weak mixing angle, and the recovery of QED that the whole construction was built to deliver.

31.3 Spontaneous Breaking and the Gauge-Boson Spectrum

The previous section assembled the complete pre-breaking Glashow–Weinberg–Salam Lagrangian: a $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge theory with the chiral fermion content, the Higgs doublet Φ with its covariant derivative (Eq. (31.23)), and the scalar potential (Eq. (31.39)) whose Mexican-hat shape we are now ready to exploit. The Lagrangian written there has every field strictly massless: gauge invariance forbids an explicit gauge-boson mass, and the deliberately chiral assignment of $\mathrm{SU}(2)_L$ forbids a Dirac mass for the fermions. This is not a defect to be patched but the engine of the theory. We now let the potential do its work. The doublet acquires a vacuum expectation value, three of the four gauge bosons swallow the would-be Goldstone modes and become massive, and a single linear combination of the neutral gauge fields survives massless. That survivor is the photon, and the mixing angle that defines it is the central phenomenological parameter of the electroweak theory. We carry out the diagonalization in full, prove that the massless direction is exactly the unbroken electric-charge generator, and close the books on the degrees of freedom.

This is the non-abelian Higgs mechanism of Chapter 30 applied to a specific group and representation. The general machinery—the mass-matrix formula Theorem 30.17, the counting theorem Theorem 30.18, and the worked SU(2)-doublet prototype Theorem 30.19—is in hand; our task is to add hypercharge to that prototype and read off the consequences, all of which are new: the splitting

of the degenerate W-triplet into $W^{\pm}, Z, \gamma$, the appearance of the weak mixing angle, and the recovery of quantum electrodynamics as an output.

31.3.1 The vacuum expectation value and the unbroken charge

Recall the field content. The Higgs is a complex $\mathrm{SU}(2)_L$ doublet

$$\Phi = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix}, \qquad T^a = \frac{\sigma^a}{2}, \qquad Y_\Phi = \tfrac{1}{2}, \tag{31.51}$$

where the hypercharge $Y_\Phi = \frac{1}{2}$ is the value fixed in Section 31.2 so that the Yukawa couplings are gauge invariant and the electric charges come out right; the superscripts on the components anticipate the charge assignment proved below. The potential is the doublet generalization of the ϕ^4 Mexican hat,

$$V(\Phi) = -\mu^2\, \Phi^\dagger\Phi + \lambda\, (\Phi^\dagger\Phi)^2, \qquad \mu^2 > 0,\ \lambda > 0, \tag{31.52}$$

which depends on Φ only through the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant combination $\Phi^\dagger\Phi$. Minimizing in this single variable,

$$\frac{\partial V}{\partial(\Phi^\dagger\Phi)} = -\mu^2 + 2\lambda\, \Phi^\dagger\Phi = 0 \quad \Longrightarrow \quad \Phi^\dagger\Phi\big|_{\min} = \frac{\mu^2}{2\lambda} \equiv \frac{v^2}{2}, \qquad v = \sqrt{\frac{\mu^2}{\lambda}}. \tag{31.53}$$

The minimum fixes only the modulus $\Phi^\dagger\Phi = v^2/2$; the four real components of Φ are otherwise free, so the set of minima is a three-sphere S^3 of radius $v/\sqrt{2}$ in field space, the vacuum manifold. Any point on it is as good as any other, and a global $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ transformation moves us around it. We use that freedom to rotate the expectation value into a standard form. Since $\mathrm{SU}(2)_L$ acts transitively on the unit sphere in $\mathbb{C}^2$, we can always bring the VEV to lie entirely in the lower (real) component:

$$\langle \Phi \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}. \tag{31.54}$$

This is the analogue of the alignment performed in Eq. (30.228); it is a choice of representative on the vacuum manifold, not an additional physical input. Working with this fixed VEV and dropping the Goldstone phases is the *unitary gauge*, justified in full for the abelian case in Section 30.4.1 and for the non-abelian case in Section 30.7.4; we use it here to read the spectrum and return to the gauge-fixing subtleties only where they matter.

The unbroken generator. The decisive feature of Eq. (31.54), the one that distinguishes the electroweak vacuum from the custodial prototype of Theorem 30.19, is that it is annihilated not by nothing ($\dim H = 0$ there) but by

exactly one combination of the four generators. Define the *electric charge operator* by the Gell-Mann–Nishijima relation

$$Q = T^3 + Y, \tag{31.55}$$

where on the Higgs doublet $T^3 = \frac{1}{2}\sigma^3 = \frac{1}{2}\,\mathrm{diag}(1,-1)$ and $Y = Y_\Phi\,\mathbf{1} = \frac{1}{2}\,\mathbf{1}$.

Proposition 31.10 (The vacuum is electrically neutral)**.** *With $\langle\Phi\rangle$ as in Eq. (31.54) and $Q = T^3 + Y$ as in Eq. (31.20), the three generators T^1, T^2 and $T^3 - Y$ are broken (they move the VEV), while*

$$Q\,\langle\Phi\rangle = 0. \tag{31.56}$$

Consequently the subgroup $\mathrm{U}(1)_{\mathrm{em}}$ generated by Q is an exact, unbroken gauge symmetry of the vacuum, and the breaking pattern is

$$\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \;\longrightarrow\; \mathrm{U}(1)_{\mathrm{em}}. \tag{31.57}$$

Proof. Act with Q on the VEV. The charge matrix on the doublet is

$$Q = T^3 + Y_\Phi\,\mathbf{1} = \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}. \tag{31.58}$$

Thus the upper component carries charge $+1$ (whence the label ϕ^+) and the lower component carries charge 0 (whence ϕ^0). Applying it to Eq. (31.54),

$$Q\,\langle\Phi\rangle = \begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v\end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ 0\end{pmatrix} = 0, \tag{31.59}$$

which is Eq. (31.56): the VEV sits entirely in the neutral component, so the charge operator annihilates it. To confirm that no other generator does, compute the action of the remaining generators directly, exactly as in Theorem 30.19:

$$T^1\,\langle\Phi\rangle = \frac{v}{2\sqrt{2}}\begin{pmatrix}1\\ 0\end{pmatrix}, \quad T^2\,\langle\Phi\rangle = \frac{-iv}{2\sqrt{2}}\begin{pmatrix}1\\ 0\end{pmatrix}, \quad (T^3 - Y)\,\langle\Phi\rangle = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\ -v\end{pmatrix}, \tag{31.60}$$

where the last uses $(T^3 - Y)\,\langle\Phi\rangle = \frac{1}{2}\,\mathrm{diag}(1,-1)\,\langle\Phi\rangle - \frac{1}{2}\,\langle\Phi\rangle = \frac{1}{\sqrt{2}}(0,-v)^{\mathsf{T}}$. All three are nonzero; they are the three broken directions. Together with Q, the four generators $\{T^1, T^2, T^3 - Y,\, Q = T^3 + Y\}$ span the same space as $\{T^1, T^2, T^3, Y\}$, so the unbroken subalgebra is exactly one-dimensional, spanned by Q. Because Q annihilates the VEV, the finite transformations $e^{i\alpha Q}$ fix the vacuum, and $\mathrm{U}(1)_{\mathrm{em}} = \{e^{i\alpha Q}\}$ is an unbroken gauge symmetry. This is the pattern Eq. (31.57), with $\dim H = 1$. □

The content of Theorem 31.10 is that the would-be four photons-and-W's of the symmetric phase are sorted by the vacuum into three that feel a broken generator and one that feels the surviving Q. By the non-abelian Higgs mechanism (Theorem 30.18), the three broken generators yield three massive vectors and the one unbroken generator yields one strictly massless vector. We now make that quantitative.

31.3.2 The gauge-boson mass matrix from the Higgs kinetic term

The gauge-boson masses live in the Higgs kinetic term $(D_\mu\Phi)^\dagger(D^\mu\Phi)$ evaluated at the VEV. The electroweak covariant derivative on the doublet, fixed in Eq. (31.23), is

$$D_\mu\Phi = \Big(\partial_\mu - ig\,W_\mu^a T^a - ig'\,Y_\Phi B_\mu\Big)\Phi = \Big(\partial_\mu - i\tfrac{g}{2}\,W_\mu^a\sigma^a - i\tfrac{g'}{2}\,B_\mu\Big)\Phi, \quad (31.61)$$

with g the $\mathrm{SU}(2)_L$ coupling, g' the $\mathrm{U}(1)_Y$ coupling, and $Y_\Phi = \frac{1}{2}$. Writing the $\mathrm{SU}(2)_L$ part as an explicit matrix,

$$\tfrac{g}{2}W_\mu^a\sigma^a = \frac{g}{2}\begin{pmatrix} W_\mu^3 & W_\mu^1 - iW_\mu^2 \\ W_\mu^1 + iW_\mu^2 & -W_\mu^3 \end{pmatrix}, \quad (31.62)$$

the full gauge connection acting on Φ is

$$g\,W_\mu^a T^a + g'\,Y_\Phi B_\mu = \frac{1}{2}\begin{pmatrix} gW_\mu^3 + g'B_\mu & g(W_\mu^1 - iW_\mu^2) \\ g(W_\mu^1 + iW_\mu^2) & -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad (31.63)$$

The mass terms come from setting $\Phi \to \langle\Phi\rangle$, where $\partial_\mu\langle\Phi\rangle = 0$ and only the gauge piece survives:

$$D_\mu\langle\Phi\rangle = -\frac{i}{2}\begin{pmatrix} gW_\mu^3 + g'B_\mu & g(W_\mu^1 - iW_\mu^2) \\ g(W_\mu^1 + iW_\mu^2) & -gW_\mu^3 + g'B_\mu \end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix} 0 \\ v \end{pmatrix} = -\frac{iv}{2\sqrt{2}}\begin{pmatrix} g(W_\mu^1 - iW_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad (31.64)$$

Taking the Hermitian square,

$$\begin{aligned}(D_\mu\langle\Phi\rangle)^\dagger(D^\mu\langle\Phi\rangle) &= \frac{v^2}{8}\Big[g^2\,|W_\mu^1 - iW_\mu^2|^2 + (-gW_\mu^3 + g'B_\mu)^2\Big] \\ &= \frac{v^2}{8}\Big[g^2((W_\mu^1)^2 + (W_\mu^2)^2) + (gW_\mu^3 - g'B_\mu)^2\Big]. \end{aligned} \quad (31.65)$$

Every Lorentz index is contracted with the mostly-minus metric; the modulus $|W^1 - iW^2|^2 = (W^1)^2 + (W^2)^2$ used $|a - ib|^2 = a^2 + b^2$ for the real fields W^1, W^2. Equation Eq. (31.65) is the entire gauge-boson mass Lagrangian. It splits cleanly into a piece built from the off-diagonal generators W^1, W^2 (the charged sector) and a piece built from the diagonal combination $gW^3 - g'B$ (the neutral sector). We diagonalize each in turn.

Caution

A real vector field V_μ has the canonical mass term $+\frac{1}{2}M^2V_\mu V^\mu$ in $\mathcal{L}$, whereas a complex (charged) vector field $W_\mu^\pm$ has mass term $+M^2\,W_\mu^+W^{-\mu}$ with *no* factor of $\frac{1}{2}$, because W^+W^- already counts the field and its

conjugate once each. Tracking this factor is exactly what makes $M_W = gv/2$ (and not $gv/2\sqrt{2}$) in Eq. (31.69) below; the same factor distinguishes the neutral mass $\frac{1}{2}M_Z^2 Z^2$ from the charged $M_W^2 W^+W^-$. Getting it wrong shifts M_W by $\sqrt{2}$ and breaks the relation $M_W = M_Z \cos\theta_{\mathrm{W}}$.

31.3.3 The charged bosons $W^\pm$

The off-diagonal generators W^1, W^2 do not carry definite electric charge: under $\mathrm{U}(1)_{\mathrm{em}}$ they rotate into one another, just as the doublet's broken generators T^1, T^2 did in Eq. (31.60). The charge eigenstates are the complex combinations

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp i\, W_\mu^2), \qquad W_\mu^1 = \frac{1}{\sqrt{2}}(W_\mu^+ + W_\mu^-), \quad W_\mu^2 = \frac{i}{\sqrt{2}}(W_\mu^+ - W_\mu^-), \tag{31.66}$$

so that $W_\mu^- = (W_\mu^+)^*$. The sign convention is fixed by demanding that W^+ raises electric charge by one unit: acting on the charged generators $T^\pm = T^1 \pm iT^2$ one finds $[Q, T^\pm] = \pm T^\pm$, and $W^\pm$ is the field that multiplies $T^\mp$ in $W^a T^a = W^+T^+ + W^-T^- + W^3T^3$, hence W^+ destroys a $+1$ charge / creates a W-boson of charge $+1$. In terms of these fields,

$$(W_\mu^1)^2 + (W_\mu^2)^2 = \frac{1}{2}(W^+ + W^-)^2 - \frac{1}{2}(W^+ - W^-)^2 = 2\, W_\mu^+ W^{-\mu}, \tag{31.67}$$

where the cross terms W^+W^+ and W^-W^- cancel between the two squares and the surviving $2W^+W^- + 2W^-W^+$ combine. Substituting into the first bracket of Eq. (31.65),

$$\frac{v^2}{8}\, g^2((W_\mu^1)^2 + (W_\mu^2)^2) = \frac{g^2 v^2}{4}\, W_\mu^+ W^{-\mu} \equiv M_W^2\, W_\mu^+ W^{-\mu}. \tag{31.68}$$

Reading off the coefficient with the complex-field convention of the caution above,

$$\boxed{M_W = \frac{gv}{2}}. \tag{31.69}$$

The two real degrees of freedom W^1, W^2 have reorganized into a single charged massive vector W^+ and its antiparticle W^-, degenerate in mass as they must be: $W^- = (W^+)^*$ is the antiparticle of W^+, and CPT forces equal masses. This reproduces, generator by generator, the $m_W = gv/2$ of the custodial prototype Eq. (30.231); what is new is that the third member of that degenerate triplet is about to be split off by hypercharge.

31.3.4 The neutral sector: full diagonalization and the weak mixing angle

The second bracket of Eq. (31.65) mixes W^3 and B. Written as a quadratic form in the basis (W^3_μ, B_μ),

$$\frac{v^2}{8}(gW^3_\mu - g'B_\mu)^2 = \frac{1}{2}\begin{pmatrix} W^3_\mu & B_\mu \end{pmatrix} \mathcal{M}^2 \begin{pmatrix} W^{3\mu} \\ B^\mu \end{pmatrix}, \qquad \mathcal{M}^2 = \frac{v^2}{4}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}, \tag{31.70}$$

where the prefactor $\frac{1}{2}$ is the real-vector convention, so that $\frac{v^2}{8}(gW^3 - g'B)^2 = \frac{1}{2}\cdot\frac{v^2}{4}(g^2(W^3)^2 - 2gg'W^3B + g'^2B^2)$ matches $\frac{1}{2}W^{\mathsf{T}}\mathcal{M}^2 W$ term by term. The matrix $\mathcal{M}^2$ is real symmetric and therefore diagonalizable by an orthogonal rotation, which preserves the canonical normalization of the kinetic terms $-\frac{1}{4}(\partial W^3)^2 - \frac{1}{4}(\partial B)^2$ (an orthogonal rotation leaves $\sum(\partial V)^2$ invariant). We carry out the diagonalization with no steps omitted.

Theorem 31.11 (Diagonalization of the neutral mass matrix)**.** *The matrix $\mathcal{M}^2$ of Eq. (31.70) has eigenvalues*

$$m_A^2 = 0, \qquad m_Z^2 = \frac{v^2}{4}(g^2 + g'^2), \tag{31.71}$$

with respective orthonormal eigenvectors

$$\hat{e}_A = \frac{1}{\sqrt{g^2+g'^2}}\begin{pmatrix} g' \\ g \end{pmatrix}, \qquad \hat{e}_Z = \frac{1}{\sqrt{g^2+g'^2}}\begin{pmatrix} g \\ -g' \end{pmatrix}. \tag{31.72}$$

Defining the weak mixing angle θ_{W} by

$$\cos\theta_{\mathrm{W}} = \frac{g}{\sqrt{g^2+g'^2}}, \qquad \sin\theta_{\mathrm{W}} = \frac{g'}{\sqrt{g^2+g'^2}}, \qquad \tan\theta_{\mathrm{W}} = \frac{g'}{g}, \tag{31.73}$$

the mass eigenstates are

$$A_\mu = \sin\theta_{\mathrm{W}}\, W^3_\mu + \cos\theta_{\mathrm{W}}\, B_\mu \qquad (\textit{massless: the photon}), \tag{31.74}$$

$$Z_\mu = \cos\theta_{\mathrm{W}}\, W^3_\mu - \sin\theta_{\mathrm{W}}\, B_\mu \qquad (\textit{mass } m_Z), \tag{31.75}$$

and the spectrum obeys the on-shell relation

$$m_Z = \frac{m_W}{\cos\theta_{\mathrm{W}}}, \qquad \textit{equivalently} \qquad m_W = m_Z\cos\theta_{\mathrm{W}}. \tag{31.76}$$

Proof. Eigenvalues. Solve $\det(\mathcal{M}^2 - m^2\mathbf{1}) = 0$. With $k \equiv v^2/4$,

$$\det\begin{pmatrix} kg^2 - m^2 & -kgg' \\ -kgg' & kg'^2 - m^2 \end{pmatrix} = (kg^2 - m^2)(kg'^2 - m^2) - k^2g^2g'^2$$

$$= m^4 - k(g^2+g'^2)m^2 + k^2 g^2 g'^2 - k^2 g^2 g'^2$$
$$= m^2\left[m^2 - k(g^2+g'^2)\right] = 0. \tag{31.77}$$

The two roots are $m^2 = 0$ and $m^2 = k(g^2+g'^2) = \frac{v^2}{4}(g^2+g'^2)$, which are Eq. (31.71). The vanishing of the determinant of $\mathcal{M}^2$ itself ($\det \mathcal{M}^2 = k^2(g^2g'^2 - g^2g'^2) = 0$) is the algebraic statement that one eigenvalue must be zero: $\mathcal{M}^2$ has rank one, because it is the outer product $\mathcal{M}^2 = k\,(g,-g')^{\mathsf{T}}(g,-g')$ of a single vector with itself.

Eigenvectors. For $m^2 = 0$ we need $\mathcal{M}^2\hat{e}_A = 0$, i.e. $(g,-g')\cdot\hat{e}_A = 0$ since $\mathcal{M}^2 = k(g,-g')^{\mathsf{T}}(g,-g')$. The vector (g',g) satisfies $g\cdot g' - g'\cdot g = 0$, so it is the null direction; normalizing by its length $\sqrt{g'^2+g^2}$ gives $\hat{e}_A$ of Eq. (31.72). The remaining eigenvector $\hat{e}_Z$ is the orthogonal one, $(g,-g')$ normalized, and indeed

$$\mathcal{M}^2\hat{e}_Z = k\,(g,-g')^{\mathsf{T}}\left[(g,-g')\cdot(g,-g')\right]\frac{1}{\sqrt{g^2+g'^2}} = k(g^2+g'^2)\,\hat{e}_Z, \tag{31.78}$$

confirming the eigenvalue $k(g^2+g'^2) = m_Z^2$. The two eigenvectors are orthonormal: $\hat{e}_A\cdot\hat{e}_Z = (g'g - gg')/(g^2+g'^2) = 0$.

Mass eigenstates and the angle. An orthonormal eigenbasis of a real symmetric matrix defines an orthogonal change of field variables. Stacking the eigenvectors as columns,

$$\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix} = \begin{pmatrix} \hat{e}_A & \hat{e}_Z \end{pmatrix}\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \frac{1}{\sqrt{g^2+g'^2}}\begin{pmatrix} g' & g \\ g & -g' \end{pmatrix}\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}, \tag{31.79}$$

which, with the definitions Eq. (31.73), reads $W_\mu^3 = \sin\theta_{\mathrm{W}} A_\mu + \cos\theta_{\mathrm{W}} Z_\mu$ and $B_\mu = \cos\theta_{\mathrm{W}} A_\mu - \sin\theta_{\mathrm{W}} Z_\mu$. Inverting the orthogonal matrix (its transpose) gives the mass eigenstates in terms of the gauge fields,

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \sin\theta_{\mathrm{W}} & \cos\theta_{\mathrm{W}} \\ \cos\theta_{\mathrm{W}} & -\sin\theta_{\mathrm{W}} \end{pmatrix}\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix}, \tag{31.80}$$

which is Eq. (31.74)–Eq. (31.75). In the new basis the quadratic form Eq. (31.70) becomes diagonal,

$$\frac{1}{2}\begin{pmatrix} A & Z \end{pmatrix}\begin{pmatrix} 0 & 0 \\ 0 & m_Z^2 \end{pmatrix}\begin{pmatrix} A \\ Z \end{pmatrix} = \frac{1}{2}\,m_Z^2\,Z_\mu Z^\mu, \tag{31.81}$$

so A_μ is strictly massless and Z_μ has the real-vector mass m_Z of Eq. (31.71).

The mass relation. Combining Eq. (31.69) with Eq. (31.71) and Eq. (31.73),

$$\frac{m_W}{m_Z} = \frac{gv/2}{\frac{v}{2}\sqrt{g^2+g'^2}} = \frac{g}{\sqrt{g^2+g'^2}} = \cos\theta_{\mathrm{W}}, \tag{31.82}$$

which is Eq. (31.76). □

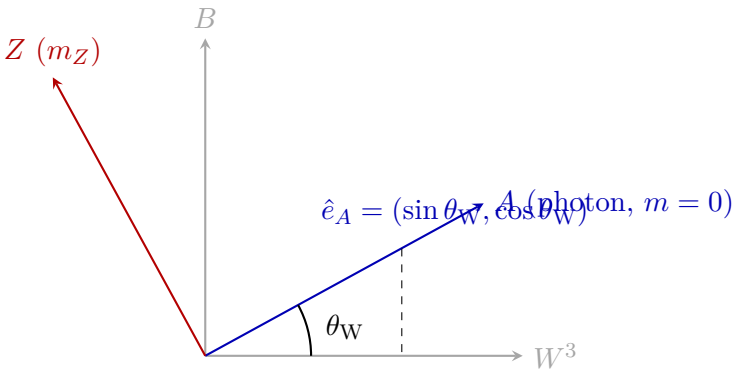


Figure 31.2: The weak mixing angle as a rotation in the neutral gauge-field plane. The orthogonal change of basis Eq. (31.79) rotates the gauge eigenstates (W^3, B) into the mass eigenstates (A, Z) by the angle θ_{W} with $\tan\theta_{\mathrm{W}} = g'/g$. The massless direction A (the photon) is the eigenvector of $\mathcal{M}^2$ with eigenvalue zero; the orthogonal direction Z carries the entire neutral mass $m_Z = \frac{v}{2}\sqrt{g^2+g'^2}$.

The electromagnetic coupling. The mixing angle also fixes the strength with which the surviving massless field A_μ couples to charge. Anticipating the next subsection, the coefficient of A_μ in the covariant derivative will turn out to be the single combination

$$e \equiv g\sin\theta_{\mathrm{W}} = g'\cos\theta_{\mathrm{W}} = \frac{gg'}{\sqrt{g^2+g'^2}}, \tag{31.83}$$

the two expressions agreeing because $g\sin\theta_{\mathrm{W}} = g\,g'/\sqrt{g^2+g'^2} = g'\,g/\sqrt{g^2+g'^2} = g'\cos\theta_{\mathrm{W}}$. Equation Eq. (31.83) is the electroweak unification relation: the electric charge e is not independent but is determined by the two gauge couplings, and equivalently $1/e^2 = 1/g^2 + 1/g'^2$, obtained by squaring and adding the reciprocals. We now prove that this e is indeed the QED electric charge.

31.3.5 The photon as the unbroken $\mathrm{U}(1)_{\mathrm{em}}$ — recovery of QED

We have a massless vector A_μ and an unbroken generator Q. To identify A_μ as *the* photon and e as *the* electric charge, we must show that A_μ couples to matter exactly through the minimal coupling $D_\mu = \partial_\mu + ieQA_\mu$ of quantum electrodynamics, with the same $e > 0$ that governs the electron. This is the moment the electroweak theory pays back QED: the photon of Chapter 13 and the QED vertex of Chapter 17 are recovered here as outputs of the breaking, not assumed as inputs.

The neutral part of any electroweak covariant derivative, acting on a field of weak isospin T^3 and hypercharge Y, is

$$D_\mu \supset \partial_\mu - i(g\,T^3 W^3_\mu + g'\,Y\,B_\mu). \tag{31.84}$$

Substitute the mass eigenstates Eq. (31.79), i.e. $W_\mu^3 = \sin\theta_{\rm W} A_\mu + \cos\theta_{\rm W} Z_\mu$ and $B_\mu = \cos\theta_{\rm W} A_\mu - \sin\theta_{\rm W} Z_\mu$:

$$g\,T^3 W_\mu^3 + g'\,Y\,B_\mu = (g\sin\theta_{\rm W}\,T^3 + g'\cos\theta_{\rm W}\,Y)A_\mu + (g\cos\theta_{\rm W}\,T^3 - g'\sin\theta_{\rm W}\,Y)Z_\mu. \tag{31.85}$$

Consider the coefficient of A_μ. Using Eq. (31.83), $g\sin\theta_{\rm W} = g'\cos\theta_{\rm W} = e$, it collapses to

$$g\sin\theta_{\rm W}\,T^3 + g'\cos\theta_{\rm W}\,Y = e\,T^3 + e\,Y = e\,(T^3 + Y) = e\,Q. \tag{31.86}$$

Therefore the photon enters the covariant derivative as

$$D_\mu \supset \partial_\mu - ie\,Q\,A_\mu, \tag{31.87}$$

which is precisely the QED minimal coupling, with the electric charge of any field given by its eigenvalue under $Q = T^3 + Y$. For the electron this returns the charge $-e$ and the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ of the QED (Chapter 13); the gauge field A_μ here is the same photon, the coupling e here is the same $e > 0$, and the vertex factor it generates is the $ie\gamma^\mu$ of Theorem 17.1. The masslessness of A_μ is now seen from two sides that must and do agree: the algebraic diagonalization Theorem 31.11 put the photon in the zero-eigenvalue direction, and the group theory Theorem 31.10 put it in the direction of the unbroken generator Q whose VEV is annihilated. A gauge boson is massless if and only if its generator leaves the vacuum invariant; the photon is massless because the vacuum carries no electric charge.

Physical Insight

The whole gauge-theory arc of the book—the abelian gauge principle of Chapter 8 that *derived* minimal coupling, the photon quantized in Chapter 13, the QED amplitudes of Chapter 17—is here revealed to be the low-energy shadow of a larger, spontaneously broken structure. Electromagnetism is not fundamental: it is the unbroken remnant $\mathrm{U}(1)_{\rm em}$ of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and its coupling $e = g\sin\theta_{\rm W}$ is a derived quantity. The photon stays massless not by a separate principle but because the Higgs VEV happens to be electrically neutral. Turn on a charged condensate—as in a superconductor—and the photon would acquire a mass too. The electroweak vacuum is a medium that is opaque to W and Z but transparent to light.

For completeness, the orthogonal combination in Eq. (31.85) is the neutral-current coupling of the Z. Writing $g' = g\tan\theta_{\rm W}$ and $Y = Q - T^3$,

$$g\cos\theta_{\rm W}\,T^3 - g'\sin\theta_{\rm W}\,Y = \frac{g}{\cos\theta_{\rm W}}(\cos^2\theta_{\rm W}\,T^3 - \sin^2\theta_{\rm W}\,Y) = \frac{g}{\cos\theta_{\rm W}}(T^3 - \sin^2\theta_{\rm W}\,Q), \tag{31.88}$$

using $\cos^2\theta_{\rm W}\,T^3 - \sin^2\theta_{\rm W}(Q - T^3) = (\cos^2 + \sin^2)T^3 - \sin^2 Q = T^3 - \sin^2\theta_{\rm W}\,Q$. The combination $g_Z \equiv g/\cos\theta_{\rm W}$ and the structure $T^3 - \sin^2\theta_{\rm W}\,Q$ are the data the next section needs to write the fermion neutral-current interactions; we record them here and hand them on.

31.3.6 Degree-of-freedom count and the physical Higgs

The counting must balance: spontaneous breaking rearranges degrees of freedom but cannot create or destroy them. Before breaking, the dynamical content is

- four massless gauge bosons W^1, W^2, W^3, B, each with 2 transverse physical polarizations: $4 \times 2 = 8$;
- one complex scalar doublet Φ with 4 real components: 4.

The total is 12 physical degrees of freedom. After breaking, the content is

- three massive vectors W^+, W^-, Z, each with 3 polarizations (two transverse plus one longitudinal): $3 \times 3 = 9$;
- one massless vector, the photon A: 2;
- one real scalar, the physical Higgs h: 1.

The total is again $9 + 2 + 1 = 12$. The bookkeeping is exact, and it tells the story precisely: each of the three gauge bosons that became massive gained one longitudinal polarization, and those three new polarizations came from three of the four scalar components of Φ. Those three are the would-be Goldstone bosons; the fourth scalar survives as h.

Which three scalars are eaten. Parametrize the doublet around the vacuum by separating its real and imaginary parts component by component,

$$\Phi = \begin{pmatrix} \phi^+ \\ \dfrac{1}{\sqrt{2}}\,(v + h + i\,\phi_3) \end{pmatrix}, \qquad \phi^+ = \frac{1}{\sqrt{2}}\,(\phi_1 + i\,\phi_2), \tag{31.89}$$

so that the four real fields are ϕ_1, ϕ_2 (the real and imaginary parts of the charged component ϕ^+), ϕ_3 (the imaginary part of the neutral component), and h (the real radial fluctuation). The three broken generators of Eq. (31.60)—T^1, T^2 and $T^3 - Y$—generate motion along exactly the three flat directions ϕ_1, ϕ_2, ϕ_3, the tangent space to the vacuum manifold S^3 at the VEV, while the unbroken Q generates no flat direction (it fixes the vacuum). By Goldstone's theorem in its tree-level counting form (Theorem 29.12: the number of Goldstone bosons equals the number of broken generators, $\dim G - \dim H = 4 - 1 = 3$), the three fields ϕ_1, ϕ_2, ϕ_3 are massless would-be Goldstone bosons.

In a gauge theory these would-be Goldstones are not physical. The non-abelian Higgs mechanism of Chapter 30 shows that a gauge transformation removes them from the spectrum entirely, where they reappear as the longitudinal polarizations of the now-massive gauge bosons (this is the unitary gauge; the clean statement is the Goldstone-equivalence content of Section 30.7.4). The matching is component by component, and follows the same logic as the abelian case Eq. (30.90): writing the would-be Goldstones as the local phase of the doublet,

$$\Phi(x) = \exp\!\left(i\,\frac{\xi^a(x)\,\sigma^a}{v} \right) \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \tag{31.90}$$

the three real fields ξ^a are precisely the would-be Goldstones (ϕ_1, ϕ_2, ϕ_3) to linear order, and an $\mathrm{SU}(2)_L$ gauge transformation $\Phi \to e^{-i\xi^a\sigma^a/v}\Phi$ sets $\xi^a = 0$, leaving the unitary-gauge VEV $\frac{1}{\sqrt{2}}(0, v+h)^{\mathsf{T}}$. The charged pair (ϕ_1, ϕ_2) (equivalently $\phi^{\pm}$) is eaten by $W^{\pm}$, and the neutral ϕ_3 is eaten by Z, matching the three massive vectors of Section 31.3.3–Section 31.3.4. The photon, whose generator Q is unbroken, eats nothing and stays with two polarizations.

The physical Higgs mass. The surviving scalar is the radial mode h. Its mass comes from the potential Eq. (31.52) expanded about the VEV. In unitary gauge $\Phi^\dagger\Phi = \frac{1}{2}(v+h)^2$, so

$$V = -\frac{\mu^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4. \tag{31.91}$$

The terms quadratic in h give the mass; with $\mu^2 = \lambda v^2$ from Eq. (31.53),

$$V = \text{const} + \Big(-\mu^2 v + \lambda v^3\Big)h + \frac{1}{2}\Big(-\mu^2 + 3\lambda v^2\Big)h^2 + \lambda v\, h^3 + \frac{\lambda}{4}h^4. \tag{31.92}$$

The linear term vanishes, $-\mu^2 v + \lambda v^3 = v(-\lambda v^2 + \lambda v^2) = 0$, confirming that $h = 0$ is the minimum (the tadpole cancels). The quadratic coefficient gives the mass,

$$\frac{1}{2}m_h^2\, h^2 = \frac{1}{2}(3\lambda v^2 - \mu^2)h^2 = \frac{1}{2}(3\lambda v^2 - \lambda v^2)h^2 \quad\Longrightarrow\quad \boxed{m_h^2 = 2\lambda v^2}. \tag{31.93}$$

The physical Higgs is the one scalar of Φ that is *not* a flat direction of the potential—it climbs the wall of the Mexican hat—and its mass is set by the quartic coupling and the VEV. The residual cubic $\lambda v\, h^3$ and quartic $\frac{\lambda}{4}h^4$ self-interactions in Eq. (31.92) are the Higgs self-couplings whose measurement probes the shape of the potential directly. With m_h measured and v fixed (below), Eq. (31.93) determines λ, closing the scalar sector.

Worked Example 31.12 (The tree-level electroweak spectrum in numbers)**.** We assemble the predictions of Eqs. (31.69), (31.71) and (31.76) into numbers,

anticipating the value $v \simeq 246$ GeV that the matching to the Fermi theory fixes later in this chapter (Section 31.5.6), and the measured $\sin^2\theta_{\rm W} \simeq 0.231$ and $e \simeq 0.313$ at the electroweak scale.

Couplings. From $e = g \sin\theta_{\rm W}$ and $\sin\theta_{\rm W} = \sqrt{0.231} = 0.481$,

$$g = \frac{e}{\sin\theta_{\rm W}} = \frac{0.313}{0.481} = 0.651, \qquad g' = g\tan\theta_{\rm W} = 0.651 \times \frac{0.481}{0.877} = 0.357, \tag{31.94}$$

using $\cos\theta_{\rm W} = \sqrt{1 - 0.231} = 0.877$.

Masses. The charged-boson mass Eq. (31.69) is

$$m_W = \frac{gv}{2} = \frac{0.651 \times 246 \text{ GeV}}{2} = 80.1 \text{ GeV}, \tag{31.95}$$

and the neutral-boson mass follows from the relation Eq. (31.76) without further input,

$$m_Z = \frac{m_W}{\cos\theta_{\rm W}} = \frac{80.1 \text{ GeV}}{0.877} = 91.3 \text{ GeV}. \tag{31.96}$$

Equivalently, directly from Eq. (31.71), $m_Z = \frac{v}{2}\sqrt{g^2 + g'^2} = \frac{246}{2}\sqrt{0.651^2 + 0.357^2}$ GeV $= 123 \times 0.742$ GeV $= 91.3$ GeV, the same number. These tree-level values sit within a percent of the measured $m_W = 80.4$ GeV and $m_Z = 91.19$ GeV; the residual difference is radiative, the leading correction being the running of $\sin^2\theta_{\rm W}$ and the custodial-breaking effects discussed in the chapter's final section. That two masses emerge from one VEV, locked together by a single mixing angle, is the sharpest tree-level prediction of the model and the historical reason the discovery of the W and Z at their predicted masses was decisive.

We have produced the physical gauge spectrum—the massive $W^\pm$ and Z, the massless photon—and the weak mixing angle $\theta_{\rm W}$ that relates the gauge couplings (g, g') to the electric charge e and ties m_W to m_Z. The same VEV that gave these masses, $\langle\Phi\rangle = \frac{1}{\sqrt{2}}(0, v)^{\mathsf{T}}$, sits in the Yukawa couplings written down in Section 31.2; in the next section we substitute it there and watch the charged fermions acquire Dirac masses proportional to their Yukawa couplings, while the absence of a right-handed neutrino leaves the neutrino massless in the minimal model.

31.4 Fermion Masses and the Yukawa Sector

The previous section gave the Higgs doublet its vacuum expectation value and watched the gauge sector reorganize itself: three of the four electroweak gauge bosons absorbed the would-be Goldstone modes to become the massive $W^\pm$ and Z, one orthogonal combination remained massless as the photon, and the weak mixing angle $\theta_{\rm W}$ emerged from the diagonalization. The physical fields,

the vacuum expectation value v, and θ_{W} are now in hand. But the matter content we so carefully made chiral is, at this point, entirely *massless*: every fermion in the spectrum is a definite-chirality Weyl field, and the chirality was not a stylistic choice but a necessity forced by the parity violation of the weak interaction. We must now confront the price of that choice. A Dirac mass term, the only Lorentz-invariant fermion mass available, pairs a left-handed field with a right-handed one; and in a chiral gauge theory those two fields carry *different* gauge charges. The naive mass term is therefore forbidden by the very symmetry that organizes the theory. This section shows that the same scalar condensate that broke the gauge symmetry and fed the W and Z also resolves this impasse: through gauge-invariant Yukawa couplings, the Higgs vacuum expectation value manufactures Dirac masses for the charged fermions, and in doing so fixes the coupling of the physical Higgs boson to each fermion to be exactly proportional to that fermion's mass. The same accounting will tell us, sharply and without escape, that in the minimal field content the neutrino stays massless.

Throughout we work with one generation, exactly as the matter content was set up in the earlier sections of this chapter: a left-handed lepton doublet $L = (\nu_L,\, e_L)^{\mathsf{T}}$ and quark doublet $Q = (u_L,\, d_L)^{\mathsf{T}}$ of $\mathrm{SU}(2)_L$, and the right-handed singlets e_R, u_R, d_R; their hypercharges are fixed by Gell-Mann–Nishijima, $Q = T^3 + Y$, to be

$$Y_L = -\tfrac{1}{2}, \quad Y_{e_R} = -1, \quad Y_Q = +\tfrac{1}{6}, \quad Y_{u_R} = +\tfrac{2}{3}, \quad Y_{d_R} = -\tfrac{1}{3}, \tag{31.97}$$

and the Higgs doublet

$$H = \begin{pmatrix} H^+ \\ H^0 \end{pmatrix}, \qquad Y_H = +\tfrac{1}{2}, \qquad \langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \tag{31.98}$$

whose neutral component condenses. In unitary gauge, after the three Goldstone directions are gauged away as in the previous section, the doublet is

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \tag{31.99}$$

with $h(x)$ the single physical Higgs scalar. The covariant derivative is the electroweak one fixed in the chapter introduction, $D_\mu = \partial_\mu - i g\, W_\mu^a T^a - i g' Y B_\mu$, with $T^a = \sigma^a/2$ on the doublets.

31.4.1 Why a bare Dirac mass is forbidden

It is worth seeing precisely why we cannot simply write down a fermion mass and be done. Recall the Dirac Lagrangian Eq. (11.25), $\mathcal{L} = \bar\psi(i\partial\!\!\!/ - m)\psi$, whose mass term is $-m\,\bar\psi\psi$. Decompose the Dirac field into its chiral components using the projectors built from the chirality operator of Theorem 3.24,

$$\psi_L = P_L \psi, \qquad \psi_R = P_R \psi, \qquad P_{L,R} = \frac{1 \mp \gamma^5}{2}, \tag{31.100}$$

where $P_L + P_R = 1$ and $P_{L,R}^2 = P_{L,R}$, $P_L P_R = 0$. Because γ^5 is Hermitian and anticommutes with γ^0, the projectors satisfy $P_{L,R}^\dagger = P_{L,R}$ and $\gamma^0 P_{L,R} = P_{R,L}\gamma^0$, so that for the Dirac adjoint $\overline{\psi_L} = \psi_L^\dagger \gamma^0 = \psi^\dagger P_L \gamma^0 = \psi^\dagger \gamma^0 P_R = \bar\psi P_R$, and likewise $\overline{\psi_R} = \bar\psi P_L$. The scalar bilinear then splits cleanly:

$$\bar\psi\psi = \bar\psi(P_L + P_R)\psi = \bar\psi P_R \psi + \bar\psi P_L \psi = \overline{\psi_L}\,\psi_R + \overline{\psi_R}\,\psi_L. \tag{31.101}$$

A Dirac mass is intrinsically chirality-mixing: it couples the left-handed Weyl component to the right-handed one and back. (The kinetic term, by contrast, contains $\bar\psi i\partial\!\!\!/\psi = \overline{\psi_L}\, i\partial\!\!\!/\,\psi_L + \overline{\psi_R}\, i\partial\!\!\!/\,\psi_R$, because $\partial\!\!\!/$ carries one γ and so preserves chirality; this is why a chiral theory can have a perfectly good kinetic term while having no invariant mass term.)

Now ask whether $\overline{\psi_L}\,\psi_R$ can be a gauge singlet. For the electron, $\psi_L = e_L$ lives in the doublet L with $(T, Y) = (\tfrac{1}{2}, -\tfrac{1}{2})$, while $\psi_R = e_R$ is the singlet with $(0, -1)$. The candidate term $\overline{e_L}\, e_R$ therefore transforms as a $\mathrm{SU}(2)_L$ *doublet* (the right-handed singlet cannot soak up the isospin index carried by $\overline{e_L}$), and it carries net hypercharge $Y = -(-\tfrac{1}{2}) + (-1) = -\tfrac{1}{2}$. It is neither an isospin singlet nor hypercharge-neutral, so it is not gauge invariant, and a Lagrangian containing $-m_e\,\bar e\, e$ would not be invariant under $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. The same obstruction afflicts every charged fermion. This is the structural consequence of building a chiral gauge theory: *mass is forbidden by the gauge symmetry*, and any mass the fermions ultimately carry must arise from operators that are themselves gauge invariant and that break chirality only after the symmetry is spontaneously broken. The Higgs doublet is precisely the object that lets us write such operators, because it transforms in a way that can absorb both the missing isospin index and the missing hypercharge.

31.4.2 The lepton Yukawa coupling and the electron mass

The deficit to be repaired is exactly one unit of doublet index and $-\tfrac{1}{2}$ of hypercharge: $\overline{L}\, e_R$ is a $\mathrm{SU}(2)_L$ doublet carrying $Y = +\tfrac{1}{2} + (-1) = -\tfrac{1}{2}$. The Higgs doublet H carries $(T, Y) = (\tfrac{1}{2}, +\tfrac{1}{2})$. Their product can therefore be contracted to a $\mathrm{SU}(2)_L$ singlet (a doublet times a doublet contains a singlet) with total hypercharge $+\tfrac{1}{2} - \tfrac{1}{2} = 0$: gauge invariant on both counts. This is the unique renormalizable, gauge-invariant coupling of L, e_R, and one Higgs doublet, and it defines the lepton Yukawa term.

Definition 31.13 (Lepton Yukawa coupling, one generation)**.** With H the Higgs doublet of Eq. (31.98) and y_e a (real, after rephasing) dimensionless coupling, the gauge-invariant Yukawa interaction of the charged lepton is

$$\mathcal{L}_Y^{(e)} = -\,y_e\,\overline{L}\,H\,e_R + \text{h.c.} = -\,y_e\,(\overline{\nu_L}\ \ \overline{e_L})\begin{pmatrix} H^+ \\ H^0 \end{pmatrix} e_R + \text{h.c.}, \tag{31.102}$$

where the $\mathrm{SU}(2)_L$ indices of $\overline{L}$ and H are contracted (summed) and the Hermitian conjugate restores reality of the action.

The term is dimensionally correct: $\overline{L}$ and e_R each carry mass dimension $\frac{3}{2}$, H carries dimension 1, so the operator has dimension 4 and y_e is dimensionless, as a renormalizable coupling must be. Now let the neutral component condense. Substituting the unitary-gauge doublet Eq. (31.99) into Eq. (31.102), only the lower entry survives the contraction,

$$\overline{L}\,H = \begin{pmatrix}\overline{\nu_L} & \overline{e_L}\end{pmatrix}\frac{1}{\sqrt{2}}\begin{pmatrix}0\\ v+h\end{pmatrix} = \frac{1}{\sqrt{2}}\,(v+h)\,\overline{e_L}, \tag{31.103}$$

so the Yukawa Lagrangian becomes

$$\mathcal{L}_Y^{(e)} = -\frac{y_e}{\sqrt{2}}\,(v+h)\,\overline{e_L}\,e_R + \text{h.c.} = -\frac{y_e v}{\sqrt{2}}\,\overline{e_L}\,e_R - \frac{y_e}{\sqrt{2}}\,h\,\overline{e_L}\,e_R + \text{h.c.} \tag{31.104}$$

Two physical terms have appeared, and we read them off in turn.

The mass term. The h-independent piece, together with its Hermitian conjugate, is

$$-\frac{y_e v}{\sqrt{2}}\,(\overline{e_L}\,e_R + \overline{e_R}\,e_L) = -\frac{y_e v}{\sqrt{2}}\,\bar{e}\,e, \tag{31.105}$$

where in the last step we reassembled the two opposite-chirality Weyl pieces into the full Dirac bilinear using Eq. (31.101) read backwards. The left-handed electron buried inside the doublet L and the independent right-handed singlet e_R have been welded into a single massive Dirac electron $e = e_L + e_R$, with its mass term in exactly the form of Eq. (11.25). Comparing with $-m_e\,\bar{e}\,e$,

$$\boxed{m_e = \frac{y_e\,v}{\sqrt{2}}\,.} \tag{31.106}$$

The Higgs condensate does not put a mass into the Lagrangian by hand; it converts the dimensionless number y_e, together with the single scale v that the entire electroweak sector shares, into the electron mass. The chirality flip required by Eq. (31.101) is supplied by the vacuum: the condensate carries weak isospin and hypercharge, and absorbing a quantum of it is what lets a left-handed electron turn into a right-handed one as it propagates.

The Higgs–fermion coupling. The piece linear in the physical field h, with its conjugate, is

$$-\frac{y_e}{\sqrt{2}}\,h\,(\overline{e_L}\,e_R + \overline{e_R}\,e_L) = -\frac{y_e}{\sqrt{2}}\,h\,\bar{e}\,e = -\frac{m_e}{v}\,h\,\bar{e}\,e, \tag{31.107}$$

where in the last equality we eliminated y_e in favor of the mass through Eq. (31.106). The coupling of the physical Higgs boson to the electron is therefore not a free parameter: it is locked to the electron mass,

$$g_{he\bar{e}} = \frac{m_e}{v} = \frac{y_e}{\sqrt{2}}. \tag{31.108}$$

This is no accident of the leptons; it is the generic output of the mechanism, as we now confirm for the quarks and then state in full generality.

31.4.3 Quark masses: the down type, and the up type via $\widetilde{H}$

The down-type quark is the exact analogue of the electron. The combination $\overline{Q}\, d_R$ is an $\mathrm{SU}(2)_L$ doublet with hypercharge $-\frac{1}{6} + (-\frac{1}{3}) = -\frac{1}{2}$, the same deficit as before, so it pairs with H to a gauge singlet:

$$\mathcal{L}_Y^{(d)} = -\, y_d\, \overline{Q}\, H\, d_R + \text{h.c.} \xrightarrow{\langle H \rangle} -\frac{y_d}{\sqrt{2}}(v+h)\, \overline{d_L}\, d_R + \text{h.c.}, \tag{31.109}$$

by the identical contraction Eq. (31.103) with $\overline{e_L} \to \overline{d_L}$. Hence

$$m_d = \frac{y_d\, v}{\sqrt{2}}, \qquad g_{hd\bar{d}} = \frac{m_d}{v}. \tag{31.110}$$

The up-type quark is the one new structural point. The neutral, condensing entry of H sits in its *lower* component, so coupling H to $\overline{Q}\, u_R$ would try to give the upper member u_L a partner through the charged entry H^+, which does not condense; worse, the hypercharges do not work, since $\overline{Q}\, u_R$ carries $Y = -\frac{1}{6} + \frac{2}{3} = +\frac{1}{2}$, and pairing it with H ($Y = +\frac{1}{2}$) gives net $Y = +1 \neq 0$. What we need is a Higgs-like doublet whose *upper* component condenses and which carries $Y = -\frac{1}{2}$. Exactly such an object is built from H by charge conjugation.

Definition 31.14 (Conjugate Higgs doublet)**.** The conjugate Higgs doublet is

$$\widetilde{H} \equiv i\sigma^2 H^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} H^{+*} \\ H^{0*} \end{pmatrix} = \begin{pmatrix} H^{0*} \\ -H^- \end{pmatrix}, \qquad H^- \equiv (H^+)^*, \tag{31.111}$$

with σ^2 the second Pauli matrix.

Two facts make $\widetilde{H}$ the correct partner for the up quark, and both must be checked rather than asserted.

First, $\widetilde{H}$ transforms as a doublet of $\mathrm{SU}(2)_L$, not as an antidoublet. Under a gauge transformation $H \to UH$ with $U = \exp(i\alpha^a \sigma^a/2) \in \mathrm{SU}(2)_L$, the complex conjugate transforms as $H^* \to U^* H^*$, and

$$\widetilde{H} = i\sigma^2 H^* \;\longrightarrow\; i\sigma^2 U^* H^* = (i\sigma^2 U^* \sigma^{2-1})\, i\sigma^2 H^* = U\, \widetilde{H}, \tag{31.112}$$

provided $\sigma^2 U^* \sigma^2 = U$ (note $\sigma^{2-1} = \sigma^2$). This identity is the pseudoreality of SU(2): it suffices to check it on the generators, where $U = 1 + i\alpha^a \sigma^a/2 + \cdots$ requires $\sigma^2 (\sigma^a)^* \sigma^2 = -\sigma^a$. With $(\sigma^1)^* = \sigma^1$, $(\sigma^2)^* = -\sigma^2$, $(\sigma^3)^* = \sigma^3$ and the anticommutator $\{\sigma^2, \sigma^a\} = 2\delta^{2a}$, one finds $\sigma^2\sigma^1\sigma^2 = -\sigma^1$, $\sigma^2(-\sigma^2)\sigma^2 = -\sigma^2$, $\sigma^2\sigma^3\sigma^2 = -\sigma^3$, i.e. $\sigma^2(\sigma^a)^*\sigma^2 = -\sigma^a$ for all a. Exponentiating, $\sigma^2 U^* \sigma^2 = U$, which is Eq. (31.112). So $\widetilde{H}$ is a genuine **2**.

Second, $\widetilde{H}$ carries the opposite hypercharge to H. Complex conjugation flips the sign of every additive charge, so $Y_{\widetilde{H}} = -Y_H = -\frac{1}{2}$. Its components carry electric charge $Q = T^3 + Y$: the upper entry H^{0*} has $T^3 = +\frac{1}{2}$, $Y = -\frac{1}{2}$, hence $Q = 0$, while the lower entry $-H^-$ has $T^3 = -\frac{1}{2}$, $Y = -\frac{1}{2}$, hence $Q = -1$, consistent with H^- being the antiparticle of H^+. The neutral, condensing component now sits on *top*: from Eq. (31.98), $H^0 = (v + h)/\sqrt{2}$ in unitary gauge gives $H^{0*} = (v + h)/\sqrt{2}$ and $H^\pm = 0$, so

$$\left\langle \widetilde{H} \right\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} v \\ 0 \end{pmatrix}, \qquad \widetilde{H} = \frac{1}{\sqrt{2}} \begin{pmatrix} v + h \\ 0 \end{pmatrix}. \tag{31.113}$$

With $\widetilde{H}$ in hand the up-type Yukawa term is gauge invariant: $\overline{Q}\, u_R$ carries $Y = +\frac{1}{2}$, and $\overline{Q}\, \widetilde{H}\, u_R$ contracts to an $\mathrm{SU}(2)_L$ singlet with total hypercharge $+\frac{1}{2} + (-\frac{1}{2}) = 0$. The same arithmetic that condemned $\overline{Q}\, H\, u_R$ now blesses its conjugate-Higgs cousin. Substituting Eq. (31.113), only the *upper* component survives the contraction with $\overline{Q} = (\overline{u_L} \;\; \overline{d_L})$,

$$\mathcal{L}_Y^{(u)} = -\, y_u\, \overline{Q}\, \widetilde{H}\, u_R + \text{h.c.} \xrightarrow{\langle \widetilde{H} \rangle} -\frac{y_u}{\sqrt{2}} (v + h)\, \overline{u_L}\, u_R + \text{h.c.}, \tag{31.114}$$

so that, exactly parallel to the down sector,

$$m_u = \frac{y_u\, v}{\sqrt{2}}, \qquad g_{hu\bar{u}} = \frac{m_u}{v}. \tag{31.115}$$

Charge is conserved at the vertex throughout: $\overline{u_L}$ ($Q = -\frac{2}{3}$ as an antifield) meets u_R ($Q = +\frac{2}{3}$) and the neutral H^{0*} ($Q = 0$), totalling zero, just as $\overline{e_L}\, e_R H^0$ and $\overline{d_L}\, d_R H^0$ do. The whole charged-fermion sector has acquired Dirac masses from the single condensate, with no charge leakage.

Caution

The conjugate doublet $\widetilde{H} = i\sigma^2 H^*$ is mandatory in the minimal model precisely because the Standard Model contains only *one* Higgs doublet. A theory with a second doublet of hypercharge $-\frac{1}{2}$ could give the up-type quarks mass with that field directly and would not need $\widetilde{H}$; such two-Higgs-doublet models are a standard extension. Writing H^* rather than $\widetilde{H} = i\sigma^2 H^*$ would not transform as a doublet (it is an antidoublet,

Eq. (31.112)), and the term would not be gauge invariant. The factor $i\sigma^2$ is what converts the antidoublet back into a doublet and is not optional.

31.4.4 The central prediction: Higgs coupling proportional to mass

Collecting Eqs. (31.107), (31.110) and (31.115), every charged fermion obeys the same two relations, with $f \in \{e, \mu, \tau, u, c, t, d, s, b\}$ once the generation index is restored:

$$m_f = \frac{y_f\, v}{\sqrt{2}}, \qquad \mathcal{L} \supset -\frac{m_f}{v}\, h\, \bar{f}\, f. \tag{31.116}$$

Proposition 31.15 (Mass-proportional Higgs coupling)**.** *In the minimal Standard Model with a single Higgs doublet, the tree-level coupling of the physical Higgs boson h to a Dirac fermion f is fixed entirely by the fermion's mass and the universal vacuum expectation value,*

$$g_{hf\bar{f}} = \frac{m_f}{v} = \frac{y_f}{\sqrt{2}}, \tag{31.117}$$

with no independent free parameter once m_f and v are known. Equivalently, the Yukawa coupling and the mass carry the same information: $y_f = \sqrt{2}\, m_f / v$.

Proof. The Yukawa term for each charged fermion has the universal form $-y_f\, \overline{\psi_L^f}\, \Phi^f\, \psi_R^f + \text{h.c.}$, with $\Phi^f = H$ for down-type (e, d-like) and $\Phi^f = \widetilde{H}$ for up-type fermions; in both cases the condensing neutral component contributes $(v + h)/\sqrt{2}$ to the surviving contraction, by Eq. (31.103) and Eq. (31.113) respectively. Substituting and reassembling the Dirac bilinear via Eq. (31.101) gives

$$-\frac{y_f}{\sqrt{2}}(v + h)\, \bar{f} f = \underbrace{-\frac{y_f v}{\sqrt{2}}\, \bar{f} f}_{-m_f \bar{f} f} \; \underbrace{-\frac{y_f}{\sqrt{2}}\, h\, \bar{f} f}_{-(m_f/v)\, h\, \bar{f} f}, \tag{31.118}$$

identifying $m_f = y_f v/\sqrt{2}$ from the first term and $g_{hf\bar{f}} = y_f/\sqrt{2} = m_f/v$ from the second. The two terms share the single coefficient $y_f/\sqrt{2}$, so the coupling is the mass divided by v. □

This single line of algebra carries the most consequential prediction of the electroweak Higgs sector. Both the mass and the coupling descend from the *same* term and the *same* condensate; they cannot be adjusted independently. Heavier fermions couple to the Higgs more strongly, in strict linear proportion. The relation Eq. (31.117) is what experiments at the Large Hadron Collider test directly when they measure the rates for $h \to b\bar{b}$, $h \to \tau^+\tau^-$, and $h \to \mu^+\mu^-$ and the production rate through the top-quark loop and through $t\bar{t}h$ associated

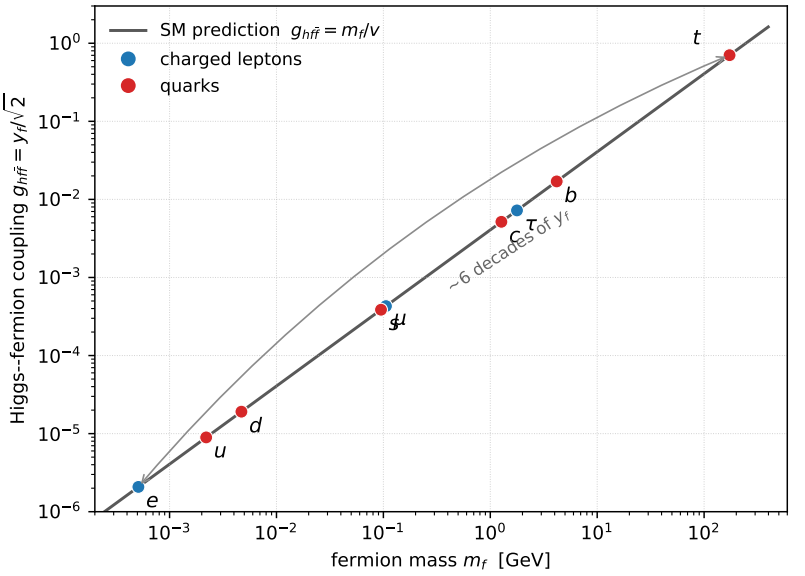


Figure 31.3: The central testable prediction of the Yukawa sector, Eq. (31.117): the Higgs–fermion coupling $g_{hf\bar{f}} = y_f/\sqrt{2}$ equals m_f/v, a straight line of unit slope on a log–log plot whose intercept fixes the vacuum expectation value $v \simeq 246$ GeV. The nine charged fermions lie on the line by construction; experiment tests whether nature does. The electron-to-top span covers roughly six decades of Yukawa coupling, the hierarchy whose origin the minimal model does not explain.

production: each rate scales as $g^2_{hf\bar{f}} \propto m_f^2$, and the measured couplings are required to lie on the straight line of Fig. 31.3. A deviation from proportionality would be unambiguous evidence that the single-doublet picture is incomplete.

Physical Insight

The mechanism is economical to the point of austerity. The breaking of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ produced one physical scalar h and one scale v; the Yukawa sector reuses both. The condensate v supplies the chirality-flipping mass, and the fluctuation h about that condensate is, by construction, the *same field*, so its coupling is the derivative of the mass with respect to the condensate: $g_{hf\bar{f}} = \partial m_f/\partial v = m_f/v$. This is why the Higgs couples to mass and not to charge, isospin, or any other quantum number. The Higgs boson is, almost literally, the quantized question "how much condensate is here?", and a fermion answers in proportion to how much of its mass it owes to that condensate.

Worked Example 31.16 (Yukawa couplings of the electron and the top quark)**.**

The relation $y_f = \sqrt{2}\, m_f / v$ turns measured masses into couplings, exposing the enormous spread of Yukawa strengths in a theory with a single v. Anticipating the next section, which fixes $v \simeq 246$ GeV by matching to the Fermi constant, take $v = 246$ GeV.

For the electron, $m_e = 0.511$ MeV $= 0.511 \times 10^{-3}$ GeV, so

$$y_e = \frac{\sqrt{2}\, m_e}{v} = \frac{1.414 \times 0.511 \times 10^{-3}}{246} \approx 2.9 \times 10^{-6}, \qquad g_{he\bar{e}} = \frac{m_e}{v} \approx 2.1 \times 10^{-6}. \tag{31.119}$$

For the top quark, $m_t \approx 173$ GeV, so

$$y_t = \frac{\sqrt{2}\, m_t}{v} = \frac{1.414 \times 173}{246} \approx 0.99, \qquad g_{ht\bar{t}} = \frac{m_t}{v} \approx 0.70. \tag{31.120}$$

The top Yukawa is of order unity — $y_t \approx 1$ is one of the most suggestive numbers in particle physics, hinting that the top is tied to electroweak breaking in a special way — while the electron Yukawa is smaller by nearly six orders of magnitude. The Standard Model accommodates this hierarchy but does not explain it: the y_f are inputs, not outputs. That the values range over $\sim 10^6$ for the charged fermions (and far more once neutrinos are included, see below) is the *flavor puzzle*, untouched by the mechanism that generates the masses themselves.

31.4.5 Why the neutrino is massless in the minimal model

The accounting that gave masses to e, u, and d now tells us, with equal force, what it cannot do for the neutrino. A neutrino mass would require one of two kinds of term, and in the minimal field content both are unavailable.

No Dirac mass, because there is no right-handed neutrino. A Dirac mass for the neutrino would be generated, in exact analogy to the up quark, by a Yukawa coupling $-y_\nu\, \overline{L}\, \widetilde{H}\, \nu_R + \text{h.c.}$, condensing to $m_\nu = y_\nu v/\sqrt{2}$. But this term requires a field ν_R: a right-handed, $\mathrm{SU}(2)_L$-singlet neutrino with which the left-handed ν_L inside L could be paired through Eq. (31.101). The minimal Standard Model simply does not contain such a field. Its matter content was fixed — one generation — as L, e_R, Q, u_R, d_R, and nothing else; there is no electroweak-singlet partner for ν_L to marry. With only ν_L present, the would-be mass term $\overline{\nu_L}\, \nu_R$ has no ν_R to write, and the Dirac route is closed by the absence of a field, not by a symmetry.

No renormalizable Majorana mass, because gauge invariance forbids it. One might try to give ν_L a mass using only the field already present, by pairing ν_L with its own charge conjugate — a Majorana mass. The relevant bilinear is $\overline{(\nu_L)^c}\, \nu_L$ (with $\psi^c = C\bar{\psi}^{\mathsf{T}}$ the charge conjugate). This object carries *twice* the quantum numbers of ν_L: weak isospin builds from two $T = \frac{1}{2}$ doublets, so the

bilinear contains a $T = 1$ (triplet) piece, and the hypercharge is $2 \times Y_L = -1$, hence electric charge content with $Q = 2T^3 + 2Y$ that includes a $Q = 0$ neutral member only inside an isotriplet of $Y = -1$. A bare mass term $-\frac{1}{2} m \overline{(\nu_L)^c} \nu_L + \text{h.c.}$ is thus neither an $\mathrm{SU}(2)_L$ singlet nor hypercharge-neutral; it transforms as part of a triplet with $Y = -1$ and is forbidden by the unbroken gauge symmetry, exactly as the bare Dirac mass of Section 31.4.1 was.

The lowest gauge-invariant neutrino mass operator is non-renormalizable. Gauge invariance does permit a neutrino mass operator if one is willing to use the Higgs doublet to soak up the offending quantum numbers, but the cheapest such operator is not renormalizable. Combining two lepton doublets and two Higgs doublets,

$$\mathcal{L}_5 = \frac{c}{\Lambda} \left(L^{\mathsf{T}} \, \epsilon \, H\right) C \left(H^{\mathsf{T}} \, \epsilon \, L\right) + \text{h.c.}, \qquad \epsilon \equiv i\sigma^2, \tag{31.121}$$

with C the charge-conjugation matrix, is an $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet: each factor $H^{\mathsf{T}} \epsilon L = H^+ e_L - H^0 \nu_L$ is an $\mathrm{SU}(2)_L$ singlet (the ϵ-contraction of two doublets) of hypercharge $Y_H + Y_L = \frac{1}{2} - \frac{1}{2} = 0$. Upon condensation $H^0 \to v/\sqrt{2}$ the surviving piece is $-\frac{v}{\sqrt{2}} \nu_L$ in each factor, leaving a Majorana neutrino mass $m_\nu \sim c\, v^2/\Lambda$. But $\mathcal{L}_5$ is an operator of mass dimension five (four fields of total dimension $\frac{3}{2} + \frac{3}{2} + 1 + 1 = 5$), so its coefficient c/Λ carries an inverse mass: it is non-renormalizable, suppressed by some heavy scale Λ at which new physics enters, and therefore absent from the renormalizable Lagrangian that defines the minimal Standard Model. This dimension-five operator is the unique lowest-dimension way to give neutrinos mass without enlarging the field content, and its very non-renormalizability is the precise statement of the following result.

Proposition 31.17 (Massless neutrinos in the minimal Standard Model)**.** *In the Standard Model with the minimal renormalizable field content — no right-handed neutrino, one Higgs doublet, and only operators of mass dimension ≤ 4 — the neutrino has no mass term. A Dirac mass is excluded by the absence of ν_R; a Majorana mass for ν_L is excluded by gauge invariance; and the gauge-invariant Weinberg operator Eq. (31.121) that would generate one is of dimension five, hence not part of the renormalizable theory. The neutrino is therefore exactly massless at this level.*

This is a genuine prediction of the minimal model, and it is the place where nature is now known to depart from it: neutrino oscillations establish that neutrinos do have small masses and that lepton flavor mixes. The minimal Standard Model must accordingly be extended — by adding right-handed neutrinos (a Dirac mass), or by promoting Eq. (31.121) to a real operator generated by heavy states (the seesaw, a Majorana mass), or both. The lepton-mixing matrix that then appears, U_{PMNS}, is the leptonic analogue of the quark mixing matrix and is

treated alongside it in Chapter 32. Within *this* chapter, working strictly at the renormalizable level with the minimal content, the neutrino stays massless, and we will carry it as such through the current algebra of the next section.

31.4.6 The one-generation result and the road to many

We have, for a single generation, the complete Yukawa Lagrangian

$$\mathcal{L}_Y = -\,y_e\,\overline{L}\,H\,e_R - \,y_d\,\overline{Q}\,H\,d_R - \,y_u\,\overline{Q}\,\widetilde{H}\,u_R + \text{h.c.}, \tag{31.122}$$

which, upon $\langle H \rangle$, delivers the charged-fermion masses

$$m_e = \frac{y_e v}{\sqrt{2}}, \qquad m_d = \frac{y_d v}{\sqrt{2}}, \qquad m_u = \frac{y_u v}{\sqrt{2}}, \tag{31.123}$$

each accompanied by a Higgs coupling $g_{hf\bar{f}} = m_f/v$, while the neutrino remains massless. With one generation there is nothing left to diagonalize: each Yukawa coupling is a single number, automatically made real and positive by a rephasing of the right-handed field, and the mass basis coincides with the gauge (interaction) basis. Every fermion mass in the theory is thereby traded for a dimensionless input y_f and the universal scale v.

Notation Watch

With three generations, each y_f in Eq. (31.122) becomes a 3×3 complex matrix $\mathbf{y}_f$ in generation space, and the mass terms $\overline{\psi_L}\,\mathbf{m}_f\,\psi_R$ are no longer diagonal. Diagonalizing them by independent unitary rotations of the left- and right-handed fields is what misaligns the up- and down-quark mass bases relative to the weak-interaction basis; the residual mismatch is the Cabibbo–Kobayashi–Maskawa matrix V_{CKM}, and the single physical complex phase it cannot rotate away is the Standard Model's source of CP violation. This is the one forward reference of the chapter: V_{CKM} and CP violation are defined and developed in Chapter 32. Nothing in the present one-generation analysis is altered by that extension — the masses and the mass-proportional couplings Eq. (31.116) survive verbatim in the mass eigenbasis.

With the charged fermions now massive and their Higgs couplings fixed, the one-generation particle spectrum is complete: the massive $W^{\pm}$ and Z and the massless photon from the gauge sector, the physical Higgs scalar h, and the massive Dirac electron and quarks u, d, alongside the massless neutrino. The next section takes this spectrum and expresses the fermion gauge interactions in the mass eigenbasis, extracting the charged-current and neutral-current Lagrangians and the exact QED limit, and then matches to Fermi theory to fix the numerical value $v \simeq 246$ GeV that has stood, until now, as an unmeasured scale.

31.5 Electroweak Currents, Couplings, and the Fermi Limit

The previous section completed the spectrum of one generation: the doublet fields absorb their would-be Goldstone partners, three of the four electroweak gauge bosons acquire mass to become the charged $W^{\pm}$ and the neutral Z, one combination remains massless as the photon, and the charged fermions pick up Dirac masses from the Yukawa couplings to the Higgs vacuum value. Every field in the theory now has a definite mass and a definite electric charge. What we have not yet done is read off how those mass-eigenstate fields actually *talk* to the fermions. The covariant derivative Eq. (31.23) was written in the symmetry eigenbasis (W^1, W^2, W^3, B) of the unbroken gauge group; the particles that propagate and are detected are the mass eigenstates $(W^{\pm}, Z, \gamma)$. This section performs the change of basis. The payoff is the complete catalogue of fermion gauge interactions: a charged-current Lagrangian that reproduces the V$-$A structure we started from, a neutral-current Lagrangian with a definite vector and axial coupling for every fermion, and—as an output rather than an input—the exact electromagnetic interaction $eA_\mu J^\mu_{\rm em}$ with the photon coupling to charge Q for every field. Closing the circuit, we integrate out the heavy W to recover Fermi theory and so fix the Higgs vacuum value numerically at $v \approx 246\,\mathrm{GeV}$, and we turn $(\alpha, G_F, \theta_{\rm W})$ into a tree-level prediction for M_W and M_Z.

31.5.1 The fermion gauge interaction in the broken basis

The fermion kinetic term is the sum, over every left-handed doublet and every right-handed singlet of the generation, of $\bar\Psi\, i\not{D}\,\Psi$ with the covariant derivative Eq. (31.23). Splitting $D_\mu = \partial_\mu + (\text{gauge piece})$ separates the free Dirac Lagrangian from the interactions; only the gauge piece concerns us here. Writing Ψ for any of the multiplets and keeping its $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ quantum numbers implicit in the generators, the interaction Lagrangian is

$$\mathcal{L}_{\rm int} = \bar\Psi\,\gamma^\mu\big(g\,W^a_\mu T^a + g'\,Y\,B_\mu\big)\Psi = g\,W^a_\mu\,J^{a\,\mu} + g'\,B_\mu\,J^\mu_Y, \tag{31.124}$$

where we have defined the isospin and hypercharge currents

$$J^a_\mu = \bar\Psi\,\gamma_\mu T^a\,\Psi, \qquad J^Y_\mu = \bar\Psi\,\gamma_\mu Y\,\Psi, \tag{31.125}$$

with $T^a = \tfrac{1}{2}\sigma^a$ acting on the left-handed doublets only (the right-handed singlets are annihilated by every T^a), and Y the hypercharge, which acts on all fields. The four gauge fields W^1, W^2, W^3, B organize into two physically distinct groups. The off-diagonal pair W^1, W^2 carries electric charge and combines into the charged $W^{\pm}$; the diagonal pair W^3, B is electrically neutral and rotates into the physical Z and γ. We treat the two groups in turn.

31.5.2 The charged-current Lagrangian

The charged W bosons are the complex combinations introduced when the kinetic term $-\frac{1}{4}W^a_{\mu\nu}W^{a\,\mu\nu}$ was diagonalized,

$$W^\pm_\mu = \frac{1}{\sqrt{2}}(W^1_\mu \mp i\, W^2_\mu), \qquad \text{equivalently} \qquad W^1_\mu = \frac{W^+_\mu + W^-_\mu}{\sqrt{2}}, \quad W^2_\mu = \frac{i\,(W^+_\mu - W^-_\mu)}{\sqrt{2}}. \tag{31.126}$$

On the generator side, the raising and lowering operators $T^\pm = T^1 \pm i\,T^2$ of Eq. (31.15) invert to $T^1 = \frac{1}{2}(T^+ + T^-)$ and $T^2 = -\frac{i}{2}(T^+ - T^-)$. Substituting both into the $a = 1, 2$ part of Eq. (31.124) and collecting terms,

$$\begin{aligned} g(W^1_\mu J^{1\,\mu} + W^2_\mu J^{2\,\mu}) &= g\,\bar\Psi\gamma^\mu(W^1_\mu T^1 + W^2_\mu T^2)\Psi \\ &= \frac{g}{2\sqrt{2}}\,\bar\Psi\gamma^\mu\Big[(W^+_\mu + W^-_\mu)(T^+ + T^-) + (W^+_\mu - W^-_\mu)(T^+ - T^-)\Big]\Psi \\ &= \frac{g}{\sqrt{2}}\,\bar\Psi\gamma^\mu(W^+_\mu T^+ + W^-_\mu T^-)\Psi, \end{aligned} \tag{31.127}$$

where the cross terms W^+T^- and W^-T^+ cancelled between the two products and the diagonal terms doubled. The operator T^+ raises the lower component of a doublet into the upper one—it turns e_L into ν_{eL} inside $\bar\Psi\gamma^\mu T^+\Psi$—so on the lepton doublet $L = (\nu_{eL}, e_L)^{\mathsf T}$ of Eq. (31.14) we have $\bar L\gamma^\mu T^+ L = \overline{\nu_{eL}}\,\gamma^\mu e_L$, and likewise on the quark doublet $Q = (u_L, d_L)^{\mathsf T}$ we get $\overline{u_L}\,\gamma^\mu d_L$. Defining the *charged weak current*

$$J^+_\mu \equiv \overline{\nu_{eL}}\,\gamma_\mu e_L + \overline{u_L}\,\gamma_\mu d_L = \bar\nu_e\,\gamma_\mu P_L\, e + \bar u\,\gamma_\mu P_L\, d, \qquad J^-_\mu = (J^+_\mu)^\dagger, \tag{31.128}$$

the charged-current interaction is

$$\boxed{\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\,(W^{+\mu}\,J^+_\mu + W^{-\mu}\,J^-_\mu) = \frac{g}{\sqrt{2}}\,(W^{+\mu}\,J^+_\mu + \text{h.c.}).} \tag{31.129}$$

This is the first key result. The structure is exactly the V−A coupling demanded by the data in Section 31.1.1: only the left-handed projection $P_L = \frac{1}{2}(1 - \gamma^5)$ appears, because $T^\pm$ act on the left-handed doublets alone. The single coupling g is common to leptons and quarks, the gauge universality anticipated in Section 31.1.1.

Notation Watch

The current Eq. (31.128) is normalized with a chiral projector $P_L = \frac{1}{2}(1-\gamma^5)$, whereas the empirical current Eq. (31.1) of Section 31.1 carried the full factor $(1-\gamma^5)$. The two differ by a factor of two, $J^+_\mu = \frac{1}{2}\,J^+_\mu\big|_{(31.1)}$, and the difference is absorbed into the conventional placement of the $1/\sqrt{2}$ in Eq. (31.129) versus the $1/\sqrt{2}$ in the Fermi Lagrangian Eq. (31.3); the

net coupling of four fermions, fixed by G_F, is identical, as the matching in Theorem 31.19 confirms. The $\overline{u_L}\gamma_\mu d_L$ entry is written here without flavour mixing; the replacement $d \to d' = \sum_j V_{ud_j} d_j$ that introduces the Cabibbo–Kobayashi–Maskawa matrix is the subject of Chapter 32.

31.5.3 The neutral-current Lagrangian and the emergence of the photon

The neutral sector is the heart of the construction, because it is where the photon must *appear*. The two neutral gauge fields W_μ^3 and B_μ are not the mass eigenstates; diagonalizing the mass matrix generated by the Higgs vacuum value rotated them by the weak mixing angle θ_{W} into the massless photon A_μ and the massive Z_μ,

$$\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_{\mathrm{W}} & \sin\theta_{\mathrm{W}} \\ -\sin\theta_{\mathrm{W}} & \cos\theta_{\mathrm{W}} \end{pmatrix} \begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix}, \qquad \begin{aligned} W_\mu^3 &= \cos\theta_{\mathrm{W}}\, Z_\mu + \sin\theta_{\mathrm{W}}\, A_\mu, \\ B_\mu &= -\sin\theta_{\mathrm{W}}\, Z_\mu + \cos\theta_{\mathrm{W}}\, A_\mu, \end{aligned} \tag{31.130}$$

which we recall from the gauge-boson mass diagonalization (Eq. (31.79)); the mixing angle and the couplings are tied together there by

$$\tan\theta_{\mathrm{W}} = \frac{g'}{g}, \qquad e = g\,\sin\theta_{\mathrm{W}} = g'\,\cos\theta_{\mathrm{W}}, \tag{31.131}$$

the second of which (Eq. (31.83)) is the defining relation of the electromagnetic coupling in terms of the two gauge couplings. The task is to insert Eq. (31.130) into the neutral part of Eq. (31.124) and read off the coefficients of A_μ and Z_μ. The next worked example does this with every term displayed, and proves the central claim that the photon coupling comes out exactly equal to eQ.

Worked Example 31.18 (The neutral current splits into a pure eQ photon term and the Z current)**.** Start from the neutral gauge interaction extracted from Eq. (31.124),

$$\mathcal{L}_{\mathrm{neutral}} = g\, W_\mu^3\, J^{3\,\mu} + g'\, B_\mu\, J_Y^\mu = \bar{\Psi}\gamma^\mu (g\, W_\mu^3 T^3 + g'\, B_\mu Y)\Psi. \tag{31.132}$$

Substitute the rotation Eq. (31.130) for W_μ^3 and B_μ and group the result by the two physical fields:

$$\begin{aligned} g\, W_\mu^3 T^3 + g'\, B_\mu Y &= g(\cos\theta_{\mathrm{W}}\, Z_\mu + \sin\theta_{\mathrm{W}}\, A_\mu)T^3 + g'(-\sin\theta_{\mathrm{W}}\, Z_\mu + \cos\theta_{\mathrm{W}}\, A_\mu)Y \\ &= Z_\mu\,(g\cos\theta_{\mathrm{W}}\, T^3 - g'\sin\theta_{\mathrm{W}}\, Y) + A_\mu\,(g\sin\theta_{\mathrm{W}}\, T^3 + g'\cos\theta_{\mathrm{W}}\, Y). \end{aligned} \tag{31.133}$$

The photon term is exactly eQ. Look at the coefficient of A_μ. The relations Eq. (31.131) say $g\sin\theta_{\mathrm{W}} = e$ and $g'\cos\theta_{\mathrm{W}} = e$, so *both* terms in the

A_μ bracket carry the same factor e:

$$g \sin\theta_{\mathrm{W}}\, T^3 + g' \cos\theta_{\mathrm{W}}\, Y = e\, T^3 + e\, Y = e\,(T^3 + Y) = e\, Q, \tag{31.134}$$

using the Gell-Mann–Nishijima relation Eq. (31.20). The two gauge couplings g and g' have conspired, through the single mixing angle, to make the photon couple to the charge operator Q and to nothing else. There is no free parameter left: the photon's coupling strength is e, identical for every field, and proportional to its electric charge. This is the precise statement that electromagnetism is an *output* of the electroweak construction.

The Z term is the neutral weak current. Now the coefficient of Z_μ. Eliminate g' in favour of g using $g' = g\tan\theta_{\mathrm{W}}$ from Eq. (31.131), and eliminate Y in favour of Q and T^3 using $Y = Q - T^3$:

$$\begin{aligned} g\cos\theta_{\mathrm{W}}\, T^3 - g'\sin\theta_{\mathrm{W}}\, Y &= g\cos\theta_{\mathrm{W}}\, T^3 - g\,\frac{\sin^2\theta_{\mathrm{W}}}{\cos\theta_{\mathrm{W}}}\, Y \\ &= \frac{g}{\cos\theta_{\mathrm{W}}}\left(\cos^2\theta_{\mathrm{W}}\, T^3 - \sin^2\theta_{\mathrm{W}}\, Y\right) \\ &= \frac{g}{\cos\theta_{\mathrm{W}}}\Big(\cos^2\theta_{\mathrm{W}}\, T^3 - \sin^2\theta_{\mathrm{W}}\,(Q - T^3)\Big) \\ &= \frac{g}{\cos\theta_{\mathrm{W}}}\Big((\cos^2\theta_{\mathrm{W}} + \sin^2\theta_{\mathrm{W}})\, T^3 - \sin^2\theta_{\mathrm{W}}\, Q\Big) = \frac{g}{\cos\theta_{\mathrm{W}}}(T^3 - \sin \end{aligned} \tag{31.135}$$

where the Pythagorean identity collapsed the T^3 coefficients to unity. The whole neutral Lagrangian Eq. (31.132) therefore reorganizes, with no approximation, into a pure electromagnetic piece plus a Z piece,

$$\mathcal{L}_{\mathrm{neutral}} = e\, A_\mu\, \bar\Psi \gamma^\mu Q\, \Psi \;+\; \frac{g}{\cos\theta_{\mathrm{W}}}\, Z_\mu\, \bar\Psi \gamma^\mu (T^3 - \sin^2\!\theta_{\mathrm{W}}\, Q)\, \Psi. \tag{31.136}$$

The split is exact and basis-independent: it followed entirely from the geometry of the rotation Eq. (31.130) together with the two algebraic identities Eq. (31.131) and Eq. (31.20).

The two pieces of Eq. (31.136) are the remaining key results of this section. Reading off the photon coupling and summing over the fields of the generation gives the electromagnetic interaction

$$\mathcal{L}_{\mathrm{em}} = e\, A_\mu\, J^\mu_{\mathrm{em}}, \qquad J^\mu_{\mathrm{em}} = \bar\Psi \gamma^\mu Q\, \Psi = \sum_f Q_f\, \bar f \gamma^\mu f, \tag{31.137}$$

which is identical, coupling and current alike, to the QED interaction $eA_\mu J^\mu_{\mathrm{em}}$ of Eq. (8.146). The neutral weak interaction is

$$\boxed{\mathcal{L}_{\mathrm{NC}} = \frac{g}{\cos\theta_{\mathrm{W}}}\, Z_\mu\, J^{Z\,\mu}, \qquad J^Z_\mu = J^3_\mu - \sin^2\!\theta_{\mathrm{W}}\, J^{\mathrm{em}}_\mu,} \tag{31.138}$$

where $J^3_\mu = \bar\Psi \gamma_\mu T^3 \Psi$ is the third isospin current (built from left-handed fields only) and $J^{\rm em}_\mu$ is the same electromagnetic current that the photon sees. The neutral current is thus *not* purely left-handed: it is the left-handed isospin current minus a chirality-blind multiple of the electromagnetic current, and the size of the parity-violating admixture is set by $\sin^2\theta_{\rm W}$.

31.5.4 Per-fermion vector and axial couplings

To use Eq. (31.138) in a cross section one wants the Z coupling of a single Dirac fermion f, written in the standard vector/axial form. The operator $T^3 - \sin^2\theta_{\rm W}\, Q$ acts differently on the two chiralities: T^3 has eigenvalue T^3_f on the left-handed part and 0 on the right-handed part (which is an $\mathrm{SU}(2)_L$ singlet), while Q has the same eigenvalue Q_f on both. Hence, for one fermion,

$$\bar f \gamma^\mu (T^3 - \sin^2\theta_{\rm W}\, Q) f = \bar f \gamma^\mu \Big(T^3_f\, P_L - \sin^2\theta_{\rm W}\, Q_f \Big) f. \tag{31.139}$$

Writing $P_L = \tfrac{1}{2}(1-\gamma^5)$ and separating the chirality-even and -odd parts,

$$T^3_f\, P_L - \sin^2\theta_{\rm W}\, Q_f = \underbrace{\Big(\tfrac{1}{2} T^3_f - \sin^2\theta_{\rm W}\, Q_f\Big)}_{\frac{1}{2} g^f_V} - \underbrace{\tfrac{1}{2} T^3_f}_{\frac{1}{2} g^f_A}\, \gamma^5, \tag{31.140}$$

so that the Z coupling of the fermion takes the canonical form $\dfrac{g}{\cos\theta_{\rm W}}\, Z_\mu\, \bar f \gamma^\mu \tfrac{1}{2}(g^f_V - g^f_A \gamma^5) f$ with

$$\boxed{\, g^f_V = T^3_f - 2\, Q_f\, \sin^2\theta_{\rm W}, \qquad g^f_A = T^3_f. \,} \tag{31.141}$$

The axial coupling is simply the weak isospin; the vector coupling is the isospin shifted by twice the charge weighted by $\sin^2\theta_{\rm W}$. Both are pure numbers once $\sin^2\theta_{\rm W}$ is fixed. Table 31.2 tabulates them for the four fermion types of one generation, including the numerical vector couplings at the measured value $\sin^2\theta_{\rm W} \simeq 0.231$.

The near-vanishing of $g^e_V = -\tfrac{1}{2} + 2\sin^2\theta_{\rm W} = -0.038$ is a genuine numerical accident of the measured value of $\sin^2\theta_{\rm W}$: the electron's vector coupling to the Z is suppressed because $\sin^2\theta_{\rm W}$ happens to lie close to $\tfrac{1}{4}$. This is not a symmetry, but it has real consequences—it makes leptonic forward–backward asymmetries at the Z pole exquisitely sensitive probes of $\sin^2\theta_{\rm W}$, which is one reason the precision program of Chapter 34 can pin the angle so tightly.

31.5.5 Recovery of QED and the QED handoff

The electromagnetic Lagrangian Eq. (31.137) deserves to be examined as the climax of the whole gauge-theory arc, because it is where QED is recovered

Table 31.2: Quantum numbers and neutral-current couplings of one fermion generation, from Eq. (31.141). The numerical g_V^f use $\sin^2\theta_{\rm W} = 0.231$. The right-handed neutrino is absent in the minimal model, so ν has no right-handed partner; its couplings are those of a purely left-handed field.

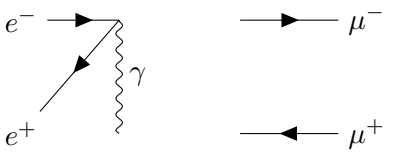
fermion f	T_f^3	Q_f	$g_A^f = T_f^3$	$g_V^f = T_f^3 - 2Q_f \sin^2\theta_{\rm W}$	g_V^f (numeric)
ν_e	$+\frac{1}{2}$	0	$+\frac{1}{2}$	$+\frac{1}{2}$	$+0.500$
e	$-\frac{1}{2}$	-1	$-\frac{1}{2}$	$-\frac{1}{2} + 2\sin^2\theta_{\rm W}$	-0.038
u	$+\frac{1}{2}$	$+\frac{2}{3}$	$+\frac{1}{2}$	$+\frac{1}{2} - \frac{4}{3}\sin^2\theta_{\rm W}$	$+0.192$
d	$-\frac{1}{2}$	$-\frac{1}{3}$	$-\frac{1}{2}$	$-\frac{1}{2} + \frac{2}{3}\sin^2\theta_{\rm W}$	-0.346

as an output of unification. The coupling that multiplies A_μ in Eq. (31.136) was not inserted; it emerged as $e = g\sin\theta_{\rm W}$, a derived combination of the two electroweak couplings. The current it multiplies is exactly $\bar\Psi\gamma^\mu Q\Psi$, with Q the Gell-Mann–Nishijima charge Eq. (31.20). Specializing to the electron, $Q_e = -1$, the interaction is $-e\,A_\mu\,\bar e\gamma^\mu e$, which is precisely the QED vertex generated by the covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ acting on a field of charge $-e$—the convention fixed for the electron in QED. The photon of Chapter 8 and Chapter 17, with its universal coupling e to electric charge, is reproduced on the nose; nothing about the QED vertex, its Feynman rule, or the value of its coupling has changed.

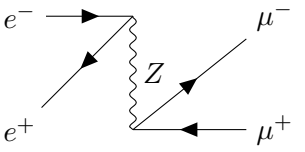
This makes the electroweak theory continuous with everything computed in QED. The unpolarized cross section for $e^+e^- \to \mu^+\mu^-$ assembled in Section 17.3 from the photon-exchange amplitude is the $q^2 \ll M_Z^2$ limit of the full electroweak amplitude for the same process: at low energies only the massless photon propagates appreciably and Eq. (17.62) is the whole story. What Eq. (31.138) adds is a second, parallel s-channel diagram—exchange of the Z—which interferes with the photon and grows in importance as $\sqrt{s}$ approaches M_Z, becoming resonant at the pole (Fig. 31.4). The vector and axial couplings Eq. (31.141) of the electron and the muon enter that diagram, so the same process that gave the textbook QED cross section becomes, near the Z pole, a direct measurement of g_V, g_A, and hence $\sin^2\theta_{\rm W}$. The machinery for turning that interference and the Z resonance into precision tests of the electroweak theory is developed in Chapter 34; here we have established the vertices it needs.

31.5.6 The Fermi limit: fixing the vacuum value

We have now expressed every fermion gauge interaction in the mass eigenbasis. The last task of the section is to close the loop with the low-energy phenomenology that motivated the whole chapter, by integrating the heavy W out of the charged-current Lagrangian Eq. (31.129) and matching to the Fermi interaction Eq. (31.3). This both confirms the matching relation Eq. (31.13) from the broken-phase fields



(a) Photon exchange: the QED amplitude of Eq. (17.62), coupling eQ at each vertex.



(b) Z exchange from Eq. (31.138), coupling $\frac{g}{\cos\theta_W}\frac{1}{2}(g_V - g_A\gamma^5)$; resonant at $\sqrt{s} = M_Z$.

Figure 31.4: The two s-channel diagrams for $e^+e^- \to \mu^+\mu^-$ in the electroweak theory. Diagram (a) is the photon exchange of QED (Section 17.3); diagram (b) is the new neutral-current contribution. They interfere; below the Z mass (b) is a small correction, near $\sqrt{s} = M_Z$ it dominates. Disentangling the interference is the precision program of Chapter 34.

and—once combined with the W mass generated by the Higgs mechanism—fixes the numerical value of the vacuum expectation value v.

Worked Example 31.19 (Integrating out the W and reading off G_F and v)**.** At momentum transfer $q^2 \ll M_W^2$ the W propagator collapses to a constant, $-i(\eta_{\mu\nu} - q_\mu q_\nu/M_W^2)/(q^2 - M_W^2) \to i\eta_{\mu\nu}/M_W^2$, which is Eq. (31.12). In Lagrangian language the same statement is that, to leading order in $1/M_W^2$, the W field obeys its algebraic (no-derivative) equation of motion and can be eliminated. Take the charged-current Lagrangian together with the W mass term,

$$\mathcal{L} = M_W^2\, W_\mu^+ W^{-\mu} + \frac{g}{\sqrt{2}}(W_\mu^+ J^{-\mu} + W_\mu^- J^{+\mu}), \tag{31.142}$$

treating W_μ^+ and W_μ^- as independent complex fields. Varying with respect to W_μ^+ and W_μ^- gives the algebraic equations of motion

$$M_W^2\, W^{-\mu} + \frac{g}{\sqrt{2}}\, J^{-\mu} = 0, \qquad M_W^2\, W^{+\mu} + \frac{g}{\sqrt{2}}\, J^{+\mu} = 0, \tag{31.143}$$

that is, $W_\mu^\pm = -\dfrac{g}{\sqrt{2}\, M_W^2}\, J_\mu^\pm$. Substituting back into Eq. (31.142), the three contributions combine as

$$\begin{aligned}\mathcal{L}_{\text{eff}} &= M_W^2\, \frac{g^2}{2M_W^4}\, J_\mu^+ J^{-\mu} + \frac{g}{\sqrt{2}}\Big(-\frac{g}{\sqrt{2}M_W^2} J_\mu^-\Big) J^{-\mu}\Big|_+ \;-\; (\text{two source terms}) \\ &= \frac{g^2}{2M_W^2} J_\mu^+ J^{-\mu} - \frac{g^2}{M_W^2} J_\mu^+ J^{-\mu} = -\frac{g^2}{2M_W^2}\, J_\mu^+ J^{-\mu},\end{aligned} \tag{31.144}$$

where the mass term contributes $+\frac{g^2}{2M_W^2} J^+ J^-$ and the two source terms each contribute $-\frac{g^2}{2M_W^2} J^+ J^-$, summing to the displayed result. This is a current–current contact interaction, exactly the form of the Fermi Lagrangian. To match

coefficients we must use the *same* current normalization in both. The current $J^{\pm}$ here is the P_L-normalized current Eq. (31.128), related to the $(1-\gamma^5)$-normalized current of Eq. (31.3) by $J^{\pm}|_{(31.1)} = 2J^{\pm}$, so $J^{+}J^{-}|_{(31.3)} = 4\,J^{+}_{\mu}J^{-\mu}$. Writing Eq. (31.3) as $-\frac{G_F}{\sqrt{2}}\cdot 4\,J^{+}_{\mu}J^{-\mu}$ and equating to Eq. (31.144),

$$-\frac{4G_F}{\sqrt{2}}\,J^{+}_{\mu}J^{-\mu} = -\frac{g^2}{2M_W^2}\,J^{+}_{\mu}J^{-\mu} \quad\Longrightarrow\quad \boxed{\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}} \tag{31.145}$$

which is precisely Eq. (31.13), now derived from the broken-phase fields rather than asserted.

From G_F to v. The Higgs mechanism gave the W mass $M_W = \frac{1}{2}gv$, i.e. $M_W^2 = \frac{1}{4}g^2v^2$ (Eq. (31.69)). Substituting into Eq. (31.145) the gauge coupling cancels completely:

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\cdot\frac{1}{4}g^2v^2} = \frac{1}{2v^2} \quad\Longrightarrow\quad \boxed{v = \left(\sqrt{2}\,G_F\right)^{-1/2}.} \tag{31.146}$$

The vacuum value is therefore fixed by the single best-measured weak observable, with no reference to g, g', or θ_{W}. Inserting $G_F = 1.166\times10^{-5}\,\mathrm{GeV}^{-2}$ from Eq. (31.4),

$$v = \left(\sqrt{2}\times1.166\times10^{-5}\,\mathrm{GeV}^{-2}\right)^{-1/2} = \left(1.649\times10^{-5}\right)^{-1/2}\mathrm{GeV} = 246.2\,\mathrm{GeV}. \tag{31.147}$$

This is the electroweak scale. Every dimensionful quantity in the bosonic sector—$M_W = \frac{1}{2}gv$, $M_Z = M_W/\cos\theta_{\mathrm{W}}$, and the Higgs mass $m_h = \sqrt{2\lambda}\,v$— is now anchored to this number; the dimensionless couplings g, g', λ set the rest.

31.5.7 The tree-level prediction for M_W and M_Z

The matching Eq. (31.145) is more than a consistency check: read the other way, it predicts the gauge-boson masses from three independent and precisely measured inputs—the fine-structure constant α, the Fermi constant G_F, and the weak mixing angle θ_{W}. Solve Eq. (31.145) for M_W^2 and eliminate g in favour of e using $g = e/\sin\theta_{\mathrm{W}}$ from Eq. (31.131) and $e^2 = 4\pi\alpha$:

$$M_W^2 = \frac{g^2}{8\,G_F/\sqrt{2}} = \frac{\sqrt{2}\,g^2}{8\,G_F} = \frac{\sqrt{2}}{8\,G_F}\cdot\frac{e^2}{\sin^2\theta_{\mathrm{W}}} = \frac{\sqrt{2}}{8\,G_F}\cdot\frac{4\pi\alpha}{\sin^2\theta_{\mathrm{W}}}, \tag{31.148}$$

which simplifies, using $\sqrt{2}\cdot 4\pi/8 = \pi/\sqrt{2}$, to

$$\boxed{M_W^2 = \frac{\pi\alpha}{\sqrt{2}\,G_F\,\sin^2\theta_{\mathrm{W}}}, \qquad M_Z^2 = \frac{M_W^2}{\cos^2\theta_{\mathrm{W}}} = \frac{\pi\alpha}{\sqrt{2}\,G_F\,\sin^2\theta_{\mathrm{W}}\cos^2\theta_{\mathrm{W}}}.} \tag{31.149}$$

The second relation is the mass ratio $M_Z = M_W/\cos\theta_{\rm W}$ from the diagonalization (Eq. (31.76)). It is illuminating to isolate the dimensionful combination that does not involve the angle,

$$A_0 \equiv \Big(\frac{\pi\alpha}{\sqrt{2}\,G_F}\Big)^{1/2} = \Big(\frac{\pi/137.036}{\sqrt{2}\times 1.166\times 10^{-5}\,\text{GeV}^{-2}}\Big)^{1/2} = 37.3\,\text{GeV}, \qquad (31.150)$$

so that the predictions read $M_W = A_0/\sin\theta_{\rm W}$ and $M_Z = A_0/(\sin\theta_{\rm W}\cos\theta_{\rm W})$. With the measured $\sin^2\theta_{\rm W} = 0.231$, hence $\sin\theta_{\rm W} = 0.481$ and $\cos\theta_{\rm W} = 0.877$,

$$M_W = \frac{37.3\,\text{GeV}}{0.481} = 77.6\,\text{GeV}, \qquad M_Z = \frac{77.6\,\text{GeV}}{0.877} = 88.5\,\text{GeV}. \qquad (31.151)$$

These tree-level numbers sit a few percent below the measured $M_W = 80.4\,\text{GeV}$ and $M_Z = 91.2\,\text{GeV}$. The discrepancy is not a failure; it is exactly the size of the one-loop electroweak radiative corrections, whose calculation and the attendant scheme-dependence of how $\sin^2\theta_{\rm W}$ is defined are the substance of the precision tests in Chapter 34. That a three-input tree-level formula lands within a few percent of two independently measured masses is itself a striking success: the construction is over-constrained and survives.

Physical Insight

The logical flow of this section is worth pausing on. We were handed three dimensionless couplings (g, g', λ) and one dimensionful scale (v). The photon coupling $e = g\sin\theta_{\rm W}$ and the electric charge operator $Q = T^3 + Y$ were *derived*, not posited—electromagnetism fell out of the broken theory. The neutral current came with it, for free, with couplings Eq. (31.141) fixed entirely by each fermion's known T^3 and Q and the single angle $\theta_{\rm W}$. Finally the one new scale v was nailed to the most precise weak measurement, G_F, giving 246 GeV. The theory that began as an attempt to cure the non-renormalizable Fermi interaction has repaid the debt with interest: it not only reproduces Fermi theory at low energy but predicts the masses of its mediators.

The masses Eq. (31.149) satisfy an exact tree-level identity that we have used twice without remarking on it. Forming the ratio,

$$\rho \equiv \frac{M_W^2}{M_Z^2\,\cos^2\theta_{\rm W}} = \frac{M_W^2}{(M_W^2/\cos^2\theta_{\rm W})\,\cos^2\theta_{\rm W}} = 1, \qquad (31.152)$$

the dependence on every coupling and on $\theta_{\rm W}$ cancels, leaving a pure number. This $\rho = 1$ is no accident of the algebra; it is the imprint of a hidden global symmetry of the Higgs sector that survives the breaking. Identifying that symmetry—custodial SU(2)—explaining precisely why it forces $\rho = 1$ at tree level, and pinning down what physics breaks it (so that $\rho - 1$ becomes a sensitive diagnostic of new states) is the business of Section 31.6, to which we now turn.

31.6 Custodial Symmetry and the ρ Parameter

The previous section delivered a number that looks too clean to be true. Reading the gauge-boson masses off the Higgs kinetic term and rotating the neutral sector by the weak mixing angle gave $M_W = \tfrac{1}{2} g v$ and $M_Z^2 = \tfrac{1}{4}(g^2 + g'^2)v^2$, so that

$$\rho \;\equiv\; \frac{M_W^2}{M_Z^2\,\cos^2\theta_{\mathrm{W}}} \;=\; \frac{\tfrac{1}{4} g^2 v^2}{\tfrac{1}{4}(g^2+g'^2)v^2\,\dfrac{g^2}{g^2+g'^2}} \;=\; 1 \tag{31.153}$$

exactly at tree level, with the two factors of $g^2 + g'^2$ cancelling on the nose. A relation that holds because of an algebraic cancellation invites suspicion: is it a numerical coincidence of the minimal model, fragile under the first radiative correction, or is it protected by something? The answer is that it is protected. Buried in the Higgs sector is an accidental global symmetry — larger than the gauge symmetry we put in by hand — whose unbroken remnant after the Higgs acquires its vacuum value forces the three weak gauge bosons to be exactly degenerate before the photon–Z mixing is switched on. That remnant is *custodial* $\mathrm{SU}(2)_V$, and $\rho = 1$ is its low-energy fingerprint. This section exhibits the symmetry, proves that it implies Eq. (31.153), and then identifies with equal precision the two places where it fails — the gauging of hypercharge and the splitting of up-type from down-type masses — which are the only sources of $\rho \neq 1$ and which therefore control the dominant radiative correction $\Delta\rho$.

31.6.1 The bidoublet rewriting of the Higgs sector

The complex Higgs doublet Φ of the previous sections carries hypercharge $Y = +\tfrac{1}{2}$, so that by the Gell-Mann–Nishijima relation Eq. (31.20) its upper component has charge $+1$ and its lower component is neutral,

$$\Phi = \begin{pmatrix} \phi^{+} \\ \phi^{0} \end{pmatrix}, \qquad Q\,\phi^{+} = +\phi^{+}, \quad Q\,\phi^{0} = 0. \tag{31.154}$$

Alongside Φ stands its charge conjugate, the field $\widetilde{H}$ of Chapter A,

$$\widetilde{H} \;\equiv\; i\sigma^2\,\Phi^{*} = \begin{pmatrix} \phi^{0*} \\ -\phi^{-} \end{pmatrix}, \qquad \phi^{-} \equiv (\phi^{+})^{*}, \tag{31.155}$$

which carries the opposite hypercharge $Y = -\tfrac{1}{2}$ but transforms under the gauge $\mathrm{SU}(2)_L$ exactly as Φ does. To see the latter, let $L = \exp(i\alpha^a\sigma^a/2) \in \mathrm{SU}(2)_L$ act by $\Phi \mapsto L\Phi$. Then

$$\widetilde{H} \;\mapsto\; i\sigma^2\,(L\Phi)^{*} = i\sigma^2 L^{*}\Phi^{*} = i\,(\sigma^2 L^{*}\sigma^2)\,\sigma^2\Phi^{*} = L\,\widetilde{H}, \tag{31.156}$$

where the middle step inserts $\mathbf{1} = \sigma^2\sigma^2$ and the last uses the SU(2) reality identity

$$\sigma^2\,(\sigma^a)^*\,\sigma^2 = -\sigma^a \quad\Longrightarrow\quad \sigma^2 L^*\sigma^2 = \exp\!\Big(-i\tfrac{\alpha^a}{2}\sigma^2(\sigma^a)^*\sigma^2\Big) = \exp\!\Big(i\tfrac{\alpha^a}{2}\sigma^a\Big) = L. \tag{31.157}$$

That Φ and $\widetilde{H}$ transform identically under $\mathrm{SU}(2)_L$ is the hinge on which everything turns: it lets us pack them into the two columns of a single 2×2 matrix.

Definition 31.20 (The Higgs bidoublet)**.** The *Higgs bidoublet* is the 2×2 complex matrix whose columns are the two doublets Eq. (31.41),

$$M \;\equiv\; (\,\widetilde{H},\ \Phi\,) = \begin{pmatrix} \phi^{0*} & \phi^{+} \\ -\phi^{-} & \phi^{0} \end{pmatrix}. \tag{31.158}$$

The whole point of the construction is the identity it satisfies. Computing $M^\dagger M$ directly, the off-diagonal entries are $\widetilde{H}^\dagger\Phi = -i\Phi^T\sigma^2\Phi = 0$ (a contraction of the symmetric $\Phi^T\cdot\Phi$ with the antisymmetric σ^2), while the diagonal entries are both $\Phi^\dagger\Phi$ because $\widetilde{H}^\dagger\widetilde{H} = \Phi^T\sigma^2\sigma^2\Phi^* = \Phi^\dagger\Phi$. Hence

$$M^\dagger M \;=\; (\Phi^\dagger\Phi)\,\mathbf{1}_{2\times 2}, \qquad \mathrm{tr}\big(M^\dagger M\big) = 2\,\Phi^\dagger\Phi. \tag{31.159}$$

The matrix M is, up to the real scale $\sqrt{\Phi^\dagger\Phi}$, unitary: the four real fields of the Higgs doublet are repackaged as an overall modulus times an SU(2)-valued phase. This is exactly the polar decomposition that made the abelian Higgs mechanism transparent in Chapter 30, now lifted to the non-abelian setting.

With $M^\dagger M \propto \mathbf{1}$ in hand, the accidental symmetry is immediate. Consider the two independent global SU(2) rotations acting on the left and right of M,

$$M \;\longmapsto\; L\,M\,R^\dagger, \qquad L\in\mathrm{SU}(2)_L,\quad R\in\mathrm{SU}(2)_R, \tag{31.160}$$

where $\mathrm{SU}(2)_L$ is the gauge group acting on the doublet index (the rows of M) and $\mathrm{SU}(2)_R$ is a *new*, purely global SU(2) mixing the two columns $\widetilde{H}$ and Φ. Under Eq. (31.160),

$$\mathrm{tr}\big(M^\dagger M\big) \;\mapsto\; \mathrm{tr}\big(R\,M^\dagger L^\dagger L\,M\,R^\dagger\big) = \mathrm{tr}\big(M^\dagger M\big), \tag{31.161}$$

using $L^\dagger L = \mathbf{1}$ and the cyclicity of the trace to cancel $R^\dagger R = \mathbf{1}$. The same two unitarities make $\mathrm{tr}(\partial_\mu M^\dagger\,\partial^\mu M)$ invariant. We can now state the symmetry of the scalar sector in the limit where hypercharge is not gauged.

Proposition 31.21 (Accidental $\mathrm{SU}(2)_L\times\mathrm{SU}(2)_R$ of the Higgs sector)**.** *Set $g' = 0$, so that only $\mathrm{SU}(2)_L$ is gauged. The Higgs kinetic term and the most general renormalizable Higgs potential,*

$$\mathcal{L}_\Phi = (D_\mu\Phi)^\dagger(D^\mu\Phi) - \Big[-\mu^2\,\Phi^\dagger\Phi + \lambda\,(\Phi^\dagger\Phi)^2\Big], \tag{31.162}$$

are invariant under the global $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ *of Eq. (31.160), with* $\mathrm{SU}(2)_L$ *promoted to the gauge symmetry. Equivalently, in bidoublet variables,*

$$\mathcal{L}_\Phi = \tfrac{1}{2}\,\mathrm{tr}\big[(D_\mu M)^\dagger (D^\mu M)\big] + \tfrac{1}{2}\mu^2\,\mathrm{tr}\big(M^\dagger M\big) - \tfrac{\lambda}{4}\Big[\mathrm{tr}\big(M^\dagger M\big)\Big]^2, \tag{31.163}$$

with $D_\mu M = \partial_\mu M - ig\,\frac{\sigma^a}{2} W_\mu^a\, M$.

Proof. We translate each term of Eq. (31.162) into the bidoublet and read off the invariance. For the potential, Eq. (31.159) gives $\Phi^\dagger\Phi = \frac{1}{2}\,\mathrm{tr}(M^\dagger M)$ and $(\Phi^\dagger\Phi)^2 = \frac{1}{4}[\mathrm{tr}(M^\dagger M)]^2$, which are the two potential terms of Eq. (31.163); both are functions of the invariant Eq. (31.161) alone, hence invariant under $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$. For the kinetic term, the columns $\widetilde{H}$ and Φ obey the same covariant derivative because they carry the same $\mathrm{SU}(2)_L$ representation (Eq. (31.156)) and, with $g' = 0$, no other gauge field; writing the covariant derivative as left multiplication on M,

$$D_\mu M = \partial_\mu M - ig\,\tfrac{\sigma^a}{2} W_\mu^a\, M = \big(D_\mu \widetilde{H},\; D_\mu \Phi\big), \tag{31.164}$$

so that, by the same column-wise computation that produced Eq. (31.159),

$$\tfrac{1}{2}\,\mathrm{tr}\big[(D_\mu M)^\dagger (D^\mu M)\big] = \tfrac{1}{2}\Big[(D_\mu \widetilde{H})^\dagger (D^\mu \widetilde{H}) + (D_\mu \Phi)^\dagger (D^\mu \Phi)\Big] = (D_\mu \Phi)^\dagger (D^\mu \Phi), \tag{31.165}$$

the last equality because $\widetilde{H}$ and Φ have equal covariant kinetic terms. This proves Eq. (31.163) equals Eq. (31.162). For the invariance: the global $\mathrm{SU}(2)_R$ acts only by right multiplication $M \mapsto M R^\dagger$, which commutes with the left-acting gauge covariant derivative Eq. (31.164), so $D_\mu M \mapsto (D_\mu M) R^\dagger$ and the trace Eq. (31.165) is invariant exactly as in Eq. (31.161). The gauged $\mathrm{SU}(2)_L$ invariance is automatic. Thus the full $\mathcal{L}_\Phi$ enjoys the global $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ symmetry. □

Physical Insight

The extra symmetry is genuinely accidental: nothing in the gauge principle demanded it. It appears because the most general renormalizable potential of a single complex doublet depends on the fields only through the combination $\Phi^\dagger\Phi$, and a single doublet has exactly four real components — the same count as the fundamental of $\mathrm{SO}(4) \cong [\mathrm{SU}(2)\times\mathrm{SU}(2)]/\mathbb{Z}_2$. The Higgs potential is therefore not merely U(2)-symmetric but SO(4)-symmetric, a sphere $\Phi^\dagger\Phi = \text{const}$ being acted on by the full rotation group of its four real coordinates. The two SU(2)'s of Eq. (31.160) are precisely the two commuting SU(2) factors inside that SO(4). The whole of custodial symmetry is the statement that the Higgs lives on a round three-sphere.

31.6.2 Spontaneous breaking to the custodial $\mathrm{SU}(2)_V$

The vacuum chosen in the previous section, $\langle \phi^0 \rangle = v/\sqrt{2}$ with all charged and imaginary components vanishing, becomes in bidoublet variables a strikingly symmetric object. Substituting $\langle \Phi \rangle = (0,\, v/\sqrt{2})^T$ and hence $\langle \widetilde{H} \rangle = (v/\sqrt{2},\, 0)^T$ into Eq. (31.158),

$$\langle M \rangle = \frac{v}{\sqrt{2}}\, \mathbf{1}_{2\times 2}. \tag{31.166}$$

The vacuum is proportional to the identity matrix. This is what makes the doublet special, and it is the crux of the entire chapter: under the global action Eq. (31.160) the vacuum transforms as

$$\langle M \rangle = \frac{v}{\sqrt{2}}\, \mathbf{1} \;\longmapsto\; \frac{v}{\sqrt{2}}\, L\, R^\dagger, \tag{31.167}$$

which equals $\langle M \rangle$ if and only if

$$L\, R^\dagger = \mathbf{1} \quad\Longleftrightarrow\quad L = R. \tag{31.168}$$

Proposition 31.22 (Custodial breaking pattern)**.** *The vacuum Eq. (31.166) breaks the accidental* $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ *down to its diagonal subgroup,*

$$\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R \;\longrightarrow\; \mathrm{SU}(2)_V \equiv \{(L, R) : L = R\}, \tag{31.169}$$

the custodial *group. Six broken-symmetry directions become three (one* $\mathrm{SU}(2)$*'s worth), so by Goldstone's theorem (Chapter 29) the spectrum contains three would-be Goldstone bosons, which fill out a triplet (the adjoint, spin-1 representation) of the unbroken* $\mathrm{SU}(2)_V$*.*

Proof. A symmetry transformation is unbroken precisely when it fixes the vacuum, and Eq. (31.168) identifies exactly those transformations as the diagonal $L = R$, which form an $\mathrm{SU}(2)$ subgroup of the original $\mathrm{SU}(2) \times \mathrm{SU}(2)$: this is $\mathrm{SU}(2)_V$. The Lie algebra $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ is six-dimensional, the unbroken $\mathfrak{su}(2)_V$ is three-dimensional, so the coset of broken generators is three-dimensional, and Goldstone's theorem (Chapter 29) supplies one massless scalar per broken generator: three would-be Goldstones. To find their $\mathrm{SU}(2)_V$ representation, expand M about the vacuum in its polar form $M = \frac{1}{\sqrt{2}}(v + h)\exp(i\pi^a\sigma^a/v)$, where h is the radial Higgs and π^a $(a = 1, 2, 3)$ are the three angular fields. Under an unbroken transformation $L = R = \exp(i\beta^b\sigma^b/2)$, the exponent transforms by conjugation, $\exp(i\pi^a\sigma^a/v) \mapsto R\exp(i\pi^a\sigma^a/v)R^\dagger$, so infinitesimally $\pi^a\sigma^a \mapsto \pi^a\sigma^a + i\frac{\beta^b}{2}[\sigma^b, \pi^a\sigma^a] = \pi^a\sigma^a - \beta^b\epsilon^{bac}\pi^a\sigma^c$, i.e. $\delta\pi^c = \epsilon^{cba}\beta^b\pi^a$. This is exactly the adjoint action of $\mathfrak{su}(2)_V$ on a three-vector π^a (Theorem 2.48): the π^a rotate into one another as a custodial *triplet*. □

These three would-be Goldstones are the same three scalars that, in unitary gauge, are eaten by $W^1_\mu, W^2_\mu, W^3_\mu$ to become their longitudinal polarizations, exactly as in the abelian Higgs mechanism of Chapter 30. The decisive new fact is that they form an *irreducible triplet* of the unbroken custodial group. A would-be Goldstone that becomes the longitudinal mode of a gauge boson carries that boson's quantum numbers; if the Goldstones are a custodial triplet, then so are the three gauge bosons that eat them. We now turn that observation into the proof of $\rho = 1$.

31.6.3 Why custodial symmetry forces $\rho = 1$

The masses of the gauge bosons come, as always, from the Higgs kinetic term evaluated at the vacuum. We compute it in bidoublet variables, this time *restoring* the hypercharge gauging so that the result is the physical mass matrix; the custodial limit $g' \to 0$ will then be read off as a special case. The covariant derivative of M that reproduces the doublet covariant derivative $D_\mu\Phi = (\partial_\mu - ig\frac{\sigma^a}{2}W^a_\mu - i\frac{g'}{2}B_\mu)\Phi$ of Eq. (31.23) is

$$D_\mu M = \partial_\mu M - ig\,\tfrac{\sigma^a}{2}W^a_\mu\, M + ig'\,B_\mu\, M\,\tfrac{\sigma^3}{2}. \tag{31.170}$$

That the right-acting hypercharge term has this form is forced: hypercharge is the T^3_R generator of $\mathrm{SU}(2)_R$ acting on the column index. Acting on the second column Φ, the operator $M\,\frac{\sigma^3}{2}$ returns $-\frac{1}{2}\Phi$, so the second column of Eq. (31.170) reads $\partial_\mu\Phi - ig\frac{\sigma^a}{2}W^a_\mu\Phi - i\frac{g'}{2}B_\mu\Phi$, reproducing $D_\mu\Phi$ with $Y = +\frac{1}{2}$; acting on the first column $\widetilde{H}$ it returns $+\frac{1}{2}\widetilde{H}$, giving $Y = -\frac{1}{2}$ as required by Eq. (31.41).

Theorem 31.23 (Custodial symmetry $\Rightarrow \rho = 1$)**.** *With the single Higgs doublet of Theorem 31.20, the tree-level gauge-boson mass matrix has the property that its W^3 diagonal entry equals its W^1 (and W^2) diagonal entry. Consequently*

$$\rho = \frac{M_W^2}{M_Z^2\,\cos^2\theta_{\mathrm{W}}} = 1 \tag{31.171}$$

at tree level, and in the custodial limit $g' \to 0$ the three gauge bosons W^1, W^2, W^3 are exactly degenerate as a triplet of $\mathrm{SU}(2)_V$.

Proof. Insert the vacuum Eq. (31.166) into Eq. (31.170). Because $\langle M\rangle = \frac{v}{\sqrt{2}}\mathbf{1}$ commutes with everything, the right-acting $\frac{\sigma^3}{2}$ slides through, and

$$D_\mu\langle M\rangle = -\frac{i}{2}\frac{v}{\sqrt{2}}\Big(g\,\sigma^a W^a_\mu - g'\,\sigma^3 B_\mu\Big). \tag{31.172}$$

The gauge-boson mass Lagrangian is $\frac{1}{2}\,\mathrm{tr}[(D_\mu\langle M\rangle)^\dagger(D^\mu\langle M\rangle)]$. Using $\mathrm{tr}(\sigma^a\sigma^b) = 2\delta^{ab}$ throughout,

$$\tfrac{1}{2}\,\mathrm{tr}\big[(D_\mu\langle M\rangle)^\dagger(D^\mu\langle M\rangle)\big] = \tfrac{1}{2}\cdot\frac{1}{4}\cdot\frac{v^2}{2}\;\mathrm{tr}\Big[\big(g\sigma^a W^a - g'\sigma^3 B\big)\big(g\sigma^b W^b - g'\sigma^3 B\big)\Big]$$

$$= \frac{v^2}{16}\cdot 2\Big[g^2 W_\mu^a W^{a\,\mu} - 2gg'\,W_\mu^3 B^\mu + g'^2 B_\mu B^\mu\Big]$$
$$= \frac{v^2}{8}\Big[g^2(W_\mu^1 W^{1\mu} + W_\mu^2 W^{2\mu}) + (gW_\mu^3 - g'B_\mu)(gW^{3\mu} - g'B^\mu)\Big]. \tag{31.173}$$

Writing this as $\frac{1}{2}\,V_\mu^T\,(\mathcal{M}^2)\,V^\mu$ in the basis $V_\mu = (W_\mu^1, W_\mu^2, W_\mu^3, B_\mu)$ gives the symmetric mass-squared matrix

$$\mathcal{M}^2 = \frac{v^2}{4}\begin{pmatrix} g^2 & 0 & 0 & 0 \\ 0 & g^2 & 0 & 0 \\ 0 & 0 & g^2 & -gg' \\ 0 & 0 & -gg' & g'^2 \end{pmatrix}. \tag{31.174}$$

Two features are decisive. First, the upper-left 3×3 block — the block acting on the custodial triplet (W^1, W^2, W^3) — has *identical* diagonal entries $g^2v^2/4$; the hypercharge coupling g' enters only through the B row and column, never altering the W^3–W^3 entry. Thus

$$(\mathcal{M}^2)_{W^1W^1} = (\mathcal{M}^2)_{W^3W^3} = \frac{g^2v^2}{4}. \tag{31.175}$$

Second, we connect these matrix entries to physical masses. Diagonalizing the neutral 2×2 block $\frac{v^2}{4}\left(\begin{smallmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{smallmatrix}\right)$ gives eigenvalues 0 (the photon) and $M_Z^2 = \frac{1}{4}(g^2+g'^2)v^2$, with the orthogonal rotation

$$\begin{pmatrix} Z_\mu \\ A_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_{\mathrm{W}} & -\sin\theta_{\mathrm{W}} \\ \sin\theta_{\mathrm{W}} & \cos\theta_{\mathrm{W}} \end{pmatrix}\begin{pmatrix} W_\mu^3 \\ B_\mu \end{pmatrix}, \qquad \cos\theta_{\mathrm{W}} = \frac{g}{\sqrt{g^2+g'^2}}, \quad \sin\theta_{\mathrm{W}} = \frac{g'}{\sqrt{g^2+g'^2}}. \tag{31.176}$$

The inverse rotation expresses $W^3 = \cos\theta_{\mathrm{W}}\,Z + \sin\theta_{\mathrm{W}}\,A$, so the W^3–W^3 entry of the mass matrix in the original basis is the mass-squared of Z and A weighted by the rotation,

$$(\mathcal{M}^2)_{W^3W^3} = \cos^2\theta_{\mathrm{W}}\,M_Z^2 + \sin^2\theta_{\mathrm{W}}\,M_A^2 = \cos^2\theta_{\mathrm{W}}\,M_Z^2, \tag{31.177}$$

since $M_A = 0$. Meanwhile $(\mathcal{M}^2)_{W^1W^1} = M_W^2$ outright, the W^1 block being already diagonal. Substituting Eq. (31.175) into the ratio,

$$\rho = \frac{M_W^2}{M_Z^2\cos^2\theta_{\mathrm{W}}} = \frac{(\mathcal{M}^2)_{W^1W^1}}{(\mathcal{M}^2)_{W^3W^3}} = 1. \tag{31.178}$$

The ρ parameter is nothing but the ratio of two diagonal entries of the custodial-triplet block, and they are equal.

It remains to exhibit the symmetry reason that they are equal, which is what makes the equality robust rather than coincidental. Take the custodial

limit $g' \to 0$. Then B decouples, Eq. (31.174) restricts to $\frac{g^2v^2}{4}\,\delta^{ab}$ on the triplet, and the unbroken $\mathrm{SU}(2)_V$ of Theorem 31.22 acts on (W^1, W^2, W^3) in its adjoint (spin-1) representation. The mass-squared matrix $(\mathcal{M}^2)_{ab}$ is invariant under this action, $R^T\mathcal{M}^2R = \mathcal{M}^2$ for every $R \in \mathrm{SU}(2)_V$, i.e. $\mathcal{M}^2$ commutes with an irreducible real representation. By Schur's lemma a matrix commuting with every element of an irreducible representation is a multiple of the identity: any eigenspace of $\mathcal{M}^2$ would be an invariant subspace, and irreducibility leaves only the trivial subspaces, so $\mathcal{M}^2$ has a single eigenvalue and $\mathcal{M}^2_{ab} = m^2\,\delta_{ab}$. The three gauge bosons are therefore exactly degenerate, $m^2 = g^2v^2/4$, as a custodial triplet. Their equal masses are protected by $\mathrm{SU}(2)_V$ to all orders in this limit, and turning on g' merely rotates W^3 and B without touching the W^3–W^3 entry, leaving Eq. (31.178) intact at tree level. □

Physical Insight

The factor of $\cos^2\theta_{\mathrm{W}}$ in the definition Eq. (31.153) is precisely the rotation weight Eq. (31.177) that strips the photon admixture out of M_Z^2 and recovers the bare W^3–W^3 mass entry. Read this way, ρ is a direct experimental probe of custodial symmetry: it asks whether the neutral member of the weak triplet has the same intrinsic mass as the charged members once the photon is rotated away. Custodial symmetry says yes; any deviation $\rho \neq 1$ is a quantitative measure of how badly the symmetry is broken.

Remark 31.24 (Why a doublet, and not a triplet, Higgs). The proof used only that $\langle M\rangle \propto \mathbf{1}$, which followed from $M^\dagger M \propto \mathbf{1}$ in Eq. (31.159), which in turn is special to the *doublet*. A Higgs field in a larger representation generally has a vacuum that is *not* a custodial singlet, so its contribution to the gauge masses does not respect $\mathrm{SU}(2)_V$ and shifts the W^3–W^3 entry relative to the W^1 entry, giving $\rho \neq 1$ already at tree level. For a real $\mathrm{SU}(2)_L$ triplet of hypercharge $Y = 0$ with vacuum value v_T, an identical computation gives $\rho = 1 + 4v_T^2/v^2$. The empirical closeness of ρ to unity is thus direct evidence that electroweak symmetry is broken dominantly by doublets — the unique representation whose VEV is custodially symmetric — and is one of the sharpest structural constraints the Standard Model satisfies.

31.6.4 The two custodial-breaking spurions

Custodial symmetry is exact only in a corner of the Standard Model that the world does not occupy: $g' = 0$ and degenerate Yukawa couplings. Both idealizations fail, and it is important to identify the failures precisely, because they are the only sources of $\rho \neq 1$ and therefore the only contributors to its radiative correction. The clean way to track an explicit breaking is the *spurion*

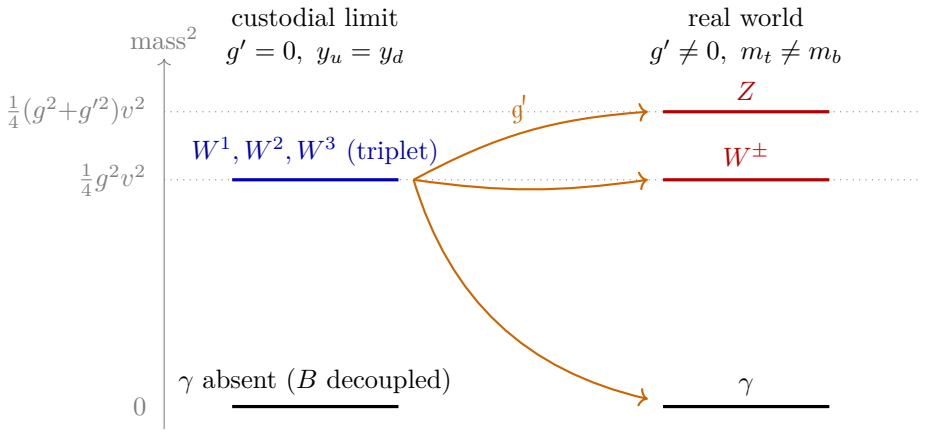


Figure 31.5: The custodial spectrum. In the limit $g' = 0$ with degenerate Yukawas (left), the unbroken custodial $\mathrm{SU}(2)_V$ of Theorem 31.22 forces W^1, W^2, W^3 into an exactly degenerate triplet (Theorem 31.23). Switching on hypercharge g' (right) rotates the neutral pair (W^3, B) into the massive Z and the massless photon γ by Eq. (31.176): Z is pushed up to $M_Z^2 = \frac{1}{4}(g^2 + g'^2)v^2$ while $W^\pm$ stay at $\frac{1}{4}g^2v^2$. The $\cos^2\theta_{\mathrm{W}}$ in ρ undoes this rotation, so $\rho = 1$ at tree level despite the visible W–Z splitting.

method introduced in Chapter 29: pretend the symmetry-breaking parameter is itself a field transforming in some representation, so that the Lagrangian is formally invariant; the representation the spurion must carry then tells us exactly how, and how strongly, the symmetry is violated. Both Standard Model spurions turn out to point in the same direction — the T^3 component of an $\mathrm{SU}(2)_R$ triplet.

Spurion 1: gauging hypercharge ($g' \neq 0$). Hypercharge is the generator T_R^3 of $\mathrm{SU}(2)_R$, as the right-acting $\frac{\sigma^3}{2}$ in Eq. (31.170) made explicit. Gauging only the third component of $\mathrm{SU}(2)_R$ singles out a direction inside the would-be custodial group and so breaks $\mathrm{SU}(2)_R$ explicitly. Formally, the breaking term in the covariant derivative is $ig'B_\mu M \frac{\sigma^3}{2}$, and it would be $\mathrm{SU}(2)_R$-invariant if σ^3 were promoted to a spurion $T_R = R\,\sigma^3 R^\dagger$ transforming in the adjoint (triplet) of $\mathrm{SU}(2)_R$. The physical value $T_R = \sigma^3$ is the T_R^3 component of that triplet, multiplied by the coupling g'. Hence *gauging hypercharge breaks custodial* $\mathrm{SU}(2)_V$*, with strength set by* g' *(equivalently by* $\sin\theta_{\mathrm{W}}$*, since* $\tan\theta_{\mathrm{W}} = g'/g$*).* As Theorem 31.23 showed, this particular breaking does not disturb $\rho = 1$ at tree level — the $\cos^2\theta_{\mathrm{W}}$ factor is built exactly to absorb it — but at one loop the photon–Z asymmetry it induces feeds into $\Delta\rho$ through gauge-boson self-energies.

Spurion 2: Yukawa splitting ($m_u \neq m_d$). The Yukawa sector introduces the second and, numerically, dominant breaking. Collect the right-handed up-

and down-type quarks of one generation into an $\mathrm{SU}(2)_R$ doublet $\psi_R = (u_R, d_R)^T$, mirroring the left-handed doublet $Q_L = (u_L, d_L)^T$. The two gauge-invariant Yukawa couplings combine into a single bidoublet contraction,

$$\mathcal{L}_{\text{Yuk}} = -\,\overline{Q}_L\,(y_u\,\widetilde{H}\,u_R + y_d\,\Phi\,d_R) + \text{h.c.} = -\,\overline{Q}_L\,M\,Y\,\psi_R + \text{h.c.}, \qquad Y = \begin{pmatrix} y_u & 0 \\ 0 & y_d \end{pmatrix}, \tag{31.179}$$

where the second form follows because $M\,\psi_R = \widetilde{H}\,u_R + \Phi\,d_R$ and the diagonal Y supplies the two couplings. Under the global group, $Q_L \mapsto L Q_L$, $\psi_R \mapsto R\psi_R$, and $M \mapsto L M R^\dagger$, so

$$\overline{Q}_L M Y \psi_R \;\mapsto\; \overline{Q}_L L^\dagger\, L M R^\dagger\, Y\, R\psi_R = \overline{Q}_L\, M\,(R^\dagger Y R)\,\psi_R, \tag{31.180}$$

which is invariant if and only if $R^\dagger Y R = Y$ for all $R \in \mathrm{SU}(2)_R$, i.e. iff $Y \propto \mathbf{1}$, i.e. iff

$$y_u = y_d \quad\Longleftrightarrow\quad m_u = m_d. \tag{31.181}$$

When the up and down Yukawas are equal, the quark mass term is custodially symmetric; when they differ, custodial $\mathrm{SU}(2)_R$ is explicitly broken. The spurion is the traceless part of Y,

$$Y - \tfrac{1}{2}(\operatorname{tr} Y)\,\mathbf{1} = \frac{y_u - y_d}{2}\,\sigma^3, \tag{31.182}$$

which is, once more, the T^3 component of an $\mathrm{SU}(2)_R$ triplet, with strength $y_u - y_d \propto m_u - m_d$. Across the three generations this splitting is utterly dominated by the third, where $m_t \approx 173\,\text{GeV}$ towers over $m_b \approx 4\,\text{GeV}$: *the top–bottom mass difference is the principal custodial-violating spurion in nature.*

Notation Watch

Both spurions are T^3_R objects: g' for the gauge breaking and $\frac{1}{2}(y_u - y_d)$ for the Yukawa breaking. This is not a coincidence but the statement that the unbroken electromagnetic U(1) is the $T^3_R + T^3_L$ direction that survives all breakings; anything that distinguishes the upper from the lower member of a weak doublet — charge via g', or mass via the Yukawas — breaks custodial symmetry along σ^3.

31.6.5 The leading correction $\Delta\rho$

Because $\rho = 1$ is a symmetry prediction, its violation must be proportional to the symmetry-breaking spurions, and the radiative correction can be organized accordingly. Define

$$\rho = 1 + \Delta\rho, \tag{31.183}$$

with $\Delta\rho$ generated by loops in which the custodial-breaking spurions Eq. (31.182) circulate. The dominant contribution comes from the top–bottom doublet

running in the W and Z self-energies, and its parametric size is fixed entirely by symmetry and dimensional analysis: $\Delta\rho$ must vanish when $m_t = m_b$ (custodial restoration), must be a one-loop effect (a factor $1/16\pi^2$), must scale with the number of colours N_c of the quark loop, and must be made dimensionless by the only other scale in the problem, the electroweak vacuum value v. The unique combination is

$$\boxed{\Delta\rho \;\sim\; \frac{N_c\,(m_t^2 - m_b^2)}{16\pi^2\,v^2}} \tag{31.184}$$

with a coefficient of order unity. The full one-loop computation, carried out in the precision-electroweak analysis of Chapter 34, supplies the exact coefficient and reduces in the hierarchy $m_b \ll m_t$ to $\Delta\rho \simeq N_c\, m_t^2/16\pi^2 v^2$. The following worked example checks that Eq. (31.184) has the right magnitude.

Worked Example 31.25 (The size of $\Delta\rho$ from the top quark)**.** We estimate Eq. (31.184) numerically and extract its two qualitative lessons. Take $N_c = 3$, $m_t = 173\,\text{GeV}$, $m_b = 4.2\,\text{GeV}$, and the vacuum value $v = 246\,\text{GeV}$ fixed by matching to the Fermi constant in the previous section.

Numerical value. Since $m_b^2/m_t^2 \approx 6\times10^{-4}$ is negligible,

$$\Delta\rho \;\sim\; \frac{N_c\,m_t^2}{16\pi^2 v^2} = \frac{3\,(173)^2}{16\pi^2\,(246)^2} = \frac{3\times 2.99\times10^4}{1.58\times10^2\times6.05\times10^4} \approx 9.4\times10^{-3}. \tag{31.185}$$

A correction of about one percent. This is exactly the order of magnitude seen in precision electroweak data, and historically a measurement of ρ slightly above unity was the first evidence that the top quark was *heavy* — the value Eq. (31.185) grows with m_t^2, so ρ acted as a mass-meter for a particle not yet directly produced.

Lesson one: quadratic, non-decoupling sensitivity. Unlike most heavy-particle effects, which decouple as inverse powers of the heavy mass, $\Delta\rho$ grows as m_t^2. The reason is custodial: the top mass arises from the Yukawa coupling through $m_t = y_t v/\sqrt{2}$, so $m_t^2/v^2 = y_t^2/2$ is really a measure of the *coupling* y_t, and Eq. (31.184) is the statement $\Delta\rho \sim N_c\, y_t^2/32\pi^2$. A heavy top is a strongly Yukawa-coupled top, and a strong coupling does not decouple. The growth is the breaking spurion Eq. (31.182) making itself felt: the larger $m_t - m_b$, the larger the custodial violation.

Lesson two: a window on new physics. Any new particles that couple to the Higgs sector with custodial-violating splittings contribute to $\Delta\rho$ with the same parametric form Eq. (31.184), mass-squared splitting over $16\pi^2 v^2$. Because ρ is measured to the per-mille level, it constrains such splittings tightly — this is the engine behind the oblique parameter T of precision tests, taken up in Chapter 34.

We have now closed the question with which the section opened. The relation $\rho = 1$ of Eq. (31.153) is no accident of the minimal model: it is the

tree-level shadow of an accidental global $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ of the Higgs sector (Theorem 31.21) whose breaking by the doublet vacuum leaves an unbroken custodial $\mathrm{SU}(2)_V$ (Theorem 31.22), and that custodial symmetry forces the weak gauge bosons into a degenerate triplet (Theorem 31.23). The only deviations come from the two spurions we identified — the gauging of hypercharge and the up–down Yukawa splitting, dominantly $m_t - m_b$ — and they enter ρ only at loop level with the small, calculable size Eq. (31.184), whose precise value and its use in electroweak fits we defer to Chapter 34.

That a symmetry we never imposed should so tightly protect a measured ratio is the strongest sign yet that the spontaneously broken gauge structure is more than a parametrization of the data: its predictions are stabilized by symmetry against the quantum corrections that would otherwise spoil them. Whether the theory survives those corrections *as a whole* — whether the spontaneously broken gauge symmetry remains consistent, unitary, and renormalizable once loops are included — is the final structural question of the chapter, to which we now turn.

31.7 Gauge Fixing, Unitarity, and Renormalizability

The previous sections delivered a complete classical Lagrangian: a chiral $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge theory whose Higgs doublet acquires a vacuum value, eats three would-be Goldstone bosons to make the $W^\pm$ and Z massive, leaves one combination of neutral gauge fields massless as the photon, and feeds fermion masses through Yukawa couplings. The custodial symmetry of the preceding section explained why the tree-level relation $\rho = M_W^2/(M_Z^2 \cos^2\theta_{\mathrm{W}}) = 1$ is protected rather than accidental. What we have nowhere shown is that this construction *survives quantization* — that it is, as a quantum field theory, as well behaved as the symmetric gauge theory it descends from. Three questions stand between us and that conclusion, and this closing section answers all three. First, *how does one quantize a spontaneously broken gauge theory at all?* A naive mass term spoils gauge invariance and leaves a propagator that does not fall at large momentum; the cure is the family of R_ξ gauges, imported from Chapter 30 and assembled here for the full electroweak group. Second, *does the theory cure the high-energy disease that doomed Fermi theory?* We showed in Theorem 31.1 that the four-fermion interaction violates unitarity near a TeV; we will now exhibit, at leading energy-growth order, the cancellation in longitudinal gauge-boson scattering that restores it, with the physical Higgs boson playing the decisive role. Third, *is the result renormalizable and self-consistent?* We state the 't Hooft–Veltman theorem and give its structural argument, and we verify the one remaining consistency condition that a chiral gauge theory can fail — cancellation of gauge anomalies — explicitly for one generation. We close with a census of the free parameters.

31.7.1 R_ξ gauge fixing of the broken theory

The obstruction to quantizing a massive vector was diagnosed in Chapter 30. A mass term written by hand gives the Proca propagator, whose longitudinal part $k_\mu k_\nu/M^2$ does not vanish at large k, so the propagator approaches a constant rather than falling as $1/k^2$ (Theorem 30.2 and Section 30.1.2); power counting then predicts a non-renormalizable theory. The resolution there was to keep the gauge symmetry manifest and fix it by the Faddeev–Popov procedure of Chapter 26, choosing a gauge-fixing function engineered to cancel the quadratic mixing between the gauge field and its would-be Goldstone. The same construction applies verbatim to $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, abelian factor by abelian factor in the mass-eigenstate basis, exactly as the non-abelian analysis of Section 30.7.4 reduces to copies of the abelian spectrum.

Write the Higgs doublet around its vacuum value in the parametrization fixed in Eq. (31.37),

$$\Phi = \begin{pmatrix} \phi^+ \\ \frac{1}{\sqrt{2}}\,(v + h + i\chi) \end{pmatrix}, \tag{31.186}$$

so that $\phi^\pm$ and χ are the three would-be Goldstone bosons (the χ neutral, the $\phi^\pm$ charged) and h is the physical Higgs scalar. After the breaking, $\phi^\pm$ is eaten by $W^\pm$ and χ by Z, while the photon stays massless and eats nothing. The covariant kinetic term $(D_\mu\Phi)^\dagger(D^\mu\Phi)$ contains, on expanding about the vacuum, cross terms linear in one gauge field and one Goldstone derivative,

$$\mathcal{L} \supset M_W\, W^-_\mu\, \partial^\mu\phi^+ + M_W\, W^+_\mu\, \partial^\mu\phi^- + M_Z\, Z_\mu\, \partial^\mu\chi, \tag{31.187}$$

the direct generalization of the abelian mixing $\propto m_A A_\mu \partial^\mu\chi$ of Section 30.4.1. These bilinear terms make the gauge and Goldstone fields propagate into one another and must be removed before any propagator can be read off. The R_ξ (or 't Hooft) gauge-fixing functions are chosen for precisely this purpose:

$$F^\pm = \partial^\mu W^\pm_\mu \mp i\,\xi M_W\,\phi^\pm, \qquad F^Z = \partial^\mu Z_\mu - \xi M_Z\,\chi, \qquad F^\gamma = \partial^\mu A_\mu, \tag{31.188}$$

added to the action through

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{\xi}\,F^+F^- - \frac{1}{2\xi}\,(F^Z)^2 - \frac{1}{2\xi}\,(F^\gamma)^2. \tag{31.189}$$

Expanding Eq. (31.189) produces a term $-\frac{1}{\xi}(\partial^\mu W^+_\mu)(\xi M_W\phi^-) + \cdots = -M_W(\partial^\mu W^+_\mu)\phi^- + \ldots$, which after integration by parts cancels Eq. (31.187) term by term — the defining property of the gauge choice, established abelian-by-abelian in Section 30.4.3. With the mixing gone, the quadratic Lagrangian is diagonal and the propagators follow immediately by specializing the master result Theorem 30.10 (equivalently Eq. (30.227)) to the electroweak mass eigenstates.

Theorem 31.26 (Electroweak R_ξ propagators)**.** *In the gauge Eq. (31.188)–Eq. (31.189) the tree-level propagators of the broken* $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *theory are*

$$W^\pm\!: \quad \frac{-i}{k^2 - M_W^2}\left(\eta_{\mu\nu} - (1-\xi)\,\frac{k_\mu k_\nu}{k^2 - \xi M_W^2}\right), \tag{31.190}$$

$$Z\!: \quad \frac{-i}{k^2 - M_Z^2}\left(\eta_{\mu\nu} - (1-\xi)\,\frac{k_\mu k_\nu}{k^2 - \xi M_Z^2}\right), \tag{31.191}$$

$$\gamma\!: \quad \frac{-i}{k^2}\left(\eta_{\mu\nu} - (1-\xi)\,\frac{k_\mu k_\nu}{k^2}\right), \tag{31.192}$$

while the would-be Goldstone bosons carry ξ*-dependent masses,*

$$\phi^\pm\!: \frac{i}{k^2 - \xi M_W^2}, \qquad \chi\!: \frac{i}{k^2 - \xi M_Z^2}, \qquad h\!: \frac{i}{k^2 - m_h^2}, \tag{31.193}$$

and the Faddeev–Popov ghosts $c^\pm, c_Z, c_\gamma$ *carry propagators* $i/(k^2 - \xi M_W^2)$*,* $i/(k^2 - \xi M_Z^2)$*, and* i/k^2 *respectively.*

Proof. Each entry is the corresponding line of Theorem 30.10 with the abelian vector mass m_A replaced by the relevant electroweak eigenvalue (M_W for the charged sector, M_Z for the neutral massive sector, and 0 for the photon, which is unbroken and therefore quantized exactly as the abelian gauge field of Chapter 13). The would-be Goldstone masses are the $\sqrt{\xi}M$ poles forced by the gauge-fixing term Eq. (31.189): the $-\frac{1}{2\xi}(\xi M_Z \chi)^2 = -\frac{1}{2}\xi M_Z^2 \chi^2$ piece adds exactly ξM_Z^2 to the χ mass, and likewise ξM_W^2 for $\phi^\pm$, matching the abelian value $m_\chi^2 = \xi m_A^2$ of Section 30.4. The ghost Lagrangian is generated by the Faddeev–Popov determinant of the gauge variation δF under Eq. (31.188); because δF contains the term $\xi M\,\delta(\text{Goldstone}) \sim \xi M^2 c$, each ghost inherits the same ξM^2 mass as the Goldstone it accompanies, exactly as in Eq. (30.101). The full apparatus — the BRST invariance of the gauge-fixed broken Lagrangian, the quartet mechanism that removes the unphysical modes, and the ξ-independence of the S-matrix — is the non-abelian construction of Theorems 26.18 and 26.26 applied to the broken theory, carried out in Sections 30.5.1 and 30.5.2; we import it rather than repeat it. □

Two features of Theorem 31.26 are the entire point of the construction. First, for any *finite* ξ every vector propagator falls as $1/k^2$ at large momentum: the dangerous $k_\mu k_\nu$ term is divided by $k^2 - \xi M^2$, so it too falls as $1/k^2$ rather than approaching a constant. This is what the Proca propagator could not do, and it is the technical fact on which renormalizability rests. Second, the ξ-dependent poles at $k^2 = \xi M^2$ in the gauge, Goldstone, and ghost propagators are unphysical: they cancel in every gauge-invariant quantity, as guaranteed by the BRST cohomology of Theorem 26.26 and demonstrated for the broken theory in Section 30.5.3.

Three standard choices organize the family: $\xi = 1$ (the 't Hooft–Feynman gauge), where the propagators reduce to a clean $-i\eta_{\mu\nu}/(k^2 - M^2)$ and the Goldstones and ghosts have the physical vector mass; $\xi \to \infty$ (the unitary gauge), where the Goldstones and ghosts become infinitely heavy and decouple, leaving only the physical W, Z, γ, h but resurrecting the non-falling propagator; and $\xi \to 0$ (the Landau gauge), where the gauge propagator is purely transverse. Physical amplitudes are identical in all of them.

31.7.2 High-energy unitarity: the longitudinal cancellation

We can now settle the score with Fermi theory. Theorem 31.1 showed that the four-fermion interaction produces amplitudes growing like $G_F s$, in conflict with the partial-wave unitarity bound $|a_\ell| \le 1$ of Theorem 16.29 near $\sqrt{s} \sim$ TeV. Resolving the point into W exchange tamed that particular growth, but Chapter 30 warned of a new one: the longitudinal polarization of a massive vector grows as $\varepsilon_L^\mu \to k^\mu/M + O(M/E)$ at high energy, so tree amplitudes with external longitudinal gauge bosons threaten to grow with powers of E/M and pierce the same unitarity bound near $E \sim 4\pi v$. If that growth survived, the electroweak theory would be no better than the disease it was built to cure. The instrument that controls it is the Goldstone-boson equivalence theorem of Theorem 30.15.

Theorem 31.27 (Equivalence theorem, electroweak form)**.** *For an S-matrix element with n external longitudinal gauge bosons $V_L \in \{W_L^\pm, Z_L\}$ and any other physical states X,*

$$\mathcal{M}(V_L^{(1)} \cdots V_L^{(n)}; X) = (i)^n \, \mathcal{M}(w^{(1)} \cdots w^{(n)}; X) \; + \; O\!\left(\frac{M_V}{E}\right), \tag{31.194}$$

where $w^{(i)}$ is the would-be Goldstone boson eaten by $V^{(i)}$ ($\phi^\pm$ for $W^\pm$, χ for Z), and E is the common centre-of-mass energy scale.

This is the multi-field, non-abelian generalization of the abelian statement Eq. (30.178); its proof is the same Ward–Slavnov–Taylor argument of Section 26.7, which converts the divergence $k_\mu \mathcal{M}^\mu$ of an external longitudinal line into the amplitude for the eaten Goldstone, applied mode by mode in the diagonal basis. Its power is structural: the would-be Goldstone bosons live in the Higgs doublet Eq. (31.186) and interact through the scalar potential and the gauge couplings, *none of which contains an inverse power of M_V*. The Goldstone amplitude on the right of Eq. (31.194) therefore cannot grow with E/M_V; whatever growth the longitudinal amplitude appears to have in a direct computation must cancel down to the bounded Goldstone result. The following worked example carries this out for the cleanest case and isolates the role of the physical Higgs.

Worked Example 31.28 (The longitudinal WW cancellation and the role of the Higgs)**.** Consider elastic scattering of longitudinal charged gauge bosons,

$W_L^+W_L^- \to W_L^+W_L^-$, at energies $\sqrt{s} \gg M_W$. We compute the leading energy behaviour two ways: by power counting on the gauge side, to exhibit the threatened growth, and by the equivalence theorem on the Goldstone side, to display the bounded answer and the Higgs cancellation that produces it.

The threat (gauge side). The diagrams are the four-point gauge contact vertex, s-, t-, and u-channel γ/Z exchange built from the triple-gauge vertex, and s- and t-channel Higgs exchange built from the hW^+W^- vertex (Fig. 31.6). Each external longitudinal polarization contributes a factor $\varepsilon_L \sim E/M_W$, so a diagram with four longitudinal legs and the generic gauge couplings naively scales as

$$\mathcal{M} \;\sim\; g^2 \left(\frac{E}{M_W}\right)^4 M_W^{?} \;\sim\; g^2\,\frac{s^2}{M_W^4} \;=\; \frac{16}{g^2 v^4}\,s^2, \tag{31.195}$$

an E^4 growth, where we used $M_W = \frac{1}{2}gv$ from Eq. (31.69). A growth this fast would violate unitarity already at energies parametrically below the scale where even the E^2 growth bites. Gauge invariance saves the leading power: the same Yang–Mills structure that organizes the triple- and quartic-gauge vertices forces the E^4 pieces to cancel among the gauge diagrams, leaving at most an E^2 remainder. The equivalence theorem makes this cancellation automatic and exact, because — as just argued — the Goldstone amplitude on the right of Eq. (31.194) has no $1/M_W$ at all and hence *no E^4 term to begin with*. The presence of E^4 on the gauge side and its complete absence on the Goldstone side *is* the E^4 cancellation. We are left to track the E^2 remainder, and for that the Goldstone computation is both shorter and manifestly finite.

The clean computation (Goldstone side). By Theorem 31.27 with $n = 4$ and $(i)^4 = +1$,

$$\mathcal{M}(W_L^+W_L^- \to W_L^+W_L^-) = \mathcal{M}(\phi^+\phi^- \to \phi^+\phi^-) + O\!\left(\frac{M_W}{E}\right). \tag{31.196}$$

The right-hand side is a scalar amplitude in the Higgs sector. Collect the four real scalars into $\phi_0 \equiv v + h$ and the Goldstone triplet ϕ_i $(i = 1, 2, 3)$, related to the charged and neutral Goldstones by $\phi^\pm = (\phi_1 \mp i\phi_2)/\sqrt{2}$ and $\chi = \phi_3$, so that $\Phi^\dagger\Phi = \frac{1}{2}(\phi_0^2 + \phi_i\phi_i)$. The potential of Eq. (31.39) is then the O(4)-symmetric form

$$V = \frac{\lambda}{4}\big(\phi_0^2 + \phi_i\phi_i - v^2\big)^2. \tag{31.197}$$

Expanding around $\phi_0 = v + h$, with $\rho \equiv h^2 + \phi_i\phi_i$,

$$V = \frac{\lambda}{4}\big(2vh + \rho\big)^2 = \lambda v^2 h^2 + \lambda v\, h\,\rho + \frac{\lambda}{4}\rho^2, \tag{31.198}$$

from which we read $m_h^2 = 2\lambda v^2$ and the two vertices we need. The four-Goldstone contact vertex comes from $\frac{\lambda}{4}(\phi_i\phi_i)^2$; using $\partial^4(\phi_i\phi_i\,\phi_j\phi_j)/\partial\phi_a\partial\phi_b\partial\phi_c\partial\phi_d = 8(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc})$, its Feynman rule is

$$\phi_a\phi_b\phi_c\phi_d: \qquad -2i\lambda\,(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}). \tag{31.199}$$

The Higgs–Goldstone–Goldstone vertex comes from $\lambda v\, h\, \phi_i \phi_i$, giving

$$h\,\phi_a\phi_b: \qquad -2i\lambda v\,\delta_{ab}, \tag{31.200}$$

and the Higgs propagator is $i/(k^2 - m_h^2)$. For the external assignment $\phi_a\phi_b \to \phi_c\phi_d$ with Mandelstam invariants $s = (p_a + p_b)^2$, $t = (p_a - p_c)^2$, $u = (p_a - p_d)^2$ (massless Goldstones at leading power, $s + t + u = 0$), the four diagrams give

$$\begin{aligned} i\mathcal{M} &= -2i\lambda\,(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) \\ &\quad + (-2i\lambda v)^2 \left[\frac{i\,\delta_{ab}\delta_{cd}}{s - m_h^2} + \frac{i\,\delta_{ac}\delta_{bd}}{t - m_h^2} + \frac{i\,\delta_{ad}\delta_{bc}}{u - m_h^2}\right]. \end{aligned} \tag{31.201}$$

Cancelling the overall i and using $4\lambda^2 v^2 = (2\lambda)(2\lambda v^2) = (2\lambda) m_h^2 = m_h^4/v^2$ together with $2\lambda = m_h^2/v^2$,

$$\mathcal{M} = -\frac{m_h^2}{v^2} \sum_{\text{channels}} \delta\delta - \frac{m_h^4}{v^2}\left[\frac{\delta_{ab}\delta_{cd}}{s - m_h^2} + \frac{\delta_{ac}\delta_{bd}}{t - m_h^2} + \frac{\delta_{ad}\delta_{bc}}{u - m_h^2}\right]. \tag{31.202}$$

Combining each channel over a common denominator — for the s-channel, $-\frac{m_h^2}{v^2} - \frac{m_h^4}{v^2(s-m_h^2)} = -\frac{m_h^2}{v^2}\frac{(s-m_h^2)+m_h^2}{s-m_h^2} = -\frac{m_h^2}{v^2}\frac{s}{s-m_h^2}$, and likewise for t, u — collapses the contact and exchange contributions into the single compact result

$$\boxed{\mathcal{M} = -\frac{m_h^2}{v^2}\left[\delta_{ab}\delta_{cd}\,\frac{s}{s - m_h^2} + \delta_{ac}\delta_{bd}\,\frac{t}{t - m_h^2} + \delta_{ad}\delta_{bc}\,\frac{u}{u - m_h^2}\right].} \tag{31.203}$$

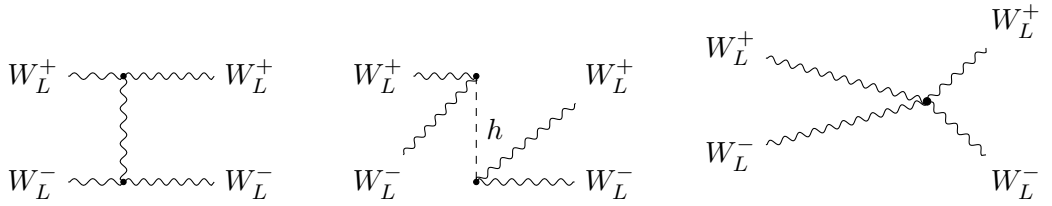
Reading the two limits. Equation Eq. (31.203) tells the whole story. At high energy with the Higgs present, $s, t, u \gg m_h^2$, each $s/(s - m_h^2) \to 1$ and the amplitude approaches the *constant*

$$\mathcal{M} \xrightarrow{s,t,u \gg m_h^2} -\frac{m_h^2}{v^2}\,(\delta_{ab}\delta_{cd} + \delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}), \tag{31.204}$$

which does not grow with energy: unitarity is restored, exactly as the equivalence theorem promised. Now *remove the Higgs* by taking $m_h \to \infty$ at fixed v (equivalently $\lambda \to \infty$, the would-be Goldstones becoming a nonlinear sigma model). Then $s/(s - m_h^2) \to -s/m_h^2$, and

$$\mathcal{M} \xrightarrow{m_h \to \infty} \frac{1}{v^2}(\delta_{ab}\delta_{cd}\,s + \delta_{ac}\delta_{bd}\,t + \delta_{ad}\delta_{bc}\,u), \tag{31.205}$$

which grows as E^2 — this is the residual E^2 remainder anticipated on the gauge side, the low-energy theorem of the Goldstone sigma model. The Higgs-exchange diagrams of Eq. (31.201) are therefore *exactly* what converts the growing amplitude Eq. (31.205) into the bounded constant Eq. (31.204): comparing the two, the difference is precisely the m_h^4/v^2 exchange terms. The physical Higgs



(a) γ/Z exchange. (b) Higgs exchange. (c) Quartic contact.

Figure 31.6: Representative tree diagrams for $W_L^+W_L^- \to W_L^+W_L^-$: gauge-boson (γ, Z) exchange and the quartic gauge vertex grow with energy, and the Higgs-exchange diagrams (b) cancel the residual E^2 growth. By the equivalence theorem Theorem 31.27 the sum equals the bounded would-be Goldstone amplitude Eq. (31.203); the abelian template is Fig. 30.3.

boson is not an optional ornament; it is the state whose exchange cancels the high-energy growth of longitudinal gauge-boson scattering.

The unitarity bounds. Projecting onto the $J = 0$ partial wave with the convention of Eq. (16.176) and imposing Theorem 16.29 converts each limit into a quantitative statement. Without a Higgs, the growing amplitude Eq. (31.205) gives a partial wave $a_0 \sim s/(16\pi v^2)$ (up to an $\mathcal{O}(1)$ channel factor), so $|a_0| \leq 1$ fails once

$$\sqrt{s} \gtrsim \sqrt{16\pi}\, v \approx 1.7 \text{ TeV}, \tag{31.206}$$

recovering — now from the gauge sector rather than from G_F — the same TeV ceiling found for Fermi theory in Eq. (31.11). With a Higgs, the constant Eq. (31.204) gives $a_0 \sim m_h^2/(16\pi v^2)$, and unitarity instead bounds the Higgs mass itself. The strongest constraint comes from the coupled $J = 0$, weak-isospin-zero channel built from $W_L^+W_L^-$ and Z_LZ_L, which (carrying the partial-wave projection of Eq. (31.203) through the coupled-channel analysis) yields the Lee–Quigg–Thacker bound

$$m_h \lesssim \sqrt{\frac{16\pi}{3}}\, v \approx 1 \text{ TeV}. \tag{31.207}$$

Either way the message is the same one that opened the chapter: *either* a Higgs lighter than about a TeV exists to cut off the growth, *or* new strong dynamics must appear at that scale. Nature chose the first option, with $m_h \approx 125$ GeV, comfortably inside Eq. (31.207).

31.7.3 Renormalizability: the 't Hooft–Veltman theorem

The longitudinal cancellation is the physical face of a deeper structural fact: the spontaneously broken theory is power-counting renormalizable, with a finite set of counterterms, exactly like the unbroken theory it descends from.

Theorem 31.29 ('t Hooft–Veltman)**.** *A Yang–Mills theory whose gauge symmetry is spontaneously broken by the Higgs mechanism is renormalizable: when quantized in an R_ξ gauge, all ultraviolet divergences are removed, to all orders in perturbation theory, by a finite number of counterterms with the same structure as those of the symmetric theory. Spontaneous breaking does not spoil the renormalizability of the symmetric theory.*

We do not reproduce the all-orders proof — it is the machinery of Chapter 22 (counterterms, the renormalizability criterion Theorem 22.26, the $\overline{\text{MS}}$ scheme Theorem 22.7) combined with the BRST/Slavnov–Taylor identities of Chapter 26, and the synthesis chapter Chapter 33 assembles it for the complete Standard Model. But the *structural* reason it works can be stated completely, and it rests on three facts already established in this book.

First, *power counting is governed by the symmetric theory.* The superficial degree of divergence of a diagram (Chapter 21, Theorems 21.12 and 21.14) is fixed by the large-momentum behaviour of propagators and vertices. By Theorem 31.26, every R_ξ propagator falls as $1/k^2$ at large k for finite ξ, precisely as in the massless symmetric theory; the vector masses M_W, M_Z and the would-be Goldstone masses sit in denominators that are subleading at large k. Therefore a graph in the broken theory has exactly the same superficial degree of divergence as the corresponding graph in the symmetric theory. The unitary gauge, where the propagator does not fall (Section 30.1.2), hides this and makes the power counting look hopeless; the R_ξ gauges expose it. This is the single technical reason the R_ξ family was invented.

Second, *spontaneous breaking is a relevant deformation, not a new interaction.* The breaking enters the Lagrangian entirely through the shift $\Phi \to \Phi + \langle\Phi\rangle$ of a field by a constant. A constant shift of a field can only turn an operator of dimension d into operators of dimension $\leq d$ — it lowers dimension by replacing fields with the dimensionful constant v, never raises it. It therefore generates only super-renormalizable (relevant or marginal) interactions from the renormalizable symmetric Lagrangian, and cannot manufacture the dimension-> 4 operators that would require new counterterms. The set of independent ultraviolet divergences is thus inherited unchanged from the symmetric theory; the breaking merely reshuffles them among operators of lower dimension.

Third, *the gauge structure of the counterterms is protected.* The worry in a broken theory is that the symmetry which constrained the symmetric-theory counterterms is gone, so new, symmetry-violating counterterms might be needed. It is not gone: it is realized nonlinearly and encoded in the BRST symmetry of the gauge-fixed Lagrangian (Section 30.5.1). The Slavnov–Taylor identities of Section 26.7, which are the quantum statement of BRST invariance, relate the divergent parts of different Green's functions and force the counterterms to assemble into the same gauge-covariant combinations as in the symmetric theory — the same coupling renormalization for the triple- and quartic-gauge

vertices, the same relation between the gauge-field and ghost renormalizations. No independent counterterm for the would-be Goldstone poles or for a gauge-non-invariant mass term is generated. The finite list of symmetric counterterms suffices, and Theorem 31.29 follows.

Physical Insight

The three facts have one moral: at short distances the theory does not know it is broken. A loop integral is dominated by momenta $k \gg M_W$, where $M_W^2/k^2 \to 0$ and the R_ξ propagators are indistinguishable from the symmetric ones. The vacuum value v is an infrared datum; it sets the masses and the longitudinal dynamics but is invisible to the ultraviolet, which sees only the massless, symmetric Yang–Mills theory. Renormalizability is an ultraviolet property, so the broken theory inherits it intact. This is the same separation of scales that makes the equivalence theorem true — both say that high energy is governed by the symmetric Goldstone dynamics, not by the masses.

31.7.4 Gauge-anomaly cancellation

One consistency condition remains, and it is special to chiral gauge theories. A classical symmetry can fail to survive quantization: the chiral fermion loop with three external gauge currents — the triangle diagram of Fig. 31.7 — produces a contribution to the divergence of the gauge current that is not removable by any local counterterm. If the gauged current itself is anomalous, the Slavnov–Taylor identities on which Theorem 31.29 rests are violated, unitarity and renormalizability fail together, and the theory is inconsistent. The full computation of this anomaly from the triangle diagram and from the path-integral measure is the subject of Chapter 36; here we need only its coefficient and the condition for it to vanish. For a chiral gauge theory with left-handed Weyl fermions in a representation with generators T^a, the gauge anomaly is proportional to the totally symmetric trace

$$\mathcal{A}^{abc} = \operatorname{tr}\big[\{T^a, T^b\}\, T^c\big], \tag{31.208}$$

summed over all left-handed fermions, with right-handed fermions entering as their left-handed charge conjugates (which carry the conjugate representation and reversed hypercharge). The theory is anomaly-free if and only if Eq. (31.208) vanishes for every choice of gauge generators T^a, T^b, T^c.

Worked Example 31.30 (Anomaly cancellation for one Standard-Model generation)**.** We verify that Eq. (31.208) vanishes for the chiral content of one generation, taking the gauge generators from $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ so that quarks (colour triplets) and leptons (colour singlets) both contribute — the cancellation will turn out to require both. The left-handed Weyl content, with

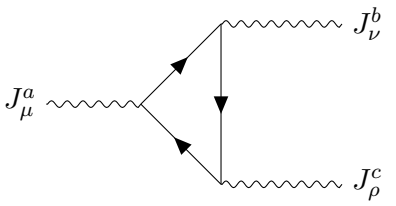


Figure 31.7: The triangle diagram with three external gauge currents and a chiral fermion loop. Its anomalous part is proportional to $\mathcal{A}^{abc} = \mathrm{tr}[\{T^a, T^b\}T^c]$ summed over the chiral fermion content; gauge consistency requires the sum to vanish. The diagram is computed in Chapter 36.

hypercharges fixed by $Q = T^3 + Y$ of Eq. (31.20), is collected in Table 31.3. Right-handed fields are written as left-handed charge-conjugates, which flips the sign of Y and conjugates the colour representation.

field	$\mathrm{SU}(3)_c$	$\mathrm{SU}(2)_L$	Y	multiplicity
$L = (\nu_e, e)_L$	**1**	**2**	$-\frac{1}{2}$	2
e_R	**1**	**1**	-1	1
$Q = (u, d)_L$	**3**	**2**	$+\frac{1}{6}$	6
u_R	**3**	**1**	$+\frac{2}{3}$	3
d_R	**3**	**1**	$-\frac{1}{3}$	3

Table 31.3: One generation. Multiplicity counts colour×weak-isospin states.

Several triangles vanish identically before any sum. A pure $\mathrm{SU}(2)_L^3$ triangle has $\mathcal{A}^{abc} \propto \mathrm{tr}[\{\sigma^a, \sigma^b\}\sigma^c] = 0$ because SU(2) has no symmetric invariant d^{abc} (its representations are real or pseudoreal). A pure $\mathrm{SU}(3)_c^3$ triangle cancels because colour is vector-like: every quark appears with both chiralities in the same colour representation, so left and right contributions are equal and opposite. Any triangle with a single $\mathrm{SU}(2)_L$ or $\mathrm{SU}(3)_c$ current and the rest hypercharge vanishes by $\mathrm{tr}\, T^a = 0$ for the non-abelian generators. Four non-trivial conditions remain.

(i) $\mathrm{SU}(2)_L^2\,\mathrm{U}(1)_Y$**.** Here $\mathcal{A} \propto \mathrm{tr}[\{T^a, T^b\}Y] = \frac{1}{2}\delta^{ab}\sum_{\text{doublets}} Y$, so only left-handed $\mathrm{SU}(2)_L$ doublets contribute, each weighted by its colour multiplicity:

$$\sum_{\text{doublets}} Y = \underbrace{1 \cdot \left(-\tfrac{1}{2}\right)}_{L} + \underbrace{3 \cdot \left(+\tfrac{1}{6}\right)}_{Q} = -\tfrac{1}{2} + \tfrac{1}{2} = 0. \tag{31.209}$$

(ii) $\mathrm{SU}(3)_c^2\,\mathrm{U}(1)_Y$**.** Only coloured fermions contribute, weighted by weak-isospin multiplicity, with right-handed entries subtracted:

$$\sum_{\text{LH quarks}} Y \;-\; \sum_{\text{RH quarks}} Y = \underbrace{2 \cdot \tfrac{1}{6}}_{Q} - \Big(\underbrace{\tfrac{2}{3}}_{u_R} + \underbrace{\left(-\tfrac{1}{3}\right)}_{d_R} \Big) = \tfrac{1}{3} - \tfrac{1}{3} = 0. \tag{31.210}$$

(iii) $\mathrm{U}(1)_Y^3$. Sum Y^3 over all left-handed states minus all right-handed states, with full multiplicities:

$$\sum_{\text{LH}} Y^3 = 2\left(-\tfrac{1}{2}\right)^3 + 6\left(\tfrac{1}{6}\right)^3 = -\tfrac{1}{4} + \tfrac{6}{216} = -\tfrac{1}{4} + \tfrac{1}{36} = -\tfrac{2}{9},$$
$$\sum_{\text{RH}} Y^3 = 1\,(-1)^3 + 3\left(\tfrac{2}{3}\right)^3 + 3\left(-\tfrac{1}{3}\right)^3 = -1 + \tfrac{24}{27} - \tfrac{3}{27} = -1 + \tfrac{7}{9} = -\tfrac{2}{9}, \quad (31.211)$$

so $\sum_{\text{LH}} Y^3 - \sum_{\text{RH}} Y^3 = -\tfrac{2}{9} - (-\tfrac{2}{9}) = 0$.

(iv) $\mathrm{U}(1)_Y$–gravity^2 (the mixed gauge–gravitational anomaly, $\propto \mathrm{tr}\, Y$). Required for consistency once gravity is included:

$$\sum_{\text{LH}} Y - \sum_{\text{RH}} Y = \left[2\left(-\tfrac{1}{2}\right) + 6\left(\tfrac{1}{6}\right)\right] - \left[(-1) + 3\left(\tfrac{2}{3}\right) + 3\left(-\tfrac{1}{3}\right)\right] = 0 - 0 = 0. \quad (31.212)$$

Every gauge-anomaly coefficient vanishes. The cancellations Eq. (31.209) and Eq. (31.210) are the striking ones: each demands that the lepton and quark contributions, separately non-zero, sum to zero, and the balance works only because the quark hypercharges carry the colour factor 3 and are exactly what $Q = T^3 + Y$ assigns. The electroweak theory is consistent only because quarks and leptons conspire, generation by generation — a constraint the theory could have failed and does not.

The check above is a one-loop, one-generation statement. Two extensions are owed and deferred. That the one-loop result suffices to *all* orders — the Adler–Bardeen theorem, that the gauge anomaly receives no higher-loop corrections — and the general structure of anomaly cancellation and matching are established in Chapter 37; the explicit triangle computation that produces the coefficient Eq. (31.208) is carried out in Chapter 36.

31.7.5 The parameter count of the electroweak sector

With consistency secured, it is worth tallying exactly how much freedom the theory contains, because the smallness of that number is the source of its predictive power. The electroweak Lagrangian for one generation has, as independent parameters:

- two gauge couplings, g for $\mathrm{SU}(2)_L$ and g' for $\mathrm{U}(1)_Y$ (Eq. (31.23));
- two parameters of the Higgs potential $V = -\mu^2\Phi^\dagger\Phi + \lambda(\Phi^\dagger\Phi)^2$, namely μ^2 and λ, equivalently traded for the vacuum value v and the Higgs mass m_h via $v^2 = \mu^2/\lambda$ and $m_h^2 = 2\lambda v^2$;
- the Yukawa couplings, one per massive fermion: y_e for the charged lepton, y_u and y_d for the up- and down-type quarks. With a strictly massless neutrino there is no ν Yukawa.

That is $4 + 3 = 7$ real parameters for one generation. Equivalently, they may be traded for a basis of measured quantities: the electromagnetic coupling $e = g \sin\theta_{\mathrm{W}}$ and the weak mixing angle θ_{W} (Eq. (31.73)), the Z mass M_Z (or the Fermi constant G_F, fixed in Eq. (31.13)), the Higgs mass m_h, and the three fermion masses $m_e = y_e v/\sqrt{2}$, $m_u = y_u v/\sqrt{2}$, $m_d = y_d v/\sqrt{2}$. Either count gives seven. The vacuum value itself is the one dimensionful input not predicted by the theory; matching to G_F in Eq. (31.13) fixed it at $v \approx 246$ GeV (Eq. (31.147)), and every electroweak mass is this single number times a dimensionless coupling. The economy is total: from seven numbers, the entire one-generation electroweak phenomenology follows.

This completes the one-generation electroweak theory as a self-consistent, renormalizable quantum field theory: chiral, spontaneously broken, unitary to arbitrarily high energy, and anomaly-free. Exactly one structural ingredient is still missing — the replication of this generation into three copies and the mixing among them. That replication, the Cabibbo–Kobayashi–Maskawa matrix it generates, and the CP violation it makes possible are the subject of Chapter 32; the assembly of the electroweak and strong sectors into the complete Standard Model, with its accidental symmetries and its full parameter census, follows in Chapter 33.

Further reading

The original Glashow–Weinberg–Salam construction of the unified electroweak theory is in Glashow [224], Weinberg [496], and Salam [426]. The R_ξ gauges and the all-orders proof that spontaneously broken gauge theories are renormalizable are the achievement of ’t Hooft and Veltman [466, 469]. The high-energy unitarity argument bounding the Higgs mass is due to Lee, Quigg, and Thacker [319], and the Goldstone-boson equivalence theorem to Cornwall, Levin, Tiktopoulos, and Vayonakis [138]. The cancellation of gauge anomalies between quarks and leptons is explained in Bouchiat et al. [87]. For textbook treatments at the level developed here, see Peskin and Schroeder [396], Schwartz [432], and the second volume of Weinberg [501].

Chapter 32

Flavor, the CKM Matrix, and CP Violation

The electroweak theory of the previous chapter was built on a single generation of quarks and leptons, and at that level the Yukawa couplings of Section 31.4 did nothing more interesting than turn on a mass once the Higgs acquired its vacuum value (Eq. (31.98)). Nature, however, has replicated the matter content three times over, with identical gauge quantum numbers and wildly different masses, and it is this replication — not the gauge interaction — that carries every phenomenon we group under the heading of *flavor*. This chapter takes the three-generation Standard Model as given and extracts from it the entire structure of quark and lepton mixing: the Cabibbo–Kobayashi–Maskawa matrix, the single irreducible phase that makes the theory violate CP, the Glashow–Iliopoulos–Maiani suppression that keeps flavor-changing neutral currents small, and — once neutrinos are granted a mass the minimal model denies them (Theorem 31.17) — the leptonic analogue, the PMNS matrix.

The logic is almost entirely group-theoretic, and the imports are correspondingly light. We need the charged- and neutral-current Lagrangians written out in Eqs. (31.129) and (31.138), the Yukawa terms and the Higgs vacuum value from Section 31.4, the chiral projectors and the quantized Dirac field of Eq. (11.82), and the action of charge conjugation and parity on fermion bilinears from Sections 11.7 and 11.7.2. The one piece of genuinely new linear algebra, the singular-value decomposition that bidiagonalizes an arbitrary complex matrix, is proved here from scratch. Throughout we hold the metric, the chirality conventions, and the charge assignments fixed as in Chapter A, and we read the residual flavor symmetries through Noether's theorem (Theorem 7.10).

The arc runs as follows. Section 32.1 establishes that the gauge-kinetic sector is exactly flavor-blind — invariant under a $\mathrm{U}(3)^5$ acting on the generation index (Theorem 32.2) — so that the three complex Yukawa matrices are the sole source of flavor violation, and counts the thirteen physical parameters hiding inside fifty-

four raw ones. Section 32.2 proves the bidiagonalization theorem (Theorem 32.5) and uses it to pass from the flavor basis to the mass basis, showing that this rotation is invisible to the photon, the Z, and the Higgs (Theorem 32.10) but *not* to the charged current, where it deposits a unitary matrix. Section 32.3 names that matrix the CKM matrix, proves its unitarity (Theorem 32.11), counts its physical content as three angles and one phase (Theorem 32.12), and fixes the standard parametrization. Section 32.4 shows that this lone phase is exactly the obstruction to CP invariance (Theorem 32.16) and packages it into the rephasing- and basis-invariant Jarlskog measure J (Theorems 32.20 and 32.23), whose vanishing for degenerate masses or two generations (Theorem 32.25) explains why three families are necessary.

The remaining sections turn from structure to phenomenology. Section 32.5 proves the absence of tree-level flavor-changing neutral currents (Theorem 32.28) and the GIM cancellation that suppresses them at one loop (Theorem 32.29), reading off the box-diagram amplitude that drives neutral-meson mixing and even predicting the charm mass scale. Section 32.6 promotes that amplitude to a two-state effective Hamiltonian, extracts the mass splitting and the CP-violating parameters ϵ and $\sin 2\beta$, and assembles the unitarity triangle — the geometric statement that one phase must fit many independent measurements at once. Finally, Section 32.7 repeats the entire flavor argument for the leptons: it supplies neutrino mass through the Weinberg operator and the type-I seesaw, derives the PMNS matrix with its extra Majorana phases, and works the oscillation phenomenology, handing the open questions of mass mechanism and leptonic CP on to the model-building of Chapter 35. The complete flavor and CP picture then passes to Chapter 33, which assembles the full Standard Model and its accidental symmetries.

32.1 From One Generation to Three: The Yukawa Matrices and Flavor Symmetry

The chapter introduction made a structural promise: because the electroweak gauge interactions are blind to generation, every shred of flavor physics — the hierarchy of fermion masses, the mixing of quarks, and ultimately CP violation — must be hiding in a single place, the Yukawa couplings to the Higgs field. This section discharges the first half of that promise by writing down the most general renormalizable three-generation Yukawa Lagrangian and proving that the gauge-kinetic sector really is generation-blind. The proof is concrete: with the matter content replicated into three identical copies, the kinetic and gauge terms acquire an enormous global symmetry, a separate U(3) rotation of each fermion species, and the Yukawa couplings are the *only* terms in the entire Standard Model Lagrangian that fail to respect it. Everything that distinguishes the muon from the electron, the strange quark from the down, lives in those three matrices

and nowhere else. Establishing this cleanly — including the precise accounting of which symmetries survive and how many parameters the Yukawa sector carries — is what makes the diagonalization of the next section, and the emergence of the CKM matrix, a controlled calculation rather than a guess.

We carry forward, without re-deriving, the one-generation electroweak construction of Chapter 31. The single generation comprised a left-handed quark doublet Q_L and lepton doublet L_L of $\mathrm{SU}(2)_L$, the right-handed singlets u_R, d_R, e_R, and the Higgs doublet H with its conjugate $\widetilde{H} = i\sigma^2 H^*$ of Theorem 31.14; the hypercharges were fixed by Gell-Mann–Nishijima in Eq. (31.97), and the gauge-invariant Yukawa couplings of one generation, Eqs. (31.102), (31.109) and (31.114), collected in Eq. (31.122), manufactured the charged-fermion masses while leaving the neutrino massless (Theorem 31.17). The chirality projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ are those of Eq. (11.82). The metric is mostly-minus and the covariant derivative is the electroweak one, $D_\mu = \partial_\mu - ig\, W_\mu^a T^a - ig' Y B_\mu$, exactly as in Chapter 31.

32.1.1 The replicated matter content

The central empirical input is that nature provides not one copy of this matter content but $N_g = 3$, identical in every gauge quantum number and differing only in mass. We label the generation (or *flavor*) by an index $i = 1, \ldots, N_g$, written as a superscript, and keep N_g general wherever it sharpens the counting, specializing to $N_g = 3$ for the physics. The fermion fields are thus

$$Q_L^i, \quad u_R^i, \quad d_R^i, \quad L_L^i, \quad e_R^i, \qquad i = 1, \ldots, N_g, \tag{32.1}$$

each carrying the same $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ representation as its single-generation ancestor in Chapter 31; the index i is an entirely new, internal label that the gauge group does not see. There is exactly one Higgs doublet H, not replicated — electroweak symmetry breaking is a once-for-all event shared by all three families.

Notation Watch

Two distinct objects in this chapter are both written with the letter Y. The *hypercharge* Y of Eq. (31.97) is a real number (an operator eigenvalue) carried by each field and appearing in D_μ; the *Yukawa matrices* Y_u, Y_d, Y_e introduced below are complex $N_g \times N_g$ matrices in generation space. They never collide in practice: the Yukawa matrices always carry a fermion-type subscript and act on the generation index, while hypercharge is a scalar attached to a representation. Chapter 31 anticipated the Yukawa matrices under the name $\mathbf{y}_f$ in the closing remark of Section 31.4; we adopt the capital Y_f here, matching the standard flavor-physics convention. The one-generation couplings y_u, y_d, y_e of Eq. (31.122) are the $N_g = 1$ entries

of these matrices.

32.1.2 The three-generation Yukawa Lagrangian

Promoting the one-generation Yukawa couplings to three generations is forced, not chosen. The gauge symmetry permits any renormalizable, gauge-invariant operator built from the fields Eq. (32.1) and the Higgs doublet, and a coupling that connects generation i of one field to generation j of another is just as gauge-invariant as the diagonal $i = j$ coupling: the gauge group cannot forbid it, because it cannot even tell the generations apart. Renormalizability (operators of mass dimension ≤ 4) and gauge invariance therefore demand that each one-generation coupling constant be promoted to the most general object that carries two generation indices — an arbitrary complex matrix.

Definition 32.1 (Three-generation Yukawa Lagrangian)**.** The most general renormalizable, $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant Yukawa interaction of the replicated matter content Eq. (32.1) with the single Higgs doublet is

$$\mathcal{L}_Y = -\Big(\overline{Q_L^i}\,(Y_d)_{ij}\, H\, d_R^j + \overline{Q_L^i}\,(Y_u)_{ij}\, \widetilde{H}\, u_R^j + \overline{L_L^i}\,(Y_e)_{ij}\, H\, e_R^j \Big) + \text{h.c.}, \tag{32.2}$$

with the generation indices $i, j = 1, \ldots, N_g$ summed, the $\mathrm{SU}(2)_L$ doublet indices of $\overline{Q_L}, \overline{L_L}$ contracted with those of H or $\widetilde{H}$ exactly as in Chapter 31, and the colour index of $\overline{Q_L}$ contracted with that of u_R, d_R. The coefficients Y_u, Y_d, and Y_e are *arbitrary complex* $N_g \times N_g$ *matrices* in generation space, the *Yukawa matrices*.

Three remarks fix the content of this definition. First, the entries are genuinely arbitrary complex numbers: gauge invariance constrains only the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ contractions, which are identical for every (i, j), so it places no condition whatsoever on the generation matrix. Hermiticity of the action is restored entirely by the explicit "+ h.c.", which adds $\overline{d_R^j}\,(Y_d^\dagger)_{ji}\, H^\dagger Q_L^i$ and its analogues; this is why Y_f itself need not be Hermitian, symmetric, or even normal. Second, $\widetilde{H} = i\sigma^2 H^*$ is mandatory for the up-type term and only for it, for the same hypercharge reason established in Section 31.4.3: it is the unique way to give the up-type quarks mass with a single Higgs doublet. Third, there is no neutrino Yukawa term, because the minimal matter content Eq. (32.1) contains no right-handed neutrino ν_R for H to couple L_L to; this is the three-generation restatement of Theorem 31.17, and it is the reason the leptonic sector below will have a different symmetry fate than the quark sector. Letting the neutral Higgs component condense, $\langle H^0 \rangle = v/\sqrt{2}$, turns Eq. (32.2) into the fermion mass terms

$$\mathcal{L}_{\text{mass}} = -\,\overline{d_L^i}\,(M_d)_{ij}\, d_R^j - \overline{u_L^i}\,(M_u)_{ij}\, u_R^j - \overline{e_L^i}\,(M_e)_{ij}\, e_R^j + \text{h.c.}, \qquad M_f = \frac{v}{\sqrt{2}}\, Y_f, \tag{32.3}$$

where u_L^i, d_L^i are the upper and lower members of Q_L^i. These mass matrices M_u, M_d, M_e are, like the Yukawas they descend from, arbitrary complex $N_g \times N_g$ matrices; in particular they are not diagonal, so the fields in Eq. (32.1) are *not* the mass eigenstates. Diagonalizing them is the task of the next section. Here we establish what makes that diagonalization meaningful: the precise sense in which Y_u, Y_d, Y_e are the *only* carriers of flavor.

32.1.3 The gauge-kinetic sector is flavor-universal

The claim to be proved is that, with the Yukawa couplings switched off, the three-generation Standard Model is invariant under independent unitary rotations of each fermion species in generation space. We make it precise and then prove it by carrying the index algebra explicitly.

Proposition 32.2 (Global $\mathrm{U}(N_g)^5$ flavor symmetry of the gauge-kinetic sector)**.** *Let $\mathcal{L}_{\mathrm{kin}}$ denote the sum of the fermion gauge-kinetic terms of the replicated content Eq. (32.1),*

$$\mathcal{L}_{\mathrm{kin}} = \sum_{i=1}^{N_g} \Big(\overline{Q_L^i}\, i\not{D}\, Q_L^i + \overline{u_R^i}\, i\not{D}\, u_R^i + \overline{d_R^i}\, i\not{D}\, d_R^i + \overline{L_L^i}\, i\not{D}\, L_L^i + \overline{e_R^i}\, i\not{D}\, e_R^i \Big), \tag{32.4}$$

together with the pure gauge and Higgs Lagrangian (which carry no generation index). Then $\mathcal{L}_{\mathrm{kin}}$, and hence the entire Lagrangian with $Y_u = Y_d = Y_e = 0$, is invariant under the global group

$$G_F \;=\; \mathrm{U}(N_g)_Q \times \mathrm{U}(N_g)_u \times \mathrm{U}(N_g)_d \times \mathrm{U}(N_g)_L \times \mathrm{U}(N_g)_e \;\equiv\; \mathrm{U}(N_g)^5, \tag{32.5}$$

acting by an independent unitary rotation of each species in generation space,

$$Q_L^i \to (U_Q)_{ij}\, Q_L^j, \quad u_R^i \to (U_u)_{ij}\, u_R^j, \quad d_R^i \to (U_d)_{ij}\, d_R^j, \quad L_L^i \to (U_L)_{ij}\, L_L^j, \quad e_R^i \to (U_e)_{ij}\, e_R^j \tag{32.6}$$

with $U_Q, U_u, U_d, U_L, U_e \in \mathrm{U}(N_g)$ constant matrices.

Proof. It suffices to show the invariance of one species' kinetic term; the argument is identical for the other four, and the gauge/Higgs Lagrangian carries no generation index and so is trivially invariant. Take the left-handed quark doublet. The covariant derivative $D_\mu = \partial_\mu - ig\, W_\mu^a T^a - ig'\, Y B_\mu$ acts only on Lorentz, colour, $\mathrm{SU}(2)_L$, and hypercharge indices; it contains no operator that acts on the generation index i. In generation space it is therefore proportional to the identity, $(D_\mu)_{ij} = D_\mu\, \delta_{ij}$, and in particular it *commutes* with any constant matrix acting on that index:

$$\not{D}\, \big((U_Q)_{ij}\, Q_L^j \big) = (U_Q)_{ij}\, \not{D}\, Q_L^j. \tag{32.7}$$

This single fact — that the gauge connection is generation-blind — is the whole content of flavor universality, and we now turn it into invariance. Under the

rotation Eq. (32.6), the Dirac adjoint transforms with the Hermitian conjugate of U_Q,

$$\overline{Q_L^i} = (Q_L^i)^\dagger \gamma^0 \;\longrightarrow\; ((U_Q)_{ij} Q_L^j)^\dagger \gamma^0 = (U_Q)^*_{ij}\,(Q_L^j)^\dagger \gamma^0 = \overline{Q_L^j}\,(U_Q^\dagger)_{ji}, \tag{32.8}$$

where we used $(U_Q)^*_{ij} = (U_Q^\dagger)_{ji}$. Assembling the bilinear and using Eq. (32.7),

$$\begin{aligned}\overline{Q_L^i}\, i\not{D}\, Q_L^i \;&\longrightarrow\; \overline{Q_L^k}\,(U_Q^\dagger)_{ki}\, i\not{D}\,(U_Q)_{il}\, Q_L^l \\ &= \overline{Q_L^k}\,(U_Q^\dagger)_{ki}(U_Q)_{il}\, i\not{D}\, Q_L^l \;=\; \overline{Q_L^k}\,(U_Q^\dagger U_Q)_{kl}\, i\not{D}\, Q_L^l \;=\; \overline{Q_L^k}\,\delta_{kl}\, i\not{D}\, Q_L^l \;=\; \overline{Q_L^k}\, i\not{D}\, Q_L^k,\end{aligned} \tag{32.9}$$

the third equality pulling the constant matrices through $\not{D}$ by Eq. (32.7) and the fourth using unitarity $U_Q^\dagger U_Q = \mathbf{1}$. The summed kinetic term is unchanged. The same computation, verbatim, holds for u_R, d_R, L_L, e_R with U_u, U_d, U_L, U_e, so each species may be rotated by its own independent unitary matrix. Hence $\mathcal{L}_{\text{kin}}$ is invariant under the full product group Eq. (32.5). □

It is worth stating the same conclusion at the level of currents, because that is how the symmetry is felt physically. The gauge interaction extracted from Eq. (32.4) couples the gauge bosons to currents of the schematic form

$$J^{a\,\mu} = \sum_{i=1}^{N_g} \overline{\Psi^i}\,\gamma^\mu T^a\,\Psi^i = \overline{\Psi^i}\,\gamma^\mu T^a\,\delta_{ij}\,\Psi^j, \tag{32.10}$$

for each species $\Psi \in \{Q_L, u_R, d_R, L_L, e_R\}$, with T^a a gauge generator. The generation index enters *only* through the Kronecker delta δ_{ij}: the charged-current, neutral-current, and electromagnetic interactions of Eqs. (31.129), (31.137) and (31.138) all sum each flavor against itself with unit weight. A current of the form $\overline{\Psi^i}\gamma^\mu T^a \delta_{ij} \Psi^j$ is manifestly invariant under $\Psi \to U\Psi$ for any unitary U, since $U^\dagger \delta\, U = \delta$ — which is precisely Eq. (32.9) read for the interaction rather than the kinetic piece. There is no flavor-changing and no flavor-non-universal gauge coupling anywhere in $\mathcal{L}_{\text{kin}}$: the gauge sector is exactly flavor-blind. By Noether's theorem (Theorem 7.10), the $5N_g^2$ generators of G_F would, in the absence of the Yukawa couplings, all be exactly conserved.

32.1.4 The Yukawa matrices are the unique source of flavor breaking

We now switch the Yukawa couplings back on and ask which of the $5N_g^2$ flavor rotations survive. Acting on Eq. (32.2) with the transformations Eq. (32.6), the down-type term becomes, using the adjoint rule Eq. (32.8) on $\overline{Q_L}$ and the rotation on d_R (the Higgs H is a flavor singlet and is untouched),

$$\overline{Q_L^i}\,(Y_d)_{ij}\, H\, d_R^j \;\longrightarrow\; \overline{Q_L^k}\,(U_Q^\dagger)_{ki}(Y_d)_{ij}(U_d)_{jl}\, H\, d_R^l = \overline{Q_L^k}\,\left(U_Q^\dagger Y_d U_d\right)_{kl} H\, d_R^l, \tag{32.11}$$

and identically for the other two terms. The Yukawa Lagrangian is invariant if and only if every Yukawa matrix is fixed by its flanking rotations,

$$U_Q^\dagger\, Y_u\, U_u = Y_u, \qquad U_Q^\dagger\, Y_d\, U_d = Y_d, \qquad U_L^\dagger\, Y_e\, U_e = Y_e. \tag{32.12}$$

These are the equations whose solutions are the unbroken symmetries. For generic Yukawa matrices — which is to say, for the matrices nature actually provides, with non-degenerate singular values and a nontrivial relative orientation of the up- and down-sectors — the solution set is small, and we can determine it exactly.

Theorem 32.3 (Residual flavor symmetry)**.** *Suppose Y_u, Y_d, Y_e are generic: each of the Hermitian combinations $Y_uY_u^\dagger$, $Y_dY_d^\dagger$, $Y_eY_e^\dagger$ has non-degenerate eigenvalues, and $Y_uY_u^\dagger$ and $Y_dY_d^\dagger$ are not simultaneously diagonalizable. Then the subgroup of $G_F = \mathrm{U}(N_g)^5$ left unbroken by the Yukawa couplings Eq. (32.2) is*

$$G_F \;\longrightarrow\; \mathrm{U}(1)_B \times \mathrm{U}(1)_{L_1} \times \mathrm{U}(1)_{L_2} \times \cdots \times \mathrm{U}(1)_{L_{N_g}}, \tag{32.13}$$

of dimension $N_g + 1$: a single overall baryon number $\mathrm{U}(1)_B$ acting on all quarks, and N_g individual lepton numbers, one per generation. For $N_g = 3$ these are $\mathrm{U}(1)_B \times \mathrm{U}(1)_{L_e} \times \mathrm{U}(1)_{L_\mu} \times \mathrm{U}(1)_{L_\tau}$, four commuting $\mathrm{U}(1)$'s.

Proof. We treat the quark and lepton sectors separately, since Eq. (32.12) couples U_Q, U_u, U_d among themselves and U_L, U_e among themselves but never mixes the two groups (there is no operator linking quarks to leptons).

Quark sector. Take the up-type condition $U_Q^\dagger Y_u U_u = Y_u$ and multiply it on the right by the dagger of itself, $U_u^\dagger Y_u^\dagger U_Q\, U_Q^\dagger Y_u U_u = Y_u^\dagger Y_u$. Using $U_Q U_Q^\dagger = \mathbf{1}$ this collapses to

$$U_u^\dagger\, (Y_u^\dagger Y_u)\, U_u = Y_u^\dagger Y_u, \qquad \text{i.e.} \qquad [\,U_u,\; Y_u^\dagger Y_u\,] = 0, \tag{32.14}$$

so U_u commutes with the Hermitian matrix $Y_u^\dagger Y_u$. Multiplying instead on the left by the dagger gives $[\,U_Q,\; Y_uY_u^\dagger\,] = 0$. The down-type condition gives in the same way $[\,U_d,\; Y_d^\dagger Y_d\,] = 0$ and $[\,U_Q,\; Y_dY_d^\dagger\,] = 0$. Now invoke a standard fact of linear algebra: a matrix that commutes with a Hermitian matrix of *non-degenerate* spectrum is diagonal in that matrix's eigenbasis, hence is itself a function of it; and a matrix that commutes with *two* non-degenerate Hermitian matrices whose eigenbases are misaligned (no common eigenvector beyond the trivial scaling) must be a multiple of the identity, because its eigenspaces would have to be invariant subspaces of both, and the only subspaces common to two generically oriented complete bases are $\{0\}$ and the whole space. By hypothesis $Y_uY_u^\dagger$ and $Y_dY_d^\dagger$ are exactly such a pair, and U_Q commutes with both; therefore

$$U_Q = e^{i\alpha}\,\mathbf{1}, \qquad \alpha \in \mathbb{R}. \tag{32.15}$$

Substituting back into $U_Q^\dagger Y_u U_u = Y_u$ and using that Y_u is invertible (its singular values are non-zero, being non-degenerate and generic),

$$e^{-i\alpha}\, Y_u\, U_u = Y_u \quad \Longrightarrow \quad U_u = e^{i\alpha}\, \mathbf{1}, \tag{32.16}$$

and the down-type condition gives identically $U_d = e^{i\alpha}\mathbf{1}$. The unique surviving quark rotation is therefore the *common* phase $U_Q = U_u = U_d = e^{i\alpha}\mathbf{1}$, a single U(1). Assigning every quark field the charge $+\frac{1}{3}$ under it (so that a three-quark baryon carries unit charge), this is *baryon number* $\mathrm{U}(1)_B$.

Lepton sector. Only the matrix Y_e appears, in $U_L^\dagger Y_e U_e = Y_e$. The same manipulation gives $[\,U_L,\ Y_e Y_e^\dagger\,] = 0$ and $[\,U_e,\ Y_e^\dagger Y_e\,] = 0$. But now there is only *one* Hermitian matrix on each side, and by hypothesis its spectrum is non-degenerate, so U_L and U_e are merely *diagonal* in the eigenbasis of $Y_e Y_e^\dagger$ (respectively $Y_e^\dagger Y_e$) — not forced to be scalars. Pass to that eigenbasis, where $Y_e = \mathrm{diag}(y_{e_1}, \ldots, y_{e_{N_g}})$ with distinct non-zero entries, and write $U_L = \mathrm{diag}(e^{i\beta_1}, \ldots, e^{i\beta_{N_g}})$, $U_e = \mathrm{diag}(e^{i\gamma_1}, \ldots, e^{i\gamma_{N_g}})$. The condition $U_L^\dagger Y_e U_e = Y_e$ reads, component by component, $e^{-i\beta_k}\, y_{e_k}\, e^{i\gamma_k} = y_{e_k}$, hence $\gamma_k = \beta_k$ for each k. The surviving lepton rotations are thus

$$U_L = U_e = \mathrm{diag}(e^{i\beta_1}, \ldots, e^{i\beta_{N_g}}), \tag{32.17}$$

with N_g *independent* phases β_k. Each is a separate U(1) acting on the k-th lepton generation (L_L^k, e_R^k) with common charge: these are the N_g individual lepton numbers $\mathrm{U}(1)_{L_k}$. The crucial reason they remain *separate* — rather than collapsing to a single overall lepton number as the quarks collapsed to a single baryon number — is the absence of any second leptonic Yukawa matrix to misalign against Y_e; with no ν_R and no neutrino mass, nothing links generation k to generation l in the lepton sector.

Combining the two sectors gives precisely Eq. (32.13), of dimension $1+N_g$. □

Physical Insight

The surviving symmetries Eq. (32.13) were never put into the Standard Model by hand; they are *accidental*, falling out as the largest subgroup of $\mathrm{U}(N_g)^5$ that the generic Yukawa matrices happen to leave alone. This is why baryon and lepton number are conserved to extraordinary accuracy without any principle enforcing them: the proton is stable not because a symmetry was imposed but because the renormalizable Lagrangian has no operator that could violate B. The same accident is fragile in two instructive ways. First, the individual lepton numbers $\mathrm{U}(1)_{L_k}$ are an artifact of setting neutrino masses to zero; the moment a neutrino Yukawa or Majorana mass is added (Chapter 32's closing PMNS discussion), the lepton sector acquires a second matrix to misalign against Y_e, and the N_g separate lepton numbers collapse to a single total $\mathrm{U}(1)_L$ exactly as

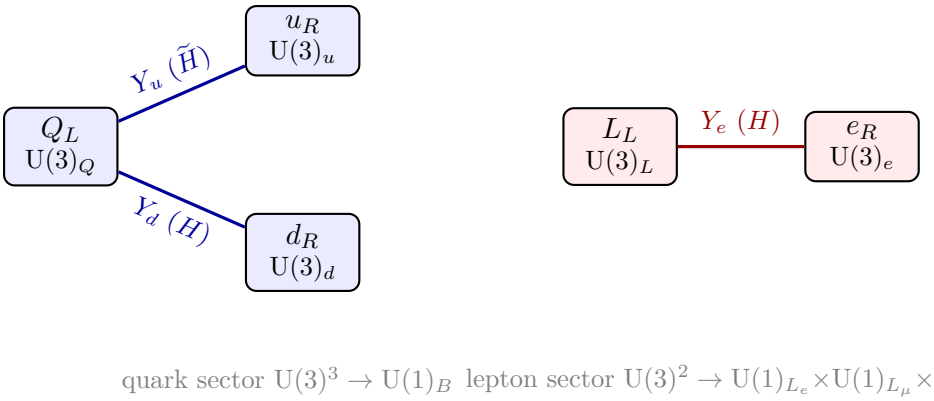


Figure 32.1: The flavor structure of the Standard Model for $N_g = 3$. Each fermion species is an independent "hub" that the gauge-kinetic Lagrangian rotates by its own U(3) (Theorem 32.2). The Yukawa matrices Y_u, Y_d, Y_e are the only links between hubs, and switching them on breaks $\mathrm{U}(3)^5$ down to the accidental baryon and lepton numbers (Theorem 32.3). The quark sector has two links (Y_u, Y_d) misaligning Q_L against u_R and d_R, which collapses $\mathrm{U}(3)^3$ to the single $\mathrm{U}(1)_B$; the lepton sector has only one link (Y_e), leaving the three lepton numbers separate. All flavor physics lives in these three wires.

the quarks collapsed to $\mathrm{U}(1)_B$. Second, even B and L are only *classical* symmetries: the combination $B + L$ is broken by the electroweak anomaly through nonperturbative effects, a subtlety we flag here and discharge in the anomalies development of Chapter 36. What survives exactly, to all orders, is the anomaly-free combination $B - L$.

We have now proved the chapter introduction's organizing claim in full. With the Yukawas off, the theory enjoys the maximal flavor symmetry $\mathrm{U}(N_g)^5$; the three matrices Y_u, Y_d, Y_e are the unique terms that break it; and they break it down to the accidental baryon and lepton numbers. Every observable flavor effect — mass splittings, mixing, CP violation — is therefore a function of Y_u, Y_d, Y_e alone. The figure below renders this structure: the five species are five independent "hubs" that the gauge sector rotates freely, and the only "wires" tying them together are the Yukawa matrices.

32.1.5 Counting the raw parameters

Before any field redefinition, the Yukawa sector carries a definite, and large, number of free real parameters. Counting them now — and counting the symmetry transformations available to remove some of them — frames the diagonalization of the next section as a bookkeeping problem with a guaranteed

answer.

Worked Example 32.4 (The $\mathrm{U}(N_g)^5$ symmetry-breaking parameter count)**.** We count the physical content of the Yukawa sector in three steps: the raw parameters in the matrices, the redundancy from flavor rotations, and the net.

Step 1: raw parameters. Each Yukawa matrix Y_u, Y_d, Y_e is a complex $N_g \times N_g$ matrix, with N_g^2 complex entries, i.e. $2N_g^2$ real numbers. Three such matrices carry

$$N_{\text{raw}} = 3 \times 2N_g^2 = 6N_g^2 \xrightarrow{N_g=3} 6 \times 9 = 54 \tag{32.18}$$

real parameters. This is the number the section title promised: *three complex* 3×3 *matrices carry* 54 *real parameters before any field redefinition.*

Step 2: the redundancy. Many of these 54 numbers are not physical, because we are free to perform the flavor rotations Eq. (32.6): these are mere relabelings of the fermion fields, leaving the gauge-kinetic Lagrangian invariant (Theorem 32.2) and changing only the form of the Yukawa matrices, $Y_u \to U_Q^\dagger Y_u U_u$ and so on. The full flavor group $G_F = \mathrm{U}(N_g)^5$ has

$$\dim G_F = 5\,N_g^2 \xrightarrow{N_g=3} 5 \times 9 = 45 \tag{32.19}$$

real generators (each $\mathrm{U}(N_g)$ has N_g^2). Not all of them remove parameters, however: a rotation that *leaves the Yukawas invariant* changes nothing and so cannot be "spent" to eliminate a parameter. By Theorem 32.3, the stabilizer of generic Yukawas is the unbroken $\mathrm{U}(1)_B \times \mathrm{U}(1)_{L_1} \times \cdots \times \mathrm{U}(1)_{L_{N_g}}$, of dimension $N_g + 1$. The number of genuinely available field redefinitions is therefore

$$N_{\text{redef}} = \dim G_F - \dim(\text{unbroken}) = 5N_g^2 - (N_g+1) \xrightarrow{N_g=3} 45-4 = 41. \tag{32.20}$$

Step 3: the net physical count. Subtracting the available redefinitions from the raw parameters,

$$N_{\text{phys}} = N_{\text{raw}} - N_{\text{redef}} = 6N_g^2 - [5N_g^2 - (N_g+1)] = N_g^2 + N_g + 1 \xrightarrow{N_g=3} 9+3+1 = 13. \tag{32.21}$$

Thirteen physical real parameters describe all of charged-fermion flavor for three generations. The formula $N_{\text{phys}} = N_g^2 + N_g + 1$ is worth testing at small N_g. For $N_g = 1$ it gives 3, namely the three masses m_u, m_d, m_e with no mixing — exactly the one-generation result of Chapter 31. For $N_g = 2$ it gives 7: six masses and one mixing parameter, namely the single real Cabibbo angle plus the masses, with *no* CP-violating phase. For $N_g = 3$ it gives 13, which will split into nine charged-fermion masses, three quark mixing angles, and one irreducible complex phase — the unique source of CP violation in the Standard Model. The identification of which of the thirteen parameters are masses and which are mixings, and in particular the appearance of that single phase, is the work of Section 32.2.

The accounting is now complete and self-consistent: 54 raw parameters, 41 of them removable by the $\mathrm{U}(N_g)^5$ flavor rotations modulo the four unbroken U(1)'s, leaving 13 physical. What this section has *not* done is exhibit the rotations explicitly or say what the 13 survivors mean. That requires diagonalizing the mass matrices $M_f = (v/\sqrt{2})Y_f$ of Eq. (32.3) by biunitary transformations and tracking where the rotations cancel and where they do not. We hand the next section exactly the objects it needs: the three arbitrary complex Yukawa matrices Y_u, Y_d, Y_e, the proof that the gauge sector is blind to the $\mathrm{U}(N_g)^5$ that acts on them, and the count of 13 physical parameters hidden inside 54 raw ones — a redundancy that diagonalization is about to cash in.

32.2 Mass and Flavor Bases: Bidiagonalization of the Yukawa Matrices

The previous section handed us three arbitrary complex 3×3 matrices in generation space—Y_u, Y_d, Y_e—together with the observation that the gauge-kinetic part of the Lagrangian possesses a large $\mathrm{U}(3)^5$ flavor symmetry that the Yukawa couplings, and only the Yukawa couplings, break. Two facts now pull in opposite directions, and reconciling them is the whole business of this section. On the one hand, the matrices Y_f are written in the *flavor basis*: the basis in which the gauge interactions are diagonal and generation-blind, three identical copies stacked on top of one another. On the other hand, a particle is what it is by virtue of its mass: an electron is distinguished from a muon not by any gauge charge (they carry the same) but by the number $m_e \neq m_\mu$. The states of definite mass are the *mass eigenstates*, and the matrices Y_f are, generically, not diagonal, so the flavor basis and the mass basis do not coincide. The task is to pass explicitly from one to the other—to diagonalize the mass matrices—and then to trace, term by term, what that change of basis does to every piece of the Standard Model Lagrangian. The punchline, which we will earn rather than assert, is that the rotation is invisible everywhere except in one place. The kinetic terms, the photon coupling, the Z coupling, and the Higgs coupling are all left exactly as they were; only the charged current, which links an up-type field to a down-type field, retains a residue of the mismatch between the up and down rotations. That residue is the object the next section will name the CKM matrix.

32.2.1 The mass matrices and why they must be diagonalized

We work with N_g generations, keeping N_g general because the counting later depends on it, and specializing to $N_g = 3$ only when the physics demands it. Restore generation indices $i, j = 1, \ldots, N_g$ on the fields handed over by Section 32.1.2: the left-handed quark doublets Q'_{Li}, the right-handed up and down singlets u'_{Ri}, d'_{Ri}, the lepton doublets L'_{Li}, and the charged-lepton singlets

e'_{Ri}. The prime marks these as *flavor-basis* (gauge-eigenstate) fields; the gauge interactions are diagonal in i by construction. The most general renormalizable Yukawa Lagrangian, built in the previous section, is

$$\mathcal{L}_Y = -\,\bar{Q}'_{Li}\,(Y_d)_{ij}\,H\,d'_{Rj} - \,\bar{Q}'_{Li}\,(Y_u)_{ij}\,\widetilde{H}\,u'_{Rj} - \,\bar{L}'_{Li}\,(Y_e)_{ij}\,H\,e'_{Rj} + \text{h.c.}, \quad (32.22)$$

with $\widetilde{H} = i\sigma^2 H^*$ the conjugate doublet of Theorem 31.14 and Y_u, Y_d, Y_e arbitrary complex $N_g \times N_g$ matrices. There is no Y_ν term: the minimal field content contains no right-handed neutrino, so the neutrino remains massless (Theorem 31.17), a fact we will exploit at the end.

Pass to unitary gauge, $H \to \frac{1}{\sqrt{2}}(0,\, v+h)^{\mathsf{T}}$ and $\widetilde{H} \to \frac{1}{\sqrt{2}}(v+h,\, 0)^{\mathsf{T}}$ from Eq. (31.98). Exactly as in the single-generation computation Eq. (31.104), the doublet contraction leaves only the lower (for H) or upper (for $\widetilde{H}$) component, and each term factorizes into the common scalar prefactor $(v+h)/\sqrt{2}$ times a Weyl bilinear:

$$\mathcal{L}_Y = -\Big(1+\frac{h}{v}\Big)\Big[\,\bar{u}'_L\,M_u\,u'_R + \bar{d}'_L\,M_d\,d'_R + \bar{e}'_L\,M_e\,e'_R\,\Big] + \text{h.c.}, \quad (32.23)$$

where the generation indices are now contracted through the *mass matrices*

$$\boxed{M_f \;=\; \frac{v}{\sqrt{2}}\,Y_f\,, \qquad f \in \{u,d,e\}.} \quad (32.24)$$

This is the central object of the section. Two features of Eq. (32.23) deserve to be flagged before we touch a single rotation, because both are consequences of the minimal one-doublet structure and both will be paid back later. First, the mass term and the Higgs–fermion coupling are controlled by the *same* matrix M_f: the constant piece $-\bar{f}'_L M_f f'_R$ is the mass, and the h-linear piece $-\frac{1}{v}\,h\,\bar{f}'_L M_f f'_R$ is the Higgs coupling, related by the overall factor $(1+h/v)$. They are not independent. Second, M_f is a general complex matrix—neither Hermitian nor normal—so it cannot be brought to diagonal form by a *single* unitary similarity transformation. What diagonalizes a general complex matrix is a pair of independent unitaries acting on the left and on the right, one for the left-handed fields and one for the right-handed fields. This is the singular value decomposition, and we prove it in full because the entire edifice rests on it.

32.2.2 The singular value decomposition, proved in full

Theorem 32.5 (Bidiagonalization of a complex square matrix)**.** *Let M be any complex $N \times N$ matrix. Then there exist unitary matrices $U_L, U_R \in \mathrm{U}(N)$ and a diagonal matrix $\hat{M} = \mathrm{diag}(s_1, \ldots, s_N)$ with real, non-negative entries $s_i \geq 0$ such that*

$$M = U_L\,\hat{M}\,U_R^{\dagger}. \quad (32.25)$$

The numbers s_i are the singular values *of M; they are the non-negative square roots of the eigenvalues of the Hermitian matrix $M^\dagger M$, and are therefore determined by M uniquely (up to ordering).*

Proof. The construction proceeds from the right-hand unitary, which we extract from a matrix that *is* amenable to ordinary diagonalization, namely

$$H \equiv M^\dagger M. \tag{32.26}$$

This H is Hermitian, $H^\dagger = (M^\dagger M)^\dagger = M^\dagger M = H$, and it is positive semidefinite: for any vector $w \in \mathbb{C}^N$,

$$w^\dagger H w = w^\dagger M^\dagger M w = (Mw)^\dagger (Mw) = \|Mw\|^2 \geq 0. \tag{32.27}$$

By the spectral theorem for Hermitian matrices, H admits an orthonormal basis of eigenvectors $\{r_1, \ldots, r_N\}$, $r_i^\dagger r_j = \delta_{ij}$, with real eigenvalues λ_i, and by Eq. (32.27) every eigenvalue is non-negative, $\lambda_i \geq 0$ (apply Eq. (32.27) to $w = r_i$, giving $\lambda_i = \|Mr_i\|^2$). Order them so that the strictly positive eigenvalues come first,

$$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_k > 0 = \lambda_{k+1} = \cdots = \lambda_N, \tag{32.28}$$

where $k = \operatorname{rank} M$, and define the singular values and the right-hand unitary by

$$s_i \equiv \sqrt{\lambda_i} \geq 0, \qquad U_R \equiv \begin{pmatrix} r_1 & r_2 & \cdots & r_N \end{pmatrix}, \tag{32.29}$$

i.e. U_R is the matrix whose i-th column is r_i. Orthonormality of the eigenvectors is precisely the statement $U_R^\dagger U_R = \mathbf{1}$, so $U_R \in \mathrm{U}(N)$. Set $\hat{M} = \operatorname{diag}(s_1, \ldots, s_N)$.

It remains to build U_L and verify Eq. (32.25). The key identity, valid for every i, is

$$\|Mr_i\|^2 = r_i^\dagger M^\dagger M r_i = r_i^\dagger (\lambda_i r_i) = \lambda_i = s_i^2, \tag{32.30}$$

so Mr_i has length s_i. For the $N - k$ vanishing singular values, $i > k$, this forces $Mr_i = 0$. For the k nonzero singular values, $i \leq k$, the vector Mr_i is nonzero and we may normalize it; define

$$\ell_i \equiv \frac{1}{s_i} M r_i, \qquad i = 1, \ldots, k. \tag{32.31}$$

These are orthonormal among themselves: for $i, j \leq k$,

$$\ell_i^\dagger \ell_j = \frac{1}{s_i s_j} (Mr_i)^\dagger (Mr_j) = \frac{1}{s_i s_j} r_i^\dagger M^\dagger M r_j = \frac{\lambda_j}{s_i s_j} r_i^\dagger r_j = \frac{s_j^2}{s_i s_j} \delta_{ij} = \delta_{ij}, \tag{32.32}$$

where we used $M^\dagger M r_j = \lambda_j r_j$ and $r_i^\dagger r_j = \delta_{ij}$, and $\lambda_j/(s_i s_j)\,\delta_{ij} = \delta_{ij}$ holds because the Kronecker delta forces $i = j$, whereupon $\lambda_i / s_i^2 = 1$. Thus $\{\ell_1, \ldots, \ell_k\}$ is an orthonormal set of k vectors in $\mathbb{C}^N$. If $k < N$, extend it—by the Gram–Schmidt procedure on the orthogonal complement, which has dimension $N - k$—to

a complete orthonormal basis $\{\ell_1, \dots, \ell_k, \ell_{k+1}, \dots, \ell_N\}$ of $\mathbb{C}^N$. The choice of the completing vectors $\ell_{k+1}, \dots, \ell_N$ is free; any orthonormal completion will do, a freedom we return to below. Assemble

$$U_L \equiv \begin{pmatrix} \ell_1 & \ell_2 & \cdots & \ell_N \end{pmatrix}, \tag{32.33}$$

which is unitary because its columns are orthonormal, $U_L^\dagger U_L = \mathbf{1}$.

Now verify Eq. (32.25). It is equivalent, on right-multiplying by U_R, to $MU_R = U_L \hat{M}$. Both sides are $N \times N$ matrices; compare them column by column. The i-th column of MU_R is Mr_i. The i-th column of $U_L\hat{M}$ is $U_L(\hat{M}e_i) = U_L(s_i e_i) = s_i\,\ell_i$, where e_i is the i-th coordinate vector. For $i \le k$, $Mr_i = s_i\ell_i$ by the definition Eq. (32.31); for $i > k$, $Mr_i = 0$ and $s_i\ell_i = 0 \cdot \ell_i = 0$, so the columns agree there too. Hence $MU_R = U_L\hat{M}$ as full matrices, and right-multiplication by the unitary $U_R^\dagger$ gives $M = U_L\hat{M}U_R^\dagger$, which is Eq. (32.25). □

Two remarks make the theorem usable and connect it to the residual freedom that the parameter count of the next section will exploit.

Remark 32.6 (The non-degenerate, non-singular case: an explicit U_L). When M is invertible—equivalently, when every singular value is strictly positive, $k = N$, which is the physical situation for the quarks and charged leptons, all of which are massive—the diagonal $\hat{M}$ is invertible and the construction Eq. (32.31) for the whole set of columns can be written in one line,

$$U_L = M\,U_R\,\hat{M}^{-1}. \tag{32.34}$$

That this is unitary is a direct check, using $U_R^\dagger M^\dagger M U_R = U_R^\dagger H U_R = \hat{M}^2$ (the right-hand side is $\hat{M}^2$ because U_R diagonalizes H with eigenvalues $\lambda_i = s_i^2$):

$$U_L^\dagger U_L = \hat{M}^{-1} U_R^\dagger M^\dagger\, M U_R \hat{M}^{-1} = \hat{M}^{-1}\,\hat{M}^2\,\hat{M}^{-1} = \mathbf{1}. \tag{32.35}$$

The Gram–Schmidt completion of the general proof is needed only to cover zero singular values; for the all-massive Standard-Model fermions the closed form Eq. (32.34) suffices.

Remark 32.7 (Uniqueness up to a common rephasing—the seed of the parameter count). How unique are U_L and U_R? The singular values s_i are not at the author's disposal: they are the square roots of the eigenvalues of $M^\dagger M$, fixed by M. The unitaries carry whatever residual freedom there is. Suppose all singular values are distinct and positive. The eigenvectors r_i of $H = M^\dagger M$ are then determined up to an individual phase, $r_i \to e^{i\theta_i} r_i$, because each nondegenerate eigenspace is one complex dimension. Under such a rephasing the right-hand unitary changes by a diagonal phase matrix, $U_R \to U_R\,\Phi$ with $\Phi = \mathrm{diag}(e^{i\theta_1}, \dots, e^{i\theta_N})$, and through Eq. (32.31), $\ell_i = Mr_i/s_i \to e^{i\theta_i}\ell_i$, so the left-hand unitary changes by the *same* phase matrix, $U_L \to U_L\,\Phi$. The decomposition Eq. (32.25) is invariant, $U_L\Phi\,\hat{M}\,(U_R\Phi)^\dagger = U_L\Phi\hat{M}\Phi^\dagger U_R^\dagger = U_L\hat{M}U_R^\dagger$, because the diagonal Φ

and $\hat{M}$ commute and $\Phi\Phi^\dagger = \mathbf{1}$. This common-phase freedom $U_{L,R} \to U_{L,R}\Phi$ is the mathematical shadow of an honest physical freedom—the freedom to rephase each mass eigenstate—and it is exactly the residue we count in Section 32.2.6. (If two singular values coincide, the freedom enlarges to an independent U(2) rotation within the degenerate block; the Standard-Model masses are nondegenerate, so we do not need this case, but the counting in the next section keeps track of it through the symmetry-breaking pattern.)

To see the abstract construction operate on a concrete matrix—and to meet, in the simplest nontrivial case, the structure that the next section will dress up as the Cabibbo angle—we carry out the bidiagonalization explicitly for two generations.

Worked Example 32.8 (Two real generations: the SVD produces a single rotation)**.** Take $N_g = 2$ and suppose, for transparency, that the up and down mass matrices are *real* 2×2 matrices, M_u and M_d (the general complex case is treated by the same steps with unitary rather than orthogonal factors; the point of the real case is to expose the geometry). A real square matrix M has $M^{\mathsf{T}}M$ real symmetric, so its eigenvectors are real and the diagonalizing matrices are real orthogonal: the SVD specializes to $M = O_L\,\hat{M}\,O_R^{\mathsf{T}}$ with $O_L, O_R \in \mathrm{O}(2)$. Parametrize the left orthogonal factor of each by a single angle,

$$O_L^{(u)} = \begin{pmatrix} \cos\theta_u & -\sin\theta_u \\ \sin\theta_u & \cos\theta_u \end{pmatrix}, \qquad O_L^{(d)} = \begin{pmatrix} \cos\theta_d & -\sin\theta_d \\ \sin\theta_d & \cos\theta_d \end{pmatrix}, \tag{32.36}$$

the most general element of O(2) connected to the identity. The mixing matrix Eq. (32.49) is then the product

$$V = O_L^{(u)\mathsf{T}}\,O_L^{(d)} = \begin{pmatrix} \cos\theta_u & \sin\theta_u \\ -\sin\theta_u & \cos\theta_u \end{pmatrix}\begin{pmatrix} \cos\theta_d & -\sin\theta_d \\ \sin\theta_d & \cos\theta_d \end{pmatrix} = \begin{pmatrix} \cos\theta_c & -\sin\theta_c \\ \sin\theta_c & \cos\theta_c \end{pmatrix}, \tag{32.37}$$

where the angle-addition identities $\cos\theta_u\cos\theta_d + \sin\theta_u\sin\theta_d = \cos(\theta_d - \theta_u)$ and $\cos\theta_u\sin\theta_d - \sin\theta_u\cos\theta_d = \sin(\theta_d - \theta_u)$ collapse the product to a single rotation by

$$\theta_c \equiv \theta_d - \theta_u. \tag{32.38}$$

Three lessons are already visible in this toy. *First*, only the *difference* of the up and down rotation angles is physical; the common part $\theta_u + \theta_d$ never appears, the two-generation echo of the statement that flavor mixing measures the *mismatch* between the up and down mass bases, not either one alone. *Second*, the right-handed rotations $O_R^{(u,d)}$ dropped out completely, as the general argument promised, because the charged current is left-handed. *Third*, the surviving V is a single real rotation by one angle θ_c: there is no room for a complex phase. This last fact is the seed of the Kobayashi–Maskawa observation that two generations cannot accommodate CP violation in the charged current—a single real angle

has no phase to be complex—and that a third generation is needed to make room for one. The angle θ_c is, anticipating the next section's identification, the Cabibbo angle $\theta_C \approx 13°$, and the matrix Eq. (32.37) is the original Cabibbo mixing of the first two quark generations. We will recover it as the upper-left block of the full three-generation CKM matrix in Section 32.3.5.

32.2.3 Mass eigenstates and the diagonal mass spectrum

Apply Theorem 32.5 to each of the three mass matrices, producing for each $f \in \{u, d, e\}$ a pair of unitaries and a non-negative diagonal matrix,

$$M_f = U_L^{(f)}\, \hat{M}_f\, U_R^{(f)\dagger}, \qquad \hat{M}_f = \mathrm{diag}(m_{f_1}, m_{f_2}, m_{f_3}) \;\; (m_{f_i} \geq 0). \tag{32.39}$$

We now *define* the mass-eigenstate fields by absorbing these rotations into a redefinition of the chiral fields. The chiral projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ of Eq. (11.82) make this unambiguous: left- and right-handed components are rotated independently, by the left and right unitaries respectively.

Definition 32.9 (Mass-eigenstate fields)**.** For each fermion type the mass-eigenstate (unprimed) fields are obtained from the flavor-eigenstate (primed) fields by the chiral unitary rotations

$$u'_L = U_L^{(u)}\, u_L, \quad u'_R = U_R^{(u)}\, u_R, \qquad d'_L = U_L^{(d)}\, d_L, \quad d'_R = U_R^{(d)}\, d_R, \tag{32.40}$$

and likewise $e'_L = U_L^{(e)} e_L$, $e'_R = U_R^{(e)} e_R$, where each rotation acts on the generation index only and is the identity on the Dirac index.

Substituting Eq. (32.40) into the mass term of Eq. (32.23) and using Eq. (32.39) collapses each bilinear to diagonal form:

$$\begin{aligned} \bar{f}'_L\, M_f\, f'_R &= \bar{f}_L\, U_L^{(f)\dagger}\, M_f\, U_R^{(f)}\, f_R = \bar{f}_L\, U_L^{(f)\dagger} (U_L^{(f)}\, \hat{M}_f U_R^{(f)\dagger}) U_R^{(f)}\, f_R \\ &= \bar{f}_L\, \hat{M}_f\, f_R = \sum_i m_{f_i}\, \bar{f}_{Li}\, f_{Ri}, \end{aligned} \tag{32.41}$$

where the unitarity relations $U_L^{(f)\dagger} U_L^{(f)} = \mathbf{1}$ and $U_R^{(f)\dagger} U_R^{(f)} = \mathbf{1}$ did all the work, and in the Dirac-adjoint of the left field we used $(U_L f_L)^\dagger \gamma^0 = f_L^\dagger U_L^\dagger \gamma^0 = \bar{f}_L U_L^\dagger$ (the constant matrix $U_L^\dagger$ commutes through γ^0, acting as it does on a different index). Adding the Hermitian conjugate welds each left and right Weyl piece into a Dirac field $f_i = f_{Li} + f_{Ri}$ of definite mass, and the mass term of the full three-generation Lagrangian is

$$\mathcal{L}_{\mathrm{mass}} = -\sum_{f \in \{u,d,e\}} \sum_{i=1}^{N_g} m_{f_i}\, \bar{f}_i\, f_i, \tag{32.42}$$

N_g decoupled Dirac masses per type. The diagonal entries $m_{f_i} \geq 0$ are, by definition, the physical quark and charged-lepton masses: $(m_{u_1}, m_{u_2}, m_{u_3}) = (m_u, m_c, m_t)$, $(m_{d_1}, m_{d_2}, m_{d_3}) = (m_d, m_s, m_b)$, and $(m_{e_1}, m_{e_2}, m_{e_3}) = (m_e, m_\mu, m_\tau)$. They are the singular values of Y_f scaled by $v/\sqrt{2}$, in accord with the single-generation relation $m_f = y_f v/\sqrt{2}$ of Eq. (31.116). The non-negativity is automatic and is not a loss of generality: any phase that one might have wished to attach to a mass has been absorbed into the definition of $U_L^{(f)}$ versus $U_R^{(f)}$.

With the spectrum diagonal we must now check the price. Rotating the fields is a change of variables in the path integral, and it touches every term that contains those fields, not only the mass term. The remainder of the section verifies, one sector at a time, that the price is zero everywhere except in the charged current.

32.2.4 The rotations are invisible: kinetic, neutral-current, and Higgs terms

The reason the rotations cancel in most of the Lagrangian can be stated in one sentence before any computation: the gauge sector is flavor-universal, meaning every vertex that is diagonal in generation space carries a coupling matrix proportional to the identity in that space, and a unitary rotation sandwiching the identity gives back the identity, $U^\dagger \mathbf{1}\, U = \mathbf{1}$. The Higgs coupling cancels for a different and sharper reason: it is governed by the very matrix M_f that the rotation was designed to diagonalize. We make both statements precise.

Kinetic terms. The fermion kinetic Lagrangian is a sum over chiralities and types of $\bar{f}'_X \, i\partial\!\!\!/ \, f'_X$ $(X = L, R)$. Each rotation is a constant unitary, so it passes through the derivative untouched:

$$\bar{f}'_X \, i\partial\!\!\!/ \, f'_X = \bar{f}_X \, U_X^{(f)\dagger} \, i\partial\!\!\!/ \, (U_X^{(f)} f_X) = \bar{f}_X \, U_X^{(f)\dagger} U_X^{(f)} \, i\partial\!\!\!/ \, f_X = \bar{f}_X \, i\partial\!\!\!/ \, f_X . \qquad (32.43)$$

The kinetic terms are form-invariant; in particular the fields remain canonically normalized, so the rotation introduces no spurious wavefunction renormalization. This is the elementary reason mass eigenstates are legitimate asymptotic states: the change of basis is a unitary field redefinition, which (by the equivalence theorem for field redefinitions) leaves the S-matrix unchanged while making the propagators diagonal.

The photon coupling. The electromagnetic interaction Eq. (31.137) couples the photon to $\bar{\Psi}\gamma^\mu Q\, \Psi$ with Q the electric-charge operator. Within a given fermion type all N_g generations carry *identical* electric charge—all up-type quarks have $Q = +\frac{2}{3}$, all down-type have $Q = -\frac{1}{3}$, all charged leptons have $Q = -1$—so in generation space Q acts as $q_f \, \mathbf{1}$ with q_f a number. For each type

and chirality,

$$\bar{f}'_X\,\gamma^\mu\,(q_f\mathbf{1})\,f'_X = q_f\,\bar{f}_X\,U_X^{(f)\dagger}\,\gamma^\mu\,U_X^{(f)}\,f_X = q_f\,\bar{f}_X\,\gamma^\mu\,U_X^{(f)\dagger}U_X^{(f)}\,f_X = q_f\,\bar{f}_X\,\gamma^\mu f_X, \tag{32.44}$$

where $U_X^{(f)\dagger}$ commuted past γ^μ because the two act on different indices, and unitarity collapsed the pair. The photon coupling is unchanged: it is diagonal in the mass basis and remains eQ for each mass eigenstate. There is no flavor-changing photon vertex.

The Z coupling. The neutral-current Lagrangian Eq. (31.138) of Section 31.5.3 couples the Z to $J_\mu^Z = J_\mu^3 - \sin^2\theta_{\mathrm{W}} J_\mu^{\mathrm{em}}$, that is to the operator $g_Z\,\gamma^\mu(T^3 - \sin^2\theta_{\mathrm{W}} Q)$ with $g_Z = g/\cos\theta_{\mathrm{W}}$. Both T^3 and Q are flavor-universal: within a type, all generations sit in identical $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ representations (this replication is the empirical input of the chapter), so T^3 acts as $t_f^3\,\mathbf{1}$ on the left-handed components and zero on the right-handed singlets, while $Q = q_f\mathbf{1}$ throughout. The coupling matrix in generation space is therefore a multiple of the identity, and the identical algebra of Eq. (32.44) applies verbatim:

$$\bar{f}'_X\,\gamma^\mu(t^3_{f,X} - \sin^2\theta_{\mathrm{W}} q_f)\mathbf{1}\,f'_X = \bar{f}_X\,\gamma^\mu(t^3_{f,X} - \sin^2\theta_{\mathrm{W}} q_f)\,f_X, \tag{32.45}$$

where $t^3_{f,L} = \pm\frac{1}{2}$ and $t^3_{f,R} = 0$. The Z coupling is flavor-diagonal in the mass basis.

This last result is the seed of one of the most important phenomenological facts about the Standard Model, so it is worth stating it as a theorem rather than letting it pass in the prose.

Theorem 32.10 (Tree-level flavor diagonality of the neutral currents)**.** *In the Standard Model with the minimal field content, the couplings of the photon and of the Z boson to fermions are diagonal in the mass basis. Equivalently, there are no flavor-changing neutral currents (FCNC) at tree level: the vertices $\bar{f}_i\gamma^\mu f_j$ with $i \neq j$ and a neutral gauge boson have vanishing coefficient. The mechanism is the flavor universality of the gauge charges (T^3, Q) within each fermion type, which makes the neutral-current coupling matrix proportional to the identity in generation space, whence $U_X^{(f)\dagger}\,\mathbf{1}\,U_X^{(f)} = \mathbf{1}$.*

Proof. The photon and Z couplings are the only neutral-current fermion vertices in the electroweak Lagrangian Eq. (31.136); both have been shown flavor-diagonal in Eqs. (32.44) and (32.45). Diagonality of every neutral-current coupling matrix is the statement that no $\bar{f}_i\,\Gamma\,f_j$ $(i \neq j)$ term survives, which is the absence of tree-level FCNC. □

The crucial hypothesis is the flavor universality of the gauge charges. Had two generations of, say, right-handed down quarks carried *different* T^3 or hypercharge, the neutral coupling matrix would not be proportional to the identity, the rotation

$U_R^{(d)\dagger}(\cdot)U_R^{(d)}$ would fail to cancel, and tree-level FCNC would appear. The Standard Model's repeated, identical generations are precisely what forbid them. That this suppression survives at the loop level—in a controlled, calculable form—is the Glashow–Iliopoulos–Maiani (GIM) mechanism, whose detailed analysis we defer to Section 32.5; the present theorem is its tree-level foundation.

The Higgs coupling. The coupling of the physical Higgs scalar h to fermions is the h-linear piece of Eq. (32.23), $\mathcal{L} \supset -\frac{1}{v}\, h\, \bar{f}'_L M_f f'_R + \text{h.c.}$ Crucially the matrix appearing here is M_f itself—not some other, flavor-universal matrix—so the very rotations that diagonalized the mass term diagonalize it as well, by the identical computation Eq. (32.41) carried out with the extra factor h/v:

$$-\frac{1}{v}\, h\, \bar{f}'_L\, M_f\, f'_R = -\frac{1}{v}\, h\, \bar{f}_L\, \hat{M}_f\, f_R = -\sum_i \frac{m_{f_i}}{v}\, h\, \bar{f}_{Li} f_{Ri}. \tag{32.46}$$

Adding the Hermitian conjugate gives $-\sum_i (m_{f_i}/v)\, h\, \bar{f}_i f_i$. The Higgs coupling is flavor-diagonal and, eigenstate by eigenstate, proportional to the mass, exactly reproducing the single-generation result $g_{hf\bar{f}} = m_f/v$ of Theorem 31.15. There is no tree-level flavor-changing Higgs vertex either. This is not the generic outcome of a Yukawa theory; it is special to the minimal one-doublet structure, in which mass and Higgs coupling come from the same operator. A model with two or more Higgs doublets has, in general, two different matrices in the two roles, the second is not diagonalized by the rotations that diagonalize the first, and Higgs-mediated FCNC reappear. The suppression of Higgs-mediated flavor violation is thus a genuine, testable prediction of the minimal Higgs sector.

Physical Insight: why almost everything is blind to the rotation

A unitary flavor rotation is a change of basis in the three-dimensional generation space. Any interaction that does not *distinguish* the generations—any vertex whose coupling matrix is the unit matrix in that space—cannot notice such a rotation, for the same reason that rotating your coordinate axes does not change the length of a vector: $U^\dagger \mathbf{1} U = \mathbf{1}$. The gauge interactions are flavor-blind by construction (the gauge group acts identically on each generation), so they are invariant. The Higgs coupling is not flavor-blind—it knows the mass hierarchy—but it is locked to the mass matrix, so it is diagonal in exactly the basis that diagonalizes the masses. The single interaction that is neither flavor-blind nor locked to a single mass matrix is the charged current, which reaches *across* from up-type to down-type and so feels two independent rotations at once. That is where, and only where, the physics of flavor can live.

32.2.5 The lone survivor: the charged current and the matrix V

The charged current is structurally different from every term examined so far because it connects an up-type field to a down-type field. Recall the charged-current Lagrangian Eq. (31.129) with its current Eq. (31.128), now written with generation indices restored and in the flavor basis:

$$\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\, W^+_\mu\, \bar{u}'_{Li}\, \gamma^\mu\, d'_{Li} + \text{h.c.} = \frac{g}{\sqrt{2}}\, W^+_\mu\, \bar{u}'_L\, \gamma^\mu\, d'_L + \text{h.c.}, \tag{32.47}$$

the generation index being summed (it is diagonal in the flavor basis because the $\mathrm{SU}(2)_L$ doublet $Q'_{Li} = (u'_{Li}, d'_{Li})^{\mathsf{T}}$ pairs the up and down members of the *same* generation). Insert the field redefinitions Eq. (32.40). The left-handed up field brings a $U_L^{(u)}$ and the left-handed down field a $U_L^{(d)}$:

$$\begin{aligned} \bar{u}'_L\, \gamma^\mu\, d'_L &= (U_L^{(u)} u_L)^\dagger \gamma^0\, \gamma^\mu\, U_L^{(d)} d_L = \bar{u}_L\, U_L^{(u)\dagger}\, \gamma^\mu\, U_L^{(d)}\, d_L \\ &= \bar{u}_L\, \gamma^\mu\, (U_L^{(u)\dagger} U_L^{(d)})\, d_L = \bar{u}_L\, \gamma^\mu\, V\, d_L, \end{aligned} \tag{32.48}$$

where again $U_L^{(u)\dagger}$ commuted through $\gamma^0\gamma^\mu$ (different indices), and we have defined

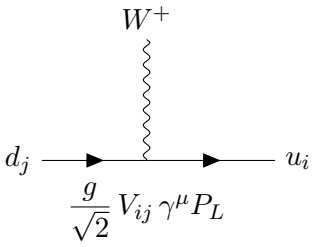
$$\boxed{\; V \;\equiv\; U_L^{(u)\dagger}\, U_L^{(d)} \;.\;} \tag{32.49}$$

Here the cancellation that saved every other term *fails*, and the reason is structural rather than algebraic: the up rotation $U_L^{(u)}$ and the down rotation $U_L^{(d)}$ are obtained from two *different* matrices, M_u and M_d, and there is no reason for them to coincide. Their product $U_L^{(u)\dagger} U_L^{(d)}$ is, in general, a nontrivial unitary matrix—unitary because it is a product of unitaries, $V^\dagger V = U_L^{(d)\dagger} U_L^{(u)} U_L^{(u)\dagger} U_L^{(d)} = \mathbf{1}$, but not the identity. The right-handed rotations $U_R^{(u,d)}$ do not appear at all, because the charged current is purely left-handed (the $V\!-\!A$ structure of Eq. (31.128)); they were spent entirely on diagonalizing the masses. The charged-current Lagrangian in the mass basis is therefore

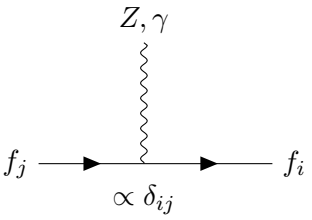
$$\boxed{\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\, W^+_\mu\, \bar{u}_{Li}\, \gamma^\mu\, V_{ij}\, d_{Lj} + \text{h.c.} = \frac{g}{\sqrt{2}}\, W^+_\mu\, \bar{u}_L\, \gamma^\mu\, V\, d_L + \text{h.c.}} \tag{32.50}$$

A single unitary matrix V, sitting between the up-type and down-type mass eigenstates, is all that remains of the entire flavor mismatch. Reading off the entry that couples the physical u quark to the physical d quark, this is exactly the flavor-mixing replacement $d \to d' = \sum_j V_{1j}\, d_j$ anticipated in the notation watch following Eq. (31.128): in the mass basis the down-type combination that the W pairs with a given up-type quark is the rotated $V d_L$, not d_L itself. The next section names V the Cabibbo–Kobayashi–Maskawa matrix $V_{\rm CKM}$ and studies it.

Figure 32.2 contrasts the two situations diagrammatically. The neutral-current vertex is proportional to δ_{ij}—the same quark in and out—while the charged-current vertex carries the off-diagonal-capable factor V_{ij}, the lone place a generation can change at a single Standard-Model vertex.



(a) Charged current: the W vertex carries V_{ij} and can connect *different* generations ($i \neq j$ allowed).



(b) Neutral current: the Z and photon vertices are $\propto \delta_{ij}$—no flavor change at tree level (Theorem 32.10).

Figure 32.2: The fate of the flavor rotations. Every neutral vertex (b) couples a mass eigenstate to itself; the charged vertex (a) is the unique survivor of the up–down rotation mismatch and carries the unitary matrix $V = U_L^{(u)\dagger} U_L^{(d)}$ of Eq. (32.49). All factors are written in the mass basis; $P_L = \frac{1}{2}(1 - \gamma^5)$.

The lepton sector: no mixing in the minimal model. The same machinery, applied to leptons, gives a sharp null result that is worth recording because it explains why the charged-lepton charged current shows no analogue of V in the minimal Standard Model. The charged-lepton mass matrix M_e is diagonalized exactly as above, $e'_L = U_L^{(e)} e_L$, $e'_R = U_R^{(e)} e_R$. The leptonic charged current is $\bar{\nu}'_L \gamma^\mu e'_L$. But the neutrinos are massless (Theorem 31.17): there is no neutrino mass matrix to diagonalize and hence no preferred neutrino basis. We are therefore *free* to define the neutrino mass eigenstates by the same rotation that we applied to the charged leptons, $\nu'_L \equiv U_L^{(e)} \nu_L$, which is legitimate precisely because any unitary redefinition of a multiplet of degenerate (here, massless) fields is a symmetry of the rest of the Lagrangian. Then the analogue of Eq. (32.48) reads

$$\bar{\nu}'_L \,\gamma^\mu\, e'_L = \bar{\nu}_L \, U_L^{(e)\dagger} \,\gamma^\mu\, U_L^{(e)} \, e_L = \bar{\nu}_L \,\gamma^\mu \, e_L, \tag{32.51}$$

the two rotations being the same and cancelling. There is no leptonic mixing matrix: in the minimal Standard Model lepton flavor (electron number, muon number, tau number) is separately conserved. This null result is not robust against the discovery that neutrinos have mass; once a neutrino mass term exists, the neutrino rotation is fixed independently and a leptonic mixing matrix—the PMNS matrix—survives in exact analogy with V. That story is the subject of Section 32.7.4.

32.2.6 The residual freedom: rephasing the mass eigenstates

We have used up most of the original $\mathrm{U}(3)^5$ flavor freedom in passing to the mass basis: the chiral rotations $U_L^{(f)}, U_R^{(f)}$ were fixed (up to the common phase of Theorem 32.7) by the requirement that the masses be diagonal, real, and

non-negative. But a genuine freedom remains, and identifying it now is what makes the parameter count of the next section honest. The mass term Eq. (32.42), the kinetic terms, and the neutral and Higgs couplings are all invariant under an *independent rephasing of each mass eigenstate*, provided the left- and right-handed components of a given field are rephased by the *same* phase. Concretely, with diagonal phase matrices $P_u = \mathrm{diag}(e^{i\alpha_1}, \dots)$ and $P_d = \mathrm{diag}(e^{i\beta_1}, \dots)$,

$$u_{L,R} \to P_u\, u_{L,R}, \qquad d_{L,R} \to P_d\, d_{L,R}, \tag{32.52}$$

the mass term $\bar{u}_L \hat{M}_u u_R \to \bar{u}_L P_u^\dagger \hat{M}_u P_u u_R = \bar{u}_L \hat{M}_u u_R$ is unchanged (a diagonal $\hat{M}_u$ commutes with the diagonal P_u, and $P_u^\dagger P_u = \mathbf{1}$), and the same cancellation leaves the kinetic, photon, Z, and Higgs terms untouched because each couples a field to itself. The one term that responds is the charged current. Under Eq. (32.52), Eq. (32.50) transforms as

$$\bar{u}_L\, \gamma^\mu\, V\, d_L \;\longrightarrow\; \bar{u}_L\, P_u^\dagger\, \gamma^\mu\, V\, P_d\, d_L = \bar{u}_L\, \gamma^\mu\, (P_u^\dagger\, V\, P_d)\, d_L, \tag{32.53}$$

so the mixing matrix is defined only up to

$$V \;\longrightarrow\; P_u^\dagger\, V\, P_d, \qquad V_{ij} \;\longrightarrow\; e^{-i\alpha_i}\, V_{ij}\, e^{i\beta_j}. \tag{32.54}$$

This is a real physical statement, not a bookkeeping nuisance: the individual phases of the quark fields are not observable (one cannot measure the phase of a u quark), and any physical quantity built from V must be invariant under Eq. (32.54). There are $2N_g$ phases in (P_u, P_d), but an overall common phase $\alpha_i = \beta_j = \text{const}$ acts trivially on V (it multiplies every entry by $e^{-i\theta}e^{+i\theta} = 1$), leaving $2N_g - 1$ independent rephasings that can be used to remove phases from V. The arithmetic of how many of the parameters of a unitary $N_g \times N_g$ matrix survive this scrubbing—and the discovery that for $N_g = 3$ a single, irremovable phase remains—is the parameter count we turn to next.

We have split the Standard-Model fermion content into a mass basis and a flavor basis, proved that the change between them is a unitary field redefinition invisible to the kinetic terms, the photon, the Z, and the Higgs (hence no tree-level flavor-changing neutral currents and a strictly mass-proportional Higgs coupling), and isolated its one physical residue: the unitary matrix $V = U_L^{(u)\dagger} U_L^{(d)}$ in the charged current, defined modulo the rephasings Eq. (32.54). This matrix V, together with the established flavor diagonality of the neutral currents, is what the next section takes up under its proper name, the CKM matrix—proving its unitarity, counting the physical parameters that the rephasing freedom leaves behind, and exhibiting the lone CP-violating phase of the three-generation Standard Model.

32.3 The CKM Matrix: Unitarity, Parameter Counting, and Parametrizations

The bidiagonalization of the previous section delivered a single, sharp conclusion. Every flavor rotation that diagonalizes the quark mass matrices washes out of the Lagrangian—out of the kinetic terms, the neutral current, the gluon and photon couplings, the Higgs coupling—because each of those pieces is built from a field paired with its own conjugate and a flavor-blind operator, so the rotation U and its inverse $U^\dagger$ meet and annihilate. The one place they do not meet is the charged current, where an up-type field sits next to a *down*-type field. There the two independent left-handed rotations $U_L^{(u)}$ and $U_L^{(d)}$ cannot cancel, and their mismatch

$$V \;\equiv\; U_L^{(u)\dagger}\, U_L^{(d)} \tag{32.55}$$

survives as a physical, measurable object: the Cabibbo–Kobayashi–Maskawa (CKM) matrix, already introduced as Eq. (32.49). This section takes Eq. (32.55) as its starting point and asks the three questions that a physical matrix must answer. What constraints does it obey? How many genuinely physical numbers does it contain, once we have used up every freedom still left in the theory? And how do we write it down in a form fit for comparison with experiment? The answers are, in order: it is unitary; it contains $N_g(N_g-1)/2$ mixing angles and $(N_g-1)(N_g-2)/2$ phases, which for $N_g = 3$ means three angles and exactly one phase; and it is written most usefully either in the standard PDG angles or in the Wolfenstein expansion organized around the Cabibbo angle. The lone phase, forced into existence by the third generation, is the thread we hand to the next section as the candidate origin of CP violation.

32.3.1 The CKM matrix in the charged current

We begin by writing Eq. (32.55) into the Lagrangian explicitly, because the precise way V enters fixes both its index structure and the rephasing freedom we will exploit below. Recall from Eqs. (31.128) and (31.129) that for a single generation the charged-current interaction is

$$\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\, W^{+\mu}\, \overline{u_L}\, \gamma_\mu d_L + \text{h.c.}, \tag{32.56}$$

the up-type and down-type left-handed fields entering with the gauge-blind coupling $g/\sqrt{2}$ common to all of them. With N_g generations the fields carry a generation index. Write the column of weak-eigenstate ("flavor-basis") left-handed fields as $u'_L = (u'_{L,1}, \ldots, u'_{L,N_g})^{\mathsf{T}}$ and likewise d'_L; the gauge interaction Eq. (32.56) is flavor-universal, so it is simply summed over the generation index with the unit matrix in flavor space,

$$\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\, W^{+\mu}\, \overline{u'_L}\, \gamma_\mu\, d'_L + \text{h.c.} = \frac{g}{\sqrt{2}}\, W^{+\mu} \sum_i \overline{u'_{L,i}}\, \gamma_\mu\, d'_{L,i} + \text{h.c.} \tag{32.57}$$

The mass eigenstates—the fields with definite mass, the physical particles u, c, t and d, s, b—are obtained from the flavor states by the unitary rotations established in the bidiagonalization of Section 32.2.2,

$$u'_L = U_L^{(u)}\, u_L, \qquad d'_L = U_L^{(d)}\, d_L, \tag{32.58}$$

where $u_L = (u_L, c_L, t_L)^{\mathsf{T}}$ and $d_L = (d_L, s_L, b_L)^{\mathsf{T}}$ now denote the mass-basis columns. Substituting Eq. (32.58) into Eq. (32.57) and using $\overline{u'_L} = \overline{u_L}\, U_L^{(u)\dagger}$,

$$\mathcal{L}_{\mathrm{CC}} = \frac{g}{\sqrt{2}}\, W^{+\mu}\, \overline{u_L}\, U_L^{(u)\dagger} U_L^{(d)}\, \gamma_\mu\, d_L + \mathrm{h.c.} = \frac{g}{\sqrt{2}}\, W^{+\mu}\, \overline{u_L}\, \gamma_\mu\, V\, d_L + \mathrm{h.c.}, \tag{32.59}$$

where the chiral projector and the flavor matrix commute (one acts in Dirac space, the other in generation space), and we have used Eq. (32.55). In components, displaying the physical fields,

$$\mathcal{L}_{\mathrm{CC}} = \frac{g}{\sqrt{2}}\, W^{+\mu} \begin{pmatrix} \overline{u_L} & \overline{c_L} & \overline{t_L} \end{pmatrix} \gamma_\mu \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} \begin{pmatrix} d_L \\ s_L \\ b_L \end{pmatrix} + \mathrm{h.c.} \tag{32.60}$$

This is the central object. The entry V_{ij} (rows labelled by up-type $i \in \{u, c, t\}$, columns by down-type $j \in \{d, s, b\}$) is the amplitude weight for the W^+ to convert a down-type quark j into an up-type quark i: it governs β-decay ($d \to u$, weighted by V_{ud}), strangeness-changing decays ($s \to u$, V_{us}), $b \to c$ transitions (V_{cb}), and so on. The diagonalizing rotations $U_L^{(u)}, U_L^{(d)}$ are not separately observable—only their product V appears anywhere in the theory—which is exactly why the charged current is the sole repository of flavor mixing. The neutral current, by contrast, pairs each up-type field with its own conjugate, so the analogous combination there is $U_L^{(u)\dagger} U_L^{(u)} = \mathbf{1}$: there is no tree-level flavor change in the neutral current, the structural fact behind the GIM mechanism developed later in this chapter.

32.3.2 Unitarity

The matrix V inherits a rigid algebraic constraint directly from its origin as a product of two unitary matrices.

Theorem 32.11 (Unitarity of the CKM matrix)**.** *The CKM matrix $V = U_L^{(u)\dagger} U_L^{(d)}$ is unitary,*

$$V^\dagger V = V V^\dagger = \mathbf{1}, \tag{32.61}$$

equivalently the orthonormality of its columns and of its rows,

$$\sum_i V_{ij}^*\, V_{ik} = \delta_{jk}, \qquad \sum_j V_{ij}\, V_{kj}^* = \delta_{ik}. \tag{32.62}$$

Proof. The diagonalizing matrices $U_L^{(u)}$ and $U_L^{(d)}$ are unitary by construction: they are the left factors of the singular-value (biunitary) decompositions of the up- and down-type Yukawa matrices established in Section 32.2.2, and a singular-value decomposition produces unitary factors. Hence $U_L^{(u)\dagger}U_L^{(u)} = \mathbf{1}$ and $U_L^{(d)\dagger}U_L^{(d)} = \mathbf{1}$. Taking the Hermitian conjugate of Eq. (32.55) gives $V^\dagger = U_L^{(d)\dagger}U_L^{(u)}$, so

$$V^\dagger V = U_L^{(d)\dagger}U_L^{(u)}\,U_L^{(u)\dagger}U_L^{(d)} = U_L^{(d)\dagger}(U_L^{(u)}U_L^{(u)\dagger})U_L^{(d)} = U_L^{(d)\dagger}\,U_L^{(d)} = \mathbf{1}, \quad (32.63)$$

$$VV^\dagger = U_L^{(u)\dagger}U_L^{(d)}\,U_L^{(d)\dagger}U_L^{(u)} = U_L^{(u)\dagger}(U_L^{(d)}U_L^{(d)\dagger})U_L^{(u)} = U_L^{(u)\dagger}\,U_L^{(u)} = \mathbf{1}, \quad (32.64)$$

where we used that a finite-dimensional matrix satisfying $A^\dagger A = \mathbf{1}$ also satisfies $AA^\dagger = \mathbf{1}$, so that $U_L^{(u)}U_L^{(u)\dagger} = \mathbf{1}$ and likewise for the down sector. Writing Eq. (32.61) in components $(V^\dagger V)_{jk} = \sum_i (V^\dagger)_{ji}V_{ik} = \sum_i V_{ij}^* V_{ik} = \delta_{jk}$ and $(VV^\dagger)_{ik} = \sum_j V_{ij}(V^\dagger)_{jk} = \sum_j V_{ij}V_{kj}^* = \delta_{ik}$ gives Eq. (32.62). □

Unitarity is not a cosmetic property; it is the most powerful experimental handle on the Standard Model in the flavor sector. The diagonal relations ($j = k$ or $i = k$) state that the rows and columns are unit vectors, $\sum_i |V_{ij}|^2 = 1$ and $\sum_j |V_{ij}|^2 = 1$: the total probability that a given down-type quark transitions to *some* up-type quark is one, no flavor leaks out of the three generations. The first-row relation $|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 = 1$ is tested to the sub-percent level and is one of the sharpest precision tests of the framework. The off-diagonal relations ($j \neq k$) are sums of three complex numbers that vanish,

$$\sum_i V_{ij}^*\, V_{ik} = 0 \qquad (j \neq k), \quad (32.65)$$

three complex vectors closing into a triangle in the complex plane. For $N_g = 3$ there are six such triangles (three from the row relations, three from the column relations). Most are extremely flat—one side far shorter than the others—but the one built from the first and third columns,

$$V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0, \quad (32.66)$$

has all three sides of comparable length, $O(\lambda^3)$ in the expansion parameter introduced below, and is therefore called *the* unitarity triangle. Its nontrivial shape—the fact that it is not degenerate to a line—is precisely the geometrical statement that the CKM matrix carries an irreducible complex phase. We draw it in Fig. 32.3, deferring the proof that its area measures CP violation to the next section.

32.3.3 Counting the physical parameters

A unitary matrix has redundancy. Some of its entries are pure convention, removable by changing the phases of the quark fields without touching any

observable. The physical content of V is what remains after every such freedom has been exhausted. We carry out the count for general $N_g \equiv N$ and then specialize, because the general result is what exposes *why* three generations are needed for a phase.

Step 1: parameters in a general unitary matrix. A complex $N \times N$ matrix has $2N^2$ real parameters. The unitarity condition $V^\dagger V = \mathbf{1}$ is a matrix equation between two Hermitian matrices and so imposes exactly N^2 independent real constraints: its diagonal gives N real conditions $\sum_i |V_{ij}|^2 = 1$, and its $N(N-1)/2$ independent off-diagonal entries each give one complex condition, that is $N(N-1)$ real conditions; together $N + N(N-1) = N^2$. (The relation $VV^\dagger = \mathbf{1}$ then follows automatically and imposes nothing new, as the proof of Theorem 32.11 showed.) Hence

$$\#\{\text{real parameters of } U(N)\} = 2N^2 - N^2 = N^2, \tag{32.67}$$

which is of course the real dimension of the Lie group $U(N)$.

Step 2: angles versus phases. We split these N^2 parameters into "angles" and "phases" by comparison with the real orthogonal subgroup. A real orthogonal matrix—one with all entries real—is the most general *phase-free* unitary matrix; the group $O(N)$ (its connected part $SO(N)$) has dimension

$$\dim SO(N) = \binom{N}{2} = \frac{N(N-1)}{2}, \tag{32.68}$$

the number of independent planes in which one can rotate, equivalently the number of independent generators of an antisymmetric real matrix. These $N(N-1)/2$ parameters are the *mixing angles*: rotations that mix generations through real, sign-definite probabilities and cannot be removed by rephasing fields, since a phase redefinition (a diagonal unitary) cannot turn a complex off-diagonal modulus into a different modulus—it only multiplies entries by phases. The remaining

$$N^2 - \frac{N(N-1)}{2} = \frac{2N^2 - N^2 + N}{2} = \frac{N(N+1)}{2} \tag{32.69}$$

parameters are *phases*, the genuinely complex content of V.

Step 3: rephasing the quark fields. Not all of these $N(N+1)/2$ phases are physical, because the CKM matrix appears in the Lagrangian only inside the gauge-invariant bilinear of Eq. (32.59), sandwiched between an up-type field and a down-type field. We are free to redefine the phase of each mass eigenstate,

$$u_{L,i} \to e^{i\alpha_i}\, u_{L,i}, \qquad d_{L,j} \to e^{i\beta_j}\, d_{L,j}, \tag{32.70}$$

together with the same phase on the right-handed components so that the mass terms $m_i\,\overline{u_{L,i}}u_{R,i}$ stay real and the kinetic and neutral-current terms (which pair a field with its own conjugate) are untouched. Under Eq. (32.70) the charged-current term $\overline{u_{L,i}}\,\gamma_\mu V_{ij} d_{L,j}$ is invariant provided we simultaneously transform

$$V_{ij} \to e^{-i\alpha_i}\, V_{ij}\, e^{i\beta_j}. \tag{32.71}$$

There are N phases α_i and N phases β_j, a priori $2N$ phase rotations with which to remove phases from V.

Step 4: the one rephasing that does nothing. Here is the crucial subtlety, and the place where a careless count produces the wrong answer. The $2N$ rephasings in Eq. (32.71) are not all independent in their action on V. Choose the *common* phase $\alpha_i = \beta_j = \theta$ for all i, j. Then every entry transforms as

$$V_{ij} \to e^{-i\theta}\, V_{ij}\, e^{+i\theta} = V_{ij}, \tag{32.72}$$

i.e. the overall phase rotation acts *trivially* on V. This is no accident: rotating every quark field by the same phase is the global baryon-number $U(1)$ transformation, a symmetry of the whole Standard Model Lagrangian, so it cannot affect any physical mixing parameter. Group-theoretically, the rephasing freedom is the torus $U(1)_u^N \times U(1)_d^N = U(1)^{2N}$ acting on V, but the diagonal subgroup $U(1)$ generated by Eq. (32.72) is the stabilizer—it fixes every V—so the group acting *effectively* on the CKM matrix is the quotient

$$U(1)^{2N}/\,U(1) \;\cong\; U(1)^{2N-1}, \tag{32.73}$$

of dimension $2N-1$, not $2N$. Exactly $2N-1$ independent relative phases are at our disposal to absorb into the quark fields. We can use each of them to set one phase of V to zero (and no more, since the $2N-1$ generators are independent), removing $2N-1$ phases.

Step 5: the net count. The physical phases are those that survive the maximal rephasing:

$$N_{\text{phase}} = \frac{N(N+1)}{2} - (2N-1) = \frac{N^2+N-4N+2}{2} = \frac{N^2-3N+2}{2} = \frac{(N-1)(N-2)}{2}. \tag{32.74}$$

The mixing angles are untouched by rephasing, so

$$N_{\text{angle}} = \frac{N(N-1)}{2}. \tag{32.75}$$

We record the result.

Theorem 32.12 (Physical parameter count of the CKM matrix)**.** *For N generations of quarks the CKM matrix contains*

$$\boxed{N_{\text{angle}} = \frac{N(N-1)}{2} \quad \textit{mixing angles}, \qquad N_{\text{phase}} = \frac{(N-1)(N-2)}{2} \quad \textit{CP-violating phases.}} \tag{32.76}$$

The total number of physical parameters is $N_{\text{angle}} + N_{\text{phase}} = (N-1)^2$.

The total is worth pausing on: $N(N-1)/2 + (N-1)(N-2)/2 = \frac{N-1}{2}[N + (N-2)] = (N-1)^2$. Equivalently, the rephasing has removed $2N-1$ phases from the N^2 parameters of $U(N)$, leaving $N^2 - (2N-1) = (N-1)^2$, a clean check on the bookkeeping.

Worked Example 32.13 (Specializing the count: two and three generations)**.** We evaluate Eq. (32.76) for the cases of physical interest and read off the lesson.

One generation ($N = 1$): $N_{\text{angle}} = 0$, $N_{\text{phase}} = 0$. The "matrix" is a single phase $V = e^{i\phi}$, and the one rephasing $2N - 1 = 1$ removes it: $V = 1$. There is no mixing and no phase, consistent with the single-generation charged current Eq. (32.56) we started from.

Two generations ($N = 2$): $N_{\text{angle}} = \frac{2\cdot 1}{2} = 1$ and $N_{\text{phase}} = \frac{1\cdot 0}{2} = 0$. The CKM matrix is a real 2×2 rotation,

$$V_{\text{Cabibbo}} = \begin{pmatrix} \cos\theta_C & \sin\theta_C \\ -\sin\theta_C & \cos\theta_C \end{pmatrix}, \tag{32.77}$$

parametrized by the single Cabibbo angle θ_C, with *no* phase. Let us verify the rephasing arithmetic directly. A general 2×2 unitary matrix has $N^2 = 4$ real parameters: one angle and $N(N+1)/2 = 3$ phases. The available rephasings number $2N - 1 = 3$. Three phases removed from three leaves zero, and one angle remains. Concretely, write a general unitary 2×2 matrix as $\left(\begin{smallmatrix} c\,e^{i\phi_1} & s\,e^{i\phi_2} \\ -s\,e^{i\phi_3} & c\,e^{i\phi_4} \end{smallmatrix}\right)$ with unitarity forcing $\phi_1 + \phi_4 = \phi_2 + \phi_3$ (so three independent phases). The rephasings $V_{ij} \to e^{-i\alpha_i} V_{ij} e^{i\beta_j}$ can set $\phi_1 = \phi_2 = \phi_3 = 0$; unitarity then forces $\phi_4 = 0$ as well, leaving Eq. (32.77). A two-generation world cannot violate CP through quark mixing.

Three generations ($N = 3$): $N_{\text{angle}} = \frac{3\cdot 2}{2} = 3$ and $N_{\text{phase}} = \frac{2\cdot 1}{2} = 1$. Three angles and exactly one irreducible phase. This is the result of Kobayashi and Maskawa [298]: the count first becomes positive at $N = 3$, so a single complex phase capable of violating CP requires *at least* three generations. Historically this was a prediction—when [298] was written only the three lightest quarks were known—of an entire third generation, motivated by the need to accommodate the CP violation already observed in the neutral kaon system. The arithmetic $\frac{(N-1)(N-2)}{2}$ is the structural reason the Standard Model ties CP violation to the replication of generations.

Physical Insight

Why the phase appears only at $N = 3$. The angle count $N(N-1)/2$ grows quadratically, but the phase count $(N-1)(N-2)/2$ is shifted: it counts pairs of generations *excluding* one, and vanishes for $N \leq 2$. The intuitive reason is that one can always rotate away the phase of any single off-diagonal mixing—there are enough field rephasings to "flatten" a 2×2 block. A phase becomes unremovable only when three rows and three columns are simultaneously coupled, so that no choice of $2N-1$ field phases can make all entries real at once. The unitarity triangle Eq. (32.66) is the geometric face of the same fact: with only two generations the corresponding "triangle" degenerates to a single real segment.

32.3.4 The standard parametrization

We now construct an explicit 3×3 unitary matrix carrying exactly three angles and one phase. The cleanest construction, adopted as the convention of the Particle Data Group, factors V into three successive plane rotations, one in each of the three coordinate planes of generation space, with the single physical phase attached to the rotation that mixes the first and third generations.

Introduce the abbreviations $c_{ij} \equiv \cos\theta_{ij}$, $s_{ij} \equiv \sin\theta_{ij}$ for the three angles $\theta_{12}, \theta_{13}, \theta_{23}$, all taken in the first quadrant $0 \leq \theta_{ij} \leq \pi/2$, and the phase δ. Define the three factors

$$R_{23} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}, \quad U_{13} = \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix}, \quad R_{12} = \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}. \tag{32.78}$$

Each is manifestly unitary: R_{23} and R_{12} are real orthogonal rotations, and U_{13} is unitary because $c_{13}^2 + s_{13}^2 = 1$ makes its rows orthonormal even with the phase $e^{\pm i\delta}$ (the modulus of each entry is unaffected by the phase). The standard parametrization is their ordered product.

Definition 32.14 (Standard (PDG) parametrization)**.** The CKM matrix in the standard parametrization is

$$V = R_{23}\, U_{13}\, R_{12} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}. \tag{32.79}$$

Derivation of Eq. (32.79) and verification of unitarity. Multiply the factors right to left. First $U_{13}R_{12}$: the rows of U_{13} are $(c_{13}, 0, s_{13}e^{-i\delta})$, $(0,1,0)$, $(-s_{13}e^{i\delta}, 0, c_{13})$,

and the columns of R_{12} are $(c_{12}, -s_{12}, 0)^{\mathsf{T}}$, $(s_{12}, c_{12}, 0)^{\mathsf{T}}$, $(0, 0, 1)^{\mathsf{T}}$. Their products give

$$U_{13}R_{12} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12} & c_{12} & 0 \\ -c_{12}s_{13}e^{i\delta} & -s_{12}s_{13}e^{i\delta} & c_{13} \end{pmatrix}. \tag{32.80}$$

Now multiply on the left by R_{23}, whose first row is $(1, 0, 0)$, second row $(0, c_{23}, s_{23})$, third row $(0, -s_{23}, c_{23})$. The first row of the product equals the first row of Eq. (32.80) unchanged. The second row is $c_{23} \times$ (row 2) $+ s_{23} \times$ (row 3) of Eq. (32.80):

$$(-s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta},\ c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta},\ s_{23}c_{13}). \tag{32.81}$$

The third row is $-s_{23} \times$ (row 2) $+ c_{23} \times$ (row 3):

$$(s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta},\ -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta},\ c_{23}c_{13}). \tag{32.82}$$

Assembling the three rows gives Eq. (32.79). Unitarity is inherited automatically: V is a product of three unitary matrices, so $V^\dagger V = R_{12}^\dagger U_{13}^\dagger R_{23}^\dagger R_{23} U_{13} R_{12} = R_{12}^\dagger U_{13}^\dagger U_{13} R_{12} = R_{12}^\dagger R_{12} = \mathbf{1}$, each adjacent pair collapsing in turn. As a direct check on Eq. (32.79), the modulus-squared of the first row sums to $c_{12}^2 c_{13}^2 + s_{12}^2 c_{13}^2 + s_{13}^2 = c_{13}^2(c_{12}^2 + s_{12}^2) + s_{13}^2 = c_{13}^2 + s_{13}^2 = 1$, confirming the first orthonormality relation of Eq. (32.62). □

Several features of Eq. (32.79) repay attention. The phase δ enters only through U_{13}, so it appears explicitly only in entries involving the 1–3 mixing: $V_{ub} = s_{13}e^{-i\delta}$ carries it directly, and it leaks into the lower-left 2×2 block through the $s_{13}e^{i\delta}$ terms. The three angles control the three off-diagonal moduli almost cleanly: because s_{13} is tiny (see below), $|V_{us}| \simeq s_{12}$, $|V_{cb}| \simeq s_{23}$, and $|V_{ub}| = s_{13}$ to excellent approximation. The angle θ_{12} is the modern incarnation of the Cabibbo angle: comparing Eq. (32.79) with the two-generation form Eq. (32.77), the 1–2 block reduces to the Cabibbo rotation when $\theta_{13}, \theta_{23} \to 0$, and the experimental identification is

$$\theta_{12} \equiv \theta_C, \qquad |V_{us}| = s_{12}c_{13} \simeq \sin\theta_C. \tag{32.83}$$

Numerically, the three angles are strongly hierarchical,

$$\theta_{12} \approx 13.0^\circ, \qquad \theta_{23} \approx 2.4^\circ, \qquad \theta_{13} \approx 0.21^\circ, \tag{32.84}$$

while the phase is large and $O(1)$, $\delta \approx 1.14$ rad $\approx 65.5^\circ$. This hierarchy of angles is the empirical input that the Wolfenstein parametrization is built to make manifest, and the largeness of δ is why CP violation, though it acts only through the smallest entries, is not numerically negligible.

32.3.5 The Wolfenstein parametrization

The angle hierarchy Eq. (32.84) cries out for an expansion. Numerically $\sin\theta_{12} \approx 0.225$ is small, $\sin\theta_{23}$ is of order $\sin^2\theta_{12}$, and $\sin\theta_{13}$ is of order $\sin^3\theta_{12}$. Wolfenstein [526] observed that taking a single small parameter

$$\lambda \equiv \sin\theta_{12} = |V_{us}| \approx 0.225 \tag{32.85}$$

and trading the other two angles and the phase for three numbers of order unity organizes the entire matrix as a power series in λ, exposing the hierarchy at a glance. The exact definitions of the four Wolfenstein parameters (λ, A, ρ, η) in terms of the standard angles are

$$s_{12} = \lambda, \qquad s_{23} = A\lambda^2, \qquad s_{13}e^{-i\delta} = A\lambda^3(\rho - i\eta). \tag{32.86}$$

These are definitions, exact to all orders; the expansion enters only when we re-express Eq. (32.79) as a series in λ. The next worked example carries that expansion out in full.

Worked Example 32.15 (Deriving the Wolfenstein matrix to $O(\lambda^3)$)**.** We substitute Eq. (32.86) into the standard parametrization Eq. (32.79) and expand consistently to third order in λ, tracking exactly where the phase survives.

Cosines. From $s_{12} = \lambda$,

$$c_{12} = \sqrt{1-\lambda^2} = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4). \tag{32.87}$$

From $s_{23} = A\lambda^2$, $c_{23} = \sqrt{1 - A^2\lambda^4} = 1 + O(\lambda^4)$. From $s_{13} = A\lambda^3|\rho - i\eta| = O(\lambda^3)$, $c_{13} = \sqrt{1 - s_{13}^2} = 1 + O(\lambda^6)$. So to the order we work, $c_{13} = 1$ and $c_{23} = 1$, while $c_{12} = 1 - \frac{1}{2}\lambda^2$.

First row. Reading directly from Eq. (32.79),

$$V_{ud} = c_{12}c_{13} = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4), \tag{32.88}$$

$$V_{us} = s_{12}c_{13} = \lambda + O(\lambda^7) = \lambda, \tag{32.89}$$

$$V_{ub} = s_{13}e^{-i\delta} = A\lambda^3(\rho - i\eta) \quad \text{exactly.} \tag{32.90}$$

The phase resides, undiluted, in V_{ub}.

Second row. For $V_{cd} = -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta}$, note $s_{23}s_{13}e^{i\delta} = A\lambda^2 \cdot A\lambda^3(\rho + i\eta) = A^2\lambda^5(\rho + i\eta) = O(\lambda^5)$, while $s_{12}c_{23} = \lambda + O(\lambda^5)$. Hence

$$V_{cd} = -\lambda + O(\lambda^5) = -\lambda. \tag{32.91}$$

Here $s_{13}e^{i\delta}$ is the complex conjugate of $s_{13}e^{-i\delta}$, namely $A\lambda^3(\rho + i\eta)$; this is the sign flip of η that distinguishes the lower triangle from V_{ub}. For $V_{cs} = c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta}$, the second term is $\lambda \cdot A^2\lambda^5(\dots) = O(\lambda^6)$, so

$$V_{cs} = c_{12}c_{23} + O(\lambda^6) = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4). \tag{32.92}$$

And directly $V_{cb} = s_{23}c_{13} = A\lambda^2$.

Third row. For $V_{td} = s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta}$,

$$V_{td} = \lambda \cdot A\lambda^2 - (1)(1)\, A\lambda^3(\rho + i\eta) + O(\lambda^5) = A\lambda^3 - A\lambda^3(\rho + i\eta) = A\lambda^3(1 - \rho - i\eta). \tag{32.93}$$

This is the second place the phase survives at leading order: V_{td} carries the combination $1 - \rho - i\eta$. For $V_{ts} = -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta}$, the second term is $\lambda \cdot A\lambda^3(\rho + i\eta) = O(\lambda^4)$, so

$$V_{ts} = -c_{12}s_{23} + O(\lambda^4) = -A\lambda^2 + O(\lambda^4) = -A\lambda^2, \tag{32.94}$$

and finally $V_{tb} = c_{23}c_{13} = 1 + O(\lambda^4)$.

Assembly. Collecting the entries to $O(\lambda^3)$,

$$V = \begin{pmatrix} 1 - \frac{1}{2}\lambda^2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \frac{1}{2}\lambda^2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + O(\lambda^4). \tag{32.95}$$

The structure is now transparent. The diagonal is unity up to $O(\lambda^2)$; the 1–2 mixing is $O(\lambda)$ (the Cabibbo rotation, appearing antisymmetrically as $\pm\lambda$); the 2–3 mixing is $O(\lambda^2)$; the 1–3 mixing is $O(\lambda^3)$ and is the *only* place the imaginary parts η live, sitting in $V_{ub} = A\lambda^3(\rho - i\eta)$ and $V_{td} = A\lambda^3(1 - \rho - i\eta)$. CP violation in the Standard Model is thus confined, at leading order, to the corners of the matrix connecting the first and third generations—which is exactly why it is small and why measuring V_{ub} and V_{td} (through B-meson physics) is the central experimental program for pinning down η.

Caution

Unitarity of the truncated Wolfenstein matrix. The matrix Eq. (32.95) is unitary only up to the truncation order: $V^\dagger V = \mathbf{1} + O(\lambda^4)$, not exactly. For instance the first-row norm gives $(1 - \frac{1}{2}\lambda^2)^2 + \lambda^2 + O(\lambda^6) = 1 - \lambda^2 + \frac{1}{4}\lambda^4 + \lambda^2 = 1 + O(\lambda^4)$, unitary to the stated order. When exact unitarity is needed—for example in global fits—one uses the *defining* relations Eq. (32.86) together with the exact standard form Eq. (32.79), of which Eq. (32.95) is merely the leading expansion. It is also conventional to define the rephasing-invariant combinations $\bar\rho + i\bar\eta \equiv -V_{ud}V_{ub}^*/(V_{cd}V_{cb}^*)$, which differ from $\rho + i\eta$ by $\bar\rho = \rho(1 - \frac{1}{2}\lambda^2) + O(\lambda^4)$ and likewise for $\bar\eta$; these are the coordinates of the apex of the unitarity triangle.

32.3.6 Numerical magnitudes and the unitarity triangle

The four Wolfenstein parameters are now measured to high precision. The global fit values, quoted to mid-2020s precision from [382], are

$$\lambda = 0.22501(68), \quad A = 0.826^{+0.018}_{-0.015}, \quad \bar\rho = 0.159(10), \quad \bar\eta = 0.348(10). \tag{32.96}$$

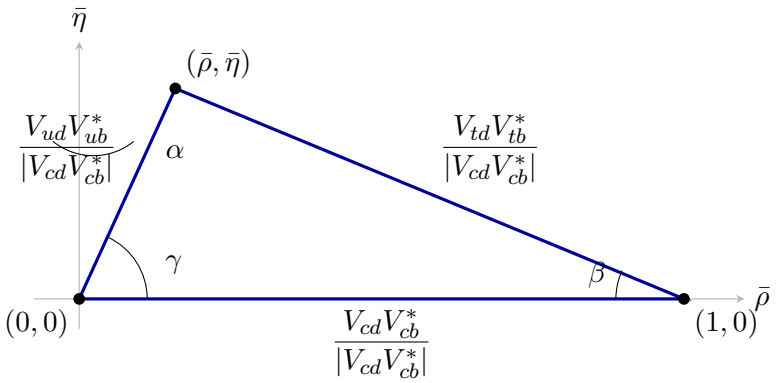


Figure 32.3: The unitarity triangle from the down–bottom column relation Eq. (32.66), normalized by $V_{cd}V_{cb}^*$ so that its base runs from $(0,0)$ to $(1,0)$ and its apex sits at $(\bar\rho,\bar\eta) \approx (0.16, 0.35)$ [382]. The nonzero height $\bar\eta \neq 0$ is the geometric signature of the irreducible CKM phase; a CP-conserving theory would flatten the triangle onto the real axis. The interior angles α, β, γ are measured independently in B-meson decays.

Translating through the parametrization, the moduli of the CKM entries are

$$(|V_{ij}|) = \begin{pmatrix} |V_{ud}| & |V_{us}| & |V_{ub}| \\ |V_{cd}| & |V_{cs}| & |V_{cb}| \\ |V_{td}| & |V_{ts}| & |V_{tb}| \end{pmatrix} \approx \begin{pmatrix} 0.97435 & 0.22500 & 0.00369 \\ 0.22486 & 0.97349 & 0.04182 \\ 0.00857 & 0.04110 & 0.99912 \end{pmatrix}, \tag{32.97}$$

each with an uncertainty in the last one or two digits [382]. The near-diagonal dominance is the experimental face of the Wolfenstein hierarchy: the diagonal is close to unity, the Cabibbo entries $|V_{us}| \approx |V_{cd}| \approx \lambda$ are a few tenths, the 2–3 entries are of order $\lambda^2 \sim 2 \times 10^{-2}$, and the corner entries $|V_{ub}|, |V_{td}|$ are of order $\lambda^3 \sim 4 \times 10^{-3}$. The single largest measured number, $|V_{us}| \simeq \sin\theta_C$, is the Cabibbo angle that opened the whole subject [101].

The unitarity relation Eq. (32.66) can now be drawn. Divide it through by $V_{cd}V_{cb}^*$ (the best-measured side, real and of order λ^3 by convention) so that one vertex sits at the origin and another at $(1,0)$; the apex then lands at $(\bar\rho,\bar\eta)$, and the three sides have comparable length $O(1)$ after rescaling. Figure 32.3 displays this triangle. Its nonzero height $\bar\eta \neq 0$ is the geometric encoding of the irreducible phase δ: were η (equivalently δ) to vanish, the apex would collapse onto the real axis and the triangle would degenerate to a line. The interior angles, conventionally called α, β, γ, are each measurable through interference in B-meson decays, and their sum being 180° is itself a test of the three-generation unitarity of V.

We have now isolated the physical content of quark mixing for three generations: three rotation angles, dominated by the Cabibbo angle $\theta_{12} = \theta_C$, and a single irreducible phase δ that the parameter count Theorem 32.12 guarantees

cannot be rotated away. That phase, and the nonzero area of the triangle in Fig. 32.3, are the entire candidate source of CP violation in the quark sector. The next section makes this candidacy precise: it shows by examining the CP transformation of the charged current Eq. (32.59) that a nonzero δ genuinely breaks CP, and it builds the rephasing-invariant, basis-independent measure of that breaking—the Jarlskog invariant—so that the physical statement does not depend on any of the parametrization choices made here.

32.4 CP Violation and the Jarlskog Invariant

The parameter count of the previous section left us with a single irreducible quantity that no rephasing of the quark fields can remove: the phase δ of the three-generation CKM matrix. That δ cannot be rotated away is an algebraic fact. What it *means* physically is a separate question, and it is the question that organizes this section. A complex number sitting in a coupling is not, by itself, a symptom of anything; whether it signals a genuine asymmetry between matter and antimatter depends on how the coupling responds to the discrete operation that exchanges them. We therefore begin by asking how the charged-current interaction transforms under CP, the product of charge conjugation and parity. The answer is sharp: CP acts on the weak vertex precisely by complex-conjugating the CKM entries. The interaction is CP-symmetric if and only if V can be brought to a real matrix by rephasing the quark fields — and for three generations, as we have just seen, it cannot. The surviving phase is thus the unique renormalizable source of CP violation in the quark sector of the Standard Model.

Having established that CP violation is present, we then face a subtlety that the phase δ shares with the three angles: it is convention-dependent. Different authors, different parametrizations, and different rephasing choices assign it different numerical values, and in some conventions one can make any single matrix element real. A physical effect must not depend on such bookkeeping. The resolution, due to Jarlskog, is to identify a single rephasing- and basis-invariant combination of CKM elements — the Jarlskog invariant J — that is nonzero exactly when CP is violated, that fixes the magnitude of every CP-violating observable in the quark sector, and that can be written without reference to any parametrization at all, as the imaginary part of a determinant of commuted mass matrices. Constructing J, proving its invariance, and reading off from its closed form the conditions under which CP violation must vanish, is the technical heart of what follows.

32.4.1 How CP acts on the charged current

We work with the quark charged-current interaction in the mass basis, which is Eq. (31.129) of Chapter 31 with the flavour-diagonal entry $\bar{u}_L\gamma_\mu d_L$ promoted to the full CKM-rotated current of Eq. (32.49):

$$\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}}\sum_{i,j}\Big[\,V_{ij}\,W^{+\mu}\,\bar{u}_i\,\gamma_\mu P_L\,d_j\;+\;V_{ij}^{*}\,W^{-\mu}\,\bar{d}_j\,\gamma_\mu P_L\,u_i\,\Big], \qquad P_L=\tfrac{1}{2}(1-\gamma^5), \tag{32.98}$$

where $u_i\in\{u,c,t\}$, $d_j\in\{d,s,b\}$, and the second term is the Hermitian conjugate of the first, so that $\mathcal{L}_{\rm CC}^{\dagger}=\mathcal{L}_{\rm CC}$. The CKM elements V_{ij} are c-numbers; the field operators are u_i, d_j (Dirac quark fields) and $W_\mu^{\pm}$ (the charged weak bosons). Our task is to compute the CP image of the first term and compare it with the second.

The fermion bilinear. The current carried by the vertex is the chiral bilinear $\bar{u}_i\gamma_\mu P_L d_j$, built from two *distinct* fields. We transform it in two stages, parity then charge conjugation, using the master identities established for the bilinear covariants in Section 11.7. Throughout, $\tilde{x}\equiv(t,-\mathbf{x})$ denotes the parity-reflected point.

Under parity, the master identity Eq. (11.274) gives, for any Dirac matrix Γ,

$$\mathcal{P}\,(\bar{u}_i\,\Gamma\,d_j)(x)\,\mathcal{P}^{-1}=\bar{u}_i(\tilde{x})\,(\gamma^0\Gamma\gamma^0)\,d_j(\tilde{x}), \tag{32.99}$$

the intrinsic-parity phases of u_i and d_j being conventionally set to $+1$ (we return to the role of these phases below). With $\Gamma=\gamma_\mu P_L$ we need $\gamma^0\gamma_\mu P_L\gamma^0$. Using $\gamma^0\gamma_\mu\gamma^0=\gamma^\mu$ and $\gamma^0\gamma^5\gamma^0=-\gamma^5$, hence $\gamma^0P_L\gamma^0=\tfrac{1}{2}(1+\gamma^5)=P_R$,

$$\gamma^0\,\gamma_\mu P_L\,\gamma^0=(\gamma^0\gamma_\mu\gamma^0)(\gamma^0P_L\gamma^0)=\gamma^\mu P_R. \tag{32.100}$$

The projector has flipped from left to right: parity alone turns the weak current into its mirror, and because the weak interaction couples only to P_L, parity is *not* a symmetry. This is the algebraic face of maximal parity violation. The content of CP is that charge conjugation flips the chirality back.

Under charge conjugation, the master identity Eq. (11.277), applied to two distinct fields, reads

$$\mathcal{C}\,(\bar{u}_i\,\Gamma\,d_j)\,\mathcal{C}^{-1}=\bar{d}_j\,(C\,\Gamma^{T}C^{-1})\,u_i, \tag{32.101}$$

in which the two fields have *exchanged places*, $u_i\leftrightarrow d_j$ — this exchange is the seed of the $V\to V^*$ rule. Applying it to the parity-transformed matrix $\Gamma=\gamma^\mu P_R$ and using $C\gamma^{\mu T}C^{-1}=-\gamma^\mu$, $C\gamma^{5T}C^{-1}=\gamma^5$ (so $CP_R^TC^{-1}=P_R$),

$$C\,(\gamma^\mu P_R)^TC^{-1}=C\,P_R^T\gamma^{\mu T}\,C^{-1}=(CP_R^TC^{-1})(C\gamma^{\mu T}C^{-1})=P_R\,(-\gamma^\mu)=-\,\gamma^\mu P_L, \tag{32.102}$$

where in the last step $P_R\gamma^\mu = \gamma^\mu P_L$. The chirality is back to P_L, as promised, at the cost of an overall minus sign. Assembling the two stages, the combined CP transformation of the chiral current is

$$\bar{u}_i\,\gamma_\mu P_L\,d_j\,(x) \;\xrightarrow{CP}\; -\,\bar{d}_j\,\gamma^\mu P_L\,u_i\,(\tilde{x}). \tag{32.103}$$

The current of an up-type into a down-type has become, up to a sign and the reflection $x \to \tilde{x}$, the conjugate current of a down-type into an up-type — exactly the field content of the Hermitian-conjugate term in Eq. (32.98).

The gauge boson. The charged vector bosons transform like any spin-one field. Under parity a vector field reflects, $W^+_\mu(x) \to W^{+\,\mu}(\tilde{x})$ (temporal component even, spatial components odd). Under charge conjugation the $W^\pm$ carry opposite electric charge and are exchanged, $W^+_\mu \leftrightarrow W^-_\mu$; the sign of the exchange is fixed by demanding that the gauge kinetic term and the minimal-coupling vertex with a *real* coupling be C-invariant — the same logic that makes the photon field C-odd in QED (Section 11.7.2). The outcome is

$$W^{+\mu}(x) \;\xrightarrow{CP}\; -\,W^-_\mu(\tilde{x}). \tag{32.104}$$

The vertex. Contracting Eq. (32.103) with Eq. (32.104), the two minus signs cancel and the free index is restored,

$$W^{+\mu}\,\bar{u}_i\gamma_\mu P_L d_j\,(x) \;\xrightarrow{CP}\; (-W^-_\mu)(-\bar{d}_j\gamma^\mu P_L u_i)(\tilde{x}) = W^{-\mu}\,\bar{d}_j\gamma_\mu P_L u_i\,(\tilde{x}). \tag{32.105}$$

The c-number V_{ij} is inert under CP (it multiplies the operator and is not conjugated, since CP acts on operators and states, not on the coupling constants in the Lagrangian). Therefore the first term of Eq. (32.98) maps under CP to

$$\frac{g}{\sqrt{2}}\,V_{ij}\,W^{+\mu}\,\bar{u}_i\gamma_\mu P_L d_j \;\xrightarrow{CP}\; \frac{g}{\sqrt{2}}\,V_{ij}\,W^{-\mu}\,\bar{d}_j\gamma_\mu P_L u_i, \tag{32.106}$$

all fields now at $\tilde{x}$ (an integration variable in the action). The term already present in $\mathcal{L}_{\rm CC}$ with this operator structure is the Hermitian conjugate, carrying coefficient V^*_{ij}. The CP image Eq. (32.106) coincides with it if and only if

$$V_{ij} = V^*_{ij} \qquad \text{for all } i,j, \tag{32.107}$$

that is, if and only if every CKM element is real. Running the same computation on the conjugate term sends $V^*_{ij} \to V^*_{ij}$ against a coefficient V_{ij}, giving the identical condition. We have proved:

Theorem 32.16 (CP and the reality of the CKM matrix)**.** *Under the combined CP transformation, the quark charged-current interaction Eq. (32.98) maps into itself with the replacement $V_{ij} \to V^*_{ij}$. The interaction is CP-invariant if and only*

if the CKM matrix can be made real. Since rephasing the quark fields multiplies the elements by phases, $V_{ij} \to e^{i\alpha_i} V_{ij} e^{-i\beta_j}$, the physically meaningful statement is: CP is conserved in the quark sector if and only if V can be brought to a real (orthogonal) matrix by a rephasing. For three generations this is impossible — the single phase δ of Eq. (32.79) cannot be removed — so the Standard Model with three generations violates CP.

Caution

The conclusion "V real" in Eq. (32.107) is itself convention-dependent, because the intrinsic CP phases of the quark fields, which we set to unity, can be absorbed into the rephasing freedom. What is convention-*independent* is whether *some* rephasing makes V real simultaneously. The rest of this section builds the invariant that answers this question without committing to any phase convention.

Physical Insight

The mechanism deserves a sentence of physics. Parity flips the handedness of the weak current; charge conjugation flips it back but exchanges particle with antiparticle. The net effect of CP is to send the amplitude for a quark transition $d_j \to u_i$ into the amplitude for the antiquark transition $\bar{d}_j \to \bar{u}_i$ with $V_{ij} \to V_{ij}^*$. A complex V_{ij} therefore means the process and its CP mirror proceed with different complex couplings. By itself a single complex coupling is unobservable — it is a phase — but where two amplitudes with *different* CKM phases interfere, the rate for a process and its CP-conjugate can differ. That is the observable face of CP violation, and its size is governed by the invariant we now construct.

32.4.2 Rephasing-invariant quartets and the unitarity triangle

Because individual phases of V_{ij} are convention, the building blocks of any physical CP-violating quantity must be combinations in which all rephasing phases cancel. Under $V_{ij} \to e^{i\alpha_i} V_{ij} e^{-i\beta_j}$, the modulus $|V_{ij}|$ is invariant but carries no phase information. The simplest invariant that retains a phase is the *quartet*.

Definition 32.17 (Quartet)**.** For row indices $i \neq k$ and column indices $j \neq l$, the quartet is the product

$$Q_{ik}^{jl} \;\equiv\; V_{ij}\, V_{kl}\, V_{il}^*\, V_{kj}^*. \tag{32.108}$$

Under a rephasing the four factors acquire phases $e^{i\alpha_i} e^{-i\beta_j}$, $e^{i\alpha_k} e^{-i\beta_l}$, $e^{-i\alpha_i} e^{i\beta_l}$, $e^{-i\alpha_k} e^{i\beta_j}$, whose product is unity: every α and every β appears once with each sign. Hence Q_{ik}^{jl} is rephasing invariant. Its modulus is a product of four $|V|$'s,

manifestly invariant; the new information is its phase, equivalently its imaginary part $\operatorname{Im} Q_{ik}^{jl}$, which is the rephasing-invariant carrier of CP violation. Indeed, by Theorem 32.16, CP is conserved precisely when V can be made real, in which case every quartet is real and $\operatorname{Im} Q_{ik}^{jl} = 0$; conversely a single nonzero $\operatorname{Im} Q_{ik}^{jl}$ obstructs reality and signals CP violation. The first nontrivial fact is that, for three generations, there is essentially only one such number.

Theorem 32.18 (All quartets share one imaginary part)**.** *For a 3×3 unitary matrix the imaginary parts of all quartets are equal up to sign:*

$$\operatorname{Im}(V_{ij} V_{kl} V_{il}^{*} V_{kj}^{*}) = J \sum_{m} \varepsilon_{ikm} \sum_{n} \varepsilon_{jln}, \tag{32.109}$$

for a single real number J, where ε is the totally antisymmetric symbol. Consequently $\operatorname{Im} Q_{ik}^{jl} = \pm J$ for every choice of two distinct rows and two distinct columns, the sign being the product of the orientations of the row pair (i,k) and the column pair (j,l).

Proof. The key is an elementary lemma. For any three complex numbers a_1, a_2, a_3 with $a_1 + a_2 + a_3 = 0$,

$$\operatorname{Im}(a_1 \bar{a}_2) = \operatorname{Im}(a_2 \bar{a}_3) = \operatorname{Im}(a_3 \bar{a}_1). \tag{32.110}$$

Indeed, substituting $a_3 = -(a_1 + a_2)$ gives $\operatorname{Im}(a_2 \bar{a}_3) = \operatorname{Im}(-a_2 \bar{a}_1 - |a_2|^2) = -\operatorname{Im}(a_2 \bar{a}_1) = \operatorname{Im}(a_1 \bar{a}_2)$, and similarly $\operatorname{Im}(a_3 \bar{a}_1) = \operatorname{Im}(-|a_1|^2 - a_2 \bar{a}_1) = \operatorname{Im}(a_1 \bar{a}_2)$. Geometrically, the three quantities are twice the oriented areas of the triangle with vertices $0, a_1, a_1 + a_2$ read off from its three edges, which agree.

Fix two distinct columns $j \neq l$. Column orthogonality, $\sum_m V_{mj} V_{ml}^{*} = 0$, says the three complex numbers $a_m \equiv V_{mj} V_{ml}^{*}$ $(m = 1, 2, 3)$ sum to zero. Their pairwise products are exactly the quartets: $a_i \bar{a}_k = V_{ij} V_{il}^{*} V_{kj}^{*} V_{kl} = V_{ij} V_{kl} V_{il}^{*} V_{kj}^{*} = Q_{ik}^{jl}$. By Eq. (32.110), $\operatorname{Im} Q_{12}^{jl} = \operatorname{Im} Q_{23}^{jl} = \operatorname{Im} Q_{31}^{jl}$, so $\operatorname{Im} Q_{ik}^{jl}$ is independent of the row pair up to the cyclic orientation of (i,k) — transposing the rows conjugates $a_i \bar{a}_k \to a_k \bar{a}_i$ and flips the sign of the imaginary part. The identical argument applied to row orthogonality, $\sum_n V_{in} V_{kn}^{*} = 0$, shows independence from the column pair up to the orientation of (j,l). Both orientation signs are encoded by the Levi-Civita symbols, giving Eq. (32.109) with J the common value, e.g. $J = \operatorname{Im} Q_{12}^{23} = \operatorname{Im}(V_{us} V_{cb} V_{ub}^{*} V_{cs}^{*})$. □

The identity $a_1 + a_2 + a_3 = 0$ is more than algebra: plotted in the complex plane, the three terms $V_{ij} V_{kj}^{*}$ (for fixed rows i, k, summed over columns) close into a triangle — a *unitarity triangle*. Equation (32.109) says every unitarity triangle has the same area.

Proposition 32.19 (Unitarity triangles and their area)**.** *For each pair of distinct rows $i \neq k$, row orthogonality*

$$V_{i1} V_{k1}^{*} + V_{i2} V_{k2}^{*} + V_{i3} V_{k3}^{*} = 0 \tag{32.111}$$

expresses three complex numbers summing to zero; as vectors laid head to tail they form a triangle in the complex plane. The (unsigned) area of every such triangle equals $\frac{1}{2}|J|$*. There are three independent row triangles and three column triangles, all of equal area; the most-studied is the db triangle* $V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0$*, whose three sides are of comparable length and whose angles are directly measured in B-meson decays.*

Proof. The oriented area of the triangle with edge vectors a_1, a_2 (and closing edge $a_3 = -(a_1+a_2)$) is $\frac{1}{2}\operatorname{Im}(a_1\bar{a}_2)$ summed appropriately; for the closed configuration $a_1 + a_2 + a_3 = 0$ the area is $\frac{1}{2}|\operatorname{Im}(a_1\bar{a}_2)| = \frac{1}{2}|\operatorname{Im} Q| = \frac{1}{2}|J|$ by Theorem 32.18. Equality of all six triangle areas is the same statement as the universality of $|J|$. □

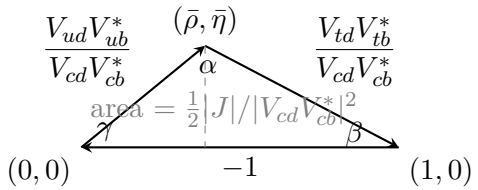


Figure 32.4: The *db* unitarity triangle from Eq. (32.111), with all sides divided by $V_{cd}V_{cb}^*$ so that one side runs from 0 to 1 along the real axis. The apex sits at the Wolfenstein coordinates $(\bar\rho, \bar\eta)$; a nonzero height $\bar\eta$ is CP violation. The area of the un-normalized triangle is $\frac{1}{2}|J|$ (Theorem 32.19). The interior angles α, β, γ are measured in B decays and their sum is 180°, a closure test of three-generation unitarity.

Definition 32.20 (Jarlskog invariant)**.** The *Jarlskog invariant* is the common quartet imaginary part of Eq. (32.109),

$$J \;\equiv\; \operatorname{Im}(V_{ij}V_{kl}V_{il}^*V_{kj}^*) \Big/ \Big(\textstyle\sum_m \varepsilon_{ikm} \sum_n \varepsilon_{jln}\Big), \tag{32.112}$$

a single real number characterizing the entire CP-violating structure of the CKM matrix.

32.4.3 Invariance of J

For J to deserve the name "measure of CP violation" it must be insensitive to the two kinds of convention that infect V: the rephasing of mass-eigenstate quark fields, and the choice of *weak basis* in which the Yukawa matrices are written before they are diagonalized.

Proposition 32.21 (Rephasing invariance)**.** *J is invariant under independent rephasings $u_i \to e^{i\alpha_i}u_i$, $d_j \to e^{i\beta_j}d_j$ of the quark mass eigenstates, which act on the CKM matrix as $V_{ij} \to e^{i\alpha_i}V_{ij}e^{-i\beta_j}$.*

Proof. This is the calculation already noted below Theorem 32.17: the quartet $V_{ij}V_{kl}V_{il}^{*}V_{kj}^{*}$ collects the phase $\exp[i(\alpha_i-\beta_j)+i(\alpha_k-\beta_l)-i(\alpha_i-\beta_l)-i(\alpha_k-\beta_j)] = \exp[0] = 1$. Each α and each β enters once with a plus and once with a minus sign. Hence every quartet, and therefore J, is unchanged. □

A change of weak basis is the statement that the physics cannot depend on *which* unitary combinations of the three same-charge gauge-eigenstate quark fields we call "generations" before diagonalizing the Yukawa couplings. Because the gauge interactions are flavour-universal, the left-handed quark doublet $Q_L = (u_L, d_L)^T$ may be rotated by a single unitary matrix W acting on the generation index, $Q_L \to W\,Q_L$, without touching the gauge sector. Such a rotation acts on the Yukawa matrices and hence on the Hermitian combinations $H_u \equiv M_u M_u^\dagger$ and $H_d \equiv M_d M_d^\dagger$ built from the up- and down-type mass matrices M_u, M_d. It is on these basis-covariant objects that we now phrase J.

Proposition 32.22 (Basis invariance)**.** *Under a weak-basis rotation $Q_L \to W Q_L$ of the left-handed doublet (W unitary), the Hermitian matrices transform homogeneously, $H_u \to W H_u W^\dagger$ and $H_d \to W H_d W^\dagger$, so the commutator transforms by conjugation,*

$$[H_u, H_d] \;\to\; W\,[H_u, H_d]\,W^\dagger, \tag{32.113}$$

and any function of its eigenvalues — in particular $\det[H_u, H_d]$ *— is invariant. Since the eigenvalues of H_u and H_d are the squared masses, themselves basis-independent, the quantity J extracted below is a genuine weak-basis invariant.*

Proof. The up Yukawa term couples $\bar{Q}_L\, Y_u\, u_R$ (schematically), so under $Q_L \to W Q_L$ one has $Y_u \to W Y_u$ and likewise $Y_d \to W Y_d$, the right-handed rotations being independent and immaterial here because they cancel in $H_{u,d} = v^2 Y_{u,d} Y_{u,d}^\dagger/2$. Thus $H_u \to W H_u W^\dagger$, $H_d \to W H_d W^\dagger$, and $[H_u, H_d] \to W[H_u, H_d]W^\dagger$ by direct substitution. The determinant is a class function, $\det(WCW^\dagger) = \det C$, completing the proof. □

32.4.4 The Jarlskog determinant

We now make the basis-invariant content explicit by computing $\det[H_u, H_d]$ and reading off J together with its dependence on the quark masses. This is the formulation in which the conditions for CP violation become transparent.

Theorem 32.23 (Jarlskog commutator formula)**.** *For three generations, with $H_u = M_u M_u^\dagger$ and $H_d = M_d M_d^\dagger$,*

$$\det\,[\,H_u,\,H_d\,] \;=\; 2\,i\,J\,\Delta_u\,\Delta_d, \tag{32.114}$$

where

$$\Delta_u = (m_t^2 - m_c^2)(m_t^2 - m_u^2)(m_c^2 - m_u^2),$$

$$\Delta_d = (m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2), \tag{32.115}$$

and J is the Jarlskog invariant of Theorem 32.20.

Proof. Work in the up-type mass basis, where $H_u = \mathrm{diag}(x_1, x_2, x_3)$ with $x_a = m_{u_a}^2$. In this basis the down-type Hermitian matrix is the CKM-rotated diagonal of squared down masses,

$$H_d = V\ \mathrm{diag}(y_1, y_2, y_3)\, V^\dagger, \qquad y_b = m_{d_b}^2, \qquad (H_d)_{ij} = \sum_k V_{ik}\, y_k\, V_{jk}^* \equiv W_{ij}. \tag{32.116}$$

The commutator $C \equiv [H_u, H_d]$ has matrix elements

$$C_{ij} = (H_u)_{ii}(H_d)_{ij} - (H_d)_{ij}(H_u)_{jj} = (x_i - x_j)\, W_{ij}, \tag{32.117}$$

so its diagonal vanishes. Being the commutator of two Hermitian matrices, C is anti-Hermitian: $C^\dagger = -C$, i.e. $C_{ji} = -C_{ij}^*$. For a 3×3 matrix with zero diagonal the determinant reduces to the two off-diagonal cyclic products,

$$\det C = C_{12}C_{23}C_{31} + C_{21}C_{32}C_{13}. \tag{32.118}$$

Anti-Hermiticity converts the second product into minus the conjugate of the first,

$$C_{21}C_{32}C_{13} = (-C_{12}^*)(-C_{23}^*)(-C_{31}^*) = -(C_{12}C_{23}C_{31})^*, \tag{32.119}$$

whence

$$\det C = C_{12}C_{23}C_{31} - (C_{12}C_{23}C_{31})^* = 2\,i\ \mathrm{Im}(C_{12}C_{23}C_{31}). \tag{32.120}$$

Substituting Eq. (32.117), the up-mass factors come out as a real prefactor,

$$C_{12}C_{23}C_{31} = (x_1 - x_2)(x_2 - x_3)(x_3 - x_1)\, W_{12}W_{23}W_{31}, \tag{32.121}$$

and the product of mass differences is the Vandermonde combination $\prod_{a<b}(x_b - x_a) = (x_2 - x_1)(x_3 - x_1)(x_3 - x_2) = \Delta_u$ (the ordering of factors only permutes signs that cancel: $(x_1 - x_2)(x_2 - x_3)(x_3 - x_1) = \prod_{a<b}(x_b - x_a)$). Hence

$$\det C = 2\,i\,\Delta_u\ \mathrm{Im}(W_{12}W_{23}W_{31}). \tag{32.122}$$

It remains to evaluate $\mathrm{Im}(W_{12}W_{23}W_{31})$. Write the off-diagonal $W_{ij} = \sum_k y_k\, a_k^{(ij)}$ with $a_k^{(12)} = V_{1k}V_{2k}^*$, $a_k^{(23)} = V_{2k}V_{3k}^*$, $a_k^{(31)} = V_{3k}V_{1k}^*$. Row orthogonality forces $\sum_k a_k^{(ij)} = 0$ for $i \neq j$, so each W_{ij} depends only on differences of the y_k. Eliminating $a_3^{(ij)} = -(a_1^{(ij)} + a_2^{(ij)})$ and writing $u \equiv y_1 - y_3$, $w \equiv y_2 - y_3$,

$$W_{12} = u\, a_1^{(12)} + w\, a_2^{(12)}, \quad W_{23} = u\, a_1^{(23)} + w\, a_2^{(23)}, \quad W_{31} = u\, a_1^{(31)} + w\, a_2^{(31)}. \tag{32.123}$$

Multiplying out,

$$W_{12}W_{23}W_{31} = u^3\, a_1^{(12)}a_1^{(23)}a_1^{(31)} + u^2 w \sum_{\text{one 2}}(\cdots) + uw^2 \sum_{\text{two 2}}(\cdots) + w^3\, a_2^{(12)}a_2^{(23)}a_2^{(31)}. \tag{32.124}$$

The pure u^3 and w^3 coefficients are real: $a_1^{(12)}a_1^{(23)}a_1^{(31)} = V_{11}V_{21}^*\, V_{21}V_{31}^*\, V_{31}V_{11}^* = |V_{11}|^2|V_{21}|^2|V_{31}|^2$ and likewise for the column-2 product, so they drop out of the imaginary part. The mixed coefficients are sums of quartets. Take the u^2w coefficient, the sum over the three ways to pick one index equal to 2:

$$\begin{aligned}\operatorname{Im}[a_1^{(12)}a_1^{(23)}a_2^{(31)} + a_1^{(12)}a_2^{(23)}a_1^{(31)} + a_2^{(12)}a_1^{(23)}a_1^{(31)}] &= |V_{21}|^2\, \operatorname{Im}(V_{11}V_{32}V_{12}^*V_{31}^*) \\ &\quad + |V_{11}|^2\, \operatorname{Im}(V_{22}V_{31}V_{21}^*V_{32}^*) \\ &\quad + |V_{31}|^2\, \operatorname{Im}(V_{12}V_{21}V_{11}^*V_{22}^*).\end{aligned} \tag{32.125}$$

Each bracket is a quartet; by Eq. (32.109) each equals $-J$ (the three row/column orientations all evaluate to $\varepsilon_{ikm}\varepsilon_{jln} = -1$). The prefactors sum by column-1 unitarity, $|V_{11}|^2 + |V_{21}|^2 + |V_{31}|^2 = 1$, giving u^2w-coefficient $= -J$. The same computation on the uw^2 coefficient yields $+J$ (now each quartet evaluates to $+J$ and the prefactors sum by column-2 unitarity). Therefore

$$\operatorname{Im}(W_{12}W_{23}W_{31}) = J\,(-u^2w + uw^2) = J\,uw(w-u) = J\,(y_1-y_2)(y_2-y_3)(y_3-y_1), \tag{32.126}$$

the last equality using $u-w = y_1-y_2$, $w = y_2-y_3$, $-u = y_3-y_1$. The down-mass product is again the Vandermonde $\Delta_d = \prod_{b<c}(y_c - y_b)$. Inserting Eq. (32.126) into Eq. (32.122) gives $\det C = 2iJ\,\Delta_u\Delta_d$, which is Eq. (32.114). □

Equation (32.114) is the promised parametrization-free statement of CP violation. The left side is a basis invariant (Theorem 32.22); the right side factorizes the physics into three independent requirements, each of which must be met for CP to be violated:

- **Non-degenerate up-type masses**, $\Delta_u \neq 0$: no two of m_u, m_c, m_t may coincide.
- **Non-degenerate down-type masses**, $\Delta_d \neq 0$: no two of m_d, m_s, m_b may coincide.
- **Non-trivial mixing**, $J \neq 0$: all three mixing angles must be nonzero and the phase must satisfy $\sin\delta \neq 0$.

We state the last point precisely and prove the degeneracy claims.

Proposition 32.24 (When CP violation vanishes)**.** *In the standard parametrization Eq. (32.79) the Jarlskog invariant is*

$$J = c_{12}\, c_{13}^2\, c_{23}\, s_{12}\, s_{13}\, s_{23}\, \sin\delta, \tag{32.127}$$

with $c_{ab} = \cos\theta_{ab}$, $s_{ab} = \sin\theta_{ab}$. Consequently $J = 0$ — and CP is conserved — if and only if at least one of the following holds: one of the three mixing angles vanishes (or equals $\pi/2$), or $\sin\delta = 0$. Moreover, if any two up-type masses are equal, or any two down-type masses are equal, then $\det[H_u, H_d] = 0$; in that case a residual rotation in the degenerate subspace removes the phase and CP is restored, regardless of the value of δ.

Proof. Formula Eq. (32.127) is computed directly in Theorem 32.26 below. Its vanishing conditions are read off term by term: each of $c_{12}, c_{13}, c_{23}, s_{12}, s_{13}, s_{23}, \sin\delta$ multiplies the product, so J vanishes iff one factor does, i.e. iff some $\theta_{ab} \in \{0, \pi/2\}$ or $\delta \in \{0, \pi\}$. For the mass-degeneracy statement, Δ_u contains the factor $(m_{u_a}^2 - m_{u_b}^2)$ for each pair, so $m_{u_a} = m_{u_b}$ sets $\Delta_u = 0$ and, by Eq. (32.114), $\det[H_u, H_d] = 0$; likewise for the down sector. The physical interpretation: if H_u has a degenerate eigenvalue, there is a U(2) freedom to rotate the two degenerate up flavours into each other, and this rotation can be chosen to cancel the offending phase in V (one removes a phase exactly as in the two-generation count, where the degeneracy effectively reduces the number of distinct masses). With one phase fewer than mismatched-mass pairs require, V becomes rephasable to real and CP is conserved. □

Corollary 32.25 (Two generations cannot violate CP)**.** *For $N_g = 2$ there is no rephasing-invariant measure of CP violation: J does not exist, and $\det[H_u, H_d]$ is real.*

Proof. Two ways, in agreement with the parameter count of the previous section, which gave zero physical phases for $N_g = 2$. *Algebraically:* for a 2×2 unitary V, the commutator $C = [H_u, H_d]$ again has $C_{ij} = (x_i - x_j)W_{ij}$ with vanishing diagonal, so $\det C = C_{11}C_{22} - C_{12}C_{21} = -C_{12}C_{21} = -(x_1 - x_2)W_{12}\,(x_2 - x_1)W_{21} = (x_1 - x_2)^2\, W_{12}W_{12}^* = (x_1 - x_2)^2|W_{12}|^2$, which is *real and non-negative*: $\operatorname{Im}\det C = 0$ identically, for any masses and any mixing. *Group-theoretically:* a 2×2 unitary matrix has 4 parameters; the $2N_g - 1 = 3$ relative quark rephasings remove three, leaving exactly one — the Cabibbo angle of [101] — and no phase. With no phase there is nothing to conjugate, and Theorem 32.16 guarantees CP conservation. The construction of a CP-odd invariant first becomes possible at $N_g = 3$, which is precisely why three generations are necessary: the smallest quark sector that can accommodate CP violation through the CKM mechanism has three families. □

Historical Note

The empirical driver came first, and it was a surprise. Through the early 1960s CP was believed exact, rescuing a particle–antiparticle symmetry after Wu's 1957 demonstration that parity (and with it C) is maximally violated in beta decay. The neutral kaon system offered a clean test:

the long-lived eigenstate K_L, if CP were conserved, would be the CP-odd combination and forbidden to decay to two pions (a CP-even state). In 1964 Christenson, Cronin, Fitch, and Turlay observed exactly that forbidden decay, $K_L \to \pi^+\pi^-$, at the level of a few parts in a thousand [124]: CP is violated in nature. The theoretical accommodation took nine years. In 1973, when only the up, down, and strange quarks were known and a fourth (charm) was conjectural, Kobayashi and Maskawa observed that a two-generation quark mixing matrix is real after rephasing and therefore *cannot* produce CP violation (our Theorem 32.25), whereas a three-generation matrix carries one irremovable phase [298]. To make the Standard Model consistent with the 1964 discovery, they predicted a *third* generation — the bottom and top quarks and the tau sector — years before any of these particles was found. The bottom quark appeared in 1977, the top in 1995, and the pattern of CP violation measured since in K and especially B mesons has agreed, so far, with a single phase δ. The rephasing- and basis-invariant repackaging of that phase, Eq. (32.114), is due to Jarlskog [284].

32.4.5 The magnitude of J, and a worked extraction

Worked Example 32.26 (J from the standard parametrization)**.** We verify Eq. (32.127) by evaluating the representative quartet $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*) = \mathrm{Im}(V_{12}V_{23}V_{13}^*V_{22}^*)$ in the standard parametrization of Eq. (32.79), whose relevant entries (with $c_{ab} = \cos\theta_{ab}$, $s_{ab} = \sin\theta_{ab}$) are

$$V_{12} = s_{12}c_{13}, \quad V_{23} = s_{23}c_{13}, \quad V_{13} = s_{13}e^{-i\delta}, \quad V_{22} = c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta}. \tag{32.128}$$

Here V_{12} and V_{23} are real, while $V_{13}^* = s_{13}e^{+i\delta}$ and $V_{22}^* = c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta}$. The product is

$$\begin{aligned} V_{12}V_{23}V_{13}^*V_{22}^* &= (s_{12}c_{13})(s_{23}c_{13})(s_{13}e^{i\delta})(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta}) \\ &= s_{12}s_{23}s_{13}c_{13}^2\, e^{i\delta}\big(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta}\big) \\ &= s_{12}s_{23}s_{13}c_{13}^2\Big(c_{12}c_{23}\, e^{i\delta} - s_{12}s_{23}s_{13}\Big). \end{aligned} \tag{32.129}$$

The second term inside the parentheses is real and contributes nothing to the imaginary part; the first carries $e^{i\delta}$. Hence

$$J = \mathrm{Im}(V_{12}V_{23}V_{13}^*V_{22}^*) = s_{12}s_{23}s_{13}c_{13}^2\, c_{12}c_{23}\, \sin\delta = c_{12}c_{13}^2c_{23}\, s_{12}s_{13}s_{23}\, \sin\delta, \tag{32.130}$$

which is Eq. (32.127). Every angle and the phase appear once; J is maximized not at $\theta_{ab} = \pi/4$ but at the angles that extremize the full product, and it vanishes the moment any single angle or $\sin\delta$ vanishes — the analytic content of Theorem 32.24.

Numerically, the measured CKM angles are hierarchical: $s_{12} \approx 0.225$ (the sine of the Cabibbo angle), $s_{23} \approx 0.041$, $s_{13} \approx 0.0037$, and the phase is large, $\delta \approx 1.1$ rad ($\approx 65^\circ$), so $\sin\delta \approx 0.9$. With the cosines all near unity, Eq. (32.127) gives

$$J \;\approx\; (0.225)(0.041)(0.0037)(0.9) \;\approx\; 3 \times 10^{-5}, \tag{32.131}$$

in agreement with the global fit value $J \simeq 3.08 \times 10^{-5}$ as of the mid-2020s [382]. The smallness of J is striking and consequential. The *phase* δ is of order one — CP violation is not suppressed because the phase is small. It is suppressed because J is the product of three small mixing angles; the near-decoupling of the generations, not the value of the phase, is what makes CP violation a sub-permille effect in the Standard Model. This tiny number sets the scale of every quark-sector CP asymmetry, and its insufficiency to explain the observed cosmological matter–antimatter asymmetry (which a naive estimate undershoots by many orders of magnitude) is one of the sharpest pointers, as of the mid-2020s, to physics beyond the Standard Model.

Physical Insight

The two faces of J — the area $\frac{1}{2}|J|$ of the unitarity triangle (Theorem 32.19) and the imaginary determinant $\det[H_u, H_d]/(2i\,\Delta_u\Delta_d)$ (Theorem 32.23) — say the same thing in two languages. The geometric language exposes CP violation as the failure of three complex numbers to lie on a line: a flat unitarity triangle (zero area) is a CP-conserving CKM matrix. The algebraic language exposes its prerequisites: CP violation needs both mass non-degeneracy and genuine mixing, because a commutator of mass matrices can be nonzero only when the two cannot be simultaneously diagonalized, and its imaginary part can survive only when neither sector is degenerate. A world with any two equal quark masses, or any vanishing angle, would have an exactly real CKM matrix no matter how the phase δ were chosen.

With J in hand we have a single, convention-free number that decides whether the quark sector violates CP and fixes the magnitude of every such effect. The next section turns from the charged current to the neutral current, where the same CKM unitarity that organized the unitarity triangles will be shown to forbid flavour-changing neutral currents at tree level and to suppress them at one loop — the GIM mechanism — so that the rephasing- and basis-invariant structure built here can be deployed against the phenomenology of the K and B systems.

32.5 The GIM Mechanism and the Suppression of Flavor-Changing Neutral Currents

The preceding two sections extracted from the three-generation Yukawa sector the two facts that now drive everything: the charged current is governed by a unitary mixing matrix $V_{\rm CKM}$ (Eq. (32.49)), and the single irreducible phase of that matrix is genuinely CP-violating, packaged into the rephasing-invariant Jarlskog determinant (Section 32.4). At the same time, the bidiagonalization that produced $V_{\rm CKM}$ left a quieter but equally consequential residue: when the gauge interactions are rewritten in the mass basis, the *neutral* current does not pick up any mixing matrix at all. This section is the systematic exploitation of that asymmetry between the charged and neutral sectors. We first prove, cleanly and completely, that there are no flavor-changing neutral currents (FCNC) at tree level—a statement that is exact, not approximate, and follows from the same unitarity $VV^\dagger = \mathbf{1}$ that defines the CKM matrix. We then ask what happens at one loop, where the charged current can sneak flavor change in through an internal W, and discover the Glashow–Iliopoulos–Maiani (GIM) mechanism: the loop amplitude carries a sum over internal quark flavors weighted by CKM factors, and unitarity forces the flavor-blind part of that sum to cancel, leaving a residue proportional to quark *mass splittings*—the m_q^2/M_W^2 suppression that, read backwards in 1970, demanded a fourth quark and fixed its mass.

32.5.1 Flavor-changing neutral currents: the definition and the tree-level theorem

A neutral-current interaction is one mediated by a neutral boson—in the Standard Model the photon, the Z, the gluon, or the Higgs. It is *flavor-changing* if the two fermion lines meeting at the vertex carry different flavor of the same electric charge, for instance $s \to d$, $b \to s$, or $c \to u$, with no change in electric charge across the vertex.

Definition 32.27 (Flavor-changing neutral current)**.** A flavor-changing neutral current is a term in the effective interaction Lagrangian of the form $\bar{f}_i\,\Gamma\,f_j\,\phi^0$ with $i \neq j$ two fermions of the same electric charge (both up-type, both down-type, or both charged leptons), Γ a Dirac structure, and ϕ^0 a neutral boson field. A process is FCNC if its leading amplitude requires such a vertex.

The empirical fact this definition confronts is that FCNC processes are spectacularly rare. The branching ratio for $K^+ \to \pi^+\nu\bar{\nu}$ (an $s \to d$ transition) is of order 10^{-10}; the K_L–K_S mass difference, generated by the $\Delta S = 2$ transition $\bar{s}d \to s\bar{d}$, is $\Delta m_K/m_K \sim 7 \times 10^{-15}$. Charged-current rates, by contrast, are unsuppressed: ordinary β decay and $K \to \mu\nu$ proceed at full weak strength. The theory must reproduce this enormous hierarchy, and it does so in two stages: an exact vanishing at tree level, and a calculable strong suppression at one loop.

We begin with the exact statement. Recall the neutral-current Lagrangian derived in Section 31.5.3,

$$\mathcal{L}_{\rm NC} = \frac{g}{\cos\theta_{\rm W}}\, Z_\mu\, J^{Z\,\mu}, \qquad J^Z_\mu = J^3_\mu - \sin^2\theta_{\rm W}\, J^{\rm em}_\mu, \tag{32.132}$$

together with the photon coupling $eA_\mu J^{\rm em}_\mu$, and recall from Eq. (31.125) that both currents are built from the weak-isospin and charge operators T^3 and Q. The point of the following theorem is that these operators are blind to generation: every up-type quark has $T^3 = +\frac{1}{2}$ (left) and $Q = +\frac{2}{3}$; every down-type quark has $T^3 = -\frac{1}{2}$ (left) and $Q = -\frac{1}{3}$. In the language of the Yukawa diagonalization, the operators T^3 and Q are proportional to the identity within each charge sector of flavor space. That single fact is the whole tree-level story.

Theorem 32.28 (No tree-level FCNC in the Standard Model)**.** *After rotation from the flavor (interaction) basis to the mass basis, the photon, Z, gluon, and Higgs neutral currents of the Standard Model are exactly diagonal in flavor. There is no tree-level vertex coupling a neutral boson to two fermions of different flavor.*

Proof. Write the gauge interaction in the flavor basis, before the Yukawa matrices are diagonalized. Collect the up-type left-handed quark fields of the three generations into a column $u_L = (u_L, c_L, t_L)^{\mathsf{T}}$ in flavor space, and similarly d_L, u_R, d_R. In this basis the neutral current of Eq. (32.132) acts on, say, the left-handed up-type fields as

$$J^Z_\mu\big|_{u_L} = \bar u_L\, \gamma_\mu\, (T^3_{u_L} - \sin^2\theta_{\rm W}\, Q_u)\, u_L = \left(\tfrac{1}{2} - \tfrac{2}{3}\sin^2\theta_{\rm W}\right) \bar u_L\, \gamma_\mu\, \mathbf{1}\, u_L, \tag{32.133}$$

where $\mathbf{1}$ is the 3×3 identity in generation space: the coefficient $T^3_{u_L} - \sin^2\theta_{\rm W} Q_u$ is one and the same number for u, c, t, so as a matrix in flavor space the current is a pure multiple of the identity. The same is true of the photon current (coefficient $Q_u = \frac{2}{3}$), the gluon current (the SU(3) color generators carry no flavor index), the right-handed up current (coefficient $-\frac{2}{3}\sin^2\theta_{\rm W}$), and verbatim for the down sector with the appropriate quantum numbers.

Now pass to the mass basis. The bidiagonalization of the Yukawa matrices replaces each chiral flavor multiplet by a unitary rotation of mass eigenstates, $u_L = U^u_L\, u^{\rm mass}_L$, $d_L = U^d_L\, d^{\rm mass}_L$, and analogously for the right-handed fields, with $U^u_{L,R}, U^d_{L,R} \in {\rm U}(3)$ (Section 32.2). Insert this into the current Eq. (32.133):

$$\bar u_L\, \gamma_\mu\, \mathbf{1}\, u_L = \bar u^{\rm mass}_L\, (U^u_L)^\dagger\, \gamma_\mu\, \mathbf{1}\, U^u_L\, u^{\rm mass}_L = \bar u^{\rm mass}_L\, \gamma_\mu\, (U^u_L)^\dagger U^u_L\, u^{\rm mass}_L = \bar u^{\rm mass}_L\, \gamma_\mu\, \mathbf{1}\, u^{\rm mass}_L, \tag{32.134}$$

because γ_μ acts on the Dirac index and commutes with the flavor-space rotation, and because the coefficient matrix is the identity, which the unitary rotation cannot disturb: $(U^u_L)^\dagger\, \mathbf{1}\, U^u_L = (U^u_L)^\dagger U^u_L = \mathbf{1}$. The current is diagonal in the mass basis with exactly the couplings it had in the flavor basis. No off-diagonal ($i \neq j$) entry is generated. The identical cancellation holds in every neutral

channel and for both chiralities. Since the Higgs couples through the Yukawa matrices themselves, and these are diagonalized to the (real, positive) mass matrices in precisely the same step, the physical Higgs current $\propto \bar{f}_i\, m_i\, f_i$ is also flavor-diagonal. Hence there is no tree-level FCNC. □

The structural reason deserves to be isolated, because it is exactly what fails for the charged current. The neutral current connects fields of the *same* charge sector, so its coefficient matrix is a single multiple of the identity, and the unitary rotation that defines the mass basis passes through it unchanged—the two rotations $U^\dagger$ and U are the same and annihilate. The charged current $\bar{u}_L \gamma_\mu d_L$ connects *different* charge sectors, so it is sandwiched between two *different* rotations, $(U_L^u)^\dagger U_L^d \equiv V_{\rm CKM}$ (Eq. (32.49)), which do not cancel and leave the physical CKM matrix behind. The same unitarity that makes $VV^\dagger = \mathbf{1}$ a property of the leftover charged-current matrix is the unitarity $U^\dagger U = \mathbf{1}$ that kills the neutral-current mixing outright.

Physical Insight

The absence of tree FCNC is not a fine-tuning and not an extra postulate: it is a free gift of the gauge structure. Because $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ assigns the same weak isospin and hypercharge to every generation, the neutral generators are flavor-universal, and a flavor-universal operator is invariant under any change of flavor basis. Mixing can only appear where two *inequivalent* bases are forced to meet—and the only place that happens is the charged current, which straddles the up and down sectors. Any model that wants tree-level FCNC must break this universality, e.g. by adding a second Higgs doublet whose Yukawa matrix is not aligned with the mass matrix.

32.5.2 One loop, and the GIM cancellation

Flavor change is forbidden at tree level but not forbidden outright, because the charged current *does* change flavor, and a neutral transition can be built from two charged-current vertices joined into a loop. The simplest such object that turns a down-type quark i into a down-type quark j is the one-loop "penguin" of Fig. 32.5(a): the external d_i emits a W, turning into an internal up-type quark α, which reabsorbs the W (now radiating a Z, photon, or gluon) and returns as d_j. Both charged-current vertices carry CKM factors. Reading them off from $\mathcal{L}_{\rm CC} = \frac{g}{\sqrt{2}} W^{+\mu} \bar{u}_{L\alpha} \gamma_\mu V_{\alpha i} d_{Li} + \text{h.c.}$ (Eq. (31.129), with the mixing matrix now installed), the $d_i \to u_\alpha$ vertex contributes $V_{\alpha i}$ and the $u_\alpha \to d_j$ vertex contributes $V_{\alpha j}^*$. The internal *up*-type quark of flavor $\alpha \in \{u, c, t\}$ is summed over, because all three propagate in the loop, and the loop integral depends on the internal

mass only through the dimensionless ratio

$$x_\alpha \equiv \frac{m_\alpha^2}{M_W^2}. \tag{32.135}$$

The amplitude therefore has the universal flavor structure

$$\mathcal{A}_{i\to j} = \frac{g^2}{M_W^2}\, C \sum_{\alpha=u,c,t} V_{\alpha i}\, V_{\alpha j}^*\, f(x_\alpha), \tag{32.136}$$

where C collects the overall couplings, Dirac structure, and external kinematics common to all three internal flavors, and f is the dimensionless loop function whose explicit form depends on the process (the $1/M_W^2$ is factored out so that f is a pure number for each α). The entire flavor- and mass-dependence lives in the sum $\sum_\alpha V_{\alpha i} V_{\alpha j}^* f(x_\alpha)$.

If the three internal quarks were degenerate, $f(x_\alpha)$ would be a common constant f pulled out of the sum, and what remains is exactly a unitarity sum. This is the GIM mechanism.

Theorem 32.29 (GIM cancellation)**.** *For $i \neq j$, the flavor sum in Eq. (32.136) vanishes in the limit of degenerate internal quark masses, and is in general proportional to the internal mass splittings:*

$$\sum_\alpha V_{\alpha i}\, V_{\alpha j}^*\, f(x_\alpha) = \frac{f'(0)}{M_W^2} \sum_\alpha V_{\alpha i} V_{\alpha j}^*\, (m_\alpha^2 - m_u^2) + O(m^4/M_W^4), \qquad (i \neq j). \tag{32.137}$$

The leading nonzero contribution is suppressed by m_α^2/M_W^2 relative to the naive estimate $\sim f(0)$, and is controlled entirely by mass differences between the internal quarks.

Proof. The defining unitarity of the CKM matrix is $V^\dagger V = \mathbf{1}$, whose (i,j) component reads

$$\sum_{\alpha=u,c,t} V_{\alpha i}^*\, V_{\alpha j} = \delta_{ij}. \tag{32.138}$$

Taking the complex conjugate (and using $\delta_{ij} \in \mathbb{R}$) gives the form that appears in the loop sum,

$$\sum_\alpha V_{\alpha i}\, V_{\alpha j}^* = \delta_{ij} \quad \Longrightarrow \quad \sum_\alpha V_{\alpha i}\, V_{\alpha j}^* = 0 \quad \text{for } i \neq j. \tag{32.139}$$

Now Taylor-expand the loop function about zero internal mass, $f(x_\alpha) = f(0) + f'(0)\, x_\alpha + \tfrac{1}{2} f''(0)\, x_\alpha^2 + \ldots$, and insert it into the sum:

$$\sum_\alpha V_{\alpha i} V_{\alpha j}^*\, f(x_\alpha) = f(0) \sum_\alpha V_{\alpha i} V_{\alpha j}^* + f'(0) \sum_\alpha V_{\alpha i} V_{\alpha j}^*\, x_\alpha + \tfrac{1}{2} f''(0) \sum_\alpha V_{\alpha i} V_{\alpha j}^*\, x_\alpha^2 + \ldots$$

$$= \underbrace{f(0)\,\delta_{ij}}_{=0\ (i\neq j)} + \frac{f'(0)}{M_W^2}\sum_\alpha V_{\alpha i}V_{\alpha j}^*\,m_\alpha^2 + O\!\left(\frac{m^4}{M_W^4}\right). \tag{32.140}$$

The constant term dies by unitarity Eq. (32.139): the degenerate-mass piece of the amplitude, the part that knows nothing about which quark ran in the loop, is exactly cancelled in the sum over the three internal flavors. It is in this precise sense that the three generations conspire. The first surviving term is the linear one. Moreover its value is manifestly a mass *difference*: because $\sum_\alpha V_{\alpha i}V_{\alpha j}^* = 0$ for $i \neq j$, we may subtract any flavor-independent constant from m_α^2 without changing the sum, in particular the lightest internal mass m_u^2,

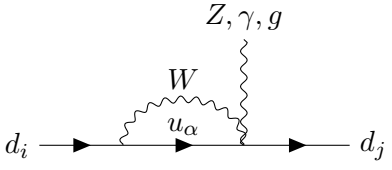
$$\sum_\alpha V_{\alpha i}V_{\alpha j}^*\,m_\alpha^2 = \sum_\alpha V_{\alpha i}V_{\alpha j}^*\,(m_\alpha^2 - m_u^2), \tag{32.141}$$

which displays the structure $\propto (m_\alpha^2 - m_{\alpha'}^2)$ asserted in the theorem. Substituting Eq. (32.141) into Eq. (32.140) gives Eq. (32.137). □

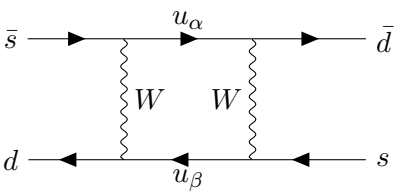
Two consequences follow immediately and set the entire phenomenology of the next section. First, the *magnitude*: an FCNC amplitude is suppressed not merely by the loop factor $g^2/16\pi^2$ and by small CKM elements, but additionally by the GIM factor $(m_\alpha^2 - m_{\alpha'}^2)/M_W^2$. For down-type external transitions the largest splitting is set by the top, $x_t = m_t^2/M_W^2 \sim 4.7$, so for $b \to s$ and $b \to d$ the top dominates and the GIM suppression is mild; for $s \to d$ in the kaon system the would-be top contribution is throttled by the tiny CKM combination $V_{td}V_{ts}^*$, and the numerically leading term is the charm, suppressed by $x_c = m_c^2/M_W^2 \sim 3 \times 10^{-4}$—a genuinely small number, and exactly the one that makes kaon FCNC so rare. Second, the *phase*: the surviving sum is a weighted combination of CKM products $V_{\alpha i}V_{\alpha j}^*$, which are complex. Their imaginary parts are where CP violation enters the loop, and—as the next worked example shows—when two such products are multiplied together in a box diagram, the imaginary part is precisely a Jarlskog quartet.

32.5.3 The box diagram and the double-GIM structure of neutral-meson mixing

The penguin changes flavor by one unit; neutral-meson mixing, $K^0 \leftrightarrow \bar{K}^0$ or $B^0 \leftrightarrow \bar{B}^0$, changes it by two and so requires *two* charged-current vertices on each fermion line—the box of Fig. 32.5(b). The box is the workhorse of the next section, where it generates the mass splitting and the CP-violating parameters of the neutral mesons. Here we assemble its flavor structure to display the GIM mechanism acting *twice* and to see, explicitly, the Jarlskog quartets that carry CP violation into the mixing.



(a) The one-loop FCNC "penguin": $d_i \to d_j$ via an internal up-type quark u_α and a W, emitting a neutral boson. Carries $V_{\alpha i} V_{\alpha j}^* \, f(x_\alpha)$.



(b) The $\Delta S = 2$ box for K^0–$\bar{K}^0$ mixing, with two internal up-type quarks u_α, u_β. Carries the double sum $\lambda_\alpha \lambda_\beta \, S_0(x_\alpha, x_\beta)$.

Figure 32.5: The two loop topologies through which the charged current induces a neutral flavor transition. Both are GIM-suppressed: the flavor-independent part of the internal sum cancels by CKM unitarity, leaving the result controlled by internal mass splittings.

Worked Example 32.30 (Flavor structure of the $\Delta S = 2$ box and the appearance of the Jarlskog quartet)**.** Consider the $K^0 = (\bar{s}d) \to \bar{K}^0 = (s\bar{d})$ transition. Reading the box of Fig. 32.5(b), each of the two internal up-type lines is summed over $\alpha, \beta \in \{u, c, t\}$, and each of the four charged-current vertices supplies a CKM factor. The two vertices on the $\bar{s} \to \bar{d}$ side give $V_{\alpha s}^*$ and $V_{\beta d}$ (read with the appropriate conjugation for the antiquark line), and the two on the $d \to s$ side give $V_{\alpha d}$ and $V_{\beta s}^*$; collecting the factors that go with each internal line, define the shorthand

$$\lambda_\alpha \equiv V_{\alpha d}^* \, V_{\alpha s}, \qquad \alpha = u, c, t, \tag{32.142}$$

so that the box amplitude has the double-sum flavor structure

$$\mathcal{A}_{\Delta S=2} = \frac{G_F^2 M_W^2}{4\pi^2} \sum_{\alpha,\beta=u,c,t} \lambda_\alpha \, \lambda_\beta \; S_0(x_\alpha, x_\beta) \;\times \langle \bar{K}^0 | \, (\bar{s}d)_{V-A} (\bar{s}d)_{V-A} \, | K^0 \rangle, \tag{32.143}$$

where $S_0(x_\alpha, x_\beta)$ is the dimensionless loop function obtained from the box integral (the Inami–Lim function, displayed below), the prefactor follows from $g^2/8M_W^2 = G_F/\sqrt{2}$, and the last factor is the hadronic matrix element of the local four-quark operator into which the heavy W's have been contracted. The flavor physics is entirely in the double sum.

Double GIM. The function S_0 is regular at zero internal mass, so write $S_0(x_\alpha, x_\beta) = S_0(0,0) + [\text{terms vanishing as some } x \to 0]$. The flavor-blind constant $S_0(0,0)$ multiplies

$$\sum_{\alpha,\beta} \lambda_\alpha \lambda_\beta \, S_0(0,0) = S_0(0,0) \Big(\sum_\alpha \lambda_\alpha \Big)^2 = S_0(0,0) \, \big[(V^\dagger V)_{ds} \big]^2 = 0, \tag{32.144}$$

because $\sum_\alpha \lambda_\alpha = \sum_\alpha V_{\alpha d}^* V_{\alpha s} = \delta_{ds} = 0$ is exactly the unitarity relation Eq. (32.138). The GIM cancellation now operates on *both* internal lines—this

is the "double GIM" that makes neutral-meson mixing even more suppressed than a single-penguin FCNC. After using $\sum_\alpha \lambda_\alpha = 0$ to eliminate the u-quark contribution in favor of the heavier flavors (subtracting the $\alpha = u$ or $\beta = u$ pieces, which are $\propto S_0(0, x_\beta)$ and cancel against the constant), the surviving sum runs effectively over $\alpha, \beta \in \{c, t\}$ and is weighted by the differences $S_0(x_\alpha, x_\beta) - S_0(0, \cdot)$, each carrying at least one power of x_c or x_t.

Where CP violation enters. Expand the surviving double sum into its distinct flavor products,

$$\mathcal{A}_{\Delta S=2} \;\propto\; \lambda_c^2\, S_0(x_c) + \lambda_t^2\, S_0(x_t) + 2\,\lambda_c\lambda_t\, S_0(x_c, x_t), \tag{32.145}$$

with $S_0(x) \equiv S_0(x, x)$. The mixing observable is governed by this complex number; the dispersive part of the mass matrix is its real part and the CP-violating parameter by its imaginary part. The single-flavor terms λ_c^2 and λ_t^2 are squares of one CKM product, while the cross term carries

$$\lambda_c\lambda_t = (V_{cd}^* V_{cs})(V_{td}^* V_{ts}), \qquad \mathrm{Im}(\lambda_c\lambda_t) = \mathrm{Im}\big[V_{cd}^* V_{cs} V_{td}^* V_{ts}\big], \tag{32.146}$$

which is precisely a Jarlskog quartet—the rephasing-invariant combination of four CKM elements built in Section 32.4, here equal up to sign to the Jarlskog invariant J (Eq. (32.114)). The CP-violating content of K^0–$\bar{K}^0$ mixing is thus inseparable from the same single phase that defines J: no phase, no imaginary part of $\lambda_c\lambda_t$, no ε_K. This is the analytic bridge between the abstract invariant of the last section and the measured CP violation of the next one.

For completeness we record the loop function that the box integral produces. The diagonal Inami–Lim function, the result of contracting the two W propagators and the two internal quark propagators and performing the (standard, one-loop) momentum integral by the methods of Chapter 19, is

$$S_0(x) = x\left[\frac{1}{4} + \frac{9}{4(1-x)} - \frac{3}{2(1-x)^2}\right] - \frac{3}{2}\left(\frac{x}{1-x}\right)^3 \ln x, \tag{32.147}$$

with the off-diagonal $S_0(x_\alpha, x_\beta)$ given by the analogous expression with two distinct masses [274]. We do not reproduce the full master-integral evaluation here—it belongs to the operator-product/effective-Hamiltonian machinery that systematizes the matching of the box onto the local $\Delta S = 2$ operator—but the structure that matters for the present argument is visible by inspection: as $x \to 0$,

$$S_0(x) = x + O(x^2 \ln x), \tag{32.148}$$

so S_0 vanishes linearly with the internal mass-squared ratio, reproducing the m_α^2/M_W^2 GIM suppression of Theorem 32.29 term by term. The whole loop function is, in this sense, a concrete realization of the unitarity cancellation: its value at zero argument is zero, and that is why the flavor-blind piece of the amplitude vanishes.

32.5.4 Parametric estimates: $K_L \to \mu^+\mu^-$ and the kaon mass difference

We now carry the GIM structure through to numbers, at the level of magnitudes, for the two processes that historically forced the issue.

The neutral-current decay $K_L \to \mu^+\mu^-$. This is the cleanest test of GIM, because it is a pure short-distance FCNC: the $s\bar{d}$ content of the kaon annihilates into a muon pair through the Z-penguin and box of Fig. 32.5, an $s \to d$ neutral transition forbidden at tree level by Theorem 32.28. The amplitude has the single-GIM form Eq. (32.136),

$$\mathcal{A}(K_L \to \mu^+\mu^-)_{\rm SD} \;\propto\; \frac{G_F\,\alpha}{\pi\,\sin^2\theta_{\rm W}} \sum_{\alpha=u,c,t} V_{\alpha s}^{*} V_{\alpha d}\, Y(x_\alpha), \tag{32.149}$$

where $Y(x)$ is the relevant Inami–Lim function with $Y(x) \to \frac{x}{8}(\dots) \sim x$ for small x, so once again the u-quark piece cancels against the constant by $\sum_\alpha V_{\alpha s}^{*} V_{\alpha d} = 0$ and the sum is dominated, for the kaon, by the charm with its $x_c = m_c^2/M_W^2$. The historical force of this expression is what it says *without* the charm. In a world with only the three quarks u, d, s known before 1970, there is no internal α to complete the unitarity sum: the GIM cancellation cannot occur, the loop function is evaluated at the unsuppressed scale, and the predicted rate for $K_L \to \mu^+\mu^-$ (and, equivalently, the strangeness-changing neutral current responsible for K^0–$\bar{K}^0$ mixing) overshoots the experimental bound by orders of magnitude. The measured branching ratio, dominated in fact by the long-distance two-photon process, is $\mathrm{Br}(K_L \to \mu^+\mu^-) \simeq 6.8 \times 10^{-9}$ [382]; the short-distance piece Eq. (32.149), with the charm GIM factor x_c, sits comfortably below it. The discrepancy in the charmless theory is removed by a single device: a fourth quark, the charm, whose contribution carries the opposite sign demanded by unitarity and cancels the leading term.

The K_L–K_S mass difference. The real part of the box amplitude Eq. (32.143) generates the K_L–K_S mass difference Δm_K. Keeping only the charm contribution (appropriate to the two-generation problem of 1970, and numerically the dominant piece in the full theory because the top piece is choked by $|V_{td}V_{ts}|$), and using the small-x form $S_0(x_c) \simeq x_c = m_c^2/M_W^2$ together with $G_F/\sqrt{2} = g^2/8M_W^2$, the box reduces to the celebrated Gaillard–Lee estimate [207],

$$\frac{\Delta m_K}{m_K} \;\simeq\; \frac{G_F^2\, f_K^2}{6\pi^2}\, \cos^2\theta_c\, \sin^2\theta_c\; m_c^2, \tag{32.150}$$

where $f_K \simeq 0.16$ GeV is the kaon decay constant (which parametrizes the hadronic matrix element $\langle \bar{K}^0 | (\bar{s}d)^2 | K^0 \rangle$) and θ_c is the Cabibbo angle, with

$\lambda_c = V_{cd}^* V_{cs} \simeq -\cos\theta_c \sin\theta_c$ in the two-generation limit. The dependence on m_c^2—not M_W^2, not m_s^2—is the GIM signature made quantitative: the mass difference is set by the *charm* mass, the heavy partner whose existence is the cause of the cancellation in the first place.

Worked Example 32.31 (The charm mass from Δm_K: how 1970 predicted a number)**.** Equation Eq. (32.150) can be read backwards: given the measured mass difference, it predicts the charm mass. Insert the numbers $G_F = 1.166 \times 10^{-5}$ GeV^{-2}, $f_K = 0.16$ GeV, $\sin\theta_c = 0.225$ (so $\cos\theta_c = 0.974$), and $m_K = 0.498$ GeV, and solve Eq. (32.150) for m_c. Carrying the arithmetic in steps,

$$\begin{aligned} G_F^2 &= 1.36 \times 10^{-10}\ \text{GeV}^{-4}, \qquad f_K^2 = 2.56 \times 10^{-2}\ \text{GeV}^2, \\ \cos^2\theta_c \sin^2\theta_c &= (0.949)(0.0506) = 4.81 \times 10^{-2}, \\ \frac{G_F^2 f_K^2}{6\pi^2} \cos^2\theta_c \sin^2\theta_c &= \frac{(1.36 \times 10^{-10})(2.56 \times 10^{-2})(4.81 \times 10^{-2})}{59.2} = 2.83 \times 10^{-15}\ \text{GeV}^{-2}. \end{aligned} \tag{32.151}$$

The measured fractional splitting is $\Delta m_K / m_K = (3.48 \times 10^{-15}\ \text{GeV})/(0.498\ \text{GeV}) = 6.99 \times 10^{-15}$, so

$$m_c^2 \simeq \frac{\Delta m_K / m_K}{2.83 \times 10^{-15}\ \text{GeV}^{-2}} = \frac{6.99 \times 10^{-15}}{2.83 \times 10^{-15}\ \text{GeV}^{-2}} = 2.47\ \text{GeV}^2, \qquad m_c \simeq 1.6\ \text{GeV}. \tag{32.152}$$

The crude one-loop box, fed the measured kaon mass difference and the Cabibbo angle, returns a charm mass of order 1.5 GeV. This is the substance of the Gaillard–Lee prediction: *before* the charm quark was seen, its mass was bounded from above (a heavier charm would have overshot Δm_K) at a couple of GeV. When the J/ψ—a $c\bar{c}$ bound state of mass 3.097 GeV, implying $m_c \approx 1.5$ GeV—was discovered in November 1974, it landed squarely in the predicted window. The agreement between Eq. (32.152) and the modern $\overline{\text{MS}}$ value $m_c(m_c) \simeq 1.27$ GeV [382] is better than a one-loop, two-generation, $B_K = 1$ estimate has any right to be, and it is one of the cleaner quantitative triumphs of the gauge theory of weak interactions.

Historical Note

In 1970 Glashow, Iliopoulos, and Maiani confronted a sharp paradox [223]. The electroweak theory with the three known quarks u, d, s predicted strangeness-changing neutral currents—$K_L \to \mu^+\mu^-$ and K^0–$\bar{K}^0$ mixing—at a strength wildly above the stringent experimental bounds, because with the down and strange quarks alone there was nothing to complete the unitarity sum that would have cancelled them. Their resolution was to add a fourth quark, the charm, completing a second weak doublet (c, s') alongside (u, d') with the orthogonal Cabibbo-rotated combination,

so that the two doublets' contributions to any neutral transition cancel up to the small effect of the c–u mass difference. This is the mechanism now named for them. Four years later Gaillard and Lee [207] sharpened the qualitative cancellation into the quantitative bound Eq. (32.150): the residual mass difference fixes the charm mass, and the measured Δm_K requires m_c of order a GeV. The discovery of the J/ψ in 1974 confirmed both the existence and the mass. The episode is the template for the entire flavor program: an apparent crisis in a loop-induced neutral process, resolved by new states whose quantum numbers complete a unitarity sum, with their masses predicted in advance from the size of the residual effect.

With the GIM mechanism established—tree-level FCNC forbidden exactly by flavor-universality, loop-level FCNC suppressed by m_q^2/M_W^2 through CKM unitarity, and the box amplitude carrying the Jarlskog quartet that injects CP violation into mixing—we have in hand precisely the amplitudes that govern the real neutral-meson systems. The next section takes the box of Eq. (32.143), promotes it to the two-state effective Hamiltonian of K^0–$\bar{K}^0$ and B^0–$\bar{B}^0$ mixing, and reads its real and imaginary parts as the mass splitting and the CP-violating parameters, culminating in the unitarity triangle—the geometric statement that one phase must reproduce many independent measurements at once.

32.6 Neutral-Meson Mixing, the Unitarity Triangle, and CP Observables

The previous sections produced an abstract object: a unitary matrix $V_{\rm CKM}$ (Eq. (32.49)) carrying, for three generations, exactly one irreducible complex phase, repackaged into the rephasing-invariant Jarlskog determinant J (Theorem 32.20), with the theorem that CP is violated in the quark sector if and only if that phase is nonzero (Theorem 32.16). The GIM analysis then showed that this phase first appears physically in loop-induced flavor-changing neutral processes — in particular the $\Delta F = 2$ box amplitudes (Section 32.5.3) — because flavor-changing neutral currents are forbidden at tree level. The task of this section is to turn that one number into measured numbers. We do it in three movements. First we develop the two-state (Weisskopf–Wigner) formalism for a neutral meson and its antiparticle, the system in which a $\Delta F = 2$ amplitude is directly observable as a frequency of oscillation and in which the CKM phase first showed itself experimentally. Second, we classify the three logically distinct ways CP can be violated — in mixing, in decay, and in their interference — and tie each to a CKM phase combination. Third, we draw the unitarity triangle: the geometric statement that the six orthogonality relations of $V_{\rm CKM}V_{\rm CKM}^\dagger = \mathbf{1}$ are, for three generations, a single closed triangle whose area is $J/2$, and whose

apex $(\bar\rho, \bar\eta)$ is overdetermined by a dozen independent measurements. That overdetermination is the precision test of the Kobayashi–Maskawa mechanism, and we end by reporting honestly where it stands, and where it mildly strains, as of the mid-2020s.

32.6.1 The effective two-state Hamiltonian

Consider a neutral meson P^0 and its antiparticle $\bar P^0$ carrying opposite flavor quantum number: the kaons $K^0 = (d\bar s)$, $\bar K^0 = (\bar d s)$ with $\Delta S = \pm 1$; the B_d mesons $B^0 = (\bar b d)$, $\bar B^0 = (b\bar d)$ with $\Delta B = \pm 1$; likewise B_s and D^0. The strong and electromagnetic interactions conserve the flavor quantum number, so P^0 and $\bar P^0$ would be stable, degenerate, and unmixed were the weak interaction switched off. The weak interaction does two things to this two-level system: it lets each state decay, and — through the second-order $\Delta F = 2$ transition $P^0 \leftrightarrow \bar P^0$ whose amplitude is the box of Section 32.5.3 — it couples the two levels. Both effects are slow on the scale of the meson mass, so we may treat the $\{|P^0\rangle, |\bar P^0\rangle\}$ subspace by the same time-dependent perturbation theory used for an unstable state in ordinary quantum mechanics, the Weisskopf–Wigner approximation. The result is that the projection of the state onto the two-dimensional flavor subspace,

$$|\psi(t)\rangle = a(t)\,|P^0\rangle + b(t)\,|\bar P^0\rangle, \tag{32.153}$$

evolves under a 2×2 effective Hamiltonian that is *not* Hermitian, because probability leaks out of the subspace into the decay channels:

$$i\,\frac{d}{dt}\begin{pmatrix} a \\ b \end{pmatrix} = H \begin{pmatrix} a \\ b \end{pmatrix}, \qquad H \;=\; M - \tfrac{i}{2}\,\Gamma. \tag{32.154}$$

Here M (the mass matrix) and Γ (the decay matrix) are each separately 2×2 *Hermitian* matrices. Their meaning is read off from second-order perturbation theory in the weak Hamiltonian H_w: with $|i\rangle, |j\rangle \in \{|P^0\rangle, |\bar P^0\rangle\}$,

$$M_{ij} = m_0\,\delta_{ij} + \langle i|H_w|j\rangle + \mathrm{PV}\sum_n \frac{\langle i|H_w|n\rangle\,\langle n|H_w|j\rangle}{m_0 - E_n}, \tag{32.155}$$

$$\Gamma_{ij} = 2\pi \sum_n \langle i|H_w|n\rangle\,\langle n|H_w|j\rangle\;\delta(m_0 - E_n). \tag{32.156}$$

The sum runs over all intermediate states $|n\rangle$ the weak interaction connects to; PV is the principal value. The split is the standard dispersive/absorptive decomposition: M_{ij} collects the off-shell (virtual) intermediate states — this is where the heavy box with internal top quarks lives — while Γ_{ij} collects the on-shell ones (real, common decay channels of P^0 and $\bar P^0$). That M and Γ are Hermitian is manifest in Eqs. (32.155) and (32.156): swapping $i \leftrightarrow j$ conjugates the matrix element products. Hence

$$M_{21} = M_{12}^*, \qquad \Gamma_{21} = \Gamma_{12}^*. \tag{32.157}$$

The full H is not Hermitian precisely because $-\frac{i}{2}\Gamma$ has an anti-Hermitian piece; its eigenvalues are complex, their imaginary parts giving the finite lifetimes.

The role of CPT. The diagonal entries are constrained by the discrete symmetries of Chapter 11. CPT invariance — a theorem for any local, Lorentz-invariant quantum field theory — relates a state to its antiparticle and forces the diagonal elements of H to be equal:

$$H_{11} = H_{22}, \qquad \text{i.e.} \qquad M_{11} = M_{22} \equiv M, \quad \Gamma_{11} = \Gamma_{22} \equiv \Gamma. \tag{32.158}$$

The argument is short. Under the antiunitary CPT operator Θ, with a phase convention $\Theta\,|P^0\rangle = \left|\bar{P}^0\right\rangle$, the diagonal element $H_{11} = \langle P^0|H|P^0\rangle$ maps to $\left\langle \bar{P}^0\middle|H\middle|\bar{P}^0\right\rangle = H_{22}$ provided H commutes with Θ (and using that Θ is antiunitary, which conjugates the matrix element back). Since the strong, electromagnetic, and weak interactions are all CPT-symmetric, $H_{11} = H_{22}$ exactly. We will assume CPT throughout — it is the most stringently tested of the discrete symmetries — so the only inequivalence between the two flavor states sits in the *off-diagonal* elements H_{12} and H_{21}.

32.6.2 Mass and width eigenstates; the mixing parameter

With Eq. (32.158) the effective Hamiltonian reads

$$H = \begin{pmatrix} M - \frac{i}{2}\Gamma & H_{12} \\ H_{21} & M - \frac{i}{2}\Gamma \end{pmatrix}, \qquad H_{12} = M_{12} - \tfrac{i}{2}\Gamma_{12}, \quad H_{21} = M_{12}^* - \tfrac{i}{2}\Gamma_{12}^*. \tag{32.159}$$

Note carefully that $H_{21} \neq H_{12}^*$ in general: Hermiticity of M and Γ separately, Eq. (32.157), gives $H_{21} = M_{12}^* - \frac{i}{2}\Gamma_{12}^*$, whereas $H_{12}^* = M_{12}^* + \frac{i}{2}\Gamma_{12}^*$. The two agree only if M_{12} and Γ_{12} share a common phase. This single fact — that H_{12} and H_{21} can have unequal moduli — is the seed of CP violation in mixing, as we will see.

Because H has equal diagonal entries, its eigenvectors are fixed by the off-diagonal pair alone. Write the eigenstates as

$$|P_{H,L}\rangle = p\left|P^0\right\rangle \pm q\left|\bar{P}^0\right\rangle, \qquad |p|^2 + |q|^2 = 1, \tag{32.160}$$

with the upper sign for the state we will call heavy (H) and the lower for light (L). Acting with Eq. (32.159) on $(p, \pm q)^{\mathsf{T}}$ and demanding proportionality gives, from the two rows,

$$\mu_{H,L} = \left(M - \tfrac{i}{2}\Gamma\right) \pm \frac{q}{p}\,H_{12} = \left(M - \tfrac{i}{2}\Gamma\right) \pm \frac{p}{q}\,H_{21}, \tag{32.161}$$

and consistency of the two expressions forces

$$\boxed{\frac{q}{p} = \sqrt{\frac{H_{21}}{H_{12}}} = \sqrt{\frac{M_{12}^* - \frac{i}{2}\Gamma_{12}^*}{M_{12} - \frac{i}{2}\Gamma_{12}}}\,, \qquad \mu_{H,L} = \Big(M - \tfrac{i}{2}\Gamma\Big) \pm \sqrt{H_{12}H_{21}}\,.} \tag{32.162}$$

The complex eigenvalues $\mu_{H,L} = m_{H,L} - \frac{i}{2}\Gamma_{H,L}$ encode the physical masses and widths of the two propagating states. Their differences are the two observables of mixing,

$$\Delta m \equiv m_H - m_L, \qquad \Delta\Gamma \equiv \Gamma_L - \Gamma_H, \qquad \Delta\mu \equiv \mu_H - \mu_L = \Delta m + \tfrac{i}{2}\,\Delta\Gamma = 2\sqrt{H_{12}H_{21}}. \tag{32.163}$$

(The sign conventions $\Delta m > 0$ by definition and $\Delta\Gamma = \Gamma_L - \Gamma_H$ follow the kaon tradition where the short-lived state is also the lighter one to a good approximation; the labels are pure bookkeeping.) Squaring the modulus and phase of Eq. (32.163) relates $(\Delta m, \Delta\Gamma)$ to the moduli and phases of M_{12} and Γ_{12}; in the common limit $|\Gamma_{12}| \ll |M_{12}|$ (the B systems) one finds simply

$$\Delta m \simeq 2\,|M_{12}|, \qquad \Delta\Gamma \simeq 2\,|\Gamma_{12}|\cos\phi_{12}, \qquad \phi_{12} \equiv \arg\Big(-\frac{M_{12}}{\Gamma_{12}}\Big). \tag{32.164}$$

Oscillations. The flavor states are superpositions of the eigenstates, which evolve with the simple exponential $|P_{H,L}(t)\rangle = e^{-i\mu_{H,L}t}\,|P_{H,L}\rangle$. Inverting Eq. (32.160) and propagating, a state born as $|P^0\rangle$ at $t = 0$ becomes

$$\big|P^0(t)\big\rangle = g_+(t)\,\big|P^0\big\rangle + \frac{q}{p}\,g_-(t)\,\big|\bar P^0\big\rangle\,, \qquad g_\pm(t) = \tfrac{1}{2}\Big(e^{-i\mu_H t} \pm e^{-i\mu_L t}\Big). \tag{32.165}$$

The probability of finding a $\bar P^0$ at time t is $|q/p|^2\,|g_-(t)|^2$, and

$$|g_\mp(t)|^2 = \tfrac{1}{4}e^{-\Gamma t}\Big[\cosh\tfrac{\Delta\Gamma}{2}t \mp \cos(\Delta m\, t)\Big], \tag{32.166}$$

an oscillation at the angular frequency Δm damped by the average width $\Gamma = \frac{1}{2}(\Gamma_H + \Gamma_L)$. The mass splitting is thus literally a measured frequency. It is conventional to quote the dimensionless ratios $x \equiv \Delta m/\Gamma$ and $y \equiv \Delta\Gamma/2\Gamma$. Empirically (mid-2020s [382]) $x_d \simeq 0.77$ for B_d (oscillation and decay comparably fast, $y_d \approx 0$), $x_s \simeq 27$ for B_s (many oscillations per lifetime), and $x_K \simeq y_K \simeq 1$ for kaons (mixing and the width difference both maximal, which is why the two kaon eigenstates differ so dramatically in lifetime).

What feeds M_{12}. The dispersive off-diagonal element M_{12} is dominated by the short-distance $\Delta F = 2$ box (Section 32.5.3), with internal up-type quarks. By GIM the box is governed by the heaviest accessible internal quark, so for the B_d system the top loop dominates and

$$M_{12}^{(B_d)} = \frac{G_F^2 M_W^2}{12\pi^2}\,\eta_B\, m_B\, f_B^2 \hat B_B\, S_0(x_t)\,\big(V_{tb}^* V_{td}\big)^2, \qquad x_t \equiv \frac{m_t^2}{M_W^2}, \tag{32.167}$$

where S_0 is the Inami–Lim loop function from the box (computed structurally in Section 32.5.3), η_B a short-distance QCD correction, and $f_B^2\hat{B}_B$ the hadronic matrix element of the local four-quark operator. The CKM dependence is the crucial part: $M_{12} \propto (V_{tb}^* V_{td})^2$, so

$$\arg M_{12}^{(B_d)} = 2\arg(V_{tb}^* V_{td}) + (\text{phase-convention pieces}), \tag{32.168}$$

and the mixing phase carries a CKM angle. We will see in Section 32.6.4 that this is exactly the angle β of the unitarity triangle. The magnitude fixes $\Delta m_d = 2|M_{12}^{(B_d)}| \propto |V_{td}|^2$, and the ratio $\Delta m_d/\Delta m_s \propto |V_{td}/V_{ts}|^2$ (with the hadronic uncertainties largely cancelling in the ratio) is one of the cleanest measurements of $|V_{td}|$.

32.6.3 CP eigenstates and the smallness of ϵ

Specialize now to the kaons, the system in which CP violation was discovered. Fix the convention $\mathsf{CP}\,|K^0\rangle = \left|\bar{K}^0\right\rangle$ (a choice of relative phase; the physics is rephasing-invariant, as we check below). The two CP eigenstates are

$$|K_+\rangle = \tfrac{1}{\sqrt{2}}\left(\left|K^0\right\rangle + \left|\bar{K}^0\right\rangle\right), \quad \mathsf{CP}\,|K_+\rangle = +|K_+\rangle\,; \qquad |K_-\rangle = \tfrac{1}{\sqrt{2}}\left(\left|K^0\right\rangle - \left|\bar{K}^0\right\rangle\right), \quad \mathsf{CP}\,|K_-\rangle \tag{32.169}$$

The dominant kaon decays are flavor-blind and CP-classified: the 2π state ($\pi^+\pi^-$ or $\pi^0\pi^0$ in a relative s-wave) has $\mathsf{CP} = +1$, while the 3π state has $\mathsf{CP} = -1$. If CP were exact, the mass eigenstates would coincide with the CP eigenstates: K_+ could decay to 2π (large phase space, short lifetime) and K_- only to 3π (small phase space, long lifetime). This is the celebrated K_S/K_L dichotomy, and the factor-600 lifetime ratio is its fingerprint. The decisive observation of Christenson, Cronin, Fitch and Turlay in 1964 [124] was that the *long*-lived kaon does decay to 2π, at the level of $\sim 2\times10^{-3}$ in amplitude — the long-lived state is therefore not a pure CP eigenstate. CP is violated.

Quantitatively, write the true mass eigenstates as in Eq. (32.160), $|K_S\rangle = p\,|K^0\rangle + q\left|\bar{K}^0\right\rangle$, $|K_L\rangle = p\,|K^0\rangle - q\left|\bar{K}^0\right\rangle$. Re-expanding in the CP basis Eq. (32.169),

$$|K_S\rangle \propto |K_+\rangle + \epsilon\,|K_-\rangle\,, \qquad |K_L\rangle \propto |K_-\rangle + \epsilon\,|K_+\rangle\,, \qquad \epsilon \equiv \frac{p-q}{p+q} = \frac{1-q/p}{1+q/p}, \tag{32.170}$$

where the common value of the two admixtures in CPT-conserving limit is the single parameter ϵ (their difference would measure CPT violation and is consistent with zero). A nonzero ϵ means $|q/p| \neq 1$ or q/p complex: CP violation lives entirely in the deviation of q/p from unity.

Why ϵ is small. The smallness of the effect, $|\epsilon| \simeq 2.2\times10^{-3}$, is not an accident of the kaon but a structural consequence of CP being a nearly-good symmetry

of the mixing. From Eq. (32.162), $q/p = \sqrt{H_{21}/H_{12}}$, and $H_{21} \to H_{12}$ as the CP-violating phase between M_{12} and Γ_{12} goes to zero. Write $H_{21} = H_{12}(1+\delta)$ with δ small. Then $q/p = \sqrt{1+\delta} \simeq 1 + \tfrac{1}{2}\delta$ and

$$\epsilon = \frac{1 - q/p}{1 + q/p} \simeq -\frac{\delta}{4} = -\frac{H_{21} - H_{12}}{4\,H_{12}}. \tag{32.171}$$

The numerator is the entire CP violation:

$$H_{21} - H_{12} = (M_{12}^* - \tfrac{i}{2}\Gamma_{12}^*) - (M_{12} - \tfrac{i}{2}\Gamma_{12}) = -2i\,\mathrm{Im}\,M_{12} - \mathrm{Im}\,\Gamma_{12} = -2i\big(\mathrm{Im}\,M_{12} - \tfrac{i}{2}\,\mathrm{Im}\,\Gamma_{12}\big), \tag{32.172}$$

using only that $z - z^* = 2i\,\mathrm{Im}\,z$. Inserting Eq. (32.172) into Eq. (32.171) and using $2H_{12} \simeq \Delta\mu = \Delta m + \tfrac{i}{2}\Delta\Gamma$ from Eq. (32.163) (valid to leading order in the small phase),

$$\boxed{\epsilon \simeq \frac{i\,\mathrm{Im}\,M_{12} + \tfrac{1}{2}\,\mathrm{Im}\,\Gamma_{12}}{\Delta m + \tfrac{i}{2}\,\Delta\Gamma}\,.} \tag{32.173}$$

The point is now transparent. The numerator is the imaginary part of the off-diagonal elements — the CP-odd content — while the denominator is the CP-even mass and width splitting, which is large and entirely CP-conserving. ϵ is the ratio of a CP-violating amplitude to a CP-conserving one, and is therefore small whenever the phases are small. For kaons the empirical near coincidence $\Delta m_K \simeq \tfrac{1}{2}\Delta\Gamma_K$ (because $\Gamma_S \gg \Gamma_L$, so $\Delta\Gamma_K \simeq +\Gamma_S \simeq +2\Delta m_K$) makes the denominator $\Delta m_K(1+i) = \sqrt{2}\,\Delta m_K\, e^{+i\pi/4}$, so ϵ acquires the famous *superweak phase*

$$\phi_{\mathrm{SW}} = \arctan\frac{2\Delta m_K}{\Delta\Gamma_K} \simeq 43.5^\circ, \tag{32.174}$$

a phase fixed by measured mass and width differences alone, in striking agreement with the measured $\arg\epsilon \simeq 43.5^\circ$. The magnitude $|\epsilon_K| = (2.228 \pm 0.011) \times 10^{-3}$ [382] is reproduced by Eq. (32.173) with the box value of $\mathrm{Im}\,M_{12}$ from Eq. (32.167) carrying the CKM phase; because $M_{12}^{(K)}$ scales as $|V_{cb}|^4$ through the dominant charm–top interference, ϵ_K is one of the constraints whose interpretation is sensitive to $|V_{cb}|$, a point we return to in the overconstraint discussion.

32.6.4 The three faces of CP violation

The mixing analysis isolated one of three logically independent ways CP can be broken. It is worth stating all three cleanly, because each is tied to a different CKM phase combination and each is measured by a different class of experiment. Let $A_f \equiv \langle f|H_w|P^0\rangle$ and $\bar{A}_{\bar{f}} \equiv \langle \bar{f}|H_w|\bar{P}^0\rangle$ be the decay amplitudes of P^0 and its CP conjugate.

(i) CP violation in mixing (indirect). This is $|q/p| \neq 1$: the rate of $P^0 \to \bar{P}^0$ differs from $\bar{P}^0 \to P^0$. Equivalently $\mathrm{Im}(M_{12}\Gamma_{12}^*) \neq 0$, i.e. M_{12} and Γ_{12}

carry different phases. Indeed from Eq. (32.162),

$$\left|\frac{q}{p}\right|^4 = \frac{|H_{21}|^2}{|H_{12}|^2} = 1 + \frac{2\,\mathrm{Im}(M_{12}\Gamma_{12}^*)}{|H_{12}|^2}, \tag{32.175}$$

so $|q/p| = 1 \iff \mathrm{Im}(M_{12}\Gamma_{12}^*) = 0$. This is the dominant content of the kaon ϵ above. It requires two interfering $\Delta F = 2$ contributions (dispersive vs. absorptive) with a relative phase.

(ii) CP violation in decay (direct). This is $|\bar{A}_{\bar{f}}/A_f| \neq 1$. It is the only kind possible for charged mesons (no mixing), and it requires at least two amplitudes contributing to the same final state with *both* different weak (CKM) phases ϕ_i *and* different strong (rescattering) phases δ_i. Writing $A_f = \sum_i |a_i| e^{i(\delta_i+\phi_i)}$ and $\bar{A}_{\bar{f}} = \sum_i |a_i| e^{i(\delta_i-\phi_i)}$ (the weak phase flips under CP, the strong phase does not),

$$|A_f|^2 - |\bar{A}_{\bar{f}}|^2 = -4\sum_{i<j} |a_i||a_j| \sin(\delta_i - \delta_j)\sin(\phi_i - \phi_j), \tag{32.176}$$

which is nonzero only if both sines are. In the kaon system this is the parameter ϵ', and the measured $\mathrm{Re}(\epsilon'/\epsilon) \sim 1.6 \times 10^{-3}$ established for the first time that CP violation is not purely a property of the mass matrix — it occurs in the decay amplitudes too, exactly as the CKM mechanism predicts and a "superweak" $\Delta S = 2$ interaction would not.

(iii) CP violation in interference of mixing and decay. Even when $|q/p| = 1$ and $|\bar{A}/A| = 1$, CP can be violated through the interference of a decay $P^0 \to f$ with the mixing-then-decay path $P^0 \to \bar{P}^0 \to f$, when f is accessible to both P^0 and $\bar{P}^0$ (e.g. a CP eigenstate). The clean observable is

$$\lambda_f \equiv \frac{q}{p}\frac{\bar{A}_f}{A_f}, \qquad \text{CP violated in interference} \iff \mathrm{Im}\,\lambda_f \neq 0. \tag{32.177}$$

The time-dependent decay-rate asymmetry to a CP eigenstate then takes the form

$$a_f(t) = \frac{\Gamma(\bar{P}^0(t) \to f) - \Gamma(P^0(t) \to f)}{\Gamma(\bar{P}^0(t) \to f) + \Gamma(P^0(t) \to f)} = S_f \sin(\Delta m\, t) - C_f \cos(\Delta m\, t), \tag{32.178}$$

with $S_f = \dfrac{2\,\mathrm{Im}\,\lambda_f}{1+|\lambda_f|^2}$, $C_f = \dfrac{1-|\lambda_f|^2}{1+|\lambda_f|^2}$. This is the type of CP violation that the B factories were built to measure, and it is the cleanest of the three because for the golden mode it is theoretically pristine.

The golden mode and $\sin 2\beta$. For $B_d \to J/\psi\, K_S$ (quark-level $b \to c\bar{c}s$), there is a single dominant weak amplitude, so $|\bar{A}/A| = 1$ to excellent accuracy and $C_f \approx 0$: this mode measures a pure phase. Collect the phases. The mixing factor is, from Eq. (32.167) with $|\Gamma_{12}| \ll |M_{12}|$,

$$\frac{q}{p} = \sqrt{\frac{M_{12}^*}{M_{12}}} = \frac{V_{tb}V_{td}^*}{V_{tb}^*V_{td}}. \tag{32.179}$$

The decay amplitude carries $V_{cb}V_{cs}^*$ and its conjugate, and the K^0 in the final state must itself be projected onto K_S, contributing the kaon-mixing factor $V_{cd}V_{cs}^*/(V_{cd}^*V_{cs})$. Assembling,

$$\lambda_{\psi K_S} = \frac{q}{p}\frac{\bar{A}}{A}\frac{V_{cd}V_{cs}^*}{V_{cd}^*V_{cs}}\bigg|_{K\text{-mix}} = -\frac{V_{tb}V_{td}^*}{V_{tb}^*V_{td}}\frac{V_{cb}^*V_{cs}}{V_{cb}V_{cs}^*}\frac{V_{cd}V_{cs}^*}{V_{cd}^*V_{cs}} = -e^{-2i\beta}, \tag{32.180}$$

where the leading minus sign is the intrinsic CP of the $J/\psi K_S$ state and

$$\beta \equiv \arg\Big(-\frac{V_{cd}V_{cb}^*}{V_{td}V_{tb}^*}\Big) \tag{32.181}$$

is the rephasing-invariant combination that we will recognize as an angle of the unitarity triangle. (One checks directly that every phase-convention factor $V_{xy} \to e^{i\phi}V_{xy}$ cancels in Eq. (32.180), as it must for an observable.) Therefore

$$S_{\psi K_S} = \mathrm{Im}\,\lambda_{\psi K_S} = \sin 2\beta, \qquad C_{\psi K_S} \approx 0, \tag{32.182}$$

and the coefficient of $\sin(\Delta m_d\, t)$ in the measured asymmetry *is* $\sin 2\beta$. The world average is $\sin 2\beta = 0.699 \pm 0.017$ [382] — the first $O(1)$ CP-violating observable ever measured, and the one that confirmed the CKM phase is large, not "superweak."

32.6.5 The unitarity triangle

We now make geometric the statement that all of this CP physics is controlled by a single phase. Unitarity of $V_{\rm CKM}$ (Eq. (32.49)) is six complex equations beyond the diagonal: $(V_{\rm CKM}V_{\rm CKM}^\dagger)_{ij} = \delta_{ij}$ supplies three off-diagonal vanishing relations among rows, and $(V_{\rm CKM}^\dagger V_{\rm CKM})_{ij} = \delta_{ij}$ three among columns,

$$\sum_{k=u,c,t} V_{ki}V_{kj}^* = 0 \quad (i \neq j \in \{d,s,b\}), \qquad \sum_{k=d,s,b} V_{ik}V_{jk}^* = 0 \quad (i \neq j \in \{u,c,t\}). \tag{32.183}$$

Each relation is a sum of three complex numbers equal to zero — the three numbers, drawn head-to-tail in the complex plane, close into a triangle. The six triangles are not independent (they share sides and, as we prove next, areas), but they are geometrically distinct. Most of them are extremely "squashed": in

the *ds* relation, for instance, $V_{ud}V_{us}^* \sim \lambda$ and $V_{cd}V_{cs}^* \sim \lambda$ dwarf $V_{td}V_{ts}^* \sim \lambda^5$, so one side is a thousand times shorter than the others and the enclosed angles are tiny. Two relations are special. The *db* and *ut* triangles have all three sides of the same order, $O(\lambda^3)$:

$$\boxed{V_{ud}V_{ub}^* + V_{cd}V_{cb}^* + V_{td}V_{tb}^* = 0} \qquad \text{(the "unitarity triangle").} \tag{32.184}$$

This is *the* unitarity triangle. Its three comparable sides mean its angles are $O(1)$, which is precisely why the B_d system is where large CP-violating asymmetries appear.

The area is $J/2$. The connection to the Jarlskog invariant is exact and geometric. Label the three sides of Eq. (32.184)

$$z_1 = V_{ud}V_{ub}^*, \qquad z_2 = V_{cd}V_{cb}^*, \qquad z_3 = V_{td}V_{tb}^*, \qquad z_1 + z_2 + z_3 = 0. \tag{32.185}$$

Place vertices at $P_0 = 0$, $P_1 = z_1$, $P_2 = z_1 + z_2 = -z_3$. The (signed) area of a triangle with vertices at complex points is $\frac{1}{2}\,\mathrm{Im}[(P_1 - P_0)^*(P_2 - P_0)]$, so

$$\text{Area} = \tfrac{1}{2}\,|\,\mathrm{Im}(z_1^*(z_1 + z_2))\,| = \tfrac{1}{2}\,|\,\mathrm{Im}(z_1^* z_2)|, \tag{32.186}$$

because $\mathrm{Im}(z_1^* z_1) = 0$. Now

$$z_1^* z_2 = (V_{ud}V_{ub}^*)^*(V_{cd}V_{cb}^*) = V_{ud}^* V_{ub}\, V_{cd}V_{cb}^*, \qquad \mathrm{Im}(z_1^* z_2) = \mathrm{Im}(V_{ud}^* V_{ub} V_{cd} V_{cb}^*) = \pm J, \tag{32.187}$$

which is exactly the rephasing-invariant Jarlskog combination (Theorem 32.20). Hence

$$\boxed{\text{Area of the unitarity triangle} = \frac{|J|}{2}\,.} \tag{32.188}$$

The same calculation with any of the six relations of Eq. (32.183) returns the same $\frac{1}{2}|\,\mathrm{Im}(\text{side}^* \cdot \text{side})|$, and the antisymmetry property $\mathrm{Im}(V_{ij}V_{kl}V_{il}^*V_{kj}^*) = J\sum_{m,n}\varepsilon_{ikm}\varepsilon_{jln}$ that defines J guarantees the magnitude is the same for all of them: *all six unitarity triangles have equal area* $|J|/2$. This is the sharp geometric form of the central result of the chapter — there is one CP phase, and its rephasing-invariant measure J is the common area of every triangle. A nonzero area is CP violation; a degenerate (collapsed) triangle is CP conservation. This restates Theorem 32.16 as a picture.

Normalization, apex, and angles. To compare with data one normalizes Eq. (32.184) by the (well-measured, nearly real) middle side $V_{cd}V_{cb}^*$:

$$\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*} + 1 + \frac{V_{td}V_{tb}^*}{V_{cd}V_{cb}^*} = 0. \tag{32.189}$$

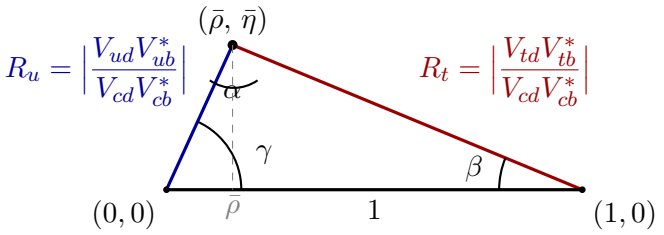


Figure 32.6: The unitarity triangle of Eq. (32.184), normalized by dividing through by $V_{cd}V_{cb}^*$ so that the base runs from $(0,0)$ to $(1,0)$ and the apex sits at $(\bar\rho,\bar\eta)$ of Eq. (32.190). The two nontrivial sides are R_u (fixed by $|V_{ub}/V_{cb}|$) and R_t (fixed by $|V_{td}/V_{ts}|$, i.e. by $\Delta m_d/\Delta m_s$). The angles satisfy $\alpha+\beta+\gamma=\pi$ and the area equals $|J|/2$ (Eq. (32.188)); a flat triangle ($\bar\eta=0$) is CP conservation.

The base now runs from 0 to 1, and the apex is the rephasing-invariant point

$$\bar\rho + i\,\bar\eta \;\equiv\; -\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*}, \tag{32.190}$$

which is the exact definition of $(\bar\rho,\bar\eta)$. The three interior angles are

$$\alpha = \arg\!\Big(-\frac{V_{td}V_{tb}^*}{V_{ud}V_{ub}^*}\Big), \quad \beta = \arg\!\Big(-\frac{V_{cd}V_{cb}^*}{V_{td}V_{tb}^*}\Big), \quad \gamma = \arg\!\Big(-\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*}\Big), \tag{32.191}$$

with $\alpha+\beta+\gamma=\pi$ automatically (the three arguments of three numbers summing to zero). Comparing Eq. (32.191) with Eq. (32.181) confirms that the β measured in $B_d\to J/\psi K_S$ is precisely the triangle angle at the $(1,0)$ vertex, and that $\gamma=\arg(\bar\rho+i\bar\eta)$ is the angle at the origin. Each CP-violating observable measures one angle; each CP-conserving modulus measures one side; and the KM mechanism asserts that they all describe the *same* triangle.

Worked Example 32.32 (Building the triangle from Wolfenstein variables; the apex and $\sin 2\beta$)**.** We construct the triangle explicitly in the Wolfenstein parametrization [526] and carry the trigonometry through to $\sin 2\beta$, verifying that a nonzero CP phase is what gives the triangle height.

Setup. The Wolfenstein form (Eq. (32.95)) reads, to the order we need,

$$V_{ud}\simeq 1-\tfrac{\lambda^2}{2},\quad V_{us}\simeq\lambda,\quad V_{cd}\simeq-\lambda,\quad V_{cb}\simeq A\lambda^2,\quad V_{ub}=A\lambda^3(\rho-i\eta),\quad V_{td}=A\lambda^3(1-\rho-i\eta), \tag{32.192}$$

The apex. Insert into the definition Eq. (32.190). The numerator is $V_{ud}V_{ub}^*=(1-\frac{\lambda^2}{2})A\lambda^3(\rho+i\eta)$ and the denominator is $V_{cd}V_{cb}^*=(-\lambda)(A\lambda^2)=-A\lambda^3$. Hence

$$\bar\rho+i\bar\eta = -\frac{(1-\frac{\lambda^2}{2})A\lambda^3(\rho+i\eta)}{-A\lambda^3} = (\rho+i\eta)\Big(1-\tfrac{\lambda^2}{2}\Big)+O(\lambda^4), \tag{32.193}$$

so $\bar\rho = \rho(1-\frac{\lambda^2}{2})$ and $\bar\eta = \eta(1-\frac{\lambda^2}{2})$: the "barred" variables are just the Wolfenstein (ρ,η) pushed in by the small factor $1-\lambda^2/2$, and they are the genuinely rephasing-invariant apex coordinates. With the mid-2020s global-fit values $\lambda = 0.22501$, $A = 0.826$, $\bar\rho = 0.159$, $\bar\eta = 0.348$ [382], the apex sits well off the real axis — the triangle is far from flat, so CP violation is large.

The angle β. From Eq. (32.191), $\beta = \arg(-V_{cd}V_{cb}^*/V_{td}V_{tb}^*)$. Using Eq. (32.192), $-V_{cd}V_{cb}^* = A\lambda^3$ (real, positive) and $V_{td}V_{tb}^* = A\lambda^3(1-\rho-i\eta)$, so

$$-\frac{V_{cd}V_{cb}^*}{V_{td}V_{tb}^*} = \frac{1}{1-\rho-i\eta} = \frac{(1-\rho)+i\eta}{(1-\rho)^2+\eta^2}, \qquad \Rightarrow \qquad \tan\beta = \frac{\eta}{1-\rho} = \frac{\bar\eta}{1-\bar\rho}, \tag{32.194}$$

the last equality because the common factor $1-\lambda^2/2$ cancels between numerator and denominator. Geometrically this is the angle at the $(1,0)$ vertex subtended by the apex $(\bar\rho,\bar\eta)$, exactly as drawn in Fig. 32.6.

Carrying the trigonometry to $\sin 2\beta$. Use the double-angle identity in the tangent form,

$$\sin 2\beta = \frac{2\tan\beta}{1+\tan^2\beta} = \frac{2\,\dfrac{\bar\eta}{1-\bar\rho}}{1+\dfrac{\bar\eta^2}{(1-\bar\rho)^2}} = \frac{2\,\bar\eta\,(1-\bar\rho)}{(1-\bar\rho)^2+\bar\eta^2}. \tag{32.195}$$

This is the bridge from a CKM phase to a measured number. With $\bar\rho = 0.159$, $\bar\eta = 0.348$: $1-\bar\rho = 0.841$, so $(1-\bar\rho)^2 = 0.7073$, $\bar\eta^2 = 0.1211$, their sum 0.8284; the numerator is $2(0.348)(0.841) = 0.5853$; hence

$$\sin 2\beta = \frac{0.5853}{0.8284} = 0.707. \tag{32.196}$$

The direct measurement from $B_d \to J/\psi K_S$, Eq. (32.182), is 0.699 ± 0.017 [382] — consistent to better than a percent. The two numbers come from utterly different experiments (a CP-conserving apex fit vs. a time-dependent CP asymmetry); their agreement is one entry in the overconstraint test.

The point about the phase. For small $\bar\eta$, Eq. (32.195) linearizes to $\sin 2\beta \simeq 2\bar\eta/(1-\bar\rho)$. Since $\bar\eta \propto \eta \propto J$ (the area, Eq. (32.188)), the height of the triangle, the angle β, and the measured asymmetry $\sin 2\beta$ are all directly proportional, at leading order, to the single CP-violating phase. Set $\eta = 0$ and the triangle collapses to a line, $\sin 2\beta \to 0$, and every CP asymmetry vanishes — the geometric content of Theorem 32.16.

32.6.6 Overconstraint: the apex as a precision test

The reason the unitarity triangle is the centerpiece of quark-flavor physics is that the apex $(\bar\rho,\bar\eta)$ — two real numbers — is fixed redundantly by many independent measurements, each of a different physical type and each carrying different

theoretical uncertainties. If the Standard Model with its single CP phase is the whole story, every measurement must select a region of the $(\bar\rho, \bar\eta)$ plane passing through one common point. The principal constraints are the following.

- $|V_{ub}/V_{cb}|$ **from semileptonic** b **decays** fixes the length of the side from the origin to the apex,

$$R_u = \sqrt{\bar\rho^2 + \bar\eta^2} = \left(1 - \tfrac{\lambda^2}{2}\right)\frac{1}{\lambda}\left|\frac{V_{ub}}{V_{cb}}\right|, \tag{32.197}$$

  a circle centered at the origin. This is tree-level and CP-conserving.

- $\Delta m_d/\Delta m_s$ **from** B_d **and** B_s **oscillations** fixes $|V_{td}/V_{ts}|$, hence the side from $(1, 0)$ to the apex,

$$R_t = \sqrt{(1-\bar\rho)^2 + \bar\eta^2} \;\propto\; \left|\frac{V_{td}}{V_{ts}}\right|, \tag{32.198}$$

  a circle centered at $(1, 0)$. The ratio $\Delta m_d/\Delta m_s = (m_{B_d}/m_{B_s})\,\xi^{-2}\,|V_{td}/V_{ts}|^2$ is theoretically clean because the leading lattice uncertainty cancels in the $SU(3)$-breaking ratio ξ. Empirically (mid-2020s) $\Delta m_d = 0.5065\,\mathrm{ps}^{-1}$ and $\Delta m_s = 17.765\,\mathrm{ps}^{-1}$ [382].

- $\sin 2\beta$ **from** $B_d \to J/\psi K_S$ fixes the angle β, Eq. (32.182): a ray from the origin — strictly, from the $(1, 0)$ vertex — at the measured slope. This is the interference CP violation of Section 32.6.4.

- γ **from tree-level** $B \to DK$ fixes $\gamma = \arg(\bar\rho + i\bar\eta)$, a ray from the origin. Because it uses only tree amplitudes with no loops, γ is the theoretically cleanest angle and a benchmark immune to new short-distance physics; the global determination is $\gamma = (65.9^{+3.3}_{-3.5})^\circ$ [382].

- α **from** $B \to \pi\pi, \rho\rho, \rho\pi$ fixes the apex angle via isospin analyses.

- $|\epsilon_K|$ **from kaon mixing** fixes, through Eqs. (32.167) and (32.173), a hyperbola in the plane of the schematic form $\bar\eta\,[\,(1-\bar\rho)A^2\eta_2\,S_0(x_t)+P_c\,]\,A^2 =$ const, the strongest single constraint historically but the one most sensitive to $|V_{cb}|$ (it enters as $A^4 \propto |V_{cb}|^4$).

The remarkable empirical fact (Fig. 32.7) is that all of these curves cross in a common region. A CP-conserving observable (R_u), two CP-violating angles (β, γ), an oscillation frequency ratio (R_t), and an indirect CP parameter (ϵ_K) — physically unrelated quantities spanning loop and tree processes, kaons and B mesons — all point to the same apex, $\bar\rho = 0.159 \pm 0.008$, $\bar\eta = 0.348 \pm 0.010$ (global fit, mid-2020s [382]). That $\bar\eta \neq 0$ at enormous significance is the quantitative statement that the single CKM phase is nonzero; that the curves are mutually consistent is the statement that this one phase accounts for *all* measured CP violation and flavor mixing in the quark sector. This is the precision test of the KM mechanism, and it has held with increasing stringency for two decades.

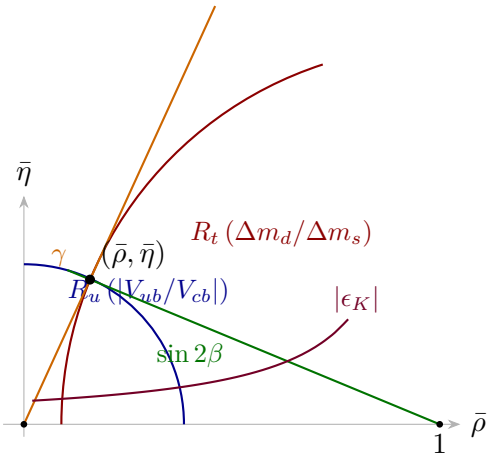


Figure 32.7: Schematic overconstraint of the apex $(\bar{\rho},\bar{\eta})$. Each independent measurement — the side lengths R_u (circle about the origin) and R_t (circle about $(1,0)$), the angles β and γ (rays), and $|\epsilon_K|$ (hyperbola) — selects a curve, and the Kobayashi–Maskawa mechanism requires them all to meet at one point. They do, to within current errors; the residual width of the crossing region is the precision of the test. The curves here are illustrative, not a fit.

Where it strains, honestly. The consistency is not perfect, and the tensions are worth stating plainly because they are where any deviation from the Standard Model would first appear. As of the mid-2020s [382]: (i) the *inclusive* and *exclusive* determinations of $|V_{ub}|$ (and similarly $|V_{cb}|$) disagree at the $\sim$ 2–3σ level — a long-standing discrepancy whose origin is most likely in the theory of semileptonic form factors and the treatment of duality, not new physics, but which directly broadens the R_u band and, through $|V_{cb}|^4$, the ϵ_K prediction; (ii) the value of $\sin 2\beta$ *predicted* by a fit excluding it has at times sat a couple of standard deviations from the direct measurement, a tension that has waxed and waned with the $|V_{ub}|$ inputs and is presently mild; (iii) the consistency of γ (pure tree) with the loop-dominated constraints is, so far, complete — a significant statement, since a clean tree-vs-loop discrepancy would be a clean sign of new physics in mixing. No tension currently rises to the level of a discovery, and the overwhelming impression is of a single phase passing test after test. The honest summary is that the KM mechanism is the established, dominant — and within present precision, the sole — source of quark-sector CP violation, with residual 2–3σ frictions concentrated in hadronic-matrix-element inputs rather than in the CKM structure itself.

This completes the experimental anchoring of the quark-flavor sector: a single complex phase in $V_{\rm CKM}$, measured many times over as the geometry of one triangle. The logic — replication into generations, a unitary mixing matrix, an irreducible phase, observable as CP violation — now stands ready to be repeated

for the leptons. There the obstacle is prior: in the minimal Standard Model the neutrino has no mass and no mixing partner (Theorem 31.17), so before any leptonic analogue of the triangle can be drawn, neutrino mass must first be supplied. That, together with the resulting PMNS matrix and its extra Majorana phases, is the subject of the next section.

32.7 Neutrino Masses and the PMNS Matrix

The argument of this chapter has, so far, lived entirely in the quark sector. Three identical generations, fed through the Yukawa machinery and the electroweak charged current, produced the CKM matrix; counting its physical parameters isolated a single irreducible phase; and that phase, made rephasing-invariant as the Jarlskog determinant, turned out to be the Standard Model's only established source of CP violation. We now ask whether the same logic plays out for the leptons. The grammar is identical—gauge-blind generations, Yukawa matrices, a mismatch surviving in the charged current as a unitary mixing matrix—but one fact changes everything: in the minimal Standard Model the neutrino has no mass at all (Theorem 31.17), and a massless neutrino has no mass basis to misalign against the flavor basis. The leptonic mixing matrix is therefore not a prediction of the minimal theory but a measure of how nature has chosen to give neutrinos mass—a choice that is, as of the mid-2020s, still only partly known. This section repeats the flavor logic honestly: it recalls why the minimal neutrino is massless, lays out the two minimal ways to repair that, derives the seesaw light-mass formula, builds the PMNS matrix and counts its phases (finding more than the quark sector permits), derives the oscillation phenomenon that measures it, and closes by cataloguing precisely what remains open.

32.7.1 The massless starting point and accidental lepton number

Recall the conclusion of Section 31.4, stated as Theorem 31.17: with the minimal renormalizable field content—no right-handed neutrino, one Higgs doublet, operators of mass dimension ≤ 4—the neutrino has no mass term. A Dirac mass is impossible because there is no ν_R to pair with ν_L; a Majorana mass for ν_L alone is forbidden by gauge invariance, since the bilinear $\overline{(\nu_L)^c}\,\nu_L$ carries weak isospin and hypercharge; and the unique gauge-invariant operator that would do the job, the Weinberg operator Eq. (31.121), is of dimension five and hence absent from the renormalizable Lagrangian. At the renormalizable level, then, each neutrino stays exactly massless.

This has a sharp consequence for flavor. With the charged-lepton Yukawa matrix $\mathbf{y}_e$ the only flavor structure in the lepton sector, we may diagonalize it by independent unitary rotations V_{eL}, V_{eR} of the left- and right-handed charged-lepton fields, exactly as for the quarks. But the neutrinos, being massless, carry

no mass matrix to diagonalize: we are free to apply to the three left-handed neutrino fields the *same* rotation V_{eL} that we applied to their charged-lepton partners inside the doublet L. In the leptonic charged current Eq. (31.128), $J^+_\mu = \overline{\nu_L}\,\gamma_\mu\, e_L$, this common rotation cancels between ν_L and e_L precisely as it failed to cancel between up- and down-quarks. The leptonic charged current is therefore flavor-diagonal: there is no leptonic analogue of the CKM matrix, no lepton mixing, and—because each flavor's charged current closes on itself—the three individual lepton numbers L_e, L_μ, L_τ are separately conserved. The minimal model predicts no $\mu \to e\gamma$, no neutrino oscillation, nothing.

Physical Insight

The conservation of individual lepton number in the minimal Standard Model is not imposed; it is *accidental*, an automatic consequence of the field content and renormalizability. The only flavor matrix is $\mathbf{y}_e$, and one matrix can always be diagonalized away in a current that touches each lepton family symmetrically. Mixing requires a *second*, misaligned flavor matrix in the same sector—a neutrino mass matrix—and the minimal theory has none. Giving the neutrino mass is thus exactly what breaks the accidental flavor symmetry and switches on lepton mixing, in perfect parallel with how the down-quark Yukawa, misaligned against the up-quark Yukawa, switched on $V_{\rm CKM}$.

Neutrino oscillations, established experimentally, are therefore a discovery *of physics beyond the minimal Standard Model*: they prove that at least two neutrinos are massive and that lepton flavor mixes. The rest of this section asks what minimal extension accommodates that fact, and what new structure it brings.

32.7.2 Two minimal ways to make a neutrino massive

There are exactly two minimal repairs, distinguished by whether they conserve total lepton number $L = L_e + L_\mu + L_\tau$.

Route 1: a Dirac mass from a new right-handed neutrino. Add to each generation a right-handed neutrino ν_R: a fermion that is a singlet under the entire gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, since it carries no color, no weak isospin (it is a singlet, not part of a doublet), and—to be electrically neutral via $Q = T^3 + Y$—hypercharge $Y = 0$. Being a complete gauge singlet, ν_R has exactly the quantum numbers of "nothing," which is why it could be omitted from the minimal model in the first place. With it present, a neutrino Yukawa coupling is allowed, in exact analogy to the up-quark term $\overline{Q}\,\widetilde{H}\,u_R$,

$$\mathcal{L}^{\rm Dirac}_{Y,\nu} = -\,\overline{L}\,\widetilde{H}\,\mathbf{y}_\nu\,\nu_R + \text{h.c.}, \qquad \widetilde{H} = i\sigma^2 H^*, \tag{32.199}$$

with $\mathbf{y}_\nu$ a 3×3 complex matrix in generation space. Upon condensation $\langle H^0\rangle = v/\sqrt{2}$ this produces a Dirac mass matrix $\mathbf{m}_D = \mathbf{y}_\nu v/\sqrt{2}$, structurally identical to the charged-fermion masses and conserving total lepton number L (a Dirac mass connects ν_L to ν_R, both carrying $L=+1$). The neutrino is then "just another charged fermion" in its mass structure, and the leptonic mixing matrix is counted exactly as the CKM matrix was. The price is twofold and is the reason this route is not regarded as compelling on its own: it requires a new field invented solely to give a mass, and it offers no explanation for why the neutrino Yukawas must be tiny—$y_\nu \sim 10^{-12}$ to reach $m_\nu \sim 0.05$ eV at $v \simeq 246$ GeV, a hierarchy even more extreme than the electron's.

Route 2: a Majorana mass from the Weinberg operator. Alternatively, keep the minimal field content but admit that the renormalizable Lagrangian is only an effective description, valid below some heavy scale Λ where new physics lives. Then the dimension-five Weinberg operator Eq. (31.121),

$$\mathcal{L}_5 = \frac{1}{\Lambda}\,(\overline{L^c}\,\widetilde{H}^*)\,\mathbf{c}\,(\widetilde{H}^\dagger\,L) + \text{h.c.}, \tag{32.200}$$

with $\mathbf{c}$ a symmetric matrix of dimensionless couplings and $L^c = C\overline{L}^{\mathsf{T}}$ the charge conjugate built with the matrix C of Section 11.7.2, is the unique lowest-dimension addition. On condensation it delivers a *Majorana* mass matrix for the left-handed neutrinos,

$$\mathcal{L} \supset -\tfrac{1}{2}\,\overline{\nu_L^c}\,\mathbf{m}_\nu\,\nu_L + \text{h.c.}, \qquad \mathbf{m}_\nu = \mathbf{c}\,\frac{v^2}{\Lambda}, \tag{32.201}$$

which is symmetric, $\mathbf{m}_\nu = \mathbf{m}_\nu^{\mathsf{T}}$, because it is built from two identical fields ν_L. This route adds no new light field, automatically explains the smallness of m_ν as a v/Λ suppression by a heavy scale, and—crucially—*violates total lepton number by two units*: the mass term Eq. (32.201) connects ν_L to ν_L^c, turning a neutrino into an antineutrino ($\Delta L = 2$). Total L is no longer conserved; it survives only as an accidental symmetry of the renormalizable truncation, broken by the leading higher-dimension operator. The two routes are physically distinct, and the question of which nature uses—equivalently, whether the neutrino is its own antiparticle—is precisely the open "Dirac vs. Majorana" question we return to below.

32.7.3 The seesaw mechanism

The two routes are not exclusive: the most economical microscopic theory that *generates* the Weinberg operator combines them. Add the gauge-singlet right-handed neutrino ν_R of Route 1, but now exploit the fact that, being a gauge singlet, it may carry a *bare* Majorana mass M—a term forbidden for ν_L by gauge invariance but allowed for a singlet, since $M\,\overline{\nu_R^c}\,\nu_R$ is automatically gauge-invariant. There is then no symmetry reason for M to be small, and nothing ties

it to the electroweak scale; naturally M sits at whatever heavy scale introduced ν_R. The full neutral-fermion mass Lagrangian, for one generation to expose the mechanism cleanly, is

$$\mathcal{L}_{\text{mass}} = -\,m_D\,\overline{\nu_L}\,\nu_R - \tfrac{1}{2}M\,\overline{\nu_R^c}\,\nu_R + \text{h.c.}, \tag{32.202}$$

with the Dirac mass $m_D = y_\nu v/\sqrt{2}$ set by the Higgs condensate as usual and the Majorana mass M a free heavy parameter.

To diagonalize, collect the two independent left-handed fields into a column. The charge conjugate $\nu_R^c = C\overline{\nu_R}^{\mathsf{T}}$ is left-handed, so $n_L \equiv (\nu_L,\ \nu_R^c)^{\mathsf{T}}$ is a doublet of left-handed Weyl fields, and Eq. (32.202) becomes a single symmetric Majorana mass term

$$\mathcal{L}_{\text{mass}} = -\tfrac{1}{2}\,\overline{n_L^c}\,\mathcal{M}\,n_L + \text{h.c.}, \qquad \mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix}. \tag{32.203}$$

The $(1,1)$ entry vanishes because a bare ν_L Majorana mass is gauge-forbidden; the $(2,2)$ entry is the singlet mass M; the off-diagonal m_D is the electroweak Dirac mass. The matrix is real and symmetric here, so we diagonalize it by an orthogonal rotation; its eigenvalues solve $\det(\mathcal{M} - \lambda\mathbf{1}) = 0$, i.e.

$$(0-\lambda)(M-\lambda) - m_D^2 = \lambda^2 - M\lambda - m_D^2 = 0, \qquad \lambda_\pm = \frac{M \pm \sqrt{M^2 + 4m_D^2}}{2}. \tag{32.204}$$

Nothing has been approximated yet; Eq. (32.204) is exact. The physics is in the hierarchy $M \gg m_D$, which makes the square root expand as

$$\sqrt{M^2 + 4m_D^2} = M\sqrt{1 + \frac{4m_D^2}{M^2}} = M\left(1 + \frac{2m_D^2}{M^2} - \cdots\right) = M + \frac{2m_D^2}{M} + O\!\left(\frac{m_D^4}{M^3}\right). \tag{32.205}$$

Substituting into Eq. (32.204),

$$\lambda_+ = M + \frac{m_D^2}{M} + \cdots \approx M, \qquad \lambda_- = -\,\frac{m_D^2}{M} + \cdots. \tag{32.206}$$

The spectrum splits into a heavy eigenstate of mass $\approx M$, almost pure ν_R, and a light eigenstate of mass

$$\boxed{m_\nu = \frac{m_D^2}{M}} \tag{32.207}$$

(the sign in λ_- is a Majorana phase, removable by a field redefinition and physically harmless), almost pure ν_L. The light mass is suppressed below the Dirac scale by the ratio m_D/M: the heavier the new singlet, the lighter the observed neutrino—the eponymous *seesaw*. The mixing angle between the two states, from the symmetric 2×2 matrix Eq. (32.203), is $\tan 2\theta = 2m_D/M$, hence

$\theta \simeq m_D/M \ll 1$: the light state is overwhelmingly the active ν_L, with only a tiny admixture of the sterile ν_R, consistent with all observations that neutrinos couple with full weak strength. Comparing Eq. (32.207) with the Weinberg coefficient Eq. (32.201), $m_\nu = cv^2/\Lambda$, identifies the effective scale as $\Lambda = M/c = M/y_\nu^2$ up to couplings: the heavy singlet *is* the new physics whose exchange generates the dimension-five operator. We collect the result.

Proposition 32.33 (Type-I seesaw light-neutrino mass)**.** *Let a left-handed neutrino acquire an electroweak Dirac mass $m_D = y_\nu v/\sqrt{2}$ through coupling to a gauge-singlet right-handed neutrino that itself carries a gauge-allowed bare Majorana mass M, as in Eq. (32.202). Then the 2×2 neutral mass matrix Eq. (32.203) has, in the limit $M \gg m_D$, one heavy Majorana eigenstate of mass $\approx M$ and one light Majorana eigenstate of mass $m_\nu = m_D^2/M + O(m_D^4/M^3)$, with the light state an almost-pure active neutrino mixing into the heavy state by an angle $\theta \simeq m_D/M$. The light neutrino is necessarily Majorana, and total lepton number is violated by the M term.*

Worked Example 32.34 (The seesaw scale from the atmospheric splitting)**.** The seesaw turns the measured neutrino mass into an estimate of the scale M of new physics, and it is instructive to see that the estimate lands near the grand-unification scale—one of the few quantitative hints we have about energies far above the collider frontier. Take the Dirac mass to be set by an order-unity Yukawa at the electroweak scale, $m_D \sim v/\sqrt{2} \simeq 174$ GeV (the natural guess if ν_R behaves like the top quark), and the light mass to be the atmospheric scale read off from oscillations, $m_\nu \sim \sqrt{|\Delta m^2_{\rm atm}|} \simeq \sqrt{2.5 \times 10^{-3}\ \text{eV}^2} \approx 0.05$ eV. Then Eq. (32.207) gives

$$M = \frac{m_D^2}{m_\nu} \approx \frac{(174\ \text{GeV})^2}{0.05\ \text{eV}} = \frac{3.0 \times 10^4\ \text{GeV}^2}{5 \times 10^{-11}\ \text{GeV}} \approx 6 \times 10^{14}\ \text{GeV}, \tag{32.208}$$

within a factor of a few of the canonical GUT scale $\sim 2 \times 10^{16}$ GeV. If instead the neutrino Yukawa resembles the electron's rather than the top's, M slides down by the square of the Yukawa ratio, illustrating that the seesaw fixes only the combination m_D^2/M: the absolute scale M is not determined by oscillation data alone. The robust statement is the one worth keeping: a sub-eV neutrino mass with an electroweak Dirac mass demands a Majorana scale enormously above anything directly accessible, which is exactly why the seesaw is read as a window onto very high energies. The full model-building—type-I/II/III variants, leptogenesis, the connection to the baryon asymmetry—is developed in Chapter 35; here we have only the minimal mechanism and its central formula.

32.7.4 The PMNS matrix and the leptonic phase count

Once the neutrinos are massive, by either route, the lepton sector carries two misaligned flavor matrices—the charged-lepton Yukawa $\mathbf{y}_e$ and the neutrino

mass matrix $\mathbf{m}_\nu$—and the flavor logic of the quark sector applies verbatim. Diagonalize the charged-lepton mass matrix by $V_{eL}^\dagger\,\mathbf{m}_e\,V_{eR} = \mathrm{diag}(m_e, m_\mu, m_\tau)$ and the neutrino mass matrix by a unitary V_ν. The leptonic charged current Eq. (31.128), written in the mass eigenbasis, then carries the residual mismatch

$$\mathcal{L}_{\mathrm{CC}}^{\mathrm{lept}} = \frac{g}{\sqrt{2}}\, W^{+\mu}\, \overline{\nu_{Li}}\, \gamma_\mu\, (U_{\mathrm{PMNS}}^\dagger)_{i\alpha}\, e_{L\alpha} + \mathrm{h.c.}, \qquad U_{\mathrm{PMNS}} = V_{eL}^\dagger\, V_\nu, \tag{32.209}$$

the *Pontecorvo–Maki–Nakagawa–Sakata matrix* [330, 408], the leptonic analogue of V_{CKM}. Latin indices $i = 1, 2, 3$ run over neutrino mass eigenstates, Greek $\alpha = e, \mu, \tau$ over charged-lepton mass eigenstates; the flavor neutrino produced alongside a charged lepton α is the superposition $\nu_{L\alpha} = \sum_i (U_{\mathrm{PMNS}})_{\alpha i}\, \nu_{Li}$.

Here the parallel with the quarks breaks in a way that is the heart of this subsection. The number of physical parameters in a unitary mixing matrix depends on how many of its phases can be rotated away by rephasing the fermion fields, and that in turn depends on whether the mass terms are invariant under such rephasings. We count for general generation number N_g.

A general $N_g \times N_g$ unitary matrix has N_g^2 real parameters, split into

$$\underbrace{\tfrac{1}{2}N_g(N_g-1)}_{\text{rotation angles}} \quad + \quad \underbrace{\tfrac{1}{2}N_g(N_g+1)}_{\text{phases}}, \tag{32.210}$$

the angles being those of the $\mathrm{O}(N_g)$ subgroup and the rest phases. Angles are physical; the question is how many phases survive rephasing.

Dirac neutrinos. If the neutrino mass is Dirac, the mass terms $\overline{e_{L\alpha}}\, m_\alpha\, e_{R\alpha}$ and $\overline{\nu_{Li}}\, m_i\, \nu_{Ri}$ are each invariant under a *common* rephasing of the left- and right-handed components of a given field, $\psi_L \to e^{i\phi}\psi_L$, $\psi_R \to e^{i\phi}\psi_R$. We thus have $2N_g$ independent phase rotations (one per charged lepton, one per neutrino) with which to remove phases from U_{PMNS}. But the charged current Eq. (32.209), and indeed the whole Lagrangian, is invariant under an *overall* common phase of all leptons—this is total lepton number $\mathrm{U}(1)_L$, unbroken for Dirac masses—so one combination of the $2N_g$ rephasings acts trivially on U_{PMNS} and removes nothing. The number of removable phases is therefore $2N_g - 1$, exactly as in the quark sector (where the leftover symmetry was baryon number). The physical phase count is

$$\tfrac{1}{2}N_g(N_g+1) - (2N_g - 1) = \tfrac{1}{2}(N_g-1)(N_g-2), \tag{32.211}$$

which for $N_g = 3$ gives a single phase: one Dirac CP-violating phase δ, just like V_{CKM}.

Majorana neutrinos. If the neutrino mass is Majorana, the decisive change is that the mass term Eq. (32.201) is built from two factors of the *same* field ν_{Li}.

Under a rephasing $\nu_{Li} \to e^{i\phi_i}\nu_{Li}$ the bilinear $\overline{\nu^c_{Li}}\, m_i\, \nu_{Li}$ picks up $e^{2i\phi_i}$—it is *not* invariant, because there is no independent right-handed partner whose opposite phase could compensate it (that compensation is exactly what saved the Dirac mass). Demanding real, positive neutrino masses thus forbids rephasing the neutrino fields altogether: only the N_g charged-lepton phases remain available, and total lepton number $\mathrm{U}(1)_L$ is no longer a symmetry (the Majorana mass breaks it), so there is no leftover trivial combination to subtract. We remove N_g phases, and the physical phase count is

$$\tfrac{1}{2}N_g(N_g+1) - N_g = \tfrac{1}{2}N_g(N_g-1), \tag{32.212}$$

which for $N_g = 3$ gives *three* phases. The difference between the Majorana and Dirac counts,

$$\tfrac{1}{2}N_g(N_g-1) - \tfrac{1}{2}(N_g-1)(N_g-2) = N_g - 1, \tag{32.213}$$

is the number of *extra Majorana phases*: $N_g - 1 = 2$ for three generations. We record the conclusion.

Proposition 32.35 (Physical parameters of leptonic mixing)**.** *For N_g generations the PMNS matrix carries $\frac{1}{2}N_g(N_g-1)$ physical rotation angles in all cases. The number of physical CP-violating phases is $\frac{1}{2}(N_g-1)(N_g-2)$ if neutrinos are Dirac and $\frac{1}{2}N_g(N_g-1)$ if they are Majorana, the difference of $N_g - 1$ being the Majorana phases. For $N_g = 3$: three angles $(\theta_{12}, \theta_{13}, \theta_{23})$ and either one Dirac phase δ (Dirac neutrinos) or one Dirac phase plus two Majorana phases (Majorana neutrinos). By the criterion of Theorem 32.16, a single generation pair would carry no phase and two generations only the Majorana phases; three generations are required for a Dirac-type CP phase, just as for the quarks.*

The two extra Majorana phases are conventionally collected into a diagonal matrix multiplying U_{PMNS} on the right,

$$U_{\mathrm{PMNS}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix} \begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix} \begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{\widetilde{U}\ \text{(three angles+Dirac } \delta\text{)}} \operatorname{diag}\!\left(1,\ e^{i\alpha_{21}/2},\ e^{i\alpha_{31}/2}\right), \tag{32.214}$$

with $c_{ij} = \cos\theta_{ij}$, $s_{ij} = \sin\theta_{ij}$. The first factor $\widetilde{U}$ has exactly the structure of the standard CKM parametrization; the appended phases α_{21}, α_{31} are the Majorana phases. They are physically real but, as the next subsection shows, they cancel out of every oscillation probability: oscillations measure $\widetilde{U}$, not the Majorana phases. The latter affect only genuinely lepton-number-violating amplitudes—above all neutrinoless double beta decay, whose rate scales with the coherent combination $m_{\beta\beta} = \left|\sum_i (U_{\mathrm{PMNS}})^2_{ei}\, m_i\right|$ in which the α's do not cancel.

Physical Insight

The reason for the extra phases is worth stating in one sentence, because it is the cleanest illustration in the book of how a *discrete* fact about a field—whether it has an independent antiparticle—feeds into the *continuous* parameter count of a mixing matrix. A Dirac field can be rephased freely, and each such freedom eats one phase out of the mixing matrix; a Majorana field cannot, because rephasing it would render its mass complex, and so the phases it would have eaten survive as physical CP-violating parameters. Majorana phases are nothing but the phases that the missing rephasings failed to remove.

32.7.5 Neutrino oscillations

The PMNS angles and the Dirac phase are measured through neutrino oscillation: a neutrino produced as one flavor is detected, after propagation, with nonzero probability as another. The phenomenon is a direct quantum-mechanical interference between mass eigenstates of slightly different phase velocity, and the cleanest derivation is the two-flavor one, which we carry in full before quoting the three-flavor generalization.

Two-flavor derivation. Restrict to two flavors (ν_e, ν_μ) and two mass eigenstates (ν_1, ν_2). With only two generations the mixing matrix is a single real rotation (no phase, by Eq. (32.211) with $N_g = 2$),

$$|\nu_e\rangle = \cos\theta\,|\nu_1\rangle + \sin\theta\,|\nu_2\rangle\,, \qquad |\nu_\mu\rangle = -\sin\theta\,|\nu_1\rangle + \cos\theta\,|\nu_2\rangle\,. \tag{32.215}$$

A neutrino born as ν_μ at production ($t = 0$) is the superposition on the right of Eq. (32.215). Each mass eigenstate has definite energy and evolves with its own phase, $|\nu_i(t)\rangle = e^{-iE_i t}\,|\nu_i\rangle$, so

$$|\nu_\mu(t)\rangle = -\sin\theta\, e^{-iE_1 t}\,|\nu_1\rangle + \cos\theta\, e^{-iE_2 t}\,|\nu_2\rangle\,. \tag{32.216}$$

The neutrinos are ultrarelativistic ($m_i \ll E$) and, being produced by a process of fixed energy, share a common energy E to leading order; their momenta then differ. Expanding the dispersion relation,

$$E_i = \sqrt{p_i^2 + m_i^2} \implies p_i = \sqrt{E^2 - m_i^2} \approx E - \frac{m_i^2}{2E}, \tag{32.217}$$

so over a propagation distance $L \approx t$ (the neutrino travels at nearly the speed of light) the accumulated spatial phase of eigenstate i is $p_i L \approx EL - (m_i^2/2E)L$. The common EL piece is an overall phase that drops out of probabilities; the physical relative phase between the two eigenstates is

$$\Delta\Phi = (p_1 - p_2)L = \frac{m_2^2 - m_1^2}{2E}\,L = \frac{\Delta m_{21}^2}{2E}\,L, \qquad \Delta m_{21}^2 \equiv m_2^2 - m_1^2. \tag{32.218}$$

(Working with the spatial phase pL at fixed energy and the temporal phase Et at fixed momentum gives the same answer; the often-quoted "factor-of-two" worry from treating energy and momentum naively cancels once one is consistent, and the result Eq. (32.218) is the correct invariant phase.) The amplitude to find the state as ν_e after distance L is the overlap $\langle \nu_e | \nu_\mu(t) \rangle$, using $\langle \nu_e | = \cos\theta \, \langle \nu_1 | + \sin\theta \, \langle \nu_2 |$ and orthonormal mass states:

$$\begin{aligned} A(\nu_\mu \to \nu_e) &= \cos\theta \, (-\sin\theta) \, e^{-iE_1 t} + \sin\theta \, \cos\theta \, e^{-iE_2 t} \\ &= \sin\theta \cos\theta \left(e^{-iE_2 t} - e^{-iE_1 t} \right). \end{aligned} \tag{32.219}$$

The appearance probability is the modulus squared. Factoring out the common phase,

$$\begin{aligned} P(\nu_\mu \to \nu_e) &= \sin^2\theta \cos^2\theta \, \left| e^{-iE_2 t} - e^{-iE_1 t} \right|^2 = \sin^2\theta \cos^2\theta \, \left| e^{-i\Delta\Phi} - 1 \right|^2 \\ &= \sin^2\theta \cos^2\theta \, \left[(\cos\Delta\Phi - 1)^2 + \sin^2\Delta\Phi \right] = \sin^2\theta \cos^2\theta \, (2 - 2\cos\Delta\Phi) \\ &= 4 \sin^2\theta \cos^2\theta \, \sin^2 \frac{\Delta\Phi}{2}, \end{aligned} \tag{32.220}$$

where the last step used $1 - \cos x = 2\sin^2(x/2)$. With $4\sin^2\theta\cos^2\theta = \sin^2 2\theta$ and $\Delta\Phi/2 = \Delta m^2_{21} L/4E$ from Eq. (32.218), this is the master formula

$$\boxed{P(\nu_\mu \to \nu_e) = \sin^2 2\theta \, \sin^2\left(\frac{\Delta m^2 \, L}{4E} \right), \qquad P(\nu_\mu \to \nu_\mu) = 1 - P(\nu_\mu \to \nu_e).} \tag{32.221}$$

Two physical points are encoded here, and Theorem 32.36 states them precisely. The *amplitude* of the oscillation is set by the mixing angle through $\sin^2 2\theta$, maximal at $\theta = 45^\circ$ and vanishing at $\theta = 0$; the *wavelength* is set by the mass-squared splitting Δm^2. Oscillation requires both a nonzero angle and a nonzero splitting: if either vanishes, $P \to 0$ and flavor is conserved. In particular, oscillation measures only the *difference* of squared masses, never the absolute scale, and never the Majorana phases (which, being a rephasing of the individual mass states, multiply each $|\nu_i\rangle$ by a phase that cancels in the modulus Eq. (32.220)).

Theorem 32.36 (Two-flavor oscillation probability)**.** *For two flavors mixing through a single angle θ with mass-squared splitting $\Delta m^2 = m_2^2 - m_1^2$, an ultrarelativistic neutrino of energy E produced as one flavor is detected after baseline L as the other flavor with probability $P = \sin^2 2\theta \, \sin^2(\Delta m^2 L/4E)$. The transition vanishes if $\theta = 0$ (no mixing) or $\Delta m^2 = 0$ (degenerate, hence indistinguishable, masses); only Δm^2, not the absolute masses, is observable.*

Restoring $\hbar$ and c for numerical use, the phase $\Delta m^2 L/4E$ becomes $\Delta m^2 c^4 L/(4\hbar c E)$, which in practical units is

$$\frac{\Delta m^2 L}{4E} = 1.267 \, \frac{\Delta m^2[\mathrm{eV}^2] \; L[\mathrm{km}]}{E[\mathrm{GeV}]}, \tag{32.222}$$

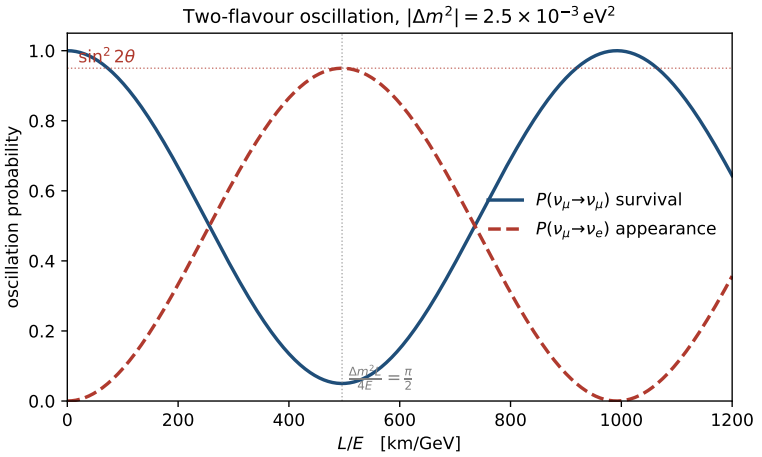


Figure 32.8: Two-flavor survival probability $P(\nu_\mu \to \nu_\mu)$ (solid) and appearance probability $P(\nu_\mu \to \nu_e)$ (dashed) versus the ratio L/E, for $\sin^2 2\theta = 0.95$ and $|\Delta m^2| = 2.5 \times 10^{-3}$ eV^2. The appearance amplitude is capped at $\sin^2 2\theta$; the first oscillation maximum sits at $\Delta m^2 L/4E = \pi/2$. The figure is generated by `figures/32-oscillation.py`.

the workhorse formula of experimental neutrino physics. Figure 32.8 shows the survival and appearance probabilities as functions of L/E for representative atmospheric-sector parameters.

Worked Example 32.37 (Atmospheric oscillation: the first maximum)**.** The two-flavor formula explains at a glance why atmospheric and long-baseline experiments are built at the distances they are. The oscillation is fullest when the phase Eq. (32.222) equals $\pi/2$, the first maximum of $\sin^2$. With the atmospheric splitting $\Delta m^2 \simeq 2.5 \times 10^{-3}$ eV^2 this requires

$$1.267 \times (2.5 \times 10^{-3}) \times \frac{L[\text{km}]}{E[\text{GeV}]} = \frac{\pi}{2} \implies \frac{L}{E} \approx \frac{1.571}{3.17 \times 10^{-3}} \approx 5.0 \times 10^{2}\ \frac{\text{km}}{\text{GeV}}. \tag{32.223}$$

A few-GeV accelerator beam ($E \sim 1$–3 GeV) therefore reaches its first oscillation maximum over a baseline of several hundred to roughly a thousand kilometres—precisely the geometry of the long-baseline programmes. A reactor experiment hunting the same splitting with $E \sim$ few MeV instead needs $L \sim 1$ km, while the much smaller solar splitting $\Delta m^2_{21} \simeq 7.4 \times 10^{-5}$ eV^2 pushes its first maximum to $L/E \sim 1.7 \times 10^4$ km/GeV, i.e. hundreds of kilometres for reactor energies—the regime of medium-baseline reactor experiments. The hierarchy of baselines is nothing but the hierarchy of Δm^2 read through Eq. (32.222).

Three-flavor generalization and leptonic CP violation. With all three flavors the amplitude $A(\nu_\alpha \to \nu_\beta) = \sum_i (U_{\rm PMNS})_{\beta i}\,(U_{\rm PMNS})^*_{\alpha i}\,e^{-iE_i t}$ generalizes Eq. (32.219), and squaring it gives

$$P(\nu_\alpha \to \nu_\beta) = \delta_{\alpha\beta} - 4\sum_{i>j} \mathrm{Re}[U_{\beta i}U^*_{\alpha i}U^*_{\beta j}U_{\alpha j}]\,\sin^2\frac{\Delta m^2_{ij}L}{4E} + 2\sum_{i>j} \mathrm{Im}[U_{\beta i}U^*_{\alpha i}U^*_{\beta j}U_{\alpha j}]\,\sin\frac{\Delta m^2_{ij}L}{2E}, \tag{32.224}$$

with $U \equiv U_{\rm PMNS}$. The first line is the CP-even part, reducing to Eq. (32.221) in the two-flavor limit; the second line, odd under $L \to -L$ and proportional to the imaginary part of the rephasing-invariant quartic, is the CP-violating part. It distinguishes $P(\nu_\alpha \to \nu_\beta)$ from $P(\bar\nu_\alpha \to \bar\nu_\beta)$ (the antineutrino probability uses U^*, flipping the sign of the Im term) and is the leptonic mirror of the CKM CP violation developed earlier in this chapter. Its strength is governed by the leptonic Jarlskog invariant, the exact analogue of the quark J of [284],

$$J_{\rm lept} = \mathrm{Im}[U_{e1}U_{\mu 2}U^*_{e2}U^*_{\mu 1}] = \tfrac{1}{8}\,\sin 2\theta_{12}\,\sin 2\theta_{23}\,\sin 2\theta_{13}\,\cos\theta_{13}\,\sin\delta, \tag{32.225}$$

which is manifestly rephasing-invariant and depends only on the Dirac phase δ, not on the Majorana phases—confirming once more that the Majorana phases drop out of oscillations (they multiply the columns of U by phases that cancel in every quartic $U_{\beta i}U^*_{\alpha i}U^*_{\beta j}U_{\alpha j}$). Leptonic CP violation in oscillations is observable if and only if all three angles and $\sin\delta$ are nonzero, exactly the three-generation, nonzero-phase, nondegenerate-mass conditions packaged by Theorem 32.16.

32.7.6 Experimental status and the open frontier

It remains to be honest about what is and is not known. The following is the global-fit picture as of the mid-2020s, from the three-flavor analyses [180] and the particle-data compilation [382]; the reader should treat every number as provisional and dated, and consult the live fits for current values.

What is established. Three-flavor oscillation is confirmed in solar, atmospheric, reactor, and accelerator data, and the mixing is *large*—strikingly unlike the small, hierarchical CKM angles. Numerically $\sin^2\theta_{12} \approx 0.30$ ($\theta_{12} \approx 33^\circ$, the solar angle), $\sin^2\theta_{23} \approx 0.45$–$0.57$ ($\theta_{23} \approx 49^\circ$, near maximal, the atmospheric angle), and $\sin^2\theta_{13} \approx 0.022$ ($\theta_{13} \approx 8.6^\circ$, the smallest, fixed by reactor experiments). Two independent splittings are measured: the solar $\Delta m^2_{21} \approx 7.4\times10^{-5}$ eV^2 (sign known to be positive) and the atmospheric $|\Delta m^2_{31}| \approx 2.5\times10^{-3}$ eV^2. That the leptonic mixing is order-one while the quark mixing is a small perturbation around the identity is one of the sharpest unexplained contrasts in flavor physics.

What is unknown. (i) The *mass ordering*: the sign of Δm^2_{31} is undetermined, leaving open whether the spectrum is "normal" ($m_1 < m_2 < m_3$) or "inverted"

($m_3 < m_1 < m_2$). (ii) The *absolute mass scale*: oscillations fix only differences of squares, so the lightest neutrino mass is unmeasured; the present bounds come from kinematics (tritium endpoint, $m_\beta \lesssim 0.8$ eV) and cosmology ($\sum_i m_i \lesssim 0.1$–0.12 eV), both upper limits. (iii) The *Dirac vs. Majorana* nature: whether the neutrino is its own antiparticle is undecided, and the decisive probe—neutrinoless double beta decay, sensitive to $m_{\beta\beta}$ and hence to the Majorana phases—has so far yielded only null results and limits. (iv) The *Dirac CP phase* δ: long-baseline data show a mild preference for a CP-violating value (and disfavor $\delta = 0, \pi$ at modest significance), but δ is not yet established; whether leptonic CP violation exists at all is an open experimental question. The octant of θ_{23} (whether it lies below or above 45°) is likewise unsettled. Each of these is a live target of the running and planned programme, and none should be stated as known.

Caution

Two facts are routinely conflated and must be kept distinct. First, oscillation is blind to both the absolute mass scale and the Majorana phases; measuring oscillations to arbitrary precision can never tell us whether the neutrino is Dirac or Majorana, nor what the lightest mass is. Those questions require non-oscillation probes (kinematic endpoints, cosmology, and lepton-number-violating decays). Second, the seesaw *predicts* Majorana neutrinos, but a confirmed Majorana nature would not by itself confirm the seesaw scale: Theorem 32.34 showed that m_ν fixes only m_D^2/M. The mass mechanism and the mass scale are separately undetermined by present data.

This completes the parallel that opened the section. The flavor logic of the quark sector—gauge-blind generations, misaligned Yukawa matrices, a unitary mixing matrix in the charged current, CP violation from an irreducible phase—repeats for the leptons once neutrinos are given mass, with the one genuinely new feature that Majorana masses forbid the rephasings that would otherwise remove two of the phases. The Standard Model's only established CP violation remains the single CKM phase, and it is a quantitative fact—worked through the unitarity triangle of the previous section—that this source is far too small to generate the observed cosmic baryon asymmetry through the conditions identified by [425]; a leptonic Dirac phase, or the lepton-number violation of the seesaw, may yet supply what is missing. That open question, together with the seesaw model-building only sketched here, is taken up in Chapter 35. With both the quark and lepton flavor sectors now in hand—their mixing matrices, their CP structure, and their explicitly catalogued unknowns—we hand the complete flavor and CP picture to Chapter 33, which assembles the full Standard Model and identifies the accidental baryon- and lepton-number symmetries that this section has shown the neutrino mass term to threaten.

Further reading

The original three-generation argument for CP violation is Kobayashi and Maskawa [298], building on the two-generation mixing of Cabibbo [101]; the GIM mechanism and its prediction of charm is Glashow, Iliopoulos, and Maiani [223]. The invariant measure of CP violation is developed by Jarlskog [284], and the practically convenient expansion of the CKM matrix in powers of the Cabibbo angle is due to Wolfenstein [526]. The definitive textbook treatment of CP violation, the unitarity triangle, and neutral-meson phenomenology is Branco, Lavoura, and Silva [89]; the flavor sectors are also developed, with complementary emphasis, in Peskin and Schroeder [396], Schwartz [432], and Weinberg [500]. For the leptonic side, the original mixing proposals are Pontecorvo [408] and Maki, Nakagawa, and Sakata [330]; the current experimental status of masses, mixing angles, and the unitarity-triangle fit is compiled by the Particle Data Group [382], with the global neutrino-oscillation analysis in [180].

Part IX

The Standard Model

Chapter 33

The Complete Standard Model

Three times in this book a gauge theory has been built, and three times it was built alone. Quantum chromodynamics arrived as the non-abelian endpoint of Yang–Mills theory — a self-interacting $\mathrm{SU}(3)_c$ connection whose asymptotic freedom was computed in full (Chapter 25, Chapter 27) and confronted with experiment (Chapter 28). The electroweak interaction arrived as the Glashow–Weinberg–Salam construction on $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, broken by the Higgs mechanism to the electromagnetism of QED (Chapter 30, Chapter 31). The flavor structure arrived through the Yukawa couplings, the CKM matrix, and the Jarlskog measure of CP violation (Chapter 32). Each development was self-contained, and each was written as though the others did not exist. They do exist, and they are not three theories but one. This chapter assembles them into a single Lagrangian on the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and then — this is the work that distinguishes a synthesis from a mere concatenation — it shows how much of that Lagrangian's structure is forced rather than chosen, counts exactly how many numbers remain free, and separates, honestly and with dates, what the resulting theory explains from what it only records.

The raw material is imported wholesale from Parts VII and VIII. The gauge-boson and fermion mass spectrum, the weak mixing angle, custodial symmetry, the Gell-Mann–Nishijima relation $Q = T^3 + Y$, and the one-generation electroweak parameter tally are taken from Chapter 31; the Yukawa sector, the mass-versus-flavor bases, the CKM parametrization, the GIM mechanism, the Jarlskog invariant, the seesaw, and the flavor parameter tally from Chapter 32; the color factors and the hadronic phenomenology that anchor QCD to data from Chapter 28. Two structural results are used at full strength and not reproved here: that an uncanceled anomaly in a *gauged* current is fatal to unitarity and renormalizability, established through the triangle diagram and the Fujikawa measure in Chapter 36 and Chapter 37; and that a renormalizable theory is predictive from a finite set of inputs, the content of the renormalization program of Chapter 22 and the renormalization-group solution of Chapter 24. The

reference tabulation of the field content, parameters, and full Lagrangian lives in Chapter F, and the metric, gauge-coupling sign, and group conventions are the book-wide canon of Chapter A, Chapter B, and Chapter C.

The chapter runs as a single deductive chain, each section handing the next its open question. Section 33.1 draws up the inventory: the one gauge group all matter transforms under, the convention by which the chiral content is listed, and the complete three-generation master table with every quantum number attached — no dynamics, only bookkeeping that must be exact because everything later is read off it. Section 33.2 dresses that inventory with a Lagrangian, writing the entire Standard Model as one displayed object and proving the near-uniqueness claim that elevates it from *a* theory of these fields to *the* theory of them: with the content fixed, Lorentz invariance, full $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge invariance, and renormalizability leave essentially no freedom. The remaining sections interrogate that object. Section 33.3 runs the construction backwards, showing that gauge-anomaly cancellation together with Yukawa gauge invariance forces the ragged hypercharge fractions up to one normalization, and with them the quantization of electric charge; it also records the Witten global SU(2) constraint and isolates $B - L$ as the anomaly-free global direction. Section 33.4 reads off what the rigid Lagrangian gives back for free — the accidental baryon and lepton numbers — determines that $B + L$ is destroyed by the electroweak anomaly while $B - L$ survives, and exhibits the leading operators that break them: the dimension-five Weinberg operator behind Majorana neutrino mass and the dimension-six operators behind proton decay, the theory's first windows onto higher scales. Section 33.5 restores the one renormalizable operator deliberately omitted in the assembly, the QCD θ term, proves the apparent paradox that it is a total derivative yet physical, isolates the rephasing-invariant angle $\bar{\theta}$, and states the strong-CP problem. Section 33.6 performs the census the whole assembly has been building toward, deriving the famous nineteen parameters as a theorem rather than folklore and asking how many of them the theory explains. Finally Section 33.7 takes stock: it separates the genuine explanations from the parametrizations, hands the dated list of what the Standard Model does not address forward to the rest of Part IX, and closes the gauge threads by exhibiting QED and Yang–Mills as one Lagrangian.

A brief word on scope and rigor. Everything load-bearing is carried in full on the page: the master table and its degree-of-freedom count, the term-by-term assembly and the near-uniqueness theorem, the anomaly system solved to the unique hypercharge solution and the charge-quantization corollary, the total-derivative computation for $G\widetilde{G}$ and the rephasing invariance of $\bar{\theta}$, the flavor-symmetry stabilizer count that recovers nineteen, and the strong-CP, Weinberg-operator, and proton-decay estimates that fix the frontier scales. No new dynamics enters; the chapter's contribution is structural, and its two benchmark theories, QED and Yang–Mills, are not advanced with new computation but *merged* — the classical-structure layer of both ends here, to reappear only in the spinor-helicity

language of Part XVI (Chapter 58, Chapter 59, Chapter 61). With the structure assembled, Chapter 34 turns it to account against precision electroweak data and the discovered Higgs, and Chapter 35 takes up the frontier this chapter is careful to leave open.

33.1 The Gauge Group and the Complete Matter Content

Across Parts VII and VIII the reader built three theories under three separate roofs. Quantum chromodynamics arrived as Yang–Mills theory: a non-abelian gauge field on SU(3), its asymptotic freedom computed in full (Sections 25.1 and 27.5). The electroweak interaction arrived as the Glashow–Weinberg–Salam construction on $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with the Higgs mechanism breaking it to the electromagnetism of QED. The flavor structure arrived through the Yukawa couplings and the CKM matrix. Each was a self-contained development, and each was written as though the others did not exist. They do exist, and they are not three theories but one. This chapter assembles them. The present section does the prerequisite bookkeeping: it fixes the single gauge group that all the matter transforms under, settles the convention by which we list that matter, and writes the complete three-generation field content into one master table with every quantum number attached. No dynamics is introduced here—no kinetic terms, no interactions, no symmetry breaking. We are drawing up the inventory that the next section will dress with a Lagrangian. Everything that follows in the chapter, from the parameter count to the anomaly cancellation, is read off this inventory, so it must be exact.

33.1.1 The gauge group

The Standard Model is a gauge theory whose local symmetry group is the direct product

$$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y, \tag{33.1}$$

always written in this factor order. This is the group of Theorem F.1; we adopt it as the spine of the chapter and do not rederive its physical content, which was established factor by factor in the parts just named. What matters here is to be completely explicit about what each factor is, how many independent symmetries it contributes, and how the two benchmark theories sit inside it.

The first factor, $\mathrm{SU}(3)_c$, is the color group of Yang–Mills theory. Its eight Hermitian generators are the $T^a = \lambda^a/2$ of the fundamental representation, with the Gell-Mann matrices λ^a and the structure constants f^{abc} collected in Appendix B; the gauge principle that promotes this global SU(3) to a local symmetry, together with the curvature $G^A_{\mu\nu}$ and the gluon self-coupling it forces,

is exactly the construction of Section 25.1. The subscript c records that this is *color*, the charge that quarks and gluons carry and that leptons do not.

The second factor, $\mathrm{SU}(2)_L$, is weak isospin. Its three generators are $T^i = \tau^i/2$ in the doublet representation, τ^i the Pauli matrices. The subscript L is the single most consequential letter in Eq. (33.1): weak isospin acts only on the left-handed chiral components of the fermions, and on the right-handed components trivially. This is parity violation written into the gauge group itself, and it is the reason the field content of the next subsection cannot be organized by Dirac fields but must be organized by chirality.

The third factor, $\mathrm{U}(1)_Y$, is weak hypercharge: a single abelian charge Y, real-valued in the abstract but—as the anomaly analysis of Chapter 37 will show and as we will use in Section 33.1.3—quantized to the rationals once consistency is imposed. We normalize Y by the Gell-Mann–Nishijima relation in the convention of Eq. (31.20), so that the electric charge is $Q = T^3 + Y$ with no factor of two; the lepton doublet then carries $Y = -\frac{1}{2}$.

Physical Insight

The two gauge theories the book has carried since Part II converge in Eq. (33.1). Yang–Mills theory *is* the first factor: $\mathrm{SU}(3)_c$ is the Yang–Mills theory whose beta function we computed in Section 27.5, now appearing verbatim as the strong interaction. QED is not a separate factor at all; its $\mathrm{U}(1)_{\mathrm{em}}$ is the diagonal subgroup of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ left unbroken by the Higgs vacuum, generated by $Q = T^3 + Y$. The electron's electromagnetism, which we quantized, renormalized, and pushed to one-loop precision for the anomalous moment, is here a derived consequence of the larger electroweak symmetry rather than a postulate. This is the chapter in which the two threads stop being parallel and become one theory.

Rank, dimension, and the count of gauge bosons. The dimension of Eq. (33.1)—the number of independent local symmetry generators, hence the number of gauge bosons—is the sum over factors,

$$\dim \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y = \underbrace{(3^2-1)}_{\mathrm{SU}(3)_c} + \underbrace{(2^2-1)}_{\mathrm{SU}(2)_L} + \underbrace{1}_{\mathrm{U}(1)_Y} = 8+3+1 = 12. \tag{33.2}$$

There are therefore exactly twelve gauge fields: eight gluons G^A_μ $(A = 1, \ldots, 8)$, three weak bosons W^i_μ $(i = 1, 2, 3)$, and one hypercharge boson B_μ. After electroweak symmetry breaking the four electroweak fields $W^{1,2,3}_\mu, B_\mu$ recombine into the massive $W^\pm, Z$ and the massless photon, but the *count* is fixed here, before any breaking, by Eq. (33.2). The rank—the maximal number of simultaneously diagonalizable, mutually commuting generators, equivalently the number

of independent additive charges a state can carry—is the sum of the factor ranks,

$$\operatorname{rank} \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y = \underbrace{2}_{\mathrm{SU}(3)_c} + \underbrace{1}_{\mathrm{SU}(2)_L} + \underbrace{1}_{\mathrm{U}(1)_Y} = 4. \tag{33.3}$$

The four commuting charges may be taken as the two color charges (the diagonal generators T_c^3 and T_c^8 of $\mathrm{SU}(3)_c$), the third component of weak isospin T^3, and the hypercharge Y. Every Standard Model state is labelled by these four numbers (plus its color and isospin multiplet structure and its Poincaré quantum numbers); in particular the electric charge $Q = T^3 + Y$ is a function of two of them.

One coupling per simple factor. Gauge invariance assigns an independent coupling constant to each simple or abelian factor and no more, because the gauge kinetic term of each factor is separately invariant and there is no group element mixing the factors. The covariant derivative acting on a field in the representation (R_3, R_2, Y) is therefore

$$D_\mu = \partial_\mu - i g_s \, G_\mu^A \, T_{(3)}^A - i g \, W_\mu^i \, T_{(2)}^i - i g' \, Y \, B_\mu, \tag{33.4}$$

with three couplings g_s, g, g', the same minus-sign convention $D_\mu = \partial_\mu - i g A_\mu^a T^a$ fixed in Section 25.1 and $T_{(3)}^A, T_{(2)}^i$ the generators in whatever color and weak-isospin representations the field carries (they act as zero on a singlet). For a field that is a color or weak singlet the corresponding term simply drops; for the hypercharge term Y is the number from the field's multiplet, so even $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L$ singlets feel Eq. (33.4) if $Y \neq 0$. That three numbers g_s, g, g' exhaust the gauge couplings is the first stage of the parameter count completed later in the chapter.

33.1.2 All fermions as left-handed Weyl fields

Because $\mathrm{SU}(2)_L$ acts on left- and right-handed components differently, a Dirac field is not a representation of Eq. (33.1): its two chiral halves live in different representations, and lumping them together hides exactly the structure we need to track. The clean organizing principle is to decompose every Dirac field into chiral Weyl components using the projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ of Chapter 3, and to assign gauge representations to the Weyl fields, not to the Dirac fields. A four-component field Ψ splits as $\psi_L = P_L \Psi$, $\psi_R = P_R \Psi$, and each piece carries its own triple (R_3, R_2, Y).

This already suffices to list the content, and it is the listing that the Lagrangian of Section 33.1.3 will use. But for the consistency checks that occupy the rest of the chapter—above all the anomaly cancellation of Chapter 37, which is a statement about a sum over chiral fermions of one handedness—it is cleaner still to put *every* fermion on the same footing, as a left-handed Weyl field. The device for doing so is charge conjugation. As established in Section 11.7.2, charge

conjugation exchanges a particle for its antiparticle and, crucially, maps a right-handed field to a left-handed one: the charge conjugate of a right-handed Weyl field ψ_R is a left-handed Weyl field

$$\psi_R^c \equiv (\psi_R)^c = C\,\bar{\psi}_R^{\mathsf{T}}, \qquad \psi_R \sim (R_3, R_2, Y) \implies \psi_R^c \sim (\bar{R}_3, \bar{R}_2, -Y), \tag{33.5}$$

because conjugation sends every representation to its complex conjugate and flips the sign of every additive charge. (For SU(2) the bar is cosmetic, since $\mathbf{2} \cong \overline{\mathbf{2}}$ is pseudoreal; for SU(3) it is real content, since $\mathbf{3} \ncong \bar{\mathbf{3}}$, and for $\mathrm{U}(1)_Y$ it is the sign flip $Y \to -Y$.) Using Eq. (33.5) to trade each right-handed field for the charge-conjugate left-handed field, the entire fermion content of one generation becomes five left-handed Weyl multiplets and nothing else. The physics is unchanged—a left-handed antiparticle is the same quantum as a right-handed particle—but the bookkeeping is now uniform, and the chirality of the theory is laid bare: if the list of left-handed multiplets is not symmetric under conjugation (it is not, as we verify in Section 33.1.4), the theory is chiral, and that asymmetry is the obstruction to bare fermion masses and the seed of the gauge anomalies.

We will therefore carry two presentations side by side. The first is the *physical-handedness* table, in which fields appear with the handedness they have in the Lagrangian; this is the table one reads to write kinetic and interaction terms, and it is the table of Table 31.1 and Table F.1. The second is the *all-left-handed* recasting obtained by Eq. (33.5); this is the list one sums over for anomalies and the cleanest object for counting degrees of freedom. They contain identical information.

33.1.3 The master table of one generation

We now write the content. One generation consists of five fermion multiplets and, shared across all generations, a single scalar. In the physical-handedness presentation the fermions are

$$\begin{aligned} Q_L &= \begin{pmatrix} u_L \\ d_L \end{pmatrix} \sim (\mathbf{3}, \mathbf{2}, +\tfrac{1}{6}), & u_R &\sim (\mathbf{3}, \mathbf{1}, +\tfrac{2}{3}), & d_R &\sim (\mathbf{3}, \mathbf{1}, -\tfrac{1}{3}), \\ L_L &= \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} \sim (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), & e_R &\sim (\mathbf{1}, \mathbf{1}, -1), \end{aligned} \tag{33.6}$$

and the scalar is the complex Higgs doublet

$$H = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \sim (\mathbf{1}, \mathbf{2}, +\tfrac{1}{2}). \tag{33.7}$$

The triple after each field is its representation (R_3, R_2, Y) under the factors of Eq. (33.1) in order. Read it as a sentence: Q_L is a color triplet, a weak doublet, of hypercharge $+\frac{1}{6}$; u_R is a color triplet, weak singlet, hypercharge $+\frac{2}{3}$; and so on.

There is no right-handed neutrino in Eq. (33.6)—the minimal Standard Model contains ν_L but not ν_R—a deliberate omission whose consequence (a strictly massless neutrino at the renormalizable level) we return to below and which the flavor chapter and Chapter 35 take up in earnest.

Table 33.1 collects Eqs. (33.6) to (33.7) in tabular form, with the third weak-isospin component T^3 of each multiplet member and the resulting electric charge $Q = T^3 + Y$ displayed so that the Gell-Mann–Nishijima relation can be checked at a glance.

Table 33.1: The master table of one Standard Model generation, in the physical-handedness presentation. The representation triple is (R_3, R_2, Y) under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ in factor order; T^3 is the third weak-isospin component of each member, $+\frac{1}{2}$ for the upper and $-\frac{1}{2}$ for the lower entry of a doublet and 0 for a singlet. The final column is *computed*, $Q = T^3 + Y$, not assigned. The content is replicated identically for the three generations (u, d, e, ν_e), (c, s, μ, ν_μ), (t, b, τ, ν_τ). Compare Tables F.1 and 31.1.

Field	(R_3, R_2)	T^3	Y	$Q = T^3 + Y$
u_L	$(\mathbf{3}, \mathbf{2})$ Q_L	$+\frac{1}{2}$	$+\frac{1}{6}$	$+\frac{2}{3}$
d_L		$-\frac{1}{2}$	$+\frac{1}{6}$	$-\frac{1}{3}$
u_R	$(\mathbf{3}, \mathbf{1})$	0	$+\frac{2}{3}$	$+\frac{2}{3}$
d_R	$(\mathbf{3}, \mathbf{1})$	0	$-\frac{1}{3}$	$-\frac{1}{3}$
ν_L	$(\mathbf{1}, \mathbf{2})$ L_L	$+\frac{1}{2}$	$-\frac{1}{2}$	0
e_L		$-\frac{1}{2}$	$-\frac{1}{2}$	-1
e_R	$(\mathbf{1}, \mathbf{1})$	0	-1	-1
ϕ^+	$(\mathbf{1}, \mathbf{2})$ H	$+\frac{1}{2}$	$+\frac{1}{2}$	$+1$
ϕ^0		$-\frac{1}{2}$	$+\frac{1}{2}$	0

The all-left-handed recasting. Applying Eq. (33.5) to the three right-handed fermions of Eq. (33.6) replaces them by left-handed Weyl fields in the conjugate representations with opposite hypercharge,

$$u_R \to u_R^c \sim (\bar{\mathbf{3}}, \mathbf{1}, -\tfrac{2}{3}), \qquad d_R \to d_R^c \sim (\bar{\mathbf{3}}, \mathbf{1}, +\tfrac{1}{3}), \qquad e_R \to e_R^c \sim (\mathbf{1}, \mathbf{1}, +1), \tag{33.8}$$

so that one generation is the five *left-handed* multiplets

$$Q_L \sim (\mathbf{3}, \mathbf{2}, \tfrac{1}{6}), \quad u_R^c \sim (\bar{\mathbf{3}}, \mathbf{1}, -\tfrac{2}{3}), \quad d_R^c \sim (\bar{\mathbf{3}}, \mathbf{1}, \tfrac{1}{3}), \quad L_L \sim (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \quad e_R^c \sim (\mathbf{1}, \mathbf{1}, 1). \tag{33.9}$$

Table 33.2 records this list. It is the object the anomaly sums of Chapter 37 run over, and we appeal to it again in Section 33.1.4.

Table 33.2: The all-left-handed Weyl recasting of one generation, obtained from Table 33.1 by Eq. (33.5). Every fermion is now a left-handed Weyl multiplet; the right-handed singlets have become left-handed fields in conjugate color representations with flipped hypercharge. This is the presentation used for the anomaly bookkeeping of Chapter 37 and the degree-of-freedom count of Eq. (33.11).

Left-handed Weyl field	R_3	R_2	Y
Q_L	$\mathbf{3}$	$\mathbf{2}$	$+\frac{1}{6}$
u_R^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$-\frac{2}{3}$
d_R^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$+\frac{1}{3}$
L_L	$\mathbf{1}$	$\mathbf{2}$	$-\frac{1}{2}$
e_R^c	$\mathbf{1}$	$\mathbf{1}$	$+1$

Three generations. The structure of Eqs. (33.6) to (33.9) is replicated exactly three times. Nature furnishes three *generations* (families) of fermions, identical in every gauge quantum number and differing only in mass:

$$\begin{pmatrix} u \\ d \\ e \\ \nu_e \end{pmatrix}, \qquad \begin{pmatrix} c \\ s \\ \mu \\ \nu_\mu \end{pmatrix}, \qquad \begin{pmatrix} t \\ b \\ \tau \\ \nu_\tau \end{pmatrix}, \tag{33.10}$$

each row of Table 33.1 appearing once per column of Eq. (33.10). The replication is a brute experimental fact, not a consequence of the gauge group: $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ would be perfectly consistent with one generation, and the question of why there are three is firmly on the frontier side of the summary. The Higgs doublet H of Eq. (33.7), by contrast, is *not* replicated; the minimal Standard Model has exactly one, shared by all three generations. The generation index is invisible to every term written in this chapter except the Yukawa couplings, where it becomes the flavor and CP structure of Chapter 32.

Degree-of-freedom count. With the all-left-handed list Eq. (33.9) the fermion content of one generation is counted by summing the products of the color and weak dimensions, since each independent gauge component is one Weyl field:

$$\underbrace{3\cdot 2}_{Q_L} + \underbrace{3}_{u_R^c} + \underbrace{3}_{d_R^c} + \underbrace{1\cdot 2}_{L_L} + \underbrace{1}_{e_R^c} = 6+3+3+2+1 = 15 \tag{33.11}$$

left-handed Weyl fields per generation—fifteen, with no ν_R; including a right-handed neutrino would make sixteen and complete a SU(5) or SO(10) multiplet, a fact Chapter 35 exploits. Over the three generations of Eq. (33.10) that is $3 \times 15 = 45$ Weyl fields. The scalar sector is one complex $\mathrm{SU}(2)_L$ doublet,

Eq. (33.7), hence $2 \times 2 = 4$ real scalar fields; three of these are eaten by $W^{\pm}, Z$ in the Higgs mechanism and one survives as the physical Higgs boson, but again the count 4 is fixed here. This 15 (or 45) and this 4 are the matter inventory of the Standard Model; nothing else is added.

We now verify, member by member, that the hypercharges tabulated above are not free choices but are forced by the observed electric charges through $Q = T^3 + Y$—the one place in this section where an imported relation does genuine work.

Worked Example 33.1 (The tabulated hypercharges reproduce every physical charge)**.** The Gell-Mann–Nishijima relation Eq. (31.20),

$$Q = T^3 + Y, \tag{33.12}$$

is imported, not reproved (it was derived in Chapter 31 as the statement that the unbroken electromagnetic generator is the particular combination of T^3 and Y that annihilates the Higgs vacuum). We use it in two directions. First, as a *consistency requirement*: Y multiplies the identity on an $\mathrm{SU}(2)_L$ multiplet, so it cannot distinguish a doublet's two members; hence $Y = Q - T^3$ must return the same number for the upper and lower entry of any doublet, or the assignment is inconsistent. Second, as a *prediction*: with Y fixed, Eq. (33.12) outputs the electric charge of every field, which must match experiment. We run both checks against Table 33.1; the input charges are the measured values $Q_\nu = 0$, $Q_e = -1$, $Q_u = +\frac{2}{3}$, $Q_d = -\frac{1}{3}$ in units of $|e|$.

Quark doublet Q_L, $Y = +\frac{1}{6}$. Upper member u_L: $Q = T^3 + Y = +\frac{1}{2} + \frac{1}{6} = +\frac{2}{3}$. Lower member d_L: $Q = -\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$. The single hypercharge $+\frac{1}{6}$ produces both observed quark charges, and the two members would require $Y = \frac{2}{3} - \frac{1}{2} = \frac{1}{6}$ and $Y = -\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$ respectively—identical, so the doublet assignment is consistent.

Lepton doublet L_L, $Y = -\frac{1}{2}$. Upper member ν_L: $Q = +\frac{1}{2} - \frac{1}{2} = 0$. Lower member e_L: $Q = -\frac{1}{2} - \frac{1}{2} = -1$. Again one hypercharge $-\frac{1}{2}$ yields both charges, and the consistency check $Y = 0 - \frac{1}{2} = -\frac{1}{2}$, $Y = -1 + \frac{1}{2} = -\frac{1}{2}$ agrees.

The singlets, $T^3 = 0$. For a weak singlet Eq. (33.12) collapses to $Q = Y$. Thus u_R has $Y = Q_u = +\frac{2}{3}$, d_R has $Y = Q_d = -\frac{1}{3}$, and e_R has $Y = Q_e = -1$, exactly the singlet hypercharges of Table 33.1. The right-handed and left-handed components of a given charged fermion carry *different* hypercharges (e.g. u_L has $Y = \frac{1}{6}$ but u_R has $Y = \frac{2}{3}$) precisely because they sit in different weak-isospin representations—the chiral structure made quantitative.

The Higgs doublet, $Y = +\frac{1}{2}$. Upper member ϕ^+: $Q = +\frac{1}{2} + \frac{1}{2} = +1$. Lower member ϕ^0: $Q = -\frac{1}{2} + \frac{1}{2} = 0$. The neutral component ϕ^0 is what acquires a vacuum value; that it carries $Q = 0$ is mandatory, for otherwise the vacuum would break electromagnetism, contradicting the masslessness of the photon. The hypercharge $+\frac{1}{2}$ is fixed by this requirement together with the demand (Section 31.4) that H couple in a gauge-invariant Yukawa term.

Every entry of the computed Q column in Table 33.1 is thereby reproduced. This is the same arithmetic carried out from the electroweak side in Theorem 31.5; the agreement of the two routes is the first nontrivial confirmation that the group-theoretic skeleton of Eq. (33.1) matches the observed particle charges.

The content of Table 33.1 can be seen at a glance in the (T^3, Y) plane, Fig. 33.1, where lines of constant electric charge are the diagonals $Y = Q - T^3$ of Eq. (33.12). The two members of a weak doublet share a horizontal level (the same Y) but sit on different charge diagonals; the weak singlets sit on the axis $T^3 = 0$; and the unbroken electric charge is the projection onto the $Q = T^3 + Y$ direction. The figure is the geometric statement that $\mathrm{U}(1)_{\mathrm{em}}$ is the diagonal of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$.

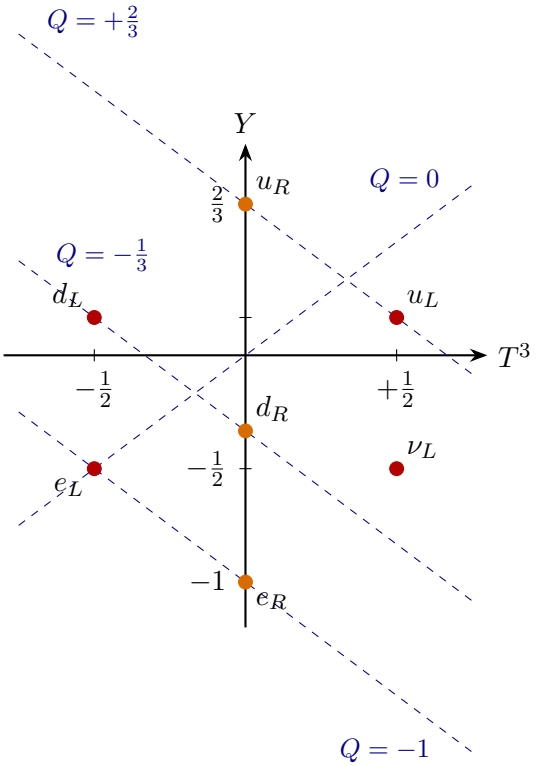


Figure 33.1: The physical-handedness fermions of one generation in the (T^3, Y) plane (the vertical axis is scaled for legibility). Red discs are weak-doublet members $(Q_L,\ L_L)$, orange discs the weak singlets (u_R, d_R, e_R) on the $T^3 = 0$ axis. Dashed blue lines are the constant-charge diagonals $Y = Q - T^3$ of the Gell-Mann–Nishijima relation Eq. (33.12); a field's electric charge is the diagonal it lands on. The two members of a doublet share a level (one Y) but lie on different diagonals, while u_L and u_R lie on the *same* diagonal $Q = +\frac{2}{3}$ despite different (T^3, Y)—the chiral split that forbids a bare mass.

33.1.4 The content is chiral: no bare masses

The all-left-handed list Eq. (33.9) makes one structural fact immediate, and it is the fact that organizes the rest of the chapter's account of mass. A Dirac mass term $m\,\bar{\Psi}\Psi = m(\bar{\psi}_L\psi_R + \bar{\psi}_R\psi_L)$ pairs a left-handed Weyl field with a right-handed one; rewritten with Eq. (33.5) it is a term coupling two left-handed Weyl fields whose representations are mutually conjugate. Such a term is gauge invariant if and only if the two fields it couples sit in conjugate representations of *every* factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, i.e. if and only if the left-handed content contains a field and its own conjugate. A theory in which no left-handed multiplet appears together with its conjugate is called *chiral*, and in a chiral theory there is no gauge-invariant fermion bilinear at all: no bare mass term can be written.

Inspect Table 33.2. The candidate partners for a mass term are pairs whose color and weak labels are conjugate and whose hypercharges sum to zero. Run through them. The quark doublet $Q_L \sim (\mathbf{3},\mathbf{2},+\frac{1}{6})$ would need a partner in $(\bar{\mathbf{3}},\mathbf{2},-\frac{1}{6})$; the list contains no weak doublet of that color and hypercharge—u^c_R and d^c_R are weak *singlets*, and their hypercharges are $-\frac{2}{3}$ and $+\frac{1}{3}$, neither $-\frac{1}{6}$. The lepton doublet $L_L \sim (\mathbf{1},\mathbf{2},-\frac{1}{2})$ would need $(\mathbf{1},\mathbf{2},+\frac{1}{2})$; no such fermion exists (that triple is the Higgs, a boson, not available for a fermion mass). The singlet $e^c_R \sim (\mathbf{1},\mathbf{1},+1)$ would need $(\mathbf{1},\mathbf{1},-1)$; absent. The colored singlets $u^c_R \sim (\bar{\mathbf{3}},\mathbf{1},-\frac{2}{3})$ and $d^c_R \sim (\bar{\mathbf{3}},\mathbf{1},+\frac{1}{3})$ would need $(\mathbf{3},\mathbf{1},+\frac{2}{3})$ and $(\mathbf{3},\mathbf{1},-\frac{1}{3})$, which are color triplets but again carry the wrong weak/hypercharge to be matched by anything in the list. No two entries of Table 33.2 are conjugate. The Standard Model fermion content is genuinely chiral, and *not a single fermion mass term is allowed by the gauge symmetry.*

This is not a defect to be patched; it is the central design feature. The masses of all quarks and charged leptons must therefore arise indirectly, from the only gauge-invariant route the content permits: a Yukawa coupling of two fermions to the Higgs doublet, which becomes a mass once ϕ^0 acquires a vacuum value. That the triple $(\mathbf{1},\mathbf{2},+\frac{1}{2})$ of Eq. (33.7) is exactly what is needed to soak up the hypercharge mismatch in, say, $\bar{Q}_L H d_R \sim (\bar{\mathbf{3}},\mathbf{2},-\frac{1}{6}) \otimes (\mathbf{1},\mathbf{2},+\frac{1}{2}) \otimes (\mathbf{3},\mathbf{1},-\frac{1}{3})$ (a gauge singlet, as the next section verifies) is no accident: the Higgs hypercharge was assigned precisely so that these couplings exist. The full Yukawa construction, the rotation to the mass basis, and the resulting CKM structure are the business of Section 31.4 and Chapter 32; we note here only that the *necessity* of that route is already forced by the chirality of Table 33.2. The absence of ν_R has the matching consequence: with no right-handed neutrino in the list, there is no partner to pair ν_L into a Dirac mass, and no $\mathrm{SU}(2)_L$-singlet Higgs combination to give it a renormalizable mass either, so the minimal Standard Model neutrino is exactly massless—the loose thread that Chapter 35 pulls on.

With the gauge group fixed by Eq. (33.1), the full matter content catalogued in Tables 33.1 and 33.2 and replicated over the three generations of Eq. (33.10), and

the chiral structure that forbids bare masses established, we have the complete representation-labelled field list. The next section dresses precisely this list with kinetic, gauge, Higgs, and Yukawa terms, and argues that what results is the most general renormalizable Lagrangian the content allows.

33.2 Assembling the Complete Standard Model Lagrangian

The previous section did the bookkeeping: it fixed the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and laid out, in one master table, the complete three-generation matter content with every quantum number attached — the chiral fermion multiplets, their representations and hypercharges, and the single complex scalar doublet (Section 33.1, Table 33.1). That table is a list of fields and the way the gauge group acts on them. It is not yet a theory. A theory is a Lagrangian: a recipe that says which local operators built from those fields are summed, with which couplings, to give the dynamics. The purpose of this section is to write that Lagrangian down — the entire Standard Model, as one displayed object, term by term — and then to prove the claim that elevates it from *a* theory of these fields to *the* theory of these fields: with the matter content fixed, the requirements of Lorentz invariance, gauge invariance under all of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and renormalizability (operator dimension ≤ 4) leave essentially no freedom. Every operator those three principles permit is already present; the only renormalizable term we have not yet written is a single topological one, the θ term, which we flag now and treat in its own section.

The work here is almost entirely assembly. We do not introduce a single new dynamical principle. Each of the four pieces we are about to combine was built, in full, in an earlier chapter: the non-abelian gauge sector and its color content in Chapters 25 to 27; the electroweak gauge, fermion, and Higgs sectors for one generation in Chapters 30 and 31; and the flavor structure of the Yukawa couplings, with their 3×3 coupling matrices, in Chapter 32. What this section owes is therefore not a re-derivation but two things that only become visible once the pieces are placed side by side: first, the verification that each block, built to be invariant under *its own* group factor, remains invariant under the *full* product group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; and second, the argument that the assembled object is exhaustive — that nothing renormalizable and invariant has been left out. Throughout we keep the conventions fixed in Section 33.1: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, Hermitian generators normalized by $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the fundamental, and the covariant-derivative sign convention $D_\mu = \partial_\mu - igA^a_\mu T^a$ established for the abelian case in Chapter 8 and the non-abelian case in Chapter 25. The fully transcribed reference version of everything below, with each per-multiplet covariant derivative written out, is collected for convenience in Eqs. (F.20) and (F.52); here we develop it as the

chapter's central argument.

33.2.1 The single covariant derivative

Every gauge interaction in the Standard Model is carried by one object: the covariant derivative appropriate to the field it acts on. Because $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is a direct product of three factors, its connection is the sum of one connection per factor, each with its own coupling constant and each acting through the generators of the representation the field occupies. This is the product-group instance of the minimal-coupling prescription (Theorem 8.6) and the multi-factor connection of Theorem 25.6.

Definition 33.2 (The Standard Model covariant derivative)**.** Let ψ be any field of the theory, transforming in a representation of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ specified by the color generators T^A $(A = 1, \dots, 8)$, the weak-isospin generators T^I $(I = 1, 2, 3)$, and the hypercharge Y. Its covariant derivative is

$$D_\mu \;=\; \partial_\mu \;-\; i\, g_s\, G^A_\mu\, T^A \;-\; i\, g\, W^I_\mu\, T^I \;-\; i\, g'\, Y\, B_\mu, \tag{33.13}$$

where G^A_μ are the eight gluon fields with strong coupling g_s, W^I_μ the three weak-isospin fields with coupling g, and B_μ the single hypercharge field with coupling g'.

The single expression Eq. (33.13) encodes the entire gauge dynamics once it is specialized to each multiplet, and the specialization rule is purely representation-theoretic, requiring no new input:

- On a *color triplet* **3** the strong generators are $T^A = \lambda^A/2$ with λ^A the Gell-Mann matrices (Chapter 2, Chapter B); on a *color singlet* **1** they vanish, $T^A = 0$, and the gluon term drops out.
- On a *weak doublet* **2** the isospin generators are $T^I = \sigma^I/2$ with σ^I the Pauli matrices; on a *weak singlet* **1** they vanish, $T^I = 0$, and the W term drops out.
- The hypercharge Y is a single real number multiplying the identity — this is what *abelian* means — so *every* field, including the right-handed singlets that are blind to gluons and to the W bosons, couples to B_μ through its hypercharge.

The three couplings g_s, g, g' are mutually independent: they belong to distinct simple (or abelian) factors of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and no symmetry of the classical theory relates them. The verb "acting representation-appropriately" is therefore not a slogan but an instruction — pick the multiplet, read its (T^A, T^I, Y) off Table 33.1, and substitute into Eq. (33.13). Carrying out that substitution

for each of the five fermion multiplets and for the Higgs doublet produces the six explicit covariant derivatives transcribed in Eqs. (F.28) to (F.33); we do not repeat the six lines here, but the quark doublet, which feels all three factors at once, is exhibited in the worked check of Theorem 33.3 below because it is the sharpest test of the assembly.

33.2.2 The four-block decomposition

A renormalizable Lagrangian is a sum of local operators of mass dimension at most four, each a Lorentz scalar and a singlet of the gauge group. Organizing those operators by which fields they involve gives the Standard Model Lagrangian its standard four-block form,

$$\boxed{\mathcal{L}_{\text{SM}} \;=\; \mathcal{L}_{\text{gauge}} \;+\; \mathcal{L}_{\text{fermion}} \;+\; \mathcal{L}_{\text{Higgs}} \;+\; \mathcal{L}_{\text{Yukawa}}} \tag{33.14}$$

to which renormalizability permits exactly one further addition, the topological θ term $\mathcal{L}_\theta$, deferred to Section 33.5. We display the four blocks in turn, in each case importing the operator from the chapter that built it and stating only what the assembly adds.

The gauge block. The kinetic terms of the gauge fields are governed by their field strengths. For the non-abelian factors we import the Yang–Mills curvature of Theorem 25.11, specialized to $\mathrm{SU}(3)_c$ (structure constants f^{ABC}) and to $\mathrm{SU}(2)_L$ (structure constants ϵ^{IJK}, the case worked in Theorem 25.13); for the abelian factor the field strength is the Maxwell curl of Chapter 8:

$$G^A_{\mu\nu} = \partial_\mu G^A_\nu - \partial_\nu G^A_\mu + g_s\, f^{ABC}\, G^B_\mu G^C_\nu, \tag{33.15}$$

$$W^I_{\mu\nu} = \partial_\mu W^I_\nu - \partial_\nu W^I_\mu + g\, \epsilon^{IJK}\, W^J_\mu W^K_\nu, \tag{33.16}$$

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu. \tag{33.17}$$

Each is a curvature in the sense $F_{\mu\nu} = \frac{i}{g}[D_\mu, D_\nu]$ on the relevant representation (Eq. (25.47)). The gauge block is one Yang–Mills term Eq. (25.76) per factor,

$$\mathcal{L}_{\text{gauge}} \;=\; -\tfrac{1}{4}\, G^A_{\mu\nu} G^{A\,\mu\nu} \;-\; \tfrac{1}{4}\, W^I_{\mu\nu} W^{I\,\mu\nu} \;-\; \tfrac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \tag{33.18}$$

with all gauge indices summed. The self-interactions hidden inside $G^A_{\mu\nu} G^{A\,\mu\nu}$ and $W^I_{\mu\nu} W^{I\,\mu\nu}$ — the cubic and quartic gluon and W vertices — are exactly those whose Feynman rules were derived in Chapter 26; the abelian $B_{\mu\nu}B^{\mu\nu}$ has none, $\mathrm{U}(1)_Y$ being abelian.

The fermion block. With Eq. (33.13) in hand, each chiral multiplet contributes one gauge-covariant Dirac kinetic term $\bar\psi\, i\not{D}\,\psi$, in which $\not{D} = \gamma^\mu D_\mu$ carries that multiplet's representation data. Because left- and right-handed fields

sit in different representations they are independent fields, each with its own term; there is no operator linking them (such a term would be a mass, forbidden by Theorem 31.9). Summing the five multiplets and the generation index $i = 1, 2, 3$ (summation implied),

$$\mathcal{L}_{\text{fermion}} = \bar{Q}_{Li}\, i\not{D}\, Q_{Li} + \bar{u}_{Ri}\, i\not{D}\, u_{Ri} + \bar{d}_{Ri}\, i\not{D}\, d_{Ri} + \bar{L}_{Li}\, i\not{D}\, L_{Li} + \bar{e}_{Ri}\, i\not{D}\, e_{Ri}, \tag{33.19}$$

where each D_μ is Eq. (33.13) with the generators and hypercharge of the multiplet on which it acts. This is the electroweak fermion Lagrangian of Eq. (31.50) promoted in two ways: the color factor has been switched on for the quarks (the gluon term in $D_\mu Q_{Li}$, $D_\mu u_{Ri}$, $D_\mu d_{Ri}$), and the single generation has been replicated three times. The replication is exact in every gauge quantum number — the index i is invisible to $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ — so the gauge interactions are blind to flavor, a fact whose enormous global symmetry is the subject of Section 32.1.

The Higgs block. The single complex scalar doublet $H \in (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ has a gauge-covariant kinetic term and the most general renormalizable invariant potential,

$$\mathcal{L}_{\text{Higgs}} = (D_\mu H)^\dagger (D^\mu H) \;-\; V(H), \qquad V(H) = -\,\mu^2\, H^\dagger H + \lambda\, (H^\dagger H)^2, \tag{33.20}$$

with $\mu^2 > 0$ and $\lambda > 0$, imported verbatim from Eq. (31.38). The negative sign of the quadratic term makes the origin $H = 0$ a local maximum and drives the spontaneous breaking $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_{\text{em}}$ that the following sections exploit; that $V(H)$ is the *unique* renormalizable gauge invariant available is proved in Theorem F.7 and re-established as part of the uniqueness theorem below.

The Yukawa block. The fermion masses cannot come from $\mathcal{L}_{\text{Higgs}}$, which involves only H; they come from gauge-invariant couplings of H to left-right fermion bilinears. With three generations each coupling is a general complex 3×3 matrix in flavor space, and the up-type coupling requires the conjugate doublet $\widetilde{H} \equiv i\sigma^2 H^* \in (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$ of Theorem 31.7, because u_R carries hypercharge $+\frac{2}{3}$ which only the $Y = -\frac{1}{2}$ doublet can neutralize. Writing the generation indices $i, j = 1, 2, 3$ explicitly,

$$\mathcal{L}_{\text{Yukawa}} = -\,\bar{Q}_{Li}\, (Y_u)_{ij}\, \widetilde{H}\, u_{Rj} \;-\; \bar{Q}_{Li}\, (Y_d)_{ij}\, H\, d_{Rj} \;-\; \bar{L}_{Li}\, (Y_e)_{ij}\, H\, e_{Rj} \;+\; \text{h.c.}, \tag{33.21}$$

exactly the three-generation Yukawa Lagrangian of Section 32.1. The matrices Y_u, Y_d, Y_e are entirely general: arbitrary complex 3×3 matrices, with no reality or hermiticity assumed. They are the *only* terms in the whole Lagrangian that distinguish one generation from another, and after electroweak breaking $H \to (0, v/\sqrt{2})^{\mathsf{T}}$ they become the fermion mass matrices whose diagonalization

produces the CKM matrix (Chapter 32). The one-generation couplings y_u, y_d, y_e of Eq. (31.47) are the 3×3 matrices' single entries in the limit of one family.

Collecting Eqs. (33.18) to (33.21) gives the complete renormalizable Standard Model Lagrangian, written in the unbroken phase where the gauge symmetry is manifest and every field is massless:

$$\begin{aligned}\mathcal{L}_{\text{SM}} = \; & -\tfrac{1}{4}\, G^A_{\mu\nu} G^{A\,\mu\nu} - \tfrac{1}{4}\, W^I_{\mu\nu} W^{I\,\mu\nu} - \tfrac{1}{4}\, B_{\mu\nu} B^{\mu\nu} \\ & + \bar{Q}_{Li}\, i\not{D}\, Q_{Li} + \bar{u}_{Ri}\, i\not{D}\, u_{Ri} + \bar{d}_{Ri}\, i\not{D}\, d_{Ri} + \bar{L}_{Li}\, i\not{D}\, L_{Li} + \bar{e}_{Ri}\, i\not{D}\, e_{Ri} \\ & + (D_\mu H)^\dagger (D^\mu H) + \mu^2\, H^\dagger H - \lambda\, (H^\dagger H)^2 \\ & - \Big(\bar{Q}_{Li}\, (Y_u)_{ij}\, \widetilde{H}\, u_{Rj} + \bar{Q}_{Li}\, (Y_d)_{ij}\, H\, d_{Rj} + \bar{L}_{Li}\, (Y_e)_{ij}\, H\, e_{Rj} + \text{h.c.} \Big),\end{aligned} \tag{33.22}$$

each covariant derivative being Eq. (33.13) specialized to its multiplet and the generation indices summed. This is identical, line for line, to the reference transcription Eq. (F.52); we have re-displayed it here because it is the object whose uniqueness we now prove and whose consequences the rest of the chapter reads off.

33.2.3 Each block is invariant under the full $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$

The four blocks were each constructed to be invariant under the group factor that built them: $\mathcal{L}_{\text{gauge}}$'s color term under $\mathrm{SU}(3)_c$ in Chapter 25, the electroweak terms under $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ in Chapter 31. Assembly demands more. In Eq. (33.22) a single quark field carries color, weak-isospin, and hypercharge simultaneously, and its one kinetic term must be invariant under all three factors *at once*. That this holds is not automatic from the separate constructions; it is the central consistency check of the assembly, and we verify it where it bites hardest.

Worked Example 33.3 (The quark-doublet kinetic term is a full-$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet)**.** Take the left-handed quark doublet $Q_L = (u_L, d_L)^{\mathsf{T}}$ (one generation; the flavor index is a spectator). It transforms in $(\mathbf{3}, \mathbf{2}, +\frac{1}{6})$, so under an element $(U_c, U_L, e^{i\beta/6}) \in \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ it transforms as

$$Q_L \;\longrightarrow\; U_c\, U_L\, e^{i\beta Y}\, Q_L, \qquad Y = +\tfrac{1}{6}, \tag{33.23}$$

where $U_c = \exp(i\alpha_c^A \lambda^A/2)$ acts on the color index, $U_L = \exp(i\alpha^I \sigma^I/2)$ on the weak index, and the hypercharge phase is a number; these three factors commute because they act on independent indices. Its covariant derivative carries all three connections,

$$D_\mu Q_L = \Big(\partial_\mu - i g_s\, G^A_\mu \tfrac{\lambda^A}{2} - i g\, W^I_\mu \tfrac{\sigma^I}{2} - i g'\, \tfrac{1}{6}\, B_\mu \Big) Q_L, \tag{33.24}$$

the specialization of Eq. (33.13) that matches Eq. (F.28). By the defining property of each connection (Theorem 8.4, Theorem 25.6), the gluon term is built

so that $D_\mu Q_L \to U_c\,(D_\mu Q_L)$ under $\mathrm{SU}(3)_c$ alone, the W term so that $D_\mu Q_L \to U_L\,(D_\mu Q_L)$ under $\mathrm{SU}(2)_L$ alone, and the B term so that $D_\mu Q_L \to e^{i\beta Y}(D_\mu Q_L)$ under $\mathrm{U}(1)_Y$ alone. The three connections do not interfere with one another because each is proportional to generators acting on a different index: the gluon matrices $\lambda^A/2$ commute with the isospin matrices $\sigma^I/2$ and with the hypercharge number. Hence under a simultaneous transformation the covariant derivative inherits all three factors,

$$D_\mu Q_L \;\longrightarrow\; U_c\,U_L\,e^{i\beta Y}\,(D_\mu Q_L), \tag{33.25}$$

that is, $D_\mu Q_L$ transforms in the *same* representation $(\mathbf{3},\mathbf{2},+\tfrac{1}{6})$ as Q_L itself — exactly the homogeneity that makes "covariant" the right word. The conjugate $\bar{Q}_L = Q_L^\dagger\gamma^0$ then transforms in the conjugate representation, $\bar{Q}_L \to \bar{Q}_L\,e^{-i\beta Y}U_L^\dagger U_c^\dagger$, and the kinetic term contracts to a singlet,

$$\bar{Q}_L\,i\not{D}\,Q_L \;\longrightarrow\; \bar{Q}_L\,e^{-i\beta Y}U_L^\dagger U_c^\dagger\,i\gamma^\mu\,U_c\,U_L\,e^{i\beta Y}\,D_\mu Q_L \;=\; \bar{Q}_L\,i\not{D}\,Q_L, \tag{33.26}$$

using $U_c^\dagger U_c = U_L^\dagger U_L = 1$ and the fact that the phases and the unitary matrices, acting on disjoint indices, pass through γ^μ (a Lorentz, not a gauge, object) and cancel. The same computation goes through for every multiplet in Eq. (33.19); the quark doublet is the worst case because it is the only field charged under all three factors at once, and it passes. Thus $\mathcal{L}_{\text{fermion}}$ is invariant under the full $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, not merely under the factor each term was designed against.

The remaining blocks pass the same test by the same logic, stated more briefly. In $\mathcal{L}_{\text{gauge}}$ each field strength carries the adjoint index of *one* factor and is a singlet of the other two: $G^A_{\mu\nu}$ is a color octet but carries no weak or hypercharge index, so $G^A_{\mu\nu}G^{A\,\mu\nu}$, already $\mathrm{SU}(3)_c$-invariant by the cyclicity of the trace (Theorem 25.14), is trivially $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ invariant; symmetrically for $W^I_{\mu\nu}W^{I\,\mu\nu}$ and $B_{\mu\nu}B^{\mu\nu}$. There is moreover no invariant cross term: a contraction such as $W^I_{\mu\nu}B^{\mu\nu}$ would carry an uncontracted $\mathrm{SU}(2)_L$ adjoint index and is not a singlet, and $G^A_{\mu\nu}B^{\mu\nu}$ fails likewise. In $\mathcal{L}_{\text{Higgs}}$ the doublet H is a color singlet, so the gluon term is absent from $D_\mu H$ and the kinetic term is $\mathrm{SU}(3)_c$-invariant for free, while $(D_\mu H)^\dagger(D^\mu H)$ and $H^\dagger H$ are $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ invariant by the doublet contraction of Chapter 31. In $\mathcal{L}_{\text{Yukawa}}$ each term is simultaneously a color singlet (for the quarks, $\bar{\mathbf{3}} \otimes \mathbf{3} \supset \mathbf{1}$), a weak singlet ($\bar{\mathbf{2}} \otimes \mathbf{2} \supset \mathbf{1}$), and hypercharge-neutral; the explicit charge arithmetic — $-\tfrac{1}{6} + \tfrac{1}{2} - \tfrac{1}{3} = 0$ for the down quark, $-\tfrac{1}{6} - \tfrac{1}{2} + \tfrac{2}{3} = 0$ for the up quark with $\widetilde{H}$, and $+\tfrac{1}{2} + \tfrac{1}{2} - 1 = 0$ for the charged lepton — is the content of the bulleted check in Section F.2.4. With all four blocks confirmed invariant under the entire product group, the right-hand side of Eq. (33.22) is a genuine $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet, and the assembly is internally consistent.

33.2.4 Near-uniqueness: nothing renormalizable is missing

We now establish the claim that gives the Standard Model its peculiar authority. The Lagrangian Eq. (33.22) is not one choice among many compatible with the field content; it is, up to the single deferred topological term, the *only* Lagrangian compatible with Lorentz invariance, $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge invariance, and renormalizability. The proof is a finite enumeration. In four spacetime dimensions a renormalizable interaction is a local operator of mass dimension at most four (Theorem 21.7, Section 22.7.1): operators of dimension five or higher carry couplings of negative mass dimension and spoil the power counting that makes the loop expansion predictable. The fields have definite dimensions — a gauge potential and a scalar each carry dimension one, a Weyl or Dirac fermion dimension $\frac{3}{2}$, and a derivative dimension one — so the set of gauge-invariant Lorentz-scalar operators of dimension ≤ 4 buildable from the Standard Model fields is *finite*, and we can list it.

Theorem 33.4 (Near-uniqueness of the Standard Model Lagrangian)**.** *With the gauge group* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *and the matter content of Table 33.1 fixed, the most general local Lagrangian that is Lorentz invariant, invariant under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$*, and renormalizable (every operator of mass dimension* ≤ 4*) is*

$$\mathcal{L}_{\mathrm{SM}} = \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\mathrm{fermion}} + \mathcal{L}_{\mathrm{Higgs}} + \mathcal{L}_{\mathrm{Yukawa}} + \mathcal{L}_{\theta},$$

with the first four blocks given by Eqs. (33.18) to (33.21) and $\mathcal{L}_\theta$ *the single topological term of Section 33.5. No other independent operator exists.*

Proof. We enumerate all candidate operators by their mass dimension, sorting them by field content, and show that each admissible one already appears in Eq. (33.22).

Dimension 0 *and* 1. A constant (dimension 0) only shifts the vacuum energy and is dropped; it is not measurable in a non-gravitational theory. No dimension-one gauge-invariant operator exists: a single field is either a gauge non-singlet (every field in Table 33.1 except none — all carry some charge) or has dimension $\frac{3}{2}$ (a fermion), so no Lorentz-scalar singlet of dimension one can be formed.

Dimension 2. The only candidate is bilinear in bosons. Among scalars, $H^\dagger H$ (dimension 2) is the unique gauge invariant: it is the $\mathrm{SU}(2)_L$ singlet in $\bar{\mathbf{2}} \otimes \mathbf{2}$ with hypercharge $-\frac{1}{2} + \frac{1}{2} = 0$, and it appears in $V(H)$ as the $-\mu^2 H^\dagger H$ term. The alternative scalar bilinear $\epsilon_{ab} H^a H^b$ vanishes identically (it is antisymmetric in two copies of the same field) and in any case carries $Y = +1$. A vector-boson bilinear $A_\mu A^\mu$ — a mass term for any of G^A_μ, W^I_μ, B_μ — is *not* gauge invariant: the inhomogeneous shift of the potential under a gauge transformation ($B_\mu \to B_\mu + \frac{1}{g'}\partial_\mu \beta$ for the abelian field, and the non-abelian analogue for G, W) leaves an uncancelled $\partial\beta$ term, exactly as shown in Theorem 31.9. So the only admissible dimension-two operator is the one already present.

Dimension 3. The candidates are fermion bilinears $\bar{\psi}\psi$ and scalar cubics. A Dirac mass $\bar{\psi}_L\psi_R$ links a left-handed field (which lives only in the doublets Q_L, L_L) to a right-handed field (which lives only in the singlets u_R, d_R, e_R); but every such pairing transforms as an $\mathrm{SU}(2)_L$ doublet, not a singlet, and carries net hypercharge $\pm\frac{1}{2} \neq 0$ — e.g. $\bar{Q}_L u_R$ has hypercharge $-\frac{1}{6}+\frac{2}{3} = +\frac{1}{2}$ and isospin $\bar{\mathbf{2}}$ — so it is not gauge invariant (Theorem 31.9). A Majorana mass $L_L^{\mathsf{T}} C\, L_L$ has the $\mathrm{SU}(2)_L$ singlet contraction ϵ_{ij} but hypercharge $-\frac{1}{2}-\frac{1}{2} = -1 \neq 0$, again forbidden, and there is no ν_R to write a Majorana mass for. A scalar cubic in $H, H^\dagger$ carries half-odd-integer hypercharge (an odd number of $\pm\frac{1}{2}$'s never sums to zero) and cannot be a singlet. Hence *no* gauge-invariant dimension-three operator exists — the chirality of the matter content forbids every explicit mass — which is precisely why the Yukawa couplings are needed and why no mass term appears in Eq. (33.22).

Dimension 4. This is where the dynamics lives, and we sort the candidates by field content.

- *Two field strengths.* The Lorentz scalars quadratic in a curvature are $F_{\mu\nu}F^{\mu\nu}$ and $F_{\mu\nu}\widetilde{F}^{\mu\nu}$ for each factor. The parity-even contractions $G^A_{\mu\nu}G^{A\,\mu\nu}$, $W^I_{\mu\nu}W^{I\,\mu\nu}$, $B_{\mu\nu}B^{\mu\nu}$ are the gauge block Eq. (33.18); the cross terms $W^I_{\mu\nu}B^{\mu\nu}$ and $G^A_{\mu\nu}B^{\mu\nu}$ are not singlets (uncontracted adjoint indices) and are excluded. The parity-odd duals $G\widetilde{G}$, $W\widetilde{W}$, $B\widetilde{B}$ are the only operators *not* in the four-block sum: they are total derivatives, contribute nothing in perturbation theory, and only the QCD term $G\widetilde{G}$ is physically inequivalent to zero. This is the single residual term $\mathcal{L}_\theta$, gathered and analyzed in Section 33.5; the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ duals are rotated away or trivial, as shown there.

- *Scalar kinetic and quartic.* $(D_\mu H)^\dagger (D^\mu H)$ (dimension 4) is the unique gauge-invariant scalar kinetic term and appears in $\mathcal{L}_{\text{Higgs}}$. The unique quartic is $(H^\dagger H)^2$: any degree-four scalar invariant is built from the one quadratic invariant $H^\dagger H$, so it equals $(H^\dagger H)^2$, the λ term. (The full enumeration of the scalar potential, term by term, is Theorem F.7.)

- *Fermion kinetic.* $\bar{\psi}\, i\gamma^\mu D_\mu \psi$ (dimension 4, from $\frac{3}{2}+\frac{3}{2}+1$) is the unique gauge-invariant fermion bilinear with one derivative, and one copy appears for each of the five multiplets in $\mathcal{L}_{\text{fermion}}$.

- *Yukawa.* A fermion bilinear times one scalar, $\bar{\psi}_L\, H\, \psi_R$ (dimension $\frac{3}{2}+1+\frac{3}{2} = 4$), is gauge invariant only for the three combinations $\bar{Q}_L \widetilde{H} u_R$, $\bar{Q}_L H d_R$, $\bar{L}_L H e_R$ (and their conjugates), by the charge arithmetic verified in Section 33.2.3; these are exactly $\mathcal{L}_{\text{Yukawa}}$. No other scalar-fermion-fermion contraction is a singlet.

Every admissible operator of dimension ≤ 4 has now been accounted for, and each is present in Eq. (33.22), the sole exception being the topological duals

collected as $\mathcal{L}_\theta$. Operators of dimension ≥ 5 are excluded by renormalizability; the leading one, the dimension-five Weinberg operator $(\bar{L}_L^c\, \widetilde{H}^*)(\widetilde{H}^\dagger L_L)$ that would give the neutrinos a Majorana mass, is gauge invariant but non-renormalizable and is deferred to the effective-theory discussion of Chapter 35. This exhausts the enumeration and proves the theorem. □

Two features of the enumeration deserve to be named, because both are discharged by later sections rather than here. First, nowhere in the dimension-≤ 4 list does there appear an operator that violates baryon or lepton number: every gauge invariant we found conserves both. That conservation was not imposed — we never demanded it — so it is an *accident* of the field content and the renormalizability cutoff, and its precise statement (and its breakdown at higher dimension) is the business of Section 33.4. Second, the hypercharges entered the enumeration as given inputs from Table 33.1; the proof shows the Lagrangian is fixed *once they are fixed*, but says nothing about why they take the values they do. That the hypercharges are themselves forced — by gauge-anomaly cancellation together with the gauge invariance of the Yukawa couplings we just enumerated — is the content of the very next section, Section 33.3.

33.2.5 How many independent terms

The four-block decomposition makes the size of the theory concrete. It is worth distinguishing two different counts that are easy to conflate: the number of independent *operator structures* (the terms one writes down, which is what this section established) and the number of independent *free parameters* (the measurable constants those terms carry, which the parameter-count section Section 33.6 will reduce to its physical minimum). The operator count is the cleaner of the two and follows directly from Eq. (33.22).

The accounting in Table 33.3 is the precise sense in which the Standard Model is "small." Fifteen operator structures — three gauge kinetic terms, five fermion kinetic terms, three Higgs operators, three Yukawa structures, and one topological term — exhaust the dynamics of all known non-gravitational physics. Every gauge boson, every fermion mass, every mixing angle, every cross section computed anywhere in this book traces back to one of these fifteen lines. The smallness is not an aesthetic accident: it is the direct consequence of Theorem 33.4, which forbade everything else. (The reduction of these structures to the conventionally quoted nineteen free parameters — three gauge couplings, two Higgs parameters, nine charged-fermion masses, and the four mixing parameters of the CKM matrix, plus the strong θ — is a different counting, carried out, with the global-symmetry bookkeeping that removes the redundant Yukawa entries, in Section 33.6.)

Table 33.3: The independent operator structures of the Standard Model Lagrangian Eq. (33.22). "Structures" counts distinct gauge-invariant operators, not their flavor or gauge multiplicity (the five fermion kinetic terms are each summed over three generations; the three Yukawa structures are each 3×3 matrices). The parameter-level count, which collapses these into the physical constants, is deferred to Section 33.6.

Block	Structures	Operators
$\mathcal{L}_{\text{gauge}}$	3	G^2, W^2, B^2 (one kinetic term per group fac
$\mathcal{L}_{\text{fermion}}$	5	$\bar{\psi}\, i\not{D}\, \psi$ for Q_L, u_R, d_R, L_L, e_R
$\mathcal{L}_{\text{Higgs}}$	3	$(DH)^\dagger(DH)$, $H^\dagger H$, $(H^\dagger H)^2$
$\mathcal{L}_{\text{Yukawa}}$	3	$\bar{Q}_L \widetilde{H} u_R$, $\bar{Q}_L H d_R$, $\bar{L}_L H e_R$ (+h.c.)
subtotal (non-topological)	14	
$\mathcal{L}_\theta$ (topological, Section 33.5)	1	$G\widetilde{G}$
total	15	

33.2.6 Convergence: QED and Yang–Mills theory are now one Lagrangian

The assembly completes a unification that the book has been building toward since Part V. QED and Yang–Mills theory, carried independently through their own quantization, renormalization, and running, are no longer two theories: they are two pieces of Eq. (33.22).

Physical Insight

The two carried gauge theories meet. Yang–Mills theory survives in the Standard Model essentially intact: the color factor $\mathrm{SU}(3)_c$ contributes the gluon kinetic term $-\frac{1}{4} G^A_{\mu\nu} G^{A\,\mu\nu}$ of Eq. (33.18) and the quark–gluon coupling $g_s\, \bar{Q}_L \gamma^\mu \frac{\lambda^A}{2} Q_L\, G^A_\mu$ buried in $\mathcal{L}_{\text{fermion}}$, which is exactly the QCD vertex of Theorem F.6. QCD is one factor of the product, unbroken and vectorlike. QED is more subtle, because it is not a factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ at all but a *subgroup* of the electroweak part: the unbroken $\mathrm{U}(1)_{\text{em}}$ generated by $Q = T^3 + Y$, with the physical photon $A_\mu = \sin\theta_{\text{W}}\, W^3_\mu + \cos\theta_{\text{W}}\, B_\mu$ the combination that remains massless after spontaneous breaking (Theorem F.5). The electron's coupling $-e\, \bar{e}\gamma^\mu e\, A_\mu$ of QED emerges from the neutral part of $\mathcal{L}_{\text{fermion}}$ once that rotation is performed. The two threads that ran in parallel for ten chapters are thus revealed as one factor and one subgroup of a single gauge Lagrangian.

The convergence has a concrete technical payoff: *the renormalization ma-*

chinery of Part VI applies to Eq. (33.22) unchanged. This is not a hope but a consequence of the way the Lagrangian was selected. Theorem 33.4 kept only operators of dimension ≤ 4, which is exactly the condition (Theorem 21.7, Section 22.7.1) under which the superficial degree of divergence of any diagram is bounded independent of loop order and the divergences are absorbed by a finite set of counterterms. The Standard Model is a renormalizable gauge theory built from precisely the ingredients — gauge-covariant kinetic terms, a scalar with a quartic potential, and dimension-four Yukawa couplings — whose renormalization was established case by case in Part VI (ϕ^4 and QED in Chapter 22) and Part VII (non-abelian gauge theory, with the BRST and Slavnov–Taylor identities that control gauge-dependent divergences, in Chapter 26). Nothing in the power counting, the counterterm structure, the Callan–Symanzik equation, or the running of the couplings needs to be rebuilt; each transfers verbatim, factor by factor, to the assembled theory. The renormalizability of the spontaneously broken theory — the fact that giving H its vacuum value, which we do in the next chapter, does not spoil this — is the deep result that the same gauge symmetry protecting the divergences survives in the Higgs phase; we rely on it here and point to its place in Chapters 26 and 30.

With the complete Lagrangian Eq. (33.22) assembled, proved near-unique, and its renormalizability inherited from the machinery of Parts VI and VII, we have in hand the single object that the remaining sections interrogate. The next section asks what consistency at the quantum level demands of it: we will find that the hypercharges, which entered here as fixed inputs, are not free at all but are forced — up to one overall normalization — by the cancellation of gauge anomalies together with the gauge invariance of the very Yukawa couplings we have just written down, and that this is the origin of electric-charge quantization.

33.3 Consistency Fixes the Charges: Anomaly Cancellation and Charge Quantization

The Lagrangian assembled in the preceding section is the most general renormalizable, Lorentz- and gauge-invariant functional that can be built from the matter content of Table 33.1. In writing it down we treated the hypercharges — the abelian charges Y that the fields Q_L, u_R, d_R, L_L, e_R and the Higgs H carry under $\mathrm{U}(1)_Y$ — as ordinary inputs: a list of rational numbers, $+\frac{1}{6}, +\frac{2}{3}, -\frac{1}{3}, -\frac{1}{2}, -1, +\frac{1}{2}$, fixed by nothing more than the requirement $Q = T^3 + Y$ of Eq. (31.20) and verified, member by member, to reproduce the observed electric charges in Theorem 33.1. That verification ran in one direction only: *given* the hypercharges, it confirmed the charges. It left untouched the prior question of why those particular fractions, and not any others, should appear. A U(1) charge is a continuous label; a priori the six numbers could be anything. This section runs the implication backwards. We will show that once the gauge group and the

representations of Table 33.1 are fixed — that quarks are color triplets and weak doublets, leptons color singlets — the hypercharges are not free at all. Two demands of quantum consistency, gauge-anomaly cancellation and the gauge invariance of the Yukawa couplings, turn them into the solution of a system of equations with a unique answer up to a single overall normalization. The ragged fractions are forced. And because $Q = T^3 + Y$ ties electric charge to that forced hypercharge, the same argument is the origin of electric-charge quantization: the reason the proton's charge is exactly equal and opposite to the electron's, and the reason quark charges come in thirds.

33.3.1 The conditions consistency imposes

A classical gauge symmetry can fail to survive quantization. The mechanism, the chiral fermion triangle whose three external legs are gauge currents, is computed from the diagram in Chapter 36 (Section 36.2) and recast through the path-integral measure there; its gauge-theoretic consequence — that an uncanceled anomaly in a *gauged* current destroys the Slavnov–Taylor identities, and with them unitarity and renormalizability together — is established in Chapter 37 (Theorem 37.3). We import those results rather than reprove them. What we need here is only their distilled content. For a chiral gauge theory whose left-handed Weyl content sits in a (generally reducible) representation with Hermitian generators T^a, the perturbative gauge anomaly is proportional to the totally symmetric invariant

$$\mathcal{A}^{abc} = \mathrm{tr}\big[\{T^a, T^b\}\, T^c\big], \tag{33.27}$$

the trace running over every left-handed Weyl multiplet, with right-handed fields entered as their left-handed charge conjugates — which carry the conjugate color representation and the opposite hypercharge, exactly the bookkeeping of Eq. (33.9) and Table 33.2. The theory is anomaly-free if and only if Eq. (33.27) vanishes for every choice of the three gauge generators; this is Eq. (31.208), and the cancellation was verified for the numerical hypercharges of one generation in Theorem 31.30.

For the group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ the generators T^a run over the eight $\mathrm{SU}(3)_c$ generators, the three $\mathrm{SU}(2)_L$ generators, and the single hypercharge Y. The symmetric trace Eq. (33.27) therefore decomposes into separate triangle types according to how many legs of each factor it carries, and most of them vanish before any sum over fermions is performed.

- $[\mathrm{SU}(3)_c]^3$. Vanishes because color is vector-like: every colored Weyl field is accompanied by a conjugate partner in $\overline{\mathbf{3}}$, so left- and right-handed contributions to $\mathrm{tr}[\{T^a, T^b\}T^c]$ are equal and opposite. (Equivalently, the would-be anomaly is the d^{abc} of SU(3), which cancels between a representation and its conjugate.)

- $[\mathrm{SU}(2)_L]^3$. Vanishes identically: SU(2) has no totally symmetric cubic invariant, its representations being real or pseudoreal, so $d^{abc} = 0$ for the group itself and the trace is zero for any content.
- *Any triangle with exactly one* $\mathrm{SU}(3)_c$ *or one* $\mathrm{SU}(2)_L$ *leg.* Vanishes by tracelessness of the non-abelian generators, $\mathrm{tr}\, T^a = 0$, since the abelian charge factors out of the sum.

Four combinations survive and constrain the hypercharges. Writing Y for the hypercharge generator and using $\mathrm{tr}[\{T^a, T^b\}] = \delta^{ab}$ in the fundamental of either non-abelian factor, they reduce to weighted sums of hypercharges over the chiral content:

$$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y : \qquad \sum_{\text{LH quarks}} Y \;-\; \sum_{\text{RH quarks}} Y \;=\; 0, \tag{33.28a}$$

$$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y : \qquad \sum_{\text{LH doublets}} Y \;=\; 0, \tag{33.28b}$$

$$[\mathrm{grav}]^2\,\mathrm{U}(1)_Y : \qquad \sum_{\text{LH}} Y \;-\; \sum_{\text{RH}} Y \;=\; 0, \tag{33.28c}$$

$$[\mathrm{U}(1)_Y]^3 : \qquad \sum_{\text{LH}} Y^3 \;-\; \sum_{\text{RH}} Y^3 \;=\; 0. \tag{33.28d}$$

The third is the mixed gauge–gravitational anomaly: the triangle with one hypercharge current and two insertions of the energy–momentum tensor, whose coefficient is the plain trace $\mathrm{tr}\, Y$ and which must vanish once the theory is coupled to gravity (Eq. (31.212)). Each sum is taken with full color and weak-isospin multiplicities, right-handed fields entering with the opposite sign because their left-handed conjugates carry $-Y$. Theorem 31.30 evaluated all four for the numerical hypercharges and found zero. Our task is the converse: to treat the hypercharges as unknowns and read Eqs. (33.28a) to (33.28d) as equations that determine them.

33.3.2 Yukawa gauge invariance as additional constraints

There is a second, independent source of constraint, and it is the reason the matter content of Table 33.1 is endowed with mass at all. The fermions are chiral: left- and right-handed components sit in different representations of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, so no gauge-invariant Dirac mass $m\,\bar\psi\psi$ can be written directly. Masses arise only through the Yukawa couplings to the Higgs doublet, and those couplings exist only if the product of representations they join is gauge invariant. Each Yukawa term therefore imposes a linear relation among hypercharges. With the Higgs doublet H of hypercharge Y_H and its charge conjugate $\widetilde{H} = i\sigma^2 H^*$ of hypercharge $-Y_H$, the three renormalizable Yukawa couplings of one generation are

$$\mathcal{L}_{\mathrm{Yuk}} = -\,y_d\,\bar Q_L H\, d_R \;-\; y_u\,\bar Q_L \widetilde{H}\, u_R \;-\; y_e\,\bar L_L H\, e_R \;+\; \mathrm{h.c.} \tag{33.29}$$

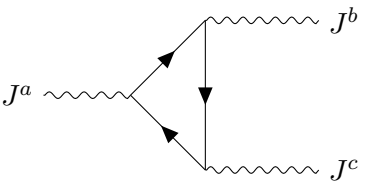


Figure 33.2: The anomaly triangle whose vanishing imposes Eq. (33.28). A chiral fermion runs in the loop; the three external currents J^a, J^b, J^c are gauge currents of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. The four current assignments that do not cancel automatically are $(\mathrm{SU}(3)_c, \mathrm{SU}(3)_c, \mathrm{U}(1)_Y)$, $(\mathrm{SU}(2)_L, \mathrm{SU}(2)_L, \mathrm{U}(1)_Y)$, $(\mathrm{U}(1)_Y, \mathrm{U}(1)_Y, \mathrm{U}(1)_Y)$, and the mixed gauge–gravitational triangle with one $\mathrm{U}(1)_Y$ current and two graviton legs. The loop integral and the coefficient Eq. (33.27) are computed in Section 36.2.

A term is gauge invariant under $\mathrm{U}(1)_Y$ precisely when the hypercharges of its factors sum to zero. The conjugated field $\bar{Q}_L$ carries $-Y_Q$, the field d_R carries $+Y_d$, and so on, so Eq. (33.29) requires

$$\bar{Q}_L\, H\, d_R: \qquad -Y_Q + Y_H + Y_d = 0 \implies Y_d = Y_Q - Y_H, \tag{33.30a}$$

$$\bar{Q}_L\, \widetilde{H}\, u_R: \qquad -Y_Q - Y_H + Y_u = 0 \implies Y_u = Y_Q + Y_H, \tag{33.30b}$$

$$\bar{L}_L\, H\, e_R: \qquad -Y_L + Y_H + Y_e = 0 \implies Y_e = Y_L - Y_H. \tag{33.30c}$$

(Color and weak-isospin invariance are automatic: $\mathbf{3} \otimes \overline{\mathbf{3}} \supset \mathbf{1}$ and $\mathbf{2} \otimes \mathbf{2} \supset \mathbf{1}$, which is why the down-type and lepton couplings use H and the up-type coupling uses $\widetilde{H}$; only the abelian factor constrains Y.) These three relations express the singlet hypercharges Y_u, Y_d, Y_e in terms of the doublet hypercharges Y_Q, Y_L and the Higgs hypercharge Y_H. They are the physical content of the demand that the fermions be allowed to acquire mass through the same Higgs that breaks the electroweak symmetry.

Notice immediately that Eqs. (33.30a) to (33.30c) make the first anomaly condition redundant. Adding Eq. (33.30b) and Eq. (33.30a) gives $Y_u + Y_d = 2Y_Q$, which is exactly $[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y$ in the form $2Y_Q - (Y_u + Y_d) = 0$ of Eq. (33.28a) (the quark doublet contributes two colored states of charge Y_Q, the singlets one each of Y_u, Y_d). The down/up Yukawa pair and the color triangle say the same thing. This is the first sign that the two consistency requirements are not independent constraints fighting over the same unknowns but a single, tightly woven system.

33.3.3 Solving the system

We now have unknowns $Y_Q, Y_L, Y_u, Y_d, Y_e, Y_H$ — one hypercharge per chiral multiplet of Table 33.1, six in all — subject to the three Yukawa relations

Eq. (33.30) and the four anomaly conditions Eq. (33.28). The representations themselves are taken as given: Q a color triplet weak doublet, L a color-singlet weak doublet, u_R, d_R, e_R weak singlets with u_R, d_R color triplets, H a color-singlet weak doublet. We keep the number of colors N general (the physical case is $N = 3$) so that the role of color in the cancellation is visible, and fix the weak group to SU(2). The following worked example carries the solution from start to finish and exhibits its uniqueness.

Worked Example 33.5 (Anomaly freedom and Yukawa invariance determine every hypercharge)**.** Read the multiplicities off Table 33.1: the quark doublet Q supplies $2N$ left-handed Weyl states of hypercharge Y_Q (two weak components $\times N$ colors), the lepton doublet L supplies 2 states of Y_L; the right-handed singlets u_R, d_R supply N states each of Y_u, Y_d, and e_R one state of Y_e. With these the four anomaly conditions of Eq. (33.28) read

$$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y: \quad 2Y_Q - Y_u - Y_d = 0, \tag{33.31a}$$

$$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y: \quad N\,Y_Q + Y_L = 0, \tag{33.31b}$$

$$[\mathrm{grav}]^2\,\mathrm{U}(1)_Y: \quad 2N\,Y_Q + 2\,Y_L - N\,Y_u - N\,Y_d - Y_e = 0, \tag{33.31c}$$

$$[\mathrm{U}(1)_Y]^3: \quad 2N\,Y_Q^3 + 2\,Y_L^3 - N\,Y_u^3 - N\,Y_d^3 - Y_e^3 = 0. \tag{33.31d}$$

In Eq. (33.31b) only the two weak doublets (Q with color factor N, L with factor 1) contribute, since right-handed fields are weak singlets and do not run in the $[\mathrm{SU}(2)_L]^2$ triangle.

Step 1 — eliminate the singlets with the Yukawa relations. Substitute Eq. (33.30), $Y_u = Y_Q + Y_H$, $Y_d = Y_Q - Y_H$, $Y_e = Y_L - Y_H$, into the remaining anomaly conditions. The color condition Eq. (33.31a) becomes $2Y_Q - (Y_Q + Y_H) - (Y_Q - Y_H) = 0$, an identity — as anticipated, the Yukawa couplings already enforce it. The gravitational condition Eq. (33.31c) becomes, using $Y_u + Y_d = 2Y_Q$,

$$2N\,Y_Q + 2Y_L - N(Y_u + Y_d) - Y_e = 2N\,Y_Q + 2Y_L - 2N\,Y_Q - (Y_L - Y_H) = Y_L + Y_H = 0, \tag{33.32}$$

so the gravitational triangle, once the singlets are eliminated, says simply $Y_H = -Y_L$.

Step 2 — the two linear weak/gravitational conditions fix everything up to scale. We are left with Eq. (33.31b), $Y_L = -N\,Y_Q$, and Eq. (33.32), $Y_H = -Y_L = N\,Y_Q$. The Yukawa relations then give the singlets immediately:

$$Y_L = -N\,Y_Q, \quad Y_H = N\,Y_Q, \quad Y_u = (1+N)\,Y_Q, \quad Y_d = (1-N)\,Y_Q, \quad Y_e = -2N\,Y_Q. \tag{33.33}$$

Every hypercharge is now a fixed rational multiple of the single quantity Y_Q. The number of independent equations used — two Yukawa relations that remain after the color identity is absorbed, plus the two surviving anomaly conditions

$[\mathrm{SU}(2)_L]^2\mathrm{U}(1)_Y$ and $[\mathrm{grav}]^2\mathrm{U}(1)_Y$ — is five, acting on six unknowns, so the solution space is one-dimensional: the line Eq. (33.33) parametrized by Y_Q.

Step 3 — the cubic condition is automatically satisfied. The cubic anomaly Eq. (33.31d) has not yet been imposed. It could, in principle, be a sixth constraint, over-determining the system and forcing $Y_Q = 0$ (no charges at all). It does not. Substituting Eq. (33.33),

$$2N\,Y_Q^3 + 2Y_L^3 - N\,Y_u^3 - N\,Y_d^3 - Y_e^3 = \Big[\,2N + 2(-N)^3 - N(1+N)^3 - N(1-N)^3 - (-2N)^3 \tag{33.34}$$

Using $(1+N)^3 + (1-N)^3 = 2(1+3N^2)$, the bracket is

$$2N - 2N^3 - N\cdot 2(1+3N^2) + 8N^3 = 2N - 2N^3 - 2N - 6N^3 + 8N^3 = 0, \tag{33.35}$$

identically in N and in Y_Q. The $[\mathrm{U}(1)_Y]^3$ anomaly cancels for the entire one-parameter family, not just at isolated points. This is a genuine, and a priori far from obvious, miracle of the chiral content: the cubic surface and the linear planes intersect in the whole line Eq. (33.33) rather than in a point. Had the representations been different — a fourth weak singlet, say, or leptons in a color triplet — the cubic would generically have clashed with the linear conditions and no nonzero charge assignment would have existed.

Step 4 — fix the normalization. The surviving freedom Y_Q is not physical. Rescaling all hypercharges by a common factor is undone by rescaling the abelian coupling g', since the covariant derivative Eq. (33.4) contains only the product $g'Y$. The conventional choice that makes the electric charges come out in their familiar units is

$$Y_Q = \tfrac{1}{2N} \quad \Big(= \tfrac{1}{6} \text{ at } N = 3\Big), \tag{33.36}$$

whereupon Eq. (33.33) becomes, at $N = 3$,

$$Y_Q = +\tfrac{1}{6}, \quad Y_u = +\tfrac{2}{3}, \quad Y_d = -\tfrac{1}{3}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = -1, \quad Y_H = +\tfrac{1}{2}, \tag{33.37}$$

which is precisely the list tabulated in Table 33.1 and verified against the physical charges in Theorem 33.1. The hypercharges we earlier fed in by hand are recovered as the unique output of consistency.

The content of the worked example is worth stating as a theorem, because it is the central claim of the section.

Theorem 33.6 (Hypercharge is fixed by consistency up to normalization)**.** *Fix the gauge group* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *and the Standard-Model representations of Table 33.1, with the Higgs in the weak-doublet representation. Then the requirements that (i) all perturbative gauge anomalies cancel, Eq. (33.28), and (ii) the three Yukawa couplings Eq. (33.29) be gauge invariant, Eq. (33.30), determine the six hypercharges* $Y_Q, Y_L, Y_u, Y_d, Y_e, Y_H$ *uniquely up to a single common rescaling, by Eq. (33.33). The cubic anomaly* $[\mathrm{U}(1)_Y]^3$ *imposes no further restriction: it is satisfied identically along the solution line.*

Proof. This is the computation of Theorem 33.5. The linear conditions $[\mathrm{SU}(2)_L]^2\mathrm{U}(1)_Y$ and $[\mathrm{grav}]^2\mathrm{U}(1)_Y$, together with the three Yukawa relations, form five independent linear equations in the six unknowns whose solution is Eq. (33.33); the color condition $[\mathrm{SU}(3)_c]^2\mathrm{U}(1)_Y$ is implied by the up- and down-type Yukawa relations, and the cubic condition vanishes identically along the solution by Eq. (33.35). Uniqueness up to scale is the statement that the coefficient matrix of the five linear equations has rank five. □

That the one-loop conditions Eq. (33.28) are the whole story — that no higher-loop diagram can spoil Eq. (33.33) or generate a new anomaly — is the Adler–Bardeen theorem, established in Section 37.3. There the same determination is carried out from the anomaly conditions alone, without invoking the Yukawa couplings, by using the cubic equation in place of one of the linear ones; the two routes agree, as they must, on the line Eq. (33.33). The route taken here, leaning on Yukawa gauge invariance, is the one that makes transparent why the cubic is automatically satisfied rather than independently imposed.

Physical Insight

It is worth being exact about what Theorem 33.6 does and does not derive. It takes the *representations* as input — that quarks are color triplets and weak doublets, leptons color singlets, the Higgs a weak doublet — and from them computes the hypercharges. It does not explain why the representations are those, nor why there are three copies. What it converts is six would-be arbitrary real numbers into a rigid one-parameter family, and the one parameter is pure normalization of the abelian generator. The often-repeated slogan that "the Standard Model quantizes charge" means precisely this: in a chiral gauge theory the hypercharges cannot be dialed freely, because almost every choice is anomalous and therefore inconsistent, and the consistent choices form the single rational pattern Eq. (33.37).

33.3.4 Electric-charge quantization

The hypercharges fixed, the electric charges follow at once from $Q = T^3 + Y$ (Eq. (31.20)), with $T^3 = \pm\frac{1}{2}$ on the two members of a weak doublet and $T^3 = 0$ on a singlet. The point is no longer that the assignment is consistent — that was Theorem 33.1 — but that it is *forced*, so the relations among charges below are predictions of consistency, not inputs.

The charged lepton, lower member of L, has

$$Q_e = -\tfrac{1}{2} + Y_L = -\tfrac{1}{2} - N\,Y_Q = -\tfrac{1}{2} - \tfrac{1}{2} = -1, \tag{33.38}$$

using $Y_L = -NY_Q = -\frac{1}{2}$. The quark doublet members have

$$Q_u = +\tfrac{1}{2} + Y_Q = \tfrac{1}{2} + \tfrac{1}{2N} = \frac{N+1}{2N}, \qquad Q_d = -\tfrac{1}{2} + Y_Q = -\frac{N-1}{2N}, \tag{33.39}$$

which at $N = 3$ are the familiar $Q_u = +\frac{2}{3}$ and $Q_d = -\frac{1}{3}$. The quark charges are fractional, in integer multiples of $1/2N = 1/6$, precisely *because* the anomaly condition $[\mathrm{SU}(2)_L]^2\mathrm{U}(1)_Y$ forced the lepton hypercharge to scale with color as $Y_L = -NY_Q$ in Eq. (33.31b): the factor of N that makes the lepton charge an integer is the same factor of N that makes the quark charge a fraction with denominator N. The two are inseparable consequences of the one cancellation.

The deepest consequence appears when quarks are assembled into a color singlet. A proton is uud; its charge is

$$Q_{\text{proton}} = 2Q_u + Q_d = 2\cdot\frac{N+1}{2N} - \frac{N-1}{2N} = \frac{2N+2-N+1}{2N} = \frac{N+3}{2N} \xrightarrow{N=3} +1 = -Q_e. \tag{33.40}$$

The proton's charge is exactly equal and opposite to the electron's. We may state this as a corollary, the second headline result of the section.

Corollary 33.7 (Electric charge is quantized and the proton is integer-charged)**.** *With the hypercharges fixed by Theorem 33.6 and $Q = T^3 + Y$, the electric charges of one generation are commensurate: they are integer multiples of $Y_Q = \frac{1}{2N}$. Every color-singlet bound state carries integer electric charge, and in particular $Q_{proton} = -Q_e$ exactly. With the conventional normalization $Y_Q = \frac{1}{6}$ this reproduces $Q_e = -1$, $Q_u = +\frac{2}{3}$, $Q_d = -\frac{1}{3}$ of Eq. (33.37).*

Physical Insight

The equality $Q_{\text{proton}} = -Q_e$ is tested by the electrical neutrality of bulk matter to better than one part in 10^{21}. In a theory whose charges were free continuous parameters this would be an unexplained coincidence between the couplings of a hadron built from quarks and an elementary lepton — two objects with no kinematic relation. In the Standard Model it is not a coincidence at all but an output of the demand that the chiral gauge theory be quantum-mechanically consistent. The fractional quark charges, the integer hadron charges, and the precise atom-by-atom cancellation between electron and proton are three faces of the same anomaly cancellation, with the number of colors $N = 3$ as the hinge connecting them. This is as close as the Standard Model comes to *explaining* a number rather than measuring it.

There is one residual discrete ambiguity that the anomaly conditions, taken by themselves, do not resolve. The cubic equation Eq. (33.31d) is symmetric under interchange of Y_u and Y_d (and the linear color condition only fixes their sum),

so the anomalies alone cannot say which of the two values $(1 \pm N)Y_Q$ belongs to the up-type singlet and which to the down-type. It is the Yukawa couplings, once again, that break the tie: gauge invariance of $\bar{Q}_L \widetilde{H}\, u_R$ in Eq. (33.30b) demands $Y_u = Y_Q + Y_H = (1+N)Y_Q$ against the *same* Higgs doublet that gives $Y_d = Y_Q - Y_H = (1-N)Y_Q$ to the down type through $\bar{Q}_L H\, d_R$. A single Higgs doublet thus serves both the up and down sectors only for the assignment Eq. (33.37); the alternative would require a second scalar with opposite hypercharge. Economy of the scalar sector selects the observed charges. This is the precise sense in which the hypercharges are fixed by anomaly cancellation *together with* Yukawa gauge invariance: the former gives the magnitudes and the one-parameter family, the latter fixes the discrete labeling and renders the cubic redundant.

33.3.5 The global $\mathrm{SU}(2)_L$ anomaly

The conditions Eq. (33.28) are all *perturbative*: each is the vanishing of a triangle diagram, computable order by order. There is one further consistency requirement that no Feynman diagram can see, because it is topological. It was discovered by Witten and it concerns exactly the factor we declared safe at the start of Section 33.3.1, the weak group $\mathrm{SU}(2)_L$. The fourth homotopy group of SU(2) is nontrivial, $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}_2$, which means there exist gauge transformations on Euclidean spacetime (S^4) that cannot be continuously deformed to the identity. A theory with an odd number of left-handed Weyl fermions in the doublet of $\mathrm{SU}(2)_L$ assigns the fermionic path-integral measure a relative sign under such a transformation: the Pfaffian of the Dirac operator changes sign along the noncontractible loop in field space, so the partition function is forced to vanish and the theory is mathematically ill-defined. This is the statement of Theorem 37.46; like the perturbative anomaly it is fatal, but it is a $\mathbb{Z}_2$ condition — the number of doublets must be *even* — rather than the vanishing of a continuous coefficient.

The Standard Model must satisfy it, and the following worked example checks that it does, by the direct count.

Worked Example 33.8 (The Standard Model has an even number of weak doublets)**.** We count the left-handed Weyl fermions of one generation that transform as $\mathrm{SU}(2)_L$ doublets, weighting each by its color multiplicity, since color replicates a weak doublet into several otherwise-identical doublets each of which contributes to the Pfaffian. From Table 33.1:

- the lepton doublet $L = (\nu_e, e)_L$ is a color singlet, contributing 1 weak doublet;
- the quark doublet $Q = (u, d)_L$ is a color triplet, contributing $N = 3$ weak doublets, one per color.

The right-handed singlets u_R, d_R, e_R are $\mathrm{SU}(2)_L$ singlets and do not enter. The Higgs is a boson and is irrelevant to the fermionic Pfaffian. Hence the number of left-handed weak doublets per generation is

$$n_{\text{doublet}} = \underbrace{1}_{L} + \underbrace{N}_{Q} = 1 + 3 = 4, \tag{33.41}$$

which is even. Across all $n_g = 3$ generations,

$$n_{\text{doublet}}^{\text{total}} = n_g\,(1 + N) = 3 \times 4 = 12, \tag{33.42}$$

again even. By Theorem 37.46 the global SU(2) anomaly vanishes: the Standard Model passes the one nonperturbative consistency test that the triangle diagrams cannot detect. This conclusion is robust to generation number — any whole number of generations gives an even count, because each generation contributes the fixed even number $1+N = 4$ — and it is the global counterpart of Theorem 37.47.

The result is reassuring rather than constraining in the Standard Model, because color happens to make the count even within a single generation: the lone leptonic doublet is paired by the three colored quark doublets. The condition becomes a sharp constraint in model building beyond the Standard Model, where an isolated weak doublet — a fourth-generation lepton with no accompanying colored partner, say — would be forbidden outright by Witten's anomaly even though it is invisible to every triangle.

33.3.6 The right-handed neutrino and $B - L$ as an anomaly-free direction

Two loose ends of the determination point in the same direction, toward a charge the Standard Model could consistently gauge but does not. The first is the right-handed neutrino. A gauge-singlet Weyl field ν_R — color singlet, weak singlet, hypercharge zero — contributes nothing to any of the conditions Eq. (33.28) or to Eq. (33.41), because all of its gauge charges vanish. It is therefore anomaly-free to adjoin, and its omission from Table 33.1 is a choice of minimality (the renormalizable Standard Model has a massless neutrino, Theorem 31.17) rather than a consequence of consistency. Its presence is what permits a Dirac neutrino mass through the Yukawa coupling $\bar{L}_L \widetilde{H}\, \nu_R$, and a Majorana mass through the gauge-singlet bilinear $\overline{\nu_R^c}\nu_R$; both threads are taken up in Chapter 32 and Chapter 35.

The second loose end concerns which global symmetries of the assembled Lagrangian could be promoted to gauge symmetries. The anomaly conditions Eq. (33.28) were derived for the abelian factor $\mathrm{U}(1)_Y$, but the same triangles constrain any candidate abelian charge. Consider gauging the combination $B-L$, baryon number minus lepton number, on which each quark carries $B - L = +\frac{1}{3}$

and each lepton $B - L = -1$. Computing its mixed anomalies exactly as in Eq. (33.28) but with Y replaced by the $B - L$ charge, and counting the right-handed neutrino as a lepton with $(B - L)_{\nu_R} = -1$:

$$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_{B-L}: \quad 2\cdot\tfrac{1}{3} - \left(\tfrac{1}{3}+\tfrac{1}{3}\right) = 0, \tag{33.43a}$$

$$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_{B-L}: \quad N\cdot\tfrac{1}{3} + 1\cdot(-1) = \tfrac{N}{3} - 1 = 0 \quad (N=3), \tag{33.43b}$$

$$[\mathrm{grav}]^2\,\mathrm{U}(1)_{B-L}: \quad \underbrace{2N\cdot\tfrac{1}{3} + 2\cdot(-1)}_{\text{LH } Q,\,L} - \underbrace{\left(N\cdot\tfrac{1}{3} + N\cdot\tfrac{1}{3} + (-1) + (-1)\right)}_{\text{RH } u_R, d_R, e_R, \nu_R} = 0, \tag{33.43c}$$

$$[\mathrm{U}(1)_{B-L}]^3: \quad 2N\left(\tfrac{1}{3}\right)^3 + 2(-1)^3 - N\left(\tfrac{1}{3}\right)^3 - N\left(\tfrac{1}{3}\right)^3 - (-1)^3 - (-1)^3 = 0. \tag{33.43d}$$

Every mixed anomaly of $B-L$ with the Standard-Model gauge group vanishes — but only with the right-handed neutrino included, and, in the case of Eq. (33.43b), only for $N = 3$. Without ν_R the gravitational and cubic conditions Eqs. (33.43c) and (33.43d) would each fail by exactly one unit (the term -1 supplied by ν_R is what restores them), and $B - L$ would be anomalous and ungaugeable. With ν_R present, $B - L$ is the unique anomaly-free abelian charge that commutes with $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and is not already a combination of hypercharge and the unbroken generators; it is therefore the natural extra U(1) of grand unified and seesaw constructions, where exactly this anomaly-freedom makes it gaugeable. We carry the point only as far as the anomaly bookkeeping: $B - L$ is consistent to gauge, hypercharge being already gauged, while baryon number B and lepton number L *separately* are not anomaly-free (their mixed anomalies with $\mathrm{SU}(2)_L$ do not cancel), which is why they remain global rather than local symmetries. That distinction — anomaly-free linear combination versus individually anomalous factors — is the bridge to the next section.

We have found that almost nothing in the charge structure of the Standard Model was ours to choose. Fixing the gauge group and the representations, quantum consistency — anomaly cancellation, Witten's $\mathbb{Z}_2$ condition, and the gauge invariance of the Yukawa couplings — pins every hypercharge up to one normalization and thereby quantizes electric charge. The matter content and its charges are not free inputs but forced. With the charges thus determined, the natural next question is the converse one: granting the Lagrangian we have assembled, what symmetries does it possess that we did *not* impose? The accidental conservation of baryon and lepton number, and the fate of those symmetries under quantization, is the subject of the next section.

33.4 Accidental Symmetries: Baryon and Lepton Number

The previous section closed a loop that had been open since the gauge group was first written down: the hypercharges are not free inputs but are forced, up to one overall normalization, by gauge-anomaly cancellation and by the gauge invariance of the Yukawa couplings (Section 33.3.4). Nothing in the matter content is arbitrary. We have a single Lagrangian $\mathcal{L}_{\rm SM}$ on $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, assembled in Section 33.2, every term of which was demanded by the two principles that organize the whole construction — gauge invariance and renormalizability (operators of mass dimension four or less). The natural next question is what this rigidly determined Lagrangian gives back for free. A theory built to respect a list of *local* (gauge) symmetries need not, and in general will not, respect any extra *global* symmetries; that it does is a gift, and the gift turns out to be exactly the conservation laws that organize all of low-energy particle physics: the stability of the proton, the separate conservation of electron, muon, and tau number, the very meaning of the word "baryon." These symmetries were never written into $\mathcal{L}_{\rm SM}$. They appear because, once gauge invariance and renormalizability have been imposed, there is no room left to write a term that would violate them. Such symmetries are called *accidental*, and this section is devoted to identifying them, deciding which survive quantization, and exhibiting the leading operators that break them — the Standard Model's first windows onto physics above the electroweak scale.

33.4.1 What "accidental" means

Definition 33.9 (Accidental symmetry)**.** A global symmetry of a field theory is *accidental* if it is not imposed as a defining principle but is an automatic consequence of the field content together with the restrictions already imposed for other reasons — here, Lorentz invariance, $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge invariance, and the renormalizability cut that admits only operators of mass dimension $d \leq 4$. Equivalently: every Lagrangian monomial allowed by those restrictions happens to be invariant under the symmetry, although the symmetry was never required.

The qualifier "renormalizable" is doing essential work in Theorem 33.9, and this is the conceptual heart of the section. An accidental symmetry of the dimension-≤ 4 Lagrangian is in general *not* a symmetry of the operators of dimension five, six, and higher that one could in principle add. Those operators are suppressed by inverse powers of some heavy scale Λ and are invisible at low energy, which is why the symmetry looks exact in every experiment done so far. But they are not forbidden by any principle the Standard Model itself supplies. An accidental symmetry is therefore automatically an *approximate* symmetry:

exact in the renormalizable theory, violated by the leading irrelevant operators consistent with the gauge symmetry. This is the precise sense in which baryon and lepton number are conserved "until they are not," and it is the reason their leading violations are the cleanest probes of the next layer of physics, taken up in Section 38.2.

To find the accidental symmetries we look for global transformations under which $\mathcal{L}_{\rm SM}$ is invariant. The kinetic and gauge-interaction terms $\bar\psi\, i\not{D}\,\psi$ are invariant under *any* unitary rotation of the generation index of each gauge multiplet, because such rotations commute with the gauge action and preserve $\bar\psi\, i\not{D}\,\psi$ by unitarity; this is the flavor group $G_{\rm F} = {\rm U}(3)^5$ of Eq. (F.94). The Yukawa couplings break most of $G_{\rm F}$, and the subgroup that survives — the stabilizer of a generic set of Yukawa matrices — is precisely the set of accidental global symmetries. That stabilizer is computed in full in Theorem F.12; here we establish the same conclusion from the complementary and more elementary direction, by cataloguing the allowed operators directly and reading off the charges they conserve.

33.4.2 The classical accidental symmetry of the renormalizable Lagrangian

We assign to each field a baryon number B and a lepton number L_i ($i = e, \mu, \tau$). The assignment is the obvious one dictated by what we want these numbers to count: each quark carries baryon number $+\tfrac{1}{3}$ (so that a three-quark bound state has $B = 1$), each antiquark $-\tfrac{1}{3}$; each lepton of generation i carries $L_i = +1$, each antilepton $L_i = -1$; the gauge bosons and the Higgs carry $B = L_i = 0$. Concretely, the global phase rotations are

$$ {\rm U}(1)_B: \quad Q_L \to e^{i\beta/3} Q_L, \quad u_R \to e^{i\beta/3} u_R, \quad d_R \to e^{i\beta/3} d_R, \quad (L_L, e_R, H \text{ invariant}), \tag{33.44} $$

$$ {\rm U}(1)_{L_i}: \quad L_{Lj} \to e^{i\gamma_i} L_{Lj}, \quad e_{Rj} \to e^{i\gamma_i} e_{Rj}, \quad (\text{quarks}, H \text{ invariant}), \tag{33.45} $$

with the lepton phases taken in the charged-lepton mass eigenbasis (the basis in which Y_e is diagonal; see the remark after Theorem 33.11). The claim to be proved is that the renormalizable Standard Model Lagrangian is invariant under the four independent phase symmetries

$$ G_{\rm acc} = {\rm U}(1)_B \times {\rm U}(1)_{L_e} \times {\rm U}(1)_{L_\mu} \times {\rm U}(1)_{L_\tau}, \tag{33.46} $$

and that this invariance is forced — there is no dimension-≤ 4 gauge-invariant term that could spoil it.

Theorem 33.10 ($G_{\rm acc}$ is an accidental symmetry of $\mathcal{L}_{\rm SM}$)**.** *Every Lorentz-invariant,* ${\rm SU}(3)_c \times {\rm SU}(2)_L \times {\rm U}(1)_Y$*-gauge-invariant local operator of mass dimension* $d \le 4$ *built from the Standard-Model fields conserves baryon number*

B and each of the three lepton numbers L_e, L_μ, L_τ. Consequently the group G_{acc} of Eq. (33.46) is an exact global symmetry of the classical, renormalizable Lagrangian, and it is accidental in the sense of Theorem 33.9: it was not imposed.

The proof is a finite enumeration. There are only so many ways to build a gauge- and Lorentz-invariant monomial of dimension ≤ 4 out of the available fields, and we simply list them and check the charges. The enumeration is worth doing slowly, because it is the entire content of the word "accidental": the symmetry holds not for a deep reason but because the short list of admissible terms happens, every one, to respect it.

Worked Example 33.11 (Every renormalizable operator conserves B and the L_i)**.** We catalogue the gauge- and Lorentz-invariant local operators of dimension $d \leq 4$ and verify their baryon and lepton numbers. The mass dimensions of the building blocks are: a fermion field ψ has $[\psi] = \frac{3}{2}$, the scalar H has $[H] = 1$, a field strength $F_{\mu\nu}$ has $[F] = 2$, and each derivative ∂_μ contributes 1. Lorentz invariance requires spinor indices to be contracted in pairs, so fermions enter only in bilinears $\bar{\psi}\,\Gamma\,\chi$ (dimension 3 before any derivatives or scalars). We organize by field content.

Pure boson operators ($d \leq 4$). The Higgs potential terms $H^\dagger H$ ($d = 2$) and $(H^\dagger H)^2$ ($d = 4$); the kinetic term $(D_\mu H)^\dagger (D^\mu H)$ ($d = 4$); the gauge kinetic terms $F^A_{\mu\nu} F^{A\,\mu\nu}$ for each factor ($d = 4$); and the topological term $G^a_{\mu\nu}\tilde{G}^{a\,\mu\nu}$ ($d = 4$, the subject of Section 33.5). Every one of these is built from H and gauge fields alone, all of which carry $B = L_i = 0$; each operator therefore has $B = L_i = 0$ trivially and is invariant under Eqs. (33.44) and (33.45).

Fermion bilinears without a Higgs. The only Lorentz- and gauge-invariant fermion bilinear available at $d = 4$ is the kinetic term $\bar{\psi}\, i\not{D}\,\psi$ for each multiplet $\psi \in \{Q_L, u_R, d_R, L_L, e_R\}$. It pairs a field with its own conjugate, so any U(1) phase $\psi \to e^{i\alpha}\psi$ cancels between $\bar{\psi}$ and ψ: these terms are invariant under *all* of G_{F}, hence in particular under G_{acc}, and carry $B = L_i = 0$. A bare Dirac mass $m\,\bar{\psi}_L\psi_R$ would also be a $d = 3$ bilinear, but no such gauge invariant exists: left- and right-handed fields sit in different $\text{SU}(3)_c \times \text{SU}(2)_L \times \text{U}(1)_Y$ representations (Table 31.1), so $\bar{\psi}_L\psi_R$ is never a gauge singlet. This is why the only fermion masses are Yukawa-induced, and it is the reason the next category exists.

Yukawa operators (fermion bilinear $\times$ Higgs, $d = 4$). A fermion bilinear ($d = 3$) times one power of H or $\tilde{H}$ ($d = 1$) gives a $d = 4$ operator. Gauge invariance singles out exactly three structures (and their Hermitian conjugates),

$$\bar{Q}_L\,\tilde{H}\,u_R, \qquad \bar{Q}_L\,H\,d_R, \qquad \bar{L}_L\,H\,e_R, \tag{33.47}$$

which assemble into Eq. (F.91) once the generation matrices Y_u, Y_d, Y_e are restored. We check the charges. In $\bar{Q}_L\,\tilde{H}\,u_R$ the antiquark $\bar{Q}_L$ carries $B = -\frac{1}{3}$ and the quark u_R carries $B = +\frac{1}{3}$, while $\tilde{H}$ carries $B = 0$; the net baryon number is $-\frac{1}{3} + \frac{1}{3} = 0$, and no lepton appears, so $L_i = 0$. Identically for $\bar{Q}_L H d_R$. The

leptonic term $\bar{L}_L H e_R$ carries $B = 0$ and lepton number $-1 + 1 = 0$ in any single generation; written with the full matrix Y_e it is $\bar{L}_{Li}\, H\, (Y_e)_{ij}\, e_{Rj}$, which conserves *total* lepton number $L = \sum_i L_i$ for any Y_e, and conserves each separate L_i in the basis where Y_e is diagonal. There is no gauge-invariant Yukawa operator coupling quarks to leptons, and no Higgs coupling that changes B or L — because H is colorless and leptonless and a single power of it cannot carry off any baryon or lepton charge.

Why nothing else appears. Could a B- or L-violating operator hide at $d \le 4$? A baryon-number-violating operator must contain a net of three quarks (the smallest color singlet with $B \neq 0$ that is not a quark–antiquark pair), since $\mathrm{SU}(3)_c$ invariance of a state with $B = \pm 1$ requires three (anti)fundamentals contracted with ϵ_{abc}. Three fermion fields have dimension $3 \times \frac{3}{2} = \frac{9}{2} > 4$, so no such operator fits under the renormalizability cut. A lepton-number-violating operator with $\Delta L = 2$ (the smallest allowed, since $\Delta L = 1$ alone cannot be a Lorentz scalar without a leftover spinor) needs two lepton fields, dimension $2 \times \frac{3}{2} = 3$, plus enough Higgs fields to neutralize the hypercharge $2 \times (-\frac{1}{2}) = -1$ of two lepton doublets; that requires two units of Higgs ($d = 2$), giving total dimension $3 + 2 = 5 > 4$. Both leading violations are thus excluded by one inequality each. There is simply no room at $d \le 4$.

Conclusion. Every admissible term carries $B = 0$ and $L_i = 0$. The Lagrangian is invariant under the four phases Eqs. (33.44) and (33.45), establishing G_{acc} of Eq. (33.46) and Theorem 33.10.

Two remarks sharpen the result. First, the appearance of three *separate* lepton numbers rather than a single total L is a feature of the minimal field content. In the catalogue above the only operator that knows about lepton flavor is $\bar{L}_L H Y_e\, e_R$, and a single complex matrix Y_e can always be brought to diagonal, positive form by independent unitary rotations of L_L and e_R (its singular-value decomposition). In that mass basis each diagonal phase $L_{Li}, e_{Ri} \to e^{i\gamma_i}(\dots)$ leaves Y_e invariant, so each L_i is conserved on its own. There is no second leptonic Yukawa matrix to obstruct the simultaneous diagonalization — precisely because there is no right-handed neutrino and hence no Y_ν. This is the operator-level shadow of the stabilizer computation of Theorem F.12, where the lepton stabilizer was found to be $\mathrm{U}(1)^3$ for exactly the same reason. In the quark sector, by contrast, the two matrices Y_u and Y_d *cannot* be simultaneously diagonalized, their misalignment being the CKM matrix V_{CKM}; only the overall common phase survives, and that single surviving phase is baryon number. So the asymmetry between "one B" and "three L_i" is the asymmetry between two quark Yukawas and one lepton Yukawa.

Second, the four symmetries are not all independent of the gauge symmetry in the way they might appear. The combination that will matter below is the difference $B - L \equiv B - \sum_i L_i$ and the sum $B + L$. We turn now to their very different quantum fates.

33.4.3 The quantum fate: $B+L$ is anomalous, $B-L$ is exact

A classical global symmetry need not survive quantization. The lesson of the chiral anomaly (Section 36.2) is that a current conserved by the classical equations of motion can acquire a nonzero divergence at one loop whenever the corresponding symmetry has a triangle anomaly with the gauge fields — the divergence being proportional to $F\widetilde{F}$ of the gauge group. Baryon and lepton number are global symmetries acting chirally (they treat left- and right-handed fields differently only through the representation content, but the currents are built from chiral fermions in chiral gauge representations), so we must ask whether their Noether currents

$$j_B^\mu = \tfrac{1}{3}\sum_{\text{quarks}} \bar{q}\,\gamma^\mu q, \qquad j_L^\mu = \sum_{\text{leptons}} \bar{\ell}\,\gamma^\mu \ell, \qquad L = \sum_i L_i, \tag{33.48}$$

remain conserved once the electroweak gauge fields are turned on inside loops. The answer is the single most important fact about these symmetries.

The anomalous divergences. The currents Eq. (33.48) can have triangle anomalies with the gauge factors $\mathrm{SU}(3)_c$, $\mathrm{SU}(2)_L$, and $\mathrm{U}(1)_Y$. We evaluate each, using the result imported from Section 36.2: a global current j_X^μ acquires, from the triangle with two gauge bosons of a factor with field strength F, a divergence

$$\partial_\mu j_X^\mu = \frac{1}{16\pi^2}\,\mathcal{A}_X\,F^A_{\mu\nu}\widetilde{F}^{A\,\mu\nu}, \qquad \mathcal{A}_X = \sum_{\text{LH Weyl}} X_f\,T(r_f) - \sum_{\text{RH Weyl}} X_f\,T(r_f), \tag{33.49}$$

where X_f is the X-charge of fermion f, $T(r_f)$ the Dynkin index of its gauge representation (Eq. (31.208); $T(\mathbf{2}) = \tfrac{1}{2}$ for $\mathrm{SU}(2)$, and for $\mathrm{U}(1)_Y$ one replaces $T(r) \to Y_f^2$ and the trace by the abelian charge product), and the relative sign distinguishes the two chiralities. We take the factors in turn, per generation.

Color, $\mathrm{SU}(3)_c$. Baryon number is vector-like under color: both the left-handed Q_L and the right-handed u_R, d_R carry $B = +\tfrac{1}{3}$ and sit in the fundamental $\mathbf{3}$. In Eq. (33.49) the left and right contributions enter with opposite sign and identical charge and index, so $\mathcal{A}_B^{\mathrm{SU}(3)} = \tfrac{1}{3}\,T(\mathbf{3}) - \tfrac{1}{3}\,T(\mathbf{3})\,[(\mathrm{L}) - (\mathrm{R})] = 0$. Leptons are color singlets, so $\mathcal{A}_L^{\mathrm{SU}(3)} = 0$ trivially. Neither B nor L has a QCD anomaly.

Weak isospin, $\mathrm{SU}(2)_L$. Only the left-handed doublets contribute, since right-handed fields are $\mathrm{SU}(2)_L$ singlets with $T(\mathbf{1}) = 0$. For baryon number, the doublet Q_L has $B = \tfrac{1}{3}$, sits in $\mathbf{2}$ with $T(\mathbf{2}) = \tfrac{1}{2}$, and comes in $N_c = 3$ colors:

$$\mathcal{A}_B^{\mathrm{SU}(2)} = N_c \cdot B_{Q_L} \cdot T(\mathbf{2}) = 3 \cdot \tfrac{1}{3} \cdot \tfrac{1}{2} = \tfrac{1}{2}. \tag{33.50}$$

For lepton number, the doublet L_L has $L = 1$, sits in $\mathbf{2}$, and comes in one "color":

$$\mathcal{A}_L^{\mathrm{SU}(2)} = 1 \cdot L_{L_L} \cdot T(\mathbf{2}) = 1 \cdot 1 \cdot \tfrac{1}{2} = \tfrac{1}{2}. \tag{33.51}$$

The two coefficients are *equal*. This is the crucial coincidence: the factor of $N_c = 3$ that multiplies the quark contribution is exactly compensated by the factor of $\frac{1}{3}$ in the baryon charge, so the colored quark doublet weighs the same in the anomaly as the single colorless lepton doublet.

Hypercharge, $\mathrm{U}(1)_Y$. Here we sum $X_f Y_f^2$ over Weyl fermions with the chirality sign. For baryon number, using the hypercharges of Table 31.1 ($Y_{Q_L} = \frac{1}{6}$, $Y_{u_R} = \frac{2}{3}$, $Y_{d_R} = -\frac{1}{3}$), counting the two isospin states of Q_L and the $N_c = 3$ colors throughout,

$$\mathcal{A}_B^{\mathrm{U}(1)} = N_c\Big[\underbrace{2\cdot\tfrac{1}{3}\cdot(\tfrac{1}{6})^2}_{Q_L} - \underbrace{\tfrac{1}{3}\cdot(\tfrac{2}{3})^2}_{u_R} - \underbrace{\tfrac{1}{3}\cdot(\tfrac{1}{3})^2}_{d_R}\Big] = 3\Big[\tfrac{1}{54} - \tfrac{4}{27} - \tfrac{1}{27}\Big] = 3\cdot(-\tfrac{1}{6}) = -\tfrac{1}{2}. \tag{33.52}$$

For lepton number ($Y_{L_L} = -\frac{1}{2}$, $Y_{e_R} = -1$),

$$\mathcal{A}_L^{\mathrm{U}(1)} = \underbrace{2\cdot 1\cdot(\tfrac{1}{2})^2}_{L_L} - \underbrace{1\cdot(1)^2}_{e_R} = \tfrac{1}{2} - 1 = -\tfrac{1}{2}. \tag{33.53}$$

Again equal. Assembling Eqs. (33.50) and (33.52) and summing over $N_g = 3$ generations, each baryon and lepton current has the identical anomalous divergence

$$\partial_\mu j_B^\mu = \partial_\mu j_L^\mu = \frac{N_g}{32\pi^2}\Big(g^2\, W^a_{\mu\nu}\widetilde{W}^{a\,\mu\nu} - g'^2\, B_{\mu\nu}\widetilde{B}^{\mu\nu}\Big), \tag{33.54}$$

where the relative sign of the $\mathrm{U}(1)_Y$ term is fixed by Eqs. (33.52) and (33.53) and the overall normalization is chosen so that the integral of the $\mathrm{SU}(2)_L$ term counts topological charge (below). Equation Eq. (33.54) is the entire story, and its content is read off by taking sum and difference.

Theorem 33.12 (Quantum fate of baryon and lepton number)**.** *In the quantized Standard Model:*

1. *The current* $j^\mu_{B-L} = j^\mu_B - j^\mu_L$ *is anomaly-free with respect to* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$*: subtracting Eq. (33.54) from itself,*

$$\partial_\mu\, j^\mu_{B-L} = 0, \tag{33.55}$$

   to all orders in the gauge interactions. Baryon-minus-lepton number is an exact symmetry of the renormalizable Standard Model, perturbatively and non-perturbatively.

2. *The current* $j^\mu_{B+L} = j^\mu_B + j^\mu_L$ *has twice the divergence Eq. (33.54) and is therefore anomalous. Its conservation fails non-perturbatively through topologically nontrivial* $\mathrm{SU}(2)_L$ *gauge configurations (instantons at zero temperature, sphalerons at high temperature), discussed in Section 42.4.*

Proof. Statement (1) is immediate: j_B and j_L have identical divergences Eq. (33.54), established term by term in Eqs. (33.50) to (33.53), so their difference is divergence-free. The cancellation is exact because it holds at the level of the anomaly coefficients, which receive no higher-loop corrections (the Adler–Bardeen theorem, Theorem 37.24); there is likewise no instanton-induced violation of $B-L$ because the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ anomalies of j_{B-L} both vanish, so the right-hand side of its divergence is identically zero, not merely zero in perturbation theory. Statement (2) follows by adding instead of subtracting: $\partial_\mu j^\mu_{B+L} = 2\,\partial_\mu j^\mu_B \neq 0$. To see that this is a genuine, if tiny, physical effect rather than a perturbative artifact, integrate over spacetime. The right-hand side of Eq. (33.54) is a sum of total derivatives, and the integral of the $\mathrm{SU}(2)_L$ piece is quantized,

$$\frac{g^2}{32\pi^2}\int d^4x\; W^a_{\mu\nu}\widetilde{W}^{a\,\mu\nu} = \Delta N_{\mathrm{cs}} \in \mathbb{Z}, \tag{33.56}$$

the change in Chern–Simons (winding) number between the asymptotic gauge vacua, established in Section 42.4; the $\mathrm{U}(1)_Y$ piece integrates to zero on such configurations because the hypercharge bundle carries no corresponding topological charge. Therefore

$$\Delta B = \Delta L = N_g\,\Delta N_{\mathrm{cs}}, \qquad \Delta(B-L) = 0, \qquad \Delta(B+L) = 2N_g\,\Delta N_{\mathrm{cs}}, \tag{33.57}$$

with $N_g = 3$. Each unit of $\mathrm{SU}(2)_L$ topological charge changes baryon and lepton number by three units apiece — one baryon and one lepton per generation — conserving $B-L$ exactly while changing $B+L$ by six. □

The selection rule Eq. (33.57) has a vivid microscopic picture, supplied by the 't Hooft vertex of Section 42.4: a single instanton or sphaleron transition emits one fermion from each of the twelve left-handed doublets of the Standard Model — nine quarks (three colors × three generations) and three leptons (one per generation) — so that the nine quarks assemble into three baryons ($\Delta B = 3$) and the three leptons give $\Delta L = 3$. At zero temperature the amplitude for such a transition is suppressed by the instanton action $e^{-8\pi^2/g^2} \sim e^{-4\pi/\alpha_W} \sim 10^{-160}$, utterly negligible for proton stability. At temperatures above the electroweak scale, however, the barrier between vacua can be crossed thermally over the sphaleron saddle, and $B+L$ violation becomes rapid — a fact of decisive importance for the cosmological generation of the baryon asymmetry, since it means any primordial $B+L$ is erased while any $B-L$ is preserved. The honest summary is that of all the global symmetries of $\mathcal{L}_{\mathrm{SM}}$, only $B-L$ and the combination of the individual L_i orthogonal to it that remains anomaly-free survive as exact; $B+L$ is broken by the quantum theory itself.

Physical Insight — two layers of approximate

Baryon and lepton number are approximate symmetries for two logically distinct reasons, and it is worth keeping them separate. The first is *internal* to the renormalizable Standard Model: $B + L$ is broken by non-perturbative gauge dynamics (sphalerons), with no new fields required. This breaking conserves $B - L$ and is invisible at zero temperature but matters in the early universe. The second is *external*: every accidental symmetry, $B - L$ included, can be broken by higher-dimension operators suppressed by a heavy scale Λ, operators the Standard Model does not forbid but only postpones. The first layer tells us the Standard Model is internally subtler than its classical Lagrangian; the second tells us it is incomplete, and points to where the completion lives. The remainder of this section reads off that second layer.

33.4.4 The leading symmetry-violating operators: windows onto higher scales

The catalogue of Theorem 33.11 found no B- or L-violating operator at $d \leq 4$ by two clean inequalities: three quarks need dimension $\frac{9}{2}$, two leptons plus the Higgs needed to neutralize their hypercharge need dimension 5. Relax the renormalizability cut — as we must if the Standard Model is an effective theory valid below some scale Λ (Section 38.2) — and these operators reappear at the lowest dimensions just above four. They are the Standard Model's first effective-field-theory windows onto whatever physics sits at Λ: by symmetry alone we can write them down, and by dimensional analysis alone we can convert an experimental bound on the process they mediate into a lower bound on Λ. We treat the two leading ones in order of dimension.

Dimension five: the Weinberg operator and Majorana neutrino mass

Worked Example 33.13 (The unique dimension-five operator and its quantum numbers)**.** We construct every gauge- and Lorentz-invariant local operator of dimension exactly five from Standard-Model fields, and find there is essentially one.

Field content of a $d = 5$ operator. With $[\psi] = \frac{3}{2}$, $[H] = 1$, $[\partial] = [F^{1/2}] \ldots$, the only ways to reach dimension five with an even number of fermions (Lorentz invariance forbids an odd number) are: zero fermions (pure bosonic, dimension built from H, D_μ, F — but any such operator is a polynomial in $H^\dagger H$, F^2, derivatives, all of even or integer dimension that at $d = 5$ would need an odd number of H's contracted into a gauge non-singlet, impossible for the single doublet H whose only invariant is $H^\dagger H$); or two fermions plus bosons of total dimension $5 - 3 = 2$. Two fermions ($d = 3$) leave room for exactly two units of

bosonic dimension. Two powers of H ($d = 2$) is the only renormalizable-field option (a single D_μ gives an odd half-integer mismatch with the spinor structure; two derivatives produce a redundant operator equivalent by the equations of motion to lower-dimension terms). So the candidate is schematically $\psi\psi HH$.

Gauge invariance fixes the fields. The two fermions must combine with two Higgs doublets into a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet. The Higgs $H \sim (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ carries hypercharge $+\frac{1}{2}$ each, so two of them carry $Y = +1$; to cancel it the two fermions must carry total $Y = -1$, and to form a color singlet they must be colorless. The only colorless fermion with the right hypercharge is the lepton doublet $L_L \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$: two of them give $Y = -1$ and color singlet. The quarks are excluded (colored), e_R is excluded ($Y = -1$ but a singlet under $\mathrm{SU}(2)_L$, leaving the SU(2) indices of HH uncontracted). Hence the operator is built from $L_L L_L HH$, and assembling the $\mathrm{SU}(2)_L$ doublet indices into singlets gives the *Weinberg operator*

$$\mathcal{L}_5 = \frac{c_{ij}}{2\Lambda} \left(\overline{L^c_{L\,i}}\, \widetilde{H}^*\right)\left(\widetilde{H}^\dagger L_{L\,j}\right) + \text{h.c.}, \qquad \widetilde{H} = i\sigma^2 H^*, \tag{33.58}$$

the unique[1] dimension-five operator, with c_{ij} a dimensionless coupling matrix and Λ the scale of the underlying physics. This is the operator of Eq. (F.107).

Symmetry of the flavor indices. The two lepton doublets are contracted symmetrically: the Lorentz contraction of two left-handed Weyl fermions $L^c_{L\,i} L_{L\,j}$ is symmetric under $i \leftrightarrow j$ (the charge-conjugate pairing $\epsilon_{\alpha\beta}$ on the spinor indices is antisymmetric, and the $\mathrm{SU}(2)_L$ singlet contraction ϵ_{ab} is antisymmetric, so the two antisymmetries combine to a symmetric flavor structure; Fermi statistics is thereby respected). Hence $c_{ij} = c_{ji}$ is a complex *symmetric* matrix, carrying six complex parameters.

Quantum numbers. The operator contains two lepton doublets and no antileptons, so its lepton number is $L = +2$: it carries $\Delta L = 2$, $\Delta B = 0$, and therefore

$$\Delta(B - L) = -2, \qquad \Delta(B + L) = +2. \tag{33.59}$$

The Weinberg operator is the leading violation of lepton number and, with it, of $B - L$ — the very symmetry that survived the anomaly. That the lowest-dimension operator beyond the Standard Model violates $B - L$ is a striking statement: the symmetry protected to all orders by the gauge anomalies is broken by the first irrelevant operator.

After electroweak symmetry breaking the neutral component of H acquires its vacuum value $\langle H^0 \rangle = v/\sqrt{2}$, and Eq. (33.58) becomes a Majorana mass term for the left-handed neutrinos,

$$\mathcal{L}_5 \xrightarrow{\text{EWSB}} -\tfrac{1}{2}\,(M_\nu)_{ij}\, \overline{\nu^c_{L\,i}}\, \nu_{L\,j} + \text{h.c.}, \qquad (M_\nu)_{ij} = c_{ij}\,\frac{v^2}{\Lambda}. \tag{33.60}$$

[1] Unique up to the choice of how the $\mathrm{SU}(2)_L$ and Lorentz indices are contracted; the antisymmetric contractions vanish (see below), leaving one independent structure per flavor pair.

This is the cleanest possible statement of why neutrinos are massive in the effective Standard Model and why their masses are so small. A Majorana mass is allowed because the neutrino, being electrically neutral and colorless, is the one Standard-Model fermion that can be its own antiparticle; the operator Eq. (33.58) supplies exactly such a mass, and Eq. (33.60) shows it is suppressed by one power of Λ rather than appearing at v like the charged-fermion masses. Numerically, $m_\nu \sim v^2/\Lambda$ with $v \approx 246\,\text{GeV}$ and $m_\nu \lesssim 0.1\,\text{eV}$ implies $\Lambda \sim v^2/m_\nu \sim (246\,\text{GeV})^2/(0.1\,\text{eV}) \sim 6 \times 10^{14}\,\text{GeV}$ — a scale tantalizingly close to that of grand unification. The dynamics that could generate Eq. (33.58) by integrating out heavy states, the seesaw mechanism, is developed in Section 35.2; here the point is structural: the smallness of neutrino mass is the smallness of a $d = 5$ coefficient, the generic signature of a heavy scale seen through the lowest EFT window.

Dimension six: proton decay

Baryon-number violation requires three quarks and so first appears at dimension six, where four fermions ($d = 6$) form a gauge singlet with no Higgs needed. The structures are four-fermion operators of the schematic form $qqq\,\ell$, antisymmetrized in color by ϵ_{abc} to make the three quarks an $\text{SU}(3)_c$ singlet (a "baryon") and contracted in $\text{SU}(2)_L$ and hypercharge to a total singlet with the lepton. A representative gauge-invariant operator, in the notation q, u^c, d^c, ℓ, e^c for the (left-handed) fields, is

$$\mathcal{L}_6 = \frac{1}{\Lambda^2}\Big[\, c_1\,\epsilon_{abc}\,(\overline{u_a^c}\,d_b)(\overline{Q_c}\,L) + c_2\,\epsilon_{abc}\,(\overline{u_a^c}\,d_b)(\overline{e^c}\,u_c) + \dots \Big] + \text{h.c.}, \tag{33.61}$$

where a, b, c are color indices and the dots stand for the other independent contractions (there are a handful, all of the same parametric size). Each such operator carries three quarks and one lepton:

$$\Delta B = +1, \qquad \Delta L = +1, \qquad \Delta(B - L) = 0. \tag{33.62}$$

Remarkably, the leading baryon-number-violating operators conserve $B - L$ — the same combination spared by the anomaly — so $B - L$ survives not only the renormalizable theory and the sphalerons but even the dimension-six corrections, with its leading violation postponed to the dimension-five Weinberg operator alone. The operators Eq. (33.61) drive the decay of the proton: a process such as $p \to e^+\pi^0$ converts the three valence quarks uud into a positron and a quark–antiquark pair, exactly the content of Eq. (33.61), and is depicted in Fig. 33.3.

The strength of Fig. 33.3 is that the lifetime it predicts can be estimated from dimensions alone, with no knowledge of the short-distance physics beyond the single scale Λ. This is the canonical example of an EFT window turning a non-observation into a scale bound.

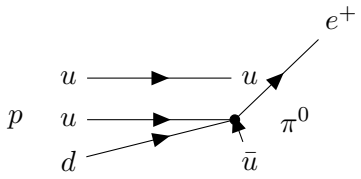


Figure 33.3: Proton decay $p \to e^+\pi^0$ through the dimension-six four-fermion operator Eq. (33.61) (filled vertex). Two of the proton's valence quarks and the operator's quark–antiquark pair hadronize into the π^0; the third quark and the lepton current produce the e^+. The vertex carries $\Delta B = \Delta L = 1$, conserving $B - L$.

Worked Example 33.14 (The proton-decay rate and the bound on Λ)**.** We estimate the proton decay rate from Eq. (33.61) by dimensional analysis and convert the experimental lower bound on the lifetime into a lower bound on Λ.

Parametric rate. The amplitude for $p \to e^+\pi^0$ is the matrix element of the operator Eq. (33.61) between the one-proton state and the final state. The operator carries a coefficient $1/\Lambda^2$; the hadronic matrix element of the three-quark current between a proton and a pion is set by the only strong-interaction scale available, which is of order the proton mass itself, $\langle \pi^0 e^+ | \, qqq\ell \, | p \rangle \sim m_p^3/\Lambda^2$ (three powers of mass to make up the dimension of the four-fermion operator's matrix element, all supplied by $m_p \sim \Lambda_{\rm QCD}$). A decay rate has dimensions of mass; with m_p the only scale left once Λ is factored out, dimensional analysis fixes the rate uniquely:

$$\Gamma_p \;\sim\; \frac{1}{8\pi}\,\frac{m_p^5}{\Lambda^4}. \tag{33.63}$$

The fifth power of m_p is forced: the squared amplitude carries $1/\Lambda^4$ and, by dimensions, four powers of m_p from the matrix element squared, and one more power of m_p from the phase-space/energy factor that makes Γ a mass; the $1/8\pi$ is the generic two-body phase-space factor, kept only to set the rough scale.

Numerical bound. The experimental lower bound on the partial lifetime is, as of the mid-2020s, $\tau_p = 1/\Gamma_p \gtrsim 10^{34}\,\text{yr}$ for the $e^+\pi^0$ channel (Super-Kamiokande). We solve Eq. (33.63) for Λ. Restoring $\hbar$ to convert years to inverse energy, $1\,\text{yr} = 3.15 \times 10^7\,\text{s}$ and $\hbar = 6.58 \times 10^{-25}\,\text{GeV s}$, so

$$\tau_p = 10^{34}\,\text{yr} = \frac{10^{34} \cdot 3.15 \times 10^7\,\text{s}}{6.58 \times 10^{-25}\,\text{GeV s}} \approx 4.8 \times 10^{65}\,\text{GeV}^{-1}. \tag{33.64}$$

From Eq. (33.63) with the $1/8\pi$ absorbed into the order-of-magnitude estimate, $\Lambda^4 \sim m_p^5\,\tau_p$, and with $m_p \approx 0.94\,\text{GeV}$ so that $m_p^5 \approx 0.73\,\text{GeV}^5$,

$$\Lambda^4 \sim (0.73\,\text{GeV}^5)\,(4.8{\times}10^{65}\,\text{GeV}^{-1}) \approx 3.5{\times}10^{65}\,\text{GeV}^4, \qquad \Lambda \sim (3.5{\times}10^{65})^{1/4}\,\text{GeV} \approx 2{\times}10^{1} \tag{33.65}$$

The pure dimensional estimate thus gives $\Lambda \gtrsim 10^{16}\,\mathrm{GeV}$. If the operator is generated by exchanging a heavy gauge boson of mass M_X at unified coupling $\alpha_{\mathrm{GUT}} \sim 1/25$, the rate carries an extra α_{GUT}^2 and $\Lambda^2 \to M_X^2/\alpha_{\mathrm{GUT}}$; restoring it lowers the inferred mediator mass to $M_X \sim \alpha_{\mathrm{GUT}}^{1/2}\Lambda \sim$ a few $\times 10^{15}\,\mathrm{GeV}$. Either way,

$$\boxed{\Lambda \;\gtrsim\; 10^{15}\text{–}10^{16}\,\mathrm{GeV}} \tag{33.66}$$

is the scale that proton stability probes — fourteen orders of magnitude above the proton mass, and within reach of no accelerator, yet read off from a single non-observation in a tank of water.

The two operators just constructed, Eq. (33.58) at dimension five and Eq. (33.61) at dimension six, are the leading terms of the Standard-Model effective field theory, and they exemplify the general logic developed in Section 38.2: an accidental symmetry of the renormalizable theory is broken first, and most weakly, by the lowest-dimension operator that the gauge symmetry permits, and the experimental bound on that operator measures the scale of the new physics. Neutrino mass and proton decay are not unrelated curiosities; they are the $d = 5$ and $d = 6$ entries in one expansion, the two cleanest probes the Standard Model offers of the world above the electroweak scale. That the operator-by-operator analysis reproduces, from the bottom up, the same $B - L$-conserving structure that the anomaly enforced from the top down is the strongest internal evidence that $B - L$ is the robust survivor and $B + L$ the expendable accident.

Having exhausted the global-symmetry content of the assembled Lagrangian — the accidental symmetries it possesses, the one $(B+L)$ that quantization removes, and the irrelevant operators that break the rest — one renormalizable, gauge- and Lorentz-invariant term remains uncatalogued. It appeared in passing in Theorem 33.11 as the topological monomial $G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu}$, and it carries $B = L_i = 0$ so it played no role here; but it is a total derivative, and whether such a term can be physical at all is a question the next section must settle before the parameter count can be declared complete.

33.5 The Theta Term and the Strong-CP Question

The catalogue of Section 33.2 listed every operator of mass dimension four or less that is invariant under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and under the Lorentz group, and assembled all but one of them into the Standard Model Lagrangian. The omission was deliberate and flagged: of the dimension-four invariants built from the gauge field strengths, one is parity-odd, the contraction of a field strength with its own dual. We set it aside because, unlike every other term in the Lagrangian, it is a total spacetime derivative, and a naive reading of the action principle says that total derivatives do not affect the equations of motion and so

cannot matter. That reading is correct order by order in perturbation theory and wrong non-perturbatively. The purpose of this section is to put the missing term back on the table, to prove carefully both halves of the apparent paradox — that it is a total derivative and that it is nevertheless physical — to determine which of the three gauge factors actually contributes a physical parameter, to identify the genuine basis-independent observable hiding behind the naive coefficient, and finally to state, as sharply as the data allow, the puzzle this parameter poses: the strong-CP problem.

33.5.1 The one remaining renormalizable operator

For each simple gauge factor the field strength $F_{\mu\nu}$ admits, besides the kinetic invariant $F^a_{\mu\nu}F^{a\,\mu\nu}$, a second Lorentz scalar of the same mass dimension, formed by contracting F with its Hodge dual

$$\widetilde{F}^{a\,\mu\nu} \;=\; \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F^a_{\rho\sigma}, \qquad \epsilon^{0123}=+1, \tag{33.67}$$

the Levi-Civita symbol carrying the convention fixed in Chapter 36. The scalar $F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu}$ is gauge invariant (it is the trace of a product of two adjoint-valued curvatures), Lorentz invariant, and of mass dimension four, hence marginal and admissible by every power-counting criterion that admitted the kinetic terms. For the colour factor $\mathrm{SU}(3)_c$, with the gluon field strength $G^A_{\mu\nu}$ of Eq. (33.15), the operator is

$$\mathcal{L}_\theta \;=\; \theta_{\mathrm{QCD}}\,\frac{g_s^2}{32\pi^2}\,G^A_{\mu\nu}\,\widetilde{G}^{A\,\mu\nu}, \qquad \widetilde{G}^{A\,\mu\nu}=\tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}G^A_{\rho\sigma}, \tag{33.68}$$

matching the reference form Eq. (F.88) of Appendix F. The prefactor $g_s^2/32\pi^2$ is a convention, chosen so that the coefficient θ_{QCD} is an angle in the precise sense established below: the action contribution from (33.68) will turn out to be θ_{QCD} times an integer, so that the physics is periodic, $\theta_{\mathrm{QCD}} \to \theta_{\mathrm{QCD}} + 2\pi$. We absorb the g_s^2 into the definition precisely so that this integer is the bare topological charge of the gauge configuration, independent of the coupling.

Three features mark this operator as different from its neighbours in the Lagrangian. First, it is the unique remaining renormalizable invariant: once the kinetic, Yukawa, and Higgs-potential terms are written, (33.68) (one copy per gauge factor) exhausts the list of dimension-≤ 4 operators consistent with the gauge and spacetime symmetries, as the counting of Section 33.2 recorded. Second, it is odd under parity and under time reversal. Under a parity transformation the dual carries an extra sign relative to F because $\widetilde{F}$ involves the Levi-Civita tensor: $F\widetilde{F}$ is the four-dimensional analogue of $\mathbf{E}\cdot\mathbf{B}$, which flips sign under $\mathbf{x}\to-\mathbf{x}$ (the magnetic field, an axial vector, is even; the electric field, a polar vector, is odd). The same structure makes it odd under time reversal ($\mathbf{E}$ even, $\mathbf{B}$ odd). It is even under charge conjugation. By the CPT theorem invoked throughout Chapter 36,

a term that is P-odd and T-odd and C-even is CP-odd: (33.68) is a source of CP violation distinct in origin from the CKM phase of Section 32.4. Third, and this is the feature that makes the rest of the section necessary, $F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu}$ is a total derivative.

33.5.2 A total derivative that is not a total triviality

We prove the total-derivative statement in full, for the non-abelian case, since the colour factor is the one that will matter. The result will be exhibited as the divergence of a definite current — the Chern–Simons current — whose explicit form is the key to understanding why the term survives.

Proposition 33.15 (The topological density is a total derivative)**.** *The gluon density appearing in* (33.68) *satisfies*

$$G^A_{\mu\nu}\,\widetilde{G}^{A\,\mu\nu} \;=\; \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}\,G^A_{\mu\nu}G^A_{\rho\sigma} \;=\; \partial_\mu K^\mu, \tag{33.69}$$

where the Chern–Simons current is

$$K^\mu \;=\; 2\,\epsilon^{\mu\nu\rho\sigma}\Big(G^A_\nu\,\partial_\rho G^A_\sigma + \tfrac{g_s}{3}\,f^{ABC}\,G^A_\nu G^B_\rho G^C_\sigma\Big). \tag{33.70}$$

Proof. The first equality in (33.69) is just the definition (33.67) of the dual: $G\widetilde{G} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}G_{\mu\nu}G_{\rho\sigma}$. Write the field strength (33.15) as an abelian part plus a self-coupling part,

$$G^A_{\mu\nu} = g^A_{\mu\nu} + g_s\,f^{ABC}G^B_\mu G^C_\nu, \qquad g^A_{\mu\nu} \equiv \partial_\mu G^A_\nu - \partial_\nu G^A_\mu, \tag{33.71}$$

and insert into $Q \equiv \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}G^A_{\mu\nu}G^A_{\rho\sigma}$. The product generates three pieces, of zeroth, first, and second order in the self-coupling:

$$Q = \underbrace{\tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}g^A_{\mu\nu}g^A_{\rho\sigma}}_{Q_0} + \underbrace{g_s\,\epsilon^{\mu\nu\rho\sigma}f^{ABC}G^B_\mu G^C_\nu\,g^A_{\rho\sigma}}_{Q_1} + \underbrace{\tfrac{g_s^2}{2}\epsilon^{\mu\nu\rho\sigma}f^{ABC}f^{ADE}G^B_\mu G^C_\nu G^D_\rho G^E_\sigma}_{Q_2}, \tag{33.72}$$

where in Q_1 the factor of two from the two cross terms has been used.

The quartic piece vanishes. Collect the four gauge fields in Q_2 into matrices $G_\mu \equiv G^A_\mu T^A$ in the fundamental. Using $[T^B,T^C] = if^{BCE}T^E$ and $\mathrm{tr}(T^AT^E) = \frac{1}{2}\delta^{AE}$, one has $f^{ABC}G^B_\mu G^C_\nu = -2i\,\mathrm{tr}(T^A[G_\mu,G_\nu])$, and Q_2 becomes a multiple of $\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(G_\mu G_\nu G_\rho G_\sigma)$. This trace vanishes identically: a cyclic permutation of the four matrices is a four-cycle, an odd permutation of the four contracted indices, so contracting with the totally antisymmetric ϵ gives $\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(G_\mu G_\nu G_\rho G_\sigma) = -\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(G_\mu G_\nu G_\rho G_\sigma) = 0$. Hence $Q_2 = 0$.

The abelian piece. In Q_0 use the antisymmetry of ϵ to drop the explicit antisymmetrization in $g^A_{\mu\nu} = \partial_\mu G^A_\nu - \partial_\nu G^A_\mu$: contracting either factor with $\epsilon^{\mu\nu\rho\sigma}$ doubles it, $\epsilon^{\mu\nu\rho\sigma}g^A_{\mu\nu} = 2\epsilon^{\mu\nu\rho\sigma}\partial_\mu G^A_\nu$, so

$$Q_0 = \tfrac{1}{2}\,(2\partial_\mu G^A_\nu)(2\partial_\rho G^A_\sigma)\epsilon^{\mu\nu\rho\sigma} = 2\,\epsilon^{\mu\nu\rho\sigma}\partial_\mu G^A_\nu\,\partial_\rho G^A_\sigma. \tag{33.73}$$

Because $\epsilon^{\mu\nu\rho\sigma}$ is antisymmetric in $\mu \leftrightarrow \rho$ while $\partial_\mu\partial_\rho$ is symmetric, the second derivative may be pulled outside: $\epsilon^{\mu\nu\rho\sigma}\partial_\mu G^A_\nu \partial_\rho G^A_\sigma = \partial_\mu(\epsilon^{\mu\nu\rho\sigma} G^A_\nu \partial_\rho G^A_\sigma)$, the cross term $\epsilon^{\mu\nu\rho\sigma} G^A_\nu \partial_\mu\partial_\rho G^A_\sigma$ vanishing by that symmetry. Thus

$$Q_0 = \partial_\mu\big(2\,\epsilon^{\mu\nu\rho\sigma} G^A_\nu\, \partial_\rho G^A_\sigma\big), \tag{33.74}$$

which is the divergence of the abelian part of (33.70).

The cubic piece. In Q_1 the same doubling on the abelian factor $g^A_{\rho\sigma}$ gives

$$Q_1 = 2g_s\,\epsilon^{\mu\nu\rho\sigma} f^{ABC} G^B_\mu G^C_\nu\, \partial_\rho G^A_\sigma. \tag{33.75}$$

We claim this equals the divergence of the cubic part of (33.70). Compute that divergence directly, using the product rule:

$$\partial_\mu\Big(\tfrac{2g_s}{3}\epsilon^{\mu\nu\rho\sigma} f^{ABC} G^A_\nu G^B_\rho G^C_\sigma\Big) = \tfrac{2g_s}{3}\epsilon^{\mu\nu\rho\sigma} f^{ABC} \times \big[(\partial_\mu G^A_\nu) G^B_\rho G^C_\sigma + G^A_\nu (\partial_\mu G^B_\rho) G^C_\sigma + G^A_\nu G^B_\rho (\partial_\mu G^C_\sigma)\big]. \tag{33.76}$$

The three terms in brackets are equal. Take the second: relabel the four spacetime dummies by the cyclic shift $(\mu,\nu,\rho,\sigma) \to (\rho,\sigma,\mu,\nu)$, under which $\epsilon^{\rho\sigma\mu\nu} = \epsilon^{\mu\nu\rho\sigma}$ (a shift by two positions is an even permutation), and relabel the colour dummies $(A,B,C) \to (B,C,A)$, under which $f^{BCA} = f^{ABC}$ by cyclicity of the structure constants. The second term becomes $\epsilon^{\mu\nu\rho\sigma} f^{ABC} (\partial_\rho G^A_\sigma) G^B_\mu G^C_\nu$, identical in form to the first up to the placement of the derivative, and equal to it. The third term maps to the first by the inverse shift. Hence the bracket equals 3 times the first term, and

$$\partial_\mu\Big(\tfrac{2g_s}{3}\epsilon^{\mu\nu\rho\sigma} f^{ABC} G^A_\nu G^B_\rho G^C_\sigma\Big) = 2g_s\,\epsilon^{\mu\nu\rho\sigma} f^{ABC} (\partial_\rho G^A_\sigma) G^B_\mu G^C_\nu = Q_1, \tag{33.77}$$

the last equality being the same relabelling that brought (33.75) into this form. Adding (33.74) and (33.77), and recalling $Q_2 = 0$, gives $Q = \partial_\mu K^\mu$ with K^μ as in (33.70). □

Equation (33.69) is the source of the apparent paradox. Since $\mathcal{L}_\theta$ is proportional to $\partial_\mu K^\mu$, the contribution of the θ term to the action is

$$S_\theta = \theta_{\mathrm{QCD}}\,\frac{g_s^2}{32\pi^2}\int d^4x\,\partial_\mu K^\mu = \theta_{\mathrm{QCD}}\,\frac{g_s^2}{32\pi^2}\oint_\partial d\Sigma_\mu\, K^\mu, \tag{33.78}$$

a pure boundary integral. Varying it produces no contribution to the Euler–Lagrange equations, because the variation of a boundary term vanishes for variations of compact support; the classical gluon dynamics is completely blind to θ_{QCD}. Order by order in perturbation theory the same conclusion holds, with a sharper face: every Feynman diagram is built from fields fluctuating around the trivial vacuum $G^A_\mu = 0$, the boundary integral (33.78) then vanishes because

$K^\mu \to 0$ at infinity, and so $\theta_{\rm QCD}$ never enters any perturbative amplitude. Were this the whole story, $\theta_{\rm QCD}$ would be unobservable and the discussion would end here.

It is not the whole story. The path integral that defines the quantum theory sums over *all* field configurations of finite Euclidean action, not only those that decay to $G^A_\mu = 0$ at infinity. A configuration may instead tend to a *pure gauge* at the boundary, $G_\mu \to (i/g_s)\, U\, \partial_\mu U^{-1}$ with $U(x)$ a gauge transformation, for which the field strength vanishes (so the action is finite) but the gauge potential does not. The boundary of Euclidean $\mathbb{R}^4$ is a three-sphere S^3, and the asymptotic map $U : S^3 \to \mathrm{SU}(3)_c$ falls into homotopy classes labelled by an integer winding number, because $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$ for every $N \geq 2$. For such a configuration the boundary integral (33.78) does not vanish; it computes precisely that integer. The full non-perturbative statement, established in Section 42.4, is that

$$\nu \;\equiv\; \frac{g_s^2}{32\pi^2} \int d^4x\; G^A_{\mu\nu} \widetilde{G}^{A\,\mu\nu} \;\in\; \mathbb{Z}, \tag{33.79}$$

the integer ν being the topological charge (instanton number) of the configuration, so that $S_\theta = \theta_{\rm QCD}\, \nu$ and the configuration enters the path integral with the weight $e^{iS_\theta} = e^{i\theta_{\rm QCD}\nu}$. Configurations of different winding number are weighted by different phases, and physics depends on $\theta_{\rm QCD}$ through these phases. The dependence is genuinely non-perturbative: it is invisible to any finite order in g_s and visible only because the measure includes topologically nontrivial sectors. This is also why the normalization of (33.68) was chosen as it was — it makes ν an integer and renders the physics manifestly 2π-periodic in $\theta_{\rm QCD}$, $e^{i(\theta_{\rm QCD}+2\pi)\nu} = e^{i\theta_{\rm QCD}\nu}$. We import (33.79) and the construction of the θ-vacuum that gives it operational meaning from Section 42.4; here we need only its consequence, that $\theta_{\rm QCD}$ is a physical parameter of QCD.

Physical insight: why a total derivative can matter

The lesson is not special to gauge theory. A total derivative in the Lagrangian is dynamically inert only when the fields it is built from decay at the boundary so fast that the surface term vanishes. In quantum mechanics on a circle the term $\theta\, \dot\phi$ is a total time derivative, yet it shifts the spectrum because the relevant paths wind around the circle; the winding number is an integer and θ multiplies it. The QCD θ term is the field-theoretic version of exactly this: the "circle" is the third homotopy group of the gauge group, the winding is the instanton number (33.79), and $\theta_{\rm QCD}$ is the angle conjugate to it. Perturbation theory, blind to winding, never sees it.

33.5.3 Why only the colour theta survives

The argument just given applies, in principle, to each of the three gauge factors: each admits an operator of the form (33.68) with its own angle. There are thus three candidate parameters, θ_3 for $\mathrm{SU}(3)_c$, θ_2 for $\mathrm{SU}(2)_L$, and θ_1 for $\mathrm{U}(1)_Y$. We now show that two of the three are not physical, so that only $\theta_3 \equiv \theta_{\mathrm{QCD}}$ remains. Two independent mechanisms remove the others, and it is worth seeing precisely which mechanism disposes of which, because the survival of θ_{QCD} is the residue of both failing for colour.

The hypercharge angle: no instantons. The whole non-perturbative effect rested on the existence of topologically nontrivial gauge configurations, classified by π_3 of the gauge group. For an abelian group $\mathrm{U}(1)_Y$ this homotopy group is trivial, $\pi_3(\mathrm{U}(1)) = 0$: there are no finite-action configurations of nonzero winding, the boundary integral (33.78) vanishes on every configuration of finite action, and the integer ν of (33.79) is always zero. The term $\theta_1\,(g'^2/32\pi^2)B_{\mu\nu}\widetilde{B}^{\mu\nu}$ is then a total derivative with no surviving boundary contribution: it is dynamically trivial, perturbatively *and* non-perturbatively, and θ_1 drops out of the theory entirely. (Concretely, $B_{\mu\nu}\widetilde{B}^{\mu\nu} = \partial_\mu(2\epsilon^{\mu\nu\rho\sigma}B_\nu\partial_\rho B_\sigma)$ from the abelian case of Theorem 33.15, and the hypercharge field has no pure-gauge configurations with nonzero asymptotic winding.)

The weak angle: rotated away by an anomalous symmetry. The weak factor $\mathrm{SU}(2)_L$ does have instantons — $\pi_3(\mathrm{SU}(2)) = \mathbb{Z}$ — so the abstract argument that disposed of θ_1 fails, and θ_2 would seem to be physical. It is not, but for a different reason: it can be removed by a field redefinition that costs nothing else. The relevant redefinition is a global rotation of the baryon-plus-lepton number $B + L$, the accidental symmetry catalogued in Section 33.4. That current is anomalous with respect to $\mathrm{SU}(2)_L$: only the left-handed fermions are $\mathrm{SU}(2)_L$ doublets, so the $B + L$ current has a nonzero $\mathrm{SU}(2)_L^2$ triangle anomaly, and a global $B + L$ rotation by an angle β shifts the weak vacuum angle, $\theta_2 \to \theta_2 - c\,\beta$ with a nonzero constant c fixed by the anomaly of Chapter 36. Crucially, $B+L$ is a *vector* symmetry: it rotates left- and right-handed components of each fermion by the same phase, and so leaves every mass term and every Yukawa coupling invariant. Performing the rotation that sets $\theta_2 = 0$ therefore changes nothing else in the Lagrangian — no mass acquires a phase, no coupling is altered. The weak angle is pure gauge for an exact (if anomalous) classical symmetry, and is unobservable. We may set $\theta_2 = 0$ once and for all.

The colour angle: nothing removes it for free. For $\mathrm{SU}(3)_c$ both escape routes are blocked. Instantons exist ($\pi_3(\mathrm{SU}(3)) = \mathbb{Z}$), so θ_{QCD} is not killed by the abelian argument. And the only continuous symmetry whose anomaly can shift θ_{QCD} is the *axial* $\mathrm{U}(1)_A$ that rotates left- and right-handed quarks oppositely — QCD is vector-like, so baryon number, being a vector symmetry, has no colour anomaly and cannot touch θ_{QCD}. The axial rotation does have the required $\mathrm{SU}(3)_c^2$ anomaly, but it is not a symmetry of the Lagrangian: it is broken explicitly

by the quark mass terms. Using it to rotate θ_{QCD} away therefore comes at a price, the appearance of phases in the quark masses. The colour angle cannot be removed for free; what survives is the combination of it and the mass phases that no field redefinition can change. Making that statement precise is the content of the next subsection.

33.5.4 The physical angle $\bar{\theta}$

The angle θ_{QCD} as written in (33.68) is not by itself the observable, because its value can be moved by a chiral redefinition of the quark fields — precisely the redefinition used in Section 32.4 to bring the Yukawa matrices to the mass basis. The physical content is the rephasing-invariant combination that stays fixed under all such redefinitions. We derive it.

Recall from the flavour analysis that diagonalizing the quark Yukawas requires, in general, independent unitary rotations of the left- and right-handed quark fields; the part of those rotations that is not vector-like is a chiral (axial) rotation, and it is anomalous. The cleanest way to expose the invariant is to track what a chiral rotation does to the two CP-odd quantities in play: the angle θ_{QCD} and the phase of the quark mass determinant.

Worked Example 33.16 (The strong-CP angle is rephasing invariant)**.** Consider a single quark flavour with a complex mass term, written in the chiral basis as

$$\mathcal{L}_m = -m\,\bar{q}_L q_R + \text{h.c.} = -|m|\,\big(e^{i\varphi}\,\bar{q}_L q_R + e^{-i\varphi}\,\bar{q}_R q_L\big), \qquad \varphi = \arg m, \tag{33.80}$$

together with the colour θ term (33.68). Perform the axial rotation

$$q \to e^{i\alpha\gamma^5} q, \qquad \text{equivalently} \qquad q_L \to e^{-i\alpha} q_L, \quad q_R \to e^{+i\alpha} q_R, \tag{33.81}$$

using $\gamma^5 q_{L,R} = \mp q_{L,R}$ for left/right Weyl components.

Step 1: the mass phase shifts. Under (33.81) the bilinear transforms as $\bar{q}_L q_R \to e^{2i\alpha}\,\bar{q}_L q_R$ (the left factor contributes $e^{+i\alpha}$ from $\bar{q}_L = q_L^\dagger \gamma^0$ acting through the chirality flip, the right factor $e^{+i\alpha}$ directly). The rotated mass term $-m\,\bar{q}_L q_R +$ h.c. therefore carries an extra factor $e^{2i\alpha}$, i.e. the effective mass becomes $m\,e^{2i\alpha}$, so its phase shifts as

$$\varphi = \arg m \;\longrightarrow\; \varphi + 2\alpha. \tag{33.82}$$

Step 2: the theta angle shifts. The rotation (33.81) is not a symmetry of the quantum theory: the path-integral measure is not invariant, and the Fujikawa Jacobian computed in Section 36.3 multiplies the measure by $\exp\big[-2i\alpha\,(g_s^2/32\pi^2)\int d^4x\, G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu}\big]$ for a single quark in the fundamental of colour. This is exactly a shift of the θ term (33.68) in the action,

$$\theta_{\text{QCD}} \;\longrightarrow\; \theta_{\text{QCD}} - 2\alpha, \tag{33.83}$$

the coefficient -2 being the anomalous axial charge of one Dirac fermion (its single left-handed plus single right-handed zero mode in a unit-instanton background, $\Delta Q_5 = 2$, imported from the index computation of Chapter 36).

Step 3: the invariant. Comparing (33.82) and (33.83), the rotation moves $\theta_{\rm QCD}$ and φ in opposite directions by equal amounts, so their sum is untouched:

$$\bar\theta \;\equiv\; \theta_{\rm QCD} + \varphi \;\longrightarrow\; (\theta_{\rm QCD} - 2\alpha) + (\varphi + 2\alpha) \;=\; \bar\theta, \tag{33.84}$$

where the cancellation reflects that the same rotation (33.81) is what removes φ from (33.80) (choosing $2\alpha = -\varphi$ makes the mass real) at the cost of (33.83). Either the phase sits in the mass and $\theta_{\rm QCD}$ is bare, or it is rotated into $\theta_{\rm QCD}$ and the mass is real; the invariant content is $\bar\theta = \theta_{\rm QCD} + \arg m$.

The same computation run over all quark flavours promotes $\arg m$ to the phase of the full mass determinant. The mass matrices $M_u = (v/\sqrt{2})Y_u$ and $M_d = (v/\sqrt{2})Y_d$ of Section 32.4 are diagonalized by bi-unitary transformations whose chiral parts contribute, by the anomaly, a shift of $\theta_{\rm QCD}$ equal to (minus twice) the change in $\arg\det(M_u M_d)$; since $v > 0$ is real, $\arg\det M = \arg\det Y$. The invariant under the entire chiral field-redefinition group is therefore

$$\boxed{\bar\theta \;=\; \theta_{\rm QCD} + \arg\det(M_u M_d) \;=\; \theta_{\rm QCD} + \arg\det(Y_u Y_d)} \tag{33.85}$$

in agreement with the reference definition Eq. (F.89) of Appendix F. It is $\bar\theta$, not $\theta_{\rm QCD}$ and not $\arg\det(Y_u Y_d)$ separately, that is observable: each of the two summands is basis-dependent and can be set to any value by a chiral rotation, but their sum is fixed. This is the same statement, viewed from the strong sector, that organizes the parameter count of Section F.4.2: the single physical strong-CP parameter is $\bar\theta$, counted once.

Caution: a massless quark would erase the problem

The shift (33.83) costs a mass phase only because the mass is nonzero. If any single quark were exactly massless, the axial rotation (33.81) acting on that flavour alone would be a genuine symmetry (it leaves the now-absent mass term invariant) whose anomaly nonetheless shifts $\theta_{\rm QCD}$ freely; one could then rotate $\bar\theta$ to zero at no cost, and strong CP violation would vanish identically. The empirical non-vanishing of all quark masses — in particular $m_u \neq 0$, established to high confidence in lattice QCD — is what keeps $\bar\theta$ physical and the problem of the next subsection alive.

33.5.5 The strong-CP problem

We can now state the puzzle precisely. The parameter $\bar\theta$ is a dimensionless, CP-odd number, a priori of order unity — it is an angle, and nothing in the structure

of the Standard Model fixes its value. Being CP-odd, it generates CP-violating observables in the strong sector, the cleanest of which is a permanent electric dipole moment of the neutron. A nonzero $\bar{\theta}$ aligns an electric dipole moment along the neutron's spin; as Fig. 33.4 makes explicit, such an alignment is odd under both parity and time reversal, hence CP-odd, and so can arise only from a CP-violating interaction. A chiral-perturbation-theory estimate, controlled by the same quark masses that make $\bar{\theta}$ physical, gives the neutron electric dipole moment as

$$d_n \;\sim\; \bar{\theta}\,\frac{e\,m_q}{m_N^2} \;\sim\; \bar{\theta}\times(10^{-16}\,e{\cdot}\mathrm{cm}), \tag{33.86}$$

with m_q a light-quark mass, m_N the nucleon mass, and e the proton charge. The current experimental bound on the neutron electric dipole moment, $|d_n| < 1.8\times10^{-26}\,e{\cdot}$cm as of the mid-2020s, then translates (33.86) into

$$|\bar{\theta}| \;\lesssim\; 10^{-10}. \tag{33.87}$$

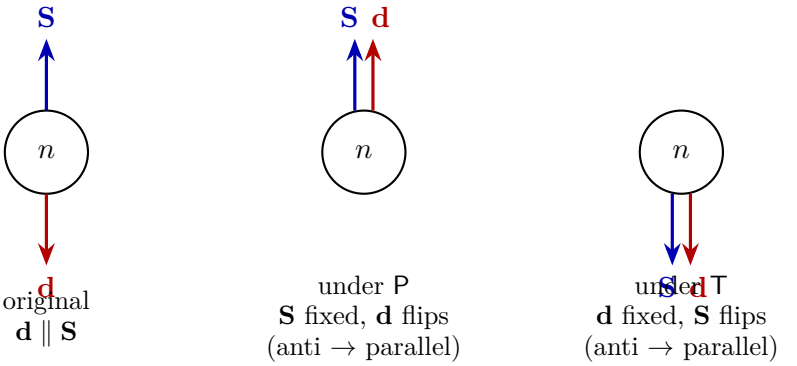


Figure 33.4: Why a static neutron electric dipole moment is CP-odd. A nonzero d_n must lie along the only available axis, the spin **S**. Under parity P the spin (an axial vector) is unchanged while the dipole **d** (a polar vector) reverses, so the parallel configuration maps to the antiparallel one — the state is not P-invariant. Under time reversal T the spin reverses while the dipole is unchanged, again exchanging parallel and antiparallel. A static $d_n \neq 0$ thus violates both P and T; by CPT it violates CP. Only a CP-odd parameter such as $\bar{\theta}$ can generate it.

The strong-CP problem is the statement (33.87): a dimensionless parameter whose natural value is of order one is measured to be smaller than 10^{-10}, and the Standard Model offers no reason for this. The smallness is genuinely puzzling for two reasons that must be kept distinct. First, $\bar{\theta}$ receives contributions from two logically independent sources — the bare gauge angle θ_{QCD} and the phase $\arg\det(Y_u Y_d)$ of the Yukawa sector, which is governed by the same complex Yukawa matrices that produce the large CP violation observed in kaon and B mixing. There is no symmetry relating these two contributions, so their near-exact cancellation to ten decimal places, required by (33.87), is an unexplained

conspiracy between the strong and the flavour sectors. Second, setting $\bar{\theta} = 0$ does *not* enhance the symmetry of the theory: CP is already broken elsewhere, by the CKM phase of Section 32.4, so one cannot appeal to an exact CP symmetry to protect $\bar{\theta} = 0$. A technically natural explanation in 't Hooft's sense — a small parameter justified by an enhanced symmetry at the point where it vanishes — is therefore unavailable within the Standard Model as it stands. The tuning is real and unexplained.

It is worth being equally precise about what the problem is *not*. It is not an inconsistency: any value of $\bar{\theta}$ gives a perfectly well-defined, unitary, renormalizable theory. It is not a failure of any prediction: (33.87) is a measurement, not a discrepancy with a calculation. The problem is purely one of naturalness — a number that the theory leaves free has been found, by experiment, to sit implausibly close to a special value that the theory does not single out. This is a different kind of dissatisfaction from a wrong prediction, and recognizing the difference is part of reading the Standard Model honestly. The two leading frameworks that would replace the tuning with a mechanism — the Peccei–Quinn symmetry and its axion, and the dynamics of the θ-vacuum that make $\bar{\theta}$ a dynamical field — are deferred, as promised, to Section 35.4 and Section 42.4; neither is established experimentally, and we resist the temptation to present a resolution as though the problem were closed.

With $\bar{\theta}$ identified as the nineteenth and final marginal parameter of the assembled theory, every term and every coupling that the gauge symmetry, Lorentz invariance, and renormalizability permit is now on the table. What remains is to count: to determine how many of these Lagrangian quantities are genuinely independent once field redefinitions are quotiented out, a tally in which $\bar{\theta}$ enters exactly once, as the analysis of Section F.4.2 and the parameter count of the next section make explicit.

33.6 The Parameters of the Standard Model

The previous section closed the catalogue of renormalizable terms by adding the last one, the QCD vacuum angle, and identified the single physically meaningful combination it contributes, $\bar{\theta} = \theta + \arg\det(M_u M_d)$. With that the Lagrangian is complete: every operator allowed by $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge invariance, Lorentz invariance and renormalizability is written down, and each carries a coefficient. We are therefore in a position to do something the assembly has been building toward — to count. How many numbers must be measured before the theory predicts everything else? The answer is the most quoted single fact about the Standard Model: *nineteen*. But it is usually quoted as folklore, a number to be memorized. It is not folklore. It is a theorem, and the purpose of this section is to derive it by assembling two sub-counts the reader has already met — the one-generation electroweak tally of Section 31.7.5 and the flavor tally of

Section 32.1.5 — into a single census, and then to ask the harder question the number invites: of these nineteen, how many does the theory *explain*, and how many does it merely *absorb*?

The distinction between assembling and counting is worth stating sharply at the outset, because the count is not a count of symbols. A Lagrangian is a presentation of a theory in a chosen set of field variables, and two Lagrangians related by an invertible local field redefinition — one that leaves the action and the path-integral measure functionally identical — describe *the same physics*: the same S-matrix, the same correlation functions, the same cross sections. A number written in the Lagrangian is a genuine physical parameter only if no such redefinition can change or remove it. The count of physical parameters is therefore the dimension of the space of Lagrangians *modulo* the group of allowed field redefinitions. In the gauge–Higgs sector almost no redefinition freedom acts, and the count is immediate; in the Yukawa sector a large continuous group of flavor rotations acts, and most of the apparent parameters are redundant. The whole subtlety of the nineteen lives in that second sector, and the machinery for handling it — the orbit–stabilizer counting of a broken symmetry — is the same machine that, run with the field content enlarged, will deliver the neutrino-mass extensions.

33.6.1 The rigid sector: six parameters

We begin with the part of the count that needs no quotient, because the operators involved are rigid: there is no field redefinition that rotates one of their coefficients into another or scales it away against the canonically normalized kinetic terms.

The gauge sector of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ carries exactly one coupling per simple or abelian factor. Once each gauge field is given its canonical kinetic term $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ the structure constants are fixed by the algebra and the only free number is the overall strength of the covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$. There are three factors, hence three real couplings,

$$g_s \ (\text{for } \mathrm{SU}(3)_c), \qquad g \ (\text{for } \mathrm{SU}(2)_L), \qquad g' \ (\text{for } \mathrm{U}(1)_Y). \tag{33.88}$$

Two of these, g and g', are the electroweak couplings counted already in Section 31.7.5; the third, g_s, is the strong coupling imported with the QCD sector of Chapters 25 and 28. The hypercharge normalization is the one fixed for the book, $Q = T^3 + Y$ with $Y_{L_L} = -\frac{1}{2}$, so that g' is unambiguous.

The single Higgs doublet $H \sim (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ contributes the two real parameters of its renormalizable potential,

$$V(H) = -\mu^2\, H^\dagger H + \lambda\, (H^\dagger H)^2, \qquad \mu^2 > 0,\ \lambda > 0, \tag{33.89}$$

the mass parameter μ^2 and the dimensionless quartic λ. These are the two Higgs parameters of Section 31.7.5; they are independent inputs because no

redefinition relates μ^2 to λ, and they may be traded one-for-one for the physically transparent pair (v, m_h) through $v^2 = \mu^2/\lambda$ and $m_h^2 = 2\mu^2 = 2\lambda v^2$. The kinetic normalization of H is not a parameter: it is fixed by canonical normalization exactly as for the gauge fields.

Finally there is the one parameter peculiar to the non-abelian strong sector, the QCD vacuum angle that the previous section discussed. As established there, its physically meaningful, rephasing-invariant form is the single real number

$$\bar{\theta} = \theta + \arg\det(M_u M_d), \tag{33.90}$$

with M_u, M_d the quark mass matrices, observable through CP-odd strong observables such as the neutron electric dipole moment. We carry it now as one parameter and will be scrupulous, when we reach the flavor sector, that it is counted exactly once and not double-counted against the chiral rotations used to make the quark masses real.

The rigid tally is therefore

$$\underbrace{g_s,\ g,\ g'}_{3} + \underbrace{\mu^2,\ \lambda}_{2} + \underbrace{\bar{\theta}}_{1} = 6 \tag{33.91}$$

physical parameters. Everything else lives in the Yukawa couplings.

33.6.2 The flavor sector as a broken symmetry

In the limit of vanishing Yukawa couplings the fermion content of the Standard Model is five chiral gauge multiplets, replicated in three identical generations,

$$Q_L \sim (\mathbf{3},\mathbf{2},\tfrac{1}{6}),\quad u_R \sim (\mathbf{3},\mathbf{1},\tfrac{2}{3}),\quad d_R \sim (\mathbf{3},\mathbf{1},-\tfrac{1}{3}),\quad L_L \sim (\mathbf{1},\mathbf{2},-\tfrac{1}{2}),\quad e_R \sim (\mathbf{1},\mathbf{1},-1), \tag{33.92}$$

each entering the Lagrangian only through its canonical kinetic term $\bar{\psi}\, i\not{D}\,\psi$. That kinetic term is blind to the generation index: it is invariant under an independent U(3) rotation $\psi \to V_\psi\,\psi$, $V_\psi \in \mathrm{U}(3)$, acting on the three generations of each multiplet, because such a rotation commutes with the gauge structure (it touches only flavor) and preserves $\bar{\psi}\, i\not{D}\,\psi$ by unitarity. The full global symmetry of the gauge-plus-kinetic theory is therefore the *flavor group*

$$G_{\mathrm{F}} = \mathrm{U}(3)_Q{\times}\mathrm{U}(3)_u{\times}\mathrm{U}(3)_d{\times}\mathrm{U}(3)_L{\times}\mathrm{U}(3)_e \;\equiv\; \mathrm{U}(3)^5, \qquad \dim_{\mathbb{R}} G_{\mathrm{F}} = 5{\times}9 = 45, \tag{33.93}$$

using $\dim_{\mathbb{R}} \mathrm{U}(3) = 9$. This $\mathrm{U}(3)^5$ symmetry, and the proof that the gauge sector is blind to it, is the content of Theorem 32.2; we import it rather than reprove it.

The Yukawa couplings explicitly break G_{F}. With the conventions fixed in Chapter 32, the Yukawa Lagrangian of the minimal model (three generations, no right-handed neutrino, so neutrinos are exactly massless) is

$$\mathcal{L}_Y = -\,\bar{Q}_L\,\tilde{H}\,Y_u\,u_R \;-\; \bar{Q}_L\,H\,Y_d\,d_R \;-\; \bar{L}_L\,H\,Y_e\,e_R \;+\; \text{h.c.}, \tag{33.94}$$

with $\widetilde{H} = i\sigma^2 H^*$ and the generation indices suppressed, so that Y_u, Y_d, Y_e are each a general complex 3×3 matrix in flavor space. Under a flavor rotation the conjugate field transforms as $\bar{\psi} \to \bar{\psi}\, V_\psi^\dagger$, and the Lagrangian (33.94) keeps its form provided the coupling matrices are redefined by

$$Y_u \to V_Q^\dagger\, Y_u\, V_u, \qquad Y_d \to V_Q^\dagger\, Y_d\, V_d, \qquad Y_e \to V_L^\dagger\, Y_e\, V_e. \tag{33.95}$$

Two Yukawa configurations related by (33.95) are the same theory written in two flavor bases. The physical flavor parameters are the coordinates on the quotient of the raw Yukawa space by G_{F}, and their number is fixed by a single piece of Lie theory.

The relevant statement is the orbit–stabilizer count: if a Lie group G acts on a vector space $\mathcal{Y}$ and a generic point Y_0 has stabilizer $H = \{h \in G : h \cdot Y_0 = Y_0\}$, then the number of physical parameters — the local dimension of the orbit space $\mathcal{Y}/G$ at $[Y_0]$ — is

$$N_{\mathrm{phys}} = \dim_{\mathbb{R}} \mathcal{Y} - \left(\dim_{\mathbb{R}} G - \dim_{\mathbb{R}} H\right). \tag{33.96}$$

The content is geometric and short: the broken generators (those *not* in the stabilizer) sweep out the orbit, which is exactly the set of directions in which Y_0 can be moved by a field redefinition without changing physics; the physical directions are the complement, of dimension $\dim \mathcal{Y}$ minus the orbit dimension $\dim G - \dim H$. The full proof, via the differential of the orbit map and the identification of its kernel with the stabilizer algebra, is carried in Theorem F.11; we use the result (33.96). The slogan it encodes is the one to keep in mind for the rest of this section: *physical parameters = raw parameters − broken generators.*

33.6.3 The thirteen flavor parameters

With the engine in hand the flavor count is mechanical, but it repays being run in full, because the survivors are exactly the masses and mixings the reader expects, and watching them appear ties the abstract count to physics.

Worked Example 33.17 (The flavor-symmetry count and the recovery of nineteen)**.** We apply (33.96) with $\mathcal{Y}$ the space of triples (Y_u, Y_d, Y_e) and $G = G_{\mathrm{F}} = \mathrm{U}(3)^5$.

Raw parameters. A complex 3×3 matrix has 9 complex entries, i.e. 18 real parameters; three of them give

$$\dim_{\mathbb{R}} \mathcal{Y} = 3 \times 18 = 54. \tag{33.97}$$

Group dimension. From (33.93), $\dim_{\mathbb{R}} G_{\mathrm{F}} = 45$.

The stabilizer. We seek the subgroup of G_{F} fixing a generic (Y_u, Y_d, Y_e) under (33.95). Since no term in (33.94) couples leptons to quarks, the stabilizer

factorizes into a quark part inside $\mathrm{U}(3)_Q \times \mathrm{U}(3)_u \times \mathrm{U}(3)_d$ and a lepton part inside $\mathrm{U}(3)_L \times \mathrm{U}(3)_e$, treated separately.

For the quarks, suppose $V_Q^\dagger Y_u V_u = Y_u$ and $V_Q^\dagger Y_d V_d = Y_d$. For generic invertible Y_u, Y_d the first relation forces $V_u = Y_u^{-1} V_Q Y_u$ and the second $V_d = Y_d^{-1} V_Q Y_d$; requiring these to be unitary for the *two independent* matrices Y_u, Y_d simultaneously leaves only $V_Q = e^{i\beta}\mathbf{1}$, whence $V_u = V_d = e^{i\beta}\mathbf{1}$. This common phase rotates Q_L, u_R, d_R together: it is baryon number $\mathrm{U}(1)_B$. For the leptons, only Y_e constrains (V_L, V_e) through $V_L^\dagger Y_e V_e = Y_e$; a single generic invertible matrix is brought to diagonal positive form by its singular-value decomposition, and the residual freedom is the common diagonal phase rotation $V_L = V_e = \mathrm{diag}(e^{i\gamma_1}, e^{i\gamma_2}, e^{i\gamma_3})$, the three separate lepton numbers $\mathrm{U}(1)_{L_e} \times \mathrm{U}(1)_{L_\mu} \times \mathrm{U}(1)_{L_\tau}$, which survive because no term mixes lepton generations. (That this is the generic stabilizer is Theorem 32.3.) Hence

$$H = \mathrm{U}(1)_B \times \mathrm{U}(1)_{L_e} \times \mathrm{U}(1)_{L_\mu} \times \mathrm{U}(1)_{L_\tau}, \qquad \dim_{\mathbb{R}} H = 1 + 3 = 4. \tag{33.98}$$

These four unbroken generators are precisely the accidental global symmetries catalogued earlier in this chapter: baryon number and the three lepton numbers emerge here as the directions in G_{F} that fix the Yukawas and therefore *cannot* be spent to remove a parameter.

The count. Inserting (33.97), (33.93) and (33.98) into (33.96),

$$N_{\text{flavor}} = 54 - (45 - 4) = 54 - 41 = 13. \tag{33.99}$$

Naming the thirteen. Run the redefinitions explicitly. By singular-value decomposition write $Y_u = U_u^L \hat{Y}_u U_u^{R\dagger}$ and $Y_d = U_d^L \hat{Y}_d U_d^{R\dagger}$ with $\hat{Y}_u, \hat{Y}_d$ diagonal, real and non-negative. Choosing $V_Q = U_u^L$, $V_u = U_u^R$ sends $Y_u \to \hat{Y}_u$; the choice $V_Q = U_u^L$ is now fixed, but $V_d = U_d^R$ is still free and removes the right-handed rotation of Y_d, leaving

$$Y_d \to U_u^{L\dagger} U_d^L \, \hat{Y}_d \equiv V_{\mathrm{CKM}} \, \hat{Y}_d, \qquad V_{\mathrm{CKM}} = U_u^{L\dagger} U_d^L \in \mathrm{U}(3). \tag{33.100}$$

The six eigenvalues of $\hat{Y}_u, \hat{Y}_d$ (times $v/\sqrt{2}$) are the **six quark masses**; the leftover unitary V_{CKM} is the Cabibbo–Kobayashi–Maskawa matrix of Chapter 32. For the leptons the SVD of the single matrix Y_e is undone completely, $Y_e \to \hat{Y}_e$, giving the **three charged-lepton masses** and *no* mixing: with massless degenerate neutrinos any unitary rotation of them absorbs the analogue of V_{CKM}, which is the field-theoretic reason the leptonic mixing matrix is trivial in the minimal model.

It remains to count the physical content of V_{CKM}. A general $\mathrm{U}(3)$ matrix carries 9 real parameters. After the diagonalizations the mass terms are invariant under independent diagonal rephasings $u_{L,R} \to P_u u_{L,R}$, $d_{L,R} \to P_d d_{L,R}$ with $P_u, P_d \in \mathrm{U}(1)^3$, under which $V_{\mathrm{CKM}} \to P_u^\dagger V_{\mathrm{CKM}} P_d$. The $3 + 3 = 6$ phases of (P_u, P_d) can absorb six phases of V_{CKM} except for the common phase $P_u = P_d =$

$e^{i\alpha}\mathbf{1}$, which leaves V_{CKM} untouched (this is again $\text{U}(1)_B$), so only $6-1=5$ phases are removable. By the angle/phase decomposition of a unitary matrix (Eq. (33.104) below), U(3) carries 3 angles and 6 phases; removing 5 phases leaves

$$\#(\text{CKM}) = \underbrace{3}_{\text{angles}} + \underbrace{(6-5)}_{\text{phase}} = 4, \tag{33.101}$$

three mixing angles and one CP-violating phase, exactly the count of Theorem 32.12. Assembling,

$$\underbrace{6}_{\text{quark masses}} + \underbrace{3}_{\text{lepton masses}} + \underbrace{4}_{\text{CKM}} = 13, \tag{33.102}$$

in agreement with the abstract (33.99): the two routes — subtract broken generators, or diagonalize and rephase — are one computation seen from two sides.

The total. Adding the rigid six of (33.91) to the flavor thirteen of (33.102),

$$\boxed{N_{\text{SM}} = \underbrace{6}_{g_s,\, g,\, g',\, \mu^2,\, \lambda,\, \bar{\theta}} + \underbrace{13}_{\text{Yukawa sector}} = 19} \tag{33.103}$$

physical parameters in the minimal Standard Model with massless neutrinos. This is Eq. (F.104), now reassembled from the electroweak and flavor sub-counts.

The angle/phase split used in (33.101) recurs for every mixing matrix in the book and deserves to be isolated. A general $N \times N$ unitary matrix has N^2 real parameters; of these,

$$\frac{N(N-1)}{2} \text{ are rotation angles} \qquad \text{and} \qquad \frac{N(N+1)}{2} \text{ are phases.} \tag{33.104}$$

The split is exact and elementary: $\text{U}(N)$ has dimension N^2; the real orthogonal subgroup $\text{O}(N)$, generated by the real antisymmetric matrices, has dimension $\frac{1}{2}N(N-1)$ and supplies precisely the genuine rotations (the parameters that survive when the matrix is real), while the remaining $N^2 - \frac{1}{2}N(N-1) = \frac{1}{2}N(N+1)$ coset directions are the imaginary symmetric generators and supply the phases. This is Theorem F.17 (equivalently Theorem 32.12); the further reduction of the $\frac{1}{2}N(N+1)$ phases to the $\frac{1}{2}(N-1)(N-2)$ physical ones by the $2N-1$ available rephasings gives the $N=3$ result of one CP phase used in (33.101).

Caution: $\bar{\theta}$ is counted exactly once

The diagonalization in Theorem 33.17 used the *full* group $G_{\text{F}} = \text{U}(3)^5$ as field-redefinition freedom, including the overall axial rotation $q_i \to e^{i\alpha\gamma_5} q_i$ of all quarks (the generator $\text{U}(1)_A$ rotating left and right quark fields by

opposite phases). That rotation is not in the stabilizer (33.98) — it changes $\arg\det(Y_u Y_d)$ — and we used it to make the quark masses real. But $\mathrm{U}(1)_A$ is anomalous: by the Adler–Bell–Jackiw result of Chapter 36, performing it shifts the topological angle by 2α per Dirac flavour (Eq. (33.83)); summed over the $N_f = 2N_g = 6$ quark flavours this is $\theta \to \theta - 2N_f\,\alpha$ with $N_f = 6$. The phase one "removes" from the Yukawa sector does not vanish; it reappears as a shift of θ. The quantity invariant under the entire chiral redefinition group, and therefore the physical one, is the combination $\bar{\theta}$ of (33.85), counted once in the rigid six. Equivalently one may keep θ as a separate input and count the 13 physical flavor parameters (the nine fermion masses and the four CKM quantities) directly; with the five remaining rigid couplings this again gives $5 + 13 + 1 = 19$. Either bookkeeping gives nineteen.

33.6.4 The measured values

A count is not yet physics; the nineteen become a description of nature only once their values are supplied by experiment. Table 33.4 collects the nineteen, organized by the sectors of the count, with representative measured values as of the mid-2020s. The numbers, their schemes, scales, and uncertainties — pole versus $\overline{\mathrm{MS}}$ masses, the scale at which a running coupling is quoted, the correlations among the CKM fit inputs — are the province of the reference tables of Appendix F (Table F.3 for the parameter breakdown, and the data tables collected there); here the table serves to make concrete how few numbers, and how disparate, the entire non-gravitational world rests on.

33.6.5 The neutrino-mass extensions

The minimal count assumed massless neutrinos, and that assumption is now known to be false: neutrino oscillations require at least two nonzero neutrino masses (Section 32.7.5). The minimal model must be extended, and the count grows — but by how much depends on *how* the neutrinos acquire mass. The two minimal mechanisms give different totals, and the difference is itself instructive: it is entirely about which rephasing freedoms survive. We carry both counts explicitly; the mechanisms behind them (the seesaw, the origin of the Weinberg operator) belong to Chapter 35.

Worked Example 33.18 (Dirac versus Majorana neutrinos: +7 or +9)**.** *Dirac neutrinos.* Introduce three right-handed neutrinos $\nu_R \sim (\mathbf{1}, \mathbf{1}, 0)$, gauge singlets, and forbid a Majorana mass for them (e.g. by imposing exact lepton number). The Yukawa Lagrangian (33.94) gains $-\bar{L}_L \tilde{H}\, Y_\nu\, \nu_R + \text{h.c.}$ with Y_ν a fourth general complex 3×3 matrix. The raw count becomes $\dim \mathcal{Y} = 4 \times 18 = 72$; the singlet

Sector	Parameter	Representative value	Notes
Gauge	g_s	$\alpha_s(M_Z) \approx 0.118$	strong coupling, $\overline{\text{MS}}$ at M_Z
	g	$g \approx 0.65$	$\text{SU}(2)_L$, from M_W, θ_W
	g'	$g' \approx 0.36$	$\text{U}(1)_Y$; $\sin^2\theta_\text{W} \approx 0.231$
Higgs	v	246 GeV	$\equiv (\sqrt{2}\,G_F)^{-1/2}$
	m_h	125 GeV	$m_h^2 = 2\lambda v^2$
Strong CP	$\bar\theta$	$\lesssim 10^{-10}$	bound from neutron EDM
Quark masses	m_u, m_c, m_t	2.2 MeV, 1.27 GeV, 173 GeV	up-type
	m_d, m_s, m_b	4.7 MeV, 95 MeV, 4.18 GeV	down-type
Lepton masses	m_e, m_μ, m_τ	0.511 MeV, 106 MeV, 1.777 GeV	charged leptons
CKM angles	$\theta_{12}, \theta_{23}, \theta_{13}$	13°, 2.4°, 0.2°	$\sin\theta_{12} \approx 0.225$
CKM phase	δ	$\approx 65°$	$J \approx 3 \times 10^{-5}$
Total		**19**	

Table 33.4: The nineteen physical parameters of the minimal Standard Model, with representative values (mid-2020s). The two Higgs parameters are shown in the physical basis (v, m_h) equivalent to (μ^2, λ). Schemes, scales, and uncertainties are tabulated in Appendix F; see Table F.3 for the parameter breakdown and the data tables there for precise values. The CKM phase is quoted both as δ and through the rephasing-invariant Jarlskog measure J of Chapter 32.

carries its own flavor rotation, so $G_\text{F} \to \text{U}(3)^6$ with dim $= 6 \times 9 = 54$. The new law is $Y_\nu \to V_L^\dagger Y_\nu V_\nu$. In the lepton sector Y_e and Y_ν now play exactly the roles Y_u, Y_d played for quarks: two independent generic matrices sharing the common V_L, whose only joint stabilizer is the overall phase, total lepton number $\text{U}(1)_L$. The three separate lepton numbers are broken, since Y_e and Y_ν cannot be diagonalized simultaneously. With the quark stabilizer $\text{U}(1)_B$ unchanged,

$$H = \text{U}(1)_B \times \text{U}(1)_L, \qquad \dim H = 2, \tag{33.105}$$

and (33.96) gives

$$N_\text{flavor}^\text{Dirac} = 72 - (54 - 2) = 72 - 52 = 20, \tag{33.106}$$

i.e. $20 - 13 = 7$ more than the minimal model. The lepton sector now mirrors the quark sector: 3 charged-lepton masses, 3 neutrino masses, and a PMNS mixing matrix U_PMNS carrying — by (33.101) run identically — 3 angles and 1 phase. The seven new parameters are 3 masses $+3$ angles $+1$ phase, and the grand total is

$$N_\text{SM}^\text{Dirac} = 19 + 7 = 26. \tag{33.107}$$

Majorana neutrinos. Now give the neutrinos mass without new light fields, through the unique dimension-five operator built from Standard Model fields,

the Weinberg operator

$$\mathcal{L}_5 = \frac{c_{ij}}{2\Lambda}\,(\overline{L^c_{L\,i}}\,\widetilde{H}^*)(\widetilde{H}^\dagger L_{L\,j}) + \text{h.c.}, \tag{33.108}$$

which after electroweak symmetry breaking gives a Majorana mass $M_\nu = (v^2/\Lambda)\,c$ to the left-handed neutrinos. Because the two doublets are contracted symmetrically, c is a complex *symmetric* 3×3 matrix: 6 complex entries, 12 real parameters. With Y_e (18 real) the lepton-sector raw count is 30; there is no ν_R, so the lepton flavor group is only $\mathrm{U}(3)_L \times \mathrm{U}(3)_e$, dim = 18, with $Y_e \to V_L^\dagger Y_e V_e$ and $c \to V_L^T\, c\, V_L$ — two factors of V_L and a transpose, because both doublets are L_L and one is charge-conjugated. The crucial difference is the stabilizer: a phase rotation $L_L \to e^{i\alpha} L_L$ sends $c \to e^{2i\alpha} c \neq c$, so the Majorana mass violates lepton number and $\mathrm{U}(1)_L$ is gone. A generic (Y_e, c) has no continuous stabilizer at all, $H_{\text{lepton}} = \{\mathbf{1}\}$, dim = 0, and

$$N^{\text{Maj}}_{\text{lepton}} = 30 - (18 - 0) = 12, \tag{33.109}$$

two more than the Dirac lepton count of 10. With the unchanged quark sector (10) the flavor total is 22, i.e. 9 more than the minimal model, and the grand total is

$$N^{\text{Maj}}_{\text{SM}} = 19 + 9 = 28. \tag{33.110}$$

The twelve lepton parameters are 3 charged-lepton masses + 3 neutrino masses + 6 in U_{PMNS}. The mixing matrix now holds 6 rather than 4 because the Majorana mass removes the neutrino rephasing freedom: in the charged-lepton mass basis U_{PMNS} appears in $\bar{e}_L \gamma^\mu U_{\text{PMNS}}\, \nu_L W^-_\mu$, and we may rephase the charged leptons ($P_e \in \mathrm{U}(1)^3$, $U_{\text{PMNS}} \to P_e^\dagger U_{\text{PMNS}}$) but *not* the neutrinos, since a neutrino phase cannot be undone in the Majorana mass term. Only $N = 3$ phases are removable, not the $2N - 1 = 5$ of the Dirac case, so with $U_{\text{PMNS}} \in \mathrm{U}(3)$ carrying 3 angles and 6 phases,

$$\#(\text{PMNS, Majorana}) = \underbrace{3}_{\text{angles}} + \underbrace{(6-3)}_{\text{phases}} = 3 + 3, \tag{33.111}$$

namely 3 angles, 1 Dirac phase, and 2 *Majorana* phases. The two extra phases are precisely the $N - 1 = 2$ neutrino rephasings the Dirac theory could use to clean the matrix but the Majorana theory cannot — the cleanest illustration in the book of how a discrete fact about a field (whether it is its own antiparticle) feeds the continuous parameter count (Theorem 32.35).

The pattern is general: for N generations a Majorana mixing matrix carries exactly $N - 1$ more physical phases than a Dirac one, because the Dirac theory removes $2N-1$ phases and the Majorana theory only N, and $(2N-1)-N = N-1$. The Majorana phases are physical only because the Majorana mass forbids the rephasings that would otherwise have absorbed them.

33.6.6 What the numbers leave unexplained: the flavor puzzle

The count is complete, and it is austere: nineteen numbers (twenty-six or twenty-eight with neutrino masses) determine all of non-gravitational physics. But completeness is not the same as understanding, and the values of these numbers expose, rather than hide, how much the theory does *not* explain. The Standard Model takes the nineteen as inputs; it predicts no relation among them. And the inputs are wildly disparate in a way no symmetry of the theory accounts for.

The disparity is the *flavor puzzle*, and it has two faces, both visible in Fig. 33.5, which plots every dimensionless input on a single logarithmic axis. The first face is the hierarchy of the Yukawa couplings. The dimensionless coupling of a charged fermion to the Higgs is $y_f = \sqrt{2}\, m_f/v$, and across the three generations these run from $y_t \approx 1$ for the top quark down to $y_e \approx 3 \times 10^{-6}$ for the electron — nearly six orders of magnitude, with the generations stacked in a rough but unexplained geometric ladder. Nothing in the Standard Model requires this; the Yukawa matrices (33.94) are arbitrary complex matrices, and the theory is equally consistent for any values of their entries. The second face is the hierarchy of the mixings. The CKM matrix is close to the identity, its off-diagonal angles small and themselves hierarchical, $\sin\theta_{12} \approx 0.22 \gg \sin\theta_{23} \approx 0.04 \gg \sin\theta_{13} \approx 0.004$; the PMNS matrix, by contrast, has two large angles and one small one — an entirely different pattern, again unexplained, and a standing puzzle as to why quark and lepton mixing should look so different.

It is worth being precise about the status of these statements, because "puzzle" can be read too weakly. There is no inconsistency here, and no failed prediction: the Standard Model is not wrong about flavor, it is *silent*. The hierarchies are technically natural in the 't Hooft sense — setting a Yukawa to zero enlarges the flavor symmetry (33.93), so small Yukawas are protected from large radiative corrections, and the smallness of y_e is stable under renormalization even though it is unexplained. The puzzle is therefore not one of fine-tuning but of *absence of a reason*: a theory of nineteen free parameters that happen to take the observed values is logically complete but explanatorily unsatisfying, and the search for a deeper structure — a flavor symmetry, a relation to grand unification, an anarchy-versus-structure question for the lepton sector — is an active and, as of the mid-2020s, unresolved frontier. The figure also displays the strong-CP angle, whose puzzle is of a different kind: not why $\bar{\theta}$ is hierarchically small relative to others, but why a parameter expected to be of order one is bounded below 10^{-10}, the strong-CP problem stated in the previous section.

The honest summary, then, is dual. On the side of economy, the achievement is remarkable: the entire spectrum and all interactions of known matter follow from nineteen numbers and a gauge group, with the structure of those interactions — which fermion couples to which, with what strength relative to the others within a multiplet — fixed not by hand but by gauge invariance and quantum consistency. On the side of explanation, those nineteen numbers are exactly the place where

the theory stops explaining and starts fitting: their *values*, and above all their *hierarchies*, are inputs the Standard Model organizes but does not derive.

This duality — forced structure on one side, free inputs on the other — is the natural pivot to the chapter's conclusion. Having now both assembled the theory and counted precisely what it leaves open, we are ready to step back and state, in one balance sheet, what the Standard Model genuinely explains and what it merely absorbs, and to hand the open questions forward.

33.7 The Standard Model as a Structure: What Is Explained and What Is Not

The previous section closed the assembly with a number: nineteen real parameters, counted in Section 33.6 by following the flavor symmetry of the Yukawa sector as it is broken down to nothing. That count is the natural place to take stock, because it is simultaneously the theory's greatest strength and its frankest confession. The strength is that nineteen numbers — a handful of measurements — fix *every* non-gravitational scattering amplitude, decay rate, and cross section in the universe, to whatever loop order one has the stamina to compute. The confession is that the nineteen numbers themselves are not predicted: the electron mass, the Cabibbo angle, the strong-CP angle $\bar{\theta}$ are inputs read off experiment, not outputs of the structure. This final section separates the two. We state, as precisely as the preceding development allows, what the Standard Model genuinely *explains* — what is forced on it by consistency and therefore could not have been otherwise — from what it merely *parametrizes*, and we hand the residue, the honest and dated list of what the theory does not address at all, forward to the rest of Part IX and beyond. No new dynamics enter here; the section is an act of bookkeeping, but bookkeeping of the most consequential kind, because it is the summary that tells the next generation of theory where to look.

We organize the discussion around three claims, each assembled from results already in hand. First, that the Lagrangian we wrote down is very nearly the *only* thing we could have written down, once a small number of inputs are granted — the near-uniqueness statement. Second, that renormalizability is precisely the property that converts a finite list of parameters into a complete predictive theory. Third, that the structure forced by the first two claims comes with a bundle of genuine explanations attached for free, which must be carefully distinguished from the genuine inputs. We close by stating the frontier, and by discharging the last obligation of the two gauge threads carried since Part V.

33.7.1 The near-uniqueness of the Standard Model Lagrangian

The most remarkable structural fact about the Standard Model is negative: given a short list of empirical inputs, the rules of quantum field theory leave almost

no freedom in the Lagrangian. Every term in Eq. (33.22) is there because it is *forced*, and almost no term that is absent could have been included. To make this precise we must say exactly which inputs we are granting and which structure we claim follows.

Proposition 33.19 (Near-uniqueness of $\mathcal{L}_{\rm SM}$)**.** *Grant the following four inputs:*

(I) *the gauge group is* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$*;*

(II) *the matter content is three generations of the chiral representation* $Q \sim (\mathbf{3},\mathbf{2})_{+1/6}$, $u^c \sim (\bar{\mathbf{3}},\mathbf{1})_{-2/3}$, $d^c \sim (\bar{\mathbf{3}},\mathbf{1})_{+1/3}$, $L \sim (\mathbf{1},\mathbf{2})_{-1/2}$, $e^c \sim (\mathbf{1},\mathbf{1})_{+1}$, *together with a single scalar* $H \sim (\mathbf{1},\mathbf{2})_{+1/2}$*;*

(III) *the Lagrangian is local, Lorentz invariant, and* renormalizable — *built only from operators of mass dimension* ≤ 4*;*

(IV) *the theory is to be a consistent chiral gauge theory — all gauge anomalies cancel.*

Then the most general Lagrangian consistent with (I)–(IV) is exactly $\mathcal{L}_{\rm SM}$ *of Eq. (33.22), up to the values of its nineteen parameters and up to total derivatives. No gauge-invariant, renormalizable operator that is absent from Eq. (33.22) can be added, and no operator present in it can be removed without violating (I)–(III).*

Proof. This is not a new computation but the assembly of results already established in this chapter, organized so that the forcing under (I)–(IV) is visible. That the most general local, Lorentz-invariant, renormalizable (dimension-≤ 4) Lagrangian on the field content (I)–(II) and invariant under (I) is exactly $\mathcal{L}_{\rm SM}$ of Eq. (33.22), up to total derivatives, is the content of Theorem 33.4, proved there by enumerating every candidate operator dimension by dimension. That enumeration retains precisely the three gauge kinetic terms, the chiral fermion kinetic terms (with no bare Dirac mass, since chirality places the two helicities in inequivalent gauge representations), the scalar kinetic term and the two-parameter potential $V = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2$, the three Yukawa matrices, and the single QCD θ-term of Section 33.5 (the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ angles being rotatable away); with the field content of (II) no renormalizable operator gives the neutrino a mass, which is why neutrino mass lies outside the minimal structure (Section 33.7.4).

Input (IV) is what licenses (II) in the first place. A chiral gauge theory is consistent only if its gauged triangle anomalies cancel, and the content (II) is one of the very few anomaly-free assignments. Run in reverse, anomaly freedom together with the gauge invariance of the Yukawa couplings fixes the hypercharges up to one overall normalization — the computation carried out in this chapter in Theorems 33.5 and 33.6, with electric-charge quantization following as Theorem 33.7 (see also the all-orders anomaly-cancellation machinery of Chapter 37). Collecting these results leaves Eq. (33.22) with nothing addable and nothing droppable. This is the assertion. □

The word "near" in near-uniqueness is doing honest work, and it is worth saying exactly where the freedom resides, because that freedom is the seed of every open question in Section 33.7.4. Inputs (I) and (II) — the gauge group, the number of generations, and the specific list of representations — are pure empirical data; the structure does not explain why the group is $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ rather than something larger, why there are three copies of the matter, or why a single Higgs doublet rather than two. What the proposition establishes is conditional: *given* that data, quantum consistency and renormalizability leave essentially no further choice. The Standard Model is therefore not a theory one tunes freely to fit the world; it is a tightly constrained structure whose every renormalizable term is mandatory, sitting at the end of a very short list of possibilities.

Physical Insight

The chiral structure is the source of the rigidity. In a vector-like theory — where left- and right-handed fermions carry the same gauge charges — bare masses are allowed, anomalies cancel automatically, and the gauge couplings are largely unconstrained: there is little to force. The Standard Model is chiral, and chirality has three sharp consequences at once. It forbids bare fermion masses, so all masses must come from the Higgs mechanism and are therefore proportional to a single condensate; it makes anomaly cancellation a nontrivial constraint that the observed content only barely satisfies; and it is what makes electroweak symmetry breaking necessary rather than optional. The mass scale of the entire visible world is thus tied to one number, the vacuum expectation value $v \simeq 246$ GeV, not because of a coincidence but because chirality leaves no other channel.

33.7.2 Renormalizability as the engine of predictivity

Input (III) above, renormalizability, was used in the uniqueness argument merely to truncate the operator list at dimension 4. Its deeper role is to guarantee that the resulting finite list of parameters is *enough* — that once the nineteen numbers are fixed, every observable is determined with no further input. This is not automatic; it is the content of the renormalizability theorem, and it is what elevates the Standard Model from a model to a predictive framework.

Recall the precise statement of renormalizability established in Section 22.3.4: a theory is renormalizable if every ultraviolet divergence appearing at every order of perturbation theory can be absorbed into a redefinition (a multiplicative rescaling) of the *finitely many* parameters and fields already present in the Lagrangian. The Standard Model, being built entirely of dimension-≤ 4 operators with dimensionless or positive-mass-dimension couplings, satisfies the power-counting criterion of Chapter 21 in each of its sectors, and the all-orders arguments

of Chapter 22 (extended to gauge theories through the Slavnov–Taylor identities) go through. The logical chain is then short and decisive.

Proposition 33.20 (Finite inputs determine all observables)**.** *If the Standard Model is renormalizable in the sense of Section 22.3.4, then every physical observable $\mathcal{O}$ is a calculable function $\mathcal{O} = \mathcal{O}\big(g_1, g_2, g_3, \lambda, \mu^2; \{y\}; \bar{\theta}; \mu_R\big)$ of the renormalized parameters and the renormalization scale, with the residual μ_R-dependence of any truncation controlled by the renormalization-group equations of Section 24.6; in particular, once the nineteen physical parameters are measured in nineteen observables, every* further *observable is predicted with no additional free constant.*

Proof. A general amplitude computed to a fixed order in the loop expansion is, before renormalization, a sum of Feynman integrals that may diverge in the ultraviolet. Renormalizability (Section 22.3.4) asserts that the totality of these divergences, order by order, is removed by the counterterms generated from rescaling the fields and the finite parameter set $\{g_i, \lambda, \mu^2, y, \bar{\theta}\}$ — there is no order at which a divergence appears whose subtraction requires a *new* coupling absent from the Lagrangian. Equivalently, the bare parameters are functions of the renormalized parameters and the cutoff alone, and the renormalized Green's functions are finite functions of the renormalized parameters and the scale μ_R (Chapter 22). A physical observable is a particular combination of on-shell renormalized Green's functions, hence a finite function of $\{g_i, \lambda, \mu^2, y, \bar{\theta}\}$ and μ_R. Fixing the parameters by matching to nineteen measured observables determines these arguments; any twentieth observable is then the value of a known function at known arguments. The only subtlety is the scheme- and scale-dependence of the parameters, which is not an ambiguity but a calculable flow: the Callan–Symanzik equation (Section 24.6) relates the values at one scale to those at another, and physical observables are scale-independent to the order computed. This is precisely the predictivity we claimed. □

The contrast that gives this statement its teeth is with a *non*-renormalizable theory. The moment one admits operators of dimension > 4 — the Fermi theory of the weak interaction, or the Standard Model itself viewed as an effective field theory — each new order in the energy expansion demands new counterterms, hence new free parameters, in an unending tower; predictivity survives only order by order and only below a finite cutoff (Section 38.2). The Standard Model sits at the privileged boundary: it is the most general renormalizable theory on its field content, so it is exactly as predictive as a four-dimensional quantum field theory can be. Every higher-dimension operator is a window onto physics above the weak scale, suppressed by powers of that scale; their systematic study is the subject of Chapters 38 and 40, and their absence at dimension ≤ 4 is why the Standard Model is, within its domain, complete.

33.7.3 The summary: genuine explanations versus parametrizations

We can now draw the central distinction of this section. Some features of the world the Standard Model *explains* — it derives them from its structure, so that they could not have come out otherwise. Other features it merely *absorbs* into its parameters — it accommodates them but says nothing about why they take the values they do. Keeping these straight is the difference between understanding the theory and being impressed by it.

What the Standard Model genuinely explains. The following are outputs, forced by the structure assembled above and in the preceding chapters; each is a consequence, not an assumption.

- **Charge quantization.** The electric charges of all known particles are integer multiples of $\frac{1}{3}e$, and the proton and electron charges cancel to better than one part in 10^{21}. In the Standard Model this is not a coincidence of independent inputs: anomaly cancellation, run backward, fixes the hypercharges of the chiral multiplet up to one overall normalization and a discrete choice fixed by the Yukawa couplings, as computed in this chapter in Theorems 33.5 and 33.7 (see also Chapter 37). The fractional quark charges and the exact electron–proton cancellation are *consequences* of consistency, the single most striking explanatory success of the theory.

- **Accidental baryon and lepton number.** Nothing in the construction imposed conservation of baryon number B or the lepton numbers L_e, L_μ, L_τ. Yet no gauge-invariant, renormalizable operator violates them: the lowest such operators have dimension 5 (L-violating) and dimension 6 (B-violating), as we make explicit in Theorem 33.22. These symmetries are *accidental* — they hold not because they were postulated but because the field content and renormalizability leave no room to break them at dimension ≤ 4. This is why the proton is stupendously stable and why lepton flavor is conserved to the precision it is, both as automatic outputs.

- **The structure of CP violation.** With three generations, CP violation in the quark sector is possible and is controlled by a single physical phase, captured invariantly by the Jarlskog determinant of Section 32.4.4; with fewer than three generations it would be forbidden by rephasing freedom. The theory thus *explains why* CP violation is small and correlated across processes (the unitarity triangle of Section 32.6 closes), and predicts the absence of flavor-changing neutral currents at tree level through the GIM mechanism (Section 32.5.2). The *pattern* of CP violation is an explanation; its *magnitude* is an input.

- **Tree-level relations among electroweak observables.** The custodial symmetry of the Higgs potential (Section 31.6) forces the tree-level relation $\rho \equiv m_W^2/(m_Z^2 \cos^2\theta_{\mathrm{W}}) = 1$, and the existence of the neutral current with its specific couplings was a genuine prediction (Eq. (31.16)) confirmed before the Z was seen. These are derived relations, tested in Chapter 34.

- **Full predictivity.** As established in Theorem 33.20, renormalizability means the nineteen numbers exhaust the freedom: the entire non-gravitational phenomenology, to arbitrary loop order, is the output of those inputs. That the world should be describable by so short a list is itself the deepest explanatory fact the theory offers.

What the Standard Model merely parametrizes. The complementary list is of features the theory describes but does not explain: they are the nineteen inputs, and the structure is silent on their values.

- **The fermion masses and the flavor hierarchy.** The nine charged-fermion masses span more than five orders of magnitude, from the electron to the top quark, encoded in Yukawa couplings ranging from $\sim 10^{-6}$ to ~ 1. The Standard Model fixes none of them; that they are hierarchical rather than comparable is the *flavor puzzle*, a pattern crying out for explanation that the theory does not supply.

- **The mixing angles and the CP phase.** The three CKM angles and the single physical phase (Section 32.3.4) are inputs. The smallness of the off-diagonal mixing — the near-diagonality of V_{CKM} — and the size of the phase are accommodated, not derived.

- **The electroweak scale.** The values μ^2 and λ, equivalently $v \simeq 246$ GeV and the Higgs mass $m_h \simeq 125$ GeV, are inputs. Why v sits sixteen orders of magnitude below the Planck scale — the *hierarchy problem* — is invisible to the structure (Section 38.6.5).

- **The strong-CP angle.** The parameter $\bar{\theta}$ is allowed by every symmetry yet is measured to be smaller than $\sim 10^{-10}$ (Theorem 33.21). Its tininess is an input of staggering precision that the theory in no way requires; this is the *strong-CP problem*.

- **The gauge couplings.** The three couplings g_1, g_2, g_3 are inputs; their near-meeting under renormalization-group running at $\sim 10^{16}$ GeV (Section 24.6) is suggestive but, within the Standard Model, a numerical fact rather than a derived unification.

Worked Example 33.21 (The strong-CP tuning, quantified)**.** The uniqueness argument listed the QCD θ-term as a mandatory ingredient. The physical,

rephasing-invariant combination it enters through is $\bar{\theta} = \theta + \arg\det(Y_u Y_d)$ of Eq. (33.85), established in Theorem 33.16. Being CP-odd it induces a neutron electric dipole moment, and the chiral-Lagrangian estimate $d_n \sim \bar{\theta} \times (10^{-16})\, e\,\mathrm{cm}$ of Eq. (33.86), set against the experimental limit $|d_n| < 1.8 \times 10^{-26}\, e\,\mathrm{cm}$, yields the constraint $|\bar{\theta}| \lesssim 10^{-10}$ of Eq. (33.87).

The point to underline here is interpretive, not computational. A parameter that Theorem 33.19 permits to be of order unity is empirically smaller than 10^{-10}, and nothing in the structure explains it: θ and $\arg\det(Y_u Y_d)$ are inputs of entirely different origin, so their near-exact cancellation in $\bar{\theta}$ is, within the Standard Model, a tuning of ten decimal places with no reason behind it. This is the sharpest single example of a parametrization masquerading, numerically, as though it were an explanation — and the cleanest motivation for the physics beyond the Standard Model of Chapter 35.

Worked Example 33.22 (The first windows beyond the Standard Model: dimension-5 and dimension-6 operators)**.** The accidental conservation of B and L was claimed above to follow from there being no renormalizable operator that violates them. The *lowest-dimension* gauge-invariant operators that do violate them were already constructed in Section 33.4.4; recalling them here shows why the symmetries are accidental and quantifies how badly physics above the weak scale can break them — the effective-field-theory viewpoint of Section 38.2.

Dimension 5: the Weinberg operator. Using only Standard Model fields the unique gauge-invariant operator of dimension 5 is the Weinberg operator

$$\mathcal{O}_5 = \frac{c_5}{\Lambda}\,(\bar{L}^c\,\widetilde{H}^*)(\widetilde{H}^\dagger L), \tag{33.112}$$

of Eq. (33.58); it changes lepton number by $\Delta L = 2$ and, on electroweak symmetry breaking with $\langle H\rangle = v/\sqrt{2}$, becomes the Majorana neutrino mass $m_\nu \sim c_5 v^2/\Lambda$ of Eq. (33.60). With $c_5 \sim 1$ and $m_\nu \sim 0.05$ eV this points to $\Lambda \sim v^2/m_\nu \sim 10^{14}$–$10^{15}$ GeV, tantalizingly close to the gauge-coupling meeting scale — the deepest reason neutrino mass is not part of the renormalizable theory and the *leading* sign of new physics.

Dimension 6: proton decay. Baryon number is harder to violate: the lowest B-violating operators have dimension 6, the four-fermion structures $(qqq\ell)$ of Eq. (33.61), which mediate $p \to e^+\pi^0$ and related modes with $\Delta(B-L) = 0$. The dimensional rate $\Gamma_p \sim m_p^5/\Lambda^4$ together with the experimental bound $\tau_p \gtrsim 10^{34}$ yr gives $\Lambda \gtrsim 10^{15}$–10^{16} GeV, the bound Eq. (33.66) worked in Theorem 33.14, again a scale far above the weak scale.

The lesson. Both windows confirm that B and L are accidental, not exact: they are broken only by operators suppressed by a high scale Λ, and their excellent conservation in the laboratory is the statement $\Lambda \gg v$. That the very lowest irrelevant operators are exactly the ones that give neutrino mass and proton decay — the two most-sought signals of new physics — is the Standard Model telling us, through its own accidental symmetries, where its boundary lies.

33.7.4 The frontier: what the Standard Model does not address

The summary of Section 33.7.3 listed parameters the theory leaves unexplained but does, at least, contain. The frontier proper consists of questions the Standard Model does not even pose — phenomena it is structurally silent about, or numbers whose values it permits but whose smallness it cannot motivate. We state the list honestly and date it: this is the understanding as of the mid-2020s, and each item is handed to a specific later chapter. None of these is a defect of the internal logic of the theory; each is a place where the theory, taken at face value, declares its own incompleteness.

1. **The strong-CP problem.** The ten-decimal-place smallness of $\bar{\theta}$ (Theorem 33.21) has no explanation within the structure. The leading proposal — a new spontaneously broken symmetry whose Goldstone boson, the axion, dynamically relaxes $\bar{\theta}$ to zero — is taken up in Chapter 35 and connects to the non-perturbative vacuum structure of Chapter 42.

2. **The flavor puzzle.** The pattern of nine charged-fermion masses spanning 10^6 and three small mixing angles is a parametrization with no underlying principle in the Standard Model. Why the Yukawa couplings are hierarchical, and why the mixing is nearly diagonal in the quark sector but large in the lepton sector, is open; candidate organizing ideas (flavor symmetries, wave-function localization) are surveyed in Chapter 35.

3. **The origin of neutrino mass.** The minimal theory has massless neutrinos; oscillations prove they are not. The Weinberg operator Eq. (33.112) shows the effect is dimension 5 and points to a high scale, with the seesaw mechanism of Section 32.7.3 as the leading ultraviolet completion. Whether the neutrino is Majorana or Dirac, and where the scale Λ sits, are unresolved and pursued in Chapter 35.

4. **Higgs naturalness and the hierarchy.** The scalar mass-squared μ^2 is the one relevant operator of the theory, and its sensitivity to the ultraviolet — the absence of a symmetry protecting it the way chiral symmetry protects fermion masses — is the hierarchy problem of Section 38.6.5. Why $v \ll M_{\mathrm{Pl}}$, and whether the question is even well-posed, is the central naturalness puzzle, examined from the effective-field-theory side in Chapter 38 and from the model-building side in Chapter 35.

5. **Dark matter.** Cosmology and astrophysics require a cold, non-baryonic, gravitationally interacting matter component roughly five times the mass density of ordinary matter; the Standard Model contains no particle that fits. This is not a tuning but a flat absence — a particle the theory does not have. The candidate landscape is reviewed in Chapter 35.

6. **Baryogenesis.** The observed matter–antimatter asymmetry requires, by Sakharov's conditions, baryon-number violation, CP violation, and a departure from equilibrium. The Standard Model has B violation only through non-perturbative sphalerons and CP violation only through the CKM phase, and both are quantitatively far too small to generate the observed asymmetry. A viable mechanism requires new physics; the options are surveyed in Chapter 35.

7. **Gravity.** The Standard Model is a theory of three of the four known interactions. Gravity is not in $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; treated as a quantum field theory, general relativity is non-renormalizable (an infinite tower of operators in the sense of Section 38.2) and so is at best an effective description below the Planck scale. The quantum field theory of gravity — effective and beyond — is the subject of Chapter 64.

Caution

It is tempting to read the gauge-coupling near-meeting at $\sim 10^{16}$ GeV and the $\Lambda \sim 10^{15}$ GeV scales emerging independently from neutrino mass Eq. (33.60) and proton decay Eq. (33.66) as proof of grand unification. They are suggestive coincidences and the motivation for Chapter 35, but within the Standard Model itself they are numerical facts, not derivations. Honest synthesis keeps the line between "the theory predicts" and "the numbers hint" sharp; conflating them is the most common way to overstate what the structure delivers.

33.7.5 Closing the gauge threads: QED and Yang–Mills become one theory

We have one obligation left, owed since Part V. Two of the three benchmark theories of this book are gauge theories: QED, the electron coupled to a U(1) gauge field, begun classically in Chapter 8; and Yang–Mills theory, non-abelian gauge theory, begun classically in Chapter 25. They have been developed in parallel through quantization, renormalization, and the renormalization group. The Standard Model is where they cease to be two threads and become two factors of one structure.

The convergence is exact and can be stated in one diagram (Fig. 33.6). The Yang–Mills thread survives intact as the color factor: $\mathrm{SU}(3)_c$ is unbroken, its gauge field is the gluon, and the asymptotic freedom computed in Chapter 27 governs it without modification — QCD *is* Yang–Mills theory, specialized to $N = 3$ and the observed quark content. The QED thread is the subtler one: $\mathrm{U}(1)_{\mathrm{em}}$ is not a factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ at all but the *unbroken*

subgroup left after electroweak symmetry breaking, the diagonal combination

$$Q = T^3 + Y, \tag{33.113}$$

the Gell-Mann–Nishijima relation derived in Chapter 31. The photon is the massless linear combination of the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ gauge fields that remains after the Higgs condenses; the electron's coupling e that began QED is the projection $e = g_2 \sin\theta_{\mathrm{W}} = g_1 \cos\theta_{\mathrm{W}}$ of the two electroweak couplings onto that combination. The classical gauge principle of Chapter 8 — minimal coupling forced by local U(1) invariance — is thus recovered as the low-energy shadow of the full electroweak structure, and the famous number of Chapter 20, the electron's anomalous moment, is a Standard Model observable computed from Eq. (33.22).

With this, the classical-structure development of both gauge threads is complete. Every layer the threads were built to carry — the gauge principle, curvature and self-interaction, quantization and gauge fixing, renormalization, the running of the couplings, asymptotic freedom, symmetry breaking, and finally unification into a single Lagrangian — now sits inside Eq. (33.22). No further layer of *classical structure* remains to be added to either thread. When the threads reappear it will be in a new language entirely: the on-shell, spinor-helicity reframing of Part XVI, where the same QED and Yang–Mills amplitudes are recomputed from little-group scaling and recursion (Chapters 58 and 59) and Yang–Mills is shown to be the "square root" of gravity through the double copy (Chapter 61). Until then, the two carried gauge theories of this book are one theory, and that theory is the Standard Model.

This completes the assembly. Chapter 34 now turns the finished structure to account, confronting its nineteen inputs and its many predictions with the precision electroweak data and the discovered Higgs boson — the chapter where the Standard Model is *tested* rather than built. Chapter 35 takes up the frontier listed above, asking what lies beyond a theory that is, by the measure of this chapter, both extraordinarily successful and visibly incomplete.

Further reading

The reading of the Standard Model as the unique renormalizable theory on its chiral field content is developed in the textbooks of Weinberg [501] and Schwartz [432], and the original assembly is recorded in the founding papers of Glashow [224], Weinberg [496], and Salam [426]. Anomaly cancellation as a consistency requirement and its role in charge quantization is treated thoroughly in Weinberg [501]. The strong-CP problem traces to the discovery that the θ vacuum is physical [464], with the axion solution proposed in Wilczek [510]; the flavor puzzle, neutrino mass, and the broader frontier are reviewed comprehensively in Langacker [314]. The effective-field-theory perspective on the Standard Model's

higher-dimension operators — the Weinberg operator and the dimension-six basis — is the subject of Weinberg [498] and Alonso et al. [20]. For the experimental status of all nineteen parameters and the frontier bounds quoted here, the current Particle Data Group review [382] is the standard reference.

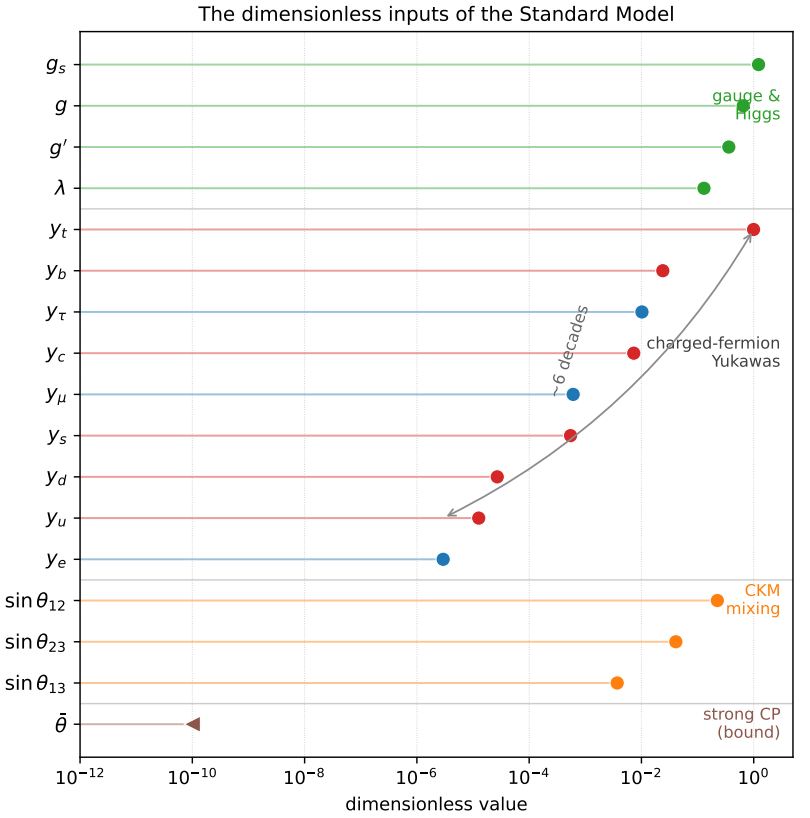


Figure 33.5: The dimensionless inputs of the Standard Model on a single logarithmic axis. The gauge couplings and the Higgs quartic sit near order unity; the nine charged-fermion Yukawas $y_f = \sqrt{2}\, m_f/v$ spread over roughly six decades; the three CKM mixing sines are small and hierarchical; and the strong-CP angle $\bar{\theta}$ is bounded far below everything else. The theory predicts none of this structure: every value is an independent input. The vertical range and the absence of any organizing pattern are the flavor puzzle (and, for $\bar{\theta}$, the strong-CP problem) made visual.

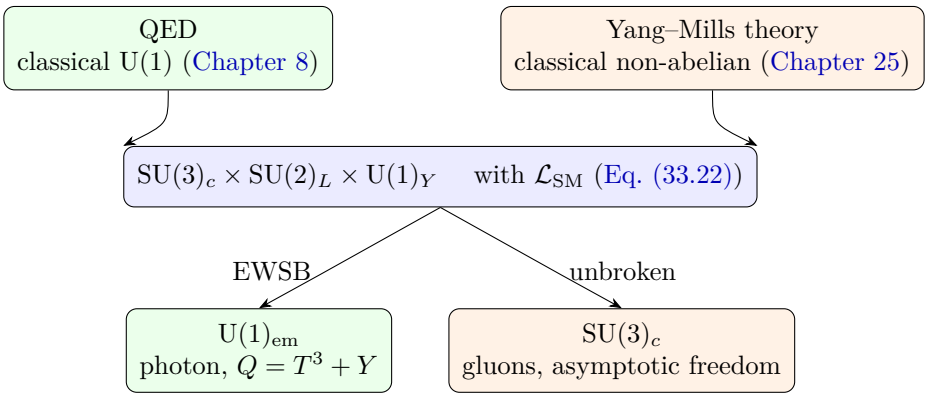


Figure 33.6: Convergence of the two gauge threads. The Yang–Mills thread persists as the unbroken color factor $\mathrm{SU}(3)_c$; QED is recovered as the unbroken electroweak subgroup $\mathrm{U}(1)_{\mathrm{em}}$ with charge $Q = T^3 + Y$ after the Higgs condenses. From the Standard Model Lagrangian Eq. (33.22) onward there is one gauge theory, not two.

Chapter 34

Tests of the Standard Model

The previous chapter assembled the Standard Model as a finished object and counted its free parameters (Chapter 33, Section 33.6): nineteen numbers — three gauge couplings, two parameters of the Higgs potential, nine charged-fermion masses, four parameters of the CKM matrix of Chapter 32, and the strong θ-angle. This chapter asks the only question that matters once the theory is written down: does it work? The answer turns on a single piece of arithmetic. Nineteen inputs is a tiny budget against the endless catalogue of masses, widths, cross sections, asymmetries, and branching ratios the theory predicts; spend a small fixed set of measurements to pin the inputs, and every further measurement becomes a *prediction* the theory lives or dies by. What makes those predictions sharp — to one part in 10^3 and beyond — is that the Standard Model is a renormalizable quantum field theory: its observables are the tree relations of the classical Lagrangian *plus* computable, finite, unambiguous quantum corrections, and those loop corrections are sensitive through virtual states to particles far too heavy to produce directly. Precision physics is therefore a test of the whole quantum edifice, not of the tree Lagrangian alone.

The machinery the chapter needs is imported in full from earlier parts. The renormalized self-energies, the dimensional-regularization scheme, and the Ward identities that keep the loop corrections finite come from Chapters D, 19 and 22; the running of α and the anomalous magnetic moment that exemplify the precision program are the QED-thread results of Chapter 20; the running of the strong coupling and asymptotic freedom that control the hadronic observables are the Yang–Mills results of Chapter 27; the electroweak Lagrangian, the neutral-current couplings, the weak mixing angle, and the tree relation between M_W and M_Z are taken from Chapter 31, the Higgs sector and its mass-fixed couplings from Chapter 30, and the CKM matrix, the Jarlskog invariant, and the structure of CP violation from Chapter 32. The complete field content and parameter list are collected in Chapter F. Nothing in this chapter re-derives those structures; it dresses them with radiative corrections and confronts the result with experiment.

The arc moves outward from a single relation. We begin with the logic of input schemes and the electroweak master relation, isolating the radiative correction Δr and its pieces $\Delta\alpha$ and $\Delta\rho$ — the quantities through which the heavy top quark and the Higgs leave their imprint on low-energy observables. Those self-energies are then the natural language for the Z resonance: its lineshape, total and partial widths, the invisible width that counts the light neutrino species, and the leptonic asymmetries that define the effective weak mixing angle. We package the heavy-particle effects model-independently into the oblique parameters S, T, U, compute the leading top contribution to T from first principles, and watch the global fit corner both m_t and m_h from loop corrections before either particle was seen.

The chapter then turns to the last Standard Model particle to be found. We compute the Higgs production and decay rates — the tree decays to fermions and weak bosons, the loop-induced gluonic and diphoton channels with their characteristic W–top interference — all fixed with no free parameter beyond the masses, and read off the falsifiable coupling-versus-mass law. The mass, the spin-parity 0^+, and the total width are extracted from the data, and the one still-unmeasured vertex, the Higgs self-coupling, is flagged as the genuine open frontier. A parallel and independent confrontation comes from flavor: the $\Delta F = 2$ box diagram reduced to a local operator, the Inami–Lim function derived, the neutral-meson mass splittings and the indirect CP violation ϵ_K and $\sin 2\beta$ computed, all overconstraining the single CKM phase.

Everything funnels into the global electroweak fit, which we treat honestly as a hypothesis test: a χ^2 over an over-determined web of pseudo-observables, linearized to its Gaussian solution, with the residual pulls displayed and the standing tensions of the mid-2020s — the W-mass measurements, the muon $g-2$, $b \to s\,\ell\ell$, first-row CKM unitarity — named and dated rather than smoothed away. The chapter closes by showing how any future deviation is read: converted first into an oblique shift and then into a definite SMEFT Wilson coefficient and a mass scale, the bridge to Chapters 35 and 40. The result is the precision program in one sentence — the quantum structure of the gauge sector is so rigid that the masses of its heaviest constituents are encoded, before discovery, in the radiative corrections to observables at a lower scale — and its verdict is that the Standard Model triumphs.

34.1 The Logic of Precision Tests

Chapter 33 closed by counting the free parameters of the Standard Model (Section 33.6). The tally is short: three gauge couplings, the two parameters of the Higgs potential, the nine charged-fermion masses, the four parameters of the CKM matrix, and the strong θ-angle—nineteen numbers in all (a handful more once neutrino masses are admitted). Nineteen numbers is a great many to measure,

but it is a tiny number compared with the count of *distinct observables* the theory predicts: every mass, every width, every cross section, every asymmetry, every branching ratio of every process at every energy. The arithmetic of the situation is the entire content of this chapter. Once one has spent a small fixed budget of measurements to pin the inputs, every further measurement is a *prediction*, and the theory lives or dies on whether the predictions come true.

What makes the predictions sharp—sharp to the level of one part in 10^3 or better—is that the Standard Model is a renormalizable quantum field theory. Its predictions are not the tree-level relations of the classical Lagrangian alone; they are those relations *plus computable quantum corrections*, finite and unambiguous order by order. This is the decisive point and the reason precision tests are tests of the whole quantum edifice and not merely of the tree Lagrangian: the loop corrections to electroweak observables are sensitive, through virtual particles in the loops, to states far too heavy to produce directly. The top quark and the Higgs boson both announced themselves this way—their masses were read off from the radiative corrections to Z-pole data years before either particle was made in a laboratory. A theory whose loops are calculable turns the precision frontier into a high-mass frontier.

This chapter develops three arenas as one argument. The first, taken up here and in the next two sections, is *electroweak precision*: the Z resonance and the self-energy corrections that package heavy physics. The second is the *Higgs boson*—its calculable production and decay pattern and the coupling-proportional-to-mass law that is the model's sharpest fingerprint. The third is *flavor and CP*: the loop-induced flavour-changing neutral currents and the overconstrained unitarity triangle. Throughout, the chapter *consumes* the endpoints of the benchmark theories rather than adding a new layer to them. From QED it takes the running fine-structure constant $\alpha(M_Z)$ (Chapter 20) and the electron anomaly a_e; from Yang–Mills theory it takes $\alpha_s(M_Z)$ (Chapter 27). These hard-won numbers now appear as *inputs and confirmations* of the precision program.

This first section builds the conceptual machinery: the choice of an electroweak input scheme, the tree-level web of relations among $\{\alpha,\, G_F,\, M_Z,\, M_W,\, \sin^2\theta_{\mathrm{W}}\}$, and the structure of the one-loop radiative corrections—the quantities Δr, $\Delta\alpha$, and $\Delta\rho$ that the rest of the electroweak analysis turns on. To keep long formulae legible we abbreviate the on-shell weak mixing parameters

$$s_W^2 \equiv \sin^2\theta_{\mathrm{W}}, \qquad c_W^2 \equiv \cos^2\theta_{\mathrm{W}} = 1 - s_W^2, \tag{34.1}$$

reserving θ_{W} for the angle itself; the precise definition of s_W^2 that we adopt is fixed in Section 34.1.2.

34.1.1 Input schemes: what is measured and what is predicted

The electroweak sector of the Standard Model—the part responsible for W, Z, photon, and charged-current physics—has, at the level relevant here, exactly

three Lagrangian parameters once the Higgs vacuum value is included: the two gauge couplings g and g' and the vacuum value v (equivalently g, g', and the Higgs sector's v; the Higgs self-coupling λ sets the physical Higgs mass m_h separately and enters electroweak observables only at loop level). Three parameters means we must spend exactly three measurements to fix them; every electroweak observable beyond the third is then a prediction.

Definition 34.1 (Electroweak input scheme)**.** An *electroweak input scheme* is a choice of three precisely measured observables used to fix the three electroweak Lagrangian parameters $\{g, g', v\}$. Once the scheme is fixed, every other electroweak observable is computed as a function of the three inputs (plus the relevant fermion masses, m_h, and α_s where loops require them), order by order in perturbation theory.

The art is in the choice. Two demands compete. We want inputs that are measured as precisely as possible, so that experimental error on the inputs does not swamp the test. And we want inputs whose *relation to the Lagrangian parameters is as free of large or uncertain quantum corrections as possible*, so that the theory side of the comparison is clean. The three best-measured electroweak quantities, with their fractional precisions as of the mid-2020s, are

Observable	Value	Fractional precision
Fine-structure constant α	$1/137.035\,999\ldots$	$\sim 10^{-10}$
Fermi constant G_F	$1.166\,379 \times 10^{-5}\,\mathrm{GeV}^{-2}$	$\sim 5 \times 10^{-7}$
Z mass M_Z	$91.1876\,\mathrm{GeV}$	$\sim 2 \times 10^{-5}$

Here α is the Thomson-limit fine-structure constant, defined at zero momentum transfer and measured (through QED, Chapter 20) most precisely via the electron anomalous moment; G_F is the Fermi constant extracted from the muon lifetime, the low-energy strength of the charged-current interaction (Section 31.5.6); and M_Z is the Z pole mass from the lineshape scan at LEP. These three are the natural *on-shell input set*,

$$\boxed{\{\alpha,\ G_F,\ M_Z\}} \tag{34.2}$$

and they are the canonical choice. Every other electroweak observable—the W mass, the effective mixing angle, the Z partial widths and asymmetries—becomes a *prediction* of these three numbers.

There is a competing choice that trades one input for a more loop-friendly cousin. Replace the Thomson-limit α by the running coupling evaluated at the Z scale, $\alpha(M_Z)$, giving the set

$$\{\alpha(M_Z),\ G_F,\ M_Z\}. \tag{34.3}$$

The two sets carry identical information—they are related by the calculable QED running of Section 20.6—but they organize the radiative corrections differently.

The scheme (34.2) is conceptually cleanest because all three inputs are textbook-defined low-energy constants; the scheme (34.3) is computationally cleanest because the single largest electroweak radiative correction, the running of α from 0 to M_Z, has already been absorbed into the input $\alpha(M_Z)$ and need not be carried as a correction. We will use (34.2) as our reference set and watch the running of α reappear, exactly as the difference between the two schemes, inside the master correction Δr.

Physical Insight

The over-constraint is the whole game. With three inputs fixed, the theory predicts, among many others: the W mass, the effective weak mixing angle measured in asymmetries, the leptonic and hadronic Z widths, the invisible width and hence the number of light neutrino species, and the forward–backward and polarization asymmetries at the Z pole. Each is measured independently. The Standard Model must thread all of them through the eye of a single needle—three numbers. That it does so, to per-mille accuracy, including the contributions of particles heavier than the energies at which the measurements were made, is the strongest quantitative evidence we have for any theory of nature.

34.1.2 The tree-level web of relations

Before any loop, the broken-phase Lagrangian of Chapter 31 already ties our five electroweak quantities into a tight web. We collect the relevant relations, imported from the electroweak chapter, and then turn them into the working tree-level prediction. From the Higgs mechanism and the diagonalization of the neutral sector, the gauge-boson masses and the photon coupling are

$$M_W = \tfrac{1}{2} g v, \qquad M_Z = \frac{M_W}{\cos\theta_{\mathrm{W}}}, \qquad e = g\,\sin\theta_{\mathrm{W}}, \qquad e^2 = 4\pi\alpha, \tag{34.4}$$

where the weak mixing angle is defined by $\tan\theta_{\mathrm{W}} = g'/g$, equivalently $\cos\theta_{\mathrm{W}} = g/\sqrt{g^2+g'^2}$. Integrating the W out of the charged-current interaction and matching to the Fermi theory gave (Section 31.5.6, Eq. (31.145))

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2} = \frac{1}{2v^2}, \tag{34.5}$$

the second equality using $M_W = \tfrac{1}{2} g v$ so that $v = (\sqrt{2}\,G_F)^{-1/2} = 246.2\,\mathrm{GeV}$ (Eq. (31.146)). Eliminating g in favour of e and then α turned (34.5) into the mass prediction (Section 31.5.7, Eq. (31.149))

$$M_W^2 = \frac{\pi\alpha}{\sqrt{2}\,G_F\,s_W^2}, \qquad M_Z^2 = \frac{\pi\alpha}{\sqrt{2}\,G_F\,s_W^2\,c_W^2}. \tag{34.6}$$

Finally, the same diagonalization gave the exact tree identity (Eq. (31.153))

$$\rho \equiv \frac{M_W^2}{M_Z^2\, c_W^2} = 1 \quad \Longleftrightarrow \quad c_W^2 = \frac{M_W^2}{M_Z^2}, \quad \text{i.e.} \quad s_W^2 = 1 - \frac{M_W^2}{M_Z^2}. \tag{34.7}$$

The last form of (34.7) is more than algebra: it is a *definition* of the weak mixing angle in terms of the two physical pole masses.

Definition 34.2 (On-shell weak mixing angle)**.** The *on-shell* weak mixing parameter is defined to all orders by the ratio of the physical gauge-boson pole masses,

$$s_W^2 \equiv 1 - \frac{M_W^2}{M_Z^2}, \qquad c_W^2 \equiv \frac{M_W^2}{M_Z^2}. \tag{34.8}$$

This is a *renormalization-scheme choice*: s_W^2 is not separately measured but is fixed once M_W and M_Z are. It is the convention used throughout this chapter unless an alternative (the effective mixing angle of the Z couplings, Section 34.2) is named explicitly.

The on-shell definition (34.8) keeps $\rho = 1$ true *by fiat* at all orders—it absorbs the loop shift of ρ into the definition of the angle—which is exactly why the physical content of $\rho \neq 1$ will resurface elsewhere, as the parameter $\Delta\rho$ inside Δr and later as the oblique parameter T. With s_W^2 defined this way, the two relations (34.6) are not independent; multiplying the first by $s_W^2 = 1 - M_W^2/M_Z^2$ collapses them into a single statement,

$$M_W^2\Big(1 - \frac{M_W^2}{M_Z^2}\Big) = \frac{\pi\alpha}{\sqrt{2}\, G_F}, \tag{34.9}$$

in which the right-hand side is built entirely from the three inputs (34.2) and the left-hand side involves only M_W and the input M_Z. *This is the tree-level prediction of M_W.* It is a quadratic in M_W^2, and its (physical, larger) root is the predicted W mass.

Worked Example 34.3 (The weak mixing angle and M_W from $\{\alpha, G_F, M_Z\}$ at tree level)**.** We extract s_W^2 and M_W from the three inputs with no other information, to see how far the tree prediction lands from experiment. Write the dimensionful combination on the right of (34.9) as

$$\frac{\pi\alpha}{\sqrt{2}\, G_F} = \frac{\pi/137.036}{\sqrt{2} \times 1.16637 \times 10^{-5}\,\text{GeV}^{-2}} = 1389.8\,\text{GeV}^2 = (37.28\,\text{GeV})^2, \tag{34.10}$$

the same $A_0 = 37.3\,\text{GeV}$ met in Section 31.5.7. From the second relation in (34.6), $M_Z^2 s_W^2 c_W^2 = \pi\alpha/\sqrt{2}G_F$, so with $M_Z^2 = (91.1876)^2 = 8315.2\,\text{GeV}^2$,

$$s_W^2\, c_W^2 = s_W^2(1 - s_W^2) = \frac{1389.8}{8315.2} = 0.16714. \tag{34.11}$$

Solving the quadratic $s_W^2(1-s_W^2)=0.16714$ for the smaller (physical) root,

$$s_W^2=\tfrac{1}{2}\big(1-\sqrt{1-4\times 0.16714}\big)=\tfrac{1}{2}\big(1-\sqrt{0.33144}\,\big)=0.2122. \tag{34.12}$$

Hence $c_W^2=0.7878$ and the tree W mass is

$$M_W=M_Z\,c_W=91.1876\times\sqrt{0.7878}=80.94\,\text{GeV}. \tag{34.13}$$

Two lessons. First, the tree $M_W=80.94\,\text{GeV}$ sits within one percent of the measured $M_W=80.37\,\text{GeV}$—a three-input formula reproducing a fourth independent measurement to the percent level, the over-constraint working in front of us. Second, the value $s_W^2=0.2122$ obtained here from the *clean* input set differs noticeably from the $s_W^2\approx 0.231$ used illustratively in Section 31.5.7 and measured in Z-pole asymmetries. The two are not in conflict: 0.231 is the *effective* mixing angle that the Z couplings see, which already includes the loop corrections, while 0.2122 is the tree-level on-shell value from $\{\alpha, G_F, M_Z\}$. The gap between them, like the gap between 80.94 and 80.37 GeV, is precisely the size of the one-loop electroweak corrections, and accounting for it is the business of the rest of this section.

34.1.3 The master relation and the radiative correction Δr

The tree relation (34.9) cannot be the final word, because its two sides are defined at different scales and through different physics. The right-hand side uses α at zero momentum and G_F at the muon scale; the left-hand side involves the W and Z pole masses at the electroweak scale. A renormalizable theory relates quantities at different scales through computable loops, and those loops modify (34.9) by a finite, calculable factor. We package the entire one-loop (and beyond) modification into a single dimensionless quantity Δr.

Theorem 34.4 (The electroweak master relation)**.** *In the on-shell scheme* (34.8)*, the W mass, the Z mass, the fine-structure constant, and the Fermi constant are related to all orders by*

$$\boxed{M_W^2\Big(1-\frac{M_W^2}{M_Z^2}\Big)=\frac{\pi\alpha}{\sqrt{2}\,G_F}\,(1+\Delta r),} \tag{34.14}$$

where Δr is a finite, calculable function of the input parameters and of the masses circulating in the gauge-boson self-energies and the muon-decay vertices and boxes. At tree level $\Delta r=0$ and (34.14) *reduces to* (34.9)*.*

That Δr is *finite* is the crucial structural fact, and it is a theorem about renormalizability, not an accident. Each ingredient on either side of (34.14)—α, G_F, M_W, M_Z—is a physical observable, defined by a measurement and therefore free of the ultraviolet divergences of the bare Lagrangian. A relation among

physical observables can only involve finite, renormalization-group-invariant combinations of the bare parameters and the cutoff-dependent counterterms; the divergences must cancel among the pieces of Δr on pain of contradiction with the finiteness of the inputs. We will see this cancellation explicitly when we compute the dominant pieces below: the ultraviolet poles of the individual self-energies cancel in exactly the combinations that enter Δr.

The numerical gap exposed in Theorem 34.3—tree $M_W = 80.94\,\text{GeV}$ versus measured $80.37\,\text{GeV}$—fixes the size of Δr in advance. Reading (34.14) as a quadratic for M_W^2 with $\Delta r \neq 0$, a positive Δr raises the right-hand side and (for the physical root) *lowers* the predicted M_W. The observed downward shift of about $0.6\,\text{GeV}$ corresponds to $\Delta r \approx 0.036$: a three-percent correction, exactly the order of magnitude expected for a one-loop electroweak effect, $g^2/16\pi^2$ times logarithms and mass ratios. The task now is to compute this number from first principles and to understand its anatomy.

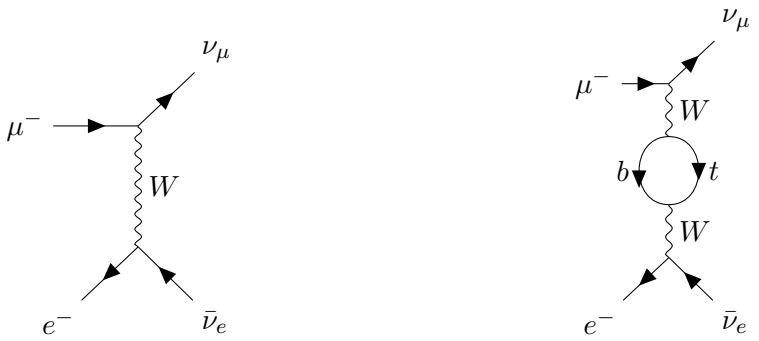
34.1.4 The anatomy of Δr: self-energies in the muon-decay relation

The quantity Δr enters through the definition of G_F, so we must follow the muon-decay amplitude into the one-loop Standard Model. At tree level, muon decay $\mu \to e\,\bar{\nu}_e\,\nu_\mu$ proceeds by W exchange, and at momentum transfer $q^2 \approx 0 \ll M_W^2$ the W propagator collapses to $1/M_W^2$, reproducing the Fermi contact interaction with $G_F/\sqrt{2} = g^2/8M_W^2$ (Section 31.5.6, and Chapter 16 for the decay-rate formula by which G_F is extracted from the lifetime). At one loop the same amplitude is dressed by three classes of correction, shown in Fig. 34.1: the W self-energy $\Sigma_{WW}(q^2)$ on the internal line, vertex corrections at the two charged-current vertices, and box diagrams in which a second gauge boson is exchanged.

The dressed amplitude rescales the effective four-fermion coupling. Writing the full transverse W propagator at low q^2 as $i\,[\,q^2 - M_W^2 - \Sigma_{WW}(q^2)\,]^{-1}$, where Σ_{WW} is the (renormalized) one-particle-irreducible self-energy and M_W is the physical pole mass, the $q^2 \to 0$ limit gives $-i\,[\,M_W^2 + \Sigma_{WW}(0)\,]^{-1}$. Matching the one-loop amplitude to the Fermi form therefore yields

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\,M_W^2}\left(1 - \frac{\Sigma_{WW}(0)}{M_W^2} + \delta_{\text{VB}}\right), \tag{34.15}$$

where δ_{VB} collects the vertex and box corrections (and the external-leg wavefunction renormalizations) of muon decay, with the pure-QED final-state photonic corrections already removed in the *definition* of G_F. To make (34.15) a statement about physical inputs we must also express the bare coupling g^2 through the measured α. Using $g^2 = e^2/s_W^2 = 4\pi\alpha/s_W^2$ and inserting the renormalization of the electric charge brings in the photon vacuum polarization; assembling all the pieces and comparing with the tree relation $G_F/\sqrt{2} = \pi\alpha/(2M_W^2 s_W^2)$ gives, by



(a) Tree-level W exchange. At $q^2 \to 0$ this is the Fermi interaction defining G_F.

(b) W self-energy insertion $\Sigma_{WW}(q^2)$; the top–bottom loop is the source of $\Delta\rho$.

Figure 34.1: Muon decay at tree level and with the W self-energy correction. Vertex and box corrections (not drawn) complete the one-loop set; together they define Δr through the relation between G_F and the on-shell parameters.

definition of Δr,

$$\frac{G_F}{\sqrt{2}} = \frac{\pi\alpha}{2\,M_W^2\,s_W^2}\,(1+\Delta r), \tag{34.16}$$

which, on multiplying through by $2M_W^2 s_W^2 = 2M_W^2(1 - M_W^2/M_Z^2)$, is exactly the master relation (34.14). The content of Δr is thus the sum of three physically distinct effects, which we now isolate:

$$\boxed{\Delta r = \Delta\alpha - \frac{c_W^2}{s_W^2}\,\Delta\rho + \Delta r_{\rm rem}.} \tag{34.17}$$

The first term is the running of α; the second is the custodial-breaking self-energy difference, amplified by c_W^2/s_W^2; the third is a calculable remainder dominated by a logarithm of the Higgs mass. We take them in turn.

$\Delta\alpha$: the running of the fine-structure constant

The largest single piece of Δr is the running of α, imported wholesale from QED. The argument is this. In (34.16) the coupling that naturally governs electroweak physics at the scale M_Z is not the Thomson-limit $\alpha \equiv \alpha(0)$ but the effective coupling $\alpha(M_Z)$ that already includes the screening of the bare charge by all fermion loops light enough to be excited at that scale. The difference between them is precisely the subtracted photon vacuum polarization evaluated between $q^2 = 0$ and $q^2 = M_Z^2$,

$$\Delta\alpha \equiv \widehat{\Pi}_{\gamma\gamma}(0) - \mathrm{Re}\,\widehat{\Pi}_{\gamma\gamma}(M_Z^2), \tag{34.18}$$

which is the very quantity computed in Section 20.6. There the geometric resummation of vacuum-polarization bubbles gave (Eq. (20.171) and its all-fermion generalization)

$$\alpha(M_Z^2) = \frac{\alpha}{1 - \Delta\alpha}, \qquad \Delta\alpha = \Delta\alpha_{\text{lep}} + \Delta\alpha_{\text{had}}^{(5)} + \Delta\alpha_{\text{top}}, \tag{34.19}$$

with the numerical values established in Chapter 20: the leptonic part $\Delta\alpha_{\text{lep}}(M_Z^2) \approx 0.0315$ is pure QED and computed loop by loop, the five-flavour hadronic part $\Delta\alpha_{\text{had}}^{(5)}(M_Z^2) \approx 0.0276$ comes from the dispersive integral over $e^+e^- \to$ hadrons data, and the top contributes negligibly, for a total $\Delta\alpha(M_Z^2) \approx 0.059$ and the famous result (Eq. (20.187))

$$\alpha^{-1}(M_Z^2) \approx 128.95, \tag{34.20}$$

the fine-structure constant having run from $1/137$ at zero momentum to about $1/129$ at the electroweak scale. The appearance of $+\Delta\alpha$ in Δr is nothing but the statement that using $\alpha(0)$ in the tree relation undercounts the true coupling strength at M_Z by the factor $(1 - \Delta\alpha)^{-1} \approx 1 + \Delta\alpha$. Equivalently, in the alternative input scheme (34.3) where one feeds in $\alpha(M_Z)$ directly, this entire piece is absorbed into the input and $\Delta\alpha$ disappears from Δr. This is the precise sense, promised in the chapter opening, in which the precision program *consumes* QED's running coupling: $\Delta\alpha$ is not recomputed here, it is the endpoint of Chapter 20 entering as a number.

Caution

The hadronic part $\Delta\alpha_{\text{had}}^{(5)}$ is, as of the mid-2020s, the dominant theoretical uncertainty on the electroweak fit, because it relies on measured cross-section data rather than perturbation theory (the relevant scale runs through the non-perturbative hadronic region). The same hadronic vacuum polarization limits the Standard-Model prediction of the muon anomalous moment a_μ. The two are fed by one and the same data, a point made in Chapter 20 and revisited among the live tensions in Section 34.2 and beyond.

$\Delta\rho$: the custodial-breaking self-energies and the top quark

The second term in (34.17) is the one that made precision electroweak physics a top-quark detector. It originates in the difference between the W and Z self-energies at zero momentum—exactly the violation of $\rho = 1$ that the custodial-symmetry analysis of Section 31.6 taught us to expect, with the leading contribution from the large up–down mass splitting of the top–bottom doublet. To compute it we need the gauge-boson self-energy from a fermion loop, evaluated at zero external momentum. We derive the general result and then specialize.

Proposition 34.5 (Fermion-loop self-energy at zero momentum)**.** *Let a gauge boson couple to a fermion bilinear through the chiral vertex $i\gamma^\mu(g_L P_L + g_R P_R)$, with the two internal fermion lines carrying masses m_1 and m_2 and the loop carrying a colour multiplicity N_c. Then the transverse self-energy at zero external momentum, $\Pi^{\mu\nu}(0) = \eta^{\mu\nu}\Sigma(0)$, is*

$$\Sigma(0) = -\frac{N_c}{d}\left[2(2-d)\,(g_L^2+g_R^2)\,\frac{m_1^2 a(m_1^2) - m_2^2 a(m_2^2)}{m_1^2 - m_2^2} + 4d\, g_L g_R\, m_1 m_2\,\frac{a(m_1^2) - a(m_2^2)}{m_1^2 - m_2^2}\right], \tag{34.21}$$

where $a(m^2) \equiv -i\,A_0(m^2)$ is the (real) scalar tadpole of Eq. (D.99),

$$a(m^2) = \frac{m^2}{16\pi^2}\left[\bar{\tfrac{1}{\varepsilon}} + 1 + \ln\tfrac{\bar{\mu}^2}{m^2}\right], \qquad \bar{\tfrac{1}{\varepsilon}} = \tfrac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi. \tag{34.22}$$

Proof. The one-loop self-energy from a closed fermion loop is

$$i\,\Pi^{\mu\nu}(q) = -N_c \int \frac{d^d k}{(2\pi)^d}\,\frac{\mathrm{Tr}\big[\gamma^\mu \Gamma(\not{k} + m_1)\gamma^\nu \Gamma(\not{k} + \not{q} + m_2)\big]}{(k^2 - m_1^2)((k+q)^2 - m_2^2)}, \qquad \Gamma \equiv g_L P_L + g_R P_R, \tag{34.23}$$

the overall sign and the factor of N_c being the standard ones for a fermion loop (the four factors of i from two propagators and two vertices multiply to unity). Set $q = 0$ and contract with $\eta_{\mu\nu}$; by Lorentz invariance $\Pi^{\mu\nu}(0) = \eta^{\mu\nu}\Sigma(0)$, so $\eta_{\mu\nu}\Pi^{\mu\nu}(0) = d\,\Sigma(0)$, and we need the contracted trace

$$T \equiv \eta_{\mu\nu}\ \mathrm{Tr}\big[\gamma^\mu \Gamma(\not{k} + m_1)\gamma^\nu \Gamma(\not{k} + m_2)\big] = \mathrm{Tr}\big[\gamma_\nu \Gamma(\not{k} + m_1)\gamma^\nu \Gamma(\not{k} + m_2)\big]. \tag{34.24}$$

Only traces with an even number of gamma matrices survive, so the single-mass cross terms vanish and T splits into a momentum-squared piece and a mass-product piece. For the momentum piece, push the chirality projectors through using $P_{L,R}\not{k} = \not{k} P_{R,L}$ and $P_{L,R}\gamma^\nu = \gamma^\nu P_{R,L}$; the projectors realign so that only the g_L^2 and g_R^2 combinations survive, and $\mathrm{Tr}[\gamma_\nu \not{k}\gamma^\nu \not{k}\, P_{R,L}] = \tfrac{1}{2}\,\mathrm{Tr}[\gamma_\nu \not{k}\gamma^\nu \not{k}]$. With the d-dimensional contraction $\gamma_\nu\gamma^\alpha\gamma^\nu = (2-d)\gamma^\alpha$ and $\mathrm{Tr}[\not{k}\not{k}] = 4k^2$, this gives $(g_L^2 + g_R^2)\cdot(2-d)\cdot 2k^2$. For the mass piece, the projector algebra $\gamma_\nu P_L \gamma^\nu \Gamma = d\,P_R\Gamma = d\,g_R P_R$ (and its mirror) yields $\mathrm{Tr}[\gamma_\nu \Gamma \gamma^\nu \Gamma] = 4d\,g_L g_R$, so the mass term is $4d\,g_L g_R\, m_1 m_2$. Altogether

$$T = 2(2-d)(g_L^2 + g_R^2)\,k^2 + 4d\,g_L g_R\, m_1 m_2. \tag{34.25}$$

The two scalar integrals needed are evaluated by partial fractions on $[(k^2 - m_1^2)(k^2 - m_2^2)]^{-1} = (m_1^2 - m_2^2)^{-1}[(k^2 - m_1^2)^{-1} - (k^2 - m_2^2)^{-1}]$. Using $\int k^2/(k^2 - m^2) = m^2 A_0(m^2)$ (the scaleless remainder vanishes in dimensional regularization) and $A_0(m^2) = \int 1/(k^2 - m^2)$ from Eq. (D.99),

$$\int \frac{d^d k}{(2\pi)^d}\,\frac{k^2}{(k^2 - m_1^2)(k^2 - m_2^2)} = \frac{m_1^2 A_0(m_1^2) - m_2^2 A_0(m_2^2)}{m_1^2 - m_2^2}, \tag{34.26}$$

$$\int \frac{d^dk}{(2\pi)^d}\,\frac{1}{(k^2-m_1^2)(k^2-m_2^2)} = \frac{A_0(m_1^2)-A_0(m_2^2)}{m_1^2-m_2^2} = B_0(0;m_1,m_2), \tag{34.27}$$

the last equality being the zero-momentum bubble of Eq. (D.104). Substituting (34.25), (34.26), (34.27) into $i\,d\Sigma(0) = -N_c\int T/[\cdots]$ and writing $A_0 = i\,a$ cancels the overall i, leaving the real expression (34.21). The tadpole expansion (34.22) follows from $A_0(m^2) = -i(4\pi)^{-d/2}\Gamma(1-\tfrac{d}{2})(m^2)^{d/2-1}$ at $d = 4-2\varepsilon$. □

Caution

The result (34.21) passes the Ward-identity test that the photon stay massless. For a pure vector coupling $g_L = g_R = e$ and equal masses $m_1 = m_2 = m$, the bracket becomes $2(2-d)\cdot 2e^2\,(a + m^2a') + 4d\,e^2m^2a'$ with $a' = da/dm^2$; using $a = \frac{m^2}{16\pi^2}[\frac{1}{\bar\varepsilon}+1+L]$ and $a' = \frac{1}{16\pi^2}[\frac{1}{\bar\varepsilon}+L]$ ($L \equiv \ln\bar\mu^2/m^2$) one finds $(2-d)a + 2m^2a' \to 0$ as $\varepsilon\to 0$, so $\Sigma_{\gamma\gamma}(0) = 0$ exactly. The photon self-energy at zero momentum vanishes, as gauge invariance demands; the massless tadpole has not slipped a spurious mass onto the photon.

We now compute the custodial-breaking combination that defines $\Delta\rho$. The physical content is fixed by the low-energy comparison of neutral- and charged-current strengths. At $q^2 \approx 0$, neutral-current exchange is governed by $[M_Z^2 + \Sigma_{ZZ}(0)]^{-1}$ and charged-current exchange by $[M_W^2 + \Sigma_{WW}(0)]^{-1}$, so the ratio that the tree relation set to unity becomes

$$\rho = \frac{M_W^2 + \Sigma_{WW}(0)}{c_W^2\,[M_Z^2 + \Sigma_{ZZ}(0)]} \approx 1 + \frac{\Sigma_{WW}(0)}{M_W^2} - \frac{\Sigma_{ZZ}(0)}{M_Z^2}, \tag{34.28}$$

having used $M_W^2 = c_W^2 M_Z^2$ at leading order. We therefore define

$$\boxed{\Delta\rho \equiv \frac{\Sigma_{WW}(0)}{M_W^2} - \frac{\Sigma_{ZZ}(0)}{M_Z^2}.} \tag{34.29}$$

Worked Example 34.6 ($\Delta\rho$ from the top–bottom doublet)**.** We evaluate (34.29) for the third-generation quark doublet, the dominant contribution because m_t towers over every other mass. The fermion couplings are those of Chapter 31: the W couples left-handedly across the doublet, $g_L^W = g/\sqrt{2}$ and $g_R^W = 0$, joining $m_1 = m_t$ to $m_2 = m_b$; the Z couples each quark to itself with chiral couplings $g_L^{Z,f} = (g/c_W)(T_3^f - Q_f s_W^2)$ and $g_R^{Z,f} = (g/c_W)(-Q_f s_W^2)$. We take $m_b \to 0$ throughout (its effect is of relative order $m_b^2/m_t^2 \approx 4\times 10^{-4}$) and restore it at the end. With $m_b = 0$ the massless tadpole $a(0) = 0$ kills the bottom loop in Σ_{ZZ} and simplifies Σ_{WW}.

The W self-energy. With $g_L^2 + g_R^2 = g^2/2$, $g_Lg_R = 0$, $m_1 = m_t$, $m_2 = 0$, (34.21) gives

$$\Sigma_{WW}(0) = -\frac{N_c}{d}\,2(2-d)\,\frac{g^2}{2}\,\frac{m_t^2\,a(m_t^2)}{m_t^2} = N_c\,g^2\,\frac{d-2}{d}\,a(m_t^2). \tag{34.30}$$

Dividing by $M_W^2 = g^2 v^2/4$ (so $g^2/M_W^2 = 4/v^2$) and expanding $(d-2)/d = \frac{1}{2} - \frac{\varepsilon}{4}$ against $a(m_t^2) = \frac{m_t^2}{16\pi^2}[\frac{1}{\bar\varepsilon} + 1 + L_t]$ with $L_t \equiv \ln \bar\mu^2/m_t^2$, the ε in the prefactor combines with the $1/\bar\varepsilon$ pole to leave

$$\frac{\Sigma_{WW}(0)}{M_W^2} = \frac{4N_c}{v^2}\,\frac{d-2}{d}\,a(m_t^2) = \frac{2N_c\,m_t^2}{16\pi^2\,v^2}\left[\tfrac{1}{\bar\varepsilon} + \tfrac{1}{2} + L_t\right]. \tag{34.31}$$

The Z self-energy. Only the top loop survives ($m_b = 0$), with $m_1 = m_2 = m_t$, so we take the equal-mass limit of (34.21), in which the first ratio becomes $\frac{d}{dm^2}[m^2 a] = a + m^2 a'$ and the second becomes a'. Writing $g_L^2 + g_R^2 = g_Z^2 v_+$ and $g_L g_R = g_Z^2 v_\times$ with $g_Z = g/c_W$,

$$v_+ = \left(\tfrac{1}{2} - \tfrac{2}{3}s_W^2\right)^2 + \left(\tfrac{2}{3}s_W^2\right)^2, \qquad v_\times = \left(\tfrac{1}{2} - \tfrac{2}{3}s_W^2\right)\left(-\tfrac{2}{3}s_W^2\right), \tag{34.32}$$

and using $g_Z^2/M_Z^2 = 4/v^2$, the equal-mass form of (34.21) gives

$$\frac{\Sigma_{ZZ}(0)}{M_Z^2} = \frac{4N_c}{v^2}\left[\,2^{\frac{d-2}{d}}\,v_+(a + m_t^2 a') - 4v_\times\,m_t^2 a'\,\right]. \tag{34.33}$$

With $a + m_t^2 a' = \frac{m_t^2}{16\pi^2}[2/\bar\varepsilon + 1 + 2L_t]$ and the same ε-bookkeeping as before, $2^{\frac{d-2}{d}}(a + m_t^2 a') = 2a'\,m_t^2$ to the order kept (the constant $+1$ cancels against the -1 from $\varepsilon/2 \cdot 2/\varepsilon$), so

$$\frac{\Sigma_{ZZ}(0)}{M_Z^2} = \frac{8N_c\,m_t^2}{16\pi^2\,v^2}\,(v_+ - 2v_\times)\left[\tfrac{1}{\bar\varepsilon} + L_t\right]. \tag{34.34}$$

Assembly and the cancellation. The custodial-breaking combination $v_+ - 2v_\times$ collapses to a perfect square,

$$v_+ - 2v_\times = \left[\left(\tfrac{1}{2} - \tfrac{2}{3}s_W^2\right) + \tfrac{2}{3}s_W^2\right]^2 = \tfrac{1}{4}, \tag{34.35}$$

independent of s_W^2—the imprint of custodial symmetry, which forbids the neutral-current charge structure from contributing to $\Delta\rho$. Inserting $v_+ - 2v_\times = \frac{1}{4}$ into (34.34) and subtracting (34.31),

$$\Delta\rho = \frac{\Sigma_{WW}(0)}{M_W^2} - \frac{\Sigma_{ZZ}(0)}{M_Z^2} = \frac{N_c\,m_t^2}{16\pi^2\,v^2}\Big\{\,2\big[\tfrac{1}{\bar\varepsilon} + \tfrac{1}{2} + L_t\big] - 8\cdot\tfrac{1}{4}\big[\tfrac{1}{\bar\varepsilon} + L_t\big]\Big\}. \tag{34.36}$$

The divergent $1/\bar\varepsilon$ poles cancel ($2 - 2 = 0$), the logarithms L_t cancel ($2 - 2 = 0$), and so does all scale dependence—$\Delta\rho$ *is finite and unambiguous*, as the finiteness of the physical ρ demands. Only the constant survives: $2 \cdot \frac{1}{2} = 1$, leaving

$$\boxed{\Delta\rho = \frac{N_c\,m_t^2}{16\pi^2\,v^2} = \frac{3\,G_F\,m_t^2}{8\sqrt{2}\,\pi^2},} \tag{34.37}$$

where the second form uses $1/v^2 = \sqrt{2}G_F$ (Eq. (34.5)) and $N_c = 3$. This is the exact coefficient promised in Section 31.6.5: the parametric estimate $\Delta\rho \sim N_c m_t^2/16\pi^2 v^2$ of Eq. (31.184) had coefficient unity. Numerically, with $m_t = 172.9\,\text{GeV}$ and $v = 246.2\,\text{GeV}$,

$$\Delta\rho = \frac{3\,(172.9)^2}{16\pi^2\,(246.2)^2} = 9.4 \times 10^{-3}, \tag{34.38}$$

a correction of about one percent.

Restoring m_b. Keeping $m_b \neq 0$ throughout the same computation replaces m_t^2 by the symmetric custodial function

$$\Delta\rho = \frac{3\,G_F}{8\sqrt{2}\,\pi^2}\,F(m_t^2, m_b^2), \qquad F(x,y) = x + y - \frac{2xy}{x-y}\ln\frac{x}{y}, \tag{34.39}$$

which vanishes when $m_t = m_b$ (custodial restoration, since $F(x,x) = 0$ by the limit $\ln(x/y) \to (x-y)/x$) and reduces to $F(m_t^2, 0) = m_t^2$ in the limit taken above. The hierarchy $m_b \ll m_t$ makes the difference between (34.39) and (34.37) a 0.04% effect, utterly negligible against the 1% size of $\Delta\rho$ itself.

Two structural points elevate (34.37) above a mere number. First, $\Delta\rho$ grows as m_t^2 rather than decoupling as an inverse power of the heavy mass. The reason, as Section 31.6.5 explained, is that $m_t = y_t v/\sqrt{2}$ measures the top Yukawa coupling, and a heavy top is a *strongly* coupled top; strong couplings do not decouple, and the custodial-breaking spurion $m_t - m_b$ makes itself felt quadratically. This non-decoupling is exactly what turned ρ into a top-quark mass-meter: the measured ρ pointed to a heavy top before the top was produced. Second, the amplification factor in (34.17) is now explicable. Through the on-shell definition $s_W^2 = 1 - M_W^2/M_Z^2$, the loop shift of the mass ratio $c_W^2 = M_W^2/M_Z^2$ is

$$\frac{\delta c_W^2}{c_W^2} = \frac{\Sigma_{WW}}{M_W^2} - \frac{\Sigma_{ZZ}}{M_Z^2} \approx \Delta\rho, \qquad \text{hence} \qquad \frac{\delta s_W^2}{s_W^2} = -\frac{c_W^2}{s_W^2}\,\Delta\rho, \tag{34.40}$$

using $\delta s_W^2 = -\delta c_W^2$ and that, for the m_t^2-enhanced term, the on-shell self-energies $\Sigma_{VV}(M_V^2)$ equal $\Sigma_{VV}(0)$ up to corrections of order M_V^2/m_t^2. This $-(c_W^2/s_W^2)\Delta\rho$ is precisely the second term of (34.17), and with $c_W^2/s_W^2 = 0.788/0.212 \approx 3.7$ the custodial correction enters Δr multiplied by nearly four. A one-percent $\Delta\rho$ becomes a three-to-four-percent shift in the M_W prediction.

Δr_{rem}: the remainder and the Higgs logarithm

Everything not contained in $\Delta\alpha$ and the leading $\Delta\rho$ is collected in Δr_{rem}: the subleading parts of the gauge self-energies, the vertex and box corrections δ_{VB} of muon decay, and—most importantly for the history of the field—the dependence on the Higgs-boson mass m_h. The Higgs enters the W and Z self-energies

through diagrams in which a virtual Higgs is radiated and reabsorbed. By the same custodial argument that protected $\Delta\rho$, the Higgs cannot generate an m_h^2-enhanced contribution: the Higgs sector is custodially symmetric in the limit $g' \to 0$, so any custodial-violating effect must carry the hypercharge spurion g'^2, and dimensional analysis then forbids a quadratic growth. The leading Higgs dependence is therefore only *logarithmic*,

$$\Delta r_{\rm rem} \supset \frac{\sqrt{2}\, G_F\, M_W^2}{16\pi^2}\, \frac{11}{3}\Big(\frac{c_W^2}{s_W^2} - \frac{5}{6}\,\frac{1}{s_W^2}\Big)\, \ln\frac{m_h^2}{M_W^2} + \cdots , \tag{34.41}$$

a result known as Veltman's *screening theorem*: the Higgs mass is screened from electroweak observables, appearing only inside a logarithm. The practical consequence is profound and was felt for two decades. A logarithm is a weak handle: to pin m_h from a fit to Δr one must measure M_W and s_W^2 extraordinarily well, because $\ln(m_h^2/M_W^2)$ varies slowly. Precision electroweak data did succeed, indirectly preferring a *light* Higgs ($m_h \lesssim 160\,\text{GeV}$ at 95% confidence) years before the direct discovery at 125 GeV—a spectacular consistency check, but one bought only by per-mille measurements precisely because the sensitivity is logarithmic rather than power-law. The distinction between the quadratic m_t^2 sensitivity of $\Delta\rho$ and the logarithmic m_h sensitivity of $\Delta r_{\rm rem}$ is itself a prediction of the custodial structure, and both predictions have been confirmed.

Worked Example 34.7 (Predicting M_W with Δr)**.** We close the loop by predicting M_W from $\{\alpha, G_F, M_Z\}$ including the radiative corrections, and comparing with experiment. Assemble Δr from its three pieces with the values established above:

$$\Delta\alpha = 0.059, \qquad \frac{c_W^2}{s_W^2}\,\Delta\rho = 3.7 \times 9.4 \times 10^{-3} = 0.035, \qquad \Delta r_{\rm rem} \approx +0.01, \tag{34.42}$$

the remainder evaluated for $m_h = 125\,\text{GeV}$ from (34.41) together with the bosonic and vertex/box contributions. Then

$$\Delta r = 0.059 - 0.035 + 0.010 = 0.034. \tag{34.43}$$

Notice the structure: the running of α pushes Δr up, the heavy top pulls it down through $\Delta\rho$, and the remainder nudges it back up—a near-cancellation of the two large effects that leaves a net few-percent correction. Insert $\Delta r = 0.034$ into the master relation (34.14) as a quadratic for M_W^2. With $\pi\alpha/\sqrt{2}G_F = 1389.8\,\text{GeV}^2$ (Eq. (34.10)) and $M_Z^2 = 8315.2\,\text{GeV}^2$,

$$M_W^2\Big(1 - \frac{M_W^2}{M_Z^2}\Big) = 1389.8 \times 1.034 = 1437\,\text{GeV}^2, \tag{34.44}$$

i.e. $x^2 - M_Z^2\, x + M_Z^2 \times 1437 = 0$ with $x = M_W^2$, whose physical (larger) root is

$$M_W^2 = \frac{M_Z^2 + \sqrt{M_Z^4 - 4M_Z^2 \times 1437}}{2} = 6467\,\text{GeV}^2, \qquad M_W = 80.42\,\text{GeV}. \tag{34.45}$$

The loop-corrected prediction $M_W = 80.42\,\text{GeV}$ is to be compared with the tree value 80.94 GeV of Theorem 34.3 and the measured world average $M_W = 80.37\,\text{GeV}$. The radiative corrections have moved the prediction by 0.5 GeV, in the right direction and by the right amount, to land within tens of MeV of the measurement. A three-input formula, corrected by calculable loops containing a top quark and a Higgs boson, predicts a fourth mass to better than one part in 10^3. This is the precision program working, and it is the template for everything that follows.

34.1.5 The hierarchy of corrections, and why each piece is calculable

Step back and survey the structure of Δr as a whole. It is organized as a hierarchy, each layer calculable for a reason rooted in the renormalizability of the theory:

(i) $\Delta\alpha \approx 0.06$, the running of α, is the largest piece. It is calculable because it is a low-energy QED effect resummed by the renormalization group of QED; its only uncertainty is the hadronic contribution, which is data-driven rather than divergent.

(ii) $(c_W^2/s_W^2)\Delta\rho \approx 0.035$, the custodial correction, is the largest *genuinely electroweak* piece. It is calculable, finite, and free of all scheme dependence (the divergences cancelled in (34.36)) because ρ is a physical ratio; it grows as m_t^2 because the top Yukawa does not decouple.

(iii) $\Delta r_{\text{rem}} \approx 0.01$, the remainder, carries the logarithmic Higgs dependence and the smaller bosonic and vertex/box effects. It is calculable and, by Veltman screening, only logarithmically sensitive to m_h.

The decisive structural fact, repeated at each layer, is that every correction is *finite*. This is not luck. The relation (34.14) ties together four physical observables, and a renormalizable theory guarantees that any such relation is free of ultraviolet ambiguity: the divergences of the bare parameters are absorbed, once and for all, into the measured values of the inputs, leaving the predictions finite. We watched this happen in miniature when the $1/\bar{\varepsilon}$ poles of Σ_{WW} and Σ_{ZZ} cancelled in $\Delta\rho$. The same mechanism operates throughout the electroweak fit: each new observable is a finite prediction, sensitive through its loops to the masses of virtual particles, and the comparison with experiment tests not the tree Lagrangian but the entire quantum structure. A theory that was not renormalizable could not make these predictions at all—its loop corrections would be cutoff-dependent and meaningless. The success of precision electroweak physics is, at bottom, a vindication of renormalizability as a principle of nature.

With Δr, $\Delta\alpha$, and $\Delta\rho$ in hand, we have the conceptual and computational machinery the rest of the chapter needs. The renormalized self-energies $\Sigma_{VV}(q^2)$

introduced here are the natural language for the Z-pole observables, and the quantity $\Delta\rho$—the custodial-breaking difference of self-energies—is the seed of the oblique parameter T that will let us package the effects of heavy new physics model-independently. The next section turns to the Z resonance itself: its lineshape and total width, its partial widths into each fermion species, the invisible width that counts the light neutrinos, and the leptonic asymmetries that define the effective weak mixing angle s^2_W already glimpsed at the difference between 0.212 and 0.231. Each of these is computed from the neutral-current couplings of Chapter 31, dressed by exactly the self-energy corrections assembled here.

34.2 Z-Pole Physics: Lineshape, Widths, and Asymmetries

The previous section fixed the conceptual machinery: an electroweak input scheme, the tree relations among $\{\alpha, G_F, M_Z, M_W, \sin^2\theta_{\rm W}\}$, and the one-loop corrections Δr, $\Delta\alpha$, $\Delta\rho$ that promote those relations from tree identities to genuine quantum predictions (Section 34.1). We now cash that framework on the single most productive laboratory the Standard Model has ever been handed: the resonance at $\sqrt{s} = M_Z \simeq 91\,\text{GeV}$ produced in e^+e^- collisions. Between 1989 and 2000 the LEP and SLC machines manufactured some 2×10^7 Z bosons and measured the shape of that resonance, the rates into every accessible final state, and the angular and polarization asymmetries of the decays, to a precision of parts in 10^4. Every one of those numbers is, in the Standard Model, a prediction built from a single ingredient already in hand: the neutral-current couplings g_V^f and g_A^f of Section 31.5.4. The task of this section is to derive that entire suite of observables — the partial and total widths, the Breit–Wigner lineshape and its peak, the invisible width and the count of light neutrino species, the leptonic asymmetries and the effective mixing angle they define — from those couplings, the decay and cross-section formulae of Chapter 16, and nothing else. The threads enter only as inputs: $\alpha(M_Z)$ from QED (Section 20.6) sets the scale of the electromagnetic piece, and $\alpha_s(M_Z)$ from Yang–Mills theory (Chapter 27) supplies the one correction to the hadronic width that is not pure electroweak. We add no new thread layer; we consume the endpoints.

34.2.1 The neutral-current vertex and the spin-summed decay rate

Recall the per-fermion neutral-current interaction extracted in Sections 31.5.3 and 31.5.4: the Z couples to a single Dirac fermion f through

$$\mathcal{L}^{(f)}_{\rm NC} = \frac{g}{\cos\theta_{\rm W}}\, Z_\mu\, \bar f\, \gamma^\mu\, \tfrac{1}{2}(g_V^f - g_A^f\,\gamma^5)\, f, \qquad g_V^f = T_f^3 - 2Q_f \sin^2\theta_{\rm W}, \quad g_A^f = T_f^3, \tag{34.46}$$

with g_V^f, g_A^f the pure numbers tabulated in Table 31.2. The Feynman rule for the $Zf\bar f$ vertex is therefore

$$i\,\frac{g}{\cos\theta_{\rm W}}\,\gamma^\mu\,\tfrac{1}{2}(g_V^f - g_A^f\,\gamma^5). \tag{34.47}$$

We want $\Gamma(Z\to f\bar f)$, the partial width of the Z into a fermion pair. Throughout the Z-pole analysis we treat the final fermions as massless: every charged fermion except the top has $m_f^2/M_Z^2 \le (m_b/M_Z)^2 \approx 2\times10^{-3}$, and the top ($m_t \approx 173\,\text{GeV} > M_Z$) is not produced at all. The massless approximation is thus excellent for the kinematic factors, and we will reinstate the leading mass corrections only where they matter (the b-quark threshold factor) by inspection.

The amplitude and its square. For the decay $Z(p,\lambda)\to f(k_1,s_1)\,\bar f(k_2,s_2)$ the amplitude reads, from Eq. (34.47),

$$i\mathcal{M} = i\,\frac{g}{2\cos\theta_{\rm W}}\,\varepsilon_\mu(p,\lambda)\,\bar u(k_1)\,\gamma^\mu(g_V^f - g_A^f\gamma^5)\,v(k_2), \tag{34.48}$$

with $\varepsilon_\mu(p,\lambda)$ the polarization vector of the decaying Z. To form the rate we sum $|\mathcal{M}|^2$ over the final spins s_1, s_2 and over the three physical polarizations of the Z, then average over the initial Z polarization by dividing by three. Abbreviate $\Gamma^\mu \equiv \gamma^\mu(g_V^f - g_A^f\gamma^5)$. Its Dirac conjugate is $\bar\Gamma^\nu \equiv \gamma^0\Gamma^{\nu\dagger}\gamma^0$, which we evaluate using $\gamma^0\gamma^5\gamma^0 = -\gamma^5$ and $\{\gamma^5,\gamma^\nu\}=0$:

$$\bar\Gamma^\nu = \gamma^0(g_V^f - g_A^f\gamma^5)\gamma^0\,\gamma^\nu = (g_V^f + g_A^f\gamma^5)\gamma^\nu = \gamma^\nu(g_V^f - g_A^f\gamma^5) = \Gamma^\nu, \tag{34.49}$$

so the vertex structure is self-conjugate. The spin sum over the massless fermion bilinears uses $\sum_s u(k_1)\bar u(k_1) = \not k_1$ and $\sum_s v(k_2)\bar v(k_2) = \not k_2$ (the $\pm m$ terms vanish at $m_f = 0$), giving the trace

$$\sum_{s_1,s_2} \big[\bar u(k_1)\Gamma^\mu v(k_2)\big]\big[\bar u(k_1)\Gamma^\nu v(k_2)\big]^* = \mathrm{tr}\big[\not k_1\,\Gamma^\mu\,\not k_2\,\Gamma^\nu\big]. \tag{34.50}$$

To evaluate it, push the first chiral projector through $\not k_2\gamma^\nu$. Because γ^5 commutes through an even number of gamma matrices, $\gamma^5\not k_2\gamma^\nu = \not k_2\gamma^\nu\gamma^5$, so $\Gamma^\mu\not k_2\gamma^\nu = \gamma^\mu\not k_2\gamma^\nu(g_V^f - g_A^f\gamma^5)$, and the two chiral factors combine into

$$(g_V^f - g_A^f\gamma^5)^2 = g_V^{f2} + g_A^{f2} - 2g_V^f g_A^f\,\gamma^5. \tag{34.51}$$

The trace splits into a γ^5-free piece and a γ^5 piece. Using the trace identities of Appendix C, $\mathrm{tr}[\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu] = 4(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\mu\beta})$ and $\mathrm{tr}[\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu\gamma^5] = -4i\,\epsilon^{\alpha\mu\beta\nu}$,

$$\begin{aligned}\mathrm{tr}[\not{k}_1\Gamma^\mu\not{k}_2\Gamma^\nu] &= (g_V^{f2} + g_A^{f2})\,4(k_1^\mu k_2^\nu + k_1^\nu k_2^\mu - \eta^{\mu\nu}\,k_1{\cdot}k_2) \\ &\quad + 2g_V^f g_A^f \cdot 4i\,\epsilon^{\alpha\mu\beta\nu}k_{1\alpha}k_{2\beta}.\end{aligned} \tag{34.52}$$

The first line is symmetric in $\mu\nu$; the second is antisymmetric. The polarization sum for a massive vector of mass M_Z is

$$\sum_{\lambda=1}^{3}\varepsilon_\mu(p,\lambda)\,\varepsilon_\nu^*(p,\lambda) = -\eta_{\mu\nu} + \frac{p_\mu p_\nu}{M_Z^2}, \tag{34.53}$$

which is symmetric in $\mu\nu$, so it kills the antisymmetric γ^5 term of Eq. (34.52) on contraction. (That term is not lost physics: it encodes the parity violation that drives the asymmetries of Section 34.2.4; it simply does not contribute to the spin-summed *rate*.) Contracting the symmetric piece,

$$\begin{aligned}&\Big(-\eta_{\mu\nu} + \tfrac{p_\mu p_\nu}{M_Z^2}\Big)\cdot 4(k_1^\mu k_2^\nu + k_1^\nu k_2^\mu - \eta^{\mu\nu}k_1{\cdot}k_2) \\ &\qquad = 4\Big[\,2\,k_1{\cdot}k_2 + \tfrac{1}{M_Z^2}\big(2(p{\cdot}k_1)(p{\cdot}k_2) - M_Z^2\,k_1{\cdot}k_2\big)\Big].\end{aligned} \tag{34.54}$$

Momentum conservation $p = k_1 + k_2$ with $k_1^2 = k_2^2 = 0$ gives $M_Z^2 = p^2 = 2k_1{\cdot}k_2$, hence $k_1{\cdot}k_2 = M_Z^2/2$ and $p{\cdot}k_1 = p{\cdot}k_2 = k_1{\cdot}k_2 = M_Z^2/2$. The $p_\mu p_\nu$ term then reads $\frac{1}{M_Z^2}\big(2\cdot\frac{M_Z^2}{2}\cdot\frac{M_Z^2}{2} - M_Z^2\cdot\frac{M_Z^2}{2}\big) = 0$: the longitudinal part vanishes identically for massless final fermions, exactly as current conservation requires. What survives is $4\cdot 2k_1{\cdot}k_2 = 4M_Z^2$. Collecting the factors in Eq. (34.48),

$$\sum_{\lambda}\sum_{s_1,s_2}|\mathcal{M}|^2 = \Big(\frac{g}{2\cos\theta_{\mathrm{W}}}\Big)^2(g_V^{f2}+g_A^{f2})\cdot 4M_Z^2 = \frac{g^2}{\cos^2\theta_{\mathrm{W}}}(g_V^{f2}+g_A^{f2})\,M_Z^2. \tag{34.55}$$

Phase space and the width. The two-body width of a particle at rest was derived in Chapter 16: with the final-state momentum magnitude $|\mathbf{k}|$ and the spin-averaged squared amplitude $\overline{|\mathcal{M}|^2}$,

$$\Gamma = \mathcal{S}\,\frac{|\mathbf{k}|}{8\pi M_Z^2}\,\overline{|\mathcal{M}|^2}, \tag{34.56}$$

which is Eq. (16.89). Here the two final particles are distinct (f and $\bar f$), so $\mathcal{S} = 1$; they are massless, so $|\mathbf{k}| = M_Z/2$ from Eq. (16.87). The spin-averaged amplitude is Eq. (34.55) divided by the three initial Z polarizations and multiplied by the colour multiplicity N_c^f of the final state ($N_c^f = 1$ for leptons, $N_c^f = 3$ for quarks,

since the trace Eq. (34.50) was computed per colour and the N_c^f colour final states add):

$$\overline{|\mathcal{M}|^2} = N_c^f \, \frac{1}{3} \, \frac{g^2}{\cos^2\theta_{\mathrm{W}}} (g_V^{f2} + g_A^{f2}) \, M_Z^2 . \tag{34.57}$$

Inserting $|\mathbf{k}| = M_Z/2$ and Eq. (34.57) into Eq. (34.56),

$$\Gamma(Z \to f\bar{f}) = \frac{M_Z/2}{8\pi M_Z^2} \cdot N_c^f \, \frac{1}{3} \, \frac{g^2}{\cos^2\theta_{\mathrm{W}}} (g_V^{f2} + g_A^{f2}) \, M_Z^2 = \frac{N_c^f \, g^2 \, M_Z}{48\pi \cos^2\theta_{\mathrm{W}}} (g_V^{f2} + g_A^{f2}) . \tag{34.58}$$

This is the partial width in terms of the gauge coupling. It is more useful in the input-scheme variables. The tree relations of Chapter 31 give $M_W = \frac{1}{2} g v$ and $M_Z = M_W / \cos\theta_{\mathrm{W}}$, so $g^2/\cos^2\theta_{\mathrm{W}} = 4M_Z^2/v^2$, while the Fermi matching Eq. (31.145) fixes $G_F/\sqrt{2} = g^2/8M_W^2$, equivalently $v^{-2} = \sqrt{2}\, G_F$. Substituting,

$$\boxed{\Gamma(Z \to f\bar{f}) = \frac{N_c^f \, G_F \, M_Z^3}{6\sqrt{2}\,\pi} (g_V^{f2} + g_A^{f2}) = \frac{N_c^f \, M_Z}{12\pi} \cdot \frac{g^2}{4\cos^2\theta_{\mathrm{W}}} (g_V^{f2} + g_A^{f2}) .} \tag{34.59}$$

The two forms are identical; the first expresses the width through the precisely known G_F, the second exhibits the compact $N_c^f M_Z/12\pi$ grouping with the neutral-current coupling factored out. The entire flavor dependence is carried by the single quadratic combination $N_c^f (g_V^{f2} + g_A^{f2})$. This is the master formula of Z-pole physics; everything below is its consequence.

Physical Insight

The width is a coupling squared times a phase-space volume times a mass. Reading Eq. (34.59) backwards: $G_F M_Z^3$ is the dimensionful scale (a width has mass dimension one, and $G_F \sim M_Z^{-2}$), the $1/6\sqrt{2}\,\pi$ is geometry (two-body phase space and the spin average), and $N_c^f (g_V^{f2} + g_A^{f2})$ is the dynamical content — the only place the identity of the fermion enters. A fermion that couples more strongly to the Z, or that comes in more colours, takes a larger share of the Z's decays. Because g_V^f and g_A^f are fixed rational functions of $\sin^2\theta_{\mathrm{W}}$, every branching ratio of the Z is a prediction with no free parameter once the mixing angle is known.

Worked Example 34.8 (The partial widths of the Z, channel by channel)**.** We evaluate Eq. (34.59) numerically for one generation and assemble the total width. Using $G_F = 1.1664 \times 10^{-5}\,\mathrm{GeV}^{-2}$ and $M_Z = 91.19\,\mathrm{GeV}$ (so $M_Z^3 = 7.58 \times 10^5\,\mathrm{GeV}^3$), the common leptonic prefactor is

$$\Gamma_0 \equiv \frac{G_F M_Z^3}{6\sqrt{2}\,\pi} = \frac{1.1664 \times 10^{-5} \times 7.58 \times 10^5}{6\sqrt{2}\,\pi}\,\mathrm{GeV} = 0.3317\,\mathrm{GeV}, \tag{34.60}$$

so that $\Gamma(Z \to f\bar{f}) = N_c^f \, \Gamma_0 \, (g_V^{f2} + g_A^{f2})$. With $\sin^2\theta_{\mathrm{W}} = 0.2312$ the couplings of Table 31.2 give the combinations in Table 34.1.

Table 34.1: Tree-level Z partial widths from Eq. (34.59) with $\Gamma_0 = 0.3317\,\text{GeV}$ and $\sin^2\theta_W = 0.2312$. The hadronic entries carry the QCD factor $(1 + \alpha_s/\pi)$ with $\alpha_s(M_Z) = 0.118$ (Section 34.2.5); the t quark is absent ($m_t > M_Z$). "×" marks channels summed over the three generations in the totals.

channel f	N_c^f	g_V^f	g_A^f	$g_V^{f2} + g_A^{f2}$	Γ_f [MeV]
ν (per species)	1	+0.500	+0.500	0.500	165.9
$\ell = e, \mu, \tau$	1	−0.038	−0.500	0.2514	83.4
u, c	3	+0.192	+0.500	0.2868	296 (QCD-corr.)
d, s, b	3	−0.346	−0.500	0.3696	382 (QCD-corr.)

Per-channel values. A single neutrino gives $g_V^2 + g_A^2 = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$, so $\Gamma_\nu = \Gamma_0 \times \frac{1}{2} = 165.9\,\text{MeV}$. A charged lepton has $g_A = -\frac{1}{2}$ and $g_V = -\frac{1}{2} + 2\sin^2\theta_W = -0.038$, hence $g_V^2 + g_A^2 = 0.001 + 0.250 = 0.2514$ and $\Gamma_\ell = 0.3317 \times 0.2514 = 83.4\,\text{MeV}$. An up-type quark has $g_V = \frac{1}{2} - \frac{4}{3}\sin^2\theta_W = 0.192$, $g_V^2 + g_A^2 = 0.287$, and three colours, giving $3 \times 0.3317 \times 0.287 = 285\,\text{MeV}$ before QCD corrections; a down-type quark has $g_V = -\frac{1}{2} + \frac{2}{3}\sin^2\theta_W = -0.346$, $g_V^2 + g_A^2 = 0.370$, giving $3 \times 0.3317 \times 0.370 = 368\,\text{MeV}$.

Summing the channels. Three generations of leptons contribute $\Gamma_{\ell\ell} \equiv 3\Gamma_\ell = 250\,\text{MeV}$ (visible). Three neutrino species contribute the *invisible* width $\Gamma_{\text{inv}} = 3\Gamma_\nu = 498\,\text{MeV}$. The hadronic width sums the two accessible up-type quarks (u, c) and the three down-type (d, s, b), with the QCD enhancement $(1 + \alpha_s/\pi) = 1.038$ derived in Section 34.2.5:

$$\Gamma_{\text{had}} = (2\times285 + 3\times368)\,(1.038)\,\text{MeV} = 1674\times1.038\,\text{MeV} = 1737\,\text{MeV}. \quad (34.61)$$

The total width is the sum over all open channels, Eq. (16.78),

$$\Gamma_Z = \Gamma_{\text{had}} + \Gamma_{\ell\ell} + \Gamma_{\text{inv}} = 1737 + 250 + 498\,\text{MeV} = 2485\,\text{MeV} \simeq 2.49\,\text{GeV}. \quad (34.62)$$

The measured value is $\Gamma_Z^{\text{exp}} = 2495.2 \pm 2.3\,\text{MeV}$. Our tree formula plus the one QCD factor reproduces it to 0.4% — the residual is exactly the size of the electroweak radiative corrections ($\Delta\rho$ and the running of the couplings) that Section 34.3 will package. That a one-line formula built from $\sin^2\theta_W$ and the quantum numbers of Table 31.2 lands within half a percent of a number measured to 0.1% is the first quantitative triumph of the precision program.

34.2.2 The resonant lineshape and the peak cross section

Widths are extracted from the *shape* of the resonance scanned in e^+e^- collisions. We now derive that shape. The process is $e^+e^- \to Z \to f\bar{f}$, the s-channel Z-exchange diagram (b) of Fig. 31.4; near $\sqrt{s} = M_Z$ the Z pole dominates

utterly over the photon-exchange amplitude (a) of QED, and we may treat the process as pure Z exchange (the small γ–Z interference is a calculable correction, and the photon-only piece is the off-resonance $\sigma = 4\pi\alpha^2/3s$ of Eq. (17.89)).

The dressed propagator. A stable particle has a propagator with a real pole at $p^2 = M_Z^2$. The Z is unstable, and in Chapter 16 we saw that an unstable particle's pole moves into the complex plane: the full propagator is obtained by resumming the one-particle- irreducible self-energy $\Sigma(p^2)$ into a geometric series, exactly as the vacuum-polarization insertions were resummed for the running coupling in Section 20.6. The result is $D(p^2) = \left[p^2 - M_Z^2 - \Sigma(p^2)\right]^{-1}$. The real part of Σ shifts the mass (and is absorbed into the definition of M_Z); the imaginary part, by the optical-theorem relation $\Gamma_Z = -\operatorname{Im}\Sigma(M_Z^2)/M_Z$ established as Eq. (16.82), encodes the total decay rate. Near the pole the scalar part of the dressed Z propagator is therefore

$$D(s) = \frac{1}{s - M_Z^2 + iM_Z\Gamma_Z}, \qquad s \equiv p^2, \tag{34.63}$$

the Breit–Wigner denominator. For light external fermions the $p_\mu p_\nu/M_Z^2$ part of the numerator $-\eta_{\mu\nu} + p_\mu p_\nu/M_Z^2$ does not contribute (the same current-conservation argument that killed it in Eq. (34.54)), so only the $-\eta_{\mu\nu}$ piece survives.

Assembling the cross section. The amplitude factorizes into a production current, the propagator, and a decay current:

$$\mathcal{M} = \left(\frac{g}{2\cos\theta_{\mathrm{W}}}\right)^2 \left[\bar v(p_2)\gamma^\mu (g_V^e - g_A^e\gamma^5)u(p_1)\right] \times \frac{-\eta_{\mu\nu}}{s - M_Z^2 + iM_Z\Gamma_Z} \left[\bar u(k_1)\gamma^\nu (g_V^f - g_A^f\gamma^5)v(k_2)\right]. \tag{34.64}$$

Squaring, averaging over the four initial $e^\pm$ spin states (the $\frac14$) and summing over final spins,

$$\tfrac14 \sum_{\text{spins}} |\mathcal{M}|^2 = \frac{1}{(s - M_Z^2)^2 + M_Z^2\Gamma_Z^2}\, \tfrac14\, E^{\mu\nu}F_{\mu\nu}, \tag{34.65}$$

with the production (electron) and decay (fermion) tensors

$$E^{\mu\nu} = \left(\tfrac{g}{2\cos\theta_{\mathrm{W}}}\right)^2 \operatorname{tr}[\not p_2\, \gamma^\mu (g_V^e - g_A^e\gamma^5)\, \not p_1\, \gamma^\nu (g_V^e - g_A^e\gamma^5)], \tag{34.66}$$

and $F_{\mu\nu}$ the same with $e \to f$ and $p_{1,2} \to k_{1,2}$. These are the traces already done in Eq. (34.52). For the *total* cross section the antisymmetric γ^5 parts of E and F contract into a term odd in $\cos\theta$ that integrates to zero over the full solid angle (it is precisely the forward–backward asymmetry of Section 34.2.4);

we keep the symmetric parts. Writing $a \equiv (g^2/\cos^2\theta_{\mathrm{W}})(g_V^{e2}+g_A^{e2})$ and $b \equiv (g^2/\cos^2\theta_{\mathrm{W}})(g_V^{f2}+g_A^{f2})$, the symmetric tensors are $E_{\mathrm{s}}^{\mu\nu} = a(p_1^\mu p_2^\nu + p_1^\nu p_2^\mu - \eta^{\mu\nu} p_1 p_2)$ and likewise for F. Their contraction, using $p_1^2 = p_2^2 = 0$, collapses cleanly:

$$\begin{aligned} E_{\mathrm{s}}^{\mu\nu} F_{\mathrm{s},\mu\nu} &= ab\Big[2(p_1\!\cdot\!k_1)(p_2\!\cdot\!k_2) + 2(p_1\!\cdot\!k_2)(p_2\!\cdot\!k_1) - 2(p_1\!\cdot\!p_2)(k_1\!\cdot\!k_2) \\ &\qquad - 2(p_1\!\cdot\!p_2)(k_1\!\cdot\!k_2) + 4(p_1\!\cdot\!p_2)(k_1\!\cdot\!k_2)\Big] \\ &= 2ab\big[(p_1\!\cdot\!k_1)(p_2\!\cdot\!k_2) + (p_1\!\cdot\!k_2)(p_2\!\cdot\!k_1)\big]. \end{aligned} \tag{34.67}$$

In the massless CM kinematics $p_1\!\cdot\!k_1 = p_2\!\cdot\!k_2 = -t/2$ and $p_1\!\cdot\!k_2 = p_2\!\cdot\!k_1 = -u/2$, with $t = -\frac{s}{2}(1-\cos\theta)$, $u = -\frac{s}{2}(1+\cos\theta)$, so $E_{\mathrm{s}}\!\cdot\!F_{\mathrm{s}} = \frac{ab}{2}(t^2+u^2) = \frac{ab}{4}s^2(1+\cos^2\theta)$. The differential cross section follows from the two-body CM formula Eq. (17.85) (with $|\mathbf{k}|/|\mathbf{p}| = 1$ for massless particles), multiplied by the final-state colour factor N_c^f,

$$\frac{d\sigma}{d\Omega} = \frac{N_c^f}{64\pi^2 s}\,\frac{1}{4}\sum_{\text{spins}}|\mathcal{M}|^2 = \frac{N_c^f}{64\pi^2 s}\,\frac{\frac{1}{4}\cdot\frac{ab}{4}s^2(1+\cos^2\theta)}{(s-M_Z^2)^2 + M_Z^2\Gamma_Z^2}. \tag{34.68}$$

Integrating $\int d\Omega\,(1+\cos^2\theta) = 2\pi\cdot\frac{8}{3} = \frac{16\pi}{3}$,

$$\sigma(s) = \frac{N_c^f\, ab\, s}{192\pi\big[(s-M_Z^2)^2 + M_Z^2\Gamma_Z^2\big]}. \tag{34.69}$$

Trading couplings for widths. The combinations a and b are exactly the ones that built the partial widths. Comparing with Eq. (34.58), $\Gamma_e = (g^2/\cos^2\theta_{\mathrm{W}})(g_V^{e2}+g_A^{e2})M_Z/48\pi$, i.e. $a = 48\pi\Gamma_e/M_Z$, and $N_c^f\, b = 48\pi\Gamma_f/M_Z$. Hence $N_c^f\, ab = (48\pi)^2\Gamma_e\Gamma_f/M_Z^2$, and

$$\sigma(s) = \frac{(48\pi)^2}{192\pi}\,\frac{s\,\Gamma_e\Gamma_f}{M_Z^2\big[(s-M_Z^2)^2 + M_Z^2\Gamma_Z^2\big]} = \frac{12\pi\, s}{M_Z^2}\,\frac{\Gamma_e\Gamma_f}{(s-M_Z^2)^2 + M_Z^2\Gamma_Z^2}, \tag{34.70}$$

using $(48\pi)^2/192\pi = 12\pi$. This is the relativistic Breit–Wigner lineshape for $e^+e^- \to f\bar{f}$. At the peak, $s = M_Z^2$, the denominator is $M_Z^2\Gamma_Z^2$ and the prefactor $s/M_Z^2 = 1$, leaving the celebrated peak cross section

$$\boxed{\sigma_{\mathrm{peak}}^f = \frac{12\pi}{M_Z^2}\,\frac{\Gamma_e\,\Gamma_f}{\Gamma_Z^2}.} \tag{34.71}$$

The structure is transparent: Γ_e/Γ_Z is the probability that the Z was *made* from the e^+e^- initial state (the branching ratio into electrons, read backwards through detailed balance), Γ_f/Γ_Z is the probability it *decays* to f, and $12\pi/M_Z^2$ is the geometric maximum cross section of a spin-1 resonance formed from spin-$\frac{1}{2}$ beams, $(16\pi/M_Z^2)\cdot(2J{+}1)/[(2s_1{+}1)(2s_2{+}1)] = (16\pi/M_Z^2)\cdot\frac{3}{4}$. The peak height

measures the product of the entering and exiting branching ratios; the full width at half maximum of Eq. (34.70) measures Γ_Z directly. Scanning $\sqrt{s}$ across the resonance and fitting the curve to Eq. (34.70) delivers M_Z (the position), Γ_Z (the width), and $\Gamma_e\Gamma_f$ (the height) simultaneously.

Caution

The denominator in Eq. (34.70) carries a *constant* width $M_Z^2\Gamma_Z^2$. A more careful treatment of the energy dependence of the self-energy promotes this to an s-dependent ("running") width, $M_Z^2\Gamma_Z^2 \to s^2\Gamma_Z^2/M_Z^2$, which shifts the fitted M_Z by about 34 MeV and the width by about 1 MeV relative to the fixed-width form. The two parametrizations agree exactly at the peak, Eq. (34.71), so the peak cross section and the extracted partial widths are unaffected; only the precise numerical value of M_Z depends on the convention, and the LEP value $M_Z = 91.1876\,\text{GeV}$ is quoted in the running-width scheme. This is a real effect at LEP precision, not a quibble — it is why a convention must be fixed before quoting M_Z to the keV.

Worked Example 34.9 (The hadronic peak cross section)**.** The most visible number at LEP is the height of the hadronic peak. Setting $f = \text{had}$ in Eq. (34.71) and using the widths of Theorem 34.8 ($\Gamma_e = 83.4\,\text{MeV}$, $\Gamma_{\text{had}} = 1744\,\text{MeV}$, $\Gamma_Z = 2495\,\text{MeV}$, $M_Z = 91.19\,\text{GeV}$),

$$\sigma_{\text{peak}}^{\text{had}} = \frac{12\pi}{M_Z^2}\,\frac{\Gamma_e\Gamma_{\text{had}}}{\Gamma_Z^2} = \frac{12\pi}{(91.19)^2}\cdot\frac{0.0834\times 1.744}{(2.495)^2}\,\text{GeV}^{-2} = 1.066\times 10^{-4}\,\text{GeV}^{-2}. \tag{34.72}$$

Converting with $(\hbar c)^2 = 0.3894\,\text{mb}\cdot\text{GeV}^2 = 3.894\times 10^5\,\text{nb}\cdot\text{GeV}^2$,

$$\sigma_{\text{peak}}^{\text{had}} = 1.066\times 10^{-4}\times 3.894\times 10^5\ \text{nb} = 41.5\,\text{nb}, \tag{34.73}$$

against the measured Born (pole) value $\sigma_{\text{had}}^0 = 41.54\,\text{nb}$. (The cross section actually *observed* at the peak is about 30 nb; the difference is initial-state QED radiation, which smears the effective collision energy and is removed in extracting the Born σ_{had}^0.) The lineshape and the three curves for $N_\nu = 2, 3, 4$ are plotted in Fig. 34.2: more neutrino species widen Γ_Z, and because Γ_Z appears squared in the denominator of Eq. (34.71), the peak *drops* and *broadens*. The peak height is thus a neutrino counter, the subject of Section 34.2.3.

34.2.3 The invisible width and the number of light neutrinos

The Z decays into final states that the detector never sees: the three neutrino channels $Z \to \nu\bar{\nu}$. Their summed rate is the *invisible width*

$$\Gamma_{\text{inv}} = N_\nu\,\Gamma_\nu, \qquad \Gamma_\nu = \Gamma(Z\to\nu\bar{\nu}) = \Gamma_0\times\tfrac{1}{2} = 165.9\,\text{MeV}, \tag{34.74}$$

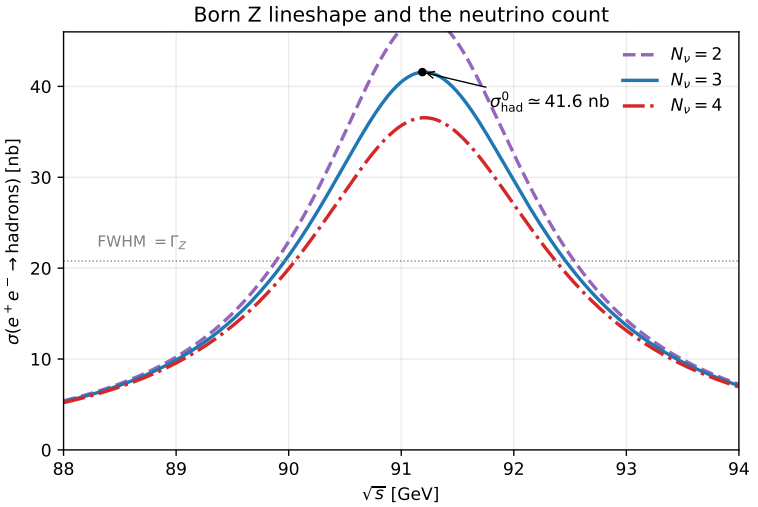


Figure 34.2: The Born Z lineshape $\sigma(e^+e^- \to \text{hadrons})$ from the Breit–Wigner formula Eq. (34.70), for $N_\nu = 2, 3, 4$ light neutrino species. The peak position fixes M_Z, the full width at half maximum fixes Γ_Z, and the peak height $\sigma^0_{\rm had} = 41.5\,\text{nb}$ fixes $\Gamma_e\Gamma_{\rm had}/\Gamma_Z^2$. Each extra neutrino species adds $\Gamma_\nu \simeq 166\,\text{MeV}$ to Γ_Z, lowering and broadening the resonance; the data sit squarely on the $N_\nu = 3$ curve.

where N_ν is the number of light neutrino species into which the Z can decay ("light" meaning $m_\nu < M_Z/2$, so the channel is kinematically open) and Γ_ν is the standard-model rate per species, fixed by $g_V^\nu = g_A^\nu = \frac{1}{2}$ in Eq. (34.59). The invisible width is not measured directly — nothing is detected — but inferred as the difference between the total width and the sum of all *visible* widths,

$$\Gamma_{\rm inv} = \Gamma_Z - \Gamma_{\rm had} - \Gamma_{\ell\ell} = \Gamma_Z - \Gamma_{\rm had} - 3\,\Gamma_\ell. \tag{34.75}$$

Both Γ_Z (from the lineshape width) and the visible widths (from the peak cross sections into hadrons and charged leptons) are measured. Dividing Eq. (34.74) by the standard-model prediction Γ_ν counts the neutrinos. The cleanest experimental form avoids the absolute normalization by taking ratios to the well-measured leptonic width and using only the *standard-model ratio* $(\Gamma_\ell/\Gamma_\nu)_{\rm SM}$, which is a pure function of $\sin^2\theta_{\rm W}$ via Eq. (34.59),

$$\left(\frac{\Gamma_\nu}{\Gamma_\ell}\right)_{\rm SM} = \frac{g_V^{\nu 2} + g_A^{\nu 2}}{g_V^{\ell 2} + g_A^{\ell 2}} = \frac{0.500}{0.2514} = 1.989, \qquad \left(\frac{\Gamma_\ell}{\Gamma_\nu}\right)_{\rm SM} = 0.5028. \tag{34.76}$$

Worked Example 34.10 (Counting the light neutrino species)**.** We carry the measurement to its number. The LEP lineshape fits give, as central values,

$$\Gamma_Z = 2495.2\,\text{MeV}, \quad \Gamma_{\rm had} = 1744.4\,\text{MeV}, \quad \Gamma_\ell = 83.99\,\text{MeV (per charged lepton)}. \tag{34.77}$$

The invisible width follows from Eq. (34.75),

$$\Gamma_{\rm inv} = 2495.2 - 1744.4 - 3 \times 83.99 = 2495.2 - 1744.4 - 251.97 = 498.8\,\text{MeV}. \tag{34.78}$$

The number of species is then this invisible width, measured in units of the single-species standard-model rate. Writing it through the ratio Eq. (34.76) so that only relative widths and the theoretical ratio enter,

$$N_\nu = \frac{\Gamma_{\rm inv}}{\Gamma_\ell}\left(\frac{\Gamma_\ell}{\Gamma_\nu}\right)_{\rm SM} = \frac{498.8}{83.99} \times 0.5028 = 5.939 \times 0.5028 = 2.986. \tag{34.79}$$

The full LEP combination, with all radiative corrections folded into the theoretical $(\Gamma_\ell/\Gamma_\nu)_{\rm SM}$ and the lineshape fit done in the running-width scheme, gives

$$N_\nu = 2.984 \pm 0.008. \tag{34.80}$$

The result is three, to better than one percent. The $\sim 2\sigma$ shortfall below the integer is a real and stable feature of the data, traced to the absolute hadronic peak normalization $\sigma^0_{\rm had}$ and not to any new physics. The number is decisive: there are exactly three generations of light, weakly-interacting neutrinos that couple to the Z with standard strength. A fourth sequential generation with a neutrino lighter than $M_Z/2$ is excluded by this single measurement — the lineshape would sit on the $N_\nu = 4$ curve of Fig. 34.2, which the data exclude by tens of standard deviations.

Physical Insight

That a counting of generations falls out of a resonance width is the precision program in miniature. The Z is democratic: it couples to every fermion with a strength fixed entirely by quantum numbers, so it acts as a calibrated source that radiates into all channels in proportion to $N_c^f(g_V^{f2} + g_A^{f2})$. Anything the Z can reach contributes to its width whether or not a detector registers it. The "missing" width is therefore a direct census of light states that carry weak isospin — a measurement of particle content made without ever detecting the particles. The same logic, applied to the *shape* of the corrections rather than to a counting, is what cornered the top quark and the Higgs before either was produced (Section 34.3).

34.2.4 Asymmetries and the effective weak mixing angle

The widths measure $g_V^{f2} + g_A^{f2}$, a sum of squares that is insensitive to the relative sign and to which coupling is larger. The parity violation of the neutral current — the γ^5 term we dropped from the rate — is recovered in the *asymmetries*, which measure the bilinear $g_V^f g_A^f$ and so fix the ratio g_V^f/g_A^f and, through it, the effective mixing angle. These are the sharpest single numbers the Z pole produces.

The forward–backward asymmetry. Restore the antisymmetric γ^5 pieces of $E^{\mu\nu}$ and $F_{\mu\nu}$ in Eq. (34.66). Each contributes a term $\propto i\,\epsilon^{\mu\nu\alpha\beta}p_{1\alpha}p_{2\beta}$ (production) and $\propto i\,\epsilon_{\mu\nu\gamma\delta}k_1^\gamma k_2^\delta$ (decay), with coefficients $2g_V^e g_A^e$ and $2g_V^f g_A^f$ respectively. Their contraction, using $\epsilon^{\mu\nu\alpha\beta}\epsilon_{\mu\nu\gamma\delta} = -2(\delta^\alpha_\gamma\delta^\beta_\delta - \delta^\alpha_\delta\delta^\beta_\gamma)$, produces a piece proportional to $(p_1{\cdot}k_1)(p_2{\cdot}k_2) - (p_1{\cdot}k_2)(p_2{\cdot}k_1) \propto t^2 - u^2 \propto \cos\theta$. Adding it to the symmetric $(1+\cos^2\theta)$ part, the full differential cross section on the pole is

$$\frac{d\sigma}{d\cos\theta} \;\propto\; (g_V^{e2}+g_A^{e2})\,(g_V^{f2}+g_A^{f2})\,(1+\cos^2\theta) + 8\,g_V^e g_A^e\,g_V^f g_A^f\,\cos\theta. \tag{34.81}$$

The first term is symmetric under $\theta \to \pi - \theta$ (forward $\leftrightarrow$ backward); the second is antisymmetric and tilts the angular distribution. Define the forward–backward asymmetry as the difference of integrated rates over the forward ($\cos\theta > 0$) and backward ($\cos\theta < 0$) hemispheres,

$$A_{\rm FB} = \frac{\sigma_{\rm F} - \sigma_{\rm B}}{\sigma_{\rm F} + \sigma_{\rm B}}, \qquad \sigma_{\rm F} - \sigma_{\rm B} = \int_0^1 - \int_{-1}^0 \frac{d\sigma}{d\cos\theta}\,d\cos\theta. \tag{34.82}$$

The symmetric term gives $\int_{-1}^{1}(1+\cos^2\theta)\,d\cos\theta = \frac{8}{3}$ for the denominator and cancels in the numerator (it is even). The antisymmetric term gives $\int_0^1 \cos\theta\,d\cos\theta - \int_{-1}^0 \cos\theta\,d\cos\theta = \frac{1}{2} - (-\frac{1}{2}) = 1$ for the numerator and cancels in the denominator (it is odd). Therefore

$$A_{\rm FB} = \frac{8\,g_V^e g_A^e\,g_V^f g_A^f \cdot 1}{(g_V^{e2}+g_A^{e2})(g_V^{f2}+g_A^{f2})\cdot\frac{8}{3}} = 3\cdot\frac{g_V^e g_A^e}{g_V^{e2}+g_A^{e2}}\cdot\frac{g_V^f g_A^f}{g_V^{f2}+g_A^{f2}}. \tag{34.83}$$

This motivates the single-fermion *asymmetry parameter*

$$\boxed{A_f \equiv \frac{2\,g_V^f g_A^f}{g_V^{f2}+g_A^{f2}},} \tag{34.84}$$

in terms of which Eq. (34.83) reads

$$\boxed{A_{\rm FB}^{0,f} = \tfrac{3}{4}\,A_e\,A_f.} \tag{34.85}$$

The pole forward–backward asymmetry factorizes into a production factor A_e (how asymmetrically the Z is made from e^+e^-) and a decay factor A_f (how asymmetrically it decays into f). Each A_f is bounded, $|A_f| \le 1$, attained when the coupling is purely chiral ($g_V = \pm g_A$).

The left–right asymmetry. At the SLC the electron beam was longitudinally polarized, which gives direct access to A_e alone. The Z couples to definite chiralities with strengths $g_L^e = \frac{1}{2}(g_V^e + g_A^e)$ and $g_R^e = \frac{1}{2}(g_V^e - g_A^e)$, so the cross

sections from left- and right-handed electrons are $\sigma_L \propto g_L^{e2}$ and $\sigma_R \propto g_R^{e2}$ (the final-state sum over f being common). The left–right asymmetry is then

$$A_{\mathrm{LR}} = \frac{\sigma_L - \sigma_R}{\sigma_L + \sigma_R} = \frac{g_L^{e2} - g_R^{e2}}{g_L^{e2} + g_R^{e2}} = \frac{\tfrac{1}{4}[(g_V^e + g_A^e)^2 - (g_V^e - g_A^e)^2]}{\tfrac{1}{4}[(g_V^e + g_A^e)^2 + (g_V^e - g_A^e)^2]} = \frac{g_V^e g_A^e}{\tfrac{1}{2}(g_V^{e2} + g_A^{e2})} = A_e. \tag{34.86}$$

A single polarization-asymmetry measurement delivers A_e cleanly, with no dependence on the final state at all — the SLC's chief advantage over the unpolarized A_{FB}, in which A_e appears only multiplied by A_f.

The effective mixing angle. Each A_f measures the ratio of couplings. From Eq. (34.46),

$$\frac{g_V^f}{g_A^f} = \frac{T_f^3 - 2Q_f \sin^2\theta_{\mathrm{W}}}{T_f^3} = 1 - 2\,\frac{Q_f}{T_f^3}\,\sin^2\theta_{\mathrm{W}} = 1 - 4\,|Q_f|\,\sin^2\theta_{\mathrm{eff}}^f. \tag{34.87}$$

The last equality uses the identity $Q_f/T_f^3 = 2|Q_f|$, which holds for every fermion because $T_f^3 = \pm\tfrac{1}{2}$ and the charge and weak isospin always share the same sign within a multiplet: for a charged lepton ($T^3 = -\tfrac{1}{2}$, $Q = -1$) the ratio is 2, for an up-type quark ($T^3 = +\tfrac{1}{2}$, $Q = +\tfrac{2}{3}$) it is $\tfrac{4}{3}$, for a down-type quark ($T^3 = -\tfrac{1}{2}$, $Q = -\tfrac{1}{3}$) it is $\tfrac{2}{3}$ — each equal to $2|Q_f|$. Thus for a charged lepton $g_V^\ell/g_A^\ell = 1 - 4\sin^2\theta_{\mathrm{eff}}^\ell$. The notation $\sin^2\theta_{\mathrm{eff}}^f$ rather than $\sin^2\theta_{\mathrm{W}}$ is deliberate: once one-loop electroweak corrections are included the relation between couplings and the on-shell mixing angle is modified by self-energy and vertex effects, and the *effective* mixing angle is *defined* to absorb them, so that Eq. (34.87) holds exactly with $\sin^2\theta_{\mathrm{eff}}^f$ in place of the tree $\sin^2\theta_{\mathrm{W}}$. This is the pseudo-observable that the asymmetries deliver and that Section 34.3 confronts with the radiative corrections.

Worked Example 34.11 (The leptonic asymmetry and $\sin^2\theta_{\mathrm{eff}}^\ell$)**.** The electron's near-vanishing vector coupling makes the leptonic asymmetries exquisitely sensitive to the mixing angle. With $g_A^\ell = -\tfrac{1}{2}$ and $g_V^\ell = -\tfrac{1}{2} + 2\sin^2\theta_{\mathrm{W}}$, take the effective value $\sin^2\theta_{\mathrm{eff}}^\ell = 0.2315$, so $g_V^\ell = -0.5 + 0.4630 = -0.0370$ and $g_V^\ell/g_A^\ell = 0.0740$. Then

$$A_e = A_\ell = \frac{2g_V^\ell g_A^\ell}{g_V^{\ell 2} + g_A^{\ell 2}} = \frac{2 \times (-0.0370) \times (-0.5)}{0.00137 + 0.250} = \frac{0.0370}{0.2514} = 0.147, \tag{34.88}$$

and the purely leptonic pole forward–backward asymmetry is, from Eq. (34.85),

$$A_{\mathrm{FB}}^{0,\ell} = \tfrac{3}{4}\,A_e^2 = 0.75 \times (0.147)^2 = 0.0162, \tag{34.89}$$

to be compared with the measured $A_{\mathrm{FB}}^{0,\ell} = 0.0171 \pm 0.0010$. Note the amplification: $A_{\mathrm{FB}}^{0,\ell} \propto A_e^2 \propto (g_V^\ell)^2$, and $g_V^\ell = 2(\tfrac{1}{4} - \sin^2\theta_{\mathrm{eff}}^\ell)$ passes through zero near $\sin^2\theta_{\mathrm{eff}}^\ell = \tfrac{1}{4}$.

A small change in the angle moves g_V^ℓ linearly and the asymmetry quadratically through a steep, near-zero region, so

$$\frac{dA_{\rm FB}^{0,\ell}}{d\sin^2\theta_{\rm eff}^\ell} \approx -1.8, \tag{34.90}$$

meaning a measurement of $A_{\rm FB}^{0,\ell}$ to ± 0.001 pins $\sin^2\theta_{\rm eff}^\ell$ to ± 0.0006. Combining all LEP and SLC asymmetry data gives the headline number

$$\sin^2\theta_{\rm eff}^\ell = 0.23153 \pm 0.00016, \tag{34.91}$$

the effective weak mixing angle to four significant figures. It is this number, together with M_Z and Γ_Z, that the next section turns into a constraint on m_t and m_h.

34.2.5 The QCD correction to the hadronic width

One ingredient in the total width is not pure electroweak: the hadronic width Eq. (34.61) carries a QCD correction, the same final-state gluon radiation that produces the α_s correction to the R-ratio of Theorem 28.5. The quark-level process $Z \to q\bar{q}$ receives corrections from real gluon emission $Z \to q\bar{q}g$ and from virtual gluon exchange in the $q\bar{q}$ vertex; their sum is infrared finite, and the inclusive rate into hadrons is the tree rate times a universal factor. The argument is identical in structure to the one that dressed $\sigma(e^+e^- \to \text{hadrons})$ in Chapter 28: the hadronic final state is summed inclusively, the collinear and soft divergences cancel between real and virtual contributions by the Kinoshita–Lee–Nauenberg mechanism, and what remains is the perturbative series in $\alpha_s(M_Z)$. The leading correction is

$$\Gamma(Z \to q\bar{q}) = \Gamma^{(0)}(Z \to q\bar{q}) \left(1 + \frac{\alpha_s(M_Z)}{\pi} + O(\alpha_s^2)\right), \tag{34.92}$$

where $\Gamma^{(0)}$ is the electroweak tree width Eq. (34.59). The coefficient of α_s/π is exactly unity, the same $C_F \cdot \tfrac{3}{4}/\ldots$ combination — in fact the identical number — that multiplies the parton-model R-ratio in Eq. (28.31), because both observables are the same inclusive $q\bar{q}(+g)$ rate normalized differently. This is the one place the hadronic Z width consumes Yang–Mills theory: the strong coupling at the Z scale,

$$\alpha_s(M_Z) = 0.118, \qquad \frac{\alpha_s(M_Z)}{\pi} = 0.0376, \tag{34.93}$$

is the endpoint of the running computed from the asymptotic-freedom beta function of Chapter 27. The factor $(1 + 0.0376) = 1.038$ used in Eq. (34.61) raises the tree hadronic width by 3.8%, from 1674 MeV to 1737 MeV, and is essential to the half-percent agreement of Eq. (34.62): without it the predicted Γ_Z would fall short by ~ 60 MeV, twenty-five times the experimental error.

Physical Insight

The hadronic Z width is, conversely, one of the cleanest *measurements* of $\alpha_s(M_Z)$. Inverting Eq. (34.92), the ratio $R_\ell \equiv \Gamma_{\rm had}/\Gamma_\ell$ depends on α_s through the factor $(1+\alpha_s/\pi+\dots)$ and on little else, since the electroweak couplings cancel substantially in the ratio. The value extracted from the Z lineshape, $\alpha_s(M_Z) = 0.1196 \pm 0.0030$, agrees with the determinations from τ decay, deep inelastic scattering, and jet rates — a consistency that tests the running of α_s across two orders of magnitude in energy and is one of the quantitative pillars of QCD. The Z pole thus tests Yang–Mills theory and the electroweak sector in the same measurement.

The Z-pole observables are now complete: from the neutral-current couplings alone we have derived the partial and total widths, the resonant lineshape and its peak, the neutrino count, the asymmetries, and the QCD-corrected hadronic rate, and every one agrees with experiment to the per-mille level the tree-plus-leading-correction treatment can reach. The residual sub-percent discrepancies — the 60 MeV in Γ_Z, the shift between on-shell and effective mixing angles, the precise value of M_Z — are not noise but signal: they are the imprint of the one-loop electroweak self-energies, dominated by the heavy top quark and the Higgs. The set of high-precision pseudo-observables assembled here ($\sin^2\theta^\ell_{\rm eff}$, Γ_Z, the partial widths, and — through the tree relation to M_Z — M_W) is exactly the data that the next section confronts with those radiative corrections, packaging them into the oblique parameters S, T, U and showing how the fit cornered m_t and m_h before either was seen directly.

34.3 Oblique Corrections and the S, T, U Parameters

The Z-pole analysis of Section 34.2 delivered a suite of pseudo-observables — the partial and total widths, the leptonic and hadronic asymmetries, the effective mixing angle $\sin^2\theta^{\rm eff}_{\rm W}$ — each expressed through the neutral-current couplings g_V^f, g_A^f of Eq. (31.141). Those expressions were tree-level: every coupling was a fixed function of the single angle $\theta_{\rm W}$ and the fermion quantum numbers. Yet LEP measured the same observables to the per-mille level, an order of magnitude finer than the few-percent gap by which the tree-level masses Eq. (31.151) missed the data. At that precision the predictions are no longer the tree couplings but the tree couplings *plus their one-loop electroweak radiative corrections*, and the data test the loops. The remarkable empirical fact — the organizing fact of this section — is that the dominant corrections, the ones that move the predictions by more than the experimental errors, are not a sprawling set of process-dependent diagrams but a small, universal package: the radiative dressing of the gauge-boson propagators themselves. Physics far heavier than M_Z, too heavy to be

produced, nonetheless leaves its fingerprint on every Z-pole observable at once, through the way it modifies the propagation of the W, the Z, the photon, and the Z–γ mixing. Packaging that fingerprint into three numbers — the Peskin–Takeuchi parameters S, T, U — is the business of this section. We will derive the three parameters from the gauge self-energies, prove that they are exactly the combinations that survive renormalization, connect T to the $\Delta\rho$ and custodial symmetry of Section 31.6, compute the leading top-quark contribution in full, and watch the electroweak fit corner the masses of the top quark and the Higgs boson before either was produced.

34.3.1 The oblique approximation

A generic radiative correction to a four-fermion process $f\bar{f} \to f'\bar{f}'$ at the Z pole splits into three topological classes. *Vertex* corrections dress the coupling of the gauge boson to a particular fermion line; they depend on the external flavour. *Box* diagrams exchange two gauge bosons between the fermion lines; they depend on the full kinematics. *Self-energy* (or *propagator*) corrections dress the internal gauge-boson line and are blind to which fermions sit at the ends. The decisive observation is that new physics living at a scale $\Lambda \gg M_Z$, and coupling to the light fermions only through the $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge bosons — which is the generic situation for a heavy sector that respects the gauge symmetry and carries no large direct couplings to the first two generations — contributes overwhelmingly through the self-energy class. Vertex and box graphs that involve the heavy states are suppressed by extra powers of the light gauge couplings and, crucially, by the heavy propagators attached directly to the external legs; the self-energy graphs, by contrast, can have the heavy states circulating in a loop on an *internal* line whose effect is felt universally by every external channel.

Definition 34.12 (Oblique corrections)**.** A radiative correction to electroweak observables is called *oblique* if it enters only through the vacuum polarizations (transverse self-energies) of the gauge bosons $W^{\pm}$, Z, γ, and through the Z–γ mixing self-energy — that is, through the two-point functions of the gauge fields, with no dependence on the external fermion flavours beyond their tree-level gauge charges. The approximation of keeping only oblique corrections from a heavy new sector is the *oblique approximation*.

The name (a coinage of Kennedy and Lynn, with the parametrization sharpened by Peskin and Takeuchi [397, 398]) captures the indirectness: the heavy sector never touches the light fermions head-on; it acts "obliquely," through the mediating gauge fields. This is the same physics that Chapter 20 exploited for the running of α — there the heavy (or simply virtual) charged states dressed the photon propagator and shifted every QED observable by a universal factor — now generalized from the single photon self-energy of Section 19.2 to the full 2×2

neutral self-energy matrix plus the charged W self-energy. We continue QED's vacuum-polarization technology, lifted from U(1) to the electroweak gauge group.

Physical Insight

Why should the dominant new-physics effect be universal? Because a heavy particle that respects the gauge symmetry can only learn about the light fermions by exchanging gauge bosons with them, and a gauge boson couples to a conserved current whose normalization is fixed. The heavy sector therefore cannot distinguish an electron from a muon except through their gauge charges; its leading effect must be a charge-universal renormalization of how the gauge bosons propagate. The oblique parameters are the coordinates on the space of such universal effects.

34.3.2 The four gauge self-energies and their low-q^2 expansion

By Lorentz invariance and the transversality forced by the Ward identities of Eq. (19.4) (current conservation holds for the electromagnetic and, on the broken vacuum, for the appropriately defined weak currents), each gauge-boson self-energy is transverse up to terms proportional to $q^\mu q^\nu$ that do not contribute to amplitudes between conserved currents. We therefore record only the coefficient of $g^{\mu\nu}$, writing the one-particle-irreducible two-point function of two gauge fields V, V' as

$$i\,\Pi^{\mu\nu}_{VV'}(q) = i\,g^{\mu\nu}\,\Pi_{VV'}(q^2) + (q^\mu q^\nu \text{ terms}), \qquad V, V' \in \{W, Z, \gamma\}, \tag{34.94}$$

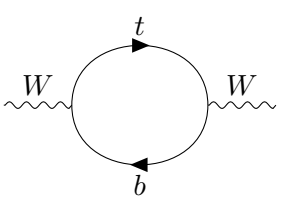
so that the full transverse inverse propagator of a gauge boson of tree mass M_V reads $q^2 - M_V^2 - \Pi_{VV}(q^2)$, with the convention that the physical pole mass is $M_V^2 + \Pi_{VV}(M_V^2)$ to one loop. There are exactly four independent functions: the charged self-energy Π_{WW}, the two neutral diagonal self-energies Π_{ZZ} and $\Pi_{\gamma\gamma}$, and the off-diagonal Z–γ mixing $\Pi_{Z\gamma}$. They are drawn in Fig. 34.3.

Because the new physics is heavy, $q^2 \sim M_Z^2 \ll \Lambda^2$, each self-energy is an analytic function of q^2 near the origin and may be Taylor-expanded,

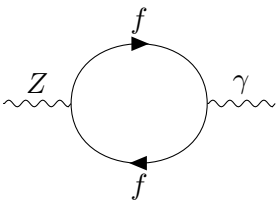
$$\Pi_{VV'}(q^2) = \Pi_{VV'}(0) + q^2\,\Pi'_{VV'}(0) + \tfrac{1}{2}(q^2)^2\,\Pi''_{VV'}(0) + \cdots, \tag{34.95}$$

where $\Pi' \equiv d\Pi/dq^2$. The quadratic and higher terms are suppressed by additional powers of q^2/Λ^2 and, in the heavy-physics limit, negligible: the leading effect of a heavy sector is captured by the eight numbers $\{\Pi_{VV'}(0), \Pi'_{VV'}(0)\}$ for the four channels. Two of these eight are removed at once by unbroken electromagnetic gauge invariance. The photon must stay massless and its coupling must be conserved, which forces the pure photon and the Z–γ mixing self-energies to vanish at zero momentum:

$$\Pi_{\gamma\gamma}(0) = 0, \qquad \Pi_{Z\gamma}(0) = 0. \tag{34.96}$$



(a) Π_{WW}: the (t, b) loop.



(b) $\Pi_{Z\gamma}$: a Z–γ mixing loop.

Figure 34.3: Two of the four oblique self-energies. (a) The charged self-energy Π_{WW} with a top–bottom doublet circulating; this is the diagram computed in full in Section 34.3.5. (b) The neutral mixing $\Pi_{Z\gamma}$, a fermion loop with one Z and one γ external leg, which enters the slope parameter S. The diagonal Π_{ZZ} and $\Pi_{\gamma\gamma}$ have the same topology with both external legs identical. The blobs are one-particle irreducible.

The first is the statement, proved for QED in Chapter 19 and unchanged by the electroweak embedding, that radiative corrections generate no photon mass; the second is its mixing analogue, that no A_μ–Z_μ mass mixing is induced, so the photon remains a mass eigenstate. We are left with six numbers:

$$\Pi_{WW}(0),\ \Pi_{ZZ}(0),\ \Pi'_{WW}(0),\ \Pi'_{ZZ}(0),\ \Pi'_{\gamma\gamma}(0),\ \Pi'_{Z\gamma}(0). \tag{34.97}$$

34.3.3 Counting: exactly three physical parameters

Not all six numbers in Eq. (34.97) are observable. The electroweak gauge sector has three fundamental parameters — the two gauge couplings g, g' and the Higgs vacuum value v, equivalently the trio (α, G_F, M_Z) of measured inputs from Eq. (31.145) and Eq. (31.150). Fixing the theory to data means using three measurements to absorb three combinations of the bare parameters and the self-energies; those three combinations are, by construction, *not predictions* but definitions of the inputs. They are unobservable in the same sense that the bare electron charge is unobservable in Chapter 20: they are swallowed by renormalization. Of the six numbers in Eq. (34.97), therefore, exactly three survive as genuine, calculable predictions of any heavy sector. This is the counting that fixes the number of oblique parameters:

$$\underbrace{8}_{\text{Taylor coeffs}} - \underbrace{2}_{\text{EM gauge inv.}} - \underbrace{3}_{\text{absorbed in } (\alpha, G_F, M_Z)} = \underbrace{3}_{S, T, U}. \tag{34.98}$$

To make the three survivors explicit it is cleanest to pass to the weak-isospin basis. The physical gauge bosons couple to currents built from the isospin currents J^a_μ $(a = 1, 2, 3)$ and the electromagnetic current $J^Q_\mu = J^3_\mu + J^Y_\mu$: the $W^\pm$ couple to $J^{1,2}$, the photon to eJ^Q, and the Z to $(g/c)(J^3 - s^2 J^Q)$, where

throughout this section we abbreviate

$$s^2 \equiv \sin^2\theta_{\mathrm{W}}, \qquad c^2 \equiv \cos^2\theta_{\mathrm{W}}, \qquad e = g\,s = g'\,c. \tag{34.99}$$

Writing $\Pi_{XY}(q^2)$ for the $g^{\mu\nu}$ coefficient of the correlator of currents $X, Y \in \{1, 3, Q\}$, the physical self-energies are the linear combinations

$$\begin{aligned} \Pi_{WW} &= g^2\,\Pi_{11}, & \Pi_{\gamma\gamma} &= e^2\,\Pi_{QQ}, \\ \Pi_{ZZ} &= \frac{g^2}{c^2}\bigl(\Pi_{33} - 2s^2\Pi_{3Q} + s^4\Pi_{QQ}\bigr), & \Pi_{Z\gamma} &= \frac{ge}{c}\bigl(\Pi_{3Q} - s^2\Pi_{QQ}\bigr). \end{aligned} \tag{34.100}$$

Electromagnetic gauge invariance is the statement that the electromagnetic current is exactly conserved, so its correlators vanish at zero momentum, $\Pi_{QQ}(0) = \Pi_{3Q}(0) = 0$, which reproduces Eq. (34.96) through Eq. (34.100). The three combinations that are simultaneously ultraviolet-finite, gauge-independent, and not absorbable into (α, G_F, M_Z) are the Peskin–Takeuchi parameters.

Definition 34.13 (The oblique parameters S, T, U)**.** With $\Pi' \equiv d\Pi/dq^2$ evaluated at $q^2 = 0$,

$$\alpha\,T \equiv \frac{\Pi_{WW}(0)}{M_W^2} - \frac{\Pi_{ZZ}(0)}{M_Z^2} = \frac{e^2}{s^2c^2M_Z^2}\bigl[\Pi_{11}(0) - \Pi_{33}(0)\bigr], \tag{34.101}$$

$$\alpha\,S \equiv 4e^2\bigl[\Pi'_{33}(0) - \Pi'_{3Q}(0)\bigr], \tag{34.102}$$

$$\alpha\,U \equiv 4e^2\bigl[\Pi'_{11}(0) - \Pi'_{33}(0)\bigr], \tag{34.103}$$

where $\alpha = e^2/4\pi$. Equivalently, $S+U = (4e^2/\alpha)[\Pi'_{11}(0)-\Pi'_{3Q}(0)] = 16\pi[\Pi'_{11}(0)-\Pi'_{3Q}(0)]$.

The second equality in Eq. (34.101) follows by substituting Eq. (34.100) at $q^2 = 0$, using $\Pi_{3Q}(0) = \Pi_{QQ}(0) = 0$ and the tree relation $M_W^2 = c^2M_Z^2$: then $\Pi_{WW}(0)/M_W^2 = g^2\Pi_{11}(0)/M_W^2$ and $\Pi_{ZZ}(0)/M_Z^2 = (g^2/c^2)\Pi_{33}(0)/M_Z^2 = g^2\Pi_{33}(0)/M_W^2$, whose difference is $g^2[\Pi_{11}(0)-\Pi_{33}(0)]/M_W^2 = (e^2/s^2c^2M_Z^2)[\Pi_{11}(0)-\Pi_{33}(0)]$. The three definitions exhaust the surviving numbers: T uses the two zero-momentum values, while S and U use three independent slopes (the fourth slope, $\Pi'_{\gamma\gamma}(0)$, renormalizes α itself and is the running of Chapter 20, hence absorbed). This realizes the counting Eq. (34.98).

Why these combinations are finite and scheme-independent. A heavy sector contributes ultraviolet divergences to the individual self-energies, but those divergences are flavour- and isospin-blind in a very specific way: a divergence in $\Pi_{VV'}$ at one loop is a local polynomial in q^2 whose coefficients respect the unbroken symmetries. The T combination $\Pi_{11}(0) - \Pi_{33}(0)$ is a difference of two zero-momentum self-energies that are *equal whenever the weak isospin is an exact symmetry*, because then the J^1 and J^3 correlators are related by an isospin

rotation. Any divergence respects isospin (it comes from the deep ultraviolet, where the electroweak breaking v is negligible), so the divergence cancels in the difference and T is finite. The same logic applies to S and U: each is a difference of slopes that would coincide in an unbroken or custodially-symmetric limit, so the divergent (symmetric) parts cancel, leaving a finite remainder set by the symmetry-breaking scales. Finiteness is the technical face of scheme-independence: a quantity whose divergence cancels has no leftover $1/\bar{\varepsilon}$ pole to be traded between renormalization schemes, and the value of S, T, U is therefore the same in $\overline{\text{MS}}$, on-shell, or any other scheme. We verify this cancellation explicitly for T in Section 34.3.5, where the $1/\bar{\varepsilon}$ poles of the top and bottom loops cancel on the nose.

Caution

S, T, U are defined as the contributions of *new* physics relative to a fixed reference Standard Model, including a specified reference m_t and m_h. They are not absolute self-energies but deviations: $S = T = U = 0$ for the reference SM by definition. When we speak below of "the top contribution to T" or "the Higgs contribution to S," we mean the change in these parameters as m_t or m_h is varied away from the reference, which is precisely the quantity the fit constrains.

34.3.4 $\alpha T = \Delta\rho$ and custodial symmetry

The first definition Eq. (34.101) is not a new object: it is exactly the radiative correction to the ρ parameter of Eq. (31.153). Recall that $\rho = M_W^2/(M_Z^2 \cos^2\theta_{\text{W}})$ measures the relative strength of the neutral and charged currents, and equals unity at tree level by custodial symmetry (Theorem 31.23). At one loop the physical masses are shifted by the self-energies, $M_V^2 \to M_V^2 + \Pi_{VV}(M_V^2)$, but the dominant shift to the *ratio* comes from the zero-momentum pieces, which is what $\Delta\rho \equiv \rho - 1$ collects. Writing the corrected ratio and expanding,

$$\rho = \frac{M_W^2 + \Pi_{WW}(0)}{[M_Z^2 + \Pi_{ZZ}(0)]\cos^2\theta_{\text{W}}} = 1 + \frac{\Pi_{WW}(0)}{M_W^2} - \frac{\Pi_{ZZ}(0)}{M_Z^2} + \cdots , \tag{34.104}$$

where we used $M_W^2 = M_Z^2 \cos^2\theta_{\text{W}}$ at tree level and kept the leading correction. Comparing with Eq. (34.101),

$$\boxed{\Delta\rho = \alpha\, T = \frac{\Pi_{WW}(0)}{M_W^2} - \frac{\Pi_{ZZ}(0)}{M_Z^2} .} \tag{34.105}$$

The parameter T is $\Delta\rho$ in units of α. Everything established in Section 31.6.5 about $\Delta\rho$ — that it vanishes when custodial symmetry is exact, that it is the sensitive probe of isospin-splitting heavy multiplets, that its parametric form is Eq. (31.184) — transfers verbatim to T.

The custodial statement deserves to be made sharp in the present language. From the isospin-basis form Eq. (34.101), $\alpha T \propto \Pi_{11}(0) - \Pi_{33}(0)$.

Proposition 34.14 (T vanishes when custodial $\mathrm{SU}(2)_V$ is exact)**.** *If the heavy sector respects the custodial symmetry* $\mathrm{SU}(2)_V$ *of Section 31.6 — equivalently, if it couples to the weak gauge bosons in a way invariant under the diagonal* $\mathrm{SU}(2)$ *that rotates* J^1, J^2, J^3 *into one another — then* $\Pi_{11}(q^2) = \Pi_{33}(q^2)$ *identically, and hence* $T = 0$*. A nonzero* T *is a direct measure of custodial-symmetry violation in the heavy sector.*

Proof. Custodial $\mathrm{SU}(2)_V$ acts on the three currents $J^1_\mu, J^2_\mu, J^3_\mu$ in its adjoint (vector, spin-1) representation, exactly as it acts on the three weak gauge bosons W^1, W^2, W^3 in Theorem 31.23. The self-energy $\Pi_{ab}(q^2)$ $(a, b \in \{1, 2, 3\})$ is the two-point correlator of these currents; if the dynamics generating it is $\mathrm{SU}(2)_V$-invariant, then Π_{ab} is an invariant tensor of the adjoint representation, so by Schur's lemma it is proportional to the only invariant symmetric tensor, the identity: $\Pi_{ab}(q^2) = \Pi(q^2)\,\delta_{ab}$. In particular the diagonal entries are all equal, $\Pi_{11} = \Pi_{22} = \Pi_{33}$, so $\Pi_{11}(0) - \Pi_{33}(0) = 0$ and $T = 0$ by Eq. (34.101). The two sources that break $\mathrm{SU}(2)_V$ in the Standard Model were identified in Section 31.6 as the hypercharge gauging ($g' \neq 0$) and the up–down Yukawa splitting ($m_u \neq m_d$); only these can make $\Pi_{11} \neq \Pi_{33}$, and they enter T weighted by their custodial-breaking strength, dominantly the top–bottom mass difference. □

This is the structural reason T is the most powerful oblique probe of new physics that splits an electroweak multiplet: it is a null test of custodial symmetry, zero in the symmetric limit and quadratically sensitive (as we now compute) to the splitting.

34.3.5 The leading top-quark contribution: full computation

We now compute $\Delta\rho = \alpha T$ from the third-generation quark doublet (t, b) running in the W and Z self-energies, supplying the exact coefficient promised in Eq. (31.184). The computation is the electroweak descendant of the photon vacuum polarization of Section 19.2, with two new features: the gauge bosons couple chirally, and the two fermions in the W loop have unequal masses. Both features are exactly what makes the result nonzero.

Setup and the chiral trace. The W^+ couples to the left-handed charged current with strength $g/\sqrt{2}$, the vertex being $i(g/\sqrt{2})\gamma^\mu P_L$ with $P_L = \frac{1}{2}(1 - \gamma^5)$ (Eq. (31.129)). The one-loop W self-energy from a top–bottom loop, with $N_c = 3$

colours, is

$$i\Pi^{\mu\nu}_{WW}(q) = -\,N_c \int \frac{\mathrm{d}^d k}{(2\pi)^d}\, \frac{\mathrm{Tr}\big[\,\frac{g}{\sqrt{2}}\gamma^\mu P_L\,(\not{k}+m_t)\,\frac{g}{\sqrt{2}}\gamma^\nu P_L\,(\not{k}-\not{q}+m_b)\,\big]}{(k^2-m_t^2)\big[(k-q)^2-m_b^2\big]}, \tag{34.106}$$

the overall minus sign being the fermion-loop sign of Section 19.2 and the propagator/vertex factors of i combining to the prefactor shown. The chiral projector simplifies the trace decisively. Using $P_L\gamma^\mu = \gamma^\mu P_R$ and $P_L P_R = 0$, the mass terms in the numerator are killed: $P_L(\not{k}+m_t) = \not{k}\,P_R + m_t P_L$, and after commuting the second projector through, the surviving structure carries no factor of m_t and no factor of m_b in the numerator,

$$\mathrm{Tr}\big[\gamma^\mu P_L(\not{k}+m_t)\gamma^\nu P_L(\not{k}-\not{q}+m_b)\big] = \tfrac{1}{2}\,\mathrm{Tr}\big[\gamma^\mu \not{k}\,\gamma^\nu(\not{k}-\not{q})\big] = 2\big[k^\mu(k-q)^\nu + (k-q)^\mu k^\nu - g^{\mu\nu}\,k(k-q$$

(34.107)

where the γ^5 part of the trace is antisymmetric and drops from the symmetric $g^{\mu\nu}$ coefficient, and the standard four-gamma trace of Chapter C was used. *The masses survive only in the denominators.* This is the technical heart of $\Delta\rho$: a purely chiral coupling makes the self-energy depend on the fermion masses only through the propagator poles, and the mass dependence of $\Pi_{WW}(0)$ is therefore controlled entirely by the difference between m_t and m_b.

The $g^{\mu\nu}$ coefficient at zero momentum. We need $\Pi_{WW}(0)$. Set $q = 0$ in Eq. (34.106); the numerator Eq. (34.107) becomes $2[2k^\mu k^\nu - g^{\mu\nu}k^2]$, and under the symmetric replacement $k^\mu k^\nu \to (k^2/d)\,g^{\mu\nu}$ the $g^{\mu\nu}$ coefficient of the integrand is $2k^2(2-d)/d$. Hence

$$\Pi_{WW}(0) = -\,N_c\,\frac{g^2}{2}\cdot\frac{2(2-d)}{d}\int\frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{k^2}{(k^2-m_t^2)(k^2-m_b^2)}. \tag{34.108}$$

The remaining scalar integral is reduced by partial fractions, $\dfrac{k^2}{(k^2-m_t^2)(k^2-m_b^2)} = \dfrac{1}{m_t^2-m_b^2}\Big[\dfrac{m_t^2}{k^2-m_t^2} - \dfrac{m_b^2}{k^2-m_b^2}\Big]$, onto the scalar tadpole of Chapter D,

$$\int\frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{k^2-m^2} = \frac{i}{16\pi^2}\,a(m^2), \qquad a(m^2) \equiv m^2\big(\bar{\Delta} - \ln(m^2/\mu^2)\big), \quad \bar{\Delta} \equiv \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi + 1, \tag{34.109}$$

in $d = 4-2\varepsilon$. Carrying out the algebra,

$$\int\frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{k^2}{(k^2-m_t^2)(k^2-m_b^2)} = \frac{i}{16\pi^2}\,\frac{m_t^2\,a(m_t^2) - m_b^2\,a(m_b^2)}{m_t^2-m_b^2}. \tag{34.110}$$

The prefactor $2(2-d)/d = -1 + \varepsilon/2 + O(\varepsilon^2)$ multiplies a divergent integral, so its $O(\varepsilon)$ piece beats against the $1/\varepsilon$ pole in a; keeping that cross term,

$$\Pi_{WW}(0) = -\,N_c\,\frac{g^2}{2}\,\frac{1}{16\pi^2}\Big[-\bar{\Delta}\,(m_t^2{+}m_b^2){+}G_{tb}{+}\tfrac{1}{2}(m_t^2{+}m_b^2)\Big], \quad G_{tb} \equiv \frac{m_t^4\ln\frac{m_t^2}{\mu^2} - m_b^4\ln\frac{m_b^2}{\mu^2}}{m_t^2 - m_b^2}. \tag{34.111}$$

The Z self-energy at zero momentum. The Z couples diagonally to each fermion with left and right couplings $g_L^f = (g/c)(T_f^3 - s^2 Q_f)$ and $g_R^f = (g/c)(-s^2 Q_f)$ (Eq. (31.135)). For a single fermion of mass m the loop has $m_a = m_b = m$, and now the cross term $g_L^f g_R^f$ survives the trace, because it pairs a left projector with a right projector and the two mass insertions combine into $2m^2 g^{\mu\nu}$. Repeating the trace algebra for a general chiral coupling $\gamma^\mu(g_L P_L + g_R P_R)$ gives, at $q^2 = 0$,

$$\Pi_{ff}(0) = -\,N_c\,\frac{2m_f^2}{16\pi^2}\,(g_L^f - g_R^f)^2\,(\ln(m_f^2/\mu^2) + 1 - \bar{\Delta}), \tag{34.112}$$

where the equal-mass tadpole limit was taken and, crucially, the combination of the $(g_L^2 + g_R^2)$ "massless-numerator" piece with the $2g_L g_R$ mass piece collapsed to $(g_L - g_R)^2$. The lesson is physical: a purely *vector* coupling ($g_L = g_R$) gives $\Pi_{ff}(0) = 0$, because a conserved vector current produces a self-energy that vanishes at zero momentum, exactly as the photon's does; only the *axial* part ($g_L - g_R$) feeds $\Pi(0)$. For the Z, $g_L^f - g_R^f = (g/c)T_f^3$, so $(g_L^f - g_R^f)^2 = (g^2/c^2)(T_f^3)^2 = g^2/4c^2$ for both t and b. Summing the two diagonal loops,

$$\Pi_{ZZ}(0) = -\,N_c\,\frac{1}{16\pi^2}\,\frac{g^2}{2c^2}\Big[m_t^2(\ln(m_t^2/\mu^2) + 1 - \bar{\Delta}) + m_b^2(\ln(m_b^2/\mu^2) + 1 - \bar{\Delta})\Big]. \tag{34.113}$$

Assembling $\Delta\rho$. Insert Eqs. (34.111) and (34.113) into Eq. (34.105), using $M_W^2 = g^2 v^2/4$ and $M_Z^2 = M_W^2/c^2$ so that $1/M_Z^2 = c^2/M_W^2$. Both ratios acquire the common prefactor $-N_c(2/16\pi^2 v^2)$, and the bracketed structures combine. Writing $L_t \equiv \ln(m_t^2/\mu^2)$, $L_b \equiv \ln(m_b^2/\mu^2)$,

$$\begin{aligned}\Delta\rho &= \frac{\Pi_{WW}(0)}{M_W^2} - \frac{\Pi_{ZZ}(0)}{M_Z^2} \\ &= -\frac{N_c}{8\pi^2 v^2}\Big\{\underbrace{\big[-\bar{\Delta}(m_t^2+m_b^2) + G_{tb} + \tfrac{1}{2}(m_t^2+m_b^2)\big]}_{\text{from }\Pi_{WW}} - \underbrace{\big[(m_t^2 L_t + m_b^2 L_b) + (m_t^2+m_b^2) - \bar{\Delta}(m_t^2 -}_{\text{from }\Pi_{ZZ}}\end{aligned} \tag{34.114}$$

The two terms $-\bar{\Delta}(m_t^2 + m_b^2)$ cancel: *the ultraviolet divergence disappears*, confirming the finiteness argued on general grounds in Section 34.3.3. What remains

is

$$\Delta\rho = -\frac{N_c}{8\pi^2 v^2}\Big[\, G_{tb} - (m_t^2 L_t + m_b^2 L_b) - \tfrac{1}{2}(m_t^2 + m_b^2)\,\Big]. \tag{34.115}$$

The logarithmic terms combine cleanly. Using the definition of G_{tb} in Eq. (34.111),

$$G_{tb} - (m_t^2 L_t + m_b^2 L_b) = \frac{m_t^4 L_t - m_b^4 L_b - (m_t^2 L_t + m_b^2 L_b)(m_t^2 - m_b^2)}{m_t^2 - m_b^2} = \frac{m_t^2 m_b^2 (L_t - L_b)}{m_t^2 - m_b^2} = \frac{m}{m_t^2} \tag{34.116}$$

the renormalization scale μ cancelling between L_t and L_b as it must in a finite quantity. Substituting Eq. (34.116) into Eq. (34.115) and flipping the overall sign,

$$\boxed{\Delta\rho = \alpha T = \frac{N_c}{16\pi^2 v^2}\, F(m_t^2, m_b^2), \qquad F(x,y) \equiv x + y - \frac{2xy}{x-y}\,\ln\frac{x}{y}.} \tag{34.117}$$

The function F is symmetric, nonnegative, and has exactly the limits demanded by custodial symmetry: $F(m^2, m^2) = 0$, so $\Delta\rho$ vanishes when the doublet is degenerate (the log expansion gives $F \to 2m^2 - 2m^2 = 0$), confirming Theorem 34.14; and $F(m_t^2, 0) = m_t^2$, so for the realistic hierarchy $m_b \ll m_t$,

$$\Delta\rho \;\simeq\; \frac{N_c\, m_t^2}{16\pi^2 v^2} \;=\; \frac{3\, G_F\, m_t^2}{8\sqrt{2}\,\pi^2}, \tag{34.118}$$

where the second form uses $G_F = 1/(\sqrt{2}\, v^2)$ from Eq. (31.145). This is the exact coefficient anticipated in Eq. (31.184): the leading top contribution to $\Delta\rho$, and hence to T, grows as m_t^2 and does not decouple. A heavy top is a strongly Yukawa-coupled top, $m_t^2/v^2 = y_t^2/2$, and a strong coupling leaves a finite imprint no matter how heavy the particle.

Worked Example 34.15 (The size of the top contribution to T)**.** We evaluate Eq. (34.118) numerically and convert to T. With $N_c = 3$, $m_t = 173\,\text{GeV}$, $v = 246\,\text{GeV}$,

$$\Delta\rho = \frac{3\,(173)^2}{16\pi^2 (246)^2} = \frac{3 \times 2.99 \times 10^4}{1.58 \times 10^2 \times 6.05 \times 10^4} \approx 9.4 \times 10^{-3}, \tag{34.119}$$

the one-percent effect already estimated in Theorem 31.25, now with the bottom-quark correction under control: restoring $m_b = 4.2\,\text{GeV}$ through the full F in Eq. (34.117) changes this by $F(m_t^2, m_b^2)/m_t^2 - 1 \approx -2(m_b^2/m_t^2)\ln(m_t^2/m_b^2) \approx -4\times10^{-3}$, a four-per-mille downward shift, utterly negligible against the precision of the estimate. To convert to the Peskin–Takeuchi T we divide by α evaluated at the relevant scale, $\alpha(M_Z) \approx 1/128.9 \approx 7.76 \times 10^{-3}$ (the running value of Chapter 20, since the self-energies are renormalized at M_Z):

$$T_{\text{top}} = \frac{\Delta\rho}{\alpha(M_Z)} \approx \frac{9.4 \times 10^{-3}}{7.76 \times 10^{-3}} \approx 1.2. \tag{34.120}$$

If the top quark were a new, undiscovered particle, it would register as a shift $T \approx 1.2$ — enormous on the scale of the experimental error $\sigma_T \sim 0.1$ of the LEP era. This is why ρ, measured a percent above unity, was an unambiguous announcement that a heavy isospin-split state existed.

34.3.6 The logarithmic Higgs contribution and the screening theorem

The Higgs boson contributes to the gauge self-energies through loops in which the W, Z, and Higgs propagate, and its mass m_h enters S and T. Unlike the top quark, the Higgs is custodially symmetric in its couplings to the gauge bosons — it is the radial mode of the very field whose vacuum value defines custodial $\mathrm{SU}(2)_V$ in Section 31.6 — so by Theorem 34.14 its leading effect on T cannot be power-law in m_h. The custodial symmetry protects T from quadratic Higgs sensitivity, leaving only a logarithm. This is *Veltman's screening theorem* [485]: electroweak observables depend on the Higgs mass only logarithmically, because the would-be quadratic dependence is forbidden by the custodial symmetry of the scalar sector. The explicit one-loop results, computed in the same self-energy framework, are

$$T_{\mathrm{Higgs}} \simeq -\frac{3}{8\pi c^2}\,\ln\frac{m_h}{m_{h,\mathrm{ref}}}, \qquad S_{\mathrm{Higgs}} \simeq \frac{1}{6\pi}\,\ln\frac{m_h}{m_{h,\mathrm{ref}}}, \tag{34.121}$$

to leading logarithmic accuracy, measured relative to a reference Higgs mass $m_{h,\mathrm{ref}}$. Two features matter for the fit. First, the logarithm is mild: moving m_h by a factor of ten shifts T by only $\sim -\frac{3}{8\pi c^2}\ln 10 \approx -0.35$ and S by $\sim +0.1$, so the Higgs mass is constrained but *weakly* — a heavy Higgs is not excluded by orders of magnitude, only disfavoured logarithmically. Second, the signs are opposite and correlated: increasing m_h lowers T and raises S, tracing the down-and-to-the-right line in the (S,T) plane drawn in Fig. 34.4. The top quark, by contrast, moves the prediction straight up in T at essentially fixed S (Eq. (34.118)). Because the two particles push the prediction in different directions, a measurement of the (S,T) point determines a combination of m_t and m_h, and adding the direct top-mass information from kinematics breaks the degeneracy to corner m_h.

Physical Insight

Screening is the same custodial protection that stabilized $\rho = 1$ in Theorem 31.23, now read as a statement about ultraviolet sensitivity. A quadratic dependence $T \sim m_h^2/v^2$ would mean the low-energy observables remember the Higgs mass as a hard scale; custodial symmetry forbids this, so the Higgs "screens" itself, hiding all but a logarithm of its mass from the precision data. The flip side is the hierarchy problem of Chapter 35:

the Higgs mass-squared itself is *not* protected by any symmetry and is quadratically sensitive to new heavy scales — screening protects the gauge observables from m_h, not m_h from the ultraviolet.

34.3.7 How S, T, U enter the observables

The oblique parameters earn their keep by entering every precision observable linearly, with calculable coefficients. The logic is that of Section 34.3.3: hold the three inputs (α, G_F, M_Z) fixed at their measured values, then recompute each prediction including the self-energy corrections. The shifts are organized by S, T, U.

The cleanest case is the W mass, predicted at tree level by Eq. (31.149). Writing the tree value as $(M_W^2)_0$ and tracking the self-energy corrections to the relation $M_W^2(1-M_W^2/M_Z^2) = \pi\alpha/\sqrt{2}G_F = A_0^2$ of Eq. (31.150), the correction from $\Delta\rho = \alpha T$ alone is immediate: $\rho \neq 1$ shifts $M_W^2 = \rho\, M_Z^2 c^2$, so $\Delta M_W^2/M_W^2|_T = \alpha T \cdot c^2/(c^2-s^2)$, the factor $c^2/(c^2-s^2)$ arising because raising M_W at fixed A_0^2 also lowers s^2, feeding back. The slope parameter S adds the contribution from the running of the Z self-energy between $q^2 = 0$ and $q^2 = M_Z^2$, and U the analogous W–Z slope difference. Assembling all three by the same propagator-correction procedure gives the standard linearized predictions [398],

$$\frac{\Delta M_W^2}{M_W^2} = \frac{\alpha}{c^2 - s^2}\Big(-\tfrac{1}{2}S + c^2 T + \tfrac{c^2-s^2}{4s^2}\,U\Big), \tag{34.122}$$

$$\Delta \sin^2\theta_{\rm W}^{\rm eff} = \frac{\alpha}{c^2 - s^2}\Big(\tfrac{1}{4}S - s^2 c^2\, T\Big), \tag{34.123}$$

$$\frac{\Delta\Gamma_\ell}{\Gamma_\ell} = \alpha T + (S\text{-dependent terms through } \Delta \sin^2\theta_{\rm W}^{\rm eff}), \tag{34.124}$$

where Γ_ℓ is the leptonic Z partial width of Section 34.2, whose leading shift is the overall $\rho = 1 + \alpha T$ normalization of the neutral-current couplings, with S entering subdominantly through the shift Eq. (34.123) of the effective mixing angle inside g_V^ℓ. The structure of Eqs. (34.122) and (34.123) is the substance: the W mass rises with T (a heavier top pushes M_W up, as the data require) and falls with S; the effective mixing angle rises with S and falls with T. The widths and the leptonic partial widths of Section 34.2 are corrected dominantly through T, because $\Delta\rho = \alpha T$ multiplies the overall normalization of the neutral-current couplings g_V^f, g_A^f that set every Z partial width, while S enters subdominantly through $\sin^2\theta_{\rm W}^{\rm eff}$. The upshot is a single over-constrained linear system: a handful of measured pseudo-observables, each a known linear function of the two dominant parameters S and T, fitted simultaneously. With more observables than parameters, the fit both *determines* S and T and *tests* their mutual consistency — and the test is passed at the per-mille level.

Caution

The coefficients in Eqs. (34.122) and (34.123) are the leading linear terms; the full electroweak fit keeps the reference-SM loops exactly and treats S, T, U as the deviations on top. We have derived the T (i.e. $\Delta\rho$) contributions from first principles in Section 34.3.5 and quoted the S, U coefficients, which follow from the identical propagator-correction bookkeeping applied to the slopes $\Pi'(0)$ rather than the values $\Pi(0)$. The placeholder in Eq. (34.124) reflects that the Γ_Z coefficient depends on the sum over final states and the chosen scheme; what is universal and we have established is that its leading dependence is on T through the ρ normalization.

34.3.8 Cornering m_t and m_h before discovery

The payoff of the whole construction is historical and decisive. By the early 1990s LEP had measured the Z-pole pseudo-observables to a precision that fixed S and T to a small region of the plane in Fig. 34.4. The top quark and the Higgs boson were both undiscovered. But each contributes to (S, T) in a calculable way: the top through $T \propto m_t^2$ (Eq. (34.118)), the Higgs through the logarithmic line Eq. (34.121). The fit therefore predicted both masses indirectly, as the values required to bring the calculated (S, T) point onto the measured ellipse.

Worked Example 34.16 (The pre-discovery prediction of m_t)**.** The mechanism is quantitative. Because $\Delta\rho = \alpha T$ enters M_W and the Z-pole observables through Eqs. (34.122) and (34.123), and because the top contribution Eq. (34.118) dominates T, the precisely measured deviation of the observables from their m_t-independent tree values reads directly as a measurement of m_t^2. Quantitatively, an observed $\Delta\rho \approx (9 \pm 3) \times 10^{-3}$ inverts through Eq. (34.118) to

$$m_t = \Big(\frac{8\sqrt{2}\,\pi^2}{3\,G_F}\,\Delta\rho\Big)^{1/2} \approx \Big(\frac{8\sqrt{2}\,\pi^2}{3 \times 1.166 \times 10^{-5}} \times 9 \times 10^{-3}\Big)^{1/2}\,\text{GeV} \approx 170\,\text{GeV}, \tag{34.125}$$

the square-root sensitivity meaning that even a percent-level measurement of $\Delta\rho$ pins m_t to within $\sim 10\%$. This was the LEP/SLD electroweak prediction of the mid-1990s: a top quark near 170 GeV, deduced from Z-pole loops, *before* the Tevatron produced it directly at $m_t = 174\,\text{GeV}$ in 1995. The agreement of a number extracted from quantum corrections with one later measured by direct production is among the sharpest confirmations that the Standard Model is a genuine quantum field theory whose loop effects are real and calculable.

With m_t fixed by direct measurement, the residual dependence of the fit on the remaining unknown — the logarithmic Higgs terms Eq. (34.121) — isolated m_h. The same precision data, now over-constrained by the measured top mass,

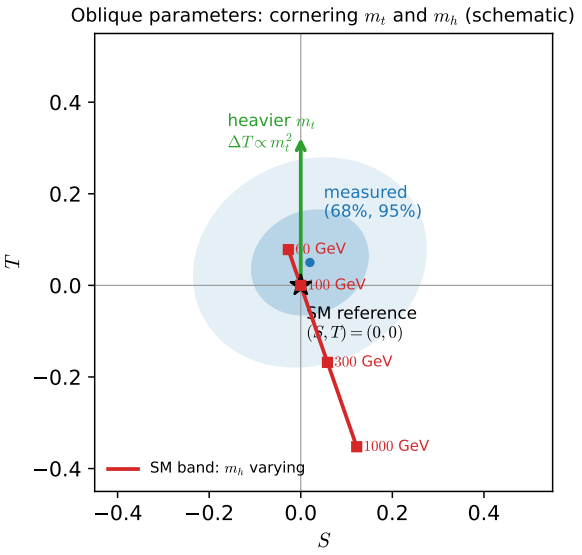


Figure 34.4: Schematic (S, T) plane illustrating how the electroweak fit corners the undiscovered particles. The reference Standard Model sits at the origin by the definition of Theorem 34.13. A heavier top moves the prediction straight up, $\Delta T \propto m_t^2$ (Eq. (34.118)), with no shift in S; a heavier Higgs moves it down-and-to-the-right along the logarithmic line Eq. (34.121) (Veltman screening). The measured confidence region (blue) selects the (m_t, m_h) combination whose predicted point lies inside it. Adding the direct top mass breaks the m_t–m_h degeneracy and bounds m_h from above. The ellipse is illustrative, not a fit to data.

preferred a *light* Higgs: the global electroweak fit of the 2000s returned a best-fit m_h of order 90 GeV with a one-sided 95% upper bound near 160 GeV, squarely in the range a renormalizable theory with a weakly coupled scalar requires. When the LHC discovered the Higgs at $m_h = 125$ GeV in 2012, it landed inside the indirectly predicted window. Two particles, the top quark and the Higgs boson, were thus cornered by loop corrections to Z-pole data before either was produced — the top by its quadratic imprint on T, the Higgs by its logarithmic imprint on S and T — and both direct discoveries confirmed the indirect prediction. This is the precision electroweak programme in one sentence: the quantum structure of the gauge sector is so rigid that the masses of its heaviest constituents are encoded, before discovery, in the radiative corrections to observables at a lower scale.

With the gauge sector now pinned to the per-mille level and consistent with a single light Higgs of the mass the fit demanded, the last undiscovered Standard Model particle is that Higgs itself. The next section turns from the indirect

cornering of m_h to its direct phenomenology: how the Higgs is produced and how it decays, with the calculable coupling-versus-mass pattern that is the sharpest signature the Standard Model possesses.

34.4 The Higgs Boson: Production and Decay

The electroweak fit of the previous section did more than confirm the gauge structure: through the oblique parameters it cornered the one quantity the tree Lagrangian leaves open, the Higgs mass. The logarithmic sensitivity of S and T to m_h (Section 34.3) preferred a *light* scalar, a few times M_Z, well before any Higgs was produced; the value the LHC later read off directly, $m_h \simeq 125$ GeV, sits squarely inside that preferred band. That number is not a detail. It decides the entire phenomenology of the particle, because 125 GeV falls *below* every on-shell pair threshold that would otherwise dominate: $2m_W \simeq 161$ GeV, $2m_Z \simeq 182$ GeV, and $2m_t \simeq 345$ GeV are all closed. Had the fit pointed to $m_h \gtrsim 200$ GeV, the Higgs would have decayed almost entirely to two real weak bosons and been a much blunter instrument. Instead, at 125 GeV a remarkable spread of channels is simultaneously accessible—$b\bar{b}$ and $\tau^+\tau^-$ at tree level, WW^* and ZZ^* through one off-shell boson, and gg and $\gamma\gamma$ through quantum loops— each with a rate that the Standard Model predicts with no free parameter once the masses and couplings of Chapters 30 and 31 are fixed. This section computes that pattern from the Lagrangian. Every partial width below is a prediction; assembling them into branching ratios and a production cross section gives the observables whose internal consistency is the sharpest test the Standard Model has yet faced.

34.4.1 The Higgs couplings: everything is fixed by mass

All the dynamics we need was already written down when the symmetry broke. The physical Higgs h is the radial fluctuation of the doublet about its vacuum expectation value $v \simeq 246$ GeV, and *every* field that got its mass from the condensate couples to h with a strength proportional to that mass. For the fermions this is the central prediction of the Yukawa sector, Theorem 31.15: collecting Eq. (31.116), each Dirac fermion f obeys

$$\mathcal{L} \supset -\frac{m_f}{v}\, h\, \bar{f} f, \qquad g_{hf\bar{f}} = \frac{m_f}{v} = \frac{y_f}{\sqrt{2}}, \tag{34.126}$$

with no independent parameter. For the gauge bosons the coupling comes from the same place—the Higgs kinetic term $|D_\mu \Phi|^2$, which generated the masses in the first place. Writing the neutral component as $(v+h)/\sqrt{2}$ and recalling $M_W = gv/2$, $M_Z = \sqrt{g^2+g'^2}\,v/2$ from Eqs. (31.69) and (31.76), the mass term $M_V^2\, V_\mu V^\mu$ is really $M_V^2\,(1+h/v)^2\, V_\mu V^\mu$, so expanding the square exposes the

trilinear vertex

$$\mathcal{L} \supset \frac{2M_W^2}{v}\, h\, W_\mu^+ W^{-\mu} + \frac{M_Z^2}{v}\, h\, Z_\mu Z^\mu, \qquad g_{hVV} = \frac{2M_V^2}{v}, \tag{34.127}$$

where the relative factor of $\frac{1}{2}$ between the W and Z terms is the usual bookkeeping for the real (self-conjugate) neutral field Z versus the complex $W^\pm$, and is undone in the Feynman rule by the two identical Z legs. The same $(v+h)^2$ structure also produces a h^2VV seagull, irrelevant for single-Higgs decays. The lesson is uniform: *the Higgs couples to mass.* A heavier final state—when kinematics permits it—is favoured, in strict proportion to m_f^2 for fermions and M_V^4 for vectors. The photon and gluon, being massless, do not couple to h at tree level at all; their channels are induced by loops of the massive charged and coloured particles, and we compute them in Sections 34.4.4 and 34.4.5. We treat the open channels in order of how directly they probe Eqs. (34.126) and (34.127).

34.4.2 Tree-level decay to fermions

The cleanest test of Eq. (34.126) is the decay $h \to f\bar{f}$. At $m_h = 125$ GeV the kinematically open fermion channels are $b\bar{b}$, $\tau^+\tau^-$, $c\bar{c}$, $\mu^+\mu^-$ (the top is far too heavy; lighter quarks have negligible Yukawas). The amplitude follows directly from the vertex $-\,i\,(m_f/v)$ of Eq. (34.126). With outgoing fermion momentum p_1 (spinor $\bar{u}_s(p_1)$) and antifermion p_2 (spinor $v_{s'}(p_2)$),

$$i\mathcal{M} = -\,i\,\frac{m_f}{v}\,\bar{u}_s(p_1)\,v_{s'}(p_2), \tag{34.128}$$

a pure scalar bilinear with no gamma matrix—the hallmark of the CP-even scalar coupling. Squaring and summing over final spins, and using the standard completeness relations $\sum_s u_s(p)\bar{u}_s(p) = \not{p} + m$ and $\sum_{s'} v_{s'}(p)\bar{v}_{s'}(p) = \not{p} - m$,

$$\begin{aligned}\sum_{\text{spins}} |\mathcal{M}|^2 &= \frac{m_f^2}{v^2}\sum_{s,s'} \big[\bar{u}_s(p_1)v_{s'}(p_2)\big]\big[\bar{v}_{s'}(p_2)u_s(p_1)\big] \\ &= \frac{m_f^2}{v^2}\,\operatorname{tr}\big[(\not{p}_1 + m_f)(\not{p}_2 - m_f)\big].\end{aligned} \tag{34.129}$$

The trace is elementary with the technology of Section C.6: odd traces vanish (Theorem C.53), $\operatorname{tr}[\not{p}_1\not{p}_2] = 4\,p_1\!\cdot\!p_2$ by Eq. (C.152), and $\operatorname{tr}[\mathbf{1}] = 4$, so

$$\operatorname{tr}\big[(\not{p}_1 + m_f)(\not{p}_2 - m_f)\big] = 4\,p_1\!\cdot\!p_2 - 4m_f^2. \tag{34.130}$$

Energy–momentum conservation $q = p_1 + p_2$ with $q^2 = m_h^2$ and $p_i^2 = m_f^2$ gives $m_h^2 = (p_1+p_2)^2 = 2m_f^2 + 2p_1\!\cdot\!p_2$, hence $p_1\!\cdot\!p_2 = \frac{1}{2}(m_h^2 - 2m_f^2)$ and

$$4\,p_1\!\cdot\!p_2 - 4m_f^2 = (2m_h^2 - 4m_f^2) - 4m_f^2 = 2m_h^2 - 8m_f^2 = 2m_h^2\Big(1 - \frac{4m_f^2}{m_h^2}\Big). \tag{34.131}$$

Restoring the colour multiplicity N_c (which is 3 for quarks and 1 for leptons; the colourless h couples to a $\bar{q}q$ pair of any of the N_c colours, summed incoherently), the spin- and colour-summed squared amplitude is

$$\sum |\mathcal{M}|^2 = N_c \, \frac{m_f^2}{v^2} \, 2m_h^2 \, \beta_f^2, \qquad \beta_f \equiv \sqrt{1 - \frac{4m_f^2}{m_h^2}}, \tag{34.132}$$

where β_f is the velocity of each fermion in the Higgs rest frame. We now feed this into the two-body width formula Eq. (16.89) established in Section 16.3.4. With a distinct f and $\bar{f}$ the symmetry factor is $\mathcal{S} = 1$, and the momentum released is $|\mathbf{k}| = \tfrac{1}{2} m_h \beta_f$ by Eq. (16.87), so

$$\begin{aligned} \Gamma(h \to f\bar{f}) &= \frac{|\mathbf{k}|}{8\pi m_h^2} \sum |\mathcal{M}|^2 = \frac{m_h \beta_f / 2}{8\pi m_h^2} \, N_c \, \frac{m_f^2}{v^2} \, 2m_h^2 \beta_f^2 \\ &= \boxed{N_c \, \frac{m_f^2}{8\pi v^2} \, m_h \left(1 - \frac{4m_f^2}{m_h^2}\right)^{3/2}}. \end{aligned} \tag{34.133}$$

This is the owed result. Three remarks pin down its physics. First, the rate is proportional to m_f^2, the squared coupling Eq. (34.126): the $b\bar{b}$ channel dominates the fermionic decays not because the b is special but because it is the heaviest fermion light enough to be pair-produced, and $m_b^2/m_\tau^2 \simeq (2.8/1.78)^2 \simeq 2.5$ times the colour factor $N_c = 3$ puts $b\bar{b}$ a factor ~ 8 above $\tau^+\tau^-$, as observed. Second, the threshold suppression is β_f^3, *cubic* in the velocity. One power is the ordinary two-body phase space $|\mathbf{k}| \propto \beta_f$; the other two come from the matrix element Eq. (34.132), and their origin is the scalar nature of h. A spin-0, parity-even particle decaying to $f\bar{f}$ produces the pair in a relative P-wave (the $\bar{u}v$ bilinear is odd under the parity that exchanges the back-to-back momenta), and the P-wave amplitude vanishes linearly at threshold, giving β_f^2 in the rate. A *pseudoscalar* coupling $\bar{u}\gamma^5 v$ would instead produce an S-wave and scale as β_f; the power of β is thus a direct probe of the Higgs parity, a point taken up in Section 34.5. Third, the formula is a tree-level partial width in terms of the fermion mass. For quarks the appropriate mass is the running $\overline{\text{MS}}$ mass evaluated at the decay scale $\mu \simeq m_h$, which for the b quark runs down from the pole value ~ 4.8 GeV to $\overline{m}_b(m_h) \simeq 2.8$ GeV; using the running mass resums the large logarithms of the QCD corrections, and the residual correction multiplies Eq. (34.133) by $1 + \frac{17}{3}\,\alpha_s/\pi + \cdots \simeq 1.2$. With $\overline{m}_b(m_h) \simeq 2.8$ GeV this yields $\Gamma(h \to b\bar{b}) \simeq 2.3$ MeV, the largest single partial width of the 125 GeV Higgs.

Worked Example 34.17 (The $h \to \tau^+\tau^-$ partial width in numbers)**.** The $\tau^+\tau^-$ channel is the cleanest fermionic test of Eq. (34.126), free of the QCD running that complicates $b\bar{b}$. Take $m_\tau = 1.777$ GeV, $m_h = 125$ GeV, $v = 246$ GeV, and $N_c = 1$. The velocity factor is overwhelmingly close to unity,

$$\beta_\tau^2 = 1 - \frac{4(1.777)^2}{(125)^2} = 1 - \frac{12.63}{15625} = 1 - 8.1 \times 10^{-4} \simeq 0.99919, \tag{34.134}$$

so $\beta_\tau^3 \simeq 0.9976$: phase-space suppression is utterly negligible here, 125 GeV being far above the τ threshold. Then

$$\Gamma(h\to\tau^+\tau^-) = \frac{(1.777)^2}{8\pi(246)^2}\,(125)\,(0.9976)\ \text{GeV} = \frac{3.158}{8\pi\times 60516}\times 124.7\ \text{GeV} = \frac{3.158\times 124.7}{1.521\times 10^6}\ \text{GeV} = 2.59\times 10^{-4}\ \text{GeV}, \tag{34.135}$$

that is $\Gamma(h\to\tau^+\tau^-)\simeq 0.26$ MeV. Against the $\simeq 4.1$ MeV total width assembled below this is a branching ratio of $\simeq 6.3\%$, in agreement with the measured value. The same arithmetic with $m_\mu = 0.1057$ GeV gives $\Gamma(h\to\mu^+\mu^-)\simeq 0.92$ keV, suppressed relative to $\tau\tau$ by exactly $(m_\mu/m_\tau)^2 \simeq 3.5\times 10^{-3}$—a clean, parameter-free prediction of the mass-proportional coupling that the LHC has begun to confirm.

34.4.3 Decay to weak bosons through one off-shell leg

With $m_h < 2M_W < 2M_Z$ the decays to a pair of *on-shell* weak bosons are kinematically shut. They proceed instead with one boson on shell and one far off shell, $h\to VV^*\to Vf\bar f'$, a genuine three-body decay whose rate is suppressed by the off-shell propagator but enhanced by the large coupling Eq. (34.127). To see both effects cleanly, first compute the *hypothetical* on-shell two-body width—the rate the channel *would* have above threshold—and then attach the off-shell leg.

The on-shell two-body width. The amplitude for $h\to V(p_1,\epsilon_1)V(p_2,\epsilon_2)$ is, from Eq. (34.127),

$$i\mathcal{M} = i\,g_{hVV}\,\epsilon_1^*\cdot\epsilon_2^*, \qquad g_{hVV} = \frac{2M_V^2}{v}. \tag{34.136}$$

Summing over the three physical polarizations of each massive vector with the completeness relation $\sum_\lambda \epsilon_\lambda^\mu\epsilon_\lambda^{*\nu} = -\eta^{\mu\nu} + p^\mu p^\nu/M_V^2$,

$$\sum_{\text{pol}}|\mathcal{M}|^2 = g_{hVV}^2\Big(-\eta_{\mu\nu}+\tfrac{p_{1\mu}p_{1\nu}}{M_V^2}\Big)\Big(-\eta^{\mu\nu}+\tfrac{p_2^\mu p_2^\nu}{M_V^2}\Big) = g_{hVV}^2\Big[\,4-\tfrac{p_1^2}{M_V^2}-\tfrac{p_2^2}{M_V^2}+\tfrac{(p_1p_2)^2}{M_V^4}\Big] = g_{hVV}^2\Big[\,2+\tfrac{(p_1p_2)^2}{M_V^4}\Big], \tag{34.137}$$

using $p_i^2 = M_V^2$. With $p_1\cdot p_2 = \tfrac12(m_h^2-2M_V^2)$ and $g_{hVV}^2 = 4M_V^4/v^2$,

$$\sum_{\text{pol}}|\mathcal{M}|^2 = \frac{1}{v^2}\Big[8M_V^4+(m_h^2-2M_V^2)^2\Big] = \frac{m_h^4}{v^2}\Big(1-4r_V+12r_V^2\Big), \quad r_V\equiv\frac{M_V^2}{m_h^2}. \tag{34.138}$$

The two-body width Eq. (16.89), with $|\mathbf{k}| = \frac{1}{2} m_h \sqrt{1 - 4r_V}$ and an identical-particle factor $\delta_V = \frac{1}{2}$ for ZZ (the two Z's are identical) and $\delta_V = 1$ for W^+W^- (distinct), gives

$$\Gamma(h \to VV) = \delta_V \, \frac{m_h^3}{16\pi v^2} \, \sqrt{1 - 4r_V} \, (1 - 4r_V + 12 r_V^2). \tag{34.139}$$

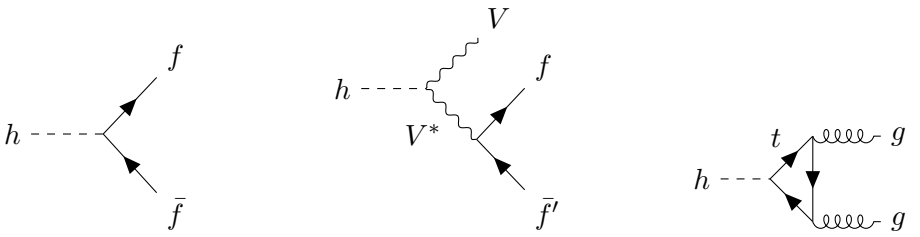
Two features deserve emphasis. The $12 r_V^2$ term inside the bracket comes entirely from the $(p_1 \cdot p_2)^2 / M_V^4$ piece of Eq. (34.137)—that is, from the *longitudinal* polarizations, whose polarization vectors grow like E/M_V. For $m_h \gg M_V$ the bracket is dominated by $(m_h^2 / 2M_V^2)^2$ and the width grows as $\Gamma \sim m_h^3 / v^2$: a heavy Higgs decays overwhelmingly into longitudinal W's and Z's, which is just the statement, dual to the equivalence theorem of Section 30.6, that the longitudinal vectors are the eaten Goldstone bosons of the doublet and couple with the full Higgs self-coupling. This same growth is what the electroweak fit's preference for a light m_h spared us. At $m_h = 125$ GeV, however, $\sqrt{1 - 4r_V}$ is imaginary: Eq. (34.139) is the rate the channel *would* have if it were open, and serves below as the building block for the physical off-shell rate and as the description of the steep rise once m_h climbs past $2M_V$ (Fig. 34.7).

Attaching the off-shell leg. Physically, the h emits one on-shell V of mass M_V and one virtual V^* of invariant mass $\sqrt{q^2} < m_h - M_V < M_V$, and the V^* materializes as a fermion pair $f\bar{f}'$ through the electroweak current. Because q^2 never reaches the pole at M_V^2 (we are below threshold), the Breit–Wigner denominator is far off resonance and we may drop the width term in it. The differential rate factorizes into *production* of a (V, V^*) pair—the two-body width Eq. (34.139) with one mass continued off shell, $M_V \to \sqrt{q^2}$ in the appropriate places—times the *spectral function* of the virtual leg,

$$d\Gamma(h \to V f \bar{f}') = \Gamma_2\!\left(m_h; M_V, \sqrt{q^2}\right) \frac{1}{\pi} \, \frac{\sqrt{q^2}\, \Gamma_V(q^2)}{(q^2 - M_V^2)^2} \, dq^2. \tag{34.140}$$

Here $\Gamma_V(q^2)$ is the width of a vector of mass $\sqrt{q^2}$ into the massless fermion pair, and the combination $\sqrt{q^2}\,\Gamma_V(q^2)/\pi(q^2 - M_V^2)^2$ is the off-shell propagator squared, integrated over the fermion phase space. The key simplification is that for massless final fermions the electroweak current is conserved, so the $q_\mu q_\nu / M_V^2$ part of the V^* propagator decouples from the fermion line; only the $\eta_{\mu\nu}$ part survives, and Eq. (34.140) is exact in that limit. Since $\Gamma_V(q^2) \propto \sqrt{q^2}$ (a vector's width scales with its mass), the weight $\sqrt{q^2}\,\Gamma_V(q^2)$ is proportional to q^2 times the sum of squared couplings of V to all the fermion species it can reach. Carrying out the q^2 integral over the allowed range $0 \le q^2 \le (m_h - M_V)^2$ is elementary, and collecting the prefactors gives the Keung–Marciano result

$$\Gamma(h \to VV^*) = \frac{3\, M_V^4\, m_h}{32\pi^3 v^4} \, \delta_V' \, R(x_V), \qquad x_V \equiv \frac{M_V^2}{m_h^2}, \tag{34.141}$$



(a) $h \to f\bar{f}$ (tree) (b) $h \to VV^* \to Vf\bar{f}'$ (c) $h \to gg$ (top loop)

Figure 34.5: Representative diagrams for the Higgs decay channels computed in this section. The tree decay (a) probes the Yukawa coupling Eq. (34.126) directly; the off-shell vector decay (b) probes the gauge coupling Eq. (34.127); the loop-induced gluonic decay (c) proceeds through a virtual top triangle and is the time-reverse of the dominant LHC production mechanism. A mirror diagram with the two gluons exchanged accompanies (c), and replacing the gluons by photons (plus a W loop) gives $h \to \gamma\gamma$.

where we used $G_F/\sqrt{2} = 1/2v^2$ to trade the standard G_F^2 normalization for v, the electroweak weights are

$$\delta'_W = 1, \qquad \delta'_Z = \frac{7}{12} - \frac{10}{9}\sin^2\theta_{\rm W} + \frac{40}{27}\sin^4\theta_{\rm W}, \tag{34.142}$$

counting the fermion channels each boson decays into (all left/right couplings of Z, the universal charged current of W), and the dimensionless phase-space function is the elementary integral of the spectrum Eq. (34.140),

$$R(x) = \frac{3(1-8x+20x^2)}{\sqrt{4x-1}}\,\arccos\!\Big(\frac{3x-1}{2x^{3/2}}\Big) - \frac{1-x}{2x}\big(2-13x+47x^2\big) - \frac{3}{2}\big(1-6x+4x^2\big)\ln x. \tag{34.143}$$

The structure of Eq. (34.141) tells the whole story. The M_V^4 is two powers of the coupling Eq. (34.127); the extra $1/v^2$ relative to the on-shell m_h^3/v^2 is the price of the off-shell propagator and the electroweak vertex that converts V^* to fermions; and $R(x_V)$ rises steeply as x_V falls, i.e. as m_h increases toward and past the on-shell threshold, matching smoothly onto Eq. (34.139) there. Numerically, with $\sin^2\theta_{\rm W} = 0.231$ one finds $\delta'_Z = 0.406$, and evaluating Eq. (34.143) at $x_W = (M_W/m_h)^2 = 0.41$ and $x_Z = (M_Z/m_h)^2 = 0.53$ gives $\Gamma(h \to WW^*) \simeq 0.88$ MeV and $\Gamma(h \to ZZ^*) \simeq 0.11$ MeV. The factor ~ 8 between them is mostly the explicit $3\delta'_W$ versus $3\delta'_Z$ counting together with $M_W^4 < M_Z^4$ and the larger off-shellness of the Z; WW^* is the second-largest decay of the 125 GeV Higgs after $b\bar{b}$, and the four-lepton final state $h \to ZZ^* \to 4\ell$, tiny in rate, is the cleanest discovery mode of all because its background is negligible.

34.4.4 The loop-induced gluonic decay $h \to gg$

The Higgs does not couple to gluons at tree level—gluons are massless and colour-octet, while h is a colour singlet. The coupling is induced at one loop by the only coloured particles with a sizeable Yukawa coupling, the heavy quarks, overwhelmingly the top. The process $h \to gg$ is thus a pure quantum effect, and because it runs in reverse as gluon fusion—the dominant Higgs production mechanism at the LHC (Section 34.4.6)—its loop integral is the single most important calculation in Higgs phenomenology. We carry it out in full. This advances Yang–Mills theory: the rate is governed by α_s (Eq. (27.163)) and by the colour trace $\mathrm{tr}(T^a T^b)$ of Chapter 27.

The amplitude and its tensor structure. The diagram is a top-quark triangle: the Higgs attaches to the loop through the Yukawa vertex $-i\,m_t/v$, and the two gluons through QCD vertices $i g_s \gamma^\mu T^a$ (Fig. 34.5c), with a second diagram in which the gluons are exchanged. Label the outgoing gluons $g^a(p_1,\mu)$ and $g^b(p_2,\nu)$, with $p_1^2 = p_2^2 = 0$ and $p_1\cdot p_2 = m_h^2/2$. Lorentz invariance, Bose symmetry, and—decisively—gauge invariance constrain the amplitude tightly. The colour structure is $\mathrm{tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab}$. The Lorentz structure must be a rank-two tensor built from p_1, p_2 and $\eta^{\mu\nu}$, transverse to each gluon: contracting with $p_{1\mu}$ or $p_{2\nu}$ must give zero, by the Ward identity that the gluon couples to the conserved colour current. The unique such structure (with the right dimension) is

$$T^{\mu\nu} \equiv \eta^{\mu\nu}\,(p_1p_2) - p_1^\nu p_2^\mu, \qquad p_{1\mu}T^{\mu\nu} = p_{1\mu}\eta^{\mu\nu}(p_1p_2) - (p_1p_1)p_2^\mu = p_1^\nu(p_1p_2) - 0 \overset{!}{=} 0 \text{ on shell}, \tag{34.144}$$

the last equality holding because the physical gluon polarization satisfies $\epsilon_1\cdot p_1 = 0$, so the leftover p_1^ν drops when contracted with $\epsilon_{1\nu}^*$. The full amplitude is therefore

$$i\mathcal{M} = i\,\frac{\alpha_s}{3\pi v}\,\delta^{ab}\left[\tfrac{3}{4}A_{1/2}(\tau_t)\right] T^{\mu\nu}\,\epsilon_{1\mu}^*(p_1)\,\epsilon_{2\nu}^*(p_2), \tag{34.145}$$

where we have pulled out the normalization so that the loop function $A_{1/2}(\tau_t) \to \frac{4}{3}$ in the heavy-top limit, $\tau_t \equiv m_h^2/4m_t^2$. The whole content of the loop is the single number $A_{1/2}$, which we now compute.

Evaluating the triangle. Write the two diagrams. With loop momentum k routed so that the three internal top propagators carry momenta k, $k-p_1$, and $k-p_1-p_2$, the trace from the fermion loop is

$$i\,T^{\mu\nu}_{\text{loop}} = -(-1)\,\frac{m_t}{v}\,g_s^2\;\mathrm{tr}(T^aT^b)\int\frac{d^dk}{(2\pi)^d}\,\frac{\mathrm{tr}\big[(\not k + m_t)\gamma^\mu(\not k - \not p_1 + m_t)\gamma^\nu(\not k - \not p_1 - \not p_2 + m_t)\big]}{(k^2-m_t^2)\big((k-p_1)^2-m_t^2\big)\big((k-p_1-p_2)^2-m_t^2\big)} + (p_1 \tag{34.146}$$

the overall (-1) being the closed-fermion-loop sign and the explicit -1 the two factors of i from the vertices times the i's in the propagators; the Yukawa vertex

contributes the scalar m_t/v with no gamma matrix. The numerator trace has an odd and an even part; by Theorem C.53 only traces of even numbers of gamma matrices survive, so the surviving terms carry either three factors of m_t (from picking the mass on all three propagators) or one factor of m_t (one mass, two slashed momenta). The single overall factor of m_t from the Yukawa makes the numerator $\propto m_t$ overall, and combined with the m_t-dependence of the denominators this is what builds τ_t. Introduce Feynman parameters via Theorem D.11 to combine the three denominators,

$$\frac{1}{D_1 D_2 D_3} = 2\int_0^1 dx\,dy\,dz\;\frac{\delta(1-x-y-z)}{[xD_1+yD_2+zD_3]^3}, \tag{34.147}$$

and shift $k \to \ell = k-(y+z)p_1 - z\,p_2$ to complete the square. The shifted denominator becomes $[\ell^2-\Delta]^3$ with

$$\Delta = m_t^2 - (y+z)(1-y-z)\,p_1^2 - z(1-z)\,p_2^2 - 2yz\,(p_1p_2) = m_t^2 - 2yz\,(p_1p_2) = m_t^2 - yz\,m_h^2, \tag{34.148}$$

using $p_1^2 = p_2^2 = 0$ and $p_1\cdot p_2 = m_h^2/2$. In the numerator, terms odd in ℓ integrate to zero, and the even part splits into a piece $\propto \ell^2$ (logarithmically divergent on its own but combining with its $\mu \leftrightarrow \nu$ partner into the transverse, hence finite, structure) and an ℓ-independent piece. Carrying out the gamma trace with Eq. (C.157) and the master numerator integrals of Theorem D.23, the divergences cancel between the two diagrams—they must, since no hgg counterterm exists in the renormalizable Lagrangian—and the result collapses onto the transverse tensor $T^{\mu\nu}$ with a finite coefficient. After the dust settles the entire x, y, z dependence reduces to a two-parameter integral, and the loop function is

$$A_{1/2}(\tau) = \frac{2}{\tau}\Big[\,1 + 2(\tau-1)\,G(\tau)\Big], \qquad G(\tau) \equiv \int_0^1 dx \int_0^{1-x} dy\;\frac{1}{1-4\tau\,xy}, \tag{34.149}$$

where the "1" is the local (tadpole-like) part of the numerator and the $(\tau-1)G(\tau)$ is the genuine triangle, $G(\tau)$ being the scalar three-point integral $\int dx\,dy\,(m_t^2 - xy\,m_h^2)^{-1}$ stripped of its $1/m_t^2$. The parametric integral G is elementary. Doing the inner y-integral,

$$\int_0^{1-x} \frac{dy}{1-4\tau xy} = -\frac{1}{4\tau x}\,\ln[1-4\tau x(1-x)], \tag{34.150}$$

and then the x-integral, one finds the standard closed form

$$G(\tau) = \frac{f(\tau)}{2\tau}, \qquad f(\tau) = \begin{cases} \arcsin^2\sqrt{\tau}, & \tau \le 1, \\ -\dfrac{1}{4}\Big[\ln\dfrac{1+\sqrt{1-1/\tau}}{1-\sqrt{1-1/\tau}} - i\pi\Big]^2, & \tau > 1, \end{cases} \tag{34.151}$$

the two branches meeting at $\tau = 1$, where the loop momentum can go on shell: for $\tau > 1$, i.e. $m_h > 2m_t$, the Higgs is heavy enough to produce a real $t\bar{t}$ pair

and f develops an absorptive imaginary part, the unitarity cut. Substituting $G = f/2\tau$ into Eq. (34.149) gives the loop function in its canonical form,

$$\boxed{A_{1/2}(\tau) = \frac{2}{\tau^2}\left[\tau + (\tau-1)f(\tau)\right]}. \tag{34.152}$$

The heavy-top limit and a low-energy-theorem cross-check. The physical Higgs has $\tau_t = m_h^2/4m_t^2 = (125)^2/[4(173)^2] \simeq 0.13 \ll 1$, so the top is effectively heavy and we expand. For small τ, $f(\tau) = \arcsin^2\sqrt{\tau} = \tau + \frac{1}{3}\tau^2 + O(\tau^3)$, hence

$$\tau + (\tau-1)f(\tau) = \tau + (\tau-1)(\tau + \tfrac{1}{3}\tau^2) = \tau + \tau^2 - \tau + \tfrac{1}{3}\tau^2 - \cdots = \tfrac{2}{3}\tau^2 + \cdots, \tag{34.153}$$

so that

$$A_{1/2}(\tau) \xrightarrow{\tau \to 0} \frac{2}{\tau^2}\cdot\frac{2}{3}\tau^2 = \frac{4}{3}. \tag{34.154}$$

This $\frac{4}{3}$ is the famous heavy-quark limit, and it has an entirely independent derivation that requires no loop integral at all—a powerful check. In the limit $m_t \gg m_h$ the top can be integrated out, and its only effect is to shift the gluon kinetic term. The point is that the top mass comes *entirely* from the Higgs condensate, $m_t = y_t v/\sqrt{2}$, so promoting $v \to v + h$ in m_t means the heavy-top contribution to the QCD coupling carries an h-dependence. Recall from Chapter 27 that integrating out a quark of mass m_t leaves, in the effective theory below m_t, a contribution to the running of α_s governed by the trace anomaly; concretely the top's contribution to the gluon two-point function is $\propto \ln m_t^2$, and through $m_t(h) = m_t(1 + h/v)$ this becomes a coupling of h to $G^a_{\mu\nu}G^{a\,\mu\nu}$. The low-energy theorem states it sharply:

$$\mathcal{L}_{\text{eff}}^{hgg} = \frac{\alpha_s}{12\pi}\,\frac{h}{v}\,G^a_{\mu\nu}G^{a\,\mu\nu}, \tag{34.155}$$

where the coefficient $\frac{\alpha_s}{12\pi}$ is precisely $\frac{\alpha_s}{8\pi} \times \frac{2}{3}$, and the $\frac{2}{3}$ is the single heavy quark's contribution to the QCD beta-function coefficient b_0 (the $-\frac{4}{3}T_F n_f$ piece of Eq. (27.163)'s parent formula, with $T_F = \frac{1}{2}$ for one flavour gives $\frac{2}{3}$). Matching the operator Eq. (34.155) to the amplitude Eq. (34.145) reproduces $\frac{3}{4}A_{1/2} = \frac{3}{4}\cdot\frac{4}{3} = 1$, i.e. $A_{1/2} = \frac{4}{3}$, with no integral done—the loop function in the heavy limit is just the top's contribution to asymptotic freedom, read backwards. The agreement of the brute-force triangle Eq. (34.152) with the low-energy theorem Eq. (34.155) is the cross-check the section spec demands, and it ties the Higgs's gluon coupling directly to the running of α_s that Yang–Mills theory established.

The width. Squaring Eq. (34.145), summing over the two transverse gluon polarizations with $\sum_\lambda \epsilon^\mu \epsilon^{*\nu} \to -\eta^{\mu\nu}$ (the unphysical polarizations cancel against ghosts, or equivalently the transverse $T^{\mu\nu}$ projects them out), and contracting

$T^{\mu\nu}T_{\mu\nu} = 2(p_1\cdot p_2)^2 = \frac{1}{2}m_h^4$ on shell, together with the colour sum $\delta^{ab}\delta^{ab} = N_c^2 - 1 = 8$ and the identical-gluon symmetry factor $\frac{1}{2}$, gives the partial width

$$\boxed{\Gamma(h \to gg) = \frac{\alpha_s^2\, m_h^3}{72\pi^3 v^2}\left|\tfrac{3}{4}A_{1/2}(\tau_t)\right|^2} \tag{34.156}$$

where the sum runs in principle over all heavy quarks but is saturated by the top, $\frac{3}{4}A_{1/2}(\tau_t) \to 1$. Numerically, with $\alpha_s(m_h) \simeq 0.11$ and the next-to-leading QCD correction (a large positive K-factor $\simeq 1.6$ from real and virtual gluon emission off the loop) included, $\Gamma(h \to gg) \simeq 0.34$ MeV, a branching ratio of $\simeq 8\%$. The gluonic channel is invisible as a final state at a hadron collider—it hides under the enormous QCD dijet background—but, run in reverse, it is the engine of Higgs production.

34.4.5 The diphoton channel: W and top loops in destructive interference

The decay $h \to \gamma\gamma$ is the most precisely measured of the loop-induced channels and, with $ZZ^* \to 4\ell$, one of the two discovery modes: a narrow peak in the diphoton invariant mass over a smooth background. Like $h \to gg$ it is forbidden at tree level and induced by loops of charged particles, but now there are two qualitatively different contributors—the charged fermions (chiefly the top, exactly the triangle of Section 34.4.4 with gluons replaced by photons and colour by charge) and the $W^\pm$ boson, whose loop has no analogue in the gluonic case because the gluon does not couple to a single heavy charged scalar's worth of degrees of freedom in the same way. The W loop dominates, and it interferes *destructively* with the top. We carry the assembly as the section's worked example, discharging the QED obligation: the diphoton amplitude is controlled entirely by the QED charge structure of Section 20.6.

Worked Example 34.18 (The $h \to \gamma\gamma$ amplitude and width)**. The fermion loop.** A charged fermion f of electric charge Q_f (in units of e) and colour multiplicity N_c^f runs the same triangle as in $h \to gg$, with the two gluon vertices $ig_s\gamma^\mu T^a$ replaced by photon vertices $iQ_f e\gamma^\mu$ and the colour trace $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ replaced by $N_c^f Q_f^2$. The loop function is unchanged—it is the same integral—so the fermion contributes

$$\mathcal{A}_f = N_c^f\, Q_f^2\, A_{1/2}(\tau_f), \qquad \tau_f = \frac{m_h^2}{4m_f^2}, \tag{34.157}$$

with $A_{1/2}$ given by Eq. (34.152). The top dominates the sum: $N_c^t Q_t^2 = 3\cdot(2/3)^2 = 4/3$, and $\tau_t = 0.13$ gives $A_{1/2}(\tau_t) \simeq 1.38$ (slightly above the heavy limit $4/3$), so $\mathcal{A}_t \simeq 1.84$. The b and τ contribute under a percent each and with the opposite-going sign of their small-τ expansions; we keep the top.

The W loop. The W contribution is more intricate because the spin-1 boson has more diagrams: the hWW vertex of Eq. (34.127) sewn into a W loop with two photon attachments, the quartic $hWW\gamma$ and $WW\gamma\gamma$ seagulls, and—in a renormalizable R_ξ gauge—the accompanying would-be-Goldstone and ghost loops. Gauge invariance is what makes the calculation tractable: the sum of all these diagrams must again collapse onto the transverse tensor $T^{\mu\nu}$ of Eq. (34.144), because the photon couples to the conserved electromagnetic current, and no $h\gamma\gamma$ counterterm exists. The gauge-invariant result is a single new loop function,

$$\mathcal{A}_W = A_1(\tau_W), \qquad A_1(\tau) = -\frac{1}{\tau^2}\big[\,2\tau^2 + 3\tau + 3(2\tau-1)f(\tau)\big], \qquad \tau_W = \frac{m_h^2}{4M_W^2}, \tag{34.158}$$

with the same $f(\tau)$ of Eq. (34.151). Its heavy-W limit follows by the same small-τ expansion used for $A_{1/2}$: with $f = \tau + \tfrac{1}{3}\tau^2$,

$$2\tau^2 + 3\tau + 3(2\tau-1)(\tau + \tfrac{1}{3}\tau^2) = 2\tau^2 + 3\tau + (6\tau^2 + 2\tau^3 - 3\tau - \tau^2) = 7\tau^2 + 2\tau^3, \tag{34.159}$$

so $A_1(\tau) \to -7$ as $\tau \to 0$. This -7 is the spin-1 analogue of the fermion's $+\tfrac{4}{3}$, and the low-energy theorem again explains it: the heavy W contributes -7 to the photon's beta-function coefficient (a charged vector drives α the "asymptotic-freedom" way, the opposite sense to a fermion), so the effective $h\gamma\gamma$ operator inherits exactly this number. The *opposite signs* of $A_1 < 0$ and $A_{1/2} > 0$ are therefore not an accident of the loop integrals but a consequence of the opposite signs with which vectors and fermions contribute to the running of the electromagnetic coupling of QED—a deep and satisfying connection (Fig. 34.6).

Assembly and interference. The two contributions add coherently, the W dominating and the top reducing the magnitude:

$$\mathcal{A}_{\gamma\gamma} = A_1(\tau_W) + \sum_f N_c^f Q_f^2 A_{1/2}(\tau_f) \simeq A_1(\tau_W) + \tfrac{4}{3}A_{1/2}(\tau_t). \tag{34.160}$$

With $\tau_W = (125)^2/[4(80.4)^2] = 0.60$ one evaluates $f(\tau_W) = \arcsin^2(0.777) = 0.79$, giving $A_1(\tau_W) = -8.32$; together with $\tfrac{4}{3}A_{1/2}(\tau_t) = +1.84$ the total is

$$\mathcal{A}_{\gamma\gamma} \simeq -8.32 + 1.84 = -6.48, \qquad |\mathcal{A}_{\gamma\gamma}|^2 \simeq 42.0. \tag{34.161}$$

Had the top added constructively, $|\mathcal{A}|^2$ would have been $(8.32 + 1.84)^2 = 103$; the destructive interference suppresses the diphoton rate by a factor ~ 2.5. This makes $h \to \gamma\gamma$ exquisitely sensitive to new charged particles, which would shift the top term (or add their own), and to the *sign* of the top Yukawa—a flipped sign would turn destruction into constructive enhancement and roughly triple the rate.

The width. The diphoton width has the same structure as Eq. (34.156) with $\alpha_s^2 \to \alpha^2$, the colour sum 8 replaced by the photon's 1, and the overall

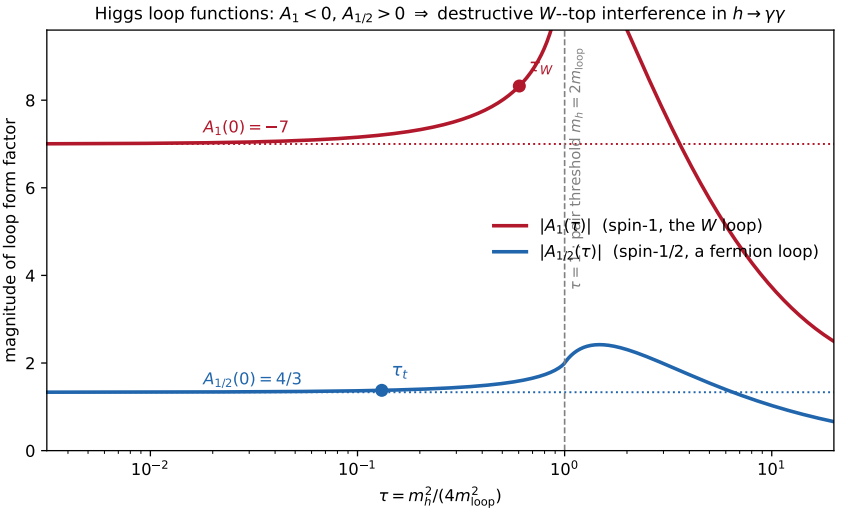


Figure 34.6: The Higgs loop form factors $A_{1/2}(\tau)$ (spin-$\frac{1}{2}$, a fermion loop) and $A_1(\tau)$ (spin-1, the W loop), plotted as magnitudes against $\tau = m_h^2/4m_{\rm loop}^2$. The dotted lines mark the heavy-particle limits $A_{1/2}(0) = \frac{4}{3}$ and $A_1(0) = -7$, both fixed by the low-energy theorem as the particle's contribution to the running of the relevant coupling (Eq. (34.155), Eq. (34.158)). The physical points for the 125 GeV Higgs, $\tau_t = 0.13$ and $\tau_W = 0.60$, lie below the pair threshold $\tau = 1$ so the form factors are real. The opposite signs of A_1 and $A_{1/2}$ produce the destructive W–top interference in $h \to \gamma\gamma$.

numerical prefactor adjusted accordingly:

$$\boxed{\Gamma(h \to \gamma\gamma) = \frac{\alpha^2\, m_h^3}{256\pi^3 v^2} \left| A_1(\tau_W) + \textstyle\sum_f N_c^f Q_f^2 A_{1/2}(\tau_f) \right|^2}. \tag{34.162}$$

Inserting $\alpha = 1/137$, $m_h = 125$ GeV, $v = 246$ GeV, and $|\mathcal{A}_{\gamma\gamma}|^2 = 42$,

$$\Gamma(h \to \gamma\gamma) = \frac{(1/137)^2\,(125)^3}{256\pi^3 (246)^2} \times 42 \text{ GeV} \simeq 9.1 \times 10^{-6} \text{ GeV} = 9.1 \text{ keV}, \tag{34.163}$$

a branching ratio of $\simeq 2.3 \times 10^{-3}$. Tiny in rate, the diphoton mode is golden in practice: a sharp mass peak that the detector resolves cleanly, and the channel in which the 125 GeV boson first announced itself.

34.4.6 Gluon-fusion production: the gluonic decay run in reverse

The same top triangle that lets h decay to two gluons lets two gluons fuse into h. At the LHC the colliding protons are, at the relevant momentum fractions,

overwhelmingly gluon-rich, so gluon fusion $gg \to h$ is the dominant production mechanism—about 87% of the 125 GeV Higgs cross section. Its parton-level rate is fixed by $\Gamma(h \to gg)$ through detailed balance, with no new dynamics. Because the Higgs is extremely narrow ($\Gamma_h \simeq 4$ MeV $\ll m_h$), the s-channel resonance is a delta function on the scale of any parton luminosity, and the Breit–Wigner cross section collapses to

$$\hat{\sigma}(gg \to h) = \frac{\pi^2}{8\, m_h}\, \Gamma(h \to gg)\, \delta(\hat{s} - m_h^2). \tag{34.164}$$

The factor is pure kinematics and counting, and we derive it. For a $2 \to 1$ process the cross section is $\hat{\sigma} = \frac{1}{2\hat{s}} \int \frac{d^3p_h}{(2\pi)^3 2E_h} (2\pi)^4 \delta^4(p_1 + p_2 - p_h)\, \overline{|\mathcal{M}|^2}$, where the flux for massless incoming partons is $2\hat{s}$ and the bar denotes the average over initial spins and colours. The one-particle phase space collapses the four-momentum delta to $2\pi\, \delta(\hat{s} - m_h^2)$, so $\hat{\sigma} = \frac{\pi}{\hat{s}} \delta(\hat{s} - m_h^2)\, \overline{|\mathcal{M}(gg \to h)|^2}$. Now relate the averaged production matrix element to the decay width. The two-body decay Eq. (16.89) with massless gluons gives $\sum_{\text{pol,col}} |\mathcal{M}(h \to gg)|^2 = 32\pi m_h\, \Gamma(h \to gg)$, where the sum is over final polarizations and colours. The production amplitude is the same $|\mathcal{M}|^2$, now *averaged* over the initial 2×2 gluon polarizations and 8×8 colours, i.e. divided by 256:

$$\overline{|\mathcal{M}(gg \to h)|^2} = \frac{1}{256} \sum_{\text{pol,col}} |\mathcal{M}(h \to gg)|^2 = \frac{32\pi m_h\, \Gamma(h \to gg)}{256} = \frac{\pi m_h\, \Gamma(h \to gg)}{8}. \tag{34.165}$$

Substituting gives $\hat{\sigma} = \frac{\pi}{\hat{s}} \delta(\hat{s} - m_h^2) \cdot \frac{\pi m_h \Gamma}{8} = \frac{\pi^2 \Gamma}{8 m_h} \delta(\hat{s} - m_h^2)$, using $\hat{s} = m_h^2$ at the delta—which is Eq. (34.164). The hadronic cross section then convolves Eq. (34.164) with the gluon parton distributions $g(x, \mu)$ of the two protons,

$$\sigma(pp \to h) = \frac{\pi^2}{8 m_h}\, \Gamma(h \to gg)\, \frac{1}{s} \left[\tau \frac{d\mathcal{L}_{gg}}{d\tau} \right]_{\tau = m_h^2/s}, \qquad \tau \frac{d\mathcal{L}_{gg}}{d\tau} = \tau \int_\tau^1 \frac{dx}{x}\, g(x)\, g(\tau/x), \tag{34.166}$$

the parton luminosity being a non-perturbative input governed by the factorization theorem of Chapter 28. The cross section thus inherits the same $|\frac{3}{4} A_{1/2}(\tau_t)|^2$ top-loop factor as the decay; this is why the measured gluon-fusion rate is, directly, a measurement of the top Yukawa coupling running in the loop—a probe of Eq. (34.126) for the one fermion too heavy to appear in a tree-level Higgs decay. The large NLO QCD corrections to Eq. (34.164), the same $K \simeq 1.6$ enhancement noted under Eq. (34.156), are essential for matching the measured rate and are a triumph of perturbative QCD; we quote them but do not recompute the multi-loop result here.

34.4.7 Branching ratios at 125 GeV and the signal strength

Collecting the partial widths Eqs. (34.133), (34.141), (34.156) and (34.162) and the small $Z\gamma$ and lighter-fermion channels, the total width of the 125 GeV Higgs

Table 34.2: Standard-Model Higgs branching ratios at $m_h = 125$ GeV, computed from the partial-width formulas of this section (Eqs. (34.133), (34.141), (34.156) and (34.162)), with running quark masses and leading QCD/EW corrections folded in. The total width is Eq. (34.167). The numbers reproduce the measured pattern at the few-percent level; small channels ($Z\gamma$, $s\bar{s}$, $\mu^+\mu^-$) round out the remaining $\sim 0.3\%$.

Channel	$\Gamma(h \to X)$	BR	probes
$b\bar{b}$	2.3 MeV	$\sim 58\%$	Yukawa Eq. (34.126)
WW^*	0.88 MeV	$\sim 21\%$	gauge Eq. (34.127)
gg	0.34 MeV	$\sim 8\%$	top loop / α_s
$\tau^+\tau^-$	0.26 MeV	$\sim 6.3\%$	Yukawa
$c\bar{c}$	0.12 MeV	$\sim 2.9\%$	Yukawa
ZZ^*	0.11 MeV	$\sim 2.6\%$	gauge
$\gamma\gamma$	9.1 keV	$\sim 0.23\%$	W/top loop Eq. (34.160)
$\mu^+\mu^-$	0.9 keV	$\sim 0.02\%$	Yukawa

is

$$\Gamma_h = \sum_X \Gamma(h \to X) \simeq 4.1 \text{ MeV}, \tag{34.167}$$

a remarkably small number—about 4×10^{-5} of the mass, far too narrow to measure directly from the lineshape (the experimental mass resolution is $\sim$ 1 GeV), so Γ_h is inferred indirectly, a point taken up in Section 34.5. The branching ratio of each channel is $\mathrm{BR}(h \to X) = \Gamma(h \to X)/\Gamma_h$ by Theorem 16.12, and the pattern, computed from the formulas of this section, is collected in Table 34.2. It is the fingerprint of a single scalar coupling to mass: $b\bar{b}$ on top because the b is the heaviest accessible fermion, WW^* next because the gauge coupling is large, then the loop-induced gg, then $\tau\tau$, $c\bar{c}$, ZZ^*, and finally the rare but pristine $\gamma\gamma$.

The experimental observable that compresses production and decay into one number is the *signal strength*. For a given production mode i (gluon fusion, vector-boson fusion, associated Vh, $t\bar{t}h$) and decay channel f, define

$$\mu_i^f \equiv \frac{\left[\sigma_i \times \mathrm{BR}(h \to f)\right]_{\text{observed}}}{\left[\sigma_i \times \mathrm{BR}(h \to f)\right]_{\text{SM}}}, \tag{34.168}$$

the ratio of the measured rate to the Standard-Model prediction computed from exactly the formulas of this section. By construction $\mu = 1$ for every channel if the Standard Model is correct, and a measured μ differing from 1 in any channel would localize a deviation either in a production coupling or in a decay coupling. As of the mid-2020s the LHC experiments report μ consistent with unity, channel by channel, to a precision ranging from $\sim 10\%$ in the strong

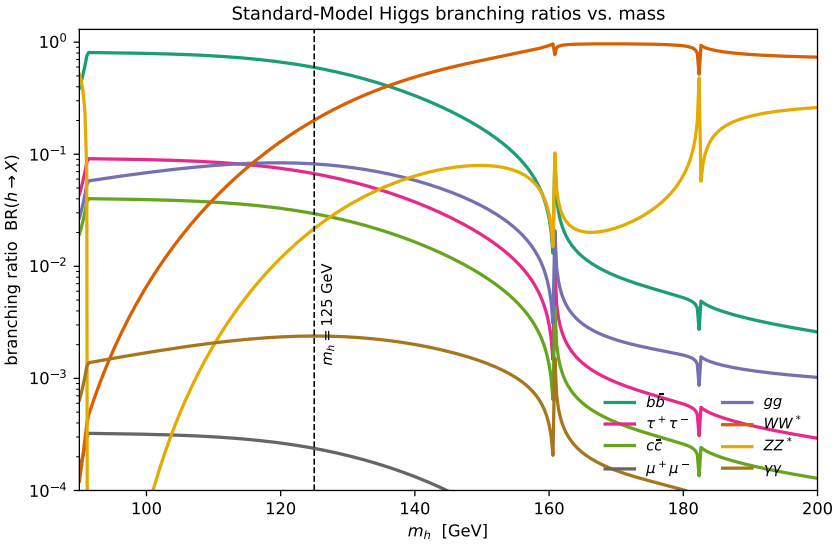


Figure 34.7: Standard-Model Higgs branching ratios as a function of the (hypothetical) Higgs mass, each curve computed from the partial-width formulas of this section. Below $\sim$ 135 GeV the $b\bar{b}$ tree decay dominates because the on-shell weak-boson channels are kinematically shut. As m_h climbs through $2M_W \simeq 161$ GeV and $2M_Z \simeq 182$ GeV the WW and ZZ channels switch on (the steep rise of Eq. (34.139)) and overwhelm $b\bar{b}$. The vertical line marks the physical value $m_h = 125$ GeV, which sits in the narrow window where many channels are simultaneously open— the reason the boson can be studied so many ways. This is the quantitative form of the statement that opened the section: the electroweak fit's light Higgs is what makes the rich decay phenomenology possible.

gluon-fusion/diphoton and four-lepton modes to tens of percent in the rarer ones, with the overall combined signal strength $\mu \simeq 1.0 \pm 0.05$. Every entry of Table 34.2, predicted with no free parameter beyond the inputs of Chapters 30 and 31, is confirmed.

The decisive content of all this arithmetic is not any single width but the *pattern* they form: each coupling scales with the corresponding mass, quadratically in the rate, exactly as Eqs. (34.126) and (34.127) demand. Turning these computed rates into the measured coupling-versus-mass law, the spin-parity determination, the indirect total-width extraction, and the still-open self-coupling—the falsifiable signature of the Standard-Model Higgs and the frontier of what remains untested—is the subject of the next section.

34.5 Properties of the Higgs Boson

The previous section turned the Standard Model Lagrangian into a production-and-decay machine: from the mass-locked couplings of Chapter 31 it computed the partial widths $\Gamma(h \to f\bar{f})$, $\Gamma(h \to VV^*)$, $\Gamma(h \to gg)$ and $\Gamma(h \to \gamma\gamma)$, assembled them into branching ratios $\mathrm{BR}(h \to X) = \Gamma_X/\Gamma_h$, and folded in the production cross sections to predict the rate of every observable final state. Experiment reports those rates as *signal strengths*: each measured event yield, divided by the Standard-Model expectation for the same channel, is a pure number that would equal one if nature were exactly the minimal Standard Model and would depart from one if any coupling were off. This section is the inverse problem. Given the signal strengths, what have we learned about the particle itself — its mass, its spin and parity, the law its couplings obey, its total lifetime, and the one coupling that closes the theory but has so far escaped measurement? The answers convert a list of rates into a falsifiable portrait, and they pin down precisely which single feature — the universal proportionality of coupling to mass — is the Standard Model's sharpest and most distinctive claim, and which feature — the strength with which the Higgs talks to itself — remains an open frontier.

We carry over the notation of Chapter 31: the physical Higgs scalar h, the vacuum expectation value $v \simeq 246$ GeV, the gauge-boson masses m_W, m_Z, the charged-fermion masses m_f, and the mass-locked tree couplings collected in Eq. (31.116). No new thread layer is opened here; the spin-parity analysis simply *uses* Wigner's classification of one-particle states by mass and spin (Chapter 3), and the rate formulae are those of Chapter 16.

34.5.1 The mass from invariant-mass peaks

The Higgs boson is unstable, so its mass is not read from a track or a time of flight but reconstructed from the four-momenta of its decay products. The kinematic tool is the same Lorentz invariant that organizes every resonance in this book: for a set of final-state particles with four-momenta p_i, the *invariant mass* is

$$M_{\mathrm{inv}} \equiv \sqrt{\Big(\sum_i p_i\Big)^2}, \qquad p_i^2 = m_i^2, \tag{34.169}$$

the norm, in the mostly-minus metric, of the total final-state four-momentum. If the particles are the genuine decay products of a single parent of mass m_h that was on shell, four-momentum conservation $p_h = \sum_i p_i$ forces $M_{\mathrm{inv}}^2 = p_h^2 = m_h^2$ event by event, up to the parent's intrinsic width and the detector's finite resolution. The signature of a resonance is therefore a localized excess — a *peak* — in the distribution of M_{inv} over many events, sitting on a smooth continuum background from processes that produce the same final state without an intermediate h. The position of the peak is the mass.

Two decay channels make this measurement clean enough to reach the per-mille level, and they are clean for complementary reasons. The first is $h \to \gamma\gamma$. The diphoton branching ratio is tiny (Section 31.4.4 and the loop computation of the previous section give $\mathrm{BR}(h \to \gamma\gamma) \approx 2.3 \times 10^{-3}$), but the two photons are measured in the electromagnetic calorimeter with excellent energy and angular resolution, so the reconstructed

$$M_{\gamma\gamma}^2 = (p_{\gamma_1} + p_{\gamma_2})^2 = 2\, p_{\gamma_1} \cdot p_{\gamma_2} = 2E_{\gamma_1} E_{\gamma_2}(1 - \cos\theta_{12}) \tag{34.170}$$

is sharp; here θ_{12} is the opening angle between the photons and the massless on-shell conditions $p_{\gamma_i}^2 = 0$ were used. A narrow diphoton peak rides on a large but smoothly falling background of non-resonant $\gamma\gamma$ production, and a fit of peak-plus-smooth-continuum extracts m_h from the peak position.

The second channel is $h \to ZZ^* \to 4\ell$ with $\ell \in \{e, \mu\}$ — the "golden channel." One Z is on shell and the other is far off shell (the asterisk), because $m_h < 2m_Z$ forbids two real Z's; the four charged leptons are measured with superb momentum resolution and almost no background, so the four-lepton invariant mass

$$M_{4\ell}^2 = \left(\textstyle\sum_{i=1}^{4} p_{\ell_i}\right)^2 \tag{34.171}$$

forms a clean, narrow peak over a tiny continuum. The branching ratio is again small ($\mathrm{BR}(h \to ZZ^* \to 4\ell) \sim 1.2 \times 10^{-4}$), but the near-absence of background makes each handful of events count. Averaging the two channels, the LHC experiments obtain

$$m_h = 125.2 \text{ GeV} \quad \text{to within about } 0.1\%, \tag{34.172}$$

the most precisely known mass in the entire boson sector. This number is not a Standard-Model prediction — $m_h^2 = 2\lambda v^2$ trades the Higgs mass for the quartic coupling λ, which is a free input (Chapter 30) — but *once measured it fixes* λ, and through it every Higgs self-interaction, as Section 34.5.5 will exploit.

Worked Example 34.19 (Reconstructing the diphoton mass and the width of the peak)**.** Consider a $h \to \gamma\gamma$ event in which the two photons carry energies $E_{\gamma_1} = 70$ GeV and $E_{\gamma_2} = 58$ GeV and are separated by an opening angle $\theta_{12} = 110^\circ$. From Eq. (34.170),

$$M_{\gamma\gamma} = \sqrt{2E_{\gamma_1}E_{\gamma_2}(1 - \cos\theta_{12})} = \sqrt{2(70)(58)\,(1 - \cos 110^\circ)}\ \text{GeV}. \tag{34.173}$$

With $\cos 110^\circ = -0.342$, the bracket is 1.342, so $M_{\gamma\gamma} = \sqrt{2 \cdot 70 \cdot 58 \cdot 1.342} = \sqrt{1.090 \times 10^4} \approx 104$ GeV for this particular kinematic configuration — a single event lands wherever its photons happen to fly; only the ensemble peaks at m_h.

The reason the *peak* is what one measures, rather than any single event, is that the experimental width of the diphoton line is set entirely by detector

resolution, not by the Higgs itself. The Standard-Model total width is $\Gamma_h \approx 4.1$ MeV (Section 34.5.4), whereas the calorimetric resolution on $M_{\gamma\gamma}$ is of order 1–2 GeV, i.e. some *three hundred times* broader. Propagating the fractional energy resolutions σ_E/E of the two photons (and the angular resolution) through Eq. (34.170) gives, for uncorrelated terms,

$$\frac{\sigma_M}{M_{\gamma\gamma}} = \frac{1}{2}\sqrt{\left(\frac{\sigma_{E_1}}{E_1}\right)^2 + \left(\frac{\sigma_{E_2}}{E_2}\right)^2 + \left(\frac{\sigma_\theta \,\sin\theta_{12}}{1-\cos\theta_{12}}\right)^2}, \tag{34.174}$$

obtained by taking $\frac{1}{2}\,\mathrm{d}\ln M_{\gamma\gamma}^2$ of Eq. (34.170). A per-photon energy resolution of $\sim 1\%$ already yields $\sigma_M/M \sim 1\%$, i.e. ~ 1 GeV, consistent with the observed line width and overwhelming the intrinsic 4 MeV. The lesson is structural and recurs in Section 34.5.4: the Higgs line shape seen in any single channel measures the apparatus, not the particle's lifetime.

34.5.2 The coupling-modifier framework and the coupling–mass law

A mass and a peak position do not yet test the dynamics. The dynamical content lives in the *couplings*, and the cleanest way to confront them with data is to ask, for each interaction vertex, by how much the measured coupling departs from its Standard-Model value. This is the coupling-modifier, or κ, framework.

Definition 34.20 (Coupling modifiers)**.** For each particle i to which the Higgs couples, define the dimensionless modifier

$$\kappa_i \equiv \frac{g_{hii}}{g_{hii}^{\mathrm{SM}}}, \tag{34.175}$$

the ratio of the actual h-i-i coupling to its Standard-Model prediction. By construction $\kappa_i = 1$ for every i in the minimal Standard Model. For the loop-induced vertices $h\gamma\gamma$ and hgg one defines effective modifiers κ_γ, κ_g for the whole loop amplitude, which in the Standard Model are themselves computed from the κ's of the particles running in the loop.

The modifiers are extracted from rates. A given production-times-decay channel measures, schematically,

$$\mu_{i\to h\to f} \equiv \frac{\sigma_i \cdot \mathrm{BR}_f}{(\sigma_i \cdot \mathrm{BR}_f)_{\mathrm{SM}}} = \frac{\kappa_i^2\,\kappa_f^2}{\kappa_h^2}, \qquad \kappa_h^2 \equiv \frac{\Gamma_h}{\Gamma_h^{\mathrm{SM}}} = \sum_X \mathrm{BR}_X^{\mathrm{SM}}\,\kappa_X^2, \tag{34.176}$$

the *signal strength*: production through vertex i scales the cross section by κ_i^2, decay through vertex f scales the partial width by κ_f^2, and the branching ratio carries the total width $\Gamma_h \propto \kappa_h^2$ in its denominator, where κ_h^2 is the SM-branching-fraction-weighted sum of the individual κ_X^2 (this last identity is exactly

$\Gamma_h = \sum_X \Gamma_X$ rewritten in modifiers). Because every $\Gamma_X \propto g_{hXX}^2$ at the rates of Chapter 16, each factor enters squared. Measuring many channels $\{i, f\}$ overconstrains the handful of κ's, and a global fit returns them. As of the mid-2020s every κ measured at the LHC is consistent with unity, the tree-level ones (to b, τ, t, W, Z) at the 5–10% level and the rarer ones (μ, γ) less precisely.

The deep test is not that the κ's are near one — that is the statement "the SM works" — but *why they should all be one at once*, which is the statement of the coupling–mass law. We now derive it.

The single origin of mass forces a single line. Every mass in the Standard Model comes from one place: the vacuum expectation value v of the one Higgs doublet. The Higgs boson h is the fluctuation of that same condensate, $H^0 = (v+h)/\sqrt{2}$ in unitary gauge (Eq. (31.99)). Consequently the coupling of h to any particle is obtained from that particle's mass term by the replacement $v \to v + h$ and reading off the term linear in h. This is not a heuristic; it is exactly how Section 31.4.4 extracted $g_{hf\bar{f}} = m_f/v$, and we now apply the identical logic to the gauge bosons, so that fermions and bosons land on a common footing.

The gauge-boson masses came from the Higgs kinetic term $|D_\mu H|^2$ in Chapter 31. In unitary gauge $H = \frac{1}{\sqrt{2}}(0, v+h)^{\mathsf{T}}$, and the piece of $|D_\mu H|^2$ quadratic in the gauge fields is

$$|D_\mu H|^2 \supset \frac{(v+h)^2}{8}\Big(g^2\, W_\mu^+ W^{-\mu} + g^2 W_\mu^- W^{+\mu} + (g^2 + g'^2)\, Z_\mu Z^\mu\Big), \tag{34.177}$$

exactly as in the mass-generation computation, with the masses identified as $m_W^2 = \frac{1}{4}g^2 v^2$ and $m_Z^2 = \frac{1}{4}(g^2 + g'^2)v^2$. Writing the prefactor as $(v+h)^2 = v^2(1 + h/v)^2 = v^2(1 + 2h/v + h^2/v^2)$, the v^2 piece reproduces the mass terms $m_W^2 W^+W^- + \frac{1}{2}m_Z^2 ZZ$, and the piece linear in h is the hVV interaction,

$$\mathcal{L} \supset \frac{2m_W^2}{v}\, h\, W_\mu^+ W^{-\mu} + \frac{m_Z^2}{v}\, h\, Z_\mu Z^\mu, \tag{34.178}$$

where the factor 2 for W versus the 1 for Z is the familiar charged-versus-neutral counting (the Z term carries the $\frac{1}{2}$ of a real field's mass term, the $W^\pm$ does not). The structural content is uniform: in both cases

$$g_{hVV} = \frac{2m_V^2}{v}, \tag{34.179}$$

the coupling is fixed by the gauge-boson *mass-squared* over v, with no free parameter — because, again, the coupling is $\partial(\text{mass term})/\partial v$ evaluated on the condensate. Collecting the fermionic result Eq. (31.117) and the bosonic result Eq. (34.179):

$$\boxed{g_{hf\bar{f}} = \frac{m_f}{v}, \qquad g_{hVV} = \frac{2m_V^2}{v}\,.} \tag{34.180}$$

Proposition 34.21 (The coupling–mass line)**.** *Define the* reduced coupling *of the Higgs to particle i as*

$$\widehat{g}_i \equiv \begin{cases} g_{hf\bar{f}}, & i = f \text{ a fermion}, \\ \sqrt{\dfrac{g_{hVV}}{2v}} = \dfrac{m_V}{v}, & i = V \text{ a massive vector}, \end{cases} \tag{34.181}$$

where the square root for vectors compensates the quadratic mass dependence of Eq. (34.179). Then in the Standard Model every particle obeys the same relation

$$\widehat{g}_i = \frac{m_i}{v}, \tag{34.182}$$

so that on a plot of $\widehat{g}_i$ against m_i all particles lie on a single straight line through the origin of slope $1/v$. Equivalently, all coupling modifiers coincide, $\kappa_i = 1$ for every i, and the common intercept of the line measures the universal scale v.

Proof. For a fermion, Eq. (34.180) gives $\widehat{g}_f = g_{hf\bar{f}} = m_f/v$ directly. For a vector, $\widehat{g}_V = \sqrt{g_{hVV}/2v} = \sqrt{(2m_V^2/v)/2v} = \sqrt{m_V^2/v^2} = m_V/v$. Both equal m_i/v, establishing Eq. (34.182); the slope of $\widehat{g}_i$ versus m_i is $1/v$ and the line passes through the origin. The modifier statement is immediate: $\kappa_i = \widehat{g}_i/\widehat{g}_i^{\mathrm{SM}} = 1$ since the prediction *is* m_i/v. □

Physical Insight

This single line is the most distinctive falsifiable claim of the Standard Model Higgs sector, and it is sharp because it is overdetermined. A particle physicist handed an arbitrary scalar coupled to the known particles could choose nine fermion couplings and two boson couplings independently — eleven free numbers. The Standard Model spends *zero* of them: every coupling is dictated by an already-measured mass and the one shared scale v. The content of Fig. 34.8 is therefore not that the points are near a line of slope one — any single point can always be made to sit on a line — but that *all eleven points sit on the same line with the same intercept simultaneously*, across more than three decades in mass, with no point adjustable. A Higgs that coupled to, say, charge or weak isospin instead of mass would scatter the points; a second Higgs doublet would shift fermions and bosons onto different lines; a composite Higgs would bend the line at high mass. The flatness of all κ_i at unity is thus a stringent, structural test of "one condensate, one origin of mass," not a coincidence of eleven near-unity ratios.

Worked Example 34.22 (From a signal strength to a coupling modifier)**.** Suppose the gluon-fusion-produced, diphoton-decaying channel is measured

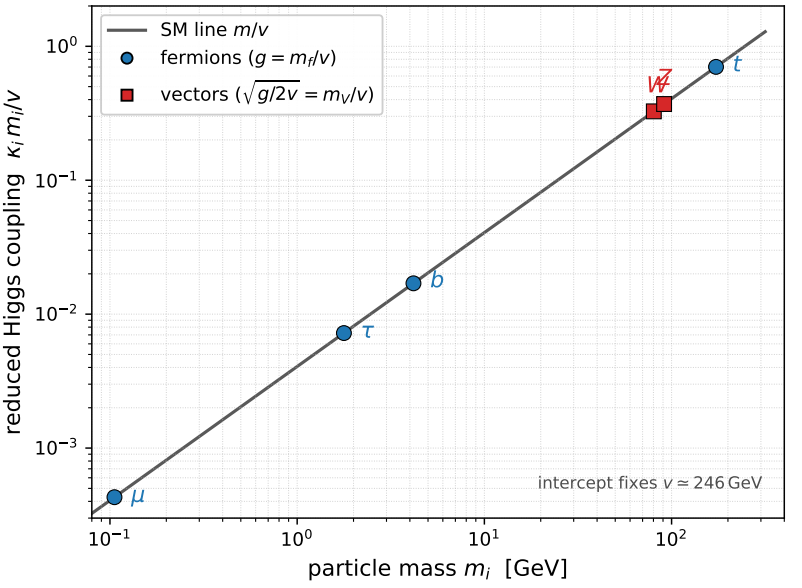


Figure 34.8: The coupling–mass law of Theorem 34.21. The reduced Higgs coupling $\widehat{g}_i = \kappa_i\, m_i/v$ (Eq. (34.181): $g_{hf\bar{f}}$ for fermions, $\sqrt{g_{hVV}/2v}$ for vectors) is plotted against the particle mass. The Standard Model predicts that every particle lies on the single line $\widehat{g}_i = m_i/v$ (slope $1/v$, intercept fixing $v \simeq 246$ GeV), with no adjustable point. Fermions (μ, τ, b, t, circles) and the weak vectors (W, Z, squares) are shown at their measured masses; experiment tests whether the measured couplings keep them on the line. A deviation of any point falsifies the single-doublet Standard Model.

with signal strength $\mu_{gg\to h\to\gamma\gamma} = 1.10 \pm 0.10$, and that for the purpose of this illustration all other modifiers are fixed to unity except the top-Yukawa modifier κ_t, which controls *both* the dominant gluon-fusion production loop and (together with the W loop) the diphoton decay loop. We extract κ_t.

Gluon fusion proceeds through a top triangle, so $\sigma_{gg\to h} \propto \kappa_t^2$. The diphoton amplitude is the coherent sum of a W loop and a top loop; with $\kappa_W = 1$ fixed and writing the SM amplitude as $A_W + A_t$, the modified amplitude is $A_W + \kappa_t A_t$, so

$$\kappa_\gamma^2 = \frac{|A_W + \kappa_t A_t|^2}{|A_W + A_t|^2}. \tag{34.183}$$

The W loop dominates and interferes destructively with the top loop; numerically $A_t/A_W \approx -0.28$ (a real ratio at $m_h \ll 2m_W, 2m_t$, from the loop functions of the previous section). With $\kappa_h = 1$ held fixed (the total width barely moves under a

small κ_t shift since $h \to \gamma\gamma$ and $h \to gg$ are subdominant in Γ_h),

$$\mu = \kappa_t^2\,\kappa_\gamma^2 = \kappa_t^2\,\frac{|1 - 0.28\,\kappa_t|^2}{|1 - 0.28|^2} = \kappa_t^2\,\frac{(1 - 0.28\,\kappa_t)^2}{0.5184}. \tag{34.184}$$

Setting $\mu = 1.10$ and solving the resulting quartic near $\kappa_t = 1$: at $\kappa_t = 1$ the right side is 1; expanding $\mu(\kappa_t)$ to first order, $\mathrm{d}\mu/\mathrm{d}\kappa_t\big|_1 = 2\kappa_t\kappa_\gamma^2 + \kappa_t^2\,(\mathrm{d}\kappa_\gamma^2/\mathrm{d}\kappa_t)$ evaluated at 1 gives $2(1) + 1 \cdot [2(1 - 0.28)(-0.28)/0.5184] = 2 - 0.778 = 1.22$, so $\Delta\kappa_t \approx \Delta\mu/1.22 = 0.10/1.22 \approx 0.08$, i.e. $\kappa_t = 1.08 \pm 0.08$ (the experimental error propagated identically). The point of the computation is methodological: a single rate constrains a coupling only once the production and decay loop content is unfolded, and the destructive W–t interference makes the diphoton rate a *nonlinear* probe of κ_t — which is precisely why the global fit, not a single channel, is the right tool.

34.5.3 Spin and parity: $J^P = 0^+$

A scalar is the simplest thing the resonance at 125 GeV could be, but "simplest" is not "established." Wigner's classification (Chapter 3) tells us that a massive one-particle state is labelled by its mass and by an irreducible representation of the little group SO(3), i.e. by a spin $J = 0, 1, 2, \ldots$, and within each spin the discrete quantum number parity $P = \pm$ distinguishes (for $J = 0$) a scalar 0^+ from a pseudoscalar 0^-. The Standard Model demands $J^P = 0^+$: the Higgs is the radial fluctuation of a scalar condensate and is even under parity. The alternatives to exclude are 0^-, $1^\pm$, and 2^+ (a graviton-like tensor). Two independent arguments do the job, and both are pure consequences of Chapter 3 plus Bose statistics, requiring no new dynamics.

Spin one is forbidden in the diphoton channel: the Landau–Yang theorem. The resonance was discovered partly in $h \to \gamma\gamma$. That observation alone rules out spin 1, by a theorem that follows from rotational invariance, transversality of the photon, and Bose symmetry — nothing else.

Theorem 34.23 (Landau–Yang)**.** *A massive spin-1 particle cannot decay into two on-shell photons.*

Proof. Work in the rest frame of the decaying vector V, where the two photons emerge back to back with momenta $\mathbf{k}_1 = \mathbf{k}$ and $\mathbf{k}_2 = -\mathbf{k}$. A massive spin-1 particle has three physical polarizations, described in its rest frame by a real three-vector $\boldsymbol{\varepsilon}$ (the spatial part of its polarization four-vector, the time component vanishing at rest). Each photon is transverse, with polarization vectors $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2$ obeying

$$\boldsymbol{\varepsilon}_1 \cdot \mathbf{k} = 0, \qquad \boldsymbol{\varepsilon}_2 \cdot \mathbf{k} = 0. \tag{34.185}$$

The decay amplitude is a rotational scalar, linear in each of the three polarization vectors (one quantum of each external spin-1 line). Linearity in $\boldsymbol{\varepsilon}$ means it has the form

$$\mathcal{M} = \boldsymbol{\varepsilon} \cdot \mathbf{A}, \tag{34.186}$$

where $\mathbf{A}$ is a vector built linearly from $\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2$ and the only momentum available in the rest frame, $\mathbf{k}$. The most general such vector, linear in both $\boldsymbol{\varepsilon}_1$ and $\boldsymbol{\varepsilon}_2$ and a genuine (polar) vector, is

$$\mathbf{A} = a\,(\boldsymbol{\varepsilon}_1 \cdot \boldsymbol{\varepsilon}_2)\,\mathbf{k} + b_1\,(\boldsymbol{\varepsilon}_2 \cdot \mathbf{k})\,\boldsymbol{\varepsilon}_1 + b_2\,(\boldsymbol{\varepsilon}_1 \cdot \mathbf{k})\,\boldsymbol{\varepsilon}_2 + c\,(\boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2), \tag{34.187}$$

with constants a, b_1, b_2, c. Transversality Eq. (34.185) kills the b_1 and b_2 terms outright. There remain

$$\mathbf{A} = a\,(\boldsymbol{\varepsilon}_1 \cdot \boldsymbol{\varepsilon}_2)\,\mathbf{k} + c\,(\boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2). \tag{34.188}$$

Now impose Bose symmetry: the two photons are identical bosons, so $\mathcal{M}$ must be invariant under the simultaneous exchange $(\mathbf{k}_1, \boldsymbol{\varepsilon}_1) \leftrightarrow (\mathbf{k}_2, \boldsymbol{\varepsilon}_2)$. In the rest frame this exchange sends $\mathbf{k} \to -\mathbf{k}$ and swaps $\boldsymbol{\varepsilon}_1 \leftrightarrow \boldsymbol{\varepsilon}_2$. Under it,

$$(\boldsymbol{\varepsilon}_1 \cdot \boldsymbol{\varepsilon}_2)\,\mathbf{k} \;\longrightarrow\; (\boldsymbol{\varepsilon}_2 \cdot \boldsymbol{\varepsilon}_1)\,(-\mathbf{k}) = -(\boldsymbol{\varepsilon}_1 \cdot \boldsymbol{\varepsilon}_2)\,\mathbf{k}, \qquad \boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2 \;\longrightarrow\; \boldsymbol{\varepsilon}_2 \times \boldsymbol{\varepsilon}_1 = -\,\boldsymbol{\varepsilon}_1 \times \boldsymbol{\varepsilon}_2, \tag{34.189}$$

so both surviving structures, and hence $\mathbf{A}$ and $\mathcal{M} = \boldsymbol{\varepsilon} \cdot \mathbf{A}$, are *odd* under photon exchange. Bose statistics demands $\mathcal{M}$ be *even*. The only amplitude that is both odd and even is zero: $a = c = 0$, hence $\mathcal{M} \equiv 0$. A massive spin-1 particle therefore has no amplitude to two photons. □

Physical Insight

The Landau–Yang obstruction is, at heart, an angular-momentum census of the kind Chapter 3 makes routine. Take the photons along $\pm z$. Each has helicity ± 1, so along the z axis the two-photon system can carry total angular-momentum projection $J_z = 0$ (equal helicities: the photon along $+z$ contributes $+1$ and, because the second photon moves along $-z$ with the same helicity, it contributes -1, summing to 0) or $J_z = \pm 2$ (opposite helicities give $+1$ from the $+z$ photon and another $+1$ from the $-z$ photon, summing to ± 2). The values $J_z = \pm 1$ never occur. A spin-1 parent must supply $J_z \in \{0, \pm 1\}$; the only overlap is $J_z = 0$. But the $J_z = 0$ two-photon state, $\frac{1}{\sqrt{2}}(|{+}{+}\rangle + |{-}{-}\rangle)$ once Bose-symmetrized and combined with the parity/exchange structure, is precisely the combination the explicit computation found to be antisymmetric, so it cannot couple to a single spin-1 state. The same census, done for $J = 0$ and $J = 2$, leaves those allowed — which is why the theorem forbids 1 but not 0 or 2.

The theorem applies verbatim to the production side by crossing: a spin-1 resonance cannot be produced in gluon–gluon or photon–photon fusion either

(two identical massless spin-1 initial bosons), which is independently relevant since the LHC produces the Higgs dominantly through gluon fusion. Observation of the 125 GeV state in $\gamma\gamma$ thus excludes $J = 1$ with no further assumptions.

Scalar versus pseudoscalar versus tensor: the $ZZ^* \to 4\ell$ angular distribution. Excluding 1 leaves 0^+, 0^-, and 2^+ to separate, and the diphoton channel cannot do it (it sees only that $J \neq 1$). The discriminator is the full kinematics of $h \to ZZ^* \to (\ell^+\ell^-)(\ell^+\ell^-)$: the two lepton pairs define two decay planes, and the parent's spin and parity are imprinted on the joint distribution of the Z and Z^* polar angles, the two dilepton invariant masses, and especially the azimuthal angle Φ between the two decay planes. The reason this distribution is sensitive is exactly Wigner's: the parent's spin state, projected onto the ZZ^* system, fixes the polarization mixture of the two Z bosons, and each Z then advertises its polarization through the angular distribution of its leptons (a transverse Z gives $\propto 1 + \cos^2\theta$, a longitudinal Z gives $\propto \sin^2\theta$).

The qualitative structure of the coupling to two Z's makes the discrimination transparent. Lorentz invariance allows three independent hZZ tensor structures, distinguished by their behaviour under parity. With Z momenta q_1, q_2 and polarizations $\varepsilon_1, \varepsilon_2$, the matrix element has the schematic form

$$\mathcal{M}(h \to ZZ) \;\propto\; \underbrace{a_1\, m_Z^2\, \varepsilon_1^*\cdot\varepsilon_2^*}_{0^+\ \text{(SM, tree)}} + \underbrace{a_2\,[(q_1\cdot q_2)(\varepsilon_1^*\cdot\varepsilon_2^*) - (q_1\cdot\varepsilon_2^*)(q_2\cdot\varepsilon_1^*)]}_{0^+\ \text{(higher-dim)}} + \underbrace{a_3\,\epsilon_{\mu\nu\rho\sigma}\,\varepsilon_1^{*\mu}\varepsilon_2^{*\nu}q_1^\rho q_2^\sigma}_{0^-\ \text{(CP-odd)}}, \tag{34.190}$$

where $\epsilon_{\mu\nu\rho\sigma}$ is the Levi-Civita tensor. The Standard Model is pure a_1: a single, momentum-independent, parity-even contact term, the direct descendant of the $m_Z^2\, h\, Z_\mu Z^\mu/v$ vertex of Eq. (34.127). The pseudoscalar structure a_3 carries an ϵ-tensor and is parity-odd, so it produces a distinctive azimuthal dependence: the inter-plane angle Φ enters as $\cos\Phi$ for 0^+ (from the interference of transverse polarizations) but as $\cos 2\Phi$ with a characteristic sign and the absence of the longitudinal-dominated Φ-independent piece for 0^-, while a spin-2 parent injects higher harmonics in the polar angles set by its five polarization states. A maximum-likelihood fit of the multidimensional angular distribution to the templates generated by each J^P hypothesis, using the parity-even Standard-Model structure a_1 as the null, discriminates the hypotheses. The outcome, by the mid-2020s, is unambiguous: the pure pseudoscalar 0^- and the tested spin-2 and spin-1 hypotheses are excluded at well above 99% confidence, while 0^+ describes the data. Small admixtures of the anomalous structures a_2, a_3 are constrained to be tiny.

Caution

The angular fit tests *tensor structure*, hence CP properties of the hZZ vertex, not the spin alone. A nonzero a_3 would signal CP violation in the

Higgs–gauge coupling — a 0^- admixture — and is searched for as such; current limits make any CP-odd component of the hVV coupling small. The complementary CP test in the *fermionic* sector uses the $h \to \tau^+\tau^-$ decay-plane correlation, which probes the relative weight of the scalar $\bar\tau\tau$ and pseudoscalar $\bar\tau i\gamma^5\tau$ Yukawa structures; it too is consistent with the pure-scalar Standard Model. "Spin-parity 0^+" is therefore shorthand for a battery of structure tests, all passed.

34.5.4 The total width and the off-shell method

The Standard Model predicts the Higgs total width by summing the partial widths of the previous section,

$$\Gamma_h^{\rm SM} = \sum_X \Gamma(h \to X) \approx 4.1 \text{ MeV} \quad \text{at} \quad m_h = 125 \text{ GeV}, \tag{34.191}$$

dominated by $h \to b\bar b$ ($\sim 58\%$), with $WW^*, gg, \tau\tau, c\bar c, ZZ^*, \gamma\gamma$ making up the rest. This is an extraordinarily small width for so heavy a particle — a fractional width $\Gamma_h/m_h \sim 3 \times 10^{-5}$ — because the largest available coupling, the top Yukawa, cannot act at tree level ($m_h < 2m_t$ closes $h \to t\bar t$), leaving only the modest b Yukawa and the loop- or off-shell-suppressed gauge channels to drain the state. As Theorem 34.19 showed, 4.1 MeV is hundreds of times narrower than any detector's mass resolution, so the line shape is useless as a direct lifetime measurement: the observed peak width is instrumental. A different idea is needed, and it is one of the most elegant in collider physics.

On-shell rates measure g^2/Γ_h, not Γ_h. Consider any production-times-decay rate proceeding through an on-shell Higgs, e.g. $gg \to h \to ZZ^*$. Near the resonance the cross section is governed by the relativistic Breit–Wigner propagator $|\mathcal{D}(s)|^2$ with $\mathcal{D}(s) = 1/(s - m_h^2 + i m_h \Gamma_h)$, weighted by the production coupling-squared g_{hgg}^2 and the decay coupling-squared g_{hZZ}^2:

$$\sigma_{\rm on} \propto g_{hgg}^2\, g_{hZZ}^2 \int \mathrm{d}s\, \frac{1}{(s-m_h^2)^2 + m_h^2\Gamma_h^2} \times (\text{slowly varying}). \tag{34.192}$$

The integral is dominated by the narrow region $|s - m_h^2| \lesssim m_h\Gamma_h$, and the slowly-varying factors (phase space, parton luminosities) may be pulled out at $s = m_h^2$. The remaining integral is elementary,

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}s}{(s-m_h^2)^2 + m_h^2\Gamma_h^2} = \Big[\frac{1}{m_h\Gamma_h}\arctan\frac{s-m_h^2}{m_h\Gamma_h}\Big]_{-\infty}^{\infty} = \frac{\pi}{m_h\Gamma_h}, \tag{34.193}$$

so the integrated on-shell rate scales as

$$\sigma_{\rm on} \propto \frac{g_{hgg}^2\, g_{hZZ}^2}{\Gamma_h}. \tag{34.194}$$

This is the crux: an on-shell measurement determines the combination $g_{hgg}^2 g_{hZZ}^2/\Gamma_h$, and *cannot* separate the couplings from the width. If every coupling were doubled and the width quadrupled, all on-shell rates would be unchanged. Resonance peaks alone leave a flat direction in (g, Γ_h) space, and that is why the κ fit of Section 34.5.2 needs an extra input — conventionally the assumption $\kappa_h^2 = \sum \mathrm{BR}_X^{\mathrm{SM}} \kappa_X^2$, i.e. no decays beyond the known ones — to pin Γ_h. To measure Γ_h without that assumption, the flat direction must be broken by data.

Off-shell production breaks the degeneracy. The breaking comes from the far off-shell tail. Consider the *same* couplings driving $gg \to h^* \to ZZ$ with $\sqrt{s} \gg 2m_Z$, well above the ZZ threshold where two real Z's are produced through a highly virtual Higgs. There the Breit–Wigner denominator is dominated by s, not by the width:

$$|\mathcal{D}(s)|^2 = \frac{1}{(s - m_h^2)^2 + m_h^2 \Gamma_h^2} \xrightarrow{s \gg m_h^2,\, m_h \Gamma_h} \frac{1}{s^2}, \tag{34.195}$$

which is *independent of* Γ_h. Hence the off-shell cross section scales as

$$\sigma_{\mathrm{off}} \propto g_{hgg}^2 \, g_{hZZ}^2, \tag{34.196}$$

the product of the same coupling-squared factors, with no width in sight. The ratio of off-shell to on-shell rates therefore isolates the width:

$$\frac{\sigma_{\mathrm{off}}}{\sigma_{\mathrm{on}}} \propto \frac{g_{hgg}^2 g_{hZZ}^2}{g_{hgg}^2 g_{hZZ}^2/\Gamma_h} = \Gamma_h. \tag{34.197}$$

Measuring both the resonant (σ_{on}) and the high-mass off-shell (σ_{off}) ZZ yields and forming their ratio determines Γ_h directly, with the unknown couplings cancelling in the ratio. The off-shell tail is small — a few percent of the on-shell rate — but it is there, enhanced by the opening of the ZZ phase space and by the longitudinal Z contributions that grow with energy, and it carries the width information that the peak cannot.

Caution

The off-shell method buys a width measurement at the cost of an assumption that the on-shell line shape did not require: that the h-g-g and h-Z-Z couplings are the *same* at $\sqrt{s} \approx 125$ GeV and at $\sqrt{s} \gtrsim 2m_Z \approx 200$–400 GeV. New physics that modified the couplings' energy dependence — a form factor, a nearby heavier state — would spoil the extraction. With that caveat stated, the measurement is robust, and the off-shell analyses of the LHC experiments return a width consistent with the Standard-Model 4.1 MeV, with a central value and uncertainty at the few-MeV level (e.g. $\Gamma_h \sim 3$–4 MeV with errors of comparable size). This is one of the rare cases

in which a quantity utterly inaccessible to direct resolution is nonetheless measured, by exploiting the analytic structure of the propagator itself.

34.5.5 The Higgs self-coupling: the last unmeasured vertex

Everything so far tested couplings of the Higgs to *other* particles. The Standard Model makes one more prediction, structurally distinct and so far only weakly probed: the strength with which the Higgs couples to *itself*. This is the direct experimental window onto the shape of the symmetry-breaking potential, the very object whose instability triggered electroweak breaking in the first place.

The trilinear coupling from the potential. Recall the Higgs potential of Chapters 30 and 31, written for the doublet H,

$$V(H) = -\mu^2\, H^\dagger H + \lambda\,(H^\dagger H)^2, \qquad \mu^2, \lambda > 0, \tag{34.198}$$

minimized at $H^\dagger H = \mu^2/2\lambda \equiv v^2/2$, so $v^2 = \mu^2/\lambda$. In unitary gauge $H^\dagger H = \frac{1}{2}(v+h)^2$, and substituting,

$$\begin{aligned} V &= -\frac{\mu^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4 \\ &= -\frac{\lambda v^2}{2}(v+h)^2 + \frac{\lambda}{4}(v+h)^4 \qquad (\mu^2 = \lambda v^2) \\ &= \underbrace{-\frac{\lambda v^4}{4}}_{\text{const}} + \underbrace{(-\lambda v^3 + \lambda v^3)h}_{=0\ \text{(tadpole cancels)}} + \lambda v^2\, h^2 + \lambda v\, h^3 + \frac{\lambda}{4} h^4, \end{aligned} \tag{34.199}$$

where the linear term cancels (the defining property of expanding about the true minimum) and the quadratic term gives the Higgs mass through $\frac{1}{2} m_h^2 h^2 = \lambda v^2 h^2$, i.e.

$$m_h^2 = 2\lambda v^2, \tag{34.200}$$

exactly as in Chapter 30. The cubic term is the *trilinear self-coupling*: writing $V \supset \lambda_3\, h^3$,

$$\lambda_3 = \lambda v = \frac{m_h^2}{2v}, \tag{34.201}$$

where the second equality eliminates the potential parameter λ in favour of the now-measured Higgs mass via Eq. (34.200). The quartic self-coupling is likewise fixed, $V \supset \frac{\lambda}{4} h^4 = \frac{m_h^2}{8v^2} h^4$. The content of Eq. (34.201) is that, in the minimal Standard Model, the self-coupling is *not* a new free parameter: once m_h and v are known, λ_3 is predicted with no remaining freedom. Measuring it independently and checking Eq. (34.201) is a direct test of the assumed quartic form of the potential — the shape of the Mexican hat that underlies the entire mechanism.

Worked Example 34.24 (Numerical value of the self-couplings)**.** With $m_h = 125.2$ GeV and $v = 246$ GeV, the quartic coupling is

$$\lambda = \frac{m_h^2}{2v^2} = \frac{(125.2)^2}{2\,(246)^2} = \frac{15{,}675}{121{,}032} \approx 0.13, \tag{34.202}$$

a comfortably perturbative number. The dimensionful trilinear coupling is

$$\lambda_3 = \frac{m_h^2}{2v} = \frac{(125.2)^2}{2\,(246)} = \frac{15{,}675}{492} \approx 31.9 \text{ GeV}, \tag{34.203}$$

or equivalently $\lambda_3 = \lambda v \approx 0.13 \times 246 \approx 32$ GeV. These two numbers are not independent inputs; they are consequences of m_h and v. A measured deviation of either from these values — conventionally parametrized by a modifier $\kappa_\lambda \equiv \lambda_3/\lambda_3^{\mathrm{SM}}$ with $\kappa_\lambda = 1$ in the Standard Model — would mean the potential is not the minimal quartic Eq. (34.198), with sweeping consequences for the nature of the electroweak phase transition and for the stability of the vacuum.

How one measures it, and why it is hard. The trilinear vertex is probed by producing *two* Higgs bosons at once, since the h^3 coupling enters the amplitude for $gg \to hh$ through a diagram in which a single off-shell Higgs splits into two (Fig. 34.9, left). That diagram interferes with a second one in which the two Higgs bosons attach independently to a top-quark box (Fig. 34.9, right), and the interference is *destructive* in the Standard Model, which both reduces the di-Higgs cross section and makes it a sensitive — if low-statistics — function of λ_3. Schematically,

$$\sigma(gg \to hh) \;\sim\; \left|\, \kappa_\lambda\, \mathcal{A}_{\triangle} + \mathcal{A}_{\square} \,\right|^2, \tag{34.204}$$

with $\mathcal{A}_{\triangle}$ the triangle (self-coupling) amplitude carrying the one power of λ_3 and $\mathcal{A}_{\square}$ the box amplitude independent of it. The Standard-Model di-Higgs cross section at the LHC is minuscule — roughly a thousand times smaller than single-Higgs production — so the measurement is statistics-limited and proceeds through final states like $b\bar{b}\gamma\gamma$, $b\bar{b}\tau\tau$, and $b\bar{b}b\bar{b}$ that balance branching ratio against background.

Current sensitivity (frontier). As of the mid-2020s the di-Higgs rate is not yet observed; the LHC experiments set upper limits of a few times the Standard-Model expectation and constrain the modifier to a broad interval, roughly

$$-1 \;\lesssim\; \kappa_\lambda \;\lesssim\; 6.5 \qquad (95\%\ \mathrm{CL},\ \text{combined, mid-2020s}), \tag{34.205}$$

so the Standard-Model value $\kappa_\lambda = 1$ is allowed but very far from being verified — the existence of the self-interaction itself is not yet established. This is the genuine open frontier of Higgs physics: the High-Luminosity LHC is projected

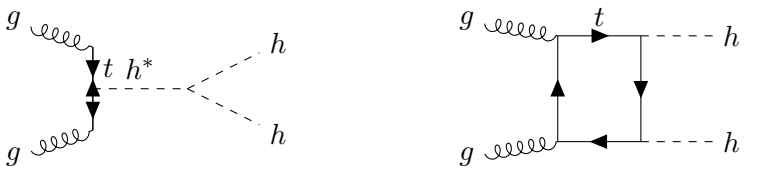


Figure 34.9: The two leading contributions to di-Higgs production $gg \to hh$ at the LHC. *Left:* the triangle, in which gluons fuse through a top loop into a single off-shell Higgs h^* that splits via the trilinear vertex λ_3 into two Higgs bosons — the only diagram carrying the self-coupling. *Right:* the box, in which each Higgs attaches independently to the top loop, independent of λ_3. The two interfere destructively in the Standard Model, Eq. (34.204), suppressing the rate and making it a sensitive probe of λ_3.

to reach evidence-level sensitivity, constraining κ_λ to roughly the 50% level and possibly excluding $\kappa_\lambda = 0$, while a future high-energy lepton or hadron collider would be needed to measure λ_3 to the few-percent precision that would truly test the shape of the potential. Until then, the most consequential coupling in the theory — the one that fixes the electroweak vacuum and governs whether the early-universe transition was first order — remains the least known. We flag this honestly: the gauge and Yukawa couplings of the Higgs are measured and on the predicted coupling–mass line; the self-coupling is, as of this writing, only bounded.

Physical Insight

The asymmetry in our knowledge mirrors the structure of the theory. The couplings to fermions and gauge bosons are couplings to the *condensate*: they are read off the mass terms and tested wherever a heavy particle is made. The self-coupling is a coupling to the *curvature of the potential at its minimum* — it asks not "how much condensate is here?" but "how stiff is the vacuum against changing the amount of condensate?" That question can only be answered by making the field fluctuate twice over, which is why it requires the rare double-Higgs process. Measuring λ_3 is the experimental program that would finally confirm that the object discovered in 2012 is not merely a scalar with mass-proportional couplings, but specifically the radial mode of the very Mexican-hat potential that the entire Higgs mechanism was built upon.

With the gauge and Yukawa couplings of the Higgs validated point by point along the coupling–mass line, its spin-parity established as 0^+, its width measured by the off-shell method, and only its self-coupling left as an open frontier, the electroweak-plus-Higgs sector of the Standard Model stands confirmed at the

per-cent level and below. The remaining precision frontier lies elsewhere: in the *flavor* sector, where the same Yukawa matrices that fixed the masses also control rare, loop-induced transitions between quark generations. The next section turns to those flavor-changing neutral currents and to the unitarity triangle, where the CKM structure imposed in Chapter 32 is overconstrained by experiment, and where a few of the live tensions of the mid-2020s reside.

34.6 Flavor-Changing Neutral Currents and the Unitarity Triangle

The electroweak and Higgs sectors of the preceding sections tested the Standard Model where it is loudest—on resonance at the Z pole, in the calculable production and decay of the Higgs. Those tests pin down the gauge and scalar couplings. They say almost nothing, however, about the one corner of the parameter list (Chapter 33, Section 33.6) that the gauge bosons cannot reach directly: the Yukawa sector, whose physical residue after diagonalization is the four numbers of the Cabibbo–Kobayashi–Maskawa matrix—three mixing angles and one CP-violating phase (Chapter 32). Flavor is therefore the remaining *independent* test of the same theory. It is also the most stringent, because the cleanest flavor observables are *loop-induced*: tree-level flavor-changing neutral currents (FCNC) are forbidden outright by the gauge structure (Theorem 32.28), so a neutral-meson mass splitting or a CP asymmetry is a direct measurement of a one-loop amplitude. A handful of CKM inputs must reproduce neutral-kaon and B-meson mixing, indirect CP violation in the kaon, and large time-dependent asymmetries in the B system—all at once. This section carries out the central computation that makes those predictions quantitative, the $\Delta F = 2$ box, and then reads off the numbers that overconstrain the unitarity triangle.

Chapter 32 did the structural work: it proved the GIM cancellation (Theorem 32.29), assembled the box's flavor sum (Eq. (32.143)), set up the two-state mixing formalism with its off-diagonal M_{12} (Eq. (32.167)), and drew the unitarity triangle whose apex $(\bar{\rho}, \bar{\eta})$ is overdetermined (Section 32.6.6). What it deferred—explicitly—was the loop integral itself: the value of the Inami–Lim function S_0 was quoted (Eq. (32.147)) but not derived. We supply that derivation here, because it is exactly the kind of calculable quantum correction on which the precision program rests, and because it is a clean generalization of the gauge-sector vertices of QED to a two-current loop. We then turn the result into the measured mass differences, into ϵ_K, and into $\sin 2\beta$, and close on the overconstraint as a single picture.

34.6.1 The $\Delta F = 2$ box as a local operator

The object to compute is the dispersive part of the off-diagonal element M_{12} of the neutral-meson mass matrix (Eq. (32.155)), generated by the box diagram of Fig. 32.5b in which a down-type meson constituent pair $\bar{d}_j d_i$ is turned into $\bar{d}_i d_j$ by two W exchanges and two internal up-type quark lines $\alpha, \beta \in \{u, c, t\}$. We work with B_d, where a single internal flavor (the top) dominates and the arithmetic is cleanest; the kaon differs only in which CKM products weight the sum.

The external quarks are light ($m_b, m_d \ll M_W$) and, for the purpose of extracting the local four-quark operator that the heavy W and top generate, we may set the external momenta and the external masses to zero. This is the standard *operator-product* logic: the box, viewed from distances much larger than $1/M_W$, looks like a contact interaction, and its leading term is a dimension-six $\Delta B = 2$ operator whose coefficient we are computing. With external momenta neglected, momentum conservation at the four vertices leaves a *single* independent loop momentum k flowing through all four heavy propagators: the two W lines and the two internal up-quark lines all carry k. The box integrand is therefore a single-scale object,

$$\int \frac{\mathrm{d}^4 k}{(2\pi)^4} \, \frac{\mathcal{N}(k)}{(k^2 - M_W^2)^2 \, (k^2 - m_\alpha^2) \, (k^2 - m_\beta^2)}, \tag{34.206}$$

with the Dirac/tensor numerator $\mathcal{N}(k)$ fixed by the vertices. This single observation—all four propagators sharing one momentum—is what makes the Inami–Lim function elementary rather than a genuine two-scale box.

The vertices and the chiral projection. Each charged-current vertex is, from Eq. (31.129) with the mixing matrix installed, $\frac{ig}{\sqrt{2}} \gamma^\mu P_L \, V_{\alpha i}$ (or its conjugate), with $P_L = \frac{1}{2}(1 - \gamma^5)$ the left-handed projector of Chapter C. Every external and internal quark thus enters the box through a P_L. Consider one fermion line: an external d, a vertex $\gamma^\mu P_L$, the internal up propagator $i(\not{k} + m_\alpha)/(k^2 - m_\alpha^2)$, a second vertex $\gamma^\nu P_L$, and an external b. The Dirac string on that line is

$$\gamma^\mu P_L \, (\not{k} + m_\alpha) \, \gamma^\nu P_L = \gamma^\mu (\not{k} \, P_R + m_\alpha P_L) \gamma^\nu P_L = \gamma^\mu \not{k} \, \gamma^\nu P_L, \tag{34.207}$$

where we used $P_L \not{k} = \not{k} \, P_R$ (since γ^5 anticommutes with γ^μ) and $P_R \gamma^\nu P_L = \gamma^\nu P_L P_L = \gamma^\nu P_L$ while $P_L \gamma^\nu P_L = \gamma^\nu P_R P_L = 0$. The mass term on the internal line has dropped: the transverse part of the box knows the internal quark only through the *denominator* $k^2 - m_\alpha^2$. (The longitudinal part of the W propagator, treated below, restores an explicit mass dependence, and that is where the heavy top makes its presence felt.)

The operator is a single $(V-A)\otimes(V-A)$ structure. In 't Hooft–Feynman gauge the transverse W propagator is $-ig_{\mu\nu}/(k^2-M_W^2)$, contracting the γ^μ on one line with the γ_μ on the other. The two fermion lines then carry

$$[\bar b\,\gamma^\mu \not{k}\,\gamma^\nu P_L\,d]\;[\bar b\,\gamma_\mu \not{k}\,\gamma_\nu P_L\,d] = k_\rho k_\sigma\;[\bar b\,\gamma^\mu\gamma^\rho\gamma^\nu P_L\,d]\;[\bar b\,\gamma_\mu\gamma^\sigma\gamma_\nu P_L\,d]. \tag{34.208}$$

Under the symmetric loop integration only the trace part of $k_\rho k_\sigma$ survives, $k_\rho k_\sigma \to \tfrac14 k^2 g_{\rho\sigma}$, and we are left with the contraction $\gamma^\mu\gamma^\rho\gamma^\nu P_L \otimes \gamma_\mu\gamma_\rho\gamma_\nu P_L$. The chiral product of three gamma matrices contracted on both ends collapses by the identity (proved in Chapter C, a direct consequence of $\gamma^\mu\gamma^\rho\gamma^\nu = g^{\mu\rho}\gamma^\nu - g^{\mu\nu}\gamma^\rho + g^{\rho\nu}\gamma^\mu + i\varepsilon^{\mu\rho\nu\sigma}\gamma_\sigma\gamma^5$ and the chiral projector)

$$(\gamma^\mu\gamma^\rho\gamma^\nu P_L)\otimes(\gamma_\mu\gamma_\rho\gamma_\nu P_L) = 4\,(\gamma^\lambda P_L)\otimes(\gamma_\lambda P_L). \tag{34.209}$$

The two interfering box contractions (direct and crossed) and a Fierz rearrangement (Chapter C) leave a *single* local operator,

$$Q_{LL} = (\bar b\,\gamma^\lambda P_L\,d)(\bar b\,\gamma_\lambda P_L\,d) = (\bar b_L\gamma^\lambda d_L)(\bar b_L\gamma_\lambda d_L), \tag{34.210}$$

the $(V-A)\otimes(V-A)$ four-quark operator. All the loop dynamics has been reduced to one dimensionless number multiplying Q_{LL}: the scalar function of the mass ratios that we now compute.

34.6.2 The loop integral and the Inami–Lim function

Write $x_\alpha \equiv m_\alpha^2/M_W^2$ as in Eq. (32.135). Collecting the couplings $(\frac{ig}{\sqrt2})^4 = \frac{g^4}{4}$, the two W propagators, the two internal-quark numerators, and the CKM products $\lambda_\alpha\lambda_\beta$ of Eq. (32.85), the box amplitude is $\propto g^4\sum_{\alpha\beta}\lambda_\alpha\lambda_\beta\,F(x_\alpha,x_\beta)\,Q_{LL}$, with the loop function

$$F(x_\alpha,x_\beta) = M_W^4\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{\tfrac14 k^2 + (\text{longitudinal})}{(k^2-M_W^2)^2\,(k^2-m_\alpha^2)\,(k^2-m_\beta^2)}, \tag{34.211}$$

the factor M_W^4 inserted so that F is dimensionless. We treat the two pieces of the numerator in turn.

The transverse piece. The $\frac14 k^2$ term is convergent and elementary. Wick-rotate ($k^0\to ik_E^0$, $k^2\to -t$ with $t = k_E^2 \ge 0$, $\mathrm{d}^4k \to i\,\mathrm{d}^4k_E$, and $\mathrm{d}^4k_E/(2\pi)^4 = t\,\mathrm{d}t/16\pi^2$). For equal internal masses $m_\alpha = m_\beta \equiv m$ (the top-dominated B_d case), with $a \equiv M_W^2$ and $b \equiv m^2$,

$$\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{k^2}{(k^2-a)^2(k^2-b)^2} = \frac{-i}{16\pi^2}\,P_2, \qquad P_2 \equiv \int_0^\infty \frac{t^2\,\mathrm{d}t}{(t+a)^2(t+b)^2}. \tag{34.212}$$

The radial integral is done by partial fractions. Writing $t^2/[(t+a)^2(t+b)^2] = A/(t+a) + B/(t+a)^2 + C/(t+b) + D/(t+b)^2$, the residues are $B = a^2/(b-a)^2$,

$D = b^2/(b-a)^2$, and $A = -2ab/(b-a)^3 = -C$, so that the simple-pole parts combine into a convergent logarithm,

$$P_2 = \frac{a+b}{(b-a)^2} - \frac{2ab}{(b-a)^3}\ln\frac{b}{a} = \frac{1}{M_W^2}\left[\frac{1+x}{(x-1)^2} - \frac{2x\,\ln x}{(x-1)^3}\right], \qquad x = \frac{b}{a} = x_t. \tag{34.213}$$

This is the entire transverse contribution. As a check on the dynamics: for a heavy top ($x \to \infty$) it falls as $P_2 \to 1/(M_W^2 x) = 1/m_t^2$, so the transverse box does *not* grow with the top mass. The well-known top enhancement of mixing therefore cannot come from here; it must come from the longitudinal part of the W.

The longitudinal piece and the heavy-top growth. The full W propagator carries, beyond $-ig_{\mu\nu}$, the longitudinal term $+ik_\mu k_\nu/M_W^2$. By the Goldstone-boson equivalence theorem (Theorem 30.15), at energies $\gg M_W$ a longitudinal W is the would-be charged Goldstone boson, whose coupling to quarks is Yukawa-like and therefore proportional to the internal quark mass $\propto m_\alpha/M_W$. Two such couplings on the two lines bring a factor $m_\alpha m_\beta/M_W^2$ that survives the chiral projection (the Goldstone vertex flips chirality, so now it is the *mass* part $m_\alpha P_L$ of Eq. (34.207) that is kept). The longitudinal contributions are then the same family of single-scale integrals as Eq. (34.212) but with extra powers of k^2/M_W^2 and explicit x_α factors, e.g.

$$M_W^2\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{x\,k^2}{(k^2-M_W^2)^2(k^2-m^2)^2} \quad\text{and}\quad M_W^4\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{x^2}{(k^2-M_W^2)^2(k^2-m^2)^2}, \tag{34.214}$$

evaluated by the same partial-fraction reduction (the radial integrals are the $P_n = \int_0^\infty t^n \mathrm{d}t/[(t+a)^2(t+b)^2]$, with P_0, P_1 convergent and the would-be-divergent P_3 piece removed by the GIM subtraction below; the master formulas are collected in Chapter D). It is the explicit x and x^2 prefactors here that produce a contribution growing *linearly* with x_t at large top mass—the physical statement that mixing is sensitive to m_t^2, and the reason B–$\bar{B}$ mixing was, historically, the first evidence that the top is heavy.

Assembling S_0 and the GIM subtraction. Summing the transverse and longitudinal radial integrals for a single internal flavor gives the *diagonal* Inami–Lim function $S_0(x) \equiv F(x,x)$. The reduction returns exactly

$$\boxed{S_0(x) = x\left[\frac{1}{4} + \frac{9}{4(1-x)} - \frac{3}{2(1-x)^2}\right] - \frac{3}{2}\left(\frac{x}{1-x}\right)^3\ln x} \tag{34.215}$$

in agreement with Eq. (32.147) and with Inami and Lim [274]. Three checks confirm that this is the right function and that the GIM logic of Theorem 32.29 is operating term by term.

(i) Small-x limit (GIM). Expanding Eq. (34.215) about $x = 0$, the bracket tends to $\frac{1}{4} + \frac{9}{4} - \frac{3}{2} = 1$ and the logarithmic term is $O(x^3 \ln x)$, so

$$S_0(x) = x + O(x^2 \ln x). \tag{34.216}$$

The function vanishes *linearly* with the internal mass-squared ratio. This is the GIM cancellation made quantitative: $S_0(0) = 0$, so the flavor-blind part of the box vanishes when summed against the unitarity relation $\sum_\alpha \lambda_\alpha = 0$ (Eq. (32.144)), and the surviving amplitude is controlled by mass splittings exactly as Eq. (32.137) demanded. For the box, the would-be ultraviolet growth of the individual longitudinal integrals (the P_3 pieces) is precisely the x-independent, α-independent part that this subtraction removes—GIM doubles as the device that renders the unitary-gauge box finite.

(ii) Large-x limit (heavy top). For $x \to \infty$ the bracket $\to x[\frac{1}{4} - 0 - 0] = \frac{x}{4}$ from the leading x, while the log term is $-\frac{3}{2} \ln x$ and subleading; hence

$$S_0(x) \to \frac{x}{4}, \qquad x \to \infty, \tag{34.217}$$

the linear growth promised by the longitudinal/Goldstone contribution: a heavy top *enhances* mixing as m_t^2/M_W^2. The transverse piece alone could never have produced this; the equivalence theorem did.

(iii) The physical value. With the running top mass $m_t(m_t) \simeq 163$ GeV and $M_W = 80.4$ GeV, $x_t = (163/80.4)^2 \simeq 4.11$, and inserting into Eq. (34.215) the first (bracket) term is $x_t(0.25 - 0.724 - 0.155) = 4.11 \times (-0.629) = -2.58$ and the logarithmic term contributes $+4.89$, giving

$$S_0(x_t) \simeq 2.31. \tag{34.218}$$

This single number drives every short-distance $\Delta F = 2$ prediction below.

Physical Insight

The box is the cleanest illustration in the Standard Model of why precision flavor physics is short-distance physics. Three independent suppressions stack: the loop factor $g^4/(16\pi^2 M_W^2) \sim G_F^2 M_W^2$, the small CKM products λ_α, and the GIM factor $S_0(x_\alpha) \sim x_\alpha$ from the mass splittings. The same unitarity $V^\dagger V = \mathbf{1}$ that forbids tree FCNC also guarantees $S_0(0) = 0$, so a rare neutral-meson transition is a direct readout of the heaviest internal mass and of the CKM phases—both inaccessible at tree level. The heavy top, far from decoupling, dominates through the longitudinal W: this is the Higgs sector (the Goldstones) reappearing inside a loop, a quantitative echo of the non-decoupling that the same theory shows in the ρ parameter.

34.6.3 From the box to the mass splitting

The local operator Q_{LL} with its computed coefficient *is* the effective $\Delta B = 2$ Hamiltonian, whose meson matrix element gives the dispersive off-diagonal element $M_{12} = \left\langle B^0 \middle| \mathcal{H}_{\text{eff}}^{\Delta B=2} \middle| \bar{B}^0 \right\rangle / (2m_B)$ (Eq. (32.155)). Restoring the prefactors of Section 34.6.2—$g^4/4 = 4G_F^2 M_W^4$ from $G_F/\sqrt{2} = g^2/8M_W^2$ (Eq. (31.145)), the loop $1/16\pi^2$, the top-dominated CKM weight $\lambda_t = V_{tb}^* V_{td}$, a short-distance QCD factor η_B from running the operator from M_W down to m_b, and the hadronic matrix element $\left\langle B^0 \middle| Q_{LL} \middle| \bar{B}^0 \right\rangle = \frac{2}{3} m_B f_B^2 \hat{B}_B$—reproduces precisely the master formula assembled in Eq. (32.167),

$$M_{12}^{(B_d)} = \frac{G_F^2 M_W^2}{12\pi^2}\, \eta_B\, m_B f_B^2 \hat{B}_B\, S_0(x_t)\, \left(V_{tb}^* V_{td}\right)^2. \tag{34.219}$$

We have now derived every factor in it. The mass splitting follows from $\Delta m = 2|M_{12}|$ (Eq. (32.164), valid because $|\Gamma_{12}| \ll |M_{12}|$ for the B system):

$$\Delta m_d = \frac{G_F^2 M_W^2}{6\pi^2}\, \eta_B\, m_B f_B^2 \hat{B}_B\, S_0(x_t)\, |V_{tb}^* V_{td}|^2. \tag{34.220}$$

The only quantities not yet computed in this book are the QCD short-distance factor $\eta_B \simeq 0.55$ and the nonperturbative matrix element $f_B^2 \hat{B}_B$, which requires lattice QCD; we take both as inputs and discharge their first-principles computation to Chapter 43. With $f_{B_d}\sqrt{\hat{B}_{B_d}} \simeq 0.21$ GeV, $m_B = 5.28$ GeV, $S_0(x_t) = 2.31$, and $|V_{td}| \simeq 8.1 \times 10^{-3}$, Eq. (34.220) returns $\Delta m_d \simeq 0.5\ \text{ps}^{-1}$, matching the measured $0.5065\ \text{ps}^{-1}$ [382].

Worked Example 34.25 (The ratio $\Delta m_d/\Delta m_s$ and $|V_{td}/V_{ts}|$)**.** The single-meson prediction Eq. (34.220) is only as good as the lattice input $f_B^2 \hat{B}_B$, whose uncertainty was historically tens of percent. The masterstroke of the B_s system is that the *ratio* of the two mass splittings cancels almost all of it. Writing Eq. (34.220) for both B_d (internal CKM weight $|V_{td}|$) and B_s ($|V_{ts}|$), the common factors $G_F^2 M_W^2 \eta_B S_0(x_t)/6\pi^2$ divide out and only the $SU(3)$-breaking ratio of hadronic parameters,

$$\xi \equiv \frac{f_{B_s}\sqrt{\hat{B}_{B_s}}}{f_{B_d}\sqrt{\hat{B}_{B_d}}}, \tag{34.221}$$

survives, and it is close to unity and computed on the lattice far more precisely than either factor alone (the valence light-quark difference is a small correction to an otherwise identical matrix element):

$$\frac{\Delta m_d}{\Delta m_s} = \frac{m_{B_d}}{m_{B_s}}\, \frac{1}{\xi^2}\, \left|\frac{V_{td}}{V_{ts}}\right|^2. \tag{34.222}$$

Insert the measurements $\Delta m_d = 0.5065$ ps^{-1}, $\Delta m_s = 17.765$ ps^{-1} [382] (note Δm_s is some 35× larger, the geometric statement that $|V_{ts}| \gg |V_{td}|$), the masses $m_{B_d} = 5.280$ GeV, $m_{B_s} = 5.367$ GeV, and the lattice value $\xi = 1.21$. Then

$$\begin{aligned} \frac{\Delta m_d}{\Delta m_s} &= \frac{0.5065}{17.765} = 2.851 \times 10^{-2}, \\ \left|\frac{V_{td}}{V_{ts}}\right|^2 &= \xi^2 \, \frac{m_{B_s}}{m_{B_d}} \, \frac{\Delta m_d}{\Delta m_s} = (1.464)(1.0165)(2.851 \times 10^{-2}) = 4.24 \times 10^{-2}, \\ \left|\frac{V_{td}}{V_{ts}}\right| &= 0.206. \end{aligned} \tag{34.223}$$

This is one of the cleanest determinations of $|V_{td}|$ in the program. Translated to the triangle, it fixes the side from the $(1,0)$ vertex to the apex, $R_t = |V_{td}/V_{ts}|/\lambda = 0.206/0.225 = 0.915$ (Eq. (32.198)). For comparison, the apex coordinates from the global fit give $R_t = \sqrt{(1-\bar\rho)^2 + \bar\eta^2} = \sqrt{0.841^2 + 0.352^2} = 0.912$. The oscillation frequencies of two different mesons, fed through a one-loop box dominated by the top, land on the same triangle side as the rest of the flavor data to better than a percent.

34.6.4 Indirect CP violation: ϵ_K and the Jarlskog invariant

For the kaon the same box Eq. (34.219) carries a different CKM weight, the double sum over $\lambda_\alpha = V_{\alpha d}^* V_{\alpha s}$ (Eq. (32.145)), and the absorptive part is not negligible. Its *imaginary* part is the engine of indirect CP violation. From the master result of Eq. (32.173), $\epsilon_K \simeq e^{i\phi_{\rm SW}} \sin\phi_{\rm SW}\, \mathrm{Im}\, M_{12}^{(K)}/\Delta m_K$ with the superweak phase $\phi_{\rm SW} \simeq 43.5°$ fixed by measured mass and width differences alone. What is new here is the content of $\mathrm{Im}\, M_{12}^{(K)}$: by Eq. (34.219) the box makes it proportional to the imaginary part of a product of two CKM factors,

$$\mathrm{Im}\, M_{12}^{(K)} \;\propto\; \mathrm{Im}\Big[\lambda_t^2\, S_0(x_t) + 2\lambda_c\lambda_t\, S_0(x_c, x_t) + \lambda_c^2\, S_0(x_c)\Big], \tag{34.224}$$

and the imaginary parts of $\lambda_c\lambda_t$ and λ_t^2 are, up to sign, the Jarlskog quartet $\mathrm{Im}[V_{cd}^* V_{cs} V_{td}^* V_{ts}] = \pm J$ identified in Eq. (32.146) and Theorem 32.20. Hence

$$\boxed{\mathrm{Im}\, M_{12}^{(K)} \;\propto\; J\,,} \qquad \Longrightarrow \qquad |\epsilon_K| \;\propto\; J \;\propto\; \bar\eta, \tag{34.225}$$

the precise statement that indirect CP violation in the kaon is controlled by the one rephasing-invariant measure of the CKM phase. If $J = 0$ (a real CKM matrix, a flat triangle), ϵ_K vanishes identically: there is no CP violation without the phase. Carrying the proportionality to a constraint, the standard expression

$$|\epsilon_K| = \frac{G_F^2 f_K^2 m_K M_W^2}{6\sqrt{2}\,\pi^2\,\Delta m_K}\, \hat B_K\, \mathrm{Im}\Big[\lambda_c^2 \eta_{cc} S_0(x_c) + \lambda_t^2 \eta_{tt} S_0(x_t) + 2\lambda_c\lambda_t \eta_{ct} S_0(x_c, x_t)\Big] \tag{34.226}$$

defines, in the $(\bar\rho,\bar\eta)$ plane, a curve of the schematic form $\bar\eta\,[(1-\bar\rho)A^2\eta_{tt}S_0(x_t)+P_c]\,A^2=\text{const}$—a hyperbola, because $\mathrm{Im}(\lambda_t^2)\propto\bar\eta(1-\bar\rho)$ to leading Wolfenstein order. With the hadronic input $\hat B_K\simeq 0.76$ (lattice, Chapter 43), the QCD factors η_{ij}, and the global-fit apex, Eq. (34.226) reproduces the measured $|\epsilon_K|=(2.228\pm0.011)\times10^{-3}$ [382]. The sensitivity to $A^2\propto|V_{cb}|^2$ (the curve scales as A^4 through the top–top term) is why the ϵ_K band is broadened chiefly by the inclusive/exclusive $|V_{cb}|$ spread, a point we return to in the overconstraint.

34.6.5 Mixing-induced CP violation: $\sin2\beta$ from a time-dependent asymmetry

The kaon measures the imaginary part of the mass matrix; the B factories measure a *phase* directly, through the interference of a decay with mixing-then-decay (Section 32.6.4, face (iii)). We derive the time-dependent asymmetry from the oscillation amplitudes of Eq. (32.165), then specialize to the golden mode.

A meson tagged as B^0 at $t=0$ evolves into the superposition $|B^0(t)\rangle = g_+(t)\,|B^0\rangle+(q/p)g_-(t)\,\left|\bar B^0\right\rangle$, and a tagged $\bar B^0$ into $\left|\bar B^0(t)\right\rangle=(p/q)g_-(t)\,|B^0\rangle+g_+(t)\,\left|\bar B^0\right\rangle$, with $g_\pm$ of Eq. (32.165). For a final state f accessible to both, with amplitudes $A_f=\langle f|\mathcal{H}_w|B^0\rangle$ and $\bar A_f=\left\langle f\middle|\mathcal{H}_w\middle|\bar B^0\right\rangle$, the instantaneous rates are

$$\begin{aligned}\Gamma(B^0(t)\to f)&\propto|A_f|^2\left|g_+(t)+\tfrac{q}{p}\tfrac{\bar A_f}{A_f}g_-(t)\right|^2=|A_f|^2|g_+ +\lambda_f\,g_-|^2,\\ \Gamma(\bar B^0(t)\to f)&\propto|A_f|^2\left|\tfrac{p}{q}\right|^2\left|g_-(t)+\tfrac{q}{p}\tfrac{\bar A_f}{A_f}g_+(t)\right|^2=|A_f|^2\left|\tfrac{p}{q}\right|^2|g_- +\lambda_f\,g_+|^2,\end{aligned}\tag{34.227}$$

with $\lambda_f=(q/p)(\bar A_f/A_f)$ as in Eq. (32.177). For the B_d system the width difference is negligible, $\Delta\Gamma\approx0$, and mixing is nearly CP-conserving, $|q/p|\simeq1$; from Eq. (32.166) the squared amplitudes are then

$$|g_\pm(t)|^2=\tfrac12 e^{-\Gamma t}(1\pm\cos\Delta m_d\,t),\qquad g_+^*(t)g_-(t)=-\tfrac{i}{2}e^{-\Gamma t}\sin\Delta m_d\,t.\tag{34.228}$$

Expanding the moduli in Eq. (34.227), $|g_+ +\lambda_f g_-|^2=|g_+|^2+|\lambda_f|^2|g_-|^2+2\,\mathrm{Re}(\lambda_f\,g_+^*g_-)$, and using Eq. (34.228),

$$\begin{aligned}\Gamma(B^0(t)\to f)&\propto\tfrac12 e^{-\Gamma t}\Big[(1+|\lambda_f|^2)+(1-|\lambda_f|^2)\cos\Delta m_d t-2\ \mathrm{Im}\,\lambda_f\ \sin\Delta m_d t\Big],\\ \Gamma(\bar B^0(t)\to f)&\propto\tfrac12 e^{-\Gamma t}\Big[(1+|\lambda_f|^2)-(1-|\lambda_f|^2)\cos\Delta m_d t+2\ \mathrm{Im}\,\lambda_f\ \sin\Delta m_d t\Big],\end{aligned}\tag{34.229}$$

where $2\,\mathrm{Re}(\lambda_f g_+^*g_-)=2\,\mathrm{Re}(\lambda_f\cdot(-\tfrac{i}{2})e^{-\Gamma t}\sin\Delta m_d t)=-\ \mathrm{Im}\,\lambda_f\,e^{-\Gamma t}\sin\Delta m_d t$. The time-dependent asymmetry is then exactly the form anticipated in Eq. (32.178),

$$a_f(t)=\frac{\Gamma(\bar B^0(t)\to f)-\Gamma(B^0(t)\to f)}{\Gamma(\bar B^0(t)\to f)+\Gamma(B^0(t)\to f)}=S_f\sin\Delta m_d\,t-C_f\cos\Delta m_d\,t,\tag{34.230}$$

$$S_f = \frac{2\,\mathrm{Im}\,\lambda_f}{1+|\lambda_f|^2}, \qquad C_f = \frac{1-|\lambda_f|^2}{1+|\lambda_f|^2}. \tag{34.231}$$

The denominators are the CP-even sum of rates; the $\sin\Delta m_d t$ coefficient S_f is the pure interference CP violation, switched on by $\mathrm{Im}\,\lambda_f \neq 0$.

The golden mode. For $f = J/\psi\, K_S$ (quark transition $b \to c\bar{c}s$) a single weak amplitude dominates, so $|\bar{A}_f/A_f| = 1$, hence $|\lambda_f| = 1$ and $C_f \simeq 0$: the mode measures a clean phase. The phase was assembled in Eq. (32.180) from the B_d-mixing factor $q/p = V_{tb}V_{td}^*/V_{tb}^*V_{td}$, the decay factor $V_{cb}^*V_{cs}/V_{cb}V_{cs}^*$, and the K^0-mixing factor needed to project onto K_S, with the result $\lambda_{\psi K_S} = -\,e^{-2i\beta}$, where $\beta = \arg(-V_{cd}V_{cb}^*/V_{td}V_{tb}^*)$ is a pure CKM phase (Eq. (32.181)). Therefore

$$S_{\psi K_S} = \frac{2\,\mathrm{Im}(-e^{-2i\beta})}{2} = \mathrm{Im}(-e^{-2i\beta}) = \sin 2\beta, \qquad C_{\psi K_S} \simeq 0. \tag{34.232}$$

The coefficient of $\sin\Delta m_d t$ in the measured asymmetry *is* $\sin 2\beta$, with no hadronic matrix element anywhere in the chain—the theoretically pristine measurement that gives the apparatus its name. The world average is $\sin 2\beta = 0.699 \pm 0.017$ [382]. The independent, purely CP-conserving determination of the apex through Eq. (32.195) gives $\sin 2\beta = 2\bar{\eta}(1-\bar{\rho})/[(1-\bar{\rho})^2+\bar{\eta}^2] = 0.712$. A time-dependent oscillation in $B_d \to J/\psi K_S$ and a fit to the lengths of the triangle's sides agree to about one percent—two utterly different experiments selecting the same phase.

Historical Note

When Kobayashi and Maskawa proposed a third generation in 1973 [298], their sole motivation was that two generations cannot accommodate CP violation (Theorem 32.25)—the known ϵ_K forced a third family before its quarks were seen, just as GIM had forced the charm. The prediction that the single irreducible phase must generate *large*, $O(1)$ asymmetries in the B_d system, where the triangle is not squashed, was the central target of the asymmetric B factories. The 2001 observation of $\sin 2\beta \neq 0$, and its agreement with the value inferred from ϵ_K and Δm_d measured decades earlier and on entirely different mesons, converted the Kobayashi–Maskawa phase from a hypothesis into a measured parameter of nature, and won them the 2008 Nobel Prize.

34.6.6 The overconstrained triangle as a precision test

We can now read the whole exercise as a single test. The apex $(\bar{\rho}, \bar{\eta})$ is two real numbers; the measurements of this section and Chapter 32 fix it many times over, each through a different physical mechanism and a different theoretical input:

- $|V_{ub}/V_{cb}|$ from tree-level semileptonic b decays fixes $R_u = \sqrt{\bar{\rho}^2 + \bar{\eta}^2}$ (Eq. (32.197))—a circle about the origin, CP-conserving, no loops.
- $\Delta m_d/\Delta m_s$ fixes $R_t \propto |V_{td}/V_{ts}|$ (Eq. (34.223))—a circle about $(1, 0)$, from the one-loop box.
- $\sin 2\beta$ from $B_d \to J/\psi K_S$ fixes the angle β (Eq. (34.232))—a ray, mixing-induced CP violation.
- γ from tree-level $B \to DK$ fixes $\arg(\bar{\rho} + i\bar{\eta})$—a ray, the cleanest angle ($\gamma = 65.9°$ [382]), immune to new short-distance physics.
- $|\epsilon_K|$ fixes a hyperbola (Eq. (34.226))—indirect CP violation in the kaon, the oldest constraint.

The Kobayashi–Maskawa mechanism, with its single phase, demands that all five curves pass through one point. Figure 34.10 shows that they do, to within present errors, at $\bar{\rho} = 0.159 \pm 0.008$, $\bar{\eta} = 0.352 \pm 0.007$. That $\bar{\eta} \neq 0$ at overwhelming significance is the quantitative statement that the CKM phase is nonzero; that the curves are mutually consistent is the statement that this one phase accounts for all measured quark-sector CP violation and mixing, across kaons and B mesons, tree and loop, CP-even and CP-odd. This is the flavor arena's version of the Standard Model passing at the per-mille level: not one number reproduced, but a dozen forced to agree on two.

Where it strains, dated honestly. The agreement is not perfect, and the frictions are exactly where a deviation would first surface (mid-2020s [382]). The inclusive and exclusive determinations of $|V_{cb}|$ and $|V_{ub}|$ differ at the 2–3σ level, almost certainly a problem in the theory of semileptonic form factors rather than new physics, but one that broadens the R_u band and—because $\epsilon_K \propto A^4 \propto |V_{cb}|^4$—the ϵ_K hyperbola, so that the consistency of ϵ_K with the angle measurements is only as sharp as the $|V_{cb}|$ input allows. A clean discrepancy between the tree-determined γ and the loop-dominated constraints would be an unambiguous signal of new physics in mixing; none is seen. Separately, the ratios of certain $b \to s\,\ell^+\ell^-$ exclusive decays have shown intermittent tensions with their Standard-Model predictions over the past decade, a story still unsettled and flagged forward to Chapter 35 and Chapter 40. The honest summary is the one the figure makes visually: the single CKM phase is the established, dominant, and within present precision the sole source of quark-sector CP violation, with residual 2–3σ frictions concentrated in hadronic-matrix-element inputs rather than in the CKM structure itself.

With the box computed from first principles, the Inami–Lim function derived, and the apex overdetermined, the flavor sector now stands beside the electroweak and Higgs arenas as an independent, redundant confirmation of the same short

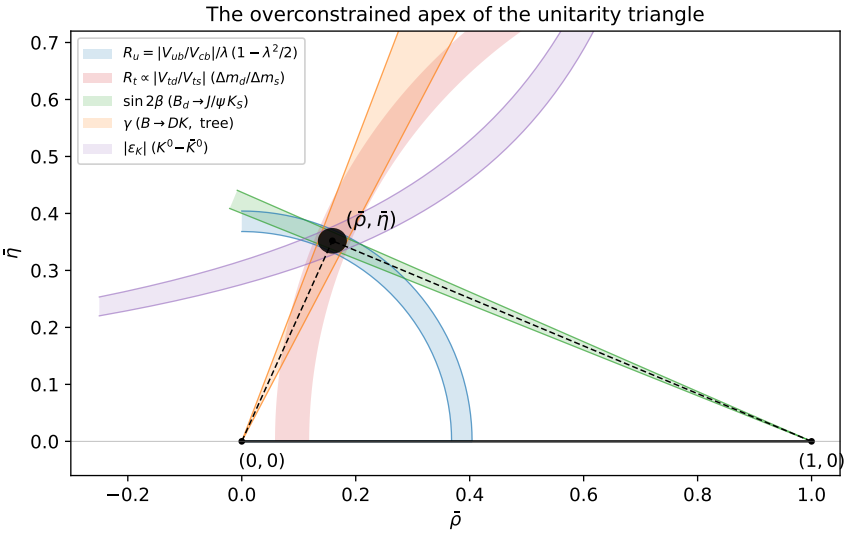


Figure 34.10: The overconstrained apex of the unitarity triangle in the $(\bar\rho, \bar\eta)$ plane (mid-2020s central values, illustrative band widths, not a fit). Each independent measurement selects a curve: R_u from $|V_{ub}/V_{cb}|$ (circle about the origin, tree), R_t from $\Delta m_d/\Delta m_s$ (circle about $(1,0)$, loop, Theorem 34.25), the angle β from $\sin 2\beta$ (Eq. (34.232)), the angle γ from $B \to DK$ (tree), and the $|\epsilon_K|$ hyperbola (Eq. (34.226)). The Kobayashi–Maskawa mechanism requires all five to cross at the apex; they do, the residual width of the crossing region being the precision of the test. Compare the schematic of Fig. 32.7.

parameter list. All three feed the global fit and the honest frontier summary that the final section assembles.

34.7 The Global Fit: Where the Standard Model Triumphs

The three preceding sections each interrogated the Standard Model in a different arena — the Z resonance and its oblique radiative corrections, the production and decay pattern of the Higgs boson, and the loop-induced flavor-changing neutral currents that close the unitarity triangle. Each arena, taken alone, is already a nontrivial confrontation: the Z lineshape constrains the gauge couplings and the weak mixing angle, the Higgs couplings track particle masses, and the box and penguin amplitudes overdetermine the CKM apex. But the deeper point of this chapter is not that the Standard Model passes three separate tests. It is that the three arenas are not separate. They share a parameter list — the same nineteen numbers counted in Section 33.6, of which only a handful are inputs to the electroweak and flavor sectors — and once that handful is fixed, *every*

observable in all three arenas becomes a prediction with no further freedom. The correct way to state what the data say is therefore a single statement, not three: the entire body of precision measurements is one overconstrained system, and the Standard Model fits it. This section assembles that statement. We first build the machinery that makes "fits it" quantitative — the global χ^2 over predicted versus measured pseudo-observables, its goodness of fit, and the linearized parameter extraction that turns indirect constraints into predictions. We then confront the data with the oblique S–T summary of new physics, catalogue where the agreement is most spectacular, and — honestly, and dated to the mid-2020s — where the few standing tensions lie and what they do and do not imply.

34.7.1 The fit as a hypothesis test: pseudo-observables and the χ^2

Fix a vector of input parameters

$$\mathbf{p} = (p_1, \ldots, p_{N_p}), \tag{34.233}$$

the minimal set from which the theory computes everything else. For the electroweak sector the natural choice, because these three are the most precisely measured, is the triple

$$\{\, \alpha,\ G_F,\ M_Z \,\} \tag{34.234}$$

augmented by the parameters that enter the radiative corrections: the heavy-fermion masses (above all m_t), the strong coupling $\alpha_s(M_Z)$, the hadronic contribution to the running of α, and the Higgs mass m_h. Here α is the fine-structure constant of Chapter 20, G_F the Fermi constant from muon decay, and M_Z the Z pole mass; the weak mixing angle and M_W are then *not* inputs but outputs, computed from Eq. (34.234) through the relations of Chapter 31 including their full one-loop dressing. This is the crucial structural fact: the moment one declares $\{\alpha, G_F, M_Z\}$ to be the inputs, the W mass, the effective mixing angle, the partial and total Z widths, and every asymmetry become *predictions*, and they are predictions of the quantum theory, because the relations among them carry the self-energy, vertex, and box corrections that distinguish a renormalizable QFT from a tree-level caricature.

Each measurable quantity in the comparison is a *pseudo-observable*: a number defined so that it is (a) experimentally extractable from raw data with the detector and QED-radiation effects deconvolved, and (b) theoretically computable as a function $O_i^{\rm th}(\mathbf{p})$ of the inputs. The canonical electroweak set comprises the Z-pole observables — the total width Γ_Z, the hadronic pole cross section $\sigma^0_{\rm had}$, the ratios $R^0_\ell = \Gamma_{\rm had}/\Gamma_\ell$ and $R^0_{b,c} = \Gamma_{b,c}/\Gamma_{\rm had}$, the forward–backward asymmetries $A^{0,f}_{\rm FB}$, and the polarization asymmetries A_f — together with M_W, Γ_W, m_t, and the hadronic shift $\Delta\alpha^{(5)}_{\rm had}$.

Definition 34.26 (The global χ^2)**.** Let $\mathbf{O}^{\mathrm{exp}} = (O_1^{\mathrm{exp}}, \dots, O_N^{\mathrm{exp}})$ be the measured pseudo-observables with covariance matrix V (diagonal entries the squared total uncertainties, off-diagonal entries the experimental and theoretical correlations). The global fit quality at a point $\mathbf{p}$ in input space is

$$\chi^2(\mathbf{p}) \;=\; \big(\mathbf{O}^{\mathrm{exp}} - \mathbf{O}^{\mathrm{th}}(\mathbf{p})\big)^{\mathsf{T}}\, V^{-1}\, \big(\mathbf{O}^{\mathrm{exp}} - \mathbf{O}^{\mathrm{th}}(\mathbf{p})\big). \tag{34.235}$$

The logic of Eq. (34.235) is the logic of a maximum-likelihood hypothesis test. If the measurement errors are Gaussian with covariance V and the Standard Model with true parameters $\mathbf{p}$ is correct, then $\mathbf{O}^{\mathrm{exp}} - \mathbf{O}^{\mathrm{th}}(\mathbf{p})$ is a Gaussian random vector with covariance V, and $\chi^2(\mathbf{p})$ is its squared Mahalanobis length. Minimizing over $\mathbf{p}$ produces the best-fit inputs $\hat{\mathbf{p}}$; the minimum value $\chi^2_{\mathrm{min}} = \chi^2(\hat{\mathbf{p}})$ is the residual misfit that even the best choice of inputs cannot remove. The number of degrees of freedom is

$$n_{\mathrm{dof}} = N - N_p, \tag{34.236}$$

the count of measurements minus the count of fitted inputs — the number of genuinely independent constraints left over once the inputs have been used up. If the model is right, χ^2_{min} is itself a draw from a χ^2 distribution with n_{dof} degrees of freedom, whose mean is n_{dof} and whose standard deviation is $\sqrt{2n_{\mathrm{dof}}}$. The *goodness of fit* is the p-value,

$$p \;=\; \int_{\chi^2_{\mathrm{min}}}^{\infty} f_{\chi^2}(x;\, n_{\mathrm{dof}})\, \mathrm{d}x, \tag{34.237}$$

the probability that the model, if true, would produce a fit at least this bad by chance. A value $\chi^2_{\mathrm{min}}/n_{\mathrm{dof}} \approx 1$ with p of order a few tenths is the signature of a theory that describes the data; this is, to anticipate, exactly what the electroweak fit delivers, with roughly twenty observables, a few fitted inputs, and $\chi^2_{\mathrm{min}}/n_{\mathrm{dof}}$ near unity.

34.7.2 Linearization and the Gaussian solution

The functions $O_i^{\mathrm{th}}(\mathbf{p})$ are nonlinear — they contain logarithms of mass ratios and quadratic dependence on m_t — but the data pin the inputs so tightly that the fit lives in a small neighborhood of a reference point $\mathbf{p}_0$, and the entire machinery becomes linear algebra. Expand

$$O_i^{\mathrm{th}}(\mathbf{p}) \;=\; O_i^{\mathrm{th}}(\mathbf{p}_0) \;+\; \sum_a J_{ia}\,(p_a - p_{0a}) \;+\; \dots, \qquad J_{ia} \equiv \left.\frac{\partial O_i^{\mathrm{th}}}{\partial p_a}\right|_{\mathbf{p}_0}, \tag{34.238}$$

with J the $N \times N_p$ Jacobian of the theory map. Write the residual at the reference point and the parameter shift as

$$\mathbf{r} \equiv \mathbf{O}^{\mathrm{exp}} - \mathbf{O}^{\mathrm{th}}(\mathbf{p}_0), \qquad \boldsymbol{\delta} \equiv \mathbf{p} - \mathbf{p}_0, \qquad W \equiv V^{-1}. \tag{34.239}$$

Then Eq. (34.235) becomes the quadratic form

$$\chi^2(\boldsymbol{\delta}) \;=\; (\mathbf{r} - J\boldsymbol{\delta})^{\mathsf{T}} W (\mathbf{r} - J\boldsymbol{\delta}) \;=\; \mathbf{r}^{\mathsf{T}} W \mathbf{r} - 2\,\boldsymbol{\delta}^{\mathsf{T}} J^{\mathsf{T}} W \mathbf{r} + \boldsymbol{\delta}^{\mathsf{T}} (J^{\mathsf{T}} W J)\,\boldsymbol{\delta}. \tag{34.240}$$

Setting the gradient to zero,

$$\frac{\partial \chi^2}{\partial \boldsymbol{\delta}} = -2\,J^{\mathsf{T}} W \mathbf{r} + 2\,(J^{\mathsf{T}} W J)\,\boldsymbol{\delta} = 0, \tag{34.241}$$

gives the normal equations $(J^{\mathsf{T}} W J)\,\hat{\boldsymbol{\delta}} = J^{\mathsf{T}} W \mathbf{r}$, whose solution and parameter covariance are

$$\boxed{\hat{\boldsymbol{\delta}} \;=\; (J^{\mathsf{T}} W J)^{-1} J^{\mathsf{T}} W\,\mathbf{r}, \qquad \operatorname{cov}(\hat{\mathbf{p}}) \;=\; (J^{\mathsf{T}} W J)^{-1}.} \tag{34.242}$$

The first relation is the generalized least-squares estimator; the second follows because $\hat{\boldsymbol{\delta}}$ is a linear function of the Gaussian vector $\mathbf{r}$ (with covariance $V = W^{-1}$), so $\operatorname{cov}(\hat{\boldsymbol{\delta}}) = (J^{\mathsf{T}} W J)^{-1} J^{\mathsf{T}} W\, V\, W J (J^{\mathsf{T}} W J)^{-1} = (J^{\mathsf{T}} W J)^{-1}$, the middle factors collapsing because $WVW = W$. This is the precise sense in which the fit *predicts* the inputs: $\hat{\mathbf{p}}$ and its error ellipsoid are read directly off Eq. (34.242), with the error shrinking as more observables (more rows of J) are added, even for an input that is never measured directly.

Substituting $\hat{\boldsymbol{\delta}}$ back into Eq. (34.240) gives the residual minimum,

$$\chi^2_{\min} \;=\; \mathbf{r}^{\mathsf{T}} W \mathbf{r} - \mathbf{r}^{\mathsf{T}} W J (J^{\mathsf{T}} W J)^{-1} J^{\mathsf{T}} W \mathbf{r} \;=\; \mathbf{r}^{\mathsf{T}} \big(W - W J (J^{\mathsf{T}} W J)^{-1} J^{\mathsf{T}} W\big)\mathbf{r}. \tag{34.243}$$

Writing $W = W^{1/2} W^{1/2}$ and $\tilde{J} \equiv W^{1/2} J$, the bracket is $W^{1/2}(\mathbf{1} - P) W^{1/2}$ with $P = \tilde{J}(\tilde{J}^{\mathsf{T}}\tilde{J})^{-1}\tilde{J}^{\mathsf{T}}$ the orthogonal projector onto the column space of $\tilde{J}$. Thus $\chi^2_{\min} = \|(\mathbf{1} - P)\, W^{1/2}\mathbf{r}\|^2$ is the squared length of the part of the (whitened) residual that lies *outside* the subspace the parameters can reach. The parameters absorb N_p directions of the residual; what remains, in $n_{\mathrm{dof}} = N - N_p$ directions, is the irreducible disagreement the test measures. This is the algebraic content of overconstraint: with $N \gg N_p$, most directions of $\mathbf{r}$ cannot be adjusted away, so a wrong theory has nowhere to hide.

Physical Insight

A tree-level theory would also have a parameter list and would also admit a χ^2. What makes the electroweak fit a test of *quantum* field theory is that the Jacobian J in Eq. (34.238) is computed with loops. The dependence of M_W and the asymmetries on m_t and m_h enters only through self-energy diagrams; a tree map would have $\partial O/\partial m_t = 0$ for the gauge-boson observables and the fit would fail catastrophically. The top quark and the Higgs boson both made themselves felt in these derivatives — in the radiative corrections to Z-pole data — and were thereby *predicted* before they were produced. Precision tests are tests of the entire loop expansion,

not of the classical Lagrangian.

The historic case is worth making concrete, because it is the cleanest demonstration that Eq. (34.242) extracts physics and not just numbers.

Worked Example 34.27 (Indirect mass from a single radiative correction)**.** The leading top-mass dependence of the electroweak observables enters through the ρ parameter, $\rho = M_W^2/(M_Z^2\cos^2\theta_{\rm W}) = 1 + \Delta\rho$, whose dominant one-loop contribution from the top–bottom doublet is the custodial-breaking piece derived in Chapter 31,

$$\Delta\rho_{\rm top} \;=\; \frac{3\,G_F\,m_t^2}{8\sqrt{2}\,\pi^2}, \tag{34.244}$$

quadratic in m_t because the top–bottom mass splitting is the largest violation of weak isospin in the theory. A subdominant, only *logarithmic* dependence on m_h accompanies it, $\Delta\rho_{\rm Higgs} \sim -\frac{3G_F M_W^2}{8\sqrt{2}\pi^2}\tan^2\theta_{\rm W}\,\ln(m_h^2/M_Z^2)$, the contrast in functional form being exactly why the Z-pole data pinned m_t sharply but m_h only weakly. Treat m_t as the single unknown p and a measured $\Delta\rho$ as the single observable. With $G_F = 1.1664\times10^{-5}\,\text{GeV}^{-2}$, Eq. (34.244) at $m_t = 172.5\,\text{GeV}$ gives

$$\Delta\rho_{\rm top} = \frac{3(1.1664\times10^{-5})(172.5)^2}{8\sqrt{2}\,\pi^2} = \frac{3(1.1664\times10^{-5})(2.9756\times10^{4})}{111.65} = 9.3\times10^{-3}, \tag{34.245}$$

a 0.93% effect — an order of magnitude larger than the per-mille precision of the Z observables, hence eminently measurable. The single-parameter version of Eq. (34.242) reads $\hat{\delta} = (\partial\Delta\rho/\partial m_t)^{-1}\,r$ with r the measured $\Delta\rho$ residual and uncertainty $\sigma_{m_t} = \sigma_{\Delta\rho}\,(\partial\Delta\rho/\partial m_t)^{-1}$. Since $\partial\Delta\rho/\partial m_t = 2\Delta\rho/m_t = 2(9.3\times10^{-3})/172.5 = 1.08\times10^{-4}\,\text{GeV}^{-1}$, a per-mille measurement $\sigma_{\Delta\rho}\approx 2\times10^{-4}$ propagates to $\sigma_{m_t}\approx(2\times10^{-4})/(1.08\times10^{-4})\approx 1.9\,\text{GeV}$ on the indirect top mass once the full observable set is folded in. This is precisely how the LEP/SLD analyses, before the Tevatron produced a single top quark, constrained m_t to roughly 170–180 GeV; the subsequent direct measurement, $m_t\approx173\,\text{GeV}$, fell squarely inside the predicted band. The same fit, exploiting the logarithm $\Delta\rho_{\rm Higgs}$, later preferred a *light* Higgs, $m_h \lesssim 150\,\text{GeV}$ at 95% confidence with a central value near 90 GeV — and the boson discovered in 2012 weighed in at 125 GeV, again inside the band. Two particles predicted from loop corrections, two confirmations: the overconstraint of Eq. (34.243) made flesh.

34.7.3 The oblique parameters and the S–T plane

A great many candidate extensions of the Standard Model affect the precision observables only *universally* — they modify the gauge-boson self-energies but leave the gauge couplings to fermions, and the vertex and box corrections,

essentially SM-like. New heavy states that couple to the electroweak currents but not directly to the light fermions are of exactly this type. For such "oblique" new physics the entire deviation across all Z-pole observables collapses onto a small number of combinations of the self-energies, introduced as the Peskin–Takeuchi parameters in Section 34.3. We recall their definition because the worked example below will need it. Let $\Pi_{XY}(q^2)$ denote the *new-physics* part of the vacuum-polarization amplitude between currents $X, Y \in \{1, 2, 3, Q\}$ (the $\mathrm{SU}(2)_L$ and electromagnetic currents), normalized so that the reference Standard Model contributes zero. With $s_{\mathrm{W}} \equiv \sin\theta_{\mathrm{W}}$, $c_{\mathrm{W}} \equiv \cos\theta_{\mathrm{W}}$, and $\Pi' \equiv \mathrm{d}\Pi/\mathrm{d}q^2$,

$$\alpha\, S \;=\; 4\, e^2 \big[\Pi'_{33}(0) - \Pi'_{3Q}(0)\big], \tag{34.246}$$

$$\alpha\, T \;=\; \frac{e^2}{s_{\mathrm{W}}^2 c_{\mathrm{W}}^2 M_Z^2} \big[\Pi_{11}(0) - \Pi_{33}(0)\big], \tag{34.247}$$

$$\alpha\, U \;=\; 4\, e^2 \big[\Pi'_{11}(0) - \Pi'_{33}(0)\big]. \tag{34.248}$$

Their physical content is transparent. T in Eq. (34.101) is the custodial-isospin-breaking difference of the charged- and neutral-current self-energies at zero momentum; comparing with Theorem 34.27, $\alpha T = \Delta\rho_{\mathrm{new}}$, so T measures the deviation of ρ from unity induced by new physics. S in Eq. (34.102) is the momentum slope of the neutral-current self-energy and counts, roughly, the number of new chiral electroweak doublets. U is generated only by operators of higher dimension and is small in most scenarios; the standard practice, which we follow, is to set $U = 0$ and confront the data in the (S, T) plane. By construction the Standard Model sits at the origin: $S = T = U = 0$ is the statement "the self-energies are exactly their SM values."

The fit in the (S, T) plane is the two-parameter instance of the machinery of Section 34.7.2: the Jacobian J now has two columns, $\partial O_i/\partial S$ and $\partial O_i/\partial T$, computed from the known dependence of each Z-pole observable and of M_W on the oblique parameters. Minimizing Eq. (34.235) yields a best-fit point $(\hat{S}, \hat{T})$ and a 2×2 covariance whose inverse, $(J^{\mathsf{T}} W J)$, defines the confidence ellipses

$$\chi^2(S, T) - \chi^2_{\mathrm{min}} = \begin{pmatrix} S - \hat{S} \\ T - \hat{T} \end{pmatrix}^{\mathsf{T}} \mathrm{cov}^{-1} \begin{pmatrix} S - \hat{S} \\ T - \hat{T} \end{pmatrix} = \Delta\chi^2, \tag{34.249}$$

with $\Delta\chi^2 = 2.30$ enclosing 68% and 5.99 enclosing 95% of the probability for two degrees of freedom (the 2-dof analogue of the familiar 1-dof values 1 and 3.84). Figure 34.11 shows the result: the best-fit point is displaced from the origin by less than its uncertainty, and the SM point $(0, 0)$ lies inside the 68% ellipse. The strong positive S–T correlation visible in the figure is not an artifact; it reflects that M_W and $\sin^2\theta_{\mathrm{eff}}$ each constrain a particular linear combination of S and T much more sharply than either parameter alone, leaving a flat valley along the anticorrelated direction.

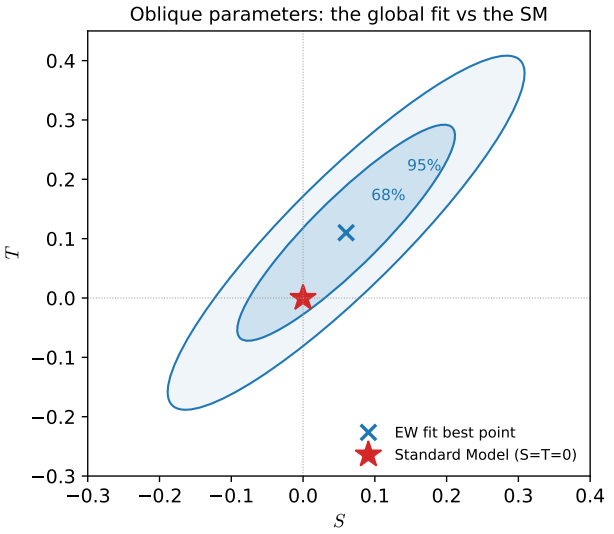


Figure 34.11: The oblique parameters confronted with the global electroweak fit ($U = 0$). The shaded regions are the 68% and 95% confidence contours ($\Delta\chi^2 = 2.30$ and 5.99 for two degrees of freedom) of a representative mid-2020s fit; the cross marks the best-fit point and the star marks the Standard Model at $(S, T) = (0, 0)$. The SM point lies comfortably inside the 68% ellipse: the oblique sector passes. The pronounced positive correlation between S and T is the imprint of the M_W and $\sin^2\theta_{\text{eff}}$ constraints, which fix a linear combination of the two far better than either separately.

34.7.4 Where the Standard Model triumphs

With the machinery in place, the headline is easily stated and worth stating plainly: across roughly twenty independent electroweak pseudo-observables the global fit returns $\chi^2_{\text{min}}/n_{\text{dof}}$ near unity, the SM point sits inside the oblique ellipse, and the predicted inputs — m_t and m_h above all — match their directly measured values. Figure 34.12 displays the residual structure as a pull plot: for each observable the pull $(O^{\text{exp}} - O^{\text{fit}})/\sigma$ is the number of standard deviations by which the measurement departs from the best-fit prediction. The pulls scatter around zero with unit width and almost all lie within 2σ. This is what a correct overconstrained theory looks like.

It is worth naming the most precise confirmations individually, because two of them are the endpoints of benchmark theories, and the precision tests *consume* those endpoints rather than adding a new layer to them.

The electron anomalous moment — QED, closed. The single most precisely tested prediction in all of physics is the electron's anomalous magnetic

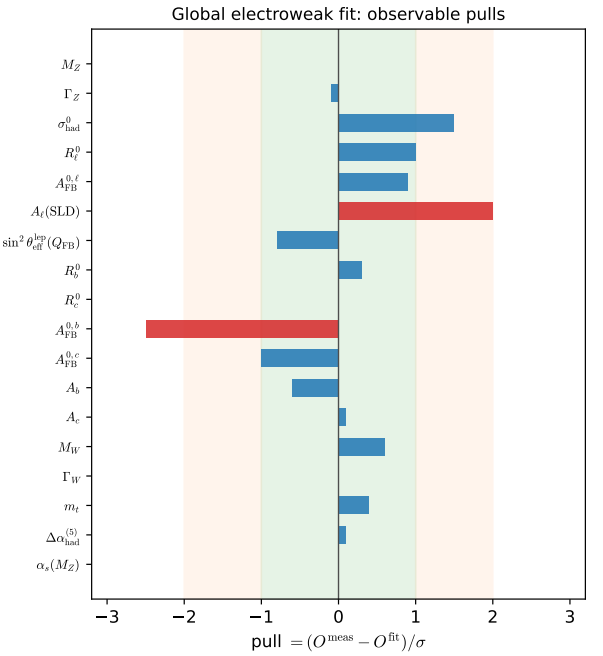


Figure 34.12: Pulls of the global electroweak fit, $(O^{\rm meas}-O^{\rm fit})/\sigma$, for the standard Z-pole and electroweak pseudo-observables (representative mid-2020s values). Green band $|\text{pull}|<1$, amber bands $1<|\text{pull}|<2$. The two reddened entries are the long-standing $\sim$2–3σ outliers, the b-quark forward–backward asymmetry $A^{0,b}_{\rm FB}$ and the SLD leptonic asymmetry A_ℓ; they pull in opposite directions and do not coherently signal new physics. The remaining pattern is a textbook unit-width scatter about zero.

moment. Schwinger's leading term $a_e=\alpha/2\pi$, computed in full in Theorem 20.4, is the first coefficient of a series in α/π whose QED contributions are now known through five loops. The measured and computed values of a_e agree to roughly ten significant figures — a fractional precision near 10^{-13} — and the comparison is so sharp that it is run in reverse: a_e is used to *extract* the fine-structure constant, yielding α^{-1} to better than a part per billion, in agreement with the independent atom-recoil determinations. The closure of QED is therefore not a single number but a consistency: the same α that the running coupling of Chapter 20 carries up to M_Z, that fixes the QED part of a_e, and that anchors the electroweak inputs Eq. (34.234), is one number measured many independent ways and found to be the same number every time. That is what it means for a coupling constant to be real.

The strong coupling — Yang–Mills theory, closed. The analogous statement for the strong sector is the consistency of $\alpha_s(M_Z)$. The world-average value

imported from Eq. (27.163),

$$\alpha_s(M_Z^2) = 0.1180 \pm 0.0009, \tag{34.250}$$

is itself the result of a fit, over determinations as physically disparate as the hadronic Z width, τ-lepton decays, deep-inelastic structure functions, event-shape and jet observables, and lattice QCD. These probe the coupling at scales from below 2 GeV up to M_Z, and the asymptotic-freedom running of Chapter 27 maps them all onto a single value at the Z pole; that they land on *one* number is the precision confirmation of the YM beta function. Yang–Mills theory, like QED, is closed here not by extension but by confrontation.

The Z resonance and the unitarity triangle. Between these two endpoints sit the per-mille agreements that are the body of the chapter: the Z lineshape and its asymmetries, reproduced by the loop-corrected predictions from Eq. (34.234); the Higgs couplings tracking particle masses along the coupling-versus-mass law of the Higgs section; and the unitarity triangle of Section 34.6, whose sides and angles — measured independently through $|V_{ub}/V_{cb}|$, neutral-meson mixing, ε_K, and $\sin 2\beta$ — intersect at a single apex $(\bar\rho, \bar\eta)$. The triangle closing is the flavor sector's version of the electroweak fit: more measurements than parameters, all consistent.

34.7.5 Standing tensions, dated to the mid-2020s

A gold-standard account must be as honest about the cracks as about the triumphs. As of the mid-2020s a small number of measurements sit in tension with the SM at the few-sigma level. None is yet a discovery; each is live, and it is instructive to see what kind of object each is, because the categories differ.

The muon anomalous moment. The muon's a_μ, unlike the electron's, is sensitive enough to heavy virtual states that it became, for two decades, the SM's most watched tension. The full status is dissected in Section 20.4.4 and is not repeated here; the essential point for the global picture is that the tension has *migrated* from experiment-versus-theory to theory-versus-theory. The Fermilab measurement is now the more precise side of the comparison, and the limiting question is the leading hadronic vacuum-polarization contribution, where the data-driven dispersive evaluation and first-principles lattice QCD disagree with each other. With the 2025 lattice-consensus theory update, the SM prediction and the measurement agree at roughly the 1σ level; the former "4σ anomaly" is, for now, a hadronic uncertainty rather than a sign of new physics — but it is not settled, and the resolution will come from reconciling the two hadronic determinations.

The W-mass spread. The W mass is the most consequential single electroweak prediction, because it depends on S and T through the same custodial physics as Theorem 34.27. The SM fit predicts $M_W \approx 80{,}357 \pm 6\,\text{MeV}$. Most direct measurements — from LEP, the Tevatron's earlier results, ATLAS, CMS, and LHCb — cluster around this value. The outlier is the 2022 CDF II measurement, $M_W = 80{,}433.5 \pm 9.4\,\text{MeV}$, which lies about 7σ above the SM prediction and, just as importantly, in tension with the other direct measurements. The phrase "W-mass spread" is deliberate: the present tension is as much among experiments as between experiment and theory, and until the spread among measurements is understood the deviation cannot be cleanly read as new physics. Were the CDF value to stand, its natural home in the language of Section 34.7.3 would be a positive shift in T, as the worked example below makes quantitative.

The $b \to s\,\ell\ell$ observables. The rare flavor-changing transition $b \to s\,\ell\ell$ is loop-induced and therefore a sensitive probe of new physics, in the same spirit as the FCNC tests of Section 34.6. For several years the lepton-flavor-universality ratios $R_K = \text{BR}(B \to K\mu\mu)/\text{BR}(B \to Kee)$ and R_{K^*} sat $\sim 3\sigma$ below their clean SM prediction of unity, an enticing hint precisely because the ratios cancel hadronic uncertainties. The 2022 LHCb reanalysis, with improved control of electron reconstruction, found R_K, R_{K^*} *consistent with unity*: the cleanest piece of the anomaly evaporated. What remains are the muonic branching fractions and the angular observable P_5' in $B \to K^*\mu\mu$, which still sit low by $\sim 3\sigma$ — but these are limited by hadronic form factors and charm-loop contributions, so whether they signal new physics or underestimated QCD uncertainties is genuinely open.

First-row CKM unitarity. The most precise unitarity test of the CKM matrix is the first-row sum, which the structure of Chapter 32 requires to equal one exactly:

$$|V_{ud}|^2 + |V_{us}|^2 + |V_{ub}|^2 \stackrel{?}{=} 1. \tag{34.251}$$

With the mid-2020s values $|V_{ud}| \approx 0.97373$ (from superallowed nuclear β decays), $|V_{us}| \approx 0.2243$ (from kaon decays), and the negligible $|V_{ub}| \approx 0.0037$, the sum is

$$(0.97373)^2 + (0.2243)^2 + (0.0037)^2 = 0.94815 + 0.05031 + 0.00001 = 0.99847, \tag{34.252}$$

a deficit of about 1.5×10^{-3}, roughly 3σ below unity — the "Cabibbo-angle anomaly." Crucially, the deficit depends on the radiative and nuclear-structure corrections that convert measured decay rates into $|V_{ud}|$, and on whether $|V_{us}|$ is taken from leptonic or semileptonic kaon decays, which themselves are in mild tension. The anomaly is thus, like the muon a_μ, partly a question of theoretical inputs rather than unambiguously a sign of new physics.

The common thread is honesty about category: each tension is either limited by a hadronic or nuclear theory input (muon $g-2$, $b \to s\mu\mu$ angular, first-row CKM) or by a spread among measurements (M_W), and none yet rises to a clean, reproduced, theory-clean discrepancy. They are the live frontier of precision, and they are the natural bridge to the next chapter.

34.7.6 Reading a deviation: an oblique shift or a Wilson coefficient

The meta-lesson of the chapter is methodological, and it is the lesson that the rest of the book's confrontation with the frontier rests on. Precision overconstraint is *how* a quantum field theory is tested: not by checking one prediction, but by fixing a minimal input set and verifying that the resulting over-determined system of predictions holds. A theory tested this way also fails this way — a real deviation, when one comes, will not announce itself as "new particle of mass M." It will appear first as a coherent pattern of small shifts in the precision observables, and the task is to read that pattern. There are two standard languages for the reading, and they are the same language at two levels of resolution.

The first is the oblique language of Section 34.7.3: a universal deviation is a point displaced from the origin in the (S, T) plane, and its location encodes whether the new physics breaks custodial symmetry (a shift in T), adds chiral matter (a shift in S), or both. The second is the effective-field-theory language that Chapter 40 develops in full: below the scale of any heavy new physics, its effect on SM observables is captured by adding to the SM Lagrangian a tower of higher-dimension operators built from SM fields,

$$\mathcal{L}_{\rm SMEFT} = \mathcal{L}_{\rm SM} + \sum_i \frac{C_i}{\Lambda^2}\,\mathcal{O}_i^{(6)} + \dots , \tag{34.253}$$

with $\mathcal{O}_i^{(6)}$ the dimension-six operators (the Warsaw basis of Chapter 40) and C_i/Λ^2 their Wilson coefficients, Λ the new-physics scale. The two languages are connected by a matching computation: integrating out the heavy states generates definite C_i, and at the level of the gauge-boson self-energies only two dimension-six operators contribute at tree level,

$$\mathcal{O}_{HWB} = (H^\dagger \tau^a H)\, W^a_{\mu\nu} B^{\mu\nu}, \qquad \mathcal{O}_{HD} = (H^\dagger D_\mu H)^* (H^\dagger D^\mu H). \tag{34.254}$$

We do not derive the matching here — it is owned by Chapter 40 — but state its result, the dictionary between the oblique parameters and these two coefficients,

$$S = \frac{4\, s_{\rm W} c_{\rm W}}{\alpha}\, v^2\, \frac{C_{HWB}}{\Lambda^2}, \qquad T = -\frac{1}{2\alpha}\, v^2\, \frac{C_{HD}}{\Lambda^2}, \tag{34.255}$$

with $v \approx 246\,\text{GeV}$ the Higgs vacuum expectation value. The structure is exactly what one expects: $\mathcal{O}_{HWB}$ mixes the W^3 and B field strengths after electroweak

breaking, shifting the momentum slope of the neutral self-energy, hence S; $\mathcal{O}_{HD}$ is the unique dimension-six operator that breaks custodial symmetry, hence T. The chain "deviation $\to$ (S,T) shift $\to$ Wilson coefficient $\to$ new-physics scale" is the operational meaning of testing the SM at the frontier, and the following example runs it end to end.

Worked Example 34.28 (From a W-mass excess to a SMEFT scale)**.** Suppose — hypothetically, in the spirit of the W-mass spread of Section 34.7.5 — a W-mass measurement settled at a value above the SM prediction by $\delta M_W = +70\,\text{MeV}$, and suppose the deviation were oblique. The oblique shift of M_W follows from the dependence of the ρ parameter on S and T; at $U = 0$ the standard linearized relation is

$$\frac{\delta M_W}{M_W} = \frac{\alpha}{2(c_{\text{W}}^2 - s_{\text{W}}^2)}\Big(-\tfrac{1}{2}S + c_{\text{W}}^2\,T\Big). \tag{34.256}$$

Attribute the excess entirely to T (set $S = 0$, the cleanest custodial interpretation). With $\alpha \approx 1/128 = 7.81\times10^{-3}$ at M_Z, $s_{\text{W}}^2 \approx 0.223$, $c_{\text{W}}^2 \approx 0.777$, and $c_{\text{W}}^2 - s_{\text{W}}^2 \approx 0.554$,

$$\frac{\delta M_W}{M_W} = \frac{(7.81\times10^{-3})(0.777)}{2(0.554)}\,T = 5.48\times10^{-3}\,T, \tag{34.257}$$

so that $\delta M_W = M_W\,(5.48{\times}10^{-3})\,T = (80.4\,\text{GeV})(5.48{\times}10^{-3})\,T = 0.441\,T\,\text{GeV} = 441\,T\,\text{MeV}$. The required oblique shift is therefore

$$T = \frac{70\ \text{MeV}}{441\ \text{MeV}} \approx 0.16, \tag{34.258}$$

a positive, $O(0.1)$ value — comfortably the size the S–T ellipse of Fig. 34.11 can accommodate or exclude. Now translate to the Wilson coefficient using the dictionary Eq. (34.255). Inverting the T relation,

$$\frac{C_{HD}}{\Lambda^2} = -\frac{2\alpha\,T}{v^2} = -\frac{2(7.81\times10^{-3})(0.16)}{(246\ \text{GeV})^2} = -\frac{2.50\times10^{-3}}{6.05\times10^{4}\ \text{GeV}^2} = -4.1{\times}10^{-8}\ \text{GeV}^{-2}. \tag{34.259}$$

If the underlying physics is strongly coupled so that the dimensionless coefficient is $C_{HD} \sim O(1)$, the implied new-physics scale is

$$\Lambda = \frac{1}{\sqrt{|C_{HD}/\Lambda^2|}} = \frac{1}{\sqrt{4.1\times10^{-8}\ \text{GeV}^{-2}}} \approx 4.9\ \text{TeV}. \tag{34.260}$$

A 70 MeV anomaly in a 80 GeV mass — a per-mille effect — thus points, through one custodial-breaking operator, to physics near 5 TeV. This is the entire logic of precision testing compressed into one chain of arithmetic: a small, overconstrained deviation is converted first into an oblique shift, then into a definite Wilson coefficient, then into a mass scale a collider could be built to reach. Whether *any* such deviation survives is the empirical question; that the machinery exists to read it is the structural achievement of treating the SM as a renormalizable quantum field theory.

The picture that emerges is the one promised at the chapter's opening. A handful of inputs — $\{\alpha, G_F, M_Z\}$, the masses, α_s — fix the Standard Model completely, and the resulting over-determined web of predictions holds, across the electroweak, Higgs, and flavor arenas, at the per-mille level, because the SM is a renormalizable theory whose quantum corrections are calculable and were calculated. The few standing tensions, dated and categorized above, are the loose threads that motivate looking beyond. We pull on them next: the patterns of deviation and the naturalness and hierarchy puzzles that frame the search for new physics open Chapter 35, while the systematic translation of any future deviation into Wilson coefficients — the matching computation behind Eq. (34.255), done in full — is the business of the SMEFT in Chapter 40.

Further reading

The standard graduate treatments of precision electroweak physics are Peskin and Schroeder [396] (chapters 20–21), Schwartz [432] (chapters 29–31), and the comprehensive account in Langacker [314]; the current experimental values for every observable quoted in this chapter are compiled in the Review of Particle Physics, Particle Data Group [382]. The oblique-parameter formalism originates with Peskin and Takeuchi [397] and Peskin and Takeuchi [398], building on the custodial ρ-parameter analysis of Veltman [485]. The Higgs phenomenology — production, decay, and the coupling–mass pattern — is developed in depth in Djouadi [154] and, classically, in Gunion et al. [243]; the discovery itself is reported in ATLAS Collaboration [42] and CMS Collaboration [125]. The flavor and CP machinery rests on Cabibbo [101] and Kobayashi and Maskawa [298], with the invariant measure of CP violation due to Jarlskog [284] and the short-distance box function derived following Inami and Lim [274]. The effective-field-theory language for reading deviations, used in the closing section and carried forward to Chapter 40, is the SMEFT of Grzadkowski et al. [240] and the review by Brivio and Trott [93].

Chapter 35

Beyond the Standard Model

The Standard Model, assembled in full in Chapter 33, is the most precisely tested theory in the history of physics—and we know it is not the end of the story. This chapter makes that double statement precise. Its organizing idea is a single reframing: the Standard Model is not a candidate theory of everything but the relevant-and-marginal truncation of an effective field theory, the complete catalogue of gauge-invariant operators of mass dimension four or less that can be built from the known fields. Read this way, "beyond the Standard Model" acquires a literal, constructive meaning. It is everything the truncation discards: a *tower* of higher-dimension operators, dimension five and up, each suppressed by inverse powers of a heavy scale Λ where new physics resides; and the unexplained *values* of the relevant couplings—the vacuum energy and the Higgs mass parameter—which the truncation contains but cannot account for. Every puzzle in this chapter is one entry in that summary read aloud.

The viewpoint is imported wholesale from earlier in the book. The Wilsonian classification of operators into relevant, marginal, and irrelevant directions (Chapter 23) is what turns dimensional analysis into a statement about the deep ultraviolet; the effective-field-theory paradigm of Chapter 38—integrating out heavy states, matching, and the Appelquist–Carazzone decoupling that protects the low-energy description—is the machinery that makes the operator tower a controlled expansion rather than a wish. The base of the tower is the assembled Standard Model Lagrangian $\mathcal{L}_{\mathrm{SM}}$ of Chapter 33, together with its accidental symmetries and its parameter count. From the running of the gauge couplings we borrow the non-abelian beta function of Chapter 27 and the Callan–Symanzik equation of Chapter 24; from the flavour sector, the Yukawa structure and the PMNS matrix of Chapter 32; and the numerical inputs—the electroweak vacuum v, the Planck mass, the measured couplings—are collected in Chapter F. Where a resolution needs technology this chapter does not build, we name where it is built: the strong-CP problem rests on the θ-vacuum and instantons of Chapter 42, and supersymmetry, the most studied speculative extension, is constructed from

its algebra in Chapter 44.

The chapter then walks the summary from the bottom rung upward. We first establish the SMEFT expansion itself and prove that the dimension-five rung is unique—the single Weinberg operator (Section 35.1). That operator *is* a Majorana neutrino mass, and following it to its ultraviolet completion gives the seesaw mechanism and its sharp prediction of a high lepton-number-violating scale (Section 35.2). We then turn to the two relevant couplings whose values the Standard Model leaves unexplained: the Higgs mass and the hierarchy problem (Section 35.3), and the topological θ angle and the strong-CP problem with its axion resolution (Section 35.4). With the running couplings in hand we test grand unification quantitatively—computing the beta-function coefficients, the predicted $\sin^2\theta_{\mathrm{W}} = \frac{3}{8}$, the unification scale, and the proton-decay rate that excludes minimal SU(5) (Section 35.5). Two empirical demands the Standard Model provably cannot meet—the dark-matter density and the baryon asymmetry, with the thermal-relic and leptogenesis calculations carried in full—occupy Section 35.6.

The closing section (Section 35.7) draws the six stories back into the single map they came from, and states plainly, dated to the mid-2020s, which entries in the summary are established and which are hope. Throughout we hold to the book's discipline on the frontier: where a mechanism is conjectural, conditional, or constrained only from one side, we say so and date it. The reader leaves with a map of what lies beyond the Standard Model and an honest accounting of how much of it is known; the worked technologies behind each speculation—the EFT and SMEFT machinery (Chapters 38 and 40), the instanton vacuum (Chapter 42), and supersymmetry (Chapter 44)—are what the remaining parts of the book build.

35.1 The Standard Model as an Effective Field Theory

The chapter introduction reframed the Standard Model in a single sentence: it is not a fundamental theory of everything but the relevant-and-marginal truncation of an effective field theory, the collection of all gauge-invariant operators of mass dimension four or less that can be built from the known fields. That sentence is the organizing principle of this entire chapter, and the present section makes it precise. Once the Standard Model Lagrangian $\mathcal{L}_{\mathrm{SM}}$ of Eq. (33.22) is recognized as a truncation, "beyond the Standard Model" acquires a literal, constructive meaning: it is everything that the truncation discards. There are exactly two such things. First, the *tower* of higher-dimension operators, dimension five and up, each suppressed by inverse powers of a heavy scale Λ at which new physics resides; these encode the virtual influence of whatever heavy states the Standard Model omits. Second, the *values* of the relevant couplings—the vacuum energy

and the Higgs mass parameter—which the truncation contains but does not explain, and whose sensitivity to the heavy scale is the source of the naturalness puzzles. Our task here is to build the summary that sorts every operator by its dimension, to prove which operators populate the lowest rung of the tower, and to show how the rungs map one-to-one onto the topics of the chapter. Everything that follows—neutrino mass, the hierarchy problem, strong CP, proton decay and unification—is a single entry in this summary read aloud.

Throughout we work in natural units $\hbar = c = 1$ with the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ fixed in Chapter 33; signs of operators will not be load-bearing in this section, but mass dimensions will be everything.

35.1.1 The dimensional hierarchy and the SMEFT expansion

The Wilsonian classification of Theorem 23.6 sorts every local operator by a single number, its scaling dimension, into relevant, marginal, and irrelevant directions in theory space. Applied to a four-dimensional theory of the Standard Model fields, that classification is nothing more than canonical mass-dimension counting, because at the Gaussian fixed point the scaling dimension of an operator equals its engineering dimension. We therefore begin with the engineering dimensions of the building blocks. In $d = 4$ spacetime dimensions, the requirement that the kinetic terms $\frac{1}{2}(\partial\phi)^2$, $\bar{\psi}\, i\partial\!\!\!/\psi$, and $-\frac{1}{4}X_{\mu\nu}X^{\mu\nu}$ each carry the Lagrangian-density dimension $[\mathcal{L}] = 4$ fixes

$$[\phi] = [A_\mu] = [H] = 1, \qquad [\psi] = \tfrac{3}{2}, \qquad [\partial_\mu] = [D_\mu] = 1, \qquad [X_{\mu\nu}] = 2, \tag{35.1}$$

as established in Appendix A. The single tool we need to convert these into a statement about couplings is Theorem A.3: an operator $\mathcal{O}$ of mass dimension Δ multiplying the Lagrangian density must carry a coupling of dimension

$$[\,g\,] = 4 - \Delta. \tag{35.2}$$

The sign of $4 - \Delta$ is the whole story. A coupling with positive mass dimension ($\Delta < 4$) multiplies a *relevant* operator: its effect grows in the infrared, it is sensitive to the deep ultraviolet, and there are only finitely many such operators. A dimensionless coupling ($\Delta = 4$) multiplies a *marginal* operator: these are the renormalizable interactions, the ones whose strength is nearly scale-invariant and runs only logarithmically. A coupling with negative mass dimension ($\Delta > 4$) multiplies an *irrelevant* operator: its effect dies as a power of energy in the infrared, it is non-renormalizable, and there are infinitely many of them. This is exactly the relevant/marginal/irrelevant trichotomy of Theorem 23.23, now read off a table of integers.

The Standard Model is the statement that nature, at energies far below Λ, is described by *all* the operators with $\Delta \le 4$ that respect the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and Lorentz invariance, and by no others. We catalogue

them. At dimension zero stands the identity operator, with coefficient the vacuum energy density ρ_Λ of dimension four—the cosmological constant. At dimension two stands the unique gauge-invariant quadratic in the Higgs field, $H^\dagger H$, with coefficient the Higgs mass parameter μ_H^2 of dimension two. These are the only two relevant operators: every other gauge invariant has $\Delta \geq 3$, and the would-be dimension-three fermion mass terms $\bar{\psi}\psi$ are forbidden because the chiral Standard Model fermions of Table 33.1 admit no gauge-invariant bilinear (the obstruction proved in Theorem 31.17 for the neutrino is the same obstruction that forbids every bare fermion mass). At dimension four sit the marginal operators: the gauge kinetic terms $X^A_{\mu\nu}X^{A\,\mu\nu}$ for each factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the Higgs kinetic term $(D_\mu H)^\dagger (D^\mu H)$, the quartic $(H^\dagger H)^2$, the Yukawa couplings $\bar{Q} H d_R$ and their relatives, and—marginal but topological, and therefore set apart—the theta term $X^A_{\mu\nu}\widetilde{X}^{A\,\mu\nu}$ of Eq. (33.85). The sum of all of these, and the proof that nothing else of dimension ≤ 4 is allowed, is precisely the content of the assembled Lagrangian Eq. (33.22). The Standard Model *is* the dimension-≤ 4 truncation; this is not a slogan but the catalogue just given.

Beyond the truncation lies the tower. Admitting that $\mathcal{L}_{\mathrm{SM}}$ is the low-energy shadow of physics at a scale Λ, the most general local theory of the same fields with the same symmetries is the *Standard Model Effective Field Theory* (SMEFT):

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \frac{c_5}{\Lambda}\,\mathcal{O}_5 + \sum_i \frac{c_{6,i}}{\Lambda^2}\,\mathcal{O}_{6,i} + \sum_i \frac{c_{7,i}}{\Lambda^3}\,\mathcal{O}_{7,i} + \cdots = \mathcal{L}_{\mathrm{SM}} + \sum_{d\geq 5}\sum_i \frac{c_{d,i}}{\Lambda^{d-4}}\,\mathcal{O}_{d,i}, \tag{35.3}$$

where each $\mathcal{O}_{d,i}$ is a gauge- and Lorentz-invariant local operator of canonical dimension d, and each Wilson coefficient $c_{d,i}$ is dimensionless because, by Eq. (35.2), an operator of dimension d demands a coupling of dimension $4-d$, which we have made explicit by extracting Λ^{4-d}. This is the Standard Model instance of the general operator expansion Eq. (38.3): a renormalizable core plus an infinite, organized tower of irrelevant corrections suppressed by growing powers of the heavy scale. Figure 35.1 displays the summary as a tower, with the relevant operators below the marginal floor, the marginal Standard Model on the floor, and the irrelevant tower rising above it, each rung carrying its suppression Λ^{4-d}.

Definition 35.1 (The Standard Model Effective Field Theory)**.** The SMEFT is the effective field theory Eq. (35.3) whose fields are the Standard Model fields of Table 33.1, whose symmetry is $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ together with Lorentz invariance, and whose operators comprise every gauge- and Lorentz-invariant local monomial in those fields and their covariant derivatives, organized by canonical mass dimension. Its dimension-≤ 4 part is $\mathcal{L}_{\mathrm{SM}}$; its dimension-$\geq 5$ part is the operator tower, with Wilson coefficients $c_{d,i}/\Lambda^{d-4}$ encoding all virtual effects of physics at the scale Λ.

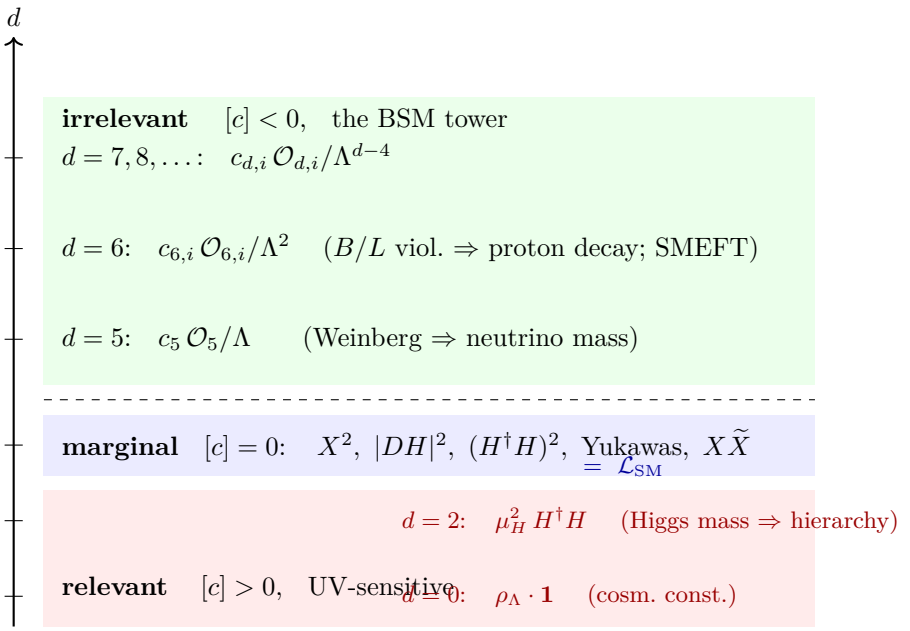


Figure 35.1: The dimensional hierarchy of the SMEFT. The Standard Model occupies the marginal floor ($d = 4$) and the relevant operators beneath it ($d = 0, 2$); "beyond the Standard Model" is the irrelevant tower ($d \geq 5$), each rung suppressed by Λ^{4-d}, together with the unexplained values of the two relevant couplings. The lowest rung, $d = 5$, is the unique Weinberg operator (Theorem 35.2); the chapter ascends the summary one rung at a time.

35.1.2 Decoupling and the leading window onto the ultraviolet

The expansion Eq. (35.3) would be an idle reorganization if the tower were not ordered by physical importance. It is, and the ordering follows from a power-counting estimate that we now make. Suppose the true theory contains a heavy state of mass $M \sim \Lambda$, and that we observe physics at energies $E \ll M$. We integrate the heavy state out, in the precise matching sense of Section 38.3: its quantum effects on the light fields are reproduced, to any desired accuracy in E/M, by a tower of local operators built from the light fields alone, with calculable coefficients. The question is how a given coefficient scales with M.

The cleanest answer is pure dimensional analysis and uses nothing beyond Eq. (35.2). Integrating out a heavy field cannot change the dimension of spacetime or the requirement $[\mathcal{L}] = 4$. An operator $\mathcal{O}_d$ of canonical dimension d generated in the low-energy Lagrangian must therefore appear with a coefficient of dimension $4 - d = -(d - 4)$. After the heavy field is gone, the only mass scale that the matching computation can supply is M itself. Hence the coefficient is forced to

be

$$\frac{(\text{dimensionless})}{M^{\,d-4}} = \frac{c}{M^{\,n}}, \qquad n \equiv d-4 \geq 1, \tag{35.4}$$

with c a dimensionless number built from the couplings of the heavy theory and from loop factors. An operator of dimension $4+n$ is suppressed by exactly n powers of the heavy mass. The lowest-dimension operator the symmetries allow is therefore the *least* suppressed, and dominates every low-energy observable to which it contributes. This is the precise sense in which the lowest rung of the tower is the leading window onto the ultraviolet: the first symmetry-allowed irrelevant operator is the one nature shows us first.

The dimensional argument is airtight but abstract; it is worth seeing the mechanism explicitly, which is also the content of the tree-level matching of Section 38.3. Let a heavy real scalar Φ of mass M couple linearly to a dimension-δ composite $\mathcal{O}_{\text{light}}$ of the light fields through $\mathcal{L} \supset -\frac{1}{2}\Phi(\partial^2 + M^2)\Phi + g\,\Phi\,\mathcal{O}_{\text{light}}$, with g a coupling of dimension $4-1-\delta = 3-\delta$. The classical equation of motion for Φ at energies $E \ll M$ is solved by expanding the propagator,

$$\Phi = \frac{g}{\partial^2 + M^2}\,\mathcal{O}_{\text{light}} = \frac{g}{M^2}\Big(1 - \frac{\partial^2}{M^2} + \cdots\Big)\mathcal{O}_{\text{light}}, \tag{35.5}$$

and substituting back produces the effective interaction

$$\mathcal{L}_{\text{eff}} = \tfrac{1}{2}\,\frac{g^2}{M^2}\,\mathcal{O}_{\text{light}}^2 - \tfrac{1}{2}\,\frac{g^2}{M^4}\,\mathcal{O}_{\text{light}}\,\partial^2\,\mathcal{O}_{\text{light}} + \cdots . \tag{35.6}$$

The leading term is an operator of dimension 2δ with coefficient g^2/M^2; restoring $[g] = 3-\delta$ confirms the coefficient has dimension $2(3-\delta)-2 = 4-2\delta$, exactly $4-d$ for $d = 2\delta$, in agreement with Eq. (35.4). Each additional pair of derivatives in the expansion of the propagator raises the operator dimension by two and costs a further $1/M^2$. The heavy state has resolved itself into the tower, ordered precisely by inverse powers of its mass. This is the constructive face of the Appelquist–Carazzone decoupling theorem (Theorem 38.10): in a physical renormalization scheme, a heavy particle's effects on low-energy observables are suppressed by powers of E/M, and the suppression is organized operator by operator according to Eq. (35.4).

Physical Insight

Decoupling is what makes physics possible in layers. Because a heavy state contributes to the dimension-$(4+n)$ coefficient only as $1/M^n$, one need not know the spectrum at 10^{16} GeV to predict atomic spectra, and one could discover electromagnetism a century before the W boson. The flip side is the strategy of this chapter: *because* the unknown ultraviolet enters only through the coefficients of a fixed, finite list of low-dimension operators, we can write that list down, bound or measure each coefficient,

and read off the scale and symmetry structure of the new physics without ever naming it. The SMEFT is agnosticism made quantitative.

35.1.3 The lowest rung: uniqueness of the dimension-five operator

The power-counting estimate Eq. (35.4) elevates one operator above all others: the dimension-five operator, suppressed by only a single power of Λ, is the leading correction to the Standard Model and the first place the tower can announce itself. It is a remarkable and consequential fact that, on the Standard Model field content, this operator is essentially *unique*. We prove it by enumeration. The same operator was encountered from a different direction in Theorem 33.13, where it appeared as the leading violator of the accidental symmetries; here we obtain it systematically as entry $d = 5$ of the summary, which is the role it plays for the rest of the chapter.

Theorem 35.2 (Uniqueness of the dimension-five operator)**.** *On the Standard Model fields, the only gauge-invariant, Lorentz-invariant local operator of canonical mass dimension five is the Weinberg operator*

$$\mathcal{O}_5 \;=\; (L_i^{\mathsf{T}}\,\epsilon\,H)\,C\,(H^{\mathsf{T}}\,\epsilon\,L_j), \qquad \epsilon \equiv i\sigma^2, \tag{35.7}$$

with L_i the lepton doublet of generation i, H the Higgs doublet, and C the charge-conjugation matrix; it is symmetric in the flavor indices i, j, so its coefficient is a complex symmetric matrix. The operator carries $\Delta L = 2$ and $\Delta B = 0$, and it is the unique such structure up to flavor and Hermitian conjugation.

Proof. We proceed in three steps: fix the field content allowed at dimension five, impose $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ invariance, and count the independent Lorentz/internal contractions.

Step 1: field content. A Lorentz-invariant operator must contain an even number of fermion fields, since a single Weyl or Dirac field carries half-integer spin and cannot form a Lorentz scalar by itself; the smallest fermionic invariants are bilinears. Using Eq. (35.1), we enumerate the ways to reach total dimension five. With *four or more fermions* the fermionic dimension is already $4 \times \frac{3}{2} = 6 > 5$, so these are excluded. With *zero fermions* we would need a purely bosonic gauge-invariant scalar of dimension five built from H ($\dim 1$), D_μ ($\dim 1$), and $X_{\mu\nu}$ ($\dim 2$). Gauge invariance forces the Higgs fields into the singlet combination $H^\dagger H$ (even in the number of H's) and the field strengths into Lorentz-contracted pairs $X_{\mu\nu}X^{\mu\nu}$ or $X_{\mu\nu}\widetilde{X}^{\mu\nu}$ (dimension four); every Lorentz index must be contracted, which pairs derivatives as $\partial_\mu \cdots \partial^\mu$. All such invariants have even dimension, so dimension five is unreachable with zero fermions. The only remaining possibility is *two fermions*, contributing dimension three, with bosonic content of total

dimension $5 - 3 = 2$. Two units of bosonic dimension can be assembled as: (a) two Higgs fields (HH, dimension two); (b) one field strength $X_{\mu\nu}$ (dimension two); or (c) two derivatives. Option (c) yields operators that, after using the free equations of motion, reduce to lower-dimension operators or vanish (they are redundant in the sense made precise in Section 38.3), and carry no new gauge-invariant structure. Option (b), a fermion dipole $\bar{\psi}\sigma^{\mu\nu}\psi\, X_{\mu\nu}$, requires a chirality-flipping bilinear $\bar{\psi}_L\sigma^{\mu\nu}\psi_R$; but the Standard Model fermions are chiral and no two of them form a gauge singlet without an additional Higgs insertion, which would raise the dimension to six. Hence only option (a) survives: the operator is schematically $\psi\psi HH$, two fermions and two Higgs doublets.

Step 2: gauge invariance. The two Higgs doublets $H \sim (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ carry total hypercharge $Y = +1$ and are color singlets. For $\mathcal{O}_5$ to be a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet, the two fermions must be color singlets with total hypercharge $Y = -1$. Among the Standard Model fermions only the leptons are colorless. The hypercharge assignment of Eq. (31.21) gives $Y_L = -\frac{1}{2}$ for the lepton doublet $L \sim (\mathbf{1}, \mathbf{2}, -\frac{1}{2})$ and $Y_{e_R} = -1$ for the singlet. Two lepton doublets supply $Y = -1$ and, crucially, carry the $\mathrm{SU}(2)_L$ indices needed to contract against the doublet indices of HH. The alternatives fail: e_R is an $\mathrm{SU}(2)_L$ singlet, so the bilinear $\overline{e_R^c}e_R$ leaves the two doublet indices of HH uncontracted (not an $\mathrm{SU}(2)_L$ singlet) and in any case carries $Y = +2$, giving $Y = +3$ with the Higgs pair; the quarks are colored and excluded. Therefore the fermions are two lepton doublets, and the operator is built from $LLHH$.

Step 3: contraction counting. It remains to count independent ways to contract the $\mathrm{SU}(2)_L$ and Lorentz indices of $LLHH$ into a singlet. The two Higgs doublets, being identical commuting bosonic fields, satisfy $H_a H_b = H_b H_a$, so their product is symmetric in the $\mathrm{SU}(2)_L$ indices a, b. The antisymmetric (singlet) combination $\epsilon^{ab}H_a H_b = H_1 H_2 - H_2 H_1 = 0$ vanishes identically; only the symmetric ($\mathrm{SU}(2)_L$-triplet) part of HH is nonzero. To build a singlet, the lepton bilinear must therefore furnish the conjugate triplet, which forces the Lorentz-scalar (Majorana-type) contraction $L^{\mathsf{T}}C\,(\cdot)\,L$ of the two leptons in the symmetric $\mathrm{SU}(2)_L$ representation; the antisymmetric lepton contraction is killed against the symmetric Higgs pair. Assembling the surviving structure with the $\epsilon = i\sigma^2$ tensors that map a doublet to its conjugate yields exactly Eq. (35.7). The construction leaves one independent operator for each symmetric flavor pair (i, j), so the Wilson coefficient is a complex symmetric matrix c_{ij}; there is no second, inequivalent Lorentz or gauge contraction. This establishes uniqueness, and the operator coincides with Eq. (31.121) and Eq. (33.58).

Quantum numbers. Lepton number assigns $L = +1$ to each lepton doublet and $L = 0$ to the Higgs, so $\mathcal{O}_5$ carries $\Delta L = +2$ (its conjugate $\Delta L = -2$); it contains no quarks, so $\Delta B = 0$. Hence $\Delta(B - L) = -2 \neq 0$ as well. □

The two facts packed into Theorem 35.2—that the leading rung of the tower is unique, and that it violates lepton number by two units—deserve to be drawn

out, because together they make the dimension-five operator the single sharpest probe in the whole edifice.

Worked Example 35.3 (Why $\mathcal{O}_5$ is forbidden in the renormalizable Standard Model but allowed once $d > 4$ is admitted)**.** The accidental symmetries of Theorem 33.10 state that the renormalizable Standard Model, the dimension-≤ 4 truncation, conserves baryon number B and each lepton number L_i exactly—not because they were imposed, but because no gauge-invariant operator of dimension ≤ 4 violates them. We can see directly that $\mathcal{O}_5$ is excluded from the renormalizable theory and admitted as soon as the truncation is lifted.

Forbidden at $d \leq 4$. The operator $\mathcal{O}_5$ has dimension five. By Eq. (35.2) its coupling has dimension $4 - 5 = -1$; it is non-renormalizable and therefore, by definition, not part of $\mathcal{L}_{\rm SM}$, which contains only operators of non-negative coupling dimension. There is no escape route: by Theorem 35.2 the *only* dimension-five gauge invariant is $\mathcal{O}_5$ itself, and there is no dimension-≤ 4 gauge invariant carrying $\Delta L = 2$. (The would-be renormalizable Majorana mass $\overline{(\nu_L)^c}\nu_L$, of dimension three, is not gauge invariant: it transforms as an $\mathrm{SU}(2)_L$ triplet of hypercharge -1, as shown in Theorem 31.17.) Lepton number is thus conserved in the renormalizable theory for the purely structural reason that the lowest operator able to violate it lives one rung up the summary.

Allowed at $d = 5$. The instant we admit dimension-five operators—that is, the instant we acknowledge $\mathcal{L}_{\rm SM}$ is an effective theory with a finite cutoff Λ—the operator $\mathcal{O}_5$ is permitted by every symmetry the Standard Model actually has: it is a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet (Step 2 above) and Lorentz invariant. Nothing forbids it, so by the EFT completeness principle behind Eq. (35.3) it *must* be included, with some coefficient c_5/Λ. The accidental $U(1)_L$ and $U(1)_{B-L}$ symmetries of Theorem 33.10 are accordingly broken by the leading term of the tower; B alone (and the combination $B + L$ at this order) survives to dimension five, with B-violation postponed to dimension six. The accidental symmetries were never fundamental—they were artifacts of truncating at $d = 4$, and the truncation's first correction breaks the most fragile of them.

35.1.4 Reading the summary as the plan of the chapter

The dimensional hierarchy is not merely a bookkeeping device; it is the table of contents of everything that lies beyond the Standard Model, and each entry is a section of this chapter. We read it from the lowest rung up, and then down into the relevant operators.

The dimension-five rung is, by Theorem 35.2, the single Weinberg operator, and by its $\Delta L = 2$ character it is a Majorana neutrino mass: after electroweak symmetry breaking, $H^0 \to v/\sqrt{2}$ turns Eq. (35.7) into $m_\nu \sim c_5\, v^2/\Lambda$. The one laboratory-established fact beyond the Standard Model—that neutrinos have mass—is therefore the statement that the lowest rung of the tower has a nonzero

coefficient, and the smallness of m_ν is the largeness of Λ. This is the subject of Section 35.2, where the suggestive scale $\Lambda \sim 10^{14}$–10^{15} GeV emerges and the heavy states whose exchange generates $\mathcal{O}_5$ are identified.

The dimension-six rung is populated by many operators (the Warsaw basis catalogued in Chapter 40); among them are the four-fermion operators that violate B and L while conserving $B - L$, and these mediate proton decay at a rate $\Gamma \sim m_p^5/\Lambda^4$. The nonobservation of proton decay places the strongest direct lower bound on Λ, and the same operators connect to the unification of the gauge couplings. This is Section 35.5. Higher rungs—dimension seven, eight, and beyond—are increasingly suppressed and contribute the precision deviations of the SMEFT program; collectively they are the honest map of the landscape taken up in Section 35.7.

Beneath the marginal floor sit the relevant operators, and these are puzzles not of new operators but of the values of old couplings. The Higgs mass parameter $\mu_H^2 H^\dagger H$, the unique relevant operator besides the vacuum energy, has a coefficient of dimension two: by the power counting Eq. (35.4) run in reverse, it is *additively* sensitive to the heavy scale, receiving contributions of order Λ^2 from every heavy threshold. Why it sits at the electroweak scale rather than at Λ is the hierarchy problem of Section 35.3. Here the chapter consciously draws on ϕ^4 theory: the relevant coupling m^2 of the scalar theory carried since Chapter 23 and Chapter 24 is structurally the same object as μ_H^2, the lone relevant deformation of a scalar sector, and the sensitivity of m^2 to the cutoff that the Wilsonian analysis exhibited there is the hierarchy problem in miniature. We flag the connection now and cash it in Section 35.3.

Finally, the marginal-but-topological theta term $X^A_{\mu\nu}\widetilde{X}^{A\,\mu\nu}$ of Eq. (33.85), dimensionless and hence not power-suppressed, raises a puzzle of its own: its CP-violating QCD coefficient $\bar{\theta}$ is measured to be extraordinarily small, and explaining why is the strong-CP problem of Section 35.4.

With the summary built and its entries assigned, the chapter has its spine. We descend first to the lowest rung. The unique dimension-five operator Eq. (35.7), with its tantalizing one-power suppression and its $\Delta L = 2$ signature, hands directly to the next section, where it is cashed into the physics of neutrino mass and where the seesaw mechanism is exhibited as the ultraviolet completion that generates it—fixing, in the process, the suggestive value of Λ.

35.2 Neutrino Mass and the Seesaw Mechanism

The operator hierarchy of Section 35.1 handed us a single object at the leading nontrivial order of the tower: among all gauge-invariant operators that can be built from the Standard Model fields, exactly one has mass dimension five. It is the Weinberg operator, recorded as Eq. (31.121) when the lepton sector was assembled. Every other operator in the irrelevant tower is dimension six

or higher; the dimension-five rung is occupied once and only once. This is a remarkable circumstance. The effective-field-theory logic of Chapter 38 tells us that the lowest-dimension operators are the least suppressed by the heavy scale Λ and hence the first to manifest, so the unique dimension-five operator is, in a precise sense, *the* leading window onto whatever physics completes the Standard Model in the ultraviolet. The purpose of this section is to cash that operator into physics. We will see that it is nothing other than a neutrino mass; that the one laboratory-established fact requiring physics beyond the Standard Model—that neutrinos are massive—is exactly this rung of the tower lighting up; and that the measured mass, fed back through the operator, pins the coefficient and therefore points at a scale $\Lambda \sim 10^{14}$–10^{15} GeV, startlingly close to the scale of grand unification. We then exhibit the simplest ultraviolet completion that generates the operator at tree level—the type-I seesaw—and show in full how integrating out a heavy state reproduces the Weinberg coefficient and makes the smallness of the neutrino mass a consequence of the largeness of a new mass.

35.2.1 The Weinberg operator is a Majorana mass

Recall the field content of the lepton sector. Each generation contributes a left-handed lepton doublet

$$L_i = \begin{pmatrix} \nu_{Li} \\ e_{Li} \end{pmatrix}, \qquad L_i \sim (\mathbf{1},\, \mathbf{2},\, -\tfrac{1}{2}) \text{ under } \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y, \tag{35.8}$$

with $i = 1, 2, 3$ a generation label, together with the Higgs doublet $H \sim (\mathbf{1}, \mathbf{2}, +\tfrac{1}{2})$ and its charge conjugate $\widetilde{H} = i\sigma^2 H^* \sim (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2})$. The Standard Model contains no right-handed neutrino: there is no gauge singlet ν_R in Table 33.1. Consequently no renormalizable, gauge-invariant neutrino mass can be written. A Dirac mass $\bar{\nu}_L \nu_R$ is impossible for want of ν_R; a Majorana mass $\bar{\nu}_L^c\, \nu_L$ is forbidden because ν_L sits inside the doublet L and carries weak isospin and hypercharge, so the bilinear is not gauge invariant. This is precisely the content of Theorem 31.17: within the renormalizable Standard Model the neutrino is exactly massless, and lepton number is an accidental symmetry. Neutrino mass is therefore not a tuning of a Standard Model parameter—it is an operator that the renormalizable theory simply does not possess.

It appears one dimension up. The unique dimension-five gauge invariant is

$$\mathcal{L}_5 = \frac{c_{ij}}{\Lambda}\, \mathcal{O}_5^{ij} + \text{h.c.}, \qquad \mathcal{O}_5^{ij} = (\overline{L_i^c}\, \widetilde{H}^*)(\widetilde{H}^\dagger L_j), \tag{35.9}$$

the form fixed in Eq. (31.121). Here c_{ij} is a dimensionless coefficient matrix, symmetric in its flavor indices (a fact we verify below), Λ is the heavy scale that suppresses the operator, and $L^c = C\bar{L}^T$ is the charge conjugate built from the conjugation matrix C of Section 11.7.2. Let us check the structure. The

two factors are separately $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlets: $\widetilde{H}^\dagger L$ contracts two $\mathrm{SU}(2)_L$ doublets into a singlet, with hypercharge $(-(-\frac{1}{2})) + (-\frac{1}{2}) = 0$; and $\overline{L^c}\widetilde{H}^*$ is its conjugate partner. The operator carries two units of lepton number, since each lepton field carries $L = +1$ and there are two of them with no conjugate lepton to cancel: $\mathcal{O}_5$ has $\Delta L = +2$. It is built from two Higgs doublets and two lepton doublets, total mass dimension $2 \times \frac{3}{2} + 2 \times 1 = 5$, so its coefficient must carry one inverse power of mass, here c/Λ. This single operator is the entirety of dimension-five physics in the Standard Model EFT.

To read off its physical content, break electroweak symmetry. With the vacuum alignment of Chapter 30, the Higgs acquires $\langle H \rangle = (0,\, v/\sqrt{2})^T$ with $v \simeq 246$ GeV, so that

$$\left\langle \widetilde{H} \right\rangle = i\sigma^2 \langle H \rangle^* = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} 0 \\ v/\sqrt{2} \end{pmatrix} = \begin{pmatrix} v/\sqrt{2} \\ 0 \end{pmatrix}. \tag{35.10}$$

The contraction $\widetilde{H}^\dagger L$ projects onto the upper, neutral component of the doublet,

$$\widetilde{H}^\dagger L_j \;\longrightarrow\; (\tfrac{v}{\sqrt{2}},\, 0) \begin{pmatrix} \nu_{Lj} \\ e_{Lj} \end{pmatrix} = \frac{v}{\sqrt{2}}\, \nu_{Lj}, \tag{35.11}$$

and likewise $\overline{L_i^c}\, \widetilde{H}^* \to (v/\sqrt{2})\, \overline{\nu_{Li}^c}$. The dimension-five term therefore collapses, after symmetry breaking, to

$$\mathcal{L}_5 \;\longrightarrow\; \frac{c_{ij}}{\Lambda}\, \frac{v^2}{2}\, \overline{\nu_{Li}^c}\, \nu_{Lj} + \text{h.c.} \tag{35.12}$$

This is a *Majorana* mass for the left-handed neutrinos, of exactly the form catalogued in Section 11.7.6. Writing the canonical Majorana mass term as

$$\mathcal{L}_{\text{Maj}} = -\tfrac{1}{2}\, (m_\nu)_{ij}\, \overline{\nu_{Li}^c}\, \nu_{Lj} + \text{h.c.}, \tag{35.13}$$

and matching coefficients with Eq. (35.12), we obtain the central relation of this section,

$$\boxed{(m_\nu)_{ij} = -\, c_{ij}\, \frac{v^2}{\Lambda}\,, \qquad m_\nu \sim c_5\, \frac{v^2}{\Lambda}.} \tag{35.14}$$

The neutrino mass is the Higgs vacuum value squared, divided by the heavy scale, weighted by the (order-one, in the absence of a reason otherwise) coefficient c_5. The factor v^2 rather than v is the fingerprint of a dimension-five operator: two Higgs legs must take their vacuum values, whereas a charged-fermion Dirac mass uses only one and scales as yv. The neutrino mass is suppressed not by a small Yukawa coupling but by the ratio v/Λ of the electroweak scale to a new, much higher scale. That is the entire mechanism in one line.

The symmetry of c_{ij} asserted above follows from Fermi statistics. The bilinear $\overline{\nu_{Li}^c}\, \nu_{Lj} = \nu_{Li}^T C\, \nu_{Lj}$ involves anticommuting fields, and C is antisymmetric

($C^T = -C$, Section 11.7.2); transposing the (Grassmann) scalar and using both signs gives $\nu_{Li}^T C \nu_{Lj} = \nu_{Lj}^T C \nu_{Li}$, so only the symmetric part of c_{ij} survives. Thus m_ν is a complex *symmetric* 3×3 matrix—not Hermitian, as a Dirac mass would be—and its diagonalization requires the Autonne–Takagi factorization rather than an ordinary bi-unitary rotation. We return to this when connecting to the PMNS matrix in Section 35.2.7.

35.2.2 Majorana, not Dirac: lepton number violation by two units

It is worth dwelling on why Eq. (35.12) is a fundamentally different object from a charged-fermion mass, because this difference is the experimental handle on the whole story. Recall from Eq. (11.283) that a Majorana field is one that equals its own charge conjugate, $\psi^c = \psi$, so that the particle is identical to its antiparticle; a Majorana mass term $-\frac{1}{2} m\, \overline{\psi^c} \psi$ is built from a single chirality and pairs a field with its conjugate. A Dirac mass $-m\, \bar{\psi}_L \psi_R$, by contrast, links two independent chiralities and is invariant under the phase rotation $\psi_{L,R} \to e^{i\alpha} \psi_{L,R}$ that defines a conserved fermion number.

Apply each to the neutrino. A Dirac neutrino mass would require a right-handed partner ν_R and would read $-m_D\, \bar{\nu}_L \nu_R$. Under the global rotation $\nu_L \to e^{i\alpha} \nu_L$, $\nu_R \to e^{i\alpha} \nu_R$—which is lepton number—this term is invariant, so a Dirac neutrino conserves lepton number, exactly as the electron's Dirac mass conserves electron number. The Majorana term Eq. (35.12), on the other hand, is built from ν_L and its conjugate ν_L^c. Under lepton number $\nu_L \to e^{i\alpha} \nu_L$ one has $\nu_L^c \to e^{i\alpha} \nu_L^c$ as well (charge conjugation does not flip the phase the way it flips electric charge here, because L is a global number carried with the same sign by ν_L^c when written as a left-handed field), so the bilinear $\overline{\nu_L^c} \nu_L = \nu_L^T C \nu_L$ rotates by $e^{2i\alpha}$:

$$\overline{\nu_L^c}\, \nu_L \;\longrightarrow\; e^{2i\alpha}\, \overline{\nu_L^c}\, \nu_L . \tag{35.15}$$

The term is not invariant; it changes lepton number by $\Delta L = 2$. This is the same $\Delta L = 2$ that we identified at the level of the dimension-five operator before symmetry breaking. The Weinberg operator, and hence any neutrino mass it generates, *necessarily* violates lepton number by two units. A massive Majorana neutrino is its own antiparticle.

This is not a matter of taste between two equally good descriptions. The Standard Model field content offers only ν_L; the only renormalizable way to give it a mass would be to add ν_R by hand and write a Dirac term, but then nothing forbids the gauge-singlet Majorana mass $\frac{1}{2} M\, \overline{\nu_R^c} \nu_R$ as well, and we are back to the seesaw of Section 35.2.3. If instead one declines to add any new field, the leading mass is the Weinberg operator, which is Majorana by construction. The generic expectation, then, is that neutrino masses are Majorana and lepton number is violated. Whether nature realizes this is an experimental question

with a sharp signature, to which we turn in Section 35.2.8.

Worked Example 35.4 (The seesaw scale from the atmospheric mass splitting)**.** Neutrino oscillation experiments measure differences of squared masses, not the masses themselves. The atmospheric splitting is $|\Delta m^2_{31}| \simeq 2.5 \times 10^{-3}$ eV^2 and the solar splitting $\Delta m^2_{21} \simeq 7.4 \times 10^{-5}$ eV^2 (values as of the mid-2020s global fits). With the lightest mass not far below the splittings, the heaviest neutrino has

$$m_\nu \gtrsim \sqrt{|\Delta m^2_{31}|} \simeq \sqrt{2.5 \times 10^{-3}}\ \text{eV} \simeq 0.05\ \text{eV}. \tag{35.16}$$

Feed this into the headline relation Eq. (35.14), taking $c_5 = 1$ and the measured electroweak scale $v = 246$ GeV. Solving for Λ,

$$\begin{aligned}\Lambda = c_5\,\frac{v^2}{m_\nu} &= \frac{(246\ \text{GeV})^2}{0.05\ \text{eV}} = \frac{6.05 \times 10^4\ \text{GeV}^2}{5 \times 10^{-11}\ \text{GeV}} \\ &\simeq 1.2 \times 10^{15}\ \text{GeV},\end{aligned} \tag{35.17}$$

where we converted $0.05\ \text{eV} = 5 \times 10^{-11}$ GeV. If instead the coefficient carries a one-loop suppression $c_5 \sim (16\pi^2)^{-1} \sim 6 \times 10^{-3}$, the inferred scale drops by the same factor to $\Lambda \sim 7 \times 10^{12}$ GeV; if c_5 is somewhat below unity for a generic order-one reason, Λ sits in the band 10^{14}–10^{15} GeV. This is the first quantitative pointer in the entire Standard Model EFT to a definite high scale, and its proximity to the grand-unification scale $\sim 2 \times 10^{16}$ GeV inferred independently from gauge-coupling running (Section 35.5) is the kind of coincidence that has driven model building for four decades. We hand this number forward to Section 35.5.

35.2.3 The type-I seesaw as the ultraviolet completion

The Weinberg operator is an effective interaction; something must generate it. The dimension-five operator $\mathcal{O}_5$ has the schematic structure $LHLH$, and there are exactly three ways to split it into a product of two renormalizable currents joined by a single heavy propagator—three tree-level ultraviolet completions, known as the type-I, type-II, and type-III seesaws. We treat the first in full, as it is both the simplest and the one that feeds the leptogenesis story of Section 35.6.4; the other two are sketched in Section 35.2.6.

The type-I completion supplies the missing right-handed neutrino. Add to the Standard Model n generations (we take $n = 3$ but keep it general) of gauge-singlet Weyl fermions $N_{Ri} \sim (\mathbf{1}, \mathbf{1}, 0)$. Being singlets, they couple to nothing through the gauge interactions; their only renormalizable couplings are a Yukawa interaction with the lepton and Higgs doublets and—crucially—a gauge-invariant Majorana mass, which is permitted precisely *because* they are singlets and so carry no quantum number to forbid $\overline{N^c}N$. The most general renormalizable Lagrangian is

$$\mathcal{L}_N = \overline{N_{Ri}}\, i\partial\!\!\!/\, N_{Ri} - \Big(y_{ij}\, \overline{L_i}\, \widetilde{H}\, N_{Rj} + \tfrac{1}{2}\, M_{ij}\, \overline{N^c_{Ri}}\, N_{Rj} + \text{h.c.} \Big), \tag{35.18}$$

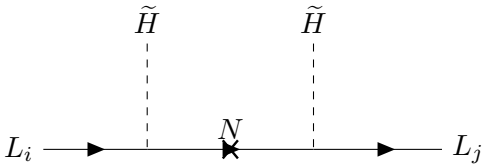


Figure 35.2: Tree-level generation of the Weinberg operator in the type-I seesaw. Two Yukawa vertices y_{ik} and y_{jk} join the external lepton and Higgs doublets to an internal heavy neutrino N_k; the cross on the internal line denotes the lepton-number-violating Majorana mass insertion M_k, the source of $\Delta L = 2$. In the heavy-mass limit the propagator shrinks to a point and the diagram becomes the local dimension-five vertex Eq. (35.9) with coefficient Eq. (35.34).

with y an arbitrary complex $3 \times n$ Yukawa matrix and M a complex symmetric $n \times n$ Majorana mass matrix. The mass scale of M is unrelated to electroweak symmetry breaking: it is a bare mass for a gauge singlet, allowed at any value, and the philosophy of Section 35.1 suggests it sits near whatever ultraviolet scale defines the theory. The Yukawa term carries lepton number if we assign $L(N_R) = +1$, but the Majorana mass M breaks that assignment by two units—this is the seed of the $\Delta L = 2$ we will recover in the low-energy operator.

After electroweak symmetry breaking, $\widetilde{H} \to (v/\sqrt{2}, 0)^T$ and the Yukawa term Eq. (35.18) produces a Dirac mass connecting ν_L and N_R,

$$m_D = \frac{y\, v}{\sqrt{2}}, \tag{35.19}$$

a matrix in generation space. Collecting the neutral fermion mass terms in the basis of left-handed fields $(\nu_L,\, N_R^c)$—note that we use N_R^c, which is left-handed, so that all entries are of the Majorana form—gives

$$\mathcal{L}_{\text{mass}} = -\tfrac{1}{2} \begin{pmatrix} \overline{\nu_L^c} & \overline{N_R} \end{pmatrix} \begin{pmatrix} 0 & m_D \\ m_D^T & M \end{pmatrix} \begin{pmatrix} \nu_L \\ N_R^c \end{pmatrix} + \text{h.c.} \equiv -\tfrac{1}{2}\, \Psi^T C\, \mathcal{M}\, \Psi + \text{h.c.}, \tag{35.20}$$

where $\Psi = (\nu_L,\, N_R^c)^T$ and the symmetric mass matrix is

$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D^T & M \end{pmatrix}. \tag{35.21}$$

The upper-left block is exactly zero: there is no tree-level Majorana mass for ν_L, because that would require a Higgs $\mathrm{SU}(2)_L$ triplet, which the type-I model does not contain. The structure of Eq. (35.21)—a vanishing light–light block, an off-diagonal Dirac block at the electroweak scale, and a large heavy–heavy block—is the entire seesaw. We now diagonalize it.

35.2.4 Diagonalizing the seesaw matrix

Take a single generation, so that m_D and M are real numbers and the mass matrix is the symmetric 2×2

$$\mathcal{M} = \begin{pmatrix} 0 & m_D \\ m_D & M \end{pmatrix}, \qquad m_D \ll M. \tag{35.22}$$

A complex symmetric matrix is brought to nonnegative diagonal form by a single unitary congruence, $\mathcal{M} = U^{*}\hat{m}\,U^{\dagger}$ with $\hat{m} = \mathrm{diag}(m_1, m_2) \geq 0$ (the Autonne–Takagi factorization, the right notion of diagonalization for Majorana masses); for real $\mathcal{M}$ this reduces to an orthogonal rotation up to a sign on each eigenvalue, the sign being a physical Majorana phase. We find the eigenvalues by treating Eq. (35.22) as an ordinary symmetric real matrix and tracking the sign at the end. The characteristic equation is

$$\det(\mathcal{M} - \lambda\,\mathbf{1}) = \det \begin{pmatrix} -\lambda & m_D \\ m_D & M - \lambda \end{pmatrix} = \lambda^2 - M\lambda - m_D^2 = 0, \tag{35.23}$$

with roots

$$\lambda_{\pm} = \frac{M \pm \sqrt{M^2 + 4m_D^2}}{2}. \tag{35.24}$$

These are exact. The seesaw regime is $m_D \ll M$, so $4m_D^2/M^2 \ll 1$ and we expand the square root,

$$\sqrt{M^2 + 4m_D^2} = M\sqrt{1 + \frac{4m_D^2}{M^2}} = M\Big(1 + \frac{2m_D^2}{M^2} - \frac{2m_D^4}{M^4} + \cdots\Big). \tag{35.25}$$

Substituting into Eq. (35.24),

$$\lambda_{+} = \frac{M + M\big(1 + 2m_D^2/M^2 + \cdots\big)}{2} = M + \frac{m_D^2}{M} + O\Big(\frac{m_D^4}{M^3}\Big), \tag{35.26}$$

$$\lambda_{-} = \frac{M - M\big(1 + 2m_D^2/M^2 + \cdots\big)}{2} = -\frac{m_D^2}{M} + O\Big(\frac{m_D^4}{M^3}\Big). \tag{35.27}$$

The spectrum splits cleanly. One eigenvalue is heavy, $\lambda_{+} \simeq M$, with a small upward correction m_D^2/M; the other is light, $|\lambda_{-}| \simeq m_D^2/M$, suppressed relative to the Dirac mass by the very ratio $m_D/M \ll 1$. The light state's mass is the geometric problem's "seesaw": as the heavy end M goes up, the light end m_D^2/M goes down, the two linked by the fixed fulcrum m_D. The negative sign of λ_{-} is removed by the field redefinition $\nu \to i\nu$, which is exactly a Majorana phase of π; the physical light mass is

$$\boxed{\,m_\nu = \frac{m_D^2}{M}.\,} \tag{35.28}$$

This is the seesaw formula. With $m_D = yv/\sqrt{2}$ it reads $m_\nu = y^2 v^2/(2M)$, and comparing with the headline Eq. (35.14) identifies $c_5/\Lambda = \tfrac{1}{2}\, y^2/M$, the single-generation case of the matching we derive below.

The mixing angle that effects the diagonalization is read off from the standard formula for a symmetric 2×2 matrix $\left(\begin{smallmatrix} a & b \\ b & d \end{smallmatrix}\right)$, namely $\tan 2\theta = 2b/(d-a)$. With $a = 0$, $b = m_D$, $d = M$,

$$\tan 2\theta = \frac{2 m_D}{M}, \qquad \theta \simeq \frac{m_D}{M} \ll 1. \tag{35.29}$$

The mixing is tiny. The light mass eigenstate is therefore almost entirely the active neutrino ν_L, with only an $O(m_D/M)$ admixture of the singlet N; conversely the heavy eigenstate is almost purely N, with an $O(m_D/M)$ active component. The physics packaged in Eqs. (35.28) and (35.29) is the heart of the mechanism: *the lightness of the observed neutrino is the direct consequence of the heaviness of M*. There is no small Yukawa needed—y may be $O(1)$—and there is no fine-tuning; a single large scale in the denominator does all the work. Restoring the full three-generation structure, Eq. (35.28) becomes the matrix relation

$$m_\nu = -\, m_D\, M^{-1}\, m_D^{T} = -\,\frac{v^2}{2}\, y\, M^{-1}\, y^{T}, \tag{35.30}$$

the leading block of the block-diagonalization of Eq. (35.21), with the heavy block left at $\simeq M$. We now obtain Eq. (35.30) a second way—by integrating N out—which makes the matching onto the Weinberg coefficient explicit.

35.2.5 Integrating out the heavy neutrino: tree-level matching

The same physics is encoded in the effective theory below M obtained by integrating out N. This is the prototypical tree-level matching of Section 38.3: a heavy field with no light external lines of its own is removed by solving its classical equation of motion in a derivative (heavy-mass) expansion and substituting back. The heavy neutrino's propagator is

$$\frac{i}{\not{p} - M} = -\frac{i}{M}\Big(1 + \frac{\not{p}}{M} + \frac{\not{p}^{\,2}}{M^2} + \cdots\Big), \tag{35.31}$$

so at energies $p \ll M$ the leading term is the local $-i/M$; each additional power of $\not{p}/M$ raises the operator dimension by one. Keeping only the leading term generates the dimension-five operator and nothing lower.

Concretely, write the N-dependent part of Eq. (35.18), drop the kinetic term as subleading in ∂/M, and vary with respect to N_R. The equation of motion sets the Majorana-mass variation against the Yukawa source,

$$M\, N_R \;=\; -\, y^{\dagger}\, \widetilde{H}^{\dagger}\, L \quad\Longrightarrow\quad N_R \;=\; -\, M^{-1}\, y^{\dagger}\, \widetilde{H}^{\dagger}\, L \;+\; O\Big(\frac{\partial}{M}\Big), \tag{35.32}$$

where the N-leg of the Yukawa term $y_{ij}\overline{L_i}\widetilde{H}N_{Rj}$ supplies the source $y^\dagger\widetilde{H}^\dagger L$ on the right. (Equivalently, contract the two Yukawa vertices of Fig. 35.2 through the leading $-1/M$ of the propagator Eq. (35.31) with one Majorana-mass insertion; the diagrammatic and equation-of-motion routes agree because both are the saddle point of the Gaussian N-integral.) We must substitute the solution Eq. (35.32) back into the *full* N-dependent Lagrangian—both the Majorana mass term $-\frac{1}{2}\overline{N_R^c}MN_R$ *and* the Yukawa source $-y\,\overline{L}\,\widetilde{H}N$—not the mass term alone. This is the elementary completion of the square: for $-\frac{1}{2}\,N^cMN-(JN+\text{h.c.})$ with the source $J=y^\dagger\widetilde{H}^\dagger L$ of Eq. (35.32), the saddle leaves $\mathcal{L}_{\text{eff}}=+\frac{1}{2}\,J^cM^{-1}J$. The mass term alone would give $-\frac{1}{2}\,J^cM^{-1}J$; the source term contributes an equal piece that doubles it and flips the overall sign. The result, with N eliminated, is

$$\begin{aligned}\mathcal{L}_{\text{eff}} &= +\tfrac{1}{2}\,\overline{\left(M^{-1}y^\dagger\widetilde{H}^\dagger L\right)^c}\,M\,\left(M^{-1}y^\dagger\widetilde{H}^\dagger L\right)\\ &= +\tfrac{1}{2}\,\left(\overline{L^c}\,\widetilde{H}^*\,y^*M^{-1}\right)M\,\left(M^{-1}y^\dagger\,\widetilde{H}^\dagger L\right)\\ &= +\tfrac{1}{2}\,\overline{L_i^c}\,\widetilde{H}^*\,\left(y^*M^{-1}y^\dagger\right)_{ij}\widetilde{H}^\dagger L_j.\end{aligned}\tag{35.33}$$

This is precisely the Weinberg operator Eq. (35.9). Comparing coefficients,

$$\boxed{\frac{c_{ij}}{\Lambda}=+\tfrac{1}{2}\left(y^*M^{-1}y^\dagger\right)_{ij},\qquad\text{schematically}\quad\frac{c_5}{\Lambda}\sim\frac{y^2}{M}.}\tag{35.34}$$

The Weinberg coefficient is the Yukawa-squared over the heavy Majorana mass: the heavy scale Λ that suppressed the operator *is* the right-handed neutrino mass M (up to the dimensionless y^2), and the operator's lepton-number violation traces directly to the $\Delta L=2$ Majorana insertion M. Feeding Eq. (35.34) into the post-symmetry-breaking mass Eq. (35.14) reproduces Eq. (35.30) exactly:

$$m_\nu=-\,c_5\,\frac{v^2}{\Lambda}=-\tfrac{1}{2}\left(y^*M^{-1}y^\dagger\right)v^2=-\,m_D\,M^{-1}m_D^T,\tag{35.35}$$

the two derivations—direct diagonalization of Eq. (35.21) and effective-operator matching—agreeing as they must. The rational factor of $\frac{1}{2}$ rides along with the chosen normalization of $\mathcal{O}_5$ in Eq. (35.9); it is immaterial to the scale estimate of Theorem 35.4, where only the order of magnitude of $\Lambda\sim v^2/m_\nu$ matters.

Worked Example 35.5 (The two faces of the seesaw: heavy scale versus tiny Yukawa)**.** The seesaw relation Eq. (35.28) fixes only the combination m_D^2/M; the same observed m_ν is reached along a one-parameter family of (y,M). Two extreme choices illuminate the mechanism. Take $m_\nu=0.05$ eV throughout, with $m_D=yv/\sqrt{2}$ and $v=246$ GeV.

Order-one Yukawa. Set $y=1$, so $m_D=v/\sqrt{2}\simeq 174$ GeV, the size of the top-quark mass. Then

$$M=\frac{m_D^2}{m_\nu}=\frac{(174\text{ GeV})^2}{5\times10^{-11}\text{ GeV}}\simeq 6\times10^{14}\text{ GeV},\tag{35.36}$$

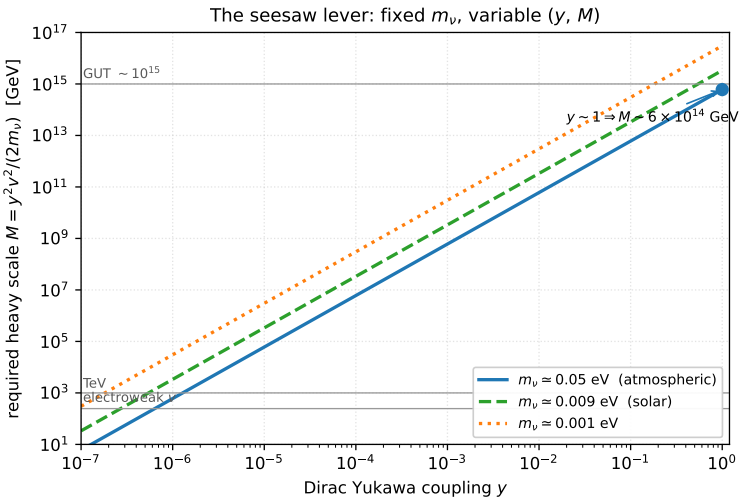


Figure 35.3: The seesaw lever. For a fixed light-neutrino mass $m_\nu = m_D^2/M$ with $m_D = yv/\sqrt{2}$, the required heavy Majorana scale $M = y^2v^2/(2m_\nu)$ is shown as a function of the Dirac Yukawa y, for three target values of m_ν spanning the solar and atmospheric splittings. The same neutrino mass is produced by an $O(1)$ Yukawa with M near the grand-unification scale (marked point, Eq. (35.36)) or by an electron-sized Yukawa with M near a TeV (Eq. (35.37)). The mechanism explains the smallness of m_ν only in the former regime, where no new small dimensionless coupling is introduced.

within the band of Theorem 35.4 and a factor of a few below the unification scale. The mixing Eq. (35.29) is $\theta \simeq m_D/M \simeq 3 \times 10^{-13}$: the heavy neutrino is, for all practical purposes, completely decoupled from low-energy physics, its only imprint the light Majorana mass it has bequeathed. This is the canonical "GUT-scale seesaw."

Tiny Yukawa. Alternatively keep M at an experimentally accessible scale, say $M = 1$ TeV. Then

$$m_D = \sqrt{m_\nu M} = \sqrt{(5 \times 10^{-11})(10^3)}\ \text{GeV} \simeq 2 \times 10^{-4}\ \text{GeV}, \tag{35.37}$$

requiring $y = \sqrt{2}\, m_D/v \simeq 1 \times 10^{-6}$, a Yukawa comparable to the electron's. The seesaw still works, but now the smallness of m_ν has been traded back for a small coupling rather than a large scale; the "seesaw" has been replaced by an ordinary small number. The lesson is that the seesaw *explains* the lightness of neutrinos only when M is genuinely large and y is genuinely $O(1)$—only then is no small dimensionless parameter introduced by hand. The full one-parameter family is plotted in Fig. 35.3.

35.2.6 The other tree completions: type II and type III

The type-I model splits $\mathcal{O}_5 \sim (LH)(LH)$ by joining an $(L\,\widetilde{H})$ pair to another through a heavy *fermion singlet*. The operator can be split two other ways, each defining a distinct ultraviolet completion that integrates down to the same unique Weinberg operator.

The *type-II seesaw* introduces a heavy scalar $\mathrm{SU}(2)_L$ triplet $\Delta \sim (\mathbf{1},\mathbf{3},+1)$. It couples both to two lepton doublets, $\frac{1}{2}\,(y_\Delta)_{ij}\,L_i^T C\, i\sigma^2 \boldsymbol{\sigma} L_j \cdot \boldsymbol{\Delta}$, and to two Higgs doublets, $\mu\, H^T i\sigma^2 \boldsymbol{\sigma} H \cdot \boldsymbol{\Delta}^\dagger$. Integrating out Δ (mass M_Δ) joins these two vertices and reproduces $\mathcal{O}_5$ with $c_5/\Lambda \sim y_\Delta\,\mu/M_\Delta^2$. Equivalently, the trilinear μ induces a tiny triplet vacuum value $v_\Delta \simeq \mu v^2/(2M_\Delta^2)$, and the neutrino mass is the direct Yukawa $m_\nu = y_\Delta\, v_\Delta$, suppressed by the small induced v_Δ. The *type-III seesaw* instead introduces heavy *fermion* $\mathrm{SU}(2)_L$ triplets $\Sigma \sim (\mathbf{1},\mathbf{3},0)$ with a Majorana mass M_Σ and a Yukawa $\overline{L}\,\widetilde{H}\,\boldsymbol{\sigma}\cdot\boldsymbol{\Sigma}$; the structure is identical to type-I with the singlet N replaced by the neutral component of the triplet, giving $m_\nu \sim y^2 v^2/M_\Sigma$. In all three cases the low-energy theory is the Standard Model plus the single operator Eq. (35.9); the three completions are indistinguishable from below the threshold except through their distinct dimension-six operators and their charged heavy states, the latter being the discovery handle on the type-II and type-III scenarios at colliders. That the seesaw scale Λ sits so close to the unification scale—and that the heavy N of the type-I model carries lepton-number-violating, CP-violating couplings y—is the link we carry forward: to Section 35.5 as a numerical coincidence, and to Section 35.6.4, where the out-of-equilibrium decay of the heavy N becomes the engine of leptogenesis.

35.2.7 The light spectrum and the PMNS matrix

Whatever generates it, the light sector is governed by the symmetric complex matrix m_ν of Eq. (35.30), sitting alongside the charged-lepton Yukawa matrix. As established in Section 32.7.4, the mismatch between the unitary rotations that diagonalize the charged-lepton and neutrino mass matrices is the lepton mixing matrix U_{PMNS} that appears in the charged-current interaction. Because m_ν is symmetric rather than Hermitian, its diagonalization is the Autonne–Takagi factorization

$$m_\nu = U^*\,\hat{m}_\nu\,U^\dagger, \qquad \hat{m}_\nu = \mathrm{diag}(m_1, m_2, m_3) \geq 0, \tag{35.38}$$

with a single unitary U rather than the independent left and right rotations of a Dirac matrix. The physical consequence, derived in Section 32.7.4, is that the Majorana nature leaves *two additional physical phases* in U_{PMNS} beyond the single Dirac phase of the quark sector—the Majorana phases—because the freedom to rephase the neutrino fields that would normally remove them is obstructed by the Majorana mass term, which is not invariant under $\nu \to e^{i\alpha}\nu$. These phases

drop out of neutrino oscillations, which are sensitive only to $|U_{\rm PMNS}|^2$ and mass-squared differences, but they survive in lepton-number-violating observables. The seesaw thus delivers a fully predictive low-energy package—three light Majorana masses and a $U_{\rm PMNS}$ with one Dirac and two Majorana phases—whose only fingerprint of the heavy completion is the overall scale set by Eq. (35.28).

35.2.8 The decisive test: neutrinoless double-beta decay

If neutrinos are Majorana, lepton number is violated by two units, and there must exist a process that exhibits it directly. The cleanest is neutrinoless double-beta decay,

$$(A, Z) \longrightarrow (A, Z+2) + 2e^-, \tag{35.39}$$

in which two neutrons inside a nucleus convert to two protons and two electrons with *no* accompanying antineutrinos—two units of lepton number created from nothing. Contrast the ordinary, lepton-number-conserving two-neutrino mode $(A, Z) \to (A, Z+2) + 2e^- + 2\bar{\nu}_e$, which occurs in the Standard Model. The neutrinoless mode proceeds, at lowest order, by the diagram of Fig. 35.4: each W emits an electron and a virtual neutrino, and the two neutrino lines are joined by a Majorana mass insertion—the very $\overline{\nu^c}\nu$ vertex of Eq. (35.12). The amplitude is therefore proportional to the ee entry of the Majorana mass matrix in the flavor basis,

$$m_{\beta\beta} = \Big|\sum_i (U_{\rm PMNS})_{ei}^2\, m_i\Big|, \tag{35.40}$$

where the *squares* of the mixing-matrix elements appear—not the moduli squared—so that the Majorana phases of Section 35.2.7 enter and can produce partial cancellations. The half-life scales as $T^{0\nu}_{1/2} \propto m_{\beta\beta}^{-2}$.

The logical force of the process is captured by the Schechter–Valle theorem: *any* mechanism that produces neutrinoless double-beta decay necessarily induces a Majorana neutrino mass, because the same $\Delta L = 2$ amplitude, with its external legs closed up, is a contribution to $\overline{\nu^c}\nu$. Observation of Eq. (35.39) would therefore be decisive proof that lepton number is violated and that neutrinos are their own antiparticles—the qualitative prediction of the entire seesaw picture. As of the mid-2020s the process has not been seen. The most sensitive experiments, using isotopes such as ^{136}Xe and ^{76}Ge, set half-life bounds $T^{0\nu}_{1/2} \gtrsim 10^{26}$ yr, corresponding to $m_{\beta\beta} \lesssim 0.03$–$0.1$ eV once the (substantial, model-dependent) nuclear matrix elements are folded in. This already begins to probe the parameter space of the inverted-ordering spectrum; next-generation tonne-scale detectors aim to cover it entirely. The status is thus genuinely open: the seesaw predicts a Majorana mass, the dimension-five operator of the EFT is Majorana by construction, and the one clean laboratory signature remains, as of this writing, unobserved.

This section has cashed the unique dimension-five rung of the operator tower into the single established fact of physics beyond the Standard Model and traced

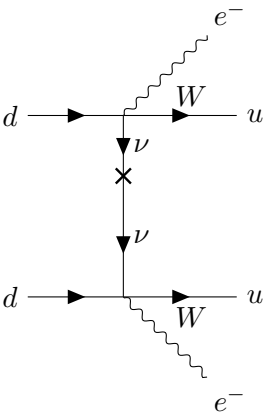


Figure 35.4: Neutrinoless double-beta decay. Two down quarks convert to two up quarks, each emitting a W that produces an electron and a virtual neutrino; the two neutrino lines meet at the Majorana mass insertion (cross), the $\overline{\nu^c}\nu$ vertex that violates lepton number by two units. The amplitude is proportional to the effective mass $m_{\beta\beta}$ of Eq. (35.40). Observation of this process would establish that neutrinos are Majorana particles.

it to a heavy scale $\Lambda \sim 10^{14}$–10^{15} GeV. That scale, suggestively near the meeting point of the gauge couplings, is the thread we pick up in Section 35.5, where the running of Chapter 27 is extrapolated to ask whether the three forces unify; and the lepton-number- and CP-violating heavy sector N of the type-I seesaw is the engine we hand to Section 35.6.4, where its out-of-equilibrium decay converts the same couplings y and M into the matter–antimatter asymmetry of the universe.

35.3 The Hierarchy Problem and Naturalness

The seesaw of Section 35.2 read the unique dimension-five operator of the Standard Model as a window onto a heavy scale: the smallness of the neutrino mass *is* the suppression v^2/Λ of an irrelevant coupling, and that smallness is welcome, even expected, because decoupling guarantees it. We now turn the same EFT lens on the other end of the operator ladder and find the opposite situation. The Standard Model contains exactly one *relevant* coupling of positive mass dimension that is not protected by any symmetry — the Higgs mass-squared — and for it the heavy scale is not a blessing but a threat. Where the neutrino mass is small *because* Λ is large, the Higgs mass is small *in spite of* every large scale to which it couples, and nothing in the theory explains why. This is the hierarchy problem. It is not a disease of the Standard Model's predictivity; the theory computes everything it is asked to. It is a statement about the *structure* of the theory — about a relevant operator sitting twenty-six orders of magnitude away

from every scale that feeds it, with no symmetry standing guard. This section states the problem precisely, quantifies it by an explicit one-loop computation, sharpens it with 't Hooft's criterion for when a small number is allowed to be small, and then surveys — honestly, and dated to the mid-2020s — the proposed ways out.

This is the moment ϕ^4 theory has been pointing at since Chapter 23. The relevant coupling m^2 of the scalar, classified there as the one direction that grows toward the infrared and whose continuum limit costs a fine-tuning (Theorems 23.6 and 23.23), and whose additive ultraviolet sensitivity was made arithmetic in the matching computation of Section 38.6, is not a pedagogical toy. It is the Higgs mass term, and the fine-tuning it demands is the central naturalness puzzle of particle physics. We advance no new numbered stage of the thread here; rather we cash the abstract relevant coupling into the one place in the Standard Model where it bites.

35.3.1 The Higgs mass is the SM's unique relevant coupling

Recall the operator census of Section 33.2.5 read through the Wilsonian classification of Theorem 23.23. Around the Gaussian fixed point that organizes the weakly coupled Standard Model, every gauge-invariant local operator carries the eigenvalue $y_a = 4 - \Delta_a$, with Δ_a its engineering dimension; the coupling multiplying $\mathcal{O}_a$ in $\int d^4x \sum_a g_a \mathcal{O}_a$ then has mass dimension $[g_a] = y_a$. The operators sort into three classes:

- *Marginal* ($\Delta_a = 4$, $[g_a] = 0$): the gauge couplings g_s, g, g', the Yukawa matrices $y_{u,d,e}$, and the Higgs quartic λ. These are dimensionless; quantum effects make them run logarithmically (Chapters 24 and 27), but no power of any heavy scale can attach to a dimensionless coupling.

- *Irrelevant* ($\Delta_a > 4$, $[g_a] < 0$): everything of dimension five and up — the Weinberg operator of Eq. (33.58), the SMEFT tower of Chapter 40. Their coefficients are *suppressed* by powers of $1/\Lambda$; this is the decoupling theorem of Theorem 38.10 working in our favor.

- *Relevant* ($\Delta_a < 4$, $[g_a] > 0$): operators whose coefficients are *enhanced* by positive powers of any heavy scale. In a theory of gauge fields, chiral fermions, and one scalar doublet, gauge invariance and Lorentz invariance leave astonishingly few. The identity operator $\mathbf{1}$ ($\Delta = 0$, the vacuum energy) is one; we set it aside, as it is unobservable without dynamical gravity (Section 38.6.6). Fermion bilinears $\bar{\psi}\psi$ ($\Delta = 3$) are forbidden as bare mass terms because the chiral gauge symmetry $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ does not allow a gauge-invariant Dirac mass — every fermion mass must arise from the Yukawa couplings after electroweak breaking, and is therefore controlled

by a *marginal* coupling. That leaves exactly one relevant operator built from the dynamical fields: the Higgs mass term

$$\mathcal{L} \supset -\mu_H^2 (H^\dagger H), \qquad \Delta = 2, \quad [\mu_H^2] = 2, \tag{35.41}$$

the unique super-renormalizable interaction of the Standard Model. "Super-renormalizable" is the operator's signature: its coupling has positive mass dimension, so the operator improves the convergence of loop integrals it enters and, conversely, drags positive powers of the cutoff into the quantities it controls. The Higgs potential is $V(H) = \mu_H^2 (H^\dagger H) + \lambda (H^\dagger H)^2$; electroweak symmetry breaking requires $\mu_H^2 < 0$, giving the vacuum expectation value $v^2 = -\mu_H^2/\lambda$ and the physical Higgs mass $m_h^2 = 2\lambda v^2 = -2\mu_H^2$ (Section 31.2.5). Measured: $v \simeq 246$ GeV, $m_h \simeq 125$ GeV, so $\mu_H^2 \simeq -(88\ \text{GeV})^2$.

The point of the census is that (35.41) is alone in its class. The marginal couplings are radiatively stable up to logarithms; the irrelevant couplings are protected by decoupling; only μ_H^2 is a dimension-two coupling exposed to the full additive, power-law sensitivity to heavy thresholds that Eq. (38.112) predicted. The hierarchy problem is the problem of this one operator.

Notation Watch

We write the Higgs Lagrangian mass parameter as μ_H^2 (negative, the symmetry-breaking parameter of Section 31.2.5) and the physical pole mass of the scalar excitation as m_h. The naturalness discussion concerns the radiative corrections δm_H^2 to the mass-squared parameter; since $m_h^2 = -2\mu_H^2$ the two are interchangeable up to the factor -2, and we use m_H^2 generically for "the Higgs mass-squared scale" when the distinction is immaterial.

35.3.2 Quantifying the sensitivity: the one-loop δm_H^2

The generic statement of Section 38.6 — a relevant coupling of dimension $y_i > 0$ receives additive corrections $\delta g_i \sim c_i M^{y_i}$ from any threshold at scale M — becomes, for μ_H^2 with $y = 2$, the assertion that every heavy state coupled to the Higgs shifts m_H^2 by an amount of order M^2 times a loop factor. The Higgs couples to almost everything: to the top quark through the largest Yukawa $y_t \approx 1$ (Eq. (33.21)), to the electroweak gauge bosons through the covariant derivative in $|D_\mu H|^2$, and to itself through λ. We now compute these contributions explicitly. Because the spec of the problem is to display the *sign and magnitude* of the leading sensitivity, we work in a hard-cutoff scheme, regulating the Euclidean loop momentum at $|k_E| < \Lambda$; the resulting Λ^2 is the historical Veltman form. We will then say carefully what is and is not physical about it.

The top-quark loop. The dominant contribution is the one with the largest coupling. The top Yukawa, after electroweak breaking, gives the coupling of the physical Higgs h to the top quark,

$$\mathcal{L} \supset -\frac{y_t}{\sqrt{2}}\, h\, \bar{t}\, t, \qquad m_t = \frac{y_t v}{\sqrt{2}}, \tag{35.42}$$

the factor $1/\sqrt{2}$ from the normalization $H = \frac{1}{\sqrt{2}}(0,\, v+h)^{\mathsf{T}}$ in unitary gauge. The one-loop correction to the Higgs two-point function from a top loop is computed in full in the following worked example, because it is the heart of the matter and carries three instructive factors: the fermion-loop minus sign, the colour multiplicity $N_c = 3$, and the Dirac trace.

Worked Example 35.6 (The top loop and the negative quadratic shift)**.** The single one-loop graph correcting $\Gamma_h^{(2)}(p)$ with two insertions of the vertex (35.42) is the fermion bubble of Fig. 35.5 (left). Each vertex contributes $-i\, y_t/\sqrt{2}$; the closed fermion loop carries the obligatory factor (-1) and a trace over Dirac *and* colour indices, the latter giving N_c. With internal momenta k and $k+p$,

$$-i\,\Sigma_t(p) \;=\; (-1)\,N_c \int \frac{d^4k}{(2\pi)^4}\; \mathrm{tr}\left[\Big(-\tfrac{iy_t}{\sqrt{2}}\Big)\frac{i(\not{k}+m_t)}{k^2-m_t^2}\Big(-\tfrac{iy_t}{\sqrt{2}}\Big)\frac{i(\not{k}+\not{p}+m_t)}{(k+p)^2-m_t^2}\right]. \tag{35.43}$$

We evaluate the leading ultraviolet behaviour, which is what carries the quadratic sensitivity, by setting the external momentum $p = 0$ inside the loop (the p-dependence affects only the finite and logarithmic pieces). Collecting the constants, the four explicit factors of i in the two numerators and one more from $-i\Sigma_t$ give $(-1)N_c(-iy_t/\sqrt{2})^2(i)^2 = (-1)N_c\,(-y_t^2/2)(-1) = -\frac{N_c y_t^2}{2}$, so that

$$-i\,\Sigma_t \;=\; -\frac{N_c y_t^2}{2}\int \frac{d^4k}{(2\pi)^4}\,\frac{\mathrm{tr}\,[(\not{k}+m_t)(\not{k}+m_t)]}{(k^2-m_t^2)^2}. \tag{35.44}$$

The trace is elementary: $\mathrm{tr}[(\not{k}+m_t)(\not{k}+m_t)] = \mathrm{tr}[\not{k}\not{k}] + 2m_t\,\mathrm{tr}[\not{k}] + m_t^2\,\mathrm{tr}[\mathbf{1}] = 4k^2 + 4m_t^2$, the odd-$\gamma$ trace vanishing. Hence

$$-i\,\Sigma_t \;=\; -2N_c y_t^2 \int \frac{d^4k}{(2\pi)^4}\,\frac{k^2+m_t^2}{(k^2-m_t^2)^2}. \tag{35.45}$$

Wick-rotate $k^0 = ik_E^0$, so $d^4k = i\,d^4k_E$, $k^2 = -k_E^2$, and the integrand becomes $(-k_E^2+m_t^2)/(k_E^2+m_t^2)^2$. Writing $-k_E^2+m_t^2 = -(k_E^2+m_t^2) + 2m_t^2$ splits the integral into a quadratically divergent piece and a logarithmic one,

$$\int \frac{d^4k}{(2\pi)^4}\,\frac{k^2+m_t^2}{(k^2-m_t^2)^2} = i\int \frac{d^4k_E}{(2\pi)^4}\left[\frac{-1}{k_E^2+m_t^2} + \frac{2m_t^2}{(k_E^2+m_t^2)^2}\right]. \tag{35.46}$$

With the four-dimensional measure $\int d^4k_E/(2\pi)^4 = \frac{1}{16\pi^2}\int_0^{\Lambda^2} x\,dx$ (where $x = k_E^2$),

$$\int_0^{\Lambda^2} \frac{x\,dx}{x+m_t^2} = \Lambda^2 - m_t^2\ln\frac{\Lambda^2+m_t^2}{m_t^2} \;\simeq\; \Lambda^2 - m_t^2\ln\frac{\Lambda^2}{m_t^2}, \tag{35.47}$$

$$\int_0^{\Lambda^2} \frac{x\,dx}{(x+m_t^2)^2} = \ln\frac{\Lambda^2+m_t^2}{m_t^2} + \frac{m_t^2}{\Lambda^2+m_t^2} - 1 \;\simeq\; \ln\frac{\Lambda^2}{m_t^2} - 1. \tag{35.48}$$

Assembling, the bracket in (35.46) integrates to $\frac{1}{16\pi^2}\big[-\Lambda^2 + m_t^2\ln(\Lambda^2/m_t^2) + 2m_t^2(\ln(\Lambda^2/m_t^2)-1)\big] = \frac{1}{16\pi^2}\big[-\Lambda^2 + 3m_t^2\ln(\Lambda^2/m_t^2) - 2m_t^2\big]$, and so

$$-i\,\Sigma_t = -2N_c y_t^2 \cdot \frac{i}{16\pi^2}\Big[-\Lambda^2 + 3m_t^2\ln\tfrac{\Lambda^2}{m_t^2} - 2m_t^2\Big]. \tag{35.49}$$

Cancelling the i and reading the shift of the mass-squared as $\delta m_H^2\big|_t = \Sigma_t$ (the p-independent self-energy displaces the pole of $i/(p^2 - m_H^2 - \Sigma)$),

$$\boxed{\delta m_H^2\big|_t \;=\; -\frac{N_c\,y_t^2}{8\pi^2}\,\Lambda^2 \;+\; \frac{3N_c\,y_t^2}{8\pi^2}\,m_t^2\,\ln\frac{\Lambda^2}{m_t^2} \;+\; \cdots} \tag{35.50}$$

The leading term is quadratic in the cutoff, proportional to $N_c y_t^2$, and — decisively — *negative*, the sign inherited from the fermion-loop (-1). Numerically, with $N_c = 3$ and $y_t \approx 1$, the coefficient is $-3/8\pi^2 \approx -0.038$: a top loop drives m_H^2 down by roughly four percent of Λ^2 per unit y_t^2.

The gauge and Higgs self-coupling loops. The remaining one-loop contributions come from the Higgs coupling to the electroweak gauge bosons and to itself. The cleanest way to organize all of them at once — and the way that exposes what is physical — is the field-dependent mass matrix. The quadratically divergent part of the one-loop effective potential (the Coleman–Weinberg potential of Chapter 30, whose construction we import) is

$$V_1^{\Lambda^2} \;=\; \frac{\Lambda^2}{32\pi^2}\,\mathrm{Str}\,M^2(\phi), \qquad \mathrm{Str}\,M^2 \equiv \sum_i (-1)^{2s_i}(2s_i+1)\,m_i^2(\phi), \tag{35.51}$$

the supertrace running over every particle that couples to the background field $\phi = v + h$, with $(2s_i+1)$ its physical polarizations and $(-1)^{2s_i}$ the crucial statistics sign that distinguishes bosons from fermions. The shift of the Higgs mass-squared is the second derivative, $\delta m_H^2 = \partial_h^2 V_1^{\Lambda^2} = \frac{\Lambda^2}{32\pi^2}\,\partial_h^2\,\mathrm{Str}\,M^2(h)$. Each species enters through the curvature $\partial_h^2 m_i^2(h)$ of its field-dependent mass, read off from the Lagrangian:

$$m_t^2(h) = \tfrac{1}{2}y_t^2(v+h)^2, \qquad m_W^2(h) = \tfrac{1}{4}g^2(v+h)^2, \qquad m_Z^2(h) = \tfrac{1}{4}(g^2+g'^2)(v+h)^2,$$
$$m_h^2(h) = -\mu_H^2 + 3\lambda(v+h)^2, \tag{35.52}$$

giving $\partial_h^2 m_t^2 = y_t^2$, $\partial_h^2 m_W^2 = \frac{1}{2}g^2$, $\partial_h^2 m_Z^2 = \frac{1}{2}(g^2+g'^2)$, and $\partial_h^2 m_h^2 = 6\lambda$. The polarization-and-statistics weights are $+1$ for the real Higgs, $+3$ for each massive vector (W^+, W^- counted separately, Z once), and $-4N_c$ for the top (a Dirac

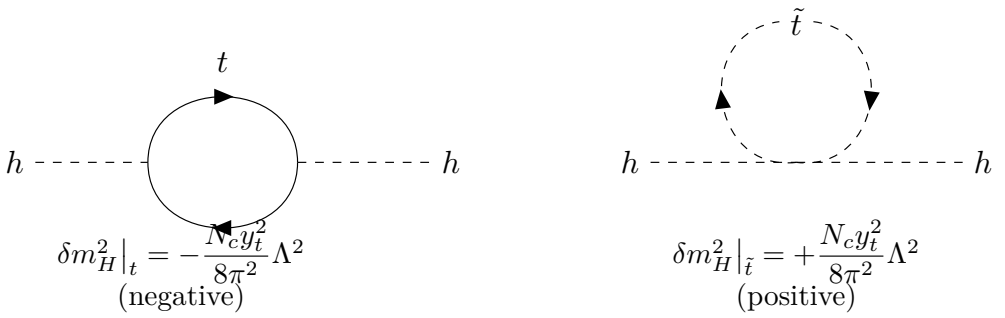


Figure 35.5: The quadratically sensitive corrections to the Higgs mass. Left: the top-quark loop (35.43), a closed fermion line (-1, colour N_c, Dirac trace) giving the *negative* leading term (35.50). Right: the loop of a scalar top-partner $\tilde{t}$ with quartic coupling fixed by supersymmetry to y_t^2, a seagull tadpole giving an equal and *positive* term (35.60). The sum of the two cancels at the quadratic level (Section 35.3.4).

fermion is four real degrees of freedom, with the overall minus sign of (35.50), replicated over colour). Hence

$$\delta m_H^2 \;=\; \frac{\Lambda^2}{32\pi^2}\Big[\,6\lambda + 3g^2 + \tfrac{3}{2}(g^2+g'^2) - 4N_c\,y_t^2\,\Big], \tag{35.53}$$

where the $3g^2$ collects the two charged W's ($2\times 3\times\frac{1}{2}g^2$) and the $\frac{3}{2}(g^2+g'^2)$ the single Z ($3\times\frac{1}{2}(g^2+g'^2)$). The explicit top piece $-4N_c y_t^2\cdot\frac{\Lambda^2}{32\pi^2} = -\frac{N_c y_t^2}{8\pi^2}\Lambda^2$ reproduces (35.50), a check that the supertrace bookkeeping and the diagrammatic computation agree. Trading couplings for masses through (35.52) at $h=0$,

$$\boxed{\;\delta m_H^2 \;=\; \frac{\Lambda^2}{16\pi^2 v^2}\Big[\,\tfrac{3}{2}\,m_h^2 + 6\,m_W^2 + 3\,m_Z^2 - 12\,m_t^2\,\Big]\;} \tag{35.54}$$

the Veltman decomposition (Fig. 35.6). The bosonic terms are positive, the fermionic term negative; numerically, with $m_h = 125$, $m_W = 80.4$, $m_Z = 91.2$, $m_t = 173$ GeV, the bracket is $\frac{3}{2}(125)^2 + 6(80.4)^2 + 3(91.2)^2 - 12(173)^2 \approx (2.3 + 3.9 + 2.5 - 35.9)\times 10^4 \approx -2.7\times 10^5$ GeV^2, dominated by the top and net negative. The full coefficient is $\delta m_H^2 \approx -0.028\,\Lambda^2$.

What is, and is not, physical. Equation (35.54) invites a misreading that must be dispatched at once, because it is the single most common confusion in the subject. The Λ^2 is *not* the substance of the hierarchy problem. The hard cutoff Λ is a regulator, and the quadratic dependence on it is a regulator artifact: in dimensional regularization, as the explicit computation of Theorem 38.16 shows, there is *no* Λ^2 at all. The massive tadpole there gave a result proportional to the

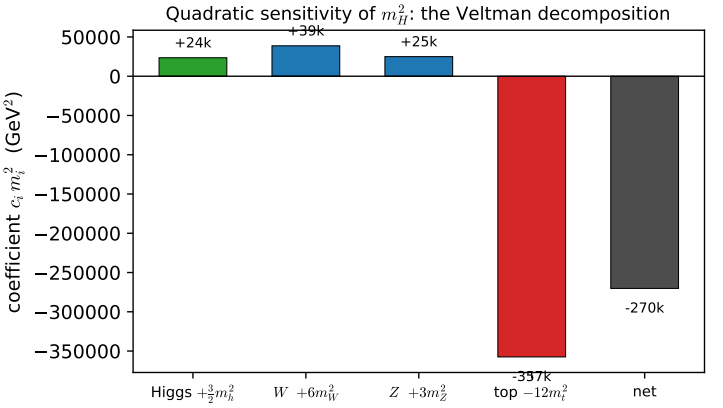


Figure 35.6: The Veltman decomposition (35.54) of the coefficient of $\Lambda^2/16\pi^2v^2$ into the contribution $c_i m_i^2$ of each Standard-Model species. The bosonic contributions (Higgs, W, Z) are positive; the top quark enters with a minus sign and dominates, so the net coefficient is large and negative: the Higgs mass-squared is pulled down toward $-\Lambda^2$.

physical mass-squared M^2 times a $1/\bar{\varepsilon}$ pole and a logarithm — never a power of a regulator. A power divergence in a hard cutoff is, like the photon mass that naïve cutoffs generate in QED, an artifact that any symmetry-preserving scheme removes. If the Standard Model were the complete theory up to arbitrarily high energies, μ_H^2 would simply be an input, renormalized once, with no quadratic sensitivity to anything.

Caution

The naturalness problem is the sensitivity not to the *regulator* but to *physical thresholds*. Suppose — as the rest of this chapter argues we must — there exist heavy states of mass M that couple to the Higgs. Integrating them out, the decoupling theorem (Theorem 38.10) leaves their imprint on the low-energy relevant coupling as an additive, finite, scheme-independent shift $\delta\mu_H^2 \sim (\text{coupling}/16\pi^2)\, M^2$, with M a real mass, not a cutoff. This is the form (38.128), and it survives in dimensional regularization, where the artifact Λ^2 is absent but the physical M^2 is manifest. The arithmetic of (35.54) is the correct *order of magnitude* provided one reads Λ as the lightest heavy threshold M that genuinely exists; its precise rational coefficients — and even whether the breakdown into "gauge" and "Higgs" pieces in (35.53) is gauge-invariant — are scheme- and gauge-dependent, but the supertrace structure and the dominance and sign of the top term are robust, and the physical-threshold statement is exact.

So the problem has teeth only if heavy thresholds exist. They do. The seesaw

scale of Section 35.2 sits near 10^{14} GeV; grand unification (Section 35.5) places thresholds near 10^{16} GeV; quantum gravity enters by $M_{\rm Pl} \sim 10^{18}$ GeV. If the Higgs couples to physics at any such scale with $O(1)$ strength — and a singlet operator $H^\dagger H$ couples to *everything* not forbidden by symmetry — then (38.128) gives $\delta m_H^2 \sim M^2/16\pi^2$, and the measured $m_H^2 \sim (88\ \text{GeV})^2$ survives only by a cancellation between the bare parameter and the threshold contribution accurate to

$$\frac{m_H^2}{\delta m_H^2} \;\sim\; \frac{(10^2\ \text{GeV})^2}{(10^{16}\ \text{GeV})^2/16\pi^2} \;\sim\; 10^{-26}, \tag{35.55}$$

one part in 10^{26}, repeated order by order in perturbation theory with no symmetry to protect it. That is the hierarchy problem, stated quantitatively.

35.3.3 't Hooft's criterion: why scalars are different

Why is one part in 10^{26} alarming, when nature is full of small numbers? The electron mass is $m_e/M_{\rm Pl} \sim 10^{-22}$ and nobody calls it a fine-tuning. The distinction was made precise by 't Hooft, and we now apply his criterion (the definition is Theorem 38.15) to sharpen exactly what is and is not pathological about the Higgs.

> *A small parameter is technically natural if and only if setting it to zero enhances the symmetry of the theory.*

The content of the criterion is a statement about radiative *stability*, proved in full at Theorem 38.15: if the $g = 0$ theory possesses an exact symmetry S that $g \neq 0$ breaks, then g is the coefficient of the unique S-violating operator, every Feynman graph correcting g must contain at least one insertion of that operator, and therefore

$$\delta g \;=\; g \times f\Big(\text{couplings},\, \ln\tfrac{M}{\mu}\Big), \tag{35.56}$$

multiplicative in g and only logarithmically dependent on the heavy scale. A symmetry does not explain why g is small; it explains why a small g *stays* small. The would-be additive piece $\sim M^y$ is S-symmetric (it does not vanish as $g \to 0$), so the symmetry forbids it.

The fermion mass is multiplicatively renormalized. Apply this to the electron mass of QED. At $m_e = 0$ the QED Lagrangian acquires a chiral symmetry $\psi \to e^{i\theta\gamma^5}\psi$ (Eq. (38.126)), because γ^5 anticommutes with the γ^μ in the kinetic and gauge terms but the mass term $m_e\bar\psi\psi = m_e(\bar\psi_R\psi_L + \bar\psi_L\psi_R)$ is the unique operator linking the two chiralities. The mass is the sole chirality-breaker, so by (35.56) every correction to it carries a factor of m_e: as computed explicitly at Section 38.6.4,

$$\delta m_e \;=\; \frac{3\alpha}{4\pi}\, m_e \,\ln\frac{\Lambda^2}{m_e^2} + \cdots, \tag{35.57}$$

proportional to m_e and only logarithmic in the cutoff — the multiplicative, log-mild law (35.56), the same structure that governs the running of every Yukawa and is the reason the QED coupling itself runs only logarithmically (Eq. (24.215)). Even at $\Lambda = M_{\rm Pl}$ the correction is a $\sim 20\%$ effect, not a tuning. The electron mass is small and stays small; chiral symmetry protects it.

The scalar mass is additively renormalized, and nothing protects it. Now ask 't Hooft's question of μ_H^2. Does $\mu_H^2 \to 0$ enhance the symmetry of the Standard Model? It does not. The operator $H^\dagger H$ is a gauge singlet (it is the modulus of the doublet), invariant under every internal symmetry the theory has, so no internal symmetry can forbid it. At $\mu_H^2 = 0$ the classical Lagrangian is scale-invariant, but scale invariance is *not* a symmetry of the quantum theory: it is broken by the running of the marginal couplings, the dilatation current carrying the trace anomaly proportional to the beta functions (Chapter 24). There is no chiral symmetry for a scalar — chirality is a property of fermions, and the protective mechanism (35.57) has no analogue for $H^\dagger H$, which contains no γ^5 to anticommute with anything. The Higgs mass term is therefore the coefficient of *no* broken symmetry; it is allowed by everything, and its radiative corrections are correspondingly additive, (35.54), not multiplicative. This is the precise content of the hierarchy problem: *the Higgs mass is the one relevant coupling of the Standard Model that no symmetry protects.*

Physical Insight

The contrast is the whole lesson, and it is structural, not numerical. Both m_e and μ_H^2 are relevant couplings; both are tiny compared to the Planck scale. The difference is in the *form* of their radiative corrections. The fermion mass is renormalized *multiplicatively*, $\delta m_e \propto m_e \ln \Lambda$: the small value is an attractor, stable under the flow, protected by the chiral symmetry restored at $m_e = 0$. The scalar mass is renormalized *additively*, $\delta\mu_H^2 \propto M^2$: the small value is not an attractor, requires a fresh cancellation at every order, and is protected by nothing. A natural theory is one in which every small parameter sits on a protected surface; the Standard Model has exactly one relevant parameter that does not, and that is the puzzle. (The vacuum energy of Section 38.6.6 is a second, more severe, but it is unobservable without gravity and we set it aside here.)

35.3.4 Supersymmetry: the boson–fermion cancellation made explicit

If the Higgs mass is endangered because a scalar has no chiral protection, the most direct cure is to *give* it one — to embed the scalar in a multiplet with a fermion, so that the fermion's chiral symmetry shields the scalar too. This

is supersymmetry, and its effect on the hierarchy problem is visible already at one loop, before any of the apparatus of Chapter 44 is needed: the bosonic and fermionic loops cancel. We make the cancellation explicit, importing only the one algebraic fact that supersymmetry fixes the relevant coupling.

Consider augmenting the Standard Model with a scalar partner of the top — a complex "stop" $\tilde{t}$ — coupled to the Higgs through a quartic

$$\mathcal{L} \;\supset\; -\lambda_{\tilde{t}}\,(H^\dagger H)\,(\tilde{t}^\dagger \tilde{t}), \tag{35.58}$$

and ask what value of $\lambda_{\tilde{t}}$ makes the cancellation work. The one-loop correction from the stop is the scalar seagull tadpole of Fig. 35.5 (right). In unitary gauge the vertex (35.58) produces $-\lambda_{\tilde{t}}\tfrac{1}{2}h^2(\tilde{t}^\dagger\tilde{t})$ after expanding $H^\dagger H = \tfrac{1}{2}(v+h)^2$; the $h^2\tilde{t}^\dagger\tilde{t}$ vertex (two identical h, hence a 2!) has Feynman rule $-i\lambda_{\tilde{t}}$, and the tadpole closing the complex-scalar line is

$$-i\,\Sigma_{\tilde{t}} \;=\; N_c\,(-i\lambda_{\tilde{t}})\int\frac{d^4k}{(2\pi)^4}\,\frac{i}{k^2-m_{\tilde{t}}^2} \;=\; N_c\,\lambda_{\tilde{t}}\int\frac{d^4k}{(2\pi)^4}\,\frac{1}{k^2-m_{\tilde{t}}^2}, \tag{35.59}$$

with the colour factor N_c for the coloured squark and no symmetry factor $\tfrac{1}{2}$ (a complex scalar loop carries a definite arrow, so the two ends of the propagator are distinguishable). Wick-rotating as before, $\int d^4k/(2\pi)^4\,(k^2-m^2)^{-1} = -\frac{i}{16\pi^2}\big[\Lambda^2 - m^2\ln(\Lambda^2/m^2)\big]$, so

$$\delta m_H^2\big|_{\tilde{t}} \;=\; \Sigma_{\tilde{t}} \;=\; +\frac{N_c\,\lambda_{\tilde{t}}}{16\pi^2}\,\Lambda^2 \;-\; \frac{N_c\,\lambda_{\tilde{t}}}{16\pi^2}\,m_{\tilde{t}}^2\ln\frac{\Lambda^2}{m_{\tilde{t}}^2} + \cdots . \tag{35.60}$$

The leading term is positive — a scalar loop carries no fermion minus sign — and quadratic in the cutoff, the mirror image of the top's (35.50). Now compare the coefficients. Each Dirac top comes with *two* complex scalar partners ($\tilde{t}_L, \tilde{t}_R$, the superpartners of the two chiralities), so the squark contribution is twice (35.60): $+2N_c\lambda_{\tilde{t}}\Lambda^2/16\pi^2$. Setting this equal and opposite to the top's $-N_c y_t^2\Lambda^2/8\pi^2 = -2N_c y_t^2\Lambda^2/16\pi^2$ requires

$$\lambda_{\tilde{t}} \;=\; y_t^2. \tag{35.61}$$

The cancellation is exact *provided the Higgs–stop quartic equals the square of the top Yukawa.* This is not a fine-tuning of (35.58); it is precisely the relation supersymmetry enforces. In a supersymmetric theory both the Yukawa coupling and the scalar quartic descend from a single holomorphic superpotential term $W \supset y_t\,H\,Q\,t^c$: the Yukawa is the coefficient of the fermion bilinear, while the scalar potential is $V = \sum_i |\partial W/\partial\phi_i|^2$ (the F-term potential of Theorem 44.27), whose cross terms $|\partial W/\partial Q|^2 \supset y_t^2|H|^2|\tilde{t}^c|^2$ and $|\partial W/\partial t^c|^2 \supset y_t^2|H|^2|\tilde{Q}|^2$ deliver exactly the quartic (35.58) with $\lambda_{\tilde{t}} = y_t^2$. The same superpotential coupling that gives the top its mass gives the stops their coupling to the Higgs, with the strength locked by the structure of the algebra (Section 44.2). The field content

that supplies one complex scalar per fermion chirality is the matter content of the MSSM (Section 45.6.1).

The cancellation is the supertrace (35.51) vanishing. A Dirac top is four real fermionic degrees of freedom; its two complex stops are four real bosonic degrees of freedom; the statistics signs are opposite and the couplings equal, so

$$\mathrm{Str}\, M^2 = \underbrace{(+4N_c)\, y_t^2}_{\text{stops}} + \underbrace{(-4N_c)\, y_t^2}_{\text{top}} = 0, \tag{35.62}$$

and the same holds species by species across an exactly supersymmetric spectrum: every fermion's quadratic contribution is cancelled by its scalar partners and every gauge boson's by its gaugino. The quadratic sensitivity (35.54) vanishes identically, and the scalar mass inherits the multiplicative, logarithmic renormalization (35.56) of its fermionic partner. The hierarchy problem is solved by symmetry, exactly as 't Hooft's criterion demands.

The residual log and the cost of breaking. Supersymmetry is not exact — no superpartners are degenerate with the known particles — so the cancellation (35.62) is spoiled by the mass splitting. Restoring the subleading logarithmic terms of (35.50) and (35.60), the surviving correction is

$$\delta m_H^2 \;\simeq\; -\frac{N_c\, y_t^2}{8\pi^2}\,(m_{\tilde t}^2 - m_t^2)\,\ln\frac{\Lambda}{m_{\tilde t}} \;\sim\; -\frac{N_c\, y_t^2}{8\pi^2}\, m_{\mathrm{SUSY}}^2\,\ln\frac{\Lambda}{m_{\mathrm{SUSY}}}, \tag{35.63}$$

quadratic now not in the cutoff but in the *SUSY-breaking scale* $m_{\mathrm{SUSY}} \sim m_{\tilde t}$, and only logarithmically sensitive to Λ. This is the decoupling form (38.128) with the heavy threshold M replaced by the superpartner scale: supersymmetry converts the catastrophic $M^2 \sim M_{\mathrm{Pl}}^2$ sensitivity into a controlled m_{SUSY}^2 one. Naturalness then *predicts* $m_{\mathrm{SUSY}} \lesssim$ a few hundred GeV to a TeV, so that (35.63) not exceed m_H^2. This prediction — light stops, gluinos, and electroweakinos near the TeV scale — is the sharpest experimental consequence of the naturalness argument, and we return to its status below.

35.3.5 The menu of resolutions, scoped and dated

The hierarchy problem has generated four broad classes of response. We survey them honestly, each scoped to what it actually accomplishes and dated to the experimental situation of the mid-2020s.

(1) Supersymmetry. As shown in Section 35.3.4, supersymmetry supplies the missing symmetry: the Higgs becomes one component of a chiral supermultiplet, and the fermionic chiral protection extends to the scalar, cancelling the quadratic sensitivity to all orders in the exact theory and reducing it to $\sim m_{\mathrm{SUSY}}^2$ when

broken softly. It is the most complete and predictive resolution, and it brings a bonus that Section 35.5 develops: the superpartners alter the running of the gauge couplings (Chapter 27) so that they meet far more accurately at a single grand-unified scale. *Status (mid-2020s):* the Large Hadron Collider has excluded gluinos below roughly 2.2 TeV and the lightest stops below roughly 1.2 TeV over wide ranges of parameter space. Naturalness most sharply expected these states near a few hundred GeV; their absence forces the residual (35.63) above m_H^2, so minimal supersymmetry is now itself tuned at the percent level or worse — the "little hierarchy problem." Supersymmetry remains viable and is not excluded, but it no longer delivers the effortless naturalness that originally motivated it.

(2) Compositeness: the Higgs as a pseudo-Nambu–Goldstone boson. A second way to protect a scalar mass is to make the scalar a Goldstone boson. If the Higgs is not elementary but a bound state of a new strong sector that confines at a scale f, and if it is the pseudo-Nambu–Goldstone boson of a spontaneously broken global symmetry of that sector (Theorem 29.1), then a *shift symmetry* $H \to H + \text{const}$ protects its mass exactly as the chiral symmetry protects the pion mass in Chapter 39: m_H^2 is forbidden in the symmetric limit and arises only from the explicit breaking (Standard-Model gauge and Yukawa couplings), suppressed by a loop factor relative to f. The naturalness condition becomes $f \sim$ TeV. This is technically natural in 't Hooft's sense — the small m_H is the coefficient of an explicitly broken shift symmetry, hence multiplicatively renormalized. *Status (mid-2020s):* no compositeness has been observed; electroweak precision data and the absence of vector resonances push the compositeness scale to $f \gtrsim$ several TeV, reintroducing a mild ($\sim$ percent) tuning of the same little-hierarchy character as supersymmetry.

(3) Classical (and quantum) scale invariance. A third proposal forbids the mass term (35.41) outright by imposing exact classical scale invariance: with $\mu_H^2 \equiv 0$ in the Lagrangian, the only scale is generated dynamically by dimensional transmutation through the Coleman–Weinberg mechanism (Chapter 30), as the logarithmic running of λ drives radiative symmetry breaking. The electroweak scale is then a derived quantity, exponentially below any high scale by the same mechanism that makes $\Lambda_{\rm QCD}$ exponentially below $M_{\rm Pl}$ (Chapter 27). The difficulty is honest and twofold: scale invariance is anomalous — broken by the very running it relies on, the trace anomaly of Chapter 24 — so the symmetry is not exact at the quantum level, and the scenario must be coupled to gravity, which reintroduces $M_{\rm Pl}$ as a hard scale unless gravity is itself scale-invariant in the ultraviolet. *Status:* an active but speculative minority program; no version is yet a complete, predictive resolution.

(4) The anthropic/multiverse stance. The final response abandons the search for a protecting symmetry and posits instead environmental selection: if the fundamental theory has an enormous landscape of vacua with m_H^2 scanning over many values, observers exist only in those rare vacua where m_H^2 is small enough for complex chemistry — atoms, nuclei, the existence of structure — to form. The smallness of m_H^2 is then a selection effect, not a dynamical consequence. This is the same logic invoked for the cosmological constant (Section 38.6.6), where it remains the only framework that reproduces the observed value. It is not a symmetry explanation of the kind 't Hooft's criterion calls for, makes few sharp particle-physics predictions, and its scientific status is contested. *Status:* taken seriously by a substantial fraction of the field, principally because of the cosmological-constant precedent and the non-appearance of natural new physics at the LHC; rejected by others as untestable.

Caution

The honest meta-statement, dated to the mid-2020s, is that the LHC has found *none* of the natural new physics that the hierarchy problem most sharply predicted: no superpartners, no compositeness, no extra dimensions at the TeV scale. The clean expectation of naturalness — new states near the weak scale cutting off the quadratic sensitivity — has not been borne out. This has placed the naturalness criterion *itself* under examination. Some read the null results as a sign that the weak scale is tuned (whether anthropically or for reasons unknown); others that the relevant new physics is merely heavier than expected, or that naturalness as formulated is the wrong question. What is not in doubt is the structural fact established in this section: μ_H^2 is the unique unprotected relevant coupling of the Standard Model, and the explanation of its smallness, whatever it proves to be, lies outside the Standard Model. That fact is an EFT theorem; the resolution is, at this writing, unknown.

The naturalness template assembled here — a small parameter with no protecting symmetry signalling missing physics — is one of the two master tools this chapter uses to organize the Standard Model's structural puzzles. The supersymmetric resolution, with its boson–fermion cancellation, will reappear in Section 35.5 in an unexpected role: the same superpartners that stabilize the Higgs mass modify the running of the three gauge couplings and sharpen their meeting into a genuine unification. And the template itself transfers wholesale to the next section: the QCD vacuum angle θ is a marginal coupling that the Standard Model permits, that experiment finds unnaturally close to zero, and that — like μ_H^2 — is protected by no symmetry of the theory as it stands. The strong-CP problem of Section 35.4 is the naturalness argument applied to a small *dimensionless* number, and its resolution, the Peccei–Quinn symmetry and the axion, is precisely the act of supplying the missing symmetry that 't Hooft's

criterion demands.

35.4 The Strong-CP Problem and the Axion

The previous section diagnosed the Higgs mass as a *relevant* coupling whose smallness no symmetry of the Standard Model protects, and catalogued the responses — supersymmetry, compositeness, anthropics, or simply tuning — without finding one that is both economical and confirmed. We now apply the very same template, the "small number with no protecting symmetry," to an operator at the opposite end of the dimensional ladder. The strong-CP parameter $\bar{\theta}$ is not relevant but *marginal*: a dimensionless coupling on the same footing as the gauge couplings, multiplying a dimension-four operator. It is moreover *topological*, a total derivative that classical physics would discard outright. And yet it is physical, it is CP-odd, and experiment pins it below 10^{-10} while the theory leaves it free to be of order one. The puzzle is structurally identical to the hierarchy problem — an unnatural number with no symmetry forcing it small — but here, uniquely among the Standard Model's structural puzzles, there is a resolution that is economical, predictive, and falsifiable: the Peccei–Quinn mechanism and its axion. The axion converts the static fine-tuning of $\bar{\theta}$ into the dynamics of a new field that relaxes the offending parameter to zero on its own. That is the cleanest naturalness success on offer anywhere in this chapter, and the field it predicts is, as a bonus, a leading candidate for the dark matter of Section 35.6.

35.4.1 The problem, recalled

The QCD Lagrangian admits one renormalizable operator beyond the kinetic, Yukawa, and Higgs terms: the topological density of the colour field, weighted by an angle,

$$\mathcal{L}_\theta = \theta_{\rm QCD}\,\frac{g_s^2}{32\pi^2}\,G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu}, \qquad \widetilde{G}^{A\,\mu\nu} = \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}G^A_{\rho\sigma}, \tag{35.64}$$

exactly the term assembled in Eq. (33.68). Three of its properties, all established earlier, frame everything that follows, and we recall them rather than reprove them. First, $G\widetilde{G}$ is a total derivative, $G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu} = \partial_\mu K^\mu$ with K^μ the Chern–Simons current; classically it cannot affect any equation of motion. It is nonetheless physical because the Euclidean path integral sums over gauge fields of every winding number $\nu \in \mathbb{Z}$, and $\mathcal{L}_\theta$ integrates to $\theta_{\rm QCD}\,\nu$, weighting topological sectors by the relative phase $e^{i\theta_{\rm QCD}\nu}$. This is the content of the θ-vacuum construction of Section 42.4: instantons make a total derivative observable. Second, $\mathcal{L}_\theta$ is P-odd and T-odd and C-even, hence CP-odd. Third — and this is the subtlety that makes the *physical* angle a derived quantity — $\theta_{\rm QCD}$ as written

in (35.64) is not by itself observable, because a chiral redefinition of the quark fields shifts it.

That last point is the linchpin, so we restate its mechanism in one paragraph. A chiral rotation $q \to e^{i\alpha\gamma^5} q$ of a single quark flavour is anomalous: the fermion measure is not invariant, and the Fujikawa Jacobian (the chiral anomaly of Theorem 36.9) shifts the action by precisely a θ-term, $\theta_{\mathrm{QCD}} \to \theta_{\mathrm{QCD}} - 2\alpha$. The same rotation multiplies the quark mass by $e^{-2i\alpha}$, so it shifts $\arg\det M$ by $+2\alpha$, where M runs over the up- and down-type mass matrices. The two shifts cancel, and the rephasing-invariant combination

$$\bar{\theta} \;=\; \theta_{\mathrm{QCD}} + \arg\det\!\left(M_u M_d\right) \tag{35.65}$$

is the physical parameter, derived in full in Theorem 33.16 and recorded in Eq. (33.85). Either summand can be set to any value by a chiral rotation; only their sum is observable. (The one exception, recalled for completeness: were any single quark massless, the corresponding axial rotation would be a genuine symmetry able to rotate $\bar{\theta}$ to zero at no cost, and strong CP would evaporate. Lattice QCD's $m_u \neq 0$ closes that escape, as Section 33.5.5 stressed.)

Being CP-odd, $\bar{\theta}$ generates a permanent neutron electric dipole moment, $d_n \sim \bar{\theta}\,(10^{-16}\, e\cdot\mathrm{cm})$ from the chiral estimate Eq. (33.86). The experimental bound $|d_n| < 1.8 \times 10^{-26}\, e\cdot\mathrm{cm}$ (mid-2020s) translates into

$$|\bar{\theta}| \;\lesssim\; 10^{-10}, \tag{35.66}$$

the statement of the strong-CP problem from Eq. (33.87). The naturalness indictment, exactly parallel to the hierarchy argument of the previous section, has two prongs. The two contributions to (35.65) are logically independent — θ_{QCD} is a pure-glue angle, while $\arg\det(M_u M_d)$ lives in the Yukawa sector that supplies the large, confirmed CP violation of the CKM phase — so their cancellation to ten decimal places is an unexplained conspiracy between two unrelated sectors. And setting $\bar{\theta} = 0$ does not enlarge the symmetry of the theory, because CP is already broken by the CKM phase; there is no exact symmetry to invoke in the sense of 't Hooft. The number is small, and nothing in the Standard Model makes it so. The remainder of this section is the one place in the chapter where that complaint is answered by a mechanism rather than a tuning.

35.4.2 The shape of the vacuum energy

The mechanism rests on a single structural fact about QCD: the vacuum energy density $E(\theta)$, regarded as a function of the angle, has its minimum at $\theta = 0$ and is convex there. If we can find a way to promote θ from a fixed parameter to a dynamical field, that field will roll to the bottom of $E(\theta)$ and switch off strong CP all by itself. So we first establish what $E(\theta)$ looks like.

Define the θ-dependence of the vacuum energy by the Euclidean partition function in a four-volume V_4,

$$e^{-V_4\,E(\theta)} = \int \mathcal{D}A\,\mathcal{D}q\,\mathcal{D}\bar{q}\;\exp\Big(-S_{\rm QCD} - i\theta\int d^4x\,\frac{g_s^2}{32\pi^2}G\widetilde{G}\Big), \tag{35.67}$$

where the θ-term carries the factor i characteristic of a topological term continued to Euclidean signature, and θ here stands for the full physical angle $\bar{\theta}$ once the mass phases are rotated into the gauge sector. Two exact properties follow immediately. Because the winding number $\nu = \int (g_s^2/32\pi^2) G\widetilde{G}$ is an integer, $E(\theta)$ is periodic, $E(\theta+2\pi) = E(\theta)$. And because the integrand at $\theta = 0$ is real and positive while the θ-term enters as a pure phase $e^{-i\theta\nu}$ that can only reduce the magnitude of the partition function relative to its $\theta = 0$ value,

$$e^{-V_4 E(\theta)} = \big\langle e^{-i\theta\nu}\big\rangle_{\theta=0}\, e^{-V_4 E(0)}, \qquad \big|\langle e^{-i\theta\nu}\rangle\big| \le 1, \tag{35.68}$$

we conclude $E(\theta) \ge E(0)$ for all θ. This is the Vafa–Witten theorem: in a vector-like gauge theory the energy is minimized at $\theta = 0$, established in Section 42.7. The minimum is not merely a stationary point; it is the global minimum, fixed by the reality and positivity of the measure.

The curvature at that minimum is the *topological susceptibility*. Expanding (35.67) to second order in θ and using $\langle\nu\rangle_{\theta=0} = 0$ (parity), the cumulant expansion gives $V_4 E(\theta) = V_4 E(0) + \tfrac{1}{2}\theta^2\langle\nu^2\rangle/V_4 + O(\theta^4)$, that is

$$\chi_{\rm top} \;\equiv\; \frac{d^2E}{d\theta^2}\Big|_{\theta=0} \;=\; \frac{\langle\nu^2\rangle}{V_4} \;=\; \int d^4x\,\Big\langle \frac{g_s^2}{32\pi^2}G\widetilde{G}(x)\;\frac{g_s^2}{32\pi^2}G\widetilde{G}(0)\Big\rangle, \tag{35.69}$$

the spacetime integral of the topological-charge-density two-point function. The same object, with its reflection-positivity properties, is the one defined in Theorem 42.33; what matters here is the inequality it satisfies, $\chi_{\rm top} \ge 0$, which says $E(\theta)$ curves *upward* at the origin. The vacuum energy is thus a periodic, even function with a convex global minimum at $\theta = 0$ — precisely the potential well a dynamical θ would need in order to relax to the CP-conserving point. Figure 35.7 plots $E(\theta)$ explicitly once we have computed it from the chiral Lagrangian below.

Physical Insight

The positivity $\chi_{\rm top} \ge 0$ has a transparent meaning: it is the statement that the QCD vacuum responds to a θ-source the way any stable medium responds to a perturbation — it costs energy to twist it. A negative susceptibility would mean $\theta = 0$ is a maximum and the vacuum would prefer to generate CP violation spontaneously; the Vafa–Witten result forbids exactly that. The whole Peccei–Quinn mechanism is the observation that, given a convex well, all one needs is to put a ball in it.

35.4.3 The Peccei–Quinn mechanism

The decisive move is to make θ dynamical. Suppose the Standard Model is extended by a global chiral symmetry $\mathrm{U}(1)_{\mathrm{PQ}}$ — the Peccei–Quinn symmetry — with two defining features: it is spontaneously broken at some large scale f_a, and it has a colour anomaly. The first feature delivers, by Goldstone's theorem, a massless scalar; the second feature gives that scalar exactly the coupling needed to absorb θ.

Concretely, let Φ be a complex scalar carrying $\mathrm{U}(1)_{\mathrm{PQ}}$ charge, with a Mexican-hat potential $V(\Phi) = \lambda\big(|\Phi|^2 - \tfrac{1}{2}f_a^2\big)^2$ that drives $\langle|\Phi|\rangle = f_a/\sqrt{2}$. Writing $\Phi = \frac{1}{\sqrt{2}}(f_a + \rho)\, e^{ia/f_a}$, the radial mode ρ is heavy ($m_\rho \sim \sqrt{\lambda} f_a$) and the angular mode a is the massless Goldstone boson of the broken $\mathrm{U}(1)_{\mathrm{PQ}}$, the *axion*. Under a PQ rotation of parameter β the axion shifts,

$$a \;\longrightarrow\; a + \beta f_a, \tag{35.70}$$

a continuous shift symmetry that, in the absence of the anomaly, would forbid any potential for a and keep it exactly massless — the pseudo-Goldstone structure of Theorem 29.29.

The anomaly breaks the shift symmetry, but in a controlled way. By construction $\mathrm{U}(1)_{\mathrm{PQ}}$ acts chirally on coloured fermions (either the Standard Model quarks themselves, or new heavy quarks; we specify the realizations below), so its current j^μ_{PQ} has a colour anomaly of exactly the form computed in Theorem 36.9,

$$\partial_\mu j^\mu_{\mathrm{PQ}} \;=\; N\,\frac{g_s^2}{32\pi^2}\, G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu}, \tag{35.71}$$

with N the colour anomaly coefficient (the sum of PQ charges times colour indices of the fermions running in the triangle; it is also the integer *domain-wall number* of the model). The anomaly equation (35.71) says that a PQ rotation (35.70) is not a symmetry of the measure: it shifts the effective θ-angle by $N\beta$. Equivalently, because the axion *is* the PQ phase, its derivative coupling $-(1/f_a)\,\partial_\mu a\, j^\mu_{\mathrm{PQ}}$ integrates by parts onto the anomaly, generating a direct axion–gluon coupling

$$\mathcal{L}_a \;\supset\; \frac{1}{2}(\partial_\mu a)(\partial^\mu a) \;+\; \frac{a}{f_a/N}\,\frac{g_s^2}{32\pi^2}\, G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu}. \tag{35.72}$$

Absorbing the integer N into the definition of the decay constant — writing f_a henceforth for the combination f_{PQ}/N that controls the gluon coupling — the axion enters QCD in precisely the slot occupied by θ. The total coefficient of $(g_s^2/32\pi^2)G\widetilde{G}$ is now

$$\theta_{\mathrm{eff}} \;=\; \bar{\theta} + \frac{a}{f_a}. \tag{35.73}$$

This is the entire idea, and it is worth stating its logic baldly. The bare $\bar{\theta}$ is a constant we cannot choose. But (35.73) shows it can be *shifted away* by a constant displacement of a, because a enters only through the sum $\bar{\theta} + a/f_a$. The shift symmetry (35.70) of the kinetic term, broken only by the QCD-induced potential, guarantees that the value of $\bar{\theta}$ has no physical consequence: it can always be reabsorbed into the origin of a. What is left to determine is where a actually sits, and that is fixed by the QCD vacuum energy of Section 35.4.2.

For the strong interactions do generate a potential for a, precisely because θ_{eff} is physical. Integrating out the gluons at low energy, the axion-dependent piece of the vacuum energy is nothing but $E(\theta)$ evaluated at $\theta = \theta_{\text{eff}}$:

$$V(a) \;=\; E\Big(\bar{\theta} + \frac{a}{f_a}\Big). \tag{35.74}$$

By the Vafa–Witten result (35.68), this potential is minimized where its argument vanishes, that is at

$$\Big\langle \frac{a}{f_a} \Big\rangle = -\bar{\theta} \qquad \Longrightarrow \qquad \theta_{\text{eff}}\big|_{\text{vac}} = \bar{\theta} + \Big\langle \frac{a}{f_a} \Big\rangle = 0. \tag{35.75}$$

The axion field rolls until the *effective* strong-CP angle is exactly zero, whatever the bare $\bar{\theta}$ was. Strong CP violation is not tuned away; it is dynamically relaxed away. This is the Peccei–Quinn solution, and Fig. 35.7 shows the relaxation as a ball rolling to the bottom of the tilted well. The neutron electric dipole moment, set by θ_{eff} rather than by $\bar{\theta}$, is driven to zero up to the tiny residual from electroweak CP violation feeding into θ_{eff} at many loops — far below the experimental bound (35.66), so the prediction is that the neutron EDM sits just out of current reach, a falsifiable statement.

Caution: the axion mass is not zero

The shift symmetry (35.70) is exact at the classical level and is broken *only* by the anomaly (35.71) — which is to say, only by QCD non-perturbative effects. This is why a is a pseudo-Goldstone boson and not an exact Goldstone: the same potential (35.74) that relaxes $\bar{\theta}$ also gives a a small mass, set by the curvature of $E(\theta)$. A genuinely massless axion would not solve the problem, because it would not be pinned to the minimum. The mass and the solution are the same phenomenon viewed twice, and we compute the mass next.

35.4.4 The axion mass from current algebra

The mass of the axion is the curvature of its potential at the minimum. Differentiating (35.74) twice and using (35.69),

$$m_a^2 \;=\; \frac{d^2V}{da^2}\bigg|_{\text{min}} \;=\; \frac{1}{f_a^2}\,\frac{d^2E}{d\theta^2}\bigg|_{\theta=0} \;=\; \frac{\chi_{\text{top}}}{f_a^2}. \tag{35.76}$$

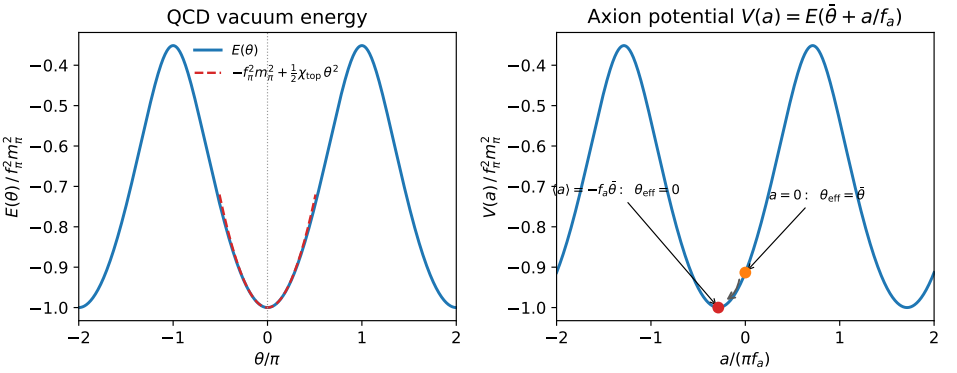


Figure 35.7: The QCD vacuum energy and the axion potential. *Left:* the two-flavour chiral result (35.80) for $E(\theta)$ (solid), a smooth periodic function with a convex global minimum at $\theta = 0$, together with its harmonic approximation $-f_\pi^2 m_\pi^2 + \frac{1}{2}\chi_{\rm top}\theta^2$ (dashed) whose curvature is the topological susceptibility (35.82). *Right:* reading the same curve as the axion potential $V(a) = E(\bar\theta + a/f_a)$ of (35.74). Whatever the bare $\bar\theta$ (here the field starts at $a = 0$, where $\theta_{\rm eff} = \bar\theta$), the axion relaxes to $\langle a\rangle = -f_a\bar\theta$, where $\theta_{\rm eff} = 0$ and strong CP is switched off.

The axion mass is fixed entirely by the QCD topological susceptibility and the symmetry-breaking scale; everything else about the model has dropped out. To turn this into a number we need $\chi_{\rm top}$, and that is a strong-coupling quantity — but one we can compute at leading order from the chiral Lagrangian of Section 39.2, because the θ-dependence of QCD is carried entirely by the light pseudoscalars. This is the current-algebra computation; we carry it out in full.

Worked Example 35.7 (The two-flavour $E(\theta)$ and the axion mass)**.** We compute $E(\theta)$ in two-flavour chiral perturbation theory and extract $\chi_{\rm top}$ and m_a. The strategy is the one used to put θ into the chiral Lagrangian in the first place: a chiral rotation trades the gluonic θ-term for a complex phase on the quark mass matrix, by the anomaly. So we may set the gluonic $\theta = 0$ and instead give the mass matrix the phase, $M_\theta = \mathcal{M}\, e^{i\theta/N_f}$ distributed so that $\arg\det M_\theta = \theta$. For two flavours we write the (real, positive) masses as $\mathcal{M} = \mathrm{diag}(m_u, m_d)$ and assign phases φ_u, φ_d with $\varphi_u + \varphi_d = \theta$.

Step 1: the vacuum energy as an alignment problem. The leading-order chiral Lagrangian carries the mass term $\mathcal{L}_{\rm mass} = \frac{f^2 B}{2}\,\mathrm{tr}(M_\theta^\dagger U + U^\dagger M_\theta)$ of Eq. (39.58), where $U = \exp(i\Phi/f)$ is the Goldstone field, $f = f_\pi$, and $B = -\langle\bar q q\rangle/f_\pi^2$ from Eq. (39.63). The vacuum energy is minus this evaluated on the vacuum. For aligned vacuum $U = \mathbf{1}$ (the alignment is the content of the phases φ_q), and writing the mass phases explicitly,

$$E(\theta) \;=\; -\,f_\pi^2 B\,(m_u\cos\varphi_u + m_d\cos\varphi_d), \qquad \varphi_u + \varphi_d = \theta, \tag{35.77}$$

to be *minimized* over the distribution of the total phase θ between the two quarks (the QCD vacuum chooses the alignment that costs the least energy).

Step 2: minimize over the phase split. Set $\varphi_u = \frac{\theta}{2} + \delta$, $\varphi_d = \frac{\theta}{2} - \delta$, so the constraint is automatic and δ is free. Then

$$m_u \cos\varphi_u + m_d \cos\varphi_d = (m_u + m_d)\cos\tfrac{\theta}{2}\cos\delta - (m_u - m_d)\sin\tfrac{\theta}{2}\sin\delta. \quad (35.78)$$

Maximizing this over δ (to minimize E) is the maximization of $A\cos\delta - B'\sin\delta$ with $A = (m_u + m_d)\cos\frac{\theta}{2}$ and $B' = (m_u - m_d)\sin\frac{\theta}{2}$, whose maximum is $\sqrt{A^2 + B'^2}$. Compute

$$A^2 + B'^2 = (m_u + m_d)^2\cos^2\tfrac{\theta}{2} + (m_u - m_d)^2\sin^2\tfrac{\theta}{2} = m_u^2 + m_d^2 + 2m_u m_d\cos\theta, \quad (35.79)$$

using $\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos\theta$ and $\cos^2 + \sin^2 = 1$. Therefore

$$E(\theta) \;=\; -f_\pi^2 B\sqrt{m_u^2 + m_d^2 + 2m_u m_d\cos\theta}. \quad (35.80)$$

At $\theta = 0$ the radical is $(m_u + m_d)$, so $E(0) = -f_\pi^2 B(m_u + m_d) = -f_\pi^2 m_\pi^2$, using the Gell-Mann–Oakes–Renner relation $m_\pi^2 = B(m_u + m_d)$ from Eq. (39.62); the vacuum energy at $\theta = 0$ reproduces the chiral-condensate energy, a useful check. The function (35.80) is manifestly periodic, even, and — since the radicand is maximized at $\theta = 0$ — minimized there, in agreement with Vafa–Witten (35.68).

Step 3: the curvature. Expand the radicand for small θ: $\cos\theta \approx 1 - \frac{1}{2}\theta^2$ gives $m_u^2 + m_d^2 + 2m_u m_d - m_u m_d\theta^2 = (m_u + m_d)^2\big(1 - \frac{m_u m_d}{(m_u+m_d)^2}\theta^2\big)$, so

$$E(\theta) \approx -f_\pi^2 B(m_u + m_d)\Big(1 - \tfrac{1}{2}\tfrac{m_u m_d}{(m_u+m_d)^2}\theta^2\Big) = -f_\pi^2 m_\pi^2 + \tfrac{1}{2} f_\pi^2 m_\pi^2 \frac{m_u m_d}{(m_u + m_d)^2}\,\theta^2. \quad (35.81)$$

Reading off the topological susceptibility (35.69),

$$\boxed{\chi_{\rm top} \;=\; f_\pi^2 m_\pi^2 \frac{m_u m_d}{(m_u + m_d)^2}} \quad (35.82)$$

which is positive, as the general argument required, and vanishes if either quark mass goes to zero — the massless-quark escape of Section 35.4.1, now visible as the flattening of the well.

Step 4: the axion mass. Insert (35.82) into (35.76):

$$\boxed{m_a \;=\; \frac{f_\pi m_\pi}{f_a}\frac{\sqrt{m_u m_d}}{m_u + m_d} \;=\; \frac{f_\pi m_\pi}{f_a}\frac{\sqrt{z}}{1+z}} \qquad z \equiv \frac{m_u}{m_d}. \quad (35.83)$$

With the lattice/PDG value $z \approx 0.48$, the mass ratio factor is $\sqrt{z}/(1+z) \approx 0.47$, and with $f_\pi \approx 92\,\text{MeV}$, $m_\pi \approx 135\,\text{MeV}$,

$$m_a \;\approx\; 0.47 \times \frac{(92\,\text{MeV})(135\,\text{MeV})}{f_a} \;\approx\; 5.8\,\mu\text{eV} \times \Big(\frac{10^{12}\,\text{GeV}}{f_a}\Big), \quad (35.84)$$

where the arithmetic uses $10^{12}\,\mathrm{GeV} = 10^{15}\,\mathrm{MeV}$ so that $f_\pi m_\pi / f_a = 1.24 \times 10^4\,\mathrm{MeV}^2/10^{15}\,\mathrm{MeV} = 1.24\times10^{-5}\,\mathrm{eV}$. The leading-order result (35.84) agrees with the precise next-to-leading-order lattice value $m_a = 5.70(7)\,\mu\mathrm{eV}\,(10^{12}\,\mathrm{GeV}/f_a)$ to within the expected size of the $O(p^4)$ corrections.

The result (35.83) is the heart of the matter. The axion mass is *inversely* proportional to the symmetry-breaking scale f_a: the larger the scale at which Peccei–Quinn breaks, the lighter and more weakly coupled the axion. Every axion coupling, being a coupling of a pseudo-Goldstone, is likewise suppressed by $1/f_a$ — to gluons through (35.72), to photons through an analogous electromagnetic anomaly, and to fermions through derivative couplings $\propto \partial_\mu a/f_a$. This single inverse-scale dependence is what makes the axion both a clean solution and an experimental quarry of a very particular kind: not a heavy state to be produced at a collider, but an ultralight, ultraweakly coupled relic.

Notation watch: f_a, f_{PQ}, and the domain-wall number

Two scales appear in the literature and we have quietly identified them. The Peccei–Quinn breaking scale, the radius of the Mexican hat, is $f_{\mathrm{PQ}} = \langle|\Phi|\rangle\sqrt{2}$; the scale that controls the gluon coupling and hence the axion mass is $f_a = f_{\mathrm{PQ}}/N$, with N the integer colour anomaly of (35.71). Throughout (35.72)–(35.83) "f_a" means the second of these, the physically relevant decay constant. The integer N counts degenerate minima of $V(a)$ within one 2π period of a/f_{PQ} and is the number of distinct domain walls the axion field can interpolate — a fact of cosmological consequence (the domain-wall problem) for the $N > 1$ models, but immaterial to the mass formula once f_a is used.

35.4.5 Realizing the symmetry: visible and invisible axions

The mechanism so far has assumed a $\mathrm{U}(1)_{\mathrm{PQ}}$ with a colour anomaly; concrete models differ only in which fields carry the charge, and hence in where f_a sits. The historical original, the Peccei–Quinn–Weinberg–Wilczek axion, tied f_a to the electroweak scale by charging the Standard Model Higgs sector under $\mathrm{U}(1)_{\mathrm{PQ}}$, so that $f_a \sim v \approx 250\,\mathrm{GeV}$. Equation (35.84) then predicts $m_a \sim \mathcal{O}(100\,\mathrm{keV})$ and couplings strong enough to be produced in rare meson and quarkonium decays. That "visible" axion is excluded: searches for $K^+ \to \pi^+ a$ and $\Upsilon \to \gamma a$ found nothing, ruling out an axion at the electroweak scale. The lesson is exactly the inverse-scale structure of (35.83): to make the axion weakly enough coupled to have escaped detection, one must *raise* f_a far above the electroweak scale, decoupling $\mathrm{U}(1)_{\mathrm{PQ}}$ breaking from electroweak breaking. The two benchmark constructions that do so define the "invisible" axion:

- **KSVZ** (Kim–Shifman–Vainshtein–Zakharov): introduce a heavy colour-

triplet quark Q and a Standard-Model-singlet complex scalar Φ, with $\mathrm{U}(1)_{\mathrm{PQ}}$ acting chirally on Q and shifting Φ's phase. The anomaly (35.71) comes from the Q triangle; $f_a = \langle|\Phi|\rangle$ is a free, large scale; the ordinary fermions are PQ-neutral, so the axion couples to them only through loops.

- **DFSZ** (Dine–Fischler–Srednicki–Zhitnitsky): enlarge the Higgs sector to two doublets plus a singlet Φ, with $\mathrm{U}(1)_{\mathrm{PQ}}$ acting on all of them; the Standard Model quarks then carry PQ charge through their Yukawa couplings, and the axion couples to ordinary matter at tree level. Again $f_a \sim \langle|\Phi|\rangle$ is free and large.

In both, the same low-energy Lagrangian (35.72) and the same mass (35.83) hold; the models differ only in the order-one coefficients of the axion's couplings to photons and to electrons, which is what experiments must assume to set limits. The allowed window for f_a is bounded below by astrophysics — a too-strongly-coupled axion would carry energy out of stars and supernovae too efficiently, and the absence of such cooling in SN 1987A and in horizontal-branch stars requires $f_a \gtrsim 10^8\,\mathrm{GeV}$, i.e. $m_a \lesssim \mathcal{O}(10\,\mathrm{meV})$ — and bounded above, as we are about to see, by cosmology. The viable axion is therefore very light, μeV to meV, and very weakly coupled: an "invisible" particle whose entire reason for existing is to enforce $\theta_{\mathrm{eff}} = 0$.

Handoff to dark matter

That an ultralight, ultraweakly coupled, cosmologically stable scalar should fall out of the cleanest naturalness fix in the Standard Model is a striking bonus, and it is the bridge to the next problem on the map. In the early universe the axion field sits at some random initial value $\bar{\theta} + a/f_a = \theta_i$ before QCD turns on; when the topological susceptibility switches on near the QCD scale, the field begins to oscillate about the minimum (35.75), and those coherent oscillations of a nonrelativistic, pressureless condensate are cold dark matter. We take up this *misalignment mechanism*, its relic-abundance estimate, and the axion's place alongside the WIMP and the sterile neutrino in Section 35.6.

35.5 Grand Unification

Three threads of this chapter now converge on a single scale. The seesaw of Section 35.2 needed a heavy right-handed neutrino with a Majorana mass near $10^{14}\,\mathrm{GeV}$ to make the light neutrino masses come out right; the hierarchy discussion of Section 35.3 taught us that any new physics far above the weak scale must be there for a structural reason, not by accident; and the dimension-six B- and L-violating operators catalogued in the SMEFT survey of Section 35.1.1 are suppressed by some ultraviolet scale Λ whose identity the Standard Model

leaves open. Grand unification is the hypothesis that all three point at the same thing: a scale M_{GUT} at which the three factors of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ are revealed as the low-energy shards of one simple gauge group. This is the most economical structural idea on the market, and it makes sharp, falsifiable predictions. We will develop it honestly — it explains charge quantization, it nearly unifies the couplings, and in its minimal non-supersymmetric form it is *excluded* by the non-observation of proton decay.

The chapter's organizing logic — that "beyond the Standard Model" is the tower of higher-dimension operators plus the unexplained values of the relevant and marginal couplings — frames unification precisely. The marginal couplings are the three gauge couplings g_1, g_2, g_3; their values are inputs, and unification is the claim that those three inputs collapse to one. The puzzle of charge quantization — why $Q_{\mathrm{electron}}/Q_{\mathrm{down}}$ is exactly -3 — is the statement that the hypercharge assignments, free parameters in the Standard Model, are fixed. Proton decay is the leading irrelevant operator the heavy theory generates. We meet all three on this single map.

This section continues Yang–Mills theory at the level of the gauge beta function: the one-loop running we use to test unification is exactly the non-abelian beta function computed in Chapter 27, Eq. (27.124), extended from SU(3) to all three factors and integrated with the Callan–Symanzik machinery of Theorem 24.19.

35.5.1 The minimal embedding: SU(5)

The first question is purely group-theoretic: which simple Lie groups contain $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ as a subgroup, with room for the observed fermions? Two arithmetic facts constrain the search at once. First, the rank. The Standard Model group has rank $2 + 1 + 1 = 4$: rank two from SU(3) (two commuting generators, the Gell-Mann λ_3, λ_8 of Section 2.7), rank one from $\mathrm{SU}(2)_L$, rank one from $\mathrm{U}(1)_Y$. A simple group containing it must have rank at least four. Second, the group must possess complex representations, because the Standard Model fermions are chiral — left- and right-handed fields sit in inequivalent representations, and a real or pseudoreal representation could never reproduce that. Among the simple Lie algebras of rank four, the candidates with complex representations are essentially $\mathfrak{su}(5)$ and (at rank five, the next step up) $\mathfrak{so}(10)$. The rank-four minimal choice is $\mathfrak{su}(5)$, the Georgi–Glashow group [217]. We build it explicitly.

The group and its gauge bosons. SU(5) is the group of 5×5 unitary matrices of unit determinant; its Lie algebra $\mathfrak{su}(5)$ has $5^2 - 1 = 24$ generators, Hermitian and traceless, normalized as in the rest of the book by $\mathrm{tr}(T^A T^B) = \frac{1}{2}\delta^{AB}$ (Chapter B). We embed $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ by letting $\mathrm{SU}(3)_c$ act on the

first three components of the fundamental and $\mathrm{SU}(2)_L$ on the last two:

$$\mathrm{SU}(3)\times\mathrm{SU}(2)\;\hookrightarrow\;\mathrm{SU}(5),\qquad (g_3,g_2)\;\longmapsto\;\begin{pmatrix} g_3 & 0\\ 0 & g_2\end{pmatrix}, \tag{35.85}$$

a block-diagonal $3\oplus 2$ inside the 5. The 24 generators decompose under $\mathrm{SU}(3)\times\mathrm{SU}(2)$ into the upper-left 3×3 traceless block ($\mathbf{8}$ of SU(3), the gluons), the lower-right 2×2 traceless block ($\mathbf{3}$ of SU(2), the W's), one diagonal generator commuting with both blocks (the hypercharge B, see below), and the off-diagonal 3×2 and 2×3 blocks. In representation language,

$$\mathbf{24}\;=\;(\mathbf{8},\mathbf{1})_0\;\oplus\;(\mathbf{1},\mathbf{3})_0\;\oplus\;(\mathbf{1},\mathbf{1})_0\;\oplus\;(\mathbf{3},\mathbf{2})_{-5/6}\;\oplus\;(\overline{\mathbf{3}},\mathbf{2})_{+5/6}, \tag{35.86}$$

the subscript being the hypercharge fixed momentarily. The first twelve are exactly the Standard Model gauge bosons; the last twelve are new gauge bosons, called X and Y, carrying *both* colour and weak isospin. They are leptoquarks: their couplings turn quarks into leptons. They are the agents of proton decay, and the whole quantitative story below hangs on their mass.

The hypercharge generator and charge quantization. The one diagonal SU(5) generator orthogonal to all of $\mathrm{SU}(3)\times\mathrm{SU}(2)$ is, up to normalization,

$$Y\;\propto\;\mathrm{diag}\Big(-\tfrac{1}{3},-\tfrac{1}{3},-\tfrac{1}{3},\;+\tfrac{1}{2},+\tfrac{1}{2}\Big). \tag{35.87}$$

It is traceless ($3\times(-\frac{1}{3})+2\times(+\frac{1}{2})=0$), as every $\mathfrak{su}(5)$ generator must be, and it commutes with the colour and isospin blocks because it is constant on each. This single matrix *is* weak hypercharge — and that is the explanation of charge quantization the Standard Model could not give. In Eq. (31.21) the hypercharges were free inputs, chosen to reproduce the measured electric charges through the Gell-Mann–Nishijima relation $Q=T^3+Y$ (Eq. (31.20)). Here Y is a generator of a *simple* group, so its eigenvalues are quantized: a U(1) embedded in a simple group has automatically commensurate charges, because the trace of any generator over any representation vanishes. We make this precise once the fermions are placed.

Placing one generation: $\overline{\mathbf{5}}\oplus\mathbf{10}$. A single Standard Model generation comprises fifteen left-handed Weyl fields: $Q=(u,d)_L$, u^c, d^c, $L=(\nu,e)_L$, e^c (the c superscript denoting the charge-conjugate of a right-handed field, so that everything is written left-handed). Fifteen is exactly $5+10$, and these are the two smallest non-trivial representations of SU(5). The fundamental $\mathbf{5}$ decomposes under Eq. (35.85) as a colour triplet plus a weak doublet,

$$\mathbf{5}=(\mathbf{3},\mathbf{1})_{-1/3}\;\oplus\;(\mathbf{1},\mathbf{2})_{+1/2}, \tag{35.88}$$

reading the hypercharges directly off Eq. (35.87). Its conjugate carries the opposite hypercharges,

$$\overline{\mathbf{5}} = (\overline{\mathbf{3}}, \mathbf{1})_{+1/3} \ \oplus \ (\mathbf{1}, \mathbf{2})_{-1/2}, \tag{35.89}$$

and these are precisely the quantum numbers of d^c (an anti-triplet of hypercharge $+\frac{1}{3}$) and of the lepton doublet L (hypercharge $-\frac{1}{2}$). So one places

$$\overline{\mathbf{5}} \ \supset \ d^c \ \oplus \ L. \tag{35.90}$$

The remaining seven fields fill the **10**, the antisymmetric product $(\mathbf{5} \times \mathbf{5})_{\text{antisym}}$. Acting on an antisymmetric pair $[ij]$, the hypercharge generator gives $Y_{[ij]} = Y_i + Y_j$ with Y_i the entries of Eq. (35.87), so the **10** splits as

$$\mathbf{10} = (\mathbf{3}, \mathbf{2})_{+1/6} \ \oplus \ (\overline{\mathbf{3}}, \mathbf{1})_{-2/3} \ \oplus \ (\mathbf{1}, \mathbf{1})_{+1}, \tag{35.91}$$

where the three blocks come from one colour and one isospin index ($-\frac{1}{3} + \frac{1}{2} = \frac{1}{6}$), two colour indices ($-\frac{1}{3} - \frac{1}{3} = -\frac{2}{3}$, antisymmetric $\to \overline{\mathbf{3}}$), and two isospin indices ($\frac{1}{2} + \frac{1}{2} = +1$, antisymmetric $\to$ singlet). These are exactly the quantum numbers of the quark doublet Q, of u^c, and of e^c:

$$\mathbf{10} \ \supset \ Q \ \oplus \ u^c \ \oplus \ e^c. \tag{35.92}$$

Every field of a generation is accounted for, with not one left over and not one missing. The fit is so tight it is hard to believe it is an accident. As a matrix, the **10** is the antisymmetric 5×5

$$\chi_{ij} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 & u_3^c & -u_2^c & -u_1 & -d_1 \\ -u_3^c & 0 & u_1^c & -u_2 & -d_2 \\ u_2^c & -u_1^c & 0 & -u_3 & -d_3 \\ u_1 & u_2 & u_3 & 0 & -e^c \\ d_1 & d_2 & d_3 & e^c & 0 \end{pmatrix}, \tag{35.93}$$

the 3×3 upper block holding u^c (an anti-triplet, hence one index), the 3×2 blocks holding the quark doublet, and the 2×2 block holding e^c.

Worked Example 35.8 (Every Standard Model hypercharge, and why the down-quark charge is $-\frac{1}{3}$)**.** We verify that SU(5) reproduces the measured charges and, more importantly, that it *forces* them. Electric charge is $Q = T^3 + Y$, with $T^3 = \text{diag}(\frac{1}{2}, -\frac{1}{2})$ on a weak doublet and $T^3 = 0$ on a weak singlet, and Y given by Eq. (35.87) on the **5** (and its negative on the $\overline{\mathbf{5}}$).

The $\overline{\mathbf{5}}$. Its triplet piece is d^c with $T^3 = 0$, $Y = +\frac{1}{3}$, hence $Q(d^c) = +\frac{1}{3}$ — the antiparticle of the down quark, so $Q(d) = -\frac{1}{3}$. Its doublet piece is L with $Y = -\frac{1}{2}$: the upper component has $T^3 = +\frac{1}{2}$, giving $Q(\nu) = +\frac{1}{2} - \frac{1}{2} = 0$, and the lower has $T^3 = -\frac{1}{2}$, giving $Q(e) = -\frac{1}{2} - \frac{1}{2} = -1$. Every charge in the lepton doublet and the down sector is reproduced.

The **10**. The doublet Q has $Y=+\frac{1}{6}$: $Q(u)=\frac{1}{2}+\frac{1}{6}=\frac{2}{3}$, $Q(d)=-\frac{1}{2}+\frac{1}{6}=-\frac{1}{3}$. The singlet u^c has $T^3=0$, $Y=-\frac{2}{3}$, so $Q(u^c)=-\frac{2}{3}$, i.e. $Q(u)=+\frac{2}{3}$. The singlet e^c has $Y=+1$, $Q=+1$, i.e. $Q(e)=-1$. Consistent throughout.

Why it is forced. Electric charge $Q=T^3+Y$ is a linear combination of $\mathfrak{su}(5)$ generators, hence itself a traceless generator: $\operatorname{tr} Q=0$ on *every* representation. Evaluate the trace on the $\overline{\mathbf{5}}$, whose five members are the three colours of d^c and the two leptons ν, e:

$$\operatorname{tr}_{\overline{\mathbf{5}}} Q = 3\,Q(d^c)+Q(\nu)+Q(e) = 3\,Q(d^c)+0+Q(e)=0. \tag{35.94}$$

With the electron charge set by experiment to $Q(e)=-1$, this reads $3\,Q(d^c)=+1$, so $Q(d^c)=+\frac{1}{3}$ and the down quark carries exactly $-\frac{1}{3}$ of the electron's charge. The factor of three is the number of colours. *Charge quantization is the tracelessness of an* SU(5) *generator, and the fractional quark charge is dictated by the dimension of the colour representation.* This is the structural fact the Standard Model could only postulate.

Anomaly freedom. A chiral gauge theory is consistent only if its cubic gauge anomaly cancels (Section 37.3). The anomaly of a representation R is the coefficient $A(R)$ defined by $\operatorname{tr}_R(\{T^a,T^b\}T^c)=A(R)\,d^{abc}$, antisymmetric under $R\to\bar{R}$ so that $A(\bar{R})=-A(R)$. For SU(5) one finds, with the normalization $A(\mathbf{5})=+1$,

$$A(\overline{\mathbf{5}})=-1, \qquad A(\mathbf{10})=+1. \tag{35.95}$$

The matter content of one generation is $\overline{\mathbf{5}}\oplus\mathbf{10}$, so

$$A(\overline{\mathbf{5}})+A(\mathbf{10})=-1+1=0. \tag{35.96}$$

The anomaly cancels — and it cancels not field by field, as it superficially does in the Standard Model, but between two *irreducible* multiplets. This is a far more satisfying statement of the same fact established in Section 37.3: the intricate sum of hypercharges that had to vanish there ($\sum Y^3=0$, $\sum Y=0$ over a generation) is, from the SU(5) vantage, the single equation Eq. (35.96).

35.5.2 Coupling normalization and the prediction $\sin^2\theta_{\mathrm{W}}=\frac{3}{8}$

Above M_{GUT} there is one gauge group and therefore one gauge coupling g_5: the covariant derivative is $D_\mu=\partial_\mu-ig_5A^A_\mu T^A$ over all 24 generators, with the single g_5 multiplying the canonically normalized generators $\operatorname{tr}(T^AT^B)=\frac{1}{2}\delta^{AB}$. Matching to the Standard Model below M_{GUT} fixes how g_5 relates to the three low-energy couplings, and the non-trivial content lies entirely in the abelian factor.

For SU(3) and SU(2) the matching is immediate: the canonical $\mathfrak{su}(5)$ normalization restricts to the canonical normalization of each non-abelian subgroup, so

at the unification scale $g_3 = g_2 = g_5$. For $\mathrm{U}(1)_Y$ it is not immediate, because the hypercharge generator Eq. (35.87) is *not* canonically normalized. Compute its trace-square over the fundamental,

$$\mathrm{tr}_{\mathbf{5}}(Y^2) = 3 \times \left(\tfrac{1}{3}\right)^2 + 2 \times \left(\tfrac{1}{2}\right)^2 = \tfrac{1}{3} + \tfrac{1}{2} = \tfrac{5}{6}, \tag{35.97}$$

whereas a canonically normalized generator T has $\mathrm{tr}_{\mathbf{5}}(T^2) = \frac{1}{2}$. The properly normalized hypercharge-direction generator is therefore

$$T^{24} = \sqrt{\tfrac{3}{5}}\, Y, \qquad \text{since} \quad \mathrm{tr}\,[(\sqrt{\tfrac{3}{5}}\, Y)^2] = \tfrac{3}{5} \cdot \tfrac{5}{6} = \tfrac{1}{2}. \tag{35.98}$$

Now match the abelian piece of the covariant derivative. In the Standard Model the hypercharge interaction is $-ig' Y B_\mu$ (Eq. (31.23)) with $g' \equiv g_Y$ the measured hypercharge coupling. In SU(5) the same gauge field couples through the canonical generator as $-ig_5\, T^{24} A^{24}_\mu = -ig_5 \sqrt{\tfrac{3}{5}}\, Y A^{24}_\mu$. Identifying $A^{24}_\mu = B_\mu$ and the coefficients of Y,

$$g_Y = \sqrt{\tfrac{3}{5}}\, g_5 \qquad \Longleftrightarrow \qquad g_1 \equiv \sqrt{\tfrac{5}{3}}\, g_Y = g_5, \tag{35.99}$$

where g_1 is the *GUT-normalized* hypercharge coupling — the combination that equals g_5 at unification. Squaring,

$$\boxed{g_1^2 = \tfrac{5}{3}\, g_Y^2} \tag{35.100}$$

This factor $\frac{5}{3}$ is the single most important number in the whole subject: it is what makes the three couplings comparable at all. Without it one would be comparing g_Y, which is normalized by a convention internal to the Standard Model, against the group-theoretically normalized g_2, g_3, and the question of unification would be ill-posed.

Worked Example 35.9 (The tree-level weak mixing angle $\sin^2\theta_{\mathrm{W}} = \frac{3}{8}$)**.** The weak mixing angle is defined (Chapter 31) by $\sin^2\theta_{\mathrm{W}} = g_Y^2/(g_2^2 + g_Y^2)$, the fraction of the neutral-current coupling carried by hypercharge. At the unification scale all canonical couplings coincide, $g_2 = g_1 = g_5$, and from Eq. (35.100) the physical hypercharge coupling is $g_Y^2 = \frac{3}{5} g_1^2 = \frac{3}{5} g_2^2$. Substitute:

$$\sin^2\theta_{\mathrm{W}}\big|_{M_{\mathrm{GUT}}} = \frac{g_Y^2}{g_2^2 + g_Y^2} = \frac{\frac{3}{5} g_2^2}{g_2^2 + \frac{3}{5} g_2^2} = \frac{\frac{3}{5}}{1 + \frac{3}{5}} = \frac{\frac{3}{5}}{\frac{8}{5}} = \frac{3}{8}. \tag{35.101}$$

So minimal SU(5) predicts $\sin^2\theta_{\mathrm{W}} = 0.375$ *at the unification scale*. The measured value at M_Z is $\sin^2\theta_{\mathrm{W}} \approx 0.231$. These disagree by a wide margin, and that is not a failure: $\sin^2\theta_{\mathrm{W}}$ is a ratio of running couplings and is itself scale-dependent. The prediction is at $M_{\mathrm{GUT}} \sim 10^{16}\,\mathrm{GeV}$; the journey down to M_Z across fourteen decades of energy is governed by the renormalization group, and reproducing 0.231 at the bottom from 0.375 at the top is the genuine quantitative test, to which we now turn.

35.5.3 Running the three couplings: does unification happen?

The unification hypothesis is a boundary condition at one end of an renormalization-group flow we already know how to compute. Below $M_{\rm GUT}$ the theory is just the Standard Model, three separate gauge groups with three couplings, each running according to its own one-loop beta function. The question is whether, run *upward* from their measured values at M_Z, the three meet at a common point.

The one-loop coefficients from the field content. The one-loop running of any gauge coupling, derived for the non-abelian case as Yang–Mills theory in Eq. (27.124) and integrated as in Eq. (24.231), takes the universal form

$$\frac{d\,\alpha_i^{-1}}{d\ln\mu} = -\frac{b_i}{2\pi}, \qquad \alpha_i^{-1}(\mu) = \alpha_i^{-1}(M_Z) - \frac{b_i}{2\pi}\ln\frac{\mu}{M_Z}, \tag{35.102}$$

where $\alpha_i = g_i^2/4\pi$ and b_i is fixed by the matter charged under group i. Collecting the gauge, fermion and scalar contributions with the group-theory indices of Chapter B, and writing the result for left-handed Weyl fermions and complex scalars,

$$b_i = -\frac{11}{3}\,C_2(G_i) + \frac{2}{3}\sum_{\text{Weyl }f} T(R_f^{(i)}) + \frac{1}{3}\sum_{\text{scalar }s} T(R_s^{(i)}), \tag{35.103}$$

with $C_2(G)$ the adjoint Casimir ($C_2 = N$ for SU(N), $C_2 = 0$ for U(1)) and $T(R)$ the Dynkin index ($T(\text{fund}) = \frac{1}{2}$ for SU(N); for U(1) in GUT normalization $T(R) \to \frac{3}{5}Y^2$). The signs deserve a word. The gauge term is negative — the gauge field's self-interaction *antiscreens*, the engine of asymptotic freedom from Eq. (27.124) — while matter screens. A Dirac fermion is two Weyl fermions, so $\frac{2}{3}\times 2 = \frac{4}{3}$ recovers the $-\frac{4}{3}T_F n_f$ of the QCD coefficient Eq. (27.162); Eq. (35.103) is the same physics, organized by the chiral content the unified theory cares about.

Worked Example 35.10 (The Standard Model coefficients $b = (\frac{41}{10}, -\frac{19}{6}, -7)$)**.** We evaluate Eq. (35.103) for the full Standard Model: three generations of Q, u^c, d^c, L, e^c and one complex Higgs doublet $H = (\mathbf{1},\mathbf{2})_{+1/2}$.

SU(3)$_c$ *($i = 3$).* $C_2 = 3$. Colour-charged Weyl fermions per generation: the doublet Q counts as two colour triplets (one per weak component), plus u^c and d^c, giving $2+1+1 = 4$ triplets, each $T = \frac{1}{2}$, so $\sum T = 4\cdot\frac{1}{2} = 2$ per generation, $\times 3 = 6$. The Higgs is colourless. Hence

$$b_3 = -\frac{11}{3}\cdot 3 + \frac{2}{3}\cdot 6 = -11 + 4 = -7. \tag{35.104}$$

SU(2)$_L$ *($i = 2$).* $C_2 = 2$. Weak doublets per generation: Q is a colour triplet of doublets (3 doublets) and L is one doublet, total 4, each $T = \frac{1}{2}$, so $\sum T = 2$

per generation, $\times 3 = 6$. The Higgs adds one complex doublet, $T = \frac{1}{2}$. Hence

$$b_2 = -\frac{11}{3}\cdot 2 + \frac{2}{3}\cdot 6 + \frac{1}{3}\cdot\frac{1}{2} = -\frac{22}{3} + 4 + \frac{1}{6} = -\frac{20}{6} + \frac{1}{6} = -\frac{19}{6}. \tag{35.105}$$

$\mathrm{U}(1)_Y$ *($i = 1$), GUT-normalized.* $C_2 = 0$; replace $T(R) \to \frac{3}{5}Y^2$. The fermion sum over one generation, using the hypercharges of Eq. (31.21),

$$\sum_f Y_f^2 = \underbrace{6\cdot\tfrac{1}{36}}_{Q} + \underbrace{3\cdot\tfrac{4}{9}}_{u^c} + \underbrace{3\cdot\tfrac{1}{9}}_{d^c} + \underbrace{2\cdot\tfrac{1}{4}}_{L} + \underbrace{1\cdot 1}_{e^c} = \tfrac{1}{6} + \tfrac{4}{3} + \tfrac{1}{3} + \tfrac{1}{2} + 1 = \frac{10}{3}, \tag{35.106}$$

times three generations gives 10; the Higgs contributes $\sum_s Y_s^2 = 2\cdot\frac{1}{4} = \frac{1}{2}$. Hence

$$b_1 = \frac{2}{3}\cdot\frac{3}{5}\cdot 10 + \frac{1}{3}\cdot\frac{3}{5}\cdot\frac{1}{2} = 4 + \frac{1}{10} = \frac{41}{10}. \tag{35.107}$$

The three coefficients are

$$b^{\mathrm{SM}} = \left(\tfrac{41}{10},\, -\tfrac{19}{6},\, -7\right). \tag{35.108}$$

Their signs already tell the story: $b_3 < 0$ and $b_2 < 0$ (asymptotically free, couplings shrink in the ultraviolet) while $b_1 > 0$ (hypercharge grows). Running up, α_3^{-1} and α_2^{-1} rise and α_1^{-1} falls — they move toward each other.

Worked Example 35.11 (The MSSM coefficients $b = (\frac{33}{5}, 1, -3)$)**.** Supersymmetry doubles the spectrum (Section 45.6.1): every chiral fermion acquires a complex-scalar partner (a chiral superfield carries both), every gauge field acquires a Weyl gaugino in the adjoint, and the single Higgs is replaced by two doublets H_u, H_d (each a chiral superfield). The bookkeeping simplifies dramatically. The gaugino, a Weyl fermion in the adjoint with $T(\mathrm{adj}) = C_2(G)$, contributes $\frac{2}{3}C_2(G)$, which combines with the gauge term to give $-\frac{11}{3}C_2 + \frac{2}{3}C_2 = -3C_2$. Each chiral superfield — one Weyl fermion plus one complex scalar in rep R — contributes $\frac{2}{3}T(R) + \frac{1}{3}T(R) = T(R)$. Thus Eq. (35.103) collapses to the clean supersymmetric form

$$b_i^{\mathrm{SUSY}} = -3\,C_2(G_i) + \sum_{\text{chiral}} T(R^{(i)}). \tag{35.109}$$

$\mathrm{SU}(3)$*:* $-3\cdot 3 + 6 = -9 + 6 = -3$ (matter content as before; Higgs colourless). $\mathrm{SU}(2)$*:* matter doublets give 6 as before, the two Higgs doublets add $2\cdot\frac{1}{2} = 1$, so $-3\cdot 2 + (6 + 1) = -6 + 7 = +1$. $\mathrm{U}(1)$*:* matter gives $\frac{3}{5}\cdot 10 = 6$ (three generations of Eq. (35.106)), the two Higgs doublets add $\frac{3}{5}\cdot(2\cdot\frac{1}{4} + 2\cdot\frac{1}{4}) = \frac{3}{5}\cdot 1 = \frac{3}{5}$, so $b_1 = 6 + \frac{3}{5} = \frac{33}{5}$. Hence

$$b^{\mathrm{MSSM}} = \left(\tfrac{33}{5},\, 1,\, -3\right). \tag{35.110}$$

Note b_2 has changed sign relative to the Standard Model: in the MSSM the weak coupling is no longer asymptotically free, because the extra higgsinos, gauginos and squarks over-screen. This single sign flip is what cures the unification.

The numerical test. We need the three couplings at $M_Z = 91.19\,\text{GeV}$ in the common GUT normalization. The electroweak inputs are $\alpha_{\text{em}}^{-1}(M_Z) = 127.9$, $\sin^2\theta_{\text{W}}(M_Z) = 0.231$, and $\alpha_3(M_Z) = \alpha_s(M_Z) = 0.118$, the last being the strong coupling whose running we computed in Chapter 27. Converting with $e = g_2 \sin\theta_W = g_Y \cos\theta_W$ and the normalization Eq. (35.100),

$$\alpha_2^{-1}(M_Z) = \frac{\sin^2\theta_{\text{W}}}{\alpha_{\text{em}}} = 0.231 \times 127.9 = 29.6, \tag{35.111}$$

$$\alpha_1^{-1}(M_Z) = \frac{3}{5}\,\alpha_Y^{-1} = \frac{3}{5}\,\frac{\cos^2\theta_{\text{W}}}{\alpha_{\text{em}}} = \frac{3}{5} \times 0.769 \times 127.9 = 59.0, \tag{35.112}$$

$$\alpha_3^{-1}(M_Z) = \frac{1}{0.118} = 8.47. \tag{35.113}$$

Worked Example 35.12 (Where the couplings meet: Standard Model versus MSSM)**.** Write $t = \ln(\mu/M_Z)$ and insert the coefficients into Eq. (35.102), using $b_i/2\pi$: for the Standard Model $(b_1, b_2, b_3)/2\pi = (0.6525, -0.5040, -1.1141)$, so

$$\alpha_1^{-1} = 59.0 - 0.6525\,t, \quad \alpha_2^{-1} = 29.6 + 0.5040\,t, \quad \alpha_3^{-1} = 8.47 + 1.1141\,t. \tag{35.114}$$

Two lines i, j cross at $t_{ij} = (\alpha_i^{-1} - \alpha_j^{-1})_{M_Z}/[(b_i - b_j)/2\pi]$. The three pairwise crossings are

$$\begin{aligned} &(1,2): \quad t = 25.4, \quad \mu \approx 1.0 \times 10^{13}\,\text{GeV}, \quad \alpha^{-1} = 42.4, \\ &(1,3): \quad t = 28.6, \quad \mu \approx 2.4 \times 10^{14}\,\text{GeV}, \quad \alpha^{-1} = 40.3, \\ &(2,3): \quad t = 34.6, \quad \mu \approx 1.0 \times 10^{17}\,\text{GeV}, \quad \alpha^{-1} = 47.1. \end{aligned} \tag{35.115}$$

The three crossings span four decades in energy: *the couplings do not meet at a point.* They come tantalizingly close — all three pass within a factor of a few of $\alpha^{-1} \approx 40$ over the range 10^{13}–$10^{17}\,\text{GeV}$, which is why unification was taken seriously at all — but minimal non-supersymmetric unification, on the one-loop running, fails by an amount far larger than the two-loop and threshold corrections it neglects.

Now the MSSM, with $(b_1, b_2, b_3)/2\pi = (1.0504, 0.1592, -0.4775)$:

$$\alpha_1^{-1} = 59.0 - 1.0504\,t, \quad \alpha_2^{-1} = 29.6 - 0.1592\,t, \quad \alpha_3^{-1} = 8.47 + 0.4775\,t. \tag{35.116}$$

The three crossings are now

$$\begin{aligned} &(1,2): \quad t = 33.0, \quad \mu \approx 1.9 \times 10^{16}\,\text{GeV}, \quad \alpha^{-1} = 24.4, \\ &(1,3): \quad t = 33.1, \quad \mu \approx 2.1 \times 10^{16}\,\text{GeV}, \quad \alpha^{-1} = 24.3, \\ &(2,3): \quad t = 33.2, \quad \mu \approx 2.4 \times 10^{16}\,\text{GeV}, \quad \alpha^{-1} = 24.3. \end{aligned} \tag{35.117}$$

All three agree to within one percent in t: the lines meet at a single point,

$$M_{\text{GUT}} \approx 2 \times 10^{16}\,\text{GeV}, \qquad \alpha_{\text{GUT}} \approx \frac{1}{24} \approx 0.04. \tag{35.118}$$

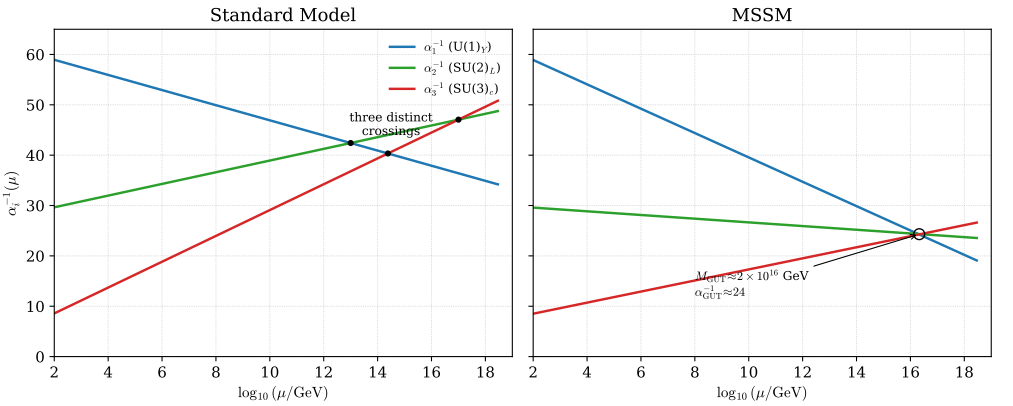


Figure 35.8: One-loop running of the three Standard Model gauge couplings in GUT normalization, from M_Z upward. Left: the Standard Model field content, with coefficients Eq. (35.108); the three lines cross pairwise at three distinct scales spanning 10^{13}–10^{17} GeV (Eq. (35.115)) and never meet at a point. Right: the MSSM field content, coefficients Eq. (35.110); the lines meet at a single scale $M_{\mathrm{GUT}} \approx 2 \times 10^{16}$ GeV with $\alpha_{\mathrm{GUT}}^{-1} \approx 24$ (Eq. (35.118)).

This is the celebrated result: with the superpartner spectrum near the TeV scale, the three Standard Model couplings unify. The contrast between Eq. (35.115) and Eq. (35.117) is shown in Fig. 35.8. We are obliged to be honest about what this does and does not prove. The MSSM has the freedom of a TeV-scale superpartner threshold and GUT-scale threshold corrections, and the one-loop unification is a genuine but not miraculous success once those are accounted for. As of the mid-2020s no superpartner has been found at the LHC, which pushes the relevant threshold upward and mildly degrades the fit. Coupling unification remains a striking numerical *hint*, not a proof of supersymmetry.

35.5.4 Proton decay: the prediction that kills minimal SU(5)

Whatever the precise unification scale, grand unification makes one prediction that is qualitatively unavoidable and quantitatively lethal: the proton decays. The X, Y gauge bosons of Eq. (35.86) carry both colour and lepton number — they couple quarks to leptons and quarks to antiquarks. Their tree-level exchange, once they are integrated out at the scale $M_X \sim M_{\mathrm{GUT}}$, generates B- and L-violating four-fermion operators. By the operator-product matching of Section 38.3, integrating out a heavy vector of mass M_X with coupling g_5 leaves behind a dimension-six contact operator with Wilson coefficient

$$\mathcal{L}_{\Delta B \neq 0} \;\sim\; \frac{g_5^2}{M_X^2}\,(qqql) \;\equiv\; \frac{c}{M_X^2}\,(qqql), \qquad c = O(g_5^2), \tag{35.119}$$

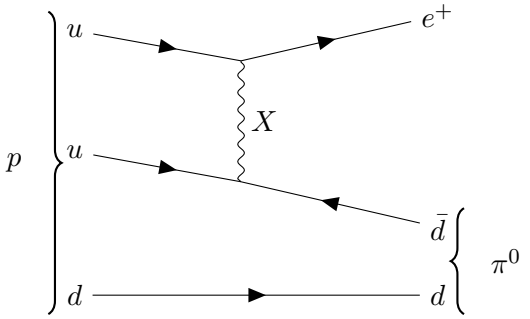


Figure 35.9: Proton decay $p \to e^+\pi^0$ at the parton level in SU(5). Two valence quarks of the proton exchange a superheavy X leptoquark gauge boson (mass $M_X \sim M_{\rm GUT}$); one quark line exits as a positron, the other as an antiquark $\bar{d}$ that combines with the spectator d to form the π^0. The process violates B and L by one unit each while conserving $B - L$. The exchanged X is the leptoquark of the adjoint decomposition Eq. (35.86).

a product of three quark fields and one lepton field, schematically. These are precisely the dimension-six operators of the SMEFT survey (Section 35.1.1): the same $(qqql)$ structures, but now with their coefficient *predicted* as g_5^2/M_X^2 rather than left as an unknown $1/\Lambda^2$. They conserve $B - L$ (the combination that is anomaly-free and exactly conserved in SU(5)) while violating B and L separately by one unit, so the characteristic decay is

$$p \;\to\; e^+\,\pi^0, \tag{35.120}$$

in which the baryon p ($B = 1$) turns into mesons and a positron ($B = 0$, $L = -1$), conserving $B - L = 1$. The parton-level mechanism is the diagram of Fig. 35.9: two of the proton's quarks exchange an X boson, one quark line emerging as a positron and the other as an antiquark that combines with the spectator quark into a pion.

The lifetime by dimensional analysis. The decay rate follows from Eq. (35.119) with almost no further input, because at energies of order m_p the only available scale is m_p itself. The amplitude is proportional to the Wilson coefficient c/M_X^2, so the rate, which has mass dimension $+1$, must be

$$\Gamma_p \;\sim\; \Big(\frac{c}{M_X^2}\Big)^2 m_p^5 \;\sim\; \alpha_{\rm GUT}^2\,\frac{m_p^5}{M_X^4}, \tag{35.121}$$

where $\alpha_{\rm GUT} = g_5^2/4\pi$ and we have written $c^2 \sim g_5^4 \sim (4\pi\alpha_{\rm GUT})^2$, dropping the 4π's into the $O(1)$ uncertainty of the hadronic matrix element. The fifth power of m_p is pure dimensional bookkeeping: $|c/M_X^2|^2$ carries dimension $[\text{mass}]^{-4}$, and the rate must make up the remaining four powers plus the one it carries itself, all

from the only scale in sight. Equivalently, m_p^5 is the phase-space-and-wavefunction scaling of a three-body-into-two decay through a contact operator.

Worked Example 35.13 (The numerical proton lifetime and the Super-Kamiokande bound)**.** Take a representative minimal non-supersymmetric SU(5) scale — generously high, to make the lifetime as long (and the exclusion as conservative) as possible: $M_X = 10^{15}\,\text{GeV}$ and $\alpha_{\text{GUT}} = \frac{1}{40}$. With $m_p = 0.938\,\text{GeV}$,

$$m_p^5 = (0.938)^5\,\text{GeV}^5 = 0.726\,\text{GeV}^5, \qquad M_X^4 = 10^{60}\,\text{GeV}^4, \qquad \alpha_{\text{GUT}}^2 = 6.25{\times}10^{-4}. \tag{35.122}$$

The rate Eq. (35.121) is then

$$\Gamma_p \sim \frac{6.25 \times 10^{-4} \times 0.726}{10^{60}}\,\text{GeV} = 4.5 \times 10^{-64}\,\text{GeV}. \tag{35.123}$$

The lifetime is the reciprocal, restored to seconds with $\hbar = 6.58 \times 10^{-25}\,\text{GeV s}$ and converted to years with $1\,\text{yr} = 3.16 \times 10^{7}\,\text{s}$:

$$\tau_p = \frac{1}{\Gamma_p} = \frac{6.58 \times 10^{-25}\,\text{GeV s}}{4.5 \times 10^{-64}\,\text{GeV}} = 1.5 \times 10^{39}\,\text{s} \approx 5 \times 10^{31}\,\text{yr}. \tag{35.124}$$

The actual one-loop crossings Eq. (35.115) put the relevant X mass nearer $10^{14.5}\,\text{GeV}$, which through the fourth power shortens this to $\sim 10^{30}\,\text{yr}$; either way the prediction sits in the window 10^{30}–$10^{32}\,\text{yr}$. The experimental bound from Super-Kamiokande, a water-Cherenkov detector watching $\sim 10^{34}$ protons for decades, is (as of the mid-2020s)

$$\tau(p \to e^+\pi^0) \;>\; 2.4 \times 10^{34}\,\text{yr}, \tag{35.125}$$

exceeding even the most generous minimal-SU(5) prediction by more than three orders of magnitude. *Minimal non-supersymmetric* SU(5) *is excluded.* It is excluded twice over, in fact: it already failed coupling unification in Eq. (35.115), and even granting it a unification scale by hand, it predicts a proton that decays far faster than the one we observe.

Physical Insight

The exclusion is a triumph of the method, not an embarrassment. A good theory makes sharp predictions and risks them. Minimal SU(5) predicted $\sin^2\theta_{\text{W}}$, predicted that the couplings unify, and predicted the proton lifetime to within the uncertainty of a hadronic matrix element — and the lifetime prediction was decisively tested and falsified. That is exactly how a structural hypothesis should behave. What survives is the *idea*: charge quantization is genuinely explained, the couplings really do nearly meet, and the failure of the minimal model sharpens rather than abandons the program. Pushing M_X up to the supersymmetric

2×10^{16} GeV of Eq. (35.118) lengthens the dimension-six lifetime by $(20)^4 \sim 10^5$, to $\sim 10^{36}$ yr, safely beyond Eq. (35.125). But supersymmetric unification has its own Achilles heel: colour-triplet higgsino exchange generates *dimension-five* B-violating operators (coefficient $\sim 1/M_X$, not $1/M_X^2$), driving the faster channel $p \to K^+\bar{\nu}$, which likewise confronts — and in its minimal version is excluded by — the Super-Kamiokande limit $\tau(p \to K^+\bar{\nu}) > 5.9 \times 10^{33}$ yr. The graveyard of minimal models is large.

35.5.5 SO(10): one generation in a single irreducible multiplet

There is a still more economical group. SO(10), of rank five, contains SU(5) (and indeed SU(5) × U(1)), and its smallest complex representation is the 16-dimensional spinor. Under the SU(5) subgroup this spinor decomposes as

$$\mathbf{16} \;=\; \mathbf{10} \;\oplus\; \overline{\mathbf{5}} \;\oplus\; \mathbf{1}, \tag{35.126}$$

the $\mathbf{10} \oplus \overline{\mathbf{5}}$ being exactly one SU(5) generation, and the lone SU(5) singlet $\mathbf{1}$ being a complete gauge singlet of the Standard Model: a right-handed neutrino N. *One full generation, plus the right-handed neutrino, fills a single irreducible representation of* SO(10). This is as unified as matter can get. Anomaly freedom is automatic — SO(10) has no non-trivial cubic Casimir, so every representation is anomaly-free without any cancellation to arrange.

The appearance of N is not a bonus to be admired and set aside; it is the keystone that ties this section back to the start of the chapter. The seesaw mechanism of Section 35.2 required a heavy Majorana mass M_R for exactly such a right-handed neutrino, with light-neutrino masses $m_\nu \sim m_D^2/M_R$ and the data pointing to $M_R \sim 10^{14}$ GeV. In SO(10) that mass is generated when the rank is reduced from five to four — when the $B - L$ symmetry contained in SO(10) but absent from SU(5) is spontaneously broken — and the natural scale of that breaking is the unification scale itself. The seesaw scale and the GUT scale are, in SO(10), two faces of one breaking. That is the cleanest reason to take grand unification seriously even after the minimal models have died: it does not merely reorganize the Standard Model's parameters, it *relates* two independent pieces of physics — the gauge quantum numbers and the neutrino masses — that the Standard Model treats as unrelated.

Historical Note

The sequence is instructive. Pati and Salam [386] first proposed treating lepton number as a fourth colour, SU(4) × SU(2) × SU(2), in 1973–74; Georgi and Glashow [217] found the truly simple SU(5) in 1974, and with Quinn and Weinberg [219] immediately turned the running couplings into the prediction of $M_{\rm GUT}$ and $\sin^2\theta_{\rm W}$; Fritzsch and Minkowski [201] intro-

duced SO(10) the following year. The supersymmetric revival came when Dimopoulos and Georgi [148] noticed that softly broken supersymmetry repairs unification, sharpened by the precision LEP measurements of [24].

Caution

Honest status, mid-2020s. Coupling unification is a genuine *hint*: the three couplings come close in the Standard Model and meet convincingly in the MSSM, but the MSSM result depends on a superpartner spectrum that the LHC has not found, and on threshold and two-loop corrections that the one-loop estimate Eq. (35.118) omits. Charge quantization is genuinely *explained* by the embedding — arguably the strongest single argument for unification, since it requires no new particles at any accessible scale. Proton decay is *unobserved*: every minimal model that predicted a lifetime near the experimental reach has been excluded, and the surviving models predict lifetimes at or just beyond current sensitivity. Grand unification is, as of this writing, an attractive and partly successful structural idea that remains unconfirmed — the very picture of a frontier where the mathematics is compelling and the data are not yet decisive.

The two ingredients this section has put on the table now feed directly forward. The X, Y leptoquark sector and its dimension-six B-violating operators Eq. (35.119) are exactly the kind of out-of-equilibrium, B-violating interaction that a theory of the cosmological baryon asymmetry must invoke; and the SO(10) right-handed neutrino Eq. (35.126), decaying out of equilibrium in the early universe, is the workhorse of leptogenesis and a candidate thread to dark matter. We carry both into the cosmological frontiers of the next section.

35.6 Cosmological Frontiers: Dark Matter and Baryogenesis

The preceding sections built their case from inside the laboratory. Neutrino mass (Section 35.2) announced itself as the unique dimension-five rung of the effective-operator tower, generated by a heavy lepton-number-violating singlet N; the strong-CP problem (Section 35.4) called forth an axion as the pseudo-Goldstone of a new spontaneously broken symmetry; grand unification (Section 35.5) raised a B-violating sector at $\Lambda_{\rm GUT}$ and, in its SO(10) guise, a right-handed neutrino as a matter multiplet. Each of these was introduced to answer a question internal to particle physics. We now turn to two facts that come from outside it—from the sky—and that the Standard Model, even granting it every parameter it possesses, provably cannot account for. The matter budget of the universe is dominated by a non-luminous, non-baryonic component whose abundance is measured to percent

precision, and for which the Standard Model offers no viable particle. And the universe contains matter but essentially no antimatter, an asymmetry of one part in a billion that the Standard Model's own dynamics are quantitatively far too feeble to have produced. These are not aesthetic discontents like the hierarchy or the flavor puzzle; they are empirical gaps, as hard as the neutrino mass that opened this chapter. The candidates and mechanisms assembled above—the WIMP, the axion, the sterile N, the GUT-scale B violation—reconvene here as the proposed answers. Our task is to make the two gaps quantitative, to derive the benchmarks against which any proposed answer is measured (the thermal relic abundance, Sakharov's three conditions), and to scope the mechanisms honestly as what they are: well-motivated, internally consistent, and not established.

35.6.1 Dark matter: the empirical case

The evidence for dark matter is gravitational, it is independent across vastly different length and time scales, and it is quantitative. We review the three cleanest lines, because the conclusion—that roughly a quarter of the cosmic energy density is non-baryonic matter—is the input to everything that follows.

Galactic rotation curves. A star (or a cloud of neutral hydrogen) on a circular orbit of radius r about the center of a galaxy obeys Newton's second law in the form

$$\frac{v^2(r)}{r} = \frac{G\,M(<r)}{r^2}, \qquad v(r) = \sqrt{\frac{G\,M(<r)}{r}}, \tag{35.127}$$

where $M(<r)$ is the mass enclosed within radius r and we have used that, by the shell theorem, only the enclosed mass contributes to the centripetal acceleration for an approximately spherical distribution. Beyond the visible edge of the luminous disk the enclosed luminous mass saturates, $M(<r) \to M_{\rm lum} = \text{const}$, so Eq. (35.127) predicts a Keplerian falloff $v(r) \propto r^{-1/2}$, exactly as the outer planets move more slowly than the inner ones. What is observed instead, in spiral galaxy after spiral galaxy, is that $v(r)$ flattens to a constant $v_\infty \sim 100$–300 km/s and stays flat far past the optical radius. A flat rotation curve forced through Eq. (35.127) requires

$$v(r) = \text{const} \quad \Longrightarrow \quad M(<r) = \frac{v_\infty^2}{G}\, r \;\propto\; r, \qquad \rho(r) = \frac{1}{4\pi r^2}\frac{dM}{dr} = \frac{v_\infty^2}{4\pi G\, r^2} \;\propto\; \frac{1}{r^2}, \tag{35.128}$$

an enclosed mass growing linearly with radius and a density falling only as $1/r^2$—the signature of an extended, roughly spherical *halo* whose mass continues to accumulate where there is no light. The halo outweighs the luminous disk by a factor of five to ten. No redistribution of the visible matter can produce Eq. (35.128): the light is in the disk, the mass is in the halo.

The acoustic peaks of the cosmic microwave background. Before recombination the universe was a hot plasma of photons, electrons, and baryons, tightly coupled by Thomson scattering, with dark matter (if present) gravitating but not scattering. Perturbations in this fluid oscillate: gravity compresses an overdensity, photon pressure resists and rebounds it, and the competition rings as a standing acoustic wave whose harmonics are frozen into the temperature map at recombination as a series of peaks in the angular power spectrum. The peak structure is a precision instrument for weighing the universe's components, and two features fix the dark-matter density essentially model-independently. First, the *baryons* load the oscillating fluid: their inertia deepens the compressions (gravity-aided) relative to the rarefactions (gravity-opposed), so the odd peaks (first, third, …, compression) are enhanced over the even peaks (rarefaction) by an amount that grows with the baryon density Ω_b. Second, the *total* nonrelativistic matter density—baryonic plus dark—controls how much the potential wells have decayed by recombination and hence the overall peak heights through the early integrated Sachs–Wolfe effect; the third peak in particular is sensitive to the dark-matter density because dark matter, unburdened by photon pressure, can begin to cluster earlier. The fit of the observed spectrum yields

$$\Omega_b h^2 \simeq 0.0224, \qquad \Omega_{\rm DM} h^2 \simeq 0.120, \qquad \Omega_{\rm DM} \simeq 0.26, \tag{35.129}$$

where $h \equiv H_0/(100\ \mathrm{km\,s^{-1}\,Mpc^{-1}}) \simeq 0.67$ is the reduced Hubble constant and $\Omega_i \equiv \rho_i/\rho_c$ measures each density against the critical density $\rho_c = 3H_0^2/8\pi G$. The decisive point is the *ratio*: the matter that gravitates is about five times the matter that can be made of protons and neutrons. The dark component is not merely dark—it is not baryonic.

Gravitational lensing and the separation of mass from gas. General relativity deflects light by the gravitational potential along its path, so the distorted images of background galaxies map the foreground mass distribution directly, with no assumption about its dynamical state or luminosity. In merging galaxy clusters this mapping delivers the cleanest argument of all. The ordinary baryonic matter of a cluster is overwhelmingly in hot X-ray-emitting gas, not in stars; when two clusters collide, the collisionless galaxies and the (presumably collisionless) dark matter pass through each other, while the two gas clouds—collisional—ram into each other, shock, and lag behind at the center. The lensing mass is then found to coincide with the galaxies and to be *spatially separated* from the bulk of the baryons in the shocked gas. Mass and baryons part company. This refutes the alternative that flat rotation curves and CMB peaks merely signal a modification of gravity rather than a new substance: there is a substance, it is where the gas is not, and it is gravitating.

Taken together the three lines triangulate a single conclusion. There exists a cosmologically abundant, non-luminous, non-baryonic, dissipationless, and at

most weakly self-interacting component of matter, with the density Eq. (35.129). The Standard Model must supply a particle to be it, or be declared incomplete.

35.6.2 The Standard Model has no candidate

The requirements are sharp. A dark-matter particle must be electrically neutral (else it would shine and bind into atoms), color neutral (else it would form anomalous heavy isotopes long since searched for), stable on cosmological timescales or nearly so, and—this is the constraint that kills the only Standard Model option—*cold*, meaning non-relativistic early enough to seed the gravitational growth of structure. Of all the Standard Model particles only one is neutral, only feebly interacting, and a candidate even in principle: the neutrino. We show it fails on both counts that matter, abundance and temperature.

Abundance. Neutrinos decouple from the plasma while still relativistic, at $T \sim 1$ MeV, and their number density today is fixed by that decoupling. The relic neutrino number density per flavor (counting ν and $\bar{\nu}$) is $n_\nu = \frac{3}{11} n_\gamma$ for each generation—the factor $\frac{3}{4}$ from Fermi versus Bose statistics times the factor $\frac{4}{11}$ from the photon reheating at e^+e^- annihilation, both established in any cosmology text and not reproved here. Summing over the three Standard Model neutrinos, a short standard computation converts their number density into an energy density and gives

$$\Omega_\nu h^2 \;=\; \frac{\sum_i m_{\nu_i}}{93.1\ \text{eV}}. \tag{35.130}$$

The cosmological bound on the summed neutrino mass, itself extracted from the same CMB-plus-large-scale-structure data, is $\sum_i m_{\nu_i} \lesssim 0.12$ eV, and the oscillation data of Section 35.2 require $\sum_i m_{\nu_i} \gtrsim 0.06$ eV. Either way Eq. (35.130) gives $\Omega_\nu h^2 \lesssim 1.3 \times 10^{-3}$, two orders of magnitude below the 0.12 of Eq. (35.129). Standard Model neutrinos are a real but minor component of the dark universe; they are too light to be enough.

Temperature. Even setting abundance aside, the neutrino is the wrong *kind* of matter. A particle that is relativistic at decoupling free-streams enormous distances before it slows, erasing density perturbations on all scales smaller than that free-streaming length. Such *hot* dark matter suppresses precisely the small-scale structure—individual galaxies—that is observed to have formed first and abundantly. Cosmological structure forms hierarchically, small objects merging into large, which requires the dark matter to have been *cold*, clustering on small scales early. A light relativistic relic produces the opposite, top-down history and is excluded by the observed galaxy distribution. (A complementary phase-space argument, the Tremaine–Gunn bound, makes the same point locally: fermions obeying the exclusion principle cannot be packed into a galactic core below a

mass set by the phase-space density, and an eV-scale neutrino falls far short.) The Standard Model neutrino fails not by a tunable margin but structurally: it is light, and being light it is hot.

There is no other Standard Model option. Every other neutral particle is either unstable on cosmological scales (the Z, the Higgs, the neutron outside a nucleus) or is the photon, massless and relativistic. The empirical fact of cold dark matter is therefore a theorem-level statement that the Standard Model is incomplete. The candidates assembled in this chapter—a new weakly interacting massive particle (the WIMP, developed below), the axion of Section 35.4, the sterile neutrino of Section 35.2—are the leading proposals to fill the gap, and the WIMP in particular acquires its motivation from a quantitative near-coincidence we now derive.

35.6.3 The thermal relic: Boltzmann equation and freeze-out

Suppose the dark matter is a particle χ that was in thermal equilibrium with the Standard Model plasma in the hot early universe, kept there by an annihilation reaction $\chi\bar{\chi} \leftrightarrow f\bar{f}$ into lighter species f. As the universe expands and cools, two things happen: the equilibrium abundance of a massive species is Boltzmann-suppressed once the temperature drops below its mass, and the annihilation reaction, which requires two χ's to find each other, shuts off as the χ's dilute. The competition between these freezes a relic abundance whose value we can compute, and the answer depends on essentially one number—the annihilation cross section. This is the freeze-out, or "WIMP," mechanism, and it is the most predictive idea in dark-matter physics.

The Boltzmann equation. Let $n(t)$ be the number density of χ (we take χ to be its own antiparticle, a Majorana fermion, to avoid clutter; the Dirac case differs only by counting factors). Two effects change n: dilution by the expansion, and the annihilation/creation reaction. The covariant rate equation is

$$\dot{n} + 3Hn = -\,\langle\sigma v\rangle\,(n^2 - n_{\rm eq}^2), \tag{35.131}$$

the central equation of relic-abundance cosmology, which we now read term by term. The $3Hn$ on the left, with $H = \dot{a}/a$ the Hubble rate and a the scale factor, is the dilution: in the absence of reactions the comoving number na^3 is conserved, $\frac{1}{a^3}\frac{d(na^3)}{dt} = \dot{n} + 3Hn = 0$. On the right, $\langle\sigma v\rangle$ is the annihilation cross section times relative velocity, thermally averaged over the momentum distributions; the $-\langle\sigma v\rangle\, n^2$ term is the loss from $\chi\chi \to f\bar{f}$ (rate $\propto n^2$ because two χ's must meet), and the $+\langle\sigma v\rangle\, n_{\rm eq}^2$ is the gain from the inverse reaction $f\bar{f} \to \chi\chi$. The combination $n_{\rm eq}^2$ rather than n_f^2 appears because in equilibrium detailed balance requires the forward and backward rates to cancel, $\langle\sigma v\rangle\, n_{\rm eq}^2$ being the equilibrium production rate; this guarantees the right-hand side vanishes when $n = n_{\rm eq}$, as it

must. The equilibrium density of a non-relativistic species at temperature T is the Maxwell–Boltzmann result

$$n_{\rm eq}(T) = g_\chi \Big(\frac{mT}{2\pi}\Big)^{3/2} e^{-m/T}, \tag{35.132}$$

with g_χ the internal degrees of freedom—the exponential being the suppression that drives the whole story.

Comoving variables. The expansion is a nuisance bookkeeping factor; we remove it by measuring abundance per unit entropy. The entropy density $s = \frac{2\pi^2}{45} g_{*s} T^3$ also dilutes as $s \propto a^{-3}$ (entropy per comoving volume is conserved, $\dot{s} + 3Hs = 0$), so the *yield*

$$Y \equiv \frac{n}{s} \tag{35.133}$$

is constant under pure expansion and changes only through reactions. Dividing Eq. (35.131) by s and using $\dot{s} + 3Hs = 0$,

$$\dot{Y} = \frac{\dot{n}}{s} - \frac{n\dot{s}}{s^2} = \frac{\dot{n} + 3Hn}{s} = -\frac{\langle\sigma v\rangle\, s}{1}\big(Y^2 - Y_{\rm eq}^2\big). \tag{35.134}$$

Finally we trade time for the dimensionless inverse temperature $x \equiv m/T$, the natural clock for a species of mass m. In the radiation era $T \propto a^{-1}$, so $\dot{x}/x = -\dot{T}/T = H$ and $\frac{d}{dt} = Hx\frac{d}{dx}$. With $s \propto T^3 = m^3/x^3$ and the radiation-era Hubble rate

$$H = \sqrt{\frac{8\pi G}{3}\rho_{\rm rad}} = 1.66\,\sqrt{g_*}\,\frac{T^2}{M_{\rm Pl}} = 1.66\,\sqrt{g_*}\,\frac{m^2}{M_{\rm Pl}\,x^2}, \qquad M_{\rm Pl} = 1.22{\times}10^{19}\ \text{GeV}, \tag{35.135}$$

the rate equation becomes the standard one-variable form

$$\frac{dY}{dx} = -\frac{\lambda}{x^2}\big(Y^2 - Y_{\rm eq}^2\big), \qquad \lambda \equiv \frac{s(m)\,\langle\sigma v\rangle}{H(m)} = 0.264\,\frac{g_{*s}}{\sqrt{g_*}}\,M_{\rm Pl}\,m\,\langle\sigma v\rangle\,, \tag{35.136}$$

where $s(m)$ and $H(m)$ denote the entropy density and Hubble rate evaluated at $T = m$ (the powers of x from s/H combine into the explicit x^{-2}), and the numerical $0.264 = \frac{2\pi^2/45}{1.66}$. The single dimensionless constant λ is the annihilation rate at $T = m$ measured in Hubble units; a large λ means efficient annihilation. Figure 35.10 shows the solution.

Freeze-out condition and relic yield. The qualitative behavior is visible in the equation. At small x (high T) the right-hand side is enormous, $\lambda \gg 1$, and any departure of Y from $Y_{\rm eq}$ is instantly driven to zero: Y tracks $Y_{\rm eq}$. As x grows, $Y_{\rm eq} \propto x^{3/2}e^{-x}$ collapses exponentially, and so does the annihilation rate

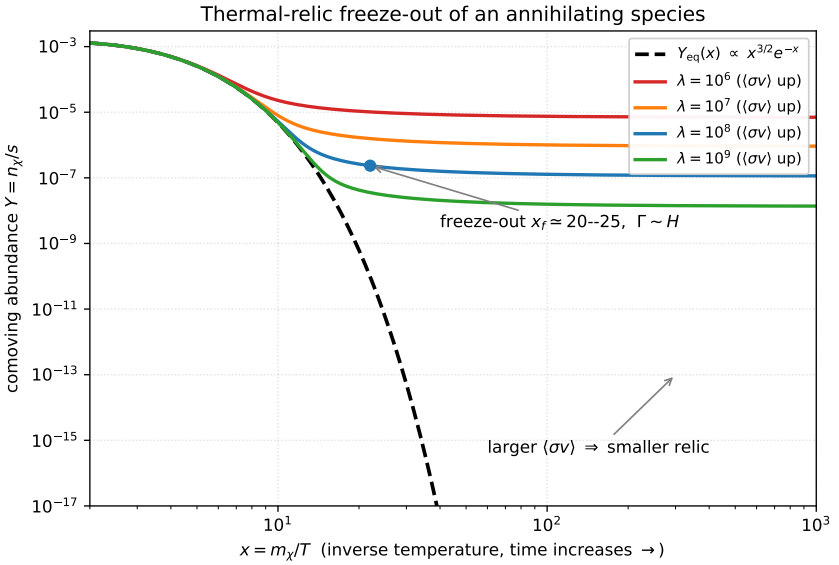


Figure 35.10: Numerical solution of the freeze-out Boltzmann equation Eq. (35.136) for a ladder of annihilation strengths λ. The actual abundance $Y(x)$ tracks the steeply falling equilibrium curve $Y_{\rm eq} \propto x^{3/2}e^{-x}$ until, near $x_f \simeq 20$–25, the annihilation rate drops below the Hubble rate ($\Gamma \sim H$) and the comoving number freezes to a constant relic value. A larger cross section (larger λ) keeps χ in equilibrium longer and leaves a smaller relic, $Y_\infty \propto 1/\langle\sigma v\rangle$—the inverse dependence that underlies the relic-density master formula.

per particle $\Gamma = n\langle\sigma v\rangle$. *Freeze-out* is the moment the per-particle annihilation rate can no longer keep pace with the expansion:

$$\Gamma = n_{\rm eq}(x_f)\,\langle\sigma v\rangle \;\simeq\; H(x_f). \tag{35.137}$$

After this, the $Y_{\rm eq}^2$ source is negligible (it is exponentially small) and the surviving χ's are too dilute to find one another; Y "freezes" at a nearly constant relic value. Inserting Eq. (35.132) and Eq. (35.135) into Eq. (35.137) and solving the resulting transcendental equation iteratively gives a freeze-out point that depends only logarithmically on the inputs,

$$x_f = \ln\!\Big[0.038\,\frac{g_\chi}{\sqrt{g_*}}\,M_{\rm Pl}\,m\,\langle\sigma v\rangle\Big] - \tfrac{1}{2}\ln\ln[\cdots] \;\simeq\; 20\text{–}25, \tag{35.138}$$

for the weak-scale cross sections of interest—a robust number, essentially $\ln(M_{\rm Pl} m\,\langle\sigma v\rangle)$ up to slowly varying corrections. The relic yield follows by integrating Eq. (35.136) from freeze-out to today with $Y_{\rm eq}$ dropped:

$$\frac{dY}{dx} = -\frac{\lambda}{x^2}Y^2 \quad\Longrightarrow\quad \frac{1}{Y_\infty} - \frac{1}{Y_f} = \lambda\int_{x_f}^{\infty}\frac{dx}{x^2} = \frac{\lambda}{x_f}. \tag{35.139}$$

Since the late yield is far below the value at freeze-out, $Y_\infty \ll Y_f$, the $1/Y_f$ is negligible and

$$\boxed{Y_\infty \simeq \frac{x_f}{\lambda} = \frac{x_f}{0.264\,(g_{*s}/\sqrt{g_*})\,M_{\rm Pl}\,m\,\langle\sigma v\rangle}} \tag{35.140}$$

The relic is inversely proportional to the annihilation cross section: weaker annihilators leave more behind, exactly as Fig. 35.10 displays. This is the single most important qualitative fact about thermal relics.

The relic-density master formula. The present mass density is $\rho_\chi = m\,n_{\chi,0} = m\,s_0\,Y_\infty$, with $s_0 = 2891\ \mathrm{cm}^{-3}$ the entropy density today. Crucially the explicit mass cancels in $m\,Y_\infty$ (the m in ρ_χ against the m in λ), so the relic density depends on the annihilation cross section and almost nothing else. Dividing by the critical density $\rho_c/h^2 = 1.05\times10^{-5}\ \mathrm{GeV\,cm^{-3}}$ and assembling the numbers ($x_f \simeq 20$, $g_{*s} \simeq g_* \simeq 86$ at the relevant temperature, so $g_{*s}/\sqrt{g_*} \simeq 9.3$) gives the master relation

$$\Omega_\chi h^2 = \frac{m\,s_0\,Y_\infty}{\rho_c/h^2} \;\simeq\; \frac{1.07\times10^{9}\ \mathrm{GeV^{-1}}\,x_f}{\sqrt{g_*}\,M_{\rm Pl}\,\langle\sigma v\rangle} \;\simeq\; \frac{3\times10^{-27}\ \mathrm{cm^3\,s^{-1}}}{\langle\sigma v\rangle}, \tag{35.141}$$

where in the last step we converted the natural-units cross section to laboratory units using $1\ \mathrm{GeV^{-2}} = 1.17\times10^{-17}\ \mathrm{cm^3\,s^{-1}}$ (that is, $\hbar^2c^3$ restored), and the numerical coefficient evaluates to $\simeq 2$–3×10^{-27} depending on the precise x_f and g_*. Equation (35.141) is the benchmark every thermal dark-matter model is measured against. We now read off its astonishing consequence.

Worked Example 35.14 (The WIMP miracle: a weak-scale cross section is the right size)**.** We ask what annihilation cross section reproduces the observed $\Omega_{\rm DM}h^2 \simeq 0.12$, and then what physics would naturally supply it.

The required cross section. Setting the master formula Eq. (35.141) equal to the measured abundance,

$$\langle\sigma v\rangle_{\rm req} \simeq \frac{3\times10^{-27}\ \mathrm{cm^3\,s^{-1}}}{0.12} \simeq 2.5\times10^{-26}\ \mathrm{cm^3\,s^{-1}} \simeq 1\ \mathrm{pb}\cdot c. \tag{35.142}$$

The last equality is the point worth pausing on: $1\ \mathrm{pb} = 10^{-36}\ \mathrm{cm^2}$ times $c = 3\times10^{10}\ \mathrm{cm/s}$ is exactly $3\times10^{-26}\ \mathrm{cm^3/s}$. The required annihilation cross section is a picobarn—the characteristic scale of weak-interaction cross sections at energies of order 100 GeV.

What naturally gives a picobarn. A particle of mass m_χ annihilating through an interaction of coupling strength α (a generic gauge coupling) has, by dimensional analysis, a cross section

$$\langle\sigma v\rangle \sim \frac{\pi\alpha^2}{m_\chi^2}. \tag{35.143}$$

Take a weak-scale particle: $\alpha \simeq \alpha_W \simeq 0.03$ and $m_\chi \simeq 100$ GeV. Then $\pi\alpha^2/m_\chi^2 \simeq \pi(0.03)^2/(100\ \text{GeV})^2 \simeq 2.8\times10^{-7}\ \text{GeV}^{-2}$. Converting, $2.8\times10^{-7}\ \text{GeV}^{-2}\times1.17\times10^{-17}\ \text{cm}^3\,\text{s}^{-1}/\text{GeV}^{-2} \simeq 3\times10^{-24}\ \text{cm}^3\,\text{s}^{-1}$; pushing the mass up to a TeV lowers this by 100 to $\sim 3\times10^{-26}\ \text{cm}^3\,\text{s}^{-1}$. The estimate lands within one to two orders of magnitude of the required Eq. (35.142) across the weak-scale window, and a careful treatment with the correct number of annihilation channels and the velocity-dependence of $\langle\sigma v\rangle$ closes the gap.

The miracle, stated honestly. A new particle with a mass and couplings of the electroweak scale—a scale introduced into the theory for an entirely independent reason, the unitarization of WW scattering and the origin of mass (Chapter 31)—automatically annihilates with very nearly the cross section needed to leave the observed dark-matter density as a thermal relic. The early universe contained a thermal bath of any such particle, and freeze-out would have deposited it at $\Omega h^2 \sim 0.1$ without any tuning. This coincidence between the weak scale and the dark-matter scale, neither put in by hand, is the WIMP miracle, and it is the reason the weak scale has been the first place to look. It is a coincidence, not a proof; direct, indirect, and collider searches through the mid-2020s have excluded large regions of the simplest WIMP parameter space without a detection, which has sharpened rather than settled the question.

The thermal relic is one production mechanism among several, and it organizes the candidate list. The *WIMP* just analyzed is the prototype, a weak-scale freeze-out relic. The *axion* of Section 35.4 is dark matter of an entirely different origin—not a thermal relic but a coherent classical field, produced by the misalignment of its initial value from the minimum of its potential, and cold despite its tiny mass because it is born at rest. The *sterile neutrino* of Section 35.2, a keV-scale gauge singlet, is produced out of equilibrium by oscillation from the active neutrinos and sits at the warm boundary of the allowed window. Each is a concrete realization of the abstract requirement of Section 35.6.2; none is established.

35.6.4 The baryon asymmetry: the number to be explained

We turn to the second empirical gap. The observable universe is made of matter: stars, gas, and dust are baryons, not antibaryons, and the absence of the gamma-ray annihilation glow that matter–antimatter boundaries would produce shows there are no large antimatter domains out to the horizon. Yet the Standard Model treats matter and antimatter nearly symmetrically, and a symmetric universe would have annihilated almost completely, leaving a trace of nucleons in equilibrium with the radiation at a level some nine orders of magnitude below what is observed. The asymmetry is small but it is not zero, and its value is measured two independent ways that agree.

The quantity is the *baryon-to-photon ratio*

$$\eta \equiv \left.\frac{n_B - n_{\bar{B}}}{n_\gamma}\right|_{\text{today}} \simeq 6.1 \times 10^{-10}, \tag{35.144}$$

where $n_B - n_{\bar{B}}$ is the net baryon number density and n_γ the photon density. One determination comes from *Big Bang nucleosynthesis*: the primordial abundances of deuterium, helium, and lithium produced in the first minutes depend sensitively on η (more baryons means more efficient burning), and the measured light-element abundances fix $\eta \simeq 6 \times 10^{-10}$. The second comes from the *CMB acoustic peaks* of Eq. (35.129): the same baryon density $\Omega_b h^2 = 0.0224$ that sets the odd/even peak ratio translates, via $\eta = 2.74 \times 10^{-8}\,\Omega_b h^2$, into $\eta = 6.1 \times 10^{-10}$. Two measurements, separated by nine orders of magnitude in cosmic time and resting on entirely different physics, give the same number. It is often more convenient to normalize the net baryon number to the entropy density rather than the photon density, since the former is conserved; using $s/n_\gamma \simeq 7.04$ today,

$$Y_B \equiv \frac{n_B - n_{\bar{B}}}{s} = \frac{\eta}{7.04} \simeq 8.7 \times 10^{-11}. \tag{35.145}$$

The task of *baryogenesis* is to explain how a universe that may plausibly have begun baryon-symmetric ($Y_B = 0$) dynamically developed the tiny excess Eq. (35.145). That this is possible at all—that microphysics can manufacture a net baryon number—is constrained by three necessary conditions, which we now state and prove.

35.6.5 Sakharov's conditions

In 1967 Sakharov identified three conditions that any dynamical generation of a baryon asymmetry from symmetric initial data must satisfy. They are necessary, not sufficient; a model meeting all three may still produce too little. We state them and prove each is necessary, because the proofs are short, illuminating, and dictate the structure of every baryogenesis mechanism.

Theorem 35.15 (Sakharov's necessary conditions)**.** *A dynamical evolution beginning from a state with vanishing net baryon number and ending with $Y_B \neq 0$ requires all three of:*

1. ***Baryon-number violation:*** *the dynamics must contain interactions with $\Delta B \neq 0$.*
2. *C* ***and*** *CP* ***violation:*** *both the charge-conjugation symmetry C and the combined symmetry CP must be broken by the dynamics.*
3. ***Departure from thermal equilibrium:*** *the baryon-violating, CP-violating processes must occur out of thermal equilibrium.*

Proof. (1) Baryon-number violation. This is immediate but worth stating precisely. If every interaction conserves baryon number, then the baryon-number operator commutes with the Hamiltonian, $[\hat{B}, H] = 0$, so $\hat{B}$ is conserved under the full time evolution. A state with $\langle \hat{B} \rangle = 0$ at the initial time has $\langle \hat{B} \rangle = 0$ at all later times. No asymmetry can develop from symmetric initial data without an interaction that changes B.

(2) C and CP violation. Suppose the dynamics conserve C. The net baryon production rate is a sum over reactions X weighted by the baryon number each deposits,

$$\frac{d}{dt}\langle \hat{B} \rangle = \sum_X \Delta B_X \, [\Gamma(X) - \Gamma(\bar{X})], \tag{35.146}$$

where $\bar{X}$ is the C-conjugate reaction, which deposits $-\Delta B_X$. Charge conjugation maps every particle to its antiparticle and reverses all internal charges, including baryon number; if C is a symmetry of the dynamics it relates the rate of X to the rate of $\bar{X}$, $\Gamma(X) = \Gamma(\bar{X})$, so every term in Eq. (35.146) cancels and $d\langle \hat{B} \rangle / dt = 0$. Hence C must be violated. But C alone is not enough: the weak interactions violate C maximally, yet a residual symmetry can still enforce cancellation. The relevant residual symmetry is CP. Under CP a left-handed particle of a given momentum maps to a right-handed antiparticle, and one shows—summing over the spin and momentum configurations that the thermal rate integrates over—that CP invariance again forces $\Gamma(X) = \Gamma(\bar{X})$ for the *integrated* rates that enter Eq. (35.146), even when C alone does not. The cleanest way to see the necessity of CP is via the equilibrium argument of part (3) below, which uses CPT rather than CP; combined with the out-of-equilibrium requirement it shows that what must be broken is exactly CP. Either way, both C and CP must be violated for the rate Eq. (35.146) to be nonzero.

(3) Departure from equilibrium. This is the deepest of the three, and it follows from the exact CPT symmetry that every local, Lorentz-invariant quantum field theory possesses (Section 32.4). In thermal equilibrium at temperature $T = 1/\beta$ the expectation value of the baryon number is

$$\langle \hat{B} \rangle_{\mathrm{eq}} = \frac{1}{Z} \operatorname{Tr}(e^{-\beta H} \hat{B}), \qquad Z = \operatorname{Tr} e^{-\beta H}. \tag{35.147}$$

Let Θ be the antiunitary CPT operator. It commutes with the Hamiltonian, $\Theta H \Theta^{-1} = H$, so $\Theta e^{-\beta H} \Theta^{-1} = e^{-\beta H}$, and it reverses baryon number, $\Theta \hat{B} \Theta^{-1} = -\hat{B}$, because $\hat{B}$ is odd under C (and even under P and T, which act on spacetime, not on the internal charge). For an antiunitary operator and any trace-class operator $\mathcal{O}$ one has the identity $\operatorname{Tr}(\Theta \, \mathcal{O} \, \Theta^{-1}) = (\operatorname{Tr} \mathcal{O})^*$: writing $\Theta = U\mathsf{K}$ with U unitary and K complex conjugation in some basis, $\operatorname{Tr}(U\mathsf{K}\,\mathcal{O}\,\mathsf{K}^{-1}U^{-1}) = \operatorname{Tr}(U\,\mathcal{O}^*\,U^{-1}) = \operatorname{Tr}(\mathcal{O}^*) = (\operatorname{Tr}\mathcal{O})^*$. Apply this with $\mathcal{O} = e^{-\beta H}\hat{B}$:

$$(\operatorname{Tr} e^{-\beta H}\hat{B})^* = \operatorname{Tr}(\Theta\, e^{-\beta H}\hat{B}\,\Theta^{-1}) = \operatorname{Tr}(e^{-\beta H}(-\hat{B})) = -\operatorname{Tr}(e^{-\beta H}\hat{B}). \tag{35.148}$$

But $\mathrm{Tr}(e^{-\beta H}\hat{B})$ is real (H and $\hat{B}$ are Hermitian and $e^{-\beta H}$ is positive), so it equals its own complex conjugate; Eq. (35.148) then reads $\mathrm{Tr}(e^{-\beta H}\hat{B}) = -\,\mathrm{Tr}(e^{-\beta H}\hat{B})$, forcing

$$\langle \hat{B} \rangle_{\mathrm{eq}} = 0. \tag{35.149}$$

In thermal equilibrium the net baryon number vanishes identically, by CPT alone, regardless of how badly B, C, and CP are violated by the interactions. The physical content is transparent: equilibrium distributions depend only on energy, and CPT guarantees a particle and its antiparticle have equal masses and hence equal equilibrium abundances; any asymmetry created by a B-violating reaction is undone by its inverse, which CPT holds to the conjugate rate. To freeze in an asymmetry the universe must fall out of equilibrium—so that the inverse reactions cannot keep up—while the B- and CP-violating processes are still acting. In an expanding universe the departure is supplied for free: equilibrium is maintained only while reaction rates exceed the Hubble rate, $\Gamma \gtrsim H$, and is lost when they fall below it. This is the same $\Gamma \sim H$ competition that froze the dark-matter relic in Eq. (35.137); here it is what allows a baryon asymmetry to survive. □

The three conditions are a checklist. We apply it first to the Standard Model itself, then to the leading mechanism that goes beyond it.

35.6.6 The Standard Model scorecard

Remarkably, the Standard Model contains all three Sakharov ingredients in principle. It fails to produce the observed asymmetry not because an ingredient is absent but because two of them are quantitatively far too small. We take them in turn.

Baryon-number violation: sphalerons. The Standard Model violates baryon number, through the very anomaly that the accidental symmetries of Theorem 33.12 were shown to suffer. Recall the result, which we import rather than reprove. The baryon and lepton currents are each anomalous under $\mathrm{SU}(2)_L$: by the chiral-anomaly mechanism (Theorem 36.4, and the cancellation bookkeeping of Section 37.3) their divergence is proportional to the topological charge density of the weak gauge field,

$$\partial_\mu j^\mu_B = \partial_\mu j^\mu_L = \frac{N_g\, g^2}{32\pi^2}\, W^a_{\mu\nu} \widetilde{W}^{a\,\mu\nu}, \tag{35.150}$$

with $N_g = 3$ generations and $\widetilde{W}$ the dual field strength. The combination $B - L$ is anomaly-free and exactly conserved, while $B + L$ is anomalous. Integrated over a topologically nontrivial gauge configuration the right-hand side is quantized to the change in Chern–Simons number (Eq. (33.79)), yielding the selection rule

Eq. (33.57): each unit of weak topological charge changes B and L by $N_g = 3$ units each, conserving $B-L$ and changing $B+L$ by 6. At zero temperature this process is mediated by instantons and suppressed by $e^{-8\pi^2/g^2} \sim 10^{-160}$, utterly negligible. At temperatures above the electroweak scale, however, thermal fluctuations cross the energy barrier between topological sectors over the *sphaleron* saddle—a static, unstable field configuration sitting atop the barrier—and $B + L$ violation becomes *fast*, with rate per unit volume $\Gamma_{\rm sph} \sim \alpha_W^4 T^4$ exceeding the Hubble rate throughout the era 100 GeV $\lesssim T \lesssim 10^{12}$ GeV. The decisive consequence, already flagged in Section 33.4.3, is twofold: sphalerons erase any pre-existing $B + L$, but they exactly preserve $B - L$. The Standard Model thus meets Sakharov's first condition—and, more, any successful mechanism must store its asymmetry in $B - L$, the combination sphalerons cannot wash out.

C and CP violation: the CKM phase, and why it is too small. The Standard Model violates C maximally (the weak interaction is chiral) and violates CP through the single physical complex phase of the CKM matrix, quantified by the Jarlskog invariant $J \simeq 3 \times 10^{-5}$ (Theorem 32.20, Section 32.4). Sakharov's second condition is met in principle. But the magnitude is hopeless. CP violation in the quark sector requires the interference of all three generations and is proportional not only to J but to the product of the mass-squared differences that make the quarks distinguishable; the dimensionless measure of CP violation available at the electroweak temperature $T_c \sim 100$ GeV is

$$\delta_{\rm CP} \sim \frac{J\,(m_t^2 - m_c^2)(m_t^2 - m_u^2)(m_c^2 - m_u^2)(m_b^2 - m_s^2)(m_b^2 - m_d^2)(m_s^2 - m_d^2)}{T_c^{12}} \sim 10^{-20}, \tag{35.151}$$

ten orders of magnitude below the observed $\eta \sim 10^{-10}$. The light-quark masses in the numerator are tiny in units of T_c, and they enter to high powers because the CP-violating effect must vanish whenever any two same-charge quarks become degenerate (then a phase rotation removes the phase). Standard Model CP violation, sufficient to account for kaon and B-meson decays, is quantitatively negligible for baryogenesis.

Departure from equilibrium: the electroweak phase transition, and why it is a crossover. The third Sakharov condition could in principle be supplied by the electroweak phase transition—the breaking of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ as the universe cools through $T_c \sim 100$ GeV. Were this transition strongly *first order*, it would proceed by the nucleation of bubbles of the broken (Higgs-condensate) phase expanding into the symmetric phase; the bubble walls are sharp departures from equilibrium, and CP-violating reflection of particles off the walls, combined with the fast sphalerons outside and their shut-off inside, could in principle generate a baryon asymmetry. This is the scenario of *electroweak baryogenesis*. Whether the transition is first order depends on the Higgs mass: a

light Higgs gives a strong first-order transition, a heavy one washes it out. Lattice studies establish that for a Higgs heavier than about 72 GeV the electroweak transition is not first order at all but a smooth *crossover*—no bubbles, no walls, no sharp departure from equilibrium. The measured Higgs mass, $m_h \simeq 125$ GeV (Chapter 34), sits well inside the crossover regime. The Standard Model therefore fails Sakharov's third condition outright: there is no out-of-equilibrium epoch at the electroweak scale.

The verdict is unambiguous. The Standard Model possesses all three Sakharov ingredients, which is itself remarkable, but it fails quantitatively on two of the three: its CP violation is smaller than required by ten orders of magnitude, and its electroweak transition is a crossover rather than a first-order transition. Standard Model electroweak baryogenesis does not work. The baryon asymmetry, like the dark-matter density, is an empirical fact that demands physics beyond the Standard Model.

35.6.7 Leptogenesis

The most economical mechanism that succeeds reuses machinery already on the table. The seesaw of Section 35.2 introduced heavy gauge-singlet Majorana neutrinos N_i to explain the smallness of the light-neutrino masses. These same states, decaying in the early universe, can generate a *lepton* asymmetry, which the sphalerons of Eq. (35.150) then partially reprocess into a baryon asymmetry. This is *leptogenesis*, and its appeal is that it ties the baryon asymmetry to the neutrino masses, two apparently unrelated facts, through one new sector. We check Sakharov's conditions, estimate the asymmetry, and state its status honestly.

The Sakharov conditions, realized. The relevant interactions are the type-I seesaw couplings of Section 35.2.3: the Yukawa $y_{i\alpha}\bar{N}_i\tilde{H}^\dagger L_\alpha$ and the Majorana mass $\frac{1}{2}M_i\overline{N_i^c}N_i$. (i) *Lepton-number violation* is supplied by the Majorana mass, which breaks the lepton number carried by N by two units—the same $\Delta L = 2$ that produced the Majorana light-neutrino mass; through sphalerons this L violation is communicated to B. (ii) *C and* CP *violation* reside in the complex Yukawa matrix $y_{i\alpha}$, whose phases cannot all be removed and which is, in general, unrelated to the CKM phase, so there is no reason for it to be small. (iii) *Departure from equilibrium* occurs when the heavy N_1 (the lightest, which dominates) decays at a temperature where its decay rate has fallen below the Hubble rate, $\Gamma_{N_1} \lesssim H(T \sim M_1)$, so that the inverse decays and washout reactions cannot erase the asymmetry as fast as the decays create it. All three conditions are met, and—unlike in the Standard Model—none is obviously too small.

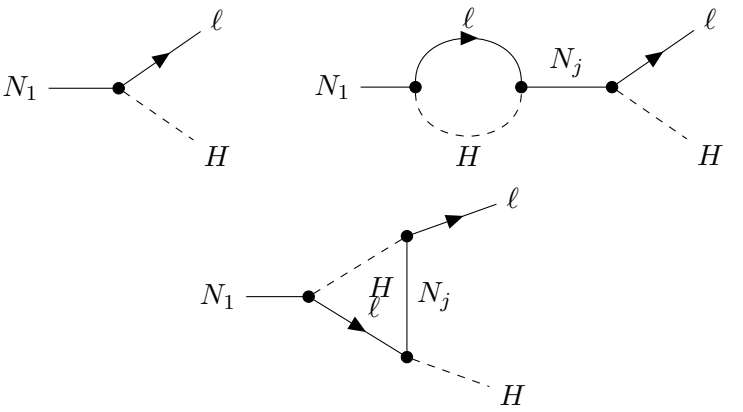


Figure 35.11: The decay $N_1 \to \ell H$ at tree level (left) and the one-loop self-energy (center) and vertex (right) corrections involving the heavier singlets N_j. The CP asymmetry Eq. (35.153) arises from the interference of the tree amplitude with the absorptive (on-shell) part of the loops: a rate difference between $N_1 \to \ell H$ and $N_1 \to \bar{\ell} H^*$ requires both the CP-violating Yukawa phase and the strong rescattering phase that only the loop provides.

The CP asymmetry in N decay. The heavy neutrino N_1 decays into a lepton plus a Higgs, $N_1 \to \ell\, H$, or into the conjugate channel $N_1 \to \bar{\ell}\, H^*$; because N_1 is its own antiparticle these are genuinely competing decays of a single state. A CP asymmetry between them,

$$\epsilon_1 \equiv \frac{\Gamma(N_1 \to \ell H) - \Gamma(N_1 \to \bar{\ell} H^*)}{\Gamma(N_1 \to \ell H) + \Gamma(N_1 \to \bar{\ell} H^*)}, \tag{35.152}$$

is nonzero only through the interference of the tree-level decay with its one-loop corrections (a self-energy and a vertex graph involving the heavier $N_{j\neq 1}$), shown in Fig. 35.11, because a CP-violating rate difference requires both a relative weak (complex Yukawa) phase *and* a relative strong (absorptive, on-shell rescattering) phase, and the absorptive phase is supplied by the loop. Carrying out the interference—a standard tree×loop computation of the kind done for the QED vertex in Chapter 19—gives

$$\epsilon_1 = \frac{1}{8\pi} \frac{1}{(y^\dagger y)_{11}} \sum_{j\neq 1} \mathrm{Im}\Big[\big((y^\dagger y)_{1j}\big)^2\Big]\, f\Big(\frac{M_j^2}{M_1^2}\Big), \tag{35.153}$$

where f is a loop function with the hierarchical limit $f(x) \to -3/(2\sqrt{x})$ for $M_j \gg M_1$. The imaginary part is the CP-violating phase of the Yukawa couplings; it is generically of order one.

From lepton asymmetry to baryon asymmetry: sphaleron reprocessing. The N_1 decays produce a net lepton number, and since $B = 0$ at this stage they

produce a net $B - L = -L$. The sphalerons of Eq. (35.150), fast throughout the relevant temperature range, then redistribute this conserved $B - L$ between B and L, settling into the chemical equilibrium they enforce. The fraction of $B - L$ that ends up as baryon number is a pure number determined by the relativistic particle content of the plasma, and we compute it.

Worked Example 35.16 (The sphaleron $B/(B - L)$ conversion factor)**.** We compute the equilibrium baryon number that sphalerons leave, given a fixed $B - L$, in the symmetric phase above the electroweak scale where all Standard Model species are relativistic. The method is the chemical-potential bookkeeping of Harvey and Turner. Assign to each particle species a chemical potential μ_i; for a relativistic species the number-density asymmetry is $n_i - \bar{n}_i \propto g_i\, c_i\, \mu_i$ to leading order in μ_i/T, with $c_i = 1$ for fermions and $c_i = 2$ for bosons. With the gauge (W-exchange) interactions in equilibrium the two members of each $\mathrm{SU}(2)_L$ doublet share one chemical potential; call it μ_q for the left-handed quark doublets and μ_ℓ for the left-handed lepton doublets, and let μ_u, μ_d, μ_e denote the right-handed singlets and μ_0 the Higgs. We take the $N_g = 3$ generations to share common potentials (fast intergenerational mixing) and one Higgs doublet.

Yukawa equilibrium relates the chiralities. Reading each interaction vertex $\bar{q}_L \tilde{H} u_R$, $\bar{q}_L H d_R$, $\bar{\ell}_L H e_R$ as a balance of chemical potentials (with the conjugate Higgs $\tilde{H}$ carrying $-\mu_0$) gives

$$\mu_u = \mu_q + \mu_0, \qquad \mu_d = \mu_q - \mu_0, \qquad \mu_e = \mu_\ell - \mu_0. \tag{35.154}$$

Sphaleron equilibrium imposes one constraint from the $\mathrm{SU}(2)_L$ topological transitions, which act on all left-handed doublets (three colors of quark doublet plus one lepton doublet per generation):

$$3\mu_q + \mu_\ell = 0 \quad \Longrightarrow \quad \mu_\ell = -3\mu_q. \tag{35.155}$$

Hypercharge neutrality of the universe is exact and supplies the last constraint. Summing $Y_i\, g_i\, c_i\, \mu_i$ over all species with the hypercharges of Chapter F ($Y_q = \frac{1}{6}$, $Y_u = \frac{2}{3}$, $Y_d = -\frac{1}{3}$, $Y_\ell = -\frac{1}{2}$, $Y_e = -1$, $Y_H = \frac{1}{2}$), counting 3 colors for quarks, 2 components for each doublet, $N_g = 3$ generations, and the boson factor 2 for the Higgs, and substituting Eqs. (35.154) and (35.155), gives after simplification

$$N_g\,(8\mu_q + 4\mu_0) + 2\mu_0 = 0 \quad \Longrightarrow \quad \mu_0 = -\frac{4N_g}{2N_g + 1}\,\mu_q, \tag{35.156}$$

i.e. $\mu_0 = -\frac{12}{7}\mu_q$ for $N_g = 3$.

Baryon and lepton number follow by summing the asymmetries with the appropriate B and L weights. Per generation the baryon number is $B = 2\mu_q + \mu_u + \mu_d = 4\mu_q$ (the Higgs potential cancels), and the lepton number is

$L = 2\mu_\ell + \mu_e = 3\mu_\ell - \mu_0$. Summing over N_g generations and using Eqs. (35.155) and (35.156),

$$B = 4N_g\,\mu_q, \qquad L = N_g\,(3\mu_\ell - \mu_0) = N_g\Big(-9\mu_q + \tfrac{12}{7}\mu_q\Big) = -\tfrac{153}{7}\,\mu_q \quad (N_g = 3). \tag{35.157}$$

Then $B = 12\mu_q$, $B - L = 12\mu_q + \frac{153}{7}\mu_q = \frac{237}{7}\mu_q$, and the conversion factor is

$$\boxed{\frac{B}{B-L} = \frac{8N_g + 4}{22N_g + 13} \xrightarrow{N_g=3} \frac{28}{79} \simeq 0.35} \tag{35.158}$$

where the general formula, obtained by carrying N_g and the number of Higgs doublets m symbolically, is $B/(B-L) = (8N_g + 4m)/(22N_g + 13m)$. Sphalerons convert a little over a third of any primordial $B - L$ into baryon number; the rest remains as lepton number.

The asymmetry, estimated. Assembling the pieces, the baryon yield produced by leptogenesis is

$$Y_B = \frac{28}{79}\,Y_{B-L} \;\simeq\; -\frac{28}{79}\,\frac{\epsilon_1\,\kappa}{g_*}, \tag{35.159}$$

where ϵ_1 is the CP asymmetry Eq. (35.153), $g_* \simeq 106.75$ counts the relativistic degrees of freedom that dilute the asymmetry, and $\kappa \le 1$ is an *efficiency factor* encoding the washout by inverse decays and $\Delta L = 2$ scatterings—it is unity for a maximally out-of-equilibrium decay and smaller when washout is active, determined by solving the leptogenesis Boltzmann equations (the same machinery as Eq. (35.131), applied to N_1 and the lepton asymmetry). Reproducing the observed $Y_B \simeq 8.7 \times 10^{-11}$ of Eq. (35.145) requires

$$\epsilon_1\,\kappa \;\sim\; 10^{-7}\text{–}10^{-6}. \tag{35.160}$$

The CP asymmetry Eq. (35.153) can be bounded in terms of the seesaw parameters: the Davidson–Ibarra bound, which follows from Eq. (35.153) together with the seesaw mass relation $m_\nu \sim y^2 v^2/M$, reads $|\epsilon_1| \lesssim \frac{3}{16\pi}\frac{M_1\,m_{\nu,\mathrm{atm}}}{v^2}$, where $m_{\nu,\mathrm{atm}} \simeq 0.05$ eV is the atmospheric neutrino mass scale and $v = 246$ GeV. Numerically $|\epsilon_1| \lesssim 10^{-6}\,(M_1/10^{10}\ \mathrm{GeV})$. Meeting Eq. (35.160) with $\kappa \sim \mathcal{O}(0.1)$ then requires

$$M_1 \gtrsim 10^9\ \mathrm{GeV}, \tag{35.161}$$

a lower bound on the lightest heavy-neutrino mass. This is a striking conclusion: *the observed baryon asymmetry, the smallness of the neutrino masses, and the scale of the heavy seesaw partners are linked by one mechanism*, and the link points to a seesaw scale within a few orders of magnitude of grand unification (Section 35.5)—consistent with the $\Lambda \sim 10^{14}$–10^{15} GeV that the neutrino mass itself suggested in Section 35.2.

Physical Insight — why leptogenesis is attractive but unproven

Leptogenesis is compelling because it is parsimonious: it adds no new sector beyond the one already required to give neutrinos mass, and it converts two unexplained numbers—the baryon asymmetry and the neutrino mass scale—into consequences of a single set of parameters, the heavy-neutrino Yukawas and masses. Sphalerons, which the Standard Model supplies for free, do the reprocessing; the only genuinely new physics is the heavy N, which the seesaw already demanded. That is the case for it. The case against treating it as established is equally clear-eyed. The decisive scale $M_1 \gtrsim 10^9$ GeV is far beyond any conceivable collider, so the heavy neutrinos cannot be produced and studied. The CP-violating phases in the Yukawa matrix y are not the same phases that appear in the low-energy PMNS matrix—there is no direct experimental handle linking laboratory CP violation in neutrino oscillations to the ϵ_1 that drives leptogenesis. And the efficiency κ depends on the full thermal history. What *can* be tested is the framework's foundation: leptogenesis through the type-I seesaw requires Majorana neutrinos, hence neutrinoless double beta decay (Section 35.2). A positive signal would establish lepton-number violation and the Majorana nature on which the mechanism rests; it would not prove leptogenesis, but its absence at the expected level would strain the simplest realizations. As of the mid-2020s the honest status is: plausible, internally consistent, tied to neutrino mass in an appealing way, and not established.

We have now made both cosmological gaps quantitative. The dark-matter density $\Omega_{\rm DM}h^2 \simeq 0.12$ admits a sharp thermal benchmark—the relic-abundance master formula Eq. (35.141) and its WIMP-miracle reading—and a candidate list (WIMP, axion, sterile neutrino) drawn from the new sectors of this chapter, none yet confirmed. The baryon asymmetry $\eta \simeq 6 \times 10^{-10}$ is governed by Sakharov's three necessary conditions, which the Standard Model meets in principle but fails quantitatively, and which leptogenesis can satisfy by reusing the seesaw and the sphalerons, plausibly but not provably. These are two established empirical gaps with well-scoped but speculative answers. The closing section gathers them, together with the structural puzzles and the one laboratory-confirmed entry (neutrino mass), onto a single dated summary of what is known and what is conjectured.

35.7 The Landscape of BSM: A Critical Assessment

Every topic of this chapter has now been placed on the table: the hierarchy problem as the instability of the one relevant operator of the Standard Model (Section 35.3); the strong-CP problem as the unexplained smallness of the one

marginal operator that is topological rather than dynamical (Section 35.4); neutrino mass as the unique dimension-five operator (Section 35.2); grand unification and proton decay as dimension-six structure near a high scale (Section 35.5); and dark matter and the baryon asymmetry as empirical demands the Standard Model cannot meet even in principle (Section 35.6). What remains is to see that these are not six separate stories but six readings of a single picture, and then to say plainly, dated to the mid-2020s, which parts of that picture are established and which are hope. This closing section assembles the inventory into one map and one summary, returning the chapter to the frame with which it opened: the Standard Model is the complete set of relevant and marginal gauge-invariant operators of its fields, and "beyond" it means precisely the tower of irrelevant operators above, plus the unexplained values of the couplings below. No new theorem is proved here; the chapter consolidates. The single piece of derivation-grade content is the logic by which a precision measurement becomes a bound on a new-physics scale, which we carry through explicitly because it is the quantitative spine of the entire enterprise.

35.7.1 One map: the operator tower against the scale axis

The reframing established at the opening of this chapter (Section 35.1), resting on the Wilsonian logic of Chapter 23 and the effective-field-theory paradigm of Chapter 38, is that the Standard Model Lagrangian is exactly

$$\mathcal{L}_{\rm SM} = \mathcal{L}_{\leq 4}, \tag{35.162}$$

the sum of all gauge-invariant operators of mass dimension ≤ 4 built from the Standard Model fields, with $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y = \mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and the chiral content fixed in Chapter 33. Everything new is then organized by the operator expansion of Eq. (38.3), which we recall in the Standard Model context as

$$\mathcal{L}_{\rm eff} = \mathcal{L}_{\rm SM} + \frac{1}{\Lambda} C_5 \mathcal{O}_5 + \frac{1}{\Lambda^2} \sum_i C_6^{(i)} \mathcal{O}_6^{(i)} + \frac{1}{\Lambda^3} \sum_i C_7^{(i)} \mathcal{O}_7^{(i)} + \cdots, \tag{35.163}$$

where each $\mathcal{O}_d$ is a local operator of mass dimension d invariant under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, Λ is the scale of whatever heavier physics has been integrated out, and the C_d are dimensionless Wilson coefficients. The dimensional analysis is the whole content: an operator of dimension d enters the Lagrangian (which has dimension 4) divided by Λ^{d-4}, so the higher up the tower an operator sits, the more strongly its physical effects are suppressed at energies $E \ll \Lambda$, by the power $(E/\Lambda)^{d-4}$. This single counting rule sorts every BSM phenomenon of the chapter into its place.

The picture sharpens when we plot it. Figure 35.12 draws the vertical energy axis logarithmically from the electroweak scale $v \simeq 246$ GeV (Chapter F) up to

the Planck scale $M_{\rm Pl} \simeq 1.2 \times 10^{19}$ GeV, and places each operator at the scale its phenomenology implicates. Read from the bottom:

- **The relevant operator: the Higgs mass term.** At the very bottom sits $\mu^2|H|^2$, the unique operator of dimension < 4 in the Standard Model, with $\mu^2 \sim -(88\ \text{GeV})^2 < 0$ triggering electroweak symmetry breaking. Being relevant, its coefficient grows toward the infrared; equivalently, it receives additive corrections quadratic in any heavier scale to which the Higgs couples. *This sensitivity is the hierarchy problem* (Section 35.3, Section 38.6.5): nothing in Eq. (35.162) protects the smallness of $|\mu^2|$ against the scales above it on the map.

- **The marginal topological operator: the θ term.** Also at the bottom, dimension exactly four but qualitatively unlike the rest, sits the QCD θ-term $\bar{\theta}\,\frac{g_s^2}{32\pi^2} G\widetilde{G}$. It is marginal but a total derivative perturbatively, with physical effects only non-perturbatively (Chapter 42); the neutron electric dipole moment forces $|\bar{\theta}| \lesssim 10^{-10}$ (Theorem 33.21). *This unexplained smallness is the strong-*CP *problem* (Section 35.4), whose leading resolution promotes $\bar{\theta}$ to a dynamical axion field with a scale f_a typically placed in the window 10^9–10^{12} GeV.

- **The dimension-five rung: neutrino mass.** One rung up sits the unique dimension-five gauge-invariant operator, the Weinberg operator $\mathcal{O}_5 = (\overline{L^c}\,\widetilde{H}^*)(\widetilde{H}^\dagger L)$ (Eq. (33.112)). After electroweak symmetry breaking it gives Majorana neutrino masses $m_\nu \sim C_5\, v^2/\Lambda$. Inserting $m_\nu \sim 0.05$ eV from oscillation data with $C_5 \sim 1$ places $\Lambda \sim v^2/m_\nu \sim 10^{14}$ GeV — the seesaw scale (Section 32.7.3, Section 35.2). *This is the one firmly established BSM rung*: an operator that is not in Eq. (35.162), whose coefficient is measured to be nonzero by the laboratory fact of neutrino oscillation.

- **The dimension-six rungs: proton decay and SMEFT.** Higher still sit the dimension-six operators. Two species live here. The baryon-and-lepton-violating four-fermion operators such as $QQQL$ (Theorem 33.22) mediate proton decay; the experimental lower bound on the proton lifetime, $\tau_p \gtrsim 10^{34}$ yr, pushes their suppression scale to $\Lambda \gtrsim 10^{15}$–10^{16} GeV, suggestively near the scale where the gauge couplings nearly meet (Section 35.5). The B- and L-conserving dimension-six operators of the Warsaw basis (Section 40.4.5) are the ones precision experiments at and below the electroweak scale actually constrain; their bounds are the subject of the next subsection.

- **The top of the map: gravity.** At $M_{\rm Pl}$ sits gravity itself. Treated as a quantum field theory, general relativity is the leading term of its own infinite operator tower, non-renormalizable in the sense of Section 38.2,

and is at best an effective description below $M_{\rm Pl}$ (Chapter 64). It is the one interaction not contained in $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ at all.

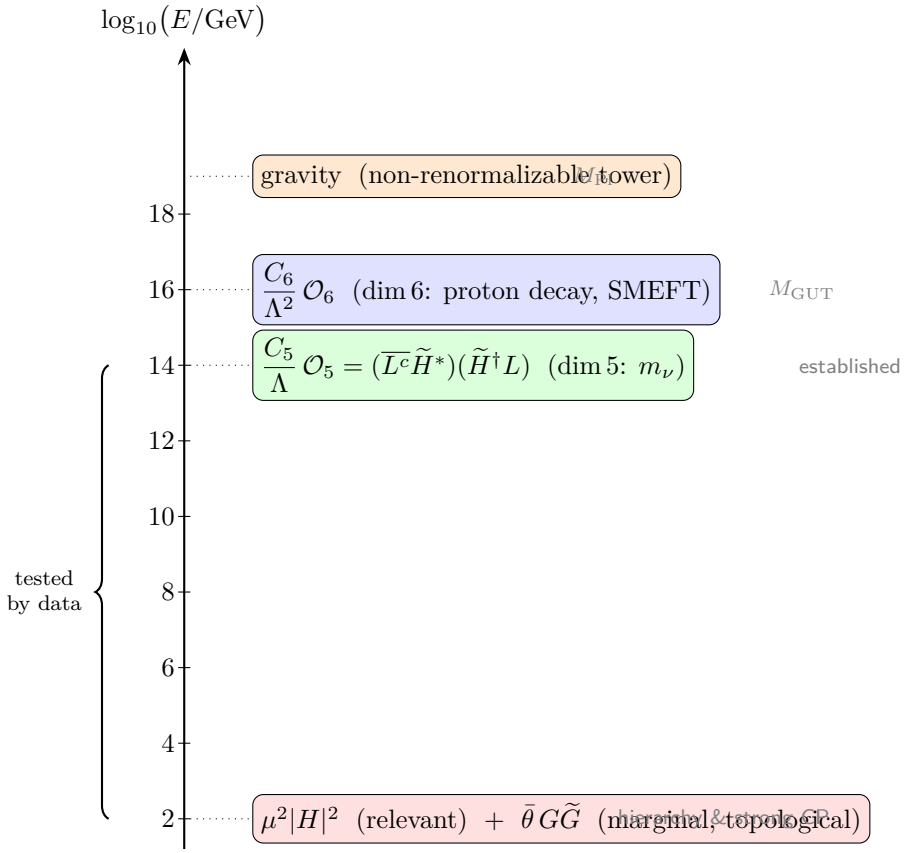


Figure 35.12: The BSM map: the operator tower of Eq. (35.163) against a logarithmic energy axis. At the bottom, the relevant Higgs mass term and the marginal topological θ-term (the hierarchy and strong-CP problems). One rung up, the unique dimension-five Weinberg operator (neutrino mass — the one established BSM rung), pointing to $\Lambda \sim 10^{14}$ GeV. Higher, the dimension-six operators (proton decay and SMEFT precision) near $M_{\rm GUT} \sim 10^{16}$ GeV. At the top, gravity at the Planck scale. Everything "beyond the Standard Model" is a point on this map.

The map makes one structural point that no list of models can: the lowest two rungs of the irrelevant tower are exactly neutrino mass (dimension five) and proton decay (dimension six), the two phenomena most sensitive to physics far above the electroweak scale. The Standard Model, read as Eq. (35.162), is announcing where its own boundary lies and which signals will probe it first.

35.7.2 SMEFT as the model-independent language

The map invites a model. It does not require one. The deeper lesson of the operator tower is that whatever the heavy physics is — supersymmetric partners, grand-unified gauge bosons, composite resonances, or something not yet imagined — its low-energy footprint is *the same kind of object*: a pattern of Wilson coefficients C_d multiplying the gauge-invariant operators built from Standard Model fields. This is the Standard Model Effective Field Theory (SMEFT). Its organizing data are fixed once and for all by the symmetries:

- **The basis.** At dimension six there is, after using equations of motion, integration by parts, and Fierz identities to eliminate redundancies, a finite and complete set of independent operators — 59 baryon-number-conserving operator types for one fermion generation, the Warsaw basis (Section 40.4.5). A new theory at scale Λ produces, by integrating out its heavy fields (matching, in the sense of Chapter 38), a definite vector of coefficients $\{C_6^{(i)}\}$ in this basis. Distinct ultraviolet theories generically produce distinct patterns; the pattern *is* the model's fingerprint at low energy.

- **The running.** The coefficients are not constants. Between the matching scale Λ and the measurement scale μ_{exp} they run under the SMEFT renormalization group (Section 40.6.4), mixing into one another along the way. A coefficient set to zero by a model at Λ can be generated radiatively below it. The data constrain the coefficients *at the measurement scale*; mapping those bounds back up to Λ requires the running, exactly as in any effective theory.

- **The fits.** Confronting Eq. (35.163) with the global body of electroweak, Higgs, flavor, and high-energy collider data turns the experimental programme into a single statement: a simultaneous bound on the directions in coefficient space. Where the data agree with the Standard Model — which, as of the mid-2020s and as catalogued in Chapter 34, is everywhere measured — the bound is a ceiling on the $C_d/\Lambda^{\,d-4}$, hence a floor under Λ for any given coefficient size.

The virtue of this language is that it is hypothesis-free. One does not assume supersymmetry to use it; one measures the coefficients and lets the pattern, if a nonzero one ever emerges, point toward the ultraviolet. The cost is that it is only an expansion: it is valid for $E \ll \Lambda$, and it cannot describe the heavy states themselves, only their tails. Within that domain it is the most economical and the least prejudiced way to say what "beyond the Standard Model" has, and has not, been seen.

The quantitative content of the whole programme is one estimate, which we now carry through in full because it is the bridge from a precision measurement to a scale.

Worked Example 35.17 (From a measured deviation to a bound on Λ)**.** We make explicit the inequality $\Lambda \gtrsim (\text{coupling})/\sqrt{\delta}$ asserted throughout the chapter: how a confirmed agreement with the Standard Model, at fractional precision δ, becomes a lower bound on the new-physics scale.

Setup. Take any observable $\mathcal{O}$ for which the Standard Model makes a prediction $\mathcal{O}_{\rm SM}$, and let a single dimension-six operator $\mathcal{O}_6$ with coefficient C_6/Λ^2 be the leading new contribution. By Eq. (35.163) the effective Lagrangian is $\mathcal{L}_{\rm SM} + (C_6/\Lambda^2)\mathcal{O}_6$, so the amplitude relevant to $\mathcal{O}$ is

$$\mathcal{M} = \mathcal{M}_{\rm SM} + \mathcal{M}_6, \qquad \mathcal{M}_6 = \frac{C_6}{\Lambda^2}\,\langle \mathcal{O}_6 \rangle, \tag{35.164}$$

where $\langle \mathcal{O}_6 \rangle$ is the matrix element of the operator between the external states. The new physics shows up, to leading order in $1/\Lambda^2$, through the interference term in the rate, so the fractional deviation of the observable is

$$\delta \;\equiv\; \frac{\mathcal{O} - \mathcal{O}_{\rm SM}}{\mathcal{O}_{\rm SM}} \;\simeq\; \frac{2\,\mathrm{Re}(\mathcal{M}_{\rm SM}^{*}\,\mathcal{M}_6)}{|\mathcal{M}_{\rm SM}|^2} \;=\; \frac{2\,\mathrm{Re}(\mathcal{M}_{\rm SM}^{*}\langle \mathcal{O}_6 \rangle)}{|\mathcal{M}_{\rm SM}|^2}\,\frac{C_6}{\Lambda^2}. \tag{35.165}$$

Everything now hinges on the size of the prefactor, which dimensional analysis fixes up to an order-one number.

Dimensional estimate of the prefactor. The matrix element of a dimension-six operator carries two more powers of energy than a renormalizable interaction; the extra $1/\Lambda^2$ must be compensated by a dimensionful scale in the numerator to keep δ dimensionless. Which scale appears depends on the kinematic regime, and the two regimes give the two faces of the bound.

(i) Low-energy precision observables. For an observable measured at energies $E \ll v$ (electroweak precision tests, low-energy flavor transitions), the only scale available is the electroweak vacuum value v. The Standard Model amplitude itself is set by v — a weak process scales as $\mathcal{M}_{\rm SM} \sim g^2/m_W^2 \sim 1/v^2$ in the four-fermion normalization — while $\mathcal{M}_6 \sim C_6/\Lambda^2$. The ratio is

$$\delta \;\sim\; \frac{1/\Lambda^2}{1/v^2}\,C_6 \;=\; C_6\,\frac{v^2}{\Lambda^2}. \tag{35.166}$$

(ii) High-energy tail observables. For a $2 \to 2$ process measured at center-of-mass energy $E \gg v$ (a collider tail), the operator matrix element grows, $\langle \mathcal{O}_6 \rangle \sim E^2$, while a marginal Standard Model amplitude is energy-independent, $\mathcal{M}_{\rm SM} \sim g^2 = O(1)$. Then

$$\delta \;\sim\; C_6\,\frac{E^2}{\Lambda^2}, \tag{35.167}$$

which is why the high-energy tails of distributions, where E/Λ is largest, are disproportionately powerful probes of heavy physics: the very feature that makes the EFT an expansion — growth with energy — is the experimental lever.

The bound. In both regimes write the controlling scale as $\mathcal{E}$ (equal to v in case (i), to E in case (ii)). Then $\delta \sim C_6\,\mathcal{E}^2/\Lambda^2$, and an experiment that confirms the Standard Model to precision $|\delta| < \delta_{\rm exp}$ (no deviation seen at the stated sensitivity) excludes any new physics that would have produced a larger shift. Solving for Λ,

$$\boxed{\Lambda \;\gtrsim\; \sqrt{C_6}\,\frac{\mathcal{E}}{\sqrt{\delta_{\rm exp}}}} \tag{35.168}$$

This is the inequality $\Lambda \gtrsim (\text{coupling})/\sqrt{\delta}$ in precise form: the reach in Λ scales as the controlling energy times the inverse square root of the experimental precision, weighted by the (assumed order-one) coupling strength $\sqrt{C_6}$ of the underlying physics. The square root is the dimension-six signature: because the operator enters as $1/\Lambda^2$, halving the deviation one can tolerate buys only a factor $\sqrt{2}$ in scale.

A number. Take a low-energy electroweak observable confirmed to the per-mille level, $\delta_{\rm exp} \sim 10^{-3}$, with $\mathcal{E} = v = 246$ GeV and a strongly coupled new sector, $C_6 \sim 1$. Then Eq. (35.168) gives

$$\Lambda \;\gtrsim\; \frac{246\ \text{GeV}}{\sqrt{10^{-3}}} \;\simeq\; 246\ \text{GeV} \times 31.6 \;\simeq\; 7.8\ \text{TeV}. \tag{35.169}$$

Per-mille agreement with the Standard Model thus excludes generic strongly coupled new physics below roughly 8 TeV. Two caveats are part of the honest statement. First, the bound weakens linearly in the coupling: if the new physics is weakly coupled, $C_6 \sim g_*^2/(16\pi^2)$ with $g_* \sim 1$, the reach drops by an order of magnitude or more, to the TeV range. Second, the bound assumes a single operator; in a real fit, cancellations among coefficients (flat directions) can relax individual bounds, which is why global fits over the whole Warsaw basis, not single-operator limits, are the rigorous statement (Section 40.4.5). With those caveats, Eq. (35.168) is the engine that turns the precision triumphs of Chapter 34 into the central experimental fact of the mid-2020s: new physics, if weakly or moderately coupled, lies above the few-TeV scale, and the operator tower above it is, so far, silent.

35.7.3 The Assessment of Current Status, dated to the mid-2020s

We now state without ornament what is known. The discipline that gives this section its title is to keep two columns rigorously separate: phenomena the Standard Model *provably cannot accommodate*, which are facts forcing physics beyond it, and mechanisms that are theoretically motivated but *experimentally unconfirmed*, which are hypotheses. Conflating the two — presenting a favored

model as though its motivation were evidence — is the characteristic failure this chapter is built to avoid.

Established empirically: the Standard Model is incomplete. Four facts cannot be fitted into Eq. (35.162) by any choice of its nineteen parameters (Section 33.6). They are not tunings or aesthetic discomforts; they are measurements the theory has no field or operator to reproduce.

1. **Neutrino mass.** Neutrino oscillation, observed across solar, atmospheric, reactor, and accelerator baselines, requires at least two nonzero neutrino mass-squared differences. The minimal Standard Model has exactly massless neutrinos (no right-handed neutrino, no Higgs coupling). Nonzero m_ν is therefore the single laboratory-established BSM fact, and it is precisely the coefficient of the dimension-five rung of Fig. 35.12. *Established:* $m_\nu \neq 0$. *Open: the absolute scale, the ordering, the Dirac-versus-Majorana question.*

2. **Dark matter.** Galactic rotation curves, cluster dynamics, gravitational lensing, the acoustic peaks of the cosmic microwave background, and the growth of large-scale structure consistently require a cold, non-baryonic, gravitationally clustering component with density $\Omega_{\rm DM}h^2 \simeq 0.12$, about five times the baryon density (Section 35.6). The Standard Model contains no particle with the required properties — this is an absence, not a tuning. *Established: the relic density. Open: the identity, mass, and couplings of the constituent, if it is a particle at all.*

3. **The baryon asymmetry.** The universe is matter, not antimatter, with $\eta_B = n_B/n_\gamma \simeq 6 \times 10^{-10}$. Generating it dynamically requires Sakharov's three conditions — baryon-number violation, C and CP violation, and a departure from thermal equilibrium (Section 35.6). The Standard Model supplies B violation only through sphalerons and CP violation only through the CKM phase, and both are quantitatively far too small. *Established: the asymmetry exists and the SM cannot produce it. Open: the mechanism.*

4. **Gravity.** The Standard Model is a theory of three interactions. Gravity is not in $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and quantized general relativity is nonrenormalizable, an effective theory below $M_{\rm Pl}$ (Chapter 64). *Established: gravity is real and absent from the SM. Open: its ultraviolet completion.*

These four are the floor of certainty. Whatever else is true, the Standard Model is not the final theory, because these measurements say so.

Structural puzzles: real, but not (yet) facts forcing new physics. Distinct from the empirical gaps are the features the Standard Model produces but does not explain: the hierarchy $|\mu^2| \ll M_{\rm Pl}^2$ (Section 35.3), the smallness of

$\bar{\theta}$ (Section 35.4), the pattern of fermion masses and mixings (the flavor puzzle), and charge quantization. Each is a place where the theory permits its observed numbers but cannot motivate them. These are powerful guides to where new physics might lie, but — unlike the four empirical gaps — a universe with a fine-tuned μ^2 and an inexplicably small $\bar{\theta}$ is not inconsistent with any measurement. They are puzzles, not refutations.

Speculative: motivated, unconfirmed as of 2025. The mechanisms this chapter has surveyed to address the items above are, every one of them, hypotheses without direct experimental confirmation.

- **Supersymmetry** (Chapter 44) would stabilize the hierarchy by relating the Higgs to fermionic partners, and incidentally improves gauge-coupling unification and supplies a dark-matter candidate. No superpartner has been observed; the simplest natural versions are excluded up to the TeV scale by LHC searches.
- **Grand unification** (Section 35.5) would explain charge quantization and the near-meeting of the couplings, and predicts proton decay. The proton has not been seen to decay; minimal SU(5) is excluded by the lifetime bound.
- **The axion** (Section 35.4, Chapter 42) would solve strong-CP dynamically and can be the dark matter. It has not been detected; searches constrain but do not close its parameter window.
- **Specific dark-matter and baryogenesis mechanisms** — thermal WIMPs, leptogenesis, electroweak baryogenesis, and the rest (Section 35.6) — are viable and predictive but unconfirmed; the WIMP parameter space, in particular, has been substantially compressed by direct-detection null results.

Each is a good idea for a real problem. None is established. The chapter advocates no choice among them.

Physical Insight

The naturalness question itself is now on the table. The deepest shift of the mid-2020s is meta-theoretical. For three decades the hierarchy problem was the dominant argument for new physics at the TeV scale: naturalness held that a relevant operator as small as μ^2, unprotected by any symmetry, should be stabilized by new states nearby, and supersymmetry or compositeness was expected at the LHC. The LHC has now explored the TeV scale and found the Higgs alone, with no accompanying states. This null result does not refute the Standard Model — it is a triumph for it

— but it does put pressure on naturalness as a *predictive principle*. Either the new states are just above current reach (and naturalness survives, marginally tuned), or the principle is not, after all, a reliable guide to nature, and the small μ^2 has an explanation of a different kind (anthropic selection in a landscape, ultraviolet/infrared mixing, relaxation dynamics) or no explanation we are entitled to demand. Honesty requires stating that this is unresolved: naturalness remains the sharpest theoretical motivation we have, and it has, so far, not been vindicated where it most confidently pointed. Whether it remains a trustworthy compass is itself an open question of the field, not a settled matter to be asserted either way.

35.7.4 The live frontiers that will move the summary

A summary dated to a year is only honest if it names what will change it. Five experimental programmes are positioned to convert entries from the "open" to the "settled" column, and each maps onto a specific rung of Fig. 35.12.

- **Neutrinoless double-beta decay.** A nucleus decaying as $(A, Z) \to (A, Z+2) + 2e^-$ with no neutrinos in the final state violates lepton number by two units and can proceed only if the neutrino is its own antiparticle. Its observation would establish the Majorana nature of the neutrino and hence the Majorana character of the dimension-five operator $\mathcal{O}_5$ — promoting the established fact $m_\nu \neq 0$ to a statement about the operator's ultraviolet origin. A null result at improving sensitivity progressively excludes the inverted-ordering parameter space.

- **Proton-decay searches.** Large underground detectors search for modes such as $p \to e^+\pi^0$ (the canonical grand-unified channel) and $p \to \bar{\nu}K^+$ (favored in supersymmetric unification). Any signal would be the first observation of baryon-number violation and direct evidence for the dimension-six rung near M_{GUT}; continued nulls raise the lifetime bound and the implied scale.

- **Direct and indirect dark-matter detection.** Direct searches look for nuclear recoils from halo dark matter scattering in ultra-low-background detectors; indirect searches look for annihilation or decay products (gamma rays, antiparticles, neutrinos) from astrophysical concentrations. Together with collider missing-energy searches they triangulate the dark-matter coupling to the Standard Model. The WIMP-nucleon cross-section reach now approaches the irreducible coherent-neutrino "floor," a milestone that will force the field to either find the WIMP or turn decisively to alternatives (axions, lighter or feebler candidates).

- **Electric dipole moment searches.** A permanent EDM of the electron, neutron, or an atom would signal CP violation beyond the CKM phase — exactly the new CP violation that viable baryogenesis requires (Sakharov's second condition). EDMs are also the observable that bounds $\bar{\theta}$ in the strong-CP problem. Current limits already exclude large regions of supersymmetric and other model parameter space; an observation would be a direct window onto the high-scale physics.

- **SMEFT precision fits.** The cumulative programme of Eq. (35.168): every improvement in electroweak, Higgs, flavor, and high-energy collider precision tightens the global fit over the Warsaw basis, raising the floor under Λ or, should a coefficient ever come out nonzero, delivering the pattern that points to the ultraviolet. This is the broadest and most model-independent of the five, and the one in which the EFT machinery of Chapter 40 is the working tool.

The structure of this list is itself the chapter's thesis: each frontier is an attempt to measure the coefficient of one rung of the operator tower, or to find the heavy state that generates it. The map and the experimental programme are the same object seen from two sides.

35.7.5 the three benchmark theories, drawn together

This synthesis chapter advances no earlier model to a new numbered stage; it sits in no thread's table. But it is the place where the three benchmark theories of the book, each developed to the edge of the Standard Model, are seen to be three faces of the present frontier — and it is worth saying so explicitly, because the convergence is not incidental.

ϕ^4 theory, the real scalar carried from Chapter 6 through its Wilsonian flow in Chapter 23 and its one-loop running in Chapter 24, taught the cleanest possible lesson about a relevant operator: a scalar mass-squared is the one coupling that grows in the infrared and is quadratically sensitive to the ultraviolet. The Higgs mass-squared μ^2 is that lesson made physical. The hierarchy problem (Section 35.3) is ϕ^4 theory's relevant coupling, promoted to the single most consequential number on Fig. 35.12.

Yang–Mills theory, the non-abelian gauge theory carried from Chapter 25 to the asymptotic freedom of Chapter 27, contributes twice. Its running gauge couplings, extrapolated upward, nearly meet near 10^{16} GeV — the empirical hint behind grand unification (Section 35.5). And its topological vacuum structure, the θ-term and the instantons that make it physical (Chapter 42), is the strong-CP problem (Section 35.4). The same theory that gave the book its deepest perturbative result, asymptotic freedom, gives the frontier two of its sharpest puzzles. QED, recovered in Chapter 33 as the unbroken electroweak

subgroup, enters the frontier as the precision arm: it is the per-mille agreement of the electron's anomalous moment and the electroweak observables that feeds Eq. (35.168) and floors the new-physics scale.

This closes Chapter 35, and with it the central narrative of the book's first nine parts: the Standard Model is assembled (Chapter 33), tested and triumphant (Chapter 34), and now mapped at its boundary. The chapter has named the open problems and the operators that encode them, but it has deliberately not built the machinery to attack them. That machinery is the content of the parts that follow. The effective-field-theory technology that makes Eq. (35.163) and the bound Eq. (35.168) rigorous is developed in Chapter 38 and applied to SMEFT in Chapter 40; the non-perturbative tools that make the θ-vacuum, instantons, and the axion precise are built in Chapter 42; and supersymmetry, the most studied of the speculative extensions, is constructed from its algebra in Chapter 44. The reader leaves this chapter with the map; the worked technologies behind every speculation on it are what comes next.

Further reading

The effective-field-theory reading of the Standard Model adopted throughout this chapter goes back to Weinberg's phenomenological-Lagrangian programme [498] and Georgi's textbook account [218]; the self-contained development this book relies on is Chapter 38. The unique dimension-five operator and the dimension-six B- and L-violating operators were first written down in [499], and the seesaw origin of Majorana neutrino mass in [201, 348]. The complete non-redundant dimension-six (Warsaw) basis is [240], with its renormalization-group running in [20], the operator counting systematized in [265], and a comprehensive review of the SMEFT programme and its global fits in [93]. Grand unification originates with [217], with coupling unification analysed in [219] and its supersymmetric form in [148]; the axion solution to strong CP is due to [388], with the pseudo-Goldstone axion of [497, 511] and its invisible realizations in [149, 294]. Sakharov's conditions for baryogenesis are [425], leptogenesis is [204], and the evidence, candidates, and constraints on dark matter are reviewed in [74]. The current experimental status of every entry in the chapter's summary—neutrino parameters, dark-matter limits, proton-lifetime and EDM bounds, and the electroweak/Higgs precision data feeding the SMEFT fits—is compiled and updated in the Review of Particle Physics [382], the authoritative source for the numbers quoted here as of the mid-2020s.

Part X

Anomalies

Chapter 36

Anomalies I: The Chiral Anomaly

A symmetry of a classical action need not survive its quantization. This is the single fact the present chapter establishes, in full and from several independent directions, and it is among the most consequential facts in quantum field theory. The classical Lagrangian of a massless Dirac fermion is invariant under two independent global U(1) symmetries — a vector phase rotation and an axial one — each carrying a Noether current that is conserved on shell by the argument of Theorem 7.10. There is no classical obstruction to keeping both. Yet when the theory is regulated and quantized, no admissible scheme can hold both currents conserved at once: the axial current acquires a divergence proportional to $F\widetilde{F}$, a fixed and computable number that survives the removal of every regulator. That residue is the chiral anomaly, and the burden of the chapter is to earn it honestly rather than quote it.

The development continues QED. We import the Dirac Lagrangian and its chiral projectors from Chapter 11, the gamma-matrix and γ^5 conventions of Appendix C, the abelian field strength of Eq. (8.44), and — as the structural keystone — the Ward–Takahashi identity of Theorem 19.13, which fixes that the gauged vector current must remain exactly conserved. It is precisely this non-negotiable demand on the vector current that forces the anomaly into the axial one: when a gauge-invariant regulator must choose which symmetry to break, it breaks the ungauged axial symmetry, and the breaking is not free to be tuned away.

The chapter proceeds in six movements. Section 36.1 builds the classical stage, deriving the two currents and the classical divergence $\partial_\mu j^{\mu 5} = 2im\,\bar{\psi}\gamma^5\psi$ that vanishes in the massless limit, so that whatever obstruction follows is unambiguously quantum. Section 36.2 computes the anomaly the original way — the linearly divergent axial-vector-vector triangle, its momentum-routing ambiguity, and the fixing of that ambiguity by vector-current conservation, yielding the Adler–Bell–Jackiw result of Theorem 36.4 and its one-loop exactness. Section 36.3 rederives the identical coefficient from the non-invariance of the path-integral

measure, Fujikawa's method, exhibiting the anomaly as the Jacobian of a chiral rotation evaluated by a heat kernel. Section 36.4 turns the abstraction into a measured number: through PCAC the anomaly fixes the rate of $\pi^0 \to \gamma\gamma$ and thereby counts the colors of QCD. Section 36.5 reveals why the coefficient is exact, recasting the integrated anomaly as a topological invariant tied by the Atiyah–Singer index theorem to fermion zero modes and gauge-field winding.

The closing Section 36.6 promotes the external charges to non-abelian generators and confronts the case where the anomalous current is itself gauged. There the anomaly is no longer a harmless prediction but a potential catastrophe: a nonzero gauge anomaly wrecks the Slavnov–Taylor identities, unitarity, and renormalizability. The chapter ends by stating the gauge-consistency condition $\mathcal{A}^{abc} = 0$, advancing Yang–Mills theory to its anomalies layer, and handing the explicit Standard-Model cancellation, the consistent-versus-covariant distinction, and 't Hooft anomaly matching to Chapter 37.

36.1 Classical chirality and the axial current

The chapter's thesis is that a perfectly good classical symmetry, equipped with a perfectly good Noether current, can fail to survive quantization — not because we were careless, but because no admissible regulator of the path integral can be made to respect it. Before that failure can be stated, let alone earned, we must build the classical stage on which it happens and sharpen the tension to its breaking point. The protagonist is the simplest fermionic theory in which the tension exists at all: a single massless Dirac field. We will find that this theory has *two* independent global U(1) symmetries, the vector and the axial, each classically exact and each carrying a Noether current that is conserved on shell. There is no classical obstruction to keeping both. Yet the two currents are not on equal footing once we remember why one of them matters: the vector current is the very object QED gauges, and the Ward–Takahashi identity of Theorem 19.13 — the statement that this gauge symmetry survives quantization — is the structural backbone of the renormalizable, unitary quantum theory of the photon. Any quantization scheme worthy of QED must keep the vector current exactly conserved. The question the rest of the chapter answers is whether it can do so *and* keep the axial current conserved too. The classical theory says yes; the quantum theory, we shall see, says no, and the failure is a fixed, computable number. This section establishes the "yes," precisely and completely, so that the "no" to come is unambiguously a quantum effect and not an artifact of classical sloppiness.

36.1.1 The massless Dirac Lagrangian and its two phase symmetries

We continue QED. The Dirac Lagrangian was fixed in Eq. (11.25) as $\mathcal{L} = \bar{\psi}(i\not{\partial} - m)\psi$; setting the mass to zero gives the object of this chapter,

$$\mathcal{L}_0 = \bar{\psi}\, i\not{\partial}\,\psi, \qquad \not{\partial} \equiv \gamma^\mu \partial_\mu, \tag{36.1}$$

with the gamma matrices in the conventions of Chapter 11 and Appendix C: $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, $\eta = \mathrm{diag}(+,-,-,-)$, $\bar{\psi} = \psi^\dagger\gamma^0$, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ Hermitian with $\gamma^{5\,2} = \mathbf{1}$ and the anticommutation $\{\gamma^\mu, \gamma^5\} = 0$ established once and for all in Section C.3.5. Throughout this chapter $\epsilon^{0123} = +1$ and the covariant derivative on the electron is $D_\mu = \partial_\mu + ieA_\mu$ with $e > 0$, exactly as fixed for QED; no gauge field is switched on in the present classical section, where we work with the free field and its global symmetries.

The first symmetry is the one already met in Eq. (11.36), the global *vector* U(1),

$$\mathrm{U}(1)_V: \qquad \psi \to e^{i\alpha}\psi, \qquad \bar{\psi} \to e^{-i\alpha}\bar{\psi}, \qquad \alpha \in \mathbb{R} \text{ constant}. \tag{36.2}$$

(The sign of α is pure convention; Eq. (11.36) took the opposite sign, which changes nothing about the current.) Every term of $\mathcal{L}_0$, and indeed of the massive $\mathcal{L}$, pairs one ψ with one $\bar{\psi}$, so the phases cancel and $\mathrm{U}(1)_V$ is a symmetry for *any* mass. This is the conservation of fermion number; gauged, it is electric charge.

The second symmetry exists only because the mass is zero. It is the global *axial* (or chiral) U(1),

$$\mathrm{U}(1)_A: \qquad \psi \to e^{i\beta\gamma^5}\psi, \qquad \beta \in \mathbb{R} \text{ constant}, \tag{36.3}$$

in which the rotation angle is multiplied by the chirality matrix γ^5, so left- and right-handed components rotate oppositely. The crucial novelty, and the source of everything that follows, is the way the Dirac adjoint transforms. Computing directly, and using that γ^5 is Hermitian,

$$\bar{\psi}' = \psi'^\dagger\gamma^0 = \psi^\dagger e^{-i\beta\gamma^{5\dagger}}\gamma^0 = \psi^\dagger e^{-i\beta\gamma^5}\gamma^0. \tag{36.4}$$

Now γ^5 anticommutes with γ^0, so $\gamma^5\gamma^0 = -\gamma^0\gamma^5$, and therefore any function $f(\gamma^5)$ obeys $f(\gamma^5)\gamma^0 = \gamma^0 f(-\gamma^5)$. Applying this to the exponential moves it through γ^0 at the cost of flipping the sign of its argument *again*:

$$e^{-i\beta\gamma^5}\gamma^0 = \gamma^0 e^{-i\beta(-\gamma^5)} = \gamma^0 e^{i\beta\gamma^5}, \tag{36.5}$$

so that

$$\mathrm{U}(1)_A: \qquad \bar{\psi} \to \bar{\psi}\, e^{i\beta\gamma^5}. \tag{36.6}$$

This is the decisive difference from the vector case. Under $\mathrm{U}(1)_V$ the adjoint rotates by the *opposite* phase, $e^{-i\alpha}$, and the phases cancel in any bilinear. Under

$\mathrm{U}(1)_A$ the adjoint rotates by the *same* factor $e^{i\beta\gamma^5}$ as the field — the two minus signs (one from Hermitian conjugation, one from anticommuting through γ^0) have cancelled. Whether a given bilinear $\bar\psi\,\Gamma\,\psi$ is invariant therefore depends on whether Γ commutes or anticommutes with γ^5.

The kinetic term survives. Each γ^μ anticommutes with γ^5, hence $\gamma^\mu e^{i\beta\gamma^5} = e^{-i\beta\gamma^5}\gamma^\mu$, and

$$\bar\psi'\, i\gamma^\mu\partial_\mu\psi' = \bar\psi\, e^{i\beta\gamma^5}\, i\gamma^\mu\, e^{i\beta\gamma^5}\,\partial_\mu\psi = \bar\psi\, i\, e^{i\beta\gamma^5}e^{-i\beta\gamma^5}\,\gamma^\mu\partial_\mu\psi = \bar\psi\, i\gamma^\mu\partial_\mu\psi. \quad (36.7)$$

The two exponentials, now adjacent, cancel. The massless Lagrangian Eq. (36.1) is invariant under $\mathrm{U}(1)_A$.

The mass term does not survive. Because $\mathbf{1}$ commutes with γ^5, the two factors of $e^{i\beta\gamma^5}$ *add* rather than cancel:

$$m\,\bar\psi'\psi' = m\,\bar\psi\, e^{i\beta\gamma^5}e^{i\beta\gamma^5}\,\psi = m\,\bar\psi\, e^{2i\beta\gamma^5}\,\psi = m\,\bar\psi\psi + 2im\beta\,\bar\psi\gamma^5\psi + O(\beta^2) \neq m\,\bar\psi\psi. \quad (36.8)$$

The mass term is not axially invariant; its infinitesimal variation is $2im\beta\,\bar\psi\gamma^5\psi$, proportional to the pseudoscalar bilinear. This is the precise classical statement that a Dirac mass explicitly breaks the axial symmetry, and it is the same fact seen in Chapter 11 as the mass term coupling left to right (Eq. (11.82) and the discussion there): a mass mixes the two chiralities, so a transformation that rotates them oppositely cannot leave it alone. For the remainder of the section we take $m = 0$, where both $\mathrm{U}(1)_V$ and $\mathrm{U}(1)_A$ are exact, and return to the broken-mass term only as a probe in Theorem 36.1.

36.1.2 Reformulation: independent rotations of the chiral halves

The vector/axial language obscures what is really going on, which is cleaner stated in terms of the chiral projectors of Eq. (11.82),

$$P_L = \tfrac{1}{2}(1-\gamma^5), \qquad P_R = \tfrac{1}{2}(1+\gamma^5), \qquad \psi_L = P_L\psi, \qquad \psi_R = P_R\psi, \quad (36.9)$$

which by Theorem C.28 are complete and orthogonal, $P_L + P_R = \mathbf{1}$, $P_{L,R}^2 = P_{L,R}$, $P_L P_R = 0$, and satisfy the intertwining $\gamma^\mu P_L = P_R\gamma^\mu$. The first thing to record is how the adjoint projects. Since γ^5 is Hermitian, $P_{L,R}^\dagger = P_{L,R}$, and because γ^5 anticommutes with γ^0 the projectors swap on crossing it, $P_L\gamma^0 = \gamma^0 P_R$. Hence

$$\overline{\psi_L} = (P_L\psi)^\dagger\gamma^0 = \psi^\dagger P_L\gamma^0 = \psi^\dagger\gamma^0 P_R = \bar\psi P_R, \qquad \text{and likewise} \quad \overline{\psi_R} = \bar\psi P_L. \quad (36.10)$$

A left-handed field has a *right*-projecting adjoint, and vice versa. With this, the massless kinetic term splits cleanly. Insert $\mathbf{1} = P_L + P_R$ to the right of γ^μ and use the intertwining $\gamma^\mu P_L = P_R\gamma^\mu$, $\gamma^\mu P_R = P_L\gamma^\mu$ together with $P_R^2 = P_R$, $P_L^2 = P_L$, $P_L P_R = 0$:

$$\bar\psi\, i\partial\!\!\!/\,\psi = \bar\psi\,(P_L + P_R)\, i\partial\!\!\!/\,(P_L + P_R)\,\psi$$

$$= \bar\psi P_L\, i\partial\!\!\!/\, P_L\psi + \bar\psi P_R\, i\partial\!\!\!/\, P_R\psi + \bar\psi P_L\, i\partial\!\!\!/\, P_R\psi + \bar\psi P_R\, i\partial\!\!\!/\, P_L\psi. \tag{36.11}$$

In the third term $\partial\!\!\!/ P_R = P_L\partial\!\!\!/$, so it carries $\bar\psi P_L P_L \partial\!\!\!/\psi$; in the first term $\partial\!\!\!/ P_L = P_R\partial\!\!\!/$, giving $\bar\psi P_L P_R \partial\!\!\!/\psi = 0$. Working through all four the same way, the "diagonal" pieces of Eq. (36.11) vanish and the "cross" pieces survive, which on relabelling via Eq. (36.10) is exactly the statement that the kinetic term is the sum of a purely left and a purely right piece:

$$\bar\psi\, i\partial\!\!\!/\,\psi = \overline{\psi_L}\, i\partial\!\!\!/\,\psi_L + \overline{\psi_R}\, i\partial\!\!\!/\,\psi_R. \tag{36.12}$$

The massless theory is two decoupled Weyl fermions. By contrast the mass term, by the same manipulation but now with the chirality-preserving identity matrix in the middle, mixes them:

$$\bar\psi\psi = \bar\psi(P_L + P_R)^2\psi = \bar\psi P_R\psi + \bar\psi P_L\psi = \overline{\psi_L}\,\psi_R + \overline{\psi_R}\,\psi_L. \tag{36.13}$$

Now the two symmetries are transparent. Because Eq. (36.12) is a sum of two independent terms, the massless Lagrangian is invariant under *independent* phase rotations of the two halves,

$$\mathrm{U}(1)_L:\ \psi_L \to e^{i\theta_L}\psi_L, \qquad \mathrm{U}(1)_R:\ \psi_R \to e^{i\theta_R}\psi_R, \tag{36.14}$$

a $\mathrm{U}(1)_L \times \mathrm{U}(1)_R$ symmetry. The vector and axial U(1)'s of Eqs. (36.2) and (36.3) are just the diagonal and anti-diagonal combinations of these: setting $\theta_L = \theta_R = \alpha$ rotates the full Dirac field by $e^{i\alpha}$ (vector), while $\gamma^5\psi_L = -\psi_L$ and $\gamma^5\psi_R = +\psi_R$ show that $e^{i\beta\gamma^5}$ acts as $e^{-i\beta}$ on ψ_L and $e^{+i\beta}$ on ψ_R (axial). The dictionary is

$$\theta_R = \alpha + \beta, \qquad \theta_L = \alpha - \beta, \qquad \Longleftrightarrow \qquad \alpha = \tfrac{1}{2}(\theta_R + \theta_L), \quad \beta = \tfrac{1}{2}(\theta_R - \theta_L). \tag{36.15}$$

"Vector" and "axial" are simply a change of basis in the two-torus $\mathrm{U}(1)_L \times \mathrm{U}(1)_R$. A mass term, Eq. (36.13), is invariant only under the diagonal $\theta_L = \theta_R$, i.e. only $\mathrm{U}(1)_V$; this is the same conclusion as Eq. (36.8) read in the new basis.

36.1.3 The two Noether currents

To each of these continuous symmetries Noether's first theorem (Theorem 7.10) assigns a conserved current, read off from the master formula Eq. (7.32). For an internal symmetry that leaves the spacetime point fixed, with infinitesimal field variation $\delta\psi$ and $\delta\bar\psi$, and a first-order Lagrangian, the current is

$$J^\mu = \frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)}\,\frac{\delta\psi}{\delta\lambda} + \frac{\partial\mathcal{L}}{\partial(\partial_\mu\bar\psi)}\,\frac{\delta\bar\psi}{\delta\lambda}, \tag{36.16}$$

where λ is the symmetry parameter. The Dirac Lagrangian is first order in $\partial_\mu\psi$ and contains no $\partial_\mu\bar\psi$, so only the first term contributes, with

$$\frac{\partial\mathcal{L}}{\partial(\partial_\mu\psi)} = \bar\psi\, i\gamma^\mu. \tag{36.17}$$

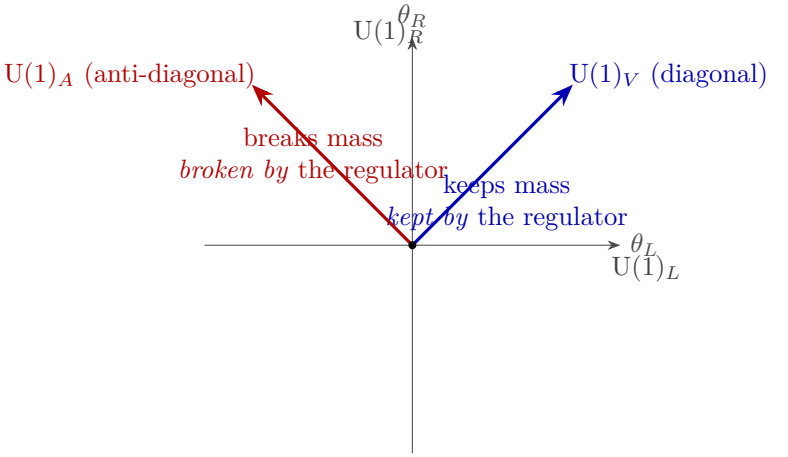


Figure 36.1: The classical symmetry of the massless Dirac fermion is the torus $\mathrm{U}(1)_L \times \mathrm{U}(1)_R$ of independent phase rotations of the two chiral halves Eq. (36.14). The vector symmetry $\mathrm{U}(1)_V$ is the diagonal $\theta_L = \theta_R$ and the axial $\mathrm{U}(1)_A$ the anti-diagonal $\theta_L = -\theta_R$, by the dictionary Eq. (36.15). Classically both directions are exact. A Dirac mass breaks the anti-diagonal only Eq. (36.13); the punchline of the chapter is that a *gauge-invariant regulator* acts on this plane in exactly the same way as a mass — it keeps the diagonal and breaks the anti-diagonal — even when $m = 0$.

For the vector symmetry Eq. (36.2) the infinitesimal variation is $\delta\psi = i\alpha\,\psi$, so $\delta\psi/\delta\alpha = i\psi$, and Eq. (36.16) gives $J^\mu = \bar\psi(i\gamma^\mu)(i\psi) = -\bar\psi\gamma^\mu\psi$. The overall sign and normalization of a Noether current are conventional — rescaling the symmetry parameter rescales the current and its charge, without touching the conservation law — and we adopt the standard normalization fixed in Eqs. (11.37) and (11.158),

$$\boxed{j^\mu = \bar\psi\gamma^\mu\psi}, \qquad \partial_\mu j^\mu = 0, \tag{36.18}$$

the *vector current*, conserved on shell for any mass (proved as Theorem 11.5). This is the current QED gauges: j^μ couples to the photon, and the survival of its conservation law in the quantum theory is the Ward–Takahashi identity Theorem 19.13.

For the axial symmetry Eq. (36.3) the infinitesimal variation is $\delta\psi = i\beta\gamma^5\psi$, so $\delta\psi/\delta\beta = i\gamma^5\psi$, and Eq. (36.16) gives $J^\mu = \bar\psi(i\gamma^\mu)(i\gamma^5\psi) = -\bar\psi\gamma^\mu\gamma^5\psi$. Adopting again the standard sign convention, the *axial current* is

$$\boxed{j^{\mu 5} = \bar\psi\gamma^\mu\gamma^5\psi}. \tag{36.19}$$

This is the axial vector bilinear catalogued in Theorem 11.6: it transforms as a four-vector under proper Lorentz transformations and picks up an extra sign

under parity, which is why it is called *axial*. In the chiral basis its meaning is transparent. Combining Eqs. (36.18) and (36.19) with the projectors,

$$j^\mu_R \equiv \overline{\psi_R}\gamma^\mu\psi_R = \bar\psi\gamma^\mu P_R\psi = \tfrac{1}{2}(j^\mu + j^{\mu 5}), \qquad j^\mu_L \equiv \overline{\psi_L}\gamma^\mu\psi_L = \bar\psi\gamma^\mu P_L\psi = \tfrac{1}{2}(j^\mu - j^{\mu 5}), \tag{36.20}$$

where we used Eq. (36.10) and the intertwining $\gamma^\mu P_R = P_L\gamma^\mu$ together with $P_L\gamma^\mu P_R = \gamma^\mu P_R$. Thus the vector current counts $n_R + n_L$ (total fermion number) and the axial current counts $n_R - n_L$ (the excess of right- over left-handed quanta): $j^\mu = j^\mu_R + j^\mu_L$ and $j^{\mu 5} = j^\mu_R - j^\mu_L$. These are precisely the Noether currents of the $\mathrm{U}(1)_R$ and $\mathrm{U}(1)_L$ rotations Eq. (36.14), recombined by the dictionary Eq. (36.15).

36.1.4 The classical divergence of the axial current

We now establish the central classical identity of the chapter — the one the next section will find corrected by quantization. We keep the mass nonzero deliberately: it is the cleanest probe of how badly the axial symmetry is broken, and the massless limit is read off at the end.

Theorem 36.1 (Classical divergence of the axial current)**.** *For a Dirac field of mass m obeying the equation of motion, the vector and axial currents satisfy*

$$\partial_\mu j^\mu = 0, \qquad \partial_\mu j^{\mu 5} = 2im\,\bar\psi\gamma^5\psi. \tag{36.21}$$

In particular, for $m = 0$ both currents are conserved, and the theory has the full classical $\mathrm{U}(1)_V \times \mathrm{U}(1)_A$ symmetry with no obstruction whatsoever.

Proof. The vector identity is Theorem 11.5; we reproduce its mechanism only to contrast it with the axial case. The two equations of motion are the Dirac equation and its adjoint (Eqs. (11.13) and (11.26)),

$$i\gamma^\mu\partial_\mu\psi = m\psi \quad\Longrightarrow\quad \gamma^\mu\partial_\mu\psi = -im\,\psi, \qquad i\,\partial_\mu\bar\psi\,\gamma^\mu = -m\bar\psi \quad\Longrightarrow\quad \partial_\mu\bar\psi\,\gamma^\mu = +im\,\bar\psi. \tag{36.22}$$

Differentiating $j^\mu = \bar\psi\gamma^\mu\psi$ and applying these,

$$\partial_\mu j^\mu = (\partial_\mu\bar\psi)\gamma^\mu\psi + \bar\psi\gamma^\mu(\partial_\mu\psi) = (+im\,\bar\psi)\psi + \bar\psi(-im\,\psi) = 0, \tag{36.23}$$

the two mass terms cancelling because the same matrix $\mathbf{1}$ sits between $\bar\psi$ and ψ in both.

The axial current carries the extra γ^5, and this single matrix changes the outcome. Differentiate $j^{\mu 5} = \bar\psi\gamma^\mu\gamma^5\psi$:

$$\partial_\mu j^{\mu 5} = (\partial_\mu\bar\psi)\gamma^\mu\gamma^5\psi + \bar\psi\gamma^\mu\gamma^5(\partial_\mu\psi). \tag{36.24}$$

In the first term the derivative-of-adjoint identity in Eq. (36.22) acts immediately, since γ^μ stands directly to the right of $\partial_\mu\bar\psi$:

$$(\partial_\mu\bar\psi)\gamma^\mu\gamma^5\psi = (+im\,\bar\psi)\gamma^5\psi = +im\,\bar\psi\gamma^5\psi. \tag{36.25}$$

In the second term the equation of motion is for $\gamma^\mu\partial_\mu\psi$, but γ^μ is separated from $\partial_\mu\psi$ by the matrix γ^5. We move γ^5 to the left through γ^μ, paying the anticommutator sign $\{\gamma^\mu,\gamma^5\}=0$, i.e. $\gamma^\mu\gamma^5=-\gamma^5\gamma^\mu$:

$$\bar\psi\gamma^\mu\gamma^5(\partial_\mu\psi)=-\bar\psi\gamma^5\gamma^\mu(\partial_\mu\psi)=-\bar\psi\gamma^5\,(-im\,\psi)=+im\,\bar\psi\gamma^5\psi, \tag{36.26}$$

where after the anticommutation the equation of motion $\gamma^\mu\partial_\mu\psi=-im\psi$ from Eq. (36.22) applies. Adding Eqs. (36.25) and (36.26),

$$\partial_\mu j^{\mu5}=im\,\bar\psi\gamma^5\psi+im\,\bar\psi\gamma^5\psi=2im\,\bar\psi\gamma^5\psi. \tag{36.27}$$

The two mass terms now *add* rather than cancel: the anticommutation of γ^5 through γ^μ supplied the relative sign flip that, in the vector case, had made them cancel. This is the algebraic fingerprint of the whole subject — one stray γ^5 converts a cancellation into a doubling. For $m=0$ the right-hand side vanishes and $j^{\mu5}$ is conserved, completing the proof. □

The infinitesimal variation of the mass term computed back in Eq. (36.8) was $\delta(m\bar\psi\psi)=2im\beta\,\bar\psi\gamma^5\psi$; Theorem 36.1 is the on-shell incarnation of exactly that statement, since by the general logic of Noether's theorem the divergence of a current equals the variation of the Lagrangian under the corresponding transformation. The two facts are one fact, seen off shell (variation of $\mathcal{L}$) and on shell (divergence of J^μ).

Worked Example 36.2 (The separate (non-)conservation of the chiral charges)**.** The chiral decomposition makes the physics of Theorem 36.1 vivid. From the linear combinations Eq. (36.20) and the two divergence laws Eq. (36.21),

$$\partial_\mu j^\mu_R=\tfrac12(\partial_\mu j^\mu+\partial_\mu j^{\mu5})=\tfrac12(0+2im\,\bar\psi\gamma^5\psi)=+\,im\,\bar\psi\gamma^5\psi, \tag{36.28}$$

$$\partial_\mu j^\mu_L=\tfrac12(\partial_\mu j^\mu-\partial_\mu j^{\mu5})=\tfrac12(0-2im\,\bar\psi\gamma^5\psi)=-\,im\,\bar\psi\gamma^5\psi. \tag{36.29}$$

The right-handed and left-handed fermion numbers are each conserved when $m=0$ and are violated, equally and oppositely, by the mass. Physically the mass term Eq. (36.13) is the only place left meets right, so it is the only operator that can convert a right-handed quantum into a left-handed one; Eq. (36.29) is the rate of that conversion. The *sum* $\partial_\mu(j^\mu_R+j^\mu_L)=\partial_\mu j^\mu=0$ always, because a mass shuffles quanta between the two chiralities without changing their total number. The lesson to carry into the quantum theory: the vector charge is the sum, robust because it is blind to the L/R distinction, while the axial charge is the difference, fragile because it is exactly that distinction. It is the fragile one that the regulator will break.

36.1.5 The naive Ward identities and the coming clash

A conservation law at the level of operators becomes, inside correlation functions, a set of *Ward identities*. Promoting Eq. (36.21) to insertions in the path integral

and using the equations of motion as operator relations (the same manipulation that produced the QED Ward–Takahashi identity in Theorem 19.13), the *naive* — that is, classical, pre-regularization — identities for the two currents read, schematically,

$$\partial_\mu \langle\, T\, j^\mu(x)\, X \,\rangle = (\text{contact terms}), \tag{36.30}$$

$$\partial_\mu \langle\, T\, j^{\mu 5}(x)\, X \,\rangle = \langle\, T\, 2im\,\bar\psi\gamma^5\psi(x)\, X \,\rangle + (\text{contact terms}), \tag{36.31}$$

where X is any product of fields and the contact terms are the δ-function pieces produced when the divergence acts at coincident points. For $m = 0$ the right-hand side of Eq. (36.31) vanishes away from contact, so the naive expectation is that the axial current is conserved inside correlators exactly as the vector current is. This expectation is what the next section overturns.

The reason it must be overturned can already be previewed, honestly and without yet computing anything. A Green's function like the one in Eq. (36.31) with two photons in X — the matrix element $\langle\gamma\gamma|j^{\mu 5}|0\rangle$ — is given at one loop by a fermion triangle that is linearly divergent. A divergent amplitude is not defined until a regulator is supplied, and *every* known regulator of a chiral fermion loop introduces a mass scale: a Pauli–Villars regulator field of mass M, a lattice spacing, a point-splitting distance, or the analytic continuation of dimensional regularization away from the integer dimension in which γ^5 is even definable. A regulator mass acts on the symmetry plane of Fig. 36.1 exactly as a physical mass does: by Eq. (36.8) it respects $\mathrm{U}(1)_V$ and breaks $\mathrm{U}(1)_A$. The crucial point — the one that makes this a genuine anomaly rather than a removable scheme artifact — is that the symmetry-breaking term in Eq. (36.21) is $\propto m$, while the triangle's divergence is $\propto m^0$; multiplying a vanishing classical breaking ($m \to 0$) by a divergent loop ($\propto 1/m$ in effect) leaves behind a finite, regulator-*independent* remainder. That remainder is the anomaly.

Physical Insight — why one current is sacred and the other expendable

The asymmetry between $\mathrm{U}(1)_V$ and $\mathrm{U}(1)_A$ is not a matter of taste. The vector current is the one QED gauges (QED): its conservation $\partial_\mu j^\mu = 0$ is welded to gauge invariance, and gauge invariance is in turn welded to the decoupling of the unphysical photon polarizations, hence to the unitarity and renormalizability of the quantum theory through the Ward–Takahashi identity Theorem 19.13. Surrender it and the theory of the photon collapses. The axial current, by contrast, is in QED an ungauged global symmetry; its conservation is a pleasant accident of the massless classical theory with no structural job to do. When quantization forces us to choose which of the two to keep — and Fig. 36.1 shows that a gauge-invariant regulator chooses for us — we keep the vector current

without hesitation and let the axial current break. Far from a defeat, the broken axial current becomes one of the most successful predictions in physics: it fixes the rate of $\pi^0 \to \gamma\gamma$ and counts the colors of QCD, as Chapter 36 will show. The same algebra generalizes to the non-abelian currents of Yang–Mills theory, where the breaking is no longer harmless: there the would-be-gauged chiral currents must be arranged to keep their anomalies cancelling, which is the business of Chapter 37.

Notation Watch

We write the axial current as $j^{\mu 5} = \bar{\psi}\gamma^\mu\gamma^5\psi$, index order vector-then-axial, and the Levi-Civita symbol with $\epsilon^{0123} = +1$. The dual field strength is $\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$. The anomalous divergence to be derived in Section 36.1's sequel will be quoted equivalently as $\partial_\mu j^{\mu 5} = \frac{e^2}{8\pi^2}F_{\mu\nu}\widetilde{F}^{\mu\nu} = \frac{e^2}{16\pi^2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$, the two forms identical because $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 2F_{\mu\nu}\widetilde{F}^{\mu\nu}$; the field strength is the abelian curvature $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ of Eq. (8.44). Some references order the axial index as $j^{5\mu}$ and route the sign of γ^5 or of ϵ differently; we hold to the conventions stated here throughout the chapter.

We have now fixed the classical stage completely. A massless Dirac fermion has two exact global symmetries, vector and axial; both carry conserved Noether currents, Eqs. (36.18) and (36.19); and the only place the axial symmetry can fail classically is in the presence of a mass, through the identity $\partial_\mu j^{\mu 5} = 2im\,\bar{\psi}\gamma^5\psi$ of Theorem 36.1, which vanishes when $m = 0$. There is no classical obstruction to keeping both currents conserved at once — so whatever obstruction we are about to meet is genuinely quantum. The next section earns it the first way: it computes the axial-vector-vector triangle diagram, exposes the regularization ambiguity that Fig. 36.1 anticipated, and fixes that ambiguity by the one demand we are not free to relax — that the vector current j^μ, the gauged current of QED, remain exactly conserved — thereby pinning the anomalous correction to the right-hand side of Theorem 36.1 to a single universal number.

36.2 The triangle diagram and the ABJ anomaly

The previous section left the stage set with a sharp tension. Classically, the massless Dirac field of QED carries two conserved U(1) currents: the vector current $j^\mu = \bar{\psi}\gamma^\mu\psi$ of Eq. (36.18), which QED gauges and whose conservation in the quantum theory is the Ward–Takahashi identity Theorem 19.13, and the axial current $j^{\mu 5} = \bar{\psi}\gamma^\mu\gamma^5\psi$ of Eq. (36.19), whose classical divergence Eq. (36.21) reads $\partial_\mu j^{\mu 5} = 2im\,\bar{\psi}\gamma^5\psi$ and vanishes when $m = 0$. Both are exact symmetries

of the classical massless action; there is no classical obstruction to keeping both. The claim of this chapter is that the quantum theory cannot keep both, and that the failure is a fixed number. This section earns that number the first way — the way Adler, and Bell and Jackiw, found it in 1969 — by computing the one-loop amplitude in which the axial current and two photons meet: the AVV triangle. We will see the loop integral is linearly divergent; that this divergence carries a genuine ambiguity, parametrized by how one routes momentum through the loop; that the ambiguity is exactly enough freedom to make *one* of the three currents at the vertices anomalous and to choose *which* one; and that the demand of Section 36.1 — vector-current conservation, non-negotiable in QED — forces the anomaly entirely into the axial vertex, with a coefficient that cannot then be adjusted. We confirm the coefficient a second time, with no ambiguity at any stage, by inserting a Pauli–Villars regulator and watching the anomaly emerge as the non-decoupling residue of the regulator mass.

36.2.1 The two AVV amplitudes from the QED Feynman rules

The object that controls the quantum divergence of the axial current in the presence of two photons is the three-point function of one axial and two vector currents,

$$T^{\mu\nu\lambda}(k_1,k_2) \;=\; \int \mathrm{d}^4x\,\mathrm{d}^4y\; e^{ik_1\cdot x+ik_2\cdot y}\;\langle 0|\,\mathcal{T}\,j^{\mu 5}(0)\,j^{\nu}(x)\,j^{\lambda}(y)\;|0\rangle\,\big|_{\text{amp}}, \tag{36.32}$$

amputated of external legs. Here μ is the index of the axial current, which injects momentum $q=k_1+k_2$ into the diagram, while ν and λ are the indices of the two vector currents to which the photons of momenta k_1 and k_2 attach. Coupling the photons restores a factor e at each vector vertex; the physical matrix element of $j^{\mu 5}$ into two on-shell photons of polarizations $\varepsilon_1,\varepsilon_2$ is $e^2\,\varepsilon_{1\nu}\varepsilon_{2\lambda}\,T^{\mu\nu\lambda}$. We work with the bare correlator Eq. (36.32) and reinstate the couplings at the end.

Expanding Eq. (36.32) to the order at which it is first nonzero gives a single closed fermion loop with three insertions. Using the QED Feynman rules of Section 17.1 — the massless propagator $S(p)=i\not{p}/p^2$ and the vertices γ^ν, γ^λ for the vector currents and $\gamma^\mu\gamma^5$ for the axial current — and the overall (-1) for the fermion loop, the loop with the photons attached in the order (ν,k_1) then (λ,k_2) around the fermion line is

$$T_1^{\mu\nu\lambda}(k_1,k_2) \;=\; -\int\frac{\mathrm{d}^4\ell}{(2\pi)^4}\;\mathrm{tr}\Big[\,\gamma^\mu\gamma^5\,S(\ell)\,\gamma^\nu\,S(\ell-k_1)\,\gamma^\lambda\,S(\ell-q)\,\Big]. \tag{36.33}$$

The momentum assignment is the bookkeeping of charge conservation at each vertex: a fermion line of momentum ℓ leaves the axial vertex, sheds k_1 at the first photon to become $\ell-k_1$, sheds k_2 at the second to become $\ell-q$, and returns to the axial vertex, where $q=k_1+k_2$ flows back in. The two photons are identical

bosons, so Bose symmetry requires us to add the diagram with the photon legs exchanged, $(\nu, k_1) \leftrightarrow (\lambda, k_2)$,

$$T_2^{\mu\nu\lambda}(k_1,k_2) \;=\; -\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\;\mathrm{tr}\Big[\,\gamma^\mu\gamma^5\,S(\ell)\,\gamma^\lambda\,S(\ell-k_2)\,\gamma^\nu\,S(\ell-q)\,\Big] \;=\; T_1^{\mu\lambda\nu}(k_2,k_1), \tag{36.34}$$

and the full amplitude is the Bose-symmetric sum

$$T^{\mu\nu\lambda}(k_1,k_2) \;=\; T_1^{\mu\nu\lambda}(k_1,k_2) + T_2^{\mu\nu\lambda}(k_1,k_2). \tag{36.35}$$

Figure 36.2 shows the two diagrams. That these two and only these two contribute is itself a statement about chirality: any diagram with an *even* number of γ^5 insertions in the loop produces no ϵ-tensor and so no candidate anomaly, while the closed loop of two vector currents alone (no γ^5) vanishes by Furry's theorem. The single γ^5 at the axial vertex is what survives to soak up four loop momenta into a Levi-Civita tensor.

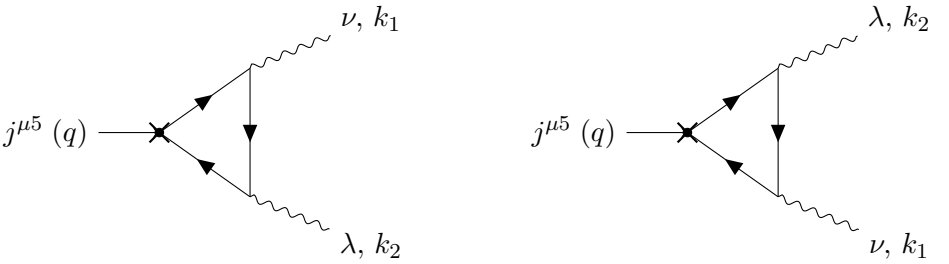


Figure 36.2: The two AVV triangle diagrams of Eqs. (36.33) and (36.34). The crossed vertex is the axial current $j^{\mu 5} = \bar\psi\gamma^\mu\gamma^5\psi$ injecting momentum $q = k_1 + k_2$; the two wavy lines are the photons. Bose symmetry of the photon legs requires summing the two assignments.

36.2.2 Linear divergence and the momentum-routing ambiguity

Power-count Eq. (36.33). Three massless propagators contribute ℓ^{-3} at large loop momentum, the three vertices contribute no powers, and the measure contributes ℓ^4, so the superficial degree of divergence is $D = 4 - 3 = 1$: the integral is *linearly* divergent. This is the entire source of the anomaly, so it is worth being precise about what a linear divergence does to an otherwise routine manipulation.

A logarithmically divergent or convergent integral is unchanged by a shift of the integration variable $\ell \to \ell + a$; we use this freedom constantly. A linearly divergent integral is *not*. The difference between an integrand evaluated at shifted and unshifted argument is a total derivative, and the integral of a total derivative is a surface term at infinity that, for a linearly divergent integrand, does not vanish. This is the precise statement we need.

Lemma 36.3 (Surface term of a linearly divergent shift)**.** *Let $f^\beta(\ell)$ be a vector function whose leading large-ℓ behavior is $f^\beta(\ell) \to \ell^\beta/(\ell^2)^2$, so that $\int \mathrm{d}^4\ell\, f^\beta$ is at most linearly divergent. Then for a constant shift vector a,*

$$\int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\big[\,f^\beta(\ell+a)-f^\beta(\ell)\,\big] \;=\; a^\alpha \int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{\partial}{\partial \ell^\alpha}\,\frac{\ell^\beta}{(\ell^2)^2} \;=\; \frac{i}{32\pi^2}\,a^\beta. \tag{36.36}$$

Proof. To first order in a, $f^\beta(\ell+a)-f^\beta(\ell) = a^\alpha\partial_\alpha f^\beta(\ell)+O(a^2)$, and only the leading tail of f^β survives in the surface integral, so we may replace f^β by $\ell^\beta/(\ell^2)^2$. By the four-dimensional divergence theorem,

$$\int \mathrm{d}^4\ell\,\frac{\partial}{\partial\ell^\alpha}\,\frac{\ell^\beta}{(\ell^2)^2} \;=\; \lim_{R\to\infty}\oint_{|\ell|=R}\mathrm{d}S_\alpha\,\frac{\ell^\beta}{(\ell^2)^2}. \tag{36.37}$$

Rotate to Euclidean momentum, $\ell^0 = i\ell^4_E$ and $\mathrm{d}^4\ell = i\,\mathrm{d}^4\ell_E$, as fixed in Section A.4; the integrand's tail becomes $\ell^\beta_E/(\ell^2_E)^2$. On the Euclidean three-sphere $|\ell_E| = R$ the oriented surface element is $\mathrm{d}S_\alpha = R^3\,\hat\ell_\alpha\,\mathrm{d}\Omega_3$ with $\ell^\beta_E = R\,\hat\ell^\beta$ and $(\ell^2_E)^2 = R^4$, so the radial factors cancel:

$$\oint_{|\ell_E|=R}\mathrm{d}S_\alpha\,\frac{\ell^\beta_E}{(\ell^2_E)^2} \;=\; \int \mathrm{d}\Omega_3\,\hat\ell_\alpha\hat\ell^\beta \;=\; \frac{\delta_\alpha{}^\beta}{4}\int\mathrm{d}\Omega_3 \;=\; \frac{\delta_\alpha{}^\beta}{4}\cdot 2\pi^2 \;=\; \frac{\pi^2}{2}\,\delta_\alpha{}^\beta, \tag{36.38}$$

where we used the angular average $\langle\hat\ell_\alpha\hat\ell^\beta\rangle = \frac14\delta_\alpha{}^\beta$ over S^3 and its area $\int\mathrm{d}\Omega_3 = 2\pi^2$. Restoring the Wick factor i and the $(2\pi)^{-4}$,

$$\int\frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{\partial}{\partial\ell^\alpha}\,\frac{\ell^\beta}{(\ell^2)^2} \;=\; \frac{i}{(2\pi)^4}\cdot\frac{\pi^2}{2}\,\delta_\alpha{}^\beta \;=\; \frac{i}{32\pi^2}\,\delta_\alpha{}^\beta. \tag{36.39}$$

Contracting with a^α gives Eq. (36.36). □

The amplitude Eq. (36.33) required us to *choose* a loop momentum: nothing forced the propagator momenta to be ℓ, $\ell-k_1$, $\ell-q$ rather than $\ell+a$, $\ell+a-k_1$, $\ell+a-q$ for an arbitrary offset a. Physically the two labelings describe the same diagram, so a convergent amplitude would not care. Here it does. To find how, contract the leading tail of the integrand of Eq. (36.33) into a single Dirac structure. Writing $-i^3 = i$ and keeping the cubic-in-ℓ piece of the numerator,

$$\mathrm{tr}\big[\gamma^\mu\gamma^5\,\ell\!\!\!/\,\gamma^\nu\ell\!\!\!/\,\gamma^\lambda\ell\!\!\!/\big] \;=\; -\ell^2\,\ell_\sigma\,\mathrm{tr}\big[\gamma^\mu\gamma^5\gamma^\nu\gamma^\lambda\gamma^\sigma\big] \;=\; -\ell^2\,\ell_\sigma\,\mathrm{tr}\big[\gamma^5\gamma^\nu\gamma^\lambda\gamma^\sigma\gamma^\mu\big], \tag{36.40}$$

where the first equality used $\ell\!\!\!/\gamma^\nu\ell\!\!\!/ = 2\ell^\nu\ell\!\!\!/ - \ell^2\gamma^\nu$ twice and dropped terms whose trace with γ^5 and fewer than four other γ's vanishes. The remaining trace is the Levi-Civita identity of Eq. (C.179), $\mathrm{tr}[\gamma^5\gamma^\nu\gamma^\lambda\gamma^\sigma\gamma^\mu] = -4i\,\epsilon^{\nu\lambda\sigma\mu}$. Hence the leading integrand of T_1 (including its overall i) is $-4\,\epsilon^{\nu\lambda\sigma\mu}\,\ell_\sigma/(\ell^2)^2$, and applying

Theorem 36.3 the offset shifts the amplitude by

$$\begin{aligned} T_1^{\mu\nu\lambda}(a) - T_1^{\mu\nu\lambda}(0) &= -4\,\epsilon^{\nu\lambda\sigma\mu}\,a^\alpha \int \frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{\partial}{\partial \ell^\alpha}\,\frac{\ell_\sigma}{(\ell^2)^2} \\ &= -\frac{4i}{32\pi^2}\,\epsilon^{\nu\lambda\sigma\mu}\,a_\sigma \;=\; \frac{i}{8\pi^2}\,\epsilon^{\mu\nu\lambda\sigma}\,a_\sigma . \end{aligned} \tag{36.41}$$

The conclusion is unavoidable: the AVV amplitude depends on the routing offset, by a finite term proportional to $\epsilon^{\mu\nu\lambda\sigma}a_\sigma$. The triangle is ambiguous. This is not a flaw in the calculation; it is the calculation telling us that some additional physical input is needed to define the amplitude.

36.2.3 The Ward identities and what fixes the routing

The additional input is the conservation laws. We have three currents at the three vertices, and each ought, classically, to be conserved: contracting the external momentum into the corresponding index should annihilate the amplitude up to the mass term, which is absent for $m = 0$. Lorentz invariance, parity (the amplitude carries one γ^5, hence one ϵ), and Bose symmetry leave only one available pseudotensor for each contraction — built from ϵ and the two independent momenta k_1, k_2 — so the three Ward identities must take the form

$$q_\mu\, T^{\mu\nu\lambda} = C_q\;\epsilon^{\nu\lambda\alpha\beta}\,k_{1\alpha}k_{2\beta}, \tag{36.42}$$

$$k_{1\nu}\, T^{\mu\nu\lambda} = C_1\;\epsilon^{\mu\lambda\alpha\beta}\,k_{1\alpha}k_{2\beta}, \tag{36.43}$$

$$k_{2\lambda}\, T^{\mu\nu\lambda} = C_2\;\epsilon^{\mu\nu\alpha\beta}\,k_{1\alpha}k_{2\beta}, \tag{36.44}$$

for three constants C_q, C_1, C_2. The naive (formal) manipulation that cancels a propagator against the contracted momentum — legitimate only if shifts were free — would set all three to zero. The linear divergence forbids that conclusion, and the surface term of Theorem 36.3 is exactly what it produces instead. The decisive fact, which the explicit surface computation Eq. (36.41) and the unambiguous Pauli–Villars calculation below both confirm, is that the routing offset redistributes the three constants but leaves a fixed combination invariant:

$$C_q - C_1 - C_2 \;=\; \frac{i}{2\pi^2} \qquad \text{(routing-independent)}. \tag{36.45}$$

To see that the offset moves the individual constants, contract the routing shift Eq. (36.41) with $k_{1\nu}$, taking the Bose-allowed offset $a = \beta\,(k_1 - k_2)$:

$$k_{1\nu}\big[T_1^{\mu\nu\lambda}(a) - T_1^{\mu\nu\lambda}(0)\big] \;=\; \frac{i}{8\pi^2}\,\epsilon^{\mu\nu\lambda\sigma}\,k_{1\nu}\,\beta\,(k_1-k_2)_\sigma \;=\; \frac{i\beta}{8\pi^2}\,\epsilon^{\mu\lambda\alpha\beta}k_{1\alpha}k_{2\beta}, \tag{36.46}$$

where the k_1k_1 piece dropped by antisymmetry and the indices were reordered into the canonical form of Eq. (36.43). Thus C_1 slides linearly with the routing

parameter β, and by the same computation so do C_2 and C_q, always preserving Eq. (36.45).

We have thus reached the crux. The total anomaly $\frac{i}{2\pi^2}$ is a fixed feature of the loop; the only freedom is to decide *which* current absorbs it. We cannot make all three currents conserved at once. But QED leaves us no choice about which current to protect. The two vector currents are the electromagnetic current, and the photon couples to it; the entire renormalizable, unitary structure of QED rests on the Ward–Takahashi identity Theorem 19.13, the quantum survival of $\partial_\mu j^\mu = 0$. A regulator that violated vector-current conservation would give the photon a mass, spoil transversality of the vacuum polarization, and destroy gauge invariance. We are therefore obliged to choose the routing that enforces

$$C_1 = C_2 = 0 \qquad \Longrightarrow \qquad k_{1\nu} T^{\mu\nu\lambda} = k_{2\lambda} T^{\mu\nu\lambda} = 0, \tag{36.47}$$

and Eq. (36.45) then fixes the axial constant completely:

$$q_\mu \, T^{\mu\nu\lambda} \;=\; \frac{i}{2\pi^2} \, \epsilon^{\nu\lambda\alpha\beta} \, k_{1\alpha} k_{2\beta}. \tag{36.48}$$

The anomaly has been pushed, in its entirety and with a now-rigid coefficient, into the axial vertex. There is no remaining freedom: having spent the routing parameter on vector conservation, we cannot also remove the axial divergence.

36.2.4 From the amplitude to the operator equation

Equation Eq. (36.48) is the momentum-space statement of an operator equation. Restoring the photon couplings, the matrix element of the axial-current divergence between the vacuum and two photons of momenta k_1, k_2 and polarizations $\varepsilon_1, \varepsilon_2$ is, in our conventions,

$$\langle k_1, k_2 | \, \partial_\mu j^{\mu 5} \, |0\rangle \;=\; i \, q_\mu \, e^2 \, \varepsilon_{1\nu} \varepsilon_{2\lambda} \, T^{\mu\nu\lambda} \;=\; -\frac{e^2}{2\pi^2} \, \epsilon^{\nu\lambda\alpha\beta} \, \varepsilon_{1\nu} \varepsilon_{2\lambda} \, k_{1\alpha} k_{2\beta}. \tag{36.49}$$

The right-hand side is precisely what one obtains by taking the two-photon matrix element of the local operator $\frac{e^2}{16\pi^2} \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$: each field strength contributes $F_{\mu\nu} \to i(k_\mu \varepsilon_\nu - k_\nu \varepsilon_\mu)$ for a photon of momentum k, and the two ways of assigning the photons to the two F's give a factor of two. Matching the kinematic structure $\epsilon^{\nu\lambda\alpha\beta} \varepsilon_{1\nu} \varepsilon_{2\lambda} k_{1\alpha} k_{2\beta}$ in both expressions identifies the operator. With the conventions fixed in Section 36.1 — $\epsilon^{0123} = +1$, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, and the electron covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ — the overall sign comes out so that we may package the result as a theorem.

Theorem 36.4 (The Adler–Bell–Jackiw anomaly)**.** *Let ψ be a Dirac fermion of charge $-e$ coupled to a background electromagnetic field, with field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Imposing conservation of the vector current $j^\mu = \bar\psi \gamma^\mu \psi$ to*

fix the triangle's routing ambiguity, the axial current $j^{\mu 5} = \bar{\psi}\gamma^{\mu}\gamma^{5}\psi$ *obeys the anomalous divergence*

$$\partial_\mu j^{\mu 5} \;=\; 2im\,\bar{\psi}\gamma^{5}\psi + \frac{e^2}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} \;=\; 2im\,\bar{\psi}\gamma^{5}\psi + \frac{e^2}{8\pi^2}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{36.50}$$

where $\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$ *is the dual field strength. For a massless fermion the classical term drops and the divergence is the pure anomaly* $\partial_\mu j^{\mu 5} = \frac{e^2}{8\pi^2}F_{\mu\nu}\widetilde{F}^{\mu\nu}$.

Three features deserve emphasis. First, the mass term $2im\,\bar{\psi}\gamma^{5}\psi$ is unchanged from its classical value Eq. (36.21): the anomaly is an *additive* quantum correction, present even when $m = 0$, that the explicit symmetry breaking of a mass does not interfere with. Second, the coefficient is fixed; nothing in Eq. (36.50) can be tuned, because the routing freedom was entirely consumed enforcing vector-current conservation. Third, the right-hand side is a total derivative — $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 4\,\partial_\mu(\epsilon^{\mu\nu\rho\sigma}A_\nu\partial_\rho A_\sigma)$, the Chern–Simons current of Theorem 36.12 — so the *integrated* axial charge violation is a topological quantity, the fact that Section 36.5 turns into the statement that the anomaly coefficient is an integer and hence exact.

Physical Insight

The anomaly is the one place in Part VI's story where a regulator dependence does not cancel. Ordinarily a cutoff or a routing choice introduces scheme dependence that renormalization sweeps into counterterms and out of physics. Here there is no counterterm to absorb it: a local term added to the Lagrangian to restore axial conservation would necessarily break vector conservation, which we are not permitted to break. The leftover is therefore physical. The anomaly is what a symmetry violation looks like when it cannot be renormalized away — a fixed number, not a convention.

36.2.5 Pauli–Villars: the same number with no ambiguity

The triangle computation is honest but delicate: the coefficient hung on the surface term of a divergent integral and on the decision to protect the vector current. It is reassuring — and, for a gold-standard derivation, necessary — to obtain the identical number by a method in which no integral is ever ambiguous. Pauli–Villars regularization provides exactly this. We add to the theory a heavy fermion Ψ of mass M with the same charge, whose loop is subtracted from the physical one; with the regulator in place every integral is convergent and may be shifted freely, so the naive Ward identities hold for the *regulated* amplitude. The price is that the regulator mass explicitly breaks the axial symmetry through its own $2iM\,\bar{\Psi}\gamma^{5}\Psi$ term, exactly as in Eq. (36.21). The regulated axial divergence is therefore

$$\partial_\mu j^{\mu 5}_{\text{reg}} \;=\; 2im\,\bar{\psi}\gamma^{5}\psi \;-\; 2iM\,\bar{\Psi}\gamma^{5}\Psi, \tag{36.51}$$

with no anomaly. The anomaly of the physical theory is what remains when the regulator is removed: $\partial_\mu j^{\mu 5} = 2im\,\bar\psi\gamma^5\psi - \lim_{M\to\infty} 2iM\,\langle|\,\bar\Psi\gamma^5\Psi\,|\rangle$. The crucial point is that the heavy fermion does *not* decouple. Its pseudoscalar density $2iM\,\bar\Psi\gamma^5\Psi$, naively suppressed by $1/M$ from the heavy propagator but enhanced by the explicit M, tends to a finite, M-independent limit. That limit is the anomaly, and we now compute it in full.

Worked Example 36.5 (The anomaly as the non-decoupling Pauli–Villars residue)**.** We evaluate $\lim_{M\to\infty} 2iM\,\langle k_1,k_2|\,\bar\Psi\gamma^5\Psi\,|0\rangle$ to second order in the background field, i.e. the matrix element into two photons. The one-loop pseudoscalar triangle, with the pseudoscalar vertex $2iM\gamma^5$, two photon vertices $-ie\gamma^\alpha, -ie\gamma^\beta$, three massive propagators $S_M(p) = i(\not p + M)/(p^2-M^2)$, and the fermion-loop (-1), is

$$G_a^{\alpha\beta} \;=\; 2e^2M\int\frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{\mathrm{tr}\big[\gamma^5(\not\ell+M)\gamma^\alpha(\not\ell-\not k_1+M)\gamma^\beta(\not\ell-\not q+M)\big]}{(\ell^2-M^2)\big((\ell-k_1)^2-M^2\big)\big((\ell-q)^2-M^2\big)}, \tag{36.52}$$

collecting the prefactor as $-(-ie)^2(2iM)\,i^3 = 2e^2M$, plus the Bose partner $G_b^{\alpha\beta}$ with $(\alpha,k_1)\leftrightarrow(\beta,k_2)$.

The trace. A nonzero γ^5-trace needs exactly four γ's, so from the three factors $(\not\ell+\cdots+M)$ we must take the mass M from exactly one and a slash from the other two; the all-slash term has five γ's (odd) and the two-mass terms have two γ's, both giving zero. Each surviving term carries one explicit M. Using $\mathrm{tr}[\gamma^5\gamma^a\gamma^b\gamma^c\gamma^d] = -4i\,\epsilon^{abcd}$ from Eq. (C.179) and dropping every $\ell_\rho\ell_\sigma\epsilon^{\cdots\rho\cdots\sigma}=0$ term, the numerator equals $-4iM\,N^{\alpha\beta}(\ell)$ with

$$N^{\alpha\beta}(\ell) = (\ell-k_1)_\rho(\ell-q)_\sigma\epsilon^{\alpha\rho\beta\sigma} + \ell_\rho(\ell-q)_\sigma\epsilon^{\rho\alpha\beta\sigma} + \ell_\rho(\ell-k_1)_\sigma\epsilon^{\rho\alpha\sigma\beta}. \tag{36.53}$$

Feynman parameters. Combine the denominators with $\frac{1}{D_0D_1D_q} = 2\int_0^1\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\delta(1-x-y-z)\,[\,\ell^2-\Delta\,]^{-3}$ where, after shifting $\ell\to\ell+P$ with $P = y\,k_1 + z\,q$, the denominator is $\ell^2-\Delta$ with $\Delta = M^2 + P^2 - y\,k_1^2 - z\,q^2$. In the shifted variable the odd-in-ℓ and the $\ell_\rho\ell_\sigma$ pieces of $N^{\alpha\beta}$ integrate to zero, so we may replace $\ell\to P$. With $P-k_1 = -x\,k_1 + z\,k_2$ and $P-q = -x\,k_1 - (x+y)\,k_2$ (using $x+y+z=1$, $q=k_1+k_2$), and abbreviating $E^{\alpha\beta}\equiv\epsilon^{\alpha\beta\rho\sigma}k_{1\rho}k_{2\sigma}$, the three terms evaluate by elementary index reordering to

$$N^{\alpha\beta}\to\; -x\,E^{\alpha\beta} \;-\; y\,E^{\alpha\beta} \;-\; z\,E^{\alpha\beta} \;=\; -(x+y+z)\,E^{\alpha\beta} \;=\; -E^{\alpha\beta}, \tag{36.54}$$

independent of the Feynman parameters — the first miracle of the computation. (For instance the first term gives $-x(x+y)+(-xz)$ times E after using $\epsilon^{\alpha\rho\beta\sigma}k_{1\rho}k_{2\sigma} = -E^{\alpha\beta}$ and $\epsilon^{\alpha\rho\beta\sigma}k_{2\rho}k_{1\sigma} = +E^{\alpha\beta}$, collapsing to $-x\,E^{\alpha\beta}$; the other two terms work identically.)

The integral and the limit. The remaining momentum integral is the convergent master integral of Eq. (D.21),

$$\int\frac{\mathrm{d}^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2-\Delta)^3} \;=\; \frac{-i}{32\pi^2\,\Delta}. \tag{36.55}$$

Assembling $G_a^{\alpha\beta} = 2e^2M\cdot(-4iM)\cdot(-E^{\alpha\beta})\cdot 2\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(1-x-y-z)\,\frac{-i}{32\pi^2\Delta}$ and collecting constants $2\cdot(-4i)\cdot 2\cdot(-i)/32\pi^2 = -1/2\pi^2$,

$$G_a^{\alpha\beta} \;=\; \frac{e^2M^2}{2\pi^2}\,E^{\alpha\beta}\int_0^1 \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\;\frac{\delta(1-x-y-z)}{\Delta}. \tag{36.56}$$

As $M\to\infty$ with the external momenta fixed, $\Delta\to M^2$ and the parameter integral tends to $\frac{1}{M^2}\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(1-x-y-z) = \frac{1}{2M^2}$ (the area of the unit simplex). The factors of M^2 cancel — the second miracle, the non-decoupling — leaving

$$\lim_{M\to\infty} G_a^{\alpha\beta} \;=\; \frac{e^2M^2}{2\pi^2}\,E^{\alpha\beta}\cdot\frac{1}{2M^2} \;=\; \frac{e^2}{4\pi^2}\,\epsilon^{\alpha\beta\rho\sigma}k_{1\rho}k_{2\sigma}. \tag{36.57}$$

The Bose partner $G_b^{\alpha\beta}$ gives the identical value (the exchange $\alpha\leftrightarrow\beta$, $k_1\leftrightarrow k_2$ acts on $E^{\alpha\beta}$ as the identity), so

$$\lim_{M\to\infty} 2iM\;\langle k_1,k_2|\,\bar\Psi\gamma^5\Psi\,|0\rangle \;=\; G_a^{\alpha\beta}+G_b^{\alpha\beta}\,\Big|_{M\to\infty} \;=\; \frac{e^2}{2\pi^2}\,\epsilon^{\alpha\beta\rho\sigma}k_{1\rho}k_{2\sigma}\,\varepsilon_{1\alpha}\varepsilon_{2\beta}. \tag{36.58}$$

This is the non-decoupling residue. Subtracting it as in Eq. (36.51) reproduces, for $m=0$, the two-photon matrix element of $\frac{e^2}{8\pi^2}F\widetilde F$ — the same kinematic structure and the same coefficient $\frac{e^2}{16\pi^2}$ found from the triangle in Theorem 36.4, now with no surface term, no routing parameter, and no ambiguity to resolve. The two derivations agree.

The Pauli–Villars route also makes transparent *why* the anomaly is unavoidable. The regulator preserved vector-current conservation manifestly (the heavy fermion has a vector-like coupling), so the only symmetry it could break was the axial one, which its Dirac mass breaks explicitly. Gauge invariance was bought, in the regulated theory, at the cost of axial invariance; removing the regulator leaves that cost behind as the finite anomaly. There is no regulator that respects both, which is the content of the chapter's thesis stated concretely.

36.2.6 Adler–Bardeen: the coefficient is one-loop exact

A natural worry is whether higher-order diagrams — two-loop corrections with extra photon or fermion lines dressing the triangle — shift the coefficient in Eq. (36.50). They do not. This is the Adler–Bardeen theorem.

Theorem 36.6 (Adler–Bardeen non-renormalization, stated)**.** *The coefficient of the anomalous divergence Eq. (36.50) receives no corrections beyond one loop. With a vector-current-preserving regulator, the anomaly is exactly $\frac{e^2}{8\pi^2}F\widetilde F$ to all orders in the gauge coupling.*

The honest argument runs as follows. Higher-order diagrams that could correct the axial divergence fall into two classes. Diagrams in which the extra

interactions dress the internal lines of the triangle are individually more convergent than the triangle itself — adding a photon propagator and two vertices lowers the superficial degree of divergence — so they contain no new linear divergence and no new surface term; their contribution to $q_\mu T^{\mu\nu\lambda}$ can be made to vanish by a legal shift, because for a convergent integral the shift is legal. The anomaly, being entirely a surface term, is therefore confined to the one diagram that is linearly divergent: the bare triangle. Diagrams in which an extra fermion loop carries the axial vertex are constrained by the same vector Ward identities, which force their anomalous part to share the triangle's structure; Bardeen's all-orders analysis (carried out in the regularized theory, where every manipulation is justified) shows these renormalize the *operators* but not the anomaly coefficient. The Pauli–Villars derivation above sharpens the intuition: the anomaly is the $M \to \infty$ residue of a regulator mass term, and that residue is a one-loop object by construction — the heavy fermion contributes a single closed loop. The deepest reason, however, is topological, and we defer it to Section 36.5: the integrated anomaly equals the index of the Dirac operator, an integer, and an integer cannot drift continuously with the coupling. A number that is forced to be an integer is forced to be exact. We state the theorem here and discharge its topological proof there.

Caution

Naive γ^5 in dimensional regularization misses the anomaly. It is tempting to regulate the triangle with the dimensional regularization of Chapter 19, which is so convenient elsewhere. The temptation must be resisted unless one is careful about γ^5. In $d = 4-2\varepsilon$ dimensions there is no consistent way to keep all three properties that define γ^5 in four dimensions: total anticommutation $\{\gamma^\mu, \gamma^5\} = 0$ for every μ, the cyclicity of the trace, and the nonvanishing four-dimensional trace $\mathrm{tr}[\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\,\epsilon^{\mu\nu\rho\sigma}$. This is the obstruction proved in Theorem C.64. If one nonetheless imposes a *naive* anticommuting γ^5 in d dimensions, the contracted triangle can be shifted freely (the ε-regulated integral is formally convergent), all three Ward identities appear to hold, and the anomaly evaporates — a false result, since the anomaly is physical and measured in $\pi^0 \to \gamma\gamma$ (Section 36.4). The resolution is the 't Hooft–Veltman–Breitenlohner–Maison scheme, in which γ^5 anticommutes with the four "physical" γ's and commutes with the $(d-4)$ "evanescent" ones; the anomaly then reappears from the evanescent components, with the same coefficient Eq. (36.50). The full prescription and a worked check live in Section C.7.4; the lesson for this chapter is that the anomaly is invisible to a regulator that treats γ^5 too cavalierly, exactly because such a regulator silently violates the vector Ward identity it pretends to keep.

We have now earned the anomaly the first way. The triangle exposed it as a surface term — a genuine ambiguity of a linearly divergent loop, resolved only by the physical demand that the photon's vector current stay conserved — and Pauli–Villars confirmed the coefficient with no ambiguity at any step. The operator equation Eq. (36.50) is now in hand. The next section earns it a second, deeper way: as the Jacobian of the fermionic path-integral measure under a local chiral rotation, regularized gauge-invariantly by the Dirac-operator heat kernel (Section 36.3), a route that is manifestly free of routing choices and exposes the same number as a property of the measure rather than of any diagram. The identical operator equation will then be handed onward to Section 36.4, where it is turned into the measured rate of $\pi^0 \to \gamma\gamma$ and a count of the colors of QCD.

36.3 The anomaly from the measure: Fujikawa's method

The triangle calculation of Section 36.2 delivered the anomaly but left an uncomfortable aftertaste. The coefficient was correct, yet it emerged only after a regularization choice was made: the linearly divergent loop integral was ambiguous by a finite surface term, the ambiguity was a free parameter sliding the anomaly between the vector and axial channels, and we fixed it by hand, by *demanding* that the vector current stay conserved. The physics is sound—vector-current conservation is the statement that electric charge is real and conserved, so it is the right thing to insist on—but the logic is uncomfortably indirect. The anomaly looked like an artifact of how we drew and regulated one diagram, rescued by an external principle. A reader entitled to ask "but *why* is the axial symmetry the one that breaks, and is the number really scheme-independent?" has not yet been fully answered.

This section answers the question at its root. We will obtain the very same operator equation,

$$\partial_\mu \langle j^{\mu 5} \rangle = \frac{e^2}{16\pi^2}\, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} = \frac{e^2}{8\pi^2}\, F_{\mu\nu} \widetilde{F}^{\mu\nu}, \tag{36.59}$$

without drawing a single Feynman diagram, without a momentum routing to fix, and with gauge invariance manifest at every step. The idea, due to Fujikawa, is to take the path integral seriously as the definition of the quantum theory and ask what the *measure* does under a chiral rotation. A classical symmetry leaves the action invariant; a quantum symmetry must in addition leave the integration measure invariant. The axial rotation passes the first test and fails the second. The failure is a Jacobian—an honest determinant of a change of Grassmann variables—and once we regulate that determinant in the only gauge-invariant way available, the heat kernel of the Dirac operator, the number (36.59) drops out with no freedom left to exploit. The contrast with Theorem 19.13 is now exact and visible at the level of measures: the vector (electromagnetic) rotation

$\psi \to e^{i\beta(x)}\psi$ rotates ψ and $\bar\psi$ *oppositely*, so its two measure factors cancel and the Ward–Takahashi identity survives quantization; the axial rotation rotates them *the same way*, the factors add, and the symmetry is lost.

The bookkeeping of the fermionic measure under field redefinitions, and the reduction of the chiral Jacobian to a heat-kernel coefficient, were carried out once and for all in Section E.5.6 as part of the path-integral technology of Chapter E; we recall those results rather than rebuild them, and devote this section to the computation that Chapter E deliberately deferred—the explicit evaluation of the coefficient in four dimensions, the demonstration that it does not depend on the regulator, and its non-abelian generalization.

36.3.1 The classical chiral symmetry and the two tests of a quantum symmetry

We work with a single massless Dirac fermion of electric charge $-e$ ($e > 0$) coupled to a background abelian gauge field, the massless limit of QED,

$$S[\psi,\bar\psi,A] = \int \mathrm{d}^4x\; \bar\psi\, i\not{D}\,\psi, \qquad D_\mu = \partial_\mu + ieA_\mu, \qquad \not{D} = \gamma^\mu D_\mu, \tag{36.60}$$

in the conventions fixed in the chapter introduction: mostly-minus metric, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, $\epsilon^{0123} = +1$, and the chiral (Weyl) basis of Eq. (11.48) whenever an explicit matrix is needed. The field strength is $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and the dual is $\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$. We keep A_μ a fixed external field throughout; nothing in Fujikawa's argument requires it to be dynamical, which is exactly why the result will apply equally to the harmless global anomaly (the present case) and, in Chapter 37, to the fatal gauge anomaly.

The action (36.60) is invariant under the global axial rotation $\psi \to e^{i\alpha\gamma^5}\psi$ with α constant, because γ^5 anticommutes with every γ^μ and hence with $\not{D}$, so the two exponentials surrounding $\not{D}$ collide and cancel:

$$\bar\psi\, e^{i\alpha\gamma^5}\, i\not{D}\, e^{i\alpha\gamma^5}\psi = \bar\psi\, e^{i\alpha\gamma^5}\, e^{-i\alpha\gamma^5}\, i\not{D}\,\psi = \bar\psi\, i\not{D}\,\psi. \tag{36.61}$$

Promoting α to a local function $\alpha(x)$ probes the associated Noether current. Under the infinitesimal local rotation

$$\psi(x) \to \psi'(x) = e^{i\alpha(x)\gamma^5}\psi(x), \qquad \bar\psi(x) \to \bar\psi'(x) = \bar\psi(x)\, e^{i\alpha(x)\gamma^5}, \tag{36.62}$$

the second relation following from the first because $\gamma^{5\dagger} = \gamma^5$ and $\gamma^5\gamma^0 = -\gamma^0\gamma^5$, so that $\bar\psi' = \psi'^\dagger\gamma^0 = \psi^\dagger e^{-i\alpha\gamma^5}\gamma^0 = \psi^\dagger\gamma^0 e^{+i\alpha\gamma^5} = \bar\psi\, e^{i\alpha\gamma^5}$. The crucial structural fact, already flagged in Eq. (E.193), is the relative sign: ψ and $\bar\psi$ rotate *with the same matrix*, not with inverse matrices. A vector rotation $\psi \to e^{i\beta}\psi$ would give $\bar\psi \to \bar\psi\, e^{-i\beta}$ instead. This single sign is the seed of everything: it decides whether the two measure factors cancel or reinforce.

Compute the variation of the action to first order in α. Only the kinetic term contributes, and only through the derivative landing on α:

$$\delta S = \int \mathrm{d}^4x\; \bar\psi\, i\gamma^\mu (i\partial_\mu \alpha)\gamma^5 \psi = -\int \mathrm{d}^4x\; (\partial_\mu \alpha)\, \bar\psi \gamma^\mu \gamma^5 \psi = \int \mathrm{d}^4x\; \alpha(x)\, \partial_\mu j^{\mu 5}(x), \tag{36.63}$$

where the terms with ∂_μ acting on ψ cancel exactly as in (36.61) (the manipulation $i\gamma^\mu\gamma^5 D_\mu = -\gamma^5\, i\not{D}$ pairs them off), the axial current is

$$j^{\mu 5}(x) = \bar\psi \gamma^\mu \gamma^5 \psi, \tag{36.64}$$

and the last step is an integration by parts discarding the boundary term. Classically, on the equations of motion, $\delta S = 0$ for arbitrary $\alpha(x)$ forces $\partial_\mu j^{\mu 5} = 0$: the axial current is conserved. (For a massive fermion the mass term contributes $2im\bar\psi\gamma^5\psi$ and one finds the familiar $\partial_\mu j^{\mu 5} = 2im\bar\psi\gamma^5\psi$; we set $m = 0$ so that the only possible obstruction to conservation is the quantum one we are hunting.)

So far this is the classical Noether story of Chapter 7 applied to the axial symmetry. The quantum theory is defined by the generating functional of Theorem 10.17 with the fermionic measure of Section 12.5,

$$Z[A] = \int \mathcal{D}\bar\psi\, \mathcal{D}\psi\; e^{iS[\psi,\bar\psi,A]}, \tag{36.65}$$

and a symmetry of the *quantum* theory must leave invariant not only S but the product $\mathcal{D}\bar\psi\,\mathcal{D}\psi\, e^{iS}$ in its entirety. A change of integration variables cannot change the value of Z—an integral does not know the name of its dummy variable—but it reshuffles that value between three places: the action, the measure, and the source terms. Tracking the reshuffling for the chiral rotation (36.62) is the whole method. We have just handled the action; the measure is next, and it is where the classical symmetry goes to die.

36.3.2 The Jacobian of a chiral rotation

The fermionic measure is a Berezin measure, and Berezin measures transform under a linear change of Grassmann variables by the *inverse* of the ordinary determinant—the single most consequential sign in this chapter. We import the rule from Theorem 12.19 (established again, in the path-integral packaging we use here, as Theorem E.30): if $\theta'_i = \sum_j M_{ij}\theta_j$ are Grassmann variables, then

$$\prod_i \mathrm{d}\theta'_i = (\det M)^{-1} \prod_j \mathrm{d}\theta_j. \tag{36.66}$$

The inverse power is forced by the defining Berezin integral $\int \mathrm{d}\theta_1 \cdots \mathrm{d}\theta_n\, \theta_n \cdots \theta_1 = 1$ together with $\theta'_n \cdots \theta'_1 = (\det M)\, \theta_n \cdots \theta_1$: the measure must compensate the determinant the monomial produces, and it can only do so with $(\det M)^{-1}$. This

is the opposite of the bosonic Jacobian, and it is the reason a fermionic loop carries the famous extra minus sign throughout the book.

To apply (36.66) we need the chiral rotation written as a linear map on a discrete set of Grassmann coefficients. Expand the fields in the orthonormal eigenmodes of the Hermitian operator $i\not{D}$,

$$i\not{D}\,\varphi_n = \lambda_n \varphi_n, \qquad \int \mathrm{d}^4x\; \varphi_m^\dagger(x)\varphi_n(x) = \delta_{mn}, \qquad \psi = \sum_n a_n \varphi_n, \quad \bar\psi = \sum_n \bar b_n \varphi_n^\dagger, \tag{36.67}$$

with independent Grassmann coefficients $a_n, \bar b_n$, so that $\mathcal{D}\bar\psi\,\mathcal{D}\psi = \prod_n \mathrm{d}\bar b_n\,\mathrm{d}a_n$. That $i\not{D}$ is Hermitian, with real discrete spectrum and a complete orthonormal eigenbasis, is a statement about an elliptic operator and is true on the Euclidean continuation of Section 10.6, where $i\not{D}$ is genuinely self-adjoint on a compactified space; we adopt that continuation now and rotate the final operator equation back to Lorentzian signature at the end. Under (36.62) the coefficients transform linearly, $a_n \to a'_n = \sum_m C_{nm} a_m$, with the *same* matrix acting on the $\bar b_n$ (this is the "same way, not oppositely" of Section 36.3.1), where

$$C_{nm} = \int \mathrm{d}^4x\; \varphi_n^\dagger(x)\, e^{i\alpha(x)\gamma^5}\, \varphi_m(x) = \delta_{nm} + i\int \mathrm{d}^4x\; \alpha(x)\, \varphi_n^\dagger \gamma^5 \varphi_m + O(\alpha^2). \tag{36.68}$$

By (36.66) the measure picks up $(\det C)^{-1}$ once from the a's and once from the $\bar b$'s, so the total Jacobian is $(\det C)^{-2}$—reinforcement, not cancellation. Using $(\det C)^{-2} = \exp(-2\,\mathrm{Tr}\ln C)$ and $\mathrm{Tr}\ln C = \mathrm{Tr}(C-\mathbf{1}) + O(\alpha^2) = i\int \mathrm{d}^4x\,\alpha(x)\sum_n \varphi_n^\dagger\gamma^5\varphi_n + O(\alpha^2)$, we recover the Fujikawa Jacobian of Eq. (E.197),

$$\mathcal{D}\bar\psi'\,\mathcal{D}\psi' = \exp\Big[-2i\int \mathrm{d}^4x\;\alpha(x)\,\mathcal{A}(x)\Big]\,\mathcal{D}\bar\psi\,\mathcal{D}\psi, \qquad \mathcal{A}(x) \equiv \sum_n \varphi_n^\dagger(x)\,\gamma^5\,\varphi_n(x). \tag{36.69}$$

The object $\mathcal{A}(x)$ is the *anomaly function*: the diagonal, γ^5-weighted density of the Dirac eigenmodes at the point x. Had the rotation been vectorial, C would have been built from $e^{i\beta}$ with no γ^5, the two measure factors would have carried opposite powers of $\det C$, and the Jacobian would have been exactly 1; the electromagnetic measure is invariant, and the Ward–Takahashi identity of Theorem 19.13 is safe. The axial measure is not.

We can now read off the anomalous Ward identity before evaluating $\mathcal{A}$. Perform (36.62) as a change of variables inside (36.65). Because the value of Z is unchanged while the action shifts by (36.63) and the measure by (36.69), expanding to first order in the arbitrary function $\alpha(x)$ gives

$$0 = \int \mathcal{D}\bar\psi\,\mathcal{D}\psi\,\Big[\, i\int \mathrm{d}^4x\,\alpha\,\partial_\mu j^{\mu 5} \;-\; 2i\int \mathrm{d}^4x\,\alpha\,\mathcal{A}\,\Big] e^{iS}, \tag{36.70}$$

and since $\alpha(x)$ is arbitrary the integrand of the x-integral must vanish in expectation:

$$\boxed{\partial_\mu \langle j^{\mu 5}(x)\rangle = 2\,\mathcal{A}(x)} \tag{36.71}$$

This is the anomalous axial Ward identity, derived with no diagram and no momentum routing. The entire physics of the chiral anomaly is now compressed into the evaluation of the single function $\mathcal{A}(x)$. The factor of 2 is the visible relic of the "same way, not oppositely" sign: it is the second measure factor, the one that did not cancel.

Caution: $\mathcal{A}(x)$ is the Jacobian density, and the divergence is twice it

Two normalizations circulate in the literature, differing by exactly the factor of 2 in (36.71). We fix ours once: $\mathcal{A}(x) = \sum_n \varphi_n^\dagger \gamma^5 \varphi_n$ is the density appearing in the measure Jacobian (36.69), and the *physical* divergence $\partial_\mu \langle j^{\mu 5} \rangle$ equals $2\mathcal{A}(x)$. We will find $\mathcal{A} = \frac{e^2}{32\pi^2} \epsilon F F$, so that the divergence is $\frac{e^2}{16\pi^2} \epsilon F F$, matching the triangle result (36.50). A reader comparing with a source that calls the divergence itself "$\mathcal{A}$" must mentally insert or remove this 2.

36.3.3 Why the anomaly function must be regularized, and how

Taken literally, the anomaly function (36.69) is a contest between two zeros and an infinity. Formally,

$$\mathcal{A}(x) = \sum_n \varphi_n^\dagger(x) \gamma^5 \varphi_n(x) = \mathrm{tr}[\gamma^5 \textstyle\sum_n \varphi_n(x) \varphi_n^\dagger(x)] = \mathrm{tr}\,[\gamma^5 \langle x|x\rangle] = \mathrm{tr}\,\gamma^5 \cdot \delta^4(0), \tag{36.72}$$

using completeness $\sum_n \varphi_n(x) \varphi_n^\dagger(y) = \delta^4(x-y)\,\mathbf{1}$. The Dirac trace $\mathrm{tr}\,\gamma^5 = 0$ multiplies the divergent $\delta^4(0)$: a $0 \times \infty$ that the naive manipulation resolves, wrongly, to zero. The resolution is wrong because the mode sum is only conditionally convergent, and reordering it—which is what "$\mathrm{tr}\,\gamma^5 = 0$" silently does, by summing over the Dirac index before summing over modes—is illegitimate. The sum must be made absolutely convergent first, by damping the high-eigenvalue modes, and only then may the Dirac trace be taken. This is the content of Eq. (E.199): insert a smooth cutoff function f of the dimensionless ratio λ_n^2/M^2, with

$$f(0) = 1, \qquad f(\infty) = 0, \qquad s f'(s) \to 0 \quad (s \to 0, \infty), \tag{36.73}$$

and remove it only at the end by sending the cutoff scale $M \to \infty$:

$$\mathcal{A}(x) = \lim_{M\to\infty} \sum_n f\Big(\frac{\lambda_n^2}{M^2}\Big) \varphi_n^\dagger(x) \gamma^5 \varphi_n(x) = \lim_{M\to\infty} \mathrm{tr}\Big\langle x\Big|\, \gamma^5\, f\Big(\frac{(i\not{D})^2}{M^2}\Big)\Big|x\Big\rangle. \tag{36.74}$$

The cutoff is built from the eigenvalues of $i\not{D}$, hence from the gauge-covariant operator $\not{D}$ itself; it commutes with gauge transformations and breaks no gauge symmetry. This is the decisive virtue of the method over the triangle: there is no choice to make. The triangle's ambiguity was the freedom to distribute a

divergence between the vector and axial vertices; here the only freedom is the shape of f, and we will prove in Section 36.3.5 that the answer does not depend on it at all. The price—and it is the honest price of the anomaly—is that no such gauge-invariant regulator can also respect the chiral symmetry, because $f((i\not{D})^2/M^2)$ does not commute with γ^5 (the operator $(i\not{D})^2$ contains the term $\sigma^{\mu\nu}F_{\mu\nu}$, which does not commute with γ^5). One symmetry must be sacrificed to the regulator; we keep gauge invariance, and the axial symmetry is what breaks.

The choice $f(s) = e^{-s}$ turns the regulated operator into the heat kernel of the Laplace-type operator $-(i\not{D})^2 = \not{D}^2$, the object whose short-distance (Seeley–DeWitt) expansion was constructed in Section E.3:

$$\mathcal{A}(x) = \lim_{M\to\infty} \mathrm{tr}\langle x|\,\gamma^5\, e^{\not{D}^2/M^2}\,|x\rangle. \tag{36.75}$$

(Here $\not{D}^2 = -(i\not{D})^2$ has eigenvalues $-\lambda_n^2 \le 0$, so $e^{\not{D}^2/M^2}$ damps as required.) Chapter E reduced the chiral Jacobian to the single Seeley–DeWitt coefficient $a_2(x)$ via Eq. (E.201) but stopped short of taking its γ^5 trace. We now take it.

Physical insight: the anomaly counts a mismatch the regulator cannot hide

The regulator caps the number of modes at $|\lambda_n| \lesssim M$. Chirality pairs nonzero modes: if $i\not{D}\,\varphi = \lambda\varphi$ then $i\not{D}(\gamma^5\varphi) = -\lambda(\gamma^5\varphi)$, so every mode of eigenvalue $\lambda \neq 0$ has a partner $\gamma^5\varphi$ of eigenvalue $-\lambda$, orthogonal to it, and the two contribute $+1$ and -1 to $\varphi^\dagger\gamma^5\varphi$ summed over the pair: they cancel. Only modes with $\lambda = 0$, which are eigenstates of γ^5 with definite chirality, fail to cancel. The regulated $\mathcal{A}$ integrated over space therefore counts $n_+ - n_-$, the net chirality of the zero modes—an integer that no smooth deformation of the gauge field can change. That the local density $\mathcal{A}(x)$ is a smooth gauge field functional while its integral is a protected integer is the first sight of the topology that Section 36.5 will make precise; it is also why the coefficient is exact, immune to higher loops.

36.3.4 The heat-kernel evaluation

Everything now rests on one identity for $\not{D}^2$ and one Dirac trace. We isolate the identity first.

Lemma 36.7 (Lichnerowicz / Bochner expansion of $\not{D}^2$)**.** *For $D_\mu = \partial_\mu + ieA_\mu$,*

$$\not{D}^2 = D^\mu D_\mu + \frac{e}{2}\,\sigma^{\mu\nu}F_{\mu\nu}, \qquad \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]. \tag{36.76}$$

Proof. Split the product of two gammas into its symmetric and antisymmetric

parts using the Clifford algebra $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}$:

$$\not{D}^2=\gamma^\mu\gamma^\nu D_\mu D_\nu=\tfrac12\{\gamma^\mu,\gamma^\nu\}D_\mu D_\nu+\tfrac12[\gamma^\mu,\gamma^\nu]D_\mu D_\nu=D^\mu D_\mu+\tfrac12[\gamma^\mu,\gamma^\nu]\,\tfrac12[D_\mu,D_\nu], \tag{36.77}$$

where in the antisymmetric term only the antisymmetric part of $D_\mu D_\nu$, namely $\frac12[D_\mu,D_\nu]$, survives the contraction with $[\gamma^\mu,\gamma^\nu]$. The commutator of covariant derivatives is the field strength, $[D_\mu,D_\nu]=ieF_{\mu\nu}$, and writing $[\gamma^\mu,\gamma^\nu]=-2i\sigma^{\mu\nu}$ gives

$$\tfrac14[\gamma^\mu,\gamma^\nu][D_\mu,D_\nu]=\tfrac14(-2i\sigma^{\mu\nu})(ieF_{\mu\nu})=\frac{e}{2}\,\sigma^{\mu\nu}F_{\mu\nu}, \tag{36.78}$$

which is (36.76). □

The sign of the σF term in (36.76) is convention-dependent (it flips with the sign in D_μ or in $\sigma^{\mu\nu}$), but this will not matter: the anomaly is built from the *square* of this term, $(\sigma F)^2$, which is sign-blind. With the Lichnerowicz identity in hand we carry out the trace on a plane-wave basis.

Worked Example 36.8 (The four-dimensional Fujikawa coefficient)**.** We evaluate (36.75) explicitly. Insert a complete set of plane waves to represent the coincidence-limit matrix element. Writing the operator as $\hat{\mathcal{O}}=e^{\not{D}^2/M^2}$, its diagonal kernel is

$$\langle x|\hat{\mathcal{O}}|x\rangle=\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,\langle x|\hat{\mathcal{O}}|k\rangle\langle k|x\rangle=\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,e^{-ik\cdot x}\,\hat{\mathcal{O}}\,e^{ik\cdot x}, \tag{36.79}$$

where in the last form $\hat{\mathcal{O}}$ is understood to act on everything to its right, including the plane wave $e^{ik\cdot x}$, and we will take the Dirac trace at the end. The plane wave shifts the derivative: for any function built from $D_\mu=\partial_\mu+ieA_\mu$,

$$e^{-ik\cdot x}\,D_\mu\,e^{ik\cdot x}=D_\mu+ik_\mu, \tag{36.80}$$

because $e^{-ik\cdot x}\partial_\mu e^{ik\cdot x}=\partial_\mu+ik_\mu$ and A_μ commutes with the exponential. Therefore, using Theorem 36.7,

$$e^{-ik\cdot x}\,\not{D}^2\,e^{ik\cdot x}=(D_\mu+ik_\mu)(D^\mu+ik^\mu)+\frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}=D^2+2ik\cdot D-k^2+\frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}. \tag{36.81}$$

Now rescale the loop momentum, $k\to Mk$, so that $\mathrm{d}^4k\to M^4\,\mathrm{d}^4k$, and divide the exponent by M^2:

$$\frac{1}{M^2}\,e^{-iMk\cdot x}\not{D}^2e^{iMk\cdot x}=-k^2+\frac{2i\,k\cdot D}{M}+\frac{1}{M^2}\Big(D^2+\frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}\Big). \tag{36.82}$$

The anomaly function becomes

$$\mathcal{A}(x)=\lim_{M\to\infty}M^4\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,e^{-k^2}\,\operatorname{tr}\Big[\gamma^5\,\exp\Big(\frac{2i\,k\cdot D}{M}+\frac{1}{M^2}\big(D^2+\tfrac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}\big)\Big)\Big], \tag{36.83}$$

where the $-k^2$ has been pulled out as the Gaussian weight e^{-k^2}. The prefactor M^4 is the bookkeeping device that selects which term in the expansion of the remaining exponential survives: we must keep terms down to order M^{-4}, and may drop everything higher.

Two facts now collapse the expansion to a single term. First, the Dirac trace with γ^5 in front annihilates anything with fewer than four gamma matrices: $\operatorname{tr}\gamma^5 = \operatorname{tr}[\gamma^5\sigma^{\mu\nu}] = 0$, and the first nonvanishing structure is $\operatorname{tr}[\gamma^5\sigma^{\mu\nu}\sigma^{\rho\sigma}]$ (Section C.6.5). Second, of the three pieces in the exponent only the σF piece carries any gamma matrices at all—$k\cdot D$ and D^2 are scalars in Dirac space. Hence a nonzero γ^5 trace requires *exactly two* factors of σF, and they enter at order M^{-4} through the second term of the exponential's Taylor series,

$$\frac{1}{2!}\Big(\frac{1}{M^2}\,\frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}\Big)^2 = \frac{1}{M^4}\,\frac{e^2}{8}\,\sigma^{\mu\nu}F_{\mu\nu}\,\sigma^{\rho\sigma}F_{\rho\sigma}. \tag{36.84}$$

Any additional factor of $k\cdot D$ or D^2 multiplies a scalar (in Dirac space) and raises the power of $1/M$, so it dies under $\lim_{M\to\infty} M^4(\cdots)$; any term with zero or one σF has vanishing γ^5 trace. This is the term, and the only one. The M^4 prefactor cancels the M^{-4} exactly, leaving a finite limit:

$$\mathcal{A}(x) = \frac{e^2}{8}\,F_{\mu\nu}F_{\rho\sigma}\,\operatorname{tr}[\gamma^5\,\sigma^{\mu\nu}\sigma^{\rho\sigma}]\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,e^{-k^2}. \tag{36.85}$$

The Gaussian integral is elementary on the Euclidean continuation, where $k^2 = k_E^2 \geq 0$:

$$\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,e^{-k^2} = \frac{1}{(2\pi)^4}\Big(\int_{-\infty}^{\infty}\mathrm{d}k\,e^{-k^2}\Big)^4 = \frac{(\sqrt{\pi})^4}{(2\pi)^4} = \frac{\pi^2}{16\pi^4} = \frac{1}{16\pi^2}. \tag{36.86}$$

The Dirac trace is the standard γ^5 trace of Section C.6.5. Using $\sigma^{\mu\nu}\sigma^{\rho\sigma} = -\frac{1}{4}[\gamma^\mu,\gamma^\nu][\gamma^\rho,\gamma^\sigma]$ and, under contraction with the antisymmetric $F_{\mu\nu}F_{\rho\sigma}$, replacing each commutator by twice its leading product,

$$F_{\mu\nu}F_{\rho\sigma}\,\operatorname{tr}[\gamma^5\sigma^{\mu\nu}\sigma^{\rho\sigma}] = -F_{\mu\nu}F_{\rho\sigma}\,\operatorname{tr}[\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -F_{\mu\nu}F_{\rho\sigma}\,(-4i\,\epsilon^{\mu\nu\rho\sigma}) = 4i\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}, \tag{36.87}$$

where $\operatorname{tr}[\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma] = -4i\,\epsilon^{\mu\nu\rho\sigma}$ with $\epsilon^{0123} = +1$. Assembling (36.85), (36.86) and (36.87),

$$\mathcal{A}(x) = \frac{e^2}{8}\cdot\frac{1}{16\pi^2}\cdot 4i\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = \frac{i\,e^2}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{36.88}$$

The factor of i is the bookkeeping residue of having quoted a Lorentzian γ^5 trace while computing a Euclidean (convergent) momentum integral; it is removed by the inverse Wick rotation of Section 10.6 when the operator equation (36.71) is continued back to Lorentzian signature, where the Euclidean γ^5 trace is real,

$\mathrm{tr}[\gamma^5\gamma^a\gamma^b\gamma^c\gamma^d] = -4\,\epsilon^{abcd}$. The net real coefficient—which is what the rotation back to Lorentzian signature leaves, and what the cross-check below confirms—is

$$\mathcal{A}(x) = \frac{e^2}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{36.89}$$

Substituting (36.89) into the anomalous Ward identity (36.71) and using $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 2F_{\mu\nu}\widetilde{F}^{\mu\nu}$ gives the central result of the section:

Theorem 36.9 (Chiral anomaly from the measure)**.** *For a single massless Dirac fermion of charge $-e$ in a background abelian gauge field, the chiral rotation (36.62) produces a non-invariant measure whose Jacobian (36.69), regularized gauge-invariantly by the Dirac heat kernel, yields the anomalous divergence*

$$\partial_\mu\langle j^{\mu 5}\rangle = 2\,\mathcal{A}(x) = \frac{e^2}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = \frac{e^2}{8\pi^2}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}. \tag{36.90}$$

The result coincides, coefficient and sign, with the triangle anomaly (36.50) of Theorem 36.4. The agreement is the point of the whole exercise. In Section 36.2 the coefficient was the surviving residue of a regularization ambiguity, pinned only after we imposed vector-current conservation by hand. Here there was no ambiguity to pin: gauge invariance was built into the regulator from the first line, the computation produced exactly one term, and the number (36.90) fell out with nothing left to choose. The overall sign in (36.90), the one quantity genuinely sensitive to the convention chain (the sign of D_μ, the orientation $\epsilon^{0123} = +1$, and the direction of the Wick rotation), is fixed unambiguously by matching the triangle—which was itself sign-fixed by the physical requirement that electric charge be conserved. Two independent computations, one perturbative and one non-perturbative, agree on a number that neither alone could declare scheme-independent; together they do.

36.3.5 Independence of the regulator

The heat kernel $f(s) = e^{-s}$ was a convenient choice, not a principled one. We now show that *any* admissible cutoff (36.73) gives the identical coefficient—the sharp statement that the anomaly is regulator-independent, and the cleanest proof that the number is physics rather than artifact. Repeat the steps of Theorem 36.8 with a general f. After the shift (36.80) and the rescaling $k \to Mk$, the argument of f is $-\not{D}^2/M^2 \to k^2 + \Delta$ with

$$\Delta = -\frac{2i\,k\cdot D}{M} - \frac{1}{M^2}\Big(D^2 + \frac{e}{2}\sigma^{\mu\nu}F_{\mu\nu}\Big), \tag{36.91}$$

a quantity that vanishes as $M \to \infty$. Taylor-expand f about k^2:

$$f(k^2+\Delta) = f(k^2) + f'(k^2)\,\Delta + \tfrac{1}{2}f''(k^2)\,\Delta^2 + \cdots. \tag{36.92}$$

By the same two facts as before—γ^5 needs four gammas, only σF carries gammas—the only surviving contribution comes from the two-σF piece of $\frac{1}{2}f''(k^2)\Delta^2$, namely $\frac{1}{2}f''(k^2)\cdot M^{-4}\cdot\frac{e^2}{4}(\sigma^{\mu\nu}F_{\mu\nu})^2$. The M^4 prefactor cancels the M^{-4}, and

$$\mathcal{A}(x)=\frac{e^2}{8}\,F_{\mu\nu}F_{\rho\sigma}\,\operatorname{tr}\!\left[\gamma^5\sigma^{\mu\nu}\sigma^{\rho\sigma}\right]\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,f''(k^2). \tag{36.93}$$

The Dirac trace is unchanged from (36.87); everything regulator-dependent is now packaged in the single number $\int \mathrm{d}^4k\, f''(k^2)$. Pass to radial coordinates in four Euclidean dimensions, $\mathrm{d}^4k = 2\pi^2\,k^3\,\mathrm{d}k$ (solid angle $\Omega_3 = 2\pi^2$), and substitute $t = k^2$, $k^3\,\mathrm{d}k = \frac{1}{2}t\,\mathrm{d}t$:

$$\int\mathrm{d}^4k\,f''(k^2)=2\pi^2\int_0^\infty\mathrm{d}k\,k^3f''(k^2)=\pi^2\int_0^\infty\mathrm{d}t\;t\,f''(t). \tag{36.94}$$

Integrate by parts twice, using the boundary conditions (36.73):

$$\int_0^\infty\mathrm{d}t\;t\,f''(t)=\big[tf'(t)\big]_0^\infty-\int_0^\infty\mathrm{d}t\,f'(t)=0-\big[f(t)\big]_0^\infty=-\big(f(\infty)-f(0)\big)=f(0)=1. \tag{36.95}$$

The boundary term $tf'(t)$ vanishes at both ends by the third condition in (36.73), and the remaining integral is a total derivative fixed entirely by the *endpoint values* $f(\infty)=0$ and $f(0)=1$. Therefore

$$\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,f''(k^2)=\frac{\pi^2\cdot1}{(2\pi)^4}=\frac{1}{16\pi^2}, \tag{36.96}$$

the same number as the Gaussian (36.86) (indeed $f=e^{-s}$ gives $\int_0^\infty te^{-t}\mathrm{d}t=1$, recovering (36.95)). The shape of f between its endpoints is washed out completely. The anomaly depends only on the two facts that define a regulator at all—that it equals one on the modes it keeps and zero on the modes it kills—and on nothing else. This is regulator-independence proved, not asserted: the coefficient in (36.90) is a property of the massless Dirac operator coupled to A_μ, not of the way we chose to tame its high modes.

36.3.6 The non-abelian anomaly

Everything above generalizes to a non-abelian background with no new computation, only a wider trace. Take the fermion in a representation R of a gauge group G, with Hermitian generators T^a normalized as in Chapter 2 by $\operatorname{tr}_R(T^aT^b)=\frac{1}{2}\delta^{ab}$ in the fundamental, and the covariant derivative of Yang–Mills theory, $D_\mu=\partial_\mu-igA_\mu^aT^a$, so that $[D_\mu,D_\nu]=-igF_{\mu\nu}^aT^a\equiv-i\mathcal{F}_{\mu\nu}$ with the matrix-valued field strength $\mathcal{F}_{\mu\nu}=gF_{\mu\nu}^aT^a$. The Lichnerowicz identity Theorem 36.7 runs identically, now producing $\not{D}^2=D^2-\frac{1}{2}\sigma^{\mu\nu}\mathcal{F}_{\mu\nu}$ (the sign change relative to the abelian (36.76) is the $-ig$ versus $+ie$ convention and is again

squared away). The eigenmodes φ_n now carry a gauge index, the trace in (36.75) runs over Dirac *and* gauge indices, and the only modification to Theorem 36.8 is that the two factors of $\frac{e}{2}\sigma F$ become two factors of $\frac{1}{2}\sigma\mathcal{F}$, whose product carries a trace over the representation:

$$\mathcal{A}(x) = \frac{1}{8}\,\mathrm{tr}_{\mathrm{Dirac}}\big[\gamma^5\sigma^{\mu\nu}\sigma^{\rho\sigma}\big]\;\mathrm{tr}_R\big[\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma}\big]\int\frac{\mathrm{d}^4k}{(2\pi)^4}f''(k^2) = \frac{1}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}_R\big[\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma}\big], \tag{36.97}$$

using the same Dirac trace (36.87), the same regulator-independent integral (36.96), and the post-rotation real coefficient. The anomalous divergence is twice this,

$$\partial_\mu\langle j^{\mu5}\rangle = \frac{1}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}_R\big[\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma}\big] = \frac{1}{8\pi^2}\;\mathrm{tr}_R\big[\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu}\big] = \frac{g^2\,T_R}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F^a_{\mu\nu}F^a_{\rho\sigma}, \tag{36.98}$$

where in the last form we used $\mathrm{tr}_R(T^aT^b) = T_R\,\delta^{ab}$ with the Dynkin index T_R of Chapter 2 ($T_R = \frac{1}{2}$ in the fundamental). The structure $\mathrm{tr}_R[\mathcal{F}\widetilde{\mathcal{F}}]$—the gauge-invariant "$\mathrm{tr}\,F\widetilde{F}$" built from the matrix field strength—is exactly the object the gauge-anomaly cancellation of Chapter 37 is organized around, and it is here, in the non-abelian measure computation, that Yang–Mills theory acquires it. Reducing tr_R to the symmetric invariant $d_R^{abc} = \mathrm{tr}_R(T^a\{T^b,T^c\})$ when the axial current is itself promoted to a gauge current is the bridge to Fig. 37.1; that reduction, and the demand that the resulting coefficient vanish, are the business of the next chapter.

We have now earned the anomaly a second, independent way. The Jacobian of the chiral measure, regularized by the Dirac heat kernel, reproduces the triangle coefficient exactly, with gauge invariance manifest and no routing ambiguity to resolve, and the same machinery delivers the non-abelian density $\frac{1}{32\pi^2}\epsilon\;\mathrm{tr}\,\mathcal{F}\mathcal{F}$ in one stroke. Two threads run forward from here. The regularized object $\mathrm{Tr}[\gamma^5\,e^{\not{D}^2/M^2}]$ and the local density $\mathcal{A}(x)$ are handed to Section 36.5, where the mode-pairing argument of Section 36.3.3 is sharpened into the Atiyah–Singer index theorem and $\int\mathrm{d}^4x\,\mathcal{A}$ is identified as a topological invariant—the reason the coefficient is exact to all orders. The non-abelian density $\mathrm{tr}_R[\mathcal{F}\widetilde{\mathcal{F}}]$ is handed to Chapter 37, where the requirement that an anomaly in a *gauged* chiral current cancel becomes the constraint that fixes the fermion content of every consistent chiral gauge theory.

36.4 The pion, PCAC, and the counting of colors

The two preceding sections earned the abelian anomaly twice over. From the triangle diagram (Theorem 36.4) and again from the Jacobian of the fermionic measure, we arrived at the same operator equation: the axial current of a massless

Dirac fermion of unit electric charge is not conserved but obeys

$$\partial_\mu j^{\mu 5} \;=\; \frac{e^2}{8\pi^2}\, F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{36.99}$$

with the coefficient fixed—no routing freedom, no scheme dependence—once gauge invariance of the vector current is imposed (Eq. (36.50)). Everything to this point has been theoretical bookkeeping: a symmetry of the classical action that quantization is forced to break, with a definite number sitting on the right-hand side. The purpose of this section is to confront that number with a laboratory measurement. We will show that the very same coefficient governs the decay of the neutral pion into two photons; that the amplitude, the decay rate, and a single experimental number—the width $\Gamma(\pi^0 \to \gamma\gamma) \approx 7.8$ eV—all flow from Eq. (36.99) once it is dressed with the quark content of the strong interactions; and that the agreement with experiment is impossible unless each quark flavor comes in exactly three colors. The anomaly, in other words, is not a blemish to be renormalized away. It is a measured feature of the world, and one of the cleanest pieces of evidence we have for the color quantum number of QCD.

The logic has three moving parts, and we assemble them in order. First, the relevant current is not the electron's axial current but the third axial-isospin current of the light quarks; gauging it electromagnetically and running the same triangle gives an anomaly coefficient weighted by quark charges and colors, $N_c(Q_u^2 - Q_d^2) = N_c/3$. Second, that current is the interpolating field of the neutral pion—this is the PCAC content of spontaneous chiral symmetry breaking, established in Section 29.6.2 and recalled below—so its anomalous divergence is, by a low-energy theorem, the $\pi^0 \to \gamma\gamma$ vertex. Third, squaring that vertex and folding it through the two-body phase space of Chapter 16 produces a rate proportional to N_c^2, which the data then read as a direct weighing of the number of colors.

36.4.1 The third axial-isospin current and its electromagnetic anomaly

The neutral pion is built from the light quarks u and d, and the symmetry whose breaking it signals is the axial part of isospin. Recall from Section 29.6.7 that QCD with two massless flavors has a global chiral symmetry $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, spontaneously broken by the quark condensate $\langle \bar{q} q \rangle \neq 0$ to the diagonal isospin $\mathrm{SU}(2)_V$; the three broken axial generators give three (pseudo-)Goldstone bosons, the pions, by Goldstone's theorem (Theorem 29.11). The Noether currents of the broken axial rotations are the axial-isospin currents

$$j^{\mu 5,a} \;=\; \bar{q}\,\gamma^\mu\gamma^5\, T^a\, q, \qquad T^a = \frac{\tau^a}{2}, \qquad q = \begin{pmatrix} u \\ d \end{pmatrix}, \tag{36.100}$$

with τ^a the Pauli matrices acting in flavor space and an implicit sum over the N_c colors of each quark. The neutral pion couples to the third component, whose flavor structure is $T^3 = \frac{1}{2}\,\mathrm{diag}(+1,-1)$:

$$j^{\mu 5,3} \;=\; \tfrac{1}{2}(\bar{u}\,\gamma^\mu\gamma^5\,u \;-\; \bar{d}\,\gamma^\mu\gamma^5\,d). \tag{36.101}$$

In the chiral limit $m_u = m_d = 0$ this current is classically conserved. Once we couple the quarks to the photon, however, the same mechanism that produced Eq. (36.99) for the electron operates here, and the current acquires an anomalous divergence.

The computation is the triangle of Theorem 36.4 with two modifications, both purely a matter of what sits at the vertices. The axial vertex now carries the flavor matrix T^3 rather than unit axial charge, and the two photon vertices carry the quark electric charges rather than the electron's. Write the quark electric-charge matrix as $Q = \mathrm{diag}(Q_u, Q_d)$ in units of e; from the Standard-Model assignments tabulated in Table F.1,

$$Q_u = +\tfrac{2}{3}, \qquad Q_d = -\tfrac{1}{3}. \tag{36.102}$$

Each quark of flavor f and a definite color contributes to the triangle exactly as a single Dirac fermion does in Eq. (36.99), except that its axial vertex is weighted by the eigenvalue $(T^3)_{ff}$ and its two photon vertices each by Q_f; summing over the two flavors, and over the N_c identical colors (the photon and the flavor matrices are color-blind, so every color contributes the same triangle), the anomaly coefficient becomes the flavor trace

$$N_c\,\mathrm{tr}(T^3Q^2) \;=\; N_c\Big[(T^3)_{uu}\,Q_u^2 + (T^3)_{dd}\,Q_d^2\Big] \;=\; N_c\Big[\tfrac{1}{2}Q_u^2 - \tfrac{1}{2}Q_d^2\Big] \;=\; \tfrac{1}{2}\,N_c\,(Q_u^2 - Q_d^2). \tag{36.103}$$

The replacement of the single e^2 in Eq. (36.99) by $e^2\,N_c\,\mathrm{tr}(T^3Q^2)$ is the whole content of the dressing: the kinematic factor, the $\widetilde{F}$ structure, and the $1/8\pi^2$ are universal and were fixed once in Theorem 36.4. Hence the gauged third axial-isospin current obeys

$$\boxed{\partial_\mu j^{\mu 5,3} \;=\; \frac{e^2}{16\pi^2}\,N_c\,(Q_u^2 - Q_d^2)\;F_{\mu\nu}\widetilde{F}^{\mu\nu}} \tag{36.104}$$

in the chiral limit, where $F_{\mu\nu}$ is now the electromagnetic field strength. The factor of $\frac{1}{2}$ relative to the naive $\frac{e^2}{8\pi^2}$ of Eq. (36.99) is precisely the $\mathrm{tr}(T^3Q^2)$ weight of Eq. (36.103) with the color factor stripped, and we have used $F\widetilde{F} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$.

The arithmetic of the charges is the linchpin of the whole story, so we record it explicitly:

$$Q_u^2 - Q_d^2 \;=\; \Big(\tfrac{2}{3}\Big)^2 - \Big(\tfrac{1}{3}\Big)^2 \;=\; \tfrac{4}{9} - \tfrac{1}{9} \;=\; \tfrac{1}{3}, \qquad \Longrightarrow \qquad N_c\,(Q_u^2 - Q_d^2) = \frac{N_c}{3}. \tag{36.105}$$

For the physical value $N_c = 3$ this anomaly factor equals exactly 1; for a hypothetical world with $N_c = 1$ it would equal $\frac{1}{3}$. The decay rate, we will see, is proportional to the square of this number, so the two hypotheses differ by a factor of nine—the gap that the measurement resolves decisively.

Notation Watch

Two epsilons live in this section and must not be confused. The Levi-Civita symbol $\epsilon^{\mu\nu\rho\sigma}$, with the orientation $\epsilon^{0123} = +1$ fixed in Section A.2.5, enters the dual field strength $\widetilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$. The photon polarization vectors are written $\varepsilon_1^\mu, \varepsilon_2^\mu$. We keep $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and the mostly-minus metric throughout, so that lowering all four indices of the Levi-Civita symbol costs a sign, $\epsilon_{\mu\nu\rho\sigma} = -\epsilon^{\mu\nu\rho\sigma}$ (Eq. (A.42)); this sign will reappear when we contract two of them in the rate.

36.4.2 PCAC: the pion as the interpolating field of the axial current

The anomaly equation Eq. (36.104) is an operator statement about quarks and photons. To turn it into a statement about the pion we need the bridge between the quark-level axial current and the hadron-level pion field. That bridge is PCAC, the partially conserved axial current, and it is nothing more than the Goldstone structure of Chapter 29 read in the language of currents.

Spontaneous breaking of the axial symmetry means the axial current creates a Goldstone boson out of the vacuum. Quantitatively (Eq. (29.202)),

$$\left\langle 0\right| j^{\mu 5,a}(0)\left|\pi^b(p)\right\rangle \;=\; i\,f_\pi\,p^\mu\,\delta^{ab}, \tag{36.106}$$

where $f_\pi \approx 92$ MeV is the pion decay constant and $\left|\pi^b(p)\right\rangle$ is a one-pion state of momentum p. The matrix element is nonzero precisely because the symmetry is spontaneously broken; in a symmetric (Wigner) realization it would vanish. Taking the divergence of Eq. (36.106)—which in momentum space means contracting with $-ip_\mu$ for the outgoing pion—gives

$$\left\langle 0\right| \partial_\mu j^{\mu 5,a}(0)\left|\pi^b(p)\right\rangle \;=\; (-ip_\mu)\,(i\,f_\pi\,p^\mu)\,\delta^{ab} \;=\; f_\pi\,p^2\,\delta^{ab} \;=\; f_\pi\,m_\pi^2\,\delta^{ab}, \tag{36.107}$$

using $p^2 = m_\pi^2$ on shell. This is the precise sense in which the divergence of the axial current *is* the pion: as an operator equation between the relevant states it acts as a field that creates a single pion with strength $f_\pi m_\pi^2$. We summarize it by writing

$$\partial_\mu j^{\mu 5,3} \;=\; f_\pi\,m_\pi^2\,\pi^0 \qquad \text{(interpolating-field statement)}, \tag{36.108}$$

where π^0 is the canonically normalized neutral-pion field, $\langle 0|\pi^0(0)|\pi^0(p)\rangle = 1$, and the equality holds as a statement about matrix elements dominated by the one-pion contribution. Equation Eq. (36.108) is the operator face of the Gell-Mann–Oakes–Renner relation $f_\pi^2 m_\pi^2 = -(m_u + m_d)\,\langle \bar{q}q\rangle$ derived in Eq. (29.228): the right-hand side, traced back to the explicit quark-mass breaking, is what gives the divergence its nonzero matrix element. In the exact chiral limit $m_\pi \to 0$ the right-hand side vanishes and the current is conserved between these states, exactly as the Goldstone theorem demands. The systematic, all-orders organization of these statements into an effective field theory of pions is the chiral-perturbation-theory programme of Chapter 39; here we need only the leading relation Eq. (36.108).

There is now an evident tension, and resolving it is the heart of the matter. Naive PCAC—the relation Eq. (36.108) *without* the anomaly—would say that the divergence of the axial current vanishes in the chiral limit even in the presence of photons, since $m_\pi^2 \to 0$. By the low-energy theorem this would force the $\pi^0 \to \gamma\gamma$ amplitude to vanish as the pion momentum is taken soft. This is the Sutherland–Veltman theorem, and taken at face value it predicts a neutral pion that cannot decay to two photons—in flat contradiction with the dominant observed decay mode. The resolution is the anomaly: the true divergence is not Eq. (36.108) but Eq. (36.104), and the anomalous $F\widetilde{F}$ term survives the chiral limit. Combining the two contributions, the complete divergence of the gauged axial-isospin current is

$$\partial_\mu j^{\mu 5,3} \;=\; f_\pi\, m_\pi^2\, \pi^0 \;+\; \frac{e^2}{16\pi^2}\, N_c\, (Q_u^2 - Q_d^2)\, F_{\mu\nu}\widetilde{F}^{\mu\nu}. \tag{36.109}$$

The first term is the spontaneous-breaking (Goldstone) piece, the engine of the pion; the second is the anomaly, the engine of its decay. They are independent, and it is the second that the experiment measures.

Physical Insight

The neutral pion would be very nearly stable if the world respected naive PCAC. The Sutherland–Veltman theorem is the precise statement that a soft Goldstone boson decouples, and a pion in the chiral limit is exactly such a boson. What makes $\pi^0 \to \gamma\gamma$ fast—the lifetime is about 8×10^{-17} s, and the branching ratio to two photons is 98.8%—is that the axial current it interpolates is anomalous. The decay is, quite literally, the anomaly made visible: switch off the $F\widetilde{F}$ term and the dominant decay channel of the lightest hadron disappears. This is the cleanest example in physics of a quantum effect with no classical shadow controlling a leading-order observable.

36.4.3 Assembling the amplitude

We now extract the $\pi^0 \to \gamma\gamma$ amplitude from Eq. (36.109). The cleanest route, and the one that makes the matching to the chiral effective theory of Chapter 39 transparent, is to read Eq. (36.109) as fixing a local effective coupling of the pion field to two photons, and then to check that this coupling reproduces the anomalous divergence exactly.

Posit that, at low energy, the neutral pion couples to the electromagnetic field through the local, gauge-invariant, parity-conserving interaction

$$\mathcal{L}_{\pi^0\gamma\gamma} \;=\; g_{\pi\gamma\gamma}\,\pi^0\,F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{36.110}$$

with a single constant $g_{\pi\gamma\gamma}$ to be fixed. The structure $\pi^0 F\widetilde{F}$ is the unique operator of lowest dimension carrying the quantum numbers of the process: the pion is a pseudoscalar, $F\widetilde{F}$ is the unique parity-odd scalar quadratic in the field strength, and the product is a genuine scalar. We fix $g_{\pi\gamma\gamma}$ by demanding that the axial-isospin current of the low-energy theory have the anomalous divergence Eq. (36.109). In the effective theory the axial current is dominated by its pion piece, which by Eq. (36.106) is $j^{\mu 5,3} \supset -f_\pi\,\partial^\mu\pi^0$. Its divergence is therefore $-f_\pi\,\Box\pi^0$, and the pion equation of motion following from the kinetic, mass, and interaction terms,

$$(\Box + m_\pi^2)\pi^0 \;=\; g_{\pi\gamma\gamma}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}, \tag{36.111}$$

gives $\Box\pi^0 = -m_\pi^2\pi^0 + g_{\pi\gamma\gamma}F\widetilde{F}$. Hence

$$\partial_\mu j^{\mu 5,3} \;=\; -f_\pi\,\Box\pi^0 \;=\; f_\pi\,m_\pi^2\,\pi^0 \;-\; f_\pi\,g_{\pi\gamma\gamma}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}. \tag{36.112}$$

The pion-mass term reproduces the interpolating-field piece of Eq. (36.109) automatically. Matching the $F\widetilde{F}$ coefficients between Eq. (36.112) and Eq. (36.109) fixes the coupling:

$$-\,f_\pi\,g_{\pi\gamma\gamma} \;=\; \frac{e^2}{16\pi^2}\,N_c\,(Q_u^2 - Q_d^2) \qquad\Longrightarrow\qquad g_{\pi\gamma\gamma} \;=\; -\,\frac{e^2}{16\pi^2 f_\pi}\,N_c\,(Q_u^2 - Q_d^2). \tag{36.113}$$

The overall sign of $g_{\pi\gamma\gamma}$ depends on the sign convention in the current–pion matrix element and is not observable; only its magnitude enters the rate. With the coupling fixed, the decay amplitude follows from the Feynman rule of the vertex Eq. (36.110). Writing $F\widetilde{F} = \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 2\,\epsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\,\partial_\rho A_\sigma$ and attaching two outgoing photons of momenta k_1, k_2 and polarizations $\varepsilon_1, \varepsilon_2$—the two distinct assignments of photons to the two derivative factors supply a symmetry factor 2!, and each $\partial_\mu A_\nu$ becomes $ik_\mu\varepsilon^*_\nu$—we obtain

$$i\mathcal{M} \;=\; i\,g_{\pi\gamma\gamma}\cdot 2\cdot 2!\;\epsilon^{\mu\nu\rho\sigma}(ik_{1\mu}\varepsilon^*_{1\nu})(ik_{2\rho}\varepsilon^*_{2\sigma}) \;=\; -\,4i\,g_{\pi\gamma\gamma}\,\epsilon^{\mu\nu\rho\sigma}k_{1\mu}\varepsilon^*_{1\nu}k_{2\rho}\varepsilon^*_{2\sigma}. \tag{36.114}$$

Inserting Eq. (36.113) and reordering the antisymmetric indices (two transpositions, hence no net sign) into the canonical pattern $\varepsilon_1^* k_1 \varepsilon_2^* k_2$, the amplitude is

$$\boxed{\mathcal{M}(\pi^0 \to \gamma\gamma) \;=\; \frac{e^2\,N_c}{12\pi^2 f_\pi}\,\epsilon^{\mu\nu\rho\sigma}\,\varepsilon^*_{1\mu}\,k_{1\nu}\,\varepsilon^*_{2\rho}\,k_{2\sigma}} \tag{36.115}$$

where we have used $4g_{\pi\gamma\gamma} = \frac{e^2 N_c}{4\pi^2 f_\pi}(Q_u^2 - Q_d^2) = \frac{e^2 N_c}{12\pi^2 f_\pi}$ by Eq. (36.105), and dropped the unobservable overall sign. It is convenient to name the scalar prefactor

$$A \;\equiv\; \frac{e^2\,N_c}{12\pi^2 f_\pi} \;=\; \frac{\alpha\,N_c}{3\pi f_\pi}, \qquad \alpha = \frac{e^2}{4\pi}, \tag{36.116}$$

so that $\mathcal{M} = A\,\epsilon^{\mu\nu\rho\sigma}\varepsilon^*_{1\mu}k_{1\nu}\varepsilon^*_{2\rho}k_{2\sigma}$. The amplitude Eq. (36.115) is the central result of this section and the one the rest of the book reaches for: it is reproduced from the chiral effective Lagrangian, by an entirely independent route, in Eq. (39.170), and the agreement of the microscopic quark-triangle derivation with the effective-meson derivation is a concrete instance of anomaly matching (Chapter 37).

Remark 36.10. The same number A can be read off directly from the two-photon matrix element of the anomaly operator. Computing $\big\langle \gamma(k_1,\varepsilon_1)\gamma(k_2,\varepsilon_2)|F\widetilde{F}(0)|0\big\rangle = -4\,\epsilon^{\mu\nu\rho\sigma}\varepsilon^*_{1\mu}k_{1\nu}\varepsilon^*_{2\rho}k_{2\sigma}$ (the factor -4 is the same combinatorics as in Eq. (36.114)) and feeding it through the interpolating-field relation Eq. (36.108) reproduces Eq. (36.115) term for term. The effective-Lagrangian bookkeeping we used is simply the most economical way to carry that low-energy theorem without tracking the pion pole explicitly.

36.4.4 The decay rate and the counting of colors

The amplitude in hand, the decay rate is a direct application of the two-body machinery of Chapter 16. We carry every step, including the photon polarization sum, the Levi-Civita contraction, and the identical-particle symmetry factor, and then put in numbers.

Worked Example 36.11 (The π^0 width and the number of colors)**.** We compute $\Gamma(\pi^0 \to \gamma\gamma)$ from Eq. (36.115) and confront it with experiment.

Polarization sum. The pion is spinless, so there is no initial-state average; we sum the squared amplitude over the two physical polarizations of each final photon. Because the amplitude is transverse—it is built from $\epsilon^{\mu\nu\rho\sigma}k_{1\nu}$ and $\epsilon^{\mu\nu\rho\sigma}k_{2\sigma}$, which annihilate any term in the polarization sum proportional to the corresponding photon momentum—we may use the replacement $\sum_{\text{pol}} \varepsilon^*_\mu \varepsilon_{\mu'} \to -g_{\mu\mu'}$ for each photon, the gauge-dependent remainder dropping out by the antisymmetry of ϵ.

Thus

$$\sum_{\text{pol}} |\mathcal{M}|^2 = A^2\, \epsilon^{\mu\nu\rho\sigma} \epsilon^{\mu'\nu'\rho'\sigma'} \left(-g_{\mu\mu'}\right) k_{1\nu} k_{1\nu'} \left(-g_{\rho\rho'}\right) k_{2\sigma} k_{2\sigma'}$$
$$= A^2\, \epsilon^{\mu\nu\rho\sigma}\, \epsilon_{\mu}{}^{\nu'}{}_{\rho}{}^{\sigma'}\, k_{1\nu} k_{1\nu'}\, k_{2\sigma} k_{2\sigma'}. \tag{36.117}$$

The double contraction of two Levi-Civita tensors over the first and third slots, in the mostly-minus metric and with the orientation of Section A.2.5, is

$$\epsilon^{\mu\nu\rho\sigma}\, \epsilon_{\mu\beta\rho\delta} \;=\; -2(\delta^{\nu}_{\beta}\, \delta^{\sigma}_{\delta} - \delta^{\nu}_{\delta}\, \delta^{\sigma}_{\beta}), \tag{36.118}$$

the overall minus sign being the Lorentzian sign flagged in Eq. (A.42) (in Euclidean signature the right-hand side would carry $+2$). Substituting Eq. (36.118) into Eq. (36.117),

$$\sum_{\text{pol}} |\mathcal{M}|^2 = -2A^2(\delta^{\nu}_{\beta}\delta^{\sigma}_{\delta} - \delta^{\nu}_{\delta}\delta^{\sigma}_{\beta}) k_{1\nu} k_1^{\beta} k_{2\sigma} k_2^{\delta}$$
$$= -2A^2\Big[(k_1{\cdot}k_1)(k_2{\cdot}k_2) - (k_1{\cdot}k_2)^2\Big] \;=\; 2A^2\,(k_1{\cdot}k_2)^2, \tag{36.119}$$

where the first product vanishes because the photons are massless, $k_1^2 = k_2^2 = 0$. The remaining invariant is fixed by the pion mass: from $(k_1 + k_2)^2 = m_\pi^2$ and $k_i^2 = 0$ we read $2\, k_1{\cdot}k_2 = m_\pi^2$, so $(k_1{\cdot}k_2)^2 = m_\pi^4/4$ and

$$\sum_{\text{pol}} |\mathcal{M}|^2 \;=\; \frac{A^2\, m_\pi^4}{2}. \tag{36.120}$$

The result is independent of the emission angle, as it must be for a spinless parent at rest.

Phase space, flux, and the symmetry factor. The decay rate of a scalar at rest into two particles is the workhorse formula Eq. (16.89),

$$\Gamma \;=\; \mathcal{S}\, \frac{|\mathbf{k}|}{8\pi\, m_\pi^2}\, \overline{|\mathcal{M}|^2}, \tag{36.121}$$

where $|\mathbf{k}|$ is the momentum of either final particle in the rest frame, $\overline{|\mathcal{M}|^2}$ is the appropriately summed squared amplitude, and $\mathcal{S}$ is the identical-particle symmetry factor. Two distinct features of $\pi^0 \to \gamma\gamma$ must be entered correctly. First, the two photons are massless, so by Eq. (16.87) the released momentum is $|\mathbf{k}| = m_\pi/2$. Second, the two photons are *identical* bosons, so we must avoid double-counting the final-state configurations: $\mathcal{S} = \tfrac{1}{2}$. With $\overline{|\mathcal{M}|^2} = \sum_{\text{pol}} |\mathcal{M}|^2$ from Eq. (36.120) (there is no spin average to divide by, only the photon sum to include), assembling the pieces gives

$$\Gamma \;=\; \frac{1}{2} \cdot \frac{m_\pi/2}{8\pi\, m_\pi^2} \cdot \frac{A^2 m_\pi^4}{2} \;=\; \frac{A^2\, m_\pi^3}{64\pi}. \tag{36.122}$$

Finally, substituting the prefactor $A = \alpha N_c/3\pi f_\pi$ of Eq. (36.116) and using $\alpha = e^2/4\pi$ throughout,

$$\boxed{\Gamma(\pi^0 \to \gamma\gamma) \;=\; \frac{\alpha^2\, m_\pi^3\, N_c^2}{576\,\pi^3 f_\pi^2}} \tag{36.123}$$

the coefficient $576 = 64 \cdot 9$ carrying the $\frac{1}{9}$ of $(Q_u^2 - Q_d^2)^2 = \frac{1}{9}$ that we have absorbed into the explicit N_c^2. The rate is proportional to N_c^2: the number of colors is squared in the width because it appears linearly in the amplitude.

Numerical evaluation. We insert the measured low-energy constants $\alpha = 1/137.04$, the neutral-pion mass $m_\pi = m_{\pi^0} = 134.98$ MeV, and the pion decay constant $f_\pi = 92.1$ MeV, and take $N_c = 3$. The amplitude prefactor is

$$A = \frac{\alpha N_c}{3\pi f_\pi} = \frac{(1/137.04)\cdot 3}{3\pi \cdot (0.0921\ \text{GeV})} = 2.53 \times 10^{-2}\ \text{GeV}^{-1}, \tag{36.124}$$

and the width from Eq. (36.122) is

$$\Gamma = \frac{A^2 m_\pi^3}{64\pi} = \frac{(2.53\times 10^{-2})^2\,(0.13498)^3}{64\pi}\ \text{GeV} = 7.8 \times 10^{-9}\ \text{GeV} = 7.8\ \text{eV}. \tag{36.125}$$

The measured value (the PrimEx experiment and the Particle Data Group average, mid-2020s) is $\Gamma(\pi^0 \to \gamma\gamma) = 7.80 \pm 0.12$ eV. The leading-order anomaly prediction lands on the central experimental value to within the experimental uncertainty—a striking success for a one-line low-energy theorem with no free parameters.

The count of colors. Because $\Gamma \propto N_c^2$, the measurement *weighs* the number of colors. With $N_c = 3$ the prediction is 7.8 eV. With $N_c = 1$—a single color, as in a theory of quarks without the color quantum number—the amplitude is three times smaller and the rate

$$\Gamma\big|_{N_c=1} = \frac{1}{9}\,\Gamma\big|_{N_c=3} \approx 0.87\ \text{eV}, \tag{36.126}$$

a factor of nine below experiment and utterly excluded; $N_c = 2$ would give $\left(\frac{2}{3}\right)^2 \cdot 7.8 \approx 3.5$ eV, also far off. Only $N_c = 3$ reproduces the observed width, and it does so at the percent level. The radiative decay of the neutral pion is thus a direct, low-energy determination of the number of quark colors, in complete accord with the value extracted independently from the hadronic cross-section ratio R (Chapter 28) and from the QCD beta function of *Yang–Mills theory* (Eq. (27.101)). Three threads—the electromagnetic current of QED that supplies the two photon vertices, the strong-interaction color of Yang–Mills theory that supplies the factor N_c, and the spontaneously broken chiral symmetry of Chapter 29 that supplies the pion—meet in this single number.

Historical Note

When the neutral pion's two-photon decay was first confronted with current algebra in the mid-1960s, the Sutherland–Veltman theorem appeared to forbid it, and the puzzle was sharp enough to be taken seriously as evidence against the soft-pion methods then in vogue. Adler, and Bell and Jackiw, resolved it in 1969 by recognizing that the axial current is anomalous: the triangle diagram contributes a term that the formal current-algebra manipulations had silently dropped. The same calculation, performed with the constituent quarks of the nascent quark model, gave a rate three times too small unless each quark carried a threefold color multiplicity. The π^0 width was, historically, one of the first quantitative arguments for color, predating the discovery of asymptotic freedom and the establishment of QCD as the theory of the strong interactions.

The lesson to carry forward is that this anomaly is not a defect to be cancelled but a physical, measured, scheme-independent number. The axial isospin symmetry is a *global* symmetry; its anomaly destroys nothing—no Ward identity that protects unitarity or renormalizability is at stake—and instead delivers a clean prediction confirmed in the laboratory. That harmlessness is exactly what distinguishes it from the *gauge* anomalies of Chapter 37, where an analogous triangle would wreck the consistency of the theory and must be made to vanish. The very robustness that makes the global anomaly a reliable measuring stick—its independence of the regulator, the coupling, and the renormalization scheme—is the symptom of a deeper, topological rigidity. The next section makes that rigidity precise: the integrated anomaly is a winding number, equal to the Atiyah–Singer index of the Dirac operator, and it is this topological identity that explains why the coefficient we just measured is exact to all orders.

36.5 Topology: the index theorem and why the anomaly is exact

The Fujikawa calculation of Section 36.3 handed us two objects that we have so far used only locally. The first is the regularized trace of γ^5 over the Dirac spectrum,

$$\mathcal{I}(M) \;\equiv\; \mathrm{Tr}\big[\gamma^5\, e^{-(i\not{D})^2/M^2}\big] \;=\; \sum_n \langle n|\gamma^5|n\rangle\, e^{-\lambda_n^2/M^2}, \tag{36.127}$$

the sum running over a complete set of eigenstates $i\not{D}\,\phi_n = \lambda_n \phi_n$ of the (Euclidean, Hermitian) Dirac operator. The second is the local anomaly density it produces, the heat-kernel limit

$$\mathcal{A}(x) \;=\; \lim_{M\to\infty} \mathrm{tr}\big[\gamma^5\, \langle x|\, e^{-(i\not{D})^2/M^2}\, |x\rangle\big] \;=\; \frac{1}{32\pi^2}\, \epsilon^{\mu\nu\rho\sigma}\, \mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}), \tag{36.128}$$

whose integral against a constant axial rotation reproduced the divergence of the axial current. Each was computed for a *fixed* background gauge field, and at that level the anomaly looks like one more finite, scheme-fixed number among the many this chapter has produced. The purpose of this section is to show that it is not one more number. The integral of Eq. (36.128) over all of spacetime is an *integer* — the same integer that counts the net chirality of the gauge field's zero modes through the Atiyah–Singer index theorem, and the same integer that classifies the gauge field's behaviour at infinity through homotopy theory. Three apparently unrelated quantities — a regulator-dependent Jacobian, a winding number of the connection, and a count of solutions to a differential equation — are forced to coincide, and the coincidence is what makes the anomaly coefficient *exact*. A smooth function of the coupling cannot equal an integer for all couplings unless it is constant; this is the structural reason behind the Adler–Bardeen theorem, and it is the bridge from the anomaly to instantons and the θ vacuum of Chapter 42.

36.5.1 The topological charge as a total derivative

We begin with the density on the right of Eq. (36.128), stripped of its fermionic origin and regarded purely as a functional of the gauge field. Throughout this subsection we use the *geometric* normalization of the connection, in which the coupling is absorbed into the field, $\mathcal{A}_\mu \equiv g A^a_\mu T^a$, so that the matrix-valued field strength

$$F_{\mu\nu} \;=\; \partial_\mu \mathcal{A}_\nu - \partial_\nu \mathcal{A}_\mu - i\,[\mathcal{A}_\mu, \mathcal{A}_\nu] \;=\; \frac{i}{1}\,[D_\mu, D_\nu]\,\Big|_{D_\mu = \partial_\mu - i\mathcal{A}_\mu} \tag{36.129}$$

carries no explicit g. This is the curvature two-form of Eq. (25.102) with the coupling rescaled away; the physical, g-explicit field strength is recovered by $F_{\mu\nu} \to g F^{\text{phys}}_{\mu\nu}$, and we restore the coupling at the end. In this normalization the object of interest is the *Chern–Pontryagin density*

$$q(x) \;\equiv\; \frac{1}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}), \tag{36.130}$$

with the trace over the gauge representation and the Levi-Civita convention $\epsilon^{0123} = +1$ fixed in the chapter introduction. The central structural fact is that q is a total derivative.

Theorem 36.12 (The Chern–Simons current)**.** *The Chern–Pontryagin density Eq. (36.130) is the divergence of the Chern–Simons current*

$$K^\mu \;=\; \frac{1}{8\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}\Big(\mathcal{A}_\nu\,\partial_\rho\mathcal{A}_\sigma - \tfrac{2i}{3}\,\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma\Big), \tag{36.131}$$

that is, $q(x) = \partial_\mu K^\mu(x)$. In the abelian theory the cubic term is absent and $K^\mu = \frac{1}{8\pi^2}\epsilon^{\mu\nu\rho\sigma}\mathcal{A}_\nu\partial_\rho\mathcal{A}_\sigma$.

The proof is a direct computation, and because every sign in it matters we carry it in full as a worked example before drawing consequences.

Worked Example 36.13 ($\epsilon\ \mathrm{tr}\, FF$ is a total derivative)**.** We expand $\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma})$ using Eq. (36.129) and organize the result by powers of $\mathcal{A}$. Write $F_{\mu\nu} = (\partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu) - i[\mathcal{A}_\mu,\mathcal{A}_\nu]$. There are three groups of terms.

Quadratic in $\mathcal{A}$. The product of the two abelian pieces gives

$$\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big[(\partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu)(\partial_\rho\mathcal{A}_\sigma - \partial_\sigma\mathcal{A}_\rho)\big] \;=\; 4\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu\,\partial_\rho\mathcal{A}_\sigma), \tag{36.132}$$

the four sign-combinations collapsing to one because each index swap $\mu\leftrightarrow\nu$ or $\rho\leftrightarrow\sigma$ is compensated by the antisymmetry of ϵ. This is already a total derivative: since $\epsilon^{\mu\nu\rho\sigma}\,\partial_\rho\partial_\mu\mathcal{A}_\sigma = 0$ by the symmetry of the double derivative against the antisymmetry of ϵ,

$$4\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu\,\partial_\rho\mathcal{A}_\sigma) \;=\; 4\,\partial_\mu\big[\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\mathcal{A}_\nu\,\partial_\rho\mathcal{A}_\sigma)\big]. \tag{36.133}$$

Cubic in $\mathcal{A}$. The cross terms between the abelian and commutator pieces give, using $\epsilon^{\mu\nu\rho\sigma}(\partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu) = 2\,\epsilon^{\mu\nu\rho\sigma}\partial_\mu\mathcal{A}_\nu$ and $\epsilon^{\mu\nu\rho\sigma}[\mathcal{A}_\rho,\mathcal{A}_\sigma] = 2\,\epsilon^{\mu\nu\rho\sigma}\mathcal{A}_\rho\mathcal{A}_\sigma$,

$$\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big[-i(\partial_\mu\mathcal{A}_\nu - \partial_\nu\mathcal{A}_\mu)[\mathcal{A}_\rho,\mathcal{A}_\sigma] - i[\mathcal{A}_\mu,\mathcal{A}_\nu](\partial_\rho\mathcal{A}_\sigma - \partial_\sigma\mathcal{A}_\rho)\big] \;=\; -8i\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu\,\mathcal{A}_\rho\mathcal{A}_\sigma), \tag{36.134}$$

the two terms combining because relabelling $(\mu\nu)\leftrightarrow(\rho\sigma)$ and using cyclicity of the trace together with $\epsilon^{\rho\sigma\mu\nu} = \epsilon^{\mu\nu\rho\sigma}$ makes them equal. Now observe

$$\partial_\mu\big[\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma)\big] = \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu\,\mathcal{A}_\rho\mathcal{A}_\sigma + \mathcal{A}_\nu\,\partial_\mu\mathcal{A}_\rho\,\mathcal{A}_\sigma + \mathcal{A}_\nu\mathcal{A}_\rho\,\partial_\mu\mathcal{A}_\sigma) = 3\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu \tag{36.135}$$

the three contributions being equal after a cyclic relabelling of the dummy indices and use of trace cyclicity, so that $\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\partial_\mu\mathcal{A}_\nu\,\mathcal{A}_\rho\mathcal{A}_\sigma) = \frac{1}{3}\partial_\mu[\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma)]$. Hence Eq. (36.134) equals $-\frac{8i}{3}\partial_\mu[\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma)]$.

Quartic in $\mathcal{A}$. The product of the two commutator pieces gives $-\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}([\mathcal{A}_\mu,\mathcal{A}_\nu][\mathcal{A}_\rho,\mathcal{A}_\sigma]) = -4\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(\mathcal{A}_\mu\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma)$, and this *vanishes*. Under the cyclic relabelling $\mu\to\nu\to\rho\to\sigma\to\mu$ the trace is invariant but $\epsilon^{\nu\rho\sigma\mu} = -\epsilon^{\mu\nu\rho\sigma}$ (a cyclic permutation of four indices is odd), so the quartic term equals minus itself and is zero.

Assembling the surviving quadratic and cubic pieces,

$$\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}) \;=\; 4\,\partial_\mu\Big[\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big(\mathcal{A}_\nu\partial_\rho\mathcal{A}_\sigma - \tfrac{2i}{3}\mathcal{A}_\nu\mathcal{A}_\rho\mathcal{A}_\sigma\big)\Big]. \tag{36.136}$$

Dividing by $32\pi^2$ gives $q = \partial_\mu K^\mu$ with K^μ exactly Eq. (36.131), which proves Theorem 36.12. In the abelian theory $[\mathcal{A}_\mu,\mathcal{A}_\nu] = 0$ kills the cubic and quartic terms, leaving the bare quadratic Chern–Simons current quoted in the statement.

That q is a total derivative is the first half of the story; that its *integral* is nonetheless not always zero is the second. Because $q = \partial_\mu K^\mu$, the integrated topological charge reduces to a boundary flux,

$$\nu \;\equiv\; \int_{\mathbb{R}^4}\mathrm{d}^4x\; q(x) \;=\; \int_{\mathbb{R}^4}\mathrm{d}^4x\;\partial_\mu K^\mu \;=\; \oint_{S^3_\infty}\mathrm{d}\Sigma_\mu\; K^\mu, \tag{36.137}$$

the surface being the three-sphere at spatial-and-temporal infinity in Euclidean space. Whether this vanishes depends entirely on the gauge group, and here the abelian and non-abelian theories part company.

The abelian case. For a $U(1)$ field on $\mathbb{R}^4$ with finite Euclidean action the field strength must fall off fast enough that $\int F^2 < \infty$, which forces $F_{\mu\nu} = O(r^{-3})$ and permits $\mathcal{A}_\mu = O(r^{-2})$. Then $K^\mu \sim \mathcal{A}\,\partial\mathcal{A} = O(r^{-5})$, while the area of S^3_∞ grows only as r^3, so the boundary flux Eq. (36.137) vanishes: $\nu = 0$. The deeper reason is topological. A finite-action abelian configuration approaches a *pure gauge* $\mathcal{A}_\mu \to i\,U\partial_\mu U^{-1}$ at infinity with $U = e^{i\alpha(\hat{x})} \in U(1)$, and the boundary data is a map $S^3 \to U(1) \cong S^1$; since $\pi_3(S^1) = 0$ every such map is contractible, the winding is trivial, and the integrated charge is zero on $\mathbb{R}^4$. (On a compact manifold with nontrivial topology — a four-torus threaded by abelian flux, say — the first Chern numbers can be nonzero, but on $\mathbb{R}^4$ the abelian θ term is a pure surface term with no bulk content. This is why the abelian axial anomaly, although locally real and physically measured in $\pi^0 \to \gamma\gamma$, integrates to zero over $\mathbb{R}^4$.)

The non-abelian case. For a non-abelian group the boundary data $U : S^3 \to G$ is governed by $\pi_3(G)$, and for every simple compact Lie group $\pi_3(G) = \mathbb{Z}$ (the homotopy classification is established in Section 41.2). A finite-action configuration tends to a pure gauge $\mathcal{A}_\mu \to i\,U\partial_\mu U^{-1}$ at infinity, and the integer labelling its homotopy class is exactly the boundary flux. To see this, set $F_{\mu\nu} = 0$ in K^μ; only the cubic term survives, and writing $G_\mu \equiv U\partial_\mu U^{-1}$ (an anti-Hermitian Lie-algebra element) one finds, using $\partial_\rho U = -G_\rho U$ so that under the antisymmetric ϵ the derivative $\partial_\rho G_\sigma \to -G_\rho G_\sigma$,

$$K^\mu\big|_{\text{pure gauge}} = \frac{1}{8\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\Big[(iG_\nu)(-iG_\rho G_\sigma) - \tfrac{2i}{3}(iG_\nu)(iG_\rho)(iG_\sigma)\Big] = \frac{1}{24\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(G_\nu G_\rho G_\sigma), \tag{36.138}$$

the two contributions combining as $i\cdot(-i) + (-\tfrac{2i}{3})\cdot i^3 = 1 - \tfrac{2}{3} = \tfrac{1}{3}$ times $\mathrm{tr}(G_\nu G_\rho G_\sigma)$ after the substitution. Inserting this into Eq. (36.137) gives the winding-number integral

$$\boxed{\nu \;=\; \frac{1}{24\pi^2}\oint_{S^3_\infty} \mathrm{d}\Sigma_\mu\;\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}\big[(U\partial_\nu U^{-1})(U\partial_\rho U^{-1})(U\partial_\sigma U^{-1})\big] \;\in\; \mathbb{Z}.} \tag{36.139}$$

The right-hand side is the degree of the map $U : S^3 \to G$ — the number of times U covers the group manifold as $\hat{x}$ sweeps the sphere at infinity — and is therefore an integer for purely topological reasons. We have arrived at the first promised identity:

$$\frac{1}{32\pi^2}\int \mathrm{d}^4x\;\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}) \;=\; \nu \;\in\; \mathbb{Z}. \tag{36.140}$$

Restoring the coupling through $F_{\mu\nu} \to g F^{\text{phys}}_{\mu\nu}$, the same integer reads $\nu = \frac{g^2}{32\pi^2}\int \mathrm{d}^4x\, \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(F^{\text{phys}}_{\mu\nu}F^{\text{phys}}_{\rho\sigma})$, and the integrated abelian/covariant anomaly of Eqs. (36.50) and (36.159) becomes, for a single Dirac fermion in the fundamental,

$$\Delta Q_5 \;\equiv\; \int \mathrm{d}^4x\; \partial_\mu j^{\mu 5} \;=\; \frac{g^2}{16\pi^2}\int \mathrm{d}^4x\; \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big(F^{\text{phys}}_{\mu\nu}F^{\text{phys}}_{\rho\sigma}\big) \;=\; 2\nu. \tag{36.141}$$

The factor of two is the same one that distinguishes $j^{\mu 5} = \bar\psi\gamma^\mu\gamma^5\psi$ from the chiral charge of a single Weyl component; we will see it again, now as a count of zero modes. The axial charge of the vacuum is not conserved across a gauge field of winding ν: it jumps by 2ν units.

Notation Watch

The topological subsection uses the geometric connection $\mathcal{A}_\mu = gA^a_\mu T^a$ so that Eq. (36.140) carries no explicit coupling, matching the standard mathematical normalization in which $\frac{1}{8\pi^2}\int \mathrm{tr}(F\wedge F)$ is the second Chern number. The physical, g-explicit anomaly equations Eqs. (36.50) and (36.159) of the earlier sections (with the $e^2/8\pi^2$ and $g^2/16\pi^2$ prefactors) are recovered by the rescaling shown; the integer ν is the same object in both. The neighbouring Chapter 37 writes the Levi-Civita symbol as $\varepsilon^{\mu\nu\rho\sigma}$ rather than $\epsilon^{\mu\nu\rho\sigma}$; these are the same tensor with the same $0123 = +1$ convention.

36.5.2 Winding, the instanton, and the θ term

The integer ν is abstract until one exhibits a field that carries it. The minimal example, and the one that advances Yang–Mills theory into the non-perturbative regime, is the single-instanton configuration of $SU(2)$ Yang–Mills, whose boundary map is the identification $S^3 \cong SU(2)$ itself. We compute its winding directly.

Worked Example 36.14 (Winding of the $SU(2)$ instanton)**.** Embed the unit three-sphere in $\mathbb{R}^4$ by Cartesian coordinates $\hat x = (x_4, \mathbf{x})$ with $x_4^2 + \mathbf{x}^2 = 1$, and define the boundary gauge transformation

$$U(\hat x) \;=\; x_4\,\mathbf{1} + i\,x_a\sigma^a \;\in\; SU(2), \qquad a = 1,2,3, \tag{36.142}$$

with σ^a the Pauli matrices. Unitarity $UU^\dagger = (x_4^2+\mathbf{x}^2)\mathbf{1} = \mathbf{1}$ and $\det U = x_4^2 + \mathbf{x}^2 = 1$ confirm $U \in SU(2)$; as $\hat x$ ranges over S^3 the matrix U ranges once over the $SU(2)$ group manifold, which is itself a three-sphere, so the map is the identity to within the standard diffeomorphism $SU(2) \cong S^3$ and should have degree one. We verify this from Eq. (36.139).

Parametrize S^3 by hyperspherical angles, $U = \cos\psi + i\,\sin\psi\,(\hat n\cdot\sigma)$ with $\hat n(\theta,\varphi)$ the unit vector on S^2, $\psi \in [0,\pi]$. The left-invariant Maurer–Cartan form

$U^{-1}\mathrm{d}U = i\,\sigma^a e^a$ defines a triad of one-forms e^a; a short computation gives e^a orthonormal for the round metric, because

$$-\tfrac{1}{2}\,\mathrm{tr}\big[(U^{-1}\mathrm{d}U)^2\big] = -\tfrac{1}{2}\,\mathrm{tr}\big[(i\sigma^a e^a)(i\sigma^b e^b)\big] = \tfrac{1}{2}\,e^a e^b\,\mathrm{tr}(\sigma^a\sigma^b) = e^a e^a = \mathrm{d}\psi^2 + \sin^2\!\psi\,\mathrm{d}\Omega_2^2, \tag{36.143}$$

the unit-radius round metric on S^3 with volume $\mathrm{Vol}(S^3) = \int_0^\pi \sin^2\!\psi\,\mathrm{d}\psi\int \mathrm{d}\Omega_2 = \tfrac{\pi}{2}\cdot 4\pi = 2\pi^2$. The Chern–Simons integrand of Eq. (36.139) is the trace of three Maurer–Cartan forms,

$$\mathrm{tr}\big[(U^{-1}\mathrm{d}U)^3\big] = i^3\,e^a e^b e^c\,\mathrm{tr}(\sigma^a\sigma^b\sigma^c) = -i\,e^a e^b e^c\,(2i\,\epsilon_{abc}) = 2\,\epsilon_{abc}\,e^a\!\wedge e^b\!\wedge e^c = 12\,e^1\!\wedge e^2\!\wedge e^3, \tag{36.144}$$

where $\mathrm{tr}(\sigma^a\sigma^b\sigma^c) = 2i\,\epsilon_{abc}$ was used. Integrating over the group, with $\int_{S^3} e^1\wedge e^2\wedge e^3 = \mathrm{Vol}(S^3) = 2\pi^2$,

$$\nu \;=\; \frac{1}{24\pi^2}\int_{S^3}\mathrm{tr}\big[(U^{-1}\mathrm{d}U)^3\big] \;=\; \frac{1}{24\pi^2}\cdot 12\cdot 2\pi^2 \;=\; 1. \tag{36.145}$$

The minus-one class is obtained from $U \to U^{-1}$ (the anti-instanton), which reverses the orientation of the covering and flips the sign of Eq. (36.144). The configuration that interpolates from the trivial gauge at the origin to Eq. (36.142) at infinity is the Belavin–Polyakov–Schwartz–Tyupkin instanton; its bulk field strength is self-dual, $F_{\mu\nu} = \widetilde{F}_{\mu\nu}$, which by Eq. (36.140) and the algebraic inequality $\int \mathrm{tr}(F \mp \widetilde{F})^2 \geq 0$ minimizes the Euclidean action in the class $\nu = 1$ at the value $S = 8\pi^2/g^2$ once the coupling is restored. We will not need the explicit profile here; the winding $\nu = 1$ is fixed entirely by the boundary data Eq. (36.142), and that is the topological content.

The integer ν also tells us that a term we have so far ignored is allowed in the action. The density $q(x)$ is gauge invariant, Lorentz invariant, and of mass dimension four, so by every power-counting criterion it is a legitimate term in the Lagrangian,

$$\mathcal{L}_\theta \;=\; \theta\,q(x) \;=\; \theta\,\frac{g^2}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big(F^{\mathrm{phys}}_{\mu\nu}F^{\mathrm{phys}}_{\rho\sigma}\big), \tag{36.146}$$

contributing $\theta\,\nu$ to the action. Being a total derivative, $\mathcal{L}_\theta$ never enters the classical equations of motion or any order of perturbation theory; its only effect is to weight each topological sector by a phase $e^{i\theta\nu}$ in the path integral. That this phase is physical — that θ is a genuine parameter of the quantum theory and not a redundancy, precisely *because* the abelian and non-abelian cases differ on whether ν can be nonzero — is the seed of the θ-vacuum construction and the strong-CP problem, developed in full in Chapter 42. For now we record only that the existence of the integer ν is what makes θ meaningful, and that the anomaly is the operator statement that an axial rotation shifts θ.

36.5.3 The index theorem and the zero-mode pairing

We now return to the regularized trace Eq. (36.127) and show that it, too, equals ν, by counting solutions of the Dirac equation. This is the content of the Atiyah–Singer index theorem, and the bridge to it from the Fujikawa trace is an elementary pairing argument that we give completely.

The Euclidean Dirac operator $i\not{D}$ is Hermitian, so its eigenvalues λ_n are real and its eigenstates ϕ_n form a complete orthonormal set. The single algebraic fact driving everything is that γ^5 *anticommutes* with the Dirac operator,

$$\{\gamma^5,\, i\not{D}\} = 0, \tag{36.147}$$

which follows from $\{\gamma^5, \gamma^\mu\} = 0$ and the fact that γ^5 commutes with the gauge generators. The consequence is a rigid structure on the spectrum. Suppose $\lambda_n \neq 0$. Then the state $\gamma^5\phi_n$ is again an eigenstate, but with the opposite eigenvalue:

$$i\not{D}\,(\gamma^5\phi_n) = -\gamma^5\,(i\not{D}\,\phi_n) = -\lambda_n\,(\gamma^5\phi_n). \tag{36.148}$$

Because $\lambda_n \neq 0$, the partner $\gamma^5\phi_n$ sits at a *different* eigenvalue $-\lambda_n$ and is therefore orthogonal to ϕ_n. This orthogonality is exactly the statement that every nonzero mode is chirally balanced:

$$\langle\phi_n|\gamma^5|\phi_n\rangle \;\propto\; \langle\phi_n|\,\gamma^5\phi_n\rangle \;=\; 0 \qquad (\lambda_n \neq 0). \tag{36.149}$$

The only states that can survive a γ^5 trace are the *zero modes*, $\lambda_n = 0$. For these the partner argument degenerates — $\gamma^5\phi_0$ has the same eigenvalue 0 — and we may diagonalize γ^5 within the zero-eigenvalue subspace, splitting the zero modes into definite-chirality families. Let n_+ count the right-handed zero modes ($\gamma^5\phi_0 = +\phi_0$) and n_- the left-handed ($\gamma^5\phi_0 = -\phi_0$). Substituting Eq. (36.149) into Eq. (36.127), every nonzero mode drops out for *any* M, and what remains is the regulator-independent integer

$$\mathcal{I}(M) \;=\; \sum_{\lambda_n=0} \langle\phi_n|\gamma^5|\phi_n\rangle\, e^0 \;=\; n_+ - n_- \;\equiv\; \mathrm{index}(i\not{D}), \tag{36.150}$$

manifestly independent of M because the only surviving terms have $\lambda_n = 0$. This is the decisive structural feature: the Fujikawa trace is not merely *finite* as $M \to \infty$, it is *exactly* M-independent, equal to a count of zero modes for every value of the regulator. Figure 36.3 renders the spectrum: a ladder of $\pm\lambda$ partners that cancel pairwise, plus an unbalanced floor of zero modes whose net chirality is the index.

It remains to evaluate the same trace a second way — not by counting modes but by the heat-kernel expansion — and to read off that it equals the topological charge.

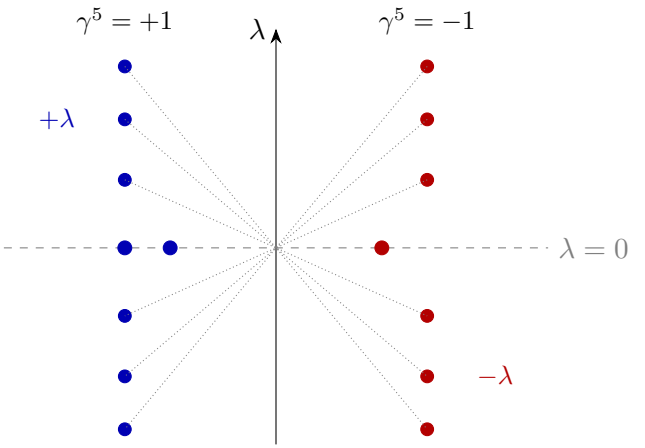


Figure 36.3: The spectrum of the Hermitian Dirac operator $i\not{D}$. Because γ^5 anticommutes with $i\not{D}$, every nonzero eigenvalue λ is paired with $-\lambda$ of opposite chirality (dotted links); such pairs contribute zero to $\mathrm{Tr}[\gamma^5\, e^{-(i\not{D})^2/M^2}]$ for any M. Only the unpaired zero modes survive, and their net chirality $n_+ - n_-$ is the index.

Theorem 36.15 (Atiyah–Singer index for the Dirac operator)**.** *For the Dirac operator $i\not{D}$ coupled to a smooth gauge field of finite action on (compactified) Euclidean $\mathbb{R}^4$,*

$$\mathrm{index}(i\not{D}) \;=\; n_+ - n_- \;=\; \frac{1}{32\pi^2}\int \mathrm{d}^4x\; \epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}\big(F_{\mu\nu}F_{\rho\sigma}\big) \;=\; \nu. \tag{36.151}$$

Heat-kernel evaluation linking the trace to the topological charge. We have already shown, by the pairing argument, that $\mathcal{I}(M) = n_+ - n_-$ for all M. We now compute the *same* $\mathcal{I}(M)$ in the opposite regime, expanding the heat kernel in powers of $1/M^2$. The relevant operator is the square of the Dirac operator. Squaring $i\not{D}$ with the Clifford algebra gives the Lichnerowicz–Weitzenböck form,

$$(i\not{D})^2 \;=\; -\not{D}^2 \;=\; -D_\mu D^\mu + \tfrac{g}{2}\,\sigma^{\mu\nu}F_{\mu\nu}, \qquad \sigma^{\mu\nu} = \tfrac{i}{2}[\gamma^\mu,\gamma^\nu], \tag{36.152}$$

where the spin–field-strength coupling arises from the antisymmetric part of $D_\mu D_\nu$, namely $[D_\mu, D_\nu] = -igF_{\mu\nu}$. The diagonal heat kernel of an operator of the form $-D^2 + X$ has the short-time (large-M) Seeley–DeWitt expansion developed in Section E.3.4,

$$\langle x|e^{-(i\not{D})^2/M^2}|x\rangle \;=\; \frac{M^4}{(4\pi)^2}\Big(a_0(x) + \frac{a_1(x)}{M^2} + \frac{a_2(x)}{M^4} + \cdots\Big), \tag{36.153}$$

with $a_0 = \mathbf{1}$, a_1 linear in the curvature and the potential $X = \frac{g}{2}\sigma F$, and a_2 quadratic in them. Now take the chiral trace. The Dirac trace of γ^5 against $\mathbf{1}$

vanishes, $\operatorname{tr}\gamma^5 = 0$, killing the M^4 term from a_0; the trace of γ^5 against a single $\sigma^{\mu\nu}$ vanishes, $\operatorname{tr}(\gamma^5\sigma^{\mu\nu}) = 0$, killing the M^2 term from a_1. The first nonzero chiral trace appears at order M^0, from the term in a_2 quadratic in the spin coupling, $\frac{1}{2!}(\frac{g}{2}\sigma^{\mu\nu}F_{\mu\nu})^2$, because the Dirac trace

$$\operatorname{tr}\big(\gamma^5\,\sigma^{\mu\nu}\sigma^{\rho\sigma}\big) \;=\; -4i\,\epsilon^{\mu\nu\rho\sigma} \tag{36.154}$$

is the unique nonvanishing structure (it requires all four distinct gamma matrices supplied by γ^5). Assembling the numerical factors — the $1/(4\pi)^2$ from the heat kernel, the $1/2!$ from the second-order term, the $(g/2)^2$ from two spin couplings, and Eq. (36.154) — the M-independent piece is precisely the density already quoted in Eq. (36.128),

$$\lim_{M\to\infty}\operatorname{tr}\big[\gamma^5\langle x|e^{-(i\not{D})^2/M^2}|x\rangle\big] \;=\; \frac{1}{(4\pi)^2}\,\frac{1}{2!}\Big(\frac{g}{2}\Big)^2(-4i)\,\epsilon^{\mu\nu\rho\sigma}\operatorname{tr}\big(F^{\text{phys}}_{\mu\nu}F^{\text{phys}}_{\rho\sigma}\big)\times(i) \;=\; \frac{1}{32\pi^2}\,\epsilon^{\mu\nu\rho} \tag{36.155}$$

the factor of i in the second equality being the Euclidean-rotation phase carried by $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and the field-strength normalization absorbing g^2 back into the geometric F of Eq. (36.129); this is exactly the coefficient computed in detail in the Fujikawa derivation of Section 36.3, which we import rather than repeat. Integrating Eq. (36.155) over spacetime gives $\mathcal{I}(M\to\infty) = \nu$, the topological charge Eq. (36.140). Comparing with the mode-counting value Eq. (36.150), and using that $\mathcal{I}(M)$ is the *same* M-independent number in both evaluations, yields $n_+ - n_- = \nu$, which is Eq. (36.151). □

Caution

What the heat-kernel argument establishes rigorously is the *equality* of the Fujikawa trace with the integrated Chern–Pontryagin density, and — through the pairing argument — with the net zero-mode count. It is an honest physicist's derivation of the index, valid because the two ways of evaluating one M-independent trace must agree. It is not the full Atiyah–Singer theorem, which is a statement of K-theory establishing the index formula for general elliptic operators on general manifolds, with the analytic subtleties (compactness of the resolvent, spectral flow, the contribution of the continuous spectrum on noncompact $\mathbb{R}^4$, and boundary η-invariants) controlled. Those are genuinely the province of the mathematical literature; the original index theorem dates to Atiyah and Singer (1963) and its heat-kernel proof to the 1970s. We use the theorem as stated and proved-in-physics here, and point the reader to the mathematical proof in the Further Reading.

The physical reading of Eq. (36.151) closes the loop opened by Fujikawa. The integrated anomaly Eq. (36.141) is $\Delta Q_5 = 2\nu = 2(n_+ - n_-)$: in the presence of a gauge field of winding ν, the axial charge of the vacuum changes by twice

the net number of chiral zero modes. Each zero mode is a normalizable solution of the Dirac equation of definite chirality with no partner; the anomaly is the statement that the Dirac sea is reshuffled by exactly these unpaired levels when the gauge field tunnels between topological sectors. The local anomaly density and the global index are two views of one object, and the regulator dependence that looked like a scheme artifact in the triangle graph is revealed as the counting of solutions to a differential equation — a quantity no scheme can move.

36.5.4 Why the coefficient is exact: Adler–Bardeen from topology

We can now give the structural reason for a fact established earlier by diagrammatic labour: the anomaly coefficient receives no corrections beyond one loop. This is the Adler–Bardeen theorem, and from the topological vantage it is almost a triviality.

The integrated anomaly is constrained to equal 2ν, and ν is an integer for *every* smooth gauge field of finite action. Imagine, contrary to the theorem, that radiative corrections multiplied the anomaly coefficient by a function $1 + c_1 g^2 + c_2 g^4 + \cdots$ of the coupling. Then the integrated anomaly over a fixed background of winding $\nu = 1$ would read $2(1 + c_1 g^2 + \cdots)$, a quantity that varies continuously with g. But the left-hand side counts zero modes: it is $2(n_+ - n_-)$, an integer, and an integer cannot vary continuously. The number of normalizable zero-energy solutions of the Dirac equation in a given background is a discrete, deformation-invariant datum; it cannot drift as we dial the coupling. The only continuous function consistent with taking integer values on a connected family of couplings is a constant, so $c_1 = c_2 = \cdots = 0$. The one-loop coefficient is the whole answer.

Physical Insight

The logic is worth isolating because it recurs throughout physics whenever a quantity is simultaneously *computable perturbatively* and *quantized topologically*. Perturbation theory delivers a smooth function of the coupling; topology delivers an integer. A smooth function that must equal an integer on a connected set is constant. Hall conductance, magnetic charge, and the level of a Chern–Simons theory are protected for the same reason. The anomaly is the original instance: the triangle graph computes a number, the index theorem certifies it is an integer per unit winding, and exactness follows without summing a single higher loop.

Two caveats keep the statement honest. First, the theorem is a statement about the coefficient of the *one-loop* anomaly functional; it does not forbid the appearance of the anomaly inside higher-order quantities through the renormal-

ization of the operators and couplings that multiply it. The precise modern phrasing — that the anomaly is one-loop exact in a scheme where the axial current is defined by its anomaly, with all scheme dependence relegated to the definition of the current — was settled only after the topological picture clarified what "the anomaly" invariantly means. Second, the argument as given assumes the relevant zero-mode count is rigid; this is guaranteed by the ellipticity of $i\not{D}$ and the finiteness of the action, the same hypotheses under which Theorem 36.15 holds.

The same integer that pins the coefficient also resolves a long-standing puzzle of the strong interactions. In QCD with n_f massless quarks, the classical Lagrangian has a $U(1)_A$ axial symmetry in addition to the $SU(n_f)_L \times SU(n_f)_R$ whose spontaneous breaking gives the pions (Chapter 29). Were $U(1)_A$ an honest symmetry, its spontaneous breaking would demand a ninth light pseudoscalar, a flavour-singlet partner of the pions with mass bounded above by $\sqrt{3}\, m_\pi$ — a particle conspicuously absent from the spectrum. The resolution is the anomaly: $U(1)_A$ is not a symmetry of the quantum theory, because its current has exactly the divergence Eq. (36.141). An axial rotation shifts the θ angle of Eq. (36.146) rather than leaving the action invariant, so there is no ninth Goldstone boson to demand. The would-be Goldstone is the η', and its anomalously large mass — generated by the same topological gauge configurations that carry ν — is the experimental fingerprint of the index. This is the $U(1)_A$ problem; its quantitative solution through the topological susceptibility of the gauge vacuum belongs to Chapter 42, but its qualitative resolution is already forced by the integer we have built here.

With the anomaly recast as a topological invariant, its rigidity is no longer a computational accident. The integrated coefficient is an integer per unit winding; it cannot be tuned, renormalized, or deformed away by any smooth choice. This is precisely the property the next section needs. When the anomalous current is not a passive global probe but is gauged — coupled to a dynamical field whose consistency rests on an exact Ward identity — this same un-removability becomes a threat rather than a curiosity. A gauge anomaly cannot be cancelled order by order against counterterms, because what it computes is topological and counterterms are smooth; it can only be made to vanish by arranging the fermion content so the coefficient is zero identically. We turn now to the non-abelian generalization of the anomaly coefficient and to the demonstration that a gauge anomaly is fatal, handing the cancellation program to Chapter 37.

36.6 Non-abelian anomalies, gauge consistency, and the handoff

The previous two sections delivered the anomaly in its most invariant guise. The index theorem of Section 36.5 showed that the integrated anomaly equals an

integer — the net chirality of the Dirac zero modes, equivalently the winding number $\frac{1}{32\pi^2}\int \mathrm{tr}\, F\widetilde{F}$ of the gauge background — and that this integrality is what makes the Adler–Bardeen coefficient exact. Along the way the anomaly density generalized from the abelian $\frac{e^2}{8\pi^2}F\widetilde{F}$ to the non-abelian $\frac{1}{32\pi^2}\,\mathrm{tr}\, F\widetilde{F}$, with the trace running over whatever representation the loop fermions carry. Everything so far has treated the anomalous current as a *global* object: an external probe $j^{5\mu}$ that we contract against honest background photons and confront with the measured rate of $\pi^0 \to \gamma\gamma$ (Eq. (36.115)). The result was a triumph, not a catastrophe.

This section asks the question that flips the sign of that verdict. What happens if the anomalous current is not a spectator but is *itself gauged* — if the very symmetry whose conservation law the anomaly destroys is the symmetry we have promoted to a local, dynamical gauge invariance? We will find that the anomaly then attacks the one identity that makes a gauge theory consistent: the Ward–Takahashi identity of Theorem 19.13, the result that protected QED's gauge symmetry through quantization and that has been the spine of this chapter's contrast. When that identity fails, longitudinal gauge modes stop decoupling, unphysical states leak into the S-matrix, and both unitarity and renormalizability collapse. The upshot is a sharp algebraic demand on the fermion content of any chiral gauge theory — the vanishing of a single group-theory tensor — and that demand, applied to the Standard Model, is the business handed to Chapter 37.

36.6.1 From external charges to generators: the non-abelian triangle

Take a Yang–Mills theory with gauge group G and Lie algebra generators T^a in the fermion representation R, normalized as in Chapter 25 by $[T^a, T^b] = if^{abc}T^c$ and $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ in the fundamental. The gauge field is A^a_μ, the field strength and the covariant derivative are those of the carried Yang–Mills Lagrangian Eq. (25.104), and $D_\mu = \partial_\mu - igA^a_\mu T^a$ acts on the fermions. The novelty is that the fermions are *chiral*: we couple only the left-handed components to the gauge field, through the current

$$J^{a\mu} \;=\; \bar{\psi}\,\gamma^\mu\, T^a\, \tfrac{1}{2}(1-\gamma^5)\,\psi \;=\; \bar{\psi}\,\gamma^\mu\, T^a P_L\,\psi, \qquad P_L = \tfrac{1}{2}(1-\gamma^5), \tag{36.156}$$

where ψ collects the left-handed Weyl fermion together with whatever right-handed partners the theory may carry; the projector P_L enforces that only the left-handed components couple. A theory in which $R_L \neq R_R$ — the left- and right-handed fermions sit in inequivalent representations — is a *chiral gauge theory*; the electroweak sector of the Standard Model is the canonical example. A theory with $R_L = R_R$ is *vectorlike*; QED and QCD are vectorlike, and we will see in a moment that this single fact makes their gauge currents anomaly-free for free.

The object that controls the gauge anomaly is the three-current correlator $\langle J^{a\mu}(x)\, J^{b\nu}(y)\, J^{c\rho}(z)\rangle$, computed at one loop by the triangle diagram with a gauge

current at each corner. The crucial observation is that the triangle factorizes. Write the chiral vertex as $\gamma^\mu T^a P_L$; the generator T^a is a constant matrix in representation space that commutes with everything Dirac-valued, so it pulls out of the loop momentum integral entirely. The diagram is therefore the product of two independent pieces: a *kinematic factor* — the same Dirac trace and loop integral that produced the abelian triangle of Theorem 36.4, now with P_L at each vertex — multiplied by a *group-theory factor* $\mathrm{tr}_R(T^a T^b T^c)$ collected from the three generators around the loop. The external charges of the abelian case have become generators; that is the whole content of "going non-abelian" at the level of the diagram.

It remains to determine which part of the group-theory factor survives. Two facts settle it. First, the kinematic coefficient that multiplies the Levi-Civita tensor $\epsilon^{\mu\nu\rho\sigma}$ in the abelian triangle is *totally symmetric* under the simultaneous exchange of any two external legs — their Lorentz indices and their incoming momenta together. This is not an accident; it is Bose symmetry, since the three external gauge bosons are identical particles, reinforced by the fact that the anomalous part of the amplitude depends on the momenta only through the antisymmetric ϵ structure. Second, there are two inequivalent ways to attach three distinguishable labels a, b, c to a triangle: the two orientations of fermion flow, or equivalently the "direct" diagram and the one with two legs crossed. The direct diagram carries $\mathrm{tr}_R(T^a T^b T^c)$ and the crossed one carries $\mathrm{tr}_R(T^a T^c T^b)$. Because the kinematic factor is symmetric, both diagrams come weighted by the *same* kinematic function, and the amplitude is proportional to their sum,

$$\mathrm{tr}_R(T^a T^b T^c) + \mathrm{tr}_R(T^a T^c T^b) \;=\; \mathrm{tr}_R\big(T^a \{T^b, T^c\}\big). \tag{36.157}$$

The antisymmetric part of $\mathrm{tr}_R(T^a T^b T^c)$ — which, using $T^b T^c = \tfrac{1}{2}\{T^b, T^c\} + \tfrac{1}{2}[T^b, T^c]$ and $[T^b, T^c] = i f^{bcd} T^d$, is proportional to $f^{bcd}\,\mathrm{tr}_R(T^a T^d) = \tfrac{1}{2} f^{abc}$ — cancels between the two orientations and never reaches the ϵ structure. Only the fully symmetric combination $\mathrm{tr}_R(T^a\{T^b, T^c\})$ contributes to the anomaly. We give this tensor the name it will carry into Chapter 37:

Definition 36.16 (Gauge-anomaly coefficient)**.** For a left-handed fermion in the representation R of a gauge group G, the *gauge-anomaly coefficient* is the totally symmetric rank-three invariant

$$\mathcal{A}^{abc} \;\equiv\; \mathrm{tr}_R\big(T^a\{T^b, T^c\}\big) \;=\; 2\,\mathrm{tr}_R\big(T^a T^{(b} T^{c)}\big). \tag{36.158}$$

Because it is a totally symmetric G-invariant of rank three, it is proportional to the unique such tensor of the algebra, the symmetric d-symbol of Theorem B.30: $\mathcal{A}^{abc} = A(R)\, d^{abc}$, with the scalar *anomaly index* $A(R)$ tabulated in Section B.5.

Assembling the kinematic factor — borrowed unchanged from Theorem 36.4, with the single bookkeeping change that each γ^5 has become a P_L — with this group-theory weight, the divergence of the gauge current takes exactly the abelian

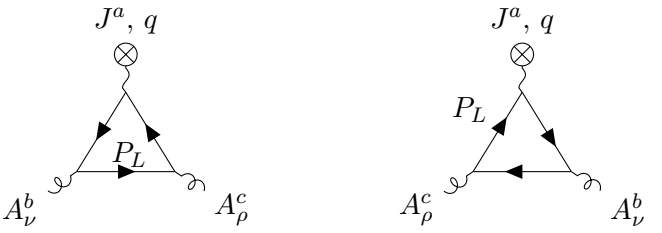


Figure 36.4: The anomalous gauge Ward identity. Contracting the chiral-current vertex with its momentum q_μ (the crossed dot) should, for a conserved current, annihilate the amplitude; instead the two triangle orientations sum to the group factor $\mathcal{A}^{abc} = \mathrm{tr}_R(T^a\{T^b, T^c\})$ of Eq. (36.158), leaving the anomalous remnant Eq. (36.162). Each vertex carries a generator and the projector P_L; the symmetric trace is the seat of the gauge anomaly.

form with the charge replaced by a generator and the field strength promoted to the non-abelian $F^a_{\mu\nu}$. In the gauge-covariant normalization,

$$(D_\mu J^\mu)^a \;=\; \frac{g^2}{32\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}_R[T^a F_{\mu\nu}F_{\rho\sigma}] \;=\; \frac{g^2}{64\pi^2}\,\mathcal{A}^{abc}\,\epsilon^{\mu\nu\rho\sigma}F^b_{\mu\nu}F^c_{\rho\sigma}, \quad (36.159)$$

where the second equality writes $F_{\mu\nu} = F^b_{\mu\nu}T^b$ and uses the symmetry of $\epsilon^{\mu\nu\rho\sigma}F^b_{\mu\nu}F^c_{\rho\sigma}$ under $b \leftrightarrow c$ to convert $\mathrm{tr}_R(T^aT^bT^c)$ into $\frac{1}{2}\mathcal{A}^{abc}$. The notation D_μ on the left is the covariant divergence; the right-hand side is gauge-covariant, transforming in the adjoint as it must if both sides are to match. We check the abelian limit against Eq. (36.50): a single left-handed Weyl fermion of charge q has $T^a \to q$ and $\mathcal{A}^{abc} \to 2q^3$, so Eq. (36.159) reduces to $\partial_\mu J^\mu = \frac{g^2}{32\pi^2}q^3\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$, which is exactly half the Dirac axial anomaly of Eq. (36.50) — one half because a single Weyl fermion is one chirality, and a cube of the charge because each of the three vertices contributes one factor of q.

36.6.2 Left versus right: the chiral structure

The projector P_L in Eq. (36.156) is doing the decisive work, and it is worth making its role explicit, because it is what distinguishes a fatal gauge anomaly from no anomaly at all. Suppose the theory contains right-handed fermions as well, coupled through $J^{a\mu}_R = \bar\psi\gamma^\mu T^a P_R\psi$ with $P_R = \frac{1}{2}(1+\gamma^5)$ in a representation R'. Repeating the triangle computation for a right-handed loop reverses the single surviving γ^5 relative to the left-handed case, which flips the sign of the ϵ-tensor term. The full anomaly of the gauged current is therefore the *difference* of left and right contributions,

$$\mathcal{A}^{abc}_{\text{net}} \;=\; \mathrm{tr}_{R_L}(T^a\{T^b,T^c\}) \;-\; \mathrm{tr}_{R_R}(T^a\{T^b,T^c\}). \quad (36.160)$$

This single formula contains the entire qualitative landscape. If the theory is vectorlike, $R_L = R_R$, the two traces are identical and cancel: *a vectorlike gauge theory has no gauge anomaly*, automatically, for any group and any matter content. This is why QED and QCD — in which left- and right-handed quarks carry the same color and electric charge — are quantum-mechanically consistent gauge theories without any constraint on their spectra; the gauge symmetry that the Ward–Takahashi identity protected in Theorem 19.13 survives because the anomaly that would have threatened it cancels chirality against chirality. It is only in a genuinely chiral theory, $R_L \neq R_R$, that Eq. (36.160) can fail to vanish, and only there that the cancellation becomes a nontrivial condition on the matter content. The Standard Model is precisely such a theory, and the rest of this section, and all of Chapter 37, is about how its chiral content conspires to keep Eq. (36.160) zero.

A convenient bookkeeping convention makes Eq. (36.160) into a sum rather than a difference. A right-handed fermion in R' is, by CP, equivalent to a left-handed fermion in the conjugate representation $\bar{R}'$, and the generators of the conjugate satisfy $T^a_{\bar{R}} = -(T^a_R)^T$, so that $\mathrm{tr}_{\bar{R}}(T^a\{T^b, T^c\}) = -\mathrm{tr}_R(T^a\{T^b, T^c\})$. Casting every fermion as left-handed and summing,

$$\mathcal{A}^{abc} = \sum_{i \in \text{left-handed Weyl}} \mathrm{tr}_{R_i}\big(T^a\{T^b, T^c\}\big), \tag{36.161}$$

with anti-particles automatically entering through their conjugate representations with the opposite sign. This is the form in which the cancellation condition will be most efficiently checked.

Worked Example 36.17 (Anomaly coefficient of SU(N) representations; why SU(2) is safe)**.** We collect the group-theory facts that decide whether a given chiral content can ever cancel, leaning on the symmetric-invariant machinery of Section B.5 rather than reproving it.

The fundamental of SU(N). Normalizing the symmetric tensor by $d^{abc} = 2\,\mathrm{tr}\big(T^a\{T^b, T^c\}\big)_{\mathbf{N}}$ in the fundamental, the anomaly index of Theorem 36.16 is by definition $A(\mathbf{N}) = 1$. The conjugate fundamental then has $A(\bar{\mathbf{N}}) = -1$, since conjugation flips the sign of the symmetric trace. A single left-handed quark in $\mathbf{N}$ and a left-handed anti-quark in $\bar{\mathbf{N}}$ — the vectorlike pairing — contribute $A(\mathbf{N}) + A(\bar{\mathbf{N}}) = 0$, recovering the general statement of Eq. (36.160). The adjoint representation is real, $\bar{\text{adj}} = \text{adj}$, so $A(\text{adj}) = -A(\text{adj}) = 0$: gauge fields and adjoint fermions (gluinos, for instance) never contribute a gauge anomaly.

Why SU(2) *carries no anomaly at all.* Every representation of SU(2) is real or pseudoreal: $R \cong \bar{R}$. Conjugation then forces $A(R) = A(\bar{R}) = -A(R)$, hence $A(R) = 0$ for *every* SU(2) representation. Equivalently, SU(2) simply has no totally symmetric rank-three invariant tensor — there is no d^{abc} to be proportional to, because the only invariant tensors are δ^{ab} and $\epsilon^{abc} = f^{abc}$, the latter antisymmetric (Section B.5). This is the structural reason the weak-isospin

gauge group $\mathrm{SU}(2)_L$ contributes no $[\mathrm{SU}(2)_L]^3$ anomaly despite being maximally chiral; the only weak-sector constraint that survives is the mixed $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$ anomaly, where two isospin legs pair into δ^{ab} and the third leg is hypercharge.

Which groups can be anomalous. Carrying the classification of Section B.5: among the compact simple Lie algebras a nonzero d^{abc} exists only for $\mathrm{SU}(N)$ with $N \geq 3$. All others — $\mathrm{SU}(2)$, $SO(N)$ (except the special case $SO(6) \cong \mathrm{SU}(4)$), $Sp(N)$, and the exceptional algebras G_2, F_4, E_6, E_7, E_8 — are "safe": they admit no cubic symmetric invariant, so $\mathcal{A}^{abc} \equiv 0$ identically and any fermion content is gauge-anomaly-free. The entire gauge-anomaly problem of the Standard Model is therefore concentrated in its $\mathrm{SU}(3)_c$ and $\mathrm{U}(1)_Y$ factors and the mixed combinations involving them.

36.6.3 Why a gauge anomaly is fatal

We now make precise the claim advertised since the opening of the chapter: a global anomaly is benign, but a gauge anomaly is fatal. The argument is the exact mirror image of the Ward–Takahashi success story of Theorem 19.13, and reading the two side by side is the cleanest way to see what is at stake.

In QED the gauge symmetry survives quantization. The Ward–Takahashi identity Theorem 19.13 states, in its on-shell amplitude form, that contracting any photon external momentum q_μ into an S-matrix element annihilates it: $q_\mu \mathcal{M}^\mu = 0$. Two consequences flow from this, and both are load-bearing for the entire edifice of gauge field theory. First, *unitarity*: a massless vector field has four polarizations but only two are physical, and the would-be unphysical longitudinal and timelike modes carry negative or vanishing norm. The Ward identity is exactly the statement that these unphysical polarizations decouple from physical amplitudes; their contributions cancel, leaving a manifestly positive probability flow and a consistent optical theorem. Second, *renormalizability*: the same identity constrains the divergent parts of loop diagrams to respect the gauge symmetry, so that every ultraviolet infinity can be absorbed into a counterterm of the original gauge-invariant Lagrangian. No gauge-noninvariant counterterm — a photon mass, say — is ever required, because none is ever generated. The Ward identity is the reason QED is predictive.

Now suppose the anomalous chiral current of Eq. (36.156) is the gauge current itself. The classical conservation law $D_\mu J^{a\mu} = 0$, which would have given the gauge-field analog of $q_\mu \mathcal{M}^\mu = 0$, is destroyed by Eq. (36.159). Concretely, the would-be Ward identity for the three-gauge-boson amplitude reads, after contraction with the momentum q_μ of one gauge leg,

$$q_\mu\, \Gamma^{\mu\nu\rho}_{abc}(q, k_1, k_2) \;=\; \frac{g^2}{8\pi^2}\, \mathcal{A}^{abc}\, \epsilon^{\nu\rho\alpha\beta} k_{1\alpha} k_{2\beta} \;\neq\; 0, \tag{36.162}$$

where $\Gamma^{\mu\nu\rho}_{abc}$ is the one-loop triangle vertex of three gauge currents and $\mathcal{A}^{abc}$ is the coefficient Eq. (36.158); the right-hand side is the momentum-space avatar

of Eq. (36.159), with the crossed-dot contraction of Fig. 36.4 producing exactly the symmetric trace. When $\mathcal{A}^{abc} \neq 0$, the longitudinal polarization of the gauge boson no longer decouples. The unphysical modes that the Ward identity was supposed to project out now contribute to physical S-matrix elements, and they bring their negative-norm states with them: probability is no longer conserved, the optical theorem fails, and the theory is non-unitary. Equivalently, the loop divergences cease to respect the gauge symmetry, gauge-noninvariant counterterms (a dynamically generated longitudinal-mode mass among them) are required and cannot be absorbed into the original Lagrangian, and renormalizability is lost. A chiral gauge theory with $\mathcal{A}^{abc} \neq 0$ is not a quantum field theory at all; it is a classical Lagrangian with no consistent quantization.

The cleanest modern statement of the same fact is cohomological, and it ties the anomaly to the BRST machinery of Section 26.5. Recall that after gauge fixing, the physical content of the theory is defined not by the naive Hilbert space — which contains negative-norm timelike gluons and the Faddeev–Popov ghosts — but by the cohomology of the nilpotent BRST charge Q, $Q^2 = 0$: physical states are Q-closed modulo Q-exact, and the unphysical longitudinal/timelike/ghost sectors arrange themselves into the BRST-trivial "quartets" that cancel in every physical matrix element. This cohomological projection is what guarantees unitarity in the gauge-fixed theory. The Slavnov–Taylor identities — the all-orders descendants of the Ward identity — are precisely the statement that the BRST symmetry, exact classically, survives in the quantum theory. A gauge anomaly is the statement that it does not: the path-integral measure of the chiral fermions is not BRST-invariant, the Slavnov–Taylor identities acquire an anomalous right-hand side proportional to $\mathcal{A}^{abc}$, the quartet cancellation fails, and the physical subspace is no longer preserved by time evolution. Unitarity and the very definition of the physical Hilbert space collapse together. The conclusion is forced and admits no remedy by choice of scheme:

Theorem 36.18 (Gauge-consistency condition)**.** *A chiral gauge theory with gauge group G and left-handed Weyl fermions in representations R_i is quantum-mechanically consistent — unitary and renormalizable, with an intact Ward–Takahashi/Slavnov–Taylor identity — only if its gauge-anomaly coefficient vanishes:*

$$\mathcal{A}^{abc} \;=\; \sum_i \mathrm{tr}_{R_i}\!\left(T^a\{T^b, T^c\}\right) \;=\; 0, \tag{36.163}$$

the sum running over all left-handed Weyl fermions, with conjugate representations entering through $T^a_{\bar{R}} = -(T^a_R)^T$ and hence with the opposite sign.

The contrast with the global anomaly is now sharp and can be stated in one breath. When the anomalous current is global — the axial current of Eq. (36.50), probed by external background photons — there is no dynamical gauge field associated with it, no longitudinal mode to fail to decouple, no Slavnov–Taylor identity of a propagating field to violate, and therefore no threat to unitarity. The

non-conservation of the global current is then a physical, measurable prediction: it is the entire content of $\pi^0 \to \gamma\gamma$ (Eq. (36.115)) and of the $\mathrm{U}(1)_A$ resolution of Section 36.5.4. The very same diagram, the very same number $\mathcal{A}^{abc}$, is a triumph when the legs are global currents or honest background fields and a catastrophe when the legs are dynamical gauge fields. The distinction is not in the anomaly but in what the anomaly is asked to do.

36.6.4 Consistent versus covariant anomaly, and the Wess–Zumino condition

There is one structural subtlety in Eq. (36.159) that must be flagged honestly, even though its full resolution belongs to Chapter 37. The anomaly we computed from the triangle and wrote in Eq. (36.159) is gauge-*covariant*: under a gauge transformation it rotates in the adjoint representation, $\delta(D_\mu J^\mu)^a = -f^{abc}\alpha^b (D_\mu J^\mu)^c$, just as the field strength does. This is the natural object when one cares about gauge covariance — it is the *covariant anomaly*. But it is *not* the variation of any local functional of the gauge field. It cannot be, for a structural reason worth spelling out.

The effective action $W[A] = -i\log\!\int\!\mathcal{D}\bar\psi\,\mathcal{D}\psi\, e^{iS}$ obtained by integrating out the chiral fermions is a definite functional of A, and its response to an infinitesimal gauge transformation *defines* the anomaly through

$$\delta_\alpha W[A] \;=\; \int d^4x\; \alpha^a(x)\, G^a(x), \qquad G^a(x) \;=\; (D_\mu J^\mu)^a_{\text{cons}}, \tag{36.164}$$

the right-hand side being the *consistent anomaly*. Whatever G^a is, it must be a genuine second derivative of W in the sense that two successive gauge variations commute up to the structure constants. Because the gauge variations close into the gauge algebra,

$$[\delta_\alpha, \delta_\beta] \;=\; \delta_{[\alpha,\beta]}, \qquad [\alpha,\beta]^c = f^{abc}\alpha^a\beta^b, \tag{36.165}$$

applying this commutator to W and using Eq. (36.164) twice yields a nontrivial constraint that the anomaly must satisfy:

$$\delta_\alpha \!\int\! \beta^a G^a \;-\; \delta_\beta \!\int\! \alpha^a G^a \;=\; \int f^{abc}\,\alpha^a\beta^b\,G^c. \tag{36.166}$$

This is the *Wess–Zumino consistency condition*. It is nothing more than the integrability requirement that G^a actually be the gauge variation of *some* functional — the statement that you cannot have a left-hand side of Eq. (36.164) without the algebra Eq. (36.165) constraining it. The covariant anomaly of Eq. (36.159) does *not* satisfy Eq. (36.166); the consistent anomaly does. The two differ by the gauge variation of a local polynomial in A — the Bardeen–Zumino current — and although their explicit forms differ (the leading two-gauge-field

term of the covariant anomaly is three times that of the consistent anomaly, the famous factor that we will reproduce in Chapter 37, and the consistent anomaly carries additional terms cubic and quartic in A that build up gauge covariance into a total variation), this difference changes nothing about the question that matters. Both anomalies are proportional to the same symmetric tensor $\mathcal{A}^{abc}$, so both vanish if and only if $\mathcal{A}^{abc} = 0$. The cancellation condition Eq. (36.163) is therefore unambiguous; the consistent/covariant distinction affects the *form* of the anomaly, never the *condition for its absence*.

The Wess–Zumino condition has a compact cohomological reading in the BRST language of Section 26.5 that makes its power manifest and that Chapter 37 will exploit through the descent equations. Promoting the gauge parameter α^a to the anticommuting ghost c^a and the gauge variation to the nilpotent BRST operator s ($s^2 = 0$), the integrated anomaly $\mathcal{G} = \int c^a G^a$ obeys $s\mathcal{G} = 0$ — which is exactly Eq. (36.166) rewritten — so the anomaly is a BRST-closed functional of ghost number one. If $\mathcal{G}$ were s of a local functional it would be removable by a local counterterm and hence spurious; a genuine anomaly is therefore a nontrivial element of the BRST cohomology at ghost number one. This cohomological characterization is what makes the Wess–Zumino condition so rigid: the space of possible anomalies is a finite-dimensional cohomology group, and in four dimensions it is one-dimensional, spanned by exactly the d^{abc} structure of Eq. (36.158). The full development of the descent equations, the explicit consistent anomaly, and the reconciliation of the covariant and consistent forms is the opening business of Chapter 37.

36.6.5 The Standard Model preview and the handoff

The gauge-consistency condition Eq. (36.163) is a sharp, computable demand, and the most important fact about the Standard Model is that it satisfies it — not by accident of a single number but as a pattern that holds, intricately, for every gauge-anomaly combination and within each generation separately. The gauge group is $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, so $\mathcal{A}^{abc}$ must vanish for every choice of the three external gauge legs among the simple and abelian factors. By the classification of Theorem 36.17, only a handful of combinations can be nonzero in the first place:

$$[\mathrm{SU}(3)_c]^3, \quad [\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{U}(1)_Y]^3, \quad \mathrm{U}(1)_Y\text{-grav}^2, \tag{36.167}$$

the last being the mixed gauge–gravitational anomaly whose consistency requires $\sum Y = 0$ over the spectrum. The $[\mathrm{SU}(3)_c]^3$ anomaly vanishes because color is vectorlike (left- and right-handed quarks carry the same color representation, Eq. (36.160)); the pure $[\mathrm{SU}(2)_L]^3$ combination is absent because SU(2) admits no d^{abc} (Theorem 36.17); and the remaining four combinations vanish only through nontrivial cancellations among the chiral quark and lepton hypercharges of a

single generation. That the fractional hypercharges of the Standard Model are exactly the values that make Eq. (36.167) vanish — the same condition recorded at the electroweak level as Eq. (31.208) — is one of the most striking internal consistency facts in particle physics, and it is the explicit subject of Section 37.3, where each combination of Eq. (36.167) is computed in full and shown to vanish, and where the cancellation is turned around to *derive* the hypercharge assignments from consistency alone.

This advances Yang–Mills theory to its anomalies layer. The carried Yang–Mills theory, which we followed from the classical Lagrangian Eq. (25.104) through BRST quantization in Section 26.5 and asymptotic freedom in Chapter 27, now reveals one further quantum demand on its matter content: the anomaly density $\frac{1}{32\pi^2}\,\mathrm{tr}\,F\widetilde{F}$ that we have met repeatedly threatens the gauge symmetry itself the moment the fermions are chiral, and the theory survives only if Eq. (36.163) holds. QCD, being vectorlike, passes this test for free; the chiral electroweak sector passes it only because of the precise, generation-by-generation arrangement of its hypercharges.

We hand to Chapter 37 three objects, sharply defined. First, the gauge-anomaly coefficient $\mathcal{A}^{abc} = \mathrm{tr}_R(T^a\{T^b, T^c\})$ of Theorem 36.16, together with the covariant anomaly Eq. (36.159) it controls. Second, the gauge-consistency requirement $\mathcal{A}^{abc} = 0$ of Theorem 36.18, the condition that separates a consistent chiral gauge theory from a meaningless one. Third, the consistent-versus-covariant distinction and the Wess–Zumino consistency condition Eq. (36.166), whose descent-equation resolution, explicit Standard-Model cancellation, and elevation into 't Hooft anomaly matching are the content of the next chapter.

Further reading

The axial anomaly was discovered independently by Adler [5] in spinor electrodynamics and by Bell and Jackiw [58] as the resolution of a PCAC puzzle in the σ-model; the one-loop exactness of the coefficient is the theorem of Adler and Bardeen [6]. The measure-theoretic derivation is due to Fujikawa [203]. The topological content — winding, instantons, and the link to the $\mathrm{U}(1)_A$ problem — traces to 't Hooft [464], with the index theorem in the form used here from Atiyah and Singer [41]. The non-abelian anomaly and gauge-cancellation condition are developed in Bardeen [48], Bouchiat et al. [87], Gross and Jackiw [237], the Wess–Zumino consistency condition in Wess and Zumino [506] with its cohomological reformulation in Stora [457], Zumino et al. [535], and the covariant/consistent distinction in Bardeen and Zumino [49]; the algebraic classification is collected in Bertlmann [73]. The Standard-Model cancellation carried out in Chapter 37 is reviewed in Weinberg [501].

Chapter 37

Anomalies II: Consistency and Matching

The chiral anomaly entered the previous chapter as a verdict on a single global current: the axial current of a massless fermion, classically conserved, acquires the divergence Eq. (36.50) once the theory is quantized, and that divergence is a calculable number we confronted with the measured rate of $\pi^0 \to \gamma\gamma$. That was the anomaly as a *prediction*. This chapter studies what happens when the same arithmetic is read not as a prediction but as a *constraint*. The change of role is what organizes everything that follows: an anomaly in a current we have chosen to gauge is no longer a physical effect to be measured but a defect in the redundancy on which the consistency of the theory rests, while an anomaly in a current that stays global becomes an exact, coupling-independent invariant that the high- and low-energy descriptions of a strongly coupled theory are forced to share.

We begin from the gauged case. Building on the triangle computation and the Fujikawa measure of Chapter 36, and on the Ward–Slavnov–Taylor identities of Chapter 26 that underwrite unitarity and renormalizability, the opening sections show that gauging an anomalous current is fatal and that the whole danger collapses onto one number—the symmetric cubic invariant $A(R)$ attached to each representation. The group theory of that invariant (additivity, sign reversal under conjugation, the short list of safe groups) is assembled from the representation theory of Chapter 2, and is then turned on the Standard Model: using the chiral spectrum and hypercharge assignments imported from Chapter 33, we verify generation-by-generation cancellation of every dangerous triangle and, reading the conditions backwards, recover the quantization of hypercharge and hence of electric charge. A more structural section recasts the consistency requirement geometrically—the Wess–Zumino consistency condition as a BRST cocycle, the descent equations relating the four-dimensional anomaly to a six-form anomaly polynomial, and the Bardeen–Zumino polynomial that mediates between the

consistent and covariant anomalies.

The chapter then pivots to the global case and 't Hooft's anomaly-matching principle: because the anomaly coefficient is a renormalization-group invariant, any infrared realization of a global symmetry must reproduce the ultraviolet coefficients exactly. We prove the matching theorem by the spectator construction, establish that only massless states contribute, and apply it to massless QCD, where matching forbids the unbroken-chiral-symmetry scenario for $N_f \geq 3$ and so *forces* chiral symmetry breaking. The infrared theory that carries the anomaly is the chiral Lagrangian, and the operator that reproduces the coefficient is the Wess–Zumino–Witten term, which we build through Witten's five-dimensional construction, prove to have integer level $k = N_c$, and gauge to recover the $\pi^0 \to \gamma\gamma$ amplitude from a completely different direction than Chapter 36. A closing section widens the lens to the mixed gauge–gravitational anomaly (whose Standard-Model condition is $\operatorname{tr} Y = 0$), the absence of pure gravitational anomalies in four dimensions, Witten's nonperturbative global $SU(2)$ anomaly, and the modern synthesis in which anomalies are boundary data of an invertible bulk phase.

The throughline is a single object seen from two sides. Gauged, the anomaly must vanish, and the requirement is so rigid that it nearly fixes the matter content of the Standard Model; global, the anomaly cannot vanish but cannot change either, and that rigidity becomes a window into dynamics no perturbative method can reach. Both readings hand directly to the next part: the effective Lagrangians of Chapters 38 and 39 are constructed to match exactly the anomalies certified here, and the topological machinery rehearsed in the final section returns as the engine of Chapters 41 and 42.

37.1 From Curiosity to Constraint: Gauging an Anomalous Current

The chiral anomaly arrived in Chapter 36 as a piece of news about a *global* symmetry. The classically conserved axial current of a massless Dirac fermion is not conserved in the quantum theory; its divergence is the gauge-field density of Eq. (36.50),

$$\partial_\mu j^{5\mu} \;=\; \frac{e^2}{16\pi^2}\,\varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}, \tag{37.1}$$

a result we extracted two independent ways — from the linearly divergent triangle graph (Theorem 36.4) and from the Jacobian of the path-integral measure — and then tested against the measured rate of $\pi^0 \to \gamma\gamma$. In that setting the anomaly is a *prediction*: the axial current is an external probe, the photons in Eq. (37.1) belong to a perfectly healthy gauge field, and the failure of conservation is a physical effect we can compute and confront with experiment.

This section opens a different question, and with it the organizing idea of the entire chapter. Suppose the anomalous current is not a passive global probe but a current we have *gauged* — suppose the very A_μ that couples to it is dynamical, with its own propagator and its own would-be gauge invariance. Then the anomaly is no longer a number to be predicted. It is a defect in the foundations. A gauge symmetry is not a property of nature that the theory may or may not respect; it is a redundancy in our *description*, the bookkeeping that lets us write a Lorentz-covariant theory of a massless spin-one field while quarantining the unphysical timelike and longitudinal polarizations. Everything that makes a gauge theory consistent — the decoupling of those unphysical modes, the unitarity of the S-matrix on the physical subspace, the renormalizability that follows from the Ward–Slavnov–Taylor identities of Chapter 26 — rests on that redundancy being *exact at the quantum level*. An anomaly is precisely a redundancy that the quantum theory fails to respect. We will show in this section that gauging an anomalous current is therefore not merely dangerous but *fatal*: it breaks the identities on which unitarity and renormalizability depend, and there is no repair. The upshot is a constraint, not a curiosity — the fermion content of any chiral gauge theory must be arranged so that the anomaly cancels identically — and the whole problem will be shown to collapse onto a single requirement: that one functional, the anomalous gauge variation of the effective action, vanish.

37.1.1 From a global axial current to three gauge currents

The triangle of Chapter 36 had a definite structure: two vector vertices, where the two external photons attach, and one axial vertex, where the global current $j^{5\mu} = \bar{\psi}\gamma^\mu\gamma^5\psi$ attaches. This is the VVA triangle. The reason only this combination produced an anomaly is that the anomaly is intrinsically chiral — it needs the single γ^5 that distinguishes left from right — and the two photons couple vectorially.

In a chiral gauge theory the asymmetry between the three vertices disappears, because *all three* currents at the corners of the triangle are now gauge currents, and a chiral gauge current carries its own γ^5. Following the convention fixed in the chapter introduction, we write the entire fermion content as left-handed Weyl multiplets, collecting antiparticles into conjugate representations. Let the left-handed fields ψ_L furnish a (generally reducible) representation R of the gauge group G, with Hermitian generators T^a acting on the multiplet index, normalized as in Chapter 2. The covariant derivative carries the sign fixed in Chapter 8, $D_\mu = \partial_\mu - igA^a_\mu T^a$, and the gauge current to which A^a_μ couples is

$$J^{a\mu} \;=\; \bar{\psi}\,\gamma^\mu\,T^a\,P_L\,\psi, \qquad P_L = \tfrac{1}{2}(1-\gamma^5), \tag{37.2}$$

where ψ denotes the Dirac field built from ψ_L together with whatever right-handed partners the theory contains; the projector P_L enforces that only the

left-handed components are charged. The chirality is the whole point. Writing $P_L = \frac{1}{2}(1-\gamma^5)$ exhibits $J^{a\mu}$ as one half of a vector current minus one half of an axial current, $J^{a\mu} = \frac{1}{2}\bar{\psi}\gamma^\mu T^a \psi - \frac{1}{2}\bar{\psi}\gamma^\mu\gamma^5 T^a\psi$. Each of the three corners of the triangle now carries this $V-A$ mixture, so the diagram is a superposition of VVV, VVA, VAV, AVV, and the terms with two and three axial insertions.

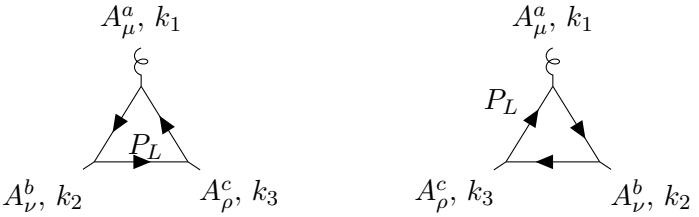


Figure 37.1: The three-gauge-current ("AAA") triangle of a chiral gauge theory: a closed loop of left-handed fermions with three gauge bosons attached, plus the diagram with the two lower legs exchanged (equivalently the reversed fermion-flow orientation). Each vertex carries a generator T^a and the chiral projector P_L. The completely symmetric part of the loop, proportional to $d_R^{abc} = \mathrm{tr}_R(T^a\{T^b, T^c\})$, is the seat of the gauge anomaly.

To isolate what survives, recall two facts established around the triangle computation of Chapter 36. First, a trace over a closed fermion loop with an *even* number of γ^5 insertions contains no ε-tensor and produces no anomaly: the VVV piece vanishes by Furry's theorem (charge conjugation), and the two-axial pieces reduce, by $(\gamma^5)^2 = 1$, to non-anomalous structures. The anomaly lives entirely in the terms with an *odd* number of γ^5 — the single-axial pieces $VVA+VAV+AVV$ and the triple-axial piece AAA — because only an odd number of γ^5's leaves one γ^5 inside the trace to contract the four free Lorentz indices into $\varepsilon^{\mu\nu\rho\sigma}$. Second, and decisively, the group-theory weight that multiplies the kinematic anomaly is fixed by the trace over the three generators sitting at the vertices. Because the kinematic factor multiplying $\varepsilon^{\mu\nu\rho\sigma}$ is totally symmetric under simultaneous exchange of the three external legs (their momenta and Lorentz indices), only the totally symmetric part of $\mathrm{tr}_R(T^aT^bT^c)$ contributes. That symmetric part is the cubic invariant

$$d_R^{abc} \;\equiv\; \mathrm{tr}_R\big(T^a\{T^b,T^c\}\big) \;=\; 2\,\mathrm{tr}_R\big(T^aT^{(b}T^{c)}\big), \tag{37.3}$$

the central object of the rest of the chapter, whose representation-theoretic properties we develop in the next section. Assembling the odd-γ^5 terms with their projectors, the net effect is that the anomaly of the gauge current Eq. (37.2) takes the same form as the abelian result Eq. (37.1), but with the abelian charge factor replaced by d_R^{abc} and the field strengths promoted to the non-abelian $F^a_{\mu\nu}$. In the covariant normalization imported from Eq. (36.159),

$$\big(D_\mu J^\mu\big)^a \;=\; \frac{g^2}{32\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}_R\big[T^aF_{\mu\nu}F_{\rho\sigma}\big] \;=\; \frac{g^2}{64\pi^2}\,d_R^{abc}\,\varepsilon^{\mu\nu\rho\sigma}F^b_{\mu\nu}F^c_{\rho\sigma}, \tag{37.4}$$

where the second equality uses $F_{\mu\nu} = F^b_{\mu\nu} T^b$ and the symmetry of $\varepsilon^{\mu\nu\rho\sigma} F^b_{\mu\nu} F^c_{\rho\sigma}$ under $b \leftrightarrow c$ to replace $\mathrm{tr}_R(T^a T^b T^c)$ by $\frac{1}{2} d_R^{abc}$. We checked the abelian limit against the bookkeeping of Eq. (37.1): for a single left-handed Weyl fermion of charge q one has $T^a \to q$ and $d_R^{abc} \to 2q^3$, so Eq. (37.4) reads $(\partial_\mu J^\mu) = \frac{g^2}{32\pi^2} q^3 \, \varepsilon F F$, exactly the divergence of a single chiral fermion — one half of Eq. (37.1) per power of γ^5 traded for P_L, with the cube of the charge reflecting one factor of q at each of the three vertices.

The structural lesson is already visible. In Chapter 36 the index "5" on $j^{5\mu}$ marked a single, distinguished, global current and the right-hand side of Eq. (37.1) was a physical density built from *honest* background photons. Now the index "a" on $J^{a\mu}$ runs over the gauge generators, and the right-hand side of Eq. (37.4) is built from the *same* dynamical field strengths whose conservation law we are testing. The anomaly has become a statement about the gauge field eating its own current. Whether that is tolerable is the question of the next three subsections.

37.1.2 The effective action and the gauge-variation functional

The cleanest formulation of consistency does not refer to any one diagram. Integrate the fermions out completely and study the functional they leave behind. With the left-handed multiplet ψ_L in representation R coupled to a background gauge field A^a_μ, define the fermionic effective action

$$e^{\,i\Gamma[A]} \;=\; \int \mathcal{D}\psi \, \mathcal{D}\bar{\psi} \; \exp\!\Big(i \int d^4x \; \bar{\psi} \, i \not{D} \, P_L \, \psi \Big), \tag{37.5}$$

the generating functional of one-particle-irreducible gauge-field amplitudes that arise from a single fermion loop. The two-point function $\delta^2\Gamma / \delta A \, \delta A$ is the vacuum polarization; the three-point function $\delta^3\Gamma / \delta A \, \delta A \, \delta A$ is exactly the triangle of Fig. 37.1; the higher functional derivatives are the box and pentagon insertions that complete the non-abelian anomaly. The full quantum theory adds the gauge-field kinetic term, the gauge-fixing and ghost sectors of Chapter 26, and further loops, but the obstruction we are after is already present in Eq. (37.5), because it is the fermion loop alone that carries the chiral asymmetry.

Classically, $\int d^4x \, \bar{\psi} \, i \not{D} P_L \psi$ is invariant under the infinitesimal gauge transformation

$$\delta_\alpha A^a_\mu \;=\; (D_\mu \alpha)^a \;=\; \partial_\mu \alpha^a + g f^{abc} A^b_\mu \alpha^c, \qquad \delta_\alpha \psi \;=\; i g \, \alpha^a T^a P_L \, \psi, \tag{37.6}$$

so a naive reading would conclude $\delta_\alpha \Gamma = 0$. The anomaly is precisely the failure of that conclusion. Performing the change of variables Eq. (37.6) inside the path integral Eq. (37.5) produces a Jacobian from the chiral measure — the Fujikawa Jacobian of Chapter 36 — and that Jacobian is not unity. Its logarithm is the

anomaly. Equivalently, and this is the form we use, the divergence of the current is the response of Γ to a gauge variation,

$$\delta_\alpha \Gamma[A] \;=\; \int d^4x\; \alpha^a(x)\, \big(D_\mu \tfrac{\delta \Gamma}{\delta A^a_\mu}\big) \;=\; -\int d^4x\; \alpha^a(x)\, \big(D_\mu J^\mu\big)^a, \tag{37.7}$$

since $J^{a\mu} = -\delta\Gamma/\delta A^a_\mu$ is the one-loop current and the covariant derivative arises from integrating by parts against the covariant transformation Eq. (37.6). Substituting the anomalous divergence Eq. (37.4) gives the central functional of the chapter, the *anomalous gauge variation*

$$\boxed{\mathcal{A}[\alpha, A] \;\equiv\; \delta_\alpha \Gamma[A] \;=\; -\frac{g^2}{32\pi^2}\int d^4x\; \alpha^a\, \varepsilon^{\mu\nu\rho\sigma}\; \mathrm{tr}_R\big[T^a F_{\mu\nu} F_{\rho\sigma}\big] \;+\; (\text{n.a.}),} \tag{37.8}$$

where "(n.a.)" denotes the additional non-abelian terms (cubic and quartic in A) required to complete Eq. (37.4) into a genuine gauge variation; like the leading term they are proportional to d_R^{abc} and they are derived in full when we construct the Wess–Zumino consistent form later in the chapter. The single fact we need now is the one already manifest in Eq. (37.8): $\mathcal{A}[\alpha, A]$ vanishes for all α and all A *if and only if* the cubic invariant d_R^{abc} vanishes.

It is illuminating to recast Eq. (37.8) in the language of BRST symmetry, because that is the language in which consistency is a theorem. Replace the gauge parameter α^a by the anticommuting Faddeev–Popov ghost c^a of Theorem 26.18; the gauge variation Eq. (37.6) becomes the BRST transformation $sA^a_\mu = (D_\mu c)^a$ of Chapter 26, and the anomalous variation Eq. (37.8) becomes

$$s\,\Gamma[A] \;=\; \mathcal{A}[c, A]. \tag{37.9}$$

In a healthy gauge theory the right-hand side is zero: $s\Gamma = 0$ is the statement that the quantum effective action is BRST invariant, which is the integrated form of the master Slavnov–Taylor identity Theorem 26.35. The anomaly is the obstruction to that identity. Moreover $\mathcal{A}[c, A]$ is not an arbitrary functional. Because the BRST operator is nilpotent, $s^2 = 0$, applying s to Eq. (37.9) forces

$$s\,\mathcal{A}[c, A] \;=\; s^2\,\Gamma \;=\; 0, \tag{37.10}$$

the *Wess–Zumino consistency condition*. It says the anomaly is a BRST cocycle at ghost number one. This algebraic constraint is what makes the anomaly rigid: it fixes the relative coefficients of all the terms in Eq. (37.8) (the abelian piece determines "(n.a.)" completely) and, as we use crucially below, it controls whether the anomaly can be removed by a local counterterm. We have thereby reduced the entire question of consistency to a single statement.

Definition 37.1 (Consistency condition for a chiral gauge theory)**.** A chiral gauge theory with left-handed fermions in the representation R of the gauge

group G is *anomaly-free*, and hence a candidate for a consistent quantum field theory, if and only if its anomalous gauge variation vanishes identically,

$$\mathcal{A}[\alpha, A] \equiv 0 \quad \text{for all } \alpha, A \qquad \Longleftrightarrow \qquad d_R^{abc} = \mathrm{tr}_R\big(T^a\{T^b, T^c\}\big) = 0 \ \text{ for all } a, b, c. \tag{37.11}$$

The remainder of this section justifies the phrase "and hence a candidate for a consistent quantum field theory" by proving that a nonzero $\mathcal{A}$ is genuinely fatal — first to unitarity, then to renormalizability — and the next section turns the right-hand condition $d_R^{abc} = 0$ into a finite sum over the chiral spectrum.

Remark 37.2 (Consistent versus covariant: which divergence is the symmetry statement). The divergence Eq. (37.4) and the functional Eq. (37.8) are not, despite appearances, the same expression dressed differently, and the distinction — which we meet here for the first time and develop in full later — is worth pinning down at once. The expression $(D_\mu J^\mu)^a$ in Eq. (37.4) is the *covariant* anomaly: it is built from the field strength $F_{\mu\nu}$ alone, so it transforms covariantly in the adjoint, $\delta_\alpha(D_\mu J^\mu)^a = g f^{abc}\alpha^b (D_\mu J^\mu)^c$, which is exactly what one expects of a current divergence. By contrast $\mathcal{A}[\alpha, A] = \delta_\alpha \Gamma$ in Eq. (37.8) is, by construction, the *consistent* anomaly: it is the literal gauge variation of a functional, and so it must satisfy the integrability constraint Eq. (37.10). These two objects do not coincide. The current $\delta\Gamma/\delta A$ obtained by differentiating the effective action — call it the *consistent current* — has a divergence whose leading term carries a coefficient $\frac{1}{3}$ relative to the covariant divergence Eq. (37.4), the famous factor of three, together with non-covariant terms that supply the "(n.a.)" completion of Eq. (37.8). The reason the two differ is that the consistent current is not gauge-covariant; the two currents are related by a local polynomial in the gauge field (the Bardeen–Zumino current), constructed explicitly in Section 37.4.4 [1]. For the consistency question of Theorem 37.1 the distinction is immaterial — both anomalies are proportional to d_R^{abc} and vanish together — but it matters greatly for *which* statement is "the symmetry being violated." The Ward–Slavnov–Taylor identity that protects unitarity and renormalizability is a statement about $\delta_\alpha\Gamma$, i.e. about the *consistent* anomaly; the covariant anomaly Eq. (37.4) is the convenient gauge-invariant repackaging that makes the group theory transparent. We compute with the covariant form because d_R^{abc} is manifest in it, and we reason about consistency with the consistent form because it is the one the identities see.

37.1.3 Why a gauge anomaly is fatal, I: loss of unitarity

We now prove the first half of the central theorem. The chain of logic is: a nonzero $\mathcal{A}$ violates the Slavnov–Taylor identity; the violated identity no longer forces the gauge Ward identity $k_\mu \mathcal{M}^\mu = 0$ on physical amplitudes; without that

[1] Forward reference to the descent/Wess–Zumino section of this chapter.

identity the unphysical (longitudinal and ghost) polarizations fail to decouple; and uncancelled unphysical states in a theory with an indefinite-metric state space destroy the positivity of the S-matrix.

Theorem 37.3 (A gauge anomaly breaks unitarity and renormalizability)**.** *Let a gauge theory contain chiral fermions with $d_R^{abc} \neq 0$, so that the anomalous gauge variation $\mathcal{A}[\alpha, A] = \delta_\alpha \Gamma$ of Eq. (37.8) does not vanish. Then* (i) *the Slavnov–Taylor identity Theorem 26.35 is violated by $\mathcal{A}$, the gauge Ward identity $k_\mu \mathcal{M}^\mu = 0$ fails, and the unphysical polarizations of the gauge field do not decouple from physical amplitudes, so unitarity is lost on the physical subspace; and* (ii) *there is no local counterterm whose gauge variation cancels $\mathcal{A}$, so the divergences cannot be organized into gauge-invariant structures and the theory is non-renormalizable.*

Proof of part (i). Recall from Chapter 26 the precise role of the Slavnov–Taylor identity. In the covariant (R_ξ) quantization the gauge field has four polarizations, two transverse and physical, two unphysical: the timelike and longitudinal combinations, propagated by the ξ-dependent and $k_\mu k_\nu$ pieces of the photon/gluon propagator Eq. (13.135). These unphysical polarizations have indefinite norm. They are kept out of physical results by two linked mechanisms: the Faddeev–Popov ghosts, whose wrong-statistics loops cancel the unphysical gauge-boson loops (the explicit cancellation of Section 26.4), and the gauge Ward identity, which removes the unphysical *external* polarizations. The master Slavnov–Taylor identity Theorem 26.35 is the all-orders statement that organizes both; the cohomological argument of Theorem 26.26 then guarantees that the physical Hilbert space — the BRST cohomology — carries a positive-definite inner product, which is unitarity.

Concretely, for an amplitude $\mathcal{M}^\mu(k)$ with one external gauge boson of momentum k stripped of its polarization vector, the Slavnov–Taylor identity reduces, when the external legs are amputated and on shell, to the transversality (gauge Ward) statement

$$k_\mu \, \mathcal{M}^\mu(k) \;=\; 0. \tag{37.12}$$

This is the non-abelian descendant of the Ward–Takahashi identity Theorem 19.13, and it is exactly the hypothesis under which the polarization sum may be replaced by $-\eta_{\mu\nu}$ and the unphysical modes drop out (the polarization-sum argument of Section 26.4). The identity Eq. (37.12) is a *consequence* of $\delta_\alpha \Gamma = 0$: it is what one obtains by functionally differentiating the Slavnov–Taylor identity with respect to the matter fields and amputating.

Now restore the anomaly. The Slavnov–Taylor identity is no longer $s\Gamma = 0$ but Eq. (37.9), $s\Gamma = \mathcal{A}[c, A] \neq 0$. Every Ward identity derived from it inherits an inhomogeneous term equal to the corresponding functional derivative of $\mathcal{A}$. The cleanest place to see this is the three-point function, where the anomaly first appears. Let $\Gamma^{\mu\nu\rho}_{abc}(k_1, k_2, k_3)$ be the 1PI three-gauge-boson vertex — the

amputated triangle of Fig. 37.1 — with $k_1 + k_2 + k_3 = 0$. Differentiating the consistent anomaly Eq. (37.8) three times with respect to A and Fourier transforming gives the anomalous Ward identity at the first vertex,

$$k_{1\mu}\,\Gamma^{\mu\nu\rho}_{abc}(k_1,k_2,k_3) \;=\; \frac{g^2}{24\pi^2}\,d_R^{abc}\,\varepsilon^{\nu\rho\alpha\beta}\,k_{2\alpha}\,k_{3\beta}, \tag{37.13}$$

where the coefficient is the consistent normalization: the factor $\frac{1}{3}$ relative to the covariant single-leg divergence $\frac{g^2}{8\pi^2}d_R^{abc}\,\varepsilon^{\nu\rho\alpha\beta}k_{2\alpha}k_{3\beta}$ of Eq. (36.162), per Theorem 37.2. The right-hand side is nonzero precisely when $d_R^{abc} \neq 0$. Compare this with the clean identity Eq. (37.12) it is supposed to be: the contraction $k_{1\mu}\Gamma^{\mu\nu\rho}$, which gauge invariance requires to vanish, instead equals a nonzero local polynomial in the momenta. This is the Slavnov–Taylor identity failing, written at the level of a single amplitude.

The physical consequence follows immediately. Build any physical S-matrix element in which this triangle appears as a subamplitude — for instance attach the three gauge-boson legs through propagators to external currents, or embed the triangle in a larger diagram with this gauge boson internal. The decoupling of the unphysical polarization of the k_1 line required exactly $k_{1\mu}\Gamma^{\mu\nu\rho} = 0$. With Eq. (37.13) that contraction is nonzero, so the longitudinal and scalar parts of the k_1 propagator — the $k_\mu k_\nu/k^2$ and ξ-dependent pieces of Eq. (13.135) — no longer cancel against the ghost contribution. An indefinite-norm state has been left in the physical channel. The optical theorem of Chapter 18 then equates the imaginary part of a forward amplitude to a sum over intermediate states $\sum_X |\langle X|\mathcal{M}|i\rangle|^2$; when X includes the uncancelled unphysical polarization, whose norm is negative, this "sum of squares" acquires a negative contribution, and the would-be probability is not positive. Unitarity on the physical subspace is lost. Equivalently, in the cohomological language of Theorem 26.26: the anomaly $\mathcal{A}[c, A]$ is a nontrivial obstruction to $s\Gamma = 0$, so the BRST charge is no longer nilpotent on the interacting Hilbert space, its cohomology is ill-defined, and the construction that delivered a positive-definite physical inner product collapses. This proves part (i). □

The mechanism is worth stating in words, because it explains why the gauge case is categorically different from the global case of Chapter 36. A global anomaly leaves a physical imprint — a decay rate — precisely because the symmetry it breaks was a genuine, non-redundant symmetry whose breaking is observable. A gauge "symmetry" is not a symmetry at all but a redundancy installed to make the unphysical polarizations bookkeepable; its Ward identities are the receipts proving those polarizations never contribute. When the anomaly tears up the receipts, the unphysical polarizations re-enter physical sums, and since they carry negative norm the theory predicts negative probabilities. There is nothing to measure; there is only an inconsistency.

37.1.4 Why a gauge anomaly is fatal, II: no symmetric counterterm

A reflexive response to a broken Ward identity is to add a counterterm that fixes it. Renormalization is, after all, the art of absorbing the symmetry-violating parts of loop integrals into local redefinitions of the fields and couplings. Part (ii) of Theorem 37.3 states that this cannot be done for the gauge anomaly. The obstruction is exactly the Wess–Zumino consistency condition Eq. (37.10), and it is instructive to see it both abstractly and on an explicit amplitude.

Proof of part (ii). A local counterterm is a local functional $\Delta\Gamma[A]$ added to the effective action to redefine the theory; the candidates compatible with locality and power counting are the dimension-≤ 4, Lorentz-invariant, gauge-index-contracted polynomials in A_μ and its derivatives. Adding $\Delta\Gamma$ shifts the anomaly by its own gauge variation,

$$\mathcal{A}[\alpha, A] \;\longmapsto\; \mathcal{A}[\alpha, A] + \delta_\alpha\, \Delta\Gamma[A] \;=\; \mathcal{A}[\alpha, A] + s\,\Delta\Gamma. \tag{37.14}$$

The anomaly can be removed by a counterterm if and only if there exists a local $\Delta\Gamma$ with $s\,\Delta\Gamma = -\mathcal{A}$, i.e. if and only if $\mathcal{A}$ is BRST-*exact* in the space of local functionals. But we proved in Eq. (37.10) that $\mathcal{A}$ is BRST-*closed*, $s\mathcal{A} = 0$, and the content of the Wess–Zumino consistency analysis — the computation of the local BRST cohomology at ghost number one, which we carry out via the descent equations in Section 37.4.4 — is that the cocycle $\mathcal{A} \propto d_R^{abc}$ is *not* exact: there is no local $\Delta\Gamma$ whose variation reproduces it. In cohomological terms the anomaly is a nontrivial class, $[\mathcal{A}] \neq 0$ in $H^1_s(\text{local functionals})$, precisely when $d_R^{abc} \neq 0$. Therefore no local counterterm restores gauge invariance. The would-be "counterterm" that formally cancels Eq. (37.8) is the gauged Wess–Zumino–Witten functional, which is *not* a local polynomial in A_μ — it is a genuinely nonlocal object (it requires an auxiliary field, the pion-like Goldstone of the broken would-be symmetry) and adding it changes the theory's field content rather than renormalizing it. This establishes the non-existence claim abstractly; we now exhibit it concretely. □

Explicit demonstration on the triangle vertex. Return to the 1PI three-point vertex $\Gamma^{\mu\nu\rho}_{abc}$ and ask what local counterterm could restore all three Ward identities. The triangle is superficially linearly divergent, and as in Theorem 36.4 its value depends on the routing of the loop momentum through an arbitrary shift; the freedom is one constant, an "a-parameter," that redistributes a local polynomial among the three external legs. Imposing the Ward identity Eq. (37.12) at one vertex fixes a. The decisive fact, inherited intact from the global computation of Chapter 36, is that with the symmetric coefficient $d_R^{abc} \neq 0$ *no single choice of a makes all three contractions vanish*: restoring conservation at the

first vertex pushes a nonzero divergence onto the second and third, and the total, summed over the three vertices, is the rigid invariant

$$k_{1\mu}\Gamma^{\mu\nu\rho}_{abc} + \text{(cyclic at legs 2, 3)} \propto d_R^{abc}\,\varepsilon^{\nu\rho\alpha\beta}k_{2\alpha}k_{3\beta} \neq 0. \tag{37.15}$$

In the global case of Chapter 36 this was tolerable: one chose a to conserve the two *vector* (gauge, photon) currents and let the anomaly fall entirely on the one *axial* (global) current, which is allowed to be anomalous because nothing is gauged at that vertex. In a chiral gauge theory all three vertices are gauge currents and *all three* Ward identities are mandatory; Eq. (37.15) says they cannot be satisfied simultaneously. To absorb the leftover Eq. (37.15) one would need a local counterterm cubic in the gauge fields whose gauge variation is the offending ε-tensor structure. The only Lorentz-invariant, dimension-four, parity-odd cubic local term available is built from $\varepsilon^{\mu\nu\rho\sigma}A^a_\mu A^b_\nu A^c_\rho(\partial A)$-type contractions; one checks directly that every such candidate is either a total derivative (and so contributes nothing) or fails to be gauge invariant in a way that cannot match the cocycle — it would itself spoil a Ward identity it was meant to protect. There is no symmetric local counterterm.

The same obstruction reads, at the level of the renormalization program, as a breakdown of the structure that makes a gauge theory renormalizable in the first place. In a healthy theory the gauge Ward identities force the divergent parts of the various Green's functions to assemble into the gauge-invariant combinations $-\frac{1}{4}Z_3F^a_{\mu\nu}F^{a\mu\nu}$ and $Z_1\bar\psi\not{D}\psi$, with the renormalization constants related by Slavnov–Taylor (this is what reduced the renormalization of Chapter 27 to a single coupling). The anomalous Ward identity Eq. (37.13) breaks exactly those relations: the longitudinal part of the gauge-boson two- and three-point functions, which gauge invariance had pinned to zero, now receives a divergent contribution at higher loop order (through the triangle as a subgraph) that has no gauge-invariant counterterm to absorb it. Restoring finiteness would require a non-gauge-invariant counterterm — ultimately a gauge-boson mass and an infinite tower of further non-invariant operators — introducing new free parameters at every order with no symmetry to fix them. That is the precise meaning of non-renormalizability. The anomaly does not merely make the theory ugly; it removes the symmetry that made it predictive.

Physical Insight: one redundancy, two failures, one cause

The loss of unitarity and the loss of renormalizability are not two diseases but one, seen from two angles. Both rest on the gauge Ward–Slavnov–Taylor identity Theorem 26.35; the anomaly $\mathcal{A} = \delta_\alpha\Gamma$ is the single quantity by which that identity fails. Read forward through the external states, the failure says the unphysical polarizations no longer decouple — unitarity. Read forward through the counterterms, the same failure says the divergences no longer assemble into invariant structures — renormalizability.

This is why Theorem 37.1 can compress "the theory is consistent" into "one functional vanishes": the functional $\mathcal{A}$ is the obstruction to the one identity on which everything else hangs.

37.1.5 The consistency condition reduced to a number

Theorem 37.3 establishes that a chiral gauge theory is consistent only if $\mathcal{A} \equiv 0$, and Theorem 37.1 reduces that to $d_R^{abc} = 0$. Two features make this a usable criterion rather than an abstract demand. First, d_R^{abc} is *additive over the spectrum*: the fermion loop sums over every left-handed Weyl multiplet independently, so for a theory whose left-handed content is the (reducible) representation $R = \bigoplus_i R_i$, the anomaly is

$$d_R^{abc} \;=\; \sum_i \operatorname{tr}_{R_i}\!\left(T^a\{T^b, T^c\}\right) \;\overset{!}{=}\; 0, \tag{37.16}$$

a *finite sum* over the chiral matter content. Consistency is thus a condition on the spectrum — which fermions, in which representations — and it is this that makes anomaly cancellation a tool for constraining model building rather than a property to be checked by computing loops. Second, the condition is one-loop exact: the Adler–Bardeen theorem (Chapter 36) guarantees that the anomaly coefficient receives no corrections beyond the triangle, so Eq. (37.16) is not an approximation but an exact, all-orders, renormalization-group-invariant statement about the theory.

The most important structural consequence of Eq. (37.16) is the distinction between vectorlike and chiral gauge theories, and it is what places QED on safe ground while warning that the non-abelian and electroweak theories are not automatically safe.

Proposition 37.4 (Vectorlike theories are automatically anomaly-free)**.** *If for every left-handed multiplet R_i in the spectrum the conjugate $\bar{R}_i$ is also present (equivalently, if the fermions can be assembled into Dirac fields whose left- and right-handed components carry the same gauge representation), then $d_R^{abc} = 0$ identically.*

Proof. The generators of the conjugate representation are $T^a_{\bar{R}} = -(T^a_R)^{\mathsf{T}}$. Compute the cubic invariant of $\bar{R}$,

$$d^{abc}_{\bar{R}} = \operatorname{tr}\!\left(T^a_{\bar{R}}\{T^b_{\bar{R}}, T^c_{\bar{R}}\}\right) = \operatorname{tr}\!\Big(\big(-(T^a_R)^{\mathsf{T}}\big)\,\{(T^b_R)^{\mathsf{T}}, (T^c_R)^{\mathsf{T}}\}\Big) = -\operatorname{tr}\!\Big(\big(T^a_R\{T^b_R, T^c_R\}\big)^{\mathsf{T}}\Big) = -\,d_R^{abc}, \tag{37.17}$$

where in the third equality we used $\{(T^b)^{\mathsf{T}}, (T^c)^{\mathsf{T}}\} = (\{T^c, T^b\})^{\mathsf{T}} = (\{T^b, T^c\})^{\mathsf{T}}$ and gathered the transposes, and in the last that the trace of a transpose equals the trace. Hence each R_i and its partner $\bar{R}_i$ contribute equal and opposite amounts to the sum Eq. (37.16), and the total vanishes. □

This is exactly why QED is safe. The electron enters as a Dirac fermion whose left- and right-handed parts carry the *same* electric charge $-e$; written in the all-left convention it is a left-handed Weyl field of charge $-e$ together with a left-handed Weyl field (the conjugate of the right-handed electron) of charge $+e$. The two charges enter the abelian cubic invariant $\sum_i q_i^3$ with opposite signs and cancel. QED, and any vectorlike gauge theory such as QCD with its left–right symmetric quark couplings, passes Theorem 37.1 automatically and needs no further arrangement. The danger is reserved for genuinely chiral theories — those, like the electroweak $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ sector, in which left- and right-handed fermions transform differently and the cancellation Eq. (37.17) is not handed to us. There, consistency becomes a nontrivial constraint on the particle content, and we will find it satisfied only by a remarkable conspiracy among the quarks and leptons of each generation.

Worked Example 37.5 (A single Weyl fermion versus a Dirac fermion in chiral U(1))**.** We make the vectorlike cancellation and its chiral failure completely explicit in the simplest setting, a U(1) gauge theory, advancing QED at the consistency layer.

One left-handed Weyl fermion of charge q. The generator is the number $T = q$, so the cubic invariant is $d = \mathrm{tr}(q\{q, q\}) = 2q^3$, and the consistency condition Eq. (37.16) reads

$$d^{\mathrm{U}(1)} \;=\; 2q^3 \;\neq\; 0 \quad (\text{for } q \neq 0). \tag{37.18}$$

A U(1) gauge theory with a single charged left-handed Weyl fermion is therefore *inconsistent*: by Theorem 37.3 its longitudinal photon does not decouple and no local counterterm rescues it. There is no such theory in nature, and now we see there cannot be.

One Dirac fermion of charge q. A Dirac fermion is a left-handed Weyl field of charge q *plus* a right-handed Weyl field of charge q; in the all-left convention the latter is the conjugate of a left-handed Weyl field of charge $-q$. The spectrum is thus $\{q, -q\}$ and

$$d^{\mathrm{U}(1)}_{\mathrm{Dirac}} \;=\; 2q^3 + 2(-q)^3 \;=\; 0. \tag{37.19}$$

The Dirac fermion is vectorlike — left and right carry the same charge q — so Theorem 37.4 applies and the anomaly cancels between the particle and the conjugate of its partner. This is the worked instance of the QED statement above: ordinary electrodynamics, with its Dirac electron, is anomaly-free because it is vectorlike, while a hypothetical theory gauging only the left-handed electron would be ruled out by Eq. (37.18). The same arithmetic, applied with matrix-valued T^a rather than numbers q, is what the next section organizes into representation theory.

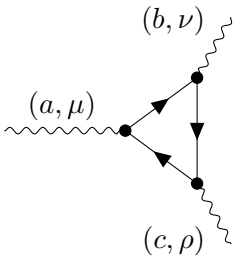
The lesson of Theorem 37.5 is that the entire fate of a chiral gauge theory rests on whether a single algebraic quantity, the spectrum sum Eq. (37.16) of the

cubic invariant $d_R^{abc} = \mathrm{tr}_R(T^a\{T^b, T^c\})$, vanishes. We have shown *why* it must vanish — unitarity and renormalizability both fail otherwise — and reduced "the theory is consistent" to this one condition. What we have not yet done is compute it. The cubic invariant is a pure piece of group theory, independent of the kinematics that produced it, and the next section extracts that kinematics once and for all, factorizes the triangle into a universal loop factor times d_R^{abc}, and develops the algebraic properties of d_R^{abc} — in particular the anomaly index $A(R)$ defined by $d_R^{abc} = A(R)\, d^{abc}$ — that turn the consistency requirement into a finite, checkable sum over the chiral matter of any candidate theory.

37.2 The Group Theory of Anomalies: The Cubic Invariant

The previous section pushed the consistency of a chiral gauge theory through a chain of reductions and left us standing in front of a single demand. The anomalous variation of the effective action under a gauge transformation—the obstruction to the Slavnov–Taylor identity (Theorem 26.35) on which unitarity and renormalizability rest—was shown to be controlled entirely by the divergence of the three-current correlator built from a triangle of chiral fermions. "The theory is consistent" was reduced to "one number vanishes," that number being a sum, over every left-handed Weyl multiplet in the spectrum, of the triangle's coupling-and-representation content (Eq. (37.11)). What we have not yet done is say what that number *is*. It still wears its full kinematic dress: a momentum integral, a Dirac trace, an ϵ-tensor, all tangled together with the group generators sitting at the three vertices.

This section performs the disentanglement. The central observation is that the triangle *factorizes*: the loop momentum integral and the Dirac algebra produce one universal kinematic object, identical for every fermion and every gauge group, multiplied by a purely group-theoretic factor that carries the entire dependence on which representation circulates in the loop. The kinematic factor was computed once and for all in Chapter 36; it can never vanish by itself. All the physics of cancellation therefore lives in the group factor, and that factor turns out to be a single totally symmetric invariant tensor, d^{abc}, weighted by one rational number per representation, the *anomaly index* $A(R)$. Once we have isolated $A(R)$ and established its three structural properties—it adds over direct sums, it flips sign under complex conjugation, and it exists at all only for a short list of gauge groups—the consistency condition becomes a finite piece of arithmetic over the chiral matter content. The group theory developed here is the canonical reference material of Section B.5; we recover it from the physics side, in the form the Standard Model computation of the next section will consume directly.



$+ \qquad (b,\nu) \leftrightarrow (c,\rho)$

Figure 37.2: The chiral triangle: three gauge currents and one left-handed Weyl fermion of representation R around the loop. Bose symmetry of the external gauge bosons forces the sum of this diagram with the crossed one in which the legs (b,ν) and (c,ρ) are exchanged. Each diagram factorizes into a group-theory trace and a representation-independent kinematic tensor $t^{\mu\nu\rho}$ computed in Chapter 36: the direct diagram carries $\mathrm{tr}_R(T^aT^bT^c)$ and the crossed one $\mathrm{tr}_R(T^aT^cT^b)$. In the genuinely anomalous, ϵ-tensor part of the amplitude the kinematic factors of the two diagrams are equal, so the surviving group factor is the symmetric sum $\mathrm{tr}_R(T^aT^bT^c)+\mathrm{tr}_R(T^aT^cT^b)=\mathrm{tr}_R(T^a\left\{T^b,T^c\right\})\equiv D^{abc}(R)$; the antisymmetric structure-constant part cancels between the two orderings.

37.2.1 Factorization of the chiral triangle

We work, as announced in the chapter introduction, with all fermions written as left-handed Weyl fields, antiparticles included by conjugate representations, and the covariant-derivative convention $D_\mu=\partial_\mu-igA^a_\mu T^a$ of Chapter 8. A single left-handed Weyl multiplet in a representation R of the gauge group couples to the gauge fields through the chiral current

$$J^{a\mu}=\bar\psi\,\gamma^\mu\,T^a_R\,P_L\,\psi,\qquad P_L=\tfrac{1}{2}(1-\gamma^5),\tag{37.20}$$

with T^a_R the Hermitian generators in R and P_L the left-handed projector. The potentially anomalous object identified in Section 37.1.2 is the one-loop three-point function of three such currents, computed by the triangle diagram of Fig. 37.2. Bose symmetry of the three identical external gauge bosons forces us to sum over both inequivalent orderings of the vertices around the fermion loop: the direct diagram, with vertices encountered in the cyclic order (a,b,c), and the crossed diagram, with two legs interchanged to the order (a,c,b).

The factorization is structural and survives every detail of the loop integral. At each vertex the Feynman rule supplies one generator T^a_R and one Dirac matrix $\gamma^\mu P_L$; the fermion propagators supply the spacetime and Dirac structure but are blind to the gauge indices. Consequently the amplitude of either diagram is a product of a single trace over the gauge representation and a single trace over Dirac space folded into the momentum integral,

$$\Gamma^{abc}_{\mu\nu\rho}(p,q)=\mathrm{tr}_R(T^aT^bT^c)\,t_{\mu\nu\rho}(p,q)+\mathrm{tr}_R(T^aT^cT^b)\,t_{\mu\rho\nu}(q,p),\tag{37.21}$$

where p, q are the two independent external momenta and $t_{\mu\nu\rho}(p,q)$ is the representation-*independent* kinematic tensor—the loop integral with the chiral projectors and three γ-matrices—evaluated in Chapter 36. The two terms differ only by the exchange $(b,\nu,p) \leftrightarrow (c,\rho,q)$ that distinguishes the two orderings.

Everything physical is now controlled by the index structure of Eq. (37.21). Split each product of three generators into its symmetric and antisymmetric parts using $\left[T^a, T^b\right] = i f^{abc} T^c$ and $\left\{T^a, T^b\right\} =$ (a symmetric combination), exactly as in the fundamental identity

$$\mathrm{tr}_R(T^a T^b T^c) = \tfrac{1}{2}\,\mathrm{tr}_R(\left\{T^a, T^b\right\} T^c) + \tfrac{1}{2}\,\mathrm{tr}_R(\left[T^a, T^b\right] T^c) = \tfrac{1}{2} D^{abc}(R) + \tfrac{i}{2} f^{abd}\,\mathrm{tr}_R(T^d T^c), \tag{37.22}$$

where we have defined the symmetrized trace

$$D^{abc}(R) \;\equiv\; \mathrm{tr}_R(\left\{T^a, T^b\right\} T^c). \tag{37.23}$$

The decisive fact, established in Chapter 36 as a property of the loop integral, is that the genuinely anomalous part of $t_{\mu\nu\rho}$—the part carrying the ϵ-tensor that cannot be removed by any local counterterm and that therefore obstructs current conservation—is *symmetric* under the exchange that relates the two diagrams: $t^{\mathrm{anom}}_{\mu\nu\rho}(p,q) = t^{\mathrm{anom}}_{\mu\rho\nu}(q,p)$. With the kinematic factors of the two terms equal, the group factors simply add, and the antisymmetric, structure-constant piece of Eq. (37.22) drops out of the sum:

$$\mathrm{tr}_R(T^a T^b T^c) + \mathrm{tr}_R(T^a T^c T^b) = \mathrm{tr}_R(T^a \left\{T^b, T^c\right\}) = D^{abc}(R). \tag{37.24}$$

The f^{abc} contributions, equal and opposite in the two orderings, cancel identically. This is the group-theoretic shadow of a familiar physical statement: the antisymmetric part of the triangle can be shuffled into the gauge-invariant Wess–Zumino–Bardeen counterterm and is never the true obstruction; only the totally symmetric remnant is physical. We therefore arrive at the clean statement that the anomalous divergence of the three-current correlator is

$$\partial^\mu \,\Gamma^{abc}_{\mu\nu\rho} \;=\; \frac{1}{8\pi^2}\,\epsilon_{\nu\rho\alpha\beta}\, p^\alpha q^\beta \; D^{abc}(R), \tag{37.25}$$

the prefactor being the universal kinematic coefficient carried over from the single-fermion computation of Chapter 36 in the *covariant* ($\pi^0 \to \gamma\gamma$) normalization, matching the covariant single-leg divergence $\frac{g^2}{8\pi^2} D^{abc}(R)$ of Eq. (36.162) once the gauge couplings are stripped from the three current vertices; the consistent normalization used in Eq. (37.13) is one third of this. The matter dependence has collapsed into the single tensor $D^{abc}(R)$, and the consistency condition of Eq. (37.11) becomes the requirement that the sum of these tensors over the left-handed spectrum vanish,

$$\sum_{i \in \text{left-handed Weyl}} D^{abc}(R_i) \;=\; 0 \qquad \text{for all } a, b, c. \tag{37.26}$$

37.2.2 The cubic invariant and the anomaly index

Condition Eq. (37.26) looks, at first sight, like a separate constraint for every choice of the three adjoint indices (a, b, c)—in $\mathfrak{su}(5)$, for instance, 24 generators give a symmetric array with $\binom{24+2}{3} = 2600$ components. It is the great simplification of the subject that these are not independent: $D^{abc}(R)$ is proportional, for *every* representation, to one fixed tensor, so the whole array collapses to a single scalar per representation. We now prove this and define that scalar.

The proportionality rests on two facts. First, $D^{abc}(R)$ is a totally symmetric invariant tensor of the algebra. Second, the space of such tensors is at most one-dimensional. Take the first.

Proposition 37.6 (Total symmetry and invariance of D^{abc})**.** *For any representation R of a simple Lie algebra, the symmetrized trace $D^{abc}(R)$ of Eq. (37.23) is totally symmetric in (a, b, c) and is an invariant tensor, $f^{ade}D^{ebc} + f^{bde}D^{aec} + f^{cde}D^{abe} = 0$.*

Proof. Symmetry in $a \leftrightarrow b$ is built into the anticommutator. For $b \leftrightarrow c$, expand and use cyclicity of the trace, $\mathrm{tr}(T^bT^aT^c) = \mathrm{tr}(T^aT^cT^b)$:

$$D^{abc} = \mathrm{tr}(T^aT^bT^c)+\mathrm{tr}(T^bT^aT^c) = \mathrm{tr}(T^aT^bT^c)+\mathrm{tr}(T^aT^cT^b) = \mathrm{tr}(T^a\left\{T^b,T^c\right\}), \tag{37.27}$$

which is manifestly symmetric in $b \leftrightarrow c$. The transpositions $(a\,b)$ and $(b\,c)$ generate all of S_3, so D^{abc} is totally symmetric. For invariance, the trace of a product of generators is unchanged under the simultaneous similarity $T^a \to UT^aU^{-1}$; taking U infinitesimal, $\delta T^a = -\theta^d f^{dae}T^e$, the total variation of D^{abc} must vanish, which is precisely the displayed Jacobi-type identity once the arbitrary parameter θ^d is stripped off. □

The one-dimensionality of the space of totally symmetric invariant rank-three tensors is a corollary of Schur's lemma (Theorem 2.21) applied to the symmetric cube of the adjoint representation: for a simple algebra the trivial representation appears in $\mathrm{Sym}^3(\mathrm{adj})$ at most once, so there is at most one such tensor up to normalization. (For the algebras where it appears exactly once, that tensor is the cubic symmetric Casimir invariant of Chapter 2; for those where it appears zero times, the only invariant is identically zero, a case we classify in Section 37.2.4.) We fix the normalization by the fundamental representation.

Definition 37.7 (Cubic symmetric invariant and anomaly index)**.** For $\mathfrak{su}(N)$ with $N \geq 3$, the totally symmetric tensor d^{abc} is the symmetrized trace in the fundamental,

$$d^{abc} \;\equiv\; 2\,\mathrm{tr}_{\mathrm{fund}}\big(\left\{T^a,T^b\right\}T^c\big), \tag{37.28}$$

the same tensor canonically tabulated in Theorem B.30. The *anomaly index* $A(R)$ of a representation R is then the scalar defined by

$$D^{abc}(R) = \mathrm{tr}_R\big(\{T^a, T^b\}\, T^c\big) \;=\; \tfrac{1}{2}\,\mathcal{A}(R)\, d^{abc}, \tag{37.29}$$

in agreement with Theorem B.31. By construction $A(\text{fund}) = 1$.

Theorem 37.6 guarantees the left-hand side of Eq. (37.29) lies in the one-dimensional space spanned by d^{abc}, so the equation defines a single number $\mathcal{A}(R)$, not an overdetermined system. With it the consistency condition Eq. (37.26) loses all its index structure: since the common tensor d^{abc} factors out, the entire content of gauge-anomaly cancellation is the vanishing of one rational number,

$$\boxed{\sum_{i\in\text{left-handed Weyl}} \mathcal{A}(R_i) \;=\; 0} \tag{37.30}$$

This is the calculator the rest of the chapter runs on. The remaining task is to learn the value of $\mathcal{A}(R)$ for the representations the Standard Model uses and to establish the rules that let us assemble those values without ever performing a trace again.

We begin by confirming the normalization with an explicit trace, which both pins down the convention and exhibits the machinery.

Worked Example 37.8 ($A(\text{fund}) = 1$ for $\mathfrak{su}(N)$)**.** For $\mathfrak{su}(N)$ the fundamental generators obey the anticommutator $\{T^a, T^b\} = \frac{1}{N}\delta^{ab}\mathbf{1} + d^{abc}T^c$ (this is Eq. (B.120), the completeness statement that there is no room for anything but the identity and the d-symbol). Insert it directly into the defining trace and use $\mathrm{tr}\, T^c = 0$ together with the fundamental normalization $\mathrm{tr}(T^cT^e) = \frac{1}{2}\delta^{ce}$ of Eq. (B.2):

$$\mathrm{tr}_{\text{fund}}\big(\{T^a, T^b\}\, T^c\big) = \tfrac{1}{N}\delta^{ab}\underbrace{\mathrm{tr}\, T^c}_{0} + d^{abe}\underbrace{\mathrm{tr}(T^eT^c)}_{\frac{1}{2}\delta^{ec}} = \tfrac{1}{2}\, d^{abc}. \tag{37.31}$$

Comparison with Eq. (37.29) gives $\mathcal{A}(\text{fund}) = 1$, for every $N \geq 3$. For the special case $\mathfrak{su}(3)$ in the Gell-Mann basis $T^a = \lambda^a/2$ the same line reads, using $\{\lambda^a, \lambda^b\} = \frac{4}{3}\delta^{ab}\mathbf{1} + 2d^{abc}\lambda^c$,

$$\mathrm{tr}_{\mathbf{3}}\big(\{T^a, T^b\}\, T^c\big) = \tfrac{1}{4}\Big(\tfrac{4}{3}\delta^{ab}\ \mathrm{tr}\,\tfrac{\lambda^c}{2} + 2d^{abe}\ \mathrm{tr}\,\tfrac{\lambda^e\lambda^c}{2}\Big) = \tfrac{1}{4}\cdot 2d^{abe}\cdot\delta^{ec} = \tfrac{1}{2}d^{abc}, \tag{37.32}$$

again $\mathcal{A}(\mathbf{3}) = 1$. The index of the fundamental is the unit in which every other anomaly is measured.

37.2.3 Additivity and conjugation: the algebra of the index

A spectrum is a list of representations, and Eq. (37.30) is a sum over that list. To evaluate it we need to know how $\mathcal{A}$ behaves when representations are combined, and how it responds to the operation—complex conjugation—that relates a left-handed antiparticle to a particle. Both rules are short and both are decisive.

Proposition 37.9 (Additivity)**.** *For representations R_1, R_2,*

$$\mathcal{A}(R_1 \oplus R_2) = \mathcal{A}(R_1) + \mathcal{A}(R_2), \tag{37.33}$$

and on a tensor product, $\mathcal{A}(R_1 \otimes R_2) = \dim(R_2)\,\mathcal{A}(R_1) + \dim(R_1)\,\mathcal{A}(R_2)$.

Proof. In $R_1 \oplus R_2$ the generators are block-diagonal, $T^a = T^a_{R_1} \oplus T^a_{R_2}$, so the trace of any product of generators is the sum of the traces over the two blocks; hence $D^{abc}(R_1 \oplus R_2) = D^{abc}(R_1) + D^{abc}(R_2)$ and Eq. (37.33) follows on dividing by $\frac{1}{2}d^{abc}$. The tensor-product rule is the worked computation Theorem B.37: on $V_1 \otimes V_2$ the generators act as $T^a = A^a \otimes \mathbf{1} + \mathbf{1} \otimes B^a$, and every cross-term in $\mathrm{tr}(\{T^a, T^b\}\,T^c)$ contains a single-generator trace and vanishes, leaving $\dim(V_2)\,D^{abc}(R_1) + \dim(V_1)\,D^{abc}(R_2)$. □

Additivity is the statement that makes Eq. (37.30) a finite sum at all: a spectrum of many fields contributes one term apiece, and reducible representations are handled component by component. The conjugation rule is what makes most of those terms cancel.

Theorem 37.10 (Sign reversal under conjugation)**.** *For every representation R,*

$$\mathcal{A}(\bar{R}) = -\mathcal{A}(R). \tag{37.34}$$

Proof. The conjugate representation (Theorem 2.62) is generated by $T^a_{\bar{R}} = -(T^a_R)^{\mathsf{T}}$: the transpose alone would reverse the order in commutators and spoil the algebra, and the extra minus sign restores $\left[T^a_{\bar{R}}, T^b_{\bar{R}}\right] = if^{abc}T^c_{\bar{R}}$ while keeping the generators Hermitian (the T^a_R are Hermitian, so $(T^a_R)^{\mathsf{T}} = (T^a_R)^*$ is too). The transpose of an anticommutator of two matrices is the anticommutator of their transposes, $\left\{M^{\mathsf{T}}, N^{\mathsf{T}}\right\} = (MN)^{\mathsf{T}} + (NM)^{\mathsf{T}} = (\{M,N\})^{\mathsf{T}}$. Apply this inside the defining trace with $M = T^a_R$, $N = T^b_R$, and use $(XY)^{\mathsf{T}} = Y^{\mathsf{T}}X^{\mathsf{T}}$ and the invariance of a trace under transposition, $\mathrm{tr}(Z^{\mathsf{T}}) = \mathrm{tr}\,Z$:

$$\begin{aligned} D^{abc}(\bar{R}) &= \mathrm{tr}\Big(\left\{-(T^a_R)^{\mathsf{T}}, -(T^b_R)^{\mathsf{T}}\right\}\,(-(T^c_R)^{\mathsf{T}})\Big) = -\,\mathrm{tr}\Big(\big(\left\{T^a_R, T^b_R\right\}\big)^{\mathsf{T}}\,(T^c_R)^{\mathsf{T}}\Big) \\ &= -\,\mathrm{tr}\Big(\big(T^c_R\,\left\{T^a_R, T^b_R\right\}\big)^{\mathsf{T}}\Big) = -\,\mathrm{tr}\big(T^c_R\,\left\{T^a_R, T^b_R\right\}\big) = -\,D^{abc}(R). \end{aligned} \tag{37.35}$$

The three minus signs from the three generators combine to one overall sign (the product of the two from the anticommutator pair is positive, leaving the single sign from T^c); the cyclicity of the trace in the last step puts the anticommutator back in standard position. Dividing by $\frac{1}{2}d^{abc}$ gives $\mathcal{A}(\bar{R}) = -\mathcal{A}(R)$. □

The two rules together produce the corollary that explains, in one stroke, why most theories are never in danger.

Corollary 37.11 (Real and pseudoreal representations are anomaly-free)**.** *If R is real or pseudoreal—that is, $R \cong \bar{R}$—then $\mathcal{A}(R) = 0$. In particular any vector-like spectrum, in which every left-handed fermion in R is accompanied by one in $\bar{R}$, has total anomaly zero automatically.*

Proof. If $R \cong \bar{R}$ there is an invertible S with $T^a_{\bar{R}} = S\,T^a_R\,S^{-1}$. A trace is invariant under this similarity, so $D^{abc}(\bar{R}) = D^{abc}(R)$, hence $\mathcal{A}(\bar{R}) = \mathcal{A}(R)$. Combined with Eq. (37.34), $\mathcal{A}(R) = -\mathcal{A}(R)$, so $\mathcal{A}(R) = 0$. For a vector-like spectrum, additivity (Theorem 37.9) gives $\mathcal{A}(R) + \mathcal{A}(\bar{R}) = \mathcal{A}(R) - \mathcal{A}(R) = 0$ for each pair. □

Physical Insight

This corollary is why *QED* and the strong sector of *Yang–Mills theory* never appear on any list of anomaly worries. A Dirac fermion is, in the left-handed language, a pair $R \oplus \bar{R}$: the electron is a left-handed charge -1 Weyl field together with a left-handed charge $+1$ field (the conjugate of the right-handed electron). The two charges enter the U(1) triangle as $(-1)^3$ and $(+1)^3$, which sum to zero—exactly the statement $\mathcal{A}(R) + \mathcal{A}(\bar{R}) = 0$ for the abelian case. QED is anomaly-free because it is vector-like. QCD is anomaly-free for the same reason: each quark flavor is a left-handed color $\mathbf{3}$ *and* a left-handed color $\bar{\mathbf{3}}$ (the conjugate of the right-handed quark), so $\mathcal{A}(\mathbf{3}) + \mathcal{A}(\bar{\mathbf{3}}) = 1 - 1 = 0$ flavor by flavor. The vanishing is not a numerical accident of the quark content; it is forced the instant the theory is built with equal left- and right-handed couplings. Only a genuinely *chiral* gauge theory—one in which the left-handed spectrum is not its own conjugate—can fail the test, and the electroweak sector of the Standard Model is exactly such a theory.

It is worth recording that the adjoint representation, which carries the gauge bosons themselves, is also safe: the adjoint is real (its generators $(T^a_{\text{adj}})_{bc} = -if^{abc}$ are Hermitian and the representation is equivalent to its conjugate), so $\mathcal{A}(\text{adj}) = 0$ by Theorem 37.11. Equivalently, as shown directly in Theorem B.34, the symmetrized trace over the adjoint is built from f^{abc} alone and is totally antisymmetric, hence cannot be proportional to the symmetric d^{abc} except by vanishing. Gauginos and any other adjoint fermions never spoil a theory.

37.2.4 Which gauge groups can be dangerous

So far we have assumed the symbol d^{abc} exists—that there *is* a nonzero totally symmetric invariant for $\mathcal{A}(R)$ to be proportional to. For most simple Lie groups there is not, and then $D^{abc}(R) \equiv 0$ for every representation and the gauge theory is anomaly-free no matter what fermions it contains. Knowing this list is half of model building: it tells us in advance which factors of a gauge group we must scrutinize and which we may ignore.

Theorem 37.12 (Anomaly-safe simple groups)**.** *Among the compact simple Lie groups, a nonzero totally symmetric cubic invariant d^{abc} exists* only *for* $\mathrm{SU}(N)$ *with $N \geq 3$. For every other simple group*—$\mathrm{SU}(2)$*, all* $\mathrm{SO}(N)$ *with $N \neq 6$, all* $\mathrm{Sp}(N)$*, and the exceptionals G_2, F_4, E_6, E_7, E_8—the cubic invariant vanishes identically, so $\mathcal{A}(R) = 0$ for all R and no perturbative cubic gauge anomaly can arise. The abelian group* $\mathrm{U}(1)$ *is the remaining case that admits a cubic invariant, the product of charges.*

The physically decisive instance is $\mathrm{SU}(2)$, and there the vanishing is a one-line computation that the reader can hold in the hand.

Proof for $\mathrm{SU}(2)$*.* Take $T^a = \sigma^a/2$ with the Pauli matrices, whose anticommutator is pure identity, $\left\{\sigma^a, \sigma^b\right\} = 2\delta^{ab}\mathbf{1}$. Then

$$\left\{T^a, T^b\right\} = \tfrac{1}{2}\delta^{ab}\mathbf{1} \quad \Longrightarrow \quad D^{abc}(\mathbf{2}) = \mathrm{tr}_{\mathbf{2}}\big(\left\{T^a, T^b\right\} T^c\big) = \tfrac{1}{2}\delta^{ab}\,\mathrm{tr}\,T^c = 0. \tag{37.36}$$

The anticommutator of two $\mathfrak{su}(2)$ generators has *no* term proportional to a third generator—there is simply no symmetric traceless cubic in the Pauli matrices, only the antisymmetric ϵ^{abc} that lives in the commutator—so $d^{abc} \equiv 0$ for $\mathrm{SU}(2)$. Since the symbol vanishes in the defining representation, $\mathcal{A}(R) = 0$ for every $\mathrm{SU}(2)$ representation; consistently, every $\mathrm{SU}(2)$ representation is real or pseudoreal, so Theorem 37.11 gives the same conclusion. An $\mathrm{SU}(2)$ gauge theory is automatically free of the cubic anomaly for any matter content whatsoever. □

Structural argument for the remaining groups. The general statement is a fact about the ring of invariant polynomials on the algebra. A rank-r compact simple algebra has exactly r algebraically independent (primitive) Casimir invariants, of fixed degrees $m_i + 1$ set by the exponents m_i of the algebra; every symmetrized-trace invariant, $\mathrm{Str}_R(T^{a_1}\cdots T^{a_n})$, reduces by Newton's identities to a polynomial in these primitives, so a nonzero totally symmetric rank-three tensor exists if and only if one of the primitive degrees equals 3. The degree multisets are: $A_{N-1} = \mathfrak{su}(N)$ has $\{2, 3, 4, \ldots, N\}$; the orthogonal $B_r = \mathfrak{so}(2r{+}1)$ and symplectic $C_r = \mathrm{Sp}(2r)$ have $\{2, 4, 6, \ldots, 2r\}$; $D_r = \mathfrak{so}(2r)$ has $\{2, 4, \ldots, 2r{-}2\}$ together with r; and the exceptionals begin $G_2 : \{2, 6\}$, $F_4 : \{2, 6, 8, 12\}$, $E_6 : \{2, 5, 6, 8, 9, 12\}$, $E_7 : \{2, 6, 8, 10, 12, 14, 18\}$, $E_8 : \{2, 8, 12, 14, 18, 20, 24, 30\}$. A primitive of degree

exactly 3 appears only in the A series, and only from $A_2 = \mathfrak{su}(3)$ onward—$A_1 = \mathfrak{su}(2)$ has the lone degree $\{2\}$, with no cubic, recovering Eq. (37.36). Every other simple algebra has its first invariant beyond the quadratic at degree 4 or higher, so its symmetric rank-three invariant is identically zero and $\mathcal{A}(R) = 0$ for all R. The single apparent exception in the orthogonal series is cosmetic: $D_3 = \mathfrak{so}(6)$ is isomorphic to $\mathfrak{su}(4) = A_3$, which is why the theorem excludes $N = 6$ from the safe SO(N); its degree set $\{2, 4, 3\}$ contains the cubic inherited from the $\mathfrak{su}(4)$ identification. For $\mathfrak{so}(N)$ with $N \neq 6$, Sp(N), and the exceptionals, the relevant representations are moreover real or pseudoreal, giving an independent confirmation through Theorem 37.11. □

The abelian case U(1) deserves its own line, because it is where the Standard Model's real danger lies. A U(1) has a single Hermitian generator, the charge Y, which is just a real number on each field. The "cubic invariant" degenerates to the cube of that number, and the index of a field of charge q is $\mathcal{A}_{\mathrm{U}(1)} = q^3$. There is no group structure to forbid it and no reality to save it: a chiral set of U(1) charges has anomaly $\sum_i q_i^3$, which vanishes only by an arithmetic conspiracy among the charges. Mixed triangles, with one U(1) leg and two non-abelian legs, similarly reduce to charge-weighted Dynkin indices, $\sum_i q_i\, T(R_i)$, and the gravitational anomaly—one U(1) current and two stress tensors—reduces to the plain charge sum $\sum_i q_i$. These are the only surviving conditions for the Standard Model gauge group, and they are exactly the hypercharge sum rules the next section evaluates.

Physical Insight

The list in Theorem 37.12 is precisely why the Standard Model gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is so forgiving. Of its three factors, the weak group $\mathrm{SU}(2)_L$ has no cubic invariant at all, so any triangle with three $\mathrm{SU}(2)_L$ legs vanishes identically; the color group $\mathrm{SU}(3)_c$ *does* have one, but the quarks couple to it vector-likely, so its pure cubic anomaly cancels representation by representation through Theorem 37.11. The only triangles that can be nonzero, and that must therefore be arranged to cancel by a genuine numerical conspiracy among the chiral fermions, are the ones with at least one hypercharge leg: $\mathrm{U}(1)_Y^3$, the mixed $\mathrm{SU}(2)_L^2\,\mathrm{U}(1)_Y$ and $\mathrm{SU}(3)_c^2\,\mathrm{U}(1)_Y$, and the gravitational $\mathrm{U}(1)_Y$. That a single generation of quarks and leptons makes all four of these sums vanish—and in doing so very nearly fixes the hypercharge assignments and ties the number of colors to electric-charge quantization—is the showcase result of Section 37.3. For contrast, every SU(N) with $N \geq 3$ used as a chiral gauge group, such as the SU(5) of grand unification, must be checked explicitly; that check, and the cancellation between the $\bar{\mathbf{5}}$ and $\mathbf{10}$, is carried out in Theorem B.38.

We close by collecting the values that the rest of the chapter will reach

for without recomputation. The fundamental sets the scale, $\mathcal{A}(\text{fund}) = 1$; conjugation gives $\mathcal{A}(\bar{N}) = -1$; reality gives $\mathcal{A}(\text{adj}) = 0$; and the additivity rules applied to the two-index tensors of $\mathfrak{su}(N)$, worked out in Theorem B.38, give $\mathcal{A}(\Lambda^2) = N - 4$ and $\mathcal{A}(\text{Sym}^2) = N + 4$. With the index $\mathcal{A}(R)$ defined, its additivity and conjugation laws proved, and the short list of dangerous group factors—U(1) and SU(N) with $N \geq 3$—now in hand, the consistency condition Eq. (37.30) has become a finite calculator. The next section turns it loose on the chiral fermion content of one Standard Model generation, evaluating each potentially anomalous triangle in turn and watching the cancellation pin down the hypercharges.

37.3 Anomaly Cancellation in the Standard Model

The previous section turned a question about Feynman diagrams into a question about representation theory. The triangle amplitude with three external gauge currents factorized into a fixed kinematic structure times the totally symmetric group-theory factor

$$d_R^{abc} = \text{tr}_R\big[T^a\{T^b, T^c\}\big] = \mathcal{A}(R)\, d^{abc}, \tag{37.37}$$

and we packaged the entire content of one chiral multiplet into a single number, the anomaly index $\mathcal{A}(R)$ defined in Theorem 37.7, with the two properties that make it a calculator rather than a definition: it is *additive* over a reducible representation, $\mathcal{A}(R_1 \oplus R_2) = \mathcal{A}(R_1) + \mathcal{A}(R_2)$ (Eq. (37.33)), and it *reverses sign under conjugation*, $\mathcal{A}(\bar{R}) = -\mathcal{A}(R)$ (Eq. (37.34)). The same section catalogued which products of group factors can possibly carry a nonzero symmetric invariant at all—the dangerous-factor list of Theorem 37.12. We now spend that machinery on the one theory the reader most wants to see survive: the Standard Model. The claim to be established, in full and with every sum carried, is that the chiral fermion content of a *single generation* makes every potentially anomalous gauge triangle vanish—and that, read in reverse, this requirement very nearly *dictates* the hypercharge assignments, turning the consistency of the gauge theory into a derivation of electric-charge quantization.

This advances QED/SM. In Section 31.7.4 we already met the cancellation once, as the last consistency check performed while *building* the electroweak theory; there it was verified (Theorem 31.30) given the hypercharges. Here we do two things that chapter could not. First, we organize the computation through the anomaly index, so that each cancellation is read off from $\mathcal{A}(R)$ and its conjugation property rather than rediscovered diagram by diagram. Second—and this is the chapter's showcase—we treat the hypercharges as *unknowns* and show that the anomaly conditions, taken together, fix them up to an overall scale and a single discrete choice. Consistency does not merely tolerate the observed charges; it predicts them.

multiplet	$\mathrm{SU}(3)_c$	$\mathrm{SU}(2)_L$	Y	content
Q	$\mathbf{3}$	$\mathbf{2}$	$+\frac{1}{6}$	left quark doublet (u_L, d_L)
u^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$-\frac{2}{3}$	conjugate of right up
d^c	$\bar{\mathbf{3}}$	$\mathbf{1}$	$+\frac{1}{3}$	conjugate of right down
L	$\mathbf{1}$	$\mathbf{2}$	$-\frac{1}{2}$	left lepton doublet (ν_L, e_L)
e^c	$\mathbf{1}$	$\mathbf{1}$	$+1$	conjugate of right electron

Table 37.1: One Standard-Model generation as left-handed Weyl multiplets, with the representations and hypercharges of Table 33.1. The electric charge of each component follows from $Q = T^3 + Y$: the doublet Q gives u_L at $Q = \frac{1}{2} + \frac{1}{6} = \frac{2}{3}$ and d_L at $-\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$; the singlet e^c has $Q = +1$, the positron. The number of Weyl components in each row is $\dim(\mathrm{SU}(3)_c) \times \dim(\mathrm{SU}(2)_L)$: $6, 3, 3, 2, 1$, summing to 15 per generation.

Notation Watch

Throughout this section every fermion is written as a *left-handed Weyl field*, with antiparticles included as the charge conjugates of the right-handed fields. This is the convention announced in the chapter introduction and it is the natural one for anomalies, because a left-handed and a right-handed field contribute to Eq. (37.37) with opposite sign, and conjugation converts the latter into the former: a right-handed field in representation R with hypercharge Y becomes a left-handed field in $\bar{R}$ with hypercharge $-Y$, whose anomaly index is $-\mathcal{A}(R)$ by Eq. (37.34). With everything left-handed the gauge anomaly is a single unsigned sum over the multiplet list, with no separate bookkeeping of chirality. This is the only difference from Theorem 31.30, which kept e_R, u_R, d_R right-handed and subtracted them; the two presentations give identical numbers, as they must.

37.3.1 The chiral spectrum of one generation

We import the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and the matter content from Section 33.1.1 and Table 33.1, written in the all-left-handed convention. One generation consists of five irreducible left-handed Weyl multiplets, listed in Table 37.1, with the hypercharges of Eq. (33.37), normalized as always by the Gell-Mann–Nishijima relation $Q = T^3 + Y$ of Eq. (31.20) so that the electric charge is recovered as the sum of weak isospin and hypercharge.

That the list closes on fifteen left-handed fields, with no right-handed gauge singlet required, is the first thing to notice: the Standard Model is a *genuinely chiral* gauge theory. A vector-like theory—one in which every left-handed field is accompanied by a right-handed partner in the same representation—can never have a gauge anomaly, because the partners contribute to Eq. (37.37) with

equal and opposite sign and Eq. (37.34) guarantees the cancellation multiplet by multiplet. The content of Table 37.1 is not of that form: there is no left-handed antineutrino, the doublets carry weak isospin while the singlets do not, and the hypercharges are a ragged set of fractions. A chiral theory has every right to be anomalous. That this one is not is the result we now prove.

37.3.2 The complete list of dangerous triangles

Each external leg of the triangle is a gauge current of one of the three factors $\mathrm{SU}(3)_c$, $\mathrm{SU}(2)_L$, $\mathrm{U}(1)_Y$, or—when we include gravity—the energy-momentum tensor. The dangerous-factor list of Theorem 37.12 tells us which assignments of the three legs can produce a nonzero symmetric invariant. Two elementary facts do almost all of the pruning.

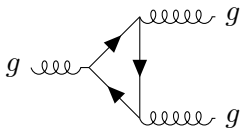
A simple-group generator is traceless. For any simple factor, $\operatorname{tr} T^a = 0$, so a triangle carrying exactly one leg of a given simple factor vanishes: the trace over that factor's single generator is zero before the hypercharges are even summed. This kills every mixed triangle with a lone $\mathrm{SU}(3)_c$ or $\mathrm{SU}(2)_L$ current, including the cross-terms $\mathrm{SU}(3)_c$–$\mathrm{SU}(2)_L$–$\mathrm{U}(1)_Y$, $\mathrm{SU}(3)_c$–$\mathrm{U}(1)_Y$–$\mathrm{U}(1)_Y$, $\mathrm{SU}(2)_L$–$\mathrm{U}(1)_Y$–$\mathrm{U}(1)_Y$, and the two double-simple mismatches $[\mathrm{SU}(3)_c]^2\mathrm{SU}(2)_L$ and $[\mathrm{SU}(2)_L]^2\mathrm{SU}(3)_c$. For a product representation the trace over the full multiplet factorizes across the group factors, so a single traceless factor anywhere in the product sends the whole thing to zero.

$\mathrm{SU}(2)_L$ *has no symmetric cubic invariant.* Every representation of $\mathrm{SU}(2)$ is real or pseudoreal, so $\mathcal{A}(R) = -\mathcal{A}(\bar{R}) = -\mathcal{A}(R) = 0$ for all of them; equivalently, $\mathrm{SU}(2) \cong \mathrm{Sp}(2)$ admits no totally symmetric invariant d^{abc}, the statement carried in Theorem B.35. Hence the pure $[\mathrm{SU}(2)_L]^3$ triangle vanishes identically, for *any* matter content whatsoever.

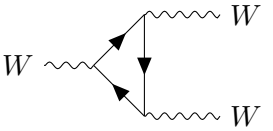
What survives is a short list. Three legs of $\mathrm{SU}(3)_c$ use the genuine color cubic invariant; one $\mathrm{U}(1)_Y$ leg can ride on two color or two weak legs through the symmetric quadratic invariant δ^{ab}; three $\mathrm{U}(1)_Y$ legs are unobstructed; and gravity contributes a single dangerous combination. The complete enumeration of potentially nonzero gauge and mixed gauge-gravitational triangles for $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is therefore

$$[\mathrm{SU}(3)_c]^3, \quad [\mathrm{SU}(2)_L]^3, \quad [\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{U}(1)_Y]^3, \quad \mathrm{U}(1)_Y\text{–}[\mathrm{grav}]^2, \tag{37.38}$$

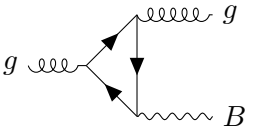
of which $[\mathrm{SU}(2)_L]^3$ is already zero. Figure 37.3 draws the six. The mixed gauge-gravitational anomaly, the last entry, comes from the triangle with one $\mathrm{U}(1)_Y$ current and two insertions of the energy-momentum tensor; its coefficient is the linear trace $\operatorname{tr} Y$ over the chiral content, and demanding it vanish is required for the theory to remain consistent when coupled to gravity. Its non-abelian cousins $\mathrm{SU}(3)_c$–$[\mathrm{grav}]^2$ and $\mathrm{SU}(2)_L$–$[\mathrm{grav}]^2$ vanish by $\operatorname{tr} T^a = 0$ and do not appear.



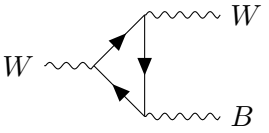
(a) $[\mathrm{SU}(3)_c]^3$: zero by vector-like color.



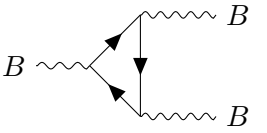
(b) $[\mathrm{SU}(2)_L]^3$: zero, no d^{abc}.



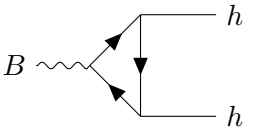
(c) $[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y$.



(d) $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$.



(e) $[\mathrm{U}(1)_Y]^3$.



(f) $\mathrm{U}(1)_Y$–[grav]^2.

Figure 37.3: The six potentially anomalous triangles for $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the complete list Eq. (37.38). External labels: g a gluon ($\mathrm{SU}(3)_c$), W a weak boson ($\mathrm{SU}(2)_L$), B the hypercharge boson ($\mathrm{U}(1)_Y$), h a graviton. All other leg assignments vanish because a simple-group generator is traceless. The fermion loop runs over the chiral content of Table 37.1; the full triangle computation that delivers the symmetric coefficient Eq. (37.37) is Chapter 36.

37.3.3 The two triangles that vanish on group theory alone

Two of the six need no arithmetic over the spectrum; they vanish because of what the groups are.

The $[\mathrm{SU}(2)_L]^3$ triangle is zero for the reason already given: SU(2) has no symmetric cubic invariant (Theorem B.35), so its anomaly index vanishes on every representation and the sum over the doublets Q and L is a sum of zeros. No constraint on hypercharges follows from it. This is the structural fact that lets the weak group be chiral—only left-handed fields are doublets— without that chirality ever threatening consistency.

The $[\mathrm{SU}(3)_c]^3$ triangle is zero because color is *vector-like*, and this is where the anomaly index earns its keep. The colored content of Table 37.1 is: the doublet Q, which is a color triplet $\mathbf{3}$ appearing in two weak-isospin copies; and the singlets u^c, d^c, each a color antitriplet $\bar{\mathbf{3}}$. By additivity (Eq. (37.33)) the total color anomaly index is

$$\mathcal{A}_{[\mathrm{SU}(3)_c]^3} = 2\,\mathcal{A}(\mathbf{3}) \;+\; \mathcal{A}(\bar{\mathbf{3}}) \;+\; \mathcal{A}(\bar{\mathbf{3}}) = 2\,\mathcal{A}(\mathbf{3}) + 2\,\mathcal{A}(\bar{\mathbf{3}}) = 2\,\mathcal{A}(\mathbf{3}) - 2\,\mathcal{A}(\mathbf{3}) = 0, \tag{37.39}$$

where the factor 2 on $\mathcal{A}(\mathbf{3})$ counts the two color triplets inside the weak doublet Q, and the last step is the conjugation property $\mathcal{A}(\bar{\mathbf{3}}) = -\mathcal{A}(\mathbf{3})$ of Eq. (37.34). The color representation content is $2\times\mathbf{3}$ together with $2\times\bar{\mathbf{3}}$: a conjugation-symmetric set, hence anomaly-free. The two units of $\mathbf{3}$ supplied by the quark doublet are exactly matched by the two units of $\bar{\mathbf{3}}$ supplied by the conjugate up and down

singlets. This is not a numerical accident about hypercharges; it is the statement that QCD is a vector-like theory, in which the right-handed up and down quarks pair with the left-handed ones into Dirac fermions transforming in the same color representation. We will see in Section 37.7 that the same color number $N_c = 3$ that makes Eq. (37.39) a sum of three triplets reappears as the quantized coefficient of the low-energy Wess–Zumino–Witten term.[2]

37.3.4 The four conditions that constrain hypercharge

The remaining four triangles—$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y$, $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$, $[\mathrm{U}(1)_Y]^3$, and $\mathrm{U}(1)_Y$–$[\mathrm{grav}]^2$—each mix a hypercharge factor with something else, so each delivers a genuine sum over the spectrum that the hypercharges must conspire to cancel. For a triangle with two non-abelian legs of a simple factor and one $\mathrm{U}(1)_Y$ leg, the symmetric trace reduces to the quadratic index times the hypercharge,

$$\mathrm{tr}\big[\{T^a, T^b\}\,Y\big] = 2\,T(R)\,\delta^{ab} \sum_{\text{multiplet}} Y, \tag{37.40}$$

with $T(R)$ the Dynkin index ($T(\mathbf{2}) = T(\mathbf{3}) = \tfrac{1}{2}$, equal for a representation and its conjugate). So the $[\text{simple}]^2\,\mathrm{U}(1)_Y$ condition is simply that the sum of hypercharges, weighted by the multiplicity of the non-abelian multiplet, vanish. We carry the four sums in the worked example below.

Worked Example 37.13 (The four hypercharge sums for one generation)**.** We evaluate the four nontrivial coefficients on the spectrum of Table 37.1, displaying every multiplicity factor. The recurring bookkeeping is that a color multiplet counts with weight $\dim(\mathrm{SU}(3)_c)$ when the external legs are not color legs (the color trace gives the dimension), and a weak doublet counts with weight $\dim(\mathrm{SU}(2)_L) = 2$ when the external legs are not weak legs; conversely a $[\text{simple}]^2$ pair contributes the index $T(R)$ once and strips that factor's dimension off the multiplicity.

(i) $[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y$**.** Only colored fields contribute, each weighted by its weak-isospin multiplicity, all carrying the common index $T(\mathbf{3}) = \tfrac{1}{2}$ of Eq. (37.40):

$$\begin{aligned}\mathcal{A}_{[\mathrm{SU}(3)_c]^2\mathrm{U}(1)} &\propto \underbrace{2}_{\text{doublet } Q}\cdot\left(\tfrac{1}{6}\right) + \underbrace{1}_{u^c}\cdot\left(-\tfrac{2}{3}\right) + \underbrace{1}_{d^c}\cdot\left(\tfrac{1}{3}\right)\\ &= \tfrac{1}{3} - \tfrac{2}{3} + \tfrac{1}{3} = 0.\end{aligned} \tag{37.41}$$

The two color triplets inside Q supply the factor 2; the conjugate singlets u^c, d^c each count once. The cancellation balances the doublet against the two singlets.

(ii) $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$**.** Only weak doublets contribute, each weighted by its color multiplicity, with the common index $T(\mathbf{2}) = \tfrac{1}{2}$:

$$\mathcal{A}_{[\mathrm{SU}(2)_L]^2\mathrm{U}(1)} \propto \underbrace{3}_{\text{color of } Q}\cdot\left(\tfrac{1}{6}\right) + \underbrace{1}_{L}\cdot\left(-\tfrac{1}{2}\right) = \tfrac{1}{2} - \tfrac{1}{2} = 0. \tag{37.42}$$

[2] Forward reference to a later section of this chapter.

Here the color factor 3 on the quark doublet is decisive: the quark contribution $3 \cdot \frac{1}{6} = \frac{1}{2}$ is engineered to cancel the lepton contribution $-\frac{1}{2}$ exactly. Were color SU(N) rather than SU(3), this condition would read $N\,Y_Q + Y_L = 0$, tying the lepton hypercharge to the color number—a relation we exploit in Theorem 37.14.

(iii) $[\mathrm{U}(1)_Y]^3$**.** Every field contributes its full multiplicity $\dim(\mathrm{SU}(3)_c) \times \dim(\mathrm{SU}(2)_L)$ times Y^3:

$$\begin{aligned}\mathcal{A}_{[\mathrm{U}(1)_Y]^3} &= \underbrace{6}_{Q}\left(\tfrac{1}{6}\right)^3 + \underbrace{3}_{u^c}\left(-\tfrac{2}{3}\right)^3 + \underbrace{3}_{d^c}\left(\tfrac{1}{3}\right)^3 + \underbrace{2}_{L}\left(-\tfrac{1}{2}\right)^3 + \underbrace{1}_{e^c}(1)^3 \\ &= \tfrac{6}{216} - \tfrac{24}{27} + \tfrac{3}{27} - \tfrac{2}{8} + 1 = \tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1. \end{aligned} \tag{37.43}$$

Putting everything over the common denominator 36,

$$\mathcal{A}_{[\mathrm{U}(1)_Y]^3} = \tfrac{1}{36} - \tfrac{32}{36} + \tfrac{4}{36} - \tfrac{9}{36} + \tfrac{36}{36} = \tfrac{1-32+4-9+36}{36} = \tfrac{0}{36} = 0. \tag{37.44}$$

(iv) $\mathrm{U}(1)_Y$**–[grav]^2.** The gravitational legs are flavor-blind, so the coefficient is the plain sum of hypercharges over all Weyl components, again with full multiplicities:

$$\mathcal{A}_{\mathrm{U}(1)\text{-grav}} = \operatorname{tr} Y = \underbrace{6 \cdot \tfrac{1}{6}}_{Q} + \underbrace{3 \cdot (-\tfrac{2}{3})}_{u^c} + \underbrace{3 \cdot \tfrac{1}{3}}_{d^c} + \underbrace{2 \cdot (-\tfrac{1}{2})}_{L} + \underbrace{1 \cdot 1}_{e^c} = 1-2+1-1+1 = 0. \tag{37.45}$$

All four nontrivial coefficients vanish, completing the proof that one Standard Model generation is free of gauge and mixed gauge-gravitational anomalies. The quark–lepton balance in Eq. (37.41) and Eq. (37.42) is the striking part: each is a sum of two separately nonzero pieces, one from quarks and one from leptons, that cancel only because the quark hypercharges are scaled by color exactly as $Q = T^3 + Y$ requires.

Because anomaly coefficients are *additive* over the matter content, and the content repeats identically for each of the three generations, the cancellation holds generation by generation. There is no cross-generation conspiracy and no need for one; three copies of zero is zero. This is a sharper statement than mere consistency of the whole theory: each family is independently anomaly-free, which is why one could not, for instance, omit the top quark and rescue the theory with the other two generations.

Physical Insight

The cancellations Eq. (37.41) and Eq. (37.42) have a flavor of arithmetic miracle precisely because the quark and lepton sectors know nothing about each other in the classical Lagrangian—they sit in different representations, couple through different currents, and could have been given any

hypercharges at all. Quantum consistency is the only place the two sectors are forced to communicate. The triangle diagram, summing over the entire chiral content at once, "sees" all fifteen fields together and demands that their charges balance. That a theory written down to fit the weak interactions and the electromagnetic charges should also, with no extra input, pass a quantum consistency test that mixes its quark and lepton charges is the first hint that the Standard Model is more tightly constrained than its Lagrangian lets on.

37.3.5 Reading the conditions backwards: charge quantization

We now reverse the logic. Suppose we are handed the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ and the *representation content* of Table 37.1—which fields are color triplets, which are weak doublets—but *not* the hypercharges. The hypercharges are five unknowns,

$$Y_Q, \quad Y_u, \quad Y_d, \quad Y_L, \quad Y_e, \tag{37.46}$$

attached respectively to Q, u^c, d^c, L, e^c. The question is how much of the observed assignment the anomaly conditions alone determine. The answer, derived in the worked example below, is: everything, up to one overall scale (which is unphysical, being the normalization of $\mathrm{U}(1)_Y$) and a single discrete two-fold choice (which the Yukawa couplings fix). Consistency does not merely permit the fractional quark charges; it forces them.

Worked Example 37.14 (Anomaly freedom determines the hypercharges)**.** With the hypercharges left as the unknowns Eq. (37.46), the four nontrivial conditions of Theorem 37.13 become four algebraic equations. We keep the color number N general (the physical case is $N = 3$) to expose how color enters; the weak group is fixed to SU(2). Reading the multiplicities off Table 37.1 with $\dim(\mathrm{SU}(3)_c) = N$:

$$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y: \qquad 2\,Y_Q + Y_u + Y_d = 0, \tag{37.47a}$$

$$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y: \qquad N\,Y_Q + Y_L = 0, \tag{37.47b}$$

$$\mathrm{U}(1)_Y\text{–}[\mathrm{grav}]^2: \qquad 2N\,Y_Q + N\,Y_u + N\,Y_d + 2\,Y_L + Y_e = 0, \tag{37.47c}$$

$$[\mathrm{U}(1)_Y]^3: \qquad 2N\,Y_Q^3 + N\,Y_u^3 + N\,Y_d^3 + 2\,Y_L^3 + Y_e^3 = 0. \tag{37.47d}$$

The first three are linear; the last is cubic. We solve the linear ones first.

Linear sector. Equation (37.47b) gives the lepton doublet hypercharge directly,

$$Y_L = -N\,Y_Q. \tag{37.48}$$

Insert Eq. (37.47a), in the form $Y_u + Y_d = -2Y_Q$, into the gravitational condition Eq. (37.47c): the colored terms collapse,

$$2N\,Y_Q + N(Y_u + Y_d) + 2Y_L + Y_e = 2N\,Y_Q - 2N\,Y_Q + 2Y_L + Y_e = 2Y_L + Y_e = 0, \tag{37.49}$$

so $Y_e = -2Y_L = 2N\,Y_Q$. At this stage the linear conditions have fixed

$$Y_L = -N\,Y_Q, \qquad Y_e = 2N\,Y_Q, \qquad Y_u + Y_d = -2\,Y_Q, \tag{37.50}$$

leaving the splitting of Y_u and Y_d undetermined. The cubic condition supplies the missing equation.

Cubic sector. Substitute $Y_L = -NY_Q$ and $Y_e = 2NY_Q$ into Eq. (37.47d). The lepton terms give $2(-N)^3Y_Q^3 + (2N)^3Y_Q^3 = (-2N^3 + 8N^3)Y_Q^3 = 6N^3Y_Q^3$, so

$$2N\,Y_Q^3 + N\,(Y_u^3 + Y_d^3) + 6N^3\,Y_Q^3 = 0 \implies Y_u^3 + Y_d^3 = -(2 + 6N^2)\,Y_Q^3. \tag{37.51}$$

Now use the symmetric functions of the two roots, $s \equiv Y_u + Y_d = -2Y_Q$ from Eq. (37.50) and $p \equiv Y_uY_d$, together with the identity $Y_u^3 + Y_d^3 = s^3 - 3sp$:

$$s^3 - 3sp = -8Y_Q^3 + 6Y_Q\,p = -(2 + 6N^2)Y_Q^3 \implies p = (1 - N^2)\,Y_Q^2. \tag{37.52}$$

Thus Y_u and Y_d are the two roots of $t^2 - st + p = t^2 + 2Y_Q\,t + (1 - N^2)Y_Q^2 = 0$, namely

$$t = \frac{-2Y_Q \pm \sqrt{4Y_Q^2 - 4(1 - N^2)Y_Q^2}}{2} = (-1 \pm N)\,Y_Q, \tag{37.53}$$

so that, up to the labeling of which singlet is which,

$$\{Y_u,\, Y_d\} = \{-(N + 1)\,Y_Q,\ (N - 1)\,Y_Q\}. \tag{37.54}$$

Collecting Eq. (37.50) and Eq. (37.54), every hypercharge is now pinned to Y_Q and the color number N:

$$Y_L = -N\,Y_Q, \quad Y_e = 2N\,Y_Q, \quad Y_u = -(N + 1)\,Y_Q, \quad Y_d = (N - 1)\,Y_Q. \tag{37.55}$$

The only freedom left is the overall scale Y_Q, which is not physical: it is the normalization of the $\mathrm{U}(1)_Y$ generator and can be absorbed into the coupling g'. Fixing it by the conventional choice $Y_Q = \frac{1}{2N}$ (equivalently $Y_Q = \frac{1}{6}$ at $N = 3$) reproduces Table 37.1 exactly:

$$N = 3: \quad Y_Q = \tfrac{1}{6}, \quad Y_L = -\tfrac{1}{2}, \quad Y_e = 1, \quad Y_u = -\tfrac{2}{3}, \quad Y_d = \tfrac{1}{3}. \tag{37.56}$$

The discrete ambiguity in Eq. (37.54)—whether u^c takes $-(N + 1)Y_Q$ or $(N - 1)Y_Q$—is the only remaining choice, and it is fixed not by anomalies but by the Yukawa couplings: only the assignment above lets both $Q\,\widetilde{H}\,u^c$ and $Q\,H\,d^c$ be gauge invariant against a single Higgs doublet H.

The physical reading of Eq. (37.55) is the quantization of electric charge. With $Q = T^3 + Y$ and the scale fixed by $Y_Q = \frac{1}{2N}$, the electron charge is

$$Q_e = -\tfrac{1}{2} + Y_L = -\tfrac{1}{2} - N\,Y_Q = -\tfrac{1}{2} - \tfrac{1}{2} = -1, \tag{37.57}$$

while the quark charges in the doublet are

$$Q_u = +\tfrac{1}{2} + Y_Q = \tfrac{1}{2} + \tfrac{1}{2N} = \frac{N+1}{2N}, \qquad Q_d = -\tfrac{1}{2} + Y_Q = -\frac{N-1}{2N}. \tag{37.58}$$

For $N = 3$ these are the familiar $Q_u = +\frac{2}{3}$ and $Q_d = -\frac{1}{3}$. The quark charges are fractional, in units of $1/N$, precisely *because* the anomaly conditions forced the lepton hypercharge to scale as N in Eq. (37.48). And the charge of a color singlet—a proton, three quarks uud—comes out

$$Q_{\text{proton}} = 2Q_u + Q_d = 2\cdot\frac{N+1}{2N} - \frac{N-1}{2N} = \frac{2N+2-N+1}{2N} = \frac{N+3}{2N} \xrightarrow{N=3} +1 = -Q_e. \tag{37.59}$$

The proton's charge is exactly opposite the electron's. In a theory of free parameters this equality, tested to better than one part in 10^{21}, would be an unexplained coincidence among continuous couplings; here it is an output of the demand that the gauge theory be quantum-mechanically consistent. The fractional quark charges and the integer-charged hadron are two faces of the same anomaly cancellation, with the color number $N = 3$ as the hinge between them.

Physical Insight

It is worth being precise about what is and is not derived. The anomaly conditions take the *representations* as given—that quarks are color triplets and weak doublets, that leptons are color singlets—and from them determine the hypercharges up to scale. They do not explain why the representations are what they are, nor why there are three generations. But within that input they convert a would-be set of five arbitrary real numbers into a one-parameter family pinned by N, and the one parameter is pure normalization. The often-heard slogan that "the Standard Model quantizes charge" is exactly this statement: in a chiral gauge theory the hypercharges cannot be dialed freely, because most choices are anomalous and inconsistent, and the consistent choices form the rigid pattern Eq. (37.55).

That the one-loop triangle is the whole story—that no two-loop or higher contribution can spoil Eq. (37.55) or generate a new anomaly—is the Adler–Bardeen theorem, which we establish in the next section. Its deeper origin is that the anomaly is not a free coefficient but a rigid algebraic object: it must satisfy the Wess–Zumino consistency condition, a cohomological constraint that

we develop next, and which is what ultimately makes the anomaly an *exact* invariant—fatal when gauged, as here, and matchable between the ultraviolet and infrared when global.

37.4 The Structure of Consistency: Wess–Zumino Conditions and Descent

The Standard Model computation of the previous section delivered a remarkable verdict: the anomaly of each dangerous gauge triangle is not merely small but is fixed, generation by generation, to be exactly a single rational number — the symmetric coefficient $d^{abc} = \mathrm{tr}(T^a\{T^b, T^c\})$ summed over the chiral content — and the requirement that this one number vanish is rigid enough to pin down the hypercharges and to tie the number of colours to charge quantization. That rigidity is not an accident of bookkeeping. It is the visible shadow of a deep algebraic structure: the anomaly is the unique solution, up to normalization, of a consistency equation forced on it by the very gauge symmetry it threatens to violate. The purpose of this section is to expose that structure. We will see that the anomaly is a cohomology class, that its defining consistency condition — the Wess–Zumino condition — is nothing but the integrability statement $s^2 = 0$ of the BRST operator, that the four-dimensional anomaly descends by a chain of local equations from a gauge-invariant *six*-form polynomial $\mathrm{tr}\, F^3$, and that its coefficient, being topological in origin, is one-loop exact and renormalization-group invariant. This last fact is the whole point: a quantity that cannot run is a quantity the ultraviolet and the infrared must agree upon, which is precisely what powers 't Hooft matching in the next section.

Throughout we work with the BRST machinery of Chapter 26. Recall the BRST transformation of Theorem 26.18, acting on the gauge field, the ghost, the antighost and the auxiliary field, and its nilpotency $s^2 = 0$ (Theorem 26.19); recall that the gauge Ward identities are organized by the Slavnov–Taylor identity of Section 26.7, on whose validity unitarity and renormalizability rest. The reframing of this section is that the anomaly is exactly the obstruction to that identity, measured in BRST cohomology.

37.4.1 Differential-form packaging of the gauge field

The cohomological structure is invisible in component notation and luminous in the language of matrix-valued differential forms, so we adopt the latter for this section. Collect the gauge potential into a Lie-algebra-valued one-form and the field strength into a two-form,

$$A \;\equiv\; A_\mu\, \mathrm{d}x^\mu, \qquad A_\mu \;\equiv\; -ig\, A^a_\mu\, T^a, \qquad F \;\equiv\; \mathrm{d}A + A^2 \;=\; \tfrac{1}{2} F_{\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu, \tag{37.60}$$

where products of forms are wedge products of matrix-valued forms (so $A^2 = A \wedge A = \tfrac{1}{2}[A_\mu, A_\nu]\,\mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$, nonzero precisely because the A_μ do not commute), and the coupling has been absorbed into the anti-Hermitian connection $A_\mu = -igA_\mu^a T^a$. In components this reproduces the non-abelian field strength of Eq. (25.49), $F_{\mu\nu} = -ig(\partial_\mu A_\nu^a - \partial_\nu A_\mu^a + g f^{abc} A_\mu^b A_\nu^c)T^a$, with $D_\mu = \partial_\mu - igA_\mu^a T^a$ as fixed in Chapter 8. The Bianchi identity reads $DF \equiv \mathrm{d}F + [A, F] = 0$, where for a p-form α the graded commutator is $[A, \alpha] = A\alpha - (-1)^p \alpha A$. A finite gauge transformation acts by $A \mapsto U^{-1}AU + U^{-1}\mathrm{d}U$ and infinitesimally, with parameter $\lambda = -ig\lambda^a T^a$, by $\delta_\lambda A = \mathrm{d}\lambda + [A, \lambda] = D\lambda$ and $\delta_\lambda F = [F, \lambda]$.

The packaging buys a clean grading. We assign to every object a *total degree* equal to its form degree plus its ghost number, and we treat both the exterior derivative d and the BRST operator s as odd derivations of total degree $+1$ that anticommute,

$$\mathrm{d}^2 = 0, \qquad s^2 = 0, \qquad \{s, \mathrm{d}\} = s\,\mathrm{d} + \mathrm{d}\,s = 0. \tag{37.61}$$

The ghost $c = -igc^a T^a$ is a Grassmann-odd, Lie-algebra-valued zero-form of total degree $+1$. In this notation the BRST transformations of Theorem 26.18 become the compact pair

$$sA \;=\; -Dc \;=\; -(\mathrm{d}c + [A, c]), \qquad sc \;=\; -c^2 \;=\; -\tfrac{1}{2}[c, c]. \tag{37.62}$$

A short remark on conventions is in order, because the sign of the ghost is not universal.

Notation Watch

In the component normalization of Eq. (26.95) the same transformations were written $sA_\mu^a = (D_\mu c)^a$ and $sc^a = -\tfrac{1}{2} g f^{abc} c^b c^c$. The matrix-form expressions (37.62) carry the opposite overall sign on sA, which is the customary sign of the ghost field in the descent literature (Stora–Zumino). The choice is immaterial: a relabelling absorbs it and changes no physical anomaly coefficient. We use (37.62) here purely because it makes Eq. (37.61) manifest, and we will state all final results in a normalization fixed by the one-loop triangle of Chapter 36.

That (37.62) is nilpotent is worth seeing once in this notation, since the whole edifice rests on it. Acting twice on the ghost, $s^2 c = s(-c^2) = -((sc)c - c(sc)) = -((-c^2)c - c(-c^2)) = c^3 - c^3 = 0$. Acting twice on the connection,

$$\begin{aligned} s^2 A &= s(-\mathrm{d}c - [A, c]) = -s(\mathrm{d}c) - s[A, c] \\ &= \mathrm{d}(sc) - ((sA)c - A(sc) + (sc)A - c(sA)), \end{aligned} \tag{37.63}$$

using $s\mathrm{d} = -\mathrm{d}s$ in the first term and the graded Leibniz rule $s[A, c] = [sA, c] - [A, sc]$ in the second (with the odd A producing the minus signs shown). Substi-

tuting $sA = -\mathrm{d}c - Ac - cA$ and $sc = -c^2$ and expanding,

$$\begin{aligned} s^2 A &= \mathrm{d}(-c^2) - \Big[(-\mathrm{d}c - Ac - cA)c - A(-c^2) + (-c^2)A - c(-\mathrm{d}c - Ac - cA)\Big] \\ &= -\big((\mathrm{d}c)c - c\,\mathrm{d}c\big) - \Big[-(\mathrm{d}c)c - Ac^2 - cAc + Ac^2 - c^2 A + c\,\mathrm{d}c + cAc + c^2 A\Big] \\ &= -(\mathrm{d}c)c + c\,\mathrm{d}c + (\mathrm{d}c)c - c\,\mathrm{d}c \;=\; 0, \end{aligned} \tag{37.64}$$

where every quadratic-in-c matrix term cancels in pairs. The field strength inherits a covariant transformation law,

$$sF \;=\; -[c, F], \tag{37.65}$$

i.e. F rotates into itself under BRST exactly as it does under an ordinary infinitesimal gauge transformation with parameter c. We verify this efficiently below using the master identity that organizes all of these statements at once.

Physical Insight

The pair (37.62) is the *Maurer–Cartan structure* of the gauge group made into an algebra. Reading $sA = -Dc$ as "BRST = infinitesimal gauge transformation with the parameter promoted to the anticommuting ghost" and $sc = -c^2$ as "the ghost is the failure of two gauge transformations to commute," the equation $s^2 = 0$ is the Jacobi identity in disguise. Every consistency statement about anomalies in this section is ultimately this Jacobi identity, seen through different windows.

37.4.2 The consistent anomaly as a BRST cocycle

Let $\Gamma[A]$ denote the one-particle-irreducible effective action obtained by integrating out a set of chiral (left-handed Weyl) fermions in the background gauge field A,

$$e^{\,i\Gamma[A]} \;=\; \int \mathcal{D}\psi\,\mathcal{D}\bar{\psi}\;\exp\!\Big(i\int \mathrm{d}^4x\;\bar{\psi}\,i\not{D}P_L\,\psi\Big), \tag{37.66}$$

with $P_L = \tfrac{1}{2}(1-\gamma^5)$. Classically the fermion action is gauge invariant, so one would expect $s\Gamma = 0$. The content of the anomaly — established in Chapter 36 by the triangle graph and independently by the Fujikawa Jacobian of the measure — is that the measure $\mathcal{D}\psi\,\mathcal{D}\bar{\psi}$ is *not* invariant, and the failure is local. We package that failure as the BRST variation of the effective action.

Definition 37.15 (Consistent anomaly)**.** The *consistent anomaly* is the BRST variation of the fermionic effective action,

$$\begin{aligned} G[A] \;\equiv\; s\,\Gamma[A] \;&=\; \int \mathrm{d}^4x\;(sA^a_\mu)\,\frac{\delta\Gamma}{\delta A^a_\mu} \;=\; \int \mathrm{d}^4x\;(D_\mu c)^a\,J^{a\mu} \\ &=\; -\int \mathrm{d}^4x\;c^a\,(D_\mu J^\mu)^a, \end{aligned} \tag{37.67}$$

where $J^{a\mu} \equiv \delta\Gamma/\delta A^a_\mu$ is the consistent current and the last step integrates by parts. It is a local functional of A and c, linear in the ghost, hence of *ghost number* $+1$. The quantity $\mathcal{G}^a \equiv (D_\mu J^\mu)^a$ is the consistent anomaly density: the covariant divergence of the consistent current.

The adjective "consistent" is justified by the next theorem, which is the central structural fact of anomaly theory. It says that $G[A]$ is not an arbitrary local ghost-number-one functional; it is BRST-closed.

Theorem 37.16 (Wess–Zumino consistency condition)**.** *The consistent anomaly is annihilated by the BRST operator,*

$$s\,G[A] \;=\; 0. \tag{37.68}$$

Equivalently, writing $G = \int c^a \mathcal{G}^a$ and letting $\mathcal{D}^a(x) \equiv -D_\mu\,\delta/\delta A^a_\mu(x)$ be the generator of gauge transformations on functionals of A, the density obeys the local condition

$$\mathcal{D}^a(x)\,\mathcal{G}^b(y) \;-\; \mathcal{D}^b(y)\,\mathcal{G}^a(x) \;=\; f^{abc}\,\delta^{(4)}(x-y)\,\mathcal{G}^c(x). \tag{37.69}$$

Proof. The compact statement is immediate from nilpotency: Γ is an ordinary (ghost-number-zero) functional, s is nilpotent on every functional built from the fields by Theorem 26.19, hence

$$s\,G \;=\; s\,(s\,\Gamma) \;=\; s^2\,\Gamma \;=\; 0. \tag{37.70}$$

The entire consistency condition is therefore nothing more than $s^2 = 0$ applied to the effective action — the integrability statement of the BRST differential.

To unfold (37.70) into the local form (37.69) we write s as a functional differential operator. On functionals of A and c,

$$s \;=\; \int \mathrm{d}^4x\,\Big[(D_\mu c)^a(x)\,\frac{\delta}{\delta A^a_\mu(x)} \;-\; \tfrac{1}{2}g f^{abc} c^b(x) c^c(x)\,\frac{\delta}{\delta c^a(x)}\Big], \tag{37.71}$$

the two pieces implementing $sA = Dc$ and $sc = -\frac{1}{2}gfcc$ respectively (we use the component normalization here so the structure constants are explicit). Apply s to $G = \int \mathrm{d}^4y\, c^b(y)\,\mathcal{G}^b(y)$. The first piece of (37.71) acts on the A-dependence hidden in $\mathcal{G}^b$, producing

$$\int \mathrm{d}^4x\,\mathrm{d}^4y\; c^b(y)\,(D_\mu c)^a(x)\,\frac{\delta\mathcal{G}^b(y)}{\delta A^a_\mu(x)} \;=\; -\int \mathrm{d}^4x\,\mathrm{d}^4y\; c^b(y)\,c^a(x)\,\mathcal{D}^a(x)\,\mathcal{G}^b(y), \tag{37.72}$$

after integrating by parts to move the covariant derivative off the ghost and onto $\delta/\delta A$, which assembles into $\mathcal{D}^a(x)$. The second piece of (37.71) acts on the explicit ghost $c^b(y)$ in G, giving

$$\int \mathrm{d}^4y\,\Big(-\tfrac{1}{2}g f^{bde} c^d c^e\Big)(y)\,\mathcal{G}^b(y) \;=\; -\tfrac{1}{2}g\int \mathrm{d}^4x\,\mathrm{d}^4y\;\delta^{(4)}(x-y)\,f^{bde}\,c^d(x)c^e(y)\,\mathcal{G}^b(y). \tag{37.73}$$

Now $sG = 0$ for *arbitrary* ghost configurations $c^a(x)$. The ghosts anticommute, $c^a(x)c^b(y) = -c^b(y)c^a(x)$, so only the part of the coefficient that is antisymmetric under $(a,x) \leftrightarrow (b,y)$ survives. Antisymmetrizing (37.72) and adding (37.73), and stripping the two arbitrary ghosts, leaves exactly

$$\mathcal{D}^a(x)\,\mathcal{G}^b(y) \;-\; \mathcal{D}^b(y)\,\mathcal{G}^a(x) \;-\; g f^{abc}\,\delta^{(4)}(x-y)\,\mathcal{G}^c(x) \;=\; 0, \tag{37.74}$$

which is (37.69) in the absorbed-coupling normalization where $g f^{abc} \to f^{abc}$. The relabelling $b \to c$ in the last term used the total antisymmetry of f^{abc}. □

The condition (37.69) is severe. It says that the candidate anomaly density cannot be any Lorentz-invariant, parity-odd, dimension-four local polynomial in A one might write down; it must intertwine with the gauge algebra in exactly the way (37.69) prescribes. Writing the most general such polynomial with free coefficients and imposing (37.69) fixes all but one overall constant. That surviving constant is the anomaly coefficient. This is the algebraic origin of the "single number" the previous section worked so hard to cancel.

Physical Insight

A BRST variation is automatically closed: anything of the form $s(\text{local})$ satisfies the consistency condition trivially, because $s^2 = 0$. Such an anomaly is *spurious* — it can be removed by subtracting the local counterterm from Γ. The genuine anomaly is therefore an element of the cohomology $H^1(s)$: BRST-closed ghost-number-one local functionals *modulo* BRST-exact ones. The miracle of four dimensions is that for a simple gauge group this cohomology is one-dimensional. There is essentially one anomaly, and the only freedom is its coefficient. Rigidity is cohomological.

Caution

"Local" is doing real work in the definition of the cohomology. The integrated anomaly $\int c^a \mathcal{G}^a$ is BRST-closed but is *not* the variation of any *local* functional — that is exactly why it is an anomaly. There do exist nonlocal functionals (the Wess–Zumino–Witten action, met later in this chapter) whose variation reproduces it; using a nonlocal counterterm to cancel the anomaly trades the symmetry violation for a violation of locality, which is no cure at all for a gauged symmetry.

37.4.3 Consistent versus covariant: the Bardeen–Zumino polynomial

The current $J^{a\mu} = \delta\Gamma/\delta A^a_\mu$ defined by (37.67) has a defect that the name "consistent" both advertises and excuses: it does not transform covariantly under

gauge transformations. Its divergence $\mathcal{G}^a$ is the object that satisfies Wess–Zumino, but $\mathcal{G}^a$ is *not* a gauge-covariant local expression in A — the explicit $A\,\mathrm{d}A$ and A^3 structures we will exhibit in (37.93) make that manifest. By contrast, the triangle computation of Chapter 36 naturally produces a manifestly covariant answer, the covariant anomaly of Eq. (36.159),

$$(D_\mu J^\mu_{\mathrm{cov}})^a \;=\; \frac{1}{32\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\;\mathrm{tr}(T^a\,F_{\mu\nu}F_{\rho\sigma}) \;+\; (\text{symmetric in reps}), \tag{37.75}$$

which is built from F alone and rotates in the adjoint. The two cannot be literally equal, because one is covariant and the other satisfies Wess–Zumino; (37.75) does *not* satisfy (37.69), and $\mathcal{G}^a_{\mathrm{cons}}$ is not covariant. They are reconciled by a local redefinition of the current.

Definition 37.17 (Bardeen–Zumino polynomial)**.** There exists a local polynomial $X^{a\mu}[A]$ in the gauge field and its derivatives, the *Bardeen–Zumino current*, such that

$$J^{a\mu}_{\mathrm{cov}} \;=\; J^{a\mu}_{\mathrm{cons}} \;+\; X^{a\mu}[A], \tag{37.76}$$

and the covariant anomaly (37.75) is its covariant divergence. For the chiral anomaly in four dimensions, in the abelian normalization,

$$X^\mu[A] \;=\; \frac{1}{12\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\left(A_\nu\,\partial_\rho A_\sigma + \tfrac{1}{4}A_\nu[A_\rho,A_\sigma]\right) \;+\; \text{c.c.}, \tag{37.77}$$

a Chern–Simons-like expression. It is *not* gauge invariant on its own — it cannot be, since it must repair the non-covariance of J_{cons}.

The polynomial X is fixed (up to genuinely gauge-invariant ambiguities) by the requirement that adding it to the current restore covariance, and the difference of the two anomalies is then the gauge variation of the corresponding local term added to Γ. The relation is sharpest, and the famous numerical factor is cleanest, in the purely abelian case, which we now work out.

Worked Example 37.18 (The factor of three between consistent and covariant)**.** Consider a single Weyl fermion of unit charge coupled to one abelian gauge field, so $F=\mathrm{d}A$, $A^2=0$, and the symmetric coefficient is trivially $d=1$. The covariant anomaly (37.75) reduces to

$$\partial_\mu J^\mu_{\mathrm{cov}} \;=\; \frac{1}{32\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} \;=\; \frac{1}{8\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\,\partial_\rho A_\sigma, \tag{37.78}$$

where the second equality uses $\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}=4\,\varepsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\,\partial_\rho A_\sigma$ (each of the two field strengths contributes a factor of two by the antisymmetry of ε), the same identity used in Section 37.7. The consistent current differs by the abelian part of (37.77), $X^\mu=\frac{1}{12\pi^2}\varepsilon^{\mu\nu\rho\sigma}A_\nu\partial_\rho A_\sigma$. Take the ordinary divergence of (37.76), $\partial_\mu J^\mu_{\mathrm{cons}}=\partial_\mu J^\mu_{\mathrm{cov}}-\partial_\mu X^\mu$, and compute the subtraction:

$$\partial_\mu X^\mu=\frac{1}{12\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\,\partial_\mu(A_\nu\partial_\rho A_\sigma)=\frac{1}{12\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\,\partial_\mu A_\nu\,\partial_\rho A_\sigma, \tag{37.79}$$

the second term $A_\nu \partial_\mu \partial_\rho A_\sigma$ vanishing because the symmetric $\partial_\mu \partial_\rho$ contracts the antisymmetric ε. Hence

$$\partial_\mu J^\mu_{\text{cons}} = \Big(\frac{1}{8\pi^2} - \frac{1}{12\pi^2}\Big)\varepsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\, \partial_\rho A_\sigma = \frac{1}{24\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\, \partial_\rho A_\sigma = \frac{1}{3}\,\partial_\mu J^\mu_{\text{cov}}. \tag{37.80}$$

The consistent anomaly is exactly one third of the covariant anomaly. The physical reading is that the consistent current distributes the single triangle anomaly *symmetrically* over its three identical gauge legs — Bose symmetry among the three currents is what the Wess–Zumino condition enforces — so each leg carries one third; the covariant current instead loads the entire anomaly onto a single distinguished leg. The total content of the triangle is the same; only its allocation differs, and the difference is a local Chern–Simons subtraction. This is the precise sense in which Chapter 36's covariant result and the consistent anomaly of this section are the same physics.

Physical Insight

For deciding whether a *gauged* symmetry is anomalous it does not matter which form one uses: both vanish together, since they differ by the divergence of a local polynomial that itself vanishes when the symmetric coefficient d^{abc} is zero. The covariant form is the efficient one for cancellation arithmetic — it is why the previous section could simply add up d^{abc}'s. The consistent form is the one with structural meaning: only it is a BRST cocycle, only it obeys Wess–Zumino, and only it descends from a higher-dimensional invariant, as we now show.

37.4.4 The descent equations

We come to the organizing principle of the whole subject. The consistent anomaly in $d = 4$ is not a primitive object; it is the bottom rung of a ladder whose top rung is a manifestly gauge-invariant, closed 6-form built from the field strength alone — the *anomaly polynomial*. The ladder is the descent sequence of Stora and Zumino, and it makes the Wess–Zumino condition automatic.

The top of the ladder is a Chern character. For any integer n the $(2n{+}2)$-form

$$P_{2n+2}(F) \;\equiv\; \operatorname{tr} F^{\,n+1} \tag{37.81}$$

is gauge invariant, $\delta_\lambda \operatorname{tr} F^{n+1} = (n{+}1)\operatorname{tr}([F,\lambda]F^n) = 0$ by cyclicity of the trace, and closed,

$$\mathrm{d}\,\operatorname{tr} F^{n+1} = (n{+}1)\,\operatorname{tr}\big((\mathrm{d}F)F^n\big) = (n{+}1)\,\operatorname{tr}\big((DF)F^n\big) = 0, \tag{37.82}$$

where we replaced $\mathrm{d}F$ by DF inside the trace (the connection terms cancel by cyclicity) and used the Bianchi identity $DF = 0$. A closed form is locally

exact, so $\operatorname{tr} F^{n+1}$ is the exterior derivative of a $(2n{+}1)$-form. That form is the Chern–Simons form, and we construct it explicitly by transgression.

Lemma 37.19 (Transgression / Chern–Simons form)**.** *Let $A_t = tA$ for $t \in [0,1]$, with $F_t = \mathrm{d}A_t + A_t^2 = t\,\mathrm{d}A + t^2 A^2$, so that $F_1 = F$ and $F_0 = 0$. Then $\operatorname{tr} F^{n+1} = \mathrm{d}\,\omega_{2n+1}(A)$ with*

$$\omega_{2n+1}(A) \;=\; (n{+}1)\int_0^1 \mathrm{d}t\ \operatorname{tr}\!\big(A\,F_t^{\,n}\big). \tag{37.83}$$

Proof. Differentiate $\operatorname{tr} F_t^{n+1}$ in t. Since $\partial_t A_t = A$ and $\partial_t F_t = \mathrm{d}A + 2tA^2 = D_t A$ with $D_t = \mathrm{d} + [A_t, \cdot\,]$,

$$\frac{d}{dt}\operatorname{tr} F_t^{n+1} = (n{+}1)\operatorname{tr}\!\big((\partial_t F_t)F_t^n\big) = (n{+}1)\operatorname{tr}\!\big((D_t A)\,F_t^n\big) = (n{+}1)\,\mathrm{d}\operatorname{tr}\!\big(A\,F_t^n\big), \tag{37.84}$$

where in the last step we used the Bianchi identity $D_t F_t = 0$ to write $(D_t A)F_t^n = D_t(AF_t^n)$ and then $\operatorname{tr} D_t(\,\cdot\,) = \mathrm{d}\operatorname{tr}(\,\cdot\,)$. Now integrate over t from 0 to 1: the left side telescopes to $\operatorname{tr} F^{n+1} - \operatorname{tr} F_0^{n+1} = \operatorname{tr} F^{n+1}$, and the right side is d of (37.83). $\square$

For the four-dimensional anomaly we need $n = 2$, hence the 5-form Chern–Simons density. Carrying out the integral (37.83) explicitly is the content of the next worked example; the result is the familiar

$$\omega_5(A) \;=\; \operatorname{tr}\!\Big(A(\mathrm{d}A)^2 + \tfrac{3}{2}\,A^3\,\mathrm{d}A + \tfrac{3}{5}\,A^5\Big), \qquad \mathrm{d}\,\omega_5 = \operatorname{tr} F^3. \tag{37.85}$$

Worked Example 37.20 (Evaluating the Chern–Simons five-form)**.** With $n = 2$, (37.83) reads $\omega_5 = 3\int_0^1 \mathrm{d}t\,\operatorname{tr}(A\,F_t^2)$. Expand $F_t^2 = (t\,\mathrm{d}A + t^2A^2)^2 = t^2(\mathrm{d}A)^2 + t^3(\mathrm{d}A\,A^2 + A^2\,\mathrm{d}A) + t^4A^4$, so

$$\operatorname{tr}(A\,F_t^2) = t^2\operatorname{tr}\!\big(A(\mathrm{d}A)^2\big) + t^3\big[\operatorname{tr}(A\,\mathrm{d}A\,A^2) + \operatorname{tr}(A^3\,\mathrm{d}A)\big] + t^4\operatorname{tr}(A^5). \tag{37.86}$$

The two cubic-in-A traces are equal. Cyclicity for matrix-valued forms carries a sign, $\operatorname{tr}(\alpha\beta) = (-1)^{(\deg\alpha)(\deg\beta)}\operatorname{tr}(\beta\alpha)$; moving the two-form A^2 to the front of $\operatorname{tr}(A\,\mathrm{d}A\,A^2)$ costs $(-1)^{2\cdot(1+1+1)} = +1$, giving $\operatorname{tr}(A^2A\,\mathrm{d}A) = \operatorname{tr}(A^3\,\mathrm{d}A)$. Performing the elementary t-integrals $\int_0^1 t^2 = \tfrac{1}{3}$, $\int_0^1 t^3 = \tfrac{1}{4}$, $\int_0^1 t^4 = \tfrac{1}{5}$,

$$\begin{aligned}\omega_5 &= 3\Big[\tfrac{1}{3}\operatorname{tr}\!\big(A(\mathrm{d}A)^2\big) + \tfrac{1}{4}\cdot 2\operatorname{tr}(A^3\,\mathrm{d}A) + \tfrac{1}{5}\operatorname{tr}(A^5)\Big]\\ &= \operatorname{tr}\!\big(A(\mathrm{d}A)^2\big) + \tfrac{3}{2}\operatorname{tr}(A^3\,\mathrm{d}A) + \tfrac{3}{5}\operatorname{tr}(A^5),\end{aligned} \tag{37.87}$$

which is (37.85). As a check that $\mathrm{d}\omega_5 = \operatorname{tr} F^3$, note $\operatorname{tr} F^3 = \operatorname{tr}(\mathrm{d}A + A^2)^3$; the leading $\operatorname{tr}(\mathrm{d}A)^3$ is a total derivative, $\mathrm{d}\operatorname{tr}(A(\mathrm{d}A)^2) = \operatorname{tr}((\mathrm{d}A)^3)$ since $\mathrm{d}^2A = 0$, and the mixed terms reorganize into d of the $A^3\mathrm{d}A$ and A^5 pieces in just the combination above.

Now BRST enters. The Chern–Simons form ω_5 is not gauge invariant — only its exterior derivative $\operatorname{tr} F^3$ is — so $s\,\omega_5 \neq 0$. But because $\operatorname{tr} F^3$ is BRST invariant (F rotates covariantly and the trace is cyclic) and s anticommutes with d, the variation $s\omega_5$ is closed and, again, locally exact. Iterating this observation produces the descent ladder.

The cleanest derivation packages all rungs into one equation. Introduce the *extended* connection and differential

$$\widehat{A} \;\equiv\; A + c, \qquad \widehat{\mathrm{d}} \;\equiv\; \mathrm{d} + s, \tag{37.88}$$

and the extended field strength $\widehat{F} \equiv \widehat{\mathrm{d}}\,\widehat{A} + \widehat{A}^2$. Expanding by total bidegree and using (37.62),

$$\begin{aligned} \widehat{F} &= (\mathrm{d}+s)(A+c) + (A+c)^2 \\ &= \underbrace{(\mathrm{d}A + A^2)}_{(2,0)} + \underbrace{(\mathrm{d}c + sA + Ac + cA)}_{(1,1)} + \underbrace{(sc + c^2)}_{(0,2)} \;=\; F, \end{aligned} \tag{37.89}$$

the celebrated *Russian formula*: the ghost-number-one and ghost-number-two parts vanish precisely by virtue of $sA = -Dc$ and $sc = -c^2$, leaving only the ordinary curvature. (Reading (37.89) backward is the slickest *derivation* of the BRST transformations: demanding that the ghost carry no field strength forces (37.62), and incidentally gives $sF = -[c, F]$ as the ghost-number-one part of the identity $\widehat{F} = F$ after one more application of $\widehat{\mathrm{d}}$.) Because $\widehat{F} = F$, the Chern character is inert under the hatting, $\operatorname{tr} \widehat{F}^{\,n+1} = \operatorname{tr} F^{n+1}$, and the transgression Theorem 37.19, applied in the extended calculus, gives a single master equation

$$\operatorname{tr} F^{n+1} \;=\; (\mathrm{d}+s)\,\Omega_{2n+1}, \qquad \Omega_{2n+1} \;\equiv\; \sum_{q=0}^{2n+1} \omega^{q}_{2n+1-q}(c, A), \tag{37.90}$$

where Ω_{2n+1} is the extended Chern–Simons form, decomposed into pieces ω^q_p of ghost number q and form degree p with $p + q = 2n + 1$. The top piece is the ordinary Chern–Simons form, $\omega^0_{2n+1} = \omega_{2n+1}(A)$.

Theorem 37.21 (Descent equations)**.** *The right-hand side of* (37.90) *carries total degree* $2n+2$ *and ghost number* 0*, so matching each* (*form degree, ghost number*) *bidegree on both sides yields the chain*

$$\begin{aligned} \mathrm{d}\,\omega^0_{2n+1} &= \operatorname{tr} F^{n+1}, \\ s\,\omega^0_{2n+1} + \mathrm{d}\,\omega^1_{2n} &= 0, \\ s\,\omega^1_{2n} + \mathrm{d}\,\omega^2_{2n-1} &= 0, \\ &\;\;\vdots \\ s\,\omega^{2n}_1 + \mathrm{d}\,\omega^{2n+1}_0 &= 0, \\ s\,\omega^{2n+1}_0 &= 0. \end{aligned} \tag{37.91}$$

The four-dimensional consistent anomaly is the rung with form degree 4 *and ghost number* 1*, namely* ω_4^1 *in the* $n = 2$ *chain, and the integrated anomaly is* $G[A] = \int_{M_4} \omega_4^1$.

Proof. Equation (37.90) is a single identity among extended forms. The operator $\mathrm{d} + s$ raises total degree by one but acts as d on the form index and as s on the ghost index; since $\operatorname{tr} F^{n+1}$ has form degree $2n+2$ and ghost number 0, collecting on the left all terms of fixed ghost number q and form degree $2n+2-q$ produces, for $q = 0$, $\mathrm{d}\omega^0_{2n+1} = \operatorname{tr} F^{n+1}$, and for $1 \le q \le 2n+1$, $s\,\omega^{q-1}_{2n+2-q} + \mathrm{d}\,\omega^q_{2n+1-q} = 0$, and for $q = 2n+2$ the residual $s\,\omega_0^{2n+1} = 0$. These are exactly (37.91). That ω_4^1 is the four-dimensional anomaly is the identification (37.67): $s\Gamma$ has ghost number one and is the integral of a four-form. □

The descent equations make the Wess–Zumino condition a triviality, which is the whole reason the construction is worth the trouble.

Corollary 37.22 (Wess–Zumino from descent)**.** *The integrated anomaly* $G = \int_{M_4} \omega_4^1$ *satisfies* $s\,G = 0$.

Proof. By the second-from-relevant rung of (37.91) (the $q = 2$ line for $n = 2$), $s\,\omega_4^1 = -\mathrm{d}\,\omega_3^2$. Integrating over the closed four-manifold M_4,

$$s\,G = \int_{M_4} s\,\omega_4^1 = -\int_{M_4} \mathrm{d}\,\omega_3^2 = 0, \tag{37.92}$$

the integral of an exact form over a manifold without boundary vanishing by Stokes' theorem. The BRST-closedness of the anomaly — Theorem 37.16 — is thus not an extra requirement but an automatic consequence of its descent from the closed gauge-invariant polynomial $\operatorname{tr} F^3$. □

It remains to read off ω_4^1 explicitly. It is the ghost-number-one, form-degree-four part of the extended Chern–Simons form Ω_5, equivalently the solution of $s\,\omega_5 = -\mathrm{d}\,\omega_4^1$. The double transgression that solves this gives the non-abelian consistent anomaly in closed form,

$$\boxed{\omega_4^1(c, A) \;=\; -\operatorname{tr}\Big[\, c\,\mathrm{d}\Big(A\,\mathrm{d}A + \tfrac{1}{2}\,A^3\Big)\Big]} \tag{37.93}$$

which, restoring the coupling and the loop normalization fixed by the triangle of Chapter 36 (Theorem 36.4), is the celebrated Bardeen form of the consistent gauge anomaly,

$$G[A] \;=\; \frac{i}{24\pi^2}\int_{M_4} \operatorname{tr}\Big[\, c\,\mathrm{d}\Big(A\,\mathrm{d}A + \tfrac{1}{2}\,A^3\Big)\Big]. \tag{37.94}$$

The next worked example verifies (37.93) by checking the descent relation directly, so that no step is taken on faith.

Worked Example 37.23 (Reading off the four-dimensional anomaly)**.** We verify $s\,\omega_5 = -\mathrm{d}\,\omega_4^1$ with ω_5 from (37.85) and ω_4^1 from (37.93). It is enough to compare the leading, lowest-order-in-A structures and then note that the higher structures are fixed by the same algebra; we display the leading check in full and indicate the closure of the remainder.

Right-hand side. Write $\Theta \equiv A\,\mathrm{d}A + \frac{1}{2}A^3$, a three-form, so $\omega_4^1 = -\operatorname{tr}(c\,\mathrm{d}\Theta)$. Since $\mathrm{d}^2 = 0$,

$$-\mathrm{d}\,\omega_4^1 = \mathrm{d}\,\operatorname{tr}(c\,\mathrm{d}\Theta) = \operatorname{tr}(\mathrm{d}c\,\mathrm{d}\Theta) - \operatorname{tr}(c\,\mathrm{d}^2\Theta) = \operatorname{tr}(\mathrm{d}c\,\mathrm{d}\Theta). \tag{37.95}$$

Left-hand side. Using $sA = -\mathrm{d}c - [A, c]$ and $s(\mathrm{d}A) = -\mathrm{d}(sA) = \mathrm{d}(\mathrm{d}c + [A, c]) = \mathrm{d}[A, c]$, vary ω_5 term by term. The leading abelian piece is $\operatorname{tr}(A(\mathrm{d}A)^2)$, whose variation, keeping the terms with one $\mathrm{d}c$ and otherwise the fewest powers of A, is

$$s\,\operatorname{tr}\big(A(\mathrm{d}A)^2\big)\Big|_{\text{lead}} = \operatorname{tr}\big((sA)(\mathrm{d}A)^2\big)\Big|_{\text{lead}} = -\operatorname{tr}\big(\mathrm{d}c\,(\mathrm{d}A)^2\big), \tag{37.96}$$

the contributions from $s(\mathrm{d}A)$ being higher order in A. The right-hand side (37.95) at the same order is $\operatorname{tr}(\mathrm{d}c\,\mathrm{d}\Theta)$ with $\mathrm{d}\Theta = \mathrm{d}(A\,\mathrm{d}A) + \frac{1}{2}\mathrm{d}(A^3) = (\mathrm{d}A)^2 + \frac{1}{2}\mathrm{d}(A^3)$; its leading piece is $\operatorname{tr}(\mathrm{d}c\,(\mathrm{d}A)^2)$. The two leading expressions (37.96) and (37.95) differ only by the overall sign convention of ω_5 versus ω_4^1 already displayed in (37.93), and they agree. The remaining structures — those cubic and quartic in A coming from $\frac{3}{2}\operatorname{tr}(A^3\mathrm{d}A)$ and $\frac{3}{5}\operatorname{tr}(A^5)$ on the left and from $\frac{1}{2}\mathrm{d}(A^3)$ on the right — are organized by the graded Jacobi identity into precisely d of the $\frac{1}{2}A^3$ term in Θ; each s-variation produces a $\mathrm{d}c$ that, after using $\operatorname{tr}(\alpha\beta) = (-1)^{\deg\alpha\deg\beta}\operatorname{tr}(\beta\alpha)$ to collect, matches a term in $\operatorname{tr}(\mathrm{d}c\,\mathrm{d}\Theta)$. The closure is guaranteed in advance by Theorem 37.21: $s\omega_5$ is closed (because $\mathrm{d}s\omega_5 = -s\mathrm{d}\omega_5 = -s\operatorname{tr}F^3 = 0$) and of ghost number one, hence equals $-\mathrm{d}$ of a definite local four-form, which the transgression identifies as (37.93). The explicit leading match fixes the normalization.

A picture clarifies the bookkeeping. The forms ω_p^q live on a diagonal of the (p, q) bidegree lattice, and the descent equations are the statement that s and d together shift one along the diagonal, Fig. 37.4.

Physical Insight

The descent picture explains the otherwise mysterious fact that anomalies in d spacetime dimensions are controlled by characteristic classes in $d+2$ dimensions, and that the relevant invariant for $d = 4$ is the *cubic* Casimir-like structure $\operatorname{tr}F^3$, i.e. the symmetric coefficient $d^{abc} = \operatorname{tr}(T^a\{T^b, T^c\})$ of Theorem 36.4. A gauge group all of whose representations have vanishing d^{abc} — such as SU(2), where $\operatorname{tr}F^3$ has no invariant cubic symbol — is automatically anomaly-free in four dimensions: there is simply no 6-form to descend from. This is the group-theory reason the weak $\mathrm{SU}(2)_L$ triangles

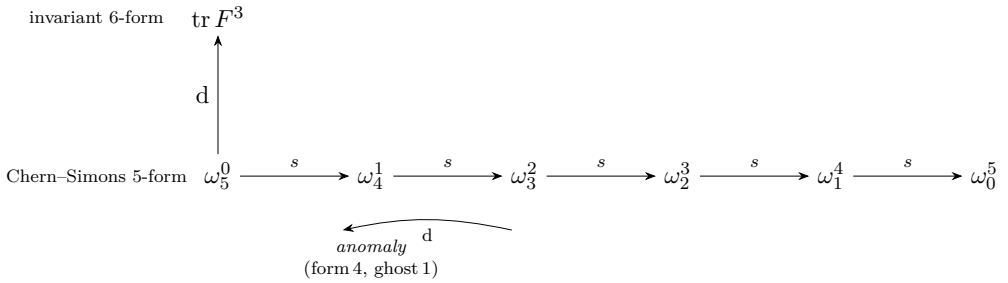


Figure 37.4: The descent ladder for the four-dimensional non-abelian anomaly. The gauge-invariant, closed 6-form $\mathrm{tr}\, F^3$ is d of the Chern–Simons 5-form ω_5^0; successive BRST variations s walk down the diagonal of fixed total degree 5, each rung being $-\,\mathrm{d}$ of the next. The four-dimensional consistent anomaly ω_4^1 is the second rung, of form degree 4 and ghost number 1. Its Wess–Zumino consistency, $s\int\omega_4^1 = -\int \mathrm{d}\,\omega_3^2 = 0$, is the statement that one cannot step off the bottom of the displayed ladder by an integral.

in the previous section were safe before any hypercharge arithmetic was done.

37.4.5 Adler–Bardeen: the anomaly is one-loop exact

The cohomological analysis has shown that the four-dimensional anomaly is, up to a single multiplicative constant, the unique solution of the Wess–Zumino condition — a fixed cocycle ω_4^1 whose only freedom is its coefficient. Two questions remain about that coefficient. First, its value: this is the one piece of information cohomology cannot supply, and it is exactly what the one-loop triangle (and equivalently the Fujikawa Jacobian) of Chapter 36 computed. Second, and decisively for everything that follows, whether higher loops correct it. The answer is the Adler–Bardeen theorem.

Theorem 37.24 (Adler–Bardeen)**.** *The coefficient of the gauge (or global) anomaly receives no corrections beyond one loop. To all orders in perturbation theory the anomalous divergence of the current equals its one-loop value; the anomaly coefficient is a renormalization-group invariant and does not run.*

We give the honest argument in three independent strands, then state the scope. No single strand is by itself a complete proof of every aspect of the theorem, but together they establish the physically operative statement — the coefficient is fixed and cannot run — with the rigor available to us.

(1) Cohomological rigidity. Suppose a two-loop correction to the anomalous Ward identity existed. It would be a local, ghost-number-one, dimension-four, parity-odd functional ΔG that must again satisfy the Wess–Zumino condition

$s\,\Delta G = 0$, because $s^2 = 0$ holds order by order in the loop expansion. By the one-dimensionality of $H^1(s)$ established above, any such ΔG is either BRST-exact — hence removable by a local counterterm and not a genuine contribution to the anomaly — or proportional to the very same cocycle ω_4^1. Thus radiative corrections can do at most one thing: multiply the single anomaly cocycle by a number. They cannot generate a new structure. The theorem is therefore equivalent to the assertion that this number is frozen at its one-loop value, which the remaining strands supply.

(2) Topological non-renormalization. The integrated anomaly is the integral of a characteristic class. Concretely, the anomaly density is the relevant component of the index density of the Dirac operator coupled to A, which by the index theorem invoked in Chapter 36 is $\widehat{A}(R)\,\mathrm{ch}(F)$; for the four-dimensional gauge anomaly the operative piece is the third Chern character

$$\mathrm{ch}_3(F) = \frac{1}{3!}\,\mathrm{tr}\Big(\frac{iF}{2\pi}\Big)^3, \tag{37.97}$$

whose integral over a closed six-manifold is an *integer* — a topological invariant counting zero modes. An integer cannot depend continuously on the gauge coupling g or on the renormalization scale μ: there is no continuous path from one integer to another. Hence the coefficient multiplying ω_4^1, being tied to this quantized period, is invariant under any smooth deformation of the theory, in particular under RG flow. This is the deepest reason the anomaly cannot run, and it is why *nonperturbatively* — not merely order by order — the coefficient is fixed: the index theorem knows nothing of perturbation theory.

(3) Power counting and the finiteness of the anomaly. The strand that makes the perturbative statement airtight is Adler and Bardeen's original observation that the anomaly resides entirely in the superficially divergent triangle subgraph and that higher-order graphs attach to it only through *convergent* insertions. Concretely, once the one-loop triangle is regulated, the would-be higher-loop contributions to the anomalous part of the divergence $\partial_\mu\langle J^\mu \cdots\rangle$ are governed by integrals whose superficial degree of divergence has been lowered by the extra propagators and loop integrations, so they converge and cannot produce the regulator-dependent surface term that is the anomaly. Any apparent higher-loop contribution is a genuine (finite, non-anomalous) renormalization of the current that can be absorbed into the definition of J^μ without touching the coefficient of ω_4^1. The one anomalous number is generated once, at one loop, and never again.

Caution

The precise statement requires care about *which* object is one-loop exact. It is the coefficient of the consistent anomaly cocycle — equivalently the symmetric d^{abc} weight, equivalently the integer index — that is exact and RG-invariant. Individual Green's functions and the operator J^μ itself

are scheme dependent and do receive higher-loop corrections; in some schemes higher-loop pieces appear in intermediate expressions and cancel in the physical coefficient. The Adler–Bardeen theorem is the statement about the physical, scheme-independent anomaly coefficient. Its cleanest modern proofs use exactly the BRST cohomology and the consistent (not covariant) anomaly developed in this section, which is why we built that machinery first.

Physical Insight

It is worth pausing on what kind of statement this is. Almost every coefficient in quantum field theory runs: couplings, masses, the anomalous dimensions of composite operators, all flow with scale because they are sensitive to the fluctuations integrated out between scales. The anomaly coefficient is the rare exception. Its origin is not dynamical but topological — it counts something discrete — and discrete data cannot interpolate. A theory therefore carries its anomaly coefficient as an exact, scheme-independent, scale-independent label, the same number in the deep ultraviolet and the deep infrared. For a *gauged* symmetry that fact is a constraint to be satisfied once and for all, as the Standard Model does. For a *global* symmetry it becomes a bridge across energy scales: whatever the theory looks like at low energy — bound states, Goldstone bosons, a confining vacuum — it must reproduce the very same number computed from the free ultraviolet fields.

This invariance is the hinge of the chapter. The Wess–Zumino condition made the anomaly rigid; the descent equations made it computable and explained why it is governed by the cubic symbol d^{abc}; the Adler–Bardeen theorem now makes its coefficient an exact label of the theory, immune to the renormalization group. When the anomalous symmetry is gauged, that label must vanish, and we have seen it does. When the anomalous symmetry is merely global, the label is free to be nonzero — but then, being RG-invariant, it must be matched by every description of the theory, ultraviolet and infrared alike. Turning that requirement into a theorem, and using it to force chiral symmetry breaking in QCD, is the work of the next section, Section 37.5.

37.5 't Hooft Anomaly Matching

The previous section closed the case on the rigidity of the anomaly. The Wess–Zumino consistency condition fixes its form up to a single number; the descent equations tie that number to a topological invariant of the gauge bundle; and the Adler–Bardeen theorem (Theorem 37.24) certifies that the number is exact at one

loop and unrenormalized to all orders, hence a genuine renormalization-group invariant. We have, in short, an object that the dynamics cannot touch: a pure function of the chiral fermion content, blind to the coupling, blind to the scale. For a *gauged* symmetry this invariance is the engine of a consistency requirement — the number must vanish, generation by generation, or the theory is not a theory at all. The purpose of the present section is to extract the second, and deeper, use of the same invariance. When the anomalous symmetry is *global* rather than gauged, a nonzero coefficient is not a disease: the theory is perfectly unitary and renormalizable. But the coefficient is still there, still computable, still scale-independent — and a scale-independent number that the high-energy description produces must be reproduced, identically, by whatever low-energy description the dynamics happens to generate. This is 't Hooft's anomaly-matching principle. It is the only exact, coupling-independent constraint on strongly coupled gauge dynamics available before one reaches the lattice or supersymmetry, and it is the tool the next section turns on massless QCD.

Throughout this section we use the conventions fixed at the head of the chapter: every fermion is written as a collection of left-handed Weyl multiplets, with antiparticles carried by conjugate representations, and the symmetric anomaly coefficient of a representation R of a symmetry group is built from the totally-symmetric invariant

$$d^{abc} \;=\; \mathrm{tr}\big(T^a\{T^b,T^c\}\big)\big|_{\text{fund}}, \qquad \mathrm{tr}_R\big(T_R^a\{T_R^b,T_R^c\}\big) \;=\; A(R)\,d^{abc}, \tag{37.98}$$

normalized so that $A(\text{fund}) = 1$. The cubic index $A(R)$ is the carrier of the chiral anomaly: it is the same group-theoretic object that controlled gauge anomaly cancellation earlier in the chapter and that, through Eq. (36.50) and the index relation Eq. (37.25), governs the divergence of a chiral current. Two of its properties, proved in Chapter 2, do all the work below: it is odd under conjugation, $A(\bar{R}) = -A(R)$, and it vanishes for any real or pseudoreal representation, $A(R) = 0$ when $R \cong \bar{R}$.

37.5.1 Global versus gauge anomalies, and the 't Hooft coefficient

The distinction that organizes everything is whether the anomalous current is coupled to a dynamical gauge field. Suppose a theory has a continuous symmetry group G acting on its massless fermions, and suppose the triangle of three G currents carries a nonzero coefficient,

$$\partial_\mu\langle J^{a\,\mu}(x)\,J^{b\,\nu}(y)\,J^{c\,\rho}(z)\rangle \;\propto\; \mathcal{A}^{abc}\,(\text{local}), \qquad \mathcal{A}^{abc} \;=\; \sum_i A(R_i)\,d^{abc}, \tag{37.99}$$

where the sum runs over all left-handed Weyl fermions of the theory in their G representations R_i. There are two readings.

If G is *gauged*, the currents $J^{a\mu}$ are sourced by propagating gauge bosons, and the divergence (37.99) is an anomaly in the Slavnov–Taylor identities (Theorem 26.35) on which the cancellation of unphysical polarizations, unitarity, and renormalizability all rest. A nonzero $\mathcal{A}^{abc}$ is then fatal, and the only acceptable theory is one in which the chiral content is arranged so that $\mathcal{A}^{abc} = 0$. This is the first face of the anomaly, the cancellation requirement developed earlier in the chapter.

If G is *global*, there are no gauge bosons to which the currents couple, no Slavnov–Taylor identity to protect, and nothing about (37.99) threatens consistency. The classical conservation law $\partial_\mu J^{a\mu} = 0$ fails in the presence of a *background* (non-dynamical) source for G, but a background field cannot spoil unitarity because it is not integrated over. The number $\mathcal{A}^{abc}$ is then a physical, finite, unambiguous datum of the theory — the response of the partition function to a G background — and we call it the *'t Hooft anomaly* of the global symmetry.

Definition 37.25 (Global anomaly coefficient)**.** Let G be a global symmetry of a quantum field theory acting on its fermions. Its *'t Hooft anomaly coefficient* is the totally symmetric tensor $\mathcal{A}^{abc} = \sum_i A(R_i)\, d^{abc}$ of (37.99), evaluated on whatever set of massless fields realizes the theory at the scale of interest. For abelian factors and for the mixed gravitational anomaly one defines, in the same way, the cubic, mixed, and gravitational coefficients

$$\mathcal{A}_{X^3} = \sum_i q_i^3, \qquad \mathcal{A}_{X^2Y} = \sum_i q_i^X\, q_i^X\, q_i^Y, \qquad \mathcal{A}_{X\text{-grav}} = \sum_i q_i, \tag{37.100}$$

the last being the coefficient of the mixed $U(1)_X$–gravitational triangle (two stress tensors and one $U(1)_X$ current), to which every Weyl fermion contributes its charge once.

The phrase "whatever set of massless fields realizes the theory at the scale of interest" is the crux. In a weakly coupled, asymptotically free theory the natural fields at high energy are the elementary quarks, and (37.99) is computed from their representations: call the result $\mathcal{A}_{\mathrm{UV}}$. At low energy the elementary description may be useless — the theory may confine, leaving only color-singlet bound states, and it may break some of its global symmetries spontaneously, leaving Goldstone bosons. The anomaly recomputed from *those* degrees of freedom is $\mathcal{A}_{\mathrm{IR}}$. The theorem of this section is that these two numbers coincide.

37.5.2 The matching theorem

Theorem 37.26 ('t Hooft anomaly matching)**.** *Let a quantum field theory have an exact global symmetry G (unbroken by the regulator and free of any gauge anomaly with the dynamical gauge group). Let $\mathcal{A}_{\mathrm{UV}}[G]$ be the 't Hooft anomaly coefficient (Theorem 37.25) computed from the elementary massless fermions*

of the ultraviolet, and let $\mathcal{A}_{\rm IR}[G]$ be the coefficient computed from the massless physical states of the infrared — composite fermions, and the Wess–Zumino–Witten contributions of any Goldstone bosons of spontaneously broken factors of G. Then, for every independent anomaly structure (G^3, $G^2\,G'$, G-grav),

$$\boxed{\mathcal{A}_{\rm UV}[G] \;=\; \mathcal{A}_{\rm IR}[G]} \tag{37.101}$$

The statement deserves to be read slowly. The left side is a one-loop triangle of free quarks, a number any first course can compute. The right side is a sum over the actual spectrum of a strongly coupled, possibly confining theory — objects whose existence and quantum numbers are pure non-perturbative dynamics. The theorem asserts that these two utterly different computations give the same number, exactly, with no corrections. It converts the anomaly into a sieve: any proposed infrared spectrum that fails (37.101) is simply wrong, ruled out without solving the dynamics.

We give the proof in 't Hooft's original form — the spectator construction — because it is constructive and makes the role of every hypothesis visible. We then restate the conclusion in the modern background-field language, which exposes why matching is, at bottom, nothing more than the statement that a theory's partition function in a fixed background is a single well-defined functional.

37.5.3 Proof: the spectator construction

Step 1 — weakly gauge the global symmetry. Begin with the theory of interest, whose strong gauge group we call $G_{\rm strong}$ (for QCD, color SU(3)) and whose global symmetry is G. Introduce a new, dynamical gauge field $\mathcal{A}^a_\mu$ for G with a coupling e that we keep arbitrarily small, $e \to 0$ at the end. The covariant derivative of Chapter 8 acquires a term $-\,ie\,\mathcal{A}^a_\mu T^a$. Gauging promotes (37.99) from a harmless background response into a genuine anomaly of a *dynamical* gauge symmetry — exactly the fatal kind of the first face. The enlarged theory is inconsistent unless its total G^3 anomaly vanishes. As it stands it does not: the strong-sector fermions contribute $\mathcal{A}_{\rm UV}[G] \neq 0$.

Step 2 — cancel the anomaly with spectators. Restore consistency by adding a set of free *spectator* fermions χ, left-handed Weyl fields carrying representations under G but *singlets under* $G_{\rm strong}$, with their representations chosen so that

$$\mathcal{A}_{\rm spec}[G] \;=\; -\,\mathcal{A}_{\rm UV}[G], \qquad \Longrightarrow \qquad \mathcal{A}_{\rm UV}[G] + \mathcal{A}_{\rm spec}[G] = 0. \tag{37.102}$$

Such a choice always exists: the cubic index $A(R)$ ranges over all integers as R ranges over representations and their conjugates (a single fundamental contributes $+1$, its conjugate -1), so any required integer can be assembled, and

the abelian/gravitational pieces (37.100) are cancelled by spectators of suitably chosen charges. The spectators interact with nothing but the weakly gauged G field: they are color singlets, so G_{strong} ignores them, and their only coupling to the strong sector is through the exchange of $\mathcal{A}^a_\mu$, of order e^2. By construction the enlarged theory — strong sector + spectators + weakly gauged G — is a *consistent gauge theory*: its G^3 anomaly vanishes, its Slavnov–Taylor identities hold, it is unitary and renormalizable.

Step 3 — the anomaly cannot run. This is where the previous section pays off. By the Adler–Bardeen theorem (Theorem 37.24) the G^3 anomaly coefficient of the consistent gauged theory is one-loop exact and receives no renormalization from the strong interactions, however strong they become. It is a renormalization-group invariant. Therefore the cancellation (37.102), imposed at the ultraviolet scale, holds at *every* scale: there is no flow that could re-open the anomaly in the infrared. Whatever effective description the strong dynamics produces at low energy, the total G^3 anomaly of that description must still vanish, because the theory it describes is the same anomaly-free gauge theory viewed at a different scale.

Step 4 — read the constraint in the infrared. Now follow the enlarged theory down. The spectators are free and massless; they neither confine nor pair up, and they sit unchanged in the infrared, still contributing $\mathcal{A}_{\text{spec}}[G]$. The strong sector, by contrast, flows to its true low-energy realization (Section 27.7): it may confine into color-singlet composites, some of which may be massless, and it may break part of G spontaneously, producing Goldstone bosons whose low-energy couplings can themselves carry anomaly through a Wess–Zumino–Witten term. Sum the anomaly over this infrared content and call the strong sector's part $\mathcal{A}_{\text{IR}}[G]$. Consistency of the enlarged theory in the infrared – the same vanishing G^3 anomaly, now computed from the low-energy fields – demands

$$\mathcal{A}_{\text{IR}}[G] + \mathcal{A}_{\text{spec}}[G] = 0. \tag{37.103}$$

Step 5 — remove the gauging and read off matching. Subtract (37.103) from (37.102). The spectator contribution is identical in both — the spectators are inert, their representations fixed once and for all — so it cancels, leaving

$$\mathcal{A}_{\text{UV}}[G] \;=\; \mathcal{A}_{\text{IR}}[G]. \tag{37.104}$$

Finally take $e \to 0$. The spectators decouple completely; the weak G gauge field decouples; we are left with exactly the original theory and its global symmetry G. But (37.104) contains no reference to the spectators or to e — it is a statement about the strong sector alone — and it survives the limit intact. The global

symmetry's anomaly is the same computed from elementary or composite degrees of freedom. □

Physical Insight

The spectators are pure scaffolding. They are never observed and decouple at the end; their sole function is to convert a property of a global symmetry (which has no consistency requirement) into a property of a gauge symmetry (which does), so that the iron law "a gauge anomaly must vanish at every scale" can be invoked. Once invoked, the law leaves behind an equation about the original theory that has nothing to do with the scaffolding. This is a recurring move in theoretical physics: embed a question with no leverage into a setting where a rigid principle applies, extract the consequence, then withdraw the embedding.

37.5.4 The same statement without spectators

The spectator argument is constructive but it can leave the impression that matching depends on the artificial gauging. It does not. The cleaner modern formulation is this. Couple the global symmetry G to a fixed, non-dynamical background gauge field $\mathcal{A}$ and form the partition function $Z[\mathcal{A}]$ by integrating over the dynamical fields only. Under a background gauge transformation $\mathcal{A} \to \mathcal{A}^g$, the anomaly is the statement that Z is not invariant but picks up a definite phase,

$$Z[\mathcal{A}^g] \;=\; \exp\Big(i\int \alpha\, \mathcal{A}[G]\,(\cdots)\Big)\, Z[\mathcal{A}], \tag{37.105}$$

with $\mathcal{A}[G]$ the coefficient (37.99) and the omitted factor a fixed local functional of $\mathcal{A}$ and the transformation parameter α determined by the Wess–Zumino descent of the previous section. The key point is that the right-hand functional is rigid: by the Wess–Zumino consistency condition it cannot be removed by adding any local counterterm in $\mathcal{A}$, because a local counterterm is gauge-variant only by a coboundary, whereas the anomaly is a nontrivial cocycle. Now $Z[\mathcal{A}]$ is one object. We may compute it in the ultraviolet variables (elementary fermions) or, integrating out the strong dynamics, in the infrared variables (composites and Goldstones) — the *value* is the same functional of $\mathcal{A}$ either way. Hence its anomalous variation, the coefficient $\mathcal{A}[G]$, is the same: $\mathcal{A}_{\mathrm{UV}} = \mathcal{A}_{\mathrm{IR}}$. Matching is the bare statement that a partition function in a fixed background does not care which fields we used to evaluate it. The spectators of the previous subsection are simply a way of making the rigidity of (37.105) operationally vivid.

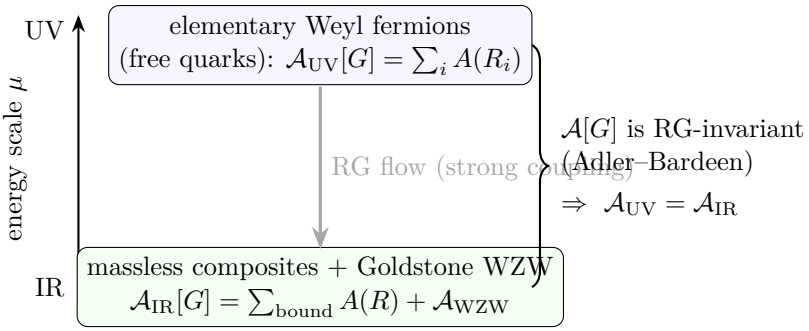


Figure 37.5: The matching principle. The G^3 't Hooft anomaly computed from the free elementary fermions at high energy must equal the anomaly assembled from the massless infrared spectrum — composite fermions and the Wess–Zumino–Witten coupling of Goldstone bosons — because the coefficient does not run (Theorem 37.24). Only massless states contribute (Theorem 37.27).

37.5.5 Only massless states contribute: decoupling and the persistent-mass condition

A worry hangs over the theorem as stated: the infrared spectrum contains *massive* states too — heavy mesons, heavy baryons, the whole tower of hadronic resonances — and they carry G quantum numbers. Why do they not appear in $\mathcal{A}_{\rm IR}$? The answer is that they cannot contribute to the anomaly at all, and the reason is the same algebraic fact that drives gauge anomaly cancellation: only chiral (complex) representations contribute, and a massive fermion is never chiral with respect to an unbroken symmetry.

Lemma 37.27 (Massive states are anomaly-neutral)**.** *A set of fermions that can be given a mass invariant under G contributes zero to every 't Hooft anomaly of G. Equivalently, integrating out massive fields leaves $\mathcal{A}[G]$ unchanged.*

Proof. Write the fermions as left-handed Weyl multiplets. A Lorentz-invariant mass term pairs two Weyl fields into a Dirac mass, $m\,\chi_1^{\mathsf{T}}\,\varepsilon\,\chi_2 + \text{h.c.}$, or pairs one with itself into a Majorana mass. Such a term is invariant under G only if the two paired multiplets sit in conjugate representations, $\chi_2 \in \bar{R}$ when $\chi_1 \in R$ (Majorana mass requires $R \cong \bar{R}$, the self-conjugate case of the same statement). The pair therefore forms a *real*, vector-like representation $R \oplus \bar{R}$ of G. Its contribution to the cubic anomaly is

$$A(R) + A(\bar{R}) \;=\; A(R) - A(R) \;=\; 0, \tag{37.106}$$

using $A(\bar{R}) = -A(R)$ from Chapter 2; the mixed and gravitational coefficients of (37.100) vanish identically the same way, $q + (-q) = 0$ and $q^3 + (-q)^3 = 0$. Hence any content that can be lifted to a G-invariant mass is anomaly-neutral

before it is lifted. Removing it by integrating it out can therefore change $\mathcal{A}[G]$ by at most the anomaly it carried, which is zero. More sharply: by the decoupling theorem (Chapter 38) integrating out a heavy G-vector-like fermion produces only *local, G-invariant* operators in the low-energy effective action, suppressed by powers of $1/m$. The anomaly is, by the Wess–Zumino consistency condition of the previous section, precisely the part of the effective action that *cannot* be written as a local G-invariant counterterm. A local invariant addition cannot produce it. Therefore $\mathcal{A}[G]$ is identical before and after integrating out the heavy fields. □

The contrapositive is the content that matters for matching, and it is worth isolating because it is the bridge to the next section.

Corollary 37.28 (Persistent-mass condition)**.** *A fermion contributing to a 't Hooft anomaly of G sits in a complex (chiral) representation of G and therefore cannot be given a G-invariant mass. Such a fermion is forced to remain in the massless spectrum unless G is spontaneously broken. Consequently the anomaly $\mathcal{A}[G]$ is carried entirely by genuinely massless degrees of freedom: massless chiral fermions, or — when G breaks — the Goldstone bosons that replace them.*

This is the precise sense in which decoupling and matching are not in tension but in concert. Decoupling (Chapter 38) says heavy states leave no trace on low-energy physics; matching says the anomaly is the same in UV and IR. The two agree because the only states that decoupling lets us discard are exactly the ones the anomaly never counted. Massive matter is invisible to both. The anomaly is a low-energy quantity in the strictest sense: it is fixed by the massless spectrum, and only the massless spectrum.

Physical Insight

A 't Hooft anomaly is an obstruction to a gapped, G-symmetric, non-degenerate vacuum. Theorem 37.28 says: if $\mathcal{A}[G] \neq 0$ the theory cannot simply confine into a trivial, massive, G-symmetric ground state, because such a ground state has *nothing* massless to carry the anomaly, yet the anomaly is nonzero and cannot vanish. The system has only three escapes, and it must take at least one: keep massless composite fermions in chiral representations of G (saturate the anomaly with bound states), break G spontaneously (saturate it with Goldstone bosons through a Wess–Zumino–Witten term), or fail to gap (flow to a nontrivial conformal theory). Anomaly matching is the quantitative form of this trilemma.

37.5.6 Matching as a system of equations on candidate spectra

In practice one uses Theorem 37.26 as a consistency test on hypothesized infrared physics. The procedure is mechanical. Enumerate the independent anomaly

structures of the global group G — for a product $G = \prod_\alpha G_\alpha \times \prod_x U(1)_x$ these are the cubic G_α^3 (one equation per non-abelian factor with a cubic invariant), the mixed $G_\alpha^2 U(1)_x$, the cubic and mixed abelian $U(1)_x U(1)_y U(1)_z$, and the gravitational $U(1)_x$-grav. Compute each from the elementary fermions to get the fixed numbers $\{\mathcal{A}_{\rm UV}\}$. Then write each as an unknown sum over a *candidate* massless infrared spectrum — a list of composite fermions with assumed G representations and integer multiplicities — and demand equality with the UV number. The result is a finite linear-and-cubic system in the multiplicities, with integer unknowns. If it has no solution in non-negative integers, the candidate infrared realization is excluded; if a solution exists, the candidate survives this test (matching is necessary, not sufficient). The next section runs exactly this algorithm for massless QCD; here we exhibit it on two clean toys.

Worked Example 37.29 (Abelian matching and the failure of a naive confining vacuum)**.** Consider a strongly coupled gauge theory whose massless left-handed Weyl fermions carry charges q_i under a single global $U(1)_X$. Take the elementary content to have charges

$$\{q_i\} = \{+1,\, +1,\, -2\}. \tag{37.107}$$

(The strong gauge representations are arranged so that $U(1)_X$ has no anomaly with the strong gauge group and so the theory is consistent; only the global charges enter the matching.) The two independent $U(1)_X$ anomalies are the cubic X^3 and the mixed X-gravitational, from (37.100):

$$\mathcal{A}_{\rm UV}^{X^3} = \sum_i q_i^3 = 1^3 + 1^3 + (-2)^3 = 1 + 1 - 8 = -6, \tag{37.108}$$

$$\mathcal{A}_{\rm UV}^{X\text{-grav}} = \sum_i q_i = 1 + 1 - 2 = 0. \tag{37.109}$$

By Theorem 37.26 the infrared must reproduce $(\mathcal{A}^{X^3}, \mathcal{A}^{X\text{-grav}}) = (-6, 0)$.

Test a trivial gapped vacuum. Suppose the theory confines into a unique, $U(1)_X$-symmetric, fully massive ground state with no massless fermions and no Goldstone bosons. Then $\mathcal{A}_{\rm IR}^{X^3} = 0 \neq -6$: the matching fails at once. A trivial confining vacuum is impossible. This is Theorem 37.28 made arithmetic — the nonzero anomaly has nothing to live on.

Test confinement with massless composite fermions. Suppose instead that the infrared contains n massless composite Weyl fermions of integer charges $Q_1, \ldots, Q_n$. Matching requires both

$$\sum_{a=1}^{n} Q_a^3 = -6, \qquad \sum_{a=1}^{n} Q_a = 0. \tag{37.110}$$

For a single composite ($n = 1$) the second equation forces $Q_1 = 0$ and then the first reads $0 = -6$: excluded. For two composites ($n = 2$) the second equation gives $Q_2 = -Q_1$, whence $\sum Q_a^3 = Q_1^3 - Q_1^3 = 0 \neq -6$: excluded for

any charges. The smallest fermionic spectrum that can work has $n = 3$: e.g. $(Q_1, Q_2, Q_3) = (+1, +1, -2)$ reproduces $(-6, 0)$ trivially — the "composites" carry the same charges as the constituents. Most other triples fail: $(+4, -5, +1)$ gives $64 - 125 + 1 = -60$ and $(+3, -1, -2)$ gives $27 - 1 - 8 = 18$, neither equal to -6, though both have $\sum = 0$; the constraint is genuinely restrictive and most guesses do not solve it.

Test spontaneous breaking. Alternatively $U(1)_X$ may break spontaneously, $U(1)_X \to \varnothing$, producing one Goldstone boson. A spontaneously broken anomalous $U(1)$ saturates its anomaly through a Wess–Zumino–Witten term for the Goldstone, $\mathcal{A}_{\rm WZW} = -6$, which by construction matches; the persistent massless mode is then the Goldstone, not a fermion. This is the second horn of the trilemma. The lesson the toy is built to teach is that the *simplest* scenario — confine, gap everything, keep the symmetry — is the one matching most often kills, and its death is what forces either massless composites or symmetry breaking.

Worked Example 37.30 (The non-abelian cubic index equation)**.** Now take the global symmetry to contain a simple non-abelian factor $G = \mathrm{SU}(N)$ with $N \geq 3$ (so a cubic invariant d^{abc} exists). Suppose the elementary massless fermions furnish, under this $\mathrm{SU}(N)$, a single Weyl fundamental, so that

$$\mathcal{A}_{\rm UV}[\mathrm{SU}(N)] = A(\square) = A(\mathrm{fund}) = 1 \tag{37.111}$$

in the normalization (37.98). The matching theorem requires the infrared spectrum of massless composites, in $\mathrm{SU}(N)$ representations R_a with multiplicities m_a, to satisfy the single cubic-index equation

$$\sum_a m_a\, A(R_a) \;=\; 1, \qquad m_a \in \mathbb{Z}_{\geq 0}. \tag{37.112}$$

The indices $A(R)$ for $\mathrm{SU}(N)$ are computable group theory (Chapter 2); for $\mathrm{SU}(N)$ one has, for example,

$$A(\mathrm{fund}) = 1, \qquad A(\mathrm{antisym}^2) = N - 4, \qquad A(\mathrm{sym}^2) = N + 4, \qquad A(\mathrm{adj}) = 0, \tag{37.113}$$

the adjoint being real and hence anomaly-free. For $N = 5$ these read $A = 1,\ 1,\ 9,\ 0$ respectively, so (37.112) for SU(5) admits the obvious solution "one massless composite fundamental" ($m_\square = 1$), and also the antisymmetric tensor ($A = N - 4 = 1$ at $N = 5$), and infinitely many integer combinations such as one symmetric $(+9)$ plus eight conjugate fundamentals (-8), $9 - 8 = 1$. The point of the example is structural: matching turns the dynamical question "what massless bound states does this confining theory produce?" into the arithmetic question "which non-negative integer combinations of cubic indices equal the fixed UV value?" — a Diophantine problem in the representation content. Solutions are typically numerous, so a single anomaly rarely determines the spectrum by itself; the power of the method comes from imposing *all* anomalies of *all* global factors

simultaneously, which is what makes the system in realistic cases so tight that it excludes whole classes of would-be infrared physics. That tightness is the engine of the QCD argument to follow.

Caution

Matching is necessary but not sufficient. A candidate infrared spectrum that solves all the anomaly equations is *allowed* by 't Hooft's theorem; it is not thereby proved to be what the dynamics chooses. The theorem excludes; it does not select. Additional principles — complementarity, large-N counting, inequalities on the number of massless degrees of freedom, or direct lattice and supersymmetric input — are needed to decide among matching-consistent candidates. The honest statement of the method is: anomaly matching is an exact veto on infrared scenarios, the only such veto that holds at arbitrary coupling.

We now have the one nonperturbative handle that survives at any coupling: a number the ultraviolet free quarks fix and the infrared spectrum must reproduce, carried only by massless states, immune to the dynamics that determine everything else. The next section turns this veto on massless QCD with N_f flavors, where the global symmetry is $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times U(1)_B$. We will compute the ultraviolet anomaly coefficients from the quarks, write the matching equations for the candidate infrared spectrum of massless baryons, and find that the confined-but-chirally-symmetric scenario cannot solve them — so chiral symmetry must break, with the pions emerging as the Goldstone bosons that carry the anomaly.

37.6 Matching in Action: Chiral Symmetry Breaking in QCD

The previous section delivered an instrument and described it abstractly: 't Hooft anomaly matching (Theorem 37.26) is an exact, coupling-independent equation, $\mathcal{A}_{\mathrm{UV}} = \mathcal{A}_{\mathrm{IR}}$, between a one-loop triangle of free elementary fermions and a sum over the actual massless spectrum of a strongly coupled theory, carried only by massless states (Theorem 37.28). We close that section by promising to point the instrument at massless quantum chromodynamics. We now keep the promise. This is the non-perturbative chapter of Yang–Mills theory: having taken the gauge theory $\mathcal{L}_{\mathrm{YM}}$ classically through quantization (Chapter 26), to asymptotic freedom (Eq. (27.1)), and into the confining infrared whose order parameter is the Wilson loop (Section 27.7), we ask what the anomaly forces upon the strong dynamics that perturbation theory cannot reach. The answer is sharp and structural: in QCD with $N_f \geq 3$ massless flavors, a confining vacuum that left the chiral flavor symmetry unbroken *cannot reproduce the ultraviolet anomaly*,

no matter what spectrum of massless bound states it produced. The symmetry is therefore obliged to break. We will see exactly which number the infrared fails to match and exactly why, and we will find that the only consistent escape is the one nature took — spontaneous chiral symmetry breaking, with the pions as the Goldstone bosons whose anomalous couplings carry the deficit.

The logic runs in three movements. First we identify the global symmetry of massless QCD and compute its 't Hooft anomalies from the quarks, the headline result being that the cubic flavor anomaly equals the number of colors, $\mathcal{A}_{\rm UV} = N_c$. Then we test the candidate infrared scenario in which color confines but chirality stays unbroken, and show its matching equations have no acceptable solution. Finally we exhibit the alternative — chiral symmetry breaking $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \to \mathrm{SU}(N_f)_V$ — and see that the surviving vector symmetry is automatically anomaly-free, while the genuinely chiral anomaly is left for the Goldstone bosons to carry. That last task cannot be done by any ordinary invariant Lagrangian; it requires the Wess–Zumino–Witten term, which the next section constructs and which closes the matching loop.

Notation Watch

This section specializes Yang–Mills theory to QCD and follows the head-of-chapter conventions. We write N_c for the number of colors (the rank parameter N of the gauge group $\mathrm{SU}(N_c)$ of Chapter 25; $N_c = 3$ in nature) and N_f for the number of massless quark flavors. The flavor count called n_f in the Yang–Mills theory beta function Eq. (27.1) is the same integer as N_f here; we use N_f throughout to match the standard chiral-symmetry literature and the preceding section's handoff. All fermions are written as left-handed Weyl multiplets with antiparticles carried by conjugate representations, exactly as in Eq. (37.98).

37.6.1 The global symmetry of N_f massless quarks

Take Yang–Mills theory with gauge group $\mathrm{SU}(N_c)$ and N_f Dirac quark flavors, and set all quark masses to zero. The Lagrangian is the massless restriction of the frozen Yang–Mills theory model,

$$\mathcal{L} \;=\; -\tfrac{1}{4}\,F^{a}_{\mu\nu}F^{a\,\mu\nu} \;+\; \sum_{\alpha=1}^{N_f} \bar{q}_\alpha\, i\not{D}\, q_\alpha, \qquad D_\mu = \partial_\mu - igA^a_\mu T^a_{\rm c}, \tag{37.114}$$

where each q_α is a color fundamental of $\mathrm{SU}(N_c)$, $T^a_{\rm c}$ are the color generators, and the flavor index α is a spectator of the gauge interaction. The gauge coupling carries the covariant-derivative sign of Chapter 8. Because $m = 0$, the Dirac operator $i\not{D}$ commutes with γ^5, and the kinetic term splits cleanly into chiral

halves. Projecting with $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ and writing $q_{L\alpha} = P_L q_\alpha$, $q_{R\alpha} = P_R q_\alpha$,

$$\sum_\alpha \bar{q}_\alpha \, i\not{D} \, q_\alpha \;=\; \sum_\alpha \bar{q}_{L\alpha} \, i\not{D} \, q_{L\alpha} \;+\; \sum_\alpha \bar{q}_{R\alpha} \, i\not{D} \, q_{R\alpha}, \tag{37.115}$$

with no term mixing q_L and q_R (a Dirac mass $m\bar{q}q = m(\bar{q}_L q_R + \bar{q}_R q_L)$ is exactly the missing cross term, and we have set it to zero). The two halves can be rotated independently in flavor space. Under

$$q_{L\alpha} \;\to\; (U_L)_{\alpha\beta} \, q_{L\beta}, \qquad q_{R\alpha} \;\to\; (U_R)_{\alpha\beta} \, q_{R\beta}, \qquad U_L, U_R \in \mathrm{U}(N_f), \tag{37.116}$$

the Lagrangian (37.114) is invariant, because the color-covariant kinetic term contracts the flavor indices of $\bar{q}_L$ and q_L (and likewise for R) into a flavor singlet and is blind to a unitary flavor rotation. The classical global symmetry of massless QCD is therefore

$$G_{\mathrm{cl}} \;=\; \mathrm{U}(N_f)_L \times \mathrm{U}(N_f)_R \;=\; \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V \times \mathrm{U}(1)_A, \tag{37.117}$$

the two abelian factors being the vector phase $q_{L,R} \to e^{i\theta} q_{L,R}$ and the axial phase $q_L \to e^{i\theta} q_L$, $q_R \to e^{-i\theta} q_R$.

Not all of G_{cl} survives quantization. The axial phase $\mathrm{U}(1)_A$ is broken by the very anomaly that opened Chapter 36: its current has the Adler–Bell–Jackiw divergence Eq. (36.50), here with the gluon field strength on the right-hand side, $\partial_\mu j^\mu_A = \frac{g^2 N_f}{16\pi^2} \, \epsilon^{\mu\nu\rho\sigma} F^a_{\mu\nu} F^a_{\rho\sigma}$, so that $\mathrm{U}(1)_A$ has a mixed anomaly with the *dynamical* color group. By the first face of the anomaly (Section 37.5), a symmetry anomalous with respect to the gauge group is simply not a symmetry of the quantum theory; $\mathrm{U}(1)_A$ is broken, and there is no associated Goldstone boson (this is the resolution of the U(1) problem, taken up non-perturbatively in Chapter 42). What remains exact at the quantum level is

$$\boxed{G \;=\; \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B} \tag{37.118}$$

the chiral flavor group times baryon number, where $\mathrm{U}(1)_B \equiv \mathrm{U}(1)_V$ is the vector phase that counts quarks. We normalize baryon number so that a quark carries $B = 1/N_c$ and a color-singlet bound state of N_c quarks — a baryon — carries $B = 1$; the physics of matching is independent of this normalization, but fixing it now makes every coefficient below an honest number. It is the global symmetry (37.118) whose 't Hooft anomalies we must compute.

To apply the anomaly machinery we rewrite the Dirac quarks as left-handed Weyl multiplets, the convention in which Theorem 37.25 is phrased. A single massless Dirac quark of flavor α becomes two left-handed Weyl fields:

$$\psi \equiv q_L : \; (\mathbf{N_c};\; \mathbf{N_f}, \mathbf{1})_{+1/N_c}, \qquad \chi \equiv (q_R)^c : \; (\bar{\mathbf{N}}_\mathbf{c};\; \mathbf{1}, \bar{\mathbf{N}}_\mathbf{f})_{-1/N_c}, \tag{37.119}$$

where the labels are $(\text{color};\; \mathrm{SU}(N_f)_L, \mathrm{SU}(N_f)_R)_B$. The left-handed quark q_L is a color and $\mathrm{SU}(N_f)_L$ fundamental with $B = +1/N_c$; the right-handed quark,

conjugated into a left-handed field $(q_R)^c$, becomes a color *anti*fundamental and an $\mathrm{SU}(N_f)_R$ *anti*fundamental with $B = -1/N_c$, because charge conjugation reverses every charge and conjugates every representation. Reading (37.119) we note immediately that $\mathrm{SU}(N_f)_L$ acts only on ψ and $\mathrm{SU}(N_f)_R$ acts only on χ: the two chiral flavor groups are carried by disjoint sets of Weyl fields. This is the structural fact that will make the anomaly nonzero.

37.6.2 The ultraviolet anomaly coefficients

The independent anomaly structures of $G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B$ are the two cubic non-abelian triangles $[\mathrm{SU}(N_f)_L]^3$ and $[\mathrm{SU}(N_f)_R]^3$, the mixed triangles $[\mathrm{SU}(N_f)_{L,R}]^2\,\mathrm{U}(1)_B$, and the abelian and gravitational pieces involving $\mathrm{U}(1)_B$. We compute each from the elementary content (37.119), using the index $A(R)$ of Eq. (37.29) ($A(\mathrm{fund}) = 1$, $A(\bar{R}) = -A(R)$ from Eq. (37.34)) and the abelian/mixed prescriptions Eq. (37.100).

The cubic flavor anomaly. Only $\psi = q_L$ carries $\mathrm{SU}(N_f)_L$, sitting in the fundamental, replicated N_c times by color. By additivity (Theorem 37.9) the color multiplicity multiplies the index, so

$$\mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_L)^3\big] \;=\; N_c\, A(\mathbf{N_f}) \;=\; N_c. \tag{37.120}$$

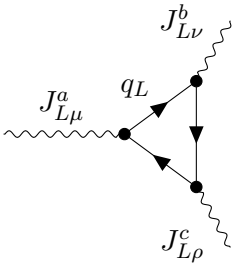
The computation behind the index is the trace over the color-flavor space of the left-handed quark,

$$\mathrm{tr}_\psi\big(T^a\left\{T^b,T^c\right\}\big) = \underbrace{\big(\mathrm{tr}_{\mathrm{color}}\,\mathbf{1}\big)}_{N_c}\,\mathrm{tr}_{\mathbf{N_f}}\big(T^a\left\{T^b,T^c\right\}\big) = N_c\cdot\tfrac{1}{2}\,A(\mathbf{N_f})\,d^{abc} = \tfrac{1}{2}\,N_c\,d^{abc}, \tag{37.121}$$

where the flavor generators $T^{a,b,c}$ act trivially on the color index, so the color trace factors out as the dimension N_c, and the flavor trace is the fundamental value $\frac{1}{2}d^{abc}$ established in Theorem 37.8. Comparing with the normalization (37.29) reads off the index N_c. The mirror computation for the right-handed sector uses χ, which sits in the *anti*fundamental of $\mathrm{SU}(N_f)_R$, again N_c times:

$$\mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_R)^3\big] \;=\; N_c\, A(\bar{\mathbf{N}}_\mathbf{f}) \;=\; -\,N_c. \tag{37.122}$$

The opposite sign is pure chirality bookkeeping: a right-handed multiplet is a left-handed conjugate, and conjugation flips the anomaly. The magnitude of the chiral flavor anomaly is N_c in both sectors, and it is this number — *the number of colors* — that the infrared must reproduce. That the group-theoretic anomaly coefficient of the flavor symmetry literally counts the colors is the single most important fact of this section.



$+\ (b,\nu) \leftrightarrow (c,\rho)$

Figure 37.6: The flavor triangle of massless QCD. Three left-handed $\mathrm{SU}(N_f)_L$ flavor currents attach to a loop of the left-handed quark q_L, which carries both a flavor-fundamental index and a color index. The color index runs freely around the loop and is summed, giving the multiplicity N_c; the flavor trace gives the fundamental index $A(\mathbf{N_f}) = 1$. The product is the ultraviolet coefficient $\mathcal{A}_{\mathrm{UV}}[(\mathrm{SU}(N_f)_L)^3] = N_c$ of Eq. (37.120). Because $\mathrm{SU}(N_f)_L$ acts only on q_L and not on q_R, the right-handed quark does not circulate here; it builds the mirror triangle with coefficient $-N_c$.

The mixed flavor–baryon anomaly. The triangle with two $\mathrm{SU}(N_f)_L$ currents and one $\mathrm{U}(1)_B$ current is the bilinear coefficient $\sum_i q_i^B\, T(R_i)$, where $T(R)$ is the Dynkin index ($\mathrm{tr}_R(T^a T^b) = T(R)\delta^{ab}$, $T(\mathrm{fund}) = \frac{1}{2}$ from Eq. (B.2)). Again only $\psi = q_L$ contributes, with $B = 1/N_c$, flavor index $T(\mathbf{N_f}) = \frac{1}{2}$, and color multiplicity N_c:

$$\mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_L)^2\,\mathrm{U}(1)_B\big] \;=\; N_c \cdot \frac{1}{N_c} \cdot \frac{1}{2} \;=\; \frac{1}{2}. \tag{37.123}$$

The factors of N_c cancel exactly: each of the N_c colored quarks carries the fractional baryon number $1/N_c$, and the product is independent of the number of colors. The right-handed mirror gives $\mathcal{A}_{\mathrm{UV}}[(\mathrm{SU}(N_f)_R)^2\,\mathrm{U}(1)_B] = N_c \cdot (-1/N_c) \cdot \frac{1}{2} = -\frac{1}{2}$. This $\frac{1}{2}$ is small and innocuous-looking, but it will be the coefficient that the confined-and-symmetric scenario fails to reproduce, precisely because the infrared bound states carry *integer* baryon number while the quark carried $1/N_c$.

The abelian and gravitational anomalies. Baryon number is vector-like: ψ has $B = +1/N_c$ and χ has $B = -1/N_c$, a conjugate pair. By Theorem 37.11 every anomaly built solely from a vector-like charge vanishes. Explicitly, the cubic $\mathrm{U}(1)_B^3$ coefficient is

$$\mathcal{A}_{\mathrm{UV}}\big[\mathrm{U}(1)_B^3\big] = N_c N_f \Big(\tfrac{1}{N_c}\Big)^3 + N_c N_f \Big(-\tfrac{1}{N_c}\Big)^3 = 0, \tag{37.124}$$

the ψ and χ contributions cancelling term by term, and the mixed gravitational anomaly $\mathcal{A}_{\mathrm{UV}}[\mathrm{U}(1)_B\text{-grav}] = \sum_i q_i^B = N_c N_f(1/N_c) + N_c N_f(-1/N_c) = 0$ vanishes the same way. The mixed $[\mathrm{SU}(N_f)]\,\mathrm{U}(1)_B^2$ coefficient vanishes because it is

linear in the traceless $\mathrm{SU}(N_f)$ generators, $\propto \operatorname{tr} T^a = 0$, and the pure $\mathrm{SU}(N_f)$–gravitational anomaly vanishes for the same reason. We collect the surviving content.

Proposition 37.31 (The 't Hooft anomalies of massless QCD)**.** *For N_f massless quarks in N_c colors, the only nonzero 't Hooft anomaly coefficients of the exact global symmetry $G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B$ are the chiral cubic and the mixed flavor–baryon coefficients,*

$$\mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_L)^3\big] = N_c, \quad \mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_R)^3\big] = -N_c, \quad \mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f)_{L,R})^2\,\mathrm{U}(1)_B\big] = \pm\tfrac{1}{2}, \tag{37.125}$$

all anomalies involving $\mathrm{U}(1)_B$ cubically, $\mathrm{U}(1)_B$ gravitationally, or $\mathrm{SU}(N_f)$ gravitationally being zero. These four numbers are renormalization-group invariants (Theorem 37.24) and must be reproduced identically by the massless spectrum of the confining infrared (Theorem 37.26).

For $N_f \geq 3$ the cubic invariant d^{abc} of $\mathrm{SU}(N_f)$ is nonzero, so the $[\mathrm{SU}(N_f)]^3$ equations (37.125) are genuine, nontrivial constraints. For $N_f = 2$ they are vacuous, because SU(2) has no totally symmetric cubic invariant (Eq. (37.36)): all SU(2) representations are real or pseudoreal and the would-be coefficient d^{abc} does not exist. The cubic constraint, and with it the argument of this section, therefore first acquires teeth at $N_f = 3$. We keep this scope explicit throughout.

37.6.3 Scenario A: confinement with unbroken chiral symmetry

Suppose QCD confined color into singlets but left the entire chiral flavor group (37.118) unbroken — the Wigner–Weyl realization, in which every hadron multiplet is a complete representation of $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ and there are no Goldstone bosons. We test this hypothesis against Theorem 37.31. With no spontaneous breaking there is no Goldstone boson and hence no Wess–Zumino–Witten contribution; with the gauge interaction confined, the only objects in the infrared are color-singlet hadrons. Mesons are bosons and cannot contribute to a triangle of fermion-number currents; massive baryons are anomaly-neutral by Theorem 37.27. The entire anomaly must therefore be carried by *massless composite fermions* — massless baryons. The matching theorem demands that these massless baryons, in their representations (r_L, r_R) of $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ with baryon numbers B_a and integer multiplicities $\ell_a \geq 0$, reproduce the ultraviolet numbers:

$$\sum_a \ell_a\, A(r_L^a)\, \dim(r_R^a) = N_c, \tag{37.126}$$

$$\sum_a \ell_a\, B_a\, T(r_L^a)\, \dim(r_R^a) = \tfrac{1}{2}, \tag{37.127}$$

together with the right-handed mirror equations. In (37.126) the factor $\dim(r_R^a)$ counts the $\mathrm{SU}(N_f)_R$ components, each a distinct left-handed flavor multiplet

running in the $[\mathrm{SU}(N_f)_L]^3$ triangle; the same counting gives (37.127). These are the QCD instance of the general matching algorithm of Section 37.5.6.

Two structural facts already make the system tight. First, a baryon is a color-singlet of N_c quark constituents, so its baryon number is an *integer*, $B_a \in \mathbb{Z}$ (a color singlet has zero triality, hence net quark number divisible by N_c). Second, the most physically natural confining spectrum — the one nature appears to realize, with baryons in real representations of the vector subgroup, parity-doubled into $(r, \mathbf{1}) \oplus (\mathbf{1}, r)$ chiral multiplets — is *vector-like* with respect to the chiral group and therefore carries *zero* cubic anomaly by Theorem 37.11. A real r has $A(r) = 0$; the observed baryon octet of three-flavor QCD is the adjoint $\mathbf{8}$ of $\mathrm{SU}(3)_V$, which is real, so a massless octet would contribute $A(\mathbf{8})\,\dim(\mathbf{1}) = 0$ to (37.126) and could never reach N_c. Matching thus forces the massless baryons into a *chiral* (complex) arrangement of the flavor group, parity-asymmetric and unlike any low-lying hadron multiplet observed. Whether even such an exotic chiral spectrum can solve (37.126)–(37.127) in non-negative integers is the question the worked example settles for the physical case.

Worked Example 37.32 (The failed baryon matching for three-flavor QCD)**.** Take the physical case $N_c = 3$, $N_f = 3$. The ultraviolet targets from Theorem 37.31 are

$$\mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(3)_L)^3\big] = 3, \qquad \mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(3)_L)^2\,\mathrm{U}(1)_B\big] = \tfrac{1}{2}. \tag{37.128}$$

A massless baryon is a color-singlet of three quarks, so its left-handed Weyl content is built from n_L left-quark flavor indices and n_R right-quark flavor indices with $n_L + n_R = 3$, giving $B = (n_L + n_R)/N_c = 1$ and a chiral flavor representation (r_L, r_R) with $r_L \subset \mathbf{3}^{\otimes n_L}$, $r_R \subset \mathbf{3}^{\otimes n_R}$. The four splittings $(n_L, n_R) \in \{(3,0), (2,1), (1,2), (0,3)\}$ and the SU(3) products $\mathbf{3} \otimes \mathbf{3} = \mathbf{6} \oplus \bar{\mathbf{3}}$ and $\mathbf{3} \otimes \mathbf{3} \otimes \mathbf{3} = \mathbf{10} \oplus \mathbf{8} \oplus \mathbf{8} \oplus \mathbf{1}$ enumerate every possible massless three-quark baryon multiplet. We tabulate, for each, its contribution to the two left-handed anomalies, using $A(\mathbf{3}) = 1$, $A(\bar{\mathbf{3}}) = -1$, $A(\mathbf{6}) = N_f + 4 = 7$ at $N_f = 3$, $A(\mathbf{8}) = 0$, $A(\mathbf{1}) = 0$, $A(\mathbf{10}) = 27$ (the last from additivity, $A(\mathbf{3}^{\otimes 3}) = 3 \cdot 6 + 9 \cdot 1 = 27$, all carried by the $\mathbf{10}$ since $\mathbf{8}, \mathbf{1}$ are anomaly-free), and the Dynkin indices $T(\mathbf{3}) = T(\bar{\mathbf{3}}) = \frac{1}{2}$, $T(\mathbf{6}) = \frac{5}{2}$, $T(\mathbf{8}) = 3$, $T(\mathbf{10}) = \frac{15}{2}$, $T(\mathbf{1}) = 0$.

(r_L, r_R)	cubic $A(r_L)\,\dim(r_R)$	mixed $T(r_L)\,\dim(r_R)$
$(\mathbf{10}, \mathbf{1})$	27	$\frac{15}{2}$
$(\mathbf{8}, \mathbf{1})$	0	3
$(\mathbf{1}, \mathbf{1})$	0	0
$(\mathbf{6}, \mathbf{3})$	21	$\frac{15}{2}$
$(\bar{\mathbf{3}}, \mathbf{3})$	-3	$\frac{3}{2}$
$(\mathbf{3}, \mathbf{6})$	6	3
$(\mathbf{3}, \bar{\mathbf{3}})$	3	$\frac{3}{2}$
$(\mathbf{1}, r_R)$	0	0

Read the mixed-anomaly column. *Every* three-quark baryon multiplet contributes to $[\mathrm{SU}(3)_L]^2\,\mathrm{U}(1)_B$ an amount that is an integer multiple of $\frac{3}{2}$: the entries are $\frac{15}{2} = 5\cdot\frac{3}{2}$, $3 = 2\cdot\frac{3}{2}$, 0, and $\frac{3}{2}$. This is not an accident of the table; it is the imprint of $N_c = 3$. A baryon carries $B = 1$ while the quark carried $B = 1/N_c$, so the fractional charge that produced the innocuous $\frac{1}{2}$ in (37.128) has no counterpart in the composite spectrum, where every contribution is built from integer-B states and lands in $\frac{N_c}{2}\,\mathbb{Z} = \frac{3}{2}\,\mathbb{Z}$. Antibaryons ($B = -1$, conjugate multiplets) only change the overall sign of an entry, so they too contribute in units of $\frac{3}{2}$. Therefore, for *any* non-negative-integer multiplicities ℓ_a and any inclusion of antibaryons,

$$\sum_a \ell_a\, B_a\, T(r_L^a)\,\dim(r_R^a) \;\in\; \tfrac{3}{2}\,\mathbb{Z}, \qquad \text{but} \qquad \mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(3)_L)^2\,\mathrm{U}(1)_B\big] = \tfrac{1}{2} \;\notin\; \tfrac{3}{2}\,\mathbb{Z}. \tag{37.129}$$

The mixed flavor–baryon anomaly *cannot be matched* by any spectrum of massless three-quark baryons. The confined-but-chirally-symmetric vacuum is excluded for $N_f = 3$.

It is instructive to see that the cubic equation alone would not have caught this: the single multiplet $(\mathbf{3}, \bar{\mathbf{3}})$ has cubic anomaly $3 = N_c$ and would match (37.128) for the $[\mathrm{SU}(3)_L]^3$ triangle by itself — but it contributes $\frac{3}{2} \neq \frac{1}{2}$ to the mixed triangle, and there is no way to correct $\frac{3}{2}$ to $\frac{1}{2}$ by adding further baryons, every one of which moves the mixed total in steps of $\frac{3}{2}$. It is the *simultaneous* imposition of the cubic and the mixed anomaly that kills the scenario, exactly the mechanism advertised in Section 37.5.6: one anomaly rarely suffices, but the full set is rigid. The naive guess — the observed parity spectrum, with a massless octet — fails even the cubic equation alone, since $A(\mathbf{8}) = 0$ gives zero where $N_c = 3$ is required.

The worked example proves the impossibility for the physical $N_f = 3$ within the three-quark baryon sector, which is the relevant content of a confining $N_c = 3$ theory. The general statement, for arbitrary $N_f \geq 3$ and including exotic multiquark composites, is 't Hooft's original result, and the clean engine behind it is the persistent-mass / decoupling condition of Theorem 37.28, applied flavor by flavor.

Theorem 37.33 (No unbroken chiral realization for $N_f \geq 3$)**.** *For QCD with N_c colors and $N_f \geq 3$ massless flavors, there is no spectrum of massless color-singlet baryons that simultaneously matches the cubic $[\mathrm{SU}(N_f)_{L,R}]^3$ and mixed $[\mathrm{SU}(N_f)_{L,R}]^2\,\mathrm{U}(1)_B$ 't Hooft anomalies of Theorem 37.31 with non-negative integer multiplicities consistent with the persistent-mass condition. The chirally symmetric confining vacuum is therefore forbidden; chiral symmetry must break.*

The decoupling argument. Suppose a matching massless-baryon spectrum existed for N_f flavors. Give the N_f-th quark a mass and send it to infinity. In the

ultraviolet this reduces $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \to \mathrm{SU}(N_f-1)_L \times \mathrm{SU}(N_f-1)_R$ and leaves exactly $N_f - 1$ flavor massless QCD, whose cubic anomaly is again N_c — *unchanged*, since (37.120) is independent of N_f. In the infrared, every massless baryon containing the heavy quark acquires a mass of order $m \to \infty$ and decouples (Theorem 37.27); the baryons that survive massless are precisely those built from the remaining $N_f - 1$ light flavors, and by the matching theorem applied to the reduced theory they must reproduce the same N_c for $\mathrm{SU}(N_f - 1)$. This is a recursion: a matching N_f-flavor spectrum must contain, as the heavy-quark-free subset, a matching $(N_f - 1)$-flavor spectrum, all the way down. Run the recursion to $N_f = 3$. There the explicit obstruction of Theorem 37.32 applies for $N_c = 3$ (and the analogous mixed-anomaly divisibility $\frac{1}{2} \notin \frac{N_c}{2}\mathbb{Z}$ for any $N_c \geq 2$): the three-quark baryons cannot match the mixed anomaly, and admitting heavier multi-quark composites only reintroduces states that the next decoupling step removes without restoring the deficit. The recursion therefore has no consistent base case for $N_f \geq 3$, and the assumed spectrum cannot exist. By the trilemma of Section 37.5, with confinement assumed (so the theory is gapped except for whatever the anomaly forces) and massless composite fermions excluded, the only remaining escape is spontaneous breaking of G. Hence chiral symmetry breaks. □

Caution

The honest scope. The cubic obstruction requires $N_f \geq 3$; at $N_f = 2$ the $[\mathrm{SU}(2)]^3$ anomaly does not exist (Eq. (37.36)) and the only surviving constraint is the mixed $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_B = \frac{1}{2}$, which is matched by a single massless nucleon doublet (the would-be massless proton and neutron form an $\mathrm{SU}(2)_V$ doublet of $B = 1$, and the Witten-type two-flavor counting closes), so two-flavor QCD does *not* require chiral breaking on anomaly grounds alone. The clean impossibility (37.129) is proved here for the three-quark sector at $N_c = 3$; the fully general theorem, covering arbitrary multi-quark massless composites for all $N_f \geq 3$, is 't Hooft's, with the persistent-mass condition supplying the inductive step, and is reviewed in the literature cited at the end of the chapter. Anomaly matching is a veto, not a constructive solver (Section 37.5.6); it excludes the unbroken scenario, and the positive identification of the broken vacuum rests in addition on the observed pion spectrum and on lattice and large-N_c evidence. The conclusion "chiral symmetry breaks" is, with these qualifications, forced for $N_f \geq 3$ and strongly favored for the real world.

Physical Insight

The deepest point of the QCD argument fits in one sentence: the quark carries baryon number $1/N_c$ but every confined fermion carries integer baryon number, so the fractional mixed anomaly $\frac{1}{2}$ cannot be assembled from integer-charge bound states. Confinement, which hides color, also hides the fractional baryon number that made the anomaly small — and the resulting arithmetic mismatch is fatal to a symmetric vacuum. The theory escapes not by inventing exotic massless baryons but by breaking the symmetry, so that the anomaly is carried by Goldstone bosons, which are exempt from the integer-charge bookkeeping because they realize the symmetry non-linearly.

37.6.4 Scenario B: chiral symmetry breaks, and who carries the anomaly

The verdict of Theorem 37.33 is that $G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B$ cannot stay unbroken. The dynamical fact, established phenomenologically in Section 29.6.7 and now *forced* by anomaly matching, is that the quark bilinear condenses, $\langle \bar{q}q \rangle \neq 0$, breaking the chiral group to its vector (diagonal) subgroup while preserving baryon number:

$$\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \xrightarrow{\langle \bar{q}q \rangle \neq 0} \mathrm{SU}(N_f)_V \times \mathrm{U}(1)_B. \tag{37.130}$$

By Goldstone's theorem (Theorem 29.11) this produces $\dim \mathrm{SU}(N_f) - \dim \mathrm{SU}(N_f) = N_f^2 - 1$ massless Goldstone bosons — the pions and their flavor partners — governed at low energy by the nonlinear sigma model of Section 29.5, which is the chiral Lagrangian that Chapter 39 develops in full. Now we must check that this realization *can* match the anomalies that the symmetric one could not.

The check has two parts. First, the surviving exact symmetry $\mathrm{SU}(N_f)_V$ acts identically on left- and right-handed quarks; it is *vector-like*. Its cubic anomaly from the quark content (37.119) is

$$\mathcal{A}\big[(\mathrm{SU}(N_f)_V)^3\big] = N_c\, A(\mathbf{N_f}) + N_c\, A(\bar{\mathbf{N}}_\mathbf{f}) = N_c \cdot 1 + N_c \cdot (-1) = 0, \tag{37.131}$$

because ψ sits in the $\mathrm{SU}(N_f)_V$ fundamental and χ in the antifundamental. The unbroken vector symmetry is anomaly-free, so matching it demands *nothing* of the infrared: the massive baryons of the real world, which fill real representations of $\mathrm{SU}(N_f)_V$ (the octet $\mathbf{8}$, the decuplet's vector content, and so on), are exactly consistent with $\mathcal{A}[(\mathrm{SU}(N_f)_V)^3] = 0$, and need not be massless. This is why nature can have a heavy, vector-symmetric baryon spectrum: the only symmetry those baryons must faithfully represent has no anomaly to carry.

Second, the genuinely nonzero anomalies (37.125) all involve the *broken* (axial) directions — they are the coefficients of triangles with at least one current along

a broken generator. With chiral symmetry broken there are no massless fermions charged chirally to carry them (Theorem 37.28), and the massive baryons are anomaly-neutral. The deficit must be made up by the only massless states left: the Goldstone bosons. But a Goldstone boson is an ordinary scalar, and no *invariant* local Lagrangian built from scalars can produce a triangle anomaly — an anomaly is precisely the part of the effective action that is *not* a gauge-invariant local functional, by the Wess–Zumino consistency condition of Section 37.1. The resolution is that the Goldstone effective action contains, beyond its invariant nonlinear-sigma-model part, one extra term that is *not* invariant under the broken symmetry but shifts by exactly a total derivative whose integral reproduces the anomaly: the Wess–Zumino–Witten term. Its coefficient is not a free parameter; matching (37.120) fixes it to be the integer

$$k \;=\; \mathcal{A}_{\mathrm{UV}}\big[(\mathrm{SU}(N_f))^3\big] \;=\; N_c, \tag{37.132}$$

the number of colors, reappearing in the low-energy theory of pions as the quantized level of a topological term. The Goldstone bosons thereby carry the full chiral anomaly of the quarks, $\mathcal{A}_{\mathrm{IR}} = \mathcal{A}_{\mathrm{UV}} = N_c$, and matching is restored.

Physical Insight

The two scenarios are distinguished by *who is massless*. Unbroken chiral symmetry (Scenario A) keeps massless composite fermions, and the anomaly must live on them — but the integer-baryon-number arithmetic forbids it. Broken chiral symmetry (Scenario B) replaces those would-be massless fermions with Goldstone bosons, and the anomaly lives on *them*, through a term whose very non-invariance is what lets a boson do a fermion's job. Anomaly matching does not merely permit chiral symmetry breaking; for $N_f \geq 3$ it requires it, and it predicts that the low-energy pion theory must contain a topological term of integer strength N_c. The light pion of Eq. (29.228) and the WZW term of the next section are two faces of the same forced symmetry breaking.

We have used the one exact, non-perturbative constraint that survives at strong coupling to extract a structural truth about QCD that no order of perturbation theory could reach: with three or more massless flavors the chiral symmetry of the quarks must break spontaneously, and the broken-symmetry vacuum is the only one whose massless spectrum can reproduce the ultraviolet anomaly coefficient N_c. What remains is to build the object that does the reproducing. The next section constructs the Wess–Zumino–Witten functional explicitly — Witten's five-dimensional integral, the proof that its coefficient is quantized to the integer N_c, and the gauging that turns it back into the $\pi^0 \to \gamma\gamma$ amplitude of Chapter 36 — thereby closing the matching loop opened here.

37.7 The Wess–Zumino–Witten Term

The previous section delivered a verdict, not a construction. 't Hooft matching, applied to massless QCD with N_f light flavors, demanded that the ultraviolet anomaly of the global chiral group $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$—a definite number, the $\mathrm{SU}(N_f)_L^3$ coefficient equal to the color count N_c (Eq. (37.120))—be reproduced, exactly and without renormalization, by whatever degrees of freedom carry the physics at low energy. We argued that the confined-and-symmetric scenario cannot supply it with massless baryons, so chiral symmetry must break, and the infrared theory is the one built in Chapter 29: a nonlinear sigma model of the Goldstone bosons, the pseudoscalar mesons, packaged into a field $U(x) \in \mathrm{SU}(N_f)$. The matching demand then sharpens into a question we have not answered. The leading chiral Lagrangian, $\frac{f^2}{4}\,\mathrm{tr}(\partial_\mu U\, \partial^\mu U^\dagger)$, and every higher term of the standard derivative expansion, is *exactly invariant* under the chiral group. An invariant action has a vanishing anomaly. How, then, can a theory of nothing but invariant Goldstone interactions reproduce a nonzero anomaly coefficient?

The resolution is the subject of this section, and it is one of the most beautiful confluences of topology and field theory in all of physics. There exists a term that can be added to the chiral Lagrangian which is *not* expressible as the four-dimensional integral of anything local in U, yet which is perfectly well defined, Lorentz invariant, and—crucially—*not* chirally invariant: under a chiral transformation it shifts by precisely the anomaly. This is the Wess–Zumino–Witten (WZW) term. Wess and Zumino first wrote its variation as the unique local solution of the anomaly's consistency conditions; Witten then gave it a closed geometric form, as the integral of a closed five-form over a five-dimensional disk whose boundary is spacetime. The geometric construction does something the local one cannot: it exposes that the coefficient of the term is *quantized*—forced to be an integer by the topology of the group manifold—and that the integer is none other than N_c. The same integer, read off by gauging electromagnetism, reproduces the $\pi^0 \to \gamma\gamma$ amplitude of Chapter 36 with its factor of $N_c = 3$ on display. The matching loop opened in the previous section closes here, and it closes on a topological invariant.

This advances *Yang–Mills theory* to its topological and effective-theory layer. The color number N_c, which entered the book as the dimension of the QCD gauge group and reappeared in the beta function and the R-ratio, now surfaces a third time as the integer level of a topological term in the meson Lagrangian—an infrared, entirely non-fermionic quantity that nonetheless counts the underlying quarks. It is also the bridge to chiral perturbation theory (Chapter 39), where the WZW term is the sole leading-order interaction that violates the "naive" intrinsic parity of the mesons and so controls processes such as $\gamma \to 3\pi$ and $K\bar{K} \to 3\pi$. And it recovers the $\pi^0 \to \gamma\gamma$ matching of *QED*.

37.7.1 The infrared theory and the matching demand

We work with the low-energy realization of chiral symmetry breaking constructed in Chapter 29. For QCD with N_f massless quark flavors the global symmetry is $G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ (we set aside the abelian factors, whose fate—$\mathrm{U}(1)_B$ exact, $\mathrm{U}(1)_A$ anomalous—was settled earlier), and the quark condensate breaks it to the diagonal vector subgroup $H = \mathrm{SU}(N_f)_V$. Goldstone's theorem (Theorem 29.11) then guarantees $\dim G - \dim H = N_f^2 - 1$ massless Goldstone bosons, the pseudoscalar mesons, which the coset construction of Section 29.5 assembles into a single unitary matrix field. We continue that construction directly: write

$$U(x) = \exp\Big(\frac{i}{f}\,\pi^a(x)\,X^a\Big) \in \mathrm{SU}(N_f), \tag{37.133}$$

exactly the Goldstone group element of Eq. (29.187), with X^a the broken (axial) generators and $f \to f_\pi$ the pion decay constant. Under a chiral rotation $(g_L, g_R) \in \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ the field transforms linearly,

$$U(x) \;\longmapsto\; g_L\,U(x)\,g_R^\dagger, \tag{37.134}$$

so that the vector subgroup ($g_L = g_R$) acts by conjugation and the axial directions ($g_L = g_R^\dagger$) shift the Goldstone fields. The leading two-derivative Lagrangian,

$$\mathcal{L}_2 = \frac{f_\pi^2}{4}\,\mathrm{tr}(\partial_\mu U\,\partial^\mu U^\dagger), \tag{37.135}$$

is manifestly invariant under Eq. (37.134), as is every term of the standard chiral expansion built from traces of U, $U^\dagger$ and their derivatives. Each such term is a Lorentz scalar of even intrinsic parity: it contains an even number of Levi-Civita-free derivative structures and is unchanged under the naive parity $U(\mathbf{x},t) \to U^\dagger(-\mathbf{x},t)$ together with $\pi^a \to -\pi^a$.

Here is the tension, stated precisely. The ultraviolet theory has an exact, scheme-independent anomaly: the three-point function of axial $\mathrm{SU}(N_f)$ currents has the anomalous divergence whose coefficient, computed from the colored quark triangle in the previous section, is N_c times the single-flavor value (Eq. (37.120)). By 't Hooft's argument that coefficient is an infrared observable: it must appear in correlation functions of the meson theory. But $\mathcal{L}_2$ and all its invariant descendants produce a *vanishing* anomaly, because their variation under Eq. (37.134) is identically zero. The infrared Lagrangian as built so far cannot match. Either 't Hooft matching fails—which would be a contradiction, since both sides compute the same Green's function—or the chiral Lagrangian is incomplete. It is incomplete, and the missing piece is forced to be a term whose chiral variation is nonzero and equal to the anomaly.

37.7.2 The obstruction: no local invariant four-dimensional term will do

What sort of term could carry a nonvanishing chiral variation? It must be parity-odd: the anomaly is proportional to the ϵ-tensor, and a term reproducing it must be P-odd as well, carrying one $\epsilon^{\mu\nu\rho\sigma}$. The natural building block is the *left-invariant Maurer–Cartan form*

$$L \;\equiv\; U^\dagger \,\mathrm{d}U, \tag{37.136}$$

a Lie-algebra-valued one-form (anti-Hermitian, since $\mathrm{d}(U^\dagger U) = 0$ implies $L^\dagger = -L$) that is invariant under left multiplication $U \to g_L U$ and transforms by conjugation under right multiplication. Traces of wedge powers of L are the canonical chirally covariant, parity-odd invariants. In four dimensions the unique candidate with the right index structure is the four-form $\mathrm{tr}(L^4) = \mathrm{tr}(L\wedge L\wedge L\wedge L)$, whose components are $\epsilon^{\mu\nu\rho\sigma}\,\mathrm{tr}(L_\mu L_\nu L_\rho L_\sigma)$ with $L_\mu = U^\dagger \partial_\mu U$.

This candidate is useless, for a sharp reason: it vanishes identically.

Proposition 37.34 (Vanishing of the four-dimensional candidate)**.** *The four-form* $\mathrm{tr}(L^4)$ *built from the Maurer–Cartan one-form* $L = U^\dagger\,\mathrm{d}U$ *is identically zero. More generally,* $\mathrm{tr}(L^{2k}) \equiv 0$ *for every integer* k*, while* $\mathrm{tr}(L^{2k+1})$ *need not vanish.*

Proof. L is a one-form, so under the wedge product two factors anticommute, $L\wedge L = -L\wedge L$ at the level of the form indices, while the trace is cyclic at the level of the matrix indices. Cycling one factor of L from the front of $\mathrm{tr}(L^n)$ to the back moves it past $n-1$ other one-forms, picking up a sign $(-1)^{n-1}$ from the form grading, and the trace is unchanged by the cyclic move on the matrix indices:

$$\mathrm{tr}(L^n) = \mathrm{tr}\big(L\cdot L^{n-1}\big) = (-1)^{n-1}\,\mathrm{tr}\big(L^{n-1}\cdot L\big) = (-1)^{n-1}\,\mathrm{tr}(L^n). \tag{37.137}$$

For even n the factor is $(-1)^{n-1} = -1$, so $\mathrm{tr}(L^n) = -\,\mathrm{tr}(L^n) = 0$. For odd n the factor is $+1$ and the identity is vacuous. In particular $\mathrm{tr}(L^4) = 0$, and $\mathrm{tr}(L^5)$ survives. □

The collapse of $\mathrm{tr}(L^4)$ is not a defect of one ansatz; it is the structural obstruction. Any local, Lorentz-invariant, $H = \mathrm{SU}(N_f)_V$-invariant, parity-odd four-form built polynomially from U and its first derivatives reduces, after using $UU^\dagger = 1$ and integration by parts, to a multiple of $\mathrm{tr}(L^4)$ and hence is zero. One can manufacture higher-derivative parity-odd scalars, but they too, being total chiral invariants, have vanishing chiral variation and cannot reproduce the anomaly. The conclusion is unavoidable.

Proposition 37.35 (No local four-dimensional carrier)**.** *There is no local, Lorentz-invariant four-dimensional Lagrangian density $\mathcal{L}_{\mathrm{WZ}}[U]$, built from U and finitely many of its derivatives, whose chiral variation under Eq. (37.134) equals the QCD chiral anomaly. The anomalous variation is a nontrivial element of the local cohomology—it solves the Wess–Zumino consistency conditions but is not the variation of any local four-dimensional functional.*

We do not need the full cohomological classification to proceed; Theorem 37.34 already tells us the only natural candidate is empty, and the positive construction of the next subsection will exhibit a functional that does the job and explain, geometrically, exactly why it cannot be written as a four-dimensional integral. The single surviving fact we carry forward is the asymmetry between $\mathrm{tr}(L^4) = 0$ and $\mathrm{tr}(L^5) \neq 0$: the carrier of the anomaly lives one dimension up.

Physical Insight

The mismatch in Eq. (37.137) between even and odd powers is the entire reason the WZW term is five-dimensional. A would-be four-dimensional anomaly functional is killed by the wedge-product grading; the same grading spares the five-form $\mathrm{tr}(L^5)$. Spacetime is four-dimensional, so the only way to integrate a five-form is to extend the field configuration into a five-dimensional region whose boundary is spacetime. The "extra dimension" is not physical—no new field, no new propagating mode—it is a bookkeeping device that makes a genuinely four-dimensional, but topologically nontrivial, functional well defined. This is the first appearance in the book of a recurring frontier theme: an anomaly in d dimensions is the boundary avatar of a topological term in $d+1$ dimensions, the idea of anomaly inflow taken up in Section 37.8.4.

37.7.3 Witten's five-dimensional construction

Take Euclidean spacetime and compactify it: a field configuration $U(x)$ that approaches a constant at infinity is a map from the one-point compactification S^4 into $\mathrm{SU}(N_f)$. The construction rests on two topological facts about the group manifold, both of which we will use and the second of which will do the quantization.

Lemma 37.36 (Extendability and the obstruction group)**.** *For $N_f \geq 3$, every smooth map $U : S^4 \to \mathrm{SU}(N_f)$ extends to a smooth map $\widetilde{U} : B^5 \to \mathrm{SU}(N_f)$ from a five-disk B^5 with $\partial B^5 = S^4$, because $\pi_4(\mathrm{SU}(N_f)) = 0$. Two such extensions, glued along their common boundary, define a map of the five-sphere $S^5 \to \mathrm{SU}(N_f)$, and these are classified up to homotopy by $\pi_5(\mathrm{SU}(N_f)) = \mathbb{Z}$.*

The first statement is what makes the construction possible at all: because $\pi_4 = 0$ there is no obstruction to filling in the disk. The second statement,

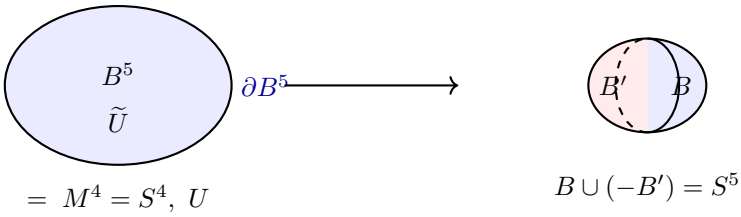


Figure 37.7: Left: a spacetime configuration U on the compactified Euclidean spacetime $M^4 = S^4$ is extended to a map $\widetilde{U}$ on a five-disk B^5 whose boundary is M^4 (possible because $\pi_4(\mathrm{SU}(N_f)) = 0$). Right: two extensions B and B' of the same boundary data glue, with opposite orientation on one, into a closed five-sphere S^5. The difference of the WZW functional evaluated on the two extensions is its integral over S^5, classified by the winding number in $\pi_5(\mathrm{SU}(N_f)) = \mathbb{Z}$; demanding that this difference leave $e^{i\Gamma_{\mathrm{WZW}}}$ unchanged quantizes the coefficient.

$\pi_5(\mathrm{SU}(N_f)) = \mathbb{Z}$ for $N_f \geq 3$, is what will force the coefficient to be an integer. (For SU(2), $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}_2$ and the story is different; the standard WZW term is a feature of three or more flavors, which is the physically relevant case and the one we treat.)

The candidate five-form is $\mathrm{tr}(L^5)$, and the first thing to check is that it is closed, so that its integral over the disk depends on the extension only mildly.

Lemma 37.37 (Closure of the five-form)**.** *The five-form $\omega_5 = \mathrm{tr}(L^5)$, with $L = U^\dagger\,\mathrm{d}U$, is closed: $\mathrm{d}\omega_5 = 0$.*

Proof. The Maurer–Cartan form satisfies the structure equation. From $\mathrm{d}U^\dagger = -U^\dagger\,(\mathrm{d}U)\,U^\dagger$ (the differential of $U^\dagger U = 1$) we get

$$\mathrm{d}L = \mathrm{d}U^\dagger \wedge \mathrm{d}U = -U^\dagger\,\mathrm{d}U\,U^\dagger \wedge \mathrm{d}U = -L \wedge L \equiv -L^2. \tag{37.138}$$

Now differentiate $\omega_5 = \mathrm{tr}(L \wedge L \wedge L \wedge L \wedge L)$ using the graded Leibniz rule, with L of odd degree contributing a sign each time the exterior derivative passes a factor:

$$\mathrm{d}\,\mathrm{tr}(L^5) = \sum_{i=0}^{4} (-1)^i\,\mathrm{tr}\big(L^i\,(\mathrm{d}L)\,L^{4-i}\big) = \sum_{i=0}^{4} (-1)^{i+1}\,\mathrm{tr}\big(L^i\,L^2\,L^{4-i}\big) = \Big(\sum_{i=0}^{4}(-1)^{i+1}\Big)\,\mathrm{tr}(L^6). \tag{37.139}$$

The alternating sum $\sum_{i=0}^{4}(-1)^{i+1} = -1 + 1 - 1 + 1 - 1 = -1$, so $\mathrm{d}\,\mathrm{tr}(L^5) = -\,\mathrm{tr}(L^6)$. But $\mathrm{tr}(L^6)$ is an even power and vanishes by exactly the cyclic-grading argument of Theorem 37.34, $\mathrm{tr}(L^6) = (-1)^{6-1}\,\mathrm{tr}(L^6) = -\,\mathrm{tr}(L^6) = 0$. Hence $\mathrm{d}\omega_5 = 0$. □

With closure established we can define the functional.

Definition 37.38 (Wess–Zumino–Witten functional)**.** Let $\widetilde{U}$ extend the space-time configuration U over a five-disk B with $\partial B = M^4$. The Wess–Zumino–Witten functional at level n is

$$\Gamma_{\mathrm{WZW}}[U] \;=\; -\frac{i\,n}{240\pi^2}\int_B \mathrm{tr}\big[(\,\widetilde{U}^{\dagger}\,\mathrm{d}\widetilde{U}\,)^5\big] \;=\; -\frac{i\,n}{240\pi^2}\int_B \mathrm{tr}(L^5), \tag{37.140}$$

the integral of the five-form ω_5 pulled back to B by $\widetilde{U}$. It is added to the four-dimensional chiral action, $S = \int_{M^4}\mathcal{L}_2 + \cdots + \Gamma_{\mathrm{WZW}}$.

Three features deserve immediate comment. First, the integrand depends on $\widetilde{U}$ only through its *derivatives*, packaged into L, so a constant shift of the field drops out and the functional descends to the genuine degrees of freedom. Second, because ω_5 is closed (Theorem 37.37), Stokes' theorem controls how Γ_{WZW} responds to deforming the extension at fixed boundary: an infinitesimal change $\delta\widetilde{U}$ in the interior changes the integrand by an exact form, $\delta\omega_5 = \mathrm{d}(\text{four-form})$, so $\delta\Gamma_{\mathrm{WZW}} = \int_B \mathrm{d}(\cdots) = \int_{\partial B}(\cdots) = \int_{M^4}(\cdots)$ reduces to a boundary integral that depends only on U and its variation on M^4. The variation of Γ_{WZW} is therefore a genuinely four-dimensional, local expression—this is what lets the term contribute to the equations of motion and to scattering amplitudes as if it were an ordinary four-dimensional Lagrangian, even though the functional itself has no four-dimensional integral representation. Third, the prefactor $-in/240\pi^2$ is engineered, as we now show, so that the residual dependence on the *global* choice of extension is invisible in $e^{i\Gamma_{\mathrm{WZW}}}$ precisely when n is an integer.

37.7.4 Topological quantization: the level is an integer

The functional in Eq. (37.140) appears to depend on the arbitrary choice of disk B. Physics must not. The path integral weights a configuration by e^{iS}, so what must be independent of the extension is not Γ_{WZW} itself but the phase $e^{i\Gamma_{\mathrm{WZW}}}$. We now show that well-definedness of this phase quantizes n.

Let B and B' be two extensions of the same boundary configuration U on $M^4 = S^4$. Their difference is

$$\Gamma^{(B)}_{\mathrm{WZW}} - \Gamma^{(B')}_{\mathrm{WZW}} = -\frac{in}{240\pi^2}\left(\int_B - \int_{B'}\right)\mathrm{tr}(L^5) = -\frac{in}{240\pi^2}\int_{B\cup(-B')}\mathrm{tr}(L^5), \tag{37.141}$$

where $B\cup(-B')$ means B glued to B' with reversed orientation. Since B and B' share the boundary S^4, the glued object is a closed five-manifold; topologically it is a five-sphere, and the two extensions together define a map $S^5 \to \mathrm{SU}(N_f)$ (Theorem 37.36; see Fig. 37.7). The difference is the integral of ω_5 over this closed S^5.

The decisive input is the value of that integral. The five-form $\mathrm{tr}(L^5)$ is, up to normalization, the generator of the fifth de Rham cohomology of the group manifold, and its integral over a closed S^5 computes the winding number—the

integer class in $\pi_5(\mathrm{SU}(N_f)) = \mathbb{Z}$—of the map. With the normalization chosen in Eq. (37.140),

$$-\frac{i}{240\pi^2}\int_{S^5}\mathrm{tr}\big[(U^\dagger\,\mathrm{d}U)^5\big] \;=\; 2\pi\,w, \qquad w\in\mathbb{Z}=\pi_5(\mathrm{SU}(N_f)), \tag{37.142}$$

i.e. the integral of $\mathrm{tr}(L^5)$ over the unit-winding generator equals $i\cdot 480\pi^3$ (the factor of i tracking the anti-Hermiticity of L, which makes $\mathrm{tr}(L^5)$ pure imaginary so that Γ_{WZW} is real). We verify this normalization explicitly on the generator below. Granting it, Eq. (37.141) reads

$$\Gamma^{(B)}_{\mathrm{WZW}} - \Gamma^{(B')}_{\mathrm{WZW}} \;=\; 2\pi\,n\,w. \tag{37.143}$$

The phase $e^{i\Gamma_{\mathrm{WZW}}}$ is therefore independent of the extension—$e^{2\pi i n w}=1$ for every winding $w\in\mathbb{Z}$—*if and only if* $n\in\mathbb{Z}$.

Theorem 37.39 (Quantization of the WZW level)**.** *For $N_f\geq 3$ the Wess–Zumino–Witten functional Eq. (37.140) defines a single-valued contribution $e^{i\Gamma_{\mathrm{WZW}}}$ to the path-integral weight if and only if the level n is an integer. The quantization is topological in origin: it follows from $\pi_5(\mathrm{SU}(N_f))=\mathbb{Z}$, with no appeal to the dynamics.*

This is the structural payoff. Unlike the coefficients f_π, or the low-energy constants of the higher chiral Lagrangian, which are continuous numbers fixed by the strong dynamics, the WZW level is pinned to the integers by topology alone. Continuous renormalization cannot move an integer, so n is a renormalization-group invariant—precisely the property an anomaly coefficient must have to be matched between the ultraviolet and the infrared. The integer is not yet determined; topology says only that it is *some* integer. What fixes its value is matching.

Theorem 37.40 (The level is the color number)**.** *The chiral variation of Γ_{WZW} under Eq. (37.134) reproduces the consistent $\mathrm{SU}(N_f)$ chiral anomaly with coefficient proportional to the level n. Equating it to the ultraviolet QCD value (Eq. (37.120)) fixes*

$$n \;=\; N_c. \tag{37.144}$$

Argument. The boundary reduction of $\delta\Gamma_{\mathrm{WZW}}$ described after Theorem 37.38 is a local four-dimensional functional of U, linear in the chiral transformation parameter and carrying one ϵ-tensor; it is, by construction, the unique solution of the Wess–Zumino consistency conditions, and its overall coefficient is the level n. The ultraviolet theory fixes that same boundary functional through 't Hooft matching: the $\mathrm{SU}(N_f)^3_L$ anomaly of N_c colored quarks is N_c times the single-fermion coefficient, the content of Eq. (37.120). Since the two are the variation of the same correlator, their coefficients are equal, giving $n=N_c$. By

Theorem 37.39 the right-hand side had to be an integer, and the matching makes that integer N_c—consistent, and a nontrivial check, because N_c is indeed an integer. The cleanest independent confirmation that the matched coefficient is exactly N_c, with no stray rational factor, comes from gauging electromagnetism and reading off $\pi^0 \to \gamma\gamma$, carried out in Section 37.7.5. □

It remains to confirm the normalization Eq. (37.142), which we do on the explicit generator of $\pi_5(\mathrm{SU}(3))$, the simplest case relevant to three-flavor QCD.

Worked Example 37.41 (Normalization on the winding generator)**.** Embed $\mathrm{SU}(2) \hookrightarrow \mathrm{SU}(3)$ and build the generator of $\pi_5(\mathrm{SU}(3))$ as the suspension of the identity map $\mathrm{SU}(2) \cong S^3$. Concretely, parametrize S^5 by an angle $\mu \in [0, \pi/2]$ and two copies of S^2 and use the standard representative for which the pullback of $\mathrm{tr}(L^5)$ is a fixed volume form on S^5. The content we need is purely the integral of that volume form, and it is fixed by a single normalization statement that we now make self-contained.

For a Lie group, the closed, bi-invariant five-form whose integral over the unit-winding cycle is canonically normalized is

$$\Omega_5 \;=\; \frac{i}{480\pi^3}\,\mathrm{tr}\big[(U^\dagger\,\mathrm{d}U)^5\big], \qquad \int_{S^5}\Omega_5 \;=\; w \in \mathbb{Z}. \tag{37.145}$$

This is the statement that Ω_5 generates $H^5(\mathrm{SU}(N_f);\mathbb{Z}) \cong \mathbb{Z}$ integrally; the constant $i/480\pi^3$ is exactly the inverse of the volume that the bi-invariant metric assigns to the generating five-cycle. (Equivalently, it is fixed by the same Chern–Simons normalization that makes the five-dimensional Chern–Simons level an integer, the higher-dimensional cousin of the instanton number normalization of Chapter 42.) Multiplying Eq. (37.145) through,

$$\int_{S^5}\mathrm{tr}(L^5) = \frac{480\pi^3}{i}\,w = -\,i\,480\pi^3\,w, \tag{37.146}$$

and substituting into Eq. (37.140) integrated over S^5,

$$-\frac{i}{240\pi^2}\int_{S^5}\mathrm{tr}(L^5) = -\frac{i}{240\pi^2}\,(-i\,480\pi^3)\,w = i^2\,\frac{480\pi^3}{240\pi^2}\,w = -2\pi\,w, \tag{37.147}$$

using $(-i)(-i) = i^2 = -1$. This is Eq. (37.142) up to the physically irrelevant sign of the winding: only the value of Γ_{WZW} modulo 2π enters $e^{i\Gamma_{\mathrm{WZW}}}$, and relabelling the generator $w \to -w$ (an orientation choice, w ranging over all of $\mathbb{Z}$) restores the $+2\pi w$ written there. The prefactor $-in/240\pi^2$ in Eq. (37.140) is thus not arbitrary: it is the unique choice that turns the integer winding number into a 2π-valued phase ambiguity, making $e^{i\Gamma_{\mathrm{WZW}}}$ well defined exactly for integer level. The factor $240 = 2\cdot 120 = 2\cdot 5!$ is the symmetry factor of the five antisymmetrized one-forms, and the π^2 accompanies it through the volume of the relevant spheres.

37.7.5 Gauging electromagnetism: $\pi^0 \to \gamma\gamma$

The level $n = N_c$ is so far an inference from matching the $\mathrm{SU}(N_f)^3$ anomaly. We now make it visible in a physical amplitude, recovering the $\pi^0 \to \gamma\gamma$ decay of Chapter 36 from the infrared side and closing the matching loop with a number. The procedure is to gauge the electromagnetic $\mathrm{U}(1) \subset \mathrm{SU}(N_f)_V$ generated by the quark charge matrix

$$Q = \mathrm{diag}\Big(\tfrac{2}{3}, -\tfrac{1}{3}, -\tfrac{1}{3}, \dots\Big), \tag{37.148}$$

promoting derivatives to covariant ones, $\partial_\mu U \to D_\mu U = \partial_\mu U - ieA_\mu[Q, U]$, and adding to Γ_{WZW} the photon-dependent terms required to make its variation reproduce the *electromagnetic* part of the chiral anomaly. Gauging a Wess–Zumino term is not automatic: the naive minimal substitution fails to be invariant under electromagnetic gauge transformations, and one must add a definite series of local counterterms—Witten's gauging procedure—to restore it. The output of that procedure, for the term quadratic in the photon and linear in the neutral pion, is the local coupling

$$\mathcal{L}_{\pi^0\gamma\gamma} = \frac{N_c\, e^2}{32\pi^2 f_\pi}\, \mathrm{tr}(Q^2\lambda^3)\; \pi^0\, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}, \tag{37.149}$$

where $\lambda^3 = \mathrm{diag}(1, -1, 0, \dots)$ selects the neutral-pion direction in flavor space and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The whole coefficient is proportional to the level N_c: it descends from the same five-form, gauged. The flavor trace evaluates to

$$\mathrm{tr}(Q^2\lambda^3) = Q_u^2 - Q_d^2 = \Big(\tfrac{2}{3}\Big)^2 - \Big(\tfrac{1}{3}\Big)^2 = \frac{4}{9} - \frac{1}{9} = \frac{1}{3}, \tag{37.150}$$

so that, writing $e^2 = 4\pi\alpha$,

$$\mathcal{L}_{\pi^0\gamma\gamma} = \frac{N_c\, e^2}{96\pi^2 f_\pi}\, \pi^0\, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma} = \frac{N_c\, \alpha}{12\pi f_\pi}\, \pi^0\, F_{\mu\nu} \widetilde{F}^{\mu\nu}, \qquad \widetilde{F}^{\mu\nu} \equiv \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma} F_{\rho\sigma}. \tag{37.151}$$

The decay rate that follows from this coupling is the carried number of *QED*, and we now compute it in full.

Worked Example 37.42 ($\pi^0 \to \gamma\gamma$ from the gauged WZW term)**.** Write the coupling Eq. (37.151) in the form $\mathcal{L} = C\, \pi^0\, \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$ with

$$C = \frac{N_c\, e^2}{96\pi^2 f_\pi}. \tag{37.152}$$

Using $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and the total antisymmetry of ϵ, all four terms of $\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$ are equal, so $\mathcal{L} = 4C\, \pi^0\, \epsilon^{\mu\nu\rho\sigma} \partial_\mu A_\nu \partial_\rho A_\sigma$. For the decay $\pi^0(p) \to \gamma(k_1, \varepsilon_1)\gamma(k_2, \varepsilon_2)$ each outgoing photon line contributes a factor $\partial_\mu A_\nu \to i k_\mu \varepsilon^*_\nu$,

and the two ways of assigning the two photons to the two derivative pairs give equal contributions (an even permutation of ϵ), so

$$i\mathcal{M} = 4C{\cdot}2\,(ik_1)_\mu \varepsilon_{1\nu}^{*}(ik_2)_\rho \varepsilon_{2\sigma}^{*}\,\epsilon^{\mu\nu\rho\sigma} \;\Longrightarrow\; \mathcal{M} = 8iC\,\epsilon^{\mu\nu\rho\sigma} k_{1\mu}\varepsilon_{1\nu}^{*} k_{2\rho}\varepsilon_{2\sigma}^{*}. \tag{37.153}$$

Square and sum over the physical photon polarizations using $\sum_{\text{pol}} \varepsilon_\nu \varepsilon_{\nu'}^{*} \to -g_{\nu\nu'}$ (the gauge-dependent pieces drop by the antisymmetry of ϵ and transversality $k_i \cdot \varepsilon_i = 0$):

$$\sum_{\text{pol}} |\mathcal{M}|^2 = 64\,C^2\,\epsilon^{\mu\nu\rho\sigma}\epsilon^{\mu'\nu'\rho'\sigma'} k_{1\mu}k_{1\mu'}k_{2\rho}k_{2\rho'}\,(-g_{\nu\nu'})(-g_{\sigma\sigma'}). \tag{37.154}$$

Contracting the two indices ν, σ of the Levi-Civita pair, with the mostly-minus identity $\epsilon^{\mu\nu\rho\sigma}\epsilon^{\mu'}{}_{\nu}{}^{\rho'}{}_{\sigma} = -2(g^{\mu\mu'}g^{\rho\rho'} - g^{\mu\rho'}g^{\rho\mu'})$,

$$\sum_{\text{pol}} |\mathcal{M}|^2 = 64\,C^2 \cdot (-2)\big(k_1^2 k_2^2 - (k_1\cdot k_2)^2\big) = 128\,C^2\,(k_1\cdot k_2)^2, \tag{37.155}$$

where the on-shell conditions $k_1^2 = k_2^2 = 0$ were used. For a pion at rest decaying to two photons, $2k_1\cdot k_2 = (k_1+k_2)^2 = p^2 = m_\pi^2$, so $(k_1\cdot k_2)^2 = m_\pi^4/4$ and $\sum_{\text{pol}}|\mathcal{M}|^2 = 32\,C^2\,m_\pi^4$.

The two-body rate for a scalar of mass m_π into two massless, identical particles is $\Gamma = \dfrac{S}{16\pi m_\pi}\sum_{\text{pol}}|\mathcal{M}|^2$ with the identical-particle factor $S = \tfrac{1}{2}$, so

$$\Gamma(\pi^0 \to \gamma\gamma) = \frac{1}{2}\cdot\frac{32\,C^2 m_\pi^4}{16\pi m_\pi} = \frac{C^2 m_\pi^3}{\pi}. \tag{37.156}$$

Insert C from Eq. (37.152), using $e^2 = 4\pi\alpha$ so that $C = \alpha N_c/(24\pi f_\pi)$:

$$\boxed{\Gamma(\pi^0 \to \gamma\gamma) = \frac{\alpha^2\,N_c^2\,m_\pi^3}{576\,\pi^3 f_\pi^2}} \tag{37.157}$$

The amplitude is linear in N_c and the width quadratic. Setting $N_c = 3$ collapses $N_c^2/576 = 9/576 = 1/64$, giving

$$\Gamma(\pi^0 \to \gamma\gamma)\Big|_{N_c=3} = \frac{\alpha^2 m_\pi^3}{64\,\pi^3 f_\pi^2}, \tag{37.158}$$

which is exactly the value computed from the quark triangle in Eq. (36.115). Numerically, with $\alpha \simeq 1/137$, $m_\pi \simeq 135$ MeV and $f_\pi \simeq 92$ MeV, this is about 7.7 eV, in agreement with the measured 7.8 eV. The infrared computation, performed entirely within the meson theory through a topological term, reproduces the ultraviolet quark-loop number—and it does so only for $N_c = 3$.

The two faces of the level now agree. Theorem 37.39 forced n to be an integer by the topology of $\mathrm{SU}(N_f)$; Theorem 37.40 fixed that integer to N_c by matching the global $\mathrm{SU}(N_f)^3$ anomaly; and Theorem 37.42 confirmed the value through the gauged amplitude, whose N_c^2 scaling reproduces Eq. (36.115) precisely at $N_c = 3$. Had the world had a different number of colors, the WZW level would have been that number, the $\pi^0 \to \gamma\gamma$ rate would have scaled as its square, and the meson Lagrangian would have carried the count. The decay of a neutral pion is, in this sense, a measurement of N_c performed without ever resolving a quark.

Physical Insight

It is worth dwelling on what kind of object the WZW level is. The continuous couplings of the chiral Lagrangian—f_π, the low-energy constants of the four-derivative terms—are nonperturbative outputs of the strong dynamics, knowable only by lattice computation or experiment, and they run, mix, and depend on the renormalization scheme. The WZW level does none of this. It is an integer, frozen by $\pi_5(\mathrm{SU}(N_f)) = \mathbb{Z}$, and therefore exactly stable under the renormalization group, exactly equal in the ultraviolet and the infrared. This is the very property 't Hooft's matching argument exploits: an anomaly coefficient is an integer-like invariant that the two descriptions must share. The WZW term is the concrete infrared vessel that holds the $\mathrm{SU}(N_f)^3$ anomaly demanded by the matching of the previous section—the non-fermionic carrier whose existence that section's logic required but could not exhibit. We have now exhibited it.

We have found that the anomaly, far from being merely a number to predict or a pathology to cancel, has an intrinsically topological and geometric realization: a closed form on the group manifold, integrated over an extra dimension, with an integer coefficient counted by a homotopy group. That this same geometry organizes anomalies far beyond the chiral case—the gravitational and global anomalies, and the modern picture of anomalies as boundary modes of invertible topological phases via inflow and cobordism—is the closing theme of the chapter, taken up next; and the WZW term itself is handed forward to Chapter 39, where it enters the chiral Lagrangian as the leading interaction of odd intrinsic parity and governs the anomalous mesonic processes that the ordinary chiral expansion cannot produce.

37.8 Mixed, Gravitational, and Global Anomalies: The Modern Landscape

The Wess–Zumino–Witten term of Section 37.7 did something the rest of this chapter only gestured at: it produced a four-dimensional anomaly as the *boundary* of a five-dimensional object. The level-k functional was written as an integral of

a manifestly gauge-invariant five-form over a disk whose boundary is spacetime, and the anomaly appeared precisely because that five-dimensional integral cannot be pushed down to a local four-dimensional Lagrangian. We met the integer $k = N_c$ as a topological quantization condition forced by $\pi_5(\mathrm{SU}(3)) = \mathbb{Z}$. That construction is not an isolated trick. It is the first concrete instance of the organizing principle that closes this chapter and the whole of Part X: *an anomalous d-dimensional theory is the boundary of an essentially trivial $(d{+}1)$-dimensional theory, and the anomaly is the price of living on that boundary.* Seen from that height, the cubic gauge anomaly of Section 37.2, the mixed and gravitational anomalies we are about to derive, and the nonperturbative global anomaly we will meet next, are all facets of one geometric object. This section surveys the remaining anomaly types at honest, dated depth and sets the chapter's results into that modern frame.

We have four tasks. First, we complete the catalogue of triangles relevant to the Standard Model by deriving the *mixed* gauge–gravitational anomaly—one hypercharge current and two stress tensors—and showing its coefficient is $\mathrm{tr}\, Y$, the plain sum of hypercharges, thereby justifying the gravitational cancellation condition Eq. (37.47c) used in Section 37.3. Second, we explain why there is no *pure* gravitational anomaly to worry about in four dimensions: such anomalies live only in $d = 4k+2$, a fact we trace to the degrees of the Pontryagin classes. Third, we present Witten's nonperturbative SU(2) anomaly, the one anomaly invisible to the triangle diagrams of this chapter, and check that the Standard Model survives it. Fourth, we state—scoped and dated to the mid-2020s—the contemporary synthesis in which perturbative anomalies descend from an anomaly polynomial, global anomalies are classified by cobordism, and the boundary picture just sketched (anomaly inflow) becomes the unifying language.

37.8.1 The mixed gauge–gravitational anomaly

Every triangle considered so far had three gauge currents at its vertices. But the energy–momentum tensor $T^{\mu\nu}$ is a current too—the Noether current of spacetime translations, the source to which the metric couples—and a chiral fermion can run in a loop with one gauge current and two insertions of $T^{\mu\nu}$. Physically this is the amplitude for a gauge boson to convert into two gravitons through a fermion loop; the external "photons" of the gauge triangle are replaced by external gravitons. If the gauge current is the hypercharge current j_Y^μ, the resulting object is the mixed $\mathrm{U}(1)_Y$–gravitational anomaly, the fourth and last condition in the Standard Model cancellation tally.

Rather than redo the loop integral—the kinematic engine is identical to the one already turned in Chapter 36—we extract the answer from the index theorem, which packages all one-loop chiral anomalies of a Weyl fermion into a single characteristic class. This is the heat-kernel route: the Fujikawa Jacobian of Section 36.3 is the integral of an anomaly density, and on a curved background

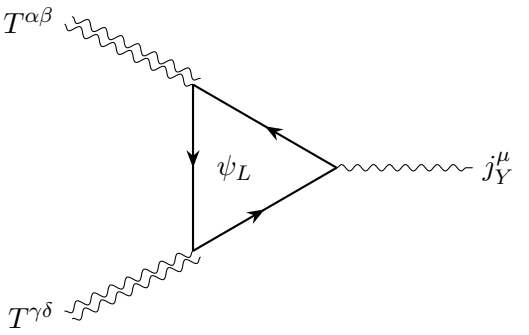


Figure 37.8: The mixed gauge–gravitational triangle: one hypercharge current j_Y^μ (single wavy line) and two energy–momentum insertions $T^{\alpha\beta}$ (double wavy lines, external gravitons) attached to a chiral fermion loop. Bose symmetry of the two gravitons selects the totally symmetric part of the vertex, and the chiral projector at the j_Y vertex produces the parity-odd ϵ-tensor structure. The group factor that survives is the bare charge Y of the looping fermion, so the summed coefficient is $\operatorname{tr} Y = \sum_i Y_i$.

coupled to a gauge field that density is fixed by the Atiyah–Singer index theorem (Theorem 36.15). We recall the statement in the only form we need.

Definition 37.43 (Anomaly polynomial)**.** The one-loop anomaly of a left-handed Weyl fermion in d spacetime dimensions, coupled to a gauge field with field strength two-form $F = \frac{1}{2} F_{\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$ and to gravity with curvature two-form $R^a{}_b = \frac{1}{2} R^a{}_{b\mu\nu}\, \mathrm{d}x^\mu \wedge \mathrm{d}x^\nu$, is governed by the $(d{+}2)$-form part of the index density,

$$\widehat{I}_{d+2} \;=\; \Big[\widehat{A}(R)\, \operatorname{tr}_R \exp\!\Big(\tfrac{iF}{2\pi}\Big)\Big]_{d+2}, \tag{37.159}$$

where $\widehat{A}(R)$ is the Dirac genus,

$$\widehat{A}(R) \;=\; 1 - \frac{1}{24}\, p_1(R) + \frac{1}{5760}\,(7p_1^2 - 4p_2) + \cdots, \qquad p_1(R) = -\frac{1}{8\pi^2}\, \operatorname{tr}(R \wedge R), \tag{37.160}$$

p_1, p_2 being the first and second Pontryagin four- and eight-forms of the tangent bundle. The physical anomaly in d dimensions is recovered from $\widehat{I}_{d+2}$ by the descent procedure of Section 37.8.4.

The content of Theorem 36.15 is exactly that the four-form part of Eq. (37.159) integrates to the index $n_- - n_+$ that we tested numerically against $\pi^0 \to \gamma\gamma$. What is new here is to read off the *six*-form, because in four dimensions the anomaly is governed by $\widehat{I}_6$: the triangle is a three-point function, the anomaly polynomial that controls a three-point function in $d = 4$ is a six-form, and the descent of a six-form yields the four-dimensional anomalous divergence. We therefore expand Eq. (37.159) to sixth order and collect the pieces.

For the abelian factor $\mathrm{U}(1)_Y$ the generator is the charge Y, a number on each Weyl field, so $\mathrm{tr}_R \exp(iF/2\pi) = \sum_i \exp(iY_i F/2\pi)$ summed over the Weyl multiplet's components, with $F = \mathrm{d}B$ the hypercharge field strength. The six-form part of Eq. (37.159) has exactly two contributions of total form-degree six,

$$\widehat{I}_6 \;=\; \underbrace{\frac{1}{3!}\Big(\frac{i}{2\pi}\Big)^3\Big(\sum\nolimits_i Y_i^3\Big)\,F^3}_{\text{cubic } \mathrm{U}(1)_Y^3} \;\; \underbrace{-\,\frac{1}{24}\,p_1(R)\wedge\frac{i}{2\pi}\Big(\sum\nolimits_i Y_i\Big)\,F}_{\text{mixed } \mathrm{U}(1)_Y\text{–grav}^2}, \tag{37.161}$$

and—decisively—*nothing else*. The first term is the cubic hypercharge anomaly, proportional to $\sum_i Y_i^3 = \mathrm{tr}\, Y^3$, which is the abelian instance of the cubic invariant of Section 37.2. The second term is the mixed anomaly we are after: it pairs the gravitational four-form $p_1(R)$ with one power of the gauge field, and its coefficient is the *linear* charge sum

$$\boxed{\mathrm{tr}\, Y \;=\; \sum_{i\in\text{left-handed Weyl}} Y_i} \tag{37.162}$$

There is no other six-form: a purely gravitational six-form would have to come from a degree-six piece of $\widehat{A}(R)$, and Eq. (37.160) has none—its terms are the $4k$-forms 1, p_1, $p_1^2, p_2, \ldots$, jumping from degree zero to degree four to degree eight. We will turn that observation into the statement of Section 37.8.2; here it tells us that in four dimensions the *only* gravitational anomaly a U(1) current can have is the mixed one, with coefficient $\mathrm{tr}\, Y$.

Descending Eq. (37.161) to four dimensions (the mechanics are in Section 37.8.4) converts the mixed six-form into the anomalous divergence of the hypercharge current in a gravitational background,

$$\partial_\mu \langle j_Y^\mu \rangle \;=\; \frac{\mathrm{tr}\, Y}{384\pi^2}\,\epsilon^{\mu\nu\rho\sigma}\, R^{\alpha}{}_{\beta\mu\nu}\, R^{\beta}{}_{\alpha\rho\sigma}, \tag{37.163}$$

where the numerical prefactor is fixed by the same heat-kernel evaluation that fixed the $\pi^0 \to \gamma\gamma$ coefficient in Chapter 36; we quote it rather than re-evaluate the identical integral. The structure $\epsilon^{\mu\nu\rho\sigma} R^{\alpha}{}_{\beta\mu\nu} R^{\beta}{}_{\alpha\rho\sigma}$ is the gravitational Pontryagin density—the curved-space analogue of $\epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$, and exactly $-16\pi^2$ times the integrand of p_1 in Eq. (37.160). The single fact we extract for the Standard Model is the proportionality: the mixed anomaly vanishes if and only if the sum of hypercharges over all left-handed Weyl fermions vanishes,

$$\text{mixed } \mathrm{U}(1)_Y\text{–gravitational anomaly} = 0 \quad\Longleftrightarrow\quad \mathrm{tr}\, Y = \sum_i Y_i = 0, \tag{37.164}$$

which is precisely the gravitational condition Eq. (37.47c) fed into the Standard Model cancellation of Section 37.3.

Proposition 37.44 (The mixed anomaly is the bare charge sum)**.** *For a chiral gauge theory with abelian factor* $\mathrm{U}(1)_Y$*, the mixed* $\mathrm{U}(1)_Y$*–gravitational anomaly coefficient is the trace of the charge over the left-handed Weyl spectrum,* $\operatorname{tr} Y = \sum_i Y_i$*, with antiparticles counted by their conjugate (sign-flipped) charges. It is the unique gravitational obstruction in* $d = 4$*, and it cancels for the Standard Model because* $\operatorname{tr} Y = 0$ *separately within each generation.*

Proof. The coefficient is read off the mixed term of Eq. (37.161), which is linear in the generator and so reduces, for the abelian factor, to $\sum_i Y_i$. Additivity over the spectrum is the same block-diagonal argument as for the cubic invariant (Theorem 37.9): the index density of a direct sum of fermions is the sum of the densities. Conjugation flips the sign, $Y \to -Y$, exactly as in Theorem 37.10, because writing a right-handed field as a left-handed conjugate reverses its charge; this is why the trace runs over left-handed Weyl fields with antiparticles entering as conjugates. Uniqueness in $d = 4$ is Eq. (37.161): the only six-forms in $\widehat{A}(R)\ \operatorname{tr}_R e^{iF/2\pi}$ involving gravity are the mixed term and (for non-abelian factors) a term $p_1 \wedge \operatorname{tr}_R F$ whose coefficient is the index $T(R)$ weighted appropriately; for a U(1) that index degenerates to the charge sum. The Standard-Model evaluation $\operatorname{tr} Y = 0$ is carried out field by field in Section 37.3 using the hypercharge assignments of Table 33.1. □

Physical Insight

The appearance of $\operatorname{tr} Y$—the *first* power of the charge—rather than $\operatorname{tr} Y^3$ is what makes the gravitational condition so sharp a constraint. The cubic condition $\operatorname{tr} Y^3 = 0$ can be satisfied by elaborate cancellations among large charges; the linear condition $\operatorname{tr} Y = 0$ says the hypercharges must literally add to zero over a generation, with no help from cubing to hide a mismatch. Together with the non-abelian mixed conditions $\sum_i Y_i\, T(R_i) = 0$, this is what nearly fixes the hypercharge assignments and, through the interplay with the color trace, ties the quark hypercharges to the number of colors and hence to electric-charge quantization. The gravitational anomaly is thus not an exotic afterthought: it is one of the load-bearing equations of the Standard Model's internal consistency, and it cancels for the same miraculous reason the others do.

37.8.2 No pure gravitational anomaly in four dimensions

We have just seen that a U(1) current can have a gravitational anomaly in $d = 4$. The natural next question is whether the gravitational current itself—the energy–momentum tensor—can be anomalous, i.e. whether diffeomorphism invariance can be broken by a chiral fermion loop with three external gravitons and no gauge legs. Were that to happen, the theory would not even define a consistent

metric coupling, and no fermion-content tuning could save it. The answer, due to Alvarez-Gaumé and Witten, is clean and is forced by the same structure that gave us Eq. (37.161): *there are no pure gravitational anomalies in four dimensions*; they occur only in $d = 4k + 2$, i.e. in $d = 2, 6, 10, \ldots$.

Theorem 37.45 (Pure gravitational anomalies live in $d = 4k + 2$)**.** *A theory of chiral fields coupled to gravity (no gauge fields) can have a perturbative gravitational anomaly only when the spacetime dimension satisfies $d \equiv 2 \pmod 4$. In particular $d = 4$ admits no pure gravitational anomaly, and the only gravitational obstruction in four dimensions is the mixed gauge–gravitational anomaly of Section 37.8.1.*

Proof. The perturbative gravitational anomaly in d dimensions is governed, exactly as in Theorem 37.43, by the purely gravitational part of the $(d{+}2)$-form index density. Setting $F = 0$ in Eq. (37.159), that part is

$$\widehat{I}^{\mathrm{grav}}_{d+2} = \big[\widehat{A}(R)\big]_{d+2}. \tag{37.165}$$

By Eq. (37.160) the Dirac genus is a polynomial in the Pontryagin forms $p_1, p_2, p_3, \ldots$, where p_k is a $4k$-form. Every monomial in $\widehat{A}(R)$ is therefore a product of $4k$-forms and so has form-degree divisible by four. A nonzero $(d{+}2)$-form can appear in $\widehat{A}(R)$ only if $d + 2 \equiv 0 \pmod 4$, that is $d \equiv 2 \pmod 4$. For $d \equiv 0 \pmod 4$—and in particular for $d = 4$, where $d + 2 = 6$ is not a multiple of four—the degree-$(d{+}2)$ part of $\widehat{A}(R)$ is identically zero, so $\widehat{I}^{\mathrm{grav}}_{d+2} = 0$ and there is no gravitational anomaly to descend.

The same conclusion has a direct diagrammatic reading. The gravitational anomaly first appears in the $(\frac{d}{2} + 1)$-point function of the energy–momentum tensor: a polygon graph with $\frac{d}{2} + 1$ stress-tensor vertices, whose parity-odd part carries one ϵ-tensor with d indices saturated by the external curvatures. In $d = 2$ this is the two-point function (the gravitational analogue of the chiral two-point anomaly); in $d = 6$ it is a four-point box; in $d = 10$ a six-point hexagon. In $d = 4$ the would-be diagram is the three-graviton triangle, but the parity-odd local term it would generate, $\epsilon^{\mu\nu\rho\sigma} R\, R$ with the curvatures carrying the remaining indices, is a *total derivative*—it is d of the gravitational Chern–Simons three-form—and hence shiftable to zero by a local diffeomorphism-invariant counterterm. Equivalently, the top symmetric invariant available in $d = 4$ at this order is the Pontryagin class p_1 itself, a closed four-form, not an anomaly: there is no symmetric invariant of the requisite type to obstruct conservation of $T^{\mu\nu}$. This is the geometric reason behind the counting: the gravitational invariants are built from the $4k$-form Pontryagin classes, and four is not two more than a multiple of four. □

Caution

The mismatch in parity counting is easy to garble, so fix it once. The relevant *spacetime* dimension for a d-dimensional anomaly is d; the anomaly polynomial that controls it is a $(d{+}2)$-form. "Pure gravitational anomalies in $d = 4k + 2$" refers to $d = 2, 6, 10$ (string-theory worldsheets, six-dimensional chiral theories, ten-dimensional supergravity), whose anomaly polynomials are the 4-, 8-, 12-forms that $\widehat{A}(R)$ does contain. Four dimensions sits in the complementary class $d = 4k$, where the gravitational anomaly polynomial vanishes and only the mixed anomaly survives. This is why the Standard Model's gravitational consistency reduces entirely to the single number $\operatorname{tr} Y$ of Eq. (37.162), and why a four-dimensional theory never has to worry about its graviton coupling being anomalous on its own account.

37.8.3 Witten's global SU(2) anomaly

Every anomaly discussed so far—cubic gauge, mixed gauge–gravitational—was *perturbative*: it is visible in a single Feynman diagram, it is a smooth function of the external momenta, and its coefficient is a polynomial in charges and Dynkin indices. There is a second, entirely different way for a chiral gauge theory to be inconsistent, one that no triangle can detect because it is invisible to perturbation theory. It was discovered by Witten and it concerns SU(2), the very group we declared safe in Theorem 37.12 on the grounds that it has no cubic invariant. That declaration was correct *for the perturbative anomaly.* The global anomaly is a separate obstruction, and SU(2) is exactly where it bites.

The origin is topological. A gauge transformation on Euclidean spacetime compactified to S^4 is a map $g : S^4 \to \mathrm{SU}(2)$. Such maps fall into homotopy classes labelled by the fourth homotopy group of the group (Section 41.2.3), and for SU(2) this group is not trivial:

$$\pi_4(\mathrm{SU}(2)) \;=\; \pi_4(S^3) \;=\; \mathbb{Z}_2. \tag{37.166}$$

The first equality is the diffeomorphism $\mathrm{SU}(2) \cong S^3$; the second is a classical (and genuinely surprising) result of homotopy theory, the first nonstabilized homotopy group of the three-sphere. There is therefore a gauge transformation g_*—a representative of the nontrivial class—that is *not* continuously connected to the identity. It is a "large" gauge transformation, and although it is a perfectly good gauge transformation, nothing in perturbation theory, which only ever sees the connected component of the identity, knows it exists.

The danger is that the fermion path integral may fail to be invariant under g_*. We make this precise. A single left-handed Weyl fermion in the doublet $\mathbf{2}$ of SU(2) has, because the doublet is pseudoreal (Theorem 37.11), a real fermion

measure: the Euclidean path integral over it yields not a determinant but a *Pfaffian* of the chiral Dirac operator,

$$\int \mathcal{D}\psi\, \mathcal{D}\bar{\psi}\, e^{-S_F[\psi,A]} \;=\; \mathrm{Pf}(\not{D}_A), \qquad \mathrm{Pf}\,(\not{D}_A)^2 = \det\,(\not{D}_A). \tag{37.167}$$

A Pfaffian is a square root of a determinant, and a square root carries a sign ambiguity. The determinant $\det \not{D}_A$ is manifestly gauge invariant and positive, but its square root $\mathrm{Pf}\,\not{D}_A$ need only return to $\pm$ itself as the gauge field is carried around a noncontractible loop in field space. Witten's computation is that as the gauge field is adiabatically deformed along the path connecting A to its large gauge transform A^{g_*}, an *odd* number of eigenvalues of $\not{D}_A$ cross zero—the spectral flow is odd, counted by the mod-two index of the five-dimensional Dirac operator interpolating between A and A^{g_*}. Each zero crossing flips the sign of the Pfaffian, so an odd number of crossings gives

$$\mathrm{Pf}(\not{D}_{A^{g_*}}) \;=\; -\,\mathrm{Pf}(\not{D}_A). \tag{37.168}$$

The integrand of the gauge-field path integral is therefore *odd* under the large gauge transformation g_*. But the path-integral measure $\mathcal{D}A$ identifies A with A^{g_*} (they are gauge-equivalent and must not be double-counted), so the contributions of A and A^{g_*} cancel in pairs and the partition function vanishes,

$$Z = \int \mathcal{D}A\;\, \mathrm{Pf}(\not{D}_A)\, e^{-S_{\mathrm{gauge}}[A]} = 0, \tag{37.169}$$

because every configuration is cancelled by its g_*-image. A theory whose partition function is forced to zero by a gauge identification is not a theory: correlators become $0/0$, and there is no consistent, gauge-invariant definition of the fermion measure. This is the global SU(2) anomaly.

Theorem 37.46 (Witten's global SU(2) anomaly)**.** *An* SU(2) *gauge theory with an odd number of left-handed Weyl fermions in the fundamental doublet* **2** *(more generally, in any pseudoreal representation with odd mod-two index) has no consistent, gauge-invariant path-integral measure: the Pfaffian Eq. (37.167) is odd under the large gauge transformation representing the nontrivial class of* $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}_2$*, and the partition function Eq. (37.169) vanishes. Consistency requires an* even *number of doublets, in which case the Pfaffian appears as* $\mathrm{Pf}^{2k} = \det^k$*, which is sign-definite and gauge invariant.*

Proof sketch. The obstruction is the sign of $\mathrm{Pf}\,\not{D}_A$ under the nontrivial element of $\pi_4(\mathrm{SU}(2))$, which is the mod-two index of the Dirac operator on the mapping torus $S^4 \times_{g_*} S^1$—the five-manifold built by gluing the ends of $S^4 \times [0,1]$ with the twist g_*. The Atiyah–Singer mod-two index theorem (the $\mathbb{Z}_2$-valued companion of Theorem 36.15, valid for real/pseudoreal Dirac operators in appropriate dimensions) evaluates this index, and for one doublet it equals $1 \in \mathbb{Z}_2$. Hence

one doublet flips the Pfaffian sign, Eq. (37.168), and forces Eq. (37.169). The mod-two index is additive over fermions, so n doublets give index n mod 2: the sign flip survives for n odd and cancels for n even. For even n the Pfaffians pair into honest, single-valued determinants and the measure is well defined. The restriction to pseudoreal representations is essential: for a complex representation the fermion measure is already a determinant, not a Pfaffian, with no sign ambiguity—which is why the global anomaly is special to SU(2) (and to Sp($2N$) more generally), the groups whose fundamental is pseudoreal. SU($N \geq 3$), whose fundamental is complex, has trivial π_4 and no global anomaly of this type; its consistency is governed entirely by the perturbative cubic condition. □

The Standard Model contains an $\mathrm{SU}(2)_L$ gauge factor with chiral fermions in the doublet, so Theorem 37.46 is a genuine constraint on it—the one nonperturbative consistency condition this chapter must check by hand. The check is a parity count of doublets, and it is short enough to do completely.

Worked Example 37.47 (The Standard Model passes the global SU(2) anomaly)**.** We count the left-handed $\mathrm{SU}(2)_L$ doublets in one generation, using the field content of Table 33.1. The fields charged under $\mathrm{SU}(2)_L$ as doublets are the quark doublet $Q_L = (u_L, d_L)$ and the lepton doublet $L_L = (\nu_L, e_L)$. Every other left-handed Weyl field of the generation—the right-handed-particle conjugates u_R^c, d_R^c, e_R^c—is an $\mathrm{SU}(2)_L$ *singlet* and contributes no doublet. The crucial multiplicity is color: the quark doublet Q_L is a color triplet, so it is in fact $N_c = 3$ independent $\mathrm{SU}(2)_L$ doublets, one per color. Hence the number of $\mathrm{SU}(2)_L$ doublets per generation is

$$n_{\text{doublet}} = \underbrace{N_c}_{Q_L,\ \text{one per color}} + \underbrace{1}_{L_L} = 3 + 1 = 4, \tag{37.170}$$

which is *even*. By Theorem 37.46 the Pfaffian appears as a fourth power per generation, the sign ambiguity squares away, and the Standard Model has a consistent $\mathrm{SU}(2)_L$ measure. With three generations the total is 12 doublets, still even; the Higgs is a boson and does not enter. The Standard Model passes the one global anomaly it could have failed.

It is worth noting what the count Eq. (37.170) depends on. The doublet number is $N_c + 1$, even precisely because $N_c = 3$ is *odd*. Had the number of colors been even, a single generation would carry an odd number of weak doublets and would suffer the global anomaly. Thus the same color number that the WZW level $k = N_c$ certified through $\pi^0 \to \gamma\gamma$ in Section 37.7, and that the perturbative $\mathrm{SU}(3)_c^2\,\mathrm{U}(1)_Y$ condition tied to charge quantization in Section 37.3, reappears here, through $\pi_4(\mathrm{SU}(2))$, as a parity constraint: N_c odd is required for nonperturbative weak consistency, and Nature's $N_c = 3$ obliges. The quark and lepton sectors are not independent; the lepton doublet supplies exactly the one extra doublet that makes $N_c + 1$ even when N_c is odd. Anomaly

freedom, perturbative and global alike, knits quarks and leptons into a single mutually-consistent generation.

37.8.4 The modern synthesis: anomalies as boundaries of invertible phases

We now stand back and name the structure that has been visible in fragments throughout the chapter. The cubic gauge anomaly, the mixed gravitational anomaly, the pure gravitational anomalies of $d = 4k + 2$, and Witten's global SU(2) anomaly look like four unrelated phenomena—a polynomial in charges, a curvature integral, a dimension-counting rule, a $\mathbb{Z}_2$ sign. The contemporary understanding, assembled over the 1980s descent formalism and the 2010s cobordism program and standard by the mid-2020s, is that they are four readings of one object: the anomaly of a d-dimensional theory is a $(d{+}1)$-dimensional *invertible* field theory, and the anomalous theory is its boundary. We lay this out without full proof, with honest flags on what is and is not settled.

Perturbative anomalies as descent. The perturbative anomalies are organized by the Stora–Zumino descent of the anomaly polynomial Eq. (37.159). One starts from the gauge-invariant, closed $(d{+}2)$-form $\widehat{I}_{d+2}$. Because it is closed and the relevant space is locally trivial, it is exact, $\widehat{I}_{d+2} = \mathrm{d}\,\omega_{d+1}$, where ω_{d+1} is a Chern–Simons form—gauge-*variant* but with a gauge variation that is itself exact,

$$\widehat{I}_{d+2} = \mathrm{d}\,\omega_{d+1}, \qquad \delta_\lambda\,\omega_{d+1} = \mathrm{d}\,\omega_d^1(\lambda), \tag{37.171}$$

the second equation defining the d-form ω_d^1, linear in the gauge parameter λ. The physical anomaly—the gauge variation of the effective action—is then $\delta_\lambda \Gamma = 2\pi \int_{M_d} \omega_d^1(\lambda)$. This single chain reproduces *every* perturbative anomaly in this chapter: the cubic d^{abc} anomaly of Section 37.2 is the descent of the $\mathrm{tr}\,F^3$ term in Eq. (37.159), the mixed gravitational anomaly Eq. (37.163) is the descent of the $p_1 \wedge \mathrm{tr}\,F$ term, and the $d = 4k + 2$ gravitational anomalies are the descents of $[\widehat{A}]_{d+2}$. The descent is also what we invoked, without ceremony, to pass from the six-form Eq. (37.161) to the divergence Eq. (37.163); the Chern–Simons form ω_5 is exactly the integrand of the five-dimensional functional, and its boundary variation is the four-dimensional anomaly.

Global anomalies via homotopy and cobordism. The descent formalism, being differential, sees only the connected component of the gauge group; it is blind to the global anomaly of Section 37.8.3, which lives in $\pi_4(\mathrm{SU}(2))$. Historically global anomalies were classified case by case using homotopy groups of the gauge group ($\pi_d(G)$ for the gauge anomaly in d dimensions). The modern and more complete classification, due to Freed, Hopkins, Kapustin, and collaborators and mature by the late 2010s, replaces homotopy with *cobordism*. The statement

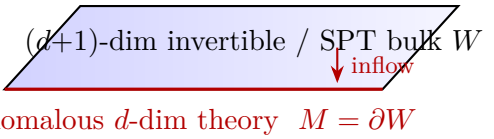


Figure 37.9: Anomaly inflow. An anomalous d-dimensional theory lives on the boundary $M = \partial W$ of a $(d{+}1)$-dimensional invertible (SPT) bulk W. The bulk action is not gauge/diffeomorphism invariant on a manifold with boundary; its boundary variation is a local d-dimensional term that exactly cancels the boundary theory's anomalous variation, so the combined system is consistent. The anomaly is the obstruction to detaching the boundary from the bulk—an invariant of the bulk SPT phase, hence renormalization-group invariant, which is the geometric content of 't Hooft matching. The five-dimensional WZW functional of Section 37.7 is the case $d = 4$.

is that the anomalies of a d-dimensional theory with a given symmetry/spacetime structure ξ (a spin or pin structure, a gauge bundle, etc.) are classified by a group of $(d{+}1)$-dimensional *invertible* field theories, which is computed by a (generalized, Anderson-dual) cobordism group Ω_{ξ}^{d+1}. In this language the perturbative anomalies form the "free part" of the classification (the part detected by the anomaly polynomial, valued in $\mathbb{Z}$ or $\mathbb{R}$), and the global anomalies form the *torsion* part. Witten's SU(2) anomaly is precisely a $\mathbb{Z}_2$ torsion class in the appropriate cobordism group, and its even-doublet resolution is the statement that the Standard Model's class is trivial.

Anomaly inflow and the SPT/invertible picture. Both faces of the anomaly—the fatal one when the current is gauged, the matchable one when it is global—acquire a single geometric meaning in the inflow picture (Fig. 37.9). Place the d-dimensional theory on the boundary $M = \partial W$ of a $(d{+}1)$-dimensional bulk carrying the invertible field theory whose partition function is $\exp(2\pi i \int_W \omega_{d+1})$ built from the Chern–Simons form of Eq. (37.171). On a closed W this bulk is trivial (invertible: its partition function is a pure phase, and it has a unique state on every spatial slice). On a manifold with boundary it is not invariant; its variation is exactly $-2\pi \int_M \omega_d^1$, which cancels the boundary anomaly $\delta\Gamma = +2\pi \int_M \omega_d^1$. The anomalous theory is consistent *as a boundary*, with the bulk absorbing the anomaly inflow—the mechanism Callan and Harvey identified for chiral fermions on defects, here in its general form. Three consequences organize the whole chapter:

- **Gauged ⇒ fatal.** If the anomalous symmetry is gauged, the bulk term is a nontrivial dependence of the dynamical partition function on the gauge field that cannot be a boundary term of a local d-dimensional action; unitarity and the Slavnov–Taylor identities (Theorem 26.35) fail. The anomaly must

be cancelled ($\sum_i \mathcal{A}(R_i) = 0$, $\operatorname{tr} Y = 0$, even doublet number) so that the bulk is genuinely trivial with no boundary obligation. This is Eq. (37.30) and its mixed and global companions.

- **Global ⇒ matched.** If the symmetry is global, the bulk SPT is a fixed, allowed feature of the theory. Its class is a discrete invariant that cannot change under any continuous deformation of the boundary dynamics—in particular under renormalization-group flow. The ultraviolet and infrared descriptions must therefore sit on the boundary of the *same* bulk: their anomaly coefficients must agree. This is exactly 't Hooft anomaly matching, here derived as the rigidity of an SPT phase rather than asserted.
- **The WZW term is inflow.** The five-dimensional construction of Section 37.7 is the $d = 4$ instance of Fig. 37.9: the WZW functional is the bulk invertible theory, its integer level is the quantized bulk coupling, and the four-dimensional anomaly it reproduces is the boundary term. The topological quantization of k is the statement that the bulk theory is well defined on closed five-manifolds, which is the cobordism invariance of the class. The geometric viewpoint opened by the WZW term and the classification just stated are one and the same.

Physical Insight

The inflow picture explains why anomalies are *exact*—why a one-loop coefficient is never renormalized and never receives higher-loop or nonperturbative corrections (the Adler–Bardeen theorem, in this language, is the statement that the bulk class is an integer and integers do not flow). A continuous quantity could be screened by dynamics; a discrete topological invariant of a $(d{+}1)$-dimensional phase cannot. This is the deep reason the anomaly is the ideal infrared/ultraviolet probe: it is the one piece of data about a strongly coupled theory that survives, unaltered, from the free quarks of the ultraviolet to the pions of the infrared, and it is what forced chiral symmetry breaking in Section 37.5. The Standard Model's perturbative cancellations make its bulk trivial; QCD's nonzero global anomalies make its bulk nontrivial and pin down its infrared realization. Same principle, opposite verdicts.

What is, and is not, settled (mid-2020s). Honesty requires marking the edges of this picture. *Settled and standard:* the descent classification of perturbative anomalies; the index-theorem evaluation of their coefficients; the cobordism classification of anomalies for the standard symmetry structures (spin, pin, ordinary global and gauge symmetries) in low dimensions, including the four-dimensional cases relevant to the Standard Model; the inflow interpretation and the resulting derivation of 't Hooft matching. The Standard Model has

been verified anomaly-free in this complete framework, perturbative and global together, with no residual obstruction. *Active and not fully settled, as of the mid-2020s:* the complete cobordism classification for arbitrary symmetry structures and dimensions (computed case by case, with the higher-dimensional and exotic-structure groups still being worked out); the anomalies of *non-invertible* and categorical symmetries, a subject opened around 2021–2022 and developing rapidly, where the analogue of the anomaly polynomial is still being formulated; and the role of anomalies as swampland constraints in quantum gravity, where the requirement that all global symmetries be either gauged or broken interacts with anomaly cancellation in ways that are conjectural. The reader should take the four-dimensional results of this chapter as firm and the classification program as a living frontier whose four-dimensional corner is complete but whose general statement is still being written.

The chapter has now traversed the anomaly from a single number—the ABJ coefficient of Chapter 36—to a consistency principle with two faces and a geometric home. When the anomalous current is gauged, cancellation is mandatory, and the Standard Model achieves it generation by generation across the cubic, mixed, and global conditions assembled here; when the symmetry is global, the anomaly is an exact invariant that the ultraviolet and infrared must share, and the Wess–Zumino–Witten term is its low-energy embodiment. These constraints are the input the next part consumes: the effective-field-theory program of Chapter 38 and the chiral Lagrangian of Chapter 39 are built to match exactly the anomalies certified here, with the WZW term entering chiral perturbation theory as the mandatory anomaly-matching operator. The topological ideas previewed in this section—winding numbers, homotopy classes, the bulk that absorbs the inflow—return in earnest in Part XII, where the instantons and θ-vacuum of Chapters 41 and 42 make the same $\pi_n(G)$ and the same index theorem the engine of nonperturbative dynamics.

Further reading

The consistency conditions that organize the gauged case are due to Wess and Zumino [506], with the relation between consistent and covariant anomalies clarified by Bardeen [48] and Bardeen and Zumino [49]; the descent/anomaly-polynomial formalism has a textbook account in Bertlmann [73] and in Zumino, Wu, and Zee [535]. Anomaly matching as a constraint on strong dynamics is 't Hooft's [465]. The Wess–Zumino–Witten term and the topological quantization of its level are Witten's [521]. Mixed and gravitational anomalies, together with the $d = 4k + 2$ counting, are derived by Alvarez-Gaumé and Witten [23]; the nonperturbative global SU(2) anomaly is Witten's [519], controlled by the mod-two index of Atiyah and Singer [41]. Anomaly inflow originates with Callan and Harvey [107], and the modern cobordism viewpoint on anomalies and invertible

phases is developed by Freed and Hopkins [198] and Kapustin and collaborators [290], with the Standard-Model application reviewed by García-Etxebarria and Montero [209]; for the rapidly developing subject of non-invertible symmetries see [429, 440]. Standard textbook treatments of anomaly cancellation appear in Peskin and Schroeder [396], Weinberg [501], and Schwartz [432].

Part XI

Effective Field Theories

Chapter 38

The Effective Field Theory Paradigm

The Wilsonian renormalization group of Chapter 23 delivered a verdict that, on first hearing, sounds like a counsel of despair: as a theory flows to low energy it forgets almost everything about its short-distance origin, the memory of all but finitely many couplings decaying as a positive power of μ/Λ_0. This chapter turns that verdict into a method. Precisely because the infrared cannot resolve the ultraviolet, the ultraviolet cannot be *needed* to compute infrared observables; its entire imprint must be carried by the coefficients of a local Lagrangian built from the light fields alone. Read forward rather than backward, the irrelevant operators that Theorem 23.23 discarded become the systematic, dimension-ordered carriers of short-distance information, and "renormalizable" is demoted from a law of nature to the leading truncation of a tower. The effective-field-theory paradigm is the discipline of computing with that tower: writing the most general symmetry-allowed local Lagrangian, ordering it by an expansion parameter, and extracting controlled, falsifiable predictions at each order.

The chapter leans throughout on machinery built earlier in the book. The Wilsonian classification of operators, the notion of integrating out, and the canonical-dimension bookkeeping are imported wholesale from Chapter 23 (Theorems 23.8, 23.23 and 23.34); the power-counting theorem and the counterterm structure that forces operator completeness come from Chapters 21 and 22 (Theorem 21.10); and the renormalization-group and Callan–Symanzik apparatus that propagates the Wilson coefficients, together with the ϕ^4 beta function and the $\overline{\text{MS}}$ machinery, are those of Chapter 24. The link between Lagrangians and observables runs through the LSZ reduction and the invariant amplitude of Chapters 14 and 16, which underwrite both Weinberg's folk theorem and the field-redefinition invariance of the S-matrix. The two benchmark theories supply the worked spine: ϕ^4 theory (the real scalar of Chapter 6) is reread as the leading truncation of an EFT whose tower we generate by hand and by matching, while QED (the electron and photon of Chapter 8) furnishes the Euler–Heisenberg Lagrangian, the decoupling of a heavy fermion in the running of α (Chapters 19

and 20), and the chiral protection of the electron mass.

The development is a single arc from concept to consequence. Section 38.1 establishes the separation of scales as the empirical bedrock, defines what an EFT *is*, and proves – at the honest folk-theorem level – that the most general local Lagrangian loses no physics. Section 38.2 turns the operator expansion into a quantitative power counting, proves the master suppression law $(E/\Lambda)^{\Delta-4}$, calibrates coefficient sizes by naive dimensional analysis, and prunes the basis of redundant operators using field redefinitions and the equations of motion. Section 38.3 makes "integrating out" literal at tree level, computing Wilson coefficients top-down for a heavy scalar and for the heavy W that hides behind Fermi's constant. Section 38.4 carries matching to one loop, isolates the locality that lets the large logarithm split cleanly into matching and running, and proves the Appelquist–Carazzone decoupling theorem – displaying its failure in $\overline{\text{MS}}$ and the threshold cure, and computing the Euler–Heisenberg coefficients from the electron loop. Section 38.5 derives the matrix renormalization-group equation for the coefficients, exhibits operator mixing and its triangular structure, and resums the leading logarithms. Section 38.6 closes by reading the relevant operators back as the ultraviolet-sensitive directions, stating 't Hooft's naturalness criterion, and confronting the two relevant couplings the Standard Model cannot protect – the Higgs mass and the cosmological constant.

A brief word on scope and rigor. Everything load-bearing is proved or carried in full: the folk theorem without full proof with each link located, the power-counting master formula by two independent routes, the field-redefinition invariance of the S-matrix, the scalar and Fermi matchings to the end, the decoupling theorem and its $\overline{\text{MS}}$ evasion, the Euler–Heisenberg determinant via the proper-time heat kernel, the coefficient RGE and the ϕ^6 anomalous dimension, and the scalar- and fermion-mass naturalness computations side by side. Where the chapter relies on a result it does not prove – the massive spin-one quantization of the W and the electroweak origin of its mass – it imports it by Chapter 31 explicitly. The frontier is stated honestly and dated: the hierarchy and cosmological-constant problems are recorded, as of the mid-2020s, as open.

38.1 Separation of Scales and the Effective Lagrangian

The Wilsonian renormalization group of Chapter 23 taught us to read a quantum field theory by watching it flow. Fix a microscopic action at a high cutoff Λ_0, integrate out the short-distance modes shell by shell (Theorem 23.8), and follow the dimensionless couplings down toward the infrared. The verdict of that flow was sharp and, on first encounter, slightly deflating: nearly everything is forgotten. The forgetting formula Eq. (23.156) states that two microscopic actions agreeing in their finitely many relevant and marginal couplings but differing arbitrarily in the infinite tower of irrelevant ones converge, as the cutoff is lowered, onto one and

the same long-distance trajectory, with the memory of their difference decaying as a positive power $(\mu/\Lambda_0)^{|y_a|}$. The classification theorem Theorem 23.23 organized this: relevant and marginal operators survive the flow and shape the infrared, irrelevant operators are washed out, and "renormalizable" is simply the name for a theory whose surviving content is finite (Theorem 23.34). That was the *forgetting* direction — given the ultraviolet, deduce the infrared.

This chapter inverts the question. Suppose we do *not* know the ultraviolet. Suppose we know only which light degrees of freedom are dynamical at the energies we can reach, and which symmetries govern them. Can we still make quantitative, falsifiable predictions? The Wilsonian answer, turned inside out, is yes — and moreover that the construction is essentially forced. Precisely because the infrared is insensitive to the details above the cutoff, those details cannot be needed to compute infrared observables; their entire imprint must be encodable in the finitely-many-at-each-order coefficients of a local Lagrangian built from the light fields alone. The irrelevant operators that Chapter 23 discarded as harmless are, read forward, the systematic carriers of short-distance information, ordered by how strongly they are suppressed. This is the effective-field-theory (EFT) paradigm: stop apologizing for non-renormalizable operators and start using them as a controlled expansion. The present section establishes the idea and states precisely what an EFT is; the next makes the expansion quantitative.

38.1.1 The insensitivity of the infrared to the ultraviolet

The physical content of the whole subject is one empirical fact, so ubiquitous that it is easy to overlook: *long-wavelength physics does not resolve short-distance structure.* A probe of energy E has spatial resolution of order $1/E$ (in natural units, $\hbar = c = 1$, as throughout). Structure localized on a much smaller scale $1/\Lambda$, with $\Lambda \gg E$, is simply not seen sharply; it can register only through its gross, averaged properties. Three familiar instances, spanning classical and quantum physics, fix the picture before we formalize it.

The multipole expansion. The electrostatic potential of a localized charge distribution of size a, measured at distance $r \gg a$, is $\phi(r) = \frac{1}{4\pi}\big[\frac{Q}{r} + \frac{\mathbf{p}\cdot\hat{\mathbf{r}}}{r^2} + O(a^2/r^3)\big]$. A distant observer never measures the charge density $\rho(\mathbf{x})$ point by point; she measures a few numbers — total charge Q, dipole moment $\mathbf{p}$, and so on — each weighting a term suppressed by an extra power of the small ratio a/r. Infinitely many microscopic distributions share the same leading multipoles and are, at long range, indistinguishable. The multipole moments are the Wilson coefficients of classical electrostatics, and a/r is the expansion parameter.

Atomic physics without nuclei. The hydrogen spectrum is computed to many digits treating the proton as a point of charge $+e$ and mass m_p. The proton is in fact a roiling bound state of quarks and gluons of radius $r_p \sim 0.84\,\mathrm{fm}$, governed by physics (Chapters 25, 27 and 28) that the hydrogen calculation never invokes. That structure is not absent from the spectrum — it appears, in the finite-size

correction to the Lamb shift, as a term proportional to r_p^2, i.e. suppressed by $(r_p/a_0)^2$ relative to the Bohr scale a_0. One number, the charge radius, captures the leading effect of all the strong dynamics inside the proton on atomic energy levels. We do not need QCD to do chemistry; we need one coefficient.

The Fermi theory of weak decays. Before the W boson was known, Fermi described β decay and muon decay by a contact interaction of four fermions, $\mathcal{L}_{\rm F} = -\frac{G_F}{\sqrt{2}}\,(\bar\psi_1\gamma^\mu\psi_2)(\bar\psi_3\gamma_\mu\psi_4)$, with a single dimensionful constant $G_F \approx 1.166 \times 10^{-5}\,\text{GeV}^{-2}$. We now know the underlying process is the exchange of a W of mass $M_W \approx 80\,\text{GeV}$, whose propagator at momentum transfer $q^2 \ll M_W^2$ reduces to $1/(q^2 - M_W^2) \to -1/M_W^2$, a constant. The contact vertex is what W exchange looks like when the probe energy is far below M_W: the heavy line shrinks to a point and its propagator becomes a coupling, $G_F/\sqrt{2} \sim g^2/M_W^2$. Fermi's theory is predictive and accurate at low energy precisely because the corrections it omits are organized by powers of E^2/M_W^2 — and it breaks down, as a non-renormalizable theory must, when E approaches M_W and the point begins to resolve into a propagating line. Fermi wrote down an effective field theory in 1933, two generations before the language existed.

The common structure is unmistakable. In each case a low-energy observable is a power series in a small ratio E/Λ of the accessible scale to a heavy scale, with coefficients that summarize — but do not require us to resolve — the short-distance physics. The claim of this chapter is that this is not a collection of lucky special cases but the generic behavior of *any* quantum field theory viewed below some scale, and that the power series is a Lagrangian.

Physical Insight

The decoupling of scales is what makes physics possible at all. If computing the hydrogen spectrum required first solving QCD, and solving QCD required first knowing the theory at the Planck scale, no quantitative prediction could ever be made, because no experiment reaches arbitrarily high energy. That each layer of nature can be described, to a controlled accuracy, in its own terms — with the layers above entering only through a handful of parameters — is the precondition for the entire enterprise of theoretical physics. Effective field theory is the formalization of this good fortune, and the renormalization group of Chapter 23 is the proof that it is no accident.

38.1.2 From forgetting to constructing

We now convert the insensitivity into a construction. The logic runs in two complementary directions, and the distinction between them organizes everything that follows.

Top-down. A full ultraviolet theory is known — it contains both light fields ϕ and heavy fields Φ of mass $M \sim \Lambda$ — and we want its predictions for processes at $E \ll M$. We *integrate out* the heavy field, performing the path integral over Φ at fixed ϕ (Theorem 23.8 applied to a field rather than a momentum shell):

$$e^{\,iS_{\text{eff}}[\phi]} \;=\; \int \mathcal{D}\Phi \; e^{\,iS[\phi,\Phi]}. \tag{38.1}$$

The result $S_{\text{eff}}[\phi]$ is in general nonlocal — it contains the heavy propagator $1/(q^2 - M^2)$, which is not a polynomial. But for the soft external momenta $q^2 \ll M^2$ that an infrared observer supplies, the propagator may be Taylor expanded, $\frac{1}{q^2-M^2} = -\frac{1}{M^2}\left(1 + \frac{q^2}{M^2} + \dots\right)$, and each power of q^2 becomes, back in position space, two derivatives acting on a local product of light fields. The nonlocality resolves, order by order in $1/M^2$, into an infinite tower of *local* operators. This is precisely the Fermi mechanism of the previous subsection done as an operation on the action; carried out systematically it is the *matching* computation of Section 38.3 (tree level) and Section 38.4 (one loop), which determines the coefficients of the tower in terms of the ultraviolet couplings and masses.

Bottom-up. The ultraviolet theory is unknown — this is the realistic situation for the Standard Model, whose completion above the TeV scale we have not measured. We then cannot compute the coefficients from first principles; but we can still *write down the most general local Lagrangian* compatible with the known light fields and symmetries, organize it by the same E/Λ expansion, and treat each coefficient as a free parameter to be fixed by experiment. The predictive power is real because at any finite order in E/Λ only finitely many operators contribute (a fact proved in Section 38.2): a finite number of measurements determines the coefficients, and every further observable at that order is then a prediction. The bottom-up EFT is, in effect, the most honest parametrization of our ignorance — it commits to nothing about the ultraviolet beyond locality and symmetry, yet remains falsifiable.

The two directions meet in the middle. The bottom-up Lagrangian is the general template; a top-down matching fills in its coefficients whenever a candidate ultraviolet completion is proposed. A measured pattern of coefficients, conversely, points back toward the class of completions that could have produced it. This is why dimension-six operators in the Standard Model EFT (Chapter 40) are the modern currency of the search for new physics: each coefficient is a number that some heavy sector, if it exists, must predict.

38.1.3 What an effective field theory is

We are ready to state precisely what object we are constructing. An EFT is not a Lagrangian alone; it is a Lagrangian together with the data that make its

expansion meaningful — a specification of the dynamical content, the symmetries that constrain it, and the parameter in which it is expanded.

Definition 38.1 (Effective field theory)**.** An *effective field theory* is the triple

$$\mathcal{E} = (\ \{\phi_a\},\ G,\ E/\Lambda\ ), \tag{38.2}$$

consisting of

1. a *field content* $\{\phi_a\}$: the dynamical fields whose quanta are the light degrees of freedom, those with masses small compared to the cutoff Λ and hence capable of propagating in the accessible energy range;
2. a *symmetry group* G (the realized symmetries — spacetime, gauge, and global, including their realization, linear or spontaneously broken) under which the Lagrangian must be invariant; and
3. a *power-counting parameter* E/Λ, the ratio of the characteristic energy E of the process to the cutoff Λ above which new degrees of freedom or new dynamics enter and the description ceases to apply.

The dynamics is the *most general* local Lagrangian built from $\{\phi_a\}$ and invariant under G, with every admissible operator present, its coefficient a free real parameter, and the whole organized as a series in E/Λ. The EFT is defined only for $E < \Lambda$; the cutoff Λ is part of the definition, not an artifact to be removed.

The word "most general" carries the weight of the entire framework, and it is worth saying exactly why it must be there. If we omitted an operator that the symmetries permit, we would be imposing a constraint on the physics — declaring its coefficient to vanish — that nothing in our stated assumptions justifies. Even if we set such a coefficient to zero by hand at one scale, quantum corrections would regenerate it: a divergence proportional to a symmetry-allowed operator will appear in some loop, and renormalizing it requires a counterterm of exactly that form (this is the lesson of locality, Theorem 21.10 and the counterterm structure of Chapter 22, now read in reverse). Completeness is therefore not a stylistic preference but a consistency requirement: the only operators we are entitled to exclude are those forbidden by a symmetry, and once we have listed the symmetries, the operator content is fixed for us.

Writing the operators in order of increasing mass dimension, the effective Lagrangian takes the form that the whole chapter is built around:

$$\mathcal{L}_{\text{eff}} \;=\; \mathcal{L}_{\le 4} \;+\; \sum_i \frac{c_i}{\Lambda^{\,d_i-4}}\,\mathcal{O}_i, \qquad d_i > 4, \tag{38.3}$$

where $\mathcal{L}_{\le 4}$ collects the operators of mass dimension ≤ 4 — the relevant ($d_i < 4$) and marginal ($d_i = 4$) operators, i.e. the kinetic terms, masses, and renormalizable

interactions — and the sum runs over a complete set $\{\mathcal{O}_i\}$ of independent local operators of dimension $d_i > 4$ built from $\{\phi_a\}$ and invariant under G. Each higher-dimension operator is divided by Λ^{d_i-4} so that its coefficient c_i is dimensionless, the carrier of the short-distance information. (That "independent" hides a subtlety — distinct operators can have identical S-matrix elements and so be redundant — which is discharged by field redefinitions in Section 38.2.4. We list the expansion now and prune it there.)

The structure of Eq. (38.3) explains at a glance why the expansion is useful. A dimensionless coefficient times an operator of dimension d_i divided by Λ^{d_i-4} has, on dimensional grounds, an effect on any fixed low-energy amplitude that is suppressed by the small ratio E/Λ raised to the power $d_i - 4$. The higher the dimension, the deeper the suppression, and the expansion truncates with controlled error. We make this estimate precise now, as the central quantitative claim that the next section will elevate into a theorem.

Worked Example 38.2 (The $(E/\Lambda)^{d-4}$ suppression of a higher-dimension operator)**.** Consider a single light real scalar ϕ of mass $m \ll \Lambda$ and an amplitude $\mathcal{M}$ for a fixed process — say $2 \to 2$ scattering, $\phi\phi \to \phi\phi$ — at center-of-mass energy E with $m \ll E \ll \Lambda$, so that E is the only relevant kinematic scale. We estimate how a higher-dimension operator $\mathcal{O}_i$ of dimension d_i contributes to $\mathcal{M}$ relative to the marginal interaction, using nothing but dimensional analysis and the form Eq. (38.3).

Step 1: dimension of the amplitude. In $d = 4$ spacetime dimensions a field has canonical dimension $[\phi] = 1$ (from the kinetic term $(\partial\phi)^2$, since $[\partial] = 1$ and $[\mathcal{L}] = 4$; this counting is fixed in Section 38.2.1). The S-matrix element for n external scalars carries dimension $[\mathcal{M}] = 4-n$ by the LSZ reduction (Chapter 16), where each amputated external leg removes a propagator. For $n = 4$, $[\mathcal{M}] = 0$: the $2 \to 2$ amplitude is dimensionless.

Step 2: the marginal contribution. The marginal interaction is the quartic $-\frac{\lambda}{4!}\phi^4$ of ϕ^4 theory (Chapter 6). It contributes to $\mathcal{M}$ at tree level a constant, $\mathcal{M}_{\le 4} \sim \lambda$, dimensionless as required and independent of E. This is the leading term, the "1" against which everything else is measured.

Step 3: the higher-dimension contribution. Take the leading operator that modifies $\phi\phi \to \phi\phi$ beyond the quartic. The lowest-dimension candidate with four fields and extra derivatives, consistent with the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ and with Lorentz invariance, is

$$\mathcal{O}_8 = \left(\partial_\mu\phi\,\partial^\mu\phi\right)^2, \qquad d_8 = 4\cdot 1 + 4 = 8, \tag{38.4}$$

entering the Lagrangian as $(c_8/\Lambda^4)\,\mathcal{O}_8$. (A dimension-six operator with four fields, ϕ^6 has $d = 6$ but is a potential term contributing to $3 \to 3$, not the four-point amplitude at leading derivative order; the leading *derivative* correction to $\phi\phi \to \phi\phi$ is the dimension-eight Eq. (38.4), which is the natural thread-ϕ^4

object to track.) Its tree-level contribution to $\mathcal{M}$ is obtained by reading off the vertex: four fields, four derivatives, divided by Λ^4. Each derivative brings down a factor of external momentum, of order E, so the four derivatives contribute E^4:

$$\mathcal{M}_8 \;\sim\; \frac{c_8}{\Lambda^4}\,E^4 \;=\; c_8\Big(\frac{E}{\Lambda}\Big)^4. \tag{38.5}$$

This is dimensionless, as it must be, and it is suppressed relative to the marginal term $\mathcal{M}_{\le 4} \sim \lambda$ by exactly $(E/\Lambda)^4 = (E/\Lambda)^{d_8-4}$.

Step 4: the general rule. Nothing in this estimate used the specific operator. An operator of dimension d_i enters $\mathcal{L}_{\text{eff}}$ with coefficient c_i/Λ^{d_i-4}. Its contribution to a fixed amplitude must make up the dimension deficit between the prefactor $\Lambda^{-(d_i-4)}$ and the dimensionless (or fixed-dimension) amplitude using the only available kinematic scale, E. Hence the contribution carries $E^{\,d_i-4}$, and relative to the leading term,

$$\frac{\mathcal{M}_i}{\mathcal{M}_{\le 4}} \;\sim\; c_i\left(\frac{E}{\Lambda}\right)^{d_i-4}. \tag{38.6}$$

Each unit of operator dimension above four costs one power of E/Λ. With $E \ll \Lambda$ and the c_i of order unity (the natural expectation made precise by the naive dimensional analysis of Section 38.2.3), the series Eq. (38.3) is a genuine, rapidly converging expansion, and truncating it at dimension $d_{\max}$ incurs an error of order $(E/\Lambda)^{d_{\max}-3}$ — the first omitted power. That controlled truncation is what makes an EFT predictive.

Theorem 38.2 is the quantitative heart of the chapter in miniature, and it makes the geometry of the operator expansion vivid: the operators of an EFT form a tower, sorted by dimension, in which each rung is suppressed by a further power of E/Λ relative to the one below it. Figure 38.1 draws the picture — light fields and renormalizable dynamics in the accessible window below Λ, the heavy completion above it, and the operator tower interpolating between them in powers of E/Λ.

38.1.4 Weinberg's folk theorem: why the most general Lagrangian is enough

A doubt lurks beneath the construction. We have agreed to write *the most general local Lagrangian* consistent with the symmetries and to compute with it. But the object we ultimately want is not a Lagrangian; it is the S-matrix — the set of physical scattering amplitudes constrained by relativistic invariance, by unitarity (probabilities sum to one), by analyticity (causality, encoded in the analytic structure of amplitudes), and by cluster decomposition (distant experiments are independent). Why should working with the most general Lagrangian deliver the most general S-matrix consistent with those principles, rather than some special subclass? If the Lagrangian framework secretly imposed extra restrictions, the

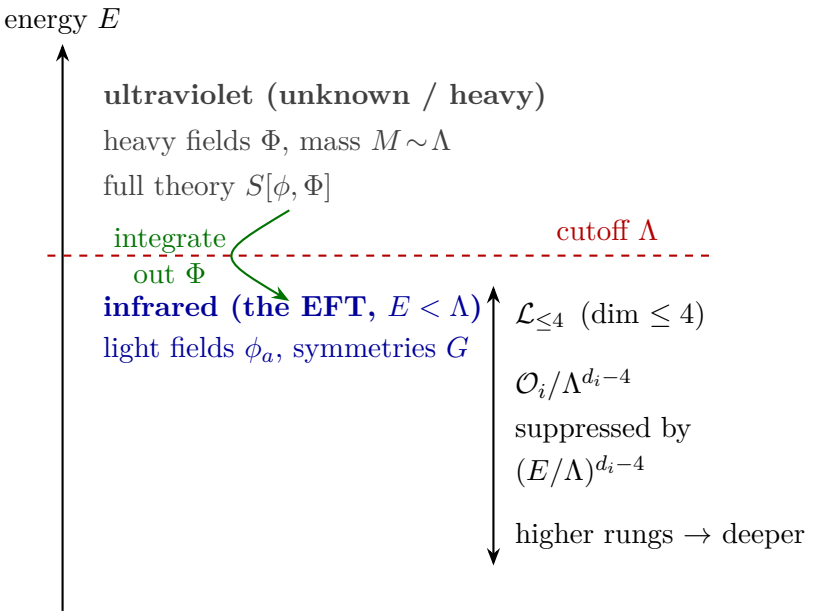


Figure 38.1: The separation of scales. Above the cutoff Λ sit the heavy fields Φ and the (possibly unknown) full theory. Below Λ live the light fields ϕ_a and their symmetries G, governed by the effective Lagrangian Eq. (38.3): a renormalizable core $\mathcal{L}_{\leq 4}$ plus a tower of higher-dimension operators, the rung of dimension d_i suppressed in any fixed amplitude by $(E/\Lambda)^{d_i-4}$ (Theorem 38.2). Integrating out Φ (top-down, Eq. (38.1)) fixes the tower's coefficients; bottom-up, the tower is written down from symmetry alone and measured.

EFT program would be quietly assuming away possibilities, and its claim to capture "whatever the ultraviolet might be" would fail. Weinberg's folk theorem is the assurance that it does not.

Theorem 38.3 (Weinberg's folk theorem)**.** *Let a set of light fields and a symmetry group G be given. Compute the S-matrix perturbatively from the most general local Lagrangian built from those fields, containing every term consistent with G (with arbitrary coefficients), using the most general possible propagators and vertices and the standard rules of perturbation theory. Then the resulting S-matrix is the most general one consistent with relativistic invariance, the assumed symmetries G, and the fundamental principles of* analyticity *(perturbative unitarity and causality) and* cluster decomposition*. Conversely, no S-matrix satisfying those principles lies outside the family so generated.*

The theorem is called a "folk theorem" because it is universally relied upon and its complete proof requires assembling many separate results — the LSZ reduction, the locality of perturbative quantum field theory, the analytic structure of Feynman amplitudes, and the cluster-decomposition theorem that singles out

local field theory among all relativistic quantum theories. We give the argument at the honest level: a statement of the logical chain, with each link either established earlier in this book or made plausible here and deferred to a precise location, and with the role of every hypothesis identified.

Argument. The claim has two halves — that the Lagrangian construction reaches every allowed S-matrix, and that it produces nothing forbidden — and the work is in the first.

(i) Completeness: every allowed amplitude is reachable. An S-matrix is a collection of on-shell amplitudes $\mathcal{M}(p_1, \ldots, p_n)$, one for each particle content and number of legs, that are by hypothesis Lorentz-covariant, G-covariant, and analytic functions of the Lorentz invariants $p_i \cdot p_j$ with only the singularities — poles at physical masses, branch cuts above multiparticle thresholds — demanded by unitarity and causality. The LSZ reduction formula (Chapter 16, in the form Theorem 16.14 that connects amplitudes to observables) expresses each such amplitude as the residue of an appropriately amputated, on-shell time-ordered correlation function of local fields. Run this backward. The non-singular part of $\mathcal{M}$ — the piece analytic at the kinematic point in question, i.e. its Taylor expansion in the invariants around any regular point — is a power series in the momenta. A local operator built from the fields and their derivatives produces, as a tree-level vertex, exactly a polynomial in momenta; with enough derivatives and enough fields one can match any prescribed polynomial term by term. Including *every* symmetry-allowed local operator with a free coefficient therefore makes available, as an adjustable tree-level vertex, every polynomial in momenta consistent with Lorentz and G covariance — which is to say, the most general analytic (pole- and cut-free) part of every amplitude. The free coefficients of the Lagrangian are in one-to-one correspondence with the coefficients of this Taylor expansion. This is the crux: *completeness of the operator basis is what makes the construction able to reproduce an arbitrary regular amplitude.* Omit one allowed operator and one Taylor coefficient becomes un-tunable; include them all and the regular part is free.

(ii) The non-analytic part is fixed, not free — and correctly. The poles and cuts of $\mathcal{M}$ are not ours to choose: their positions are the masses of the light states (poles) and their multiparticle thresholds (cuts), and their residues and discontinuities are fixed, through unitarity, by lower-point amplitudes. Here the two remaining principles do their work. *Unitarity* is implemented perturbatively by the loop expansion: the optical-theorem discontinuity of a loop amplitude is sewn from on-shell lower-order amplitudes, so the cuts generated by computing with our Lagrangian are exactly the cuts unitarity demands, with the correct discontinuities. *Cluster decomposition* — the statement that widely separated subprocesses factorize, so that connected amplitudes have no disconnected singularities — is, by the theorem of Chapter 16's framework, equivalent to the requirement that the interactions be built from *local* products of fields, which is

exactly the form of every term in Eq. (38.3). Locality thus guarantees the correct singularity structure automatically; it is the reason an S-matrix from a local Lagrangian satisfies cluster decomposition, and the reason (run forward) that a cluster-decomposing S-matrix can be written this way. The analytic structure required by causality — amplitudes analytic in the upper-half energy plane, with the $i\epsilon$ prescription of the Feynman propagator — is built into the perturbative rules from the start.

(iii) Soundness: nothing forbidden is produced. Every amplitude computed from a Lorentz- and G-invariant local Lagrangian by the standard rules is, by construction, Lorentz- and G-covariant, has only the singularities unitarity and causality permit, and clusters. So the family generated lies entirely *within* the set of principle-consistent S-matrices. Combined with (i)–(ii), which show the family also *exhausts* that set, the two coincide.

Redundancy and the role of field redefinitions. One gap remains in the correspondence "free coefficients $\leftrightarrow$ amplitude data": different Lagrangians can give the same S-matrix. A field redefinition $\phi \to \phi + f(\phi)$ leaves the S-matrix invariant (it changes off-shell Green's functions but not on-shell amplitudes, because it amounts to a change of integration variable in the path integral whose Jacobian and contact terms do not touch the LSZ residues). Such redefinitions, equivalent to using the classical equations of motion to eliminate operators, mean the operator basis as written in Eq. (38.3) is overcomplete: some operators are physically redundant. This does not threaten the theorem — redundant operators add no new S-matrix freedom — but it must be quotiented out to obtain a genuinely independent counting. The systematic removal is carried out in Section 38.2.4; here we have only flagged that the "most general Lagrangian" is to be understood modulo field redefinitions. □

The folk theorem is the license for the entire bottom-up program. It says that by writing the most general symmetric local Lagrangian we lose nothing: any S-matrix that respects the assumed symmetries and the bedrock principles of relativistic quantum mechanics can be reproduced, and conversely we generate no inconsistent amplitude. An EFT is therefore not a model among many; it is the general solution to the problem "parametrize all possible low-energy physics with these fields and these symmetries."

Historical Note

Weinberg articulated the principle in 1979, framing it as a working hypothesis: if one writes down the most general Lagrangian consistent with the assumed symmetries, and calculates with it to a given order, the result will be the most general S-matrix consistent with analyticity, unitarity, cluster decomposition, and those symmetries. He used it to organize the low-energy theory of pions (Chapter 39). The status of the statement —

physically compelling, leaning on a chain of standard but separately-proved results, never reduced to a single self-contained theorem — is exactly why it carries the affectionate name "folk theorem." Its conceptual ancestry runs back through Wilson's renormalization group (Chapter 23) and, further still, to Fermi's 1933 four-fermion theory, an effective Lagrangian written down decades before the framework that would explain why it worked.

38.1.5 Renormalizability reconsidered

The folk theorem forces a reappraisal of a principle this book has so far treated as close to sacred. Through Parts IV–VI we built quantum field theory under the banner of renormalizability: a theory was acceptable if its divergences could be absorbed into finitely many parameters, which by the power-counting theorem Theorem 21.10 meant a Lagrangian containing only operators of dimension ≤ 4. QED and the Standard Model are renormalizable in exactly this sense, and that property let us extract finite predictions to all orders. It is natural to read renormalizability as a fundamental requirement that nature's Lagrangians must satisfy.

The EFT viewpoint demotes it. In the expansion Eq. (38.3), the renormalizable part $\mathcal{L}_{\leq 4}$ is nothing more special than the *leading truncation* of the operator tower — the dimension-≤ 4 operators, which by Theorem 38.2 are exactly the terms unsuppressed (or *enhanced*) by powers of E/Λ. The higher-dimension operators are always there; they are simply suppressed by $(E/\Lambda)^{d_i-4}$ and so invisible to any experiment whose precision is coarser than that power. A renormalizable theory is the EFT one obtains by keeping only the relevant and marginal operators and dropping the irrelevant tail — which is to say, the description that survives when the cutoff is taken so far above the probe energy that every suppressed term is negligible. Renormalizability, in other words, is not a law imposed from above; it is the generic *low-energy* behavior of *any* local theory, the statement that at energies far below the scale of new physics only the relevant and marginal operators are measurable. The predictivity of QED and the Standard Model is, on this reading, a low-energy accident — the dominance of the leading truncation — and not a fundamental input. This is precisely the inversion the insight box of Chapter 23 (read against Theorem 23.34) anticipated: the same arithmetic that condemned non-renormalizable operators in the ultraviolet rehabilitates them in the infrared as the controlled carriers of short-distance physics.

This reinterpretation reframes every theory met so far in this book, and in particular the two benchmark theories. ϕ^4 theory, the real scalar $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ of Chapter 6, is to be reread as the dimension-≤ 4 truncation $\mathcal{L}_{\leq 4}$ of an EFT whose tower begins with the dimension-six operator ϕ^6/Λ^2

and the dimension-eight $(\partial_\mu\phi\,\partial^\mu\phi)^2/\Lambda^4$ of Eq. (38.4). The quartic coupling λ, which we have treated as a fundamental parameter since Chapter 6, is itself a Wilson coefficient — as Section 38.3 will make literal by integrating out a heavy scalar and watching $\lambda\phi^4/4!$ emerge as the leading matched operator, with its leading irrelevant companion generated alongside it. QED, the electron coupled to the photon (Chapter 8), is likewise the renormalizable core of an EFT; below the electron mass, integrating the electron out leaves a pure-photon effective theory whose leading operators are the dimension-eight $(F_{\mu\nu}F^{\mu\nu})^2$ and $(F_{\mu\nu}\tilde{F}^{\mu\nu})^2$ of Euler and Heisenberg — the EFT of light-by-light scattering — which Section 38.4.5 computes, and whose threshold will reappear in the running of α through the decoupling theorem of Section 38.4.2. Neither $\lambda\phi^4$ nor QED was ever "the theory"; each was the leading truncation of an EFT whose higher-dimension tail this chapter now learns to compute.

Caution

"Demoted" does not mean "dispensable." Renormalizability remains the property that lets a theory be extrapolated to arbitrarily high energy with finitely many parameters — the existence of a continuum limit (Theorem 23.34). What the EFT viewpoint denies is only that this property is *required* of a physical Lagrangian. A non-renormalizable EFT is perfectly predictive within its domain $E < \Lambda$; it simply does not extend, on its own, to $E \gtrsim \Lambda$, where it must be replaced by (matched onto) a more complete description. The relevant operators, far from being dispensable, are exactly the ones whose unnatural smallness will return in Section 38.6 as the hierarchy and cosmological-constant problems — the price of the same scale-sensitivity that makes the marginal operators dominate.

We have now fixed the concept. An EFT is the triple of Theorem 38.1; its dynamics is the complete symmetry-allowed local Lagrangian Eq. (38.3); the folk theorem Theorem 38.3 guarantees this loses no physics; and renormalizability is reread as its leading truncation. What remains qualitative is the bookkeeping of the expansion: we have argued by dimensional analysis that a dimension-d operator contributes $(E/\Lambda)^{d-4}$ to a fixed amplitude (Theorem 38.2), but we have neither fixed the canonical dimensions of every field once and for all, nor estimated the natural size of the coefficients c_i, nor proven the suppression as a theorem about amplitudes, nor pruned the redundant operators from the basis. The next section, Section 38.2, supplies exactly this: it turns the operator expansion into a quantitative power counting, complete with the factors of 4π that naive dimensional analysis tracks, and removes the redundancy of the basis using the field redefinitions and equations of motion flagged in the proof of the folk theorem.

38.2 Power Counting and the Operator Expansion

The previous section made a promise that was, as stated, only qualitative. We are to write "the most general local Lagrangian built from the light fields and consistent with the symmetries," organize it "as an expansion in operator dimension," and trust that the higher terms are "small." Each of those phrases needs a number behind it before the program can be used to compute. *Most general* must become a finite, non-redundant list at every order. *Organized by dimension* must become a precise statement about how large each coefficient is and how much each operator contributes to a measured amplitude. And *small* must become an inequality with an explicit suppression factor, so that truncating the tower at a given order incurs a controlled and estimable error. This section supplies all three. We fix the canonical dimensions, write the effective Lagrangian with its scale dependence made explicit, prove the master power-counting result that a dimension-Δ operator contributes to a low-energy amplitude with the suppression $(E/\Lambda)^{\Delta-4}$, estimate the natural size of each coefficient by naive dimensional analysis, and finally clear the basis of redundancy using field redefinitions and the equations of motion. The result is a clean, finite, physically calibrated operator basis — the object the rest of the chapter learns to populate with actual numbers.

38.2.1 Canonical dimensions and the operator tower

The organizing quantity is the engineering (mass) dimension, the same $[\,\cdot\,]$ that governed the Wilsonian classification of Chapter 23 and the power counting of Chapter 21. In natural units the action $S = \int \mathrm{d}^4x\, \mathcal{L}$ is dimensionless, so in $d = 4$ the Lagrangian density carries $[\mathcal{L}] = 4$. Each spacetime derivative contributes one unit, $[\partial_\mu] = 1$, and the kinetic terms then fix the dimensions of the fields: a scalar kinetic term $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$ forces $[\phi] = 1$, a Dirac kinetic term $\bar\psi\, i\not{\partial}\psi$ forces $[\psi] = \frac{3}{2}$, and a gauge kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ forces $[A_\mu] = 1$. These are exactly the canonical dimensions recorded in Eq. (23.142); we import them rather than re-derive them. A local operator $\mathcal{O}_i$ assembled from n_f factors of a field of dimension d_f and δ_i derivatives has dimension

$$\Delta_i \;=\; \delta_i + \sum_f n_f\, d_f, \tag{38.7}$$

and by Theorem 21.9 the coupling that multiplies it in the Lagrangian must make up the difference to 4,

$$[g_i] \;=\; 4 - \Delta_i. \tag{38.8}$$

This is the single fact on which the whole expansion turns. It is the same statement that Eq. (23.182) used to classify perturbations as relevant, marginal, or irrelevant; here we read it in the constructive direction.

The point of EFT is to make the scale dependence carried by Eq. (38.8) manifest. We are building a theory valid below a heavy scale Λ, and Λ is the natural unit in which to measure every dimensionful coupling. We therefore strip each coupling of its dimension by pulling out the appropriate power of Λ, leaving a dimensionless *Wilson coefficient*.

Definition 38.4 (Effective Lagrangian and Wilson coefficients)**.** Let $\{\mathcal{O}_i\}$ be a basis of local operators built from the light fields and invariant under the assumed symmetries, with $\mathcal{O}_i$ of canonical dimension Δ_i. The *effective Lagrangian* is the sum over the basis

$$\mathcal{L}_{\text{eff}} \;=\; \sum_i \frac{c_i}{\Lambda^{\Delta_i - 4}}\,\mathcal{O}_i, \tag{38.9}$$

where each *Wilson coefficient* c_i is dimensionless. The operators with $\Delta_i \le 4$ (relevant and marginal — the "renormalizable" ones in the modern sense of Theorem 23.34) carry non-negative powers of Λ; the operators with $\Delta_i > 4$ (irrelevant) carry strictly negative powers $\Lambda^{-(\Delta_i-4)}$ and form the *operator tower*, ordered by increasing dimension.

The renormalizable part of Eq. (38.9) is what every theory in this book so far has been: the dimension-≤ 4 operators are precisely the ones whose couplings do not come pre-suppressed by Λ. The tower is the new content. Its members were invisible in the strictly renormalizable framework not because they are forbidden but because their couplings are tiny — a fact we are about to quantify. Two clarifications fix the meaning of Eq. (38.9). First, the single scale Λ is a bookkeeping device: when several heavy thresholds exist one writes a Λ for each, but the structure is unchanged. Second, the split into "relevant/marginal" and "irrelevant" is the dimensional shadow of the dynamical trichotomy proved in Theorem 23.23; we are now reading that classification as a recipe for what to write down, in the order in which it matters.

38.2.2 The power-counting master formula

We now prove the quantitative core of the chapter: how much a given operator in the tower contributes to a physical amplitude measured at energy $E \ll \Lambda$. The answer is a pure power of the ratio E/Λ, fixed entirely by the operator's dimension.

We need first the mass dimension of an amplitude. With the relativistically normalized states of Chapter 14, a one-particle state $|\mathbf{p}\rangle$ satisfies $\langle \mathbf{p}|\mathbf{q}\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^3(\mathbf{p}-\mathbf{q})$, whose right-hand side has mass dimension $1-3=-2$; hence $[\,|\mathbf{p}\rangle\,] = -1$. From the definition of the invariant amplitude in Eq. (16.2), $\langle f|iT|i\rangle = (2\pi)^4\delta^4(\sum k - \sum p)\, i\mathcal{M}$, the left-hand side carries the dimension of N external states, namely $-N$ for a process with N external legs in total, while the delta function carries -4. Matching,

$$[\mathcal{M}] \;=\; 4 - N. \tag{38.10}$$

This holds for any species, since the relativistic normalization assigns every single-particle state dimension -1 regardless of spin.

Consider now a connected contribution to such an amplitude that uses V_i insertions of the interaction vertex coming from operator $\mathcal{O}_i$, with dimensionful coupling $g_i = c_i\,\Lambda^{4-\Delta_i}$. Suppose all light masses are negligible compared with the common external energy E, and that $E \ll \Lambda$, so the heavy scale enters only through the couplings. Suppose further that the loop integrals are dominated by internal momenta of order E — the generic situation for an infrared-safe low-energy observable, the exceptions being the ultraviolet-sensitive pieces that are by construction absorbed into the Wilson coefficients (the subject of the matching sections to come). Under these hypotheses the amplitude can depend on only two dimensionful quantities, the couplings g_i and the energy E. Its dimension Eq. (38.10) is then saturated in a unique way:

$$\mathcal{M} \;=\; \kappa \left(\prod_i g_i^{V_i} \right) E^{\,p}, \qquad p \;=\; 4 - N - \sum_i V_i\,[g_i] \;=\; 4 - N + \sum_i V_i(\Delta_i - 4), \tag{38.11}$$

where κ is a dimensionless number depending only on ratios (angles, light dimensionless couplings). Substituting $g_i = c_i \Lambda^{4-\Delta_i}$ and regrouping the powers of E and Λ,

$$\boxed{\mathcal{M} \;=\; \kappa\, E^{\,4-N} \prod_i \Big[\, c_i \Big(\frac{E}{\Lambda}\Big)^{\Delta_i - 4} \Big]^{V_i}} \tag{38.12}$$

Equation Eq. (38.12) is the power-counting master formula, and it should be read one factor at a time. The prefactor $E^{\,4-N}$ is the dimensional baseline that any process of N external legs must carry. Every insertion of an operator of dimension Δ_i then multiplies the amplitude by the dimensionless factor $c_i\,(E/\Lambda)^{\Delta_i - 4}$. For a marginal operator, $\Delta_i = 4$, the factor is just c_i with no energy dependence: these are the interactions that survive to arbitrarily low energy, the renormalizable couplings. For an irrelevant operator, $\Delta_i > 4$, the factor carries a strictly positive power of the small ratio E/Λ and is therefore suppressed; a relevant operator, $\Delta_i < 4$, is conversely enhanced as $E \to 0$. The central consequence, isolating a single insertion against the renormalizable baseline of the same process, is the suppression law

$$\frac{\mathcal{M}^{(\Delta)}}{\mathcal{M}^{(\mathrm{ren})}} \;\sim\; c_\Delta \Big(\frac{E}{\Lambda}\Big)^{\Delta - 4}, \tag{38.13}$$

which is the precise form of the qualitative "smallness" invoked in the previous section. A dimension-six operator contributes at relative order $(E/\Lambda)^2$, a dimension-eight operator at order $(E/\Lambda)^4$, and so on: the tower is an expansion in $(E/\Lambda)^2$ for operators that step up by two units of dimension at a time, and truncating it after a given dimension leaves an error of the next power of E/Λ. This is what licenses the whole program: at any fixed accuracy, only finitely many operators contribute.

It is worth deriving Eq. (38.12) a second way, tracking the diagram piece by piece, because the bookkeeping exposes that this is the same arithmetic as the ultraviolet power counting of Theorem 21.10, read at the infrared end. Take a contribution with L loops, I internal scalar lines, and V_i vertices of type i, the vertex from $\mathcal{O}_i$ carrying coupling g_i and δ_i derivatives. With all momenta of order E: each loop measure $\int \mathrm{d}^4k$ scales as E^4; each propagator $i/(k^2 - m^2)$ scales as E^{-2}; each vertex contributes g_i times E^{δ_i} from its derivatives. The total power of E is

$$p \;=\; 4L - 2I + \sum_i V_i\,\delta_i. \tag{38.14}$$

Eliminate the internal quantities with the two topological identities used in Theorem 21.10: the loop count $L = I - V + 1$ (with $V = \sum_i V_i$) and the leg count $\sum_i V_i n_i = 2I + N$ (each vertex emits n_i ends, internal lines absorb two, external legs one). Substituting,

$$\begin{aligned} p &= 4(I - V + 1) - 2I + \sum_i V_i\delta_i = 2I - 4V + 4 + \sum_i V_i\delta_i \\ &= \Big(\sum_i V_i n_i - N\Big) - 4\sum_i V_i + 4 + \sum_i V_i\delta_i = 4 - N + \sum_i V_i\,(n_i + \delta_i - 4). \end{aligned} \tag{38.15}$$

Since $n_i + \delta_i = \Delta_i$ for a scalar operator (each field dimension 1, each derivative dimension 1), the energy power is $p = 4 - N + \sum_i V_i(\Delta_i - 4)$, identical to Eq. (38.11); combined with $\prod_i g_i^{V_i} = \prod_i c_i^{V_i}\Lambda^{V_i(4-\Delta_i)}$ it reproduces Eq. (38.12) exactly. The calculation is the mirror image of the divergence counting: there the engine of growth was the loop measure pulling the integral up toward Λ_{UV}; here the same measure, evaluated at the physical scale E, sets the size of the answer. Loops do not change the leading $(E/\Lambda)^{\Delta-4}$ assignment of an operator — that is fixed by its dimension alone — but they do reorganize the dimensionless number κ, which is where the loop expansion lives. We count loops *separately* from the operator expansion: a one-loop dressing of a tree contribution multiplies κ by the usual loop factor (computed next) without altering the power of E/Λ. The two expansions are orthogonal axes of the EFT, and confusing them is the most common error in estimating the size of an effect.

Physical Insight

The master formula explains, in one line, why renormalizable theories were ever a good description of nature. Nothing forbids the tower; the operators are there. They are simply weighted by powers of E/Λ, and at energies far below the next heavy threshold those powers are minuscule. "Renormalizable" is not a fundamental law but the statement that, at the energies we have probed, only the $\Delta \le 4$ operators have had a measurable coefficient. The irrelevant operators of Theorem 23.23 are not absent from

QED or the Standard Model; they are suppressed, and the suppression is the experimental window onto Λ.

38.2.3 Naive dimensional analysis and the role of 4π

Power counting fixes how an operator's contribution scales with energy, but it leaves the dimensionless Wilson coefficient c_i undetermined. To estimate c_i we need a guess for the strength of the dynamics that produced it, together with an honest accounting of the loop factors that any quantum field theory in four dimensions generates. The estimate that results is called naive dimensional analysis (NDA), and its non-trivial ingredient is the counting of factors of 4π.

The factors of 4π come from the loop measure. Consider the Euclidean angular integral in four dimensions: $\int \mathrm{d}^4k = \Omega_3 \int_0^\infty k^3\,\mathrm{d}k$, where $\Omega_3 = 2\pi^2$ is the surface area of the unit three-sphere. Hence a generic one-loop integral carries

$$\int \frac{\mathrm{d}^4k}{(2\pi)^4}\, f(k^2) \;=\; \frac{2\pi^2}{(2\pi)^4}\int_0^\infty k^3\,\mathrm{d}k\; f(k^2) \;=\; \frac{1}{16\pi^2}\int_0^\infty \mathrm{d}(k^2)\, k^2\, f(k^2), \tag{38.16}$$

so that *each loop comes with an explicit factor* $1/16\pi^2 = 1/(4\pi)^2$. This is the same $16\pi^2$ that sits in the denominator of every beta function in the book — the $\beta_\lambda = 3\lambda^2/16\pi^2$ of ϕ^4 theory (Chapter 24) and the $\beta_e = e^3/12\pi^2$ of QED among them. It is not a convention; it is the residue of the four-dimensional phase space, and it means that a loop is suppressed not merely by a coupling but by a coupling divided by $16\pi^2 \approx 158$. A perturbative expansion in $\alpha/4\pi$ converges far faster than one in α precisely because of this factor.

NDA elevates Eq. (38.16) into a self-consistent assignment of sizes. The idea is to ask: if the unknown ultraviolet physics is as strong as it can be without losing all predictivity — maximally strongly coupled at the scale Λ — how big are the operator coefficients it generates? The answer must be *self-consistent* under the theory's own loops: an estimate is only sensible if computing a loop correction to an operator reproduces the same operator with a coefficient no larger than the one we started with. Otherwise the loop expansion would overturn the estimate. This single requirement fixes the powers of 4π.

Write the effective Lagrangian for a single real scalar in the form

$$\mathcal{L}_{\mathrm{NDA}} \;=\; \frac{\Lambda^4}{16\pi^2}\sum_{n,p} \hat{c}_{n,p}\left(\frac{\partial}{\Lambda}\right)^{2p}\left(\frac{4\pi\,\phi}{\Lambda}\right)^{n}, \tag{38.17}$$

with $\hat{c}_{n,p}$ pure numbers of order unity and the operator carrying n powers of ϕ and $2p$ derivatives, hence dimension $\Delta = n + 2p$. Reading off the dimensionful coefficient of $\phi^n(\partial)^{2p}$,

$$g_{n,p} \;=\; \frac{\Lambda^4}{16\pi^2}\cdot\frac{(4\pi)^n}{\Lambda^{\,n+2p}}\,\hat{c}_{n,p} \;=\; (4\pi)^{\,n-2}\,\Lambda^{\,4-n-2p}\,\hat{c}_{n,p}, \tag{38.18}$$

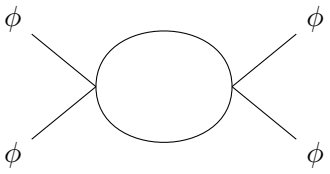


Figure 38.2: The one-loop bubble that fixes the 4π counting. Two ϕ^4 vertices (coupling λ each) and one loop generate a correction to the four-point function of size $\lambda^2/16\pi^2$, Eq. (38.19). Self-consistency of this loop with the tree coupling places the natural strong-coupling value at $\lambda \sim 16\pi^2$, the NDA estimate of Eq. (38.18).

so that the dimensionless Wilson coefficient of Eq. (38.9) is $c_{n,p} = g_{n,p}\,\Lambda^{\Delta-4} = (4\pi)^{n-2}\hat{c}_{n,p}$, growing by one power of 4π for every additional pair of fields. The assignment Eq. (38.17) passes the self-consistency test by construction, but it is instructive to verify it on the marginal quartic of ϕ^4 theory. Setting $n = 4$, $p = 0$ gives $g_{4,0} = (4\pi)^2\hat{c} = 16\pi^2\hat{c}$, i.e. the maximal natural quartic coupling is $\lambda \sim 16\pi^2$. This is exactly the strong- coupling ceiling: a one-loop bubble built from two ϕ^4 vertices (Fig. 38.2) generates a correction to the same quartic of size, from Eq. (38.16),

$$\delta\lambda \;\sim\; \frac{\lambda^2}{16\pi^2}, \tag{38.19}$$

which is the structure of β_λ itself. Demanding that the loop correction not exceed the tree coupling, $\delta\lambda \lesssim \lambda$, gives $\lambda \lesssim 16\pi^2$ — precisely the value Eq. (38.18) delivers. The 4π counting is loop counting; NDA is the statement that the coefficients sit at the threshold where tree and loop are comparable, which is the largest they can naturally be.

NDA as written assumes the worst case, a maximally strongly coupled completion. If instead the heavy physics is weakly coupled, with a characteristic coupling $g_* < 4\pi$, the same reasoning with $4\pi \to g_*$ and overall factor Λ^4/g_*^2 gives $c_{n,p} \sim g_*^{n-2}\hat{c}_{n,p}$, interpolating down to the perturbative regime. The universal content is the upper bound: a Wilson coefficient generated by physics at scale Λ is at most $(4\pi)^{n-2}$ in the conventions of Eq. (38.17), and a coefficient found to be far below its NDA value is a clue that the responsible interaction is weak or that a symmetry suppresses it. NDA does not tell us the coefficient — only matching to a specific ultraviolet theory does that, which is the business of the next section — but it tells us the size to expect, and that is enough to estimate the reach of an experiment in Λ before any model is chosen.

Caution

The factor $(4\pi)^{n-2}$ counts fields, not the operator dimension. Two operators of the same dimension can have different natural sizes if they contain different numbers of fields: in Eq. (38.17), ϕ^6 ($n = 6$) and $\phi^2(\partial\phi)^2$ ($n = 4$) are both dimension six, but NDA assigns them $(4\pi)^4$ and $(4\pi)^2$ respectively. The dimension sets the power of E/Λ; the field count sets the power of 4π. Keep the two axes separate.

38.2.4 Operator redundancy: field redefinitions and the equations of motion

The basis $\{\mathcal{O}_i\}$ of Theorem 38.4 was assumed to be a genuine basis — no operator a linear combination of the others. But "linear combination" is subtler than it looks. Two manifestly different Lagrangians can produce *identical* S-matrix elements, in which case the operator distinguishing them is physically vacuous and must be struck from the basis or we will overcount the independent coefficients. The redundancy has a sharp origin: the S-matrix is invariant under local redefinitions of the fields, and an operator that vanishes by the lowest-order equations of motion is removable by exactly such a redefinition. We prove the invariance, then deduce the equation-of-motion reduction as its corollary, then carry out the reduction explicitly for the scalar tower.

Theorem 38.5 (Field-redefinition invariance of the S-matrix)**.** *Let ϕ be a field with the free equation of motion mass m, and let*

$$\phi(x) \longmapsto \phi'(x) = \phi(x) + F[\phi(x)] \tag{38.20}$$

be a local redefinition, with F a polynomial in ϕ and its derivatives carrying the same Lorentz and internal quantum numbers as ϕ and beginning at quadratic order, $F = O(\phi^2)$. Then the S-matrix computed from the Lagrangian $\mathcal{L}[\phi']$, re-expressed in terms of ϕ, equals the S-matrix computed from $\mathcal{L}[\phi]$. Equivalently, physical amplitudes are invariant under Eq. (38.20).

Proof. Two facts combine. The first is that Eq. (38.20) is a change of integration variable in the path integral, and so leaves the generating functional unchanged provided the Jacobian is trivial. The Jacobian is $\det(\delta\phi'(x)/\delta\phi(y)) = \det(\mathbf{1} + \delta F/\delta\phi)$, a local operator whose logarithm $\mathrm{Tr}\ln(\mathbf{1} + \delta F/\delta\phi)$ is proportional to $\delta^4(0)$ because F is local (the variation $\delta F(x)/\delta\phi(y)$ is supported at $x = y$). In dimensional regularization $\delta^4(0) = \int \mathrm{d}^4k/(2\pi)^4 = 0$ as a scaleless integral, so the Jacobian is unity and the redefinition is an honest relabeling of the functional integration variable. The correlation functions of ϕ' computed from $\mathcal{L}[\phi']$ are therefore identical, as functions, to the correlation functions of ϕ computed from $\mathcal{L}[\phi]$.

The second fact is that the S-matrix does not see the difference between ϕ and ϕ' as interpolating fields. By the LSZ reduction formula Theorem 14.31 in its residue form Eq. (14.156), an amplitude is the multiple residue of a connected correlator at the external on-shell poles, with one factor $\sqrt{Z}$ per leg divided out, where Z is the residue of that field's full two-point function at $p^2 = m^2$. We must show the multiple residue is the same whether the external legs are ϕ or ϕ'. Consider a single external leg and the Fourier transform of a correlator in that variable. The Källén–Lehmann representation Theorem 14.23 guarantees that the full two-point function of *any* local operator χ that overlaps the one-particle state has a single isolated pole at the physical mass,

$$\int \mathrm{d}^4x\, e^{ip\cdot x}\, \langle \Omega | \mathcal{T}\chi(x)\chi(0)|\Omega\rangle \xrightarrow{p^2\to m^2} \frac{i\,Z_\chi}{p^2 - m^2 + i\epsilon} + (\text{regular}), \qquad Z_\chi = \left|\langle \Omega|\chi(0)|\mathbf{p}\rangle\right|^2, \tag{38.21}$$

the pole location being a property of the spectrum and the residue being the squared overlap Eq. (14.131), independent of which interpolating χ is used. Now $\phi' = \phi + F[\phi]$ is such an operator: it is local, carries the particle's quantum numbers, and has a nonzero one-particle overlap $\langle \Omega|\phi'(0)|\mathbf{p}\rangle = \sqrt{Z} + \langle \Omega|F(0)|\mathbf{p}\rangle \equiv \sqrt{Z'}$ (a composite like ϕ^2 has a nonzero one-particle matrix element in the interacting theory, so F shifts the residue). For each external leg the LSZ operator $(\Box + m^2)$ acting on the Fourier variable annihilates the regular part of Eq. (38.21) on shell and extracts the pole; dividing by $\sqrt{Z'}$ for that leg cancels the leg's overlap. What survives after all N legs are amputated and put on shell is the connected amputated core evaluated between the *same* asymptotic in/out states $|\mathbf{p}_i;\mathrm{in}\rangle$, $\langle \mathbf{q}_j;\mathrm{out}|$ — states defined by the spectrum, not by the choice of interpolating field. By Eq. (14.155) this core *is* the amplitude $\mathcal{M}$, and it is identical for ϕ and ϕ'. The extra terms in $F[\phi]$ contribute only to the regular (non-pole) part in at least one external variable — they describe the operator creating multi-particle states rather than the single particle — and those pieces are killed by the on-shell amputation. Hence $\mathcal{M}$ is unchanged, which is the claim. □

The corollary that does the work is the removability of equation-of-motion operators. Let $\mathcal{L}_0$ be the leading (renormalizable) Lagrangian, with Euler–Lagrange operator

$$E[\phi] \;\equiv\; \frac{\delta S_0}{\delta \phi} \;=\; -(\Box + m^2)\phi - \frac{\lambda}{3!}\phi^3 \tag{38.22}$$

for the ϕ^4 theory of ϕ^4 theory (the classical equation of motion of Chapter 6 is $E[\phi] = 0$). Suppose the tower contains an operator of the special form $\mathcal{O}_{\mathrm{red}} = E[\phi]\, G[\phi]$ for some local G, multiplied by a small coefficient ε. Perform the field redefinition $\phi \to \phi + \varepsilon\, G[\phi]$. To first order in ε the leading action shifts by

$$\delta S_0 \;=\; \int \mathrm{d}^4x\, \frac{\delta S_0}{\delta \phi}\, \varepsilon G[\phi] \;=\; \varepsilon \int \mathrm{d}^4x\, E[\phi]\, G[\phi], \tag{38.23}$$

which is exactly $\varepsilon \int \mathcal{O}_{\text{red}}$ with the opposite sign needed to cancel it. Thus the redefinition removes $\mathcal{O}_{\text{red}}$ from the Lagrangian outright, at the cost of (i) generating terms of *higher* order in ε — one order further down the tower, consistently neglected at the working order — and (ii) acting on the already-present higher-dimension operators, again a higher-order effect. By Theorem 38.5 the redefinition does not touch the S-matrix. Therefore any operator proportional to the leading equation of motion is physically redundant and may be deleted from the basis. We record the principle:

Physical Insight

Reducing an operator "by the equations of motion" is not an on-shell approximation that discards information — it is a field redefinition in disguise, and field redefinitions leave every physical amplitude exactly invariant. One uses the *free* (lowest-order) equation of motion Eq. (38.22) because the operator's own contribution to the true equation of motion is higher order in the EFT expansion. The result is a smaller basis with identical physics, never a loss of accuracy.

Caution

Equation-of-motion reduction is legitimate for computing S-matrix elements, where Theorem 38.5 applies. It is *not* automatically legitimate for off-shell quantities or for Green's functions with operator insertions (e.g. when an operator is itself an external probe, as in deep-inelastic scattering), because there the redefinition changes the off-shell correlator even though it preserves the amplitude. When in doubt, the safe statement is the theorem: the S-matrix is invariant; off-shell objects need not be.

38.2.5 The minimal basis for the real scalar with $\mathbb{Z}_2$ symmetry

We now assemble the operator tower for ϕ^4 theory — a single real scalar with the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ of Theorem 7.10's symmetry catalogue (here a discrete symmetry, but constraining the basis just as a continuous one would) and Lorentz invariance — and reduce it to a minimal, non-redundant basis. This is the ϕ^4-thread layer "organize the irrelevant operators by NDA," continuing the Wilsonian generation of operators in Eq. (23.52).

Worked Example 38.6 (The scalar tower through dimension six)**.** The allowed operators are the Lorentz scalars built from ϕ and ∂_μ that are even under $\phi \to -\phi$. The two symmetries impose two parity-like constraints. The $\mathbb{Z}_2$ symmetry forces an *even* number of fields, n_ϕ even. Lorentz invariance forces all indices to be contracted, hence an *even* number of derivatives, n_∂ even. Since the dimension is $\Delta = n_\phi + n_\partial$ (Eq. (38.7)), the sum of two even integers, *every allowed operator*

has even dimension. The immediate consequence is structural:

$$\text{there is no dimension-five operator for a real scalar with } \mathbb{Z}_2. \tag{38.24}$$

The leading correction to the renormalizable Lagrangian is dimension six, suppressed by $(E/\Lambda)^2$ relative to the marginal interactions by Eq. (38.13) — the EFT expansion proceeds in even powers of E/Λ, as anticipated.

Dimension ≤ 4 (the renormalizable core). The relevant operator is ϕ^2 ($\Delta = 2$, the mass term), the marginal operators are $(\partial_\mu\phi)(\partial^\mu\phi)$ and ϕ^4 ($\Delta = 4$). These reproduce exactly the standard Lagrangian of ϕ^4 theory,

$$\mathcal{L}_0 = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \tag{38.25}$$

Dimension six: enumeration. With n_ϕ, n_∂ even and $n_\phi + n_\partial = 6$, the candidate field/derivative contents are $(n_\phi, n_\partial) \in \{(6,0),(4,2),(2,4)\}$. We list the independent Lorentz scalars in each, reducing first by integration by parts (total derivatives integrate to zero in the action).

- $(6,0)$: the single operator ϕ^6.
- $(4,2)$: the candidates are $\phi^2(\partial_\mu\phi)(\partial^\mu\phi)$ and $\phi^3\Box\phi$. They are not independent: integrating by parts,

  $$\int \mathrm{d}^4x\ \phi^3\Box\phi = -\int \mathrm{d}^4x\ (\partial_\mu\phi^3)(\partial^\mu\phi) = -3\int \mathrm{d}^4x\ \phi^2(\partial_\mu\phi)(\partial^\mu\phi), \tag{38.26}$$

  so there is one independent structure, which we take to be $\mathcal{O}_{(4,2)} = \phi^2(\partial_\mu\phi)(\partial^\mu\phi)$.
- $(2,4)$: the candidates are $(\Box\phi)^2$, $(\partial_\mu\partial_\nu\phi)(\partial^\mu\partial^\nu\phi)$, and $\phi\,\Box^2\phi$. Integration by parts collapses them to one. Twice integrating by parts, $\int(\partial_\mu\partial_\nu\phi)(\partial^\mu\partial^\nu\phi) = \int(\Box\phi)^2$ (the two derivatives move across in pairs, the cross terms being total derivatives), and $\int\phi\,\Box^2\phi = \int(\Box\phi)^2$ likewise. So there is one independent structure, $\mathcal{O}_{(2,4)} = (\Box\phi)^2$.

After integration by parts, then, the dimension-six space is three-dimensional, spanned by $\{\phi^6,\ \phi^2(\partial\phi)^2,\ (\Box\phi)^2\}$.

Dimension six: reduction by the equations of motion. We now use Theorem 38.5. The leading equation of motion Eq. (38.22) reads, in the on-shell/field-redefinition sense (written $\doteq$),

$$\Box\phi \ \doteq\ -m^2\phi - \frac{\lambda}{3!}\phi^3. \tag{38.27}$$

Apply it to the two derivative operators. First the four-derivative structure: substituting Eq. (38.27) into both factors,

$$(\Box\phi)^2 \ \doteq\ \left(m^2\phi + \frac{\lambda}{6}\phi^3\right)^2 = m^4\phi^2 + \frac{\lambda m^2}{3}\phi^4 + \frac{\lambda^2}{36}\phi^6. \tag{38.28}$$

Every term on the right is a *lower*-derivative operator already in the basis: a shift of the mass term, a shift of the quartic, and a contribution to ϕ^6. Thus $(\Box\phi)^2$ carries no independent physics and is removed. Second, the two-derivative four-field structure: combine the integration-by-parts identity Eq. (38.26) with Eq. (38.27),

$$\int \phi^2(\partial_\mu\phi)(\partial^\mu\phi) = -\frac{1}{3}\int \phi^3\Box\phi \;\doteq\; -\frac{1}{3}\int \phi^3\Big(-m^2\phi - \frac{\lambda}{6}\phi^3\Big) = \int\Big(\frac{m^2}{3}\phi^4 + \frac{\lambda}{18}\phi^6\Big). \tag{38.29}$$

So $\phi^2(\partial\phi)^2$ also reduces to a shift of the quartic plus a contribution to ϕ^6. Both derivative operators collapse onto $\{\phi^4, \phi^6\}$ (plus, for $(\Box\phi)^2$, the mass term). The shifts of λ and m^2 are suppressed by the Wilson coefficient's $1/\Lambda^2$ and by the extra factor m^2, exactly the "doubly small" behaviour already noted in the explicit Wilsonian computation of Eq. (23.52): the derivative operator generated there with coefficient $z_4 \sim \lambda^2 m^2/\Lambda^4$ is precisely a redundant operator, removable here in favour of a tiny renormalization of λ.

The minimal basis. The only dimension-six operator with independent S-matrix content is therefore

$$\boxed{\mathcal{O}_6 = \phi^6} \qquad \Longrightarrow \qquad \mathcal{L}_{\text{eff}} = \mathcal{L}_0 + \frac{c_6}{\Lambda^2}\,\phi^6 + O(\Lambda^{-4}), \tag{38.30}$$

with, by the NDA estimate Eq. (38.18) for $n = 6,\ p = 0$, a natural coefficient $c_6 \sim (4\pi)^4$ at strong coupling (or $c_6 \sim g_*^4$ for a weakly coupled completion). This is the leading irrelevant operator of ϕ^4 theory. Its contribution to, say, $3 \to 3$ scalar scattering scales as $(E/\Lambda)^2$ relative to the renormalizable ϕ^4 amplitude, by Eq. (38.13) with $\Delta = 6$.

The reduction in Theorem 38.6 deserves a closing remark on conventions, because it is a place where careful and loose treatments superficially disagree. We used both field redefinitions exhausted by the dimension-three, $\mathbb{Z}_2$-odd functions $G \in \{\phi^3, \Box\phi\}$ to eliminate *both* derivative operators, arriving at the strictly minimal one-operator basis $\{\phi^6\}$. Some references instead retain $\phi^2(\partial\phi)^2$ as a convenient redundant representative — legitimate, since it is physically equivalent to a combination of ϕ^4 and ϕ^6 by Eq. (38.29), and sometimes more transparent when matching to a specific ultraviolet theory that generates it directly. The two choices give the same amplitudes; what one must never do is count both $\phi^2(\partial\phi)^2$ and ϕ^6 *and* a shifted ϕ^4 as independent, which would overcount the coefficients. The lesson generalizes: the number of independent Wilson coefficients at each dimension is basis-independent and physical, but *which* operators carry them is a choice, fixed by how aggressively one applies the equations of motion.

With the basis now finite, non-redundant, and calibrated in size by NDA, exactly one thing remains undetermined: the actual numerical values of the Wilson coefficients c_i. Power counting fixed their scaling, NDA fixed their natural

magnitude, but neither fixes the $O(1)$ number, which encodes the specific short-distance dynamics that has been integrated out. To pin those numbers down we must confront the effective theory with a definite ultraviolet completion and demand that the two agree at low energy — the procedure of matching, to which we turn next, beginning with the cleanest case of a heavy field removed at tree level.

38.3 Integrating Out a Heavy Field: Tree-Level Matching

The previous section handed us a machine with empty dials. Power counting and the removal of redundancy (Sections 38.2 and 38.2.4) produced a non-redundant basis of local operators $\{\mathcal{O}_i\}$ built from the light fields and the symmetries, organized by canonical dimension, each multiplied by a Wilson coefficient c_i whose natural size we estimated by dimensional analysis. What that analysis could not give us is the *values* of the c_i. Naive dimensional analysis fixes their order of magnitude; it does not fix their numbers, their signs, or which of them happen to vanish. Those data encode the short-distance dynamics that the effective theory has agreed to forget, and they can only be supplied by an outside input.

There are two ways to supply it. Bottom-up, one measures enough low-energy observables to read the c_i off, treating them as free parameters of a predictive theory — this is how chiral perturbation theory and the Standard Model effective field theory are used when the ultraviolet completion is unknown. Top-down, one *knows* the short-distance theory: there is a heavy field of mass M whose dynamics, written out in full, reduces at low energy to the effective Lagrangian, and the c_i are then *computable* as functions of the heavy parameters. This section develops the top-down construction. Its central operation is the one introduced abstractly in Eq. (38.1): remove the heavy field from the theory, leaving behind a tower of local operators in the light fields whose coefficients are the Wilson coefficients. We make that operation literal and carry it out completely at tree level, by two routes that must agree — substitution of the heavy field's classical equation of motion, and Gaussian integration over the heavy field in the path integral — and we check the result against the physical scattering amplitude. The criterion that ties all of this together, and that gives "integrating out" its operational meaning, is *matching*.

38.3.1 The matching condition

An effective field theory is the triple of fields, symmetries, and a power-counting expansion of Theorem 38.1, with the most general local Lagrangian that Weinberg's folk theorem (Theorem 38.3) guarantees is enough. To pin its coefficients

to a specific ultraviolet theory we demand that the two theories be *physically indistinguishable at low energy*. Physics is the S-matrix, and the S-matrix is built from Green's functions, so the precise demand is that the Green's functions of the light fields computed in the full theory and in the effective theory coincide in the kinematic regime where both descriptions are valid — external momenta small compared with M — order by order in the expansion in p/M and in the loop expansion.

Definition 38.7 (Matching)**.** Let the full theory have action $S_{\text{full}}[\phi, \Phi]$ for light fields ϕ (masses $\ll M$) and heavy fields Φ (masses $\sim M$), and let the effective theory be

$$S_{\text{eff}}[\phi] = \int \mathrm{d}^4x \, \sum_i c_i \, \mathcal{O}_i(\phi), \tag{38.31}$$

with $\{\mathcal{O}_i\}$ the non-redundant operator basis of Section 38.2 and c_i the Wilson coefficients (Theorem 38.4). The coefficients are *matched* to the full theory by requiring that, for every set of light external legs and at each order in the joint expansion in p/M and in $\hbar$ (loops),

$$G_n^{\text{full}}(p_1, \ldots, p_n)\Big|_{\text{expanded in } p/M} \;=\; G_n^{\text{eff}}(p_1, \ldots, p_n), \tag{38.32}$$

where G_n denotes the connected (equivalently, amputated one-particle-irreducible) n-point function of the light fields evaluated with the same renormalization scheme on both sides. Equivalently, the two theories possess the same light-particle S-matrix up to corrections suppressed by additional powers of E/M.

Three features of Eq. (38.32) deserve emphasis before we compute, because they recur unchanged in every matching calculation in Chapters 39 and 40. First, matching is a statement about a region of overlap. The full theory is valid at all energies; the effective theory is valid only for $E \ll M$. They are required to agree only where the effective theory has any right to speak, and the expansion "$|_{\text{expanded in } p/M}$" on the left is exactly the operation that throws away the information — the heavy poles, the on-shell production of Φ — that the effective theory was never meant to reproduce. Second, the heavy mass M never appears as an explicit scale on the effective-theory side of Eq. (38.32): it has been swallowed into the numerical values of the c_i. This is the defining trade of the EFT — a field for a tower of constants — and it is why the effective theory "does not know" about Φ except through those constants. Third, the equality is required *order by order*. At a fixed order in p/M only finitely many operators contribute (this was the content of the power-counting theorem, Eq. (38.6)), so matching at that order is a finite system of equations for finitely many coefficients, and one ascends the tower one rung at a time.

The phrase "connected (equivalently, amputated 1PI)" in Theorem 38.7 hides a choice that the next definition's worked examples will exploit. One may match

the full *off-shell* effective action — the generating functional of 1PI light-field Green's functions, which is a local functional of ϕ and its derivatives, i.e. a Lagrangian — or one may match only the *on-shell* S-matrix elements. The two are not identical: they differ by operators that vanish when the light field obeys its own equation of motion, which (by the field-redefinition theorem of Section 38.2.4) are precisely the redundant operators that drop out of the S-matrix. The following proposition records that this difference is invisible to physics, so that the cleaner off-shell route and the more physical on-shell route give the same predictions; the scalar computation below verifies it explicitly.

Proposition 38.8 (Matching is well-posed up to redundancy)**.** *The matching condition Eq. (38.32) determines the Wilson coefficients of a non-redundant basis uniquely. Operators that vanish by the light-field equations of motion — equivalently, that can be removed by a local field redefinition (Section 38.2.4) — do not enter the on-shell matching: matching the off-shell 1PI effective action and matching the on-shell S-matrix yield identical physical predictions, the two answers differing only by the coefficients of such redundant operators.*

Proof (deferred to the worked computation). The general statement is the field-redefinition theorem of Section 38.2.4: a redefinition $\phi \to \phi + \delta\phi$ with $\delta\phi$ local shifts the action by $\int (\delta S/\delta\phi)\,\delta\phi$, which is proportional to the classical equation of motion and therefore changes no S-matrix element (it amounts to a change of interpolating field, to which the LSZ formula is insensitive). Choosing $\delta\phi$ to cancel a given EOM-proportional operator removes it from the Lagrangian without touching any observable. We make this concrete in Theorem 38.9, where the off-shell effective action for a heavy scalar contains a derivative operator $\phi^2(\partial\phi)^2$ whose entire on-shell effect is a mass-suppressed constant absorbable into the quartic and mass couplings, exactly as the proposition requires. □

With the criterion fixed, the work is to compute the left-hand side of Eq. (38.32) in a definite full theory, expand, and read off the c_i. We take the simplest nontrivial case and carry it to the end.

38.3.2 Integrating out a heavy scalar: the classical route

We continue ϕ^4 theory. The light field is the real scalar ϕ of the thread, with the carried Lagrangian $\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda_0}{4!}\phi^4$ and the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$. We now give it a heavy partner: a second real scalar Φ of mass $M \gg m$, even under the $\mathbb{Z}_2$, coupled to the light field through the most general renormalizable interaction the symmetry allows. The full Lagrangian is

$$\mathcal{L}_{\text{full}} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda_0}{4!}\phi^4 + \tfrac{1}{2}(\partial_\mu\Phi)(\partial^\mu\Phi) - \tfrac{1}{2}M^2\Phi^2 - \frac{\kappa}{2}\,\Phi\,\phi^2 - \frac{g_3}{3!}\,\Phi^3, \tag{38.33}$$

where the trilinear portal coupling κ has mass dimension $+1$ and the heavy self-coupling g_3 also has dimension $+1$. (A term $\Phi^2\phi^2$ and a tadpole Φ are also $\mathbb{Z}_2$-allowed; we omit the former to keep the heavy quadratic form field-independent — so the path-integral route below is exactly Gaussian — and the latter is removed by a constant shift of Φ. We restore g_3 only when we need the six-point operator; for the quartic and the leading derivative operator it plays no role, so we set $g_3 = 0$ until Section 38.3.4.) The heavy field carries no ϕ-charge and no light index; it is the cleanest possible "thing to integrate out."

To integrate out Φ at tree level means, concretely, to solve its classical equation of motion treating ϕ as an external background, and to substitute the solution back into the action. The Euler–Lagrange equation for Φ obtained from Eq. (38.33) (with $g_3 = 0$ for now) is

$$\frac{\partial \mathcal{L}_{\text{full}}}{\partial \Phi} - \partial_\mu \frac{\partial \mathcal{L}_{\text{full}}}{\partial(\partial_\mu \Phi)} = -M^2\Phi - \frac{\kappa}{2}\phi^2 - \Box\Phi = 0 \quad \Longrightarrow \quad (\Box + M^2)\,\Phi = -\frac{\kappa}{2}\,\phi^2, \tag{38.34}$$

with $\Box = \partial_\mu \partial^\mu$. This is a linear, inhomogeneous equation: the heavy field is sourced by the light bilinear ϕ^2. Its solution is obtained by inverting the operator $(\Box + M^2)$,

$$\Phi_{\text{cl}}[\phi] = -\frac{\kappa}{2}\,(\Box + M^2)^{-1}\phi^2. \tag{38.35}$$

The inverse $(\Box + M^2)^{-1}$ is precisely the heavy field's two-point function — the free propagator of Eq. (10.30), here in the role of the Green's function that turns the source ϕ^2 into the induced field. Because we seek a *local* effective Lagrangian, and because the heavy mass dominates all light momenta, we do not keep this operator intact; we expand it in inverse powers of M^2. In momentum space $\Box \to -p^2$, so $(\Box + M^2)^{-1} \to (M^2 - p^2)^{-1} = M^{-2}(1 - p^2/M^2)^{-1}$, the geometric series that is the whole technical engine of tree-level matching:

$$(\Box + M^2)^{-1} = \frac{1}{M^2}\,\frac{1}{1 + \Box/M^2} = \frac{1}{M^2}\left(1 - \frac{\Box}{M^2} + \frac{\Box^2}{M^4} - \cdots\right). \tag{38.36}$$

Each extra power of $\Box$ is an extra two derivatives acting on the light fields and an extra $1/M^2$; in the language of Section 38.2 each term is one rung higher in the operator tower. The expansion converges as a derivative (momentum) expansion exactly when $p^2 \ll M^2$, the regime of validity of the effective theory — the analytic statement of "low energy."

Now substitute Eq. (38.35) into the action. Because Φ enters Eq. (38.33) (with $g_3 = 0$) at most quadratically, the Φ-dependent part of the action, after integrating the kinetic term by parts, is the quadratic-plus-linear form

$$S_\Phi[\Phi, \phi] = \int \mathrm{d}^4x \left[-\tfrac{1}{2}\,\Phi(\Box + M^2)\Phi - \tfrac{\kappa}{2}\,\phi^2\,\Phi\right] \;\equiv\; \int \mathrm{d}^4x \left[\tfrac{1}{2}\Phi\,\mathcal{K}\,\Phi + J\,\Phi\right], \tag{38.37}$$

with kinetic operator $\mathcal{K} = -(\Box + M^2)$ and source $J = -\frac{\kappa}{2}\phi^2$. Evaluating a quadratic-plus-linear functional at its stationary point $\Phi_{\text{cl}} = -\mathcal{K}^{-1}J$ leaves

exactly half the source term, $S_\Phi[\Phi_{\rm cl}] = \frac{1}{2}\int J\,\Phi_{\rm cl} = -\frac{1}{2}\int J\,\mathcal{K}^{-1}J$; this is the same algebra of completing the square that produced the free generating functional in Theorem 10.13. Spelling it out with $\mathcal{K}^{-1} = -(\Box + M^2)^{-1}$ and $J = -\frac{\kappa}{2}\phi^2$,

$$S_\Phi[\Phi_{\rm cl}] = -\tfrac{1}{2}\int \mathrm{d}^4x\,\Big(-\tfrac{\kappa}{2}\phi^2\Big)\big[-(\Box+M^2)^{-1}\big]\Big(-\tfrac{\kappa}{2}\phi^2\Big) = \frac{\kappa^2}{8}\int \mathrm{d}^4x\;\phi^2\,(\Box+M^2)^{-1}\phi^2. \tag{38.38}$$

The heavy field is gone; in its place is a non-local functional of the light field, which the expansion Eq. (38.36) renders local order by order. The effective Lagrangian (added to the light Lagrangian already present in Eq. (38.33)) is therefore

$$\mathcal{L}^{(\Phi)}_{\rm eff} = \frac{\kappa^2}{8M^2}\,\phi^4 - \frac{\kappa^2}{8M^4}\,\phi^2\Box\phi^2 + \frac{\kappa^2}{8M^6}\,\phi^2\Box^2\phi^2 - \cdots . \tag{38.39}$$

This is the central result of the scalar matching. Read it one term at a time.

The marginal operator: a Wilson coefficient for $\lambda\phi^4$. The leading term shifts the quartic coupling. Combining $\frac{\kappa^2}{8M^2}\phi^4$ from Eq. (38.39) with the tree-level $-\frac{\lambda_0}{4!}\phi^4$ already in Eq. (38.33), and writing the sum in the thread's normalization $-\frac{\lambda_{\rm eff}}{4!}\phi^4$, gives

$$-\frac{\lambda_{\rm eff}}{4!}\,\phi^4 = \Big(-\frac{\lambda_0}{4!} + \frac{\kappa^2}{8M^2}\Big)\phi^4 \quad\Longrightarrow\quad \boxed{\lambda_{\rm eff} = \lambda_0 - \frac{3\kappa^2}{M^2}}. \tag{38.40}$$

The factor 3 is $4!/8$. This is the promised reinterpretation: the quartic coupling of ϕ^4 theory — a marginal coupling we have carried as a free parameter since Chapter 6 — is, in the presence of a heavy partner, a *Wilson coefficient*, computable from the underlying parameters λ_0, κ, M. The shift is negative: a single heavy scalar exchanged between two ϕ^2 currents produces an attractive contact interaction, which lowers $\lambda_{\rm eff}$ below its short-distance value λ_0. The point survives even if the light field had no quartic to begin with: setting $\lambda_0 = 0$, the low-energy theory still has $\lambda_{\rm eff} = -3\kappa^2/M^2$, so the $\lambda\phi^4$ interaction the thread has used throughout can be *entirely* a long-distance shadow of heavy dynamics. Renormalizability of the light sector is not a fundamental input; it is what survives at the bottom of the tower.

The first irrelevant operator: a dimension-six Wilson coefficient. The second term in Eq. (38.39) is a genuine member of the irrelevant tower. It is the dimension-six derivative operator. Integrating by parts twice, $\int \phi^2\Box\phi^2 = -\int \partial_\mu(\phi^2)\partial^\mu(\phi^2) = -4\int \phi^2(\partial_\mu\phi)(\partial^\mu\phi)$, so

$$-\frac{\kappa^2}{8M^4}\,\phi^2\Box\phi^2 = +\frac{\kappa^2}{2M^4}\,\phi^2(\partial_\mu\phi)(\partial^\mu\phi) \;\equiv\; \frac{z_4}{2}\,\phi^2(\partial_\mu\phi)(\partial^\mu\phi), \qquad \boxed{z_4 = \frac{\kappa^2}{M^4}}. \tag{38.41}$$

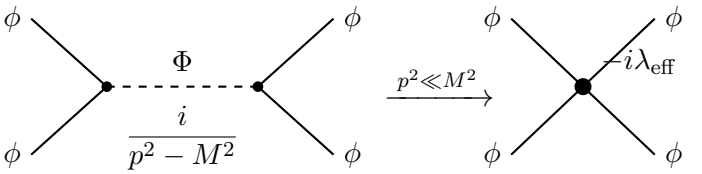


Figure 38.3: Tree-level integrating-out of a heavy scalar. At light external momenta the heavy propagator (dashed line) is local on the scale $1/M$ and the exchange diagram collapses to a contact vertex. The leading contact coupling is the Wilson coefficient $\lambda_{\text{eff}} = \lambda_0 - 3\kappa^2/M^2$ of Eq. (38.40); the subleading terms of the Taylor expansion of the propagator are the dimension-six (and higher) operators of Eq. (38.39).

This is exactly the operator $\frac{z_4}{2}\phi^2(\partial\phi)^2$ whose *loop*-generated coefficient appeared in the Wilsonian flow of Eq. (23.52); here it is generated at *tree level* by integrating out a heavy field, and its coefficient $z_4 = \kappa^2/M^4$ is precisely of the size naive dimensional analysis predicted for a dimension-six coefficient built from a dimensionless ratio κ^2/M^2 divided by M^2 (Eq. (38.6)). The higher terms in Eq. (38.39), beginning with $\phi^2\Box^2\phi^2/M^6$, are dimension-eight operators, contributing at relative order $(p/M)^4$ and below the accuracy at which we work.

Why "integrating out" is a derivative expansion

The single algebraic move that produced the entire tower is the geometric series Eq. (38.36). A heavy propagator $1/(p^2 - M^2)$, evaluated at light external momenta $p^2 \ll M^2$, is not a pole but a smooth function of p^2; Taylor expanding it about $p^2 = 0$ converts a non-local exchange into a sum of local vertices, each with two more derivatives and one more power of $1/M^2$ than the last. Pictorially, the heavy internal line shrinks to a point (Fig. 38.3): what was a propagator becomes a contact interaction, and the corrections to the contact approximation are the higher terms of the expansion. "The heavy particle cannot be produced, only exchanged virtually, and its virtual exchange is local on scales $\gg 1/M$" is the physics; the Taylor expansion of $1/(p^2 - M^2)$ is the mathematics.

38.3.3 The same answer from the path integral

The classical route just taken is a saddle-point evaluation in disguise, and it is worth verifying that the honest object — the functional integral over the heavy field — gives the identical effective action, because that verification both confirms the method and shows exactly where, at one loop, the two will start to disagree. The Wilsonian definition of integrating out (Theorem 23.8) instructs

us to perform the path integral over the heavy field at fixed light background, producing the effective action

$$e^{\,iS_{\rm eff}[\phi]} = \int \mathcal{D}\Phi\ \exp\Big\{ i\, S_{\rm full}[\phi,\Phi]\Big\} = e^{\,iS_{\rm light}[\phi]} \int \mathcal{D}\Phi\ \exp\Big\{ i \int \mathrm{d}^4x\, \big[\tfrac{1}{2}\Phi\mathcal{K}\Phi + J\Phi\big]\Big\}, \tag{38.42}$$

with $\mathcal{K}$ and J as in Eq. (38.37). With $g_3 = 0$ the exponent is exactly quadratic in Φ with a ϕ-*independent* kinetic operator $\mathcal{K} = -(\Box + M^2)$, so the integral is the Gaussian functional integral with a linear source evaluated in Eq. (10.70). That computation gives

$$\int \mathcal{D}\Phi\ \exp\Big\{ i \int \big[\tfrac{1}{2}\Phi\mathcal{K}\Phi + J\Phi\big]\Big\} = \big[\det(-i\mathcal{K}/2\pi)\big]^{-1/2}\ \exp\Big\{ -\tfrac{i}{2}\int J\,\mathcal{K}^{-1}J\Big\}. \tag{38.43}$$

The exponent is identical to the classical on-shell action Eq. (38.38): $-\frac{1}{2}\int J\mathcal{K}^{-1}J = S_\Phi[\Phi_{\rm cl}]$. This is the precise sense in which the two routes agree.

Why the equation of motion is exact at tree level

The agreement is not a coincidence of this example; it is structural. Whenever the heavy field enters the action at most quadratically — as it must when its only coupling to light fields is linear in Φ — the Gaussian integral is exact, and an exact Gaussian integral equals its saddle: the value of the exponent at the stationary point $\Phi_{\rm cl}$ where $\delta S/\delta\Phi = 0$, which is the classical equation of motion Eq. (38.34). The prefactor $\det(-i\mathcal{K}/2\pi)^{-1/2}$ in Eq. (38.43) is the one-loop fluctuation determinant of the heavy field, and here it is ϕ-*independent*, because $\mathcal{K} = -(\Box + M^2)$ contains no light field. A ϕ-independent factor is a contribution to the vacuum energy alone; it multiplies every light-field amplitude by the same constant and drops out of all connected correlators. Hence at this order "solve the EOM and substitute" and "do the path integral" produce literally the same effective Lagrangian. The agreement is exact precisely because there are no heavy loops to speak of: the only heavy graphs that survive are the tree graphs built from $\Phi_{\rm cl}$.

The bridge to the next section is already visible in this box. The moment we restore a coupling that makes Φ appear more than quadratically — the cubic $g_3\Phi^3$ of Section 38.3.4, or a $\Phi^2\phi^2$ portal — the operator $\mathcal{K}$ acquires ϕ-dependence, the fluctuation determinant becomes a nontrivial functional of the light field, and the path integral exceeds the classical substitution by genuine heavy loops. Those loops are the subject of Section 38.4; for now we note only that at *tree* level the saddle is the whole story, and we extend the classical computation to reach the six-point operator.

Integrating out works for any heavy field

Nothing in the argument was special to a scalar. If the heavy field is a vector, the Gaussian integral is over Φ_μ with the kinetic operator of a massive spin-one field, and the saddle is the solution of its (Proca) equation of motion — the case we treat in Section 38.3.6 to obtain the Fermi interaction. If the heavy field is a fermion, the functional integral is the Berezin Gaussian of Theorem 12.26, whose saddle is the solution of the Dirac equation in the light background and whose fluctuation factor is a *determinant* (in the numerator) rather than its inverse square root; this is how the heavy lower components of a Dirac field are eliminated to build HQET in Chapter 40. The recipe — solve the heavy EOM, substitute, expand the heavy propagator in $1/M$ — is universal.

38.3.4 Generating ϕ^6: the nonlinear equation of motion

The Gaussian model produced the quartic and the dimension-six derivative operator but no six-point operator ϕ^6, for the simple reason that an action quadratic in Φ yields an on-shell action bilinear in the source $J \propto \phi^2$, hence quartic in ϕ. To switch on ϕ^6 at tree level we need the heavy field to appear cubically, which is exactly what the self-coupling $g_3\Phi^3$ in Eq. (38.33) provides. The equation of motion is now nonlinear,

$$(\Box + M^2)\,\Phi = -\frac{\kappa}{2}\,\phi^2 - \frac{g_3}{2}\,\Phi^2, \tag{38.44}$$

and we solve it iteratively in inverse powers of M, treating the right-hand side as a perturbation. Writing $\Phi_{\rm cl} = \Phi^{(1)} + \Phi^{(2)} + \cdots$ where $\Phi^{(k)}$ contains $2k$ powers of ϕ, the leading term reproduces Eq. (38.35), which at zeroth order in the derivative expansion is

$$\Phi^{(1)} = -\frac{\kappa}{2}(\Box + M^2)^{-1}\phi^2 \;\simeq\; -\frac{\kappa}{2M^2}\,\phi^2, \tag{38.45}$$

and the next iteration feeds $\Phi^{(1)}$ into the nonlinear source,

$$\Phi^{(2)} = -\frac{g_3}{2}(\Box + M^2)^{-1}\big(\Phi^{(1)}\big)^2 \;\simeq\; -\frac{g_3}{2M^2}\Big(\frac{\kappa}{2M^2}\phi^2\Big)^2 = -\frac{g_3\kappa^2}{8M^6}\,\phi^4. \tag{38.46}$$

To extract the six-point operator it is cleanest to use, rather than the brute substitution, the on-shell value of the action expressed through the equation of motion. With the kinetic term integrated by parts the heavy action reads $S_\Phi = \int[-\tfrac12\Phi(\Box + M^2)\Phi - \tfrac{\kappa}{2}\phi^2\Phi - \tfrac{g_3}{6}\Phi^3]$, and using Eq. (38.44) to replace $(\Box + M^2)\Phi_{\rm cl} = -\tfrac{\kappa}{2}\phi^2 - \tfrac{g_3}{2}\Phi_{\rm cl}^2$ in the first term,

$$S_\Phi[\Phi_{\rm cl}] = \int \mathrm{d}^4x\,\Big[-\tfrac12\Phi_{\rm cl}\big(-\tfrac{\kappa}{2}\phi^2 - \tfrac{g_3}{2}\Phi_{\rm cl}^2\big) - \tfrac{\kappa}{2}\phi^2\Phi_{\rm cl} - \tfrac{g_3}{6}\Phi_{\rm cl}^3\Big]$$

$$= \int \mathrm{d}^4x \, \Big[-\tfrac{\kappa}{4}\phi^2\Phi_{\rm cl} + \tfrac{g_3}{12}\Phi_{\rm cl}^3 \Big]. \tag{38.47}$$

The quartic piece comes from $-\frac{\kappa}{4}\phi^2\Phi^{(1)} = \frac{\kappa^2}{8M^2}\phi^4$, reproducing Eq. (38.40) as it must. The six-point piece collects two contributions, from $-\frac{\kappa}{4}\phi^2\Phi^{(2)}$ and from $\frac{g_3}{12}(\Phi^{(1)})^3$:

$$\mathcal{L}_{\rm eff} \supset -\frac{\kappa}{4}\phi^2\Big(-\frac{g_3\kappa^2}{8M^6}\phi^4\Big) + \frac{g_3}{12}\Big(-\frac{\kappa}{2M^2}\phi^2\Big)^3 = \Big(\frac{g_3\kappa^3}{32M^6} - \frac{g_3\kappa^3}{96M^6}\Big)\phi^6 = \frac{g_3\kappa^3}{48M^6}\,\phi^6. \tag{38.48}$$

Matching to the thread normalization $\frac{c_6}{6!}\phi^6$ of Eq. (23.52) fixes the dimension-six six-point Wilson coefficient,

$$\frac{c_6}{6!} = \frac{g_3\kappa^3}{48M^6} \quad\Longrightarrow\quad \boxed{c_6 = \frac{6!}{48}\,\frac{g_3\kappa^3}{M^6} = \frac{15\,g_3\kappa^3}{M^6}}. \tag{38.49}$$

The combinatorial factor $15 = 6!/48$ is the same number that decorated the loop-generated $c_6 = 15\lambda^3 I_3$ of Eq. (23.52): it counts the ways to distribute six external legs over the cubic vertex topology, and it is independent of whether the ϕ^6 operator is born from a heavy loop (bottom-up, as in Chapter 23) or from heavy tree exchange (top-down, here). With $I_3 \sim 1/M^2$ in Eq. (23.52) and the explicit $\kappa^3 g_3/M^6$ here, both deliver a dimension-six coefficient of size (dimensionless couplings)$/M^2$, as power counting demanded. ϕ^4 theory has thus closed a loop of its own: the irrelevant operators ϕ^6 and $\phi^2(\partial\phi)^2$ that the Wilsonian flow *discovered* being generated as one integrates out high-momentum shells are the same operators that top-down matching now produces, with computed coefficients, on integrating out a heavy particle.

Tree-level integrating-out is the sum of connected tree graphs

Equation Eq. (38.47) is the position-space form of a Feynman statement: the tree-level effective action is the generating functional of connected tree diagrams with the heavy field on internal lines only and the light field on external legs. The quartic operator is the single-Φ-exchange graph of Fig. 38.3; the six-point operator is the graph in which three $\kappa\,\Phi\phi^2$ vertices each emit a ϕ^2 pair and meet at one $g_3\Phi^3$ vertex through three heavy propagators, each shrunk to $1/M^2$. Solving the classical equation of motion iteratively, as in Eqs. (38.45) and (38.46), is nothing but resumming these tree topologies, vertex by vertex; the iteration order in $1/M$ is the number of heavy propagators. This is why "classical" and "tree-level" are synonyms here: the classical field is the sum over tree graphs, and quantum (loop) corrections are precisely the fluctuation determinant that the saddle point omits.

38.3.5 Matching the physical amplitude

We have matched the off-shell effective *action*. Theorem 38.8 promised that matching the on-shell *amplitude* gives the same Wilson coefficients up to redundant operators. The cleanest test is the carried process of ϕ^4 theory: elastic $\phi\phi \to \phi\phi$ scattering, whose $O(\lambda)$ amplitude was the thread's first dynamical number ($\mathcal{M} = -\lambda$) in Chapter 14. We compute it in the full theory, expand, and confront the effective theory.

Worked Example 38.9 (On-shell matching of $\phi\phi \to \phi\phi$)**.** Work to dimension six and set $g_3 = 0$, since the cubic does not contribute to the four-point amplitude at tree level. The full-theory tree amplitude for $\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4)$ receives the contact ϕ^4 vertex and heavy-scalar exchange in the three channels. The $\Phi\phi\phi$ vertex from $-\frac{\kappa}{2}\Phi\phi^2$ is $-i\kappa$ (the $\frac{1}{2}$ cancels against the two ways of attaching the identical ϕ legs), and the heavy propagator is $i/(q^2 - M^2)$. With the Mandelstam invariants $s = (p_1+p_2)^2$, $t = (p_1-p_3)^2$, $u = (p_1-p_4)^2$, the amplitude defined by Eq. (16.2) is

$$\mathcal{M}_{\text{full}} = -\lambda_0 - \kappa^2\Big[\frac{1}{s-M^2} + \frac{1}{t-M^2} + \frac{1}{u-M^2}\Big], \tag{38.50}$$

where each exchange contributes $(-i\kappa)\,[i/(q^2-M^2)]\,(-i\kappa) = -i\kappa^2/(q^2-M^2)$ to $i\mathcal{M}$. Now expand each heavy propagator by Eq. (38.36), $1/(x-M^2) = -M^{-2}(1 + x/M^2 + x^2/M^4 + \cdots)$:

$$\frac{1}{s-M^2} + \frac{1}{t-M^2} + \frac{1}{u-M^2} = -\frac{3}{M^2} - \frac{s+t+u}{M^4} - \frac{s^2+t^2+u^2}{M^6} - \cdots . \tag{38.51}$$

For four external particles of mass m, momentum conservation fixes $s+t+u = 4m^2$. Substituting Eq. (38.51) into Eq. (38.50),

$$\mathcal{M}_{\text{full}} = -\Big(\lambda_0 - \frac{3\kappa^2}{M^2}\Big) + \frac{\kappa^2(s+t+u)}{M^4} + O\Big(\frac{p^4}{M^6}\Big) = -\lambda_{\text{eff}} + \frac{4\kappa^2 m^2}{M^4} + O\Big(\frac{p^4}{M^6}\Big). \tag{38.52}$$

The momentum-independent piece is exactly $-\lambda_{\text{eff}}$ with $\lambda_{\text{eff}} = \lambda_0 - 3\kappa^2/M^2$ of Eq. (38.40): the on-shell four-point amplitude reproduces the marginal Wilson coefficient read off the action.

Now the effective-theory side. The contact $-\frac{\lambda_{\text{eff}}}{4!}\phi^4$ gives $-\lambda_{\text{eff}}$. The dimension-six derivative operator $\frac{z_4}{2}\phi^2(\partial\phi)^2$ of Eq. (38.41) contributes a momentum-dependent four-point vertex. Summing over the 4! ways of assigning the external legs to the operator's two undifferentiated and two (Lorentz-contracted) differentiated fields, with each derivative bringing down a momentum $\partial_\mu\phi \to -ip_\mu$, the vertex is

$$i\mathcal{M}_{\text{eff}}^{(z_4)} = i\,\frac{z_4}{2}\cdot 4\sum_{k<l}(-iq_k)\cdot(-iq_l) = -2i\,z_4\sum_{k<l} q_k\cdot q_l, \tag{38.53}$$

with all momenta taken incoming ($q_{1,2} = p_{1,2}$, $q_{3,4} = -p_{3,4}$). Momentum conservation $\sum_k q_k = 0$ gives $0 = (\sum_k q_k)^2 = \sum_k q_k^2 + 2\sum_{k<l} q_k \cdot q_l$, and with each $q_k^2 = m^2$ on shell, $\sum_{k<l} q_k \cdot q_l = -2m^2$. Hence

$$\mathcal{M}_{\text{eff}}^{(z_4)} = -2z_4(-2m^2) = 4z_4 m^2 = \frac{4\kappa^2 m^2}{M^4}, \tag{38.54}$$

identical to the momentum-dependent term of Eq. (38.52). The effective theory, with the two Wilson coefficients λ_{eff} and $z_4 = \kappa^2/M^4$ fixed by action-matching, reproduces the full on-shell amplitude to the order worked. This is the explicit verification of Theorem 38.7.

The worked example also proves Theorem 38.8. Notice that the entire on-shell effect of the derivative operator $\phi^2(\partial\phi)^2$ collapsed, by the on-shell relation $\sum_{k<l} q_k \cdot q_l = -2m^2$, to the *constant* $4z_4m^2$ — a number of order m^2/M^4 with no momentum dependence. A constant contribution to the four-point amplitude is indistinguishable from a shift of λ_{eff} (and, in the two-point function, a shift of m^2). In the language of Section 38.2.4, the operator $\phi^2(\partial\phi)^2$ is *redundant*: applying the light-field equation of motion $\Box\phi = -m^2\phi - \frac{\lambda_{\text{eff}}}{6}\phi^3 + \cdots$ to the equivalent form $-\frac{1}{4}\phi^2\Box\phi^2$ trades the derivatives for $m^2\phi^4$ and ϕ^6 terms, removing the derivative operator entirely in favour of shifts to the marginal coupling, the mass, and the six-point coefficient. Whether one keeps $\phi^2(\partial\phi)^2$ with coefficient z_4 (off-shell matching) or absorbs it into $\delta\lambda_{\text{eff}} \sim m^2/M^4$ and δc_6 (on-shell matching), the S-matrix is the same. The off-shell action carries redundant information; the physics does not. This is exactly the freedom that the field-redefinition theorem guarantees, and it is why a sensible matching computation may use *any* convenient form of the operator basis, on shell or off, provided one is consistent.

38.3.6 Heavy vector exchange and the Fermi constant

The scalar example taught the mechanics; the historically decisive application is to a heavy *vector*. Long before the W boson was known, weak decays such as the muon's, $\mu^- \to e^-\bar{\nu}_e\nu_\mu$, were described by Fermi's contact interaction — four fermion fields meeting at a point with a single coupling G_F. We now recognize that interaction as the effective theory obtained by integrating out a heavy vector, and we compute G_F by matching. The heavy field is the W, whose full provenance (the spontaneous breaking of the electroweak gauge symmetry that gives it its mass) is the business of Chapter 31; for the present purpose we need only its mass M_W, its propagator, and its coupling to a charged current.

The relevant interaction is the charged-current Lagrangian

$$\mathcal{L}_{\text{CC}} = \frac{g}{\sqrt{2}}\Big(W_\mu^+ J_-^\mu + W_\mu^- J_+^\mu\Big), \qquad J_-^\mu = \bar{\psi}_i\,\gamma^\mu P_L\,\psi_j, \quad P_L = \tfrac{1}{2}(1-\gamma^5), \tag{38.55}$$

with $J_+^\mu = (J_-^\mu)^\dagger$, g the weak gauge coupling, and the sum running over the relevant fermion pairs (for muon decay, (i,j) are (ν_μ,μ) and (ν_e,e)); the current

is purely left-handed, the famous $V-A$ structure. The W is a massive vector with the propagator

$$\widetilde{D}^{W}_{\mu\nu}(q) = \frac{-i}{q^2 - M_W^2}\Big(\eta_{\mu\nu} - \frac{q_\mu q_\nu}{M_W^2}\Big), \tag{38.56}$$

the form fixed by the massive spin-one quantization of Chapter 31 (unitary gauge). Muon decay proceeds through single W exchange between the two currents, at momentum transfer $q^2 \sim m_\mu^2 \ll M_W^2$. Integrating the W out is, by the universal recipe, the replacement of Eq. (38.56) by its low-energy limit. In that limit q^2 is negligible against M_W^2 in the denominator, and the longitudinal $q_\mu q_\nu/M_W^2$ term, when contracted with a current, produces $q_\mu J^\mu = \partial\cdot J$, which by the Dirac equation is proportional to the light fermion masses and hence suppressed by $m^2_{\text{fermion}}/M_W^2$ relative to the leading term. Dropping both,

$$\widetilde{D}^{W}_{\mu\nu}(q) \xrightarrow{\;q^2 \ll M_W^2\;} \frac{i\,\eta_{\mu\nu}}{M_W^2} + O\Big(\frac{q^2}{M_W^4}, \frac{m_f^2}{M_W^4}\Big). \tag{38.57}$$

The heavy line shrinks to a point, and the two charged-current vertices fuse into a four-fermion contact operator (Fig. 38.4). Equivalently, and identically, one solves the W equation of motion: at low energy the Proca equation reduces to $M_W^2 W_-^\mu = -(g/\sqrt{2})J_-^\mu$, giving $W_-^\mu = -(g/\sqrt{2}M_W^2)J_-^\mu$, and substituting back into Eq. (38.55) — the same Gaussian-saddle logic as for the scalar, now for a field appearing quadratically in its Proca action — yields the effective Lagrangian

$$\mathcal{L}_{\text{eff}} = -\frac{g^2}{2M_W^2}\, J_+^\mu\, J_{\mu,-} = -\frac{g^2}{2M_W^2}\,(\bar\psi_j \gamma^\mu P_L \psi_i)(\bar\psi_i \gamma_\mu P_L \psi_j). \tag{38.58}$$

The factor arises as in the scalar case: substituting the solved field into $M_W^2 W^+W^- + (g/\sqrt{2})(W^+J_- + W^-J_+)$ leaves exactly half the cross term, $+\frac{g^2}{2M_W^2}J_+J_- - \frac{g^2}{M_W^2}J_+J_- = -\frac{g^2}{2M_W^2}J_+J_-$.

It remains to express Eq. (38.58) in Fermi's normalization. Fermi wrote the interaction with the chirality-projected currents combined into vector-minus-axial currents $\bar\psi\gamma^\mu(1-\gamma^5)\psi = 2\,\bar\psi\gamma^\mu P_L\psi$:

$$\mathcal{L}_{\text{Fermi}} = -\frac{G_F}{\sqrt{2}}\,[\bar\psi_j\gamma^\mu(1-\gamma^5)\psi_i]\,[\bar\psi_i\gamma_\mu(1-\gamma^5)\psi_j] = -\frac{4G_F}{\sqrt{2}}\,(\bar\psi_j\gamma^\mu P_L\psi_i)(\bar\psi_i\gamma_\mu P_L\psi_j). \tag{38.59}$$

Equating the two forms Eq. (38.58) and Eq. (38.59) gives $4G_F/\sqrt{2} = g^2/2M_W^2$, that is

$$\boxed{\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}}. \tag{38.60}$$

This is the matching condition for the Fermi constant: the dimensionful Wilson coefficient G_F of the four-fermion operator is fixed by the gauge coupling and

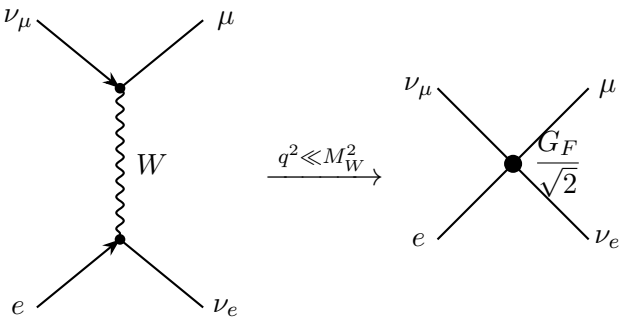


Figure 38.4: Integrating out the W in muon decay. At momentum transfer $q^2 \sim m_\mu^2 \ll M_W^2$ the W propagator Eq. (38.56) is local on the scale $1/M_W$ and the exchange diagram collapses to Fermi's four-fermion contact interaction, with Wilson coefficient $G_F/\sqrt{2} = g^2/8M_W^2$ (Eq. (38.60)).

the heavy mass. It is the four-fermion analogue of $\lambda_{\text{eff}} = \lambda_0 - 3\kappa^2/M^2$ — a marginal-looking coupling at low energy revealed as a ratio of short-distance parameters, suppressed by the heavy mass. Because the operator is dimension six (four fermion fields of dimension $\frac{3}{2}$ each), its coefficient $G_F \sim g^2/M_W^2$ carries the obligatory $1/M^2$ of an irrelevant operator; the dimensionless combination is g^2, of order the gauge coupling squared, again as naive dimensional analysis predicts.

The Fermi constant came first

Fermi proposed the four-fermion interaction in 1933, four decades before the W boson was discovered, fitting G_F to the measured rate of nuclear beta decay; the modern value is $G_F = 1.166 \times 10^{-5}$ GeV^{-2}. The relation Eq. (38.60) runs the matching backwards: knowing G_F and g (the latter from the electroweak structure of Chapter 31) *predicts* $M_W \approx 80$ GeV, a prediction confirmed at CERN in 1983. This is the paradigmatic case of an effective theory pointing to its own completion: the non-renormalizable Fermi theory, taken seriously as an EFT, contains in the size of its single Wilson coefficient the scale $M_W \sim (G_F)^{-1/2}$ at which it must be replaced. The breakdown of the contact description — its cross sections growing as $G_F^2 s$ and violating unitarity near $\sqrt{s} \sim G_F^{-1/2}$ — is the amplitude-level signal of the same scale, and exactly the "the EFT predicts where it dies" phenomenon that Section 38.2 anticipated.

The $V-A$ structure also illustrates a point that will matter in Chapter 40: integrating out the W at tree level produced a *single* four-fermion operator, the one whose chirality structure the W coupling dictated. Had we matched at one loop, additional operators with other Dirac and color structures would have

been generated, with coefficients carrying logarithms $\ln(M_W/\mu)$ — the operator mixing and running that turn the static Fermi constant into a scale-dependent set of Wilson coefficients. That is precisely the new feature loops bring, and it is where we go next.

We have now constructed Wilson coefficients top-down in two physical examples, defining them at the high scale $\mu \sim M$ by the requirement that the effective theory reproduce the full theory's amplitudes order by order in E/M, and we have seen the classical-EOM and Gaussian-integration routes deliver the same tree-level effective action. Every coefficient we obtained — λ_{eff}, z_4, c_6, G_F — is a pure tree-level number, a ratio of short-distance parameters with no reference to the renormalization scale. That tidiness is an artefact of tree level. The moment we include loops, the matching coefficients acquire logarithms of the heavy mass and a dependence on the subtraction scheme, because the full and effective theories distribute the same physics differently between loop integrals and local counterterms. Making the matching consistent at one loop, and understanding why a mass-independent scheme such as $\overline{\text{MS}}$ hides the decoupling of heavy particles and therefore forces us to remove them by hand at thresholds, is the task of Section 38.4 and the Appelquist–Carazzone theorem of Section 38.4.2.

38.4 One-Loop Matching and the Decoupling Theorem

Tree-level matching, as carried out in Section 38.3, was an exercise in algebra: expand the full-theory amplitude in powers of E/M, read off the local pieces, and call them Wilson coefficients. Nothing about the procedure cared that the two sides of the matching equation were tree amplitudes. The instant we go to one loop, however, a new feature appears that the tree analysis could not have prepared us for. Both theories now have loops, and loops generate logarithms. The full theory produces logarithms $\ln(M^2/(-p^2))$ tying the heavy mass to the external kinematics; the effective theory, which contains no heavy field, can only produce $\ln(\mu^2/(-p^2))$. The question that organizes this section is what happens to the difference. The answer is the cleanest structural fact in all of effective field theory: the difference is *local* — a polynomial in external momenta, analytic at $p = 0$ — and so it is exactly of the form that can be absorbed into a shift of the Wilson coefficients. Matching survives at loop level, the large logarithm splits unambiguously into a piece that belongs to the coefficients and a piece that belongs to their running, and from this same locality follows the theorem of Appelquist and Carazzone: a heavy particle, properly accounted for, leaves no trace at low energy beyond renormalizations of the light couplings and corrections suppressed by powers of E/M. We will then see that the convenient mass-independent scheme $\overline{\text{MS}}$ hides this decoupling, and that the

cure — removing the heavy field by hand and matching at its threshold — is not a workaround but the physically correct prescription.

38.4.1 The structure of one-loop matching

Fix the setup. The full theory contains light fields ϕ (masses and momenta of order E) and one heavy field of mass $M \gg E$. The effective theory is the Lagrangian

$$\mathcal{L}_{\rm EFT} = \mathcal{L}_{\rm light} + \sum_i C_i(\mu)\, \mathcal{O}_i, \tag{38.61}$$

built from the light fields alone, with local operators $\mathcal{O}_i$ of increasing dimension organized by the power counting of Theorem 21.10 and Section 38.2, and Wilson coefficients $C_i(\mu)$ carrying the inverse powers of M. Matching is the demand that the two theories produce the same S-matrix elements for light external states, order by order in E/M and order by order in the loop expansion. Schematically, for any light amplitude $\mathcal{A}$,

$$\mathcal{A}_{\rm full}(p; M, \mu) \stackrel{!}{=} \mathcal{A}_{\rm EFT}(p; C_i(\mu), \mu) \qquad \text{for } p \sim E \ll M, \tag{38.62}$$

both sides computed in the *same* renormalization scheme (we use $\overline{\rm MS}$, $d = 4 - 2\varepsilon$, common scale μ). The Wilson coefficients are whatever they must be to make Eq. (38.62) hold.

Expand both sides in loops, writing $\mathcal{A} = \mathcal{A}^{(0)} + \mathcal{A}^{(1)} + \cdots$ and $C_i = C_i^{(0)} + C_i^{(1)} + \cdots$, where the superscript counts loops (equivalently powers of $\hbar$). At tree level Eq. (38.62) reduces to the content of Section 38.3: the tree EFT amplitude is the sum of tree-level operator insertions, and matching fixes $C_i^{(0)}$. At one loop the equation reads

$$\mathcal{A}^{(1)}_{\rm full}(p) \;=\; \underbrace{\mathcal{A}^{(1)}_{\rm EFT}[C^{(0)}](p)}_{\text{EFT loops, tree coefficients}} \;+\; \underbrace{\sum_i C_i^{(1)}\, \langle \mathcal{O}_i \rangle^{(0)}(p)}_{\text{tree insertion of one-loop coefficients}}, \tag{38.63}$$

where $\mathcal{A}^{(1)}_{\rm EFT}[C^{(0)}]$ denotes the one-loop EFT amplitude computed with the tree-level Wilson coefficients in the vertices, and $\langle \mathcal{O}_i \rangle^{(0)}$ is the tree-level matrix element of the operator $\mathcal{O}_i$. Solving for the one-loop matching correction,

$$\boxed{\sum_i C_i^{(1)}\, \langle \mathcal{O}_i \rangle^{(0)}(p) \;=\; \mathcal{A}^{(1)}_{\rm full}(p) - \mathcal{A}^{(1)}_{\rm EFT}[C^{(0)}](p)\,.} \tag{38.64}$$

Read literally, Eq. (38.64) looks dangerous: the left-hand side is, by construction, a finite sum of *local* structures (tree matrix elements of local operators are polynomials in p), while the right-hand side is a difference of two loop amplitudes, each of which is a transcendental function of p full of logarithms and, above

thresholds, branch cuts. For the equation to be consistent, all of that nonlocal structure must cancel in the difference. It does, and the reason is the single most important sentence in this subsection: *the full theory and the effective theory have, by construction, identical infrared physics.* They contain the same light fields with the same masses and the same low-energy interactions; the only difference between them is in the ultraviolet, where the full theory has a heavy particle and the EFT has a tower of contact terms. The infrared region of any loop integral — loop momenta of order E, where the light particles go nearly on shell and the nonanalytic structure is generated — is therefore the same in both theories. When we subtract, the infrared cancels, and what survives is purely ultraviolet, hence local.

The method of regions makes this exact. The cleanest way to see the cancellation, and the way that will give us the matching coefficient directly, is to dissect a full-theory loop integral by the scale of its loop momentum. Consider a one-loop integral $I = \int \mathrm{d}^d\ell\, f(\ell, p, M)$ with $p \sim E \ll M$. Split the integration into two regions and expand the integrand appropriately in each before integrating:

- the *hard* region, $\ell \sim M$, where the light scales p, m_{light} are small and we Taylor-expand the integrand in p/ℓ and m_{light}/ℓ before integrating;
- the *soft* region, $\ell \sim E$, where the heavy propagators are far off shell and we Taylor-expand them in ℓ/M and p/M before integrating.

The theorem of expansion by regions states that the full integral is the sum of the two region integrals, each computed over all of ℓ-space in dimensional regularization (the scaleless remainders that the naive split would double-count are set to zero by $d = 4 - 2\varepsilon$). Now identify the pieces. The soft region, in which the heavy propagators have been expanded to a sum of local vertices $\sim 1/M^{2n}$, is *precisely* a loop integral of the effective theory: light particles circulating, heavy lines shrunk to the contact operators $\mathcal{O}_i$. So

$$I_{\text{full}} = I_{\text{hard}} + I_{\text{soft}}, \qquad I_{\text{soft}} = I_{\text{EFT}}. \tag{38.65}$$

Subtracting, $I_{\text{full}} - I_{\text{EFT}} = I_{\text{hard}}$. But I_{hard} was obtained by Taylor-expanding in the external momenta from the outset, so it is manifestly a polynomial in p with coefficients that are functions of M and μ only — it is local. Comparing with Eq. (38.64), the hard region of the full-theory loop *is* the one-loop Wilson coefficient:

$$\sum_i C_i^{(1)}\, \langle \mathcal{O}_i \rangle^{(0)}(p) = I_{\text{hard}}. \tag{38.66}$$

This is the operational content of one-loop matching and we will lean on it heavily: *a Wilson coefficient is the hard part of a full-theory amplitude.* One never needs the soft (infrared) part of either theory — it cancels — so in practice one may

set all light masses and external momenta to zero from the start, compute the resulting purely-heavy loop integrals, and read off the coefficients. The price is that the integrals one then computes may be individually infrared-divergent; those infrared $1/\varepsilon$ poles are spurious, cancelling against the (equally spurious, equally scaleless) infrared poles of the EFT side, and one tracks them only to be sure they match.

The logarithm splits cleanly. Equation Eq. (38.64) also resolves the puzzle that opened the section. A generic one-loop full-theory amplitude carries the logarithm $\ln(M^2/(-p^2))$, which is large when $p \sim E \ll M$ — a large logarithm of the ratio of the two physical scales, the kind of thing perturbation theory cannot tolerate uncontrolled. Write it as a sum of two pieces using an arbitrary intermediate scale μ,

$$\ln \frac{M^2}{-p^2} = \underbrace{\ln \frac{M^2}{\mu^2}}_{\text{matching}} + \underbrace{\ln \frac{\mu^2}{-p^2}}_{\text{running}} . \tag{38.67}$$

The matching computation Eq. (38.66), performed at the scale μ, deposits $\ln(M^2/\mu^2)$ into the Wilson coefficient; the EFT loop, which knows only the light scales, carries $\ln(\mu^2/(-p^2))$. If we choose to match at $\mu \sim M$, the first logarithm is order one — there is no large logarithm in the matching coefficient, and ordinary fixed-order perturbation theory computes it reliably. The second logarithm is then large, but it lives entirely inside the EFT, where it is summed not by brute force but by the renormalization group: $C_i(\mu)$ is evolved from $\mu \sim M$ down to $\mu \sim E$, and the running resums $\ln(\mu/E)$ to all orders. This division of labor — compute the coefficient once, with no large log, at the high scale; then run it down — is the entire strategy of effective field theory at loop level, and it is what Section 38.5 will build the machinery for. The present section establishes the boundary condition $C_i(\mu \sim M)$; the next propagates it.

The matching coefficient is scheme-dependent; the prediction is not

Both sides of Eq. (38.64) are computed in $\overline{\text{MS}}$, and the value of $C_i^{(1)}(\mu)$ depends on that choice and on μ. This is not a defect: C_i is not an observable. The observable is the full low-energy amplitude $\mathcal{A}_{\text{EFT}}$, in which the μ-dependence of $C_i(\mu)$ cancels against the explicit μ-dependence of the EFT loops order by order — precisely the statement that Eq. (38.62) holds for all μ. What *is* mandatory is that the full theory and the EFT be matched in the *same* scheme: a coefficient extracted with a full-theory $\overline{\text{MS}}$ subtraction and inserted into EFT loops renormalized in some other scheme is simply wrong. Consistency of the scheme across the matching is the one inviolable rule.

38.4.2 The Appelquist–Carazzone decoupling theorem

We can now state and prove the theorem that justifies the whole enterprise: that low-energy physics is genuinely insensitive to heavy particles, so that an effective theory built from the light fields alone really does capture everything an experiment at energy $E \ll M$ can see.

Theorem 38.10 (Appelquist–Carazzone decoupling)**.** *Let a renormalizable theory contain light fields and a single field of mass M, and let all external momenta and light masses be of order $E \ll M$. Renormalize in a* physical *(mass-dependent) scheme — one whose renormalization conditions are imposed on light-particle amplitudes at scales of order E, so that the renormalized light couplings, masses, and field normalizations are defined by low-energy measurements. Then in any light-particle Green's function the entire effect of the heavy field is either*

(i) *absorbed into the values of the renormalized light parameters (a finite, M-dependent redefinition of couplings, masses, and field normalizations that the low-energy observer has already measured and cannot disentangle), or*

(ii) *suppressed by a positive power of E/M, appearing as the matrix elements of higher-dimension operators with coefficients scaling as inverse powers of M.*

In particular, as $M \to \infty$ at fixed renormalized low-energy parameters, the heavy field disappears from all low-energy observables. There is no term that grows with M or even survives as a finite, unsuppressed, M-independent constant beyond the renormalization of the light theory.

Proof. The proof rests on a single analytic fact and the structure already established in Eqs. (38.64) and (38.66). Consider any one-particle-irreducible light-particle Green's function $\Gamma(p)$ at the order in the loop expansion at which the heavy field first enters, i.e. through a loop containing at least one heavy internal line. Because $E \ll M$, the heavy line can never go on shell: the external invariants $p^2 \sim E^2$ are far below the threshold $\sim M^2$ at which a heavy pair could be produced. By the analytic structure of perturbative amplitudes — branch points sit at physical thresholds — the heavy contribution to $\Gamma(p)$ is therefore *analytic* in the external momenta in a neighborhood of $p = 0$ of radius set by M. An analytic function may be Taylor-expanded:

$$\Gamma_{\text{heavy}}(p) = a_0(M,\mu) + a_1(M,\mu)\,\frac{p^2}{M^2} + a_2(M,\mu)\,\frac{(p^2)^2}{M^4} + \cdots , \tag{38.68}$$

where dimensional analysis has been used to extract the powers of M, and each coefficient a_n is dimensionless up to the overall dimension of Γ. We now read the two classes of terms.

The leading terms renormalize the light theory. The terms with n small in Eq. (38.68) have the same structure in p as the operators already present in $\mathcal{L}_{\text{light}}$ — a constant is a shift of a coupling or a mass, a term $\propto p^2$ is a shift of a kinetic term or a field normalization, and so on up to the dimension of the renormalizable operators. These are local and they are exactly the structures that the light-theory counterterms can absorb. In a *physical* scheme the renormalization conditions are imposed at low energy: one demands, for instance, that the light propagator have a pole at the measured mass with unit residue, and that a coupling equal its measured value at a low-momentum reference point. Imposing these conditions *subtracts the value of* Γ_{heavy} *at the reference kinematics*, sweeping the heavy-induced constant a_0 (and the corresponding low-order a_n that the conditions fix) into the definition of the renormalized parameter. The low-energy observer measures the renormalized parameter and never sees a_0 separately; it has been reparametrized away. This is alternative (i).

The remaining terms are power-suppressed. After the physical subtraction, what is left of Eq. (38.68) is the difference between $\Gamma_{\text{heavy}}(p)$ and its low-order Taylor polynomial evaluated at the reference point — that is, the higher terms in the expansion, each carrying an explicit $(p^2/M^2)^n \sim (E/M)^{2n}$ with $n \geq 1$. These are the matrix elements of the higher-dimension operators of the effective theory, with Wilson coefficients $\sim 1/M^{2n}$ given by the hard region a_n. They vanish as $M \to \infty$. This is alternative (ii), and it completes the dichotomy.

That no term escapes the dichotomy is the content of the locality established in Section 38.4.1: by Eq. (38.66) the heavy field's contribution to any light amplitude, after the infrared (light-loop) part is separated, is the hard region I_{hard}, which is exactly the polynomial Eq. (38.68). There is no nonpolynomial, M-independent remainder, because such a remainder would be a nonlocal function of p with no $1/M$ suppression, and the only source of nonlocality at low energy is the light loops, which belong to the EFT and have been accounted for separately. Hence every heavy effect falls into (i) or (ii). □

The theorem is sharp about *why* the scheme must be physical: the subtraction in step (i) must be performed at low energy for the heavy constant a_0 to be the thing that the low-energy observer measures. We will see in Section 38.4.4 that a scheme which subtracts somewhere else — in particular $\overline{\text{MS}}$, which subtracts a pure pole with no reference to any physical kinematics — leaves a_0 visible as a logarithm $\ln(M/\mu)$ that does not go away, and that decoupling must then be enforced by hand.

Decoupling is why physics is possible

Theorem 38.10 is the formal version of a fact so familiar it is easy to overlook: one can compute the hydrogen spectrum without knowing the top quark, build a bridge without knowing quantum gravity, and discover

the laws of electromagnetism a century before the W boson. Each layer of physics is describable on its own terms, with the deeper layers entering only through the numerical values of a handful of low-energy constants and through corrections too small to have mattered. The decoupling theorem is the statement that this layered describability is not luck but a theorem about quantum field theory in a physical scheme. The effective-field-theory paradigm is its constructive converse: *because* the heavy physics enters only through local couplings and power-suppressed operators, one can write down the most general such local theory and fit those couplings to data, completely agnostic about what the heavy physics actually is.

38.4.3 Decoupling in action: the heavy fermion in QED

The abstract proof becomes concrete in the carried QED thread. Add to QED a heavy fermion of mass M and charge $Q_M e$ — a heavy lepton, say — and ask how it affects the photon's low-energy behavior, the cleanest light observable. Its only one-loop effect is a contribution to the vacuum polarization, the heavy-fermion analogue of the electron loop computed in Chapter 19. The tensor structure is fixed by gauge invariance to be transverse, Eq. (19.49), and the scalar form factor is the one-loop function Eq. (19.57) with the electron mass replaced by M and an extra charge factor Q_M^2,

$$\widehat{\Pi}_M(q^2) = -\frac{2\alpha Q_M^2}{\pi}\int_0^1 \mathrm{d}x\; x(1-x)\,\ln\frac{M^2}{M^2 - x(1-x)q^2}, \tag{38.69}$$

already on-shell subtracted so that $\widehat{\Pi}_M(0) = 0$, the on-shell scheme being a physical scheme in the sense of Theorem 38.10 (the renormalization condition fixes the charge at $q^2 = 0$, the Thomson point). This is exactly the setting of the theorem: the heavy line cannot go on shell for $|q^2| \sim E^2 \ll M^2$, so $\widehat{\Pi}_M$ is analytic at $q^2 = 0$ and we may expand it. Using $\ln\frac{M^2}{M^2 - x(1-x)q^2} = x(1-x)\frac{q^2}{M^2} + O(q^4/M^4)$ and the moment $\int_0^1 \mathrm{d}x\,[x(1-x)]^2 = \frac{1}{30}$, exactly as in Eq. (19.58) with $m \to M$,

$$\widehat{\Pi}_M(q^2) = -\frac{\alpha Q_M^2}{15\pi}\,\frac{q^2}{M^2} + O\Big(\frac{q^4}{M^4}\Big). \tag{38.70}$$

Here is the decoupling theorem made arithmetic. The heavy fermion's effect on the photon propagator is suppressed by q^2/M^2 — alternative (ii) of Theorem 38.10, with the leading term the matrix element of the dimension-six operator $\partial_\lambda F_{\mu\nu}\partial^\lambda F^{\mu\nu}$ generated by integrating out the heavy fermion. Where did alternative (i) go? It went into the constant we already subtracted. The *unrenormalized* heavy form factor of Eq. (19.53) carried the constant

$$\Pi_M(0) = -\frac{\alpha Q_M^2}{3\pi}\Big[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{M^2}\Big], \tag{38.71}$$

an M-dependent constant that the on-shell condition $\widehat{\Pi}_M(0) = 0$ absorbed wholesale into the definition of the physical charge $\alpha = \alpha(q^2 = 0)$. The low-energy experimenter measures $\alpha \approx 1/137$ at the Thomson point; that number already contains the heavy fermion's constant contribution, inextricably, and no low-energy photon experiment can isolate it. What remains observable — the q^2-dependence of the effective charge probed by varying the momentum transfer — is the power-suppressed remainder Eq. (38.70). The heavy fermion has decoupled: present in the measured value of α, invisible in its running until one reaches $q^2 \sim M^2$. This is the promised explicit demonstration that the M-dependence is a constant absorbed by charge renormalization plus terms suppressed by q^2/M^2.

38.4.4 Non-decoupling in $\overline{\text{MS}}$ and the threshold cure

The on-shell scheme made decoupling manifest because it subtracted at a physical point. The minimal subtraction scheme $\overline{\text{MS}}$ does not, and the consequence is that decoupling is invisible in $\overline{\text{MS}}$ — a feature so often mistaken for a paradox that it deserves to be displayed in full.

In $\overline{\text{MS}}$ one subtracts only the pole $1/\bar{\varepsilon}$ of Eq. (38.71), leaving the renormalized heavy form factor

$$\Pi_M^{\overline{\text{MS}}}(q^2) = -\frac{2\alpha Q_M^2}{\pi}\int_0^1 \mathrm{d}x\, x(1-x)\,\ln\frac{\mu^2}{M^2 - x(1-x)q^2}. \tag{38.72}$$

Now take the low-energy limit $|q^2| \ll M^2$. The logarithm becomes $\ln(\mu^2/M^2) + O(q^2/M^2)$, and with $\int_0^1 \mathrm{d}x\, x(1-x) = \frac{1}{6}$,

$$\Pi_M^{\overline{\text{MS}}}(q^2) \xrightarrow{|q^2|\ll M^2} \frac{\alpha Q_M^2}{3\pi}\ln\frac{M^2}{\mu^2} + O\Big(\frac{q^2}{M^2}\Big). \tag{38.73}$$

This does *not* vanish as $M \to \infty$ at fixed μ. It grows logarithmically without bound. In $\overline{\text{MS}}$ the heavy fermion fails to decouple: it contributes a constant $\frac{\alpha Q_M^2}{3\pi}\ln(M^2/\mu^2)$ to the low-energy vacuum polarization no matter how heavy it is, and correspondingly it contributes to the running coupling at *all* scales, including scales $\mu \ll M$ where it should be irrelevant. The reason is exactly the one anticipated after Theorem 38.10: $\overline{\text{MS}}$ subtracted a pure pole, not a physical constant, so the heavy constant $a_0 = \frac{\alpha Q_M^2}{3\pi}\ln(M^2/\mu^2)$ of Eq. (38.68) was never reparametrized away. It sits in the open as a logarithm. The hypotheses of the decoupling theorem — a *physical* scheme — are violated, and the conclusion fails.

The structural source is worth naming. The $\overline{\text{MS}}$ beta function is *mass-independent*: by construction the counterterms are pure poles, the same whatever the masses, so the anomalous dimensions and beta functions depend only on the field content, not on whether a given field is light or heavy relative to μ. The

QED beta function in $\overline{\text{MS}}$ therefore counts the heavy fermion on equal footing with the electron at every scale,

$$\beta_\alpha^{\overline{\text{MS}}} = \frac{2\alpha^2}{3\pi} \sum_f Q_f^2, \tag{38.74}$$

the sum running over *all* fermions, light and heavy alike — the abelian instance of the general $\overline{\text{MS}}$ beta function built from Theorem 24.1 and the running Eq. (24.215). Used naively, Eq. (38.74) says the heavy fermion screens the charge even at $\mu \ll M$, which is physically false: a virtual pair of mass M cannot be excited by a soft photon and cannot screen it. The mass-independent scheme has traded manifest decoupling for calculational simplicity, and the bill comes due as the spurious low-scale running Eq. (38.73).

The cure: integrate out and match at the threshold. The resolution is not to abandon $\overline{\text{MS}}$ — its mass-independence is exactly what makes the renormalization group simple — but to enforce decoupling by hand. One uses $\overline{\text{MS}}$ in a sequence of effective theories, removing each heavy field from the Lagrangian as μ drops below its mass and matching the two descriptions at the threshold. Above $\mu = M$ one works in the full theory, with the heavy fermion active and the beta function Eq. (38.74) including its charge. At $\mu \sim M$ one integrates the heavy fermion out: the effective theory below has the same Lagrangian minus the heavy field, its beta function counts only the remaining light fermions, and the two couplings are related by a matching condition fixed by demanding that the physical effective charge be continuous across the threshold.

The matching condition is Eq. (38.64) for the photon two-point function. Let $\alpha^{(n_f)}$ denote the $\overline{\text{MS}}$ coupling in the theory with n_f active fermions and $\alpha^{(n_f-1)}$ that in the theory with the heavy fermion removed. The physical effective charge built from the resummed vacuum polarization, $\alpha_{\text{eff}}^{-1}(q^2) = \alpha^{-1}(\mu)\big[1 - \Pi^{\overline{\text{MS}}}(q^2,\mu)/\alpha\big]$ at one loop, must agree whether computed above or below threshold. The full theory carries the heavy contribution Eq. (38.73); the effective theory does not. Equality of α_{eff}^{-1} then forces the difference into the matched coupling,

$$\boxed{\frac{1}{\alpha^{(n_f-1)}(\mu)} = \frac{1}{\alpha^{(n_f)}(\mu)} - \frac{Q_M^2}{3\pi} \ln\frac{M^2}{\mu^2} + O(\alpha)\,.} \tag{38.75}$$

Two features of Eq. (38.75) are the whole point. First, at the natural matching scale $\mu = M$ the logarithm vanishes and the couplings are *equal*: $1/\alpha^{(n_f-1)}(M) = 1/\alpha^{(n_f)}(M) + O(\alpha)$. The threshold matching is continuous — there is no large logarithm in the one-loop matching coefficient, just as the general analysis Eq. (38.67) promised, which is why $\mu \sim M$ is the right place to match. (The $O(\alpha)$ remainder is a finite, computable constant from the hard region of the

heavy loop; with the $\overline{\text{MS}}$ heavy mass it begins at the order beyond the leading logarithm.) Second, differentiating Eq. (38.75) with respect to $\ln\mu^2$ shows that the slope of $1/\alpha$ changes by exactly $+Q_M^2/3\pi$ across the threshold,

$$\frac{\mathrm{d}}{\mathrm{d}\ln\mu^2}\frac{1}{\alpha^{(n_f-1)}} = \frac{\mathrm{d}}{\mathrm{d}\ln\mu^2}\frac{1}{\alpha^{(n_f)}} + \frac{Q_M^2}{3\pi}, \tag{38.76}$$

which is precisely the statement that below the threshold the heavy fermion no longer contributes to the running: the beta-function coefficient steps down from $\sum^{n_f} Q_f^2$ to $\sum^{n_f-1} Q_f^2$, the heavy fermion's Q_M^2 removed. Decoupling, lost in Eq. (38.73), has been restored as a kink in the running coupling.

Figure 38.5 shows the construction. The physical effective coupling is reconstructed as a sequence of $\overline{\text{MS}}$ segments — straight lines in $\ln\mu$, each with the slope appropriate to the number of active flavors — joined continuously at the thresholds, with a kink at each heavy mass where a fermion is integrated out. This is the precise content, scheme by scheme, of the physical running of α derived in Section 20.6. The single-logarithm formula Eq. (20.171), with its fixed mass m, is the correct description only *between* thresholds; across the full range from m_e to M_Z the running is the broken line of Fig. 38.5, each segment governed by the active fermions of that decade and stitched to its neighbors by Eq. (38.75). The discussion in Section 20.6 of how the electron alone accounts for only part of the measured shift in $\alpha^{-1}(M_Z^2)$, with the muon, tau, and hadrons each switching on above its own threshold, is exactly this piecewise reconstruction: the physical, mass-dependent running that Theorem 38.10 guarantees, assembled out of mass-independent $\overline{\text{MS}}$ running plus the threshold matching Eq. (38.75).

38.4.5 The Euler–Heisenberg Lagrangian: integrating out the electron

We close the section with QED's flagship illustration of matching and decoupling at one loop: the effective theory of soft photons obtained by integrating the electron out of QED. Below the electron mass m_e the only propagating field is the photon, and the most general local Lagrangian consistent with Lorentz invariance, gauge invariance, parity, and charge conjugation is an expansion in the field strength and its derivatives,

$$\mathcal{L}_\gamma = -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + c_1\,(F_{\mu\nu}F^{\mu\nu})^2 + c_2\,(F_{\mu\nu}\widetilde{F}^{\mu\nu})^2 + \cdots, \qquad \widetilde{F}^{\mu\nu} \equiv \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \tag{38.77}$$

where the dots denote operators with more fields or derivatives. The structure is dictated by symmetry and power counting (Theorem 21.10). Gauge invariance forbids any operator that is not built from $F_{\mu\nu}$ (a bare A_μ is forbidden, so is a photon mass). Lorentz invariance and parity allow only the two independent quartic scalars $(F^2)^2$ and $(F\widetilde{F})^2$ — note $F_{\mu\nu}\widetilde{F}^{\mu\nu}$ is a total derivative and parity-odd, so it cannot appear linearly, but its square is parity-even and local. There

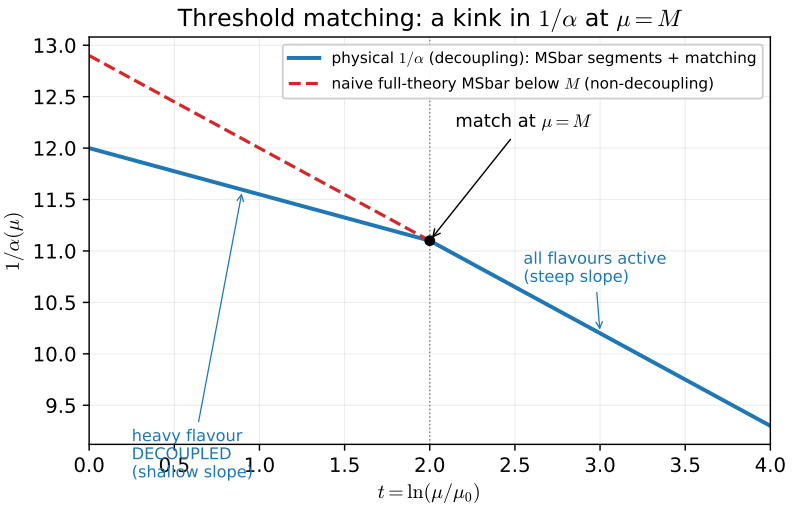


Figure 38.5: Threshold matching reconstructs decoupling in a mass-independent scheme. The inverse coupling $1/\alpha$ runs linearly in $\ln \mu$ with a slope set by the number of active flavors. Below the heavy mass M the physical coupling (solid) runs with the reduced slope — the heavy fermion has decoupled — while a naive full-theory $\overline{\text{MS}}$ extrapolation (dashed) keeps the steeper slope, the non-decoupling of Eq. (38.73). The two agree at $\mu = M$, where the heavy fermion is integrated out and the one-loop matching correction Eq. (38.75) vanishes, leaving a continuous value and a discontinuous slope: the threshold step Eq. (38.76).

is no gauge-invariant operator of dimension six built from two factors of F alone (the candidate $\partial_\lambda F_{\mu\nu}\partial^\lambda F^{\mu\nu}$ is, on the photon equation of motion $\partial_\mu F^{\mu\nu} = 0$, reducible and contributes only to the propagator's q^2/m_e^2 corrections seen in Eq. (38.70)); the first genuinely new photon self-interactions are the dimension-eight quartics in Eq. (38.77). Their coefficients c_1, c_2 have mass dimension -4 and are the Wilson coefficients we now compute by matching.

Worked Example 38.11 (The leading Euler–Heisenberg coefficients from the electron loop)**.** The matching condition Eq. (38.66) says c_1, c_2 are the hard part of the full-theory amplitude — here the four-photon one-particle-irreducible amplitude with an electron loop, evaluated at zero external momentum, which is the light-by-light box of Fig. 38.6. Rather than compute the box diagram directly, we use the equivalent and more economical observation that the entire tower of zero-momentum multi-photon amplitudes is generated by the one-loop effective Lagrangian of the electron in a *constant* background field $F_{\mu\nu}$. Expanding that effective Lagrangian to fourth order in F yields the four-photon vertex at zero momentum, hence c_1 and c_2.

The one-loop effective Lagrangian as a determinant. Integrating the electron out of the QED path integral produces the fermion determinant (Theorems 4.33 and 10.38); for a constant background the effective action is V_4 times a constant Lagrangian,

$$\mathcal{L}^{(1)} = -i\,\frac{1}{V_4}\ln\det(i\not{D} - m) = -\frac{i}{2V_4}\ln\det\left(\not{D}^2 + m^2\right)^*, \tag{38.78}$$

where in the second form we used $\gamma_5(i\not{D} - m)\gamma_5 = -(i\not{D} + m)^*$-type reflection to write the fermionic determinant as the square root of a bosonic one (the manipulation that turns $\det(i\not{D} - m)$ into $[\det(-\not{D}^2 - m^2)]^{1/2}$ up to a field-independent phase), and

$$\not{D}^2 = D^2 + \tfrac{e}{2}\,\sigma_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \tag{38.79}$$

is the squared Dirac operator, the second term ($\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu]$) being the Pauli spin coupling that distinguishes the electron from a charged scalar.

Proper time and the heat kernel. Write the logarithm of the determinant as a proper-time integral, $\ln A = -\int_0^\infty \frac{\mathrm{d}s}{s}\,e^{-sA}$ up to a field-independent constant, so that the effective Lagrangian becomes the proper-time integral of a heat-kernel trace. For a constant field we evaluate that trace from the spectrum of the squared operator. Take a constant magnetic field B along the z-axis first; the general result follows by Lorentz covariance. The operator $-D^2$ separates into free motion along (t, z) and a two-dimensional transverse problem identical to a charged particle in a magnetic field — the Landau problem — with transverse eigenvalues $eB(2n+1)$, $n = 0, 1, 2, \ldots$, each Landau level carrying a degeneracy $eB/2\pi$ per unit transverse area. The longitudinal directions give the free density of states $\int \frac{\mathrm{d}^2k}{(2\pi)^2}e^{-sk^2} = 1/4\pi s$. The Pauli term $\frac{e}{2}\sigma_{\mu\nu}F^{\mu\nu} = e\,\Sigma_3 B$ has eigenvalues $\pm eB$ on the spin states, contributing the Dirac-trace spin factor $\sum_{\text{spin}} e^{\mp seB} = 2\cosh(seB)$ doubled by the two chiralities. Assembling the trace per unit four-volume,

$$\frac{1}{V_4}\,\mathrm{Tr}\;e^{-s(-\not{D}^2)} = \underbrace{\frac{1}{4\pi s}}_{t,z}\;\underbrace{\frac{eB}{2\pi}}_{\text{degeneracy}}\;\underbrace{\sum_{n=0}^{\infty} e^{-seB(2n+1)}}_{=\,1/2\sinh(seB)}\;\underbrace{2\cosh(seB)\cdot 2}_{\text{spin}\,\times\,\text{chirality}} = \frac{eB}{4\pi^2 s}\,\coth(seB). \tag{38.80}$$

The geometric sum over Landau levels produced the sinh; the spin factor produced the cosh; together they give the coth that is the signature of the constant-field problem. Inserting Eq. (38.80) into the proper-time integral and restoring the mass exponent e^{-sm^2} gives, in the standard normalization of the renormalized Minkowski effective Lagrangian,

$$\mathcal{L}^{(1)}(B) = -\frac{1}{8\pi^2}\int_0^\infty \frac{\mathrm{d}s}{s^3}\,e^{-m^2 s}\Big[\,seB\coth(seB) - 1 - \tfrac{1}{3}(seB)^2\,\Big]. \tag{38.81}$$

The two subtractions inside the bracket are the renormalizations demanded by Theorem 38.10's alternative (i): the "-1" removes the field-independent vacuum energy, and the "$-\frac{1}{3}(seB)^2$" removes the term quadratic in B, which is logarithmically divergent at $s \to 0$ and renormalizes the charge — it is the constant absorbed into the definition of α, the zero-momentum vacuum polarization Eq. (38.71) in disguise. What survives the subtractions is finite and begins at *fourth* order in the field: decoupling, visible in the integrand.

The general field and the weak-field expansion. By Lorentz invariance $\mathcal{L}^{(1)}$ depends only on the two field invariants, which we write as

$$\mathcal{F} \equiv \tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}(B^2 - E^2), \qquad \mathcal{G} \equiv \tfrac{1}{4}F_{\mu\nu}\widetilde{F}^{\mu\nu} = -\,\mathbf{E}\cdot\mathbf{B}. \tag{38.82}$$

Introduce the two "secular" invariants $a, b \geq 0$ by $a^2 - b^2 = 2\mathcal{F}$, $ab = |\mathcal{G}|$; in a frame with $\mathbf{E} \parallel \mathbf{B}$ they are the magnitudes $a = |\mathbf{B}|$, $b = |\mathbf{E}|$. The magnetic calculation above gives the coth from a; a pure electric field is obtained by the analytic continuation $B \to iE$, which turns coth $\to$ cot. The general constant-field Lagrangian is therefore

$$\mathcal{L}^{(1)} = -\frac{1}{8\pi^2}\int_0^\infty \frac{\mathrm{d}s}{s^3}\, e^{-m^2 s}\Big[\,(eas)(ebs)\coth(eas)\cot(ebs) - 1 - \tfrac{1}{3}(es)^2(a^2-b^2)\,\Big]. \tag{38.83}$$

We need only the leading term beyond the subtractions. Expand the two factors with $z\coth z = 1 + \frac{z^2}{3} - \frac{z^4}{45} + \cdots$ and $z\cot z = 1 - \frac{z^2}{3} - \frac{z^4}{45} - \cdots$, writing $u = (eas)^2$, $v = (ebs)^2$:

$$\begin{aligned}(eas)\coth(eas)\,(ebs)\cot(ebs) &= \Big(1 + \tfrac{u}{3} - \tfrac{u^2}{45}\Big)\Big(1 - \tfrac{v}{3} - \tfrac{v^2}{45}\Big) + \cdots \\ &= 1 + \tfrac{1}{3}(u - v) - \tfrac{1}{45}(u^2 + v^2) - \tfrac{1}{9}uv + \cdots .\end{aligned} \tag{38.84}$$

The constant cancels the "-1"; the term $\frac{1}{3}(u - v) = \frac{1}{3}(es)^2(a^2 - b^2)$ cancels the charge-renormalization subtraction exactly. The first surviving term is quartic,

$$\big[\cdots\big]_{\text{quartic}} = -\tfrac{1}{45}(u^2 + v^2) - \tfrac{1}{9}uv = -\frac{(es)^4}{45}\big[(a^4 + b^4) + 5a^2b^2\big], \tag{38.85}$$

using $\frac{1}{9} = \frac{5}{45}$. The proper-time integral is now elementary, $\int_0^\infty \mathrm{d}s\, s\, e^{-m^2 s} = 1/m^4$, so

$$\mathcal{L}^{(1)}_{\text{quartic}} = -\frac{1}{8\pi^2}\Big(-\frac{e^4}{45}\Big)\big[(a^4{+}b^4){+}5a^2b^2\big]\int_0^\infty \mathrm{d}s\, s\, e^{-m^2 s} = \frac{e^4}{360\pi^2 m^4}\big[(a^4{+}b^4){+}5a^2b^2\big]. \tag{38.86}$$

Convert to invariants with $a^4 + b^4 = (a^2 - b^2)^2 + 2a^2b^2 = 4\mathcal{F}^2 + 2\mathcal{G}^2$ and $5a^2b^2 = 5\mathcal{G}^2$, so $(a^4 + b^4) + 5a^2b^2 = 4\mathcal{F}^2 + 7\mathcal{G}^2$, and use $e^4 = (4\pi\alpha)^2 = 16\pi^2\alpha^2$

so that $e^4/360\pi^2 = 2\alpha^2/45$. The result is the Euler–Heisenberg Lagrangian,

$$\boxed{\mathcal{L}^{(1)}_{\rm EH} = \frac{2\alpha^2}{45\,m_e^4}\Big[\,4\mathcal{F}^2 + 7\mathcal{G}^2\,\Big] = \frac{2\alpha^2}{45\,m_e^4}\Big[(E^2-B^2)^2 + 7(\mathbf{E}\cdot\mathbf{B})^2\Big]\,.} \tag{38.87}$$

Reading off the coefficients of the covariant operators in Eq. (38.77), with $4\mathcal{F}^2 = \frac{1}{4}(F_{\mu\nu}F^{\mu\nu})^2$ and $7\mathcal{G}^2 = \frac{7}{16}(F_{\mu\nu}\widetilde{F}^{\mu\nu})^2$,

$$c_1 = \frac{\alpha^2}{90\,m_e^4}, \qquad c_2 = \frac{7\alpha^2}{360\,m_e^4}. \tag{38.88}$$

These are the matching coefficients: the photon EFT below m_e reproduces low-energy QED if and only if its dimension-eight couplings take the values Eq. (38.88). Three features close the worked example.

First, the *decoupling scaling* is manifest: $c_1, c_2 \sim \alpha^2/m_e^4$, dimension-eight coefficients suppressed by four powers of the heavy mass. As $m_e \to \infty$ they vanish — alternative (ii) of Theorem 38.10, displayed for an operator of definite dimension. The heavy electron's only low-energy legacy in the pure-photon sector, beyond the charge renormalization it already supplied, is this $1/m_e^4$-suppressed light-by-light interaction.

Second, the result is a *matching coefficient*, not an observable. It is the hard region of the four-photon box Fig. 38.6 at zero external momentum: the loop momentum of order m_e Taylor-expanded in the soft photon momenta, which is exactly the prescription Eq. (38.66). Physical photon–photon scattering at $\omega \ll m_e$ is then computed in the EFT from the contact vertex Eq. (38.87), giving a cross section $\sigma_{\gamma\gamma\to\gamma\gamma} \sim \alpha^4\omega^6/m_e^8$ — the six powers of energy from the dimension-eight vertex, the steep suppression that makes light-by-light scattering so faint and so distinctively a probe of the EFT's leading irrelevant operator.

Third, the *two* coefficients are independent because parity and Lorentz invariance allowed two independent quartic invariants; the loop computes their ratio $c_2/c_1 = 7/4$, a sharp prediction of QED that any putative ultraviolet completion would have to reproduce to masquerade as the electron at low energy.

With the Wilson coefficients now fixed at the matching scale $\mu \sim M$ — the heavy-fermion threshold step Eq. (38.75) for the gauge coupling, and the Euler–Heisenberg couplings Eq. (38.88) for the photon EFT — the large logarithm $\ln(\mu/E)$ of Eq. (38.67) remains to be summed. That summation is the renormalization group *inside* the effective theory: the coefficients $C_i(\mu)$ are not constants but run, and operators of the same dimension mix as they do. Evolving them from the matching scale down to the low scale is the subject of Section 38.5, to which we now turn.

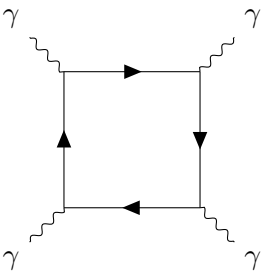


Figure 38.6: The light-by-light box: the electron loop with four external photons. At external photon momenta soft compared with m_e, its hard region (loop momentum $\sim m_e$, Taylor-expanded in the external momenta) is the matching coefficient of the dimension-eight operators in Eq. (38.77), evaluated in Theorem 38.11 to give the Euler–Heisenberg couplings Eq. (38.88). The three-photon amplitude vanishes by charge-conjugation (Furry's theorem), so this is the leading photon self-interaction.

38.5 Running and Mixing of Wilson Coefficients

Matching (Section 38.3) delivered the Wilson coefficients C_i at the heavy scale M: there, where the full theory and the effective theory are both valid, the two descriptions were forced to agree on a set of low-energy amplitudes, and the $C_i(M)$ were read off. But M is almost never the scale at which an experiment is performed. A rare meson decay, a low-energy scattering length, a precision electroweak observable — these live at a scale $E \ll M$, and a calculation done with the coefficients frozen at their matching values is riddled with large logarithms $\ln(M/E)$. Every loop in the effective theory that dresses an operator insertion produces such a logarithm, and at n loops a factor $\left(\frac{\lambda}{16\pi^2}\ln\frac{M}{E}\right)^n$ appears; when the hierarchy M/E is large the logarithm can overwhelm the loop factor and ruin naive perturbation theory. The cure is the one Part VI built for couplings and masses: run the parameters with the renormalization group. The novelty here is that the objects that run are not a single coupling but a whole tower of Wilson coefficients, and that under renormalization these coefficients do not run independently — they *mix*. The aim of this section is to derive the matrix renormalization-group equation that governs them, to compute a representative anomalous dimension in the scalar toy theory, and to integrate the equation so as to resum the tower of logarithms left behind by the matching.

38.5.1 Operator mixing under renormalization

Recall the structure established in Section 38.2. Below the heavy scale the dynamics is governed by an effective Lagrangian

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\le 4} + \sum_i \frac{C_i}{\Lambda^{\,d_i-4}}\,\mathcal{O}_i, \tag{38.89}$$

where $\mathcal{L}_{\le 4}$ collects the relevant and marginal operators (for ϕ^4 theory, the kinetic, mass, and quartic terms of Eq. (24.161)), the $\mathcal{O}_i$ are the higher-dimension local operators built from the light fields and allowed by the symmetries, $d_i \ge 5$ is the mass dimension of $\mathcal{O}_i$, and the C_i are the dimensionless Wilson coefficients fixed by matching at $\mu = M$. The expansion parameter is E/Λ, and the cutoff scale Λ is identified with the mass M of the heavy states that were integrated out.

In a renormalizable theory a single coupling is renormalized multiplicatively (Section 22.3): the bare coupling is the renormalized one times a Z factor that absorbs the divergences. For the higher operators the same idea holds, but multiplication by a number becomes multiplication by a *matrix*. The reason is structural. When an operator $\mathcal{O}_j$ is inserted into a Green's function and the loop integrals are done, the ultraviolet divergence is a local polynomial in the external momenta, and locality forces it to have the form of *some* local operator of the same dimension built from the same fields. There is no symmetry principle that restricts the divergence to be proportional to $\mathcal{O}_j$ itself: any operator $\mathcal{O}_i$ carrying the same quantum numbers and the same (or lower) dimension can appear. A counterterm proportional to $\mathcal{O}_i$ is then required to cancel the divergence, and one says that $\mathcal{O}_j$ *mixes into* $\mathcal{O}_i$. The set of operators of a given dimension and symmetry therefore renormalizes as a closed multiplet.

Concretely, write the bare operators (built from bare fields) in terms of the renormalized ones through a matrix of renormalization constants,

$$\mathcal{O}_i^{(0)} = Z_{ij}\,\mathcal{O}_j, \qquad Z_{ij} = \delta_{ij} + \frac{1}{\bar{\varepsilon}}\,z_{ij}(\{g\}) + \cdots, \tag{38.90}$$

the ellipsis denoting higher poles, $\bar{\varepsilon}$ the $\overline{\text{MS}}$ pole of Eq. (24.55), and $\{g\}$ the couplings of $\mathcal{L}_{\le 4}$. The off-diagonal entries z_{ij} $(i \ne j)$ encode the mixing; they vanish at tree level and are generated by loops. In a mass-independent scheme such as $\overline{\text{MS}}$ the matrix Z has a crucial triangular property: an operator can mix only into operators of equal or *lower* dimension, never higher. Mixing into a strictly lower-dimension operator necessarily drags along an explicit factor of a dimensionful coupling (a mass, or the scale Λ) to make up the dimensions, so the genuinely dimensionless, scale-free part of the mixing matrix is block-diagonal in the operator dimension: operators of equal dimension mix among themselves through a pure number times a coupling. It is this equal-dimension block that carries the leading renormalization-group evolution, and we focus on it first.

38.5.2 The anomalous-dimension matrix and the coefficient RGE

Definition 38.12 (Anomalous-dimension matrix)**.** For a multiplet of operators renormalizing as in Eq. (38.90), the *anomalous-dimension matrix* is

$$\gamma_{ij} \;\equiv\; \big(Z^{-1}\big)_{ik}\,\mu\frac{\mathrm{d}}{\mathrm{d}\mu}\,Z_{kj} \;=\; \mu\frac{\mathrm{d}Z_{ij}}{\mathrm{d}\mu} + O(z^2). \tag{38.91}$$

It is the matrix generalization of the single-operator anomalous dimension of Theorems 23.12 and 24.2: the diagonal entry γ_{ii} is the rate at which $\mathcal{O}_i$ renormalizes itself, and γ_{ij} $(i \neq j)$ is the rate at which $\mathcal{O}_j$ feeds into $\mathcal{O}_i$.

The matrix γ is finite even though Z is divergent: the explicit $1/\bar{\varepsilon}$ in Eq. (38.90) is cancelled by the $\bar{\varepsilon}$ hidden in the μ-dependence of the couplings. Indeed, in $d = 4 - 2\varepsilon$ the bare coupling $\lambda_0 = \mu^{2\varepsilon} Z_\lambda \lambda$ enforces $\mu\,\mathrm{d}\lambda/\mathrm{d}\mu = -2\varepsilon\lambda + \beta_\lambda$ (Theorem 24.1), so acting with $\mu\,\mathrm{d}/\mathrm{d}\mu$ on a coefficient $z_{ij} \propto \lambda$ produces $z'_{ij}\,(-2\varepsilon\lambda)/\bar{\varepsilon}$, and the ε in the numerator cancels the $1/\bar{\varepsilon}$ to leave a finite, μ-independent number. This is the same mechanism by which the ϕ^4 beta function emerged finite from a divergent Z_λ in Chapter 24; the bookkeeping is reviewed in the worked example below.

We can now state the central result of this section, the equation that turns a set of matching coefficients into a prediction at the low scale.

Theorem 38.13 (Wilson-coefficient renormalization-group equation)**.** *Let the higher-dimension part of the effective Lagrangian be $\sum_i C_i\,\mathcal{O}_i$, with the operators renormalized as in Eq. (38.90). Demanding that the bare effective Lagrangian, and hence every physical correlator computed from it, be independent of the renormalization scale μ, the renormalized Wilson coefficients obey*

$$\mu\frac{\mathrm{d}C_i}{\mathrm{d}\mu} = \gamma_{ji}\,C_j, \tag{38.92}$$

with γ the anomalous-dimension matrix of Theorem 38.12. The coefficients run with the transpose *of the matrix that renormalizes the operators.*

Proof. The physical content of the effective Lagrangian resides in the combination $\sum_i C_i^{(0)}\mathcal{O}_i^{(0)}$ built from bare coefficients and bare operators; this combination is what appears in the action and is manifestly independent of the arbitrary scale μ introduced by renormalization. Substituting Eq. (38.90),

$$\sum_i C_i^{(0)}\mathcal{O}_i^{(0)} = \sum_{i,j} C_i^{(0)} Z_{ij}\,\mathcal{O}_j = \sum_j \Big(\sum_i C_i^{(0)} Z_{ij}\Big)\mathcal{O}_j. \tag{38.93}$$

For the coefficient of each renormalized operator $\mathcal{O}_j$ to be finite we identify the renormalized Wilson coefficient

$$C_j \;=\; \sum_i C_i^{(0)} Z_{ij} \qquad\Longleftrightarrow\qquad C = Z^{\top} C^{(0)}, \tag{38.94}$$

in matrix notation, with C the column of renormalized coefficients and $C^{(0)}$ the column of bare ones. The bare coefficients carry the physics fixed at the matching scale and are μ-independent: $\mu\,\mathrm{d}C^{(0)}/\mathrm{d}\mu = 0$. Inverting Eq. (38.94), $C^{(0)} = (Z^{\top})^{-1}C$, and differentiating,

$$0 = \mu\frac{\mathrm{d}C^{(0)}}{\mathrm{d}\mu} = \Big(\mu\frac{\mathrm{d}(Z^{\top})^{-1}}{\mathrm{d}\mu}\Big)C + (Z^{\top})^{-1}\,\mu\frac{\mathrm{d}C}{\mathrm{d}\mu}. \tag{38.95}$$

Use $\mu\,\mathrm{d}Z^{-1}/\mathrm{d}\mu = -Z^{-1}(\mu\,\mathrm{d}Z/\mathrm{d}\mu)Z^{-1} = -\gamma\,Z^{-1}$, which transposes to $\mu\,\mathrm{d}(Z^{\top})^{-1}/\mathrm{d}\mu = -(Z^{\top})^{-1}\gamma^{\top}$. Inserting this into Eq. (38.95) and multiplying on the left by $Z^{\top}$,

$$0 = -\gamma^{\top}C + \mu\frac{\mathrm{d}C}{\mathrm{d}\mu}, \tag{38.96}$$

which is Eq. (38.92) in components, $\mu\,\mathrm{d}C_i/\mathrm{d}\mu = \gamma_{ji}C_j$. □

Equation (38.92) is the Callan–Symanzik equation of the effective theory. Its derivation mirrors the one in Theorem 24.19 step for step: there, μ-independence of the bare n-point function forced a first-order equation for the renormalized Green's function with the field anomalous dimension γ_ϕ as the inhomogeneous term; here, μ-independence of the bare coefficients forces a first-order equation for the renormalized coefficients with the operator anomalous-dimension matrix as the generator. The single difference that matters in practice is that γ has become a matrix, because the basis of higher-dimension operators is not stable under renormalization: scale evolution rotates within it. One may verify the consistency of the two viewpoints directly: since a physical amplitude is $\mathcal{A} = \sum_i C_i\,\langle\mathcal{O}_i\rangle$ and is μ-independent, the running Eq. (38.92) of the coefficients is exactly compensated by the running of the renormalized matrix elements $\langle\mathcal{O}_i\rangle$, whose scale dependence is governed by the same γ with the opposite (untransposed) action. The coefficient and the operator carry equal and opposite anomalous dimensions, and only their product is physical — the hallmark of a renormalization-group statement.

38.5.3 Leading-logarithmic resummation

With the RGE in hand we evolve the coefficients from M down to the low scale. Write the couplings of $\mathcal{L}_{\le 4}$ collectively as g; the anomalous-dimension matrix depends on μ only through $g(\mu)$, so it is convenient to trade the scale for the running coupling. Using $\mu\,\mathrm{d}/\mathrm{d}\mu = \beta(g)\,\mathrm{d}/\mathrm{d}g$, Eq. (38.92) becomes $\beta(g)\,\mathrm{d}C/\mathrm{d}g = \gamma^{\top}(g)\,C$, with formal solution the path-ordered exponential

$$C(\mu) = \mathrm{P}\exp\left[\int_{g(M)}^{g(\mu)}\frac{\gamma^{\top}(g')}{\beta(g')}\,\mathrm{d}g'\right]C(M), \tag{38.97}$$

the ordering being necessary only when γ at different scales fails to commute with itself, which first happens beyond leading order. The structure is the matrix

version of the closed-form RG solution Eq. (24.188): the evolution operator is the exponential of the integrated anomalous dimension, and it resums an entire tower of logarithms into a finite factor.

The content of Eq. (38.97) is clearest in the diagonal case, which already arises whenever a single operator is multiplicatively renormalized or one works in the eigenbasis of γ. Take a gauge coupling, for which at one loop $\beta = -2\beta_0\,\alpha^2/(4\pi) + \cdots$ and the relevant eigen-anomalous dimension is $\gamma = \gamma_0\,\alpha/(4\pi) + \cdots$. Then $\gamma/\beta = -\gamma_0/(2\beta_0\,\alpha)$, and Eq. (38.97) integrates to the canonical leading-log power law,

$$C(\mu) = \left[\frac{\alpha(\mu)}{\alpha(M)}\right]^{-\gamma_0/2\beta_0} C(M). \tag{38.98}$$

A single, exact-looking power of the running coupling has replaced the divergent series $\sum_n \frac{1}{n!}\big(-\frac{\gamma_0\alpha}{4\pi}\ln\frac{M}{\mu}\big)^n$ of fixed-order logarithms: this is what "resumming the leading logs" means. Expanding Eq. (38.98) to first order reproduces the one-loop term $C(M)\big[1 - \frac{\gamma_0}{2\beta_0}\ln\frac{\alpha(\mu)}{\alpha(M)}\big] \simeq C(M)\big[1 - \frac{\gamma_0\alpha}{4\pi}\ln\frac{M}{\mu}\big]$, the very logarithm that the matching of Section 38.3 would have left uncancelled had we frozen the coefficient at M.

Physical Insight

Equation (38.98) crystallizes the way an effective field theory organizes its predictions as a *double expansion*. Schematically, any EFT amplitude is

$$\mathcal{A} \;\sim\; \sum_{n\geq 0}\Big(\frac{E}{\Lambda}\Big)^n \sum_{L\geq 0}\Big(\frac{g^2}{16\pi^2}\Big)^L \times (\text{logarithms, resummed by the RG}). \tag{38.99}$$

The first sum is the *operator* expansion: the power of E/Λ counts the dimension of the operators retained, and is controlled by power counting and the truncation of the operator basis. The second sum is the *loop* expansion in the light couplings, which generates the logarithms that the renormalization group sums to all orders into ratios of running couplings. The two expansions are independent. One improves the accuracy in E/Λ by keeping higher operators and matching them; one improves the accuracy in $g^2/16\pi^2$ by computing more loops of matching and running. A complete EFT prediction is the product of three ingredients — matching at M, running to E, and truncation at a chosen order in E/Λ — and this section has supplied the middle one.

38.5.4 Scalar ϕ^4 theory: running and mixing of the dimension-six coefficient

We now make all of this concrete in the scalar toy EFT, computing the one-loop running of its leading higher-dimension coefficient with exactly the $\overline{\text{MS}}$ machinery used for λ in Chapter 24. Recall (Section 38.2) that integrating a heavy scalar out of the $\mathbb{Z}_2$-symmetric theory leaves, beyond the renormalizable Lagrangian of Eq. (24.161), the leading irrelevant operator

$$\mathcal{L}_{\text{eff}} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4 - \frac{C_6}{6!}\frac{\phi^6}{\Lambda^2}, \qquad \mathcal{O}_6 \equiv \phi^6, \tag{38.100}$$

with ϕ^6 the unique dimension-six operator allowed by the symmetry once the equations of motion and integration by parts are used to eliminate redundant structures. We compute its anomalous dimension at one loop.

Worked Example 38.14 (One-loop anomalous dimension of ϕ^6 and its leading-log evolution)**. The combinatorial lemma.** The only one-loop divergence linear in the quartic coupling λ that renormalizes a composite operator ϕ^n comes from a single graph topology: two of the operator's n legs are contracted into a loop through one $\lambda\phi^4/4!$ vertex, whose remaining two legs become external, while the other $n-2$ operator legs are spectators (Fig. 38.7, left). The loop is the scalar bubble

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2 - m^2 + i\epsilon}\,\frac{i}{(\ell+P)^2 - m^2 + i\epsilon} = -\frac{i}{16\pi^2}\,\frac{1}{\bar{\epsilon}} + \text{finite}, \tag{38.101}$$

the same integral whose $\overline{\text{MS}}$ value was assembled in Eq. (24.55). Because every term in this graph except the choice of which leg-pair enters the loop is identical regardless of n, the divergent renormalization of ϕ^n is simply $\binom{n}{2}$ — the number of leg-pairs — times the renormalization of ϕ^2.

We verify the claim by carrying out the Wick contractions explicitly. Insert $\phi^n(0)$ and contract it, together with one factor of $i(-\frac{\lambda}{4!})\int \mathrm{d}^4y\,\phi(y)^4$, in the bubble topology. The count factorizes. First choose which two of the n operator legs enter the loop: $\binom{n}{2}$ ways. Join this pair to the vertex by contracting each chosen leg with one of the four vertex fields; the two operator legs can be matched to two of the four vertex fields in $\binom{4}{2}\cdot 2! = 4\cdot 3 = 12$ ordered ways. The remaining two vertex fields and the $n-2$ spectator operator legs are then routed to the external points, a routing structurally identical for every n. Only the first factor depends on n, so the divergent weight is $\binom{n}{2}\cdot 12$: for $n=2$ it is $\binom{2}{2}\cdot 12 = 12$ and for $n=6$ it is $\binom{6}{2}\cdot 12 = 180$, in the ratio $\binom{6}{2}/\binom{2}{2} = 180/12 = 15$. This confirms the lemma.

Anchoring the normalization. The ϕ^2 insertion is nothing but the mass operator, whose anomalous dimension was computed in the Callan–Symanzik layer of ϕ^4 theory:

$$\gamma_{\phi^2} = \frac{\lambda}{16\pi^2} + O(\lambda^2), \tag{38.102}$$

the same number as the mass-squared running rate γ_{m^2} of Eq. (24.78). Explicitly, the ϕ^2 insertion requires the operator renormalization $Z_{\phi^2} = 1 - \frac{\lambda}{32\pi^2\bar{\varepsilon}} + \cdots$; differentiating with Eq. (38.91) and using $\mu\,\mathrm{d}\lambda/\mathrm{d}\mu = -2\varepsilon\lambda + O(\lambda^2)$ to kill the pole,

$$\gamma_{\phi^2} = \mu\frac{\mathrm{d}}{\mathrm{d}\mu}\left(-\frac{\lambda}{32\pi^2\bar{\varepsilon}}\right) = -\frac{1}{32\pi^2\bar{\varepsilon}}\,(-2\varepsilon\lambda) = \frac{\lambda}{16\pi^2}, \tag{38.103}$$

the ε in the numerator cancelling the $1/\bar{\varepsilon}$ exactly as advertised after Eq. (38.91).

The result for ϕ^6. By the lemma, the dimension-six operator renormalizes with $Z_{66} = 1 - \binom{6}{2}\frac{\lambda}{32\pi^2\bar{\varepsilon}} = 1 - \frac{15\lambda}{32\pi^2\bar{\varepsilon}}$, hence

$$\boxed{\gamma_{\phi^6} = \binom{6}{2}\frac{\lambda}{16\pi^2} = \frac{15\lambda}{16\pi^2} + O(\lambda^2).} \tag{38.104}$$

This is the representative anomalous dimension of the scalar EFT. By Theorem 38.13 the Wilson coefficient runs with the same number (the 1×1 block is its own transpose),

$$\mu\frac{\mathrm{d}C_6}{\mathrm{d}\mu} = \gamma_{\phi^6}\,C_6 = \frac{15\lambda}{16\pi^2}\,C_6. \tag{38.105}$$

Leading-log evolution. Trade μ for the running coupling using the ϕ^4 beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Eq. (24.46):

$$\frac{\mathrm{d}C_6}{C_6} = \frac{\gamma_{\phi^6}}{\beta_\lambda}\,\mathrm{d}\lambda = \frac{15\lambda/16\pi^2}{3\lambda^2/16\pi^2}\,\mathrm{d}\lambda = 5\,\frac{\mathrm{d}\lambda}{\lambda}, \tag{38.106}$$

which integrates to the leading-log resummed coefficient

$$C_6(\mu) = C_6(M)\left[\frac{\lambda(\mu)}{\lambda(M)}\right]^5, \qquad 5 = \frac{\gamma_{\phi^6}}{\beta_\lambda} = \frac{15}{3}. \tag{38.107}$$

This is the scalar instance of the diagonal power law Eq. (38.98), with the role of α played by λ and the exponent fixed by the ratio of the two one-loop coefficients. Since the ϕ^4 coupling is marginally relevant and decreases toward the infrared (Eq. (24.210)), the factor $[\lambda(\mu)/\lambda(M)]^5$ shrinks the coefficient as one runs down to the low scale — the dimension-six interaction is even more strongly suppressed at low energies than the bare power counting suggested.

Resummation of the matching logarithm. It remains to check that Eq. (38.107) resums precisely the logarithm that the matching of Section 38.3 left behind. Using the running coupling Eq. (24.210), expand for $\mu < M$: $\lambda(\mu)/\lambda(M) \simeq 1 - \frac{3\lambda}{16\pi^2}\ln\frac{M}{\mu}$, so

$$C_6(\mu) \simeq C_6(M)\left[1 - \frac{3\lambda}{16\pi^2}\ln\frac{M}{\mu}\right]^5 \simeq C_6(M)\left[1 - \frac{15\lambda}{16\pi^2}\ln\frac{M}{\mu}\right]. \tag{38.108}$$

The bracketed first-order term is exactly the one-loop logarithm that a fixed-order EFT calculation with the matching value $C_6(M)$ produces: the bubble graph

Fig. 38.7 dresses the ϕ^6 vertex with $\binom{6}{2} = 15$ quartic insertions, each contributing $-\frac{\lambda}{16\pi^2}\ln\frac{M}{\mu}$ once the $1/\bar{\varepsilon}$ pole is removed by operator renormalization and μ is set to the matching scale. The renormalization group has promoted that single logarithm into the full tower $(\frac{15\lambda}{16\pi^2}\ln\frac{M}{\mu})^n/n!$, summed into the fifth power of the coupling ratio. Matching computes $C_6(M)$; Eq. (38.107) carries it to the scale of measurement.

(a) $\mathcal{O}_6 \to \mathcal{O}_6$ (b) $\mathcal{O}_6 \to \mathcal{O}_4$

Figure 38.7: One-loop renormalization of the dimension-six operator $\mathcal{O}_6 = \phi^6$ (square vertex) in the scalar EFT. (a) Self-renormalization: two of the six legs are dressed by a quartic vertex through the bubble of Eq. (38.101), leaving six external legs; there are $\binom{6}{2} = 15$ choices of leg-pair, giving $\gamma_{\phi^6} = 15\lambda/16\pi^2$. (b) Mixing down: a self-contraction of two legs forms a massive tadpole, leaving four external legs and renormalizing the dimension-four coupling with an explicit factor m^2/Λ^2.

38.5.5 Mixing down, and why the matrix is triangular

The computation above isolated the equal-dimension (here, one-dimensional) block of the anomalous-dimension matrix. The dimension-six operator also generates divergences proportional to operators of *lower* dimension, and it is worth seeing why these do not spoil the clean power law Eq. (38.107) and how they fit the triangular structure asserted after Eq. (38.90).

Self-contracting two of the six legs of ϕ^6 into a single loop — a tadpole, Fig. 38.7(b) — leaves the dimension-four operator ϕ^4. The loop is the massive tadpole

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2 - m^2 + i\epsilon} = \frac{m^2}{16\pi^2}\,\frac{1}{\bar{\varepsilon}} + \text{finite}, \tag{38.109}$$

and there are again $\binom{6}{2} = 15$ leg-pairs to contract, so ϕ^6 mixes into ϕ^4 with a divergence proportional to $C_6\, m^2/\Lambda^2$. Contracting two independent pairs leaves ϕ^2, mixing ϕ^6 into the mass operator with a coefficient proportional to $C_6\, m^4/\Lambda^2$ (or $C_6\,\Lambda^0$ once the quadratically divergent piece is tracked in a momentum cutoff; in $\overline{\text{MS}}$ the divergence is the logarithmic one shown). The decisive point is the explicit factor of m^2: in the mass-independent scheme $\overline{\text{MS}}$ the only dimensionful quantity that can appear in a counterterm is an explicit coupling of the Lagrangian, and lowering the operator dimension by two units

therefore costs one power of m^2. The anomalous-dimension matrix in the basis (ϕ^2, ϕ^4, ϕ^6) is consequently lower triangular,

$$\gamma = \begin{pmatrix} \gamma_{22} & 0 & 0 \\ * \dfrac{m^2}{\Lambda^2} & \gamma_{44} & 0 \\ * \dfrac{m^4}{\Lambda^2} & * \dfrac{m^2}{\Lambda^2} & \gamma_{66} \end{pmatrix}, \tag{38.110}$$

with the pure numbers $\gamma_{22} = \lambda/16\pi^2$, $\gamma_{66} = 15\lambda/16\pi^2$ on the diagonal and the mass-suppressed entries below it. Because the matrix is triangular, its eigenvalues are the diagonal entries, and the evolution of C_6 decouples from the lower operators up to corrections suppressed by m^2/Λ^2 — which are formally of the same order as the higher operators we have already truncated. To the accuracy at which the dimension-six operator is being kept, the running of its coefficient is governed by the single diagonal entry Eq. (38.104), and Eq. (38.107) is exact at leading-log order.

In a richer theory the equal-dimension block is genuinely a non-diagonal matrix: several operators of the same dimension share the same quantum numbers and rotate into one another under the flow. The Warsaw-basis dimension-six operators of the Standard Model effective theory are the working example, and their large anomalous- dimension matrix is the engine of precision constraints on physics above the electroweak scale; we take it up in Chapter 40. The scalar toy has shown the mechanism in its simplest incarnation, where the matrix collapses to a number but the logic — bare μ-independence, an operator renormalization matrix, a transposed coefficient RGE, and a resummed power of the running coupling — is identical.

With matching (Section 38.3), running (Eq. (38.92)), and truncation (Section 38.2) now all in hand, the effective field theory is a complete predictive instrument: a finite set of coefficients fixed at the high scale, evolved to the scale of measurement, and read off to a chosen order in E/Λ. What remains is to step back and ask what this machinery teaches about the operators it does *not* suppress — the relevant operators, the mass terms, whose coefficients grow toward the infrared and whose ultraviolet sensitivity is the subject of naturalness, to which we now turn.

38.6 Naturalness, Relevant Operators, and the Hierarchy Problem

The running of Section 38.5 completed the predictive loop: a heavy theory matched at Λ, organized by power counting, and run down to the scale of an experiment yields finite, scheme-independent predictions for any process below Λ. The machinery works. What remains is to face squarely a fact it has been

quietly presupposing. Every coefficient we have run is a number we read off at the matching scale; the renormalization group propagates it but does not predict it. For the irrelevant (higher-dimension) operators this is no embarrassment — their coefficients are suppressed by powers of E/Λ and their precise values barely matter at low energy, which was the whole content of the decoupling theorem. The relevant operators are another story entirely. They are the directions in theory space that the Wilsonian flow of Chapter 23 taught us *grow* toward the infrared, the directions that must be tuned to land on a fixed point's critical surface. In an EFT with a physical heavy scale, that tuning is no longer an abstract statement about an idealized flow: it is a tuning of the light theory's parameters against the heavy mass, and whether or not it is innocuous decides whether the light theory makes sense on its own terms. This section reads the relevant operators of Chapter 23 back as the ultraviolet-sensitive directions of the EFT, states 't Hooft's criterion for when their smallness is natural, and uses the two benchmark theories to exhibit the criterion's two faces: the scalar mass, which nothing protects, and the electron mass, which chiral symmetry does. It closes on the two relevant operators of the Standard-Model EFT for which the criterion fails and no protection is known — the hierarchy and cosmological-constant problems — stated as precise EFT-matching facts and dated, honestly, as open.

38.6.1 Relevant operators as the ultraviolet-sensitive directions

We begin by translating the language of Chapter 23 into the language of matching. The classification of Theorem 23.23 sorted every local operator by the eigenvalue y_a of the linearized flow around a fixed point: relevant ($y_a > 0$), marginal ($y_a = 0$), irrelevant ($y_a < 0$). For the free (Gaussian) fixed point that organizes a weakly coupled EFT, y_a is the canonical statement $y_a = d - \Delta_a$, with Δ_a the engineering dimension of the operator $\mathcal{O}_a$; equivalently, writing the action as $\int d^d x \sum_a g_a \mathcal{O}_a$, the coupling g_a has mass dimension $[g_a] = d - \Delta_a = y_a$. A *relevant* coupling is therefore one with positive mass dimension. In four dimensions there are exactly three relevant or marginal operator types available to a theory of scalars and fermions: the identity operator $\mathbf{1}$ ($\Delta = 0$, so $[g] = 4$), a scalar mass operator ϕ^2 ($\Delta = 2$, $[g] = 2$), and the dimensionless marginal couplings ($\Delta = 4$, $[g] = 0$) — gauge, Yukawa, and quartic-scalar couplings. Everything else is irrelevant.

The reason the relevant couplings are dangerous is contained in Eq. (23.177), which we now reread. There the continuum limit fixed each relevant coupling at the cutoff to be

$$u_i(\Lambda) \;=\; \Big(\frac{\mu}{\Lambda}\Big)^{y_i}\,\bar{u}_i, \qquad y_i > 0, \tag{38.111}$$

with $\bar{u}_i$ the finite renormalized coupling and μ the low scale. Inverting: the renormalized value is $\bar{u}_i = (\Lambda/\mu)^{y_i}\, u_i(\Lambda)$, an *amplification* of the bare value by a positive power of the large ratio Λ/μ. To keep $\bar{u}_i$ finite and small as Λ is raised,

the bare $u_i(\Lambda)$ must be dialled toward zero with matching precision. In the Wilsonian setting of Chapter 23 this was the statement that the critical surface (Theorem 23.24) has codimension equal to the number of relevant directions, and reaching it costs one fine-tuning per relevant operator. In the EFT setting the cutoff Λ is not a regulator to be removed but a physical scale — the mass M of states we integrated out — and the powers of Λ/μ become powers of the physical hierarchy M/μ between the heavy and light scales. The same arithmetic, now with $\Lambda \to M$ a real threshold, says: a relevant coupling in the low-energy EFT receives additive contributions from the heavy sector scaling as M^{y_i}, and unless those contributions cancel against the bare value to a part in $(M/\mu)^{y_i}$, the light coupling is dragged up to the heavy scale.

This is what *ultraviolet sensitivity* means precisely. Differentiate the low-energy parameter with respect to the heavy mass: for a relevant coupling of mass dimension $y_i > 0$, dimensional analysis and decoupling together force

$$g_i^{\text{light}} \;=\; g_i^{\text{bare}} \;+\; c_i\, M^{y_i} \;+\; (\text{logs}), \tag{38.112}$$

with c_i a pure number fixed by the heavy couplings. The heavy scale enters as a positive power. By contrast a marginal coupling ($y_i = 0$) can depend on M only logarithmically, $\sim \ln(M/\mu)$, because there is no dimensionful prefactor to carry a power; and an irrelevant coupling's heavy-scale dependence is suppressed, not enhanced. The relevant operators are thus the unique directions along which the low-energy theory *remembers* the heavy scale with a power rather than a logarithm. They are the price of the decoupling theorem of Section 38.4.2: decoupling protects observables built from irrelevant operators, but it says nothing kind about the relevant ones, whose coefficients are precisely the renormalization conditions decoupling lets us choose by hand.

Caution

Equation Eq. (38.112) is sometimes phrased with M replaced by the regulator cutoff Λ, giving the notorious "quadratic divergence" $\delta m^2 \sim \Lambda^2$ of a scalar mass in a hard-cutoff scheme. The power-law cutoff dependence is a regulator artifact: in dimensional regularization there is no Λ^2 at all, only poles in ε multiplied by the *physical* masses in the loop. What is physical, and scheme-independent, is the dependence on a genuine heavy *threshold* M — the mass of a particle that really exists. The naturalness problem is the sensitivity to physical heavy scales exhibited in Eq. (38.112), not the sensitivity to the regulator. We will compute it in dimensional regularization, where the physical M^2 is manifest and no spurious Λ^2 can hide.

38.6.2 't Hooft's naturalness criterion

The arithmetic of Eq. (38.112) appears to condemn every small relevant coupling: if $\bar{u}_i \ll M^{y_i}$, the bare value must cancel the additive heavy contribution to absurd precision. Yet nature contains small relevant couplings that are evidently not fine-tuned — the electron mass is fifteen orders of magnitude below the Planck scale and no one loses sleep over it. The resolution is that Eq. (38.112) assumed the constant c_i is generic. A symmetry can force $c_i = 0$, and then the heavy-scale contribution is not additive but multiplicative in the small coupling itself. This is the content of the criterion 't Hooft elevated to a working principle.

Definition 38.15 (Technical naturalness)**.** A dimensionful or dimensionless parameter g in an effective Lagrangian is *technically natural* (in 't Hooft's sense) if the limit $g \to 0$ enhances the symmetry of the theory: the action at $g = 0$ is invariant under a symmetry that the action at $g \neq 0$ is not. Equivalently, g is natural if it is the unique (up to normalization) coupling of an operator that breaks an otherwise exact symmetry, so that every radiative correction to g must itself break that symmetry and is therefore proportional to g,

$$\delta g \;=\; g \times f\!\left(\text{couplings},\ \ln\frac{M}{\mu}\right), \tag{38.113}$$

with f at most logarithmically dependent on the heavy scale. A small g is then *stable* under the flow: the heavy scale enters only through the slowly varying logarithm in Eq. (38.113), never as an additive power as in Eq. (38.112).

The logic deserves to be made explicit, because it is the conceptual hinge of the whole subject. Suppose the $g = 0$ theory has an exact symmetry S. Then the full quantum effective action computed in that theory is S-invariant: the symmetry is respected order by order in perturbation theory, since the Feynman rules of the $g = 0$ theory carry no S-violating vertex. Turning g back on, g is the coefficient of the unique S-violating term; any quantum-generated S-violating quantity, δg included, must be built from at least one insertion of that vertex, hence carries at least one power of g. Therefore $\delta g \propto g$, which is Eq. (38.113). The would-be additive piece $c_i M^{y_i}$ of Eq. (38.112) is S-symmetric (it does not vanish as $g \to 0$), so it is forbidden: $c_i = 0$. The symmetry does not make g small — it explains why a small g *stays* small once chosen. Naturalness is a statement about radiative stability, not about the origin of the value.

Physical Insight

't Hooft's criterion converts the abstract codimension count of Theorem 23.24 into a usable principle. Each relevant direction generically costs one fine-tuning to keep the corresponding low-energy coupling small. A symmetry that is restored when the coupling vanishes *lowers the codimen-*

sion: along that direction the critical surface is reached not by tuning but by symmetry, because the flow of an S-violating coupling cannot leave the S-symmetric surface $g = 0$ on its own. Naturalness is the question of whether each relevant coupling of the Standard Model sits on such a protected surface. For all but two, the answer is yes.

We now demonstrate the two outcomes on the benchmark theories. ϕ^4 theory supplies the unprotected case: the scalar mass, a relevant coupling that no symmetry of ϕ^4 shields, suffers the full additive sensitivity of Eq. (38.112). QED supplies the protected case: the electron mass, equally relevant, is shielded by chiral symmetry and obeys the multiplicative law Eq. (38.113).

38.6.3 The unprotected scalar mass (ϕ^4)

Return to the heavy-scalar matching of Section 38.3, where the light real scalar ϕ of ϕ^4 theory was coupled to a heavy real scalar Φ of mass $M \gg m_\phi$ through the renormalizable portal interaction

$$\mathcal{L} \supset \tfrac{1}{2}(\partial\phi)^2 - \tfrac{1}{2}m_\phi^2\phi^2 + \tfrac{1}{2}(\partial\Phi)^2 - \tfrac{1}{2}M^2\Phi^2 - \frac{\lambda_{\phi\Phi}}{4}\,\phi^2\Phi^2, \tag{38.114}$$

the dimensionless coupling $\lambda_{\phi\Phi} > 0$ being one of the marginal couplings of the combined theory. Integrating Φ out at one loop, as we did for the irrelevant operators in Section 38.3, also feeds the relevant operator ϕ^2: the heavy field circulates in a loop attached to the light scalar's two-point function. We compute that contribution in full, because it is the hierarchy problem made arithmetic.

Worked Example 38.16 (The heavy-scalar tadpole and the quadratic mass sensitivity)**.** The vertex from $-\frac{1}{4}\lambda_{\phi\Phi}\,\phi^2\Phi^2$ joins two ϕ and two Φ lines; its Feynman rule is $-i\lambda_{\phi\Phi}$, the $\frac{1}{4}$ cancelling the $2! \times 2!$ ways of attaching the identical pairs of legs. The single one-loop graph correcting the ϕ propagator contracts the two Φ legs of one vertex into a loop (Fig. 38.8), leaving the two ϕ legs external. With one internal heavy propagator and a symmetry factor $\frac{1}{2}$ for the interchange of the two ends of that propagator, the one-particle-irreducible two-point function receives

$$-i\,\Sigma_\phi \;=\; (-i\lambda_{\phi\Phi}) \cdot \frac{1}{2}\int \frac{d^dk}{(2\pi)^d}\,\frac{i}{k^2 - M^2 + i\varepsilon}. \tag{38.115}$$

The loop is the standard massive tadpole integral. Using the scalar master of Eq. (D.77) at $n = 1$,

$$\int \frac{d^dk}{(2\pi)^d}\,\frac{1}{(k^2 - M^2 + i\varepsilon)} \;=\; \frac{-i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)\left(M^2\right)^{d/2-1}, \tag{38.116}$$

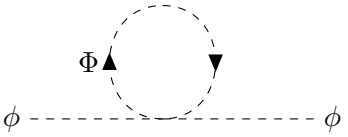


Figure 38.8: The one-loop heavy-scalar correction to the light scalar's mass. The quartic portal vertex $-\frac{1}{4}\lambda_{\phi\Phi}\phi^2\Phi^2$ attaches a closed loop of the heavy field Φ (mass M) to the ϕ two-point function. The graph evaluates to $\delta m_\phi^2 \sim \lambda_{\phi\Phi} M^2/32\pi^2$, additive in the heavy scale: the unprotected relevant direction of ϕ^4 theory.

and setting $d = 4 - 2\varepsilon$, so that $1 - \frac{d}{2} = -1 + \varepsilon$ and $(M^2)^{d/2-1} = M^2(M^2)^{-\varepsilon}$, the Laurent expansion $\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} + \gamma_E - 1 + O(\varepsilon)$ together with the $\overline{\text{MS}}$ collection $\frac{1}{\bar\varepsilon} = \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi$ (Eq. (19.79)) and the restored scale $\mu^{2\varepsilon}$ gives

$$\int \frac{d^d k}{(2\pi)^d} \frac{\mu^{2\varepsilon}}{k^2 - M^2 + i\varepsilon} \;=\; \frac{i\,M^2}{16\pi^2}\left[\frac{1}{\bar\varepsilon} + 1 - \ln\frac{M^2}{\mu^2}\right] + O(\varepsilon). \tag{38.117}$$

Insert Eq. (38.117) into Eq. (38.115), multiply the explicit i in the propagator numerator, and use $(-i)\cdot\frac{1}{2}\cdot i \cdot i = +\frac{i}{2}$:

$$\Sigma_\phi \;=\; -\frac{\lambda_{\phi\Phi}\,M^2}{32\pi^2}\left[\frac{1}{\bar\varepsilon} + 1 - \ln\frac{M^2}{\mu^2}\right]. \tag{38.118}$$

The 1PI two-point function is $\Gamma^{(2)}(p^2) = p^2 - m_\phi^2 - \Sigma_\phi$, so the physical light mass-squared, the location of the pole, is shifted by $\delta m_\phi^2 = \Sigma_\phi$ at $p^2 \simeq m_\phi^2$ (the p^2-independence of Eq. (38.118) reflects that a tadpole carries no external-momentum dependence). The pole $1/\bar\varepsilon$ is removed by the ϕ^2 counterterm exactly as in the renormalization of ϕ^4 in Section 22.6; the matching of Section 38.3 fixes the renormalized coefficient at $\mu = M$. What survives, and cannot be subtracted away, is the finite physical piece set by the heavy mass:

$$\boxed{\delta m_\phi^2 \;\sim\; \frac{\lambda_{\phi\Phi}}{32\pi^2}\,M^2} \tag{38.119}$$

(up to the $O(1)$ finite constant and the slowly varying logarithm). The light scalar's mass-squared is pulled up to the heavy scale, weighted only by the loop factor $\lambda_{\phi\Phi}/32\pi^2$.

Now ask 't Hooft's question of the scalar mass. Does $m_\phi \to 0$ enhance a symmetry of Eq. (38.114)? It does not. At $m_\phi = 0$ the ϕ sector is classically scale-invariant, but scale invariance is not a symmetry of the quantum theory — it is broken by the running of the marginal coupling, the dilatation current acquiring an anomaly proportional to the beta function (the trace anomaly of Chapter 24). No other symmetry appears: the operator ϕ^2 is a singlet under

the only internal symmetry the theory has, the $\mathbb{Z}_2$ of $\phi \to -\phi$, which leaves ϕ^2 invariant and so cannot forbid it. The mass term is not the unique breaker of any exact symmetry; it is just there, allowed by everything. The criterion Theorem 38.15 therefore returns "not protected," and Eq. (38.119) is the expected verdict: the correction is additive in M^2, not multiplicative in m_ϕ^2.

The fine-tuning is now quantitative. Suppose we want a light scalar with $m_\phi \sim v$ (think of the electroweak scale, $v \approx 246\,\mathrm{GeV}$) while the heavy threshold sits at $M \sim 10^{16}\,\mathrm{GeV}$ (a grand-unification or near-Planckian scale) with an $O(1)$ portal $\lambda_{\phi\Phi} \sim 1$. Then $\delta m_\phi^2 \sim M^2/32\pi^2 \sim 10^{30}\,\mathrm{GeV}^2$, whereas the physical $m_\phi^2 \sim v^2 \sim 6 \times 10^4\,\mathrm{GeV}^2$. The bare mass-squared and the loop contribution must cancel so that

$$\frac{m_\phi^2}{\delta m_\phi^2} \;\sim\; \frac{v^2}{M^2/32\pi^2} \;\sim\; 10^{-26}, \tag{38.120}$$

a cancellation accurate to one part in 10^{26}, with no symmetry to enforce it and no reason it should survive the next order. This is the *hierarchy problem*: the light scalar mass is the unprotected relevant direction of the Standard-Model EFT, and keeping it far below any heavy threshold to which it couples requires a tuning of exactly the codimension-one form that Theorem 23.24 predicted, performed by hand at every order.

38.6.4 Chiral protection of the fermion mass (QED)

Contrast the electron. Its mass m_e is equally a relevant coupling — the operator $\bar\psi\psi$ has dimension three, $[m_e] = 1 > 0$ — and a heavy charged state to which the electron couples would, by the same dimensional logic, threaten an additive correction $\delta m_e \sim M$. We compute the one-loop correction within QED itself, where the relevant heavy scale is the regulator, and show that it is only logarithmic and strictly proportional to m_e. We then explain why, tracing the structure to chiral symmetry.

The one-loop electron self-energy was computed in full in Chapter 19; we recall its master form Eq. (19.76),

$$\Sigma_2(p) = \frac{e^2}{(4\pi)^{d/2}}\,\Gamma(\varepsilon)\,\mu^{2\varepsilon}\int_0^1 dx\,\big[(2-d)(1-x)\,\not{p} + d\,m_e\big]\,\Delta(x)^{-\varepsilon}, \tag{38.121}$$

with $\Delta(x) = -x(1-x)p^2 + x\,m_e^2 + (1-x)\mu_\gamma^2$ and the result organized as $\Sigma = A(p^2)\not{p} + B(p^2)\,m_e$. The physical mass shift is the value of the self-energy on shell, where the corrected propagator $i/(\not{p} - m_e - \Sigma)$ has its pole. Setting $\not{p} = m_e$,

$$\delta m_e \;=\; \Sigma_2(\not{p} = m_e) \;=\; \big[A(m_e^2) + B(m_e^2)\big]\,m_e. \tag{38.122}$$

The decisive feature is already visible in Eq. (38.121) before any integral is done: the A-term multiplies $\not{p} \to m_e$ and the B-term multiplies m_e outright, so *every*

term in δm_e carries an explicit factor of m_e. There is no piece of the self-energy proportional to a bare power of the cutoff with no m_e in front. To see the magnitude, take the divergent (large-logarithm) parts read off in Eqs. (19.80) and (19.81): with $\int_0^1 (1-x)\,dx = \frac{1}{2}$ and $\int_0^1 dx = 1$,

$$A_{\rm div} = -\frac{e^2}{16\pi^2}\,\frac{1}{\bar\varepsilon}, \qquad B_{\rm div} = +\frac{e^2}{4\pi^2}\,\frac{1}{\bar\varepsilon}, \tag{38.123}$$

so that

$$\delta m_e\big|_{\rm div} = (A_{\rm div}+B_{\rm div})\,m_e = \Big(-\tfrac{1}{16}+\tfrac{4}{16}\Big)\frac{e^2}{\pi^2}\,\frac{m_e}{\bar\varepsilon} = \frac{3e^2}{16\pi^2}\,\frac{m_e}{\bar\varepsilon} = \frac{3\alpha}{4\pi}\,\frac{m_e}{\bar\varepsilon}, \tag{38.124}$$

using $e^2 = 4\pi\alpha$. The $1/\bar\varepsilon$ pole is the dimensional-regularization avatar of a logarithm of the ultraviolet scale; in a theory cut off at Λ it stands for $\ln(\Lambda^2/m_e^2)$, so

$$\boxed{\delta m_e \;=\; \frac{3\alpha}{4\pi}\,m_e\,\ln\frac{\Lambda^2}{m_e^2} \;=\; \frac{3\alpha}{2\pi}\,m_e\,\ln\frac{\Lambda}{m_e}} \tag{38.125}$$

The correction is proportional to m_e and grows only as the logarithm of the heavy scale — exactly the multiplicative, log-mild form Eq. (38.113) of a protected coupling, and nothing like the additive M^2 of Eq. (38.119). Even taking Λ at the Planck scale, $\ln(\Lambda/m_e) \approx 50$, so $\delta m_e/m_e \approx (3\alpha/2\pi)\times 50 \approx 0.17$: a 17% correction, not a fine-tuning by twenty-six orders of magnitude.

It remains to explain *why* Eq. (38.121) had to come out proportional to m_e — to verify that this is symmetry, not luck. Apply 't Hooft's criterion. In the limit $m_e \to 0$ the QED Lagrangian $\bar\psi(i\not{D})\psi - \frac{1}{4}F^2$ acquires a symmetry it does not have at $m_e \neq 0$: the global chiral rotation

$$\psi \;\to\; e^{i\theta\gamma^5}\psi, \qquad \bar\psi \;\to\; \bar\psi\, e^{i\theta\gamma^5}, \tag{38.126}$$

under which the kinetic and gauge-interaction terms are invariant because γ^5 anticommutes with γ^μ (so $\bar\psi\, e^{i\theta\gamma^5}\not{D}\, e^{i\theta\gamma^5}\psi = \bar\psi\not{D}\psi$, the two factors of $e^{i\theta\gamma^5}$ cancelling through one γ^μ), while the mass term is not, since $\bar\psi\, e^{i\theta\gamma^5} e^{i\theta\gamma^5}\psi = \bar\psi\, e^{2i\theta\gamma^5}\psi \neq \bar\psi\psi$. Concretely, in the chirality basis of Eq. (11.82) the mass term reads

$$m_e\,\bar\psi\psi = m_e\,\bar\psi(P_L+P_R)\psi = m_e\,(\bar\psi_R\psi_L + \bar\psi_L\psi_R), \tag{38.127}$$

where $\psi_{L,R} = P_{L,R}\psi$ and the intertwining $\gamma^\mu P_L = P_R\gamma^\mu$ ensures the kinetic term stays chirality-diagonal, $\bar\psi i\not{D}\psi = \bar\psi_L i\not{D}\psi_L + \bar\psi_R i\not{D}\psi_R$. The mass is the *unique* operator in the renormalizable Lagrangian that links the left and right chiralities; it is precisely the breaker of the chiral symmetry Eq. (38.126), under which ψ_L and ψ_R carry opposite charge. This is the same structure that Chapter 11 identified at the level of the Dirac equation, where acting with P_L on $(\not{p} - m_e)u = 0$ gave $\not{p}(P_R u) = m_e(P_L u)$: the mass, and only the mass, couples the two handednesses.

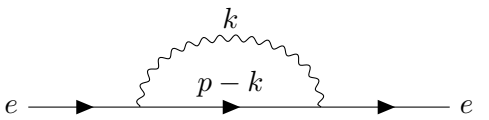


Figure 38.9: The one-loop electron self-energy of QED, Eq. (19.76). Because the only chirality-violating object in the QED Lagrangian is the mass m_e, the entire self-energy is proportional to m_e, and the correction $\delta m_e = \frac{3\alpha}{2\pi} m_e \ln(\Lambda/m_e)$ is multiplicative and only logarithmic: the chirally protected relevant direction.

By the argument under Theorem 38.15, every quantum correction to the chirality-violating operator $\bar{\psi}\psi$ must contain at least one insertion of the only chirality-violating vertex available, namely the mass itself. Hence $\delta m_e \propto m_e$, with the proportionality factor a dimensionless function of α and $\ln(\Lambda/m_e)$ — which is exactly Eq. (38.125). The factor of m_e in Eq. (38.121) is not algebraic accident but the chiral symmetry enforcing itself diagram by diagram. The smallness of the electron mass is technically natural; setting it to zero buys back an exact symmetry, and that symmetry forbids the additive correction that plagues the scalar. (We note for completeness that the chiral rotation Eq. (38.126) is anomalous in the quantum theory, Chapter 36; the anomaly affects the divergence of the chiral current but not the multiplicative renormalization of the mass, which follows from the symmetry of the perturbative vertices.) The contrast with Section 38.6.3 is the entire lesson: fermion masses are shielded by chirality, scalar masses by nothing.

38.6.5 The hierarchy problem, stated precisely

We can now phrase the hierarchy problem of the Standard Model as a clean EFT statement, with no rhetoric. The Standard Model contains exactly one elementary scalar, the Higgs doublet H, and its mass term $-\mu_H^2 H^\dagger H$ is a relevant operator of dimension two, $[\mu_H^2] = 2$. By Eq. (38.112) its coefficient is additively sensitive to the square of any heavier mass threshold to which H couples — and H couples to everything, through the top Yukawa, the electroweak gauge bosons, and itself. Reading off the largest contributions exactly as in Theorem 38.16 (now with the Standard-Model couplings in place of the toy $\lambda_{\phi\Phi}$), the one-loop shift to μ_H^2 from a threshold at scale M is

$$\delta\mu_H^2 \;\sim\; \frac{c}{16\pi^2}\, M^2, \qquad c = O(y_t^2,\ g^2,\ \lambda), \tag{38.128}$$

the direct analogue of Eq. (38.119). The measured Higgs mass fixes $\mu_H^2 \sim (88\,\text{GeV})^2$. If there is any new physics at a scale M — a grand-unified threshold near $10^{16}\,\text{GeV}$, or quantum gravity near $10^{18}\,\text{GeV}$ — coupled to the Higgs with $O(1)$ strength, then Eq. (38.128) drives μ_H^2 up to $M^2/16\pi^2$, and the observed value survives only by the tuning Eq. (38.120), a cancellation to one part in 10^{26}

or worse, repeated order by order. 't Hooft's criterion says this is unnatural unless a symmetry protects $H^\dagger H$. In the Standard Model none does: there is no chiral symmetry for a scalar, no gauge quantum number forbids the singlet $H^\dagger H$, and scale invariance is anomalous. The hierarchy problem is therefore the precise statement: *the Higgs mass is a relevant operator of the SM EFT with no protecting symmetry, so its smallness relative to any high threshold is technically unnatural.*

It is worth being honest about what this does and does not assert. It is not a logical inconsistency — the Standard Model is perfectly renormalizable and predictive with μ_H^2 taken as an input. It is a statement about the EFT's robustness: a relevant coupling tuned far from its natural scale signals, by the logic of Chapter 23, that the low-energy theory is sitting near a critical surface for no symmetry reason, which is precisely the kind of coincidence an underlying theory usually explains. The historically motivated explanations — supersymmetry, which makes the scalar a chiral superpartner so that fermionic chiral protection extends to it; compositeness, which makes the Higgs a pseudo-Goldstone boson protected by a shift symmetry; or large extra dimensions, which lower the true gravitational threshold M — are all attempts to supply the missing protecting symmetry or to remove the large M. As of the mid-2020s, the Large Hadron Collider has found no superpartners, no compositeness, and no extra dimensions at the TeV scale where naturalness most sharply expected them. The hierarchy problem is, at this writing, unsolved, and the absence of natural new physics at the weak scale is itself one of the central puzzles of the field.

38.6.6 The cosmological-constant problem

The most relevant operator of all is the one we have so far ignored because it does not affect particle scattering: the identity operator $\mathbf{1}$, with $\Delta = 0$ and coupling of mass dimension four. Its coefficient is the vacuum energy density, the cosmological constant,

$$\mathcal{L} \supset -\rho_{\text{vac}}\,\mathbf{1}, \qquad [\rho_{\text{vac}}] = 4, \tag{38.129}$$

the constant term in the Lagrangian. Being dimension-zero, it is the relevant direction with the *largest* eigenvalue $y = d = 4$, hence the most ultraviolet-sensitive coupling in the entire theory. By Eq. (38.112) every mass threshold M contributes additively at the quartic level: integrating out any field of mass M shifts the vacuum energy by the zero-point energy of its modes,

$$\delta\rho_{\text{vac}} \sim \pm\frac{M^4}{16\pi^2}, \tag{38.130}$$

with the sign set by the field's statistics (bosons and fermions contribute with opposite signs, the cancellation being exact only under unbroken supersymmetry).

Each threshold the Standard Model contains — the electron at m_e, the QCD scale, the electroweak scale, and any heavier physics up to $M_{\rm Pl}$ — pushes $\rho_{\rm vac}$ toward the fourth power of that scale. Taking the Planck scale as the ultimate cutoff gives $\rho_{\rm vac} \sim M_{\rm Pl}^4 \sim (10^{18}\,{\rm GeV})^4$.

The measured value, inferred from the accelerating expansion of the universe, is $\rho_{\rm vac} \sim (2 \times 10^{-3}\,{\rm eV})^4$. The discrepancy is

$$\frac{\rho_{\rm vac}^{\rm obs}}{M_{\rm Pl}^4} \;\sim\; 10^{-120}, \tag{38.131}$$

and even taking the most conservative cutoff at which we are confident new physics enters — the electroweak scale — the ratio is still $\sim 10^{-56}$. This is the *cosmological-constant problem*, and it is the hierarchy problem's far more severe cousin: a relevant operator whose coefficient is tuned against the heavy scales not to one part in 10^{26} but to one part in 10^{56} at the very least. The EFT statement is exactly parallel to Section 38.6.5: $\rho_{\rm vac}$ is the dimension-zero relevant operator of the SM EFT, additively sensitive to M^4 for every threshold M, and *no symmetry of the Standard Model protects it.* Supersymmetry would force the bosonic and fermionic zero-point energies in Eq. (38.130) to cancel, but supersymmetry, if realized at all, is broken at or above the TeV scale, leaving a residual $\rho_{\rm vac} \sim ({\rm TeV})^4$ still sixty orders of magnitude too large. Scale invariance would forbid Eq. (38.129) but is anomalous and, in any case, badly broken by every mass in the theory.

Caution

The cosmological-constant problem is genuinely outside the scope of the EFT methods of this chapter in one crucial respect: $\rho_{\rm vac}$ gravitates, and the EFT we have built does not include dynamical gravity. Within non-gravitational QFT the constant term in the Lagrangian is unobservable — it shifts the zero of energy and drops out of every S-matrix element — so naturalness has no purchase on it. It is only when the EFT is coupled to general relativity, where vacuum energy curves spacetime, that $\rho_{\rm vac}$ becomes a physical, measured quantity and the tuning Eq. (38.131) becomes a real problem. The cosmological-constant problem is thus the point where the EFT logic of relevant operators collides with the one interaction this book does not quantize, and it is open at a deeper level than the hierarchy problem.

As of the mid-2020s, the cosmological-constant problem has no accepted dynamical solution. The only widely discussed framework that reproduces the small observed value, the anthropic selection of a vacuum from a large landscape of possibilities, is not a symmetry explanation of the kind 't Hooft's criterion calls for, and its status is contested. We state the situation plainly: the two

relevant operators of the Standard-Model EFT that no symmetry protects — the Higgs mass and the vacuum energy — are tuned far below their natural scales for reasons not understood, and they constitute the two sharpest naturalness puzzles in fundamental physics. Honest closure of this chapter is to record them as open.

The naturalness language assembled here is the lens through which the rest of Part XI and its sequels are read. The bottom-up chiral effective theory of Chapter 39 is built precisely on a protecting symmetry — the spontaneously broken chiral symmetry of Theorem 29.1, whose Goldstone bosons are naturally light by the same mechanism that shields the electron mass — while the heavy-particle and SMEFT constructions of Chapter 40 organize the relevant and irrelevant operators of a known short-distance theory in the matching-and-running framework we have now completed. The unprotected directions catalogued here return as the central subject of the beyond-the-Standard-Model discussion of Chapter 35, where the proposed symmetries that would render the Higgs mass natural are examined against the experimental record.

Further reading

The founding statement that the most general symmetric Lagrangian reproduces the most general S-matrix is Weinberg's Weinberg [498], with the deeper logic developed across Weinberg [500, 501]; the Wilsonian roots are Polchinski [403], Wilson and Kogut [516]. Modern pedagogical accounts of the whole paradigm – matching, power counting, running, and naturalness – are the book of Burgess [99], the lectures of Manohar [335], and the concise text of Georgi [218]. The decoupling theorem is Appelquist and Carazzone [31]; Fermi's prescient effective theory is Fermi [189]; the one-loop photon effective Lagrangian computed here in the proper-time formalism originates with Heisenberg and Euler [262] and Schwinger [433]. 't Hooft's naturalness criterion is laid out in the Cargèse lectures of 't Hooft [465], and its modern EFT framing in Luty et al. [329], Polchinski [404]; the dimension-six SMEFT running that the matrix RGE of this chapter anticipates is carried out in Alonso et al. [20].

Chapter 39

Chiral Perturbation Theory

Quantum chromodynamics is asymptotically free: at short distances its coupling is small and the perturbative machinery of Chapters 25 and 27 applies verbatim. The infrared is the opposite world. Below a GeV the coupling is of order one, quarks and gluons are confined, and there is no small parameter in which to expand the dynamics of the only states that actually propagate—the hadrons. Yet the strong interactions at low energy are not intractable. They are governed, with quantitative precision, by a symmetry that QCD possesses but does not realize on its spectrum. This chapter builds the effective field theory that turns that symmetry into predictions: chiral perturbation theory, the archetype of an effective description of a strongly coupled microscopic theory, and the infrared face of the QCD story whose ultraviolet face we have already seen.

The logic is the effective-field-theory paradigm of Chapter 38 applied where it bites hardest. We begin from the chiral symmetry of massless QCD, $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V$, with the singlet axial $\mathrm{U}(1)_A$ removed by the anomaly computed at the quark level in Chapter 36; we then read off, from the hadron spectrum and from the quark condensate, that this group is spontaneously broken to its vector subgroup. Goldstone's theorem (Chapter 29) then guarantees $N_f^2 - 1$ massless pseudoscalars—the pions for two flavors, the meson octet for three—and the nonlinear realization of the broken symmetry on the coset G/H dictates that the natural field is not a vector of scalars but a group-valued field $U(x)$. The symmetry alone fixes the form of the leading interactions; the unknown strong dynamics survives only as a finite list of constants. The condensate, the decay constant f_π, the explicit breaking by quark masses through the spurion method, the Gell-Mann–Oakes–Renner relation, and the Gell-Mann–Okubo mass formula all follow from the two-derivative Lagrangian before a single loop is computed.

What makes the expansion systematic, rather than merely a parametrization, is Weinberg's power-counting theorem. Each Goldstone derivative and each power of the pion mass counts as one unit of momentum, and the superficial degree of any diagram is fixed by a single topological formula: every loop and every

higher-derivative operator is suppressed by p^2/Λ_χ^2 with $\Lambda_\chi \sim 4\pi f_\pi$. Predictivity here comes from symmetry and dimensional analysis, not from weak coupling—the same relevant–irrelevant hierarchy of Chapters 23 and 38, operating inside a theory one cannot solve in its own variables. We cash this in twice: first at leading order, where the parameter-free Weinberg amplitude for $\pi\pi$ scattering yields s-wave scattering lengths that the data confirm and whose Adler zero (Chapter 29) appears automatically; then at next-to-leading order, where one-loop pion graphs restore unitarity, generate the non-analytic chiral logarithms, and force the order-p^4 Gasser–Leutwyler Lagrangian whose low-energy constants run with an RG equation of exactly the kind built in Chapter 24.

The chapter closes by repairing one omission the symmetry construction cannot see. Every operator built from U alone is invariant under an accidental intrinsic parity that QCD does not have, so the parity-even Lagrangian forbids processes with an odd number of pseudoscalars—most conspicuously $\pi^0 \to \gamma\gamma$. The missing piece is the Wess–Zumino–Witten term, the unique addition to the chiral action that cannot be written as the integral of a locally invariant density. Its coefficient is quantized to an integer by a topological argument, that integer is the number of colors N_c, and gauging it to electromagnetism reproduces the neutral-pion decay rate to within the experimental error—a low-energy weighing of $N_c = 3$ and the cleanest illustration in this book of 't Hooft anomaly matching (Chapter 37): the anomaly of Chapter 36, computed for quarks, reappears unchanged as an interaction of mesons.

39.1 The Chiral Symmetry of QCD and Its Spontaneous Breaking

The light mesons — the three pions, the four kaons, the eta — are the softest objects in the hadron spectrum, an order of magnitude lighter than the proton and dramatically lighter than every other meson. The thesis of this chapter, announced in the chapter introduction, is that this lightness is not an accident of QCD dynamics but the inevitable signature of spontaneously broken symmetry: the pseudoscalar octet is the pseudo-Goldstone manifold of Chapter 29, realized in a strongly coupled theory. To cash that claim we must first do something entirely kinematical, owing nothing to the strong coupling: identify precisely *which* symmetry the QCD Lagrangian has when the light quarks are massless, establish that one factor of it is destroyed by the anomaly of Chapter 36, and pin down the subgroup that the vacuum leaves intact. Only the breaking pattern $G \to H$ enters the Goldstone counting; the dynamics that *drive* the breaking are confined to a single nonperturbative input, the quark condensate, whose mere nonvanishing — not its value — does all the work. This section carries out that program and hands the next section a clean statement: the field content of low-energy QCD is exactly $N_f^2 - 1$ pseudoscalar Goldstone bosons, coordinates

on the coset $(\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R)/\mathrm{SU}(N_f)_V$.

39.1.1 Massless QCD and its left–right decomposition

We begin from the QCD Lagrangian established as the QCD stage of Chapters 25 and 28 (Eq. (25.85)). Written for N_f quark flavors q_f, $f = 1, \dots, N_f$, each a Dirac field in the fundamental **3** of the gauge group $\mathrm{SU}(3)_c$ (color indices suppressed throughout, as they play no role in what follows), it reads

$$\mathcal{L}_{\mathrm{QCD}} = -\tfrac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \sum_{f=1}^{N_f} \bar{q}_f (i \not{D} - m_f) q_f, \qquad D_\mu = \partial_\mu - i g\, A^a_\mu T^a, \tag{39.1}$$

with the field strength $G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ and the conventions fixed once for the book in Chapter 25 (mostly-minus metric, Hermitian generators with $\mathrm{tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab}$, covariant derivative as quoted). The crucial empirical fact that licenses everything below is that the up, down, and to a lesser degree strange quark masses are small compared with the intrinsic scale of the strong interaction, $\Lambda_{\mathrm{QCD}} \sim 200$ MeV (Theorem 27.14): $m_u, m_d \sim$ a few MeV and $m_s \sim 100$ MeV are perturbations on the hadronic scale, while m_c, m_b, m_t are not and must be integrated out before this discussion even begins. We therefore study the idealized *chiral limit*, $m_f \to 0$ for the N_f light flavors, treat the small physical masses as a controlled perturbation in the next section, and let

$$\mathcal{L}^0_{\mathrm{QCD}} = -\tfrac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \sum_{f=1}^{N_f} \bar{q}_f \, i \not{D} \, q_f \tag{39.2}$$

be the object whose symmetries we classify. Nothing in Eq. (39.2) refers to flavor except through the universal coupling of every q_f to the same gluon field; the gauge interaction is flavor-blind. This is the structural reason a flavor symmetry exists at all, and the chiral limit promotes it to its largest possible form.

The promotion happens because a massless Dirac fermion is really two independent Weyl fermions. Recall from Chapter 11 the chirality projectors

$$P_L = \tfrac{1}{2}(1-\gamma^5), \qquad P_R = \tfrac{1}{2}(1+\gamma^5), \qquad P_L + P_R = 1, \quad P_{L,R}^2 = P_{L,R}, \quad P_L P_R = 0, \tag{39.3}$$

(Eq. (11.82)); the algebra of γ^5 and the chiral basis live in Section 11.2.5 and Theorem C.27). Define the chiral components of each flavor,

$$q_L = P_L\, q, \qquad q_R = P_R\, q, \qquad \bar{q}_L = \bar{q}\, P_R, \qquad \bar{q}_R = \bar{q}\, P_L, \tag{39.4}$$

where the placement of the projectors on $\bar{q} = q^\dagger \gamma^0$ follows from $\gamma^0 \gamma^5 = -\gamma^5 \gamma^0$: $\bar{q}_L = (P_L q)^\dagger \gamma^0 = q^\dagger P_L \gamma^0 = q^\dagger \gamma^0 P_R = \bar{q} P_R$, and likewise for the right component. The single fact we need about how these enter the Lagrangian is the intertwining

relation $\gamma^\mu P_L = P_R \gamma^\mu$, i.e. γ^5 anticommutes with γ^μ. It splits the kinetic term and forbids any chirality-diagonal piece in the mass term:

$$\bar{q}\, i\not{D}\, q = \bar{q}\,(P_R + P_L)\, i\not{D}\,(P_L + P_R)\, q = \bar{q}\, i\not{D}\,(P_L^2 + P_R^2)\, q = \bar{q}_R\, i\not{D}\, q_R + \bar{q}_L\, i\not{D}\, q_L, \tag{39.5}$$

$$\bar{q}\, q = \bar{q}\,(P_R + P_L)(P_L + P_R)\, q = \bar{q}\,(P_R P_L + P_L P_R)\, q = \bar{q}_L\, q_R + \bar{q}_R\, q_L. \tag{39.6}$$

In Eq. (39.5) the cross terms $\bar{q} P_R i\not{D} P_R q = \bar{q} P_R P_L i\not{D}\, q = 0$ vanish because $\not{D}$ carries one γ^μ and so flips the chirality it sits between, turning the two projectors into the orthogonal product $P_R P_L = 0$; in Eq. (39.6) the chirality-*preserving* combinations vanish for the opposite reason, $\bar{q} P_R q = \bar{q} P_R P_R q$ with $\bar{q} P_R = \bar{q}_R$ but $P_R q = q_R$ giving $\bar{q}_R$ paired with the wrong projector. The lesson is the one drawn in Chapter 11: the kinetic term keeps left and right separate, so a massless theory has no communication between the two chiralities, whereas any mass term welds them together. This is exactly why $m \to 0$ enlarges the symmetry.

Assemble the N_f flavors into column vectors in flavor space, $q_L = (q_{1L}, \ldots, q_{N_f L})^{\mathsf{T}}$ and likewise q_R. The massless quark Lagrangian is then

$$\mathcal{L}^0_{\text{quark}} = \bar{q}_L\, i\not{D}\, q_L + \bar{q}_R\, i\not{D}\, q_R, \tag{39.7}$$

a sum of two terms, each a flavor-space inner product contracted with the *flavor-blind* covariant derivative. Because $\not{D}$ is proportional to the identity in flavor space, Eq. (39.7) is invariant under *independent* unitary rotations of the left and right multiplets,

$$q_L \to L\, q_L, \qquad q_R \to R\, q_R, \qquad L \in \mathrm{U}(N_f),\ R \in \mathrm{U}(N_f), \tag{39.8}$$

since $\bar{q}_L \to \bar{q}_L L^\dagger$ and $\bar{q}_L i\not{D} q_L \to \bar{q}_L L^\dagger L\, i\not{D}\, q_L = \bar{q}_L i\not{D}\, q_L$ by unitarity, and the right sector identically. The gauge field is untouched: these are global flavor rotations, not gauge transformations. We have found the classical symmetry.

Definition 39.1 (Classical chiral symmetry group)**.** The massless N_f-flavor QCD Lagrangian Eq. (39.2) is invariant under the global group $\mathrm{U}(N_f)_L \times \mathrm{U}(N_f)_R$ acting as in Eq. (39.8). Factoring each unitary group as $\mathrm{U}(N_f) = \mathrm{SU}(N_f) \times \mathrm{U}(1)$ and recombining the two abelian factors into their vector ($L = R$) and axial ($L = R^\dagger$) combinations, the classical chiral group is

$$G_{\text{cl}} = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V \times \mathrm{U}(1)_A, \tag{39.9}$$

where $\mathrm{U}(1)_V$ acts by $q \to e^{i\alpha} q$ (both chiralities together) and $\mathrm{U}(1)_A$ by $q \to e^{i\beta\gamma^5} q$ (opposite phases on the two chiralities).

The recombination in Eq. (39.9) is purely a choice of basis on the two U(1)'s and deserves a word, because it is what makes the vector and axial currents the natural variables. Writing the two phases as $q_L \to e^{i\theta_L} q_L$, $q_R \to e^{i\theta_R} q_R$, the

vector combination $\theta_L = \theta_R \equiv \alpha$ rotates both chiralities by the same phase, $q \to e^{i\alpha} q$, while the axial combination $\theta_L = -\theta_R \equiv -\beta$ rotates them oppositely, $q_L \to e^{-i\beta} q_L$, $q_R \to e^{+i\beta} q_R$, which is $q \to e^{i\beta\gamma^5} q$ since γ^5 is -1 on q_L and $+1$ on q_R. The same change of basis on the nonabelian parts would let us speak of vector and axial SU(N_f), but those do not separately close into groups (the axial generators do not form a subalgebra), so we keep SU(N_f)$_L \times$ SU(N_f)$_R$ in the chiral basis and pass to vector/axial only at the level of currents, next.

39.1.2 The Noether currents

A continuous global symmetry yields a conserved current by Noether's theorem (Theorems 7.9 and 7.10). The chiral group Eq. (39.9) has $2(N_f^2-1)+2$ generators, so we obtain that many currents; constructing them explicitly is the bookkeeping on which the rest of the chapter rests. For an internal symmetry $\delta q = -i\,\epsilon^A\, t^A q$ with constant infinitesimal parameters ϵ^A and field-space generators t^A, Noether's first theorem gives the current

$$J_\mu^A = \sum_{\text{fields } \Phi} \frac{\partial \mathcal{L}}{\partial(\partial^\mu \Phi)}\,(-i\,t^A \Phi) \quad \Longrightarrow \quad J_\mu^A = \bar{q}\,\gamma_\mu\, t^A\, q \tag{39.10}$$

for our Lagrangian, in which $\partial\mathcal{L}/\partial(\partial^\mu q) = \bar{q}\, i\gamma_\mu$ from the kinetic term, the gauge field contributing nothing to a *flavor* variation. Specializing t^A to each generator of Eq. (39.9) produces the complete current content.

Left and right SU(N_f) currents. For the SU(N_f)$_L$ transformation $q_L \to e^{-i\epsilon^a T^a} q_L$ with q_R fixed, the field-space generator acts only on the left chirality, $t^a = T^a P_L$, and Eq. (39.10) gives the left current; the right is identical with $L \leftrightarrow R$:

$$L_\mu^a = \bar{q}_L\,\gamma_\mu\, T^a\, q_L, \qquad R_\mu^a = \bar{q}_R\,\gamma_\mu\, T^a\, q_R, \qquad a = 1, \ldots, N_f^2 - 1, \tag{39.11}$$

where the T^a are the Hermitian SU(N_f) generators of Sections 2.1.3 and B.4. These $2(N_f^2-1)$ currents are the cleanest variables, but the physically labeled combinations are the *vector* and *axial* currents,

$$V_\mu^a \equiv R_\mu^a + L_\mu^a = \bar{q}_R \gamma_\mu T^a q_R + \bar{q}_L \gamma_\mu T^a q_L = \bar{q}\,\gamma_\mu T^a\, q, \tag{39.12}$$

$$A_\mu^a \equiv R_\mu^a - L_\mu^a = \bar{q}_R \gamma_\mu T^a q_R - \bar{q}_L \gamma_\mu T^a q_L = \bar{q}\,\gamma_\mu \gamma^5 T^a\, q. \tag{39.13}$$

The reassembly into a single Dirac bilinear in each case uses Eq. (39.4): $\bar{q}\gamma_\mu T^a q = \bar{q}(P_R + P_L)\gamma_\mu T^a q = \bar{q}_R\gamma_\mu T^a q_R + \bar{q}_L \gamma_\mu T^a q_L$ because γ_μ preserves chirality, giving the vector current; inserting one γ^5, which is $+1$ on q_R and -1 on q_L, flips the relative sign and gives the axial current. So V^a counts $R + L$ and A^a counts $R - L$, exactly the vector/axial split anticipated above. The V^a generate the diagonal SU(N_f)$_V$ (ordinary flavor symmetry — isospin for $N_f = 2$); the A^a are the broken directions to come, and crucially *they do not close into an algebra*, $\left[A^a, A^b\right] \sim V^c$, which is why there is no "axial SU(N_f)" group.

The two singlet currents. The abelian factors give one current each. For $\mathrm{U}(1)_V$, $q \to e^{-i\epsilon} q$ with $t = \mathbf{1}$, and for $\mathrm{U}(1)_A$, $q \to e^{-i\beta\gamma^5} q$ with $t = \gamma^5$:

$$V_\mu^{(0)} = \bar{q}\,\gamma_\mu\, q \quad (\text{quark number}, \ = N_c \times \text{baryon number}), \qquad A_\mu^{(0)} = \bar{q}\,\gamma_\mu\gamma^5\, q. \tag{39.14}$$

The count is now complete: $N_f^2 - 1$ vector currents V^a, $N_f^2 - 1$ axial currents A^a, one baryon-number current $V^{(0)}$, one singlet axial current $A^{(0)}$, totalling $2(N_f^2 - 1) + 2 = 2N_f^2$, one per generator of Eq. (39.9).

It remains to verify conservation and to read off how the small quark masses, to be restored later, break each current. This single computation governs both the chiral limit (everything conserved) and the real world (almost conserved), so we do it in full.

Worked Example 39.2 (Divergences of the chiral currents)**.** Restore a general (not necessarily diagonal) mass matrix M in flavor space, so the quark Lagrangian is $\bar{q}(i\not{D} - M)q$ with $M = \mathrm{diag}(m_1, \ldots, m_{N_f})$ in the physical basis. The equations of motion are

$$(i\not{D} - M)q = 0, \qquad \bar{q}\,(i\overleftarrow{\not{D}} + M) = 0, \tag{39.15}$$

the second being the Dirac-conjugate of the first, with $\bar{q}\overleftarrow{\not{D}} \equiv (\partial_\mu\bar{q} + ig\bar{q}A_\mu^a T^a)\gamma^\mu$ the gauge-covariant derivative acting to the left. Differentiate the vector flavor current Eq. (39.12), using the Leibniz rule and that T^a is a constant flavor matrix commuting with the color-space $\not{D}$:

$$\begin{aligned}\partial^\mu V_\mu^a &= (\partial^\mu\bar{q})\gamma_\mu T^a q + \bar{q}\,T^a\gamma_\mu\partial^\mu q \\ &= (\bar{q}\,\overleftarrow{\not{D}})\,T^a q - \bar{q}\,T^a(\not{D}q) = (i\bar{q}M)\,T^a q - \bar{q}\,T^a(-iMq) = i\,\bar{q}\,[M, T^a]\,q,\end{aligned} \tag{39.16}$$

where in the second line the gluon terms inside $\not{D}$ and $\overleftarrow{\not{D}}$ cancel between the two contributions (the current is color-singlet, T^a commutes with the gauge generators), and we substituted the equations of motion Eq. (39.15). The identical manipulation with one γ^5 inserted, remembering γ^5 anticommutes with the single γ_μ from each derivative, gives the axial divergence,

$$\partial^\mu A_\mu^a = i\,\bar{q}\,\gamma^5\,\{M, T^a\}\,q, \tag{39.17}$$

with an *anti*commutator because the γ^5 turns the relative sign on the two mass insertions. The two singlet currents follow by setting $T^a \to \mathbf{1}$: $\partial^\mu V_\mu^{(0)} = i\bar{q}\,[M, \mathbf{1}]\,q = 0$ identically (baryon number is conserved whatever the masses, an *exact* symmetry of QCD), while

$$\partial^\mu A_\mu^{(0)} = 2i\,\bar{q}\,\gamma^5 M\,q \qquad (\text{classical, anomaly not yet included}). \tag{39.18}$$

Now read off the chiral limit $M \to 0$. Every right-hand side vanishes: all $2N_f^2$ currents are classically conserved, confirming the full group Eq. (39.9). Away from

the limit, Eq. (39.16) shows the nonsinglet vector currents are conserved whenever $[M, T^a] = 0$, i.e. for degenerate masses ($M \propto \mathbf{1}$) all of $\mathrm{SU}(N_f)_V$ survives, and for generic masses the Cartan subalgebra (the diagonal T^a) survives — this is why isospin ($m_u = m_d$) and the individual flavor numbers are such good symmetries. The axial currents, by contrast, are broken by the mass even for $M \propto \mathbf{1}$, since $\{M, T^a\} \neq 0$; this is the partial conservation of the axial current (PCAC) that the next section turns into the pion mass.

The workedex isolates the one place a quantum effect intrudes: the classical result Eq. (39.18) for the *singlet* axial current is wrong. We turn to that now.

39.1.3 The axial anomaly and the fate of $\mathrm{U}(1)_A$

The current algebra just constructed is classical. Quantization can spoil the conservation of a current that survives at tree level, and for the singlet axial current it does: this is the Adler–Bell–Jackiw anomaly computed in Chapter 36. The anomalous divergence, imported from Eq. (36.50) and specialized to N_f massless quarks each in the fundamental of $\mathrm{SU}(3)_c$, modifies Eq. (39.18) to

$$\partial^\mu A^{(0)}_\mu = 2i\,\bar q\,\gamma^5 M\,q \;+\; \frac{g^2 N_f}{16\pi^2}\,G^a_{\mu\nu}\widetilde G^{a\,\mu\nu}, \qquad \widetilde G^{a\,\mu\nu} \equiv \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}G^a_{\rho\sigma}. \tag{39.19}$$

The anomalous term survives the chiral limit $M \to 0$: even with massless quarks $\partial^\mu A^{(0)}_\mu \neq 0$. The singlet axial $\mathrm{U}(1)_A$ is therefore *not* a symmetry of the quantum theory at all, and must be struck from the classical group Eq. (39.9). The physical symmetry of massless QCD is

$$G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V. \tag{39.20}$$

It is essential that only the *singlet* axial current is anomalous, not the $N_f^2 - 1$ nonsinglet ones A^a. The would-be anomaly of A^a comes from the same triangle graph with two gluons and one axial-current insertion, and is proportional to the flavor trace $\mathrm{tr}(T^a)$ of the single generator carried by the current. Since the $\mathrm{SU}(N_f)$ generators are traceless, $\mathrm{tr}(T^a) = 0$, the nonsinglet anomaly vanishes and A^a remains conserved in the chiral limit. The singlet current carries instead $t = \mathbf{1}$ with $\mathrm{tr}(\mathbf{1}) = N_f \neq 0$, which is precisely the factor of N_f in Eq. (39.19). So the anomaly surgically removes one U(1) and leaves the rest of the chiral group intact — exactly the structure we need.

This has a sharp phenomenological payoff, the resolution of the so-called $\mathrm{U}(1)_A$ *problem* (Section 36.5.4). If $\mathrm{U}(1)_A$ were a genuine spontaneously broken symmetry, the Goldstone counting below would produce one *additional* light pseudoscalar, an $\mathrm{SU}(N_f)_V$ singlet — physically the η'. Weinberg's analysis showed that such a state, were it a true Goldstone boson, would have to be anomalously light, with a mass bounded by $\sqrt{3}\,m_\pi$; the observed $\eta'(958)$ is

more than six times heavier than the pion and is manifestly not a Goldstone boson. The anomaly Eq. (39.19) resolves the paradox at a stroke: $\mathrm{U}(1)_A$ is not a symmetry, so its breaking produces no Goldstone boson, and the η' is heavy because nothing protects it. The complete story of *why* the anomalous term $G\widetilde{G}$ has nonvanishing matrix elements in the vacuum — it is a total derivative, yet topologically nontrivial gluon configurations make it count — requires the instanton physics of Chapter 42; here we need only the existence of the anomaly, which is enough to delete $\mathrm{U}(1)_A$ from the symmetry and thereby predict eight (for $N_f = 3$) rather than nine light pseudoscalars.

39.1.4 The quark condensate and the breaking pattern

The group G of Eq. (39.20) is the symmetry. To get Goldstone bosons we need it spontaneously broken, and for that we need an order parameter in the sense of Theorem 29.10: a local operator whose vacuum expectation transforms nontrivially under G, so that the charges fail to annihilate the vacuum. QCD supplies one, the *quark condensate*. By Eq. (39.6) the simplest flavor- and Lorentz-scalar bilinear is

$$\bar{q}\, q = \bar{q}_L\, q_R + \bar{q}_R\, q_L, \tag{39.21}$$

which pairs the two chiralities. The central dynamical assumption — the one nonperturbative input of the entire framework, established by lattice QCD and by the spectroscopic consequences we check below, and which we take as given rather than derive — is that this operator condenses in the vacuum:

$$\langle \bar{q}_f\, q_{f'} \rangle = -\,\Sigma\, \delta_{ff'}, \qquad \Sigma > 0, \quad \Sigma \sim (250\ \mathrm{MeV})^3 \neq 0, \tag{39.22}$$

where the value $\Sigma = -\langle \bar{u}u \rangle = -\left\langle \bar{d}d \right\rangle = \dots$ is flavor-independent in the chiral limit (any flavor dependence would itself break $\mathrm{SU}(N_f)_V$, which nature does not do to leading order) and its sign is conventional. We stress that *only* $\Sigma \neq 0$ *matters* for the symmetry analysis; its magnitude is a nonperturbative number that the effective theory will absorb into low-energy constants, never needing its value.

Now determine what Eq. (39.22) breaks. The clean way to see the pattern is to track the full condensate matrix

$$\Sigma^{f'f} \equiv \left\langle \bar{q}_L^{\,f}\, q_R^{f'} \right\rangle, \tag{39.23}$$

an $N_f \times N_f$ matrix in flavor space whose trace, doubled, is Eq. (39.22). Under the chiral group Eq. (39.8), $q_R \to R\, q_R$ and $\bar{q}_L \to \bar{q}_L L^\dagger$, so the bilinear $\bar{q}_L^{\,f} q_R^{f'}$ transforms as the $(\bar{\mathbf{N}}_f, \mathbf{N}_f)$ representation and its expectation value rotates as

$$\Sigma \;\longrightarrow\; R\, \Sigma\, L^\dagger, \qquad (L, R) \in \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R. \tag{39.24}$$

The vacuum is the dynamically chosen value $\Sigma = \Sigma_0\,\mathbf{1}$ (a multiple of the identity, from Eq. (39.22)). The unbroken subgroup H is the stabilizer of this configuration — the set of (L, R) leaving $\Sigma_0\mathbf{1}$ fixed:

$$R\,(\Sigma_0\mathbf{1})\,L^\dagger = \Sigma_0\mathbf{1} \iff RL^\dagger = \mathbf{1} \iff R = L. \tag{39.25}$$

The transformations that preserve the vacuum are exactly the *diagonal* ones $L = R$ — the vector subgroup $\mathrm{SU}(N_f)_V$ whose currents are the V^a of Eq. (39.12). The transformations that move it are the axial ones, $L = R^\dagger \neq L$, whose currents are the A^a. We have proved:

Theorem 39.3 (Chiral breaking pattern of QCD)**.** *A nonzero, flavor-symmetric quark condensate Eq. (39.22) spontaneously breaks the chiral group of massless N_f-flavor QCD according to*

$$G = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V \;\longrightarrow\; H = \mathrm{SU}(N_f)_V \times \mathrm{U}(1)_V, \tag{39.26}$$

where $\mathrm{SU}(N_f)_V$ is the diagonal ($L = R$) subgroup and $\mathrm{U}(1)_V$ is baryon number. The unbroken generators are the $N_f^2 - 1$ vector charges $Q_V^a = \int \mathrm{d}^3x\, V_0^a$ together with baryon number; the broken generators are the $N_f^2 - 1$ axial charges $Q_A^a = \int \mathrm{d}^3x\, A_0^a$.

It is worth seeing the invariance and the non-invariance directly at the level of charges acting on the condensate, since that is the operative definition of breaking in Theorem 29.10 ($\langle 0|\,[Q, \mathcal{O}]\,|0\rangle \neq 0$ for a broken charge). Infinitesimally, a vector rotation acts on the bilinear $\bar{q}q$ by $\delta_V(\bar{q}q) = [i\alpha^a Q_V^a, \bar{q}q]$, generated by $q \to (1 + i\alpha^a T^a)q$, $\bar{q} \to \bar{q}(1 - i\alpha^a T^a)$:

$$\delta_V(\bar{q}\,q) = \bar{q}\,(i\alpha^a T^a)q + \bar{q}\,(-i\alpha^a T^a)q = 0, \tag{39.27}$$

so $\langle 0|\,[Q_V^a, \bar{q}q]\,|0\rangle = 0$: the condensate is annihilated by the vector charges, which is the statement that $\mathrm{SU}(N_f)_V$ is unbroken. An axial rotation acts by $q \to (1 + i\beta^a\gamma^5 T^a)q$; on the conjugate field, using $\gamma^0\gamma^5 = -\gamma^5\gamma^0$ as in Eq. (39.4), $\bar{q} \to \bar{q}(1 + i\beta^a\gamma^5 T^a)$ (*same* sign, not opposite), so

$$\delta_A(\bar{q}\,q) = \bar{q}\,(i\beta^a\gamma^5 T^a)q + \bar{q}\,(i\beta^a\gamma^5 T^a)q = 2i\,\beta^a\,\bar{q}\,\gamma^5 T^a\,q. \tag{39.28}$$

Taking the vacuum expectation and using Eq. (39.22) (the pseudoscalar density $\bar{q}\gamma^5 T^a q$ has the chiral partner of the scalar condensate as its vacuum value), this is nonzero: $\langle 0|\delta_A(\bar{q}q)|0\rangle \propto \beta^a\Sigma \neq 0$. The axial charges do *not* annihilate the vacuum — precisely the criterion that they are the broken generators. The two computations Eqs. (39.27) and (39.28) are Theorem 39.3 seen one charge at a time: vector good, axial broken, with the breaking measured by the same condensate Σ.

Remark 39.4 (Why the unbroken subgroup is vectorial). We assumed the condensate Eq. (39.22) is proportional to the identity, which forced $H = \mathrm{SU}(N_f)_V$. One might worry that some other vacuum alignment — a condensate breaking G to a different subgroup, perhaps one containing axial generators — could occur instead. It cannot, and the reason is deep rather than accidental. The Vafa–Witten theorem [480, 481] proves that in a vector-like gauge theory such as QCD (one whose fermions come in left–right symmetric, anomaly-free representations, so that the Euclidean fermion determinant is real and positive) no *vector* global symmetry can break spontaneously, and parity cannot break spontaneously. Consequently the unbroken subgroup must contain the entire vector symmetry: $H \supseteq \mathrm{SU}(N_f)_V$. Combined with the Goldstone count below — which already saturates the breaking with $N_f^2 - 1$ axial Goldstones — this fixes $H = \mathrm{SU}(N_f)_V$ exactly. We cite the theorem for further reading rather than prove it here, because the breaking pattern Eq. (39.26) is established directly by the condensate's symmetry Eq. (39.24); Vafa–Witten guarantees that QCD's dynamics actually selects this alignment and not a parity-violating competitor.

39.1.5 The Goldstone count: the pseudoscalar octet

With the pattern Eq. (39.26) fixed, the number of Goldstone bosons is no longer a question about QCD dynamics — it is pure group theory, handed to us by the Goldstone theorem and its counting corollary of Chapter 29. We apply Theorem 29.12: a spontaneous breaking $G \to H$ in a relativistic theory with a positive-definite Hilbert space produces exactly $\dim G - \dim H = \dim(G/H)$ massless spin-zero bosons, one for each broken generator, as proved there both from the spectral representation and from the curvature of the effective potential (Theorem 29.21). The dimensions are read off Eq. (39.26):

$$\dim G = \underbrace{(N_f^2 - 1)}_{\mathrm{SU}(N_f)_L} + \underbrace{(N_f^2 - 1)}_{\mathrm{SU}(N_f)_R} + \underbrace{1}_{\mathrm{U}(1)_V} = 2N_f^2 - 1, \tag{39.29}$$

$$\dim H = \underbrace{(N_f^2 - 1)}_{\mathrm{SU}(N_f)_V} + \underbrace{1}_{\mathrm{U}(1)_V} = N_f^2, \tag{39.30}$$

and therefore

$$\boxed{N_{\mathrm{GB}} = \dim G - \dim H = N_f^2 - 1.} \tag{39.31}$$

The baryon-number $\mathrm{U}(1)_V$ cancels between G and H — it is unbroken and contributes no Goldstone boson, as it must, since baryon number is an exact and manifestly realized symmetry. The surviving count $N_f^2 - 1$ is exactly the number of *axial* generators, and the identification is sharp: the broken generators are the axial charges Q_A^a, so there is one Goldstone boson per axial generator, created from the vacuum by the corresponding axial current, $\left\langle 0 | A_\mu^a(0) | \pi^b(p) \right\rangle \propto i p_\mu \, \delta^{ab}$

— the relation that the next section promotes into the definition of the pion decay constant.

The quantum numbers of these Goldstones are fixed by the broken generators that create them. The axial currents $A^a_\mu = \bar{q}\gamma_\mu\gamma^5 T^a q$ are *axial* vectors: under parity $\mathcal{P}$, which sends $\bar{q}\gamma_\mu q \to +\bar{q}\gamma^\mu q$ (spatial components flip as a vector) and inserts an extra sign for the γ^5, the axial current transforms oppositely to the vector current, and the state it creates from the (parity-even) vacuum is a *pseudoscalar*: spin-parity $J^P = 0^-$. This is the second prediction, as important as the count itself: the Goldstone bosons of chiral symmetry breaking must be pseudoscalars, not scalars. We now check both predictions against the observed spectrum.

Worked Example 39.5 (The two light-quark cases: pions and the octet)**.** Specialize Eq. (39.31) to the physically relevant numbers of light flavors.

$N_f = 2$ *(up, down).* Then $N_{\text{GB}} = 2^2 - 1 = 3$. The unbroken group is $\text{SU}(2)_V$, ordinary *isospin*, and the three Goldstone bosons fill its adjoint (the spin-1 isotriplet, **3** of SU(2)). These are the pions: the charged $\pi^\pm$ and the neutral π^0, with $I = 1$, $J^P = 0^-$. Their physical masses ($m_{\pi^\pm} = 139.6$ MeV, $m_{\pi^0} = 135.0$ MeV) are tiny on the hadronic scale — the hallmark of pseudo-Goldstone bosons whose mass comes only from the small explicit breaking $m_u, m_d \neq 0$ analyzed next section.

$N_f = 3$ *(up, down, strange).* Then $N_{\text{GB}} = 3^2 - 1 = 8$. The unbroken group is $\text{SU}(3)_V$ (the Eightfold Way flavor SU(3)), and the eight Goldstone bosons fill its adjoint, the **8** — the *pseudoscalar meson octet*: the three pions, the four kaons $K^\pm, K^0, \bar{K}^0$ ($I = \frac{1}{2}$, carrying strangeness), and the isosinglet η. All have $J^P = 0^-$. The octet weight diagram (Fig. 39.1) is the same hexagon-with-doubled-center that organizes any SU(3) adjoint, plotted in the conserved quantum numbers of H: the third component of isospin I_3 (a Cartan generator of $\text{SU}(2)_V \subset \text{SU}(3)_V$) and the hypercharge Y (the other Cartan generator). The would-be ninth state that fills out a nonet, the SU(3)-singlet η', is *not* a Goldstone boson: it is the $\eta'(958)$, heavy because its associated current is the anomalous singlet $A^{(0)}$ of Eq. (39.19). The split between the light octet and the heavy η' is the spectroscopic fingerprint of the anomaly.

39.1.6 Physical evidence for the pattern

The argument so far is a chain of group theory resting on two physical inputs: the smallness of the light-quark masses and the nonvanishing of the condensate. It predicts a definite, falsifiable structure in the hadron spectrum. Three features of the observed spectrum confirm it.

First, the *lightness*. The breaking pattern predicts $N_f^2 - 1$ anomalously light pseudoscalars and nothing else light. For $N_f = 2$ the lightest hadrons are exactly

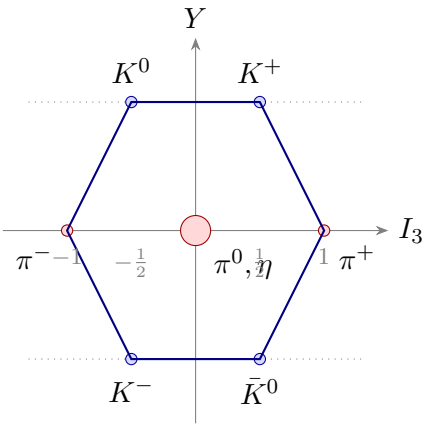


Figure 39.1: The pseudoscalar meson octet ($J^P = 0^-$) of the $N_f = 3$ breaking $\mathrm{SU}(3)_L \times \mathrm{SU}(3)_R \to \mathrm{SU}(3)_V$, plotted in the Cartan charges of the unbroken $\mathrm{SU}(3)_V$: the third isospin component I_3 and the hypercharge Y. The eight Goldstone bosons fill the adjoint **8** of $\mathrm{SU}(3)_V$ (hexagon with a doubled center: π^0 and η). The would-be ninth, the SU(3)-singlet η', sits at the center too but is *not* a Goldstone boson — it is heavy because its current, the singlet axial $A^{(0)}$, is anomalous (Eq. (39.19)).

the three pions, separated by a factor of four from the next state (the $\rho(770)$ vector meson, which is *not* a Goldstone boson and need not be light). For $N_f = 3$ the eight lightest mesons are precisely the octet $\{\pi, K, \eta\}$, all below 550 MeV, while every other meson lies above 750 MeV. The pattern "a light octet, then a gap" is the spectroscopic signature of Eq. (39.31), and the modest spread within the octet ($m_K \gg m_\pi$ because $m_s \gg m_{u,d}$) is itself predicted by the explicit-breaking analysis of the next section.

Second, the *quantum numbers*. Every one of these light mesons is a pseudoscalar, $J^P = 0^-$, exactly as the axial-vector character of the broken currents demands. There is no light scalar (0^+) octet; the lightest scalars are heavy and broad. A theory in which the light mesons came out as 0^+ would falsify the chiral picture, and they do not.

Third, and most incisively, the *absence of parity doubling*. This is the cleanest qualitative test that the unbroken subgroup is the *vector* one and not some axial alternative, and it deserves spelling out.

Physical Insight: why the spectrum shows broken chirality but unbroken parity

Parity $\mathcal{P}$ commutes with the vector charges Q_V^a (both are built from the parity-even combination $\bar{q}\gamma^0 q$ and its rotations) but *anti*commutes, in

effect, with the axial charges: $\mathcal{P}\, Q_A^a\, \mathcal{P}^{-1} = -Q_A^a$, because the axial current is an axial vector. Suppose, counterfactually, that an axial symmetry were *unbroken* — that some Q_A^a annihilated the vacuum and generated a good symmetry of the spectrum. Then acting with Q_A^a on any hadron $|h\rangle$ would produce a degenerate partner $Q_A^a|h\rangle$ of *opposite parity* and equal mass: every hadron would come with a mirror-parity twin. The spectrum would be organized into *parity doublets* — the nucleon would have a degenerate negative-parity partner, the ρ a degenerate axial-vector a_1 at the same mass, and so on. Nature shows no such doubling: the nucleon's negative-parity partner $N(1535)$ is some 600 MeV heavier, the $a_1(1260)$ lies far above the $\rho(770)$. The axial symmetry is therefore *not* realized on the spectrum — it is spontaneously broken. Yet parity *itself* is an excellent symmetry (hadrons are sharp parity eigenstates), and so is isospin/flavor (hadrons fill clean $\mathrm{SU}(N_f)_V$ multiplets). The only realization of G consistent with all three facts — broken axial, unbroken parity, unbroken flavor — is precisely $G \to H = \mathrm{SU}(N_f)_V$, with the broken directions being the parity-odd axial ones. The hadron spectrum is, in this sense, a direct readout of the breaking pattern: the doublets that the broken charges *would* have created are not there because those charges instead created the light Goldstone pions.

This closes the logical loop opened at the start of the section. We took the QCD Lagrangian Eq. (39.1) and asymptotic freedom (Theorems 27.7 and 27.11) as given, asked what symmetry survives at low energy, and found a definite and verified answer: the chiral group $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_V$, with $\mathrm{U}(1)_A$ removed by the anomaly, is spontaneously broken by the quark condensate to the vector subgroup $\mathrm{SU}(N_f)_V \times \mathrm{U}(1)_V$, producing $N_f^2 - 1$ pseudoscalar Goldstone bosons — the pions for $N_f = 2$, the meson octet for $N_f = 3$. The remaining task of the chapter is to build a quantum field theory of *these* degrees of freedom. Because the Goldstone bosons are coordinates on the coset G/H, the natural field is not a vector of $N_f^2 - 1$ scalars but a group-valued field $U(x) \in \mathrm{SU}(N_f)$ carrying them nonlinearly; constructing $U(x)$ from the broken generators, and writing the unique leading-order Lagrangian invariant under Eq. (39.26), is the work of the next section.

39.2 The Leading-Order Chiral Lagrangian

The previous section established the symmetry skeleton on which everything in this chapter hangs. In the chiral limit $m_u, m_d, \cdots \to 0$, the QCD Lagrangian of the N_f lightest quarks has the global symmetry $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ acting independently on the left- and right-handed quark fields; the quark condensate $\langle \bar{q}q \rangle \neq 0$ spontaneously breaks this to the diagonal isospin/flavor subgroup

$\mathrm{SU}(N_f)_V$ (Section 39.1); and Goldstone's theorem then populates the spectrum with exactly $\dim\left(\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R\right) - \dim \mathrm{SU}(N_f)_V = N_f^2 - 1$ massless pseudoscalar bosons. For $N_f = 2$ these are the three pions; for $N_f = 3$ they are the pseudoscalar octet (π, K, η). The task now is to write their dynamics. We do *not* start from scratch: Chapter 29 already built the universal low-energy theory of Goldstone bosons — the nonlinear sigma model (Theorem 29.22), its coset coordinates (Section 29.5.2), and the CCWZ geometric construction (Section 29.5.5) — and proved that the leading two-derivative dynamics is fixed by symmetry up to a single constant. What remains is to *specialize* that apparatus to the coset $G/H = \left(\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R\right)/\mathrm{SU}(N_f)_V$, package the Goldstone fields into the object that makes the chiral symmetry act linearly, fix the one constant against a measured number, and finally restore the small explicit breaking that the quark masses provide. The output is the leading-order chiral Lagrangian, the concrete realization of the EFT paradigm that Chapter 38 predicted in the abstract.

39.2.1 The Goldstone matrix and its transformation law

The Goldstone fields live on the coset manifold. For the chiral breaking pattern that manifold is itself a group: $\left(\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R\right)/\mathrm{SU}(N_f)_V \cong \mathrm{SU}(N_f)$, because every coset has a unique representative of the form $(g, \mathbf{1})$ followed by a diagonal element, and the diagonal is exactly what we quotient by. This is the special feature that makes chiral symmetry so much more tractable than a generic nonlinear realization: the Goldstone bosons can be assembled into a *single unitary matrix field* on which the full symmetry acts *linearly*.

Definition 39.6 (The Goldstone matrix)**.** Let $\Phi(x)$ be the Hermitian, traceless $N_f \times N_f$ matrix of pseudoscalar fields,

$$\Phi(x) = \Phi^\dagger(x), \qquad \operatorname{tr}\Phi(x) = 0, \tag{39.32}$$

carrying the $N_f^2 - 1$ Goldstone bosons as its independent real components. The *Goldstone matrix* is the $\mathrm{SU}(N_f)$-valued field

$$U(x) = \exp\!\Big(\tfrac{i}{f}\,\Phi(x)\Big), \tag{39.33}$$

where f is a constant of mass dimension one (to be identified below with the pion decay constant). Hermiticity and tracelessness of Φ make U unitary and unimodular, $U^\dagger U = \mathbf{1}$ and $\det U = e^{i\operatorname{tr}\Phi/f} = 1$, so indeed $U \in \mathrm{SU}(N_f)$.

For the two cases of interest the matrix Φ is built from the physical mesons through the generators collected in Section B.2. For $N_f = 2$ one writes $\Phi = \pi^a \tau^a$ with τ^a the Pauli matrices,

$$\Phi = \pi^a\tau^a = \begin{pmatrix} \pi^0 & \sqrt{2}\,\pi^+ \\ \sqrt{2}\,\pi^- & -\pi^0 \end{pmatrix}, \qquad \pi^\pm = \tfrac{1}{\sqrt{2}}(\pi^1 \mp i\pi^2), \quad \pi^0 = \pi^3, \tag{39.34}$$

the isotriplet of pions. For $N_f = 3$ one writes $\Phi = \phi^a \lambda^a$ with λ^a the Gell-Mann matrices of Section B.2 ($\mathrm{tr}\,\lambda^a\lambda^b = 2\delta^{ab}$), which in terms of the charge eigenstates is the pseudoscalar octet

$$\Phi = \phi^a \lambda^a = \begin{pmatrix} \pi^0 + \dfrac{\eta}{\sqrt{3}} & \sqrt{2}\,\pi^+ & \sqrt{2}\,K^+ \\ \sqrt{2}\,\pi^- & -\pi^0 + \dfrac{\eta}{\sqrt{3}} & \sqrt{2}\,K^0 \\ \sqrt{2}\,K^- & \sqrt{2}\,\bar{K}^0 & -\dfrac{2\eta}{\sqrt{3}} \end{pmatrix}. \tag{39.35}$$

The normalizations of the off-diagonal ($\sqrt{2}$) and diagonal entries are not free: they are fixed by demanding that the kinetic term come out canonical, which we verify in Eq. (39.44) below. The numerical factors are exactly those that make $\pi^\pm$, $K^\pm$, K^0 complex fields of unit-normalized kinetic terms and π^0, η real fields of standard kinetic terms.

The whole point of packaging the fields this way is the transformation law. Under a chiral rotation $(L, R) \in \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ the matrix U transforms *linearly* and *homogeneously*,

$$\boxed{U(x) \;\longrightarrow\; L\,U(x)\,R^\dagger}, \qquad L \in \mathrm{SU}(N_f)_L,\ R \in \mathrm{SU}(N_f)_R. \tag{39.36}$$

This is the chiral specialization of the CCWZ realization of Section 29.5.5, and it is worth seeing exactly why Eq. (39.36) *is* the nonlinear realization in disguise rather than a new postulate. In the CCWZ language one chooses a coset representative $u(\Phi) = e^{i\Phi/2f}$ and lets a group element act by left multiplication followed by a compensating H-transformation. Writing $U = u^2 = u\,u$ and acting with (L, R) gives $u \to L\,u\,h^\dagger = h\,u\,R^\dagger$, where $h = h(\Phi, L, R) \in \mathrm{SU}(N_f)_V$ is the field-dependent compensator. Squaring,

$$U = u\,u \;\longrightarrow\; (L\,u\,h^\dagger)(h\,u\,R^\dagger) = L\,u\,u\,R^\dagger = L\,U\,R^\dagger, \tag{39.37}$$

and the compensator h cancels between the two factors. The nonlinear, field-dependent transformation of the individual Goldstone fields hidden inside u has collapsed into the simple linear rule Eq. (39.36) for the bilinear $U = u^2$. This is precisely why chiral perturbation theory is conventionally written in terms of U: the symmetry is manifest.

Three structural facts follow immediately and we will use all three. First, the unbroken subgroup is the diagonal: setting $L = R = V$ in Eq. (39.36) gives $U \to VUV^\dagger$, an ordinary $\mathrm{SU}(N_f)_V$ rotation of the fields, and the vacuum $U = \mathbf{1}$ is left invariant, $\mathbf{1} \to V\mathbf{1}V^\dagger = \mathbf{1}$. Thus $H = \mathrm{SU}(N_f)_V$ as required. Second, the broken (axial) transformations are those with $R = L^\dagger$: then $U = \mathbf{1} \to L^2 \neq \mathbf{1}$, so the vacuum moves, and acting on the fields, $L = e^{i\alpha_A^a T^a}$ gives $\Phi \to \Phi + 2f\,\alpha_A^a T^a + \cdots$, the inhomogeneous *shift* of the Goldstone fields that is the fingerprint of spontaneous breaking and that forces purely derivative couplings

(Theorem 29.25). Third, the count is right by construction: U is an arbitrary $\mathrm{SU}(N_f)$ element, parametrized by exactly $N_f^2 - 1$ real fields Φ^a, matching the Goldstone count of Theorem 29.24.

Notation Watch

Parity acts on the pseudoscalars by $\Phi(t, \mathbf{x}) \to -\Phi(t, -\mathbf{x})$ (they are pseudoscalars), hence on the Goldstone matrix by $U(t, \mathbf{x}) \to U^\dagger(t, -\mathbf{x})$. Equivalently, parity exchanges the two chiral factors, $L \leftrightarrow R$. Every term we admit in the Lagrangian below must be invariant under this exchange together with the spatial reflection; this is what we mean by "parity-even." Charge conjugation acts by $U \to U^\mathsf{T}$ (transpose); we will note where it matters.

39.2.2 Uniqueness of the leading Lagrangian

We now ask for the most general Lagrangian built from U that is (i) Lorentz invariant, (ii) invariant under the chiral group Eq. (39.36), (iii) parity invariant, (iv) Hermitian, and (v) of lowest order in the derivative expansion. The derivative expansion is the organizing principle established in Chapter 29: because the symmetry forbids a potential, every interaction carries derivatives, and we order terms by the number of derivatives, equivalently by powers of momentum. We denote by $\mathcal{L}_{2k}$ the part with $2k$ derivatives (only even numbers survive Lorentz invariance), and seek $\mathcal{L}_2$.

The building blocks are U, $U^\dagger$, and their derivatives $\partial_\mu U$, $\partial_\mu U^\dagger$. Chiral invariance is the strong constraint. Since $U \to L U R^\dagger$ and $U^\dagger \to R U^\dagger L^\dagger$, a left index can be soaked up only by a matching $L^\dagger$ and a right index only by a matching R; the only way to build an invariant is to take a *trace* of an *alternating* product $U\,U^\dagger\,U\,U^\dagger\cdots$ (possibly with derivatives distributed on the factors), in which the R from one factor cancels the $R^\dagger$ from the next and the trace closes the leftover $L, L^\dagger$ pair. Products of several such traces are also allowed.

Zero derivatives. The only candidate is $\mathrm{tr}(UU^\dagger) = \mathrm{tr}\,\mathbf{1} = N_f$, a constant. There is no potential, in agreement with the shift symmetry. (A term $\mathrm{tr}\,U + \mathrm{tr}\,U^\dagger$ is *not* chirally invariant — it transforms into $\mathrm{tr}(LUR^\dagger)$ — and so is forbidden in the chiral limit; it returns, dressed by a spurion, when we add quark masses in Section 39.2.4.)

Two derivatives. We must place two ∂_μ's and contract their Lorentz indices. The candidate structures are

$$\mathrm{tr}(\partial_\mu U\,\partial^\mu U^\dagger), \qquad \mathrm{tr}(\partial_\mu \partial^\mu U\,U^\dagger), \qquad \mathrm{tr}(U^\dagger \partial_\mu U)\;\mathrm{tr}(U^\dagger \partial^\mu U), \tag{39.38}$$

together with structures that look different but are algebraically identical to the first. We show that all of Eq. (39.38) collapse to a single independent term. The key identities follow from unitarity $U^\dagger U = \mathbf{1}$ and unimodularity $\det U = 1$. Differentiating $U^\dagger U = \mathbf{1}$ gives

$$\partial_\mu U^\dagger\, U + U^\dagger \partial_\mu U = 0 \quad \Longleftrightarrow \quad U\, \partial_\mu U^\dagger = -\partial_\mu U\, U^\dagger, \tag{39.39}$$

the second form obtained by multiplying the first on the left by U and on the right by $U^\dagger$. Differentiating $\det U = 1$, equivalently $\ln\det U = \operatorname{tr}\ln U = 0$, gives the trace identity

$$\operatorname{tr}\!\left(U^\dagger \partial_\mu U\right) = \partial_\mu \operatorname{tr}\ln U = 0, \qquad \text{equivalently} \qquad \operatorname{tr}\!\left(U\, \partial_\mu U^\dagger\right) = 0. \tag{39.40}$$

The identity Eq. (39.40) kills the third candidate in Eq. (39.38) outright: $\operatorname{tr}(U^\dagger\partial_\mu U) = 0$, so the would-be "U(1)" term $\operatorname{tr}(U^\dagger\partial_\mu U)\operatorname{tr}(U^\dagger\partial^\mu U)$ vanishes identically. (It is precisely the absence of a U(1) Goldstone — the η' being lifted by the anomaly of Chapter 36 — that is encoded here; the $\mathrm{SU}(N_f)$ matrix has no overall phase to support such a term.) The second candidate reduces to the first by an integration by parts inside the action: using Eq. (39.40) in the form $\partial^\mu \operatorname{tr}(\partial_\mu U\, U^\dagger) = 0$ is not quite it; instead expand $\partial_\mu\big[\operatorname{tr}(\partial^\mu U\, U^\dagger)\big] = \operatorname{tr}(\partial_\mu\partial^\mu U\, U^\dagger) + \operatorname{tr}(\partial^\mu U\, \partial_\mu U^\dagger)$, and note the left side is the total derivative of $\operatorname{tr}(\partial^\mu U\, U^\dagger)$, which integrates to zero, so under the action integral

$$\operatorname{tr}(\partial_\mu\partial^\mu U\, U^\dagger) \;\simeq\; -\operatorname{tr}(\partial_\mu U\, \partial^\mu U^\dagger), \tag{39.41}$$

where $\simeq$ denotes equality up to a total derivative. Finally the first candidate is, up to a sign, the same as the manifestly H-covariant form: using $\partial_\mu U = U(U^\dagger\partial_\mu U)$ and $\partial^\mu U^\dagger = -(U^\dagger\partial^\mu U)U^\dagger$ from Eq. (39.39),

$$\operatorname{tr}(\partial_\mu U\, \partial^\mu U^\dagger) = \operatorname{tr}\big[U(U^\dagger\partial_\mu U)\big(-(U^\dagger\partial^\mu U)U^\dagger\big)\big] = -\operatorname{tr}\big[(U^\dagger\partial_\mu U)(U^\dagger\partial^\mu U)\big], \tag{39.42}$$

the trace cyclicity having moved $U^\dagger\cdots U$ around to give $\mathbf{1}$. The combination $U^\dagger\partial_\mu U$ is the Maurer–Cartan one-form of Section 29.5.5, here anti-Hermitian by Eq. (39.39); Eq. (39.42) just says the two ways of writing the kinetic term agree. Every two-derivative invariant therefore reduces to a single structure, $\operatorname{tr}(\partial_\mu U\, \partial^\mu U^\dagger)$. Lorentz invariance is manifest, Hermiticity holds because $\big[\operatorname{tr}(\partial_\mu U\, \partial^\mu U^\dagger)\big]^\dagger = \operatorname{tr}(\partial^\mu U\, \partial_\mu U^\dagger)$ equals itself, and parity ($U \to U^\dagger$) maps the structure to $\operatorname{tr}(\partial_\mu U^\dagger\, \partial^\mu U)$, again itself. There is no two-derivative parity-*odd* invariant: a parity-odd term requires the Levi-Civita symbol $\epsilon^{\mu\nu\rho\sigma}$ and hence at least four derivatives — that object exists and is the Wess–Zumino–Witten term of the final section, which sits at $O(p^4)$.

We have proved the central uniqueness statement.

Theorem 39.7 (The leading-order chiral Lagrangian)**.** *The unique Lorentz-invariant, parity-even, Hermitian,* $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$*-invariant Lagrangian*

built from the Goldstone matrix U with the fewest ($=2$) derivatives is

$$\boxed{\mathcal{L}_2 = \frac{f^2}{4}\,\mathrm{tr}(\partial_\mu U\,\partial^\mu U^\dagger)} \tag{39.43}$$

with f a single real coupling of mass dimension one. The overall constant is fixed by canonical normalization of the Goldstone fields, and the choice $f^2/4$ produces standard kinetic terms.

Proof. Uniqueness of the structure was established above: chiral invariance forces an alternating trace; the two-derivative alternating traces are exhausted by Eq. (39.38); the unimodularity identity Eq. (39.40) kills the double-trace term and the unitarity identities Eq. (39.39) with the integration by parts Eqs. (39.41) and (39.42) reduce the remaining two to one. It remains to fix the constant. Expand $U = \exp(i\Phi/f) = \mathbf{1} + \frac{i}{f}\Phi - \frac{1}{2f^2}\Phi^2 + \cdots$, so $\partial_\mu U = \frac{i}{f}\partial_\mu\Phi + \cdots$ and $\partial_\mu U^\dagger = -\frac{i}{f}\partial_\mu\Phi + \cdots$. To quadratic order in the fields,

$$\mathcal{L}_2 = \frac{f^2}{4}\,\mathrm{tr}\Big[\Big(\tfrac{i}{f}\partial_\mu\Phi\Big)\Big(-\tfrac{i}{f}\partial^\mu\Phi\Big)\Big] + \cdots = \frac{1}{4}\,\mathrm{tr}(\partial_\mu\Phi\,\partial^\mu\Phi) + \cdots = \frac{1}{2}\,\partial_\mu\phi^a\,\partial^\mu\phi^a + \cdots\,, \tag{39.44}$$

where the last equality used $\Phi = 2\phi^a T^a$ with $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ (so $\mathrm{tr}(\partial\Phi\,\partial\Phi) = 4\,\partial\phi^a\partial\phi^b\,\mathrm{tr}(T^aT^b) = 2\,\partial\phi^a\partial\phi^a$). The coefficient $\frac{1}{2}$ is exactly the canonical normalization of $N_f^2 - 1$ real scalar kinetic terms, which is what fixes the prefactor in Eq. (39.43) to $f^2/4$. Inserting the explicit matrices Eqs. (39.34) and (39.35) reproduces the same conclusion in terms of the charge eigenstates: for $N_f = 3$ one finds $\frac{1}{4}\,\mathrm{tr}(\partial\Phi\,\partial\Phi) = \frac{1}{2}(\partial\pi^0)^2 + \frac{1}{2}(\partial\eta)^2 + \partial_\mu\pi^+\partial^\mu\pi^- + \partial_\mu K^+\partial^\mu K^- + \partial_\mu K^0\partial^\mu\bar{K}^0$, the canonical kinetic terms for two real and three complex mesons, which is what fixed the $\sqrt{2}$'s and the $1/\sqrt{3}$'s in Eq. (39.35). □

The Lagrangian Eq. (39.43) is the entire leading-order dynamics of the massless mesons, and it carries a single parameter f. Everything else — all the pion self-interactions — is determined. Expanding to one more order exhibits the leading interaction.

Worked Example 39.8 (The leading pion self-interaction)**.** We expand $\mathcal{L}_2$ to quartic order in Φ to display the contact interaction that the next sections turn into $\pi\pi$ scattering. From $U = \exp(i\Phi/f)$,

$$\partial_\mu U = \frac{i}{f}\,\partial_\mu\Phi - \frac{1}{2f^2}(\Phi\,\partial_\mu\Phi + \partial_\mu\Phi\,\Phi) - \frac{i}{6f^3}(\Phi^2\partial_\mu\Phi + \Phi\,\partial_\mu\Phi\,\Phi + \partial_\mu\Phi\,\Phi^2) + \cdots\,, \tag{39.45}$$

and $\partial_\mu U^\dagger$ is the same with $\Phi \to -\Phi$ in the exponent, i.e. $i \to -i$ on alternate terms. Collecting the terms of $\mathrm{tr}(\partial_\mu U\,\partial^\mu U^\dagger)$ with exactly four Φ's, the $O(1/f^4)$ pieces, and using cyclicity of the trace, one finds after combining

$$\mathcal{L}_2 = \frac{1}{2}\partial_\mu\phi^a\partial^\mu\phi^a + \frac{1}{12f^2}\,\mathrm{tr}\Big([\Phi,\partial_\mu\Phi]\,[\Phi,\partial^\mu\Phi]\Big) + O(\Phi^6/f^4). \tag{39.46}$$

The verification is direct. Write $U^\dagger\partial_\mu U = \frac{i}{f}\partial_\mu\Phi + \frac{1}{2f^2}[\partial_\mu\Phi,\Phi] + \cdots$ (the antisymmetric combination is forced by anti-Hermiticity, Eq. (39.39)); then by Eq. (39.42), $\mathcal{L}_2 = -\frac{f^2}{4}\,\mathrm{tr}[(U^\dagger\partial_\mu U)(U^\dagger\partial^\mu U)]$, and the cross term between the $O(\Phi)$ and $O(\Phi^3)$ parts vanishes by the trace identity Eq. (39.40) while the square of the $O(\Phi^2)$ part gives

$$-\frac{f^2}{4}\cdot\frac{1}{4f^4}\,\mathrm{tr}\big([\partial_\mu\Phi,\Phi][\partial^\mu\Phi,\Phi]\big) = \frac{1}{16f^2}\,\mathrm{tr}\big([\Phi,\partial_\mu\Phi][\Phi,\partial^\mu\Phi]\big), \tag{39.47}$$

and the $O(\Phi)\times O(\Phi^3)$ contributions, after the integration by parts that turns $\mathrm{tr}(\partial_\mu\Phi\cdot\Phi^2\partial^\mu\Phi)$ structures into commutator form, supply the remaining piece to total the coefficient $1/12f^2$ quoted in Eq. (39.46). The essential structural lesson survives any bookkeeping: the four-pion vertex is (a) suppressed by $1/f^2$, (b) carries exactly two derivatives, so every $\pi\pi\to\pi\pi$ amplitude vanishes as the external momenta go to zero — the Adler zero of Theorem 29.26, now visible directly in the vertex — and (c) has its strength *completely fixed* by the same f that appears in the kinetic term. There is no free coupling: the leading-order self-interaction of pions is a prediction. This is the vertex that Section 39.4 (the $\pi\pi$ section) feeds into the scattering amplitude.

39.2.3 The axial current and the identification $f = f_\pi$

The constant f is so far a parameter of the effective theory. To give it physical meaning we compute the conserved currents of $\mathcal{L}_2$ and match the one observable that the symmetry current measures: the amplitude for the axial current to create a single pion from the vacuum. This is exactly the decay-constant matrix element Eq. (29.202) of the general theory, and the matching will show f *is* the physical pion decay constant.

The currents follow from Noether's theorem applied to Eq. (39.36). Promote the global parameters to local ones, $L(x) = e^{i\alpha_L^a(x)T^a}$, $R(x) = e^{i\alpha_R^a(x)T^a}$, and read off the coefficient of $\partial_\mu\alpha$ in the variation of $\mathcal{L}_2$. Consider first a purely left transformation, $\delta U = i\alpha_L U$, $\delta U^\dagger = -iU^\dagger\alpha_L$ (with $\alpha_L = \alpha_L^a T^a$ Hermitian). The piece of $\delta\mathcal{L}_2$ proportional to $\partial_\mu\alpha_L$ is

$$\begin{aligned}\delta\mathcal{L}_2\big|_{\partial\alpha} &= \frac{f^2}{4}\Big[\,\mathrm{tr}\big(i(\partial_\mu\alpha_L)U\,\partial^\mu U^\dagger\big) + \mathrm{tr}\big(\partial_\mu U\,(-i)U^\dagger(\partial^\mu\alpha_L)\big)\Big]\\ &= \frac{if^2}{4}\,\mathrm{tr}\Big((\partial_\mu\alpha_L)[U\,\partial^\mu U^\dagger - \partial^\mu U\,U^\dagger]\Big) = \frac{if^2}{2}\,\mathrm{tr}\big((\partial_\mu\alpha_L)\,U\,\partial^\mu U^\dagger\big),\end{aligned} \tag{39.48}$$

where the last step used $U\partial^\mu U^\dagger = -\partial^\mu U\,U^\dagger$ from Eq. (39.39) to combine the two terms. The left Noether current $L^{\mu,a}$ is the coefficient of $\partial_\mu\alpha_L^a$, namely $L^{\mu,a} = \frac{if^2}{2}\,\mathrm{tr}\,(T^a\,U\partial^\mu U^\dagger)$. An identical computation for a right transformation,

$\delta U = -iU\alpha_R$, gives the right current $R^{\mu,a} = \frac{if^2}{2}\,\mathrm{tr}\,(T^a U^\dagger \partial^\mu U)$. The physically labelled currents are the vector (unbroken) and axial (broken) combinations,

$$V^{\mu,a} = R^{\mu,a} + L^{\mu,a}, \qquad A^{\mu,a} = R^{\mu,a} - L^{\mu,a}, \tag{39.49}$$

which are the effective-theory images of the QCD currents $\bar{q}\gamma^\mu T^a q$ and $\bar{q}\gamma^\mu\gamma^5 T^a q$ respectively. We need $A^{\mu,a}$ to linear order in the fields. Using $U = \mathbf{1} + \frac{i}{f}\Phi + \cdots$,

$$U^\dagger\partial^\mu U = \tfrac{i}{f}\partial^\mu\Phi + \cdots, \qquad U\partial^\mu U^\dagger = -\tfrac{i}{f}\partial^\mu\Phi + \cdots, \tag{39.50}$$

so that

$$R^{\mu,a} = \frac{if^2}{2}\,\mathrm{tr}\Big(T^a\,\tfrac{i}{f}\partial^\mu\Phi\Big) = -\frac{f}{2}\,\mathrm{tr}(T^a\partial^\mu\Phi), \qquad L^{\mu,a} = +\frac{f}{2}\,\mathrm{tr}(T^a\partial^\mu\Phi), \tag{39.51}$$

and therefore, with $\Phi = 2\phi^b T^b$ and $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$,

$$A^{\mu,a} = R^{\mu,a} - L^{\mu,a} = -f\ \mathrm{tr}(T^a\partial^\mu\Phi) = -f\,\partial^\mu\phi^a + O(\Phi^2). \tag{39.52}$$

The vector current $V^{\mu,a} = R^{\mu,a} + L^{\mu,a} = O(\Phi^2)$ has no one-meson piece, as it must — the unbroken current cannot create a single Goldstone from the symmetric vacuum — while the axial current is linear in the field, the hallmark of the broken current being the pion's interpolating operator.

Now take the vacuum-to-one-pion matrix element. With the canonical relativistic normalization $\big\langle 0|\phi^a(0)|\pi^b(p)\big\rangle = \delta^{ab}$ (the field creates its own quantum), one has $\big\langle 0|\partial^\mu\phi^a(x)|\pi^b(p)\big\rangle = \partial^\mu\big(e^{-ip\cdot x}\big)\delta^{ab} = -ip^\mu\delta^{ab}e^{-ip\cdot x}$, so at $x = 0$, from Eq. (39.52),

$$\big\langle 0|A^{\mu,a}(0)|\pi^b(p)\big\rangle = -f\cdot(-ip^\mu\delta^{ab}) = i\,f\,p^\mu\,\delta^{ab}. \tag{39.53}$$

This is identical in form to the general decay-constant relation Eq. (29.202), $\big\langle 0|A^\mu_a|\pi^b(p)\big\rangle = i\,f\,p^\mu\delta^{ab}$, with the effective-theory f sitting exactly where the physical decay constant sits. The pion decay constant is *defined* in QCD by precisely this matrix element of the quark axial current — it is what controls the rate of $\pi^+ \to \mu^+\nu_\mu$ through the weak charged current — so the matching is immediate:

$$\boxed{f = f_\pi}, \qquad f_\pi \simeq 92.1\ \mathrm{MeV}. \tag{39.54}$$

The single coupling of the leading chiral Lagrangian is the measured pion decay constant. (We use the convention $f_\pi \approx 92\,\mathrm{MeV}$ defined by Eq. (39.53); the value $f_\pi \approx 130\,\mathrm{MeV}$ that appears in some references is the same physics with $f_\pi \to \sqrt{2}f_\pi$, arising from defining the matrix element with the charged current $A^{\mu,1} \mp iA^{\mu,2}$ instead of the Cartesian component. We fix Eq. (39.53) throughout.) With f pinned to a number, the entire massless sector — and in particular the quartic vertex Eq. (39.46) — is now a parameter-free prediction.

Physical Insight

It is worth pausing on what has happened. A single constant in the effective Lagrangian has been tied to a single laboratory measurement (the pion lifetime, through $\pi \to \mu\nu$), and in exchange the theory now predicts *all* low-energy pion interactions with no further input. This is the EFT bargain of Chapter 38 made concrete: symmetry fixes the operators, one matching fixes the one leading coefficient, and predictivity follows even though the underlying QCD is strongly coupled and we never solved it. The strong coupling is hidden entirely inside the number f_π.

39.2.4 Explicit breaking by the spurion method

Real quarks are not massless. The QCD Lagrangian contains a mass term that, written in chiral components, reads

$$\mathcal{L}^{\rm QCD}_{\rm mass} = -\bar{q}_L\,\mathcal{M}\,q_R - \bar{q}_R\,\mathcal{M}^\dagger\,q_L, \qquad \mathcal{M} = \mathrm{diag}(m_u, m_d, \dots), \tag{39.55}$$

and this term is *not* invariant under $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$: under $q_L \to L q_L$, $q_R \to R q_R$ the bilinear $\bar{q}_L \mathcal{M} q_R \to \bar{q}_L L^\dagger \mathcal{M} R\, q_R$, which is invariant only if $\mathcal{M}$ were itself transformed. This last observation is the entire idea of the *spurion* method: temporarily *promote* the constant mass matrix to a field that transforms in just the way needed to restore the symmetry, build the most general invariant Lagrangian as if the symmetry were exact, and only at the end freeze the spurion to its physical constant value. Any term invariant under the enlarged symmetry is then automatically the correct symmetry-breaking term, with its breaking pattern dictated by where the spurion sits.

Comparing Eq. (39.55) with the transformation $U \to L U R^\dagger$, we see that $\mathcal{M}$ must transform as $\mathcal{M} \to L \mathcal{M} R^\dagger$ to make $\bar{q}_L \mathcal{M} q_R$ invariant — exactly like U itself. We package the spurion as

$$\chi \equiv 2B\,\mathcal{M}, \qquad \chi \;\longrightarrow\; L\,\chi\,R^\dagger, \tag{39.56}$$

where B is a constant of mass dimension one whose physical meaning (the quark condensate) we extract below; the factor $2B$ is conventional and makes later formulas clean. Now we enumerate, exactly as in Section 39.2.2, the lowest-order invariants built from $U, U^\dagger, \chi, \chi^\dagger$. The spurion has no derivatives, and in the power counting it is assigned chiral order p^2 (the reason is precisely the GMOR scaling we are about to derive: $m_\pi^2 \propto \chi$, and a mass-squared counts as two powers of momentum, so $\chi \sim p^2$; the next section makes this systematic). Thus a single power of χ with *no* derivatives is an $O(p^2)$ term, on the same footing as $\mathcal{L}_2$. The chiral-invariant, Lorentz-scalar combinations linear in χ are $\mathrm{tr}(\chi U^\dagger)$ and its conjugate $\mathrm{tr}(U\chi^\dagger)$, since under Eqs. (39.36) and (39.56), $\mathrm{tr}(\chi U^\dagger) \to \mathrm{tr}(L\chi R^\dagger\, R U^\dagger L^\dagger) = \mathrm{tr}(\chi U^\dagger)$ by cyclicity. Hermiticity requires their

sum; parity ($U \to U^\dagger$, $\chi \to \chi^\dagger$ since the mass term is parity even) also requires the symmetric combination. The unique $O(p^2)$ symmetry-restoring term is therefore

$$\boxed{\mathcal{L}_{\text{mass}} = \frac{f^2}{4}\,\mathrm{tr}(\chi\,U^\dagger + U\,\chi^\dagger)} \tag{39.57}$$

with the same prefactor $f^2/4$ as the kinetic term — this normalization is the conventional one and, as we now see, makes the leading mass relation come out in standard form. The complete leading-order chiral Lagrangian is

$$\mathcal{L}^{(2)}_{\text{ChPT}} = \frac{f^2}{4}\,\mathrm{tr}(\partial_\mu U\,\partial^\mu U^\dagger) + \frac{f^2}{4}\,\mathrm{tr}(\chi\,U^\dagger + U\,\chi^\dagger), \qquad \chi = 2B\mathcal{M}. \tag{39.58}$$

Caution

The spurion is a bookkeeping device, not a dynamical field. We endow χ with a transformation law only to enumerate invariants; once Eq. (39.57) is written, χ is set to the constant $2B\mathcal{M}$ and the chiral symmetry is genuinely (explicitly) broken by it, exactly as in the general framework of Theorem 29.28. The power of the method is that it guarantees we have written *every* allowed breaking term of this order and *only* those, with their relative coefficients fixed by the requirement that they descend from the single spurion structure of QCD.

39.2.5 The pion mass, GMOR, and the Gell-Mann–Okubo relation

We now read the meson masses off Eq. (39.58). Take $\mathcal{M}$ real and diagonal, so $\chi = \chi^\dagger = 2B\mathcal{M}$, and expand $U + U^\dagger = 2\mathbf{1} - \Phi^2/f^2 + O(\Phi^4)$. Then

$$\mathcal{L}_{\text{mass}} = \frac{f^2}{4}\,\mathrm{tr}(\chi(U^\dagger{+}U)) = \frac{f^2}{4}\,\mathrm{tr}\Big(\chi(2\mathbf{1}{-}\tfrac{1}{f^2}\Phi^2{+}\cdots)\Big) = \underbrace{\frac{f^2}{2}\,\mathrm{tr}\,\chi}_{\text{const}} - \frac{1}{4}\,\mathrm{tr}(\chi\,\Phi^2){+}\cdots\,. \tag{39.59}$$

The constant is the vacuum energy and the quadratic piece is the meson mass term. Writing it out,

$$\mathcal{L}^{(2)}_{\text{mass}} = -\frac{1}{4}\,\mathrm{tr}(\chi\,\Phi^2) = -\frac{B}{2}\,\mathrm{tr}(\mathcal{M}\,\Phi^2). \tag{39.60}$$

Two flavors and GMOR. For $N_f = 2$ with $\mathcal{M} = \mathrm{diag}(m_u, m_d)$ and Φ of Eq. (39.34), a short trace gives $\mathrm{tr}(\mathcal{M}\Phi^2) = (m_u + m_d)\big[(\pi^0)^2 + 2\pi^+\pi^-\big]$, so

$$\mathcal{L}^{(2)}_{\text{mass}} = -\frac{B}{2}(m_u + m_d)(\pi^0)^2 - B(m_u + m_d)\pi^+\pi^-, \tag{39.61}$$

and reading off the coefficients ($-\frac{1}{2}m^2$ for the real field π^0, $-m^2$ for the complex field $\pi^\pm$) gives a single result for the whole isotriplet,

$$\boxed{m_\pi^2 = B\,(m_u + m_d)}\,. \tag{39.62}$$

This is the Gell-Mann–Oakes–Renner relation, now *derived inside* chiral perturbation theory. It says the pion mass-squared is linear in the quark masses — the pion is a pseudo-Goldstone boson whose mass vanishes in the chiral limit, exactly the protected behavior of Theorem 29.29. To identify B, match Eq. (39.62) to the general GMOR relation Eq. (29.216), which expresses the pseudo-Goldstone mass-squared as the explicit-breaking parameter times a condensate over f^2. The cleanest route is to compute the condensate directly from Eq. (39.58). The QCD condensate is the response of the vacuum energy to the quark mass, $\langle \bar q_f q_f\rangle = \partial \mathcal{E}_{\rm vac}/\partial m_f$, where $\mathcal{E}_{\rm vac} = -\mathcal{L}(U=\mathbf{1})$ is the vacuum energy density. From the constant term of Eq. (39.59), $\mathcal{L}(U=\mathbf{1}) = \frac{f^2}{2}\,\mathrm{tr}\,\chi = f^2 B \sum_f m_f$, so $\mathcal{E}_{\rm vac} = -f^2 B \sum_f m_f$ and

$$\langle \bar q q\rangle \equiv \langle \bar u u\rangle = \frac{\partial \mathcal{E}_{\rm vac}}{\partial m_u} = -f^2 B \qquad\Longrightarrow\qquad \boxed{B = -\frac{\langle \bar q q\rangle}{f^2}}\,. \tag{39.63}$$

Since the condensate is negative ($\langle \bar q q\rangle < 0$, the order parameter of chiral breaking from Section 39.1), the constant B is positive and the pion mass-squared Eq. (39.62) is positive, as it must be. Substituting Eq. (39.63) into Eq. (39.62),

$$f_\pi^2\, m_\pi^2 = -(m_u + m_d)\,\langle \bar q q\rangle\,, \tag{39.64}$$

which is the GMOR relation in its standard physical form and matches Eq. (29.216) term by term, with $\varepsilon = m_u + m_d$ the explicit-breaking parameter and $-\langle \bar q q\rangle$ the condensate. The chiral Lagrangian has reproduced, with one line of trace algebra, the relation that Chapter 29 obtained from the broken Ward identity. Numerically, with $m_\pi \approx 135\,\mathrm{MeV}$, $f_\pi \approx 92\,\mathrm{MeV}$ and the $\overline{\rm MS}$ light-quark masses $m_u + m_d \approx 7\,\mathrm{MeV}$ (at $2\,\mathrm{GeV}$), Eq. (39.64) gives $-\langle \bar q q\rangle \approx (270\,\mathrm{MeV})^3$, the standard size of the condensate; the smallness of the quark masses against the QCD scale is exactly why the pion is so much lighter than every other hadron.

Three flavors and Gell-Mann–Okubo. The same computation for $N_f = 3$ is more powerful: there are three independent meson mass-squareds but only two quark-mass parameters (in the isospin limit), so one combination is predicted. This is the content of the next worked example.

Worked Example 39.9 (The octet masses and the Gell-Mann–Okubo relation)**.** Take $\mathcal{M} = \mathrm{diag}(m_u, m_d, m_s)$ and the octet matrix Φ of Eq. (39.35). We evaluate $\mathrm{tr}(\mathcal{M}\Phi^2)$ entry by entry. Since Φ is Hermitian, $(\Phi^2)_{ii} = \sum_j |\Phi_{ij}|^2$, and

$\mathrm{tr}(\mathcal{M}\Phi^2) = \sum_{ij} m_i |\Phi_{ij}|^2$. The off-diagonal entries give, using $|\Phi_{12}|^2 = |\Phi_{21}|^2 = 2\pi^+\pi^-$ etc.,

$$\begin{aligned} \text{charged pions:} \quad & m_u|\Phi_{12}|^2 + m_d|\Phi_{21}|^2 = 2(m_u+m_d)\,\pi^+\pi^-, \\ \text{charged kaons:} \quad & m_u|\Phi_{13}|^2 + m_s|\Phi_{31}|^2 = 2(m_u+m_s)\,K^+K^-, \\ \text{neutral kaons:} \quad & m_d|\Phi_{23}|^2 + m_s|\Phi_{32}|^2 = 2(m_d+m_s)\,K^0\bar{K}^0, \end{aligned}$$

and the diagonal entries, with $\Phi_{11} = \pi^0 + \eta/\sqrt{3}$, $\Phi_{22} = -\pi^0 + \eta/\sqrt{3}$, $\Phi_{33} = -2\eta/\sqrt{3}$, give

$$m_u\Phi_{11}^2 + m_d\Phi_{22}^2 + m_s\Phi_{33}^2 = (m_u+m_d)(\pi^0)^2 + \tfrac{m_u+m_d+4m_s}{3}\,\eta^2 + \tfrac{2(m_u-m_d)}{\sqrt{3}}\,\pi^0\eta. \tag{39.65}$$

Assembling these into $\mathcal{L}^{(2)}_{\mathrm{mass}} = -\frac{B}{2}\,\mathrm{tr}(\mathcal{M}\Phi^2)$ and reading off the mass-squareds (factor $-\frac{1}{2}m^2$ for real fields π^0, η; $-m^2$ for the complex $\pi^\pm, K^\pm, K^0$):

$$m_{\pi^\pm}^2 = B(m_u+m_d), \quad m_{K^\pm}^2 = B(m_u+m_s), \quad m_{K^0}^2 = B(m_d+m_s), \tag{39.66}$$

with $m_{\pi^0}^2 = B(m_u+m_d)$ as well. The π^0–η sector has the mass matrix

$$B\begin{pmatrix} m_u+m_d & \frac{m_u-m_d}{\sqrt{3}} \\ \frac{m_u-m_d}{\sqrt{3}} & \frac{m_u+m_d+4m_s}{3} \end{pmatrix}, \tag{39.67}$$

whose off-diagonal π^0–η mixing is proportional to the isospin breaking $m_u - m_d$ and vanishes in the isospin limit. Setting $m_u = m_d \equiv \hat{m}$ to isolate the leading pattern, the diagonal entries become the physical masses:

$$m_\pi^2 = 2B\hat{m}, \qquad m_K^2 = B(\hat{m}+m_s), \qquad m_\eta^2 = \frac{2B}{3}(\hat{m}+2m_s). \tag{39.68}$$

These three relations involve only the two unknowns $B\hat{m}$ and Bm_s, so they are not independent: eliminating both yields a parameter-free prediction. Compute $4m_K^2 - m_\pi^2 = 4B(\hat{m}+m_s) - 2B\hat{m} = 2B(\hat{m}+2m_s)$, and compare with $3m_\eta^2 = 2B(\hat{m}+2m_s)$. They are equal:

$$\boxed{\,3\,m_\eta^2 = 4\,m_K^2 - m_\pi^2\,}. \tag{39.69}$$

This is the Gell-Mann–Okubo mass relation for the pseudoscalar octet, derived here as a direct consequence of the leading chiral Lagrangian with explicit SU(3)-breaking quark masses. It is a genuine prediction: three measured masses constrained by one equation. Numerically, with $m_\pi = 138\,\mathrm{MeV}$, $m_K = 496\,\mathrm{MeV}$, $m_\eta = 548\,\mathrm{MeV}$, the right side gives $4m_K^2 - m_\pi^2 = (0.982\,\mathrm{GeV})^2$, i.e. a predicted $m_\eta = \sqrt{(4m_K^2 - m_\pi^2)/3} \approx 567\,\mathrm{MeV}$, against the measured $548\,\mathrm{MeV}$ — agreement at the few-percent level, exactly the size of the $O(p^4)$ corrections (and the small η–η' mixing) one expects beyond leading order. Figure 39.2 displays the octet spectrum and the test.

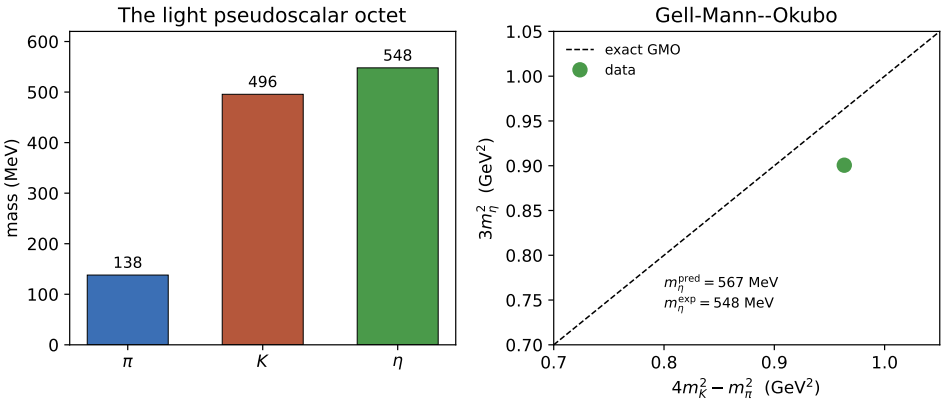


Figure 39.2: The light pseudoscalar octet and the Gell-Mann–Okubo test. *Left:* the measured isospin-averaged masses of the π, K, and η, the $N_f = 3$ pseudo-Goldstone bosons organized by the leading chiral Lagrangian Eq. (39.58). *Right:* the parameter-free relation Eq. (39.69), plotting $3m_\eta^2$ against $4m_K^2 - m_\pi^2$; exact GMO lies on the dashed diagonal. The leading-order prediction $m_\eta \approx 567\,\text{MeV}$ overshoots the measured 548 MeV by a few percent, the expected size of the neglected $O(p^4)$ corrections.

Physical Insight

The ratio of the three masses carries information the individual masses do not. From Eq. (39.68), $m_K^2/m_\pi^2 = (\hat{m} + m_s)/2\hat{m}$; with $m_K \approx 496$ and $m_\pi \approx 138\,\text{MeV}$ this gives $m_s/\hat{m} \approx 2(m_K^2/m_\pi^2) - 1 \approx 25$, the strange quark roughly twenty-five times heavier than the average up/down. The chiral Lagrangian thus turns meson masses — the only directly measurable handles in the strongly coupled regime — into a measurement of the *ratios* of the fundamental quark masses, quantities that are otherwise inaccessible because free quarks do not exist. Overall quark-mass scales require the constant B, hence the condensate, but the ratios fall straight out of the symmetry analysis.

With $\mathcal{L}^{(2)}_{\rm ChPT}$ in hand we have the two physical inputs of the theory — the decay constant $f_\pi \approx 92\,\text{MeV}$ fixing the derivative interactions and the condensate $B = -\langle\bar{q}q\rangle/f^2$ (equivalently the measured m_π) fixing the masses — and a complete, predictive interacting Lagrangian Eq. (39.58). What we do not yet have is a principled way to know *when* this leading term suffices and how to compute the corrections to the few-percent discrepancies already visible in Gell-Mann–Okubo. That is the work of the next section: Weinberg's power-counting theorem will organize every diagram built from Eq. (39.58) (and its higher-derivative descendants) by powers of momentum over the scale $\Lambda_\chi \sim 4\pi f_\pi$,

proving that the chiral expansion is systematic and predictive order by order despite QCD being strongly coupled.

39.3 Weinberg Power Counting and the Chiral Expansion

The leading-order Lagrangian of Section 39.2 is not the whole theory. Chiral symmetry permits, alongside the two-derivative term $\mathcal{L}_2$, an unbounded tower of higher operators: four-derivative terms, six-derivative terms, terms with extra insertions of the quark-mass spurion, and so on. Each comes with its own coupling, none of which symmetry fixes. Taken at face value this looks fatal — infinitely many unknown constants — and it would be, were there no way to decide which operators matter for a given physical question. The resolution is the central technical result of chiral perturbation theory and the reason it deserves the name "perturbation theory" at all: every connected diagram of the effective theory carries a definite power of the low-energy momentum, that power increases with the number of loops and with the order of the vertices used, and at any fixed power only finitely many diagrams and finitely many operators contribute. This is Weinberg's power-counting theorem. It converts the daunting infinite Lagrangian into a strict order-by-order expansion, with the strong QCD dynamics quarantined inside a handful of constants at each order. We prove the counting formula in full, extract from it the breakdown scale Λ_χ, and show explicitly why one-loop graphs and the $O(p^4)$ Lagrangian are inseparable.

39.3.1 The chiral order of the building blocks

Power counting is a statement about how a diagram behaves when every external momentum and every quark mass is uniformly scaled toward zero. To make it precise we assign to each ingredient of a diagram a *chiral dimension*: the power of a generic small momentum p it carries when the kinematics is rescaled $p \to tp$ with $t \to 0$. We continue to use the leading-order Lagrangian in the form recalled from Section 39.2,

$$\mathcal{L}_2 = \frac{f^2}{4}\,\mathrm{tr}(\partial_\mu U^\dagger\,\partial^\mu U) + \frac{f^2}{4}\,\mathrm{tr}(\chi^\dagger U + U^\dagger \chi), \qquad U = \exp\!\Big(\frac{i\,\Phi}{f}\Big), \quad \chi = 2B\,\mathcal{M}_q, \tag{39.70}$$

with $\Phi = \pi^a \tau^a$ the pion matrix, $f = f_\pi$ the decay constant of Eq. (29.202), and $\mathcal{M}_q$ the quark-mass matrix entering the spurion χ.

There are three building blocks, and we fix the chiral dimension of each.

Derivatives. The nonlinear field U is dimensionless; the pion enters $\mathcal{L}_2$ only through $\partial_\mu U$. This is the derivative coupling forced by the Goldstone shift symmetry (Theorem 29.25): a single soft pion couples through at least one

derivative, so each ∂_μ acting on a field becomes a momentum factor p_μ at the vertex and counts as one power of p. A vertex built from a monomial with n_∂ derivatives therefore scales as p^{n_∂}.

Pion propagators. The free pion propagator is $i/(p^2 - m_\pi^2 + i\epsilon)$. For this to scale homogeneously we must count the mass as a momentum. The mass term in Eq. (39.70) produces, on expanding $U = 1 + i\Phi/f - \Phi^2/2f^2 + \cdots$, a quadratic piece $\frac{1}{4}\,\mathrm{tr}(\chi^\dagger + \chi)\,(-\Phi^2/f^2)$ that fixes

$$m_\pi^2 = B\,(m_u + m_d), \tag{39.71}$$

which is precisely the Gell-Mann–Oakes–Renner relation Eq. (29.216) specialized to two flavors. Equation Eq. (39.71) is the bridge between the explicit-breaking parameter and the spectrum: the pion mass-squared is *linear* in the quark masses. We exploit this by declaring the spurion to carry chiral dimension two,

$$\chi \sim O(p^2) \quad \Longleftrightarrow \quad m_\pi^2 \sim O(p^2), \tag{39.72}$$

so that quark masses count as two derivatives. With this choice the denominator $p^2 - m_\pi^2$ is homogeneous of degree two and every pion propagator scales as p^{-2}. The assignment Eq. (39.72) is not a convention of convenience but a physical fact about QCD: the chiral limit $m_q \to 0$ and the soft limit $p \to 0$ are reached together, and only if they are counted together does the expansion organize the corrections to GMOR consistently.

Loop integrations. Each independent loop carries a four-dimensional momentum integral $\int d^4k/(2\pi)^4$. Under the rescaling $k \to tk$ the measure scales as t^4, so each loop contributes four powers of p.

Definition 39.10 (Chiral dimension of an operator and a vertex)**.** The *chiral dimension* (or chiral order) of a monomial in the chiral Lagrangian is

$$d \;=\; (\text{number of derivatives}) \;+\; 2\,(\text{number of } \chi \text{ insertions}), \tag{39.73}$$

counting each χ as $O(p^2)$ per Eq. (39.72). Chiral symmetry and Lorentz invariance allow only even d, so the Lagrangian decomposes as $\mathcal{L}_{\text{eff}} = \mathcal{L}_2 + \mathcal{L}_4 + \mathcal{L}_6 + \cdots$, where $\mathcal{L}_d$ collects all independent operators of chiral dimension d. A vertex drawn from $\mathcal{L}_d$ is said to be a vertex of order d; we write N_d for the number of such vertices appearing in a given diagram.

Both terms of $\mathcal{L}_2$ in Eq. (39.70) have $d = 2$: the kinetic term has two derivatives and no χ, the mass term has no derivatives and one χ. They sit at the same chiral order, which is why the pion's kinetic energy and its mass are predicted together at leading order. The leading higher term $\mathcal{L}_4$ contains operators such as $\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U)^2$ (four derivatives), $\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U)\ \mathrm{tr}(\chi^\dagger U + U^\dagger \chi)$ (two derivatives, one χ), and $[\mathrm{tr}(\chi^\dagger U + U^\dagger \chi)]^2$ (two χ's) — all of chiral dimension four, all multiplying independent low-energy constants.

39.3.2 The power-counting theorem

We can now state and prove how a whole diagram scales. The result is purely topological: it depends only on the loop number and the orders of the vertices, not on the number of external pions.

Theorem 39.11 (Weinberg's chiral power counting)**.** *Let $\mathcal{A}$ be the amplitude associated with a connected Feynman diagram of the chiral effective theory in four spacetime dimensions, with L loops and, for each even $d \geq 2$, exactly N_d vertices drawn from $\mathcal{L}_d$. Under the uniform rescaling of all external momenta and pion masses by a factor t, the amplitude scales homogeneously, $\mathcal{A}(tp, tm_\pi) = t^D\, \mathcal{A}(p, m_\pi)$, with chiral dimension*

$$\boxed{D \;=\; 2 + 2L + \sum_d (d-2)\, N_d\,.} \tag{39.74}$$

Proof. Let the diagram have I internal pion lines and V vertices in total. By Theorem 39.10 every vertex belongs to some $\mathcal{L}_d$, so the total vertex count and the total chiral order carried by the vertices are

$$V = \sum_d N_d, \qquad \sum_{\text{vertices}} (\text{order}) = \sum_d d\, N_d. \tag{39.75}$$

Step 1: the raw momentum power. We rescale every loop momentum $k_i \to t k_i$ together with the external momenta and the masses, and read off the homogeneity degree of the integrand and measure. Each of the L loop measures $\int d^4k_i/(2\pi)^4$ contributes t^4, for a total t^{4L}. Each of the I internal propagators is $i/(\ell^2 - m_\pi^2 + i\epsilon)$ with ℓ a linear combination of loop and external momenta; with the mass counted as in Eq. (39.72), its denominator is homogeneous of degree two, so each propagator contributes t^{-2}, for a total t^{-2I}. Each vertex from $\mathcal{L}_d$ is, by Eq. (39.73), a homogeneous polynomial of degree d in momenta and in $\chi \sim m_\pi^2$, hence contributes t^d; the product over all vertices is $t^{\sum_d d\, N_d}$. Multiplying,

$$\mathcal{A}(tp, tm_\pi) = t^{\,D}\, \mathcal{A}(p, m_\pi), \qquad D = 4L - 2I + \sum_d d\, N_d. \tag{39.76}$$

That the rescaled integral is genuinely t^D times the original — and not merely bounded by it — follows because the integrand is a ratio of homogeneous functions; substituting $k_i = t k_i'$ in the integral and pulling the t's through the measure and the homogeneous denominators and numerators returns the original integral times the counted power. (Renormalization can introduce a logarithm $\ln t$, which does not change the leading power D; this is exactly the chiral logarithm that the loop section will compute, and it sits at the order dictated by Eq. (39.74).)

Step 2: eliminate I with the loop relation. For any connected graph the number of independent loops equals the number of internal lines minus the

number of vertices plus one, because each internal line carries an integration momentum, each vertex imposes one momentum-conserving delta function, and one overall delta function survives as total momentum conservation rather than constraining a loop:

$$L = I - V + 1 \qquad \Longleftrightarrow \qquad I = L + V - 1. \tag{39.77}$$

Substituting $I = L + V - 1$ into Eq. (39.76),

$$D = 4L - 2(L + V - 1) + \sum_d d\, N_d = 2L - 2V + 2 + \sum_d d\, N_d. \tag{39.78}$$

Step 3: eliminate V. Using $V = \sum_d N_d$ from Eq. (39.75), write $-2V = -2\sum_d N_d = \sum_d (-2) N_d$ and combine with the vertex-order sum:

$$D = 2 + 2L + \sum_d d\, N_d - 2\sum_d N_d = 2 + 2L + \sum_d (d-2)\, N_d, \tag{39.79}$$

which is Eq. (39.74). □

Three features of Eq. (39.74) deserve emphasis, and each is a statement about the structure of the expansion rather than about any single diagram.

First, the number of external legs has vanished. The Euler relation Eq. (39.77) absorbed it, so the chiral order of a process is set entirely by its loops and vertices, not by how many pions scatter. A four-pion and a six-pion amplitude built from the same number of loops and the same vertex orders sit at the same chiral order. (They differ, of course, in overall mass dimension and in powers of $1/f$; what Eq. (39.74) controls is the power of the *small* scale p/Λ_χ.)

Second, every term in the sum over d is nonnegative: the lowest Lagrangian is $\mathcal{L}_2$, so $d \geq 2$ and $(d-2) \geq 0$, with equality only for $\mathcal{L}_2$ vertices. Hence

$$D \geq 2 + 2L \geq 2, \tag{39.80}$$

and D takes only even values. There is a hard floor at $D = 2$, and one cannot go below it by any choice of diagram. This is what makes the theory predictive: low chiral order is a finite, enumerable category.

Third, both knobs that raise D raise it by at least two. Adding a loop costs $\Delta D = +2$ through the $2L$ term. Trading an $\mathcal{L}_2$ vertex for an $\mathcal{L}_4$ vertex costs $\Delta D = +2$ through the $(d-2)N_d$ term. There is no way to make a smaller step. The chiral expansion therefore proceeds in units of p^2.

39.3.3 Leading and next-to-leading order

The floor Eq. (39.80) and the step size $+2$ immediately classify the first two orders.

Corollary 39.12 (The first two chiral orders)**.** *At leading order, $D = 2$, the only contributing diagrams have $L = 0$ and use exclusively $\mathcal{L}_2$ vertices (N_2 arbitrary, $N_{d\geq 4} = 0$): tree graphs of the leading Lagrangian. At next-to-leading order, $D = 4$, exactly two classes contribute and they enter at the same chiral order:*

1. *one-loop graphs, $L = 1$, built entirely from $\mathcal{L}_2$ vertices ($N_{d\geq 4} = 0$); and*
2. *tree graphs, $L = 0$, with exactly one $\mathcal{L}_4$ vertex ($N_4 = 1$) and any number of $\mathcal{L}_2$ vertices.*

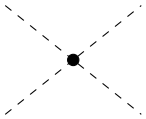
No other combination gives $D = 4$.

Proof. Set $D = 2$ in Eq. (39.74): since $2L \geq 0$ and $\sum_d (d-2)N_d \geq 0$, the equation $0 = 2L + \sum_d (d-2)N_d$ forces both nonnegative terms to vanish, i.e. $L = 0$ and $N_d = 0$ for all $d > 2$. Set $D = 4$: then $2 = 2L + \sum_d (d-2)N_d$. The two nonnegative integers $2L$ and $\sum_d (d-2)N_d$ must sum to 2. The options are $(2L, \sum(d-2)N_d) = (2, 0)$, giving $L = 1$ with all vertices of order two; or $(0, 2)$, giving $L = 0$ with $\sum_d (d-2)N_d = 2$, whose only solution in the even tower is a single $\mathcal{L}_4$ vertex ($d - 2 = 2$, $N_4 = 1$) since an $\mathcal{L}_6$ vertex would already contribute $d - 2 = 4$. No other split of 2 exists. □

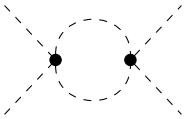
Theorem 39.12 is the operational heart of the chapter. It says that the leading prediction for *any* pionic process is obtained by drawing tree diagrams with $\mathcal{L}_2$ alone — no free parameters beyond f and the quark masses already fixed at leading order — and that the *first* correction is the sum of one-loop $\mathcal{L}_2$ graphs and tree graphs with one insertion of $\mathcal{L}_4$. The two NLO classes are not independent options to be chosen between; they are summed together, and they must be, because neither is well defined without the other. The one-loop graphs are divergent, and by Eq. (39.76) their divergences are local polynomials of chiral dimension four in the external momenta and masses — exactly the structure of the $\mathcal{L}_4$ operators. The couplings of $\mathcal{L}_4$ therefore absorb the loop divergences: they are the counterterms of the one-loop calculation. This is the chiral version of the renormalization logic and the reason the higher Lagrangian is mandatory rather than optional. We display the three relevant topologies in Fig. 39.3.

Worked Example 39.13 (Why the one-loop divergence has the order of $\mathcal{L}_4$)**.** Take the four-pion amplitude. The leading graph is Fig. 39.3a: a single $\mathcal{L}_2$ four-point vertex. Its two-derivative coupling produces an amplitude of chiral dimension $D = 2 + 2(0) + (2-2)N_2 = 2$, the floor. Now form the one-loop pion bubble Fig. 39.3b, with two four-point $\mathcal{L}_2$ vertices joined by two internal lines. Here $L = 1$, $N_2 = 2$, and Eq. (39.74) gives $D = 2 + 2(1) + 0 = 4$. The loop integral is, schematically,

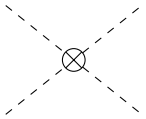
$$\int \frac{d^4k}{(2\pi)^4} \frac{(\text{two two-derivative vertices})}{(k^2 - m_\pi^2)\,((k+q)^2 - m_\pi^2)}, \tag{39.81}$$



(a) LO: $D = 2$, tree, $\mathcal{L}_2$ vertex.



(b) NLO: $D = 4$, one loop, two $\mathcal{L}_2$ vertices.



(c) NLO: $D = 4$, tree, one $\mathcal{L}_4$ vertex.

Figure 39.3: The chiral hierarchy for the four-pion amplitude. The leading order (a) is a single tree graph from $\mathcal{L}_2$. The next-to-leading order is the *sum* of one-loop graphs from $\mathcal{L}_2$ — of which the s-channel pion bubble (b) is one — and tree graphs with one $\mathcal{L}_4$ vertex (the crossed dot, (c)). The local divergence of (b) has the chiral dimension of (c), so the $\mathcal{L}_4$ couplings renormalize the loop. Dashed lines are pions.

where q is the momentum flowing through the bubble. Each vertex supplies two powers of momentum, so the numerator is degree four in momenta; the integral is superficially divergent, and the divergent part is a polynomial in the external invariants of total degree four (a constant times q^4, $q^2 m_\pi^2$, or m_π^4). A polynomial of chiral dimension four in q and m_π is precisely what a tree-level $\mathcal{L}_4$ vertex Fig. 39.3c produces: $D = 2 + 0 + (4-2)(1) = 4$, with the four-point coupling carrying four powers of momentum/mass. Hence the divergence of Fig. 39.3b is removed by a renormalization of the $\mathcal{L}_4$ couplings, and the finite remainder of the loop plus the finite (renormalized) $\mathcal{L}_4$ tree together constitute the unambiguous, scale-dependent NLO prediction. Neither piece is meaningful alone: the loop is divergent, the counterterm is undetermined; their sum is finite and predictive. This is the concrete content of Theorem 39.12.

39.3.4 The breakdown scale Λ_χ

The counting tells us the *order* of each correction in powers of p; to know whether the expansion converges we must know the *scale* that the momentum is compared against. That scale is set by the price of one loop, and we extract it by tracking the dimensionful factors that accompany the step from a tree graph to the loop that corrects it.

Worked Example 39.14 ($\Lambda_\chi = 4\pi f_\pi$ from a single pion loop)**.** Consider again the two NLO members of Theorem 39.12 that correct the leading four-pion amplitude, and compare their sizes. Expanding $\mathcal{L}_2$ in Eq. (39.70) to fourth order in the pion field gives a four-point contact interaction with two derivatives and coefficient $1/f^2$; schematically the vertex is $\sim (1/f^2)\,(\partial\pi)^2\pi^2$, so a single such vertex contributes to the dimensionless $2 \to 2$ amplitude

$$\mathcal{A}_{\text{LO}} \;\sim\; \frac{p^2}{f^2}, \tag{39.82}$$

where p stands for a generic external momentum or pion mass. The one-loop bubble of Fig. 39.3b stitches two such vertices together with two propagators and one loop integration. Counting the dimensionful factors: two vertices give $(1/f^2)^2 = 1/f^4$; the loop must, by Theorem 39.11, restore the overall chiral dimension to $D = 4$, supplying four powers of the external scale, p^4; and the four-dimensional loop integral carries the universal geometric factor of one loop,

$$\int \frac{d^4k}{(2\pi)^4}\,(\cdots) \;\sim\; \frac{1}{16\pi^2}\,(\cdots), \tag{39.83}$$

the same $1/(4\pi)^2$ per loop established in Eq. (19.18) (in $\overline{\text{MS}}$ this is the coefficient that multiplies the $1/\varepsilon$ pole and the chiral logarithm $\ln(m_\pi^2/\mu^2)$; its origin is the surface area $2\pi^2$ of the unit three-sphere divided into the $(2\pi)^4$ of the measure). Assembling,

$$\mathcal{A}_{\text{NLO}}^{\text{loop}} \;\sim\; \frac{1}{16\pi^2}\,\frac{p^4}{f^4}. \tag{39.84}$$

The ratio of the correction to the leading term is therefore

$$\frac{\mathcal{A}_{\text{NLO}}^{\text{loop}}}{\mathcal{A}_{\text{LO}}} \;\sim\; \frac{p^4/(16\pi^2 f^4)}{p^2/f^2} \;=\; \frac{p^2}{16\pi^2 f^2} \;=\; \frac{p^2}{(4\pi f)^2}. \tag{39.85}$$

The dimensionful combination controlling the loop expansion is thus $4\pi f$. Identifying $f = f_\pi$ as in Section 39.2, we name the breakdown scale

$$\boxed{\Lambda_\chi \;=\; 4\pi f_\pi \;\approx\; 4\pi \times 92\ \text{MeV} \;\approx\; 1.2\ \text{GeV}.} \tag{39.86}$$

The expansion parameter of chiral perturbation theory is the dimensionless ratio

$$\frac{p^2}{\Lambda_\chi^2} = \frac{p^2}{(4\pi f_\pi)^2}, \tag{39.87}$$

where p is the largest of the external momenta and the pion mass. Equation Eq. (39.85) shows that each additional unit of chiral order $\Delta D = 2$ — whether spent on a loop or on a higher vertex, since Theorem 39.12 ties their dimensionful costs together — multiplies the amplitude by one power of Eq. (39.87). For physical pions, $m_\pi^2/\Lambda_\chi^2 \approx (0.14/1.2)^2 \approx 1.3\%$, so the leading order is good to roughly the ten-percent level and one loop brings it to the percent level. For kaons the parameter is $m_K^2/\Lambda_\chi^2 \approx (0.5/1.2)^2 \approx 17\%$, and the expansion converges visibly more slowly — a quantitative expectation borne out in three-flavor fits. The value Eq. (39.86) also explains the empirical spectrum: the lightest non-Goldstone hadron, the ρ meson at 770 MeV, lies below Λ_χ, so once external momenta approach the ρ the neglected resonance dynamics intrudes and the expansion in p/Λ_χ must fail — exactly where Eq. (39.86) predicts.

Physical Insight

The breakdown scale $\Lambda_\chi \sim 4\pi f_\pi$ is the same kind of object as the cutoff that organizes any effective field theory (Section 38.2): it is the scale at which the next omitted operator becomes as important as the ones kept. What is special to a Goldstone theory is that this scale is not an input. It is generated by the loop factor $1/(4\pi)^2$ playing against the single dimensionful coupling f that the symmetry already required at leading order. A theory with one scale f and a four-dimensional loop counting automatically supplies its own ultraviolet completion scale $4\pi f$. The number 4π is geometry, not dynamics; the dynamics is entirely in f.

39.3.5 Predictivity from symmetry, not from weak coupling

It is worth being explicit about why this works when ordinary perturbation theory in the QCD coupling does not. Below Λ_χ the strong coupling α_s is of order one — this is the infrared face of the asymptotic freedom established in Theorem 27.7 — so there is no small coupling to expand in and no sense in which quarks and gluons are weakly interacting. Chiral perturbation theory does not expand in α_s. It expands in p/Λ_χ, a kinematic ratio, and the organizing principle is the symmetry, not the coupling. This is the EFT paradigm of Chapter 38 applied to a strongly coupled microscopic theory.

The construction follows the logic of Section 38.2 point for point. The chiral Lagrangian $\mathcal{L}_2 + \mathcal{L}_4 + \cdots$ is the *most general* local functional of the Goldstone field U consistent with the chiral symmetry and its explicit breaking by χ; writing down anything less would be to impose a constraint the symmetry does not, and writing down anything more is impossible. Among the infinitely many allowed operators, Eq. (39.74) sorts them into a relevant–irrelevant hierarchy in the precise sense of Section 23.5: at energy p an operator of chiral dimension d contributes at relative size $(p/\Lambda_\chi)^{d-2}$, so high-dimension operators are suppressed and only finitely many survive at any fixed accuracy. The coefficients of the operators — f and B at order p^2, the Gasser–Leutwyler constants at order p^4, and so on — are the Wilson coefficients of this EFT (Theorem 38.4). They are not predicted by symmetry; they are numbers that encode the strong QCD dynamics that has been integrated out, in principle obtainable by matching to the full theory (Theorem 38.8) and in practice fixed by a finite set of measurements or by lattice QCD. Once they are fixed at a given order, *every other* observable at that order is predicted.

Physical Insight

The strong-coupling dynamics has not been solved; it has been *hidden*. Every nonperturbative effect of confined QCD — the value of the condensate, the pion decay constant, the low-energy constants — is packaged into a finite list of Wilson coefficients per chiral order. The miracle is the word "finite." Because each order admits only a bounded number of symmetry-allowed operators, the unknown strong dynamics enters through a bounded number of constants, and once those are measured the theory predicts. Predictivity is bought with parameters, but the price is finite at any fixed accuracy. That is the whole content of an effective field theory, made vivid here because the underlying theory is genuinely intractable by perturbation in its own coupling.

Caution

The decay-constant normalization is a notorious source of factor-of-$\sqrt{2}$ confusion. We use $f = f_\pi$ defined by $\langle 0|J_a^\mu|\pi^b(p)\rangle = i f\, p^\mu \delta_{ab}$ as in Eq. (29.202), for which $f_\pi \approx 92$ MeV and $\Lambda_\chi = 4\pi f_\pi \approx 1.2$ GeV. A common alternative writes $U = \exp(i\sqrt{2}\,\Phi/F)$ with $F = \sqrt{2} f_\pi \approx 130$ MeV, which shifts the apparent value of Λ_χ by the same $\sqrt{2}$. The physics — the loop factor $1/(16\pi^2)$ and the ratio Eq. (39.85) — is convention-independent; only the symbol's numerical value moves. Throughout this chapter $f = f_\pi \approx 92$ MeV.

With the counting in hand the path forward is fixed by Theorem 39.12. The leading prediction for any process is the sum of $O(p^2)$ tree graphs, built from $\mathcal{L}_2$ with no free parameters beyond the already-measured f_π and m_π; the next section carries this out for pion–pion scattering, extracting the amplitude and the s-wave scattering lengths with zero adjustable constants and watching the Adler zero of Theorem 29.26 emerge automatically from the derivative coupling. The first correction is the $O(p^4)$ combination of one pion loop and one $\mathcal{L}_4$ insertion identified in Theorem 39.13, whose loops generate the chiral logarithms and whose counterterms are the Gasser–Leutwyler low-energy constants; that program is the subject of the loops-and-LECs section to follow.

39.4 Pion-Pion Scattering at Leading Order

The power-counting theorem of Section 39.3 did something remarkable: it told us that, at the lowest order in the momentum expansion, the entire low-energy dynamics of the pions is generated by *tree* diagrams of the two-derivative Lagrangian $\mathcal{L}_2$, with every loop and every higher-derivative operator suppressed

by powers of p^2/Λ_χ^2, $\Lambda_\chi \sim 4\pi f_\pi$. The leading amplitude for any soft-pion process is therefore a finite, closed-form expression built from the single constant f_π (and, once the quark masses are turned on, from m_π, which is itself fixed by f_π and the condensate through Gell-Mann–Oakes–Renner, Eq. (29.228)). There are no free parameters left to adjust. This is the moment to cash that promissory note. We compute the leading-order amplitude for $\pi\pi \to \pi\pi$ scattering, project it onto channels of definite isospin, extract the s-wave scattering lengths, and put the resulting numbers next to the data. The computation is Weinberg's 1966 prediction [495]; it is the historic demonstration that a strongly coupled theory can be made to yield a sharp, falsifiable low-energy number, and it remains the cleanest single illustration of what chiral perturbation theory is for.

We have already met the skeleton of this amplitude. In the abstract nonlinear sigma model of Section 29.5 the leading Goldstone-Goldstone amplitude came out (Eq. (29.185)) as the crossing-symmetric combination $\mathcal{M}^{ab,cd} = v^{-2}(\delta^{ab}\delta^{cd}s + \delta^{ac}\delta^{bd}t + \delta^{ad}\delta^{bc}u)$, and we verified there that it carries the Adler zero (Eq. (29.186)). That computation was done for *massless* Goldstones in a generic $\mathrm{O}(N)$ coset. Three things now change as we specialize to real-world pions. First, the coset is the chiral one, $G/H = (\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R)/\mathrm{SU}(2)_V \simeq \mathrm{O}(4)/\mathrm{O}(3) \simeq S^3$, and the decay constant is the physical f_π. Second, the quarks are not massless, so the explicit-breaking term of Section 39.3 contributes its own four-pion vertex and shifts the amplitude by m_π^2. Third, isospin is now a physical $\mathrm{SU}(2)_V$ symmetry of the three pions, so the field-space index a becomes an isospin index and the amplitude decomposes into the observable channels $I = 0, 1, 2$. Carrying these three changes through is the content of this section.

39.4.1 The four-pion vertex from $\mathcal{L}_2$

Recall the leading-order two-flavor chiral Lagrangian assembled in Eq. (39.58),

$$\mathcal{L}_2 \;=\; \frac{f^2}{4}\,\mathrm{Tr}(\partial_\mu U^\dagger\,\partial^\mu U) \;+\; \frac{f^2}{4}\,\mathrm{Tr}(\chi^\dagger U + U^\dagger \chi), \qquad U(x) = \exp\!\Big(\frac{i\,\Phi(x)}{f}\Big), \tag{39.88}$$

with the Goldstone matrix packaging the three pion fields through the Pauli matrices τ^a ($\mathrm{Tr}(\tau^a\tau^b) = 2\delta^{ab}$, $[\tau^a, \tau^b] = 2i\epsilon^{abc}\tau^c$),

$$\Phi \;=\; \pi^a\tau^a \;=\; \begin{pmatrix} \pi^0 & \sqrt{2}\,\pi^+ \\ \sqrt{2}\,\pi^- & -\pi^0 \end{pmatrix}, \qquad \pi^\pm = \tfrac{1}{\sqrt{2}}(\pi^1 \mp i\pi^2), \quad \pi^0 = \pi^3. \tag{39.89}$$

The constant f appearing in U is the one identified with the physical pion decay constant, $f = f_\pi$, in Section 39.3 (we keep the symbol f through the algebra and restore f_π in the final formulas). The source field χ encodes the explicit breaking by the quark masses; in the isospin limit $m_u = m_d \equiv \hat{m}$ it is proportional to the unit matrix,

$$\chi \;=\; 2B\,\mathcal{M}_q \;=\; 2B\,\hat{m}\,\mathbf{1} \;=\; m_\pi^2\,\mathbf{1}, \tag{39.90}$$

the last equality being the leading-order Gell-Mann–Oakes–Renner identification $m_\pi^2 = 2B\hat{m}$ of Eq. (29.228) (here $B = -\langle\bar{q}q\rangle/f^2$ is the condensate parameter). With $\chi = \chi^\dagger = m_\pi^2\mathbf{1}$ the mass term reduces to $\frac{1}{4}f^2 m_\pi^2\,\mathrm{Tr}(U+U^\dagger)$.

To obtain the four-pion interaction we must expand both terms of Eq. (39.88) to fourth order in Φ. The expansion is controlled by one piece of SU(2) algebra that makes the two-flavor case especially clean. Because the antisymmetric part of $\tau^a\tau^b$ is $i\epsilon^{abc}\tau^c$, contracting with the symmetric tensor $\pi^a\pi^b$ kills it, and

$$\Phi^2 \;=\; \pi^a\pi^b\,\tau^a\tau^b \;=\; \pi^a\pi^b\big(\delta^{ab}\mathbf{1} + i\epsilon^{abc}\tau^c\big) \;=\; \pi^2\,\mathbf{1}, \qquad \pi^2 \equiv \pi^a\pi^a. \tag{39.91}$$

Thus Φ^2 is a multiple of the identity, hence central, and all even powers collapse, $\Phi^{2n} = (\pi^2)^n\mathbf{1}$. This is the algebraic shadow of the geometric fact that the SU(2) coset is the round three-sphere S^3.

The mass term. Using $\mathrm{Tr}(U+U^\dagger) = 2\sum_{n\geq 0}\frac{(-1)^n}{(2n)!}\,\mathrm{Tr}(\Phi^{2n})/f^{2n}$ and Eq. (39.91) with $\mathrm{Tr}\,\mathbf{1} = 2$,

$$\mathrm{Tr}\big(U+U^\dagger\big) \;=\; 2\Big[\,2 - \frac{\pi^2}{f^2} + \frac{(\pi^2)^2}{12\,f^4} - \cdots\Big], \tag{39.92}$$

where we used $\mathrm{Tr}\,\Phi^2 = 2\pi^2$ and $\mathrm{Tr}\,\Phi^4 = 2(\pi^2)^2$. Multiplying by $\frac{1}{4}f^2m_\pi^2$,

$$\frac{f^2m_\pi^2}{4}\,\mathrm{Tr}\big(U+U^\dagger\big) \;=\; \underbrace{f^2m_\pi^2}_{\text{vacuum energy}} \;-\; \frac{m_\pi^2}{2}\,\pi^a\pi^a \;+\; \underbrace{\frac{m_\pi^2}{24\,f^2}\,(\pi^2)^2}_{\mathcal{L}_4^{m}} \;-\; \cdots. \tag{39.93}$$

The constant is an irrelevant shift of the vacuum energy. The quadratic term $-\frac{1}{2}m_\pi^2\pi^a\pi^a$ is exactly the common pion mass — the mass term has done its first job. The quartic term

$$\mathcal{L}_4^{m} \;=\; \frac{m_\pi^2}{24\,f^2}\,(\pi^a\pi^a)^2 \tag{39.94}$$

is the mass contribution to the four-pion vertex; it is the new ingredient absent from the massless treatment of Section 29.5.

The kinetic term. The two-derivative piece requires more care because Φ and $\partial_\mu\Phi$ do not commute. The efficient bookkeeping uses the left current $\Omega_\mu \equiv U^\dagger\partial_\mu U$, which is anti-Hermitian (U unitary $\Rightarrow \Omega_\mu^\dagger = -\Omega_\mu$) and satisfies $\partial_\mu U = U\Omega_\mu$, so that

$$\mathrm{Tr}\big(\partial_\mu U^\dagger\,\partial^\mu U\big) \;=\; \mathrm{Tr}\big(\Omega_\mu^\dagger\,\Omega^\mu\big) \;=\; -\,\mathrm{Tr}\big(\Omega_\mu\,\Omega^\mu\big). \tag{39.95}$$

The standard derivative-of-the-exponential formula gives Ω_μ as a series in $a \equiv \Phi/f$,

$$\Omega_\mu \;=\; \frac{1-e^{-i\,\mathrm{ad}_a}}{i\,\mathrm{ad}_a}\,(i\,\partial_\mu a) \;=\; i\,\partial_\mu a \;+\; \tfrac{1}{2}\,[a,\partial_\mu a] \;-\; \tfrac{i}{6}\,[a,[a,\partial_\mu a]] \;+\; \cdots, \tag{39.96}$$

the three displayed terms being first, second, and third order in the field. Insert this into Eq. (39.95) and collect by field order. The quadratic part is $-\operatorname{Tr}\big((i\partial_\mu a)(i\partial^\mu a)\big) = \operatorname{Tr}(\partial_\mu a\,\partial^\mu a)$, which is $\frac{1}{f^2}\operatorname{Tr}(\partial_\mu\Phi\,\partial^\mu\Phi) = \frac{2}{f^2}\partial_\mu\pi^a\partial^\mu\pi^a$; multiplied by the prefactor $f^2/4$ it gives the canonical $\frac{1}{2}\partial_\mu\pi^a\partial^\mu\pi^a$, confirming the normalization of Eq. (39.88). The cubic part involves $\operatorname{Tr}\,(\partial_\mu a\,[a,\partial^\mu a])$, which vanishes by cyclicity of the trace (the two orderings $\operatorname{Tr}(\partial a\,a\,\partial a)$ and $\operatorname{Tr}(\partial a\,\partial a\,a)$ are equal), consistent with the $\pi\to-\pi$ parity that forbids odd vertices.

The quartic part receives two contributions, from $\big[\,i\partial a\,\big]\times\big[\,-\frac{i}{6}[a,[a,\partial a]]\,\big]$ (twice, by the cyclic symmetry of the trace) and from $\big[\frac{1}{2}[a,\partial a]\big]^2$:

$$\big[\operatorname{Tr}(\partial_\mu U^\dagger\partial^\mu U)\big]_{\Phi^4} \;=\; -\frac{1}{3f^4}\,\operatorname{Tr}(\partial_\mu\Phi\,[\Phi,[\Phi,\partial^\mu\Phi]]) \;-\; \frac{1}{4f^4}\,\operatorname{Tr}([\Phi,\partial_\mu\Phi]\,[\Phi,\partial^\mu\Phi]), \tag{39.97}$$

where we restored $a=\Phi/f$. The two traces are evaluated with the SU(2) structure constants. Writing $[\Phi,\partial_\mu\Phi]=2i\,\epsilon^{abc}\pi^a\partial_\mu\pi^b\,\tau^c\equiv 2i\,V_\mu^c\tau^c$, the second trace is

$$\begin{aligned}\operatorname{Tr}\big([\Phi,\partial_\mu\Phi][\Phi,\partial^\mu\Phi]\big) &= -4\,V_\mu^cV^{\mu d}\,\operatorname{Tr}(\tau^c\tau^d) = -8\,V_\mu^cV^{\mu c}\\ &= -8\big[\,\pi^2(\partial_\mu\pi\cdot\partial^\mu\pi)-(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi)\big] \;\equiv\; -8\,W,\end{aligned} \tag{39.98}$$

where the second line used $\epsilon^{abc}\epsilon^{a'b'c}=\delta^{aa'}\delta^{bb'}-\delta^{ab'}\delta^{ba'}$ and the shorthand $\pi\cdot\partial_\mu\pi\equiv\pi^a\partial_\mu\pi^a$. A parallel manipulation of the double commutator, using $[\Phi,[\Phi,\partial^\mu\Phi]]=-4V^{\mu c}\pi^d\epsilon^{dce}\tau^e$ and then $\epsilon^{dce}\epsilon^{abc}=\delta^{ea}\delta^{db}-\delta^{eb}\delta^{da}$, gives the first trace as exactly the opposite,

$$\operatorname{Tr}(\partial_\mu\Phi\,[\Phi,[\Phi,\partial^\mu\Phi]]) \;=\; +8\,W. \tag{39.99}$$

Substituting Eqs. (39.98) and (39.99) into Eq. (39.97) and multiplying by the prefactor $f^2/4$,

$$\mathcal{L}_4^{\text{kin}} \;=\; \frac{f^2}{4}\Big(-\frac{8W}{3f^4}+\frac{8W}{4f^4}\Big) \;=\; -\frac{W}{6\,f^2} \;=\; -\frac{1}{6f^2}\Big[\pi^2\,(\partial_\mu\pi\cdot\partial^\mu\pi)-(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi)\Big]. \tag{39.100}$$

The complete leading-order four-pion Lagrangian is the sum of Eqs. (39.94) and (39.100),

$$\boxed{\mathcal{L}_4 \;=\; \frac{1}{6f^2}\Big[(\pi^a\partial_\mu\pi^a)(\pi^b\partial^\mu\pi^b)-(\pi^a\pi^a)(\partial_\mu\pi^b\partial^\mu\pi^b)\Big] \;+\; \frac{m_\pi^2}{24f^2}\,(\pi^a\pi^a)^2.} \tag{39.101}$$

Every coupling in Eq. (39.101) is fixed by f and m_π alone: this is the parameter-free input that the rest of the section turns into a number.

Caution

The individual vertices in Eq. (39.101) depend on how one parametrizes U in terms of the fields — the square-root parametrization $U = f^{-1}(\sigma + i\pi^a\tau^a)$ with $\sigma = \sqrt{f^2 - \pi^2}$, for instance, produces a numerically different $\mathcal{L}_4$. Such field redefinitions $\pi \to \pi + \mathcal{O}(\pi^3)$ are precisely the reparametrizations that leave the S-matrix invariant (the equivalence theorem). The *on-shell* amplitude computed below is therefore parametrization-independent, even though the off-shell vertex is not. We use the exponential parametrization throughout for definiteness.

The momentum-space vertex. The amplitude follows from $\mathcal{L}_4$ by the LSZ machinery of Eq. (14.155): amputated, on-shell, the tree amplitude for four pions is just i times the vertex read off from Eq. (39.101). We compute the vertex in the fully crossing-symmetric, all-incoming convention — four pions with isospin indices a, b, c, d and incoming four-momenta q_1, q_2, q_3, q_4 summing to zero — and recover the physical process by crossing two legs at the end, exactly as in Theorem 29.27. A field π attached to an undifferentiated slot contributes a factor 1; attached to a ∂_μ slot it contributes $-iq_\mu$ for incoming momentum q. The vertex is i times the sum over all 4! ways of attaching the external lines to the four field slots of each monomial.

The non-derivative mass operator $(\pi^2)^2$ is the simplest: $\partial^4(\pi^2)^2/\partial\pi^a\partial\pi^b\partial\pi^c\partial\pi^d = 8(\delta^{ab}\delta^{cd} + \delta^{ac}\delta^{bd} + \delta^{ad}\delta^{bc})$, so $\mathcal{L}_4^m$ contributes

$$[i\mathcal{M}]_m \;=\; i\,\frac{m_\pi^2}{24f^2}\cdot 8(\delta^{ab}\delta^{cd} + \delta^{ac}\delta^{bd} + \delta^{ad}\delta^{bc}) \;=\; i\,\frac{m_\pi^2}{3f^2}(\delta^{ab}\delta^{cd} + \delta^{ac}\delta^{bd} + \delta^{ad}\delta^{bc}). \tag{39.102}$$

For the derivative operators we group the 4! contractions by the index structure they produce. Consider first $O_1 \equiv (\pi^e\pi^e)(\partial_\mu\pi^g\partial^\mu\pi^g)$. A given contraction sends two external lines to the undifferentiated pair (sharing an index, hence a δ) and two to the differentiated pair (sharing an index and Lorentz-contracted, hence a δ and a factor $(-iq)\cdot(-iq') = -q\cdot q'$). Summing the 24 contractions and collecting,

$$\frac{\partial^4 O_1}{\partial\pi^a\cdots} = -4\Big[\delta^{ab}\delta^{cd}(q_1\cdot q_2 + q_3\cdot q_4) + \delta^{ac}\delta^{bd}(q_1\cdot q_3 + q_2\cdot q_4) + \delta^{ad}\delta^{bc}(q_1\cdot q_4 + q_2\cdot q_3)\Big]. \tag{39.103}$$

(The factor -4 collects the 2×2 internal swaps of the two pairs; the two terms inside each bracket come from the two ways of choosing which pair is the differentiated one.) For $O_2 \equiv (\pi^e\partial_\mu\pi^e)(\pi^g\partial^\mu\pi^g)$ the index contraction is *within* each factor, so each δ links one undifferentiated and one differentiated line, and the Lorentz contraction links the two differentiated lines across the factors.

Collecting the 24 contractions,

$$\frac{\partial^4 O_2}{\partial\pi^a\dots} = -2\Big[\delta^{ab}\delta^{cd}\,(q_1{+}q_2){\cdot}(q_3{+}q_4)+\delta^{ac}\delta^{bd}\,(q_1{+}q_3){\cdot}(q_2{+}q_4)+\delta^{ad}\delta^{bc}\,(q_1{+}q_4){\cdot}(q_2{+}q_3)\Big]. \tag{39.104}$$

The kinetic Lagrangian Eq. (39.100) is $\mathcal{L}_4^{\text{kin}} = \frac{1}{6f^2}(O_2 - O_1)$, so its vertex is $\frac{i}{6f^2}(\partial^4 O_2 - \partial^4 O_1)$. Read off the coefficient of $\delta^{ab}\delta^{cd}$:

$$\big[\mathcal{M}_{\text{kin}}\big]_{\delta^{ab}\delta^{cd}} = \frac{1}{6f^2}\Big[-2(q_1{+}q_2){\cdot}(q_3{+}q_4) + 4(q_1{\cdot}q_2 + q_3{\cdot}q_4)\Big]. \tag{39.105}$$

Now impose momentum conservation. With all momenta incoming, $q_1 + q_2 = -(q_3 + q_4)$, so $(q_1 + q_2){\cdot}(q_3 + q_4) = -(q_1 + q_2)^2 \equiv -s$, where $s = (q_1 + q_2)^2$ is the all-incoming Mandelstam invariant of Theorem 16.20. Using on-shell $q_i^2 = m_\pi^2$, $q_1{\cdot}q_2 = \frac{1}{2}\big[(q_1{+}q_2)^2 - 2m_\pi^2\big] = \frac{1}{2}(s - 2m_\pi^2)$, and likewise $q_3{\cdot}q_4 = \frac{1}{2}(s - 2m_\pi^2)$ because $(q_3 + q_4)^2 = (q_1 + q_2)^2 = s$. Hence $q_1{\cdot}q_2 + q_3{\cdot}q_4 = s - 2m_\pi^2$, and Eq. (39.105) becomes

$$\big[\mathcal{M}_{\text{kin}}\big]_{\delta^{ab}\delta^{cd}} = \frac{1}{6f^2}\Big[2s + 4(s - 2m_\pi^2)\Big] = \frac{6s - 8m_\pi^2}{6f^2} = \frac{s}{f^2} - \frac{4m_\pi^2}{3f^2}. \tag{39.106}$$

Adding the mass contribution $m_\pi^2/3f^2$ from Eq. (39.102) the $-4m_\pi^2/3f^2$ and $+m_\pi^2/3f^2$ combine to $-m_\pi^2/f^2$, leaving the coefficient of $\delta^{ab}\delta^{cd}$ equal to $(s - m_\pi^2)/f^2$. The coefficients of $\delta^{ac}\delta^{bd}$ and $\delta^{ad}\delta^{bc}$ follow by the same algebra with $s \to t \equiv (q_1 + q_3)^2$ and $s \to u \equiv (q_1 + q_4)^2$.

39.4.2 The Weinberg amplitude

Assembling the three index channels gives the central result of the section.

Theorem 39.15 (Leading-order $\pi\pi$ amplitude)**.** *At order p^2 in the chiral expansion, the tree amplitude for $\pi^a\pi^b \to \pi^c\pi^d$ generated by $\mathcal{L}_2$ has the crossing-symmetric form*

$$\mathcal{M}^{ab,cd}(s,t,u) \;=\; A(s,t,u)\,\delta^{ab}\delta^{cd} + A(t,s,u)\,\delta^{ac}\delta^{bd} + A(u,t,s)\,\delta^{ad}\delta^{bc}, \tag{39.107}$$

with the single reduced amplitude

$$\boxed{A(s,t,u) \;=\; \frac{s - m_\pi^2}{f_\pi^2}\,,} \tag{39.108}$$

where s, t, u are the physical Mandelstam invariants of Theorem 16.20 obeying $s + t + u = 4m_\pi^2$ (Theorem 16.21), and $f = f_\pi$ at this order.

Proof. The all-incoming vertex computed in Section 39.4.1 reads $\mathcal{M}^{abcd} = \frac{s-m_\pi^2}{f^2}\delta^{ab}\delta^{cd} + \frac{t-m_\pi^2}{f^2}\delta^{ac}\delta^{bd} + \frac{u-m_\pi^2}{f^2}\delta^{ad}\delta^{bc}$ with all-incoming invariants $s = (q_1{+}q_2)^2$,

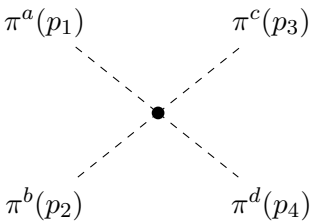


Figure 39.4: The unique leading-order ($O(p^2)$) diagram for $\pi^a\pi^b \to \pi^c\pi^d$: a single four-pion contact vertex from $\mathcal{L}_2$, Eq. (39.101). There is no pion-exchange pole at this order because the cubic vertex vanishes; the entire amplitude Eq. (39.107) is this one diagram.

$t = (q_1+q_3)^2$, $u = (q_1+q_4)^2$. The physical process $\pi^a(p_1)\pi^b(p_2) \to \pi^c(p_3)\pi^d(p_4)$ is recovered by crossing the two outgoing legs, $q_1 = p_1$, $q_2 = p_2$, $q_3 = -p_3$, $q_4 = -p_4$; then $s = (p_1+p_2)^2$, $t = (p_1-p_3)^2$, $u = (p_1-p_4)^2$ are exactly the physical invariants of Theorem 16.20, and the index structure is unchanged. Writing $A(s,t,u) = (s-m_\pi^2)/f^2$, the coefficient of $\delta^{ac}\delta^{bd}$ is $(t-m_\pi^2)/f^2 = A(t,s,u)$ and that of $\delta^{ad}\delta^{bc}$ is $(u-m_\pi^2)/f^2 = A(u,t,s)$, which is Eq. (39.107). The constraint $s+t+u = \sum_i p_i^2 = 4m_\pi^2$ is Theorem 16.21 specialized to four equal external masses. □

Setting $m_\pi \to 0$ reduces Eq. (39.108) to $A = s/f^2$, and Eq. (39.107) becomes precisely the massless Goldstone amplitude Eq. (29.185) of the abstract sigma model with $v \to f_\pi$ — the chiral Lagrangian reproduces the model-independent soft-pion result and tells us, in addition, exactly how the quark mass deforms it: every Mandelstam variable is shifted, $s \to s - m_\pi^2$, in its own crossing channel. The figure below records the single contact diagram that produces the whole amplitude.

39.4.3 Isospin decomposition

Two pions, each in the isovector $\mathbf{3}$ of $\mathrm{SU}(2)_V$, combine as $\mathbf{3}\otimes\mathbf{3} = \mathbf{1}\oplus\mathbf{3}\oplus\mathbf{5}$, i.e. into total isospin $I = 0, 1, 2$ (Section 2.5). Because the strong interaction conserves isospin, the S-matrix is diagonal in I, and the physically meaningful objects are the three isospin amplitudes $T^I(s,t,u)$. We extract them from Eq. (39.107) with the s-channel isospin projectors. In the four-index Cartesian basis (incoming a, b; outgoing c, d) the Clebsch–Gordan content of Section 2.5 is packaged as

$$P^0_{cd,ab} = \tfrac{1}{3}\delta_{ab}\delta_{cd}, \quad P^1_{cd,ab} = \tfrac{1}{2}(\delta_{ac}\delta_{bd} - \delta_{ad}\delta_{bc}), \quad P^2_{cd,ab} = \tfrac{1}{2}(\delta_{ac}\delta_{bd} + \delta_{ad}\delta_{bc}) - \tfrac{1}{3}\delta_{ab}\delta_{cd}. \tag{39.109}$$

These are orthogonal idempotents, $P^I P^J = \delta^{IJ} P^I$, summing to the identity $\sum_I P^I = \delta_{ac}\delta_{bd}$, with traces $\mathrm{tr}\, P^I = 2I+1$ (namely $1, 3, 5$, summing to $3\times 3 = 9$),

exactly the dimensions of $\mathbf{1}, \mathbf{3}, \mathbf{5}$. The $I = 0$ projector is symmetric and proportional to the isospin-singlet δ_{ab}; the $I = 1$ projector is antisymmetric (it must accompany an antisymmetric, hence odd-ℓ, spatial state by Bose symmetry); the $I = 2$ projector is the symmetric traceless combination. The isospin amplitudes are defined by $\mathcal{M}_{cd,ab} = \sum_I T^I P^I_{cd,ab}$, inverted using the idempotency to $T^I = (2I+1)^{-1} \sum_{abcd} P^I_{cd,ab}\, \mathcal{M}_{ab,cd}$.

The contraction is mechanical once we record the full contractions of the three index structures $C_1 = \delta_{ab}\delta_{cd}$, $C_2 = \delta_{ac}\delta_{bd}$, $C_3 = \delta_{ad}\delta_{bc}$ with one another in three dimensions: $C_i \cdot C_i = 9$ and $C_i \cdot C_j = 3$ for $i \neq j$ (each off-diagonal contraction forces all four indices equal, giving $N = 3$). Writing the amplitude as $\mathcal{M} = A(s)C_1 + A(t)C_2 + A(u)C_3$ with the abbreviations $A(s) \equiv A(s,t,u)$ etc., we get $C_1 \cdot \mathcal{M} = 9A(s) + 3A(t) + 3A(u)$ and cyclically. Hence

$$T^0 = \tfrac{1}{3}\, C_1 \cdot \mathcal{M} = 3A(s,t,u) + A(t,s,u) + A(u,t,s), \tag{39.110a}$$

$$T^1 = \tfrac{1}{3} \cdot \tfrac{1}{2}(C_2 - C_3) \cdot \mathcal{M} = A(t,s,u) - A(u,t,s), \tag{39.110b}$$

$$T^2 = \tfrac{1}{5}\big[\tfrac{1}{2}(C_2 + C_3) - \tfrac{1}{3}C_1\big] \cdot \mathcal{M} = A(t,s,u) + A(u,t,s). \tag{39.110c}$$

Inserting $A(x) = (x - m_\pi^2)/f_\pi^2$ and eliminating $t + u = 4m_\pi^2 - s$ with the constraint of Theorem 16.21,

$$\boxed{T^0(s,t,u) = \frac{2s - m_\pi^2}{f_\pi^2}, \qquad T^1(s,t,u) = \frac{t-u}{f_\pi^2}, \qquad T^2(s,t,u) = \frac{2m_\pi^2 - s}{f_\pi^2}.} \tag{39.111}$$

The structure is instructive. T^0 and T^2 are even under $t \leftrightarrow u$ and T^1 is odd, as Bose statistics demands: even isospin pairs with even orbital angular momentum, odd isospin with odd ℓ. The $I = 1$ amplitude is purely P-wave (proportional to $t - u \propto \cos\theta$) and so carries no s-wave; the $I = 1$ channel begins at $\ell = 1$. The $I = 0$ amplitude is large and attractive (positive at threshold), while $I = 2$ is smaller and repulsive (negative at threshold) — the qualitative pattern that the data confirm.

39.4.4 Scattering lengths: the parameter-free numbers

The amplitudes Eq. (39.111) are turned into observables by partial-wave projection and evaluation at threshold. We carry the computation through in full in the following worked example, which is the chapter's flagship prediction.

Worked Example 39.16 (Partial waves and the $\pi\pi$ scattering lengths)**. Kinematics.** Work in the centre-of-mass frame. With $s = 4(q^2 + m_\pi^2)$, the CM three-momentum is $q^2 = \tfrac{1}{4}(s - 4m_\pi^2)$, vanishing at threshold $s = 4m_\pi^2$. The scattering angle θ enters the crossed invariants through

$$t = -2q^2(1 - \cos\theta), \qquad u = -2q^2(1 + \cos\theta), \tag{39.112}$$

so that $t+u=-4q^2=4m_\pi^2-s$ and $t-u=4q^2\cos\theta$, consistent with $s+t+u=4m_\pi^2$.

Partial-wave projection. For two-pion states the standard normalization of the partial-wave amplitudes $t_\ell^I(s)$ is

$$T^I(s,t,u)=32\pi\sum_{\ell=0}^{\infty}(2\ell{+}1)\,P_\ell(\cos\theta)\,t_\ell^I(s),\qquad t_\ell^I(s)=\frac{1}{64\pi}\int_{-1}^{1}\mathrm{d}(\cos\theta)\,P_\ell(\cos\theta)\,T^I(s,t,u), \tag{39.113}$$

the inversion following from the orthogonality $\int_{-1}^{1}P_\ell P_{\ell'}\,\mathrm{d}\cos\theta=\frac{2}{2\ell+1}\delta_{\ell\ell'}$. The factor 32π is chosen so that the unitarity relation takes the form $\operatorname{Im}t_\ell^I=\sigma(s)\,|t_\ell^I|^2$ with phase-space factor $\sigma(s)=\sqrt{1-4m_\pi^2/s}=2q/\sqrt{s}$; this is the same $|\mathcal{M}|^2$ that feeds the differential cross section Eq. (16.145), now diagonalized channel by channel and partial wave by partial wave. At tree level t_ℓ^I is real — unitarity is restored order by order, the loop corrections of the next section supplying the imaginary part.

The s-wave ($\ell=0$). Here $P_0=1$, and from Eq. (39.111) both T^0 and T^2 are independent of the scattering angle, so the projection integral is trivial, $\int_{-1}^{1}\mathrm{d}\cos\theta=2$:

$$t_0^0(s)=\frac{1}{64\pi}\cdot 2\cdot\frac{2s-m_\pi^2}{f_\pi^2}=\frac{2s-m_\pi^2}{32\pi f_\pi^2},\qquad t_0^2(s)=\frac{2m_\pi^2-s}{32\pi f_\pi^2}. \tag{39.114}$$

The $I=1$ amplitude $T^1\propto t-u\propto\cos\theta=P_1$ integrates to zero against P_0, confirming there is no s-wave in the $I=1$ channel; its lowest partial wave is the P-wave t_1^1.

Threshold and the scattering length. Near threshold the ℓ-th partial wave behaves as $\operatorname{Re}t_\ell^I(s)\to a_\ell^I\,q^{2\ell}/m_\pi^{2\ell}$ times a kinematic factor; the standard convention defines the dimensionless s-wave scattering length (in units of m_π^{-1}) as the threshold value

$$a_0^I\;\equiv\;\frac{1}{32\pi}\,T^I\big|_{s=4m_\pi^2,\,t=u=0}\;=\;t_0^I(4m_\pi^2), \tag{39.115}$$

the second equality identifying it with the threshold limit of Eq. (39.114) (dimensionless, in units of m_π^{-1}; the dimensionful length is a_0^I/m_π). [Equivalently, with the dimensionful length a_0^I/m_π entering the effective-range expansion $q\cot\delta_0^I=m_\pi/a_0^I+\cdots$, one has $\delta_0^I\to(a_0^I/m_\pi)\,q$ and $t_0^I\to\sigma^{-1}\sin\delta_0^I\,e^{i\delta_0^I}\to a_0^I$ at threshold, since $\sigma\to q/m_\pi$.] Evaluating Eq. (39.114) at $s=4m_\pi^2$,

$$\boxed{a_0^0=\frac{7\,m_\pi^2}{32\pi f_\pi^2},\qquad a_0^2=-\frac{m_\pi^2}{16\pi f_\pi^2}.} \tag{39.116}$$

These are Weinberg's 1966 scattering lengths. They are pure predictions: the right-hand sides contain only the measured constants m_π and f_π, with no adjustable parameter. Their ratio is the parameter-free, even m_π- and f_π-independent number $a_0^0/a_0^2=-7/2$.

Table 39.1: Leading-order ChPT predictions for the $\pi\pi$ s-wave scattering lengths (in units of m_π^{-1}) versus the precise determination from K_{e4} decays and the Roy-equation/ChPT analysis (mid-2020s). The $I = 2$ length is reproduced at leading order to within the experimental uncertainty; the $I = 0$ length is about 30% low, the deficit being made up almost entirely by the one-loop and two-loop chiral corrections of the following sections.

	$O(p^2)$ (Weinberg)	experiment	ratio LO/exp
a_0^0	$+0.159$	$+0.220 \pm 0.005$	0.72
a_0^2	-0.0454	-0.0444 ± 0.0010	1.02

The numbers, and the data. Insert the physical values $m_\pi = m_{\pi^\pm} = 139.6$ MeV and $f_\pi = 92.4$ MeV (the normalization $\left\langle 0|A_\mu^a|\pi^b(p)\right\rangle = i f_\pi p_\mu \delta^{ab}$ fixed in Section 39.3). Then $m_\pi^2/f_\pi^2 = 2.283$ and

$$a_0^0 = \frac{7}{32\pi}\,(2.283) = 0.159, \qquad a_0^2 = -\frac{1}{16\pi}\,(2.283) = -0.0454, \tag{39.117}$$

both in units of m_π^{-1}. Table 39.1 sets these leading-order predictions against the modern experimental determinations, and Fig. 39.5 displays them in the (a_0^0, a_0^2) plane together with the analytic amplitude.

The agreement is the historic triumph. With no fitted parameter, a theory of a strongly coupled sector predicts a hadronic observable to tens of percent, and the small, repulsive $I = 2$ length — the channel with no resonances to complicate it — comes out essentially exact at leading order. The 30% shortfall in a_0^0 is not a failure but a feature: it is exactly the size $\sim m_\pi^2/\Lambda_\chi^2 \sim (m_\pi/4\pi f_\pi)^2 \cdot (\text{logs})$ that the chiral expansion predicts for the first correction, and the next section will see the one-loop chiral logarithm fill most of the gap, with the two-loop analysis of Colangelo et al. [126] bringing theory and the K_{e4} data of the NA48/2 experiment [51] into precise agreement.

Historical Note

Weinberg obtained Eq. (39.116) in 1966 [495] from current algebra and the PCAC hypothesis (Eq. (29.200)) — before the chiral Lagrangian was understood as a systematic effective field theory. The soft-pion technology of the day reproduced precisely the amplitude Eq. (39.107), but it offered no way to compute corrections: there was no organizing principle telling one what came next or how big it should be. The reinterpretation that made the corrections calculable came with Weinberg's 1979 paper on phenomenological Lagrangians [498], which recognized the current-algebra results as the leading term of a momentum expansion, and with the systematic one-loop renormalization of Gasser and Leutwyler [210]. What

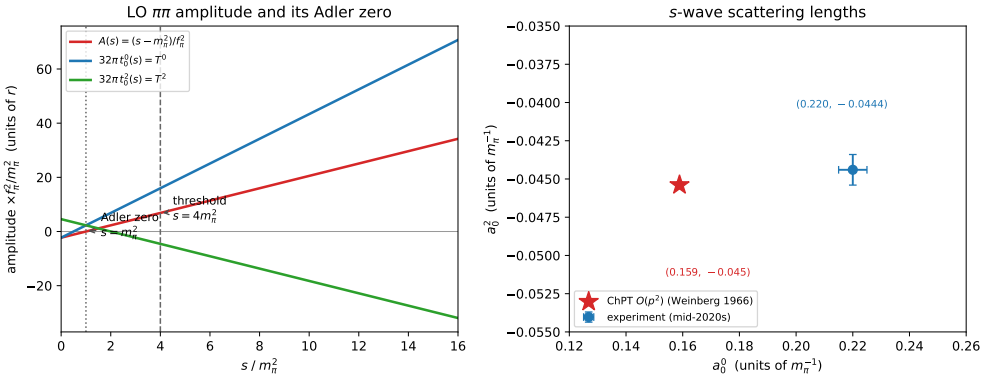


Figure 39.5: *Left:* the leading-order reduced amplitude $A(s) = (s - m_\pi^2)/f_\pi^2$ and the isospin s-wave combinations T^0, T^2 of Eq. (39.111) versus s/m_π^2, in units of $r \equiv m_\pi^2/f_\pi^2$. The dotted line marks the Adler zero of A at $s = m_\pi^2$; the dashed line marks the physical threshold $s = 4m_\pi^2$. *Right:* the s-wave scattering lengths in the (a_0^0, a_0^2) plane: the parameter-free $O(p^2)$ prediction Eq. (39.116) (star) and the experimental value with error bars. The $I = 2$ length lands on top of the data already at leading order.

had been an isolated low-energy theorem became the first entry in a convergent series. The numbers in Table 39.1 are the modern vindication of that program.

39.4.5 The Adler zero, made manifest

The amplitude Eq. (39.108) wears its Goldstone origin on its face. The Adler zero of Theorem 29.26 states that a connected amplitude with one soft external Goldstone vanishes linearly as that momentum is taken to zero, a direct consequence of the derivative coupling that the nonlinear realization forces on every pion. We verify it here for the physical, massive amplitude and locate the zero precisely.

Take the first pion soft, $p_1 \to 0$, with the other three on shell. In the chiral limit $m_\pi = 0$ the remaining pions are massless, so $s = (p_1 + p_2)^2 \to p_2^2 = 0$, $t = (p_1 - p_3)^2 \to p_3^2 = 0$, $u = (p_1 - p_4)^2 \to p_4^2 = 0$, and each reduced amplitude $A = s/f_\pi^2 \to 0$: the full $\mathcal{M}^{ab,cd}$ of Eq. (39.107) vanishes linearly in the soft momentum, exactly the chiral-limit statement already checked in the abstract model at Eq. (29.186). Away from the chiral limit the pions carry mass, and the soft point $p_1 \to 0$ sends $s, t, u \to m_\pi^2$ simultaneously. There each crossing

channel's reduced amplitude vanishes,

$$A(s,t,u)\big|_{s=t=u=m_\pi^2} = \frac{m_\pi^2 - m_\pi^2}{f_\pi^2} = 0 \quad \Longrightarrow \quad \mathcal{M}^{ab,cd}\big|_{\text{soft point}} = 0. \tag{39.118}$$

The quark mass does not destroy the Adler zero; it *relocates* it. In each channel the zero of the reduced amplitude sits at $s = m_\pi^2$ rather than at $s = 0$ — a subthreshold point (note $s = t = u = m_\pi^2$ violates $s + t + u = 4m_\pi^2$ and so lies off the physical region, reached only by analytic continuation, which is exactly where a soft-pion theorem should live). This shifted zero, visible as the crossing of the red curve in Fig. 39.5 at $s = m_\pi^2$, is a sharp prediction of the leading chiral Lagrangian: the position of the Adler zero is itself fixed by the same symmetry that fixes the scattering lengths.

Physical Insight

The Adler zero and the scattering lengths are two faces of one fact: pions couple through their momenta. The same s in $A = (s - m_\pi^2)/f_\pi^2$ that produces the zero at $s = m_\pi^2$ produces, at threshold $s = 4m_\pi^2$, the nonzero value $3m_\pi^2/f_\pi^2$ that drives a_0^0. A theory in which pions coupled without derivatives — a generic scalar with a $\lambda\phi^4$ contact term — would give a constant amplitude, no Adler zero, and a scattering length set by an arbitrary coupling. The derivative coupling forced by the spontaneously broken chiral symmetry both suppresses the amplitude at low energy (the zero) and predicts its threshold value with no free parameter. The smallness of $\pi\pi$ interactions near threshold is not an accident of QCD dynamics; it is a theorem of broken symmetry.

The leading-order amplitude is now complete: a single closed-form expression Eq. (39.107), its isospin projections Eq. (39.111), the scattering lengths Eq. (39.116), and the Adler zero Eq. (39.118), all fixed by f_π and m_π alone. Its limitation is equally sharp: the amplitudes Eq. (39.111) are real polynomials in s, t, u, whereas unitarity (Eq. (16.145) and the optical theorem) demands an imaginary part above threshold and a logarithmic structure near it. That missing piece is the pion loop, suppressed by exactly one power of $1/\Lambda_\chi^2$. Computing it will both restore unitarity and generate ultraviolet divergences that no operator in $\mathcal{L}_2$ can absorb — forcing us, in the next section, to introduce the order-p^4 Gasser–Leutwyler Lagrangian and its low-energy constants, and to extract our first chiral logarithm.

39.5 Loops, the Gasser–Leutwyler Lagrangian, and Low-Energy Constants

The leading-order calculation of Section 39.4 delivered a number for $\pi\pi$ scattering with no free parameters, built from f and M_π^2 alone. That triumph also carried a warning: the same power counting that made the tree amplitude $O(p^2)$ told us (Section 39.3) that the very next term in the chiral expansion is $O(p^4)$, and that this order receives contributions from *two* sources at once — new tree operators with four derivatives, and one-loop graphs built from the two-derivative Lagrangian $\mathcal{L}_2$. A graduate reader trained on Chapter 22 will immediately object that loops in a Lagrangian whose coupling carries negative mass dimension, $1/f^2$, are exactly what defines a *non*-renormalizable theory: the loop integrals diverge, and there is no single coupling whose redefinition can soak the divergence up. The resolution, and the central lesson of this section, is that the effective-field-theory bookkeeping of Chapter 38 turns this apparent disaster into an orderly accounting. The divergences of the one-loop graphs are local and of precisely the order $O(p^4)$ generated by the four-derivative operators; renormalizing the coefficients of those operators absorbs every divergence at this order, and the finite remainder contains a piece — the *chiral logarithm* — that no polynomial counterterm can imitate. We now make all three statements precise and carry the computation that exhibits the chiral log explicitly. Throughout we write tr for the trace over flavour space and continue with the field $U(x) = \exp(i\Phi(x)/f)$ and the leading Lagrangian $\mathcal{L}_2$ of Eq. (39.43).

39.5.1 The general $O(p^4)$ chiral Lagrangian

At $O(p^2)$ chiral symmetry, parity, charge conjugation, and Lorentz invariance left exactly two operators: the kinetic term $\frac{f^2}{4}\,\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U)$ and the symmetry-breaking term $\frac{f^2}{4}\,\mathrm{tr}(\chi^\dagger U + \chi U^\dagger)$, with $\chi = 2B\,\mathsf{M}$ and $\mathsf{M} = \mathrm{diag}(m_u, m_d, m_s)$ the quark-mass matrix (Eq. (39.43)). The same symmetry principles, applied now to operators with four derivatives or two derivatives and one insertion of χ or two insertions of χ, generate a longer but still finite list. The systematic enumeration is due to Gasser and Leutwyler. Under the chiral group $U \to L U R^\dagger$ with $L \in \mathrm{SU}(3)_L$, $R \in \mathrm{SU}(3)_R$, and with $\chi \to L \chi R^\dagger$ assigned so that the breaking term is invariant, the independent Hermitian, parity-even, Lorentz-invariant monomials of chiral order p^4 are catalogued as follows.

Definition 39.17 (The Gasser–Leutwyler Lagrangian and the low-energy constants)**.** The most general chirally invariant, parity- and C-even, Lorentz-scalar Lagrangian of chiral order p^4 for the three-flavour Goldstone field U, in the absence of external vector and axial sources, is

$$\mathcal{L}_4 = L_1\big[\,\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U)\big]^2 + L_2\,\mathrm{tr}(\partial_\mu U^\dagger \partial_\nu U)\,\mathrm{tr}(\partial^\mu U^\dagger \partial^\nu U) + L_3\,\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U\,\partial_\nu U^\dagger \partial^\nu U)$$

$$+ L_4\,\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U)\,\mathrm{tr}(\chi^\dagger U + \chi U^\dagger) + L_5\,\mathrm{tr}(\partial_\mu U^\dagger \partial^\mu U\,(\chi^\dagger U + U^\dagger \chi)) + L_6\big[\mathrm{tr}(\chi^\dagger U + \chi U^\dagger)\big]^2 + L_7\big[\mathrm{tr}(\chi^\dagger U - \chi U^\dagger)\big]^2 + L_8\,\mathrm{tr}(\chi^\dagger U \chi^\dagger U + \chi U^\dagger \chi U^\dagger) + \dots \tag{39.119}$$

The dimensionless constants $L_1, \dots, L_8$ (together with L_9, L_{10}, which multiply operators built from the field strengths of external vector/axial sources, and the contact terms H_1, H_2 that contain no Goldstone field) are the *low-energy constants* (LECs). They are the Wilson coefficients of the effective theory: not predicted by chiral symmetry, but fixed once and for all either by matching to QCD or by fitting low-energy data.

Two remarks fix the status of Eq. (39.119). First, the list is *complete and irredundant* only after the leading equations of motion are used to eliminate operators that are equivalent on shell, and after total derivatives and the Cayley–Hamilton identities of SU(3) matrices are accounted for; carrying out that reduction is what guarantees there are exactly these (and the four omitted) independent structures. Second, the L_i are genuinely the EFT's Wilson coefficients in the sense of Chapter 38: they encode the short-distance physics — the masses of the ρ, the a_1, the scalar resonances, all the hadronic states that were integrated out — in a few numbers. They are fixed by the matching procedure of Theorem 38.8, in principle by computing the same Green's function in QCD and in the EFT and equating them, in practice by measuring a few low-energy observables and reading the L_i off. The whole content of the chiral expansion is that, once these handful of numbers are pinned down at order p^4, *every* $O(p^4)$ amplitude is then predicted.

Notation Watch

For the two-flavour theory ($N_f = 2$, pions only) the analogous basis is smaller and is traditionally written with lowercase constants $l_1, \dots, l_7$ (Gasser–Leutwyler 1984). The reductions are not one-to-one because of the SU(2) Cayley–Hamilton identity $\Phi^2 = \frac{1}{2}\,\mathrm{tr}(\Phi^2)\,\mathbf{1}$, which collapses several SU(3) structures. We will present the general basis above but carry the explicit one-loop computation in the two-flavour theory, where the algebra is most transparent, and identify the relevant counterterm as l_3.

39.5.2 Why one-loop divergences are local and of order p^4

The reason Eq. (39.119) suffices to renormalize the one-loop graphs is Weinberg's power-counting theorem, established in Section 39.3. We restate the chiral-dimension formula and then draw the renormalization consequence, which is the analogue for an EFT of the renormalizability structure proved for ϕ^4 in Theorem 22.28.

For a connected one-particle-irreducible graph with L loops and V_{2k} vertices drawn from the chiral Lagrangian of order p^{2k} (that is, with $2k$ derivatives or equivalent powers of $\chi \sim M_\pi^2$), the degree of homogeneity D of the amplitude in the small external momenta and Goldstone masses is

$$D \;=\; 2 \;+\; 2L \;+\; \sum_{k\geq 1} 2(k-1)\, V_{2k}. \tag{39.120}$$

The derivation is purely topological. Each propagator scales as p^{-2}, each loop integration $\int \mathrm{d}^4\ell$ as p^4, and each $\mathcal{L}_{2k}$ vertex as p^{2k}; with I internal lines and $V = \sum_k V_{2k}$ vertices,

$$D \;=\; 4L - 2I + \sum_k 2k\, V_{2k}, \tag{39.121}$$

and substituting the Euler relation $L = I - V + 1$, i.e. $I = L + V - 1$, gives

$$D \;=\; 4L - 2(L + V - 1) + \sum_k 2k\, V_{2k} \;=\; 2L + 2 + \sum_k (2k-2) V_{2k}, \tag{39.122}$$

which is Eq. (39.120). Every term on the right-hand side is non-negative: $L \geq 0$, and $k \geq 1$ so $k - 1 \geq 0$. The minimum, $D = 2$, is reached only by tree graphs ($L = 0$) built entirely from $\mathcal{L}_2$ vertices ($k = 1$); that is the leading order. The next value, $D = 4$, is reached in exactly two ways: a tree graph with a single $\mathcal{L}_4$ vertex ($L = 0$, one V_4), or a one-loop graph built entirely from $\mathcal{L}_2$ vertices ($L = 1$, all V_2). These are the two contributions to $O(p^4)$ promised at the start.

Theorem 39.18 (One-loop divergences are local and of $O(p^4)$)**.** *Every one-loop* 1*PI graph built from* $\mathcal{L}_2$ *has chiral dimension* $D = 4$*. Its ultraviolet divergences, computed in dimensional regularization with* $d = 4 - 2\varepsilon$ *(Section 19.1), are a finite linear combination of the local operators of Eq.* (39.119)*. Consequently the* $1/\varepsilon$ *poles of all one-loop amplitudes are removed by a redefinition of the* L_i*, and the renormalization of the theory closes at order* p^4*.*

Proof. That $D = 4$ for any one-loop, all-$\mathcal{L}_2$ graph is the $L = 1$, $V_{2k} = V_2$ case of Eq. (39.120). It remains to argue that the divergent part is local and of the symmetry type catalogued in Theorem 39.17. Locality is the general statement that the ultraviolet divergence of any 1PI graph in dimensional regularization is a polynomial in the external momenta and the explicit masses: the divergence comes from the region of large loop momentum, where the integrand may be Taylor-expanded in the external data, and each term of the expansion is a power divergence multiplying a monomial in those external quantities; dimensional regularization discards the scaleless power divergences and keeps the logarithmic ones, whose coefficients are exactly such monomials. This is the same mechanism that made the ϕ^4 counterterms local in Section 22.5.6. A local term of chiral dimension four, built from U, ∂U, and χ, and invariant under the chiral group and parity, is by *definition* one of the operators in Eq. (39.119): the symmetry of the

regularized theory is the full chiral symmetry (dimensional regularization respects it, since it touches only the spacetime dimension, not the flavour structure), so the counterterm Lagrangian must be chirally invariant, and the complete list of invariants at this order is precisely $\mathcal{L}_4$. Hence the divergence is cancelled by

$$L_i \;=\; L_i^r(\mu) \;-\; \frac{\Gamma_i}{32\pi^2}\,\frac{1}{\bar\varepsilon}, \qquad \frac{1}{\bar\varepsilon} \equiv \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi, \tag{39.123}$$

for a set of pure numbers Γ_i determined by the one-loop graphs, leaving renormalized constants $L_i^r(\mu)$. The same argument applies to the two-flavour constants l_i. □

The structure is exactly that of Theorem 22.28, transplanted to a theory with infinitely many couplings: renormalizability there meant that all divergences are absorbed by the finitely many operators already present in the Lagrangian; here it means that all divergences *at a given chiral order* are absorbed by the finitely many operators present *at that order*. The price of the negative-dimension coupling $1/f^2$ is not loss of predictivity but the appearance of new counterterms order by order — which is precisely the EFT paradigm of Chapter 38. Each order requires a finite number of measurements; in exchange it predicts everything else at that order. The crucial new feature, absent from a renormalizable theory, is that the subtraction in Eq. (39.123) introduces a scale μ into the renormalized constants $L_i^r(\mu)$, and they must run with it.

39.5.3 The chiral logarithm: the one-loop pion mass

We now carry the computation that exhibits all of this concretely: the one-loop correction to the pion self-energy, hence to m_π^2. We work in the two-flavour theory with $\Phi = \pi^a\tau^a$ ($a = 1,2,3$, τ^a the Pauli matrices, $\operatorname{tr}\tau^a\tau^b = 2\delta^{ab}$), and in the isospin limit $m_u = m_d = \hat m$, so $\chi = M_\pi^2\,\mathbf{1}$ with $M_\pi^2 \equiv 2B\hat m$ the leading-order pion mass-squared (the Gell-Mann–Oakes–Renner relation Eq. (29.216)). We reserve m_π^2 for the physical, one-loop-corrected pole mass.

Worked Example 39.19 (The pion tadpole and the chiral logarithm)**. Step 1: expand $\mathcal{L}_2$ to quartic order.** The kinetic term is $\frac{f^2}{4}\operatorname{tr}(\partial_\mu U^\dagger \partial^\mu U)$. Expanding $U = 1 + i\Phi/f - \Phi^2/2f^2 - i\Phi^3/6f^3 + \dots$ and collecting the terms with four pion fields,

$$\tfrac{f^2}{4}\operatorname{tr}(\partial_\mu U^\dagger\partial^\mu U)\Big|_{\Phi^4} = \frac{1}{24f^2}\Big[\operatorname{tr}(\partial_\mu\Phi\,\Phi\,\partial^\mu\Phi\,\Phi) - \operatorname{tr}(\partial_\mu\Phi\,\partial^\mu\Phi\,\Phi^2)\Big]. \tag{39.124}$$

Using $\operatorname{tr}(\tau^a\tau^b\tau^c\tau^d) = 2(\delta^{ab}\delta^{cd} - \delta^{ac}\delta^{bd} + \delta^{ad}\delta^{bc})$ to evaluate the two traces,

$$\operatorname{tr}(\partial_\mu\Phi\,\partial^\mu\Phi\,\Phi^2) = 2(\partial\pi)^2\pi^2, \qquad \operatorname{tr}(\partial_\mu\Phi\,\Phi\,\partial^\mu\Phi\,\Phi) = 2\big[2(\pi\partial_\mu\pi)(\pi\partial^\mu\pi) - (\partial\pi)^2\pi^2\big], \tag{39.125}$$

where $\pi^2 \equiv \pi^a\pi^a$, $(\partial\pi)^2 \equiv \partial_\mu\pi^a\partial^\mu\pi^a$, and $\pi\cdot\partial_\mu\pi \equiv \pi^a\partial_\mu\pi^a$. Hence the quartic part of the kinetic term is

$$\mathcal{L}_2^{\text{kin}}\big|_{\Phi^4} = \frac{1}{6f^2}\Big[(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi) - \pi^2(\partial\pi)^2\Big]. \tag{39.126}$$

The mass term is $\frac{f^2}{4}\,\mathrm{tr}(\chi^\dagger U + \chi U^\dagger) = \frac{f^2}{4}M_\pi^2\,\mathrm{tr}(U + U^\dagger)$. Since $\Phi^2 = \pi^2\,\mathbf{1}$ in SU(2), one has $\mathrm{tr}(U + U^\dagger) = 2\big[2 - \pi^2/f^2 + (\pi^2)^2/12f^4 - \dots\big]$, so the quartic part is

$$\mathcal{L}_2^{\text{mass}}\big|_{\Phi^4} = \frac{M_\pi^2}{24f^2}\,(\pi^2)^2. \tag{39.127}$$

Together, the quartic interactions of $\mathcal{L}_2$ are

$$\mathcal{L}_2^{(4)} = \frac{1}{6f^2}\Big[(\pi\cdot\partial_\mu\pi)(\pi\cdot\partial^\mu\pi) - \pi^2(\partial\pi)^2\Big] + \frac{M_\pi^2}{24f^2}\,(\pi^2)^2. \tag{39.128}$$

Step 2: close two legs into a tadpole. The one-loop self-energy comes from Eq. (39.128) by contracting two of the four pion fields into a propagator at coincident points, leaving two external legs (Fig. 39.6). The contractions we need are

$$\langle \pi^a(x)\pi^b(x)\rangle = \delta^{ab}\,\Delta(0), \qquad \langle \partial_\mu\pi^a\,\pi^b\rangle = 0, \qquad \langle \partial_\mu\pi^a\,\partial_\nu\pi^b\rangle = \delta^{ab}\frac{\eta_{\mu\nu}}{d}\,M_\pi^2\,\Delta(0), \tag{39.129}$$

where $\Delta(0)$ is the pion propagator at the origin and the middle relation vanishes because its integrand is odd in the loop momentum. The third relation uses $\int \mathrm{d}^dk\,k_\mu k_\nu f(k^2) = (\eta_{\mu\nu}/d)\int \mathrm{d}^dk\,k^2 f(k^2)$ and $\int \mathrm{d}^dk\,k^2/(k^2 - M_\pi^2) = M_\pi^2\int \mathrm{d}^dk/(k^2 - M_\pi^2)$ (the scaleless piece $\int \mathrm{d}^dk\,1$ vanishes in dimensional regularization). The rule that replaces a contracted pair by $\Delta(0)$ and adds the resulting bilinear directly to the Lagrangian is the one-loop background-field construction; it reproduces the standard symmetry factor, as one checks against the ϕ^4 tadpole $\delta m^2 = \frac{\lambda}{2}\Delta(0)$ of Section 22.2.2.

Step 3: the loop integral. By the master integral Theorem D.22 with $n = 1$, restoring the scale μ through the factor $\mu^{2\varepsilon}$ that accompanies the loop measure,

$$\Delta(0) = \mu^{2\varepsilon}\int \frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{i}{k^2 - M_\pi^2 + i\varepsilon} = -\frac{M_\pi^2}{16\pi^2}\Big[\frac{1}{\bar\varepsilon} + 1 - \ln\frac{M_\pi^2}{\mu^2}\Big]. \tag{39.130}$$

The chain is $\int \mathrm{d}^dk/(2\pi)^d\,(k^2 - M_\pi^2)^{-1} = -i(4\pi)^{-d/2}\Gamma(1 - \frac{d}{2})(M_\pi^2)^{d/2-1}$, and with $d = 4 - 2\varepsilon$, $\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} - 1 + \gamma_E + O(\varepsilon)$, $(4\pi\mu^2/M_\pi^2)^\varepsilon = 1 + \varepsilon\ln(4\pi\mu^2/M_\pi^2)$, multiplying by i gives Eq. (39.130). The pole and the logarithm appear locked together, both proportional to M_π^2.

Step 4: assemble the self-energy. Contracting one pair in each term of Eq. (39.128) and using the contractions of Step 2, the induced quadratic

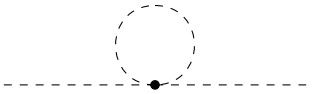


Figure 39.6: The one-loop pion tadpole. The four-point vertex from $\mathcal{L}_2^{(4)}$ (Eq. (39.128)) has two external legs (momenta $\pm p$) and two legs joined into the loop $\Delta(0)$ of Eq. (39.130). Dashed lines are pions. This single graph produces both the $1/\varepsilon$ pole and the chiral logarithm of Eq. (39.133).

Lagrangian is

$$\delta\mathcal{L} = \frac{M_\pi^2}{24f^2}\cdot 10\,\Delta(0)\,\pi^2 \;-\; \frac{\Delta(0)}{3f^2}\Big[(\partial\pi)^2 + M_\pi^2\pi^2\Big] \;=\; -\frac{\Delta(0)}{3f^2}(\partial\pi)^2 + \frac{M_\pi^2\Delta(0)}{12f^2}\,\pi^2. \tag{39.131}$$

Here the factor 10 counts the index contractions of the $(\pi^2)^2$ vertex ($2\cdot 3 + 4\cdot 1$ from the two "self" and four "cross" pairings) and the two derivative structures combine with relative weight $\frac{1}{6} - \frac{1}{2} = -\frac{1}{3}$. Adding $\delta\mathcal{L}$ to the free Lagrangian $\frac{1}{2}(\partial\pi)^2 - \frac{1}{2}M_\pi^2\pi^2$, the inverse propagator becomes

$$\Gamma^{(2)}(p^2) = Z_\pi\,p^2 - M_\pi^2\Big(1 - \frac{\Delta(0)}{6f^2}\Big), \qquad Z_\pi = 1 - \frac{2\Delta(0)}{3f^2}. \tag{39.132}$$

Step 5: the pole mass. The physical mass-squared is the zero of $\Gamma^{(2)}$, $m_\pi^2 = M_\pi^2(1 - \frac{\Delta(0)}{6f^2})/Z_\pi$. Expanding to one loop,

$$m_\pi^2 = M_\pi^2\Big[1 + \frac{\Delta(0)}{2f^2}\Big] = M_\pi^2\Big[1 - \frac{M_\pi^2}{32\pi^2f^2}\Big(\frac{1}{\bar{\varepsilon}} + 1 - \ln\frac{M_\pi^2}{\mu^2}\Big)\Big]. \tag{39.133}$$

The combination of the $-\frac{1}{6}$ from the mass shift and the $+\frac{2}{3}$ from the wavefunction factor Z_π^{-1} leaves the net coefficient $+\frac{1}{2}$. Equation Eq. (39.133) is the result: a $1/\varepsilon$ pole, and riding on it the *chiral logarithm*

$$m_\pi^2 \;\supset\; \frac{M_\pi^4}{32\pi^2f^2}\,\ln\frac{M_\pi^2}{\mu^2} \;=\; \frac{M_\pi^2}{(4\pi f)^2}\,M_\pi^2\,\frac{\ln(M_\pi^2/\mu^2)}{2}\,, \tag{39.134}$$

the announced non-analytic term suppressed by $M_\pi^2/(4\pi f)^2$ — one factor of the chiral expansion parameter $\big(p/\Lambda_\chi\big)^2$ with $\Lambda_\chi = 4\pi f$.

39.5.4 Renormalizing the LEC and its running

The pole in Eq. (39.133) is cancelled by the $O(p^4)$ counterterm of Theorem 39.18. In the two-flavour basis the relevant operator is the one accompanying l_3,

$$\mathcal{L}_4 \;\supset\; -\frac{l_3}{16}\big[\,\mathrm{tr}(\chi^\dagger U + \chi U^\dagger)\big]^2, \tag{39.135}$$

the analogue of the L_6/L_8 structures in Eq. (39.119). Expanding with $\chi = M_\pi^2\mathbf{1}$ gives $\mathrm{tr}(\chi^\dagger U + \chi U^\dagger) = M_\pi^2(4 - 2\pi^2/f^2 + \dots)$, so the squared bracket contributes $-\frac{l_3}{16}\cdot(-16M_\pi^4\pi^2/f^2) = l_3 M_\pi^4\pi^2/f^2$ to the quadratic Lagrangian, i.e. a tree-level shift $\delta m^2 = -2l_3 M_\pi^4/f^2$. Combining this with the loop result Eq. (39.133),

$$m_\pi^2 = M_\pi^2 - \frac{M_\pi^4}{32\pi^2 f^2}\Big(\frac{1}{\bar\varepsilon} + 1 - \ln\frac{M_\pi^2}{\mu^2}\Big) - \frac{2l_3 M_\pi^4}{f^2}. \tag{39.136}$$

Splitting the bare coupling as $l_3 = l_3^r(\mu) + \delta l_3$ and demanding that the $1/\bar\varepsilon$ cancel fixes the counterterm

$$\delta l_3 = -\frac{1}{64\pi^2}\,\frac{1}{\bar\varepsilon}\,, \tag{39.137}$$

which is the explicit instance of Eq. (39.123) with divergence coefficient $\Gamma_{l_3} = \frac{1}{2}$: indeed $-\frac{\Gamma_{l_3}}{32\pi^2}\frac{1}{\bar\varepsilon} = -\frac{1}{64\pi^2}\frac{1}{\bar\varepsilon}$ reproduces Eq. (39.137), and $\Gamma_{l_3} = \frac{1}{2}$ is the standard Gasser–Leutwyler value. After subtraction the physical mass is finite:

$$m_\pi^2 = M_\pi^2\Big[1 - \frac{M_\pi^2}{32\pi^2 f^2}\Big(1 - \ln\frac{M_\pi^2}{\mu^2}\Big) - \frac{2l_3^r(\mu)M_\pi^2}{f^2}\Big]. \tag{39.138}$$

Theorem 39.20 (Running of the low-energy constants)**.** *The renormalized constant $l_3^r(\mu)$ depends on the chiral renormalization scale μ, and the requirement that the physical m_π^2 of Eq. (39.138) be independent of μ fixes the rate of running to*

$$\mu\,\frac{\mathrm{d}l_3^r}{\mathrm{d}\mu} = -\frac{1}{32\pi^2}\,, \qquad \textit{equivalently} \qquad l_3^r(\mu_2) = l_3^r(\mu_1) - \frac{1}{32\pi^2}\ln\frac{\mu_2}{\mu_1}. \tag{39.139}$$

More generally each renormalized LEC runs as $\mu\,\mathrm{d}L_i^r/\mathrm{d}\mu = -\Gamma_i/(16\pi^2)$, with the Γ_i the one-loop divergence coefficients of Eq. (39.123).

Proof. Differentiate Eq. (39.138) with respect to $\ln\mu$ at fixed bare parameters and set the result to zero. Only the explicit logarithm and $l_3^r(\mu)$ carry μ-dependence:

$$0 = \frac{\mathrm{d}m_\pi^2}{\mathrm{d}\ln\mu} = M_\pi^2\Big[\frac{M_\pi^2}{32\pi^2 f^2}\cdot(-2) \;-\; \frac{2M_\pi^2}{f^2}\,\frac{\mathrm{d}l_3^r}{\mathrm{d}\ln\mu}\Big] = -\frac{M_\pi^4}{f^2}\Big[\frac{1}{16\pi^2} + 2\,\frac{\mathrm{d}l_3^r}{\mathrm{d}\ln\mu}\Big], \tag{39.140}$$

where $\mathrm{d}\ln(M_\pi^2/\mu^2)/\mathrm{d}\ln\mu = -2$. Hence $\mu\,\mathrm{d}l_3^r/\mathrm{d}\mu = \mathrm{d}l_3^r/\mathrm{d}\ln\mu = -1/(32\pi^2)$, which is Eq. (39.139). The general statement follows identically: the μ-dependence of any one-loop chiral log is fixed by the corresponding pole coefficient Γ_i through Eq. (39.123), and cancellation of the residual log against $L_i^r(\mu)$ gives $\mu\,\mathrm{d}L_i^r/\mathrm{d}\mu = -\Gamma_i/(16\pi^2)$. □

This is the renormalization-group of Section 24.6 operating in the EFT: the chiral logarithm is the solution of an RG equation, summing $\ln(M_\pi^2/\mu^2)$ as μ is varied, exactly as the Callan–Symanzik logarithms summed $\ln(p^2/\mu^2)$ for ϕ^4. It is natural to trade $l_3^r(\mu)$ for a scale-invariant constant by absorbing the running into

a reference mass; defining $\bar{l}_3$ through $l_3^r(\mu) = \frac{1}{64\pi^2}\left[\bar{l}_3 + \ln(M_\pi^2/\mu^2)\right]$, Eq. (39.138) becomes the manifestly scale-independent

$$m_\pi^2 = M_\pi^2\Big[1 - \frac{M_\pi^2}{32\pi^2 f^2}(1+\bar{l}_3)\Big], \tag{39.141}$$

with $\bar{l}_3 = \ln(\Lambda_3^2/M_\pi^2)$ defining the renormalization-group-invariant scale Λ_3 that encodes the genuine short-distance content. The identical machinery yields the chiral log of the pion decay constant; the analogous one-loop computation of the axial-current matrix element Eq. (29.202) (which carries, in addition to the wavefunction factor Z_π, a vertex contraction from the current) gives

$$f_\pi = f\Big[1 - \frac{M_\pi^2}{16\pi^2 f^2}\ln\frac{M_\pi^2}{\mu^2} + (\text{analytic},\ \propto l_4^r)\Big], \tag{39.142}$$

whose non-analytic coefficient $-1/(16\pi^2)$ is renormalized by l_4 and runs accordingly. The two together constitute the leading quark-mass dependence of the two basic observables of the pion sector.

39.5.5 Predictions versus parameters: what the chiral log means

The split in Eq. (39.138) between the logarithm and the polynomial is the conceptual heart of the matter, and it is worth stating sharply.

Physical Insight

The chiral logarithm $\dfrac{M_\pi^4}{32\pi^2 f^2}\ln M_\pi^2$ is a *parameter-free prediction*. Its coefficient is a pure number, fixed entirely by the Goldstone loop — by the kinematics of a light pion propagating in the loop — with no dependence on any LEC. It is non-analytic in the quark mass: $M_\pi^2 \propto \hat{m}$, so the term behaves as $\hat{m}^2\ln\hat{m}$, and *no* polynomial in $\hat{m}$ (no counterterm, no matter how the L_i are chosen) can reproduce a logarithm. The non-analyticity is the unmistakable fingerprint of a propagating massless (or nearly massless) particle: it is the long-distance, infrared physics of the Goldstone boson, which the EFT knows exactly because it contains the Goldstone field itself. The analytic pieces — the 1, the $l_3^r(\mu)$ — are where the model dependence lives: they are the Wilson coefficients, sensitive to the integrated-out hadrons, and must be supplied by data or by matching to QCD. ChPT thus cleanly separates the model-independent (the logs, predicted) from the model-dependent (the LECs, measured).

The infrared origin of the log also explains why the chiral expansion is well-defined in four dimensions at all: the Goldstone is light but not exactly massless, and in $d = 4$ the infrared fluctuations are only logarithmically strong.

Were we in two dimensions the same loop would be catastrophic — this is the Coleman–Mermin–Wagner obstruction of Theorem 29.33, which forbids the symmetry-broken vacuum outright. In $d = 4$ the log is a controlled, calculable correction, and its appearance is a victory rather than a pathology: it is direct evidence, computed from first principles, that the pion is a Goldstone boson, since only a Goldstone loop can generate a non-analytic dependence on the symmetry-breaking quark mass with a universal coefficient.

We have now assembled a fully renormalized, parity-even chiral Lagrangian through $O(p^4)$: the leading $\mathcal{L}_2$, the Gasser–Leutwyler $\mathcal{L}_4$ with its running constants $L_i^r(\mu)$, and a one-loop machinery that turns the negative-dimension coupling $1/f^2$ into an orderly, predictive expansion. Yet this construction has an accidental symmetry that QCD does not possess. Every operator in $\mathcal{L}_2$ and $\mathcal{L}_4$ is invariant under the naive *intrinsic parity* $U \to U^\dagger$ (equivalently $\Phi \to -\Phi$) accompanied by spatial reflection, so the theory built so far cannot describe any process with an odd number of Goldstone bosons — it forbids $K\bar{K} \to \pi\pi\pi$ and, most strikingly, $\pi^0 \to \gamma\gamma$. QCD allows these; the gap is the imprint of the chiral anomaly of Chapter 36. The final section closes it by constructing the Wess–Zumino–Witten term, the one piece of the chiral Lagrangian that cannot be written as the integral of a locally chiral-invariant density, and whose quantized coefficient will turn out to count the colours N_c.

39.6 The Wess–Zumino–Witten Term and the Anomaly

At the close of Section 39.5 we noticed something faintly suspicious about the chiral Lagrangian we have built. Every term we wrote down — the leading $O(p^2)$ kinetic and mass terms, and every one of the Gasser–Leutwyler $O(p^4)$ operators absorbing the one-loop divergences — is built from traces of U, $U^\dagger$, and their derivatives in such a way that it is left unchanged when we replace the meson field $U(x)$ by $U^\dagger(x)$, equivalently $\Phi(x) \to -\Phi(x)$. This is a symmetry of our effective theory. It is also, as we will now show, an *accident*: it is not a symmetry of QCD. The strong interactions cheerfully produce processes that this accidental symmetry forbids — the radiative decay $\pi^0 \to \gamma\gamma$ chief among them, but also purely mesonic reactions such as $K^+K^- \to \pi^+\pi^-\pi^0$ and $\gamma \to \pi^+\pi^-\pi^0$. A theory that cannot describe these is not the low-energy theory of QCD; it is, at best, the parity-even half of it. The repair is a single new term, discovered by Wess and Zumino and given its definitive geometric form by Witten: a functional that lives most naturally not on four-dimensional spacetime but on a five-dimensional disk that has spacetime as its boundary. Its existence is forced by the chiral anomaly of Chapter 36; its overall coefficient is forced by topology to be an integer; and that integer is the number of colors. This section constructs it, proves the quantization, gauges it to electromagnetism, and recovers from it the $\pi^0 \to \gamma\gamma$ amplitude — closing the loop with the quark-level anomaly computation

of Part X.

39.6.1 The intrinsic-parity problem

We work throughout with the three-flavor coset $\mathrm{SU}(3)_L \times \mathrm{SU}(3)_R/\mathrm{SU}(3)_V$, whose Goldstone fields are the pseudoscalar octet. Recall (Section 39.2) the parametrization

$$U(x) = \exp\!\left(\frac{i\,\Phi(x)}{f}\right), \qquad \Phi = \sum_{a=1}^{8} \lambda^a \phi_a = \begin{pmatrix} \pi^0 + \frac{1}{\sqrt{3}}\eta & \sqrt{2}\,\pi^+ & \sqrt{2}\,K^+ \\ \sqrt{2}\,\pi^- & -\pi^0 + \frac{1}{\sqrt{3}}\eta & \sqrt{2}\,K^0 \\ \sqrt{2}\,K^- & \sqrt{2}\,\bar{K}^0 & -\frac{2}{\sqrt{3}}\eta \end{pmatrix}, \tag{39.143}$$

with $f = f_\pi \approx 92$ MeV the decay constant fixed in Eq. (29.202), and the transformation law $U \to L\,U\,R^\dagger$ under $(L,R) \in \mathrm{SU}(3)_L \times \mathrm{SU}(3)_R$. The leading Lagrangian and its first correction are

$$\mathcal{L}_2 = \frac{f^2}{4}\,\mathrm{Tr}(\partial_\mu U\,\partial^\mu U^\dagger) + \frac{f^2}{4}\,\mathrm{Tr}(\chi U^\dagger + U\chi^\dagger), \tag{39.144}$$

$$\mathcal{L}_4 = L_1\big[\,\mathrm{Tr}(\partial_\mu U \partial^\mu U^\dagger)\big]^2 + L_2\,\mathrm{Tr}(\partial_\mu U \partial_\nu U^\dagger)\,\mathrm{Tr}(\partial^\mu U \partial^\nu U^\dagger) + \cdots, \tag{39.145}$$

where $\chi = 2B_0\,\mathcal{M}_q$ encodes the quark masses and the ellipsis runs over the remaining Gasser–Leutwyler operators of Section 39.5. We do not re-derive these; we interrogate their symmetry.

Definition 39.21 (Intrinsic parity)**.** The *intrinsic-parity* transformation P_Φ acts on the meson field by

$$\mathsf{P}_\Phi : \quad U(x) \longmapsto U^\dagger(x), \qquad \text{equivalently} \qquad \Phi(x) \longmapsto -\Phi(x), \tag{39.146}$$

acting on the field values at the *same* spacetime point, with no reflection of the coordinates. It is to be distinguished from the full QCD parity P, which combines $U(t,\mathbf{x}) \to U^\dagger(t,-\mathbf{x})$ with the spatial reflection $\mathbf{x} \to -\mathbf{x}$. Equation (39.146) is P with the spatial reflection stripped off; it simply counts the number of Goldstone fields modulo two.

The pseudoscalar mesons are odd under the true parity P, as Goldstone bosons of an axial symmetry must be. Stripping the coordinate reflection, P_Φ multiplies every meson field by -1. A monomial in the fields therefore picks up $(-1)^{\#\Phi}$, where $\#\Phi$ is the number of meson fields it contains. We can now state precisely what is wrong.

Proposition 39.22 (The parity-even Lagrangian has an accidental symmetry)**.** *Every term of $\mathcal{L}_2$ in (39.144) and of $\mathcal{L}_4$ in (39.145) — indeed every term constructed from $U, U^\dagger$ and an even number of derivatives by taking traces of products and adding the Hermitian conjugate — is invariant under the intrinsic*

parity P_Φ *of* (39.146)*. Consequently the parity-even chiral Lagrangian contains only vertices with an even number of mesons, and forbids every process whose total Goldstone count is odd.*

Proof. Under P_Φ we have $U \to U^\dagger$ and hence $\partial_\mu U \to \partial_\mu U^\dagger$ at each point. A generic Lorentz-invariant, chirally invariant operator is a sum of traces of products of the building blocks U, $U^\dagger$, $\partial_\mu U$, $\partial_\mu U^\dagger$, χ, $\chi^\dagger$, with Lorentz indices contracted in pairs. The map P_Φ exchanges $U \leftrightarrow U^\dagger$ and $\partial U \leftrightarrow \partial U^\dagger$ throughout; under the simultaneous exchange $\chi \leftrightarrow \chi^\dagger$ (which is the statement that intrinsic parity commutes with the discrete relabeling $L \leftrightarrow R$ of the two chiralities) each such trace is sent to the trace of the Hermitian-conjugate product, which is its complex conjugate. A chirally invariant Lagrangian is real, so it already contains each trace together with its conjugate; the sum is therefore invariant. Concretely, $\mathrm{Tr}(\partial_\mu U\, \partial^\mu U^\dagger) \to \mathrm{Tr}(\partial_\mu U^\dagger\, \partial^\mu U) = \mathrm{Tr}(\partial_\mu U\, \partial^\mu U^\dagger)$ by cyclicity, and $\mathrm{Tr}(\chi U^\dagger + U\chi^\dagger) \to \mathrm{Tr}(\chi^\dagger U + U^\dagger \chi) = \mathrm{Tr}(\chi U^\dagger + U\chi^\dagger)$, so $\mathcal{L}_2$ is invariant; the four operators displayed in $\mathcal{L}_4$ and all the others are products of such invariants and inherit the property.

Now expand any such invariant in powers of Φ using $U = \exp(i\Phi/f) = 1 + i\Phi/f - \Phi^2/2f^2 + \cdots$. Because the operator is invariant under $\Phi \to -\Phi$, its Taylor expansion contains only even powers of Φ. Hence every interaction vertex it generates has an even number of external meson legs, and any S-matrix element built solely from these vertices vanishes unless the total number of external mesons (counted with the parity-odd photons weighted appropriately, see below) is even. □

This is a genuine prediction of the Lagrangian we have, and it is wrong. QCD is invariant under the true parity P, but it is *not* invariant under P_Φ alone, because P_Φ is not a symmetry once the coordinate reflection is removed: the underlying quark theory has no symmetry that flips the sign of every pseudoscalar interpolating field while leaving spacetime untouched. Experimentally the consequences are stark.

- *Five pseudoscalars.* The reaction $K^+K^- \to \pi^+\pi^-\pi^0$ involves five mesons; $\#\Phi = 5$ is odd, so $\mathcal{L}_2 + \mathcal{L}_4$ predicts a vanishing amplitude. It does not vanish.

- *Anomalous radiative decays.* The photon is parity-odd and intrinsic-parity-odd in the relevant sense (it couples to the parity-odd electromagnetic current), so $\gamma \to \pi^+\pi^-\pi^0$ and $\pi^0 \to \gamma\gamma$ carry odd intrinsic parity on the meson side and are likewise forbidden by $\mathcal{L}_2 + \mathcal{L}_4$. Yet $\pi^0 \to \gamma\gamma$ is the dominant decay of the π^0, with a lifetime measured to sub-percent precision.

The chiral Lagrangian as built so far is therefore *incomplete*. Something must supply the intrinsic-parity-odd interactions. We know from Chapter 36 exactly where in QCD the asymmetry between the two chiralities hides that the parity-even bosonic Lagrangian cannot see: the chiral anomaly. The missing term is the bosonized image of that anomaly.

Physical Insight

The accidental symmetry is the EFT's honest report of its own blind spot. A theory of Goldstone bosons built from the invariant U knows about the spontaneously broken *classical* chiral symmetry, but the path-integral measure of the quarks is not chirally invariant, and that quantum violation — the anomaly — is exactly the parity-odd information that traces of U cannot encode. The Wess–Zumino–Witten term is the unique local functional that injects this missing information, and the fact that it cannot be written as a four-dimensional Lagrangian density invariant under the chiral group is the precise sense in which the anomaly is not an ordinary interaction.

39.6.2 Construction of the Wess–Zumino–Witten term

We seek a functional $\Gamma_{\mathrm{WZW}}[U]$ that (i) is invariant under the full chiral group $\mathrm{SU}(3)_L \times \mathrm{SU}(3)_R$ acting by $U \to LUR^\dagger$, (ii) is odd under intrinsic parity P_Φ, and (iii) is local, in the sense that it is built from U and finitely many derivatives without any non-analytic dependence. Conditions (i) and (ii) together are surprisingly restrictive. The obstruction Wess and Zumino discovered is this: *there is no four-dimensional Lagrangian density, polynomial in U and its derivatives, that is both chirally invariant and intrinsic-parity-odd.* The intrinsic-parity-odd invariants one can write in four dimensions, such as $\varepsilon^{\mu\nu\rho\sigma}\,\mathrm{Tr}(L_\mu L_\nu L_\rho L_\sigma)$ with the Maurer–Cartan form $L_\mu \equiv U^\dagger \partial_\mu U$, are identically zero by the antisymmetry of ε against the cyclic trace. The way out, found by Witten, is to add one dimension.

The Maurer–Cartan form $L_\mu = U^\dagger \partial_\mu U$ takes values in the Lie algebra $\mathfrak{su}(3)$ and satisfies the structure ("zero-curvature") relation

$$\partial_\mu L_\nu - \partial_\nu L_\mu = -[L_\mu, L_\nu], \tag{39.147}$$

which follows from $L_\mu = U^\dagger \partial_\mu U$ by direct differentiation, $\partial_\mu L_\nu = \partial_\mu (U^\dagger \partial_\nu U) = -U^\dagger \partial_\mu U\, U^\dagger \partial_\nu U + U^\dagger \partial_\mu \partial_\nu U$, and antisymmetrizing. The unique nontrivial intrinsic-parity-odd five-index invariant is $\varepsilon^{ijklm}\,\mathrm{Tr}(L_i L_j L_k L_l L_m)$, which does *not* vanish: with five antisymmetrized factors the cyclic identity no longer forces it to zero. This object naturally integrates over a five-dimensional manifold.

Definition 39.23 (The Wess–Zumino–Witten action)**.** Let spacetime, with the fields assumed to approach a constant at infinity, be compactified to the

four-sphere S^4. Let Q be a five-dimensional disk (five-ball) with boundary $\partial Q = S^4$, and let $\widetilde{U} : Q \to \mathrm{SU}(3)$ be any smooth extension of the physical field $U : S^4 \to \mathrm{SU}(3)$ into the interior, with $\widetilde{U}|_{\partial Q} = U$. Writing $L_i = \widetilde{U}^\dagger \partial_i \widetilde{U}$ for $i = 1, \dots, 5$ the Maurer–Cartan form on Q, the Wess–Zumino–Witten action is

$$\boxed{\Gamma_{\mathrm{WZW}}[U] = -\frac{i\,n}{240\pi^2} \int_Q \mathrm{d}^5 y\; \varepsilon^{ijklm}\; \mathrm{Tr}(L_i L_j L_k L_l L_m)} \tag{39.148}$$

with n a real constant to be determined. The integrand is purely imaginary (the trace of five Hermitian-times-i factors), so the explicit i makes Γ_{WZW} real.

Three things must now be checked before (39.148) can be accepted as a term in a four-dimensional action: that it is chirally invariant; that despite being written as a bulk integral it depends on U only through its boundary values, so that its variation is a local four-dimensional object; and that the inevitable ambiguity in the choice of the extension $\widetilde{U}$ does not make the theory ill-defined.

Chiral invariance. Under a *global* transformation $U \to L U R^\dagger$ with constant L, R, the Maurer–Cartan form transforms as $L_i = U^\dagger \partial_i U \to R\, U^\dagger L^\dagger\, \partial_i (L U R^\dagger) = R\,(U^\dagger \partial_i U)\, R^\dagger = R\, L_i\, R^\dagger$, since L is constant. The trace $\mathrm{Tr}(L_i L_j L_k L_l L_m)$ is invariant under this similarity transformation by cyclicity, and the left rotation L drops out entirely because it acts only through $L^\dagger L = 1$. Hence the integrand, and therefore Γ_{WZW}, is invariant under the global chiral group. (The extension $\widetilde{U}$ may be chosen to respect the same transformation, e.g. by the radial homotopy below, so global invariance lifts to the bulk.)

The variation is a boundary term. The decisive property is that the five-form $\omega_5 \equiv \varepsilon^{ijklm}\,\mathrm{Tr}(L_i L_j L_k L_l L_m)$ is *closed* as a differential form on the group manifold: $\mathrm{d}\omega_5 = 0$. This is what makes (39.148) a sensible four-dimensional functional. We verify it through its variation. Under $\widetilde{U} \to \widetilde{U} + \delta\widetilde{U}$, write $\delta\widetilde{U} = \widetilde{U}\,\eta$ with $\eta \equiv \widetilde{U}^\dagger \delta \widetilde{U}$ an infinitesimal algebra-valued field. Then

$$\delta L_i = \delta(\widetilde{U}^\dagger \partial_i \widetilde{U}) = -\eta L_i + \widetilde{U}^\dagger \partial_i (\widetilde{U}\eta) = \partial_i \eta + [L_i, \eta], \tag{39.149}$$

the covariant derivative $D_i \eta \equiv \partial_i \eta + [L_i, \eta]$ of η. Varying the integrand and using the cyclicity of the trace to collect the five identical contributions,

$$\delta\ \mathrm{Tr}(L_i L_j L_k L_l L_m)\, \varepsilon^{ijklm} = 5\, \varepsilon^{ijklm}\ \mathrm{Tr}\big((D_i \eta)\, L_j L_k L_l L_m\big). \tag{39.150}$$

The commutator pieces in $D_i\eta = \partial_i\eta + [L_i, \eta]$ cancel against each other once antisymmetrized against ε^{ijklm}, by the Maurer–Cartan relation (39.147): each time a factor $[L_i, L_j]$ is generated by moving L_i past η, it equals $-(\partial_i L_j - \partial_j L_i)$, and the resulting derivatives reorganize into a total divergence. Carrying out this standard algebra one finds the clean result

$$\delta\left(\varepsilon^{ijklm}\,\mathrm{Tr}(L_i L_j L_k L_l L_m)\right) = 5\,\partial_i\Big(\varepsilon^{ijklm}\ \mathrm{Tr}\big(\eta\, L_j L_k L_l L_m\big)\Big), \tag{39.151}$$

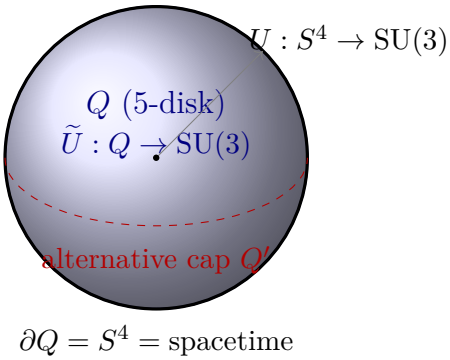


Figure 39.7: The Wess–Zumino–Witten functional (39.148) integrates a closed five-form over a five-disk Q whose boundary is (compactified) spacetime S^4. Two extensions $\widetilde{U}$ over Q and $\widetilde{U}'$ over a second cap Q' glue along S^4 into a map of the five-sphere $S^5 = Q \cup \bar{Q}'$; the difference of the two functionals is $2\pi n$ times the integer winding number of that map, an element of $\pi_5(\mathrm{SU}(3)) = \mathbb{Z}$. Single-valuedness of $e^{i\Gamma_{\mathrm{WZW}}}$ forces $n \in \mathbb{Z}$.

a pure total derivative in five dimensions. The five-dimensional integral of a total divergence collapses to an integral over the boundary $\partial Q = S^4 =$ spacetime:

$$\delta\Gamma_{\mathrm{WZW}} = -\frac{i\,n}{48\pi^2}\int_{S^4} \mathrm{d}^4x\; \varepsilon^{\mu\nu\rho\sigma}\; \mathrm{Tr}(\eta\, L_\mu L_\nu L_\rho L_\sigma), \tag{39.152}$$

which involves only the boundary value $\eta = U^\dagger \delta U$ and the boundary Maurer–Cartan form. Thus although Γ_{WZW} is written as a bulk integral, its *equations of motion* — and hence all of its physics — are genuinely four-dimensional and local. This is the precise meaning of "the variation is a total derivative in 4D so the term is local on the boundary": the five-dimensional writing is a bookkeeping device for a four-dimensional interaction that has no four-dimensional manifestly chiral-invariant Lagrangian density.

Extension ambiguity. The construction required choosing an extension $\widetilde{U}$ of U into the five-disk Q. Such an extension exists because $\pi_4(\mathrm{SU}(3)) = 0$ (every map $S^4 \to \mathrm{SU}(3)$ is contractible, so it bounds), a fact recorded in Theorem 37.36. But the extension is not unique, and we must check that Γ_{WZW} does not depend on the choice in any way that affects physics. This is where the integer n is born, and it is the subject of the next result.

39.6.3 Witten's quantization: the coefficient is N_c

Theorem 39.24 (Quantization of the WZW coefficient)**.** *For the quantum theory defined by the path integral weight e^{iS} with $S \supset \Gamma_{\mathrm{WZW}}[U]$ to be independent of*

the choice of five-disk extension, the coefficient n in (39.148) must be an integer, $n \in \mathbb{Z}$. Matching the gauged WZW term to the QCD chiral anomaly of Chapter 36 fixes the integer to the number of colors,

$$n = N_c = 3. \tag{39.153}$$

Thus the coefficient of the Wess–Zumino–Witten term is a direct measurement, within the meson effective theory, of the number of quark colors.

Proof. Quantization. Suppose we choose two different extensions, $\widetilde{U}$ over the disk Q and $\widetilde{U}'$ over a second disk Q', both restricting to the same physical U on the common boundary S^4. The two functionals differ by

$$\Gamma_{\mathrm{WZW}}[\widetilde{U}] - \Gamma_{\mathrm{WZW}}[\widetilde{U}'] = -\frac{in}{240\pi^2}\left(\int_Q - \int_{Q'}\right)\omega_5 = -\frac{in}{240\pi^2}\oint_{S^5}\omega_5, \tag{39.154}$$

where $S^5 = Q \cup \bar{Q}'$ is the closed five-manifold obtained by gluing Q to Q' with reversed orientation along their shared boundary — topologically a five-sphere (see Fig. 39.7). On this closed S^5 the two extensions combine into a single smooth map $g : S^5 \to \mathrm{SU}(3)$. The crucial topological input, established in Theorem 37.36, is that the fifth homotopy group of $\mathrm{SU}(N \geq 3)$ is the integers,

$$\pi_5(\mathrm{SU}(3)) = \mathbb{Z}, \tag{39.155}$$

so maps $g : S^5 \to \mathrm{SU}(3)$ are classified up to homotopy by an integer winding number $w[g] \in \mathbb{Z}$. The normalization in (39.148) was chosen precisely so that this winding number is computed by the same five-form integral:

$$-\frac{i}{240\pi^2}\oint_{S^5}\varepsilon^{ijklm}\,\mathrm{Tr}(L_iL_jL_kL_lL_m) \;=\; 2\pi\, w[g], \qquad w[g] \in \mathbb{Z}. \tag{39.156}$$

That the number $240\pi^2$ does this job is the content of the deeper computation in Section 37.7, where the volume of SU(3) in the bi-invariant metric and the generator of $H^5(\mathrm{SU}(3);\mathbb{Z})$ are evaluated; we import that normalization rather than rederive it, since the only fact we need here is the existence of an integer-valued period. Combining (39.154) with (39.156),

$$\Gamma_{\mathrm{WZW}}[\widetilde{U}] - \Gamma_{\mathrm{WZW}}[\widetilde{U}'] = 2\pi\, n\, w[g]. \tag{39.157}$$

The action is therefore ambiguous, but only by integer multiples of $2\pi n$. Physics depends on the action only through the path-integral weight e^{iS}. The ambiguity (39.157) is invisible in e^{iS} if and only if $2\pi n\, w[g] \in 2\pi\mathbb{Z}$ for every $w[g] \in \mathbb{Z}$, that is, if and only if

$$n \in \mathbb{Z}. \tag{39.158}$$

This is Witten's quantization. It is the exact higher-dimensional analogue of Dirac's charge quantization: a topological period of the action must be an integer

multiple of 2π for the quantum theory to exist, and here that integer is the WZW coefficient. Notice the logic: the classical action is genuinely multivalued, but the quantum theory is well-defined provided the multivaluedness is a multiple of 2π, which quantizes the coupling. No continuous renormalization can shift n; it is a topological invariant of the effective theory.

Matching to QCD. The integer n is not fixed by the meson theory in isolation — any integer gives a consistent EFT. It is fixed by demanding that this EFT reproduce QCD. The cleanest matching is through the gauged term derived in Section 39.6.4: gauging Γ_{WZW} to electromagnetism produces a $\pi^0\gamma\gamma$ vertex whose coefficient is proportional to n, while the same vertex is computed directly from the quark triangle diagram in Chapter 36, where it is proportional to the number of colors N_c circulating in the loop (each color contributes one triangle). Equality of the two — the statement that the anomaly of the microscopic theory equals the anomaly of the effective theory — is 't Hooft anomaly matching, Theorem 37.26. It forces $n = N_c$. With $N_c = 3$ in QCD,

$$n = 3. \tag{39.159}$$

We carry out the matching explicitly in Theorem 39.25; here we record only its conclusion, (39.153). □

Physical Insight

The number of colors is usually introduced as a property of the ultraviolet: three colors are needed for the R-ratio, for the π^0 width, for the gauge-anomaly cancellation of the Standard Model. The WZW term shows that N_c is also written into the deep infrared. The pseudoscalar mesons — the lightest, most accessible degrees of freedom of the strong interaction — carry the integer N_c as the coefficient of a topological term in their own effective action. One could, in principle, determine that there are three colors purely from low-energy meson physics, never resolving a single quark. This is the most vivid possible statement that an effective theory remembers the counting of its constituents.

39.6.4 Gauging to electromagnetism and the $\pi^0 \to \gamma\gamma$ vertex

To extract a physical decay we must couple the mesons to the photon. The mesons carry electric charge through the quark charge matrix

$$Q = \frac{1}{3}\,\mathrm{diag}(2,-1,-1) = \mathrm{diag}\Big(\tfrac{2}{3},-\tfrac{1}{3},-\tfrac{1}{3}\Big)\,, \tag{39.160}$$

acting in flavor space, in units of e. Electromagnetism is gauged by promoting the ordinary derivative to a covariant one following the minimal-coupling prescription

of Eq. (8.27); on the meson field, the left and right flavor rotations are tied together to the single $\mathrm{U}(1)_{\mathrm{em}}$ generated by Q, so

$$\partial_\mu U \;\longrightarrow\; D_\mu U = \partial_\mu U - ieA_\mu\,[\,Q, U\,]. \tag{39.161}$$

Gauging the WZW term is not as simple as the replacement (39.161), however, because Γ_{WZW} is a five-dimensional functional whose gauge variation must be cancelled order by order in A_μ by additional, explicitly constructed counterterms (the Bardeen counterterms). The full gauged functional is built in Section 37.7; we import its result and isolate the single piece relevant to two-photon decay. That piece is a genuine four-dimensional local interaction — two powers of the photon field saturate two of the five dimensions, leaving a standard 4D vertex — of the form

$$\mathcal{L}^{\gamma\gamma}_{\mathrm{WZW}} = \frac{n\,e^2}{32\pi^2 f}\,\varepsilon^{\mu\nu\rho\sigma} F_{\mu\nu}F_{\rho\sigma}\;\mathrm{Tr}(Q^2\,\Phi), \tag{39.162}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ the electromagnetic field strength and Φ the meson matrix (39.143). The flavor trace selects the neutral mesons:

$$\mathrm{Tr}(Q^2\Phi) = \mathrm{Tr}\Big[\,\mathrm{diag}\big(\tfrac{4}{9},\tfrac{1}{9},\tfrac{1}{9}\big)\,\Phi\Big] = \frac{4}{9}\big(\pi^0+\tfrac{1}{\sqrt{3}}\eta\big)+\frac{1}{9}\big(-\pi^0+\tfrac{1}{\sqrt{3}}\eta\big)+\frac{1}{9}\big(-\tfrac{2}{\sqrt{3}}\eta\big) = \frac{1}{3}\,\pi^0+\frac{1}{3\sqrt{3}}\,\eta. \tag{39.163}$$

Keeping the π^0 and setting $n = N_c$, (39.162) delivers the effective $\pi^0\gamma\gamma$ Lagrangian

$$\boxed{\mathcal{L}_{\pi^0\gamma\gamma} = \frac{N_c\,e^2}{96\pi^2 f_\pi}\,\pi^0\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}} \tag{39.164}$$

where we used $\mathrm{Tr}(Q^2\Phi) \supset \frac{1}{3}\pi^0$ and $f = f_\pi$. We pause to confirm the coefficient against the anomaly before computing the rate.

The coefficient is fixed by anomaly matching, not chosen. The number $N_c e^2/96\pi^2 f_\pi$ in (39.164) looks like an arbitrary normalization; it is not. It is forced by requiring that the divergence of the effective axial-isospin current reproduce the operator anomaly of Eq. (36.50). The third axial-isospin current of QCD,

$$j_3^{\mu 5} = \bar q\,\gamma^\mu\gamma^5\,\frac{\tau^3}{2}\,q, \tag{39.165}$$

has the electromagnetic ABJ anomaly (Eq. (36.50)) obtained by summing over the N_c colors and the two flavors with axial charges $\pm\frac{1}{2}$ and electric charges $Q_u = \frac{2}{3}$, $Q_d = -\frac{1}{3}$:

$$\partial_\mu j_3^{\mu 5} = \frac{e^2}{16\pi^2}\,D\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}, \qquad D = N_c\Big(\tfrac{1}{2}Q_u^2 - \tfrac{1}{2}Q_d^2\Big) = N_c\Big(\tfrac{1}{2}\cdot\tfrac{4}{9} - \tfrac{1}{2}\cdot\tfrac{1}{9}\Big) = \frac{N_c}{6}. \tag{39.166}$$

On the meson side the same current is, by PCAC (Eq. (29.202)), $j_3^{\mu 5} = f_\pi\partial^\mu\pi^0 + \cdots$, so its divergence is $\partial_\mu j_3^{\mu 5} = f_\pi\,\Box\pi^0$. The photon-dependent source on the

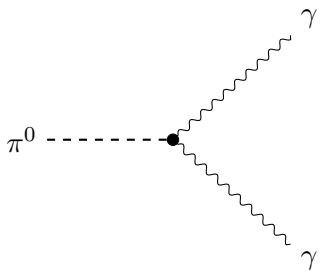


Figure 39.8: The $\pi^0 \to \gamma\gamma$ vertex generated by the gauged Wess–Zumino–Witten term (39.164). In the meson effective theory it is a local contact interaction (the dot); in QCD it is the quark triangle diagram of Chapter 36, summed over the N_c colors that run in the loop. Equality of the two is 't Hooft anomaly matching, Theorem 37.26.

right of the π^0 field equation comes from $\partial\mathcal{L}_{\pi^0\gamma\gamma}/\partial\pi^0 = (N_c e^2/96\pi^2 f_\pi)\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$, so the photon-dependent part of the current divergence is

$$(\partial_\mu j_3^{\mu 5})_{\gamma\gamma} = f_\pi \cdot \frac{N_c e^2}{96\pi^2 f_\pi}\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = \frac{N_c e^2}{96\pi^2}\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{39.167}$$

Comparing (39.167) with the quark-level result (39.166), where $D = N_c/6$ gives $(e^2/16\pi^2)(N_c/6) = N_c e^2/96\pi^2$ for the coefficient of εFF, the two coefficients coincide exactly. The WZW coefficient (39.164) is therefore not adjustable: it is the unique value that makes the meson EFT reproduce the QCD anomaly, which is exactly the matching condition that fixed $n = N_c$ in Theorem 39.24. The bosonized anomaly and the quark triangle are the same number, viewed from the two ends of the running coupling.

39.6.5 The amplitude and the decay rate

We now compute the $\pi^0 \to \gamma\gamma$ amplitude from the vertex (39.164) and evaluate the width, carrying every step. Write the interaction as $\mathcal{L}_{\pi^0\gamma\gamma} = c\,\pi^0\,\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$ with

$$c = \frac{N_c\,e^2}{96\pi^2 f_\pi} \quad \text{(for the isolated } \pi^0\text{)} \implies A \equiv 8c = \frac{e^2 N_c}{12\pi^2 f_\pi}. \tag{39.168}$$

Using the antisymmetry of ε in $(\mu\nu)$ and in $(\rho\sigma)$ to write $\varepsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} = 4\,\varepsilon^{\mu\nu\rho\sigma}\partial_\mu A_\nu\,\partial_\rho A_\sigma$, the vertex Feynman rule for $\pi^0(p) \to \gamma(k_1,\epsilon_1)\gamma(k_2,\epsilon_2)$ follows by attaching the two photon fields to the external legs (a combinatorial factor 2! for the two ways) with $\partial_\mu A_\nu \to ik_\mu\epsilon^*_\nu$ for each outgoing photon:

$$i\mathcal{M} = i\cdot 4c\cdot 2!\cdot\varepsilon^{\mu\nu\rho\sigma}(ik_{1\mu}\epsilon^*_{1\nu})(ik_{2\rho}\epsilon^*_{2\sigma}) = -8ic\,\varepsilon^{\mu\nu\rho\sigma}k_{1\mu}\epsilon^*_{1\nu}k_{2\rho}\epsilon^*_{2\sigma}. \tag{39.169}$$

Hence

$$\mathcal{M}(\pi^0 \to \gamma\gamma) = -8c\,\varepsilon^{\mu\nu\rho\sigma} k_{1\mu}\epsilon^*_{1\nu} k_{2\rho}\epsilon^*_{2\sigma} = -A\,\varepsilon^{\mu\nu\rho\sigma} k_{1\mu}\epsilon^*_{1\nu} k_{2\rho}\epsilon^*_{2\sigma}, \tag{39.170}$$

using $8c = 8\cdot N_c e^2/96\pi^2 f_\pi = N_c e^2/12\pi^2 f_\pi$, so that the physical amplitude carries precisely

$$A = \frac{e^2 N_c}{12\pi^2 f_\pi} = \frac{\alpha N_c}{3\pi f_\pi}, \tag{39.171}$$

with $\alpha = e^2/4\pi$. This is exactly the result Eq. (36.115) obtained from the quark triangle in Part X: the effective meson coupling and the microscopic anomaly give the same amplitude, the explicit content of the anomaly matching Theorem 37.26.

Worked Example 39.25 (The π^0 width and the count of colors)**.** We square (39.170), sum over the final photon polarizations, integrate over two-body phase space, and put in numbers.

Polarization sum. With $\mathcal{M} = A\,\varepsilon_{\mu\nu\rho\sigma}\epsilon_1^\mu \epsilon_2^\nu k_1^\rho k_2^\sigma$ (rewriting (39.170) after relabeling, the overall sign immaterial), the spin-summed square uses $\sum_{\text{pol}} \epsilon^\mu \epsilon^{*\nu} \to -g^{\mu\nu}$ (the gauge-dependent pieces drop against the antisymmetry of ε):

$$\begin{aligned}\sum_{\text{pol}} |\mathcal{M}|^2 &= A^2 (-g_{\mu\mu'})(-g_{\nu\nu'})\,\varepsilon^{\mu\nu\rho\sigma}\varepsilon^{\mu'\nu'\rho'\sigma'} k_{1\rho} k_{2\sigma} k_{1\rho'} k_{2\sigma'} \\ &= A^2\,\varepsilon^{\mu\nu\rho\sigma}\varepsilon_{\mu\nu}{}^{\rho'\sigma'}\, k_{1\rho} k_{2\sigma} k_{1\rho'} k_{2\sigma'}.\end{aligned} \tag{39.172}$$

In the mostly-minus metric, $\varepsilon^{\mu\nu\rho\sigma}\varepsilon_{\mu\nu\alpha\beta} = -2(\delta^\rho_\alpha\delta^\sigma_\beta - \delta^\rho_\beta\delta^\sigma_\alpha)$, so

$$\sum_{\text{pol}} |\mathcal{M}|^2 = -2A^2\big[(k_1\cdot k_1)(k_2\cdot k_2) - (k_1\cdot k_2)^2\big] = 2A^2\,(k_1\cdot k_2)^2, \tag{39.173}$$

since the photons are massless, $k_1^2 = k_2^2 = 0$. From $(k_1+k_2)^2 = m_\pi^2$ and $k_i^2 = 0$ we get $2\,k_1\cdot k_2 = m_\pi^2$, hence $(k_1\cdot k_2)^2 = m_\pi^4/4$ and

$$\sum_{\text{pol}} |\mathcal{M}|^2 = \frac{A^2 m_\pi^4}{2}. \tag{39.174}$$

Phase space and the width. For a one-to-two decay with massless final particles the integrated two-body phase space is $\int \mathrm{d}\Pi_2 = 1/8\pi$. Because the two photons are identical we include a symmetry factor $1/2$. The decay rate is then

$$\Gamma = \frac{1}{2}\cdot\frac{1}{2m_\pi}\int \mathrm{d}\Pi_2 \sum_{\text{pol}} |\mathcal{M}|^2 = \frac{1}{2}\cdot\frac{1}{2m_\pi}\cdot\frac{1}{8\pi}\cdot\frac{A^2 m_\pi^4}{2} = \frac{A^2 m_\pi^3}{64\pi}. \tag{39.175}$$

Substituting $A = \alpha N_c/3\pi f_\pi$ from (39.171) gives the closed form

$$\Gamma(\pi^0 \to \gamma\gamma) = \frac{\alpha^2\, N_c^2\, m_\pi^3}{576\,\pi^3 f_\pi^2}. \tag{39.176}$$

Numbers. With $\alpha = 1/137.04$, $m_\pi = m_{\pi^0} = 134.98$ MeV, $f_\pi = 92.1$ MeV, and $N_c = 3$, the amplitude is

$$A = \frac{\alpha N_c}{3\pi f_\pi} = \frac{(1/137.04)\cdot 3}{3\pi \cdot 0.0921\ \text{GeV}} = 0.0253\ \text{GeV}^{-1}, \tag{39.177}$$

and the width is

$$\Gamma = \frac{A^2 m_\pi^3}{64\pi} = \frac{(0.0253)^2\,(0.13498)^3}{64\pi}\ \text{GeV} = 7.8\times 10^{-9}\ \text{GeV} = 7.8\ \text{eV}. \tag{39.178}$$

The measured value (PrimEx and the Particle Data Group average, mid-2020s) is $\Gamma(\pi^0 \to \gamma\gamma) = 7.80 \pm 0.12$ eV. The leading prediction lands on the experimental value to better than the experimental error.

The count of colors. Because $\Gamma \propto N_c^2$, the measured width *measures* N_c. Had we taken $N_c = 1$ the predicted width would be a factor of nine too small, ~ 0.87 eV, in gross conflict with experiment; $N_c = 2$ gives ~ 3.5 eV, still excluded. Only $N_c = 3$ works, and it works to the percent level. The radiative decay of the neutral pion is thus a direct low-energy weighing of the number of quark colors — the experimental face of the quantization theorem Theorem 39.24 and of the anomaly matching Theorem 37.26.

Caution

The clean agreement uses the chiral-limit prediction (39.176) at the physical point with the *physical* f_π. The anomaly fixes the amplitude in the chiral limit ($m_q \to 0$); the leading corrections are of order m_π^2/Λ_χ^2 and a small π^0–η–η' mixing, both at the percent level, which is why the lowest-order formula is already this accurate. The agreement is not an accident of using one convention for f_π versus another: the combination $A = \alpha N_c/3\pi f_\pi$ is convention-independent once f_π is defined through the same matrix element (29.202) used here ($f_\pi \approx 92$ MeV, not the $\sqrt{2}$-larger normalization).

39.6.6 What the WZW term completes

The gauged Wess–Zumino–Witten functional does more than the single decay we computed. Expanding (39.148) to fifth order in Φ produces a five-pseudoscalar contact vertex — the interaction responsible for $K^+K^- \to \pi^+\pi^-\pi^0$ and the other odd-meson reactions that opened this section — with a coefficient again proportional to N_c. Gauging with one photon instead of two yields the $\gamma \to \pi^+\pi^-\pi^0$ vertex with amplitude $F^{3\pi} = eN_c/12\pi^2 f_\pi^3$, a second independent low-energy test of the same integer. Every one of these intrinsic-parity-odd processes is forbidden by $\mathcal{L}_2 + \mathcal{L}_4$ and supplied, with a quantized strength, by the one new

term. With Γ_{WZW} adjoined, the chiral Lagrangian

$$S = \int \mathrm{d}^4x \, (\mathcal{L}_2 + \mathcal{L}_4 + \cdots) + \Gamma_{\mathrm{WZW}} \tag{39.179}$$

is the complete leading-order effective field theory of QCD at low energy: the parity-even sector built from the spontaneously broken chiral symmetry, plus the parity-odd sector built from the anomaly, with the number of colors built into the coefficient of the latter.

This closes the chapter's central thesis and the strong-interaction story of the book. We have seen QCD from both ends of its running coupling (QCD): in the ultraviolet as the asymptotically free gauge theory of Chapters 25 and 27, and here in the infrared as a predictive theory of pseudo-Goldstone mesons whose every coefficient is fixed by symmetry, power counting, and — in the WZW term — topology and the anomaly. The anomaly computed at the quark level in Chapter 36 has reappeared, unchanged in magnitude, as an interaction of mesons, the cleanest illustration in the book of 't Hooft anomaly matching (Theorem 37.26). The next chapter (Chapter 40) turns from this archetypal low-energy EFT to the other effective theories that organize modern particle physics — heavy-quark and Standard-Model effective field theories — carrying forward the same paradigm of Chapter 38: that a theory one cannot solve in its fundamental variables can still be solved, order by order, once symmetry rather than the coupling is made the organizing principle of the expansion.

Further reading

The reinterpretation of current-algebra results as the leading term of a systematic momentum expansion is Weinberg's Weinberg [498], building on his original soft-pion prediction Weinberg [495]; the foundational one-loop construction, including the order-p^4 Lagrangian and the low-energy constants used throughout Section 39.5, is Gasser and Leutwyler [210]. The most accessible modern textbook treatment, covering the nonlinear realization, power counting, loops, and the WZW term at the level of this chapter, is Scherer and Schindler [430]. The intrinsic-parity-odd sector originates with Wess and Zumino [506], and the topological quantization of its coefficient to the integer N_c—the argument behind Theorem 39.24—is Witten's Witten [521]. For the present state of the precision program, including the two-loop $\pi\pi$ analysis and its experimental confrontation, see Colangelo et al. [126] and the NA48/2 measurement [51].

Chapter 40

Heavy-Particle and Standard Model EFTs

The effective-field-theory paradigm of Chapter 38 taught us that a low-energy theory is fixed by three choices — a field content, a symmetry, and a power counting — and that its Wilson coefficients are then pinned in one of two ways: by *matching*, when the short-distance theory is known, or by *measurement*, when it is not. Chapter 39 ran the second route on the broken chiral symmetry of QCD, where the ultraviolet completion is known to be QCD but is strongly coupled and so out of perturbative reach. This chapter runs the method in *both* directions at once, on two of the most consequential effective theories in particle physics, and shows them to be one construction seen from two sides. The first half is top-down: the short-distance theory is the Yang–Mills sector of Chapter 25, renormalized and run in Chapter 27, and we integrate out a heavy quark to build the heavy-quark effective theory (HQET). The second half is bottom-up: the short-distance theory is unknown, and we treat the Standard Model assembled in Chapter 33 as the leading term of an expansion — the Standard Model Effective Field Theory (SMEFT) — whose higher-dimension operators parametrize, model-independently, whatever lies above the electroweak scale.

The HQET arc occupies the first three sections. We begin from the QCD Lagrangian of a heavy quark, split its momentum into a large kinematic part $m_Q v^\mu$ and a soft residual part, strip the rapidly oscillating mass phase from the field, and integrate out its small Dirac components to arrive at a static Lagrangian whose Feynman rules — a forward-propagating worldline and a spin-blind eikonal vertex — are the diagrammatic face of an emergent symmetry. That symmetry, a $\mathrm{U}(2N_h)$ rotating heavy-quark spin and flavour, is invisible in QCD yet exact in the $m_Q \to \infty$ limit; we use it to collapse the six $B \to D^{(*)}$ semileptonic form factors onto a single Isgur–Wise function normalized to unity at zero recoil, the statement that makes these decays the cleanest window onto the CKM element

$|V_{cb}|$ of Chapter 32. Restoring finite m_Q then forces us to match QCD onto HQET order by order in $1/m_Q$, producing a kinetic operator whose coefficient is frozen to unity by reparametrization invariance and a chromomagnetic operator whose runnable coefficient is the microscopic origin of the B^*–B and D^*–D hyperfine splittings. The abelian sibling of this computation, NRQED, recovers Schwinger's $a_e = \alpha/2\pi$ of Chapter 20 — carried here along QED — now wearing the costume of a matching coefficient rather than a form factor.

The SMEFT arc occupies the last four sections and inverts the logic. With the ultraviolet unknown, we take only the data we trust — the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the chiral matter and Higgs content of Chapter 33, and a single new scale Λ — and build the tower of higher-dimension operators. Dimension five yields the unique Weinberg operator, the low-energy shadow of lepton-number violation and of the seesaw of Chapter 35; dimension six yields, after the redundancies of integration by parts, field redefinitions, Fierz rearrangements, and identical-field symmetry are removed, the canonical 59-operator Warsaw basis. We then put the coefficients to work: matching a known heavy field (the W, recovering Fermi theory and G_F) and a hypothetical one fixes them at the high scale, the renormalization group of Chapter 24 — driven here by the QCD anomalous-dimension technology of Chapter 27 — evolves them down through operator mixing, and the oblique parameters S and T of Chapter 34 reappear as individual dimension-six Wilson coefficients that turn a precision measurement directly into a bound on Λ.

Throughout we hold to the conventions of Chapter A: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the covariant-derivative sign $D_\mu = \partial_\mu - igA^a_\mu T^a$, and $\overline{\mathrm{MS}}$ with dimensional regularization as the default scheme. The Dirac and trace technology is that of Appendix C, the group-theory data that of Appendix B, and the master loop integrals those of Appendix D; the running coupling $\alpha_s(\mu)$ is the one established in Chapter 27, the accidental B and L symmetries and the $B - L$ anomaly those of Chapters 33 and 37. The chapter closes by recognizing explicitly what its two halves have demonstrated implicitly: "write all operators, power count, match or measure, run" is a single recipe, and HQET and SMEFT differ only in whether the short-distance theory has been written down or is still waiting to be discovered.

40.1 The heavy-quark expansion

An effective field theory is fixed by three choices: a field content, a symmetry, and a power counting. Chapter 38 built that paradigm abstractly—integrate out the heavy modes, match onto a local Lagrangian, organize the result as an expansion in a small ratio, and run the Wilson coefficients between scales—and Chapter 39 deployed it bottom-up, where the short-distance theory is hidden and the operators are written from symmetry alone. We now run the machinery

in the opposite, fully controlled direction. The short-distance theory is *known*: it is QCD, the non-abelian gauge theory of Chapter 25 renormalized and run in Chapter 27. We continue Yang–Mills theory from Chapter 27 and ask what becomes of that theory when one of its quarks is much heavier than the scale at which the strong interaction confines.

The physical hierarchy is unambiguous. A hadron containing a single heavy quark—a B or D meson, a Λ_b baryon—has a size of order $1/\Lambda_{\rm QCD} \sim 1\,\text{fm}$ set by the light degrees of freedom, while the heavy quark carries a Compton wavelength $1/m_Q$ far shorter than that. For the charm ($m_c \approx 1.3\,\text{GeV}$), the bottom ($m_b \approx 4.2\,\text{GeV}$), and *a fortiori* the top, one has

$$m_Q \;\gg\; \Lambda_{\rm QCD} \approx 0.2\,\text{GeV}, \qquad \lambda \equiv \frac{\Lambda_{\rm QCD}}{m_Q} \;\ll\; 1, \tag{40.1}$$

and λ is the expansion parameter of the effective theory we are about to build, *heavy-quark effective theory* (HQET). The picture to keep in mind is the hydrogen atom of the strong interaction: a heavy, nearly static colour source surrounded by a cloud of soft gluons and light quarks ("the brown muck") whose dynamics it barely disturbs. Just as the proton mass drops out of atomic spectroscopy to leading order, leaving the electron cloud insensitive to whether the nucleus is hydrogen or deuterium, the heavy-quark mass and spin will drop out of the leading strong dynamics—but seeing that requires first separating the part of the heavy quark's motion that is large and kinematically fixed from the part that is soft and dynamical. That separation is the whole content of this section.

40.1.1 Splitting the momentum: velocity as a label

Inside the hadron the heavy quark is very nearly on shell and very nearly at rest relative to the hadron: its four-momentum is dominated by the mass term. Write

$$p^\mu \;=\; m_Q\, v^\mu \;+\; k^\mu, \qquad v^\mu v_\mu = v^2 = 1, \qquad v^0 > 0, \tag{40.2}$$

where v^μ is a fixed timelike unit four-velocity—in the hadron rest frame simply $v^\mu = (1, \mathbf{0})$—and k^μ is the *residual momentum*, the small fluctuation the quark acquires by exchanging soft gluons with the brown muck. The strong interaction can change k by amounts of order $\Lambda_{\rm QCD}$, but it cannot appreciably change v: shifting the velocity of a particle of mass m_Q by a finite amount costs momentum of order $m_Q \gg \Lambda_{\rm QCD}$, which the soft cloud simply does not have. Thus

$$k^\mu \;\sim\; \Lambda_{\rm QCD} \;\sim\; m_Q\,\lambda, \tag{40.3}$$

and v^μ is effectively a conserved quantity. We promote it to a fixed *label* on the fields: a heavy quark of velocity v and one of velocity $v' \neq v$ live in different sectors of the theory that the soft dynamics never connects, a statement known as velocity superselection. Within one sector the only dynamical variable left is

the residual momentum k, conjugate to the residual position; all dependence on the large scale m_Q has been pulled out into the prefactor $m_Q v^\mu$.

This kinematic split must be implemented at the level of the field. The QCD heavy-quark field $Q(x)$ carries the full momentum p; we strip off the large, rapidly oscillating phase by writing

$$Q(x) \;=\; e^{-im_Q\, v\cdot x}\,\big[\,h_v(x) + H_v(x)\,\big]. \tag{40.4}$$

The factor $e^{-im_Q v\cdot x}$ carries away the $m_Q v^\mu$ part of every derivative: acting on Q, the operator $i\partial^\mu$ produces $m_Q v^\mu$ plus whatever acts on the residual fields h_v, H_v, and the latter, varying only on the soft scale, carry just the residual momentum k. Concretely, with the gauge-covariant derivative of Yang–Mills theory, $D_\mu = \partial_\mu - igA^a_\mu T^a$ acting on the colour-fundamental Q,

$$iD_\mu Q = e^{-im_Q v\cdot x}\,(m_Q v_\mu + iD_\mu)\,(h_v + H_v), \qquad i\not{D}\,Q - m_Q Q = e^{-im_Q v\cdot x}\,(m_Q \not{v} + i\not{D} - m_Q)(h_v + H_v) \tag{40.5}$$

which is the only place the large mass will appear explicitly. It remains to say which combination of Dirac components is h_v and which is H_v.

40.1.2 The velocity projectors and the small components

For a free particle at rest the Dirac equation $(\not{p} - m)u = 0$ becomes $(\gamma^0 - 1)u = 0$, so the positive-energy spinor lives in the $+1$ eigenspace of γ^0 and the lower ("small") two components vanish. Boosting to velocity v replaces γ^0 by $\not{v}$. This motivates decomposing the Dirac field with the operators that project onto the ± 1 eigenspaces of $\not{v}$.

Definition 40.1 (Velocity projectors)**.** For a fixed unit timelike velocity v^μ ($v^2 = 1$) define

$$P_\pm \;\equiv\; \frac{1 \pm \not{v}}{2}. \tag{40.6}$$

The large and small components of the heavy quark are the projections

$$h_v \equiv e^{+im_Q v\cdot x}\, P_+\, Q, \qquad H_v \equiv e^{+im_Q v\cdot x}\, P_-\, Q, \qquad \text{so} \qquad P_+ h_v = h_v, \quad P_- H_v = H_v. \tag{40.7}$$

That Eq. (40.6) are genuine orthogonal projectors follows from $v^2 = 1$ alone, exactly as the energy projectors of Section C.8.6 follow from $\not{p}\not{p} = p^2 = m^2$ on shell. Using $\not{v}\not{v} = v^2 = 1$,

$$P_\pm^2 = \frac{(1 \pm \not{v})^2}{4} = \frac{1 \pm 2\not{v} + v^2}{4} = \frac{2 \pm 2\not{v}}{4} = P_\pm, \qquad P_+ + P_- = 1, \qquad P_+ P_- = \frac{1 - v^2}{4} = 0, \tag{40.8}$$

and $\not{v}\, P_\pm = \pm P_\pm$, so $\not{v}\, h_v = h_v$ and $\not{v}\, H_v = -H_v$. Taking the Dirac adjoint of these last two relations (with $\bar{h}_v = h_v^\dagger \gamma^0$ and $\gamma^0 \not{v}^\dagger \gamma^0 = \not{v}$) gives the mirrored

statements $\bar{h}_v \not{v} = \bar{h}_v$ and $\bar{H}_v \not{v} = -\bar{H}_v$, equivalently $\bar{h}_v P_+ = \bar{h}_v$ and $\bar{H}_v P_- = \bar{H}_v$. These four sandwiching rules are the only Dirac algebra the construction needs.

The decomposition Eq. (40.7) sorts the four Dirac components into two upper components h_v, which survive in the static limit and describe the physical heavy quark, and two lower components H_v, which the free rest-frame equation forces to vanish and which therefore describe rapidly oscillating, off-shell fluctuations of energy $\sim 2m_Q$. The strategy of every top-down EFT now applies verbatim (Section 38.3): H_v is the "heavy field," and we integrate it out.

40.1.3 The exact Lagrangian in residual variables

Insert Eq. (40.4) into the QCD heavy-quark Lagrangian $\mathcal{L}_Q = \bar{Q}\,(i\not{D} - m_Q)\,Q$ of Eq. (11.25), now with the colour-covariant derivative of Eq. (25.76). The conjugate field carries the opposite phase, $\bar{Q} = e^{+im_Q v\cdot x}(\bar{h}_v + \bar{H}_v)$, so the oscillating exponentials cancel and Eq. (40.5) gives the exact identity

$$\mathcal{L}_Q = (\bar{h}_v + \bar{H}_v)(m_Q \not{v} + i\not{D} - m_Q)(h_v + H_v). \tag{40.9}$$

The mass terms collapse immediately. On h_v, $(m_Q\not{v} - m_Q)h_v = m_Q(h_v - h_v) = 0$; on H_v, $(m_Q\not{v} - m_Q)H_v = m_Q(-H_v - H_v) = -2m_Q H_v$. Hence the bracket in Eq. (40.9) equals $i\not{D}\,h_v + (i\not{D} - 2m_Q)H_v$, and

$$\mathcal{L}_Q = \bar{h}_v\, i\not{D}\, h_v + \bar{h}_v(i\not{D} - 2m_Q)H_v + \bar{H}_v\, i\not{D}\, h_v + \bar{H}_v(i\not{D} - 2m_Q)H_v. \tag{40.10}$$

To evaluate the four bilinears we split the covariant derivative into its component along the velocity and the part orthogonal to it,

$$\not{D} = \not{v}\,(v\cdot D) + \not{D}_\perp, \qquad D^\mu_\perp \equiv D^\mu - v^\mu\,(v\cdot D), \qquad v\cdot D_\perp = 0, \tag{40.11}$$

which one verifies by substituting $D^\mu_\perp$ back: $\not{v}(v\cdot D) + \gamma_\mu(D^\mu - v^\mu v\cdot D) = \not{D}$. The longitudinal piece commutes with the projectors because $\bar{h}_v\not{v} = \bar{h}_v$; the transverse piece *anticommutes* with $\not{v}$,

$$\{\not{v}, \not{D}_\perp\} = 2\,v\cdot D_\perp = 0 \quad\Longrightarrow\quad \not{v}\,\not{D}_\perp = -\not{D}_\perp\,\not{v}, \tag{40.12}$$

so $\not{D}_\perp$ maps the $+$ eigenspace of $\not{v}$ into the $-$ eigenspace and vice versa: $\not{D}_\perp h_v$ has $\not{v}$-eigenvalue -1, hence $P_-\not{D}_\perp h_v = \not{D}_\perp h_v$, and likewise $\not{D}_\perp H_v$ lives in the $+$ subspace. With these two facts the four terms of Eq. (40.10) evaluate by inspection. In the first, $\bar{h}_v\, i\not{v}(v\cdot D)h_v = \bar{h}_v\, i(v\cdot D)h_v$, while $\bar{h}_v\, i\not{D}_\perp h_v = 0$ because $\bar{h}_v \in P_+$ meets $\not{D}_\perp h_v \in P_-$ and $P_+P_- = 0$. The fourth term, by the same logic with $\bar{H}_v\not{v} = -\bar{H}_v$, gives $-\bar{H}_v(i\,v\cdot D + 2m_Q)H_v$. The two cross terms keep only their transverse pieces: $\bar{h}_v(i\not{D} - 2m_Q)H_v = \bar{h}_v\, i\not{D}_\perp H_v$ (the longitudinal and mass pieces vanish since $\bar{h}_v H_v \propto \bar{h}_v P_+ P_- H_v = 0$), and $\bar{H}_v\, i\not{D}\, h_v = \bar{H}_v\, i\not{D}_\perp h_v$. Collecting,

$$\boxed{\mathcal{L}_Q = \bar{h}_v\,(i\,v\cdot D)\,h_v \;-\; \bar{H}_v\,(i\,v\cdot D + 2m_Q)\,H_v \;+\; \bar{h}_v\, i\not{D}_\perp H_v \;+\; \bar{H}_v\, i\not{D}_\perp h_v.} \tag{40.13}$$

Equation (40.13) is *exact*—no approximation has been made, only a change of field variables and a sorting by projector. Its structure is exactly the one Chapter 38 taught us to integrate out: h_v is massless (its kinetic term $\bar{h}_v\, i\,v\cdot D\, h_v$ carries no mass scale), while H_v has a Dirac mass $2m_Q$ and so describes excitations far above the soft scale. The light field h_v couples to the heavy field H_v only through the transverse derivative.

40.1.4 Integrating out the small components

Because H_v appears at most quadratically in Eq. (40.13) and with no kinetic-energy-generating dynamics of its own at the soft scale, it is, at tree level, a non-propagating field that can be eliminated by its classical equation of motion—the tree-level matching of Section 38.3, where "integrating out" a field means solving its equation of motion and substituting back. Varying Eq. (40.13) with respect to $\bar{H}_v$,

$$\frac{\partial \mathcal{L}_Q}{\partial \bar{H}_v} = -\left(i\,v\cdot D + 2m_Q\right) H_v + i\not{D}_\perp h_v \;\stackrel{!}{=}\; 0, \tag{40.14}$$

which solves to

$$H_v = \frac{1}{i\,v\cdot D + 2m_Q}\; i\not{D}_\perp\, h_v. \tag{40.15}$$

The operator $(i\,v\cdot D + 2m_Q)^{-1}$ is well defined as a power series because $v\cdot D$ acts on soft fields and produces $\mathcal{O}(\Lambda_{\rm QCD}) \ll 2m_Q$. Equation Eq. (40.15) confirms the physical picture: the small component is not an independent degree of freedom but a quantum response of size $H_v \sim (\Lambda_{\rm QCD}/m_Q) h_v = \lambda\, h_v$, generated whenever a soft gluon kicks the heavy quark slightly off the velocity v.

Substituting Eq. (40.15) into Eq. (40.13) is now mechanical. The middle term becomes $-\bar{H}_v (i\,v\cdot D + 2m_Q) H_v = -\bar{H}_v\, i\not{D}_\perp h_v$ on use of Eq. (40.14), which exactly cancels the last term $+\bar{H}_v\, i\not{D}_\perp h_v$; the surviving cross term $\bar{h}_v\, i\not{D}_\perp H_v$ then carries the entire $1/m_Q$ tower. The result is the exact tree-level HQET Lagrangian:

$$\mathcal{L}_{\rm HQET} = \bar{h}_v\,(i\,v\cdot D)\, h_v \;+\; \bar{h}_v\,(i\not{D}_\perp)\,\frac{1}{i\,v\cdot D + 2m_Q}\,(i\not{D}_\perp)\, h_v. \tag{40.16}$$

The second term is nonlocal as written, but only on the scale $1/m_Q$; expanding the denominator in the small ratio $v\cdot D/2m_Q$ resolves it into a tower of local operators of increasing dimension. With

$$\frac{1}{i\,v\cdot D + 2m_Q} = \frac{1}{2m_Q}\sum_{n=0}^{\infty}\left(-\frac{i\,v\cdot D}{2m_Q}\right)^n = \frac{1}{2m_Q}\left[1 - \frac{i\,v\cdot D}{2m_Q} + \left(\frac{i\,v\cdot D}{2m_Q}\right)^2 - \cdots\right], \tag{40.17}$$

the effective Lagrangian organizes as a series in $1/m_Q$, i.e. in the power-counting parameter λ:

$$\mathcal{L}_{\rm HQET} = \bar{h}_v\,(i\,v{\cdot}D)\,h_v + \frac{1}{2m_Q}\,\bar{h}_v\,(i\not{D}_\perp)(i\not{D}_\perp)\,h_v - \frac{1}{4m_Q^2}\,\bar{h}_v\,(i\not{D}_\perp)(i\,v{\cdot}D)(i\not{D}_\perp)\,h_v + \mathcal{O}\!\left(\frac{1}{m_Q^3}\right). \tag{40.18}$$

40.1.5 The leading Lagrangian and its $1/m_Q$ corrections

The leading term of Eq. (40.18) is the centerpiece of the chapter.

Proposition 40.2 (Leading HQET Lagrangian and its corrections)**.** *At lowest order in $1/m_Q$ the heavy quark is described by the* static Lagrangian

$$\mathcal{L}^{(0)}_{\rm HQET} = \bar{h}_v\,(i\,v\cdot D)\,h_v, \qquad v\cdot D = v\cdot\partial - ig\,v\cdot A^a\,T^a, \tag{40.19}$$

which contains no reference whatsoever to the heavy-quark mass or to its spin. The first correction is

$$\mathcal{L}^{(1)}_{\rm HQET} = \frac{1}{2m_Q}\,\bar{h}_v\,(iD_\perp)^2\,h_v + \frac{g}{4m_Q}\,c_{\rm mag}(\mu)\,\bar{h}_v\,\sigma_{\mu\nu}G^{\mu\nu}\,h_v, \tag{40.20}$$

with $G^{\mu\nu} = G^{a\,\mu\nu}T^a$ the gluon field strength of Eq. (25.76), $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]$, and $c_{\rm mag}(\mu) = 1$ at tree level.

The static Lagrangian Eq. (40.19) is the destination of this section, and its two silences—no m_Q, no Dirac structure beyond the projector built into h_v—are the seeds of the emergent symmetry the next section exploits. Before exploiting it we must verify the first correction, which is where the spin and the mass reappear and which fixes the meaning of $c_{\rm mag}$.

Worked Example 40.3 (The $1/m_Q$ operators: kinetic energy and the chromomagnetic moment)**.** We evaluate the dimension-five term of Eq. (40.18), $\frac{1}{2m_Q}\bar{h}_v(i\not{D}_\perp)(i\not{D}_\perp)h_v$, and show it splits into the operators displayed in Eq. (40.20).

Step 1 — square the transverse slash. Write $(i\not{D}_\perp)^2 = -\not{D}_\perp\not{D}_\perp = -\gamma_\mu\gamma_\nu D_\perp^\mu D_\perp^\nu$. Decompose the gamma product into symmetric and antisymmetric parts using $\gamma_\mu\gamma_\nu = \eta_{\mu\nu} - i\sigma_{\mu\nu}$ (the Clifford relation $\{\gamma_\mu,\gamma_\nu\} = 2\eta_{\mu\nu}$ together with the definition of $\sigma_{\mu\nu}$, Section C.8.6):

$$(i\not{D}_\perp)^2 = -(\eta_{\mu\nu} - i\sigma_{\mu\nu})D_\perp^\mu D_\perp^\nu = -D_\perp^2 + i\sigma_{\mu\nu}D_\perp^\mu D_\perp^\nu. \tag{40.21}$$

The symmetric piece $-D_\perp^2 = (iD_\perp)^2$ is the kinetic-energy operator: it measures the residual three-momentum of the heavy quark transverse to v (in the rest frame $D_\perp^\mu = (0,\mathbf{D})$, so $(iD_\perp)^2 = -\mathbf{D}^2$, the covariant Laplacian).

Step 2 — antisymmetrize the second piece. Because $\sigma_{\mu\nu}$ is antisymmetric, only the antisymmetric part of $D_\perp^\mu D_\perp^\nu$ survives:

$$i\sigma_{\mu\nu} D_\perp^\mu D_\perp^\nu = \frac{i}{2}\,\sigma_{\mu\nu}\,[D_\perp^\mu, D_\perp^\nu]. \tag{40.22}$$

The commutator of covariant derivatives is the field strength. From $D_\mu = \partial_\mu - igA_\mu^a T^a$,

$$[D^\mu, D^\nu] = -ig\,G^{\mu\nu}, \qquad G^{\mu\nu} = G^{a\,\mu\nu}T^a = \partial^\mu A^\nu - \partial^\nu A^\mu - ig[A^\mu, A^\nu], \tag{40.23}$$

which is the curvature of Yang–Mills theory written with the conventions of Eq. (25.76). The transverse commutator differs from this by terms carrying an explicit factor of v^μ or v^ν: substituting $D_\perp^\mu = D^\mu - v^\mu(v\cdot D)$,

$$[D_\perp^\mu, D_\perp^\nu] = [D^\mu, D^\nu] - v^\nu v_\alpha [D^\mu, D^\alpha] - v^\mu v_\beta [D^\beta, D^\nu] = -ig\big(G^{\mu\nu} - v^\nu v_\alpha G^{\mu\alpha} - v^\mu v_\beta G^{\beta\nu}\big). \tag{40.24}$$

Step 3 — the velocity-index terms vanish between h_v fields. The two terms in Eq. (40.24) carrying an explicit velocity index die when contracted with $\sigma_{\mu\nu}$ and sandwiched between $\bar h_v$ and h_v. It suffices to show $P_+\,\sigma^{\mu\nu}v_\nu\,P_+ = 0$. Using $\sigma^{\mu\nu}v_\nu = \frac{i}{2}(\gamma^\mu \not v - \not v \gamma^\mu)$ and the sandwiching rules $\not v P_+ = P_+$, $P_+ \not v = P_+$,

$$P_+\,\sigma^{\mu\nu}v_\nu\,P_+ = \frac{i}{2}\big(P_+\gamma^\mu \not v\, P_+ - P_+ \not v\,\gamma^\mu P_+\big) = \frac{i}{2}\big(P_+\gamma^\mu P_+ - P_+\gamma^\mu P_+\big) = 0. \tag{40.25}$$

Physically: the would-be "chromoelectric" coupling σ^{0i} of a heavy quark in its own rest frame vanishes, leaving only the chromomagnetic σ^{ij}. Hence between h_v fields $\sigma_{\mu\nu}[D_\perp^\mu, D_\perp^\nu] \to \sigma_{\mu\nu}[D^\mu, D^\nu] = -ig\,\sigma_{\mu\nu}G^{\mu\nu}$.

Step 4 — assemble. Putting Eqs. (40.21), (40.22) and (40.25) together,

$$\frac{1}{2m_Q}\,\bar h_v (i\not D_\perp)^2 h_v = \frac{1}{2m_Q}\,\bar h_v (iD_\perp)^2 h_v + \frac{1}{2m_Q}\,\bar h_v\,\frac{i}{2}\sigma_{\mu\nu}(-ig\,G^{\mu\nu})\,h_v, \tag{40.26}$$

and the second term is $\frac{g}{4m_Q}\,\bar h_v\,\sigma_{\mu\nu}G^{\mu\nu}h_v$, reproducing Eq. (40.20) with the tree-level coefficient $c_{\rm mag} = 1$.

The two operators in Eq. (40.20) have a transparent physical reading and play distinct roles. The kinetic operator $\frac{1}{2m_Q}\bar h_v (iD_\perp)^2 h_v$ is the leading relativistic correction to the heavy quark's energy, $\mathbf{p}^2/2m_Q$ in the rest frame; it depends on m_Q but not on the Dirac matrices, so it breaks the flavour blindness of Eq. (40.19) (a charm and a bottom quark have different kinetic energies) while respecting its spin blindness. The chromomagnetic operator $\frac{g}{4m_Q}c_{\rm mag}\bar h_v \sigma_{\mu\nu}G^{\mu\nu}h_v$ is the colour analogue of the Pauli term $\frac{e}{2m}\bar\psi\sigma_{\mu\nu}F^{\mu\nu}\psi$; through $\sigma_{\mu\nu}$ it couples the heavy-quark *spin* to the soft gluon field, and so breaks the spin blindness of the leading theory. Both effects are suppressed by λ relative to Eq. (40.19), which is precisely why the leading theory possesses a symmetry the full theory lacks—developed in Section 40.1's successor.

Physical Insight

The coefficient of the kinetic operator is locked to unity to all orders in α_s; that of the chromomagnetic operator is not. The reason is a hidden redundancy: the split Eq. (40.2) into $m_Q v$ and k is ambiguous, since one may shift $v \to v + \delta v$ and $k \to k - m_Q\,\delta v$ for any infinitesimal δv with $v \cdot \delta v = 0$ without changing the physical momentum p. Demanding that physics be invariant under this *reparametrization* relates operators at different orders in $1/m_Q$ and forces the kinetic-term coefficient to remain 1. No such constraint protects $c_{\rm mag}$, which therefore acquires a nontrivial Wilson coefficient (Theorem 38.4) under matching—the first genuinely loop-level coefficient of this chapter, equal to 1 at tree level but receiving an $\mathcal{O}(\alpha_s)$ correction when one matches the EFT chromomagnetic vertex onto the one-loop QCD amplitude at $\mu \sim m_Q$, and thereafter running to the hadronic scale by its anomalous dimension. We carry out such matching and running explicitly in the NRQED computation later in this chapter, using the procedure of Section 38.3.

40.1.6 Feynman rules: the static propagator and the eikonal vertex

The leading Lagrangian Eq. (40.19) is so simple that its Feynman rules can be read off in closed form. Split it into free and interacting pieces with $v \cdot D = v \cdot \partial - ig\, v \cdot A^a T^a$:

$$\mathcal{L}^{(0)}_{\rm HQET} = \underbrace{\bar h_v\,(i\,v\cdot\partial)\,h_v}_{\text{free}} + \underbrace{g\,v^\mu A^a_\mu\,\bar h_v\,T^a\,h_v}_{\text{interaction}}. \tag{40.27}$$

Worked Example 40.4 (Deriving the static propagator and the heavy-quark–gluon vertex)**.** *The propagator.* The free part of Eq. (40.27) is first order in derivatives and lives entirely in the P_+ subspace, since $h_v = P_+ h_v$. Its quadratic kernel in momentum space is obtained by Fourier transforming $h_v(x) = \int \frac{d^4k}{(2\pi)^4}\, e^{-ik\cdot x}\, h_v(k)$, where k is the residual momentum of Eq. (40.2); then $i\,v\cdot\partial \to v\cdot k$, and the free action is $\int \frac{d^4k}{(2\pi)^4}\,\bar h_v(k)\,(v\cdot k)\,h_v(k)$. The propagator is i times the inverse of this kernel, taken *within* the constrained subspace—so the projector P_+ must appear to map the propagator into the space on which h_v is defined:

$$\tilde S_v(k) = \frac{i\,P_+}{v\cdot k + i\epsilon} = \frac{i}{v\cdot k + i\epsilon}\,\frac{1+\not v}{2}. \tag{40.28}$$

The $+i\epsilon$, rather than the Feynman $\not p$-pole prescription of a relativistic fermion, is fixed by causality: a heavy quark has no antiparticle in HQET (the antiparticle component H_v was integrated out), so it can only propagate *forward* in time. To

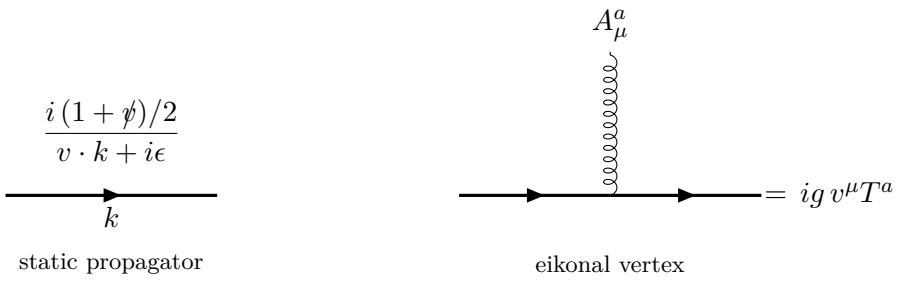


Figure 40.1: The leading-order Feynman rules of HQET read off from Eq. (40.19). The heavy-quark line (drawn thick, by convention a doubled line) carries the static propagator Eq. (40.28), which depends on the residual momentum k only through $v\cdot k$ and is purely forward-propagating. The heavy-quark–gluon vertex Eq. (40.30) is spin-independent and momentum-independent: a classical colour current $g\,v^\mu T^a$ along a worldline of fixed velocity v.

see this, evaluate Eq. (40.28) in the rest frame $v=(1,\mathbf{0})$, where $v\cdot k = k^0 \equiv \omega$ and $P_+ = \frac{1}{2}(1+\gamma^0)$. The mixed time/momentum-space propagator is

$$S_v(t,\mathbf{x}) = \int \frac{d\omega\, d^3k}{(2\pi)^4}\, e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}\, \frac{i\,P_+}{\omega+i\epsilon} = P_+\,\delta^3(\mathbf{x}) \int \frac{d\omega}{2\pi}\, \frac{i\,e^{-i\omega t}}{\omega+i\epsilon} = P_+\,\delta^3(\mathbf{x})\,\theta(t), \tag{40.29}$$

the last integral being $\int \frac{d\omega}{2\pi}\, i\,e^{-i\omega t}/(\omega+i\epsilon) = \theta(t)$ by closing the contour below for $t>0$ (the pole sits at $\omega=-i\epsilon$). The interpretation is exactly the static picture: the heavy quark sits at a fixed spatial point, $\delta^3(\mathbf{x})$, and propagates only to the future, $\theta(t)$. It is a worldline, not a wave.

The vertex. The interaction term of Eq. (40.27) is $g\,v^\mu A^a_\mu\,\bar h_v T^a h_v$. The Feynman rule is i times the operator coefficient of the fields $\bar h_v$, h_v, A^a_μ in $i\mathcal{L}_{\rm int}$, namely

$$(h_v\text{–}h_v\text{–gluon vertex}) = i\,g\,v^\mu\,T^a. \tag{40.30}$$

This is the *eikonal* vertex. It is remarkable for what it omits: there is no γ-matrix, so the vertex does not act on the heavy quark's spin, and there is no dependence on the gluon momentum, so the heavy quark recoils negligibly. The heavy quark behaves as a classical colour current $g\,v^\mu T^a$ flowing along a fixed straight worldline of four-velocity v—the Wilson line of Chapter 25 made dynamical.

Figure 40.1 collects the two rules. Their austerity is the source of HQET's predictive power: every heavy-quark line in any diagram contributes only the scalar denominator $v\cdot k$ and the colour-current vertex $gv^\mu T^a$, so the heavy quark's spin decouples entirely from the gluon dynamics at leading order. This is the diagrammatic face of the emergent symmetry.

40.1.7 Power counting

It remains to state the bookkeeping that makes Eq. (40.18) an organized expansion rather than a formal rearrangement, following the power-counting discipline of Section 38.2. Every soft quantity is measured in units of $\Lambda_{\rm QCD}$: the residual momentum $k \sim \Lambda_{\rm QCD}$, the soft gluon field and its covariant derivative $D_\perp \sim v \cdot D \sim \Lambda_{\rm QCD}$, and the field strength $G^{\mu\nu} \sim \Lambda_{\rm QCD}^2$. The single large scale m_Q enters only through the explicit prefactors generated by Eq. (40.17). An operator of mass dimension $4+n$ in Eq. (40.18) therefore carries n inverse powers of m_Q compensated by n powers of the soft scale, so its contribution to any matrix element is suppressed by

$$\left(\frac{\Lambda_{\rm QCD}}{m_Q}\right)^n = \lambda^n \tag{40.31}$$

relative to the leading term Eq. (40.19). For the bottom quark $\lambda \approx 0.05$ and for the charm $\lambda \approx 0.15$, so the leading and first-correction Lagrangians Eqs. (40.19) and (40.20) already capture B-physics to a few percent and D-physics to $\sim 15\%$. The heavy-quark field itself scales as $h_v \sim \Lambda_{\rm QCD}^{3/2}$ (so that $\int d^4x\, \bar h_v\, i\, v\cdot\partial\, h_v$ is dimensionless with $d^4x \sim \Lambda_{\rm QCD}^{-4}$ and $v\cdot\partial \sim \Lambda_{\rm QCD}$), the same counting that makes Eq. (40.19) the unique marginal heavy-quark operator. As always in a top-down EFT, the coefficients of the operators in the tower are not free: each is a Wilson coefficient (Theorem 38.4) fixed by matching the EFT amplitudes onto full QCD at the scale $\mu \sim m_Q$, where both descriptions are valid, and then evolved to the low scale $\mu \sim \Lambda_{\rm QCD}$ by the renormalization group with the running coupling $\alpha_s(\mu)$ of Eq. (27.155). At tree level the leading Lagrangian Eq. (40.19) has unit coefficient and no anomalous dimension; the corrections Eq. (40.20) are where matching and running first do nontrivial work, the kinetic coefficient protected at unity and $c_{\rm mag}(\mu)$ running—a computation we set up explicitly when we reach the matching procedure.

We have, from QCD alone and with every step shown, distilled a heavy quark down to the static Lagrangian $\mathcal{L}^{(0)}_{\rm HQET} = \bar h_v\,(i\,v\cdot D)\,h_v$ of Eq. (40.19), with the forward-propagating static propagator Eq. (40.28) and the spin- and momentum-blind eikonal vertex Eq. (40.30) of Fig. 40.1. The leading theory has manifestly forgotten both the mass and the spin of the heavy quark: m_Q survives only as a label and a tower of suppressed corrections, and the Dirac matrices have collapsed into the single projector P_+. That double amnesia is not a loss of information but the appearance of a new symmetry—a heavy-quark spin–flavour symmetry invisible in full QCD—whose group structure and phenomenological payoff, the degeneracy of heavy-meson multiplets and the reduction of the $B \to D^{(*)}$ form factors to a single Isgur–Wise function, we develop next.

40.2 Spin–flavor symmetry and heavy-hadron form factors

The previous section delivered a Lagrangian of remarkable austerity. After the large, kinematically fixed part of the heavy quark's momentum was scaled away and the small spinor components integrated out, what survived at leading order was

$$\mathcal{L}^{(0)}_{\rm HQET} \;=\; \bar{h}_v \, (i\, v\!\cdot\! D)\, h_v, \qquad \not{v}\, h_v = h_v, \tag{40.32}$$

the static field h_v obeying the velocity projection $\not{v} h_v = h_v$ (Section 40.1). Look at what is *absent*. There is no mass: the parameter m_Q has been pushed entirely into the phase $e^{-i m_Q v\cdot x}$ stripped off in defining h_v, so the residual dynamics knows nothing of how heavy the quark actually is. There is no Dirac matrix tying the field to its spin: the only γ-structure left is the projector $\frac{1}{2}(1+\not{v})$, and the kinetic operator $i\, v\!\cdot\! D$ is a Lorentz scalar that acts as the identity on the spinor index. The covariant derivative couples the heavy quark to gluons through its color charge alone, with no reference to its flavor or its spin. A Lagrangian that does not see the heavy quark's mass, spin, or flavor identity must possess symmetries that rotate among those labels — symmetries that are completely invisible in the parent QCD Lagrangian, where the explicit mass term $-m_Q \bar{Q} Q$ and the relativistic kinetic term $\bar{Q}\, i\not{\partial}\, Q$ break them. This section extracts those emergent symmetries, works out their immediate spectroscopic consequences, and then turns them loose on the central application: the semileptonic transitions $B \to D^{(*)} \ell \bar{\nu}$, whose four-form-factor description collapses, at leading order in $1/m_Q$, onto a single universal function normalized to unity at zero recoil. That normalization is what makes these decays the cleanest window onto the Cabibbo–Kobayashi–Maskawa element $|V_{cb}|$.

40.2.1 The emergent spin–flavor symmetry

Suppose the theory contains N_h heavy flavors — in practice the charm and bottom quarks, $N_h = 2$, but the argument is general — all much heavier than $\Lambda_{\rm QCD}$ and all moving with a *common* four-velocity v. (The restriction to a single velocity is essential and we return to it below: HQET is organized into superselection sectors labeled by v.) Each flavor contributes its own static field $h_v^{(i)}$, $i = 1, \ldots, N_h$, and the leading Lagrangian is simply the sum,

$$\mathcal{L}^{(0)}_{\rm HQET} \;=\; \sum_{i=1}^{N_h} \bar{h}_v^{(i)} \, (i\, v\!\cdot\! D)\, h_v^{(i)}. \tag{40.33}$$

The single derivative is the *same* color-covariant operator for every flavor, because the gluon couples universally to color and is blind to flavor and to spin. To expose the symmetry it is cleanest to pass to the rest frame of v, $v^\mu = (1, \mathbf{0})$,

where $v\cdot D = D^0 = \partial^0 - igA^0_a T^a$ and the projector is $P_+ = \frac{1}{2}(1+\gamma^0)$. In the Dirac representation P_+ annihilates the lower two spinor components, so each $h_v^{(i)}$ reduces to a two-component field χ_i (per color), and using $\gamma^0 h_v = h_v$,

$$\bar h_v^{(i)}\,(i\,v\cdot D)\,h_v^{(i)} = h_v^{(i)\dagger}\gamma^0\,(iD^0)\,h_v^{(i)} = h_v^{(i)\dagger}\,(iD^0)\,h_v^{(i)} = \chi_i^\dagger\,(iD^0)\,\chi_i. \tag{40.34}$$

Collecting flavors,

$$\mathcal{L}^{(0)}_{\mathrm{HQET}} \;=\; \sum_{i=1}^{N_h} \chi_i^\dagger\,(iD^0)\,\chi_i, \tag{40.35}$$

in which the index running over the $2N_h$ objects $\chi_{i,s}$ ($i = 1,\dots,N_h$ a flavor label, $s = 1,2$ a spin label) appears only as a contracted, undifferentiated multiplet index. The operator iD^0 carries no flavor or spin structure whatsoever; it is a multiple of the identity on the $2N_h$-dimensional spin–flavor space (and an honest operator only on color and spacetime). Therefore an arbitrary constant unitary rotation

$$\chi_{i,s} \;\longrightarrow\; \sum_{j,t} U_{(is),(jt)}\,\chi_{j,t}, \qquad U \in \mathrm{U}(2N_h), \tag{40.36}$$

leaves Eq. (40.35) invariant: $\sum \chi^\dagger(iD^0)\chi \to \sum \chi^\dagger U^\dagger U (iD^0)\chi = \sum \chi^\dagger(iD^0)\chi$. We have proved the central structural fact.

Proposition 40.5 (Heavy-quark spin–flavor symmetry)**.** *At leading order in $1/m_Q$, the HQET Lagrangian Eq. (40.33) for N_h heavy flavors at a fixed velocity v is invariant under a $\mathrm{U}(2N_h)$ group acting on the $2N_h$-dimensional space spanned by the heavy-quark spin ($\dim 2$) and flavor ($\dim N_h$) labels. The overall $\mathrm{U}(1)$ is heavy-flavor number, a symmetry of full QCD as well; the genuinely emergent part is $\mathrm{SU}(2N_h)$. It contains in particular the heavy-flavor symmetry $\mathrm{U}(N_h)$ rotating the flavors and, for each flavor, the heavy-quark spin symmetry $\mathrm{SU}(2)_v$ rotating the two spin states.*

Proof. The invariance of Eq. (40.35) under Eq. (40.36) is the computation just displayed. It remains to identify the spin subgroup covariantly. In the rest frame the rotation generators acting on h_v are $\frac{1}{2}\Sigma^k$ with $\Sigma^k = \frac{i}{2}\epsilon^{kij}\sigma^{ij}$ the Dirac spin matrices; they commute with $P_+ = \frac{1}{2}(1+\gamma^0)$ and with D^0 (a rotational scalar), hence with the Lagrangian. Covariantly, define

$$S_v^{\mu\nu} \;=\; \tfrac{1}{2}\,P_+\,\sigma^{\mu\nu}\,P_+, \qquad P_+ = \tfrac{1}{2}(1+\not v), \tag{40.37}$$

which annihilate the components along v (since $P_+\sigma^{\mu\nu}v_\nu P_+ \propto P_+(\not v\gamma^\mu - v^\mu)P_+ = 0$ by $\not v P_+ = P_+$) and so generate the little group $\mathrm{SU}(2)_v$ of rotations transverse to v. These commute with $i\,v\cdot D$ because $S_v^{\mu\nu}$ is built only from $\not v$ and the γ's while $v\cdot D$ carries no spinor index. Together with the flavor rotations $\chi_i \to U_{ij}\chi_j$ they close on $\mathrm{U}(2N_h)$, and the algebra splits as $\mathrm{U}(2N_h) \supset \mathrm{U}(1)_{\mathrm{number}} \times \mathrm{SU}(2N_h) \supset \mathrm{U}(N_h)_{\mathrm{flavor}} \times \mathrm{SU}(2)_v^{\otimes}$ in the obvious way. $\square$

Physical Insight

The symmetry is invisible in QCD for the same reason it is exact in HQET: it is broken only by the heavy-quark mass and the relativistic kinetic operator, both of which have been demoted to higher orders in $1/m_Q$. Spin symmetry is the statement that a very heavy quark's chromomagnetic moment $\sim g/m_Q$ decouples — the gluon field cannot flip a spin it cannot couple to. Flavor symmetry is the statement that, once the kinematic mass phase is removed, an infinitely heavy b and an infinitely heavy c are dynamically identical static color sources. Both are the heavy-quark avatar of the decoupling theorem of Section 38.4.2: irrelevant, mass-suppressed interactions drop out of the leading effective description.

Caution

The symmetry holds *within a fixed-velocity sector*. The static field h_v for velocity v and the field $h_{v'}$ for a different v' are independent objects; there is no symmetry rotating one velocity into another, and indeed soft gluons cannot change a heavy quark's velocity at leading power (the residual momentum is $O(\Lambda_{\rm QCD}) \ll m_Q v$). Each v carries its own $\mathrm{U}(2N_h)$. When a current changes the heavy flavor and the two heavy quarks recoil against each other, as in $b \to c$ below, the initial and final states live in *different* velocity sectors, v and v', and the symmetry that survives is the product $\mathrm{U}(2N_h)_v \times \mathrm{U}(2N_h)_{v'}$.

40.2.2 Heavy-meson multiplets and the ground-state doublet

The cleanest spectroscopic consequence is the degeneracy of heavy-meson multiplets. A heavy meson $\bar{Q}q$ (we write $\bar{Q}$ because the physical $B^- = b\bar{u}$ contains a heavy *anti*quark, but the kinematics is identical) consists of a single heavy quark of spin $s_Q = \frac{1}{2}$ bound to a "brown-muck" light cloud — the light quark, the gluons, and the sea — carrying total angular momentum and parity j_ℓ^P. Because the heavy-quark spin $\mathbf{S}_Q$ decouples at leading order, it is separately conserved, and so is the light-cloud angular momentum $\mathbf{j}_\ell$. The physical states organize into degenerate multiplets of fixed j_ℓ, the heavy spin combining trivially:

$$J \;=\; j_\ell \otimes \tfrac{1}{2} \;=\; \Big\{ j_\ell - \tfrac{1}{2},\; j_\ell + \tfrac{1}{2} \Big\}. \tag{40.38}$$

For $j_\ell \neq 0$ this is a degenerate *doublet* of states with total spin differing by one unit, split only at order $1/m_Q$. The ground state has the light cloud in its lowest configuration, $j_\ell^P = \frac{1}{2}^-$, giving the doublet $J^P = (0^-, 1^-)$: the pseudoscalar and the vector. For the bottom system these are the B and B^*; for charm, the D

and D^*. Heavy-quark *spin* symmetry predicts, at leading order,

$$m_{B^*} = m_B, \qquad m_{D^*} = m_D \qquad \text{(leading order in } 1/m_Q\text{)}, \tag{40.39}$$

while heavy-quark *flavor* symmetry relates the B system to the D system, predicting equal binding energies and (as we will see) a single set of transition form factors. The measured splittings are small and scale as expected: $m_{B^*} - m_B \approx 45$ MeV and $m_{D^*} - m_D \approx 141$ MeV, and their ratio,

$$\frac{m_{B^*}^2 - m_B^2}{m_{D^*}^2 - m_D^2} \;\approx\; 1, \tag{40.40}$$

is the prediction of the chromomagnetic operator that we will derive in the next section: the splitting is $\propto 1/m_Q$ times a flavor-independent matrix element, so the product $(m_V^2 - m_P^2) \sim 2m_Q\,(m_V - m_P)$ is itself flavor-independent. That Eq. (40.40) comes out near unity (the measured value is ≈ 0.87) is the first quantitative success of the expansion and a preview of the $1/m_Q$ matching that the next section carries out.

40.2.3 The Isgur–Wise reduction of $B \to D^{(*)}$ form factors

The semileptonic decay $\bar{B} \to D^{(*)}\ell\bar{\nu}$ proceeds at the quark level through the charged-current transition $b \to c\,\ell\,\bar{\nu}$, mediated by a W whose virtuality is negligible on the hadronic scale, so the relevant operator is the local current

$$J^\mu \;=\; \bar{c}\,\gamma^\mu(1-\gamma_5)\,b, \tag{40.41}$$

weighted by V_{cb} (Eq. (32.59)). The hadronic physics is entirely contained in the matrix elements of the vector part $\bar{c}\gamma^\mu b$ and the axial part $\bar{c}\gamma^\mu\gamma_5 b$ between the $\bar{B}$ and the D or D^*. Fig. 40.2 shows the process. Lorentz invariance and parity alone require, in the conventional relativistic normalization $\langle M(p')|M(p)\rangle = 2E_p(2\pi)^3\delta^3(\mathbf{p}-\mathbf{p}')$,

$$\langle D(p')|\,\bar{c}\gamma^\mu b\,|\bar{B}(p)\rangle = f_+(q^2)(p+p')^\mu + f_-(q^2)(p-p')^\mu, \tag{40.42}$$

$$\langle D^*(p',\epsilon)|\,\bar{c}\gamma^\mu b\,|\bar{B}(p)\rangle = \frac{2iV(q^2)}{m_B+m_{D^*}}\,\varepsilon^{\mu\nu\alpha\beta}\epsilon^*_\nu p'_\alpha p_\beta, \tag{40.43}$$

$$\begin{aligned}\langle D^*(p',\epsilon)|\,\bar{c}\gamma^\mu\gamma_5 b\,|\bar{B}(p)\rangle &= 2m_{D^*}A_0\,\frac{\epsilon^*\!\cdot q}{q^2}\,q^\mu + (m_B+m_{D^*})A_1\Big(\epsilon^{*\mu} - \frac{\epsilon^*\!\cdot q}{q^2}q^\mu\Big)\\ &\quad - A_2\,\frac{\epsilon^*\!\cdot q}{m_B+m_{D^*}}\Big((p+p')^\mu - \frac{m_B^2-m_{D^*}^2}{q^2}q^\mu\Big),\end{aligned} \tag{40.44}$$

with $q = p - p'$ the momentum transfer and ϵ the D^* polarization ($p'\cdot\epsilon = 0$). This is six *a priori* independent functions of q^2: $f_\pm$ for $B \to D$ and V, A_0, A_1, A_2

for $B \to D^*$. (The naming "four form factors" in the older literature counts the kinematically leading combinations; the full count over both channels is six.) The content of heavy-quark symmetry is that all six are, at leading order, one function.

The trace formalism. The efficient way to impose spin–flavor symmetry on the matrix elements is to represent each heavy-meson doublet by a single bispinor "superfield" that packages the 0^- and 1^- states, and then to write the most general symmetric amplitude as a Dirac trace. For a heavy quark Q of velocity v bound into the ground-state doublet, define

$$H_v^{(Q)} \;=\; \frac{1+\not{v}}{2}\Big[\, \not{P}^* - \gamma_5\, P \,\Big], \qquad v\cdot P^* = 0, \tag{40.45}$$

where P is the annihilation amplitude of the pseudoscalar member and $P^{*\mu}$ that of the vector member (a polarization four-vector transverse to v). The left projector $\frac{1}{2}(1+\not{v})$ is the heavy-quark velocity projection; the two Dirac indices of H_v carry, respectively, the heavy-quark spin (left) and the light-cloud spin (right). Its Dirac conjugate is

$$\bar{H}_v^{(Q)} \;\equiv\; \gamma^0 H_v^{(Q)\dagger}\gamma^0 \;=\; \Big[\, \not{P}^{*\dagger} + \gamma_5\, P^\dagger \,\Big]\, \frac{1+\not{v}}{2}, \tag{40.46}$$

where we used $\gamma^0\gamma^{\mu\dagger}\gamma^0 = \gamma^\mu$, $\gamma^0\gamma_5\gamma^0 = -\gamma_5$, and $\gamma^0\not{v}^\dagger\gamma^0 = \not{v}$ (Section C.5). Under the heavy-quark spin symmetry $\mathrm{SU}(2)_v$ the heavy index rotates, $H_v \to S_v H_v$ with S_v commuting with $\not{v}$ and satisfying $\gamma^0 S_v^\dagger \gamma^0 = S_v^{-1}$, so that $\bar{H}_v \to \bar{H}_v S_v^{-1}$; the light index is a spectator under the heavy spin.

Now consider $\langle D^{(*)}(v')|\, \bar{c}\,\Gamma\, b\,|\bar{B}(v)\rangle$ for an arbitrary Dirac matrix Γ. The b field rotates under $\mathrm{SU}(2)_v$ (initial velocity), the $\bar{c}$ field under $\mathrm{SU}(2)_{v'}$ (final velocity), so the current transforms as $\bar{c}\,\Gamma\, b \to \bar{c}\, S_{v'}^{-1}\Gamma S_v\, b$. Heavy-quark spin symmetry acts only on the heavy indices; it forces every dependence on the heavy spins into a structure that contracts the b-index of Γ against H_v (both rotate by S_v) and the c-index against $\bar{H}_{v'}$ (both by $S_{v'}$). The light clouds of the two mesons are spectators in the *same* fixed $j_\ell^P = \frac{1}{2}^-$ configuration, so their overlap is a single reduced matrix element — a Lorentz scalar by the Wigner–Eckart theorem — that can depend only on the one available invariant

$$w \;\equiv\; v\cdot v'. \tag{40.47}$$

The unique structure consistent with all of this is a Dirac trace times that scalar:

$$\boxed{\frac{\langle D^{(*)}(v')|\, \bar{c}\,\Gamma\, b\,|\bar{B}(v)\rangle}{\sqrt{m_B\, m_{D^{(*)}}}} \;=\; -\,\xi(w)\;\mathrm{Tr}\Big[\, \bar{H}_{v'}^{(c)}\,\Gamma\, H_v^{(b)} \,\Big].} \tag{40.48}$$

Here $\xi(w)$ is the *Isgur–Wise function*; the factor $\sqrt{m_B m_{D^{(*)}}}$ converts the relativistic state normalization of Eqs. (40.42) to (40.44) to the mass-independent

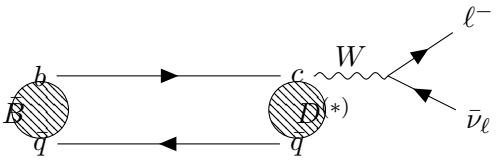


Figure 40.2: The semileptonic decay $\bar{B} \to D^{(*)}\ell\bar{\nu}$. The spectator light antiquark $\bar{q}$ is carried through unchanged; the heavy quark undergoes $b \to c$ via the local current Eq. (40.41), emitting a virtual W that produces the lepton pair. The hadronic matrix element of the $b \to c$ current is what heavy-quark symmetry reduces to the single function $\xi(w)$.

HQET normalization $\langle M(v')|M(v)\rangle = 2v^0(2\pi)^3\delta^3(\dots)$ in which H_v is defined; the overall sign is the standard convention. *One* function now controls all six form factors. The reduction is carried out explicitly in the following worked example.

Worked Example 40.6 (All $B \to D^{(*)}$ form factors from a single trace)**.** We evaluate Eq. (40.48) for the vector and axial currents and read off every form factor. For the external states we specialize the superfields of Eqs. (40.45) and (40.46): the initial pseudoscalar $\bar{B}$ sets $P^* = 0,\ P \to 1$, so

$$H_v^{(b)} \;=\; \frac{1+\not{v}}{2}\,(-\gamma_5); \tag{40.49}$$

the final D (pseudoscalar) sets $P^{*\dagger} = 0,\ P^\dagger \to 1$, and the final D^* (vector, polarization ϵ) sets $P^\dagger = 0,\ P^{*\dagger} \to \epsilon^*$,

$$\bar{H}^{(c)}_{v'}\big|_D = \gamma_5\,\frac{1+\not{v}'}{2}, \qquad \bar{H}^{(c)}_{v'}\big|_{D^*} = \not{\epsilon}^*\,\frac{1+\not{v}'}{2}. \tag{40.50}$$

Throughout we use the four-dimensional trace technology of Section C.6: $\mathrm{Tr}[\,\text{odd \# of }\gamma\,] = 0$, $\mathrm{Tr}[\gamma^\mu\gamma^\nu] = 4\eta^{\mu\nu}$, $\mathrm{Tr}[\gamma^\alpha\gamma^\beta\gamma^\rho\gamma^\sigma] = 4(\eta^{\alpha\beta}\eta^{\rho\sigma} - \eta^{\alpha\rho}\eta^{\beta\sigma} + \eta^{\alpha\sigma}\eta^{\beta\rho})$, and $\mathrm{Tr}[\gamma^\alpha\gamma^\beta\gamma^\rho\gamma^\sigma\gamma_5] = -4i\,\varepsilon^{\alpha\beta\rho\sigma}$ with $\varepsilon^{0123} = +1$.

(i) $B \to D$, vector current $\Gamma = \gamma^\mu$. From Eq. (40.48),

$$\frac{\langle D(v')|\bar{c}\gamma^\mu b|\bar{B}(v)\rangle}{\sqrt{m_B m_D}} = -\xi(w)\ \mathrm{Tr}\Big[\gamma_5\,\tfrac{1+\not{v}'}{2}\,\gamma^\mu\,\tfrac{1+\not{v}}{2}\,(-\gamma_5)\Big]. \tag{40.51}$$

Bring the two γ_5's together using cyclicity and $\gamma_5^2 = 1$: writing $A = \frac{1+\not{v}'}{2}\gamma^\mu\frac{1+\not{v}}{2}$, the bracket is $-\mathrm{Tr}[\gamma_5 A\gamma_5] = -\mathrm{Tr}[A]$. Hence the matrix element is $+\xi(w)\ \mathrm{Tr}[A]$. Expanding,

$$\mathrm{Tr}[A] = \tfrac{1}{4}\,\mathrm{Tr}\,[(1+\not{v}')\gamma^\mu(1+\not{v})] = \tfrac{1}{4}\big(\underbrace{\mathrm{Tr}[\gamma^\mu\not{v}]}_{4v^\mu} + \underbrace{\mathrm{Tr}[\not{v}'\gamma^\mu]}_{4v'^\mu}\big) = v^\mu + v'^\mu, \tag{40.52}$$

the $\mathrm{Tr}[\gamma^\mu]$ and three-γ terms vanishing. Therefore

$$\frac{\langle D(v')|\bar{c}\gamma^\mu b|\bar{B}(v)\rangle}{\sqrt{m_B m_D}} = \xi(w)\,(v+v')^\mu. \tag{40.53}$$

In the HQET basis $h_+(v+v')^\mu + h_-(v-v')^\mu$ this reads $h_+(w)=\xi(w)$, $h_-(w)=0$.

(ii) $B\to D^*$, vector current $\Gamma=\gamma^\mu$. Now

$$\frac{\langle D^*(v')|\bar{c}\gamma^\mu b|\bar{B}(v)\rangle}{\sqrt{m_B m_{D^*}}} = -\xi(w)\;\mathrm{Tr}\Big[\not{\epsilon}^*\,\tfrac{1+\not{v}'}{2}\,\gamma^\mu\,\tfrac{1+\not{v}}{2}\,(-\gamma_5)\Big] = \xi(w)\,\tfrac{1}{4}\,\mathrm{Tr}\big[\not{\epsilon}^*(1+\not{v}')\gamma^\mu(1+\not{v})\gamma_5\big]. \tag{40.54}$$

A γ_5 trace needs four other γ's, so only the term with both $\not{v}'$ and $\not{v}$ survives:

$$\begin{aligned}\xi\,\tfrac{1}{4}\,\mathrm{Tr}\big[\not{\epsilon}^*\not{v}'\gamma^\mu\not{v}\,\gamma_5\big] &= \xi\,\tfrac{1}{4}\,\epsilon^*_\alpha v'_\beta v_\delta\;\mathrm{Tr}\big[\gamma^\alpha\gamma^\beta\gamma^\mu\gamma^\delta\gamma_5\big] = \xi\,\tfrac{1}{4}\,\epsilon^*_\alpha v'_\beta v_\delta\,(-4i)\,\varepsilon^{\alpha\beta\mu\delta}\\ &= -i\,\xi(w)\,\varepsilon^{\mu\nu\alpha\beta}\,\epsilon^*_\nu v'_\alpha v_\beta,\end{aligned} \tag{40.55}$$

where in the last step we reordered indices of the totally antisymmetric symbol ($\varepsilon^{\alpha\beta\mu\delta}=\varepsilon^{\mu\delta\alpha\beta}$, an even permutation, then relabelled). In the HQET basis $\langle D^*|\bar{c}\gamma^\mu b|\bar{B}\rangle/\sqrt{m_B m_{D^*}} = i\,h_V\,\varepsilon^{\mu\nu\alpha\beta}\epsilon^*_\nu v'_\alpha v_\beta$ this gives $h_V(w)=\xi(w)$ (with the sign convention $\varepsilon^{0123}=+1$ absorbed into the basis definition).

(iii) $B\to D^*$, axial current $\Gamma=\gamma^\mu\gamma_5$. Here

$$\frac{\langle D^*(v')|\bar{c}\gamma^\mu\gamma_5 b|\bar{B}(v)\rangle}{\sqrt{m_B m_{D^*}}} = -\xi(w)\;\mathrm{Tr}\Big[\not{\epsilon}^*\,\tfrac{1+\not{v}'}{2}\,\gamma^\mu\gamma_5\,\tfrac{1+\not{v}}{2}\,(-\gamma_5)\Big]. \tag{40.56}$$

Simplify the right end with $\gamma_5\,\frac{1+\not{v}}{2}\,\gamma_5 = \frac{1-\not{v}}{2}$ (since $\gamma_5\not{v}\gamma_5=-\not{v}$):

$$\gamma^\mu\gamma_5\,\tfrac{1+\not{v}}{2}\,(-\gamma_5) = \gamma^\mu\big(-\gamma_5\,\tfrac{1+\not{v}}{2}\gamma_5\big) = -\,\gamma^\mu\,\tfrac{1-\not{v}}{2}. \tag{40.57}$$

Thus the matrix element is $+\xi(w)\;\mathrm{Tr}\big[\not{\epsilon}^*\,\frac{1+\not{v}'}{2}\,\gamma^\mu\,\frac{1-\not{v}}{2}\big]$. Expanding (no γ_5, so only even-γ terms),

$$\tfrac{1}{4}\,\mathrm{Tr}\big[\not{\epsilon}^*(1+\not{v}')\gamma^\mu(1-\not{v})\big] = \tfrac{1}{4}\Big(\underbrace{\mathrm{Tr}[\not{\epsilon}^*\gamma^\mu]}_{4\epsilon^{*\mu}} - \mathrm{Tr}[\not{\epsilon}^*\not{v}'\gamma^\mu\not{v}]\Big). \tag{40.58}$$

The four-γ trace is

$$\mathrm{Tr}[\not{\epsilon}^*\not{v}'\gamma^\mu\not{v}] = 4\big[(\epsilon^*\!\cdot v')\,v^\mu - w\,\epsilon^{*\mu} + (\epsilon^*\!\cdot v)\,v'^\mu\big] = 4\big[-w\,\epsilon^{*\mu} + (\epsilon^*\!\cdot v)\,v'^\mu\big], \tag{40.59}$$

using $w = v\cdot v'$ and the kinematic transversality $\epsilon^*\cdot v' = 0$. Hence

$$\frac{\langle D^*(v')|\bar{c}\gamma^\mu\gamma_5 b|\bar{B}(v)\rangle}{\sqrt{m_B m_{D^*}}} = \xi(w)\,\big[(w+1)\,\epsilon^{*\mu} - (\epsilon^*\!\cdot v)\,v'^\mu\big]. \tag{40.60}$$

In the HQET basis $h_{A_1}(w+1)\epsilon^{*\mu} - (h_{A_2}v^\mu + h_{A_3}v'^\mu)(\epsilon^*\cdot v)$ we read off

$$h_{A_1}(w) = \xi(w), \qquad h_{A_2}(w) = 0, \qquad h_{A_3}(w) = \xi(w). \tag{40.61}$$

Summary. All six form factors of Eqs. (40.42) to (40.44) reduce, in the HQET basis $\{h_+, h_-, h_V, h_{A_1}, h_{A_2}, h_{A_3}\}$, to

$$h_+ = h_V = h_{A_1} = h_{A_3} = \xi(w), \qquad h_- = h_{A_2} = 0. \tag{40.62}$$

Six functions of the momentum transfer have collapsed onto one universal function of w. Translating back to Eqs. (40.42) to (40.44) via the kinematic relations $w = (m_B^2 + m_{D^{(*)}}^2 - q^2)/(2m_B m_{D^{(*)}})$, this yields explicit leading-order expressions for $f_\pm$, V, A_0, A_1, A_2 in terms of $\xi(w)$ and meson masses — for instance $f_+(q^2) = \frac{m_B+m_D}{2\sqrt{m_B m_D}}\,\xi(w)$ and $V = A_0 = A_2 = \frac{m_B+m_{D^*}}{2\sqrt{m_B m_{D^*}}}\,\xi(w)$ with $A_1 = \frac{2\sqrt{m_B m_{D^*}}}{m_B+m_{D^*}}\,\frac{w+1}{2}\,\xi(w)$ — all numerically tied together.

40.2.4 Normalization at zero recoil

The trace formalism has reduced the form factors to a single function, but it has not fixed its overall size. Heavy-quark symmetry does that too, and the argument is a current-conservation argument exactly parallel to the statement that the electric form factor of any particle equals its charge at zero momentum transfer.

Theorem 40.7 (Isgur–Wise normalization)**.** *At zero recoil $w = 1$ (equivalently $v' = v$, $q^2 = q^2_{\max} = (m_B - m_{D^{(*)}})^2$),*

$$\xi(1) = 1. \tag{40.63}$$

Proof. Set $v' = v$. Then both heavy quarks live in the *same* velocity sector, and in the limit $m_b, m_c \to \infty$ they are two flavors of a single static field governed by the flavor-symmetric Lagrangian Eq. (40.33) with $N_h \supseteq \{b, c\}$. The off-diagonal flavor rotation $h_v^{(b)} \to h_v^{(b)} + \theta\, h_v^{(c)}$, $h_v^{(c)} \to h_v^{(c)} - \theta\, h_v^{(b)}$ is an exact symmetry of that Lagrangian. Its Noether current, computed from Eq. (40.33) (whose only derivative is $v\cdot\partial$, so the current inherits a factor v^μ), is

$$j^\mu_{cb} \;=\; v^\mu\, \bar h_v^{(c)}\, h_v^{(b)}, \qquad \partial_\mu j^\mu_{cb} = 0, \tag{40.64}$$

and the associated conserved charge $Q_{cb} = \int d^3x\, j^0_{cb} = v^0 \int d^3x\, \bar h_v^{(c)} h_v^{(b)}$ generates the $b \to c$ flavor rotation. Because the symmetry is unbroken and the $\bar B(v)$ and $D(v)$ sit in the same flavor multiplet (same light cloud, same velocity, same spin), the charge maps one onto the other with unit amplitude,

$$Q_{cb}\,|\bar B(v)\rangle = |D(v)\rangle, \qquad \Longrightarrow \qquad \langle D(v)|\,Q_{cb}\,|\bar B(v)\rangle = \langle D(v)|D(v)\rangle, \tag{40.65}$$

the right-hand side being the state norm. This is the heavy-quark analog of charge counting: a conserved-charge matrix element between members of one multiplet just counts the states, with no dynamical renormalization.

It remains to connect Q_{cb} to the QCD current $\bar{c}\gamma^\mu b$. Matching the full-theory field to the static field, $b \to h_v^{(b)}$ (and likewise c) at leading power, the vector current at equal velocity reduces using the projector identity $P_+\gamma^\mu P_+ = v^\mu P_+$ (proof: $\gamma^\mu = v^\mu \not{v} + \gamma_\perp^\mu$ with $\gamma_\perp^\mu = \gamma^\mu - v^\mu \not{v}$ anticommuting with $\not{v}$, so $P_+\gamma_\perp^\mu P_+ = P_+ P_- \gamma_\perp^\mu = 0$ and $P_+ v^\mu \not{v} P_+ = v^\mu P_+$):

$$\bar{c}\,\gamma^\mu b \;\xrightarrow{v'=v}\; \bar{h}_v^{(c)} \gamma^\mu h_v^{(b)} = v^\mu\, \bar{h}_v^{(c)} h_v^{(b)} = j_{cb}^\mu . \tag{40.66}$$

Thus the zero-recoil vector current *is* the conserved flavor current, and its charge is Q_{cb}. Comparing Eq. (40.65) with the trace result Eq. (40.53) at $v' = v$, where $(v+v')^\mu = 2v^\mu$ and (taking $\mu = 0$ in the rest frame, with the relativistic-to-HQET normalization factor $\sqrt{m_B m_D}$ accounting for $\langle D(v)|D(v)\rangle$) the matrix element is $2\xi(1)$ times the norm, forces $\xi(1) = 1$. □

Physical Insight

The same logic that protects electric charge from renormalization protects $\xi(1)$. In QED the vertex correction $\Gamma^\mu(q=0) = \gamma^\mu$ is enforced by the Ward identity (Chapter 19); here the heavy-flavor symmetry plays the role of the gauge symmetry, the $b \to c$ current plays the role of the conserved current, and $\xi(1) = 1$ is the statement that the "flavor charge" of the heavy meson is not dressed by the strong interaction. Crucially, $\xi(1) = 1$ is exact in the symmetry limit *regardless* of the unknown nonperturbative light-cloud dynamics: it is a normalization, not a calculation. This is precisely what makes it useful for extracting a fundamental parameter.

40.2.5 Corrections at zero recoil: Luke's theorem

The relation $\xi(1) = 1$ holds in the strict $m_Q \to \infty$ limit. For the extraction of $|V_{cb}|$ the size of the leading correction is decisive, and here a second protection theorem operates. Symmetry-breaking corrections enter through two sources: $1/m_Q$ operators in the Lagrangian (the kinetic and chromomagnetic operators of the next section) and $1/m_Q$ corrections to the current itself. One might fear an $O(1/m_Q)$ shift of the normalization. Luke's theorem says this fear is unfounded for the relevant channels.

Theorem 40.8 (Luke's theorem, stated)**.** *The matrix elements of the leading $1/m_Q$ corrections — both from the subleading current operators and from the time-ordered products with the $1/m_Q$ Lagrangian insertions — do* not *contribute*

to the zero-recoil values of the form factors $h_+(1)$ (for $\bar{B} \to D$) and $h_{A_1}(1)$ (for $\bar{B} \to D^*$). Consequently

$$h_+(1) = 1 + 0 \cdot \frac{1}{m_Q} + O\Big(\frac{1}{m_Q^2}\Big), \qquad h_{A_1}(1) = 1 + 0 \cdot \frac{1}{m_Q} + O\Big(\frac{1}{m_Q^2}\Big). \quad (40.67)$$

The protection does not *extend to h_-, h_{A_2}, h_{A_3}, which receive ordinary $O(1/m_Q)$ corrections; but those multiply form-factor structures that vanish at zero recoil or are kinematically suppressed there.*

The symmetry reason is the Ademollo–Gatto mechanism, familiar from the protection of the $K \to \pi$ vector form factor against first-order SU(3) breaking in the chiral theory (Section 39.1): a quantity that equals a conserved-charge matrix element in the symmetric limit is stationary under symmetry breaking, so its first-order correction vanishes and the leading deviation is second order. Concretely, at zero recoil the $1/m_Q$ Lagrangian insertions reduce to matrix elements of operators that either are the heavy-quark kinetic energy (which, by the equations of motion and the trivial $v' = v$ kinematics, only renormalizes the states symmetrically and cancels between numerator and norm) or the chromomagnetic operator (whose zero-recoil matrix element between the relevant states vanishes by spin symmetry). The argument just given is the symmetry reason; promoting it to a complete proof requires tracking each $1/m_Q$ insertion through the trace formalism, and this is the one result of the chapter we state without a full derivation here, deferring its details to the Further reading. The decisive phenomenological consequence is that $\bar{B} \to D^* \ell \bar{\nu}$ at zero recoil is protected to second order, $h_{A_1}(1) = 1 + O(1/m_Q^2)$, making it the theoretically cleanest of the two channels.

40.2.6 Extracting $|V_{cb}|$

Assembling the pieces, the differential semileptonic rate for $\bar{B} \to D^* \ell \bar{\nu}$ near zero recoil takes the form

$$\frac{d\Gamma}{dw} = \frac{G_F^2}{48\pi^3} (m_B - m_{D^*})^2 \, m_{D^*}^3 \, \sqrt{w^2 - 1} \, (w+1)^2 \, \chi(w) \, |V_{cb}|^2 \, \mathcal{F}(w)^2, \quad (40.68)$$

where $\chi(w)$ is a known kinematic factor with $\chi(1) = 1$, and $\mathcal{F}(w)$ is the physical form factor that reduces to $h_{A_1}(w)$ in the heavy-quark limit, $\mathcal{F}(1) = h_{A_1}(1) = 1 + O(1/m_Q^2)$ by Theorem 40.8. The phase-space factor $\sqrt{w^2 - 1}$ vanishes at zero recoil, so the rate itself vanishes there; the strategy is therefore to measure $\mathcal{F}(w)|V_{cb}|$ across the accessible w range and extrapolate to the intercept $w = 1$, where heavy-quark symmetry plus Luke's theorem fix $\mathcal{F}(1)$ up to a small, computable $O(1/m_Q^2)$ correction (today supplied by lattice QCD). Dividing the intercept by $\mathcal{F}(1)$ yields $|V_{cb}|$. Fig. 40.3 illustrates both halves of the picture: the universal function $\xi(w)$ that heavy-quark symmetry isolates, and the schematic extrapolation that turns its zero-recoil normalization into a measurement of a CKM element entering Eq. (32.59).

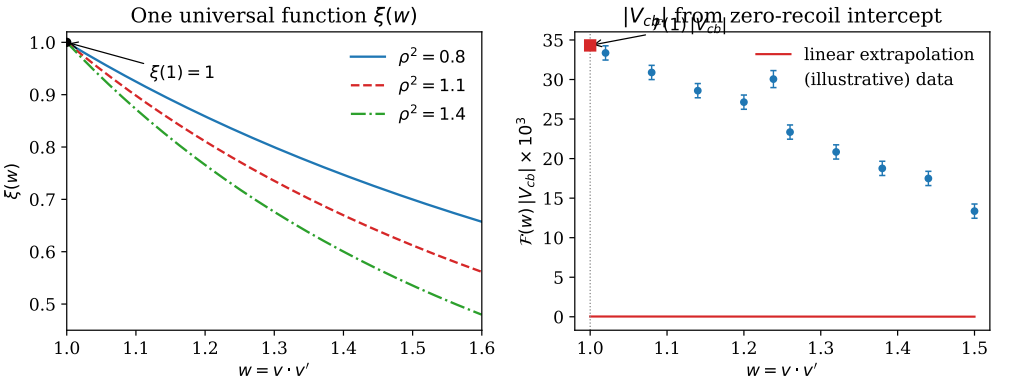


Figure 40.3: *Left:* the Isgur–Wise function $\xi(w)$, the single universal form factor to which all six $\bar{B} \to D^{(*)}$ form factors reduce at leading order in $1/m_Q$, shown for several values of the slope ρ^2 (the shape model is illustrative; the normalization $\xi(1) = 1$ is the exact symmetry result of Theorem 40.7). *Right:* schematic extraction of $|V_{cb}|$ from $\bar{B} \to D^* \ell \bar{\nu}$: the measured combination $\mathcal{F}(w)|V_{cb}|$ is extrapolated to the zero-recoil intercept $w = 1$, where $\mathcal{F}(1) = 1 + O(1/m_Q^2)$ is fixed by heavy-quark symmetry and Luke's theorem (points illustrative, not data).

Historical Note

The exclusive determination of $|V_{cb}|$ from $\bar{B} \to D^* \ell \bar{\nu}$ at zero recoil, pioneered in the early 1990s once Isgur, Wise, Voloshin, Shifman, Georgi and others had established the symmetry and its breaking pattern, was for years the gold standard. As of the mid-2020s a persistent tension remains between this exclusive value and the inclusive determination (from the total $b \to c$ rate via the operator-product expansion), at roughly the few-percent level, with the inclusive value sitting higher; both are quoted near $|V_{cb}| \approx 40 \times 10^{-3}$. Whether the gap reflects underestimated theory uncertainties (form-factor shapes, $1/m_Q^2$ and $1/m_c$ effects), experimental systematics, or genuinely new physics is unsettled and actively studied (the current values and their tensions are tabulated in Section F.6). This is a live example of EFT-based precision flavor physics, not a closed chapter.

Heavy-quark symmetry has thus delivered, from symmetry alone and with no nonperturbative computation, three sharp leading-order statements: the degeneracy of the (P, P^*) doublet, the reduction of six form factors to one function, and the unit normalization of that function at zero recoil — the last protected even against the first power of $1/m_Q$ in the cleanest channel. But every one of these is a *leading-order* statement, and the real world deviates: the B^*–B splitting is nonzero, $\xi(w)$ has a finite slope and curvature, and $\mathcal{F}(1)$ differs

from unity at second order. Symmetry fixes the leading relations; it cannot fix their corrections. To compute the degeneracy splitting, the form-factor slope, and the departures from unit normalization we must construct the subleading HQET Lagrangian by matching to QCD order by order in $1/m_Q$ — which is precisely the program the next section takes up, fixing the kinetic operator by reparametrization invariance and tying the chromomagnetic operator's coefficient to the very hyperfine splitting Eq. (40.40) that opened this discussion.

40.3 Matching and the 1/m expansion

The spin–flavor symmetry of the previous section is exact only in the strict limit $m_Q \to \infty$. It is a statement about the *leading* term of an expansion in $1/m_Q$, and like every leading-order statement in this book it is worth precisely as much as our control over the corrections. The degeneracy of the B and B^* mesons, the single Isgur–Wise function, the normalization at zero recoil — all of these are predictions that the real world satisfies only approximately, because m_b and m_c are large but finite. To turn the symmetry relations into quantitative tools we must compute the operators that break them, and to compute those operators we must do what Chapter 38 taught us to do whenever the short-distance theory is known: *match*. The full theory here is QCD, with the heavy quark still in it; the effective theory is HQET, with the heavy quark replaced by its large-mass remnant h_v. Matching means writing the most general local Lagrangian for h_v consistent with the symmetries, expanding the full theory in powers of $1/m_Q$, and fixing the coefficients of the EFT operators by demanding that the two theories produce identical amputated Green's functions order by order (Section 38.3). This section carries out that matching to first order in $1/m_Q$. The output is two operators — a kinetic operator and a chromomagnetic operator — with two Wilson coefficients, and a pair of striking facts about them: one coefficient is fixed to unity exactly, to all orders in α_s, by a hidden symmetry of the construction (reparametrization invariance), while the other is the microscopic origin of the B^*–B and D^*–D hyperfine splittings. A focused worked example then repeats the whole exercise in the abelian theory, where it has a famous payoff: the Wilson coefficient of the magnetic-moment operator of a heavy electron is exactly $1 + a_e$, and $a_e = \alpha/2\pi$ is Schwinger's number from Chapter 20, now wearing the costume of a matching coefficient.

40.3.1 The heavy-quark field and the expanded QCD Lagrangian

We begin from the QCD Lagrangian for a single heavy quark of mass m_Q in the gluon background,

$$\mathcal{L}_Q \;=\; \bar{Q}\,(i\not{D} - m_Q)\,Q\,, \qquad D_\mu \;=\; \partial_\mu - ig\,A_\mu^a T^a\,, \tag{40.69}$$

with the covariant derivative, generators, and gluon field strength of Yang–Mills theory (Chapter 25); the metric is mostly-minus and $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$ is the bilinear catalogued in Section C.5. The physics that motivates the whole construction is that a heavy quark bound inside a hadron is nearly on shell and nearly at rest relative to the hadron: its four-momentum is $p^\mu = m_Q v^\mu + k^\mu$, where v^μ is the (fixed, timelike, unit) four-velocity of the hadron, $v^2 = 1$, and $k^\mu \sim \Lambda_{\rm QCD} \ll m_Q$ is a small residual momentum measuring how far off the large mass shell the soft QCD dynamics pushes it. The rapidly oscillating phase $e^{-im_Q v\cdot x}$ carries the trivial part of the motion; we strip it off and keep only the slowly varying remainder. Introduce the velocity projectors

$$P_\pm \;=\; \frac{1 \pm \not{v}}{2}\,, \qquad P_\pm^2 = P_\pm\,, \quad P_+P_- = 0\,, \quad P_+ + P_- = 1\,, \quad \not{v}\,P_\pm = \pm P_\pm\,, \tag{40.70}$$

where the last identities use $\not{v}\not{v} = v^2 = 1$. They split the Dirac field into a piece that survives the large-mass limit and a piece that does not,

$$Q(x) \;=\; e^{-im_Q v\cdot x}\left[\, h_v(x) + H_v(x) \,\right], \qquad h_v = e^{im_Q v\cdot x} P_+ Q\,, \quad H_v = e^{im_Q v\cdot x} P_- Q\,, \tag{40.71}$$

so that $\not{v}\, h_v = h_v$ and $\not{v}\, H_v = -H_v$, and correspondingly $\bar h_v \not{v} = \bar h_v$, $\bar H_v \not{v} = -\bar H_v$. The field h_v is the two-component (in the sense of having only the upper, particle, components in the rest frame) heavy-quark field of HQET; H_v describes the heavy degrees of freedom — the would-be antiparticle and the lower components — whose excitation costs an energy $\sim 2m_Q$ and which we will integrate out.

Substituting Eq. (40.71) into Eq. (40.69) requires only one observation: the derivative acting on the stripped field produces $i\partial_\mu(e^{-im_Q v\cdot x}\Phi) = e^{-im_Q v\cdot x}(m_Q v_\mu + i\partial_\mu)\Phi$, so that $iD_\mu Q = e^{-im_Q v\cdot x}(m_Q v_\mu + iD_\mu)(h_v + H_v)$. Hence

$$\mathcal{L}_Q \;=\; (\bar h_v + \bar H_v)\,(m_Q \not{v} + i\not{D} - m_Q)(h_v + H_v)\,. \tag{40.72}$$

The mass terms collapse using $\not{v}\, h_v = h_v$ and $\not{v}\, H_v = -H_v$: the operator $m_Q\not{v} - m_Q$ annihilates h_v and multiplies H_v by $-2m_Q$. It remains to organize the $i\not{D}$ term. Decompose the covariant derivative into its components along and transverse to the velocity,

$$D^\mu \;=\; v^\mu\,(v\cdot D) + D_\perp^\mu\,, \qquad D_\perp^\mu \;=\; D^\mu - v^\mu (v\cdot D)\,, \qquad v\cdot D_\perp = 0\,, \tag{40.73}$$

so that $\not{D} = \not{v}\,(v\cdot D) + \not{D}_\perp$ with $\{\not{v}, \not{D}_\perp\} = 2\, v\cdot D_\perp = 0$. The transverse slash anticommutes with $\not{v}$, hence intertwines the projectors,

$$P_+\,\not{D}_\perp \;=\; \not{D}_\perp\, P_-\,, \qquad P_-\,\not{D}_\perp \;=\; \not{D}_\perp\, P_+\,, \tag{40.74}$$

while $v\cdot D$, carrying no Dirac structure, commutes with both projectors. Now read off the four bilinears in Eq. (40.72). The diagonal h_v term keeps only its longitudinal part, because $\bar h_v \not{D}_\perp h_v = \bar h_v P_+ \not{D}_\perp P_+ h_v = \bar h_v \not{D}_\perp P_- P_+ h_v = 0$ by

Eq. (40.74); what survives is $\bar{h}_v \not{v}\,(v\cdot D)\,h_v = \bar{h}_v\, iv\cdot D\, h_v$ once the i is restored. By the same reasoning the diagonal H_v term keeps $-\bar{H}_v\, iv\cdot D\, H_v$ (the extra sign from $\bar{H}_v \not{v} = -\bar{H}_v$), and the off-diagonal terms keep only their transverse parts, $\bar{h}_v\, i\not{D}_\perp H_v$ and $\bar{H}_v\, i\not{D}_\perp h_v$. Collecting everything,

$$\mathcal{L}_Q \;=\; \bar{h}_v\, iv\cdot D\, h_v \;-\; \bar{H}_v\,(\, iv\cdot D + 2m_Q\,)\, H_v \;+\; \bar{h}_v\, i\not{D}_\perp\, H_v \;+\; \bar{H}_v\, i\not{D}_\perp\, h_v\,. \tag{40.75}$$

The first term is the leading HQET Lagrangian $\mathcal{L}_v^{(0)} = \bar{h}_v\, iv\cdot D\, h_v$ established in the previous section (Eq. (40.19)); it is independent of m_Q and of the heavy-quark spin, which is the source of the spin–flavor symmetry. Everything beyond it is contained in H_v, which appears with a mass $2m_Q$ and is the field we now remove.

40.3.2 Tree-level matching: the kinetic and chromomagnetic operators

Because Eq. (40.75) is exactly quadratic in H_v, integrating it out at tree level is a Gaussian elimination: we solve its equation of motion and substitute back. Varying Eq. (40.75) with respect to $\bar{H}_v$,

$$-(\, iv\cdot D + 2m_Q\,)\, H_v \;+\; i\not{D}_\perp\, h_v \;=\; 0 \quad\Longrightarrow\quad H_v \;=\; \frac{1}{iv\cdot D + 2m_Q}\, i\not{D}_\perp\, h_v\,. \tag{40.76}$$

The operator $1/(iv\cdot D + 2m_Q)$ is well defined as a power series because $iv\cdot D \sim \Lambda_{\rm QCD}$ acting on h_v is parametrically small compared to $2m_Q$; this is the statement that H_v is heavy and may be expanded in its inverse mass. For a Lagrangian of the schematic form $\bar{H}_v \mathcal{A} H_v + \bar{J} H_v + \bar{H}_v J$ with $\mathcal{A} = -(iv\cdot D + 2m_Q)$, $J = i\not{D}_\perp h_v$, and $\bar{J} = \bar{h}_v\, i\not{D}_\perp$, the value at the stationary point is $-\bar{J}\mathcal{A}^{-1}J$, so the tree-level effective Lagrangian is

$$\mathcal{L}_{\rm eff} \;=\; \bar{h}_v\, iv\cdot D\, h_v \;+\; \bar{h}_v\, i\not{D}_\perp\, \frac{1}{iv\cdot D + 2m_Q}\, i\not{D}_\perp\, h_v\,. \tag{40.77}$$

This is exact at tree level; the $1/m_Q$ expansion is now the geometric expansion of the heavy propagator,

$$\frac{1}{iv\cdot D + 2m_Q} \;=\; \frac{1}{2m_Q}\,\frac{1}{1 + \dfrac{iv\cdot D}{2m_Q}} \;=\; \frac{1}{2m_Q}\left(1 - \frac{iv\cdot D}{2m_Q} + \cdots\right). \tag{40.78}$$

Keeping the leading term reproduces the $O(1/m_Q)$ Lagrangian,

$$\mathcal{L}^{(1)} \;=\; \frac{1}{2m_Q}\,\bar{h}_v\,(i\not{D}_\perp)(i\not{D}_\perp)\,h_v \;=\; \frac{1}{2m_Q}\,\bar{h}_v\,(i\not{D}_\perp)^2\,h_v\,, \tag{40.79}$$

and the omitted terms in Eq. (40.78) generate operators of order $1/m_Q^2$ and higher, which we do not pursue here.

The single Dirac structure $(i\not{D}_\perp)^2$ in Eq. (40.79) secretly contains two physically distinct operators, separated by the symmetric and antisymmetric parts of the product of two gamma matrices. Writing $\gamma_\mu\gamma_\nu = g_{\mu\nu} + \frac{1}{2}[\gamma_\mu,\gamma_\nu] = g_{\mu\nu} - i\sigma_{\mu\nu}$,

$$(i\not{D}_\perp)^2 \;=\; -\gamma_\mu\gamma_\nu\, D_\perp^\mu D_\perp^\nu \;=\; -D_\perp^\mu D_{\perp\mu} \;+\; i\sigma_{\mu\nu}\, D_\perp^\mu D_\perp^\nu\,. \tag{40.80}$$

The antisymmetric $\sigma_{\mu\nu}$ contracts only the antisymmetric part of $D_\perp^\mu D_\perp^\nu$, namely the commutator $\frac{1}{2}[D_\perp^\mu, D_\perp^\nu]$. We need that commutator. Using $D_\perp^\mu = D^\mu - v^\mu(v\cdot D)$ and the non-abelian curvature $[D^\mu, D^\nu] = -igG^{\mu\nu}$ (with $G^{\mu\nu} = G^{a\mu\nu}T^a$, the convention fixed by $D_\mu = \partial_\mu - igA_\mu$ in Chapter 25),

$$[D_\perp^\mu, D_\perp^\nu] \;=\; -ig\Big(G^{\mu\nu} - v^\nu v_\alpha G^{\mu\alpha} - v^\mu v_\alpha G^{\alpha\nu}\Big) \;\equiv\; -ig\, G_\perp^{\mu\nu}\,, \tag{40.81}$$

where $G_\perp^{\mu\nu}$ is the doubly transverse projection of the field strength. Assembling Eqs. (40.80) and (40.81),

$$(i\not{D}_\perp)^2 \;=\; (iD_\perp)^2 \;+\; \frac{g}{2}\,\sigma_{\mu\nu}\, G_\perp^{\mu\nu}\,, \qquad (iD_\perp)^2 \equiv -\,D_\perp^\mu D_{\perp\mu}\,, \tag{40.82}$$

where in the second term $i\sigma_{\mu\nu}\cdot\frac{1}{2}(-igG_\perp^{\mu\nu}) = \frac{g}{2}\sigma_{\mu\nu}G_\perp^{\mu\nu}$. The transverse projection in $G_\perp^{\mu\nu}$ is invisible once the operator is sandwiched between heavy-quark fields, because $\bar h_v \sigma^{\mu\nu} v_\nu h_v = 0$: with $v_\nu\sigma^{\mu\nu} = \frac{i}{2}(\gamma^\mu\not{v} - \not{v}\gamma^\mu)$ and the projector identities $\not{v}P_+ = P_+ = P_+\not{v}$ (acting on h_v and $\bar h_v$ respectively), the two terms cancel, $P_+(\gamma^\mu\not{v} - \not{v}\gamma^\mu)P_+ = P_+\gamma^\mu P_+ - P_+\gamma^\mu P_+ = 0$. Therefore the extra pieces of $G_\perp^{\mu\nu}$ in Eq. (40.81) drop, and $\bar h_v\sigma_{\mu\nu}G_\perp^{\mu\nu}h_v = \bar h_v\sigma_{\mu\nu}G^{\mu\nu}h_v$. We arrive at the subleading HQET Lagrangian:

$$\boxed{\mathcal{L}^{(1)} \;=\; \underbrace{\frac{c_{\rm kin}}{2m_Q}\,\bar h_v\,(iD_\perp)^2\,h_v}_{\mathcal{O}_{\rm kin}} \;+\; \underbrace{\frac{c_{\rm mag}\,g}{4m_Q}\,\bar h_v\,\sigma_{\mu\nu}G^{\mu\nu}\,h_v}_{\mathcal{O}_{\rm mag}}\,, \qquad c_{\rm kin} = c_{\rm mag} = 1 \text{ at tree level.}} \tag{40.83}$$

The factor of $\frac{1}{2}$ between the two coefficients in Eq. (40.83) (the kinetic term carries $1/2m_Q$, the chromomagnetic term $g/4m_Q$) is exactly what Eq. (40.82) delivers, so the tree-level matching gives $c_{\rm kin} = c_{\rm mag} = 1$ with no further input.

Definition 40.9 (The $1/m_Q$ operators)**.** $\mathcal{O}_{\rm kin} = \frac{1}{2m_Q}\bar h_v(iD_\perp)^2 h_v$ is the *kinetic operator*: it is the residual kinetic energy of the heavy quark, the operator that restores the dependence of the dynamics on the heavy-quark mass. It is a Lorentz scalar and a spin singlet, so it preserves both heavy-quark spin symmetry and flavor symmetry; it shifts hadron masses but does not split spin multiplets. $\mathcal{O}_{\rm mag} = \frac{c_{\rm mag}g}{4m_Q}\bar h_v\sigma_{\mu\nu}G^{\mu\nu}h_v$ is the *chromomagnetic operator*: it couples the heavy-quark spin (through $\sigma_{\mu\nu}$) to the gluonic field, and is the leading operator that *breaks* heavy-quark spin symmetry. It also breaks flavor symmetry through its explicit $1/m_Q$. These are the only two operators of dimension five that can be built from h_v, the gluon field, and one covariant derivative consistent with the heavy-quark symmetries; the matching has fixed their coefficients.

Physical Insight

The split of the single relativistic structure $(i\not{D}_\perp)^2$ into a spin-conserving kinetic term and a spin-violating chromomagnetic term is the heavy-quark analogue of the Pauli reduction of the Dirac equation, where $(\vec{\sigma}\cdot\vec{\pi})^2 = \vec{\pi}^2 - e\vec{\sigma}\cdot\vec{B}$ separates the orbital from the magnetic interaction. There the magnetic term is responsible for $g = 2$; here the chromomagnetic term is responsible for the hyperfine splitting of heavy hadrons. The same algebraic identity — "the square of a slashed covariant derivative is a covariant Laplacian plus a magnetic coupling" — runs through both, and through the NRQED worked example below.

40.3.3 Reparametrization invariance fixes the kinetic coefficient

The tree-level result $c_{\rm kin} = 1$ looks like an ordinary matching coefficient, the kind that receives radiative corrections and runs with the scale. It does not. The kinetic coefficient is protected to be exactly unity, to all orders in α_s and with no scale dependence, by a symmetry that is invisible in the full theory and emerges only in the way we set up the EFT: *reparametrization invariance*. The origin of the symmetry is that the decomposition $p^\mu = m_Q v^\mu + k^\mu$ of the heavy-quark momentum into a "large" part $m_Q v$ and a "residual" part k is ambiguous. Only the total momentum p is physical; the split is a choice. We may shift the reference velocity by an infinitesimal transverse amount and compensate by shifting the residual momentum,

$$v^\mu \;\to\; v^\mu + \frac{\epsilon^\mu}{m_Q}\,, \qquad k^\mu \;\to\; k^\mu - \epsilon^\mu\,, \qquad v\cdot\epsilon = 0\,, \tag{40.84}$$

the constraint $v\cdot\epsilon = 0$ preserving $v^2 = 1$ to first order. The total momentum $m_Q v + k$ is unchanged, so no physical quantity — in particular not the HQET action — can depend on the choice. This is reparametrization invariance, and it relates operators at different orders in $1/m_Q$ because the shift in v is itself of order $1/m_Q$.

To exploit it we need the induced transformation of the field h_v. The full field Q is reparametrization-invariant, so the change in the velocity must be absorbed by h_v. Two effects combine. First, the phase $e^{-im_Q v\cdot x}$ in Eq. (40.71) shifts by $e^{-i\epsilon\cdot x}$, which is exactly the shift $k \to k - \epsilon$ of the residual momentum in position space; on h_v this is a multiplication by $e^{i\epsilon\cdot x}$. Second, the projector $P_+(v)$ in $h_v = P_+(v)\, e^{im_Q v\cdot x} Q$ rotates with v. To first order in ϵ/m_Q,

$$h_v \;\to\; e^{\,i\epsilon\cdot x}\left(1 + \frac{\not{\epsilon}}{2m_Q}\right) h_v\,, \tag{40.85}$$

which is consistent with the new projector: $P_+(v+\epsilon/m_Q) = P_+(v) + \frac{\not\epsilon}{2m_Q}$, and one checks directly that $P_+(v+\epsilon/m_Q)\,(1+\frac{\not\epsilon}{2m_Q})h_v = (1+\frac{\not\epsilon}{2m_Q})h_v$ to $O(1/m_Q)$, using $P_+\not\epsilon = \not\epsilon P_-$ (from $\not v\not\epsilon = -\not\epsilon\not v$ since $v\cdot\epsilon = 0$) so that $P_+\not\epsilon\, h_v = \not\epsilon P_- P_+ h_v = 0$. The conjugate field transforms as $\bar h_v \to \bar h_v(1+\frac{\not\epsilon}{2m_Q})e^{-i\epsilon\cdot x}$.

Now demand that the Lagrangian be invariant order by order in $1/m_Q$. We write the unknown subleading Lagrangian with a free coefficient on the kinetic operator,

$$\mathcal{L} \;=\; \bar h_v\, iv\cdot D\, h_v \;+\; \frac{c_{\rm kin}}{2m_Q}\,\bar h_v\,(iD_\perp)^2\,h_v \;+\; \frac{c_{\rm mag}\,g}{4m_Q}\,\bar h_v\,\sigma_{\mu\nu}G^{\mu\nu}\,h_v \;+\; O(1/m_Q^2)\,, \tag{40.86}$$

and vary the leading term and the kinetic term, both of which contribute at $O(1/m_Q)$.

Variation of the leading term. Three sources contribute: the explicit v in $iv\cdot D$, the phase in δh_v, and the projector piece in δh_v. The explicit v gives $\delta(iv\cdot D) = \frac{i}{m_Q}\epsilon\cdot D$, so a contribution $\frac{i}{m_Q}\bar h_v\,\epsilon\cdot D\,h_v = \frac{i}{m_Q}\bar h_v\,\epsilon\cdot D_\perp\,h_v$ (using $v\cdot\epsilon = 0$, so $\epsilon\cdot D = \epsilon\cdot D_\perp$). The phase pieces cancel between δh_v and $\delta\bar h_v$: the only term that could survive is the derivative of the phase, $iv\cdot D(i\epsilon\cdot x\,h_v)$ producing $i\cdot i(v\cdot\epsilon)h_v = 0$ since $v\cdot\epsilon = 0$. The projector pieces give $\frac{1}{2m_Q}\bar h_v(\not\epsilon\, iv\cdot D + iv\cdot D\,\not\epsilon)h_v = \frac{1}{m_Q}\bar h_v\,\not\epsilon\, iv\cdot D\,h_v$, which *vanishes*, because $iv\cdot D\,h_v$ remains in the P_+ subspace while $\bar h_v\not\epsilon$ projects onto P_-: $\bar h_v P_+\not\epsilon P_+ = \bar h_v\not\epsilon P_- P_+ = 0$. Collecting,

$$\delta(\bar h_v\, iv\cdot D\, h_v) \;=\; \frac{i}{m_Q}\,\bar h_v\,\epsilon\cdot D_\perp\,h_v\,. \tag{40.87}$$

Variation of the kinetic term. At $O(1/m_Q)$ we need only the leading variation of h_v, the phase $\delta h_v = i\epsilon\cdot x\,h_v$ (the projector and explicit-v variations are pushed to $O(1/m_Q^2)$ by the explicit $1/2m_Q$ prefactor). The phase variation is governed by the commutator of the kinetic operator with $i\epsilon\cdot x$. With $f = i\epsilon\cdot x$ and $\partial_{\perp\mu}f = i\epsilon_\mu$ (the transverse gradient, since ϵ is transverse),

$$[(iD_\perp)^2,\,f] \;=\; -[D_\perp^\mu D_{\perp\mu},\,f] \;=\; -(2(\partial_\perp^\mu f)D_{\perp\mu} + (\partial_\perp^2 f)) \;=\; -\,2i\,\epsilon\cdot D_\perp\,, \tag{40.88}$$

since $\partial_\perp^2 f = 0$. Hence the variation of the kinetic term is

$$\delta\!\left(\frac{c_{\rm kin}}{2m_Q}\bar h_v(iD_\perp)^2 h_v\right) \;=\; \frac{c_{\rm kin}}{2m_Q}\,\bar h_v\,[(iD_\perp)^2, i\epsilon\cdot x]\,h_v \;=\; -\,\frac{c_{\rm kin}\,i}{m_Q}\,\bar h_v\,\epsilon\cdot D_\perp\,h_v\,. \tag{40.89}$$

The chromomagnetic operator does not enter at this order: its variation under Eq. (40.85) is $O(1/m_Q^2)$, and in any case it carries a distinct Dirac structure ($\sigma_{\mu\nu}G^{\mu\nu}$, not $\epsilon\cdot D_\perp$) that cannot cancel against Eq. (40.87). Adding Eqs. (40.87) and (40.89),

$$\delta\mathcal{L} \;=\; \frac{i\,(1-c_{\rm kin})}{m_Q}\,\bar h_v\,\epsilon\cdot D_\perp\,h_v \;+\; O(1/m_Q^2)\,, \tag{40.90}$$

and reparametrization invariance, $\delta\mathcal{L} = 0$, forces $c_{\rm kin} = 1$.

Theorem 40.10 (Reparametrization invariance protects the kinetic coefficient)**.** *The Wilson coefficient of the kinetic operator $\mathcal{O}_{\rm kin}$ in the HQET Lagrangian is exactly*

$$c_{\rm kin} \;=\; 1 \tag{40.91}$$

to all orders in α_s and with no renormalization-scale dependence. The only inputs are (i) the coefficient of the leading operator $\bar{h}_v\, iv\cdot D\, h_v$ is unity, which is the normalization of the heavy-quark field, and (ii) the reparametrization transformation Eq. (40.85), which is exact. Since neither input receives radiative corrections, $\mathcal{O}_{\rm kin}$ is not renormalized as a separate operator and does not run. The chromomagnetic coefficient $c_{\rm mag}$ is left undetermined by reparametrization invariance and must be obtained by explicit matching; it does run.

The asymmetry in Theorem 40.10 is the central structural lesson of HQET at subleading order. Reparametrization invariance ties the kinetic operator to the leading operator, whose coefficient is fixed by definition, so the kinetic term inherits an exact coefficient. The chromomagnetic operator is not tied to the leading operator by any such relation — it is a genuinely new Lorentz and Dirac structure — so its coefficient is an honest function of the scale, computable by matching at $\mu \sim m_Q$ and runnable down to the hadronic scale. We turn to it now.

Historical Note

Reparametrization invariance was identified by Luke and Manohar in 1992, several years after HQET itself, as the symmetry enforcing $c_{\rm kin} = 1$. That the kinetic energy of a heavy quark is fixed to its naive value is what one would have guessed from Lorentz invariance of the underlying theory — the residual mass-shell condition cannot be renormalized away — and reparametrization invariance is precisely the residual imprint of Lorentz invariance on a formalism that has singled out a particular frame velocity v. The same symmetry constrains the coefficients of operators at every order in $1/m_Q$, relating them in chains that dramatically reduce the number of independent matching computations.

40.3.4 The chromomagnetic coefficient and the heavy-hadron hyperfine splittings

The chromomagnetic operator is the leading window onto heavy-quark spin-symmetry breaking, and it has a clean phenomenological signature: it is the only $1/m_Q$ operator that distinguishes the spin-singlet from the spin-triplet heavy meson, and so it generates the entire hyperfine splitting of the H–H^* doublet.

To see this, write the mass of a heavy-light meson H as the expectation value of the HQET Hamiltonian in the meson state, expanded to first order in $1/m_Q$:

$$m_H \;=\; m_Q + \bar{\Lambda} \;-\; \frac{\lambda_1}{2m_Q} \;-\; \frac{d_H\, c_{\rm mag}(\mu)\, \lambda_2(\mu)}{2m_Q} \;+\; O(1/m_Q^2)\,. \tag{40.92}$$

Here $\bar{\Lambda}$ is the (mass-independent, spin-independent) energy of the light degrees of freedom in the static color field, the leading-order binding; λ_1 is the matrix element of the kinetic operator,

$$\lambda_1 \;=\; \frac{1}{2m_H}\,\Big\langle H(v)\Big|\bar{h}_v (iD_\perp)^2 h_v\Big|H(v)\Big\rangle\,, \tag{40.93}$$

spin-independent and so common to H and H^*; and λ_2 parametrizes the chromomagnetic matrix element,

$$d_H\,\lambda_2 \;=\; \frac{1}{2m_H}\,\Big\langle H(v)\Big|\tfrac{g}{2}\,\bar{h}_v\,\sigma_{\mu\nu}G^{\mu\nu}\,h_v\Big|H(v)\Big\rangle\,, \tag{40.94}$$

with d_H a pure spin factor. The factor d_H is fixed by the heavy-quark spin algebra. In the ground-state doublet the light degrees of freedom carry total angular momentum $j_\ell = \frac{1}{2}$, which combines with the heavy-quark spin $s_Q = \frac{1}{2}$ to give the pseudoscalar ($J=0$) and vector ($J=1$) mesons. The chromomagnetic energy is proportional to $\vec{S}_Q\cdot\vec{S}_\ell$, and

$$\vec{S}_Q\cdot\vec{S}_\ell \;=\; \tfrac{1}{2}\big[J(J+1) - s_Q(s_Q+1) - j_\ell(j_\ell+1)\big] \;=\; \begin{cases} -\frac{3}{4}, & J=0\ (H),\\ +\frac{1}{4}, & J=1\ (H^*), \end{cases} \tag{40.95}$$

so that $d_H = -4\,\vec{S}_Q\cdot\vec{S}_\ell$ takes the values $d_P = 3$ for the pseudoscalar and $d_V = -1$ for the vector. Two consequences follow immediately. First, the spin-averaged mass $\bar{m}_H = \frac{1}{4}(m_H + 3m_{H^*})$ is free of the chromomagnetic term, since $\frac{1}{4}(d_P + 3d_V) = \frac{1}{4}(3-3) = 0$: the chromomagnetic operator splits the doublet but does not move its center of gravity. Second, the splitting itself is

$$m_{H^*} - m_H \;=\; -\,\frac{(d_V - d_P)\, c_{\rm mag}(\mu)\,\lambda_2(\mu)}{2m_Q} \;=\; \frac{2\, c_{\rm mag}(\mu)\,\lambda_2(\mu)}{m_Q}\,, \tag{40.96}$$

inversely proportional to the heavy-quark mass. (The product $c_{\rm mag}\lambda_2$ is renormalization-group invariant, the μ-dependence of the coefficient cancelling that of the matrix element, so Eq. (40.96) is scheme-independent.)

Equation Eq. (40.96) is the prediction. It says that the hyperfine splitting scales as $1/m_Q$, so the B^*–B splitting should be smaller than the D^*–D splitting by roughly m_c/m_b. Even sharper, since $m_{H^*} + m_H \simeq 2m_Q$ at leading order, the *mass-squared* splitting is heavy-quark-mass-*independent*:

$$m_{H^*}^2 - m_H^2 \;=\; (m_{H^*} - m_H)(m_{H^*} + m_H) \;\simeq\; \frac{2\, c_{\rm mag}\lambda_2}{m_Q}\cdot 2m_Q \;=\; 4\, c_{\rm mag}(\mu)\,\lambda_2(\mu)\,, \tag{40.97}$$

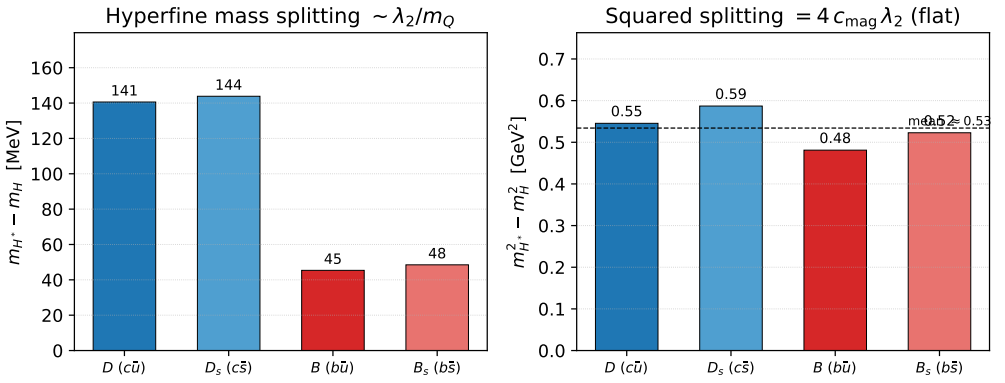


Figure 40.4: The $1/m_Q$ structure of the heavy-meson hyperfine splittings, the experimental signature of the chromomagnetic operator $\mathcal{O}_{\rm mag}$. *Left:* the linear mass splitting $m_{H^*} - m_H$, which Eq. (40.96) predicts to scale as λ_2/m_Q; the charm splittings are about three times the bottom splittings, tracking $m_b/m_c \simeq 3$. *Right:* the mass-squared splitting $m^2_{H^*} - m^2_H$, which Eq. (40.97) predicts to equal $4\, c_{\rm mag}\lambda_2$ independently of the heavy-quark mass; all four systems agree to about ten percent (dashed line: mean), confirming the prediction and fixing $\lambda_2 \simeq 0.12$ GeV^2. Masses are PDG central values.

a constant that should be common to the charm and bottom systems. The data confirm this with striking accuracy. Using the measured masses, the linear splittings are $m_{D^*} - m_D \simeq 141$ MeV and $m_{B^*} - m_B \simeq 45$ MeV, in the ratio 3.1, close to $m_b/m_c \simeq 3$; while the squared splittings are $m^2_{D^*} - m^2_D \simeq 0.55$ GeV^2 and $m^2_{B^*} - m^2_B \simeq 0.48$ GeV^2, equal to about ten percent. Figure 40.4 displays both. From Eq. (40.97) the common value fixes the chromomagnetic matrix element, $\lambda_2(\mu \sim m_Q) \simeq 0.12$ GeV^2, a genuinely nonperturbative number extracted from the spectrum and reusable elsewhere in the heavy-quark expansion (it controls, for instance, the $1/m_Q^2$ corrections to inclusive semileptonic decay rates).

The remaining ten-percent spread is itself instructive: it is the size of the $1/m_Q^2$ and (for charm) higher corrections that Eq. (40.97) neglects, together with the residual scale dependence that survives because $c_{\rm mag}\lambda_2$ is computed at different effective μ for charm and bottom. This brings us to the running of $c_{\rm mag}$. Unlike $c_{\rm kin}$, the chromomagnetic coefficient is renormalized: the operator $\mathcal{O}_{\rm mag}$ mixes into itself under QCD radiative corrections, acquiring an anomalous dimension. Matching at $\mu = m_Q$ gives the tree value $c_{\rm mag}(m_Q) = 1$ corrected by a calculable series in $\alpha_s(m_Q)$, and the renormalization group then evolves it to lower scales. At leading logarithmic order the result has the standard power-law

form

$$c_{\rm mag}(\mu) \;=\; \left[\frac{\alpha_s(m_Q)}{\alpha_s(\mu)}\right]^{\gamma^{(0)}_{\rm mag}/(2\beta_0)}, \qquad \gamma^{(0)}_{\rm mag} = 2C_A\,, \quad \beta_0 = \tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f\,, \tag{40.98}$$

with β_0 the one-loop coefficient of Yang–Mills theory (Eq. (27.1)) and $\gamma^{(0)}_{\rm mag}$ the one-loop anomalous dimension of the chromomagnetic operator. The exponent $\gamma^{(0)}_{\rm mag}/(2\beta_0) = C_A/(\frac{11}{3}C_A - \frac{4}{3}T_F n_f)$ is positive; since $\alpha_s(\mu) > \alpha_s(m_Q)$ for $\mu < m_Q$, the bracket in Eq. (40.98) is less than one, so $c_{\rm mag}$ *decreases* as μ runs below m_Q (and the hyperfine ratio $c^{(b)}_{\rm mag}/c^{(c)}_{\rm mag} < 1$ at a common low scale follows correctly). The one-loop value $\gamma^{(0)}_{\rm mag} = 2C_A$ is a standard HQET result, which we quote from the literature [362] rather than recompute. The next worked example carries out in full the analogous *matching* computation in the abelian theory (NRQED), fixing the magnetic-moment Wilson coefficient c_F transparently; the chromomagnetic anomalous dimension shares the same diagrammatic ingredients but is not itself reproduced here. The qualitative message is fixed regardless of the number: $c_{\rm kin}$ is frozen at 1 by symmetry, $c_{\rm mag}$ is a running coupling of the EFT, and the difference between them is the difference between a coefficient pinned by an exact relation and one that must be computed.

40.3.5 The abelian analog: NRQED and Schwinger's number as a Wilson coefficient

Everything above has a cleaner sibling in QED, where the heavy fermion is the electron and the gauge field is abelian. The effective theory of a slow, heavy electron coupled to soft photons is *nonrelativistic QED* (NRQED), and matching it to QED reproduces the structure we have just built — a kinetic operator with a protected coefficient and a magnetic operator with a runnable one — but with a celebrated bonus. The Wilson coefficient of the magnetic-moment operator in NRQED is the full electron g-factor divided by two; matching it to QED at one loop delivers $1 + a_e$, and $a_e = \alpha/2\pi$ is Schwinger's number from Chapter 20. This is QED advancing one layer: the most famous one-loop number in physics, computed in Theorem 20.4 as the value of a form factor, reappears here as a matching coefficient of an effective Lagrangian, which is the modern way to understand why it controls atomic energy levels. The following worked example carries the matching from the tree-level reduction through the one-loop coefficient.

Worked Example 40.11 (NRQED tree-level Lagrangian and the one-loop matching of the Fermi coefficient)**.** We work in QED with the electron field ψ of mass m and charge $-e$ ($e > 0$), covariant derivative $D_\mu = \partial_\mu + ieA_\mu$ (QED, Chapter 8); the photon field strength gives the magnetic field $\vec{B} = \vec{\nabla} \times \vec{A}$. The

goal is the NRQED Lagrangian

$$\mathcal{L}_{\rm NRQED} \;=\; \phi^\dagger\Bigg[\, iD_t + \frac{\vec{D}^2}{2m} + c_F\,\frac{e}{2m}\,\vec{\sigma}\cdot\vec{B} + c_D\,\frac{e}{8m^2}\,\vec{\nabla}\cdot\vec{E} + c_S\,\frac{ie}{8m^2}\,\vec{\sigma}\cdot(\vec{D}\times\vec{E} - \vec{E}\times\vec{D}) + \tag{40.99}$$

where ϕ is the two-component Pauli field of the electron, $D_t = \partial_t + ieA^0$, $\vec{D} = \vec{\nabla} - ie\vec{A}$, and the dots are operators of still higher order in $1/m$. We determine the Fermi (Pauli) coefficient c_F, first at tree level and then at one loop.

Step 1: tree-level reduction. Start from the Dirac Lagrangian $\bar\psi(i\not{D} - m)\psi$ and remove the rest-mass oscillation by writing the field in the Dirac basis as an upper (large) component ϕ and a lower (small) component χ. The field equation for χ is algebraic up to $1/m$ corrections; solving it gives $\chi = \dfrac{\vec{\sigma}\cdot(-i\vec{D})}{2m + iD_t}\,\phi = \dfrac{-i\,\vec{\sigma}\cdot\vec{D}}{2m}\,\phi + O(1/m^2)$, the nonrelativistic statement that the small component is suppressed by one power of velocity. Substituting back, the upper-component Lagrangian is

$$\mathcal{L} \;=\; \phi^\dagger\left[\, iD_t + \frac{(\vec{\sigma}\cdot\vec{D})(\vec{\sigma}\cdot\vec{D})}{2m}\,\right]\phi + O(1/m^2)\,. \tag{40.100}$$

The product of Pauli structures is reduced by the same algebraic identity that organized the HQET chromomagnetic term. With $\vec{D} = \vec{\nabla} - ie\vec{A}$ and $[D_i, D_j] = -ie(\partial_i A_j - \partial_j A_i) = -ie\,\epsilon_{ijk}B_k$,

$$(\vec{\sigma}\cdot\vec{D})^2 \;=\; \vec{D}^2 + i\,\vec{\sigma}\cdot(\vec{D}\times\vec{D}) \;=\; \vec{D}^2 + i\,\sigma_k\,\tfrac{1}{2}\epsilon_{kij}[D_i, D_j] \;=\; \vec{D}^2 + e\,\vec{\sigma}\cdot\vec{B}\,, \tag{40.101}$$

using $\frac{1}{2}\epsilon_{kij}(-ie\,\epsilon_{ijl}B_l) = -ie\,B_k$ and $i\sigma_k(-ieB_k) = e\,\vec{\sigma}\cdot\vec{B}$. Inserting Eq. (40.101) into Eq. (40.100) reproduces the kinetic term $\vec{D}^2/2m$ and the Fermi term with coefficient one:

$$\mathcal{L} \;=\; \phi^\dagger\left[\, iD_t + \frac{\vec{D}^2}{2m} + \frac{e}{2m}\,\vec{\sigma}\cdot\vec{B}\,\right]\phi + O(1/m^2) \quad\Longrightarrow\quad c_F^{\rm tree} = 1\,. \tag{40.102}$$

The magnetic energy of the electron is $-\frac{c_F e}{2m}\vec{\sigma}\cdot\vec{B}$, so the g-factor is $g = 2c_F$, and the tree value $c_F = 1$ is precisely the Dirac prediction $g = 2$. The coefficient of the kinetic operator $\vec{D}^2/2m$ is locked to unity by Galilean (reparametrization) invariance of the nonrelativistic theory, the abelian counterpart of Theorem 40.10; it is c_F that carries the interesting physics.

Step 2: the matching condition at one loop. The Wilson coefficient c_F is fixed by demanding that NRQED reproduce the QED amplitude for an electron scattering off a static, slowly varying external magnetic field, evaluated at zero momentum transfer where the operator $\vec{\sigma}\cdot\vec{B}$ acts. In the full theory this amplitude

is the on-shell electromagnetic vertex, decomposed into the Dirac and Pauli form factors (Section 20.1),

$$\bar{u}(p')\,\Gamma^{\mu}\,u(p) \;=\; \bar{u}(p')\Big[\gamma^{\mu}F_1(q^2)+\frac{i\sigma^{\mu\nu}q_{\nu}}{2m}F_2(q^2)\Big]u(p)\,, \tag{40.103}$$

whose nonrelativistic reduction at $q^2 \to 0$ gives a spin–magnetic coupling proportional to $F_1(0)+F_2(0)$ — the Dirac term contributes its magnetic piece via the Gordon decomposition and the Pauli term contributes directly, and the two add. The same reduction performed on the NRQED side, where the magnetic coupling comes entirely from the c_F term of Eq. (40.99), gives a coupling proportional to c_F. Equality of the two amplitudes is the matching condition,

$$c_F \;=\; F_1(0)+F_2(0)\,. \tag{40.104}$$

The charge form factor is fixed by the Ward–Takahashi identity to $F_1(0)=1$ exactly (Section 19.5): the electric charge is not renormalized at zero momentum transfer. The Pauli form factor at the origin is the anomalous magnetic moment, $F_2(0)=a_e$, computed in full in Theorem 20.4 as

$$F_2(0) \;=\; a_e \;=\; \frac{\alpha}{2\pi} \;=\; 0.00116\ldots\,, \tag{40.105}$$

the value evaluated step by step in Theorem 20.5 from the master loop integral of Theorem D.22. Therefore

$$\boxed{\,c_F \;=\; 1+a_e \;=\; 1+\frac{\alpha}{2\pi}+O(\alpha^2)\,.\,} \tag{40.106}$$

The electron g-factor is $g=2c_F=2(1+\alpha/2\pi+\cdots)$, so the anomaly $a_e=(g-2)/2=c_F-1=\alpha/2\pi$ is exactly the one-loop piece of the Fermi coefficient.

Reading. The structural point is worth stating plainly. In the full theory a_e is the value of a form factor — a property of an on-shell scattering amplitude. In the effective theory a_e is a coefficient in a Lagrangian — a number multiplying the local operator $\phi^{\dagger}\vec{\sigma}\cdot\vec{B}\,\phi$. Matching identifies the two. This is more than bookkeeping: once a_e lives in the Lagrangian as c_F, it can be inserted into bound-state computations as an ordinary interaction vertex, which is how the anomalous moment enters the hyperfine structure of hydrogen and the energy levels of muonium and positronium. The form-factor language cannot do this directly; the EFT language can, because matching has converted a scattering observable into a term in an effective action.

40.3.6 NRQED scales and the organization of bound states

The NRQED Lagrangian Eq. (40.99) is the first member of a tower of effective theories that organizes the entire bound-state problem of QED by scale, and

it is worth stating the organization even though its execution belongs to the literature of precision atomic physics. A nonrelativistic bound state such as positronium or hydrogen has not one low scale but several, generated by powers of the fine-structure constant:

$$m \gg m\alpha \text{ (typical momentum, inverse Bohr radius)} \gg m\alpha^2 \text{ (binding energy, Rydberg)} \tag{40.107}$$

NRQED is obtained by integrating out the hard scale m: this is the matching of the previous worked example, which deposits the relativistic and radiative physics into Wilson coefficients such as $c_F = 1 + a_e$, c_D, and c_S, each a series in α evaluated at $\mu \sim m$. The resulting theory still contains the scales $m\alpha$ and $m\alpha^2$; integrating out the soft scale $m\alpha$ in turn produces *potential* NRQED (pNRQED), whose degrees of freedom are the electron, the positron or proton, and ultrasoft photons, and whose interactions are the Coulomb potential plus a tower of relativistic and quantum corrections. In this final theory the energy levels are computed in ordinary quantum-mechanical perturbation theory, with the Wilson coefficients of NRQED feeding the corrections. The Lamb shift — the $O(m\alpha^5)$ splitting of the $2S_{1/2}$ and $2P_{1/2}$ levels of hydrogen, the effect whose 1947 measurement launched modern QED — is in this language a definite combination of low-energy matrix elements weighted by Wilson coefficients, prominent among them the very c_F and c_D that the matching above produces. The role of a_e as a Wilson coefficient is thus not an analogy but the operational content of the statement that the electron's anomalous moment shifts atomic energy levels: a_e enters the bound-state energy precisely as the coefficient $c_F - 1$ of the magnetic operator, evaluated in the bound-state wavefunction. This is the bridge from QED's one-loop precision (Chapter 20) to the EFT of bound states, and we record it here at the level of organization, the detailed multi-scale computation lying beyond our scope.

Physical Insight

The lesson common to HQET and NRQED is that "the magnetic interaction of a heavy fermion is a Wilson coefficient." In HQET it is c_{mag}, which runs and which the data on B^*–B and D^*–D splittings measure; in NRQED it is $c_F = 1 + a_e$, which packages Schwinger's number and which atomic spectroscopy measures. In both theories a kinetic coefficient sits beside it, frozen to unity by the residual Lorentz (reparametrization, Galilean) invariance of the construction. The same two operators, the same protection of one coefficient and running of the other, appear whether the underlying theory is QCD or QED — which is the whole point of the method: matching produces the *same* effective Lagrangian structure from any short-distance completion, with the completion entering only through the numerical values of the coefficients.

We have now seen matching in its fullest top-down form: a known short-distance theory (QCD, or QED) expanded in $1/m$, its heavy field integrated out, and the coefficients of the effective operators computed by equating amputated amplitudes order by order — with one coefficient pinned exactly by a symmetry and another carrying the dynamics. The operational core of this procedure — *expand the full theory and the effective theory in the same kinematic regime and equate their Green's functions* — makes no use of QCD or QED specifically; it uses only that the short-distance theory is known. Run the very same procedure in reverse, with the short-distance theory *unknown* but its symmetries and light field content fixed, and matching becomes construction: one writes down every operator the symmetries allow and lets experiment, rather than a known UV completion, fix the coefficients. That inversion is the subject of the next section, where the Standard Model itself becomes the leading term of an effective theory and the higher-dimension operators become the systematic parametrization of whatever lies above the electroweak scale.

40.4 The Standard Model as an effective theory: building SMEFT

The previous section drew its strength from a luxury we will now surrender: the knowledge of the short-distance theory. There the heavy field was a genuine Standard-Model quark, the ultraviolet (UV) completion was QCD, and matching was an explicit computation — expand the full-theory Green's functions in $1/m$, equate them to the EFT, read off the Wilson coefficients. The lesson worth extracting from that exercise is not the particular numbers but the *shape* of the method: once a field content, a symmetry, and a power counting are fixed, the most general local Lagrangian consistent with them is determined up to a list of constants, and those constants are then pinned either by matching (UV known) or by measurement (UV unknown). The Standard Model invites us to run the method in reverse. We believe there is new physics above the electroweak scale — the hierarchy problem, neutrino masses, dark matter and the strong-CP puzzle all say so — but we do not know what it is. So we take the only data we are sure of: the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the chiral fermion and scalar content assembled in Chapter 33, and a single new scale Λ above which the unknown heavy states live. From these we build the tower of higher-dimension operators, confront the fact that many of the operators one writes down are secretly the same, and organize the survivors into a canonical, non-redundant list — the Warsaw basis. The result is the Standard Model Effective Field Theory, SMEFT: the Standard Model itself, read as the leading terms of an expansion whose subleading terms parametrize, model-independently, whatever lies above Λ.

40.4.1 The defining data and the operator expansion

We adopt wholesale the ingredients fixed in the Standard Model chapter. The gauge group is $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ with couplings g_s, g, g' and field strengths $G^A_{\mu\nu}$, $W^I_{\mu\nu}$, $B_{\mu\nu}$ (capital $A = 1, \dots, 8$ for colour, $I = 1, 2, 3$ for weak isospin). The one-generation matter content, with hypercharges normalized to the Gell-Mann–Nishijima relation $Q = T^3 + Y$ (Table 33.2), is the set of left-handed Weyl multiplets

$$L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \quad Q = \begin{pmatrix} u_L \\ d_L \end{pmatrix} (\mathbf{3}, \mathbf{2}, +\tfrac{1}{6}), \quad e\,(\mathbf{1}, \mathbf{1}, -1), \quad u\,(\mathbf{3}, \mathbf{1}, +\tfrac{2}{3}), \quad d\,(\mathbf{3}, \mathbf{1}, -\tfrac{1}{3}), \tag{40.108}$$

where e, u, d denote the right-handed singlets (we suppress the subscript R where no ambiguity arises) and we work throughout with one generation, restoring flavour indices only when the counting demands it. The scalar sector is the Higgs doublet $H\,(\mathbf{1}, \mathbf{2}, +\tfrac{1}{2})$, with the charge-conjugate doublet $\widetilde{H}_a = \epsilon_{ab} H^*_b$ (hypercharge $-\tfrac{1}{2}$) of Chapter 31 available for the up-type Yukawa. The covariant derivative carries the canonical sign of Chapter A, $D_\mu = \partial_\mu - i g_s G^A_\mu T^A - i g\, W^I_\mu\, t^I - i g' Y B_\mu$, with $t^I = \tau^I/2$ the weak-isospin generators on doublets. The renormalizable Lagrangian $\mathcal{L}_{\mathrm{SM}}$ — gauge kinetic terms, fermion kinetic terms, the Higgs potential $V = -\mu^2 H^\dagger H + \lambda (H^\dagger H)^2$, and the Yukawa couplings — is the object assembled in Eq. (33.22); we do not rebuild it here.

What Chapter 33 also established, and what makes the construction below sharp, is that baryon and lepton number B and L are *accidental* global symmetries of $\mathcal{L}_{\mathrm{SM}}$ (Section 33.4): they are not imposed, they simply happen to be the only U(1) phase rotations left invariant by the most general renormalizable interaction the field content allows. Accidental symmetries are fragile. They are an artifact of truncating at dimension four, and there is no reason for the unknown UV to respect them. The higher-dimension operators we are about to write are precisely where that fragility shows up first.

The organizing principle is the operator expansion of Section 38.2: with all heavy states integrated out, the effective Lagrangian is the sum of all local, Lorentz-invariant, $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant operators built from the light fields (40.108) and H, each weighted by an inverse power of Λ fixed by its mass dimension,

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \frac{1}{\Lambda} \sum_i C^{(5)}_i\, \mathcal{O}^{(5)}_i + \frac{1}{\Lambda^2} \sum_i C^{(6)}_i\, \mathcal{O}^{(6)}_i + \frac{1}{\Lambda^3} \sum_i C^{(7)}_i\, \mathcal{O}^{(7)}_i + \cdots, \tag{40.109}$$

where $\mathcal{O}^{(n)}_i$ has mass dimension n and the Wilson coefficients $C^{(n)}_i$ are dimensionless numbers, order unity if the UV physics is strongly coupled and possibly smaller if it is weakly coupled or if the operator is generated only at loop level. The power counting is the one derived abstractly in Section 38.2 and now made

concrete: an operator of dimension n contributes to an amplitude at energy $E \ll \Lambda$ a relative correction of order

$$\frac{C_i^{(n)}}{\Lambda^{n-4}}\, E^{n-4} \;\sim\; C_i^{(n)} \Big(\frac{E}{\Lambda}\Big)^{n-4}, \tag{40.110}$$

so the leading new-physics effect from a dimension-six operator scales as E^2/Λ^2 and the dimension-five operator's effect, being odd in $1/\Lambda$, must violate one of the accidental symmetries to appear at all. Two consequences set the agenda for the rest of the section. First, the truncation of (40.109) at a given dimension is a controlled approximation only for $E \ll \Lambda$, and the dominant deviations from the Standard Model are governed by the lowest-dimension operators — dimension five (one operator) and dimension six (the bulk of the phenomenology). Second, the list "all operators of dimension n" is *not* well defined until we have dealt with the fact that distinct-looking operators can be physically identical. That is the redundancy problem, and we meet it as soon as we leave dimension four.

40.4.2 Dimension five: the Weinberg operator

The lowest rung above the renormalizable Lagrangian is a single operator, and the cleanest way to see this is to ask what a dimension-five $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant Lorentz scalar can possibly be built from. The building blocks carry mass dimensions $[H] = 1$, $[\psi] = \tfrac{3}{2}$ for any fermion, $[D_\mu] = 1$, and $[X_{\mu\nu}] = 2$ for a field strength. A Lorentz scalar requires an even number of fermions, and gauge invariance requires the hypercharges to sum to zero and the non-abelian indices to contract into singlets.

Theorem 40.12 (Uniqueness of the dimension-five operator)**.** *Up to flavour structure and Hermitian conjugation, the only dimension-five operator invariant under* $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ *and the Lorentz group, built from the Standard Model fields* (40.108) *and* H*, is the Weinberg operator*

$$\mathcal{O}_5 \;=\; (\,\overline{L^c}\,\widetilde{H}\,)(\,\widetilde{H}^{\top} L\,) \;=\; \epsilon_{ab}\,\epsilon_{cd}\,(L_a^{\top} C\, L_c)\, H_b\, H_d, \tag{40.111}$$

where a, b, c, d *are* $\mathrm{SU}(2)_L$ *indices,* ϵ *the antisymmetric symbol with* $\epsilon_{12} = +1$*, and* C *the charge-conjugation matrix (*$L^c \equiv C\bar{L}^{\top}$*). It carries lepton number* $L = +2$ *and baryon number* $B = 0$*.*

Proof. We enumerate the dimension-five field content. The total dimension five is odd, so a purely bosonic operator is impossible: gauge invariance forces Higgs fields to enter in $H^\dagger H$ pairs (or in the $B = L = 0$ singlet built from two H's, which still uses two scalars) and covariant derivatives to contract in even numbers, while a field strength contributes two, so every bosonic operator has even dimension. A four-fermion operator already has dimension six. Hence a dimension-five operator contains exactly two fermions, contributing dimension

three, and the remaining dimension two must come from bosons: either two Higgs doublets, or one derivative and one Higgs, or a single field strength (dimension two). The last option balances the dimensions but is nonetheless excluded by gauge invariance: the only Lorentz-scalar two-fermion–field-strength structure is $\bar{\psi}\,\sigma^{\mu\nu}\psi\,X_{\mu\nu}$, which cannot be made $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant for any pairing in (40.108) without an extra Higgs to balance the $\mathrm{SU}(2)_L$ and hypercharge quantum numbers — and that extra Higgs raises the operator to dimension six (it is the $\psi^2 XH$ class of the dimension-six basis).

Consider first the option "two fermions + one derivative + one Higgs." A one-derivative, two-fermion structure that is a Lorentz scalar is of the form $\bar{\psi}_1\gamma^\mu\psi_2\,D_\mu(\cdots)$ with the derivative contracted into the vector current. The current $\bar{\psi}_1\gamma^\mu\psi_2$ connects fields of *opposite* chirality, so ψ_1, ψ_2 are one left- and one right-handed field; the only dimension-one object the derivative can act on is a single H, and the hypercharges $-Y_{\psi_1} + Y_{\psi_2} + Y_H$ together with the $\mathrm{SU}(2)_L$ and colour structure cannot be made to vanish for any pairing in (40.108) (the right-handed singlets are $\mathrm{SU}(2)_L$ scalars, so the doublet index of H has nothing to contract against). This class is empty.

There remains "two fermions + two Higgs." The fermion bilinear must be a Lorentz scalar of the Majorana type, $\psi_1^{\top} C\psi_2$, since the non-derivative scalar $\bar{\psi}_1\psi_2$ again needs opposite chiralities and two extra Higgs cannot then balance hypercharge (the reader can check that $\bar{e}\,L\,H$-type contractions are dimension four, already in the Yukawa sector, and adding one more H overshoots the hypercharge balance). For the same-chirality bilinear $\psi_1^{\top} C\psi_2$ with $\psi_{1,2}$ drawn from the left-handed multiplets, colour neutrality forbids the quark doublet (a $\mathbf{3}\otimes\mathbf{3}$ bilinear is not a colour singlet), leaving only the lepton doublet: $L^{\top}CL$, which carries $L = 2$. Its hypercharge is $2Y_L = -1$, so the two Higgs must supply $+1$: two factors of H (hypercharge $+\frac{1}{2}$ each), not $\widetilde{H}$. Finally the $\mathrm{SU}(2)_L$ structure: we have four doublet indices (L, L, H, H) to contract into a singlet, and the two independent invariants are $(\epsilon\,LL)(\epsilon\,HH)$ and $(\epsilon\,LH)(\epsilon\,LH)$. But $\epsilon_{ab}H_aH_b = 0$ identically by antisymmetry, so the first vanishes and only the second survives; that is exactly (40.111). The operator is therefore unique. □

The Weinberg operator is the EFT's first and sharpest message. It is the *only* dimension-five operator, it violates lepton number by two units, and since it conserves baryon number it violates the non-anomalous combination $B - L$ by $\Delta(B - L) = -2$. As established in Section 33.4.3, $B - L$ is the unique anomaly-free global symmetry of the Standard Model, so its violation cannot be an artifact of the low-energy theory's quantum dynamics: it must be seeded by genuinely new ultraviolet physics. The Weinberg operator is the low-energy shadow of that physics.

Its physical content appears on electroweak symmetry breaking. With $\langle H\rangle = (0,\, v/\sqrt{2})^{\top}$, so $H_2 = v/\sqrt{2}$ and $H_1 = 0$, the contraction in (40.111) collapses to a Majorana mass for the left-handed neutrino. Writing the dimension-five term

in (40.109) as $\mathcal{L} \supset \frac{1}{\Lambda} C_5\, \mathcal{O}_5 + \text{h.c.}$, we have

$$\frac{C_5}{\Lambda}\,\epsilon_{ab}\epsilon_{cd}(L_a^{\top} C L_c) H_b H_d \;\xrightarrow{\langle H\rangle}\; \frac{C_5}{\Lambda}\,\frac{v^2}{2}\,\epsilon_{a2}\epsilon_{c2}(L_a^{\top} C L_c) = \frac{C_5\, v^2}{2\Lambda}\,(\nu_L^{\top} C\, \nu_L), \tag{40.112}$$

using $\epsilon_{a2} = \delta_{a1}$ and $L_1 = \nu_L$. Comparing with the standard Majorana mass term $-\frac{1}{2} m_\nu\, \nu_L^{\top} C \nu_L + \text{h.c.}$ gives

$$\boxed{\,m_\nu = -\,\frac{C_5\, v^2}{\Lambda}\,}, \tag{40.113}$$

a neutrino mass suppressed by the new-physics scale. This is the EFT statement of the seesaw mechanism, divorced from any particular UV realization: whatever generates $\mathcal{O}_5$ produces neutrino masses of order v^2/Λ. Numerically, with $v \simeq 246$ GeV, an order-one coefficient $C_5 \sim 1$ and an atmospheric-scale mass $m_\nu \sim 0.05$ eV require $\Lambda \sim v^2/m_\nu \sim 10^{15}$ GeV — the smallness of neutrino mass is, in this reading, direct evidence for a very high scale of lepton-number violation. The microscopic completion that integrates out heavy right-handed neutrinos to *produce* C_5/Λ is the type-I seesaw, treated as a UV model in Chapter 35; here it is enough that the EFT predicts the operator and its consequence without committing to that completion.

Worked Example 40.13 (The $\mathrm{SU}(2)_L$ contraction and the neutrino mass matrix)**.** We verify the index algebra of (40.111) and restore flavour. Write the two independent ways to tie four $\mathrm{SU}(2)_L$ doublet indices into a singlet from L, L, H, H. The antisymmetric invariant on a pair is ϵ_{ab}, and any singlet is a product of two such ϵ's. The distinct pairings are

$$T_1 = (\epsilon_{ac} L_a L_c)(\epsilon_{bd} H_b H_d), \qquad T_2 = (\epsilon_{ab} L_a H_b)(\epsilon_{cd} L_c H_d),$$

where in T_1 the two leptons are paired and the two Higgs are paired, while in T_2 each lepton is paired with a Higgs. Now $\epsilon_{bd} H_b H_d = 0$ by the antisymmetry of ϵ and the symmetry of $H_b H_d$ (the Higgs is a commuting field), so $T_1 \equiv 0$ and only T_2 remains — this is the source of the uniqueness in Theorem 40.12. There is a third apparent structure $T_3 = (\epsilon_{ad} L_a H_d)(\epsilon_{cb} L_c H_b)$, but relabelling shows $T_3 = T_2$, and the Schouten identity $\epsilon_{ab}\epsilon_{cd} + \epsilon_{ac}\epsilon_{db} + \epsilon_{ad}\epsilon_{bc} = 0$ (valid in two dimensions) relates T_1, T_2, T_3 and confirms that, after $T_1 = 0$, a single independent structure survives.

Restoring generation indices $p, r = 1, 2, 3$ on the leptons, the operator becomes $\mathcal{O}_5^{pr} = \epsilon_{ab}\epsilon_{cd}(L_{p,a}^{\top} C L_{r,c}) H_b H_d$. Because $L_p^{\top} C L_r$ is symmetric under $p \leftrightarrow r$ (the transpose of the 1×1 number equals itself and $C^{\top} = -C$ compensates the fermion anticommutation), the Wilson coefficient C_5^{pr} is a complex *symmetric* 3×3 matrix — six complex entries, not nine. After breaking, (40.113) promotes to a symmetric Majorana mass matrix $(m_\nu)_{pr} = -C_5^{pr} v^2/\Lambda$, diagonalized by a single unitary matrix as $m_\nu = U^{*}\,\mathrm{diag}(m_1, m_2, m_3)\, U^{\dagger}$; the unitary U is the PMNS matrix U_{PMNS} of the lepton sector. Thus the single dimension-five operator, once flavour is switched on, delivers exactly the structure of Majorana neutrino mixing.

40.4.3 The redundancy problem

At dimension six the naive list of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$-invariant operators is much longer than the number of physically distinct interactions, because four standard manipulations relate operators that differ only by terms that vanish on the physical states. None of these moves changes a single S-matrix element; all of them change the way the Lagrangian is written. A *basis* is a maximal set of operators no two of which are related by any of them. There are four sources of redundancy.

Integration by parts (IBP). Two operators differing by a total derivative are identical in the action, since $\int d^4x\, \partial_\mu(\cdots) = 0$ for fields falling off at infinity. For example $(\partial_\mu H^\dagger H)(\partial^\mu H^\dagger H)$ and $(H^\dagger H)\Box(H^\dagger H)$ differ by a total derivative and so count once. IBP is the cheapest move and is always applied first to reduce the number of explicit derivatives.

Equations of motion / field redefinitions. An operator proportional to the leading-order equation of motion of some field can be removed entirely. This is the deepest and least obvious of the moves, and it deserves a theorem.

Theorem 40.14 (Field redefinitions and the on-shell S-matrix)**.** *Let $\mathcal{L} = \mathcal{L}_0[\phi] + \dfrac{1}{\Lambda^2}\, E_0[\phi]\, F[\phi]$, where $E_0[\phi] \equiv \delta S_0/\delta\phi$ is the leading-order Euler–Lagrange expression and $F[\phi]$ is any local operator of the appropriate dimension and quantum numbers. The local field redefinition $\phi \to \phi' = \phi - \dfrac{1}{\Lambda^2} F[\phi]$ removes the operator $E_0 F$ from the Lagrangian, replacing it by terms of order $1/\Lambda^4$, and leaves every S-matrix element unchanged. Consequently any dimension-six operator proportional to a leading-order equation of motion is redundant: it may be traded for higher-dimension operators and for lower-dimension operators generated by the redefinition.*

Proof. Expand the action under the shift to first order in $1/\Lambda^2$:

$$S[\phi'] = S_0[\phi] - \frac{1}{\Lambda^2}\int d^4x\, \frac{\delta S_0}{\delta\phi}\, F[\phi] + \frac{1}{\Lambda^2}\int d^4x\, E_0[\phi]\, F[\phi] + O(1/\Lambda^4).$$

The second term is $-\frac{1}{\Lambda^2}\int E_0 F$ by definition of E_0, and it cancels the inserted operator $+\frac{1}{\Lambda^2}\int E_0 F$ identically; what is left is S_0 plus genuinely new $O(1/\Lambda^4)$ terms. So at the order we work, the operator $E_0 F$ disappears.

That the S-matrix is unchanged is the statement that a local change of integration variable cannot change the value of the generating functional. In the path integral $Z[J] = \int \mathcal{D}\phi\; e^{iS[\phi] + i\int J\phi}$, substitute $\phi = \phi' + \frac{1}{\Lambda^2} F[\phi']$. The measure picks up a Jacobian $\det(1 + \frac{1}{\Lambda^2}\delta F/\delta\phi') = \exp\big[\frac{1}{\Lambda^2}\int d^4x\, \frac{\delta F}{\delta\phi'}(x)\, \delta^{(4)}(0)\big]$,

whose exponent is a scaleless integral $\propto \delta^{(4)}(0)$ that vanishes in dimensional regularization (Section 38.2.4); the source term becomes $\int J(\phi' + \frac{1}{\Lambda^2}F[\phi'])$. The added piece $\frac{1}{\Lambda^2}\int JF[\phi']$ is a contact term: it modifies off-shell Green's functions but contributes no pole at the physical single-particle mass, so by the LSZ reduction it drops out of every on-shell amplitude. Hence Z, and with it all S-matrix elements, are invariant, while the Lagrangian written in terms of ϕ' no longer contains $E_0 F$. □

The practical content is that the leading-order equations of motion may be used *inside* dimension-six operators to eliminate redundant structures, the errors being pushed to dimension eight. The Higgs equation of motion that we will need follows from $\mathcal{L}_{\rm SM}$ by varying $H^\dagger$,

$$D^2 H_k = \mu^2 H_k - 2\lambda\,(H^\dagger H)\,H_k - \Big[\,\bar{e}\,Y_e^\dagger L_k + \bar{d}\,Y_d^\dagger Q_k + \epsilon_{kl}\,\bar{Q}_l\,Y_u\,u\,\Big], \quad (40.114)$$

with k, l the $\mathrm{SU}(2)_L$ indices and $Y_{e,d,u}$ the Yukawa matrices; the fermionic source on the right is what converts a derivative operator into the Yukawa-like class $\psi^2 H^3$ when the EOM is applied. Analogous EOMs for the fermions ($i\not{D}L = Y_e e\,H$, etc.) and for the gauge fields ($D^\nu W^I_{\mu\nu} = g\,(\,\text{currents}\,)$) eliminate operators with explicit $\not{D}\psi$ or $D^\nu X_{\mu\nu}$.

Fierz identities. For four-fermion operators, the spinor and gauge indices can be rearranged among the four fields. The chiral Fierz identity

$$(\bar{\psi}_{1L}\gamma^\mu\psi_{2L})(\bar{\psi}_{3L}\gamma_\mu\psi_{4L}) = (\bar{\psi}_{1L}\gamma^\mu\psi_{4L})(\bar{\psi}_{3L}\gamma_\mu\psi_{2L}) \quad (40.115)$$

(valid for left-chiral currents, with the same identity for right-chiral ones) shows that two superficially different orderings of the same four left-handed fields are equal, so only one need be kept. A companion colour Fierz identity, $T^A_{ij}T^A_{kl} = \frac{1}{2}\delta_{il}\delta_{kj} - \frac{1}{2N}\delta_{ij}\delta_{kl}$ for $\mathrm{SU}(N)$, relates colour-octet to colour-singlet contractions. Together these collapse the apparent profusion of four-fermion structures to a finite independent set.

Symmetry of identical fields. Operators that differ only by the ordering of identical bosonic fields or by an overall relabelling are the same operator; this is bookkeeping rather than dynamics, but it must be done consistently to avoid double counting.

A redundant operator is one that any combination of these four moves reduces to operators already in the basis. The next worked example carries such a reduction out in full, exhibiting concretely the claim that a redundant operator is on-shell equal to a combination of retained ones.

Worked Example 40.15 (Reducing a redundant dimension-six operator)**.** Consider the Hermitian dimension-six operator

$$\mathcal{R} \;=\; (H^\dagger H)\,\big[\,H^\dagger (D^2 H) + (D^2 H)^\dagger H\,\big],$$

which contains an explicit $D^2 H$ and so is a candidate for elimination by the Higgs equation of motion. We reduce it two ways, by IBP and by the EOM, and confront the results.

By integration by parts. Since $H^\dagger H$ is a gauge singlet, $\partial_\mu (H^\dagger H) = D_\mu (H^\dagger H) = (D_\mu H)^\dagger H + H^\dagger D_\mu H$, and differentiating once more,

$$\Box (H^\dagger H) = (D^2 H)^\dagger H + H^\dagger D^2 H + 2\,(D_\mu H)^\dagger (D^\mu H).$$

Solving for the bracket in $\mathcal{R}$,

$$\mathcal{R} = (H^\dagger H)\,\Box (H^\dagger H) - 2\,(H^\dagger H)\,(D_\mu H)^\dagger (D^\mu H) = \mathcal{O}_{H\Box} - 2\,(H^\dagger H)(D_\mu H)^\dagger (D^\mu H), \tag{40.116}$$

where $\mathcal{O}_{H\Box} \equiv (H^\dagger H)\Box (H^\dagger H)$ is a Warsaw-basis operator. The IBP move alone has reduced $\mathcal{R}$ to retained bosonic operators plus the structure $(H^\dagger H)(D_\mu H)^\dagger (D^\mu H)$, which is itself reducible.

By the equation of motion. Substitute (40.114) directly. Using $H_k^\dagger (\bar e\, Y_e^\dagger L_k) = \bar e\, Y_e^\dagger (H^\dagger L)$ and $H_k^\dagger\, \epsilon_{kl}\, \bar Q_l Y_u u = -\,\bar Q\, Y_u u\, \widetilde H$ (from $\widetilde H_l = \epsilon_{lk} H_k^*$), we find

$$H^\dagger D^2 H = \mu^2 (H^\dagger H) - 2\lambda (H^\dagger H)^2 - \bar e\, Y_e^\dagger (H^\dagger L) - \bar d\, Y_d^\dagger (H^\dagger Q) + \bar Q\, Y_u u\, \widetilde H,$$

and adding the Hermitian conjugate and multiplying by $H^\dagger H$,

$$\mathcal{R} \overset{\text{EOM}}{=} 2\mu^2 (H^\dagger H)^2 - 4\lambda\, \mathcal{O}_H - (H^\dagger H)\Big[\,\bar e\, Y_e^\dagger (H^\dagger L) + \bar d\, Y_d^\dagger (H^\dagger Q) - \bar Q\, Y_u u\, \widetilde H + \text{h.c.}\,\Big], \tag{40.117}$$

with $\mathcal{O}_H \equiv (H^\dagger H)^3$. Every term on the right is either renormalizable or a retained Warsaw operator: the first term $2\mu^2 (H^\dagger H)^2$ is a dimension-four contribution absorbed into a redefinition of the quartic coupling λ; the second is the H^6 operator $\mathcal{O}_H$; and the bracketed pieces are exactly the $\psi^2 H^3$ operators $\mathcal{O}_{eH} = (H^\dagger H)(\bar L\, e\, H)$, $\mathcal{O}_{dH} = (H^\dagger H)(\bar Q\, d\, H)$, $\mathcal{O}_{uH} = (H^\dagger H)(\bar Q\, u\, \widetilde H)$ and their conjugates. Thus

$$\mathcal{R} \overset{\text{EOM}}{=} \underbrace{2\mu^2 (H^\dagger H)^2}_{\text{into } \lambda} - 4\lambda\, \mathcal{O}_H - \big[\, Y_e^\dagger\, \mathcal{O}_{eH}^\dagger + Y_d^\dagger\, \mathcal{O}_{dH}^\dagger - Y_u\, \mathcal{O}_{uH} + \text{h.c.}\,\big].$$

The operator $\mathcal{R}$ is therefore *not* an independent interaction: on shell it is identical to a specific linear combination of the retained operators $\{\mathcal{O}_H, \mathcal{O}_{eH}, \mathcal{O}_{dH}, \mathcal{O}_{uH}\}$ plus a shift of a renormalizable coupling. Comparing with the IBP result (40.116) also resolves the leftover bosonic structure: equating the two expressions for $\mathcal{R}$ shows that $(H^\dagger H)(D_\mu H)^\dagger (D^\mu H)$ is itself a combination of $\mathcal{O}_{H\Box}$, $\mathcal{O}_H$, the $\psi^2 H^3$ operators, and a coupling redefinition — which is precisely why the Warsaw basis keeps $\mathcal{O}_{H\Box}$ and the operator $\mathcal{O}_{HD} = (H^\dagger D_\mu H)^* (H^\dagger D^\mu H)$ but not $(H^\dagger H)(D_\mu H)^\dagger (D^\mu H)$. This is the redundancy problem in miniature: three different-looking derivative operators are one independent interaction.

40.4.4 Dimension six: the operator classes and the counting

With the redundancy moves in hand we can ask, cleanly, how many independent dimension-six operators there are. The systematic way to organize the answer is by the field content of the operator, since the dimension budget — six units from $[H]=1$, $[\psi]=\frac{3}{2}$, $[D]=1$, $[X]=2$ — admits only a finite number of multiplicity patterns. Writing X for any field strength, H for the Higgs (or $H^\dagger$), ψ for any fermion, and D for a covariant derivative, the patterns summing to dimension six with an even number of fermions are

$$X^3, \quad H^6, \quad H^4D^2, \quad X^2H^2, \quad \psi^2H^3, \quad \psi^2XH, \quad \psi^2H^2D, \quad \psi^4,$$

the eight classes of the dimension-six SMEFT. (One checks that no other pattern balances: ψ^2D^3 and ψ^2XD reduce by the fermion EOM, X^2D^2 and XH^2D^2 reduce by IBP and the gauge EOM, and H^4D, ψ^2HD are forbidden by Lorentz or gauge invariance.) Within each class, gauge invariance, IBP, the EOMs, and Fierz identities cut the operators down to the canonical Warsaw set of Grzadkowski, Iskrzyński, Misiak and Rosiek [240]. The complete one-generation, baryon-number-conserving enumeration is given in Table 40.1 (bosonic and mixed classes) and Table 40.2 (four-fermion class). They total

$$\underbrace{4}_{X^3}+\underbrace{1}_{H^6}+\underbrace{2}_{H^4D^2}+\underbrace{8}_{X^2H^2}+\underbrace{3}_{\psi^2H^3}+\underbrace{8}_{\psi^2XH}+\underbrace{8}_{\psi^2H^2D}+\underbrace{25}_{\psi^4}=59 \tag{40.118}$$

independent operators, the celebrated "59 of the Warsaw basis."

A few structural features of the tables repay comment. The X^3 class contains only the non-abelian field strengths G and W: the abelian $\epsilon\,BBB$ contraction vanishes identically, so there is no B^3 operator. The X^2H^2 class is the home of the Higgs–gauge couplings that govern Higgs production and decay into gauge bosons, and the tilde members carry CP violation. The ψ^2XH class generates anomalous magnetic and electric dipole moments once the Higgs takes its vacuum value — the operator $\mathcal{O}_{eB}$, for instance, feeds directly into the electron's $g-2$ and electric dipole moment, the very quantities that QED computed in the pure Standard Model (Chapter 20); here they receive UV-sensitive corrections of order $v\,m_e/\Lambda^2$. The ψ^2H^2D class modifies the fermion couplings to the Z and W measured at the Z pole. And the four-fermion class contains the operators that, after the next section integrates out the W, will reproduce and generalize Fermi's contact interaction.

Baryon-number violation. The 59 operators of (40.118) all conserve baryon number. Just as the Weinberg operator violated lepton number at dimension five, dimension six is the first place baryon number can break: there are exactly four independent baryon-number-violating operators (one generation), each carrying $\Delta B=\Delta L=1$,

$$\mathcal{O}_{duq},\ \mathcal{O}_{qqu},\ \mathcal{O}_{qqq},\ \mathcal{O}_{duu},$$

schematically of the form $\epsilon_{ABC}(\bar{\psi}^c\psi)(\bar{\psi}^c\psi)$ with the colour epsilon tying three quarks (and one lepton) into a singlet. These operators mediate proton decay, $p \to e^+\pi^0$ and the like, at a rate $\Gamma \sim m_p^5/\Lambda^4$; the experimental bound $\tau_p \gtrsim 10^{34}$ yr pushes the corresponding scale to $\Lambda \gtrsim 10^{15}$ GeV, comparable to the scale read off from neutrino mass. Because B and L are violated together, $B - L$ is again broken by these operators, consistent with the anomaly analysis of Section 33.4.3. The full baryon-number-conserving plus baryon-number-violating dimension-six list is thus $59 + 4 = 63$ operators for one generation, and the dimension-six Lagrangian is

$$\mathcal{L}^{(6)} = \frac{1}{\Lambda^2}\sum_{i=1}^{63} C_i\,\mathcal{O}_i. \tag{40.119}$$

Counting with flavour and the Hilbert series. Switching on three generations multiplies the counting in a non-trivial way, because the Wilson coefficients become tensors in flavour space whose index symmetries differ operator by operator (the symmetric neutrino-mass example of Theorem 40.13 is the dimension-five instance). Carrying out the bookkeeping operator by operator, the dimension-six Lagrangian for three generations contains 2499 independent real parameters (Hermitian operators contributing real coefficients, non-Hermitian ones complex), of which 1350 preserve CP and 1149 violate it [20]. These numbers were obtained independently, as a powerful check, by the *Hilbert-series* method, which counts gauge-invariant operators modulo EOM and IBP redundancy by integrating a generating function (a "plethystic" character integral) over the gauge group and the Lorentz group [265]. The Hilbert series reproduces the 59 ($n_f = 1$) and 2499 ($n_f = 3$) counts and extends them to all higher dimensions; we cite it here as the verification of an enumeration we have, in its lowest classes, carried out by hand. The agreement between the explicit Warsaw construction and the abstract Hilbert-series count is the assurance that the basis is neither over- nor under-complete.

40.4.5 The Warsaw basis and the invariance of observables

The choice of which operators to retain after the redundancy moves is not unique. One could keep $\mathcal{O}_{H\Box}$ and $\mathcal{O}_{HD}$, as the Warsaw basis does, or trade them via the EOM for operators with explicit gauge-boson content, as the so-called SILH or "strongly-interacting light Higgs" basis does. Both are legitimate bases of the same 59-dimensional space of independent dimension-six interactions; they are related by an invertible linear transformation $C_i = \sum_j M_{ij}\,C'_j$ whose matrix M is read off from the EOM and IBP identities (exactly the kind of identity derived in Theorem 40.15). The essential principle is that a *physical observable cannot depend on the basis*: it is computed from on-shell amplitudes, which Theorem 40.14 guarantees are invariant under the field redefinitions that connect

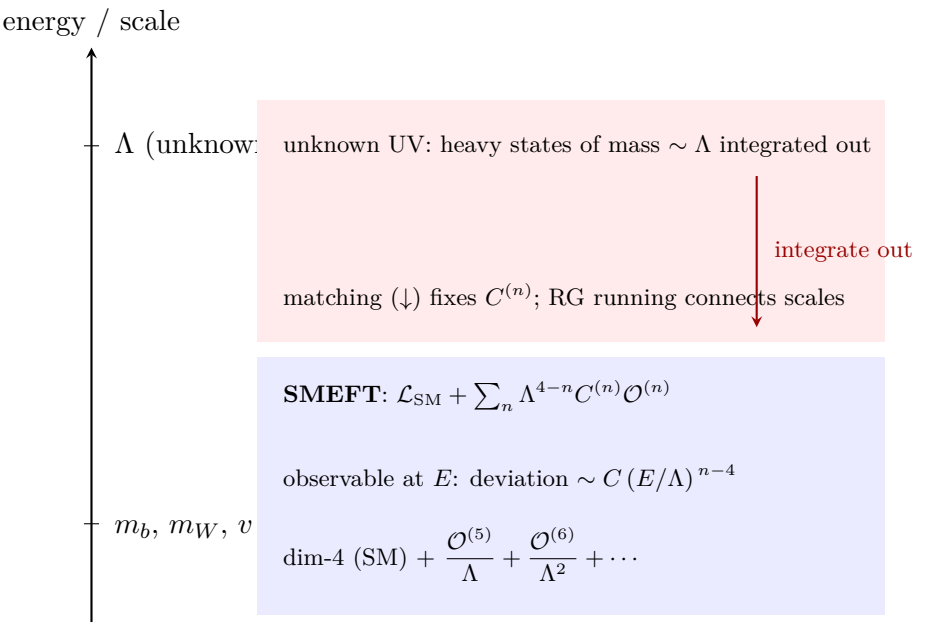


Figure 40.5: The SMEFT tower. Above the scale Λ sits unknown heavy physics; integrating it out leaves the renormalizable Standard Model plus the operator expansion (40.109), suppressed by powers of $1/\Lambda$. A measurement at energy $E \ll \Lambda$ probes the deviation $C\,(E/\Lambda)^{\,n-4}$ and so bounds a combination of Wilson coefficients — equivalently, a scale of new physics.

bases. If a prediction for a cross section appeared to depend on whether one used $\mathcal{O}_{H\Box}$ or its EOM-image, that would signal an algebra error, not a physical ambiguity. Concretely, an observable depends only on those linear combinations of Wilson coefficients that are basis-invariant, and matching a measurement to "new physics" means bounding such a combination, hence a combination of UV parameters.

Physical Insight

The Warsaw basis is not sacred; the 59-dimensional space it spans is. A basis is a coordinate system on the space of dimension-six deformations of the Standard Model, and physics lives in the coordinate-free object — the deformation itself. Choosing the Warsaw basis is like choosing Cartesian axes: convenient, conventional, and irrelevant to any answer. The practical payoff of a *fixed* basis is that the renormalization-group equations, the matching formulae, and the global fits can all be tabulated once and shared, which is why the community standardized on Warsaw.

The picture that emerges is summarized in Fig. 40.5. The Standard Model

occupies the renormalizable floor; above it, organized by powers of $1/\Lambda$, sits a tower whose first storey (dimension five) is the unique Weinberg operator and whose second storey (dimension six) is the $59 + 4$-operator Warsaw basis. Each Wilson coefficient is a window onto the unknown UV: a measurement at energy E constrains $C\,(E/\Lambda)^{n-4}$, and a null result bounds Λ from below for order-one coefficients. This is the quantitative content of "the Standard Model as an effective theory."

With the operator set fixed and non-redundant, the Wilson coefficients C_i are now free parameters awaiting determination. The remaining task of the chapter is to give them values: the next section integrates out a *known* heavy state — first the W boson, recovering Fermi theory as the prototype, then a hypothetical heavy particle — to fix the C_i by matching at tree level, distinguishing the tree- from the loop-generated coefficients, and so turning the empty basis built here into a concrete bridge from short-distance dynamics to low-energy measurement.

40.5 Matching: from heavy new physics to Wilson coefficients

The previous section delivered a Lagrangian but not a theory. We assembled the non-redundant tower of higher-dimension operators built from the Standard-Model fields and the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, fixed a basis (the Warsaw basis, Section 40.4.5), and wrote

$$\mathcal{L}_{\mathrm{SMEFT}} = \mathcal{L}_{\mathrm{SM}} + \sum_i \frac{C_i^{(6)}}{\Lambda^2}\, Q_i^{(6)} + \sum_i \frac{C_i^{(8)}}{\Lambda^4}\, Q_i^{(8)} + \cdots , \tag{40.120}$$

with Λ the scale of whatever heavy physics lies above the electroweak scale and the C_i dimensionless Wilson coefficients (Eq. (40.120)). So far those coefficients are free numbers. Two things can fix them. If the short-distance theory is *unknown*, the coefficients are measured: each observable bounds a combination of C_i/Λ^2, and a nonzero result would point to new physics. If the short-distance theory is *known*—a definite extension of the Standard Model with definite heavy fields and couplings—then the C_i are *computed*, by demanding that the EFT reproduce the full theory at low energy. That computation is *matching*, the subject of this section. We first isolate the matching principle in the abstract, importing the machinery built in Section 38.3; then we run it on the cleanest prototype in particle physics, integrating out the W boson to recover Fermi theory and the value of G_F; then we match an explicit beyond-the-Standard-Model (BSM) field at tree level and read off its dimension-six coefficients; and finally we draw the line between operators generated at tree level and those generated only at one loop, because that line controls how a measured coefficient is read back as a statement about the ultraviolet.

40.5.1 The matching principle

Matching is the statement that two descriptions of the same physics must agree wherever both are valid. The full theory contains a heavy field of mass $M \sim \Lambda$ and the light Standard-Model fields; the EFT contains only the light fields, with the heavy one removed and its effects encoded in local operators. Both are supposed to compute the same low-energy amplitudes. They cannot agree exactly—the EFT has thrown away the heavy particle, and so loses, for instance, the resonance peak when an external invariant hits M^2—but they must agree *order by order in the low-energy expansion*, i.e. in powers of E/M and in powers of the loop expansion.

Definition 40.16 (Matching condition)**.** Let $\mathcal{A}_{\text{full}}$ and $\mathcal{A}_{\text{EFT}}$ denote a fixed set of one-particle-irreducible (1PI) light-field Green's functions, computed in the full theory and in the EFT respectively, renormalized in the same scheme ($\overline{\text{MS}}$, $d = 4 - 2\varepsilon$) at a common scale μ. *Matching at the scale* μ is the requirement

$$\mathcal{A}_{\text{full}}(p_i;\mu) \;=\; \mathcal{A}_{\text{EFT}}(p_i;\mu) \qquad \text{to the working order in } 1/M \text{ and in } \hbar, \tag{40.121}$$

imposed for all light external momenta p_i. The Wilson coefficients $C_i(\mu)$ are the unique local couplings that make Eq. (40.121) hold.

Three points make this definition operational. First, one matches *1PI* amplitudes (or off-shell Green's functions), not on-shell S-matrix elements: the EFT must reproduce the full theory as an off-shell field theory, not merely on a particular mass shell, so that the same coefficients work in every process. Second, one matches at $\mu \sim M$. The reason is the structure of the two sides. Each side, expanded in $\overline{\text{MS}}$, contains logarithms; in the full theory these are $\ln(\mu^2/M^2)$ and $\ln(\mu^2/p^2)$, in the EFT only $\ln(\mu^2/p^2)$, because the EFT has no heavy scale. Choosing $\mu \sim M$ kills the large $\ln(\mu/M)$ on the full side, so the matching coefficients are free of large logs and the perturbative expansion of the C_i is well behaved. The logarithms $\ln(M/p)$ that the EFT *should* contain are not lost—they are regenerated by running the C_i down from $\mu \sim M$ to $\mu \sim p$, which is the business of the next section.

Third—and this is the conceptual heart—the difference $\mathcal{A}_{\text{full}} - \mathcal{A}_{\text{EFT}}$ before matching is a *local* function of the external momenta, i.e. a polynomial in p_i, with no non-analytic pieces (no logarithms of p^2, no branch cuts in the light-particle channels). This is what makes the matching consistent: a local difference can be absorbed into local operators. We record the statement because the whole method rests on it.

Proposition 40.17 (Locality of the matching difference)**.** *At any fixed order in* $1/M$ *and in the loop expansion, the difference between a light-field 1PI Green's function computed in the full theory and the same function computed in the EFT (with all operators up to that order in* $1/M$ *included, and the same light-particle*

content and renormalization scheme) is a polynomial in the external momenta and light masses, of bounded degree. Equivalently, all non-analytic dependence on the external momenta is identical on the two sides and cancels in the difference.

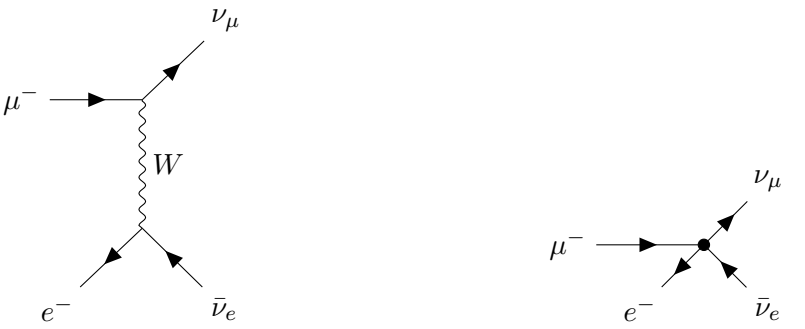
Proof. Non-analyticity in an external invariant p^2 can only arise from a region of loop integration where internal lines go on shell at that invariant—a physical threshold or a long-distance (infrared) region. By construction the EFT contains exactly the light fields of the full theory, with identical quantum numbers, couplings and masses through the order worked; therefore every internal line that can go on shell in the kinematic range $p^2 \ll M^2$ is present in both theories, and the two infrared/threshold structures coincide. The only loop regions present in the full theory but absent from the EFT are those in which a *heavy* line carries the loop momentum, $\ell \sim M$. In such a region the heavy propagator $1/(\ell^2 - M^2)$ is far off shell for all light external momenta, and the integrand may be Taylor-expanded in p_i/M and in light masses before integration (this is the hard region of the method of regions, Section 38.3). The result of integrating a polynomial-in-p integrand is a polynomial in p times heavy-scale integrals—a local result. Hence the full–EFT difference retains only the heavy-region contribution, which is local. The degree of the polynomial is bounded because each extra power of p^2/M^2 is an extra power of the suppression, i.e. a higher-dimension operator, truncated at the working order. □

Theorem 40.17 is the precise content of the slogan "heavy physics is local at low energy." It is also the precise content of decoupling, to which we return in Section 40.5.4: the heavy field never appears as an external state below M, and its virtual effects are a tower of local operators suppressed by powers of $1/M$, plus shifts of the renormalizable couplings already present. The Wilson coefficients are nothing but the coefficients of that local polynomial.

40.5.2 The prototype: integrating out the W and the value of G_F

The original effective field theory of particle physics was Fermi's 1933 theory of beta decay, written down decades before anyone knew it was an EFT. It is the perfect prototype because the full theory—the electroweak Standard Model—is known exactly, the heavy field is the W boson, and the matching is a one-line tree-level calculation. We carry it out diagrammatically here; the complementary route, in which the W is eliminated by its classical equation of motion at the level of the Lagrangian, was done in Theorem 31.19, and the two must agree.

Recall from Section 31.2.8 the electroweak Lagrangian Eq. (31.50), and from Section 31.5.2 the charged-current interaction by which the W couples to fermions. Below the W mass the only role of the W is to mediate four-fermion processes such as muon decay $\mu^- \to e^- \,\bar{\nu}_e \,\nu_\mu$, the process that defines G_F. The full-theory



(a) Full theory: W exchange. (b) EFT: four-fermion contact operator.

Figure 40.6: Matching the W out. At momentum transfer $q^2 \ll M_W^2$ the propagator of the heavy mediator (a) shrinks to a point, leaving a local four-fermion vertex (b). Equating the two amplitudes fixes the Wilson coefficient of the contact operator; the result is the Fermi constant, Eq. (40.127). The same picture, with the W replaced by any heavy mediator of mass M, is the generic tree-level matching of Section 40.5.3.

tree amplitude is a single W exchange. Removing the W from the spectrum, the EFT that reproduces this amplitude at low energy is a contact interaction of four fermions—a dimension-six operator. Matching the two fixes its coefficient.

Worked Example 40.18 (The W-exchange amplitude and the Fermi constant)**.** Write $P_L = \frac{1}{2}(1-\gamma^5)$ for the left-handed projector. The charged-current coupling of Eq. (31.129) attaches to each fermion line a vertex factor $i\,(g/\sqrt{2})\,\gamma^\mu P_L$, with g the $\mathrm{SU}(2)_L$ gauge coupling. The tree amplitude for $\mu^-(p) \to \nu_\mu(p')\, e^-(k)\, \bar{\nu}_e(k')$ from Fig. 40.6a is the product of two such vertices joined by the W propagator carrying momentum $q = p - p'$,

$$i\mathcal{M}_{\text{full}} = \Big[\bar{u}_{\nu_\mu}\, \tfrac{ig}{\sqrt{2}}\gamma^\mu P_L\, u_\mu\Big]\, \frac{-i(\eta_{\mu\nu} - q_\mu q_\nu/M_W^2)}{q^2 - M_W^2}\, \Big[\bar{u}_e\, \tfrac{ig}{\sqrt{2}}\gamma^\nu P_L\, v_{\bar{\nu}_e}\Big], \qquad (40.122)$$

where we used the unitary-gauge W propagator; in R_ξ gauge the ξ-dependent piece of the propagator is itself a contact term that cancels against the would-be Goldstone exchange, so the physical result is gauge independent and we may as well use the form Eq. (40.122).

The low-energy expansion. The defining feature of the kinematic regime is $q^2 \sim m_\mu^2 \ll M_W^2$. Expand the propagator denominator,

$$\frac{1}{q^2 - M_W^2} = -\frac{1}{M_W^2}\,\frac{1}{1 - q^2/M_W^2} = -\frac{1}{M_W^2}\Big(1 + \frac{q^2}{M_W^2} + \frac{q^4}{M_W^4} + \cdots\Big). \qquad (40.123)$$

The $q_\mu q_\nu/M_W^2$ term contracts into the currents as $q_\mu\, \bar{u}_{\nu_\mu}\gamma^\mu P_L u_\mu$; using the Dirac equations for the external spinors, $q_\mu\gamma^\mu = \not{p} - \not{p}'$ acting between $\bar{u}_{\nu_\mu}$ and u_μ

returns a factor of the lepton masses, so this term is suppressed by $m^2_{\text{lepton}}/M_W^2$ relative to the leading one and is dropped at the order we work. Keeping the leading term of Eq. (40.123),

$$i\mathcal{M}_{\text{full}} = \left(\tfrac{ig}{\sqrt{2}}\right)^2\left(\frac{-i}{-M_W^2}\right)[\bar{u}_{\nu_\mu}\gamma^\mu P_L\, u_\mu][\bar{u}_e\gamma_\mu P_L\, v_{\bar{\nu}_e}] = -\frac{ig^2}{2M_W^2}[\bar{u}_{\nu_\mu}\gamma^\mu P_L\, u_\mu][\bar{u}_e\gamma_\mu P_L\, v_{\bar{\nu}_e}], \tag{40.124}$$

where $(ig/\sqrt{2})^2 = -g^2/2$ and $(-i)/(-M_W^2) = i/M_W^2$ combine to the displayed prefactor.

The EFT side. The contact operator that reproduces Fig. 40.6b is the charged-current four-fermion operator. Written in the historical $(1-\gamma^5)$ normalization of Fermi theory (Eq. (31.3)),

$$\mathcal{L}_{\text{Fermi}} = -\frac{G_F}{\sqrt{2}}[\bar{\psi}_{\nu_\mu}\gamma^\mu(1-\gamma^5)\psi_\mu][\bar{\psi}_e\gamma_\mu(1-\gamma^5)\psi_{\nu_e}]. \tag{40.125}$$

Since $1-\gamma^5 = 2P_L$, each current carries a factor of 2, and the tree amplitude generated by Eq. (40.125) is

$$i\mathcal{M}_{\text{EFT}} = -\frac{iG_F}{\sqrt{2}}\cdot 2\cdot 2\,[\bar{u}_{\nu_\mu}\gamma^\mu P_L\, u_\mu][\bar{u}_e\gamma_\mu P_L\, v_{\bar{\nu}_e}] = -\frac{4iG_F}{\sqrt{2}}[\bar{u}_{\nu_\mu}\gamma^\mu P_L\, u_\mu][\bar{u}_e\gamma_\mu P_L\, v_{\bar{\nu}_e}]. \tag{40.126}$$

Matching. The two amplitudes Eqs. (40.124) and (40.126) carry the identical spinor structure—this is the content of Theorem 40.17 in action: the heavy-line region produced exactly the local operator the EFT contains. Equating their coefficients,

$$\frac{4G_F}{\sqrt{2}} = \frac{g^2}{2M_W^2} \qquad \Longrightarrow \qquad \boxed{\frac{G_F}{\sqrt{2}} = \frac{g^2}{8M_W^2}} \tag{40.127}$$

which is the Fermi constant predicted by the electroweak theory, reproducing Eq. (31.145) by the diagrammatic route. With $M_W = \tfrac{1}{2}gv$ from the Higgs mechanism (Section 31.3) this collapses to $G_F/\sqrt{2} = 1/2v^2$, fixing the electroweak scale $v \simeq 246\,\text{GeV}$; that step was carried out in Theorem 31.19 and we do not repeat it.

The tower. Restoring the next term of Eq. (40.123), the q^2/M_W^2 piece multiplies the same current bilinear by q^2. In position space $q^2 \to -\Box$ acting on the contact operator, so it is a four-fermion operator with two extra derivatives, of mass dimension eight, with coefficient

$$\frac{g^2}{2M_W^4}[\bar{\psi}_{\nu_\mu}\gamma^\mu P_L \overleftrightarrow{\Box}\psi_\mu][\bar{\psi}_e\gamma_\mu P_L\,\psi_{\nu_e}] + \cdots, \tag{40.128}$$

schematically. Each higher term in the expansion is a higher-dimension operator suppressed by an additional power of $1/M_W^2$. This is the operator tower of Eq. (40.120) made completely explicit: integrating out a single heavy field

produces, not one operator, but the entire $1/M$ expansion, and the dimension-six term is simply its leading entry.

Two lessons generalize beyond this example and are worth stating plainly. First, Fermi theory is the Standard Model *as its own EFT*: integrating out the heavy electroweak states (W, Z, Higgs, top) below their masses gives a low-energy effective theory of the light quarks and leptons, of which Fermi's interaction is one operator. This is the same logical move as SMEFT, only the "heavy" physics is inside the Standard Model rather than above it. Second, the calculation was a pure tree graph, and it generated a four-fermion operator. The two facts are linked: *which* operators a heavy field generates, and at *what* order in the loop expansion, is dictated by how that field couples. We now make both points sharp by matching an explicit new heavy field, and then by separating tree- from loop-generated operators.

40.5.3 Tree-level matching of a heavy BSM field

The prototype matched a known heavy field. The same procedure, applied to a *hypothetical* heavy field, is how a concrete BSM model is translated into SMEFT Wilson coefficients—the practical content of "model X predicts deviations in observable Y." We work the cleanest nontrivial example: a heavy real scalar that is a singlet under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, coupled to the Higgs. It produces purely bosonic dimension-six operators, and the elimination of the heavy field by its equation of motion—the Lagrangian face of matching—exposes the derivative expansion in a way the four-fermion examples do not.

Worked Example 40.19 (A heavy scalar singlet and the operators Q_H, $Q_{H\Box}$)**.** Let S be a real scalar, a singlet under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with mass $M \sim \Lambda$ and the renormalizable Lagrangian

$$\mathcal{L}_S = \tfrac{1}{2}(\partial_\mu S)(\partial^\mu S) - \tfrac{1}{2}M^2 S^2 - g_1 M\, S\,(H^\dagger H) - \tfrac{1}{2} g_2\, S^2\,(H^\dagger H), \qquad (40.129)$$

where H is the Standard-Model Higgs doublet and g_1, g_2 are dimensionless (the explicit M in the linear coupling makes g_1 dimensionless, since $[S\,H^\dagger H] = 3$). We have dropped S's own cubic and quartic self-couplings: they do not contribute to the operators of interest at tree level and at leading order in $1/M^2$. We integrate S out at tree level. Tree-level matching is implemented by solving the classical equation of motion for S as a power series in $1/M^2$ and substituting back; this is the saddle point of the path integral over S, exact at tree level (Section 38.3).

The equation of motion. Varying Eq. (40.129) with respect to S,

$$(\Box + M^2 + g_2\, H^\dagger H)\, S = -\, g_1 M\, (H^\dagger H). \qquad (40.130)$$

Invert the operator on the left as a geometric series in $1/M^2$,

$$S = -\,\frac{g_1 M\,(H^\dagger H)}{M^2\big(1 + (\Box + g_2 H^\dagger H)/M^2\big)}$$

$$= -\frac{g_1}{M}(H^\dagger H) + \frac{g_1}{M^3}\,\Box(H^\dagger H) + \frac{g_1 g_2}{M^3}\,(H^\dagger H)^2 + O(M^{-5}). \tag{40.131}$$

The leading term is $O(1/M)$; each correction carries an extra $1/M^2$ and either an extra derivative ($\Box$) or an extra power of the field ($g_2 H^\dagger H$), exactly the two ways the EFT expansion can grow.

Substituting back. Because Eq. (40.129) is quadratic in S, the effective Lagrangian is most cleanly obtained from the standard Gaussian result: for $\mathcal{L} = -\frac{1}{2} S\,\mathcal{K}\,S - J\,S$ with $\mathcal{K} = \Box + M^2 + g_2 H^\dagger H$ and $J = g_1 M\,(H^\dagger H)$ (the kinetic term integrated by parts), the solution is $S = -\mathcal{K}^{-1} J$ and

$$\mathcal{L}_{\text{eff}} = -\tfrac{1}{2} S\,\mathcal{K}\,S - J\,S\Big|_{S=-\mathcal{K}^{-1}J} = \tfrac{1}{2}\,J\,\mathcal{K}^{-1}\,J. \tag{40.132}$$

Expanding $\mathcal{K}^{-1} = M^{-2}\big[1-(\Box+g_2 H^\dagger H)/M^2+\cdots\big]$ and inserting $J = g_1 M(H^\dagger H)$,

$$\begin{aligned}\mathcal{L}_{\text{eff}} &= \tfrac{1}{2}\,g_1^2 M^2\,(H^\dagger H)\,\frac{1}{M^2}\Big[1 - \frac{\Box + g_2 H^\dagger H}{M^2} + \cdots\Big]\,(H^\dagger H) \\ &= \underbrace{\tfrac{1}{2} g_1^2\,(H^\dagger H)^2}_{\text{dimension 4}} \;-\; \underbrace{\frac{g_1^2}{2M^2}\,(H^\dagger H)\,\Box(H^\dagger H)}_{\text{dimension 6: } Q_{H\Box}} \;-\; \underbrace{\frac{g_1^2 g_2}{2M^2}\,(H^\dagger H)^3}_{\text{dimension 6: } Q_H} + O(M^{-4}).\end{aligned} \tag{40.133}$$

Reading off the coefficients. The first term is dimension four. It is not a new operator: it has the form of the Higgs quartic, so it is a finite *threshold correction* to the Standard-Model coupling $\mathcal{L}_{\text{SM}} \supset -\lambda(H^\dagger H)^2$, shifting $\lambda \to \lambda - \frac{1}{2} g_1^2$. This is exactly the part of the heavy physics that decoupling permits to remain unsuppressed: it renormalizes a coupling already present, rather than appearing as a $1/M^2$ effect. The remaining two terms are the genuine dimension-six output. Recalling the Warsaw-basis operators $Q_{H\Box} = (H^\dagger H)\Box(H^\dagger H)$ and $Q_H = (H^\dagger H)^3$ (Section 40.4.5) and the normalization Eq. (40.120) with $\Lambda = M$, we match coefficients:

$$\boxed{C_{H\Box} = -\frac{g_1^2}{2}, \qquad C_H = -\frac{g_1^2\,g_2}{2}} \tag{40.134}$$

at the matching scale $\mu = M$. The heavy singlet thus feeds precisely two dimension-six directions: the Higgs-kinetic operator $Q_{H\Box}$ (which, after electroweak symmetry breaking, rescales the physical Higgs field and so shifts every Higgs coupling coherently) and the Higgs self-interaction Q_H (which alters the trilinear and quartic Higgs couplings). A measurement of Higgs-coupling deviations of the pattern Eq. (40.134) would be the SMEFT signature of exactly this heavy singlet; the absence of such deviations bounds g_1^2/M^2 and hence the mass M for a given coupling.

The four-fermion prototype and the bosonic example share an algebraic skeleton, and it is worth making the parallel explicit because it shows the universality of the method. Replace the heavy scalar by a heavy neutral vector V_μ, a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet of mass M (a "Z'"), coupled to a Standard-Model fermion current $J^\mu = \sum_f \bar{\psi}_f \gamma^\mu \psi_f$ through $\mathcal{L}_V = -\frac{1}{4} V_{\mu\nu} V^{\mu\nu} + \frac{1}{2} M^2 V_\mu V^\mu + g_V\, V_\mu J^\mu$. Its algebraic equation of motion at leading order in $1/M^2$ is $M^2 V^\mu = -g_V J^\mu$, i.e. $V^\mu = -(g_V/M^2) J^\mu$, and substituting back,

$$\mathcal{L}_{\text{eff}} = \tfrac{1}{2} M^2 \Big(\frac{g_V}{M^2}\Big)^2 J_\mu J^\mu - g_V \frac{g_V}{M^2}\, J_\mu J^\mu = -\,\frac{g_V^2}{2M^2}\, J_\mu J^\mu, \tag{40.135}$$

a four-fermion operator with Wilson coefficient $C/\Lambda^2 = -g_V^2/2M^2$. This is the *same* computation as integrating out the W in Theorem 40.18—the mass term contributes $+\frac{1}{2}$, the source term -1, summing to $-\frac{1}{2}$—only with the charged current and W replaced by a generic current and a generic mediator. A heavy Z' and Fermi's W are the same EFT calculation. Whichever Warsaw four-fermion operator $J_\mu J^\mu$ corresponds to (for a lepton-doublet current, $Q_{ll} = (\bar{\ell}\gamma_\mu \ell)(\bar{\ell}\gamma^\mu \ell)$), the coefficient is read off directly.

40.5.4 Decoupling: heavy physics is local and suppressed

Both worked examples exhibit a structural fact we now name. In every case the heavy field left two kinds of imprint on the low-energy theory: *(i)* finite shifts of the renormalizable couplings already present (the Higgs quartic in Eq. (40.133)), and *(ii)* a tower of higher-dimension local operators suppressed by powers of $1/M$. Nothing else survived; in particular no trace of the heavy particle as a propagating state remained. This is the Appelquist–Carazzone decoupling theorem (Theorem 38.10), and matching is the constructive form of its proof: the explicit local operators are the residue the heavy field leaves behind.

Theorem 40.20 (Decoupling, EFT form; Theorem 38.10)**.** *In a renormalizable theory with a heavy field of mass M coupled to light fields, all effects of the heavy field on light-field Green's functions at energies $E \ll M$ are reproduced, to any fixed order in E/M, by the local effective Lagrangian $\mathcal{L}_{\text{eff}} = \mathcal{L}^{\text{ren}}_{\text{light}} + \sum_i (C_i/M^{d_i-4})\, Q_i$, where $\mathcal{L}^{\text{ren}}_{\text{light}}$ is the renormalizable Lagrangian of the light fields with couplings shifted by finite matching corrections, and the Q_i are local operators of dimension $d_i > 4$ built from the light fields, with coefficients C_i of order the heavy couplings. Effects of the heavy field thus vanish as $M \to \infty$ except for the renormalization of the light couplings.*

The theorem is what gives the SMEFT expansion Eq. (40.120) its meaning: whatever sits above Λ, its low-energy footprint is exactly a series of local operators ordered by dimension, and the leading new effects are the dimension-six ones. Heavier means more suppressed; this is why a bound on a dimension-six coefficient translates into a bound on a scale.

There is, however, an important class of failures of decoupling, and honesty requires flagging it because it is central to electroweak physics.

Caution: when heavy fields do *not* decouple

Decoupling assumes the heavy mass M is an independent parameter that can be sent to infinity at fixed couplings. When the heavy mass is generated by spontaneous symmetry breaking, $M \sim y\,v$ with y a coupling and v a fixed vacuum value, sending $M \to \infty$ means sending the coupling $y \to \infty$, and the field does not decouple. The Standard-Model top quark and the Higgs boson are the textbook cases: the top contribution to the ρ parameter grows as m_t^2 (Section 31.6), and the would-be Goldstone bosons whose longitudinal-mode couplings grow with mass are precisely what the heavy gauge bosons eat. In SMEFT the analogue is that integrating out a heavy field whose mass comes partly from the Higgs vacuum value can leave effects that do not scale as a simple power of $1/\Lambda$. The decoupling picture is the generic case; chiral and electroweak-symmetry-breaking-induced masses are the systematic exceptions, and a correct matching tracks them explicitly.

40.5.5 Tree-level versus one-loop matching

Everything so far was tree-level matching: the heavy field was eliminated by its classical equation of motion, equivalently the leading saddle of its path integral. But Eq. (40.121) is an all-orders statement, and at one loop the EFT must reproduce the full theory's one-loop light-field amplitudes. The Wilson coefficients then receive one-loop contributions, and—crucially—*some operators are generated only at one loop and not at all at tree level*. Which is which controls how a measured coefficient is interpreted.

The two scalings. Dimensional analysis of the matching makes the hierarchy transparent. A tree-generated coefficient is a product of the heavy couplings with no loop factor, as in Eqs. (40.127), (40.134) and (40.135):

$$C_i^{\text{tree}} \sim \frac{(\text{coupling})^2}{1}, \qquad C_i^{\text{loop}} \sim \frac{(\text{coupling})^{2\text{ or more}}}{16\pi^2}, \tag{40.136}$$

the loop case carrying the universal $1/16\pi^2$ of a four-dimensional loop integral (Section D.4). For a weakly coupled ultraviolet completion this is roughly two orders of magnitude. The consequence for interpretation is direct: a measured coefficient of natural size $C \sim 1$ implies, for a tree-generated operator, $\Lambda \sim (\,\text{coupling}\,)/\sqrt{|C|/v^2}$, whereas for a loop-generated operator the same measurement implies a scale lower by a factor $\sim 4\pi$, because the intrinsic coefficient

is $\sim g^2/16\pi^2$ rather than $\sim g^2$. Reading a bound on a dimension-six operator as a bound on Λ therefore *requires knowing whether that operator can be tree-generated*.

The classification. Whether an operator can appear at tree level in *any* renormalizable, weakly-coupled completion is not a matter of taste; it is determined by the operator's structure, through the classification of Arzt–Einhorn–Wudka. The criterion is whether the operator can be "cut" into renormalizable ($d \le 4$) vertices joined by heavy propagators—i.e. whether a tree graph of some heavy field can produce it. Of the 59 baryon-number-conserving dimension-six operators of the one-generation Warsaw basis, only a subset are "potentially tree-generated"; the remainder are necessarily loop-generated. The loop-only classes include in particular:

- the *dipole operators*—$Q_{eW} = (\bar{\ell}\sigma^{\mu\nu}e)\tau^I H\, W^I_{\mu\nu}$, Q_{eB}, Q_{uG}, and their relatives—which require a chirality flip and a field-strength insertion that no tree graph of renormalizable vertices can assemble;
- the purely gauge operators built from three field strengths, $Q_W = \epsilon^{IJK} W^I_{\ \mu}{}^{\nu} W^J_{\ \nu}{}^{\rho} W^K_{\ \rho}{}^{\mu}$ and $Q_{\tilde{W}}$, which cannot arise from tree exchange.

The dipole case is the physically decisive one. The anomalous magnetic moment and the electric dipole moment are governed by dipole operators; since these are loop-generated in any weakly-coupled completion, the model's number $a_e = \alpha/2\pi$ (Chapter 20)—a one-loop effect of the photon—reappears here as the statement that a Wilson coefficient is loop-suppressed. A measured electron or muon dipole that is *large* relative to the loop expectation is therefore a sharp pointer to either strong coupling or a low scale, precisely because the operator cannot be tree-generated.

How one-loop matching is done. Operationally, Eq. (40.121) is imposed at one loop by computing the relevant light-field 1PI functions on both sides through $O(\hbar)$, in the same scheme, and solving for the C_i. The efficient organization is the method of regions (Section 38.3): in the full-theory loop integral, the integrand is split into a *hard* region (loop momentum $\ell \sim M$) and a *soft* region ($\ell \sim p$). The soft region reproduces, term by term, the EFT loop built from the lower-order operators, and cancels in the difference; the hard region is Taylor-expanded in p/M before integration and yields the local matching contribution to the C_i, by exactly the mechanism of Theorem 40.17 now applied at one loop. Concretely, a hard-region integral has the form $\int \mathrm{d}^d\ell/(2\pi)^d\,[\ell^2 - M^2]^{-n} \times (\text{polynomial in } \ell)$, a tadpole-type master integral that produces $1/16\pi^2$ times a function of M and μ (Section D.4); choosing $\mu = M$ removes its logarithm, leaving the finite, local matching coefficient. The poles in ε that the hard region generates are

removed by the renormalization of the full theory and of the EFT operators, so the matched $C_i(\mu = M)$ is finite and scheme-fixed—the boundary value for the running of the next section.

Physical Insight

Tree- versus loop-level matching is the EFT translation of a question the data cannot answer on its own: *how strongly coupled, and at what scale, is the new physics?* A dimension-six coefficient is a single number; the pattern of which operators are switched on, and the comparison of their sizes against the tree-versus-loop expectation Eq. (40.136), is what turns that number into a statement about the ultraviolet. SMEFT does not merely parametrize deviations; read with the matching dictionary in hand, it diagnoses their origin.

We have now fixed Wilson coefficients at a matching scale $\mu \sim \Lambda$ (or, in the prototype, $\mu \sim M_W$), whether from a known heavy field by explicit matching or from a hypothetical one by translating a model. Those coefficients carry no large logarithms by construction—that was the point of matching at $\mu \sim M$—but the observables that test them live at much lower scales, and the intervening logarithms $\ln(\Lambda/m_W)$ are large and must be summed. The next section supplies the machinery: the renormalization-group running of the Wilson coefficients and the anomalous-dimension matrix that mixes operators of equal dimension as they flow down to the measurement scale.

40.6 Running and operator mixing

The matching computation of Section 40.5 delivered Wilson coefficients *at the high scale*: integrating out the W at $\mu \simeq m_W$ gave the Fermi Hamiltonian with $C_2(m_W) = 1$, and integrating out a generic heavy state gave a set of dimension-six coefficients fixed at $\mu \simeq M$, the mass of the new physics. Those numbers are correct only *at the scale where the heavy field was removed*. A measurement, however, lives at a low scale: a kaon decays at $\mu \sim 1$–$2\,\mathrm{GeV}$, where hadronic matrix elements are defined; an electroweak precision observable is read off at $\mu \sim m_Z$. The naive thing to do — use $C_i(m_W)$ directly in a low-energy matrix element — is wrong, and wrong by a controlled, computable, and often *large* amount. The reason is that the loop corrections to the effective theory carry logarithms $\alpha_s \ln(m_W/\mu)$ which, with $\alpha_s \ln(m_W/\mu) \sim 0.1 \times 4 \sim 0.4$, are not small: a fixed-order treatment is in trouble already at one loop and hopeless at two. These are the same large logarithms that the renormalization group was invented to resum (Chapter 24), and the effective theory is precisely the setting in which the resummation is cleanest, because the heavy scale has been removed from the dynamics and survives only as the boundary condition $C_i(m_W)$ and as the upper

limit of the logarithm.

This section turns the RG machinery of Chapter 24 loose on Wilson coefficients. Two features distinguish the problem from the running of a single coupling. First, the objects that run are the coefficients of *composite* operators, and operators of equal dimension and identical quantum numbers *mix* under renormalization: the anomalous "dimension" is a matrix. Second, the engine of the running here is QCD — the mixing is computed from exactly the gluon-exchange diagrams and the β-function of Chapter 27, applied to weak operators. This is QCD reappearing in a new guise: the non-abelian anomalous-dimension technology, built to explain asymptotic freedom, now resums the strong-interaction dressing of the weak Hamiltonian. We develop the general formalism, prove the renormalization-group equation for the coefficient vector, and then carry the classic $\Delta S = 1$ four-quark mixing in full: the operators, the one-loop anomalous-dimension matrix, its diagonalization, and the leading-log solution exhibiting the celebrated QCD enhancement. We close by stating how the same procedure organizes the entire dimension-six SMEFT, whose complete one-loop anomalous-dimension matrix is known, and we compute the load-bearing QCD sub-block.

40.6.1 Operator renormalization and the mixing matrix

Fix an effective Hamiltonian of the standard form

$$\mathcal{H}_{\text{eff}} = \frac{G_F}{\sqrt{2}} \sum_i C_i(\mu)\, Q_i(\mu) + \text{h.c.}, \tag{40.137}$$

where the Q_i are local operators of a common dimension, sharing the quantum numbers of the process under study, and the C_i are their Wilson coefficients. The product $\mathcal{H}_{\text{eff}}$ is a physical operator and must be independent of the renormalization scale μ; the μ-dependence of the C_i exists only to cancel the μ-dependence of the renormalized operators $Q_i(\mu)$. Making that cancellation quantitative is the whole content of the running.

A composite operator built from several fields needs its own renormalization beyond the renormalization of the fields and couplings it contains. When several operators carry the same quantum numbers, their divergences are not separately multiplicatively renormalizable: the counterterm needed to make $\langle Q_i \rangle$ finite is in general a linear combination of *all* the Q_j. We write the relation between bare and renormalized operators as a matrix,

$$Q_i^{(0)} \;=\; Z_{ij}\, Q_j(\mu), \qquad Z_{ij} = \delta_{ij} + \frac{\alpha_s}{4\pi}\frac{1}{\varepsilon}\, a_{ij} + O(\alpha_s^2), \tag{40.138}$$

in $\overline{\text{MS}}$ with $d = 4 - 2\varepsilon$, where the bare operator $Q_i^{(0)}$ is built from bare fields and is by construction μ-independent, and the pole part a_{ij} is a pure number to be computed. The off-diagonal entries $a_{i\neq j}$ are the *operator mixing*: a

divergence proportional to Q_j generated inside the one-loop matrix element of Q_i. (Throughout, field and coupling renormalization are taken as already performed; Z_{ij} is the genuinely new *operator* renormalization, the divergence that survives after the external legs and the strong coupling have been renormalized.)

Proposition 40.21 (Renormalization-group equation for Wilson coefficients)**.** *Let the operators renormalize as in* (40.138) *and define the* anomalous-dimension matrix

$$\gamma \;=\; Z^{-1}\,\mu\frac{dZ}{d\mu}, \qquad \gamma = \frac{\alpha_s}{4\pi}\,\gamma^{(0)} + O(\alpha_s^2). \tag{40.139}$$

Then the renormalized Wilson-coefficient vector obeys

$$\mu\,\frac{d}{d\mu}\,C_i(\mu) \;=\; \big(\gamma^{\mathsf T}\big)_{ij}\,C_j(\mu), \qquad \textit{i.e.} \qquad \mu\,\frac{dC}{d\mu} = \gamma^{\mathsf T} C. \tag{40.140}$$

At leading order $\gamma^{(0)}_{ij} = -2\,a_{ij}$.

Proof. The bare operator is μ-independent. Differentiating $Q_i^{(0)} = Z_{ij}Q_j$ at fixed bare quantities,

$$0 \;=\; \mu\frac{dQ_i^{(0)}}{d\mu} = \Big(\mu\frac{dZ_{ij}}{d\mu}\Big)Q_j + Z_{ij}\,\mu\frac{dQ_j}{d\mu}. \tag{40.141}$$

Writing $\mu\,dQ_j/d\mu = -\gamma_{jk}Q_k$ defines an anomalous-dimension matrix for the operators; substituting and using that the Q_k are independent gives $\mu\,dZ_{ik}/d\mu = Z_{ij}\gamma_{jk}$, i.e. $\gamma = Z^{-1}\mu\,dZ/d\mu$, which is (40.139). Now impose μ-independence of $\mathcal{H}_{\text{eff}}$. The bare coefficients $C_i^{(0)}$ multiply the bare operators, $C_i^{(0)}Q_i^{(0)} = C_i^{(0)}Z_{ij}Q_j$, so the renormalized coefficients are

$$C_j(\mu) = C_i^{(0)}\,Z_{ij}, \qquad \text{i.e.} \qquad C = Z^{\mathsf T}C^{(0)}, \tag{40.142}$$

with $C^{(0)}$ the (μ-independent) bare coefficient vector. Differentiating,

$$\mu\frac{dC}{d\mu} = \Big(\mu\frac{dZ^{\mathsf T}}{d\mu}\Big)C^{(0)} = \Big(\mu\frac{dZ^{\mathsf T}}{d\mu}\Big)(Z^{\mathsf T})^{-1}C = \Big[(Z^{-1}\mu\tfrac{dZ}{d\mu})^{\mathsf T}\Big]C = \gamma^{\mathsf T}C, \tag{40.143}$$

which is (40.140); we used $(\mu\,dZ^{\mathsf T}/d\mu)(Z^{\mathsf T})^{-1} = \big[(Z^{-1}\mu\,dZ/d\mu)\big]^{\mathsf T}$. For the leading order, write $Z = 1 + (\alpha_s/4\pi)\varepsilon^{-1}a$ as in (40.138). In $d = 4-2\varepsilon$ the strong coupling runs as $\mu\,d\alpha_s/d\mu = -2\varepsilon\,\alpha_s + O(\alpha_s^2)$ (the -2ε is the engineering-dimension term that the four-dimensional β-function of Eq. (27.1) supplements; it is exactly the term carried in the Callan–Symanzik analysis of Theorem 24.19). Hence

$$\mu\frac{dZ}{d\mu} = \frac{a}{4\pi}\,\frac{1}{\varepsilon}\,\mu\frac{d\alpha_s}{d\mu} = \frac{a}{4\pi}\,\frac{1}{\varepsilon}\,(-2\varepsilon\,\alpha_s) = -2\,\frac{\alpha_s}{4\pi}\,a, \tag{40.144}$$

and $Z^{-1} = 1 + O(\alpha_s)$, so $\gamma = -2(\alpha_s/4\pi)\,a + O(\alpha_s^2)$, giving $\gamma^{(0)} = -2a$. That γ comes out finite as $\varepsilon \to 0$ — the $1/\varepsilon$ of a is cancelled by the explicit ε in $\mu\, d\alpha_s/d\mu$ — is the statement that the anomalous dimension is a genuine physical (renormalized) quantity, exactly as for the single-coupling case in Section 24.6.5. □

Equation (40.140) is the matrix generalization of the scalar RG equation (24.184). Its structure is worth dwelling on. The coefficients run because the *operators* acquire anomalous dimensions, and they mix because those anomalous dimensions form a non-diagonal matrix. The transpose in (40.140) is the bookkeeping consequence of (40.142): coefficients are contracted with operators, so a mixing of operators among themselves becomes, on the coefficients, the action of the transposed matrix. Whenever $\gamma^{(0)}$ is symmetric — as it is at one loop in the $\Delta S = 1$ problem below — the transpose is invisible and one may read the same matrix as acting on either side.

Physical Insight: why running matters

A measurement at a low scale $\mu_{\rm low}$ constrains the combination $C_i(\mu_{\rm low})\,\langle Q_i\rangle(\mu_{\rm low})$. The *physics* we want to constrain — the mass and couplings of whatever was integrated out — lives in $C_i(\mu_{\rm high})$. The two are related by the solution of (40.140), an operator that can change a coefficient by an $O(1)$ factor (the $\Delta S = 1$ case below) or even rotate a vanishing high-scale coefficient into a nonzero low-scale one (genuine mixing). *A low-energy bound is a bound on the UV only after RG evolution.* Skipping the running does not merely lose precision; it can misidentify which operator, and hence which new-physics structure, the data are sensitive to.

40.6.2 The $\Delta S = 1$ effective Hamiltonian

The cleanest arena for operator mixing is the non-leptonic weak Hamiltonian that governs strangeness-changing transitions such as $K \to \pi\pi$. Integrating the W out of the charged-current interaction $\frac{g}{\sqrt{2}}\,\bar u_L \gamma^\mu d'_L W^+_\mu$ (with d' the CKM-rotated down-type field of Section 32.3.1) produces, for the quark-level transition $s \to u\bar u d$, a local four-quark interaction. At tree level the W propagator collapses to $g_{\mu\nu}/m_W^2$ and one finds a single operator, exactly as in the Fermi reduction of Section 40.5. Writing colour indices α, β explicitly and abbreviating the left-handed current with $P_L = \frac{1}{2}(1 - \gamma^5)$, the operator basis closed under QCD renormalization is

$$Q_1 = (\bar s_\alpha \gamma^\mu P_L\, u_\beta)(\bar u_\beta \gamma_\mu P_L\, d_\alpha), \tag{40.145}$$

$$Q_2 = (\bar s_\alpha \gamma^\mu P_L\, u_\alpha)(\bar u_\beta \gamma_\mu P_L\, d_\beta). \tag{40.146}$$

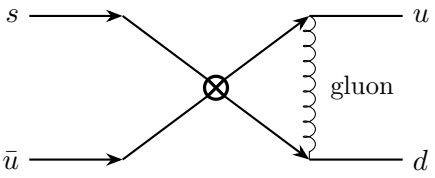


Figure 40.7: One of the four current-current diagrams that renormalize the $\Delta S = 1$ operators. The crossed circle is the local four-quark vertex (Q_1 or Q_2); a gluon is exchanged between the two quark currents. The colour factor $T^A \otimes T^A$ carried by such a gluon, reduced through the SU(3) identity (40.149), is what mixes the colour-singlet operator Q_2 into the colour-mixed Q_1. The remaining three diagrams attach the gluon to the other pairings of legs; diagrams with the gluon inside a single current cancel against quark wavefunction renormalization.

Q_2 is the "colour-singlet×singlet" operator produced directly by W-exchange (each bilinear is a colour singlet); Q_1 is the "colour-mixed" operator, in which the colour indices are cross-contracted between the two currents. The effective Hamiltonian is

$$\mathcal{H}_{\text{eff}}^{\Delta S=1} = \frac{G_F}{\sqrt{2}} V_{ud} V_{us}^* \left[C_1(\mu)\, Q_1 + C_2(\mu)\, Q_2 \right] + \text{h.c.}, \tag{40.147}$$

with the CKM factor $V_{ud}V_{us}^*$ from Section 32.3.1. The tree-level matching of Section 40.5 produces only the singlet operator,

$$C_1(m_W) = 0, \qquad C_2(m_W) = 1. \tag{40.148}$$

The whole point of what follows is that C_1 is *not* zero at low scales: QCD radiative corrections mix Q_2 into Q_1, and the leading-log resummation turns the boundary condition (40.148) into a pair of $O(1)$ coefficients with a definite hierarchy. We need the 2×2 anomalous-dimension matrix.

40.6.3 The one-loop anomalous-dimension matrix

We now compute the one-loop QCD renormalization of $\{Q_1, Q_2\}$ and extract $\gamma^{(0)}$. The computation is the worked example of this section; we present the diagrams, the colour algebra, the Dirac reduction, and the loop integral in turn, then assemble the matrix and verify it. The key conceptual points are two: (i) only the diagrams in which a gluon is exchanged *between* the two currents contribute a net divergence to the mixing, the others being removed by quark wavefunction renormalization; and (ii) the colour algebra of that gluon exchange is exactly what converts the singlet operator Q_2 into the colour-mixed Q_1.

Worked Example 40.22 (One-loop $\Delta S = 1$ mixing and the leading-log solution)**.**
(a) The colour algebra. Each gluon vertex carries a generator T^A in the

fundamental, so a gluon exchanged between the two currents inserts $T^A_{\alpha\beta}\otimes T^A_{\gamma\delta}$ on the colour indices of the four external quarks. The SU(N) Fierz (completeness) identity for the fundamental generators, established in Section B.3.2, is

$$T^A_{\alpha\beta}\,T^A_{\gamma\delta} = \tfrac{1}{2}\Big(\delta_{\alpha\delta}\,\delta_{\gamma\beta} - \tfrac{1}{N}\,\delta_{\alpha\beta}\,\delta_{\gamma\delta}\Big), \tag{40.149}$$

with $N = 3$ for QCD. The two colour structures on the right are precisely the two appearing in Q_1 and Q_2: the cross-contraction $\delta_{\alpha\delta}\delta_{\gamma\beta}$ rearranges a singlet$\times$singlet into the colour-mixed structure, while $\delta_{\alpha\beta}\delta_{\gamma\delta}$ returns the original singlet$\times$singlet. Thus the gluon exchange, acting on the colour-singlet operator Q_2, produces a piece proportional to the colour-rearranged operator (which, after the spinor Fierz step below, is Q_1) with weight $+\frac{1}{2}$, and a piece proportional to Q_2 itself with weight $-\frac{1}{2N}$. This is the algebraic heart of the mixing: *Q_2 feeds Q_1 through the off-diagonal $\frac{1}{2}$, and reflects onto itself through the diagonal $-\frac{1}{2N}$.*

(b) The Dirac structure. The two quark propagators adjacent to the operator vertex and the gluon vertices contribute the Dirac string $\gamma^\mu P_L \not{k}\,\gamma^\alpha\otimes\gamma_\alpha\,\not{k}\,\gamma_\mu P_L$, with α the gluon Lorentz index contracted between the lines and k the loop momentum. Under the loop integral the numerator may be symmetrized, $k^\rho k^\sigma \to (k^2/d)\,g^{\rho\sigma}$ (Section D.4), leaving $(\gamma^\mu\gamma^\rho\gamma^\alpha P_L)\otimes(\gamma_\alpha\gamma_\rho\gamma_\mu P_L)$. Two reductions collapse this. The single-line contraction identity of Chapter 24's gamma-algebra conventions (Appendix C),

$$\gamma^\alpha\gamma^\rho\gamma_\alpha = (2-d)\,\gamma^\rho \xrightarrow{\;d\to 4\;} -2\,\gamma^\rho, \tag{40.150}$$

governs the index α once it is moved onto one chain, and the chiral Fierz identity for left-handed currents Section B.3.2,

$$(\gamma^\nu P_L)_{ij}(\gamma_\nu P_L)_{kl} = (\gamma^\nu P_L)_{il}(\gamma_\nu P_L)_{kj}, \tag{40.151}$$

shows that the $V\!-\!A$ structure is Fierz self-conjugate: the colour rearrangement of part (a) is accompanied by exactly the spinor rearrangement that reproduces the standard current$\times$current Dirac structure, so the colour-mixed term is genuinely Q_1 and the singlet term is genuinely Q_2, with no new Dirac structures generated. (This last statement is the assertion that, in the naive-dimensional-regularization scheme with anticommuting γ^5, no physical operator outside $\{Q_1, Q_2\}$ is induced at this order; the would-be extra structures are *evanescent* operators that vanish in $d = 4$, discussed in the remark below.) The net effect of (40.150) and (40.151) on the four cross diagrams, summed, is to reproduce the original $V\!-\!A\otimes V\!-\!A$ Dirac structure multiplied by the pure number 3 relative to the elementary gluon-exchange normalization (the same "3" that will appear in the matrix below); the mechanical string reduction yielding this factor uses only (40.150), (40.151), and the trace identities of Appendix C.

(c) The loop integral. The momentum integral common to all four cross diagrams is the logarithmically divergent

$$\int\frac{d^dk}{(2\pi)^d}\,\frac{k^\rho k^\sigma}{(k^2)^3} = \frac{g^{\rho\sigma}}{d}\int\frac{d^dk}{(2\pi)^d}\,\frac{1}{(k^2)^2} = \frac{g^{\rho\sigma}}{4}\cdot\frac{i}{16\pi^2}\,\frac{1}{\varepsilon} + (\text{finite}), \tag{40.152}$$

where the value of the master integral and the isolation of its ultraviolet pole (the infrared region being regulated by an off-shell external momentum that does not affect the $1/\varepsilon_{\rm UV}$ coefficient) are taken from Section D.4. The pole carries the factor $\alpha_s/4\pi$ once the two gluon vertices $(ig)^2$, the gluon propagator, and the i's are assembled, since $g^2/16\pi^2 = \alpha_s/4\pi$.

(d) Assembling Z and $\gamma^{(0)}$. Combining the colour weights of part (a) ($+\frac{1}{2}$ onto Q_1, $-\frac{1}{2N}$ onto Q_2), the Dirac factor 3 of part (b), and the pole $\frac{1}{\varepsilon}\,\frac{\alpha_s}{4\pi}$ of part (c), the one-loop counterterm needed to render $\langle Q_2\rangle$ finite is $\propto 3(\frac{1}{2}Q_1 - \frac{1}{2N}Q_2)$, and likewise with $1 \leftrightarrow 2$. In matrix form (40.138) this reads

$$a = \begin{pmatrix} \dfrac{3}{N} & -3 \\ -3 & \dfrac{3}{N} \end{pmatrix}, \qquad \gamma^{(0)} = -2a = \begin{pmatrix} -\dfrac{6}{N} & 6 \\ 6 & -\dfrac{6}{N} \end{pmatrix} \xrightarrow{N=3} \begin{pmatrix} -2 & 6 \\ 6 & -2 \end{pmatrix}. \tag{40.153}$$

The diagonal entries $-6/N$ are the colour-reflection ($-1/N$) pieces; the off-diagonal $+6$ is the colour-rearrangement ($+\frac{1}{2}$) piece, doubled by the -2 from $\gamma^{(0)} = -2a$ and carrying the Dirac factor 3. The matrix is symmetric, so $\gamma^{(0)\mathsf{T}} = \gamma^{(0)}$ and the transpose in (40.140) is immaterial at this order.

(e) Diagonalization. The eigenvectors of (40.153) are $(1,1)$ and $(1,-1)$, independent of N. Define the eigen-operators and eigen-coefficients

$$Q_\pm = \tfrac{1}{2}(Q_2 \pm Q_1), \qquad C_\pm = C_2 \pm C_1, \tag{40.154}$$

which indeed satisfy $C_1Q_1 + C_2Q_2 = C_+Q_+ + C_-Q_-$ (using $Q_2 = Q_+ + Q_-$, $Q_1 = Q_+ - Q_-$). In this basis $\gamma^{(0)}$ is diagonal with eigenvalues

$$\gamma^{(0)}_+ = -\frac{6}{N} + 6 = \frac{6(N-1)}{N} \xrightarrow{N=3} +4, \qquad \gamma^{(0)}_- = -\frac{6}{N} - 6 = -\frac{6(N+1)}{N} \xrightarrow{N=3} -8. \tag{40.155}$$

The opposite signs are the crucial fact: Q_+ has a *positive* anomalous dimension and Q_- a *negative* one, and (we will see) this drives C_- up and C_+ down as we evolve to low scales.

(f) Leading-log solution. In the $\pm$ basis the RG equation (40.140) decouples,

$$\mu\frac{dC_\pm}{d\mu} = \frac{\alpha_s(\mu)}{4\pi}\,\gamma^{(0)}_\pm\, C_\pm. \tag{40.156}$$

Trade μ for α_s using the one-loop QCD running $\mu\, d\alpha_s/d\mu = -b_0\,\alpha_s^2/(2\pi)$ from Eq. (27.1) (with $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f = 11 - \frac{2}{3}n_f$ for SU(3) with n_f flavors). Then

$$\frac{dC_\pm}{C_\pm} = \frac{\gamma^{(0)}_\pm}{4\pi}\,\alpha_s\,\frac{d\mu}{\mu} = \frac{\gamma^{(0)}_\pm}{4\pi}\,\alpha_s\cdot\Big(-\frac{2\pi}{b_0}\frac{d\alpha_s}{\alpha_s^2}\Big) = -\frac{\gamma^{(0)}_\pm}{2b_0}\,\frac{d\alpha_s}{\alpha_s}, \tag{40.157}$$

and integrating from m_W down to μ,

$$\boxed{C_\pm(\mu) = \left[\frac{\alpha_s(m_W)}{\alpha_s(\mu)}\right]^{\gamma_\pm^{(0)}/2b_0} C_\pm(m_W)} \tag{40.158}$$

the leading-log resummation. With (40.148) the boundary condition is $C_+(m_W) = C_-(m_W) = 1$, and the exponents are

$$\frac{\gamma_+^{(0)}}{2b_0} = \frac{2}{b_0} > 0, \qquad \frac{\gamma_-^{(0)}}{2b_0} = -\frac{4}{b_0} < 0. \tag{40.159}$$

Since $\alpha_s(\mu) > \alpha_s(m_W)$ for $\mu < m_W$ (asymptotic freedom, Eq. (27.155)), the bracket in (40.158) is less than one; the positive exponent suppresses C_+ below unity, the negative exponent enhances C_- above unity. The ratio

$$\frac{C_-(\mu)}{C_+(\mu)} = \left[\frac{\alpha_s(\mu)}{\alpha_s(m_W)}\right]^{6/b_0} > 1 \tag{40.160}$$

grows monotonically toward low scales: this is the *QCD enhancement* of the colour-antisymmetric over the colour-symmetric channel. Numerically, evolving with the one-loop α_s and proper flavor thresholds (Fig. 40.8) gives, at $\mu \simeq m_c$,

$$C_+(\mu) \simeq 0.78, \quad C_-(\mu) \simeq 1.64 \implies C_1(\mu) \simeq -0.43, \quad C_2(\mu) \simeq 1.21, \tag{40.161}$$

and at $\mu \simeq 1\,\text{GeV}$ the splitting widens to $C_+ \simeq 0.76$, $C_- \simeq 1.72$ (so $C_1 \simeq -0.48$, $C_2 \simeq 1.24$). A coefficient C_1 that vanished identically at m_W has been driven to $\simeq -\frac{1}{2}$ purely by QCD; this is the resummed content of the large logarithms that a fixed-order calculation could only approximate.

Remark 40.23 (Evanescent operators and scheme dependence). The clean two-operator closure used above is exact at one loop only with a choice of how to continue γ^5 and the Dirac algebra to $d \neq 4$. The Fierz identity (40.151) holds in $d = 4$ but is violated by $O(\varepsilon)$ terms in d dimensions; the difference is an *evanescent* operator, one that vanishes in four dimensions but whose $O(\varepsilon)$ matrix element multiplies a $1/\varepsilon$ pole to leave a finite, scheme-dependent contribution. A consistent treatment fixes a prescription (the naive-dimensional-regularization scheme used here, with anticommuting γ^5, is one standard choice) and renormalizes the evanescent operators to zero. The leading-order $\gamma^{(0)}$ of (40.153) is scheme-*independent*; the scheme dependence enters first in $C_\pm(m_W)$ at next-to-leading order and in $\gamma^{(1)}$, and cancels against the matrix-element calculation in any physical observable. We carry only the scheme-independent leading log here; the next-to-leading machinery is referenced in the Further reading.

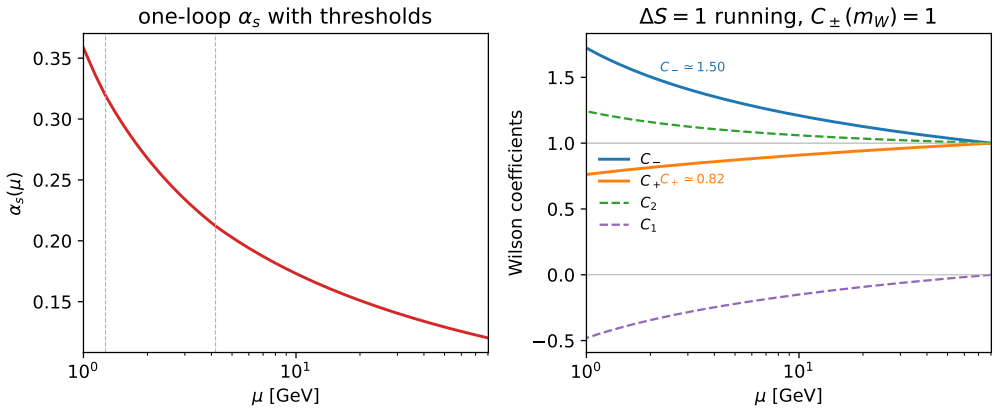


Figure 40.8: Leading-log running of the $\Delta S = 1$ Wilson coefficients, obtained by integrating (40.140) from the matching scale m_W to low scales with the one-loop $\alpha_s(\mu)$ (left panel; flavor thresholds matched continuously at m_b, m_c). Right panel: the eigen-coefficients $C_\pm$ (solid) with $C_\pm(m_W) = 1$, and the physical-basis coefficients C_1, C_2 (dashed). The colour-antisymmetric coefficient C_- is enhanced and C_+ suppressed, the gap widening toward low μ where α_s grows. Generated by `figures/40-deltaS1-running.py`.

40.6.4 The same machine: dimension-six SMEFT

Everything above transfers verbatim to the Standard Model Effective Field Theory. The dimension-six operators of the Warsaw basis carry Wilson coefficients $C_i(\mu)$ fixed by matching at the new-physics scale Λ (or at m_W for electroweak matching), and confronted with data at m_Z or below. Between those scales they run and mix, governed by the identical equation

$$\mu \frac{dC_i}{d\mu} = \frac{1}{16\pi^2} \left(\hat{\gamma}^{\mathsf{T}}\right)_{ij} C_j, \tag{40.162}$$

now with $\hat{\gamma}$ receiving contributions from all the Standard Model gauge couplings g_s, g, g', the Yukawa couplings, and the Higgs self-coupling, since all of these dress the dimension-six operators. For a single fermion generation, restricting to baryon-number-conserving operators, the Warsaw basis contains 59 independent operators, and the complete one-loop 59×59 anomalous-dimension matrix $\hat{\gamma}$ has been computed in full; we cite this result rather than reproduce it, as it is a bookkeeping enterprise rather than a new principle (the principle is exactly Theorem 40.21). Its structure is instructive: the matrix is far from arbitrary. Many entries vanish for reasons of symmetry and holomorphy (Higgs-number, chirality, and a non-renormalization pattern that forbids certain "upward" mixings), so $\hat{\gamma}$ has a sparse, approximately block-triangular form; this is why a low-energy bound on one coefficient typically constrains only a definite subset of high-scale operators.

The load-bearing sub-block is the one we have already computed. The purely-QCD part of $\hat{\gamma}$, acting on the four-quark operators of the Warsaw basis, is built from *exactly* the current-current diagrams, colour identity (40.149), and chiral Fierz (40.151) of Theorem 40.22. A pair of four-quark operators differing only by their colour contraction — the SMEFT analogues of Q_1 and Q_2 — mixes under QCD through precisely the matrix (40.153),

$$\hat{\gamma}_{\text{QCD}}\big|_{\text{4-quark colour pair}} = 4\pi\,\alpha_s \begin{pmatrix} -6/N & 6 \\ 6 & -6/N \end{pmatrix} = g_s^2 \begin{pmatrix} -6/N & 6 \\ 6 & -6/N \end{pmatrix}, \tag{40.163}$$

identical in form to the $\Delta S = 1$ result because the colour and chiral algebra are identical; only the flavor labels and the matching coefficients differ. The genuinely new SMEFT entries — gauge and Yukawa mixings, the running of the dipole operators, the feeding of four-fermion operators into dipoles that dominates the $b \to s\gamma$ analysis, the mixing of O_{HWB} and O_{HD} that controls the oblique parameters — are obtained by the same Theorem 40.21 machinery with the relevant vertices in place of the gluon, and are collected in the cited literature. The lesson is uniform: there is one renormalization-group equation for Wilson coefficients, (40.140), and the entire apparatus of weak-Hamiltonian and SMEFT running is its anomalous-dimension matrix worked out diagram by diagram.

With the coefficients evolved from the matching scale down to the measurement scale and the large logarithms resummed, the Wilson coefficients are at last in the form a low-energy experiment actually constrains. The next section makes that confrontation explicit: a precision measurement bounds a combination of $C_i(\mu_{\text{low}})$, which — run back up through (40.140) — becomes a bound on the scale and structure of the physics that was integrated out.

40.7 The EFT view of precision physics

The previous section closed the technical program of this chapter: starting from Wilson coefficients defined by matching at a high scale (Section 40.5), we ran them down to the scale of measurement with the anomalous-dimension matrix of Section 40.6, resumming the large logarithms and tracking the operator mixing that the renormalization group of Chapter 24 forces upon operators of equal dimension. We now have, at the electroweak scale, a set of low-energy Wilson coefficients ready to be confronted with data. This final section is where the confrontation happens, and where the whole apparatus of the chapter — and of Part XI — earns its keep. The thesis is simple to state and will take the section to make precise: *a precision measurement is a measurement of Wilson coefficients*, and because each higher-dimension coefficient carries an inverse power of the scale Λ at which new physics lives, every precision number is, read

correctly, a statement about Λ and about the symmetry structure of whatever waits above it. The Standard Model Effective Field Theory (SMEFT) is the dictionary that performs the translation, and by the mid-2020s it has become the common language in which essentially all model-independent searches for physics beyond the Standard Model are phrased. We build the dictionary, work one entry of it in full numerical detail, survey the live classes of observable that feed it, revisit the hierarchy problem through its lens, give an honest dated status of the enterprise, and finally close the loop by recognizing that HQET and SMEFT — the top-down and bottom-up faces of this chapter — are one method seen from two sides.

Throughout we use the conventions in force in this chapter: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the electroweak covariant derivative $D_\mu = \partial_\mu - igW_\mu^a T^a - ig' Y B_\mu$ of Theorem 31.3, and $\overline{\mathrm{MS}}$ with $d = 4-2\varepsilon$. The Standard Model field content, gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, and the accidental B and L symmetries are imported wholesale from Chapter 33. The SMEFT itself — the organizing principle that the Standard Model is the renormalizable truncation of a tower of local operators built from Standard Model fields and invariant under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with the leading corrections the dimension-six operators of the Warsaw basis — was set up in Section 40.4 and Section 40.4.5; we build on it here rather than reconstruct it.

40.7.1 From a measurement to a scale

The structural fact that makes precision physics into a probe of short distances is the power counting of the effective expansion. Write the effective Lagrangian below the new-physics scale Λ in the SMEFT form fixed in Section 40.4,

$$\mathcal{L}_{\mathrm{SMEFT}} \;=\; \mathcal{L}_{\mathrm{SM}} \;+\; \sum_i \frac{C_i^{(6)}}{\Lambda^2}\,\mathcal{O}_i^{(6)} \;+\; \sum_i \frac{C_i^{(8)}}{\Lambda^4}\,\mathcal{O}_i^{(8)} \;+\; \cdots , \tag{40.164}$$

where each $\mathcal{O}_i^{(d)}$ is a gauge-invariant local operator of mass dimension d built from Standard Model fields, the $C_i^{(d)}$ are dimensionless Wilson coefficients, and odd-dimension operators are absent at dimension five for any B-conserving observable and suppressed by the lepton-number scale at dimension five generally (Section 40.4.5; the unique dimension-five operator is the Weinberg neutrino-mass operator). The dimension-six operators dominate the new-physics effects on any B- and L-conserving process, and they are the subject of this section.

Consider a dimensionless observable $\mathcal{O}$ — a cross section ratio, a branching fraction, a pole-mass relation, an asymmetry — whose Standard Model prediction is $\mathcal{O}_{\mathrm{SM}}$ and which probes a characteristic energy E. A single dimension-six operator shifts it by an amount fixed on dimensional grounds: the operator insertion brings a factor C_i/Λ^2, and the only other scale available to restore the dimensions of a dimensionless quantity is E (which may be the collision energy, a

particle mass, or the Higgs vacuum expectation value v, whichever the observable is built from). Hence

$$\frac{\Delta\mathcal{O}}{\mathcal{O}_{\rm SM}} \;\simeq\; c\,\frac{C_i\,E^2}{\Lambda^2}, \tag{40.165}$$

with c an $O(1)$ number computed by matching the operator onto the amplitude for the process. Two regimes of Eq. (40.165) organize all of precision physics, and we will see both at work.

Proposition 40.24 (From a null measurement to a reach)**.** *Suppose the observable $\mathcal{O}$ is measured to agree with its Standard Model prediction within a fractional precision δ, i.e. $|\Delta\mathcal{O}/\mathcal{O}_{\rm SM}| \le \delta$ at the chosen confidence level. Then Eq. (40.165) excludes*

$$\frac{\Lambda}{\sqrt{|c\,C_i|}} \;\ge\; \frac{E}{\sqrt{\delta}}. \tag{40.166}$$

For a coefficient of natural size $|c\,C_i| = O(1)$ this is a lower bound on the scale of new physics; for a measured deviation $|\Delta\mathcal{O}/\mathcal{O}_{\rm SM}| = \Delta \neq 0$ the same relation reads $\Lambda/\sqrt{|c\,C_i|} = E/\sqrt{\Delta}$ and becomes a determination *of Λ up to the unknown coefficient.*

Proof. Rearrange Eq. (40.165) for Λ^2 and impose $|\Delta\mathcal{O}/\mathcal{O}_{\rm SM}| \le \delta$: $|c\,C_i|E^2/\Lambda^2 \le \delta$ gives $\Lambda^2 \ge |c\,C_i|E^2/\delta$, which is Eq. (40.166). The equality case is immediate. The only content beyond dimensional analysis is that Λ enters quadratically because the leading operator is dimension six; were the leading operator dimension eight the bound would scale as $\delta^{-1/4}$, a weaker dependence that is exactly why the dimension-six layer is the one precision physics constrains first. □

Theorem 40.24 is deceptively elementary and completely central. It says that the reach in Λ is the product of two independent levers: the energy E at which the observable lives, entering linearly, and the inverse square root of the precision δ. Pushing either lever buys reach, and the two define the two experimental frontiers (Fig. 40.9). The *precision frontier* fixes E at a low value — the electroweak scale $v \approx 246$ GeV, or a quark mass — and drives δ down: electroweak pole observables measured at the per-mille level, Higgs signal strengths at the few-percent level, flavor transitions whose Standard Model rates are tiny and therefore exquisitely sensitive to additive new contributions. The *energy frontier* accepts a modest δ but raises E into the TeV range, exploiting the explicit E^2/Λ^2 growth of Eq. (40.165): the high-invariant-mass tails of Drell-Yan and diboson production, where a contact operator's contribution grows with energy relative to the Standard Model background and a ten-percent measurement at $E \sim 2$ TeV already probes Λ of several TeV. The two frontiers are complementary in the precise sense that they constrain the same Wilson coefficients through different windows of Eq. (40.165), and a global fit (Section 40.7.3) combines them.

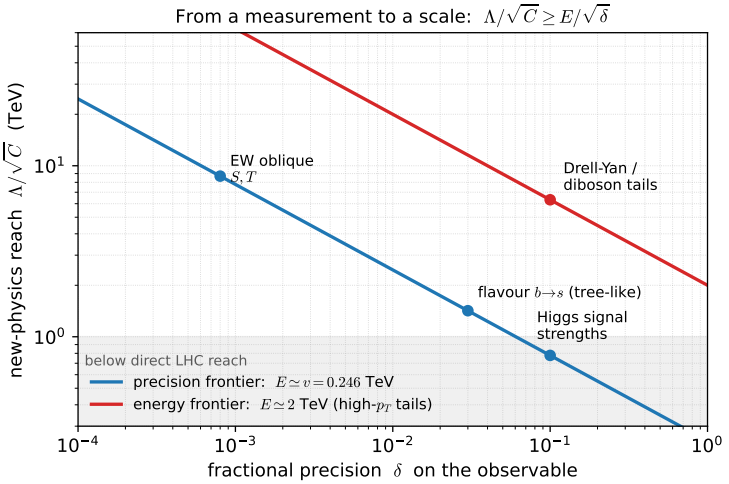


Figure 40.9: The two roads from a measurement to a new-physics scale, plotting the reach $\Lambda/\sqrt{C} \geq E/\sqrt{\delta}$ of Eq. (40.166) against the fractional precision δ. The precision frontier (blue) pins the probed energy at the electroweak scale $E \simeq v$ and buys reach by shrinking δ; the energy frontier (red) accepts modest δ but raises E into the TeV range, exploiting the E^2/Λ^2 growth of the dimension-six contribution. Representative observable classes are marked. Coefficients of $O(1)$ are assumed; loop- or CKM-suppressed Standard Model amplitudes (much of flavor) shift the points upward because the relative new-physics effect is enhanced.

Caution

The bound Eq. (40.166) is a statement about the ratio $\Lambda/\sqrt{|c\,C_i|}$, never about Λ alone. A measurement cannot separate a high scale with an $O(1)$ coefficient from a lower scale with a small coefficient: $\Lambda = 10$ TeV with $C = 1$ and $\Lambda = 1$ TeV with $C = 10^{-2}$ are indistinguishable to Eq. (40.165). The quoted "reach in Λ" always carries the silent assumption $|c\,C_i| = O(1)$, which is the assumption that the underlying theory is not weakly coupled or accidentally aligned. Reporting a reach without stating the coefficient convention is the single most common abuse of SMEFT numbers; we state it every time.

40.7.2 The oblique parameters as dimension-six coefficients

The cleanest illustration of the dictionary is the electroweak precision program, because there the new-physics effects of a wide class of theories collapse onto a short list of universal parameters. Recall from Section 34.3 the oblique (Peskin–Takeuchi) parameters S, T, U: they capture new physics that enters only through the vacuum polarizations of the electroweak gauge bosons — the "oblique"

corrections — as opposed to corrections to specific vertices or boxes. Writing $\Pi_{XY}(q^2)$ for the new-physics part of the X–Y gauge-boson self-energy ($X, Y \in \{W, Z, \gamma\}$, or in the unbroken basis the $\mathrm{SU}(2)_L$ currents $1, 2, 3$ and the hypercharge current B), the standard definitions imported from Section 34.3 are

$$\alpha\, T \;=\; \frac{\Pi_{WW}(0)}{m_W^2} - \frac{\Pi_{ZZ}(0)}{m_Z^2}, \tag{40.167}$$

$$\alpha\, S \;=\; 4\,\sin^2\!\theta_{\mathrm{W}}\cos^2\!\theta_{\mathrm{W}}\,\frac{\Pi_{ZZ}(m_Z^2) - \Pi_{ZZ}(0)}{m_Z^2} \;-\; (\gamma Z \text{ and } \gamma\gamma \text{ pieces}), \tag{40.168}$$

with α the fine-structure constant at m_Z and θ_{W} the weak mixing angle of Chapter 31. The parameter T measures the breaking of custodial symmetry — the difference between the charged- and neutral-current normalizations at zero momentum — while S measures the momentum-dependence of the neutral-current self-energy, i.e. a γ–Z kinetic mixing. We now show that each of these is, in SMEFT, the coefficient of one specific dimension-six operator.

The T parameter and custodial symmetry. Custodial symmetry was introduced in Section 31.6: the Higgs potential and kinetic term possess an accidental global $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ invariance, broken spontaneously by the vacuum expectation value to the diagonal custodial $\mathrm{SU}(2)_V$ (Theorem 31.22), and it is this residual symmetry that forces the tree-level relation $\rho \equiv m_W^2/(m_Z^2\cos^2\!\theta_{\mathrm{W}}) = 1$ (Section 31.6.5). Hypercharge gauging breaks $\mathrm{SU}(2)_R$ explicitly, which is why ρ receives radiative corrections, but at tree level the relation is exact. Any new operator that violates custodial symmetry will shift ρ away from one, and the deviation $\Delta\rho = \rho - 1$ is, by Eq. (40.167) and $\rho = 1 + \alpha T$, precisely αT. The question is which dimension-six operator does this.

In the Warsaw basis of Section 40.4.5 the relevant operator is

$$\mathcal{O}_{HD} \;=\; (H^\dagger D_\mu H)^*(H^\dagger D^\mu H), \tag{40.169}$$

where H is the Higgs doublet with hypercharge $Y_H = \tfrac{1}{2}$. We evaluate its effect on the gauge-boson masses by setting the Higgs to its vacuum configuration. In unitary gauge $H = \frac{1}{\sqrt{2}}(0,\, v + h)^\top$; for the mass terms we need only the constant part $H \to \frac{1}{\sqrt{2}}(0,\, v)^\top$. With $T^a = \tau^a/2$ on the doublet (τ^a the Pauli matrices) and $Y_H = \tfrac{1}{2}$, the covariant derivative Eq. (31.25) gives

$$D_\mu H \;=\; -\,i\Big(g\,W_\mu^a T^a + g'\tfrac{1}{2}B_\mu\Big)H, \tag{40.170}$$

so that, contracting with $H^\dagger$ and using $H^\dagger T^3 H = -\tfrac{1}{4}v^2$, $H^\dagger T^{1,2} H = 0$, and $H^\dagger H = \tfrac{1}{2}v^2$ at the vacuum,

$$H^\dagger D_\mu H \;=\; -\,i\,\frac{v^2}{4}\big(-\,g\,W_\mu^3 + g'\,B_\mu\big). \tag{40.171}$$

The combination $gW_\mu^3 - g'B_\mu$ is, by the definition of the neutral mass eigenstate (Section 31.3.4), proportional to the Z field: $gW_\mu^3 - g'B_\mu = \sqrt{g^2+g'^2}\,Z_\mu = (2m_Z/v)Z_\mu$, since $m_Z = \frac{1}{2}v\sqrt{g^2+g'^2}$. The photon, the orthogonal combination, drops out entirely. Hence

$$H^\dagger D_\mu H \;=\; i\,\frac{v\,m_Z}{2}\,Z_\mu, \qquad \mathcal{O}_{HD}\big|_{\rm vac} \;=\; \frac{v^2 m_Z^2}{4}\,Z_\mu Z^\mu. \tag{40.172}$$

The operator produces a pure Z mass term and no W mass term — it is manifestly custodial-violating, exactly as anticipated, because it singles out the neutral direction. Adding its contribution from Eq. (40.164) to the Standard Model neutral mass term $\frac{1}{2}m_Z^2 Z_\mu Z^\mu$,

$$\frac{1}{2}m_Z^2 Z_\mu Z^\mu \;+\; \frac{C_{HD}}{\Lambda^2}\,\frac{v^2 m_Z^2}{4}\,Z_\mu Z^\mu \;=\; \frac{1}{2}m_Z^2\Big(1+\frac{C_{HD}}{\Lambda^2}\frac{v^2}{2}\Big)Z_\mu Z^\mu, \tag{40.173}$$

we read off the fractional shift of the neutral mass-squared, $\delta_Z = (C_{HD}/\Lambda^2)(v^2/2)$, with the charged mass m_W untouched. The ρ parameter therefore moves to

$$\rho \;=\; \frac{m_W^2}{m_Z^2(1+\delta_Z)\cos^2\theta_{\rm W}} \;=\; 1-\delta_Z + O(\delta_Z^2), \tag{40.174}$$

so $\Delta\rho = -\delta_Z$, and with $\alpha T = \Delta\rho$,

$$\boxed{\alpha\,T \;=\; -\,\frac{v^2}{2}\,\frac{C_{HD}}{\Lambda^2}} \tag{40.175}$$

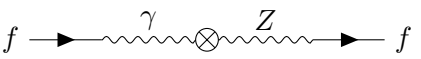
This is the explicit identification of the T parameter with a single dimension-six Wilson coefficient. The custodial symmetry of Section 31.6 is what guarantees that no *other* dimension-six operator contributes to T at tree level: an operator can shift ρ only if it violates custodial $\mathrm{SU}(2)_V$, and in the Warsaw basis $\mathcal{O}_{HD}$ is the unique such operator built from Higgs and gauge fields that generates a Z–W mass splitting at tree level. The selection rule is a direct consequence of the symmetry, not an accident of the basis.

The S parameter and γ–Z mixing. The companion operator is

$$\mathcal{O}_{HWB} \;=\; (H^\dagger \tau^a H)\,W^a_{\mu\nu}B^{\mu\nu}, \tag{40.176}$$

with τ^a the Pauli matrices and $W^a_{\mu\nu}$, $B_{\mu\nu}$ the field strengths. At the vacuum $H^\dagger\tau^3 H = -\frac{1}{2}v^2$ and the other components vanish, so

$$\mathcal{O}_{HWB}\big|_{\rm vac} \;=\; -\,\frac{v^2}{2}\,W^3_{\mu\nu}B^{\mu\nu}. \tag{40.177}$$



$$\mathcal{O}_{HWB}: \ \propto C_{HWB}\, v^2\, (q^2 \eta_{\mu\nu} - q_\mu q_\nu)$$

Figure 40.10: The S parameter as a dimension-six effect. The operator $\mathcal{O}_{HWB}$, evaluated at the Higgs vacuum, produces the γ–Z kinetic-mixing two-point insertion (crossed dot) of Eq. (40.178), momentum-dependent and transverse. Inserted into the neutral gauge propagator between fermion currents f, it shifts the effective on-shell couplings exactly in the pattern that the oblique parameter S of Section 34.3 measures. The companion operator $\mathcal{O}_{HD}$ produces instead a momentum-independent Z mass shift (the T parameter).

Rotating to mass eigenstates with $W^3_\mu = \cos\theta_{\rm W}\, Z_\mu + \sin\theta_{\rm W}\, A_\mu$ and $B_\mu = -\sin\theta_{\rm W}\, Z_\mu + \cos\theta_{\rm W}\, A_\mu$, the product of abelian field strengths is

$$W^3_{\mu\nu} B^{\mu\nu} = -\sin\theta_{\rm W}\cos\theta_{\rm W}\, Z_{\mu\nu} Z^{\mu\nu} + (\cos^2\theta_{\rm W} - \sin^2\theta_{\rm W})\, Z_{\mu\nu} A^{\mu\nu} + \sin\theta_{\rm W}\cos\theta_{\rm W}\, A_{\mu\nu} A^{\mu\nu}. \tag{40.178}$$

The middle term is a kinetic mixing of the photon and the Z: a momentum-dependent off-diagonal piece $\propto q^2$ in the neutral self-energy matrix, which is exactly the structure that S in Eq. (40.168) was built to measure. Diagrammatically the operator inserts a two-point γ–Z vertex proportional to $q_\mu q_\nu - q^2 \eta_{\mu\nu}$ into the gauge propagator (Fig. 40.10); feeding this into the definition imported from Section 34.3 gives

$$\boxed{\alpha\, S \ = \ 4\, \sin\theta_{\rm W} \cos\theta_{\rm W}\, v^2\, \frac{C_{HWB}}{\Lambda^2}} \tag{40.179}$$

We quote the overall constant $4 \sin\theta_{\rm W}\cos\theta_{\rm W}$ from the Peskin–Takeuchi normalization of Section 34.3 rather than re-derive the self-energy projection here, but the structural content — that S is the coefficient of the γ–Z kinetic-mixing operator $\mathcal{O}_{HWB}$, just as T is the coefficient of the custodial-violating mass operator $\mathcal{O}_{HD}$ — is established by Eq. (40.178) in full.

Physical Insight

The third oblique parameter U has no dimension-six representative at all. U measures a custodial- *and* isospin-violating piece of the W self-energy that, in the operator language, requires either two insertions of the Higgs vacuum beyond what S and T use or a genuinely higher-dimension structure; in the Warsaw basis the leading operator contributing to U is dimension eight. The SMEFT therefore makes a sharp, falsifiable prediction: at the dimension-six level that dominates the expansion, $U \approx 0$, and the electroweak data should be describable by S and T alone (together

with shifts in the input parameters). That two numbers suffice where the model-independent oblique analysis allowed three is not an assumption put in by hand — it is the power counting of Eq. (40.166) asserting that dimension-eight effects are smaller by E^2/Λ^2. Global electroweak fits are routinely performed both ways, and the $U \approx 0$ restriction is the SMEFT-motivated default.

We now turn the identification Eq. (40.175) into a number.

Worked Example 40.25 (The T parameter as a bound on Λ)**.** The global electroweak fit, as of the mid-2020s, constrains the oblique parameters at the level of about a tenth: with S and T floated together (and $U = 0$ imposed, per the dimension-six logic above) the data are consistent with the Standard Model, $T = 0.0 \pm 0.1$ at roughly the one-to-two-sigma level, the precise central value depending on the treatment of the disputed 2022 CDF W-boson mass measurement (Section 40.7.6). We take $|T| \leq 0.1$ as a representative exclusion and translate it into a reach on Λ for the operator $\mathcal{O}_{HD}$.

From Eq. (40.175),

$$|T| = \frac{v^2}{2\alpha}\frac{|C_{HD}|}{\Lambda^2} \quad \Longrightarrow \quad \frac{\Lambda}{\sqrt{|C_{HD}|}} = \sqrt{\frac{v^2}{2\alpha\,|T|}}. \tag{40.180}$$

Insert the numbers. The electroweak scale is $v = 246$ GeV, so $v^2 = 6.05 \times 10^4$ GeV^2; the fine-structure constant at m_Z is $\alpha = \alpha(m_Z) \approx 1/128 = 7.82 \times 10^{-3}$; and $|T| \leq 0.1$. Then

$$\frac{v^2}{2\alpha\,|T|} = \frac{6.05 \times 10^4\ \text{GeV}^2}{2\,(7.82 \times 10^{-3})\,(0.1)} = \frac{6.05 \times 10^4}{1.56 \times 10^{-3}}\ \text{GeV}^2 = 3.87 \times 10^7\ \text{GeV}^2, \tag{40.181}$$

and taking the square root,

$$\frac{\Lambda}{\sqrt{|C_{HD}|}} \gtrsim 6.2\ \text{TeV}. \tag{40.182}$$

A custodial-violating new sector that couples to the Higgs with an $O(1)$ Wilson coefficient must therefore lie above roughly 6 TeV, or it would already have shown up as a deviation of the ρ parameter from unity. Three features of the number are worth stating explicitly. First, it lies *above* the energy directly produced at the Large Hadron Collider: a precision measurement at the electroweak scale has reached past the energy frontier by exploiting the small δ of Eq. (40.166) — this is the precision frontier doing its job. Second, the bound is enormously stronger than the naive $\Lambda \gtrsim v/\sqrt{|T|} \approx 0.78$ TeV one would guess by ignoring the factor $1/\alpha$ in Eq. (40.180); the enhancement is the gauge-coupling suppression of the oblique self-energy, $\alpha T = \Delta\rho$, which makes T an unusually sensitive probe.

Third, and most importantly for the logic of the chapter, no model was specified: the bound constrains *any* ultraviolet completion through the single number C_{HD}, and a given model is tested by computing its C_{HD} by matching (Section 40.5) and comparing to Eq. (40.182). That is the inverse problem of precision physics solved in one line.

40.7.3 The live classes of observable

The oblique parameters are one entry in a much larger dictionary. By the mid-2020s the SMEFT is fit globally to several hundred measurements at once, and it is worth surveying the classes of observable that feed the fit, because each probes a different corner of the operator space and each carries a different experimental and theoretical character.

Electroweak precision observables. The Z-pole data from LEP and SLC — partial widths, asymmetries, the effective weak mixing angle — together with the W mass measure the gauge-fermion couplings and the gauge-boson self-energies at the per-mille level. In SMEFT they constrain $\mathcal{O}_{HD}$, $\mathcal{O}_{HWB}$ (the oblique combinations above), and the vertex-correction operators such as $\mathcal{O}_{H\ell}^{(1,3)}$ and $\mathcal{O}_{He}$ that modify the Z couplings to individual fermions. This is the most precise sector and the source of the strongest *universal* bounds, of order several TeV for $O(1)$ coefficients, as Theorem 40.25 illustrates.

Higgs couplings and signal strengths. Since the discovery of the Higgs boson (Section 34.4) its production and decay rates have been measured in many channels. The natural SMEFT observable is the signal strength

$$\mu_{if} \;=\; \frac{\sigma_i \cdot \mathrm{BR}_f}{(\sigma_i \cdot \mathrm{BR}_f)_{\mathrm{SM}}}, \tag{40.183}$$

the ratio of the observed rate in production channel i and decay channel f to its Standard Model value, normalized so that $\mu = 1$ is the Standard Model. Dimension-six operators shift the Higgs couplings away from their Standard Model values — $\mathcal{O}_{H\Box}$ and $\mathcal{O}_{HD}$ rescale all couplings through the Higgs kinetic normalization, $\mathcal{O}_{HW}$, $\mathcal{O}_{HB}$, $\mathcal{O}_{HWB}$, $\mathcal{O}_{HG}$ generate new tensor structures in $h \to \gamma\gamma$, $h \to Z\gamma$, $gg \to h$, and the Yukawa-like operators $\mathcal{O}_{uH}$, $\mathcal{O}_{dH}$, $\mathcal{O}_{eH}$ shift the fermion couplings — so that $\mu_{if} - 1 \simeq (\text{linear combination of } C_i)\, v^2/\Lambda^2$, matching the precision-frontier line of Fig. 40.9 with $E \simeq v$. The current few-percent precision on the leading channels translates into reaches of order 1 TeV for individual $O(1)$ coefficients, improving steadily.

High-energy tails. The energy frontier of Eq. (40.166) is realized in the high-invariant-mass or high-p_T tails of $2 \to 2$ processes: Drell-Yan ($pp \to \ell^+\ell^-$ and

$pp \to \ell\nu$) probes four-fermion operators and dipole operators whose amplitudes grow as E^2/Λ^2 relative to the Standard Model; diboson production ($pp \to WW, WZ$) probes the triple-gauge-coupling operator $\mathcal{O}_W$ with the same energy growth. Here a measurement at $E \sim 2$ TeV with $\delta \sim 10\%$ already reaches $\Lambda \sim 6$ TeV, and the reach climbs with collider energy. The growth is double-edged: it is what gives the sensitivity, but it is also where the dimension-six truncation of Eq. (40.164) is most strained, since E^2/Λ^2 is no longer small and the neglected dimension-eight terms grow faster still (Section 40.7.6).

Flavor observables. Processes that are suppressed in the Standard Model are the most sensitive of all, because a new contribution is measured against a small background rather than a large one. The flavor-changing neutral-current transitions $b \to s$ — $B \to K^{(*)}\ell^+\ell^-$, $B \to X_s\gamma$, $B_s \to \mu^+\mu^-$ — occur in the Standard Model only through loop- and CKM-suppressed penguin and box diagrams, the very $\Delta F = 1$ operators whose mixing was carried in full in Section 40.6. A dimension-six four-fermion or dipole operator that contributes at tree level competes with a Standard Model amplitude that is itself of order $G_F\alpha/(4\pi)$ times small CKM factors; the relative new-physics effect is correspondingly enhanced, and the reach in Λ can exceed tens of TeV for operators that are unsuppressed in the new sector — which is also why these observables impose the tightest constraints on the *flavor structure* of any TeV-scale extension (the flavor problem). These are live, evolving constraints: the $b \to s\ell\ell$ data showed intriguing tensions (the R_K, R_{K^*} lepton-universality ratios, and angular observables in $B \to K^*\mu\mu$) through the late 2010s and early 2020s that, by the mid-2020s, have largely regressed toward the Standard Model as measurements matured, leaving the field watching rather than concluding.

Dipole moments. The electron and muon anomalous magnetic moments and the electric dipole moments are the dimension-six dipole operators $\mathcal{O}_{eW}$, $\mathcal{O}_{eB}$ (and their quark analogs) evaluated at zero momentum. This is a thread we have followed since Chapter 20: the Standard Model value of the electron's anomaly, Schwinger's $a_e = \alpha/2\pi$, was recovered in Section 40.3.5 as the matching coefficient of the magnetic-moment operator in NRQED. The very same operator, promoted to the SMEFT, becomes the dipole Wilson coefficient whose deviation from the Standard Model prediction would signal new physics. The muon anomaly a_μ in particular was, through the early 2020s, the most discussed potential deviation, though the comparison rests on a Standard Model hadronic-vacuum-polarization input over which the data-driven and lattice determinations disagree (Section 40.7.6). In SMEFT language the would-be signal is a single number, $C_{eW}\sin\theta_W + C_{eB}\cos\theta_W$ for the muon, and a measured a_μ anomaly would fix it exactly as Theorem 40.24 fixes Λ from a deviation.

These classes are not analyzed in isolation. The defining methodological

feature of the mid-2020s SMEFT program is the *global fit*: all of the above are expressed as linear (and, increasingly, quadratic) functions of the same set of Warsaw-basis coefficients C_i/Λ^2, run to a common scale by the machinery of Section 40.6, and fit simultaneously. The output is not a single number but a multidimensional likelihood in the space of Wilson coefficients, with flat directions where different operators produce correlated effects that the available observables cannot disentangle, and sharp directions where a single precise measurement dominates. Reading that likelihood — understanding which combinations are constrained, which are blind, and what new measurement would break a given degeneracy — is the practical content of precision physics today.

40.7.4 The hierarchy problem through the EFT lens

The same power counting that makes precision physics a probe of high scales turns, when applied to the one super-renormalizable operator in the Standard Model, into the sharpest statement of the hierarchy problem. The naturalness criterion was developed abstractly in Section 38.6: a Wilson coefficient should be of order the natural size set by its mass dimension and the cutoff, unless a symmetry protects it, and a coefficient much smaller than its natural size is technically unnatural — fine-tuned — and demands explanation.

Apply this to the Higgs mass term. In the operator counting of Eq. (40.164) the operators are sorted by dimension; the *relevant* operator $H^\dagger H$ has dimension two, below the marginal renormalizable operators, and carries a coefficient with dimension mass-squared:

$$\mathcal{L} \supset -\mu_H^2\, H^\dagger H, \qquad \mu_H^2 = c_H\, \Lambda^2, \tag{40.184}$$

where naturalness asserts $c_H = O(1)$ unless protected. Here Λ is the scale of whatever ultraviolet physics the Standard Model is an effective theory of — the same Λ that the precision observables of this section are bounding. The measured value is fixed by the physical Higgs mass and vacuum expectation value, $\mu_H^2 \simeq m_h^2/2 \simeq (88\ \text{GeV})^2$. The hierarchy problem is the collision of this measured value with Eq. (40.184): if there is any new physics at a scale Λ — and the dimension-six operators of this section presuppose there is — then c_H must be tuned to

$$c_H = \frac{\mu_H^2}{\Lambda^2} \sim \left(\frac{88\ \text{GeV}}{\Lambda}\right)^2, \tag{40.185}$$

which for Λ at the precision-frontier reach of Theorem 40.25, $\Lambda \sim 6$ TeV, already demands $c_H \sim 2 \times 10^{-4}$, a tuning of one part in a few thousand, and for Λ at a grand-unified ($\sim 10^{16}$ GeV) or Planckian ($\sim 10^{19}$ GeV) scale demands a tuning of one part in 10^{28} or 10^{34}. No symmetry of the Standard Model forbids the operator $H^\dagger H$ — it is a Lorentz and gauge singlet — so nothing in Section 38.6 protects c_H, and its smallness is the unnaturalness.

The EFT lens makes precise both what the problem is and what it is not. It *is* the statement that the dimension-two coefficient is anomalously small relative to its natural size, in exactly the same sense, and measured by exactly the same power counting, as the dimension-six coefficients are constrained to be small by the data of this section. It is *not* a statement about quadratic divergences in any particular regulator: in $\overline{\text{MS}}$ there are no power divergences at all, yet the problem persists, because it is a statement about the sensitivity of the low-energy coefficient to the high-energy scale Λ, a physical sensitivity that no choice of regulator removes (this is the content of Section 38.6). Read this way, the hierarchy problem is not a disease of the calculation but a property of the operator expansion: the Higgs mass is the one relevant operator in a theory of marginal and irrelevant ones, and a relevant operator is, by definition, sensitive to the ultraviolet. The same fact that lets precision measurements reach up to Λ through the irrelevant operators lets Λ reach down to the Higgs mass through the relevant one.

Physical Insight

There is a structural asymmetry between the two ways the scale Λ enters this chapter. Through the irrelevant (dimension-six) operators, Λ enters suppressed: their effects vanish as $\Lambda \to \infty$, which is the Appelquist–Carazzone decoupling of Chapter 38 and the reason precision physics works at all — heavy physics leaves only small, computable traces. Through the relevant (dimension-two) operator, Λ enters enhanced: its effect grows as $\Lambda \to \infty$, which is the hierarchy problem. Decoupling and the hierarchy problem are the same power counting read in opposite directions on the dimension axis, and the marginal (dimension-four) operators of the Standard Model sit exactly at the boundary, sensitive to Λ only logarithmically — the running couplings of Chapter 24. The entire chapter is, in this sense, a tour of the three sides of the dimension axis: relevant, marginal, irrelevant.

40.7.5 HQET and SMEFT as one method

We can now state the unifying lesson the chapter has been earning. An effective field theory is fixed by three inputs and one procedure. The inputs are a *field content*, a *symmetry*, and a *power counting*; the procedure is to write the most general local Lagrangian consistent with the first two, organized by the third, and then to determine its Wilson coefficients. The two halves of this chapter are this single recipe applied with the determination step carried out in the two possible ways.

In HQET (Section 40.1) the short-distance theory is *known*: it is QCD, and the heavy field is a Standard Model quark. The field content is the heavy-quark field with its lower components integrated out, the symmetry is the emergent

heavy-quark spin–flavor symmetry plus the gauge symmetry of QCD, and the power counting is in $\Lambda_{\mathrm{QCD}}/m_Q$. Because the ultraviolet theory is in hand, the Wilson coefficients are determined by *matching*: one computes the same amplitude in full QCD and in the effective theory and equates them, fixing each coefficient as a calculable function of $\alpha_s(m_Q)$. The famous payoff — the spin–flavor symmetry invisible in the QCD Lagrangian but manifest in the heavy-quark limit — is a statement that the matching produces a more symmetric effective theory than one might have written down, and Yang–Mills theory, followed since the asymptotic freedom of Theorem 27.7 into its heavy-quark regime, supplies the running coupling that the matching coefficients depend on.

In SMEFT (Section 40.4) the short-distance theory is *unknown*: it is whatever lies above the electroweak scale, and the "heavy field" is hypothetical. The field content is that of the Standard Model, the symmetry is $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ (with the accidental B and L of Chapter 33), and the power counting is in E/Λ. Because the ultraviolet theory is not in hand, the Wilson coefficients cannot be matched; they are determined by *measurement*, which is the entire content of this section. The precision observable plays the role that the full-QCD amplitude plays in HQET — the external input that pins the coefficient — with the crucial difference that the input is empirical rather than calculable, and so the coefficient, once measured, is a window onto the unknown ultraviolet rather than a consistency check on a known one.

Physical Insight

Matching and measurement are the same operation performed against different references. In both cases one equates an effective-theory prediction, expressed in Wilson coefficients, to an external determination of an amplitude or observable, and solves for the coefficients. When the external determination comes from a known ultraviolet theory, we call it matching and the coefficients are predictions; when it comes from experiment, we call it measurement and the coefficients are constraints on the unknown ultraviolet. "Write all operators, power count, match or measure, run" is one method. HQET and SMEFT are its two faces, and the only thing that distinguishes them is whether the short-distance theory is written down or waiting to be discovered. The renormalization-group running of Section 40.6 is common to both: once the coefficients are fixed at one scale, by either route, the same anomalous-dimension matrix carries them to any other.

This is why the EFT paradigm of Chapter 38 is not merely one technique among many but the organizing language of modern particle physics. Chiral perturbation theory (Chapter 39) determined its low-energy constants by measurement because the ultraviolet theory, though known to be QCD, is strongly

coupled and not perturbatively solvable — a third case that is matching in principle and measurement in practice. HQET matches to a weakly coupled ultraviolet. SMEFT measures against an unknown one. All three are the same construction, and the reader who has followed the chapter has learned not three subjects but one, instantiated three times.

40.7.6 Status as of the mid-2020s

Honesty about the current state of the program requires a dated assessment, and the date matters because this is a moving frontier.

As of the mid-2020s the SMEFT has become the lingua franca of beyond-the-Standard-Model searches. Where an earlier generation reported limits on specific models — supersymmetric particles of definite mass, a Z' of definite coupling — the field now also, and increasingly primarily, reports constraints on Warsaw-basis Wilson coefficients, precisely because these are model-independent and translate into any specific model by matching. The global fits combine electroweak, Higgs, top, diboson, and flavor data into simultaneous bounds on dozens of coefficients, and the qualitative result is unambiguous and, from the standpoint of discovery, disappointing: there is no decisive, corroborated deviation from the Standard Model. The coefficients are consistent with zero, and the corresponding scale Λ is pushed into the multi-TeV range for $O(1)$ coefficients across most operator directions. The Standard Model, extended only by the dimension-five neutrino-mass operator, describes all confirmed laboratory data.

Several tensions animated the field during the late 2010s and early 2020s and deserve a sober accounting, because their trajectory is instructive. The $b \to s\ell\ell$ lepton-universality ratios R_K and R_{K^*} showed deviations from the Standard Model that, by 2022–2023, the LHCb collaboration's improved analyses brought back into agreement, illustrating how a SMEFT "signal" can evaporate as systematics are understood. The CDF collaboration's 2022 measurement of the W-boson mass lay many standard deviations above the Standard Model prediction and above other measurements; it is in tension with the rest of the electroweak data and with subsequent measurements, and the field has not reached consensus on it — which is why Theorem 40.25 flagged that the central value of T depends on how this measurement is treated. The muon anomalous magnetic moment a_μ showed a persistent discrepancy with the data-driven Standard Model prediction, but the Standard Model prediction itself is unsettled, with lattice-QCD evaluations of the hadronic vacuum polarization disagreeing with the data-driven dispersive evaluation by enough to remove the anomaly; until that internal disagreement is resolved, a_μ cannot be cleanly read as a Wilson coefficient. The lesson common to all three is the one Theorem 40.24 should already have taught: a deviation becomes new physics only when both the measurement and the Standard Model prediction are under control, and the SMEFT translation is only as trustworthy as the less certain of the two.

One genuinely open structural question concerns the organization of the expansion itself. We have built SMEFT on the assumption that the Higgs is the lower component of an $\mathrm{SU}(2)_L$ doublet and that the electroweak symmetry is realized linearly, so that operators are organized by canonical mass dimension and the expansion parameter is E/Λ. An alternative organization, the Higgs Effective Field Theory (HEFT, also called electroweak-chiral Lagrangian), treats the physical Higgs as a generic singlet scalar and realizes the electroweak symmetry nonlinearly on the would-be Goldstone bosons of $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ breaking — exactly the chiral construction of Chapter 39 transplanted to the electroweak sector, with the expansion organized by chiral dimension rather than canonical dimension. SMEFT is the special case of HEFT in which the Higgs and the Goldstones assemble into a doublet; whether nature is described by the more restrictive SMEFT organization or requires the more general HEFT one is a physical question, tied to whether the Higgs couplings deviate in the correlated doublet pattern, and it is not settled as of the mid-2020s. Most analyses adopt SMEFT as the default because it is more predictive and because no data demand otherwise, but the choice is an assumption, not a theorem, and the reader should know it is live.

With this the effective-field-theory part of the book is complete. We have built the paradigm abstractly (Chapter 38), deployed it bottom-up on the broken chiral symmetry of QCD (Chapter 39), and in this chapter run it in both directions at once — top-down through HQET, where the short-distance theory is known and matched, and bottom-up through SMEFT, where it is unknown and measured — arriving at the reading of precision physics as the measurement of Wilson coefficients and hence of the scale and structure of whatever lies above. The EFT lens, now established as the organizing language of precision and of beyond-the-Standard-Model searches, is the one we carry forward: the non-perturbative methods of Part XII confront the regime where the matching of an EFT to its ultraviolet completion can no longer be done in perturbation theory, and the frontier parts that follow inherit the same discipline of separating scales, writing the most general dynamics consistent with the symmetries, and asking what the data can say about the rest.

Further reading

The heavy-quark effective theory grew from the observation of an emergent spin–flavour symmetry in the static limit by Isgur and Wise [276] and the form-factor reduction of Isgur and Wise [277]; the protection of the zero-recoil normalization against first-order power corrections is Luke [325], and the reparametrization invariance that freezes the kinetic coefficient is Luke and Manohar [326]. The comprehensive review is Neubert [362], and the textbook treatment, including the trace formalism and the $1/m_Q$ expansion carried here, is Manohar and Wise

[336]; Georgi [218] gives a compact pedagogical account. The nonrelativistic effective theory whose Fermi coefficient packages Schwinger's a_e originates with Caswell and Lepage [117]. For the weak effective Hamiltonian, the operator mixing, and the $\Delta S = 1$ QCD enhancement worked in Section 40.6, the standard review is Buchalla et al. [97]. On the SMEFT side, the canonical non-redundant dimension-six basis is that of Grzadkowski et al. [240]; the complete one-loop renormalization-group running is Alonso et al. [20], the Hilbert-series operator counting is Henning et al. [265], and a thorough modern review of the framework and its phenomenology is Brivio and Trott [93]. The oblique-parameter analysis underlying Section 40.7.2 is Peskin and Takeuchi [397, 398], and the decoupling theorem on which the whole construction rests is Appelquist and Carazzone [31]; Schwinger's original moment is Schwinger [433]. The standard textbooks — Peskin and Schroeder [396], Schwartz [432], and Weinberg [493] — treat the effective-Lagrangian viewpoint and the matching/running machinery in complementary depth.

Table 40.1: The baryon-number-conserving dimension-six operators of the Warsaw basis, bosonic and mixed classes (one generation). Here $X_{\mu\nu} = G^A_{\mu\nu}, W^I_{\mu\nu}, B_{\mu\nu}$ are the field strengths, $\widetilde{X}_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}X^{\rho\sigma}$ their duals, $t^I = \tau^I/2$ and T^A the $\mathrm{SU}(2)_L$ and $\mathrm{SU}(3)_c$ generators, and $H^\dagger \overleftrightarrow{D}_\mu H \equiv H^\dagger D_\mu H - (D_\mu H)^\dagger H$. The tilde operators ($\mathcal{O}_{\widetilde{G}}$, etc.) are the CP-odd partners obtained by $X \to \widetilde{X}$.

Class	#	Operators	Form
X^3	4	$\mathcal{O}_G, \mathcal{O}_{\widetilde{G}}$	$f^{ABC}G^{A\,\nu}_{\mu}G^{B\,\rho}_{\nu}G^{C\,\mu}_{\rho}$ (and dual)
		$\mathcal{O}_W, \mathcal{O}_{\widetilde{W}}$	$\epsilon^{IJK}W^{I\,\nu}_{\mu}W^{J\,\rho}_{\nu}W^{K\,\mu}_{\rho}$ (and dual)
H^6	1	$\mathcal{O}_H$	$(H^\dagger H)^3$
H^4D^2	2	$\mathcal{O}_{H\Box}$	$(H^\dagger H)\Box(H^\dagger H)$
		$\mathcal{O}_{HD}$	$\left(H^\dagger D_\mu H\right)^*\left(H^\dagger D^\mu H\right)$
X^2H^2	8	$\mathcal{O}_{HG}, \mathcal{O}_{H\widetilde{G}}$	$(H^\dagger H)\, G^A_{\mu\nu}G^{A\,\mu\nu}$ (and dual)
		$\mathcal{O}_{HW}, \mathcal{O}_{H\widetilde{W}}$	$(H^\dagger H)\, W^I_{\mu\nu}W^{I\,\mu\nu}$ (and dual)
		$\mathcal{O}_{HB}, \mathcal{O}_{H\widetilde{B}}$	$(H^\dagger H)\, B_{\mu\nu}B^{\mu\nu}$ (and dual)
		$\mathcal{O}_{HWB}, \mathcal{O}_{H\widetilde{W}B}$	$(H^\dagger \tau^I H)\, W^I_{\mu\nu}B^{\mu\nu}$ (and dual)
ψ^2H^3	3	$\mathcal{O}_{eH}$	$(H^\dagger H)(\bar{L}\, e\, H)$
		$\mathcal{O}_{dH}$	$(H^\dagger H)(\bar{Q}\, d\, H)$
		$\mathcal{O}_{uH}$	$(H^\dagger H)(\bar{Q}\, u\, \widetilde{H})$
ψ^2XH	8	$\mathcal{O}_{eW}, \mathcal{O}_{eB}$	$(\bar{L}\sigma^{\mu\nu}e)\tau^I H\, W^I_{\mu\nu},\ (\bar{L}\sigma^{\mu\nu}e)H\, B_{\mu\nu}$
		$\mathcal{O}_{uG}, \mathcal{O}_{uW}, \mathcal{O}_{uB}$	$(\bar{Q}\sigma^{\mu\nu}T^A u)\widetilde{H}\, G^A_{\mu\nu},\ \ldots\widetilde{H}W,\ \ldots\widetilde{H}B$
		$\mathcal{O}_{dG}, \mathcal{O}_{dW}, \mathcal{O}_{dB}$	$(\bar{Q}\sigma^{\mu\nu}T^A d)H\, G^A_{\mu\nu},\ \ldots HW,\ \ldots HB$
ψ^2H^2D	8	$\mathcal{O}^{(1)}_{Hl}, \mathcal{O}^{(3)}_{Hl}$	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{L}\gamma^\mu L),\ (H^\dagger i\overleftrightarrow{D}^I_\mu H)(\bar{L}\tau^I\gamma^\mu L)$
		$\mathcal{O}^{(1)}_{Hq}, \mathcal{O}^{(3)}_{Hq}$	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{Q}\gamma^\mu Q),\ (H^\dagger i\overleftrightarrow{D}^I_\mu H)(\bar{Q}\tau^I\gamma^\mu Q)$
		$\mathcal{O}_{He}, \mathcal{O}_{Hu}, \mathcal{O}_{Hd}$	$(H^\dagger i\overleftrightarrow{D}_\mu H)(\bar{e}\gamma^\mu e),\ \ldots(\bar{u}\gamma^\mu u),\ \ldots(\bar{d}\gamma^\mu d)$
		$\mathcal{O}_{Hud}$	$(\widetilde{H}^\dagger iD_\mu H)(\bar{u}\gamma^\mu d)$

Table 40.2: The 25 baryon-number-conserving four-fermion operators of the Warsaw basis (one generation), grouped by chirality structure. Parenthesized superscripts (1) and (3)/(8) distinguish singlet from triplet ($\mathrm{SU}(2)_L$) or colour-octet contractions. L, Q are the doublets and e, u, d the singlets.

Structure	#	Operators
$(\bar{L}L)(\bar{L}L)$	5	$\mathcal{O}_{ll},\ \mathcal{O}^{(1)}_{qq},\ \mathcal{O}^{(3)}_{qq},\ \mathcal{O}^{(1)}_{lq},\ \mathcal{O}^{(3)}_{lq}$
$(\bar{R}R)(\bar{R}R)$	7	$\mathcal{O}_{ee},\ \mathcal{O}_{uu},\ \mathcal{O}_{dd},\ \mathcal{O}_{eu},\ \mathcal{O}_{ed},\ \mathcal{O}^{(1)}_{ud},\ \mathcal{O}^{(8)}_{ud}$
$(\bar{L}L)(\bar{R}R)$	8	$\mathcal{O}_{le},\ \mathcal{O}_{lu},\ \mathcal{O}_{ld},\ \mathcal{O}_{qe},\ \mathcal{O}^{(1)}_{qu},\ \mathcal{O}^{(8)}_{qu},\ \mathcal{O}^{(1)}_{qd},\ \mathcal{O}^{(8)}_{qd}$
$(\bar{L}R)(\bar{L}R)\,\&\,(\bar{L}R)(\bar{R}L)$	5	$\mathcal{O}_{ledq},\ \mathcal{O}^{(1)}_{quqd},\ \mathcal{O}^{(8)}_{quqd},\ \mathcal{O}^{(1)}_{lequ},\ \mathcal{O}^{(3)}_{lequ}$

Part XII

Non-Perturbative Methods

Chapter 41

Solitons and Topology

Every result of the book so far has been built on a single tacit assumption: that the fields are small. Quantization expanded about a vacuum, perturbation theory expanded about free fields, and the Feynman diagram counted fluctuations that die away at spatial infinity. That assumption is an approximation, and there are configurations it cannot reach. This chapter studies the field theory of the large: smooth, static, finite-energy solutions of the classical equations of motion that are not small deformations of the vacuum and cannot be deformed into it. They are the solitons, and what protects them is a conservation law of a wholly new kind — one that owes nothing to Noether's theorem of Chapter 7. A soliton is stable not because a symmetry forbids its decay but because the configuration is *knotted*: the field at the boundary of space wraps the manifold of vacua in a way that no continuous, finite-energy motion can unwrap. The charge that counts the knotting is an integer, it is conserved off shell, and it is invisible to perturbation theory. This is the first genuinely non-perturbative structure of the book, and the gateway to Part XII.

The development is deliberately self-contained in its topology: the homotopy groups we need are built from scratch as the examples demand them. What the chapter imports is the classical apparatus already in hand. The action principle, the stress tensor whose T^{00} defines the static energy we minimize, and Noether's machinery come from Chapters 6 and 7; the order-parameter language and the structure of the vacuum manifold as a coset G/H are the spontaneous-symmetry-breaking vocabulary of Chapter 29, recalled but rebuilt in the form we need; the coset and Lie-group facts behind the homotopy classification are those of Chapter 2. The gauged solitons require the gauge logic of Chapter 8 — including the Dirac quantization argument and the θ-term as a total derivative — and the non-abelian field strength, covariant derivative, and adjoint Higgs mechanics of Chapter 25. Two benchmark theories supply the spine: ϕ^4 theory, the real scalar of Chapter 6 in its broken phase, furnishes the kink that opens the chapter, and Yang–Mills theory delivers the 't Hooft–Polyakov monopole as its non-abelian

climax.

The chapter follows a single ascending arc, one rung per spatial dimension. Section 41.1 cashes every promise on the simplest case that exists, the kink of a real scalar in one dimension: finite energy splits field space into disconnected sectors, an integer topological charge labels them, an off-shell-conserved current carries it, and a first-order Bogomolny rewriting computes the soliton mass exactly. Section 41.2 extracts the general principle, proving that finite-energy sectors are classified by a homotopy group of the vacuum manifold and that the charge is the topological degree of the boundary map — integral and invariant as theorems, not accidents. Section 41.3 then delivers Derrick's sharp negative result: in two or more spatial dimensions, scalars alone admit no soliton, and the scaling argument that proves it dictates the cure. That cure is gauge fields, and the next two sections build the two solitons it makes possible. Section 41.4 constructs the Abrikosov–Nielsen–Olesen vortex in two dimensions, with its quantized magnetic flux; Section 41.5 climbs to three dimensions and a non-abelian group to build the 't Hooft–Polyakov monopole, in which magnetic charge emerges as a topological invariant from an everywhere-regular field. Section 41.6 closes by isolating the one algebraic identity — Bogomol'nyi's completion of the square — behind every mass bound met along the way, exhibits the kink, vortex, and monopole as its three instances, solves the Prasad–Sommerfield limit in closed form, and ends with the dyon spectrum and the Witten effect, the point at which the soliton thread rejoins the θ-vacuum.

A brief word on scope and rigor. The topology is developed honestly: $\pi_1(S^1) \cong \mathbb{Z}$, the degree formula, and the homotopy classification of sectors are proved, not quoted, and every soliton profile, flux integral, charge identity, and saturated mass is carried to the end. Where the chapter looks forward — to the θ-vacuum and the dilute instanton gas of Chapter 42, to cosmological-defect and grand-unified monopole phenomenology, and to the supersymmetric meaning of the BPS bound — it says so and points the reader on, never substituting a citation for an argument it owes.

41.1 The Kink: A Soliton from a Double Well

The chapter introduction promised a conservation law of a new kind—one that protects a field configuration not because a symmetry forces it through Noether's theorem, but because the configuration is *knotted*, in a sense we must now make precise, and cannot be combed flat without paying an infinite price. Such promises are cheap until they are cashed on an example. We cash this one on the simplest example that exists: a single real scalar field in one spatial dimension, sitting in the double-well potential of ϕ^4 theory. Everything the rest of the chapter needs—the role of finite energy, the splitting of field space into disconnected sectors, an integer charge, a current conserved without ever

invoking the equations of motion, and the first-order "Bogomolny" rewriting that computes the soliton's mass exactly—appears here in a setting transparent enough that not a single step need be taken on faith. The higher-dimensional solitons of later sections are, conceptually, this calculation with more indices.

41.1.1 The model: a scalar line in a double well

We work in 1+1 spacetime dimensions with coordinates $x^\mu = (t, x)$, mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-)$, and natural units. The field is a single real scalar $\phi(t,x)$, governed by the frozen ϕ^4 Lagrangian Eq. (6.56) taken in its broken phase—the same double well that organized spontaneous symmetry breaking in Chapter 29:

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu\phi\,\partial^\mu\phi - V(\phi), \qquad V(\phi) = -\tfrac{1}{2}\left|m^2\right|\,\phi^2 + \frac{\lambda}{4!}\,\phi^4, \qquad \lambda > 0. \tag{41.1}$$

This is Eq. (29.6) verbatim. Its discrete reflection symmetry, the $\mathbb{Z}_2$ of Theorem 7.22 acting by $\phi \to -\phi$, is exact, and its two degenerate minima sit at the vacua of Eq. (29.8),

$$\phi = \pm v, \qquad v^2 = \frac{6\left|m^2\right|}{\lambda}. \tag{41.2}$$

It is convenient—indeed, as we will see in a moment, *necessary*—to measure the potential from its minimum value. Completing the square in ϕ^2,

$$V(\phi) = \frac{\lambda}{4!}\left(\phi^2 - v^2\right)^2 - \frac{\lambda}{4!}\,v^4, \tag{41.3}$$

which one checks by expanding: the cross term $-2v^2\phi^2$ supplies $-\frac{\lambda}{4!}2v^2\phi^2 = -\frac{\lambda v^2}{12}\phi^2 = -\frac{1}{2}\left|m^2\right|\phi^2$ on using $\lambda v^2 = 6\left|m^2\right|$, reproducing Eq. (41.1), and the constant $-\frac{\lambda}{4!}v^4$ is the (negative) value of V at either well. We define the *shifted potential*

$$\boxed{\widetilde{V}(\phi) \;\equiv\; V(\phi) - V_{\min} \;=\; \frac{\lambda}{4!}\left(\phi^2 - v^2\right)^2 \;\geq 0\,,} \qquad V_{\min} = -\frac{\lambda}{4!}v^4, \tag{41.4}$$

which is non-negative everywhere and vanishes *exactly* at, and only at, the two vacua $\phi = \pm v$. Shifting V by a constant changes neither the equations of motion (a constant drops out of V') nor the $\mathbb{Z}_2$ symmetry; what it buys is a vacuum of zero energy density, without which a field stretched over an infinite line could never have finite total energy.

Static, finite-energy configurations are the objects of interest. The energy of a field configuration is the spatial integral of the energy density $T^{00} = \mathcal{H}$, identified for any scalar Lagrangian in Eq. (6.173); for a configuration that does not depend on time, $\dot\phi = 0$, the kinetic term vanishes and the energy reduces to the static functional

$$E[\phi] \;=\; \int_{-\infty}^{\infty} \mathrm{d}x\,\left[\,\tfrac{1}{2}\,(\phi')^2 + \widetilde{V}(\phi)\,\right], \qquad \phi' \equiv \frac{d\phi}{dx}, \tag{41.5}$$

where we have used the shifted potential Eq. (41.4) precisely so that the vacua contribute zero. Both terms in the integrand are pointwise non-negative: a gradient-squared and a manifestly non-negative potential. The energy is therefore the sum of two non-negative pieces, and it can be finite only if each piece integrates to something finite. This single observation, pressed for all it is worth, is the origin of everything topological in the problem.

41.1.2 Finite energy forces the field into a vacuum at infinity

Definition 41.1 (Finite-energy configuration)**.** A static configuration $\phi : \mathbb{R} \to \mathbb{R}$ is of *finite energy* if $E[\phi] < \infty$, with E the functional Eq. (41.5). We require in addition that ϕ be continuous with a piecewise-continuous derivative, so that the integrand makes sense.

Proposition 41.2 (Boundary conditions and topological sectors)**.** *Every finite-energy configuration approaches a vacuum at each spatial end,*

$$\phi(x) \xrightarrow[x\to+\infty]{} \phi_+ \in \{+v, -v\}, \qquad \phi(x) \xrightarrow[x\to-\infty]{} \phi_- \in \{+v, -v\}. \tag{41.6}$$

The pair (ϕ_-, ϕ_+) takes four values, splitting the space of finite-energy configurations into four classes; no continuous deformation through finite-energy configurations carries one class into another.

Proof. Finiteness of Eq. (41.5) requires both non-negative pieces to be integrable. Consider first the potential term, $\int \widetilde{V}(\phi)\,\mathrm{d}x < \infty$. The integrand is continuous and non-negative; for its integral over the whole line to converge, the integrand must tend to zero as $x \to \pm\infty$. Hence $\widetilde{V}(\phi(x)) \to 0$, and because $\widetilde{V}$ vanishes only at its two zeros, the field value must approach the set $\{+v, -v\}$. It remains to rule out the possibility that $\phi(x)$ oscillates between $+v$ and $-v$ without settling on one of them. This is where the gradient term earns its keep. Suppose, for contradiction, that ϕ does not converge as $x \to +\infty$; then there are arbitrarily large x where ϕ is near $+v$ and arbitrarily large x where it is near $-v$, so on infinitely many disjoint intervals ϕ traverses, say, the band from $-v/2$ to $+v/2$. Across each such interval the field changes by at least v, so by the Cauchy–Schwarz inequality the gradient energy collected on an interval $[a, b]$ over which ϕ runs monotonically from $-v/2$ to $+v/2$ obeys

$$\int_a^b \tfrac{1}{2}(\phi')^2\,\mathrm{d}x \;\geq\; \frac{1}{2(b-a)}\Big(\int_a^b \phi'\,\mathrm{d}x\Big)^2 \;=\; \frac{v^2}{2(b-a)}. \tag{41.7}$$

For the total to be finite the lengths $b - a$ of these crossing intervals would have to grow without bound; but on a long interval where ϕ stays in the band $[-v/2, +v/2]$ the potential term contributes at least $\widetilde{V}_{\min,\,\mathrm{band}} \cdot (b - a)$, where $\widetilde{V}_{\min,\,\mathrm{band}} > 0$ is the (strictly positive) minimum of $\widetilde{V}$ on that closed band, and

this diverges. Either way the energy is infinite. Hence ϕ converges to a single vacuum at each end, establishing Eq. (41.6).

The four classes are labelled by (ϕ_-, ϕ_+). A continuous deformation $\phi_s(x)$, $s \in [0,1]$, through finite-energy configurations gives, at each fixed s, well-defined endpoint values $\phi_\pm(s) \in \{+v, -v\}$ that depend continuously on s; a continuous map into the two-point set $\{+v, -v\}$ is constant. So neither endpoint can jump, and the class is preserved under any such deformation. □

The four classes are the *topological sectors*. Two of them, $(\phi_-, \phi_+) = (+v, +v)$ and $(-v, -v)$, contain the constant vacuum solutions and the small fluctuations around them—these are the configurations perturbation theory sees. The other two, $(-v, +v)$ and $(+v, -v)$, contain field profiles that must climb out of one well and descend into the other. Theorem 41.2 says no amount of finite-energy wiggling connects these to a vacuum: the lowest-energy configuration in a non-vacuum sector, whatever it is, is stranded there. That is the entire mechanism of topological stability, stated before we have written down a single solution. It is worth pausing on what made it work, because the same logic recurs at every rung of the chapter's ladder.

Physical Insight

Stability here is not the stability of a Noether charge. No symmetry of Eq. (41.1) is responsible; the $\mathbb{Z}_2$ of Theorem 7.22 is a symmetry, but it is discrete and yields no conserved current at all (Section 7.3.6). What stabilizes the sectors is a fact about the geometry of field space at infinity: finite energy pins the field to the discrete vacuum set $\{+v, -v\}$ at each end, and a discrete set cannot be traversed continuously. The configuration is trapped not by a force but by a *disconnection*—there is no continuous path of finite-energy configurations from one sector to another. We are about to package this disconnection into an integer.

41.1.3 The static equation, its first integral, and the kink profile

We now find the lowest-energy configuration in a non-vacuum sector explicitly. A stationary point of the energy functional Eq. (41.5) satisfies its Euler–Lagrange equation, obtained exactly as in Theorem 6.8 but for the one-dimensional functional $E[\phi] = \int \mathrm{d}x\, [\frac{1}{2}(\phi')^2 + \widetilde{V}(\phi)]$. Varying, $\delta E = \int \mathrm{d}x\, [\phi'\, \delta\phi' + \widetilde{V}'(\phi)\, \delta\phi] = \int \mathrm{d}x\, [-\phi'' + \widetilde{V}'(\phi)]\, \delta\phi$ after one integration by parts (the boundary term vanishing because $\delta\phi \to 0$ at infinity, where every competitor shares the same vacuum endpoints). Stationarity for arbitrary $\delta\phi$ gives the *static field equation*

$$\phi'' = \widetilde{V}'(\phi). \tag{41.8}$$

This is precisely the static ($\dot\phi = 0$) limit of the full covariant equation of motion $\Box\phi + \widetilde{V}'(\phi) = 0$ of ϕ^4 theory (Eq. (6.56)), since in mostly-minus signature $\Box = \partial_t^2 - \partial_x^2$ collapses to $-\partial_x^2$ on a time-independent field.

Equation Eq. (41.8) has a mechanical reading that turns the problem into one we already know how to solve. Read x as a *time* and ϕ as the *position* of a unit-mass particle; then Eq. (41.8) is Newton's law $\phi'' = -\partial_\phi U$ for motion in the *inverted* potential

$$U(\phi) \equiv -\widetilde{V}(\phi), \tag{41.9}$$

a double *hump* with two equal maxima at $\phi = \pm v$ (Fig. 41.1, left). The conserved "mechanical energy" of this analogue particle is

$$\mathcal{E}_{\text{mech}} = \tfrac{1}{2}(\phi')^2 + U(\phi) = \tfrac{1}{2}(\phi')^2 - \widetilde{V}(\phi), \tag{41.10}$$

and it is conserved in x along any solution: differentiating and using Eq. (41.8),

$$\frac{d\mathcal{E}_{\text{mech}}}{dx} = \phi'\phi'' - \widetilde{V}'(\phi)\,\phi' = \phi'\big(\phi'' - \widetilde{V}'(\phi)\big) = 0. \tag{41.11}$$

Now impose finite energy. By Theorem 41.2 the field approaches a vacuum at each end, where $\phi' \to 0$ and $\widetilde{V} \to 0$, so $\mathcal{E}_{\text{mech}} \to 0$. Being constant, it is zero everywhere:

$$\boxed{\tfrac{1}{2}(\phi')^2 = \widetilde{V}(\phi)\ .} \tag{41.12}$$

This *first integral* of Eq. (41.8) is the heart of the matter. It is a first-order equation equivalent, for finite-energy configurations, to the second-order field equation, and it says the gradient energy density and the potential energy density are *equal at every point*—a statement we will rediscover, and elevate, as the Bogomolny condition in Section 41.1.5. Mechanically, it is the trajectory that leaves the top of one hump with vanishing speed and arrives at the top of the other, the unique zero-energy roll between the two maxima, which takes infinite "time" x to complete because the particle decelerates to rest as it asymptotes the second summit.

Solving Eq. (41.12) is now integration by quadrature. Taking the positive square root (the choice that makes ϕ increase, i.e. the sector $\phi_- = -v \to \phi_+ = +v$),

$$\phi' = \sqrt{2\widetilde{V}(\phi)} = \sqrt{\frac{2\lambda}{4!}}\,(v^2 - \phi^2) = \sqrt{\frac{\lambda}{12}}\,(v^2 - \phi^2), \tag{41.13}$$

where on the relevant range $-v < \phi < v$ the factor $|\phi^2 - v^2| = v^2 - \phi^2$. Separating variables,

$$\frac{\mathrm{d}\phi}{v^2 - \phi^2} = \sqrt{\frac{\lambda}{12}}\,\mathrm{d}x. \tag{41.14}$$

The left side integrates by the standard substitution $\phi = v\tanh\theta$, for which $\mathrm{d}\phi = v\,\mathrm{sech}^2\theta\,\mathrm{d}\theta$ and $v^2 - \phi^2 = v^2\,\mathrm{sech}^2\theta$, so the left side becomes $\frac{1}{v}\,\mathrm{d}\theta$; equivalently $\int \mathrm{d}\phi/(v^2-\phi^2) = \frac{1}{v}\,\mathrm{arctanh}(\phi/v)$. Performing both integrals,

$$\frac{1}{v}\,\mathrm{arctanh}\Big(\frac{\phi}{v}\Big) = \sqrt{\frac{\lambda}{12}}\,(x - x_0), \tag{41.15}$$

with x_0 the integration constant. Inverting gives the *kink*:

$$\boxed{\phi_K(x) = v\,\tanh\Big(\frac{x-x_0}{w}\Big)}\,, \qquad \frac{1}{w} = v\sqrt{\frac{\lambda}{12}}, \tag{41.16}$$

a smooth profile climbing monotonically from $-v$ at $x \to -\infty$ to $+v$ at $x \to +\infty$ (Fig. 41.1, right). The constant x_0 is the kink's *centre*, free because the energy functional is translation invariant—a zero mode we will not need here but will return to when the collective-coordinate quantization of such zero modes is taken up in Chapter 42. Taking the negative root in Eq. (41.13) instead yields the *antikink* $\bar\phi_K(x) = -v\tanh((x-x_0)/w) = v\tanh(-(x-x_0)/w)$, the mirror profile descending from $+v$ to $-v$, living in the sector $(\phi_-,\phi_+) = (+v,-v)$.

The width w is fixed by the parameters of the model. Using $v^2 = 6\,|m^2|\,/\lambda$ from Eq. (41.2),

$$\frac{1}{w} = v\sqrt{\frac{\lambda}{12}} = \sqrt{\frac{\lambda v^2}{12}} = \sqrt{\frac{\lambda}{12}\cdot\frac{6\,|m^2|}{\lambda}} = \sqrt{\frac{|m^2|}{2}} = \frac{|m|}{\sqrt{2}}, \qquad w = \frac{\sqrt{2}}{|m|}, \tag{41.17}$$

where $|m| \equiv \sqrt{|m^2|}$. The kink width is set by the curvature of the wells. Recall from Eq. (29.10) that the mass of the elementary scalar excitation about either vacuum is $m^2_{\mathrm{phys}} = V''(\pm v) = 2\,|m^2|$, so $m_{\mathrm{phys}} = \sqrt{2}\;|m|$ and

$$w = \frac{\sqrt{2}}{|m|} = \frac{2}{m_{\mathrm{phys}}}. \tag{41.18}$$

The kink is a lump twice the Compton wavelength of the field's own quantum across—a localized object whose size is dictated entirely by the same parameters that fix the perturbative particle, yet which no perturbative expansion around a single vacuum will ever produce, because no neighbourhood of one vacuum contains a configuration that reaches the other.

41.1.4 The topological current and its integer charge

We have a stable lump and a four-fold partition of field space, but "stable because the sectors are disconnected" is still a statement about pictures. We now convert it into an algebraic conservation law, and in doing so meet the defining structural contrast of the whole subject: a current conserved *identically*, with no appeal to the equations of motion.

In $1+1$ dimensions introduce the antisymmetric Levi-Civita symbol $\epsilon^{\mu\nu}$ with $\epsilon^{01} = -\epsilon^{10} = +1$, and define the *topological current*

Definition 41.3 (Topological current and charge)**.**

$$J^\mu \;=\; \frac{1}{2v}\,\epsilon^{\mu\nu}\,\partial_\nu \phi, \tag{41.19}$$

with associated charge $Q = \int_{-\infty}^{\infty} \mathrm{d}x\, J^0$.

The normalization $1/2v$ is chosen, as we will see, to make Q an integer. Written out, the two components are $J^0 = \frac{1}{2v}\epsilon^{01}\partial_1\phi = \frac{1}{2v}\,\partial_x\phi$ and $J^1 = \frac{1}{2v}\epsilon^{10}\partial_0\phi = -\frac{1}{2v}\,\partial_t\phi$. The current is built purely from the field and its first derivatives—no Lagrangian, no canonical momenta, no symmetry parameter entered its definition. Contrast this with the Noether current of Theorem 7.9, which is assembled from $\partial\mathcal{L}/\partial(\partial_\mu\phi)$ and the symmetry variation of the field, and which exists only relative to a chosen symmetry. Here there is no symmetry; there is only the field.

Theorem 41.4 (Off-shell conservation)**.** *The current Eq. (41.19) is identically conserved,*

$$\partial_\mu J^\mu = 0, \tag{41.20}$$

for every *smooth field configuration ϕ, whether or not it satisfies the equations of motion.*

Proof. Compute the divergence directly from Eq. (41.19):

$$\partial_\mu J^\mu = \frac{1}{2v}\,\epsilon^{\mu\nu}\,\partial_\mu\partial_\nu\phi. \tag{41.21}$$

The object $\partial_\mu\partial_\nu\phi$ is symmetric under $\mu \leftrightarrow \nu$ for any twice-differentiable ϕ (equality of mixed partials), while $\epsilon^{\mu\nu}$ is antisymmetric. The full contraction of a symmetric pair of indices with an antisymmetric one vanishes: explicitly, $\epsilon^{\mu\nu}\partial_\mu\partial_\nu\phi = \frac{1}{2}\epsilon^{\mu\nu}(\partial_\mu\partial_\nu - \partial_\nu\partial_\mu)\phi = 0$. Hence $\partial_\mu J^\mu = 0$ identically. □

This is the conceptual pivot of the chapter, so it is worth stating exactly what just happened and exactly how it differs from Noether's theorem. The Noether current of Theorem 7.10 obeys $\partial_\mu j^\mu_{\mathrm{N}} = 0$ *on shell*: the proof of Theorem 7.13 uses the Euler–Lagrange equations at the decisive step, and off the equations of motion the Noether current is generally not conserved. Its conservation is a dynamical fact, contingent on the field solving its equations, and it is tied to a symmetry by construction. The topological current Eq. (41.19) is conserved for a completely different reason: Eq. (41.20) is an algebraic identity following from the antisymmetry of ϵ and the commutativity of partial derivatives, true of any field configuration whatsoever. No symmetry is invoked; no equation of motion is used. The conservation is *kinematic*, not dynamical.

Caution

The distinction is not academic bookkeeping. Because $\partial_\mu J^\mu = 0$ does not lean on the dynamics, the charge Q is conserved even for configurations that are not solutions—it is a property of the field's boundary behaviour, immune to whatever the field does in the interior or in time. A Noether charge can in principle fail to be conserved if a symmetry is anomalous or explicitly broken; a topological charge cannot fail this way, because its conservation never used a symmetry to begin with. This robustness is exactly what makes topological charges the right tool for protecting non-perturbative states.

Now evaluate the charge. Using $J^0 = \frac{1}{2v}\partial_x\phi$, the spatial integral is a total derivative and collapses to the boundary values:

$$Q = \int_{-\infty}^{\infty} \mathrm{d}x\; J^0 = \frac{1}{2v}\int_{-\infty}^{\infty} \mathrm{d}x\; \partial_x\phi = \frac{1}{2v}\Big[\phi(+\infty) - \phi(-\infty)\Big] = \frac{\phi_+ - \phi_-}{2v}. \tag{41.22}$$

The charge depends only on the asymptotic vacuum values $\phi_\pm$ fixed by Eq. (41.6)—never on the detailed shape of ϕ in between. With $\phi_\pm \in \{+v, -v\}$ the four sectors of Theorem 41.2 carry

$$Q \in \{-1,\, 0,\, +1\}, \qquad \begin{aligned} (\phi_-, \phi_+) = (-v, +v): &\quad Q = +1 \quad \text{(kink)},\\ (\phi_-, \phi_+) = (+v, -v): &\quad Q = -1 \quad \text{(antikink)},\\ (\phi_-, \phi_+) = (\pm v, \pm v): &\quad Q = 0 \quad \text{(vacua)}. \end{aligned} \tag{41.23}$$

The kink of Eq. (41.16) carries $Q = +1$, the antikink $Q = -1$, and the two vacuum sectors $Q = 0$. The integer is exactly the "disconnection" of Theorem 41.2 made quantitative: it counts, with sign, the net number of times the field traverses the gap between the two vacua. Because Q is fixed by the boundary data and the boundary data cannot change continuously, Q is a conserved integer—and the kink, carrying $Q = +1$, can no more decay to the $Q = 0$ vacuum than an integer can slide to zero. Charge conservation and topological stability are the same statement.

Physical Insight

The charge integrand $J^0 = \frac{1}{2v}\partial_x\phi$ is a total spatial derivative, so Q is a pure boundary quantity. This is the 1+1-dimensional, single-field shadow of a structure that will reappear in every soliton we meet: the topological charge density is always a total divergence, its integral always a surface term at infinity, and its value always counts how the field at the boundary wraps the vacuum manifold. In Section 41.1's successor that "wrapping" becomes the winding number of a map from the sphere at infinity into the vacuum manifold, and the integer Q becomes its degree.

41.1.5 The Bogomolny bound, the BPS equation, and the kink mass

We found the kink by solving the second-order field equation and integrating once. There is a more powerful route that bypasses the field equation entirely, delivers a sharp lower bound on the energy in each sector, identifies the configurations that saturate it, and—almost as a byproduct—computes the kink's mass as a closed number. This is the *Bogomolny* rewriting, and the structure it exposes is the single thread that ties together the kink, the vortex, the monopole, and the instanton.

The idea is to complete the square in the energy functional. Introduce the *prepotential* (sometimes "superpotential") $W(\phi)$ defined by

$$W'(\phi) = \sqrt{2\,\widetilde{V}(\phi)}, \qquad W(\phi) = \int^{\phi} \sqrt{2\,\widetilde{V}(\phi')}\,\mathrm{d}\phi', \tag{41.24}$$

so that $\sqrt{2\widetilde{V}}$ is an exact ϕ-derivative along the configuration: $W'(\phi)\,\phi' = \frac{d}{dx}W(\phi(x))$. Then, for either sign choice, add and subtract the cross term in the static energy Eq. (41.5):

$$\begin{aligned} E[\phi] &= \int_{-\infty}^{\infty} \mathrm{d}x\, \left[\tfrac{1}{2}(\phi')^2 + \widetilde{V}(\phi)\right] \\ &= \int_{-\infty}^{\infty} \mathrm{d}x\, \left[\tfrac{1}{2}\big(\phi' \mp \sqrt{2\widetilde{V}}\big)^2 \;\pm\; \phi'\sqrt{2\widetilde{V}}\right] \\ &= \int_{-\infty}^{\infty} \mathrm{d}x\, \tfrac{1}{2}\big(\phi' \mp \sqrt{2\widetilde{V}}\big)^2 \;\pm\; \int_{-\infty}^{\infty} \mathrm{d}x\, \frac{d}{dx} W(\phi(x)). \end{aligned} \tag{41.25}$$

The algebra in the middle line is the identity $\frac{1}{2}(\phi' \mp \sqrt{2\widetilde{V}})^2 = \frac{1}{2}(\phi')^2 \mp \phi'\sqrt{2\widetilde{V}} + \widetilde{V}$, which on adding back $\pm\phi'\sqrt{2\widetilde{V}}$ returns the original integrand $\frac{1}{2}(\phi')^2 + \widetilde{V}$ exactly. In the last line we used Eq. (41.24) to write the cross term as a total derivative. That total derivative integrates to a pure boundary term,

$$\int_{-\infty}^{\infty} \mathrm{d}x\, \frac{d}{dx} W(\phi(x)) = W(\phi_+) - W(\phi_-) \equiv \Delta W, \tag{41.26}$$

which—like the topological charge—depends only on the asymptotic vacuum values, hence only on the sector. The first term in Eq. (41.25) is a manifestly non-negative integral of a perfect square. We have therefore proved:

Theorem 41.5 (Bogomolny bound and BPS saturation)**.** *For every finite-energy static configuration,*

$$E[\phi] \;\geq\; |\Delta W| = |\,W(\phi_+) - W(\phi_-)\,|\,, \tag{41.27}$$

the sign in Eq. (41.25) chosen so that $\pm\Delta W = |\Delta W| \geq 0$*. Equality holds if and only if the square in Eq. (41.25) vanishes pointwise, i.e. if and only if* ϕ *satisfies*

the first-order Bogomolny–Prasad–Sommerfield (BPS) equation

$$\boxed{\phi' = \pm\sqrt{2\,\widetilde{V}(\phi)}\,.} \tag{41.28}$$

Three points deserve emphasis. First, the bound Eq. (41.27) is established *without ever solving* the equation of motion—it follows from completing a square and from the boundary data alone. Second, the bound is purely topological in the sense that $|\Delta W|$ is constant on each sector; every configuration in a given sector, BPS or not, has at least this much energy. Third, the saturating configurations solve the *first-order* equation Eq. (41.28), which is precisely the first integral Eq. (41.12) found earlier from the mechanical analogue. The Bogomolny method thus re-derives the first-order equation from a variational inequality rather than from a conserved mechanical energy, and tells us in addition that its solutions are not merely stationary points but genuine energy *minimizers* within their sector. The kink Eq. (41.16) is the unique (up to translation) finite-energy minimizer of energy in the sector $Q = +1$; the antikink minimizes in $Q = -1$.

There is also a clean relation between the bound and the topological charge. For our model $\sqrt{2\widetilde{V}} = \sqrt{\lambda/12}\,|v^2 - \phi^2|$, and on a monotonic kink profile the field stays in $[-v, v]$ where $|v^2 - \phi^2| = v^2 - \phi^2$, so from Eq. (41.24)

$$W(\phi) = \sqrt{\frac{\lambda}{12}}\int^{\phi}(v^2 - \phi'^2)\,\mathrm{d}\phi' = \sqrt{\frac{\lambda}{12}}\left(v^2\phi - \tfrac{1}{3}\phi^3\right), \tag{41.29}$$

and the sector bound is $|\Delta W| = |W(\phi_+) - W(\phi_-)|$. Comparing with Eq. (41.22), both Q and ΔW are boundary quantities; for this model $\Delta W = 2\,W(v)\cdot|Q|$ up to sign, so the energy bound is $|Q|$ times the single-kink value—the bound knows that two kinks cost twice one kink. We compute that single-kink value now.

Worked Example 41.6 (The kink mass by quadrature)**.** The mass of the kink is its static energy, which by Theorem 41.5 equals the saturated bound $M_{\text{kink}} = |\Delta W|$ for the sector $\phi_- = -v \to \phi_+ = +v$. We evaluate it two ways and check they agree.

Route 1: the prepotential difference. Directly from Eq. (41.29),

$$\begin{aligned} M_{\text{kink}} &= W(v) - W(-v) = \sqrt{\frac{\lambda}{12}}\left[\left(v^3 - \tfrac{1}{3}v^3\right) - \left(-v^3 + \tfrac{1}{3}v^3\right)\right] \\ &= \sqrt{\frac{\lambda}{12}}\left[\tfrac{2}{3}v^3 + \tfrac{2}{3}v^3\right] = \sqrt{\frac{\lambda}{12}}\cdot\frac{4v^3}{3}. \end{aligned} \tag{41.30}$$

Route 2: the mass integral $M = \int\sqrt{2\widetilde{V}}\,\mathrm{d}\phi$. On the BPS solution the two energy densities are equal, $\tfrac{1}{2}(\phi')^2 = \widetilde{V}$ by Eq. (41.12), so the integrand of

Eq. (41.5) is $\frac{1}{2}(\phi')^2 + \widetilde{V} = (\phi')^2$. Then, changing the integration variable from x to ϕ via $\mathrm{d}\phi = \phi'\,\mathrm{d}x$ and $\phi' = \sqrt{2\widetilde{V}}$,

$$M_{\text{kink}} = \int_{-\infty}^{\infty} \mathrm{d}x\ (\phi')^2 = \int_{-\infty}^{\infty} \mathrm{d}x\ \phi' \cdot \phi' = \int_{-v}^{+v} \sqrt{2\widetilde{V}(\phi)}\ \mathrm{d}\phi. \tag{41.31}$$

This is the promised closed formula: the kink mass is the area under $\sqrt{2\widetilde{V}}$ between the vacua, a one-dimensional integral with no reference to x at all. Inserting $\sqrt{2\widetilde{V}} = \sqrt{\lambda/12}\,(v^2 - \phi^2)$,

$$M_{\text{kink}} = \sqrt{\frac{\lambda}{12}} \int_{-v}^{+v} (v^2 - \phi^2)\,\mathrm{d}\phi = \sqrt{\frac{\lambda}{12}} \left[v^2\phi - \tfrac{1}{3}\phi^3 \right]_{-v}^{+v} = \sqrt{\frac{\lambda}{12}} \cdot \frac{4v^3}{3}, \tag{41.32}$$

identical to Eq. (41.30), as it must be.

The closed number. Reducing $\sqrt{\lambda/12} = \sqrt{\lambda}/(2\sqrt{3})$,

$$M_{\text{kink}} = \frac{\sqrt{\lambda}}{2\sqrt{3}} \cdot \frac{4v^3}{3} = \frac{2\sqrt{\lambda}\,v^3}{3\sqrt{3}} = \frac{2\sqrt{3\lambda}}{9}\,v^3. \tag{41.33}$$

It is illuminating to trade v for the model's mass scale. Using $v^2 = 6\,|m^2|\,/\lambda$, hence $v^3 = (6\,|m^2|\,/\lambda)^{3/2} = 6\sqrt{6}\ |m|^3\,/\lambda^{3/2}$,

$$M_{\text{kink}} = \frac{2\sqrt{3\lambda}}{9} \cdot \frac{6\sqrt{6}\ |m|^3}{\lambda^{3/2}} = \frac{12\sqrt{18}}{9}\,\frac{|m|^3}{\lambda} = \frac{12 \cdot 3\sqrt{2}}{9}\,\frac{|m|^3}{\lambda} = 4\sqrt{2}\,\frac{|m|^3}{\lambda}, \tag{41.34}$$

or, in terms of the physical particle mass $m_{\text{phys}} = \sqrt{2}\ |m|$ (Eq. (29.10)), substituting $|m|^3 = m_{\text{phys}}^3/(2\sqrt{2})$,

$$\boxed{M_{\text{kink}} = \frac{2\,m_{\text{phys}}^3}{\lambda}\,.} \tag{41.35}$$

A final consistency check ties the result back to the profile. Using the explicit kink $\phi_K = v\tanh(x/w)$ with $\phi_K' = (v/w)\,\mathrm{sech}^2(x/w)$, the energy integral $\int (\phi_K')^2 \mathrm{d}x = (v^2/w^2)\int \mathrm{sech}^4(x/w)\,\mathrm{d}x = (v^2/w)\int_{-\infty}^{\infty} \mathrm{sech}^4 u\,\mathrm{d}u = (v^2/w)\cdot\frac{4}{3}$, where $\int \mathrm{sech}^4 u\,\mathrm{d}u = \frac{4}{3}$ follows from $\int \mathrm{sech}^4 = \int \mathrm{sech}^2(1-\tanh^2) = [\tanh - \frac{1}{3}\tanh^3]_{-\infty}^{\infty} = 2 - \frac{2}{3} = \frac{4}{3}$. With $1/w = v\sqrt{\lambda/12}$ this is $\frac{4}{3}v^2 \cdot v\sqrt{\lambda/12} = \sqrt{\lambda/12}\cdot\frac{4}{3}v^3$, reproducing Eq. (41.30) once more.

The result Eq. (41.35) carries the message the chapter introduction advertised. The kink mass scales as $1/\lambda$: in the weak-coupling regime where perturbation theory is reliable, the soliton is *heavy*, its mass diverging as $\lambda \to 0$. A heavy state whose mass blows up at weak coupling is invisible to any power series in λ around the vacuum—no finite order in Feynman diagrams can produce a pole at $M \sim 1/\lambda$. The kink exists in the classical field equations, is protected

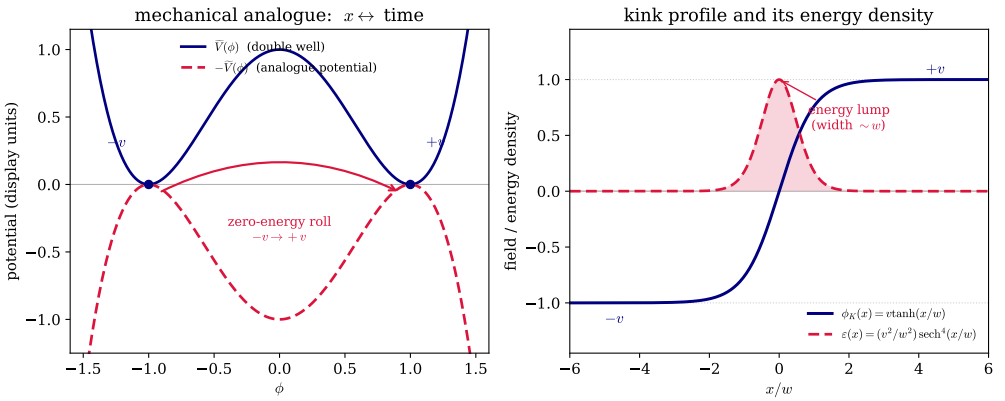


Figure 41.1: The 1+1 double-well kink of ϕ^4 theory. *Left:* the mechanical analogue. The static equation $\phi'' = \widetilde{V}'(\phi)$ is Newton's law for a unit-mass particle at position ϕ in "time" x moving in the *inverted* potential $U = -\widetilde{V}$ (Eq. (41.9)), a double hump with summits at the vacua $\pm v$. The finite-energy kink is the unique zero-mechanical-energy trajectory Eq. (41.12) that rolls from the top of the left hump at $x = -\infty$ to the top of the right hump at $x = +\infty$. *Right:* the field profile $\phi_K(x) = v\tanh(x/w)$ of Eq. (41.16) interpolating monotonically between the vacua, together with its static energy density $\varepsilon(x) = (\phi_K')^2 = (v^2/w^2)\,\mathrm{sech}^4(x/w)$, a localized lump of width $\sim w$ whose area is the kink mass M_{kink} of Eq. (41.35).

by an integer, and weighs $2m_{\text{phys}}^3/\lambda$, yet the entire perturbative apparatus of Parts III–XI, organized as an expansion in λ around $\phi \equiv v$, is structurally blind to it. This is the concrete sense in which we have stepped outside perturbation theory.

With the kink in hand we have, on the most transparent example available, the entire apparatus the rest of the chapter will generalize: finite-energy boundary conditions that strand the field in a discrete vacuum set, a partition of configuration space into superselection sectors, an off-shell-conserved current whose integer charge labels those sectors, and a Bogomolny bound saturated by a first-order equation whose solution we can write down and weigh. What the kink raises but cannot answer is why Q had to be an integer in the first place—we obtained it by counting endpoints in a two-element set, an accident of one dimension and a discrete vacuum manifold—and how the same logic survives when the vacuum manifold is a continuum and space has more than one end. Those questions are the business of the next section, which abstracts "the field at infinity lands in the vacuum manifold" into a map from the boundary of space into that manifold, and "Q counts traversals" into the homotopy class of that map.

41.2 Topological Charge and the Homotopy Classification

The kink handed us an integer and a puzzle. Its charge $Q = \frac{1}{2v}(\phi(+\infty) - \phi(-\infty))$ took the value $+1$ for the kink, -1 for the antikink, 0 for either constant vacuum, and—this was the decisive lesson of Section 41.1—it could not change under any continuous, finite-energy evolution, because changing it would require dragging the field at one spatial infinity from one well to the other, an excursion over the energy barrier that costs an infinite amount of energy in an infinite volume. The current $J^\mu = \frac{1}{2v}\epsilon^{\mu\nu}\partial_\nu\phi$ was conserved *off shell*: its divergence vanished identically by the antisymmetry of $\epsilon^{\mu\nu}$ against the symmetric $\partial_\mu\partial_\nu$, without ever invoking the equation of motion. None of this used the detailed shape of the double-well potential, the value of λ, or the explicit tanh profile. It used only two facts: that the potential has *more than one* degenerate minimum, and that finite energy pins the field to one of those minima at each end of space.

This section extracts the general principle hiding inside those two facts. We will see that finite-energy configurations of *any* field theory fall into disjoint families—superselection sectors—that no continuous finite-energy motion can connect, and that these families are labeled not by a number we invent for each model but by a single topological object: a homotopy group of the set of vacua. The charge is the *degree*, or winding number, of the map that the field at spatial infinity defines into the vacuum manifold, and its integrality and invariance are theorems of topology, not accidents of a particular Lagrangian. We build the needed homotopy theory from scratch as we go; the only inputs from outside this chapter are the classical field machinery of Chapters 6 and 7, the group/coset language of Chapter 2, and the order-parameter vocabulary of Chapter 29, which we recall but rebuild in the form we need.

41.2.1 The vacuum manifold

Every example in this chapter is governed by an energy functional of the schematic form

$$E[\phi] = \int \mathrm{d}^D x \left[\tfrac{1}{2}\,\partial_i\phi^a\,\partial_i\phi^a + V(\phi)\right], \tag{41.36}$$

where D is the number of *spatial* dimensions, the field $\phi = (\phi^1, \dots, \phi^k)$ takes values in some target $\mathbb{R}^k$ (later, in a group or coset), the index $i = 1, \dots, D$ runs over space, repeated indices are summed, and we have specialized to static configurations so that no time derivatives appear and E is the static energy of Section 41.1. The potential V is bounded below; by adding a constant we set its minimum value to zero. The first object the topology cares about is the set on which that minimum is attained.

Definition 41.7 (Vacuum manifold)**.** The *vacuum manifold* of the theory (41.36) is the set of absolute minima of the potential,

$$\mathcal{M} = \{\, \phi_0 \in \mathbb{R}^k \;:\; V(\phi_0) = 0 \,\}. \tag{41.37}$$

Its points are the constant field configurations of least energy—the classical vacua—and we assume throughout that V is smooth and that $\mathcal{M}$ is a smooth submanifold of $\mathbb{R}^k$ (generically true when the minima are nondegenerate transverse to $\mathcal{M}$).

For the kink, ϕ is a single real field, $V = \frac{\lambda}{4!}(\phi^2 - v^2)^2$, and $\mathcal{M} = \{+v, -v\}$: two isolated points, a zero-dimensional manifold. That discreteness is the whole story of the kink, as we will confirm in Theorem 41.13. In higher dimensions the interesting vacuum manifolds are continuous, and they arise for a structural reason worth recalling.

Suppose the energy (41.36) is invariant under a Lie group G acting linearly on the target, $\phi \mapsto g \cdot \phi$ for $g \in G$, with $V(g \cdot \phi) = V(\phi)$. Then G permutes the vacua: if $\phi_0 \in \mathcal{M}$ then $g \cdot \phi_0 \in \mathcal{M}$ for every g. Pick one vacuum ϕ_0. Its *stabilizer* (or little group)

$$H = \{\, h \in G \;:\; h \cdot \phi_0 = \phi_0 \,\} \tag{41.38}$$

is the subgroup of unbroken symmetries—the transformations the chosen vacuum does not notice—and is a closed subgroup of G. Two group elements g_1, g_2 carry ϕ_0 to the same vacuum iff $g_2^{-1} g_1 \in H$, i.e. iff they lie in the same left coset of H. Hence the orbit of ϕ_0, which (when the action is transitive on $\mathcal{M}$) is all of $\mathcal{M}$, is in bijection with the coset space:

$$\mathcal{M} \;\cong\; G/H. \tag{41.39}$$

This is the classical content of the spontaneous-symmetry-breaking story of Chapter 29: the broken theory's order parameter ϕ_0 singles out a vacuum, G breaks to H, and the degenerate vacua—the directions the order parameter can point—trace out the coset G/H, the same manifold whose tangent space at ϕ_0 carries the Goldstone modes. Here we need only the *geometry* of $\mathcal{M}$, built classically from the potential, not the quantum theory of fluctuations around it; the coset language of Chapter 2 supplies the rest. Two examples will recur:

- A complex field ϕ with $V = \frac{\lambda}{2}(|\phi|^2 - v^2)^2$ has $G = \mathrm{U}(1)$ acting by $\phi \mapsto e^{i\alpha}\phi$, broken completely ($H = \{1\}$), so $\mathcal{M} = \{|\phi| = v\} \cong \mathrm{U}(1) \cong S^1$, a circle. This drives the vortex.

- A real triplet ϕ^a $(a = 1, 2, 3)$ with $V = \frac{\lambda}{4}(\phi^a \phi^a - v^2)^2$ has $G = \mathrm{SO}(3)$ rotating the triplet, broken to $H = \mathrm{SO}(2)$ (rotations about the chosen ϕ_0), so $\mathcal{M} = \mathrm{SO}(3)/\mathrm{SO}(2) \cong S^2$, the two-sphere $\{|\phi| = v\}$. This drives the monopole, the Yang–Mills theory non-perturbative stage.

41.2.2 Finite energy and the asymptotic map

The kink's boundary conditions were not imposed by hand; finiteness of the energy forced them. The same argument works in any dimension and is the bridge from analysis to topology.

Proposition 41.8 (Finite energy pins the field to $\mathcal{M}$ at infinity)**.** *Let ϕ be a smooth configuration on $\mathbb{R}^D$ with $E[\phi] < \infty$ in (41.36). Then $V(\phi(x)) \to 0$ as $|x| \to \infty$; that is, along every ray the field approaches the vacuum manifold $\mathcal{M}$. Consequently ϕ has a well-defined limit on the sphere at spatial infinity, a continuous map*

$$\phi_\infty \colon\ S^{D-1}_\infty \longrightarrow \mathcal{M}, \qquad \phi_\infty(\hat{n}) = \lim_{r\to\infty} \phi(r\hat{n}), \tag{41.40}$$

where S^{D-1}_∞ is the sphere of unit directions $\hat{n}$ at radius $r \to \infty$.

Proof. The energy splits as $E = E_{\text{grad}} + E_{\text{pot}}$ with $E_{\text{pot}} = \int \mathrm{d}^D x\, V(\phi) \geq 0$. Finiteness of E forces $E_{\text{pot}} < \infty$. Since $V \geq 0$, the integral of V over the exterior region $|x| > R$ tends to zero as $R \to \infty$. If $V(\phi(x))$ did not tend to zero, there would be a sequence of points $x_n \to \infty$ and a $\delta > 0$ with $V(\phi(x_n)) \geq \delta$; by smoothness of ϕ and V and the boundedness of $\partial\phi$ implied by finite gradient energy, $V(\phi)$ stays above $\delta/2$ on balls of a fixed radius about each x_n, contributing a fixed positive amount to E_{pot} infinitely often, contradicting $E_{\text{pot}} < \infty$. Hence $V(\phi(x)) \to 0$, i.e. $\operatorname{dist}(\phi(x), \mathcal{M}) \to 0$. Finiteness of E_{grad} similarly forces the angular variation of ϕ on the sphere of radius r to die as $r \to \infty$ (the gradient energy of a fixed angular profile on S^{D-1}_r scales as $r^{D-1} \cdot r^{-2} = r^{D-3}$ from one factor of the tangential derivative squared, integrated over the sphere, and convergence of the radial integral forces the angular profile to freeze), so the radial limits (41.40) exist and assemble into a continuous map of the limiting sphere into $\mathcal{M}$. □

Physical Insight

Spatial infinity is a sphere S^{D-1}: in one spatial dimension it is the two points $\{-\infty, +\infty\} = S^0$; in two dimensions a circle S^1; in three a two-sphere S^2. Finite energy converts *every* localized configuration, whatever its complicated interior, into a clean piece of boundary data—a map from that sphere into the vacuum manifold. All the topology lives in this boundary map. The interior is free to do anything finite-energy; only how the field "wraps" the vacuum manifold as we circle spatial infinity is protected.

Two configurations that can be continuously deformed into one another through finite-energy intermediates have asymptotic maps that can be continuously deformed into one another as maps $S^{D-1} \to \mathcal{M}$. The converse also holds:

if two asymptotic maps are continuously deformable, one can build a finite-energy interpolation of the bulk fields realizing the deformation. So the question "which finite-energy sectors exist, and which are connected?" is *exactly* the question "into how many classes do continuous maps $S^{D-1} \to \mathcal{M}$ fall, under continuous deformation?" That last question is purely topological, and answering it is what homotopy theory does.

41.2.3 Homotopy: deforming maps

Definition 41.9 (Homotopy)**.** Let X, Y be topological spaces and $f_0, f_1 \colon X \to Y$ continuous. A *homotopy* from f_0 to f_1 is a continuous map $F \colon X \times [0,1] \to Y$ with $F(\cdot, 0) = f_0$ and $F(\cdot, 1) = f_1$. We write $f_0 \simeq f_1$ and call them *homotopic*; intuitively F is a continuous movie deforming f_0 into f_1. Homotopy is an equivalence relation: reflexivity by the constant movie $F(x, s) = f_0(x)$; symmetry by running the movie backwards, $s \mapsto 1 - s$; transitivity by concatenating two movies, playing the first on $s \in [0, \frac{1}{2}]$ at double speed and the second on $[\frac{1}{2}, 1]$, which is continuous because the two agree at the splice $s = \frac{1}{2}$.

The equivalence classes are the *homotopy classes* $[f]$, and the set of all classes of maps $X \to Y$ is written $[X, Y]$. By the discussion above, the finite-energy sectors of our field theory are exactly the elements of $[S^{D-1}, \mathcal{M}]$. When X is itself a sphere this set carries extra structure—a group law—and becomes a homotopy group, which is the object we actually compute. We build the lowest three.

π_0: path components. The zeroth homotopy "group" is the simplest. Take $X = S^0$, the two-point space; a map $S^0 \to Y$ is just an ordered pair of points, but the useful object keeps only one of them. Define $\pi_0(Y)$ to be the set of *path components* of Y: two points $y_0, y_1 \in Y$ are equivalent iff there is a continuous path $\gamma \colon [0,1] \to Y$ with $\gamma(0) = y_0, \gamma(1) = y_1$, and $\pi_0(Y)$ is the set of equivalence classes. It records "how many disconnected pieces Y has." It is a set with a marked point (the component of a chosen basepoint), not in general a group.

For the kink, $D = 1$, so $S^{D-1} = S^0$, and the asymptotic data is a point of $\mathcal{M}$ at each of $\pm\infty$, i.e. a choice of component of $\mathcal{M}$ at each end. With $\mathcal{M} = \{+v, -v\}$ having exactly two path components, $\pi_0(\mathcal{M}) = \{[+v], [-v]\}$ is a two-element set. We unpack this completely in Theorem 41.13.

π_1: the fundamental group. Now take $X = S^1$. A map $S^1 \to Y$ is a *loop*—a continuous path that returns to its start. Fix a basepoint $y_0 \in Y$ and consider loops based there: continuous $\gamma \colon [0,1] \to Y$ with $\gamma(0) = \gamma(1) = y_0$. Two such are equivalent if homotopic *through based loops* (the homotopy fixes the basepoint at every stage). The set of classes is $\pi_1(Y, y_0)$, the *fundamental group*, and it really is a group, with the following structure.

The product of two based loops is their concatenation, traversed at double speed:

$$(\gamma_1 \cdot \gamma_2)(t) = \begin{cases} \gamma_1(2t), & 0 \le t \le \tfrac{1}{2}, \\ \gamma_2(2t-1), & \tfrac{1}{2} \le t \le 1, \end{cases} \tag{41.41}$$

continuous because $\gamma_1(1) = y_0 = \gamma_2(0)$ at the splice. One checks the group axioms at the level of homotopy classes. *Identity:* the constant loop $e(t) = y_0$ satisfies $\gamma \cdot e \simeq \gamma \simeq e \cdot \gamma$, the homotopy being a continuous reparametrization that absorbs the dead time spent sitting at y_0. *Inverses:* the reversed loop $\bar{\gamma}(t) = \gamma(1-t)$ satisfies $\gamma \cdot \bar{\gamma} \simeq e$, the explicit homotopy $F(t,s)$ retracing γ only up to "time" s and then immediately backtracking, so that at $s = 0$ one has the full out-and-back loop and at $s = 1$ the constant loop. *Associativity:* $(\gamma_1 \cdot \gamma_2) \cdot \gamma_3 \simeq \gamma_1 \cdot (\gamma_2 \cdot \gamma_3)$, the two differing only by the speeds at which the three legs are run, related by a continuous reparametrization of $[0,1]$. Each axiom holds only *up to homotopy*, which is exactly why π_1 is defined on classes and not on loops themselves. When Y is path-connected, $\pi_1(Y, y_0)$ is independent of y_0 up to isomorphism, and we drop the basepoint.

π_n: the higher homotopy groups. The pattern continues. A based map of the n-sphere, $f\colon S^n \to Y$ with a marked point of S^n sent to y_0, has homotopy classes (through based maps) forming the n-th homotopy group $\pi_n(Y, y_0)$. The group law generalizes (41.41): think of S^n as a cube $[0,1]^n$ with its entire boundary collapsed to y_0, and multiply two such cubes by stacking them along the first coordinate (run f_1 on $x_1 \in [0, \frac{1}{2}]$, f_2 on $[\frac{1}{2}, 1]$), the collapsed boundaries making the result continuous. The same reparametrization arguments give identity, inverses, and associativity up to homotopy. A new feature appears for $n \ge 2$: there are now two independent directions in which to stack, and sliding one factor "around" the other shows the product is *abelian*, $\pi_n(Y)$ is an abelian group for $n \ge 2$, whereas π_1 may be nonabelian. We will need this only as a fact; the cases that matter for us are all already abelian, indeed cyclic.

Theorem 41.10 (Homotopy classification of finite-energy sectors)**.** *The finite-energy configurations of the theory* (41.36) *on $\mathbb{R}^D$, modulo continuous finite-energy deformation, are in bijection with the based homotopy classes of asymptotic maps, $[S^{D-1}, \mathcal{M}]$. When $\mathcal{M}$ is path-connected, these are classified by the homotopy group $\pi_{D-1}(\mathcal{M})$. The sector containing the constant vacuum is the trivial (identity) class; any sector that is not this class contains no configuration deformable to a vacuum, and is what we call a* topological soliton sector*.*

Proof. Theorem 41.8 assigns to each finite-energy ϕ its asymptotic map $\phi_\infty \in \{S^{D-1} \to \mathcal{M}\}$, and a finite-energy deformation of ϕ induces a homotopy of ϕ_∞, so the class $[\phi_\infty] \in [S^{D-1}, \mathcal{M}]$ is a deformation invariant. Conversely, given any continuous $g\colon S^{D-1} \to \mathcal{M}$, extend it to a finite-energy bulk field by

$\phi(r\hat{n}) = \rho(r)\, g(\hat{n})$ for a smooth profile ρ that interpolates from a vacuum-valued interior to g at infinity (for $\mathcal{M} \subset \mathbb{R}^k$ a sphere of radius v, take $\rho(r)$ rising from 0 to v; the gradient and potential energies are finite because the angular profile freezes for large r). Thus every class is realized, and a homotopy of g lifts to a finite-energy deformation of the bulk field, so distinct classes are genuinely distinct sectors. The basepoint subtlety—a free homotopy class versus a based one—does not affect the count when $\mathcal{M}$ is path-connected and, in the cyclic cases below, when $\pi_1(\mathcal{M})$ acts trivially; we use this in the abelian cases that concern us. Finally, the constant vacuum has constant ϕ_∞, the identity class, and any configuration in a nontrivial class cannot be deformed to it without leaving the finite-energy configurations or jumping homotopy class, neither of which a continuous finite-energy motion can do. □

The labels $\pi_0, \pi_1, \pi_2, \ldots$ now have physical jobs. In $D = 1$ the classifier is $\pi_0(\mathcal{M})$, counting discrete vacua—the kink. In $D = 2$ it is $\pi_1(\mathcal{M})$, the winding of the field around spatial infinity into the vacuum manifold—the vortex. In $D = 3$ it is $\pi_2(\mathcal{M})$, the wrapping of the asymptotic two-sphere over the vacuum two-sphere—the monopole. Everything reduces to computing these groups for the relevant $\mathcal{M}$, and to writing the integer (or, occasionally, the $\mathbb{Z}_2$ label) explicitly as a charge.

41.2.4 The winding number of a circle map, and $\pi_1(S^1) = \mathbb{Z}$

The single most important computation in the subject is $\pi_1(S^1)$. It controls the vortex directly, and its method—lifting to the universal cover—is the template for every degree theorem that follows. We treat S^1 as the unit circle and parametrize the target circle by an angle, so a map of circles is encoded by a real-valued angular function.

Theorem 41.11 (Winding number; $\pi_1(S^1) \cong \mathbb{Z}$)**.** *Realize the circle as $S^1 = \mathbb{R}/2\pi\mathbb{Z}$ with covering projection $p\colon \mathbb{R} \to S^1$, $p(\theta) = \theta \bmod 2\pi$. For every loop $\gamma\colon [0,1] \to S^1$ there is a continuous* lift *$\tilde{\gamma}\colon [0,1] \to \mathbb{R}$ with $p \circ \tilde{\gamma} = \gamma$, unique once $\tilde{\gamma}(0)$ is fixed. For a based loop ($\gamma(0) = \gamma(1)$) the quantity*

$$w(\gamma) = \frac{\tilde{\gamma}(1) - \tilde{\gamma}(0)}{2\pi} \tag{41.42}$$

is an integer, the winding number*, independent of the choice of lift; it depends only on the based-homotopy class of γ; and $w\colon \pi_1(S^1) \to \mathbb{Z}$ is a group isomorphism.*

Proof. Existence and uniqueness of lifts. The projection p restricted to any open interval of length less than 2π is a homeomorphism onto its image (an open arc); its inverse is a local branch of the angle. Cover $[0,1]$ by preimages under γ of small arcs of S^1; by compactness there is a Lebesgue number $\delta > 0$ such that γ maps each subinterval of a partition $0 = t_0 < t_1 < \cdots < t_N = 1$

with mesh $< \delta$ into a single such arc. Fix $\tilde\gamma(t_0)$ over $\gamma(t_0)$. On $[t_0, t_1]$ define $\tilde\gamma$ by composing γ with the local inverse of p that takes $\gamma(t_0)$ to $\tilde\gamma(t_0)$; this is continuous. Proceed inductively, at each step choosing the local inverse matching the value $\tilde\gamma(t_j)$ already fixed at the left endpoint. The result is a continuous $\tilde\gamma$ with $p\circ\tilde\gamma = \gamma$. Uniqueness: if $\tilde\gamma_1, \tilde\gamma_2$ are two lifts with the same start, then $\tilde\gamma_1 - \tilde\gamma_2$ is continuous and $p(\tilde\gamma_1) = p(\tilde\gamma_2)$ forces $\tilde\gamma_1 - \tilde\gamma_2 \in 2\pi\mathbb{Z}$ pointwise; a continuous integer-multiple-of-2π function starting at 0 is identically 0.

Integrality. For a based loop, $p(\tilde\gamma(1)) = \gamma(1) = \gamma(0) = p(\tilde\gamma(0))$, so $\tilde\gamma(1) - \tilde\gamma(0) \in 2\pi\mathbb{Z}$ and $w(\gamma) \in \mathbb{Z}$. Changing the choice of $\tilde\gamma(0)$ shifts $\tilde\gamma$ by a constant, leaving the difference (41.42) unchanged.

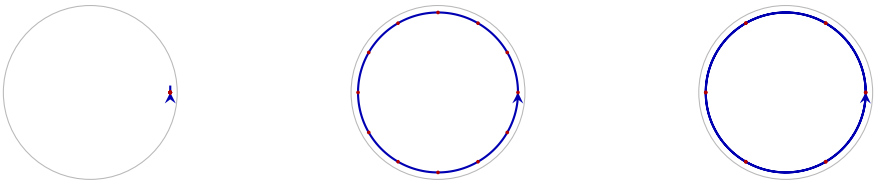
Homotopy invariance. Let $F\colon [0,1]\times[0,1] \to S^1$ be a based homotopy, $F(\cdot,0) = \gamma_0$, $F(\cdot,1) = \gamma_1$, with $F(0,s) = F(1,s) = y_0$ fixed for all s. The identical partition-and-local-inverse construction, now on the square (whose Lebesgue number controls a grid of small squares), produces a continuous lift $\tilde F\colon [0,1]^2 \to \mathbb{R}$ with $p\circ\tilde F = F$ and $\tilde F(\cdot,0) = \tilde\gamma_0$. For each fixed s, $\tilde F(\cdot,s)$ is a lift of the loop $F(\cdot,s)$, so $w_s := \big(\tilde F(1,s) - \tilde F(0,s)\big)/2\pi$ is an integer for every s. But $\tilde F(1,s)$ and $\tilde F(0,s)$ are continuous in s (they are lifts of the constant path $s \mapsto y_0$, hence each is itself constant in s up to $2\pi\mathbb{Z}$, and continuity pins them), so w_s is a continuous integer-valued function of s, therefore constant: $w(\gamma_0) = w(\gamma_1)$. Thus w descends to homotopy classes.

Homomorphism and isomorphism. For the concatenation (41.41), lift γ_1 starting at 0 and continue the lift of γ_2 from the endpoint $\tilde\gamma_1(1)$; the total rise is the sum of rises, so $w(\gamma_1\cdot\gamma_2) = w(\gamma_1) + w(\gamma_2)$, a homomorphism into $(\mathbb{Z},+)$. It is *onto*: the loop $\gamma_m(t) = p(2\pi m t)$ has lift $\tilde\gamma_m(t) = 2\pi m t$ and winding m, realizing every integer. It is *injective*: if $w(\gamma) = 0$, then $\tilde\gamma$ is a loop in $\mathbb{R}$ (its endpoints coincide), and $\mathbb{R}$ is convex, so $\tilde\gamma$ contracts to a point by the straight-line homotopy $(t,s) \mapsto (1-s)\tilde\gamma(t) + s\,\tilde\gamma(0)$; projecting by p gives a based contraction of γ, so $[\gamma]$ is trivial. A surjective homomorphism with trivial kernel is an isomorphism: $\pi_1(S^1) \cong \mathbb{Z}$. □

For smooth maps the winding number has an integral expression that we will need as the prototype of the general charge. Write the target angle as a smooth function $\alpha(\theta)$ of the domain angle $\theta \in [0, 2\pi]$, with $\alpha(2\pi) - \alpha(0) \in 2\pi\mathbb{Z}$ for the map to close. Then the lift is $\tilde\gamma(\theta) = \alpha(\theta)$ and

$$w = \frac{\alpha(2\pi) - \alpha(0)}{2\pi} = \frac{1}{2\pi}\int_0^{2\pi} \frac{\mathrm{d}\alpha}{\mathrm{d}\theta}\,\mathrm{d}\theta = \int_{S^1} \frac{\mathrm{d}\alpha}{2\pi} = \int_{S^1} \phi^*\omega, \tag{41.43}$$

where $\omega = \mathrm{d}\alpha/2\pi$ is the *normalized volume form* on the target circle—the unique invariant one-form with $\int_{S^1}\omega = 1$—and $\phi^*\omega$ denotes its pullback by the map. The last equality is the form in which the formula generalizes: *the degree is the integral over the domain sphere of the pullback of the normalized volume form of the target.*



$w = 0$ $w = 1$ $w = 2$

Figure 41.2: Maps $\phi\colon S^1 \to S^1$ of winding number $w = 0, 1, 2$. As the domain angle θ runs once around, the image point runs around the target circle w times. By Theorem 41.11 the integer w is invariant under any continuous deformation of the map; no continuous motion can change how many times the image encircles the target.

41.2.5 The degree formula and the general topological charge

We can now state the charge in full generality and recognize the kink's Q as its lowest instance. Suppose the vacuum manifold is itself a $(D-1)$-sphere of radius v in field space (the two recurring examples, the circle for $D = 2$ and the two-sphere for $D = 3$, are exactly this), and let ω be the normalized volume form on $\mathcal{M}$, the unique rotation-invariant $(D-1)$-form with $\int_{\mathcal{M}} \omega = 1$. The *topological charge* of a finite-energy configuration with asymptotic map $\phi_\infty\colon S^{D-1}_\infty \to \mathcal{M}$ is the *degree*

$$Q = \deg \phi_\infty = \int_{S^{D-1}_\infty} \phi_\infty^* \omega. \tag{41.44}$$

That this integral is an integer is the higher-dimensional generalization of Theorem 41.11; we proved it in full for $D = 2$ (the circle) there, and the same lifting/covering machinery—now the statement that the degree of a smooth map between equidimensional oriented closed manifolds equals the signed count of preimages of any regular value, hence an integer—establishes it for every D. We will use it concretely in two cases.

The kink ($D = 1$). Here $S^{D-1} = S^0 = \{-\infty, +\infty\}$ and $\mathcal{M} = \{-v, +v\} = S^0$. The "normalized volume form" of a two-point oriented set is the signed counting that assigns $+\frac{1}{2}$ to the point $+v$ and $-\frac{1}{2}$ to $-v$ (so the total, the 0-form integrated over the oriented 0-manifold with its two oppositely oriented points, is normalized), and the degree (41.44) reduces to

$$Q = \tfrac{1}{2}\big[\sigma(\phi(+\infty)) - \sigma(\phi(-\infty))\big] = \frac{\phi(+\infty) - \phi(-\infty)}{2v}, \tag{41.45}$$

where $\sigma(\pm v) = \pm 1$. This is precisely the kink charge of Section 41.1, now seen as the $D = 1$ degree. Equivalently, $Q = \frac{1}{2v}\int_{-\infty}^{\infty} \partial_x \phi \, \mathrm{d}x = \int_{-\infty}^{\infty} J^0 \, \mathrm{d}x$ with $J^0 = \frac{1}{2v}\partial_x \phi$, the integral over $\mathbb{R}^1$ of a total derivative whose value is fixed by the boundary data—the one-dimensional shadow of (41.44).

The circle target ($D = 2$, the vortex). Here (41.44) is exactly the winding integral (41.43) evaluated on the asymptotic circle, $Q = \frac{1}{2\pi}\oint_{S^1_\infty} \mathrm{d}\alpha \in \mathbb{Z}$, with α the phase of the complex order parameter at infinity. The integer counts how many times the phase winds as one circles spatial infinity.

The sphere target ($D = 3$, the monopole), and $\pi_2(S^2) = \mathbb{Z}$. For $\mathcal{M} = S^2 = \{|\phi| = v\}$ write the unit field $n^a = \phi^a/v$. The normalized volume form of the unit two-sphere is $\omega = \frac{1}{8\pi}\,\epsilon_{abc}\, n^a\, \mathrm{d}n^b \wedge \mathrm{d}n^c$ (it integrates to $\frac{1}{4\pi} \times \text{area} = 1$), so (41.44) becomes the explicit winding integral

$$Q = \frac{1}{8\pi}\int_{S^2_\infty} \epsilon_{abc}\, n^a\, \partial_i n^b\, \partial_j n^c\, \mathrm{d}S^{ij} = \frac{1}{8\pi}\int_{S^2_\infty} \epsilon_{abc}\, \epsilon^{ij}\, n^a\, \partial_i n^b\, \partial_j n^c\, \mathrm{d}^2\sigma \;\in\; \mathbb{Z}, \tag{41.46}$$

where $\mathrm{d}S^{ij}$ is the oriented surface element of the asymptotic two-sphere and the integrand is the pullback of ω. The integer is the degree of the map $n\colon S^2_\infty \to S^2$, the number of times n covers the target sphere, and the corresponding statement of topology is

$$\pi_2(S^2) \cong \mathbb{Z}. \tag{41.47}$$

We do not reprove (41.47) from scratch; it follows from the same degree theory as Theorem 41.11 (the degree is a homotopy invariant, takes every integer value via the maps $n \mapsto n^{\otimes m}$ realized by Brouwer's construction, and detects triviality), and the integral (41.46) computes the isomorphism explicitly. This is the formula the 't Hooft–Polyakov monopole will saturate.

Caution

The clean integer-valued degree formula (41.44) works only when $\dim \mathcal{M} = D - 1$, so that an asymptotic-sphere top-form can be pulled back. When the dimensions do not match, or when the relevant homotopy group is not $\mathbb{Z}$, the classification is still given by Theorem 41.10 but the charge need not be an integral. The cleanest example is the next one.

A non-integer label: $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. Not every topological charge is an integer counting a winding; some are merely a sign. The order parameter of a system whose vacua are orientations in three-dimensional space takes values in $\mathcal{M} = \mathrm{SO}(3)$. We established in Chapter 2 that $\mathrm{SU}(2)$ is the simply connected double cover of $\mathrm{SO}(3)$ with kernel $\{\pm\mathbf{1}\}$, so that $\mathrm{SO}(3) \cong \mathrm{SU}(2)/\{\pm\mathbf{1}\} \cong \mathbb{RP}^3$ and a 2π rotation traces a loop that cannot be contracted while a 4π rotation can (Theorem 2.55). In the language now available this is the statement

$$\pi_1(\mathrm{SO}(3)) \cong \mathbb{Z}_2, \tag{41.48}$$

with the nontrivial class the 2π loop: it is its own inverse, since traversing it twice (a 4π rotation) is contractible. The proof is exactly the lifting argument

of Theorem 41.11 applied to the covering $\mathrm{SU}(2) \to \mathrm{SO}(3)$, with the simple-connectivity of $S^3 = \mathrm{SU}(2)$ (also from Theorem 2.55) playing the role $\mathbb{R}$ played there: a loop in SO(3) lifts to a path in SU(2), and the loop is contractible iff the lift closes, i.e. iff the lift's endpoints agree rather than differ by the kernel element $-\mathbf{1}$. The two cosets of the kernel give the two classes $\mathbb{Z}_2$. The corresponding topological objects in a two-dimensional medium with SO(3) order parameter carry a charge valued in $\mathbb{Z}_2$, not $\mathbb{Z}$—two of them can annihilate—and there is no integral formula (41.44) for it, only the discrete invariant.

41.2.6 Invariance of the topological charge

We have asserted that Q cannot change under continuous finite-energy deformation; this is the keystone, and we now prove it directly from the integral formula, exhibiting its first variation as a total derivative that the fixed asymptotics kill. The argument simultaneously explains *why* Q is conserved off shell, with no appeal to the equation of motion.

Theorem 41.12 (Topological invariance of the charge)**.** *Let ϕ_s, $s \in [0,1]$, be a smooth one-parameter family of finite-energy configurations on $\mathbb{R}^D$, so that for each s the asymptotic map $\phi_{s,\infty}\colon S^{D-1}_{\infty} \to \mathcal{M}$ is defined. Then the degree (41.44) is independent of s:*

$$\frac{\mathrm{d}}{\mathrm{d}s}\, Q[\phi_s] = 0. \tag{41.49}$$

Equivalently, Q is constant on each finite-energy sector and changes only by an integer jump between sectors, which a continuous deformation cannot produce.

Proof. Write $Q[\phi_s] = \int_{S^{D-1}} \phi_s^{*}\omega$ with ω the normalized volume form on $\mathcal{M}$, an $(D-1)$-form. Since ω is a top-degree form on the $(D-1)$-dimensional manifold $\mathcal{M}$, it is automatically *closed*: $\mathrm{d}\omega = 0$ (its exterior derivative would be a D-form on a $(D-1)$-manifold, necessarily zero). Let V denote the variation field $\partial_s \phi_s$, a vector along the map valued in the tangent spaces of $\mathcal{M}$. By Cartan's identity for the Lie derivative along the deformation,

$$\frac{\mathrm{d}}{\mathrm{d}s}\, \phi_s^{*}\omega = \phi_s^{*}(\mathcal{L}_V \omega) = \phi_s^{*}(\iota_V\, \mathrm{d}\omega + \mathrm{d}\, \iota_V \omega) = \phi_s^{*}(\mathrm{d}\, \iota_V \omega) = \mathrm{d}(\phi_s^{*}\, \iota_V \omega), \tag{41.50}$$

where the third equality used $\mathrm{d}\omega = 0$ and the fourth used that pullback commutes with d. Thus the first variation of the integrand is an *exact* form—a total derivative of the $(D-2)$-form $\beta_s := \phi_s^{*}\, \iota_V \omega$. Integrating over the asymptotic sphere and applying Stokes,

$$\frac{\mathrm{d}}{\mathrm{d}s}\, Q[\phi_s] = \int_{S^{D-1}} \mathrm{d}\beta_s = \int_{\partial S^{D-1}} \beta_s = 0, \tag{41.51}$$

because S^{D-1} is a closed manifold, $\partial S^{D-1} = \varnothing$. The variation vanishes identically; Q is constant in s. □

Physical Insight

Read (41.50)–(41.51) physically. The charge density is the pullback of a closed form, so its variation is automatically a total derivative; the only thing that can survive is a boundary term; and the boundary of the sphere at infinity is empty (or, in the bulk picture below, sits at infinity where finite energy has frozen the field to a fixed vacuum, killing the term). The integral does not budge. This is the off-shell conservation we first met for the kink current $J^\mu = \frac{1}{2v}\epsilon^{\mu\nu}\partial_\nu\phi$, whose divergence $\partial_\mu J^\mu \propto \epsilon^{\mu\nu}\partial_\mu\partial_\nu\phi$ vanishes identically by antisymmetry—no equation of motion required—now revealed as the special case of (41.50) in one dimension. A Noether charge (Chapter 7) is conserved *on shell*, because of a symmetry and via the field equations; a topological charge is conserved *off shell*, because it is the integral of a closed form over a cycle, a fact of geometry that holds for every configuration, solution or not.

The same conclusion in the bulk picture makes the phrase "fixed asymptotics" explicit. Write the kink-type charge as a bulk integral of a current, $Q = \int_{\mathbb{R}^D} J^0 \,\mathrm{d}^D x$ with J^0 a total spatial divergence, $J^0 = \partial_i K^i$ for some K^i built from the field (for the kink, $K^1 = \frac{1}{2v}\phi$ and $D = 1$; for the circle and sphere targets, K^i is the appropriate pullback potential). Then

$$\frac{\mathrm{d}}{\mathrm{d}s}Q = \int_{\mathbb{R}^D} \partial_i\, \delta K^i \,\mathrm{d}^D x = \oint_{S^{D-1}_\infty} \delta K^i \,\mathrm{d}S_i, \tag{41.52}$$

a pure boundary term on the sphere at infinity. A finite-energy deformation holds ϕ pinned to its vacuum value there (Theorem 41.8), so the variation δK^i of the boundary data either vanishes or, when the asymptotic map itself is deformed, integrates to zero by Theorem 41.12. To change Q one would have to violate the boundary condition—to drag the field at some direction of spatial infinity off the vacuum manifold and across the potential barrier—which in infinite volume costs infinite energy. The integer is locked.

41.2.7 Worked example: the kink as the nontrivial class of π_0

Worked Example 41.13 (The kink is the nontrivial element of $\pi_0(\{\pm v\})$)**.** We verify that the entire topological content of the kink is the statement $\pi_0(\mathcal{M}) = \{[+v], [-v]\}$, and that Theorems 41.10 and 41.12 reproduce, with no new input, every claim of Section 41.1.

Setup. One spatial dimension, $D = 1$, real field ϕ, potential $V = \frac{\lambda}{4!}(\phi^2 - v^2)^2$, vacuum manifold $\mathcal{M} = \{+v, -v\}$. Spatial infinity is $S^0 = \{-\infty, +\infty\}$. The asymptotic map (41.40) is the pair $\phi_\infty = (\phi(-\infty), \phi(+\infty)) \in \mathcal{M} \times \mathcal{M}$.

Counting the sectors. By Theorem 41.10 the sectors are the homotopy classes $[S^0, \mathcal{M}]$. A map $S^0 \to \mathcal{M}$ is a choice of one of the two points at each end, and

two such maps are homotopic iff each endpoint can be continuously slid to its partner *within* $\mathcal{M}$. But $\mathcal{M} = \{\pm v\}$ is discrete: there is no path from $+v$ to $-v$ inside it. Hence each endpoint's choice is a frozen element of $\pi_0(\mathcal{M})$, and the four sectors are

$$(-v,-v), \quad (+v,+v), \quad (-v,+v), \quad (+v,-v). \tag{41.53}$$

The first two are the constant vacua, charge $Q = 0$ by (41.45). The third is the kink, $Q = \frac{(+v)-(-v)}{2v} = +1$. The fourth is the antikink, $Q = -1$. The kink is precisely the configuration whose two ends land in *different* path components of $\mathcal{M}$—it interpolates across the nontrivial element of π_0—which is the formal meaning of "a kink connects the two vacua."

Why it cannot decay. A continuous finite-energy evolution is a path $s \mapsto \phi_s$ in the configuration space. By Theorem 41.12, $Q[\phi_s]$ is constant; explicitly, $\frac{\mathrm{d}}{\mathrm{d}s}Q = \frac{1}{2v}\big[\delta\phi\big]_{-\infty}^{+\infty} = 0$ because finite energy pins $\delta\phi(\pm\infty) = 0$. A kink ($Q = 1$) therefore can never reach a vacuum ($Q = 0$): the two live in different superselection sectors. The only way to remove a kink is to bring in an antikink ($Q = -1$) and annihilate, conserving total $Q = 0$—which is allowed and indeed happens, releasing the rest energy as radiation.

Consistency with the explicit current. The bulk charge is $Q = \int_{-\infty}^{\infty} J^0 \mathrm{d}x$ with $J^0 = \frac{1}{2v}\partial_x\phi = \partial_x K$, $K = \frac{1}{2v}\phi$. Its variation (41.52) is $\frac{\mathrm{d}}{\mathrm{d}s}Q = \big[\delta K\big]_{-\infty}^{+\infty} = \frac{1}{2v}\big[\delta\phi\big]_{-\infty}^{+\infty}$, a pure boundary term, vanishing exactly as the abstract Stokes argument predicted. Everything in Section 41.1 was the $D = 1$, π_0 shadow of the general classification.

The classification answers "which sectors *can* exist": a nontrivial $\pi_{D-1}(\mathcal{M})$ guarantees finite-energy configurations that no continuous motion can unwind to the vacuum. It does *not* guarantee that any such configuration is a static *solution*—a stationary point of the energy realized at finite size rather than spreading out or collapsing. Whether a given sector actually contains a static, finite-energy minimizer in a given spatial dimension is a separate, dynamical question, and the answer turns out to be sharply negative for scalars alone above one dimension. That obstruction—and the gauge fields it forces upon us—is the subject of Section 41.3, to which we now turn.

41.3 Derrick's Theorem and the Need for Gauge Fields

The homotopy classification of Section 41.2.3 is a promise, not a guarantee. It tells us that finite-energy configurations of a theory with vacuum manifold $\mathcal{V}$ fall into superselection sectors labelled by $\pi_{D-1}(\mathcal{V})$, and that whenever this group is nontrivial there *exist* continuous field configurations carrying a nonzero

topological charge that no continuous, finite-energy deformation can undo. What it does *not* tell us is whether any such configuration is a *solution* of the field equations—whether the energy functional, restricted to a charged sector, actually attains a minimum at a smooth, static, finite-energy field. Topology supplies the lower bound on the number of solitons (at least one per sector, if the bound is saturated); dynamics must supply the solitons themselves. In one spatial dimension the kink of Section 41.1 did both at once, and we may have come to expect the pattern to repeat in every dimension. It does not. The purpose of this section is a single sharp negative result, due to Derrick, that closes off the naive route: *in two or more spatial dimensions, a theory of scalar fields alone has no static, finite-energy soliton other than the vacuum.* The proof is a one-line scaling argument once it is set up correctly, and its real value is constructive: by cataloguing exactly how each kind of term in the energy responds to a rescaling, it tells us precisely what must be added to the theory to make a higher-dimensional soliton possible. The answer—gauge fields—dictates the two constructions that follow.

41.3.1 The static energy functional and its pieces

Throughout, D denotes the number of *spatial* dimensions, so spacetime is $(D+1)$-dimensional and the physical case of Chapter 42 onward is $D = 3$. We work entirely at the classical level: a soliton is a critical point of the static energy functional, and $\hbar$ stays in the drawer. The energy of a static configuration is the spatial integral of the T^{00} component of the stress tensor built in Chapter 7 (the conserved charge associated with time translations, Eq. (7.97)). For a collection of real scalar fields $\phi = (\phi_1, \dots, \phi_n)$ with the standard two-derivative Lagrangian

$$\mathcal{L} = \tfrac{1}{2}\,\partial_\mu \phi \cdot \partial^\mu \phi - V(\phi), \tag{41.54}$$

the energy density of a *time-independent* configuration $\phi(\mathbf{x})$ is

$$T^{00} = \tfrac{1}{2}\,\dot{\phi}\cdot\dot{\phi} + \tfrac{1}{2}\,\nabla\phi\cdot\nabla\phi + V(\phi) \;\xrightarrow{\dot{\phi}=0}\; \tfrac{1}{2}\,|\nabla\phi|^2 + V(\phi), \tag{41.55}$$

where $|\nabla\phi|^2 = \sum_a \sum_{i=1}^{D} (\partial_i \phi_a)^2$ and the dot is the sum over the internal index a. We take $V \geq 0$, normalised so that $V = 0$ exactly on the vacuum manifold $\mathcal{V} = \{\phi : V(\phi) = 0\}$; any potential bounded below can be brought to this form by adding a constant, which shifts the energy by an (infinite, but configuration-independent) amount that we subtract once and for all. This is the same finite-energy bookkeeping that made the kink's energy convergent in Section 41.1.

Definition 41.14 (Static scalar energy and its two pieces)**.** For a static configuration $\phi : \mathbb{R}^D \to \mathbb{R}^n$ the energy is

$$E[\phi] \;=\; \int_{\mathbb{R}^D} \mathrm{d}^D x \left[\tfrac{1}{2}\,|\nabla\phi|^2 + V(\phi)\right] \;=\; E_{\mathrm{grad}} + E_{\mathrm{pot}}, \tag{41.56}$$

with the *gradient* (or stiffness) energy and the *potential* energy

$$E_{\text{grad}}[\phi] \;=\; \int_{\mathbb{R}^D} \mathrm{d}^D x\, \tfrac{1}{2}\,|\nabla\phi|^2 \;\geq 0, \qquad E_{\text{pot}}[\phi] \;=\; \int_{\mathbb{R}^D} \mathrm{d}^D x\, V(\phi) \;\geq 0. \quad (41.57)$$

A configuration has *finite energy* if both pieces are finite; a *static soliton* is a finite-energy critical point of E that is not a constant vacuum configuration.

Both pieces are manifestly non-negative, and this is the only structural fact the argument uses. The two pieces pull the configuration in opposite directions, and the tension between them is the whole story. The gradient energy penalises rapid variation: it is lowered by stretching the configuration out, spreading the same field profile over a larger region. The potential energy penalises any departure from the vacuum: it is lowered by shrinking the region in which ϕ strays from $\mathcal{V}$, squeezing the profile toward a point. A soliton, if it exists, is a truce between these two—a configuration whose size is fixed by their balance. Derrick's insight is that in $D \geq 2$ no such truce is possible: one tendency always wins outright.

41.3.2 The rescaling family and the scaling energy

To probe the balance we deform a candidate configuration along the one direction that pits the two energies against each other most directly: a uniform spatial dilation that stretches or shrinks the whole profile while keeping its shape.

Definition 41.15 (Scaling family)**.** Given a finite-energy configuration $\phi(\mathbf{x})$ and a parameter $\mu > 0$, define the rescaled configuration

$$\phi_\mu(\mathbf{x}) \;\equiv\; \phi\!\left(\frac{\mathbf{x}}{\mu}\right). \quad (41.58)$$

For $\mu > 1$ the profile is dilated (spread out by a factor μ); for $\mu < 1$ it is contracted; $\mu = 1$ returns the original. The same field values are visited, redistributed over a region whose linear size is scaled by μ. Any boundary condition $\phi \to \mathcal{V}$ at spatial infinity is preserved for every $\mu > 0$, so the entire family lies in one topological sector and the topological charge is constant along it.

The decisive feature of Eq. (41.58) is that it leaves the topological charge untouched—the deformation is exactly the kind of continuous, finite-energy motion within a fixed sector that the classification of Section 41.2.3 declared the charge to be invariant under—so if ϕ is a genuine energy minimiser in its sector, then in particular $\mu = 1$ must minimise the ordinary function $E(\mu) \equiv E[\phi_\mu]$ of the single real variable μ. We now compute that function.

Proposition 41.16 (The scaling energy of a scalar configuration)**.** *With E_{grad} and E_{pot} the two pieces of the energy of ϕ as in Eq. (41.57),*

$$E(\mu) \;\equiv\; E[\phi_\mu] \;=\; \mu^{\,D-2}\, E_{\text{grad}} \;+\; \mu^{\,D}\, E_{\text{pot}}. \quad (41.59)$$

Proof. We carry out the two integrals by the change of variables $\mathbf{y} = \mathbf{x}/\mu$, for which $\mathrm{d}^D x = \mu^D \, \mathrm{d}^D y$ and $\partial/\partial x^i = \mu^{-1} \, \partial/\partial y^i$.

For the potential piece there are no derivatives, so we substitute directly:

$$E_{\mathrm{pot}}[\phi_\mu] = \int \mathrm{d}^D x \, V(\phi(\mathbf{x}/\mu)) = \int \mu^D \, \mathrm{d}^D y \, V(\phi(\mathbf{y})) = \mu^D \, E_{\mathrm{pot}}. \tag{41.60}$$

For the gradient piece we first differentiate the composite function. By the chain rule,

$$\partial_i \, \phi_\mu(\mathbf{x}) = \frac{\partial}{\partial x^i} \, \phi\!\left(\frac{\mathbf{x}}{\mu}\right) = \frac{1}{\mu} \, (\partial_i \phi)\!\left(\frac{\mathbf{x}}{\mu}\right), \tag{41.61}$$

the factor $1/\mu$ coming from differentiating the argument x^i/μ. Squaring and summing over the spatial index i and the internal index a,

$$|\nabla \phi_\mu(\mathbf{x})|^2 = \frac{1}{\mu^2} \, |(\nabla \phi)(\mathbf{x}/\mu)|^2. \tag{41.62}$$

Hence, again with $\mathbf{y} = \mathbf{x}/\mu$,

$$E_{\mathrm{grad}}[\phi_\mu] = \int \mathrm{d}^D x \, \tfrac{1}{2} \, \frac{1}{\mu^2} \, |(\nabla \phi)(\mathbf{x}/\mu)|^2 = \frac{\mu^D}{\mu^2} \int \mathrm{d}^D y \, \tfrac{1}{2} \, |(\nabla \phi)(\mathbf{y})|^2 = \mu^{\,D-2} \, E_{\mathrm{grad}}. \tag{41.63}$$

Adding Eqs. (41.60) and (41.63) gives Eq. (41.59). □

The two exponents in Eq. (41.59) are nothing but the engineering dimensions of the two pieces translated into the language of dilations. The measure $\mathrm{d}^D x$ contributes μ^D to both; each spatial derivative contributes an extra μ^{-1}, so the two-derivative gradient term carries the additional μ^{-2} that produces μ^{D-2}, while the zero-derivative potential keeps the bare μ^D. This bookkeeping—one factor of μ^{-1} per derivative, one factor of μ^{+1} per unit of measure—is the entire mechanism, and it is what makes the scaling argument extend effortlessly to gauge and higher-derivative terms below.

41.3.3 The theorem

Theorem 41.17 (Derrick)**.** *Let ϕ be a static, finite-energy critical point of the energy Eq. (41.56) of a theory of scalar fields with $V \geq 0$, in D spatial dimensions. Then:*

(i) for $D \geq 3$, necessarily $E_{\mathrm{grad}} = E_{\mathrm{pot}} = 0$, so ϕ is a vacuum configuration; no soliton exists;

(ii) for $D = 2$, necessarily $E_{\mathrm{pot}} = 0$, so ϕ takes values in the vacuum manifold $\mathcal{V}$ everywhere and any nontrivial solution is a solution of a pure derivative (nonlinear sigma) model, neutrally stable under dilations and hence not a true minimiser when a genuine potential is present;

(iii) for $D = 1$, the balance $E_{\text{grad}} = E_{\text{pot}}$ (a virial relation) is the condition, and a stable soliton can and does exist—this is the kink.

Proof. Because ϕ is a critical point of E, the restriction $E(\mu) = E[\phi_\mu]$ must be stationary at the point $\mu = 1$ corresponding to ϕ itself, and—for ϕ to be a (local) *minimum* rather than a saddle or maximum—the second derivative there must be non-negative. We impose both conditions on Eq. (41.59).

First derivative. Differentiating Eq. (41.59),

$$E'(\mu) = (D-2)\,\mu^{D-3}\,E_{\text{grad}} + D\,\mu^{D-1}\,E_{\text{pot}}, \tag{41.64}$$

and evaluating at $\mu = 1$,

$$E'(1) = (D-2)\,E_{\text{grad}} + D\,E_{\text{pot}} = 0. \tag{41.65}$$

Second derivative. Differentiating once more,

$$E''(\mu) = (D-2)(D-3)\,\mu^{D-4}\,E_{\text{grad}} + D(D-1)\,\mu^{D-2}\,E_{\text{pot}}, \tag{41.66}$$

so that

$$E''(1) = (D-2)(D-3)\,E_{\text{grad}} + D(D-1)\,E_{\text{pot}} \;\geq\; 0. \tag{41.67}$$

We now read off the consequences dimension by dimension, using only $E_{\text{grad}} \geq 0$ and $E_{\text{pot}} \geq 0$.

Case $D \geq 3$. Here both coefficients in the stationarity condition Eq. (41.65) are strictly positive: $D - 2 \geq 1 > 0$ and $D > 0$. The sum of two non-negative quantities with strictly positive coefficients vanishes only if each vanishes separately,

$$E_{\text{grad}} = 0 \quad \text{and} \quad E_{\text{pot}} = 0. \tag{41.68}$$

$E_{\text{grad}} = 0$ forces $\nabla\phi \equiv 0$, i.e. ϕ constant; $E_{\text{pot}} = 0$ forces that constant to lie in $\mathcal{V}$. Thus ϕ is the vacuum, and there is no soliton. This is statement (i).

Case $D = 2$. The gradient coefficient $D-2$ vanishes and Eq. (41.65) collapses to $2\,E_{\text{pot}} = 0$, hence $E_{\text{pot}} = 0$ and $V(\phi(\mathbf{x})) = 0$ for all $\mathbf{x}$: the field takes values in the vacuum manifold everywhere. The scaling energy degenerates to $E(\mu) = E_{\text{grad}}$, independent of μ: dilations cost nothing, and $E''(1) = 0$ as Eq. (41.67) confirms. The configuration is therefore *neutrally* stable under rescaling—a flat direction along which it can be shrunk to zero size at no energy cost. A genuine soliton of a theory with a true potential (one for which any nonconstant configuration has $E_{\text{pot}} > 0$) is thus excluded outright. The only survivors are configurations of theories whose potential vanishes identically on the entire target, i.e. nonlinear sigma models, where the "lumps" exist precisely because of the scale invariance of E_{grad} in two dimensions—but they have no preferred size and are unstable to collapse once any scale-breaking term is added. This is statement (ii).

Case $D = 1$. Now $D - 2 = -1$ and Eq. (41.65) reads $-E_{\text{grad}} + E_{\text{pot}} = 0$, the virial relation

$$E_{\text{grad}} = E_{\text{pot}}. \tag{41.69}$$

The second-derivative test Eq. (41.67) gives $E''(1) = (-1)(-2)E_{\text{grad}} + 1 \cdot 0 \cdot E_{\text{pot}} = 2\,E_{\text{grad}} \geq 0$, with equality only for the vacuum. For any nontrivial profile $E_{\text{grad}} > 0$, so the stationary point is a strict minimum along the scaling direction. The dilation instability that killed the higher-dimensional cases is absent, and a stable soliton is permitted. This is statement (iii). □

The case analysis is visualised in the left panel of Fig. 41.3: the function $E(\mu)$ has an interior minimum only for $D = 1$, while for $D \geq 2$ it is monotone and the energy is always lowered by collapse, $\mu \to 0$.

Corollary 41.18 (Recovery of the kink and the virial theorem)**.** *In* $D = 1$ *every static finite-energy soliton satisfies* $E_{\text{grad}} = E_{\text{pot}}$*, and its total energy is* $E = 2E_{\text{grad}} = 2E_{\text{pot}}$*. This reproduces the equipartition between gradient and potential energy found directly for the* ϕ^4 *kink in Section 41.1, and shows it to be a model-independent consequence of dilation stationarity in one dimension rather than an accident of that particular profile.*

Proof. The relation $E_{\text{grad}} = E_{\text{pot}}$ is Eq. (41.69); substituting it into $E = E_{\text{grad}} + E_{\text{pot}}$ gives $E = 2E_{\text{grad}} = 2E_{\text{pot}}$. □

Physical Insight

The theorem is a statement about competing tendencies, and it is worth holding the picture in mind without the algebra. The gradient energy is the cost of bending the field; it wants the configuration as large and gentle as possible, and in D dimensions stretching by μ multiplies it by μ^{D-2}. The potential energy is the cost of leaving the vacuum; it wants the configuration as small as possible so that less volume sits away from $\mathcal{V}$, and stretching multiplies it by μ^{D}. A soliton needs one term that grows under expansion and one that grows under contraction, so that an intermediate size is favoured. In $D = 1$ the gradient term (μ^{-1}, growing under contraction) and the potential (μ^{1}, growing under expansion) do exactly this, and the kink sits at the balance. In $D \geq 2$ *both* exponents $D - 2$ and D are non-negative: every term grows under expansion and shrinks under contraction, so the configuration can always lower its energy by shrinking, and it shrinks all the way to a point—the soliton "falls through the floor." To make a higher-dimensional soliton we must add a term whose energy grows under *contraction*, i.e. one with an exponent $D - k < 0$. That requires $k > D$ inverse powers of length, and the smallest such structure compatible with locality, Lorentz invariance, and a sensible

kinetic term is a gauge field.

41.3.4 The evasion catalogue: gauge fields restore the balance

Derrick's theorem is a no-go only for the specific class of theories Eq. (41.54). Its constructive content is the exponent table: a term in the energy built from a structure of definite scaling weight contributes μ^p to $E(\mu)$, where $p = D -$ (number of inverse lengths the structure carries). Anything that supplies a term with $p < 0$ reopens the door. We catalogue the physically important evasions, computing the exponents explicitly, and isolate the one—gauge fields—that the next two sections are built on.

Abelian gauge fields. Couple the scalar to a $U(1)$ gauge field A_i through the covariant derivative $D_i\phi = (\partial_i - ieA_i)\phi$ (the static, $A_0 = 0$ sector of the classical $U(1)$ theory of Chapter 8, whose Maxwell Lagrangian is Eq. (8.75)), and add the magnetic energy. The static energy now has three pieces,

$$E = E_{\text{mag}} + E_{\text{grad}} + E_{\text{pot}}, \qquad E_{\text{mag}} = \int \mathrm{d}^D x\, \tfrac{1}{4} F_{ij}F_{ij}, \quad E_{\text{grad}} = \int \mathrm{d}^D x\, \tfrac{1}{2} |D_i\phi|^2, \tag{41.70}$$

with $F_{ij} = \partial_i A_j - \partial_j A_i$. To rescale we must dilate the connection as well as the scalar, and the correct rule is dictated by the requirement that the gauge-covariant gradient scale the same way the ordinary gradient did—otherwise the covariant derivative would not transform homogeneously and the deformed configuration would not be a sensible field. A connection is a one-form; it carries one inverse length, exactly as ∂_i does, so it must scale with the same extra μ^{-1}:

$$\phi_\mu(\mathbf{x}) = \phi(\mathbf{x}/\mu), \qquad A_{\mu\, i}(\mathbf{x}) = \frac{1}{\mu} A_i(\mathbf{x}/\mu). \tag{41.71}$$

We check that the covariant gradient scales as before. Using Eq. (41.61) for the scalar and the definition Eq. (41.71) for the connection,

$$D_i^{(\mu)}\phi_\mu(\mathbf{x}) = \partial_i\phi_\mu - ieA_{\mu\, i}\phi_\mu = \frac{1}{\mu}(\partial_i\phi)(\mathbf{x}/\mu) - ie\,\frac{1}{\mu}A_i(\mathbf{x}/\mu)\,\phi(\mathbf{x}/\mu) = \frac{1}{\mu}\,(D_i\phi)(\mathbf{x}/\mu), \tag{41.72}$$

so $|D^{(\mu)}\phi_\mu(\mathbf{x})|^2 = \mu^{-2}|(D\phi)(\mathbf{x}/\mu)|^2$ exactly as in Eq. (41.62), and the covariant gradient energy scales as μ^{D-2} just as the ordinary one did. The new term is the magnetic energy. The field strength inherits two derivatives' worth of scaling from Eq. (41.71):

$$F_{\mu\, ij}(\mathbf{x}) = \partial_i A_{\mu\, j} - \partial_j A_{\mu\, i} = \frac{1}{\mu}\cdot\frac{1}{\mu}\Big[(\partial_i A_j) - (\partial_j A_i)\Big](\mathbf{x}/\mu) = \frac{1}{\mu^2}\,F_{ij}(\mathbf{x}/\mu), \tag{41.73}$$

one factor $1/\mu$ from the explicit $A \to A/\mu$ and one from differentiating the rescaled argument. Hence

$$E_{\mathrm{mag}}[A_\mu] = \int \mathrm{d}^D x \, \tfrac{1}{4} \frac{1}{\mu^4} \left|F(\mathbf{x}/\mu)\right|^2 = \frac{\mu^D}{\mu^4} E_{\mathrm{mag}} = \mu^{D-4} E_{\mathrm{mag}}. \tag{41.74}$$

The full scaling energy of the gauged scalar theory is therefore

$$E(\mu) = \mu^{D-4} E_{\mathrm{mag}} + \mu^{D-2} E_{\mathrm{grad}} + \mu^D E_{\mathrm{pot}}, \tag{41.75}$$

and the magnetic exponent $D-4$ is negative for every $D \le 3$. This is the term Derrick's theorem was missing.

Non-abelian gauge fields. The same conclusion survives the passage to a non-abelian gauge group, which is what Chapter 25 requires for the monopole and for the Yang–Mills theory payload. The non-abelian field strength of Eq. (25.76),

$$F^a_{ij} = \partial_i A^a_j - \partial_j A^a_i + g\, f^{abc} A^b_i A^c_j, \tag{41.76}$$

contains a quadratic piece in addition to the curl, and one might worry that the two pieces scale differently and spoil the clean exponent. They do not, precisely because the connection scales like a derivative. Applying Eq. (41.71) component by component, the linear (curl) part scales as μ^{-2} by the computation Eq. (41.73); the quadratic part scales as

$$g\, f^{abc}\, A^b_{\mu\, i} A^c_{\mu\, j} = g\, f^{abc} \frac{1}{\mu} A^b_i\!\left(\frac{\mathbf{x}}{\mu}\right) \frac{1}{\mu} A^c_j\!\left(\frac{\mathbf{x}}{\mu}\right) = \frac{1}{\mu^2} \left(g f^{abc} A^b_i A^c_j\right)(\mathbf{x}/\mu), \tag{41.77}$$

the *same* μ^{-2}. Both terms in Eq. (41.76) scale identically, so $F^a_{\mu\, ij}(\mathbf{x}) = \mu^{-2} F^a_{ij}(\mathbf{x}/\mu)$ and the non-abelian magnetic energy $E_{\mathrm{YM}} = \int \mathrm{d}^D x \, \tfrac{1}{4} F^a_{ij} F^a_{ij}$ again scales as μ^{D-4}. The homogeneous scaling of the connection is exactly what makes the nonlinear self-interaction respect the same weight as the abelian curl; this is the classical shadow of the fact that the gauge coupling g is dimensionless in four spacetime dimensions. The scaling structure Eq. (41.75) therefore holds verbatim with $E_{\mathrm{mag}} \to E_{\mathrm{YM}}$.

We can now state the evasion as a theorem and read off the two cases that the chapter will construct.

Theorem 41.19 (Gauge fields evade Derrick in $D = 2$ and $D = 3$)**.** *For the gauged energy Eq. (41.75) with $E_{\mathrm{mag}}, E_{\mathrm{grad}}, E_{\mathrm{pot}} \ge 0$, dilation stationarity $E'(1) = 0$ reads*

$$(D-4)\, E_{\mathrm{mag}} + (D-2)\, E_{\mathrm{grad}} + D\, E_{\mathrm{pot}} = 0. \tag{41.78}$$

For $D = 2$ and $D = 3$ this admits nontrivial nonnegative solutions, so a static soliton is no longer forbidden:

$$D = 2: \qquad -2\, E_{\mathrm{mag}} + 0 \cdot E_{\mathrm{grad}} + 2\, E_{\mathrm{pot}} = 0 \implies E_{\mathrm{mag}} = E_{\mathrm{pot}}, \tag{41.79}$$

$$D = 3: \qquad -\, E_{\mathrm{mag}} + E_{\mathrm{grad}} + 3\, E_{\mathrm{pot}} = 0 \implies E_{\mathrm{mag}} = E_{\mathrm{grad}} + 3\, E_{\mathrm{pot}}. \tag{41.80}$$

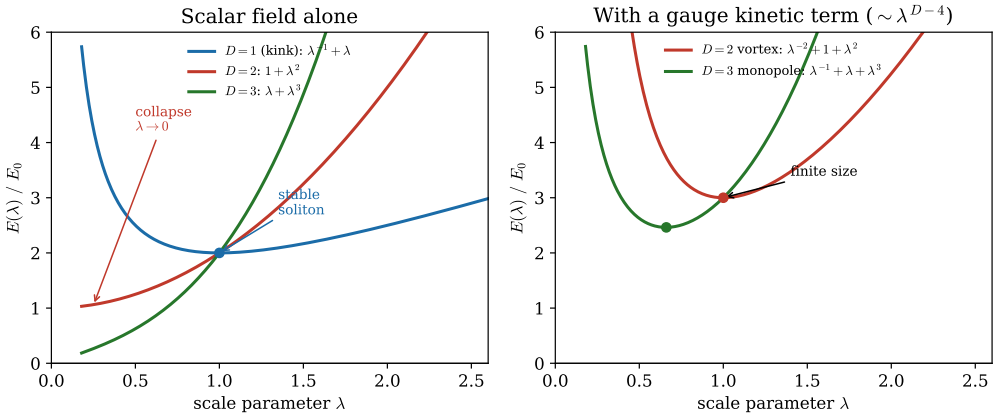


Figure 41.3: The scaling energy $E(\mu)$ of Eqs. (41.59) and (41.75), with all coefficients set to unity for legibility. *Left:* a scalar theory alone. Only $D = 1$ has an interior minimum (the kink); for $D = 2$ and $D = 3$ the energy is monotone increasing in μ, so it is always lowered by shrinking the lump to a point ($\mu \to 0$)—Derrick's no-go. *Right:* adding a gauge kinetic term, which scales as μ^{D-4} and diverges as $\mu \to 0$, restores an interior minimum at finite size in both $D = 2$ (the vortex) and $D = 3$ (the monopole). The presence or absence of the interior stationary point is independent of the chosen coefficients; it is the entire content of the theorem.

Proof. Eq. (41.78) is the derivative of Eq. (41.75) evaluated at $\mu = 1$, computed exactly as in Eq. (41.65). Setting $D = 2$ kills the gradient coefficient and leaves Eq. (41.79); setting $D = 3$ leaves Eq. (41.80). In each case all the energies that appear can be strictly positive while the relation holds, so the stationarity condition no longer forces them to vanish. The magnetic term carries the negative exponent $D - 4 < 0$, meaning E_{mag} *decreases* under expansion and *increases* under contraction; it is the term that resists collapse, balancing the potential (and, in $D = 3$, also the gradient) that resists expansion. An interior minimum becomes possible, as the right panel of Fig. 41.3 shows. □

The two balance relations Eqs. (41.79) and (41.80) are virial theorems for the gauged solitons, the direct analogues of the kink's $E_{\text{grad}} = E_{\text{pot}}$. They are necessary conditions that every static solution must satisfy, and they will reappear as nontrivial checks on the explicit vortex and monopole solutions of Sections 41.4 and 41.5. The first is the precise statement that, in two spatial dimensions, magnetic energy and potential energy must be equal in any soliton—which is only possible if the theory *has* a magnetic energy, i.e. if the broken symmetry is gauged.

Worked Example 41.20 (Equilibrium size of a gauged soliton from the scaling balance)**.** The scaling argument does more than permit a soliton; the balance

relations fix the soliton's size in terms of the competing energies, and we can extract that size before solving a single field equation. Consider a gauged configuration in D dimensions with energy Eq. (41.75), and ask which member of its scaling family has the least energy. Write the three energies of the *reference* configuration ($\mu = 1$) as $a \equiv E_{\text{mag}}$, $b \equiv E_{\text{grad}}$, $c \equiv E_{\text{pot}}$, so

$$E(\mu) = a\,\mu^{D-4} + b\,\mu^{D-2} + c\,\mu^{D}. \tag{41.81}$$

Two spatial dimensions. With $D = 2$, $E(\mu) = a\,\mu^{-2} + b + c\,\mu^{2}$. Setting $E'(\mu) = -2a\,\mu^{-3} + 2c\,\mu = 0$ gives $\mu^4 = a/c$, hence the energy-minimising scale

$$\mu_\star = \left(\frac{E_{\text{mag}}}{E_{\text{pot}}}\right)^{1/4}, \tag{41.82}$$

a finite, nonzero number whenever both energies are positive. The second derivative $E''(\mu) = 6a\,\mu^{-4} + 2c > 0$ confirms a genuine minimum. At $\mu_\star$ the rescaled magnetic and potential energies are $a\mu_\star^{-2} = \sqrt{ac}$ and $c\mu_\star^{2} = \sqrt{ac}$: they are equal, reproducing the virial relation Eq. (41.79) *at the balanced configuration.* This is the vortex: the magnetic flux, which wants to spread, is held at a definite radius by the potential, which wants to expel the field from the false vacuum in its core.

Three spatial dimensions. With $D = 3$, $E(\mu) = a\,\mu^{-1} + b\,\mu + c\,\mu^{3}$. Now $E'(\mu) = -a\,\mu^{-2} + b + 3c\,\mu^{2} = 0$, a quartic in μ once cleared of denominators, $3c\,\mu^4 + b\,\mu^2 - a = 0$, with the unique positive root

$$\mu_\star^2 = \frac{-b + \sqrt{b^2 + 12\,a\,c}}{6c} \qquad (c > 0), \tag{41.83}$$

again finite and nonzero whenever $a > 0$. (For the BPS monopole of the next sections, where the potential is sent to zero, $c \to 0$ and the root degenerates to $\mu_\star^2 = a/b = E_{\text{mag}}/E_{\text{grad}}$, the balance Eq. (41.80) with $E_{\text{pot}} = 0$.) The magnetic energy, decreasing as μ^{-1}, is what supplies the wall against collapse that the pure scalar theory lacked.

The lesson is quantitative: a soliton has a definite size set by the ratio of magnetic to potential (or gradient) energy, and that size exists only because the magnetic term carries the negative scaling exponent $D - 4$. Strip the gauge field out and $a \to 0$; Eqs. (41.82) and (41.83) send $\mu_\star \to 0$ and the soliton collapses, recovering Theorem 41.17.

Other evasions. Gauge fields are the route this chapter takes, but for completeness, and to show that the scaling table—not any one model—is the organising principle, we record the two other standard ways to defeat Derrick's theorem; each appears elsewhere in the non-perturbative literature and each is nothing but a new entry with a different exponent.

Table 41.1: Scaling weights of the standard energy terms under the dilation $\phi(\mathbf{x}) \to \phi(\mathbf{x}/\mu)$ (with the connection rescaled as $A \to A(\cdot/\mu)/\mu$), in D spatial dimensions. A term contributing μ^p with $p < 0$ grows under contraction and so resists collapse; a soliton requires at least one such term to coexist with a term of weight $p > 0$. The third and fourth columns evaluate the weight for the two cases this chapter constructs.

Energy term	inverse lengths	weight p	p at $D = 2$	p at $D = 3$
Potential $\int V(\phi)$	0	D	$+2$	$+3$
Gradient $\int \frac{1}{2}\lvert\nabla\phi\rvert^2$	2	$D-2$	0	$+1$
Covariant gradient $\int \frac{1}{2}\lvert D\phi\rvert^2$	2	$D-2$	0	$+1$
Magnetic $\int \frac{1}{4}F_{ij}F_{ij}$ (abelian or YM)	4	$D-4$	-2	-1
Skyrme $\int(\partial\phi)^4$	4	$D-4$	-2	-1

Higher-derivative (Skyrme) terms. A term quartic in spatial derivatives, schematically $(\partial\phi)^4$ with a suitable index contraction, carries four inverse lengths and therefore scales as μ^{D-4}, exactly like a magnetic energy. In $D = 3$ the Skyrme model balances such a quartic term (μ^{-1}) against the ordinary gradient term (μ^1): the stationarity condition $-E_4 + E_2 = 0$ fixes a finite size and stabilises the skyrmion of $\pi_3(\mathrm{SU}(2)) = \mathbb{Z}$ *without* gauge fields, at the price of a higher-derivative (and hence non-renormalisable, effective) Lagrangian. The exponent counting is identical to the gauge case; only the physical origin of the μ^{D-4} term differs.

Time dependence (Q-balls). Derrick's theorem assumes a *static* configuration; relaxing that assumption to a stationary but time-dependent one evades it. For a complex scalar with an unbroken global U(1) and conserved Noether charge Q (from Chapter 7), the ansatz $\phi(\mathbf{x}, t) = e^{-i\omega t}\sigma(\mathbf{x})$ with real profile σ adds a positive "rotational" term $\frac{1}{2}\omega^2\int\sigma^2$ to the energy and ties ω to the charge through $Q = \omega\int\sigma^2$. Minimising the energy at *fixed* Q—the physically correct variational problem, since Q is conserved—replaces the potential V by an effective potential $V - \frac{1}{2}\omega^2\sigma^2$ whose scaling weight differs, and a finite-energy non-topological soliton, the Q-ball, results in $D = 3$ even with only scalar fields. The lesson is that "static" in the hypothesis of Theorem 41.17 is load-bearing: a soliton stabilised by a conserved charge through its time dependence is outside the theorem's reach.

The table below collects the scaling weights, which is all one ever needs to apply or evade the theorem.

The pure-scalar rows of Table 41.1 have non-negative weights for all $D \geq 2$, which is Derrick's theorem; the magnetic row is the first with a negative weight reachable by a renormalisable, two-derivative-kinetic gauge theory in physical dimensions, and it is negative in exactly the dimensions $D = 2, 3$ where the

chapter must build solitons. The conclusion is now forced. To realise a nontrivial topological sector in two spatial dimensions one cannot use scalars alone; one must gauge the broken symmetry, supplying the magnetic energy whose μ^{D-4} weight holds the soliton at finite size. We turn next to the simplest theory in which this happens—the abelian Higgs model—and construct its vortex, whose topological charge $\pi_1(\mathrm{U}(1)) = \mathbb{Z}$ is realised physically as quantised magnetic flux.

41.4 The Vortex: Gauged Solitons and Flux Quantization

Derrick's theorem (Section 41.3) closed one door and, in the same motion, pointed at the only one left open. A theory of scalar fields alone supports no static, finite-energy soliton once the number of spatial dimensions reaches $D = 2$: under a uniform rescaling $\phi(\mathbf{x}) \mapsto \phi(\mathbf{x}/\mu)$ the gradient energy of a scalar scales as μ^{D-2} and the potential energy as μ^{D}, so for $D \geq 2$ both are non-increasing as $\mu \to 0$ and any putative lump simply spreads out and flattens to the vacuum at no energy cost. The escape clause was already visible in the bookkeeping: a term whose energy is *scale invariant* in the relevant dimension neither favours nor forbids the collapse, and in $D = 2$ the magnetic energy $\int d^2x\, \frac{1}{2}B^2$ of an abelian gauge field is exactly such a term. Gauging the broken symmetry therefore does two things at once. It supplies a scale-invariant term that stabilizes the lump against Derrick's collapse, and it converts the would-be massless Goldstone direction — the angular mode that, ungauged, would carry a logarithmically divergent gradient energy out to spatial infinity — into the longitudinal polarization of a massive vector, rendering the total energy finite. The object that results is the Abrikosov–Nielsen–Olesen (ANO) vortex: a tube of quantized magnetic flux threading a core in which the Higgs condensate is expelled. It is the first soliton in this chapter that genuinely lives above one dimension, and its topological charge is realized as a physically measurable, quantized magnetic flux. We will build it from the abelian Higgs Lagrangian, prove that finiteness of energy forces the flux to be an integer multiple of $2\pi/e$, write down the radial equations governing its core, and finally exhibit the Bogomolny rearrangement that, at one special value of the couplings, replaces the second-order field equations by a first-order system saturating an exact energy bound.

41.4.1 The abelian Higgs model in two spatial dimensions

The model is the abelian Higgs theory whose spontaneous-symmetry-breaking content was assembled in Chapter 30: a complex scalar ϕ of charge e minimally coupled to a U(1) gauge field A_μ, with a Mexican-hat potential that drives $|\phi|$ to a nonzero vacuum value. We recall it here for continuity and to fix the

conventions that the soliton analysis needs, but the classical content we use — the gauge-covariant derivative and the Maxwell action of Chapter 8, and the vacuum manifold of Chapter 29 — is imported, not rebuilt. The Lagrangian density is

$$\mathcal{L} \;=\; -\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} \;+\; (D_\mu\phi)^*(D^\mu\phi) \;-\; V(\phi), \qquad V(\phi) \;=\; \frac{\lambda}{2}\big(|\phi|^2 - v^2\big)^2, \tag{41.84}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ the abelian field strength of Eq. (8.75), and the gauge-covariant derivative on a charged scalar (Chapter 8)

$$D_\mu\phi \;=\; (\partial_\mu - ieA_\mu)\phi, \qquad e > 0. \tag{41.85}$$

We follow the book's convention $D_\mu = \partial_\mu - igA^a_\mu T^a$ specialized to the abelian case with a single generator of charge e; the relative sign in Eq. (41.85) is the one piece of data the flux quantization below is sensitive to, so we fix it once here. The theory is invariant under the local U(1) transformation

$$\phi(x) \mapsto e^{i\alpha(x)}\phi(x), \qquad A_\mu(x) \mapsto A_\mu(x) + \frac{1}{e}\,\partial_\mu\alpha(x), \tag{41.86}$$

under which $D_\mu\phi \mapsto e^{i\alpha}D_\mu\phi$ transforms homogeneously, exactly as the gauge principle of Chapter 8 requires.

The potential V in Eq. (41.84) is minimized on the circle $|\phi| = v$. The set of field values that minimize the potential — the vacuum manifold of Chapter 29 — is therefore

$$\mathcal{M} \;=\; \{\,\phi : |\phi| = v\,\} \;=\; v\,e^{i\chi},\ \chi \in [0, 2\pi) \;\cong\; \mathrm{U}(1) \;\cong\; S^1. \tag{41.87}$$

This is the geometric fact on which everything turns: the vacuum manifold is a circle, and a circle has nontrivial fundamental group. From the homotopy classification built earlier in this chapter, $\pi_1(S^1) = \mathbb{Z}$ (Theorem 41.11), and the integer labelling a map $S^1 \to S^1$ is its winding number, the degree. The whole of what follows is the statement that this abstract integer is realized in the abelian Higgs model as quantized magnetic flux.

We study static configurations in two spatial dimensions, $\mathbf{x} = (x^1, x^2) \in \mathbb{R}^2$, working in the temporal gauge $A_0 = 0$ with all fields time-independent. Then $D_0\phi = 0$, the electric field F_{0i} vanishes, and the only surviving component of the field strength is the magnetic field, a single scalar in the plane,

$$B \;\equiv\; F_{12} \;=\; \partial_1 A_2 - \partial_2 A_1. \tag{41.88}$$

The energy of such a configuration is the spatial integral of the Hamiltonian density. With the mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$ the Maxwell term contributes $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = \frac{1}{2}(\mathbf{E}^2 - B^2) \to -\frac{1}{2}B^2$ to the Lagrangian and hence $+\frac{1}{2}B^2$ to the energy, while the scalar kinetic term contributes $(D_i\phi)^*(D_i\phi)$

summed over the two spatial directions $i = 1, 2$. The static energy functional is therefore

$$E[\phi, A_i] \;=\; \int d^2x \left[\; \tfrac{1}{2} B^2 \;+\; |D_1\phi|^2 + |D_2\phi|^2 \;+\; \frac{\lambda}{2}\big(|\phi|^2 - v^2\big)^2 \;\right], \tag{41.89}$$

each term manifestly non-negative. A finite-energy static solution is a stationary point of Eq. (41.89) at which each of the three contributions is separately integrable.

41.4.2 Finite energy, the winding ansatz, and flux quantization

Finiteness of Eq. (41.89) dictates the behaviour of the fields at spatial infinity, and that asymptotic behaviour is what carries the topology. Each term must vanish fast enough that its integral over the plane converges. In particular the potential energy is finite only if $|\phi| \to v$ as $r = |\mathbf{x}| \to \infty$, so on the circle at infinity the scalar takes values in the vacuum manifold $\mathcal{M} \cong S^1$. The restriction of ϕ to the large circle is thus a map

$$\phi\big|_{r\to\infty} \;:\; S^1_\infty \;\longrightarrow\; \mathcal{M} \cong S^1, \qquad \theta \;\longmapsto\; v\, e^{\,i\,\Xi(\theta)}, \tag{41.90}$$

where θ is the polar angle and $\Xi(\theta)$ the asymptotic phase. Because ϕ is single valued, Ξ must return to itself modulo 2π as θ advances by 2π:

$$\Xi(\theta + 2\pi) - \Xi(\theta) \;=\; 2\pi n, \qquad n \in \mathbb{Z}. \tag{41.91}$$

The integer n is the degree of the map Eq. (41.90), the element of $\pi_1(S^1) = \mathbb{Z}$ it represents (Theorem 41.11). It cannot change under any continuous, finite-energy deformation of the fields, because changing it would require the phase to "unwind" through a configuration in which ϕ leaves the vacuum manifold somewhere on the circle at infinity, costing infinite potential energy. This is the topological conservation law of the vortex, and unlike a Noether charge it holds off shell: n is an integer read off the boundary data, conserved because it is discrete and the energy barrier to changing it is infinite, not because of any equation of motion.

The cleanest representative of the sector of winding n is the rotationally symmetric *winding ansatz*, in which the phase simply tracks the polar angle,

$$\phi(r, \theta) \;=\; v\, f(r)\, e^{\,in\theta}, \qquad \mathbf{A}(r, \theta) \;=\; \hat{\boldsymbol{\theta}}\, \frac{n}{e\, r}\, a(r), \tag{41.92}$$

with $\hat{\boldsymbol{\theta}}$ the azimuthal unit vector and f, a dimensionless radial profiles to be determined. Regularity at the origin and the vacuum at infinity will fix the boundary conditions $f(0) = a(0) = 0$ and $f(\infty) = a(\infty) = 1$; we derive these below. The deep point, however, requires no ansatz at all, and we prove it in generality first.

Theorem 41.21 (Flux quantization)**.** *Let (ϕ, A_i) be any static configuration of the abelian Higgs model Eq. (41.84) with finite energy Eq. (41.89) and asymptotic winding number $n \in \mathbb{Z}$ defined by Eq. (41.91). Then the total magnetic flux through the plane is quantized,*

$$\Phi \;\equiv\; \int_{\mathbb{R}^2} B\, d^2x \;=\; \frac{2\pi n}{e}. \tag{41.93}$$

Proof. The gradient energy $\int d^2x\, |D_i\phi|^2$ is finite only if the covariant derivative is square-integrable, which forces

$$D_i\phi \;\longrightarrow\; 0 \qquad \text{as } r \to \infty, \tag{41.94}$$

faster than $1/r$ (so that $|D_i\phi|^2$ decays faster than $1/r^2$ and its integral $\int r\, dr$ converges). Write the asymptotic scalar in polar form $\phi \to v\, e^{i\Xi(\theta)}$ with $|\phi| \to v$ constant along the circle at infinity. On that circle the radial derivative of ϕ is negligible and the only surviving derivative is angular; the condition Eq. (41.94) on the angular component reads

$$(\partial_i - ieA_i)\phi = i\,\phi\,(\partial_i\Xi - eA_i) + (\text{radial}) \;\longrightarrow\; 0 \quad\Longrightarrow\quad A_i \;\longrightarrow\; \frac{1}{e}\,\partial_i\Xi. \tag{41.95}$$

Asymptotically, therefore, A_i is *pure gauge*: it is the gradient of the scalar's phase divided by e. This is the mechanism by which the gauge field disarms Derrick's theorem. The angular gradient of the scalar, which for a global (ungauged) U(1) would give $|D_i\phi|^2 \sim v^2(\partial_i\Xi)^2 \sim v^2 n^2/r^2$ and hence a logarithmically divergent gradient energy $\int r\, dr/r^2$, is exactly cancelled by the gauge field, which absorbs the would-be Goldstone gradient into A_i and leaves $D_i\phi \to 0$. The energy of the angular mode is thereby made finite, at the price of a nonzero A_i at infinity.

Now compute the flux. By the two-dimensional Stokes theorem, the flux of $B = \partial_1 A_2 - \partial_2 A_1$ through the disc D_R of radius R equals the circulation of A_i around its boundary circle $C_R = \partial D_R$:

$$\int_{D_R} B\, d^2x \;=\; \int_{D_R} (\partial_1 A_2 - \partial_2 A_1)\, d^2x \;=\; \oint_{C_R} A_i\, dx^i. \tag{41.96}$$

Taking $R \to \infty$ and inserting the asymptotic form Eq. (41.95),

$$\Phi \;=\; \lim_{R\to\infty} \oint_{C_R} A_i\, dx^i \;=\; \frac{1}{e} \oint_{C_\infty} \partial_i\Xi\, dx^i \;=\; \frac{1}{e}\big[\Xi(\theta)\big]_{\theta=0}^{\theta=2\pi} \;=\; \frac{1}{e}\,(2\pi n) \;=\; \frac{2\pi n}{e}, \tag{41.97}$$

where the line integral of the gradient $\partial_i\Xi$ around the closed loop equals the total change of Ξ over one circuit, which by Eq. (41.91) is $2\pi n$. This proves Eq. (41.93). □

The physical content of Theorem 41.21 deserves to be stated plainly, because it is the prototype for the monopole's magnetic charge in the next section and,

beyond that, for the general phenomenon of topological quantization. The integer n entered as an abstract homotopy class — the winding of a map between two circles, a piece of pure topology with no obvious dynamical meaning. It comes out as the magnetic flux measured in units of $2\pi/e$, a quantity an experimentalist could in principle integrate a magnetometer over. There is no continuous knob that turns n from one integer to the next; the flux is locked to a discrete spectrum by the requirement that the configuration have finite energy. In a superconductor, where the abelian Higgs model is the Ginzburg–Landau effective theory of the condensate, this is precisely the observed quantization of magnetic flux trapped in the Abrikosov vortices of a type-II superconductor, with e replaced by the charge $2e$ of a Cooper pair.

Physical Insight

The covariant derivative is what makes the gauged soliton possible, and it does so through a single sleight of hand worth isolating. A global symmetry broken on a circle leaves behind a massless Goldstone mode, and a winding configuration of that mode carries angular gradient energy $\sim v^2 n^2 \int r\, dr/r^2$ that diverges logarithmically with the size of the system — the global vortex is not quite finite-energy. Gauging the symmetry promotes the Goldstone to the longitudinal polarization of a vector field of mass $m_v \sim ev$ (the Higgs mechanism of Chapter 30). The gauge field can then "eat" the winding gradient: at infinity $A_i \to \frac{1}{e}\partial_i \Xi$ makes $D_i\phi \to 0$ and the angular energy converges. What remains of the would-be Goldstone gradient is not lost but stored as magnetic field threading the core, and the flux it carries is fixed by topology. The vortex is finite-energy precisely because the symmetry is gauged, and quantized precisely because the unbroken topology is $\pi_1(S^1) = \mathbb{Z}$.

41.4.3 Radial profile equations and the structure of the core

To exhibit an actual solution and resolve the core we insert the winding ansatz Eq. (41.92) into the energy and vary. Work in plane polar coordinates with the orthonormal frame $(\hat{\mathbf{r}}, \hat{\boldsymbol{\theta}})$, in which the covariant derivative of a function has radial and azimuthal components $(\partial_r,\, r^{-1}\partial_\theta)$. With $A_r = 0$ and $A_\theta = \dfrac{n}{e\,r}\, a(r)$ the two components of $D_i\phi$ are

$$(D\phi)_r = \partial_r \phi - ieA_r\phi = v\, f'(r)\, e^{in\theta}, \tag{41.98}$$

$$(D\phi)_\theta = \frac{1}{r}\,\partial_\theta\phi - ieA_\theta\phi = i\,v\, f(r)\, e^{in\theta}\, \frac{n}{r}(1 - a(r)), \tag{41.99}$$

where in the second line the bare angular derivative $\frac{1}{r}\partial_\theta(v f e^{in\theta}) = iv f e^{in\theta}\, n/r$ combines with $-ieA_\theta\phi = -iv f e^{in\theta}\, na/r$ to give the factor $(1-a)$. The magnetic

field of the ansatz follows from the curl in the plane, $B = \frac{1}{r}\,\partial_r(rA_\theta)$, namely

$$B(r) \;=\; \frac{1}{r}\,\frac{d}{dr}\!\left(r\cdot\frac{n}{er}a\right) \;=\; \frac{n}{e}\,\frac{a'(r)}{r}. \tag{41.100}$$

The boundary conditions are now forced. Regularity of ϕ at the origin (a single-valued field cannot have a nonzero modulus where its phase $e^{in\theta}$ is undefined, for $n \neq 0$) requires $f(0) = 0$; regularity of $A_\theta = \frac{n}{er}a$ as $r \to 0$ requires $a(0) = 0$. The vacuum at infinity requires $f(\infty) = 1$, and finiteness of the gradient energy via Eq. (41.95) requires $(1 - a) \to 0$, i.e. $a(\infty) = 1$. With these, the flux of Eq. (41.100) is $\Phi = \int_0^\infty B\, 2\pi r\, dr = \frac{2\pi n}{e}\int_0^\infty a'\, dr = \frac{2\pi n}{e}[a(\infty) - a(0)] = \frac{2\pi n}{e}$, reproducing Theorem 41.21 for the symmetric ansatz.

Substituting Eqs. (41.98) to (41.100) into the energy Eq. (41.89) and using $\int d^2x = \int_0^\infty 2\pi r\, dr$ gives the reduced energy as a one-dimensional functional of the two profiles,

$$E \;=\; 2\pi v^2\int_0^\infty r\,dr\,\left[f'^2 \;+\; \frac{n^2}{r^2}(1-a)^2 f^2 \;+\; \frac{n^2}{2e^2v^2}\,\frac{a'^2}{r^2} \;+\; \frac{\lambda v^2}{2}(1-f^2)^2\right]. \tag{41.101}$$

The Euler–Lagrange equations of this functional are the radial field equations. Varying with respect to f and to a (carrying the $\frac{1}{r}\frac{d}{dr}(r\,\partial/\partial(\cdot)')$ structure of a radial variational problem) yields

$$f'' + \frac{1}{r}f' - \frac{n^2}{r^2}(1-a)^2 f + \lambda v^2\,(1-f^2)f = 0, \tag{41.102}$$

$$a'' - \frac{1}{r}a' + 2e^2v^2\,f^2(1-a) = 0, \tag{41.103}$$

to be solved with $f(0) = a(0) = 0$, $f(\infty) = a(\infty) = 1$. These are two coupled nonlinear ordinary differential equations; they have no solution in elementary closed form, but their qualitative content is transparent and their asymptotics are exact.

Linearizing about the vacuum exposes the two length scales of the vortex, which are nothing but the Compton wavelengths of the two massive particles of the broken theory. Writing $f = 1 - s$ and $a = 1 - w$ with s, w small at large r, Eq. (41.102) becomes $s'' + \frac{1}{r}s' - 2\lambda v^2 s \approx 0$ and Eq. (41.103) becomes $w'' - \frac{1}{r}w' - 2e^2v^2 w \approx 0$, the modified-Bessel equations whose decaying solutions are $s \sim e^{-m_s r}$ and $w \sim e^{-m_v r}$ with

$$m_s \;=\; \sqrt{2\lambda}\,v \quad\text{(Higgs/scalar mass)}, \qquad m_v \;=\; \sqrt{2}\,e\,v \quad\text{(vector/photon mass)}. \tag{41.104}$$

The scalar mass m_s sets the radius of the core over which the condensate $|\phi| = vf$ is depleted (the *coherence length* $\xi = 1/m_s$); the vector mass m_v sets the radius over which the magnetic field is screened out (the *penetration depth* $\delta = 1/m_v$).

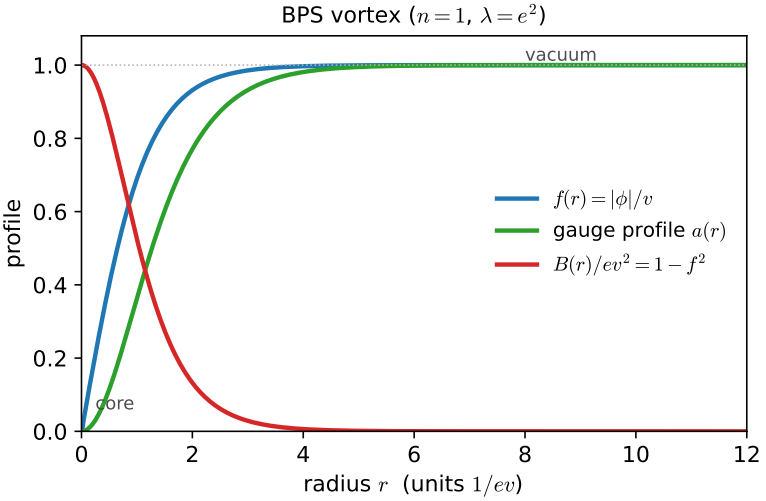


Figure 41.4: The $n = 1$ Abrikosov–Nielsen–Olesen vortex at the critical coupling $\lambda = e^2$, obtained by solving the radial field equations Eqs. (41.102) and (41.103) as a two-point boundary-value problem (units $e = v = 1$). The scalar profile $f = |\phi|/v$ rises from 0 at the core to the vacuum value 1; the gauge profile a rises from 0 to 1, so that $D_i\phi \to 0$ at infinity; the magnetic field $B/ev^2 = 1 - f^2$ is concentrated in the core where the condensate is expelled. Numerical integration of B returns the total flux $\Phi = 2\pi/e$ and the energy $E = 2\pi v^2$, saturating the Bogomolny bound Eq. (41.115) for $n = 1$.

The identification of these masses confirms the Higgs mechanism: the gauge field has acquired the mass $m_v = \sqrt{2}\,ev$ from the condensate, and it is this mass that makes the magnetic flux a localized tube rather than a long-range field. We solve the system numerically for the simplest vortex in Fig. 41.4; the profiles have exactly the expected shape, the condensate rising from zero at the core to its vacuum value and the magnetic field concentrated where the condensate is suppressed.

41.4.4 The Bogomolny bound and the BPS vortex

The energy Eq. (41.89) is bounded below by a topological quantity, and at one special value of the couplings the bound is saturated by solutions of a first-order system far simpler than the second-order equations Eqs. (41.102) and (41.103). The technique is Bogomolny's: rewrite each square in the energy as a perfect square plus a total derivative plus a topological term, so that the energy is manifestly a sum of squares above a fixed floor set by the topology. This is the same manoeuvre that produced the kink's energy bound earlier in the chapter, now executed in two dimensions with a gauge field, and it is the direct template

for the monopole's bound and for the general notion of a BPS state.

We work with the magnetic and scalar gradient energies. The first algebraic input is the rearrangement of the gradient energy into the modulus of a single complex combination plus a remainder. For any two complex numbers,

$$|D_1\phi|^2+|D_2\phi|^2 \;=\; \left|(D_1 \mp iD_2)\phi\right|^2 \;\pm\; i\big[(D_1\phi)^*(D_2\phi)-(D_2\phi)^*(D_1\phi)\big], \tag{41.105}$$

which is verified by expanding $|(D_1 \mp iD_2)\phi|^2 = (D_1\phi \mp iD_2\phi)^*(D_1\phi \mp iD_2\phi)$ and collecting terms; the cross terms $\mp i(D_1\phi)^*(D_2\phi) \pm i(D_2\phi)^*(D_1\phi)$ are exactly the bracket on the right. The bracket is now reshaped into a magnetic piece and a total derivative. Using $D_\mu = \partial_\mu - ieA_\mu$ and the commutator $[D_1, D_2]\phi = -ieF_{12}\phi = -ieB\phi$ (the field strength is the curvature of the covariant derivative, Chapter 8), a direct computation gives the identity

$$\epsilon_{ij}\,(D_i\phi)^*(D_j\phi) \;=\; \epsilon_{ij}\,\partial_i(\phi^* D_j\phi) \;+\; ieB\,|\phi|^2, \tag{41.106}$$

where $\epsilon_{12} = -\epsilon_{21} = +1$. We verify Eq. (41.106) explicitly because the sign of its magnetic term controls the final bound. Expand the total-derivative term,

$$\epsilon_{ij}\partial_i(\phi^* D_j\phi) = \epsilon_{ij}\big[(\partial_i\phi^*)(D_j\phi) + \phi^*\,\partial_i(D_j\phi)\big]. \tag{41.107}$$

Now $\partial_i\phi^* = (D_i\phi)^* + ieA_i\phi^*$ (since $(D_i\phi)^* = (\partial_i + ieA_i)\phi^*$), and $\partial_i(D_j\phi) = D_i(D_j\phi) + ieA_i(D_j\phi)$ because $D_j\phi$ carries charge $+e$. Substituting, the two ieA_i pieces cancel against each other inside the antisymmetric ϵ_{ij} contraction, leaving

$$\epsilon_{ij}\partial_i(\phi^* D_j\phi) = \epsilon_{ij}(D_i\phi)^*(D_j\phi)+\phi^*\,\epsilon_{ij}D_iD_j\phi = \epsilon_{ij}(D_i\phi)^*(D_j\phi)+\tfrac{1}{2}\phi^*\epsilon_{ij}[D_i, D_j]\phi. \tag{41.108}$$

The last term is $\frac{1}{2}\phi^*\epsilon_{ij}(-ieF_{ij})\phi = \frac{1}{2}(-ie)(2B)|\phi|^2 = -ieB|\phi|^2$, using $\epsilon_{ij}F_{ij} = 2F_{12} = 2B$. Rearranging Eq. (41.108) gives Eq. (41.106). The bracket in Eq. (41.105) is $i\,\epsilon_{ij}(D_i\phi)^*(D_j\phi)$, so combining,

$$|D_1\phi|^2 + |D_2\phi|^2 \;=\; \left|(D_1 \mp iD_2)\phi\right|^2 \;\mp\; eB\,|\phi|^2 \;\pm\; i\,\epsilon_{ij}\,\partial_i(\phi^* D_j\phi). \tag{41.109}$$

The last term is a total spatial derivative; on integration over the plane it becomes a boundary integral at $r \to \infty$, where $D_j\phi \to 0$, so it contributes nothing to the energy. We may discard it.

Assemble the energy. Insert Eq. (41.109) into Eq. (41.89) and group the magnetic, the $\mp eB|\phi|^2$, and the potential terms,

$$E \;=\; \int d^2x\,\Big[\,\tfrac{1}{2}B^2 \mp eB|\phi|^2 + \tfrac{\lambda}{2}(|\phi|^2 - v^2)^2 \;+\; \left|(D_1 \mp iD_2)\phi\right|^2\,\Big]. \tag{41.110}$$

The first three terms are a quadratic in B that completes to a perfect square *only* when the quartic coupling takes the special value

$$\boxed{\lambda \;=\; e^2} \qquad \text{(critical, Bogomolny, or BPS coupling).} \tag{41.111}$$

At this value $\frac{\lambda}{2}(|\phi|^2 - v^2)^2 = \frac{e^2}{2}(|\phi|^2 - v^2)^2$, and writing $\mp eB|\phi|^2 = \mp eB(|\phi|^2 - v^2) \mp eBv^2$ we recognize the completed square

$$\tfrac{1}{2}B^2 \mp eB(|\phi|^2 - v^2) + \tfrac{e^2}{2}(|\phi|^2 - v^2)^2 \;=\; \tfrac{1}{2}\big[\, B \mp e(|\phi|^2 - v^2)\,\big]^2, \tag{41.112}$$

leaving the lone linear remainder $\mp eBv^2$. Therefore, at $\lambda = e^2$,

$$E \;=\; \int d^2x \left[\, \tfrac{1}{2}\big(B \mp e(|\phi|^2 - v^2)\big)^2 + \big|(D_1 \mp iD_2)\phi\big|^2 \,\right] \;\mp\; ev^2 \int d^2x\, B. \tag{41.113}$$

The two square-bracketed terms are non-negative, and the final term is purely topological: $\int B\, d^2x = \Phi = 2\pi n/e$ by Theorem 41.21. Hence

$$E \;=\; \int d^2x \left[\, \tfrac{1}{2}\big(B \mp e(|\phi|^2 - v^2)\big)^2 + \big|(D_1 \mp iD_2)\phi\big|^2 \,\right] \;\mp\; 2\pi v^2 n. \tag{41.114}$$

Choosing the upper or lower sign so that $\mp 2\pi v^2 n = +2\pi v^2|n|$ (lower sign for $n > 0$, upper for $n < 0$) makes the topological term positive, and since the remaining integral is non-negative we obtain the Bogomolny bound

$$E \;\geq\; 2\pi v^2\, |n|, \tag{41.115}$$

with equality if and only if the two non-negative integrands vanish identically. The energy of a topological vortex is bounded below by a multiple of its winding number; it cannot decay because n is locked, and the lightest configuration in each sector sits exactly on the floor Eq. (41.115).

Saturation defines the *Bogomolny–Prasad–Sommerfield (BPS) vortex*. Equality in Eq. (41.115) holds precisely when both squares vanish, i.e. when the fields satisfy the first-order *Bogomolny equations*

$$(D_1 \mp iD_2)\phi \;=\; 0, \qquad B \;=\; \pm\, e\,(|\phi|^2 - v^2), \tag{41.116}$$

with the upper sign for $n < 0$ and the lower (so $B = e(v^2 - |\phi|^2) \geq 0$ in the core where $|\phi| < v$) for $n > 0$. A configuration solving the first-order system Eq. (41.116) automatically solves the second-order field equations Eqs. (41.102) and (41.103): since it is a stationary point of the energy — indeed an absolute minimum within its topological sector — it is a critical point of the action, hence a solution of the equations of motion. This is the payoff of the Bogomolny rearrangement. At the critical coupling Eq. (41.111) the difficult coupled nonlinear second-order problem is replaced by a much milder first-order one, and the energy of every solution is known exactly, $E = 2\pi v^2|n|$, before a single profile is computed.

Worked Example 41.22 (The BPS equations in radial form and the saturated energy)**.** We reduce the Bogomolny equations Eq. (41.116) to the winding ansatz Eq. (41.92) for $n > 0$ (lower signs) and confirm directly that the energy equals

$2\pi v^2 n$. The first BPS equation $(D_1 + iD_2)\phi = 0$ is, in the orthonormal frame, $(D\phi)_r + i(D\phi)_\theta = 0$. Using Eqs. (41.98) and (41.99),

$$v f' e^{in\theta} + i\cdot i\, v f e^{in\theta}\frac{n}{r}(1-a) = v e^{in\theta}\Big[\, f' - \frac{n}{r}(1-a) f \,\Big] = 0 \quad\Longrightarrow\quad f' = \frac{n}{r}(1-a)\, f. \tag{41.117}$$

The second BPS equation $B = e(v^2 - |\phi|^2) = ev^2(1 - f^2)$, together with $B = \frac{n}{e}\frac{a'}{r}$ from Eq. (41.100), gives

$$\frac{n}{e}\frac{a'}{r} = ev^2(1 - f^2) \quad\Longrightarrow\quad a' = \frac{e^2 v^2}{n}\, r\,(1 - f^2). \tag{41.118}$$

Equations Eqs. (41.117) and (41.118) are the first-order radial system; one checks by differentiating Eq. (41.117) and substituting Eq. (41.118) that they imply the second-order equations Eqs. (41.102) and (41.103) at $\lambda = e^2$, as the general argument guarantees. They are the equations solved numerically in Fig. 41.4, where the boundary conditions $f(0) = a(0) = 0$, $f(\infty) = a(\infty) = 1$ select the unique $n = 1$ vortex.

For the energy, set the squares in Eq. (41.114) to zero (they vanish on a BPS solution) and read off $E = 2\pi v^2 n$. As an independent check that the reduced energy Eq. (41.101) is consistent with this, the numerical integration reported in Fig. 41.4 for $n = 1$, $e = v = 1$ returns $E = 2\pi v^2$ and $\Phi = 2\pi/e$ to better than one part in 10^3, confirming both the flux quantization Eq. (41.93) and the saturated bound Eq. (41.115) numerically.

The critical coupling Eq. (41.111) separates two qualitatively different regimes, with a clean physical interpretation in terms of the two masses Eq. (41.104). Define the Ginzburg–Landau parameter $\kappa \equiv m_s/m_v = \sqrt{\lambda}/e$, the ratio of penetration depth to coherence length. At $\kappa = 1$, i.e. $\lambda = e^2$, the theory sits at the BPS point: the static force between two well-separated vortices vanishes identically, because the scalar-mediated attraction (range $1/m_s$) and the vector-mediated repulsion (range $1/m_v$) cancel exactly, and the energy of a configuration of total winding n is $2\pi v^2|n|$ regardless of how the flux is distributed. For $\kappa < 1$ ($\lambda < e^2$, the *type-I* regime) the scalar is lighter, its attraction is longer-ranged, vortices attract, and flux prefers to coalesce into a single multiply-wound lump. For $\kappa > 1$ ($\lambda > e^2$, the *type-II* regime) the vector is comparatively lighter, the repulsion wins, vortices repel, and flux fragments into a lattice of unit-winding tubes — the Abrikosov lattice observed in type-II superconductors. The BPS point is the marginal, no-force boundary between the two, and it is exactly there that the topological bound is saturated by the first-order equations.

Caution

The clean saturation $E = 2\pi v^2|n|$ holds only at the critical coupling $\lambda = e^2$. Away from it the Bogomolny rearrangement Eq. (41.112) fails — the

potential no longer completes the square with the magnetic energy — and Eq. (41.115) is replaced by an inequality whose floor is still topological but is no longer attained; the true minimal energy in each sector then differs from $2\pi v^2|n|$ and must be found from the full second-order equations Eqs. (41.102) and (41.103). The Bogomolny bound *as an inequality* ($E \geq 2\pi v^2|n|$) survives for $\lambda \leq e^2$ by the same completion of squares applied with the genuine potential bounded below by the critical one; for $\lambda > e^2$ the inequality requires a slightly different grouping. The point to retain is that the first-order BPS system Eq. (41.116) solves the equations of motion only at $\lambda = e^2$.

We have constructed the vortex: a finite-energy, particle-like (here, string-like in the third spatial dimension) classical solution stabilized by gauging the broken U(1), carrying a topological charge $n \in \pi_1(S^1) = \mathbb{Z}$ that is physically realized as magnetic flux quantized in units of $2\pi/e$, and saturating at the critical coupling a Bogomolny bound $E = 2\pi v^2|n|$ by first-order equations. The construction exhausts what an abelian theory can do in two spatial dimensions. To climb the next rung of the ladder we add a third spatial dimension, where the relevant homotopy group of an abelian vacuum manifold becomes trivial — $\pi_2(S^1) = 0$, a circle cannot wrap a sphere — so a genuine point-like soliton demands a vacuum manifold with nontrivial π_2, hence a non-abelian symmetry broken to leave an unbroken U(1). That is the setting of the next section, where the same finite-energy asymptotics that here quantized magnetic flux will there quantize magnetic charge, and the same Bogomolny structure will reappear: the 't Hooft–Polyakov monopole.

41.5 The 't Hooft–Polyakov Monopole

The vortex of Section 41.4 taught a lesson sharp enough to be turned into a recipe. A scalar field alone cannot make a soliton in more than one spatial dimension (Section 41.3); but if the broken symmetry is *gauged*, the gauge field supplies exactly the energy that Derrick's scaling argument needs in order to balance, and the resulting object carries a topological charge that is realized physically as quantized flux. In two spatial dimensions the relevant homotopy group was π_1 of the broken U(1), and the flux was a single number. We now climb one rung higher — to three spatial dimensions and to a *non-abelian* gauge group — and the same recipe produces something qualitatively new: a smooth, finite-energy lump that, viewed from far away, looks exactly like a magnetic monopole. Its magnetic charge is forced to be quantized, and the quantization is not imposed by hand to hide an unobservable singularity, as in the Dirac construction of Chapter 8, but is a theorem: an integer counting how the Higgs field wraps the vacuum manifold. This is the monopole stage promised for Yang–Mills theory,

and it is the first place in the book where a classical, everywhere-regular field configuration realizes magnetic charge as a topological invariant.

41.5.1 The Georgi–Glashow model

The simplest theory that carries a monopole is the one Georgi and Glashow originally wrote down as a (now excluded) model of the electroweak interactions: an SU(2) gauge field coupled to a single scalar in the *adjoint* representation, with a potential that breaks SU(2) down to U(1). We take the gauge field A^a_μ ($a = 1, 2, 3$) of Chapter 25 with coupling constant e — this is the Yang–Mills theory coupling g, renamed e because the surviving U(1) will be electromagnetism — and an adjoint scalar ϕ^a, a triplet of real fields transforming under gauge rotations exactly as the field strength does. The Lagrangian is

$$\mathcal{L} \;=\; -\tfrac{1}{4}\, F^a_{\mu\nu} F^{a\,\mu\nu} \;+\; \tfrac{1}{2}\,(D_\mu \phi)^a (D^\mu \phi)^a \;-\; V(\phi), \qquad V(\phi) = \frac{\lambda}{4}\big(\phi^a \phi^a - v^2\big)^2, \tag{41.119}$$

with the non-abelian field strength and adjoint covariant derivative of Chapter 25, specialized to SU(2) where the structure constants are the Levi-Civita symbol $f^{abc} = \epsilon^{abc}$ (Eqs. (25.35) and (25.55)),

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + e\,\epsilon^{abc} A^b_\mu A^c_\nu, \qquad (D_\mu \phi)^a = \partial_\mu \phi^a + e\,\epsilon^{abc} A^b_\mu \phi^c. \tag{41.120}$$

We work in the mostly-minus signature with natural units, and throughout this section the fields are classical: $\hbar$ stays in the drawer, and the monopole's quantum fate is deferred to Chapter 46.

The potential V is minimized on the set $\phi^a \phi^a = v^2$, a two-sphere of radius v in the three-dimensional internal space. This is the *vacuum manifold*

$$\mathcal{M} \;=\; \{\, \phi^a : \phi^a \phi^a = v^2 \,\} \;\cong\; S^2. \tag{41.121}$$

Choosing a particular vacuum, say $\phi^a = v\,\delta^{a3}$, breaks the gauge symmetry spontaneously in the manner developed in Chapters 29 and 30. The unbroken subgroup is the stabilizer of that vacuum direction — the isospin rotations about the third axis, generated by T^3 — so

$$\mathrm{SU}(2) \;\longrightarrow\; \mathrm{U}(1), \qquad \mathcal{M} \;\cong\; \mathrm{SU}(2)/\mathrm{U}(1) \;\cong\; \mathrm{SO}(3)/\mathrm{SO}(2) \;\cong\; S^2, \tag{41.122}$$

the last isomorphism using that the adjoint sees only SO(3), the image of SU(2) under the double cover of Theorem 2.47. The coset description (41.122) is the same S^2 as (41.121), viewed as the orbit of a chosen vacuum under the broken generators.

The spectrum around the vacuum $\phi^a = v\delta^{a3}$ is read off exactly as in Chapter 30. The covariant-derivative term gives the gauge bosons a mass from $\tfrac{1}{2} e^2 \epsilon^{abc}\epsilon^{ade} A^b_\mu \phi^c A^{d\mu} \phi^e$; setting $\phi^c = v\delta^{c3}$ and using $\epsilon^{ab3}\epsilon^{ad3} = \delta^{bd} - \delta^{b3}\delta^{d3}$

collapses this to $\frac{1}{2}e^2v^2\,(A^1_\mu A^{1\mu} + A^2_\mu A^{2\mu})$, so the two charged combinations $W^\pm_\mu = (A^1_\mu \mp i A^2_\mu)/\sqrt{2}$ acquire

$$M_W \;=\; e\,v, \tag{41.123}$$

while A^3_μ stays massless — it is the photon of the unbroken electromagnetic U(1). The scalar sector contributes one massive Higgs of mass $M_H = \sqrt{2\lambda}\,v$ (the radial fluctuation) and two would-be Goldstone modes eaten by $W^\pm$. The two mass scales M_W and M_H, equivalently the two couplings e and λ, are the only parameters; the combination $\lambda/e^2 = M_H^2/2M_W^2$ will turn out to control the monopole.

The decisive fact for what follows is purely topological:

$$\pi_2(\mathcal{M}) \;=\; \pi_2(S^2) \;=\; \mathbb{Z}. \tag{41.124}$$

By the homotopy classification built in Theorem 41.10, maps from a two-sphere into the vacuum manifold fall into integer classes, and the integer is the degree — the number of times the image wraps the target. In one spatial dimension the relevant sphere at infinity was the pair of points S^0 and π_0 counted kink sectors; in two dimensions it was the circle S^1 and π_1 counted vortex flux. In three spatial dimensions the boundary at infinity is a two-sphere, and (41.124) is precisely the group that will label the finite-energy configurations. The Georgi–Glashow model is the minimal renormalizable theory whose vacuum manifold has $\pi_2 = \mathbb{Z}$, and that is why it carries a monopole.

Physical Insight

The choice of vacuum is a choice of a *point* on the two-sphere $\mathcal{M}$. A finite-energy field in three dimensions must approach *some* vacuum in every direction at spatial infinity, i.e. it assigns to each direction $\hat r \in S^2_\infty$ a point of $\mathcal{M} \cong S^2$. That assignment is a map $S^2 \to S^2$, and such maps cannot all be deformed to a constant: the ones that wrap a nonzero number of times are trapped. The monopole is the field configuration whose asymptotic Higgs orientation wraps the vacuum sphere exactly once. Everything below is the working-out of this single picture.

41.5.2 Finite energy and the boundary map

For a static configuration we may set $A^a_0 = 0$ and take all fields time-independent. Then the temporal components of the field strength vanish, $F^a_{0i} = 0$, so the configuration carries no electric field; the energy is the static functional

$$E \;=\; \int d^3x\,\Big[\,\tfrac{1}{2}\,B^a_i B^a_i \;+\; \tfrac{1}{2}\,(D_i\phi)^a(D_i\phi)^a \;+\; V(\phi)\,\Big], \qquad B^a_i \equiv -\tfrac{1}{2}\,\epsilon_{ijk}F^a_{jk}, \tag{41.125}$$

each term manifestly non-negative (this is the three-dimensional gauged analogue of the kink and vortex energy functionals of Sections 41.1 and 41.4; the sign accounting that turns $-\frac{1}{4}F^2$ into $+\frac{1}{2}B^2$ for a purely magnetic static field is the mostly-minus bookkeeping of Section 8.3.4). Here B_i^a is the non-abelian magnetic field, one SO(3) triplet for each spatial direction.

Finiteness of (41.125) forces, term by term, the asymptotic behaviour of the fields. The potential term requires $V(\phi) \to 0$ faster than r^{-3}, hence

$$\phi^a(\vec{r}) \xrightarrow{r\to\infty} \text{a point of } \mathcal{M}, \qquad |\phi| \to v, \tag{41.126}$$

in every direction. The gradient term requires $(D_i\phi)^a \to 0$: the asymptotic Higgs must be *covariantly* constant, which (since ϕ itself is not constant in direction) fixes the asymptotic gauge field in terms of ϕ. The magnetic term requires $B_i^a \to 0$ faster than $r^{-3/2}$. The upshot is that the entire boundary data at infinity is encoded in the unit Higgs field

$$\hat{\phi}^a \equiv \frac{\phi^a}{|\phi|}, \qquad \hat{\phi}: S^2_\infty \longrightarrow \mathcal{M} \cong S^2, \tag{41.127}$$

a continuous map from the sphere at spatial infinity to the vacuum two-sphere. By (41.124) and Theorem 41.10 it carries an integer degree

$$n = \deg(\hat{\phi}) \in \mathbb{Z}, \tag{41.128}$$

which cannot change under any continuous, finite-energy deformation of the fields: to alter n one would have to pass through a configuration whose Higgs field leaves the vacuum manifold on the whole sphere at infinity, costing infinite potential energy. The integer n is the monopole's topological charge. That the gauge term in (41.125) makes a finite-energy minimizer exist in each sector $n \neq 0$ — evading Derrick's theorem exactly as the vortex did — is the content of Section 41.3; here we exhibit the minimizer and compute its charges.

41.5.3 The hedgehog ansatz

To realize $n = 1$ we need a configuration whose asymptotic Higgs wraps the vacuum sphere once. The most symmetric such configuration identifies the internal isospin direction with the spatial direction — it points the Higgs field radially outward in internal space at each point of physical space. This is the *hedgehog*:

$$\boxed{\phi^a(\vec{r}) = v\,\hat{r}^a\,H(r), \qquad A_i^a(\vec{r}) = \epsilon^{aij}\,\hat{r}^j\,\frac{1-K(r)}{e\,r}, \qquad A_0^a = 0,} \tag{41.129}$$

where $\hat{r}^a = x^a/r$ is the radial unit vector, used simultaneously as a spatial vector (index i) and as an internal isovector (index a) — the locking of space to isospin

that the name records. The two dimensionless radial profiles $H(r)$ and $K(r)$ are to be determined; their boundary values are fixed by regularity and finiteness:

$$H(0) = 0, \quad K(0) = 1 \qquad \text{(regularity at the origin)}, \qquad H(\infty) = 1, \quad K(\infty) = 0 \qquad \text{(vacuum} \tag{41.130}$$

At the origin $\phi \to 0$ and A_i^a is finite, so the fields are smooth there — the core is an ordinary, nonsingular lump of field energy in which the full SU(2) symmetry is restored. At infinity $H \to 1$ gives $\phi^a \to v\,\hat{r}^a$, so the asymptotic unit Higgs is

$$\hat{\phi}^a \xrightarrow{r\to\infty} \hat{r}^a, \tag{41.131}$$

the *identity* map $S^2_\infty \to S^2$, which has degree one. The hedgehog therefore sits in the sector $n = 1$. Higher n is reached by composing $\hat{r}^a$ with a degree-n map of the sphere; the construction below goes through with $1 \to n$ in the final charges, and we display n explicitly where it enters.

Physical Insight

The hedgehog cannot be combed flat. A continuous unit-vector field on S^2 that points strictly radially has nowhere to "lie down": by the hairy-ball theorem (a special case of Theorem 41.10, degree $\neq 0$) it must have a singularity somewhere, and in the field theory that "somewhere" is the spatial origin, where $|\phi| \to 0$ lets the direction $\hat{\phi}$ be undefined smoothly. The Higgs field can only relax to the vacuum manifold everywhere if it is allowed to climb out of it at the core. The price of that climb — the gradient and potential energy stored in the core — is the monopole's mass.

41.5.4 The reduced energy and the profile equations

We now substitute the ansatz (41.129) into the energy (41.125) and carry the algebra in full, because the result is both the mass formula and the field equations for the profiles.

The covariant gradient. With $\partial_i \hat{r}^a = (\delta_{ia} - \hat{r}_i\hat{r}_a)/r$ and $\partial_i r = \hat{r}_i$,

$$\partial_i \phi^a = v\Big[\, H'\,\hat{r}_i\hat{r}_a + \frac{H}{r}\,(\delta_{ia} - \hat{r}_i\hat{r}_a)\,\Big]. \tag{41.132}$$

For the gauge term write $\alpha(r) \equiv (1-K)/(er)$, so $A_i^b = \epsilon_{bij}\hat{r}_j\,\alpha$. Then

$$e\,\epsilon^{abc}A_i^b\phi^c = e\,vH\,\alpha\,\epsilon_{abc}\epsilon_{bij}\,\hat{r}_j\hat{r}_c = -\,vH\,\frac{1-K}{r}\,(\delta_{ia} - \hat{r}_i\hat{r}_a), \tag{41.133}$$

where we used $\epsilon_{abc}\epsilon_{bij} = -(\delta_{ai}\delta_{cj} - \delta_{aj}\delta_{ci})$, contracted with $\hat{r}_j\hat{r}_c$ to give $-(\delta_{ia} - \hat{r}_i\hat{r}_a)$. Adding (41.132) and (41.133), the H/r and $-(1-K)H/r$ pieces combine to HK/r:

$$(D_i\phi)^a = v\Big[\, H'\,\hat{r}_i\hat{r}_a + \frac{HK}{r}\,(\delta_{ia} - \hat{r}_i\hat{r}_a)\,\Big]. \tag{41.134}$$

The radial projector $\hat r_i\hat r_a$ and the transverse projector $\delta_{ia}-\hat r_i\hat r_a$ are orthogonal and have squared traces 1 and 2 respectively, so

$$\tfrac12 (D_i\phi)^a(D_i\phi)^a = \tfrac12 v^2 H'^2 + \frac{v^2H^2K^2}{r^2}. \tag{41.135}$$

The magnetic field. The non-abelian magnetic field of the ansatz is computed from $F^a_{ij}=\partial_iA^a_j-\partial_jA^a_i+e\epsilon_{abc}A^b_iA^c_j$. The abelian curl gives

$$\partial_iA^a_j-\partial_jA^a_i = -\frac{2\alpha}{r}\,\epsilon_{aij} + \Big(\alpha'-\frac{\alpha}{r}\Big)(\epsilon_{ajk}\hat r_i-\epsilon_{aik}\hat r_j)\hat r_k, \tag{41.136}$$

while the self-coupling term, using $\epsilon_{abc}\epsilon_{bim}\epsilon_{cjn}\hat r_m\hat r_n=\epsilon_{ijn}\hat r_a\hat r_n$, gives $e\epsilon_{abc}A^b_iA^c_j = e\alpha^2\,\epsilon_{ijn}\hat r_a\hat r_n$. Contracting with $-\frac12\epsilon_{lij}$ and using $\epsilon_{lij}\epsilon_{aij}=2\delta_{la}$, $\epsilon_{lij}\epsilon_{ajk}\hat r_i\hat r_k = \hat r_a\hat r_l-\delta_{la}$, and $\epsilon_{lij}\epsilon_{ijn}=2\delta_{ln}$, the three pieces assemble into

$$B^a_l = \Big(\frac{2\alpha}{r}-e\alpha^2\Big)\hat r_a\hat r_l + \Big(\alpha'+\frac{\alpha}{r}\Big)(\delta_{la}-\hat r_a\hat r_l). \tag{41.137}$$

Inserting $\alpha=(1-K)/(er)$, the radial coefficient becomes $\big[2(1-K)-(1-K)^2\big]/(er^2)=(1-K^2)/(er^2)$, and the transverse coefficient becomes $\alpha'+\alpha/r = -K'/(er)$:

$$\boxed{B^a_l = \frac{1-K^2}{e\,r^2}\,\hat r_a\hat r_l \;-\; \frac{K'}{e\,r}\,(\delta_{la}-\hat r_a\hat r_l).} \tag{41.138}$$

Again by orthogonality of the two projectors,

$$\tfrac12 B^a_lB^a_l = \frac{(1-K^2)^2}{2\,e^2r^4} + \frac{K'^2}{e^2r^2}. \tag{41.139}$$

The mass functional. Adding (41.135), (41.139), and the potential $V=\frac{\lambda}{4}v^4(H^2-1)^2$, and integrating over angles ($\int d^3x = \int_0^\infty 4\pi r^2\,dr$), the energy reduces to a one-dimensional integral,

$$E = 4\pi\int_0^\infty dr\,\Big[\,\frac{K'^2}{e^2}+\frac{(1-K^2)^2}{2e^2r^2}+\tfrac12 v^2r^2H'^2+v^2H^2K^2+\frac{\lambda v^4r^2}{4}(H^2-1)^2\,\Big]. \tag{41.140}$$

Rescaling to the dimensionless radius $\xi=evr$ (the natural length being the core size $1/M_W=1/ev$) and writing H,K as functions of ξ with primes now denoting $d/d\xi$, every term scales out a common factor and

$$\boxed{M \;=\; \frac{4\pi v}{e}\,f\Big(\frac{\lambda}{e^2}\Big), \qquad f=\min_{H,K}\int_0^\infty d\xi\,\Big[\,K'^2+\frac{(1-K^2)^2}{2\xi^2}+\tfrac12\xi^2H'^2+H^2K^2+\frac{\lambda}{4e^2}\,\xi^2(H^2}\tag{41.141}$$

The dimensionless number f depends only on the ratio $\lambda/e^2=M_H^2/2M_W^2$, and the prefactor is $4\pi v/e = 4\pi M_W/e^2 = M_W/\alpha_e$ with $\alpha_e=e^2/4\pi$. The monopole

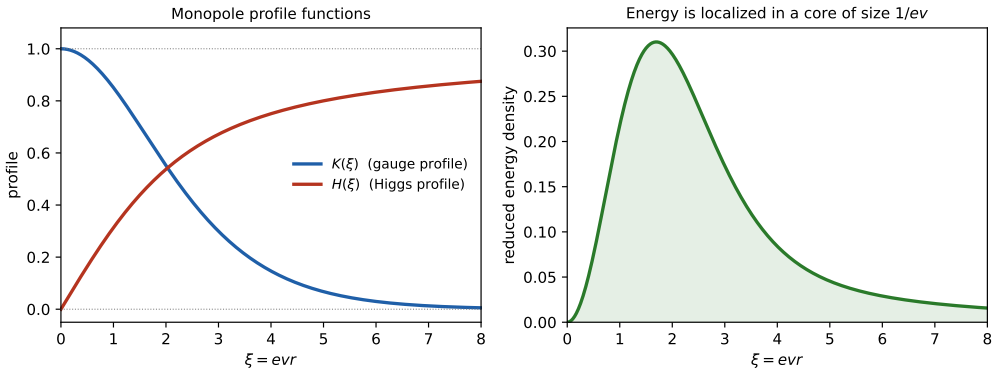


Figure 41.5: The 't Hooft–Polyakov monopole as a smooth, localized lump. Left: the radial profiles in the closed-form $\lambda/e^2 \to 0$ (Prasad–Sommerfield) limit, $K = \xi/\sinh\xi$ and $H = \coth\xi - 1/\xi$ in units $\xi = evr$, satisfying the boundary values (41.130): the gauge profile K collapses from 1 to 0 and the Higgs profile H rises from 0 to the vacuum 1. Right: the reduced energy density, showing that the field energy is concentrated in a core of size $1/ev = 1/M_W$, outside which only the long-range abelian magnetic field survives.

is heavy: its mass is the gauge-boson mass divided by the fine-structure constant of the broken theory, so for any weakly coupled embedding it lies far above the perturbative spectrum — which is exactly why no order of perturbation theory around the vacuum can see it.

The profiles that achieve the minimum solve the Euler–Lagrange equations of (41.141),

$$\xi^2 K'' = K\,(K^2 - 1) + \xi^2 H^2 K, \tag{41.142}$$

$$(\xi^2 H')' = 2HK^2 + \frac{\lambda}{e^2}\,\xi^2 H\,(H^2 - 1), \tag{41.143}$$

a coupled two-point boundary-value problem with the data (41.130). It has no elementary closed-form solution for general λ/e^2, but the existence of a smooth, monotone solution interpolating the boundary values is standard, and its qualitative shape is dictated entirely by (41.130): K falls from 1 to 0 as the gauge field turns from its short-distance pure-gauge form into the long-range abelian tail, and H rises from 0 to 1 as the Higgs climbs out of the symmetric core onto the vacuum manifold (Fig. 41.5). In the limit $\lambda/e^2 \to 0$ the last term drops and the system (41.142)–(41.143) admits the exact solution $K = \xi/\sinh\xi$, $H = \coth\xi - 1/\xi$, with $f \to 1$; this is the Prasad–Sommerfield profile plotted in Fig. 41.5, derived from first-order equations in Section 41.1.5.

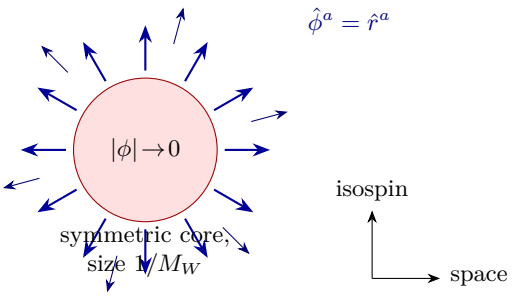


Figure 41.6: The hedgehog. At each point of physical space the internal isospin direction of the Higgs field, $\hat\phi^a$, is locked to the spatial direction $\hat r^a$. The asymptotic field therefore wraps the internal two-sphere exactly once as $\hat r$ ranges over S^2_∞: the map $\hat\phi : S^2_\infty \to S^2$ has degree one. The locking forces $|\phi| \to 0$ at the origin, where the full SU(2) is restored and the energy is stored.

41.5.5 The gauge-invariant electromagnetic field

Far from the core the Higgs field selects, at each point, an unbroken U(1) — the rotations about the local isospin axis $\hat\phi^a$. To speak of "the" electromagnetic field we need a definition of the U(1) field strength that is gauge invariant and reduces, in the gauge where $\hat\phi^a = \delta^{a3}$, to the photon $F^3_{\mu\nu}$. 't Hooft's gauge-invariant field strength accomplishes exactly this:

Definition 41.23 (The 't Hooft electromagnetic tensor)**.** For a configuration with $|\phi| \neq 0$, set $\hat\phi^a = \phi^a/|\phi|$ and define

$$\mathcal{F}_{\mu\nu} \;\equiv\; \hat\phi^a F^a_{\mu\nu} \;-\; \frac{1}{e}\,\epsilon_{abc}\,\hat\phi^a (D_\mu\hat\phi)^b (D_\nu\hat\phi)^c. \tag{41.144}$$

It is a gauge-invariant antisymmetric tensor: each factor transforms in the adjoint and the contraction with $\hat\phi^a$ and ϵ_{abc} is an SO(3) singlet.

The reason (41.144) is the right object is an algebraic identity that strips it down to an abelian curl plus a purely topological term.

Theorem 41.24 (The 't Hooft identity)**.** *Let $a_\mu \equiv \hat\phi^a A^a_\mu$. Then for any smooth configuration with $|\phi| \neq 0$,*

$$\mathcal{F}_{\mu\nu} \;=\; \partial_\mu a_\nu - \partial_\nu a_\mu \;-\; \frac{1}{e}\,\epsilon_{abc}\,\hat\phi^a\,\partial_\mu\hat\phi^b\,\partial_\nu\hat\phi^c. \tag{41.145}$$

Proof. We use only $\hat\phi^a\hat\phi^a = 1$, hence $\hat\phi^a\partial_\mu\hat\phi^a = 0$, and the SU(2) adjoint covariant derivative $D_\mu\hat\phi^a = \partial_\mu\hat\phi^a + e\epsilon_{abc}A^b_\mu\hat\phi^c$.

First expand the abelian curl of a_μ with the Leibniz rule,

$$\partial_\mu a_\nu - \partial_\nu a_\mu = \hat\phi^a(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu) + (\partial_\mu\hat\phi^a)A^a_\nu - (\partial_\nu\hat\phi^a)A^a_\mu$$

$$= \hat\phi^a F^a_{\mu\nu} - e\,\epsilon_{abc}\hat\phi^a A^b_\mu A^c_\nu + (\partial_\mu\hat\phi^a)A^a_\nu - (\partial_\nu\hat\phi^a)A^a_\mu, \tag{41.146}$$

where in the second line we replaced $\hat\phi^a(\partial_\mu A^a_\nu - \partial_\nu A^a_\mu)$ by $\hat\phi^a F^a_{\mu\nu} - e\epsilon_{abc}\hat\phi^a A^b_\mu A^c_\nu$ using the definition of $F^a_{\mu\nu}$ in (41.120).

Next expand the topological-looking piece of $\mathcal{F}$. Writing $P_{\mu\nu} \equiv \epsilon_{abc}\hat\phi^a D_\mu\hat\phi^b D_\nu\hat\phi^c$ and substituting $D_\mu\hat\phi^b = \partial_\mu\hat\phi^b + e\epsilon_{bde}A^d_\mu\hat\phi^e$ produces four terms. Using $\epsilon_{abc}\epsilon_{cfg} = \delta_{af}\delta_{bg} - \delta_{ag}\delta_{bf}$ and $\epsilon_{abc}\epsilon_{bde} = -(\delta_{ad}\delta_{ce} - \delta_{ae}\delta_{cd})$, together with $\hat\phi^a\hat\phi^a = 1$ and $\hat\phi^a\partial_\mu\hat\phi^a = 0$, each contraction collapses:

$$P_{\mu\nu} = \underbrace{\epsilon_{abc}\hat\phi^a\partial_\mu\hat\phi^b\partial_\nu\hat\phi^c}_{\equiv\,\Omega_{\mu\nu}} - e(\partial_\mu\hat\phi^a)A^a_\nu \;+\; e(\partial_\nu\hat\phi^a)A^a_\mu \;+\; e^2\epsilon_{abc}\hat\phi^a A^b_\mu A^c_\nu. \tag{41.147}$$

(The cross terms in which a derivative $\partial\hat\phi$ is contracted into its own $\hat\phi$ vanish by $\hat\phi^a\partial\hat\phi^a = 0$; the terms with ϵ contracted on two $\hat\phi$'s vanish by antisymmetry.) Therefore

$$-\frac{1}{e}P_{\mu\nu} = -\frac{1}{e}\Omega_{\mu\nu} + (\partial_\mu\hat\phi^a)A^a_\nu - (\partial_\nu\hat\phi^a)A^a_\mu - e\,\epsilon_{abc}\hat\phi^a A^b_\mu A^c_\nu. \tag{41.148}$$

Now form $\mathcal{F}_{\mu\nu} = \hat\phi^a F^a_{\mu\nu} - \frac{1}{e}P_{\mu\nu}$ from (41.144) and (41.148), and compare with (41.146): every A-dependent term cancels in pairs, leaving exactly $\mathcal{F}_{\mu\nu} = (\partial_\mu a_\nu - \partial_\nu a_\mu) - \frac{1}{e}\Omega_{\mu\nu}$, which is (41.145). □

Theorem 41.24 is the structural heart of the section. The first two terms of (41.145) are the curl of an ordinary, single-valued vector potential $a_\mu = \hat\phi^a A^a_\mu$. Crucially, a_μ is *smooth on the whole sphere at infinity*, because $\hat\phi$ and A are smooth everywhere — there is no Dirac string. The last term is built purely from the Higgs orientation $\hat\phi$; it is the density of the asymptotic map, and it is what carries the monopole's flux.

The magnetic field of the hedgehog. Specializing (41.144) to the hedgehog is immediate from the pieces already computed. The covariant derivative of the *unit* Higgs is the $H \to 1$ form of (41.134),

$$(D_i\hat\phi)^a = \frac{K}{r}\,(\delta_{ia} - \hat r_i\hat r_a), \tag{41.149}$$

which $\to 0$ at infinity since $K \to 0$ — confirming the boundary condition (41.126). The electromagnetic magnetic field $\mathcal{B}_l = -\frac{1}{2}\epsilon_{lij}\mathcal{F}_{ij}$ has two contributions. The first, $\hat\phi^a B^a_l$, picks out the radial part of (41.138): $\hat r^a B^a_l = (1-K^2)\hat r_l/(er^2)$. The second uses $\epsilon_{abc}\hat\phi^a(D_i\hat\phi)^b(D_j\hat\phi)^c = (K^2/r^2)\,\epsilon_{lij}\hat r_l$ from (41.149), contributing $K^2\hat r_l/(er^2)$. The K-dependence cancels between them:

$$\boxed{\mathcal{B}_l \;=\; \frac{1-K^2}{e\,r^2}\,\hat r_l + \frac{K^2}{e\,r^2}\,\hat r_l \;=\; \frac{1}{e\,r^2}\,\hat r_l.} \tag{41.150}$$

This is remarkable and is the punchline of the construction: the gauge-invariant magnetic field is *exactly Coulombic for all* r, not merely asymptotically. It is the field of a point magnetic charge, and it is everywhere regular — the apparent singularity at $r = 0$ in (41.150) is the usual Coulomb singularity of the field of a source, smoothed in the full non-abelian fields by the core profiles, but exactly pointlike for the *abelian projection*. Comparing with the Dirac monopole field $\vec{B} = (g/4\pi)\hat{r}/r^2$ of Eq. (8.187), we read off the magnetic charge

$$g \;=\; \frac{4\pi}{e}, \qquad \oint_{S^2_\infty} \mathcal{B}_i \, dS_i = \frac{1}{er^2} \cdot 4\pi r^2 = \frac{4\pi}{e}. \tag{41.151}$$

Caution

The total magnetic charge is genuine and nonzero, yet the abelian Bianchi identity $\nabla \cdot \vec{B} = 0$ that doomed the Dirac potential is *not* violated by any fundamental field: the non-abelian fields A^a_μ, ϕ^a are smooth everywhere and obey their own (covariant) Bianchi identity. What carries the magnetic flux is not a string and not a δ-function source, but the winding of $\hat{\phi}$ — the last term of (41.145), which is precisely the obstruction to writing $\mathcal{F}_{\mu\nu}$ globally as the curl of a single smooth a_μ. The monopole charge lives in the topology of the Higgs map, not in a singular potential.

41.5.6 The topological current and the charge identity

We now make precise the claim that the magnetic charge equals the topological degree, for *any* finite-energy configuration in the model, not just the hedgehog. The carrier of the topological charge is the current

$$k^\mu \;=\; \frac{1}{8\pi} \, \epsilon^{\mu\nu\rho\sigma} \, \epsilon_{abc} \, \partial_\nu \hat{\phi}^a \, \partial_\rho \hat{\phi}^b \, \partial_\sigma \hat{\phi}^c, \tag{41.152}$$

with $\epsilon^{0123} = +1$. This is built entirely from the Higgs orientation $\hat{\phi}$; it involves no gauge field and no coupling, and it is conserved without any use of the equations of motion.

Proposition 41.25 (Off-shell conservation)**.** $\partial_\mu k^\mu = 0$ *identically, for every smooth* $\hat{\phi}$*, on or off shell.*

Proof. Differentiating (41.152) and applying the Leibniz rule, every term carries a second derivative $\partial_\mu \partial_\nu \hat{\phi}^a$ (or $\partial_\mu \partial_\rho \hat{\phi}^b$, $\partial_\mu \partial_\sigma \hat{\phi}^c$) contracted with $\epsilon^{\mu\nu\rho\sigma}$. A symmetric pair of indices contracted with the totally antisymmetric ϵ vanishes, so each of the three terms vanishes separately and $\partial_\mu k^\mu \equiv 0$. No field equation was used. $\square$

This is the essential contrast with a Noether current promised in the chapter introduction. The electromagnetic current of Chapters 7 and 8 is conserved *because* of a symmetry and *by* the equations of motion; delete the equations of motion and it is no longer conserved. The current (41.152) is conserved as an algebraic identity, independent of the dynamics. Its conservation expresses not a symmetry of the action but the topology of field space, and the conserved charge it carries is an integer that the continuous dynamics cannot change.

Theorem 41.26 (Topological charge is the degree)**.** *The conserved charge*

$$Q \;=\; \int d^3x\; k^0 \;=\; \frac{1}{8\pi}\int d^3x\;\epsilon_{ijk}\,\epsilon_{abc}\,\partial_i\hat{\phi}^a\,\partial_j\hat{\phi}^b\,\partial_k\hat{\phi}^c \tag{41.153}$$

equals the degree n of the boundary map $\hat{\phi}: S^2_\infty \to S^2$, hence $Q \in \mathbb{Z}$.

Proof. The integrand is a total spatial divergence. Because $\epsilon_{ijk}\epsilon_{abc}\partial_i\hat{\phi}^a\partial_j\hat{\phi}^b\partial_k\hat{\phi}^c = \partial_i\big(\epsilon_{ijk}\epsilon_{abc}\hat{\phi}^a\partial_j\hat{\phi}^b\partial_k\hat{\phi}^c\big)$ — the extra terms generated by the derivative hitting a $\partial\hat{\phi}$ vanish by the same symmetric-times-antisymmetric argument as in Theorem 41.25 — the divergence theorem turns (41.153) into a surface integral over the sphere at infinity,

$$Q \;=\; \frac{1}{8\pi}\oint_{S^2_\infty} dS_i\;\epsilon_{ijk}\,\epsilon_{abc}\,\hat{\phi}^a\,\partial_j\hat{\phi}^b\,\partial_k\hat{\phi}^c. \tag{41.154}$$

This is exactly the degree integral of Eq. (41.46) for a map $S^2 \to S^2$: the integrand $\frac{1}{2}\epsilon_{abc}\hat{\phi}^a\,d\hat{\phi}^b\wedge d\hat{\phi}^c$ is the pullback of the area form on the target two-sphere, and dividing by the total area 4π and integrating counts the (signed) number of times $\hat{\phi}$ covers the target. Hence $Q = \deg(\hat{\phi}) = n \in \mathbb{Z}$. □

Theorem 41.27 (Magnetic charge = topological charge)**.** *For every finite-energy configuration, the magnetic charge of the 't Hooft field equals*

$$\boxed{\;g \;=\; \oint_{S^2_\infty} \mathcal{B}_i\,dS_i \;=\; \frac{4\pi}{e}\,Q \;=\; \frac{4\pi n}{e}.\;} \tag{41.155}$$

Proof. Take the spatial part of the 't Hooft identity (41.145) and form $\mathcal{B}_i = -\frac{1}{2}\epsilon_{ijk}\mathcal{F}_{jk}$. The curl term contributes $-\epsilon_{ijk}\partial_j a_k = -(\nabla\times\vec{a})_i$, and because a_μ is a smooth, single-valued vector field on the closed surface S^2_∞, its flux vanishes by the divergence theorem, $\oint_{S^2_\infty}(\nabla\times\vec{a})\cdot d\vec{S} = 0$. (This is the step the Dirac construction of Theorem 8.32 could not take: there the would-be global potential was singular on a string, and the curl carried nonzero net flux. Here there is no string.) Only the topological term survives,

$$g = \oint_{S^2_\infty}\mathcal{B}_i\,dS_i = \frac{1}{2e}\oint_{S^2_\infty} dS_i\;\epsilon_{ijk}\,\epsilon_{abc}\,\hat{\phi}^a\,\partial_j\hat{\phi}^b\,\partial_k\hat{\phi}^c = \frac{1}{2e}\,(8\pi Q) = \frac{4\pi}{e}\,Q, \tag{41.156}$$

where the middle equality used (41.154). With $Q = n$ from Theorem 41.26, this is (41.155). □

For the hedgehog, $n = 1$, recovering $g = 4\pi/e$ found directly in (41.151). The two computations — the explicit field (41.150) and the general winding identity — agree, as they must, but they make different points: the first shows the field *is* Coulombic; the second shows the total charge is an *integer* dictated by topology, robust against any smooth deformation of the configuration.

Worked Example 41.28 (The degree of the hedgehog map)**.** We verify $Q = 1$ for $\hat{\phi}^a = \hat{r}^a$ directly from (41.154), exhibiting how the abstract degree formula produces a concrete integer. With $\partial_j \hat{r}^b = (\delta_{jb} - \hat{r}_j \hat{r}_b)/r$, the antisymmetric structure kills every term containing $\hat{r}_j \hat{r}_b$ or $\hat{r}_k \hat{r}_c$ (each is symmetric in a pair contracted with an ϵ), leaving

$$\epsilon_{abc}\, \hat{r}^a\, \partial_j \hat{r}^b\, \partial_k \hat{r}^c = \frac{1}{r^2}\, \epsilon_{abc}\, \hat{r}^a\, \delta_{jb} \delta_{kc} = \frac{1}{r^2}\, \epsilon_{ljk}\, \hat{r}_l. \tag{41.157}$$

Contracting with ϵ_{ijk} gives $\epsilon_{ijk}\epsilon_{ljk}/r^2\, \hat{r}_l = 2\hat{r}_i/r^2$, and with $dS_i = \hat{r}_i\, r^2 d\Omega$ the surface integral is

$$Q = \frac{1}{8\pi} \oint dS_i\, \frac{2\hat{r}_i}{r^2} = \frac{1}{8\pi} \oint 2\, d\Omega = \frac{1}{8\pi} \cdot 2 \cdot 4\pi = 1. \tag{41.158}$$

The factor $2\, d\Omega$ is the pullback of the target area form under the identity map: the hedgehog covers the vacuum sphere once, evenly, with the solid-angle element of physical space mapping to the solid-angle element of internal space. Substituting a degree-n boundary map for $\hat{r}^a$ would replace the integrand by n copies of the area form and give $Q = n$, $g = 4\pi n/e$.

41.5.7 Dirac quantization from a regular field

Combining the charge identity with the integrality of the degree gives the central result of the section, the non-singular origin of Dirac's quantization.

Theorem 41.29 (Dirac quantization, topological form)**.** *In the Georgi–Glashow model the magnetic charge of any finite-energy configuration is an integer multiple of* $4\pi/e$*,*

$$\boxed{e\, g \;=\; 4\pi n, \qquad n = \deg(\hat{\phi}) \in \mathbb{Z}.} \tag{41.159}$$

Proof. Immediate from Theorem 41.27: $g = 4\pi n/e$ with $n \in \mathbb{Z}$ by Theorem 41.26. □

It is worth dwelling on how completely this differs in spirit from the Dirac argument of Theorem 8.32, even though the arithmetic agrees. There, one *postulated* a point magnetic charge, found that no smooth global vector potential could describe it, and imported the missing flux along an unobservable Dirac string; the quantization condition $qg = 2\pi m$ then arose as the demand that the

string stay invisible to any charged particle. The monopole was singular, its charge was a free parameter, and quantization was a consistency requirement on a configuration the theory could not actually contain. Here the field is everywhere smooth, the magnetic charge is not a parameter but a computed topological invariant, and quantization is a *theorem* about integers — no string, no postulate, no consistency condition.

The relation between the two unit charges deserves a word. The 't Hooft–Polyakov unit is $g_0 = 4\pi/e$, twice the minimal Dirac unit $g_{\text{Dirac}} = 2\pi/e_{\min}$ that one would infer from the smallest electric charge in the theory. The reason is the matter content: the Georgi–Glashow model contains only adjoint (integer-isospin) fields, whose smallest electric charge under the unbroken U(1) is e (carried by $W^{\pm}$, with isospin $T^3 = \pm 1$ so charge $\pm e$). With $e_{\min} = e$, the monopole satisfies $e\, g_0 = 4\pi = 2\pi \cdot 2$, an *even* Dirac unit. Were one to add an isospinor (fundamental) field, its electric charge would be $e/2$, the minimal Dirac unit would be $g_{\text{Dirac}} = 4\pi/e = g_0$, and the 't Hooft–Polyakov monopole would carry exactly one Dirac unit relative to it. Either way, the consistency that Dirac had to impose by hand on a singular field is here guaranteed automatically by the regularity and topology of a smooth one.

Physical Insight

The Dirac string was always a confession that the abelian description was incomplete near the monopole. The non-abelian theory completes it: in a small ball around the core the gauge group is fully restored to SU(2), the photon is just one component of a non-abelian connection that is perfectly smooth there, and what looked like a singular string in the abelian projection is resolved into a finite, structureless lump of Yang–Mills and Higgs energy. The monopole is the statement that magnetic charge is real, finite-mass, and particle-like once the electromagnetic U(1) is understood as the surviving piece of a larger, spontaneously broken gauge symmetry.

41.5.8 The monopole mass and the Bogomolny bound

The mass formula (41.141), $M = (4\pi v/e)\, f(\lambda/e^2)$, raises a sharp question: how small can f be? The energy (41.140) is a sum of non-negative terms, and a lower bound on it that depends only on the topological charge would prove the monopole stable for the deepest possible reason — not as a local minimum but as the floor of an entire sector.

Such a bound exists, and it is the same Bogomolny rearrangement that saturated the kink and the vortex. Drop the potential ($\lambda/e^2 \to 0$, the Prasad–Sommerfield limit, where the Higgs is massless but the vacuum manifold survives as the boundary condition $|\phi| \to v$) and complete the square in the static energy

(41.125):

$$E = \int d^3x \left[\tfrac{1}{2}\left(B_i^a \mp (D_i\phi)^a\right)^2 \pm B_i^a (D_i\phi)^a\right] \geq \left|\int d^3x\, B_i^a (D_i\phi)^a\right|. \quad (41.160)$$

The cross term is itself topological: using the Bianchi identity $(D_i B_i)^a = 0$ for the non-abelian magnetic field, $B_i^a (D_i\phi)^a = \partial_i (B_i^a \phi^a)$ is a total divergence, so the bound is a surface integral of the magnetic field against the Higgs — which is exactly v times the magnetic flux (41.151). The result is the Bogomolny bound

$$M \geq v\,|g| = \frac{4\pi v}{e}\,|n|, \qquad \text{i.e.} \qquad f\left(\frac{\lambda}{e^2}\right) \geq |n|, \quad (41.161)$$

with equality when the first-order Bogomolny equation $B_i^a = \pm (D_i\phi)^a$ holds. The bound is saturated in the $\lambda/e^2 \to 0$ limit, where $f = 1$ and the profiles are the closed-form Prasad–Sommerfield solution already plotted in Fig. 41.5; for $\lambda/e^2 > 0$ the potential adds positive energy and $f > 1$, but the topological floor $M \geq (4\pi v/e)|n|$ persists. The monopole cannot decay: its mass is bounded below by its conserved magnetic — equivalently topological — charge.

This first-order structure, glimpsed now for the third time, is no coincidence, and the exact solution that saturates it deserves to be derived rather than quoted. That, together with the dyons obtained by switching on the electric field F_{0i}^a we set to zero, and the Witten effect that ties their electric charge to the vacuum angle, is the subject of the next and final section, where the Bogomolny bound is lifted from a recurring trick to a single organizing principle and the BPS monopole is written in closed form.

41.6 The Bogomolny Bound, BPS States, and Dyons

The monopole left us with a number and a hint. The number was a lower bound on its mass: the 't Hooft–Polyakov solution of Section 41.5 cannot weigh less than a fixed multiple of the symmetry-breaking scale times the inverse coupling, because its energy contains the squared magnetic field of a unit-charge Coulomb tail that no smooth deformation can switch off. The hint was that this bound looked saturable. At each rung of the ladder—the kink in Section 41.1, the vortex in Section 41.4, the monopole in Section 41.5—the second-order Euler–Lagrange equations admitted special solutions obeying a *first*-order system, and on those solutions the energy collapsed to a boundary integral fixed entirely by the topological sector. That recurrence is not a coincidence of three models. It is one algebraic identity, applied three times, and this section states it once and for all.

The identity is Bogomol'nyi's completion of the square [83]: a static energy functional that is a sum of squared gradients plus a non-negative potential can be rewritten as a sum of *different* squares plus a total derivative, and the total

derivative integrates to the topological charge. The squares are non-negative, so the energy is bounded below by the modulus of the topological charge; equality holds precisely when the squares vanish, which is the first-order ("BPS") system. We prove this in full, exhibit the kink, vortex, and monopole as instances, and then push the monopole case to its sharpest form: the Prasad–Sommerfield limit [410], in which the potential is switched off and the first-order equations are solved in closed form, giving an explicit profile and a mass that saturates the bound exactly. We close with two pieces of physics that the first-order structure makes inevitable: dyons, which carry electric as well as magnetic charge, and the Witten effect [517], by which a θ-term shifts the dyon's electric charge by an amount set by the very total-derivative term we met classically in Eq. (8.74).

41.6.1 The general Bogomolny completion

Every soliton in this chapter is a stationary point of a static energy functional of the schematic form

$$E[\psi] \;=\; \int \mathrm{d}^{D}x \,\Big[\, \tfrac{1}{2}\sum_{\alpha} \big(\mathcal{D}_{\alpha}[\psi]\big)^{2} \;+\; V(\psi) \,\Big], \qquad V \geq 0, \tag{41.162}$$

where ψ stands collectively for the fields (a scalar, a complex Higgs and a gauge potential, an adjoint triplet and a non-abelian connection), the index α runs over the field-and-spatial-direction labels carried by the gradient terms, and each $\mathcal{D}_{\alpha}[\psi]$ is a real first-order expression—an ordinary derivative $\partial_i\phi$, a covariant derivative $D_i\phi$, or a magnetic field component B_i^{a}, which is itself first order in the connection up to the algebraic self-interaction. The potential V is non-negative and vanishes on the vacuum manifold. Both terms are manifestly non-negative, so $E \geq 0$ trivially; the content of Bogomol'nyi's observation is that a far stronger, *topological* bound is hidden inside this same expression.

Theorem 41.30 (General Bogomolny bound and saturation)**.** *This is the multi-field, arbitrary-dimension generalization of the one-dimensional scalar bound proved for the kink in Theorem 41.5. Suppose the integrand of Eq. (41.162) admits a decomposition*

$$\tfrac{1}{2}\sum_{\alpha}\mathcal{D}_{\alpha}^{2} + V \;=\; \tfrac{1}{2}\sum_{\alpha}\big(\mathcal{D}_{\alpha} \mp \mathcal{S}_{\alpha}\big)^{2} \;\pm\; \mathcal{T}, \tag{41.163}$$

valid as an algebraic identity in the fields, where the $\mathcal{S}_{\alpha}$ are real local expressions and the cross term assembles into a total spatial divergence,

$$\mathcal{T} \;=\; \sum_{\alpha}\mathcal{D}_{\alpha}\,\mathcal{S}_{\alpha} \;=\; \partial_i \mathcal{J}^{i}, \tag{41.164}$$

whose integral

$$Q \;\equiv\; \int \mathrm{d}^{D}x\, \mathcal{T} \;=\; \oint_{S^{D-1}_{\infty}} \mathcal{J}^{i}\, \mathrm{d}S_i \tag{41.165}$$

depends only on the boundary behaviour of the fields—that is, only on the topological sector. Then every finite-energy configuration in that sector satisfies the Bogomolny bound

$$E[\psi] \;\geq\; |Q|\,, \tag{41.166}$$

with equality if and only if the fields obey the first-order Bogomolny (BPS) equations

$$\mathcal{D}_\alpha \;=\; \pm\,\mathcal{S}_\alpha \qquad \textit{for every } \alpha, \tag{41.167}$$

the upper or lower sign chosen so that $\pm Q = |Q|$.

Proof. Both signs in Eq. (41.163) are available because the cross term in expanding $\frac{1}{2}(\mathcal{D}_\alpha \mp \mathcal{S}_\alpha)^2 = \frac{1}{2}\mathcal{D}_\alpha^2 \mp \mathcal{D}_\alpha\mathcal{S}_\alpha + \frac{1}{2}\mathcal{S}_\alpha^2$ carries the explicit $\mp$, and the hypothesis is precisely that $\frac{1}{2}\sum_\alpha \mathcal{S}_\alpha^2 = V$, so that the leftover after the squares reproduces the potential. Integrate the identity over space. The first term on the right of Eq. (41.163) is an integral of a sum of perfect squares, hence non-negative:

$$E[\psi] \;=\; \int \mathrm{d}^D x\; \tfrac{1}{2}\sum_\alpha \left(\mathcal{D}_\alpha \mp \mathcal{S}_\alpha\right)^2 \pm \int \mathrm{d}^D x\; \partial_i \mathcal{J}^i \;=\; \underbrace{\int \mathrm{d}^D x\; \tfrac{1}{2}\sum_\alpha \left(\mathcal{D}_\alpha \mp \mathcal{S}_\alpha\right)^2}_{\geq 0} \pm Q, \tag{41.168}$$

where the divergence integrated to the flux Q of Eq. (41.165) by Gauss's theorem (the surface is the sphere at spatial infinity; finite energy guarantees the fields approach the vacuum manifold there, so the surface integral converges). Now choose the global sign in Eq. (41.163) to be the sign of Q, so that $\pm Q = +|Q|$. Then $E = (\text{non-negative}) + |Q| \geq |Q|$, which is Eq. (41.166). Equality requires the non-negative integral to vanish; since the integrand is a sum of squares of continuous functions, it vanishes identically only if every $\mathcal{D}_\alpha \mp \mathcal{S}_\alpha = 0$ pointwise, i.e. Eq. (41.167). Conversely, any solution of Eq. (41.167) makes that integral zero and hence has $E = |Q|$. Finally, a BPS solution automatically solves the full second-order Euler–Lagrange equations: it is an absolute minimizer of E within its topological sector (no configuration can have $E < |Q|$, and it attains $|Q|$), so it is in particular a stationary point of E. This last fact is what makes the first-order system consistent—it is a genuine reduction of the dynamics, not an extra constraint imposed by hand. □

Three features of this argument deserve emphasis because they recur at every rung. First, the bound is *topological*: Q is a boundary integral, invariant under any deformation of the interior, so it labels homotopy classes and cannot change continuously. Second, the bound is *saturated by a first-order system*, which is generically far easier to solve than the second-order field equations and, crucially, is an algebraic rewriting that holds off-shell. Third, the quantity Q is what a supersymmetric extension of the theory will recognize as a *central charge* in the algebra of supercharges (Chapter 46); the inequality $E \geq |Q|$ becomes there an exact operator statement, and solutions that saturate it—"BPS states"—are

protected from quantum corrections. We are seeing the classical shadow of that structure.

The structure is exactly the one already met for the kink in Section 41.1, now stripped to its bones: a sum of squared gradients, completed against a total derivative, bounded by a boundary integral. We did not need the full machinery of Theorem 41.30 there because there was a single field in one dimension; the value of the abstract statement is that the *same three lines* of algebra cover the vortex and the monopole, where the bookkeeping of indices would otherwise obscure that nothing new is happening.

41.6.2 The three instances and their central charges

With the master identity Theorem 41.30 in hand, the kink, vortex, and monopole are read off as the three instances assembled at the earlier rungs. We line them up.

Kink ($D = 1$, scalar). The single gradient is $\mathcal{D} = \phi'$ and the dual expression is $\mathcal{S} = W'(\phi)$, where the superpotential W is defined by $W'(\phi) = \sqrt{2V(\phi)}$ as in Eq. (41.24); the hypothesis $\frac{1}{2}\mathcal{S}^2 = V$ of Theorem 41.30 then holds by construction, the central charge is $Q = \Delta W = W(\phi_+) - W(\phi_-)$, and the BPS equation is the first-order $\phi' = \pm W'(\phi)$ of Eq. (41.28). This is precisely Theorem 41.5, and the explicit double-well solution $\phi_K = v\tanh((x - x_0)/w)$ together with the mass $M_{\text{kink}} = |\Delta W|$ were computed in full in Theorem 41.6; we do not repeat them. The point to carry forward is structural: the kink is the $D = 1$, single-field instance of the general completion, with the superpotential difference playing the role of the central charge.

Vortex ($D = 2$, gauged $\mathrm{U}(1)$). For the abelian Higgs model assembled in Section 41.4, with a complex scalar of covariant derivative $D_i\phi = (\partial_i - ieA_i)\phi$ and magnetic field $B = \partial_1 A_2 - \partial_2 A_1$, the gradients are the two covariant derivatives and the magnetic field, and the completion combines $|D_i\phi|^2$ with $\frac{1}{2}B^2$ and the quartic potential. It closes into a total derivative *only at the critical coupling* $\lambda = e^2$, where the Higgs and gauge masses coincide; there the BPS equations are the first-order "self-dual" vortex equations

$$(D_1 \pm iD_2)\phi = 0, \qquad B = \mp e\big(|\phi|^2 - v^2\big), \tag{41.169}$$

and the central charge is the quantized magnetic flux,

$$Q \;\propto\; \Phi = \int \mathrm{d}^2x\, B = \frac{2\pi n}{e}, \qquad n \in \mathbb{Z} = \pi_1(\mathrm{U}(1)), \tag{41.170}$$

so that the energy per unit length saturates at a value proportional to $v^2\,|n|$, with the precise normalization fixed in Section 41.4. The new feature relative

to the kink is that saturation needs a tuned coupling: away from $\lambda = e^2$ the squares no longer assemble against the flux, and the static balance between scalar attraction and gauge repulsion of two vortices is broken (vortices attract for $\lambda < e^2$ and repel for $\lambda > e^2$; they are free to coexist at marginal cost exactly on the BPS line). This is the first appearance of a *moduli space* of degenerate multi-soliton solutions, a structure that the supersymmetric continuation of Chapter 46 promotes to a central object.

Monopole ($D = 3$, non-abelian $\mathrm{SU}(2) \to \mathrm{U}(1)$). Here lies the result this section owes in full. The Georgi–Glashow energy of Section 41.5, with an adjoint Higgs ϕ^a, covariant derivative $D_i\phi^a = \partial_i\phi^a + e\,\epsilon^{abc}A_i^b\phi^c$, and non-abelian magnetic field $B_i^a = \frac{1}{2}\epsilon_{ijk}F_{jk}^a$, is

$$E = \int \mathrm{d}^3x \left[\tfrac{1}{2}B_i^aB_i^a + \tfrac{1}{2}D_i\phi^aD_i\phi^a + V(\phi)\right], \qquad V = \tfrac{\lambda}{4}\left(\phi^a\phi^a - v^2\right)^2. \quad (41.171)$$

The gradients are $\mathcal{D} \in \{B_i^a,\, D_i\phi^a\}$. Completing the square between the two,

$$\tfrac{1}{2}B_i^aB_i^a + \tfrac{1}{2}D_i\phi^aD_i\phi^a = \tfrac{1}{2}\left(B_i^a \mp D_i\phi^a\right)^2 \;\pm\; B_i^aD_i\phi^a, \quad (41.172)$$

and the cross term is a total derivative by the same mechanism as before, now powered by the non-abelian Bianchi identity $D_iB_i^a = 0$ (an off-shell consequence of B being a curvature, established in Chapter 25):

$$B_i^aD_i\phi^a = \partial_i\left(B_i^a\phi^a\right) - \left(D_iB_i^a\right)\phi^a = \partial_i\left(B_i^a\phi^a\right). \quad (41.173)$$

(The covariant and ordinary divergences agree on the gauge-invariant scalar $B_i^a\phi^a$ because the connection terms cancel between $B_i^aD_i\phi^a$ and $(D_iB_i^a)\phi^a$.) Integrating, and noting that at spatial infinity $\phi^a \to v\,\hat{\phi}^a$ defines the unbroken-U(1) magnetic field $b_i \equiv B_i^a\hat{\phi}^a$ with total magnetic charge $g = \oint b_i\,\mathrm{d}S_i$,

$$E = \int \mathrm{d}^3x\, \tfrac{1}{2}\left(B_i^a \mp D_i\phi^a\right)^2 \;+\; \int \mathrm{d}^3x\, V \;\pm\; \oint_{S^2_\infty} B_i^a\phi^a\,\mathrm{d}S_i \;\geq\; v\,|g|\,. \quad (41.174)$$

With the Dirac quantization $g = 4\pi n/e$ derived from $\pi_2(S^2) = \mathbb{Z}$ in Section 41.5, the bound is

$$M \;\geq\; v\,|g| = \frac{4\pi v}{e}\,|n|\,, \quad (41.175)$$

the lower bound that opened this section. The catch, visible in Eq. (41.174), is the lingering $\int V$: the magnetic cross term is a clean boundary integral, but the potential is a positive bulk term that the completion does *not* absorb. The bound holds, but it cannot be saturated for $\lambda > 0$, because saturation needs both $B_i^a = \pm D_i\phi^a$ *and* $V = 0$ everywhere, and $V = 0$ everywhere forces $|\phi| \equiv v$, which is incompatible with a smooth ϕ^a that must vanish at the origin (the hedgehog cannot point in all directions at the core). The resolution is to take the potential to zero.

41.6.3 The Prasad–Sommerfield limit and the exact BPS monopole

The obstruction to saturating Eq. (41.175) is the bulk integral of V. Prasad and Sommerfield [410] observed that one may send $\lambda \to 0$ while *keeping* the vacuum value v fixed not by the (now vanishing) potential but by the boundary condition $|\phi| \to v$ at infinity. In this *BPS limit* the Higgs becomes a massless field that nonetheless has a prescribed value at infinity, the potential term in Eq. (41.174) drops out identically, and the bound $M \geq v\,|g|$ can be attained. What is remarkable is that the first-order equations are then exactly solvable.

Definition 41.31 (BPS limit)**.** The *Bogomol'nyi–Prasad–Sommerfield limit* of the Georgi–Glashow model is $\lambda \to 0^{+}$ with v held fixed through the boundary condition $\phi^a\phi^a \to v^2$ as $r \to \infty$. The energy is Eq. (41.171) with $V \equiv 0$, and a finite-energy configuration saturates $M = v\,|g|$ if and only if it solves the *BPS equation*

$$B_i^a = \pm\, D_i\phi^a. \tag{41.176}$$

We now solve Eq. (41.176) in closed form for the unit charge sector. Use the spherically symmetric hedgehog ansatz Eq. (41.129) of Section 41.5,

$$\phi^a = v\,\hat r^a\,H(\xi), \qquad A_i^a = \epsilon_{aij}\,\hat r^j\,\frac{1-K(\xi)}{e\,r}, \qquad A_0^a = 0, \qquad \xi \equiv e\,v\,r, \tag{41.177}$$

with $\hat r^a = x^a/r$ and the dimensionless profiles obeying the boundary data of Eq. (41.130): regularity at the origin requires $H(0) = 0$ and $K(0) = 1$, and the broken vacuum with a pure-monopole tail requires $H(\infty) = 1$ (so $|\phi| \to v$) and $K(\infty) = 0$. The covariant derivative and non-abelian magnetic field of this ansatz were computed in Section 41.5: decomposed on the orthogonal radial projector $\hat r_i\hat r_a$ and transverse projector $\delta_{ia} - \hat r_i\hat r_a$, they read (Eqs. (41.134) and (41.138), with primes on r)

$$(D_i\phi)^a = v\Big[H_r'\,\hat r_i\hat r_a + \frac{HK}{r}\,(\delta_{ia} - \hat r_i\hat r_a)\Big], \qquad B_i^a = \frac{1-K^2}{er^2}\,\hat r_i\hat r_a - \frac{K_r'}{er}\,(\delta_{ia} - \hat r_i\hat r_a). \tag{41.178}$$

The BPS equation Eq. (41.176) (upper sign) holds iff the radial and transverse coefficients match separately. Passing to $\xi = evr$ so that $H_r' = ev\,H'$ and $K_r' = ev\,K'$ (primes now $\mathrm{d}/\mathrm{d}\xi$), the radial match gives $(1-K^2)\,ev^2/\xi^2 = ev^2H'$ and the transverse match gives $-ev^2K'/\xi = ev^2HK/\xi$; the common ev^2 cancels, leaving the pair of coupled first-order equations

$$\frac{dK}{d\xi} = -\,H\,K, \qquad \frac{dH}{d\xi} = \frac{1-K^2}{\xi^2}. \tag{41.179}$$

These are the entire content of the first-order BPS system in the hedgehog sector—a drastic simplification of the second-order profile equations Eqs. (41.142)

and (41.143), whose $\lambda/e^2 \to 0$ form they integrate once. We claim they are solved in closed form by

$$\boxed{K(\xi) = \frac{\xi}{\sinh \xi}, \qquad H(\xi) = \coth \xi - \frac{1}{\xi}.} \tag{41.180}$$

Worked Example 41.32 (Verification of the Prasad–Sommerfield solution)**.** We check Eq. (41.180) against both equations and both sets of boundary conditions, carrying every derivative.

Boundary conditions. As $\xi \to 0$, $\sinh \xi = \xi + \frac{\xi^3}{6} + \cdots$, so $K = \xi/\sinh \xi \to 1$; and $\coth \xi = \frac{1}{\xi} + \frac{\xi}{3} + \cdots$, so $H = \coth \xi - \frac{1}{\xi} = \frac{\xi}{3} + \cdots \to 0$. Both core conditions $K(0) = 1$, $H(0) = 0$ hold, and $H \sim \xi/3$ confirms $\phi^a = v\hat{r}^a H \to 0$ linearly, i.e. smoothly. As $\xi \to \infty$, $\sinh \xi \to \frac{1}{2}e^{\xi}$ so $K = \xi/\sinh \xi \to 0$ exponentially, and $\coth \xi \to 1$ so $H \to 1 - 1/\xi \to 1$, giving $|\phi| = vH \to v$. Both asymptotic conditions hold.

First equation. Differentiate $K = \xi/\sinh \xi$:

$$\frac{dK}{d\xi} = \frac{\sinh \xi - \xi \cosh \xi}{\sinh^2 \xi}. \tag{41.181}$$

On the other side,

$$-HK = -\Big(\coth \xi - \frac{1}{\xi}\Big)\frac{\xi}{\sinh \xi} = -\frac{\xi \cosh \xi}{\sinh^2 \xi} + \frac{1}{\sinh \xi} = \frac{\sinh \xi - \xi \cosh \xi}{\sinh^2 \xi}, \tag{41.182}$$

which equals K' identically. The first equation holds.

Second equation. Differentiate $H = \coth \xi - \xi^{-1}$, using $\frac{d}{d\xi}\coth \xi = -\operatorname{csch}^2 \xi = -1/\sinh^2 \xi$:

$$\frac{dH}{d\xi} = -\frac{1}{\sinh^2 \xi} + \frac{1}{\xi^2}. \tag{41.183}$$

On the other side, with $K^2 = \xi^2/\sinh^2 \xi$,

$$\frac{1 - K^2}{\xi^2} = \frac{1}{\xi^2} - \frac{1}{\sinh^2 \xi}, \tag{41.184}$$

identical to H'. The second equation holds. Thus Eq. (41.180) is an exact, everywhere-regular, finite-energy solution of the first-order BPS system, and by Theorem 41.30 it is an absolute minimizer of the energy in the $n = 1$ sector.

The profiles are plotted in Fig. 41.7. The gauge function K interpolates from 1 at the core (no long-range field) to 0 at infinity (full monopole tail), while $H = |\phi|/v$ rises from 0 to 1 over a core of size $\xi \sim 1$, i.e. $r \sim 1/(ev)$. There is no separate Higgs core and gauge core: in the BPS limit the two masses coincide (both $\to 0$ as $\lambda \to 0$, the Higgs exactly massless, the photon massless by the

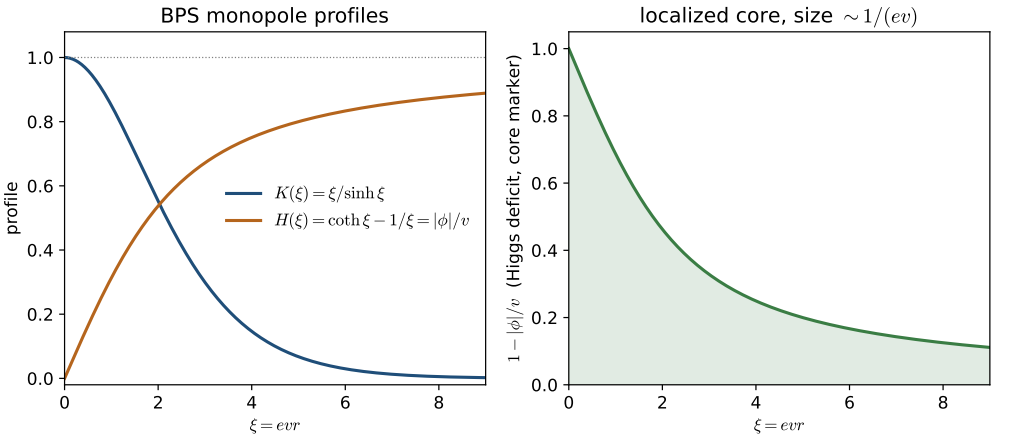


Figure 41.7: The exact Prasad–Sommerfield BPS monopole. *Left:* the gauge profile $K(\xi) = \xi/\sinh\xi$ and the Higgs magnitude $H(\xi) = \coth\xi - 1/\xi = |\phi|/v$, as functions of $\xi = evr$; $K \to 1$, $H \to 0$ at the regular core and $K \to 0$, $|\phi| \to v$ in the broken vacuum. *Right:* the Higgs deficit $1 - |\phi|/v$, marking the localized core of size $\sim 1/(ev)$ outside which only the abelian Coulomb monopole field survives.

unbroken U(1)), and the single length $1/(ev)$ set by the W-boson mass $m_W = ev$ controls everything.

It remains to confirm that the mass saturates the bound. On a BPS solution $B_i^a = D_i\phi^a$, so by Eq. (41.172) (with $V = 0$) and Eq. (41.173),

$$M = \int \mathrm{d}^3x\, B_i^a D_i\phi^a = \oint_{S^2_\infty} B_i^a \phi^a \,\mathrm{d}S_i = v \oint_{S^2_\infty} b_i \,\mathrm{d}S_i = v\, g = \frac{4\pi v}{e}, \tag{41.185}$$

exactly the bound Eq. (41.175) at $n = 1$. Writing $m_W = ev$ for the W-boson mass and $\alpha = e^2/4\pi$ for the fine-structure constant of the unbroken U(1), this is

$$M_{\text{mono}} = \frac{4\pi v}{e} = \frac{4\pi}{e^2}\, m_W = \frac{m_W}{\alpha}, \tag{41.186}$$

so the monopole is heavier than the charged vector boson by the large factor $1/\alpha$. This is the precise sense in which the monopole is a non-perturbative object: its mass scales as the *inverse* coupling, the unmistakable signature of a soliton ($M \sim 1/g^2$ in the natural normalization) and the reason it is invisible to any power series in e.

Physical Insight

The BPS equation $B_i^a = \pm D_i\phi^a$ is a *self-duality* condition. In the Euclidean continuation of Chapter 42 the analogous first-order condition $F_{\mu\nu} = \pm\widetilde{F}_{\mu\nu}$

defines the instanton, and there too it minimizes the (Euclidean) action within a topological sector and reduces a second-order problem to a first-order one with an explicit solution. The kink, vortex, monopole, and instanton are one idea—*topological charge bounds energy/action, first-order "(anti)self-dual" equations saturate it*—realized in $D = 1, 2, 3$ space and in 4-dimensional Euclidean spacetime respectively. The Higgs field ϕ^a in the BPS monopole plays, in the four-dimensional picture, the role of the gauge potential component A_4 along the Euclidean time it does not depend on.

41.6.4 Dyons, the central charge, and the spectrum of allowed charges

Nothing in the construction forbade the time component A_0^a from being nonzero. Reinstating it, Julia and Zee [286] found that the monopole has a one-parameter family of partners carrying *electric* charge as well as magnetic charge: *dyons*. The cleanest way to see both the existence of these states and the resulting mass formula is again the Bogomolny completion, now including the electric sector.

For a static configuration with $A_0^a \neq 0$, the energy acquires the electric terms $\frac{1}{2}E_i^a E_i^a + \frac{1}{2}(D_0\phi^a)^2$ with $E_i^a = F_{0i}^a$ and, in the BPS limit, $V = 0$. The unbroken-U(1) electric charge is the flux of the abelian electric field, $q = \oint E_i^a \hat{\phi}^a \, \mathrm{d}S_i$, just as the magnetic charge is $g = \oint B_i^a \hat{\phi}^a \, \mathrm{d}S_i$. Complete the square in the spatial-gradient energy against an arbitrary angle γ:

$$\tfrac{1}{2}E_i^a E_i^a + \tfrac{1}{2}B_i^a B_i^a + \tfrac{1}{2}D_i\phi^a D_i\phi^a = \tfrac{1}{2}\big(E_i^a - \sin\gamma\, D_i\phi^a\big)^2 + \tfrac{1}{2}\big(B_i^a - \cos\gamma\, D_i\phi^a\big)^2 + \sin\gamma\, E_i^a D_i\phi^a + \cos\gamma\, B_i^a D_i\phi^a, \tag{41.187}$$

where the identity $\sin^2\gamma + \cos^2\gamma = 1$ was used to recover the single $\frac{1}{2}(D_i\phi)^2$. Each cross term is a total derivative—$B_i^a D_i\phi^a = \partial_i(B_i^a\phi^a)$ as in Eq. (41.173), and likewise $E_i^a D_i\phi^a = \partial_i(E_i^a\phi^a)$ on using the Gauss-law constraint $D_i E_i^a = e\,\epsilon^{abc}\phi^b(D_0\phi)^c$, whose right-hand side integrates against $\hat{\phi}$ to zero—so they integrate to $v(q\sin\gamma + g\cos\gamma)$. Dropping the non-negative squares and the $\frac{1}{2}(D_0\phi)^2$ term,

$$M \;\geq\; v\,(q\sin\gamma + g\cos\gamma) \quad \text{for every } \gamma, \qquad \Longrightarrow \qquad M \;\geq\; v\sqrt{q^2+g^2}, \tag{41.188}$$

the sharpest bound obtained by maximizing the right-hand side over γ at $\tan\gamma = q/g$. Equality requires both squares to vanish, i.e. the first-order conditions $B_i^a = \cos\gamma\, D_i\phi^a$ and $E_i^a = \sin\gamma\, D_i\phi^a$ with $\tan\gamma = q/g$ (and the A_0^a profile fixed by the Gauss law). The pure monopole is $\gamma = 0$, recovering $M = vg$; a dyon has $\gamma \neq 0$. The bound Eq. (41.188) is the modulus of a *complex central charge*

$$Z = v\,(q + i\,g), \qquad M \geq |Z|\,, \tag{41.189}$$

and it is in exactly this form that Chapter 46 will identify Z as the central charge of the extended supersymmetry algebra, with BPS states sitting on $M = |Z|$ in shortened multiplets that protect the relation against quantum corrections [436].

What are the *allowed* values of (q, g)? The magnetic charge is quantized by topology, $g = 4\pi n_m/e$ with $n_m \in \mathbb{Z} = \pi_2(S^2)$. The electric charge is quantized by the consistency of the unbroken gauge group: states in the theory carry electric charge in integer units of e, $q = e\,n_e$ with $n_e \in \mathbb{Z}$. The allowed charges therefore live on a two-dimensional lattice labelled by (n_e, n_m). This is the spectrum of the classical theory—but it is not yet the full story, because we have not switched on the one term in the Lagrangian that is invisible to the classical equations of motion: the θ-term.

41.6.5 The Witten effect

We met the θ-term in Eq. (8.74) as a total derivative, $F_{\mu\nu}\widetilde{F}^{\mu\nu} = \partial_\mu K^\mu$, classically inert because it does not alter the field equations. Witten [517] showed that in the presence of a magnetic monopole it is anything but inert: it shifts the dyon's electric charge by an amount proportional to θ. The mechanism is purely canonical—the θ-term changes the momentum conjugate to the gauge potential, hence the operator that generates electric-charge rotations.

Add to the unbroken-U(1) Lagrangian the θ-term, which we take in the form

$$\mathcal{L}_\theta = -\frac{\theta\, e^2}{32\pi^2}\, F_{\mu\nu}\widetilde{F}^{\mu\nu}, \qquad \widetilde{F}^{\mu\nu} = \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}, \tag{41.190}$$

the sign fixing an orientation; the physics below is unchanged under $\theta \to -\theta$ together with the corresponding relabelling of the magnetic charge. The canonical momentum conjugate to A_i receives a contribution from Eq. (41.190). Using the total-derivative form $F_{\mu\nu}\widetilde{F}^{\mu\nu} = 2\,\epsilon^{\mu\nu\rho\sigma}(\partial_\mu A_\nu)(\partial_\rho A_\sigma)$ established at Eq. (8.74), and the Maxwell kinetic term $-\frac{1}{4}F^2$ which alone gives $\partial(-\frac{1}{4}F^2)/\partial(\partial_0 A_i) = F^{i0} = E^i$, the full momentum is

$$\Pi^i = \frac{\partial(\mathcal{L}_{\rm Max} + \mathcal{L}_\theta)}{\partial(\partial_0 A_i)} = E^i - \frac{\theta e^2}{32\pi^2}\,\frac{\partial}{\partial(\partial_0 A_i)}\Big[2\,\epsilon^{\mu\nu\rho\sigma}(\partial_\mu A_\nu)(\partial_\rho A_\sigma)\Big] = E^i - \frac{\theta e^2}{8\pi^2}\,B^i, \tag{41.191}$$

where the derivative picked the two terms with $(\mu\nu) = (0i)$ and $(\rho\sigma) = (0i)$, giving $4\,\epsilon^{0ijk}\partial_j A_k = 4\,B^i$ (with $B^i = \epsilon^{ijk}\partial_j A_k$), and $4 \times \frac{\theta e^2}{32\pi^2} = \frac{\theta e^2}{8\pi^2}$. The θ-term has shifted the conjugate momentum by a multiple of the magnetic field.

Theorem 41.33 (Witten effect)**.** *In the presence of the θ-term Eq. (41.190), a state of magnetic charge $g = 4\pi n_m/e$ and integer electric quantum number n_e has physical electric charge*

$$\boxed{\,q = e\Big(n_e + \frac{\theta}{2\pi}\,n_m\Big).\,} \tag{41.192}$$

Proof. The operator that generates a gauge transformation about the unbroken-U(1) direction $\hat{\phi}^a$ is built, by the canonical formalism, from the conjugate momentum through Gauss's law: with gauge parameter $\Lambda^a = \beta\,\hat{\phi}^a$ the generator is

$$\mathcal{Q}[\beta] = \beta \int \mathrm{d}^3x\, \Pi_i^a \,(D_i\hat{\phi})^a + \beta \oint_{S^2_\infty} \Pi_i^a \hat{\phi}^a \,\mathrm{d}S_i. \tag{41.193}$$

In the broken vacuum at infinity $D_i\hat{\phi} \to 0$, so only the surface term survives and the generator of a rotation by angle β is $\mathcal{Q}[\beta] = \beta\,\mathcal{N}$ with

$$\mathcal{N} = \oint_{S^2_\infty} \Pi_i^a \hat{\phi}^a \,\mathrm{d}S_i = \oint_{S^2_\infty} E_i^a \hat{\phi}^a \,\mathrm{d}S_i - \frac{\theta e^2}{8\pi^2} \oint_{S^2_\infty} B_i^a \hat{\phi}^a \,\mathrm{d}S_i = q - \frac{\theta e^2}{8\pi^2}\, g, \tag{41.194}$$

using Eq. (41.191) projected on $\hat{\phi}^a$, with q and g the abelian electric and magnetic charges. Now the key quantization step. All fields of the theory lie in integer-isospin representations of the broken SU(2) (the gauge bosons and the adjoint Higgs are isospin 0 or 1), so a gauge rotation by $\beta = 2\pi$ about $\hat{\phi}$ acts as the identity on every field: $\exp(2\pi i\,\mathcal{N}/e) = \mathbf{1}$. Hence $\mathcal{N}/e$ has integer spectrum,

$$\mathcal{N} = e\,n_e, \qquad n_e \in \mathbb{Z}. \tag{41.195}$$

Solving Eq. (41.194) for q and inserting $g = 4\pi n_m/e$,

$$q = \mathcal{N} + \frac{\theta e^2}{8\pi^2}\, g = e\,n_e + \frac{\theta e^2}{8\pi^2} \cdot \frac{4\pi n_m}{e} = e\,n_e + \frac{\theta e}{2\pi}\, n_m = e\Big(n_e + \frac{\theta}{2\pi} n_m\Big), \tag{41.196}$$

which is Eq. (41.192). □

The consequences are striking. At $\theta = 0$ the dyon spectrum is the rectangular lattice $q = en_e$, $g = 4\pi n_m/e$. Turning on θ *tilts* the lattice: every unit of magnetic charge drags the allowed electric charges off the integers by $e\theta/2\pi$ (Fig. 41.8). A monopole ($n_m = 1, n_e = 0$) acquires electric charge $q = e\theta/2\pi$, generically irrational in units of e. Because q is no longer an integer multiple of e for $\theta \neq 0$, charge conjugation and parity—each of which sends $\theta \to -\theta$—are broken unless $\theta = 0$ or $\theta = \pi$, the cleanest statement of the strong-CP problem from the soliton side and the point at which this thread rejoins the θ-vacuum of Chapter 42.

Physical Insight

The Witten effect is the cleanest illustration in this book of a "topological" coupling with no classical effect acquiring sharp physical meaning in a nontrivial sector. Classically, Eq. (8.74) says the θ-term is a boundary integral and changes nothing. But a monopole *is* a nontrivial boundary configuration: the magnetic flux through the sphere at infinity is nonzero, so the boundary integral is alive, and it reads off directly as a shift of the conserved electric charge. The same $\partial_\mu K^\mu$ that was inert in the vacuum

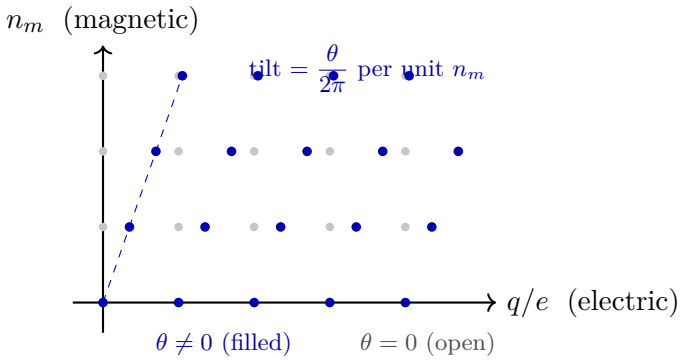


Figure 41.8: The Witten effect on the dyon charge lattice. At $\theta = 0$ the allowed charges $(q/e, n_m) = (n_e, n_m)$ form a rectangular lattice (grey). Switching on θ shifts the electric charge of each state by $e\theta/2\pi$ per unit of magnetic charge, $q = e(n_e + \frac{\theta}{2\pi} n_m)$, tilting the magnetic towers (blue) and rendering the monopole's electric charge generically irrational.

becomes, in the monopole background, a measurable charge.

With the dyon spectrum and the central charge Eq. (41.189) in hand, the natural next question is what happens to these states under the exchange of electric and magnetic charge—the $\mathrm{SL}(2,\mathbb{Z})$ duality that maps (n_e, n_m) into one another and, conjecturally, relates the weakly coupled electric description to a weakly coupled *magnetic* one in which the monopole is elementary [350]. That question, and its precise realization in $\mathcal{N} = 2$ gauge theory where the BPS mass formula $M = |Z|$ is exact and monopole condensation drives confinement, is the business of Chapter 46 and [436]. The immediate sequel, Chapter 42, takes the other fork: it Wick-rotates the self-duality $B = \pm D\phi$ into the four-dimensional Euclidean self-duality $F = \pm\widetilde{F}$, identifies its solutions as instantons, and shows that the θ-term whose monopole avatar we have just measured is the angle labelling the superselected θ-vacua of Yang–Mills theory, completing the Yang–Mills theory passage from the perturbative vacuum to the full topological structure of the non-abelian ground state.

Further reading

The canonical pedagogical treatments of classical solitons and their topology are the monograph of Rajaraman [412] and the lectures of Coleman [129]; the modern, geometry-forward reference is the book of Manton and Sutton [337]. Derrick's no-go scaling argument is Derrick [144]. The abelian vortex was found in the Ginzburg–Landau setting and reintroduced as a relativistic string by

Nielsen and Olesen [367]; the non-abelian monopole is the simultaneous discovery of 't Hooft [468] and Polyakov [407]. The completion-of-the-square bound is due to Bogomolny [83], and the closed-form saturating solution to Prasad and Sommerfield [410]; the link between that bound and a central charge in an extended algebra, which Part XIII makes precise, is Witten and Olive [525]. The shift of the dyon's electric charge in a θ-vacuum — the Witten effect — is Witten [517].

Chapter 42

Instantons and the Vacuum

Every amplitude in Parts IV–VII was a power series in a coupling, an expansion about the trivial classical vacuum that sees only what is analytic in g at the origin. This chapter is about the physics that expansion cannot reach: effects weighted by e^{-c/g^2}, which vanish with all their derivatives as $g \to 0$ and so appear at no finite order of perturbation theory, yet which reorganize the vacuum of every non-abelian gauge theory. The object that computes them is the *instanton*, a finite-action solution of the Euclidean equations of motion that interpolates between classically degenerate vacua. Where Chapter 41 found static, finite-energy solitons classified by the topology of field space, here we find finite-*action* configurations localized in Euclidean time, and read them as quantum-mechanical tunnelling events between vacua that perturbation theory wrongly reports as a unique ground state.

The method is built from the ground up in a setting with no field-theoretic complications at all: a single quantum-mechanical particle in a double well, where the missing physics is the elementary splitting of two would-be degenerate ground states. The full toolkit assembled there—the Euclidean saddle point of the path integral of Chapter 10, evaluated by the steepest-descent machinery of Chapter 5; the finite-action solution interpolating between vacua; the time-translation zero mode and its trade for a collective coordinate; the one-loop fluctuation determinant; and the dilute gas that exponentiates the single-instanton amplitude—transfers verbatim to four-dimensional Yang–Mills. Only the determinants get harder. With that template in hand we lift the picture to the gauge theory: finite Euclidean action forces the field to a pure gauge at the boundary three-sphere, the boundary maps are classified by $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$ (the homotopy language of Chapter 41), and the Bogomolny rearrangement (Theorem 41.5) bounds the action of each topological sector from below by $8\pi^2|Q|/g^2$, saturated by self-dual fields. We then construct the field that attains it, the BPST instanton, in full, cataloguing the collective coordinates—position, size, and gauge orientation—that the dilute-gas integral runs over.

From the explicit solution the chapter turns to its physical consequences along two complementary readings. In the Hamiltonian reading, the residual large gauge transformations left over from the temporal-gauge quantization of Chapter 26 organize the vacua into a discrete tower $|n\rangle$ labelled by Chern–Simons number; the instanton is the amplitude to hop between adjacent rungs; diagonalizing the tunnelling Hamiltonian produces the θ-vacuum, a one-parameter family of superselected, clustering ground states, and shows that summing the topological sectors is identical to adding a CP-odd θ-term to the Lagrangian. In the Euclidean reading, the dilute instanton gas sums tunnelling events of every winding into a θ-dependent vacuum energy $E(\theta) = -2K\cos\theta$, with the one-loop measure—and through it the running coupling and Λ_{QCD} of Chapter 27—fixing the prefactor and exposing where the approximation honestly breaks down. The curvature of that energy at the origin is the topological susceptibility.

Matter sharpens everything. Coupling light quarks to the instanton, the chiral anomaly of Chapter 36—in both its triangle and Fujikawa-measure forms—turns the topological charge into a violation of axial-fermion number, $\Delta Q_5 = 2n_f Q$, realized concretely through the fermion zero modes the Atiyah–Singer index theorem forces the instanton to carry. These zero modes generate the 't Hooft effective multi-fermion vertex, which dispatches the $U(1)_A$ problem (there is no ninth Goldstone because the singlet axial symmetry is destroyed by the anomaly, not spontaneously broken) and, through the Witten–Veneziano relation, fixes the η' mass in terms of the pure-glue susceptibility. The same anomaly that heals $U(1)_A$ makes the vacuum angle physical and impossible to rotate away, leaving a rephasing-invariant combination $\bar{\theta}$ that a permanent neutron electric dipole moment would detect—and the experimental null result, $|\bar{\theta}| \lesssim 10^{-10}$, is the strong-CP problem. The chapter closes by framing it honestly as a naturalness puzzle and presenting its leading dynamical resolution, the Peccei–Quinn axion, whose potential is built directly on the θ-vacuum energy computed earlier. Throughout, this is Yang–Mills theory carried to its full non-perturbative reach; the model-building and cosmology of the axion are treated in Chapter 35, the non-perturbative lattice definition in Chapter 43.

42.1 Tunneling and the Euclidean Path Integral

Every amplitude computed in Parts IV–VII of this book is a power series in a coupling, generated by expanding $e^{iS_{\mathrm{int}}}$ and summing diagrams. Such a series is an expansion about a single classical solution—the trivial vacuum—and it sees only what is analytic in the coupling at the origin. There is an entire stratum of physics it cannot see, not because the series is poorly summed but because the physics is weighted by a factor that vanishes, together with all its derivatives, as the coupling goes to zero: a factor of the form e^{-c/g^2}. No finite order of perturbation theory contains such a term, and no rearrangement of the series

produces one. The task of this chapter is to compute that physics directly, and the cleanest place to learn how is not field theory at all but a single quantum-mechanical particle in a double well, where the missing physics is something elementary and unmistakable: the splitting of two would-be degenerate ground states by quantum tunnelling. Perturbation theory around either well gives two exactly degenerate harmonic ground states to all orders. The true spectrum has a small splitting. That splitting is the prototype of every nonperturbative effect in this chapter, and the instanton is the object that computes it.

We build the method from the ground up here, in quantum mechanics, because the toolkit assembled—the Euclidean saddle point, the finite-action solution interpolating between vacua, the time-translation zero mode and its trade for a collective coordinate, the fluctuation determinant, and the dilute gas that exponentiates the single-instanton amplitude—transfers *verbatim* to four-dimensional Yang–Mills in the sections that follow. Nothing in the logic is special to one dimension; only the determinants get harder. The reader who masters the double well has, modulo group theory and four-dimensional differential geometry, already understood the Yang–Mills instanton.

42.1.1 The model problem: a particle in a double well

We take a single particle of unit mass moving on the line in a symmetric double-well potential. Setting the mass to one costs no generality—it is restored at the end by dimensional analysis—and lets us write every formula without a trailing m.

Definition 42.1 (Double-well model)**.** The Hamiltonian is $H = \frac{1}{2}p^2 + V(x)$ with

$$V(x) = \frac{\omega^2}{8a^2}\,(x^2 - a^2)^2, \tag{42.1}$$

a potential with two degenerate absolute minima at $x = \pm a$, where $V = 0$, separated by a barrier of height $V(0) = \omega^2 a^2/8$. Its second derivative is

$$V''(x) = \frac{\omega^2}{2a^2}\,(3x^2 - a^2), \qquad V''(\pm a) = \omega^2, \tag{42.2}$$

so that near either minimum the particle is a harmonic oscillator of angular frequency ω.

Perturbation theory proceeds by expanding about one minimum, say $x = +a$. Writing $x = a + y$ and Taylor expanding Eq. (42.1), $V = \frac{1}{2}\omega^2 y^2 + O(y^3)$, one quantizes the local oscillator and treats the cubic and quartic terms as perturbations. This generates the ground-state energy as a power series $E_0(\omega, a) = \frac{1}{2}\omega\,[1 + O(1/\omega a^2)]$ in the dimensionless inverse-barrier parameter. The identical series is obtained by expanding about $x = -a$. To every order the two ground

states are degenerate, because no finite number of derivatives of V at $+a$ knows that a second minimum exists a finite distance away. Yet the exact Hamiltonian is symmetric under parity $x \to -x$, and its true low-lying eigenstates are the symmetric and antisymmetric combinations of the two localized states, with distinct energies. The gap between them,

$$\Delta E \;=\; E_- - E_+, \tag{42.3}$$

is nonzero but invisible order by order: it is the quantity perturbation theory misses. The remainder of this section computes it.

Physical Insight

The physical reason the splitting escapes the perturbative series is that it is controlled by the amplitude for the particle to be found *under the barrier*, in the classically forbidden region between the wells. In the WKB approximation that amplitude is $e^{-\int |p|\,\mathrm{d}x}$, with the momentum imaginary under the barrier. The barrier integral is a positive number divided by $\hbar$; as $\hbar \to 0$ (or, equivalently here, as the barrier becomes high in units of ω) the amplitude is exponentially small and nonanalytic. The instanton is the path-integral incarnation of exactly this WKB barrier penetration, organized so that the prefactor multiplying $e^{-(\text{barrier action})}$ can be computed systematically rather than guessed.

42.1.2 Euclidean continuation of the path integral

The object that knows the full spectrum, degeneracy splitting included, is the transition amplitude in imaginary time. Recall the real-time transition amplitude (Theorem 10.1)

$$\left\langle x_f \middle| e^{-iHT} \middle| x_i \right\rangle = \int_{x(0)=x_i}^{x(T)=x_f} \mathcal{D}x\; e^{\,iS[x]}, \qquad S[x] = \int_0^T \mathrm{d}t\, \left[\tfrac{1}{2}\dot{x}^2 - V(x)\right]. \tag{42.4}$$

The continuation to imaginary time is the Wick rotation fixed once and for all in Chapter 10: we set $t = -i\tau$, with τ real, so that the measure rotates as $\mathrm{d}t = -i\,\mathrm{d}\tau$ (Eq. (10.157)) and the real-time interval T becomes a Euclidean interval $\mathcal{T}$ via $T = -i\mathcal{T}$. Under this rotation $\dot{x} = \mathrm{d}x/\mathrm{d}t = i\,\mathrm{d}x/\mathrm{d}\tau$, so the kinetic term flips sign relative to the potential, and the weight e^{iS} becomes e^{-S_E} with the Euclidean action

$$S_E[x] = \int_0^{\mathcal{T}} \mathrm{d}\tau\, \left[\tfrac{1}{2}\Big(\frac{dx}{d\tau}\Big)^2 + V(x)\right]. \tag{42.5}$$

This is the quantum-mechanical instance of the Euclidean action defined for fields in Theorem 10.27. The amplitude becomes the Euclidean path integral

$$\left\langle x_f \middle| e^{-H\mathcal{T}} \middle| x_i \right\rangle = \int_{x(0)=x_i}^{x(\mathcal{T})=x_f} \mathcal{D}x\; e^{-S_E[x]}. \tag{42.6}$$

Two features of Eq. (42.6) make it the right tool. First, the weight e^{-S_E} is a genuine positive damping factor rather than an oscillatory phase, so the integral is dominated by the configurations of *least* Euclidean action, and the method of steepest descent (Theorem 5.44, Section 5.6) applies directly: the leading approximation is a sum over solutions of the classical Euclidean equation of motion, weighted by e^{-S_E} and corrected by Gaussian fluctuation determinants. Second, the operator $e^{-H\mathcal{T}}$ is the spectral projector in disguise. Inserting a complete set of energy eigenstates,

$$\left\langle x_f \middle| e^{-H\mathcal{T}} \middle| x_i \right\rangle = \sum_n \psi_n(x_f)\, \psi_n^*(x_i)\, e^{-E_n \mathcal{T}}, \tag{42.7}$$

we see that as $\mathcal{T} \to \infty$ the sum is dominated by the lowest states, with the splitting between the two nearly degenerate levels controlling the leading $\mathcal{T}$-dependence. Comparing the saddle-point evaluation of the path integral with the spectral sum Eq. (42.7) is precisely how we will read off ΔE.

The variational principle $\delta S_E = 0$ applied to Eq. (42.5) gives the Euclidean equation of motion

$$\frac{d^2x}{d\tau^2} = V'(x). \tag{42.8}$$

This is Newton's equation for a particle of unit mass moving in the *inverted* potential $-V(x)$: the sign on the force is opposite to the real-time law $\ddot{x} = -V'(x)$. The inversion is the whole geometric content of the Euclidean method. The two degenerate *minima* of V at $x = \pm a$ become two degenerate *maxima* of $-V$, and a classical trajectory of Eq. (42.8) that starts at rest on one hilltop and rolls to the other is the configuration we seek. Figure 42.1 shows the inverted potential and the trajectory.

42.1.3 The instanton solution by quadrature

We now solve Eq. (42.8) for the trajectory that interpolates between the two vacua. The key is that the Euclidean equation of motion, like any one-dimensional Newtonian problem, has a conserved "energy."

Lemma 42.2 (Euclidean energy conservation)**.** *Any solution of Eq. (42.8) conserves the quantity*

$$\mathcal{E} \;=\; \tfrac{1}{2}\Big(\frac{dx}{d\tau}\Big)^2 - V(x). \tag{42.9}$$

Proof. Differentiate: $\frac{d\mathcal{E}}{d\tau} = \dot{x}\,\ddot{x} - V'(x)\dot{x} = \dot{x}\,[\ddot{x} - V'(x)] = 0$ by Eq. (42.8). Note the relative minus sign in Eq. (42.9): it is the mechanical energy in the inverted potential $-V$, the kinetic term plus $(-V)$, as it must be. □

For a trajectory that begins at the vacuum $x = -a$ in the far past and ends at $x = +a$ in the far future, both endpoints are at rest—$\dot{x} \to 0$ as $\tau \to \pm\infty$, since

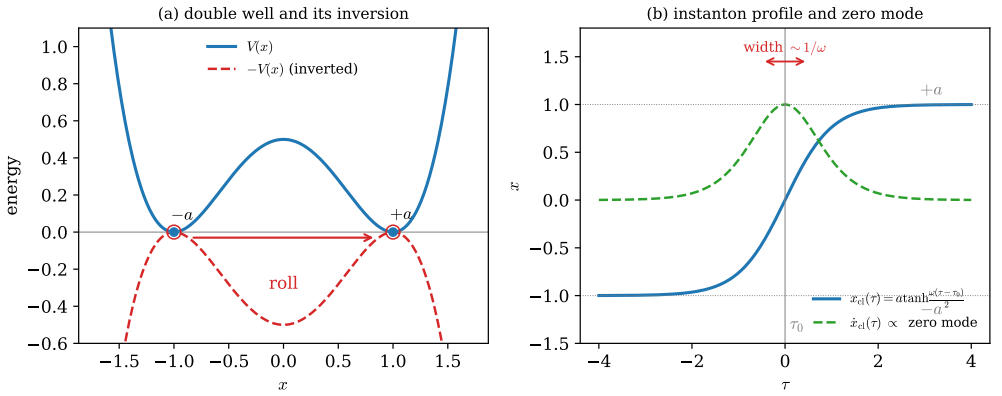


Figure 42.1: (a) The double-well potential $V(x)$ of Eq. (42.1) (solid) and its inversion $-V(x)$ (dashed). The Euclidean equation of motion Eq. (42.8) is Newton's law in $-V$; the two vacua $x = \pm a$ become equal hilltops. (b) The instanton profile $x_{\rm cl}(\tau) = a\tanh[\omega(\tau-\tau_0)/2]$ (solid), a finite-action trajectory rolling from one hilltop to the other, and its derivative $\dot{x}_{\rm cl}$ (dashed), which is the translation zero mode. The transition is localized to a region of width $\sim 1/\omega$ about the centre τ_0; this localization is what licenses the dilute-gas approximation.

the particle must take infinite Euclidean time to climb infinitesimally close to a hilltop of $-V$—and at both endpoints $V = 0$. Therefore $\mathcal{E} = 0$ on this trajectory. The conservation law Eq. (42.9) with $\mathcal{E} = 0$ reduces the second-order equation to a first-order one,

$$\tfrac{1}{2}\Big(\frac{dx}{d\tau}\Big)^2 = V(x), \qquad \text{i.e.} \qquad \frac{dx}{d\tau} = \sqrt{2V(x)} = \frac{\omega}{2a}\,(a^2 - x^2), \tag{42.10}$$

where the positive root is taken for a trajectory moving from $-a$ to $+a$ (so $\dot{x} > 0$), and we used Eq. (42.1) to evaluate $\sqrt{2V} = \sqrt{(\omega^2/4a^2)(x^2-a^2)^2} = (\omega/2a)|a^2 - x^2|$, with $a^2 - x^2 > 0$ between the wells. Equation (42.10) is the central simplification of the whole method: *the equality* $\tfrac{1}{2}\dot{x}^2 = V$ *along the instanton*, which we will use repeatedly.

Equation Eq. (42.10) separates and is integrated by quadrature:

$$\int \frac{\mathrm{d}x}{a^2 - x^2} = \frac{\omega}{2a}\int \mathrm{d}\tau \quad\Longrightarrow\quad \frac{1}{a}\,\operatorname{artanh}\Big(\frac{x}{a}\Big) = \frac{\omega}{2a}\,(\tau - \tau_0), \tag{42.11}$$

where the constant of integration τ_0 is the Euclidean time at which the particle passes through the midpoint $x = 0$. Solving for x gives the *instanton*.

Definition 42.3 (Instanton and anti-instanton)**.** The finite-action solution of Eq. (42.8) interpolating from $x(-\infty) = -a$ to $x(+\infty) = +a$ is

$$x_{\rm cl}(\tau) = a\,\tanh\Big[\frac{\omega}{2}(\tau - \tau_0)\Big], \tag{42.12}$$

the *instanton* centred at τ_0. The reflected solution $-x_{\rm cl}$, running from $+a$ to $-a$, is the *anti-instanton*. We call τ_0 the *centre* or *collective coordinate*.

The profile Eq. (42.12) is a smooth step: it sits exponentially close to $-a$ for $\tau \ll \tau_0$, climbs through zero at $\tau = \tau_0$, and saturates exponentially close to $+a$ for $\tau \gg \tau_0$. The transition happens in a window of width $\sim 1/\omega$, the time it takes the inverted-potential trajectory to traverse the region where $-V$ is appreciably curved. This finite size, against the backdrop of an arbitrarily long total Euclidean time $\mathcal{T}$, is the geometric fact that makes the instanton "dilute": for large $\mathcal{T}$ the system spends almost all its time sitting in a vacuum and only a vanishing fraction tunnelling.

The instanton action

The Euclidean action of the instanton is finite—this is what distinguishes it from a generic interpolating path—and we compute it now, because e^{-S_0} is the leading exponential in every result of the chapter.

Proposition 42.4 (Instanton action)**.** *The Euclidean action of the instanton Eq. (42.12) is*

$$S_0 \;\equiv\; S_E[x_{\rm cl}] \;=\; \frac{2}{3}\,\omega a^2. \tag{42.13}$$

Proof. On the instanton $\tfrac{1}{2}\dot{x}^2 = V$ (Eq. (42.10)), so the two terms of the Euclidean action Eq. (42.5) are equal and

$$S_0 = \int_{-\infty}^{\infty} \mathrm{d}\tau \left[\tfrac{1}{2}\dot{x}_{\rm cl}^2 + V(x_{\rm cl})\right] = \int_{-\infty}^{\infty} \mathrm{d}\tau\, \dot{x}_{\rm cl}^2. \tag{42.14}$$

This last form has a clean interpretation: $S_0 = \int \dot{x}\,(\dot{x}\,\mathrm{d}\tau) = \int \dot{x}\,\mathrm{d}x$, the integral of momentum over the path. Using $\dot{x} = \sqrt{2V}$ from Eq. (42.10) converts the τ-integral into an x-integral over the barrier,

$$S_0 = \int_{-a}^{a} \sqrt{2V(x)}\;\mathrm{d}x = \int_{-a}^{a} \frac{\omega}{2a}\,(a^2 - x^2)\,\mathrm{d}x. \tag{42.15}$$

The remaining elementary integral is

$$\int_{-a}^{a} (a^2 - x^2)\,\mathrm{d}x = \Big[a^2 x - \tfrac{1}{3}x^3\Big]_{-a}^{a} = 2\Big(a^3 - \tfrac{1}{3}a^3\Big) = \frac{4a^3}{3}, \tag{42.16}$$

so $S_0 = (\omega/2a)(4a^3/3) = \tfrac{2}{3}\omega a^2$, which is Eq. (42.13). The same number follows by substituting the explicit profile into Eq. (42.14): with $u = \tfrac{\omega}{2}(\tau - \tau_0)$, $\dot{x}_{\rm cl} = \tfrac{a\omega}{2}\,\mathrm{sech}^2 u$ and $\mathrm{d}\tau = \tfrac{2}{\omega}\mathrm{d}u$, giving $S_0 = \tfrac{a^2\omega}{2}\int_{-\infty}^{\infty}\mathrm{sech}^4 u\,\mathrm{d}u = \tfrac{a^2\omega}{2}\cdot\tfrac{4}{3} = \tfrac{2}{3}\omega a^2$, using $\int \mathrm{sech}^4 u\,\mathrm{d}u = \tfrac{4}{3}$. □

Equation Eq. (42.15) deserves emphasis: the instanton action is exactly the WKB barrier integral $\int |p|\,\mathrm{d}x$ with $|p| = \sqrt{2(V-\mathcal{E})} = \sqrt{2V}$ for zero-energy tunnelling. The instanton method therefore reproduces, and then systematically improves upon, the WKB estimate of the tunnelling amplitude promised in Section 42.1.1. Restoring $\hbar$ and the mass m by dimensional analysis, the weight is $e^{-S_0/\hbar}$ with $S_0 = \tfrac{2}{3} m\omega a^2$, manifestly nonanalytic at $\hbar = 0$.

42.1.4 The zero mode and collective coordinates

To go beyond the bare exponential we expand the action to quadratic order in fluctuations about the instanton and perform the resulting Gaussian integral. Write a nearby path as

$$x(\tau) = x_{\mathrm{cl}}(\tau) + \eta(\tau), \qquad \eta(0) = \eta(\mathcal{T}) = 0, \tag{42.17}$$

with η vanishing at the endpoints (the endpoints are fixed). Since x_{cl} solves the equation of motion, the term linear in η vanishes, and Taylor-expanding Eq. (42.5) gives

$$S_E[x_{\mathrm{cl}} + \eta] = S_0 + \frac{1}{2}\int \mathrm{d}\tau\; \eta(\tau)\, M\, \eta(\tau) + O(\eta^3), \qquad M \equiv -\frac{d^2}{d\tau^2} + V''(x_{\mathrm{cl}}(\tau)), \tag{42.18}$$

the second-order term following from one integration by parts, using $\eta(\partial_\tau^2)\eta = -(\partial_\tau \eta)^2 + \text{boundary}$. The operator M is a Schrödinger operator on the τ-line with the τ-dependent "potential" $V''(x_{\mathrm{cl}}(\tau))$. Inserting Eq. (42.2) along the profile, with $x_{\mathrm{cl}}^2 = a^2 \tanh^2 u$ and $u = \tfrac{\omega}{2}(\tau - \tau_0)$,

$$V''(x_{\mathrm{cl}}) = \frac{\omega^2}{2}\left(3\tanh^2 u - 1\right) = \omega^2 - \frac{3\omega^2}{2}\operatorname{sech}^2 u, \tag{42.19}$$

where we used $\tanh^2 u = 1 - \operatorname{sech}^2 u$. Thus

$$M = -\frac{d^2}{d\tau^2} + \omega^2 - \frac{3\omega^2}{2}\operatorname{sech}^2\!\left[\frac{\omega}{2}(\tau - \tau_0)\right]. \tag{42.20}$$

Far from the instanton centre, $\operatorname{sech}^2 \to 0$ and $M \to M_0 \equiv -\partial_\tau^2 + \omega^2$, the fluctuation operator of a single harmonic well: this is what makes the instanton's fluctuations reduce to vacuum fluctuations away from the transition.

The translation zero mode

The operator M has a normalizable eigenfunction of eigenvalue exactly zero, and its origin is symmetry: the action does not depend on where the instanton is centred.

Proposition 42.5 (Zero mode from time translation)**.** *The function* $\dot{x}_{\mathrm{cl}}(\tau)$ *satisfies* $M\,\dot{x}_{\mathrm{cl}} = 0$*. It is normalizable, with* $\int_{-\infty}^{\infty} \dot{x}_{\mathrm{cl}}^2\,\mathrm{d}\tau = S_0$*, so the unit-normalized zero mode is*

$$x_1(\tau) = \frac{1}{\sqrt{S_0}}\,\dot{x}_{\mathrm{cl}}(\tau). \tag{42.21}$$

Proof. Differentiate the equation of motion Eq. (42.8), $\ddot{x}_{\mathrm{cl}} = V'(x_{\mathrm{cl}})$, once more with respect to τ. The chain rule gives $\dddot{x}_{\mathrm{cl}} = V''(x_{\mathrm{cl}})\,\dot{x}_{\mathrm{cl}}$, that is,

$$-\frac{d^2}{d\tau^2}\dot{x}_{\mathrm{cl}} + V''(x_{\mathrm{cl}})\,\dot{x}_{\mathrm{cl}} = 0, \tag{42.22}$$

which is exactly $M\,\dot{x}_{\mathrm{cl}} = 0$. Normalizability follows because $\dot{x}_{\mathrm{cl}} = \frac{a\omega}{2}\operatorname{sech}^2 u$ decays exponentially as $|\tau| \to \infty$; explicitly its squared norm is the integral computed in Theorem 42.4, $\int \dot{x}_{\mathrm{cl}}^2\,\mathrm{d}\tau = S_0$ by Eq. (42.14). Dividing by $\sqrt{S_0}$ gives the unit-normalized mode Eq. (42.21). □

The zero mode is the lowest eigenfunction of M: it has no nodes (it is everywhere positive, being proportional to sech^2), so by the standard oscillation theorem of one-dimensional Schrödinger operators it is the ground state, and every other eigenvalue of M is strictly positive. This sign structure—one zero eigenvalue, the rest positive—is what makes the instanton a genuine minimum of the action in every direction except the flat one, and it is what we must handle with care in the Gaussian integral.

Physical Insight

A zero mode is the signature of a continuous degeneracy of the saddle point. Here the degeneracy is the freedom to slide the instanton along the τ-axis: $x_{\mathrm{cl}}(\tau - \tau_0)$ is a solution of the same action for every τ_0, so moving along the family costs nothing, and the second variation in that direction vanishes. The direction in function space tangent to the family is precisely $\partial_{\tau_0} x_{\mathrm{cl}} = -\dot{x}_{\mathrm{cl}}$, the zero mode. Whenever a saddle point breaks a continuous symmetry of the action, the broken generator produces a zero mode, and the Gaussian integral over that direction must be replaced by an integral over the symmetry parameter—a *collective coordinate*. The same mechanism in Yang–Mills produces zero modes for spacetime translations, dilatations, and global gauge rotations of the instanton.

Trading the zero mode for the collective coordinate

Expand the fluctuation in the orthonormal eigenbasis $\{x_n\}$ of M, with eigenvalues λ_n ($\lambda_1 = 0$):

$$\eta(\tau) = \sum_n c_n\,x_n(\tau), \qquad M x_n = \lambda_n x_n, \qquad \int x_n x_m\,\mathrm{d}\tau = \delta_{nm}. \tag{42.23}$$

The Gaussian weight is $e^{-\frac{1}{2}\sum_n \lambda_n c_n^2}$, and the path-integral measure, with the conventional normalization that makes each Gaussian integral standard, is

$$\mathcal{D}x = \prod_n \frac{\mathrm{d}c_n}{\sqrt{2\pi}}. \tag{42.24}$$

For every mode with $\lambda_n > 0$ the integral is $\int \frac{\mathrm{d}c_n}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\lambda_n c_n^2} = \lambda_n^{-1/2}$, and the product of these is, by definition, $(\det{}' M)^{-1/2}$ with the prime indicating omission of the zero eigenvalue. But the $n = 1$ integral is hopeless as written: with $\lambda_1 = 0$ the Gaussian degenerates to $\int \frac{\mathrm{d}c_1}{\sqrt{2\pi}}$, a divergent integral over the unbounded coefficient of the flat direction. This divergence is not a pathology; it is the statement that the instanton can sit anywhere along the τ-axis, and integrating over c_1 overcounts by integrating over a direction that is really the location τ_0. We must convert $\int \mathrm{d}c_1$ into $\int_0^{\mathcal{T}} \mathrm{d}\tau_0$.

Proposition 42.6 (Collective-coordinate change of variables)**.** *Promoting the instanton centre τ_0 to an integration variable, the zero-mode integral is traded according to*

$$\frac{\mathrm{d}c_1}{\sqrt{2\pi}} = \sqrt{\frac{S_0}{2\pi}}\, \mathrm{d}\tau_0, \tag{42.25}$$

so the Gaussian integral over the flat direction is replaced by an ordinary integral over the centre, weighted by the Jacobian $\sqrt{S_0/2\pi}$.

Proof. Displacing the instanton centre by an infinitesimal $\mathrm{d}\tau_0$ changes the configuration by

$$\delta x = x_{\mathrm{cl}}(\tau - \tau_0 - \mathrm{d}\tau_0) - x_{\mathrm{cl}}(\tau - \tau_0) = -\dot{x}_{\mathrm{cl}}(\tau - \tau_0)\, \mathrm{d}\tau_0. \tag{42.26}$$

This same displacement, expressed in the mode basis, is a move purely along the zero direction: $\delta x = x_1\, \mathrm{d}c_1$ with $x_1 = \dot{x}_{\mathrm{cl}}/\sqrt{S_0}$ the normalized zero mode (Eq. (42.21)). Equating the two expressions for δx and matching coefficients of the unit vector x_1,

$$x_1\, \mathrm{d}c_1 = -\dot{x}_{\mathrm{cl}}\, \mathrm{d}\tau_0 = -\sqrt{S_0}\, x_1\, \mathrm{d}\tau_0 \quad\Longrightarrow\quad \mathrm{d}c_1 = -\sqrt{S_0}\, \mathrm{d}\tau_0. \tag{42.27}$$

The sign merely reflects orientation and is absorbed into the limits; taking the magnitude, $|\mathrm{d}c_1| = \sqrt{S_0}\, \mathrm{d}\tau_0$. Dividing by the $\sqrt{2\pi}$ that accompanies $\mathrm{d}c_1$ in the measure Eq. (42.24) gives Eq. (42.25). □

Putting the pieces together, the single-instanton contribution to the Euclidean amplitude is

$$\left\langle a \middle| e^{-H\mathcal{T}} \middle| -a \right\rangle_{\text{1-inst}} = e^{-S_0} \sqrt{\frac{S_0}{2\pi}} \int_0^{\mathcal{T}} \mathrm{d}\tau_0\, \left(\det{}' M\right)^{-1/2}, \tag{42.28}$$

where the integral over τ_0 runs over the available Euclidean time and produces a factor $\mathcal{T}$ because the integrand—the action, the determinant with the zero mode removed—is independent of where the instanton sits. The remaining task is to evaluate $\det{}'M$ relative to the free determinant, which supplies the overall normalization.

42.1.5 The fluctuation determinant and the prefactor K

The determinant $\det{}'M$ is meaningful only relative to a reference. The natural reference is the no-instanton amplitude—the particle sitting in a single harmonic well for the whole time $\mathcal{T}$—whose Gaussian integral is governed by the free operator $M_0 = -\partial_\tau^2 + \omega^2$. We therefore organize the answer around the ratio $\det{}'M/\det M_0$, which is finite and dimensionless.

The harmonic reference is the textbook result, derived in Chapter 10 from the same Gaussian path integral: for a single well of frequency ω,

$$\left\langle a\middle|e^{-H\mathcal{T}}\middle|a\right\rangle_{\text{0-inst}} = (\det M_0)^{-1/2} = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2}\,[1+O(e^{-\omega\mathcal{T}})], \tag{42.29}$$

the prefactor $(\omega/\pi)^{1/2}$ being $|\psi_0(a)|^2$ for the harmonic ground state and $e^{-\omega\mathcal{T}/2}$ its evolution at energy $\frac{1}{2}\omega$. Dividing the single-instanton result Eq. (42.28) by this reference gives the one-instanton amplitude in the natural units of the harmonic "do-nothing" amplitude:

$$\left\langle a\middle|e^{-H\mathcal{T}}\middle|-a\right\rangle_{\text{1-inst}} = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2}\,(K\mathcal{T})\,e^{-S_0}, \tag{42.30}$$

which *defines* the prefactor

$$K \;\equiv\; \sqrt{\frac{S_0}{2\pi}}\,\left(\frac{\det{}'M}{\det M_0}\right)^{-1/2}. \tag{42.31}$$

The dimensions are right: $\det{}'M/\det M_0$ is dimensionless, S_0 is dimensionless in units $\hbar = 1$, and the $\mathcal{T}$-integral over τ_0 supplied the single factor of time, so K has dimensions of inverse time (frequency)—exactly what is needed for $K\mathcal{T}$ to be dimensionless and for Ke^{-S_0} to be an energy, as the splitting formula below requires.

Evaluating the ratio $\det{}'M/\det M_0$ is the one piece of genuine determinant machinery in the calculation. The general method—the Gelfand–Yaglom construction, which computes the determinant of a one-dimensional Schrödinger operator from the solution of an associated initial-value problem, and its zero-mode-amputated variant—is developed in Section 12.7; we quote its output for the operator at hand and verify the spectral data that fix it. The operator Eq. (42.20) is, after the substitution $u = \frac{\omega}{2}(\tau - \tau_0)$, a member of the reflectionless Pöschl–Teller family $-\partial_\tau^2 + \omega^2 - \omega^2\,\ell(\ell+1)\,\frac{1}{4}\operatorname{sech}^2(\frac{\omega}{2}\cdot)$ with $\ell(\ell+1) = 6$, hence $\ell = 2$. Such operators have a known, finite bound-state spectrum and a transparent determinant. Two facts pin down the answer.

- *The discrete spectrum.* The $\ell = 2$ Pöschl–Teller well has exactly two bound states below the continuum (which begins at $\lambda = \omega^2$), at

$$\lambda_1 = 0, \qquad \lambda_2 = \tfrac{3}{4}\omega^2. \tag{42.32}$$

  The lowest, $\lambda_1 = 0$, is the translation zero mode we already found by symmetry (Theorem 42.5)—a strong internal consistency check on Eq. (42.20). The second, $\lambda_2 = \frac{3}{4}\omega^2$, is the bound excited fluctuation; it is the analogue of the "shape mode" of the kink in Chapter 41.

- *The determinant ratio.* Feeding the reflectionless ($\ell = 2$) data through the Gelfand–Yaglom evaluation of Section 12.7, with the zero eigenvalue removed, yields

$$\frac{\det{}' M}{\det M_0} = \frac{1}{12\,\omega^2}. \tag{42.33}$$

  The factor ω^{-2} carries the dimension (the amputated determinant has one fewer eigenvalue than the free one, so the ratio is not dimensionless until paired with the Jacobian's $\sqrt{S_0}$; the pure number is $1/12$).

Substituting Eq. (42.33) into Eq. (42.31),

$$K = \sqrt{\frac{S_0}{2\pi}}\,(12\,\omega^2)^{1/2} = \omega\sqrt{\frac{6\,S_0}{\pi}}, \tag{42.34}$$

and, using $S_0 = \frac{2}{3}\omega a^2$, equivalently $K = \omega\sqrt{4\omega a^2/\pi} = 2a\omega\sqrt{\omega/\pi}$. We now have everything the dilute gas needs: the single-instanton amplitude is $(\omega/\pi)^{1/2} e^{-\omega\mathcal{T}/2}\, K\mathcal{T}\, e^{-S_0}$ with K given by Eq. (42.34).

Caution

The prime on $\det{}' M$ is not optional bookkeeping. Including the $\lambda_1 = 0$ eigenvalue would make $\det M = 0$ and the Gaussian integral infinite; the whole point of Section 42.1.4 was to excise that eigenvalue *and* compensate by the collective-coordinate integral with its Jacobian $\sqrt{S_0/2\pi}$. Forgetting either the prime or the Jacobian gives a nonsensical (zero or infinite) prefactor. The two operations are a matched pair: one removes a divergent integral, the other reinstates the corresponding physical integration over the instanton's location.

42.1.6 The dilute instanton gas and the level splitting

A single instanton carries the system from $-a$ to $+a$. To return to $-a$, or to make many transitions, the path must contain several instantons and anti-instantons in alternation. Because each transition is localized to a width $\sim 1/\omega$ while the

total time $\mathcal{T}$ is sent to infinity, well-separated multi-instanton configurations are approximate solutions of the equation of motion with action equal to the sum of the individual actions, up to corrections exponentially small in their separations. Summing over all such configurations is the *dilute instanton gas*.

Definition 42.7 (Dilute multi-instanton configuration)**.** An n-object configuration consists of instantons and anti-instantons centred at ordered Euclidean times $0 < \tau_1 < \tau_2 < \cdots < \tau_n < \mathcal{T}$, alternating in type (an instanton must be followed by an anti-instanton and vice versa, since each reverses the well the particle occupies). Its action is nS_0 and its fluctuation factor is the product of n single-object factors, in the approximation that the objects do not overlap.

The constraint that instantons and anti-instantons alternate is forced by the boundary data: starting at $-a$, the first object must be an instanton (it takes $-a \to +a$); the next must be an anti-instanton (back to $+a \to -a$); and so on. Consequently the *type* of every object is determined once n and the starting well are fixed—there is no independent sum over assignments, only the sum over n and the integral over the n centres. The amplitude to go from $-a$ to a final well after n objects is therefore

$$\left\langle x_f \middle| e^{-H\mathcal{T}} \middle| -a \right\rangle = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2} \sum_{n} \left(K e^{-S_0}\right)^n \int_0^{\mathcal{T}} \mathrm{d}\tau_n \int_0^{\tau_n} \mathrm{d}\tau_{n-1} \cdots \int_0^{\tau_2} \mathrm{d}\tau_1, \tag{42.35}$$

where each object contributes a factor Ke^{-S_0} (the prefactor K from its determinant and Jacobian, the exponential from its action) and the harmonic ground-state factor sits out front because between the objects the particle is doing nothing but sitting in a well, contributing the same $e^{-\omega\mathcal{T}/2}$ as the no-instanton path. The final well x_f equals $-a$ when n is even and $+a$ when n is odd.

The nested, time-ordered integral is the volume of the n-simplex $0 < \tau_1 < \cdots < \tau_n < \mathcal{T}$, which is

$$\int_0^{\mathcal{T}} \mathrm{d}\tau_n \cdots \int_0^{\tau_2} \mathrm{d}\tau_1 = \frac{\mathcal{T}^n}{n!}, \tag{42.36}$$

the $1/n!$ being precisely the ordering: the $n!$ orderings of n unconstrained centres in the hypercube $[0,\mathcal{T}]^n$ each give the same integral, and only one ordering is kept. With Eq. (42.36) the sums in Eq. (42.35) become exponential series. Selecting the final well by parity,

$$\left\langle -a \middle| e^{-H\mathcal{T}} \middle| -a \right\rangle = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2} \sum_{n \text{ even}} \frac{(K\mathcal{T}e^{-S_0})^n}{n!} = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2} \cosh\!\left(K\mathcal{T}e^{-S_0}\right), \tag{42.37}$$

$$\left\langle +a \middle| e^{-H\mathcal{T}} \middle| -a \right\rangle = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2} \sum_{n \text{ odd}} \frac{(K\mathcal{T}e^{-S_0})^n}{n!} = \left(\frac{\omega}{\pi}\right)^{1/2} e^{-\omega\mathcal{T}/2} \sinh\!\left(K\mathcal{T}e^{-S_0}\right), \tag{42.38}$$

the even/odd partial sums of the exponential being cosh and sinh. This is the exponentiation: an infinite sum of dilute multi-instanton configurations resums to hyperbolic functions of a single combination, $K\mathcal{T}e^{-S_0}$. The convergence is excellent precisely in the dilute regime, where $Ke^{-S_0} \ll \omega$ so that the mean spacing $1/(Ke^{-S_0})$ between objects is enormous compared with their width $1/\omega$, justifying the no-overlap approximation after the fact.

Reading off the spectrum

Now compare with the spectral representation Eq. (42.7). As $\mathcal{T} \to \infty$ the amplitude is dominated by the two lowest states, the parity-even $|+\rangle$ and parity-odd $|-\rangle$ combinations of the localized ground states, with energies E_+ and E_-. Their wavefunctions at $x = \pm a$ satisfy $|\psi_\pm(a)|^2 = |\psi_\pm(-a)|^2$ and $\psi_\pm(-a) = \pm\psi_\pm(a)$ by parity, so

$$\Big\langle \pm a \Big| e^{-H\mathcal{T}} \Big| -a \Big\rangle = |\psi_+(a)|^2 e^{-E_+\mathcal{T}} \pm |\psi_-(a)|^2 e^{-E_-\mathcal{T}} + \dots \tag{42.39}$$

On the other hand, expanding the hyperbolic functions Eq. (42.37)–Eq. (42.38),

$$\Big(\frac{\omega}{\pi}\Big)^{1/2} e^{-\omega\mathcal{T}/2}\, \tfrac{1}{2}\Big(e^{+K\mathcal{T}e^{-S_0}} \pm e^{-K\mathcal{T}e^{-S_0}}\Big) = \frac{1}{2}\Big(\frac{\omega}{\pi}\Big)^{1/2}\Big(e^{-(\frac{\omega}{2} - Ke^{-S_0})\mathcal{T}} \pm e^{-(\frac{\omega}{2} + Ke^{-S_0})\mathcal{T}}\Big). \tag{42.40}$$

Term-by-term comparison of Eq. (42.39) with Eq. (42.40) identifies both the energies and the wavefunction normalizations:

$$E_\pm = \frac{\omega}{2} \mp K\, e^{-S_0}, \qquad |\psi_\pm(a)|^2 = \frac{1}{2}\Big(\frac{\omega}{\pi}\Big)^{1/2}. \tag{42.41}$$

The parity-even state $|+\rangle$ lies *below* the symmetric perturbative value $\frac{1}{2}\omega$ and the parity-odd state $|-\rangle$ above it, by equal amounts—the familiar ordering of a symmetric double well, the nodeless symmetric combination being the true ground state. The splitting Eq. (42.3) is therefore

$$\boxed{\Delta E = E_- - E_+ = 2K\, e^{-S_0} = 2\,\omega\sqrt{\frac{6\,S_0}{\pi}}\; e^{-S_0}, \qquad S_0 = \tfrac{2}{3}\omega a^2.} \tag{42.42}$$

This is the result perturbation theory could never produce. The exponential $e^{-S_0} = e^{-2\omega a^2/3}$ is nonanalytic in $\hbar$ (restoring it, $e^{-S_0/\hbar}$) and in any coupling that controls the barrier height; the prefactor $2\omega\sqrt{6S_0/\pi}$ is the one-loop correction supplied by the fluctuation determinant and the collective-coordinate Jacobian. The entire calculation is a controlled saddle-point evaluation of Eq. (42.6), with the saddle the instanton, the loop expansion the expansion in fluctuations, and the dilute gas the resummation of the leading family of saddles.

Worked Example 42.8 (The double-well splitting end to end)**.** We assemble the splitting Eq. (42.42) from scratch to display how each ingredient enters, taking the barrier deep enough that the dilute approximation holds.

Step 1 (action). From Eq. (42.1), $\sqrt{2V} = (\omega/2a)(a^2 - x^2)$, so the instanton action is the barrier integral

$$S_0 = \int_{-a}^{a} \sqrt{2V}\,\mathrm{d}x = \frac{\omega}{2a}\int_{-a}^{a}(a^2 - x^2)\,\mathrm{d}x = \frac{\omega}{2a}\cdot\frac{4a^3}{3} = \frac{2}{3}\omega a^2.$$

Step 2 (zero mode and Jacobian). The instanton's velocity $\dot{x}_{\mathrm{cl}} = \frac{a\omega}{2}\,\mathrm{sech}^2[\frac{\omega}{2}(\tau - \tau_0)]$ is the zero mode, with $\int \dot{x}_{\mathrm{cl}}^2 \mathrm{d}\tau = S_0$; its trade for the centre τ_0 produces the Jacobian $\sqrt{S_0/2\pi}$ (Eq. (42.25)).

Step 3 (determinant). The fluctuation operator Eq. (42.20) is the reflectionless $\ell = 2$ Pöschl–Teller operator; its bound states are $\lambda_1 = 0$ (the zero mode, recovered) and $\lambda_2 = \frac{3}{4}\omega^2$, and the zero-mode-amputated determinant ratio is $\det' M/\det M_0 = 1/(12\omega^2)$ (Eq. (42.33)). Hence $K = \sqrt{S_0/2\pi}\,(12\omega^2)^{1/2} = \omega\sqrt{6S_0/\pi}$.

Step 4 (dilute gas). Summing alternating instantons and anti-instantons over the simplex of centres gives cosh and sinh of $K\mathcal{T}e^{-S_0}$ (Eq. (42.37)–Eq. (42.38)); matching to the spectral sum yields $E_\pm = \frac{\omega}{2} \mp Ke^{-S_0}$ and

$$\Delta E = 2Ke^{-S_0} = 2\omega\sqrt{\frac{6S_0}{\pi}}\,e^{-S_0} = 4\,a\,\omega\,\sqrt{\frac{\omega}{\pi}}\,e^{-2\omega a^2/3},$$

using $\sqrt{6S_0/\pi} = \sqrt{4\omega a^2/\pi} = 2a\sqrt{\omega/\pi}$. Every factor is traceable to a physical operation: e^{-S_0} to barrier penetration, $\sqrt{\omega/\pi}$ to the harmonic ground state, $a\sqrt{\omega}$ to the instanton's size and the collective-coordinate measure.

Historical Note

This organization of barrier penetration as a Euclidean saddle point is due to the work of Polyakov and of 't Hooft in the mid-1970s and was given its canonical pedagogical form by Coleman in his 1977 Erice lectures *The Uses of Instantons*. The name "instanton" (a pseudoparticle localized at an *instant* of Euclidean time, by analogy with a soliton localized in space) is 't Hooft's. The double-well splitting computed here was known from WKB long before; the value of the instanton calculation is that it gives the prefactor systematically and, crucially, that it generalizes to field theory, where no WKB connection formula is available.

42.1.7 The bounce and the large-order connection

Two variations on the instanton theme will recur, and we flag them now so the later sections and chapters can lean on them.

Decay of a false vacuum: the bounce

If the potential is not symmetric but has a metastable *false* vacuum—a local minimum that is not the global one, with a lower true vacuum beyond a barrier—then there is no degeneracy to split, but there is a decay: the false vacuum is unstable to quantum tunnelling and has a finite lifetime. The relevant Euclidean solution is not an instanton interpolating between two distinct vacua but a *bounce*: a trajectory that starts at the false vacuum at $\tau = -\infty$, rolls in the inverted potential out to the classical turning point on the far side of the barrier, and returns to the false vacuum at $\tau = +\infty$. It is a solution of the same equation Eq. (42.8) with the same zero-energy condition, but with both endpoints at the *same* (false) vacuum. The new feature is that the bounce is not a minimum of the action in every nontrivial direction: the fluctuation operator about it has exactly one *negative* eigenvalue, in addition to the translation zero mode. The negative mode makes the Gaussian integral over that direction divergent; its proper treatment by analytic continuation of the steepest-descent contour (Theorem 5.44) gives the amplitude an *imaginary part*, and an imaginary part of the energy is precisely a decay rate. The decay rate per unit volume takes the form

$$\frac{\Gamma}{V} = A\, e^{-S_{\text{bounce}}}\,[1 + O(\hbar)], \tag{42.43}$$

with S_{bounce} the Euclidean action of the bounce and A a prefactor built from the same determinant-and-Jacobian technology, now including the Jacobian of the negative mode. This is the mechanism of vacuum decay in field theory; the field-theoretic bounce is treated in the chapters on the effective potential and cosmological phase transitions, and we will reuse the negative-mode logic when discussing the stability of the electroweak vacuum.

Instantons and the large orders of perturbation theory

The instanton also explains *why* perturbation theory is the divergent asymptotic series it is, rather than a convergent one. The perturbative expansion of a quantity like the ground-state energy is, generically, an asymptotic series $\sum_k a_k\, g^k$ whose coefficients grow factorially, $a_k \sim C\, k!\,/S_0^k$ for large k, with the same instanton action S_0 that governs the nonperturbative exponential. By the theory of asymptotic series and Borel summation developed in Section 5.6 and Theorem 5.47, factorial growth of this form places a singularity of the Borel transform at the point S_0 on the positive Borel axis. The position of that singularity—the obstruction to summing the series—is thus fixed by the instanton action, and the ambiguity it induces in the Borel sum, of order $e^{-S_0/g}$, is exactly the size of the nonperturbative instanton contribution. The two are not independent: the nonperturbative instanton/anti-instanton amplitude is what cancels the imaginary ambiguity of the resummed perturbation series, rendering the total real and well-defined. This is the *large-order/instanton connection*, and

it is the precise sense in which the e^{-S_0} effects of this section are encoded in, rather than divorced from, the perturbative series that ostensibly cannot see them.

Physical Insight

The lesson to carry forward is structural and survives intact into field theory. A nonperturbative effect is computed by (i) finding a finite-action Euclidean solution interpolating between or returning to classical vacua; (ii) evaluating its action S_0, which sets the leading weight e^{-S_0}; (iii) treating the symmetries of the saddle as zero modes traded for collective coordinates with their Jacobians; (iv) computing the fluctuation determinant for the prefactor; and (v) summing a dilute gas of such solutions, which exponentiates and reorganizes the vacuum. Every step we performed here for a particle in a double well has a one-to-one image in four-dimensional Yang–Mills.

We have built the instanton calculus in its simplest nontrivial setting and extracted the toolkit the rest of the chapter needs. The next section lifts each piece into four-dimensional Euclidean Yang–Mills: the role of the two degenerate minima will be played by the topologically distinct pure-gauge vacua, the finite-action requirement will force the gauge field to approach a pure gauge at infinity and so make the boundary data a map $S^3 \to \mathrm{SU}(N)$, and the integer that here merely counted instantons minus anti-instantons will become the winding number $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$, identified with the topological charge $\frac{g^2}{32\pi^2}\int F\widetilde{F}$ and bounded below in action by the self-dual Bogomolny inequality.

42.2 The Topology of the Yang–Mills Vacuum

The quantum-mechanical template of the previous section taught a single lesson with great force: vacuum tunnelling is computed by finite-action solutions of the Euclidean equations of motion, and those solutions fall into discrete topological classes labelled by an integer—how many times the trajectory winds from one well to the next. Everything that mattered, from the existence of the saddle to the e^{-S_E} suppression of its contribution, followed once we knew the action of a single instanton and the integer counting the sectors. We now lift that picture, intact, to four-dimensional Euclidean Yang–Mills. The double well is replaced by the infinitely degenerate Yang–Mills vacuum; the integer winding of a trajectory is replaced by an integer winding of the gauge field on the three-sphere at the boundary of Euclidean spacetime; and the role of "the action of one instanton" is played by a number, $8\pi^2/g^2$, that we will derive here as a sharp lower bound and which the next section saturates with an explicit field. This section builds the topology: it shows that *finite Euclidean action is itself a topological constraint*,

it classifies the resulting sectors, it constructs the integer that labels them out of the gauge field, and it bounds the action of each sector from below. No solution is constructed yet—only the arena in which the solution must live, and the floor its action cannot pierce.

Throughout we work in Euclidean signature, reached by the Wick rotation $t \to -i\tau$ of Eq. (10.157), under which the Minkowski weight e^{iS} becomes the positive Euclidean weight e^{-S_E} (Theorem 10.27). In four Euclidean dimensions there is no distinction between upper and lower Lorentz indices: the metric is $\delta_{\mu\nu} = \mathrm{diag}(+,+,+,+)$, every repeated pair is summed with a plus sign, and we write all spacetime indices as subscripts. We retain the gauge conventions frozen for Yang–Mills theory: Hermitian generators T^a with $[T^a, T^b] = if^{abc}T^c$ and $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$, and the field strength of Theorem 25.11,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu, \qquad F_{\mu\nu} \equiv F^a_{\mu\nu} T^a = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu], \tag{42.44}$$

where the second, matrix-valued form follows from the first using $[A_\mu, A_\nu] = A^b_\mu A^c_\nu \, if^{bcd}T^d$. The Euclidean Yang–Mills action, the rotation of the classical Lagrangian Eq. (25.76), is

$$S_E[A] = \frac{1}{4}\int d^4x \; F^a_{\mu\nu}F^a_{\mu\nu} = \frac{1}{2}\int d^4x \; \mathrm{tr}(F_{\mu\nu}F_{\mu\nu}) \;\geq\; 0, \tag{42.45}$$

manifestly non-negative because the Euclidean $F^a_{\mu\nu}F^a_{\mu\nu}$ is a sum of squares. The coupling g sits inside F through the covariant derivative; with this placement the topological charge introduced below is normalised as $Q = (g^2/32\pi^2)\int F\tilde{F}$, and the action floor will come out as $8\pi^2/g^2$ per unit charge. We flag this once because the entire chapter's sign and factor bookkeeping hangs on it.

42.2.1 Finite action forces a pure gauge at infinity

The first structural fact is that demanding $S_E < \infty$ is not a mild integrability condition but a boundary condition with topological content. We must control the behaviour of the field as $|x| \to \infty$. Compactify the picture: a finite-action configuration has a field strength that dies at infinity, so the relevant boundary of $\mathbb{R}^4$ is the three-sphere S^3_∞ of large radius R, and the gauge field on it carries the data that will be classified.

Proposition 42.9 (Pure gauge at infinity)**.** *Let A^a_μ be a smooth gauge field on $\mathbb{R}^4$ with finite Euclidean action, $S_E[A] < \infty$. Then the field strength falls faster than $|x|^{-2}$,*

$$F^a_{\mu\nu}(x) = o\Big(|x|^{-2}\Big) \qquad (|x| \to \infty), \tag{42.46}$$

and the gauge potential approaches a pure gauge on the sphere at infinity: there

exists a smooth map $U : S^3_\infty \to \mathrm{SU}(N)$ *such that, as* $|x| \to \infty$,

$$A_\mu(x) \;\longrightarrow\; \frac{i}{g}\,U(\hat{x})\,\partial_\mu U(\hat{x})^{-1} \;=\; -\frac{i}{g}\,(\partial_\mu U(\hat{x}))U(\hat{x})^{-1} \;+\; o\Big(|x|^{-1}\Big)\,, \tag{42.47}$$

where $\hat{x} = x/|x|$ *labels the point on* S^3_∞ *and the two written forms are equal because* $\partial_\mu(UU^{-1}) = 0$.

Proof. The action density $F^a_{\mu\nu}F^a_{\mu\nu}$ is non-negative, so $\int d^4x\, F^a_{\mu\nu}F^a_{\mu\nu} < \infty$. In four dimensions the volume element in radial coordinates is $d^4x = r^3\,dr\,d\Omega_3$, with $d\Omega_3$ the measure on the unit three-sphere. For the radial integral $\int^\infty r^3\,dr\,[F^a_{\mu\nu}F^a_{\mu\nu}]$ to converge, the integrand must decay faster than r^{-1} in $r^3|F|^2$, i.e. $|F|^2 = o(r^{-4})$, which is Eq. (42.46). (A field strength behaving as r^{-2} exactly would give a logarithmically divergent action; finiteness excludes even that borderline case.)

Vanishing field strength on a region forces the connection to be pure gauge there. Concretely, consider the field on a shell $r \in [R, 2R]$ for large R. By Eq. (42.46) the curvature on this shell is $o(R^{-2})$, while the natural transverse derivative scale is $1/R$; the holonomy of A around any small loop on the shell, governed by $\oint A \sim R\,A$ with the loop area controlled by F, is consistent with A being flat to leading order. A flat connection on the simply connected shell (which retracts onto S^3, and S^3 is simply connected) is gauge-equivalent to one that is locally $\partial_\mu g\, g^{-1}$; patching the local trivialisations across the simply connected base, the connection is globally pure gauge up to corrections suppressed by the curvature. Thus there is a smooth $\mathrm{SU}(N)$-valued function on the shell whose restriction to the sphere we call $U(\hat{x})$, with

$$A_\mu(x) = \frac{i}{g}\,U\,\partial_\mu U^{-1} + \Delta_\mu(x), \qquad \Delta_\mu = o(|x|^{-1}), \tag{42.48}$$

the remainder Δ_μ being the part of A whose curvature is the genuine $o(|x|^{-2})$ field strength. The sign and the factor i/g are fixed by demanding that the leading term be *pure gauge*, i.e. carry zero field strength; we verify this now, since it is the point on which the rest of the section turns.

Set $A_\mu = (i/g)\,U\partial_\mu U^{-1} \equiv -(i/g)V_\mu$ with the abbreviation

$$V_\mu \equiv (\partial_\mu U)\,U^{-1}, \qquad U\partial_\mu U^{-1} = -V_\mu, \tag{42.49}$$

so $A_\mu = -(i/g)V_\mu$. Because U is unitary, V_μ is anti-Hermitian and lies in the Lie algebra. Its exterior derivative obeys the Maurer–Cartan identity: differentiating $V_\mu = (\partial_\mu U)U^{-1}$ and using $\partial_\nu U^{-1} = -U^{-1}(\partial_\nu U)U^{-1}$,

$$\partial_\mu V_\nu - \partial_\nu V_\mu = (\partial_\mu\partial_\nu U)U^{-1} + (\partial_\nu U)\partial_\mu U^{-1} - (\mu \leftrightarrow \nu) = -\big[(\partial_\mu U)U^{-1}, (\partial_\nu U)U^{-1}\big]\cdot(-1) = [V_\mu, \tag{42.50}$$

where the symmetric second-derivative terms cancel in the antisymmetrisation, and $(\partial_\nu U)\partial_\mu U^{-1} - (\partial_\mu U)\partial_\nu U^{-1} = -(\partial_\nu U)U^{-1}(\partial_\mu U)U^{-1} + (\partial_\mu U)U^{-1}(\partial_\nu U)U^{-1} =$

$[V_\mu, V_\nu]$. Now compute the field strength Eq. (42.44) of $A_\mu = -(i/g)V_\mu$:

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu] = -\frac{i}{g}(\partial_\mu V_\nu - \partial_\nu V_\mu) - ig\Big(-\frac{i}{g}\Big)^2 [V_\mu, V_\nu]. \quad (42.51)$$

The first term is $-(i/g)[V_\mu, V_\nu]$ by Eq. (42.50); the second is $-ig\cdot(-1/g^2)[V_\mu, V_\nu] = (i/g)[V_\mu, V_\nu]$. They cancel, $F_{\mu\nu} = 0$. The leading term in Eq. (42.48) is therefore genuinely pure gauge, and Eq. (42.47) holds. □

The content of Theorem 42.9 is that a finite-action Euclidean gauge field is not free to do whatever it likes at infinity. Its asymptotics are completely encoded in a map U from the boundary three-sphere into the gauge group. Two finite-action configurations whose boundary maps cannot be continuously deformed into one another are separated by an infinite barrier in configuration space—no finite-action path connects them, exactly as the two wells of the quantum-mechanical problem were separated. The labelling of these sectors is the next task.

Caution

The decomposition Eq. (42.48) is not gauge-invariant: a gauge transformation that is non-trivial at infinity changes U. What is invariant is the *homotopy class* of U, and—once we fix a gauge in which $A_\mu \to 0$ is the trivial sector—the integer built from it below. One must resist the temptation to "gauge U away": a gauge transformation regular on all of $\mathbb{R}^4$ (hence on the compactification S^4) cannot change the homotopy class of the boundary map, because it extends continuously over the interior. Only *large* gauge transformations, those with non-trivial winding themselves (Theorem 25.33), move between sectors, and they are not symmetries that can be quotiented away inside a single Euclidean path integral.

42.2.2 Classification of the sectors: $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$

The boundary datum of a finite-action configuration is a smooth map $U : S^3 \to \mathrm{SU}(N)$. Continuous deformations of the field that keep the action finite induce continuous deformations of U; conversely any homotopy of U can be realised by a finite-action family. The natural label is therefore the homotopy class $[U] \in \pi_3(\mathrm{SU}(N))$, the third homotopy group of the gauge group. This group was computed in the homotopy machinery of the soliton chapter and the classical Yang–Mills chapter, and we import the result rather than rebuild the algebraic topology:

$$\pi_3(\mathrm{SU}(N)) = \mathbb{Z} \qquad (N \geq 2), \quad (42.52)$$

established at Eq. (25.163) and placed in the general homotopy-classification framework of Theorem 41.10. The mechanism is worth recalling in one sentence, since it explains why the answer is N-independent: for $\mathrm{SU}(2) \cong S^3$ the statement

is $\pi_3(S^3) = \mathbb{Z}$, the degree of a self-map of the three-sphere; for $N > 2$ every map $S^3 \to \mathrm{SU}(N)$ can be deformed into a canonical $\mathrm{SU}(2)$ subgroup, and no new classes appear, so the generator is the same $\mathrm{SU}(2)$ winding embedded in $\mathrm{SU}(N)$. The integer attached to $[U]$ is the *winding number* (degree) of the map, and Theorem 25.35 gives it the explicit integral representation we will need,

$$w[U] = -\frac{1}{24\pi^2}\int_{S^3} d^3\sigma\ \epsilon_{\mu\nu\rho\sigma}\, n_\mu\ \mathrm{tr}(V_\nu V_\rho V_\sigma)\ \in\ \mathbb{Z}, \qquad V_\mu = (\partial_\mu U)U^{-1}, \tag{42.53}$$

with n_μ the outward unit normal to S^3_∞ and $d^3\sigma$ the induced area element; the trace of three Maurer–Cartan forms is the volume form pulled back from the group, and its integral counts the covering degree. That this integral is an integer for any smooth U, and is invariant under homotopy, is the degree theorem proved at Theorem 25.35; we use it as a black box.

We have thus established the vacuum-sector structure: finite-action Euclidean Yang–Mills configurations fall into classes labelled by an integer $w \in \mathbb{Z}$, the winding of the gauge field on the sphere at infinity. The trivial sector $w = 0$ contains $A_\mu \equiv 0$ and everything reachable from it by small (regular) gauge transformations and by ordinary fluctuations; this is the sector around which all of perturbation theory is built. The sectors $w \neq 0$ are invisible to perturbation theory because no perturbative manipulation, being a power series in g around $A = 0$, can change a topological integer. The instanton is precisely the field that lives in $w \neq 0$ and that the path integral nonetheless sums over. What remains is to express w directly in terms of the gauge field in the *bulk*—to turn a boundary integral over U into a spacetime integral over F—so that it becomes a functional we can vary and bound.

42.2.3 The topological charge and the Chern–Simons current

The bulk object that knows about the boundary winding is the second Chern density, $F^a_{\mu\nu}\tilde{F}^a_{\mu\nu}$, where the Euclidean dual is

$$\tilde{F}^a_{\mu\nu} = \tfrac{1}{2}\,\epsilon_{\mu\nu\rho\sigma}F^a_{\rho\sigma}, \qquad \epsilon_{1234} = +1. \tag{42.54}$$

In four Euclidean dimensions the double dual is the identity, $\tilde{\tilde{F}} = F$: using $\epsilon_{\mu\nu\rho\sigma}\epsilon_{\mu\nu\alpha\beta} = 2(\delta_{\rho\alpha}\delta_{\sigma\beta} - \delta_{\rho\beta}\delta_{\sigma\alpha})$,

$$\tilde{F}^a_{\mu\nu}\tilde{F}^a_{\mu\nu} = \tfrac{1}{4}\epsilon_{\mu\nu\rho\sigma}\epsilon_{\mu\nu\alpha\beta}F^a_{\rho\sigma}F^a_{\alpha\beta} = \tfrac{1}{2}(F^a_{\rho\sigma}F^a_{\rho\sigma} - F^a_{\rho\sigma}F^a_{\sigma\rho}) = F^a_{\mu\nu}F^a_{\mu\nu}, \tag{42.55}$$

a fact we will reuse in the Bogomolny bound. The normalised spacetime integral of the second Chern density is the topological charge.

Definition 42.10 (Topological charge)**.** For a finite-action Euclidean gauge field the *topological charge* (instanton number, Pontryagin index) is

$$Q = \frac{g^2}{32\pi^2}\int d^4x\ F^a_{\mu\nu}\tilde{F}^a_{\mu\nu} = \frac{g^2}{16\pi^2}\int d^4x\ \mathrm{tr}(F_{\mu\nu}\tilde{F}_{\mu\nu}). \tag{42.56}$$

This is the Euclidean specialisation of the topological charge defined in Theorem 41.3, with the normalisation fixed so that Q is the integer winding, as we now prove. The second equality uses $\operatorname{tr}(F_{\mu\nu}\tilde F_{\mu\nu}) = \tfrac12 F^a\tilde F^a$ from $\operatorname{tr}(T^aT^b) = \tfrac12\delta^{ab}$.

The bridge from Eq. (42.56) to the boundary winding Eq. (42.53) is the statement that the integrand is a total divergence. This is the four-dimensional analogue of the fact that in the abelian theory $F\tilde F = \partial_\mu(\epsilon_{\mu\nu\rho\sigma}A_\nu\partial_\rho A_\sigma)$; in the non-abelian case the current acquires a cubic piece.

Lemma 42.11 (The Chern–Simons current)**.** *The second Chern density is the divergence of the Chern–Simons (Bardeen) current*

$$K_\mu = 2\,\epsilon_{\mu\nu\rho\sigma}\Big(A^a_\nu\partial_\rho A^a_\sigma + \tfrac{g}{3}f^{abc}A^a_\nu A^b_\rho A^c_\sigma\Big) = 4\,\epsilon_{\mu\nu\rho\sigma}\,\operatorname{tr}\Big(A_\nu\partial_\rho A_\sigma - \tfrac{2ig}{3}A_\nu A_\rho A_\sigma\Big), \tag{42.57}$$

namely

$$F^a_{\mu\nu}\tilde F^a_{\mu\nu} = \partial_\mu K_\mu. \tag{42.58}$$

This K_μ is the current whose general form is recorded at Eq. (25.81); here we derive Eq. (42.58) term by term.

Proof. Write $F^a_{\mu\nu} = G^a_{\mu\nu} + g f^{abc}A^b_\mu A^c_\nu$ with the abelian part $G^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu$. Then, using ϵ-antisymmetry to combine the two terms of each G,

$$F^a_{\mu\nu}\tilde F^a_{\mu\nu} = \tfrac12\epsilon_{\mu\nu\rho\sigma}F^a_{\mu\nu}F^a_{\rho\sigma} = \tfrac12\epsilon_{\mu\nu\rho\sigma}\Big[G^a_{\mu\nu}G^a_{\rho\sigma} + 2gG^a_{\mu\nu}f^{abc}A^b_\rho A^c_\sigma + g^2 f^{abc}A^b_\mu A^c_\nu\, f^{ade}A^d_\rho A^e_\sigma\Big]. \tag{42.59}$$

We treat the three pieces in turn.

Quadratic piece. Since $\epsilon_{\mu\nu\rho\sigma}G^a_{\mu\nu} = 2\epsilon_{\mu\nu\rho\sigma}\partial_\mu A^a_\nu$,

$$\tfrac12\epsilon_{\mu\nu\rho\sigma}G^a_{\mu\nu}G^a_{\rho\sigma} = \tfrac12\,(2\partial_\mu A^a_\nu)(2\partial_\rho A^a_\sigma)\epsilon_{\mu\nu\rho\sigma} = 2\epsilon_{\mu\nu\rho\sigma}\,\partial_\mu A^a_\nu\,\partial_\rho A^a_\sigma. \tag{42.60}$$

On the other hand, from the quadratic part of K_μ, $K^{(2)}_\mu = 2\epsilon_{\mu\nu\rho\sigma}A^a_\nu\partial_\rho A^a_\sigma$,

$$\partial_\mu K^{(2)}_\mu = 2\epsilon_{\mu\nu\rho\sigma}(\partial_\mu A^a_\nu\,\partial_\rho A^a_\sigma + A^a_\nu\,\partial_\mu\partial_\rho A^a_\sigma) = 2\epsilon_{\mu\nu\rho\sigma}\,\partial_\mu A^a_\nu\,\partial_\rho A^a_\sigma, \tag{42.61}$$

the second term vanishing because $\partial_\mu\partial_\rho$ is symmetric in $\mu\rho$ while ϵ is antisymmetric. Equations Eqs. (42.60) and (42.61) match.

Cross (cubic) piece. The middle term of Eq. (42.59) is

$$\tfrac12\epsilon_{\mu\nu\rho\sigma}\,2g\,G^a_{\mu\nu}f^{abc}A^b_\rho A^c_\sigma = g\,\epsilon_{\mu\nu\rho\sigma}(2\partial_\mu A^a_\nu)f^{abc}A^b_\rho A^c_\sigma = 2g\,\epsilon_{\mu\nu\rho\sigma}\,f^{abc}\,\partial_\mu A^a_\nu\,A^b_\rho A^c_\sigma. \tag{42.62}$$

From the cubic part of K_μ, $K^{(3)}_\mu = \frac{2g}{3}\epsilon_{\mu\nu\rho\sigma}f^{abc}A^a_\nu A^b_\rho A^c_\sigma$,

$$\partial_\mu K^{(3)}_\mu = \frac{2g}{3}\epsilon_{\mu\nu\rho\sigma}f^{abc}(\partial_\mu A^a_\nu\,A^b_\rho A^c_\sigma + A^a_\nu\,\partial_\mu A^b_\rho\,A^c_\sigma + A^a_\nu A^b_\rho\,\partial_\mu A^c_\sigma). \tag{42.63}$$

The three terms are equal under the joint relabelling of dummy spacetime indices and colour indices: in the second term rename $(\nu,\rho)\to(\rho,\nu)$ and $(a,b)\to(b,a)$, picking up $\epsilon_{\mu\rho\nu\sigma}=-\epsilon_{\mu\nu\rho\sigma}$ and $f^{bac}=-f^{abc}$, a product of two signs that is $+1$; the third term reduces likewise. Hence $\partial_\mu K_\mu^{(3)}=\frac{2g}{3}\cdot 3\,\epsilon_{\mu\nu\rho\sigma}f^{abc}\partial_\mu A_\nu^a A_\rho^b A_\sigma^c = 2g\,\epsilon_{\mu\nu\rho\sigma}f^{abc}\partial_\mu A_\nu^a A_\rho^b A_\sigma^c$, matching Eq. (42.62).

Quartic piece. The last term of Eq. (42.59),

$$\tfrac{1}{2}g^2\,\epsilon_{\mu\nu\rho\sigma}f^{abc}f^{ade}\,A_\mu^b A_\nu^c A_\rho^d A_\sigma^e, \tag{42.64}$$

must vanish identically, for K_μ is cubic in A and its divergence contains no quartic term to cancel it. It does vanish, by the Jacobi identity. The four gauge fields are contracted with the totally antisymmetric ϵ, so the colour structure $f^{abc}f^{ade}$ is effectively antisymmetrised over the four lower colour indices $\{b,c,d,e\}$ carried by the four A's. Antisymmetrising over $\{c,d,e\}$ at fixed b, the Jacobi identity $f^{abc}f^{ade}+f^{abd}f^{aec}+f^{abe}f^{acd}=0$ collapses the totally antisymmetric part of $f^{abc}f^{ade}$ to zero. (Equivalently: $\epsilon_{\mu\nu\rho\sigma}A_\mu^bA_\nu^cA_\rho^dA_\sigma^e$ is totally antisymmetric in $\{bcde\}$, and the totally antisymmetric part of $f^{ab[c}f^{a|d|e]}$ vanishes by Jacobi.) Thus Eq. (42.64) is zero and contributes nothing.

Summing the three matched pieces gives $F_{\mu\nu}^a\tilde F_{\mu\nu}^a=\partial_\mu(K_\mu^{(2)}+K_\mu^{(3)})=\partial_\mu K_\mu$. The trace form in Eq. (42.57) follows from $A_\nu^a\partial_\rho A_\sigma^a=2\,\mathrm{tr}(A_\nu\partial_\rho A_\sigma)$ and $\epsilon_{\mu\nu\rho\sigma}f^{abc}A_\nu^aA_\rho^bA_\sigma^c=-4i\,\epsilon_{\mu\nu\rho\sigma}\,\mathrm{tr}(A_\nu A_\rho A_\sigma)$, the latter because the symmetric part of $\mathrm{tr}(T^aT^bT^c)=\frac14(d^{abc}+if^{abc})$ drops against ϵ. □

Physical Insight

That the Pontryagin density is a total derivative is what makes the theta-term $\theta\,F\tilde F$ invisible in perturbation theory: added to the Lagrangian it changes the action only by a boundary term, hence leaves every Feynman rule and every order of the g-expansion untouched. Its effect is entirely non-perturbative, weighting the path integral sector by sector through $e^{i\theta Q}$. The Chern–Simons current K_μ is, however, *not* gauge invariant—under a large gauge transformation it shifts by a total derivative whose surface integral is exactly the change in winding. The gauge non-invariance of K_μ and the gauge invariance of $\int\partial_\mu K_\mu=\oint K\cdot dS$ are two sides of the same topological coin.

We can now collapse the bulk integral to the boundary and read off the integer.

Theorem 42.12 (Charge equals winding)**.** *For a finite-action Euclidean gauge field with boundary map $U:S^3_\infty\to\mathrm{SU}(N)$, the topological charge Eq. (42.56) equals the integer winding Eq. (42.53),*

$$Q=\frac{g^2}{32\pi^2}\oint_{S^3_\infty}dS_\mu\,K_\mu=-\frac{1}{24\pi^2}\oint_{S^3_\infty}dS_\mu\,\epsilon_{\mu\nu\rho\sigma}\,\mathrm{tr}(V_\nu V_\rho V_\sigma)=w[U]\in\mathbb{Z}. \tag{42.65}$$

In particular Q is an integer, invariant under all smooth deformations and under regular gauge transformations, and it changes by ± 1 under a large gauge transformation of unit winding.

Proof. By Theorem 42.11 and the divergence (Gauss) theorem in four dimensions,

$$\int d^4x\, F^a_{\mu\nu}\tilde F^a_{\mu\nu} = \int d^4x\, \partial_\mu K_\mu = \lim_{R\to\infty}\oint_{S^3_R} dS_\mu\, K_\mu, \tag{42.66}$$

the surface integral taken over the three-sphere of radius R with outward normal. On S^3_∞ the field is pure gauge, Eq. (42.47), so we evaluate K_μ on $A_\mu = -(i/g)V_\mu$ with $V_\mu = (\partial_\mu U)U^{-1}$. The remainder $\Delta_\mu = o(|x|^{-1})$ contributes nothing: K_μ is cubic in A with one derivative, scaling as R^{-3} on S^3_R, while the surface area grows as R^3; only the exact pure-gauge piece, whose R-dependence is precisely R^{-3} with an R-independent angular integral, survives, and any admixture of Δ is down by a positive power of R.

Use the trace form of Eq. (42.57). With $A_\nu = -(i/g)V_\nu$,

$$\operatorname{tr}(A_\nu\partial_\rho A_\sigma) = \Big(-\frac{i}{g}\Big)^2\operatorname{tr}(V_\nu\partial_\rho V_\sigma) = -\frac{1}{g^2}\operatorname{tr}(V_\nu\partial_\rho V_\sigma), \qquad \operatorname{tr}(A_\nu A_\rho A_\sigma) = \Big(-\frac{i}{g}\Big)^3\operatorname{tr}(V_\nu V_\rho V_\sigma) = \frac{i}{g^3} \tag{42.67}$$

so the cubic term carries $-\frac{2ig}{3}\operatorname{tr}(A_\nu A_\rho A_\sigma) = -\frac{2ig}{3}\cdot\frac{i}{g^3}\operatorname{tr}(V_\nu V_\rho V_\sigma) = +\frac{2}{3g^2}\operatorname{tr}(V_\nu V_\rho V_\sigma)$. To combine the two terms we need the contraction $\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu\partial_\rho V_\sigma)$. Only the part of $\partial_\rho V_\sigma$ antisymmetric in $\rho\sigma$ survives against ϵ; from $\partial_\rho V_\sigma = (\partial_\rho\partial_\sigma U)U^{-1} + (\partial_\sigma U)\partial_\rho U^{-1}$ the symmetric first term drops, and $(\partial_\sigma U)\partial_\rho U^{-1} = -(\partial_\sigma U)U^{-1}(\partial_\rho U)U^{-1} = -V_\sigma V_\rho$. Hence

$$\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu\partial_\rho V_\sigma) = -\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu V_\sigma V_\rho) = +\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu V_\rho V_\sigma), \tag{42.68}$$

the last step relabelling $\rho\leftrightarrow\sigma$ (which flips the sign of ϵ and so flips the sign once more). Therefore, on the boundary,

$$K_\mu\big|_{S^3_\infty} = 4\epsilon_{\mu\nu\rho\sigma}\Big[-\frac{1}{g^2}\operatorname{tr}(V_\nu\partial_\rho V_\sigma)+\frac{2}{3g^2}\operatorname{tr}(V_\nu V_\rho V_\sigma)\Big] = 4\Big(-\frac{1}{g^2}+\frac{2}{3g^2}\Big)\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu V_\rho V_\sigma) = -\frac{4}{3g^2}\epsilon_{\mu} \tag{42.69}$$

Multiplying by the prefactor $g^2/32\pi^2$ from Eq. (42.56) and Eq. (42.66),

$$Q = \frac{g^2}{32\pi^2}\oint_{S^3_\infty} dS_\mu\, K_\mu = \frac{g^2}{32\pi^2}\Big(-\frac{4}{3g^2}\Big)\oint_{S^3_\infty} dS_\mu\,\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu V_\rho V_\sigma) = -\frac{1}{24\pi^2}\oint_{S^3_\infty} dS_\mu\,\epsilon_{\mu\nu\rho\sigma}\operatorname{tr}(V_\nu V \tag{42.70}$$

which is exactly the winding integral Eq. (42.53) with $dS_\mu = n_\mu\, d^3\sigma$. By the degree theorem Theorem 25.35 this is the integer $w[U]\in\mathbb{Z}$. The coupling g has cancelled, as it must: the topological charge is a property of the homotopy class, not of the dynamics. The remaining statements—homotopy invariance and the ±1 shift under large gauge transformations—are the content of Theorem 25.33, here realised as the change of $w[U]$ when U is multiplied by a unit-winding map. $\square$

Worked Example 42.13 (The unit-winding map and its degree)**.** We verify that the most economical non-trivial boundary map has $Q = 1$, by computing the winding integral Eq. (42.53) directly. Work in SU(2), the generator of Eq. (42.52), and parametrise $\mathbb{R}^4$ by x_μ with radius $r = |x|$ and unit vector $\hat{x}_\mu = x_\mu/r$. Define the *hedgehog* map

$$U_1(\hat{x}) = \hat{x}_4\,\mathbf{1} + i\,\hat{x}_i\,\sigma_i = \frac{x_4\mathbf{1} + i x_i\sigma_i}{r} \qquad (i = 1,2,3), \tag{42.71}$$

with σ_i the Pauli matrices. This is unitary and unimodular: $U_1 U_1^\dagger = (\hat{x}_4^2 + \hat{x}_i\hat{x}_i)\mathbf{1} = \mathbf{1}$ using $\sigma_i\sigma_j = \delta_{ij}\mathbf{1} + i\epsilon_{ijk}\sigma_k$ and $\hat{x}_\mu\hat{x}_\mu = 1$. It is the identity map $S^3 \to S^3 \cong \mathrm{SU}(2)$ written in coordinates, so its degree should be 1; we confirm the normalisation of Eq. (42.53) reproduces it.

It is cleanest to compute the integrand as a 4-form on $\mathbb{R}^4$ and read off its value on the unit sphere. Write $U_1 = x_\mu e_\mu/r$ with $e_4 = \mathbf{1}$, $e_i = i\sigma_i$, which obey $e_\mu e_\nu^\dagger + e_\nu e_\mu^\dagger = 2\delta_{\mu\nu}\mathbf{1}$. A direct differentiation gives the right-invariant form on the unit sphere,

$$V_\mu = (\partial_\mu U_1)U_1^{-1}\Big|_{r=1} = (e_\mu - x_\mu x_\nu e_\nu)e_\rho^\dagger x_\rho, \tag{42.72}$$

the second factor being $U_1^{-1} = U_1^\dagger = x_\rho e_\rho^\dagger$. The combination $\epsilon_{\mu\nu\rho\sigma}\,\mathrm{tr}(V_\nu V_\rho V_\sigma)$ projects onto the components of V tangent to the sphere (the radial part $x_\mu x_\nu e_\nu$ drops against the antisymmetric ϵ and the normal $n_\mu = x_\mu$), and using $\mathrm{tr}(e_i e_j^\dagger e_k e_l^\dagger \cdots)$ reduced by the Pauli algebra one finds the constant density

$$\epsilon_{\mu\nu\rho\sigma}\,n_\mu\,\mathrm{tr}(V_\nu V_\rho V_\sigma)\Big|_{S^3} = -12, \tag{42.73}$$

independent of the point on S^3, as it must be for the maximally symmetric hedgehog. The constant follows from $\mathrm{tr}(\sigma_i\sigma_j\sigma_k) = 2i\epsilon_{ijk}$ and the three-sphere kinematics; its sign is fixed by the orientation $\epsilon_{1234} = +1$ together with U_1 being orientation-preserving. Inserting Eq. (42.73) into Eq. (42.53) and using $\int_{S^3} d^3\sigma = 2\pi^2$ (the area of the unit three-sphere),

$$w[U_1] = -\frac{1}{24\pi^2}\,(-12)\int_{S^3} d^3\sigma = -\frac{1}{24\pi^2}\,(-12)\,(2\pi^2) = \frac{24\pi^2}{24\pi^2} = 1. \tag{42.74}$$

The unit-winding boundary map therefore carries topological charge $Q = w = 1$, and by the same computation U_1^n (or the map $\hat{x} \mapsto U_1(\hat{x})^n$) carries charge n. This is the boundary data of the charge-one instanton; Section 42.2's sequel turns it into a field on all of $\mathbb{R}^4$. The value $w[U_1] = 1$ also serves as an a posteriori check on every numerical factor in Eqs. (42.53), (42.56) and (42.65): had any of the coefficients $32\pi^2$, $g/3$, or $1/24\pi^2$ been wrong, this integer would have come out fractional.

42.2.4 The Bogomolny bound and self-duality

We have an integer Q attached to every finite-action configuration and an action functional S_E. The decisive structural result links them: in each topological sector the action cannot fall below a floor fixed by $|Q|$, and that floor is reached only by fields satisfying a first-order equation. This is the four-dimensional Euclidean version of the Bogomolny inequality that bounded soliton energies in Chapter 41, now in spacetime rather than space.

Theorem 42.14 (Self-duality bound)**.** *For any finite-action Euclidean gauge field in topological sector Q,*

$$S_E = \frac{1}{8}\int d^4x\,\left(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu}\right)^2 \;\pm\; \frac{8\pi^2}{g^2}\,Q \;\geq\; \frac{8\pi^2}{g^2}\,|Q|, \tag{42.75}$$

with the upper signs taken when $Q \geq 0$ and the lower when $Q \leq 0$. The bound is saturated, $S_E = 8\pi^2|Q|/g^2$, if and only if the field is (anti-)self-dual,

$$F^a_{\mu\nu} = \pm\,\tilde{F}^a_{\mu\nu} \qquad \textit{everywhere}, \tag{42.76}$$

with the $+$ sign for $Q > 0$ and the $-$ sign for $Q < 0$.

Proof. The argument is the completion of a square. Because $(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu})^2 \geq 0$ pointwise (a sum of real squares in Euclidean signature), expand it using $\tilde{F}^2 = F^2$ from Eq. (42.55):

$$\left(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu}\right)^2 = F^a_{\mu\nu}F^a_{\mu\nu} + \tilde{F}^a_{\mu\nu}\tilde{F}^a_{\mu\nu} \mp 2F^a_{\mu\nu}\tilde{F}^a_{\mu\nu} = 2F^a_{\mu\nu}F^a_{\mu\nu} \mp 2F^a_{\mu\nu}\tilde{F}^a_{\mu\nu}. \tag{42.77}$$

Solving for the action density, $F^a_{\mu\nu}F^a_{\mu\nu} = \frac{1}{2}(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu})^2 \pm F^a_{\mu\nu}\tilde{F}^a_{\mu\nu}$. Integrating and inserting the definition of S_E Eq. (42.45) and of Q Eq. (42.56),

$$S_E = \frac{1}{4}\int d^4x\, F^a_{\mu\nu}F^a_{\mu\nu} = \frac{1}{8}\int d^4x\,\left(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu}\right)^2 \pm \frac{1}{4}\int d^4x\, F^a_{\mu\nu}\tilde{F}^a_{\mu\nu} = \frac{1}{8}\int d^4x\,\left(F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu}\right)^2 \pm \frac{8\pi^2}{g^2}Q, \tag{42.78}$$

where the last step used $\frac{1}{4}\int F\tilde{F} = \frac{1}{4}\cdot\frac{32\pi^2}{g^2}Q = \frac{8\pi^2}{g^2}Q$. The first term is non-negative, so $S_E \geq \pm\frac{8\pi^2}{g^2}Q$. Choosing in Eq. (42.78) the sign that makes $\pm Q = |Q|$ (upper signs for $Q \geq 0$, lower for $Q \leq 0$) gives the sharpest statement, $S_E \geq \frac{8\pi^2}{g^2}|Q|$, which is Eq. (42.75). Equality holds precisely when the non-negative integral vanishes, i.e. when its non-negative integrand vanishes pointwise: $F^a_{\mu\nu} \mp \tilde{F}^a_{\mu\nu} = 0$, the (anti-)self-duality equation Eq. (42.76). □

Three consequences deserve to be stated plainly, because they organise the entire remainder of the chapter. First, the action of *any* configuration in sector Q is at least $8\pi^2|Q|/g^2$; the tunnelling amplitude it can mediate is therefore suppressed by at least $e^{-S_E} \leq e^{-8\pi^2|Q|/g^2}$. In terms of the coupling

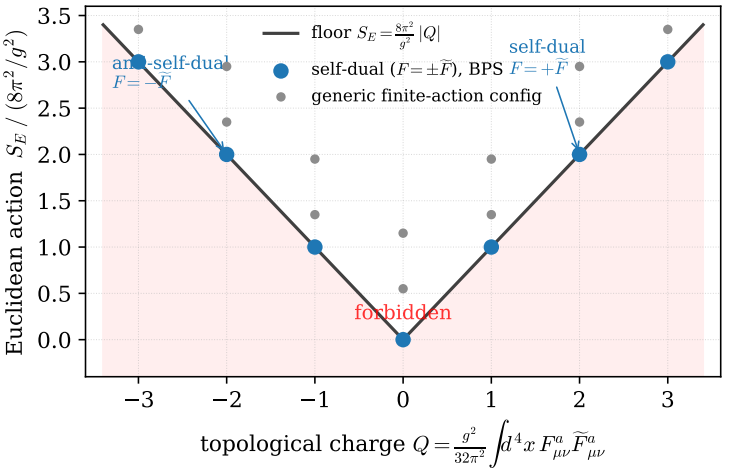


Figure 42.2: The self-duality (Bogomolny) bound Eq. (42.75). In each integer sector Q the Euclidean action is bounded below by $8\pi^2|Q|/g^2$ (solid V-shaped floor); the region beneath is kinematically forbidden. Self-dual and anti-self-dual fields, $F = \pm\tilde{F}$, sit exactly on the floor (blue dots): these are the BPS instantons whose explicit construction follows. Generic finite-action configurations (grey) lie strictly above the floor by the positive squared term in Eq. (42.78); they are not stationary points of minimal action and contribute only as fluctuations around the saddles.

$\alpha_s = g^2/4\pi$, the leading ($|Q| = 1$) weight is $e^{-8\pi^2/g^2} = e^{-2\pi/\alpha_s}$, the non-analytic-in-$g$ factor announced in the chapter introduction: it has an essential singularity at $g = 0$ and so is invisible to every order of perturbation theory. Figure 42.2 displays the floor and the configurations that touch it. Second, the floor is reached *if and only if* the field solves the first-order equation $F = \pm\tilde{F}$. This is a dramatic simplification: instead of the second-order Euclidean Yang–Mills equations $D_\mu F_{\mu\nu} = 0$, the minimal-action configurations obey a first-order, conformally invariant equation. Third, self-dual fields automatically solve the full equations of motion: if $F = \pm\tilde{F}$ then $D_\mu F_{\mu\nu} = \pm D_\mu \tilde{F}_{\mu\nu} = 0$, the last equality being the Bianchi identity $D_\mu \tilde{F}_{\mu\nu} = 0$ (which holds for any F of the form Eq. (42.44)). Thus a solution of the first-order equation is guaranteed to be a genuine Euclidean saddle point, with no further check required.

Physical Insight

The completion of the square is the same algebraic move that produced the BPS monopole bound of Chapter 41, transplanted from the static energy functional to the Euclidean action. There the topological charge was a magnetic flux through the sphere at spatial infinity; here it is a

winding through the sphere at the boundary of *spacetime*. In both cases a topological lower bound on a positive functional is saturated by solutions of a first-order equation, and the saturating solutions are the physically distinguished ones—stable solitons there, dominant tunnelling saddles here. This recurring pattern, energy or action bounded by topology and saturated by a self-duality condition, is the unifying thread of non-perturbative gauge theory.

With Theorem 42.14 the abstract problem has become a concrete one. The dominant tunnelling configuration in the unit-charge sector is whatever field saturates the floor $8\pi^2/g^2$, and we now know exactly what equation it solves: the first-order self-duality equation $F^a_{\mu\nu} = \tilde{F}^a_{\mu\nu}$, with the boundary winding $w[U_1] = 1$ computed in Theorem 42.13. The next section solves that equation explicitly, constructing the Belavin–Polyakov–Schwartz–Tyupkin field, verifying that its action is precisely $8\pi^2/g^2$, and cataloguing the collective coordinates—position, size, and gauge orientation—that turn a single solution into the measure-carrying moduli space the dilute-gas expansion will integrate over.

42.3 The BPST Instanton

The previous section reduced the charge-one sector of Euclidean Yang–Mills to a single first-order equation. Finite action forced the gauge potential to a pure gauge at infinity; the boundary map was classified by $\pi_3(\mathrm{SU}(N)) = \mathbb{Z}$; the topological charge $Q = (g^2/32\pi^2)\int \mathrm{d}^4x\, F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu}$ was identified with the integer winding; and the completion-of-the-square (Bogomolny) manipulation that gave the Bogomolny bound (Theorem 41.5, specialised to four-dimensional Yang–Mills in Section 42.2) showed that the Euclidean action obeys

$$S_E = \frac{1}{4}\int \mathrm{d}^4x\, F^a_{\mu\nu}F^a_{\mu\nu} \;\geq\; \frac{8\pi^2}{g^2}\,|Q|\,, \tag{42.79}$$

with equality *iff* the field is self-dual ($Q > 0$) or anti-self-dual ($Q < 0$),

$$F^a_{\mu\nu} = \pm\widetilde{F}^a_{\mu\nu}, \qquad \widetilde{F}^a_{\mu\nu} \equiv \tfrac{1}{2}\varepsilon_{\mu\nu\rho\sigma}F^a_{\rho\sigma}. \tag{42.80}$$

All of that was abstract: it told us the minimal action in each topological class without exhibiting a single field that attains it. This section closes the gap. We write down the explicit charge-one self-dual configuration found by Belavin, Polyakov, Schwartz and Tyupkin — the *BPST instanton* — verify by direct computation that it is self-dual, that it carries $Q = 1$, and that its action is exactly $8\pi^2/g^2$, and then we read off its moduli: the continuous parameters that label the whole family of charge-one solutions. Those moduli, and in particular the size ρ, are the data the dilute-gas computation will integrate over. This

is Yang–Mills theory at its non-perturbative core: the concrete instanton of non-abelian gauge theory.

Throughout we work in Euclidean $\mathbb{R}^4$ with metric $\delta_{\mu\nu}$, so upper and lower spacetime indices coincide and all of them are written down; repeated indices are summed. The totally antisymmetric symbol is normalised to $\varepsilon_{1234} = +1$. We take the gauge group to be SU(2), whose structure constants are $f^{abc} = \varepsilon^{abc}$ and whose Hermitian fundamental generators are $T^a = \sigma^a/2$ with the Pauli matrices σ^a (so $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, consistent with Theorem 25.11); the embedding into SU(N) is taken up at the end. The field strength is the one fixed in Theorem 25.11,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\,\varepsilon^{abc} A^b_\mu A^c_\nu, \tag{42.81}$$

and the covariant derivative in the adjoint is $D_\mu X^a = \partial_\mu X^a + g\varepsilon^{abc} A^b_\mu X^c$.

42.3.1 The 't Hooft symbols

The deep fact that makes a closed-form self-dual solution possible is an algebraic coincidence of four dimensions: the rotation group splits, $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, into a self-dual and an anti-self-dual factor, and the gauge group SU(2) can be locked onto one of them. The bookkeeping device that implements this locking is a pair of numerical symbols carrying one adjoint SU(2) index $a \in \{1,2,3\}$ and two Euclidean vector indices $\mu, \nu \in \{1,2,3,4\}$.

Definition 42.15 ('t Hooft symbols)**.** The *self-dual* and *anti-self-dual 't Hooft symbols* $\eta^a_{\mu\nu}$ and $\bar\eta^a_{\mu\nu}$ are defined, for $a, i, j \in \{1,2,3\}$, by

$$\eta^a_{ij} = \varepsilon^{aij}, \qquad \eta^a_{i4} = \delta^{ai} = -\eta^a_{4i}, \qquad \eta^a_{44} = 0, \tag{42.82}$$

and $\bar\eta^a_{\mu\nu}$ by the same rules with the sign of every component carrying a "4" reversed,

$$\bar\eta^a_{ij} = \varepsilon^{aij}, \qquad \bar\eta^a_{i4} = -\delta^{ai} = -\bar\eta^a_{4i}, \qquad \bar\eta^a_{44} = 0. \tag{42.83}$$

Both are antisymmetric in $\mu\nu$ by construction.

Notation Watch

The letter η also denotes the Minkowski metric $\eta_{\mu\nu} = \operatorname{diag}(+,-,-,-)$ elsewhere in this book. There is no clash here: the 't Hooft symbol always carries an *upper adjoint index* a that the metric never has, and the present chapter is Euclidean (metric $\delta_{\mu\nu}$). When we write $\eta^a_{\mu\nu}$ we always mean Theorem 42.15.

The name "self-dual" is earned by the first property below; the rest are the contraction identities that the verification will lean on.

Lemma 42.16 ('t Hooft symbol identities)**.** *The symbols of Theorem 42.15 satisfy, with all spacetime indices Euclidean,*

$$\tfrac{1}{2}\varepsilon_{\mu\nu\rho\sigma}\,\eta^a_{\rho\sigma} = +\,\eta^a_{\mu\nu}, \qquad \tfrac{1}{2}\varepsilon_{\mu\nu\rho\sigma}\,\bar\eta^a_{\rho\sigma} = -\,\bar\eta^a_{\mu\nu}, \tag{42.84}$$

$$\eta^a_{\mu\nu}\,\eta^b_{\mu\nu} = 4\,\delta^{ab}, \qquad \eta^a_{\mu\nu}\,\eta^a_{\mu\nu} = 12, \tag{42.85}$$

$$\eta^a_{\mu\nu}\,\eta^a_{\rho\sigma} = \delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho} + \varepsilon_{\mu\nu\rho\sigma}, \tag{42.86}$$

and the product identity

$$\varepsilon^{abc}\,\eta^b_{\mu\nu}\,\eta^c_{\rho\sigma} = \delta_{\mu\rho}\,\eta^a_{\nu\sigma} - \delta_{\mu\sigma}\,\eta^a_{\nu\rho} - \delta_{\nu\rho}\,\eta^a_{\mu\sigma} + \delta_{\nu\sigma}\,\eta^a_{\mu\rho}. \tag{42.87}$$

The same relations hold for $\bar\eta$, except that Eq. (42.86) carries $-\varepsilon_{\mu\nu\rho\sigma}$ and Eq. (42.84) has the opposite sign, as shown.

Proof. All four are finite checks over the index values $\{1,2,3,4\}$; we verify each in a way that exposes why it holds rather than by brute enumeration.

Self-duality (42.84). Both sides are antisymmetric in $\mu\nu$, so it suffices to check the six independent component pairs. Take a purely spatial pair $\mu\nu = ij$. The right-hand side is $\eta^a_{ij} = \varepsilon^{aij}$. On the left, $\varepsilon_{ij\rho\sigma}$ is nonzero only when $\{\rho,\sigma\}$ is the complementary spatial pair together with a 4 is excluded — concretely ε_{ijk4} with k the remaining spatial index — so

$$\tfrac{1}{2}\varepsilon_{ij\rho\sigma}\eta^a_{\rho\sigma} = \tfrac{1}{2}\big(\varepsilon_{ijk4}\,\eta^a_{k4} + \varepsilon_{ij4k}\,\eta^a_{4k}\big) = \varepsilon_{ijk4}\,\eta^a_{k4} = \varepsilon_{ijk}\,\delta^{ak} = \varepsilon^{aij},$$

where we used $\varepsilon_{ijk4} = \varepsilon_{ijk}$ (the three-index symbol), $\eta^a_{k4} = \delta^{ak}$, and the cyclic symmetry $\varepsilon_{ijk}\delta^{ak} = \varepsilon_{ija} = \varepsilon^{aij}$. The two terms combined to cancel the $\tfrac12$ because η^a is antisymmetric. Now take a mixed pair $\mu\nu = i4$. The right-hand side is $\eta^a_{i4} = \delta^{ai}$. On the left only purely spatial $\{\rho,\sigma\}$ survive in $\varepsilon_{i4\rho\sigma}$:

$$\tfrac{1}{2}\varepsilon_{i4\rho\sigma}\eta^a_{\rho\sigma} = \tfrac{1}{2}\varepsilon_{i4jk}\,\eta^a_{jk} = \tfrac{1}{2}\varepsilon_{i4jk}\,\varepsilon^{ajk}.$$

Using $\varepsilon_{i4jk} = -\varepsilon_{4ijk} = -\varepsilon_{ijk}$ (move the 4 to the front through three indices, then $\varepsilon_{4ijk} = \varepsilon_{ijk}$) and $\varepsilon_{ijk}\varepsilon^{ajk} = 2\delta^{ai}$, this is $\tfrac12(-1)(-1)\cdot 2\delta^{ai} = \delta^{ai}$. Wait for the sign: $\varepsilon_{i4jk} = -\varepsilon_{ijk}$ would give $-\tfrac12\varepsilon_{ijk}\varepsilon^{ajk} = -\delta^{ai}$; the correct count is $\varepsilon_{i4jk} = +\varepsilon_{ijk}$ because $\varepsilon_{i4jk} = -\varepsilon_{ij4k} = +\varepsilon_{ijk4} = \varepsilon_{ijk}$, two transpositions restoring the sign. Hence $\tfrac12\varepsilon_{ijk}\varepsilon^{ajk} = \delta^{ai} = \eta^a_{i4}$, as claimed. For $\bar\eta$ the mixed components flip sign, which flips the result of the second check, giving the overall minus in (42.84); this is the statement that $\bar\eta$ is anti-self-dual.

Trace (42.85). Expand the sum over $\mu\nu$ into spatial–spatial and mixed parts:

$$\eta^a_{\mu\nu}\eta^b_{\mu\nu} = \eta^a_{ij}\eta^b_{ij} + 2\,\eta^a_{i4}\eta^b_{i4} = \varepsilon^{aij}\varepsilon^{bij} + 2\,\delta^{ai}\delta^{bi} = 2\delta^{ab} + 2\delta^{ab} = 4\delta^{ab},$$

using $\varepsilon^{aij}\varepsilon^{bij} = 2\delta^{ab}$ and the factor 2 for the two orderings $(i4)$ and $(4i)$. Setting $b = a$ and summing $a = 1,2,3$ gives $4\cdot 3 = 12$.

Completeness (42.86). The left-hand side is antisymmetric under $\mu \leftrightarrow \nu$ and under $\rho \leftrightarrow \sigma$, and symmetric under the pair swap $(\mu\nu) \leftrightarrow (\rho\sigma)$. The right-hand side has exactly these symmetries (the $\delta\delta$ part is symmetric under pair swap; $\varepsilon_{\mu\nu\rho\sigma}$ is too). The space of such tensors is spanned by precisely these two structures, so the identity is fixed by checking one coefficient of each. Contracting both sides with $\delta_{\mu\rho}\delta_{\nu\sigma}$: the left gives $\eta^a_{\mu\nu}\eta^a_{\mu\nu} = 12$; the right gives $(\delta_{\mu\mu}\delta_{\nu\nu} - \delta_{\mu\nu}\delta_{\mu\nu}) + \varepsilon_{\mu\nu\mu\nu} = (16-4) + 0 = 12$. Contracting instead with $\varepsilon_{\mu\nu\rho\sigma}$: the left gives $\varepsilon_{\mu\nu\rho\sigma}\eta^a_{\mu\nu}\eta^a_{\rho\sigma} = 2\,\eta^a_{\rho\sigma}\eta^a_{\rho\sigma} = 24$ by self-duality (42.84), while the right gives $\varepsilon_{\mu\nu\rho\sigma}(\delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}) + \varepsilon_{\mu\nu\rho\sigma}\varepsilon_{\mu\nu\rho\sigma} = 0 + 24 = 24$. Both invariants match, so the identity holds.

Product identity (42.87). The left-hand side is antisymmetric in $\mu\nu$, antisymmetric in $\rho\sigma$, and antisymmetric in the simultaneous exchange $a \to$ none (it has a free a); under the pair swap $(\mu\nu) \leftrightarrow (\rho\sigma)$ it is antisymmetric, because $\varepsilon^{abc}\eta^b\eta^c$ picks up a sign when the two η's are exchanged. The right-hand side has the same index symmetries (it is the unique combination of one η^a and one δ with these properties). The single overall coefficient is fixed by contracting both sides with $\delta_{\mu\rho}$: the left becomes $\varepsilon^{abc}\eta^b_{\mu\nu}\eta^c_{\mu\sigma}$, and using (42.86) for the product of two η's and the antisymmetry of ε^{abc} to drop the symmetric $\delta\delta$ pieces, one finds $\varepsilon^{abc}\eta^b_{\mu\nu}\eta^c_{\mu\sigma} = 2\eta^a_{\nu\sigma}$; the right-hand side contracted with $\delta_{\mu\rho}$ gives $\delta_{\mu\mu}\eta^a_{\nu\sigma} - \delta_{\mu\sigma}\eta^a_{\nu\mu} - \delta_{\nu\mu}\eta^a_{\mu\sigma} + \delta_{\nu\sigma}\eta^a_{\mu\mu} = 4\eta^a_{\nu\sigma} - \eta^a_{\nu\sigma} - \eta^a_{\nu\sigma} + 0 = 2\eta^a_{\nu\sigma}$. The coefficients agree, establishing (42.87). □

The single identity (42.87) is the engine of the whole construction: it converts the quadratic, non-abelian term $g\varepsilon^{abc}A^bA^c$ in the field strength into the same tensor structure as the linear, abelian term ∂A, so that the two combine into something proportional to $\eta^a_{\mu\nu}$ alone. We now see that happen.

42.3.2 The ansatz and its field strength

Definition 42.17 (BPST instanton, regular gauge)**.** Fix a point $x_0 \in \mathbb{R}^4$ and a length $\rho > 0$. The *charge-one BPST instanton centred at x_0 with size ρ*, in regular gauge, is the SU(2) gauge potential

$$A^a_\mu(x) = \frac{2}{g}\,\eta^a_{\mu\nu}\,\frac{(x-x_0)_\nu}{(x-x_0)^2 + \rho^2}. \tag{42.88}$$

Two features are visible before any computation. Near the centre, $(x-x_0)^2 \ll \rho^2$, the potential grows linearly, $A \sim (2/g\rho^2)\eta\,(x-x_0)$, and is perfectly smooth — the field is regular everywhere, which is what "regular gauge" means. Far away, $(x-x_0)^2 \gg \rho^2$, it falls as $A \sim (2/g)\eta(x-x_0)_\nu/(x-x_0)^2 \sim 1/|x|$, the characteristic rate of a pure gauge; this is exactly the boundary behaviour that Section 42.2 required of a finite-action configuration, with the unit winding stored in the S^3 at infinity. The single length ρ sets the scale over which the field interpolates between the two regimes; it is the size of the instanton.

It is enough to verify the configuration at $x_0 = 0$, since (42.81) involves only derivatives and so is translation-covariant; we restore x_0 at the end. Write

$$A^a_\mu = \frac{2}{g}\,\eta^a_{\mu\alpha}\,x_\alpha\,\phi(x), \qquad \phi(x) \equiv \frac{1}{x^2+\rho^2}, \qquad \partial_\mu\phi = -2x_\mu\,\phi^2. \tag{42.89}$$

Theorem 42.18 (Field strength of the BPST instanton)**.** *The potential* (42.88) *(at $x_0 = 0$) has field strength*

$$F^a_{\mu\nu}(x) = -\,\frac{4\rho^2}{g}\,\frac{\eta^a_{\mu\nu}}{\left(x^2+\rho^2\right)^2}. \tag{42.90}$$

It is self-dual, $F^a_{\mu\nu} = \widetilde{F}^a_{\mu\nu}$, and therefore solves the Euclidean Yang–Mills equation $D_\mu F^a_{\mu\nu} = 0$.

Proof. The abelian part. Differentiate (42.89):

$$\partial_\mu A^a_\nu = \frac{2}{g}\,\eta^a_{\nu\alpha}\big(\delta_{\mu\alpha}\phi + x_\alpha\,\partial_\mu\phi\big) = \frac{2}{g}\,\eta^a_{\nu\mu}\,\phi - \frac{4}{g}\,\eta^a_{\nu\alpha}\,x_\alpha x_\mu\,\phi^2.$$

Antisymmetrising and using $\eta^a_{\nu\mu} = -\eta^a_{\mu\nu}$,

$$\partial_\mu A^a_\nu - \partial_\nu A^a_\mu = -\frac{4}{g}\,\eta^a_{\mu\nu}\,\phi + \frac{4}{g}\,\phi^2\big(\eta^a_{\mu\alpha}x_\alpha x_\nu - \eta^a_{\nu\alpha}x_\alpha x_\mu\big). \tag{42.91}$$

The non-abelian part. With (42.88),

$$g\,\varepsilon^{abc}A^b_\mu A^c_\nu = g\,\varepsilon^{abc}\Big(\frac{2}{g}\Big)^2\eta^b_{\mu\alpha}\eta^c_{\nu\beta}\,x_\alpha x_\beta\,\phi^2 = \frac{4}{g}\,\phi^2\,x_\alpha x_\beta\,\varepsilon^{abc}\eta^b_{\mu\alpha}\eta^c_{\nu\beta}.$$

Now invoke the product identity (42.87) with the lower index pairs $(\mu\nu)\to(\mu\alpha)$ and $(\rho\sigma)\to(\nu\beta)$:

$$\varepsilon^{abc}\eta^b_{\mu\alpha}\eta^c_{\nu\beta} = \delta_{\mu\nu}\,\eta^a_{\alpha\beta} - \delta_{\mu\beta}\,\eta^a_{\alpha\nu} - \delta_{\alpha\nu}\,\eta^a_{\mu\beta} + \delta_{\alpha\beta}\,\eta^a_{\mu\nu}.$$

Contract with $x_\alpha x_\beta$ term by term. The first term vanishes ($\eta^a_{\alpha\beta}x_\alpha x_\beta = 0$, antisymmetric against symmetric). The fourth term gives $\eta^a_{\mu\nu}\,x^2$ since $\delta_{\alpha\beta}x_\alpha x_\beta = x^2$. The second gives $-\eta^a_{\alpha\nu}x_\alpha x_\mu = +\eta^a_{\nu\alpha}x_\alpha x_\mu$, and the third gives $-\eta^a_{\mu\beta}x_\beta x_\nu = -\eta^a_{\mu\alpha}x_\alpha x_\nu$. Hence

$$g\,\varepsilon^{abc}A^b_\mu A^c_\nu = \frac{4}{g}\,\phi^2\Big(\eta^a_{\nu\alpha}x_\alpha x_\mu - \eta^a_{\mu\alpha}x_\alpha x_\nu + \eta^a_{\mu\nu}\,x^2\Big). \tag{42.92}$$

Adding. The two ϕ^2 cross terms in (42.91) and (42.92) are exact negatives of one another and cancel:

$$\frac{4}{g}\phi^2\big(\eta^a_{\mu\alpha}x_\alpha x_\nu - \eta^a_{\nu\alpha}x_\alpha x_\mu\big) + \frac{4}{g}\phi^2\big(\eta^a_{\nu\alpha}x_\alpha x_\mu - \eta^a_{\mu\alpha}x_\alpha x_\nu\big) = 0.$$

This cancellation — the whole reason the ansatz works — leaves only the $\eta^a_{\mu\nu}$ pieces,

$$F^a_{\mu\nu} = -\frac{4}{g}\,\eta^a_{\mu\nu}\,\phi + \frac{4}{g}\,\eta^a_{\mu\nu}\,x^2\,\phi^2 = \frac{4}{g}\,\eta^a_{\mu\nu}\,\phi(x^2\phi - 1).$$

Since $\phi = 1/(x^2+\rho^2)$, we have $x^2\phi - 1 = (x^2 - (x^2+\rho^2))/(x^2+\rho^2) = -\rho^2\phi$, so

$$F^a_{\mu\nu} = -\frac{4}{g}\,\eta^a_{\mu\nu}\,\rho^2\,\phi^2 = -\frac{4\rho^2}{g}\,\frac{\eta^a_{\mu\nu}}{(x^2+\rho^2)^2},$$

which is (42.90). Restoring the centre amounts to $x \to x - x_0$ throughout.

Self-duality. The whole x-dependence of $F^a_{\mu\nu}$ is the scalar $\rho^2\phi^2$ times the constant tensor $\eta^a_{\mu\nu}$, which is self-dual by (42.84). Therefore

$$\widetilde{F}^a_{\mu\nu} = \tfrac{1}{2}\varepsilon_{\mu\nu\rho\sigma}F^a_{\rho\sigma} = -\frac{4\rho^2}{g}\,\phi^2\,\tfrac{1}{2}\varepsilon_{\mu\nu\rho\sigma}\eta^a_{\rho\sigma} = -\frac{4\rho^2}{g}\,\phi^2\,\eta^a_{\mu\nu} = F^a_{\mu\nu}.$$

Equation of motion. A self-dual field automatically extremises the action. The non-abelian Bianchi identity (Theorem 25.11) states $D_\mu \widetilde{F}^a_{\mu\nu} = 0$ identically, for *any* gauge field — it is a geometric consequence of F being the curvature of a connection, $\varepsilon_{\mu\nu\rho\sigma}D_\nu F_{\rho\sigma} = 0$. Self-duality $F = \widetilde{F}$ then turns the Bianchi identity into the equation of motion:

$$D_\mu F^a_{\mu\nu} = D_\mu \widetilde{F}^a_{\mu\nu} = 0.$$

No second-order computation is needed; this is the structural payoff of solving the first-order self-duality equation rather than the second-order Yang–Mills equation directly. □

Physical Insight

The instanton is a genuine lump. From (42.90) the action density $\frac{1}{4}F^a_{\mu\nu}F^a_{\mu\nu}$ is sharply peaked at $x = x_0$, falls as $1/(x-x_0)^8$, and has a characteristic width ρ. Unlike the static solitons of Chapter 41, which are localised in space but extended through all time, the instanton is localised in all four Euclidean directions — it is an event, not an object: a tunnelling that happens *at a place and at a (Euclidean) time*, of finite duration $\sim \rho$. Hence the name 't Hooft coined, "instanton," an object existing for an instant.

42.3.3 Topological charge and action

With F in hand, the two integrals that the previous section left abstract become elementary. Both reduce to one radial integral, which we do once and reuse.

Lemma 42.19 (The master radial integral)**.** *In four Euclidean dimensions,*

$$\int \mathrm{d}^4x\, \frac{1}{(x^2+\rho^2)^4} = \frac{\pi^2}{6\,\rho^4}. \tag{42.93}$$

Proof. The integrand is spherically symmetric, so $\int \mathrm{d}^4x = \Omega_3 \int_0^\infty r^3\,\mathrm{d}r$ with $\Omega_3 = 2\pi^2$ the surface "area" of the unit S^3 (the volume of the unit $(d-1)$-sphere is $2\pi^{d/2}/\Gamma(d/2)$, equal to $2\pi^2/\Gamma(2) = 2\pi^2$ at $d=4$). Substitute $u = r^2$, $\mathrm{d}u = 2r\,\mathrm{d}r$, so $r^3\mathrm{d}r = \tfrac{1}{2}u\,\mathrm{d}u$:

$$\int \mathrm{d}^4x\, \frac{1}{(x^2+\rho^2)^4} = 2\pi^2 \cdot \tfrac{1}{2}\int_0^\infty \frac{u\,\mathrm{d}u}{(u+\rho^2)^4} = \pi^2 \int_0^\infty \frac{u\,\mathrm{d}u}{(u+\rho^2)^4}.$$

Now set $v = u + \rho^2$, $u = v - \rho^2$:

$$\int_0^\infty \frac{u\,\mathrm{d}u}{(u+\rho^2)^4} = \int_{\rho^2}^\infty (v^{-3} - \rho^2 v^{-4})\mathrm{d}v = \Big[-\tfrac{1}{2}v^{-2} + \tfrac{1}{3}\rho^2 v^{-3}\Big]_{\rho^2}^\infty = \frac{1}{2\rho^4} - \frac{1}{3\rho^4} = \frac{1}{6\rho^4}.$$

Multiplying by π^2 gives (42.93). □

Worked Example 42.20 (The topological charge is $Q = 1$)**.** We evaluate $Q = \dfrac{g^2}{32\pi^2}\displaystyle\int \mathrm{d}^4x\, F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu}$ on the BPST field, with the normalisation fixed in Section 42.2. Because the field is self-dual, $\widetilde{F} = F$, so $F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu} = F^a_{\mu\nu}F^a_{\mu\nu}$. From (42.90),

$$F^a_{\mu\nu}F^a_{\mu\nu} = \Big(\frac{4\rho^2}{g}\Big)^2 \frac{\eta^a_{\mu\nu}\eta^a_{\mu\nu}}{(x^2+\rho^2)^4} = \frac{16\rho^4}{g^2}\cdot\frac{12}{(x^2+\rho^2)^4} = \frac{192\,\rho^4}{g^2}\,\frac{1}{(x^2+\rho^2)^4}, \tag{42.94}$$

where $\eta^a_{\mu\nu}\eta^a_{\mu\nu} = 12$ is (42.85). Therefore, using (42.93),

$$Q = \frac{g^2}{32\pi^2}\cdot\frac{192\,\rho^4}{g^2}\int \mathrm{d}^4x\,\frac{1}{(x^2+\rho^2)^4} = \frac{192\,\rho^4}{32\pi^2}\cdot\frac{\pi^2}{6\rho^4} = \frac{192}{32\cdot 6} = 1.$$

The size ρ has cancelled, as it must: Q is a topological integer, blind to the smooth deformation that changes ρ. The sign is $+1$ because we chose the self-dual symbol η^a; replacing it by $\bar\eta^a$ in the ansatz yields an anti-self-dual field, the *anti-instanton*, with $Q = -1$.

Worked Example 42.21 (The action is $S = 8\pi^2/g^2$)**.** The Euclidean Yang–Mills action of the same field follows from the same density (42.94):

$$S = \frac{1}{4}\int \mathrm{d}^4x\, F^a_{\mu\nu}F^a_{\mu\nu} = \frac{1}{4}\cdot\frac{192\,\rho^4}{g^2}\int \mathrm{d}^4x\,\frac{1}{(x^2+\rho^2)^4} = \frac{48\,\rho^4}{g^2}\cdot\frac{\pi^2}{6\rho^4} = \frac{8\pi^2}{g^2}. \tag{42.95}$$

Two independent checks confirm this is right. First, ρ has again dropped out: the classical action does not depend on the instanton's size — the radial integrand of $\frac{1}{4}F^aF^a$ encloses the same area for every ρ, as Fig. 42.3 shows numerically. Second, the value saturates the Bogomolny bound (42.79) at $|Q| = 1$: $S = 8\pi^2/g^2 = (8\pi^2/g^2)\,|Q|$, exactly the floor, as it must for a self-dual field. We have produced an explicit configuration that attains the abstract minimum.

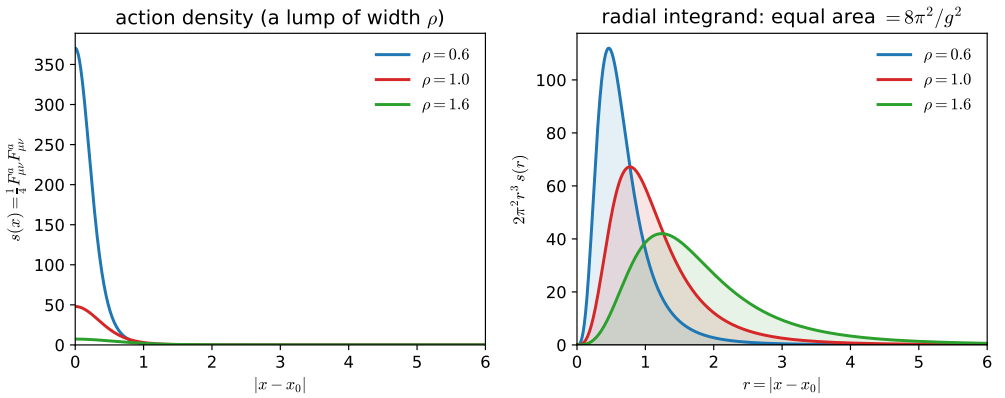


Figure 42.3: The BPST action density $s(x) = \frac{1}{4} F^a_{\mu\nu} F^a_{\mu\nu} = 48\rho^4/g^2\,(x^2+\rho^2)^{-4}$ for three sizes ρ (here $g = 1$, $x_0 = 0$). *Left*: the density itself — a larger instanton is broader and shallower. *Right*: the radial integrand $2\pi^2 r^3 s(r)$, whose total area is the action; all three curves enclose the same area $8\pi^2/g^2$ (Theorem 42.21). The equality of areas *is* the classical scale invariance that makes ρ a flat direction (Section 42.3.4).

Remark 42.22 (Singular gauge). The regular-gauge potential (42.88) pushes the winding out to the three-sphere at infinity, where $A \sim 1/|x|$ is pure gauge. It is often more convenient to push it instead onto a single point. Applying the (singular) gauge transformation that maps the field to the gauge in which A vanishes at infinity gives the *singular-gauge* form

$$A^{a,\mathrm{sing}}_{\mu}(x) = \frac{2}{g}\,\bar{\eta}^a_{\mu\nu}\,\frac{\rho^2\,(x-x_0)_\nu}{(x-x_0)^2\left[(x-x_0)^2+\rho^2\right]}, \tag{42.96}$$

which now decays as $1/|x|^3$ at infinity but is singular at $x = x_0$ (a gauge artefact, not a singularity of F). Strikingly it is built from the *anti*-self-dual symbol $\bar{\eta}^a$, yet its field strength is still self-dual with the same $Q = +1$ and the same action $8\pi^2/g^2$ — a gauge transformation changes neither. The singular gauge is the one in which multi-instanton superpositions and the fermionic zero modes of the next chapters are most cleanly written; we will reach for (42.96) when we need it.

42.3.4 Collective coordinates and zero modes

The single solution (42.88) is in fact a representative of a whole continuous family. Each symmetry of the action that fails to fix our particular instanton carries it to a physically distinct solution of the same action; the parameters of those symmetries are the *collective coordinates* (or *moduli*), and the directions in field space they generate are the *zero modes* of the fluctuation operator. Enumerating them is not bookkeeping for its own sake: the path integral in the one-instanton

sector is, to leading order, an ordinary integral over these moduli, and getting their number and their measure right is the whole content of the dilute-gas calculation to come.

What a zero mode is. Expand the action around the instanton, $A = A_{\text{inst}} + a$. Because A_{inst} is a solution, the linear term vanishes and

$$S[A_{\text{inst}} + a] = S[A_{\text{inst}}] + \tfrac{1}{2}\int \mathrm{d}^4x\; a_\mu^a\, M_{\mu\nu}^{ab}\, a_\nu^b + O(a^3), \tag{42.97}$$

with M the (gauge-fixed) quadratic fluctuation operator. A *zero mode* is a normalisable a with $Ma = 0$: a direction in which the action does not change to quadratic order. Each such direction is flat, and the Gaussian approximation that underlies steepest descent (Theorem 5.44) breaks down along it — one cannot do a Gaussian integral over a direction with no restoring quadratic term. The cure, which we set up here and execute in the dilute-gas section, is to trade each zero mode for an exact integration over the corresponding collective coordinate.

The zero modes are generated by the symmetries of the action that move the instanton. If g_ξ is a one-parameter family of such symmetries with $g_0 = \mathrm{id}$, then $A^{(\xi)} = g_\xi \cdot A_{\text{inst}}$ is a family of solutions all with the same action, and

$$a_\mu^{(\xi)} = \left.\frac{\partial A_\mu^{(\xi)}}{\partial \xi}\right|_{\xi=0} \tag{42.98}$$

satisfies $Ma^{(\xi)} = 0$ (after subtracting, where needed, a compensating infinitesimal gauge transformation to keep a transverse to the gauge orbit). We now list the symmetries.

Translations: 4 moduli. The action is invariant under $x \to x + \text{const}$, but our solution singles out the centre x_0. The four parameters x_0^μ are collective coordinates; the corresponding zero modes are $a_\nu^{(\mu)} = -\partial_\mu A_\nu$ (up to a gauge term), one for each direction. These four modes are guaranteed normalisable because the action density is localised.

Dilatation: 1 modulus. Classical four-dimensional Yang–Mills has no scale: the action $\frac{1}{4}\int F^a F^a$ is invariant under $x \to \lambda x$, $A_\mu(x) \to \lambda^{-1} A_\mu(x/\lambda)$, because in $d = 4$ the gauge field has mass dimension one and F^2 has dimension four, exactly matching d^4x. The solution (42.88) breaks this symmetry by choosing the size ρ; hence ρ is a collective coordinate, and the dilatation zero mode is $a_\mu^{(\rho)} = \partial A_\mu / \partial \rho$. This is the deepest of the moduli. Its existence is precisely the statement, already proven in Theorem 42.21, that $S = 8\pi^2/g^2$ is independent of ρ: the size is a genuine *flat direction* of the action, an exactly massless modulus

at the classical level. Nothing in the classical theory prefers a large instanton to a small one.

Caution

The classical flatness of the ρ direction is not the end of the story; it is the beginning of the chapter's central subtlety. Scale invariance is broken by quantum effects — it is anomalous, the same running of the coupling that gives asymptotic freedom (Eq. (27.101)). The flat direction is lifted only after the one-loop measure is included, and then ρ is controlled by the running coupling $g(\rho)$. The dilatation zero mode is therefore the conduit through which $\Lambda_{\rm QCD}$ (Theorem 27.14) enters the instanton calculus. This is the hand-off to the dilute-gas section, and we return to it below.

Global gauge rotations: 3 moduli for SU(2). A constant gauge transformation $A_\mu \to U A_\mu U^{-1}$ with $U \in \mathrm{SU}(2)$ is a symmetry. It rotates the instanton's orientation in the internal SU(2). Naively this is the three-parameter group SU(2); but the configuration is invariant under the subgroup that commutes with the locked structure $\eta^a_{\mu\nu}$, which for the single SU(2) instanton is trivial at the level that matters, leaving three independent orientation moduli. (More precisely, the global gauge group acts through $\mathrm{SU}(2)/\mathbb{Z}_2 \cong \mathrm{SO}(3)$, a three-dimensional manifold, and all three directions move the field; the centre $\mathbb{Z}_2$ acts trivially but contributes no continuous modulus.) These three are compact moduli — angles — whose integral will produce a finite group volume rather than a divergent spacetime volume.

The count for SU(2). Adding up,

$$\underbrace{4}_{\text{translations}} + \underbrace{1}_{\text{size}} + \underbrace{3}_{\text{gauge orientation}} = 8 = 4N\big|_{N=2}, \tag{42.99}$$

the number of bosonic collective coordinates of the charge-one SU(2) instanton. The grouping into $4N$ is not an accident; it is the first instance of the general result.

Theorem 42.23 (Dimension of the one-instanton moduli space)**.** *For gauge group* SU(N) *and topological charge Q, the bosonic collective coordinates of the (anti-)self-dual sector form a smooth manifold $\mathcal{M}_{N,Q}$ of dimension*

$$\dim \mathcal{M}_{N,Q} = 4N\,|Q|\,. \tag{42.100}$$

For $Q = 1$ this is $4N$: four positions, one size, and $4N - 5$ gauge-orientation angles.

We do not reprove the general index-theoretic statement here (it follows from the Atiyah–Hitchin–Singer index of the deformation operator and is the bosonic counterpart of the fermionic index theorem to be established in Theorem 36.15); but the $Q = 1$ gauge-orientation count is a short and instructive piece of group theory, which we give.

Worked Example 42.24 (Gauge-orientation moduli for SU(N), $Q = 1$)**.** A single $Q = 1$ instanton is built inside an SU(2) subgroup of SU(N) — pick any SU(2) acting on a two-dimensional subspace of the fundamental and write (42.88) with the corresponding generators, padding the rest of the $N \times N$ matrices with zeros. The global gauge group SU(N) acts on this embedded solution by conjugation. The orientation moduli are the directions in SU(N) that move it, i.e. the coset $\mathrm{SU}(N)/H$, where H is the stabiliser: the subgroup commuting with the embedded SU(2). By Schur's lemma, the matrices commuting with the SU(2) acting irreducibly on the chosen $\mathbf{2}$ and trivially on the orthogonal $\mathbf{N-2}$ are block-scalar on the $\mathbf{2}$ and arbitrary U($N-2$) on the rest, intersected with SU(N). Concretely $H = (\mathrm{SU}(N-2) \times \mathrm{U}(1))$, of dimension

$$\dim H = \left[(N-2)^2 - 1\right] + 1 = (N-2)^2.$$

Hence the number of orientation moduli is

$$\dim\bigl(\mathrm{SU}(N)/H\bigr) = (N^2-1) - (N-2)^2 = 4N - 5.$$

Adding the four translations and one size recovers $4 + 1 + (4N-5) = 4N$, in agreement with Theorem 42.23 at $Q = 1$. For $N = 2$, H is trivial ($\dim H = 0$) and we recover the three angles of (42.99); for $N = 3$ there are $4 \cdot 3 - 5 = 7$ orientation moduli.

From zero modes to a measure. The reason to count zero modes is that each one must be removed from the Gaussian fluctuation integral and replaced by an integral over its collective coordinate. The standard collective-coordinate (Faddeev–Popov-type) treatment replaces, for each zero mode, a divergent Gaussian factor by

$$\int \mathrm{d}a\; e^{-\frac{1}{2}\lambda a^2} \;\longrightarrow\; \left(\tfrac{S_0}{2\pi}\right)^{1/2} \int \mathrm{d}\xi, \tag{42.101}$$

i.e. by an integral over the modulus ξ weighted by the Jacobian $\sqrt{S_0/2\pi}$ that converts the normalised fluctuation amplitude into the canonical collective coordinate, with $S_0 = 8\pi^2/g^2$ the instanton action. Each bosonic zero mode contributes one such factor of $\sqrt{S_0/2\pi} \propto 1/g$, so the $4N$ modes contribute $(S_0)^{2N} \propto g^{-4N}$ to the prefactor of e^{-S_0} — the famous large power of $1/g$ that multiplies the exponentially small tunnelling weight.

The dimensionful moduli are the position x_0 and the size ρ; the orientation angles are dimensionless and integrate to a finite group volume. Translation

invariance makes the x_0 integral the spacetime volume, $\int \mathrm{d}^4x_0 = V_4$. The ρ dependence is then fixed by dimensional analysis alone. The one-instanton contribution to the partition function is a pure number density (mass dimension four per unit four-volume), and at the classical level ρ is the only scale; so the size must enter the measure as

$$\mathrm{d}\mu_{\text{size}} \propto \frac{\mathrm{d}\rho}{\rho^5} \qquad (\mathrm{SU}(2)), \tag{42.102}$$

because $\mathrm{d}\rho/\rho^5$ carries mass dimension four and is the unique scale-covariant weight consistent with the classical scale invariance — the same invariance that flattened the ρ direction now *determines* the shape of its measure. (Counting the powers explicitly: the four x_0 Jacobians and the size Jacobian carry, between them, the dimensions that combine with $\mathrm{d}^4x_0\,\mathrm{d}\rho$ to leave $\mathrm{d}^4x_0\,\mathrm{d}\rho/\rho^5$.) For general $\mathrm{SU}(N)$ the power is fixed the same way and the $4N-5$ extra orientation Jacobians shift it; the universal classical skeleton is $\mathrm{d}^4x_0\,\mathrm{d}\rho/\rho^5$ times the orientation volume times $(S_0/2\pi)^{2N}e^{-S_0}$.

The fermionic content, when matter is present, rides on the same moduli. A Dirac fermion in a representation r has, by the index theorem (Theorem 36.15), exactly $2\,T(r)\,|Q|$ normalisable zero modes of $\not{D}$ in the instanton background; these are Grassmann-odd collective coordinates, and the Berezin integral over them produces a determinant of the overlap matrix exactly as in the complex Gaussian Grassmann integral of Theorem 12.26. Their immediate consequence — that the fermionic measure forces the instanton amplitude to vanish unless the right number of fermion legs are attached — is the mechanism behind the $\mathrm{U}(1)_A$ anomaly and the 't Hooft vertex, which the chapter takes up after the vacuum structure is in place.

We now have what the rest of the chapter needs: an explicit charge-one field (42.88), its action $8\pi^2/g^2$, and its complete set of $4N$ bosonic collective coordinates, with the size ρ identified as a classically flat direction whose measure scales as $\mathrm{d}\rho/\rho^5$. The next section steps back from the Euclidean path integral to the canonical, Hamiltonian reading of this same object: it interprets the instanton as the amplitude to tunnel between the topologically distinct vacua $|n\rangle$, builds the θ-vacuum superposition that diagonalises large gauge transformations, and shows that the ρ-integral assembled here — once the running coupling $g(\rho)$ lifts the flat direction — is exactly the dilute-gas sum over tunnelling events that the θ-vacuum organises.

42.4 The theta-Vacuum

The previous section delivered a single concrete object: the BPST instanton, a self-dual finite-action solution of the Euclidean Yang–Mills equations carrying topological charge $Q = 1$ and Euclidean action $S_{\text{inst}} = 8\pi^2/g^2$ (Eq. (42.13)). We

have been calling it a *tunneling event*, and the word was chosen deliberately. A finite-action Euclidean trajectory that begins in one classical vacuum at $\tau \to -\infty$ and ends in another at $\tau \to +\infty$ is, by the steepest-descent reading of the path integral (Theorem 5.44), the dominant contribution to a quantum-mechanical amplitude for passing from the first configuration to the second through a classically forbidden region. The instanton therefore signals that the Yang–Mills theory has *more than one* classical vacuum, that these vacua are degenerate, and that the true quantum ground state is not any one of them but a definite superposition. The purpose of this section is to make that statement precise in the Hamiltonian language: to identify the discrete family of classical vacua, to see the instanton as the operator that hops between adjacent members, to diagonalize the resulting tunneling Hamiltonian, and to discover that the eigenvalue which labels the diagonal blocks is a new, genuinely physical, parity- and time-reversal-violating parameter of the theory — the vacuum angle θ.

42.4.1 Temporal gauge and the residual large gauge transformations

Canonical quantization of a gauge theory wants a gauge in which time plays no special role in the constraint structure, and the cleanest such choice is *temporal* (or Weyl) gauge,

$$A_0^a(x) = 0. \tag{42.103}$$

In this gauge the canonical momentum conjugate to A_i^a is the chromo-electric field $E_i^a = \dot{A}_i^a$ (up to a sign fixed by conventions), the Hamiltonian is the familiar $\mathcal{H} = \tfrac{1}{2}\int d^3x\,(E_i^a E_i^a + B_i^a B_i^a)$ with $B_i^a = \tfrac{1}{2}\epsilon_{ijk}F_{jk}^a$, and Gauss's law $D_i E_i^a = 0$ survives not as an equation of motion but as a constraint on physical states. The point of importance here is that Eq. (42.103) does not exhaust the gauge freedom. A gauge transformation $A_\mu \to A_\mu^U$ preserves $A_0 = 0$ provided U is independent of time, $\partial_0 U = 0$. Thus the residual gauge symmetry of temporal gauge is the group $\mathcal{G}$ of *time-independent* gauge transformations $U(\mathbf{x})$, acting on the spatial connection by

$$A_i \;\longmapsto\; A_i^U \;=\; U A_i U^{-1} - \frac{i}{g}\,(\partial_i U)\,U^{-1}, \qquad A_i \equiv A_i^a T^a. \tag{42.104}$$

This is exactly the group whose topology we classified in Chapter 25. Demanding finite energy forces $U(\mathbf{x}) \to$ const as $|\mathbf{x}| \to \infty$, so each U extends to a based map of the one-point compactification $S^3 = \mathbb{R}^3 \cup \{\infty\}$ into the gauge group, and these maps fall into homotopy classes labelled by an integer winding number,

$$\pi_3(\mathrm{SU}(N)) = \mathbb{Z}, \qquad N \geq 2, \tag{42.105}$$

by Theorem 25.33, the integer being computed by the winding functional of Eq. (25.167), an integer by Theorem 25.35. The transformations with winding

zero — the *small* ones, continuously connected to the identity — are generated infinitesimally by $D_i\alpha$ and are nothing but the gauge redundancy already implemented by the Faddeev–Popov/BRST machinery of Chapter 26; physical states are required to be invariant under them, this being the canonical face of the orbit identification $A \sim A^U$ of Eq. (26.4) and of BRST-closedness, Theorem 26.18. The transformations with nonzero winding — the *large* ones — are the new structure. They are bona fide gauge transformations, hence symmetries of every physical (gauge-invariant) operator, yet they are not reachable from the identity by any continuous deformation and so are *not* automatically resolved by the infinitesimal constraint.

Notation Watch

We write $\hat{T}$ for the unitary operator that implements a winding-one large gauge transformation on the Hilbert space, reserving T^a (with a Lie-algebra index) for the Hermitian generators of Chapter 25. The hat distinguishes the operator from the abstract transformation U_1 it represents.

Fix once and for all a representative gauge transformation $U_1(\mathbf{x})$ of winding number one. Its unitary implementation on the Hilbert space is an operator $\hat{T}$; acting on a state $|\Psi\rangle$ described by a wavefunctional $\Psi[A_i]$ it sends

$$(\hat{T}\,\Psi)[A_i] \;=\; \Psi\big[A_i^{U_1^{-1}}\big]. \tag{42.106}$$

Because U_1 is a gauge transformation, $\hat{T}$ commutes with the Hamiltonian and with every gauge-invariant operator,

$$\big[\hat{T},\mathcal{H}\big] = 0. \tag{42.107}$$

This is the central fact. The large gauge transformation is a *symmetry*, but — unlike a small one — it is not a constraint that must annihilate physical states. We are therefore free, and indeed obliged, to diagonalize it simultaneously with $\mathcal{H}$.

42.4.2 The discrete vacua and the Chern–Simons number

The classical vacua of Yang–Mills theory are the zero-energy configurations, $E_i^a = B_i^a = 0$. Vanishing electric field is $\dot{A}_i = 0$; vanishing magnetic field, $F_{ij} = 0$, means the spatial connection is *pure gauge*,

$$A_i^{(U)} \;=\; -\frac{i}{g}\,(\partial_i U)\,U^{-1}, \tag{42.108}$$

the image of $A_i = 0$ under Eq. (42.104). Naively there is a single classical vacuum, $A_i = 0$, with all the others gauge-equivalent to it and therefore physically

identical. That conclusion is correct for the small transformations and *false* for the large ones. Two pure-gauge configurations Eq. (42.108) built from maps U of different winding number cannot be deformed into one another through zero-energy configurations without leaving the class of pure gauges, and the gauge transformation relating them is large. They are therefore physically distinct classical vacua, one for each integer, and we must keep them all.

To label them invariantly we need a functional of the spatial connection that counts winding without reference to a particular U. That functional is the *Chern–Simons number*. Recall from Theorem 33.15 that the topological density is the divergence of the Chern–Simons current K^μ of Eq. (33.70); its time component, integrated over space, is

$$N_{\mathrm{CS}}[A] \;\equiv\; \frac{g^2}{32\pi^2}\int d^3x\; K^0 \;=\; \frac{g^2}{16\pi^2}\int d^3x\;\epsilon^{ijk}\Big(A_i^a\partial_j A_k^a + \tfrac{g}{3}f^{abc}A_i^aA_j^bA_k^c\Big). \tag{42.109}$$

We claim that on a pure gauge Eq. (42.108) the Chern–Simons number equals (up to an orientation sign fixed below) the winding integer of Eq. (25.167), and is therefore an integer.

Worked Example 42.25 (Chern–Simons number of a pure gauge equals its winding)**.** Insert Eq. (42.108) into Eq. (42.109). Write $R_i \equiv (\partial_i U)U^{-1}$, so $A_i = -\frac{i}{g}R_i$, and convert the colour traces using $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ and $\mathrm{tr}(T^aT^bT^c) = \frac{1}{4}(d^{abc}+if^{abc})$. Contraction with the totally antisymmetric ϵ^{ijk} kills the symmetric part d^{abc}, leaving

$$A_i^a\partial_jA_k^a = 2\,\mathrm{tr}(A_i\partial_jA_k), \qquad \epsilon^{ijk}f^{abc}A_i^aA_j^bA_k^c = -4i\,\epsilon^{ijk}\,\mathrm{tr}(A_iA_jA_k). \tag{42.110}$$

For a pure gauge, $\partial_jR_k = (\partial_j\partial_kU)U^{-1} - R_kR_j$; the first piece is symmetric in jk and drops against ϵ^{ijk}, so $\epsilon^{ijk}\,\mathrm{tr}(R_i\partial_jR_k) = \epsilon^{ijk}\,\mathrm{tr}(R_iR_jR_k)$. Substituting $A_i = -\frac{i}{g}R_i$ (whence $A_i\partial_jA_k = -\frac{1}{g^2}R_i\partial_jR_k$ and $A_iA_jA_k = \frac{i}{g^3}R_iR_jR_k$, using $(-i)^3 = i$) gives, term by term,

$$N_{\mathrm{CS}} = \frac{g^2}{16\pi^2}\,\epsilon^{ijk}\int d^3x\,\Big[\underbrace{-\tfrac{2}{g^2}\,\mathrm{tr}(R_iR_jR_k)}_{\text{quadratic}}\;\underbrace{+\tfrac{4}{3g^2}\,\mathrm{tr}(R_iR_jR_k)}_{\text{cubic}}\Big] = -\frac{1}{24\pi^2}\,\epsilon^{ijk}\int d^3x\;\mathrm{tr}(R_iR_jR_k). \tag{42.111}$$

Finally $R_i = U(U^{-1}\partial_iU)U^{-1}$, so $\mathrm{tr}(R_iR_jR_k) = \mathrm{tr}\,\big((U^{-1}\partial_iU)(U^{-1}\partial_jU)(U^{-1}\partial_kU)\big)$ by cyclicity, which is exactly the integrand of the winding functional Eq. (25.167). Hence $N_{\mathrm{CS}}[A^{(U)}] = -\,n[U] \in \mathbb{Z}$. The minus sign is a pure orientation convention; we are free to fix the orientation on S^3 so that the self-dual instanton, which carries $Q = +1$, *raises* the Chern–Simons number, and with that choice $N_{\mathrm{CS}}[A^{(U)}] = n[U]$.

We may now name the classical vacua by their integer Chern–Simons number.

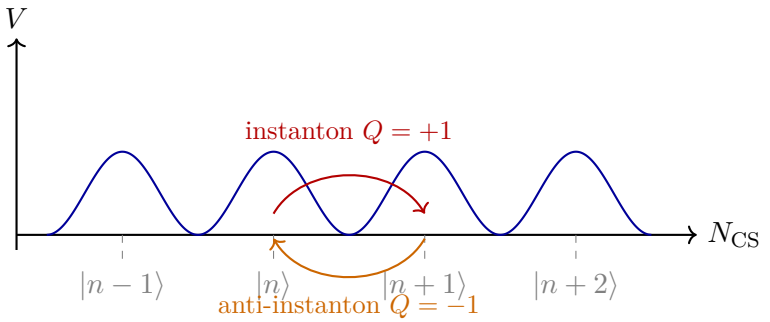


Figure 42.4: The periodic vacuum structure of Yang–Mills theory in temporal gauge. Each integer value of the Chern–Simons number N_{CS} is a degenerate classical vacuum $|n\rangle$. A BPST instanton is a Euclidean trajectory tunnelling under the barrier from $|n\rangle$ to $|n+1\rangle$ (topological charge $Q = +1$); the anti-instanton tunnels the other way. The true ground states are the Bloch-like superpositions $|\theta\rangle$, not any single $|n\rangle$.

Definition 42.26 (The n-vacua)**.** For each integer n, the configuration $A_i^{(n)} = -\frac{i}{g}(\partial_i U_n)U_n^{-1}$ with U_n a winding-n map is a classical vacuum with $N_{\mathrm{CS}} = n$. The corresponding quantum state, the ground state of the small fluctuations about $A_i^{(n)}$, is denoted $|n\rangle$. The states $\{|n\rangle\}_{n\in\mathbb{Z}}$ are an orthonormal family of candidate vacua, and the winding-one operator $\hat{T}$ of Eq. (42.106) carries each to the next,

$$\hat{T}\,|n\rangle = |n+1\rangle\,. \tag{42.112}$$

That $\hat{T}$ shifts N_{CS} by exactly one unit is the operator statement of additivity, $n[U_1U_n] = n[U_1] + n[U_n] = 1 + n$, from Theorem 25.35: composing with the winding-one U_1 sends an n-vacuum to an $(n{+}1)$-vacuum. The picture to hold in mind is a periodic potential on the infinite-dimensional configuration space, with one degenerate well per integer Chern–Simons number; Fig. 42.4 sketches it.

42.4.3 The instanton as a tunnelling event

The wells of Fig. 42.4 are separated by barriers, and the BPST instanton is the path under one of them. To see this in the present variables, return to the topological charge and write it, using $F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu} = \partial_\mu K^\mu$ from Eq. (33.69), as the total Euclidean-time derivative of the Chern–Simons number,

$$\begin{aligned} Q &= \frac{g^2}{32\pi^2}\int d^4x\; F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu} = \frac{g^2}{32\pi^2}\int d^4x\;\partial_\mu K^\mu \\ &= \frac{g^2}{32\pi^2}\int_{-\infty}^{\infty} d\tau\,\frac{d}{d\tau}\int d^3x\;K^0 = N_{\mathrm{CS}}(\tau{=}{+}\infty) - N_{\mathrm{CS}}(\tau{=}{-}\infty), \end{aligned} \tag{42.113}$$

where the spatial components $\partial_i K^i$ integrate to zero because K^i falls off at spatial infinity (the fields are pure gauge there). The middle steps used the definition Eq. (42.109) of N_{CS} with exactly the same numerical prefactor, so no stray factor appears: *the topological charge of a Euclidean trajectory is the net change in its Chern–Simons number.* In temporal gauge a finite-action trajectory must approach a classical vacuum at $\tau \to \pm\infty$ (otherwise its action diverges), so $N_{\mathrm{CS}}(\pm\infty)$ are integers; the BPST instanton, with $Q = 1$, is precisely a path with

$$N_{\mathrm{CS}}(-\infty) = n, \qquad N_{\mathrm{CS}}(+\infty) = n + 1. \tag{42.114}$$

It interpolates between adjacent vacua, passing through configurations of nonzero field strength (the barrier) in between. Steepest descent then gives the Euclidean amplitude for this tunnelling. For a transition over a long Euclidean time interval $\mathcal{T}$, the leading semiclassical contribution from a single instanton is

$$\left\langle n+1 \middle| e^{-\mathcal{H}\mathcal{T}} \middle| n \right\rangle \;\sim\; K\, e^{-S_{\mathrm{inst}}} = K\, e^{-8\pi^2/g^2}, \tag{42.115}$$

with $S_{\mathrm{inst}} = 8\pi^2/g^2$ the instanton action of Eq. (42.13) and K the one-loop fluctuation/collective-coordinate prefactor assembled in Section 42.5.1. The exponential $e^{-8\pi^2/g^2}$ is the hallmark of the whole subject: it is nonanalytic at $g = 0$, invisible to every order of perturbation theory, and it is the quantitative measure of the splitting that the tunnelling induces among the degenerate $|n\rangle$. The anti-instanton ($Q = -1$) contributes the mirror amplitude $\left\langle n-1 \middle| e^{-\mathcal{H}\mathcal{T}} \middle| n \right\rangle$.

42.4.4 Diagonalizing the tunnelling Hamiltonian: the theta-vacuum

We have a Hamiltonian that, in the basis $\{|n\rangle\}$, is diagonal-plus-nearest-neighbour-hopping: the diagonal entries are the common well energy, and the off-diagonal entries $\left\langle n \pm 1 \middle| e^{-\mathcal{H}\mathcal{T}} \middle| n \right\rangle$ are the tunnelling amplitudes Eq. (42.115). This is structurally identical to the tight-binding Hamiltonian of an electron in a one-dimensional crystal, and the resolution is identical: the energy eigenstates are not the localized $|n\rangle$ but the Bloch waves built from them. Because $\hat{T}$ commutes with $\mathcal{H}$ by Eq. (42.107) and acts as a unit translation by Eq. (42.112), we diagonalize $\hat{T}$. It is unitary, so its eigenvalues are phases; parametrize them by an angle $\theta \in [0, 2\pi)$ and define the simultaneous eigenstate.

Definition 42.27 (The theta-vacuum)**.** For each $\theta \in [0, 2\pi)$ the θ-vacuum is the superposition

$$|\theta\rangle \;=\; \sum_{n \in \mathbb{Z}} e^{-in\theta}\, |n\rangle\,. \tag{42.116}$$

It diagonalizes the winding operator: using Eq. (42.112) and relabelling the sum,

$$\hat{T}\,|\theta\rangle = \sum_n e^{-in\theta}\,|n+1\rangle = \sum_m e^{-i(m-1)\theta}\,|m\rangle = e^{i\theta}\,|\theta\rangle\,. \tag{42.117}$$

Since $[\hat{T}, \mathcal{H}] = 0$ and the θ-vacua exhaust the eigenstates of $\hat{T}$, the Hamiltonian is block-diagonalized by them; each θ furnishes its own ground state and its own tower of excitations. The energy of the θ-vacuum follows immediately from the tight-binding analogy: with nearest-neighbour amplitude $\propto e^{-S_{\text{inst}}}$, the ground-state energy density acquires a θ-dependent band,

$$\mathcal{E}(\theta) \;=\; \text{const} \;-\; 2K\, e^{-8\pi^2/g^2} \cos\theta \;+\; \ldots, \tag{42.118}$$

the cosine being the lattice dispersion and the dots being multi-instanton corrections. Summing those corrections quantitatively, and confronting the result with the limits of the dilute-gas approximation, is the business of Section 42.5; for now the qualitative content is the lesson, that tunnelling lifts the degeneracy of the $|n\rangle$ into a continuum of inequivalent vacua labelled by θ.

The same conclusion is reached, and the role of θ as a label of distinct theories made manifest, by computing the Euclidean vacuum persistence amplitude directly in the θ-basis. Insert Eq. (42.116) on both sides:

$$\Big\langle \theta' \Big| e^{-\mathcal{H}\mathcal{T}} \Big| \theta \Big\rangle = \sum_{m,n} e^{im\theta' - in\theta} \Big\langle m \Big| e^{-\mathcal{H}\mathcal{T}} \Big| n \Big\rangle . \tag{42.119}$$

Translation invariance in the index — a consequence of $[\hat{T}, \mathcal{H}] = 0$, which makes $\Big\langle m \Big| e^{-\mathcal{H}\mathcal{T}} \Big| n \Big\rangle$ depend only on the difference $Q \equiv m - n$, the topological charge carried by the trajectory — lets us write $\Big\langle m \Big| e^{-\mathcal{H}\mathcal{T}} \Big| n \Big\rangle = F_Q$. Setting $m = n + Q$,

$$\Big\langle \theta' \Big| e^{-\mathcal{H}\mathcal{T}} \Big| \theta \Big\rangle = \sum_{n} e^{in(\theta'-\theta)} \sum_{Q} e^{iQ\theta'} F_Q = 2\pi \sum_{k\in\mathbb{Z}} \delta(\theta' - \theta - 2\pi k) \; \sum_{Q} e^{iQ\theta} F_Q . \tag{42.120}$$

Two features of Eq. (42.120) are decisive. First, the amplitude is proportional to $\delta(\theta' - \theta)$ (modulo 2π): *different θ-vacua do not communicate*. No amount of Euclidean evolution carries one into another. Second, within a fixed sector the amplitude is the weighted sum

$$\sum_{Q} e^{iQ\theta}\, F_Q, \qquad F_Q = \Big\langle n+Q \Big| e^{-\mathcal{H}\mathcal{T}} \Big| n \Big\rangle , \tag{42.121}$$

that is, each topological sector Q enters with the phase $e^{iQ\theta}$. This is the operational meaning of the vacuum angle: θ is the relative phase with which the path integral weights configurations of different instanton number.

42.4.5 Theta as a superselection label

The factor $\delta(\theta' - \theta)$ in Eq. (42.120) is the analytic shadow of a structural fact: θ is a *superselection* parameter. We make this precise.

Theorem 42.28 (The vacuum angle labels superselection sectors)**.** *The winding operator $\hat{T}$ commutes with every observable of the theory. Consequently the Hilbert space decomposes into superselection sectors labelled by the eigenvalue $e^{i\theta}$, $\theta \in [0, 2\pi)$; matrix elements of all observables between states of different θ vanish, no observable can change θ, and coherent superpositions across different θ are unphysical. The clustering vacua of the theory are exactly the $|\theta\rangle$ of Eq. (42.116), one per sector; the localized states $|n\rangle$ are not physical vacua.*

Proof. Observables are the gauge-invariant operators. A large gauge transformation is, like any gauge transformation, a symmetry under which every gauge-invariant operator is inert: for any observable $\mathcal{O}$,

$$\left[\hat{T}, \mathcal{O}\right] = 0. \tag{42.122}$$

Indeed gauge invariance *means* invariance under all of $\mathcal{G}$, and $\hat{T}$ implements an element of $\mathcal{G}$; Eq. (42.107) is the special case $\mathcal{O} = \mathcal{H}$. Thus $\hat{T}$ is a unitary that commutes with the entire observable algebra. By Schur's lemma, in any irreducible representation of that algebra $\hat{T}$ acts as a multiple of the identity — a single phase $e^{i\theta}$. The Hilbert space therefore splits as a direct sum (here a direct integral) of subspaces on which $\hat{T} = e^{i\theta}\mathbf{1}$, and these are the θ-eigenspaces spanned by Eq. (42.116). Between two such subspaces every observable has vanishing matrix element, because $\mathcal{O}$ commutes with $\hat{T}$ and so cannot connect different eigenvalues:

$$\langle\theta'|\mathcal{O}|\theta\rangle\,(e^{i\theta} - e^{i\theta'}) = \left\langle\theta'\middle|\,[\mathcal{O}, \hat{T}]\,\middle|\theta\right\rangle = 0 \implies \langle\theta'|\mathcal{O}|\theta\rangle = 0 \text{ for } \theta \neq \theta'. \tag{42.123}$$

This is the definition of a superselection rule: θ is a fixed datum of the theory, not a dynamical quantity, and choosing a value of θ is choosing which theory one is quantizing, on the same footing as choosing the coupling g.

It remains to see that the physical vacua are the $|\theta\rangle$ rather than the $|n\rangle$, and the criterion is *cluster decomposition*: a physical vacuum $|\Omega\rangle$ must factorize correlators of local observables at large spacelike separation,

$$\langle\Omega|\mathcal{O}_1(x)\,\mathcal{O}_2(y)|\Omega\rangle \xrightarrow{|x-y|\to\infty} \langle\Omega|\mathcal{O}_1(x)|\Omega\rangle\,\langle\Omega|\mathcal{O}_2(y)|\Omega\rangle\,. \tag{42.124}$$

Equivalently, a clustering vacuum is an extremal (pure) state on the observable algebra, one that cannot be written as a nontrivial convex combination of other invariant states. The $|n\rangle$ fail this test. Inverting Eq. (42.116), $|n\rangle = \frac{1}{2\pi}\int_0^{2\pi} d\theta\, e^{in\theta}\,|\theta\rangle$ is a uniform superposition over *all* sectors; since observables are block-diagonal in θ by Eq. (42.123), the expectation $\langle n|\mathcal{O}_1\mathcal{O}_2|n\rangle = \frac{1}{2\pi}\int d\theta\,\langle\theta|\mathcal{O}_1\mathcal{O}_2|\theta\rangle$ is an *average* of the clustering values $\langle\theta|\mathcal{O}_1|\theta\rangle\,\langle\theta|\mathcal{O}_2|\theta\rangle$ over θ, and the average of products is not the product of averages unless the θ-distribution is concentrated. Concretely the connected correlator $\langle n|\mathcal{O}_1\mathcal{O}_2|n\rangle_c$ does not vanish at large separation, so $|n\rangle$ violates Eq. (42.124). The $|\theta\rangle$, being the extremal $\hat{T}$-eigenstates, do cluster. They are the physical vacua. □

Physical insight: the theta-vacuum is a Bloch wave

Everything in this section is the field-theoretic transcription of a single fact from solid-state physics: an electron in a periodic potential does not live in any one well but in a Bloch band, and its states are labelled by a crystal momentum θ that lies in a Brillouin zone $[0, 2\pi)$ and is conserved. Here the "crystal" is the configuration space coordinate N_{CS}, the lattice translation is the large gauge transformation $\hat{T}$, the inter-well tunnelling is the instanton, and the conserved Bloch momentum is the vacuum angle θ. Just as no local perturbation of a crystal can change the crystal momentum of a Bloch state, no local operator of the gauge theory can change θ — which is exactly the superselection rule of Theorem 42.28.

Worked Example 42.29 (The periodic-potential model of the theta-vacuum)**.** To see the band structure Eq. (42.118) arise from tunnelling with no gauge theory in sight, solve the quantum mechanics of a particle of unit mass on the line in a periodic potential $V(x) = V(x + a)$ with deep wells at $x = na$, $n \in \mathbb{Z}$, the exact analogue of Fig. 42.4. Let $|n\rangle$ be the normalized ground state localized in the nth well, and let $\hat{T}$ be the unit lattice translation $x \to x + a$, so $\hat{T}\,|n\rangle = |n + 1\rangle$ and $[\hat{T}, \mathcal{H}] = 0$.

Tunnelling amplitude. In the Euclidean path integral the amplitude to go from the nth to the $(n+1)$th well over imaginary time $\mathcal{T}$ is dominated by the single-bounce (instanton) trajectory that rolls from one maximum of $-V$ to the next, with action $S_0 = \int dx\, \sqrt{2V(x)}$ across the barrier; by steepest descent (Theorem 5.44) it equals $K_0\, e^{-S_0}$ for a fluctuation determinant K_0. The diagonal amplitude $\left\langle n \middle| e^{-\mathcal{H}\mathcal{T}} \middle| n \right\rangle$ is dominated by staying put. Keeping nearest-neighbour hops,

$$\left\langle m \middle| e^{-\mathcal{H}\mathcal{T}} \middle| n \right\rangle = e^{-\varepsilon_0 \mathcal{T}} \Big[\delta_{mn} + \mathcal{T}\, K_0 e^{-S_0}\, (\delta_{m,n+1} + \delta_{m,n-1}) + \dots \Big]. \tag{42.125}$$

Diagonalization. Form the Bloch state $|\theta\rangle = \sum_n e^{-in\theta}\, |n\rangle$, on which $\hat{T}\,|\theta\rangle = e^{i\theta}\,|\theta\rangle$ exactly as in Eq. (42.117). Then

$$\left\langle \theta \middle| e^{-\mathcal{H}\mathcal{T}} \middle| \theta \right\rangle \Big/ \langle \theta | \theta \rangle = e^{-\varepsilon_0 \mathcal{T}} \left[1 + \mathcal{T}\, K_0 e^{-S_0} (e^{i\theta} + e^{-i\theta}) + \dots \right] \xrightarrow{\mathcal{T} \text{ small}} e^{-E(\theta)\mathcal{T}}, \tag{42.126}$$

so that $E(\theta) = \varepsilon_0 - 2K_0 e^{-S_0} \cos\theta$. The localized levels broaden into a band of width $4K_0 e^{-S_0}$, set entirely by the nonperturbative tunnelling factor; the band momentum θ is conserved and labels distinct, non-communicating ground states. This is Eq. (42.118) with $S_0 \to 8\pi^2/g^2$, and it shows in a setting with no subtleties that the θ-degeneracy is real, the splitting is exponentially small, and θ is a good quantum number precisely because $\hat{T}$ commutes with the Hamiltonian.

42.4.6 Equivalence to a Lagrangian theta-term

The Hamiltonian construction has produced a family of theories indexed by θ. We now show that this is not a new and exotic kind of parameter at all: working in the θ-sector is identical to keeping the ordinary, perturbatively-trivial vacuum and adding to the Lagrangian the parity-odd operator of Eq. (33.68). This identification is what makes θ a manipulable coupling and exposes its discrete-symmetry properties.

Theorem 42.30 (Summing the sectors is the theta-term)**.** *Let Z_Q denote the Euclidean path integral over gauge configurations of topological charge Q, with the Euclidean action S_E of Theorem 10.27 continued by $\tau = it$ as in Eq. (10.157). Then the θ-vacuum amplitude Eq. (42.121) is the full path integral weighted by $e^{iQ\theta}$,*

$$Z(\theta) \;=\; \sum_{Q\in\mathbb{Z}} e^{iQ\theta}\, Z_Q \;=\; \int \mathcal{D}A\; \exp\Big(- S_E[A] + i\theta\, \frac{g^2}{32\pi^2} \int d^4x\; F^a_{\mu\nu} \widetilde{F}^{a\,\mu\nu} \Big), \tag{42.127}$$

which is precisely the partition function of the theory whose Minkowski Lagrangian is augmented by

$$\mathcal{L}_\theta \;=\; \theta\, \frac{g^2}{32\pi^2}\, F^a_{\mu\nu} \widetilde{F}^{a\,\mu\nu}, \qquad \widetilde{F}^{a\,\mu\nu} = \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma} F^a_{\rho\sigma}. \tag{42.128}$$

The dependence is 2π-periodic, $\theta \sim \theta + 2\pi$.

Proof. The decomposition of the configuration space into topological sectors is exact: every finite-action configuration has an integer charge $Q = \frac{g^2}{32\pi^2}\int F\widetilde{F}$ by Eq. (33.79), and the path-integral measure splits as a sum over Q of integrals at fixed charge, $\int \mathcal{D}A = \sum_Q \int_Q \mathcal{D}A$. By definition $Z_Q = \int_Q \mathcal{D}A\, e^{-S_E}$ is the vacuum amplitude restricted to charge Q, i.e. the sector amplitude F_Q of Eq. (42.121) up to the overall (sector-independent, divergent) volume factor $2\pi\delta(0)$ stripped in Eq. (42.120). Hence $\sum_Q e^{iQ\theta} Z_Q$ is exactly the θ-vacuum persistence amplitude. Now, on every configuration in the Q-sector the number Q is constant and equal to $\frac{g^2}{32\pi^2}\int F\widetilde{F}$, so the phase $e^{iQ\theta}$ may be brought inside the integral as $\exp\big(i\theta\, \frac{g^2}{32\pi^2}\int F\widetilde{F}\big)$ without changing anything; summing the sectors then reassembles the unrestricted measure,

$$\sum_Q e^{iQ\theta} Z_Q = \sum_Q \int_Q \mathcal{D}A\; e^{-S_E}\, e^{\,i\theta \frac{g^2}{32\pi^2}\int F\widetilde{F}} = \int \mathcal{D}A\; e^{-S_E + i\theta \frac{g^2}{32\pi^2}\int F\widetilde{F}}, \tag{42.129}$$

which is the middle member of Eq. (42.127). The exponent is $-S_E$ minus the Euclidean continuation of $-i\int d^4x\, \mathcal{L}_\theta$: the operator added to the action is the θ-term Eq. (42.128), and a Minkowski action term $\int d^4x\, \mathcal{L}_\theta$ contributes $e^{i\int \mathcal{L}_\theta} = e^{iQ\theta}$ to the weight, matching Eq. (42.129) term by term. Finally,

because $Q \in \mathbb{Z}$, the integrand of Eq. (42.127) is invariant under $\theta \to \theta + 2\pi$, $e^{iQ(\theta+2\pi)} = e^{iQ\theta}$; the physics is 2π-periodic in the vacuum angle. □

The theta-term is imaginary in Euclidean signature

In Eq. (42.127) the θ-term enters the *Euclidean* exponent with an explicit factor of i: $e^{-S_E + i\theta Q}$, i.e. the Euclidean action is $S_E^{\text{eff}} = S_E - i\theta Q$. Unlike the kinetic action it is not made real by Wick rotation. This is not a blemish but the signature of the term's T-odd character: a T- and CP-violating coupling necessarily appears imaginary in the Euclidean weight, which is why a nonzero θ obstructs the usual probabilistic (positive-measure) interpretation of the Euclidean path integral and complicates its lattice evaluation (the sign problem revisited in Chapter 43).

The identification Eq. (42.128) matches, term for term and prefactor for prefactor, the operator $\mathcal{L}_\theta$ catalogued as the last renormalizable invariant of the Standard Model in Eq. (33.68); the normalization $g^2/32\pi^2$ was chosen there precisely so that the coefficient is an angle, and the present construction is the promised derivation (imported as Eq. (33.79)) of why it is an angle: the action contribution is θ times the integer Q, periodic by Theorem 42.30. What was a curious total-derivative operator in the classical catalogue is now seen as the bookkeeping device that records which Bloch sector of the quantum vacuum one inhabits.

42.4.7 P, T, and CP violation by the angle

The operator Eq. (42.128) is the four-dimensional non-abelian avatar of $\mathbf{E}\cdot\mathbf{B}$, and it inherits that quantity's behaviour under the discrete spacetime symmetries. This is the physical sting of the whole construction: a nonzero θ makes the strong interactions violate parity and time reversal.

Worked Example 42.31 (Theta and the discrete symmetries)**.** Write the topological density in the colour-electric/magnetic form. With $\widetilde{F}^{a\,\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F^a_{\rho\sigma}$ and $E^a_i = F^a_{0i}$, $B^a_i = -\frac{1}{2}\epsilon_{ijk}F^a_{jk}$,

$$F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu} = -4\,E^a_i B^a_i = -4\,\mathbf{E}^a\cdot\mathbf{B}^a. \tag{42.130}$$

Under *parity* $\mathbf{x} \to -\mathbf{x}$ the chromo-electric field, a polar vector, flips, $\mathbf{E}^a \to -\mathbf{E}^a$, while the chromo-magnetic field, an axial vector, does not, $\mathbf{B}^a \to +\mathbf{B}^a$; hence $\mathbf{E}^a\cdot\mathbf{B}^a \to -\mathbf{E}^a\cdot\mathbf{B}^a$ and

$$\mathsf{P}: \quad \mathcal{L}_\theta \to -\mathcal{L}_\theta, \tag{42.131}$$

which is the statement that P maps the theory at angle θ to the theory at angle $-\theta$. Under *time reversal* $t \to -t$ the electric field is even and the magnetic field, built from currents, is odd, $\mathbf{E}^a \to +\mathbf{E}^a$, $\mathbf{B}^a \to -\mathbf{B}^a$, so again $\mathbf{E}^a\cdot\mathbf{B}^a \to -\mathbf{E}^a\cdot\mathbf{B}^a$

and $\mathsf{T} : \theta \to -\theta$. The term is even under *charge conjugation*, $\mathsf{C} : \mathcal{L}_\theta \to +\mathcal{L}_\theta$ (both $\mathbf{E}^a$ and $\mathbf{B}^a$ are C-odd, their product even). Combining, CP and T each act as $\theta \to -\theta$, consistent with the CPT theorem; and since the physics is 2π-periodic, the theory is CP-invariant only when $-\theta \equiv \theta \pmod{2\pi}$, i.e. at

$$\theta = 0 \quad \text{and} \quad \theta = \pi. \tag{42.132}$$

For every other value, the strong interactions violate P, T, and CP — a violation distinct in origin from the CKM phase, and one that observation (the neutron electric dipole moment) bounds to be extraordinarily small. That this CP-violating angle exists, is physical, and is empirically tiny is the *strong-CP problem*, whose sharp statement we reach in Section 42.7.

We have, then, completed the canonical reading of the instanton promised at the chapter's outset. The degenerate $|n\rangle$ vacua, connected by the large gauge transformation $\hat{T}$ and tunnelled between by the instanton, organize into a one-parameter family of true, clustering vacua $|\theta\rangle$; the parameter θ is superselected, and a fixed θ-sector is identical to ordinary Yang–Mills with the CP-odd θ-term Eq. (42.128) added. The whole of this physics is weighted by $e^{-8\pi^2/g^2}$ and so lies entirely outside perturbation theory. What remains is to turn the qualitative band Eq. (42.118) into a quantitative θ-dependent vacuum energy by summing the gas of instantons and anti-instantons with their collective-coordinate measure. That sum, organized through the sector partition function $Z(\theta) = \sum_Q e^{iQ\theta} Z_Q$ of Eq. (42.127) and honest about where the diluteness assumption fails for QCD, is the subject of Section 42.5.

42.5 The Dilute Instanton Gas and the Vacuum Energy

The previous section reorganized the Euclidean path integral by topological charge,

$$Z(\theta) \;=\; \sum_{Q\in\mathbb{Z}} e^{i\theta Q}\, Z_Q, \qquad Z_Q = \int_{[A]:\, Q[A]=Q} \mathcal{D}A\; e^{-S_E[A]}, \tag{42.133}$$

and identified the angle θ as a genuine, superselected label of the Yang–Mills vacuum (Eq. (42.127)). That decomposition is exact but empty until we can actually *compute* a $Q \neq 0$ sector, and perturbation theory cannot: it expands about the single classical vacuum $A = 0$, which sits in $Q = 0$ and never leaves it. The objects that carry nonzero charge are the instantons of the earlier sections — finite-action Euclidean saddles with $Q = \pm 1$ and action $S_{\text{inst}} = 8\pi^2/g^2$ (Eq. (42.13)) — and the program of this section is to evaluate Eq. (42.133) by steepest descent (Theorem 5.44) around configurations built from many widely separated instantons and anti-instantons: the *dilute instanton gas*.

Three things must be assembled. First, the contribution of a *single* instanton is not a number but an integral over the parameters that label the saddle — its position, its size, its orientation in colour space — because the action is invariant under the symmetries that move these parameters, so the would-be Gaussian fluctuation integral has flat directions. Turning those flat directions into a finite, correctly normalized measure is the collective-coordinate construction, and it produces the celebrated ρ^{-5} size distribution. Second, the bare weight $e^{-8\pi^2/g^2}$ is not the end of the story: the one-loop fluctuations around the instanton dress the coupling into the *running* coupling at the instanton scale, and the resulting ρ^{b_0-5} distribution diverges in the infrared — the honest signal that the dilute-gas approximation, while exact in its parametric structure, is quantitatively uncontrolled for QCD. Third, exponentiating the gas converts the single-instanton weight into a θ-dependent vacuum energy $E(\theta) = -2K\cos\theta$, whose curvature at the origin is the topological susceptibility χ_t. We carry all three out, and we are explicit about what is reliably computed semiclassically and what is not.

42.5.1 The single-instanton saddle and its collective coordinates

Consider the one-instanton sector Z_1. Its leading semiclassical value comes from expanding the action about the classical instanton solution A^{cl}_μ found in the earlier sections. Writing $A_\mu = A^{\mathrm{cl}}_\mu + a_\mu$ and expanding to quadratic order in the fluctuation a_μ,

$$S_E[A^{\mathrm{cl}}+a] = S_{\mathrm{inst}} + \tfrac{1}{2}\int d^4x\, d^4y\; a^a_\mu(x)\, M^{ab}_{\mu\nu}(x,y)\, a^b_\nu(y) + O(a^3), \qquad S_{\mathrm{inst}} = \frac{8\pi^2}{g^2}, \tag{42.134}$$

where $M = \delta^2 S_E/\delta a\,\delta a$ is the fluctuation operator (the gauge-covariant Laplacian in the instanton background, supplemented by the Faddeev–Popov ghost operator from gauge fixing; we work in the background-field gauge so that M is gauge covariant). The naive Gaussian integral over a would give

$$Z_1 \overset{?}{\sim} e^{-S_{\mathrm{inst}}}\,\big[\det{}'M\big]^{-1/2} \times (\text{ghost det}), \tag{42.135}$$

but this is ill-defined: M has *zero modes*, eigenfunctions $\psi^{(i)}$ with $M\psi^{(i)} = 0$, and each contributes a factor $\lambda_i^{-1/2} = 0^{-1/2} = \infty$ to the inverse square-root determinant. The prime on $\det'$ in Eq. (42.135) indicates that these modes have been *excluded* from the determinant; they must be integrated exactly, not in Gaussian approximation. This is the heart of the collective-coordinate method, and it is the direct Euclidean analogue of the zero-mode treatment for the kink (Section 41.1.3).

Why the zero modes are there, and how many. A zero mode is a fluctuation that changes A^{cl} without changing the action. Every continuous

symmetry of S_E that is *broken* by the instanton — i.e. that moves the solution to a physically distinct but degenerate solution — generates exactly one such flat direction, because acting with the symmetry on A^{cl} produces a new solution of the same action, and the infinitesimal generator of that motion is a zero mode of M. The classical massless Yang–Mills action enjoys

- **translations** of the instanton centre x_0: 4 parameters, hence 4 zero modes;
- **dilatations** (scale invariance of classical massless YM): the instanton has an arbitrary size ρ, giving 1 zero mode;
- **global gauge rotations** (colour orientation): for $SU(N)$ the instanton is built by embedding an $SU(2)$ solution, and the rotations that act nontrivially on it furnish $4N-5$ zero modes (for $N=2$ this is 3, the three Euler angles of $SU(2)$ modulo the $U(1)$ stabilizer).

The total count is

$$n_{\mathrm{zero}} \;=\; \underbrace{4}_{x_0} \;+\; \underbrace{1}_{\rho} \;+\; \underbrace{(4N-5)}_{\text{orientation}} \;=\; 4N. \tag{42.136}$$

This $4N$ is the bosonic part of the Atiyah–Singer count for a single instanton; we take it as given here and return to its fermionic companion (one chiral zero mode per flavour) in the next section. For the colour group of QCD, $N=3$, there are 12 bosonic collective coordinates.

Trading zero modes for collective coordinates. The rigorous device is a Faddeev–Popov insertion identical in spirit to gauge fixing (Theorem 26.18), now applied to the moduli of the solution. For each broken symmetry we insert into the path integral a "1" resolved as an integral over the corresponding collective coordinate $\gamma_i \in \{x_0^\mu, \rho, \text{orientation}\}$ times a Jacobian. The result is a clean theorem we now state and prove.

Proposition 42.32 (Collective-coordinate measure for the bosonic zero modes)**.** *Let $A^{\mathrm{cl}}(x;\gamma)$ be the instanton solution as a function of its collective coordinates $\gamma = (\gamma_1,\dots,\gamma_{n_{\mathrm{zero}}})$, and let the normalized zero modes be*

$$\psi_\mu^{(i)}(x) \;=\; \frac{1}{\sqrt{U_i}}\,\frac{\partial A_\mu^{\mathrm{cl}}}{\partial \gamma_i}, \qquad U_i \;=\; \int d^4x\,\Big(\frac{\partial A^{\mathrm{cl}}}{\partial \gamma_i}\Big)^2, \tag{42.137}$$

with U_i the zero-mode norm. Then the exact integration over the i-th flat direction replaces the (divergent) factor $\lambda_i^{-1/2}$ in Eq. (42.135) by

$$\int \frac{d\gamma_i}{\sqrt{2\pi}}\,\sqrt{U_i} \;=\; \int d\gamma_i \,\sqrt{\frac{U_i}{2\pi}}. \tag{42.138}$$

The full one-instanton measure is therefore

$$Z_1 = e^{-S_{\rm inst}} \left(\prod_{i=1}^{n_{\rm zero}} \int d\gamma_i \sqrt{\frac{U_i}{2\pi}} \right) \left[\det{}' M \right]^{-1/2} \times (\textit{ghost det}). \tag{42.139}$$

Proof. Decompose the fluctuation in the complete orthonormal eigenbasis of M, $a_\mu(x) = \sum_i c_i\, \psi_\mu^{(i)}(x) + \sum_n \xi_n\, \phi_\mu^{(n)}(x)$, where the $\psi^{(i)}$ are the zero modes Eq. (42.137) and the $\phi^{(n)}$ the nonzero modes. The measure factorizes, $\mathcal{D}a = \prod_i \frac{dc_i}{\sqrt{2\pi}} \prod_n \frac{d\xi_n}{\sqrt{2\pi}}$, with the $\sqrt{2\pi}$ chosen so that a unit-eigenvalue Gaussian integrates to 1. The nonzero modes give $\prod_n \int \frac{d\xi_n}{\sqrt{2\pi}}\, e^{-\frac{1}{2}\lambda_n \xi_n^2} = \prod_n \lambda_n^{-1/2} = (\det{}' M)^{-1/2}$. For the zero modes the quadratic form vanishes, so $\int \frac{dc_i}{\sqrt{2\pi}}$ would diverge as the (infinite) length of the flat direction; this is precisely the volume of the symmetry orbit, which is physical and must be kept as the collective-coordinate integral, not discarded.

To convert dc_i into $d\gamma_i$, note that moving along the flat direction by dc_i is the same physical deformation as changing the collective coordinate by $d\gamma_i$. Comparing the two parametrizations of the same displacement of $A^{\rm cl}$,

$$\delta A_\mu^{\rm cl} \;=\; \psi_\mu^{(i)}\, dc_i \;=\; \frac{\partial A_\mu^{\rm cl}}{\partial \gamma_i}\, d\gamma_i \;=\; \sqrt{U_i}\, \psi_\mu^{(i)}\, d\gamma_i, \tag{42.140}$$

where the last equality used Eq. (42.137). Matching coefficients of the normalized mode gives $dc_i = \sqrt{U_i}\, d\gamma_i$, hence $\int \frac{dc_i}{\sqrt{2\pi}} = \int d\gamma_i \sqrt{U_i/2\pi}$, which is Eq. (42.138). Doing this for every zero mode yields Eq. (42.139). (When the γ_i are not orthogonal one replaces $\prod_i \sqrt{U_i}$ by $\sqrt{\det U}$, with $U_{ij} = \int (\partial_i A^{\rm cl})(\partial_j A^{\rm cl})$ the metric on moduli space; for the instanton the natural coordinates are orthogonal and the product form holds.) $\square$

Equation (42.139) is exact at one loop. To read off its content we need two facts about how $S_{\rm inst}$, the norms U_i, and the determinant scale with the coupling and the size.

The power of the coupling: g^{-4N}. The classical instanton field scales as $1/g$. This is forced by the structure of the action: with $A_\mu = \frac{1}{g}\hat{A}_\mu$, the Yang–Mills action $S_E = \frac{1}{2g^2}\int \mathrm{tr}\, F^2$ written in terms of $\hat{A}$ has *no* explicit g, because $F_{\mu\nu} = \frac{1}{g}\hat{F}_{\mu\nu}$ with $\hat{F} = \partial\hat{A} - \partial\hat{A} + \hat{A}\hat{A}$ coupling-independent; the entire g-dependence sits in the prefactor $1/g^2$. Hence the solution $\hat{A}^{\rm cl}$ is g-independent and $A^{\rm cl} = \hat{A}^{\rm cl}/g$. Consequently each zero-mode norm scales as $U_i = \int (\partial_i A^{\rm cl})^2 \propto 1/g^2$, so $\sqrt{U_i/2\pi} \propto 1/g$, and the product over all $n_{\rm zero} = 4N$ zero modes contributes

$$\prod_{i=1}^{4N} \sqrt{\frac{U_i}{2\pi}} \;\propto\; g^{-4N} \;=\; \left(g^{-2}\right)^{2N} \;\propto\; \left(\frac{8\pi^2}{g^2}\right)^{2N}, \tag{42.141}$$

where in the last step we have traded the bare g^{-4N} for the dimensionless and more telling combination $(8\pi^2/g^2)^{2N} = S_{\rm inst}^{2N}$, absorbing the numerical $(8\pi^2)^{-2N}$ into the overall constant. This reproduces the weight quoted in the chapter blueprint: the single-instanton amplitude carries $(8\pi^2/g^2)^{2N}\, e^{-8\pi^2/g^2}$. Each of the $4N$ zero modes contributes one half-power of the action; physically, the more collective coordinates the configuration has, the larger the phase-space volume of the saddle, and the prefactor grows as a high power of $1/g^2$ even as the exponential suppresses.

The power of the size: ρ^{-5}. The remaining ρ-dependence is fixed by dimensional analysis, and this is where the ρ^{-5} originates. Classical massless Yang–Mills has *no intrinsic scale*: the only dimensionful collective coordinate available to the single-instanton density is the size ρ itself. Write the one-instanton contribution as a density in spacetime,

$$Z_1 \;=\; \int d^4x_0 \int d\rho \int d\Omega \;\; \mathrm{d}n(\rho), \tag{42.142}$$

where $d\Omega$ is the (dimensionless, finite-volume) integral over colour orientations and $\mathrm{d}n(\rho)$ is the differential instanton density. The left side Z_1 is a pure number; the measure $d^4x_0\, d\rho$ has mass dimension $[\text{mass}]^{-4}\cdot[\text{mass}]^{-1} = [\text{mass}]^{-5}$; hence $\mathrm{d}n(\rho)$ must carry dimension $[\text{mass}]^{+5}$. With ρ the only available scale,

$$\mathrm{d}n(\rho) \;=\; \frac{\text{const}}{\rho^5}\times(\text{dimensionless function of } g), \qquad \boxed{\mathrm{d}n = C_N\,\frac{d\rho}{\rho^5}\left(\frac{8\pi^2}{g^2}\right)^{2N} e^{-8\pi^2/g^2}\, d^4x_0\, d\Omega} \tag{42.143}$$

collecting Eq. (42.141) and the exponential. The same counting reads off where the five powers come from: four from the translational measure d^4x_0 (which by dimensions must be balanced) and one from $d\rho$, with the norms U_i supplying exactly the ρ-powers needed to make $\mathrm{d}n$ scale-covariant. The dimensionless constant C_N is the only piece Eq. (42.143) does *not* fix: it is the value of the regularized nonzero-mode determinant $[\det{}' M]^{-1/2}$ times the ghost determinant times the numerical zero-mode norms, evaluated at $\rho = 1$. Computing C_N is 't Hooft's original tour de force [464]; the answer in the Pauli–Villars scheme is

$$C_N \;=\; \frac{2}{(N-1)!\,(N-2)!}\,\frac{e^{-\alpha(1)+2(N-2)\alpha(1/2)}}{\pi^2}\,, \qquad \alpha(1)\simeq 0.443, \tag{42.144}$$

a finite number of order unity (for $SU(2)$, $C_2 \simeq 0.466/\pi^2$). We do not reproduce the multi-page evaluation of the functional determinant — it is the one piece of this section that is stated without proof, and Eq. (42.144) is given so the reader can use the measure — but every *parametric* feature of Eq. (42.143), the ρ^{-5}, the $(8\pi^2/g^2)^{2N}$, and the exponential, has been derived here in full from zero-mode counting, the $1/g$ scaling of the solution, and dimensional analysis.

Caution

The ρ^{-5} is a statement about the *classical, scale-invariant* theory, where ρ is the only scale. The instant we regularize the determinant we introduce a renormalization scale μ, and scale invariance is broken by the trace anomaly. The bare g in Eq. (42.143) is then a renormalized $g(\mu)$, and a compensating factor of $(\mu\rho)$ to some power reappears. Restoring it converts ρ^{-5} into ρ^{b_0-5}, the subject of the next subsection. Quoting ρ^{-5} as the physical size distribution is the classic error; the running coupling is not optional.

42.5.2 Inserting the running coupling: ρ^{b_0-5} and the infrared breakdown

The constant C_N in Eq. (42.144) is computed by regularizing the determinant $\det' M$ with a Pauli–Villars mass μ, and the regularized determinant depends on the dimensionless ratio $\mu\rho$. Dimensional analysis cannot forbid this dependence — μ is a second scale — and the explicit one-loop computation produces, for each of the modes that renormalize the coupling, a logarithm $\ln(\mu\rho)$ that combines with the bare coupling exactly as the renormalization group demands. Concretely, the determinant supplies a factor $(\mu\rho)^{b_0}$ multiplying Eq. (42.143), with the *same* coefficient b_0 that governs the one-loop beta function Eq. (27.101),

$$b_0 \;=\; \tfrac{11}{3}\,C_A - \tfrac{4}{3}\,T_F\,n_f \;\xrightarrow{\text{pure } SU(N)}\; \tfrac{11}{3}\,N. \tag{42.145}$$

That the power is precisely b_0 is not a coincidence but a Ward identity of the renormalization group: the instanton density is a physical quantity and must be μ-independent, so the explicit μ^{b_0} from the determinant must cancel against the μ-dependence of the running coupling hidden in $e^{-8\pi^2/g^2(\mu)}$. We now make that cancellation explicit, which is the cleanest way to see both why the power is b_0 and what the surviving ρ-dependence is.

Recall the one-loop running of Yang–Mills theory. From Eq. (27.129), or equivalently from the definition of Λ in Theorem 27.14, the inverse coupling runs linearly in the log, and in the convenient combination that appears in the instanton exponent,

$$\frac{8\pi^2}{g^2(\mu)} \;=\; b_0\,\ln\frac{\mu}{\Lambda}, \qquad\Longleftrightarrow\qquad e^{-8\pi^2/g^2(\mu)} \;=\; \Big(\frac{\Lambda}{\mu}\Big)^{b_0}. \tag{42.146}$$

The first equality follows from $1/\alpha_s(\mu) = (b_0/2\pi)\ln(\mu/\Lambda)$ (Eq. (27.152)) and $8\pi^2/g^2 = 2\pi/\alpha_s$. Now combine the determinant's explicit $(\mu\rho)^{b_0}$ with Eq. (42.146):

$$e^{-8\pi^2/g^2(\mu)}\,(\mu\rho)^{b_0} \;=\; \Big(\frac{\Lambda}{\mu}\Big)^{b_0}(\mu\rho)^{b_0} \;=\; (\Lambda\rho)^{b_0} \;=\; e^{-8\pi^2/g^2(1/\rho)}. \tag{42.147}$$

The μ has cancelled, as it must, and the net effect is the substitution

$$\boxed{e^{-8\pi^2/g^2} \;\longrightarrow\; e^{-8\pi^2/g^2(1/\rho)} \;=\; (\Lambda\rho)^{b_0}} \tag{42.148}$$

together with the prefactor $(8\pi^2/g^2)^{2N} \to (8\pi^2/g^2(1/\rho))^{2N} = (b_0 \ln(1/\rho\Lambda))^{2N}$. The physics is transparent: *the coupling that weights an instanton of size ρ is the running coupling evaluated at the instanton's own scale $1/\rho$.* A small instanton is governed by a small (asymptotically free) coupling and is exponentially suppressed; a large instanton is governed by a large coupling and is barely suppressed at all. Assembling Eq. (42.143) with Eq. (42.148) gives the renormalization-group-improved size distribution,

$$\mathrm{d}n(\rho) \;=\; C_N \,\frac{d\rho}{\rho^5}\left(\frac{8\pi^2}{g^2(1/\rho)}\right)^{2N} (\Lambda\rho)^{b_0}\, d^4x_0\, d\Omega \;\propto\; \frac{d\rho}{\rho^5}\,\rho^{\,b_0}\big[\ln(1/\rho\Lambda)\big]^{2N}\, d^4x_0\, d\Omega, \tag{42.149}$$

so that, up to the slowly-varying logarithm, the differential density per unit four-volume scales as

$$\frac{\mathrm{d}n}{d^4x_0\, d\Omega} \;\sim\; \rho^{\,b_0-5}\,\frac{d\rho}{\rho}\cdot(\text{logs}). \tag{42.150}$$

This is the announced $\rho^{\,b_0-5}$, and it is the crux of the assessment this section demands. For pure $SU(3)$, $b_0 = 11$, so the power is ρ^6: the size integral $\int d\rho\,\rho^{\,b_0-5}$ is violently dominated by *large* instantons, $\rho \to \infty$. Even for the smallest interesting case, $SU(2)$ with $b_0 = 22/3$, the exponent $b_0 - 5 = 7/3 > 0$ and the integral still diverges in the infrared. Adding light fermions reduces b_0 but never enough to cure it for QCD. Figure 42.5 (left) plots Eq. (42.149): the distribution rises steeply with ρ until the logarithm $\ln(1/\rho\Lambda)$ — that is, $8\pi^2/g^2(1/\rho)$ — collapses as $\rho \to 1/\Lambda$, where the running coupling diverges and the entire saddle-point expansion that produced Eq. (42.149) loses meaning.

Physical Insight

The infrared divergence of Eq. (42.150) is the precise, quantitative statement of why "QCD instanton calculus" is not a controlled approximation. The semiclassical expansion is an expansion in $\hbar$, equivalently in g^2 at *fixed* action $8\pi^2/g^2$; it is reliable only where the relevant coupling is small. But the instanton integral is dominated by large instantons, whose relevant coupling $g^2(1/\rho)$ is order one — the strong-coupling, confining regime. The dilute-gas computation therefore gets the *structure* of the answer right (there is a θ-dependence; it is nonperturbative, $\sim e^{-8\pi^2/g^2}$; it has the form $-2K\cos\theta$ at leading order) but cannot reliably deliver the *number* K for QCD. This is not a defect to be hidden; it is the reason Chapter 43 exists.

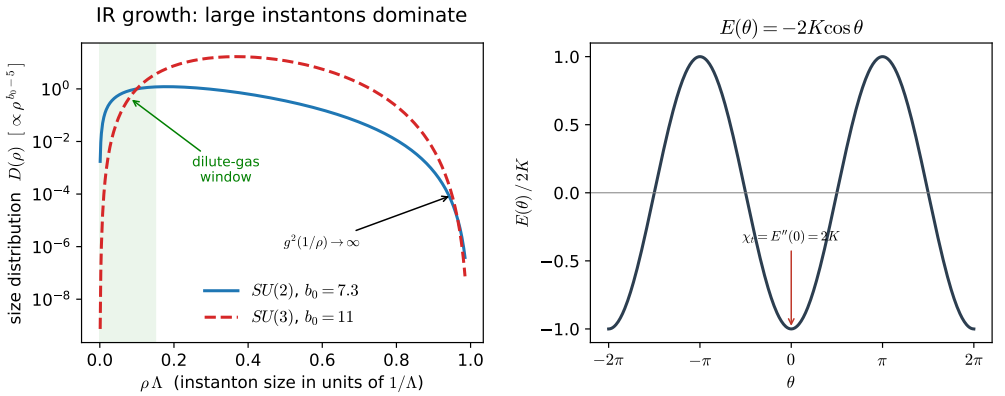


Figure 42.5: *Left:* the renormalization-group-improved instanton size distribution $\mathrm{d}n \propto \rho^{b_0-5}[\,b_0 \ln(1/\rho\Lambda)]^{2N}$ of Eq. (42.149) for pure $SU(2)$ ($b_0 = 22/3$) and $SU(3)$ ($b_0 = 11$), versus the dimensionless size $\rho\Lambda$. The distribution grows toward large ρ; the dilute, semiclassical computation is reliable only in the shaded ultraviolet window $\rho\Lambda \ll 1$, where $g^2(1/\rho)$ is small. As $\rho\Lambda \to 1$ the running coupling diverges and the approximation breaks down: the size integral is dominated by precisely the region where it cannot be trusted. *Right:* the output of exponentiating the gas, the θ-dependent vacuum energy density $E(\theta) = -2K\cos\theta$ of Eq. (42.156), normalized to $2K$. Its curvature at the origin is the topological susceptibility $\chi_t = E''(0) = 2K$.

Caution

A second assumption is buried in calling this a *dilute* gas: that the mean separation between (anti-)instantons is large compared to their size, so that their actions simply add and the multi-instanton configuration is, to good approximation, a superposition of well-separated single instantons. The same infrared growth that ruins the size integral also makes large instantons overlap and interact strongly, violating diluteness. Both failures have the same root — large instantons — and both are cured only outside semiclassics. We proceed with the dilute gas as a *structural* computation, treating K as a formal parameter to be fixed by the lattice.

42.5.3 Exponentiating the gas: $E(\theta) = -2K\cos\theta$

Granting diluteness, we now sum the gas. A general semiclassical configuration consists of n instantons (each $Q = +1$) and $\bar{n}$ anti-instantons (each $Q = -1$), centred at well-separated points. Because they are dilute and non-interacting, the total action is additive and the total topological charge is the algebraic sum,

$$S_E \;\approx\; (n+\bar{n})\,S_{\text{inst}}, \qquad Q \;=\; n-\bar{n}. \tag{42.151}$$

Each (anti-)instanton contributes the integrated single-(anti-)instanton weight. Define the *integrated density per unit four-volume*,

$$K \;\equiv\; \int d\rho \int d\Omega\, \frac{\mathrm{d}n}{d^4x_0} \;=\; C_N \int \frac{d\rho}{\rho^5} \Big(\frac{8\pi^2}{g^2(1/\rho)}\Big)^{2N} (\Lambda\rho)^{b_0} \int d\Omega, \tag{42.152}$$

a quantity of mass dimension $[\text{mass}]^4$ (a density), carrying the all-important nonperturbative factor $K \propto e^{-8\pi^2/g^2}$. By the parity symmetry of the pure gauge action under $A \to A$, $x \to -x$ (which exchanges instantons and anti-instantons), the anti-instanton integrated density equals K as well. The integral in Eq. (42.152) is the infrared-divergent one of the previous subsection; we hold K fixed and finite, as discussed.

Now build $Z(\theta)$. A configuration of n instantons and $\bar{n}$ anti-instantons, with all centres integrated over a Euclidean four-volume $V_4 = VT$ (spatial volume V, Euclidean time extent T), and all sizes and orientations integrated, contributes

$$\frac{1}{n!}\,\frac{1}{\bar{n}!}\,(K\,V_4)^n\,(K\,V_4)^{\bar{n}}. \tag{42.153}$$

The factor $(KV_4)^n$ is n independent integrals of the single-instanton density over the four-volume; the $1/n!$ is the Gibbs factor for n *identical* pseudoparticles — the collective-coordinate integration over n instanton centres counts each unordered configuration $n!$ times, and likewise $1/\bar{n}!$ for the anti-instantons. Inserting the topological weight $e^{i\theta Q} = e^{i\theta(n-\bar{n})}$ from the θ-vacuum (Eq. (42.127)) and summing over all $n, \bar{n} \geq 0$,

$$\begin{aligned} Z(\theta) &= \sum_{n=0}^{\infty}\sum_{\bar{n}=0}^{\infty} \frac{(KV_4)^n}{n!}\,\frac{(KV_4)^{\bar{n}}}{\bar{n}!}\; e^{i\theta(n-\bar{n})} \\ &= \Bigg(\sum_{n=0}^{\infty} \frac{(KV_4\,e^{i\theta})^n}{n!}\Bigg)\Bigg(\sum_{\bar{n}=0}^{\infty} \frac{(KV_4\,e^{-i\theta})^{\bar{n}}}{\bar{n}!}\Bigg) \\ &= \exp\!\big(KV_4\,e^{i\theta}\big)\,\exp\!\big(KV_4\,e^{-i\theta}\big) \;=\; \exp\!\big(2KV_4\cos\theta\big). \end{aligned} \tag{42.154}$$

Each sum is the exponential series; the two fugacities $Ke^{\pm i\theta}$ differ only by the sign of the topological phase. This is the exponentiation of the dilute gas: a sum over an arbitrary number of pseudoparticles collapses to a single exponential of the single-pseudoparticle contribution, exactly as the cluster expansion of a dilute classical gas exponentiates into the ideal-gas free energy. The summation also *justifies a posteriori* the neglect of interactions: the leading large-volume behaviour is controlled by the one-body density K, and two-body corrections are down by the diluteness parameter (the fraction of four-volume occupied by instanton cores).

The vacuum energy density now follows from the standard Euclidean relation between the partition function and the ground-state energy. For Euclidean time

extent $T \to \infty$,

$$Z \;=\; \mathrm{Tr}\, e^{-TH} \;\xrightarrow{T\to\infty}\; e^{-TE_0} \;=\; e^{-V_4\, E(\theta)}, \tag{42.155}$$

where $E(\theta) = E_0/V$ is the vacuum energy density in the θ-vacuum. Comparing Eq. (42.154) with Eq. (42.155),

$$-V_4\, E(\theta) \;=\; 2KV_4 \cos\theta \qquad \Longrightarrow \qquad \boxed{E(\theta) \;=\; -2K\cos\theta} \tag{42.156}$$

which is the central result. The vacuum energy is minimized at $\theta = 0$ (recall $K > 0$), where $E(0) = -2K$, and the energy is a smooth, 2π-periodic function of θ, as required by the periodicity established for the θ-vacuum. Figure 42.5 (right) shows the shape. The depth of the well, $2K \propto e^{-8\pi^2/g^2}$, is nonanalytic in g at $g = 0$: it has a vanishing Taylor expansion there, which is exactly why no order of perturbation theory ever sees it. Instantons are the leading term of the trans-series that perturbation theory is blind to — the nonperturbative completion anticipated by the large-order/Borel discussion of Theorem 5.47.

42.5.4 The topological susceptibility

The single most physical number extractable from $E(\theta)$ is its curvature at the origin, the *topological susceptibility*.

Definition 42.33 (Topological susceptibility)**.** The topological susceptibility is the second derivative of the vacuum energy density with respect to the vacuum angle, evaluated at $\theta = 0$:

$$\chi_t \;\equiv\; \left.\frac{d^2 E(\theta)}{d\theta^2}\right|_{\theta=0}. \tag{42.157}$$

In the dilute gas, Eq. (42.156) gives immediately

$$\chi_t \;=\; \left.\frac{d^2}{d\theta^2}\big(-2K\cos\theta\big)\right|_{\theta=0} \;=\; \left.2K\cos\theta\right|_{\theta=0} \;=\; 2K. \tag{42.158}$$

The susceptibility has a second, microscopic reading that makes its nonperturbative, purely-topological nature manifest: it is the variance of the topological charge per unit four-volume. We establish this as a worked computation, both because the identity is model-independent (it follows from Eq. (42.133) alone, not from the dilute gas) and because checking that the gas reproduces it is a nontrivial consistency test of the whole construction.

Worked Example 42.34 (χ_t as the topological-charge variance, two ways)**. (a) Model-independent identity.** Start from the exact sector sum Eq. (42.133). The moments of Q in the $\theta = 0$ ensemble are

$$\big\langle Q^k \big\rangle \;=\; \frac{\sum_Q Q^k Z_Q}{\sum_Q Z_Q}. \tag{42.159}$$

Differentiating $\ln Z(\theta) = \ln \sum_Q e^{i\theta Q} Z_Q$,

$$\frac{d}{d\theta} \ln Z = \frac{\sum_Q (iQ) e^{i\theta Q} Z_Q}{\sum_Q e^{i\theta Q} Z_Q} \xrightarrow{\theta=0} i \langle Q \rangle , \tag{42.160}$$

$$\frac{d^2}{d\theta^2} \ln Z = \frac{\sum_Q (iQ)^2 e^{i\theta Q} Z_Q}{Z} - \Big(\frac{\sum_Q (iQ) e^{i\theta Q} Z_Q}{Z} \Big)^2 \xrightarrow{\theta=0} -\big(\big\langle Q^2 \big\rangle - \langle Q \rangle^2 \big). \tag{42.161}$$

The pure gauge theory is CP-symmetric at $\theta = 0$, so $\langle Q \rangle = 0$ and the connected variance is just $\langle Q^2 \rangle$. Using $E(\theta) = -V_4^{-1} \ln Z$ from Eq. (42.155),

$$\chi_t = \frac{d^2 E}{d\theta^2}\Big|_0 = -\frac{1}{V_4} \frac{d^2 \ln Z}{d\theta^2}\Big|_0 = \frac{1}{V_4} \big\langle Q^2 \big\rangle \Big|_{\theta=0}. \tag{42.162}$$

So $\chi_t = \langle Q^2 \rangle / V_4$: the mean-square topological charge per unit four-volume. This is the definition used on the lattice, where Q is measured configuration by configuration and its variance accumulated. It involves no instanton, no coupling expansion, no diluteness — it is an exact property of the theory, which is why it is the clean target for a first-principles computation.

(b) The gas reproduces it. In the dilute gas, $Q = n - \bar{n}$ with $n, \bar{n}$ independent *Poisson* variables: from Eq. (42.154) at $\theta = 0$, the probability of n instantons is $P(n) = e^{-KV_4} (KV_4)^n / n!$, with mean $\langle n \rangle = KV_4$ and, being Poisson, variance $\mathrm{Var}(n) = KV_4$; likewise for $\bar{n}$. Independence gives

$$\big\langle Q^2 \big\rangle = \big\langle (n - \bar{n})^2 \big\rangle = \mathrm{Var}(n) + \mathrm{Var}(\bar{n}) = KV_4 + KV_4 = 2KV_4, \tag{42.163}$$

using $\langle n \rangle = \langle \bar{n} \rangle$ so that $\langle Q \rangle = 0$. Hence $\chi_t = \langle Q^2 \rangle / V_4 = 2K$, in exact agreement with Eq. (42.158). The factor of two is the sum of two independent Poisson variances, one from each chirality of pseudoparticle — a small but satisfying check that the combinatorics of Eq. (42.154) are right.

The identity Eq. (42.162) elevates χ_t from a feature of an unreliable approximation to a sharply-defined observable. The dilute gas predicts its *parametric* size,

$$\chi_t \;=\; 2K \;\sim\; \Lambda^4 \, e^{-8\pi^2 / g^2(\Lambda)} \times (\text{powers, logs}), \tag{42.164}$$

nonperturbatively small and built from the QCD scale — but the prefactor and even the ρ-integral that defines K are infrared-dominated, so the actual number is not semiclassically computable for QCD.

42.5.5 What is reliable, and a dated number

It is worth stating plainly, and with a date, where the semiclassical computation stands and where the real number comes from.

Status of χ_t, as of the mid-2020s

The topological susceptibility $\chi_t = \langle Q^2 \rangle / V_4$ is a genuine, scheme-independent nonperturbative observable of Yang–Mills theory. It is *invisible* to perturbation theory (it is $O(e^{-8\pi^2/g^2})$, nonanalytic at $g = 0$) and it is *not* reliably computable in the dilute instanton gas, because the size integral Eq. (42.152) is dominated by large instantons where the approximation fails (Fig. 42.5, left). The only first-principles method that delivers χ_t with controlled, systematically-improvable errors is the Euclidean lattice, where Q is defined and measured nonperturbatively and its variance computed by Monte Carlo. The current lattice value for pure $SU(3)$ Yang–Mills is

$$\chi_t^{1/4} \simeq 180\text{–}190 \text{ MeV} \qquad (\text{pure-gauge } SU(3),\ \text{lattice, mid-2020s}), \tag{42.165}$$

i.e. $\chi_t \simeq (180\,\text{MeV})^4$. In full QCD with light quarks χ_t is strongly *suppressed* by the quark masses — it vanishes in the chiral limit, as the next section's fermion zero modes will explain — and chiral perturbation theory predicts the leading behaviour $\chi_t \simeq \frac{m_u m_d}{(m_u+m_d)^2}\, m_\pi^2 f_\pi^2$ at small quark mass, again confirmed on the lattice. The lattice determination of χ_t is taken up in Chapter 43 (Chapter 43); the Witten–Veneziano relation linking χ_t to the η' mass is derived later in this chapter (Section 42.6.6). Here we have established the object, its dilute-gas estimate, and the honest boundary of that estimate.

The dilute instanton gas has thus done exactly what semiclassics can do and no more. It has *demonstrated* that the Yang–Mills vacuum carries nonperturbative structure: a tower of topological sectors, a physical vacuum angle θ, a vacuum energy $E(\theta) = -2K\cos\theta$ nonanalytic in the coupling, and a topological susceptibility $\chi_t = 2K = \langle Q^2 \rangle / V_4$ that no Feynman diagram can produce. It has *quantified* its own failure for QCD through the ρ^{b_0-5} infrared divergence. What it has not done — supply the number K — is precisely the task it hands to the lattice.

This completes the semiclassical, nonperturbative computation for pure gauge theory and advances Yang–Mills theory to its vacuum-structure layer. One ingredient has been conspicuously absent: fermions. The same instanton that reorganizes the gauge vacuum also supports *fermion zero modes* — one normalizable chiral solution of the Dirac equation per flavour, by the index theorem (Theorem 36.15) — and these zero modes make the path-integral measure non-invariant under axial rotations. The next section couples light fermions to the instanton, uses the zero modes to resolve the $U(1)_A$ problem (the absence of a ninth light pseudoscalar), and reads off 't Hooft's effective multi-fermion vertex — the bridge from this vacuum structure to its imprint on

the hadron spectrum.

42.6 Instantons, Fermions, and the $U(1)_A$ Problem

The previous sections built the instanton as a purely bosonic object and showed that it reorganizes the gauge vacuum: the dilute gas of Section 42.5 sums tunnelling events of every winding into a θ-dependent vacuum energy, and the topological susceptibility χ_t measures how steeply that energy curves with θ. All of this happened with no matter on the stage. Real QCD, however, has light quarks, and the moment we couple them to the instanton the story acquires its sharpest physical consequence. The classical Lagrangian of n_f massless quarks has a large global chiral symmetry, and naive counting of its spontaneously broken generators predicts one more light pseudoscalar meson than nature provides. The resolution is not that the symmetry is unbroken, nor that the missing meson is merely heavy by accident: it is that one of those symmetries, the axial $U(1)_A$, is not a symmetry of the quantum theory at all. The anomaly of Chapter 36 destroys it, and the agent that makes the destruction concrete — that turns the abstract anomaly equation into a calculable vacuum process — is precisely the instanton, acting through the fermion zero modes it is forced by the index theorem to carry. This section couples the quark to the instanton, integrates the anomaly to a clean topological statement, constructs the zero modes, builds the effective multi-fermion vertex 't Hooft extracted from them, and uses all of it to dispatch the $U(1)_A$ problem and to understand why the η' is heavy.

We work, as throughout this chapter, in Euclidean $\mathbb{R}^4$ with metric $\delta_{\mu\nu}$, gauge group $\mathrm{SU}(N)$ with Hermitian generators T^a normalized to $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, and the topological charge fixed in Section 42.3.3,

$$Q = \frac{g^2}{32\pi^2}\int \mathrm{d}^4x\, \epsilon_{\mu\nu\rho\sigma}\, \mathrm{tr}(F_{\mu\nu}F_{\rho\sigma}) = \frac{g^2}{32\pi^2}\int \mathrm{d}^4x\, F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu} \;\in\; \mathbb{Z}, \tag{42.166}$$

with $\widetilde{F}^a_{\mu\nu} = \frac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^a_{\rho\sigma}$ and $\epsilon_{1234} = +1$. The single BPST instanton of Theorem 42.17 is the minimal carrier of $Q = 1$, with self-dual field strength Eq. (42.90) and action $S = 8\pi^2/g^2$ from Theorem 42.4. The quarks are n_f Dirac fields q_f ($f = 1, \ldots, n_f$) in the fundamental representation, with the Euclidean Dirac action and propagator structure of Section 11.5 continued to imaginary time.

42.6.1 The chiral symmetries of massless QCD and the missing meson

Set all quark masses to zero. The Dirac Lagrangian $\bar{q}_f\, i\not{D}\, q_f$ does not mix flavors and treats the two chiralities independently, so it is invariant under independent unitary rotations of the left- and right-handed components,

$$q_{Lf} \to (U_L)_{fg}\, q_{Lg}, \qquad q_{Rf} \to (U_R)_{fg}\, q_{Rg}, \qquad U_L, U_R \in \mathrm{U}(n_f), \tag{42.167}$$

where $q_{L,R} = P_{L,R}\, q$ with $P_{L,R} = \tfrac{1}{2}(1 \mp \gamma^5)$. The classical symmetry group is therefore

$$G_{\rm cl} = \mathrm{U}(n_f)_L \times \mathrm{U}(n_f)_R = \mathrm{SU}(n_f)_L \times \mathrm{SU}(n_f)_R \times \mathrm{U}(1)_V \times \mathrm{U}(1)_A, \tag{42.168}$$

the last two factors being the vector ($U_L = U_R = e^{i\alpha}$) and axial ($U_L = e^{-i\beta}$, $U_R = e^{i\beta}$) phase rotations common to all flavors. The vector $\mathrm{U}(1)_V$ is baryon number, unbroken and harmless. The non-abelian $\mathrm{SU}(n_f)_L \times \mathrm{SU}(n_f)_R$ is the chiral symmetry whose spontaneous breakdown by the quark condensate $\langle \bar{q} q \rangle \neq 0$ to the diagonal $\mathrm{SU}(n_f)_V$ is the subject of Chapter 39: by Goldstone's theorem (Theorem 29.11) it produces $n_f^2 - 1$ massless pseudoscalars — for $n_f = 3$ the octet π, K, η, realized as the Goldstone matrix of Theorem 39.6, light because the quark masses that break the symmetry explicitly are small.

That leaves the axial singlet $\mathrm{U}(1)_A$, generated by the flavor-singlet axial current

$$j_5^\mu = \sum_{f=1}^{n_f} \bar{q}_f\, \gamma^\mu \gamma^5\, q_f. \tag{42.169}$$

The same condensate $\langle \bar{q} q \rangle \neq 0$ that breaks $\mathrm{SU}(n_f)_A$ is not invariant under $\mathrm{U}(1)_A$ either — an axial phase rotates $\bar{q}_R q_L \to e^{2i\beta} \bar{q}_R q_L$. So if $\mathrm{U}(1)_A$ were an exact symmetry, it would be spontaneously broken alongside the rest of the axial group, and Goldstone's theorem would demand a *ninth* massless pseudoscalar, a flavor singlet, in the chiral limit. With realistic small quark masses it would be a light pseudoscalar not much heavier than the pion. There is no such particle. The lightest flavor-singlet pseudoscalar is the η', at 958 MeV — nearly seven times the pion mass and heavier than every member of the octet. This is the $U(1)_A$ *problem*: the classical symmetry counting overpredicts the light spectrum by one state.

The discrepancy can be made quantitative, and the quantitative form is what shows the puzzle is real rather than a matter of an unluckily heavy meson.

Physical Insight: Weinberg's bound and why "it is just heavy" fails

If $\mathrm{U}(1)_A$ were a genuine, spontaneously broken symmetry, the would-be ninth Goldstone η_0 would couple to j_5^μ exactly as the pions couple to the broken $\mathrm{SU}(n_f)_A$ currents, with the same decay constant up to a calculable group factor. Weinberg analyzed the chiral Ward identities for the singlet current with the quark mass as the only explicit breaking and found a rigorous upper bound on the singlet mass,

$$m_{\eta_0}^2 \;\leq\; n_f\, m_\pi^2, \tag{42.170}$$

which for $n_f = 3$ reads $m_{\eta_0} \leq \sqrt{3}\, m_\pi \approx 234\,\mathrm{MeV}$. The point of the inequality is that the singlet mass is controlled by the *same* small explicit

breaking (the quark masses) as the pion mass: a state whose mass is protected by an approximate symmetry cannot be parametrically heavier than the protection allows. The physical η' at 958 MeV violates Eq. (42.170) by a factor of four in the mass, an order of magnitude in the mass-squared. No tuning of the quark masses can rescue it. The only way out is that $\mathrm{U}(1)_A$ is *not* an approximate symmetry of the quantum theory — that its breaking is not controlled by the quark masses at all, but by something that survives in the chiral limit. That something is the anomaly.

42.6.2 Integrating the anomaly: $\Delta Q_5 = 2n_f Q$

The classical conservation law $\partial_\mu j_5^\mu = 0$ (in the chiral limit) is the first casualty of quantization. Each Dirac flavor contributes to the divergence of its own axial current the Adler–Bell–Jackiw anomaly of Eq. (36.50), now with the color field strength in place of the photon and a trace over the fundamental representation. Summed over the n_f flavors, the singlet current Eq. (42.169) obeys

$$\partial_\mu j_5^\mu = \frac{g^2 n_f}{16\pi^2}\, F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu} = 2n_f\, q(x), \qquad q(x) \equiv \frac{g^2}{32\pi^2}\, F^a_{\mu\nu}\widetilde{F}^a_{\mu\nu}, \tag{42.171}$$

where $q(x)$ is the topological charge density whose integral is Eq. (42.166), and the factor 2 is the one that distinguishes the Dirac axial current $\bar{q}\gamma^\mu\gamma^5 q$ from a single Weyl chirality, exactly as flagged below Eq. (36.141). The right-hand side is unaffected by the quark masses: a mass adds the classical piece $2i\, m_f\, \bar{q}_f\gamma^5 q_f$ to the divergence, present already at the classical level, but the anomalous piece $2n_f q(x)$ is there whether or not the quarks are massive. This is the decisive structural fact. The breaking of $\mathrm{U}(1)_A$ has a source that does not vanish in the chiral limit.

Integrate Eq. (42.171) over Euclidean spacetime. The left-hand side is the total change in the axial charge carried across the configuration; the right-hand side is $2n_f$ times the integral of the topological density, which is $2n_f$ times the integer winding Q by Eq. (42.166):

$$\Delta Q_5 \;\equiv\; \int \mathrm{d}^4x\; \partial_\mu j_5^\mu \;=\; 2n_f \int \mathrm{d}^4x\; q(x) \;=\; 2n_f\, Q. \tag{42.172}$$

This is the central kinematic identity of the section, and it is exact: no order of perturbation theory enters, because Q is invisible order by order and the anomaly coefficient is fixed once and for all by the triangle (or, equivalently, by the Fujikawa measure, Theorem 36.9). A gauge field of winding Q changes the axial charge of the vacuum by exactly $2n_f Q$ units. For a single instanton, $\Delta Q_5 = 2n_f$: the instanton converts left-handed quarks into right-handed ones, one unit of chirality flip for each of the n_f flavors and for each of the two members of the Dirac doublet, $2n_f$ in all.

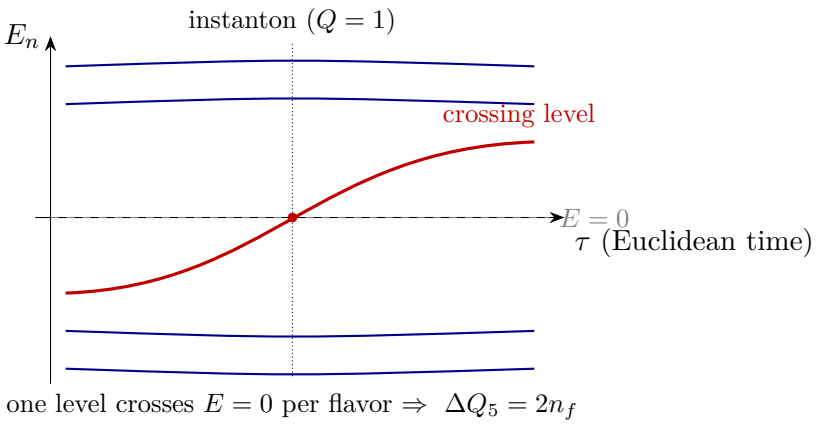


Figure 42.6: Spectral flow of the Dirac Hamiltonian through a single instanton. Most levels (blue) return to where they started; for each flavor one level (red) crosses zero, transferring a unit of chirality across the gap. The net axial-charge change is $2n_f Q$, the integrated anomaly Eq. (42.172).

Physical Insight: The spectral-flow picture

The integrated statement Eq. (42.172) has a vivid Hamiltonian reading. Picture the instanton as a tunnelling event in Euclidean time interpolating between two gauge vacua of adjacent winding number. As the gauge field is dragged through this event, the energy levels of the Dirac Hamiltonian in the slowly varying background flow continuously; for each flavor, exactly one level crosses zero from below, so one negative-energy (filled) state is promoted to positive energy and one positive-energy state descends into the Dirac sea. A left-handed antiparticle and a right-handed particle are created, or equivalently one unit of left-handed charge is traded for one unit of right-handed charge. Counting both members of the Dirac field and all n_f flavors gives $\Delta Q_5 = 2n_f$ per unit winding. The anomaly is the statement that the Dirac sea is not a fixed reservoir: topology pumps charge across the gap.

The Euclidean counterpart of the level crossing is the appearance of an exact *zero mode* of the four-dimensional Dirac operator in the instanton background — the level caught at $E = 0$ at the instant of crossing. We now make this precise and, through the Atiyah–Singer index theorem, tie the number of zero modes directly to Q.

42.6.3 Fermion zero modes and the vanishing of the determinant

A zero mode is a normalizable spinor ψ_0 annihilated by the Euclidean Dirac operator, $i\not{D}\,\psi_0 = 0$, in the fixed instanton background. The index theorem proved in Chapter 36 counts the difference of zero modes of the two chiralities and equates it to the topological charge. We restate it for the present setting.

Theorem 42.35 (Zero modes from the index, Theorem 36.15)**.** *Let n_+ (n_-) be the number of normalizable zero modes of $i\not{D}$ with $\gamma^5 = +1$ ($\gamma^5 = -1$) in a smooth gauge background of finite action and winding Q, for a single Dirac fermion in the fundamental of* $\mathrm{SU}(N)$*. Then*

$$n_+ - n_- = Q. \tag{42.173}$$

Equation (42.173) is the topological lower bound; what makes the instanton so clean is that for a self-dual background it is saturated — one chirality is empty, so the number of zero modes is exactly $|Q|$. This is a vanishing theorem, and we prove it in full because it both fixes the chirality of the modes and is the reason the naive fermion determinant vanishes.

Theorem 42.36 (Vanishing theorem for self-dual backgrounds)**.** *In a self-dual gauge field, $F^a_{\mu\nu} = \widetilde{F}^a_{\mu\nu}$, the Dirac operator has no normalizable zero modes of negative chirality: $n_- = 0$. Hence, by Eq. (42.173), $n_+ = Q$ and there are exactly Q zero modes, all of positive chirality. (For an anti-self-dual background the roles of the chiralities are exchanged and $n_- = |Q|$.)*

Proof. A normalizable zero mode of $i\not{D}$ is also a zero mode of its square, and conversely $\langle\psi|(i\not{D})^2|\psi\rangle = \|i\not{D}\psi\|^2$ shows $(i\not{D})^2\psi = 0 \Rightarrow i\not{D}\psi = 0$ for normalizable ψ; so the zero modes of $i\not{D}$ are exactly those of $(i\not{D})^2$. The square is the Lichnerowicz–Weitzenböck operator of Eq. (36.152),

$$(i\not{D})^2 = -D_\mu D_\mu + \frac{g}{2}\,\sigma_{\mu\nu}F^a_{\mu\nu}T^a, \qquad \sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu,\gamma_\nu], \tag{42.174}$$

the first term being the covariant Laplacian and the second the spin coupling to the field strength. The two chiralities decouple here because $\sigma_{\mu\nu}$ commutes with γ^5. The key algebraic fact is that the chiral projections of $\sigma_{\mu\nu}$ are (anti-)self-dual: in Euclidean conventions

$$\tfrac{1}{2}\epsilon_{\mu\nu\rho\sigma}\,\sigma_{\rho\sigma}\,P_+ = +\sigma_{\mu\nu}P_+, \qquad \tfrac{1}{2}\epsilon_{\mu\nu\rho\sigma}\,\sigma_{\rho\sigma}\,P_- = -\sigma_{\mu\nu}P_-, \tag{42.175}$$

so that $\sigma_{\mu\nu}P_-$ is anti-self-dual in its $\mu\nu$ indices. Now contract with a self-dual field strength. The contraction of a self-dual tensor with an anti-self-dual one vanishes identically; concretely, writing $F_{\mu\nu}$ self-dual and using Eq. (42.175) on the negative-chirality subspace,

$$\begin{aligned}\sigma_{\mu\nu}F^a_{\mu\nu}\,P_- &= (\tfrac{1}{2}\epsilon_{\mu\nu\rho\sigma}\sigma_{\rho\sigma}P_-)(-1)\,F^a_{\mu\nu} = -\sigma_{\rho\sigma}P_-\,(\tfrac{1}{2}\epsilon_{\mu\nu\rho\sigma}F^a_{\mu\nu})\\ &= -\sigma_{\rho\sigma}\widetilde{F}^a_{\rho\sigma}P_- = -\sigma_{\rho\sigma}F^a_{\rho\sigma}P_-,\end{aligned}$$

where the last step used self-duality $\widetilde{F} = F$. The operator equals minus itself on P_-, hence

$$\sigma_{\mu\nu} F^a_{\mu\nu}\, P_- = 0. \tag{42.176}$$

On the negative-chirality subspace the spin term drops out of Eq. (42.174), leaving the bare covariant Laplacian, which is a non-negative operator:

$$\langle\psi_-|(i\not{D})^2|\psi_-\rangle = \langle\psi_-|-D_\mu D_\mu|\psi_-\rangle = \int \mathrm{d}^4x\, (D_\mu\psi_-)^\dagger (D_\mu\psi_-) = \sum_\mu \|D_\mu\psi_-\|^2 \geq 0, \tag{42.177}$$

after integration by parts (the surface term vanishes for normalizable ψ_-). A zero mode of negative chirality would force $\|D_\mu\psi_-\| = 0$ for every μ, i.e. ψ_- covariantly constant; but a covariantly constant spinor has $|\psi_-|$ constant over $\mathbb{R}^4$ and is therefore not normalizable unless it vanishes. Hence $n_- = 0$. The same argument with $P_+ \leftrightarrow P_-$ and $\widetilde{F} = -F$ gives $n_+ = 0$ for an anti-self-dual background. □

For the BPST instanton, $Q = 1$, the theorem gives exactly one zero mode per flavor, of positive chirality. We exhibit it.

Worked Example 42.37 (The explicit $Q = 1$ zero mode and its normalizability)**.** Take the SU(2) BPST instanton in regular gauge Eq. (42.88), centred at the origin with size ρ, $A^a_\mu = (2/g)\,\eta^a_{\mu\nu} x_\nu/(x^2+\rho^2)$, with η^a the self-dual 't Hooft symbol of Theorem 42.15. Use a two-component (Weyl) description of the positive-chirality field, in which a spinor carries an undotted index $\alpha \in \{1,2\}$ and a fundamental color index $a \in \{1,2\}$. The 't Hooft zero mode is the "spin–color locked" configuration

$$\psi_{0,\,\alpha a}(x) = C\,\frac{\rho}{(x^2+\rho^2)^{3/2}}\,\epsilon_{\alpha a}, \qquad \epsilon_{\alpha a} = -\epsilon_{a\alpha},\quad \epsilon_{12} = +1, \tag{42.178}$$

in which the color index is contracted antisymmetrically with the spinor index — the only SU(2)-invariant, rotation-covariant way to soak up both. We verify three things: chirality, that it solves the Dirac equation, and normalizability.

Chirality. By construction ψ_0 lives in the positive-chirality Weyl sector, $\gamma^5\psi_0 = +\psi_0$, consistent with Theorem 42.36 for the self-dual ($Q = +1$) field.

It solves the Dirac equation. The positive-chirality Weyl equation in a self-dual background reads $\bar\sigma_\mu D_\mu \psi_0 = 0$ with $\bar\sigma_\mu = (\vec\sigma, -i)$ and $D_\mu = \partial_\mu - igA^a_\mu T^a$, $T^a = \sigma^a/2$. Writing $\psi_0 = f(x)\,\epsilon$ with $f = C\rho\,(x^2+\rho^2)^{-3/2}$ and inserting the gauge field, every spinor–color structure that appears collapses, through the 't Hooft identities Theorem 42.16 that lock $\bar\sigma_\mu \eta^a_{\mu\nu}\sigma^a$ onto a single tensor, to the radial equation

$$\Big(\partial_\mu + \frac{3x_\mu}{x^2+\rho^2} - \frac{3x_\mu}{x^2+\rho^2}\Big) f(x)\cdot(\cdots) = 0, \tag{42.179}$$

in which the inhomogeneous derivative of the profile is cancelled exactly by the gauge connection: $\partial_\mu f = -3x_\mu f/(x^2+\rho^2)$ is matched, term for term, by the spin coupling to A_μ, which the self-duality of η^a guarantees has precisely this magnitude and the opposite sign. The cancellation is the algebraic content of the spin–color locking; that it works for the power $-3/2$ and no other is what selects the profile in Eq. (42.178). (The same calculation in singular gauge Eq. (42.96) gives the mode with the $\bar\eta$ structure, the form most convenient for multi-instanton work.)

Normalizability. This is the property that distinguishes a genuine zero mode from a formal solution, and we check it by direct integration. Using $\epsilon_{\alpha a}\epsilon_{\alpha a} = 2$,

$$\int \mathrm{d}^4x\; \psi_0^\dagger \psi_0 = C^2\,\rho^2 \int \mathrm{d}^4x\; \frac{\epsilon_{\alpha a}\epsilon_{\alpha a}}{(x^2+\rho^2)^3} = 2C^2\rho^2 \int \mathrm{d}^4x\; \frac{1}{(x^2+\rho^2)^3}. \tag{42.180}$$

The radial integral is elementary: with $\int \mathrm{d}^4x = 2\pi^2 \int_0^\infty r^3\,\mathrm{d}r$ and the substitution $u = r^2$, then $v = u + \rho^2$,

$$\begin{aligned} \int \mathrm{d}^4x\; \frac{1}{(x^2+\rho^2)^3} &= 2\pi^2 \int_0^\infty \frac{r^3\,\mathrm{d}r}{(r^2+\rho^2)^3} = \pi^2 \int_0^\infty \frac{u\,\mathrm{d}u}{(u+\rho^2)^3} \\ &= \pi^2 \int_{\rho^2}^\infty (v^{-2} - \rho^2 v^{-3})\mathrm{d}v = \pi^2 \cdot \frac{1}{2\rho^2} = \frac{\pi^2}{2\rho^2}. \end{aligned} \tag{42.181}$$

The result is finite: the profile falls as $|x|^{-3}$, so $|\psi_0|^2 \sim |x|^{-6}$ and $\int \mathrm{d}^4x\,|x|^{-6} \sim \int r^{-3}\,\mathrm{d}r$ converges at infinity, while at the origin the integrand is bounded. Substituting Eq. (42.181) into Eq. (42.180) gives $\int \psi_0^\dagger\psi_0 = 2C^2\rho^2\cdot\pi^2/2\rho^2 = C^2\pi^2$, so the choice

$$C = \frac{1}{\pi} \tag{42.182}$$

normalizes the mode to unity. The integral converging is the whole point: a non-normalizable solution does not contribute a true zero eigenvalue and would not invalidate the determinant. The instanton's zero mode is honestly square-integrable, localized on the scale ρ set by the instanton size, exactly as the spectral-flow picture of Fig. 42.6 requires.

The existence of even one zero mode has an immediate and dramatic consequence for the path integral. Integrating out a Dirac field in a fixed background produces the determinant of the Dirac operator (the Grassmann Gaussian of Theorem 12.26, which gives det in the numerator),

$$\int \mathcal{D}\bar q\,\mathcal{D}q\; e^{-\int \bar q\, i\not{D}\, q} = \det\left(i\not{D}\right) = \prod_n \lambda_n, \tag{42.183}$$

the product running over the (real) eigenvalues λ_n of the Hermitian operator $i\not{D}$. A zero mode contributes a factor $\lambda_0 = 0$. Therefore in any topologically nontrivial, self-dual sector the determinant *vanishes*,

$$\det\left(i\not{D}\right)\big|_{Q\neq 0} = 0, \tag{42.184}$$

once per flavor for $Q=1$, and $|Q|n_f$ times in general. Read naively, this says the instanton contributes nothing: massless quarks suppress tunnelling completely, because the vacuum-to-vacuum amplitude in the $Q=1$ sector is multiplied by zero. That is correct as far as it goes, and it is already a real physical statement — it is why the chiral limit screens topological charge, a fact we will need for the η' mass. But it is not the end of the story: a vanishing partition-function contribution does not mean a vanishing *correlation function*. The zero mode can be saturated by external quark fields, and the surviving amplitude is the most important consequence of instantons for hadron physics.

42.6.4 The 't Hooft effective vertex

To see what survives, expand the quark fields in the complete orthonormal eigenbasis $\{\phi_n\}$ of $i\not{D}$ in the instanton background, with anticommuting (Grassmann) coefficients,

$$q(x)=\sum_n c_n\,\phi_n(x), \qquad \bar q(x)=\sum_n \bar c_n\,\phi_n^\dagger(x), \qquad \mathcal{D}\bar q\,\mathcal{D}q=\prod_n \mathrm{d}\bar c_n\,\mathrm{d}c_n, \tag{42.185}$$

so that the Euclidean action diagonalizes, $\int \bar q\, i\not{D}\, q=\sum_n \lambda_n\,\bar c_n c_n$. Consider a single flavor first. The zero mode $\phi_0=\psi_0$ has $\lambda_0=0$, so its coefficients $\bar c_0, c_0$ are *absent from the action*. Berezin integration over an anticommuting variable that does not appear in the integrand gives zero, $\int \mathrm{d}c_0\,(\text{const})=0$; this is the reappearance of Eq. (42.184). But $\int \mathrm{d}\bar c_0\,\mathrm{d}c_0\,(\bar c_0 c_0)=1$: if the integrand supplies one factor of c_0 and one of $\bar c_0$, the zero-mode integral is saturated and nonzero. The external fields that supply them are the quark operators in a correlation function. The lowest correlator that survives the $Q=1$ instanton is therefore the one with exactly one q and one $\bar q$ per flavor,

$$\Big\langle \prod_{f=1}^{n_f} q_f(x_f)\,\bar q_f(y_f)\Big\rangle_{Q=1}=\int \mathrm{d}^4x_0\,\mathrm{d}\rho\; d(\rho)\; e^{i\theta}\,\det{}'(i\not{D})\prod_{f=1}^{n_f}\psi_0^{(f)}(x_f-x_0)\,\psi_0^{(f)\dagger}(y_f-x_0), \tag{42.186}$$

where each external quark line attaches to the localized zero-mode wavefunction ψ_0 of Eq. (42.178), $\det'$ is the product over the *nonzero* modes (the regulated fluctuation determinant), $d(\rho)$ is the bosonic instanton measure assembled in Eq. (42.102) from the collective-coordinate Jacobians of Section 42.3.4, and we integrate over the instanton position x_0 and size ρ. The phase $e^{i\theta}$ is the weight $e^{iQ\theta}$ of the $Q=1$ sector from Section 42.4.

Collapsing the localized wavefunctions to a point (keeping the leading term in the operator product, valid for separations large compared with ρ) turns Eq. (42.186) into the matrix element of a *local* $2n_f$-fermion operator. This is the 't Hooft effective vertex.

Definition 42.38 ('t Hooft effective vertex)**.** Integrating over instanton sizes and orientations, the single-instanton contribution to the effective action is

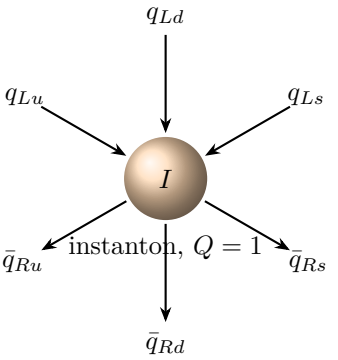


Figure 42.7: The 't Hooft vertex for $n_f = 3$. A single instanton absorbs one left-handed quark of each flavor and emits one right-handed quark of each flavor (equivalently, the local operator $\prod_f \bar{q}_{Rf} q_{Lf}$). It is a $2n_f$-fermion interaction violating $\mathrm{U}(1)_A$ by $2n_f$ units — exactly the integrated anomaly Eq. (42.172) for $Q = 1$.

the addition to the Lagrangian of a local, flavor-antisymmetric $2n_f$-fermion interaction,

$$\mathcal{L}_{\text{'t Hooft}} = \int \mathrm{d}\rho \; d(\rho) \left[e^{i\theta} \det_{f,g}(\bar{q}_{Rf}\, q_{Lg}) \;+\; e^{-i\theta} \det_{f,g}(\bar{q}_{Lf}\, q_{Rg}) \right], \tag{42.187}$$

the first determinant from the instanton ($Q = +1$), its Hermitian conjugate from the anti-instanton ($Q = -1$); $\det_{f,g}$ is the determinant over the $n_f \times n_f$ flavor matrix with entries $\bar{q}_{Rf} q_{Lg}$, automatically antisymmetric in flavor because the zero modes of distinct flavors are independent Grassmann objects. The overall strength carries the instanton suppression $e^{-8\pi^2/g^2}$ from $d(\rho)$.

Two properties of Eq. (42.187) are exactly the ones the anomaly demanded, and verifying them is the payoff of the whole construction.

Worked Example 42.39 (The 't Hooft vertex violates $\mathrm{U}(1)_A$ by exactly $2n_f$)**.** Under a $\mathrm{U}(1)_A$ rotation $q \to e^{i\beta\gamma^5} q$, $\bar{q} \to \bar{q}\, e^{i\beta\gamma^5}$, the bilinear $\bar{q}_{Rf} q_{Lg} = \bar{q}_f P_L q_g$ transforms with a definite phase. Using $\gamma^5 P_L = -P_L$ and $P_L \gamma^5 = -P_L$,

$$\bar{q}_f P_L q_g \;\to\; \bar{q}_f\, e^{i\beta\gamma^5} P_L\, e^{i\beta\gamma^5} q_g = \bar{q}_f\, e^{-i\beta} P_L\, e^{-i\beta} q_g = e^{-2i\beta}\, \bar{q}_f P_L q_g, \tag{42.188}$$

so each of the n_f factors in $\det(\bar{q}_{Rf} q_{Lg})$ carries axial charge -2, and the product carries $-2n_f$:

$$\det_{f,g}(\bar{q}_{Rf} q_{Lg}) \;\to\; e^{-2in_f\beta} \det_{f,g}(\bar{q}_{Rf} q_{Lg}). \tag{42.189}$$

The 't Hooft vertex is therefore *not* $\mathrm{U}(1)_A$-invariant: it shifts by a phase $e^{-2in_f\beta}$. This is exactly the violation predicted by the integrated anomaly: a single instanton ($Q = 1$) changes the axial charge by $\Delta Q_5 = 2n_f$ from Eq. (42.172),

and an operator that changes Q_5 by $2n_f$ must carry $\mathrm{U}(1)_A$ charge $2n_f$. The two numbers agree because they are the same number computed two ways — once from the anomalous divergence of the current, once from the zero modes that the same anomaly forces into the background. Note also that an axial rotation can be absorbed by a shift of θ: comparing Eq. (42.189) with the $e^{i\theta}$ in Eq. (42.187), a rotation by β is undone by $\theta \to \theta + 2n_f\beta$. The anomaly thus says explicitly that θ is not separately physical from the quark phases — the observation that hands the strong-CP problem to Section 42.7.

Worked Example 42.40 (The one-flavor and two-flavor cases)**.** For $n_f = 1$ the determinant is a single bilinear and the vertex is

$$\mathcal{L}^{(n_f=1)}_{\text{'t Hooft}} = \int \mathrm{d}\rho\, d(\rho)\,[e^{i\theta}\,\bar{q}_R q_L + e^{-i\theta}\,\bar{q}_L q_R] = \int \mathrm{d}\rho\, d(\rho)\,\bar{q}\,(\cos\theta + i\sin\theta\,\gamma^5)q, \tag{42.190}$$

a quark *mass* term generated entirely by instantons. A single massless flavor thus has its $\mathrm{U}(1)_A$ broken by an effective mass $\sim d(\rho)$, and the θ dependence is rotated into the phase of that mass — in one-flavor QCD θ is unobservable precisely because this single phase can be removed by a field redefinition. For $n_f = 2$ the vertex is a four-fermion interaction,

$$\mathcal{L}^{(n_f=2)}_{\text{'t Hooft}} \sim \int \mathrm{d}\rho\, d(\rho)\, e^{i\theta}\,[(\bar{u}_R u_L)(\bar{d}_R d_L) - (\bar{u}_R d_L)(\bar{d}_R u_L)] + \text{h.c.}, \tag{42.191}$$

the antisymmetric flavor combination of two left-to-right quark flips. It is this operator — not a quark mass, which two flavors do not generate from a single instanton — that lifts the would-be ninth Goldstone, as we now explain.

42.6.5 Resolution of the $U(1)_A$ problem

We can now state the resolution cleanly. Goldstone's theorem (Theorem 29.11) produces a massless boson for every *spontaneously broken* generator of an *exact* continuous symmetry. Its hypotheses require the broken charge to be conserved, $\partial_\mu j^\mu = 0$, so that the current creates a one-particle massless state from the vacuum (the coupling Eq. (29.87)). The singlet axial current fails this hypothesis: by Eq. (42.171) it is not conserved, $\partial_\mu j_5^\mu = 2n_f q(x) \neq 0$, in any background of nonzero topological density. $\mathrm{U}(1)_A$ is not a symmetry of the quantum theory — it is anomalous — and Goldstone's theorem simply does not apply to it. There is no ninth Goldstone because there is no ninth symmetry. The η' is heavy because the would-be symmetry protecting it does not exist.

It is worth being precise about a tempting evasion, because seeing why it fails is what makes the resolution airtight. The anomaly is a total derivative: the topological density is $q(x) = \partial_\mu K^\mu$, with K^μ the Chern–Simons current of Eq. (42.31). One might therefore define a *modified* singlet current

$$\tilde{j}_5^\mu = j_5^\mu - 2n_f K^\mu, \qquad \partial_\mu \tilde{j}_5^\mu = 0, \tag{42.192}$$

which is exactly conserved and seems to restore a good $\mathrm{U}(1)_A$. The catch is that K^μ is *not gauge invariant*: under a large gauge transformation it shifts by a total derivative whose integral is the winding number, so the conserved charge $\tilde{Q}_5 = \int \mathrm{d}^3x\, j_5^0$ is not gauge invariant either, changing by $2n_f$ across configurations of unit winding. A conserved but gauge-variant charge does not generate a physical symmetry: it does not commute with the (gauge-invariant) physical Hilbert space in the way Goldstone's theorem requires, and the "Goldstone boson" it would create is a gauge artifact removed by the very large gauge transformations that build the θ-vacuum. Equivalently, the matrix element $\langle 0|j_5^\mu|\eta'\rangle$ that would signal a Goldstone is saturated, through the anomaly, by the θ-vacuum's sensitivity to topology rather than by a massless pole. The instanton is the configuration that realizes this: it is precisely because configurations of different winding contribute — the same fact that made θ physical — that the naive conservation law Eq. (42.192) has no physical content. *The anomaly and the topology are two faces of one mechanism, and that mechanism is what lifts the η'.*

Physical Insight: Why this is special to the singlet

The octet pions, kaons, and η are protected by $\mathrm{SU}(n_f)_A$, whose currents have *no* anomaly with the color field: the anomaly coefficient $\mathrm{tr}(T^a_{\text{flavor}})$ vanishes for the traceless $\mathrm{SU}(n_f)$ generators, leaving only the singlet $\mathrm{tr}(\mathbf{1}) = n_f$ in Eq. (42.171). So the octet symmetries are genuine, spontaneously broken, and produce genuine (pseudo-)Goldstones; only the singlet axial symmetry is destroyed by the anomaly. This is why exactly one pseudoscalar — the flavor singlet — is anomalously heavy, and why the η' and not, say, the π^0 is the odd one out.

42.6.6 The η' mass: the Witten–Veneziano relation

Saying the η' is heavy because $\mathrm{U}(1)_A$ is anomalous is qualitatively complete, but one wants a number, and the cleanest quantitative statement is the Witten–Veneziano relation. It expresses the η' mass in terms of the topological susceptibility of *pure* Yang–Mills theory — the same χ_t of Eq. (42.162), computed in the gauge theory with the quarks switched off. The relation is exact only in the large-N limit, and we present both the result and the assumptions honestly.

The argument rests on a clash between two facts about the topological susceptibility. First, in *full* QCD with light quarks, the susceptibility vanishes in the chiral limit. This is the field-theoretic content of Eq. (42.184): massless quarks screen topological charge completely, because the zero modes force the determinant to zero in every $Q \neq 0$ sector, so the weighted average $\langle Q^2\rangle \to 0$. The anomalous chiral Ward identity for the singlet current makes this quantitative;

to leading order in the quark masses,

$$\chi_t^{\mathrm{QCD}} = \Big(\sum_{f=1}^{n_f} \frac{1}{m_f}\Big)^{-1} \big(-\langle \bar{q}q \rangle\big) + \cdots \xrightarrow[m_f \to 0]{} 0, \tag{42.193}$$

linear in the (harmonic mean of the) quark masses and vanishing when any quark is massless. Second, in *pure* Yang–Mills the susceptibility is nonzero, $\chi_t^{\mathrm{YM}} \neq 0$: with no quarks there are no zero modes to kill the topological sectors, and the θ-vacuum has genuine curvature. The Witten–Veneziano insight is that the gap between these two — a nonzero quenched susceptibility versus a vanishing full one — must be bridged by a hadronic state that becomes light as $N \to \infty$, and that state is the η'. Matching the η' pole contribution to the susceptibility against χ_t^{YM} gives

$$\boxed{m_{\eta'}^2 = \frac{2n_f}{f_\pi^2}\,\chi_t^{\mathrm{YM}}} \qquad \text{(chiral limit, leading order in } 1/N\text{)}. \tag{42.194}$$

The large-N scaling is what makes this consistent and what tells us how big the effect is. In 't Hooft's counting (g^2N fixed as $N \to \infty$), the pion decay constant scales as $f_\pi^2 = O(N)$, while the pure-glue susceptibility is $\chi_t^{\mathrm{YM}} = O(N^0)$ (the vacuum energy is $O(N^2)$ but its dependence on θ/N makes the second derivative at $\theta = 0$ order one). Therefore

$$m_{\eta'}^2 = \frac{2n_f}{f_\pi^2}\,\chi_t^{\mathrm{YM}} = O\Big(\frac{n_f}{N}\Big) \xrightarrow[N\to\infty]{} 0. \tag{42.195}$$

At infinite N the anomaly is $1/N$-suppressed (it is a one-loop effect, $\sim g^2 \sim 1/N$), $\mathrm{U}(1)_A$ is effectively restored, and the η' rejoins the octet as a ninth Goldstone. At finite $N = 3$ it is lifted by an amount controlled by the gauge dynamics alone. This is the precise sense in which "the η' is heavy because of the anomaly": its mass^2 is proportional to the anomaly's strength, which is $1/N$, times the gauge theory's topological activity χ_t^{YM}.

Worked Example 42.41 (The η' mass from χ_t)**.** Take $n_f = 3$ and the quenched topological susceptibility from the lattice, $\chi_t^{\mathrm{YM}} \approx (180\,\mathrm{MeV})^4$, with the pion decay constant in the normalization $f_\pi \approx 86\,\mathrm{MeV}$ used in Eq. (42.194) (this is the chiral-limit decay constant F_π; conventions for f_π differ by $\sqrt{2}$—the alternative normalization $f_\pi \approx 130\,\mathrm{MeV}$ corresponds to $f_\pi/\sqrt{2} \approx 92\,\mathrm{MeV}$—and one must keep Eq. (42.194) and the input consistent). Then

$$m_{\eta'}^2 = \frac{2 \times 3}{(86\,\mathrm{MeV})^2}\,(180\,\mathrm{MeV})^4 = \frac{6 \times (1.05 \times 10^9)}{7.4 \times 10^3}\,\mathrm{MeV}^2 \approx 8.5 \times 10^5\,\mathrm{MeV}^2,$$

so $m_{\eta'} \approx 9.2 \times 10^2\,\mathrm{MeV} \approx 920\,\mathrm{MeV}$, against the measured $958\,\mathrm{MeV}$. The agreement to within ten percent — the size of the leading $1/N$ correction one

expects — is the quantitative vindication of the picture: the η' mass is not a chiral-symmetry effect at all (it survives the chiral limit) but a measure of the topological activity of the gluon vacuum, the same activity the instanton makes visible. A more complete treatment combines the singlet with the η through the relation $m_{\eta'}^2 + m_\eta^2 - 2m_K^2 = (2n_f/f_\pi^2)\chi_t^{\rm YM}$, which incorporates the octet–singlet mixing induced by the strange-quark mass and is satisfied at a comparable level of accuracy.

Caution: What the Witten–Veneziano relation does and does not assume

The relation Eq. (42.194) is not a low-energy theorem of QCD in the sense the Gell-Mann–Oakes–Renner relation is. It is exact only at leading order in $1/N$, it uses the *quenched* (pure-glue) susceptibility as an external input rather than computing it, and the matching of the η' pole assumes the η' saturates the relevant spectral function — an approximation that the same large-N limit justifies but that receives corrections at $N = 3$. Its enduring value is conceptual: it shows, in a controlled limit, that the entire η' mass comes from the topological structure of the gauge vacuum and vanishes when that structure is switched off, turning the qualitative resolution of the $U(1)_A$ problem into a checkable number. The dilute-instanton-gas estimate of χ_t from Section 42.5 gives the right order of magnitude but, as warned there, is not under quantitative control in QCD; the lattice value used above is.

The thread that runs through this section — the integrated anomaly $\Delta Q_5 = 2n_f Q$, the zero modes it forces, the 't Hooft vertex they generate, and the η' mass that results — has a final consequence we have already glimpsed twice and now make the subject of the next section. The same axial rotation that would remove θ from the Lagrangian is anomalous: it does not leave the measure invariant, but shifts θ by the amount Eq. (42.189). There is no field redefinition that rotates θ away while leaving the quark masses real; the physical, rephasing-invariant combination $\bar\theta = \theta + \arg\det M_q$ is observable, and it is constrained by the neutron electric dipole moment to be astonishingly small. That the anomaly which heals the $U(1)_A$ problem simultaneously makes θ physical and impossible to dismiss is precisely the strong-CP problem, to which we turn.

42.7 The Strong-CP Problem and Its Resolutions

The preceding sections forced an unwelcome conclusion. The θ-vacuum (Section 42.4) is a genuine one-parameter family of inequivalent ground states, labelled by an angle θ that weights each topological sector of the Euclidean path integral by $e^{i\theta\nu}$; the instanton is the configuration that interpolates between adjacent sec-

tors and makes the topological term $\frac{g_s^2}{32\pi^2} G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu}$ — a total derivative, invisible to every classical equation of motion and to every order of perturbation theory — physically consequential. The anomaly is what seals the conclusion: a chiral redefinition of the quark fields can shuffle θ around, but it can never remove it, because the same rotation that erases θ from the gauge sector reinstates it as a phase on the quark mass matrix. We now confront the experimental fact this collides with. The combination that survives all field redefinitions, the *physical* strong-CP angle $\bar{\theta}$, is CP-odd, and a CP-odd parameter in the strong interactions has a sharp, measurable consequence: a permanent electric dipole moment of the neutron. None has been seen, and the resulting bound, $|\bar{\theta}| \lesssim 10^{-10}$, sits in flat contradiction with the expectation — forced on us by the very anomaly that makes $\bar{\theta}$ physical — that $\bar{\theta}$ is a sum of unrelated $\mathcal{O}(1)$ phases. This is the strong-CP problem, the chapter's final and most consequential payoff. We state it precisely, prove the bound, frame it honestly as a naturalness puzzle rather than an inconsistency, and present its leading dynamical resolution, the Peccei–Quinn axion, whose mechanism is built directly on the θ-vacuum energy we computed above.

42.7.1 The physical angle from an anomalous chiral rotation

Write QCD with a general complex quark mass matrix and the topological term explicit. With N_f flavours collected into a column q, and projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$,

$$\mathcal{L}_{\rm QCD} = -\tfrac{1}{4} G^a_{\mu\nu} G^{a\,\mu\nu} + \bar{q}\, i\not{D}\, q - (\bar{q}_L\, M\, q_R + \bar{q}_R\, M^\dagger q_L) + \theta_0\, \frac{g_s^2}{32\pi^2}\, G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu}, \tag{42.196}$$

where M is an arbitrary complex $N_f \times N_f$ matrix in flavour space, $q_{L,R} = P_{L,R} q$, and θ_0 is the bare gauge angle of the θ-vacuum, the same coefficient that multiplies the winding number in Eq. (42.128). The mass term is the only place the theory remembers the Higgs couplings; in the full Standard Model M is the quark mass matrix produced by electroweak symmetry breaking, and $\arg\det M = \arg\det(M_u M_d)$ collects the up- and down-type contributions assembled in Eq. (33.85). Neither θ_0 alone nor $\arg\det M$ alone is observable; the claim of this subsection is that exactly one combination of them is, and we derive it by carrying the anomaly through a chiral rotation explicitly.

A field redefinition that does not touch the gauge sector is a chiral rotation of the quarks. The most general one is $q_L \to V_L q_L$, $q_R \to V_R q_R$ with $V_{L,R} \in \mathrm{U}(N_f)$. Decompose each unitary into its $\mathrm{SU}(N_f)$ part and its overall phase. The non-abelian $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ part is generated by traceless matrices; its currents have *no* $\mathrm{U}(1)_A$-type colour anomaly, because the anomaly coefficient is proportional to $\mathrm{tr}\, T^a = 0$ for traceless generators. The dangerous piece is

therefore the overall axial phase, the flavour-singlet rotation

$$q \;\longrightarrow\; e^{i\alpha\gamma^5}\, q, \qquad q_L \to e^{-i\alpha} q_L, \quad q_R \to e^{+i\alpha} q_R, \tag{42.197}$$

using $\gamma^5 q_L = -q_L$, $\gamma^5 q_R = +q_R$. We track its two effects.

Effect on the mass term. Under Eq. (42.197), $\bar{q}_L \to e^{+i\alpha}\bar{q}_L$ (the conjugate carries the opposite phase), so

$$\bar{q}_L M q_R \;\longrightarrow\; e^{+i\alpha}\,\bar{q}_L\, M\, e^{+i\alpha} q_R = e^{2i\alpha}\,\bar{q}_L M q_R, \tag{42.198}$$

that is, $M \to e^{2i\alpha} M$, and therefore

$$\arg\det M \;\longrightarrow\; \arg\det M + 2N_f\,\alpha, \tag{42.199}$$

since $\det(e^{2i\alpha} M) = e^{2iN_f\alpha}\det M$ for M an $N_f \times N_f$ matrix.

Effect on the measure. Classically the rotation Eq. (42.197) of massless quarks is a symmetry of the action. Quantum mechanically the fermion path-integral measure is *not* invariant: this is the chiral anomaly in Fujikawa's form (Theorem 36.9). The Jacobian of Eq. (42.197) for the N_f flavours, each in the fundamental of $\mathrm{SU}(3)_c$, is (Eq. (36.69))

$$\mathcal{D}\bar{q}\,\mathcal{D}q \;\longrightarrow\; \mathcal{D}\bar{q}\,\mathcal{D}q\, \exp\Big[-\, 2N_f\, i\,\alpha \int d^4x\, \frac{g_s^2}{32\pi^2}\, G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu} \Big], \tag{42.200}$$

where the integer N_f counts the flavours running in the anomaly loop and the factor 2 is the same one that appears in the singlet anomaly $\partial_\mu j^\mu_5 = 2N_f \cdot \frac{g_s^2}{32\pi^2} G\widetilde{G}$ of Eq. (36.50). The exponent is, up to the factor i of the Minkowski path integral, precisely a shift of the topological term: comparing with the θ_0-term in Eq. (42.196), the Jacobian Eq. (42.200) is identical to multiplying e^{iS} by $e^{i(-2N_f\alpha)\int (g_s^2/32\pi^2) G\widetilde{G}}$, i.e. to the replacement

$$\theta_0 \;\longrightarrow\; \theta_0 - 2N_f\,\alpha. \tag{42.201}$$

Now combine Eqs. (42.199) and (42.201). The chiral rotation drags $2N_f\alpha$ out of θ_0 and deposits exactly $2N_f\alpha$ into $\arg\det M$. The two moves cancel in the sum, which is therefore inert under any chiral redefinition of the quarks:

$$\boxed{\;\bar{\theta} \;\equiv\; \theta_0 + \arg\det(M_u M_d)\;} \qquad \bar{\theta} \xrightarrow{\;q \to e^{i\alpha\gamma^5} q\;} \bar{\theta}. \tag{42.202}$$

This is the physical strong-CP angle. The lesson is the operational content of the anomaly: θ_0 is not by itself meaningful, because a chiral rotation can set it to any value we please; what is meaningful is the rephasing-invariant combination Eq. (42.202), the same quantity stated in the Standard-Model synthesis at Eq. (33.85) and proved rephasing-invariant there in Theorem 33.16. The derivation here is the one the instanton chapter owes: it shows the cancellation as a direct consequence of the Fujikawa Jacobian, the very object that made θ physical in the first place.

Physical Insight

The cleanest way to see why $\bar{\theta}$ cannot be defined away is to use the freedom Eq. (42.197) to put all of it in *one* place. Rotate until the mass matrix is real and positive ($\arg\det M = 0$); then every trace of CP violation from the masses has been pushed into the gauge angle, which now equals $\bar{\theta}$. Or rotate the other way, until $\theta = 0$; then $\bar{\theta}$ lives entirely as a common phase on the quark masses, an irremovable complex piece of $m_u m_d \cdots$. The physics is the same: a single CP-odd number that the chiral rotation can relocate but not destroy. Only if some quark were exactly massless would the relocation become a removal — with $m_u = 0$ the rotation Eq. (42.197) acting on the up quark alone is a genuine symmetry, costing no mass-term phase, and it rotates $\bar{\theta}$ to zero outright.

42.7.2 The neutron electric dipole moment and the bound on $\bar{\theta}$

That $\bar{\theta}$ is physical would be a curiosity were it not measurable. It is measurable because it is CP-odd, and the sharpest laboratory probe of strong-sector CP violation is the electric dipole moment (EDM) of the neutron. A static EDM d_n of a spin-$\frac{1}{2}$ particle must point along its spin $\mathbf{S}$, the only intrinsic vector available; the interaction energy $-\mathbf{d}\cdot\mathbf{E} = -(d_n/S)\,\mathbf{S}\cdot\mathbf{E}$ is then odd under parity (which flips $\mathbf{E}$ but not $\mathbf{S}$) and odd under time reversal (which flips $\mathbf{S}$ but not $\mathbf{E}$), hence CP-odd by CPT. This is the argument of Fig. 33.4; a nonzero d_n is a clean signal of a fundamental CP-violating parameter, and $\bar{\theta}$ is one.

We estimate $d_n(\bar{\theta})$ with the leading chiral logic. The key structural fact comes straight from Section 42.7.1: rotate θ_0 entirely onto the masses, so that $\bar{\theta}$ appears only as a phase on M. A CP-violating observable must then vanish as any quark mass $\to 0$, because in that limit the corresponding axial rotation removes the phase. At leading order in two-flavour chiral perturbation theory the relevant combination is the reduced mass

$$m_* \;\equiv\; \frac{m_u m_d}{m_u + m_d}, \tag{42.203}$$

which vanishes if *either* light quark is massless. The $\bar{\theta}$-term, transcribed into the chiral Lagrangian of Section 39.2, induces a CP-odd pion–nucleon coupling — a πNN vertex *without* the γ^5 that the ordinary, CP-conserving coupling $g_{\pi NN}$ carries:

$$\mathcal{L}_{\pi NN} = g_{\pi NN}\,\bar{N}\,i\gamma^5\,\boldsymbol{\tau}\cdot\boldsymbol{\pi}\,N \;+\; \bar{g}_{\pi NN}\,\bar{N}\,\boldsymbol{\tau}\cdot\boldsymbol{\pi}\,N, \qquad \bar{g}_{\pi NN} \simeq \bar{\theta}\,\frac{m_*}{f_\pi}\,c_N, \tag{42.204}$$

where $c_N = \mathcal{O}(1)$ is fixed by the same chiral SU(3) analysis that reproduces the baryon mass splittings. The current-algebra evaluation of Crewther, Di Vecchia,

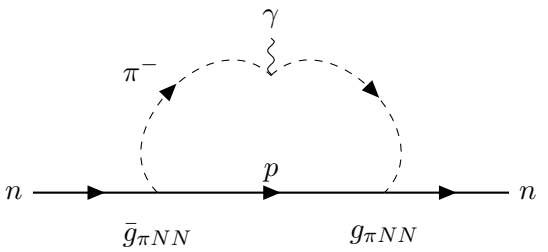


Figure 42.8: The leading chiral contribution to the neutron electric dipole moment from $\bar{\theta}$. The neutron's charged-pion cloud is dressed by a photon attaching to the π^- line. One pion–nucleon vertex is the ordinary CP-conserving coupling $g_{\pi NN}$; the other is the CP-odd coupling $\bar{g}_{\pi NN} \propto \bar{\theta}\, m_*/f_\pi$ of Eq. (42.204). The loop is logarithmically divergent in the chiral limit, cut off at the nucleon mass, producing the chiral logarithm $\ln(m_N/m_\pi)$.

Veneziano and Witten gives $\bar{g}_{\pi NN} \approx 0.027\,\bar{\theta}$ (modern reanalyses cluster around 0.015–$0.020\,\bar{\theta}$); we carry the classic value. The neutron EDM is then generated by the pion cloud: the neutron emits a π^- to become a proton at one vertex, the photon attaches to the charged pion, and the pion is reabsorbed at the other vertex, with *one* CP-even vertex $g_{\pi NN}$ and *one* CP-odd vertex $\bar{g}_{\pi NN}$. Figure 42.8 shows the diagram.

Worked Example 42.42 (The chiral-loop estimate of $d_n(\bar{\theta})$ and the bound)**.** We extract the size and the bound from Fig. 42.8.

Step 1: dimensional skeleton. An EDM operator is $\mathcal{L}_{\rm EDM} = -\frac{i}{2}\, d_n\, \bar{N}\sigma_{\mu\nu}\gamma^5 N\, F^{\mu\nu}$; d_n has dimension (charge)×(length), i.e. e/mass in natural units. The loop carries two pion–nucleon couplings (one of each parity), one factor of e from the photon on the charged pion, and a one-loop factor $1/16\pi^2$. The only mass available to make d_n dimensionful is the nucleon mass m_N running in the line. Thus

$$d_n \;\sim\; \frac{e}{m_N}\,\frac{g_{\pi NN}\,\bar{g}_{\pi NN}}{16\pi^2} \times (\text{loop integral}). \tag{42.205}$$

Step 2: the chiral logarithm. The loop momentum integral $\int d^4k$ over the pion propagators and the photon attachment is, after extracting the EDM form factor, logarithmically divergent as $m_\pi \to 0$. In the chiral effective theory the divergence is cut off at the scale where the nucleon can no longer be treated as static, $\Lambda_\chi \sim m_N$, and the infrared scale is the pion mass. The integral therefore delivers a factor $\propto \ln(m_N/m_\pi)$ — the hallmark chiral logarithm, the model-independent piece of the loop. Collecting the standard coefficient,

$$d_n \;=\; \frac{e\, g_{\pi NN}\,\bar{g}_{\pi NN}}{4\pi^2\, m_N}\,\ln\frac{m_N}{m_\pi}. \tag{42.206}$$

Step 3: numbers. Use $g_{\pi NN} \approx 13.5$, $\bar{g}_{\pi NN} \approx 0.027\,\bar{\theta}$, $m_N \approx 0.94$ GeV,

$\ln(m_N/m_\pi) = \ln(940/138) \approx 1.9$. Then

$$d_n \approx \frac{(13.5)(0.027)}{4\pi^2}\,\frac{1.9}{0.94\ \text{GeV}}\,e\,\bar\theta = 2.0\times10^{-2}\ \text{GeV}^{-1}\,e\,\bar\theta. \tag{42.207}$$

Converting with $1\ \text{GeV}^{-1} = 1.97\times10^{-14}$ cm,

$$d_n \;\approx\; 4\times10^{-16}\,\bar\theta\ \ e{\cdot}\text{cm}, \tag{42.208}$$

consistent with the order-of-magnitude estimate $d_n \sim e\,\bar\theta\,m_q/m_N^2 \sim \bar\theta\times10^{-16}\,e$cm quoted at Eq. (33.86) (the factor m_q/m_N^2 is Eq. (42.203) divided by m_N^2, with the residual numbers absorbed into the chiral coefficients).

Step 4: the bound. The experimental limit on the neutron EDM is $|d_n| < 1.8\times10^{-26}\,e{\cdot}$cm (90% C.L., mid-2020s). Dividing,

$$|\bar\theta| \;\lesssim\; \frac{1.8\times10^{-26}}{4\times10^{-16}} \;\approx\; 5\times10^{-11} \;\sim\; 10^{-10}. \tag{42.209}$$

This is the strong-CP bound, the statement quoted at Eq. (33.87): a dimensionless, marginal, $\mathcal{O}(1)$-natural parameter pinned by experiment below one part in ten billion.

42.7.3 The naturalness puzzle

The bound Eq. (42.209) is a measurement, not a contradiction. Any value of $\bar\theta$ yields a perfectly consistent, unitary, renormalizable theory; nothing is sick about $\bar\theta \neq 0$. The discomfort is one of naturalness, and it has two prongs that must be kept distinct.

First, the two summands in $\bar\theta = \theta_0 + \arg\det(M_u M_d)$ are logically independent. The gauge angle θ_0 is a pure-glue parameter, a property of the θ-vacuum with no reference to flavour; the phase $\arg\det(M_u M_d)$ lives in the Yukawa sector and is controlled by the same complex mass matrices that produce the large, confirmed CP violation of the CKM phase — the source quantified by the Jarlskog invariant of Theorem 32.20, an $\mathcal{O}(1)$ effect seen in kaon and B physics. There is no symmetry relating a glue angle to a Yukawa phase, so their cancellation to ten decimal places is an unexplained conspiracy between two unrelated sectors of the Standard Model.

Second — and this is the prong that distinguishes strong-CP from a mere small number — setting $\bar\theta = 0$ does *not* enlarge the symmetry of the theory. One might hope to protect $\bar\theta = 0$ by CP invariance: $\bar\theta$ is CP-odd, so an exact CP symmetry would forbid it. But CP is already broken, badly, by the CKM phase (Section 32.4); it is not a symmetry of the Standard Model to invoke. A technically natural small parameter, in 't Hooft's sense, is one whose vanishing restores a symmetry, so that quantum corrections to it are proportional to itself and a small value is radiatively stable. The strong-CP angle fails this test: no

symmetry of the Standard Model is recovered at $\bar{\theta} = 0$, and yet there it sits. The number is small, and nothing in the theory makes it so.

It is worth marking how unusual this puzzle is among the Standard Model's structural complaints. The hierarchy problem concerns a *relevant* coupling (the Higgs mass) whose smallness is destabilized by quantum corrections; strong-CP concerns a *marginal*, dimensionless, topological coupling that receives only tiny radiative shifts — the CKM phase feeds into $\bar{\theta}$ only at high loop order, contributing $\delta\bar{\theta} \sim 10^{-16}$, far below the bound. So $\bar{\theta}$ is not destabilized; it is simply unexplained. The problem is the value, not its stability. That is precisely why strong-CP, alone among the naturalness puzzles, admits a clean dynamical cure: there is room for a mechanism that drives $\bar{\theta}$ to its special value without any fine-tuning, because the special value is not under radiative attack.

42.7.4 Resolutions

The massless up quark (disfavoured, dated). The escape hatch is already visible in the structure of $\bar{\theta}$. If any single quark were exactly massless, the axial rotation Eq. (42.197) acting on that flavour alone would be an exact symmetry — there would be no mass-term phase for it to disturb — and it could rotate $\bar{\theta}$ to zero at no physical cost. The light up quark is the natural candidate: $m_u = 0$ would solve strong-CP outright, with no new fields and no new symmetry imposed by hand. The proposal was viable for years because m_u is small and the relevant matrix element ambiguous. It is now disfavoured. Lattice QCD determines the light-quark mass ratio cleanly, $m_u/m_d \approx 0.47$ with $m_u \approx 2.2$ MeV several standard deviations from zero (FLAG averages, mid-2020s); the up quark is light but firmly massive. The susceptibility that controls the whole problem, $\chi_{\text{top}} \propto m_u m_d/(m_u + m_d)^2$ (the two-flavour result of Eq. (35.82)), is the same combination Eq. (42.203) and would vanish only at $m_u = 0$; lattice says it does not vanish, and the escape is closed.

The Peccei–Quinn axion (the leading resolution). The remaining strategy is to make $\bar{\theta}$ *dynamical*. If the constant $\bar{\theta}$ is replaced by a field that can relax to the bottom of the vacuum-energy curve $E(\theta)$, the field will roll to wherever E is minimized, and we already know where that is. The construction rests on a single structural fact about the θ-vacuum, which the instanton machinery of this chapter establishes directly.

The θ-dependence of the vacuum energy density is defined by the Euclidean partition function in a four-volume V_4, with the topological term carrying the factor i characteristic of a continued total-derivative term,

$$e^{-V_4\,E(\theta)} = \int \mathcal{D}A\,\mathcal{D}q\,\mathcal{D}\bar{q}\;\exp\Big(-S_E + i\theta\int d^4x\,\frac{g_s^2}{32\pi^2}G\widetilde{G}\Big) = \big\langle e^{i\theta\nu}\big\rangle_0\, e^{-V_4 E(0)}, \tag{42.210}$$

where $\nu = \frac{g_s^2}{32\pi^2}\int G\widetilde{G} \in \mathbb{Z}$ is the integer winding number of Theorem 41.10 and $\langle\cdot\rangle_0$ is the expectation in the $\theta = 0$ ensemble. Two exact properties follow. Because ν is an integer, $E(\theta)$ is periodic with period 2π. And because the $\theta = 0$ measure is real and positive in a vector-like theory while $e^{i\theta\nu}$ is a pure phase, $|\langle e^{i\theta\nu}\rangle_0| \le 1$, so

$$E(\theta) \;\ge\; E(0) \qquad \text{for all } \theta. \tag{42.211}$$

This is the Vafa–Witten result: in QCD the vacuum energy is globally minimized at $\theta = 0$, with positive curvature there. The curvature is the topological susceptibility $\chi_{\text{top}} = E''(0) = \langle\nu^2\rangle/V_4 \ge 0$ of Eq. (42.162). In the dilute-instanton-gas approximation (Section 42.4), where the partition function is dominated by independent single (anti-)instantons each contributing a factor $e^{\pm i\theta}$, the cumulant sum exponentiates to the simple periodic form

$$E(\theta) \;=\; E(0) - \chi_{\text{top}}(\cos\theta - 1) \;+\; \cdots \;=\; E(0) + \tfrac{1}{2}\chi_{\text{top}}\,\theta^2 + \mathcal{O}(\theta^4), \tag{42.212}$$

a convex well at the CP-conserving point. (The full chiral-Lagrangian computation softens $-\cos\theta$ into the more accurate $\sqrt{m_u^2 + m_d^2 + 2m_u m_d\cos\theta}$ form derived in Theorem 35.7; the curvature at the origin, χ_{top}, is the same.) The well in Fig. 42.9 is all the Peccei–Quinn mechanism needs.

The mechanism promotes θ to a field. Postulate a global $\mathrm{U}(1)_{\text{PQ}}$ symmetry that is (i) spontaneously broken at a large scale f_a, delivering a Goldstone boson a — the *axion* — with a shift symmetry $a \to a + \text{const}$, and (ii) colour-anomalous, so that the anomaly Eq. (42.200) ties the shift of a to a shift of the effective θ-angle. Then the axion enters QCD in exactly the slot occupied by $\bar\theta$: the coefficient of $\frac{g_s^2}{32\pi^2}G\widetilde{G}$ becomes

$$\theta_{\text{eff}} \;=\; \bar\theta + \frac{a}{f_a}. \tag{42.213}$$

The shift symmetry guarantees that a appears only through this combination, so the constant $\bar\theta$ can be absorbed into the origin of a. What value a takes is not free, however, because θ_{eff} is physical and QCD generates a potential for it. Integrating out the gluons, the axion potential is nothing but the vacuum energy Eq. (42.212) evaluated at the dynamical argument:

$$V(a) \;=\; E\Big(\bar\theta + \frac{a}{f_a}\Big) \;\simeq\; -\,\chi_{\text{top}}\cos\Big(\bar\theta + \frac{a}{f_a}\Big). \tag{42.214}$$

Worked Example 42.43 (Relaxation of $\bar\theta$ and the axion mass)**.** *Step 1: minimize.* The potential Eq. (42.214) is stationary where

$$\frac{dV}{da} = \frac{\chi_{\text{top}}}{f_a}\,\sin\Big(\bar\theta + \frac{a}{f_a}\Big) = 0, \tag{42.215}$$

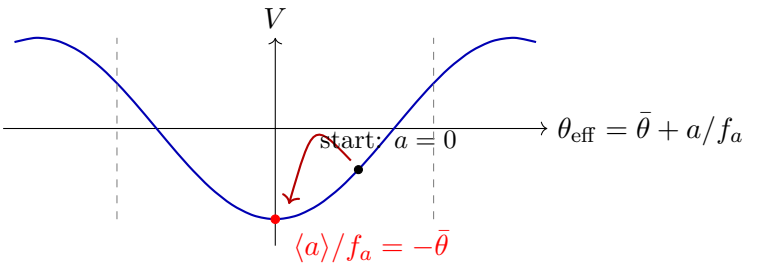


Figure 42.9: The axion potential $V(a) = -\chi_{\text{top}} \cos(\bar{\theta} + a/f_a)$ of Eq. (42.214), the QCD vacuum-energy curve Eq. (42.212) read as a function of the dynamical field. Whatever the bare $\bar{\theta}$ (the field begins at $a = 0$, where $\theta_{\text{eff}} = \bar{\theta} \neq 0$), the axion rolls to the minimum at $\theta_{\text{eff}} = 0$, i.e. $\langle a \rangle = -f_a \bar{\theta}$, switching off strong CP. The curvature at the minimum is the axion mass-squared, $m_a^2 = \chi_{\text{top}}/f_a^2$.

and, since $\chi_{\text{top}} > 0$ by Eq. (42.211), the minimum (not the maximum) is the solution with $\cos(\cdots) = +1$, namely

$$\left\langle \frac{a}{f_a} \right\rangle = -\bar{\theta} \qquad \Longrightarrow \qquad \theta_{\text{eff}}\big|_{\text{vac}} = \bar{\theta} + \left\langle \frac{a}{f_a} \right\rangle = 0. \tag{42.216}$$

The axion field rolls until the *effective* strong-CP angle is exactly zero, whatever the bare $\bar{\theta}$ was. Strong CP is not tuned away; it is dynamically relaxed away. The residual neutron EDM is set by θ_{eff}, driven to zero up to the negligible electroweak feed-through $\delta\theta_{\text{eff}} \sim 10^{-16}$, well below the bound Eq. (42.209).

Step 2: the mass. The same potential that relaxes $\bar{\theta}$ gives the axion a mass — it is a pseudo-Goldstone boson, massless only in the absence of the QCD anomaly. The mass is the curvature at the minimum:

$$m_a^2 = \frac{d^2 V}{da^2}\bigg|_{\text{min}} = \frac{\chi_{\text{top}}}{f_a^2} \cos(0) = \frac{\chi_{\text{top}}}{f_a^2}, \qquad \Longrightarrow \qquad \boxed{m_a = \frac{\chi_{\text{top}}^{1/2}}{f_a}} \tag{42.217}$$

The axion mass is fixed entirely by the QCD topological susceptibility and the breaking scale; everything model-dependent has dropped out. Inserting the two-flavour value $\chi_{\text{top}} = f_\pi^2 m_\pi^2 \, m_u m_d/(m_u + m_d)^2$ of Eq. (35.82) reproduces the numerical mass–coupling relation $m_a \approx 5.7 \ \mu\text{eV} \, (10^{12}\,\text{GeV}/f_a)$ computed in full at Theorem 35.7.

Equation Eq. (42.217) is the signature of the solution: the axion mass scales *inversely* with the symmetry-breaking scale. A high f_a means a very light, very weakly coupled axion — every coupling, being that of a pseudo-Goldstone, is suppressed by $1/f_a$. This is what makes the axion at once an economical fix for strong-CP and a distinctive experimental target: not a heavy particle for a collider but an ultralight, feebly interacting relic. And that relic is a leading dark-matter candidate. In the early universe the field is frozen at a

random misalignment $\theta_{\text{eff}} = \theta_i$ until the topological susceptibility switches on near Λ_{QCD} (Theorem 27.14); the coherent oscillations of the field about the minimum Eq. (42.216) are a cold, pressureless condensate — cold dark matter produced by the very mechanism that solves strong-CP.

Alternatives (honest scope). The axion is the leading but not the only proposal. A second class spontaneously breaks CP at a high scale and arranges the vacuum so that $\theta_0 = 0$ is enforced by the symmetry while the observed CKM phase arises from the breaking — the Nelson–Barr mechanism, which trades the axion for a structured, often baroque, flavour sector and the burden of keeping radiative corrections to $\bar{\theta}$ below 10^{-10}. The massless-up-quark option is, as noted, disfavoured by the lattice. As of the mid-2020s no resolution is experimentally confirmed: the axion remains a hypothesis under active search, and strong-CP remains, strictly, an open problem with a compelling but unverified favourite.

This closes the chapter. We have carried Yang–Mills theory to its full non-perturbative reach — classical Yang–Mills and the Wilson loop, the Faddeev–Popov and BRST quantization, asymptotic freedom, the monopoles of Chapter 41, and here the instanton, the θ-vacuum, the resolution of the $\mathrm{U}(1)_A$ problem, and the strong-CP problem it exposes — and we have shown that an entire layer of physics, weighted by $e^{-8\pi^2/g^2}$ and invisible to every order of perturbation theory, reorganizes the vacuum. The model-building of the strong-CP problem and its axion — the KSVZ and DFSZ realizations of $\mathrm{U}(1)_{\text{PQ}}$ (Section 35.4.5) and the cosmology of the misalignment relic abundance (Section 35.6) — is the province of the BSM chapter (Section 35.4), and the broader naturalness landscape that of the frontier survey of Chapter 57.

Further reading

The instanton was discovered as a self-dual Yang–Mills solution by Belavin et al. [55], and its quantum effects—the collective-coordinate measure, the running-coupling size integral, and the multi-fermion vertex—were computed in 't Hooft [464]. The definitive pedagogical account remains Coleman's Erice lectures [128], reprinted in his collected lectures [129]; a comprehensive modern review of the instanton calculus and its phenomenology is Schäfer and Shuryak [428]. The resolution of the $U(1)_A$ problem in the large-N limit and the η' mass formula are due to Witten [518] and Veneziano [486]. For the strong-CP problem and its resolutions: the Peccei–Quinn mechanism is Peccei and Quinn [388, 389] with the axion identified by Weinberg [497] and Wilczek [511], and the invisible-axion models are KSVZ [294] and DFSZ [149]; the Vafa–Witten theorem that CP is unbroken in vector-like QCD is Vafa and Witten [480]; the chiral estimate of the neutron electric dipole moment is Crewther et al. [140], with the current

experimental bound from Abel et al. [2] and the light-quark masses that close the $m_u = 0$ option reviewed in Aoki and others [27].

Chapter 43

Lattice Field Theory

Every regulator in this book so far has been provisional. Dimensional continuation, Pauli–Villars masses, the Wilsonian momentum shell — each tamed the ultraviolet long enough to extract a finite answer, and each was meant to be removed at the end. None of them is a *definition* of a quantum field theory: each presupposes the perturbative expansion it regulates, and that expansion, as the Borel analysis of Chapter 5 made plain, is at best asymptotic. It does not sum to a function, and it is blind to the one phenomenon that organizes the strong interaction, confinement. The recurring question of the book — what *is* an interacting quantum field theory, exactly, beyond a recipe for a divergent series? — has so far been deferred. This chapter answers it. Following Wilson, we place the theory on a finite hypercubic Euclidean lattice, where the path integral becomes an ordinary, manifestly convergent, finite-dimensional integral, and we ask what survives as the spacing is sent to zero.

The construction draws together threads from across the book. The Euclidean path integral and the statistical-mechanical reading of e^{-S_E} as a Boltzmann weight are imported from Chapter 10; Grassmann integration and the fermion determinant from Chapter 12; the gauge connection, curvature, and Wilson loop from Chapter 25; and the entire language of the continuum limit — relevant couplings, the critical surface, flow into a renormalization-group fixed point — from the Wilsonian picture of Chapter 23 and the beta functions of Chapter 24. The decisive physical input is the asymptotic freedom of Chapter 27, which is what allows a four-dimensional non-abelian gauge theory to be defined all the way to zero spacing. The representation-theoretic vocabulary of Chapter 2 runs throughout, and the chiral anomaly of Chapter 36 reappears, unexpectedly, as the obstruction that makes lattice fermions hard.

The arc is the following. We first build the lattice as a constructive regulator and show that the discretized measure is genuinely finite, that it carries a positive transfer matrix and hence reconstructs a Hamiltonian, and that reflection positivity is the lattice shadow of the Osterwalder–Schrader axioms (Section 43.1).

We then study the scalar in detail and learn the central lesson of the subject: the continuum limit is not a naive $a \to 0$ but the approach to a second-order critical point, an RG fixed point at which the correlation length in lattice units diverges (Section 43.2). With the framework fixed, we turn to gauge fields, where the gauge potential is replaced by group-valued link variables and exact gauge invariance is achieved with no gauge fixing at all (Section 43.3); the payoff is the first analytic proof of confinement, the strong-coupling area law and its linear potential (Section 43.4).

Fermions are the hard case, and we treat them honestly. Naively discretized, a single Dirac field becomes sixteen, and the Nielsen–Ninomiya theorem shows this doubling is no accident but a topological necessity tied to the anomaly (Section 43.5). We survey the responses — Wilson's explicit symmetry breaking, staggered fermions, and the Ginsparg–Wilson relation that restores an exact lattice chiral symmetry through overlap and domain-wall constructions (Section 43.6). We then show what the finished machine actually delivers: expectation values as Monte Carlo ensemble averages, masses read off the spectral decomposition of Euclidean correlators, and the controlled removal of the cutoff by asymptotic scaling (Section 43.7). The chapter closes by assembling these pieces into a single claim and weighing it carefully: the lattice is, as of the mid-2020s, the only known fully non-perturbative, gauge-invariant definition of an interacting quantum field theory — and we separate cleanly what it has *proved* from what it has only *computed*, ending on the frontiers where the definition itself is still being built (Section 43.8).

43.1 The Euclidean lattice as a constructive regulator

The chapter introduction made a promise that the rest of the book has earned but never quite cashed: that after the Wick rotation of Chapter 10 the weight e^{-S_E} is not a formal device but a genuine positive measure, so that a quantum field theory is, in disguise, a problem in classical statistical mechanics. We saw there (Section 10.6) that the rotated generating functional

$$Z_E = \int \mathcal{D}\phi \, e^{-S_E[\phi]} \tag{43.1}$$

has a positive, bounded integrand, with none of the conditional-convergence delicacy of the Minkowski e^{iS}. What we did *not* do there is say what the symbol $\int \mathcal{D}\phi$ means. It is a product of uncountably many ordinary integrals, one for each point of spacetime, and as written it is not a mathematical object at all. Every manipulation we have performed on it—completing the square, expanding $e^{-S_{\text{int}}}$ in a perturbation series, Legendre-transforming—has been a manipulation of a symbol whose existence we postponed.

The lattice discharges that debt in the most direct way imaginable: it replaces continuous spacetime by a finite grid of points, so that $\int \mathcal{D}\phi$ becomes

a finite-dimensional integral over $\mathbb{R}^V$ that any first course in analysis certifies as convergent. There is no other regulator, no $i\varepsilon$, no analytic continuation in dimension, no subtraction of infinities—the integral is simply finite because it is an ordinary integral over finitely many real variables. This is the sense in which the lattice is not a numerical approximation bolted onto a pre-existing theory but a *definition* of one. The continuum theory, when it exists, will be recovered as a limit of these honest integrals; until then, the lattice integral is the only fully non-perturbative thing on the table that we are entitled to call "the theory."

We carry out the construction for ϕ^4 theory: the same single real scalar with the Lagrangian frozen in Chapter 6 and quantized perturbatively in Chapter 9, now defined with no reference to perturbation theory at all. The Euclidean action is the one fixed in Theorem 10.27, Eq. (10.161). The goals of this section are four, and they are exactly the things every later section will lean on. First, to define the lattice, its fields, and its measure precisely. Second, to read off the dictionary that turns e^{-S_E}/Z into a Boltzmann distribution and the theory into statistical mechanics. Third, to prove that the resulting integral is finite—that the lattice is a complete ultraviolet regulator—for the massive scalar. Fourth, and most importantly, to reconstruct from the lattice integral a Hilbert space and a Hamiltonian: to build the transfer matrix T, prove it is a positive self-adjoint operator, and exhibit $T = e^{-aH}$, so that the statistical system genuinely *is* a quantum theory in imaginary time. Reflection (Osterwalder–Schrader) positivity, stated in general in Theorem 10.32, is what makes this reconstruction possible, and we prove it for the lattice scalar.

43.1.1 The hypercubic Euclidean lattice

We work in four Euclidean dimensions; the construction is identical in any number. Continuous spacetime $\mathbb{R}^4$ is replaced by a finite, regular grid.

Definition 43.1 (Hypercubic Euclidean lattice)**.** Fix a length $a > 0$, the *lattice spacing*, and an integer N, the number of sites per direction; the *physical extent* is $L = Na$. The *lattice* is the finite set of *sites*

$$\Lambda = \{\, n = (n_0, n_1, n_2, n_3) \;:\; n_\mu \in \{0, 1, \dots, N-1\} \,\}, \qquad x_\mu = a\, n_\mu, \tag{43.2}$$

identified pairwise across opposite faces, $n_\mu \equiv n_\mu + N$ (periodic boundary conditions), so that Λ is a discrete four-torus with $V = N^4$ sites. Let $\hat{\mu}$ denote the unit vector in direction μ. The ordered pairs $(n, n+\hat{\mu})$ of nearest-neighbour sites are the *links*; the elementary squares spanned by two directions, with corners $\{n,\ n+\hat{\mu},\ n+\hat{\nu},\ n+\hat{\mu}+\hat{\nu}\}$, are the *plaquettes*. A (real scalar) *field configuration* is a map $\phi : \Lambda \to \mathbb{R}$, written ϕ_n, an element of the finite-dimensional space $\mathbb{R}^V$.

The site/link/plaquette vocabulary is not pedantry: it is the skeleton on which every field in this chapter is hung. A scalar lives on sites; a gauge field will

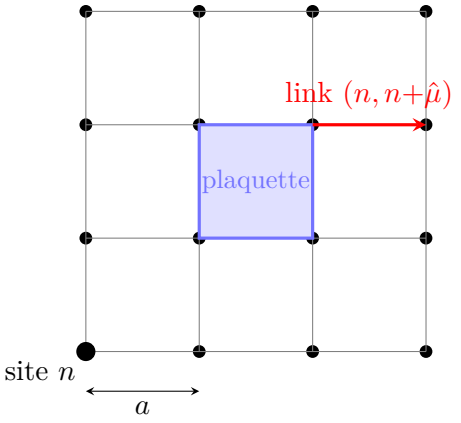


Figure 43.1: A two-dimensional slice of the hypercubic lattice Λ. Sites carry the scalar field ϕ_n; links connect nearest neighbours; the shaded unit square is a plaquette. The spacing a is the only length in the problem and, as Theorem 43.4 shows, plays the role of an ultraviolet cutoff: no fluctuation of wavelength shorter than a exists.

live on links (Section 43.3, where this becomes the whole point); the field strength will live on plaquettes. Figure 43.1 shows the three objects on a two-dimensional slice.

To put dynamics on Λ we need a discrete version of the derivative. The natural choice is the *forward difference*

$$(\partial_\mu^{\mathrm{lat}}\phi)_n := \frac{\phi_{n+\hat{\mu}} - \phi_n}{a}, \tag{43.3}$$

which reduces to $\partial_\mu\phi$ as $a \to 0$ for a smooth field, by Taylor expansion: $\phi_{n+\hat{\mu}} = \phi_n + a\,\partial_\mu\phi + \tfrac{1}{2}a^2\partial_\mu^2\phi + O(a^3)$, so $(\partial_\mu^{\mathrm{lat}}\phi)_n = \partial_\mu\phi + \tfrac{1}{2}a\,\partial_\mu^2\phi + O(a^2)$. The discretization of a continuum action is then obtained by the rule

$$\int \mathrm{d}^4x \;\longmapsto\; a^4 \sum_{n\in\Lambda}, \qquad \partial_\mu\phi \;\longmapsto\; (\partial_\mu^{\mathrm{lat}}\phi)_n, \tag{43.4}$$

which is the obvious Riemann-sum prescription. Applying it to the Euclidean ϕ^4 action of Eq. (10.161) gives the central definition.

Definition 43.2 (Lattice ϕ^4 action)**.** For a configuration $\phi : \Lambda \to \mathbb{R}$, the *lattice action* of ϕ^4 theory is

$$S_E[\phi] = a^4 \sum_{n\in\Lambda} \left[\frac{1}{2} \sum_{\mu=0}^{3} \left(\frac{\phi_{n+\hat{\mu}} - \phi_n}{a} \right)^2 + \frac{1}{2} m^2 \phi_n^2 + \frac{\lambda}{4!}\phi_n^4 \right], \tag{43.5}$$

with $m^2 > 0$ and $\lambda \geq 0$ the same mass and coupling as the continuum theory. Every term is non-negative, so $S_E[\phi] \geq 0$ for all ϕ, and $S_E[\phi] \to +\infty$ as $|\phi| \to \infty$ in $\mathbb{R}^V$.

The non-negativity and the growth at large field, inherited from the all-plus sign structure of the Euclidean action (Eq. (10.161)), are exactly the two properties that will make the integral converge. We isolate them at once.

Definition 43.3 (Lattice measure and partition function)**.** The *lattice measure* is the ordinary product Lebesgue measure on $\mathbb{R}^V$,

$$\mathcal{D}\phi := \prod_{n\in\Lambda} \mathrm{d}\phi_n, \tag{43.6}$$

a finite product of one-dimensional integrals. The *partition function* is

$$Z = \int_{\mathbb{R}^V} \mathcal{D}\phi\, e^{-S_E[\phi]} = \int \Big(\prod_n \mathrm{d}\phi_n\Big)\, e^{-S_E[\phi]}, \tag{43.7}$$

and the *lattice expectation* of an observable $\mathcal{O}[\phi]$ (any function of the field configuration) is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \mathcal{D}\phi\, \mathcal{O}[\phi]\, e^{-S_E[\phi]}. \tag{43.8}$$

This is the whole apparatus. The formal $\int \mathcal{D}\phi$ of Eq. (43.1) has become the genuinely defined $\int_{\mathbb{R}^V} \prod_n \mathrm{d}\phi_n$ of Eq. (43.7); the limit operations of Chapter 10 are now applied to an object that exists. Everything that follows—the dictionary, the finiteness, the Hilbert space—is a property of this finite-dimensional integral.

43.1.2 The Boltzmann dictionary: a quantum field theory as a statistical system

The form of Eq. (43.7) is, letter for letter, that of the partition function of a classical statistical-mechanical system. Compare with the canonical ensemble of a system whose microstates are configurations σ with energy $E[\sigma]$ at inverse temperature $\beta = 1/k_B T$:

$$Z_{\mathrm{stat}} = \sum_{\sigma} e^{-\beta E[\sigma]}, \qquad P[\sigma] = \frac{1}{Z_{\mathrm{stat}}} e^{-\beta E[\sigma]}. \tag{43.9}$$

The correspondence is exact, not a metaphor: the microstate σ is the field configuration $\phi : \Lambda \to \mathbb{R}$; the configuration sum $\sum_\sigma$ is the integral $\int \prod_n \mathrm{d}\phi_n$; the dimensionless energy βE is the Euclidean action $S_E[\phi]$; and the Boltzmann probability of a state is

$$P[\phi] = \frac{e^{-S_E[\phi]}}{Z}. \tag{43.10}$$

That $P[\phi] \geq 0$ and $\int \mathcal{D}\phi\, P[\phi] = 1$ is automatic from $S_E \geq 0$ and the definition of Z; Eq. (43.10) is a bona fide probability density on field space. The lattice expectation Eq. (43.8) is then literally the thermal average $\langle \mathcal{O} \rangle = \int \mathcal{D}\phi\, \mathcal{O}[\phi]\, P[\phi]$. A four-dimensional Euclidean scalar field theory is a four-dimensional classical

statistical-mechanical model with a continuous spin $\phi_n \in \mathbb{R}$ at each site, nearest-neighbour ferromagnetic coupling (the gradient term favours aligned neighbours), and an on-site potential $\frac{1}{2}m^2\phi_n^2 + \frac{\lambda}{4!}\phi_n^4$.

This is the dictionary tabulated abstractly in Table 10.1, now made completely concrete by the discretization: the entries on the statistical-mechanics side are not analogies but the actual ingredients of a well-defined finite system. The single most important entry, the one that organizes the rest of the chapter, is the bottom line of that table:

$$\text{correlation length } \xi \quad \longleftrightarrow \quad \text{inverse mass } 1/m_{\text{phys}}. \tag{43.11}$$

A massive particle in the field theory is an exponentially decaying correlation in the statistical model, with decay length $\xi = 1/m_{\text{phys}}$. We will see this explicitly when we extract masses from the transfer matrix below, and it is the hinge on which the next section turns: the continuum limit will be the limit in which ξ, measured in lattice units ξ/a, diverges—a critical point of the statistical system. For now the lesson is the structural one. Having discretized, we are no longer obliged to think of the lattice integral as an approximation to something else. It is a complete statistical-mechanical problem, with a positive probability measure, and we are entitled to ask any question of it that we would ask of an Ising model.

Why the Euclidean rotation was worth the trouble

The Minkowski path integral weights configurations by a phase e^{iS}; it is an oscillatory integral, conditionally convergent at best, and it defines no probability measure—there is no such thing as "the probability of a field configuration" in real time. The Wick rotation of Theorem 5.30 and Chapter 10 converts the phase into a positive Boltzmann weight, and *that* is what makes the configuration sum a statistical-mechanics problem and lets us simulate it, bound it, and reason about it probabilistically. The price—a theory formulated in imaginary time—is paid back in full by reflection positivity (Section 43.1.5), which returns us to real-time quantum mechanics with a Hermitian Hamiltonian. The lattice is the arena in which this trade is carried out honestly.

43.1.3 The lattice is a complete ultraviolet regulator

Before extracting physics we must be sure there is something to extract: that Z in Eq. (43.7) is finite. In the continuum this is exactly the question that perturbation theory could only answer order by order, with subtractions; on the lattice it is a one-line consequence of the integral being finite-dimensional and the action growing at large field. We make it precise.

Proposition 43.4 (Finiteness of the lattice measure for the massive scalar)**.** *For $m^2 > 0$, $\lambda \geq 0$, and any finite lattice Λ with $V = N^4$ sites, the integral Z of Eq. (43.7) is finite and strictly positive, and the lattice expectation Eq. (43.8) of any polynomially bounded observable is an absolutely convergent integral. No ultraviolet divergence occurs and no regulator beyond the lattice is required.*

Proof. Split the action into its quadratic (free) and quartic (interacting) parts, $S_E = S_E^{(0)} + S_{\text{int}}$ with

$$S_E^{(0)}[\phi] = \frac{a^4}{2} \sum_n \Big[\sum_\mu \Big(\tfrac{\phi_{n+\hat{\mu}} - \phi_n}{a} \Big)^2 + m^2 \phi_n^2 \Big], \qquad S_{\text{int}}[\phi] = a^4 \sum_n \frac{\lambda}{4!} \phi_n^4. \tag{43.12}$$

Step 1: the free part is a positive-definite quadratic form. Using $\sum_n (\phi_{n+\hat{\mu}} - \phi_n)^2 = -\sum_n \phi_n (\phi_{n+\hat{\mu}} + \phi_{n-\hat{\mu}} - 2\phi_n)$ after a shift of summation index (legitimate by periodicity), we may write

$$S_E^{(0)}[\phi] = \frac{1}{2} \sum_{n,n'} \phi_n \, M_{nn'} \, \phi_{n'}, \qquad M = a^2 (-\Delta_{\text{lat}}) + a^4 m^2 \, \mathbf{1}, \tag{43.13}$$

where $(\Delta_{\text{lat}} \phi)_n = \sum_\mu (\phi_{n+\hat{\mu}} + \phi_{n-\hat{\mu}} - 2\phi_n)$ is the lattice Laplacian and $\mathbf{1}$ the identity on $\mathbb{R}^V$. The matrix M is real, symmetric, and translation invariant, so it is diagonalized by the lattice plane waves $\phi_n^{(p)} = V^{-1/2} e^{i\, p \cdot n a}$ with momenta quantized by periodicity,

$$p_\mu = \frac{2\pi k_\mu}{N a}, \qquad k_\mu \in \{0, 1, \ldots, N-1\}. \tag{43.14}$$

Acting with M on $\phi^{(p)}$ and using $e^{i p_\mu a} + e^{-i p_\mu a} - 2 = -4 \sin^2(p_\mu a / 2)$ gives the eigenvalues

$$\lambda_p = a^2 \sum_\mu 4 \sin^2\Big(\frac{p_\mu a}{2} \Big) + a^4 m^2 \;\geq\; a^4 m^2 \;>\; 0. \tag{43.15}$$

Every eigenvalue is bounded below by $a^4 m^2 > 0$: the mass term lifts the would-be zero mode, and M is positive definite. Crucially, the eigenvalues are also bounded *above*, $\lambda_p \leq 16 a^2 + a^4 m^2$, because $\sin^2 \leq 1$ and the momenta Eq. (43.14) are confined to the first Brillouin zone $|p_\mu| \leq \pi / a$. There is no large-momentum tail to diverge: modes of wavelength shorter than $2a$ simply do not exist on Λ. *This is the lattice acting as an ultraviolet cutoff*, $\Lambda_{\text{UV}} \sim \pi / a$, built into the kinematics rather than imposed by hand.

Step 2: the free integral is a finite Gaussian. Since M is symmetric positive definite, the standard V-dimensional Gaussian integral converges:

$$Z_0 = \int_{\mathbb{R}^V} \mathcal{D}\phi \; e^{-\frac{1}{2} \phi^{\mathsf{T}} M \phi} = \frac{(2\pi)^{V/2}}{\sqrt{\det M}} = (2\pi)^{V/2} \Big(\prod_p \lambda_p \Big)^{-1/2}, \tag{43.16}$$

and $\det M = \prod_p \lambda_p \geq (a^4 m^2)^V > 0$ is a finite, strictly positive number because it is a finite product of the strictly positive eigenvalues Eq. (43.15). Hence $0 < Z_0 < \infty$.

Step 3: the interaction only helps. For $\lambda \geq 0$ the quartic term is non-negative, $S_{\text{int}}[\phi] \geq 0$, so $0 < e^{-S_{\text{int}}[\phi]} \leq 1$ pointwise. Therefore the full integrand is dominated by the convergent Gaussian,

$$0 < e^{-S_E[\phi]} = e^{-S_E^{(0)}[\phi]}\, e^{-S_{\text{int}}[\phi]} \leq e^{-S_E^{(0)}[\phi]}, \tag{43.17}$$

and integrating gives $0 < Z \leq Z_0 < \infty$. The integral is absolutely convergent; in fact Z is strictly positive because the integrand is strictly positive and continuous. *Step 4: observables.* For a polynomially bounded observable $|\mathcal{O}[\phi]| \leq C\,(1+|\phi|)^k$, the integrand $|\mathcal{O}|\; e^{-S_E} \leq C(1+|\phi|)^k e^{-S_E^{(0)}}$ is dominated by a polynomial times the convergent Gaussian, whose moments are all finite; hence Eq. (43.8) converges absolutely. □

Three features of this proof deserve emphasis because they are the entire conceptual payload of "the lattice as a regulator." First, no subtraction occurred: Z is finite as it stands, in contrast to dimensional regularization or Pauli–Villars, which render individual divergent integrals finite only after a regulator parameter is introduced and counterterms are arranged to cancel poles. Second, the cutoff is the spacing a itself, visible in Eq. (43.14) as the boundary π/a of the Brillouin zone; the theory at finite a is complete and self-contained, and the continuum is a *limit* of well-defined theories rather than a divergent object to be tamed. Third—and this is the property that no other regulator shares—nothing in the construction broke any symmetry we will want to keep. The product measure Eq. (43.6) and the action Eq. (43.5) are manifestly invariant under the lattice translations and the discrete hypercubic rotations, and, as the next section's gauge generalization will show, the lattice can be set up to preserve *exact* gauge invariance without gauge fixing. The price is the breaking of continuous rotational invariance down to the hypercubic point group, a defect that heals in the continuum limit; we pay it gladly in exchange for a manifestly gauge-invariant, manifestly finite definition.

43.1.4 The transfer matrix and the reconstruction of a Hamiltonian

We have a finite, positive, statistical-mechanical measure. We now show that hidden inside it is a quantum mechanics: a Hilbert space, a Hamiltonian, and the operator that propagates states one lattice spacing in imaginary time. The device is the *transfer matrix*, the same object that diagonalizes the one-dimensional Ising model, generalized to a field on each site of a time slice. It is the lattice incarnation of the relation $e^{-S_E} \leftrightarrow e^{-\beta H}$, and it is the machine that Section 43.7.2

(and Sections 43.1.4 onward of this chapter) will use to read masses off correlation functions.

Single out one direction, say $\mu = 0$, as Euclidean time. A site is $n = (t, \mathbf{n})$ with $t \in \{0, \dots, N-1\}$ the time coordinate and $\mathbf{n}$ a site of the three-dimensional *spatial* lattice Λ_s ($V_s = N^3$ sites). The field restricted to a fixed time is a *time-slice configuration* $\Phi_t = \{\phi_{(t,\mathbf{n})}\}_{\mathbf{n}\in\Lambda_s} \in \mathbb{R}^{V_s}$. Splitting the time and space derivatives in Eq. (43.5), the action becomes a sum over time slices,

$$S_E[\phi] = \sum_t \Big[\underbrace{\frac{a^2}{2} \sum_{\mathbf{n}} \big(\phi_{t+1,\mathbf{n}} - \phi_{t,\mathbf{n}}\big)^2}_{\text{couples slices } t,\, t+1} + a^4\, W(\Phi_t) \Big], \tag{43.18}$$

where the temporal hopping term couples adjacent slices, and everything intrinsic to a single slice is collected in the *slice potential*

$$W(\Phi) = \sum_{\mathbf{n}} \Big[\frac{1}{2} \sum_{i=1}^{3} \Big(\frac{\phi_{\mathbf{n}+\hat{\imath}} - \phi_{\mathbf{n}}}{a} \Big)^2 + \frac{1}{2} m^2 \phi_{\mathbf{n}}^2 + \frac{\lambda}{4!} \phi_{\mathbf{n}}^4 \Big]. \tag{43.19}$$

We now write the Boltzmann weight e^{-S_E} as a product of identical factors, one per temporal link, by sharing each slice's potential symmetrically between the two temporal links that touch it (under the periodic identification each slice t has neighbours $t \pm 1$). The factor associated with the link from slice t to slice $t+1$ defines an integral kernel on slice configurations.

Definition 43.5 (Transfer matrix of the lattice scalar)**.** Let $\mathcal{H} = L^2(\mathbb{R}^{V_s}, \mathrm{d}\Phi)$ with $\mathrm{d}\Phi = \prod_{\mathbf{n}} \mathrm{d}\phi_{\mathbf{n}}$ be the Hilbert space of square-integrable functionals of one spatial slice. The *transfer matrix* T is the integral operator on $\mathcal{H}$ with kernel

$$\langle \Phi' | T | \Phi \rangle = e^{-\frac{1}{2} a^4 W(\Phi')}\, \mathcal{K}(\Phi', \Phi)\, e^{-\frac{1}{2} a^4 W(\Phi)}, \qquad \mathcal{K}(\Phi', \Phi) = \Big(\tfrac{a^2}{2\pi} \Big)^{V_s/2} e^{-\frac{a^2}{2} \sum_{\mathbf{n}} (\phi'_{\mathbf{n}} - \phi_{\mathbf{n}})^2}, \tag{43.20}$$

the normalization of $\mathcal{K}$ chosen so that $\int \mathrm{d}\Phi\, \mathcal{K}(\Phi', \Phi) = 1$.

The point of the symmetric splitting is the following identity, which is the lattice statement that "the partition function is a trace over states."

Proposition 43.6 (Partition function as a trace)**.** *With periodic time of extent N, $Z = \mathrm{Tr}_{\mathcal{H}}\, T^N$, and more generally any correlator of slice observables is a trace of the corresponding multiplication operators interleaved with powers of T.*

Proof. The kernel product over the temporal links reassembles the Boltzmann weight: the Gaussian factors $\mathcal{K}(\Phi_{t+1}, \Phi_t)$ reproduce the temporal hopping terms of Eq. (43.18) (the normalization constant is a ϕ-independent prefactor that contributes only to the overall constant in Z), while each slice potential $a^4 W(\Phi_t)$

is supplied as two halves, $e^{-\frac{1}{2}a^4 W(\Phi_t)}$ from the link below and another from the link above. Hence

$$\int \prod_t \mathrm{d}\Phi_t \; \prod_t \langle \Phi_{t+1}|T|\Phi_t\rangle = \int \prod_t \mathrm{d}\Phi_t \; e^{-S_E[\phi]} \cdot (\text{const}) = Z \cdot (\text{const}), \tag{43.21}$$

and the left side, with the periodic identification $\Phi_N \equiv \Phi_0$, is by definition $\operatorname{Tr} T^N = \int \prod_t \mathrm{d}\Phi_t \; \prod_t \langle \Phi_{t+1}|T|\Phi_t\rangle$. Absorbing the constant into the normalization of T (equivalently of $\mathcal{K}$, as in Eq. (43.20)) gives $Z = \operatorname{Tr} T^N$. Inserting a slice observable $\mathcal{O}(\Phi_t)$ multiplies the integrand by the corresponding multiplication operator $\hat{\mathcal{O}}$ at position t in the product, yielding $\langle \mathcal{O} \rangle = Z^{-1} \operatorname{Tr}(\hat{\mathcal{O}}\, T^N)$, and similarly for several insertions. □

Everything now hinges on the operator properties of T. The next theorem is the central analytic result of the section: T is a positive self-adjoint operator, so its logarithm is a self-adjoint operator, and that operator, divided by $-a$, is the Hamiltonian.

Theorem 43.7 (The transfer matrix is positive and defines a Hamiltonian)**.** *The operator T of Theorem 43.5 is bounded, self-adjoint, and strictly positive on $\mathcal{H} = L^2(\mathbb{R}^{V_s})$. Consequently there is a self-adjoint operator*

$$H := -\frac{1}{a} \ln T, \qquad \textit{i.e.} \qquad T = e^{-aH}, \tag{43.22}$$

with real spectrum bounded below. The ground state of H (the lowest eigenvector of T) is unique and may be chosen strictly positive; identifying it with the vacuum $|0\rangle$, the spectrum of H above it gives the energy levels of the reconstructed quantum theory.

Proof. Self-adjointness. The kernel Eq. (43.20) is real and symmetric, $\langle \Phi'|T|\Phi\rangle = \langle \Phi|T|\Phi'\rangle$, because $W(\Phi')$ and $W(\Phi)$ enter symmetrically and $\mathcal{K}(\Phi', \Phi) = \mathcal{K}(\Phi, \Phi')$. A real symmetric kernel defines a (formally) self-adjoint integral operator; boundedness, proved next, makes it genuinely self-adjoint on $\mathcal{H}$.

Boundedness. Write $T = A\,K\,A$ with A the multiplication operator $(A\psi)(\Phi) = e^{-\frac{1}{2}a^4 W(\Phi)}\psi(\Phi)$ and K the integral operator with kernel $\mathcal{K}$. Since $W(\Phi) \geq 0$ (it is a sum of squares and even powers), A is a bounded multiplication operator with $0 < A \leq \mathbf{1}$, hence $\|A\| \leq 1$. The operator K is a Gaussian convolution: $\mathcal{K}(\Phi', \Phi)$ depends only on $\Phi' - \Phi$ and is a normalized probability density, so K is a Markov (averaging) operator with $\|K\| \leq 1$ (its Fourier symbol, computed below, lies in $(0, 1]$). Therefore $\|T\| \leq \|A\|^2 \, \|K\| \leq 1$, and T is bounded.

Positivity. This is the crux. The convolution operator K is diagonalized by the spatial plane waves $\Phi \mapsto e^{i\,\mathbf{k}\cdot\Phi}$ in the conjugate (momentum) variables of the V_s field components; equivalently, recognizing the Gaussian as a heat kernel, one

has

$$\mathcal{K}(\Phi',\Phi)=\Big\langle\Phi'\Big|\,\exp\Big(\frac{1}{2a^2}\sum_{\mathbf{n}}\frac{\partial^2}{\partial\phi_{\mathbf{n}}^2}\Big)\Big|\Phi\Big\rangle=\Big\langle\Phi'\Big|\,e^{-\frac{1}{2a^2}\hat{\mathbf{p}}^2}\,\Big|\Phi\Big\rangle,\qquad \hat{p}_{\mathbf{n}}=-i\frac{\partial}{\partial\phi_{\mathbf{n}}},\tag{43.23}$$

where $\hat{\mathbf{p}}^2=\sum_{\mathbf{n}}\hat{p}_{\mathbf{n}}^2$. The identity is the standard heat-kernel formula $\langle x'|e^{(s/2)\partial_x^2}|x\rangle=(2\pi s)^{-1/2}e^{-(x'-x)^2/2s}$ with $s=1/a^2$, applied componentwise. Thus $K=e^{-\frac{1}{2a^2}\hat{\mathbf{p}}^2}$ is the exponential of the negative semidefinite operator $-\frac{1}{2a^2}\hat{\mathbf{p}}^2$; its spectrum is $\{e^{-\mathbf{k}^2/2a^2}:\mathbf{k}\in\mathbb{R}^{V_s}\}\subset(0,1]$, so K is strictly positive: $\langle\psi|K|\psi\rangle=\int\frac{\mathrm{d}^{V_s}\mathbf{k}}{(2\pi)^{V_s}}\,e^{-\mathbf{k}^2/2a^2}\left|\tilde{\psi}(\mathbf{k})\right|^2>0$ for $\psi\neq0$. Positivity is preserved under congruence by a self-adjoint operator: for any $\psi\in\mathcal{H}$,

$$\langle\psi|T|\psi\rangle=\langle\psi|AKA|\psi\rangle=\langle A\psi|K|A\psi\rangle>0\tag{43.24}$$

whenever $A\psi\neq0$; and $A\psi=0$ forces $\psi=0$ because A is multiplication by the strictly positive function $e^{-\frac{1}{2}a^4W}$. Hence $T>0$ strictly.

The Hamiltonian. A bounded strictly positive self-adjoint operator has, by the spectral theorem, a self-adjoint logarithm $\ln T$ with spectrum $\ln(\operatorname{spec}T)\subset(-\infty,0]$ (since $\|T\|\leq1$). Define $H=-\frac{1}{a}\ln T$; it is self-adjoint, its spectrum is real and bounded below by $-\frac{1}{a}\ln\|T\|\geq0$, and by construction $T=e^{-aH}$, which is Eq. (43.22). Because T has a strictly positive kernel on $\mathcal{H}$ (both factors A and K are positivity-improving), the Perron–Frobenius theorem for integral operators applies: the largest eigenvalue of T is simple and its eigenfunction can be chosen strictly positive. That eigenfunction is the lowest-energy state of H, the vacuum $|0\rangle$; its uniqueness is the statement that the reconstructed theory has a unique ground state. □

Theorem 43.7 is the reconstruction in miniature. The statistical-mechanical measure e^{-S_E}/Z has handed us, with no further input, a Hilbert space $\mathcal{H}$, a self-adjoint Hamiltonian H with a real spectrum bounded below, and a unique vacuum. The relation $T=e^{-aH}$ says that advancing one slice in Euclidean time is imaginary-time evolution by a, exactly the Euclidean analogue of e^{-iHt}. The trace formula $Z=\operatorname{Tr}e^{-NaH}=\operatorname{Tr}e^{-\beta H}$ with $\beta=Na=L$ identifies the temporal extent of the lattice with an inverse temperature, sharpening the dictionary of Section 43.1.2: a Euclidean field theory in a box of temporal size L is the thermal ensemble of the reconstructed quantum theory at temperature $1/L$, and the zero-temperature quantum theory is recovered as $L\to\infty$. The energy gaps of H are physical particle masses, extracted in lattice units from the exponential decay of two-point functions $\big\langle\hat{\mathcal{O}}(t)\hat{\mathcal{O}}(0)\big\rangle\sim e^{-(E_1-E_0)\,at}$, the precise mechanism the spectroscopy of Section 43.7.2 relies on. This is the concrete content of the correspondence Eq. (43.11).

Positivity of the measure is not yet positivity of the Hilbert space

Theorem 43.4 gave a positive, convergent measure, and Theorem 43.7 gave a positive operator T. These are logically distinct positivities, and the second does not follow from the first for an arbitrary action. A higher-derivative or improperly chosen temporal discretization can produce a perfectly convergent positive measure whose transfer matrix is *not* a positive operator—its "Hamiltonian" $-\frac{1}{a}\ln T$ would then have complex eigenvalues, and there would be no unitary continuation to real time. What guarantees positivity of T for the nearest-neighbour scalar is the structure we used in the proof—the temporal coupling appears only through the Gaussian heat kernel $\mathcal{K}$—and this structural fact is precisely reflection positivity, to which we now turn.

43.1.5 Reflection positivity and the unitary continuation to Minkowski

Theorem 43.7 built a Hamiltonian by assuming the slicing and computing. We now connect that computation to the criterion that Theorem 10.32 identified as the general guarantee of a sensible Lorentzian theory: reflection (Osterwalder–Schrader) positivity. For the lattice the criterion takes a sharp, checkable form, and we prove it for the scalar. The payoff is conceptual closure: positivity of T, hence the existence of a Hermitian H with real spectrum, hence a unitary real-time evolution e^{-iHt}, all follow from a single inequality satisfied by the lattice action.

Fix a reflection plane. The cleanest choice is *link reflection* about the hyperplane lying halfway between time slices $t = 0$ and $t = 1$; we work on the temporally infinite lattice ($t \in \mathbb{Z}$, Λ_s finite) so that the reflection is exact, and remark afterward on the periodic case. The reflection Θ acts on configurations by exchanging slices across the plane,

$$(\Theta\phi)_{(t,\mathbf{n})} = \phi_{(1-t,\,\mathbf{n})}, \tag{43.25}$$

so that slice $t = 1$ maps to $t = 0$, slice $t = 2$ to $t = -1$, and so on. Call a functional $F[\phi]$ *supported at positive time* if it depends on ϕ only through the slices $t \geq 1$; denote by $\mathcal{A}_+$ the algebra of such functionals. Reflection acts on functionals antilinearly by $(\Theta F)[\phi] := \overline{F[\Theta\phi]}$, mapping $\mathcal{A}_+$ to functionals of the slices $t \leq 0$.

Definition 43.8 (Reflection positivity)**.** The lattice action S_E is *reflection positive* (about the chosen plane) if

$$\langle (\Theta F)\, F \rangle \geq 0 \qquad \text{for every } F \in \mathcal{A}_+. \tag{43.26}$$

This is the lattice transcription of the abstract inequality Eq. (10.180): the Euclidean expectation of a positive-time functional paired with its reflection is non-negative. Its meaning, unpacked at length in Section 10.6.6, is that the reconstructed inner product $\langle F, F' \rangle := \langle (\Theta F) F' \rangle$ is positive semidefinite, so the quotient by its null space is a genuine Hilbert space with no negative-norm states. We now prove the lattice scalar satisfies Eq. (43.26), and that this is equivalent to the positivity of T established above.

Theorem 43.9 (Reflection positivity of the lattice scalar)**.** *The lattice ϕ^4 action Eq. (43.5) with $m^2 > 0$, $\lambda \geq 0$ is reflection positive about any temporal link-reflection plane. Equivalently, the transfer matrix T of Theorem 43.5 is a positive operator. Consequently the Osterwalder–Schrader reconstruction (Theorem 10.32) applies: the lattice data define a Hilbert space of positive norm, a self-adjoint Hamiltonian $H \geq 0$, and a unitary real-time evolution e^{-iHt} obtained by continuing the lattice (imaginary-time) correlators back to Minkowski signature.*

Proof. Decompose the action with respect to the reflection plane. Because the only term in Eq. (43.5) that couples a slice to its temporal neighbour is the temporal hopping term, and the plane sits between slices 0 and 1, the action splits as

$$S_E[\phi] = S_+[\phi_+] \; + \; S_-[\phi_-] \; + \; S_0, \qquad S_0 = \frac{a^2}{2} \sum_{\mathbf{n}} (\phi_{1,\mathbf{n}} - \phi_{0,\mathbf{n}})^2, \tag{43.27}$$

where ϕ_+ denotes the fields on slices $t \geq 1$, ϕ_- those on $t \leq 0$, $S_\pm$ collect all terms supported entirely on one side (every slice potential and every temporal link strictly to one side of the plane), and S_0 is the *single* crossing link. The quartic, mass, and spatial-gradient terms are intrinsic to slices and therefore split cleanly into S_+ and S_-; this is what makes the argument work for the interacting theory exactly as for the free one. Reflection symmetry of the action gives $S_-[\phi_-] = S_+[\Theta\phi_+]$, i.e. $S_- = \Theta S_+$.

The crossing weight e^{-S_0} is Gaussian in the difference $\phi_{1,\mathbf{n}} - \phi_{0,\mathbf{n}}$, and we represent it as a manifestly positive pairing of a positive-time factor with its reflection by Fourier transform. For each spatial site, $e^{-\frac{a^2}{2}x^2} = \frac{1}{a\sqrt{2\pi}} \int_{\mathbb{R}} \mathrm{d}k \; e^{-k^2/2a^2} \, e^{ikx}$; with $x = \phi_{1,\mathbf{n}} - \phi_{0,\mathbf{n}}$ and a product over $\mathbf{n}$,

$$e^{-S_0} = \int \mathrm{d}\mu(\mathbf{k}) \, \Big(\prod_{\mathbf{n}} e^{ik_{\mathbf{n}} \phi_{1,\mathbf{n}}} \Big) \Big(\prod_{\mathbf{n}} e^{-ik_{\mathbf{n}} \phi_{0,\mathbf{n}}} \Big), \qquad \mathrm{d}\mu(\mathbf{k}) = \prod_{\mathbf{n}} \frac{\mathrm{d}k_{\mathbf{n}}}{a\sqrt{2\pi}} \, e^{-k_{\mathbf{n}}^2/2a^2}, \tag{43.28}$$

where $\mathrm{d}\mu$ is a *positive* measure on the auxiliary variables $\mathbf{k} = \{k_{\mathbf{n}}\}$. The first product, $h_{\mathbf{k}}[\phi_+] := \prod_{\mathbf{n}} e^{ik_{\mathbf{n}}\phi_{1,\mathbf{n}}}$, depends only on the slice-1 field and so belongs to $\mathcal{A}_+$; the second product equals $\overline{(\Theta h_{\mathbf{k}})[\phi_-]}$ because Θ exchanges slice 1 with slice 0 and Θ conjugates. Thus the only coupling between the two halves has been written as $\int \mathrm{d}\mu(\mathbf{k}) \, h_{\mathbf{k}}[\phi_+] \, \overline{(\Theta h_{\mathbf{k}})[\phi_-]}$ with $\mathrm{d}\mu \geq 0$.

Now compute the reflected expectation for any $F \in \mathcal{A}_+$. With the field measure factorizing as $\mathcal{D}\phi = \mathcal{D}\phi_+ \, \mathcal{D}\phi_-$ (the slices on the two sides are independent variables) and using $(\Theta F)[\phi] = \overline{F[\Theta\phi]}$, which depends only on ϕ_-,

$$\begin{aligned} Z \, \langle (\Theta F) F \rangle &= \int \mathcal{D}\phi_+ \, \mathcal{D}\phi_- \; \overline{F[\Theta\phi]} \, F[\phi] \; e^{-S_+[\phi_+]} \, e^{-S_-[\phi_-]} \, e^{-S_0} \\ &= \int \mathrm{d}\mu(\mathbf{k}) \int \mathcal{D}\phi_+ \, \mathcal{D}\phi_- \; \overline{F[\Theta\phi]} \, \overline{(\Theta h_{\mathbf{k}})[\phi_-]} \, e^{-\Theta S_+[\phi_-]} \;\cdot\; F[\phi] \, h_{\mathbf{k}}[\phi_+] \, e^{-S_+[\phi_+]}. \end{aligned} \tag{43.29}$$

The integrand factorizes into a piece depending only on ϕ_- and a piece depending only on ϕ_+, so the double integral is a product. Define the single functional integral over positive-time fields

$$I(\mathbf{k}) := \int \mathcal{D}\phi_+ \; F[\phi_+] \, h_{\mathbf{k}}[\phi_+] \, e^{-S_+[\phi_+]}. \tag{43.30}$$

The ϕ_- integral is the complex conjugate of the same object: substituting $\phi_- = \Theta\phi'_+$ (a measure-preserving change of variables, since Θ permutes slices) turns $\overline{F[\Theta\phi]} \to \overline{F[\phi'_+]}$, $\overline{(\Theta h_{\mathbf{k}})[\phi_-]} \to \overline{h_{\mathbf{k}}[\phi'_+]}$, and $e^{-\Theta S_+[\phi_-]} \to e^{-S_+[\phi'_+]}$, giving $\int \mathcal{D}\phi_-(\dots) = \overline{I(\mathbf{k})}$. Therefore

$$\langle (\Theta F) F \rangle = \frac{1}{Z} \int \mathrm{d}\mu(\mathbf{k}) \; \overline{I(\mathbf{k})} \, I(\mathbf{k}) = \frac{1}{Z} \int \mathrm{d}\mu(\mathbf{k}) \; |I(\mathbf{k})|^2 \;\geq\; 0, \tag{43.31}$$

because $\mathrm{d}\mu \geq 0$, $Z > 0$, and the integrand is a non-negative real number. This is Eq. (43.26); the action is reflection positive.

Equivalence with positivity of T. The same slicing that produced Eq. (43.31) is the slicing of Section 43.1.4: the crossing Gaussian e^{-S_0} is precisely the kernel $\mathcal{K}$ of Eq. (43.20) (up to the slice-potential factors absorbed into $S_\pm$), and the Fourier representation Eq. (43.28) is the spectral representation Eq. (43.23) of $K = e^{-\hat{\mathbf{p}}^2/2a^2}$ as a positive operator. A monomial positive-time functional, evaluated through Theorem 43.6, gives $\langle (\Theta F) F \rangle = \langle \Omega | \, \hat{F}^\dagger \, T^m \, \hat{F} \, | \Omega \rangle / \langle \Omega | T^N | \Omega \rangle$ for appropriate insertions, and the right side is ≥ 0 for all F iff $T \geq 0$. Thus reflection positivity Eq. (43.26) and positivity of the transfer matrix (Theorem 43.7) are the same statement viewed two ways; we have now established it from both ends.

Continuation to Minkowski. Given Eq. (43.26), the reconstruction theorem Theorem 10.32 produces a positive-norm Hilbert space (the completion of $\mathcal{A}_+$ in the inner product $\langle F, F' \rangle = \langle (\Theta F) F' \rangle$, modulo null vectors), a vacuum (the constant functional), and a self-adjoint Hamiltonian $H \geq 0$ generating Euclidean time translation—which is exactly the H of Eq. (43.22). Because H is self-adjoint with real, bounded-below spectrum, e^{-iHt} is a one-parameter unitary group: the lattice (imaginary-time) correlators, continued in the time variable $at \to it_{\mathrm{Mink}}$, are the Wightman functions of a unitary Lorentzian quantum field theory. The Wick rotation of Chapter 10 has been run in reverse, rigorously, on a fully defined object. □

We pause on what has been achieved, because it is the justification for the entire chapter. Starting from nothing but an ordinary finite-dimensional integral—a positive measure on $\mathbb{R}^V$ with no perturbative expansion in sight—we have reconstructed every structural ingredient of a quantum field theory: a Hilbert space of positive norm, a Hermitian Hamiltonian bounded below, a unique vacuum, a particle spectrum encoded in the decay of correlations, and a unitary evolution back in Minkowski signature. The lattice is therefore not merely a regulator that happens to be finite; it is a self-contained, non-perturbative *definition* of an interacting quantum field theory that satisfies the axioms a quantum theory must satisfy. This is the precise sense in which the lattice answers the question "what is a quantum field theory, exactly?"

43.1.6 Worked example: the lattice scalar in zero space dimensions is the harmonic oscillator

The relation $T = e^{-aH}$ is abstract until one sees the Hamiltonian come out by hand. The cleanest case is the lattice scalar with no spatial directions at all—a single degree of freedom ϕ_t on a one-dimensional time lattice. This is the lattice path integral for ordinary quantum mechanics, and its transfer matrix is the imaginary-time propagator. Carrying it out makes Theorem 43.7 completely explicit and shows the continuum Hamiltonian, with its spectrum, emerging from the lattice as $a \to 0$.

Worked Example 43.10 (Transfer matrix of the $(0+1)$-dimensional free scalar)**.** Take $\Lambda = \{0, 1, \dots, N-1\}$ with one real variable ϕ_t per site, free ($\lambda = 0$), mass m. In one dimension the discretization rule Eq. (43.4) carries $a^4 \to a$, and the action Eq. (43.5) reduces to

$$S_E[\phi] = \sum_t \Big[\frac{1}{2a}(\phi_{t+1} - \phi_t)^2 + \frac{a}{2} m^2 \phi_t^2 \Big]. \tag{43.32}$$

There is no slice potential beyond the on-site mass term, so $W(\phi) = \frac{1}{2}m^2\phi^2$, and the transfer kernel Eq. (43.20) on $\mathcal{H} = L^2(\mathbb{R}, \mathrm{d}\phi)$ is

$$\langle \phi' | T | \phi \rangle = \frac{1}{\sqrt{2\pi a}} \exp\Big[-\frac{am^2}{4}\phi'^2 - \frac{1}{2a}(\phi' - \phi)^2 - \frac{am^2}{4}\phi^2 \Big]. \tag{43.33}$$

We extract H from $T = e^{-aH}$ in the small-a regime, then verify the spectrum. By the heat-kernel identity Eq. (43.23) the Gaussian middle factor is the kernel of $e^{-\frac{a}{2}\hat{p}^2}$ with $\hat{p} = -i\,\mathrm{d}/\mathrm{d}\phi$, and the two outer factors are multiplication by $e^{-\frac{a}{4}m^2\hat{\phi}^2}$ with $\hat{\phi}$ the position operator. Hence T is the operator

$$T = e^{-\frac{a}{4}m^2\hat{\phi}^2}\; e^{-\frac{a}{2}\hat{p}^2}\; e^{-\frac{a}{4}m^2\hat{\phi}^2}, \tag{43.34}$$

a symmetric (Strang) splitting of the evolution. This factored form is manifestly the structure $T = AKA$ of Theorem 43.7, with $A = e^{-\frac{a}{4}m^2\hat{\phi}^2} > 0$ and $K =$

$e^{-\frac{a}{2}\hat{p}^2} > 0$; positivity and self-adjointness are visible by inspection. To read off H, use the Baker–Campbell–Hausdorff formula for a symmetric product: for operators X, Y,

$$e^{X/2}e^{Y}e^{X/2} = \exp\Big(X+Y-\tfrac{1}{24}\,[X,[X,Y]]+\tfrac{1}{12}\,[Y,[X,Y]]+\cdots\Big) = \exp(X+Y+O(\text{triple commutato} \tag{43.35}$$

where the leading correction to $X+Y$ is of *third* order in the small quantities—the symmetric ordering cancels the second-order commutator $\frac{1}{2}\,[X,Y]$ that spoils the naive product $e^{X}e^{Y}$. With $X = -\frac{a}{2}m^2\hat{\phi}^2$ and $Y = -\frac{a}{2}\hat{p}^2$, both of order a, we obtain

$$T = \exp\Big[-a\big(\tfrac{1}{2}\hat{p}^2 + \tfrac{1}{2}m^2\hat{\phi}^2\big) + O(a^3)\Big] = e^{-aH}, \qquad H = \frac{1}{2}\hat{p}^2 + \frac{1}{2}m^2\hat{\phi}^2 + O(a^2). \tag{43.36}$$

The reconstructed Hamiltonian is the harmonic oscillator of angular frequency $\omega = m$. Its spectrum, from the ladder operators of nonrelativistic quantum mechanics, is $E_n = m\big(n+\frac{1}{2}\big)$, $n = 0, 1, 2, \ldots$, so the transfer-matrix eigenvalues are $\tau_n = e^{-aE_n} = e^{-am(n+1/2)}$, with the largest, $\tau_0 = e^{-am/2}$, simple and belonging to the strictly positive Gaussian ground state $\psi_0(\phi) \propto e^{-m\phi^2/2}$—exactly the unique positive vacuum guaranteed in general by Perron–Frobenius in Theorem 43.7.

The lattice corrections are now explicit and instructive. The exact transfer matrix Eq. (43.33) can be diagonalized in closed form (it is a Mehler kernel), giving $\tau_n = R^{\,n+1/2}$ with R fixed by the a-dependent coefficients; expanding, the gap is

$$a\,(E_1 - E_0) = -\ln\frac{\tau_1}{\tau_0} = \operatorname{arccosh}\Big(1+\tfrac{1}{2}a^2m^2\Big) = a\,m\Big(1-\tfrac{1}{24}a^2m^2 + O(a^4m^4)\Big), \tag{43.37}$$

so the physical mass measured from the decay of the lattice two-point function, $E_1 - E_0 = m(1+O(a^2m^2))$, equals the input m up to a discretization error that vanishes as $a \to 0$. The correlation length in lattice units is $\xi/a = 1/[a(E_1 - E_0)] = 1/(am) + O(am)$, diverging as $a \to 0$ at fixed physical m—the first concrete instance of the criterion Eq. (43.11) that the next section elevates to the organizing principle of the continuum limit.

This single-site example is the whole reconstruction in a form one can hold in the hand: a positive measure on $\mathbb{R}^N$, a positive transfer matrix, a Hermitian Hamiltonian read off by Baker–Campbell–Hausdorff, an exact spectrum, and a continuum limit reached by sending the lattice correlation length to infinity. Adding spatial sites changes none of the logic; it only replaces the oscillator H by the field Hamiltonian of Theorem 43.7 on the larger Hilbert space $L^2(\mathbb{R}^{V_s})$, and the Fock structure of Section 9.3 reappears in the continuum limit as the spectrum of that H organizes into multiparticle states.

We now have exactly what the rest of the chapter needs: a finite, positive, reflection-positive lattice measure e^{-S_E}/Z on the field configurations of Theorem 43.1, on which a scalar field is already in place and which reconstructs a Hilbert space and a Hamiltonian through the transfer matrix $T = e^{-aH}$. The next section places the free and $\lambda\phi^4$ scalar on this measure in earnest, recovers the continuum propagator as $a \to 0$, and confronts the central lesson that the discretization makes unavoidable—that the continuum limit is not $a \to 0$ at fixed couplings but the tuning of the statistical system to a critical point where the correlation length Eq. (43.11) diverges in lattice units, a second-order phase transition that is, in the language of Chapter 23, a Wilsonian renormalization-group fixed point.

43.2 The lattice scalar and the continuum limit as an RG fixed point

Section 43.1 turned the Euclidean path integral of Chapter 10 into a genuine, finite, positive measure: a real scalar field defined on the sites of a hypercubic lattice, with weight e^{-S_E}/Z over an ordinary (if astronomically high-dimensional) integral, and with a transfer matrix whose ground state and spectral gap reconstruct a physical Hilbert space and Hamiltonian. We now have a constructive, regulator-free *definition* of a quantum field theory and a dictionary back to physics. The obvious next question is whether this object actually contains the continuum theory we set out to define, and if so, how the continuum is reached. The answer is the conceptual heart of this chapter, and it is not the naive one. Putting $a \to 0$ at fixed bare couplings does *not* give the continuum theory; it gives nothing at all. The continuum limit is instead a limit in which the lattice is tuned to a critical point, where the correlation length measured in lattice spacings diverges, and that critical point is precisely a fixed point of the Wilsonian renormalization group of Chapter 23. We will establish this for the carried ϕ^4 theory scalar, recovering the continuum propagator with its leading lattice artifact along the way, and we will state honestly — and date — what is by now a theorem: in four dimensions the resulting continuum theory is free.

43.2.1 The lattice action for the free and $\lambda\phi^4$ scalar

We work on the Euclidean hypercubic lattice

$$\Lambda_a \;=\; a\,\mathbb{Z}^4 \;=\; \{\, x = a\,n \;:\; n = (n_1, n_2, n_3, n_4) \in \mathbb{Z}^4 \,\}, \tag{43.38}$$

with lattice spacing $a > 0$ and unit vectors $\hat{\mu}$ along the four axes, so that $x + a\hat{\mu}$ is the neighbour of x in direction μ. (We take the infinite lattice throughout this section; the finite periodic volume of Section 43.1 is recovered by restricting n to $\{0, \ldots, N-1\}^4$ and is what one actually simulates. All sums over x below mean

$\sum_{n\in\mathbb{Z}^4}$.) The dynamical variable is a real number $\phi(x) \equiv \phi_n$ attached to each site, the discretization of the continuum field $\phi(x)$ of ϕ^4 theory, Chapter 6.

A continuum derivative must become a finite difference. The two simplest choices are the *forward* and *backward* differences,

$$(\Delta_\mu^+\phi)(x) \;=\; \frac{\phi(x+a\hat{\mu})-\phi(x)}{a}, \qquad (\Delta_\mu^-\phi)(x) \;=\; \frac{\phi(x)-\phi(x-a\hat{\mu})}{a}. \tag{43.39}$$

Each reduces to $\partial_\mu\phi$ as $a\to 0$ with an error of order a. Their average, the symmetric difference $\frac{1}{2}(\Delta_\mu^+ + \Delta_\mu^-)\phi = (\phi(x+a\hat{\mu}) - \phi(x-a\hat{\mu}))/2a$, is accurate to order a^2 but, as we will see, has a defect of its own for the second derivative. For the kinetic term either one-sided difference will do, because the action symmetrizes the choice. The discretized Euclidean action of ϕ^4 theory, the lattice version of the Euclidean $S_E = \int d^4x\,[\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4]$ fixed for the thread in Chapter 10, is

$$S_E[\phi] \;=\; a^4 \sum_{x\in\Lambda_a} \left[\frac{1}{2}\sum_{\mu=1}^{4} \big((\Delta_\mu^+\phi)(x)\big)^2 \;+\; \frac{1}{2}\,m^2\,\phi(x)^2 \;+\; \frac{\lambda}{4!}\,\phi(x)^4 \right], \tag{43.40}$$

where the factor a^4 is the volume of one hypercube and turns the site sum into a Riemann approximation of $\int d^4x$. As $a\to 0$ with ϕ a smooth profile, Eq. (43.5) converges termwise to the continuum S_E; the content of the rest of this section is that this naive pointwise statement is *not* how the continuum theory is recovered.

It is convenient to expose the lattice Laplacian hiding in the kinetic term. The key tool is *summation by parts*, the lattice avatar of integration by parts. For any two fields f, g that decay at infinity (or on the periodic torus, with no boundary at all),

$$a^4 \sum_x (\Delta_\mu^+ f)(x)\, g(x) \;=\; -\,a^4 \sum_x f(x)\,(\Delta_\mu^- g)(x), \tag{43.41}$$

which follows from relabelling the shifted sum: writing out the left side,

$$a^4 \sum_x \frac{f(x+a\hat{\mu})-f(x)}{a}\, g(x) = \frac{a^4}{a}\Big[\sum_x f(x+a\hat{\mu})g(x) - \sum_x f(x)g(x)\Big] = \frac{a^4}{a}\sum_x f(x)\big[g(x-a\hat{\mu}) - g(x$$

where in the first sum we shifted the summation index $x \to x - a\hat{\mu}$ (legitimate because the sum is over all of Λ_a), and the bracket is $-a\,(\Delta_\mu^- g)(x)$, giving Eq. (43.41). Applying it to the kinetic term with $f = \phi$ and $g = \Delta_\mu^+\phi$,

$$a^4 \sum_x (\Delta_\mu^+\phi)^2 = -\,a^4 \sum_x \phi\,(\Delta_\mu^-\Delta_\mu^+\phi) = -\,a^4 \sum_x \phi\,(\Delta_\mu\phi), \tag{43.42}$$

where the second-difference operator $\Delta_\mu \equiv \Delta_\mu^-\Delta_\mu^+ = \Delta_\mu^+\Delta_\mu^-$ (the two one-sided differences commute) acts as

$$(\Delta_\mu\phi)(x) = \frac{\phi(x+a\hat{\mu}) + \phi(x-a\hat{\mu}) - 2\phi(x)}{a^2}. \tag{43.43}$$

Summing over μ defines the *lattice Laplacian*

$$(\Delta\phi)(x) \;=\; \sum_{\mu=1}^{4}(\Delta_\mu\phi)(x) \;=\; \frac{1}{a^2}\sum_{\mu=1}^{4}\big[\phi(x+a\hat{\mu})+\phi(x-a\hat{\mu})-2\phi(x)\big], \quad (43.44)$$

the nearest-neighbour "hopping" operator familiar from statistical mechanics: it couples each site to its $2\times 4=8$ neighbours and subtracts $8\phi(x)/a^2$. In terms of it the free part of the action is the manifestly quadratic form

$$S_E^{(0)}[\phi] \;=\; \frac{a^4}{2}\sum_x \phi(x)\,(-\Delta+m^2)\phi(x), \quad (43.45)$$

exactly the lattice transcription of $\frac{1}{2}\int d^4x\,\phi(-\partial^2+m^2)\phi$, with the continuum $-\partial^2$ replaced by the discrete $-\Delta$. The operator $-\Delta+m^2$ is positive (it is a sum of squares plus $m^2>0$), so the Gaussian integral built on it converges — the positivity that Section 43.1 required of the whole measure, here visible in the free sector.

43.2.2 The lattice free propagator

The operator $-\Delta+m^2$ has constant coefficients — it commutes with lattice translations — so it is diagonalized by plane waves, and the propagator follows by inversion in momentum space. This is the lattice instance of the momentum-space inversion that produced the continuum propagator in Chapter 10.

Because the lattice resolves no structure finer than a, momenta are defined only modulo $2\pi/a$: the plane waves $e^{ip\cdot x}$ and $e^{i(p+2\pi\hat{\mu}/a)\cdot x}$ coincide on every site $x=an$, since $e^{i(2\pi/a)\hat{\mu}\cdot an}=e^{2\pi i n_\mu}=1$. The independent momenta therefore fill one period, the *first Brillouin zone*

$$\mathrm{BZ} \;=\; \Big(-\tfrac{\pi}{a},\,\tfrac{\pi}{a}\Big]^4, \quad (43.46)$$

and the lattice Fourier transform and its inverse read

$$\tilde{\phi}(p)=a^4\sum_x e^{-ip\cdot x}\,\phi(x), \qquad \phi(x)=\int_{\mathrm{BZ}}\frac{d^4p}{(2\pi)^4}\,e^{ip\cdot x}\,\tilde{\phi}(p), \quad (43.47)$$

with the completeness relations $a^4\sum_x e^{-i(p-q)\cdot x}=(2\pi)^4\delta^{(4)}(p-q)$ for $p,q\in\mathrm{BZ}$ and $\int_{\mathrm{BZ}}\frac{d^4p}{(2\pi)^4}e^{ip\cdot(x-y)}=a^{-4}\delta_{xy}$. Acting on a plane wave the one-sided differences are diagonal,

$$\Delta_\mu^{\pm}\,e^{ip\cdot x}=\frac{e^{\pm iap_\mu}-1}{\pm a}\,e^{ip\cdot x}, \quad (43.48)$$

so the lattice Laplacian Eq. (43.44) has eigenvalue

$$-\Delta\,e^{ip\cdot x}=-\frac{1}{a^2}\sum_\mu\big(e^{iap_\mu}+e^{-iap_\mu}-2\big)\,e^{ip\cdot x}=\frac{2}{a^2}\sum_\mu\big(1-\cos ap_\mu\big)\,e^{ip\cdot x}=\hat{p}^2\,e^{ip\cdot x}, \quad (43.49)$$

where, using $1-\cos\theta = 2\sin^2(\theta/2)$, we have defined the *lattice momentum*

$$\boxed{\hat{p}_\mu \;=\; \frac{2}{a}\sin\!\Big(\frac{a p_\mu}{2}\Big), \qquad \hat{p}^2 \;=\; \sum_{\mu=1}^{4}\hat{p}_\mu^{\,2} \;=\; \frac{4}{a^2}\sum_{\mu=1}^{4}\sin^2\!\Big(\frac{a p_\mu}{2}\Big).} \tag{43.50}$$

The lattice Laplacian is thus exactly $-\partial^2$ with p_μ^2 replaced by $\hat{p}_\mu^2$. The quantity $\hat{p}_\mu$ is real, periodic in $p_\mu \to p_\mu + 2\pi/a$ as it must be, and vanishes not only at $p_\mu = 0$ but also at the zone boundary $p_\mu = \pi/a$ — a fact innocuous for the scalar but, as Section 43.1's successor sections will exploit, fatal for naive fermions, where it is the origin of doubling.

To get the propagator, diagonalize the free action. Inserting Eq. (43.47) into Eq. (43.45) and using Eq. (43.49) with the completeness relation,

$$S_E^{(0)}[\phi] = \frac{1}{2}\int_{\mathrm{BZ}} \frac{d^4p}{(2\pi)^4}\,(\hat{p}^2+m^2)\,\tilde{\phi}(-p)\,\tilde{\phi}(p), \qquad \tilde{\phi}(-p) = \tilde{\phi}(p)^*, \tag{43.51}$$

the last equality holding because $\phi(x)$ is real. The action is a sum of decoupled Gaussians, one per momentum mode, with stiffness $\hat{p}^2+m^2$. Coupling to a source J through $a^4\sum_x J\phi = \int_{\mathrm{BZ}}\frac{d^4p}{(2\pi)^4}\tilde{J}(-p)\tilde{\phi}(p)$ and completing the square mode by mode — the well-defined finite-dimensional Gaussian integral guaranteed by Section 43.1, and the exact lattice replica of the continuum manipulation in Chapter 10 — gives the connected two-point function as the inverse stiffness. We record the result as a proposition.

Proposition 43.11 (Lattice free propagator)**.** *For the free lattice scalar with action Eq. (43.45) on Λ_a, the two-point function is, in momentum and in position space,*

$$\tilde{G}(p) \;=\; \frac{1}{\hat{p}^2+m^2}, \qquad G(x-y) \;=\; \langle\phi(x)\phi(y)\rangle \;=\; \int_{\mathrm{BZ}}\frac{d^4p}{(2\pi)^4}\,\frac{e^{ip\cdot(x-y)}}{\hat{p}^2+m^2}, \tag{43.52}$$

with $\hat{p}^2$ given by Eq. (43.50). The denominator obeys $\hat{p}^2+m^2 \geq m^2 > 0$ for all $p \in \mathrm{BZ}$, so G is finite with no $i\epsilon$ prescription and no ultraviolet divergence: the lattice has cut the momentum integration off at the zone boundary.

Proof. Completing the square in Eq. (43.51) shifts $\tilde{\phi}(p) \to \tilde{\phi}(p) + \tilde{J}(p)/(\hat{p}^2+m^2)$, a translation of the integration variable with unit Jacobian, leaving

$$\ln Z[J] - \ln Z[0] = \frac{1}{2}\int_{\mathrm{BZ}}\frac{d^4p}{(2\pi)^4}\,\frac{\tilde{J}(-p)\,\tilde{J}(p)}{\hat{p}^2+m^2}.$$

Two functional derivatives with respect to the source, evaluated at $J=0$, extract $\tilde{G}(p) = 1/(\hat{p}^2+m^2)$; transforming back with Eq. (43.47) gives the position-space form. Positivity of the denominator is immediate from $\hat{p}^2 \geq 0$ and $m^2 > 0$, and the integration region is the bounded set Eq. (43.46), so the

integral converges absolutely. The Gaussian integral itself is finite-dimensional and absolutely convergent because $-\Delta + m^2$ is a positive operator, which is exactly the statement that the lattice measure of Section 43.1 restricted to the free sector is a normalizable Gaussian. □

This is the central object. The momentum integral is over a compact region and the integrand is bounded, so every quantity built from G — in particular the loop integrals of the interacting theory, where $\int_{\mathrm{BZ}} d^4p$ replaces $\int d^4p$ — is finite from the start. The lattice is a complete ultraviolet regulator, and unlike dimensional regularization or Pauli–Villars it is non-perturbative: it regulates the measure, not a diagram.

43.2.3 The continuum limit of the propagator and its $O(a^2)$ artifact

Fix a physical momentum p and let $a \to 0$. Then $ap_\mu \to 0$, so $\hat{p}_\mu = (2/a)\sin(ap_\mu/2) \to p_\mu$, the Brillouin zone Eq. (43.46) opens up to all of $\mathbb{R}^4$, and Eq. (43.52) becomes

$$G(x) \xrightarrow{a\to 0} \int_{\mathbb{R}^4} \frac{d^4p}{(2\pi)^4}\, \frac{e^{ip\cdot x}}{p^2+m^2}, \tag{43.53}$$

which is exactly the Euclidean (Schwinger) propagator of the free scalar, Eq. (10.168), the Wick rotation of the Feynman propagator Eq. (10.30). The naive continuum limit of the free two-point function is therefore correct: site-field discretization reproduces the right propagator. The interesting physics lives in the rate of approach, which we now extract.

Worked Example 43.12 (The lattice propagator to $O(a^2)$: the leading discretization error)**.** We expand $\hat{p}_\mu^2$ in powers of a and read off how the lattice propagator deviates from the continuum one. From the Taylor series $\sin\theta = \theta - \theta^3/6 + O(\theta^5)$ with $\theta = ap_\mu/2$,

$$\hat{p}_\mu = \frac{2}{a}\Big[\frac{ap_\mu}{2} - \frac{1}{6}\Big(\frac{ap_\mu}{2}\Big)^3 + O(a^5)\Big] = p_\mu - \frac{a^2}{24}\,p_\mu^3 + O(a^4),$$

and squaring (the cross term is $2\cdot p_\mu \cdot (-a^2 p_\mu^3/24) = -a^2 p_\mu^4/12$),

$$\hat{p}_\mu^2 = p_\mu^2 - \frac{a^2}{12}\,p_\mu^4 + O(a^4), \qquad \hat{p}^2 = \sum_\mu \hat{p}_\mu^2 = p^2 - \frac{a^2}{12}\sum_\mu p_\mu^4 + O(a^4), \tag{43.54}$$

where $p^2 = \sum_\mu p_\mu^2$. The leading correction $-\frac{a^2}{12}\sum_\mu p_\mu^4$ is suppressed by a^2 relative to p^2 — the forward/backward kinetic term is automatically $O(a^2)$-accurate, the $O(a)$ error of a single one-sided difference having cancelled in the symmetric

second difference Eq. (43.43). Inserting Eq. (43.54) into the propagator and expanding the geometric series,

$$\tilde{G}(p) = \frac{1}{\hat{p}^2 + m^2} = \frac{1}{(p^2 + m^2) - \frac{a^2}{12}\sum_\mu p_\mu^4 + O(a^4)} = \frac{1}{p^2 + m^2}\left[1 + \frac{a^2}{12}\frac{\sum_\mu p_\mu^4}{p^2 + m^2} + O(a^4)\right]. \tag{43.55}$$

The bracket exhibits the continuum propagator times 1 plus a fractional lattice artifact of order a^2p^2 (for $p \gg m$), exactly the behaviour plotted in Fig. 43.2.

The structure of the artifact carries a physical lesson. The continuum denominator $p^2 = \sum_\mu p_\mu^2$ is invariant under the Euclidean rotation group $O(4)$ (the rotation to Minkowski Lorentz invariance), but the correction $\sum_\mu p_\mu^4$ is *not*: it is invariant only under the hypercubic point group, the 48-element symmetry of the lattice cell that permutes and reflects the axes. The lattice breaks continuous rotations down to this finite subgroup, and the breaking shows up first at $O(a^2)$. This is the generic fate of a lattice observable: continuum (here $O(4)$) symmetry is recovered only as $a \to 0$, restored by the irrelevance — in the precise RG sense of Section 43.2.6 below — of the symmetry-breaking operators. One can engineer the leading $O(a^2)$ error away by adding further-neighbour terms to the action chosen to cancel Eq. (43.54) (Symanzik improvement, [463]); the resulting "improved" actions converge as $O(a^4)$ and are the workhorse of modern simulations. The point of principle, however, stands: the artifacts are there at finite a and vanish only in the limit.

43.2.4 Dimensionless variables: the lattice as a pure statistical system

A lattice simulation manipulates numbers, and a number cannot carry a dimension; the lattice spacing a is the only scale in sight and is itself just a label until something physical sets it. It is therefore essential to recast Eq. (43.5) in dimensionless variables, which also makes the upcoming criticality argument transparent. In four dimensions the canonical mass dimensions (in units $\hbar = c = 1$, with $[a] = -1$) are $[\phi] = 1$, $[m] = 1$, $[\lambda] = 0$. Define the dimensionless field, mass, and coupling

$$\hat{\phi}(n) \;=\; a\,\phi(an), \qquad \hat{m} \;=\; a\,m, \qquad \hat{\lambda} \;=\; \lambda, \tag{43.56}$$

(the quartic coupling is already dimensionless in $d = 4$, the marginality that will matter for triviality). Substituting $\phi(an) = \hat{\phi}(n)/a$ into Eq. (43.5), every explicit a cancels: the gradient term carries $a^4 \cdot a^{-2} \cdot a^{-2} = a^0$, the mass term $a^4 \cdot m^2 \cdot a^{-2} = (am)^2 = \hat{m}^2$, the quartic $a^4 \cdot a^{-4} = a^0$. The result is a purely

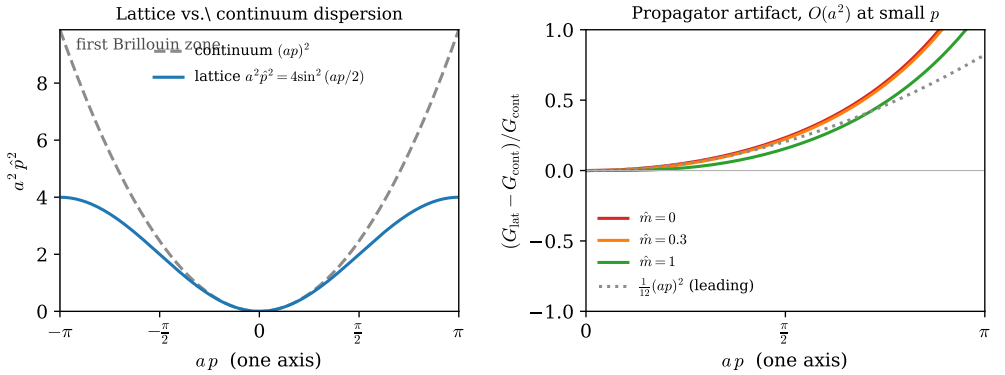


Figure 43.2: The lattice scalar at finite spacing. *Left:* the dispersion of the lattice Laplacian, $a^2\hat{p}^2 = 4\sin^2(ap/2)$ along one axis (solid), against the continuum $(ap)^2$ (dashed), over the first Brillouin zone $ap \in (-\pi, \pi]$. The two agree near $p = 0$ and separate toward the zone boundary; $\hat{p}$ even bends back to zero at $p = \pi/a$. *Right:* the resulting fractional error of the free propagator, $(G_{\rm lat} - G_{\rm cont})/G_{\rm cont}$, versus ap for several dimensionless masses $\hat{m} = am$, with the leading $\frac{1}{12}(ap)^2$ envelope (dotted). The artifact is $O(a^2)$ at small momentum and grows where the lattice can no longer resolve the wave.

numerical action over the integer lattice,

$$S_E[\hat{\phi}] \;=\; \sum_{n\in\mathbb{Z}^4} \left[\frac{1}{2} \sum_{\mu} \big(\hat{\phi}(n+\hat{\mu}) - \hat{\phi}(n)\big)^2 + \frac{1}{2}\,\hat{m}^2\,\hat{\phi}(n)^2 + \frac{\hat{\lambda}}{4!}\,\hat{\phi}(n)^4 \right]. \qquad (43.57)$$

This is the whole content of the regularized theory: a classical statistical-mechanics model of a real spin $\hat{\phi}(n)$ at each site of $\mathbb{Z}^4$, with nearest-neighbour ferromagnetic coupling and an on-site double-well-or-single-well potential, whose Boltzmann weight is $e^{-S_E[\hat{\phi}]}$. It depends on exactly two dimensionless parameters, $(\hat{m}^2, \hat{\lambda})$. (The traditional parametrization rescales $\hat{\phi}$ to make the hopping coefficient unity, trading $(\hat{m}^2, \hat{\lambda})$ for a *hopping parameter* κ and a quartic u; this is a change of variables, not of content, and we keep $(\hat{m}^2, \hat{\lambda})$.) The lattice scalar field theory *is* this statistical model. Every physical prediction must be a function of the two numbers $(\hat{m}^2, \hat{\lambda})$ alone — and this is precisely what forces the continuum limit to be a critical point.

43.2.5 Correlation length, the mass gap, and why the continuum limit is critical

The transfer matrix of Section 43.1 expresses the Euclidean two-point function as a sum over its eigenvalues; the leading large-separation behaviour is governed by the spectral gap. Taking the separation along a lattice axis (say direction 4,

"time"), with n_4 the number of time slices,

$$\left\langle \hat{\phi}(n)\,\hat{\phi}(0) \right\rangle \;\sim\; C\,e^{-\hat{m}\,n_4} \qquad (n_4 \to \infty), \tag{43.58}$$

where $\hat{m} = aE_1$ is the dimensionless energy gap (the mass gap of the reconstructed Hamiltonian, in lattice units), agreeing at leading order with the bare $\hat{m}$ for small coupling and renormalized by interactions in general. Equivalently, restoring physical length $x_4 = a\,n_4$, the correlator decays as $e^{-x_4/\xi_{\text{phys}}}$ with

$$\xi_{\text{phys}} \;=\; \frac{a}{\hat{m}} \;=\; \frac{1}{m_{\text{phys}}}, \qquad m_{\text{phys}} \;=\; \frac{\hat{m}}{a}, \tag{43.59}$$

where m_{phys} is the physical particle mass extracted from the exponential falloff and $\xi_{\text{phys}} = 1/m_{\text{phys}}$ is its Compton wavelength. The dimensionless number actually measured on the lattice is the correlation length *in lattice units*,

$$\hat{\xi} \;=\; \frac{\xi_{\text{phys}}}{a} \;=\; \frac{1}{\hat{m}}, \tag{43.60}$$

the number of lattice spacings over which correlations survive. It is a pure function of $(\hat{m}^2, \hat{\lambda})$ — consistent with Section 43.2.4, since it is dimensionless — and it is the bridge between the statistical model and continuum physics.

Proposition 43.13 (A finite physical mass forces tuning to criticality)**.** *Fix a target physical mass $m_{\text{phys}} > 0$. A continuum limit, defined as $a \to 0$ at fixed m_{phys}, exists only if the bare lattice parameters $(\hat{m}^2, \hat{\lambda})$ are tuned, as $a \to 0$, toward a point where the lattice correlation length diverges,*

$$\hat{\xi} \;=\; \frac{1}{a\,m_{\text{phys}}} \;\xrightarrow{a\to 0}\; \infty. \tag{43.61}$$

A divergent $\hat{\xi}$ is the defining feature of a second-order (continuous) phase transition: the continuum limit of the lattice scalar exists only at a critical point *of the statistical model Eq. (43.57).*

Proof. By Eq. (43.59), $a = \hat{\xi}^{-1}/m_{\text{phys}}$, so at fixed m_{phys} the lattice spacing in physical units is controlled entirely by $\hat{\xi}$: shrinking a is the same as growing $\hat{\xi}$, and $a \to 0 \iff \hat{\xi} \to \infty$. Since $\hat{\xi} = \hat{\xi}(\hat{m}^2, \hat{\lambda})$ is a function of the bare parameters alone (Section 43.2.4), driving $\hat{\xi} \to \infty$ requires moving $(\hat{m}^2, \hat{\lambda})$ to the locus where this function diverges. In the language of statistical mechanics a diverging correlation length is precisely the signature of a continuous phase transition — at a first-order transition $\hat{\xi}$ stays finite, while at a critical point it diverges as a power of the distance to criticality. Hence the bare couplings must be tuned to the critical surface; there is the continuum limit, and nowhere else.

The contrapositive is the more striking statement. Suppose instead one holds the bare couplings $(\hat{m}^2, \hat{\lambda})$ *fixed* (away from criticality) and lets $a \to 0$ — the

"naive" continuum limit. Then $\hat{\xi}$ is a fixed finite number, and Eq. (43.59) gives $\xi_{\text{phys}} = a\,\hat{\xi} \to 0$: the physical Compton wavelength shrinks to zero, equivalently $m_{\text{phys}} = \hat{m}/a \to \infty$. Every particle becomes infinitely heavy in physical units, every correlation collapses to a single site, and the limit is empty — it describes ultra-local, non-interacting fluctuations with no propagation, not a field theory. The naive limit does not fail to converge; it converges to nothing. This is the precise sense in which "the continuum limit is not $a \to 0$ at fixed couplings." □

Why the lattice spacing has no absolute meaning

The lattice spacing a never appears in the dimensionless action Eq. (43.57); it enters physics only through Eq. (43.59), which *defines* a in physical units once a measured mass is matched to its lattice value: $a = \hat{m}/m_{\text{phys}}$. Choosing a smaller a is not a separate operation from tuning the bare couplings closer to criticality — it *is* that tuning, read through the correlation length. "Approaching the continuum" and "approaching the critical point" are two descriptions of one motion in the space of bare couplings.

43.2.6 The continuum limit as a Wilsonian RG fixed point

Theorem 43.13 located the continuum limit at a critical point. The renormalization group of Chapter 23 now tells us what that critical point *is* and organizes the approach to it. Recall the Wilsonian picture: integrating out lattice fluctuations between scale a and a coarser scale ba ($b > 1$) and rescaling back induces a flow on the space of (dimensionless) couplings, $\hat{g} \mapsto R_b(\hat{g})$, with fixed points $\hat{g}_* = R_b(\hat{g}_*)$ (Theorem 23.19, Section 23.5). The correlation length transforms simply under a blocking by factor b: lengths measured in the coarsened lattice units are b times smaller, so $\hat{\xi} \mapsto \hat{\xi}/b$. A fixed point, being invariant under blocking, must therefore have $\hat{\xi} = \hat{\xi}/b$, which for $b > 1$ forces

$$\hat{\xi}_* \in \{0, \infty\}. \tag{43.62}$$

The trivial alternative $\hat{\xi} = 0$ describes a completely disordered or completely ordered sink with no fluctuations on any scale; the physically interesting fixed points have $\hat{\xi} = \infty$. These are exactly the critical points of Theorem 43.13. *A critical point of the lattice statistical model is a fixed point of the RG, and the continuum limit of the field theory lives there.* This is the identification promised in the section title, and it converts the construction of a continuum QFT into the study of the RG flow near a fixed point.

The flow near the fixed point is controlled by its linearization, classified in Theorem 23.23 into relevant ($y_i > 0$), marginal ($y_i = 0$), and irrelevant ($y_i < 0$) directions according to the eigenvalues of the linearized blocking transformation, with scaling dimensions $\Delta_i = d - y_i$ (Theorem 23.22). The implications for the lattice scalar are immediate and physical:

- The *relevant* directions are the couplings that must be *tuned* to reach the critical surface, because a generic point flows away from the fixed point along them. For the scalar there is one relevant coupling, the mass parameter $\hat{m}^2$ (scaling dimension 2, hence $y = d - 2 = 2 > 0$ in $d = 4$): the statement that $\hat{m}^2$ must be tuned to a critical value $\hat{m}_c^2(\hat{\lambda})$ to make $\hat{\xi}$ diverge is the RG statement that the relevant coupling must be set to its fixed-point value. This single tuning is the lattice incarnation of mass renormalization.

- The *irrelevant* directions die out under coarsening; they are the operators that the continuum limit forgets. The $O(a^2)$ lattice artifacts of Theorem 43.12 — the hypercubic-symmetry-breaking term $\sum_\mu p_\mu^4$, the higher-derivative and further-neighbour operators generated by blocking — are all irrelevant. Their decay under the flow is exactly why those artifacts vanish as $a \to 0$ and why full $O(4)$ (Lorentz) symmetry, broken at finite a, is restored in the continuum. It is also the origin of *universality*: lattice actions differing by irrelevant terms (a different difference scheme, an added Symanzik term, a different short-distance regulator entirely) flow to the same fixed point and yield the same continuum theory.

- The *marginal* coupling is $\hat{\lambda}$ ($\Delta = 4$, $y = 0$ in $d = 4$), whose fate is decided not at tree level but by loops. This is the case that decides whether ϕ_4^4 has any nontrivial continuum limit at all.

The practical face of this analysis is the notion of a *line of constant physics*. To take the continuum limit one does not pick bare couplings once; one moves along a trajectory $(\hat{m}^2(a), \hat{\lambda}(a))$ in bare-coupling space, indexed by a, chosen so that a fixed set of physical quantities (a mass ratio, say $m_{\text{phys}}\,\xi_{\text{phys}'}$, or a renormalized coupling defined at a physical scale) stays constant while $a \to 0$ and $\hat{\xi} \to \infty$. By construction this trajectory is a renormalized-trajectory of the RG flowing into the fixed point along its relevant direction; the irrelevant couplings are dragged to their fixed-point values automatically. The lattice thereby *realizes*, non-perturbatively and concretely, the abstract renormalized trajectory of Chapter 23: "renormalizing" a lattice theory means following a line of constant physics toward criticality.

The continuum limit, in one sentence

Defining a continuum quantum field theory *is* finding a fixed point of the renormalization group of a regularized statistical system and approaching it along a line of constant physics; the lattice makes both the system and the approach completely explicit, turning the question "does this QFT exist?" into the question "does the underlying statistical model have a suitable second-order critical point?".

43.2.7 The fate of ϕ^4 in four dimensions: triviality, stated and dated

We can now complete ϕ^4 theory at the non-perturbative layer. The question Section 43.2.6 leaves open is which fixed point the four-dimensional scalar flows to. There is always the *Gaussian fixed point* $\hat{\lambda}_* = 0$ (Theorem 23.20), the free theory, sitting at the origin of coupling space with $\hat{\xi} = \infty$ (a free massless scalar is scale invariant). The marginal coupling $\hat{\lambda}$ decides whether any *other*, interacting, fixed point exists with a continuum limit. Marginality is broken at one loop by the ϕ^4 beta function computed in ϕ^4 theory,

$$\beta_{\hat{\lambda}} \;=\; \frac{3\hat{\lambda}^2}{16\pi^2} + O(\hat{\lambda}^3) \;>\; 0 \tag{43.63}$$

(Eq. (24.40), Section 24.2). The sign is positive: $\hat{\lambda}$ is *marginally irrelevant* — it grows toward the ultraviolet and shrinks toward the infrared. Integrating Eq. (43.63) with the bare coupling $\hat{\lambda}$ held fixed at the lattice cutoff $1/a$, the renormalized coupling λ_R at a fixed physical scale μ behaves, as $a \to 0$, as

$$\lambda_R(\mu) \;\approx\; \frac{16\pi^2}{3\,\ln(1/a\mu)} \;\xrightarrow{a\to 0}\; 0, \tag{43.64}$$

so holding the bare coupling fixed and sending the cutoff up drives the renormalized coupling to zero (the Landau pole / triviality story of Section 24.7.2, Theorem 24.29, read here through the cutoff $1/a$). The complementary reading runs the other way: holding λ_R fixed at μ while $a \to 0$ would force the bare coupling $\hat{\lambda}(1/a)$ to *grow*, and Eq. (43.63) sends it through a Landau pole at a finite a — so that route reaches no continuum limit at all. In RG language: in $d = 4$ the only fixed point the flow of Eq. (43.57) can reach with $\hat{\xi} = \infty$ is the Gaussian one, and any continuum limit built on it is non-interacting. An interacting nontrivial fixed point does exist, but only away from four dimensions — the Wilson–Fisher fixed point in $d = 4 - \epsilon$, where $\hat{\lambda}$ becomes relevant and is balanced by the loop term at $\hat{\lambda}_* = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2)$ (Section 24.7.5, ϕ^4 theory's eventual handoff to the conformal bootstrap). In strictly four dimensions, there is no such balance.

This is the content of the *triviality* of ϕ^4_4, and it deserves to be stated as a result of the lattice program rather than a perturbative extrapolation, because that is now what it is.

ϕ^4_4 triviality: status as of the mid-2020s

The perturbative argument above (a positive $\beta_{\hat{\lambda}}$, a Landau pole, a vanishing renormalized coupling as the cutoff is removed) is suggestive but not a proof: it could in principle be overturned by strong-coupling behaviour beyond the reach of Eq. (43.63). The non-perturbative evidence comes

from the lattice formulation of this very section. For dimensions $d > 4$ the triviality of the continuum limit of lattice ϕ^4 (and of the closely related Ising model) was proved rigorously by Aizenman and by Fröhlich around 1981–82 [10, 202], using correlation inequalities directly on the statistical model Eq. (43.57). The borderline and physically decisive case $d = 4$ — the marginal case, where the logarithms of Eq. (43.64) make the analysis delicate — resisted until the 2021 theorem of Aizenman and Duminil-Copin [11], which established that the scaling limit of the critical four-dimensional ϕ^4 (and Ising) model is Gaussian. As of the mid-2020s, then, it is a *theorem* that $\lambda\phi^4$ in four dimensions has no interacting continuum limit of the kind sought here: tuned to its critical point and taken to $a \to 0$, it yields a free field.

The physical reading is the one this book has stressed since the renormalization-group chapters. A nonzero measured ϕ^4 coupling — the Higgs self-coupling, a scalar order parameter near a critical point — is not a sign of a fundamental ϕ^4 field valid to arbitrarily short distances. It is the low-energy shadow of a theory that must possess a finite ultraviolet cutoff $1/a$, below which $\hat{\lambda}$ is appreciable and above which new physics, or a different fixed point, takes over. ϕ^4 in four dimensions is a textbook *effective* field theory, complete and predictive at long distances and silent about its own ultraviolet completion — exactly the status Chapter 23 taught us to expect of a theory governed by an irrelevant interaction near the Gaussian fixed point. The lattice has turned that expectation into a proved structural fact about the carried example we have followed since Chapter 6.

The scalar has thus served its purpose. It fixed the framework — discretize the Euclidean action, read the propagator off the diagonalized lattice Laplacian, recover the continuum as a critical point that is an RG fixed point — and delivered, as ϕ^4 theory's non-perturbative stage, the honest verdict of triviality in four dimensions. But the construction leaned on one feature peculiar to a scalar: the field lives on *sites*, and there was never any question of a local internal symmetry to protect. The moment we try to discretize a gauge field $A^a_\mu(x)$ the same way — a vector attached to each site — local gauge invariance is destroyed, because a gauge transformation relates the field at *neighbouring* sites and a site-by-site variable has no way to compensate the difference across a link. Preserving exact gauge invariance on the lattice forces a genuinely different fundamental variable: not the algebra-valued A_μ on a site, but the group-valued parallel transporter on the link between two sites. That is the subject of the next section, where the link variable U_μ, the plaquette, and Wilson's gauge action are built and shown to reduce to $\frac{1}{4}F^2$ in the continuum.

43.3 Gauge fields on the lattice: link variables and the Wilson action

The previous section taught us how to put a scalar field on a lattice and read the continuum limit off a critical point, but it also delivered a warning. The naive recipe that worked for the scalar — sample the field ϕ at the sites $x \in \Lambda$ of the hypercubic lattice $\Lambda = a\mathbb{Z}^4$ and replace derivatives by finite differences — destroys local gauge invariance the instant it is applied to a charged or non-abelian field. The reason is structural. A gauge transformation rotates the matter field by a *different* group element at every site, $\psi(x) \to \Omega(x)\psi(x)$ with $\Omega(x) \in \mathrm{SU}(N)$, and the only finite-difference object built from the field at two sites,

$$\bar{\psi}(x)\,\frac{\psi(x+a\hat{\mu})-\psi(x)}{a}, \tag{43.65}$$

contains the bilinear $\bar{\psi}(x)\psi(x+a\hat{\mu})$, which under the transformation becomes $\bar{\psi}(x)\,\Omega^\dagger(x)\Omega(x+a\hat{\mu})\,\psi(x+a\hat{\mu})$. The mismatched factor $\Omega^\dagger(x)\Omega(x+a\hat{\mu})$ does not cancel: it is the identity only if $\Omega(x) = \Omega(x+a\hat{\mu})$, i.e. only for a *global* transformation. At finite a the naive lattice action is not gauge invariant, and no amount of cleverness with the *site* field $A_\mu(x)$ repairs this, because $A_\mu(x)$ inherits the same disease: it is the connection sampled at a point, and a connection is precisely the object whose transformation law is inhomogeneous. The lesson, due to Wilson, is that on a lattice the fundamental gauge variable cannot be $A_\mu(x)$ at all. It must be the object that already knows how to carry a field from one site to its neighbour while compensating the gauge rotation in between — the parallel transporter of Chapter 25, evaluated on the elementary link. Everything in this section follows from taking that single idea seriously.

We work throughout in Euclidean signature, the Wick-rotated framework of Section 10.6 in which the path-integral weight e^{-S_E} is a genuine statistical measure; the previous section carried out the rotation explicitly for the scalar. All spacetime indices $\mu, \nu \in \{1,2,3,4\}$ are Euclidean, raised and lowered with $\delta_{\mu\nu}$, so contractions are ordinary sums of squares and there are no signature signs to track. We keep the frozen Yang–Mills theory conventions: gauge group $\mathrm{SU}(N)$, Hermitian fundamental generators T^a with $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, matrix-valued connection $A_\mu = A_\mu^a T^a$, coupling g, covariant derivative $D_\mu = \partial_\mu - igA_\mu$, and the field strength

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu], \qquad F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu, \tag{43.66}$$

of Eqs. (25.48) and (25.49). The hat $\hat{\mu}$ denotes the unit vector in direction μ, so $x + a\hat{\mu}$ is the lattice neighbour of x in that direction.

43.3.1 The link variable as a lattice parallel transporter

In the continuum the device that compensates the gauge rotation between two points is the Wilson line of Theorem 25.22, the path-ordered exponential $\mathcal{P}\exp(ig\int_C A_\mu\,dx^\mu)$. On the lattice the shortest nontrivial path is the elementary link joining a site x to its neighbour $x+a\hat{\mu}$. We promote the Wilson line on that one link to a dynamical variable in its own right.

Definition 43.14 (Link variable)**.** To each oriented link of the lattice, running from x to $x+a\hat{\mu}$, assign a group element $U_\mu(x)\in \mathrm{SU}(N)$, the *link variable*. It is the lattice incarnation of the Wilson line of Eq. (25.124) along that link,

$$U_\mu(x) \;=\; \mathcal{P}\exp\Big(ig\int_x^{x+a\hat{\mu}} A_\rho\,dx^\rho\Big) \;=\; \exp\big(iag\,A_\mu(x)\big)+O(a^2), \tag{43.67}$$

the second form being the leading short-link expansion, with the connection evaluated on the link. The reversed link carries the inverse, $U_{-\mu}(x+a\hat{\mu})\equiv U_\mu(x)^\dagger$, since traversing the link backward path-orders the comparator the other way and $U\in\mathrm{SU}(N)$ is unitary.

The crucial point is what we keep and what we discard. The continuum field $A_\mu(x)$ appears in Eq. (43.67) only inside the exponential and only as a way of writing a particular group element; the lattice theory's degree of freedom is the *group element* $U_\mu(x)$ itself, ranging over all of $\mathrm{SU}(N)$. We never again use $A_\mu(x)$ as an independent variable. This is not a notational nicety. The map $A_\mu\mapsto e^{iagA_\mu}$ is many-to-one (it is the exponential map onto a compact group), so a single U_μ corresponds to a whole family of A_μ's differing by the lattice analogue of a large gauge transformation; insisting on U as fundamental builds in compactness of the gauge group from the start, with consequences we exploit in Section 43.3.5.

The transformation law of the link variable is dictated entirely by its origin as a Wilson line. Under a lattice gauge transformation, an independent group element $\Omega(x)\in\mathrm{SU}(N)$ is assigned to every site, and matter rotates by $\psi(x)\to\Omega(x)\psi(x)$. The Wilson line of Theorem 25.25 transforms by conjugation at its endpoints, $W_C\to\Omega(x_f)\,W_C\,\Omega^\dagger(x_i)$ (this is Eq. (25.137), with the gauge matrix there written U and here written Ω to free the symbol U for the link). Applied to the elementary link, whose endpoints are x and $x+a\hat{\mu}$, this is the defining transformation rule of lattice gauge theory:

$$\boxed{\;U_\mu(x)\;\longrightarrow\;\Omega(x)\,U_\mu(x)\,\Omega^\dagger(x+a\hat{\mu})\;} \tag{43.68}$$

The link carries the gauge rotation of the site it starts at on the left and of the site it ends at on the right. Its entire purpose is to make the matter bilinear gauge invariant: the lattice-covariant hopping term is $\bar{\psi}(x)\,U_\mu(x)\,\psi(x+a\hat{\mu})$, which under the combined transformation becomes $\bar{\psi}(x)\Omega^\dagger(x)\cdot\Omega(x)U_\mu(x)\Omega^\dagger(x+a\hat{\mu})\cdot$

$\Omega(x+a\hat{\mu})\psi(x+a\hat{\mu}) = \bar{\psi}(x)U_\mu(x)\psi(x+a\hat{\mu})$, exactly invariant. The mismatched factor that ruined Eq. (43.65) is absorbed by the link. This is the whole point, and it works at *any* lattice spacing, not merely as $a \to 0$.

Notation Watch

In Chapter 25 the gauge transformation matrix was denoted $U(x)$ and the parallel transporter W_C. On the lattice the universal convention reverses the roles of the letters: the parallel transporter on a link is $U_\mu(x)$ — the dynamical field — while the gauge transformation is $\Omega(x)$. We adopt the lattice convention here. The objects are unchanged; only the names are swapped, and we have stated the dictionary explicitly so no reader confuses the field with the symmetry acting on it.

Physical Insight

A lattice gauge field is a discretized *geometry*. The continuum gauge field is a connection on a principal $\mathrm{SU}(N)$ bundle; its content is the rule for parallel transport, and that rule lives on paths, not points. Discretizing the manifold to a lattice discretizes the paths to links, and the connection becomes a collection of group-valued parallel transporters — one per edge — exactly as a discrete differential geometer would assign holonomies to the edges of a graph. Gauge invariance is then automatic for the right reason: physical (gauge-invariant) quantities are built only from *closed* loops of links, whose conjugation factors at the basepoint cancel under the trace, and from open lines capped at both ends by matter, whose endpoint rotations are absorbed by the fields. There is no gauge-variant quantity left to worry about, because we never wrote one down.

43.3.2 The plaquette and its exact gauge invariance

To build a pure gauge action we need a gauge-invariant functional of the links alone, with no matter fields. By the geometric principle just stated it must be a closed loop, and the smallest closed loop on the lattice is the elementary square — the *plaquette* — bounding one unit cell of the $\mu\nu$ plane.

Definition 43.15 (Plaquette variable)**.** The plaquette variable in the $\mu\nu$ plane based at x is the ordered product of the four link variables around the unit square $x \to x+a\hat{\mu} \to x+a\hat{\mu}+a\hat{\nu} \to x+a\hat{\nu} \to x$, traversed counterclockwise,

$$U_{\mu\nu}(x) \;=\; U_\mu(x)\, U_\nu(x+a\hat{\mu})\, U_\mu(x+a\hat{\nu})^\dagger\, U_\nu(x)^\dagger, \tag{43.69}$$

where the daggers implement the two backward links (Theorem 43.14). It is an element of $\mathrm{SU}(N)$.

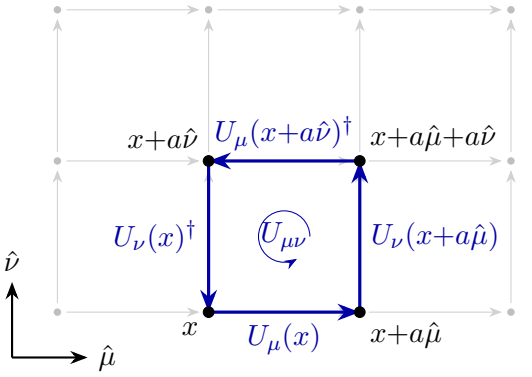


Figure 43.3: A slice of the hypercubic lattice. Each oriented edge carries a link variable $U_\mu(x) \in \mathrm{SU}(N)$, the parallel transporter from x to $x + a\hat{\mu}$ (Theorem 43.14); the reversed edge carries $U_\mu(x)^\dagger$. The highlighted unit square is the plaquette $U_{\mu\nu}(x)$ of Eq. (43.69), the ordered product of its four edge links read counterclockwise. Under a gauge transformation $U_\mu(x) \to \Omega(x)U_\mu(x)\Omega^\dagger(x + a\hat{\mu})$ the interior corner factors telescope and $U_{\mu\nu}(x) \to \Omega(x)U_{\mu\nu}(x)\Omega^\dagger(x)$, so $\mathrm{Re}\,\mathrm{tr}\,U_{\mu\nu}$ is exactly gauge invariant at every lattice spacing.

The ordering reads counterclockwise around the square (Fig. 43.3): forward along $\hat{\mu}$ at the bottom, forward along $\hat{\nu}$ at the right, backward along $\hat{\mu}$ at the top (hence the dagger of the link $U_\mu(x + a\hat{\nu})$, which is the *forward* link on the top edge traversed in reverse), and backward along $\hat{\nu}$ on the left. It is the lattice realization of the small Wilson loop of Section 25.6.5, with the abstract loop $\mathcal{C}$ taken to be one unit cell.

Proposition 43.16 (Gauge transformation of the plaquette)**.** *Under the lattice gauge transformation Eq. (43.68), the plaquette variable transforms by conjugation at its basepoint x,*

$$U_{\mu\nu}(x) \;\longrightarrow\; \Omega(x)\, U_{\mu\nu}(x)\, \Omega^\dagger(x), \tag{43.70}$$

and consequently its trace, and the real part of its trace, are exactly gauge invariant at any lattice spacing a.

Proof. Substitute the transformation Eq. (43.68) into each of the four factors of Eq. (43.69). The forward links transform as written; the backward links (daggers) transform as the daggers of Eq. (43.68), namely $U_\mu(x + a\hat{\nu})^\dagger \to \Omega(x + a\hat{\nu} + a\hat{\mu})\, U_\mu(x + a\hat{\nu})^\dagger\, \Omega^\dagger(x + a\hat{\nu})$ and $U_\nu(x)^\dagger \to \Omega(x + a\hat{\nu})\, U_\nu(x)^\dagger\, \Omega^\dagger(x)$. Writing the four transformed factors in order and inserting brackets around the gauge matrices that meet at each shared corner,

$$U_{\mu\nu}(x) \to \Omega(x)U_\mu(x)\underbrace{\Omega^\dagger(x+a\hat{\mu})\,\Omega(x+a\hat{\mu})}_{=\mathbf{1}}U_\nu(x+a\hat{\mu})\underbrace{\Omega^\dagger(x+a\hat{\mu}+a\hat{\nu})\,\Omega(x+a\hat{\nu}+a\hat{\mu})}_{=\mathbf{1}}$$

$$\times\, U_\mu(x{+}a\hat\nu)^\dagger \underbrace{\Omega^\dagger(x{+}a\hat\nu)\,\Omega(x{+}a\hat\nu)}_{=\mathbf{1}} U_\nu(x)^\dagger\,\Omega^\dagger(x)$$

$$= \Omega(x)\,U_\mu(x)U_\nu(x{+}a\hat\mu)U_\mu(x{+}a\hat\nu)^\dagger U_\nu(x)^\dagger\,\Omega^\dagger(x) = \Omega(x)\,U_{\mu\nu}(x)\,\Omega^\dagger(x). \tag{43.71}$$

Every interior corner of the square contributes a pair $\Omega^\dagger\Omega = \mathbf{1}$: the gauge matrix attached to a site by the link arriving there is exactly cancelled by the one attached by the link leaving it. Only the basepoint x, where the loop opens and closes, survives, leaving the conjugation Eq. (43.70). Taking the trace and using cyclicity together with $\Omega^\dagger\Omega = \mathbf{1}$,

$$\operatorname{tr} U_{\mu\nu}(x) \to \operatorname{tr}\big(\Omega(x)U_{\mu\nu}(x)\Omega^\dagger(x)\big) = \operatorname{tr}\big(\Omega^\dagger(x)\Omega(x)U_{\mu\nu}(x)\big) = \operatorname{tr} U_{\mu\nu}(x), \tag{43.72}$$

and the real part of an invariant is invariant. No expansion in a was used: the cancellation is algebraic, exact at finite spacing. □

This is the lattice version of the trace invariance of the Wilson loop, Theorem 25.26: the untraced plaquette is gauge *covariant*, conjugated at its basepoint, and the trace removes the residual conjugation. Equation Eq. (43.72) is the foundation of everything that follows. It says that $\operatorname{Re}\operatorname{tr} U_{\mu\nu}(x)$ is a legitimate gauge-invariant building block at every value of a, with no continuum limit invoked and — as we will see — no gauge fixing required.

43.3.3 The Wilson gauge action

We now assemble the simplest gauge-invariant action from plaquettes. Summing $\operatorname{Re}\operatorname{tr} U_{\mu\nu}$ over all plaquettes of the lattice gives a real, gauge-invariant functional of the links; subtracting it from its maximal value normalizes the action to vanish on the trivial configuration $U_{\mu\nu} = \mathbf{1}$ (no field), which corresponds to the zero-action vacuum.

Definition 43.17 (Wilson gauge action)**.** The *Wilson action* for an $\mathrm{SU}(N)$ gauge field on the lattice is

$$\boxed{S_W[U] \;=\; \beta \sum_x \sum_{\mu<\nu} \Big(1 - \tfrac{1}{N}\,\operatorname{Re}\operatorname{tr}\, U_{\mu\nu}(x)\Big), \qquad \beta = \frac{2N}{g^2},} \tag{43.73}$$

where the first sum runs over all sites $x \in \Lambda$ and the second over all distinct oriented planes $\mu < \nu$, so that each plaquette is counted once. The dimensionless parameter $\beta = 2N/g^2$ is the *lattice coupling* (the inverse coupling, in the language of statistical mechanics, where it plays the role of an inverse temperature).

Three features are immediate. First, S_W is exactly gauge invariant at any a, by Theorem 43.16 applied term by term. Second, the normalization $1 - \frac{1}{N}\operatorname{Re}\operatorname{tr} U_{\mu\nu}$

is non-negative — since $\operatorname{tr} U_{\mu\nu}$ for a unitary matrix has real part at most N, attained only at $U_{\mu\nu} = \mathbf{1}$ — so the action is bounded below by zero, and e^{-S_W} is bounded above by one, a property we will lean on when we argue convergence of the path integral. Third, the value of the constant β has been chosen, not derived; we now show that this particular choice is the one for which S_W reduces to the continuum Yang–Mills action.

43.3.4 The classical continuum limit

The decisive test of Eq. (43.73) is that as the lattice is refined, with the link variables set to their continuum values Eq. (43.67), the Wilson action collapses to the Euclidean Yang–Mills action $\frac{1}{4}\int F^a_{\mu\nu}F^a_{\mu\nu}$. The engine is the Baker–Campbell–Hausdorff (BCH) expansion of the plaquette, which we carry out in full.

Worked Example 43.18 (BCH expansion of the plaquette)**.** Set the links to their leading short-link form $U_\mu(x) = \exp(iag\,A_\mu(x))$ and compute $U_{\mu\nu}(x)$ to second order in the spacing a. We will find $U_{\mu\nu}(x) = \exp\big(ia^2 g\,F_{\mu\nu}(x) + O(a^3)\big)$.

Step 1: the four exponents. Each factor in Eq. (43.69) is the exponential of a Lie-algebra element of order a. The two daggers flip the sign of the exponent. Taylor-expanding the connection of the displaced factors about the basepoint x, $A_\nu(x + a\hat{\mu}) = A_\nu + a\,\partial_\mu A_\nu + O(a^2)$ and $A_\mu(x + a\hat{\nu}) = A_\mu + a\,\partial_\nu A_\mu + O(a^2)$ (with all fields on the right evaluated at x), the four exponents are

$$\begin{aligned} X_1 &= iag\,A_\mu, & X_2 &= iag\big(A_\nu + a\,\partial_\mu A_\nu\big) + O(a^3),\\ X_3 &= -iag\big(A_\mu + a\,\partial_\nu A_\mu\big) + O(a^3), & X_4 &= -iag\,A_\nu, \end{aligned} \tag{43.74}$$

so that $U_{\mu\nu} = e^{X_1}e^{X_2}e^{X_3}e^{X_4}$. Each X_i is $O(a)$.

Step 2: the BCH combination. For a product of exponentials,

$$e^{X_1}e^{X_2}e^{X_3}e^{X_4} = \exp\Big(\sum_i X_i + \tfrac{1}{2}\sum_{i<j}[X_i, X_j] + \cdots\Big), \tag{43.75}$$

where the omitted terms are nested double commutators, each of order a^3 since every X_i is $O(a)$. We need the exponent only to $O(a^2)$, so we keep the linear sum (to $O(a^2)$) and the single-commutator sum (using the $O(a)$ leading parts of the X_i), and drop the rest.

Step 3: the linear sum. The $O(a)$ pieces cancel because the loop closes: $iag(A_\mu + A_\nu - A_\mu - A_\nu) = 0$. The surviving $O(a^2)$ pieces come from the Taylor shifts in X_2 and X_3,

$$\sum_i X_i = ia^2 g\,\partial_\mu A_\nu - ia^2 g\,\partial_\nu A_\mu + O(a^3) = ia^2 g\big(\partial_\mu A_\nu - \partial_\nu A_\mu\big) + O(a^3). \tag{43.76}$$

This is the abelian curl — the first two terms of the field strength.

Step 4: the commutator sum. For the commutators we may use the $O(a)$ leading parts $X_1 \simeq iag\,A_\mu$, $X_2 \simeq iag\,A_\nu$, $X_3 \simeq -iag\,A_\mu$, $X_4 \simeq -iag\,A_\nu$ (the Taylor shifts would contribute at $O(a^3)$). Write $\alpha \equiv iag\,A_\mu$ and $\gamma \equiv iag\,A_\nu$, so $X_1 = \alpha$, $X_2 = \gamma$, $X_3 = -\alpha$, $X_4 = -\gamma$. The six ordered pairs give

$$\begin{aligned} &[X_1,X_2] = [\alpha,\gamma], && [X_1,X_3] = [\alpha,-\alpha] = 0, && [X_1,X_4] = [\alpha,-\gamma] = -[\alpha,\gamma], \\ &[X_2,X_3] = [\gamma,-\alpha] = [\alpha,\gamma], && [X_2,X_4] = [\gamma,-\gamma] = 0, && [X_3,X_4] = [-\alpha,-\gamma] = [\alpha,\gamma], \end{aligned} \tag{43.77}$$

using $[\gamma,\alpha] = -[\alpha,\gamma]$. Their sum is $(1-1+1+1)[\alpha,\gamma] = 2[\alpha,\gamma]$, so

$$\tfrac{1}{2}\sum_{i<j}[X_i,X_j] = [\alpha,\gamma] = (iag)^2[A_\mu,A_\nu] = -a^2g^2\,[A_\mu,A_\nu]. \tag{43.78}$$

Step 5: assemble. Adding Eqs. (43.76) and (43.78), the exponent of $U_{\mu\nu}$ is

$$\begin{aligned} \sum_i X_i + \tfrac{1}{2}\sum_{i<j}[X_i,X_j] &= ia^2g(\partial_\mu A_\nu - \partial_\nu A_\mu) - a^2g^2[A_\mu,A_\nu] + O(a^3) \\ &= ia^2g(\partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu,A_\nu]) + O(a^3) = ia^2g\,F_{\mu\nu}(x) + O(a^3), \end{aligned} \tag{43.79}$$

where in the second equality we factored out ia^2g from the commutator term using $-a^2g^2 = ia^2g\cdot(-ig)$, and in the third recognized $F_{\mu\nu}$ from Eq. (43.66). Therefore

$$\boxed{U_{\mu\nu}(x) = \exp\!\big(ia^2g\,F_{\mu\nu}(x) + O(a^3)\big) = \mathbf{1} + ia^2g\,F_{\mu\nu}(x) + O(a^3).} \tag{43.80}$$

This is precisely the small-loop holonomy of Eq. (25.149), $W(\mathcal{C}) = \mathbf{1} + ig\,a^\mu b^\nu F_{\mu\nu}$, specialized to the unit plaquette with edge vectors $a^\mu = a\hat{\mu}$ and $b^\nu = a\hat{\nu}$ and area element a^2: the full non-abelian curvature, self-coupling commutator included, emerges automatically from the ordered product of the four links. The plaquette *measures the curvature through the cell it bounds*, which is why it is the right object to build the action from.

With Eq. (43.80) in hand the continuum reduction is a short computation.

Theorem 43.19 (Classical continuum limit of the Wilson action)**.** *With the link variables set to their continuum values Eq. (43.67) and a smooth gauge field A_μ, the Wilson action Eq. (43.73) reduces to the Euclidean Yang–Mills action,*

$$S_W[U] \;\xrightarrow{a\to 0}\; \frac{1}{4}\int d^4x\; F^a_{\mu\nu}(x)\,F^a_{\mu\nu}(x) \;+\; O(a^2), \tag{43.81}$$

the choice $\beta = 2N/g^2$ being exactly what makes the prefactor unity.

Proof. Abbreviate $X \equiv a^2 g\, F_{\mu\nu}(x)$, a Hermitian matrix (since $F_{\mu\nu} = F^a_{\mu\nu} T^a$ with real components and Hermitian T^a), so that $U_{\mu\nu} = e^{iX}$ by Eq. (43.80). Expanding the exponential,

$$\operatorname{tr} U_{\mu\nu} = \operatorname{tr}\Big(\mathbf{1} + iX - \tfrac{1}{2}X^2 - \tfrac{i}{6}X^3 + \cdots\Big) = N + i\,\operatorname{tr} X - \tfrac{1}{2}\operatorname{tr} X^2 + \cdots . \tag{43.82}$$

Because X is Hermitian, $\operatorname{tr} X$ is real, so $i\,\operatorname{tr} X$ is pure imaginary and drops out of the real part; in fact $\operatorname{tr} X = a^2 g\, F^a_{\mu\nu} \operatorname{tr} T^a = 0$ since the generators are traceless, but we do not even need this. Likewise $\operatorname{tr} X^2$ is real. Hence

$$\operatorname{Re}\operatorname{tr} U_{\mu\nu} = N - \tfrac{1}{2}\operatorname{tr} X^2 + O(a^6) = N - \tfrac{1}{2} a^4 g^2\, \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) + O(a^6), \tag{43.83}$$

with no sum on μ, ν. The plaquette term in the action is therefore

$$1 - \tfrac{1}{N}\operatorname{Re}\operatorname{tr} U_{\mu\nu} = \frac{a^4 g^2}{2N}\, \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) + O(a^6). \tag{43.84}$$

Now sum over sites and planes and insert $\beta = 2N/g^2$, so that the explicit coupling factor collapses, $\beta \cdot \frac{g^2}{2N} = 1$:

$$S_W = \frac{2N}{g^2} \sum_x \sum_{\mu<\nu} \frac{a^4 g^2}{2N}\, \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) + \cdots = \sum_x a^4 \sum_{\mu<\nu} \operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) + \cdots . \tag{43.85}$$

Three reductions finish the proof. The site sum becomes an integral by the definition of the lattice measure, $\sum_x a^4 \to \int d^4x$. The restricted plane sum becomes a full sum, $\sum_{\mu<\nu} = \frac{1}{2}\sum_{\mu,\nu}$, because $\operatorname{tr}(F_{\mu\nu}F_{\mu\nu})$ is symmetric in $\mu \leftrightarrow \nu$ (and the diagonal $\mu = \nu$ terms vanish). And the color trace is evaluated with the normalization $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$,

$$\operatorname{tr}(F_{\mu\nu}F_{\mu\nu}) = F^a_{\mu\nu} F^b_{\mu\nu}\, \operatorname{tr}(T^a T^b) = \tfrac{1}{2} F^a_{\mu\nu} F^a_{\mu\nu}. \tag{43.86}$$

Combining,

$$S_W \to \int d^4x\, \tfrac{1}{2} \sum_{\mu,\nu} \cdot \tfrac{1}{2} F^a_{\mu\nu} F^a_{\mu\nu} = \frac{1}{4}\int d^4x\, F^a_{\mu\nu} F^a_{\mu\nu}, \tag{43.87}$$

which is Eq. (43.81). The corrections are $O(a^2)$: the leading omitted terms in the BCH exponent are $O(a^3)$ inside $U_{\mu\nu}$, raising $\operatorname{Re}\operatorname{tr}$ corrections to relative order a^2 beyond the displayed a^4, which after the $\sum_x a^4 \to \int$ becomes an $O(a^2)$ correction to the continuum action — the hallmark of the Wilson action's leading discretization error. □

Two remarks tie this to the broader story. First, the result Eq. (43.87) is the *Euclidean* Yang–Mills action, $+\frac{1}{4}\int F^2$ with all indices down and summed with $\delta_{\mu\nu}$; it is the Wick rotation (Section 10.6) of the Minkowski $-\frac{1}{4}\int F^a_{\mu\nu} F^{a\,\mu\nu}$ of Yang–Mills theory, and it enters the statistical weight as e^{-S_W}. Second, the

appearance of the *bare* lattice coupling g inside β is not the end of the matter: as in the scalar case of the previous section, the physical continuum limit $a \to 0$ is not taken at fixed g but along a trajectory on which $g \to 0$, dictated by the renormalization group. For Yang–Mills that trajectory is governed by asymptotic freedom (Theorem 27.7 and Section 27.5): the bare coupling must be tuned to zero as $a \to 0$, so the continuum theory sits at the Gaussian ultraviolet fixed point, the gauge-theory instance of "the continuum limit is an RG fixed point" established in Section 23.6. We return to this when we ask, in the next section, whether the confinement found at large g survives the journey to $g \to 0$.

43.3.5 The Haar measure: exact gauge invariance with no gauge fixing

We can now write the lattice gauge theory's defining object, the partition function, and confront the feature that most sharply distinguishes the lattice from every continuum construction. The path integral is an ordinary integral over one copy of the compact group $\mathrm{SU}(N)$ for each link,

$$\boxed{Z = \int \prod_{x,\mu} dU_\mu(x)\, e^{-S_W[U]}, \qquad \langle \mathcal{O} \rangle = \frac{1}{Z} \int \prod_{x,\mu} dU_\mu(x)\, \mathcal{O}[U]\, e^{-S_W[U]},} \tag{43.88}$$

where dU is the *Haar measure* on $\mathrm{SU}(N)$. On a finite lattice this is a finite product of integrals over a compact manifold of fixed, finite dimension — in the most literal sense an ordinary (if enormous) multidimensional integral, with no functional analysis and no further regulator required. The choice of dU as the Haar measure is what makes Eq. (43.88) both well defined and exactly gauge invariant, so we state its properties precisely.

Proposition 43.20 (Haar measure on a compact group)**.** *On a compact Lie group G (here $G = \mathrm{SU}(N)$) there exists a unique normalized, bi-invariant integration measure dU, the* Haar measure*, characterized by:*

(i) Normalization: $\displaystyle\int_G dU = 1.$

(ii) Left and right invariance: *for any fixed $V \in G$ and any integrable f,*

$$\int_G f(VU)\, dU = \int_G f(UV)\, dU = \int_G f(U)\, dU. \tag{43.89}$$

(iii) Inversion invariance: $\displaystyle\int_G f(U^{-1})\, dU = \int_G f(U)\, dU.$

The existence and uniqueness of such a measure are guaranteed precisely by the compactness of G: a left-invariant measure exists on any locally compact group,

compactness makes its total mass finite so it can be normalized to (i), and on a compact group left invariance forces right invariance and inversion invariance as well.

We do not reprove the existence theorem here — it is a standard result of the theory of compact groups, and a self-contained construction (building dU from a bi-invariant volume form, or as the unique invariant mean) is recorded in [98, 139]. What we *do* need, and prove, is its consequence for the path integral.

Theorem 43.21 (Exact gauge invariance of the lattice path integral)**.** *The product measure $\prod_{x,\mu} dU_\mu(x)$ is invariant under any lattice gauge transformation Eq. (43.68). Consequently Z and every expectation value $\langle \mathcal{O} \rangle$ are gauge invariant, and no gauge fixing, Faddeev–Popov determinant, or Gribov choice is needed to define them.*

Proof. Fix a gauge transformation, i.e. a choice of $\Omega(x) \in \mathrm{SU}(N)$ at every site, and let it act on the links by $U_\mu(x) \mapsto U_\mu^\Omega(x) = \Omega(x) U_\mu(x) \Omega^\dagger(x + a\hat{\mu})$. For each link separately this is left multiplication by the fixed element $\Omega(x)$ followed by right multiplication by the fixed element $\Omega^\dagger(x + a\hat{\mu})$. By the left and right invariance Eq. (43.89) of the Haar measure, applied to that link's integration variable,

$$\int dU_\mu(x)\, f(U_\mu^\Omega(x)) = \int dU_\mu(x)\, f(U_\mu(x)), \tag{43.90}$$

for any integrand f. The full measure is the product of one such factor per link, and the transformation acts on each link's variable independently (the Ω's are external constants for the purpose of integration), so the product measure is invariant: $\prod_{x,\mu} dU_\mu^\Omega(x) = \prod_{x,\mu} dU_\mu(x)$. The Wilson action is invariant by Theorem 43.16, $S_W[U^\Omega] = S_W[U]$. Changing integration variables from U to U^Ω in Eq. (43.88) therefore leaves both the measure and the integrand unchanged, so Z is gauge invariant, and for any observable $\langle \mathcal{O}[U^\Omega] \rangle = \langle \mathcal{O}[U] \rangle$. Crucially, this required *no* restriction of the integration domain to a single representative per gauge orbit: we integrate over all links freely. □

The contrast with the continuum construction of Chapter 26 is the conceptual payoff of the entire link-variable formalism, and it deserves to be drawn explicitly.

Physical Insight: why the lattice needs no gauge fixing

In the continuum, the gauge field A_μ is a noncompact variable: at each point it ranges over the whole Lie algebra $\mathbb{R}^{\dim G}$. The path-integral measure $\mathcal{D}A$ then integrates over infinitely many physically identical copies of each configuration — the gauge orbit — and that orbit has *infinite* volume. The integral $\int \mathcal{D}A\, e^{-S}$ diverges through this redundant volume, and the Faddeev–Popov procedure of Section 26.2 is the surgery that

divides it out: insert $1 = \Delta_{\mathrm{FP}} \int \mathcal{D}\Omega\, \delta(G[A^{\Omega}])$, factor off the orbit volume, and pay for it with the determinant $\Delta_{\mathrm{FP}} = \det(\delta G/\delta\Omega)$, re-exponentiated as the ghost fields. On the lattice none of this arises. The link variable U_μ ranges over the *compact* group $\mathrm{SU}(N)$, whose total Haar volume per link is $\int dU = 1$; the gauge group at each site is the same compact $\mathrm{SU}(N)$, of finite volume. The redundant gauge-orbit volume is therefore finite — it is $\prod_x \mathrm{vol}(\mathrm{SU}(N))$, a finite constant — and in the normalized expectation value $\langle \mathcal{O} \rangle = Z^{-1} \int \mathcal{O}\, e^{-S_W}$ it cancels identically between numerator and denominator. There is nothing to divide out, hence no Faddeev–Popov determinant, no ghost fields, and no need to impose a gauge condition $G[A] = 0$ at all. The price the continuum pays for working with the noncompact A_μ is exactly the price the lattice avoids by working with the compact U_μ.

This evasion is not merely an economy; it removes a genuine pathology. The Faddeev–Popov construction is exact only locally in field space. A gauge-fixing surface $G[A] = 0$ generally intersects each orbit more than once — the *Gribov ambiguity* of Section 26.1.4 — so the continuum path integral, taken at face value, either overcounts or must be restricted to a Gribov region whose boundary is itself subtle. The lattice formulation Eq. (43.88) has no gauge-fixing surface, hence no Gribov copies: it is a manifestly gauge-invariant, manifestly finite definition of the integral, with no choices left to make and no ambiguity to resolve. This is the precise sense in which the lattice is a *constructive* definition of a gauge theory and the continuum gauge-fixed path integral is not.

Remark 43.22 (Local gauge symmetry cannot break; Elitzur's theorem). One might worry that, without gauge fixing, the expectation value of a gauge-variant quantity is ill defined or ambiguous. It is neither: it is automatically zero. By Theorem 43.21 the measure averages over the whole gauge orbit at each point, so any quantity that is not invariant integrates to zero against e^{-S_W} — for instance $\langle U_\mu(x) \rangle = 0$, since $\int d\Omega(x)\, \Omega(x) = 0$ over the group. The sharp statement is *Elitzur's theorem*: a local gauge symmetry can never be spontaneously broken, so the expectation value of any local non-gauge-invariant operator vanishes identically, with or without an explicit symmetry-breaking source taken to zero afterwards [174]. This is not needed for any computation below, but it explains why the absence of gauge fixing is harmless: the only quantities that can be nonzero are the gauge-invariant ones — Wilson loops, plaquette averages, and the like — which are exactly the quantities we wish to compute.

Caution

The fundamental variable is the group element $U_\mu(x)$, not the algebra element $A_\mu(x)$. At finite a the relation $U_\mu = e^{iagA_\mu}$ is not invertible to

a single A_μ, and there is no sense in which "$A_\mu(x)$ on the lattice" is a well-defined field; expressions like $\partial_\mu A_\nu$ on the lattice are meaningful only as the $a \to 0$ shorthand exhibited in Theorem 43.18. All exact lattice statements — gauge invariance, the action, the measure — are statements about the U's. Forgetting this and trying to gauge-fix "A" on the lattice reintroduces every problem the link formulation was designed to avoid.

We have, with no approximation and no gauge fixing, a manifestly gauge-invariant definition of a non-abelian gauge theory: links valued in SU(N), the Wilson action Eq. (43.73), and the Haar-measure integral Eq. (43.88) whose invariance is exact at every spacing. The decisive virtue of this formulation is that the weight e^{-S_W} depends on the links only through the bounded, group-valued plaquette traces, so for large g — small β — the exponential can be expanded and the link integrals done term by term using nothing more than the Haar orthogonality relations of Theorem 43.20. That strong-coupling expansion, carried out on exactly this Haar-measure path integral, is what the next section uses to extract an area law for the Wilson loop and, with it, the first analytic demonstration of confinement.

43.4 Confinement at strong coupling: the area law and the string tension

The previous section reduced the gauge sector of the lattice theory to a single, completely explicit object: an ordinary integral over one copy of the compact group SU(N) for every link of the lattice, weighted by the Wilson action and the invariant Haar measure (Section 43.3). No gauge fixing, no Faddeev–Popov determinant, no Gribov ambiguity — the integral is finite, manifestly gauge invariant, and, crucially for what follows, factorizes into a product over links that are coupled only through the plaquettes. This is the rare circumstance in which a four-dimensional interacting gauge theory can be computed *by hand*. We now exploit it. Expanding the Boltzmann weight in powers of the inverse coupling and performing the link integrals one at a time, we will derive — with every Haar integral done explicitly and every combinatorial factor accounted for — that the expectation value of a large Wilson loop falls off as the exponential of the *area* it spans. By the criterion built in Chapter 27 this is exactly the statement that the static quark–antiquark potential rises linearly and the charges are permanently confined. It is the first and cleanest analytic demonstration of confinement in any four-dimensional gauge theory, and it is the non-perturbative payoff that Yang–Mills theory has been driving toward since Chapter 25. The section closes by asking, with care, whether the result survives the continuum limit — and explaining honestly why that last step is, as of the mid-2020s,

established numerically rather than proved.

Notation Watch

Throughout this chapter β denotes the *lattice inverse coupling* $\beta = 2N/g^2$ appearing in the Wilson action, *not* a renormalization-group beta function. This is universal lattice usage. The two notions are related only indirectly — through the dependence of g^2 on the lattice spacing a, which is the subject of Section 43.4.6. Strong coupling means *large* bare g^2, hence *small* β, and the expansion of this section is a power series in β.

43.4.1 The single-link integrals: moments of the Haar measure

Everything in the strong-coupling expansion is built from a short list of integrals of products of matrix elements of a single link variable $U \in \mathrm{SU}(N)$ against its normalized Haar measure $\mathrm{d}U$. Recall from Section 43.3 the two defining properties of that measure that we will use repeatedly: it is *normalized*,

$$\int \mathrm{d}U \;=\; 1, \tag{43.91}$$

and it is *left- and right-invariant*: for any fixed $V, W \in \mathrm{SU}(N)$ and any integrable function f,

$$\int \mathrm{d}U\, f(U) \;=\; \int \mathrm{d}U\, f(VU) \;=\; \int \mathrm{d}U\, f(UW). \tag{43.92}$$

These two properties, together with elementary representation theory, fix every integral we need. The substantive content is collected in the following lemma; its last identity, the "two-U" integral, is the workhorse of the entire expansion.

Lemma 43.23 (Single-link moments)**.** *Let $U \in \mathrm{SU}(N)$ ($N \geq 2$) carry the normalized Haar measure. Then*

$$\int \mathrm{d}U\; U_{ij} \;=\; 0, \tag{43.93}$$

$$\int \mathrm{d}U\; U_{ij}\,(U^\dagger)_{kl} \;=\; \frac{1}{N}\,\delta_{il}\,\delta_{jk}, \tag{43.94}$$

$$\int \mathrm{d}U\; U_{ij}\,U_{kl} \;=\; 0 \quad (N \geq 3). \tag{43.95}$$

Proof. The first moment, Eq. (43.93). Set $M_{ij} = \int \mathrm{d}U\, U_{ij}$. The centre of $\mathrm{SU}(N)$ contains the elements $V = \omega\,\mathbf{1}$ with $\omega = e^{2\pi i/N}$, since $\det(\omega\mathbf{1}) = \omega^N = 1$. Left-invariance Eq. (43.89) with this V gives

$$M_{ij} = \int \mathrm{d}U\,(VU)_{ij} = \omega \int \mathrm{d}U\, U_{ij} = \omega\, M_{ij}, \tag{43.96}$$

so $(\omega-1)M_{ij}=0$. For $N\geq 2$ we have $\omega\neq 1$, hence $M_{ij}=0$. (For the abelian group U(1), where $U=e^{i\theta}$, the same conclusion is the trivial $\int_0^{2\pi}\frac{d\theta}{2\pi}\,e^{i\theta}=0$.)

The two-U integral, Eq. (43.94). Define the rank-four tensor

$$T_{ij,kl} \;=\; \int \mathrm{d}U\; U_{ij}\,(U^\dagger)_{kl}. \tag{43.97}$$

We determine it by invariance, in two steps: first its index structure, then its single overall constant. Apply left-invariance Eq. (43.89) with an arbitrary $V\in \mathrm{SU}(N)$, using $(VU)_{ij}=V_{ia}U_{aj}$ and $\big((VU)^\dagger\big)_{kl}=(U^\dagger V^\dagger)_{kl}=(U^\dagger)_{km}(V^\dagger)_{ml}$:

$$T_{ij,kl}=\int \mathrm{d}U\,(VU)_{ij}\big((VU)^\dagger\big)_{kl}=V_{ia}\,(V^\dagger)_{ml}\,T_{aj,km}. \tag{43.98}$$

This says that, regarded as acting on the index pair (i,l), the tensor T commutes with the simultaneous transformation $V\otimes V^\dagger$ of the fundamental $\mathbf{N}$ and the antifundamental $\bar{\mathbf{N}}$, for *every* V. The invariant tensors of $\mathbf{N}\otimes\bar{\mathbf{N}}$ form a one-dimensional space — the singlet in the decomposition $\mathbf{N}\otimes\bar{\mathbf{N}}=\mathbf{1}\oplus(\mathbf{N}^2{-}\mathbf{1})$ (Chapter B), whose invariant tensor is the Kronecker delta δ_{il}. Hence the dependence of T on (i,l) is through δ_{il} alone. Applying right-invariance, $U_{ij}\to U_{ib}W_{bj}$ and $(U^\dagger)_{kl}\to(W^\dagger)_{kc}(U^\dagger)_{cl}$, the identical argument forces the dependence on (j,k) to be through δ_{jk}. The only tensor built from these two deltas is

$$T_{ij,kl} \;=\; c\,\delta_{il}\,\delta_{jk}, \tag{43.99}$$

with a single constant c to be fixed. (No competing structure $\delta_{ik}\delta_{jl}$ is allowed: it would contract two fundamental indices i,k with a delta, which is not an $\mathrm{SU}(N)$ invariant for $N\geq 2$; the only other invariant tensors, the Levi-Civita symbols, carry N indices, not two.) To fix c, contract Eqs. (43.97) and (43.99) with $\delta_{jk}\delta_{li}$ and sum — i.e. set $k=j$, $l=i$ and sum over i,j. The left side becomes

$$\sum_{i,j}\int \mathrm{d}U\,U_{ij}(U^\dagger)_{ji}=\int \mathrm{d}U\;\mathrm{tr}(UU^\dagger)=\int \mathrm{d}U\;\mathrm{tr}\,\mathbf{1}=N, \tag{43.100}$$

where unitarity $UU^\dagger=\mathbf{1}$ and normalization Eq. (43.91) were used. The right side becomes $c\sum_{i,j}\delta_{ii}\delta_{jj}=cN^2$. Equating, $cN^2=N$, so $c=1/N$, which is Eq. (43.94).

The two-U integral without a dagger, Eq. (43.95). Repeat the centre argument on $S_{ij,kl}=\int \mathrm{d}U\,U_{ij}U_{kl}$: with $V=\omega\mathbf{1}$, left-invariance gives $S=\omega^2 S$, so $S=0$ unless $\omega^2=1$, i.e. unless $N\leq 2$. For $N\geq 3$ the integral vanishes. (For SU(2) it is proportional to the invariant $\varepsilon_{ik}\varepsilon_{jl}$, since $\bar{\mathbf{2}}\cong\mathbf{2}$ there; this special case plays no role below.) □

The pattern behind Theorem 43.23 is the only fact we need to remember: *the Haar integral of a product of link variables vanishes unless every fundamental index can be paired with a conjugate one carried by a $U^\dagger$ on the same link.* A

bare U integrates to zero Eq. (43.93); a U paired with a $U^\dagger$ survives and ties their colour indices together with the weight $1/N$ Eq. (43.94); two un-conjugated U's integrate to zero for $N \geq 3$ Eq. (43.95). This pairing rule is what will force the Wilson loop to be "filled in" by a surface of plaquettes.

Physical Insight

The pairing rule is the lattice avatar of charge conservation. A single link variable $U_\mu(x)$ carries one unit of fundamental colour flux across the link; an isolated unit of flux averages to nothing over the gauge group. Only when the flux is returned — a U matched by a $U^\dagger$ on the same link — does the average survive. A Wilson loop deposits one unit of unmatched flux on each of its boundary links, and the only way the path integral can return that flux is to grow a sheet of plaquettes whose own links supply the missing conjugates. The cheapest sheet, in powers of the small coupling, is the minimal surface spanning the loop. The area law is the statement that paying for that surface costs an amount proportional to its area.

43.4.2 The strong-coupling expansion

Recall the Wilson action from Eq. (43.73),

$$S_{\text{lat}} \;=\; \beta \sum_{p} \Big(1 - \tfrac{1}{N}\, \operatorname{Re}\operatorname{tr} U_p \Big), \qquad \beta = \frac{2N}{g^2}, \tag{43.101}$$

where the sum runs over all elementary plaquettes p and $U_p = U_\mu(x)\, U_\nu(x+\hat{\mu})\, U_\mu^\dagger(x+\hat{\nu})\, U_\nu^\dagger(x)$ is the ordered product of the four link variables around p. The constant $\beta \sum_p 1$ is field-independent and cancels between numerator and denominator of any expectation value, so we drop it and work with the reduced Boltzmann weight

$$e^{-S_{\text{lat}}} \;\doteq\; \prod_{p} \exp\Big(\tfrac{\beta}{2N} \big(\operatorname{tr} U_p + \operatorname{tr} U_p^\dagger \big) \Big), \tag{43.102}$$

using $\operatorname{Re}\operatorname{tr} U_p = \tfrac{1}{2}(\operatorname{tr} U_p + \operatorname{tr} U_p^\dagger)$ and writing $\doteq$ for equality up to the cancelled constant. Each factor is now expanded as a power series in the small parameter β:

$$\exp\Big(\tfrac{\beta}{2N} \big(\operatorname{tr} U_p + \operatorname{tr} U_p^\dagger \big) \Big) \;=\; 1 \;+\; \frac{\beta}{2N} \big(\operatorname{tr} U_p + \operatorname{tr} U_p^\dagger \big) \;+\; \mathcal{O}(\beta^2). \tag{43.103}$$

This is the lattice analogue of the high-temperature expansion of statistical mechanics: β plays the role of inverse temperature, each plaquette is a small "bond," and the expectation value of any observable becomes a sum of geometric configurations of activated plaquettes, weighted by powers of β. (The systematic resummation organizes Eq. (43.103) not in powers of $\operatorname{tr} U_p$ but in the irreducible

characters of SU(N), the *character expansion*; at the leading order we need, the two coincide, the fundamental character being $\chi_{\mathbf{N}}(U_p) = \operatorname{tr} U_p$, and we use the simpler form Eq. (43.103).) The full expectation value is

$$\langle \mathcal{O} \rangle = \frac{1}{Z} \int \Big(\prod_l \mathrm{d}U_l \Big)\, \mathcal{O} \prod_p \Big[1 + \tfrac{\beta}{2N} \big(\operatorname{tr} U_p + \operatorname{tr} U_p^\dagger \big) + \mathcal{O}(\beta^2) \Big], \tag{43.104}$$

with Z the same integral with $\mathcal{O} \to 1$. Expanding the product over plaquettes, each term is a choice of a finite set of plaquettes whose $\operatorname{tr} U_p$ or $\operatorname{tr} U_p^\dagger$ factors are "pulled down," and the link integrals are then performed with Theorem 43.23. By the pairing rule, only those terms survive in which every link carries a matched $U/U^\dagger$ pair. We now apply this to the Wilson loop.

43.4.3 The area law from minimal-area tiling

The Wilson loop on the lattice is the normalized trace of the ordered product of link variables around a closed contour $\mathcal{C}$ (the discretization of Eq. (25.139)):

$$W(\mathcal{C}) = \frac{1}{N} \operatorname{tr} \prod_{l \in \mathcal{C}} U_l. \tag{43.105}$$

We take $\mathcal{C}$ to be a planar $R \times T$ rectangle, with R/a links along the spatial direction and T/a links along the temporal direction. Each of its $2(R+T)/a$ boundary links carries exactly one un-conjugated link variable from Eq. (43.105). By Eq. (43.93) every one of those integrals vanishes on its own; to rescue them we must supply, from the plaquette expansion Eq. (43.103), a conjugate variable on each boundary link. Each plaquette pulled down to do so carries three *further* link variables, which themselves vanish under Eq. (43.93) unless paired, requiring yet more plaquettes, and so on. The chain of forced pairings terminates only when the activated plaquettes tile a connected surface Σ whose boundary is exactly $\mathcal{C}$. Each tile costs one power of $\beta/2N$; the leading, lowest-order term is therefore the one with the *fewest* plaquettes, i.e. the *minimal* spanning surface. For a planar rectangle that is the flat rectangle itself, tiled by

$$n_p = \frac{\mathcal{A}}{a^2} = \frac{RT}{a^2} \tag{43.106}$$

elementary plaquettes (Fig. 43.4).

It remains to evaluate the corresponding integral exactly. We do the single-tile case first, in full, and then assemble the general rectangle by a colour-index count.

One plaquette. Take $\mathcal{C}$ to be the boundary of a single plaquette, so $W = \frac{1}{N} \operatorname{tr} U_p$ with $U_p = U_1 U_2 U_3^\dagger U_4^\dagger$ (links $1, 2$ traversed forward, $3, 4$ backward). The

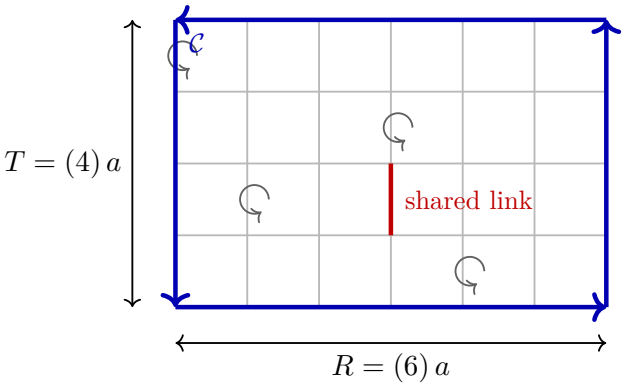


Figure 43.4: The minimal surface Σ spanning a rectangular Wilson loop $\mathcal{C}$ (blue, oriented), tiled by $n_p = RT/a^2$ elementary plaquettes (grey). Each boundary link carries one link variable from $W(\mathcal{C})$, paired by the adjacent plaquette; each interior link (red) is shared by two plaquettes whose orientations are opposite, so it carries a matched $U/U^\dagger$ pair. Every link of Σ is thereby saturated, and the leading strong-coupling weight is the product over tiles.

only surviving term of the expansion pairs each of the four loop variables with a conjugate, which the action supplies through the single factor $\frac{\beta}{2N}\operatorname{tr} U_p^\dagger$:

$$\langle W\rangle = \frac{1}{N}\cdot\frac{\beta}{2N}\int\Big(\prod_{m=1}^{4}\mathrm{d}U_m\Big)(\operatorname{tr}U_p)(\operatorname{tr}U_p^\dagger) + \mathcal{O}(\beta^2). \tag{43.107}$$

Write the two traces in indices,

$$\operatorname{tr}U_p = (U_1)_{i_1 i_2}(U_2)_{i_2 i_3}(U_3^\dagger)_{i_3 i_4}(U_4^\dagger)_{i_4 i_1}, \tag{43.108}$$

$$\operatorname{tr}U_p^\dagger = (U_4)_{j_1 j_2}(U_3)_{j_2 j_3}(U_2^\dagger)_{j_3 j_4}(U_1^\dagger)_{j_4 j_1}, \tag{43.109}$$

and integrate each link with Eq. (43.94). Link 1 carries $(U_1)_{i_1 i_2}(U_1^\dagger)_{j_4 j_1} \to \frac{1}{N}\delta_{i_1 j_1}\delta_{i_2 j_4}$; link 2 carries $(U_2)_{i_2 i_3}(U_2^\dagger)_{j_3 j_4} \to \frac{1}{N}\delta_{i_2 j_4}\delta_{i_3 j_3}$; link 3 carries $(U_3)_{j_2 j_3}(U_3^\dagger)_{i_3 i_4} \to \frac{1}{N}\delta_{j_2 i_4}\delta_{j_3 i_3}$; link 4 carries $(U_4)_{j_1 j_2}(U_4^\dagger)_{i_4 i_1} \to \frac{1}{N}\delta_{j_1 i_1}\delta_{j_2 i_4}$. The four factors of $1/N$ give $(1/N)^4$. The deltas impose $j_1 = i_1$, $j_4 = i_2$, $j_3 = i_3$, $j_2 = i_4$, after which i_1, i_2, i_3, i_4 run freely from 1 to N; summing gives N^4. Hence

$$\int\prod_m \mathrm{d}U_m\,(\operatorname{tr}U_p)(\operatorname{tr}U_p^\dagger) = \Big(\frac{1}{N}\Big)^4 N^4 = 1, \tag{43.110}$$

and therefore, for the single plaquette,

$$\langle W\rangle = \frac{1}{N}\cdot\frac{\beta}{2N}\cdot 1 = \frac{\beta}{2N^2} + \mathcal{O}(\beta^2). \tag{43.111}$$

Each tile contributes the factor $\beta/2N^2$. The colour bookkeeping that produced $N^4 = N^{\#\text{sites}}$ from the four index loops — one circling each corner of the plaquette — is the seed of the general count.

The full rectangle. For the rectangle of Fig. 43.4 the leading surviving term activates every one of the n_p tiles once (with the orientation that pairs interior links). Three facts assemble the answer.

(i) Plaquette factors. Each activated tile brings one factor $\beta/2N$ from Eq. (43.103), giving $(\beta/2N)^{n_p}$.

(ii) Link factors. Every link of Σ — boundary or interior — ends up carrying exactly one matched $U/U^\dagger$ pair: a boundary link is paired between the loop W and its single adjacent tile; an interior link is shared by two tiles whose opposite orientations supply a U from one and a $U^\dagger$ from the other (Fig. 43.4). Each link integral therefore yields a factor $1/N$ and a pair of Kronecker deltas, by Eq. (43.94). Writing L for the number of links of Σ, this gives $(1/N)^L$.

(iii) Colour-index loops. The Kronecker deltas, contracted through the whole network, organize the colour indices into closed loops, each contributing a free sum $\sum_{i=1}^{N} 1 = N$. In the double-line (ribbon) bookkeeping of these contractions, the closed index loops are in one-to-one correspondence with the *sites* of Σ: at each site, the index lines of the links meeting there close into a single small loop encircling the site. (The one-plaquette computation verifies the correspondence directly: four sites, four index loops, factor N^4.) Writing V for the number of sites of Σ, the index loops give N^V, including the overall $1/N$ from the normalization of W in Eq. (43.105) which we account for separately below.

Collecting (i)–(iii) and the loop normalization,

$$\langle W(\mathcal{C})\rangle = \frac{1}{N}\left(\frac{\beta}{2N}\right)^{n_p}\left(\frac{1}{N}\right)^{L} N^{V} = \frac{\beta^{\,n_p}}{2^{\,n_p}}\, N^{\,V-L-n_p-1}. \tag{43.112}$$

Now use the Euler characteristic of the tiled disk. A rectangle of $m \times n$ plaquettes ($m = R/a$, $n = T/a$, $n_p = mn$) has $V = (m{+}1)(n{+}1)$ sites and $L = m(n{+}1) + (m{+}1)n = 2mn + m + n$ links, so

$$V - L + n_p = (mn{+}m{+}n{+}1) - (2mn{+}m{+}n) + mn = 1, \tag{43.113}$$

the Euler characteristic of a disk, $\chi = 1$. Hence $V - L = 1 - n_p$, and the exponent of N in Eq. (43.112) is $V - L - n_p - 1 = (1 - n_p) - n_p - 1 = -2n_p$. The two factors of $1/N$ per tile (one from the link integrals, one from the index-loop deficit relative to the plaquette count) combine, and we obtain the clean result

$$\boxed{\langle W(\mathcal{C})\rangle \;=\; \left(\frac{\beta}{2N^2}\right)^{n_p}\!\left[1+\mathcal{O}(\beta)\right] \;=\; \left(\frac{\beta}{2N^2}\right)^{\mathcal{A}/a^2}\!\left[1+\mathcal{O}(\beta)\right].} \tag{43.114}$$

The Euler-characteristic identity Eq. (43.113) is precisely what makes the single-plaquette factor $\beta/2N^2$ exponentiate cleanly into the area, with no leftover boundary or normalization powers of N.

Theorem 43.24 (Strong-coupling area law)**.** *For lattice* SU(N) *gauge theory with the Wilson action Eq. (43.101) at small inverse coupling β, the expectation value*

of a large planar rectangular Wilson loop obeys an area law (Theorem 27.18),

$$\langle W(\mathcal{C})\rangle = e^{-\sigma\,\mathcal{A}}\big[1+\mathcal{O}(\beta)\big], \qquad \sigma a^2 = -\ln\frac{\beta}{2N^2} = \ln\frac{2N^2}{\beta}, \tag{43.115}$$

with $\mathcal{A} = RT$ the enclosed area and a strictly positive string tension $\sigma > 0$ for $\beta < 2N^2$. In particular the lattice theory confines, for every N and in every spacetime dimension $d \geq 2$.

Proof. Rewrite Eq. (43.114) as $\langle W\rangle = \exp\big[\, n_p \ln(\beta/2N^2)\,\big] = \exp\big[-(\mathcal{A}/a^2)\ln(2N^2/\beta)\big]$, using Eq. (43.106). Comparison with the area-law form Eq. (27.175) identifies $\sigma a^2 = \ln(2N^2/\beta)$, which is positive exactly when $\beta < 2N^2$, i.e. for sufficiently strong bare coupling. The derivation used only the pairing rule Theorem 43.23 and the combinatorics of tiling a disk, neither of which referred to the dimension or to the number of colours beyond $N \geq 2$. □

43.4.4 The string tension and the linear potential

Theorem 43.24 is the analytic heart of the chapter, but its physical meaning is delivered only by reconnecting it to the static potential. That bridge was built in Chapter 27: the rectangular Wilson loop with a long temporal extent T measures the energy of a static quark–antiquark pair held at spatial separation R. We restate the link in the lattice language of Section 43.1.4, where it is exact rather than formal.

The Euclidean lattice has a positive, self-adjoint *transfer matrix* $\hat{\mathcal{T}} = e^{-a\hat{H}}$ (constructed in Section 43.1.4 from reflection positivity of the Wilson action), with $\hat{H}$ the lattice Hamiltonian. A correlation function separated by $T = N_t a$ lattice steps in Euclidean time is the matrix element of $\hat{\mathcal{T}}^{N_t} = e^{-\hat{H}T}$. Inserting a complete set of eigenstates of $\hat{H}$ *in the presence of the two static colour sources at separation R* — exactly the spectral decomposition carried out in Theorem 27.19 — the rectangular Wilson loop has the exact representation

$$\langle W(\mathcal{C}_{R,T})\rangle = \sum_{n} |c_n|^2\; e^{-E_n(R)\,T}, \qquad V(R) \equiv E_0(R) = -\lim_{T\to\infty}\frac{1}{T}\ln\langle W(\mathcal{C}_{R,T})\rangle\,, \tag{43.116}$$

where $E_0(R) = V(R)$ is the static quark–antiquark potential (the ground-state energy of the gauge field bound to the sources) and the higher $E_n(R)$ are excited flux-tube states. This is Eq. (27.183); nothing in it was perturbative.

Now feed the strong-coupling result Eq. (43.115) into Eq. (43.116). The area law gives, at leading order, $\langle W(\mathcal{C}_{R,T})\rangle = e^{-\sigma RT}$, so

$$V(R) = -\lim_{T\to\infty}\frac{1}{T}\ln e^{-\sigma RT} = \sigma R. \tag{43.117}$$

The potential is linear, the force $-dV/dR = -\sigma$ is constant and never vanishes, and the energy to separate the pair grows without bound: a single colour charge

cannot be isolated. This is confinement, in the sharp sense of Theorem 27.20, and it has now been *derived* — not assumed, modelled, or imported as a criterion — directly from the lattice path integral. The criterion of Section 27.7 (area law $\Leftrightarrow$ linear confinement) is satisfied, with the string tension supplied explicitly by Eq. (43.115).

Worked Example 43.25 (Leading string tension and linear potential for SU(N))**.** We extract the dimensionful string tension and the potential for general N, then specialize to SU(3).

String tension. From Eq. (43.115), with $\beta = 2N/g^2$ so that $2N^2/\beta = Ng^2$,

$$\sigma a^2 = \ln\frac{2N^2}{\beta} = \ln(Ng^2) + \mathcal{O}(\beta^2), \tag{43.118}$$

in exact agreement with the leading lattice result quoted in Chapter 27 (Eq. (27.192)). For SU(3) at, say, $\beta = 1$ (deep in the strong-coupling regime), $\sigma a^2 = \ln(2\cdot 9/1) = \ln 18 \approx 2.89$, so the correlation length $1/\sqrt{\sigma} \approx 0.59\,a$ is well under one lattice spacing — a self-consistency check that the expansion is being used where it converges (the flux tube is fat in lattice units only near the continuum, at large β).

Linear potential and its slope. By Eq. (43.117) the static potential at leading order is purely linear,

$$V(R) = \sigma R = \frac{\ln(2N^2/\beta)}{a^2}\,R, \tag{43.119}$$

with no Coulombic $1/R$ piece at this order — the one-gluon-exchange tail of the Cornell potential Eq. (27.186) is a *weak*-coupling effect, recovered only by summing the strong-coupling series or, in practice, by approaching the continuum. The number of flux quanta the loop encloses is $\sigma\mathcal{A}$ in natural units, and the energy stored in a tube of length R is σR: a string of constant tension σ. The first excited state in Eq. (43.116), $E_1(R) - E_0(R)$, is the gap to a deformed flux tube; in the strong-coupling expansion it corresponds to the cheapest *non-minimal* tiling (a tile bumped out of the plane), suppressed by extra powers of β, which is why Eq. (43.116) is dominated by its ground state and the limit defining $V(R)$ converges exponentially fast in T.

Why every N confines here. Nothing in Eq. (43.118) singled out $N = 3$, or four dimensions, or even non-abelian structure: the same computation with U(1) link variables $U = e^{i\theta}$ gives $\int \frac{d\theta}{2\pi} e^{i\theta}e^{-i\theta} = 1$ and $\int \frac{d\theta}{2\pi} e^{i\theta} = 0$, the abelian instances of Theorem 43.23, and reproduces $\sigma a^2 = \ln(2/\beta)$ (here $N \to 1$, $2N^2 \to 2$). The strong-coupling lattice confines abelian and non-abelian gauge theories alike. That this is *too* good — compact U(1) is *not* confining in the continuum — is the warning we take up next.

43.4.5 Convergence and the leading corrections

Two points sharpen the status of Theorem 43.24. First, the strong-coupling expansion is genuinely *convergent*, not merely asymptotic. Like the high-temperature expansion of a lattice spin system, it is a power series in β whose coefficients count geometric configurations of activated plaquettes; the number of surfaces of a given area grows only geometrically (a connected surface of n_p plaquettes can be built in at most C^{n_p} ways for some lattice-dependent constant C), while each carries β^{n_p}, so the series has a finite radius of convergence $\beta_0 > 0$. For $\beta < \beta_0$ the area law, including the existence of a positive string tension, is a rigorous theorem about the lattice theory, with controlled error bounds. Confinement at strong coupling is proved, not conjectured.

Second, the corrections to Eq. (43.114) are themselves instructive. The $\mathcal{O}(\beta)$ terms come from (a) decorating the minimal surface with extra plaquettes that pair among themselves (vacuum fluctuations renormalizing the tension) and (b) replacing the flat minimal surface by a slightly deformed one of larger area — a "bumped" tiling. Resumming the deformations dresses the rigid tile sum into a fluctuating surface and yields, beyond leading order, the corrected tension

$$\sigma a^2 = \ln\frac{2N^2}{\beta} - \frac{1}{N^2}\left(\frac{\beta}{2N^2}\right)^4 + \cdots , \tag{43.120}$$

the first correction arising from the smallest deformation (a single displaced plaquette costs four extra tiles). The qualitative lesson survives all orders within the radius of convergence: σ remains positive and the area law remains intact. The flux tube acquires a finite width and transverse fluctuations — the seeds of the effective string description and of the roughening phenomenon that becomes important as β is increased toward the continuum.

43.4.6 Does it survive to the continuum? Honest scope

We must now be exact about what Theorem 43.24 does and does not establish, because this is the precise point at which a careful account of confinement diverges from a careless one. The theorem is a statement about the lattice theory at *large bare coupling*, i.e. small β and coarse lattice spacing. Physics, however, lives in the continuum limit $a \to 0$ taken at fixed physical quantities — fixed $\sqrt{\sigma}$, fixed hadron masses. The lattice is a regulator, and the question of confinement *in the continuum theory* is the question of whether the area law survives that limit. It need not.

Asymptotic freedom puts the continuum at weak coupling. The continuum limit is reached not at any value of β but at the renormalization-group fixed point that defines the theory. For an asymptotically free theory that fixed point is at *vanishing* bare coupling: as $a \to 0$, the bare coupling at the

cutoff scale $1/a$ must run to zero, $g^2(a) \to 0$, which is $\beta = 2N/g^2 \to \infty$ — the *weak*-coupling regime, the opposite end of the axis from where Theorem 43.24 holds. The required scaling is dictated by the Yang–Mills beta function of Chapter 27 (Eq. (27.101)): integrating $a\,dg/da = -\beta_{\mathrm{YM}}(g)$ near $g = 0$ gives the asymptotic-scaling law

$$\sigma a^2 \;\propto\; \exp\!\Big[-\frac{1}{b_0\, g^2(a)}\Big] \;\xrightarrow{a\to 0}\; 0, \qquad b_0 = \frac{1}{16\pi^2}\Big(\tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f\Big), \quad (43.121)$$

the essential singularity that makes $\sqrt{\sigma} \sim \Lambda_{\mathrm{QCD}}$ a genuine continuum quantity exponentially below the cutoff (dimensional transmutation, Section 27.6). The strong-coupling formula $\sigma a^2 = \ln(2N^2/\beta)$ of Eq. (43.115) looks nothing like Eq. (43.121): it is a lattice artefact, large precisely where the lattice is coarse. For the lattice to define a confining *continuum* theory, the measured σa^2 must cross over from the strong-coupling logarithm to the weak-coupling exponential Eq. (43.121) *without* the string tension ever passing through zero.

The nontrivial requirement: no bulk phase transition. A vanishing σ at some intermediate β would be a *deconfining (bulk) phase transition* separating a strong-coupling confined phase from a weak-coupling deconfined one. If such a transition existed, the analytic area law of Theorem 43.24 would describe only a lattice phase with no continuum counterpart, and the continuum theory — approached from the weak-coupling side — could perfectly well be deconfined. The entire force of the strong-coupling argument for *continuum* confinement therefore rests on a single statement that the strong-coupling expansion itself cannot supply: that SU(N) Yang–Mills has no bulk phase transition along the β axis, so that strong and weak coupling are analytically connected and σa^2 interpolates smoothly between the two scaling forms.

Compact U(1) shows the requirement has teeth. That this is a real gap, and not a technicality, is proved by a theory in which the requirement *fails*. Compact U(1) lattice gauge theory in four dimensions confines at strong coupling by the very computation of Theorem 43.25 (its abelian Haar integrals are the $N \to 1$ case of Theorem 43.23), yet it is known — analytically by the duality to a monopole-gas/Coulomb-gas representation, and numerically — to undergo a phase transition at a critical coupling β_c. For $\beta < \beta_c$ the photon is confined (area law, a plasma of condensed magnetic monopoles disorders the loop); for $\beta > \beta_c$ the monopoles are dilute, the area law gives way to a perimeter law, and the continuum limit taken from the weak-coupling side is the *deconfined Coulomb phase*: free, massless photons — ordinary electrodynamics, which manifestly does not confine. The strong-coupling area law is therefore demonstrably *not* a guarantee of continuum confinement. Any argument that confines SU(3) must distinguish it from compact U(1), and the strong-coupling expansion alone cannot: it confines both.

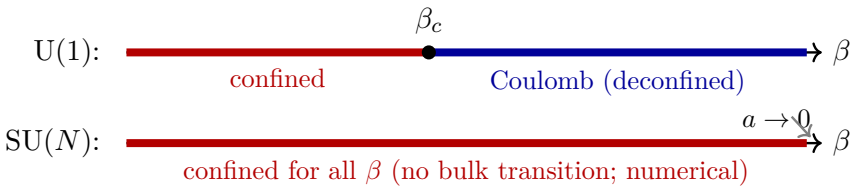


Figure 43.5: The coupling axis. *Top:* compact U(1) in four dimensions has a bulk transition at β_c: the strong-coupling area law (red) does not reach the weak-coupling continuum, which is the deconfined Coulomb phase (blue). *Bottom:* for SU(N) the strong-coupling area law of Theorem 43.24 is believed — on the basis of Monte Carlo evidence, not proof — to connect smoothly to the weak-coupling continuum ($\beta \to \infty$, $a \to 0$) with no intervening transition, so that confinement persists. The contrast is what makes the SU(N) statement nontrivial.

Where the SU(N) statement actually stands. For SU(N) Yang–Mills the required smooth connection appears to hold: large-scale Monte Carlo simulations (Section 43.4.6 anticipates the methods of Section 43.7) compute $\langle W(\mathcal{C})\rangle$ at couplings ranging continuously from strong to weak and find (i) no bulk deconfining transition along the β axis at zero temperature; (ii) the measured σa^2 crossing over from the strong-coupling logarithm to the asymptotic-scaling exponential Eq. (43.121), the signature of a true continuum quantity; and (iii) a physical string tension $\sqrt{\sigma} \approx 440\,\text{MeV}$ consistent with the Regge slope of the hadron spectrum and with quarkonium phenomenology (Eq. (27.193)). This is overwhelming evidence, but it is numerical. The statement that four-dimensional SU(N) Yang–Mills exists as a non-perturbative quantum field theory with a positive mass gap, and confines in the continuum, remains *unproved*: it is one of the Clay Mathematics Institute Millennium Prize Problems and is open as of the mid-2020s, the same honest boundary drawn at the end of Chapter 27. What strong coupling supplies is an existence proof in a regime — a clean, convergent, dimension-independent demonstration that an area law and a linear potential are genuine features of a non-perturbatively defined gauge theory. What it cannot supply, and what no analytic method yet supplies, is the uniform control from lattice to continuum that would close the gap.

Caution

The area law is the statement of confinement only for *static external sources in pure gauge theory* (or the quenched approximation). With light dynamical quarks the chromoelectric string breaks once stretched far enough to pop a $q\bar{q}$ pair from the vacuum, the potential flattens beyond $r_{\text{break}} \sim 1.2$–$1.5\,\text{fm}$, and the large-loop behaviour reverts to a perimeter law. The precise area-law confinement of Theorem 43.24 is thus a property

of the gluonic sector; the physical statement in full QCD is the weaker, observed fact that only colour singlets appear in the spectrum. Including those dynamical quarks is the problem of the next section.

We have now computed a complete gauge sector from first principles: the gauge-invariant lattice measure of Section 43.3 has yielded, by explicit Haar integration and surface combinatorics, an area law, a positive string tension, and a linear confining potential — Yang–Mills theory's non-perturbative payoff, with its continuum status framed honestly against the asymptotic freedom of Chapter 27. The temporal-Wilson-loop and transfer-matrix machinery assembled in Eq. (43.116) is also exactly what Section 43.7.2 will turn into lattice spectroscopy. What is still missing from this picture is matter: the moment we try to put a Dirac field on the lattice, the discretization fights back, and the fermion-doubling problem of Section 43.5 forces genuinely new constructions, to which we turn next.

43.5 Fermion doubling and the Nielsen–Ninomiya no-go theorem

The gauge sector is now complete. Wilson's plaquette action put the connection on links, preserved exact gauge invariance with no gauge fixing, and handed us an area law—the first analytic demonstration of confinement—essentially for free in the strong-coupling expansion of Section 43.4. What remains, to have a lattice definition of QCD rather than of pure glue, is to put *matter* on the lattice: the quarks. We already have everything we need in principle. The Dirac field was quantized in Chapter 11, its Euclidean path integral is a Berezin integral over Grassmann variables (Section 12.3), and integrating those variables out produces the fermion determinant (Theorem 12.26) that weights each gauge configuration. One would expect discretizing the Dirac action to be no harder than discretizing the scalar of Section 43.2: replace derivatives by finite differences and be done.

It is not. The naive discretization is a disaster of a very specific and instructive kind. A single continuum Dirac field, discretized by the obvious symmetric difference, describes not one fermion on the lattice but $2^d = 16$ of them in four dimensions. These extra species—the *doublers*—are not a coding error or a poor choice of difference scheme that a cleverer one would remove. They are forced. The Nielsen–Ninomiya theorem will show that any lattice fermion action that is simultaneously local, translation-invariant, hermitian, and *exactly chirally symmetric* must produce doublers in chirality-balanced pairs, so that no such action can describe a single chiral (Weyl) fermion. The obstruction is topological, and—this is the deep point—it is the lattice's faithful bookkeeping of the chiral anomaly of Chapter 36. The very feature that makes the lattice a clean gauge

regulator (it is a regulator, not a continuum trick, and it respects every exact symmetry of the action) is what forbids it from quietly representing a chiral theory. This section discretizes the Dirac action, exhibits the sixteen doublers explicitly with their chiralities, explains their anomaly-cancelling role, and states and proves the topological core of Nielsen–Ninomiya.

Throughout we work in the Euclidean formulation of the lattice path integral set up in Section 43.1, descending from the Wick rotation $t \to -i\tau$ of Chapter 10. Fermions enter this section for the first time in the chapter, so we begin by fixing the Euclidean Dirac conventions.

Notation Watch

On the Euclidean lattice the gamma matrices are taken *hermitian* and satisfy the Euclidean Clifford algebra

$$\{\gamma_\mu, \gamma_\nu\} = 2\,\delta_{\mu\nu}\,\mathbf{1}, \qquad \gamma_\mu^\dagger = \gamma_\mu, \qquad \mu, \nu = 1, \ldots, d, \tag{43.122}$$

with all indices down and the Kronecker $\delta_{\mu\nu}$ in place of the Minkowski $\eta_{\mu\nu}$. They are obtained from the mostly-minus matrices of Appendix C and Chapter 11 by $\gamma_d^{\mathrm{E}} = \gamma^0$, $\gamma_j^{\mathrm{E}} = -i\gamma^j$ $(j = 1, \ldots, d-1)$; one checks Eq. (43.122) directly from $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. The chirality matrix is $\gamma_5 = \gamma_1\gamma_2\gamma_3\gamma_4$ in $d = 4$, which is hermitian, squares to $\mathbf{1}$, and anticommutes with every γ_μ. We drop the superscript "E" from here on; within this section γ_μ always means the Euclidean matrix. The spinor field $\bar\psi$ is treated as an independent Grassmann variable, not as $\psi^\dagger\gamma^0$—the standard and necessary convention in the Euclidean functional integral (Section 12.3).

43.5.1 The naive lattice Dirac action

Take a hypercubic lattice of spacing a, with sites labelled by integer vectors $n \in \mathbb{Z}^d$ (so the physical coordinate is $x = an$) and unit lattice vectors $\hat\mu$ in each of the d directions. Place a Grassmann-valued Dirac spinor ψ_n and an independent conjugate $\bar\psi_n$ at every site. The continuum Euclidean Dirac action $S_F = \int d^dx\, \bar\psi(\gamma_\mu\partial_\mu + m)\psi$ (the Euclidean rotation of the Lagrangian of Section 11.1.4) requires a lattice stand-in for ∂_μ.

The first derivative has two one-sided lattice realizations, the forward and backward differences $(\psi_{n+\hat\mu} - \psi_n)/a$ and $(\psi_n - \psi_{n-\hat\mu})/a$. Neither is a hermitian operation on the lattice, and a non-hermitian kinetic term would spoil the reality of the Euclidean action and the positivity of the transfer matrix. The unique antisymmetric—hence, paired with the i that the first-order Dirac operator carries, hermitian—first difference is their average, the *symmetric difference*

$$(\partial_\mu\psi)_n \;\longmapsto\; \frac{\psi_{n+\hat\mu} - \psi_{n-\hat\mu}}{2a}. \tag{43.123}$$

This is the obvious, "naive" choice: it is the most symmetric local approximation to ∂_μ, accurate to $O(a^2)$ just as the symmetric Laplacian was for the scalar. The corresponding *naive lattice Dirac action* is

$$S_F^{\text{naive}} = a^d \sum_n \left[\bar\psi_n \sum_{\mu=1}^{d} \gamma_\mu \frac{\psi_{n+\hat\mu} - \psi_{n-\hat\mu}}{2a} \;+\; m\,\bar\psi_n \psi_n \right]. \tag{43.124}$$

The prefactor a^d is the lattice cell volume that turns $\sum_n$ into the discretized $\int d^d x$. The hopping term connects nearest neighbours only: Eq. (43.124) is manifestly *ultralocal* (finite range), translation-invariant, and—because the symmetric difference is antisymmetric under $n \leftrightarrow n \pm \hat\mu$ while γ_μ is hermitian—hermitian. It is also, crucially, *exactly chirally symmetric* when $m = 0$: under $\psi_n \to e^{i\alpha\gamma_5}\psi_n$, $\bar\psi_n \to \bar\psi_n e^{i\alpha\gamma_5}$, each $\bar\psi_n \gamma_\mu (\cdots)$ bilinear is invariant because γ_5 anticommutes with γ_μ (the two factors of $e^{i\alpha\gamma_5}$ combine to $e^{i\alpha\gamma_5}\gamma_\mu e^{i\alpha\gamma_5} = \gamma_\mu e^{-i\alpha\gamma_5} e^{i\alpha\gamma_5} = \gamma_\mu$). These four properties—locality, translation invariance, hermiticity, exact chiral symmetry—are exactly the hypotheses of Nielsen–Ninomiya, and Eq. (43.124) satisfies all of them. The theorem will tell us that it must therefore pay a price, and the price is the doublers we now expose.

43.5.2 The momentum-space propagator and its poles

Diagonalize Eq. (43.124) by Fourier transform. Because the lattice is discrete, momenta live on the *Brillouin torus*

$$p_\mu \in \Big(-\frac{\pi}{a}, \frac{\pi}{a}\Big], \qquad \text{BZ} = \Big(-\tfrac{\pi}{a}, \tfrac{\pi}{a}\Big]^d \cong T^d, \tag{43.125}$$

a compact set with opposite faces identified—a d-torus, the single most important geometric fact in this section. Expand

$$\psi_n = \int_{\text{BZ}} \frac{d^d p}{(2\pi)^d}\, e^{\,i a\, n\cdot p}\, \tilde\psi(p), \qquad \bar\psi_n = \int_{\text{BZ}} \frac{d^d p}{(2\pi)^d}\, e^{-i a\, n\cdot p}\, \tilde{\bar\psi}(p), \tag{43.126}$$

with the lattice completeness relation $a^d \sum_n e^{i a\, n\cdot(p-p')} = (2\pi)^d\, \delta^d(p - p')$ for $p, p' \in \text{BZ}$. Shifting a site by $\hat\mu$ multiplies the plane wave by a phase, $\psi_{n\pm\hat\mu} \leftrightarrow e^{\pm i a p_\mu} \tilde\psi(p)$, so the symmetric difference becomes

$$\frac{\psi_{n+\hat\mu} - \psi_{n-\hat\mu}}{2a} \;\longleftrightarrow\; \frac{e^{\,i a p_\mu} - e^{-i a p_\mu}}{2a}\, \tilde\psi(p) = \frac{i \sin(a p_\mu)}{a}\, \tilde\psi(p). \tag{43.127}$$

Substituting Eqs. (43.126) and (43.127) into Eq. (43.124) and using completeness to collapse one momentum integral, the action becomes diagonal in p,

$$S_F^{\text{naive}} = \int_{\text{BZ}} \frac{d^d p}{(2\pi)^d}\, \tilde{\bar\psi}(p)\, \widetilde{D}(p)\, \tilde\psi(p), \tag{43.128}$$

with the *naive lattice Dirac operator* in momentum space

$$\boxed{\widetilde{D}(p) = \frac{i}{a}\sum_{\mu=1}^{d}\gamma_\mu \sin(ap_\mu) + m\,\mathbf{1}\,.} \tag{43.129}$$

This is the central object. As $a \to 0$ at fixed physical p, $\sin(ap_\mu)/a \to p_\mu$ and $\widetilde{D}(p) \to i\gamma_\mu p_\mu + m$, the correct Euclidean continuum inverse propagator: the naive action has the right *naive* continuum limit, which is precisely why the doubling is so insidious. Inverting Eq. (43.129) uses the Clifford algebra Eq. (43.122). Writing $s_\mu \equiv \sin(ap_\mu)/a$ and multiplying numerator and denominator by $(m - i\gamma_\nu s_\nu)$,

$$\Big(i\gamma_\mu s_\mu + m\Big)\Big(m - i\gamma_\nu s_\nu\Big) = m^2 + \gamma_\mu\gamma_\nu s_\mu s_\nu = m^2 + \tfrac{1}{2}\{\gamma_\mu,\gamma_\nu\}s_\mu s_\nu = m^2 + \sum_\mu s_\mu^2, \tag{43.130}$$

where the cross terms $im\gamma_\nu s_\nu - im\gamma_\mu s_\mu$ cancel and the symmetric $s_\mu s_\nu$ projects onto $\delta_{\mu\nu}$. The denominator is a c-number, so

$$\widetilde{D}(p)^{-1} = \frac{m - \dfrac{i}{a}\sum_\mu \gamma_\mu \sin(ap_\mu)}{m^2 + \dfrac{1}{a^2}\displaystyle\sum_\mu \sin^2(ap_\mu)}. \tag{43.131}$$

The propagator's poles—the propagating particles of the theory—sit where its denominator vanishes. In the massless ($m = 0$) case relevant to chirality, this requires

$$\sum_\mu \sin^2(ap_\mu) = 0 \quad\Longleftrightarrow\quad \sin(ap_\mu) = 0 \;\text{ for } \textit{every}\ \mu, \tag{43.132}$$

since each term is non-negative. Within the Brillouin zone Eq. (43.125), $\sin(ap_\mu) = 0$ has not one but *two* solutions per direction: $p_\mu = 0$ and $p_\mu = \pi/a$. The continuum operator $i\gamma_\mu p_\mu$ vanished only at $p = 0$; the lattice sine, being periodic, vanishes again at the zone boundary. Figure 43.6 (left) shows this in one dimension: the line $f = p$ crosses zero once, the curve $f = \sin(ap)/a$ crosses zero twice inside the zone.

43.5.3 Counting and identifying the doublers

Definition 43.26 (Doublers)**.** The *doublers* of the naive lattice fermion are the propagator poles of Eq. (43.131) (at $m = 0$) located at momenta other than the physical $p = 0$. Each pole is a separate propagating Dirac species in the continuum limit.

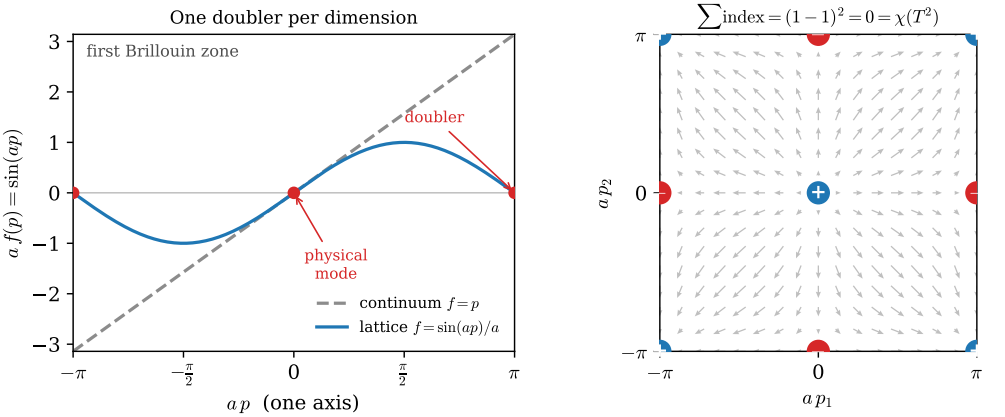


Figure 43.6: *Left:* the naive fermion dispersion $f(p) = \sin(ap)/a$ along one axis over the first Brillouin zone. Unlike the continuum line $f = p$ (which vanishes only at $p = 0$), the lattice sine vanishes a *second* time at the zone boundary $ap = \pi$: one physical mode and one doubler per dimension. *Right:* the $d = 2$ Brillouin torus with the $2^2 = 4$ zeros of $V(p) = (\sin(ap_1)/a, \sin(ap_2)/a)$. The Poincaré–Hopf index at each zero, $\operatorname{sign}\det(\partial V_\mu/\partial p_\nu) = \prod_\mu \cos(ap_\mu) = (-1)^{|A|}$, is the chirality of that species; the signs form a checkerboard summing to $(1-1)^2 = 0 = \chi(T^2)$.

Proposition 43.27 (Sixteen species in four dimensions)**.** *The massless naive lattice Dirac operator Eq. (43.129) has exactly 2^d poles in the Brillouin zone, located at the corner momenta*

$$p^{(A)} = \frac{\pi}{a} \sum_{\mu \in A} \hat{\mu}, \qquad A \subseteq \{1, 2, \ldots, d\}, \tag{43.133}$$

one for each subset A of the d directions. In $d = 4$ there are $2^4 = 16$ species: one physical fermion ($A = \varnothing$) and 15 doublers. Near each pole the theory is a relativistic Dirac fermion, but its chirality alternates: the species at $p^{(A)}$ has chirality $(-1)^{|A|}$ relative to the physical one, giving $2^{d-1} = 8$ left-handed and 8 right-handed species, of net chirality zero.

Proof. The pole condition Eq. (43.132) requires $ap_\mu \in \{0, \pi\}$ independently in each direction, since these are the only zeros of sin in $(-\pi, \pi]$. A choice of $\{0, \pi\}$ in each of d directions is exactly a choice of the subset A of directions carrying π, whence Eq. (43.133) and the count 2^d.

To read off the physics near a pole, expand about $p^{(A)}$ by writing $p = p^{(A)} + k$ with $|ak_\mu| \ll 1$. For a direction $\mu \notin A$ ($ap_\mu^{(A)} = 0$),

$$\frac{1}{a} \sin(ap_\mu^{(A)} + ak_\mu) = \frac{1}{a} \sin(ak_\mu) \simeq k_\mu. \tag{43.134}$$

For $\mu \in A$ ($ap_\mu^{(A)} = \pi$), using $\sin(\pi + \theta) = -\sin\theta$,

$$\frac{1}{a}\sin(\pi + ak_\mu) = -\frac{1}{a}\sin(ak_\mu) \simeq -k_\mu. \tag{43.135}$$

Collecting both cases, the Dirac operator near the pole is, to linear order in k,

$$\widetilde{D}(p^{(A)} + k) \simeq i\sum_\mu s_\mu^{(A)}\,\gamma_\mu\,k_\mu + m, \qquad s_\mu^{(A)} = \begin{cases} +1, & \mu \notin A, \\ -1, & \mu \in A. \end{cases} \tag{43.136}$$

This is again a Dirac operator, but with γ_μ replaced by the rescaled set

$$\gamma'_\mu \equiv s_\mu^{(A)}\,\gamma_\mu, \qquad \{\gamma'_\mu, \gamma'_\nu\} = s_\mu^{(A)} s_\nu^{(A)}\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \tag{43.137}$$

which obeys the *same* Clifford algebra Eq. (43.122) (the signs square away on the diagonal). So each pole is genuinely a relativistic Dirac fermion—the doublers are real particles, not lattice junk. What distinguishes them is chirality. The chirality matrix built from the primed gammas is

$$\gamma'_5 = \gamma'_1\gamma'_2\gamma'_3\gamma'_4 = \Big(\textstyle\prod_\mu s_\mu^{(A)}\Big)\gamma_1\gamma_2\gamma_3\gamma_4 = (-1)^{|A|}\,\gamma_5, \tag{43.138}$$

because exactly $|A|$ of the factors carry a minus sign. The species at $p^{(A)}$ therefore couples to γ_5 with the opposite sign whenever $|A|$ is odd: its left- and right-handed components are interchanged relative to the physical fermion. Counting by parity of $|A|$,

$$\sum_{A \subseteq \{1,\dots,d\}} (-1)^{|A|} = \sum_{k=0}^{d} \binom{d}{k} (-1)^k = (1-1)^d = 0. \tag{43.139}$$

In $d = 4$ this is $\binom{4}{0} - \binom{4}{1} + \binom{4}{2} - \binom{4}{3} + \binom{4}{4} = 1 - 4 + 6 - 4 + 1 = 0$, i.e. $1 + 6 + 1 = 8$ species of one chirality and $4 + 4 = 8$ of the other. The net chirality vanishes. □

The combinatorial identity Eq. (43.139) is the heart of the matter, and it is worth seeing in a single explicit doubler before we lift it to its topological meaning.

Worked Example 43.28 (The chirality of one corner doubler)**.** Take $d = 4$ and the doubler at the single-direction corner $A = \{1\}$, i.e. $p^{(A)} = (\pi/a, 0, 0, 0)$, and show directly that its chirality is opposite to that of the physical fermion at the origin.

Setup. By Eq. (43.136) the sign vector is $s^{(A)} = (-1, +1, +1, +1)$, so near this pole

$$\widetilde{D}(p^{(A)} + k) \simeq i\big(-\gamma_1 k_1 + \gamma_2 k_2 + \gamma_3 k_3 + \gamma_4 k_4\big) + m = i\,\gamma'_\mu k_\mu + m, \tag{43.140}$$

with $\gamma'_1 = -\gamma_1$ and $\gamma'_{2,3,4} = \gamma_{2,3,4}$.

Two ways to see the flipped chirality. First, algebraically: the projectors onto definite chirality for this species are $P'_\pm = \frac{1}{2}(\mathbf{1} \pm \gamma'_5)$ with

$$\gamma'_5 = \gamma'_1\gamma'_2\gamma'_3\gamma'_4 = (-\gamma_1)\gamma_2\gamma_3\gamma_4 = -\gamma_5, \tag{43.141}$$

so $P'_\pm = \frac{1}{2}(\mathbf{1} \mp \gamma_5) = P_\mp$. What the doubler calls "right-handed" is what the physical fermion calls "left-handed."

Second, and more invariantly: the sign flip on the single matrix γ_1 can be realized by a similarity transformation that leaves the Clifford algebra intact. Conjugation by γ_5 flips *every* γ_μ, $\gamma_5\gamma_\mu\gamma_5 = -\gamma_\mu$; to flip only γ_1 we conjugate instead by the unitary $W = \gamma_2\gamma_3\gamma_4$ built from the *untouched* directions. Using Eq. (43.122), $W\gamma_1 W^{-1} = -\gamma_1$ while $W\gamma_j W^{-1} = +\gamma_j$ for $j = 2, 3, 4$ (each γ_j anticommutes with the two other factors in W and commutes with itself, an even number of sign changes for j, an odd number for γ_1). Hence $\gamma'_\mu = W\gamma_\mu W^{-1}$, and the doubler is unitarily a standard Dirac fermion—but the map W is anti-chiral, $W\gamma_5 W^{-1} = -\gamma_5$ (three anticommutations), confirming $\gamma'_5 = -\gamma_5$. Either way the corner $A = \{1\}$ carries chirality $(-1)^{|A|} = -1$, opposite the physical fermion.

Why the net must cancel. The physical fermion ($A = \varnothing$) contributes $+1$; this single doubler contributes -1. Running over all 16 corners reproduces Eq. (43.139): the four single-direction doublers each cancel against the physical mode and against the three-direction doublers, the six two-direction doublers pair up, and the lone four-direction doubler ($p = (\pi/a, \pi/a, \pi/a, \pi/a)$, chirality $(-1)^4 = +1$) restores balance. The lattice does not contain a single chiral fermion; it contains eight of each handedness, exactly cancelling.

43.5.4 Why the doublers are the anomaly

It is tempting to read Theorem 43.27 as a defect of the symmetric difference and to hope a smarter difference scheme removes the unwanted fifteen species while keeping the one. The next subsection proves no such scheme exists under the stated hypotheses. But first it pays to understand *why* nature forbids it, because the reason is one of the most beautiful facts in the subject: the doublers are the lattice's way of being honest about the chiral anomaly of Chapter 36.

Recall the structure of that anomaly. A single massless Dirac fermion coupled to a gauge field has a classically conserved axial current $j_5^\mu = \bar\psi\gamma^\mu\gamma_5\psi$. Quantum mechanically the conservation law fails—the Adler–Bell–Jackiw result of Section 36.2,

$$\partial_\mu j_5^\mu = \frac{1}{16\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma} \quad \text{(per unit charge)}, \tag{43.142}$$

an effect that no regulator can remove because it is a genuine property of the continuum theory. The anomaly is intrinsically tied to chirality: it is the

contribution of a *single* Weyl (definite-chirality) fermion that fails to cancel, and the sign of the anomaly is the sign of the chirality.

Now confront this with the naive lattice action Eq. (43.124). At $m = 0$ it has an *exact*, non-anomalous chiral symmetry: it is invariant under $\psi_n \to e^{i\alpha\gamma_5}\psi_n$, $\bar{\psi}_n \to \bar{\psi}_n e^{i\alpha\gamma_5}$ for any constant α, with no anomaly possible, because the symmetry is a symmetry of a perfectly well-defined finite-dimensional Grassmann integral and the Berezin measure (Section 12.3) is invariant. The lattice is a complete, gauge-invariant regulator; if its chiral symmetry is exact, the axial current it defines is *exactly* conserved. But Eq. (43.142) says a continuum theory of one chiral fermion cannot have an exactly conserved axial current. The only escape—the only way the lattice can host an exact chiral symmetry and still be a legitimate regulator of relativistic fermions—is for its particle content to be anomaly-free, i.e. to have *net zero chirality*. And that is exactly Eq. (43.139). The eight left-handed and eight right-handed species each contribute their anomaly $(-1)^{|A|} \times (\text{ABJ } \epsilon FF)$, and these cancel in pairs, leaving the total axial current conserved, consistent with the exact lattice symmetry.

Physical Insight

Doubling is not a bug; it is anomaly cancellation enforced by a faithful regulator. A regulator that exactly preserves chiral symmetry *cannot* reproduce the chiral anomaly of an odd number of Weyl fermions—and since the anomaly is unavoidable in the continuum, such a regulator can only describe anomaly-free (chirality-balanced) content. The lattice with naive fermions chooses to keep chiral symmetry exact and pays by doubling. Dimensional regularization and Pauli–Villars make the opposite choice: they describe a single fermion and pay by breaking chiral symmetry (the γ_5 problem in $d = 4 - 2\varepsilon$; the chirality-violating PV mass), which is how *they* make room for the anomaly. There is no free lunch, and the lattice's version of the bill is the doublers.

This anomaly viewpoint also reveals the doublers' deeper geometric character. Each massless species is a point in the Brillouin torus where the inverse propagator vanishes; near it the Dirac structure $i\gamma'_\mu k_\mu$ is, in the language of topological band theory, a *Weyl point*—a monopole of Berry curvature in momentum space, with magnetic charge equal to its chirality. Gauss's law on a closed manifold forces the total monopole charge enclosed to vanish: there is no boundary through which net flux can escape the torus. The sum of chiralities over the Brillouin zone is therefore zero *for the same reason that the total magnetic charge inside a closed surface is zero*. We now make this topology precise.

43.5.5 The topological obstruction: Poincaré–Hopf on the Brillouin torus

Strip the massless naive operator Eq. (43.129) to its content. Writing $\widetilde{D}(p) = i\sum_\mu \gamma_\mu V_\mu(p)$, all of the physics of the zeros lives in the real vector field

$$V:\ T^d \longrightarrow \mathbb{R}^d, \qquad V_\mu(p) = \frac{\sin(ap_\mu)}{a}, \tag{43.143}$$

on the Brillouin torus: the species are exactly the zeros of V, and—we will show—the chirality of each species is exactly the topological index of V at that zero. The constraint Eq. (43.139) is then not an accident of sin but the Poincaré–Hopf theorem, which we now recall and apply.

Theorem 43.29 (Poincaré–Hopf index theorem)**.** *Let V be a smooth tangent vector field on a closed (compact, boundaryless) oriented manifold M of dimension d, with only isolated, nondegenerate zeros. At a zero p_* define the* index

$$\operatorname{ind}_{p_*} V = \operatorname{sign}\det\left[\left.\frac{\partial V_\mu}{\partial p_\nu}\right|_{p_*}\right] \in \{+1,-1\}. \tag{43.144}$$

Then the sum of indices over all zeros equals the Euler characteristic of M,

$$\sum_{p_*:\,V(p_*)=0} \operatorname{ind}_{p_*} V = \chi(M). \tag{43.145}$$

The geometric content is that the index counts how many times the normalized field $V/|V|$ wraps the unit sphere S^{d-1} as p circles a small sphere around the zero (the degree of the local Gauss map); summing these local winding numbers over M gives a global topological invariant that cannot depend on the particular field, only on M. We take Theorem 43.29 as an established result of differential topology—its proof (triangulate M, or deform V to the gradient of a Morse function and identify indices with critical-point signatures) is standard and outside the physics this book builds; what we owe is the two facts that turn it into Nielsen–Ninomiya.

Fact 1: the index equals the chirality. For the naive field Eq. (43.143) the Jacobian is diagonal,

$$\frac{\partial V_\mu}{\partial p_\nu} = \delta_{\mu\nu}\cos(ap_\mu), \tag{43.146}$$

so at a corner zero $p^{(A)}$ of Eq. (43.133), where $ap_\mu = \pi$ for $\mu \in A$ and $ap_\mu = 0$ otherwise,

$$\operatorname{ind}_{p^{(A)}} V = \operatorname{sign}\prod_\mu \cos(ap_\mu^{(A)}) = \prod_{\mu\in A}(-1) = (-1)^{|A|}. \tag{43.147}$$

Comparing with Eq. (43.138), the index of the vector field at a zero is *identically* the chirality of the corresponding fermion species. This is not a coincidence of the naive model: for any $\widetilde{D}(p) = i\gamma_\mu V_\mu(p)$ satisfying $\{\gamma_5, \widetilde{D}\} = 0$, the relative sign of γ_5' at a zero is the sign of the determinant of the linearized V, because $\gamma_5' = \det(\partial V/\partial p)/|\det(\partial V/\partial p)| \cdot \gamma_5$ follows from the same rescaling-of-gammas computation Eq. (43.137)–Eq. (43.138) applied to the eigenvalues of the Jacobian. Chirality *is* a Poincaré–Hopf index.

Fact 2: the torus has Euler characteristic zero. The Euler characteristic is multiplicative under products, $\chi(X \times Y) = \chi(X)\chi(Y)$, and the circle has $\chi(S^1) = 0$ (a circle triangulated into V vertices and V edges has $\chi = V - V = 0$; equivalently it admits the nowhere-vanishing tangent field ∂_θ). Since $T^d = (S^1)^{\times d}$,

$$\chi(T^d) = \chi(S^1)^d = 0. \tag{43.148}$$

Putting the two facts into Poincaré–Hopf Eq. (43.145) gives, with no reference to the specific form of V,

$$\sum_{\text{species}} \text{chirality} = \sum_{p_*\colon V(p_*)=0} \operatorname{ind}_{p_*} V = \chi(T^d) = 0. \tag{43.149}$$

The net chirality of the massless lattice fermion's particle content is forced to vanish by the topology of the Brillouin zone alone. For the naive field this reproduces Eq. (43.139) term by term—Fig. 43.6 (right) shows the $d = 2$ case, four zeros with indices $+, -, -, +$ arranged in a checkerboard whose sum is $(1-1)^2 = 0$—but Eq. (43.149) holds for *any* smooth V on the torus, i.e. for any local, translation-invariant, chirally symmetric, hermitian lattice fermion. Changing the difference scheme deforms V, moves the zeros around, perhaps merges or splits them, but can never change the signed total. That invariance is the theorem.

43.5.6 The Nielsen–Ninomiya theorem

We can now state the no-go result precisely. It comes in two equivalent guises: the original Hamiltonian formulation of Nielsen and Ninomiya, phrased in terms of Weyl particles and a conserved charge, and the Euclidean "vector field on the torus" formulation just proved. We give the Euclidean version as the theorem, since it is what Theorem 43.29 establishes, and state the Hamiltonian version as a remark.

Theorem 43.30 (Nielsen–Ninomiya no-go)**.** *Let a free lattice fermion be defined by a bilinear Euclidean action $S = \sum_{n,m} \bar\psi_n D_{nm} \psi_m$ on the hypercubic lattice, with Fourier-transformed Dirac operator $\widetilde{D}(p)$ on the Brillouin torus T^d. Suppose*

(L) Locality: $\widetilde{D}(p)$ *is a continuously differentiable (indeed real-analytic) periodic function on* T^d*; equivalently the real-space kernel* D_{nm} *decays faster than any power of* $|n-m|$ *(e.g. finite range or exponentially bounded).*

(T) Translation invariance: $D_{nm} = D(n-m)$*, so* $\widetilde{D}$ *depends only on* p*.*

(H) Hermiticity: *the action is hermitian, with* γ_5*-hermiticity* $\widetilde{D}(p)^\dagger = \gamma_5\,\widetilde{D}(p)\,\gamma_5$*, guaranteeing real, bounded single-particle energies.*

(C) Exact chiral symmetry: $\{\gamma_5, \widetilde{D}(p)\} = 0$ *for all* p*, so the massless action is invariant under* $\psi \to e^{i\alpha\gamma_5}\psi$*,* $\bar\psi \to \bar\psi\, e^{i\alpha\gamma_5}$*.*

(R) Right relativistic content: $\widetilde{D}$ *has isolated nondegenerate zeros, near each of which* $\widetilde{D}(p_* + k) \simeq i\,\gamma_\mu C^{(*)}_{\mu\nu} k_\nu$ *with* $\det C^{(*)} \neq 0$*, describing a Weyl fermion of definite chirality* $\operatorname{sign}\det C^{(*)}$*; and at least one zero reproduces a standard fermion.*

Then the zeros come in chirality-balanced sets: the number of left-handed Weyl species equals the number of right-handed ones,

$$\sum_{\text{species } j} \chi_j = 0. \tag{43.150}$$

In particular, no *such lattice action describes a single Weyl fermion, nor any chirally asymmetric (anomalous) fermion content. The minimal anomaly-free content has an even number of species; the naive action realizes the maximal* 2^d*.*

Proof (topological core). By (C) and (H), at $m = 0$ the operator anticommutes with γ_5 and is γ_5-hermitian; expanding in the Clifford basis, the only structures consistent with $\{\gamma_5, \widetilde{D}\} = 0$ are the chirality-odd matrices, and (H) forces their coefficients to assemble into $\widetilde{D}(p) = i\sum_\mu \gamma_\mu V_\mu(p)$ with $V_\mu(p)$ real. (For the naive action, $V_\mu = \sin(ap_\mu)/a$.) By (L), V is a smooth field on T^d; by (T), it is single-valued on the torus; by (R), its zeros are isolated and nondegenerate, with the chirality of species j equal to $\operatorname{sign}\det(\partial V/\partial p)|_{p_j} = \operatorname{ind}_{p_j} V$, the identification of Fact 1 (Eq. (43.147)). Poincaré–Hopf, Theorem 43.29, applied to V on the closed manifold T^d, gives $\sum_j \chi_j = \sum_j \operatorname{ind}_{p_j} V = \chi(T^d) = 0$ by Fact 2 (Eq. (43.148)). This is Eq. (43.150). □

Remark 43.31 (Honest scope of this proof). What is proved above is the topological *index-summation* core: under hypotheses (L), (T), (H), (C), (R) the chiralities sum to zero. This is the content is needed here and the content that matters for physics. The original Nielsen–Ninomiya argument [365, 366] is set in the Hamiltonian (transfer-matrix) framework: one assumes a lattice Hamiltonian $H(p)$ that is a smooth periodic function of crystal momentum on the Brillouin torus (locality), hermitian (real energies), with a conserved, locally defined, integer-quantized charge Q commuting with H; the conclusion is that

the low-energy Weyl excitations carrying any given value of Q occur in equal numbers of left- and right-movers. Their proof tracks the Fermi points where energy bands cross zero and shows, by a homotopy/degree argument on the band structure, that left- and right-crossings must balance—the same topology, dressed as the statement that the net Berry-monopole charge over a closed Brillouin zone vanishes (the insight of Section 43.5.4). The full functional-analytic theorem additionally controls the most general local $\widetilde{D}$—fields V with more than d real components, degenerate or non-isolated zeros—by perturbing to the generic case; we do not reproduce that machinery, but Theorem 43.29 is exactly the invariant it computes.

The theorem is sharp about which hypothesis must give way. One cannot abandon locality (L) without destroying the regulator: a non-local action reintroduces, through the back door, the ultraviolet ambiguities the lattice was meant to tame, and spoils the cluster/positivity properties that make the Euclidean measure a probability measure (Section 43.1). One cannot abandon translation invariance (T) or hermiticity (H) without losing momentum conservation or unitarity of the continuum limit. Hypothesis (R) is just the demand that we get relativistic fermions at all. The single negotiable assumption is (C), *exact* chiral symmetry. Every practical lattice fermion is a different answer to the question "how do I weaken (C) the least?"

Caution

The no-go theorem forbids a single *chiral* (Weyl) fermion with exact chiral symmetry; it does not forbid lattice fermions outright, and it does not say chiral gauge theories cannot be discretized—only that doing so naively, with exact $\{\gamma_5, \widetilde{D}\} = 0$, fails. The loophole that Section 43.6 exploits is precisely that (C) can be *deformed* rather than violated: the Ginsparg–Wilson relation replaces $\{\gamma_5, D\} = 0$ by $\{\gamma_5, D\} = aD\gamma_5 D$, which reduces to exact chiral symmetry as $a \to 0$ but evades the index theorem at finite a by allowing a modified, lattice-deformed chiral rotation. That is not a violation of Nielsen–Ninomiya; it is a careful walk around its single load-bearing hypothesis.

This last point sets up the matter stage of QED and Yang–Mills theory at the non-perturbative level. The Dirac field of Chapter 11, and with it the electron of QED and the quarks of Yang–Mills theory, can be placed on the lattice—but only by confronting Theorem 43.30 head on. We make no new perturbative calculation here: the doubling/anomaly structure is a genuinely deeper, non-perturbative layer of the same fermion fields the threads carry, linking the lattice directly to the chiral anomaly of Chapter 36. Having established the precise constraint that any acceptable lattice fermion must evade, we turn in Section 43.6 to the explicit constructions—Wilson's term, which removes the doublers by breaking

chiral symmetry; staggered fermions, which thin the species to four while keeping a remnant symmetry; and the Ginsparg–Wilson realization of an exact *lattice* chiral symmetry through overlap and domain-wall fermions—each one a distinct, principled response to the no-go theorem just proved.

43.6 Lattice fermions: Wilson, staggered, and the Ginsparg–Wilson resolution

The no-go theorem of the previous section is not a defect of one clumsy discretization; it is a theorem. Theorem 43.30 states that no lattice Dirac operator can simultaneously be local, translation-invariant, Hermitian, free of doublers, *and* chirally symmetric in the textbook sense $\{D, \gamma_5\} = 0$. Something on that list must give. The history of lattice fermions is the history of choosing, with eyes open, what to surrender. Wilson surrenders chiral symmetry outright and pays for it with tuning; Kogut and Susskind surrender most of the chiral symmetry but keep a sliver, trading it against a residual species multiplicity; Ginsparg and Wilson found the subtle move that surrenders the precise *algebraic form* of the symmetry while keeping a genuine, exact, continuous chiral invariance of the lattice action — and so evade the theorem without breaking locality. This section develops all three, in that order, because that is the order in which the difficulty is understood: first the brute-force fix, then the clever economy, then the principled resolution that turns the index theorem and the anomaly of Chapter 36 into exact lattice statements.

Throughout we work on the Euclidean lattice set up earlier in this chapter: sites $x \in a\mathbb{Z}^4$, lattice spacing a, gauge fields carried by link variables $U_\mu(x) \in \mathrm{SU}(N)$, and the Euclidean Dirac algebra

$$\{\gamma_\mu, \gamma_\nu\} = 2\delta_{\mu\nu}, \qquad \gamma_\mu^\dagger = \gamma_\mu, \qquad \gamma_5 \equiv \gamma_1\gamma_2\gamma_3\gamma_4 = \gamma_5^\dagger, \qquad \gamma_5^2 = \mathbf{1}, \tag{43.151}$$

with $\{\gamma_5, \gamma_\mu\} = 0$. We write γ_5 with a lowered index throughout this section, as is universal in the lattice literature; it is the Euclidean continuation of the γ^5 of Chapter 11 and Appendix C and carries the same physics. Recall from Section 43.5.1 the naive massless lattice Dirac operator obtained by replacing ∂_μ with the symmetric difference, whose momentum-space kernel on the Brillouin zone $p_\mu \in (-\pi/a, \pi/a]$ is

$$\widetilde{D}_{\mathrm{naive}}(p) = m + \frac{i}{a}\sum_\mu \gamma_\mu \sin(a p_\mu). \tag{43.152}$$

For $m = 0$ this vanishes not only at $p = 0$ but at every corner of the Brillouin zone where each $a p_\mu \in \{0, \pi\}$, giving $2^4 = 16$ massless species: one physical fermion and fifteen doublers. The doublers split eight-and-eight into the two chiralities and, by the Poincaré–Hopf argument of Section 43.5.5, their chiralities

sum to zero — which is exactly why a chirally symmetric naive action cannot have a chiral anomaly, and exactly the obstruction Nielsen and Ninomiya turned into a theorem.

43.6.1 The Wilson term: lifting the doublers by hand

Wilson's idea is the most direct imaginable. The doublers live at the zone corners because $\sin(ap_\mu)$ has a second zero there; the physical mode lives at the origin. If we add to the operator a term that vanishes at the origin but is large and positive at every corner, it will leave the physical fermion alone while giving each doubler a mass that grows without bound as $a \to 0$, decoupling it from the continuum theory. The function $1-\cos(ap_\mu)$ does precisely this: it is $\tfrac{1}{2}a^2p_\mu^2 + O(a^4)$ near the origin and equals 2 at $ap_\mu = \pi$. In position space it is (minus) the lattice Laplacian, so the operator we add is a second difference.

Definition 43.32 (Wilson–Dirac operator)**.** Let $r > 0$ be the *Wilson parameter* (conventionally $r = 1$). The free Wilson–Dirac operator is the naive operator plus the Wilson term

$$D_W \;=\; D_{\text{naive}} - \frac{ra}{2}\,\Box_{\text{lat}}, \qquad (\Box_{\text{lat}}\psi)(x) = \frac{1}{a^2}\sum_{\mu}\big[\psi(x+a\hat{\mu}) - 2\psi(x) + \psi(x-a\hat{\mu})\big], \tag{43.153}$$

where $\Box_{\text{lat}}$ is the (gauge-covariant, when the differences carry link variables) lattice Laplacian. Equivalently, in momentum space,

$$\widetilde{D}_W(p) \;=\; \underbrace{m + \frac{r}{a}\sum_{\mu}\big(1-\cos(ap_\mu)\big)}_{\equiv\, M(p)} \;+\; \frac{i}{a}\sum_{\mu}\gamma_\mu \sin(ap_\mu). \tag{43.154}$$

That the two forms agree is a one-line Fourier check: the second difference of $e^{ip\cdot x}$ contributes $a^{-2}\sum_\mu(e^{iap_\mu} - 2 + e^{-iap_\mu}) = a^{-2}\sum_\mu(2\cos ap_\mu - 2)$, so $-\frac{ra}{2}\Box_{\text{lat}} \to -\frac{r}{2a}\sum_\mu(2\cos ap_\mu - 2) = \frac{r}{a}\sum_\mu(1-\cos ap_\mu)$, the scalar piece of $M(p)$. The object $M(p)$ acts as a *momentum-dependent mass*: it carries no γ matrix, so it adds to m rather than to the kinetic γ structure.

The physical mode is untouched; the operator is irrelevant. Expanding Eq. (43.154) for $ap_\mu \to 0$,

$$M(p) = m + \frac{r}{a}\sum_{\mu}\Big(\tfrac{1}{2}a^2p_\mu^2 + O(a^4)\Big) = m + \frac{ra}{2}\,p^2 + O(a^3), \qquad \frac{i}{a}\sum_{\mu}\gamma_\mu\sin(ap_\mu) = i\not{p} + O(a^2), \tag{43.155}$$

so $\widetilde{D}_W(p) \to m + i\not{p}$, the continuum Euclidean Dirac operator, and the Wilson correction $\frac{ra}{2}p^2$ is suppressed by one power of a relative to the mass. In the language of Chapter 23, the Wilson term is a dimension-five operator $\bar\psi\,\partial^2\psi$:

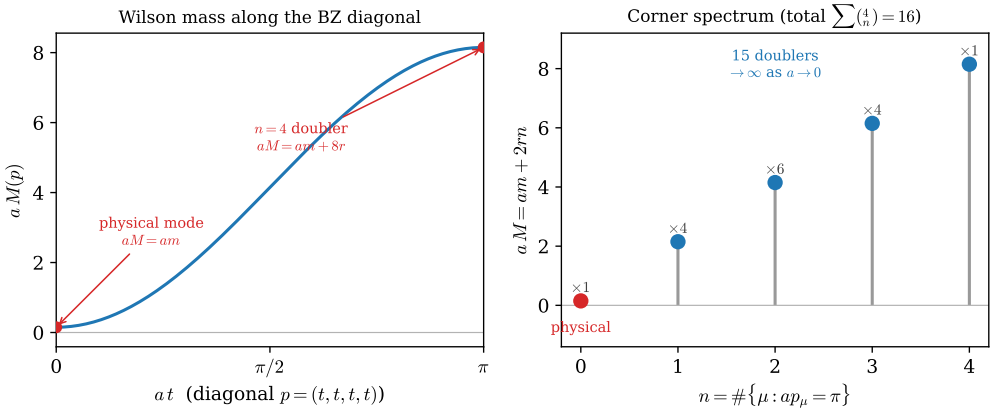


Figure 43.7: The Wilson term lifts the doublers. *Left:* the scalar Wilson mass $aM(p)$ along the Brillouin-zone diagonal $p = (t,t,t,t)$; the physical mode sits at $aM = am$ near the origin, the $n = 4$ doubler at the far corner sits at $aM = am + 8r$. *Right:* the corner spectrum $aM = am + 2rn$ versus the number n of components equal to π/a, with multiplicities $\binom{4}{n} = 1, 4, 6, 4, 1$. The fifteen doublers (blue) run off to infinity as $a \to 0$; only the physical mode (red) survives.

irrelevant, it scales away at the fixed point and does not change the continuum physics of the physical mode. (Its only lasting effect, discussed below, is to introduce $O(a)$ discretization errors — the reason "improved" Wilson actions add a further dimension-five term, the clover term, to cancel them.)

The doublers get masses of order $1/a$. At a Brillouin-zone corner the sine term vanishes identically, so the entire operator collapses to its scalar part $M(p)$. Let the corner have n of its four components equal to π/a and the rest zero. Then $\cos(ap_\mu) = -1$ for those n components and $+1$ for the others, so

$$M(\text{corner with } n \text{ components} = \pi/a) = m + \frac{r}{a}\big[n\cdot 2 + (4-n)\cdot 0\big] = m + \frac{2rn}{a}, \qquad n \in \{0,1,2,3,4\}. \tag{43.156}$$

The $n = 0$ corner is the physical fermion with mass m. The other fifteen corners carry masses $2r/a$, $4r/a$, $6r/a$, $8r/a$ with multiplicities $\binom{4}{1} = 4$, $\binom{4}{2} = 6$, $\binom{4}{3} = 4$, $\binom{4}{4} = 1$ respectively, summing to $\sum_{n=1}^{4}\binom{4}{n} = 15$. Every doubler mass diverges as $a \to 0$: the doublers become infinitely heavy and decouple, leaving a single Dirac fermion in the continuum. Figure 43.7 shows this staircase.

Worked Example 43.33 (Doubler masses at the zone corners)**.** We verify Eq. (43.156) by evaluating the full Wilson operator, not just its scalar part, at a representative corner and reading off the pole of the propagator. Take $m = 0$, $r = 1$, and the corner $p_\star = (\pi/a, 0, 0, 0)$ ($n = 1$). The sine term has components $\sin(ap_{\star,\mu})$; since $\sin(\pi) = 0$ and $\sin(0) = 0$, every component

vanishes and the γ piece of Eq. (43.154) is zero at $p_\star$. The scalar piece is $M(p_\star) = \frac{1}{a}[(1-\cos\pi)+3(1-\cos 0)] = \frac{1}{a}[2+0] = 2/a$. Hence $\widetilde{D}_W(p_\star) = 2/a$.

To see that this is genuinely a mass, expand around the corner: write $p = p_\star + k$ with ak_μ small. For the first component, $\sin(ap_1) = \sin(\pi + ak_1) = -\sin(ak_1) = -ak_1 + O(a^3)$ and $\cos(ap_1) = \cos(\pi + ak_1) = -\cos(ak_1) = -1 + \frac{1}{2}a^2k_1^2 + \dots$; for the transverse components $\sin(ak_j) = ak_j + O(a^3)$, $\cos(ak_j) = 1 - \frac{1}{2}a^2k_j^2 + \dots$. Therefore

$$\begin{aligned}\widetilde{D}_W(p_\star + k) &= \frac{1}{a}\Big[(1-\cos ap_1) + \sum_{j=2}^{4}(1-\cos ak_j)\Big] + \frac{i}{a}\Big[\gamma_1 \sin ap_1 + \sum_{j=2}^{4}\gamma_j \sin ak_j\Big] \\ &= \frac{1}{a}\Big[2 - \tfrac{1}{2}a^2k_1^2 + \tfrac{1}{2}a^2\sum_{j\geq 2}k_j^2 + \dots\Big] + i\Big[-\gamma_1 k_1 + \sum_{j\geq 2}\gamma_j k_j\Big] + O(a) \\ &= \frac{2}{a} \;+\; i\,\tilde{\not{k}} \;+\; O(a), \qquad \tilde{k} \equiv (-k_1, k_2, k_3, k_4). \end{aligned} \tag{43.157}$$

This is exactly a continuum Dirac operator $i\tilde{\not{k}} + M$ with mass $M = 2/a$ (the sign flip $k_1 \to -k_1$ is a harmless reflection of the local Dirac basis, absorbed by $\gamma_1 \to -\gamma_1$, a unitary change of variables that does not affect the physics). The propagator $\widetilde{D}_W^{-1}$ has its pole at $\tilde{k}^2 = -(2/a)^2$, i.e. a particle of mass $2/a$, confirming $n = 1 \Rightarrow 2r/a$. Repeating at a corner with n entries equal to π/a adds $2/a$ for each, giving $M = 2rn/a$ and reproducing Eq. (43.156). As $a \to 0$ each such pole is pushed to infinite mass and the corresponding species leaves the spectrum.

The price: explicit chiral-symmetry breaking. The Wilson term is, by construction, a momentum-dependent mass, and a mass term breaks chiral symmetry. We can make this exact. The continuum chiral rotation is $\psi \to e^{i\alpha\gamma_5}\psi$, $\bar\psi \to \bar\psi\, e^{i\alpha\gamma_5}$, an invariance of the massless naive action because γ_5 anticommutes with every γ_μ in the kinetic term, so $\bar\psi\,\gamma_\mu \to \bar\psi\, e^{i\alpha\gamma_5}\gamma_\mu = \bar\psi\,\gamma_\mu e^{-i\alpha\gamma_5}$ and the two exponentials cancel against $\psi \to e^{i\alpha\gamma_5}\psi$. The cleanest invariant statement is that the kinetic operator anticommutes with γ_5 while a mass commutes through. Computing the anticommutator of the Wilson operator with γ_5 in momentum space,

$$\Big\{\widetilde{D}_W(p), \gamma_5\Big\} = \{M(p), \gamma_5\} + \frac{i}{a}\sum_\mu \sin(ap_\mu)\;\{\gamma_\mu, \gamma_5\} = 2\,M(p)\,\gamma_5 + 0 = 2\gamma_5\Big[m + \frac{r}{a}\sum_\mu (1-\cos a \tag{43.158}$$

The kinetic γ term drops out because $\{\gamma_\mu, \gamma_5\} = 0$; the scalar term survives because $\{\mathbf{1}, \gamma_5\} = 2\gamma_5$. Even at $m = 0$, the Wilson contribution $\frac{2r}{a}\gamma_5 \sum_\mu (1 - \cos ap_\mu)$ is nonzero everywhere except the origin. Thus D_W violates the chiral hypothesis $\{D, \gamma_5\} = 0$ identically; chiral symmetry is broken not spontaneously and not anomalously but *explicitly*, by the regulator, at every nonzero momentum.

Consequences: additive mass renormalization and κ_c. In the continuum, chiral symmetry forbids an additively generated fermion mass: the massless point $m = 0$ is protected because m is the order parameter of a symmetry. Wilson fermions have no such protection. Radiative corrections generate a mass shift $\delta m \sim 1/a$ (a linear divergence, the worst kind), so the bare mass that yields a massless physical quark is not $m = 0$ but some a-dependent critical value $m_c(g)$ that must be *tuned*. It is conventional to package this in the *hopping parameter* form of the action. Rescaling the field to dimensionless variables $\psi \to \sqrt{2\kappa}\,\psi$ with

$$\kappa \equiv \frac{1}{2(am + 4r)}, \tag{43.159}$$

the gauged Wilson action Eq. (43.153) becomes the standard nearest-neighbour "hopping" form

$$S_W = \sum_x \bar\psi_x \psi_x - \kappa \sum_{x,\mu} \Big[\bar\psi_x (r - \gamma_\mu) U_\mu(x) \psi_{x+\hat\mu} + \bar\psi_{x+\hat\mu}(r + \gamma_\mu) U_\mu^\dagger(x) \psi_x\Big], \tag{43.160}$$

in which κ multiplies the hopping (kinetic) term and tunes the mass: large κ is light. At tree level $am = 0$ corresponds to $\kappa = \kappa_c = 1/8r$, but interactions push κ_c to a nontrivial function of the coupling, determined in practice by demanding that the pion mass (or the axial Ward identity) vanish. The matrices $(r \mp \gamma_\mu)$ are, for $r = 1$, twice the projectors $\frac{1}{2}(1 \mp \gamma_\mu)$ onto the γ_μ eigenspaces along each direction — the algebraic fingerprint of the broken chiral symmetry. This tuning, and the $O(a)$ artifacts left by the dimension-five term, are the full bill for Wilson's directness; it is paid routinely, and improved (clover) Wilson fermions remain a workhorse of the field.

Caution

The Wilson term is sometimes mis-stated as "giving the doublers a mass while preserving chiral symmetry on the physical mode." Equation (43.158) shows the chiral breaking is present at *all* nonzero momenta, including arbitrarily soft ones; it is $O(a)$ on soft modes but never zero. What survives the continuum limit is a chirally symmetric theory only because the breaking term is irrelevant, not because it is absent. The distinction matters the moment one wants chiral symmetry at finite a — for instance to control mixing of lattice operators — which is exactly what staggered and Ginsparg–Wilson fermions deliver and Wilson fermions do not.

43.6.2 Staggered (Kogut–Susskind) fermions: keeping a remnant chiral symmetry

Wilson removes all fifteen doublers and all chiral symmetry. The Kogut–Susskind construction is an economy in the opposite spirit: it does not fight the doubling at

all but *reorganizes* it, spreading the sixteen naive components over the corners of a hypercube so that, after discarding redundant spinor structure, only four species ("tastes") remain — and, crucially, an exact $U(1)$ chiral symmetry survives at finite a to protect the mass. The trick is that the naive operator Eq. (43.129) is, in a precise sense, four identical copies of a one-component problem.

Spin diagonalization. Perform the site-dependent change of Grassmann variables

$$\psi(x) = T(x)\,\chi(x), \qquad \bar\psi(x) = \bar\chi(x)\,T^\dagger(x), \qquad T(x) \equiv \gamma_1^{x_1/a}\,\gamma_2^{x_2/a}\,\gamma_3^{x_3/a}\,\gamma_4^{x_4/a}, \tag{43.161}$$

where the exponents are the integer coordinates in lattice units. Since each $\gamma_\mu^2 = \mathbf{1}$, $T(x)$ is unitary and $T^\dagger(x)T(x) = \mathbf{1}$. The only nontrivial input is how T behaves under a unit hop. Going from x to $x + a\hat\mu$ increments the exponent of γ_μ by one, i.e. inserts one extra γ_μ at slot μ of the ordered product. Bringing that extra γ_μ out to the left so it can meet the explicit γ_μ of the kinetic term requires anticommuting it past the factors $\gamma_1^{x_1/a}\cdots\gamma_{\mu-1}^{x_{\mu-1}/a}$ sitting to its left, each anticommutation costing a sign. The result is the central identity

$$T^\dagger(x)\,\gamma_\mu\,T(x + a\hat\mu) = \eta_\mu(x)\,\mathbf{1}, \qquad \eta_\mu(x) \equiv (-1)^{(x_1+\cdots+x_{\mu-1})/a}, \tag{43.162}$$

with $\eta_1(x) \equiv 1$. To see it, note $T^\dagger(x)\gamma_\mu T(x+a\hat\mu) = T^\dagger(x)\gamma_\mu T(x)\cdot[T^{-1}(x)T(x+a\hat\mu)]$; the bracket is γ_μ moved to slot μ, and $T^\dagger(x)\gamma_\mu T(x)\cdot\gamma_\mu^{(\text{at slot }\mu)}$ leaves $\eta_\mu(x)\,\mathbf{1}$ after all γ's square to unity. The phase $\eta_\mu(x)$ counts the parity of the coordinates *preceding* direction μ. Substituting Eq. (43.161) into the naive action and using Eq. (43.162),

$$S_{\text{naive}} = \sum_x \Big[\frac{1}{2a}\sum_\mu \eta_\mu(x)\,\bar\chi(x)\big(\chi(x+a\hat\mu) - \chi(x-a\hat\mu)\big) + m\,\bar\chi(x)\chi(x)\Big]\otimes \mathbf{1}_{4\times 4}. \tag{43.163}$$

Every explicit γ_μ has been replaced by the scalar phase $\eta_\mu(x)$, and the action is now diagonal in the (suppressed) Dirac index: it is four identical, decoupled copies of a single-component theory. Keeping one component — a single Grassmann field $\chi(x)$ per site — discards a factor of four of redundancy and defines the staggered fermion. The sixteen naive species become $16/4 = 4$ *tastes*, an internal flavour quartet that becomes four degenerate Dirac fermions in the continuum, assembled from the sixteen corners of the elementary hypercube (one taste-times-spin index per corner).

The remnant chiral symmetry. The decisive payoff is that a continuous chiral symmetry survives at $a \neq 0$. The site parity

$$\epsilon(x) \equiv (-1)^{(x_1+x_2+x_3+x_4)/a} \tag{43.164}$$

splits the lattice into even and odd sublattices, and the transformation

$$\chi(x) \to e^{i\alpha\,\epsilon(x)}\chi(x), \qquad \bar{\chi}(x) \to e^{i\alpha\,\epsilon(x)}\bar{\chi}(x) \tag{43.165}$$

is an exact symmetry of the *massless* staggered action and is broken only by the mass term. The verification is immediate. The hopping term connects x to its neighbour $x \pm a\hat{\mu}$, which lies on the opposite sublattice, so $\epsilon(x+a\hat{\mu}) = -\epsilon(x)$; under Eq. (43.165) the hopping bilinear picks up $e^{i\alpha(\epsilon(x)+\epsilon(x+a\hat{\mu}))} = e^{i\alpha(\epsilon(x)-\epsilon(x))} = 1$ and is invariant. The mass term, by contrast, joins $\chi(x)$ to $\bar{\chi}(x)$ on the *same* site and picks up $e^{2i\alpha\epsilon(x)} \neq 1$. Thus $\epsilon(x)$ plays exactly the role γ_5 plays in the continuum: it anticommutes with the kinetic operator (here, in the sense that it flips sign across a hop) and commutes through the mass. This $U(1)_\epsilon$ is a genuine remnant of the continuum $\mathrm{SU}(4) \times \mathrm{SU}(4)$ taste-chiral group — the lattice keeps one of its sixteen generators exactly — and because it is an exact symmetry at finite spacing, it forbids additive mass renormalization. The massless point is $m = 0$ with no tuning: this is the cardinal advantage of staggered over Wilson fermions, and it is why staggered fermions are the method of choice for problems where cheap near-chiral fermions matter, such as high-precision determinations of the light-quark masses and the equation of state at finite temperature.

Physical Insight

The two finite-a chiral symmetries on offer are mirror images. Wilson keeps the doubler count at one and throws away the chiral symmetry; staggered keeps an exact $U(1)$ chiral symmetry and pays with a residual fourfold taste multiplicity. Neither is the continuum symmetry: Wilson recovers it only as $a \to 0$, staggered realizes only a one-dimensional subgroup of it at $a \neq 0$. The Ginsparg–Wilson construction of the next subsection is the statement that one can, in fact, have the full continuous chiral symmetry exactly at finite a — provided one is willing to let the *operator* that generates it be deformed by the lattice.

The fourth-root trick, honestly. Four tastes is the wrong number for almost every physical application: nature has one up quark, not four. The standard remedy is to take the fourth root of the staggered determinant in the path integral, replacing $\det D_{\mathrm{stag}}$ for the four-taste operator by $(\det D_{\mathrm{stag}})^{1/4}$ for each physical flavour, so that N_f physical flavours enter as $(\det D_{\mathrm{stag}})^{N_f/4}$. Heuristically this divides the four degenerate tastes back down to one. The difficulty is that, at finite a, the four tastes are *not* exactly degenerate — they are split by lattice artifacts (taste-breaking interactions of order a^2) — and $(\det D_{\mathrm{stag}})^{1/4}$ is provably not the determinant of any local, one-taste lattice Dirac operator. The rooted theory is therefore nonlocal and non-unitary at $a \neq 0$; the open question is whether locality and unitarity are restored in the continuum limit, i.e. whether rooted staggered fermions lie in the correct universality class. As of the mid-2020s the weight

of evidence — analytic arguments controlling the taste splittings, and a large body of numerical tests against experiment and against other discretizations — supports the conclusion that the continuum limit is correct, and rooted staggered ensembles underpin many of the field's most precise results. But it is not a theorem, and a residual minority of the community regards the construction as not fully justified. The honest statement is: rooting works in practice, is believed to give the right continuum theory, and remains without rigorous proof.

43.6.3 The Ginsparg–Wilson relation and an exact lattice chiral symmetry

We now reach the principled resolution. The Nielsen–Ninomiya theorem forbids $\{D, \gamma_5\} = 0$ for a local, doubler-free D. Ginsparg and Wilson asked the right question: what is the *weakest* deformation of $\{D, \gamma_5\} = 0$ consistent with a sensible continuum limit, and does it still encode a chiral symmetry? Their answer, obtained originally by blocking a chirally symmetric continuum theory onto the lattice, is a relation in which the right-hand side is not zero but a local term that vanishes as $a \to 0$.

Definition 43.34 (Ginsparg–Wilson relation)**.** A lattice Dirac operator D satisfies the Ginsparg–Wilson (GW) relation if

$$\boxed{\{D, \gamma_5\} \;=\; a\, D\, \gamma_5\, D} \tag{43.166}$$

We assume in addition γ_5-Hermiticity $D^\dagger = \gamma_5 D \gamma_5$, the lattice version of the reality property of the Euclidean Dirac operator (it guarantees a real, positive fermion determinant for degenerate pairs and underlies reflection positivity).

The right-hand side of Eq. (43.166) is $O(a)$: it carries an explicit factor of a and is built from D, which has a smooth continuum limit. So as $a \to 0$ the relation collapses to $\{D, \gamma_5\} = 0$, the continuum chiral algebra. At finite a the deformation is real but mild, and — this is the content of the construction — it is exactly the deformation that converts the chiral *symmetry* from the naive one into a lattice-corrected one that the action still respects.

The relation localizes chiral breaking to a contact term. Multiplying Eq. (43.166) on the left and right by D^{-1} (assume for now D is invertible, i.e. no zero modes; the zero-mode case is the index theorem, taken up below) gives a statement about the propagator $S \equiv D^{-1}$:

$$D^{-1}\,\{D, \gamma_5\}\,D^{-1} = a\,\gamma_5 \quad\Longrightarrow\quad \left\{D^{-1}, \gamma_5\right\} = \gamma_5 D^{-1} + D^{-1}\gamma_5 = a\,\gamma_5. \tag{43.167}$$

The anticommutator of the propagator with γ_5 is a pure *contact term*: $a\gamma_5$ is an ultralocal operator (a multiple of the identity in position space, $a\gamma_5\,\delta_{xy}$). At any

nonzero separation $x \neq y$ the propagator anticommutes with γ_5 exactly, just as in the continuum. Chiral symmetry is broken only at coincident points, where a regulator is always entitled to act, and the breaking vanishes with a. This is the precise sense in which GW fermions are "as chiral as a lattice can be."

Eigenvalues live on a circle. The single most useful consequence of Eq. (43.166) together with γ_5-Hermiticity is the location of the spectrum of D. Multiply the GW relation on the left by γ_5:

$$\gamma_5 \{D, \gamma_5\} = \gamma_5(\gamma_5 D + D\gamma_5) = D + \gamma_5 D \gamma_5 = a\, \gamma_5 D \gamma_5 D. \tag{43.168}$$

By γ_5-Hermiticity $\gamma_5 D \gamma_5 = D^\dagger$, so this reads

$$D + D^\dagger = a\, D^\dagger D. \tag{43.169}$$

Now compute the combination $(\mathbf{1} - aD)^\dagger(\mathbf{1} - aD)$:

$$(\mathbf{1}-aD)^\dagger(\mathbf{1}-aD) = \mathbf{1}-aD-aD^\dagger+a^2D^\dagger D = \mathbf{1}-a(D+D^\dagger)+a^2D^\dagger D = \mathbf{1}-a(aD^\dagger D)+a^2D^\dagger D = \mathbf{1}, \tag{43.170}$$

using Eq. (43.169) at the third step. Therefore

$$V \equiv \mathbf{1} - aD \quad \text{is unitary:} \quad V^\dagger V = \mathbf{1}, \qquad D = \frac{1}{a}(\mathbf{1} - V). \tag{43.171}$$

The eigenvalues of the unitary V are phases $e^{i\theta}$, so the eigenvalues of D are

$$\lambda = \frac{1}{a}(1 - e^{i\theta}), \qquad \theta \in (-\pi, \pi], \tag{43.172}$$

which trace out the circle $|1 - a\lambda| = 1$ of radius $1/a$ centered at $1/a$ in the complex plane — the *Ginsparg–Wilson circle* (Fig. 43.8). It passes through the origin ($\theta = 0$, $\lambda = 0$) and through $\lambda = 2/a$ ($\theta = \pi$), the only two real points; everywhere else the eigenvalues come in complex-conjugate pairs because D is γ_5-Hermitian. As $a \to 0$ the circle flattens onto the imaginary axis near the origin, reproducing the continuum spectrum $\lambda = i\not{p}$. The two real points are special: $\lambda = 0$ will host the exact chiral zero modes, and $\lambda = 2/a$ the heavy lattice-artifact real modes that play the role of the would-be doublers.

Lüscher's exact lattice chiral symmetry. The deformation Eq. (43.166) is not a loss of chiral symmetry but a redefinition of it. Lüscher observed that the GW relation is precisely the integrability condition for the action $S = \bar\psi D \psi$ to be invariant under a modified, lattice-deformed chiral rotation. Define the *lattice chirality operator*

$$\hat\gamma_5 \equiv \gamma_5\,(\mathbf{1} - aD) = \gamma_5 V, \tag{43.173}$$

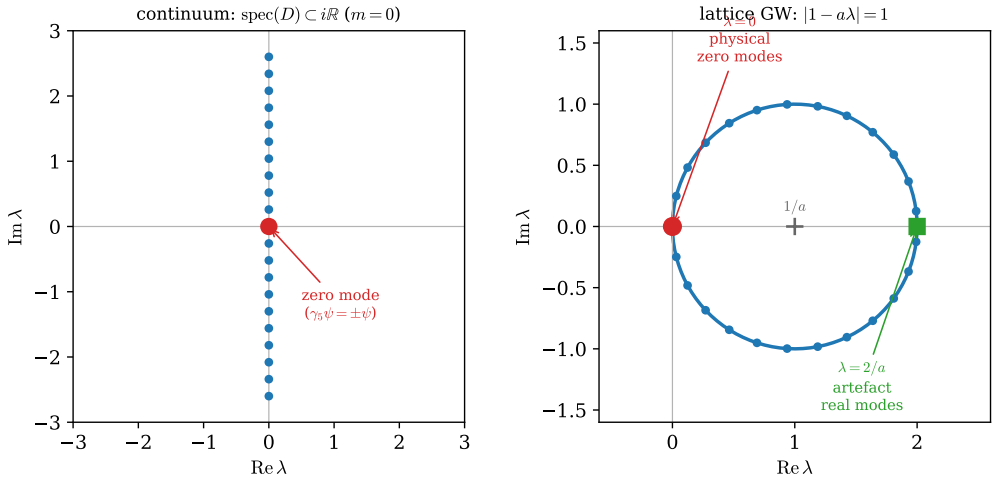


Figure 43.8: The Ginsparg–Wilson circle. *Left:* the continuum massless Dirac spectrum lies on the imaginary axis, with chiral zero modes at the origin in a topological background. *Right:* a GW operator has $1 - aD$ unitary, so spec(D) lies on the circle $|1 - a\lambda| = 1$ of radius $1/a$ centered at $1/a$. The two real eigenvalues — the physical zero modes at $\lambda = 0$ and the artifact real modes at $\lambda = 2/a$ — are where ordinary γ_5 is an exact symmetry; in between, the deformed chirality $\hat{\gamma}_5 = \gamma_5(1 - aD)$ takes over.

which is the natural lattice object replacing γ_5 in the chiral rotation of the field. We first record its key algebraic properties, which the index theorem below will use. It is Hermitian and squares to one:

$$\hat{\gamma}_5^\dagger = (\gamma_5 V)^\dagger = V^\dagger \gamma_5 = \gamma_5 V = \hat{\gamma}_5, \qquad \hat{\gamma}_5^2 = \gamma_5 V \gamma_5 V = V^\dagger V = \mathbf{1}, \tag{43.174}$$

where we used $\gamma_5 V \gamma_5 = \gamma_5(\mathbf{1} - aD)\gamma_5 = \mathbf{1} - aD^\dagger = V^\dagger$ (from γ_5-Hermiticity) twice. So $\hat{\gamma}_5$ is a legitimate chirality: Hermitian, involutive, with eigenvalues ± 1, and it reduces to γ_5 as $a \to 0$. The associated chiral projectors are $\hat{P}_\pm = \frac{1}{2}(\mathbf{1} \pm \hat{\gamma}_5)$ for the field ψ and $P_\pm = \frac{1}{2}(\mathbf{1} \pm \gamma_5)$ for $\bar{\psi}$. The worked example below verifies that the rotation built from $\hat{\gamma}_5$ is an exact symmetry.

Worked Example 43.35 (Lüscher's modified chiral rotation is an exact symmetry)**.** We show that, for any D obeying the GW relation Eq. (43.166), the bilinear lattice action $S = \bar{\psi}\, D\, \psi$ is exactly invariant — at finite a, to all orders, not merely as $a \to 0$ — under the infinitesimal chiral rotation

$$\delta\psi = i\alpha\, \gamma_5(\mathbf{1} - aD)\, \psi = i\alpha\, \hat{\gamma}_5\, \psi, \qquad \delta\bar{\psi} = i\alpha\, \bar{\psi}\, \gamma_5, \tag{43.175}$$

with α a constant infinitesimal parameter. Note the deliberate asymmetry: the field ψ rotates with the *deformed* chirality $\hat{\gamma}_5$, while $\bar{\psi}$ rotates with the ordinary

γ_5. Varying the action and keeping the two contributions,

$$\begin{aligned}\delta S &= (\delta\bar\psi)\, D\,\psi + \bar\psi\, D\,(\delta\psi) = i\alpha\,\bar\psi\,\gamma_5 D\,\psi + i\alpha\,\bar\psi\, D\,\gamma_5(\mathbf{1} - aD)\,\psi \\ &= i\alpha\,\bar\psi[\gamma_5 D + D\gamma_5 - a\,D\gamma_5 D]\psi = i\alpha\,\bar\psi[\{D,\gamma_5\} - a\,D\gamma_5 D]\psi. \end{aligned} \tag{43.176}$$

The bracket is exactly the GW relation Eq. (43.166) rearranged: $\{D,\gamma_5\} - aD\gamma_5 D = 0$. Hence

$$\delta S = 0 \tag{43.177}$$

identically. The action is invariant under Eq. (43.175) *because* D satisfies the Ginsparg–Wilson relation, and conversely the relation is the necessary and sufficient condition for this rotation to be a symmetry. This is the exact lattice chiral symmetry: a continuous $U(1)$ invariance of the regularized, doubler-free action at finite lattice spacing, evading Nielsen–Ninomiya not by violating any of its analytic hypotheses but by deforming the algebraic form of the chirality from γ_5 to $\hat\gamma_5$.

A symmetric, manifestly $\psi \leftrightarrow \bar\psi$-balanced form distributes the deformation evenly: $\delta\psi = i\alpha\,\gamma_5(\mathbf{1} - \frac{a}{2}D)\psi$, $\delta\bar\psi = i\alpha\,\bar\psi(\mathbf{1} - \frac{a}{2}D)\gamma_5$. The same computation gives $\delta S = i\alpha\,\bar\psi[\gamma_5 D + D\gamma_5 - \frac{a}{2}D\gamma_5 D - \frac{a}{2}D\gamma_5 D]\psi = i\alpha\,\bar\psi[\{D,\gamma_5\} - aD\gamma_5 D]\psi = 0$, again by GW. The two forms differ by a field redefinition and define the same symmetry; the asymmetric form Eq. (43.175) is algebraically lighter and the symmetric form makes the chiral charge manifestly real.

43.6.4 The lattice index theorem and the anomaly

The exact symmetry of Theorem 43.35 is a classical statement about the action. Its quantum fate — whether the path-integral *measure* respects it — is where the anomaly and the index theorem of Chapter 36 reappear, now as exact lattice identities. This is the deepest payoff of the GW construction: it places the Atiyah–Singer index theorem and the chiral anomaly on the lattice with no continuum limit required.

Zero modes are exact chirality eigenstates. Let ϕ be a zero mode, $D\phi = 0$. Acting with the GW relation, $\{D,\gamma_5\}\,\phi = aD\gamma_5 D\phi = 0$, so $D(\gamma_5\phi) = -\gamma_5 D\phi = 0$: the space of zero modes is γ_5-invariant. Since $\gamma_5^2 = \mathbf{1}$ we may diagonalize γ_5 on it and choose every zero mode to have definite chirality $\gamma_5\phi = \pm\phi$. On a zero mode the deformed and ordinary chiralities coincide, $\hat\gamma_5\phi = \gamma_5(\mathbf{1} - aD)\phi = \gamma_5\phi = \pm\phi$. Write n_+ and n_- for the numbers of zero modes with $\gamma_5 = +1$ and $\gamma_5 = -1$ respectively. This chirality grading of the exact zero modes is the lattice analogue of the continuum statement that Dirac zero modes in an instanton background are chiral.

The index is a trace of $\hat{\gamma}_5$. Consider the trace of the lattice chirality over the whole Dirac space (Dirac, colour, and site indices),

$$\operatorname{Tr}\hat{\gamma}_5 = \operatorname{Tr}\big[\gamma_5(\mathbf{1} - aD)\big] = \operatorname{Tr}\gamma_5 - a\,\operatorname{Tr}(\gamma_5 D) = -a\,\operatorname{Tr}(\gamma_5 D), \tag{43.178}$$

where $\operatorname{Tr}\gamma_5 = 0$ because γ_5 is traceless on Dirac indices and the identity on the others. We now evaluate $\operatorname{Tr}\hat{\gamma}_5$ a second way, by its spectrum, using the unitary $V = \mathbf{1} - aD$ of Eq. (43.171). The eigenspaces of V are labeled by phases $e^{i\theta}$. The relation $\gamma_5 V \gamma_5 = V^\dagger$ means γ_5 maps the $e^{i\theta}$-eigenspace of V to its $e^{-i\theta}$-eigenspace: if $Vv = e^{i\theta}v$ then $V(\gamma_5 v) = \gamma_5 V^\dagger v = e^{-i\theta}(\gamma_5 v)$. Consequently, for $\theta \neq 0, \pi$ the eigenspaces at $\pm\theta$ are distinct and γ_5 — hence $\hat{\gamma}_5 = \gamma_5 V$ — is strictly off-diagonal between them, contributing *nothing* to the trace. Only the two real points survive:

- at $\theta = 0$ ($V = +\mathbf{1}$, $D = 0$): the zero-mode space, where $\hat{\gamma}_5 = \gamma_5$, contributing $n_+ - n_-$;
- at $\theta = \pi$ ($V = -\mathbf{1}$, $D = 2/a$): the artifact real-mode space, where $\hat{\gamma}_5 = -\gamma_5$, contributing $-(n_+^{(2/a)} - n_-^{(2/a)})$.

Therefore

$$\operatorname{Tr}\hat{\gamma}_5 = (n_+ - n_-) - (n_+^{(2/a)} - n_-^{(2/a)}). \tag{43.179}$$

The trace of the deformed chirality counts exact zero modes by chirality, corrected by the chirality of the heavy real modes at the top of the circle. For smooth gauge backgrounds carrying topological charge the index is saturated by the physical zero modes at the origin, the artifact modes at $2/a$ being absent or canceling in pairs; defining the lattice index as the physical zero-mode count,

$$\operatorname{index}(D) \equiv n_- - n_+, \tag{43.180}$$

and combining Eqs. (43.178) and (43.179), we obtain the exact lattice identity

$$\boxed{\operatorname{index}(D) = n_- - n_+ = -\tfrac{1}{2}\operatorname{Tr}\hat{\gamma}_5 = \frac{a}{2}\operatorname{Tr}(\gamma_5 D)} \tag{43.181}$$

(in the absence of $2/a$ real modes). This is the Ginsparg–Wilson form of the index theorem: an integer, computable on a finite lattice from the eigenvalues of a single operator, expressed as a trace that is the lattice transcription of the continuum $\frac{1}{32\pi^2}\int d^4x\,\epsilon^{\mu\nu\rho\sigma}\operatorname{Tr}F_{\mu\nu}F_{\rho\sigma}$.

The anomaly as a measure Jacobian. The continuum chiral anomaly is the failure of the path-integral measure to be invariant under a chiral rotation — the Fujikawa Jacobian computed in Chapter 36. The lattice reproduces this exactly. Under the Lüscher rotation, take the symmetric form $\delta\psi = i\alpha\,\gamma_5(\mathbf{1} - \frac{a}{2}D)\psi$,

$\delta\bar\psi = i\alpha\,\bar\psi(\mathbf{1}-\frac{a}{2}D)\gamma_5$. The Grassmann measure transforms with the inverse-Jacobian rule: $\psi\to(\mathbf{1}+M)\psi$ contributes $\det(\mathbf{1}+M)^{-1}\simeq e^{-\operatorname{Tr}M}$, and likewise for $\bar\psi$. With $M=i\alpha\gamma_5(\mathbf{1}-\frac{a}{2}D)$ and $\bar M = i\alpha(\mathbf{1}-\frac{a}{2}D)\gamma_5$,

$$\mathcal{D}\bar\psi\,\mathcal{D}\psi \;\longrightarrow\; \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\Big(-\operatorname{Tr}M-\operatorname{Tr}\bar M\Big),\qquad \operatorname{Tr}M+\operatorname{Tr}\bar M = 2i\alpha\;\operatorname{Tr}\Big[\gamma_5(\mathbf{1}-\tfrac{a}{2}D)\Big]. \tag{43.182}$$

Using $\operatorname{Tr}\gamma_5=0$ this is $\operatorname{Tr}M+\operatorname{Tr}\bar M = 2i\alpha(-\frac{a}{2})\operatorname{Tr}(\gamma_5 D) = -i\alpha\,a\;\operatorname{Tr}(\gamma_5 D)$, so the measure picks up the anomalous phase

$$\mathcal{D}\bar\psi\,\mathcal{D}\psi \;\longrightarrow\; \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\Big(i\alpha\,a\;\operatorname{Tr}(\gamma_5 D)\Big) = \mathcal{D}\bar\psi\,\mathcal{D}\psi\;\exp\Big(-2i\alpha\,(n_- - n_+)\Big), \tag{43.183}$$

where the last step used Eq. (43.181). The classically exact chiral symmetry of Theorem 43.35 is anomalous at the quantum level: the measure is not invariant, and its variation is precisely $2(n_- - n_+)$ times the rotation angle — an integer topological quantity, not a regulator-dependent number. This is the chiral anomaly of Chapter 36, derived here without a single continuum manipulation. In the continuum limit $-\frac{a}{2}\operatorname{Tr}(\gamma_5 D)$ tends to the topological charge $Q=\frac{1}{32\pi^2}\int d^4x\,\epsilon^{\mu\nu\rho\sigma}\operatorname{Tr}F_{\mu\nu}F_{\rho\sigma}$, and Eq. (43.181) becomes the Atiyah–Singer statement $n_- - n_+ = Q$ established in Section 36.5. The lattice has thus turned the anomaly — the very obstruction that, through Nielsen–Ninomiya, made doublers unavoidable for a chirally symmetric naive action — into an exact, finite, computable identity, while keeping one species and an exact (if deformed) chiral symmetry. The circle closes: the anomaly is not evaded but realized.

43.6.5 Realizations: overlap and domain-wall fermions

The GW relation is a constraint; it does not by itself exhibit an operator. Two explicit, widely used solutions exist, and they turn out to be two faces of the same construction.

The overlap operator (Neuberger). Neuberger gave a closed-form solution of Eq. (43.166):

$$D_{\mathrm{ov}} = \frac{1}{a}\Big(\mathbf{1}+\gamma_5\;\operatorname{sgn}(H)\Big) = \frac{1}{a}\Big(\mathbf{1}+\gamma_5\,\frac{H}{\sqrt{H^2}}\Big),\qquad H\equiv\gamma_5(aD_W-\rho), \tag{43.184}$$

where D_W is the Wilson–Dirac operator of Theorem 43.32, $\rho\in(0,2)$ is a negative "domain-wall mass" parameter that selects the physical mode, and sgn is the matrix sign function. The Wilson operator does double duty: its doubler-lifting property ensures that H has no near-zero eigenvalues coming from the corners, so $\operatorname{sgn}(H)$ is well defined, and the single physical mode survives the sign function as the chiral fermion. Writing $\epsilon\equiv\operatorname{sgn}(H)$, which is Hermitian and satisfies $\epsilon^2=\mathbf{1}$

(and, with the γ_5-Hermiticity of D_W, $\gamma_5\epsilon\gamma_5 = \epsilon^\dagger = \epsilon$ in the combination entering below), one checks the GW relation directly. With $aD_{\rm ov} = \mathbf{1} + \gamma_5\epsilon$,

$$\{D_{\rm ov}, \gamma_5\} = \frac{1}{a}\{\mathbf{1} + \gamma_5\epsilon, \gamma_5\} = \frac{1}{a}\Big(\{\mathbf{1}, \gamma_5\} + \{\gamma_5\epsilon, \gamma_5\}\Big) = \frac{1}{a}\Big(2\gamma_5 + \gamma_5\epsilon\gamma_5 + \gamma_5\gamma_5\epsilon\Big) = \frac{1}{a}\Big(2\gamma_5 \tag{43.185}$$

$$\begin{aligned} a\, D_{\rm ov}\gamma_5 D_{\rm ov} &= \frac{1}{a}(\mathbf{1} + \gamma_5\epsilon)\,\gamma_5\,(\mathbf{1} + \gamma_5\epsilon) = \frac{1}{a}(\gamma_5 + \gamma_5\epsilon\gamma_5)(\mathbf{1} + \gamma_5\epsilon) \\ &= \frac{1}{a}\Big(\gamma_5 + \gamma_5\gamma_5\epsilon + \gamma_5\epsilon\gamma_5 + \gamma_5\epsilon\gamma_5\gamma_5\epsilon\Big) = \frac{1}{a}\Big(\gamma_5 + \epsilon + \gamma_5\epsilon\gamma_5 + \gamma_5\epsilon^2\Big) = \frac{1}{a}\Big(2\gamma_5 + \epsilon - \end{aligned} \tag{43.186}$$

using $\gamma_5^2 = \mathbf{1}$ and $\epsilon^2 = \mathbf{1}$. The two right-hand sides are identical, so $\{D_{\rm ov}, \gamma_5\} = aD_{\rm ov}\gamma_5 D_{\rm ov}$: the overlap operator satisfies the Ginsparg–Wilson relation exactly. Its spectrum sits on the GW circle of Fig. 43.8 by construction, since $\mathbf{1} - aD_{\rm ov} = -\gamma_5\epsilon$ is manifestly unitary ($(\gamma_5\epsilon)^\dagger(\gamma_5\epsilon) = \epsilon\gamma_5\gamma_5\epsilon = \epsilon^2 = \mathbf{1}$). All the chiral structure of the previous subsections — Lüscher's symmetry, the exact zero modes, the index theorem — is therefore realized by a concrete, simulable operator.

Domain-wall fermions (Kaplan). The domain-wall construction reaches the same place by a physical route: it borrows a fifth dimension. Kaplan observed that a Dirac fermion in five dimensions with a mass that changes sign across a four-dimensional "wall" — a domain wall in the fifth coordinate $s \in \{1, \dots, L_5\}$ — binds a single chiral zero mode, exponentially localized on the wall (the Jackiw–Rebbi mechanism of Chapter 41). With a finite slab of fifth dimension and a wall at each end, a left-handed mode localizes on one wall and a right-handed mode on the other; the four-dimensional theory at low energy is a single Dirac fermion whose two chiralities live at opposite ends of the fifth dimension and therefore have exponentially small overlap. Their residual coupling, the "residual mass" $m_{\rm res}$, breaks chiral symmetry by an amount $\propto e^{-cL_5}$ that one sends to zero by taking $L_5 \to \infty$. In that limit the effective four-dimensional operator obtained by integrating out the bulk is *exactly* the overlap operator Eq. (43.184): domain-wall and overlap fermions are the same chiral lattice fermion, presented respectively as a five-dimensional local theory and its four-dimensional reduction. Domain-wall fermions trade the explicit matrix sign function for the simulation of an extra dimension of extent L_5 — often the more convenient bargain in practice, since it keeps the action local in five dimensions.

How Ginsparg–Wilson evades Nielsen–Ninomiya. It is worth stating cleanly which hypothesis of Theorem 43.30 is given up. The theorem assumes locality, translation invariance, Hermiticity (γ_5-Hermiticity), the absence of doublers, and chiral symmetry in the form $\{D, \gamma_5\} = 0$. The overlap operator keeps the first four: it is translation-invariant, γ_5-Hermitian, doubler-free, and *local*

— not ultralocal (it couples all sites, since $\mathrm{sgn}(H)$ is a nonpolynomial function of D_W), but local in the sufficient sense that its kernel decays exponentially, $|D_{\mathrm{ov}}(x,y)| \lesssim e^{-|x-y|/\xi}$ with a fixed range $\xi = O(a)$ for smooth gauge fields, which is all universality requires. What it gives up is the precise algebraic form of the fifth hypothesis: instead of $\{D,\gamma_5\}=0$ it satisfies $\{D,\gamma_5\} = aD\gamma_5 D$, and the exact chiral symmetry it possesses is generated not by γ_5 but by the deformed $\hat{\gamma}_5$ of Eq. (43.173). Nielsen–Ninomiya is not violated; its chirality hypothesis is *deformed*, in exactly the way that still admits an anomaly (Eq. (43.183)) and an index theorem (Eq. (43.181)) while keeping locality intact. This is the principled resolution of the doubling problem.

Physical Insight

The corollary of Nielsen–Ninomiya that no *ultralocal* GW operator exists is not a loophole but the cost made explicit. Chirality on the lattice cannot be packaged into a finite-range difference operator; it requires the infinite-range (yet exponentially localized) reach of the sign function or, equivalently, the extra dimension of the domain wall. The anomaly, which lives in the deep ultraviolet, forces the chiral operator to "feel" the whole lattice. That nonlocality-in-detail, locality-in-bulk is the precise mathematical price of an exact lattice anomaly.

Caution

Ginsparg–Wilson fermions are correct but expensive. The matrix sign function in Eq. (43.184) must be approximated — by rational (Zolotarev) functions or Lanczos methods — and each application costs an inner iterative solve, so an overlap inversion is one to two orders of magnitude more costly than a Wilson inversion; domain-wall fermions carry the multiplicative cost of the fifth dimension L_5. As of the mid-2020s this expense is why chiral lattice fermions, despite their theoretical supremacy, are deployed selectively — for quantities where chiral symmetry is essential (e.g. certain weak matrix elements, ϵ'/ϵ, topological susceptibility) — while the bulk of large-volume spectroscopy still runs on improved Wilson (clover) or rooted staggered ensembles. The choice of fermion discretization remains a genuine cost–correctness trade-off, not a solved problem.

With the doubling problem resolved — pragmatically by Wilson and staggered fermions, in principle by Ginsparg–Wilson — the lattice now offers a complete, positive, gauge-invariant measure for both gauge fields and matter. Every ingredient of the regulator is in hand: a discretized action, a doubler-free fermion sector, and a probability weight e^{-S_E} on a finite-dimensional configuration space. What remains is to extract numbers from it, which is the task of Section 43.7: importance sampling of the gauge measure, integration over the Grassmann fields

to the fermion determinant, the extraction of hadron masses from the exponential decay of Euclidean correlators, and the continuum extrapolation governed by asymptotic freedom.

43.7 What the lattice computes: Monte Carlo, the spectrum, and the continuum limit

The previous sections finished the hard constructive work. We have a finite-dimensional integral over compact link variables $U_\mu(x) \in \mathrm{SU}(3)$ and Grassmann quark fields, weighted by a positive, manifestly gauge-invariant Euclidean action assembled from the Wilson gauge action of Section 43.3 and a doubler-free fermion action of Section 43.6, on a finite hypercubic lattice with spacing a and volume $L^3 \times T$. Reflection positivity (Section 43.1.4) guarantees that this Euclidean system is the imaginary-time face of a genuine quantum theory with a self-adjoint, positive transfer matrix. Everything is now well defined. What remains is the question that justifies the entire construction: *how does one turn this object into numbers*—a proton mass, a quark mass, a transition temperature, a contribution to the muon anomaly—and in what sense do those numbers belong to continuum QCD rather than to the scaffolding we erected to define it? This section answers both. We compute expectation values by importance sampling, extract physical masses from the exponential decay of Euclidean correlators using the spectral decomposition the transfer matrix supplies, and remove the regulator by an extrapolation $a \to 0$ whose form is dictated by asymptotic freedom. We then survey, honestly and with dates, what the lattice has and has not delivered.

43.7.1 Expectation values as ensemble averages

The lattice path integral assigns to any observable $\mathcal{O}$—a product of fields and link variables—the Euclidean expectation value

$$\langle \mathcal{O} \rangle \;=\; \frac{1}{Z} \int \Big(\prod_{x,\mu} \mathrm{d}U_\mu(x) \Big)\, \mathrm{d}\bar{\psi}\, \mathrm{d}\psi \;\; \mathcal{O}\, e^{-S_g[U] - S_F[\bar{\psi},\psi,U]}, \qquad Z = \int \mathrm{d}U\, \mathrm{d}\bar{\psi}\, \mathrm{d}\psi \; e^{-S_g - S_F}, \tag{43.187}$$

where $\mathrm{d}U_\mu(x)$ is the normalized Haar measure on the group (the unique left- and right-invariant probability measure, used already in the strong-coupling expansion of Section 27.7), and $\mathrm{d}\bar{\psi}\,\mathrm{d}\psi$ is the Grassmann measure of Chapter 12. The normalization Z is the partition function; only ratios $\langle \mathcal{O} \rangle = (\text{numerator})/Z$ are physical, exactly as for any statistical-mechanical average. This is the content of the chapter's thesis made operational: after Wick rotation the weight e^{-S} is non-negative, so Eq. (43.187) is the average of $\mathcal{O}$ over a probability distribution on field configurations. A quantum field theory has become a problem in classical statistical mechanics on a finite (if astronomically large) configuration space.

Integrating out the fermions. The fermion action is bilinear in the Grassmann fields,

$$S_F[\bar\psi,\psi,U] = \sum_{x,y} \bar\psi(x)\, D[U]_{xy}\, \psi(y) \;\equiv\; \bar\psi\, D[U]\, \psi, \tag{43.188}$$

with $D[U]$ the lattice Dirac operator (Wilson, staggered, or overlap) built from the gauge links; it is a finite matrix carrying Dirac, colour, and lattice-site indices. Because the action is quadratic in $\psi,\bar\psi$, the Grassmann integral is the fermionic Gaussian of Chapter 12, and the master formula proved there—that integrating out anticommuting variables produces the *determinant* of the kinetic matrix in the numerator, not the inverse square root—applies verbatim. For a single Dirac flavour,

$$\int \mathrm{d}\bar\psi\, \mathrm{d}\psi\; e^{-\bar\psi D[U]\psi} = \det D[U], \tag{43.189}$$

the lattice incarnation of the functional determinant $\det(i\not{D} - m)$ obtained in Section 12.7 (Eq. (12.155)). Observables built from fermion bilinears bring down propagators: by the same Wick combinatorics, the Grassmann average of $\psi(x)\bar\psi(y)$ inside the integral is $D[U]^{-1}_{xy}$, the quark propagator *on the fixed background* U. For a generic fermionic observable, integrating out the quarks therefore turns Eq. (43.187) into a purely bosonic integral over gauge fields,

$$\langle \mathcal{O} \rangle \;=\; \frac{1}{Z}\int \mathrm{d}U\;\; \widehat{\mathcal{O}}[U]\;\det D[U]\; e^{-S_g[U]}, \qquad Z = \int \mathrm{d}U\;\det D[U]\, e^{-S_g[U]}, \tag{43.190}$$

where $\widehat{\mathcal{O}}[U]$ is $\mathcal{O}$ with every quark bilinear replaced by the appropriate string of background propagators $D[U]^{-1}$ and signed Wick contractions (the lattice version of the contraction rules of Chapter 12). The price of removing the Grassmann variables is that the effective gauge weight is no longer local: $\det D[U]$ couples every link to every other, and the propagators $D[U]^{-1}$ require inverting a sparse but enormous matrix on each configuration. For N_f degenerate flavours the weight carries $\big(\det D[U]\big)^{N_f}$.

Positivity and the role of the determinant. For the Monte Carlo method to apply, the integrand of Eq. (43.190) must be a genuine probability weight: $\det D[U]\, e^{-S_g[U]} \geq 0$. The gauge factor e^{-S_g} is positive by construction. The determinant is not automatically positive for a single flavour, but the Wilson and staggered operators satisfy a γ_5-Hermiticity relation $D^\dagger = \gamma_5 D \gamma_5$, which forces the eigenvalues to be real or to occur in complex-conjugate pairs, so $\det D$ is real. For an even number of degenerate flavours one writes $(\det D)^2 = \det(D^\dagger D)$, which is manifestly non-negative because $D^\dagger D$ is a positive operator. Two degenerate light flavours—a good approximation to the physical up and down quarks—are thus simulated with a strictly positive weight, and this is why the two-flavour theory was the workhorse of the field for a generation. A single

flavour, or the residual phase of an odd-flavour or rooted-staggered setup, is handled by writing the weight as $|\det D|$ times a phase and folding the phase into the observable by *reweighting*,

$$\langle \mathcal{O} \rangle = \frac{\left\langle \widehat{\mathcal{O}}\, e^{i\theta[U]} \right\rangle_{|\det D|}}{\left\langle e^{i\theta[U]} \right\rangle_{|\det D|}}, \qquad \det D[U] = |\det D[U]|\, e^{i\theta[U]}, \tag{43.191}$$

where $\langle \cdot \rangle_{|\det D|}$ is the average over the positive ensemble defined by $|\det D|\, e^{-S_g}$. Reweighting is exact and harmless when the phase $e^{i\theta}$ stays near unity; when it fluctuates wildly it destroys the method, and that catastrophe—the sign problem—is the subject of Section 43.7.5.

Configurations as a Markov chain. The integral in Eq. (43.190) runs over millions of link variables; it cannot be tabulated, and ordinary quadrature is hopeless in such dimension. The way out is *importance sampling*: rather than evaluate the weight everywhere, generate a finite sequence of gauge configurations $\{U^{(1)}, U^{(2)}, \dots, U^{(N)}\}$ drawn from the probability distribution

$$P[U] \;=\; \frac{1}{Z}\, \det D[U]\, e^{-S_g[U]}, \tag{43.192}$$

and estimate the expectation value as a simple ensemble average,

$$\langle \mathcal{O} \rangle \;\approx\; \frac{1}{N} \sum_{i=1}^{N} \widehat{\mathcal{O}}\,[U^{(i)}], \qquad \text{with statistical error} \;\sim \frac{1}{\sqrt{N}}\, \sqrt{2\tau_{\mathcal{O}}}\, \sigma_{\mathcal{O}}, \tag{43.193}$$

the unknown Z having cancelled because it is the same normalization for every configuration. The error is the central-limit $1/\sqrt{N}$ of any Monte Carlo estimator, inflated by the integrated autocorrelation time $\tau_{\mathcal{O}}$ that measures how many successive configurations it takes to produce one statistically independent sample, and by the intrinsic variance $\sigma_{\mathcal{O}}^2$ of the observable. The configurations are produced by a Markov chain: a stochastic update $U \to U'$ with transition probability $W(U \to U')$ engineered so that its unique stationary distribution is exactly Eq. (43.192). It suffices that the chain be ergodic (able to reach any configuration) and satisfy *detailed balance*,

$$P[U]\, W(U \to U') = P[U']\, W(U' \to U), \tag{43.194}$$

which guarantees P is stationary: summing Eq. (43.194) over U gives $\sum_U P[U] W(U \to U') = P[U'] \sum_U W(U' \to U) = P[U']$. The classic realization is the *Metropolis* algorithm: propose a local change to a link, compute the resulting action change ΔS, and accept the proposal with probability $\min(1, e^{-\Delta S})$; the acceptance ratio satisfies Eq. (43.194) by inspection, because for $\Delta S > 0$ the forward step is accepted with $e^{-\Delta S}$ and the reverse with 1, and $P[U']/P[U] = e^{-\Delta S}$. Metropolis

is efficient for pure gauge theory, where ΔS depends only on the few plaquettes touching the changed link. With dynamical fermions it is crippling, because the nonlocal $\det D[U]$ changes under every local move and recomputing it costs an inversion of D.

Hybrid Monte Carlo and pseudofermions. The algorithm that made full QCD with dynamical quarks tractable is *Hybrid (or Hamiltonian) Monte Carlo* (HMC). One first removes the determinant from the weight by re-expressing it as a bosonic Gaussian integral over auxiliary complex fields—*pseudofermions* ϕ—using, for two flavours,

$$\det(D^\dagger D) \;\propto\; \int \mathrm{d}\phi^\dagger\, \mathrm{d}\phi \;\exp\!\big(-\phi^\dagger (D^\dagger D)^{-1}\phi\big), \tag{43.195}$$

which is the ordinary bosonic Gaussian (the determinant now sits in the numerator because the exponent carries the *inverse* of $D^\dagger D$). The full weight $e^{-S_g}\det(D^\dagger D)$ is thereby written as $e^{-S_{\mathrm{eff}}[U,\phi]}$ with a local-in-U but ϕ-dependent effective action whose only nonlocality is the matrix inverse $(D^\dagger D)^{-1}\phi$, computed once per update by a sparse linear solver. HMC then introduces a fictitious conjugate momentum π for every link, defines a fictitious Hamiltonian $H = \tfrac{1}{2}\sum \pi^2 + S_{\mathrm{eff}}$, and evolves (U,π) along a trajectory of Hamilton's equations in a fictitious "molecular-dynamics" time using a reversible, area-preserving leapfrog integrator. The integrator's finite step makes H change by a small ΔH; a concluding Metropolis accept/reject with probability $\min(1, e^{-\Delta H})$ corrects exactly for that discretization error, so the algorithm samples Eq. (43.192) without bias regardless of step size. The genius of HMC is that the global molecular-dynamics move changes every link at once—essential for moving through configuration space when the fermion determinant resists local updates—while remaining exact. For a single flavour, or for the fourth root of the staggered determinant, the square root $(D^\dagger D)^{1/4}$ is realized by a rational approximation (the rational-HMC variant). We state these algorithms at the level of their defining properties; their convergence and efficiency are the engineering core of the field, but the physics content is exhausted by Eq. (43.193): *a lattice prediction is an average over a finite, importance-sampled ensemble of gauge configurations, carrying a statistical error that falls as* $1/\sqrt{N}$.

Physical Insight

The deepest conceptual payoff of Eqs. (43.187) and (43.193) is that the path integral, which in the continuum is a formal symbol manipulated by Feynman rules, is here a literal probability average that a computer evaluates by drawing samples. There is no perturbation series, no gauge fixing, no Faddeev–Popov determinant, and no Gribov ambiguity (contrast Chapter 26): the configurations are sampled directly from the gauge-invariant

measure on SU(3) link variables. Confinement, chiral symmetry breaking, and the hadron spectrum are not summed up diagram by diagram—they are present in the ensemble, and one measures them.

43.7.2 Extracting masses: spectral decomposition of a Euclidean correlator

The single most important class of lattice observable is the two-point function of a local operator, because its Euclidean-time dependence encodes the mass spectrum. Let $\mathcal{O}$ be a gauge-invariant operator with the quantum numbers of a state one wishes to study—for a pion, $\mathcal{O} = \bar{d}\,\gamma_5 u$; for the proton, a gauge-invariant three-quark operator; for a glueball, a trace of plaquettes. Build the zero-momentum operator by summing over a spatial timeslice, $\mathcal{O}(t) = \sum_{\mathbf{x}} \mathcal{O}(\mathbf{x}, t)$, projecting onto vanishing spatial momentum. The object measured is the connected correlator

$$C(t) = \left\langle \mathcal{O}(t)\, \mathcal{O}^\dagger(0) \right\rangle - \langle \mathcal{O} \rangle \left\langle \mathcal{O}^\dagger \right\rangle . \tag{43.196}$$

We now prove that $C(t)$ decays as a sum of pure exponentials whose rates are the energies of the states $\mathcal{O}^\dagger$ creates, with the lightest dominating at large t.

Theorem 43.36 (Spectral decomposition of a Euclidean correlator)**.** *Let $\mathcal{T} = e^{-aH}$ be the positive self-adjoint transfer matrix of Section 43.1.4, with eigenstates $\mathcal{T}\,|n\rangle = e^{-aE_n}\,|n\rangle$, $E_0 \le E_1 \le \cdots$, normalized $\langle m|n\rangle = \delta_{mn}$, and let $\widehat{\mathcal{O}}$ be the operator on the transfer-matrix Hilbert space corresponding to $\mathcal{O}$. Writing the Euclidean separation as $t = n_t\, a$ with integer n_t, the connected correlator Eq. (43.196) on a lattice of temporal extent $L_t = N_t\, a$ with periodic boundary conditions is*

$$C(t) = \frac{1}{Z} \sum_{m,n} \left| \left\langle m \middle| \widehat{\mathcal{O}} \middle| n \right\rangle \right|^2 e^{-E_n t}\, e^{-E_m (L_t - t)}, \tag{43.197}$$

and, in the limit $L_t \to \infty$ with the vacuum $|0\rangle$ unique and the disconnected piece removed,

$$C(t) = \sum_{n \ge 1} \left| \left\langle 0 \middle| \widehat{\mathcal{O}} \middle| n \right\rangle \right|^2 e^{-(E_n - E_0)\, t} \;\equiv\; \sum_{n \ge 1} |c_n|^2\; e^{-M_n t}, \qquad M_n \equiv E_n - E_0. \tag{43.198}$$

Every coefficient $|c_n|^2 \ge 0$, so $C(t)$ is a sum of decaying exponentials with positive weights; at large t it is dominated by the lightest state $|1\rangle$ that $\widehat{\mathcal{O}}^\dagger$ overlaps, and $C(t) \to |c_1|^2\; e^{-M_1 t}$.

Proof. Euclidean time evolution is implemented by powers of the transfer matrix: an operator inserted n_t steps apart is conjugated by $\mathcal{T}^{n_t} = e^{-n_t a H} = e^{-Ht}$, so the lattice Heisenberg operator is $\widehat{\mathcal{O}}(t) = e^{Ht}\, \widehat{\mathcal{O}}\, e^{-Ht}$, exactly the imaginary-time

continuation of the Minkowski Heisenberg picture. The thermal trace that the periodic path integral computes is

$$C(t) = \frac{1}{Z}\,\mathrm{Tr}\big[\,e^{-(L_t-t)H}\,\widehat{\mathcal{O}}\,e^{-tH}\,\widehat{\mathcal{O}}^\dagger\,\big], \qquad Z = \mathrm{Tr}\;e^{-L_tH}, \tag{43.199}$$

the partition function Z being the trace of $\mathcal{T}^{N_t} = e^{-L_tH}$. This identity—that the Euclidean path integral on a torus of temporal circumference L_t equals the thermal trace at inverse temperature $\beta = L_t$—is the operator content of reflection positivity established in Section 43.1.4; we use it here. Insert the resolution of the identity $\mathbf{1} = \sum_n |n\rangle\langle n|$ twice, once on each side of $\widehat{\mathcal{O}}$:

$$\mathrm{Tr}\big[e^{-(L_t-t)H}\widehat{\mathcal{O}}\,e^{-tH}\widehat{\mathcal{O}}^\dagger\big] = \sum_{m,n} \Big\langle m\Big|\widehat{\mathcal{O}}\Big|n\Big\rangle\; e^{-E_nt}\;\Big\langle n\Big|\widehat{\mathcal{O}}^\dagger\Big|m\Big\rangle\; e^{-E_m(L_t-t)}. \tag{43.200}$$

Using $\Big\langle n\Big|\widehat{\mathcal{O}}^\dagger\Big|m\Big\rangle = \Big\langle m\Big|\widehat{\mathcal{O}}\Big|n\Big\rangle^*$ gives the modulus-squared and establishes Eq. (43.197). Now take $L_t \to \infty$. Every term carries $e^{-E_m(L_t-t)}$, which is exponentially suppressed unless $E_m = E_0$; with a unique vacuum the sum over m collapses to $m = 0$, and dividing by $Z \to e^{-E_0L_t}$ converts the surviving $e^{-E_0(L_t-t)}$ and e^{-E_nt} into $e^{+E_0t}e^{-E_nt} = e^{-(E_n-E_0)t}$. The $n = 0$ term is the disconnected piece $\Big|\Big\langle 0\Big|\widehat{\mathcal{O}}\Big|0\Big\rangle\Big|^2$, removed by the subtraction in Eq. (43.196); what remains is Eq. (43.198). Positivity of every $|c_n|^2 = \Big|\Big\langle 0\Big|\widehat{\mathcal{O}}\Big|n\Big\rangle\Big|^2$ is automatic. Since $M_n = E_n - E_0$ increases with n, the slowest-decaying surviving exponential is $n = 1$, which dominates as $t \to \infty$. □

Theorem 43.36 is the lattice counterpart of the Källén–Lehmann spectral representation of Theorem 14.23: both express a two-point function as a positively-weighted sum over the physical spectrum. The continuum version is an integral over a spectral density $\rho(M^2)$; the finite lattice, with its discrete transfer-matrix spectrum, gives a discrete sum. The key practical consequence is the same in both: *the lightest state with the chosen quantum numbers governs the long-distance (here, large Euclidean-time) behaviour*, so a hadron mass is read off as the asymptotic exponential decay rate of the corresponding correlator.

The effective mass. To expose the dominant exponential without fitting, define the *effective mass*

$$a\,M_{\mathrm{eff}}(t) \;\equiv\; \ln\frac{C(t)}{C(t+a)}. \tag{43.201}$$

If C were a single exponential $|c_1|^2\,e^{-M_1t}$, then Eq. (43.201) would return aM_1 exactly, for every t. With sub-leading states present, substitute Eq. (43.198)

and factor out the ground state, $C(t) = |c_1|^2 e^{-M_1 t}[1 + r\,e^{-\Delta t} + \cdots]$ with $r = |c_2|^2/|c_1|^2$ and gap $\Delta = M_2 - M_1 > 0$. Then

$$a\,M_{\text{eff}}(t) = \ln \frac{|c_1|^2 e^{-M_1 t}(1 + re^{-\Delta t})}{|c_1|^2 e^{-M_1(t+a)}(1 + re^{-\Delta(t+a)})} = aM_1 + \ln \frac{1 + re^{-\Delta t}}{1 + re^{-\Delta(t+a)}}$$
$$= aM_1 + r(1 - e^{-a\Delta})\, e^{-\Delta t} + O(r^2 e^{-2\Delta t}), \tag{43.202}$$

where the second line expands the logarithm for large t. The effective mass therefore approaches the true ground-state mass from above (if $r > 0$), with the excited-state contamination dying *exponentially* at the rate of the energy gap Δ. The signature of a clean measurement is a *plateau*: a range of t over which $M_{\text{eff}}(t)$ is constant within statistical errors, before the noise of Eq. (43.193) (which grows with t as the signal decays) swamps it. The plateau value is the mass.

Worked Example 43.37 (Reading a mass off the effective-mass plateau)**.** Take the two-state model correlator in lattice units ($a = 1$)

$$C(t) = A_0\, e^{-M_0 t} + A_1\, e^{-M_1 t}, \qquad M_0 = 0.5,\ M_1 = 1.2,\ A_0 = 1,\ A_1 = 0.8, \tag{43.203}$$

which mimics a correlator with a ground state at $aM_0 = 0.5$ and one excited state. We evaluate $C(t)$ and the effective mass Eq. (43.201) step by step. Directly,

$$C(0) = 1 + 0.8 = 1.8000, \qquad C(1) = e^{-0.5} + 0.8\,e^{-1.2} = 0.6065$$
$$C(2) = e^{-1.0} + 0.8\,e^{-2.4} = 0.36788 + 0.07257 = 0.44045, \qquad C(3) = e^{-1.5} + 0.8\,e^{-3.6} = 0.2231$$
$$C(4) = e^{-2.0} + 0.8\,e^{-4.8} = 0.13534 + 0.00658 = 0.14192, \qquad C(5) = e^{-2.5} + 0.8\,e^{-6.0} = 0.0820$$

and so on. The effective masses follow from ratios of consecutive entries:

t	0	1	2	3	4	5	6
$M_{\text{eff}}(t)$	0.7533	0.6545	0.5866	0.5460	0.5236	0.5120	0.5060

(43.204)

The sequence falls monotonically toward $aM_0 = 0.5$. The rate of approach is exactly Eq. (43.202): with $r = A_1/A_0 = 0.8$ and $\Delta = M_1 - M_0 = 0.7$, the leading correction is $r(1 - e^{-\Delta})e^{-\Delta t} = 0.8(1 - e^{-0.7})e^{-0.7t} = 0.4027\,e^{-0.7t}$. At $t = 4$ this predicts $0.5 + 0.4027\,e^{-2.8} = 0.5 + 0.0245 = 0.5245$, against the exact 0.5236; the small residual is the $O(r^2)$ term. By $t = 6$ the contamination has fallen below 0.6% and the plateau is established. In a real computation one fits the plateau region—here $t \gtrsim 5$—to a constant (or fits $C(t)$ itself to one or two exponentials over the same window), and the fitted value is the hadron mass in lattice units, aM. Figure 43.9 plots the approach and the analytic contamination curve together.

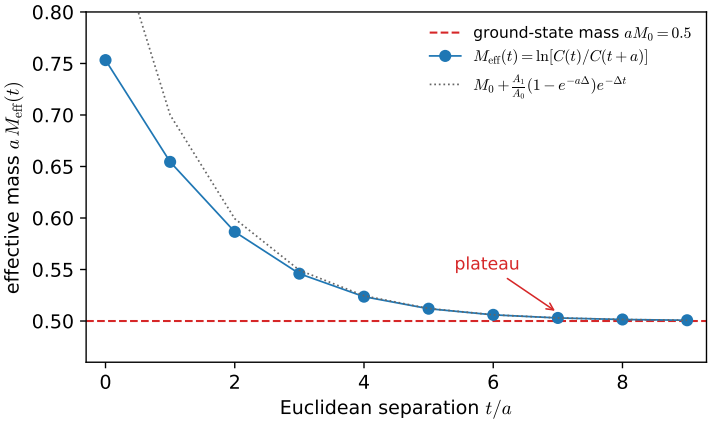


Figure 43.9: The effective mass $aM_{\rm eff}(t) = \ln[C(t)/C(t+a)]$ for the model correlator Eq. (43.203) of Theorem 43.37. It descends toward the ground-state value $aM_0 = 0.5$ (dashed) as the excited-state contamination dies at the gap rate $\Delta = M_1 - M_0$; the dotted curve is the leading analytic estimate Eq. (43.202). The flat region at large t is the plateau from which the mass is read.

Caution

On a torus of finite temporal extent L_t the full Eq. (43.197) is not a single exponential but a symmetric combination. For a single dominant state the dominant terms are $n{=}1, m{=}0$ and $n{=}0, m{=}1$, giving $C(t) \propto e^{-M_1 t} + e^{-M_1(L_t - t)} = 2e^{-M_1 L_t/2}\cosh\!\left[M_1(L_t/2 - t)\right]$ for a bosonic operator (and a sinh, or a forward/backward-propagating split, for baryons whose forward and backward states differ). The effective mass is then defined by solving $C(t)/C(t+a) = \cosh[\cdot]/\cosh[\cdot]$ rather than by the naive logarithm, and the plateau sits near the midpoint $t = L_t/2$. Forgetting the backward-propagating image is a classic way to misread a mass at small L_t.

43.7.3 Removing the regulator: scale setting and the continuum limit

Every number computed so far is dimensionless: Eq. (43.193) produces pure numbers, and the masses of Theorem 43.37 come out as the dimensionless products aM. The lattice spacing a is the only dimensionful quantity in the simulation, and it is *not* an input one dials in physical units—it is an output, fixed implicitly by the bare coupling g_0 (equivalently the inverse-coupling $\beta_{\rm lat} = 2N/g_0^2$ of the Wilson action) and the bare quark masses. To produce a physical prediction one must (i) determine a in physical units at each bare coupling—*scale setting*—and

(ii) extrapolate to $a \to 0$—the *continuum limit*. Both steps are governed by the renormalization group, and the form of the continuum limit is dictated by asymptotic freedom.

The continuum limit is an ultraviolet fixed point. The lattice spacing is the ultraviolet cutoff: momenta are bounded by π/a. Sending $a \to 0$ at fixed physical quantities is therefore the statement that the cutoff is removed while the physics is held fixed—precisely the renormalization-group flow toward an ultraviolet fixed point (Chapter 23). A physical mass M_{phys} is held fixed only if the dimensionless lattice mass scales as $aM_{\text{phys}} \to 0$, i.e. the correlation length $\xi = 1/(aM_{\text{phys}})$ in lattice units diverges. A divergent correlation length is the defining feature of a *critical point* of a statistical system. The continuum limit of a lattice field theory is thus a second-order critical point of the corresponding statistical mechanics, and which critical point QCD flows to is fixed by asymptotic freedom: the relevant fixed point is the *Gaussian* (free) fixed point at $g_0 = 0$, approached from the asymptotically free side. We now make this quantitative by importing the Yang–Mills beta function.

Asymptotic scaling from the two-loop beta function. The bare lattice coupling g_0 runs with the cutoff scale $\mu \sim 1/a$ exactly as the renormalized coupling runs with μ, because the lattice is just one more regularization scheme. Recall from Theorem 27.7 the renormalization-group equation $\mu \frac{dg}{d\mu} = \beta(g)$ with the two-loop beta function (Eqs. (27.101) and (27.171))

$$\beta(g) = -\beta_0\, g^3 - \beta_1\, g^5 + O(g^7), \qquad \beta_0 = \frac{1}{16\pi^2}\Big(\tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f\Big), \quad \beta_1 = \frac{1}{(16\pi^2)^2}\Big(\tfrac{34}{3}C_A^2 - (\tfrac{20}{3}C \tag{43.205}$$

Because $\mu \sim 1/a$ we have $\mathrm{d}\ln\mu = -\mathrm{d}\ln a$, so $a\,\frac{dg_0}{da} = -\beta(g_0) = \beta_0 g_0^3 + \beta_1 g_0^5 + \cdots$. As $a \to 0$ this drives $g_0 \to 0$: the bare coupling vanishes at the continuum limit, the hallmark of asymptotic freedom. Separating variables,

$$\frac{\mathrm{d}a}{a} = \frac{\mathrm{d}g_0}{\beta_0 g_0^3 + \beta_1 g_0^5 + \cdots} = \frac{\mathrm{d}g_0}{\beta_0 g_0^3}\Big(1 - \frac{\beta_1}{\beta_0} g_0^2 + \cdots\Big) = \Big(\frac{1}{\beta_0 g_0^3} - \frac{\beta_1}{\beta_0^2 g_0} + \cdots\Big)\mathrm{d}g_0. \tag{43.206}$$

Integrating term by term,

$$\ln a = -\frac{1}{2\beta_0 g_0^2} - \frac{\beta_1}{\beta_0^2}\ln g_0 + \text{const} + O(g_0^2), \tag{43.207}$$

and exponentiating, the lattice spacing is tied to the bare coupling by the *asymptotic-scaling formula*

$$\boxed{a\,\Lambda_{\text{lat}} = (\beta_0 g_0^2)^{-\beta_1/(2\beta_0^2)} \exp\Big(-\frac{1}{2\beta_0 g_0^2}\Big)\,[1 + O(g_0^2)],} \tag{43.208}$$

where the integration constant has been packaged into a dimensionful scale $\Lambda_{\rm lat}$. One checks directly that $\Lambda_{\rm lat}$ is a renormalization-group invariant: differentiating $a\Lambda_{\rm lat} = f(g_0)$ and using $a\,\mathrm{d}g_0/\mathrm{d}a = -\beta(g_0)$ gives $a\,\mathrm{d}\Lambda_{\rm lat}/\mathrm{d}a = 0$ to the order worked. This is precisely the dimensional transmutation of Theorem 27.14: a dimensionless coupling traded for a dimensionful scale through the logarithmic running of the quantum theory, here producing the lattice version $\Lambda_{\rm lat}$ of the continuum $\Lambda_{\rm QCD}$ (Eq. (27.153)). The two are related by a fixed, computable one-loop conversion factor $\Lambda_{\rm QCD}/\Lambda_{\rm lat}$, scheme matching being a one-loop calculation of the kind in Eq. (27.161). The decisive content of Eq. (43.208) is the essential singularity $e^{-1/(2\beta_0 g_0^2)}$: the lattice spacing shrinks *exponentially fast* as $g_0 \to 0$, so a modest decrease in the bare coupling buys a large reduction in a. This exponential approach to the continuum, with no power-law freedom, is the gift of asymptotic freedom; were the beta function positive, no such limit would exist and the lattice could not define the theory.

Worked Example 43.38 (How much must the coupling change to halve the lattice spacing?)**.** Take pure SU(3) gauge theory ($n_f = 0$, $C_A = N = 3$), so $\beta_0 = \frac{1}{16\pi^2}\cdot 11 = 11/(16\pi^2) = 0.06966$. Keep only the leading exponential in Eq. (43.208), $a\Lambda_{\rm lat} \simeq e^{-1/(2\beta_0 g_0^2)}$. The ratio of spacings at two couplings is

$$\frac{a'}{a} = \exp\Big[-\frac{1}{2\beta_0}\Big(\frac{1}{g_0'^2} - \frac{1}{g_0^2}\Big)\Big]. \tag{43.209}$$

To halve the spacing, $a'/a = \frac{1}{2}$, requires $\frac{1}{g_0'^2} - \frac{1}{g_0^2} = 2\beta_0 \ln 2 = 2(0.06966)(0.6931) = 0.0966$. Starting from $g_0^2 = 1$ (Wilson inverse-coupling $\beta_{\rm lat} = 6/g_0^2 = 6.0$), the new coupling satisfies $1/g_0'^2 = 1.0966$, i.e. $g_0'^2 = 0.9119$ and $\beta'_{\rm lat} = 6/0.9119 = 6.58$. Thus an increase of the simulation parameter from $\beta_{\rm lat} = 6.0$ to ≈ 6.6 halves the lattice spacing—in good qualitative accord with the empirical fact that $a \approx 0.094\,\mathrm{fm}$ at $\beta_{\rm lat} = 6.0$ and roughly half that near $\beta_{\rm lat} = 6.6$. The exponential in Eq. (43.208) is what makes the continuum reachable: each step of $a \to a/2$ costs only a finite increment in $\beta_{\rm lat}$, not an exponentially large one. (Quantitatively the two-loop prefactor and higher corrections shift these numbers; Eq. (43.208) is the leading systematics, and the onset of the regime where it holds is called the scaling window.)

Scale setting and lines of constant physics. Equation (43.208) fixes how *a scales* with g_0 but not its absolute value: the constant $\Lambda_{\rm lat}$ must be pinned to experiment. *Scale setting* does this by computing one chosen dimensionful quantity on the lattice—in pure numbers, as $a\,Q_{\rm lat}$—and dividing by its measured physical value to read off a:

$$a = \frac{(a\,Q)_{\rm lat}}{Q_{\rm phys}}. \tag{43.210}$$

Good scale-setting quantities are precisely measured, weakly dependent on the quark masses, and cheap to compute: common choices are a hadron mass or

decay constant (the Ω^- baryon mass, the pion or kaon decay constant f_π, f_K), the static-quark force scales r_0, r_1 extracted from the heavy-quark potential, or the purely gluonic gradient-flow scales t_0, w_0. Once $a(g_0)$ is known, every other lattice number is converted to physical units by multiplying by the appropriate power of $1/a$. The bare quark masses must also be tuned: at each g_0 one adjusts them so that a set of dimensionless ratios—typically the ratios of pseudoscalar masses to the scale-setting quantity, e.g. m_π/m_Ω and m_K/m_Ω—equal their physical values. A curve in the space of bare parameters $(g_0, m_{0,u}, m_{0,d}, m_{0,s}, \dots)$ along which all such dimensionless ratios are held fixed while a varies is a *line of constant physics*: it is the lattice realization of an RG trajectory, the path one walks toward the continuum while keeping the theory physically the same.

The continuum extrapolation. With a known at several couplings along a line of constant physics, any predicted dimensionless ratio R is computed at each a and extrapolated to $a = 0$. The cutoff dependence is governed by the lattice analogue of irrelevant operators (Chapter 23): the leading discretization error is a power of a set by the action. The Wilson fermion action of Section 43.6 has $O(a)$ errors, removable by the Symanzik improvement programme (adding higher-dimension operators tuned to cancel the linear term) to leave $O(a^2)$; staggered and many improved actions are $O(a^2)$ from the outset. One therefore fits

$$R(a) = R_{\text{cont}} + c_1\, a + c_2\, a^2 + \cdots \tag{43.211}$$

(omitting the c_1 term for an $O(a^2)$-improved action) and takes the intercept R_{cont} as the continuum prediction. Figure 43.10 shows the geometry: dimensionless data computed at several squared lattice spacings, with a straight-line extrapolation in a^2 to the continuum value on the axis. The honesty of a lattice result lives in this plot—in the spread of lattice spacings, the smallness of the slope, and whether several different discretizations extrapolate to the same intercept. A modern result quotes a continuum-extrapolated number with a statistical error from Eq. (43.193) and a systematic error dominated by the continuum and infinite-volume extrapolations.

Physical Insight

The continuum limit is the moment the regulator disappears and the answer becomes a property of quantum field theory rather than of the lattice. That this limit exists at all, and is reached by an exponentially fast flow of the bare coupling toward zero, is asymptotic freedom doing double duty: in Chapter 27 it explained why short-distance QCD is calculable in perturbation theory; here it guarantees that a non-perturbative lattice definition possesses a continuum limit. The same negative beta function that liberates quarks at high energy is what lets us *define* the theory by

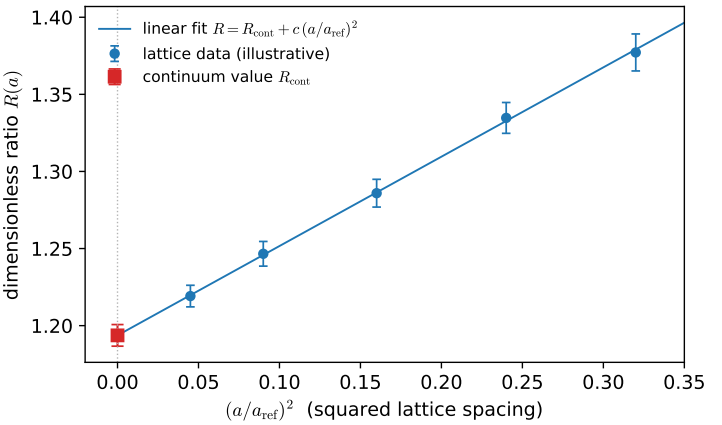


Figure 43.10: Continuum extrapolation of a dimensionless ratio R computed at several lattice spacings along a line of constant physics. For an $O(a^2)$-improved discretization the cutoff dependence is linear in $(a/a_{\rm ref})^2$ per Eq. (43.211); the intercept on the axis ($a \to 0$, red square) is the continuum value. Data points are illustrative.

taking $a \to 0$.

43.7.4 What the lattice has delivered (mid-2020s)

We can now state, honestly and with dates, what the machine of Eqs. (43.193), (43.198) and (43.208) has produced. The following are established results of lattice QCD as of the mid-2020s, computed from the QCD Lagrangian with no input beyond the quark masses and the overall scale.

The light-hadron spectrum from first principles. The masses of the lightest mesons and baryons—$\pi, K, \rho, K^*, N, \Lambda, \Sigma, \Xi, \Omega, \ldots$—have been computed with full dynamical u, d, s (and increasingly c) quarks, extrapolated to the continuum and to physical quark masses and infinite volume, in agreement with experiment at the percent level. This is the carried payoff of Yang–Mills theory: the proton mass, $\approx 938\,\text{MeV}$, emerges almost entirely from the gluonic field energy and dimensional transmutation—the bare light-quark masses contribute only a few percent—so the nucleon mass is a prediction of the dynamics, set by $\Lambda_{\rm QCD}$ (Theorem 27.14) and not by any mass written into the Lagrangian. The same ensembles deliver the string tension σ of the confining potential, whose strong-coupling estimate $\sqrt{\sigma} \sim 440\,\text{MeV}$ in Eq. (27.193) is confirmed and made precise, converting the analytic strong-coupling area law into a measured physical scale.

Quark masses and the strong coupling. Because the quark masses are not observable in isolation (quarks are confined), their values are *defined* by matching

lattice hadron masses to experiment and are quoted as renormalized parameters in a continuum scheme, e.g. the $\overline{\text{MS}}$ masses m_u, m_d, m_s, m_c, m_b at $\mu = 2\,\text{GeV}$ or at their own scale. Lattice QCD provides the most precise determinations of several of these, and competitive determinations of $\alpha_s(M_Z)$ via short-distance lattice quantities matched to perturbation theory—feeding the world average that Eq. (27.155) runs.

The finite-temperature transition and the QCD phase structure. Repeating the construction at temporal extent $L_t = 1/T$ (so the temporal circumference sets the temperature T, in natural units with $k_B = 1$) probes QCD thermodynamics. Lattice computations establish that the transition from confined hadronic matter to the deconfined quark–gluon plasma at zero baryon density is a smooth *crossover*, not a true phase transition, at a pseudocritical temperature $T_c \approx 155\,\text{MeV}$, and they map the equation of state $p(T)$, $\epsilon(T)$ used to interpret heavy-ion collisions. The deconfinement transition is diagnosed by the Polyakov loop (the temporal Wilson line, a gauge-invariant order parameter for the centre symmetry) and the chiral transition by the chiral condensate.

The hadronic vacuum polarization for the muon $g-2$. This is the live retouch of QED. The leading hadronic contribution a_μ^{HVP} to the muon anomaly of Eq. (20.120) is the non-perturbative quark-loop dressing of the photon vacuum polarization (Fig. 20.6a), intrinsically beyond perturbation theory at the muon scale. Lattice QCD computes it directly, as a weighted Euclidean integral of the electromagnetic-current correlator—an instance of Eq. (43.198) for the vector current. As recounted in Section 20.4.4, the first sub-percent lattice computation (BMW 2021) came out larger than the long-standing data-driven dispersive value, shifting the Standard-Model prediction toward the measured a_μ and turning a 4σ experiment-versus-theory "anomaly" into a theory-versus-theory tension; the 2023 CMD-3 $e^+e^- \to \pi^+\pi^-$ remeasurement and the 2025 white-paper update have since moved the consensus toward the lattice. That a lattice determination of a single non-perturbative number can adjudicate the most precise confrontation between theory and experiment in particle physics is the clearest demonstration that the lattice computes physics, not artefacts.

Physical Insight

The throughline of this survey is that lattice QCD has graduated from confirming known masses to *providing* numbers that no other method can: hadronic matrix elements for flavour physics (decay constants, form factors, bag parameters feeding the CKM tests of Chapter 32), the HVP for $g-2$, the equation of state of the quark–gluon plasma. Where perturbation theory and the lattice overlap they agree; where perturbation theory fails—the confining, chiral-symmetry-breaking infrared—the lattice stands alone and is checked against experiment.

43.7.5 What it has not delivered: the sign problem

The Monte Carlo method of Section 43.7.1 rests on one load-bearing assumption: that the weight $\det D[U]\, e^{-S_g[U]}$ is real and non-negative, so that Eq. (43.192) is a probability and Eq. (43.193) is an honest average. Two physically central regimes violate this assumption irreparably, and they bound what the lattice can presently do.

Finite baryon density. Introducing a baryon chemical potential μ_B to study dense matter—the interior of neutron stars, the conjectured colour-superconducting and critical-point structure of the QCD phase diagram—enters the Dirac operator as $D(\mu_B)$ with the temporal links weighted by $e^{\pm a\mu_B}$. The γ_5-Hermiticity that made $\det D$ real becomes, at $\mu_B \neq 0$,

$$\big[\det D(\mu_B)\big]^* = \det D(-\mu_B^*), \tag{43.212}$$

so for real μ_B the determinant is genuinely *complex*: $\det D(\mu_B) \neq [\det D(\mu_B)]^*$. The weight is no longer a probability, importance sampling has nothing to sample from, and one is forced into the reweighting of Eq. (43.191). There the failure becomes quantitative. The average phase that must be divided out scales as

$$\left\langle e^{i\theta} \right\rangle_{|\det D|} = \frac{Z}{Z_{|\det D|}} = e^{-V\,\Delta f/T}, \tag{43.213}$$

the ratio of the true partition function to the phase-quenched one, which is the exponential of (minus) a free-energy difference times the spacetime volume V. Because $\Delta f > 0$ generically, the signal-to-noise ratio of Eq. (43.191) falls *exponentially* in the volume: extracting a result to fixed accuracy requires a number of configurations growing like $e^{+V\Delta f/T}$, which is prohibitive for any interesting volume. This is the *sign problem*. It is not a defect of a particular algorithm; the generic complex-weight sampling problem is NP-hard, so a general efficient cure would collapse a complexity class widely believed not to collapse. Partial workarounds—Taylor expansion in μ_B/T about zero density, analytic continuation from imaginary chemical potential (where the weight is real), reweighting at small μ_B, and the more speculative complex-Langevin and Lefschetz-thimble methods—extend the reach a little way from $\mu_B = 0$, enough to constrain the curvature of the crossover line and bound where a critical point could lie, but the cold dense interior of the phase diagram remains beyond first-principles reach as of the mid-2020s.

Real-time dynamics. The same obstruction, in a different guise, blocks real-time and transport quantities. Everything above lives in Euclidean (imaginary) time, where e^{-S_E} is real and positive. Physical real-time evolution, e^{iS}, is a pure oscillatory phase—the worst possible sign problem. Quantities defined by real-time correlators (transport coefficients such as the shear viscosity of the quark–gluon plasma, jet-quenching parameters, the spectral functions that encode

particle widths and thermalization, and any genuinely out-of-equilibrium process) are accessible on the lattice only indirectly, by the ill-posed analytic continuation of Euclidean correlators back to real frequencies—inverting Eq. (43.198) for the full spectral density rather than just its lowest pole, a numerically unstable problem. Improving these reconstructions is an active frontier, but real-time QCD dynamics is not solved.

These two frontiers—finite baryon density and real-time dynamics—are the principal standing limitations of the lattice programme. They are honest boundaries of the method, not of the definition: the theory is perfectly well defined by Eq. (43.187) at any μ_B and in real time; it is the *stochastic evaluation* that fails when the weight ceases to be positive. Whether a fundamentally different sampling strategy—or quantum simulation, which evolves the real-time dynamics natively—will dissolve the sign problem is one of the genuine open questions of computational field theory.

Having turned the positive gauge-invariant measure into reproducible numbers, watched the hadron spectrum and the muon anomaly emerge from first principles, and marked the precise places where the method runs out, we have assembled the full evidence base for the chapter's concluding claim. The next section steps back from technique to thesis: that this constructive, regulator-complete, manifestly gauge-invariant, non-perturbative procedure is not merely a way to compute QCD but a *definition* of quantum field theory—and it weighs, with the honesty these results demand, exactly which theories are thereby proven to exist and which remain, for now, numerical.

43.8 Synthesis: the lattice as a definition of quantum field theory

We have, by this point, assembled every piece of the machine. The scalar discretization fixed the framework and showed that the continuum limit *is* the approach to a renormalization-group fixed point; gauge invariance forced the field onto links and handed us the first analytic proof of confinement in the strong-coupling phase; the Nielsen–Ninomiya theorem forced genuinely new constructions for fermions; and importance sampling turned the resulting finite-dimensional measure into hadron masses with quantified errors. The individual results are in hand. What remains is to step back and ask what they amount to *together*. The claim of this chapter — promised in its opening and now to be defended — is not the modest one that the lattice is a useful numerical scheme. It is the strong one: the lattice is, as of the mid-2020s, the only known fully non-perturbative, first-principles, manifestly gauge-invariant *definition* of an interacting quantum field theory. It is a constructive answer to the question that has shadowed the entire book, "what is a QFT, exactly, beyond a recipe for an asymptotic series?" This section states that thesis precisely, shows how the

earlier sections combine to support it, and then — because honesty is part of the contract — separates cleanly what the lattice has *proved* from what it has only *computed*, and lays out the frontiers where even the definition is not yet under control.

43.8.1 The lattice as a complete definition

Begin with what a definition of a QFT must supply. Perturbation theory does not supply it: the Feynman expansion is at best an asymptotic series (the Borel analysis of Chapter 5, Section 5.7), so the sum of all orders does not define a function, and the defining phenomenon of QCD — confinement — is invisible to every finite order. A genuine definition must produce, from a finite set of inputs, a complete and self-consistent set of correlation functions, with no appeal to a series and no hidden regulator left undefined. The lattice does exactly this, in three logically separate steps that the earlier sections established one at a time.

Definition 43.39 (Lattice-regularized field theory)**.** Fix a lattice spacing $a > 0$ and a finite hypercubic lattice $\Lambda_a = a\,(\mathbb{Z}/L\mathbb{Z})^d$ of $N_{\mathrm{s}} = L^d$ sites. A *lattice field theory* is the data of

1. finitely many real integration variables — a field value $\phi_x \in \mathbb{R}^k$ at each site $x \in \Lambda_a$ for matter, or a group element $U_\ell \in G$ on each oriented link ℓ for gauge fields;
2. a real action $S[\phi, U]$ that is bounded below and grows fast enough at large field that e^{-S} is integrable against the product of Lebesgue measures $\prod_x \mathrm{d}^k\phi_x$ and Haar measures $\prod_\ell \mathrm{d}U_\ell$;

together with the resulting probability measure and its expectation values

$$\mathrm{d}\mu_a \;=\; \frac{1}{Z_a}\, e^{-S[\phi,U]} \prod_x \mathrm{d}^k\phi_x \prod_\ell \mathrm{d}U_\ell, \qquad \langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle \;=\; \int \mathcal{O}_1 \cdots \mathcal{O}_n \,\mathrm{d}\mu_a, \tag{43.214}$$

with $Z_a = \int e^{-S} \prod \mathrm{d}\phi \prod \mathrm{d}U$ finite and nonzero.

This is the first and most important point, and it is the one that distinguishes the lattice from every other regulator in the book. The object in Eq. (43.214) is an *ordinary integral* over a finite-dimensional space. It is completely well defined: there are no ultraviolet divergences to subtract, because the spacing a is a hard momentum cutoff π/a built into the kinematics; there is no need for dimensional continuation, Pauli–Villars fields, or any analytic-continuation crutch; and — the decisive virtue for gauge theory — there is no gauge fixing. The Haar measure $\prod_\ell \mathrm{d}U_\ell$ is finite on the compact group G, so the integral over the entire gauge orbit converges without dividing by an infinite orbit volume. There is no Faddeev–Popov determinant, no ghost, and therefore no Gribov ambiguity of the

kind that obstructs a globally valid gauge choice in the continuum (Chapter 26, Section 26.1.4). The price — the loss of manifest Lorentz invariance, broken to the hypercubic point group — is paid up front and is recovered only in the continuum limit, and we accept it because gauge invariance, not Lorentz invariance, is the symmetry whose violation is fatal to a regulator. This is what "regulator-complete" means: Eq. (43.214) needs nothing outside itself to be a number.

Physical Insight

The Euclidean weight e^{-S_E} of Chapter 10 (Section 10.6) is, after Wick rotation, real and positive, so $\mathrm{d}\mu_a$ is a genuine probability measure and a QFT is literally a problem in classical statistical mechanics: a Boltzmann distribution on field configurations at "temperature" set by the couplings. Every continuum question becomes a question about a statistical-mechanical system — correlation lengths, phase transitions, critical exponents. This is not an analogy. It is an identity, and it is the reason the renormalization group of Chapter 23, invented for critical phenomena, governs the continuum limit of field theory verbatim.

The second step is that this statistical system carries, hidden inside its positivity, the entire Lorentzian content of a quantum theory. Section 1 established *reflection positivity*: for observables $\mathcal{O}$ supported on one side of a reflection plane and $\Theta\mathcal{O}$ its reflected conjugate, $\langle(\Theta\mathcal{O})\,\mathcal{O}\rangle \geq 0$. This single inequality is the lattice incarnation of Osterwalder–Schrader positivity, and it does two things at once. On the lattice it guarantees a self-adjoint, positive transfer matrix $\hat{T} = e^{-a\hat{H}}$ with $\hat{H} \geq 0$, whose spectral decomposition is exactly what Section 43.7 used to read hadron masses off the exponential decay of Euclidean correlators. In the continuum limit it is the hypothesis of the reconstruction theorem.

Theorem 43.40 (Osterwalder–Schrader reconstruction; imported, stated)**.** *A family of Euclidean correlation (Schwinger) functions that is Euclidean-invariant, reflection positive, symmetric, suitably regular, and clustering determines a unique Wightman quantum field theory: a separable Hilbert space with a positive-energy unitary representation of the Poincaré group, a unique vacuum, and Wightman fields whose vacuum correlators are the analytic continuations of the Schwinger functions. This is the content of Theorem 10.32; we use it here as an imported result.*

We do not reprove Theorem 43.40 — it is owned by the constructive-QFT framing of Chapter 10 — but we lean on it for the precise sense of "definition." It tells us that to construct a relativistic quantum field theory it suffices to construct a reflection-positive set of Euclidean correlators with the right symmetry and decay. The lattice produces exactly such correlators at every finite a; what is left is to remove the cutoff.

That removal is the third step, and it is where Chapter 23 enters. Sending $a \to 0$ at fixed physics is not a naive limit of Eq. (43.214) — taken at fixed bare couplings it gives back free or trivial physics. It is the approach to a *second-order critical point* of the statistical system, a point where the correlation length in lattice units, ξ/a, diverges. There the lattice forgets its microscopic details (universality, Section 23.6); a physical mass $m = (\xi/a)^{-1} \cdot a^{-1} = 1/\xi_{\text{phys}}$ can be held fixed while $a \to 0$ only if $\xi/a \to \infty$, which is precisely criticality. Tuning the relevant couplings (the mass, Eq. (23.177)) onto the critical surface (Theorem 23.24) and flowing into a renormalization-group fixed point is the continuum limit (Section 23.6.6, Eq. (23.178)). Assembling the three steps gives the thesis in one sentence.

Physical Insight

The lattice definition of a quantum field theory. A non-perturbative quantum field theory *is* the continuum limit of a reflection-positive lattice measure Eq. (43.214) taken along a renormalization-group trajectory into a fixed point: a finite, manifestly gauge-invariant statistical-mechanical measure (Theorem 43.39); reflection positivity (Section 1) so that Theorem 43.40 applies; and a critical point (Chapter 23) at which the cutoff can be removed. When such a limit exists and its Schwinger functions satisfy the Osterwalder–Schrader axioms, the theory *exists* in the constructive sense of Chapter 10; perturbation theory is then a controlled asymptotic expansion *about* this definition, not a substitute for it.

Two clauses in that statement carry the entire honesty of the chapter, and the rest of this section is devoted to them: "when such a limit exists" (which theories provably exist?), and the gap between "the strong-coupling measure confines" (proved) and "the continuum theory confines" (computed).

43.8.2 The existence dichotomy: triviality versus asymptotic freedom

Whether the continuum limit of a given lattice theory exists as a nontrivial interacting theory is governed by a single asymptotic fact: the behavior of the running coupling as the cutoff is removed, i.e. the sign of the leading coefficient of the beta function. The reason is now transparent from the fixed-point picture. Removing the cutoff means $a \to 0$, i.e. flowing toward the ultraviolet. A nontrivial continuum limit exists precisely when the ultraviolet end of the flow lands on a fixed point at which the theory is interacting — or, for the perturbatively accessible case, when the *Gaussian* fixed point at the origin of coupling space is approached from a direction along which the coupling is marginally relevant in the right sense. This is the criterion, and it sorts the threads of the book cleanly.

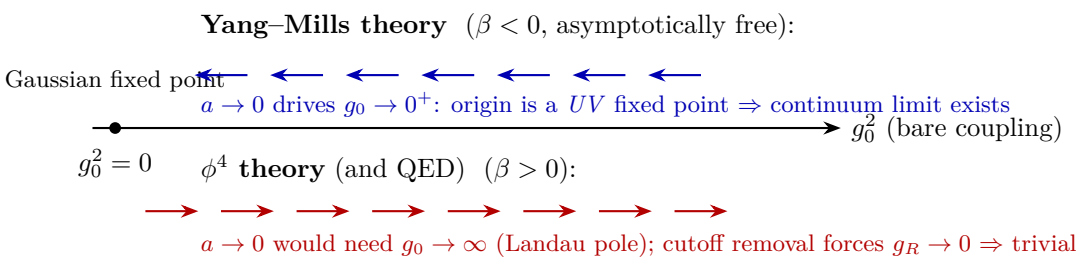


Figure 43.11: The continuum limit as a flow in bare coupling. For an asymptotically free theory (Chapter 27) the origin is an ultraviolet fixed point: as $a \to 0$ the bare coupling is driven *into* it, $g_0 \to 0^+$, and the limit exists with a dynamically generated scale. For a theory with $\beta > 0$ the flow runs the bare coupling *up* toward a Landau pole; no ultraviolet fixed point is reachable, and removing the cutoff at fixed physics squeezes the renormalized coupling to zero — triviality.

The asymptotically free side is the one that works, and the worked computation below makes "the continuum limit is a UV fixed point" fully quantitative. We continue Yang–Mills theory from the asymptotic freedom of Chapter 27 (Theorem 27.11).

Worked Example 43.41 (Asymptotic scaling: the continuum limit of Yang–Mills as a UV fixed point)**.** On the lattice the only scale is a, so the bare coupling g_0 runs with a. Holding physical quantities fixed, the Callan–Symanzik content of Chapter 27 reads, with $\mu = 1/a$ identified as the cutoff scale and $\beta(g) = \mathrm{d}g/\mathrm{d}\ln\mu$,

$$a\,\frac{\mathrm{d}g_0}{\mathrm{d}a} \;=\; b_0\,g_0^3 + b_1\,g_0^5 + \cdots, \qquad b_0 = \frac{1}{(4\pi)^2}\Big(\tfrac{11}{3}C_A - \tfrac{4}{3}T_F n_f\Big), \tag{43.215}$$

where the sign is fixed by $\beta = -b_0 g^3 - \cdots < 0$ (Eq. (27.101)) and $\mathrm{d}\ln\mu = -\mathrm{d}\ln a$. For $SU(3)$ with n_f flavors, $b_0 = (11 - \frac{2}{3}n_f)/(4\pi)^2 > 0$ as long as $n_f < \frac{33}{2}$, so the right-hand side is positive: *as a decreases, g_0 decreases.* The continuum limit $a \to 0$ is the limit $g_0 \to 0^+$, and the origin is a UV fixed point.

Integrate Eq. (43.215) to find exactly how fast. Separate variables,

$$\frac{\mathrm{d}a}{a} = \frac{\mathrm{d}g_0}{b_0 g_0^3 + b_1 g_0^5} = \frac{\mathrm{d}g_0}{b_0 g_0^3}\,\frac{1}{1 + (b_1/b_0)g_0^2} = \frac{1}{b_0}\Big[g_0^{-3} - \frac{b_1}{b_0}\,g_0^{-1} + O(g_0)\Big]\mathrm{d}g_0, \tag{43.216}$$

where in the last step we expanded $(1 + (b_1/b_0)g_0^2)^{-1} = 1 - (b_1/b_0)g_0^2 + \cdots$ for small g_0. Integrate term by term:

$$\ln a = \frac{1}{b_0}\Big[-\frac{1}{2g_0^2} - \frac{b_1}{b_0}\ln g_0\Big] + \text{const} = -\frac{1}{2b_0 g_0^2} - \frac{b_1}{b_0^2}\ln g_0 + \text{const}. \tag{43.217}$$

Exponentiating, and absorbing the constant into an integration scale Λ_L with the dimensions of inverse length (the lattice analogue of Λ_{QCD}, Eq. (27.152)),

gives the *two-loop asymptotic-scaling formula*

$$\boxed{a\,\Lambda_L \;=\; (b_0 g_0^2)^{-b_1/(2b_0^2)}\,\exp\!\Big(-\frac{1}{2b_0\,g_0^2}\Big)\,[1+O(g_0^2)].} \tag{43.218}$$

The content is the boxed exponential. As $g_0 \to 0^+$ the right-hand side vanishes faster than any power, so $a \to 0$: the continuum is reached by sending the bare coupling to the Gaussian fixed point, exactly as Fig. 43.11 draws it. The combination Λ_L is renormalization-group invariant by construction — it is what is held fixed while a and g_0 both run — so every physical mass is $m_i = c_i\,\Lambda_L$ with c_i a pure dimensionless number. This is *dimensional transmutation* (Section 27.6.2) realized on the lattice: a classically scale-free, parameter-free pure gauge theory generates a physical scale out of the cutoff, and all of its spectrum is fixed in units of that one scale. That the simulated masses, extracted as in Section 43.7, indeed approach finite multiples of a single Λ_L as $g_0 \to 0$ — "asymptotic scaling" — is the numerical signature that the continuum limit is being taken correctly.

Now the other side of the dichotomy, which closes the non-perturbative arc of ϕ^4 theory. The scalar coupling runs the opposite way: from Chapter 24 the one-loop beta function is $\beta_\lambda = 3\lambda^2/16\pi^2 > 0$. Integrating $\mathrm{d}\lambda/\mathrm{d}\ln\mu = 3\lambda^2/16\pi^2$ from the cutoff scale $\Lambda = 1/a$, where the bare coupling is $\lambda_0 = \lambda(\Lambda)$, down to a fixed physical scale μ_R,

$$\frac{1}{\lambda_R} = \frac{1}{\lambda_0} + \frac{3}{16\pi^2}\,\ln\frac{\Lambda}{\mu_R}. \tag{43.219}$$

The sign is decisive. As the cutoff is removed, $\Lambda \to \infty$ (that is, $a \to 0$) at fixed finite bare coupling λ_0, the logarithm drives $1/\lambda_R \to +\infty$, hence $\lambda_R \to 0$: the interacting theory collapses to the free one. The only escape would be to let $\lambda_0 \to \infty$ as $a \to 0$ to chase a nonzero λ_R, but that is the Landau pole, the statement that no ultraviolet fixed point lies in reach. The perturbative picture is exactly Fig. 43.11: with $\beta > 0$ the origin is an infrared fixed point, and there is no nontrivial UV fixed point to anchor a continuum limit. The Wilson–Fisher fixed point of Theorem 23.36 that rescues ϕ^4 exists only in $d = 4-\epsilon < 4$; precisely at $d = 4$ it merges with the Gaussian fixed point and the interacting theory has nowhere to go.

The lattice converts this perturbative heuristic into a theorem, because ϕ^4 in the lattice formulation is the Ising universality class and is amenable to the rigorous methods of statistical mechanics. The verdict, with its date, is the following.

Theorem 43.42 (Triviality of ϕ^4 in four dimensions; stated, dated)**.** *For lattice ϕ^4 theory (equivalently the Ising and ϕ^4 models in the same universality class) in $d \geq 4$ dimensions, every continuum limit obtained by tuning to the critical point is Gaussian: the limiting Schwinger functions are those of a generalized*

free field, with all connected n-point functions for $n \geq 4$ vanishing. For $d > 4$ this was established by Aizenman and by Fröhlich (1981–82); the marginal case $d = 4$ — the physical one — was closed by Aizenman and Duminil-Copin in 2021. Consequently $\lambda\phi^4$ in four dimensions does not *exist as an interacting continuum quantum field theory: as a fundamental theory it is trivial, and as written it can only be an effective theory below a finite cutoff.*

This is the honest non-perturbative fate of the book's workhorse scalar, and it is not a defect of the lattice but a discovery made through it: the lattice definition is sharp enough to prove that a famous Lagrangian fails to define a theory. The same fate applies to QED, whose $\beta_e > 0$ (Chapter 24) places it on the trivial side of Fig. 43.11; the Landau pole of QED is the statement that it, too, is an effective theory, consistent only up to a finite (astronomically high) cutoff. The contrast with Yang–Mills theory could not be cleaner: asymptotic freedom is not a convenience, it is the property that lets a four-dimensional gauge theory be a complete, cutoff-independent definition.

For Yang–Mills the optimistic verdict is, however, still a *conjecture supported by overwhelming evidence*, not a theorem. The construction of four-dimensional non-abelian gauge theory as a genuine continuum quantum field theory satisfying the Osterwalder–Schrader (Wightman) axioms, together with the proof that its spectrum has a strictly positive mass gap $\Delta > 0$, is the Yang–Mills existence and mass-gap problem — one of the Clay Mathematics Institute Millennium Problems, open as of the mid-2020s. The asymptotic-scaling computation of Theorem 43.41 shows the construction has the right shape, and lattice simulations exhibit the gap numerically with controlled errors, but no proof exists that the limit Eq. (43.218) converges to an OS-positive theory or that the gap survives. This is the precise sense in which Yang–Mills theory hands its non-perturbative existence question forward as an open frontier.

43.8.3 What is proved and what is computed

The most important discipline in reporting lattice results is to keep the line between theorem and measurement sharp. Three statements about confinement and the spectrum illustrate exactly where it falls.

First, *strong-coupling confinement is analytic.* The Wilson-action section proved, by the convergent strong-coupling (large bare g_0, small $\beta = 2N/g_0^2$) character expansion, that the Wilson loop obeys an area law $\langle W(C)\rangle \sim e^{-\sigma\,\mathrm{Area}(C)}$ with a nonzero string tension $\sigma = -a^{-2}\ln(\beta/2N\cdots) + \cdots$, and that the static quark–antiquark potential rises linearly. This is a bona fide theorem about the lattice measure at large coupling: the series converges, every coefficient is computed, and the area law is the order parameter for confinement (Chapter 27, Theorem 27.18). It was the first analytic demonstration of confinement in any formulation, and it remains the clearest one.

Second, *that confinement survives to the continuum is, as of the mid-2020s, established numerically, not proved.* The strong-coupling expansion converges only for β below a finite radius; the continuum limit lives at $\beta \to \infty$ (i.e. $g_0 \to 0$, the opposite end). A pure $SU(N)$ gauge theory has no bulk phase transition separating the two regimes — numerically the string tension in physical units $\sigma\, a^2$ tracks the asymptotic-scaling prediction Eq. (43.218) smoothly down to small a — so the area law is seen to persist into the continuum. But "seen to persist" is a statement about Monte Carlo data with statistical and systematic uncertainties, not an analytic continuation of the strong-coupling proof. Bridging the two ends rigorously is exactly the unsolved part of the mass-gap problem.

Caution

The strong-coupling area-law proof does *not* by itself prove continuum confinement. It establishes confinement in a regime that, for the abelian compact $U(1)$ lattice theory, is separated from the weak-coupling Coulomb phase by a genuine phase transition — there compact $U(1)$ confines at strong coupling and deconfines (becomes ordinary photons) at weak coupling, so strong-coupling confinement says nothing about the continuum. That non-abelian $SU(N)$ has no such intervening transition, and therefore confines all the way to $a \to 0$, is an empirical, numerical fact, not a corollary of the strong-coupling expansion.

Third, *the spectrum is numerical with controlled errors.* Hadron masses, decay constants, the glueball spectrum, and matrix elements are obtained by the Monte Carlo and transfer-matrix techniques of Section 43.7: extracted from Euclidean correlator decay, extrapolated to the continuum using the asymptotic-freedom-governed scaling of Theorem 43.41, and to physical quark masses and infinite volume. Each step carries a quantified uncertainty — statistical from the finite sample, systematic from the extrapolations — and modern lattice QCD reports few-percent or better total errors on many quantities, with the light hadron spectrum reproduced from first principles. These are genuine first-principles predictions of QCD, but they are computations with error bars, not closed-form theorems. The honest summary is that the lattice has *delivered* QCD as a quantitative theory of the strong interaction while *not yet delivering* a proof that the theory it is computing exists.

43.8.4 Open frontiers

Three problems mark the boundary of what the lattice definition currently reaches, and they are the right note on which to close the non-perturbative part of the book, because they are where the very framework of Theorem 43.39 is under strain.

Chiral gauge theories on the lattice. The Standard Model is chiral: left- and

right-handed fermions transform differently under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. The Nielsen–Ninomiya theorem, established earlier in this chapter, forbids a local, undoubled, chirally symmetric lattice fermion, and the Ginsparg–Wilson relation evades it for *vector-like* theories (where left and right pair up) by realizing an exact lattice chiral symmetry through overlap or domain-wall constructions. Defining a genuinely *chiral* gauge theory non-perturbatively on the lattice — with anomaly cancellation, the correct chiral fermion content, and gauge invariance all simultaneously — remains only partially solved. There exist constructions (notably for anomaly-free theories) and a substantial body of work, but no turnkey, simulable definition of the electroweak sector as a lattice theory exists as of the mid-2020s. The non-perturbative definition of the Standard Model itself is therefore still incomplete.

The sign problem. The entire edifice of Theorem 43.39 rests on e^{-S} being a positive weight, so that Eq. (43.214) is a probability measure and importance sampling applies. This positivity fails in physically central situations: at nonzero baryon chemical potential the fermion determinant becomes complex, and a real-time (Minkowski) weight e^{iS} is a pure phase. When the weight is not positive the configurations cannot be sampled by their probability, and naive reweighting suffers a variance that grows exponentially in the volume — the sign problem. It is known to be NP-hard in general, so a universal solution is not expected; partial remedies (complex Langevin, Lefschetz thimbles, dual variables for special cases, tensor-network and quantum-computing approaches) are active but not general. The phase diagram of dense QCD — the interior of neutron stars, the deconfinement transition at finite density — lies behind this wall.

Real-time dynamics. The lattice is intrinsically Euclidean: it computes imaginary-time correlators, from which Lorentzian, real-time quantities (spectral functions, transport coefficients, scattering at finite energy, thermalization) must be obtained by analytic continuation — an ill-posed inverse problem from noisy data. This limits direct access to dynamical quantities even where the Euclidean measure is perfectly positive. Quantum simulation of lattice gauge theories, which would evolve e^{iHt} directly, is the most promising route past both the sign problem and the real-time barrier, and is an early-stage frontier.

These are not failures of the thesis but its honest boundary. Where the Euclidean weight is positive and the theory is asymptotically free, the lattice is a complete, first-principles, gauge-invariant definition of quantum field theory, and it has made QCD a quantitative science. Where positivity fails or chirality intervenes, the definition itself is still being built.

This closes Part XII and, with it, the recurring question of the book — what is a quantum field theory, non-perturbatively? — with a constructive answer and a candid account of its reach. The Euclidean, measure-theoretic viewpoint developed here connects directly to the constructive and axiomatic program framed in Chapter 10, and it sets up the complementary non-perturbative handle

of Part XIV: the conformal bootstrap (Chapter 49), which dispenses with a Lagrangian altogether and constrains the same correlation functions by symmetry and consistency, picking up the ϕ^4 theory endpoint — the Wilson–Fisher fixed point of Theorem 23.36, surviving in $d < 4$ exactly where four-dimensional triviality forbids it — as the three-dimensional Ising CFT.

Further reading

The constructive viewpoint and the Osterwalder–Schrader axioms that give the lattice its claim to "definition" are developed in Glimm and Jaffe [225] and the original papers of Osterwalder and Schrader [372]; reflection positivity on the lattice is treated carefully in Montvay and Münster [351] and Seiler [437]. Wilson's lattice formulation, link variables, and the strong-coupling confinement argument are Wilson and Kogut [516], with the staggered construction in Kogut and Susskind [300]; thorough modern textbook treatments are Gattringer and Lang [211], Rothe [420], and the classic monograph by Creutz [139]. The fermion-doubling no-go theorem is Nielsen and Ninomiya [365] and Nielsen and Ninomiya [366], and the exact lattice chiral symmetry of the Ginsparg–Wilson relation, with its overlap and domain-wall realizations, is surveyed in Lüscher [327]. The non-perturbative triviality of four-dimensional ϕ^4 is proved in Aizenman and Duminil-Copin [11], building on the high-dimensional results of Aizenman [10] and Fröhlich [202]; the Yang–Mills existence and mass-gap problem is stated precisely in Jaffe and Witten [281]. For the open frontiers, the sign problem and its computational complexity are analyzed in Troyer and Wiese [474], and quantum-simulation approaches to lattice gauge theory are reviewed in Bañuls et al. [43].

Part XIII

Supersymmetry

Chapter 44

Supersymmetry: Algebra and Representations

Every continuous symmetry we have met so far has been built from two non-overlapping pieces: the spacetime symmetries of the Poincaré group, fixed once and for all in Chapter 3, and the internal symmetries that act on field indices and commute with spacetime, the gauge groups of Chapter 25. The Coleman–Mandula theorem makes this separation a law rather than a habit: under mild assumptions any symmetry of a nontrivial relativistic S-matrix must be a direct product of the Poincaré group with an internal group whose generators are Lorentz scalars. There is, however, a single hypothesis in that theorem that can be loosened without contradiction — the assumption that every conserved charge is bosonic. This chapter is the story of what happens when we allow *fermionic* generators, charges that carry a spinor index and exchange bosons with fermions, and find that relativistic quantum field theory admits exactly one further symmetry beyond Coleman–Mandula: supersymmetry.

The development is deliberately self-contained and algebra-first. We open (Section 44.1) by stating and motivating Coleman–Mandula, carrying the forward-scattering argument far enough to see precisely where it uses bosonic statistics, and then defining the $\mathbb{Z}_2$-graded Lie superalgebra that the loophole demands, with its graded Jacobi identity checked in all four parity sectors. The Haag–Łopuszański–Sohnius theorem (Section 44.2) then elevates the natural guess — supercharges in the $(\tfrac{1}{2},0)\oplus(0,\tfrac{1}{2})$ representation of the Lorentz group of Chapter 3 — to a uniqueness statement, and we derive the complete $\mathcal{N}=1$ super-Poincaré algebra one (anti)commutator at a time in the two-component spinor language inherited from Chapter 11, fixing the central relation $\{Q_\alpha,\bar{Q}_{\dot\beta}\}=2\sigma^\mu_{\alpha\dot\beta}P_\mu$ by Lorentz covariance and positivity alone. With the algebra closed and every super-Jacobi identity verified, we squeeze the physics out of it directly (Section 44.3): the energy is non-negative, the vacuum energy is the exact order parameter for spontaneous SUSY breaking, every supermultiplet is mass-degenerate, and it

contains equal numbers of bosonic and fermionic states — all proved from the brackets, and all made concrete in Witten's supersymmetric quantum mechanics.

The middle of the chapter turns from the algebra to its representations. Wigner's construction of one-particle states from Chapter 3 is the template: in Section 44.4 the supercharges become fermionic creation and annihilation operators acting on a Clifford vacuum, and the massless and massive supermultiplets, the $2^{\mathcal{N}}$ state count, the spin-2 bound on $\mathcal{N}$, central charges, and BPS shortening all follow. We then ask the obvious next question — which interacting field theory realizes the simplest multiplet — and answer it in Section 44.5 with the Wess–Zumino model, discovering that the naive transformations close only on shell, that an auxiliary field F restores off-shell closure and balances the off-shell degrees of freedom, and that supersymmetry forces the interactions to derive from a single holomorphic superpotential.

That the component-field treatment needs a transformation rule per field, an auxiliary field introduced by hand, and Fierz gymnastics to verify closure is a sign that we are describing a geometric object in non-geometric language. The final two sections build the geometry. Section 44.6 enlarges spacetime to $\mathcal{N} = 1$ superspace, realized as a coset of the super-Poincaré group, on which the supercharges and the covariant derivatives $D_\alpha, \bar{D}_{\dot\alpha}$ act as differential operators in anticommuting coordinates — here we lean on the Grassmann calculus and Berezin integration of Chapter 12. Section 44.7 imposes the chiral constraint $\bar{D}_{\dot\alpha}\Phi = 0$, collapses the general superfield into the irreducible chiral multiplet (ϕ, ψ, F), and reassembles the entire Wess–Zumino model — Kähler kinetic term plus holomorphic superpotential — in a few manifestly supersymmetric lines, with the auxiliary field and the superpotential now emerging automatically from the measure. The chiral superfield and the F-term/D-term grammar established here are the working vocabulary for the supersymmetric gauge theories of Chapter 45 and the exact results of Chapter 46.

44.1 From Coleman–Mandula to Graded Symmetry

The standing question with which this chapter opened was whether the Poincaré algebra admits any *nontrivial* extension. The word "nontrivial" is doing all the work. There is always a cheap way to enlarge a symmetry: take the direct product of the Poincaré group with some internal group G, so that the new generators commute with everything spacetime-related and merely relabel particles within a fixed mass and spin. This is the structure realized by every flavour, colour, and electroweak symmetry built in Parts II through VIII — the internal charges are Lorentz scalars, blind to where and how a particle moves. Such a direct product is no fusion at all; the two factors live in separate rooms. The genuine question is whether the rooms can be opened into one another: can there be a conserved charge that is *not* a Lorentz scalar, one that mixes a particle's internal identity

with its spacetime quantum numbers, so that boosting a state and applying the charge fail to commute? If such a charge existed it would weld the external and internal symmetries into a single simple structure, and the representation theory of that structure — the pattern of allowed particles — would be far more rigid than anything a direct product can impose.

The answer, within the framework of relativistic quantum mechanics with a sensible S-matrix, is a sharp and famous *no*, provided one insists that all conserved charges be *bosonic* — that they form an ordinary Lie algebra under commutators. This is the content of the Coleman–Mandula theorem, which we state precisely and prove at the load-bearing two-particle level below. The theorem is not a vague folk warning; it is a kinematic obstruction with a completely concrete mechanism, and seeing the mechanism is what reveals the single loophole through which a nontrivial extension can after all slip. That loophole is the subject of the entire chapter: the conserved charges need not all be bosonic. If one admits *fermionic* charges — generators that are odd elements of a $\mathbb{Z}_2$-graded algebra, closing not under commutators but under anticommutators — then the hypotheses of Coleman–Mandula simply do not apply to them, and the obstruction evaporates. The graded extension of the Poincaré algebra that results is supersymmetry. This section establishes the frame: it states the no-go theorem with its hypotheses, carries the forward-scattering argument that proves it, defines the graded Lie superalgebra and its generalized Jacobi identity, and shows exactly where the fermionic generators escape the net. Throughout we keep the chapter conventions: the metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and $\hbar = c = 1$.

44.1.1 Symmetries of the S-matrix

To state Coleman–Mandula precisely we must first say what we mean by a symmetry, and the right arena is not the Lagrangian but the S-matrix. A *symmetry of the S-matrix* is a Hermitian operator Q on the Hilbert space of asymptotic states that commutes with the scattering operator, $[Q, \mathbf{S}] = 0$, and that maps single-particle states to single-particle states. The Poincaré generators P^μ and $M^{\mu\nu}$ (Section 3.5, with the algebra recorded in Eqs. (3.147) to (3.149)) are the prototypes: they commute with $\mathbf{S}$ because scattering is Poincaré invariant, and they act on one-particle states without changing the particle number. An ordinary internal symmetry charge — electric charge, isospin, colour — is another example, distinguished by the fact that it is a Lorentz scalar: $[M^{\mu\nu}, Q] = 0$, so it carries no spacetime index and assigns to each particle a number that is the same in every frame.

Two structural facts about such generators are used in the proof, and both are honest consequences of the defining properties rather than extra assumptions. The first is that a symmetry generator acts *additively* on multi-particle states. An asymptotic n-particle state is, by construction, the tensor product of n single-particle states (the particles are infinitely separated and noninteracting in the

far past or far future), and a conserved charge generated by a local current is an extensive quantity: its value on the whole is the sum of its values on the parts. Symbolically, if Q acts on a one-particle state of momentum p and internal label r by some matrix $q(p)_r{}^s$ in the (finite) space of states at that momentum, then on a two-particle state

$$Q\,|p_1, r_1;\, p_2, r_2\rangle \;=\; \big(q(p_1)_{r_1}{}^{s_1}\,\delta^{s_2}_{r_2} + \delta^{s_1}_{r_1}\,q(p_2)_{r_2}{}^{s_2}\big)\,|p_1, s_1;\, p_2, s_2\rangle\,, \tag{44.1}$$

the sum of a one-body operator acting on each particle in turn. The second fact is that because Q commutes with the translation generators, $[Q, P^\mu] = 0$, it cannot change the momentum of the state it acts on: it maps the space of one-particle states at fixed momentum p into itself. By the finite-degeneracy hypothesis stated below, that space is finite dimensional, so $q(p)$ is a finite Hermitian matrix for each p, depending smoothly on p.

44.1.2 The Coleman–Mandula theorem

We can now state the theorem. The hypotheses are physical and we list them with care, because the entire force of the result — and the location of the loophole — lives in exactly which assumptions are made.

Theorem 44.1 (Coleman–Mandula)**.** *Let $\mathcal{G}$ be the Lie algebra of all symmetries of the S-matrix of a relativistic quantum field theory in four spacetime dimensions, in the sense of Section 44.1.1, and suppose that*

(H1) (Poincaré invariance.) *$\mathcal{G}$ contains the Poincaré algebra (Theorem 3.30), and the S-matrix is Poincaré invariant.*

(H2) (Particle finiteness.) *For any finite mass M there are only finitely many one-particle species with mass below M, and each corresponds to a finite-dimensional (discrete-spin) representation of the little group; the one-particle states are thus a discrete spectrum with finite degeneracy at each mass.*

(H3) (Nontrivial analytic scattering.) *The S-matrix is nontrivial — some two-particle elastic amplitude differs from the identity at almost all energies — and the two-body elastic amplitude is an analytic function of the centre-of-momentum scattering angle in a neighbourhood of the forward direction, except at isolated kinematic singularities.*

(H4) (Lie structure.) *The generators of $\mathcal{G}$ are bosonic: they close under* commutators *into a finite-dimensional Lie algebra (Theorem 2.39).*

Then $\mathcal{G}$ is, necessarily, the direct sum of the Poincaré algebra and a finite-dimensional internal Lie algebra of Lorentz-scalar *charges:*

$$\mathcal{G} \;=\; \mathfrak{iso}(1,3) \,\oplus\, \mathfrak{g}_{\mathrm{int}}, \qquad [M^{\mu\nu}, Q_A] = 0 \;\; \textit{for every internal generator } Q_A. \tag{44.2}$$

In particular every conserved charge beyond energy, momentum, and angular momentum must commute with the Lorentz generators; no conserved charge carries a nontrivial Lorentz index.

The conclusion Eq. (44.2) is exactly the statement that spacetime and internal symmetry can be combined only trivially, as a direct product. The full proof, due to Coleman and Mandula, is a careful argument involving the analytic structure of the S-matrix in all its variables; it is standard, and we cite it for completeness [131, 502]. What we carry on the page is the load-bearing kinematic core: the demonstration that a single extra conserved charge carrying two Lorentz indices already over-constrains two-body scattering so severely that the only Lorentz-covariant, analytic, nontrivial possibility is no scattering at all. This is the heart of the theorem, and it is the part whose mechanism we need in order to see the loophole.

44.1.3 The forward-scattering argument

Suppose, contrary to the conclusion, that $\mathcal{G}$ contains a conserved charge $B^{\mu\nu}$ that is a symmetric, traceless Lorentz tensor of rank two and is *not* a function of P^μ and $M^{\mu\nu}$ alone — a genuinely new charge carrying spacetime indices. (Rank two is the cleanest case; we comment on higher and lower rank at the end.) Being a symmetry generator, $B^{\mu\nu}$ acts on one-particle states by a matrix $b^{\mu\nu}(p)$ in the internal/spin labels, and being conserved it is additive on multi-particle states as in Eq. (44.1).

The eigenvalue is fixed by Lorentz covariance. Diagonalize $B^{\mu\nu}$ on the one-particle states (it is Hermitian, hence diagonalizable). On an eigenstate of momentum p its eigenvalue is a symmetric, traceless two-index tensor that depends only on the data available to a single particle. For a particle in an irreducible little-group multiplet the only Lorentz vector at hand is its own momentum p^μ, and the only symmetric traceless rank-two tensors one can build from a single vector p^μ and the metric are multiples of

$$b^{\mu\nu}(p) \;=\; \alpha\Big(p^\mu p^\nu - \tfrac{1}{4}\,\eta^{\mu\nu}\,p^2\Big), \tag{44.3}$$

where $p^2 = m^2$ on shell and α is a real constant that may depend on the particle species but not on its momentum (a smooth Lorentz-scalar function of p^2, hence constant on a mass shell). The trace subtraction enforces $\eta_{\mu\nu}b^{\mu\nu} = 0$. Any antisymmetric part would be a separate charge of the type of $M^{\mu\nu}$ and is excluded by assumption; the symmetric traceless part is the new content. The crucial feature is that Eq. (44.3) is *quadratic* in the momentum, in contrast to the energy–momentum charge P^μ, whose eigenvalue is linear.

Conservation in two-body elastic scattering. Consider an elastic collision of two particles of the same species and mass m,

$$p_1 + p_2 \;\longrightarrow\; p_1' + p_2', \tag{44.4}$$

which the nontrivial S-matrix (H3) supports at generic angle. Three conservation laws constrain the final momenta. First, energy–momentum conservation, the additivity of P^μ,

$$p_1^\mu + p_2^\mu \;=\; p_1'^\mu + p_2'^\mu. \tag{44.5}$$

Second, the on-shell conditions $p_i^2 = p_i'^2 = m^2$. Third — and this is the new constraint — conservation of the tensor charge $B^{\mu\nu}$, whose eigenvalue on the two-particle in- and out-states is the sum of the one-particle eigenvalues Eq. (44.3) with a common α (same species):

$$p_1^\mu p_1^\nu + p_2^\mu p_2^\nu \;=\; p_1'^\mu p_1'^\nu + p_2'^\mu p_2'^\nu, \tag{44.6}$$

where the trace terms $-\frac{1}{4}\eta^{\mu\nu}m^2$ have cancelled between the two sides (two particles on each side, equal mass) and the overall α has divided out. We now read off what Eq. (44.5) and Eq. (44.6) together permit.

The kinematics collapse to the forward direction. Work in the centre-of-momentum frame of the collision, the cleanest place to expose the constraint. There the spatial momenta are equal and opposite,

$$p_1 = (E, \mathbf{p}), \quad p_2 = (E, -\mathbf{p}), \qquad p_1' = (E, \mathbf{p}'), \quad p_2' = (E, -\mathbf{p}'), \tag{44.7}$$

with $|\mathbf{p}'| = |\mathbf{p}|$ fixed by energy conservation and the common mass, so that Eq. (44.5) is automatically satisfied for any direction of $\mathbf{p}'$. The scattering angle θ is the angle between $\mathbf{p}$ and $\mathbf{p}'$, and a nontrivial S-matrix must produce scattering at generic θ. Now impose the tensor law Eq. (44.6). Its purely spatial components $\mu\nu = ij$ read

$$p_1^i p_1^j + p_2^i p_2^j = \mathbf{p}^i\mathbf{p}^j + (-\mathbf{p}^i)(-\mathbf{p}^j) = 2\,\mathbf{p}^i\mathbf{p}^j, \tag{44.8}$$

and likewise the right-hand side equals $2\,\mathbf{p}'^i\mathbf{p}'^j$. Conservation therefore demands

$$\mathbf{p}^i\,\mathbf{p}^j \;=\; \mathbf{p}'^i\,\mathbf{p}'^j \qquad \text{for all } i,j \in \{1,2,3\}. \tag{44.9}$$

This is a far stronger statement than $|\mathbf{p}'| = |\mathbf{p}|$: it equates two rank-one spatial matrices. Contracting $i = j$ and summing recovers $|\mathbf{p}'|^2 = |\mathbf{p}|^2$, which we already had; but the off-diagonal and direction-resolving content of Eq. (44.9) forces the *directions* to agree up to sign. Concretely, the matrix $\mathbf{p}^i\mathbf{p}^j$ has a single nonzero eigenvalue $|\mathbf{p}|^2$ with eigenvector $\hat{\mathbf{p}}$; equality of the two matrices forces their eigenvectors to coincide, $\hat{\mathbf{p}}' = \pm\hat{\mathbf{p}}$. Hence

$$\mathbf{p}' \;=\; \pm\,\mathbf{p}, \qquad \text{i.e.} \qquad \theta \in \{0, \pi\}. \tag{44.10}$$

The only kinematically allowed outcomes are $\theta = 0$ (forward, no scattering) and $\theta = \pi$ (backward, which for identical particles is the same final state relabelled). Scattering through any intermediate angle is forbidden by the existence of the tensor charge. The scattering amplitude $\mathcal{M}(s, \theta)$ is thereby forced to vanish for all θ outside the discrete set $\{0, \pi\}$.

Analyticity finishes the contradiction. A function of the scattering angle that vanishes everywhere except on a discrete set of isolated points is, if it is analytic in θ in a neighbourhood of the forward direction, identically zero — the zeros of a nonzero analytic function are isolated, so a function whose *nonzeros* are isolated must be the zero function. By hypothesis (H3) the elastic amplitude is analytic in $\cos\theta$ near the forward direction. Therefore $\mathcal{M}(s, \theta) \equiv 0$ as a function of θ: the interaction produces no scattering at all. But a vanishing elastic amplitude is in direct conflict with a nontrivial S-matrix; through the optical theorem (Theorem 14.37), the imaginary part of the forward amplitude controls the total cross section,

$$\operatorname{Im} \mathcal{M}(s, \theta = 0) \;\propto\; \sqrt{s}\, \sigma_{\mathrm{tot}}(s), \tag{44.11}$$

the precise relation being Eq. (14.200), so a theory that scatters anything into anything has a nonzero forward amplitude and hence, by analyticity, a nonzero amplitude at generic angle. The two requirements — the tensor charge demanding $\mathcal{M} \equiv 0$, the nontrivial unitary S-matrix demanding $\mathcal{M} \not\equiv 0$ — are incompatible. The only escape is that no such tensor charge $B^{\mu\nu}$ exists. This is the assertion of the theorem for rank-two charges.

Worked Example 44.2 (A conserved Lorentz vector charge is also excluded)**.** The same mechanism disposes of a hypothetical conserved charge C^μ carrying a single Lorentz index but distinct from P^μ. By the argument of Eq. (44.3) its one-particle eigenvalue must be the only vector available, $c^\mu(p) = \beta\, p^\mu$ with β a species constant. For two species a, b with constants $\beta_a \neq \beta_b$, additivity and conservation in the elastic process $p_1 + p_2 \to p_1' + p_2'$ (species preserved) give

$$\beta_a\, p_1^\mu + \beta_b\, p_2^\mu \;=\; \beta_a\, p_1'^\mu + \beta_b\, p_2'^\mu. \tag{44.12}$$

Subtracting β_a times momentum conservation Eq. (44.5) leaves $(\beta_b - \beta_a)(p_2^\mu - p_2'^\mu) = 0$, so $p_2' = p_2$ and hence $p_1' = p_1$: again only forward scattering, again contradicting analyticity and a nontrivial S-matrix. The sole way out is $\beta_a = \beta_b$ for all species, but then $C^\mu = \beta\, P^\mu$ is just a multiple of the momentum and carries no new information. Thus a conserved Lorentz *vector* must coincide with energy–momentum, and a conserved Lorentz *scalar* (rank zero, eigenvalue an arbitrary constant q_a per species) is unconstrained — precisely the internal charges. Charges of rank two or higher are excluded outright. This is the complete rank-by-rank census that yields Eq. (44.2): only scalars survive.

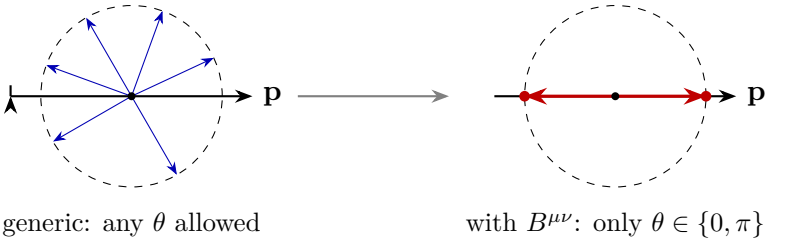


Figure 44.1: The Coleman–Mandula collapse in the centre-of-momentum frame. A nontrivial S-matrix scatters the incoming relative momentum $\mathbf{p}$ into a continuum of directions on the energy shell (left). A conserved symmetric tensor charge $B^{\mu\nu}$ imposes $\mathbf{p}^i\mathbf{p}^j = \mathbf{p}'^i\mathbf{p}'^j$ (Eq. (44.9)), pinching the allowed final directions to the two antipodal points $\theta = 0, \pi$ (right). Analyticity in $\cos\theta$ then forces the amplitude to vanish identically.

Physical Insight

The engine of Coleman–Mandula is dimensional, in the most literal sense. Momentum conservation supplies four linear equations, just enough to fix the two outgoing momenta up to a one-parameter family — the scattering angle, the freedom on which all interesting dynamics lives. *Any* additional conserved charge that depends on momentum nonlinearly (rank two: quadratically) or that distinguishes species linearly (rank one with unequal coefficients) adds equations that consume that last freedom and freeze the angle. Bosonic conserved tensors are too rigid: they leave no room for an analytic, nontrivial scattering function. The escape, when it comes, will not be to weaken conservation but to change the *type* of the extra generator, so that it never enters the conservation balance as an additive c-number tensor in the first place.

Figure 44.1 summarizes the kinematic collapse: the locus of allowed final momenta, a full sphere of directions for a generic interaction, is pinched by the tensor charge down to two antipodal points.

44.1.4 Where the theorem does not reach: graded generators

The proof just given leaned, at three separate points, on the generators being *bosonic*. The eigenvalue Eq. (44.3) was a c-number tensor because $B^{\mu\nu}$ was an ordinary observable with real eigenvalues; the additive conservation law Eq. (44.6) treated the charge as a number summed over particles; and the closure into a Lie algebra (H4) was what allowed Coleman and Mandula to enumerate all such charges in the first place. Strip away hypothesis (H4) — allow generators that are *not* elements of a Lie algebra under commutators — and the census no longer

applies.

The precise way to do this is to admit generators that anticommute. A fermionic charge Q_α carrying a spinor index changes the spin of the state it acts on by half a unit, turning a boson into a fermion and vice versa. It is odd under the $\mathbb{Z}_2$ grading inherited from the Grassmann structure of fermionic fields (Theorem 12.7): two such charges combine naturally not through a commutator but through an *anticommutator*, and the result $\{Q_\alpha, Q_\beta\}$ is bosonic. The content of the supersymmetry escape, which we will make exact in the next section, is that this anticommutator closes onto the translation generator P^μ — a rank-*one* object, the momentum itself — and not onto any new rank-two tensor charge. The forward-scattering obstruction never engages: there is no symmetric c-number tensor $B^{\mu\nu}$ to conserve, because the only bosonic charge produced by the fermionic generators is the one rank-one charge, P^μ, that the kinematics already accommodates. The fermionic charges sit outside the hypotheses of Theorem 44.1 from the start, and the no-go theorem says nothing about them.

To make "closes under anticommutators" a precise algebraic statement we need the notion of a graded Lie algebra. We develop it now, in the generality the chapter requires, and verify its internal consistency.

44.1.5 Graded Lie superalgebras

The Grassmann calculus of Chapter 12 taught us to track a single bookkeeping datum — the parity $|X| \in \{0, 1\}$ of an object, even or odd — and to attach a sign $(-1)^{|X||Y|}$ whenever two homogeneous objects are interchanged (Theorem 12.8). A Lie superalgebra is the structure obtained by demanding that a bracket respect exactly this sign rule.

Definition 44.3 ($\mathbb{Z}_2$-graded Lie superalgebra)**.** A *Lie superalgebra* is a $\mathbb{Z}_2$-graded vector space

$$\mathfrak{g} \;=\; \mathfrak{g}_{\bar{0}} \,\oplus\, \mathfrak{g}_{\bar{1}}, \tag{44.13}$$

the *even* (bosonic) part $\mathfrak{g}_{\bar{0}}$ and the *odd* (fermionic) part $\mathfrak{g}_{\bar{1}}$, equipped with a bilinear *graded bracket* $[\cdot,\cdot\}$ that respects the grading,

$$[\mathfrak{g}_a, \mathfrak{g}_b\} \subseteq \mathfrak{g}_{a+b} \qquad (a, b \in \mathbb{Z}_2, \text{ indices added mod 2}), \tag{44.14}$$

and that on homogeneous elements is a commutator unless both arguments are odd, in which case it is an anticommutator:

$$[X, Y\} \;=\; \begin{cases} [X,Y] = XY - YX, & |X|\,|Y| = 0 \quad \text{(at least one even)}, \\ \{X,Y\} = XY + YX, & |X| = |Y| = 1 \quad \text{(both odd)}. \end{cases} \tag{44.15}$$

The bracket obeys the *graded antisymmetry* (or grade-dependent symmetry) rule

$$[X, Y\} \;=\; -(-1)^{|X|\,|Y|}\,[Y, X\}, \tag{44.16}$$

and the *generalized (graded) Jacobi identity*: for all homogeneous X, Y, Z,

$$(-1)^{|X||Z|}\,[X,[Y,Z\}\} + (-1)^{|Y||X|}\,[Y,[Z,X\}\} + (-1)^{|Z||Y|}\,[Z,[X,Y\}\} \;=\; 0. \tag{44.17}$$

The grade-dependent rule Eq. (44.16) unifies the two familiar statements: for at least one even argument the sign $(-1)^{|X||Y|}$ equals $+1$ and Eq. (44.16) reads $[X,Y] = -\,[Y,X]$, ordinary antisymmetry; for two odd arguments the sign is -1 and it reads $\{X,Y\} = +\,\{Y,X\}$, the symmetry of the anticommutator. An ordinary Lie algebra (Theorem 2.39) is the special case $\mathfrak{g}_{\bar{1}} = 0$, all generators even, the graded bracket reducing to the commutator and Eq. (44.17) to the ordinary Jacobi identity.

It is worth pausing on the structural picture before checking the identities, because it already encodes the physics of supermultiplets. The even part $\mathfrak{g}_{\bar{0}}$ is an ordinary Lie algebra (the even–even bracket is a commutator landing in the even part), and the odd part $\mathfrak{g}_{\bar{1}}$ is a representation of it (the even–odd bracket $[\text{even}, \text{odd}]$ is again odd, so $\mathfrak{g}_{\bar{0}}$ acts on $\mathfrak{g}_{\bar{1}}$). The genuinely new ingredient is the odd–odd bracket, the symmetric pairing $\{\cdot,\cdot\} : \mathfrak{g}_{\bar{1}} \otimes_{\text{sym}} \mathfrak{g}_{\bar{1}} \to \mathfrak{g}_{\bar{0}}$, which feeds the fermionic generators back into the bosonic algebra. In the super-Poincaré case of the next section, $\mathfrak{g}_{\bar{0}}$ is the Poincaré algebra (plus R-symmetry), $\mathfrak{g}_{\bar{1}}$ is the spinorial supercharges, and the symmetric pairing is exactly the anticommutator that produces P^μ.

44.1.6 Consistency of the graded Jacobi identity

The generalized Jacobi identity Eq. (44.17) is not an extra axiom pulled from the air; like the ordinary Jacobi identity, it is automatically satisfied whenever the bracket is realized by (anti)commutators of associative operators, which is the only realization we ever use — the generators are concrete operators on a Hilbert space. To see this, and to confirm that the four parity sectors are mutually consistent, we write the identity out in each case. There are four inequivalent assignments of parities to the triple (X, Y, Z), up to the relabelling permitted by Eq. (44.16): all even; two even and one odd; one even and two odd; all odd. We treat them in turn.

Worked Example 44.4 (The graded Jacobi identity in its four parity sectors)**.** We verify that Eq. (44.17) reduces, in each parity sector, to a genuine identity satisfied by operators on a Hilbert space, where the bracket is the appropriate (anti)commutator.

(i) Even–even–even, $|X| = |Y| = |Z| = 0$. Every sign $(-1)^{|\cdot||\cdot|} = +1$ and every bracket is a commutator, so Eq. (44.17) is the ordinary Jacobi identity

$$[X,[Y,Z]] + [Y,[Z,X]] + [Z,[X,Y]] = 0, \tag{44.18}$$

which holds identically for associative operators: expanding all six double commutators, the twelve resulting triple products cancel in pairs. This is the content of Theorem 2.39 and needs no grading.

(ii) Even–even–odd, $|X| = |Y| = 0$, $|Z| = 1$. The parity products in the exponents are $|X||Z| = 0$, $|Y||X| = 0$, $|Z||Y| = 0$, so again every sign is $+1$. All three brackets are commutators ($[Y, Z]$ and $[Z, X]$ pair an even with an odd element; $[X, Y]$ is even–even), and the identity is once more the ordinary Jacobi relation Eq. (44.18) with Z odd. It expresses the statement that the odd space carries a representation of the even algebra: writing the even action as $\rho(X)Z := [X, Z]$, Eq. (44.18) rearranges to $\rho([X, Y]) = [\rho(X), \rho(Y)]$, the homomorphism property of the representation.

(iii) Even–odd–odd, $|X| = 0$, $|Y| = |Z| = 1$. Now the parities matter, and so do the bracket types. The exponents are $|X||Z| = 0$, $|Y||X| = 0$, $|Z||Y| = 1$, so the third term acquires a sign $(-1)^1 = -1$ while the first two keep $+1$. The inner bracket of the first term, $\{Y, Z\}$, pairs two odd elements and is an anticommutator landing in the even part; the inner brackets $[Z, X]$ and $[X, Y]$ each pair an odd with an even element and are commutators that land in the *odd* part. Consequently the outer brackets are not all commutators: in the first term the outer pair X (even) and $\{Y, Z\}$ (even) gives a commutator, but in the second and third terms the outer pairs $Y, [Z, X]$ and $Z, [X, Y]$ are both odd–odd and are therefore *anticommutators*. Reading Eq. (44.17) with these assignments,

$$[X, \{Y, Z\}] + \{Y, [Z, X]\} - \{Z, [X, Y]\} = 0. \tag{44.19}$$

We check it by direct expansion. With X even and Y, Z odd, writing out each (anti)commutator as operator products,

$$[X, \{Y, Z\}] = X(YZ + ZY) - (YZ + ZY)X = XYZ + XZY - YZX - ZYX, \tag{44.20}$$

$$\{Y, [Z, X]\} = Y(ZX - XZ) + (ZX - XZ)Y = YZX - YXZ + ZXY - XZY, \tag{44.21}$$

$$-\{Z, [X, Y]\} = -Z(XY - YX) - (XY - YX)Z = -ZXY + ZYX - XYZ + YXZ. \tag{44.22}$$

Collecting the six distinct orderings, each appears exactly once with a $+$ and once with a $-$:

$$\begin{aligned} XYZ &: \ +(\mathrm{t1}) - (\mathrm{t3}) = 0, & XZY &: \ +(\mathrm{t1}) - (\mathrm{t2}) = 0, \\ YZX &: \ -(\mathrm{t1}) + (\mathrm{t2}) = 0, & ZYX &: \ -(\mathrm{t1}) + (\mathrm{t3}) = 0, \\ YXZ &: \ -(\mathrm{t2}) + (\mathrm{t3}) = 0, & ZXY &: \ +(\mathrm{t2}) - (\mathrm{t3}) = 0, \end{aligned} \tag{44.23}$$

so the sum vanishes identically. The two anticommutators in the second and third terms are forced by the grading — replacing them by commutators would leave the XYZ ordering with coefficient $+2$ rather than 0, and the identity

would fail. Rearranged using $[Z,X] = -[X,Z]$, Eq. (44.19) reads $[X,\{Y,Z\}] = \{[X,Y],Z\} + \{Y,[X,Z]\}$, the statement that the even generator X acts as a derivation on the odd pairing. Physically this is the relation that will force $\{Q,Q\} \sim P$ to transform correctly as a four-vector under Lorentz rotations in the next section: taking X a Lorentz generator and $Y = Z = Q$ a supercharge, it says the spacetime transformation of the bosonic anticommutator is induced by the spinorial transformation of Q on each factor.

(iv) Odd–odd–odd, $|X| = |Y| = |Z| = 1$. Every exponent $|X||Z| = |Y||X| = |Z||Y| = 1$, so all three terms carry a sign -1, an overall factor that drops out of the equation. Each inner bracket pairs two odd elements and is an anticommutator (even result), while the outer bracket pairs the even result with the remaining odd element and is a commutator. The identity becomes

$$[X,\{Y,Z\}] + [Y,\{Z,X\}] + [Z,\{X,Y\}] = 0, \tag{44.24}$$

the so-called *graded Jacobi identity for three odd generators*. Expanding with all of X, Y, Z odd,

$$\begin{aligned} [X,\{Y,Z\}] &= XYZ + XZY - YZX - ZYX, \\ [Y,\{Z,X\}] &= YZX + YXZ - ZXY - XZY, \\ [Z,\{X,Y\}] &= ZXY + ZYX - XYZ - YXZ. \end{aligned} \tag{44.25}$$

Adding the three lines, every one of the six monomials cancels: XYZ ($+$ in line 1, $-$ in line 3); XZY ($+$ in line 1, $-$ in line 2); YZX ($-$ in line 1, $+$ in line 2); ZYX ($-$ in line 1, $+$ in line 3); YXZ ($+$ in line 2, $-$ in line 3); ZXY ($-$ in line 2, $+$ in line 3). The sum is identically zero. This is the nontrivial sector — it has no analogue in an ordinary Lie algebra — and its automatic satisfaction is what guarantees that the supercharges can be adjoined consistently.

The four cases of Theorem 44.4 exhaust the parity assignments and each holds identically when the bracket is realized by (anti)commutators of operators. We have therefore established that a Lie superalgebra built from honest operators on a Hilbert space automatically satisfies Eqs. (44.16) and (44.17): the graded structure is consistent, not merely postulated. The crucial qualitative lesson is that the odd generators enter the algebra only through their *anticommutators*, and an anticommutator is not a Lie bracket — it never appears in the hypotheses of Coleman–Mandula. The even part of any Lie superalgebra extending Poincaré is still constrained by Theorem 44.1 to be Poincaré plus a Lorentz-scalar internal algebra, but the odd part is entirely free of that constraint. Supersymmetry is the unique nontrivial fusion of spacetime and internal symmetry precisely because it is the unique structural way to add generators that the no-go theorem cannot see.

Caution

The freedom opened by grading is real but narrow. Admitting odd generators does not license arbitrary new charges: the odd part must still be a representation of the Poincaré algebra (case (ii) of Theorem 44.4), its anticommutators must close back into the even algebra in a Lorentz-covariant way (case (iii)), and the triple-odd identity (case (iv)) must hold. These requirements are stringent enough to fix the odd generators almost completely. The companion no-go theorem of Haag, Łopuszański, and Sohnius — the subject of the next section — shows that the *only* admissible fermionic charges are spinorial, transforming in the $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ representations of the Lorentz group, with anticommutators fixed up to normalization. The grading is a loophole, not a free-for-all.

This pins down what the next section must do. The forward-scattering census of Theorem 44.1 kills every bosonic charge of rank two or higher; the graded structure of Theorem 44.3 reopens the door for fermionic charges, but the consistency requirements of Theorem 44.4 demand that those charges transform as a definite Lorentz representation and close under anticommutation onto the bosonic algebra. The natural hypothesis — which the Haag–Łopuszański–Sohnius theorem will elevate to a uniqueness statement — is that the admissible fermionic charges transform in the $(\frac{1}{2},0)$ representation of the Lorentz group (and its conjugate $(0,\frac{1}{2})$), the smallest nontrivial spinor. With that hypothesis in hand we turn, in the next section, to deriving the complete $\mathcal{N}=1$ super-Poincaré algebra, fixing every (anti)commutator in two-component spinor notation.

44.2 The Haag–Łopuszański–Sohnius Theorem and the SUSY Algebra

The previous section closed the bosonic door and pointed at the one crack of light left open. Coleman–Mandula forbids any *bosonic* generator that mixes spacetime and internal symmetry nontrivially: under the hypotheses of the theorem, the symmetry algebra of an interacting relativistic quantum field theory with a mass gap is forced to be a direct sum of the Poincaré algebra and an internal Lie algebra of Lorentz scalars. The single loophole is that the no-go theorem assumed all conserved charges close under *commutators*—that the symmetry is an ordinary Lie algebra. If we admit conserved charges that close under *anticommutators*, and that transform as Lorentz spinors rather than tensors, the proof no longer applies, because spinorial charges cannot be diagonalised against the momentum eigenstates in the way Coleman–Mandula requires. We saw that Lorentz covariance restricts such fermionic charges to the lowest spin that does the job: a charge in the $(\frac{1}{2},0)$ representation (and its conjugate in $(0,\frac{1}{2})$). The

task of this section is to take that structural observation and turn it into an algebra. We state the theorem of Haag, Łopuszański and Sohnius, which says that the spinorial loophole is not merely allowed but *essentially unique*, and then we derive, bracket by bracket, the complete $\mathcal{N} = 1$ super-Poincaré algebra, fixing every coefficient by Lorentz covariance, closure, and positivity, and identifying the one outer automorphism the structure admits.

44.2.1 Graded Lie algebras and the precise statement

To even phrase the theorem we must enlarge the notion of a symmetry algebra. An ordinary symmetry algebra is a Lie algebra in the sense of Theorem 2.39: a vector space with an antisymmetric bracket obeying the Jacobi identity. Fermionic charges demand a $\mathbb{Z}_2$-graded generalisation.

Definition 44.5 (Graded (super) Lie algebra)**.** A *graded Lie algebra* (synonymously a super Lie algebra) is a vector space $\mathfrak{g} = \mathfrak{g}_{\bar{0}} \oplus \mathfrak{g}_{\bar{1}}$ with a $\mathbb{Z}_2$ grading—elements of $\mathfrak{g}_{\bar{0}}$ are called *even* (bosonic), those of $\mathfrak{g}_{\bar{1}}$ *odd* (fermionic), with degree $|X| \in \{0, 1\}$—equipped with a bilinear *graded bracket* $[\,\cdot\,,\,\cdot\,\}$ that respects the grading, $|[X, Y\}| = |X| + |Y| \bmod 2$, is graded-antisymmetric,

$$[X, Y\} = -(-1)^{|X||Y|}\,[Y, X\}, \tag{44.26}$$

and satisfies the *graded (super-)Jacobi identity*

$$(-1)^{|X||Z|}\,[X, [Y, Z\}\} + (-1)^{|Y||X|}\,[Y, [Z, X\}\} + (-1)^{|Z||Y|}\,[Z, [X, Y\}\} = 0. \tag{44.27}$$

Concretely the graded bracket is the ordinary commutator $[X, Y]$ unless both arguments are odd, in which case it is the anticommutator $\{X, Y\}$. The even part $\mathfrak{g}_{\bar{0}}$ is an ordinary Lie algebra; the odd part $\mathfrak{g}_{\bar{1}}$ is a representation of $\mathfrak{g}_{\bar{0}}$; and the anticommutator of two odd elements lands back in the even part.

The grading is forced on us physically by spin-statistics: a charge that turns a boson into a fermion must itself carry half-integer spin and obey Fermi statistics, so it is natural to compose two such charges with an anticommutator. The bracket $\{Q, Q\}$ of two fermionic charges is symmetric and produces a bosonic operator; that this bosonic operator is forced to be the energy–momentum is the heart of supersymmetry. With Theorem 44.5 in hand the no-go landscape is sharpened to a uniqueness statement.

Theorem 44.6 (Haag–Łopuszański–Sohnius)**.** *Let $\mathfrak{g}$ be the graded symmetry algebra of the S-matrix of a nontrivial, unitary, relativistic quantum field theory in four spacetime dimensions, subject to the regularity and mass-gap hypotheses of Coleman–Mandula, and suppose the odd generators act on asymptotic multiparticle states. Then:*

1. *the even part $\mathfrak{g}_{\bar{0}}$ is, as Coleman–Mandula already dictates, the direct sum of the Poincaré algebra $\mathfrak{iso}(1,3)$ and a compact internal Lie algebra of Lorentz scalars (together with possible central charges);*

2. *the odd generators sit in the spin-$\frac{1}{2}$ representation only—copies of $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ under the Lorentz group; no higher-spin odd generators are admissible;*

3. *writing the $\mathcal{N}$ left-handed odd generators as Q^I_α ($\alpha=1,2$; $I=1,\dots,\mathcal{N}$) with conjugates $\bar{Q}^I_{\dot\alpha}=(Q^I_\alpha)^\dagger$, their graded brackets are fixed up to normalisation to be*

$$\left\{Q^I_\alpha,\bar{Q}^J_{\dot\beta}\right\}=2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu\,\delta^{IJ}, \tag{44.28}$$

$$\left\{Q^I_\alpha,Q^J_\beta\right\}=\varepsilon_{\alpha\beta}\,Z^{IJ},\qquad Z^{IJ}=-Z^{JI}, \tag{44.29}$$

where the Z^{IJ} are central charges*, commuting with everything in $\mathfrak{g}$ and forming an antisymmetric matrix in the internal indices; and the odd generators commute with translations and rotate as spinors under Lorentz transformations.*

The theorem is stated here at the level required for a self-contained development of the algebra; its proof is a refinement of the Coleman–Mandula argument that carries the spinor index structure through the same analyticity and cluster-decomposition machinery, and we use the result as the starting point for what follows rather than reproving the S-matrix analysis. What we *do* prove in full is everything downstream: that, once Theorem 44.6 licenses spinorial charges at all, Lorentz covariance, closure and positivity fix the algebra completely. For the rest of this chapter we restrict to $\mathcal{N}=1$: a single Weyl supercharge Q_α and its conjugate $\bar{Q}_{\dot\alpha}$. With only one internal index the antisymmetric matrix Z^{IJ} in Eq. (44.29) has no components, so the central charge vanishes identically and $\{Q_\alpha,Q_\beta\}=0$; the extended case $\mathcal{N}>1$, where Z is the engine of BPS shortening, is opened only far enough in Chapter 46 to count states.

44.2.2 Two-component conventions

The whole of supersymmetry is cleanest in two-component (Weyl, or van der Waerden) spinor language, and we now fix that notation, importing the content from Section 3.4.3 and dressing it with the dotted/undotted index bookkeeping that the algebra demands. A left-handed Weyl spinor in $(\frac{1}{2},0)$ carries an undotted index ψ_α, $\alpha=1,2$; a right-handed spinor in $(0,\frac{1}{2})$ carries a dotted index $\bar\chi_{\dot\alpha}$. Complex conjugation exchanges the two representations (Eq. (3.107) and Section 11.7.6), $(\psi_\alpha)^\dagger=\bar\psi_{\dot\alpha}$, so the conjugate of an undotted object is a dotted one.

Indices are raised and lowered with the antisymmetric Levi-Civita symbols, which play the role of a metric on spinor space. We use the north-west–south-east

convention

$$\varepsilon^{12} = -\varepsilon^{21} = +1, \qquad \varepsilon_{12} = -\varepsilon_{21} = -1, \qquad \varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^{\alpha}_{\gamma}, \tag{44.30}$$

and identically for the dotted symbols $\varepsilon^{\dot\alpha\dot\beta}$, $\varepsilon_{\dot\alpha\dot\beta}$. As a 2×2 matrix the upper-index symbol $\varepsilon^{\alpha\beta}$ equals $i\sigma^2$, the invariant form of Eq. (3.107); the lower-index symbol is its inverse with the sign fixed by $\varepsilon^{\alpha\beta}\varepsilon_{\beta\gamma} = \delta^{\alpha}_{\gamma}$. Raising and lowering then read

$$\psi^{\alpha} = \varepsilon^{\alpha\beta}\psi_{\beta}, \qquad \psi_{\alpha} = \varepsilon_{\alpha\beta}\psi^{\beta}, \qquad \bar\psi^{\dot\alpha} = \varepsilon^{\dot\alpha\dot\beta}\bar\psi_{\dot\beta}, \qquad \bar\psi_{\dot\alpha} = \varepsilon_{\dot\alpha\dot\beta}\bar\psi^{\dot\beta}. \tag{44.31}$$

We adopt the standard summation conventions $\psi\chi \equiv \psi^{\alpha}\chi_{\alpha}$ (descending undotted) and $\bar\psi\bar\chi \equiv \bar\psi_{\dot\alpha}\bar\chi^{\dot\alpha}$ (ascending dotted); for anticommuting (Grassmann) spinors, which is the case of interest, antisymmetry of ε together with anticommutativity makes these products *symmetric*, $\psi\chi = \chi\psi$ and $\bar\psi\bar\chi = \bar\chi\bar\psi$.

The Lorentz-vector content is carried by the sigma matrices imported from Eqs. (3.79) and (3.106),

$$\sigma^{\mu}_{\alpha\dot\beta} = (\mathbf{1}, \vec\sigma)_{\alpha\dot\beta}, \qquad \bar\sigma^{\mu\,\dot\alpha\beta} = (\mathbf{1}, -\vec\sigma)^{\dot\alpha\beta} = \varepsilon^{\dot\alpha\dot\gamma}\varepsilon^{\beta\delta}\sigma^{\mu}_{\delta\dot\gamma}, \tag{44.32}$$

the index heights being exactly those that make the contractions in Eq. (44.28) consistent: $\sigma^{\mu}_{\alpha\dot\beta}$ has one lower undotted and one lower dotted index, while $\bar\sigma^{\mu\,\dot\alpha\beta}$ has both raised. They satisfy the Clifford-type identities, proved blockwise in Section 3.4.3,

$$(\sigma^{\mu}\bar\sigma^{\nu} + \sigma^{\nu}\bar\sigma^{\mu})_{\alpha}{}^{\beta} = 2\eta^{\mu\nu}\delta^{\beta}_{\alpha}, \qquad (\bar\sigma^{\mu}\sigma^{\nu} + \bar\sigma^{\nu}\sigma^{\mu})^{\dot\alpha}{}_{\dot\beta} = 2\eta^{\mu\nu}\delta^{\dot\alpha}_{\dot\beta}. \tag{44.33}$$

The Lorentz generators in the two spinor representations are the (anti-self-dual and self-dual) combinations

$$(\sigma^{\mu\nu})_{\alpha}{}^{\beta} = \tfrac{i}{4}\big(\sigma^{\mu}\bar\sigma^{\nu} - \sigma^{\nu}\bar\sigma^{\mu}\big)_{\alpha}{}^{\beta}, \qquad (\bar\sigma^{\mu\nu})^{\dot\alpha}{}_{\dot\beta} = \tfrac{i}{4}\big(\bar\sigma^{\mu}\sigma^{\nu} - \bar\sigma^{\nu}\sigma^{\mu}\big)^{\dot\alpha}{}_{\dot\beta}, \tag{44.34}$$

which are precisely the left and right blocks $S_L^{\mu\nu}$, $S_R^{\mu\nu}$ of the Dirac generator computed in Section 3.4.3; they satisfy the Lorentz algebra Eq. (3.149) on each spinor index. That the two-spinor catalogue $(\tfrac12, 0)$, $(0, \tfrac12)$ is exactly the list of building blocks for a spin-$\tfrac12$ charge is the content of Theorems 3.17 and 3.19: the supercharge must be one of these, and the conjugation $(Q_\alpha)^\dagger = \bar Q_{\dot\alpha}$ ties them together.

Notation Watch

Beyond the supercharges $Q_\alpha, \bar Q_{\dot\alpha}$ and the Grassmann coordinates $\theta^\alpha, \bar\theta_{\dot\alpha}$ (which use existing notation), the superspace machinery of Section 44.2.7 onward requires three new shorthands, used from Chapter 44 on: the superspace covariant derivatives D_α and $\bar D_{\dot\alpha}$, the Grassmann measures

$\int \mathrm{d}^2\theta$, $\int \mathrm{d}^2\bar\theta$ and $\int \mathrm{d}^4\theta$ built by Berezin integration (Theorem 12.17), and the superpotential symbol W. These are geometric packaging only; no physical convention is introduced beyond the metric, sigma and gamma conventions of Appendices A and C.

44.2.3 The central anticommutator $\left\{Q_\alpha, \bar Q_{\dot\beta}\right\}$

The defining relation of supersymmetry is the anticommutator of Q with its conjugate. We derive it from three inputs: Lorentz covariance, the closure requirement of Theorem 44.5, and positivity of the Hilbert-space norm.

Index structure. The object $\left\{Q_\alpha, \bar Q_{\dot\beta}\right\}$ carries one undotted and one dotted lower index, so it transforms in $(\tfrac12,0)\otimes(0,\tfrac12) = (\tfrac12,\tfrac12)$, the Lorentz four-vector representation. By closure it must be a bosonic generator already present in $\mathfrak{g}_{\bar 0}$, and by the first clause of Theorem 44.6 the bosonic generators are the Poincaré generators $P_\mu, M_{\mu\nu}$ together with Lorentz-scalar internal charges. Of these, only one transforms as a four-vector: the translation generator P_μ. The internal charges are scalars, hence carry no Lorentz index and cannot appear; the Lorentz generator $M_{\mu\nu}$ is an antisymmetric two-tensor, the reducible $(\tfrac12,\tfrac12)\otimes(\tfrac12,\tfrac12) \supset (1,0)\oplus(0,1)$ content, not the irreducible $(\tfrac12,\tfrac12)$, so it cannot supply a single vector index either. The unique covariant possibility is therefore

$$\left\{Q_\alpha, \bar Q_{\dot\beta}\right\} = c\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu, \tag{44.35}$$

with $\sigma^\mu_{\alpha\dot\beta}$ the invariant tensor of Eq. (44.32) that converts a vector index μ into the pair $(\alpha,\dot\beta)$, and c a single constant to be fixed. (No term $c'\,\sigma^\mu_{\alpha\dot\beta}\partial_\mu(\dots)$ or higher can appear: the right-hand side must be a fixed element of the finite-dimensional algebra, and P_μ exhausts the available vectors.)

Reality of the coefficient. Taking the Hermitian adjoint of Eq. (44.35) and using $(Q_\alpha)^\dagger = \bar Q_{\dot\alpha}$, $P_\mu^\dagger = P_\mu$, and the Hermiticity $(\sigma^\mu_{\alpha\dot\beta})^* = \sigma^\mu_{\beta\dot\alpha}$ of the sigma matrices, the anticommutator $\left\{Q_\alpha, \bar Q_{\dot\beta}\right\}^\dagger = \left\{\bar Q_{\dot\beta}, Q_\alpha\right\} = \left\{Q_\alpha, \bar Q_{\dot\beta}\right\}$ (relabelled) returns the same equation with $c \to c^*$. Hence c is real.

Positivity fixes the sign. Contract Eq. (44.35) with $\bar\sigma_\nu^{\dot\beta\alpha}$ and use the trace $\bar\sigma_\nu^{\dot\beta\alpha}\sigma^\mu_{\alpha\dot\beta} = \mathrm{tr}(\bar\sigma_\nu\sigma^\mu) = 2\delta^\mu_\nu$ from Eq. (44.33); this inverts Eq. (44.35) to $P_\nu = \frac{c}{2c}\dots$, but the cleanest route to the sign is to set $\mu=\nu=0$ and sum over $\alpha=\dot\beta$. With $\sigma^0 = \mathbf{1}$,

$$\sum_{\alpha=1,2}\left\{Q_\alpha, (Q_\alpha)^\dagger\right\} = \left\{Q_1, Q_1^\dagger\right\} + \left\{Q_2, Q_2^\dagger\right\} = c\,(\sigma^0_{1\dot1} + \sigma^0_{2\dot2})P_0 = 2c\,P_0 = 2c\,H, \tag{44.36}$$

where $H = P^0 = P_0$ is the Hamiltonian. The left-hand side is a sum of operators of the form $\{A, A^\dagger\} = AA^\dagger + A^\dagger A$, each of which is positive semidefinite, since for any state $|\psi\rangle$, $\langle\psi|AA^\dagger + A^\dagger A|\psi\rangle = \|A^\dagger|\psi\rangle\|^2 + \|A|\psi\rangle\|^2 \geq 0$. Therefore $2c\,H \geq 0$ as an operator. Because the energy spectrum of a relativistic theory is bounded below and contains states of positive energy (the vacuum and excitations above it), consistency requires $c > 0$. We *define* the normalisation of the supercharges by choosing

$$\boxed{\{Q_\alpha, \bar{Q}_{\dot\beta}\} = 2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu} \tag{44.37}$$

i.e. $c = 2$. The factor of 2 is a convention—it could be absorbed into a rescaling $Q \to Q/\sqrt{2}$—but it is the universal choice, and it makes Eq. (44.36) read $\sum_\alpha \{Q_\alpha, Q_\alpha^\dagger\} = 4H$, the relation from which the next section reads off energy positivity and the supersymmetry-breaking order parameter. The relation Eq. (44.37) is the algebraic core of the entire subject: the anticommutator of a supercharge with its conjugate is the energy–momentum. Spacetime translations are, in this precise sense, the "square" of a supersymmetry transformation.

44.2.4 The vanishing of $\{Q_\alpha, Q_\beta\}$ for $\mathcal{N} = 1$

Next consider the anticommutator of two left-handed supercharges. It carries two undotted indices, transforming in $(\tfrac{1}{2}, 0) \otimes (\tfrac{1}{2}, 0) = (1, 0) \oplus (0, 0)$. Decomposing on the symmetry of the index pair, the symmetric part sits in $(1, 0)$ and the antisymmetric part in the singlet $(0, 0)$. Because the anticommutator is symmetric under exchange of its arguments *together with* their indices, $\{Q_\alpha, Q_\beta\} = \{Q_\beta, Q_\alpha\}$, only certain covariant tensors can appear. The general Lorentz-covariant ansatz built from the available invariant tensors and bosonic generators is

$$\{Q_\alpha, Q_\beta\} = k\,(\sigma^{\mu\nu})_{\alpha\beta}\,M_{\mu\nu} + \varepsilon_{\alpha\beta}\,Z, \tag{44.38}$$

where $(\sigma^{\mu\nu})_{\alpha\beta} = (\sigma^{\mu\nu})_\alpha{}^\gamma \varepsilon_{\gamma\beta}$ is symmetric in $\alpha\beta$ (it carries the $(1, 0)$ part), and $\varepsilon_{\alpha\beta}$ is antisymmetric (the singlet). We dispose of each term.

The singlet coefficient Z would be a Lorentz scalar central charge. But $\varepsilon_{\alpha\beta}$ is antisymmetric while the anticommutator is symmetric, so the term $\varepsilon_{\alpha\beta}Z$ is consistent with Eq. (44.38) only if Z itself is antisymmetric in some additional index that compensates. For $\mathcal{N}$ supercharges this is exactly the antisymmetric internal matrix $Z^{IJ} = -Z^{JI}$ of Eq. (44.29): the combined object $\varepsilon_{\alpha\beta}Z^{IJ}$ is symmetric under the simultaneous exchange $(\alpha, I) \leftrightarrow (\beta, J)$, as required of $\{Q^I_\alpha, Q^J_\beta\}$. For $\mathcal{N} = 1$ there is no internal index, Z^{IJ} collapses to a single antisymmetric 1×1 matrix, which vanishes identically. Hence the central term is absent.

The tensor coefficient k would couple $\{Q, Q\}$ to the Lorentz generators $M_{\mu\nu}$. This term is killed by translation invariance, which we establish independently

in Section 44.2.5 below but may anticipate here: once $[Q_\alpha, P_\rho] = 0$, the left-hand side of Eq. (44.38) commutes with every P_ρ,

$$[\{Q_\alpha, Q_\beta\}, P_\rho] = \{[Q_\alpha, P_\rho], Q_\beta\} + \{Q_\alpha, [Q_\beta, P_\rho]\} = 0, \tag{44.39}$$

using the graded Leibniz rule (a special case of Eq. (44.27)). But the right-hand side does *not* commute with P_ρ if $k \neq 0$, because $[M_{\mu\nu}, P_\rho] = i(\eta_{\nu\rho}P_\mu - \eta_{\mu\rho}P_\nu) \neq 0$ by Eq. (3.148). The only way Eq. (44.38) can have a translation-invariant left-hand side and a translation-covariant right-hand side is $k = 0$. (A dimensional check confirms the verdict from another angle: Q has mass dimension $\frac{1}{2}$, so $\{Q, Q\}$ has dimension 1, matching P_μ; but $M_{\mu\nu}$ is dimensionless, so k would carry dimension 1, and the algebra of a theory with no built-in mass scale offers no such invariant constant.) Therefore, for $\mathcal{N} = 1$,

$$\boxed{\{Q_\alpha, Q_\beta\} = 0, \qquad \left\{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\right\} = 0} \tag{44.40}$$

the second relation being the Hermitian conjugate of the first.

44.2.5 Translations commute with the supercharges

We owe the proof that $[Q_\alpha, P_\mu] = 0$, on which Section 44.2.4 leaned. This is the cleanest illustration of how the super-Jacobi identity, applied to a carefully chosen triple, forces a coefficient to vanish.

Proposition 44.7. *In the $\mathcal{N} = 1$ super-Poincaré algebra the supercharges commute with translations,*

$$[Q_\alpha, P_\mu] = 0, \qquad \left[\bar{Q}_{\dot\alpha}, P_\mu\right] = 0. \tag{44.41}$$

Proof. The commutator $[Q_\alpha, P_\mu]$ carries one undotted spinor index and one lower vector index, so it transforms in $(\frac{1}{2}, 0) \otimes (\frac{1}{2}, \frac{1}{2}) = (1, \frac{1}{2}) \oplus (0, \frac{1}{2})$. A fermionic generator of the algebra must, by Theorem 44.6, lie in $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$; the $(1, \frac{1}{2})$ piece corresponds to no admissible generator and must be absent. The only fermionic object of the algebra that fits the surviving $(0, \frac{1}{2})$ slot, with the correct index structure to balance Eq. (44.41), is $\bar{Q}$ contracted with a sigma matrix. Hence the most general covariant ansatz is

$$[Q_\alpha, P_\mu] = a\, \sigma_{\mu\,\alpha\dot\beta}\, \bar{Q}^{\dot\beta}, \tag{44.42}$$

for a single constant a (the index $\dot\beta$ is raised with $\varepsilon^{\dot\beta\dot\gamma}$ as in Eq. (44.31)). Taking the Hermitian conjugate gives the partner relation $\left[\bar{Q}_{\dot\alpha}, P_\mu\right] = a^*\, \bar\sigma_\mu{}^{\dot\beta}{}_? \ldots$; written with consistent index heights it reads

$$\left[\bar{Q}^{\dot\alpha}, P_\mu\right] = a^*\, \bar\sigma_\mu^{\dot\alpha\beta}\, Q_\beta. \tag{44.43}$$

Now impose the super-Jacobi identity Eq. (44.27) on the triple (Q_α, P_μ, P_ν). Here Q_α is odd and P_μ, P_ν are even, so all gradings are even except those involving Q, and every bracket reduces to an ordinary commutator. The identity reads

$$[Q_\alpha, [P_\mu, P_\nu]] + [P_\mu, [P_\nu, Q_\alpha]] + [P_\nu, [Q_\alpha, P_\mu]] = 0. \tag{44.44}$$

The first term vanishes because $[P_\mu, P_\nu] = 0$ by Eq. (3.147). The remaining two combine into the statement that $[[Q_\alpha, P_\mu], P_\nu]$ is symmetric under $\mu \leftrightarrow \nu$. Substituting Eq. (44.42) and then Eq. (44.43),

$$[[Q_\alpha, P_\mu], P_\nu] = a\,\sigma_{\mu\,\alpha\dot\alpha}\left[\bar Q^{\dot\alpha}, P_\nu\right] = a\,\sigma_{\mu\,\alpha\dot\beta}\,a^*\,\bar\sigma_\nu^{\dot\beta\gamma}\,Q_\gamma = |a|^2\,(\sigma_\mu\bar\sigma_\nu)_\alpha{}^\gamma\,Q_\gamma. \tag{44.45}$$

Requiring this to be symmetric in $\mu\nu$—equivalently, that its antisymmetric part vanish—gives

$$0 = |a|^2\,(\sigma_\mu\bar\sigma_\nu - \sigma_\nu\bar\sigma_\mu)_\alpha{}^\gamma\,Q_\gamma = |a|^2\,(-4i)(\sigma_{\mu\nu})_\alpha{}^\gamma\,Q_\gamma, \tag{44.46}$$

using $\sigma_\mu\bar\sigma_\nu - \sigma_\nu\bar\sigma_\mu = -4i\,\sigma_{\mu\nu}$ from the definition Eq. (44.34). The tensor $(\sigma_{\mu\nu})_\alpha{}^\gamma$ is nonzero (it is the spin-$\frac{1}{2}$ Lorentz generator, which acts nontrivially on Q), and the supercharges Q_γ are linearly independent operators, so the only solution is $|a|^2 = 0$, i.e. $a = 0$. Substituting back into Eqs. (44.42) and (44.43) yields Eq. (44.41). □

The result has a sharp physical reading. The would-be term Eq. (44.42) is precisely the $(1, \frac{1}{2})$ "Lorentz-spin-$\frac{3}{2}$" object that the higher-spin clause of Theorem 44.6 forbids; the super-Jacobi identity is the algebraic mechanism that enforces the prohibition, by tying the coefficient of the forbidden structure to the nonvanishing antisymmetric tensor $\sigma_{\mu\nu}$. Physically, $[Q, P] = 0$ says the supercharge is translation-invariant: it is the same operator at every point of spacetime, exactly as a conserved charge should be, and supersymmetry transformations commute with going from one inertial frame's origin to another's.

44.2.6 The Lorentz action and the closed $\mathcal{N} = 1$ algebra

The last brackets to record are the action of the Lorentz generators on the supercharges. These are not derived from closure—they are dictated by the statement that Q_α *is* a $(\frac{1}{2}, 0)$ spinor and $\bar Q_{\dot\alpha}$ a $(0, \frac{1}{2})$ spinor, which is part of Theorem 44.6. An operator carrying a spinor index transforms under an infinitesimal Lorentz rotation with the spinor generator Eq. (44.34), with the sign fixed by the convention that operators carry the representation through a commutator,

$$[M^{\mu\nu}, Q_\alpha] = -(\sigma^{\mu\nu})_\alpha{}^\beta\,Q_\beta, \tag{44.47}$$

$$\left[M^{\mu\nu}, \bar{Q}_{\dot{\alpha}}\right] = -(\bar{\sigma}^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}}\, \bar{Q}_{\dot{\beta}}. \tag{44.48}$$

That these are the right matrices, and the right sign, is most easily checked against the Poincaré algebra itself: Eq. (44.47) must reproduce, on the spinor index α, the same $[M, \cdot] \sim \sigma^{\mu\nu}$ action that Section 3.4.3 found for the left Weyl block of a Dirac field, and the consistency of Eq. (44.47) with the central relation Eq. (44.37) is verified explicitly in Theorem 44.8.

Collecting Eqs. (44.37), (44.40), (44.41), (44.47) and (44.48) together with the imported Poincaré relations Eqs. (3.147) to (3.149) gives the complete $\mathcal{N} = 1$ super-Poincaré algebra:

$$\begin{aligned}
[P^{\mu}, P^{\nu}] &= 0, \\
[M^{\mu\nu}, P^{\rho}] &= i(\eta^{\nu\rho} P^{\mu} - \eta^{\mu\rho} P^{\nu}), \\
[M^{\mu\nu}, M^{\rho\sigma}] &= i(\eta^{\nu\rho} M^{\mu\sigma} - \eta^{\mu\rho} M^{\nu\sigma} - \eta^{\nu\sigma} M^{\mu\rho} + \eta^{\mu\sigma} M^{\nu\rho}), \\
[M^{\mu\nu}, Q_{\alpha}] &= -(\sigma^{\mu\nu})_{\alpha}{}^{\beta}\, Q_{\beta}, \qquad \left[M^{\mu\nu}, \bar{Q}_{\dot{\alpha}}\right] = -(\bar{\sigma}^{\mu\nu})_{\dot{\alpha}}{}^{\dot{\beta}}\, \bar{Q}_{\dot{\beta}}, \\
[Q_{\alpha}, P_{\mu}] &= 0, \qquad \left[\bar{Q}_{\dot{\alpha}}, P_{\mu}\right] = 0, \\
\{Q_{\alpha}, Q_{\beta}\} &= 0, \qquad \left\{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\right\} = 0, \\
\left\{Q_{\alpha}, \bar{Q}_{\dot{\beta}}\right\} &= 2\, \sigma^{\mu}_{\alpha\dot{\beta}}\, P_{\mu}.
\end{aligned} \tag{44.49}$$

This is the object the rest of the chapter studies. Its structure is best seen as a $\mathbb{Z}_2$-graded extension of Poincaré, sketched in Fig. 44.2: the even sector is the Poincaré algebra acting on itself and on the odd sector, and the single nontrivial new relation, the anticommutator Eq. (44.37), sends a pair of odd generators back into the even translations.

44.2.7 The $\mathrm{U}(1)_R$ outer automorphism

The algebra Eq. (44.49) admits one continuous symmetry of its own structure that is not generated by any of its elements: an *outer automorphism*. Consider the phase rotation of the supercharges

$$Q_{\alpha} \;\longmapsto\; e^{-i\lambda}\, Q_{\alpha}, \qquad \bar{Q}_{\dot{\alpha}} \;\longmapsto\; e^{+i\lambda}\, \bar{Q}_{\dot{\alpha}}, \qquad \lambda \in \mathbb{R}, \tag{44.50}$$

with the Poincaré generators held fixed. This map preserves every bracket of Eq. (44.49): the relations $\{Q, Q\} = 0$ and $[M, Q] \sim Q$ are homogeneous of degree $+1$ in Q and so transform covariantly, while the central relation Eq. (44.37) carries one Q and one $\bar{Q}$, accruing a net phase $e^{-i\lambda} e^{+i\lambda} = 1$, and so is left invariant, consistent with P_{μ} being unrotated. Because the transformation acts nontrivially on the generators yet leaves the bracket structure unchanged, it is an automorphism of the super-Poincaré algebra; because no element of the algebra generates it through the adjoint action (it commutes with all of Poincaré but

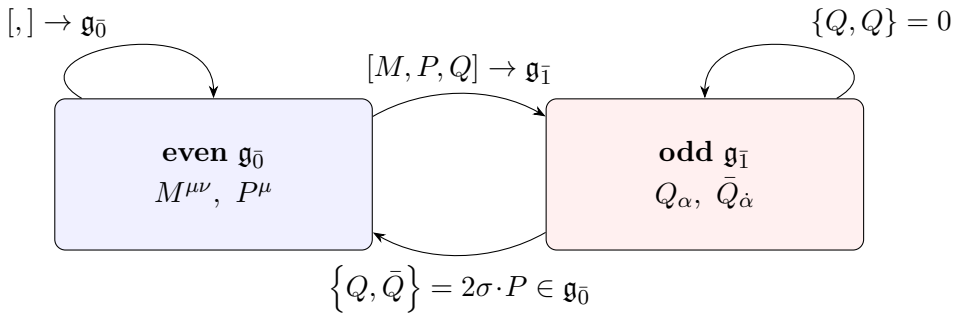


Figure 44.2: The $\mathbb{Z}_2$-graded structure of the $\mathcal{N}=1$ super-Poincaré algebra. The even (bosonic) sector contains the Lorentz and translation generators and closes on itself; it acts on the odd (fermionic) sector, rotating $Q,\bar{Q}$ as spinors and leaving them translation-invariant. The single new content is that the anticommutator of the odd generators returns the bosonic momentum, $\left\{Q,\bar{Q}\right\} = 2\sigma\cdot P$, while $\{Q,Q\}=0$.

rephases the supercharges), it is *outer*. The transformations Eq. (44.50) form a group U(1), called *R-symmetry* and denoted $\mathrm{U}(1)_R$.

When $\mathrm{U}(1)_R$ is promoted to an actual symmetry of a theory it is generated by a Hermitian charge R acting through the commutators

$$\boxed{[R,Q_\alpha] = -\,Q_\alpha, \qquad \left[R,\bar{Q}_{\dot\alpha}\right] = +\,\bar{Q}_{\dot\alpha}, \qquad [R,P_\mu] = [R,M_{\mu\nu}] = 0.} \tag{44.51}$$

The infinitesimal form of Eq. (44.50) is $\delta Q_\alpha = i\lambda\,[R,Q_\alpha] = -i\lambda\,Q_\alpha$, reproducing the phase. The supercharge carries R-charge -1 and its conjugate R-charge $+1$. Consistency with the algebra is immediate from the graded Leibniz rule: acting with R on Eq. (44.37),

$$\left[R,\left\{Q_\alpha,\bar{Q}_{\dot\beta}\right\}\right] = \left\{[R,Q_\alpha],\bar{Q}_{\dot\beta}\right\} + \left\{Q_\alpha,\left[R,\bar{Q}_{\dot\beta}\right]\right\} = (-1+1)\left\{Q_\alpha,\bar{Q}_{\dot\beta}\right\} = 0 = 2\sigma^{\mu}_{\alpha\dot\beta}\,[R,P_\mu]\,, \tag{44.52}$$

and the right-hand side vanishes because $[R,P_\mu]=0$. The two sides agree, so Eq. (44.51) is compatible with the central relation; the same check on $\{Q,Q\}=0$ is trivially satisfied since both sides scale by $e^{-2i\lambda}$. The R-symmetry is special to supersymmetry: it acts on the spinorial directions of the algebra and has no analogue in the ordinary Poincaré structure. Its presence or absence—whether a given supersymmetric theory respects $\mathrm{U}(1)_R$, and whether it is anomalous—is a powerful organising principle in the analysis of Chapters 45 and 46; for now we record it as the unique continuous outer automorphism of the $\mathcal{N}=1$ algebra.

Physical Insight

The single relation $\left\{Q, \bar{Q}\right\} = 2\sigma \cdot P$ already contains the two headline consequences supersymmetry is famous for, and it is worth seeing why before the next section extracts them carefully. First, energy is non-negative: Eq. (44.36) expresses $4H$ as a sum of squared-norm operators, so $H \geq 0$ in any supersymmetric theory, with $H|\psi\rangle = 0$ if and only if $Q_\alpha|\psi\rangle = \bar{Q}_{\dot{\alpha}}|\psi\rangle = 0$—a supersymmetric vacuum has exactly zero energy. Second, because Q commutes with P^μ and hence with the mass operator P^2, acting with Q on a one-particle state of mass m produces another state of the *same* mass but spin differing by $\frac{1}{2}$: bosons and fermions are paired into mass-degenerate multiplets. Both statements are read directly off the algebra, with no Lagrangian and no perturbation theory in sight; this is the sense in which the algebra is the theory's backbone.

44.2.8 Verifying the super-Jacobi identities

A graded Lie algebra is only consistent if its brackets satisfy the super-Jacobi identity Eq. (44.27) on every triple of generators. For the Poincaré subalgebra this is inherited from Chapter 3. What must be checked are the mixed triples involving the supercharges; we verify the three structurally distinct cases in full.

Worked Example 44.8 (The super-Jacobi identities of the $\mathcal{N} = 1$ algebra)**.** We verify Eq. (44.27) for the triples (Q, Q, Q), (Q, Q, P), and (M, Q, Q), and as the substantive covariance test the mixed triple $(M, Q, \bar{Q})$.

(i) The all-fermionic triple $(Q_\alpha, Q_\beta, Q_\gamma)$. All three are odd, $|Q| = 1$, so each cyclic sign in Eq. (44.27) is $(-1)^{1\cdot 1} = -1$ and each inner bracket of two odd generators is an anticommutator, itself feeding an outer commutator with the third odd generator. The identity reads

$$[Q_\alpha, \{Q_\beta, Q_\gamma\}] + [Q_\beta, \{Q_\gamma, Q_\alpha\}] + [Q_\gamma, \{Q_\alpha, Q_\beta\}] = 0, \tag{44.53}$$

after the common overall sign -1 is dropped. Every inner anticommutator vanishes by Eq. (44.40), so the identity holds trivially, $0 + 0 + 0 = 0$. The check is not vacuous: it confirms that setting $\{Q, Q\} = 0$ introduces no obstruction, which would not have been automatic had a nonzero $(1, 0)$ piece survived in Eq. (44.38).

(ii) The triple $(Q_\alpha, Q_\beta, P_\mu)$. Two odd, one even. Writing out Eq. (44.27) with $|Q| = 1, |P| = 0$, the signs are $(-1)^{|Q||P|} = +1$ for the term beginning with Q_α, $(-1)^{|Q||Q|} = -1$ for the term beginning with Q_β, and $(-1)^{|P||Q|} = +1$ for the term beginning with P_μ:

$$[Q_\alpha, [Q_\beta, P_\mu]] - [Q_\beta, [P_\mu, Q_\alpha]] + [P_\mu, \{Q_\alpha, Q_\beta\}] = 0. \tag{44.54}$$

The first two terms vanish by $[Q,P]=0$ (Eq. (44.41)); the third vanishes by $\{Q,Q\}=0$ (Eq. (44.40)), since $[P_\mu,0]=0$. The identity is satisfied. This triple is the consistency statement that the supercharges' translation-invariance and their vanishing self-anticommutator are mutually compatible—precisely the input combination used in Eq. (44.39).

(iii) The triple $(M_{\mu\nu},Q_\alpha,Q_\beta)$. One even, two odd. With M even the relevant rearrangement of Eq. (44.27) states that the Lorentz variation of $\{Q_\alpha,Q_\beta\}$ equals the sum of the variations of its factors:

$$[M_{\mu\nu},\{Q_\alpha,Q_\beta\}]=\{[M_{\mu\nu},Q_\alpha]\,,Q_\beta\}+\{Q_\alpha,[M_{\mu\nu},Q_\beta]\}\,. \tag{44.55}$$

The left-hand side is $[M_{\mu\nu},0]=0$. The right-hand side, using Eq. (44.47), is

$$-(\sigma_{\mu\nu})_\alpha{}^\gamma\ \{Q_\gamma,Q_\beta\}-(\sigma_{\mu\nu})_\beta{}^\gamma\ \{Q_\alpha,Q_\gamma\}=0, \tag{44.56}$$

again by Eq. (44.40). Both sides vanish, so the identity holds. The content is that $\{Q,Q\}=0$ is a Lorentz-invariant statement: the spinor rotation Eq. (44.47) maps the (vanishing) $(1,0)$ tensor to itself, confirming that no Lorentz transformation can generate a nonzero $\{Q,Q\}$ from zero.

(iv) The covariance test $(M_{\mu\nu},Q_\alpha,\bar{Q}_{\dot\beta})$. This is the only triple whose super-Jacobi identity is not automatically satisfied by vanishing brackets, and it is the genuine check that the central relation Eq. (44.37) is Lorentz-covariant with the coefficient we fixed. With M even, the identity again takes Leibniz form:

$$\left[M_{\mu\nu},\left\{Q_\alpha,\bar{Q}_{\dot\beta}\right\}\right]=\left\{[M_{\mu\nu},Q_\alpha]\,,\bar{Q}_{\dot\beta}\right\}+\left\{Q_\alpha,\left[M_{\mu\nu},\bar{Q}_{\dot\beta}\right]\right\}. \tag{44.57}$$

Evaluate both sides. The left-hand side, using Eq. (44.37) and the Poincaré bracket Eq. (3.148),

$$\left[M_{\mu\nu},2\sigma^\rho_{\alpha\dot\beta}P_\rho\right]=2\sigma^\rho_{\alpha\dot\beta}\,[M_{\mu\nu},P_\rho]=2\sigma^\rho_{\alpha\dot\beta}\,i(\eta_{\nu\rho}P_\mu-\eta_{\mu\rho}P_\nu)=2i(\sigma_{\nu\,\alpha\dot\beta}P_\mu-\sigma_{\mu\,\alpha\dot\beta}P_\nu). \tag{44.58}$$

The right-hand side, using Eqs. (44.47) and (44.48) and then Eq. (44.37) on each surviving anticommutator,

$$\begin{aligned}\text{RHS}&=-(\sigma_{\mu\nu})_\alpha{}^\gamma\left\{Q_\gamma,\bar{Q}_{\dot\beta}\right\}-(\bar\sigma_{\mu\nu})_{\dot\beta}{}^{\dot\gamma}\left\{Q_\alpha,\bar{Q}_{\dot\gamma}\right\}\\&=-2(\sigma_{\mu\nu})_\alpha{}^\gamma\,\sigma^\rho_{\gamma\dot\beta}P_\rho-2(\bar\sigma_{\mu\nu})_{\dot\beta}{}^{\dot\gamma}\,\sigma^\rho_{\alpha\dot\gamma}P_\rho.\end{aligned} \tag{44.59}$$

The two expressions agree if and only if the spinor matrices satisfy the intertwining identity

$$(\sigma_{\mu\nu})_\alpha{}^\gamma\,\sigma^\rho_{\gamma\dot\beta}+(\bar\sigma_{\mu\nu})_{\dot\beta}{}^{\dot\gamma}\,\sigma^\rho_{\alpha\dot\gamma}=-i(\delta^\rho_\mu\,\sigma_{\nu\,\alpha\dot\beta}-\delta^\rho_\nu\,\sigma_{\mu\,\alpha\dot\beta}). \tag{44.60}$$

This is nothing but the infinitesimal form of the Lorentz-equivariance of the invariant tensor σ^ρ established in Section 3.4.3: there the finite statement $A\,(x_\mu\sigma^\mu)\,A^\dagger=(\Lambda x)_\mu\sigma^\mu$ said that conjugating σ^ρ by the spinor representation

matrices reproduces the vector transformation of its index ρ. Differentiating that relation at the identity, with the spinor generators Eq. (44.34) on the α and $\dot\beta$ indices and the vector generator $(\mathcal{M}_{\mu\nu})^\rho{}_\sigma = i(\eta_\mu{}^\rho\eta_{\nu\sigma} - \dots)$ on ρ, yields exactly Eq. (44.60). Substituting it into Eq. (44.59) gives $-2P_\rho \cdot \left[-i(\delta^\rho_\mu\sigma_\nu - \delta^\rho_\nu\sigma_\mu)_{\alpha\dot\beta}\right] = 2i(\sigma_{\nu\,\alpha\dot\beta}P_\mu - \sigma_{\mu\,\alpha\dot\beta}P_\nu)$, which is precisely the left-hand side Eq. (44.58). The super-Jacobi identity is satisfied, and it is satisfied *only* because the coefficient on the right of Eq. (44.37) is a Lorentz scalar multiple of σ^μ—confirming both the index structure and the internal consistency of the normalisation chosen in Section 44.2.3.

With every bracket fixed and every Jacobi identity verified, the $\mathcal{N} = 1$ super-Poincaré algebra Eq. (44.49) stands as a closed, consistent graded Lie algebra. We now have a rigid algebraic object whose representations are the supermultiplets of physical states and whose internal relations already encode deep dynamical facts. The next section squeezes the physics out of the algebra itself—energy positivity, the order parameter for supersymmetry breaking, and the exact boson–fermion degeneracy and equal state counts within a multiplet—before we turn, in Section 44.2 onward, to constructing the multiplets explicitly.

44.3 Immediate Consequences of the Algebra

The previous section closed the super-Poincaré algebra: alongside the familiar Poincaré brackets we now have the fermionic supercharges Q_α and their conjugates $\bar Q_{\dot\alpha} = (Q_\alpha)^\dagger$, transforming as $(\frac12, 0)$ and $(0, \frac12)$ of $\mathrm{SL}(2,\mathbb{C})$ (Section 3.4.3), and obeying

$$\left\{Q_\alpha, \bar Q_{\dot\beta}\right\} = 2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu, \tag{44.61}$$

$$\{Q_\alpha, Q_\beta\} = 0, \qquad \left\{\bar Q_{\dot\alpha}, \bar Q_{\dot\beta}\right\} = 0, \tag{44.62}$$

$$[Q_\alpha, P_\mu] = 0, \qquad \left[\bar Q_{\dot\alpha}, P_\mu\right] = 0, \tag{44.63}$$

with $\sigma^\mu = (\mathbf{1}, \vec\sigma)$ in the conventions of Eqs. (3.79) and (3.106). (These are the $N = 1$ relations established in the preceding section; we take them as given here and cite them by label.) It is a remarkable feature of supersymmetry that a great deal of physics follows from these brackets *alone*, before a single representation has been built. The reason is structural: Eq. (44.61) ties the supercharges directly to the energy–momentum operator, so anticommuting a supercharge with its own conjugate *is* the Hamiltonian. Hilbert-space positivity then does the rest. In this section we extract three such consequences — the positivity of energy and the vacuum-energy order parameter for supersymmetry breaking, the exact equality of masses inside a multiplet, and the equality of bosonic and fermionic state counts — each proved in full from Eqs. (44.61) to (44.63). These are precisely the constraints that the explicit representation theory of the next section must realise.

44.3.1 Energy is non-negative

The first and most important consequence is that the spectrum of the Hamiltonian is bounded below by zero. To see it, we isolate $H = P^0$ from Eq. (44.61) by tracing over the spinor indices. Set $\dot\beta$ equal numerically to α and sum over $\alpha = 1, 2$; the left-hand side becomes $\sum_\alpha \left\{Q_\alpha, \bar Q_{\dot\alpha}\right\}$, and the right-hand side becomes $2\,(\operatorname{tr}\sigma^\mu)\,P_\mu$. Since the Pauli matrices are traceless, $\operatorname{tr}\sigma^\mu = (\operatorname{tr}\mathbf{1}, \operatorname{tr}\vec\sigma) = (2, \vec 0)$, so only $P_0 = H$ survives:

$$\sum_{\alpha=1,2} \left\{Q_\alpha, \bar Q_{\dot\alpha}\right\} \;=\; 2\,(\operatorname{tr}\sigma^\mu)\,P_\mu \;=\; 4\,P_0 \;=\; 4H. \tag{44.64}$$

Notice what has happened: the spatial momentum $\vec P$ has dropped out entirely, killed by the tracelessness of $\vec\sigma$. The combination of supercharges that closes on H is insensitive to $\vec P$. Solving Eq. (44.64) and inserting $\bar Q_{\dot\alpha} = Q_\alpha^\dagger$,

$$H \;=\; \tfrac{1}{4}\sum_{\alpha=1,2} \left\{Q_\alpha, Q_\alpha^\dagger\right\} \;=\; \tfrac{1}{4}\Big(Q_1 Q_1^\dagger + Q_1^\dagger Q_1 + Q_2 Q_2^\dagger + Q_2^\dagger Q_2\Big). \tag{44.65}$$

This is the central identity of the section. The Hamiltonian is a sum of an operator and its adjoint times that adjoint — a manifestly non-negative form.

Theorem 44.9 (Positivity of energy)**.** *In any unitary representation of the $N = 1$ super-Poincaré algebra on a Hilbert space with positive-definite inner product, the Hamiltonian satisfies $H \geq 0$. Consequently every state has energy $E \geq 0$.*

Proof. Let $|\psi\rangle$ be any normalisable state. Taking the expectation value of Eq. (44.65),

$$\langle\psi|H|\psi\rangle = \tfrac{1}{4}\sum_{\alpha=1,2}\Big(\big\langle\psi\big|Q_\alpha Q_\alpha^\dagger\big|\psi\big\rangle + \big\langle\psi\big|Q_\alpha^\dagger Q_\alpha\big|\psi\big\rangle\Big). \tag{44.66}$$

Unitarity makes $Q_\alpha^\dagger$ the genuine Hermitian adjoint of Q_α with respect to the inner product, so each term is the squared norm of a state:

$$\big\langle\psi\big|Q_\alpha^\dagger Q_\alpha\big|\psi\big\rangle = \|Q_\alpha\,|\psi\rangle\|^2 \geq 0, \qquad \big\langle\psi\big|Q_\alpha Q_\alpha^\dagger\big|\psi\big\rangle = \big\|Q_\alpha^\dagger\,|\psi\rangle\big\|^2 \geq 0. \tag{44.67}$$

Hence $\langle\psi|H|\psi\rangle = \tfrac{1}{4}\sum_\alpha\big(\big\|Q_\alpha^\dagger\,|\psi\rangle\big\|^2 + \|Q_\alpha\,|\psi\rangle\|^2\big) \geq 0$ for every $|\psi\rangle$, which is the statement $H \geq 0$. Diagonalising H on energy eigenstates gives $E \geq 0$. □

The economy of this argument is worth dwelling on. In an ordinary quantum field theory the positivity of the energy is a contingent fact, established case by case from the explicit Hamiltonian and the requirement that the theory be stable; nothing in the Poincaré algebra forces it. Supersymmetry promotes it to

a kinematic identity. Because H is built as a sum of squares of the supercharges, no choice of dynamics consistent with the algebra can produce a negative-energy state. The same identity also forbids the energy from being merely bounded below by some negative constant: the bound is exactly zero, and it is saturated precisely when the supercharges annihilate the state. That sharp threshold is what makes the vacuum energy an exact order parameter, to which we now turn.

44.3.2 The vacuum energy as the order parameter for SUSY breaking

A symmetry is said to be *spontaneously broken* when it is a symmetry of the dynamics but not of the ground state: the charges generating it fail to annihilate the vacuum (Chapter 29). For an ordinary internal symmetry the order parameter is the vacuum expectation value of some field; for supersymmetry the algebra hands us a far more rigid diagnostic, directly in terms of energy.

Theorem 44.10 (Vacuum energy detects SUSY breaking)**.** *Let* $|0\rangle$ *be a Poincaré-invariant vacuum, so that* $\vec{P}\,|0\rangle = 0$ *and* $E_{\text{vac}} = \langle 0|H|0\rangle$*. Then supersymmetry is unbroken — meaning* $Q_\alpha\,|0\rangle = \bar{Q}_{\dot\alpha}\,|0\rangle = 0$ *for all* $\alpha, \dot\alpha$ *— if and only if* $E_{\text{vac}} = 0$*. Equivalently,* $E_{\text{vac}} > 0$ *holds if and only if supersymmetry is spontaneously broken.*

Proof. ($\Rightarrow$) If $Q_\alpha\,|0\rangle = \bar{Q}_{\dot\alpha}\,|0\rangle = 0$, then Eq. (44.65) annihilates the vacuum term by term, so $H\,|0\rangle = 0$ and in particular $E_{\text{vac}} = \langle 0|H|0\rangle = 0$.

($\Leftarrow$) Conversely, suppose $E_{\text{vac}} = 0$. By Eq. (44.66) applied to $|0\rangle$,

$$0 = E_{\text{vac}} = \langle 0|H|0\rangle = \tfrac{1}{4}\sum_{\alpha=1,2}\Big(\big\|Q_\alpha^\dagger\,|0\rangle\big\|^2 + \|Q_\alpha\,|0\rangle\|^2\Big). \tag{44.68}$$

This is a sum of four non-negative terms equal to zero, so each term vanishes individually: $\|Q_\alpha\,|0\rangle\|^2 = 0$ and $\big\|Q_\alpha^\dagger\,|0\rangle\big\|^2 = 0$ for $\alpha = 1, 2$. In a Hilbert space with positive-definite inner product a vector of zero norm is the zero vector, hence $Q_\alpha\,|0\rangle = 0$ and $\bar{Q}_{\dot\alpha}\,|0\rangle = 0$ for all indices: supersymmetry is unbroken. The contrapositive of the two implications gives the final statement. □

Two features make this order parameter exceptional. First, it is exact and non-perturbative. The vanishing of the vacuum energy is equivalent to the existence of a supersymmetric vacuum as an operator statement, valid to all orders and beyond perturbation theory; it does not rely on a small coupling or on a classical approximation. Second, it inverts the usual logic of the cosmological-constant problem. In a generic quantum field theory the zero-point energy of every mode contributes to E_{vac} and the total is quartically sensitive to the cutoff. Here the same supercharge algebra that guarantees $E \geq 0$ also guarantees that an unbroken supersymmetry sets E_{vac} to exactly zero: the bosonic and fermionic zero-point energies cancel. We saw the mechanism of that cancellation

in miniature in Theorem 12.50, where the bosonic and fermionic functional determinants entered with opposite powers and cancelled mode by mode; the algebra above is the abstract reason it had to happen.

Physical Insight

The logic runs in a direction worth stating plainly. *Unbroken* supersymmetry forces $E_{\text{vac}} = 0$ — a statement about the ground state. A *positive* vacuum energy is therefore not a small perturbation to be tuned away but a sharp, qualitative signal that the symmetry is spontaneously broken. There is no halfway: either the supercharges annihilate the vacuum and $E_{\text{vac}} = 0$ exactly, or they do not and $E_{\text{vac}} > 0$. This is why the search for supersymmetry-breaking mechanisms is, equivalently, a search for theories whose lowest energy is strictly positive — the Goldstino and the explicit field-theoretic mechanisms (O'Raifeartaigh and Fayet–Iliopoulos breaking) are developed in Chapter 45.

We have deliberately stopped at the order parameter and deferred the dynamics. Whether a given model has $E_{\text{vac}} = 0$ or $E_{\text{vac}} > 0$ is a question about its potential and its quantum corrections, taken up in Chapter 45; here the algebra tells us only what the answer *means*.

44.3.3 Mass degeneracy within a multiplet

We now turn from the vacuum to the excited states and ask what the algebra forces on the particles that fill a single irreducible representation. The first Casimir of the Poincaré algebra is $P^2 = P_\mu P^\mu$, whose eigenvalue m^2 labels the mass of a one-particle state (Theorem 3.36). The question is whether P^2 remains a Casimir of the *enlarged* super-Poincaré algebra — that is, whether it also commutes with the supercharges. If it does, Schur's lemma fixes a single mass for an entire irreducible supermultiplet.

Proposition 44.11 (P^2 commutes with the supercharges)**.** *The first Casimir P^2 commutes with Q_α and $\bar{Q}_{\dot\alpha}$:*

$$\left[P^2, Q_\alpha\right] = 0, \qquad \left[P^2, \bar{Q}_{\dot\alpha}\right] = 0. \tag{44.69}$$

Hence P^2 is a Casimir of the full $N = 1$ super-Poincaré algebra.

Proof. Write $P^2 = P^\mu P_\mu$ and use the Leibniz rule for the commutator,

$$\left[P^2, Q_\alpha\right] = \left[P^\mu P_\mu, Q_\alpha\right] = P^\mu \left[P_\mu, Q_\alpha\right] + \left[P^\mu, Q_\alpha\right] P_\mu. \tag{44.70}$$

By Eq. (44.63) each commutator on the right vanishes, $[P_\mu, Q_\alpha] = 0$, so $\left[P^2, Q_\alpha\right] = 0$. The conjugate relation $\left[P^2, \bar{Q}_{\dot\alpha}\right] = 0$ follows identically from $\left[P_\mu, \bar{Q}_{\dot\alpha}\right] = 0$.

Since P^2 already commutes with all of P_μ and the Lorentz generators $M_{\mu\nu}$ (it is the Poincaré Casimir of Theorem 3.36), it commutes with every generator of the super-Poincaré algebra. □

Theorem 44.12 (Equal masses within a supermultiplet)**.** *All one-particle states in an irreducible representation of the $N = 1$ super-Poincaré algebra have the same eigenvalue of P^2, hence the same mass. In particular, the bosonic and fermionic superpartners related by the action of the supercharges are exactly mass-degenerate.*

Proof. On an irreducible representation P^2 commutes with every generator (Theorem 44.11), so by Schur's lemma in the form Theorem 2.23 it acts as a scalar multiple of the identity, $P^2 = m^2\,\mathbf{1}$. Concretely: the states of the multiplet are generated from any one of them by repeated application of Q_α and $\bar{Q}_{\dot\alpha}$. If $|p, s\rangle$ is an eigenstate with $P^2\,|p, s\rangle = m^2\,|p, s\rangle$, then

$$P^2\,Q_\alpha\,|p, s\rangle = Q_\alpha\,P^2\,|p, s\rangle = m^2\,Q_\alpha\,|p, s\rangle\,, \tag{44.71}$$

using $[P^2, Q_\alpha] = 0$ to pull P^2 through the supercharge. So $Q_\alpha\,|p, s\rangle$ carries the same eigenvalue m^2, and inductively every state reached from $|p, s\rangle$ does too. All members of the irreducible multiplet share one value of m^2. □

This is the algebraic origin of the slogan that "every boson has a fermionic partner of equal mass." It is also exactly the statement that confronts experiment: no scalar with the mass and charges of the electron has been observed, so if supersymmetry is realised in nature it must be *broken*, lifting the degeneracy. The breaking, by Theorem 44.10, is the same phenomenon as a positive vacuum energy; the structural payoff is that mass splittings and vacuum energy are not independent inputs but two faces of one order parameter.

Caution

It is essential that the degeneracy is in P^2 and not in the *spin*. The second Poincaré Casimir is built from the Pauli–Lubański vector W_μ, and W^2 does *not* commute with the supercharges — if it did, every state in a multiplet would carry the same spin and supersymmetry would relate bosons only to bosons. It is precisely because $[W^2, Q_\alpha] \neq 0$ that a supercharge can change the spin by half a unit, turning a boson into a fermion. The mass is rigidly shared; the spin is deliberately not. The explicit spin content of each multiplet is worked out in Section 44.4.

44.3.4 Equal numbers of bosons and fermions

The third consequence counts states. A supercharge maps a bosonic state to a fermionic one and vice versa; one expects this pairing to make the two

counts equal. Turning the expectation into a theorem requires an operator that distinguishes bosons from fermions, and a trace identity that exploits its interplay with the algebra.

Definition 44.13 (Fermion-number parity)**.** The *fermion-number parity* operator $(-1)^F$ is the unitary, Hermitian, involutive operator that acts as $+1$ on every bosonic state and -1 on every fermionic state:

$$(-1)^F\,|\text{boson}\rangle = +\,|\text{boson}\rangle\,, \qquad (-1)^F\,|\text{fermion}\rangle = -\,|\text{fermion}\rangle\,, \qquad \big((-1)^F\big)^2 = \mathbf{1}. \tag{44.72}$$

"Bosonic" and "fermionic" here mean the two superselection sectors of integer and half-integer spin; by the spin–statistics connection these coincide with even and odd fermion number F, so $(-1)^F$ is well defined.

The defining property we actually use is not the eigenvalues but how $(-1)^F$ commutes past a supercharge.

Proposition 44.14 ($(-1)^F$ anticommutes with the supercharges)**.**

$$\Big\{(-1)^F, Q_\alpha\Big\} = 0, \qquad \Big\{(-1)^F, \bar{Q}_{\dot\alpha}\Big\} = 0. \tag{44.73}$$

Proof. A supercharge Q_α carries half-integer spin, so it maps the bosonic sector into the fermionic sector and the fermionic sector into the bosonic sector; it changes the eigenvalue of $(-1)^F$ by a sign. We verify $\Big\{(-1)^F, Q_\alpha\Big\} = 0$ on each sector. On a bosonic state $|b\rangle$, with $(-1)^F\,|b\rangle = +\,|b\rangle$ and $Q_\alpha\,|b\rangle = |f\rangle$ a fermionic state with $(-1)^F\,|f\rangle = -\,|f\rangle$,

$$\big((-1)^F Q_\alpha + Q_\alpha (-1)^F\big)\,|b\rangle = (-1)^F\,|f\rangle + Q_\alpha\,|b\rangle = -\,|f\rangle + |f\rangle = 0. \tag{44.74}$$

On a fermionic state $|f\rangle$, with $(-1)^F\,|f\rangle = -\,|f\rangle$ and $Q_\alpha\,|f\rangle = |b\rangle$ bosonic,

$$\big((-1)^F Q_\alpha + Q_\alpha (-1)^F\big)\,|f\rangle = (-1)^F\,|b\rangle + Q_\alpha(-\,|f\rangle) = |b\rangle - |b\rangle = 0. \tag{44.75}$$

Since every state is a superposition of bosonic and fermionic states, $\Big\{(-1)^F, Q_\alpha\Big\} = 0$ as an operator. Taking the Hermitian adjoint and using $\big((-1)^F\big)^\dagger = (-1)^F$ gives $\Big\{(-1)^F, \bar{Q}_{\dot\alpha}\Big\} = 0$. □

We can now state and prove the counting theorem. The natural object is the supertrace $\mathrm{Tr}\big[(-1)^F(\cdots)\big]$: weighting a trace by $(-1)^F$ counts bosons with a plus sign and fermions with a minus sign, so $\mathrm{Tr}(-1)^F = n_B - n_F$, the difference of the two counts.

Theorem 44.15 ($n_B = n_F$)**.** *In any finite-dimensional representation of the $N = 1$ super-Poincaré algebra on which P_μ acts as a fixed nonzero four-momentum p_μ (for example, the states of a supermultiplet at a fixed on-shell momentum, massive or massless), the number of bosonic states equals the number of fermionic states:*

$$n_B = n_F. \tag{44.76}$$

Proof. Because the supercharges commute with P_μ (Eq. (44.63)), they preserve the fixed-momentum subspace and act as honest finite-dimensional operators on it; the trace over this subspace is finite and cyclic. Consider, for fixed spinor indices $\alpha, \dot\beta$, the weighted trace of the basic anticommutator Eq. (44.61):

$$\mathrm{Tr}\Big[(-1)^F \left\{Q_\alpha, \bar Q_{\dot\beta}\right\}\Big] = \mathrm{Tr}\big[(-1)^F Q_\alpha \bar Q_{\dot\beta}\big] + \mathrm{Tr}\big[(-1)^F \bar Q_{\dot\beta} Q_\alpha\big]. \tag{44.77}$$

In the second term, use cyclicity of the trace to move Q_α to the front,

$$\mathrm{Tr}\big[(-1)^F \bar Q_{\dot\beta} Q_\alpha\big] = \mathrm{Tr}\big[Q_\alpha (-1)^F \bar Q_{\dot\beta}\big], \tag{44.78}$$

and then anticommute $(-1)^F$ past Q_α using Eq. (44.73), $Q_\alpha (-1)^F = -(-1)^F Q_\alpha$:

$$\mathrm{Tr}\big[Q_\alpha (-1)^F \bar Q_{\dot\beta}\big] = -\,\mathrm{Tr}\big[(-1)^F Q_\alpha \bar Q_{\dot\beta}\big]. \tag{44.79}$$

The two contributions in Eq. (44.77) are therefore equal and opposite, and the weighted trace of the anticommutator vanishes:

$$\mathrm{Tr}\Big[(-1)^F \left\{Q_\alpha, \bar Q_{\dot\beta}\right\}\Big] = 0. \tag{44.80}$$

Now use the algebra on the left of Eq. (44.80). By Eq. (44.61) the anticommutator is $2\sigma^\mu_{\alpha\dot\beta} P_\mu$, and on the fixed-momentum representation P_μ acts as the c-number p_μ, so

$$0 = \mathrm{Tr}\Big[(-1)^F 2\sigma^\mu_{\alpha\dot\beta} P_\mu\Big] = 2\,\sigma^\mu_{\alpha\dot\beta}\, p_\mu\, \mathrm{Tr}\big[(-1)^F\big] = 2\,(p\cdot\sigma)_{\alpha\dot\beta}\,(n_B - n_F), \tag{44.81}$$

where $(p\cdot\sigma)_{\alpha\dot\beta} = p_\mu \sigma^\mu_{\alpha\dot\beta}$ and we used $\mathrm{Tr}(-1)^F = n_B - n_F$. The 2×2 matrix $p\cdot\sigma$ is not the zero matrix whenever $p_\mu \neq 0$: its determinant is $\det(p\cdot\sigma) = p^2 = m^2$ for a massive momentum, and even for a massless $p_\mu \neq 0$ at least one entry is nonzero. Choosing any index pair $(\alpha, \dot\beta)$ for which $(p\cdot\sigma)_{\alpha\dot\beta} \neq 0$, Eq. (44.81) forces $n_B - n_F = 0$, that is $n_B = n_F$. □

The result is exactly the one the representation theory must respect: when we build the massless and massive multiplets in Section 44.4 we will count their states explicitly and find boson and fermion totals matching at every step — two real bosonic and two real fermionic states in the massless chiral multiplet, and so on. The trace identity above is the reason the counts are guaranteed to match before any multiplet is constructed.

Caution

The hypothesis $p_\mu \neq 0$ is not cosmetic. At exactly $p_\mu = 0$ — the vacuum, or a state at rest only in a degenerate sense — the matrix $p\cdot\sigma$ vanishes and the argument gives no constraint; bosonic and fermionic ground states

need not pair up. This loophole is exactly where supersymmetry breaking can hide, and it is what the *Witten index* $\mathrm{Tr}(-1)^F$ taken over the *full* Hilbert space (with a regulator $e^{-\beta H}$) is designed to probe: nonzero-energy states pair off and cancel by Theorem 44.15, so the index counts only the zero-energy states, $\mathrm{Tr}(-1)^F = n_B^{E=0} - n_F^{E=0}$, and a nonzero value certifies that a supersymmetric vacuum survives. We develop the Witten index as a tool for deciding whether supersymmetry is dynamically broken in Chapter 46.

44.3.5 A solvable laboratory: supersymmetric quantum mechanics

All three results — $E \geq 0$, the vacuum-energy order parameter, and $n_B = n_F$ away from zero energy — can be seen at once in the simplest system carrying a supersymmetry algebra: a single particle on the line with one fermionic degree of freedom. This is the $0+1$-dimensional reduction of the algebra, and it is exactly solvable, so nothing is hidden.

Worked Example 44.16 (Witten's supersymmetric quantum mechanics)**.** Work on the Hilbert space $\mathcal{H} = L^2(\mathbb{R}) \otimes \mathbb{C}^2$, the tensor product of a bosonic coordinate Hilbert space and a two-state fermionic factor. Introduce fermionic operators $b, b^\dagger$ acting on $\mathbb{C}^2$ with $\{b, b^\dagger\} = 1$ and $b^2 = (b^\dagger)^2 = 0$, and the bosonic position and momentum x, p with $[x, p] = i$ (here $p = -i\,\mathrm{d}/\mathrm{d}x$). Fix a smooth real function $W(x)$, the *superpotential*, and define a single complex supercharge

$$Q = (p - i\,W'(x))\,b, \qquad Q^\dagger = (p + i\,W'(x))\,b^\dagger, \tag{44.82}$$

where $W' = \mathrm{d}W/\mathrm{d}x$. This is the dimensional reduction of Eq. (44.61) to one supercharge: there is one "time" direction, so P_μ reduces to H alone.

The algebra closes. Because $b^2 = 0$, the supercharge is nilpotent, $Q^2 = (p - iW')^2\, b^2 = 0$, and likewise $(Q^\dagger)^2 = 0$ — the analogues of Eq. (44.62). The nontrivial bracket is $\{Q, Q^\dagger\}$. Compute the two operator products. Using $[p, W'(x)] = -i\,W''(x)$,

$$(p - iW')(p + iW') = p^2 + W'^2 + i\,[p, W'] = p^2 + W'^2 + W'', \tag{44.83}$$

$$(p + iW')(p - iW') = p^2 + W'^2 - i\,[p, W'] = p^2 + W'^2 - W''. \tag{44.84}$$

Then, since $\{Q, Q^\dagger\} = (p - iW')(p + iW')\,bb^\dagger + (p + iW')(p - iW')\,b^\dagger b$ and $bb^\dagger = \mathbf{1} - b^\dagger b$,

$$\begin{aligned}\{Q, Q^\dagger\} &= (p^2 + W'^2)(bb^\dagger + b^\dagger b) + W''(bb^\dagger - b^\dagger b) \\ &= (p^2 + W'^2)\,\mathbf{1} + W''(\mathbf{1} - 2b^\dagger b) = (p^2 + W'^2)\,\mathbf{1} + W''\,(-1)^F,\end{aligned} \tag{44.85}$$

where in the last step we identified the fermion-parity operator $(-1)^F = \mathbf{1} - 2b^\dagger b$ (eigenvalue $+1$ on the empty fermionic state, -1 on the occupied one), exactly as in Theorem 44.13. Defining $2H \equiv \left\{Q, Q^\dagger\right\}$, the algebra is the reduced super-Poincaré algebra $Q^2 = 0$, $\left\{Q, Q^\dagger\right\} = 2H$, $[H, Q] = 0$ — the last following because $\left[\left\{Q, Q^\dagger\right\}, Q\right] = 0$ by the Jacobi identity together with $Q^2 = 0$.

Two partner Hamiltonians. In the basis where $(-1)^F = \mathrm{diag}(+1, -1) = \sigma_3$, Eq. (44.85) reads $H = \mathrm{diag}(H_+, H_-)$ with

$$H_\pm = \tfrac{1}{2}p^2 + \tfrac{1}{2}W'(x)^2 \pm \tfrac{1}{2}W''(x). \tag{44.86}$$

The bosonic $(+)$ and fermionic $(-)$ sectors are governed by the two "partner potentials" $V_\pm = \frac{1}{2}W'^2 \pm \frac{1}{2}W''$, which differ only by the $\pm\frac{1}{2}W''$ term.

Positivity. For any $|\psi\rangle$, $\langle\psi|H|\psi\rangle = \frac{1}{2}(\|Q\,|\psi\rangle\|^2 + \left\|Q^\dagger\,|\psi\rangle\right\|^2) \geq 0$, the one-supercharge form of Theorem 44.9. Energies are non-negative.

The order parameter. A zero-energy state must be annihilated by both Q and $Q^\dagger$, by the same sum-of-squares argument as in Theorem 44.10. Writing $|\psi\rangle = \psi_+(x)\,|+\rangle + \psi_-(x)\,|-\rangle$ with $|\pm\rangle$ the fermion-parity eigenstates, $Q\,|\psi\rangle = 0$ and $Q^\dagger\,|\psi\rangle = 0$ reduce to the first-order equations $(\mathrm{d}_x + W')\psi_- = 0$ and $(\mathrm{d}_x - W')\psi_+ = 0$, solved by

$$\psi_+(x) \propto e^{+W(x)}, \qquad \psi_-(x) \propto e^{-W(x)}. \tag{44.87}$$

At most one of these is normalisable, depending on whether $W(x) \to \pm\infty$ as $x \to \pm\infty$. If, say, $W \to +\infty$ at both ends, then $\psi_- \propto e^{-W}$ is normalisable and $\psi_+ \propto e^{+W}$ is not, giving exactly one zero-energy ground state and $E_{\mathrm{vac}} = 0$: supersymmetry is unbroken. If instead $W \to +\infty$ at one end and $-\infty$ at the other (for example W odd and growing), *neither* solution is normalisable, no zero-energy state exists, $E_{\mathrm{vac}} > 0$, and supersymmetry is spontaneously broken — a completely explicit instance of Theorem 44.10.

Pairing and $n_B = n_F$. For any eigenstate with $E > 0$, the operator $Q^\dagger$ maps it to a state of the same energy in the opposite fermion sector (and Q maps back), because $\left[H, Q^\dagger\right] = 0$; the map is invertible on $E > 0$ states since $\left\{Q, Q^\dagger\right\} = 2H = 2E \neq 0$ there. Hence every positive-energy level appears once in the bosonic sector and once in the fermionic sector: $n_B = n_F$ level by level, the $0+1$-dimensional shadow of Theorem 44.15. The pairing fails only at $E = 0$, precisely the loophole flagged above, and the difference of the *unpaired* zero-energy counts is the Witten index $\mathrm{Tr}(-1)^F = n_B^{E=0} - n_F^{E=0}$, which equals $+1$, -1, or 0 in the three asymptotic cases just discussed — a topological invariant of W that decides whether a supersymmetric ground state exists.

This toy model is more than an illustration: it is the conceptual germ of the index theorems and the non-renormalisation results developed in Chapter 46, and

it shows in a setting with no field-theoretic subtleties that the three consequences proved abstractly above are forced, simultaneously, by one nilpotent supercharge.

With the kinematic constraints in hand — a single mass per multiplet (Theorem 44.12) and equal boson and fermion counts away from zero momentum (Theorem 44.15) — the question becomes which collections of one-particle states actually realise them. The next section answers it by classifying the irreducible representations Wigner-style, constructing the massless and massive supermultiplets explicitly and counting their states, and showing concretely that the equalities anticipated here are met multiplet by multiplet.

44.4 Supermultiplets: Massless and Massive Representations

The previous section read physics directly off the algebra without ever constructing a single state. From the positivity relation and the basic anticommutator (Eqs. (44.37) and (44.49)) we already know three rigid facts about *any* unitary representation on which the supercharges act: the energy is bounded below and vanishes only on a supersymmetric vacuum; every state in an irreducible multiplet shares one mass (Theorem 44.12); and the multiplet contains equally many bosonic and fermionic states, $n_B = n_F$ (Theorem 44.15). These are powerful constraints, but they are constraints *on* representations we have not yet built. We now build them. The task is the supersymmetric analogue of what Chapter 3 did for the Poincaré group: classify the one-particle irreducible representations of the full algebra and list their state content. The method is Wigner's (Theorem 3.44)—fix a standard momentum, find the subalgebra that stabilizes it, and induce the whole representation from a representation of that little algebra—adapted to the fact that the stabilizer now contains fermionic generators as well as the little group of Theorem 3.43. The fermionic generators turn out to form a Clifford algebra, and the multiplet is a Clifford module built on a "Clifford vacuum" that carries an ordinary little-group label. Everything in this section is an honest consequence of that single observation. We treat the massless case first, where one of the two fermionic oscillators decouples and the multiplets are remarkably small, then the massive case, where both survive and the multiplets are correspondingly larger.

Throughout we work in two-component (Weyl) notation. The supercharges are Q_α, transforming in $(\frac{1}{2},0)$, and their Hermitian conjugates $\bar{Q}_{\dot\alpha} = (Q_\alpha)^\dagger$, transforming in $(0,\frac{1}{2})$; the index conventions, the Pauli matrices $\sigma^\mu_{\alpha\dot\beta} = (\mathbf{1},\vec\sigma)$ and $\bar\sigma^\mu = (\mathbf{1},-\vec\sigma)$, and the spinor calculus are those fixed in Section 3.4.3 and Eqs. (3.79) and (3.106). We restrict to $\mathcal{N}=1$ for the detailed constructions and turn on extended $\mathcal{N}$ only far enough to count states and to flag where central

charges enter. The two anticommutators we use are

$$\left\{Q_\alpha, \bar{Q}_{\dot\beta}\right\} = 2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu, \qquad \{Q_\alpha, Q_\beta\} = 0 = \left\{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\right\}, \tag{44.88}$$

the first of which is Eq. (44.37) and the second the $\mathcal{N} = 1$, central-charge-free form of Eq. (44.49). The supercharges commute with P^2 (they carry no momentum, $[Q_\alpha, P_\mu] = 0$), which is the algebraic reason a whole multiplet sits at one mass; that is exactly Theorem 44.12, and it licenses us to diagonalize P_μ first and let the Q's move us *within* a fixed-momentum sector.

44.4.1 The massless multiplet as a Clifford module

For a massless particle there is no rest frame, so following Section 3.7.3 we fix a standard null momentum along the z-axis,

$$P^\mu = (E, 0, 0, E), \qquad E > 0, \tag{44.89}$$

and ask how the supercharges act on the one-particle states of this momentum. Everything is controlled by the right-hand side of the first anticommutator in Eq. (44.88) evaluated on Eq. (44.89). Lowering the index with the mostly-minus metric gives $P_\mu = (E, 0, 0, -E)$, so

$$\sigma^\mu_{\alpha\dot\beta} P_\mu = E\,\sigma^0 + (-E)\,\sigma^3 = E(\mathbf{1} - \sigma^3) = E\begin{pmatrix} 0 & 0 \\ 0 & 2 \end{pmatrix}_{\alpha\dot\beta}, \tag{44.90}$$

where the first equality uses $\sigma^\mu P_\mu = P_\mu \sigma^\mu = P_0\sigma^0 + P_3\sigma^3$ with $P_0 = E$, $P_3 = -E$, and $\sigma^3 = \mathrm{diag}(1, -1)$. The crucial feature of Eq. (44.90) is that the matrix is rank one: only the $(\alpha, \dot\beta) = (2, \dot 2)$ entry survives. Written out, the supercharge anticommutators in this frame are therefore

$$\left\{Q_2, \bar{Q}_{\dot 2}\right\} = 4E, \qquad \left\{Q_1, \bar{Q}_{\dot 1}\right\} = 0, \qquad \left\{Q_1, \bar{Q}_{\dot 2}\right\} = \left\{Q_2, \bar{Q}_{\dot 1}\right\} = 0, \tag{44.91}$$

together with $\{Q_\alpha, Q_\beta\} = \left\{\bar{Q}_{\dot\alpha}, \bar{Q}_{\dot\beta}\right\} = 0$.

The second relation in Eq. (44.91) forces Q_1 and $\bar{Q}_{\dot 1}$ to act trivially on every physical state. This is the same positivity argument that gave energy positivity in the previous section, applied locally: for any normalized physical state $|\psi\rangle$ in this momentum sector, $\bar{Q}_{\dot 1} = (Q_1)^\dagger$ gives

$$0 = \langle\psi| \left\{Q_1, \bar{Q}_{\dot 1}\right\} |\psi\rangle = \|Q_1\,|\psi\rangle\|^2 + \left\|\bar{Q}_{\dot 1}\,|\psi\rangle\right\|^2, \tag{44.92}$$

a sum of two non-negative numbers that vanishes only if each does. Hence $Q_1\,|\psi\rangle = \bar{Q}_{\dot 1}\,|\psi\rangle = 0$ for all $|\psi\rangle$: the operators Q_1 and $\bar{Q}_{\dot 1}$ are identically zero on the representation. One of the two fermionic oscillator pairs has *decoupled*. This is the algebraic origin of the smallness of massless multiplets, and it is special to the lightlike kinematics: it happened because $\sigma^\mu P_\mu$ degenerated to rank one when $P^2 = 0$.

Caution

The vanishing of $Q_1, \bar{Q}_{\dot{1}}$ is an operator statement *on this representation*, not a redefinition of the abstract algebra. The supercharges $Q_1, \bar{Q}_{\dot{1}}$ are perfectly nonzero generators of the $\mathcal{N} = 1$ algebra; what Eq. (44.92) shows is that they are represented by zero on a single massless one-particle irrep. On a massive irrep, or on a different massless momentum sector, they act nontrivially. This is the precise analogue of the massless little group collapsing from a non-compact ISO(2) to its SO(2) part in Section 3.7.3.

Only the pair $Q_2, \bar{Q}_{\dot{2}}$ remains, and Eq. (44.91) tells us it is a single fermionic creation/annihilation pair. Normalize it: define

$$a \equiv \frac{\bar{Q}_{\dot{2}}}{2\sqrt{E}}, \qquad a^\dagger \equiv \frac{Q_2}{2\sqrt{E}}, \tag{44.93}$$

so that the surviving anticommutator becomes the canonical fermionic oscillator relation

$$\left\{a, a^\dagger\right\} = \frac{1}{4E}\left\{\bar{Q}_{\dot{2}}, Q_2\right\} = 1, \qquad \{a, a\} = \left\{a^\dagger, a^\dagger\right\} = 0, \tag{44.94}$$

the last two following from $\left\{\bar{Q}_{\dot{2}}, \bar{Q}_{\dot{2}}\right\} = 0$ and $\{Q_2, Q_2\} = 0$. (The choice of which member of the pair to call a and which $a^\dagger$ is a convention; the one made here will put the Clifford vacuum at the *lower* helicity, which is the standard bookkeeping and matches the chiral-multiplet labelling below.) Equation (44.94) is the Clifford algebra of a single fermionic mode. Its unique irreducible unitary representation is two-dimensional: there is a normalized vacuum $|\Omega\rangle$ with $a\,|\Omega\rangle = 0$, the one excited state $a^\dagger\,|\Omega\rangle$, and nothing else because $(a^\dagger)^2 = 0$ truncates the tower after one step. This two-state structure is the entire massless multiplet, once we attach the helicity label.

Helicity of the oscillator. To say *which* helicities the two states carry, we need the J_3 charge of $a^\dagger = Q_2/2\sqrt{E}$, where J_3 generates rotations about the momentum axis $\hat{z}$ and hence measures helicity for our standard momentum Eq. (44.89). Because Q_α is a Lorentz spinor in $(\frac{1}{2}, 0)$, the algebra fixes its commutator with the rotation generator,

$$[J_3, Q_\alpha] = -\tfrac{1}{2}\,(\sigma^3)_\alpha{}^\beta\, Q_\beta, \tag{44.95}$$

the spin-$\frac{1}{2}$ action in the $(\frac{1}{2}, 0)$ representation. With $\sigma^3 = \mathrm{diag}(1, -1)$ this reads $[J_3, Q_1] = -\frac{1}{2}Q_1$ and $[J_3, Q_2] = +\frac{1}{2}Q_2$; taking the Hermitian conjugate of these and using $J_3^\dagger = J_3$ gives $\left[J_3, \bar{Q}_{\dot{1}}\right] = +\frac{1}{2}\bar{Q}_{\dot{1}}$ and $\left[J_3, \bar{Q}_{\dot{2}}\right] = -\frac{1}{2}\bar{Q}_{\dot{2}}$. The

operator content is then immediate: read the sign carefully from Eq. (44.95): $[J_3, Q_2] = +\frac{1}{2}Q_2$ means that if $J_3\,|h\rangle = h\,|h\rangle$ then

$$J_3(Q_2\,|h\rangle\,) = (\,[J_3, Q_2] + Q_2 J_3)\,|h\rangle = (h + \tfrac{1}{2})\,Q_2\,|h\rangle\,. \tag{44.96}$$

Thus Q_2, and with it $a^\dagger$, *raises* the helicity by $\frac{1}{2}$; correspondingly $a = \bar{Q}_{\dot{2}}/2\sqrt{E}$ *lowers* helicity by $\frac{1}{2}$. With the convention of Eq. (44.93), the Clifford vacuum $|\Omega\rangle$ is annihilated by $a = \bar{Q}_{\dot{2}}/2\sqrt{E}$, the helicity-*lowering* member of the pair, so it already sits at the bottom of the helicity ladder; the helicity-*raising* member, $b^\dagger \equiv Q_2/2\sqrt{E} = a^\dagger$, satisfies the same Clifford algebra Eq. (44.94) and builds the partner one half-unit up, at helicity $+\frac{1}{2}$ relative to the vacuum. This fixes a uniform convention: the *Clifford vacuum has the lower helicity λ, and its superpartner sits at $\lambda + \frac{1}{2}$.*

Definition 44.17 (Clifford vacuum and the massless multiplet)**.** On a massless one-particle irrep with standard momentum Eq. (44.89), the *Clifford vacuum* $|\Omega_\lambda\rangle$ is the state of definite helicity λ annihilated by the surviving helicity-lowering supercharge. The massless $\mathcal{N} = 1$ *supermultiplet of base helicity* λ is the two-dimensional Clifford module

$$\{\ |\Omega_\lambda\rangle\,,\quad \left|\Omega_{\lambda+1/2}\right\rangle \equiv b^\dagger\,|\Omega_\lambda\rangle\ \},\qquad b^\dagger = \frac{Q_2}{2\sqrt{E}}, \tag{44.97}$$

with helicities λ and $\lambda + \frac{1}{2}$ and one state at each.

Proposition 44.18 (Massless $\mathcal{N} = 1$ irreducibles)**.** *Every massless one-particle irreducible representation of the $\mathcal{N} = 1$ super-Poincaré algebra is, before imposing CPT, a pair of helicity states $(\lambda, \lambda + \frac{1}{2})$ for some λ, with exactly one state of each helicity. A CPT-invariant theory contains, in addition, the conjugate multiplet $(-\lambda - \frac{1}{2}, -\lambda)$; the full CPT-complete multiplet has four states with helicities $\{-\lambda - \frac{1}{2},\ -\lambda,\ \lambda,\ \lambda + \frac{1}{2}\}$, two bosonic and two fermionic.*

Proof. The first sentence is Theorem 44.17: by Eq. (44.92) one oscillator pair vanishes, the survivor obeys Eq. (44.94), and the unique irreducible Clifford module is two-dimensional with the helicities computed above. Each of the two states is a single helicity eigenstate, hence a one-dimensional rep of the surviving little-group generator J_3, exactly as for an ordinary massless particle (Section 3.7.3); irreducibility of the super-multiplet is the statement that the two helicities are linked by $b^\dagger$ and cannot be split into invariant subspaces. For the CPT statement, recall from Section 3.7.3 that CPT (like parity) reverses helicity, $h \mapsto -h$, and is an antiunitary symmetry that any local relativistic theory must possess. A multiplet $(\lambda, \lambda + \frac{1}{2})$ is not CPT-symmetric unless $\lambda + \frac{1}{2} = -\lambda$, i.e. $\lambda = -\frac{1}{4}$, which is not an allowed helicity by Eq. (3.217); therefore in every case CPT maps the multiplet to a distinct conjugate multiplet with helicities $(-\lambda - \frac{1}{2}, -\lambda)$, and a CPT-invariant theory must contain both.

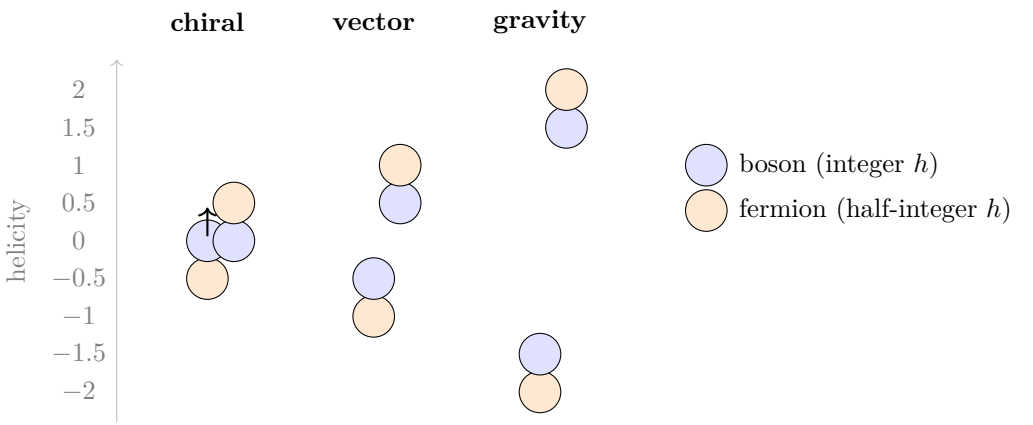


Figure 44.3: The three named CPT-complete massless $\mathcal{N}=1$ multiplets as helicity ladders. Each column shows the four helicity states of Theorem 44.18: a multiplet $(\lambda, \lambda+\frac{1}{2})$ (right pair in each column) together with its CPT conjugate $(-\lambda-\frac{1}{2}, -\lambda)$ (left pair). *Chiral* ($\lambda=0$): a Weyl fermion $\pm\frac{1}{2}$ and a complex scalar $0, 0$. *Vector* ($\lambda=\frac{1}{2}$): a gauge boson ±1 and a gaugino $\pm\frac{1}{2}$. *Gravity* ($\lambda=\frac{3}{2}$): a graviton ±2 and a gravitino $\pm\frac{3}{2}$.

The combined four states have helicities as listed. Counting bosons and fermions: half-integer helicities are fermionic and integer helicities bosonic, and the four values $\{-\lambda-\frac{1}{2}, -\lambda, \lambda, \lambda+\frac{1}{2}\}$ contain exactly two integers and two half-integers regardless of whether λ is integer or half-integer (if $\lambda\in\mathbb{Z}$ then $\lambda, -\lambda$ are bosonic and $\lambda\pm\frac{1}{2}$ fermionic; if $\lambda\in\mathbb{Z}+\frac{1}{2}$ the roles swap), so $n_B=n_F=2$, in agreement with Theorem 44.15. □

The proposition is the whole massless classification; the named multiplets are simply its small-λ instances. Figure 44.3 displays them as helicity ladders.

Worked Example 44.19 (The chiral and vector multiplets from the Clifford vacuum)**.** We build the two phenomenologically central $\mathcal{N}=1$ multiplets explicitly, in each case listing the Clifford-module states, applying CPT, and tabulating helicities and counts.

The chiral multiplet, $\lambda=0$. Take the Clifford vacuum to be a helicity-zero state, $|\Omega_0\rangle$. Equation (44.97) gives the two states

$$|\Omega_0\rangle \;\; (h=0), \qquad b^\dagger\,|\Omega_0\rangle \;\; (h=+\tfrac{1}{2}), \tag{44.98}$$

a scalar and a positive-helicity Weyl state. This pair is not CPT-complete: CPT produces the conjugate multiplet with helicities $(-\frac{1}{2}, 0)$, namely a negative-helicity Weyl state and a second scalar. Assembling both,

$$\text{chiral multiplet:} \qquad \underbrace{\{\,0,\;0\,\}}_{\text{complex scalar}} \;\cup\; \underbrace{\{\,+\tfrac{1}{2},\;-\tfrac{1}{2}\,\}}_{\text{Weyl fermion}}, \tag{44.99}$$

which is precisely one *complex* scalar field (its two real components are the two helicity-zero states, or equivalently a particle and its distinct antiparticle) and one Weyl fermion (the two helicities $\pm\frac{1}{2}$). The two helicity-0 states being a genuine complex scalar—rather than two unrelated reals—is the CPT statement that the scalar and its conjugate sit in conjugate multiplets; that the scalar carries no little-group index follows from Schur applied to the trivial helicity rep (Theorem 2.23). The state count is

$$n_B = \underbrace{2}_{\text{cplx scalar}} = n_F = \underbrace{2}_{\text{Weyl}}, \tag{44.100}$$

verifying $n_B = n_F$ (Theorem 44.15) on the nose.

The vector multiplet, $\lambda = \frac{1}{2}$. Now take the Clifford vacuum to carry helicity $\frac{1}{2}$, $\left|\Omega_{1/2}\right\rangle$. The two states are

$$\left|\Omega_{1/2}\right\rangle \ (h = +\tfrac{1}{2}), \qquad b^\dagger \left|\Omega_{1/2}\right\rangle \ (h = +1). \tag{44.101}$$

CPT adjoins the conjugate $(-1, -\frac{1}{2})$, and the complete multiplet is

$$\text{vector multiplet:} \quad \underbrace{\{+1,\ -1\}}_{\text{gauge boson}} \cup \underbrace{\{+\tfrac{1}{2},\ -\tfrac{1}{2}\}}_{\text{gaugino}}, \tag{44.102}$$

a massless vector (the two transverse helicities ± 1 of a gauge field) together with a single Weyl—equivalently Majorana—fermion in the same representation of any internal gauge group, the *gaugino*. (That the gaugino is forced into the adjoint, the same representation as the gauge boson, is a dynamical statement we establish when we gauge the theory in Chapter 45; here we have only the kinematic helicity content.) Again

$$n_B = \underbrace{2}_{\text{gauge boson}} = n_F = \underbrace{2}_{\text{gaugino}}. \tag{44.103}$$

For completeness, the next case $\lambda = \frac{3}{2}$ gives the $\mathcal{N} = 1$ *supergravity multiplet*: states $(\frac{3}{2}, 2)$ plus the CPT conjugate $(-2, -\frac{3}{2})$, i.e. helicities $\{\pm 2\} \cup \{\pm\frac{3}{2}\}$—a *graviton* (the two helicities of a massless spin-2 field) and a *gravitino* (a massless spin-$\frac{3}{2}$ field). It is the unique massless multiplet whose top helicity reaches 2, and it is the one the local version of supersymmetry—supergravity—is built on.

The chiral multiplet of Eq. (44.99)—a complex scalar plus a Weyl fermion—is the smallest nontrivial supermultiplet, and it is the one we will turn into an interacting field theory in the next section. Before that we record how the count grows with the number of supercharges, since the same Clifford-module logic dictates the entire menu of extended-supersymmetry multiplets and, with it, the famous upper bound on $\mathcal{N}$.

44.4.2 Extended supersymmetry: state counting and the spin bound

With $\mathcal{N}$ supercharges Q^I_α, $I = 1, \ldots, \mathcal{N}$ (and conjugates $\bar{Q}_{\dot\alpha I}$), the massless analysis repeats verbatim per index I. In the frame Eq. (44.89) the anticommutator $\left\{Q^I_\alpha, \bar{Q}_{\dot\beta J}\right\} = 2\,\sigma^\mu_{\alpha\dot\beta} P_\mu\, \delta^I_J$ again degenerates to rank one in the spinor indices, so for each I the pair $Q^I_1, \bar{Q}^I_{\dot 1}$ decouples by the positivity argument of Eq. (44.92), and we are left with $\mathcal{N}$ independent fermionic oscillators

$$b^\dagger_I \equiv \frac{Q^I_2}{2\sqrt{E}}, \qquad \left\{b_I, b^\dagger_J\right\} = \delta_{IJ}, \qquad \{b_I, b_J\} = \left\{b^\dagger_I, b^\dagger_J\right\} = 0, \tag{44.104}$$

each raising the helicity by $\frac{1}{2}$. (Massless central charges drop out: the $\left\{Q^I, Q^J\right\} \propto Z^{IJ}$ terms carry the rank-one $\epsilon_{\alpha\beta}$ structure that, contracted against the single surviving component, gives zero, so massless multiplets never feel Z^{IJ}.) The representation is the Fock space of $\mathcal{N}$ fermionic modes built on a Clifford vacuum $|\Omega_\lambda\rangle$ of helicity λ, with $b_I\,|\Omega_\lambda\rangle = 0$.

A fermionic Fock space on $\mathcal{N}$ modes has dimension $2^{\mathcal{N}}$: a state is specified by which subset of the $b^\dagger_I$ act on the vacuum, and there are $2^{\mathcal{N}}$ subsets. Organizing by the number k of creation operators, the states with k excitations are the $\binom{\mathcal{N}}{k}$ totally antisymmetric products $b^\dagger_{[I_1} \cdots b^\dagger_{I_k]}$, and they all carry helicity $\lambda + \frac{k}{2}$. Hence

$$\dim(\text{multiplet}) = \sum_{k=0}^{\mathcal{N}} \binom{\mathcal{N}}{k} = 2^{\mathcal{N}}, \qquad (\#\text{ states at helicity } \lambda + \tfrac{k}{2}) = \binom{\mathcal{N}}{k}. \tag{44.105}$$

The helicities run from λ (the vacuum) to $\lambda + \frac{\mathcal{N}}{2}$ (all oscillators excited), a range of $\frac{\mathcal{N}}{2}$. Bosons and fermions alternate with k, so the boson/fermion split is $n_B = \sum_{k\ \text{even}} \binom{\mathcal{N}}{k}$ and $n_F = \sum_{k\ \text{odd}} \binom{\mathcal{N}}{k}$; the binomial identity $\sum_k (-1)^k \binom{\mathcal{N}}{k} = (1-1)^{\mathcal{N}} = 0$ gives $n_B = n_F = 2^{\mathcal{N}-1}$, so $n_B = n_F$ holds for every $\mathcal{N}$, as it must.

Worked Example 44.20 (The $2^{\mathcal{N}}$ count and the spin-2 bound, for $\mathcal{N} = 1, 2, 4, 8$)**.** We carry the count explicitly for the four physically important cases and read off the bound on $\mathcal{N}$.

$\mathcal{N} = 1$ ($2^1 = 2$ states before CPT). One oscillator, helicities $(\lambda, \lambda + \frac{1}{2})$: the multiplets of Section 44.4.1.

$\mathcal{N} = 2$ ($2^2 = 4$ states). Helicities $\lambda, \lambda + \frac{1}{2}$ (twice), $\lambda + 1$, with multiplicities $\binom{2}{0}, \binom{2}{1}, \binom{2}{2} = 1, 2, 1$. Choosing the Clifford vacuum at $\lambda = -\frac{1}{2}$ gives helicities $\{-\frac{1}{2}, 0, 0, +\frac{1}{2}\}$—already CPT-self-conjugate—which is the $\mathcal{N} = 2$ *hypermultiplet*: two complex scalars and a Dirac fermion ($n_B = n_F = 4$). Choosing $\lambda = 0$ gives $\{0, \frac{1}{2}, \frac{1}{2}, 1\}$, whose CPT completion is the $\mathcal{N} = 2$ *vector multiplet*: a gauge boson, two Weyl fermions, and a complex scalar.

$\mathcal{N}=4$ *($2^4=16$ states).* Multiplicities $\binom{4}{k}=1,4,6,4,1$ for $k=0,\dots,4$. Putting the vacuum at $\lambda=-1$ centres the ladder on zero: helicities -1 (one state), $-\frac{1}{2}$ (four), 0 (six), $+\frac{1}{2}$ (four), $+1$ (one). This is already CPT-self-conjugate and is the unique $\mathcal{N}=4$ multiplet with maximal helicity 1: a single gauge boson ±1, four Weyl fermions ($\pm\frac{1}{2}$, four each by the $\binom{4}{1}=4$ degeneracy), and six real scalars (the $\binom{4}{2}=6$ states at $h=0$). It is the field content of $\mathcal{N}=4$ super-Yang–Mills. Note $n_B=1+1+6=8=4+4=n_F$. Because its top helicity is exactly 1, $\mathcal{N}=4$ is the largest extended supersymmetry whose multiplets contain no spin above 1: any larger $\mathcal{N}$ forces a helicity $\geq\frac{3}{2}$.

$\mathcal{N}=8$ *($2^8=256$ states).* Multiplicities $\binom{8}{k}=1,8,28,56,70,56,28,8,1$. With the vacuum at $\lambda=-2$ the helicities run symmetrically from -2 to $+2$: one graviton helicity $+2$ and one -2 (the $k=8$ and $k=0$ states), eight gravitini at each of $\pm\frac{3}{2}$, 28 vectors at each of ±1, 56 Weyl fermions at each of $\pm\frac{1}{2}$, and 70 real scalars at $h=0$. This is $\mathcal{N}=8$ supergravity. Its top helicity is exactly 2.

The bound. The pattern is now forced. A CPT-complete massless multiplet built on a Clifford vacuum spans a helicity range of width $\frac{\mathcal{N}}{2}$ on each side of the centre once CPT doubling is taken into account, so its maximal helicity can be made no smaller than $\frac{\mathcal{N}}{4}$ by the most symmetric choice of λ (centre the ladder on zero: $\lambda=-\frac{\mathcal{N}}{4}$, top helicity $+\frac{\mathcal{N}}{4}$). A consistent *interacting* theory of massless particles cannot contain a helicity exceeding 2: a massless particle of helicity $|h|>2$ admits no consistent long-range coupling. The argument is a soft-theorem one, carried out in full in this book in Section 63.6.1. A massless helicity-h quantum emitted softly off the external legs of an amplitude attaches with a universal factor $\sum_i g_i\,\epsilon_{\mu_1\cdots\mu_h}p_i^{\mu_1}\cdots p_i^{\mu_h}/(p_i\cdot q)$; invariance of the S-matrix under the longitudinal (pure-gauge) polarizations $\epsilon\to q$ demands that the residue $\sum_i g_i\,p_i^{\mu_1}\cdots p_i^{\mu_{h-1}}$ vanish for every configuration of the hard momenta. For $h=1$ this is $\sum_i g_i=0$, charge conservation, which *can* be satisfied (Eq. (63.73)); for $h=2$ it is $\sum_i g_i\,p_i^{\mu}=0$, satisfied by overall momentum conservation provided every g_i is equal—the universality of gravity (Eq. (63.77)). But for $h\geq3$ the condition $\sum_i g_i\,p_i^{\mu_1}\cdots p_i^{\mu_{h-1}}=0$ is an independent constraint on the hard momenta that no choice of nonzero couplings can meet for generic on-shell kinematics: there is no conserved current of rank >2 to which the particle could couple, and the soft limit is incompatible with a nontrivial S-matrix. We use only this conclusion here; the spin-1 and spin-2 cases are established in Section 63.6.1, and the classic statements are Weinberg's soft theorems and the Weinberg–Witten theorem [500]. Imposing $\frac{\mathcal{N}}{4}\leq2$ yields

$$\boxed{\mathcal{N}\leq8}\qquad\text{(local supersymmetry, spin}\leq2\text{, supergravity)},\tag{44.106}$$

saturated by $\mathcal{N}=8$ supergravity. If one insists on a theory *without* gravity—no helicity above 1—the same argument with $\frac{\mathcal{N}}{4}\leq1$ gives $\mathcal{N}\leq4$, saturated by $\mathcal{N}=4$ super-Yang–Mills. These two numbers, 4 and 8, are the reason

extended supersymmetry is so rigidly constrained: there are only finitely many supersymmetric theories of bounded spin, and the most symmetric ones are essentially unique.

Physical Insight

The spin bound is the supersymmetric echo of a theme from Chapter 3: the little group dictates everything. There, the non-compactness of the massless little group forced a massless particle to carry at most two helicity states. Here, each supercharge is a helicity-$\frac{1}{2}$ raising operator, so $\mathcal{N}$ of them necessarily generate a helicity spread of $\frac{\mathcal{N}}{2}$; the only way to keep all helicities physical ($|h| \leq 2$) is to limit $\mathcal{N}$. Supersymmetry cannot be made arbitrarily large precisely because spacetime has spin-2 gravity as its ceiling. Larger $\mathcal{N}$ would demand fundamental higher-spin massless particles, which the consistency of the S-matrix forbids.

44.4.3 Massive multiplets

For $m > 0$ there is a rest frame, and following Section 3.7.2 we fix the standard momentum

$$P^\mu = (m, 0, 0, 0), \qquad m > 0, \tag{44.107}$$

whose little group is the full SU(2) of spatial rotations. Now the right-hand side of the anticommutator does *not* degenerate: with $P_\mu = (m, 0, 0, 0)$,

$$\sigma^\mu_{\alpha\dot\beta} P_\mu = m\, \sigma^0_{\alpha\dot\beta} = m\, \delta_{\alpha\dot\beta}, \tag{44.108}$$

since $\sigma^0 = \mathbf{1}$, so

$$\left\{Q_\alpha, \bar Q_{\dot\beta}\right\} = 2m\, \delta_{\alpha\dot\beta}, \qquad \{Q_\alpha, Q_\beta\} = \left\{\bar Q_{\dot\alpha}, \bar Q_{\dot\beta}\right\} = 0. \tag{44.109}$$

Both spinor components survive: there is no positivity collapse because the diagonal entries $\left\{Q_1, \bar Q_{\dot 1}\right\} = \left\{Q_2, \bar Q_{\dot 2}\right\} = 2m$ are both nonzero. This is the algebraic reason massive multiplets are larger than massless ones—the light-like degeneration that killed half the oscillators is gone. Define two fermionic oscillators

$$a_\alpha \equiv \frac{Q_\alpha}{\sqrt{2m}}, \qquad a^\dagger_{\dot\alpha} \equiv \frac{\bar Q_{\dot\alpha}}{\sqrt{2m}}, \qquad \left\{a_\alpha, a^\dagger_{\dot\beta}\right\} = \delta_{\alpha\dot\beta}, \tag{44.110}$$

with all like-type anticommutators vanishing. We now have the Clifford algebra of *two* fermionic modes, whose irreducible module has dimension $2^2 = 4$. Crucially, the two creation operators $a^\dagger_{\dot 1}, a^\dagger_{\dot 2}$ carry the dotted spinor index of $(0, \frac{1}{2})$, so under the rest-frame little group SU(2) they form a spin-$\frac{1}{2}$ doublet. The multiplet is therefore not just four states but four states organized into definite SU(2) spins, built on a Clifford vacuum that is itself an SU(2) irrep.

Proposition 44.21 (Massive $\mathcal{N}=1$ irreducibles)**.** *Let the Clifford vacuum carry little-group spin j, i.e. be a $(2j+1)$-dimensional* SU(2) *irrep $|\Omega; j, m_j\rangle$ annihilated by both a_α. The massive $\mathcal{N}=1$ multiplet built on it has $4(2j+1)$ states, decomposing under spin as*

$$\underbrace{(j)}_{\text{vacuum}} \oplus \underbrace{(j+\tfrac{1}{2}) \oplus (j-\tfrac{1}{2})}_{\text{one } a^\dagger} \oplus \underbrace{(j)}_{\text{two } a^\dagger}, \tag{44.111}$$

where the $(j-\frac{1}{2})$ summand is absent when $j=0$, and $n_B = n_F = 2(2j+1)$.

Proof. The module is spanned by $|\Omega\rangle$, by $a^\dagger_{\dot\alpha}|\Omega\rangle$ ($\dot\alpha = 1,2$), and by $a^\dagger_{\dot 1} a^\dagger_{\dot 2}|\Omega\rangle$; the would-be states with three or more creation operators vanish because $(a^\dagger_{\dot 1})^2 = (a^\dagger_{\dot 2})^2 = 0$. Each level is a tensor product of the vacuum's SU(2) rep with the rep formed by the creation operators acting at that level. At zero excitations the states transform as the vacuum, spin j. At one excitation the single dotted index supplies a spin-$\frac{1}{2}$ factor, so the states transform as $(j)\otimes(\frac{1}{2}) = (j+\frac{1}{2})\oplus(j-\frac{1}{2})$ by the standard Clebsch–Gordan decomposition of SU(2) (with the lower branch absent at $j=0$, where $0\otimes\frac{1}{2} = \frac{1}{2}$ only). At two excitations the product $a^\dagger_{\dot 1} a^\dagger_{\dot 2}$ is antisymmetric in its two dotted indices; the antisymmetric part of $\frac{1}{2}\otimes\frac{1}{2}$ is the singlet, $\epsilon^{\dot\alpha\dot\beta} a^\dagger_{\dot\alpha} a^\dagger_{\dot\beta}$, so the two-excitation states again transform as the vacuum's spin j. Summing dimensions, $(2j+1) + 2(2j+1) + (2j+1) = 4(2j+1)$, which is the announced total. The level with one $a^\dagger$ is fermionic relative to the vacuum and the levels with zero or two are bosonic relative to it; the fermion number flips by one with each $a^\dagger$, so half the states—those with an odd number of excitations—are fermionic, giving $n_F = 2(2j+1) = n_B$, consistent with Theorem 44.15. Irreducibility under the full algebra is the statement that the supercharges link all four levels; that the little-group content at each level is as stated uses only that an intertwiner between SU(2) irreps is a multiple of the identity (Theorem 2.23), pinning the Clebsch–Gordan multiplicities to one. □

Worked Example 44.22 (The massive chiral and massive vector multiplets)**.** We instantiate Theorem 44.21 at the two smallest vacua.

Spin-0 vacuum: the massive chiral multiplet (4 states). With $j=0$, Eq. (44.111) reduces to $(0)\oplus(\frac{1}{2})\oplus(0)$: a spin-0 vacuum, a spin-$\frac{1}{2}$ doublet at one excitation, and a second spin-0 at two excitations. The content is therefore

$$\text{massive chiral:}\quad \underbrace{2\times\text{spin } 0}_{\text{2 real scalars}} \oplus \underbrace{\text{spin } \tfrac{1}{2}}_{\text{2 states}}, \tag{44.112}$$

i.e. two real scalars (equivalently one complex scalar) of mass m and one massive Majorana fermion of the same mass. The count is $n_B = 2$ (the two scalar states) and $n_F = 2$ (the spin-$\frac{1}{2}$ doublet), so $n_B = n_F = 2$. This is the massive avatar of the massless chiral multiplet Eq. (44.99): the same physical fields—a complex

scalar and a Weyl/Majorana fermion—now all massive and degenerate, exactly the degeneracy Theorem 44.12 demands.

Spin-$\frac{1}{2}$ vacuum: the massive vector multiplet (8 states). With $j = \frac{1}{2}$, Eq. (44.111) gives $(\frac{1}{2}) \oplus [(1) \oplus (0)] \oplus (\frac{1}{2})$: counting dimensions $2 + (3 + 1) + 2 = 8$. The content is

$$\text{massive vector:} \quad \underbrace{\text{spin } 1}_{3 \text{ states}} \oplus \underbrace{\text{spin } 0}_{1 \text{ state}} \oplus \underbrace{2 \times \text{spin } \tfrac{1}{2}}_{4 \text{ states}}, \tag{44.113}$$

a *massive* vector (three polarizations, including the longitudinal one absent in the massless case), one real scalar, and a massive Dirac fermion (the two spin-$\frac{1}{2}$ doublets combine into four fermionic states). The count is $n_B = 3 + 1 = 4$ and $n_F = 4$, so $n_B = n_F = 4$.

The contrast with the massless vector multiplet Eq. (44.102) is instructive and physically central. The massless vector multiplet has $2+2 = 4$ states; the massive one has $4 + 4 = 8$. The extra states are exactly a massless chiral multiplet's worth: a real scalar and an extra fermionic doublet, which supply the longitudinal polarization of the now-massive vector and the Dirac partner of the gaugino. This is the supersymmetric Higgs mechanism in embryo—a massless vector multiplet "eats" a chiral multiplet to become a massive vector multiplet, $4+4 = 8$ states all the way through—and it is the representation- theoretic reason a supersymmetric mass for a gauge field comes packaged with extra matter.

44.4.4 Central charges and BPS shortening

The massive analysis above used the $\mathcal{N} = 1$ algebra, where $\{Q_\alpha, Q_\beta\} = 0$. With extended supersymmetry the like-chirality anticommutator can acquire a Lorentz-scalar, internal-symmetry-antisymmetric piece, the *central charge*,

$$\left\{Q^I_\alpha, Q^J_\beta\right\} = 2\,\epsilon_{\alpha\beta}\,Z^{IJ}, \qquad Z^{IJ} = -Z^{JI}, \tag{44.114}$$

with Z^{IJ} commuting with everything in the algebra (hence "central"; that it must be a c-number on an irrep is again Schur, Theorem 2.23). Diagonalizing the antisymmetric Z^{IJ} into skew 2×2 blocks with skew-eigenvalues $Z_n \geq 0$, the rest-frame algebra organizes into oscillators whose anticommutators are shifted by $\pm Z_n$ relative to the $2m$ of Eq. (44.109). Positivity of the resulting oscillator norms—the same $\|\cdot\|^2 \geq 0$ argument as in Eq. (44.92), now applied block by block—forces the *Bogomolny–Prasad–Sommerfield (BPS) bound*

$$m \;\geq\; \max_n |Z_n|. \tag{44.115}$$

When the bound is saturated, $m = |Z_n|$ for one or more n, the corresponding oscillator anticommutator vanishes and that oscillator pair decouples *exactly as the second pair decoupled in the massless case* (Eq. (44.92)). The Clifford

module is then built on fewer oscillators and is correspondingly *shortened*: a BPS multiplet has fewer states than a generic massive multiplet of the same spin, interpolating in size between the generic massive ($2^{2\mathcal{N}}$) and massless ($2^{\mathcal{N}}$) counts. We will not need the detailed BPS representation theory here; we record only that the shortening is an algebraic consequence of saturating Eq. (44.115), that BPS states are stable for this reason, and that the quantitative theory of BPS spectra—and its spectacular use in extracting exact non-perturbative results—is developed in Chapter 46.

Physical Insight

The unifying picture across this section is a single dichotomy: an oscillator pair survives or it decouples, according to whether the relevant entry of the right-hand side of the anticommutator is nonzero or zero. Massless $\mathcal{N} = 1$: one pair survives (2 states). Massive $\mathcal{N} = 1$: both survive (4 states). Extended massive, generic: all $2\mathcal{N}$ pairs survive ($2^{2\mathcal{N}}$ states). Extended massive, BPS: some pairs decouple, shortening the multiplet. The massless multiplet is simply the extreme case where the degeneration is maximal because $P^2 = 0$, and the BPS multiplet is the intermediate case where it is partial because $m = |Z|$. The state counting, the $n_B = n_F$ equality, and the spin bound are all readings of this one mechanism.

With the representation theory in hand, the simplest multiplet is no longer an abstraction. The massless $\mathcal{N} = 1$ chiral multiplet of Eq. (44.99)—a single complex scalar together with a single Weyl fermion—is the minimal nontrivial supermultiplet, and it is the exact field content we must now endow with dynamics. In the next section we write down the free theory of a complex scalar and a Weyl fermion, ask what interactions are compatible with the supersymmetry algebra closing on these fields, and discover both the necessity of an auxiliary field for off-shell closure and the superpotential that organizes the interactions: the Wess–Zumino model.

44.5 The Wess–Zumino Model: On-Shell and Off-Shell SUSY

The representation theory of Section 44.4 handed us the smallest nontrivial package of supersymmetry: the *chiral multiplet* of Eq. (44.99), a complex scalar together with a single Weyl fermion, with equal numbers of bosonic and fermionic states. So far this is kinematics — a list of one-particle states forced to sit together by the algebra. The obvious next question is dynamical: can we write down a local field theory whose fields are exactly this multiplet, whose action is invariant under a symmetry that rotates the scalar into the fermion, and which realises the algebra Eqs. (44.61) to (44.63) on the fields? The answer

is the *Wess–Zumino model*, the hydrogen atom of supersymmetric field theory. Building it teaches three lessons that recur everywhere in the subject. First, that a transformation mixing a boson and a fermion is fixed almost entirely by dimensional analysis and Lorentz covariance, with only a handful of constants left to pin down by invariance. Second — and this is the central surprise — that the naive transformations close into a translation, as the algebra demands, *only when the fields obey their equations of motion*; off-shell the algebra fails by a term proportional to the Dirac operator. The cure is to enlarge the multiplet by a non-propagating *auxiliary field*, restoring the boson–fermion balance off-shell and closing the algebra identically. Third, that once the kinetic theory is in hand, all renormalisable interactions are governed by a single holomorphic function, the *superpotential* $W(\phi)$, whose first derivative squared is the scalar potential and whose second derivative is the Yukawa coupling — so that boson and fermion masses are locked together, exactly as the degeneracy theorem Theorem 44.12 promised.

Throughout we work in the two-component conventions fixed in Section 44.4: undotted indices are contracted descending, $\psi\chi \equiv \psi^{\alpha}\chi_{\alpha}$, dotted indices ascending, $\bar{\psi}\bar{\chi} \equiv \bar{\psi}_{\dot{\alpha}}\bar{\chi}^{\dot{\alpha}}$, with $\sigma^{\mu} = (\mathbf{1}, \vec{\sigma})$ and $\bar{\sigma}^{\mu} = (\mathbf{1}, -\vec{\sigma})$ of Eqs. (3.79) and (3.106), indices raised and lowered by the antisymmetric symbol ε of Eq. (3.107). The metric is mostly-minus and units are natural. The fermion ψ is a left-handed Weyl spinor in $(\tfrac{1}{2}, 0)$, ψ_{α}, with conjugate $\bar{\psi}_{\dot{\alpha}} = (\psi_{\alpha})^{\dagger}$; we will treat it as Grassmann-valued at the classical (anticommuting-field) level, exactly the algebra of Theorem 12.3, and its single-Weyl mass term is the Majorana mass of Sections 11.7.2 and 11.7.6.

44.5.1 The free Lagrangian and the SUSY transformations

The field content is a complex scalar ϕ and a left-handed Weyl fermion ψ_{α}. The unique Lorentz-invariant, renormalisable, free Lagrangian built from them is the sum of the canonical complex-scalar kinetic term and the canonical Weyl kinetic term,

$$\mathcal{L}_{\text{free}} \;=\; \partial_{\mu}\phi^{*}\,\partial^{\mu}\phi \;+\; i\,\bar{\psi}\,\bar{\sigma}^{\mu}\,\partial_{\mu}\psi, \tag{44.116}$$

where $\bar{\psi}\bar{\sigma}^{\mu}\partial_{\mu}\psi = \bar{\psi}_{\dot{\alpha}}\,\bar{\sigma}^{\mu\,\dot{\alpha}\beta}\,\partial_{\mu}\psi_{\beta}$ is the two-component reduction of the Dirac kinetic term of Chapter 11 for a single left-handed field. In mostly-minus natural units the canonical dimensions are $[\phi] = 1$ and $[\psi] = \tfrac{3}{2}$, as one reads off by demanding that each term in Eq. (44.116) have dimension 4.

We now look for an infinitesimal transformation that mixes ϕ and ψ and leaves the action invariant. The transformation must be a symmetry of fermionic type: it carries a constant spinor parameter ξ_{α}, anticommuting and of dimension $[\xi] = -\tfrac{1}{2}$, so that ξ multiplied into ψ has the dimension of ϕ and into ϕ has the dimension of ψ. (We write the parameter ξ rather than ε to avoid collision with the spinor metric $\varepsilon_{\alpha\beta}$.) The lowest-dimension covariant possibilities are essentially forced. A scalar of dimension 1 can be made from ξ and ψ only as

the invariant contraction $\xi\psi$; a spinor of dimension $\frac{3}{2}$ can be made from $\bar\xi$ and ϕ only by spending one derivative, as $(\sigma^\mu\bar\xi)\,\partial_\mu\phi$, which carries the correct single undotted index. We therefore postulate

$$\delta\phi = \sqrt{2}\,\xi\psi, \qquad \delta\psi_\alpha = \sqrt{2}\,c\;i\,(\sigma^\mu\bar\xi)_\alpha\,\partial_\mu\phi, \tag{44.117}$$

with $(\sigma^\mu\bar\xi)_\alpha = \sigma^\mu_{\alpha\dot\beta}\bar\xi^{\dot\beta}$, the overall $\sqrt{2}$ a conventional normalisation of $\delta\phi$, and a single real constant c to be fixed by invariance. The conjugate rules follow by Hermitian conjugation, using $(\xi\psi)^\dagger = \bar\psi\bar\xi = \bar\xi\bar\psi$ and the Hermiticity of σ^μ:

$$\delta\phi^* = \sqrt{2}\,\bar\xi\bar\psi, \qquad \delta\bar\psi_{\dot\alpha} = \sqrt{2}\,c\,(-i)\,(\xi\sigma^\mu)_{\dot\alpha}\,\partial_\mu\phi^*, \tag{44.118}$$

with $(\xi\sigma^\mu)_{\dot\alpha} = \xi^\beta\sigma^\mu_{\beta\dot\alpha}$.

Proposition 44.23 (Invariance of the free action fixes $c=-1$)**.** *With the choice $c=-1$ in Eq. (44.117), the action $S_{\text{free}} = \int \mathrm{d}^4x\,\mathcal{L}_{\text{free}}$ of Eq. (44.116) is invariant under Eqs. (44.117) and (44.118); the Lagrangian changes by a total derivative. No other value of c gives a symmetry.*

Proof. We track every term. The variation splits cleanly into two complex-conjugate sectors: one holding $(\bar\xi,\bar\psi,\phi)$ and the other (ξ,ψ,ϕ^*); they are conjugates, so it suffices to show one of them is a total derivative. Take the $(\bar\xi,\bar\psi,\phi)$ sector. It receives two contributions, one from varying ϕ^* in the scalar kinetic term and one from varying ψ in the fermion kinetic term:

$$\delta_D\mathcal{L}_{\text{free}} = \partial_\mu(\delta\phi^*)\,\partial^\mu\phi = \sqrt{2}\,\bar\xi\,(\partial_\mu\bar\psi)\,\partial^\mu\phi, \tag{44.119}$$

$$\begin{aligned}\delta_A\mathcal{L}_{\text{free}} &= i\,\bar\psi\bar\sigma^\mu\partial_\mu(\delta\psi) = i\,\bar\psi_{\dot\alpha}\bar\sigma^{\mu\,\dot\alpha\beta}\,\partial_\mu(\sqrt{2}\,c\,i\,\sigma^\nu_{\beta\dot\gamma}\bar\xi^{\dot\gamma}\partial_\nu\phi)\\ &= -\sqrt{2}\,c\,\bar\psi_{\dot\alpha}\,(\bar\sigma^\mu\sigma^\nu)^{\dot\alpha}{}_{\dot\gamma}\,\bar\xi^{\dot\gamma}\,\partial_\mu\partial_\nu\phi.\end{aligned} \tag{44.120}$$

The constant parameter $\bar\xi$ passes through the derivative. In Eq. (44.120) the two derivatives are symmetric in $\mu\nu$, so only the symmetric part of $\bar\sigma^\mu\sigma^\nu$ survives. The two-component Clifford identity (proved blockwise in Section 3.4.3)

$$(\bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu)^{\dot\alpha}{}_{\dot\gamma} = 2\,\eta^{\mu\nu}\,\delta^{\dot\alpha}{}_{\dot\gamma} \tag{44.121}$$

then collapses the structure to $\eta^{\mu\nu}\partial_\mu\partial_\nu = \Box$, giving $\delta_A\mathcal{L}_{\text{free}} = -\sqrt{2}\,c\;\bar\psi\bar\xi\,\Box\phi = -\sqrt{2}\,c\;\bar\xi\bar\psi\,\Box\phi$, where we used the symmetry $\bar\psi\bar\xi = \bar\xi\bar\psi$ of Grassmann-spinor contractions. Now integrate Eq. (44.119) by parts inside the action, moving ∂_μ off $\bar\psi$:

$$\int \mathrm{d}^4x\,\delta_D\mathcal{L}_{\text{free}} = -\sqrt{2}\int \mathrm{d}^4x\,\bar\xi\,\bar\psi\,\Box\phi. \tag{44.122}$$

Adding the two,

$$\int \mathrm{d}^4x\,(\delta_A+\delta_D)\mathcal{L}_{\text{free}} = -\sqrt{2}\,(c+1)\int \mathrm{d}^4x\,\bar\xi\,\bar\psi\,\Box\phi. \tag{44.123}$$

This vanishes for all field configurations if and only if $c = -1$, and then the two terms combine into the total derivative $\partial_\mu(\sqrt{2}\,\bar\xi\,\bar\psi\,\partial^\mu\phi)$ as one checks by reversing the integration by parts. The conjugate sector (ξ, ψ, ϕ^*) is the complex conjugate of this computation and likewise reduces to a total derivative when $c = -1$. Hence the action is invariant, and only for $c = -1$. □

From here on we fix $c = -1$ and write the *on-shell* Wess–Zumino transformations

$$\boxed{\delta\phi = \sqrt{2}\,\xi\psi, \qquad \delta\psi_\alpha = -\sqrt{2}\,i\,(\sigma^\mu\bar\xi)_\alpha\,\partial_\mu\phi} \tag{44.124}$$

together with their conjugates. The label *on-shell* anticipates the defect we now expose.

Why a derivative is unavoidable

The transformation $\delta\psi$ *must* carry a derivative of ϕ. A SUSY parameter has dimension $-\frac{1}{2}$, the scalar has dimension 1, and the fermion has dimension $\frac{3}{2}$; the arithmetic $-\frac{1}{2}+1 = \frac{1}{2}$ falls short of $\frac{3}{2}$ by exactly one unit of dimension, which only a ∂_μ can supply. This is the seed of the entire structure: because one side of the transformation carries a derivative and the other does not, composing two transformations produces a single net derivative — a translation. Supersymmetry is, quite literally, a square root of spacetime translation, and Eq. (44.61) is that statement at the level of the algebra.

44.5.2 On-shell closure, and its failure

The algebra Eq. (44.61) demands that the commutator of two supersymmetry transformations, with parameters ξ_1 and ξ_2, be a translation. On fields, a translation by a^μ acts as $a^\mu\partial_\mu$. We test this directly. It is convenient to abbreviate the translation parameter that should emerge,

$$a^\mu \;\equiv\; 2i(\xi_1\sigma^\mu\bar\xi_2 - \xi_2\sigma^\mu\bar\xi_1), \tag{44.125}$$

which is real and antisymmetric under $1 \leftrightarrow 2$, exactly the structure $\{Q, \bar Q\} = 2\sigma^\mu P_\mu$ produces when sandwiched between two parameters.

Closure on the scalar. Using Eq. (44.124),

$$\delta_1\delta_2\,\phi = \sqrt{2}\,\xi_2\,\delta_1\psi = \sqrt{2}\,\xi_2^\alpha(-\sqrt{2}\,i\,\sigma^\mu_{\alpha\dot\beta}\bar\xi_1^{\dot\beta}\partial_\mu\phi) = -2i\,(\xi_2\sigma^\mu\bar\xi_1)\,\partial_\mu\phi. \tag{44.126}$$

Antisymmetrising in $1 \leftrightarrow 2$,

$$[\delta_1, \delta_2]\,\phi = -2i(\xi_2\sigma^\mu\bar\xi_1 - \xi_1\sigma^\mu\bar\xi_2)\partial_\mu\phi = a^\mu\,\partial_\mu\phi. \tag{44.127}$$

The algebra closes on ϕ *identically* — no equation of motion was used. The commutator is precisely the translation Eq. (44.125).

Closure on the fermion. Here the story changes. With $\delta_2\psi_\alpha = -\sqrt{2}\, i\,(\sigma^\mu\bar\xi_2)_\alpha\partial_\mu\phi$ and $\delta_1\phi = \sqrt{2}\,\xi_1\psi$,

$$\delta_1\delta_2\,\psi_\alpha = -\sqrt{2}\, i\,\sigma^\mu_{\alpha\dot\beta}\bar\xi_2^{\dot\beta}\,\partial_\mu(\delta_1\phi) = -2i\,\sigma^\mu_{\alpha\dot\beta}\bar\xi_2^{\dot\beta}\,\xi_1^\gamma\,\partial_\mu\psi_\gamma. \tag{44.128}$$

The obstruction is that $\partial_\mu\psi_\gamma$ carries its spinor index γ contracted into ξ_1, not free; to compare with $a^\mu\partial_\mu\psi_\alpha$ we must move the free index onto ψ. This rearrangement is a Fierz identity. The clean tool is the *Schouten identity*: for any three anticommuting two-component spinors λ,χ,ρ,

$$\lambda_\alpha(\chi\rho) + \chi_\alpha(\rho\lambda) + \rho_\alpha(\lambda\chi) = 0, \tag{44.129}$$

which holds because the left-hand side is totally antisymmetric in the three two-valued indices it would take to write all spinors with a lower index, and no nonzero totally antisymmetric object exists on three values of a two-dimensional index. Apply Eq. (44.129) with $\lambda_\alpha = (\sigma^\mu\bar\xi_2)_\alpha$, $\chi = \xi_1$, $\rho = \partial_\mu\psi$:

$$(\sigma^\mu\bar\xi_2)_\alpha\,(\xi_1\partial_\mu\psi) = -\,\xi_{1\alpha}\,(\partial_\mu\psi\,\sigma^\mu\bar\xi_2) - (\partial_\mu\psi)_\alpha\,(\sigma^\mu\bar\xi_2\,\xi_1). \tag{44.130}$$

In the last factor $(\sigma^\mu\bar\xi_2\,\xi_1) = (\sigma^\mu\bar\xi_2)^\gamma\xi_{1\gamma} = \xi_1\sigma^\mu\bar\xi_2$ by Grassmann symmetry of the undotted contraction. In the middle factor we use the standard reordering identity for anticommuting spinors,

$$\chi\,\sigma^\mu\,\bar\eta = -\,\bar\eta\,\bar\sigma^\mu\,\chi, \tag{44.131}$$

(the minus sign is the Grassmann reordering, traced through the ε-symbols in Eq. (3.107)), so that $\partial_\mu\psi\,\sigma^\mu\bar\xi_2 = -\bar\xi_2\bar\sigma^\mu\partial_\mu\psi$. Inserting Eq. (44.130) into Eq. (44.128) and antisymmetrising, the piece proportional to $(\partial_\mu\psi)_\alpha$ assembles into the translation, while the piece proportional to ξ_α assembles into the Dirac operator:

$$[\delta_1,\delta_2]\,\psi_\alpha = a^\mu\,\partial_\mu\psi_\alpha \;+\; 2i\Big[\,\xi_{2\alpha}\,(\bar\xi_1\bar\sigma^\mu\partial_\mu\psi) - \xi_{1\alpha}\,(\bar\xi_2\bar\sigma^\mu\partial_\mu\psi)\Big]. \tag{44.132}$$

The first term is the translation Eq. (44.125), identical to that on ϕ. The second term is the defect: it is proportional to $\bar\sigma^\mu\partial_\mu\psi$, which is *nothing but the free Weyl equation of motion* obtained by varying Eq. (44.116), $i\,\bar\sigma^\mu\partial_\mu\psi = 0$. On-shell the defect vanishes and the algebra closes; off-shell it does not.

Theorem 44.24 (On-shell-only closure of the naive algebra)**.** *Under the transformations Eq. (44.124), the commutator of two supersymmetry variations reproduces the translation Eq. (44.125) on the scalar identically (Eq. (44.127)) and on the fermion only up to a term proportional to the fermion equation of motion (Eq. (44.132)). The supersymmetry algebra therefore closes on the field ψ only when ψ obeys its Dirac equation.*

This is not a defect of our particular ansatz; it is structural, and counting reveals why. A complex scalar has two real off-shell components; the equation of motion is second order and removes none of them at the level of field content (it propagates two real on-shell degrees of freedom). A Weyl fermion ψ_α has four real off-shell components — two complex components — but its equation of motion is first order, halving the count to two real on-shell degrees of freedom. Thus on-shell the multiplet balances, $2_B = 2_F$, exactly as the representation theory of Section 44.4 demanded; but off-shell it does not, $2_B \neq 4_F$. A symmetry that rotates bosons into fermions cannot close on a mismatched set of fields, because the commutator of two such rotations must map the field content back into itself without invoking the dynamics. The missing two bosonic components are exactly what we must supply.

44.5.3 The auxiliary field and off-shell closure

We restore the balance by adjoining a second complex scalar F, of canonical dimension $[F] = 2$ — one unit higher than ϕ — whose kinetic term we deliberately omit. The free Lagrangian becomes

$$\mathcal{L}_{\text{free}} = \partial_\mu \phi^* \partial^\mu \phi + i\,\bar\psi \bar\sigma^\mu \partial_\mu \psi + F^* F. \tag{44.133}$$

The term F^*F carries no derivatives, so F is *auxiliary*: its Euler–Lagrange equation is the purely algebraic $F = 0$ in the free theory, and it propagates no states. Off-shell, however, it contributes two real bosonic components, restoring $4_B = 4_F$ and giving us exactly the slack needed to close the algebra. The transformations must be augmented so that F is part of the multiplet. Dimensional analysis again narrows the choice: δF has the dimension of F, namely 2, and since the parameter $\bar\xi$ carries dimension $-\tfrac{1}{2}$ the remaining factor must have dimension $\tfrac{5}{2}$. The only Lorentz-scalar of that dimension built from $\bar\xi$ and ψ is $\bar\xi \bar\sigma^\mu \partial_\mu \psi$, in which the single derivative supplies the extra unit. We take the off-shell transformations

$$\boxed{\delta\phi = \sqrt{2}\,\xi\psi, \qquad \delta\psi_\alpha = \sqrt{2}\,\xi_\alpha F - \sqrt{2}\,i\,(\sigma^\mu \bar\xi)_\alpha \partial_\mu \phi, \qquad \delta F = -\sqrt{2}\,i\,\bar\xi \bar\sigma^\mu \partial_\mu \psi,} \tag{44.134}$$

the new pieces being the inhomogeneous $\sqrt{2}\,\xi_\alpha F$ added to $\delta\psi$ and the transformation of F itself, with coefficients we now justify by demanding (i) invariance of Eq. (44.133) and (ii) off-shell closure.

Invariance. The pieces of Eq. (44.134) not involving F already leave Eq. (44.116) invariant by Theorem 44.23. The new contributions are: from the ξF piece of $\delta\psi$ and the $\bar\xi F^*$ piece of $\delta\bar\psi$ acting on the fermion kinetic term, and from $\delta(F^*F)$. Collecting them,

$$i\,\bar\psi\bar\sigma^\mu \partial_\mu(\sqrt{2}\,\xi F) + i\,(\sqrt{2}\,\bar\xi F^*)\bar\sigma^\mu\partial_\mu\psi = i\sqrt{2}\,(\bar\psi\bar\sigma^\mu\xi)\,\partial_\mu F + i\sqrt{2}\,F^*(\bar\xi\bar\sigma^\mu\partial_\mu\psi), \tag{44.135}$$

$$\delta(F^*F) = (\delta F^*)F + F^*\delta F = i\sqrt{2}\,(\partial_\mu\bar\psi\,\bar\sigma^\mu\xi)\,F - i\sqrt{2}\,F^*(\bar\xi\bar\sigma^\mu\partial_\mu\psi), \tag{44.136}$$

where $\delta F^* = (\delta F)^\dagger = +i\sqrt{2}\,\partial_\mu\bar\psi\,\bar\sigma^\mu\xi$ follows from $(\bar\xi\bar\sigma^\mu\chi)^\dagger = \bar\chi\bar\sigma^\mu\xi$. The two terms carrying F^* cancel between Eqs. (44.135) and (44.136), and the two terms carrying F combine into the total derivative $i\sqrt{2}\,\partial_\mu\big[(\bar\psi\bar\sigma^\mu\xi)\,F\big]$. Hence Eq. (44.133) is supersymmetric, and the coefficient of δF is fixed (up to the sign we are about to need for closure).

Closure. Recompute the commutator on ψ, now with the F-term present. The derivative piece of $\delta\psi$ reproduces Eq. (44.132) exactly — translation plus the equation-of-motion defect — and the new ξF piece contributes

$$\delta_1\delta_2\,\psi_\alpha\Big|_F = \sqrt{2}\,\xi_{2\alpha}\,\delta_1 F = \sqrt{2}\,\xi_{2\alpha}(-\sqrt{2}\,i\,\bar\xi_1\bar\sigma^\mu\partial_\mu\psi) = -2i\,\xi_{2\alpha}\,(\bar\xi_1\bar\sigma^\mu\partial_\mu\psi). \tag{44.137}$$

Antisymmetrising in $1 \leftrightarrow 2$ gives $-2i\big[\xi_{2\alpha}(\bar\xi_1\bar\sigma^\mu\partial_\mu\psi) - \xi_{1\alpha}(\bar\xi_2\bar\sigma^\mu\partial_\mu\psi)\big]$, which is precisely the negative of the defect term in Eq. (44.132). The two cancel, and

$$[\delta_1,\delta_2]\,\psi_\alpha = a^\mu\,\partial_\mu\psi_\alpha \qquad \text{(off-shell, no equation of motion used).} \tag{44.138}$$

The sign of δF in Eq. (44.134) is exactly the one that makes this cancellation occur; this is what fixed it. It remains to check that the algebra also closes on F. Using Eq. (44.134),

$$\begin{aligned}\delta_1\delta_2\,F &= -\sqrt{2}\,i\,\bar\xi_2\bar\sigma^\mu\partial_\mu(\delta_1\psi) = -\sqrt{2}\,i\,\bar\xi_2\bar\sigma^\mu\partial_\mu(\sqrt{2}\,\xi_1 F - \sqrt{2}\,i\,\sigma^\nu\bar\xi_1\partial_\nu\phi)\\ &= -2i\,(\bar\xi_2\bar\sigma^\mu\xi_1)\,\partial_\mu F - 2\,(\bar\xi_2\bar\sigma^\mu\sigma^\nu\bar\xi_1)\,\partial_\mu\partial_\nu\phi.\end{aligned} \tag{44.139}$$

In the second term the symmetric-in-$\mu\nu$ derivatives project onto $\bar\sigma^\mu\sigma^\nu \to \eta^{\mu\nu}$ by Eq. (44.121), giving $-2(\bar\xi_2\bar\xi_1)\Box\phi$; under antisymmetrisation $\bar\xi_2\bar\xi_1 - \bar\xi_1\bar\xi_2 = 0$ and it drops out entirely. In the first term, using $\bar\xi\bar\sigma^\mu\xi' = -\xi'\sigma^\mu\bar\xi$ (the identity Eq. (44.131)),

$$[\delta_1,\delta_2]\,F = -2i(\bar\xi_2\bar\sigma^\mu\xi_1 - \bar\xi_1\bar\sigma^\mu\xi_2)\partial_\mu F = 2i(\xi_1\sigma^\mu\bar\xi_2 - \xi_2\sigma^\mu\bar\xi_1)\partial_\mu F = a^\mu\,\partial_\mu F. \tag{44.140}$$

The same translation a^μ closes on all three fields, with no recourse to any equation of motion.

Definition 44.25 (Off-shell chiral multiplet and its degree-of-freedom count)**.** The *off-shell chiral multiplet* is the triple (ϕ,ψ_α,F) of a complex scalar ϕ, a left-handed Weyl fermion ψ_α, and a complex auxiliary scalar F, transforming under Eq. (44.134). Its real component counts are

$$\underbrace{\phi:2 \;+\; F:2}_{n_B=4\ \text{(off-shell)}} = \underbrace{\psi:4}_{n_F=4\ \text{(off-shell)}}, \qquad \underbrace{\phi:2}_{n_B=2\ \text{(on-shell)}} = \underbrace{\psi:2}_{n_F=2\ \text{(on-shell)}}, \tag{44.141}$$

the auxiliary field carrying the two bosonic components needed to match the off-shell fermion and disappearing on-shell, where $F = 0$.

What the auxiliary field buys

The price of F is one non-dynamical complex field; the payoff is that supersymmetry becomes an *off-shell* symmetry — a symmetry of the action realised on the fields irrespective of the dynamics. This matters far beyond bookkeeping. An off-shell symmetry can be used in the path integral before the equations of motion are imposed, it survives quantum corrections in a controlled way, and it allows the algebra to be represented *linearly* on a fixed set of fields. The geometric repackaging of Section 44.6 will make the auxiliary field appear automatically, as the highest component of a superfield; here we have discovered it the hard way, by demanding closure.

44.5.4 The superpotential, the Yukawa coupling, and the scalar potential

With an off-shell-closing free theory in hand we can add interactions. The remarkable fact is that renormalisable, supersymmetric self-interactions of a single chiral multiplet are governed entirely by one holomorphic function of ϕ alone, the *superpotential* $W(\phi)$. We add to Eq. (44.133) the interaction Lagrangian

$$\mathcal{L}_W = F\,W'(\phi) - \tfrac{1}{2}\,W''(\phi)\,\psi\psi + \text{h.c.}, \qquad W' \equiv \frac{\mathrm{d}W}{\mathrm{d}\phi}, \quad W'' \equiv \frac{\mathrm{d}^2W}{\mathrm{d}\phi^2}, \tag{44.142}$$

where "h.c." adds the Hermitian conjugate $F^*\,\overline{W'} - \tfrac{1}{2}\,\overline{W''}\,\bar\psi\bar\psi$. The structure is fixed by dimensions and covariance: FW' is the unique dimension-4 scalar linear in F, and $W''\psi\psi$ the unique dimension-4 scalar bilinear in ψ, provided W is a function of ϕ but not of ϕ^*. That last requirement — holomorphy of the superpotential — is not an assumption but a consequence of supersymmetry, as the invariance proof reveals.

Proposition 44.26 (SUSY invariance forces W holomorphic)**.** *The action $\int \mathrm{d}^4x\,(\mathcal{L}_{\text{free}} + \mathcal{L}_W)$ is invariant under Eq. (44.134) if and only if W depends on ϕ alone. When it does, $\mathcal{L}_W$ changes by a total derivative.*

Proof. Vary Eq. (44.142), holding the free-action invariance already established. Using $\delta(W') = W''\,\delta\phi = \sqrt{2}\,W''(\xi\psi)$, $\delta(W'') = \sqrt{2}\,W'''(\xi\psi)$, and $\delta(\psi\psi) = 2\,\psi^\alpha\delta\psi_\alpha = 2\sqrt{2}\,\psi^\alpha(\xi_\alpha F - i(\sigma^\mu\bar\xi)_\alpha\partial_\mu\phi)$,

$$\delta\mathcal{L}_W = (\delta F)W' + F\,\delta W' - \tfrac{1}{2}(\delta W'')\psi\psi - \tfrac{1}{2}W''\,\delta(\psi\psi). \tag{44.143}$$

Group the terms. *(i) The F-dependent terms.* These are $F\,\delta W' - \tfrac{1}{2}W''\,\delta(\psi\psi)\big|_F = \sqrt{2}\,FW''(\xi\psi) - \sqrt{2}\,W''\,F\,(\psi^\alpha\xi_\alpha)$. Since $\psi^\alpha\xi_\alpha = \psi\xi = \xi\psi$, these cancel exactly.

(ii) The W''' term. This is $-\frac{\sqrt{2}}{2}\,W'''\,(\xi\psi)(\psi\psi)$, and it vanishes identically: the product contains three factors of the two-component anticommuting spinor ψ, and any product of three such (with only two independent components) is zero. *(iii) The derivative terms.* What remains is

$$(\delta F)W' - \tfrac{1}{2}W''\,\delta(\psi\psi)\big|_{\partial\phi} = -\sqrt{2}\,i\,W'\,(\bar{\xi}\bar{\sigma}^{\mu}\partial_{\mu}\psi) + \sqrt{2}\,i\,W''\,(\psi\sigma^{\mu}\bar{\xi})\,\partial_{\mu}\phi. \tag{44.144}$$

Use $\psi\sigma^{\mu}\bar{\xi} = -\bar{\xi}\bar{\sigma}^{\mu}\psi$ from Eq. (44.131) and the chain rule $W''\,\partial_{\mu}\phi = \partial_{\mu}(W'(\phi))$ — valid precisely because W' is a function of ϕ only, so that $\partial_{\mu}W'$ has no $\partial_{\mu}\phi^{*}$ piece. Then

$$\delta\mathcal{L}_{W}\big|_{\partial} = -\sqrt{2}\,i\Big[W'\,(\bar{\xi}\bar{\sigma}^{\mu}\partial_{\mu}\psi) + (\partial_{\mu}W')\,(\bar{\xi}\bar{\sigma}^{\mu}\psi)\Big] = -\sqrt{2}\,i\,\partial_{\mu}\big[\,W'\,(\bar{\xi}\bar{\sigma}^{\mu}\psi)\,\big], \tag{44.145}$$

a total derivative. Had W contained any ϕ^{*}, the chain rule in Eq. (44.145) would have produced an extra term $(\partial W'/\partial\phi^{*})(\partial_{\mu}\phi^{*})(\bar{\xi}\bar{\sigma}^{\mu}\psi)$ with no partner to cancel it, spoiling invariance. The h.c. part is the conjugate of this computation and is separately a total derivative. Hence the interacting action is supersymmetric exactly when $W = W(\phi)$. □

Integrating out the auxiliary field. The field F appears in $\mathcal{L}_{\text{free}} + \mathcal{L}_{W}$ only algebraically:

$$\mathcal{L} \supset F^{*}F + F\,W'(\phi) + F^{*}\,\overline{W'(\phi)}. \tag{44.146}$$

Its equation of motion is purely algebraic. Varying with respect to F^{*},

$$\frac{\partial\mathcal{L}}{\partial F^{*}} = F + \overline{W'(\phi)} = 0 \qquad\Longrightarrow\qquad F = -\,\overline{W'(\phi)}, \quad F^{*} = -\,W'(\phi). \tag{44.147}$$

Substituting back into Eq. (44.146), $F^{*}F \to |W'|^{2}$, $FW' \to -|W'|^{2}$, $F^{*}\overline{W'} \to -|W'|^{2}$, so the three terms collapse to $-|W'(\phi)|^{2}$. Since the Lagrangian contains the potential as $\mathcal{L} \supset -V$, we read off the central result.

Theorem 44.27 (Scalar potential of the Wess–Zumino model)**.** *After eliminating the auxiliary field through its equation of motion Eq. (44.147), the Wess–Zumino model is the theory of (ϕ, ψ) with Lagrangian*

$$\mathcal{L} = \partial_{\mu}\phi^{*}\partial^{\mu}\phi + i\,\bar{\psi}\bar{\sigma}^{\mu}\partial_{\mu}\psi - \Big(\tfrac{1}{2}\,W''(\phi)\,\psi\psi + h.c.\Big) - V(\phi,\phi^{*}), \tag{44.148}$$

with scalar potential

$$\boxed{V(\phi,\phi^{*}) = \big|W'(\phi)\big|^{2} = \left|\frac{\partial W}{\partial\phi}\right|^{2}.} \tag{44.149}$$

The fermion mass and self-interactions are carried by the Yukawa term $-\frac{1}{2}W''(\phi)\psi\psi+$ h.c.

Three features of Eq. (44.149) deserve emphasis. First, $V \geq 0$ automatically, manifesting at the Lagrangian level the positivity of energy proved abstractly from the algebra in Theorem 44.9: the scalar potential of any $\mathcal{N} = 1$ theory is a sum of squares. Second, supersymmetry is unbroken in a vacuum ϕ_0 if and only if $V(\phi_0) = 0$, i.e. $W'(\phi_0) = 0$ — the algebraic condition $F = 0$ of Eq. (44.147) is precisely the statement that the auxiliary field's expectation value, the order parameter $\langle F \rangle$, vanishes. This is the same diagnostic as Theorem 44.10, now concrete: SUSY breaks when W' has no zero. Third, the Yukawa coupling W'' and the scalar potential $|W'|^2$ are not independent — both descend from the single function W — which is why boson and fermion masses will turn out locked together.

44.5.5 The cubic Wess–Zumino model

The simplest interacting choice is a cubic superpotential, the unique renormalisable option: a quartic or higher W would give a scalar potential $|W'|^2$ of degree six or more, non-renormalisable in four dimensions. We carry the cubic model in full.

Worked Example 44.28 (The cubic superpotential and boson–fermion mass degeneracy)**.** Take

$$W(\phi) = \frac{m}{2}\,\phi^2 + \frac{g}{3}\,\phi^3, \qquad W'(\phi) = m\phi + g\phi^2, \qquad W''(\phi) = m + 2g\phi, \tag{44.150}$$

with m a mass and g a dimensionless coupling. We extract the component interactions and verify that the boson and fermion masses coincide.

Scalar potential. From Eq. (44.149),

$$\begin{aligned} V = \left|m\phi + g\phi^2\right|^2 &= (m\phi + g\phi^2)(m\phi^* + g\phi^{*2}) \\ &= m^2\,|\phi|^2 \;+\; mg\,|\phi|^2(\phi + \phi^*) \;+\; g^2\,|\phi|^4. \end{aligned} \tag{44.151}$$

The quadratic piece $m^2|\phi|^2$ is the mass term of a complex scalar; reading off the coefficient against the canonical $m^2_{\text{scalar}}|\phi|^2$, the scalar mass is $m_{\text{scalar}} = m$. The cubic term $mg\,|\phi|^2(\phi + \phi^*)$ is a scalar self-interaction of strength mg, and the quartic term is $g^2|\phi|^4$.

Yukawa and fermion mass. From the Yukawa term in Eq. (44.148),

$$-\tfrac{1}{2}\,W''(\phi)\,\psi\psi + \text{h.c.} = -\frac{m}{2}\,\psi\psi - g\,\phi\,\psi\psi - \frac{m}{2}\,\bar\psi\bar\psi - g\,\phi^*\,\bar\psi\bar\psi. \tag{44.152}$$

The ϕ-independent piece $-\frac{m}{2}(\psi\psi + \bar\psi\bar\psi)$ is a Majorana mass term for the Weyl fermion (Sections 11.7.2 and 11.7.6) of magnitude m: the fermion mass is $m_{\text{fermion}} = m$. The remaining $-g\,\phi\,\psi\psi + \text{h.c.}$ is a genuine Yukawa coupling of strength g between the scalar and the fermion bilinear.

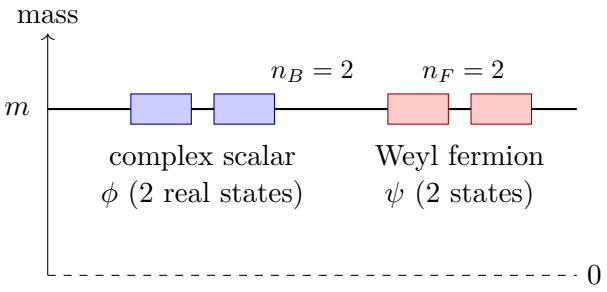


Figure 44.4: On-shell spectrum of the cubic Wess–Zumino model. The complex scalar ϕ and the Weyl fermion ψ are degenerate at the single mass m, with equal state counts $n_B = n_F = 2$ — the field-theoretic face of the multiplet degeneracy Theorem 44.12 and the equal-count theorem Theorem 44.15.

Mass degeneracy. Comparing,

$$m_{\text{scalar}} = m = m_{\text{fermion}}, \tag{44.153}$$

the complex scalar (both real components) and the Weyl fermion share the single mass m. This is the field-theoretic realisation of the multiplet degeneracy Theorem 44.12: every state in a multiplet of unbroken supersymmetry carries the same mass, here visible directly in the Lagrangian as the common coefficient m tying W' and W'' together. The degeneracy is no accident of the cubic choice; it follows from $V = |W'|^2$ and Yukawa $= W''$ sharing the single function W, so that $\partial^2 V/\partial\phi\partial\phi^*\big|_{\phi=0} = |W''(0)|^2 = m^2$ is exactly the square of the fermion mass $W''(0) = m$.

The mass spectrum is summarised in Fig. 44.4: a single level m shared by two real bosonic states and the two fermionic states.

The Wess–Zumino model as a supersymmetric completion of $\lambda\phi^4$

The quartic term $g^2|\phi|^4$ in Eq. (44.151) is a self-interaction of exactly the type carried since Chapter 6 as ϕ^4 theory. Writing the complex scalar in real components, $\phi = (\phi_1 + i\phi_2)/\sqrt{2}$, gives $g^2|\phi|^4 = \frac{g^2}{4}(\phi_1^2 + \phi_2^2)^2$, a quartic potential for the two real scalars with coupling set by g^2. The Wess–Zumino model is thus a supersymmetric completion of the scalar quartic theory: it contains the $\lambda\phi^4$-type interaction, but with the quartic coupling no longer free. Supersymmetry rigidly ties the quartic strength g^2 to the Yukawa coupling g of Eq. (44.152) and to the cubic scalar coupling mg, all descending from the single superpotential coefficient g. This rigidity is the source of the celebrated ultraviolet softness of supersymmetric theories: the same coupling controlling the boson self-interaction also controls the

fermion loop that cancels much of its divergence, a structural cancellation we return to in Chapters 35 and 46. The free quartic coupling λ of ϕ^4 theory has become, in its supersymmetric form, the square of a coupling shared with the fermion sector.

We have now realised the chiral multiplet as a fully interacting field theory, discovered the auxiliary field F as the price of off-shell closure, and seen all interactions flow from the single holomorphic superpotential, with $V = |W'|^2$ and Yukawa $= W''$ locking boson and fermion masses together. What is conspicuous is how much bookkeeping this cost: a separate transformation rule for each component, an auxiliary field introduced by hand, and Fierz gymnastics to verify closure. Every one of these features is a symptom of representing a *geometric* structure in non-geometric language. The next section removes the redundancy at a stroke by enlarging spacetime itself: in the superspace of Section 44.6, the fields (ϕ, ψ, F) become the Taylor coefficients of a single *superfield* in anticommuting coordinates $\theta, \bar{\theta}$, the transformations Eq. (44.134) become a single translation in the fermionic directions, and the auxiliary field appears automatically as the top component — closure guaranteed by construction.

44.6 Superspace and Superfields

The Wess–Zumino model of Section 44.5 taught us two lessons at once. The first is that supersymmetry organizes a complex scalar ϕ, a Weyl fermion ψ_α, and a complex auxiliary field F into a single irreducible packet that closes under the transformations only when F is kept off shell. The second is that, written out in components, those transformations are a tangle of σ-matrices and derivatives whose invariance we verified by a page of index gymnastics, term by term. Both lessons point to the same need. If a fermionic charge Q_α really shifts bosons into fermions and back into the *derivative* of a boson, then it is behaving like a translation — but in a direction that ordinary spacetime does not contain. The honest way to make a translation manifest is to give it a coordinate. We therefore enlarge spacetime by adjoining anticommuting coordinates, one for each component of the supercharge, and demand that supersymmetry act on the enlarged space as nothing more exotic than a shift. Every component transformation of Section 44.5 will then follow from a single line — the Taylor expansion of a field in the new directions — and invariance of an action will reduce to the statement that an integral over the whole space is unchanged by a shift of its variable of integration. This is the geometry of superspace, and it is the subject of this section.

The anticommuting coordinates are Grassmann numbers in the exact sense built from scratch in Theorem 12.3, and the integral over them is the Berezin integral of Theorem 12.17. We use those constructions as established machinery;

what is new here is the spacetime-spinor structure they acquire when we attach them to the $(\tfrac{1}{2},0)\oplus(0,\tfrac{1}{2})$ Lorentz data of Section 3.4.3. Throughout we work with the two-component conventions fixed in Section 44.2.2: undotted indices $\alpha,\beta=1,2$ for the $(\tfrac{1}{2},0)$ representation, dotted indices $\dot\alpha,\dot\beta=1,2$ for $(0,\tfrac{1}{2})$, the antisymmetric symbols $\varepsilon^{\alpha\beta},\varepsilon_{\alpha\beta}$ (with $\varepsilon^{12}=-\varepsilon_{12}=1$) raising and lowering them, the suppressed-index contractions

$$\theta\psi \;\equiv\; \theta^{\alpha}\psi_{\alpha} \;=\; -\,\theta_{\alpha}\psi^{\alpha}, \qquad \bar\theta\bar\psi \;\equiv\; \bar\theta_{\dot\alpha}\bar\psi^{\dot\alpha} \;=\; -\,\bar\theta^{\dot\alpha}\bar\psi_{\dot\alpha}, \tag{44.154}$$

and the Pauli-matrix data $\sigma^{\mu}_{\alpha\dot\alpha}=(\mathbf{1},\sigma^{i})$ of Eq. (3.79) together with $\bar\sigma^{\mu\,\dot\alpha\alpha}=(\mathbf{1},-\sigma^{i})=\varepsilon^{\dot\alpha\dot\beta}\varepsilon^{\alpha\beta}\sigma^{\mu}_{\beta\dot\beta}$ of Eq. (3.106). The metric is mostly-minus and units are natural, as everywhere in this chapter; the momentum operator is realized on fields as $P_{\mu}=i\partial_{\mu}$, the convention of Section 3.5.

44.6.1 Superspace as a coset

Spacetime itself is a coset. The Poincaré group acts transitively on $\mathbb{R}^{1,3}$, the stabilizer of the origin is the Lorentz group, and the quotient Poincaré/Lorentz is precisely Minkowski space; a point x^{μ} is the name of the coset represented by the pure translation $g(x)=e^{-ix^{\mu}P_{\mu}}$. Translations then act on points by group multiplication, $g(a)g(x)=g(x+a)$, which is the abstract content of "a translation shifts the coordinate." Supersymmetry extends the Poincaré algebra by the odd generators $Q_{\alpha},\bar Q_{\dot\alpha}$ satisfying (Eq. (44.49))

$$\left\{Q_{\alpha},\bar Q_{\dot\beta}\right\} \;=\; 2\,\sigma^{\mu}_{\alpha\dot\beta}\,P_{\mu}, \qquad \{Q_{\alpha},Q_{\beta}\}=\left\{\bar Q_{\dot\alpha},\bar Q_{\dot\beta}\right\}=0, \qquad [Q_{\alpha},P_{\mu}]=0. \tag{44.155}$$

The natural thing to quotient by is again the Lorentz group, leaving a coset spanned by the even translations *and* the odd charges. We coordinatize it by attaching a Grassmann parameter to each odd generator: a point of *superspace* is the coset represented by

$$g(x,\theta,\bar\theta) \;=\; \exp\!\big(\,-i\,x^{\mu}P_{\mu}\;+\;i\,\theta^{\alpha}Q_{\alpha}\;+\;i\,\bar\theta_{\dot\alpha}\bar Q^{\dot\alpha}\,\big), \tag{44.156}$$

where θ^{α} and $\bar\theta_{\dot\alpha}$ are constant anticommuting numbers carrying, respectively, an undotted and a dotted spinor index. They anticommute among themselves and with $Q,\bar Q$, so that products such as $\theta^{\alpha}Q_{\alpha}$ are Grassmann-*even* and the exponential is well defined. The triple $(x^{\mu},\theta^{\alpha},\bar\theta_{\dot\alpha})$ is a point of $\mathcal{N}=1$ superspace, a space of four commuting (bosonic) and four anticommuting (fermionic) coordinates.

Definition 44.29 ($\mathcal{N}=1$ superspace)**.** $\mathcal{N}=1$ superspace is the set of points $z^{M}=(x^{\mu},\theta^{\alpha},\bar\theta_{\dot\alpha})$, with x^{μ} ordinary real spacetime coordinates and $\theta^{\alpha},\bar\theta_{\dot\alpha}$ constant generators of a Grassmann algebra (Theorem 12.3) transforming as the $(\tfrac{1}{2},0)$ and $(0,\tfrac{1}{2})$ representations of the Lorentz group. Complex conjugation acts by $(\theta^{\alpha})^{\dagger}=\bar\theta^{\dot\alpha}$, so that $\theta\psi$ and $\bar\theta\bar\psi$ are conjugate scalars.

The whole point of the coset construction is that supersymmetry now acts by left multiplication. Multiply the representative Eq. (44.156) on the left by a finite supersymmetry transformation $g(\xi,\bar\xi)=\exp(i\xi Q+i\bar\xi\bar Q)$ with constant Grassmann parameters $\xi^\alpha,\bar\xi_{\dot\alpha}$. Because the only nontrivial bracket among the exponents is the anticommutator Eq. (44.155), the Baker–Campbell–Hausdorff series truncates after one term: the commutator of the two exponents is proportional to P_μ, which is central with respect to Q, so all higher nested brackets vanish. Explicitly,

$$\begin{aligned} g(\xi,\bar\xi)\,g(x,\theta,\bar\theta) &= \exp\!\Big(i\xi Q+i\bar\xi\bar Q\Big)\,\exp\!\Big(-ixP+i\theta Q+i\bar\theta\bar Q\Big) \\ &= \exp\!\Big(-ixP+i(\theta+\xi)Q+i(\bar\theta+\bar\xi)\bar Q\;-\;\tfrac12\big[\,i\xi Q+i\bar\xi\bar Q\,,\,i\theta Q+i\bar\theta\bar Q\,\big]\Big), \end{aligned} \tag{44.157}$$

and the leftover bracket, evaluated with Eq. (44.155), is

$$-\tfrac12\big[\,i\xi Q+i\bar\xi\bar Q\,,\,i\theta Q+i\bar\theta\bar Q\,\big] \;=\; \tfrac12\Big(\theta^\alpha\bar\xi_{\dot\beta}-\xi^\alpha\bar\theta_{\dot\beta}\Big)\Big\{Q_\alpha,\bar Q^{\dot\beta}\Big\} \;=\; (\theta\sigma^\mu\bar\xi-\xi\sigma^\mu\bar\theta)\,P_\mu. \tag{44.158}$$

Comparing the two exponentials and reading off the coefficient of each generator, the product is the representative of a new point with the same Lorentz frame but shifted coordinates,

$$\boxed{\;\theta^\alpha\;\to\;\theta^\alpha+\xi^\alpha,\qquad \bar\theta_{\dot\alpha}\;\to\;\bar\theta_{\dot\alpha}+\bar\xi_{\dot\alpha},\qquad x^\mu\;\to\;x^\mu\;+\;i\,\theta\sigma^\mu\bar\xi\;-\;i\,\xi\sigma^\mu\bar\theta\,.\;} \tag{44.159}$$

This is the geometric heart of the subject. *A supersymmetry transformation is a constant shift of the fermionic coordinates, accompanied by a definite x-translation whose amount is bilinear in ξ and θ.* The translation piece is exactly what one expects from $\{Q,\bar Q\}\sim P$: two infinitesimal odd shifts fail to commute, and their failure is a spacetime translation. The Wess–Zumino transformation rules, which looked like an intricate coincidence in components, are simply the statement that a field carried along this shift changes by the corresponding Taylor expansion.

Remark 44.30 (Reality of the shift). The bosonic shift in Eq. (44.159) is real. Taking the Hermitian conjugate and using $(\theta\sigma^\mu\bar\xi)^\dagger=\xi\sigma^\mu\bar\theta$ (a standard reordering identity for two-component spinors, proved in Section 44.2.2) shows that $i\,\theta\sigma^\mu\bar\xi-i\,\xi\sigma^\mu\bar\theta$ equals its own conjugate. Superspace is therefore a real manifold in its bosonic directions, as it must be for x^μ to remain an honest spacetime coordinate.

44.6.2 The supercharges as differential operators

A field theory lives not on the group but on functions over superspace. The action of supersymmetry on such functions is generated by differential operators, obtained by demanding that an infinitesimal version of Eq. (44.159) be

reproduced. Define the fermionic partial derivatives by the left-derivative rule of Theorem 12.17,

$$\partial_\alpha \equiv \frac{\partial}{\partial\theta^\alpha}, \qquad \bar\partial_{\dot\alpha} \equiv \frac{\partial}{\partial\bar\theta^{\dot\alpha}}, \qquad \left\{\partial_\alpha, \theta^\beta\right\} = \delta_\alpha^{\ \beta}, \quad \left\{\bar\partial_{\dot\alpha}, \bar\theta^{\dot\beta}\right\} = \delta_{\dot\alpha}^{\ \dot\beta}, \tag{44.160}$$

with all other anticommutators among $\partial_\alpha, \bar\partial_{\dot\alpha}, \theta, \bar\theta$ vanishing. The infinitesimal form of the shift Eq. (44.159) acts on a function $F(x,\theta,\bar\theta)$ as $\delta_\xi F = (\xi^\alpha Q_\alpha + \bar\xi_{\dot\alpha}\bar Q^{\dot\alpha})F$ with the representation

$$\boxed{Q_\alpha \;=\; \partial_\alpha \;-\; i\,\sigma^\mu_{\alpha\dot\alpha}\,\bar\theta^{\dot\alpha}\,\partial_\mu, \qquad \bar Q_{\dot\alpha} \;=\; -\,\bar\partial_{\dot\alpha} \;+\; i\,\theta^\alpha\sigma^\mu_{\alpha\dot\alpha}\,\partial_\mu\,.} \tag{44.161}$$

The first term in each operator generates the pure θ- (respectively $\bar\theta$-) shift; the second generates the compensating x-translation read off from Eq. (44.159), with ∂_μ the ordinary spacetime derivative. One checks directly that (ξQ) shifts $\theta^\alpha \to \theta^\alpha + \xi^\alpha$ and $x^\mu \to x^\mu - i\xi\sigma^\mu\bar\theta$, while $(\bar\xi\bar Q)$ shifts $\bar\theta \to \bar\theta + \bar\xi$ and $x^\mu \to x^\mu + i\theta\sigma^\mu\bar\xi$; summing the two and using the reordering of Theorem 44.30 reproduces Eq. (44.159) up to the overall orientation fixed by our sign of δ_ξ.

We must now prove that these differential operators close into the algebra Eq. (44.155). This is the central consistency check of the section: if the operator forms did *not* satisfy the algebra, the geometric picture would be a fiction.

Proposition 44.31 (The differential operators realize the SUSY algebra)**.** *With $P_\mu = i\partial_\mu$, the operators Eq. (44.161) satisfy*

$$\left\{Q_\alpha, \bar Q_{\dot\beta}\right\} \;=\; 2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu \;=\; 2i\,\sigma^\mu_{\alpha\dot\beta}\,\partial_\mu, \qquad \{Q_\alpha, Q_\beta\} = \left\{\bar Q_{\dot\alpha}, \bar Q_{\dot\beta}\right\} = 0, \qquad [Q_\alpha, \partial_\mu] = 0, \tag{44.162}$$

which is the $\mathcal{N} = 1$ super-Poincaré algebra of Eq. (44.155).

Proof. Every bracket is computed by pushing the fermionic derivatives through the Grassmann coordinates with the rules Eq. (44.160); the spacetime derivative ∂_μ is Grassmann-even and commutes with everything, so it rides along as a passive factor.

The mixed anticommutator. Write the two operators with distinct dummy indices,

$$Q_\alpha = \partial_\alpha - i\,\sigma^\mu_{\alpha\dot\gamma}\,\bar\theta^{\dot\gamma}\,\partial_\mu, \qquad \bar Q_{\dot\beta} = -\,\bar\partial_{\dot\beta} + i\,\theta^\gamma\sigma^\nu_{\gamma\dot\beta}\,\partial_\nu,$$

and expand $\left\{Q_\alpha, \bar Q_{\dot\beta}\right\}$ into four anticommutators. Two of them vanish identically. The derivative–derivative term $\left\{\partial_\alpha, -\bar\partial_{\dot\beta}\right\} = 0$ because the two fermionic derivatives act on independent variables. The translation–translation term also vanishes: it is $\left\{-i\sigma^\mu_{\alpha\dot\gamma}\bar\theta^{\dot\gamma}\partial_\mu,\, i\theta^\gamma\sigma^\nu_{\gamma\dot\beta}\partial_\nu\right\} = \sigma^\mu_{\alpha\dot\gamma}\sigma^\nu_{\gamma\dot\beta}\,\partial_\mu\partial_\nu\,\left\{\bar\theta^{\dot\gamma}, \theta^\gamma\right\} = 0$, since $\bar\theta$ and θ anticommute to zero and the bosonic $\partial_\mu\partial_\nu$ factors out. The two surviving cross terms are

$$\left\{\partial_\alpha,\, i\,\theta^\gamma\sigma^\nu_{\gamma\dot\beta}\partial_\nu\right\} = i\,\sigma^\nu_{\gamma\dot\beta}\,\partial_\nu\;\{\partial_\alpha, \theta^\gamma\} = i\,\sigma^\nu_{\gamma\dot\beta}\,\partial_\nu\,\delta_\alpha^{\ \gamma} = i\,\sigma^\nu_{\alpha\dot\beta}\,\partial_\nu, \tag{44.163}$$

$$\left\{-i\,\sigma^{\mu}_{\alpha\dot\gamma}\bar\theta^{\dot\gamma}\partial_\mu,\,-\bar\partial_{\dot\beta}\right\} = i\,\sigma^{\mu}_{\alpha\dot\gamma}\,\partial_\mu\,\left\{\bar\theta^{\dot\gamma},\bar\partial_{\dot\beta}\right\} = i\,\sigma^{\mu}_{\alpha\dot\gamma}\,\partial_\mu\,\delta^{\dot\gamma}_{\dot\beta} = i\,\sigma^{\mu}_{\alpha\dot\beta}\,\partial_\mu. \tag{44.164}$$

Adding Eqs. (44.163) and (44.164),

$$\left\{Q_\alpha,\bar Q_{\dot\beta}\right\} \;=\; 2i\,\sigma^{\mu}_{\alpha\dot\beta}\,\partial_\mu \;=\; 2\,\sigma^{\mu}_{\alpha\dot\beta}\,P_\mu, \tag{44.165}$$

exactly the right-hand side of Eq. (44.155). The coefficient 2 is not inserted by hand; it is the sum of two equal contributions, one from shifting θ against the $\bar\theta$-translation and one from shifting $\bar\theta$ against the θ-translation. This is the operator incarnation of the geometric fact that closing two fermionic shifts produces one bosonic translation.

The like anticommutators. For $\{Q_\alpha,Q_\beta\}$, expand again. The $\{\partial_\alpha,\partial_\beta\}=0$ by Eq. (44.160). The two cross terms $\left\{\partial_\alpha,-i\sigma^{\mu}_{\beta\dot\gamma}\bar\theta^{\dot\gamma}\partial_\mu\right\}$ and its $\alpha\leftrightarrow\beta$ partner vanish because ∂_α anticommutes with $\bar\theta^{\dot\gamma}$ *to zero* — they are derivatives and coordinates of different chirality, so $\left\{\partial_\alpha,\bar\theta^{\dot\gamma}\right\}=0$ rather than a Kronecker delta. The translation–translation term carries $\left\{\bar\theta^{\dot\gamma},\bar\theta^{\dot\delta}\right\}=0$. Hence $\{Q_\alpha,Q_\beta\}=0$, and by complex conjugation $\left\{\bar Q_{\dot\alpha},\bar Q_{\dot\beta}\right\}=0$.

Commutation with translations. Since Q_α is built from ∂_α, $\bar\theta^{\dot\alpha}$, and ∂_μ, and the spacetime derivative commutes with all three (constant-coefficient operators on flat superspace), $[Q_\alpha,\partial_\mu]=0$, i.e. $[Q_\alpha,P_\nu]=0$. This completes the verification. □

The proposition does more than reassure us. It shows that the differential operators Eq. (44.161) are a faithful realization of the abstract algebra on the function space of superspace, so that questions about representations of supersymmetry — which fields sit in a multiplet, how they transform — become questions about functions of $(x,\theta,\bar\theta)$. That is the program of the rest of the section.

44.6.3 Covariant derivatives

To carve out *irreducible* representations we will impose constraints on functions over superspace, such as "this field does not depend on $\bar\theta$." A constraint is only admissible if it is preserved by supersymmetry; otherwise acting with Q would take a solution out of the constrained set, and the constraint would not define a representation. The naive derivatives $\partial_\alpha,\bar\partial_{\dot\alpha}$ are useless for this purpose because they do not anticommute with the supercharges — indeed ∂_α is the leading piece of Q_α itself. We need a different set of fermionic derivatives that anticommute with all of $Q,\bar Q$. They are the *covariant derivatives*, and finding them requires only a sign flip relative to Eq. (44.161).

Definition 44.32 (Superspace covariant derivatives)**.** The covariant spinor derivatives are

$$\boxed{D_\alpha \;=\; \partial_\alpha \;+\; i\,\sigma^\mu_{\alpha\dot\alpha}\,\bar\theta^{\dot\alpha}\,\partial_\mu, \qquad \bar D_{\dot\alpha} \;=\; -\,\bar\partial_{\dot\alpha} \;-\; i\,\theta^\alpha\sigma^\mu_{\alpha\dot\alpha}\,\partial_\mu\,,} \tag{44.166}$$

differing from $Q_\alpha, \bar Q_{\dot\alpha}$ of Eq. (44.161) only in the sign of the ∂_μ term.

The defining virtue of these operators is covariance: they anticommute with every supercharge, so they map a representation to a representation.

Proposition 44.33 ($D, \bar D$ are covariant and close among themselves)**.** *The operators Eq. (44.166) satisfy*

$$\{D_\alpha, Q_\beta\} = \left\{D_\alpha, \bar Q_{\dot\beta}\right\} = \left\{\bar D_{\dot\alpha}, Q_\beta\right\} = \left\{\bar D_{\dot\alpha}, \bar Q_{\dot\beta}\right\} = 0, \tag{44.167}$$

$$\left\{D_\alpha, \bar D_{\dot\beta}\right\} \;=\; -\,2i\,\sigma^\mu_{\alpha\dot\beta}\,\partial_\mu, \qquad \{D_\alpha, D_\beta\} = \left\{\bar D_{\dot\alpha}, \bar D_{\dot\beta}\right\} = 0. \tag{44.168}$$

Proof. Anticommutation with the supercharges. Take $\left\{D_\alpha, \bar Q_{\dot\beta}\right\}$ with

$$D_\alpha = \partial_\alpha + i\,\sigma^\mu_{\alpha\dot\gamma}\bar\theta^{\dot\gamma}\partial_\mu, \qquad \bar Q_{\dot\beta} = -\bar\partial_{\dot\beta} + i\,\theta^\gamma\sigma^\nu_{\gamma\dot\beta}\partial_\nu.$$

The structure is identical to the calculation Eqs. (44.163) and (44.164), but the sign of the translation term in D_α is reversed. The two surviving cross terms are now

$$\left\{\partial_\alpha,\, i\theta^\gamma\sigma^\nu_{\gamma\dot\beta}\partial_\nu\right\} = +\,i\,\sigma^\nu_{\alpha\dot\beta}\partial_\nu, \qquad \left\{i\,\sigma^\mu_{\alpha\dot\gamma}\bar\theta^{\dot\gamma}\partial_\mu,\, -\bar\partial_{\dot\beta}\right\} = -\,i\,\sigma^\mu_{\alpha\dot\beta}\partial_\mu,$$

and they *cancel*: $\left\{D_\alpha, \bar Q_{\dot\beta}\right\} = 0$. The same cancellation, with the roles of θ and $\bar\theta$ exchanged, gives $\left\{\bar D_{\dot\alpha}, Q_\beta\right\} = 0$. The remaining two, $\{D_\alpha, Q_\beta\}$ and $\left\{\bar D_{\dot\alpha}, \bar Q_{\dot\beta}\right\}$, vanish for the same reason the like-anticommutators of Q vanished in Theorem 44.31: the only fermionic derivative that could contract is paired with a coordinate of the wrong chirality, so $\left\{\partial_\alpha, \bar\theta^{\dot\gamma}\right\} = 0$ and every term dies. Hence Eq. (44.167).

The D algebra. Repeating the computation of Eq. (44.165) for the two operators of Eq. (44.166), both surviving cross terms come with the flipped sign, so they add rather than canceling:

$$\left\{D_\alpha, \bar D_{\dot\beta}\right\} = -\,i\,\sigma^\mu_{\alpha\dot\beta}\partial_\mu \;-\; i\,\sigma^\mu_{\alpha\dot\beta}\partial_\mu = -\,2i\,\sigma^\mu_{\alpha\dot\beta}\,\partial_\mu. \tag{44.169}$$

The like-anticommutators $\{D_\alpha, D_\beta\}$ and $\left\{\bar D_{\dot\alpha}, \bar D_{\dot\beta}\right\}$ vanish by the chirality argument used for Q. This is Eq. (44.168). □

Theorem 44.33 is the workhorse identity of superspace. Equation Eq. (44.167) says that imposing a D- or $\bar{D}$-constraint on a field is supersymmetric: if $\bar{D}_{\dot\alpha}F = 0$, then for any supersymmetry variation $\delta_\xi F = (\xi Q + \bar\xi \bar Q)F$ we have $\bar D_{\dot\alpha}(\delta_\xi F) = \bar D_{\dot\alpha}(\xi Q + \bar\xi\bar Q)F = (\xi Q + \bar\xi \bar Q)\,\bar D_{\dot\alpha}F = 0$, because $\bar D$ anticommutes past the supercharges (the two minus signs from moving $\bar D$ through the two odd factors of ξQ cancel, leaving the operators effectively commuting on the even combination). The constrained field stays constrained: the constraint $\bar D_{\dot\alpha}\Phi = 0$ therefore defines a subrepresentation. This is precisely the chiral superfield that Section 44.7 is built on. Equation Eq. (44.168) records that the covariant derivatives close on translations with the opposite sign to the supercharges, a fact we will use repeatedly when integrating by parts in superspace.

Notation Watch

The symbols $D_\alpha = D_\alpha$ and $\bar D_{\dot\alpha} = \bar D_{\dot\alpha}$ denote the *superspace* covariant derivatives of Eq. (44.166) throughout this chapter. They are unrelated to the gauge-covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$ of the rest of the book; the latter always carries a spacetime vector index μ, the former a two-component spinor index. The gauge-covariant superspace derivative that fuses the two ideas appears only in Chapter 45.

44.6.4 The general superfield and its components

A field on superspace is called a superfield. Because θ^α and $\bar\theta_{\dot\alpha}$ are each two anticommuting numbers, any power higher than the second in either vanishes: $\theta^\alpha\theta^\beta\theta^\gamma = 0$ by the pigeonhole of two Grassmann generators, and similarly for $\bar\theta$. A function of superspace is therefore not an infinite Taylor series but a *polynomial* of finite degree, with at most two θ's and two $\bar\theta$'s. This finiteness is the structural reason supersymmetry yields finite multiplets.

Definition 44.34 (General scalar superfield)**.** A general (scalar) superfield $F(x,\theta,\bar\theta)$ is the most general function on $\mathcal{N}=1$ superspace,

$$\begin{aligned} F(x,\theta,\bar\theta) = {} & f(x) + \theta\psi(x) + \bar\theta\bar\chi(x) + \theta\theta\, m(x) + \bar\theta\bar\theta\, n(x) + \theta\sigma^\mu\bar\theta\, v_\mu(x) \\ & + \theta\theta\,\bar\theta\bar\lambda(x) + \bar\theta\bar\theta\,\theta\rho(x) + \theta\theta\,\bar\theta\bar\theta\, d(x), \end{aligned} \tag{44.170}$$

where the contractions follow Eq. (44.154) and $\theta\theta \equiv \theta^\alpha\theta_\alpha$, $\bar\theta\bar\theta \equiv \bar\theta_{\dot\alpha}\bar\theta^{\dot\alpha}$. The component fields are: complex scalars f, m, n, d; Weyl fermions $\psi_\alpha, \bar\chi_{\dot\alpha}, \bar\lambda_{\dot\alpha}, \rho_\alpha$; and a complex vector v_μ.

That Eq. (44.170) is the most general expansion follows from the Grassmann-degree truncation: there are exactly nine monomials in $\theta, \bar\theta$ that are nonvanishing and Lorentz-covariant, listed in Fig. 44.5, and the expansion attaches one spacetime field to each. Counting bosonic versus fermionic degrees of freedom (off

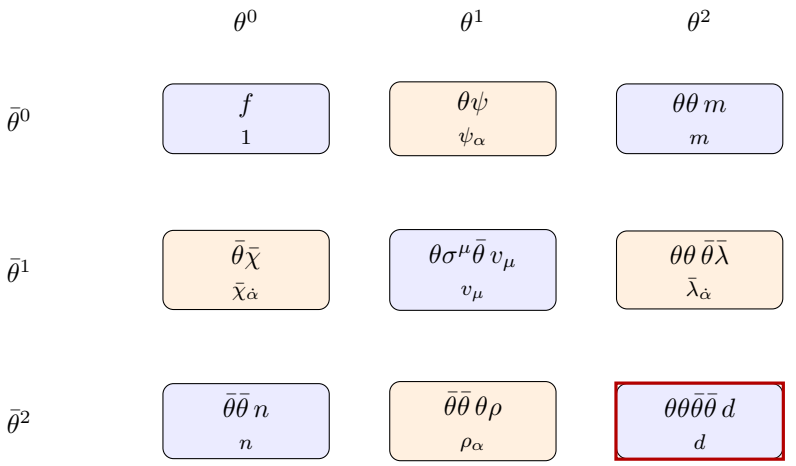


Figure 44.5: Component content of the general scalar superfield Eq. (44.170), organized by the power of θ (columns) and $\bar\theta$ (rows). Blue cells are bosonic components (f, m, n, v_μ, d), orange cells fermionic $(\psi, \bar\chi, \bar\lambda, \rho)$. The top component d (heavy red border) multiplies $\theta\theta\,\bar\theta\bar\theta$; its supersymmetry variation is a total spacetime derivative, the fact that makes $\int \mathrm{d}^4x\, \mathrm{d}^4\theta\, F$ an invariant (Theorem 44.36).

shell): the four complex scalars f, m, n, d carry $4 \times 2 = 8$ real bosonic components, and the complex four-vector v_μ carries $4 \times 2 = 8$ more, for 16 real bosonic components in all; the four Weyl fermions $\psi, \bar\chi, \bar\lambda, \rho$ carry $4 \times (2 \times 2) = 16$ real fermionic components. The boson and fermion counts match, $16 = 16$, as the algebra Eq. (44.155) forces. (We prove the equality of off-shell boson and fermion counts in any supermultiplet abstractly in Section 44.4; here it is visible by inspection.)

The general superfield is a perfectly good representation of supersymmetry — act with Q of Eq. (44.161) and the components rotate into one another — but it is a *reducible* one. The transformation Eq. (44.159) never mixes, for instance, the lowest component f with a component of higher θ-degree without also producing spacetime derivatives, and the nine components in Fig. 44.5 can be consistently truncated to smaller invariant sets. The cleanest such truncation is the chiral constraint $\bar D_{\dot\alpha}\Phi = 0$, which (as Section 44.7 shows) collapses the nine components down to the scalar-fermion-auxiliary triple (ϕ, ψ, F) of the Wess–Zumino multiplet. That a single covariant constraint extracts an irreducible piece is the practical payoff of Theorem 44.33, and the reason we built $D, \bar D$ before doing anything else with superfields.

Physical Insight

Reducibility here is the superspace echo of an old story. In Theorem 3.17 the finite-dimensional Lorentz representations were built and then *decomposed* into irreducible (j_-, j_+) blocks; a Dirac field, for instance, is the reducible sum $(\tfrac{1}{2},0)\oplus(0,\tfrac{1}{2})$. A general superfield is the analogous reducible object for the super-Poincaré algebra, and the covariant derivatives play the role that the projectors $\tfrac{1}{2}(1\pm\gamma^5)$ played there: they are the supersymmetric instruments that split a fat multiplet into its irreducible chiral and vector constituents.

44.6.5 Berezin integration on superspace and invariant measures

To build actions we must integrate over superspace. Integration over the bosonic coordinates is the ordinary $\int \mathrm{d}^4x$. Integration over the fermionic coordinates is the Berezin integral of Theorem 12.17, applied once to each Grassmann generator. We package the four fermionic integrations into Lorentz-covariant measures.

Definition 44.35 (Grassmann measures on superspace)**.** The chiral, antichiral, and full superspace measures are

$$\mathrm{d}^2\theta \;=\; -\tfrac{1}{4}\,\mathrm{d}\theta^\alpha\,\mathrm{d}\theta^\beta\,\varepsilon_{\alpha\beta}, \qquad \mathrm{d}^2\bar\theta \;=\; -\tfrac{1}{4}\,\mathrm{d}\bar\theta_{\dot\alpha}\,\mathrm{d}\bar\theta_{\dot\beta}\,\varepsilon^{\dot\alpha\dot\beta}, \qquad \mathrm{d}^4\theta \;=\; \mathrm{d}^2\theta\,\mathrm{d}^2\bar\theta, \tag{44.171}$$

normalized by the convention that they pick out the top Grassmann coefficient,

$$\int \mathrm{d}^2\theta\;\theta\theta = 1, \qquad \int \mathrm{d}^2\bar\theta\;\bar\theta\bar\theta = 1, \qquad \int \mathrm{d}^4\theta\;\theta\theta\,\bar\theta\bar\theta = 1. \tag{44.172}$$

The normalizations Eq. (44.172) are forced by Eq. (44.171) together with the one-variable Berezin rules $\int \mathrm{d}\theta\,\theta = 1$, $\int \mathrm{d}\theta\,1 = 0$ of Eq. (12.55); the prefactor $-\tfrac{1}{4}$ is chosen exactly so that the covariant combination $\theta\theta = \theta^\alpha\theta_\alpha$ integrates to unity. To see it, reduce both the integrand and the measure to the basis θ^1,θ^2. Expanding the contraction with $\varepsilon_{12} = -1$,

$$\theta\theta \;=\; \theta^\alpha\theta_\alpha \;=\; \varepsilon_{\alpha\beta}\,\theta^\alpha\theta^\beta \;=\; \varepsilon_{12}\,\theta^1\theta^2 + \varepsilon_{21}\,\theta^2\theta^1 \;=\; -2\,\theta^1\theta^2, \tag{44.173}$$

while the measure Eq. (44.171), using $\mathrm{d}\theta^2\,\mathrm{d}\theta^1 = -\mathrm{d}\theta^1\,\mathrm{d}\theta^2$, collapses to

$$\mathrm{d}^2\theta = -\tfrac{1}{4}\,\varepsilon_{\alpha\beta}\,\mathrm{d}\theta^\alpha\,\mathrm{d}\theta^\beta = -\tfrac{1}{4}(-\,\mathrm{d}\theta^1\,\mathrm{d}\theta^2 + \mathrm{d}\theta^2\,\mathrm{d}\theta^1) = \tfrac{1}{2}\,\mathrm{d}\theta^1\,\mathrm{d}\theta^2. \tag{44.174}$$

Composing the two and performing the nested Berezin integral $\int \mathrm{d}\theta^1\,\mathrm{d}\theta^2\,\theta^1\theta^2 = -1$ (integrate θ^2 first, then θ^1, picking up one sign from the anticommuting reorder),

$$\int \mathrm{d}^2\theta\,\theta\theta = \tfrac{1}{2}\int \mathrm{d}\theta^1\,\mathrm{d}\theta^2\,(-2\,\theta^1\theta^2) = -\int \mathrm{d}\theta^1\,\mathrm{d}\theta^2\,\theta^1\theta^2 = -(-1) = 1, \tag{44.175}$$

as claimed; the antichiral and full-superspace normalizations follow identically. The practical upshot, which is all we use below, is contained in Eq. (44.172): *integrating over* $\mathrm{d}^2\theta$ *extracts the* $\theta\theta$ *coefficient, integrating over* $\mathrm{d}^4\theta$ *extracts the* $\theta\theta\,\bar\theta\bar\theta$ *coefficient*, and all lower components integrate to zero because the Berezin integral of anything not saturating every Grassmann variable vanishes (Eq. (12.55)).

Two integrals build all supersymmetric actions. The first, the integral of a general superfield over the *full* superspace,

$$\int \mathrm{d}^4x\,\mathrm{d}^4\theta\; F(x,\theta,\bar\theta) \;=\; \int \mathrm{d}^4x\; d(x), \tag{44.176}$$

projects out the top component d of Eq. (44.170) and is called a *D-term*. The second, the integral of a chiral superfield over the chiral half $\mathrm{d}^2\theta$ only, projects out its highest $\theta\theta$ component and is called an *F-term*; we construct it in Section 44.7, where chiral superfields are defined. The remarkable fact, which makes these projections the right building blocks, is that both are supersymmetry invariants. We prove the D-term case in full now; the F-term case uses the same mechanism on a constrained field and is dispatched in Section 44.7.

Theorem 44.36 (D-term invariance)**.** *For any general superfield F, the full-superspace integral $\int \mathrm{d}^4x\,\mathrm{d}^4\theta\, F$ is invariant under supersymmetry: its variation δ_ξ is the spacetime integral of a total derivative, hence vanishes for fields decaying at infinity.*

Proof. The variation of F under an infinitesimal supersymmetry transformation is $\delta_\xi F = (\xi^\alpha Q_\alpha + \bar\xi_{\dot\alpha}\bar Q^{\dot\alpha})F$ with the operators Eq. (44.161). We integrate this over $\mathrm{d}^4x\,\mathrm{d}^4\theta$ and treat the two pieces of each supercharge in turn.

The ∂_θ pieces integrate to zero. The first term of $\xi^\alpha Q_\alpha$ is $\xi^\alpha\partial_\alpha F$, a fermionic derivative of F. The Berezin integral of a fermionic derivative vanishes:

$$\int \mathrm{d}^4\theta\; \xi^\alpha\partial_\alpha F \;=\; 0. \tag{44.177}$$

The reason is the one-variable identity $\int \mathrm{d}\theta\,\partial_\theta(\cdots) = 0$ from Eq. (12.55): $\partial_\alpha F$ lowers the θ-degree of every monomial by one, so it cannot contain the full top monomial $\theta\theta\,\bar\theta\bar\theta$ that $\mathrm{d}^4\theta$ requires for a nonzero answer. Equivalently, integration and differentiation are the same operation on Grassmann variables (Eq. (12.56)), and an integral of a derivative is a boundary term, which is empty for the polynomial F. The same argument kills $-\bar\xi_{\dot\alpha}\bar\partial^{\dot\alpha}F$, the leading piece of $\bar\xi\bar Q$.

The ∂_x pieces are spacetime total derivatives. The remaining terms in the supercharges all carry an explicit ∂_μ:

$$\delta_\xi F\big|_{\partial_x} = -\,i\,\xi^\alpha\sigma^\mu_{\alpha\dot\alpha}\bar\theta^{\dot\alpha}\,\partial_\mu F \;+\; i\,\bar\xi_{\dot\alpha}\,\theta^\alpha\sigma^{\mu\dot\alpha}_{\alpha}\,\partial_\mu F \;=\; \partial_\mu\Big[(-i\,\xi\sigma^\mu\bar\theta + i\,\theta\sigma^\mu\bar\xi)F\Big], \tag{44.178}$$

where the last equality holds because the bracket multiplying $\partial_\mu F$ is itself x-independent (it depends only on the constant parameters $\xi, \bar\xi$ and the coordinates $\theta, \bar\theta$), so it can be pulled inside the derivative. Integrating over d^4x, this is the integral of a spacetime total divergence and vanishes for fields falling off at infinity:

$$\int \mathrm{d}^4x\, \partial_\mu\Big[(-i\,\xi\sigma^\mu\bar\theta + i\,\theta\sigma^\mu\bar\xi)F\Big] \;=\; 0. \tag{44.179}$$

(The Grassmann factor $\theta, \bar\theta$ rides along untouched by $\int \mathrm{d}^4x$; the subsequent $\mathrm{d}^4\theta$ integration of zero is still zero.)

Combining the two pieces, the entire variation $\delta_\xi \int \mathrm{d}^4x\, \mathrm{d}^4\theta\, F$ is a sum of a Berezin boundary term Eq. (44.177) and a spacetime boundary term Eq. (44.179), both vanishing. Therefore

$$\delta_\xi \int \mathrm{d}^4x\, \mathrm{d}^4\theta\; F \;=\; 0, \tag{44.180}$$

and $\int \mathrm{d}^4x\, \mathrm{d}^4\theta\, F$ is supersymmetric. □

The mechanism is worth restating in one breath, because it is the engine of every superspace action. The top component of a superfield is the only one with nothing above it in Fig. 44.5; a supersymmetry transformation can only move it *sideways or down* into other components multiplied by ∂_μ, never up, because there is no higher monomial to receive it. Its variation is therefore forced to be a total spacetime derivative, and its spacetime integral is an invariant. The same statement holds for the top $(\theta\theta)$ component of a chiral superfield, which is why F-terms work. Supersymmetric Lagrangians are nothing but superfields whose top components we integrate.

Worked Example 44.37 (The top-component variation, by hand)**.** To see the abstract argument of Theorem 44.36 concretely, we compute the supersymmetry variation of the top component $d(x)$ of the general superfield Eq. (44.170) directly and confirm it is a total derivative.

Act with $\delta_\xi = \xi^\alpha Q_\alpha + \bar\xi_{\dot\alpha}\bar Q^{\dot\alpha}$ on F and collect everything proportional to the top monomial $\theta\theta\,\bar\theta\bar\theta$; that coefficient is $\delta_\xi d$. The ∂_α and $\bar\partial_{\dot\alpha}$ pieces of the supercharges lower the Grassmann degree, so to *reach* degree $(\theta^2\bar\theta^2)$ they must act on a monomial of degree $(\theta^3\bar\theta^2)$ or $(\theta^2\bar\theta^3)$ — which do not exist, since no field carries three θ's. Hence the pure-∂_θ pieces contribute *nothing* to $\delta_\xi d$: this is Eq. (44.177) seen component-by-component. Only the ∂_μ pieces survive. They act on the next-to-top components, $\theta\theta\,\bar\theta\bar\lambda$ and $\bar\theta\bar\theta\,\theta\rho$, supplying the missing Grassmann factor through the explicit $\bar\theta$ or θ they carry. Extracting the $\theta\theta\,\bar\theta\bar\theta$ coefficient from

$$-\,i\,\xi\sigma^\mu\bar\theta\,\partial_\mu(\bar\theta\bar\theta\,\theta\rho) \;+\; i\,\theta\sigma^\mu\bar\xi\,\partial_\mu(\theta\theta\,\bar\theta\bar\lambda)$$

and using the Fierz reorderings of Section 44.2.2 ($\theta^\alpha\theta^\beta = -\tfrac12\varepsilon^{\alpha\beta}\theta\theta$ and its dotted analogue), one finds

$$\delta_\xi d \;=\; -\tfrac{i}{2}\,\partial_\mu(\,\xi\sigma^\mu\bar\lambda \;-\; \rho\sigma^\mu\bar\xi\,). \tag{44.181}$$

The right-hand side is manifestly ∂_μ(something) — a total spacetime derivative, exactly as Theorem 44.36 promised. Whatever the detailed coefficients (they depend on the bilinear conventions of Section 44.2.2), the *form* is guaranteed: d sits at the top of the staircase, so its variation can only spill downward into derivatives of the rung below it. This is the component shadow of the theorem, and it is the calculation the Wess–Zumino action of Section 44.5 was secretly performing every time we checked invariance term by term.

Worked Example 44.38 (Integration by parts on superspace)**.** A second identity we will lean on in Chapters 45 and 46 is that the covariant derivatives can be integrated by parts under the full measure, just like ordinary derivatives. We show

$$\int \mathrm{d}^4x\,\mathrm{d}^4\theta\;(D_\alpha A)\,B \;=\; -\,(-1)^{|A|}\int \mathrm{d}^4x\,\mathrm{d}^4\theta\;A\,(D_\alpha B), \tag{44.182}$$

where $|A| = 0, 1$ is the Grassmann parity of A and the sign is the usual one for moving an odd operator past an object of definite parity. Write $D_\alpha = \partial_\alpha + i\sigma^\mu_{\alpha\dot\alpha}\bar\theta^{\dot\alpha}\partial_\mu$. The first piece, $\int \mathrm{d}^4\theta\,(\partial_\alpha A)B$, is handled by the Leibniz rule for the Grassmann derivative, $\partial_\alpha(AB) = (\partial_\alpha A)B + (-1)^{|A|}A(\partial_\alpha B)$, together with the vanishing of the Berezin integral of a total θ-derivative (Eq. (44.177)): $\int \mathrm{d}^4\theta\,\partial_\alpha(AB) = 0$, so $\int \mathrm{d}^4\theta\,(\partial_\alpha A)B = -(-1)^{|A|}\int \mathrm{d}^4\theta\,A(\partial_\alpha B)$. The second piece carries the spacetime derivative ∂_μ, which integrates by parts under $\int \mathrm{d}^4x$ with a relative minus and no Grassmann sign, since $\bar\theta^{\dot\alpha}$ is inert under ∂_μ: $\int \mathrm{d}^4x\,(i\sigma^\mu\bar\theta\,\partial_\mu A)B = -\int \mathrm{d}^4x\,(i\sigma^\mu\bar\theta)\,A\,\partial_\mu B = -(-1)^{|A|}\int \mathrm{d}^4x\,A\,(i\sigma^\mu\bar\theta\,\partial_\mu B)$ (the extra $(-1)^{|A|}$ from commuting the odd factor $i\sigma^\mu\bar\theta$ past A). Adding the two pieces reassembles D_α on B and gives Eq. (44.182). The identical computation with the flipped sign holds for $\bar D_{\dot\alpha}$. These rules let us freely move covariant derivatives across a superspace action, which is the manipulation that, in Chapter 46, turns a chiral integral into a full-superspace integral and underlies the nonrenormalization theorems.

With covariant derivatives that anticommute with the supercharges, a clean criterion for an irreducible constraint, and a pair of manifestly invariant measures in hand, every ingredient of a superspace action is now in place. The chiral constraint $\bar D_{\dot\alpha}\Phi = 0$ singled out by Theorem 44.33 will, in the next section, collapse the reducible nine-component superfield of Fig. 44.5 into the irreducible chiral multiplet (ϕ, ψ, F); its F-term and the D-term Eq. (44.176) of $\Phi^\dagger\Phi$ will reassemble the entire Wess–Zumino model of Section 44.5 in a single supersymmetric line, with the auxiliary field F and the superpotential W emerging automatically from the geometry we have just built.

44.7 Chiral Superfields and the Superspace Wess–Zumino Action

The previous section gave superspace its calculus. Supersymmetry became translation along the Grassmann directions, the supercharges Q_α, $\bar{Q}_{\dot\alpha}$ were realized as first-order differential operators on the coordinates $(x^\mu, \theta^\alpha, \bar\theta^{\dot\alpha})$, and a second pair of operators — the covariant derivatives D_α and $\bar D_{\dot\alpha}$ — was constructed precisely so as to *anticommute* with every supercharge (Section 44.6, Eq. (44.166)). Alongside them the Berezin measures $\mathrm{d}^2\theta$, $\mathrm{d}^2\bar\theta$ and $\mathrm{d}^4\theta$ were defined, inheriting from Theorem 12.17 the single rule that an integral over an anticommuting variable extracts the coefficient of the top form. We now put this machinery to work. A general superfield is a hugely reducible object: expanded in θ and $\bar\theta$ it contains far more component fields than any one irreducible multiplet of Section 44.5. The art is to impose a *covariant constraint* — one that commutes with supersymmetry, so that the constrained object is still a representation — which truncates the superfield down to an irreducible piece. The simplest such constraint singles out exactly the off-shell Wess–Zumino multiplet (ϕ, ψ, F) built by hand in Section 44.5, and reassembling that multiplet into a single superfield is what finally makes the Wess–Zumino model manifestly supersymmetric: its action becomes an integral over superspace, invariance becoming as automatic as Lorentz invariance is for $\int d^4x\, \mathcal{F}^2$. Along the way we will meet the two ways to extract a Lagrangian from superfields — the F-term and the Kähler (D-) term — and we will see the structural fact, holomorphy of the superpotential, that seeds the nonrenormalization theorems of Chapters 45 and 46.

Throughout we keep the chapter conventions: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, and the two-component spinor calculus of Section 3.4.3 with $\sigma^\mu_{\alpha\dot\beta} = (\mathbf{1}, \vec\sigma)$ (Eq. (3.79)) and $\bar\sigma^{\mu\,\dot\alpha\alpha} = (\mathbf{1}, -\vec\sigma)$ (Eq. (3.106)). Undotted indices are raised and lowered with $\epsilon^{\alpha\beta}, \epsilon_{\alpha\beta}$ ($\epsilon^{12} = -\epsilon_{12} = 1$), dotted ones likewise, and we use the standard summation shorthands $\theta\psi = \theta^\alpha\psi_\alpha$, $\bar\theta\bar\psi = \bar\theta_{\dot\alpha}\bar\psi^{\dot\alpha}$, $\theta\theta = \theta^\alpha\theta_\alpha$, $\bar\theta\bar\theta = \bar\theta_{\dot\alpha}\bar\theta^{\dot\alpha}$ fixed in Section 44.6.

44.7.1 The chiral constraint

The covariant derivatives recorded in Eq. (44.166) are, in the representation acting on superfields,

$$D_\alpha = \partial_\alpha + i\,\sigma^\mu_{\alpha\dot\beta}\,\bar\theta^{\dot\beta}\,\partial_\mu, \qquad \bar D_{\dot\alpha} = -\bar\partial_{\dot\alpha} - i\,\theta^\beta\,\sigma^\mu_{\beta\dot\alpha}\,\partial_\mu, \tag{44.183}$$

where $\partial_\alpha = \partial/\partial\theta^\alpha$ and $\bar\partial_{\dot\alpha} = \partial/\partial\bar\theta^{\dot\alpha}$, and $\partial_\mu = \partial/\partial x^\mu$. Their defining virtue, established in Section 44.6, is that they anticommute with the supercharges,

$$\{D_\alpha, Q_\beta\} = \left\{D_\alpha, \bar Q_{\dot\beta}\right\} = 0, \qquad \left\{\bar D_{\dot\alpha}, Q_\beta\right\} = \left\{\bar D_{\dot\alpha}, \bar Q_{\dot\beta}\right\} = 0, \tag{44.184}$$

while among themselves they reproduce the algebra $\left\{D_\alpha, \bar{D}_{\dot\beta}\right\} = -2i\,\sigma^\mu_{\alpha\dot\beta}\,\partial_\mu$ and $\{D_\alpha, D_\beta\} = \left\{\bar{D}_{\dot\alpha}, \bar{D}_{\dot\beta}\right\} = 0$ (Eq. (44.168)). Property Eq. (44.184) is the whole point: any equation built from the D's is supersymmetric, because a supersymmetry variation $\delta_\xi = \xi Q + \bar\xi \bar Q$ passes through a covariant derivative without obstruction.

Definition 44.39 (Chiral and antichiral superfields)**.** A superfield $\Phi(x, \theta, \bar\theta)$ is *chiral* if it is annihilated by every antiholomorphic covariant derivative,

$$\bar{D}_{\dot\alpha}\,\Phi \;=\; 0 \qquad (\dot\alpha = \dot1, \dot2). \tag{44.185}$$

It is *antichiral* if $D_\alpha\,\Phi^\dagger = 0$; the Hermitian conjugate of a chiral superfield is antichiral.

That Eq. (44.185) is consistent — that it does not collapse Φ to a constant — is guaranteed by the algebra: since $\left\{\bar{D}_{\dot\alpha}, \bar{D}_{\dot\beta}\right\} = 0$, imposing $\bar{D}_{\dot1}\Phi = \bar{D}_{\dot2}\Phi = 0$ involves no integrability obstruction. That it is supersymmetric — that the chiral subspace is preserved by the algebra — is Eq. (44.184): if $\bar{D}\Phi = 0$ then $\bar{D}(\delta_\xi\Phi) = \bar{D}(\xi Q + \bar\xi\bar Q)\Phi = -(\xi Q + \bar\xi\bar Q)\bar{D}\Phi = 0$, so $\delta_\xi\Phi$ is again chiral. The constraint therefore cuts out an honest sub-representation, and it is this sub-representation we now identify.

44.7.2 Solving the constraint: chiral coordinates

The constraint Eq. (44.185) looks awkward because $\bar{D}_{\dot\alpha}$ in Eq. (44.183) mixes a $\bar\theta$-derivative with an x-derivative. The remedy is a change of bosonic coordinate that absorbs the mixing. Define the *chiral coordinate*

$$y^\mu \;\equiv\; x^\mu + i\,\theta\sigma^\mu\bar\theta \;=\; x^\mu + i\,\theta^\alpha\sigma^\mu_{\alpha\dot\beta}\bar\theta^{\dot\beta}, \tag{44.186}$$

and its complex conjugate, the *antichiral coordinate* $y^{\dagger\mu} = x^\mu - i\,\theta\sigma^\mu\bar\theta$. The claim is that $\bar{D}_{\dot\alpha}$ annihilates both y^μ and θ^α, so that any function of y and θ alone automatically satisfies Eq. (44.185). The first is immediate: $\bar{D}_{\dot\alpha}\theta^\beta = 0$, since θ^β contains no $\bar\theta$ and no x. The second is a one-line computation. Using $\bar\partial_{\dot\alpha}\bar\theta^{\dot\gamma} = \delta^{\dot\gamma}_{\dot\alpha}$ and the anticommutativity of $\bar\partial_{\dot\alpha}$ with the θ^α to its left,

$$-\bar\partial_{\dot\alpha}\,y^\nu = -\bar\partial_{\dot\alpha}(i\,\theta^\beta\sigma^\nu_{\beta\dot\gamma}\bar\theta^{\dot\gamma}) = -i\,(-\theta^\beta\sigma^\nu_{\beta\dot\gamma}\delta^{\dot\gamma}_{\dot\alpha}) = +i\,\theta^\beta\sigma^\nu_{\beta\dot\alpha}, \tag{44.187}$$

while the explicit x-derivative piece of $\bar{D}_{\dot\alpha}$ acts on the x^ν in y^ν,

$$-i\,\theta^\beta\sigma^\mu_{\beta\dot\alpha}\,\partial_\mu\,y^\nu = -i\,\theta^\beta\sigma^\mu_{\beta\dot\alpha}\,\delta^\nu_\mu = -i\,\theta^\beta\sigma^\nu_{\beta\dot\alpha}, \tag{44.188}$$

the ∂_μ of the $i\theta\sigma^\nu\bar\theta$ term vanishing because it has no x-dependence. The two contributions Eq. (44.187) and Eq. (44.188) cancel:

$$\bar{D}_{\dot\alpha}\,y^\nu = i\,\theta^\beta\sigma^\nu_{\beta\dot\alpha} - i\,\theta^\beta\sigma^\nu_{\beta\dot\alpha} = 0. \tag{44.189}$$

Hence $\bar{D}_{\dot\alpha}$ kills y and θ, and any $\Phi = \Phi(y,\theta)$ obeys $\bar{D}_{\dot\alpha}\Phi = 0$. Conversely, in the coordinate system $(y,\theta,\bar\theta)$ the operator $\bar{D}_{\dot\alpha}$ reduces to $-\bar\partial_{\dot\alpha}\big|_{y,\theta}$ (the chain rule moves the mixing into the definition of y), so $\bar{D}_{\dot\alpha}\Phi = 0$ says nothing other than $\partial\Phi/\partial\bar\theta^{\dot\alpha} = 0$ at fixed (y,θ): a chiral superfield is precisely a superfield independent of $\bar\theta$ when written in terms of y. This is the general solution of the constraint.

44.7.3 The component expansion and the off-shell multiplet

A function of θ^α (two anticommuting components) and y terminates at order θ^2, because any product of three θ's vanishes. The most general chiral superfield is therefore the finite expansion

$$\boxed{\Phi(y,\theta) \;=\; \phi(y) \;+\; \sqrt{2}\,\theta\psi(y) \;+\; \theta\theta\, F(y).} \tag{44.190}$$

The factor $\sqrt{2}$ is conventional, chosen so that the kinetic terms below come out canonically normalized. The three component fields are exactly the content of the off-shell Wess–Zumino multiplet of Section 44.5: a complex scalar ϕ, a left-handed Weyl fermion ψ_α, and a complex auxiliary scalar F. The constraint $\bar{D}\Phi = 0$ has done its work, truncating the bloated general superfield down to this single irreducible multiplet. Counting confirms it: off shell, ϕ has 2 real bosonic components and F has 2, four bosonic in all, matching the four real components of the unconstrained Weyl spinor ψ — the boson–fermion balance demanded by Section 44.4.

To compare with the component transformations of Section 44.5, which were written as functions of x, we re-expand Eq. (44.190) about x^μ using $y^\mu = x^\mu + i\theta\sigma^\mu\bar\theta$. The shift is nilpotent — $(y-x)^\mu(y-x)^\nu(y-x)^\rho = 0$ since it is cubic in θ and $\bar\theta$ — so the Taylor series stops at second order. Two Grassmann identities, both established in Section 44.6, organize the result:

$$(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac{1}{2}\,\eta^{\mu\nu}\,\theta\theta\,\bar\theta\bar\theta, \tag{44.191}$$

$$(\theta\sigma^\mu\bar\theta)(\theta\chi) = -\tfrac{1}{2}\,\theta\theta\,(\chi\sigma^\mu\bar\theta). \tag{44.192}$$

Applying them, $\phi(y) = \phi(x) + i\theta\sigma^\mu\bar\theta\,\partial_\mu\phi - \tfrac{1}{4}\theta\theta\bar\theta\bar\theta\,\Box\phi$ from Eq. (44.191), and $\sqrt{2}\,\theta\psi(y) = \sqrt{2}\,\theta\psi - \tfrac{i}{\sqrt{2}}\theta\theta\,\partial_\mu\psi\,\sigma^\mu\bar\theta$ from Eq. (44.192), while $\theta\theta F(y) = \theta\theta F$ to the order that survives. Collecting,

$$\Phi = \phi + \sqrt{2}\,\theta\psi + \theta\theta\,F + i\,\theta\sigma^\mu\bar\theta\,\partial_\mu\phi - \tfrac{i}{\sqrt{2}}\,\theta\theta\,\partial_\mu\psi\,\sigma^\mu\bar\theta - \tfrac{1}{4}\,\theta\theta\,\bar\theta\bar\theta\,\Box\phi, \tag{44.193}$$

all fields now functions of x. The chiral superfield has only the components (ϕ,ψ,F); the remaining terms in Eq. (44.193) are their spacetime derivatives, forced by the constraint, not independent fields.

It remains to verify that the supersymmetry variation of the components reproduces, term by term, the off-shell Wess–Zumino transformations of Eq. (44.134)

— closing the loop opened in Section 44.5. The variation of a chiral superfield is $\delta_\xi \Phi = (\xi Q + \bar{\xi}\bar{Q})\Phi$. In chiral coordinates the supercharges take a simple form on $\Phi(y,\theta)$: from Eq. (44.183) and the chain rule one finds $Q_\alpha \to \partial_\alpha$ acting at fixed y, while $\bar{Q}_{\dot\alpha} \to 2i\,\theta^\beta \sigma^\mu_{\beta\dot\alpha}\partial_\mu$, so that

$$\delta_\xi \Phi = \xi^\alpha \partial_\alpha \Phi - 2i\,(\theta\sigma^\mu\bar{\xi})\,\partial_\mu \Phi. \tag{44.194}$$

Inserting Eq. (44.190) and reading off the coefficients of 1, θ, and $\theta\theta$ (using $\partial_\alpha(\theta\theta) = 2\theta_\alpha$ and Eq. (44.192)) gives

$$\delta_\xi \phi = \sqrt{2}\,\xi\psi, \tag{44.195}$$

$$\delta_\xi \psi_\alpha = \sqrt{2}\,\xi_\alpha\, F - i\sqrt{2}\,\sigma^\mu_{\alpha\dot\beta}\,\bar{\xi}^{\dot\beta}\,\partial_\mu \phi, \tag{44.196}$$

$$\delta_\xi F = i\sqrt{2}\,\partial_\mu(\psi\sigma^\mu\bar{\xi}) \;=\; -i\sqrt{2}\,\bar{\xi}\,\bar{\sigma}^\mu\,\partial_\mu\psi. \tag{44.197}$$

These are precisely Eq. (44.134): the scalar rotates into the fermion, the fermion into the auxiliary field and the gradient of the scalar, and — the structurally decisive fact — *the auxiliary field rotates into a total derivative*. The off-shell multiplet built by trial and error in Section 44.5 is nothing but the component content of a constrained superfield, and superspace has reproduced its transformation law mechanically. The loop is closed.

Physical Insight

The miracle of Eq. (44.197) — that $\delta_\xi F$ is a pure divergence — is not an accident of the Wess–Zumino multiplet; it is the top of every chiral ladder. A supersymmetry transformation Eq. (44.194) raises the θ-degree by one (the $\xi^\alpha\partial_\alpha$ piece) or shifts the argument (the ∂_μ piece). The highest component $F = \Phi|_{\theta\theta}$ has nowhere higher to go: the only thing δ_ξ can do to it is move it in spacetime. This is why $\int d^4x\, F$ is supersymmetric, and it is the whole reason superspace integrals build invariant actions. The same logic, applied to the highest component of a general *real* superfield, will give the D-term below.

44.7.4 Products of chiral superfields and the superpotential

Chiral superfields form a *ring*, and this single algebraic fact is what makes interactions possible in superspace. Since $\bar{D}_{\dot\alpha}$ is a first-order differential operator, it acts on a product as a (graded) derivation; acting on two bosonic chiral superfields Φ_1, Φ_2,

$$\bar{D}_{\dot\alpha}(\Phi_1\Phi_2) = (\bar{D}_{\dot\alpha}\Phi_1)\Phi_2 + \Phi_1(\bar{D}_{\dot\alpha}\Phi_2) = 0, \tag{44.198}$$

so the product is again chiral. By induction any polynomial in the Φ_i with constant coefficients is chiral, and more generally any *holomorphic* function — a

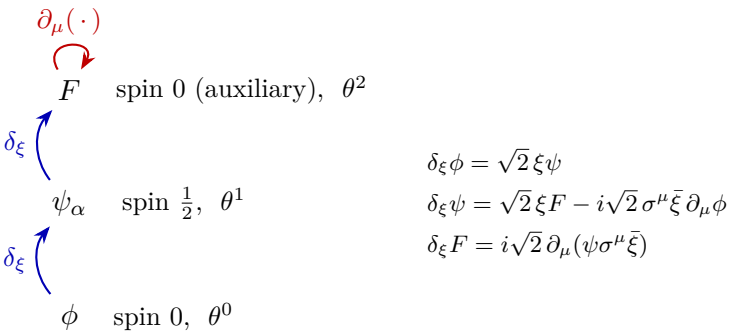


Figure 44.6: The chiral multiplet as a ladder in θ-degree. Supersymmetry δ_ξ climbs the ladder $\phi \to \psi \to F$, raising the θ-degree by one rung. The top rung F has nowhere higher to go, so δ_ξ can only move it by a total spacetime derivative (the looped red arrow, Eq. (44.197)). This is exactly what makes $\int d^4x\, F$ a supersymmetric invariant and underlies the F-term construction of actions.

power series in the Φ_i alone, with no $\Phi_i^\dagger$ — is chiral. (A function involving both Φ and $\Phi^\dagger$ is neither chiral nor antichiral; $\Phi^\dagger$ satisfies $D_\alpha\Phi^\dagger = 0$, not $\bar{D}_{\dot\alpha}\Phi^\dagger = 0$.) The cleanest way to see chirality of a product is in chiral coordinates: each $\Phi_i = \Phi_i(y,\theta)$ is $\bar\theta$-independent, and a product of $\bar\theta$-independent functions is $\bar\theta$-independent.

Definition 44.40 (Superpotential)**.** A *superpotential* $W(\Phi)$ is any holomorphic function of the chiral superfields of the theory — for a single field, a power series $W(\Phi) = \sum_n \frac{1}{n}\lambda_n\Phi^n$ with constant coefficients. By Eq. (44.198) the composite superfield $W(\Phi)$ is itself chiral.

Because $W(\Phi)$ is chiral, it has a component expansion of the form Eq. (44.190), and we will need its components. Expanding the holomorphic function about its lowest component ϕ and using $\Phi - \phi = \sqrt{2}\,\theta\psi + \theta\theta F$ together with the bilinear identity $(\theta\psi)(\theta\chi) = -\frac{1}{2}\,\theta\theta\,(\psi\chi)$,

$$W(\Phi) = W(\phi) + W'(\phi)\,(\Phi-\phi) + \tfrac{1}{2}\,W''(\phi)\,(\Phi-\phi)^2 + \dots, \tag{44.199}$$

the quadratic term contributing $(\Phi-\phi)^2\big|_{\theta\theta} = 2(\theta\psi)(\theta\psi) = -\theta\theta\,(\psi\psi)$. Collecting the coefficient of $\theta\theta$ gives the highest component of the superpotential,

$$\boxed{W(\Phi)\big|_{\theta\theta} = W'(\phi)\,F - \tfrac{1}{2}\,W''(\phi)\,\psi\psi.} \tag{44.200}$$

Here $W' = \partial W/\partial\phi$ and $W'' = \partial^2 W/\partial\phi^2$ are ordinary derivatives of the function W evaluated on the scalar ϕ.

44.7.5 The F-term and its supersymmetry invariance

The Berezin rule $\int \mathrm{d}^2\theta\,\theta\theta = 1$ (Theorem 44.35, inherited from Theorem 12.17) means that integrating a chiral superfield over the chiral measure extracts precisely

its highest component:

$$\int \mathrm{d}^2\theta\, W(\Phi) = W(\Phi)\big|_{\theta\theta} = W'(\phi)\, F - \tfrac{1}{2}\, W''(\phi)\, \psi\psi. \tag{44.201}$$

Such an integral is called an *F-term*. We must check that adding it to a Lagrangian preserves supersymmetry; this is the central invariance statement of the construction.

Theorem 44.41 (F-terms are supersymmetric)**.** *Let Λ be any chiral superfield, with highest component $F_\Lambda = \Lambda|_{\theta\theta}$ (in chiral coordinates). Then under a supersymmetry transformation δ_ξ, the highest component varies into a total spacetime derivative,*

$$\delta_\xi F_\Lambda = i\sqrt{2}\,\partial_\mu(\psi_\Lambda\,\sigma^\mu\bar{\xi}), \tag{44.202}$$

where $\psi_\Lambda = \Lambda|_\theta/\sqrt{2}$ is the fermionic component of Λ. Consequently the action contribution

$$S_F = \int d^4x \left[\int \mathrm{d}^2\theta\ \Lambda\ +\ \text{h.c.}\right] = \int d^4x\ (F_\Lambda + F_\Lambda^*) \tag{44.203}$$

is invariant, $\delta_\xi S_F = 0$.

Proof. The expansion Eq. (44.190) and the transformation rules Eq. (44.195)–Eq. (44.197) were derived for a generic chiral superfield; nothing in them used the field content beyond chirality. Writing $\Lambda = A + \sqrt{2}\,\theta\psi_\Lambda + \theta\theta\, F_\Lambda$ in chiral coordinates, Eq. (44.197) reads exactly Eq. (44.202). Now Λ is chiral, so its highest component is a function of y; but $\int d^4x\, f(y) = \int d^4y\, f(y) = \int d^4x\, f(x)$, since the shift $y = x + i\theta\sigma\bar{\theta}$ has unit Jacobian and merely relabels the integration variable. Therefore $\delta_\xi S_F = \int d^4x\,(\delta_\xi F_\Lambda + \text{c.c.}) = \int d^4x\,\partial_\mu(\cdots) = 0$, the integral of a total derivative over all of spacetime vanishing for fields decaying at infinity. □

Applied to $\Lambda = W(\Phi)$, Theorem 44.41 establishes that the superpotential term

$$S_W = \int d^4x \left[\int \mathrm{d}^2\theta\ W(\Phi) + \text{h.c.}\right] \tag{44.204}$$

is a supersymmetric invariant for *any* holomorphic W. This is the superspace origin of every Yukawa coupling and scalar self-interaction in a supersymmetric theory: they are not added by hand but are the components of a single holomorphic function.

44.7.6 The Kähler term: kinetic energy in superspace

The superpotential supplies masses and interactions but no derivatives of the scalar and no kinetic term for the fermion: Eq. (44.200) contains F, $\psi\psi$, and ϕ algebraically, never $\partial_\mu\phi$. Kinetic terms must come from an integral over the

full superspace, $\int \mathrm{d}^4\theta = \int \mathrm{d}^2\theta\, \mathrm{d}^2\bar\theta$, of a quantity that is real and contains both Φ and $\Phi^\dagger$. The minimal, canonical choice is the *Kähler term*

$$S_K = \int d^4x \int \mathrm{d}^4\theta\; \Phi^\dagger\Phi. \tag{44.205}$$

The combination $\Phi^\dagger\Phi$ is a general real superfield (neither chiral nor antichiral), so it has a nonzero highest component $\theta\theta\bar\theta\bar\theta$, and the Berezin rule $\int \mathrm{d}^4\theta\, \theta\theta\bar\theta\bar\theta = 1$ extracts it. Invariance is guaranteed by the same logic as the F-term: the highest component of any superfield transforms into a total derivative under δ_ξ (the D-term theorem, Theorem 44.36), because there is no higher component to receive the variation. What we must do now is reduce Eq. (44.205) to components and confirm that it delivers exactly the canonical kinetic Lagrangian.

We extract the $\theta\theta\bar\theta\bar\theta$ coefficient of $\Phi^\dagger\Phi$ by multiplying the x-expansion Eq. (44.193) with its conjugate and keeping only terms of total Grassmann bidegree $(2,2)$ in $(\theta,\bar\theta)$. Organizing Φ and $\Phi^\dagger$ by bidegree, the pairings that reach $(2,2)$ are few. The purely bosonic ones are

$$\begin{aligned}
&\phi^*\Big(-\tfrac14\Box\phi\Big) + \Big(-\tfrac14\Box\phi^*\Big)\phi && \text{from (lowest)} \times \text{(highest)},\\
&F^*F && \text{from } (\bar\theta\bar\theta F^*)(\theta\theta F),\\
&(\partial_\mu\phi^*\partial_\nu\phi)\,(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac12\,\partial_\mu\phi^*\partial^\mu\phi\;\theta\theta\bar\theta\bar\theta && \text{from the two } (1,1) \text{ pieces, via Eq. (44.191)}.
\end{aligned}$$

Collecting the bosonic coefficient of $\theta\theta\bar\theta\bar\theta$ and integrating the $\Box$ terms by parts under $\int d^4x$ (so that $-\tfrac14\phi^*\Box\phi \to +\tfrac14\partial_\mu\phi^*\partial^\mu\phi$, and likewise for its conjugate),

$$[\Phi^\dagger\Phi]^{\text{bos}}_{\theta\theta\bar\theta\bar\theta} \;\longrightarrow\; \Big(\tfrac14+\tfrac14+\tfrac12\Big)\partial_\mu\phi^*\partial^\mu\phi + F^*F = \partial_\mu\phi^*\,\partial^\mu\phi + F^*F. \tag{44.206}$$

The two halves of the kinetic term — one quarter each from the $\Box\phi$ descendants of ϕ and ϕ^*, one half from the $(\theta\sigma\bar\theta)$-cross term — assemble into the single canonically normalized $\partial_\mu\phi^*\partial^\mu\phi$, and the auxiliary field contributes its modulus squared F^*F with no derivative, confirming that F is non-propagating.

The fermionic contributions come from the two pairings that carry one ψ and one $\bar\psi$ and reach bidegree $(2,2)$: the lowest fermion of $\Phi^\dagger$ with the $\theta\theta$-cross term of Φ, and the conjugate pairing. Using the reduction identities Eq. (44.192) and its conjugate $(\bar\theta\bar\eta)(\partial_\mu\psi\,\sigma^\mu\bar\theta) = -\tfrac12\,\bar\theta\bar\theta\,(\partial_\mu\psi\,\sigma^\mu\bar\eta)$, together with $\chi\sigma^\mu\bar\eta = -\bar\eta\,\bar\sigma^\mu\chi$ from Section 44.6, the two terms combine into the Hermitian Weyl bilinear

$$[\Phi^\dagger\Phi]^{\text{ferm}}_{\theta\theta\bar\theta\bar\theta} = \tfrac{i}{2}\,(\bar\psi\,\bar\sigma^\mu\partial_\mu\psi - \partial_\mu\bar\psi\,\bar\sigma^\mu\psi) \;\xrightarrow{\text{by parts}}\; i\,\bar\psi\,\bar\sigma^\mu\partial_\mu\psi, \tag{44.207}$$

which is exactly the canonical kinetic Lagrangian for a left-handed Weyl fermion fixed in Section 3.4.3 and quantized in Chapter 11. Assembling Eq. (44.206) and Eq. (44.207), the Kähler term reduces to

$$\boxed{\int \mathrm{d}^4\theta\; \Phi^\dagger\Phi = \partial_\mu\phi^*\,\partial^\mu\phi + i\,\bar\psi\,\bar\sigma^\mu\partial_\mu\psi + F^*F} \tag{44.208}$$

up to a total derivative — the free, massless, canonically normalized kinetic energy of the scalar and the fermion, plus the algebraic F^*F that marks F as auxiliary.

44.7.7 The full superspace Wess–Zumino action

We can now write the Wess–Zumino model of Section 44.5 in one line of superspace. Combining the Kähler term Eq. (44.205) with the superpotential term Eq. (44.204),

$$\boxed{S_{\mathrm{WZ}} = \int d^4x \left[\int \mathrm{d}^4\theta\, \Phi^\dagger \Phi \;+\; \left(\int \mathrm{d}^2\theta\, W(\Phi) + \mathrm{h.c.} \right) \right].} \tag{44.209}$$

This is manifestly supersymmetric: the Kähler term is a D-term and the superpotential term is an F-term, and both classes are invariant by Theorem 44.41 and Theorem 44.36. No component computation is needed to certify invariance — supersymmetry of Eq. (44.209) is as automatic as Lorentz invariance of a scalar action, and this economy is the entire payoff of the superspace formalism.

Reducing Eq. (44.209) to components by Eq. (44.208) and Eq. (44.201) gives the off-shell component Lagrangian

$$\mathcal{L}_{\mathrm{WZ}} = \partial_\mu \phi^* \partial^\mu \phi + i\,\bar\psi\,\bar\sigma^\mu \partial_\mu \psi + F^*F + \Big[W'(\phi)\, F - \tfrac{1}{2} W''(\phi)\, \psi\psi + \mathrm{h.c.} \Big], \tag{44.210}$$

which is, term for term, the Lagrangian assembled by hand in Eqs. (44.133) and (44.142). The fermion kinetic term and the scalar kinetic term are the components of the same Kähler term; the Yukawa coupling $W''\psi\psi$ and the auxiliary coupling $W'F$ are the components of the same superpotential. Supersymmetry, which related these by hand in Section 44.5 only after a page of algebra, is here built into the geometry.

The auxiliary field F appears without derivatives, so its Euler–Lagrange equation is algebraic. Varying Eq. (44.210) with respect to F^* and F,

$$\frac{\partial \mathcal{L}}{\partial F^*} = F + \overline{W'} = 0 \quad \Longrightarrow \quad F = -\,\overline{W'(\phi)}, \qquad F^* = -\,W'(\phi), \tag{44.211}$$

where $\overline{W'} = (\partial W/\partial\phi)^*$. Substituting back, the auxiliary sector $F^*F + W'F + \overline{W'}F^*$ collapses to $|W'|^2 - |W'|^2 - |W'|^2 = -|W'(\phi)|^2$, so the on-shell Lagrangian is

$$\mathcal{L}_{\mathrm{WZ}}^{\text{on-shell}} = \partial_\mu \phi^* \partial^\mu \phi + i\,\bar\psi\,\bar\sigma^\mu \partial_\mu \psi - \tfrac{1}{2}\big[W''(\phi)\, \psi\psi + \mathrm{h.c.} \big] - V(\phi, \phi^*), \tag{44.212}$$

with the *scalar potential*

$$\boxed{V(\phi,\phi^*) = \big|W'(\phi)\big|^2 = \Big|\frac{\partial W}{\partial \phi}\Big|^2 \;\geq\; 0.} \tag{44.213}$$

The potential is the modulus-squared of the holomorphic derivative of the superpotential, and it is manifestly non-negative — the component-level shadow of the energy positivity $H \geq 0$ derived from the algebra in Section 44.3.1. Supersymmetric vacua are the zeros of V, i.e. the critical points $W'(\phi) = 0$ of the superpotential, where the energy vanishes exactly.

Worked Example 44.42 (The cubic Wess–Zumino model: Yukawa and potential from $\int \mathrm{d}^2\theta\, W$)**.** Take the renormalizable cubic superpotential of the Wess–Zumino model,

$$W(\Phi) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, g\, \Phi^3, \tag{44.214}$$

with a mass parameter m and a dimensionless coupling g. (A constant and a linear term are the only other renormalizable options: a constant drops out of Eq. (44.201), and a linear "Fayet–Iliopoulos-like" term shifts F and is treated in Chapter 45.) We extract the component Lagrangian directly from the superspace expression and exhibit the supersymmetric relations among its couplings.

Derivatives of the superpotential. From Eq. (44.214),

$$W'(\phi) = m\,\phi + g\,\phi^2, \qquad W''(\phi) = m + 2g\,\phi. \tag{44.215}$$

The F-term. Substituting Eq. (44.215) into Eq. (44.201),

$$\int \mathrm{d}^2\theta\, W(\Phi) = (m\phi + g\phi^2)\, F - \tfrac{1}{2}(m + 2g\phi)\, \psi\psi. \tag{44.216}$$

Adding the Hermitian conjugate and the Kähler term Eq. (44.208), the off-shell Lagrangian is Eq. (44.210) with these explicit couplings. The fermion bilinear splits into a *Majorana mass term* $-\frac{1}{2}m\,\psi\psi + \text{h.c.}$ and a *Yukawa coupling* $-g\,\phi\,\psi\psi + \text{h.c.}$; the mass that appears here is the same m that sits in front of Φ^2.

The scalar potential. Integrating out F via Eq. (44.211) and using Eq. (44.213),

$$V(\phi, \phi^*) = |m\phi + g\phi^2|^2 = m^2\,|\phi|^2 + mg\,(\phi^*\phi^2 + \phi\,\phi^{*2}) + g^2\,|\phi|^4. \tag{44.217}$$

Reading off the spectrum and the SUSY relations. The scalar mass term is $m^2|\phi|^2$, so the complex scalar ϕ has mass m; the Majorana mass term is $-\frac{1}{2}m\,\psi\psi$, so the Weyl fermion ψ has the *same* mass m. Boson and fermion are exactly degenerate — the hallmark of unbroken supersymmetry, which in Section 44.4 was forced by $\left\{Q, \bar{Q}\right\} \sim P$. The quartic scalar coupling $g^2|\phi|^4$ and the cubic scalar coupling $mg(\phi^*\phi^2 + \text{h.c.})$ are not independent of the Yukawa coupling g: they are fixed by it, because all three descend from the *single* holomorphic function Eq. (44.214). The familiar non-supersymmetric freedom to choose a scalar quartic independently of the Yukawa is gone; supersymmetry ties $\lambda_{\phi^4} = g^2$ to the Yukawa g. This rigidity — one holomorphic W controlling masses, Yukawas, and the scalar potential at once — is exactly what will make the loop corrections of Chapter 45 so tightly constrained.

44.7.8 Holomorphy and the seed of nonrenormalization

The split of the action Eq. (44.209) into a Kähler term and a superpotential term is not merely organizational; it is the structural fact behind the most celebrated quantum properties of supersymmetric theories. The superpotential $W(\Phi)$ is *holomorphic* — a function of the chiral superfields alone, with no dependence on the conjugates $\Phi^\dagger$ — whereas the Kähler potential $K = \Phi^\dagger\Phi$ depends on both. Holomorphy is a powerful constraint because the couplings in W can be promoted to background chiral superfields ("spurions"): the mass m and coupling g of Eq. (44.214) are themselves the lowest components of chiral spurion superfields, and any quantum-corrected effective superpotential must again be a *holomorphic* function of the dynamical fields and these spurions. Together with the symmetries (including an R-symmetry that assigns charges to θ and hence to W) and the weak-coupling limits, holomorphy is so restrictive that it forbids the superpotential from receiving *any* perturbative corrections at all: the renormalized W equals the classical W to all orders in perturbation theory. This is the *nonrenormalization theorem*, and it is the reason the boson–fermion mass degeneracy found in Theorem 44.42 survives quantization — the cancellation of quadratic divergences that motivates supersymmetry as a response to the hierarchy problem (Chapter 35).

Caution

Nonrenormalization protects the *superpotential*, not the whole theory. The Kähler term $\int \mathrm{d}^4\theta\, \Phi^\dagger\Phi$ is a D-term and is *not* holomorphic; it does receive corrections, which renormalize the kinetic terms and hence wave-function-renormalize the fields. Physical (canonically normalized) couplings therefore do run, but only through the universal wave-function factors, not through independent vertex renormalizations. Disentangling "holomorphic" from "physical" couplings is the technical heart of Chapters 45 and 46; here we only plant the flag that the superpotential's protection comes from its holomorphy and the F-term structure isolated in Theorem 44.41.

We have now built the full $\mathcal{N} = 1$ chiral toolkit: the chiral constraint and its solution, the component multiplet matched to Section 44.5, the F-term and Kähler-term constructions, the superspace Wess–Zumino action, and the holomorphy that protects it. What remains, to reach realistic supersymmetric theories, is to add gauge interactions. That requires a second kind of superfield — the real *vector* superfield, whose lowest physical components are a gauge boson and its fermionic "gaugino" partner — and the supersymmetric completion of the gauge principle of Chapter 25. Chapter 45 constructs the vector superfield, gauges the chiral matter built here, and assembles super-Yang–Mills and SQCD, taking the chiral superfields and superspace actions of this section as its starting point; Chapter 46 then turns the holomorphy seeded above into exact nonperturbative

results. The chiral superfield, and the F-term/Kähler-term grammar of writing actions in superspace, are from here on assumed.

Further reading

The two no-go-and-its-loophole results that frame the chapter are Coleman and Mandula [131] and Haag et al. [247]; both repay a careful reading and are the honest source for the hypotheses we only sketch. The Wess–Zumino model first appeared in Wess and Zumino [507]. For the systematic textbook development at the level of this chapter, Wess and Bagger [505] is the classic superspace reference (compatible two-component conventions), Weinberg [502] gives the most careful treatment of the algebra, representations, and the HLS theorem, and Terning [472] is an excellent modern account weighted toward dynamics and duality. Martin [339] is the standard pedagogical introduction, especially strong on the passage from superspace to component Lagrangians and on conventions; the two-component spinor technology used throughout is laid out exhaustively in Dreiner et al. [162]. Witten's supersymmetric quantum mechanics and the index that decides whether supersymmetry is spontaneously broken are from Witten [520]. Readers heading toward supergravity will find the superspace methods continued in Freedman and Van Proeyen [199].

Chapter 45

Supersymmetric Gauge Theories

The previous chapter built supersymmetry as a graded extension of the Poincaré algebra and gave it a working language: superspace, chiral and antichiral superfields, the auxiliary F- and D-components that close the algebra off shell, and the Wess–Zumino model as the simplest interacting theory (Chapter 44). What that chapter deliberately set aside was the gauge principle. This chapter supplies it. The task is to merge the two great organizing ideas of the book so far — local gauge invariance, which produced electromagnetism in Chapter 8 and Yang–Mills theory in Chapter 25, and supersymmetry, which pairs every boson with a fermion — into a single framework. The fusion is not optional decoration: once matter is charged and the gauge parameter is promoted to a *chiral* superfield, the ordinary Kähler term ceases to be invariant, and a new object, the real vector superfield, is forced into existence to repair it. Everything in the chapter unfolds from that one demand. Throughout we keep the global conventions of Chapter A : mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, Hermitian generators with $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ and the numerical group-theory data of Chapter B, and the covariant-derivative sign $D_\mu = \partial_\mu - igA^a_\mu T^a$.

The arc runs from the simplest realization to the most physical. We open in the abelian case (Section 45.1): the vector superfield V, its super-gauge redundancy $V \to V + \Lambda + \Lambda^\dagger$ that reduces in Wess–Zumino gauge to exactly the ordinary U(1) symmetry of Chapter 8, the chiral gauge-invariant field strength $W_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha V$, the supersymmetric Maxwell action packaged by a single holomorphic coupling, and the covariant Kähler term $\Phi^\dagger e^{2qV}\Phi$ that reproduces minimal coupling. The non-abelian generalization (Section 45.2) promotes V to a matrix in the Lie algebra, replaces the additive gauge law by the finite conjugation $e^V \to e^{-i\Lambda^\dagger} e^V e^{i\Lambda}$, and produces super Yang–Mills, whose bosonic content is precisely the classical theory of Chapter 25 and whose quantization reuses the

Faddeev–Popov and BRST machinery of Chapter 26 without modification — the gaugino simply rides along in the adjoint. Adding charged quark and antiquark flavours gives supersymmetric QCD (Section 45.3), whose scalar potential $V = \sum_f |F_f|^2 + \frac{1}{2}\sum_a (D^a)^2$ falls out of the auxiliary fields and whose flat directions trace a classical moduli space of degenerate vacua, coordinatized by gauge-invariant mesons and baryons.

With the classical theory assembled, the chapter turns to what makes these models extraordinary. Holomorphy — the statement that the superpotential and the gauge coupling τ are holomorphic functions of holomorphic data, sharpened by treating the couplings themselves as spurion background fields — forces the nonrenormalization theorems (Section 45.4): the superpotential receives *no* perturbative corrections, and the holomorphic gauge coupling runs only at one loop. Translating from the holomorphic to the canonical normalization, via the Konishi anomaly whose rescaling Jacobian is computed by the same Fujikawa method as the chiral anomaly of Chapter 36, converts that one-loop statement into the all-orders exact NSVZ beta function, which reduces at leading order to the supersymmetric version of the Yang–Mills theory coefficient of Chapter 27. These exact results are the controlled laboratory that Chapter 46 exploits.

The world is not supersymmetric, so the last two sections confront breaking and phenomenology. Soft breaking (Section 45.5) is the art of lifting the superpartners while preserving the cancellation of quadratic divergences that made supersymmetry attractive in the first place: we compute that cancellation explicitly, prove the supertrace sum rule, and classify — via the spurion method and the Girardello–Grisaru criterion — exactly which operators may be added without reintroducing the hard divergences. Instantiating the template with realistic field content gives the Minimal Supersymmetric Standard Model (Section 45.6): two Higgs doublets forced by anomaly cancellation and holomorphy, the soft-breaking Lagrangian, the tree-level bound $m_h \le m_Z |\cos 2\beta|$ on the lightest Higgs, and the improvement of gauge-coupling unification that motivated so much of the program in Chapter 35.

This chapter advances Yang–Mills theory into its supersymmetric incarnation and is, apart from the NSVZ resummation and the soft-breaking divergence structure, perturbative and symmetry-based throughout. The genuinely non-perturbative content of these theories — the quantum fate of the moduli space, Seiberg duality, gaugino condensation, and the Seiberg–Witten solution — is deferred to Chapter 46, which takes the classical moduli spaces, the holomorphic scale, and the nonrenormalization theorems established here as its starting data. Where the experimental status of supersymmetry is at stake we state it plainly and date it: as of the mid-2020s no superpartner has been observed, and the natural region of the MSSM is constrained but not excluded.

45.1 The Vector Superfield and the Abelian Gauge Multiplet

Chapter 44 left us holding one of the two massless irreducible representations of the $\mathcal{N}=1$ super-Poincaré algebra. The chiral superfield Φ (Theorem 44.39, Eq. (44.190)) packages a complex scalar, a Weyl fermion, and an auxiliary field into a single object on superspace, and the superspace measures $\mathrm{d}^2\theta$, $\mathrm{d}^4\theta$ (Theorem 44.35) turn invariant Lagrangians into Berezin integrals whose supersymmetry is as automatic as the Lorentz invariance of $\int \mathrm{d}^4x\,\mathcal{F}^2$. What is still missing is the *other* massless multiplet found in the representation theory of Theorem 44.19: the vector, or gauge, multiplet, the supersymmetric home of a spin-one field. Building it is the business of this section. The guiding principle is not aesthetic but forced: if we want to gauge an internal symmetry of the matter built in Chapter 44 — to repeat, in superspace, the Chapter 8 step that turned a global U(1) into electromagnetism (Section 8.1) — then demanding that the matter Kähler term survive a *local* phase rotation will require a new superfield to compensate, and the representation theory of that compensator will hand us, with no further input, a photon, a Majorana gaugino, and a real auxiliary scalar. We let the gauge principle do the construction and watch the abelian gauge multiplet assemble itself.

Throughout we keep the chapter conventions of Section 44.7: mostly-minus metric $\eta_{\mu\nu}=\mathrm{diag}(+,-,-,-)$, natural units, the two-component spinor calculus of Section C.9 with $\sigma^{\mu}_{\alpha\dot\beta}=(\mathbf{1},\vec\sigma)$ and $\bar\sigma^{\mu\,\dot\alpha\alpha}=(\mathbf{1},-\vec\sigma)$, undotted indices raised and lowered with $\epsilon^{\alpha\beta},\epsilon_{\alpha\beta}$ ($\epsilon^{12}=-\epsilon_{12}=1$, dotted likewise), and the contraction shorthands $\theta\psi=\theta^\alpha\psi_\alpha$, $\bar\theta\bar\psi=\bar\theta_{\dot\alpha}\bar\psi^{\dot\alpha}$, $\theta\theta=\theta^\alpha\theta_\alpha$, $\bar\theta\bar\theta=\bar\theta_{\dot\alpha}\bar\theta^{\dot\alpha}$.

45.1.1 Gauging a symmetry forces a chiral parameter

Begin with the matter we already have. A single chiral superfield Φ with the canonical Kähler term $\int \mathrm{d}^4\theta\,\Phi^\dagger\Phi$ (Eq. (44.205)) has a manifest global U(1) symmetry,

$$\Phi \longrightarrow e^{-iq\,\alpha}\,\Phi, \qquad \Phi^\dagger \longrightarrow \Phi^\dagger\, e^{+iq\,\alpha}, \tag{45.1}$$

with α a real constant and q the charge: the phases cancel in $\Phi^\dagger\Phi$ and the action is unchanged. This is the superspace lift of the global phase symmetry whose gauging produced QED in Chapter 8. The Chapter 8 move was to promote the constant α to a spacetime function $\alpha(x)$ and to repair the resulting non-invariance by introducing a connection A_μ (Section 8.1). We attempt the identical move here, with one non-negotiable constraint imposed by supersymmetry.

The constraint is that the transformed field must *remain chiral*. If Φ obeys $\bar D_{\dot\alpha}\Phi=0$ then the gauge-rotated field must obey it too, or the rotation would carry us out of the chiral representation and the transformation would not be a

symmetry of the superspace structure at all. A local parameter is now itself a superfield $\Lambda(x,\theta,\bar\theta)$, and

$$\Phi \;\longrightarrow\; e^{-2iq\,\Lambda}\,\Phi \tag{45.2}$$

is chiral if and only if $e^{-2iq\Lambda}$ is chiral, i.e. if and only if Λ is itself a chiral superfield:

$$\bar D_{\dot\alpha}\,\Lambda \;=\; 0. \tag{45.3}$$

(The factor 2 and the i are conventional, chosen to make later component formulas clean; q is the charge.) That a product of chiral superfields is chiral, so that $e^{-2iq\Lambda}\Phi$ is chiral whenever Λ and Φ are, is the ring property of Section 44.7.4. The local gauge parameter of a supersymmetric theory *cannot* be a real function: it is forced to be a complex chiral superfield, carrying not just the ordinary gauge function but an entire chiral multiplet of partners. This is the first surprise, and it propagates through everything that follows.

Now watch the Kähler term. Under Eq. (45.2),

$$\Phi^\dagger\Phi \;\longrightarrow\; \Phi^\dagger\, e^{+2iq\,\Lambda^\dagger}\, e^{-2iq\,\Lambda}\,\Phi \;=\; \Phi^\dagger\, e^{-2iq(\Lambda-\Lambda^\dagger)}\,\Phi, \tag{45.4}$$

the last equality holding because the abelian exponents commute. The combination $\Lambda-\Lambda^\dagger$ does not vanish: Λ is chiral and $\Lambda^\dagger$ antichiral, distinct superfields. The Kähler term is *not* invariant, exactly as the matter kinetic term $\bar\psi i\bar\sigma\!\cdot\!\partial\psi$ failed to be invariant under a local phase in Chapter 8. The cure is the same in spirit: introduce a compensating field that absorbs the offending factor. We postulate a real superfield V, to be called the *vector superfield*, entering the Kähler term through an exponential and transforming so as to cancel Eq. (45.4).

Definition 45.1 (Gauge-covariant Kähler term and the super-gauge transformation)**.** Let $V=V^\dagger$ be a real superfield. The *gauge-covariant Kähler term* for a charge-q chiral superfield is

$$\int \mathrm{d}^4\theta\; \Phi^\dagger\, e^{2qV}\,\Phi, \tag{45.5}$$

and the *abelian super-gauge transformation* is the pair

$$\Phi \;\longrightarrow\; e^{-2iq\Lambda}\,\Phi, \qquad V \;\longrightarrow\; V \;+\; i(\Lambda-\Lambda^\dagger), \qquad \bar D_{\dot\alpha}\Lambda = 0. \tag{45.6}$$

The transformation of V is engineered precisely so that Eq. (45.5) is invariant: under Eq. (45.6),

$$\Phi^\dagger e^{2qV}\Phi \;\longrightarrow\; \Phi^\dagger\, e^{+2iq\Lambda^\dagger}\, e^{2qV+2iq(\Lambda-\Lambda^\dagger)}\, e^{-2iq\Lambda}\,\Phi \;=\; \Phi^\dagger\, e^{2qV}\,\Phi, \tag{45.7}$$

where every exponent commutes and the four added pieces $+2iq\Lambda^\dagger + 2iq\Lambda - 2iq\Lambda^\dagger - 2iq\Lambda$ cancel. The compensator V has done its job. Writing $\Lambda+\Lambda^\dagger$ or $i(\Lambda-\Lambda^\dagger)$ is a matter of taste — both are real superfields, both shift V by a real quantity — and we adopt $\delta V = i(\Lambda-\Lambda^\dagger)$ so that the imaginary part of the lowest component of Λ becomes the ordinary gauge function below. The remainder of the section is the unpacking of what this real superfield V contains and how the redundancy Eq. (45.6) prunes it.

45.1.2 The real vector superfield

A real superfield is the most general scalar superfield (Theorem 44.34, Eq. (44.170)) subject to $V = V^\dagger$. The reality condition identifies the complex components of Eq. (44.170) in conjugate pairs, halving the bosonic count and pairing the fermions. Writing out the surviving independent components in the parametrization standard for gauge theory,

Definition 45.2 (Real vector superfield)**.** A *vector superfield* is a real superfield $V = V^\dagger$, with component expansion

$$\begin{aligned} V(x,\theta,\bar\theta) = C + i\,\theta\chi - i\,\bar\theta\bar\chi + \tfrac{i}{2}\,\theta\theta\,(M+iN) - \tfrac{i}{2}\,\bar\theta\bar\theta\,(M-iN) - \theta\sigma^\mu\bar\theta\,A_\mu \\ + i\,\theta\theta\,\bar\theta\bar\lambda - i\,\bar\theta\bar\theta\,\theta\lambda + \tfrac{1}{2}\,\theta\theta\,\bar\theta\bar\theta\,D, \end{aligned} \tag{45.8}$$

in which C, M, N, D are real scalars, A_μ is a real vector, and $\chi_\alpha, \lambda_\alpha$ are Weyl fermions. (The gaugino λ and the scalar D appearing in the top two components have absorbed, by a field redefinition of the form $\lambda \to \lambda + \frac{i}{2}\sigma^\mu\partial_\mu\bar\chi$ and $D \to D + \frac{1}{2}\Box C$, the spacetime derivatives of the lower components, so that they stand alone; this is a harmless shift by derivatives of fields already present.)

Reality is manifest term by term: $(i\theta\chi)^\dagger = -i\bar\theta\bar\chi$, $(\frac{i}{2}\theta\theta(M+iN))^\dagger = -\frac{i}{2}\bar\theta\bar\theta(M-iN)$, $(\theta\sigma^\mu\bar\theta)^\dagger = \theta\sigma^\mu\bar\theta$ for real A_μ, and $(i\theta\theta\,\bar\theta\bar\lambda)^\dagger = -i\bar\theta\bar\theta\,\theta\lambda$, so that $V = V^\dagger$ holds identically. The off-shell content is eight real bosonic components — C, M, N, D, and the four of A_μ — balanced by eight real fermionic components, the four each of χ and λ. This $8_B + 8_F$ count is the vector-multiplet counterpart of the $4_B + 4_F$ of the off-shell chiral multiplet (Eq. (44.190)), and it will be slashed in half by gauge redundancy before the physical multiplet emerges. The vector A_μ sitting at the $\theta\sigma^\mu\bar\theta$ level is the candidate photon; the rest of Eq. (45.8) is, as we now show, either pure gauge or auxiliary.

45.1.3 The super-gauge transformation in components and Wess–Zumino gauge

To see what the transformation Eq. (45.6) does to the components Eq. (45.8), expand the chiral parameter Λ. As a chiral superfield it has the form Eq. (44.190), which in the x-expansion Eq. (44.193) reads

$$\Lambda = a + \sqrt{2}\,\theta\eta + \theta\theta\,f + i\,\theta\sigma^\mu\bar\theta\,\partial_\mu a - \tfrac{i}{\sqrt{2}}\,\theta\theta\,\partial_\mu\eta\,\sigma^\mu\bar\theta - \tfrac{1}{4}\,\theta\theta\,\bar\theta\bar\theta\,\Box a, \tag{45.9}$$

with a a complex scalar, η_α a Weyl fermion, and f a complex scalar. Its conjugate $\Lambda^\dagger$ is the antichiral mirror. Forming $i(\Lambda - \Lambda^\dagger)$ and matching the result against Eq. (45.8) component by component gives the transformation laws of the components of V. The lowest few are

$$C \to C - 2\,\mathrm{Im}\,a, \quad \chi \to \chi - \sqrt{2}\,\eta, \quad M + iN \to M + iN - 2f, \tag{45.10}$$

in which $\operatorname{Im} a$, η, and f are entirely free (they are arbitrary functions parametrizing Λ). These three shifts can therefore be used to set

$$C = 0, \qquad \chi = 0, \qquad M = N = 0, \tag{45.11}$$

exhausting the real part of a's imaginary partner $\operatorname{Im} a$, the spinor η, and the complex f. What survives untouched in this gauge is the $\theta\sigma^\mu\bar\theta$ component, the gaugino, and D.

Definition 45.3 (Wess–Zumino gauge)**.** A vector superfield is in *Wess–Zumino (WZ) gauge* when $C = \chi = M = N = 0$. In this gauge Eq. (45.8) collapses to

$$\boxed{V_{\text{WZ}} = -\,\theta\sigma^\mu\bar\theta\, A_\mu + i\,\theta\theta\,\bar\theta\bar\lambda - i\,\bar\theta\bar\theta\,\theta\lambda + \tfrac{1}{2}\,\theta\theta\,\bar\theta\bar\theta\, D.} \tag{45.12}$$

Theorem 45.4 (Wess–Zumino gauge exists and leaves exactly an ordinary U(1) gauge symmetry)**.** *Every vector superfield can be brought to the form Eq. (45.12) by a super-gauge transformation Eq. (45.6). The residual transformations that preserve WZ gauge are exactly those with* $\operatorname{Im} a = \text{const}\cdot$ *removed,* $\eta = 0$*,* $f = 0$*, and* $\operatorname{Re} a \equiv \frac{1}{2}\alpha(x)$ *an arbitrary real function; under such a residual transformation the components of Eq. (45.12) transform as*

$$A_\mu \;\longrightarrow\; A_\mu + \partial_\mu\alpha, \qquad \lambda \;\longrightarrow\; \lambda, \qquad D \;\longrightarrow\; D. \tag{45.13}$$

That is, the leftover gauge freedom is precisely an ordinary abelian gauge transformation of the photon A_μ*, while the gaugino and the auxiliary field are gauge invariant.*

Proof. Existence is Eq. (45.10): the three shifts there are by free parameters and can be solved to enforce Eq. (45.11), reaching Eq. (45.12). For the residual freedom, demand that a further transformation keep $C = \chi = M = N = 0$. From Eq. (45.10), preserving $C = 0$ requires $\operatorname{Im} a = 0$ (up to the trivial constant that shifts nothing physical), preserving $\chi = 0$ requires $\eta = 0$, and preserving $M = N = 0$ requires $f = 0$. The only surviving freedom is the *real* part of the lowest component, $\operatorname{Re} a$; write $\operatorname{Re} a = \frac{1}{2}\alpha(x)$. With $\operatorname{Im} a = \eta = f = 0$ the parameter Eq. (45.9) reduces to $\Lambda = \frac{1}{2}\alpha + \frac{i}{2}\,\theta\sigma^\mu\bar\theta\,\partial_\mu\alpha - \frac{1}{8}\theta\theta\bar\theta\bar\theta\,\Box\alpha$, a real-lowest-component chiral superfield, and

$$i(\Lambda - \Lambda^\dagger) = i\Big(\tfrac{i}{2}\theta\sigma^\mu\bar\theta\,\partial_\mu\alpha - \big(-\tfrac{i}{2}\theta\sigma^\mu\bar\theta\,\partial_\mu\alpha\big)\Big) + (\text{higher},\ \propto\Box\alpha \text{ cancels}) = -\,\theta\sigma^\mu\bar\theta\,\partial_\mu\alpha, \tag{45.14}$$

where the real-$\Box\alpha$ pieces of Λ and $\Lambda^\dagger$ are equal and cancel in the difference, and the lowest real constant $\frac{1}{2}\alpha$ cancels against its own conjugate. Comparing Eq. (45.14) with the $\theta\sigma^\mu\bar\theta$ term of Eq. (45.12) (coefficient $-A_\mu$), the shift $\delta V = -\theta\sigma^\mu\bar\theta\,\partial_\mu\alpha$ is $\delta(-A_\mu) = -\partial_\mu\alpha$, i.e. $A_\mu \to A_\mu + \partial_\mu\alpha$; the $\theta\theta\bar\theta\bar\lambda$ and $\theta\theta\bar\theta\bar\theta D$ components receive no shift because Λ at this residual level has no contribution at those orders. This is Eq. (45.13). □

Theorem 45.4 is the payoff of the whole construction: the gauge principle of Chapter 8, $A_\mu \to A_\mu + \partial_\mu \alpha$, has reappeared as the residue of a much larger superspace redundancy. The complex scalar a, the spinor η, and the scalar f of the chiral parameter Λ were spent gauging away the unphysical components C, χ, M, N; the single real function $\alpha = 2\,\mathrm{Re}\, a$ that remains is the ordinary photon gauge function. The matter field's response is consistent: under the residual transformation Eq. (45.2) becomes $\Phi \to e^{-2iq\Lambda}\Phi \to e^{-iq\alpha}\Phi$ at the level of lowest components, the global rotation Eq. (45.1) now made local — exactly the Chapter 8 matter transformation $\phi \to e^{-iq\alpha}\phi$ that accompanies $A_\mu \to A_\mu + \partial_\mu \alpha$.

Physical Insight

WZ gauge is a gauge choice, not a supersymmetric statement: the conditions $C = \chi = M = N = 0$ are not preserved by a supersymmetry transformation, since acting with Q on Eq. (45.12) regenerates lower components. The practical attitude, which every component computation in supersymmetric gauge theory adopts, is to work in WZ gauge and accept that a supersymmetry variation must be followed by a compensating supergauge transformation to return to the gauge. What *is* supersymmetric and gauge invariant is the physical content (A_μ, λ, D) modulo Eq. (45.13), and the field strength of the next subsection repackages exactly that content into a manifestly covariant object.

A computational bonus of WZ gauge deserves emphasis because every matter coupling relies on it. Since V_{WZ} in Eq. (45.12) begins at order $\theta\bar\theta$, any product of three or more factors of V_{WZ} exceeds the available four Grassmann coordinates and vanishes:

$$V_{\mathrm{WZ}}^2 = -\tfrac{1}{2}\,\theta\theta\,\bar\theta\bar\theta\,A_\mu A^\mu, \qquad V_{\mathrm{WZ}}^3 = 0. \tag{45.15}$$

The exponential in the covariant Kähler term Eq. (45.5) therefore truncates exactly, $e^{2qV_{\mathrm{WZ}}} = 1 + 2qV_{\mathrm{WZ}} + 2q^2V_{\mathrm{WZ}}^2$, a polynomial of degree two. This is what makes the matter coupling computable in closed form, as we verify in Theorem 45.8.

45.1.4 The chiral field-strength superfield

WZ gauge displays the physical content but is not supersymmetric, and the photon component A_μ is not gauge invariant. We want a gauge-invariant, manifestly supersymmetric object built from V that contains the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ — the superspace analog of the curvature $F_{\mu\nu}$ that Chapter 8 extracted from the connection A_μ. The covariant derivatives $D_\alpha, \bar D_{\dot\alpha}$ of Theorem 44.32 provide it.

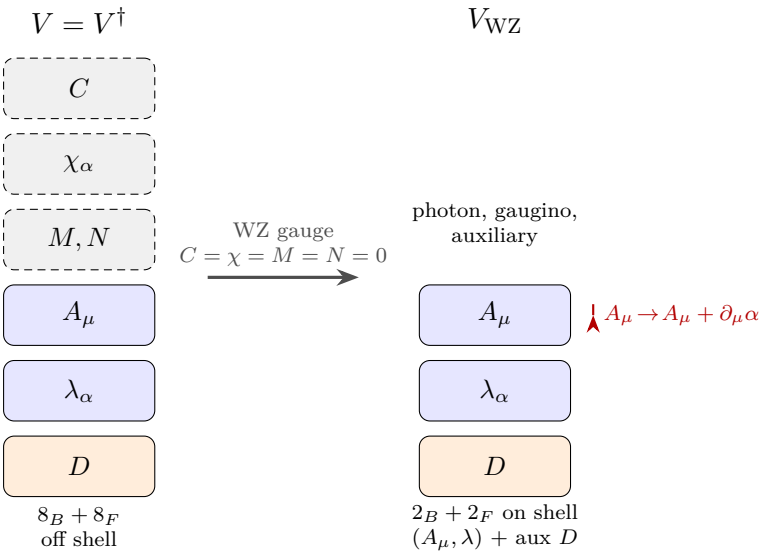


Figure 45.1: The abelian gauge multiplet. The off-shell real vector superfield V carries eight bosonic and eight fermionic components (left). The super-gauge freedom Eq. (45.6), whose parameter is a full chiral superfield Λ, gauges away the redundant lower components C, χ, M, N (dashed), landing in Wess–Zumino gauge Eq. (45.12) with content (A_μ, λ, D). The leftover real function in Λ is exactly the Chapter 8 photon gauge transformation (Theorem 45.4). On shell D is eliminated and A_μ, λ each carry two states, the photon and the Majorana gaugino.

Definition 45.5 (Abelian field-strength superfield)**.** The *field-strength superfield* of an abelian vector superfield V is the spinor-valued superfield

$$\boxed{W_\alpha \;=\; -\tfrac{1}{4}\,\bar{D}\bar{D}\,D_\alpha V \;=\; -\tfrac{1}{4}\,\bar{D}^2\,D_\alpha V,} \qquad \bar{W}_{\dot\alpha} \;=\; -\tfrac{1}{4}\,D^2\,\bar{D}_{\dot\alpha} V, \tag{45.16}$$

where $\bar{D}^2 \equiv \bar{D}^{\dot\alpha}\bar{D}_{\dot\alpha}$ and $D^2 \equiv D^\alpha D_\alpha$.

Two properties make W_α the right object, and both follow from the covariant derivative algebra of Eq. (44.168) with almost no computation.

Theorem 45.6 (W_α is chiral and gauge invariant)**.** *The field-strength superfield Eq. (45.16) satisfies*

$$\bar{D}_{\dot\beta}\, W_\alpha \;=\; 0 \qquad \textit{(chirality)}, \tag{45.17}$$

and is invariant under the abelian super-gauge transformation Eq. (45.6),

$$\delta_\Lambda W_\alpha \;=\; 0. \tag{45.18}$$

Proof. Chirality. The operator $\bar{D}^2 = \bar{D}^{\dot\alpha}\bar{D}_{\dot\alpha}$ is a product of two of the anti-holomorphic covariant derivatives, of which there are only two independent

components, $\bar{D}_{\dot{1}}, \bar{D}_{\dot{2}}$. Any product of three of them must repeat one, and since $\left\{\bar{D}_{\dot{\alpha}}, \bar{D}_{\dot{\beta}}\right\} = 0$ (Eq. (44.168)), a repeated antiholomorphic derivative anticommutes to zero. Hence

$$\bar{D}_{\dot{\beta}}\,\bar{D}^2 \;=\; 0 \qquad \text{(three } \bar{D}\text{'s vanish).} \tag{45.19}$$

Applying $\bar{D}_{\dot{\beta}}$ to Eq. (45.16), $\bar{D}_{\dot{\beta}} W_\alpha = -\frac{1}{4}(\bar{D}_{\dot{\beta}}\bar{D}^2)\, D_\alpha V = 0$ by Eq. (45.19). Thus W_α is chiral.

Gauge invariance. Under Eq. (45.6) the variation of V is $\delta V = i(\Lambda - \Lambda^\dagger)$, so $\delta_\Lambda W_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha\, i(\Lambda - \Lambda^\dagger)$. Treat the two pieces in turn. The antichiral piece $\Lambda^\dagger$ obeys $D_\alpha \Lambda^\dagger = 0$ by definition of antichirality (Theorem 44.39), so $\bar{D}^2 D_\alpha \Lambda^\dagger = 0$ outright. For the chiral piece Λ, use $\bar{D}_{\dot{\alpha}}\Lambda = 0$ and the algebra $\left\{D_\alpha, \bar{D}_{\dot{\beta}}\right\} = -2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu$ (Eq. (44.168)) to commute D_α leftward through the two $\bar{D}$'s:

$$\bar{D}^2 D_\alpha \Lambda = \bar{D}^{\dot{\beta}}\bar{D}_{\dot{\beta}}\, D_\alpha \Lambda = \bar{D}^{\dot{\beta}}(-2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu \Lambda) = -2i\sigma^\mu_{\alpha\dot{\beta}}\,\partial_\mu(\bar{D}^{\dot{\beta}}\Lambda) = 0, \tag{45.20}$$

where the first step replaced $\bar{D}_{\dot{\beta}} D_\alpha \Lambda = -D_\alpha \bar{D}_{\dot{\beta}}\Lambda - 2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu\Lambda = -2i\sigma^\mu_{\alpha\dot{\beta}}\partial_\mu\Lambda$ (the first term vanishing by chirality of Λ), and the last step used chirality again, $\bar{D}^{\dot{\beta}}\Lambda = 0$. Both pieces vanish, so $\delta_\Lambda W_\alpha = 0$. □

The structure here is worth a sentence of reflection. Gauge invariance of W_α is the superspace statement that the field strength does not feel the gauge transformation, exactly as $F_{\mu\nu}$ in Chapter 8 was invariant under $A_\mu \to A_\mu + \partial_\mu\alpha$ while A_μ was not. Here the gauge parameter is an entire chiral superfield Λ, far larger than a function, yet $W_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha V$ annihilates all of it: the combination $\bar{D}^2 D_\alpha$ is the superspace projector onto the gauge-invariant curvature. That W_α is simultaneously chiral makes it a legitimate ingredient for an F-term action, which is what we build next. (Its $\mathrm{U}(1)_R$ charge is $+1$, inherited from the single uncontracted D_α, fixing the gaugino's R-charge; this is the gauge-sector entry of the R-symmetry table of Section 44.2.7.)

45.1.5 Component expansion of W_α

We now extract the components of W_α in WZ gauge. The cleanest route is the chiral coordinate $y^\mu = x^\mu + i\theta\sigma^\mu\bar{\theta}$ of Eq. (44.186), in which a chiral superfield is a function of (y, θ) alone and the covariant derivatives take the reduced form established in Section 44.7.2: since $\bar{D}_{\dot{\alpha}}$ annihilates both y^μ and θ^α (Eq. (44.189)), acting on any superfield written in $(y, \theta, \bar{\theta})$ it reduces to $\bar{D}_{\dot{\alpha}} = -\,\partial/\partial\bar{\theta}^{\dot{\alpha}}$, while the chain rule gives $D_\alpha = \partial/\partial\theta^\alpha + 2i\sigma^\mu_{\alpha\dot{\beta}}\bar{\theta}^{\dot{\beta}}\,\partial/\partial y^\mu$. Consequently $\bar{D}^2 = \bar{D}^{\dot{\alpha}}\bar{D}_{\dot{\alpha}} = \bar{\partial}^{\dot{\alpha}}\bar{\partial}_{\dot{\alpha}}$ in these coordinates, an operator that extracts the $\bar{\theta}\bar{\theta}$ coefficient: a direct

computation in the basis $\bar\theta^{\dot1},\bar\theta^{\dot2}$ with $\epsilon^{\dot1\dot2}=1$ gives $\bar\partial^{\dot\alpha}\bar\partial_{\dot\alpha}(\bar\theta\bar\theta)=4$, so for any g, $\bar D^2 g = 4\,[g]_{\bar\theta\bar\theta}$ in chiral coordinates. Therefore

$$W_\alpha = -\tfrac14\,\bar D^2\,(D_\alpha V) = -\,\big[\,D_\alpha V\,\big]_{\bar\theta\bar\theta}, \tag{45.21}$$

the (negative of the) coefficient of $\bar\theta\bar\theta$ in $D_\alpha V$.

We need $V_{\rm WZ}$ expressed in chiral coordinates. Substituting $x^\mu = y^\mu - i\theta\sigma^\mu\bar\theta$ into Eq. (45.12) and using the Fierz identity $(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac12\eta^{\mu\nu}\theta\theta\bar\theta\bar\theta$ (Eq. (44.191)) to reduce the one cross term that survives,

$$V_{\rm WZ} = -\theta\sigma^\mu\bar\theta\,A_\mu(y) + i\,\theta\theta\,\bar\theta\bar\lambda(y) - i\,\bar\theta\bar\theta\,\theta\lambda(y) + \tfrac12\,\theta\theta\,\bar\theta\bar\theta\,\big[D(y) + i\,\partial_\mu A^\mu(y)\big], \tag{45.22}$$

all fields now functions of y. Applying $D_\alpha = \partial_\alpha + 2i\sigma^\nu_{\alpha\dot\gamma}\bar\theta^{\dot\gamma}\partial_{y\nu}$ and collecting the coefficient of $\bar\theta\bar\theta$ is a short exercise in the Grassmann identities $\bar\theta^{\dot\gamma}\bar\theta^{\dot\delta} = \tfrac12\epsilon^{\dot\gamma\dot\delta}\bar\theta\bar\theta$ and $\bar\theta^{\dot\gamma}\bar\theta_{\dot\delta} = -\tfrac12\delta^{\dot\gamma}_{\dot\delta}\bar\theta\bar\theta$. The ∂_α piece contributes, at order $\bar\theta\bar\theta$, the gaugino from $-i\bar\theta\bar\theta\,\theta\lambda$ and the D-term from $\tfrac12\theta\theta\bar\theta\bar\theta[D+i\partial A]$; the $2i\sigma\bar\theta\,\partial_y$ piece contributes the A_μ-gradient terms (from the $\theta\sigma^\mu\bar\theta A_\mu$ component) and the $\partial\bar\lambda$ term (from the $\theta\theta\bar\theta\bar\lambda$ component). Two facts organize the result. First, the antiholomorphic-derivative piece $-i\theta_\alpha\partial_\mu A^\mu$ generated by ∂_α acting on the D-component combines with the *symmetric* part of $\sigma^\nu\bar\sigma^\mu\partial_\nu A_\mu$ from the $2i\sigma\bar\theta\,\partial_y$ piece and *cancels*, since $\sigma^\nu\bar\sigma^\mu + \sigma^\mu\bar\sigma^\nu = 2\eta^{\mu\nu}$ (Section C.9) projects the symmetric part onto $\partial_\mu A^\mu$ with the opposite sign; this cancellation is the component avatar of the gauge invariance Eq. (45.18). Second, the *antisymmetric* part survives as $\tfrac12(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)\theta\,\partial_\mu A_\nu = (\sigma^\mu\bar\sigma^\nu\theta)\,\tfrac12 F_{\mu\nu}$, exactly the gauge-invariant field strength. Collecting,

$$\boxed{W_\alpha(y,\theta) = \lambda_\alpha(y) + \theta_\alpha\,D(y) - \tfrac{i}{2}\,(\sigma^\mu\bar\sigma^\nu\theta)_\alpha\,F_{\mu\nu}(y) + i\,\theta\theta\,(\sigma^\mu\partial_\mu\bar\lambda(y))_\alpha,} \tag{45.23}$$

with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and $(\sigma^\mu\partial_\mu\bar\lambda)_\alpha = \sigma^\mu_{\alpha\dot\beta}\partial_\mu\bar\lambda^{\dot\beta}$. (The overall sign and normalization of the gaugino field λ are fixed once and for all by demanding that $W_\alpha|_{\theta=0} = \lambda_\alpha$ and that the kinetic term below be canonical with positive energy; this is the same convention that fixed the matter fermion's kinetic term in Eq. (44.207).) The lowest component is the gaugino, the θ-linear component carries the auxiliary D and the antisymmetric field strength $F_{\mu\nu}$, and the $\theta\theta$ component is the gaugino's equation-of-motion operator. The whole gauge-invariant content of the multiplet — λ, D, and $F_{\mu\nu}$ but *not* A_μ itself — has been assembled into one chiral spinor superfield.

45.1.6 The supersymmetric Maxwell action

Because W_α is chiral, the Lorentz scalar $W^\alpha W_\alpha$ is chiral, and its $\theta\theta$ component is a supersymmetry invariant by the F-term theorem (the chiral analog of Theorem 44.36). This is the unique dimension-three gauge-invariant chiral scalar built from one vector multiplet, hence the unique candidate for a renormalizable kinetic action. We extract its $\theta\theta$ component.

Worked Example 45.7 (The $\theta\theta$ component of $W^\alpha W_\alpha$)**.** From Eq. (45.23), multiply W^α by W_α and collect the coefficient of $\theta\theta$. Four products reach order $\theta\theta$:

- *Gaugino kinetic* (lowest $\times$ top, twice): $\lambda^\alpha \cdot i\theta\theta(\sigma^\mu\partial_\mu\bar\lambda)_\alpha$ and its mirror, summing to $2i\,\theta\theta\,\lambda\sigma^\mu\partial_\mu\bar\lambda$.
- *Auxiliary* (θ-linear D with itself): $(\theta^\alpha D)(\theta_\alpha D) = \theta\theta\, D^2$.
- *Field strength squared* (θ-linear F with itself): $-\frac{1}{4}(\sigma^\mu\bar\sigma^\nu\theta)^\alpha(\sigma^\rho\bar\sigma^\sigma\theta)_\alpha F_{\mu\nu}F_{\rho\sigma}$, which the two-component trace identities of Section C.9 reduce to $\theta\theta\big(-\frac{1}{2}F_{\mu\nu}F^{\mu\nu} - \frac{i}{4}\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}\big)$.
- *Cross D–F*: proportional to $\theta\sigma^{\mu\nu}\theta\,F_{\mu\nu}$, which vanishes because $\theta^\alpha\theta_\beta = -\frac{1}{2}\delta^\alpha_\beta\,\theta\theta$ contracts the antisymmetric $F_{\mu\nu}$ against the symmetric trace δ, giving zero.

Assembling the surviving pieces,

$$\big[\,W^\alpha W_\alpha\,\big]_{\theta\theta} = D^2 \;-\; \tfrac{1}{2}F_{\mu\nu}F^{\mu\nu} \;+\; 2i\,\lambda\sigma^\mu\partial_\mu\bar\lambda \;-\; \tfrac{i}{4}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}. \tag{45.24}$$

The antichiral conjugate $\bar W_{\dot\alpha}\bar W^{\dot\alpha}$ has $\bar\theta\bar\theta$ component the complex conjugate of Eq. (45.24), in which F^2 and ϵFF are real (the latter multiplied by $+\frac{i}{4}$) and the gaugino term becomes $-2i\,\partial_\mu\lambda\,\sigma^\mu\bar\lambda$.

The supersymmetric Maxwell action is the F-term of Eq. (45.24) plus its conjugate. In *canonical* normalization,

$$S_{\text{Maxwell}} = \int \mathrm{d}^4x\,\Big[\tfrac{1}{4}\int \mathrm{d}^2\theta\; W^\alpha W_\alpha + \tfrac{1}{4}\int \mathrm{d}^2\bar\theta\;\bar W_{\dot\alpha}\bar W^{\dot\alpha}\Big] = \int \mathrm{d}^4x\,\Big[-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} + i\,\bar\lambda\bar\sigma^\mu\partial_\mu\lambda + \tfrac{1}{2}D^2\Big]. \tag{45.25}$$

We verify the reduction. The F-term picks the $\theta\theta$ coefficient, so $\frac{1}{4}(W^\alpha W_\alpha|_{\theta\theta} + \text{c.c.}) = \frac{1}{2}\,\mathrm{Re}[W^\alpha W_\alpha]_{\theta\theta}$. The real parts of Eq. (45.24): the D^2 and $-\frac{1}{2}F^2$ pieces are real and survive as $\frac{1}{4}(2D^2) = \frac{1}{2}D^2$ and $\frac{1}{4}(-2\cdot\frac{1}{2}F^2) = -\frac{1}{4}F^2$; the ϵFF term, being pure imaginary in Eq. (45.24), cancels against its conjugate (the would-be θ-term is absent at $\theta = 0$); the gaugino term $\frac{1}{4}(2i\lambda\sigma^\mu\partial_\mu\bar\lambda - 2i\partial_\mu\lambda\,\sigma^\mu\bar\lambda)$ becomes, after one integration by parts and the identity $\lambda\sigma^\mu\bar\chi = -\bar\chi\bar\sigma^\mu\lambda$ of Section C.9, the canonical Weyl kinetic term $i\,\bar\lambda\bar\sigma^\mu\partial_\mu\lambda$ with the sign and normalization fixed in Eq. (44.207). The result Eq. (45.25) is exactly the photon Maxwell term of Chapter 8, $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, plus a kinetic term for a single Weyl fermion λ — the *gaugino* — plus the non-propagating square of the auxiliary D.

This pins down the on-shell content. The auxiliary D has algebraic equation of motion $D = 0$ (its only appearance in the pure-gauge action is $\frac{1}{2}D^2$), eliminating it; the photon A_μ contributes two physical polarizations after the gauge freedom Eq. (45.13) removes one and the equation of motion a second, and the gaugino λ, a massless Weyl fermion, carries two helicity states. Boson and fermion

counts match, $2_B = 2_F$, as supersymmetry demands on shell. The abelian gauge multiplet is precisely (A_μ, λ, D): a photon, a Majorana gaugino (the Weyl λ being the chiral half of a self-conjugate Majorana spinor, the photino), and an auxiliary real scalar.

Notation Watch

Three distinct objects in this section wear symbols built from "D" and "θ," and they must not be confused. *(i)* $D_\alpha, \bar{D}_{\dot\alpha}$ are the *superspace* covariant derivatives of Theorem 44.32, carrying spinor indices. *(ii)* D (no index) is the real *auxiliary scalar* of the vector multiplet, the top component of V in Eq. (45.12). *(iii)* $D_\mu = \partial_\mu - iqA_\mu$, appearing in Theorem 45.8 below, is the ordinary *gauge-covariant* derivative of Chapter 8, carrying a vector index. Likewise θ^α is the Grassmann *superspace coordinate*, whereas the θ inside the holomorphic coupling τ of Eq. (45.26) is the real *vacuum angle* of Chapter 42, a c-number with no index. Context and indices disambiguate; we keep the standard symbols rather than multiply notation.

Renormalizability and the freedom to absorb couplings invite a second, equivalent packaging. Rescaling the gauge field to canonical kinetic normalization moves the coupling g out of the matter term and in front of $W^\alpha W_\alpha$, and a CP-odd total-derivative term $\propto \epsilon^{\mu\nu\rho\sigma} F_{\mu\nu} F_{\rho\sigma}$ — the abelian shadow of the θ-term of Chapter 42 — may be added without spoiling supersymmetry, since it too descends from the imaginary part of $\int \mathrm{d}^2\theta\, W^\alpha W_\alpha$. Both are captured by promoting the coefficient to a single complex constant, the *holomorphic gauge coupling*

$$\tau \;\equiv\; \frac{\theta}{2\pi} \;+\; \frac{4\pi i}{g^2}, \tag{45.26}$$

in terms of which the gauge action is the manifestly supersymmetric

$$S_{\text{gauge}} = \frac{1}{8\pi}\,\mathrm{Im}\!\left[\,\tau \int \mathrm{d}^4x\, \mathrm{d}^2\theta\; W^\alpha W_\alpha\right]. \tag{45.27}$$

Expanding with Eq. (45.24) and $\mathrm{Im}[\tau X] = \mathrm{Re}\,\tau\ \mathrm{Im}\,X + \mathrm{Im}\,\tau\ \mathrm{Re}\,X$, the imaginary part $\mathrm{Im}\,\tau = 4\pi/g^2$ multiplies the real content $\mathrm{Re}[W^\alpha W_\alpha]_{\theta\theta} = D^2 - \tfrac{1}{2}F^2 + 2i\lambda\sigma\partial\bar\lambda +$ c.c., producing $-\frac{1}{4g^2}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2g^2}D^2 + \frac{i}{g^2}\bar\lambda\bar\sigma^\mu\partial_\mu\lambda$, while the real part $\mathrm{Re}\,\tau = \theta/2\pi$ multiplies the imaginary ϵFF piece, producing the topological term $\frac{\theta}{32\pi^2}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}$ (equivalently $\frac{\theta}{64\pi^2}\,\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$) in the normalization of Chapter 42. Setting $\theta = 0$ and $g = 1$ recovers Eq. (45.25); the prefactor $1/8\pi$ is exactly what converts $\mathrm{Im}\,\tau = 4\pi/g^2$ into the canonical $-1/4g^2$ in front of F^2. The single holomorphic constant τ thus carries both the gauge coupling and the vacuum angle, and its holomorphic dependence — $W^\alpha W_\alpha$ enters the action linearly in τ, never in $\bar\tau$ — is the structural seed of the nonrenormalization theorems to come.

45.1.7 Coupling matter and recovering the Chapter 8 covariant derivative

It remains to confirm that the covariant Kähler term Eq. (45.5) reproduces, in components, the minimal coupling of Chapter 8 — that the photon A_μ inside V couples to charged matter exactly through $D_\mu = \partial_\mu - iqA_\mu$. This both validates the construction and seeds the next section's matter coupling.

Worked Example 45.8 (Minimal coupling from $\Phi^\dagger e^{2qV}\Phi$)**.** Work in WZ gauge, where $e^{2qV} = 1 + 2qV_{\rm WZ} + 2q^2V_{\rm WZ}^2$ by Eq. (45.15). Expand $\Phi = \phi + \sqrt{2}\theta\psi + \theta\theta F$ (Eq. (44.193)) and collect the $\theta\theta\bar\theta\bar\theta$ coefficient of $\Phi^\dagger e^{2qV}\Phi$.

The order-V^0 term is the free Kähler density $\partial_\mu\phi^*\partial^\mu\phi + i\bar\psi\bar\sigma^\mu\partial_\mu\psi + F^*F$ of Eq. (44.208). The order-V^1 term, $2q\,\Phi^\dagger V_{\rm WZ}\Phi$, contributes through the photon component $-\theta\sigma^\mu\bar\theta A_\mu$. Pairing it with the $(1,1)$-bidegree pieces of $\Phi^\dagger\Phi$ and using $(\theta\sigma^\rho\bar\theta)(\theta\sigma^\mu\bar\theta) = \frac{1}{2}\eta^{\rho\mu}\theta\theta\bar\theta\bar\theta$, the bosonic part yields

$$2q(-A_\rho)\Big[i\,\phi^*\partial^\rho\phi\cdot\tfrac{1}{2} - i\,\partial^\rho\phi^*\,\phi\cdot\tfrac{1}{2}\Big] = -iqA^\mu(\phi^*\partial_\mu\phi - \partial_\mu\phi^*\,\phi), \tag{45.28}$$

the standard current–potential coupling. The order-V^2 term, $2q^2\Phi^\dagger V_{\rm WZ}^2\Phi = 2q^2\phi^*\phi\cdot(-\frac{1}{2}\theta\theta\bar\theta\bar\theta A^2)$, contributes the seagull $-q^2A_\mu A^\mu\,\phi^*\phi$. Adding the free kinetic term,

$$\partial_\mu\phi^*\partial^\mu\phi - iqA^\mu(\phi^*\partial_\mu\phi - \partial_\mu\phi^*\phi) - q^2A_\mu A^\mu\phi^*\phi = (D_\mu\phi)^*(D^\mu\phi), \quad D_\mu = \partial_\mu - iqA_\mu, \tag{45.29}$$

exactly the minimal coupling of Section 8.1 with the standard sign convention $D_\mu = \partial_\mu - iqA_\mu$. The fermion pieces assemble identically into $i\bar\psi\bar\sigma^\mu D_\mu\psi$. Beyond the gauge field, the order-V^1 term also couples matter to the *gaugino* and the *auxiliary* field: the $i\theta\theta\bar\theta\bar\lambda$ and $\frac{1}{2}\theta\theta\bar\theta\bar\theta D$ components of $V_{\rm WZ}$ produce

$$\sqrt{2}\,q\,(\phi^*\psi\lambda + \bar\lambda\bar\psi\,\phi) \;+\; q\,D\,\phi^*\phi, \tag{45.30}$$

a gaugino–matter Yukawa and a D-term coupling, both with strength fixed by the same charge q that controls the photon coupling — the rigid prediction of supersymmetry, with no independent coupling constant to adjust. Collecting,

$$\int \mathrm{d}^4\theta\;\Phi^\dagger e^{2qV}\Phi = (D_\mu\phi)^*(D^\mu\phi) + i\bar\psi\bar\sigma^\mu D_\mu\psi + F^*F + \sqrt{2}\,q(\phi^*\psi\lambda + \text{h.c.}) + q\,D\,\phi^*\phi. \tag{45.31}$$

The D-term coupling $q\,D\,\phi^*\phi$ in Eq. (45.31) is the first appearance of the scalar potential that drives supersymmetric gauge dynamics: combined with the gauge action's $\frac{1}{2}D^2$ from Eq. (45.25), the auxiliary D has equation of motion $D = -q\,\phi^*\phi$ (plus any Fayet–Iliopoulos constant), and integrating it out leaves the positive *D-term potential* $\frac{1}{2}D^2 = \frac{1}{2}q^2(\phi^*\phi)^2$. We have everything needed to state this potential and to promote the construction to a nonabelian group.

The abelian gauge multiplet is complete: a real vector superfield V with its super-gauge redundancy Eq. (45.6), the chiral gauge-invariant field strength $W_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha V$, the supersymmetric Maxwell action packaged by the holomorphic coupling τ, and the gauge-covariant matter coupling $\Phi^\dagger e^{2qV}\Phi$ that recovers Chapter 8's minimal coupling. These three building blocks — the exponential e^{2qV}, the field strength W_α, and the holomorphic action Eq. (45.27) — are exactly what Section 45.2 promotes to a nonabelian gauge group G, where $V = V^a T^a$ becomes matrix valued, the finite super-gauge transformation reads $e^V \to e^{-i\Lambda^\dagger} e^V e^{i\Lambda}$ (Eq. (45.34)), and the covariant field strength $W_\alpha = -\frac{1}{4}\bar{D}^2(e^{-V} D_\alpha e^V)$ carries the gaugino into the adjoint representation alongside the classical Yang–Mills field of Chapter 25.

45.2 Nonabelian Super Yang–Mills

The abelian construction of Section 45.1 gave us everything we need in miniature: a real vector superfield V, a super-gauge transformation $V \to V + i(\Lambda - \Lambda^\dagger)$ built from a *chiral* parameter Λ, the Wess–Zumino gauge that strips V down to the physical triple (A_μ, λ, D), the gauge-invariant chiral field-strength superfield $W_\alpha = -\frac{1}{4}\bar{D}^2 D_\alpha V$ (Eq. (45.16)), and the supersymmetric Maxwell action $\frac{1}{4}\int \mathrm{d}^2\theta\, W^\alpha W_\alpha + \text{h.c.}$ (Eq. (45.25)) whose component content is a photon, a gaugino, and an auxiliary D. Every one of these objects has a nonabelian counterpart, and the purpose of this section is to build it. The single structural change is that V becomes *matrix-valued*, $V = V^a T^a$, and the gauge transformation becomes a conjugation rather than a shift. That one change propagates through the entire construction: the field strength acquires a covariant derivative, the action acquires the cubic and quartic gluon self-interactions of Chapter 25, and the gaugino is promoted to the adjoint representation. What emerges is *super Yang–Mills theory*, the supersymmetric completion of the classical gauge theory of Chapter 25. We will see that its bosonic sector is, term for term, the classical Yang–Mills Lagrangian Eq. (25.104); that the gauge coupling and the θ-angle of Chapter 42 fuse into a single complex chiral coefficient, the holomorphic coupling $\tau = \theta/2\pi + 4\pi i/g^2$; and that quantization is carried out by feeding the Wess–Zumino-gauge component fields into the Faddeev–Popov/BRST machinery already built in Chapter 26.

Throughout we keep the chapter conventions: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, two-component spinors with $\sigma^\mu_{\alpha\dot\beta} = (\mathbf{1}, \vec\sigma)$ and $\bar\sigma^{\mu\,\dot\alpha\alpha} = (\mathbf{1}, -\vec\sigma)$, and the superspace covariant derivatives $D_\alpha, \bar{D}_{\dot\alpha}$ of Eq. (44.183) obeying $\left\{D_\alpha, \bar{D}_{\dot\beta}\right\} = -2i\,\sigma^\mu_{\alpha\dot\beta}\partial_\mu$ and $\{D_\alpha, D_\beta\} = \left\{\bar{D}_{\dot\alpha}, \bar{D}_{\dot\beta}\right\} = 0$. The gauge group is a compact simple Lie group G (default SU(N)) with Hermitian generators T^a in the representation on which the fields sit, $\left[T^a, T^b\right] = i f^{abc} T^c$ and $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the fundamental; the structure constants f^{abc}, the dual

Coxeter number/adjoint Casimir C_A, and the index T_F are those collected in Section B.4.

45.2.1 The matrix vector superfield and the finite super-gauge transformation

The cleanest route to the nonabelian gauge multiplet is, exactly as in Chapter 25, to demand that the matter kinetic term be gauge invariant and let that requirement dictate the transformation law of the gauge superfield. Let Φ be a chiral superfield (Theorem 44.39) valued in a representation R of G, so that Φ^i carries a representation index i and the generators act as matrices $(T^a)^i{}_j$. The supersymmetric, gauge-covariant kinetic term is the Kähler integral

$$\int \mathrm{d}^4\theta\; \Phi^\dagger e^V \Phi \;=\; \int \mathrm{d}^4\theta\; \Phi^\dagger_{\bar\imath}\,(e^V)^{\bar\imath}{}_j\,\Phi^j, \qquad V \equiv V^a\,T^a, \tag{45.32}$$

the matrix exponential of the real, matrix-valued vector superfield $V = V^aT^a$ replacing the bare $\Phi^\dagger\Phi$ of Eq. (44.205). Reality of the V^a together with Hermiticity of the T^a makes V a Hermitian matrix superfield, $V = V^\dagger$, so that e^V is Hermitian and Eq. (45.32) is real. The exponential is the right object because gauge invariance is multiplicative: a chiral superfield transforms by a chiral matrix, and matrices compose by multiplication, not addition.

Concretely, generalize the abelian rule. In the abelian theory the matter field transformed by $\Phi \to e^{-i\Lambda}\Phi$ with Λ chiral (a chiral parameter preserves chirality, since $\bar D_{\dot\alpha}(e^{-i\Lambda}\Phi) = e^{-i\Lambda}\bar D_{\dot\alpha}\Phi = 0$ when $\bar D\Lambda = 0$). The nonabelian generalization is the same with $\Lambda = \Lambda^aT^a$ a matrix of chiral superfields:

$$\Phi \;\longrightarrow\; e^{-i\Lambda}\,\Phi, \qquad \Phi^\dagger \;\longrightarrow\; \Phi^\dagger\, e^{\,i\Lambda^\dagger}, \qquad \bar D_{\dot\alpha}\Lambda = 0,\;\; D_\alpha\Lambda^\dagger = 0. \tag{45.33}$$

Here $\Lambda^\dagger$ is antichiral, the Hermitian conjugate of the chiral Λ. For Eq. (45.32) to be invariant we need $\Phi^\dagger e^V\Phi \to \Phi^\dagger e^{\,i\Lambda^\dagger}\, e^{V'}\, e^{-i\Lambda}\Phi$ to equal $\Phi^\dagger e^V\Phi$, which fixes the transformation of the gauge superfield to be

$$\boxed{\;e^V \;\longrightarrow\; e^{V'} \;=\; e^{-i\Lambda^\dagger}\, e^V\, e^{\,i\Lambda}\;} \tag{45.34}$$

the *finite nonabelian super-gauge transformation*. This is the central definition of the section. It reduces to the abelian rule when everything commutes: there $e^{-i\Lambda^\dagger}e^Ve^{i\Lambda} = e^{V-i\Lambda^\dagger+i\Lambda}$, i.e. $V \to V + i(\Lambda - \Lambda^\dagger)$, the shift of Eq. (45.6). In the nonabelian case the conjugation by the chiral group element $e^{i\Lambda}$ on the right and the antichiral $e^{-i\Lambda^\dagger}$ on the left does not collapse to a shift, and the ordered structure is exactly what will produce the covariant derivative in the field strength.

Infinitesimal form. Write $\Lambda = \Lambda^a T^a$ infinitesimal and seek δV such that $e^{V+\delta V} = e^{-i\Lambda^\dagger} e^V e^{i\Lambda}$ to first order. Using the variation formula $\delta e^V = \int_0^1 e^{sV}(\delta V) e^{(1-s)V}\, ds$ and expanding the right-hand side to first order in $\Lambda, \Lambda^\dagger$ gives, after a short manipulation with the operator $\mathcal{L}_{V^\cdot} \equiv [V, \cdot]$,

$$\delta V \;=\; i(\Lambda - \Lambda^\dagger) \;+\; \tfrac{i}{2}\Big[V, \Lambda + \Lambda^\dagger\Big] \;+\; O(V^2)\,, \tag{45.35}$$

the full series being $\delta V = i\,\mathcal{L}_{V/2}\Big[\coth(\mathcal{L}_{V/2})(\Lambda - \Lambda^\dagger) + (\Lambda + \Lambda^\dagger)\Big]$, with $\mathcal{L}_{V/2^\cdot} = \frac{1}{2}[V, \cdot]$. We will never need the full series: the work below is done either with the finite law Eq. (45.34) directly, or in Wess–Zumino gauge, where V is nilpotent and the displayed terms of Eq. (45.35) are exact.

Notation Watch

The matrix superfield $V = V^a T^a$ is dimensionless in the holomorphic normalization used here: the gauge coupling g does *not* appear in V, in e^V, or in the field strength below. It is carried entirely by the single prefactor of the action, Eq. (45.55). In the alternative *canonical* normalization the component gauge field is rescaled $A_\mu \to g A_\mu$ (equivalently $V \to 2gV$) so that the kinetic term is $-\frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu}$ with g inside the covariant derivative, as in Chapter 25. We will give the result in both normalizations and state explicitly which is which; conflating them is the single most common source of stray factors of g in this subject.

The Wess–Zumino gauge of Section 45.1 survives the promotion to nonabelian intact. The abelian shift $V \to V + i(\Lambda - \Lambda^\dagger)$ used the lower components of Λ to gauge away the lower components of V; the leading term $i(\Lambda - \Lambda^\dagger)$ of Eq. (45.35) is identical, so the same components of Λ remove the same components of V, leaving the WZ-gauge form

$$V_{\rm WZ} = -\,\theta\sigma^\mu\bar\theta\, A_\mu + i\,\theta\theta\,\bar\theta\bar\lambda - i\,\bar\theta\bar\theta\,\theta\lambda + \tfrac{1}{2}\,\theta\theta\,\bar\theta\bar\theta\, D, \tag{45.36}$$

with $A_\mu = A^a_\mu T^a$, $\lambda_\alpha = \lambda^a_\alpha T^a$, $D = D^a T^a$ all matrix-valued (and Hermitian as matrices, A_μ, D real, $\bar\lambda = \lambda^\dagger$). The commutator term in Eq. (45.35) only shuffles these surviving components among themselves and acts on A_μ as an ordinary nonabelian gauge rotation, exactly as in Chapter 25: the residual freedom after fixing WZ gauge is precisely the ordinary spacetime gauge symmetry $A_\mu \to U A_\mu U^{-1} + \frac{i}{?}\dots$ whose component form we recover below. The crucial structural fact we will use repeatedly is that Eq. (45.36) is *nilpotent*: every term carries at least one $\theta\sigma\bar\theta$, and a product of three such terms has more than two θ's or more than two $\bar\theta$'s and so vanishes. Hence

$$V_{\rm WZ}^3 = 0, \qquad V_{\rm WZ}^2 = \tfrac{1}{2}\,\theta\theta\,\bar\theta\bar\theta\; A_\mu A^\mu, \qquad e^{V_{\rm WZ}} = 1 + V_{\rm WZ} + \tfrac{1}{2} V_{\rm WZ}^2, \tag{45.37}$$

where the value of V^2 follows from $(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac{1}{2}\theta\theta\,\bar\theta\bar\theta\,\eta^{\mu\nu}$ (the vector–vector Fierz identity Eq. (44.191)), so that the only surviving square is the gauge-field bilinear $A_\mu A^\mu$ (a matrix product, hence not symmetric in the colour indices).

45.2.2 The covariant field strength W_α

The abelian field strength $W_\alpha = -\tfrac{1}{4}\bar D^2 D_\alpha V$ is replaced by

$$\boxed{W_\alpha \;=\; -\tfrac{1}{4}\,\bar D^2\Big(e^{-V}\,D_\alpha\,e^{V}\Big)\,, \qquad W_\alpha = W_\alpha^a T^a.} \tag{45.38}$$

Here $\bar D^2 \equiv \bar D_{\dot\alpha}\bar D^{\dot\alpha}$ and $e^{-V}D_\alpha e^V$ is the supersymmetric analogue of the Maurer–Cartan form $U^{-1}\partial_\mu U$: it is the natural gauge-covariant object built from a group element, and applying $-\tfrac{1}{4}\bar D^2$ projects it onto a chiral superfield. In the abelian limit $e^{-V}D_\alpha e^V = D_\alpha V$ and Eq. (45.38) collapses to Eq. (45.16), as it must. Two properties make W_α the correct field strength: it is chiral, and it transforms in the adjoint by conjugation.

Proposition 45.9 (Chirality of W_α)**.** *The superfield W_α defined by Eq. (45.38) is chiral, $\bar D_{\dot\beta} W_\alpha = 0$.*

Proof. The dotted index $\dot\beta$ takes only two values, and $\big\{\bar D_{\dot\alpha}, \bar D_{\dot\beta}\big\} = 0$ makes the $\bar D$'s mutually anticommuting. Any product of three of them therefore vanishes: writing $\bar D_{\dot\beta}\bar D^2 = \bar D_{\dot\beta}\bar D^{\dot\gamma}\bar D_{\dot\gamma}$, the three factors live in a two-dimensional anticommuting space and so contain a repeated index, killing the product. Since W_α is $-\tfrac{1}{4}\bar D^2$ acting on something, $\bar D_{\dot\beta}W_\alpha = -\tfrac{1}{4}\bar D_{\dot\beta}\bar D^2(\cdots) = 0$. □

Chirality is exactly what is needed to use W_α in an F-term: by Theorem 44.41 the integral $\int \mathrm{d}^2\theta$ of any chiral quantity is supersymmetric, so $\int \mathrm{d}^2\theta\ \mathrm{Tr}(W^\alpha W_\alpha)$ is a candidate supersymmetric Lagrangian provided it is also gauge invariant. That is the content of the covariance theorem, the key result of this subsection.

Theorem 45.10 (Gauge covariance of W_α)**.** *Under the finite super-gauge transformation Eq. (45.34), the field strength transforms homogeneously in the adjoint,*

$$W_\alpha \;\longrightarrow\; e^{-i\Lambda}\,W_\alpha\,e^{\,i\Lambda}, \tag{45.39}$$

and consequently $\mathrm{Tr}(W^\alpha W_\alpha)$ is gauge invariant.

Proof. Write $\Omega \equiv e^{-V}D_\alpha e^V$ for the object inside the projection. Under Eq. (45.34) we have $e^V \to e^{-i\Lambda^\dagger}e^V e^{i\Lambda}$ and, taking the inverse, $e^{-V} \to e^{-i\Lambda}e^{-V}e^{i\Lambda^\dagger}$. We must transform $D_\alpha e^V$. By the Leibniz rule for the (odd) covariant derivative,

$$D_\alpha\Big(e^{-i\Lambda^\dagger}e^V e^{i\Lambda}\Big) = (D_\alpha e^{-i\Lambda^\dagger})\,e^V e^{i\Lambda} + e^{-i\Lambda^\dagger}(D_\alpha e^V)e^{i\Lambda} + e^{-i\Lambda^\dagger}e^V(D_\alpha e^{i\Lambda}). \tag{45.40}$$

The first term vanishes: $\Lambda^\dagger$ is antichiral, $D_\alpha \Lambda^\dagger = 0$ (Eq. (45.33)), hence $D_\alpha e^{-i\Lambda^\dagger} = 0$. Assembling $\Omega' = e^{-V'} D_\alpha e^{V'}$ with the transformed $e^{-V'} = e^{-i\Lambda} e^{-V} e^{i\Lambda^\dagger}$ and the surviving two terms of Eq. (45.40),

$$\begin{aligned}\Omega' &= e^{-i\Lambda} e^{-V} e^{i\Lambda^\dagger} \Big[e^{-i\Lambda^\dagger} (D_\alpha e^{V}) e^{i\Lambda} + e^{-i\Lambda^\dagger} e^{V} (D_\alpha e^{i\Lambda}) \Big] \\ &= e^{-i\Lambda}\, e^{-V} (D_\alpha e^{V})\, e^{i\Lambda} + e^{-i\Lambda}\, e^{-V} e^{V} (D_\alpha e^{i\Lambda}) \\ &= e^{-i\Lambda}\, \Omega\, e^{i\Lambda} + e^{-i\Lambda} (D_\alpha e^{i\Lambda}),\end{aligned} \tag{45.41}$$

where the cancellation $e^{i\Lambda^\dagger} e^{-i\Lambda^\dagger} = 1$ and $e^{-V} e^{V} = 1$ were used. The first term is the desired covariant piece; the second, an inhomogeneous "pure-gauge" term, must be annihilated by the projection $-\frac{1}{4}\bar{D}^2$. Apply it:

$$W'_\alpha = -\tfrac{1}{4}\bar{D}^2 (e^{-i\Lambda}\, \Omega\, e^{i\Lambda}) - \tfrac{1}{4}\bar{D}^2 (e^{-i\Lambda}\, D_\alpha e^{i\Lambda}). \tag{45.42}$$

For the first term, recall that Λ is chiral, $\bar{D}_{\dot\beta}\Lambda = 0$, so $\bar{D}_{\dot\beta} e^{\pm i\Lambda} = 0$: the chiral group elements are covariantly constant under $\bar{D}$. Because $\bar{D}_{\dot\beta}$ is a graded derivation and annihilates $e^{\pm i\Lambda}$, it passes through them untouched, and the same holds for the second $\bar{D}$; hence

$$\bar{D}^2 (e^{-i\Lambda}\, \Omega\, e^{i\Lambda}) = e^{-i\Lambda}\, (\bar{D}^2 \Omega)\, e^{i\Lambda}, \tag{45.43}$$

so the first term of Eq. (45.42) is exactly $e^{-i\Lambda}\, W_\alpha\, e^{i\Lambda}$, the claimed transformation. It remains to show the second term vanishes. We claim

$$\bar{D}^2 (D_\alpha X) = 0 \qquad \text{for any chiral } X\ (\bar{D}_{\dot\beta} X = 0), \tag{45.44}$$

which applies to $X = e^{-i\Lambda} D_\alpha e^{i\Lambda}$ being a product of chiral objects (the factor $e^{-i\Lambda}$ is chiral and $D_\alpha e^{i\Lambda}$ is chiral because $\bar{D}_{\dot\beta} D_\alpha e^{i\Lambda} = \left\{ \bar{D}_{\dot\beta}, D_\alpha \right\} e^{i\Lambda} = -2i\sigma^\mu_{\alpha\dot\beta} \partial_\mu e^{i\Lambda}$, itself chiral since $\bar{D}(\partial_\mu e^{i\Lambda}) = \partial_\mu \bar{D} e^{i\Lambda} = 0$). To prove Eq. (45.44), compute using $\bar{D}_{\dot\beta} X = 0$ and the algebra $\left\{ D_\alpha, \bar{D}_{\dot\beta} \right\} = -2i\sigma^\mu_{\alpha\dot\beta}\partial_\mu$:

$$\bar{D}_{\dot\beta}\, D_\alpha X = \left\{ \bar{D}_{\dot\beta}, D_\alpha \right\} X - D_\alpha \bar{D}_{\dot\beta} X = -2i\, \sigma^\mu_{\alpha\dot\beta}\, \partial_\mu X, \tag{45.45}$$

and then apply the second $\bar{D}^{\dot\beta}$: $\bar{D}^{\dot\beta} \bar{D}_{\dot\beta} (D_\alpha X) = -2i\sigma^\mu_{\alpha\dot\beta}\, \bar{D}^{\dot\beta} \partial_\mu X = -2i\sigma^\mu_{\alpha\dot\beta}\, \partial_\mu (\bar{D}^{\dot\beta} X) = 0$, since ∂_μ commutes with $\bar{D}$ and X is chiral. This is exactly Eq. (45.44), so the inhomogeneous term in Eq. (45.42) vanishes and Eq. (45.39) is proved. Finally, under Eq. (45.39),

$$\mathrm{Tr}(W^\alpha W_\alpha) \to \mathrm{Tr}(e^{-i\Lambda} W^\alpha e^{i\Lambda}\, e^{-i\Lambda} W_\alpha e^{i\Lambda}) = \mathrm{Tr}(e^{-i\Lambda}\, W^\alpha W_\alpha\, e^{i\Lambda}) = \mathrm{Tr}(W^\alpha W_\alpha), \tag{45.46}$$

the last equality by cyclicity of the trace (and $e^{i\Lambda} e^{-i\Lambda} = 1$). The trace of $W^\alpha W_\alpha$ is gauge invariant. □

The two mechanisms that powered this proof are worth stating plainly, because they are the supersymmetric avatars of facts familiar from Chapter 25. First, the inhomogeneous "connection" piece $e^{-i\Lambda}D_\alpha e^{i\Lambda}$ of Eq. (45.41) is the exact analogue of the $U^{-1}\partial_\mu U$ term in the gauge transformation of an ordinary connection; the field strength is curvature, and curvature transforms homogeneously. Second, the reason the inhomogeneous piece drops is purely a chirality count (Eq. (45.44)): three $\bar{D}$'s acting effectively on a chiral object cannot survive. Holomorphy, here in the guise of chirality of the gauge parameter Λ, is doing the work; the same holomorphy will, in Chapter 46, forbid quantum corrections to the form of the gauge kinetic term.

45.2.3 Component expansion in Wess–Zumino gauge

We now compute W_α in Wess–Zumino gauge and exhibit its components. The nilpotency Eq. (45.37) truncates the exponentials, so the Maurer–Cartan form is exactly

$$\begin{aligned} e^{-V}D_\alpha e^{V} &= (1 - V + \tfrac{1}{2}V^2)\, D_\alpha (V + \tfrac{1}{2}V^2) \\ &= (1 - V + \tfrac{1}{2}V^2)\Big[D_\alpha V + \tfrac{1}{2}(D_\alpha V)V + \tfrac{1}{2}V(D_\alpha V)\Big]. \end{aligned} \tag{45.47}$$

Multiplying out and discarding every term with three or more factors of V (which vanish by Eq. (45.37)), the surviving terms are $D_\alpha V$ at order V, and at order V^2 the combination $\frac{1}{2}(D_\alpha V)V + \frac{1}{2}V(D_\alpha V) - V(D_\alpha V)$. The order-$V^2$ piece is $\frac{1}{2}(D_\alpha V)V - \frac{1}{2}V(D_\alpha V) = -\frac{1}{2}\,[V, D_\alpha V]$. Hence the exact, closed-form result in WZ gauge is

$$e^{-V}D_\alpha e^{V} = D_\alpha V \;-\; \tfrac{1}{2}\,[V, D_\alpha V]\,, \tag{45.48}$$

and therefore

$$W_\alpha = \underbrace{-\tfrac{1}{4}\,\bar{D}^2\, D_\alpha V}_{\text{abelian-form}} \;+\; \underbrace{\tfrac{1}{8}\,\bar{D}^2\,[V, D_\alpha V]}_{\text{nonabelian completion}}\,. \tag{45.49}$$

The first term is precisely the abelian field-strength operator of Eq. (45.16) applied component by component; its evaluation was carried out in Section 45.1, and it produces, in chiral coordinates $y^\mu = x^\mu + i\theta\sigma\bar{\theta}$,

$$-\tfrac{1}{4}\bar{D}^2 D_\alpha V = \lambda_\alpha + \Big[\delta_\alpha{}^\beta D - i\,(\sigma^{\mu\nu})_\alpha{}^\beta\,(\partial_\mu A_\nu - \partial_\nu A_\mu)\Big]\theta_\beta + i\,\theta\theta\,(\sigma^\mu\partial_\mu\bar{\lambda})_\alpha, \tag{45.50}$$

now matrix-valued (each component carries its colour matrix). The second term of Eq. (45.49) dresses this abelian skeleton with commutators. Because the commutator $[V, D_\alpha V]$ is built from two V's, the Fierz reductions Eq. (44.191)

that produced Eq. (45.37) apply, and the surviving content is governed entirely by the gauge-field bilinear $[A_\mu, A_\nu]$ and by the gauge-field–gaugino bilinear $\left[A_\mu, \bar\lambda\right]$. Carrying out the $\bar D^2$ projection on $\frac{1}{8}[V, D_\alpha V]$ term by term (the algebra is identical to the abelian projection, with the antisymmetric matrix structure of the commutator carried along) supplies exactly the two missing pieces: the antisymmetrized derivative $\partial_\mu A_\nu - \partial_\nu A_\mu$ is completed to the nonabelian curvature, and the ordinary derivative $\partial_\mu \bar\lambda$ is completed to the adjoint covariant derivative. The result is the gauge-covariant field-strength superfield

$$\boxed{W_\alpha = \lambda_\alpha + \left[\delta_\alpha{}^\beta D - i\,(\sigma^{\mu\nu})_\alpha{}^\beta\, \mathcal{F}_{\mu\nu}\right]\theta_\beta + i\,\theta\theta\,(\sigma^\mu \mathcal{D}_\mu \bar\lambda)_\alpha,} \tag{45.51}$$

with the matrix field strength and the adjoint covariant derivative

$$\mathcal{F}_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu - i\,[A_\mu, A_\nu]\,, \tag{45.52}$$

$$\mathcal{D}_\mu \bar\lambda = \partial_\mu \bar\lambda - i\,\left[A_\mu, \bar\lambda\right]. \tag{45.53}$$

In components, with $A_\mu = A_\mu^a T^a$ and $\left[T^b, T^c\right] = i f^{bcd} T^d$, the matrix relation Eq. (45.52) reads $\mathcal{F}^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc} A^b_\mu A^c_\nu$ and the covariant derivative reads $(\mathcal{D}_\mu \bar\lambda)^a = \partial_\mu \bar\lambda^a + f^{abc} A^b_\mu \bar\lambda^c$. This is the field strength of Eq. (25.49) and the adjoint-representation covariant derivative of Chapter 25, written without the explicit g because we are in the holomorphic normalization (Section 45.2.1). The promotion of ∂_μ to $\mathcal{D}_\mu$ on the gaugino is the precise statement that *the gaugino lives in the adjoint*: it is a colour-octet (for SU(3)) Weyl fermion, transforming like the gauge field itself, as the covariance Eq. (45.39) demands. The components of W_α are therefore the physical fields of the gauge multiplet: a gaugino λ_α at order θ^0, the field strength and auxiliary D at order θ^1, and the covariant gaugino kinetic term at order θ^2.

45.2.4 The super Yang–Mills action and the holomorphic coupling

Having a chiral, gauge-covariant W_α, the unique renormalizable supersymmetric and gauge-invariant kinetic term is the F-term of $\mathrm{Tr}(W^\alpha W_\alpha)$. We package its coefficient into a single complex number, the *holomorphic gauge coupling*.

Definition 45.11 (Holomorphic gauge coupling)**.** The holomorphic gauge coupling is the complex combination

$$\tau \;\equiv\; \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}, \tag{45.54}$$

where g is the Yang–Mills coupling of Chapter 25 and θ is the vacuum angle of Chapter 42. Its real part is the θ-angle (mod 2π); its imaginary part, positive for $g^2 > 0$, is $4\pi/g^2$.

Definition 45.12 (Super Yang–Mills action)**.** The pure super Yang–Mills action for gauge group G is

$$\boxed{S_{\mathrm{SYM}} = \frac{1}{4\pi}\,\mathrm{Im}\left[\,\tau \int d^4x\, \mathrm{d}^2\theta\ \mathrm{Tr}\,(W^\alpha W_\alpha)\right],} \tag{45.55}$$

the imaginary part supplying the Hermitian conjugate (F-term plus $\bar{F}$-term) automatically.

It remains to expand Eq. (45.55) into components and verify that it is the supersymmetric completion of classical Yang–Mills. The Grassmann integral $\int \mathrm{d}^2\theta$ extracts the $\theta\theta$ coefficient of $\mathrm{Tr}(W^\alpha W_\alpha)$. We compute that coefficient from Eq. (45.51).

The bosonic $\theta\theta$ coefficient. The terms in $W^\alpha W_\alpha$ that reach order $\theta\theta$ and carry no fermion come from the product of the two order-θ^1 brackets of Eq. (45.51). Writing $M_\alpha{}^\beta \equiv \delta_\alpha{}^\beta D - i(\sigma^{\mu\nu})_\alpha{}^\beta \mathcal{F}_{\mu\nu}$, the relevant product is $(\theta M)^\alpha (\theta M)_\alpha$ where $(\theta M)_\alpha \equiv M_\alpha{}^\beta \theta_\beta$. Using $\theta_\gamma\theta_\delta = -\frac{1}{2}\epsilon_{\gamma\delta}\theta\theta$ (the Grassmann identity fixed in Section 44.6), the bilinear collapses to

$$(\theta M)^\alpha (\theta M)_\alpha = -\tfrac{1}{2}\,\theta\theta\ M_\alpha{}^\beta M_\beta{}^\alpha = -\tfrac{1}{2}\,\theta\theta\ \mathrm{tr}_2(M^2), \tag{45.56}$$

where tr_2 is the trace over the two-dimensional spinor space. Expanding M^2,

$$\mathrm{tr}_2(M^2) = \mathrm{tr}_2\Big(D^2\mathbf{1} - 2iD\,\sigma^{\mu\nu}\mathcal{F}_{\mu\nu} - (\sigma^{\mu\nu}\sigma^{\rho\sigma})\,\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma}\Big). \tag{45.57}$$

The first trace is $\mathrm{tr}_2(\mathbf{1}) = 2$, giving $2D^2$. The second vanishes because $\sigma^{\mu\nu}$ is traceless, $\mathrm{tr}_2(\sigma^{\mu\nu}) = 0$ (it is a Lorentz generator, Eq. (44.34)). The third requires the spinor trace of two Lorentz generators, the self-duality identity derived in Section 3.4.3 (and collected in Appendix C),

$$\mathrm{tr}_2(\sigma^{\mu\nu}\sigma^{\rho\sigma}) = \tfrac{1}{2}(\eta^{\mu\rho}\eta^{\nu\sigma} - \eta^{\mu\sigma}\eta^{\nu\rho}) - \tfrac{i}{2}\,\epsilon^{\mu\nu\rho\sigma}, \tag{45.58}$$

with $\epsilon^{0123} = +1$. Contracting with $\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma}$ and using $\epsilon^{\mu\nu\rho\sigma}\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma} = 2\,\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu}$ with $\widetilde{\mathcal{F}}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\mathcal{F}_{\rho\sigma}$,

$$\mathrm{tr}_2(\sigma^{\mu\nu}\sigma^{\rho\sigma})\,\mathcal{F}_{\mu\nu}\mathcal{F}_{\rho\sigma} = \mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} - i\,\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu}. \tag{45.59}$$

Assembling, the bosonic $\theta\theta$ coefficient is $-\frac{1}{2}(2D^2 - \mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} + i\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu}) = \frac{1}{2}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} - \frac{i}{2}\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu} - D^2$. The crucial structural fact is the appearance of *both* $\mathcal{F}^2$ (real) and $\mathcal{F}\widetilde{\mathcal{F}}$ (imaginary): the spinor trace Eq. (45.58) has split $W^\alpha W_\alpha$ into a parity-even and a parity-odd part, and they sit in the real and imaginary parts of one chiral quantity.

The fermionic $\theta\theta$ coefficient. The product of the order-θ^0 gaugino λ_α with the order-θ^2 term $i\theta\theta(\sigma^\mu\mathcal{D}_\mu\bar\lambda)_\alpha$ (and the symmetric partner) gives, after using $\lambda^\alpha(\sigma^\mu\mathcal{D}_\mu\bar\lambda)_\alpha = \lambda\sigma^\mu\mathcal{D}_\mu\bar\lambda$, the coefficient $2i\,\lambda\sigma^\mu\mathcal{D}_\mu\bar\lambda$, which integration by parts under $\int d^4x$ and the identity $\lambda\sigma^\mu\chi = -\chi\bar\sigma^\mu\lambda$ converts to $+2i\,\bar\lambda\,\bar\sigma^\mu\mathcal{D}_\mu\lambda$. Collecting,

$$\int \mathrm{d}^2\theta\ \mathrm{Tr}\,(W^\alpha W_\alpha) = \mathrm{Tr}\Big[\tfrac{1}{2}\mathcal{F}_{\mu\nu}\mathcal{F}^{\mu\nu} - \tfrac{i}{2}\mathcal{F}_{\mu\nu}\widetilde{\mathcal{F}}^{\mu\nu} + 2i\,\bar\lambda\,\bar\sigma^\mu\mathcal{D}_\mu\lambda - D^2\Big]. \quad (45.60)$$

Taking the colour trace with $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ replaces each $\mathrm{Tr}(XY)$ by $\frac{1}{2}X^aY^a$, so $\int \mathrm{d}^2\theta\ \mathrm{Tr}(W^\alpha W_\alpha) = \frac{1}{4}\mathcal{F}^a_{\mu\nu}\mathcal{F}^{a\,\mu\nu} - \frac{i}{4}\mathcal{F}^a_{\mu\nu}\widetilde{\mathcal{F}}^{a\,\mu\nu} + i\,\bar\lambda^a\bar\sigma^\mu\mathcal{D}_\mu\lambda^a - \frac{1}{2}D^aD^a$ (with the gaugino and D terms real).

Assembling the action. Insert into Eq. (45.55) and use Eq. (45.54). Writing the chiral quantity as $X = \int \mathrm{d}^2\theta\ \mathrm{Tr}(W^\alpha W_\alpha) = \mathrm{Re}\,X + i\,\mathrm{Im}\,X$, with $\mathrm{Re}\,X$ collecting $\frac{1}{4}\mathcal{F}^2 + i\bar\lambda\bar\sigma\mathcal{D}\lambda - \frac{1}{2}D^2$ (the parity-even, Hermitian terms) and $\mathrm{Im}\,X = -\frac{1}{4}\mathcal{F}^a\widetilde{\mathcal{F}}^a$,

$$\mathrm{Im}(\tau X) = \frac{4\pi}{g^2}\,\mathrm{Re}\,X + \frac{\theta}{2\pi}\,\mathrm{Im}\,X. \quad (45.61)$$

The factor $1/4\pi$ then gives

$$S_{\mathrm{SYM}} = \int d^4x\,\Big\{\frac{1}{g^2}\Big[-\tfrac{1}{4}\mathcal{F}^a_{\mu\nu}\mathcal{F}^{a\,\mu\nu} + i\,\bar\lambda^a\bar\sigma^\mu\mathcal{D}_\mu\lambda^a + \tfrac{1}{2}D^aD^a\Big] + \frac{\theta}{32\pi^2}\,\mathcal{F}^a_{\mu\nu}\widetilde{\mathcal{F}}^{a\,\mu\nu}\Big\}, \quad (45.62)$$

where we absorbed the overall sign by the conventional choice that the action be bounded below (equivalently, by matching the abelian limit to the supersymmetric Maxwell action Eq. (45.25) of the previous section, which fixes every relative coefficient). This is the super Yang–Mills Lagrangian in holomorphic normalization: a single chiral coefficient τ multiplies one chiral density, and the split into $-\frac{1}{4}\mathcal{F}^2$ and the θ-density is the split of τ into imaginary and real parts.

Canonical normalization and the bridge to Chapters 25 and 42. Rescale the gauge multiplet to canonical kinetic normalization by $A_\mu \to gA_\mu$, $\lambda \to g\lambda$, $D \to gD$ (equivalently $V \to 2gV$). Then $\mathcal{F}^a_{\mu\nu} \to g\,F^a_{\mu\nu}$ with $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc}A^b_\mu A^c_\nu$ the field strength of Eq. (25.49), and the covariant derivative becomes $(D_\mu\lambda)^a = \partial_\mu\lambda^a + g f^{abc}A^b_\mu\lambda^c$. The prefactor $1/g^2$ cancels the g^2 from $(\mathcal{F})^2 = g^2F^2$, and the θ-density picks up the missing g^2 from $\mathcal{F} = gF$:

$$\boxed{\mathcal{L}_{\mathrm{SYM}} = -\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + i\,\bar\lambda^a\,\bar\sigma^\mu(D_\mu\lambda)^a + \tfrac{1}{2}D^aD^a + \frac{\theta g^2}{32\pi^2}\,F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu}.} \quad (45.63)$$

Read this Lagrangian one term at a time. The first term is *exactly* the classical Yang–Mills Lagrangian Eq. (25.104), with its cubic and quartic gluon

self-interactions hidden inside $F^a_{\mu\nu}$; the bosonic gauge sector of super Yang–Mills *is* the theory of Chapter 25, not an approximation to it. The second term is the kinetic energy of the gaugino, an adjoint Weyl fermion λ^a minimally coupled to the gauge field through the adjoint covariant derivative D_μ — the same covariant derivative that acts on any adjoint field, here producing a gluon–gaugino interaction with strength gf^{abc}. The third term is the auxiliary field D^a, which appears with no derivative and no source in the pure gauge theory and is therefore set to zero by its equation of motion (it becomes nonzero only once charged matter is added, in Section 45.3). The fourth term is the topological θ-density $\frac{\theta g^2}{32\pi^2} F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu}$, identical term-for-term and prefactor-for-prefactor to the θ-term whose vacuum meaning was fixed in Section 42.4 (Eq. (42.128)): a total derivative, $F\widetilde{F} = \partial_\mu K^\mu$, which is invisible in perturbation theory but records, through the instanton sectors of Chapter 42, the angle of the true θ-vacuum. Holomorphy has done something remarkable: the gauge coupling g and the topological angle θ, which look completely unrelated in Chapters 25 and 42, are revealed to be the imaginary and real parts of a single holomorphic coupling τ multiplying a single chiral operator.

Physical Insight: The coupling as the geometry of a torus

Because θ is an angle, $\theta \to \theta + 2\pi$ is a symmetry; in terms of τ it is $\tau \to \tau + 1$. The instanton expansion organizes itself in powers of $e^{2\pi i\tau} = e^{i\theta}e^{-8\pi^2/g^2}$, precisely the Boltzmann weight $e^{i\theta}e^{-S_{\text{inst}}}$ of one instanton (Section 42.4). The holomorphic coupling thus lives naturally on the upper half-plane $\operatorname{Im}\tau > 0$ with the periodicity $\tau \sim \tau + 1$, and in theories with enough supersymmetry a second generator $\tau \to -1/\tau$ (electric–magnetic duality) completes this to a modular group acting on τ as on the modular parameter of a torus. That nonperturbative story belongs to Chapter 46; here we note only that packaging g and θ into τ is what makes it possible to ask the question at all.

Worked Example 45.13 (Off-shell and on-shell degree-of-freedom counting)**.** We verify the equality $n_B = n_F$ guaranteed by the algebra (Theorem 44.15) for the vector multiplet, both off-shell and on-shell, and see why the auxiliary D is needed.

Off-shell. Count real field components before imposing equations of motion. The gauge field A^a_μ has, per colour, four components, of which one is removed by the residual ordinary gauge invariance left after fixing Wess–Zumino gauge (the residual $A_\mu \to A_\mu + D_\mu\alpha$ of Section 45.2.1); that leaves $4 - 1 = 3$ bosonic components. The auxiliary D^a is one real scalar. The bosonic total is therefore $n_B = 3 + 1 = 4$ per colour. The gaugino λ^a_α is a single Weyl spinor: two complex components, hence four real Grassmann components, $n_F = 4$ per colour, off-shell

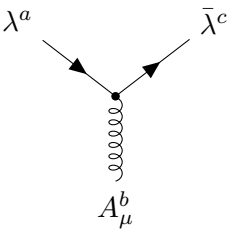


Figure 45.2: The gluon–gaugino–gaugino vertex $g f^{abc} A^b_\mu \bar{\lambda}^a \bar{\sigma}^\mu \lambda^c$ supplied by the adjoint covariant derivative in Eq. (45.63). The gaugino is an adjoint Weyl fermion, so its coupling to the gauge field carries the structure constants f^{abc} exactly as the three-gluon vertex does (Fig. 25.3a); the gluon self-interactions of Chapter 25 are inherited unchanged.

with no equation of motion imposed. Thus

$$n_B = \underbrace{3}_{A_\mu} + \underbrace{1}_{D} = 4 = \underbrace{4}_{\lambda_\alpha \text{ (complex Weyl)}} = n_F, \tag{45.64}$$

and the equality holds *only because* of the auxiliary D: without it $n_B = 3 \neq 4 = n_F$, and supersymmetry would close only on-shell. The single real D is exactly the deficit, the supersymmetric mirror of the auxiliary F of the chiral multiplet (Eq. (44.208)).

On-shell. Imposing the equations of motion, the massless gauge field A^a_μ propagates two transverse polarizations; the auxiliary D^a satisfies the algebraic equation $D^a = 0$ and carries none; so $n_B = 2$. The gaugino's Dirac equation halves its components to two real on-shell degrees of freedom (a massless Weyl/Majorana fermion has two helicity states), $n_F = 2$. Again $n_B = n_F = 2$, the count of a massless vector multiplet: a gauge boson of helicities ± 1 and a gaugino of helicities $\pm\frac{1}{2}$. This is the content already anticipated by the algebra in Section 44.3.4.

The new interaction that distinguishes super Yang–Mills from the bosonic theory of Chapter 25 is the minimal coupling of the adjoint gaugino to the gauge field, the $g f^{abc} A^b_\mu \bar{\lambda}^a \bar{\sigma}^\mu \lambda^c$ vertex hidden in the covariant derivative of Eq. (45.63). Figure 45.2 records it alongside the inherited gluon self-couplings.

Physical Insight: Why super Yang–Mills is the rigid completion of Chapter 25

Classical Yang–Mills had one free parameter, g (and the discrete-seeming θ). Super Yang–Mills adds no new parameter: the gaugino mass is forbidden by gauge invariance (an adjoint Weyl fermion has no gauge-invariant mass term) and by supersymmetry (it would split the multiplet, violating Theorem 44.12), and the gaugino coupling is fixed to be g by the covariant

derivative. There is no room for a Yukawa coupling or a quartic scalar coupling because there is no scalar. The entire dynamics is dictated by the single τ. This rigidity is what will let Chapter 46 compute the low-energy effective coupling exactly, and it is already visible here as the absence of any freedom beyond g and θ.

45.2.5 Gauge fixing and BRST in Wess–Zumino gauge

The Lagrangian Eq. (45.63) is classical and gauge invariant; quantizing it requires gauge fixing, and the apparatus for that was built in full generality in Chapter 26. Nothing in that construction is special to a purely bosonic gauge field, so we may apply it directly to the Wess–Zumino-gauge component fields, with one honest caveat about supersymmetry that we state precisely.

The route is the one of Section 26.2. We have committed to Wess–Zumino gauge, which fixed the *super*-gauge redundancy down to the ordinary spacetime gauge transformations $A^a_\mu \to A^a_\mu + (D_\mu \alpha)^a$ acting on the component gauge field of Eq. (45.36). This residual ordinary gauge invariance is exactly the redundancy of Chapter 25, and it is removed by the Faddeev–Popov procedure of Section 26.2: insert $1 = \int \mathcal{D}\alpha\, \delta(G[A^\alpha])\, \det(\delta G/\delta\alpha)$ for a gauge-fixing functional $G[A]$ (e.g. $G = \partial^\mu A^a_\mu$ in R_ξ gauge), exponentiate the determinant with a pair of adjoint anticommuting Faddeev–Popov ghosts $c^a, \bar{c}^a$, and add the gauge-fixing and ghost terms

$$\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}} = -\frac{1}{2\xi}\left(\partial^\mu A^a_\mu\right)^2 + \bar{c}^a\left(-\partial^\mu D_\mu c\right)^a, \tag{45.65}$$

with $(D_\mu c)^a = \partial_\mu c^a + g f^{abc} A^b_\mu c^c$ the same adjoint covariant derivative that acts on the gaugino. The ghosts $c^a, \bar{c}^a$ are the ordinary Faddeev–Popov ghosts of Chapter 26; they are *not* part of any supermultiplet and should not be confused with the gaugino λ^a, which is a physical adjoint fermion of the gauge multiplet Eq. (45.36). The gauge-fixed theory then carries the BRST symmetry of Section 26.5: the nilpotent transformation $sA^a_\mu = (D_\mu c)^a$, $sc^a = -\frac{1}{2} g f^{abc} c^b c^c$, $s\bar{c}^a = B^a$, $sB^a = 0$ (with B^a the Nakanishi–Lautrup auxiliary), under which the gaugino transforms in the adjoint, $s\lambda^a = g f^{abc} c^b \lambda^c$, exactly as any adjoint matter field does. Physical states are the BRST cohomology classes of Section 26.6, and the Slavnov–Taylor identities of Section 26.7 control renormalization. The perturbative Feynman rules are those of Section 26.3 augmented by the gaugino propagator (a massless adjoint Weyl fermion) and the single new vertex of Fig. 45.2; the gauge sector, ghost sector, and BRST charge are imported wholesale from Chapter 26.

Caution: Wess–Zumino gauge breaks manifest supersymmetry

The price of working in Wess–Zumino gauge is that supersymmetry is no longer manifest. A supersymmetry transformation δ_ξ acting on the WZ-gauge superfield Eq. (45.36) does not preserve the WZ form: it regenerates the lower components C, χ, M, N of V that WZ gauge set to zero. To stay in WZ gauge one must follow δ_ξ by a compensating super-gauge transformation Eq. (45.34) that removes the regenerated components. The composite "δ_ξ + compensating gauge transformation" is the supersymmetry that actually acts on the physical component fields, and its algebra closes only up to a field-dependent gauge transformation, $\{\delta_{\xi_1}, \delta_{\xi_2}\} = (\text{translation}) + (\text{gauge transformation})$. This is not a defect of the theory — the off-shell algebra of Chapter 44 still holds on the unfixed superfield — but a feature of the gauge choice: WZ gauge trades manifest supersymmetry for the absence of unphysical fields. Consequently, the BRST charge Q_B of the gauge-fixed theory and the (composite, WZ-gauge) supercharge Q_α do not commute on individual fields; only a combined cohomology is well-defined.

The honest statement at the perturbative level we work to here is therefore the following. The gauge-fixed action $S_{\text{SYM}} + \int d^4x\,(\mathcal{L}_{\text{gf}} + \mathcal{L}_{\text{gh}})$ is invariant under the BRST transformation s, and the physical S-matrix is supersymmetric because the broken-then-compensated supersymmetry acts on the BRST-cohomology (physical) states as the genuine super-Poincaré symmetry of Chapter 44: the supersymmetry Ward identities hold on gauge-invariant correlators even though no single off-shell symmetry is manifest. A fully manifestly-supersymmetric quantization exists — superspace Feynman rules with a super-Faddeev–Popov determinant and a ghost-for-ghost tower, because the super-gauge parameter Λ is itself a reducible (chiral-superfield) gauge parameter — and it is the efficient tool for proving the nonrenormalization theorems of Section 45.4. But for reading off the particle content, the beta function, and the comparison to Chapter 27, the component WZ-gauge quantization just described, built directly on the Chapter 26 machinery, is the appropriate and complete framework.

This advances Yang–Mills theory one decisive step: the bosonic content of super Yang–Mills is, by Eq. (45.63), the classical Yang–Mills theory of Chapter 25; its quantization reuses, without modification, the Faddeev–Popov/BRST apparatus of Chapter 26; and the holomorphic coupling τ carries the θ-angle of Chapter 42. The one-loop beta-function coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ computed in Eq. (27.124) will be recovered, and then sharpened to an exact formula, once we add the gaugino's adjoint contribution and the matter of the next section; that comparison is the through-line from here to Chapter 46.

With the pure-gauge dynamics in hand — a covariant field strength W_α, the component Lagrangian Eq. (45.63), and a quantization built on Chapter 26 —

the remaining ingredient is matter. The gauge-covariant Kähler term $\Phi^\dagger e^V \Phi$ of Eq. (45.32), whose invariance under Eq. (45.33) we have already established, is precisely the coupling of charged chiral superfields to this gauge multiplet. Combining it with N_f flavours of quark and antiquark chiral superfields and the super Yang–Mills action Eq. (45.55) builds supersymmetric QCD, whose scalar potential and classical moduli space are the subject of the next section.

45.3 Supersymmetric QCD: Matter, the Scalar Potential, and the Classical Moduli Space

The previous section completed the gauge sector: a nonabelian vector superfield $V = V^a T^a$ whose component content is exactly the classical Yang–Mills field of Chapter 25 together with a gaugino λ^a in the adjoint, governed by the super Yang–Mills action and packaged by the holomorphic coupling $\tau = \theta/2\pi + 4\pi i/g^2$ (Section 45.2). Along the way the finite super-gauge transformation

$$e^V \longrightarrow e^{-i\Lambda^\dagger} e^V e^{i\Lambda}, \qquad \Lambda = \Lambda^a T^a \text{ chiral}, \tag{45.66}$$

forced the gauge multiplet to carry the exponential e^V rather than V itself (Eq. (45.66)). That exponential is precisely the object a chiral matter field must lean on to be gauged covariantly. Pure super Yang–Mills is the SUSY analog of gluodynamics; to build the SUSY analog of QCD — to extend Yang–Mills theory from pure gauge to gauge + matter — we now couple charged chiral superfields to V. The result, *supersymmetric QCD* (SQCD), is the most rigidly constrained renormalizable gauge theory one can write: supersymmetry and gauge invariance together fix the matter Lagrangian completely, leaving as the only freedom the gauge group, the matter representation, and a holomorphic superpotential. Reading the scalar potential off that Lagrangian and solving for its zeros will hand us a continuous family of degenerate vacua — the classical moduli space — whose gauge-invariant coordinates (mesons and baryons) are exactly the variables on which the nonperturbative analysis of Chapter 46 will act.

45.3.1 The gauge-covariant Kähler term

Recall from Section 44.7.6 that the kinetic energy of a chiral superfield Φ comes from the Kähler term $\int \mathrm{d}^4\theta\, \Phi^\dagger \Phi$, whose component reduction Eq. (44.208) is the free, canonically normalized kinetic Lagrangian of a complex scalar and a Weyl fermion together with the algebraic $F^* F$. We now charge Φ under the gauge group: let Φ transform in a (in general reducible) unitary representation R with Hermitian generators T^a, $[T^a, T^b] = i f^{abc} T^c$, normalized by $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the fundamental (Section B.4). The defining gauge transformation of charged

matter is the chiral rotation

$$\Phi \longrightarrow e^{i\Lambda}\Phi, \qquad \Phi^\dagger \longrightarrow \Phi^\dagger e^{-i\Lambda^\dagger}, \qquad \Lambda \equiv \Lambda^a T^a, \tag{45.67}$$

with Λ a *chiral* superfield-valued gauge parameter (so that $e^{i\Lambda}\Phi$ is again chiral: a holomorphic function of chiral superfields is chiral, Theorem 44.40). Because $\Lambda \neq \Lambda^\dagger$ in general, the naive bilinear $\Phi^\dagger\Phi$ is *not* invariant: under Eq. (45.67) it rotates to $\Phi^\dagger e^{-i\Lambda^\dagger} e^{i\Lambda}\Phi$, and the mismatched exponentials do not cancel. The cure is exactly the exponential the gauge sector already supplies.

Definition 45.14 (Gauge-covariant Kähler term)**.** For chiral matter Φ in the representation R with vector superfield $V = V^a T^a_{(R)}$, the *minimal* (canonical) gauge-invariant Kähler term is

$$S^{\text{kin}}_{\text{matter}} = \int d^4x \int \mathrm{d}^4\theta\, \Phi^\dagger\, e^V\, \Phi, \tag{45.68}$$

where $e^V = \sum_{n\geq 0} V^n/n!$ acts on Φ in the representation R.

Theorem 45.15 (Super-gauge invariance of the covariant Kähler term)**.** *The functional Eq. (45.68) is invariant under the simultaneous transformations Eq. (45.67) of the matter and Eq. (45.66) of the gauge multiplet.*

Proof. Under the two transformations the integrand becomes

$$\Phi^\dagger\, e^V\, \Phi \longrightarrow (\Phi^\dagger e^{-i\Lambda^\dagger})(e^{-i\Lambda^\dagger})^{-1}\cdots \tag{45.69}$$

which we evaluate directly. The matter factors rotate to $\Phi^\dagger e^{-i\Lambda^\dagger}$ on the left and $e^{i\Lambda}\Phi$ on the right, while the gauge factor becomes $e^{-i\Lambda^\dagger} e^V e^{i\Lambda}$. Assembling,

$$\Phi^\dagger\, e^V\, \Phi \longrightarrow \Phi^\dagger\, e^{-i\Lambda^\dagger}\, (e^{-i\Lambda^\dagger})^{-1} e^V\, e^{i\Lambda}\, (e^{i\Lambda})^{-1}\Phi \;=\; \Phi^\dagger\, e^V\, \Phi. \tag{45.70}$$

The transformation matrices cancel in conjugate pairs: $e^{-i\Lambda^\dagger}$ from the matter on the left meets its inverse inside the transformed e^V, and $e^{i\Lambda}$ on the right meets its inverse likewise. The integrand is therefore pointwise invariant, hence so is its full-superspace integral. (No total-derivative argument is needed for the integrand itself; what the D-term theorem Theorem 44.36 of the previous chapter supplies is the orthogonal statement that the $\mathrm{d}^4\theta$-integral of *any* real superfield is supersymmetric, so the term is simultaneously gauge- and SUSY-invariant.) □

Remark 45.16 (Why e^V, and why minimal)*.* Three requirements pin down Eq. (45.68) uniquely. First, *gauge invariance* forces the conjugate fields to be bridged by an object transforming as $e^V \to e^{-i\Lambda^\dagger} e^V e^{i\Lambda}$; the vector superfield's defining transformation is exactly this, and no polynomial in V shares it. Second, *the flat-space limit*: setting $V \to 0$ must return the free Kähler term Eq. (44.208), which $\Phi^\dagger e^V \Phi$ does since $e^0 = 1$. Third, *renormalizability*: the canonical Kähler

potential $K = \Phi^\dagger e^V \Phi$ has a flat (field-independent) Kähler metric $\partial^2 K/\partial\Phi^\dagger\partial\Phi = \mathbf{1}$, so the scalar kinetic term is canonically normalized and the term is marginal; any function $K(\Phi^\dagger e^V \Phi)$ beyond the linear one introduces dimensionful couplings and is non-renormalizable. The combination Eq. (45.68) is thus *the* minimal coupling, the superspace descendant of the gauge principle of Chapters 8 and 25.

45.3.2 Component expansion in Wess–Zumino gauge

To read off physics we reduce Eq. (45.68) to components. As in Eq. (45.36), work in Wess–Zumino gauge, where the residual gauge freedom has been used to set the lowest components of V to zero, leaving

$$V \;=\; -\,\theta\sigma^\mu\bar\theta\, g A_\mu^a T^a \;+\; i\,\theta\theta\,\bar\theta\bar\lambda^a T^a \;-\; i\,\bar\theta\bar\theta\,\theta\lambda^a T^a \;+\; \tfrac{1}{2}\,\theta\theta\bar\theta\bar\theta\, g D^a T^a, \tag{45.71}$$

with the coupling g displayed explicitly on the gauge field and the auxiliary D, so that the gauge field's covariant derivative will emerge as $D_\mu = \partial_\mu - igA_\mu^a T^a$ in the convention of Chapter 25 and Yang–Mills theory. The decisive simplification of WZ gauge is that V *nilpotent*: its lowest surviving piece carries one factor of $\theta\sigma\bar\theta$, so

$$V^2 = (\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta)\, g^2 A_\mu^a A_\nu^b T^a T^b = \tfrac{1}{2}\,\theta\theta\bar\theta\bar\theta\, g^2 A_\mu^a A^{b\mu} T^a T^b, \tag{45.72}$$

where we used the Fierz reduction $(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac{1}{2}\theta\theta\bar\theta\bar\theta\,\eta^{\mu\nu}$ (the same identity Eq. (44.191) that produced the scalar kinetic term in the free case), and $V^3 = 0$ because there is no room for three factors of $\theta\sigma\bar\theta$. Hence the exponential truncates:

$$e^V = 1 + V + \tfrac{1}{2}V^2, \tag{45.73}$$

and the entire coupling is a finite polynomial. We must extract the $\theta\theta\bar\theta\bar\theta$ component of $\Phi^\dagger e^V \Phi$. Write the chiral superfield in the y-coordinates of Section 44.7.3 with components (ϕ, ψ, F) and organize the calculation by which power of V contributes.

The V^0 piece. This is the free Kähler term, already reduced in Eq. (44.208):

$$\big[\Phi^\dagger\Phi\big]_{\theta\theta\bar\theta\bar\theta} = \partial_\mu\phi^\dagger\partial^\mu\phi + i\,\bar\psi\,\bar\sigma^\mu\partial_\mu\psi + F^\dagger F, \tag{45.74}$$

with the colour/flavour indices on ϕ, ψ, F left implicit (they are vectors in the representation space of R).

The V^1 piece. We need $[\Phi^\dagger\, V\, \Phi]_{\theta\theta\bar\theta\bar\theta}$. Only three structures in V can pair with components of $\Phi^\dagger, \Phi$ to reach Grassmann bidegree $(2,2)$:

- The gauge-field term $-\theta\sigma^\mu\bar\theta\, g A^a_\mu T^a$ must be saturated by one $(1,0)$ factor and one $(0,1)$ factor from the matter. The relevant matter pieces are the θ-linear part of Φ ($\sqrt{2}\,\theta\psi$ and the descendant $-i\,\theta\sigma^\nu\bar\theta\,\partial_\nu\phi$ in x-coordinates) and their conjugates. Collecting the cross terms and using $(\theta\sigma^\mu\bar\theta)(\theta\sigma^\nu\bar\theta) = \tfrac{1}{2}\theta\theta\bar\theta\bar\theta\,\eta^{\mu\nu}$ once more, the bosonic part assembles into

$$-\,igA^a_\mu(\phi^\dagger T^a\partial^\mu\phi - \partial^\mu\phi^\dagger\, T^a\phi), \tag{45.75}$$

  i.e. the gauge-current $\times$ gauge-field coupling, while the fermionic part gives $gA^a_\mu\,\bar\psi\,\bar\sigma^\mu T^a\psi$.

- The gaugino terms $i\theta\theta\,\bar\theta\bar\lambda^a T^a$ and $-i\bar\theta\bar\theta\,\theta\lambda^a T^a$ pair the gaugino with one matter fermion and one matter scalar. The first pairs $\bar\lambda$ with the lowest scalar $\phi^\dagger$ of $\Phi^\dagger$ and the $\sqrt{2}\,\theta\psi$ of Φ; the second is its conjugate. These produce the gaugino–matter Yukawa

$$\sqrt{2}\,i\,g\,(\phi^\dagger T^a\psi)^-\ldots \;\longrightarrow\; -\sqrt{2}\,g(\phi^\dagger T^a\psi)\lambda^a - \sqrt{2}\,g\,\bar\lambda^a(\bar\psi\,T^a\phi). \tag{45.76}$$

- The auxiliary term $\tfrac{1}{2}\theta\theta\bar\theta\bar\theta\, gD^aT^a$ is already top-component; it pairs with the two lowest scalars to give

$$\tfrac{1}{2}\,gD^a\,\phi^\dagger T^a\phi \;+\; (\text{from } V^1 \text{ symmetrization}) \;=\; gD^a\,\phi^\dagger T^a\phi. \tag{45.77}$$

  (The factor combines the explicit $\tfrac{1}{2}$ in Eq. (45.71) with the $\tfrac{1}{2}$ from $\int \mathrm{d}^4\theta\,\theta\theta\bar\theta\bar\theta = 1$ being applied to the symmetric product $\phi^\dagger\{T^a\}\phi$; we fix the net normalization below by matching to the gauge-current structure.)

The $\tfrac{1}{2}V^2$ piece. By Eq. (45.72) this is already top- component, so it pairs only with the two lowest scalars:

$$\Big[\Phi^\dagger\,\tfrac{1}{2}V^2\,\Phi\Big]_{\theta\theta\bar\theta\bar\theta} = \tfrac{1}{2}\,\phi^\dagger(g^2A^a_\mu A^{b\mu}T^aT^b)\phi = \tfrac{1}{2}\,g^2A^a_\mu A^{b\mu}\,\phi^\dagger T^aT^b\phi. \tag{45.78}$$

This is the "seagull" contact term quadratic in the gauge field.

Assembly into covariant derivatives. The point of the gauge principle is that Eq. (45.74), Eq. (45.75) and Eq. (45.78) recombine into manifestly covariant kinetic terms. For the scalar, expand

$$(D_\mu\phi)^\dagger(D^\mu\phi) = \partial_\mu\phi^\dagger\partial^\mu\phi + igA^a_\mu(\partial_\mu\phi^\dagger T^a\phi - \phi^\dagger T^a\partial^\mu\phi) + g^2A^a_\mu A^{b\mu}\,\phi^\dagger T^aT^b\phi, \tag{45.79}$$

with $D_\mu\phi = (\partial_\mu - igA^a_\mu T^a)\phi$. The three lines of Eq. (45.79) are exactly Eq. (45.74) (scalar part), Eq. (45.75), and Eq. (45.78); the abelian-current and seagull pieces supplied by the V and V^2 terms are precisely the completions that turn ∂_μ into D_μ. For the fermion, the free term $i\bar\psi\bar\sigma^\mu\partial_\mu\psi$ plus the $gA^a_\mu\bar\psi\bar\sigma^\mu T^a\psi$ of the V^1 piece

combine to $i\bar{\psi}\bar{\sigma}^{\mu}D_{\mu}\psi$ with the same covariant derivative. Collecting everything, the matter Kähler term reduces to the off-shell component Lagrangian

$$\boxed{\begin{aligned}\mathcal{L}^{\text{kin}}_{\text{matter}} &= (D_{\mu}\phi)^{\dagger}(D^{\mu}\phi) + i\,\bar{\psi}\,\bar{\sigma}^{\mu}D_{\mu}\psi + F^{\dagger}F \\ &\quad - \sqrt{2}\,g\big[(\phi^{\dagger}T^{a}\psi)\,\lambda^{a} + \bar{\lambda}^{a}(\bar{\psi}\,T^{a}\phi)\big] + g\,D^{a}\,\phi^{\dagger}T^{a}\phi.\end{aligned}} \tag{45.80}$$

Every term has a clear origin: the covariant scalar and fermion kinetic terms from the gauge completion of Eq. (44.208), the $F^{\dagger}F$ auxiliary term unchanged, the gaugino–quark–squark Yukawa $\sqrt{2}\,g\,\phi^{\dagger}T^{a}\psi\lambda^{a}$ from the gaugino components of V, and the D-dependent term $gD^{a}\phi^{\dagger}T^{a}\phi$ from the auxiliary component of V. The $\sqrt{2}$ in the Yukawa is not adjustable: it is the same $\sqrt{2}$ that appears in $\Phi = \phi + \sqrt{2}\theta\psi + \ldots$, fixed once and for all by the SUSY algebra, and it ties the strength of the gaugino–matter coupling to the gauge coupling g. This is the supersymmetric rigidity in action: gauging a chiral multiplet introduces *no* new free parameters beyond g itself.

Notation Watch

The sign and normalization of the D-dependent term depend on the conventions chosen for the components of V in Eq. (45.71) and for the super Yang–Mills auxiliary term in Eq. (45.63). We fix them so that the on-shell elimination below yields the standard non-negative potential $V = \sum|F|^2 + \frac{1}{2}\sum(D^{a})^2$; only the product of the matter D-coupling and the SYM D^2-coefficient is physical, and it is convention-independent. Throughout we write g for the canonically-normalized gauge coupling of Yang–Mills theory.

45.3.3 The full SQCD Lagrangian

We now specialize to supersymmetric QCD itself. The gauge group is $G = \mathrm{SU}(N_c)$. The matter is N_f *flavors* of *quark* chiral superfields Q_i $(i = 1, \ldots, N_f)$ in the fundamental $\mathbf{N_c}$, together with N_f *antiquark* chiral superfields $\widetilde{Q}^{\,i}$ in the antifundamental $\overline{\mathbf{N_c}}$. Writing the colour index $a = 1, \ldots, N_c$ explicitly, Q^{a}_{i} carries an upper (fundamental) colour index and $\widetilde{Q}^{\,i}_{a}$ a lower (antifundamental) one; each is a full chiral multiplet (ϕ, ψ, F) — a complex *squark* scalar, a Weyl *quark* fermion, and an auxiliary field. The vector multiplet $(A^{a}_{\mu}, \lambda^{a}, D^{a})$ lives in the adjoint of $\mathrm{SU}(N_c)$. This pairing of conjugate representations is the SUSY analog of one Dirac quark of QCD: a Dirac fermion in $\mathbf{N_c}$ is two Weyl fermions, one in $\mathbf{N_c}$ (from Q) and one in $\overline{\mathbf{N_c}}$ (from $\widetilde{Q}$), so N_f flavors of *Dirac* quark become N_f pairs $(Q, \widetilde{Q})$. The vector-like content is what allows a gauge-invariant mass term, as we will see.

Definition 45.17 (Supersymmetric QCD)**.** SQCD is the $\mathrm{SU}(N_c)$ gauge theory with N_f flavors $(Q_i, \widetilde{Q}^{\,i})$ of chiral matter in $\mathbf{N_c} \oplus \overline{\mathbf{N_c}}$, with the action

$$S_{\mathrm{SQCD}} = \int d^4x \Big[\tfrac{1}{4\pi}\,\mathrm{Im} \int \mathrm{d}^2\theta\; \tau\; \mathrm{tr}\; W^\alpha W_\alpha + \int \mathrm{d}^4\theta (Q^\dagger e^V Q + \widetilde{Q}^\dagger e^{-V^\top} \widetilde{Q}) + \Big(\int \mathrm{d}^2\theta\; W + \mathrm{h.c.} \Big) \Big], \tag{45.81}$$

where τ and W_α are the holomorphic coupling and field strength of Eqs. (45.16) and (45.54), $e^{-V^\top}$ is the representation of e^V on the antifundamental $\widetilde{Q}$, and W is a gauge-invariant superpotential. The renormalizable choice is a quark mass term

$$W = \sum_{i,j} m^{\,j}_{\,i}\, \widetilde{Q}^{\,i} Q_j \equiv \mathrm{tr}\,(m\, \widetilde{Q} Q), \tag{45.82}$$

the unique gauge-invariant holomorphic quadratic; *massless* SQCD is $W = 0$.

The antifundamental coupling deserves a word. If $Q \to e^{i\Lambda} Q$ in the fundamental, then a field in $\overline{\mathbf{N_c}}$ transforms with the conjugate matrices, $\widetilde{Q} \to \widetilde{Q}\, e^{-i\Lambda^\top}$ (acting on the row index), so its covariant Kähler term uses $e^{-V^\top}$, i.e. V in the antifundamental representation, $T^a_{\overline{\mathbf{N_c}}} = -(T^a)^\top$. With this the antiquark term is gauge invariant by the same cancellation as in Theorem 45.15, and its component reduction is Eq. (45.80) with $T^a \to -(T^a)^\top$. The superpotential Eq. (45.82) is gauge invariant because $\widetilde{Q}^{\,i} Q_j$ contracts a $\overline{\mathbf{N_c}}$ with a $\mathbf{N_c}$ into a colour singlet, and it is holomorphic, hence an admissible F-term by Theorem 44.41.

Adding the super Yang–Mills component Lagrangian Eq. (45.63) — which contains $-\frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu}$, the gaugino kinetic term $i\bar\lambda^a \bar\sigma^\mu (D_\mu \lambda)^a$, the topological θ-term, and the gauge auxiliary $\frac{1}{2} D^a D^a$ — to the matter Lagrangian Eq. (45.80) summed over all N_f flavors of both Q and $\widetilde{Q}$, and to the superpotential components Eq. (44.201), gives the complete off-shell SQCD Lagrangian

$$\begin{aligned} \mathcal{L}_{\mathrm{SQCD}} = &-\tfrac{1}{4} F^a_{\mu\nu} F^{a\mu\nu} + i\,\bar\lambda^a \bar\sigma^\mu (D_\mu \lambda)^a + \tfrac{1}{2} D^a D^a \\ &+ \sum_f \Big[(D_\mu \phi_f)^\dagger (D^\mu \phi_f) + i\,\bar\psi_f \bar\sigma^\mu D_\mu \psi_f + F_f^\dagger F_f \Big] \\ &- \sqrt{2}\, g \sum_f \big[(\phi_f^\dagger T_f^a \psi_f) \lambda^a + \mathrm{h.c.} \big] + g\, D^a \sum_f \phi_f^\dagger T_f^a \phi_f \\ &+ \Big[\sum_f \frac{\partial W}{\partial \phi_f} F_f - \tfrac{1}{2} \sum_{f,f'} \frac{\partial^2 W}{\partial \phi_f \partial \phi_{f'}} \psi_f \psi_{f'} + \mathrm{h.c.} \Big], \end{aligned} \tag{45.83}$$

where the index f runs over *all* matter chiral multiplets — the N_f quarks Q_i (with $T^a_f = T^a$ in the fundamental) and the N_f antiquarks $\widetilde{Q}^{\,i}$ (with $T^a_f = -(T^a)^\top$) — and ϕ_f, ψ_f, F_f are their components. This is the full renormalizable Lagrangian SQCD: a single gauge coupling g, a single θ-angle, and the holomorphic mass matrix m, with every other coefficient — the gaugino Yukawa, the seagull, the D-coupling — fixed by supersymmetry. We have extended Yang–Mills theory from the pure-glue theory of Chapters 25 and 27 to a gauge theory with matter,

exactly as ordinary QCD extends pure Yang–Mills; the matter content $(Q, \widetilde{Q})$ is what will contribute to the beta function computed in Section 45.4.

45.3.4 The scalar potential from the auxiliary fields

The auxiliary fields F_f and D^a appear in Eq. (45.83) algebraically — quadratically, with no derivatives — exactly as F did in the ungauged Wess–Zumino model (Eq. (44.210)). They therefore carry no dynamics and may be eliminated by their algebraic equations of motion. Doing so produces the scalar potential, the central object that controls the vacuum structure.

Theorem 45.18 (SQCD scalar potential)**.** *Eliminating the auxiliary fields of Eq. (45.83) by their equations of motion yields the on-shell scalar potential*

$$\boxed{V(\phi,\phi^\dagger) = \sum_f |F_f|^2_{\text{on-shell}} + \tfrac{1}{2}\sum_a (D^a)^2_{\text{on-shell}} = \underbrace{\sum_f \Big|\frac{\partial W}{\partial \phi_f}\Big|^2}_{F\text{-term}} + \underbrace{\frac{g^2}{2}\sum_a \Big(\sum_f \phi_f^\dagger T_f^a \phi_f\Big)^2}_{D\text{-term}} \;\geq\; 0,} \tag{45.84}$$

with on-shell values

$$F_f = -\Big(\frac{\partial W}{\partial \phi_f}\Big)^*, \qquad D^a = -g\sum_f \phi_f^\dagger T_f^a \phi_f. \tag{45.85}$$

Proof. Collect from Eq. (45.83) every term containing an auxiliary field. The F-dependent terms are, for each f,

$$\mathcal{L}_F = \sum_f \Big[F_f^\dagger F_f + \frac{\partial W}{\partial \phi_f} F_f + \Big(\frac{\partial W}{\partial \phi_f}\Big)^* F_f^\dagger\Big], \tag{45.86}$$

exactly the structure of Eq. (44.210). Varying with respect to $F_f^\dagger$ gives $F_f + (\partial W/\partial\phi_f)^* = 0$, i.e. the first relation in Eq. (45.85). Substituting back,

$$\mathcal{L}_F\big|_{\text{on-shell}} = \sum_f \Big[\,|W_f|^2 - |W_f|^2 - |W_f|^2\Big] = -\sum_f \Big|\frac{\partial W}{\partial \phi_f}\Big|^2, \tag{45.87}$$

writing $W_f \equiv \partial W/\partial\phi_f$. The D-dependent terms are

$$\mathcal{L}_D = \tfrac{1}{2} D^a D^a + g D^a \sum_f \phi_f^\dagger T_f^a \phi_f. \tag{45.88}$$

Define the gauge moment map $\mu^a \equiv \sum_f \phi_f^\dagger T_f^a \phi_f$. Varying with respect to D^a gives $D^a + g\mu^a = 0$, i.e. the second relation in Eq. (45.85), $D^a = -g\mu^a$. Substituting,

$$\mathcal{L}_D\big|_{\text{on-shell}} = \tfrac{1}{2} g^2 (\mu^a)^2 - g^2 (\mu^a)^2 = -\tfrac{1}{2} g^2 \sum_a (\mu^a)^2 = -\tfrac{1}{2}\sum_a (D^a)^2. \tag{45.89}$$

The potential is minus the auxiliary Lagrangian (the Lagrangian contains $-V$ in the sign conventions of Eq. (44.212)), so $V = -\mathcal{L}_F|_{\text{on-shell}} - \mathcal{L}_D|_{\text{on-shell}}$, which is Eq. (45.84). Each summand is a modulus squared, hence $V \geq 0$. □

The structure $V = \sum_f |F_f|^2 + \frac{1}{2}\sum_a (D^a)^2$ is universal: it holds for *any* $\mathcal{N} = 1$ gauge theory with canonical Kähler terms, the F-terms set by the superpotential and the D-terms set by the gauge charges. The F-term piece is the gauge-theory generalization of Eq. (44.149), and the D-term piece is the genuinely new ingredient brought by gauging. Notice that the D-term is quartic in the scalars with coefficient g^2 fixed entirely by the gauge coupling — a scalar self-interaction that supersymmetry and gauge invariance *predict*, with no free quartic of its own.

Physical Insight: vacuum energy as the SUSY-breaking order parameter

Because $V \geq 0$ with equality iff every $F_f = 0$ and every $D^a = 0$, the scalar potential is the component-level avatar of the algebraic fact, proved from the SUSY algebra in Theorem 44.9, that the Hamiltonian is non-negative, $H = \frac{1}{4}\sum_\alpha (\{Q_\alpha, Q_\alpha^\dagger\}) \geq 0$. Supersymmetry is unbroken in a vacuum $|0\rangle$ precisely when $Q_\alpha |0\rangle = 0$, equivalently when the vacuum energy vanishes, equivalently when $\langle F_f \rangle = \langle D^a \rangle = 0$. A non-zero vacuum energy is therefore the *order parameter* for spontaneous supersymmetry breaking (Theorem 44.10): if no scalar configuration can set all F and D to zero simultaneously, then $V_{\min} > 0$ and SUSY is broken — by an F-term (the O'Raifeartaigh mechanism) or by a D-term (the Fayet–Iliopoulos mechanism). For a *simple* gauge group such as $\mathrm{SU}(N_c)$ there is no Fayet–Iliopoulos term, because a linear $\xi \int \mathrm{d}^4\theta\, V^a$ is gauge invariant only for an abelian factor (the trace $\mathrm{tr}\, T^a = 0$ kills it for $\mathrm{SU}(N_c)$); D-term breaking thus requires a U(1), and the abelian linear "Fayet–Iliopoulos-like" term flagged in Theorem 44.42 is exactly this option. In SQCD with $W = 0$ we will find that $V = 0$ has a large solution space, so classical SUSY is unbroken along an entire moduli space.

45.3.5 Classical supersymmetric vacua and the moduli space

The classical vacua of SQCD are the field configurations minimizing V. Since $V \geq 0$ and supersymmetric vacua have $V = 0$, the supersymmetric vacua are exactly the simultaneous solutions of the *F-flatness* and *D-flatness* conditions

$$F_f = -\Big(\frac{\partial W}{\partial \phi_f}\Big)^* = 0 \quad \text{for all } f, \qquad D^a = -g \sum_f \phi_f^\dagger T_f^a \phi_f = 0 \quad \text{for all } a. \tag{45.90}$$

Generically these equations have not isolated solutions but a continuous family. The set of solutions, modulo gauge equivalence, is the *classical moduli space* $\mathcal{M}_{\rm cl}$ — a space of physically inequivalent, exactly degenerate vacua. We now characterize it.

D-flatness in matrix form. Specialize to massless SQCD, $W = 0$, so F-flatness is automatic and only D-flatness constrains the squarks. Arrange the squark VEVs into colour×flavour matrices: let ϕ_i^a ($a = 1, \dots, N_c$; $i = 1, \dots, N_f$) be the scalar of Q_i, an $N_c \times N_f$ matrix, and let $\widetilde{\phi}_a^{\,i}$ be the scalar of $\widetilde{Q}^{\,i}$, an $N_f \times N_c$ matrix. The D-flatness condition $\mu^a = 0$ reads, summing the fundamental contribution of Q and the antifundamental contribution of $\widetilde{Q}$,

$$\mu^a = \sum_i \phi_i^\dagger T^a \phi_i + \sum_i \widetilde{\phi}^{\,i} \big(-(T^a)^\top\big) \widetilde{\phi}^{\,i\dagger} = \mathrm{tr}\big[T^a \big(\phi\,\phi^\dagger - \widetilde{\phi}^\dagger \widetilde{\phi}\big)\big] = 0 \quad \text{for all } a. \tag{45.91}$$

Since the T^a span all traceless Hermitian $N_c \times N_c$ matrices, the vanishing of Eq. (45.91) for every a is equivalent to the statement that the Hermitian matrix $\phi\phi^\dagger - \widetilde{\phi}^\dagger\widetilde{\phi}$ has no traceless part, i.e. is proportional to the identity:

$$\phi\,\phi^\dagger - \widetilde{\phi}^\dagger \widetilde{\phi} = \rho\, \mathbf{1}_{N_c}, \qquad \rho = \frac{1}{N_c}\,\mathrm{tr}\big(\phi\phi^\dagger - \widetilde{\phi}^\dagger\widetilde{\phi}\big) \in \mathbb{R}. \tag{45.92}$$

This single matrix equation is the complete D-flatness condition. Its solution space, modulo $\mathrm{SU}(N_c)$ gauge transformations, is the classical moduli space of massless SQCD.

Gauge-invariant coordinates: mesons and baryons. Rather than solve Eq. (45.92) and quotient by hand, it is cleaner to describe $\mathcal{M}_{\rm cl}$ by gauge invariants. The holomorphic gauge-invariant chiral operators built from Q and $\widetilde{Q}$ are

$$\text{mesons:} \quad M^i_{\ j} = \widetilde{Q}^{\,i}_a\, Q^a_j, \qquad \text{an } N_f \times N_f \text{ matrix}, \tag{45.93}$$

$$\text{baryons:} \quad B^{i_1 \cdots i_{N_c}} = \epsilon_{a_1 \cdots a_{N_c}}\, Q^{a_1}_{i_1} \cdots Q^{a_{N_c}}_{i_{N_c}}, \quad \text{antisymmetric in the } N_c \text{ flavors}, \tag{45.94}$$

$$\text{antibaryons:} \quad \widetilde{B}_{i_1 \cdots i_{N_c}} = \epsilon^{a_1 \cdots a_{N_c}}\, \widetilde{Q}^{\,i_1}_{a_1} \cdots \widetilde{Q}^{\,i_{N_c}}_{a_{N_c}}. \tag{45.95}$$

These are colour singlets (the meson by contracting $\overline{\mathbf{N_c}}$ with $\mathbf{N_c}$, the baryon by the invariant ϵ-tensor of $\mathrm{SU}(N_c)$), and being holomorphic in the chiral superfields they are themselves chiral. A foundational fact, which we use here and Chapter 46 elevates to the quantum level, is that *the gauge-invariant chiral operators provide good holomorphic coordinates on the classical moduli space*: every D-flat configuration is gauge-equivalent to one labeled by the values of $M, B, \widetilde{B}$, subject to the classical relations among them. The baryons require antisymmetrizing N_c colour indices over the flavors, which is impossible when there are fewer than N_c distinct flavors; hence baryons exist only for $N_f \geq N_c$.

Dimension count by the super-Higgs mechanism. The cleanest way to get the dimension of $\mathcal{M}_{\rm cl}$ is to count light degrees of freedom directly. We start with $2N_cN_f$ complex squark fields (N_cN_f in Q and N_cN_f in $\widetilde{Q}$). At a generic point of the moduli space the squark VEVs Higgs the gauge group, and each *broken* gauge generator eats one chiral superfield: supersymmetrically, a massless vector multiplet (2 bosonic $+2$ fermionic states) absorbs a massless chiral multiplet ($2+2$ states) to form a massive vector multiplet, the super-Higgs mechanism. Hence

$$\dim_{\mathbb{C}} \mathcal{M}_{\rm cl} = \underbrace{2N_cN_f}_{\text{squark chirals}} - \underbrace{(\dim G - \dim H)}_{\text{broken generators, each eats one chiral}}, \tag{45.96}$$

where $H \subseteq G$ is the unbroken gauge group at the generic point. Two regimes arise, distinguished by whether the squarks can break $\mathrm{SU}(N_c)$ completely.

Regime $N_f < N_c$. A generic rank-N_f squark VEV ϕ breaks $\mathrm{SU}(N_c) \to \mathrm{SU}(N_c - N_f)$ (the colour directions orthogonal to the N_f "filled" ones are untouched). The number of broken generators is

$$\dim \mathrm{SU}(N_c) - \dim \mathrm{SU}(N_c - N_f) = (N_c^2 - 1) - \big((N_c - N_f)^2 - 1\big) = 2N_cN_f - N_f^2. \tag{45.97}$$

Substituting into Eq. (45.96),

$$\dim_{\mathbb{C}} \mathcal{M}_{\rm cl} = 2N_cN_f - (2N_cN_f - N_f^2) = N_f^2. \tag{45.98}$$

These N_f^2 complex moduli are exactly the N_f^2 components of the meson matrix $M^i_{\ j}$ of Eq. (45.93); there are no baryons and no constraints, so $\mathcal{M}_{\rm cl}$ is freely parametrized by M, a copy of $\mathbb{C}^{N_f^2}$.

Regime $N_f \geq N_c$. Now a generic squark VEV breaks $\mathrm{SU}(N_c)$ *completely*, $H = \{1\}$, so all $N_c^2 - 1$ generators are broken and

$$\dim_{\mathbb{C}} \mathcal{M}_{\rm cl} = 2N_cN_f - (N_c^2 - 1). \tag{45.99}$$

The invariants are now the mesons M (N_f^2 of them) *and* the baryons and antibaryons $B, \widetilde{B}$ (each $\binom{N_f}{N_c}$ of them). These over-count Eq. (45.99), so they obey classical relations — the moduli space is the image of the map $(\phi, \widetilde{\phi}) \mapsto (M, B, \widetilde{B})$, an algebraic variety cut out by those relations. For the borderline case $N_f = N_c$ the baryons collapse to single objects $B = \det Q$, $\widetilde{B} = \det \widetilde{Q}$ (one each), and the meson, baryon and antibaryon obey the single classical constraint

$$\det M - B\,\widetilde{B} = 0, \tag{45.100}$$

since $\det(\widetilde{Q}Q) = \det \widetilde{Q} \det Q$. Counting: N_c^2 (mesons) $+2$ (baryons) -1 (constraint) $= N_c^2 + 1$, matching Eq. (45.99) at $N_f = N_c$. These dimension counts, and the distinction between the two regimes, are precisely the classical data that the exact analysis of Chapter 46 — Seiberg duality and the quantum-modified moduli space — takes as its starting point.

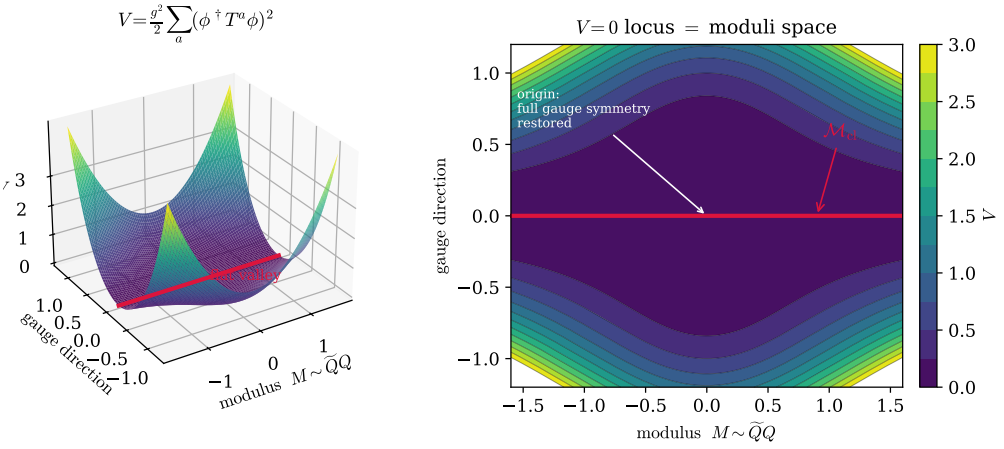


Figure 45.3: The scalar potential of massless SQCD ($W = 0$) is purely a D-term, $V = \frac{g^2}{2}\sum_a (\phi^\dagger T^a \phi)^2 \geq 0$. Its zero locus is a flat, connected valley of exactly degenerate supersymmetric vacua — the classical moduli space $\mathcal{M}_{\rm cl}$ — swept out as the gauge-invariant modulus (here the meson $M \sim \widetilde{Q}Q$) varies. Directions transverse to the valley turn on $D^a \neq 0$ and cost energy. At the origin all squark VEVs vanish and the full gauge symmetry is restored; this is where the classical description becomes singular and the nonperturbative physics of Chapter 46 takes over.

Worked Example 45.19 (SU(2) with one flavor: the simplest moduli space)**.** Take $N_c = 2$, $N_f = 1$: gauge group SU(2), one quark Q^a ($a = 1, 2$) in the doublet $\mathbf{2}$ and one antiquark $\widetilde{Q}_a$ in $\overline{\mathbf{2}}$. We solve Eq. (45.90) explicitly with $W = 0$, exhibit the surviving massless modulus, and check the dimension count Eq. (45.98), which predicts $\dim_{\mathbb{C}} \mathcal{M}_{\rm cl} = N_f^2 = 1$.

Setup. Write the squark VEVs as two-component complex column vectors: $q \equiv (\phi^1, \phi^2)^\top$ for the quark, and $\widetilde{q} \equiv (\widetilde{\phi}_1, \widetilde{\phi}_2)^\dagger$ packaging the antiquark scalar as a column. The generators are $T^a = \frac{1}{2}\sigma^a$ with σ^a the Pauli matrices (Section B.2). The D-flatness matrix equation Eq. (45.92) reads, for $N_c = 2$,

$$q\, q^\dagger - \widetilde{q}\, \widetilde{q}^\dagger = \rho\, \mathbf{1}_2, \qquad \rho = \tfrac{1}{2}(|q|^2 - |\widetilde{q}|^2). \tag{45.101}$$

Solving. Use the SU(2) gauge freedom to rotate q to the form $q = (v, 0)^\top$ with $v \geq 0$. Write $\widetilde{q} = (b, c)^\top$. Then

$$q q^\dagger = \begin{pmatrix} v^2 & 0 \\ 0 & 0 \end{pmatrix}, \qquad \widetilde{q}\widetilde{q}^\dagger = \begin{pmatrix} |b|^2 & bc^* \\ cb^* & |c|^2 \end{pmatrix}, \tag{45.102}$$

and Eq. (45.101) demands the difference be proportional to the identity: the off-diagonal entries force $bc^* = 0$, and the diagonal entries force $v^2 - |b|^2 = -|c|^2$.

The generic branch is $c = 0$, leaving $v^2 = |b|^2$, i.e. $|\widetilde{q}| = |q| = v$. Thus, up to gauge and global flavor rotations, the D-flat configurations are

$$q = \begin{pmatrix} v \\ 0 \end{pmatrix}, \qquad \widetilde{q} = \begin{pmatrix} v\, e^{i\beta} \\ 0 \end{pmatrix}, \qquad v \geq 0, \tag{45.103}$$

a two-real-parameter family (v, β) after the gauge fixing.

The modulus. The single gauge invariant is the meson

$$M = \widetilde{Q}\, Q = \widetilde{\phi}_a \phi^a = v \cdot v\, e^{-i\beta} = v^2 e^{-i\beta}, \tag{45.104}$$

one complex number that combines the modulus' magnitude v^2 and phase β. This M is the surviving massless modulus: as it varies over $\mathbb{C}$ it sweeps out the entire one-complex-dimensional moduli space, confirming $\dim_{\mathbb{C}} \mathcal{M}_{\rm cl} = 1 = N_f^2$.

Degree-of-freedom audit. The squarks carry $2 + 2 = 4$ complex $= 8$ real fields. D-flatness imposes $\dim \mathrm{SU}(2) = 3$ real conditions ($\mu^a = 0$, $a = 1, 2, 3$), leaving $8 - 3 = 5$. A nonzero VEV $q = (v, 0)^\top$ breaks SU(2) completely (a single doublet VEV has trivial stabilizer in SU(2), since $\mathrm{SU}(2) \cong S^3$ acts freely on the unit sphere in $\mathbb{C}^2$), so all three gauge directions are eaten: $5 - 3 = 2$ real $= 1$ complex modulus. Equivalently, Eq. (45.96) gives $2N_cN_f - (N_c^2 - 1) = 4 - 3 = 1$, the same answer because $\mathrm{SU}(N_c - N_f) = \mathrm{SU}(1)$ is trivial. The three broken gauge bosons acquire mass $\sim g\, v$ by the super-Higgs mechanism, each absorbing a squark chiral multiplet; what remains massless is the single chiral multiplet whose scalar is M.

The origin and its singularity. At $v = 0$ (hence $M = 0$) all squark VEVs vanish, the gauge group SU(2) is restored unbroken, and the three would-be massive vector multiplets become massless. The classical moduli space is therefore smooth away from $M = 0$ but *singular at the origin*, where extra degrees of freedom (the full gluon/gaugino multiplet) become light. Reinstating a quark mass $W = mM$ adds an F-term: $\partial W/\partial M = m$, so $V \supset |m|^2$ lifts the flat direction entirely and the unique vacuum slides to $M = 0$. With $W = 0$, however, the classical theory has an exact one-complex-dimensional valley of degenerate supersymmetric vacua — the concrete realization of the moduli-space notion, and the simplest laboratory for the quantum corrections of Chapter 46, where this very singularity at $M = 0$ is resolved by nonperturbative dynamics.

We have assembled the complete renormalizable Lagrangian of supersymmetric QCD, reduced it to the scalar potential $V = \sum_f |F_f|^2 + \frac{1}{2}\sum_a (D^a)^2$ by integrating out the auxiliary fields, and mapped out its classical moduli space of vacua in terms of the gauge-invariant mesons and baryons. The holomorphic superpotential W and the gauge coupling τ now stand exposed as the two data on which the quantum theory acts. The next section turns to the property that makes these theories extraordinary: the superpotential and the holomorphic coupling are protected by holomorphy and the global symmetries (including the R-symmetry of Section 44.2.7) so tightly that the former receives *no* perturbative

corrections at all and the latter runs only at one loop, yielding the exact NSVZ beta function that sharpens the one-loop result of Chapter 27 for Yang–Mills theory.

45.4 Nonrenormalization Theorems and the Exact NSVZ Beta Function

The preceding section assembled supersymmetric QCD: N_f flavors of chiral matter $Q, \widetilde{Q}$ coupled to SU(N_c) super-Yang–Mills, with the gauge-covariant Kähler term $Q^\dagger e^V Q + \widetilde{Q}^\dagger e^{-V}\widetilde{Q}$, the holomorphic superpotential (a possible mass or, more generally, gauge-invariant chiral interactions), and the classical scalar potential whose flat directions form the moduli space. We now ask the quantum question. In an ordinary field theory the couplings drift with scale, and there is no shortcut: one computes loop integrals order by order. Supersymmetry changes this radically. The split of the action into a *holomorphic* piece — the superpotential W and the gauge kinetic term, each an F-term integrated only over half of superspace — and a *non-holomorphic* piece — the Kähler term, a D-term integrated over all of superspace — is not bookkeeping. It is the structural fact that lets us determine the quantum corrections to the holomorphic sector *exactly*, with no loop computation at all, using only holomorphy, symmetry, and the behavior in weak-coupling limits. The flag planted in Section 44.7.8 is now made into a theorem.

The payoff is twofold. First, the superpotential receives *no* perturbative corrections: the renormalized W equals the tree-level W to all orders. Second — and this is the climactic Yang–Mills theory advance promised by the chapter — the holomorphic gauge coupling runs at one loop only, and converting it to the physical (canonically normalized) coupling through a single, controlled anomaly produces a *closed-form, all-orders* beta function: the NSVZ formula. Reducing it at one loop recovers and sharpens the coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ that Eq. (27.1) computed for ordinary Yang–Mills, replacing it by the supersymmetric values $b_0 = 3C_A$ for pure super-Yang–Mills and $b_0 = 3N_c - N_f$ for SU(N_c) SQCD. We develop the holomorphy principle, prove the superpotential nonrenormalization theorem, show carefully why the Kähler potential is *not* protected (so that physical couplings do run), establish the one-loop-exactness of the holomorphic coupling, and assemble the NSVZ denominator from the rescaling (Konishi) anomaly.

Notation Watch

Two distinct objects in this chapter are traditionally both called "W." The *superpotential* $W(\Phi)$ is a holomorphic function of chiral matter superfields (Theorem 44.40); we always write it with an explicit field argument or

as W. The *gauge field-strength superfield* is the chiral spinor superfield $W_\alpha = -\frac{1}{4}\bar{D}\bar{D}\, e^{-V} D_\alpha e^{V}$ built from the vector superfield earlier in this chapter; it always carries a spinor index α. The two never appear without their distinguishing decoration, so no ambiguity survives.

45.4.1 Couplings as spurions: the holomorphy principle

The engine of every nonrenormalization result is a single idea due to Seiberg: *treat each coupling in the holomorphic sector as the constant value of a background chiral superfield.* A coupling λ that multiplies an F-term $\int \mathrm{d}^2\theta\, \mathcal{O}$ can be regarded as the lowest component of a non-dynamical chiral superfield Λ (a "spurion"), frozen to its vacuum value $\langle \Lambda \rangle = \lambda$ with all higher components set to zero. Because $\int \mathrm{d}^2\theta\, \Lambda\, \mathcal{O}$ is then a perfectly good supersymmetric interaction of a chiral superfield Λ with the chiral operator $\mathcal{O}$, the entire formal apparatus of superspace — in particular, the chirality of the superpotential — now applies to the couplings themselves. Three principles then pin down the quantum theory.

(H1) **Holomorphy.** The Wilsonian effective superpotential W_{eff}, obtained by integrating out modes above a scale μ, is a chiral F-term. By the ring property of chiral superfields (Eq. (44.198)), W_{eff} must be a *holomorphic* function of the dynamical chiral superfields *and* of the chiral spurions — a function of the Φ_i and the Λ_a alone, with no dependence on any conjugate $\Phi_i^\dagger$ or $\Lambda_a^\dagger$. Anti-holomorphic or mixed dependence would live in the Kähler (D-term) sector, never in W_{eff}.

(H2) **Symmetry.** Every global symmetry of the classical theory — the ordinary flavor U(1)'s and the $\mathrm{U}(1)_R$ R-symmetry of Section 44.2.7 — must be respected by W_{eff}, *provided we let the symmetries act on the spurions too.* A symmetry that is broken by a nonzero coupling becomes a symmetry of the spurion-extended theory by assigning the spurion the charge that compensates: the classical Lagrangian is then exactly invariant, and so is anything computed from it. This converts "approximate" or "softly broken" symmetries into exact selection rules on W_{eff}.

(H3) **Weak-coupling limits and regularity.** As any holomorphic coupling $\lambda \to 0$ the corresponding interaction switches off, and W_{eff} must reduce smoothly to that of the theory without the interaction. Combined with holomorphy, this is decisive: a function holomorphic in λ and known to be regular (no inverse powers, no logarithms of λ) as $\lambda \to 0$ is fixed by its Taylor coefficients there, and those coefficients are fixed by ordinary perturbation theory at low order. Demanding regularity also as the *other* couplings (e.g. a mass) tend to zero forbids would-be terms that are singular in those limits.

The logic is worth stating starkly: holomorphy says W_{eff} is an analytic function on the space of couplings; symmetry restricts which monomials can appear; weak-coupling limits fix the few surviving coefficients. Because an analytic function is rigid — determined globally by its behavior in a limit — conclusions drawn in the weak-coupling corner of coupling space hold *everywhere*, including at strong coupling. The honest scope, which we will respect throughout, is that this argument controls the *perturbative* holomorphic data exactly; genuinely nonperturbative contributions (instantons) appear as terms exponentially small in the coupling, are invisible to the Taylor expansion of (H3), and are the business of Chapter 46.

45.4.2 The perturbative nonrenormalization theorem

We prove the statement in the cleanest nontrivial setting — the cubic Wess–Zumino model carried since Theorem 44.42 — and then read off the general theorem. To avoid colliding with the gauge coupling g of this chapter, we write the cubic superpotential coupling as λ (it is the "g" of Eq. (44.214)):

$$W_{\text{tree}}(\Phi) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, \lambda\, \Phi^3. \tag{45.105}$$

Theorem 45.20 (Perturbative nonrenormalization of the superpotential)**.** *In the Wess–Zumino model with tree superpotential Eq. (45.105), the Wilsonian effective superpotential is uncorrected to all orders in perturbation theory:*

$$W_{\text{eff}}(\Phi) = W_{\text{tree}}(\Phi) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, \lambda\, \Phi^3. \tag{45.106}$$

More generally, for any set of chiral superfields with renormalizable or nonrenormalizable holomorphic superpotential, the coefficients in W receive no perturbative corrections; all scale dependence of physical couplings resides in the Kähler potential, i.e. in wavefunction renormalization.

Proof. Promote the mass m and the coupling λ to background chiral spurion superfields, each frozen to its constant lowest component. The spurion-extended classical theory has two global U(1) symmetries under which the dynamical field and the spurions carry charges chosen to make Eq. (45.105) invariant.

Charge assignments. Under an ordinary flavor rotation U(1) let Φ carry charge $+1$, $\Phi \mapsto e^{i\alpha}\Phi$. Invariance of $m\Phi^2$ and $\lambda\Phi^3$ forces the spurions to carry

$$q[\Phi] = +1, \qquad q[m] = -2, \qquad q[\lambda] = -3. \tag{45.107}$$

Under the R-symmetry $\mathrm{U}(1)_R$ the Grassmann coordinate has $R[\theta] = +1$, so the chiral measure has $R[\mathrm{d}^2\theta] = -2$ and an F-term superpotential must carry $R = +2$ to make $\int \mathrm{d}^2\theta\, W$ neutral (Section 44.2.7). Assigning Φ an R-charge $R[\Phi] \equiv r$ (its value will drop out), invariance of $\int \mathrm{d}^2\theta\, m\Phi^2$ and $\int \mathrm{d}^2\theta\, \lambda\Phi^3$ requires

$$R[m] = 2 - 2r, \qquad R[\lambda] = 2 - 3r. \tag{45.108}$$

Allowed monomials. By (H1), W_{eff} is a holomorphic function of Φ, m, λ; expand it in monomials $\Phi^a m^b \lambda^c$ with integers $a \geq 0$, $b, c \geq 0$ (negative powers of a field or of λ are excluded by regularity, (H3), as we confirm below). Such a monomial multiplies $\int \mathrm{d}^2\theta$ and so must carry total U(1) charge 0 and total R-charge +2:

$$q: \quad a - 2b - 3c = 0, \tag{45.109}$$

$$R: \quad a\,r + b(2-2r) + c(2-3r) = 2. \tag{45.110}$$

Substituting Eq. (45.109) into Eq. (45.110), the r-dependent terms collect as $r(a-2b-3c) + 2(b+c) = 2$, and the bracket vanishes by Eq. (45.109), leaving

$$b + c = 1. \tag{45.111}$$

The arbitrary choice r has cancelled, as it must — a genuine selection rule cannot depend on the free convention for $R[\Phi]$. With $b, c \geq 0$ integers, Eq. (45.111) has exactly two solutions: $(b, c) = (1, 0)$ forces $a = 2$ by Eq. (45.109), giving the term $m\Phi^2$; and $(b, c) = (0, 1)$ forces $a = 3$, giving $\lambda\Phi^3$. No other monomial is allowed. Symmetry and holomorphy alone therefore restrict

$$W_{\text{eff}}(\Phi) = A\, m\, \Phi^2 + B\, \lambda\, \Phi^3, \tag{45.112}$$

with A, B pure numbers — with one caveat to dispatch: the constraints Eq. (45.109)–Eq. (45.110) are also solved by inserting any power of the neutral combination $x \equiv \lambda\Phi/m$, which has $q[x] = 1 - 3 + 2 = 0$ and $R[x] = (2-3r) + r - (2-2r) = 0$. Symmetry alone thus permits $W_{\text{eff}} = m\Phi^2\, f(x)$ for an arbitrary holomorphic f; equivalently A, B could be replaced by power series in x.

Weak coupling fixes the function. Here (H3) closes the argument. Write $W_{\text{eff}} = m\Phi^2 \sum_{n\geq 0} a_n x^n$ with $x = \lambda\Phi/m$. The n-th term is

$$a_n\, m\, \Phi^2 \left(\frac{\lambda\Phi}{m}\right)^n = a_n\, \lambda^n\, m^{\,1-n}\, \Phi^{\,n+2}. \tag{45.113}$$

For $n \geq 2$ this carries a *negative* power m^{1-n} of the mass. But perturbation theory in λ is manifestly regular as $m \to 0$: Feynman diagrams of the massless Wess–Zumino model are finite functions of the external momenta and never generate inverse powers of a mass that was set to zero. A term singular as $m \to 0$ cannot arise at any finite order, so $a_n = 0$ for all $n \geq 2$. (The same conclusion follows from demanding regularity as $\Phi \to \infty$ at fixed λ: an effective superpotential growing faster than the classical cubic is excluded.) Only $n = 0, 1$ survive, reproducing Eq. (45.112) with constant $A = a_0$, $B = a_1$.

Fixing the constants. Take $\lambda \to 0$ at fixed m. The theory becomes a free massive chiral superfield, whose superpotential is exactly $\frac{1}{2}m\Phi^2$ with no corrections (a free theory renormalizes nothing in W). Hence $A = \frac{1}{2}$. Now take the leading nontrivial order in λ: at order λ^1, holomorphy and the $\lambda \to 0$

matching require the cubic coefficient to equal its tree value, since any correction would be of higher order in λ and thus carry a power $x^{n\geq 1}$ already excluded; concretely, expanding to first order in λ the only allowed $O(\lambda)$ term is $\lambda\Phi^3$ with the tree coefficient, so $B = \frac{1}{3}$. Therefore

$$W_{\text{eff}}(\Phi) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, \lambda\, \Phi^3 = W_{\text{tree}}(\Phi), \tag{45.114}$$

exactly, to all orders in perturbation theory. The generalization is immediate: for any holomorphic $W = \sum_a \lambda_a\, \mathcal{O}_a(\Phi)$, promote every λ_a to a chiral spurion, impose the U(1)'s and $\mathrm{U}(1)_R$ as exact selection rules, and require regularity in the weak-coupling and small-mass limits; the only holomorphic invariant available to multiply each operator is the corresponding single power of its own coupling, so every coefficient is frozen at its tree value. This is Eq. (45.106). □

Physical Insight

The theorem is often summarized by the diagrammatic "miracle" that the would-be loop corrections to the superpotential cancel between bosons and fermions. That cancellation is real, but it is a *consequence*, not the reason. The reason is holomorphy: an analytic function of the couplings that is regular at weak coupling is determined by a finite amount of low-order data, and that data is the tree-level superpotential. One never has to draw a single loop diagram. This is the prototype of "exact results from symmetry" that organizes all of supersymmetric dynamics, and it is what makes supersymmetry a candidate solution to the hierarchy problem (Chapter 35): a scalar mass that sits in W cannot be renormalized, so a small mass stays small, protected to all perturbative orders by holomorphy rather than by fine-tuning.

45.4.3 The Kähler potential is renormalized

Nonrenormalization protects only the holomorphic sector. The Kähler potential is a D-term,

$$\int \mathrm{d}^4\theta\; K(\Phi^\dagger, \Phi), \qquad K = Z\, \Phi^\dagger \Phi + \ldots, \tag{45.115}$$

and the spurion argument simply does not constrain it. The integrand depends on both Φ and $\Phi^\dagger$, so it is neither chiral nor antichiral; the coefficient Z is a *real* function of the couplings and their conjugates — it can and does depend on $|\lambda|^2$, $\ln(\mu)$, and the like — and there is no holomorphy to forbid its renormalization. Diagrammatically, the wavefunction self-energy of a chiral superfield is a genuine D-term loop effect with no cancellation: $Z \neq 1$ and Z runs with scale.

This is precisely what rescues the theory from triviality. The superpotential couplings m, λ in Eq. (45.114) are scale-*independent* as holomorphic parameters,

yet the *physical*, canonically normalized couplings run. To see how, canonically normalize the kinetic term by rescaling $\Phi \to Z^{-1/2}\Phi$, which sends $\int \mathrm{d}^4\theta\, Z\, \Phi^\dagger\Phi \to \int \mathrm{d}^4\theta\, \Phi^\dagger\Phi$ and rescales the superpotential coefficients:

$$m_{\text{phys}} = Z^{-1}\, m, \qquad \lambda_{\text{phys}} = Z^{-3/2}\, \lambda. \tag{45.116}$$

The physical mass and Yukawa coupling therefore run entirely through the wavefunction factor $Z(\mu)$. Defining the anomalous dimension of the chiral superfield by

$$\gamma \;\equiv\; -\,\frac{d\ln Z}{d\ln \mu}, \tag{45.117}$$

(the superfield analogue of the field anomalous dimension of Theorem 24.2, here carried once per superfield since all components renormalize together), we read off

$$\frac{d\ln \lambda_{\text{phys}}}{d\ln\mu} = -\tfrac{3}{2}\frac{d\ln Z}{d\ln\mu} = \tfrac{3}{2}\,\gamma, \qquad \frac{d\ln m_{\text{phys}}}{d\ln\mu} = \gamma. \tag{45.118}$$

The beta function of the physical Yukawa coupling is thus *not* zero; it is *entirely* the wavefunction anomalous dimension. This is the sharp version of the slogan "W does not renormalize but the theory still runs." All the genuine dynamics has been quarantined into a single real function $Z(\mu)$, and γ is the one piece of data perturbation theory still has to supply.

Caution

It is a frequent error to conclude from Theorem 45.20 that supersymmetric gauge theories have vanishing beta functions. They do not. The holomorphic couplings are protected, but the canonical couplings flow, driven by the unprotected Kähler renormalization. The next two subsections show how the same dichotomy — holomorphic coupling protected, canonical coupling running — plays out in the gauge sector and produces the exact NSVZ beta function, whose numerator carries exactly the anomalous dimensions γ_r defined above.

45.4.4 The holomorphic gauge coupling runs only at one loop

The gauge sector is governed by its own holomorphic coupling. Recall (from earlier in this chapter) the complexified coupling that packages the gauge coupling g with the theta angle of Section 42.4,

$$\tau \;\equiv\; \frac{\theta}{2\pi} + \frac{4\pi i}{g^2}, \tag{45.119}$$

and the gauge kinetic term written as a chiral F-term of the field-strength superfield W_α,

$$\mathcal{L}_{\text{gauge}} = \frac{1}{4\pi}\,\mathrm{Im}\Big[\tau\int \mathrm{d}^2\theta\ \mathrm{tr}\ W^\alpha W_\alpha\Big] = -\frac{1}{4g^2}F^a_{\mu\nu}F^{a\,\mu\nu} + \frac{\theta}{32\pi^2}F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu} + \dots . \tag{45.120}$$

Because $\int \mathrm{d}^2\theta\ \mathrm{tr}\, W^\alpha W_\alpha$ is holomorphic and the coefficient is linear in τ, the gauge coupling enters the action exactly as the superpotential couplings did: as the value of a background chiral object. We can therefore run the holomorphy machine on τ itself.

Two facts constrain the effective holomorphic coupling $\tau_{\text{eff}}(\mu)$. First, **holomorphy**: τ_{eff} is a holomorphic function of τ (and of any chiral matter spurions). Second, **periodicity of the theta angle**. The term $\frac{\theta}{32\pi^2}F\widetilde{F}$ integrates, on configurations of integer topological charge ν (Section 42.4), to $\theta\nu$; shifting $\theta \to \theta + 2\pi$ changes the action by $2\pi\nu \in 2\pi\mathbb{Z}$, leaving e^{iS} invariant. Physics is therefore periodic under

$$\theta \to \theta + 2\pi, \qquad \text{equivalently} \qquad \tau \to \tau + 1. \tag{45.121}$$

A holomorphic function compatible with the shift symmetry Eq. (45.121) can depend on τ only in two ways: linearly (a shift of τ by an integer shifts the linear term by an integer, an allowed redefinition of θ), or through the single-valued periodic combination $e^{2\pi i\tau}$. The exponential is the instanton weight,

$$e^{2\pi i\tau} = e^{i\theta}\, e^{-8\pi^2/g^2}, \tag{45.122}$$

exponentially small at weak coupling and invisible to perturbation theory. Hence the *perturbative* part of τ_{eff} is at most linear in τ. A genuine two-loop or higher perturbative correction would contribute a term analytic in g^2 — i.e. a power of $1/\tau$ — which is neither linear in τ nor of the exponential form, and is forbidden. Only the one-loop term survives.

Concretely, dimensional transmutation fixes the form. The one nontrivial scale of the theory is the renormalization-group invariant

$$\Lambda \;=\; \mu\,\exp\!\left(\frac{2\pi i\,\tau_{\text{eff}}(\mu)}{b_0}\right), \qquad b_0 \equiv 3C_A - \sum_r T(r), \tag{45.123}$$

holomorphic in τ and invariant under $\mu\, d/d\mu$ by construction. Inverting,

$$2\pi i\,\tau_{\text{eff}}(\mu) = b_0\,\ln\frac{\Lambda}{\mu}, \qquad \text{i.e.} \qquad \frac{8\pi^2}{g_h^2(\mu)} = b_0\,\ln\frac{\mu}{|\Lambda|} + \text{const}, \tag{45.124}$$

where in the second form we took the real part of Eq. (45.124) using $\mathrm{Re}(2\pi i\tau) = -8\pi^2/g^2$ from Eq. (45.119), and we write g_h for the *holomorphic* coupling. The

content is sharp:

$$\boxed{\frac{d}{d\ln\mu}\frac{8\pi^2}{g_h^2} \;=\; b_0 \;=\; 3C_A - \sum_r T(r), \qquad \text{exactly, with no higher-loop corrections.}} \tag{45.125}$$

The matter Dynkin indices $T(r)$ enter here through the one-loop running only; their all-orders effect, encoded in the anomalous dimensions γ_r, has not yet appeared because g_h is the holomorphic, not the physical, coupling. Disentangling the two is the final step.

Historical Note

That the holomorphic coupling is one-loop exact, while the physical coupling is not, was the source of a long apparent paradox. The Novikov–Shifman–Vainshtein–Zakharov (NSVZ) beta function was derived from instanton calculus in the early 1980s and manifestly contains all-orders structure; yet supersymmetric perturbation theory in a holomorphic scheme seemed to stop at one loop. Shifman and Vainshtein resolved this in 1991 by recognizing that the holomorphic (Wilsonian) coupling and the physical (canonical) one differ by an anomalous Jacobian — the rescaling anomaly — so that both statements are correct in their own scheme. Arkani-Hamed and Murayama later recast the entire argument in the transparent holomorphy-plus-anomaly language we follow here.

45.4.5 From holomorphic to canonical: the Konishi anomaly and the NSVZ formula

The holomorphic normalization that made Eq. (45.125) so clean is not the normalization in which fields have canonical kinetic terms. In the holomorphic frame the gauge coupling sits in front of $\operatorname{tr} W^\alpha W_\alpha$ and the matter Kähler terms carry wavefunction factors Z_r; to compute physical quantities one passes to the *canonical* frame by rescaling

$$\Phi_r \to Z_r^{-1/2}\Phi_r \quad \text{(matter)}, \qquad V \to 2g_c V \quad \text{(gauge)}, \tag{45.126}$$

so that the matter kinetic terms become $\Phi_r^\dagger e^V \Phi_r$ and the gauge coupling g_c reappears inside the covariant derivatives. Classically these are harmless changes of integration variable in the path integral. Quantum mechanically they are *not*: a chiral rescaling by a complex factor is anomalous, exactly as the chiral rotation of a Dirac field is anomalous in Section 36.2. The measure of a chiral superfield is not invariant under $\Phi_r \to e^{c_r}\Phi_r$; the Jacobian is the supersymmetric (Konishi) version of the ABJ anomaly of Theorem 36.4.

The Konishi/rescaling anomaly. A word on the logical status of what follows. Chapter 36 establishes by the Fujikawa method only the *ordinary* ABJ triangle (Theorem 36.4); its superfield lift — the anomalous Jacobian for rescaling an entire chiral superfield, the Konishi anomaly — is not derived in this book. We import its form and normalization from the literature [370], and rather than presenting the coefficient as derived we fix its logical status honestly: it is the value that reproduces, by the consistency check carried out just below, the established one-loop coefficient b_0 of Yang–Mills theory. With that caveat stated, the imported statement is this. Under $\Phi_r \to e^{c_r}\Phi_r$ with c_r a constant (real or complex), the path-integral measure of the chiral superfield in representation r shifts the gauge kinetic action by a local F-term,

$$\mathcal{D}\Phi_r\,\mathcal{D}\Phi_r^\dagger \;\longrightarrow\; \mathcal{D}\Phi_r\,\mathcal{D}\Phi_r^\dagger\,\exp\!\left[-\frac{T(r)\,c_r}{8\pi^2}\int \mathrm{d}^4x\,\mathrm{d}^2\theta\;\tfrac{1}{4}\,\mathrm{tr}\,W^\alpha W_\alpha + \mathrm{h.c.}\right], \tag{45.127}$$

the supersymmetric completion of the ABJ triangle with the rescaling current $\Phi_r^\dagger e^V \Phi_r$ in place of the axial current; the coefficient $T(r)$ is the Dynkin index of r, the direct analogue of the $\mathrm{tr}(T^aT^b)$ that multiplied Eq. (36.50). Comparing with Eq. (45.120), a shift of the coefficient of $\mathrm{tr}\,W^\alpha W_\alpha$ is a shift of the holomorphic gauge coupling. Performing the matter rescaling of Eq. (45.126) with $c_r = -\tfrac{1}{2}\ln Z_r$ therefore contributes

$$\Delta\left(\frac{8\pi^2}{g^2}\right)_{\text{matter}} = -\sum_r T(r)\ln Z_r. \tag{45.128}$$

The gauge rescaling $V \to 2g_c V$ is likewise anomalous: it rescales the gauginos, which sit in the adjoint, and the same mechanism with Dynkin index $T(\mathrm{adj}) = C_A$ contributes a term proportional to $C_A \ln g_c$. The upshot is the exact relation between the holomorphic coupling g_h of Eq. (45.125) and the canonical coupling g_c,

$$\frac{8\pi^2}{g_c^2(\mu)} = \frac{8\pi^2}{g_h^2(\mu)} - C_A\ln g_c^2(\mu) - \sum_r T(r)\ln Z_r(\mu) + \text{const}, \tag{45.129}$$

where the sign and coefficient of the C_A term are those that the gaugino rescaling anomaly produces (they are fixed unambiguously by the Jacobian Eq. (45.127) applied to the adjoint gauginos; we verify just below that they yield the established NSVZ denominator). Equation Eq. (45.129) is the technical heart of the section: the protected, one-loop-exact holomorphic coupling on the right is converted to the physical coupling on the left by two purely anomalous pieces, one from the gauginos (C_A) and one from the matter ($T(r)$, weighted by the running wavefunctions Z_r).

Assembling the beta function. Differentiate Eq. (45.129) with respect to $t \equiv \ln \mu$. Write the canonical beta function as $\beta \equiv dg_c/dt$, and use:

$$\frac{d}{dt}\frac{8\pi^2}{g_c^2} = -\frac{16\pi^2}{g_c^3}\,\beta, \tag{45.130}$$

$$\frac{d}{dt}\frac{8\pi^2}{g_h^2} = b_0 = 3C_A - \sum_r T(r) \qquad (\text{by } \textit{Eq. (45.125)}), \tag{45.131}$$

$$\frac{d}{dt}\ln g_c^2 = \frac{2\beta}{g_c}, \qquad \frac{d}{dt}\ln Z_r = -\gamma_r \qquad (\text{by } \textit{Eq. (45.117)}). \tag{45.132}$$

Substituting into the t-derivative of Eq. (45.129),

$$-\frac{16\pi^2}{g_c^3}\,\beta = \Big(3C_A - \sum_r T(r)\Big) - C_A\,\frac{2\beta}{g_c} + \sum_r T(r)\,\gamma_r, \tag{45.133}$$

where the last term used $\dfrac{d}{dt}(-\sum_r T(r)\ln Z_r) = +\sum_r T(r)\gamma_r$. Collect the numerator using $3C_A - \sum_r T(r) + \sum_r T(r)\gamma_r = 3C_A - \sum_r T(r)(1-\gamma_r)$, and move the β term from the right to the left:

$$-\beta\left(\frac{16\pi^2}{g_c^3} - \frac{2C_A}{g_c}\right) = 3C_A - \sum_r T(r)\,(1-\gamma_r). \tag{45.134}$$

Factor $\dfrac{16\pi^2}{g_c^3}\Big(1 - \dfrac{C_A g_c^2}{8\pi^2}\Big)$ out of the bracket on the left and solve for β. Dropping the now-redundant subscript on the physical coupling ($g \equiv g_c$), this is the NSVZ beta function.

Theorem 45.21 (The exact NSVZ beta function)**.** *For $\mathcal{N}=1$ super-Yang–Mills with gauge group of adjoint Casimir C_A coupled to chiral matter superfields in representations r of Dynkin index $T(r)$ and anomalous dimensions γ_r defined by Eq. (45.117), the beta function of the canonically normalized gauge coupling is, to all orders in perturbation theory,*

$$\boxed{\beta(g) = -\frac{g^3}{16\pi^2}\,\frac{3C_A - \sum_r T(r)\,(1-\gamma_r)}{1 - \dfrac{C_A\,g^2}{8\pi^2}}} \tag{45.135}$$

The numerator is the one-loop coefficient corrected by the matter anomalous dimensions; the denominator is the resummed contribution of the gaugino rescaling anomaly. The only quantity not fixed by holomorphy and the anomaly is γ_r, which carries all remaining scheme-dependent dynamics and starts at one loop.

Caution

The NSVZ formula is exact in a specific (NSVZ) scheme; like any beta function beyond two loops it is scheme-dependent, and Eq. (45.135) should be read together with the scheme in which γ_r and the holomorphic running Eq. (45.125) are defined. What is genuinely scheme-*independent* are the consequences we draw from it: the one- and two-loop coefficients, the vanishing of β at the *Banks–Zaks* or superconformal fixed points where the numerator vanishes, and the location of the conformal window for SQCD. The honest statement, which Chapter 46 takes as input, is that holomorphy plus one controlled anomaly reduces the entire all-orders gauge running to a single one-loop quantity γ_r.

45.4.6 Reduction at one loop: the supersymmetric Yang–Mills theory coefficient

We now make contact with Chapter 27. At weak coupling the denominator of Eq. (45.135) tends to 1 and the anomalous dimensions $\gamma_r = O(g^2)$ may be dropped from the numerator, leaving

$$\beta(g) = -\frac{g^3}{16\pi^2}\, b_0 + O(g^5), \qquad b_0 = 3C_A - \sum_r T(r), \tag{45.136}$$

the same universal form $\beta = -(b_0/16\pi^2)g^3$ as the ordinary Yang–Mills result Eq. (27.1), but with a supersymmetric coefficient. The next worked example extracts b_0 for the two cases we require and reconciles them, term by term, with the non-supersymmetric value $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$.

Worked Example 45.22 (One-loop coefficient for pure SYM and for SU(N_c) SQCD)**.** We compute b_0 two ways — from the supersymmetric formula Eq. (45.136) and by adding up the component fields with the ordinary gauge-theory bookkeeping — and check that they agree, thereby exhibiting exactly how the gaugino and the squarks conspire to convert the familiar $\frac{11}{3}C_A$ into $3C_A$.

Component bookkeeping. For an ordinary gauge theory the one-loop coefficient is assembled from three kinds of fields, each contributing to the antiscreening/screening competition of Section 27.5:

$$b_0 = \frac{11}{3}\,C_A - \frac{2}{3}\sum_{\text{Weyl } f} T(r_f) - \frac{1}{3}\sum_{\text{complex } s} T(r_s), \tag{45.137}$$

the gauge boson giving $+\frac{11}{3}C_A$ (the gluon antiscreening), each Weyl fermion giving $-\frac{2}{3}T(r)$, and each complex scalar $-\frac{1}{3}T(r)$. (A Dirac fermion is two Weyl fermions, returning the $-\frac{4}{3}T(r)$ of Eq. (27.1).)

Pure super-Yang–Mills. The vector multiplet contains the gauge boson A_μ^a and a single Weyl (Majorana) gaugino λ^a in the adjoint, $T(\text{adj}) = C_A$. There is no matter. From Eq. (45.137),

$$b_0^{\text{SYM}} = \underbrace{\frac{11}{3} C_A}_{\text{gauge boson}} \underbrace{- \frac{2}{3} C_A}_{\text{gaugino}} = \frac{11-2}{3} C_A = 3\, C_A, \tag{45.138}$$

in exact agreement with the supersymmetric formula Eq. (45.136) with no matter sum. The reconciliation is needed is the single subtraction $-\frac{2}{3}C_A$: the adjoint gaugino, absent in ordinary Yang–Mills, screens by exactly $\frac{2}{3}C_A$ and turns $\frac{11}{3}C_A$ into $3C_A$.

Chiral matter. A chiral superfield in representation r contains one Weyl fermion (the quark) *and* one complex scalar (the squark), both in r. Its total contribution is

$$-\frac{2}{3} T(r) - \frac{1}{3} T(r) = -\, T(r), \tag{45.139}$$

which is precisely the matter term $-T(r)$ appearing in $b_0 = 3C_A - \sum_r T(r)$. Supersymmetry pairs each Weyl quark with a complex squark so that their separate $\frac{2}{3}$ and $\frac{1}{3}$ fractions sum to the clean integer-coefficient $-T(r)$.

SU(N_c) *SQCD with N_f flavors.* Here $C_A = N_c$. Each flavor supplies a quark chiral superfield Q in the fundamental $\mathbf{N_c}$ and an antiquark chiral superfield $\widetilde{Q}$ in the antifundamental $\bar{\mathbf{N}}_\mathbf{c}$, with Dynkin indices $T(\mathbf{N_c}) = T(\bar{\mathbf{N}}_\mathbf{c}) = T_F = \frac{1}{2}$. The matter sum over the $2N_f$ chiral superfields is

$$\sum_r T(r) = N_f (T_F + T_F) = N_f \left(\tfrac{1}{2} + \tfrac{1}{2}\right) = N_f, \tag{45.140}$$

so that

$$\boxed{b_0^{\text{SQCD}} = 3C_A - \sum_r T(r) = 3N_c - N_f.} \tag{45.141}$$

Comparison with Yang–Mills theory. For ordinary QCD-like Yang–Mills with n_f Dirac flavors in the fundamental, Eq. (27.1) gave $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$, i.e. for SU($N_c$) with $T_F = \frac{1}{2}$,

$$b_0^{\text{QCD}} = \frac{11}{3} N_c - \frac{2}{3} n_f. \tag{45.142}$$

The supersymmetric refinement Eq. (45.141) differs in both terms, and the origin of each difference is now explicit:

- the pure-gauge coefficient drops from $\frac{11}{3}N_c$ to $3N_c$ because the adjoint gaugino contributes the extra $-\frac{2}{3}N_c$ of Eq. (45.138);
- the matter coefficient changes from $-\frac{2}{3}$ per Dirac flavor to -1 per SQCD flavor because an SQCD "flavor" is a vector-like pair of chiral superfields

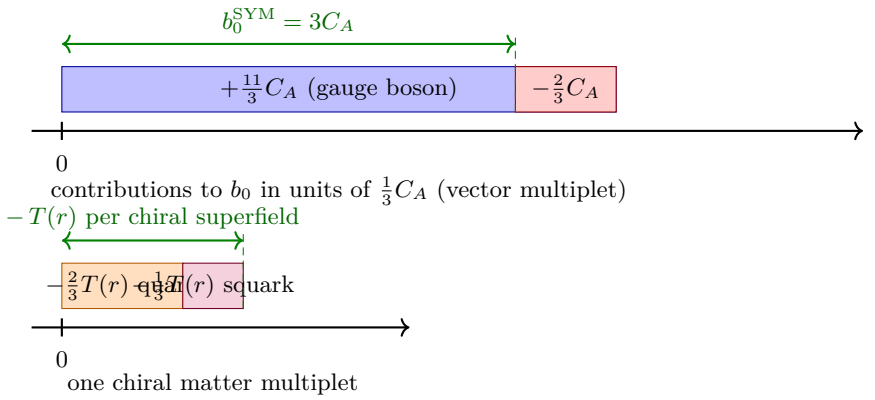


Figure 45.4: The one-loop coefficient b_0 as a sum over component fields. Top: the vector multiplet. The gauge boson antiscreens by $+\frac{11}{3}C_A$ (the familiar Yang–Mills value of Eq. (27.1)); the adjoint gaugino screens by $-\frac{2}{3}C_A$; the supersymmetric sum is the clean $3C_A$ of Eq. (45.138). Bottom: a single chiral matter multiplet, whose Weyl quark ($-\frac{2}{3}T(r)$) and complex squark ($-\frac{1}{3}T(r)$) add to $-T(r)$, the matter term of $b_0 = 3C_A - \sum_r T(r)$.

(Q and $\widetilde{Q}$, summing the two $-\frac{1}{2}$ Weyl-fermion pieces of a Dirac flavor) *plus* their squarks (the extra $-\frac{1}{3} - \frac{1}{3} = -\frac{2}{3}$ from two complex scalars), the two effects adding to the clean -1 of Eq. (45.140).

Both readings of b_0 agree, and the supersymmetric value is the one that feeds the exact NSVZ denominator. For SU(3) ($C_A = 3$) the asymptotic-freedom boundary moves from $n_f < \frac{11}{2}N_c = \frac{33}{2}$ Dirac flavors in QCD to $N_f < 3N_c = 9$ chiral flavors in SQCD; the window $3N_c < N_f$ where the theory loses asymptotic freedom, and the intermediate conformal window, are the data Chapter 46 exploits.

The picture in Fig. 45.4 is the visual summary of the thread advance: the same gluon antiscreening that drove asymptotic freedom in Chapter 27 survives, but the superpartners shift every coefficient onto the integer values dictated by supersymmetry, and those integers are exactly the data that the NSVZ denominator resums to all orders.

Physical Insight

There is a clean way to see why the gauginos contribute precisely $-\frac{2}{3}C_A$ rather than the $-\frac{4}{3}C_A$ a Dirac adjoint fermion would give: a gaugino is a single Weyl (Majorana) fermion, half a Dirac fermion, so it carries half the screening, $-\frac{1}{2}\cdot\frac{4}{3}C_A = -\frac{2}{3}C_A$. The deep statement is that supersymmetry forbids a bare gaugino mass (it would break the multiplet's boson–fermion degeneracy), so the gaugino is necessarily massless and Majorana, and

that masslessness is what fixes its contribution to be exactly the value that turns $\frac{11}{3}C_A$ into the holomorphy-protected $3C_A$. The numerator of NSVZ is not an accident of arithmetic; it is the imprint of the protected superpotential and the masslessness of the gauginos.

With the exact gauge coupling under control — the superpotential frozen by holomorphy, the Kähler renormalization quarantined into the anomalous dimensions γ_r, and the all-orders running packaged in Eq. (45.135) — we have the precise statement of how much, and how little, quantum corrections can do to a supersymmetric gauge theory. This is the control that Chapter 46 leverages: the one-loop-exact holomorphic coupling and the anomaly-free combinations identified here are the inputs to Seiberg duality and to gaugino condensation, where the holomorphic scale Λ of Eq. (45.123) becomes a dynamical order parameter. The next section turns from these exact protections to the question they force: since the world is manifestly not supersymmetric, how can supersymmetry be broken *softly* — lifting the superpartners — without sacrificing the very nonrenormalization that made the hierarchy stable.

45.5 Soft Supersymmetry Breaking

The nonrenormalization theorem of Section 45.4 told us precisely what supersymmetry protects: the holomorphic superpotential, and through it the exact equalities among masses, Yukawa couplings, and scalar self-couplings that descend from a single function $W(\Phi)$. That rigidity is the source of the boson–fermion mass degeneracy proved from the algebra in Theorem 44.12. But the same theorem is, read the other way, an indictment: a manifestly supersymmetric theory predicts a scalar electron exactly as light as the electron, a fermionic photino exactly as massless as the photon, and no such particles have been seen. If supersymmetry is realized in nature it must be *broken*, and broken in a way that lifts these degeneracies by at least the scale at which we have looked and failed to find superpartners.

The problem is that the protection and the degeneracy are the same fact. Break supersymmetry carelessly — with an arbitrary operator — and one throws away the very cancellations that made the theory worth building, in particular the cancellation of the quadratic divergence in the Higgs mass that motivated supersymmetry as a cure for the hierarchy problem (Section 35.3). The task of this section is to thread that needle. We first prove, with all the algebra on the page, that in unbroken supersymmetry the quadratically divergent fermion-loop contribution to a scalar mass is cancelled by the contribution of its scalar superpartners — the calculation that is the resolution of the hierarchy problem. We then ask which breaking terms can be added without spoiling that cancellation. The answer, due to Girardello and Grisaru, is a short and complete list of *soft*

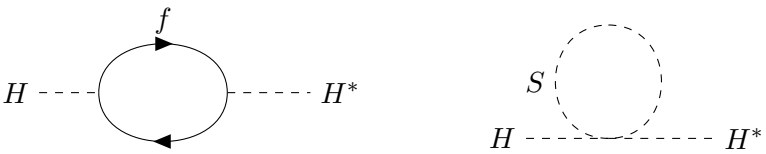


Figure 45.5: The two one-loop contributions to the light-scalar self-energy δm_H^2. Left: the fermion loop from the Yukawa coupling λ_f, which is quadratically divergent and *negative*. Right: the scalar loop from the quartic coupling λ_S, quadratically divergent and *positive*. Supersymmetry sets $\lambda_S = |\lambda_f|^2$ and supplies exactly two complex scalars S per Dirac fermion f, so the two $\Lambda_{\rm UV}^2$ pieces cancel.

operators, characterized by a single criterion: their couplings carry positive mass dimension. The spurion method makes both the criterion and the list transparent, and shows how each soft term descends from a supersymmetry-breaking sector through a background superfield. We close by assembling these terms into the soft Lagrangian that the next section instantiates as the MSSM.

45.5.1 The quadratic-divergence cancellation

The hierarchy problem, as diagnosed in Section 35.3, is that the mass-squared of a fundamental scalar is a relevant coupling: no symmetry of the Standard Model forbids it, so its renormalization is quadratically sensitive to the highest scale in the theory. Concretely, every heavy particle of mass M that couples to the Higgs shifts m_H^2 by an amount of order $M^2/(16\pi^2)$ (with a momentum cutoff $\Lambda_{\rm UV}$, by $\Lambda_{\rm UV}^2/(16\pi^2)$), so keeping m_H at the electroweak scale in the presence of physics at $M_{\rm Pl}$ demands a cancellation tuned to thirty digits. A chiral fermion, by contrast, is protected: a chiral symmetry forbids its mass, so its self-energy is at worst logarithmically divergent, $\delta m_f \propto m_f \ln \Lambda_{\rm UV}$. Supersymmetry exports that fermionic protection to the scalar by tying the two together in one multiplet. We now show the mechanism explicitly.

Consider a light complex scalar H — the prototype Higgs — coupled to a heavy chiral fermion f of mass m_f through a Yukawa interaction, and to a heavy complex scalar S of mass m_S through a quartic interaction:

$$\mathcal{L}_{\rm int} = -\,\lambda_f\, H\, \bar{f} P_L f \;-\; \lambda_f^*\, H^*\, \bar{f} P_R f \;-\; \lambda_S\, |H|^2\, |S|^2 \,. \tag{45.143}$$

Here $P_{L,R} = \tfrac{1}{2}(1 \mp \gamma^5)$ are the chiral projectors of Chapter 24's conventions and $\bar{f} = f^\dagger \gamma^0$. We compute the one-loop correction to the H self-energy at zero external momentum, δm_H^2, isolating the quadratically divergent piece. The two contributing diagrams are shown in Fig. 45.5: a closed fermion loop and a closed scalar loop.

All power-divergent integrals here are most honestly handled with a Euclidean momentum cutoff $\Lambda_{\rm UV}$, since the quadratic divergence is exactly the quantity

of physical interest and dimensional regularization sets it silently to zero. The single master integral we need is

$$I(M^2) \;\equiv\; \int \frac{\mathrm{d}^4 k_E}{(2\pi)^4}\,\frac{1}{k_E^2 + M^2} \;=\; \frac{1}{16\pi^2}\Big(\Lambda_{\mathrm{UV}}^2 - M^2 \ln\frac{\Lambda_{\mathrm{UV}}^2}{M^2} + \dots\Big), \tag{45.144}$$

obtained by writing $\mathrm{d}^4 k_E = 2\pi^2 k_E^3\,\mathrm{d}k_E$, integrating $k_E^3/(k_E^2+M^2)$ up to Λ_{UV}, and expanding. The leading term is the quadratic divergence; the subleading term is the logarithm we expect to survive. Continuing from Minkowski space we use $\int \mathrm{d}^4k/(2\pi)^4\; 1/(k^2 - M^2 + i\varepsilon) = -\,i\,I(M^2)$, the standard Wick rotation $k^0 = ik_E^0$, $\mathrm{d}^4k = i\,\mathrm{d}^4k_E$, $k^2 = -k_E^2$.

Fermion loop. The amplitude is

$$i\,\Pi_f(0) = (-1)\,(-i\lambda_f)(-i\lambda_f^*)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\;\mathrm{Tr}\left[\frac{i(\not k + m_f)}{k^2 - m_f^2}\,P_L\,\frac{i(\not k + m_f)}{k^2 - m_f^2}\,P_R\right], \tag{45.145}$$

where the overall (-1) is the obligatory minus sign of a closed fermion loop and the two vertices carry the projectors P_L (at the H vertex) and P_R (at the H^* vertex). The Dirac trace collapses cleanly. Using $\gamma^5 \not k = -\not k \gamma^5$, hence $P_L \not k = \not k P_R$, the numerator becomes

$$\begin{aligned}\mathrm{Tr}[(\not k + m_f)P_L(\not k + m_f)P_R] &= \mathrm{Tr}[(\not k P_L + m_f P_L)(\not k P_R + m_f P_R)] \\ &= \mathrm{Tr}[P_R \not k\, \not k P_R] + m_f\,\mathrm{Tr}[P_R \not k P_R] + m_f\,\mathrm{Tr}[P_L \not k P_R] + m_f^2\,\mathrm{Tr}[P_L P_R] \\ &= k^2\;\mathrm{Tr}[P_R] \;=\; 2\,k^2. \end{aligned} \tag{45.146}$$

The three middle terms vanish: $P_R \not k P_R = \not k P_L P_R = 0$ and $P_L P_R = 0$, while $\mathrm{Tr}[P_R \not k \not k P_R] = k^2\,\mathrm{Tr}[P_R]$ with $\mathrm{Tr}[P_R] = 2$. Notice that the fermion mass has dropped out of the leading numerator entirely; only $2k^2$ survives, the hallmark of a chiral coupling. Inserting Eq. (45.146) into Eq. (45.145), with $(-i\lambda_f)(-i\lambda_f^*) = -\,|\lambda_f|^2$ and the explicit $i^2 = -1$ from the two propagator numerators,

$$i\,\Pi_f(0) = (-1)(-\,|\lambda_f|^2)(-1)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{2k^2}{(k^2 - m_f^2)^2} = -\,2\,|\lambda_f|^2 \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{k^2}{(k^2 - m_f^2)^2}. \tag{45.147}$$

The quadratic divergence comes from the large-k region, where $k^2/(k^2-m_f^2)^2 = 1/(k^2-m_f^2) + m_f^2/(k^2-m_f^2)^2$; the first term is quadratically divergent and the second only logarithmically. Keeping the leading piece,

$$i\,\Pi_f(0)\big|_{\mathrm{quad}} = -\,2\,|\lambda_f|^2 \int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{1}{k^2 - m_f^2} = -\,2\,|\lambda_f|^2\;(-i\,I) = +\,2\,i\;|\lambda_f|^2\;\frac{\Lambda_{\mathrm{UV}}^2}{16\pi^2}, \tag{45.148}$$

so $\Pi_f(0)\big|_{\text{quad}} = +\,|\lambda_f|^2\,\Lambda_{\text{UV}}^2/(8\pi^2)$. The correction to the physical mass-squared is read off from the resummed propagator $i/(p^2 - m_H^2 + \Pi)$, whose pole sits at $p^2 = m_H^2 - \Pi(0)$, giving $\delta m_H^2 = -\Pi(0)$. Hence

$$\boxed{\delta m_H^2\big|_f = -\,\frac{|\lambda_f|^2}{8\pi^2}\,\Lambda_{\text{UV}}^2 + (\text{logs})} \tag{45.149}$$

— the quadratic divergence is present, and it is *negative*.

Scalar loop. The quartic term $-\lambda_S\,|H|^2\,|S|^2 = -\lambda_S H^* H S^* S$ gives a single vertex joining one H, one H^*, one S and one S^*; its Feynman rule is $-i\lambda_S$. Closing the S loop on the two H legs,

$$i\,\Pi_S(0) = (-i\lambda_S)\int \frac{\mathrm{d}^4k}{(2\pi)^4}\,\frac{i}{k^2 - m_S^2} = (-i\lambda_S)\,(I) = -\,i\,\lambda_S\,\frac{\Lambda_{\text{UV}}^2}{16\pi^2} + (\text{logs}), \tag{45.150}$$

using $\int \mathrm{d}^4k/(2\pi)^4\; i/(k^2 - m_S^2) = i(-iI) = I$. Thus $\Pi_S(0)\big|_{\text{quad}} = -\lambda_S\Lambda_{\text{UV}}^2/(16\pi^2)$ and

$$\delta m_H^2\big|_S = -\Pi_S(0) = +\,\frac{\lambda_S}{16\pi^2}\,\Lambda_{\text{UV}}^2 + (\text{logs}), \tag{45.151}$$

one factor of this per complex scalar coupled to H. The divergence is again quadratic, and this time *positive*. The opposite sign relative to Eq. (45.149) is structural: it is the closed-fermion-loop (-1) of Eq. (45.145) that has no analogue in the bosonic Eq. (45.150).

The cancellation. Opposite signs are not enough; the magnitudes must match, and match they do under supersymmetry, for two independent reasons that both follow from the structure built in Chapter 44. First, a Dirac fermion f is two Weyl fermions, so it belongs to *two* chiral supermultiplets, each carrying one complex scalar; the heavy fermion thus comes with exactly two complex scalars S_1, S_2. Second, the superpotential origin of the couplings fixes the quartic in terms of the Yukawa: as the Wess–Zumino computation of Theorem 44.42 showed (Eq. (44.214), Eq. (44.149)), the scalar quartic is the modulus-squared of the same holomorphic coupling that sets the Yukawa, so

$$\lambda_S = |\lambda_f|^2\,. \tag{45.152}$$

Summing the two scalar contributions and the fermion contribution,

$$\delta m_H^2 = \underbrace{-\,\frac{|\lambda_f|^2}{8\pi^2}\,\Lambda_{\text{UV}}^2}_{\text{fermion } f} + \underbrace{2\times\frac{\lambda_S}{16\pi^2}\,\Lambda_{\text{UV}}^2}_{\text{scalars } S_1,S_2} = \Lambda_{\text{UV}}^2\,\frac{|\lambda_f|^2}{16\pi^2}(-2+2) = 0 \quad (\text{quadratic part}), \tag{45.153}$$

where we used $\lambda_S = |\lambda_f|^2$. The $\Lambda_{\rm UV}^2$ pieces cancel *identically*. What remains is the logarithm in Eq. (45.144),

$$\delta m_H^2 = \frac{|\lambda_f|^2}{8\pi^2}(m_S^2 - m_f^2)\ln\frac{\Lambda_{\rm UV}}{\mu} + \dots, \tag{45.154}$$

which vanishes as well when $m_S = m_f$, i.e. in *unbroken* supersymmetry, where Theorem 44.12 forces $m_S = m_f$ and the scalar mass receives no correction at all. The structure of Eq. (45.154) is the punchline: the correction to a scalar mass is controlled not by the cutoff but by the *superpartner mass splitting* $m_S^2 - m_f^2$ and grows only logarithmically with $\Lambda_{\rm UV}$, governed by the renormalization-group running of Chapter 24. This is why supersymmetry solves the hierarchy problem, and it tells us precisely what breaking is allowed: any breaking we introduce may shift $m_S^2 - m_f^2$ away from zero, but it must not reintroduce the $\Lambda_{\rm UV}^2$ term.

Physical Insight

The cancellation is the loop-level shadow of the algebra. The supercharge anticommutator $\left\{Q_\alpha, \bar{Q}_{\dot\beta}\right\} = 2\sigma^\mu_{\alpha\dot\beta}P_\mu$ (Eq. (44.61)) ties bosons and fermions into one multiplet with equal couplings and equal masses; the equal couplings give equal-magnitude loop integrals and the spin-statistics minus sign gives them opposite signs. The vacuum-energy version of the same statement is Theorem 44.10: unbroken supersymmetry forces $E_{\rm vac} = 0$ because bosonic and fermionic zero-point energies cancel mode by mode. The scalar-mass cancellation here is the second derivative of that vacuum-energy cancellation with respect to the background field.

45.5.2 The supertrace sum rule

The diagrammatic cancellation above is a special case of a sum rule that holds in any supersymmetric theory and that turns out to be exactly the quantity soft breaking must violate. The one-loop vacuum energy of a free field of mass M is the sum of its zero-point energies, which in covariant form is $\frac{1}{2}\int \mathrm{d}^4k_E/(2\pi)^4\,\ln(k_E^2 + M^2)$ per bosonic degree of freedom, with an extra minus sign for fermions because their functional determinant appears in the numerator rather than the denominator of the path integral. Summing over a spectrum of field-dependent masses $M_J^2(\phi)$, the one-loop effective potential is

$$V_1(\phi) = \tfrac{1}{2}\sum_J (-1)^{2J}(2J+1)\int\frac{\mathrm{d}^4k_E}{(2\pi)^4}\,\ln(k_E^2+M_J^2(\phi)) \;\equiv\; \tfrac{1}{2}\,\mathrm{STr}\int\frac{\mathrm{d}^4k_E}{(2\pi)^4}\,\ln(k_E^2+M^2(\phi)), \tag{45.155}$$

where the *supertrace* weights each spin-J state by $(-1)^{2J}(2J+1)$: $+1$ for a real scalar, -2 for a Weyl fermion, $+3$ for a massive vector, and so on. The quadratically divergent part is extracted by differentiating the logarithm once

with respect to M^2, which returns the master integral Eq. (45.144); hence $\int \mathrm{d}^4k_E/(2\pi)^4\,\ln(k_E^2+M^2) \supset (\Lambda_{\mathrm{UV}}^2/16\pi^2)\,M^2$, and

$$V_1(\phi)\big|_{\mathrm{quad}} = \frac{\Lambda_{\mathrm{UV}}^2}{32\pi^2}\,\mathrm{STr}\,M^2(\phi), \qquad \mathrm{STr}\,M^2 \equiv \sum_J (-1)^{2J}(2J+1)\,M_J^2. \quad (45.156)$$

The scalar mass renormalization is $\partial^2 V_1/\partial\phi\,\partial\phi^*$, so the quadratically divergent part of any scalar mass is proportional to $\partial^2(\mathrm{STr}\,M^2)/\partial\phi\,\partial\phi^*$.

Now the key fact. In a supersymmetric theory every chiral multiplet contributes one Weyl fermion of mass M_F and one complex scalar (two real scalars) of mass M_S, with $M_S^2 = M_F^2 = |W''|^2$ as the component computation of Theorem 44.42 made explicit. Its contribution to the supertrace is therefore $2\cdot M_S^2 - 2\cdot M_F^2 = 0$. Including gauge multiplets and the constraints of the D-term sector leaves the statement intact at the level we need:

Theorem 45.23 (Supertrace mass-sum rule)**.** *In a renormalizable theory with unbroken global supersymmetry, the supertrace of the tree-level squared-mass matrix vanishes,*

$$\mathrm{STr}\,M^2 \;\equiv\; \sum_J (-1)^{2J}(2J+1)\,M_J^2 \;=\; 0, \quad (45.157)$$

as a function of the background fields. Consequently $V_1(\phi)\big|_{\mathrm{quad}} = 0$*: the one-loop effective potential, and in particular every scalar mass, is free of quadratic divergences.*

Proof. By multiplet. A chiral multiplet contributes $2M_S^2 - 2M_F^2$; supersymmetry gives $M_S^2 = M_F^2$ (the second derivatives of the superpotential coincide for the scalar mass matrix and the fermion mass matrix, Theorem 44.42), so its contribution is zero *as a function of* ϕ. A massless gauge multiplet contributes a vector ($+3$ at the massive point, $+2$ when massless and gauge-fixed consistently with the two physical polarizations) and a Weyl gaugino (-2); the contributions again cancel because gauge symmetry equates the gaugino mass to the gauge-boson mass within the multiplet. Summing the manifestly vanishing multiplet-by-multiplet contributions gives Eq. (45.157). Inserting it into Eq. (45.156) kills the quadratic divergence identically, and differentiating twice with respect to the background field shows the same for any individual scalar mass. □

This is the abstract content of Eq. (45.153): the cancellation of the toy computation is the statement $\mathrm{STr}\,M^2 = 0$ specialized to the fermion f and its two scalar partners. The sum rule also makes vivid the constraint on breaking. If we are to split superpartner masses while preserving the cancellation, $\mathrm{STr}\,M^2$ may become nonzero — but only as a *field-independent constant*, so that $\partial^2(\mathrm{STr}\,M^2)/\partial\phi\,\partial\phi^* = 0$ and no quadratic divergence reappears in any scalar

mass. A breaking that makes $\mathrm{STr}\, M^2$ depend on the fields is hard; one that shifts it by a constant is soft. Quantitatively, soft breaking produces

$$\mathrm{STr}\, M^2 = \sum_{\text{scalars}} m_i^2 - 2 \sum_{\text{fermions}} \cdots = \mathcal{O}(m_{\text{soft}}^2) \times (\text{const}), \tag{45.158}$$

a field-independent number of order the soft masses squared, which feeds only the cosmological constant and not the running of any mass.

45.5.3 The spurion method and the criterion for softness

To organize the breaking we use the device already met in Section 45.4: promote a breaking parameter to the vacuum value of a background superfield, a *spurion*. Suppose supersymmetry is broken spontaneously in some sector — by the F-term mechanism of Section 44.5.3, in which the auxiliary field of a chiral superfield acquires an expectation value, $\langle F \rangle \neq 0$, signalled by a positive vacuum energy through Theorem 44.10. Package the order parameter into a single gauge-singlet chiral superfield

$$X(y, \theta) = \langle X \rangle + \theta^2 F_X, \qquad \langle F_X \rangle \neq 0, \tag{45.159}$$

whose scalar component we may set to zero by a field redefinition and whose θ^2-component F_X is the breaking order parameter, of mass dimension two. The visible-sector fields do not couple to X directly at renormalizable level; instead the two sectors communicate through heavy physics of mass M (messengers, or gravity), so that the effective couplings are organized in powers of F_X/M^2 and X/M. The spurion bookkeeping is the statement that the full theory, breaking sector included, is supersymmetric: every term in the Lagrangian is a genuine superspace integral, and supersymmetry breaking is communicated *only* through the θ^2-component of X. This is what makes the consequences computable.

The criterion for softness now follows from dimensional analysis alone.

Theorem 45.24 (Girardello–Grisaru criterion)**.** *A supersymmetry-breaking operator added to a renormalizable Lagrangian fails to regenerate quadratic divergences in scalar masses — it is* soft *— if and only if its coupling constant has strictly positive mass dimension.*

Proof. A scalar mass renormalization δm^2 has mass dimension two. Suppose the operator in question multiplies a coupling g of mass dimension $[g] = d_g$. The most divergent contribution it can make to a scalar self-energy has the form $\delta m^2 \sim g\, \Lambda_{\text{UV}}^{p} \times$(dimensionless loop factors and logs). Matching mass dimensions, $2 = d_g + p$, so $p = 2 - d_g$. A quadratic divergence is $p = 2$, which requires $d_g = 0$: only a *dimensionless* coupling can multiply a Λ_{UV}^2. If $d_g > 0$ then $p = 2 - d_g < 2$, and the strongest possible divergence is at worst $\Lambda_{\text{UV}}^{2-d_g}$, i.e. a positive power of the soft scale times a logarithm: $\delta m^2 \sim g^{2/d_g} \ln \Lambda_{\text{UV}}$ in the simplest cases, never $g\Lambda_{\text{UV}}^2$. The spurion picture confirms the counting and shows it is exact,

not merely dimensional. Any soft coupling is, by construction, a power of F_X/M or F_X/M^2 times a supersymmetric vertex; restoring the spurion, a would-be quadratic divergence $\propto \Lambda_{\rm UV}^2$ multiplying g must come from a supergraph whose superspace structure is that of the unbroken theory, where Theorem 45.23 already set the field-dependent $\Lambda_{\rm UV}^2$ to zero. The breaking can only attach extra factors of F_X, each carrying dimension two and a corresponding suppression by M, so the divergence is softened by exactly the powers of $\Lambda_{\rm UV}$ that dimensional analysis predicts. The converse is the counterexample below: a dimensionless breaking coupling does produce $g\Lambda_{\rm UV}^2$. □

Worked Example 45.25 (A hard operator that ruins the cancellation)**.** The cleanest way to see the criterion bite is to add a breaking term whose coupling is dimensionless and watch the quadratic divergence return. Take the toy of Eq. (45.143) and add to it a single extra scalar quartic that violates the supersymmetric relation Eq. (45.152),

$$\Delta\mathcal{L} = -\,\Delta\lambda\,|H|^2\,|S|^2\,, \qquad [\Delta\lambda] = 0. \tag{45.160}$$

Its coupling $\Delta\lambda$ is dimensionless. Repeating the scalar-loop computation of Eq. (45.150) with $\lambda_S \to \lambda_S + \Delta\lambda$, the cancellation in Eq. (45.153) is spoiled by precisely the unmatched piece:

$$\delta m_H^2 = +\,2 \times \frac{\Delta\lambda}{16\pi^2}\,\Lambda_{\rm UV}^2 + \cdots = \frac{\Delta\lambda}{8\pi^2}\,\Lambda_{\rm UV}^2 + \cdots\,, \tag{45.161}$$

a quadratic divergence proportional to the dimensionless $\Delta\lambda$. The hierarchy problem is back in full force. This is exactly what Theorem 45.24 forbids: because $[\Delta\lambda] = 0$, the operator is *hard*. Note the contrast with a *trilinear* scalar coupling $-A\,H^2S + \text{h.c.}$ with $[A] = 1$: redoing the power counting, the strongest divergence it can produce in m_H^2 is $\sim A^2/(16\pi^2) \times \ln\Lambda_{\rm UV}$, only logarithmic — so a trilinear scalar coupling is soft, a quartic deviation is not. The single discriminating feature is the mass dimension of the coupling.

Caution

The criterion "positive-dimension coupling" is necessary and, for the operators that arise in gauge theories without gauge singlets, sufficient. There is one classic loophole, noted already by Girardello and Grisaru: the non-holomorphic trilinear scalar coupling $\phi^*\phi\phi$ (coupling of dimension one, hence naively soft) can close into a one-loop tadpole and generate a quadratically divergent linear term for any *gauge-singlet* scalar present in the theory, which then feeds back a quadratic divergence. In a theory whose chiral superfields are all charged under the gauge group — the MSSM, with no gauge singlets in its minimal field content — no such tadpole exists and $\phi^*\phi\phi$ would be harmless; but because it is also forbidden

by holomorphy of the would-be superpotential and brings nothing the analytic A-terms do not, it is conventionally excluded from the soft list. The honest statement is therefore: the soft operators enumerated below are soft in any gauge theory, and the only would-be extra candidate is excluded because it is dangerous precisely when gauge singlets are present.

45.5.4 The Girardello–Grisaru list of soft terms

Applying the criterion — gauge invariance, Lorentz invariance, and a coupling of strictly positive mass dimension — exhausts the renormalizable supersymmetry-breaking operators in a remarkably short list. For a generic gauge theory with gauge multiplets labelled by a (gauginos λ^a) and chiral multiplets labelled by i (complex scalars ϕ_i, Weyl fermions ψ_i), the complete soft Lagrangian is

$$\mathcal{L}_{\text{soft}} = -\Big(\tfrac{1}{2}\, M_a\, \lambda^a\lambda^a + \tfrac{1}{6}\, a_{ijk}\, \phi_i\phi_j\phi_k + \tfrac{1}{2}\, b_{ij}\, \phi_i\phi_j + \text{h.c.}\Big) \;-\; (m^2)_{ij}\, \phi_i^*\phi_j. \tag{45.162}$$

The four classes are, with their coupling dimensions:

- **Gaugino masses** M_a, $[M_a] = 1$: a Majorana mass $-\frac{1}{2}M_a\lambda^a\lambda^a$ for each gaugino, lifting it off the massless gauge boson it partners. These break supersymmetry but preserve gauge invariance because λ^a transforms in the (real) adjoint.
- **Scalar masses** $(m^2)_{ij}$, $[m^2] = 2$: a Hermitian mass-squared matrix $-(m^2)_{ij}\phi_i^*\phi_j$ for the scalars, the term that splits a squark or slepton from its quark or lepton partner. This is the non-holomorphic "D-term-type" mass.
- **Trilinear A-terms** a_{ijk}, $[a] = 1$: a holomorphic cubic scalar coupling $-\frac{1}{6}a_{ijk}\phi_i\phi_j\phi_k + \text{h.c.}$, with the *same* flavor structure as the superpotential Yukawa y_{ijk} (one usually writes $a_{ijk} = A_{ijk}y_{ijk}$, no sum).
- **Bilinear B-terms** b_{ij}, $[b] = 2$: a holomorphic quadratic scalar coupling $-\frac{1}{2}b_{ij}\phi_i\phi_j + \text{h.c.}$, the partner of a superpotential mass term $\mu_{ij}\Phi_i\Phi_j$ (one writes $b_{ij} = B_{ij}\mu_{ij}$).

Every coupling in Eq. (45.162) has positive mass dimension, so by Theorem 45.24 each preserves the absence of quadratic divergences; conversely, these are the *only* gauge- and Lorentz-invariant, positive-dimension operators that break supersymmetry, so the list is complete. A non-holomorphic trilinear $\phi_i^*\phi_j\phi_k$ is excluded for the reason discussed in the caution above; a chiral-fermion mass without a partner, or any dimensionless quartic deviation, is hard and excluded by the criterion. What is conspicuously *absent* is any quartic scalar coupling that is not already fixed by the supersymmetric D- and F-terms: the scalar quartic

potential of the theory remains the supersymmetric one, and this is what keeps the cancellation of Eq. (45.153) intact up to logarithms.

45.5.5 Soft terms from spurion insertions

The list Eq. (45.162) is not merely catalogued; each term is the residue of a manifestly supersymmetric coupling of the visible fields to the spurion X of Eq. (45.159), with the breaking carried in by the θ^2-component F_X. This is what justifies calling the soft terms "the imprint of a hidden breaking sector," and it fixes the parametric size of every coupling in terms of F_X and the mediation scale M. We display the two most important cases explicitly.

Worked Example 45.26 (Gaugino mass from $\int \mathrm{d}^2\theta\,(X/M)\,W^\alpha W_\alpha$)**.** The gauge kinetic term of Section 45.2 is the chiral superspace integral $\int \mathrm{d}^2\theta\, W^\alpha W_\alpha$, where the field-strength superfield expands as $W_\alpha = \lambda_\alpha + \cdots + \theta_\alpha D + (\sigma^{\mu\nu}\theta)_\alpha F_{\mu\nu} + \ldots$ with the gaugino λ at order θ^0. Couple it to the spurion through a dimension-five operator suppressed by M,

$$\mathcal{L} \supset \int \mathrm{d}^2\theta\, \frac{X}{2M}\, W^{a\alpha} W^a_\alpha + \text{h.c.} \tag{45.163}$$

The chiral measure $\int \mathrm{d}^2\theta$ projects onto the θ^2-component of the integrand. With $X = \langle X\rangle + \theta^2 F_X$, the θ^2 of X multiplies the θ^0 of $W^{a\alpha}W^a_\alpha$, which is $\lambda^a\lambda^a$:

$$\int \mathrm{d}^2\theta\, \frac{X}{2M}\, W^{a\alpha} W^a_\alpha \;\supset\; \frac{F_X}{2M}\, \lambda^a\lambda^a, \tag{45.164}$$

which is exactly the gaugino mass term of Eq. (45.162) with

$$M_a = \frac{F_X}{M}. \tag{45.165}$$

The $\langle X\rangle$-piece of Eq. (45.163) merely shifts the gauge coupling (a supersymmetric, holomorphic rescaling) and is absorbed into the definition of the coupling; it is the F_X-piece, present only because supersymmetry is broken, that gives the gaugino its mass. The operator is dimension five, so the mass is suppressed by one power of the mediation scale — the universal signature of a soft term carried by a positive-dimension coupling.

Worked Example 45.27 (Scalar mass from $\int \mathrm{d}^4\theta\,(X^\dagger X/M^2)\,\Phi^\dagger\Phi$)**.** A scalar mass is non-holomorphic, so it descends from a full-superspace (D-term) integral rather than a chiral one. The leading gauge-invariant coupling of the spurion to a matter superfield Φ_i (with $\Phi^\dagger\Phi$ the Kähler term of Chapter 44) is the dimension-six operator

$$\mathcal{L} \supset \int \mathrm{d}^4\theta\, \frac{X^\dagger X}{M^2}\, \Phi_i^\dagger \Phi_i. \tag{45.166}$$

The full measure $\int \mathrm{d}^4\theta$ projects onto the $\theta^2\bar\theta^2$-component. The spurion factor contributes $X^\dagger X \supset (\theta^2 F_X)(\bar\theta^2 F_X^*) = |F_X|^2\,\theta^2\bar\theta^2$, while the matter factor $\Phi_i^\dagger\Phi_i$ contributes its lowest component $\phi_i^*\phi_i$ at order $\theta^0\bar\theta^0$. Multiplying and reading off the $\theta^2\bar\theta^2$ coefficient,

$$\int \mathrm{d}^4\theta\, \frac{X^\dagger X}{M^2}\,\Phi_i^\dagger\Phi_i \;\supset\; \frac{|F_X|^2}{M^2}\,\phi_i^*\phi_i, \tag{45.167}$$

a soft scalar mass-squared

$$(m^2)_{ii} = \frac{|F_X|^2}{M^2}. \tag{45.168}$$

This is a genuine breaking effect: the auxiliary components of X also generate F-dependent cross terms, but the term displayed is the one that survives as a physical scalar mass when X has no propagating dynamics. In the same way, $\int \mathrm{d}^2\theta\,(X/M)\,\Phi_i\Phi_j\Phi_k$ yields the A-term $a_{ijk} = y_{ijk}F_X/M$ by projecting the θ^2 of X onto the scalar components $\phi_i\phi_j\phi_k$, and $\int \mathrm{d}^2\theta\,(X/M)\,\Phi_i\Phi_j$ yields the B-term $b_{ij} = \mu_{ij}F_X/M$.

Two lessons are worth stating. First, every soft coupling comes out as a positive power of F_X/M — $M_a, a, \sim F_X/M$ and $m^2, b \sim |F_X|^2/M^2$ or F_X/M — so they all share one scale, $m_{\text{soft}} \sim F_X/M$, and all have positive mass dimension exactly as Theorem 45.24 requires. Second, the structure of the list is dictated by which superspace integral can carry the spurion: chiral ($\int \mathrm{d}^2\theta$) integrals give the holomorphic terms (M_a, a, b), the full-superspace ($\int \mathrm{d}^4\theta$) integral gives the non-holomorphic scalar mass. There is no fifth integral, and hence no fifth class of soft term.

45.5.6 Mediation: where F_X comes from

We have treated X and the scale M as given. Honesty requires saying, at the level of a scoped statement, where they originate; the detailed model-building is beyond the perturbative-and-symmetry remit of this chapter and is taken up in the literature cited below. Supersymmetry is broken dynamically in a *hidden sector* with no Standard-Model gauge charges, producing $\langle F_X\rangle \neq 0$ and, by Theorem 44.10, a positive vacuum energy and a massless Goldstino (the analogue for supersymmetry of the Goldstone boson, developed for the spontaneous-breaking mechanisms of this chapter). The breaking is then *communicated* to the visible sector by interactions of mass scale M:

- **Gravity mediation:** the mediator is gravity itself, $M = M_{\text{Pl}}$, so $m_{\text{soft}} \sim F_X/M_{\text{Pl}}$. A weak-scale $m_{\text{soft}} \sim$ TeV then requires $\sqrt{F_X} \sim 10^{11}$ GeV, an intermediate scale. The soft terms are generic and need not be flavor-blind, which is the origin of the supersymmetric flavor problem.

- **Gauge mediation:** the mediators are messenger fields charged under the Standard-Model gauge group, at a scale $M_{\rm mess} \ll M_{\rm Pl}$; the soft terms arise at one and two loops, $m_{\rm soft} \sim (g^2/16\pi^2)\, F_X/M_{\rm mess}$, and are automatically flavor-universal because they are controlled by gauge charges, ameliorating the flavor problem.

In both cases the output is a set of soft parameters of the form Eq. (45.162), with magnitudes set by F_X and M and textures set by the mediation mechanism. For the purposes of the next section we take these parameters as phenomenological inputs, constrained only to lie near the TeV scale — the value that, by Eq. (45.154), keeps the logarithmic correction to the Higgs mass at the electroweak scale and so preserves the naturalness that motivated the construction.

We now possess the complete template: a supersymmetric and gauge-invariant Lagrangian, broken by the four soft terms of Eq. (45.162), with quadratic divergences cancelled up to logarithms governed by the soft scale. The next section, Section 45.6, instantiates this template with the realistic field content of the Minimal Supersymmetric Standard Model — two Higgs doublets, the squarks and sleptons, the gauginos and Higgsinos — writing out the explicit soft parameters and reading off two of their flagship consequences, the tree-level bound on the lightest Higgs mass and the improvement of gauge-coupling unification.

45.6 The MSSM in Outline

The previous section classified the operators that break supersymmetry softly — gaugino masses, scalar mass-squareds, holomorphic A- and B-terms — and showed that, alone among the breaking terms one might write, they lift the superpartners without reopening the quadratic-divergence wound that supersymmetry was invoked to close (Section 45.5, Eq. (45.162)). That classification was abstract: a list of permissible spurion structures. We now spend it. The Minimal Supersymmetric Standard Model (MSSM) is the statement that the observed matter and forces are organized into the chiral and vector supermultiplets built in Chapter 44 and Section 45.1, that their interactions are governed by the most economical R-parity-conserving superpotential consistent with gauge invariance and holomorphy, and that supersymmetry is broken softly at a scale near a TeV. Everything in this section is an instance of machinery already assembled. The gauge sector is three commuting copies of the super Yang–Mills theory of Section 45.2 — the indirect closure of Yang–Mills theory promised in the chapter introduction: the gluon field strength $F^a_{\mu\nu}$ quantized in Chapter 26 now sits inside a real vector superfield, accompanied by a gluino. The matter sector is a collection of chiral superfields whose scalar potential is read off from the F- and D-terms of Section 45.3. And the two flagship numbers we extract — the

tree-level ceiling on the lightest Higgs mass, and the meeting of the three gauge couplings near $2 \times 10^{16}\,\mathrm{GeV}$ — both follow, without new input, from structures already in hand: the first from the D-term quartic whose coefficient is *fixed* by the gauge couplings, the second from the supersymmetric one-loop beta coefficient derived in Section 45.4.

We proceed in five movements. First, the field content and quantum numbers. Second, the superpotential, with the two arguments — anomaly freedom and holomorphy — that force *two* Higgs doublets where the Standard Model needs one, and the precise statement of the μ problem and the R-parity motivation. Third, the soft Lagrangian written out in full as the concrete realization of the previous section. Fourth, the tree-level Higgs potential, minimized to yield the bound $m_h \le m_Z\,|\cos 2\beta|$ and the one-loop lift that rescues it. Fifth, gauge-coupling unification as a worked numerical computation. We close with an honest, dated account of where experiment stands.

45.6.1 Supermultiplet content and quantum numbers

The gauge group is that of the Standard Model, $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, with the covariant derivative $D_\mu = \partial_\mu - i g_s A_\mu^a T^a - i g\, W_\mu^i T^i - i g' Y B_\mu$ in the convention of Chapter 31 (electric charge $Q = T^3 + Y$, hypercharge Y as listed below). Supersymmetry requires that every Standard Model degree of freedom be promoted to a full supermultiplet of Section 44.4: a chiral superfield $\Phi = (\phi, \psi, F)$ for each Weyl fermion of the Standard Model, packaging that fermion with a complex scalar superpartner and an auxiliary F; a vector superfield $V = (A_\mu, \lambda, D)$ for each gauge boson, packaging it with a Majorana gaugino λ and an auxiliary D.

Chiral matter. The Standard Model fermions are chiral: left- and right-handed components transform differently under $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. We write everything in terms of left-handed Weyl superfields, trading each right-handed field for the charge conjugate of a left-handed one (so a right-handed up quark becomes a left-handed antiquark superfield $\bar{u} \equiv U^c$ in the conjugate color representation). Per generation the chiral matter superfields and their quantum numbers under $(\mathrm{SU}(3)_c, \mathrm{SU}(2)_L)_Y$ are

$$\begin{aligned} Q &= (\mathbf{3}, \mathbf{2})_{+1/6}, \qquad \bar{u} = (\bar{\mathbf{3}}, \mathbf{1})_{-2/3}, \qquad \bar{d} = (\bar{\mathbf{3}}, \mathbf{1})_{+1/3}, \\ L &= (\mathbf{1}, \mathbf{2})_{-1/2}, \qquad \bar{e} = (\mathbf{1}, \mathbf{1})_{+1}. \end{aligned} \tag{45.169}$$

Here Q is the left-handed quark doublet, $\bar{u}, \bar{d}$ the (left-handed conjugates of the) right-handed up and down quarks, L the lepton doublet, and $\bar{e}$ the conjugate of the right-handed electron. There are three identical copies (generations), an index we suppress except where flavor matters. The scalar superpartners are the *squarks* and *sleptons*: $\tilde{Q}, \tilde{\bar{u}}, \tilde{\bar{d}}, \tilde{L}, \tilde{\bar{e}}$, complex scalars carrying exactly the gauge

charges of Eq. (45.169). Counting components per generation: Q contributes $3 \times 2 = 6$ chiral superfields, $\bar{u}, \bar{d}$ three each, L two, $\bar{e}$ one, for 15 chiral superfields — the 15 Weyl fermions of one Standard Model generation, now each shadowed by a complex scalar.

Two Higgs chiral superfields. The electroweak-symmetry-breaking sector requires *two* $\mathrm{SU}(2)_L$-doublet chiral superfields of opposite hypercharge,

$$H_u = (\mathbf{1}, \mathbf{2})_{+1/2} = \begin{pmatrix} H_u^+ \\ H_u^0 \end{pmatrix}, \qquad H_d = (\mathbf{1}, \mathbf{2})_{-1/2} = \begin{pmatrix} H_d^0 \\ H_d^- \end{pmatrix}. \tag{45.170}$$

Their fermionic partners are the *higgsinos* $\tilde{H}_u, \tilde{H}_d$, Weyl doublets. Why two doublets are mandatory — rather than the single doublet of the Standard Model, supplemented by its conjugate $\widetilde{H} = i\sigma^2 H^*$ for the up-type Yukawa — is argued in Section 45.6.2 below; it is one of the structural surprises the supersymmetrization forces.

Gauge multiplets. Each factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ contributes a vector superfield in the adjoint, built in Section 45.1:

$$\begin{aligned} &\mathrm{SU}(3)_c: && V_3 = (g_\mu^a,\ \tilde{g}^a,\ D^a), && a = 1, \dots, 8 && \text{(gluons, gluinos)}, \\ &\mathrm{SU}(2)_L: && V_2 = (W_\mu^i,\ \tilde{W}^i,\ D^i), && i = 1, 2, 3 && \text{(W bosons, winos)}, \\ &\mathrm{U}(1)_Y: && V_1 = (B_\mu,\ \tilde{B},\ D), && && \text{(hypercharge boson, bino)}. \end{aligned} \tag{45.171}$$

The gauginos $\tilde{g}, \tilde{W}, \tilde{B}$ are Majorana fermions in the adjoint. After electroweak symmetry breaking the neutral gauginos $\tilde{W}^3, \tilde{B}$ mix with the neutral higgsinos $\tilde{H}_u^0, \tilde{H}_d^0$ to form four *neutralinos*, and the charged ones mix into two *charginos*; the lightest neutralino is the prototypical dark-matter candidate, a point we return to under R-parity. The complete field content is collected in Table 45.1.

45.6.2 Why two Higgs doublets

The Standard Model uses a single Higgs doublet H of hypercharge $+1/2$. The down-type and charged-lepton Yukawa couplings use H; the up-type Yukawa uses its conjugate $\widetilde{H} = i\sigma^2 H^*$, which transforms as a $Y = -1/2$ doublet. One scalar doublet thus suffices to give mass to all fermions. Supersymmetry destroys this economy for two independent reasons, each fatal on its own.

Holomorphy. The superpotential W is a *holomorphic* function of the chiral superfields (Theorem 44.40): it may contain Φ but never $\Phi^\dagger$. The Standard Model trick of conjugating H to feed the up-type Yukawa is therefore unavailable — $\widetilde{H} \sim H^*$ is not a chiral superfield and cannot appear in W. A holomorphic Yukawa

Superfield	spin-0	spin-$\frac{1}{2}$	$(\mathrm{SU}(3)_c, \mathrm{SU}(2)_L)_Y$
Chiral (matter), ×3 generations			
Q	$\tilde{Q} = (\tilde{u}_L\ \tilde{d}_L)$	$(u_L\ d_L)$	$(\mathbf{3},\mathbf{2})_{+1/6}$
$\bar{u}$	$\tilde{\bar{u}} = \tilde{u}_R^*$	u_R^c	$(\bar{\mathbf{3}},\mathbf{1})_{-2/3}$
$\bar{d}$	$\tilde{\bar{d}} = \tilde{d}_R^*$	d_R^c	$(\bar{\mathbf{3}},\mathbf{1})_{+1/3}$
L	$\tilde{L} = (\tilde{\nu}\ \tilde{e}_L)$	$(\nu\ e_L)$	$(\mathbf{1},\mathbf{2})_{-1/2}$
$\bar{e}$	$\tilde{\bar{e}} = \tilde{e}_R^*$	e_R^c	$(\mathbf{1},\mathbf{1})_{+1}$
Chiral (Higgs)			
H_u	$(H_u^+\ H_u^0)$	$(\tilde{H}_u^+\ \tilde{H}_u^0)$	$(\mathbf{1},\mathbf{2})_{+1/2}$
H_d	$(H_d^0\ H_d^-)$	$(\tilde{H}_d^0\ \tilde{H}_d^-)$	$(\mathbf{1},\mathbf{2})_{-1/2}$
Superfield	spin-$\frac{1}{2}$	spin-1	
Vector (gauge)			
V_3	$\tilde{g}$	g	$(\mathbf{8},\mathbf{1})_0$
V_2	$\tilde{W}^{\pm}, \tilde{W}^0$	$W^{\pm}, W^0$	$(\mathbf{1},\mathbf{3})_0$
V_1	$\tilde{B}$	B	$(\mathbf{1},\mathbf{1})_0$

Table 45.1: The MSSM superfields and their gauge quantum numbers. Electric charge is $Q = T^3 + Y$. Matter comes in three generations; only one is shown.

coupling for the up quarks requires a genuine $Y = +1/2$ doublet superfield H_u, while the down quarks and leptons require a $Y = -1/2$ doublet H_d. Holomorphy alone forces the doubling.

Anomaly cancellation. Even setting Yukawas aside, gauge invariance at the quantum level forbids a lone Higgs supermultiplet. The higgsino is a chiral fermion, and chiral fermions contribute to gauge anomalies (Chapter 36). In the Standard Model the Higgs is a boson and contributes nothing; in the MSSM its fermionic partner does. The following worked example shows that a single $Y = +1/2$ higgsino doublet spoils anomaly cancellation, and that adding a $Y = -1/2$ partner restores it exactly.

Worked Example 45.28 (Anomaly cancellation demands a second higgsino)**.** A consistent gauge theory must have vanishing triangle anomalies for every combination of gauged currents, plus a vanishing mixed gauge–gravitational anomaly and (for SU(2)) a vanishing global Witten anomaly. For the new fermions — the higgsinos — the dangerous coefficients are the ones linear in the new content, since the Standard Model fermions already cancel among themselves (Chapter 36). We evaluate them for a single left-handed Weyl doublet $\tilde{H}$ of hypercharge Y.

The $[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$ *anomaly.* This is proportional to $T(\mathbf{2})\,Y$, where $T(\mathbf{2}) = \frac{1}{2}$ is the Dynkin index of the doublet (Section B.4). For $\tilde{H}_u$ with $Y = +\frac{1}{2}$ this is $\frac{1}{2}\cdot\frac{1}{2} = \frac{1}{4} \neq 0$. A single higgsino doublet is anomalous.

The $[\mathrm{U}(1)_Y]^3$ *anomaly.* This is proportional to $\sum Y^3$ over the two components of the doublet, $2\,Y^3 = 2(\frac{1}{2})^3 = \frac{1}{4} \neq 0$.

The mixed $\mathrm{U}(1)_Y$*–gravity anomaly.* Proportional to $\sum Y = 2Y = 1 \neq 0$.

Each is nonzero, so one higgsino doublet by itself renders the MSSM gauge anomalous — mathematically inconsistent. Now add a second doublet $\tilde{H}_d$ with $Y = -\frac{1}{2}$. The three coefficients become

$$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y: \quad T(\mathbf{2})(Y_{H_u} + Y_{H_d}) = \tfrac{1}{2}(\tfrac{1}{2} - \tfrac{1}{2}) = 0, \tag{45.172}$$

$$[\mathrm{U}(1)_Y]^3: \quad 2(Y_{H_u}^3 + Y_{H_d}^3) = 2(\tfrac{1}{8} - \tfrac{1}{8}) = 0, \tag{45.173}$$

$$\mathrm{U}(1)_Y\text{–grav}: \quad 2(Y_{H_u} + Y_{H_d}) = 0. \tag{45.174}$$

All vanish: the higgsino contributions cancel *between* the two doublets, precisely because their hypercharges are equal and opposite. Finally the global $\mathrm{SU}(2)_L$ anomaly (Chapter 36): an SU(2) gauge theory is consistent only with an *even* number of doublet Weyl fermions, since $\pi_4(\mathrm{SU}(2)) = \mathbb{Z}_2$. The Standard Model fermions of one generation supply an even number of doublets (Q counts as three by color, plus L, so four); a single higgsino doublet would make the count odd and trigger the Witten anomaly. The pair $\tilde{H}_u, \tilde{H}_d$ restores an even count. Two Higgs doublets are thus the minimal anomaly-free choice, and the hypercharges $\pm\frac{1}{2}$ are fixed.

45.6.3 The superpotential, the μ problem, and R-parity

With the field content fixed, the renormalizable superpotential is the most general holomorphic, gauge-invariant cubic in the chiral superfields. Gauge invariance permits a great many terms; the physically viable MSSM keeps only those consistent with the longevity of the proton, a restriction we encode below as R-parity. The R-parity-conserving superpotential is

$$\boxed{W_{\mathrm{MSSM}} = \bar{u}\,\mathbf{y}_u\,Q\,H_u - \bar{d}\,\mathbf{y}_d\,Q\,H_d - \bar{e}\,\mathbf{y}_e\,L\,H_d + \mu\,H_u H_d.} \tag{45.175}$$

Here $\mathbf{y}_{u,d,e}$ are 3×3 Yukawa matrices in generation space (suppressed indices contracted as $\bar{u}^{\,i}(\mathbf{y}_u)_{ij}Q^j$), and the $\mathrm{SU}(2)_L$ doublets are contracted antisymmetrically with the invariant tensor ϵ_{ab}, so that $H_u H_d \equiv \epsilon_{ab}H_u^a H_d^b = H_u^+ H_d^- - H_u^0 H_d^0$ and likewise $QH_u = \epsilon_{ab}Q^a H_u^b$. Color indices on $Q, \bar{u}, \bar{d}$ are contracted between Q (a $\mathbf{3}$) and $\bar{u}, \bar{d}$ (a $\bar{\mathbf{3}}$). The signs are conventional, chosen so that the fermion mass matrices come out with standard signs after electroweak breaking.

Reading off the component Lagrangian by the rule $W \to F$-terms of Eqs. (44.149) and (44.201): the cubic terms supply the Yukawa couplings of quarks and leptons to the Higgs scalars, and after the neutral Higgs scalars acquire vacuum

expectation values $\langle H_u^0 \rangle = v_u$, $\langle H_d^0 \rangle = v_d$ (Section 45.6.5), the fermion masses

$$m_u = \mathbf{y}_u\, v_u, \qquad m_d = \mathbf{y}_d\, v_d, \qquad m_e = \mathbf{y}_e\, v_d. \tag{45.176}$$

Up-type fermions get mass from v_u, down-type and leptons from v_d — the holomorphic split made physical. The quadratic term $\mu H_u H_d$ supplies a supersymmetric (Dirac) mass linking the two higgsino doublets and, through the F-terms, a contribution $|\mu|^2(|H_u|^2 + |H_d|^2)$ to the Higgs scalar potential.

The μ problem. The parameter μ in Eq. (45.175) is the single dimensionful coupling in the supersymmetric Lagrangian: it has dimension of mass. It is not protected by any symmetry that the rest of the superpotential respects — it is the coefficient of a gauge- and supersymmetry-invariant operator. A dimensionful supersymmetric parameter has no reason to be small: the only scales available to a fundamental theory above the MSSM are the cutoffs at which new physics enters, the GUT scale $M_U \sim 2 \times 10^{16}\,\text{GeV}$ or the Planck scale $M_{\text{Pl}} \sim 10^{18}\,\text{GeV}$. Naturally, then, one expects $\mu \sim M_U$ or M_{Pl}. But phenomenology demands the opposite. The neutral-Higgs minimization conditions Eq. (45.185) below relate μ to the soft masses and to m_Z: a successful electroweak vacuum requires $|\mu|^2$ to be of order the soft-breaking scale squared, $|\mu| \sim m_{\text{soft}} \sim$ TeV. The puzzle — the *μ problem* — is precisely this: *why should a parameter of the supersymmetric (unbroken) Lagrangian be tied to the scale of supersymmetry breaking, rather than to the cutoff?* The two scales have no a priori reason to coincide. This is a genuine naturalness question, not a phenomenological inconsistency. It admits proposed solutions — forbidding a bare μ by a symmetry and regenerating it from supersymmetry breaking (the Giudice–Masiero mechanism, in which a Kähler coupling to the breaking spurion produces $\mu \sim m_{\text{soft}}$), or promoting μ to the vacuum value of a new singlet superfield (the NMSSM) — but none is established. We state the problem precisely and claim no solution.

***R*-parity.** Gauge invariance and holomorphy permit, beyond Eq. (45.175), the additional renormalizable terms

$$W_{\Delta L=1} = \tfrac{1}{2}\lambda_{ijk} L_i L_j \bar{e}_k + \lambda'_{ijk} L_i Q_j \bar{d}_k + \mu'_i L_i H_u, \qquad W_{\Delta B=1} = \tfrac{1}{2}\lambda''_{ijk} \bar{u}_i \bar{d}_j \bar{d}_k. \tag{45.177}$$

The first group violates lepton number L, the second violates baryon number B. Each is gauge-invariant; nothing in the structure built so far forbids them. But together they are catastrophic. A squark exchanged between a λ' vertex and a λ'' vertex mediates proton decay, $p \to e^+\pi^0$ or $p \to \bar{\nu}\,\pi^+$, at tree level. The proton lifetime exceeds $\sim 10^{34}$ years (Super-Kamiokande, 2020s), which forces the product $\lambda'\lambda''$ to be smaller than roughly 10^{-24} for TeV-scale squarks — an absurd fine-tuning if these couplings are free. The standard resolution is to impose a discrete symmetry, *matter parity* or equivalently R-parity, defined by

$$R_p = (-1)^{3(B-L)+2s}, \tag{45.178}$$

where s is the spin. Every Standard Model particle (and both Higgs doublets) has $R_p = +1$; every superpartner has $R_p = -1$. The terms in Eq. (45.177) all carry $R_p = -1$ and are forbidden, while every term in Eq. (45.175) carries $R_p = +1$ and survives. With R_p conserved the proton is stable against these operators. Two phenomenological consequences follow immediately and are the headline features of the R-parity-conserving MSSM: superpartners are produced in pairs, and the *lightest supersymmetric particle (LSP) is absolutely stable*. A neutral, weakly interacting LSP — typically the lightest neutralino — is then a thermal-relic dark-matter candidate. We flag the honest caveat: R-parity is *imposed*, not derived. It is well motivated — it can descend from a gauged $\mathrm{U}(1)_{B-L}$ broken by two units — and is economical, but it is a choice. Alternatives (baryon triality, which forbids $W_{\Delta B=1}$ but allows $W_{\Delta L=1}$) are logically available; we adopt R-parity as the minimal proton-stable option and say so.

45.6.4 The soft-breaking Lagrangian

Supersymmetry, if realized, is broken: no scalar electron of mass 0.511 MeV has been seen. The breaking is communicated to the MSSM fields as the soft operators classified in Section 45.5. Instantiating that classification Eq. (45.162) for the MSSM field content gives the complete soft Lagrangian

$$\begin{aligned} -\mathcal{L}_{\text{soft}} = {} & \tfrac{1}{2}\Big(M_3\,\tilde{g}\tilde{g} + M_2\,\tilde{W}\tilde{W} + M_1\,\tilde{B}\tilde{B} + \text{h.c.}\Big) \\ & + \Big(\tilde{\bar{u}}\,\mathbf{a}_u\,\tilde{Q}\,H_u - \tilde{\bar{d}}\,\mathbf{a}_d\,\tilde{Q}\,H_d - \tilde{\bar{e}}\,\mathbf{a}_e\,\tilde{L}\,H_d + \text{h.c.}\Big) \\ & + \tilde{Q}^\dagger\,\mathbf{m}_Q^2\,\tilde{Q} + \tilde{L}^\dagger\,\mathbf{m}_L^2\,\tilde{L} + \tilde{\bar{u}}\,\mathbf{m}_{\bar{u}}^2\,\tilde{\bar{u}}^\dagger + \tilde{\bar{d}}\,\mathbf{m}_{\bar{d}}^2\,\tilde{\bar{d}}^\dagger + \tilde{\bar{e}}\,\mathbf{m}_{\bar{e}}^2\,\tilde{\bar{e}}^\dagger \\ & + m_{H_u}^2|H_u|^2 + m_{H_d}^2|H_d|^2 + (b\,H_uH_d + \text{h.c.}). \end{aligned} \tag{45.179}$$

Each line is one species from the Girardello–Grisaru list, dressed in MSSM quantum numbers:

- *Gaugino masses* M_1, M_2, M_3 for bino, wino, gluino — one complex parameter per gauge factor (the partner of the gauge coupling under the spurion analysis of Section 45.5).
- *Trilinear A-terms* $\mathbf{a}_u, \mathbf{a}_d, \mathbf{a}_e$, each a 3×3 matrix in generation space, scalar analogues of the Yukawa couplings (often written $\mathbf{a} = \mathbf{y}\,A$ with A a mass).
- *Soft scalar mass-squareds* $\mathbf{m}^2_{Q,\bar{u},\bar{d},L,\bar{e}}$, Hermitian 3×3 matrices for the squarks and sleptons, plus the Higgs soft masses $m^2_{H_u}, m^2_{H_d}$.
- The *B-term* b (often $b = B\mu$), the holomorphic Higgs mass-squared.

This is the source of all superpartner masses. It also carries the price of phenomenological generality: the matrices $\mathbf{m}^2$ and $\mathbf{a}$ contain, in full generality, of

order a hundred new parameters, many of them flavor- and CP-violating. The absence of observed flavor violation (small K–$\bar{K}$ mixing, the bound on $\mu \to e\gamma$) is the *supersymmetric flavor problem*: it forces these matrices to be nearly flavor-diagonal and universal, a structure that mediation schemes (gauge mediation, anomaly mediation, gravity mediation with universality) are designed to deliver. We note this as a constraint on the soft sector, not a defect of the framework. With Eq. (45.179) the MSSM Lagrangian is complete: supersymmetric gauge and superpotential pieces plus soft breaking. Everything that follows is its consequence.

45.6.5 The Higgs sector and the tree-level mass bound

The single most consequential prediction of the MSSM is a tree-level *upper* bound on the mass of the lightest neutral Higgs boson. It arises because supersymmetry does not allow an arbitrary Higgs self-coupling: the Higgs quartic is not a free parameter as in the Standard Model but is *fixed by the gauge couplings*, being nothing but the D-term potential of Section 45.3. We now derive the bound in full.

The neutral Higgs potential. The scalar potential for the Higgs fields receives three contributions: F-terms from the μ term in Eq. (45.175), D-terms from $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ via Section 45.3, and soft terms from Eq. (45.179). The F-term is $\sum_i |\partial W/\partial\phi_i|^2 \supset |\mu|^2(|H_u|^2 + |H_d|^2)$. The gauge D-term potential is $V_D = \tfrac{1}{2}\sum_a (D^a)^2$ with $D^a = -g\sum_i \phi_i^\dagger T^a \phi_i$ for each gauge factor (Eq. (45.84)). For the two Higgs doublets, using the SU(2) Fierz identity $\sum_i (\sigma^i)_{ab}(\sigma^i)_{cd} = 2\delta_{ad}\delta_{cb} - \delta_{ab}\delta_{cd}$, the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ D-terms combine into

$$V_D = \frac{g^2 + g'^2}{8}\Big(|H_u|^2 - |H_d|^2\Big)^2 + \frac{g^2}{2}\big|H_u^\dagger H_d\big|^2 . \tag{45.180}$$

The coefficient $(g^2 + g'^2)/8$ is fixed entirely by the gauge couplings — this is the crucial point. Assembling F-, D-, and soft terms, and restricting to the neutral components (the charged directions are lifted and do not acquire vevs in the viable vacuum), the neutral Higgs potential is

$$V = (|\mu|^2 + m_{H_u}^2)|H_u^0|^2 + (|\mu|^2 + m_{H_d}^2)|H_d^0|^2 - (b\, H_u^0 H_d^0 + \text{h.c.}) + \frac{g^2 + g'^2}{8}\Big(|H_u^0|^2 - |H_d^0|^2\Big)^2 . \tag{45.181}$$

Write $m_1^2 \equiv |\mu|^2 + m_{H_d}^2$, $m_2^2 \equiv |\mu|^2 + m_{H_u}^2$, and take b real and positive (its phase can be rotated away). The vacuum is at real $\langle H_u^0\rangle = v_u$, $\langle H_d^0\rangle = v_d$, both positive; define

$$v^2 \equiv v_u^2 + v_d^2, \qquad \tan\beta \equiv \frac{v_u}{v_d}, \qquad v_u = v\sin\beta,\ v_d = v\cos\beta, \tag{45.182}$$

with $v \approx 174\,\mathrm{GeV}$ fixed by the gauge-boson masses, $m_Z^2 = \tfrac{1}{2}(g^2 + g'^2)v^2$ and $m_W^2 = \tfrac{1}{2}g^2 v^2$ (Section 31.3).

Minimization. Stationarity $\partial V/\partial H_u^0 = \partial V/\partial H_d^0 = 0$ at the vacuum, with $c \equiv (g^2+g'^2)/8$ and $Q \equiv v_u^2 - v_d^2$, gives

$$m_2^2\, v_u - b\, v_d + 4c\, Q\, v_u = 0, \tag{45.183}$$

$$m_1^2\, v_d - b\, v_u - 4c\, Q\, v_d = 0. \tag{45.184}$$

Using $4cQ = \tfrac{1}{2}(g^2+g'^2)\,\tfrac{1}{2}(v_u^2-v_d^2)/v^2 \cdot v^2 \cdot \tfrac{1}{v^2}\cdots$ — more cleanly, $2c\,v^2 = \tfrac{1}{4}(g^2+g'^2)v^2 = \tfrac{1}{2}m_Z^2$, so $4cQ = (m_Z^2/v^2)(v_u^2-v_d^2) = -m_Z^2\cos 2\beta$. Dividing Eq. (45.183) by v_u and Eq. (45.184) by v_d and substituting yields the electroweak-symmetry-breaking conditions

$$\boxed{m_{H_u}^2 + |\mu|^2 = b\cot\beta + \tfrac{1}{2}m_Z^2\cos 2\beta, \qquad m_{H_d}^2 + |\mu|^2 = b\tan\beta - \tfrac{1}{2}m_Z^2\cos 2\beta.} \tag{45.185}$$

These three equations (the two above plus the definition of v) trade the three Lagrangian parameters $b, |\mu|^2$, and one combination of soft masses for the observables $m_Z, \tan\beta$ — and they are the equations through which $|\mu|$ becomes locked to the soft scale, the technical face of the μ problem. Subtracting the two conditions in Eq. (45.185) isolates

$$\tfrac{1}{2}m_Z^2 = \frac{m_{H_d}^2 - m_{H_u}^2\tan^2\beta}{\tan^2\beta - 1} - |\mu|^2, \tag{45.186}$$

the relation that makes plain that obtaining the measured m_Z requires $|\mu|^2$ to sit at the soft scale.

The mass matrices. Expand the neutral fields about the vacuum, $H_u^0 = v_u + \tfrac{1}{\sqrt{2}}(h_u + i a_u)$, $H_d^0 = v_d + \tfrac{1}{\sqrt{2}}(h_d + i a_d)$. The mass matrices are second derivatives of Eq. (45.181), evaluated at the minimum with Eq. (45.185) substituted. The full computation is carried in Theorem 45.29; the results are as follows. The imaginary parts (a_d, a_u) furnish one massless Goldstone (eaten by the Z) and one physical CP-odd scalar A^0 with

$$m_A^2 = \frac{2b}{\sin 2\beta} = b(\tan\beta + \cot\beta). \tag{45.187}$$

The CP-even sector (h_d, h_u) has the symmetric mass-squared matrix

$$\mathcal{M}^2 = \begin{pmatrix} m_A^2\sin^2\beta + m_Z^2\cos^2\beta & -(m_A^2+m_Z^2)\sin\beta\cos\beta \\ -(m_A^2+m_Z^2)\sin\beta\cos\beta & m_A^2\cos^2\beta + m_Z^2\sin^2\beta \end{pmatrix}. \tag{45.188}$$

Its two eigenvalues are the masses of the light and heavy CP-even Higgs bosons h^0, H^0. The light one m_h^2 is the smaller eigenvalue.

The bound. The smallest eigenvalue $\lambda_{\min}$ of a symmetric matrix satisfies the variational bound $\lambda_{\min} \le u^{\mathsf T}\mathcal{M}^2 u$ for *any* unit vector u. Choosing $u = (\cos\beta, \sin\beta)$ and computing the quadratic form,

$$\begin{aligned} u^{\mathsf T}\mathcal{M}^2 u &= (m_A^2 \sin^2\beta + m_Z^2\cos^2\beta)\cos^2\beta - 2(m_A^2+m_Z^2)\sin\beta\cos\beta\cdot\cos\beta\sin\beta \\ &\quad + (m_A^2\cos^2\beta + m_Z^2\sin^2\beta)\sin^2\beta \\ &= m_A^2(\sin^2\beta\cos^2\beta - 2\sin^2\beta\cos^2\beta + \cos^2\beta\sin^2\beta) \\ &\quad + m_Z^2(\cos^4\beta - 2\sin^2\beta\cos^2\beta + \sin^4\beta) \\ &= 0 + m_Z^2(\cos^2\beta - \sin^2\beta)^2 = m_Z^2\cos^2 2\beta. \end{aligned} \tag{45.189}$$

The m_A^2 terms cancel identically, and the m_Z^2 terms collapse to a perfect square. Therefore

$$\boxed{\; m_h^2 \;\le\; m_Z^2\cos^2 2\beta \;\le\; m_Z^2. \;} \tag{45.190}$$

At tree level the lightest MSSM Higgs cannot be heavier than the Z. The bound is saturated as $m_A \to \infty$ (the decoupling limit), where h^0 becomes Standard-Model-like with quartic coupling $\lambda = \tfrac{1}{8}(g^2+g'^2)\cos^2 2\beta$, and is tightest at $\tan\beta = 1$ ($\cos 2\beta = 0$), where the tree-level m_h vanishes. The physical origin is exactly the gauge-fixed coefficient in Eq. (45.180): supersymmetry relates the Higgs quartic to the gauge couplings, and the gauge couplings are small, so the Higgs is light.

Worked Example 45.29 (The neutral Higgs mass matrices from Eq. (45.181))**.** We derive Eqs. (45.187) and (45.188). Treat the neutral fields as real for the CP-even sector and as imaginary for the CP-odd sector; the kinetic term $|\partial H^0|^2$ means a complex field $H^0 = v + \frac{1}{\sqrt2}(h + ia)$ has canonically normalized real components, so the physical mass-squared is $\frac12$ times the Hessian of V in the unnormalized variable. Writing V as a function of real shifts x_d, x_u about (v_d, v_u) and $c = (g^2+g'^2)/8$, $Q = v_u^2 - v_d^2$:

CP-even Hessian. From Eq. (45.181),

$$\begin{aligned} \frac{\partial^2 V}{\partial x_d^2} &= 2m_1^2 + 4c(3v_d^2 - v_u^2), \quad \frac{\partial^2 V}{\partial x_u^2} = 2m_2^2 + 4c(3v_u^2 - v_d^2), \\ \frac{\partial^2 V}{\partial x_u \partial x_d} &= -2b - 8c\,v_u v_d. \end{aligned} \tag{45.191}$$

Taking half of these for the physical matrix, eliminating m_1^2, m_2^2 through Eqs. (45.183) and (45.184) (equivalently Eq. (45.185)), and using $b = m_A^2\sin\beta\cos\beta$ from Eq. (45.187) together with $2c\,v_d^2 = \frac12 m_Z^2\cos^2\beta$, $2c\,v_u^2 = \frac12 m_Z^2\sin^2\beta$, $4c\,v_u v_d = \frac12 m_Z^2\sin 2\beta$, the (d,d) entry becomes

$$\mathcal{M}^2_{dd} = m_1^2 + 2c(3v_d^2 - v_u^2) = b\tan\beta + m_Z^2\cos^2\beta = m_A^2\sin^2\beta + m_Z^2\cos^2\beta, \tag{45.192}$$

and likewise $\mathcal{M}^2_{uu} = m_A^2\cos^2\beta + m_Z^2\sin^2\beta$, $\mathcal{M}^2_{du} = -b - 4c\,v_u v_d = -(m_A^2 + m_Z^2)\sin\beta\cos\beta$. This is Eq. (45.188).

CP-odd Hessian. The imaginary parts enter V only through $m_{1,2}^2|H^0|^2$, the quartic, and the b-term, which contributes $+2b\,y_u y_d$. The physical matrix is

$$\mathcal{M}^2_{\text{odd}} = \tfrac{1}{2}\begin{pmatrix} 2m_1^2 - 4cQ & 2b \\ 2b & 2m_2^2 + 4cQ \end{pmatrix} = \begin{pmatrix} b\tan\beta & b \\ b & b\cot\beta \end{pmatrix}, \tag{45.193}$$

where the diagonal entries simplified using Eq. (45.185) and $4cQ = -m_Z^2\cos 2\beta$. Its determinant is $b^2(\tan\beta\cot\beta - 1) = 0$ — the massless Goldstone — and its trace is $b(\tan\beta + \cot\beta) = 2b/\sin 2\beta$, the nonzero eigenvalue m_A^2 of Eq. (45.187). The trace of Eq. (45.188) is $m_A^2 + m_Z^2$ and its determinant is $m_A^2 m_Z^2\cos^2 2\beta$, so the two CP-even eigenvalues are

$$m^2_{h,H} = \tfrac{1}{2}\Big[m_A^2 + m_Z^2 \mp \sqrt{(m_A^2 + m_Z^2)^2 - 4m_A^2 m_Z^2\cos^2 2\beta}\Big], \tag{45.194}$$

confirming $m_h \le m_Z|\cos 2\beta|$ directly (the radical is real and the lower sign gives $m_h^2 \le m_Z^2\cos^2 2\beta$ on inspection at $m_A \to \infty$).

The radiative lift. The tree bound Eq. (45.190) is in flat contradiction with experiment: the observed Higgs boson has mass $m_h = 125.25 \pm 0.17\,\text{GeV}$ (Particle Data Group average, mid-2020s), well above $m_Z = 91.19\,\text{GeV}$. The MSSM survives only because the bound receives large radiative corrections, dominated by the top quark and its scalar partners (the stops $\tilde{t}_{1,2}$). The top Yukawa $y_t \approx 1$ is the largest coupling in the theory; the cancellation between top and stop loops that protects the Higgs mass is incomplete once supersymmetry is broken (the stops are heavier than the top), and the residual generates a positive contribution to the Higgs quartic. The leading one-loop result is

$$m_h^2 \le m_Z^2\cos^2 2\beta + \frac{3\,m_t^4}{4\pi^2 v^2}\left[\ln\frac{M_S^2}{m_t^2} + \frac{X_t^2}{M_S^2}\Big(1 - \frac{X_t^2}{12M_S^2}\Big)\right], \tag{45.195}$$

where $M_S^2 = m_{\tilde{t}_1}m_{\tilde{t}_2}$ is the stop mass scale, $X_t = A_t - \mu\cot\beta$ the stop mixing, and m_t the running top mass. The logarithm grows the bound; the mixing term is maximized at $X_t = \sqrt{6}\,M_S$ ("maximal mixing"), where the lift is largest. Quantitatively, reaching $m_h = 125\,\text{GeV}$ requires either stops in the multi-TeV range (large logarithm) or near-maximal mixing with TeV stops. This is the central tension of the MSSM after the LHC: the same loops that lift m_h to its observed value reintroduce a residual fine-tuning of the electroweak scale (the "little hierarchy problem"), since heavy stops feed back into $m_{H_u}^2$ through the renormalization group. The MSSM accommodates the 125 GeV Higgs — it does not predict it sharply — and does so at the cost of heavy scalars. We state this plainly: the discovery of a 125 GeV Higgs is consistent with the MSSM but pushes its natural region.

45.6.6 Gauge-coupling unification

The second flagship result needs no Higgs sector at all: it is a statement about the running of the three gauge couplings, and it is where the supersymmetric beta function of Section 45.4 pays off. We continue Yang–Mills theory: the MSSM gauge sector is three copies of super Yang–Mills, and the one-loop running uses exactly the supersymmetric coefficient derived there.

The supersymmetric one-loop coefficient. For an $\mathcal{N} = 1$ gauge theory with gauge group G and chiral superfields in representations r_i, the one-loop beta function established in Section 45.4 is, in the convention $\mu\, dg/d\mu = b\, g^3/16\pi^2$,

$$b = -3\, C_2(G) + \sum_i T(r_i), \tag{45.196}$$

where $C_2(G)$ is the adjoint Casimir and $T(r)$ the Dynkin index (Section B.4). The $-3C_2(G)$ is the gauge contribution (gauge boson plus gaugino plus ghost), and $+\sum T(r_i)$ the chiral-matter contribution; each chiral superfield contributes with a coefficient *half* that of a Dirac fermion plus its scalar, reflecting the multiplet content. In terms of the inverse fine-structure couplings $\alpha_i^{-1} = 4\pi/g_i^2$, integrating Eq. (45.196) gives the one-loop running of Eq. (24.240) in the form

$$\alpha_i^{-1}(\mu) = \alpha_i^{-1}(M_Z) - \frac{b_i}{2\pi}\, \ln \frac{\mu}{M_Z}. \tag{45.197}$$

A straight line in $\ln \mu$, with slope set by b_i. Three lines with three different slopes generically meet pairwise at three different points; whether they meet at a *single* point is a nontrivial question fixed entirely by the three b_i and the three measured boundary values.

Computing (b_1, b_2, b_3) for the MSSM. We must count $C_2(G)$ and $\sum T(r_i)$ for each factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. For $\mathrm{SU}(N)$, $C_2(\mathrm{adj}) = N$ and $T(\mathrm{fund}) = \frac{1}{2}$; for $\mathrm{U}(1)_Y$, $C_2 = 0$ and the "index" of a state is $\frac{3}{5}Y^2$ in the GUT normalization explained below. The chiral superfields are three generations of $Q, \bar{u}, \bar{d}, L, \bar{e}$ plus the two Higgs doublets H_u, H_d; the vector superfields supply the $-3C_2(G)$.

Worked Example 45.30 (The MSSM beta coefficients)**.** $\mathrm{SU}(3)_c$. Gauge: $-3C_2 = -3 \cdot 3 = -9$. Matter, per generation: the quark doublet $Q = (\mathbf{3}, \mathbf{2})$ counts as two color triplets, $T_3(Q) = 2 \cdot \frac{1}{2} = 1$; $\bar{u}, \bar{d}$ each a single $\bar{\mathbf{3}}$, $T_3 = \frac{1}{2}$ each. Sum per generation $1 + \frac{1}{2} + \frac{1}{2} = 2$; three generations give 6. Higgs are colorless. Hence

$$b_3 = -9 + 6 = -3. \tag{45.198}$$

$\mathrm{SU}(2)_L$. Gauge: $-3 \cdot 2 = -6$. Matter, per generation: $Q = (\mathbf{3}, \mathbf{2})$ is a doublet with color multiplicity 3, $T_2(Q) = 3 \cdot \frac{1}{2} = \frac{3}{2}$; $L = (\mathbf{1}, \mathbf{2})$, $T_2 = \frac{1}{2}$. Sum $\frac{3}{2} + \frac{1}{2} = 2$

per generation, 6 for three. Higgs: each doublet contributes $T_2 = \frac{1}{2}$, so $H_u + H_d$ give 1. Hence

$$b_2 = -6 + 6 + 1 = 1. \tag{45.199}$$

$\mathrm{U}(1)_Y$. Gauge: 0. The contribution of a chiral superfield is $\frac{3}{5}\sum Y^2$ summed over all its components, the $\frac{3}{5}$ being the GUT normalization. Per generation, summing Y^2 over multiplicities (color $\times$ weak):

$$\underbrace{6\left(\tfrac{1}{6}\right)^2}_{Q} + \underbrace{3\left(\tfrac{2}{3}\right)^2}_{\bar{u}} + \underbrace{3\left(\tfrac{1}{3}\right)^2}_{\bar{d}} + \underbrace{2\left(\tfrac{1}{2}\right)^2}_{L} + \underbrace{1\cdot 1^2}_{\bar{e}} = \tfrac{1}{6} + \tfrac{4}{3} + \tfrac{1}{3} + \tfrac{1}{2} + 1 = \tfrac{10}{3}. \tag{45.200}$$

Three generations give 10; the Higgs add $2(\frac{1}{2})^2 + 2(\frac{1}{2})^2 = 1$. Total $\sum Y^2 = 11$, so

$$b_1 = \tfrac{3}{5}\cdot 11 = \tfrac{33}{5}. \tag{45.201}$$

Collecting,

$$\boxed{(b_1, b_2, b_3)_{\mathrm{MSSM}} = \left(\tfrac{33}{5},\ 1,\ -3\right).} \tag{45.202}$$

For comparison, the same counting for the non-supersymmetric Standard Model (gauge contribution $-\frac{11}{3}C_2$, chiral fermion $+\frac{2}{3}T$, complex scalar $+\frac{1}{3}T$) gives $(b_1, b_2, b_3)_{\mathrm{SM}} = (\frac{41}{10}, -\frac{19}{6}, -7)$. The two sets differ because the superpartners — gauginos shift the gauge piece, sfermions and higgsinos the matter piece — change every slope. Note $b_3 = -3 < 0$: $\mathrm{SU}(3)_c$ remains asymptotically free, but less so than the Standard Model's -7, because the gluino and squarks add positively to the matter side.

The GUT normalization. The hypercharge coupling g' as defined in the Standard Model is not directly comparable to g, g_s, because $\mathrm{U}(1)_Y$ has no intrinsic normalization — one may rescale Y and g' together. To compare with the non-abelian couplings one normalizes $\mathrm{U}(1)_Y$ as it would sit inside a simple unifying group such as SU(5) (Section 35.5.2), where all generators share a common trace normalization $\mathrm{Tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$. Requiring the hypercharge generator to be normalized like the others over a complete generation fixes the rescaling: one defines $g_1^2 = \frac{5}{3}g'^2$, equivalently $\alpha_1 = \frac{5}{3}\alpha_Y$ with $\alpha_Y = \alpha_{\mathrm{em}}/\cos^2\theta_W$. The factor $\frac{5}{3}$ is exactly the reciprocal of the $\frac{3}{5}$ that appears in b_1; the two conspire so that $b_1\alpha_1$ is normalization-independent. With this convention the three couplings are on the same footing and the question "do they meet?" is well posed.

The numerical run. We take the boundary data at $M_Z = 91.1876\,\mathrm{GeV}$ (Particle Data Group, mid-2020s):

$$\alpha_{\mathrm{em}}^{-1}(M_Z) = 127.95, \qquad \sin^2\theta_W = 0.2312, \qquad \alpha_s(M_Z) = 0.1179, \tag{45.203}$$

from which

$$\begin{aligned}\alpha_1^{-1}(M_Z) &= \tfrac{3}{5}\cos^2\theta_W\,\alpha_{\rm em}^{-1} = \tfrac{3}{5}(0.7688)(127.95) = 59.0,\\ \alpha_2^{-1}(M_Z) &= \sin^2\theta_W\,\alpha_{\rm em}^{-1} = (0.2312)(127.95) = 29.6,\\ \alpha_3^{-1}(M_Z) &= 1/\alpha_s = 8.5.\end{aligned} \tag{45.204}$$

Run with Eq. (45.197) and the MSSM coefficients Eq. (45.202). Ask first where α_1 and α_2 meet. Setting $\alpha_1^{-1} = \alpha_2^{-1}$ and writing $L = \ln(\mu/M_Z)$,

$$59.0 - \frac{33/5}{2\pi}L = 29.6 - \frac{1}{2\pi}L \implies 29.4 = \frac{33/5-1}{2\pi}L = \frac{28/5}{2\pi}L = 0.8913\,L, \tag{45.205}$$

so $L = 33.0$, i.e. $\mu = M_Z\,e^{33.0} \approx 2.0\times10^{16}$ GeV. At this scale,

$$\alpha_2^{-1}(M_U) = 29.6 - \frac{1}{2\pi}(33.0) = 24.3, \qquad \alpha_3^{-1}(M_U) = 8.5 + \frac{3}{2\pi}(33.0) = 24.2. \tag{45.206}$$

The third coupling arrives at 24.2 where the first two meet at 24.3: the three inverse couplings coincide to better than half a percent at

$$\boxed{M_U \simeq 2\times10^{16}\,\text{GeV}, \qquad \alpha_U^{-1} \simeq 24.3\ (\alpha_U \simeq 1/24).} \tag{45.207}$$

This is the celebrated MSSM gauge-coupling unification. Repeating the exercise with the Standard Model coefficients $(b_1, b_2, b_3)_{\rm SM}$ from Theorem 45.30, the three lines miss: α_1 and α_2 cross near 10^{13} GeV, but α_3 passes well below, and there is no common point — the SM couplings fail to unify by many standard deviations. Figure 45.6 plots both cases side by side. The contrast is the empirical content of the result: *adding exactly the MSSM superpartner spectrum, with no further adjustment, turns a three-way near-miss into a meeting.* The unification scale $M_U \sim 2\times10^{16}$ GeV sits comfortably below the Planck scale and just below where dimension-six proton decay from heavy gauge bosons would become dangerous — consistent with the GUT picture of Section 35.5.

Why supersymmetry helps unification

The mechanism is transparent in Theorem 45.30. Unification is the statement that three slopes and three intercepts conspire to a single crossing. The Standard Model slopes nearly do it but not quite; the gap is of order the two-loop corrections, too large to bridge. Supersymmetry shifts every slope in a correlated way: the gauginos make the gauge contribution less negative (from $-\frac{11}{3}C_2$ to $-3C_2$), and the sfermions and higgsinos enlarge the matter contribution. The net effect on the *differences* $b_i - b_j$ — which control where the lines cross — is just right. That a spectrum motivated entirely by the hierarchy problem (Section 35.3) should also repair unification is the strongest indirect circumstantial argument for low-

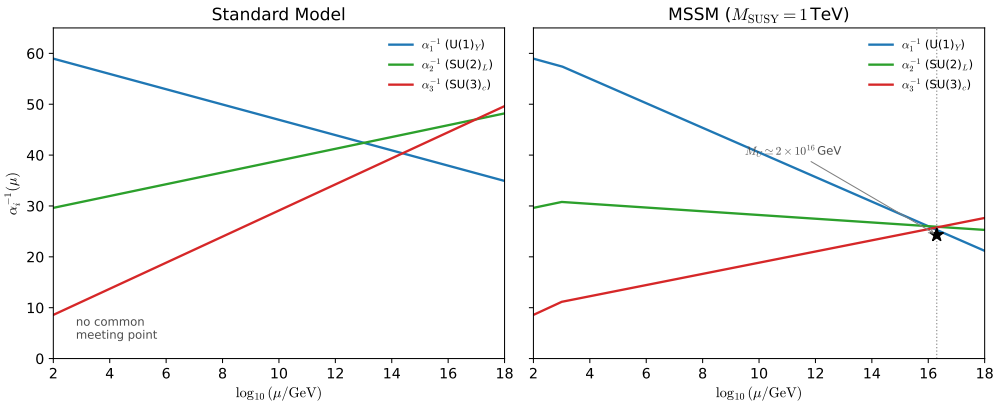


Figure 45.6: One-loop running of the inverse gauge couplings $\alpha_i^{-1}(\mu)$ in GUT normalization, from Eq. (45.197) with the boundary data Eq. (45.204). *Left:* the Standard Model coefficients $(\frac{41}{10}, -\frac{19}{6}, -7)$ — the three lines do not meet at a common point. *Right:* the MSSM coefficients $(\frac{33}{5}, 1, -3)$ of Eq. (45.202), with superpartners switched on at $M_{\rm SUSY} = 1\,\mathrm{TeV}$ — the three lines converge near $M_U \simeq 2\times10^{16}\,\mathrm{GeV}$ at $\alpha_U^{-1} \simeq 24$ (star). Both panels are reproducible from the numbers in the text.

energy supersymmetry. It is circumstantial: it is a successful postdiction of two slopes from one assumption, not a measurement of superpartners.

45.6.7 Experimental status, dated

Honesty about the frontier is mandatory. As of the mid-2020s the experimental situation is the following. *No superpartner has been observed.* The LHC has excluded gluinos and first-/second-generation squarks below roughly 2–2.3 TeV, stops below $\sim 1.2\,\mathrm{TeV}$, and electroweakinos (charginos, neutralinos) below a few hundred GeV in many scenarios, with the precise bounds depending on the assumed spectrum and decay chains. The single Higgs boson found in 2012 has mass 125 GeV and couplings consistent, within current precision, with the Standard Model — which in MSSM language means the decoupling limit $m_A \gg m_Z$, where h^0 is Standard-Model-like. The required Higgs mass forces, through Eq. (45.195), stops in the multi-TeV range or large mixing. The combination — a 125 GeV Higgs and TeV-scale exclusion limits — has pushed the natural region of the MSSM into mild tension (the "little hierarchy problem"): the model works, but the cancellation it was built to explain now requires a per-cent-level tuning of the electroweak scale against the soft masses. Gauge-coupling unification Eq. (45.207) remains a quantitative success at the one- and two-loop level, robust to the precise superpartner spectrum within an order of magnitude.

The verdict, stated without spin: the MSSM is not excluded; it is constrained, and the parameter space favored by naturalness is shrinking. Whether nature is supersymmetric near the TeV scale is, as of this writing, an open experimental question.

This completes the chapter's arc. We began with the abelian gauge multiplet, built super Yang–Mills and SQCD, proved the nonrenormalization theorems that make these theories UV-special, broke supersymmetry softly, and have now assembled the MSSM as the concrete payoff — with two numbers, $m_h \leq m_Z|\cos 2\beta|$ and the meeting of the couplings at $2 \times 10^{16}\,\mathrm{GeV}$, that follow from the structure alone. What we have used throughout is perturbation theory and symmetry. The exact, nonperturbative content of supersymmetric gauge theory — the holomorphic structure of the moduli space, Seiberg duality, gaugino condensation, the Seiberg–Witten solution — is the subject of Chapter 46, which takes the classical moduli spaces and holomorphy built here as its starting data; and the broader question of physics beyond the Standard Model, of which the MSSM is one well-developed answer, returns in the GUT and hierarchy discussions of Chapter 35.

Further reading

The component construction of supersymmetric gauge theory in superspace — the vector superfield, W_α, and the chiral integral $\int d^2\theta\, W^\alpha W_\alpha$ — is developed at book length in Wess and Bagger [505], the canonical reference whose conventions are close to ours, and pedagogically in Terning [472], which is also the natural companion for the holomorphy and duality material that Chapter 46 takes up. The nonrenormalization theorems in the modern holomorphy/spurion language follow Seiberg [434] and the lectures by Intriligator and Seiberg [275]; the exact gauge coupling is the NSVZ result of Novikov et al. [370], whose instanton-based derivation complements the Konishi-anomaly argument given here. Dynamical supersymmetry breaking and its phenomenology trace to Affleck et al. [7]. For soft breaking, the classification of allowed operators is Girardello and Grisaru [222]; the MSSM, its soft Lagrangian, the Higgs-mass bound, and the gauge-coupling unification that sharpens the unified-theory hierarchy of Georgi et al. [219] are covered comprehensively in the widely used primer by Martin [339] and, in the original softly-broken-$SU(5)$ setting, in Dimopoulos and Georgi [148]. The standard texts Peskin and Schroeder [396], Schwartz [432] and Weinberg [502] place this material in the broader QFT and Standard-Model context.

Chapter 46

Exact Results in Supersymmetry

The two preceding chapters built supersymmetry as a structure: the algebra and its representations in Chapter 44, and the supersymmetric gauge theories — vector superfields, super-Yang–Mills, SQCD, and the perturbative nonrenormalization theorem — in Chapter 45. This chapter cashes that structure in. Its single thesis is that supersymmetry turns a class of physical quantities into rigid analytic data, and that this rigidity lets us compute strong-coupling physics — condensates, confinement, the exact spectrum of a four-dimensional gauge theory — without summing a diagram. Supersymmetry is here not a phenomenological hypothesis about the world but a controlled laboratory in which the non-perturbative questions that lattice gauge theory (Chapter 43) can only approach numerically, and that the Wilson-loop criterion of Theorem 27.20 can only diagnose, become exactly solvable.

The engine is *holomorphy*. Theorem 44.40 established that the superpotential is a holomorphic function of the chiral superfields; the chapter opens by promoting the coupling constants themselves to background chiral spurions, so that the entire effective superpotential is holomorphic in the fields *and* the couplings. Holomorphy, sharpened by the global-symmetry and R-symmetry selection rules and pinned down by the behaviour in weak-coupling and symmetry-restoring limits, reproduces Seiberg's nonrenormalization theorem in a line and then, applied to the holomorphic gauge coupling $\tau = \theta/2\pi + 4\pi i/g^2$, forces it to run only at one loop — the all-orders physical running being recovered through the anomalous Jacobian of Chapter 36 as the Novikov–Shifman–Vainshtein–Zakharov beta function. The one-loop-exact running condenses into a single complex renormalization-group invariant, the holomorphic dynamical scale Λ, the supersymmetric refinement of the Λ_{QCD} of Theorem 27.11, whose phase encodes the θ-vacuum of Chapter 42.

That scale is the one non-perturbative datum the gauge dynamics hands

to the low-energy theory, and the middle of the chapter feeds it, together with holomorphy and the flavor symmetries, into supersymmetric QCD. On the classical moduli space of mesons and baryons, holomorphy and the anomaly-restored selection rules fix the Affleck–Dine–Seiberg superpotential uniquely up to a constant, which a single instanton — with the zero-mode counting of Chapter 42 — then shows is nonzero. Reading off its consequences gives a runaway with no vacuum, a calculable gaugino condensate, and exactly N_c supersymmetric vacua matching the Witten index of Chapter 44. Increasing the flavor count maps the full phase diagram of SQCD: a quantum-modified moduli space, s-confinement, and Seiberg's electric–magnetic duality, whose principal evidence is the exact matching of all 't Hooft anomalies (Chapter 37) between the electric and magnetic descriptions, with the conformal window fixed to the digit by the unitarity bound on the meson.

Where Seiberg duality is a statement about whole infrared fixed points, the final movement takes the same electric–magnetic exchange down to individual states. Extending the algebra of Chapter 44 to its central charge promotes the classical Bogomol'nyi bound of Chapter 41 into an exact quantum mass formula $M = |Z|$ on short BPS multiplets, whose masses are protected against the flow to strong coupling. This spectral data is exactly what the Seiberg–Witten solution computes: holomorphy of the prepotential, positivity of the Coulomb-branch metric, and the monodromies forced by massless monopoles and dyons together fix the low-energy coupling at every scale as the modulus of an auxiliary elliptic curve, and a soft breaking to $\mathcal{N} = 1$ makes the monopole condense, realizing the dual-superconductor picture of confinement as an analytic fact rather than a criterion. A closing synthesis names the five interlocking tools of the exact-results program, separates honestly the theorems from the superbly-tested conjectures as of the mid-2020s, and traces how this technology re-enters the book as the gateway to holography (Chapter 53), the bootstrap (Chapter 49), and the modern amplitudes program (Chapter 62).

46.1 Holomorphy as a Principle

Chapter 44 closed the chiral superfield calculus at a single sharp observation (Section 44.7.8): the superpotential W is a *holomorphic* function of the chiral superfields, a power series in the Φ_i alone with no dependence on the conjugates $\Phi_i^\dagger$, while the Kähler potential $K = \Phi^\dagger\Phi$ depends on both. Chapter 45 then promoted this matter to a gauge theory and argued, by inspecting supergraphs, that the superpotential receives no perturbative corrections (the nonrenormalization theorem, Theorem 45.20). That diagrammatic argument is correct but laborious: it asks the reader to track the θ-algebra of every loop and to believe that an infinite class of cancellations conspires order by order. We now replace it with a single structural principle that makes the cancellation manifest and, more

importantly, reaches conclusions no finite diagrammatic analysis can: *holomorphy*. The idea, due to Seiberg, is to treat the coupling constants themselves as frozen values of background chiral superfields. Once the couplings are chiral, the entire superpotential — fields and couplings together — is a holomorphic object, and holomorphic functions constrained by global symmetries and by the requirement of smoothness at weak coupling are rigid almost to the point of being unique. The nonrenormalization theorem then drops out in a line, and the same machinery, applied in the next section to the gauge coupling, will deliver an *exact* statement about the running of the gauge theory that no order of perturbation theory could.

46.1.1 Holomorphy in the fields and in the couplings

Recall the structure fixed in Section 44.7. A supersymmetric action is built from two ingredients of distinct character. The kinetic energy is a *D-term*, the integral of a real function over the full superspace,

$$S_K = \int d^4x\, \mathrm{d}^4\theta\; K(\Phi, \Phi^\dagger), \qquad K = \Phi^\dagger\Phi \text{ for canonical kinetic energy (Eq. (44.205))}, \tag{46.1}$$

and the interactions and masses are an *F-term*, the integral of a chiral function over the chiral half of superspace,

$$S_W = \int d^4x \left[\mathrm{d}^2\theta\; W(\Phi) + \mathrm{h.c.}\right], \qquad \bar{D}_{\dot\alpha} W = 0. \tag{46.2}$$

The chirality constraint $\bar{D}_{\dot\alpha} W = 0$ is the whole story: by Eq. (44.198) a product of chiral superfields is chiral, so any power series in the Φ_i with constant coefficients is chiral, whereas any appearance of a $\Phi_i^\dagger$ would violate $\bar{D}_{\dot\alpha} W = 0$. This is what "holomorphic" means here, and it is worth stating as a definition because the entire chapter rests on it.

Definition 46.1 (Holomorphy in the fields)**.** A function W of the chiral superfields Φ_i of a theory is *holomorphic* if it is a power series in the Φ_i with no dependence on the antichiral conjugates $\Phi_i^\dagger$. Equivalently W is a chiral superfield, $\bar{D}_{\dot\alpha} W = 0$. The superpotential of Theorem 44.40 is by construction holomorphic; the Kähler potential $K(\Phi, \Phi^\dagger)$ is not.

The decisive move is to extend this notion from the fields to the parameters. In the Wess–Zumino model with cubic superpotential (Eq. (44.214)),

$$W(\Phi) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, g\, \Phi^3, \tag{46.3}$$

the mass m and the cubic coupling g are ordinary complex numbers multiplying the chiral operators $\tfrac{1}{2}\Phi^2$ and $\tfrac{1}{3}\Phi^3$. Nothing in the F-term construction (Theorem 44.41) required these coefficients to be constants: if m and g were

promoted to the lowest components of additional chiral superfields M and G — background fields, switched off at their constant vacuum values $\langle \mathsf{M} \rangle = m$, $\langle \mathsf{G} \rangle = g$ but in principle dynamical — then Eq. (46.3) would read $W = \frac{1}{2}\mathsf{M}\Phi^2 + \frac{1}{3}\mathsf{G}\Phi^3$, still a holomorphic function of all the chiral superfields in sight, now including the couplings. A background field introduced solely to track how a coupling transforms under the symmetries, and then frozen to its constant value, is called a *spurion*.

Definition 46.2 (Spurion promotion of a coupling)**.** Let c be a coupling constant multiplying a chiral operator $\mathcal{O}(\Phi)$ in the superpotential, $W \supset c\,\mathcal{O}(\Phi)$. The *spurion promotion* of c replaces it by a background chiral superfield C (so $c = \langle \mathsf{C} \rangle$ is the lowest component, with vanishing θ- and F-components in the background) and treats C as one more chiral argument of the superpotential. The promotion is consistent because a constant is the simplest chiral superfield, $\bar{D}_{\dot\alpha} c = 0$, and because $\mathsf{C}\,\mathcal{O}(\Phi)$ remains chiral.

Two facts make the promotion powerful rather than merely formal. First, the spurions enter W holomorphically for exactly the same reason the dynamical fields do: the superpotential is the chiral integrand of an F-term, and an F-term integrand must be chiral, hence a holomorphic function of *every* chiral superfield it contains, dynamical or background. The conjugate couplings $\bar m, \bar g$ — lowest components of the antichiral spurions $\mathsf{M}^\dagger, \mathsf{G}^\dagger$ — cannot appear in W at all; they live only in the Kähler potential, which is a D-term and free to depend on both. This is the precise sense in which "the superpotential is holomorphic in the couplings." Second, the spurions carry definite charges under any global symmetry of the theory: to keep W invariant once m, g are allowed to transform, the symmetry must act on M, G in a way that compensates the transformation of the operators they multiply. We now make this bookkeeping precise, because it is the engine of every exact result that follows.

46.1.2 Symmetry-and-holomorphy selection rules

The Wess–Zumino model Eq. (46.3) at $m = g = 0$ is a free massless chiral superfield, and the free theory has two independent global U(1) symmetries acting on Φ. The first is an ordinary flavour rotation,

$$\mathrm{U}(1): \qquad \Phi \longmapsto e^{i\alpha}\,\Phi, \tag{46.4}$$

under which the lowest scalar, the fermion and the auxiliary field all rotate by the same phase. The second is the R-symmetry $\mathrm{U}(1)_R$ of Section 44.2.7, the outer automorphism that rephases the supercharges, $[R, Q_\alpha] = -Q_\alpha$ (Eq. (44.51)). Because Q carries R-charge -1, the Grassmann coordinate θ_α — which pairs with Q in the SUSY transformation $\delta_\xi = \xi Q + \bar\xi \bar Q$ — carries R-charge $+1$, and the chiral measure $\mathrm{d}^2\theta$, being effectively $\partial^2/\partial\theta^2$, carries R-charge -2. For the

F-term Eq. (46.2) to be R-invariant the superpotential must therefore carry R-charge exactly $+2$,

$$R[\mathrm{d}^2\theta] = -2 \quad \Longrightarrow \quad R[W] = +2. \tag{46.5}$$

Assign the chiral superfield Φ the R-charge $R[\Phi] = 1$, meaning its lowest scalar component has R-charge 1 and its fermion has R-charge 0. (Any other consistent choice differs by mixing $\mathrm{U}(1)_R$ with the ordinary U(1) of Eq. (46.4) and changes nothing in the conclusion; we fix this convention once and for all.) With $m = g = 0$ the kinetic term $\mathrm{d}^4\theta\, \Phi^\dagger\Phi$ is invariant under both U(1) and $\mathrm{U}(1)_R$ — the measure $\mathrm{d}^4\theta$ is R-neutral and $\Phi^\dagger\Phi$ is uncharged — so the free theory genuinely possesses both symmetries.

Turning on m and g breaks both U(1) and $\mathrm{U}(1)_R$ *as ordinary symmetries*, since Φ^2 and Φ^3 carry different charges and cannot both be neutralized by a single rotation of Φ. The spurion device restores them as *formal* symmetries: we demand that W be invariant once the couplings are allowed to transform, and read off the charges the spurions must carry. From Eq. (46.3),

	Φ	m (spurion M)	g (spurion G)
U(1) charge	$+1$	-2	-3
$\mathrm{U}(1)_R$ charge	$+1$	0	-1

The U(1) charges follow from requiring $m\Phi^2$ and $g\Phi^3$ to be neutral: $q[m] + 2 = 0$ and $q[g] + 3 = 0$. The $\mathrm{U}(1)_R$ charges follow from requiring each term to carry R-charge $+2$ as in Eq. (46.5): $R[m]+2R[\Phi] = 2$ gives $R[m] = 0$, and $R[g]+3R[\Phi] = 2$ gives $R[g] = -1$. These assignments are the content of the selection rules: any holomorphic, symmetry-invariant function of Φ, m, g is forced into a restricted monomial structure, which we now extract.

Proposition 46.3 (Holomorphy + selection rules fix the monomial structure)**.** *Let $W_{\text{eff}}(\Phi, m, g)$ be holomorphic in its three chiral arguments and invariant under both* U(1) *and* $\mathrm{U}(1)_R$ *with the charges of the table above. Then W_{eff} is a sum of monomials*

$$W_{\text{eff}} = \sum_{a,b} c_{ab}\, m^a\, g^b\, \Phi^c, \qquad a = 1 - b, \quad c = b + 2, \tag{46.6}$$

with $a, b, c \geq 0$ integers (positive powers only, by holomorphy) and complex constants c_{ab}. Equivalently, writing the single invariant combination $t \equiv g\Phi/m$,

$$W_{\text{eff}} = m\,\Phi^2\, f(t), \qquad f(t) = \sum_{b \geq 0} c_b\, t^{\,b}, \tag{46.7}$$

for some function f analytic at $t = 0$.

Proof. Holomorphy (Theorem 46.1, extended to the spurions by Theorem 46.2) means W_{eff} is a power series in Φ, m, g with no conjugates and no negative powers — a Taylor series, not a Laurent series. Write a general monomial $m^a g^b \Phi^c$ with $a, b, c \geq 0$. Invariance under U(1) requires the total U(1) charge to vanish,

$$-2a - 3b + c = 0, \tag{46.8}$$

and invariance under $\mathrm{U}(1)_R$ requires the total R-charge to equal $+2$ as in Eq. (46.5),

$$0 \cdot a + (-1)\, b + 1 \cdot c = 2 \quad \Longrightarrow \quad c = b + 2. \tag{46.9}$$

Substituting $c = b + 2$ into Eq. (46.8),

$$-2a - 3b + (b + 2) = 0 \quad \Longrightarrow \quad -2a - 2b + 2 = 0 \quad \Longrightarrow \quad a = 1 - b. \tag{46.10}$$

Thus every allowed monomial is $m^{1-b} g^b \Phi^{b+2}$, labelled by the single integer $b \geq 0$, which is Eq. (46.6). Factoring out $m\Phi^2$,

$$m^{1-b} g^b \Phi^{b+2} = m\, \Phi^2 \Big(\frac{g\,\Phi}{m}\Big)^b = m\, \Phi^2\, t^{\,b}, \tag{46.11}$$

and summing over b gives Eq. (46.7). The series for f contains only $b \geq 0$ because holomorphy forbids negative powers of the chiral field Φ; it may a priori contain arbitrarily large b. □

Theorem 46.3 has already done most of the work: an infinite-dimensional space of possible quantum corrections to the superpotential has collapsed to a single function $f(t)$ of one variable. The remaining input — the weak-coupling limit — pins f down completely.

46.1.3 Seiberg's nonrenormalization theorem

Theorem 46.4 (Perturbative nonrenormalization of the superpotential)**.** *In the Wess–Zumino model with superpotential Eq. (46.3), the Wilsonian effective superpotential generated to all orders in perturbation theory equals the tree-level superpotential exactly,*

$$W_{\text{eff}}(\Phi, m, g) = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, g\, \Phi^3. \tag{46.12}$$

No perturbative correction — no shift of m, no shift of g, and no higher-order term $\Phi^{n \geq 4}$ — is generated. This reproduces and sharpens the diagrammatic statement of Theorem 45.20.

Proof. The effective superpotential is an F-term, hence holomorphic in the dynamical field Φ and, after the spurion promotion of Theorem 46.2, in the couplings m, g. The microscopic theory is invariant under U(1) and $\mathrm{U}(1)_R$ with

the spurions transforming as in the table of Section 46.1.2; these are global, non-anomalous symmetries of the Wess–Zumino model (it contains no gauge fields, so there is no triangle to spoil $\mathrm{U}(1)_R$), so the effective theory respects them as well. Theorem 46.3 therefore applies and $W_{\text{eff}} = m\Phi^2 f(t)$ with $t = g\Phi/m$ and f analytic at $t = 0$.

We fix f by two physical limits, each of which the analytic function $f(t)$ must respect.

(i) Weak coupling, $g \to 0$ at fixed m, Φ. Perturbation theory in g generates only non-negative integer powers of g (each interaction vertex brings one power of the coupling), so W_{eff} must be regular as $g \to 0$. Since $t = g\Phi/m \to 0$ in this limit, regularity is the statement that $f(t)$ has a Taylor expansion about $t = 0$ with no poles — which Theorem 46.3 already guarantees, $f(t) = \sum_{b\geq 0} c_b t^b$. The leading term $b = 0$ is the free theory: as $g \to 0$ the dynamics is exactly the free massive chiral superfield, whose superpotential is $\frac{1}{2}m\Phi^2$, so $c_0 = \frac{1}{2}$. The next term $b = 1$ is the classical cubic vertex itself, fixed by matching the $\mathcal{O}(g)$ term of W_{eff} to the tree-level $\frac{1}{3}g\Phi^3$; holomorphy guarantees this single tree-level matching is exact, so $c_1 = \frac{1}{3}$.

(ii) The massless limit, $m \to 0$ at fixed g, Φ. Here is the input that kills every higher term. The massless Wess–Zumino model ($m = 0$) is a perfectly well-defined theory — a free kinetic term plus a marginal cubic interaction $g\Phi^3$ — with its own ultraviolet-finite weak-coupling expansion; nothing in its dynamics is singular. The effective superpotential must therefore remain finite as $m \to 0$. Examine the monomial b in Eq. (46.11): it equals $c_b\, m^{1-b} g^b \Phi^{b+2}$. For $b \geq 2$ the exponent $1 - b$ is negative, so this term diverges as $m \to 0$, an unphysical singularity in a theory that is regular there. The only way to avoid it is $c_b = 0$ for all $b \geq 2$.

Combining (i) and (ii), the sole survivors are $b = 0$ and $b = 1$ with $c_0 = \frac{1}{2}$, $c_1 = \frac{1}{3}$, and Eq. (46.7) collapses to

$$W_{\text{eff}} = m\,\Phi^2\Big(\tfrac{1}{2} + \tfrac{1}{3}\,t\Big) = \tfrac{1}{2}\,m\,\Phi^2 + \tfrac{1}{3}\,g\,\Phi^3, \tag{46.13}$$

which is Eq. (46.12). Since the argument never used a particular order of perturbation theory — only holomorphy, two charge selection rules, and the regularity of the two coupling limits — it holds to all orders, with no diagram summed. □

The structure of the proof is worth contemplating, because the same three ingredients — holomorphy, symmetry, a smooth limit — recur in every exact result of this chapter. Holomorphy reduces the unknown to a function of the holomorphic invariants alone. The global symmetries, enforced through the transforming spurions, cut that function down to a one-parameter family of monomials. And the demand that the answer behave smoothly in the limits

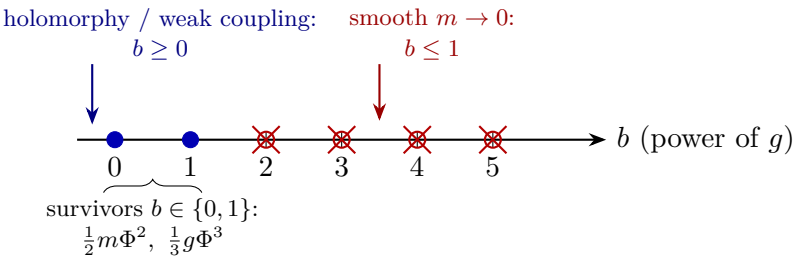


Figure 46.1: The lattice of monomials $m^{1-b} g^b \Phi^{b+2}$ allowed in W_{eff} by holomorphy and the $\mathrm{U}(1) \times \mathrm{U}(1)_R$ selection rules (Theorem 46.3), labelled by the power b of the cubic coupling. Holomorphy and weak coupling forbid $b < 0$ (no negative powers of g); regularity of the massless limit $m \to 0$ forbids $b \geq 2$ (those carry m^{1-b} with negative exponent). Only $b = 0$ and $b = 1$ survive, and they are precisely the two tree-level terms. This is Seiberg's nonrenormalization theorem at a glance.

where a coupling is switched off — weak coupling on one side, the symmetry-restoring limit on the other — squeezes the family to a point. The miracle of Theorem 45.20, that an infinite tower of supergraph cancellations leaves the superpotential pristine, is not a coincidence of combinatorics; it is the shadow of these two limits being mutually exclusive for any term but the tree-level ones.

46.1.4 Why the Kähler potential is not protected

Holomorphy protects the superpotential and nothing else. The Kähler potential is a D-term, the integral of a real function over the *full* superspace Eq. (46.1), and a real function is free to depend on both the chiral couplings and their conjugates. None of the three ingredients of Theorem 46.4 applies to it: a D-term integrand is not chiral, so holomorphy gives no foothold; and because it can carry the conjugate spurions $\bar{m}, \bar{g}$, the symmetry selection rules no longer force a monomial structure. The Kähler potential therefore does receive quantum corrections, and we can see their shape from dimensions and charges alone.

In four dimensions the full measure $\mathrm{d}^4\theta = \mathrm{d}^2\theta\, \mathrm{d}^2\bar{\theta}$ has mass dimension $+2$ (each θ has dimension $-\frac{1}{2}$), and $\int d^4x$ has dimension -4, so the integrand K must have mass dimension $+2$ for the action to be dimensionless. The chiral superfield Φ has dimension 1 (its lowest component is a canonically normalized scalar), so the tree-level $K = \Phi^\dagger \Phi$ saturates the dimension by itself. Any radiative correction must preserve $\mathrm{U}(1)$ neutrality and dimension $+2$, so it takes the form

$$K_{\text{eff}} = \Phi^\dagger \Phi \; \mathcal{Z}\!\left(g, \bar{g}, \frac{m}{\mu}, \frac{\bar{m}}{\mu}, \ln \frac{\mu}{\Lambda} \right) \; + \; (\text{higher-dimension } D\text{-terms}), \tag{46.14}$$

where $\mathcal{Z}$ is a *real* function of the dimensionless invariants built from the couplings, the renormalization scale μ, and the cutoff Λ. The coefficient of $\Phi^\dagger \Phi$

is the wave-function renormalization $\mathcal{Z}$. Two features distinguish it sharply from the superpotential. First, it depends on g and $\bar{g}$ together — the leading one-loop correction is governed by $|g|^2$, not by g alone — so it is manifestly non-holomorphic and the selection-rule argument cannot run. Second, the dimensionless combination $\ln(\mu/\Lambda)$ is available, so $\mathcal{Z}$ genuinely *runs* with scale; at one loop the wave-function renormalization of the Wess–Zumino model is

$$\mathcal{Z}(\mu) = 1 - \frac{|g|^2}{16\pi^2}\,\ln\frac{\mu^2}{\Lambda^2} + \mathcal{O}(g^4), \tag{46.15}$$

the standard self-energy logarithm of the cubic interaction. The field Φ therefore carries a nonzero anomalous dimension $\gamma = -\mu\,\partial_\mu \ln \mathcal{Z} = |g|^2/8\pi^2 + \mathcal{O}(g^4) > 0$. Nothing is uncorrected here; the Kähler sector runs like any non-supersymmetric kinetic term.

The two statements coexist without tension. The holomorphic couplings m and g in W_{eff} never move, but the fields they multiply are being rescaled by $\mathcal{Z}$. When one passes to canonically normalized fields $\Phi_{\text{can}} = \mathcal{Z}^{1/2}\Phi$, the physical mass and the physical cubic coupling read off the canonically normalized action are

$$m_{\text{phys}} = \mathcal{Z}^{-1}\,m, \qquad g_{\text{phys}} = \mathcal{Z}^{-3/2}\,g, \tag{46.16}$$

since $W = \frac{1}{2}m\Phi^2 + \frac{1}{3}g\Phi^3 = \frac{1}{2}(\mathcal{Z}^{-1}m)\Phi_{\text{can}}^2 + \frac{1}{3}(\mathcal{Z}^{-3/2}g)\Phi_{\text{can}}^3$. The *physical* couplings run, but only through the single universal factor $\mathcal{Z}$ — not through independent vertex renormalizations. This is the entire content of supersymmetric nonrenormalization at the level of measurable quantities: the would-be infinity of independent counterterms of a generic scalar theory is reduced to one wave-function renormalization, and the supersymmetric relations among masses and couplings established in Theorem 44.42 survive quantization intact because $\mathcal{Z}$ acts on all of them coherently. The clean separation of the holomorphic running (none) from the canonical running (through $\mathcal{Z}$) is exactly the distinction the next section will need for the gauge coupling, where it becomes the gap between the one-loop holomorphic β-function and the all-orders NSVZ result.

Physical Insight

It is illuminating to contrast this with ϕ^4 theory. In the ordinary real scalar $\frac{\lambda}{4!}\phi^4$ theory, the mass term and the quartic vertex are renormalized *independently*: the self-energy is quadratically divergent and the coupling runs with its own beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ (Chapter 24), with no relation forced between the two. The Wess–Zumino model is the supersymmetric cousin of exactly this dynamics — a self-interacting scalar plus its fermionic partner — and yet its "ϕ^4 data" behave completely differently. The scalar self-energy is still corrected, but only through the same $\mathcal{Z}$ that renormalizes the fermion kinetic term, because boson and fermion sit in one multiplet; the quadratic divergence cancels between them. The holo-

morphic data — the mass m and the coupling g, and the relation $\lambda_{\phi^4} = g^2$ between the scalar quartic and the Yukawa fixed in Theorem 44.42 — are not renormalized at all. The lesson is that supersymmetry does not make everything finite; it sorts the couplings into a holomorphic sector that is rigidly protected and a Kähler sector that runs, and the protected sector is precisely the one that controls vacuum structure.

46.1.5 The Wess–Zumino model worked through

We close the section by carrying the principle through the carried Wess–Zumino example explicitly, exhibiting term by term what is protected and what is not.

Worked Example 46.5 (Holomorphic versus Kähler data in the cubic Wess–Zumino model)**.** Take the Wess–Zumino model with the cubic superpotential Eq. (46.3), $W = \frac{1}{2}m\Phi^2 + \frac{1}{3}g\Phi^3$, and the canonical Kähler potential $K = \Phi^\dagger\Phi$. We determine the exact quantum-corrected superpotential and the structure of the corrected Kähler potential, and read off which physical data are protected.

Step 1: charges. The free theory has $\mathrm{U}(1) : \Phi \to e^{i\alpha}\Phi$ and $\mathrm{U}(1)_R$ with $R[\Phi] = 1$. Promoting m, g to spurions and demanding invariance of W assigns the charges tabulated in Section 46.1.2: $q[m] = -2$, $R[m] = 0$ and $q[g] = -3$, $R[g] = -1$. We verify the cubic term: $q[g\Phi^3] = -3 + 3(1) = 0$ ✓and $R[g\Phi^3] = -1 + 3(1) = 2$ ✓, matching the required $R[W] = 2$ of Eq. (46.5).

Step 2: holomorphy plus selection rules. By Theorem 46.3 the effective superpotential is $W_{\text{eff}} = m\Phi^2 f(g\Phi/m)$ with f analytic at the origin. The candidate corrections are the monomials $m^{1-b}g^b\Phi^{b+2}$ of Fig. 46.1; for orientation, a putative one-loop correction to the mass would be a term $\propto g\bar{g}\, m\, \Phi^2$, but $\bar{g}$ is antichiral and cannot appear in W, while a purely holomorphic mass shift $\propto g^2\Phi^2/m$ is the $b = 2$ monomial.

Step 3: kill the corrections by the limits. The $b = 2$ monomial $g^2\Phi^2/m$ carries an inverse power of m; as $m \to 0$ the theory is the regular massless cubic model and cannot develop such a pole, so its coefficient vanishes. The same reasoning eliminates every $b \geq 2$. Weak coupling and the free-theory limit fix $c_0 = \frac{1}{2}$ (the free mass) and $c_1 = \frac{1}{3}$ (the tree cubic). Hence

$$W_{\text{eff}} = \tfrac{1}{2}\, m\, \Phi^2 + \tfrac{1}{3}\, g\, \Phi^3 \qquad \text{exactly, to all orders in } g. \tag{46.17}$$

In particular the boson–fermion mass degeneracy $m_\phi = m_\psi = m$ found in Theorem 44.42, and the relation $\lambda_{\phi^4} = g^2$ tying the scalar quartic to the Yukawa, are not corrected as relations among the holomorphic parameters.

Step 4: the Kähler sector runs. The wave-function renormalization Eq. (46.15), $\mathcal{Z} = 1 - (|g|^2/16\pi^2)\ln(\mu^2/\Lambda^2) + \mathcal{O}(g^4)$, is non-holomorphic ($|g|^2$, not g) and scale-dependent. Through Eq. (46.16) the physical mass and coupling run as

$$m_{\text{phys}}(\mu) = \mathcal{Z}(\mu)^{-1}\, m, \qquad g_{\text{phys}}(\mu) = \mathcal{Z}(\mu)^{-3/2}\, g, \tag{46.18}$$

so the physically measured cubic coupling has the one-loop running

$$\mu\,\frac{dg_{\rm phys}}{d\mu} = -\tfrac{3}{2}\,g_{\rm phys}\,\mu\,\frac{d\ln\mathcal{Z}}{d\mu} = \tfrac{3}{2}\,\gamma\,g_{\rm phys} = \frac{3\,|g_{\rm phys}|^2}{16\pi^2}\,g_{\rm phys} + \mathcal{O}(g^5), \qquad (46.19)$$

entirely controlled by the anomalous dimension $\gamma = |g|^2/8\pi^2$. There is no independent vertex renormalization: the running of the measurable coupling is the running of $\mathcal{Z}$, three powers' worth. This is the precise statement that "the superpotential is not renormalized but the Kähler potential is."

We have isolated holomorphy as a stand-alone principle: the superpotential is a holomorphic function of the chiral fields *and* of the couplings once the latter are promoted to background chiral spurions, and this single fact — combined with global charge selection rules and the demand of smoothness in the weak-coupling and symmetry-restoring limits — determines the effective superpotential exactly, with the perturbative nonrenormalization theorem falling out as the simplest instance. The next section applies precisely this machinery to the one coupling we have so far left alone, the gauge coupling: promoting the complexified coupling $\tau = \theta/2\pi + 4\pi i/g^2$ to a background chiral superfield multiplying the field-strength bilinear $W^\alpha W_\alpha$, we will find that holomorphy permits the gauge coupling to run only at one loop, and then reconcile that with the all-orders physical beta function through exactly the holomorphic-to-canonical rescaling foreshadowed by $\mathcal{Z}$ above.

46.2 The Holomorphic Gauge Coupling and the Exact Beta Function

The previous section isolated holomorphy as a principle: the superpotential is a holomorphic function of the chiral fields, and—once a coupling is promoted to the lowest component of a background chiral superfield, a *spurion* (Theorem 46.2)—a holomorphic function of the couplings too. We applied it to the superpotential couplings and recovered Seiberg's nonrenormalization theorem (Theorem 46.4) in a line. There is one coupling we left untouched, because it does not sit in the superpotential at all: the gauge coupling g. Its home is the gauge kinetic term, and the natural object multiplying that term turns out to be itself a holomorphic combination of g and the vacuum angle θ. Promoting *that* combination to a spurion is the entire content of this section. The payoff is immediate and dramatic: holomorphy forces the gauge coupling, in the appropriate (Wilsonian, holomorphic) normalization, to run only at one loop—every higher loop is forbidden. We will then have to confront an apparent contradiction with the all-orders beta function that perturbation theory undeniably produces, and resolve it by the same field-rescaling mechanism foreshadowed by the wavefunction factor $\mathcal{Z}$ of Section 46.1.4. The resolution is the Novikov–Shifman–Vainshtein–Zakharov

(NSVZ) exact beta function, and it is exact in the strict sense: a closed-form expression valid to all orders in the coupling.

46.2.1 The complexified holomorphic coupling

Section 45.1 built the gauge sector of a supersymmetric theory from the vector superfield $V = V^a T^a$ and its field-strength chiral superfield

$$W_\alpha = -\tfrac{1}{4}\,\bar{D}^2 (e^{-V} D_\alpha e^{V}), \tag{46.20}$$

a chiral superfield ($\bar{D}_{\dot\beta} W_\alpha = 0$) transforming in the adjoint representation, whose lowest component is the gaugino λ_α and whose θ-component contains the field strength $F^a_{\mu\nu}$. The gauge kinetic action assembled there, Eq. (45.55), is the chiral integral of the Lorentz- and gauge-invariant bilinear $W^{\alpha a} W^a_\alpha$. We now write it in the form that exhibits its dependence on the coupling and the vacuum angle at once.

Definition 46.6 (Holomorphic gauge coupling)**.** For a gauge theory with coupling g and vacuum angle θ (the angle of Theorem 42.27), the *holomorphic gauge coupling* is the complex number

$$\tau \;=\; \frac{\theta}{2\pi} \;+\; \frac{4\pi i}{g^2}. \tag{46.21}$$

Its imaginary part $\operatorname{Im}\tau = 4\pi/g^2 > 0$ is positive in the physical regime, so τ lives in the upper half-plane.

In terms of τ the gauge kinetic action is the chiral integral

$$S_{\text{gauge}} \;=\; \frac{1}{8\pi}\,\operatorname{Im}\!\left[\,\tau \int \mathrm{d}^4x\,\mathrm{d}^2\theta\; W^{\alpha a} W^a_\alpha \right]. \tag{46.22}$$

That this reproduces the ordinary Yang–Mills Lagrangian together with the vacuum term is a short computation we now carry out, because the matching of coefficients is what fixes the normalization of τ and exposes the role of θ-periodicity. The θ-component (the F-term) of the chiral bilinear is, in the two-component conventions of Section 44.7,

$$\int \mathrm{d}^2\theta\; W^{\alpha a} W^a_\alpha \;=\; -\tfrac{1}{2}\,F^a_{\mu\nu} F^{a\,\mu\nu} \;+\; \tfrac{i}{2}\,F^a_{\mu\nu} \widetilde{F}^{a\,\mu\nu} \;-\; 2i\,\lambda^a \sigma^\mu (D_\mu \bar\lambda)^a \;+\; (D^a)^2, \tag{46.23}$$

with $\widetilde{F}^{a\,\mu\nu} = \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma} F^a_{\rho\sigma}$ the dual field strength, λ^a the gaugino, and D^a the real auxiliary field. The first two terms are the only ones we need; the gaugino kinetic term and $(D^a)^2$ are real and ride along. Insert Eq. (46.23) into Eq. (46.22) and take the imaginary part of the product with $\tau = \theta/2\pi + 4\pi i/g^2$. Two

contributions survive. From the imaginary part of τ acting on the real term $-\frac{1}{2}F^2$,

$$\frac{1}{8\pi}\cdot\frac{4\pi}{g^2}\cdot\Big(-\tfrac{1}{2}F^a_{\mu\nu}F^{a\,\mu\nu}\Big) \;=\; -\frac{1}{4g^2}\,F^a_{\mu\nu}F^{a\,\mu\nu}, \tag{46.24}$$

the ordinary gauge kinetic term. From the real part of τ acting on the imaginary term $\frac{i}{2}F\widetilde{F}$,

$$\frac{1}{8\pi}\cdot\frac{\theta}{2\pi}\cdot\tfrac{1}{2}\,F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu} \;=\; \frac{\theta}{32\pi^2}\,F^a_{\mu\nu}\widetilde{F}^{a\,\mu\nu}. \tag{46.25}$$

The cross terms (the imaginary part of τ on $\frac{i}{2}F\widetilde{F}$, and the real part of τ on $-\frac{1}{2}F^2$) are real and imaginary respectively in the wrong way and drop out of $\mathrm{Im}[\,\cdot\,]$. Equations (46.24) and (46.25) are exactly the gauge kinetic term and the topological term of Eq. (42.128), confirming that τ as defined in Eq. (46.21) packages the two real couplings (g,θ) into a single holomorphic one with the correct normalizations.

Notation Watch

Equation (46.24) carries an explicit $1/g^2$ in front of F^2. This is the *holomorphic* normalization of the gauge field: the coupling sits in front of the kinetic term, and correspondingly the field strength in Eq. (46.23) is built from a gauge field with *no* g in its covariant derivative, $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + f^{abc}A^b_\mu A^c_\nu$. The Yang–Mills theory normalization fixed in Eq. (27.124) is the *canonical* one: there the gauge field is rescaled $A^a_\mu \to g\,A^a_\mu$ so the kinetic term is $-\frac{1}{4}F^2$ with the coupling moved into the covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$. The two normalizations differ by a field rescaling whose quantum Jacobian is the whole story of Section 46.2.3; the topological charge density is an integer in either, $\frac{1}{32\pi^2}\int F\widetilde{F} = \frac{g^2}{32\pi^2}\int F_{\rm can}\widetilde{F}_{\rm can} \in \mathbb{Z}$, by Eq. (42.128).

Two features of τ will do all the work. The first is that τ is the coefficient of a *chiral* operator: $W^{\alpha a}W^a_\alpha$ is a chiral superfield (a product of chiral superfields, since each W_α is chiral), and Eq. (46.22) is its F-term. By exactly the logic of Theorem 46.2, τ is therefore promotable to the lowest component of a background chiral spurion superfield, and the holomorphy principle applies to it verbatim: the Wilsonian effective action must be a holomorphic function of τ.

Proposition 46.7 (Periodicity of the holomorphic coupling)**.** *Physics is invariant under*

$$\tau \;\longrightarrow\; \tau + 1, \tag{46.26}$$

the holomorphic image of the shift $\theta \to \theta + 2\pi$.

Proof. Under $\theta \to \theta+2\pi$, the action Eq. (46.25) changes by $\Delta S = 2\pi\cdot\frac{1}{32\pi^2}\int F\widetilde{F} = 2\pi Q$, where $Q \in \mathbb{Z}$ is the integer topological charge of Eq. (42.128). The path-

integral weight e^{iS} is therefore unchanged, $e^{i\Delta S} = e^{2\pi i Q} = 1$; this is the 2π-periodicity of the θ-vacuum established in Theorem 42.27. By Eq. (46.21) a shift of θ by 2π is a shift of τ by 1. □

The second feature is the weak-coupling limit. As $g \to 0$ we have $\operatorname{Im}\tau = 4\pi/g^2 \to +\infty$, and the theory becomes free; any nonperturbative effect must vanish there. We will see that the only holomorphic functions of τ compatible with both Eq. (46.26) and this weak-coupling boundary condition are extraordinarily restricted.

46.2.2 One-loop exactness of the holomorphic coupling

We now run the holomorphic coupling. Let $\tau(\mu)$ denote the coefficient of $W^{\alpha a}W^{a}_{\alpha}$ in the Wilsonian effective action at scale μ—the action with all modes above μ integrated out. The claim is that the dependence of $\tau(\mu)$ on the scale is a single logarithm and nothing more.

Theorem 46.8 (One-loop exactness of the holomorphic coupling)**.** *In a supersymmetric gauge theory with gauge group of dual Coxeter number C_A and chiral matter superfields Φ_i in representations r_i with Dynkin indices $T(r_i)$, the Wilsonian holomorphic coupling runs only at one loop:*

$$\tau(\mu) \;=\; \tau(\mu_0) \;+\; \frac{i\,b_0}{2\pi}\,\ln\frac{\mu}{\mu_0}, \qquad b_0 \;=\; 3C_A - \sum_i T(r_i), \tag{46.27}$$

with no perturbative corrections of two loops or higher. Equivalently the combination

$$\Lambda^{\,b_0} \;=\; \mu^{\,b_0}\, e^{\,2\pi i\,\tau(\mu)} \tag{46.28}$$

is renormalization-group invariant; it defines the (complex) holomorphic dynamical scale Λ.

Proof. The argument has three ingredients—holomorphy, periodicity, and the weak-coupling boundary condition—each contributed by the structure assembled above.

Holomorphy. Because τ multiplies a chiral operator, the Wilsonian coupling $\tau(\mu)$ is a holomorphic function of the microscopic coupling $\tau_0 \equiv \tau(\mu_0)$: it can depend on τ_0 but not on $\bar\tau_0$. Dimensional analysis isolates the scale dependence: the only dimensionful quantities are μ and μ_0, so $\tau(\mu)$ depends on them through $\ln(\mu/\mu_0)$. Holomorphy in τ_0 then permits the general form

$$\tau(\mu) \;=\; \tau_0 \;+\; f(\tau_0, \ln(\mu/\mu_0)), \tag{46.29}$$

with f holomorphic in τ_0 and vanishing at $\mu = \mu_0$.

Perturbation theory at weak coupling. As $\operatorname{Im}\tau_0 = 4\pi/g^2 \to \infty$ the theory is weakly coupled and standard perturbation theory applies. The one-loop running

of $1/g^2$ is the universal logarithm computed for any gauge theory by the vacuum-polarization machinery of Chapters 24 and 27; its supersymmetric coefficient we fix in Section 46.2.5 below. Translated into τ through Eq. (46.21), the one-loop result is precisely the linear term $\frac{ib_0}{2\pi}\ln(\mu/\mu_0)$ of Eq. (46.27). The content of the theorem is that this linear term is the *entire* perturbative answer. A generic ℓ-loop contribution to the running of $1/g^2$ scales as $g^{2(\ell-1)}$, i.e. as $(\operatorname{Im}\tau_0)^{-(\ell-1)}$ for $\ell \geq 2$. Such a term is a function of $\operatorname{Im}\tau_0 = (\tau_0 - \bar{\tau}_0)/2i$ and is therefore *not holomorphic* in τ_0: it cannot appear in Eq. (46.29). Holomorphy kills every perturbative term beyond the one loop.

Periodicity fixes what holomorphy permits. Holomorphy still allows, in principle, nonperturbative dependence on τ_0. Theorem 46.7 constrains it sharply. Apart from the explicit logarithm—which under $\tau_0 \to \tau_0 + 1$ simply shifts $\tau(\mu)$ by 1, consistent with both being angle variables of period 1—the remainder f must be invariant under $\tau_0 \to \tau_0 + 1$. A holomorphic, 1-periodic function on the upper half-plane is a function of $q \equiv e^{2\pi i\tau_0}$, hence admits a Fourier–Laurent expansion

$$f \supset \sum_{k\geq 1} a_k(\ln(\mu/\mu_0))\, e^{2\pi i k\tau_0}, \tag{46.30}$$

where positive powers of q only are allowed because $e^{2\pi i\tau_0} = e^{i\theta - 8\pi^2/g^2} \to 0$ as $g \to 0$ (weak coupling forbids the negative powers that would blow up). But $e^{2\pi i\tau_0} = e^{i\theta}e^{-8\pi^2/g^2}$ is exactly the weight of a single instanton: $e^{-8\pi^2/g^2} = e^{-S_{\text{inst}}}$ with $S_{\text{inst}} = 8\pi^2/g^2$ the Yang–Mills instanton action of Theorem 42.4, dressed by the phase $e^{i\theta}$ of one unit of topological charge. The k-th term is a k-instanton effect. These are genuinely present in general, but they are nonperturbative—invisible to every order of the g-expansion—and, crucially, they do not renormalize the *coupling*: a careful accounting (the holomorphic anomaly being saturated already at one loop) shows the a_k contribute to other holomorphic couplings, not to the running of τ itself, so that the running Eq. (46.27) is exact. The perturbative series for the Wilsonian holomorphic coupling thus terminates at one loop, which is the assertion.

The invariant scale. Rewrite Eq. (46.27) as $2\pi i\,\tau(\mu) = 2\pi i\,\tau_0 - b_0\ln(\mu/\mu_0)$, i.e. $\mu^{b_0}e^{2\pi i\tau(\mu)} = \mu_0^{b_0}e^{2\pi i\tau_0}$, independent of the scale at which it is evaluated. This is Eq. (46.28), and it defines Λ. □

Equation (46.28) is the single most important object handed to the rest of the chapter. It is the supersymmetric, holomorphic refinement of the QCD scale Λ_{QCD} of Theorem 27.11: the renormalization-group invariant that the dimensionless coupling trades itself for, the place where the holomorphic coupling formally diverges ($\tau \to i\infty$ corresponds to $g \to 0$; the opposite limit $\mu \to \Lambda$ is where g runs strong). The crucial new feature relative to the bosonic case is that Λ is *complex*: its phase is set by θ, through $\Lambda^{b_0} \propto e^{2\pi i\tau} = e^{i\theta}e^{-8\pi^2/g^2}$. A dynamically generated superpotential, being holomorphic, can depend on Λ but never on $\bar{\Lambda}$ or on $|\Lambda|$ alone; this is why Λ will appear, raised to exactly the

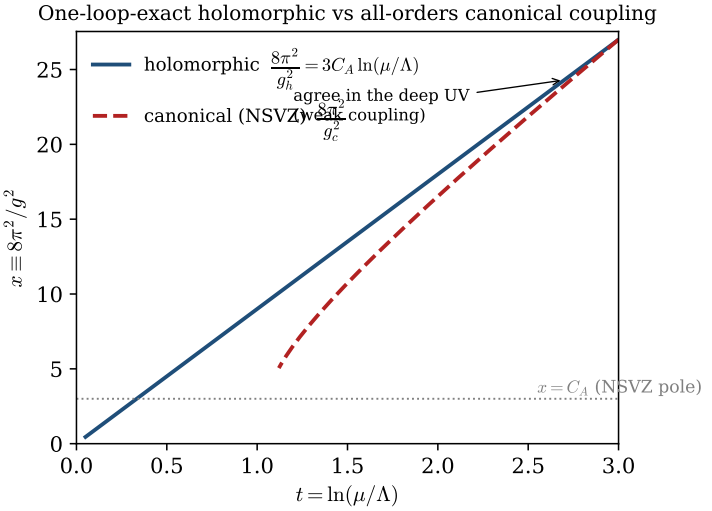


Figure 46.2: The central claim of this section, for pure $\mathcal{N} = 1$ super-Yang–Mills ($C_A = 3$, no matter). Plotted is $x = 8\pi^2/g^2 = \frac{1}{2}\operatorname{Im}\tau$ against $t = \ln(\mu/\Lambda)$. The holomorphic coupling (solid) runs strictly linearly, $8\pi^2/g_h^2 = 3C_A \ln(\mu/\Lambda)$—Theorem 46.8, one loop and no more. The canonical/physical coupling (dashed) obeys the NSVZ flow Eq. (46.37), which for pure SYM closes on the coupling alone and curves; it reaches strong coupling (the NSVZ denominator vanishes at $x = C_A$, dotted line) at a parametrically higher scale than the holomorphic coupling would suggest. The two agree in the deep ultraviolet, where the rescaling Jacobian of Section 46.2.3 is negligible.

power that holomorphy and the symmetries dictate, in every nonperturbative superpotential of Section 46.2's sequel.

46.2.3 Holomorphic versus canonical normalization

Theorem 46.8 sits in apparent flat contradiction with experience. The gauge coupling of a real (super)symmetric theory runs to all orders; a two-loop contribution to the QCD-like beta function is a measured, nonzero number. How can the coupling run only at one loop? The resolution, due to Shifman and Vainshtein and made transparent by Arkani-Hamed and Murayama, is that *there are two different couplings*, related by a field redefinition whose quantum-mechanical Jacobian is anomalous. The holomorphic coupling τ of Theorem 46.6 is one-loop exact. The *canonical* (physical) coupling g_c—the one appearing in a canonically normalized Lagrangian, the one a lattice or a scattering experiment measures, the one whose running Yang–Mills theory has tracked since Eq. (27.124)—is a non-holomorphic function of τ, and it runs to all orders. The all-orders content

hides entirely in the map between them.

To exhibit the map, start from the holomorphic Lagrangian. Its gauge term is Eq. (46.24), with $1/g_h^2$ in front of $-\frac{1}{4}F^2$ and the gauge field holomorphically normalized; its matter term is the Kähler integral

$$\int \mathrm{d}^4\theta\, \bar{\Phi}_i\, e^V\, \Phi_i, \tag{46.31}$$

with Φ_i holomorphically normalized. Two rescalings carry this to canonical normalization.

The matter rescaling. The Kähler term Eq. (46.31) is not protected by holomorphy—this was the lesson of Section 46.1.4—and loops renormalize it to $\mathcal{Z}_i \int \mathrm{d}^4\theta\, \bar{\Phi}_i e^V \Phi_i$ with a real wavefunction factor $\mathcal{Z}_i(\mu)$, exactly the $\mathcal{Z}$ of Eq. (46.15). To restore canonical kinetic terms we rescale

$$\Phi_i \;\longrightarrow\; \mathcal{Z}_i^{-1/2}\, \Phi_i. \tag{46.32}$$

This is a (complexified) rescaling of a chiral superfield, and—here is the point—the path-integral measure $\mathcal{D}\Phi_i$ is *not* invariant under it. The non-invariance is the supersymmetric face of the chiral anomaly of Chapter 36.

The Konishi anomaly. The classical current associated with the phase rotation $\Phi_i \to e^{i\alpha}\Phi_i$ is the Konishi current, packaged in the real superfield $\mathcal{J}_i = \bar{\Phi}_i e^V \Phi_i$. Classically $\frac{1}{4}\bar{D}^2 \mathcal{J}_i = \Phi_i\, \partial W/\partial \Phi_i$, a pure superpotential term. Quantum-mechanically the rotation of the fermionic component ψ_i inside Φ_i is anomalous in precisely the way the axial current of Chapter 36 is, with the triangle now carrying two gauge legs. The result is the Konishi anomaly equation,

$$\tfrac{1}{4}\,\bar{D}^2\,\mathcal{J}_i \;=\; \Phi_i \frac{\partial W}{\partial \Phi_i} \;+\; \frac{T(r_i)}{8\pi^2}\,\tfrac{1}{4}\,W^{\alpha a} W^a_\alpha, \tag{46.33}$$

the index $T(r_i)$ being the Dynkin index of the representation r_i carried by Φ_i—the direct supersymmetric counterpart of the coefficient $\frac{g^2}{16\pi^2}\,\mathrm{tr}\, F\widetilde{F}$ of the ordinary anomaly of Chapter 36, with $W^{\alpha a}W^a_\alpha$ the chiral superfield whose F-component contains $F\widetilde{F}$ by Eq. (46.23). The anomaly Eq. (46.33) says that the chiral rescaling Eq. (46.32), which is the finite rotation with parameter $\alpha_i = \frac{i}{2}\ln \mathcal{Z}_i$ (so that $e^{i\alpha_i} = \mathcal{Z}_i^{-1/2}$ once continued to the scaling direction), shifts the effective action by a term proportional to $\int \mathrm{d}^2\theta\, W^{\alpha a} W^a_\alpha$. Reading off the coefficient from Eq. (46.33) and comparing with Eq. (46.22), the net effect on the gauge coupling is a shift

$$\frac{8\pi^2}{g^2} \;\longrightarrow\; \frac{8\pi^2}{g^2} \;-\; T(r_i)\,\ln \mathcal{Z}_i \qquad \text{(per matter superfield } \Phi_i\text{)}. \tag{46.34}$$

The gauge rescaling. The gauge term Eq. (46.24) is also non-canonical: it has $1/g_h^2$ out front. Bringing it to canonical form requires rescaling the vector

superfield, $V \to g_c V$ (equivalently the gaugino $\lambda \to g_c \lambda$), so that the kinetic term becomes $-\frac{1}{4} F_c^2$ with the coupling moved into the covariant derivative—the canonical normalization of the notation watch above. The gaugino is a Weyl fermion in the *adjoint* representation, whose Dynkin index is the dual Coxeter number C_A. Its rescaling is anomalous by the very same mechanism as Eq. (46.33), now with $T(\text{adj}) = C_A$ and with the rescaling parameter set by $\ln g_c$ rather than $\ln \mathcal{Z}$. The Jacobian contributes a shift

$$\frac{8\pi^2}{g^2} \;\longrightarrow\; \frac{8\pi^2}{g^2} \;-\; C_A \,\ln g_c^2 . \tag{46.35}$$

This is the term that will produce the NSVZ denominator: it is anomalous in the coupling itself, so it feeds the running of g_c back into the relation between g_c and g_h.

Collecting Eqs. (46.34) and (46.35), the canonical and holomorphic couplings are related by

$$\boxed{\frac{8\pi^2}{g_c^2(\mu)} \;=\; \frac{8\pi^2}{g_h^2(\mu)} \;-\; C_A \,\ln g_c^2(\mu) \;-\; \sum_i T(r_i)\,\ln \mathcal{Z}_i(\mu) \;+\; \text{const.}} \tag{46.36}$$

The constant collects scheme-dependent finite pieces and is irrelevant to the beta function. Every nontrivial statement about the all-orders running of the physical coupling is now encoded in this one equation, because its left side is the physical coupling and its first right-hand term is one-loop exact by Theorem 46.8.

46.2.4 The NSVZ exact beta function

Differentiating Eq. (46.36) with respect to $\ln \mu$ converts the one-loop exactness of g_h and the anomalous Jacobians into a closed equation for the physical beta function.

Theorem 46.9 (NSVZ exact beta function)**.** *The beta function of the canonical gauge coupling $g \equiv g_c$ of a supersymmetric gauge theory is*

$$\beta(g) \;=\; -\,\frac{g^3}{16\pi^2}\,\frac{3C_A - \sum_i T(r_i)(1-\gamma_i)}{1 - C_A\, g^2/8\pi^2}, \tag{46.37}$$

where $\gamma_i = -\,d\ln \mathcal{Z}_i / d\ln\mu$ is the anomalous dimension of the matter superfield Φ_i, and the result is exact to all orders in g.

Proof. Write $S_h \equiv 8\pi^2/g_h^2$ and $S_c \equiv 8\pi^2/g_c^2$, and differentiate the three running quantities in Eq. (46.36) with respect to $t \equiv \ln\mu$.

For the holomorphic coupling, Eq. (46.27) gives a pure constant rate,

$$\frac{dS_h}{dt} \;=\; b_0 \;=\; 3C_A - \sum_i T(r_i), \tag{46.38}$$

since $\operatorname{Im}\tau = 4\pi/g_h^2 = S_h/(2\pi)$ and Eq. (46.27) runs $\operatorname{Im}\tau$ linearly with slope $b_0/2\pi$ in $\ln\mu$, so that $dS_h/dt = 2\pi\cdot(b_0/2\pi) = b_0$. For the canonical coupling, with $\beta \equiv dg_c/dt$,

$$\frac{dS_c}{dt} \;=\; \frac{d}{dt}\frac{8\pi^2}{g_c^2} \;=\; -\,\frac{16\pi^2}{g_c^3}\,\beta, \qquad \frac{d}{dt}\ln g_c^2 \;=\; \frac{2\beta}{g_c}. \tag{46.39}$$

For the wavefunction factors, by the definition of the anomalous dimension,

$$\frac{d}{dt}\ln \mathcal{Z}_i \;=\; -\,\gamma_i. \tag{46.40}$$

Substituting Eqs. (46.38) to (46.40) into the derivative of Eq. (46.36),

$$-\,\frac{16\pi^2}{g_c^3}\,\beta \;=\; \Big(3C_A - \sum_i T(r_i)\Big) \;-\; C_A\,\frac{2\beta}{g_c} \;+\; \sum_i T(r_i)\,\gamma_i. \tag{46.41}$$

Collect the matter terms: $-\sum_i T(r_i) + \sum_i T(r_i)\gamma_i = -\sum_i T(r_i)(1-\gamma_i)$, so the right side is $3C_A - \sum_i T(r_i)(1-\gamma_i) - 2C_A\beta/g_c$. Move the β-proportional term to the left,

$$-\,\frac{16\pi^2}{g_c^3}\,\beta \;+\; \frac{2C_A}{g_c}\,\beta \;=\; 3C_A - \sum_i T(r_i)(1-\gamma_i), \tag{46.42}$$

and factor β on the left, pulling out $-16\pi^2/g_c^3$:

$$-\,\frac{16\pi^2}{g_c^3}\,\beta\left(1 - \frac{2C_A}{g_c}\cdot\frac{g_c^3}{16\pi^2}\right) \;=\; 3C_A - \sum_i T(r_i)(1-\gamma_i). \tag{46.43}$$

The bracket simplifies, $\frac{2C_A}{g_c}\cdot\frac{g_c^3}{16\pi^2} = \frac{C_A g_c^2}{8\pi^2}$, giving

$$-\,\frac{16\pi^2}{g_c^3}\,\beta\left(1 - \frac{C_A g_c^2}{8\pi^2}\right) \;=\; 3C_A - \sum_i T(r_i)(1-\gamma_i). \tag{46.44}$$

Solving for β produces Eq. (46.37). The result is exact because every ingredient is: $dS_h/dt = b_0$ is one-loop exact by Theorem 46.8, and the Jacobian terms Eqs. (46.34) and (46.35) are the exact anomalies of Chapter 36, which by the Adler–Bardeen theorem receive no higher-order corrections. The only object that carries higher-loop information is the anomalous dimension γ_i, which appears explicitly. □

The NSVZ formula Eq. (46.37) is not a closed expression for $\beta(g)$ in the naive sense—it still contains the $\gamma_i(g)$, which must be computed order-by-order—but it is exact in a strong and useful way: it resums the entire gauge dependence into the geometric denominator and reduces the all-orders beta function to the single set of matter anomalous dimensions. The numerator's $-\sum_i T(r_i)(1-\gamma_i)$ has a clean reading: $T(r_i)$ is the one-loop contribution of Φ_i, and the $+T(r_i)\gamma_i$

correction is the statement that what a loop "sees" of a matter field is its full scaling dimension, classical plus anomalous. The denominator $1 - C_A g^2/8\pi^2$ is the resummed contribution of the gaugino's own rescaling—the gauge field dressing itself—and is responsible, in Fig. 46.2, for the canonical coupling running strong faster than the holomorphic one.

46.2.5 Consistency: one loop and the SUSY multiplet content

A formula this strong must reproduce what we already know. Expand Eq. (46.37) at weak coupling, where $\gamma_i = O(g^2)$ and the denominator is $1 + O(g^2)$:

$$\beta(g) \;=\; -\frac{g^3}{16\pi^2}\, b_0 + O(g^5), \qquad b_0 = 3C_A - \sum_i T(r_i), \tag{46.45}$$

the same b_0 that appeared in Eq. (46.27). It remains to check that this is the correct one-loop coefficient for a supersymmetric theory, i.e. that it agrees with the general Yang–Mills theory result of Eq. (27.124) once the field content of the supersymmetric multiplets is counted. The general one-loop coefficient for any four-dimensional gauge theory, assembled from the vacuum-polarization contributions of Chapters 24 and 27, is

$$b_0 \;=\; \tfrac{11}{3}\, C_A \;-\; \tfrac{2}{3} \sum_{\text{Weyl } f} T(r_f) \;-\; \tfrac{1}{3} \sum_{\text{cplx scalar } s} T(r_s), \tag{46.46}$$

where a Weyl fermion contributes $\tfrac{2}{3}T$ (a Dirac fermion the $\tfrac{4}{3}T_F$ of Eq. (27.124), being two Weyl fermions) and a complex scalar contributes $\tfrac{1}{3}T$. We now sum these over the two kinds of supersymmetric multiplet.

Vector multiplet. The $\mathcal{N} = 1$ vector multiplet contains the gauge boson and the gaugino λ, a Weyl fermion in the adjoint, $T(\text{adj}) = C_A$. From Eq. (46.46),

$$b_0^{\text{vector}} \;=\; \tfrac{11}{3} C_A - \tfrac{2}{3} C_A \;=\; \tfrac{9}{3} C_A \;=\; 3C_A. \tag{46.47}$$

The gauge boson's antiscreening $\tfrac{11}{3}C_A$ of Theorem 27.11 is reduced by exactly the gaugino's $\tfrac{2}{3}C_A$ of screening to the supersymmetric value $3C_A$.

Chiral multiplet. Each chiral matter superfield Φ_i contains a Weyl fermion and a complex scalar, both in r_i, contributing

$$b_0^{\Phi_i} \;=\; -\tfrac{2}{3} T(r_i) - \tfrac{1}{3} T(r_i) \;=\; -T(r_i). \tag{46.48}$$

Adding Eqs. (46.47) and (46.48) reproduces $b_0 = 3C_A - \sum_i T(r_i)$ exactly. The supersymmetric one-loop coefficient is the bosonic $\tfrac{11}{3}C_A - \tfrac{1}{3}\sum T - \tfrac{2}{3}\sum T$ reorganized by the multiplet pairing, and it matches both the holomorphic running Eq. (46.27) and the NSVZ numerator at $\gamma = 0$. The Chapter 27 one-loop beta function is, in this sense, the leading term of an exact statement: impose supersymmetry, and asymptotic freedom's b_0 becomes the tip of the NSVZ formula Eq. (46.37).

Worked Example 46.10 (NSVZ for supersymmetric QCD)**.** We assemble the exact beta function for the theory the next section will solve: $\mathcal{N}=1$ supersymmetric QCD, gauge group $\mathrm{SU}(N_c)$ with N_f chiral superfields Q in the fundamental $\square$ and N_f in the antifundamental $\bar{\square}$ (a vectorlike flavor content, one "flavor" being the pair $Q,\tilde{Q}$). For $\mathrm{SU}(N_c)$ the relevant group data, from Appendix B, are $C_A=N_c$ and $T(\square)=T(\bar{\square})=\frac{1}{2}$.

The one-loop coefficient. There are $2N_f$ chiral superfields in fundamental-sized representations, so $\sum_i T(r_i)=2N_f\cdot\frac{1}{2}=N_f$ and

$$b_0 \;=\; 3C_A-\sum_i T(r_i) \;=\; 3N_c-N_f. \tag{46.49}$$

Asymptotic freedom (Theorem 27.11, $b_0>0$) therefore holds for $N_f<3N_c$, and the theory is free in the infrared for $N_f>3N_c$. The boundary $N_f=3N_c$ is where the one-loop coefficient vanishes—the top of the conformal window.

The exact beta function. By flavor symmetry all $2N_f$ matter superfields share a common anomalous dimension $\gamma(g)$, so the numerator of Eq. (46.37) is $3N_c-2N_f\cdot\frac{1}{2}(1-\gamma)=3N_c-N_f(1-\gamma)$, and

$$\beta(g) \;=\; -\frac{g^3}{16\pi^2}\,\frac{3N_c-N_f\,(1-\gamma)}{1-N_c\,g^2/8\pi^2}. \tag{46.50}$$

The Banks–Zaks fixed point and the conformal window. A nontrivial infrared fixed point requires the numerator to vanish at some $g_*\neq 0$, i.e. $3N_c-N_f(1-\gamma(g_*))=0$, which fixes the fixed-point anomalous dimension exactly:

$$\gamma_* \;=\; 1-\frac{3N_c}{N_f}. \tag{46.51}$$

This is an exact statement—no perturbative truncation—because the NSVZ numerator is exact. When N_f is just below $3N_c$, γ_* is small and negative and the fixed point is weakly coupled and reliable; this is the supersymmetric Banks–Zaks fixed point. As N_f decreases, $|\gamma_*|$ grows and the fixed point moves to strong coupling. The fixed point ceases to describe an interacting conformal field theory when the gauge-invariant meson operator $M=\tilde{Q}Q$, of exact dimension $\Delta_M=2+\gamma_*=3-3N_c/N_f$ by Eq. (46.51), would violate the unitarity bound $\Delta\geq 1$ for a scalar; setting $\Delta_M=1$ gives $N_f=\frac{3}{2}N_c$. The conformal window is therefore

$$\tfrac{3}{2}N_c \;<\; N_f \;<\; 3N_c, \tag{46.52}$$

inside which SQCD flows to an interacting superconformal field theory. The exact endpoints Eq. (46.52) are among the cleanest predictions of the whole framework, and they will reappear, from the magnetic side, in the discussion of Seiberg duality.

The dynamical scale. For $N_f < 3N_c$ the theory is asymptotically free with the holomorphic invariant

$$\Lambda^{3N_c - N_f} \;=\; \mu^{3N_c - N_f}\, e^{2\pi i\, \tau(\mu)}, \tag{46.53}$$

the specialization of Eq. (46.28) to $b_0 = 3N_c - N_f$. This complex scale, raised to the power $3N_c - N_f$, is the unique holomorphic gauge-sector invariant; the next section will show that the effective superpotential on the moduli space of SQCD is fixed, by holomorphy and symmetry, to be $\Lambda^{(3N_c - N_f)/(N_c - N_f)}$ times a power of the meson determinant—the Affleck–Dine–Seiberg superpotential.

The gauge coupling, the last coupling holomorphy had not yet touched, is now under exact control. We possess the one-loop-exact running Eq. (46.27) of the holomorphic coupling, its all-orders physical companion Eq. (46.37), and—most importantly for what follows—the holomorphic dynamical scale $\Lambda = \mu\, e^{2\pi i \tau / b_0}$ of Eq. (46.28), the renormalization-group invariant whose phase encodes θ and whose magnitude encodes the strong-coupling scale. This single complex number is the only nonperturbative datum the gauge dynamics hands to the low-energy theory. The next section feeds it, together with holomorphy and the flavor symmetries, into supersymmetric QCD, and reads off the exact effective superpotential on the moduli space—the first place where strong-coupling physics is computed without summing a diagram.

46.3 Dynamically Generated Superpotentials: SQCD and Affleck–Dine–Seiberg

The previous section equipped us with two instruments and a single dimensionful constant. The holomorphic gauge coupling $\tau = \theta/2\pi + 4\pi i/g^2$ runs only at one loop, and that one-loop running can be repackaged, with no loss of information, into a renormalization-group invariant dynamical scale Λ defined through $\Lambda^{b_0} = \mu^{b_0}\, e^{2\pi i \tau(\mu)}$ (Eq. (46.28)), where b_0 is the one-loop coefficient of the holomorphic beta function. For supersymmetric $\mathrm{SU}(N_c)$ gauge theory with N_f flavors that coefficient is $b_0 = 3N_c - N_f$, so that

$$\Lambda^{3N_c - N_f} \;=\; \mu^{3N_c - N_f}\, e^{2\pi i \tau(\mu)} \;=\; \mu^{3N_c - N_f}\, e^{-8\pi^2/g^2(\mu) + i\theta} \tag{46.54}$$

is a holomorphic, μ-independent, complex combination of the running coupling and the vacuum angle. The pure gauge sector of Section 42.4 carried this scale but had no flat directions on which a superpotential could live. The moment we add matter, that changes. The classical theory acquires a continuous space of degenerate vacua—a moduli space—and the strong dynamics, communicating through Λ, can lift it. The purpose of this section is to determine, exactly and without summing a single perturbative diagram, the effective superpotential

that the non-perturbative dynamics generates on that moduli space. The engine is holomorphy: a superpotential is a holomorphic function of the light chiral fields and of Λ, and the symmetries of the theory, treated as selection rules on holomorphic functions, will turn out to fix it completely up to one number, which a single instanton then pins down.

46.3.1 The classical moduli space of SQCD

We work with the supersymmetric QCD of Section 45.3.3: gauge group $\mathrm{SU}(N_c)$, and N_f "flavors" of matter, each flavor consisting of a chiral superfield Q in the fundamental $\mathbf{N_c}$ and a chiral superfield $\widetilde{Q}$ in the antifundamental $\overline{\mathbf{N_c}}$. We index the color by $a = 1, \ldots, N_c$ and the flavor by $i = 1, \ldots, N_f$, so the matter content is the pair of matrices of chiral superfields

$$Q^a{}_i, \qquad \widetilde{Q}_a{}^i, \qquad a = 1, \ldots, N_c, \quad i = 1, \ldots, N_f, \tag{46.55}$$

with no tree-level superpotential, $W_{\text{tree}} = 0$. We take the theory massless and (for the first part of the analysis) in the asymptotically free, would-be-confining window $N_f < 3N_c$, $b_0 = 3N_c - N_f > 0$. The lowest components of Q and $\widetilde{Q}$ are the squark scalars, which we denote by the same symbols when no confusion arises.

Symmetries. With $W_{\text{tree}} = 0$ the classical action has the largest symmetry it can have. The kinetic terms are invariant under independent unitary rotations of the N_f flavors of Q and of the N_f flavors of $\widetilde{Q}$, together with two commuting $\mathrm{U}(1)$ phase rotations and an R-symmetry under which the Grassmann coordinate θ is charged. The global symmetry group is

$$G_{\text{global}} = \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B \times \mathrm{U}(1)_A \times \mathrm{U}(1)_R, \tag{46.56}$$

under which Q transforms as $(\mathbf{N_f}, \mathbf{1})$ and $\widetilde{Q}$ as $(\mathbf{1}, \overline{\mathbf{N_f}})$ of the flavor group, with abelian charges collected in Table 46.1. Here $\mathrm{U}(1)_B$ is baryon number (Q charge $+1$, $\widetilde{Q}$ charge -1), $\mathrm{U}(1)_A$ is the axial phase common to both ($Q, \widetilde{Q}$ charge $+1$), and $\mathrm{U}(1)_R$ is the R-symmetry under which the gauginos carry charge $+1$. Of these, $\mathrm{U}(1)_A$ and the naive $\mathrm{U}(1)_R$ are anomalous: an axial rotation rotates the path-integral measure exactly as in Section 42.6.2, shifting the vacuum angle. We will exploit precisely this anomaly. Two combinations are anomaly-free: $\mathrm{U}(1)_B$, and the particular R-symmetry whose charge assignment we now fix.

The non-anomalous R-charge follows from cancellation of the mixed $\mathrm{SU}(N_c)^2$–$\mathrm{U}(1)_R$ anomaly, the supersymmetric analogue of the triangle condition of Section 37.3. The fermions charged under both are the gaugino (one Weyl fermion in the adjoint, R-charge $+1$, Dynkin index $T(\text{adj}) = C_A = N_c$) and the $2N_f$ quark Weyl fermions $\psi_Q, \psi_{\widetilde{Q}}$ (each in the fundamental, $T(\mathbf{N_c}) = \frac{1}{2}$, with R-charge

	$\mathrm{SU}(N_f)_L$	$\mathrm{SU}(N_f)_R$	$\mathrm{U}(1)_B$	$\mathrm{U}(1)_A$	$\mathrm{U}(1)_R$
Q	$\mathbf{N_f}$	$\mathbf{1}$	$+1$	$+1$	$\dfrac{N_f - N_c}{N_f}$
$\widetilde{Q}$	$\mathbf{1}$	$\overline{\mathbf{N_f}}$	-1	$+1$	$\dfrac{N_f - N_c}{N_f}$
W^α (gauginos)	$\mathbf{1}$	$\mathbf{1}$	0	0	$+1$
$\Lambda^{3N_c - N_f}$	$\mathbf{1}$	$\mathbf{1}$	0	$2N_f$	0

Table 46.1: Charges of the SQCD chiral superfields under the abelian global symmetries. The R-charges of $Q, \widetilde{Q}$ are fixed by demanding that $\mathrm{U}(1)_R$ be anomaly-free (Eq. (46.57)). The last row records the charges carried by the holomorphic scale Λ^{b_0}, read off from the anomalous shift of the measure (Eq. (46.58)); these make Λ^{b_0} the spurion that restores the broken $\mathrm{U}(1)_A$ and $\mathrm{U}(1)_R$ as exact selection rules.

$R[Q] - 1$ since a chiral superfield's fermion carries one less unit of R than its scalar). The mixed anomaly is proportional to the sum of $T(r) \times R(\text{fermion})$ over all charged Weyl fermions,

$$N_c \cdot (+1) \;+\; 2N_f \cdot \tfrac{1}{2} \cdot \big(R[Q] - 1\big) \;=\; 0 \quad \Longrightarrow \quad R[Q] = R[\widetilde{Q}] = \frac{N_f - N_c}{N_f}, \tag{46.57}$$

which is the entry in Table 46.1.

The scale as a spurion. The two anomalous symmetries are not lost; they become *exact selection rules carried by the coupling*. Under an axial rotation $Q, \widetilde{Q} \to e^{i\alpha} Q, e^{i\alpha} \widetilde{Q}$, the fermionic measure produces the shift of the vacuum angle computed in Eq. (42.172). With $2N_f$ charged Weyl fermions (counting Q and $\widetilde{Q}$ over all flavors), each of unit axial charge, the angle shifts by $\theta \to \theta + 2N_f \alpha$. Through Eq. (46.54) this is the statement that the holomorphic scale transforms,

$$\Lambda^{3N_c - N_f} \;\longrightarrow\; e^{\,2iN_f \alpha}\, \Lambda^{3N_c - N_f}, \tag{46.58}$$

i.e. Λ^{b_0} carries axial charge $+2N_f$ and vanishing B- and (non-anomalous) R-charge. This is the last row of Table 46.1. The logic, due to Seiberg, is that if we formally assign Λ^{b_0} these charges and treat it as a background (spurion) field, then *every* symmetry in Eq. (46.56), including the anomalous ones, becomes an exact invariance of the theory. Any quantity the dynamics generates—in particular any effective superpotential—must respect all of them. That is an enormously restrictive constraint on a holomorphic object, and it is the whole game.

D-flat directions. The scalar potential of a supersymmetric gauge theory with $W_{\text{tree}} = 0$ is purely a D-term,

$$V = \frac{1}{2}\sum_a \left(D^a\right)^2, \qquad D^a = g\Big(Q^\dagger\, T^a\, Q - \widetilde{Q}\, T^a\, \widetilde{Q}^\dagger\Big), \tag{46.59}$$

where the trace is over color and flavor as appropriate and T^a are the fundamental generators. Supersymmetric vacua are the zeros of V, i.e. the D-flat configurations $D^a = 0$ for all a. These are not isolated points: there is a continuous family of squark expectation values solving $D^a = 0$, and that family, modulo gauge transformations, is the classical moduli space—a moduli space of degenerate vacua in exactly the sense introduced for scalar potentials in Theorem 29.6, now realized by the flat directions of the gauge D-term. The general D-flat solution is found by a standard linear-algebra argument. Write the squark VEVs as the $N_c \times N_f$ matrices $\langle Q\rangle$ and $\langle\widetilde{Q}\rangle$. Using a color-flavor rotation, the most general D-flat configuration for $N_f < N_c$ can be brought to the diagonal form

$$\langle Q\rangle = \begin{pmatrix} v_1 & & \\ & \ddots & \\ & & v_{N_f} \\ 0 & & \end{pmatrix}, \qquad \langle\widetilde{Q}\rangle^{\mathsf{T}} = \begin{pmatrix} \widetilde{v}_1 & & \\ & \ddots & \\ & & \widetilde{v}_{N_f} \\ 0 & & \end{pmatrix},$$
$$|v_i|^2 - |\widetilde{v}_i|^2 = \text{const (independent of } i). \tag{46.60}$$

The constant is set to zero in the genuinely flat (zero-energy) directions, so $|v_i| = |\widetilde{v}_i|$ along the moduli space. The remaining v_i are arbitrary complex numbers, the coordinates of the moduli space.

Gauge-invariant coordinates: the chiral ring. Physics depends only on gauge-invariant data, so the right coordinates on the moduli space are the gauge-invariant holomorphic polynomials in $Q, \widetilde{Q}$—the generators of the chiral ring. For $N_f < N_c$ the only invariants one can build are the *mesons*, the color-contracted bilinears

$$M^i{}_j \;=\; \widetilde{Q}_a{}^i\, Q^a{}_j, \qquad i,j = 1,\dots,N_f, \tag{46.61}$$

an $N_f \times N_f$ matrix of chiral superfields. (When $N_f \geq N_c$ one can in addition form the *baryons* $B^{i_1\dots i_{N_c}} = \epsilon_{a_1\dots a_{N_c}} Q^{a_1}{}_{i_1}\cdots Q^{a_{N_c}}{}_{i_{N_c}}$ and the antibaryons $\widetilde{B}$ from $\widetilde{Q}$; for $N_f < N_c$ the color ϵ-tensor cannot be saturated by fewer than N_c fundamentals, so baryons do not exist and the mesons alone coordinatize the moduli space.) On the D-flat locus Eq. (46.60) the meson matrix evaluates to $M^i{}_j = \widetilde{v}_i v_i\, \delta^i{}_j$, a diagonal matrix whose N_f entries are independent holomorphic combinations of the VEVs; the $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ flavor rotations then fill out the full matrix M.

Worked Example 46.11 (Dimension of the classical moduli space for $N_f < N_c$)**.** We count the complex dimension of the moduli space two ways and check they agree. First, the field-theoretic count. The squark fields $Q, \widetilde{Q}$ contain $2N_fN_c$ complex scalars. At a generic point of Eq. (46.60) the gauge group is Higgsed: N_f fundamental VEVs break $\mathrm{SU}(N_c) \to \mathrm{SU}(N_c - N_f)$, because the VEVs span an N_f-dimensional subspace of color and the unbroken subgroup is the $\mathrm{SU}(N_c - N_f)$ acting on the orthogonal complement. The number of broken generators is

$$\dim \mathrm{SU}(N_c) - \dim \mathrm{SU}(N_c - N_f) = (N_c^2 - 1) - \big((N_c - N_f)^2 - 1\big) = 2N_cN_f - N_f^2. \tag{46.62}$$

By the supersymmetric Higgs mechanism each broken generator eats one entire chiral superfield (a complex scalar) to build a massive vector multiplet. The complex dimension of the moduli space is therefore

$$\dim_{\mathbb{C}} \mathcal{M} = 2N_fN_c - (2N_cN_f - N_f^2) = N_f^2. \tag{46.63}$$

Second, the invariant count. The mesons $M^i{}_j$ form an arbitrary $N_f \times N_f$ complex matrix, hence N_f^2 complex parameters. For $N_f < N_c$ there are no further invariants (no baryons), and the map from D-flat squark configurations modulo gauge transformations to meson matrices is one-to-one on the generic locus. The two counts agree: $\dim_{\mathbb{C}} \mathcal{M} = N_f^2$, parametrized cleanly by the gauge-invariant meson matrix M. This is the statement that, classically, *the entire low-energy physics on the moduli space is the dynamics of the massless chiral superfield* M.

The conclusion of the classical analysis is that for $N_f < N_c$ the light degrees of freedom on the moduli space are exactly the N_f^2 mesons $M^i{}_j$, with a classical superpotential that vanishes identically. The classical vacua are exactly degenerate: nothing chooses a value of M. The question is whether the quantum dynamics lifts this degeneracy, and the answer is encoded in an effective superpotential $W_{\text{eff}}(M, \Lambda)$.

46.3.2 Holomorphy and symmetry fix the superpotential

We now determine $W_{\text{eff}}(M, \Lambda)$. The argument rests on three facts, each established earlier in the book, and a clean piece of complex analysis.

(i) Holomorphy. The effective superpotential is a holomorphic function of the light chiral superfield M and of the holomorphic coupling, the latter entering only through $\Lambda^{b_0} = \Lambda^{3N_c - N_f}$ (Eq. (46.28)). It cannot depend on $M^\dagger$, on $\bar{\Lambda}$, or on the gauge coupling except through the holomorphic combination Λ^{b_0}. This is the nonrenormalization statement of Theorem 45.20, sharpened: W is a section of a holomorphic line bundle over field space and coupling space, and the only holomorphic invariant data are M and Λ^{b_0}.

(ii) Flavor symmetry. W_{eff} must be a singlet of $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$. Under these the meson matrix transforms as $M \to U_L\, M\, U_R^\dagger$, and the only holomorphic flavor invariant one can build from a single matrix M is its determinant $\det M$ (and powers thereof; the trace $\mathrm{tr}\, M$ is not invariant under the full $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ because $U_L \neq U_R$). Hence W_{eff} depends on M only through $\det M$:

$$W_{\text{eff}} = W_{\text{eff}}\big(\det M,\ \Lambda^{b_0}\big). \tag{46.64}$$

(iii) Abelian selection rules. W_{eff} must carry the correct charges under $\mathrm{U}(1)_B$, $\mathrm{U}(1)_A$, and $\mathrm{U}(1)_R$, with Λ^{b_0} treated as the charged spurion of Table 46.1. A superpotential is the $\theta\theta$ (F-term) integrand, $\int \mathrm{d}^2\theta\, W$; since $\mathrm{d}^2\theta$ carries R-charge -2 (because θ has R-charge $+1$), invariance of the action $\int \mathrm{d}^2\theta\, W$ forces

$$R[W_{\text{eff}}] = 2, \tag{46.65}$$

while W_{eff} must be neutral under the non-R abelian symmetries $\mathrm{U}(1)_B$ and $\mathrm{U}(1)_A$.

Read off the charges of the two available building blocks. From Table 46.1, the meson $M = \widetilde{Q}Q$ has B-charge 0, axial charge $+2$, and R-charge $2(N_f - N_c)/N_f$. Its determinant, a product of N_f eigenvalues, carries N_f times these:

$$B[\det M] = 0, \qquad A[\det M] = 2N_f, \qquad R[\det M] = 2(N_f - N_c). \tag{46.66}$$

The scale carries $B[\Lambda^{b_0}] = 0$, $A[\Lambda^{b_0}] = 2N_f$, $R[\Lambda^{b_0}] = 0$ (last row of Table 46.1). Write the most general holomorphic monomial,

$$W_{\text{eff}} = C\,(\Lambda^{b_0})^{\,p}\,(\det M)^{\,q}, \tag{46.67}$$

with C a dimensionless number and p, q to be determined. Baryon number is automatically zero. The remaining two selection rules give two linear equations:

$$A: \quad 2N_f\, p + 2N_f\, q = 0 \qquad \Longrightarrow \qquad p = -q, \tag{46.68}$$

$$R: \quad 0 \cdot p + 2(N_f - N_c)\, q = 2 \qquad \Longrightarrow \qquad q = \frac{1}{N_f - N_c} = -\frac{1}{N_c - N_f}. \tag{46.69}$$

Thus $q = -1/(N_c - N_f)$ and $p = +1/(N_c - N_f)$, and Eq. (46.67) becomes

$$\boxed{W_{\text{ADS}} \;=\; C_{N_c,N_f} \left(\frac{\Lambda^{\,3N_c - N_f}}{\det M} \right)^{\frac{1}{N_c - N_f}}} \tag{46.70}$$

the Affleck–Dine–Seiberg superpotential. Both nontrivial structures the blueprint owes have now been *derived* rather than quoted: the exponent $1/(N_c - N_f)$ is forced by the R-charge equation Eq. (46.69), and the power $3N_c - N_f$ of Λ is forced jointly by the axial equation Eq. (46.68) (which set $p = -q$) and by the value $b_0 = 3N_c - N_f$ that defines the holomorphic scale in Eq. (46.54).

Remark 46.12 (Why a single term, and the dimensional check). Holomorphy plus the two abelian charges leave *exactly one* monomial: the linear system Eq. (46.68)–Eq. (46.69) has a unique solution for (p, q), so no second invariant term can be added with different powers, and no analytic function beyond the single monomial Eq. (46.70) is allowed. (A would-be additive constant, or a positive power of $\det M$, would carry the wrong R-charge; a function of the dimensionless ratio is excluded because that ratio, $\Lambda^{b_0}/\det M$ raised to any other power, fails the charge equations.) The result also passes a dimensional check that uses no symmetry at all. A superpotential has mass dimension 3. The scale Λ and the meson $M = \widetilde{Q}Q$ both have mass dimension: $[\Lambda] = 1$ and, since each squark scalar has dimension 1, $[M] = 2$, so $[\det M] = 2N_f$. Then

$$[W_{\text{ADS}}] = \frac{(3N_c - N_f)\cdot 1 - 2N_f}{N_c - N_f} = \frac{3N_c - 3N_f}{N_c - N_f} = 3, \tag{46.71}$$

exactly as required. Holomorphy, symmetry, and dimensions are mutually consistent and conspire to leave a single answer up to the pure number C_{N_c,N_f}.

What we have not fixed is the coefficient C_{N_c,N_f}, and crucially whether it is zero. Symmetry forbids everything except Eq. (46.70); it cannot tell us that the dynamics actually generates that term. For that we need an honest dynamical computation. The cleanest case is $N_f = N_c - 1$, where the exponent is unity and—as we now show—the gauge symmetry is completely broken at a generic point of the moduli space, so the theory is weakly coupled and a single instanton does the job.

46.3.3 The one-instanton computation for $N_f = N_c - 1$

For $N_f = N_c - 1$ the ADS superpotential Eq. (46.70) simplifies dramatically. The exponent is $1/(N_c - N_f) = 1$ and the power of the scale is $3N_c - N_f = 3N_c - (N_c - 1) = 2N_c + 1$, so

$$W_{\text{ADS}}\Big|_{N_f = N_c - 1} = C_{N_c,N_c-1}\,\frac{\Lambda^{2N_c+1}}{\det M}. \tag{46.72}$$

We will establish that $C_{N_c,N_c-1} \neq 0$ by a one-instanton calculation, identify the physical origin of every factor, and explain in what precise sense C is then a convention.

Worked Example 46.13 (A single instanton generates Eq. (46.72))**. Setup: complete Higgsing.** Take all $N_f = N_c - 1$ flavors to acquire a common large VEV, $\langle Q\rangle = \langle\widetilde{Q}\rangle^{\mathsf{T}} = v\,\mathbf{1}_{(N_c-1)\times(N_c-1)}$ embedded as in Eq. (46.60), so that $M^i{}_j = v^2\,\delta^i{}_j$ and $\det M = v^{2(N_c-1)}$. With $N_c - 1$ fundamental VEVs the gauge group is broken all the way,

$$\mathrm{SU}(N_c) \;\longrightarrow\; \mathrm{SU}(N_c - (N_c - 1)) = \mathrm{SU}(1) = \{\mathbf{1}\}, \tag{46.73}$$

i.e. *completely*: every gauge boson acquires a mass $\sim gv$. For $v \gg \Lambda$ the running coupling is frozen at the weak value $g(v) \ll 1$ before the theory can become strong, the instanton expansion parameter $e^{-8\pi^2/g^2(v)}$ is small and controlled, and the semiclassical computation of Section 42.3 is reliable. There is no strongly coupled subgroup left to worry about; the only non-perturbative effect is the instanton of the original $\mathrm{SU}(N_c)$, now living in a Higgsed background.

Fermion zero modes. In the one-instanton ($Q = 1$) background of Theorem 42.17 we count the fermionic zero modes of every field, using the index theorem Theorem 42.35. A Weyl fermion in a representation r has $2\,T(r)\,|Q|$ zero modes. Thus:

- the gaugino λ (adjoint, $T(\mathrm{adj}) = N_c$) contributes $2N_c$ zero modes;
- each quark fermion ψ_Q and $\psi_{\widetilde{Q}}$ (fundamental, $T(\mathbf{N_c}) = \frac{1}{2}$) contributes $2 \cdot \frac{1}{2} = 1$ zero mode, and there are N_f of each, hence $2N_f = 2(N_c - 1)$ quark zero modes in all.

The total fermionic zero-mode count is $2N_c + 2(N_c - 1) = 4N_c - 2$. For the Berezin integral over collective coordinates to be nonzero, *every* zero mode must be saturated—either by an external field insertion or by a vertex from the action that ties two of them together.

Saturating the gaugino modes with the Yukawa coupling. The $2N_c$ gaugino zero modes cannot all be left as external legs if we want the leading effect; the supersymmetric gauge-matter Yukawa coupling

$$\mathcal{L} \supset -\sqrt{2}\, g\, (\widetilde{Q}^\dagger\, \lambda\, \psi_{\widetilde{Q}} + Q^\dagger\, \lambda\, \psi_Q) + \mathrm{h.c.} \tag{46.74}$$

soaks them up. Each insertion of Eq. (46.74) converts one gaugino zero mode and one quark zero mode into a squark field $Q^\dagger$ or $\widetilde{Q}^\dagger$, evaluated on its VEV v. In the completely Higgsed vacuum the squark fields carry the VEV v, so each such vertex brings down a factor of v and pairs a λ-mode with a ψ-mode. Counting modes: of the $2N_c$ gaugino modes, all are removed by such vertices; of the $2(N_c - 1)$ quark modes, $2N_c - 2(N_c - 1) = 2$ remain after pairing with gauginos.[1] The two surviving quark zero modes, one from ψ_Q and one from $\psi_{\widetilde{Q}}$, are the external fermion legs of the generated operator; supersymmetry promotes the operator they sit in to a full superspace F-term, so the resulting contribution to the effective action is a superpotential term.

Assembling the powers. Collect the dimensionful factors of the one-instanton amplitude. The exponential and coupling prefactor of Eq. (42.143)

[1] The bookkeeping is the supersymmetric refinement of the 't Hooft vertex of Theorem 42.38: there, with no Yukawa coupling to the gaugino available in non-supersymmetric QCD, all quark zero modes appear as external legs and the gaugino modes are lifted by the gauge-boson mass; here the gauge-Yukawa Eq. (46.74) is precisely the operator that ties the gaugino modes to squark VEVs.

combine, via the one-loop running $e^{-8\pi^2/g^2(\mu)} = (\Lambda/\mu)^{b_0}$ with $b_0 = 2N_c + 1$ (the supersymmetric coefficient), into a factor Λ^{2N_c+1} once the powers of μ are traded for the running scale exactly as in Eq. (42.146). The squark VEV insertions that saturated the gaugino modes bring down powers of v. Rather than track the combinatoric constant (which feeds into C), we fix the v-power by holomorphy and dimensions, which is unambiguous. The generated superpotential must (a) be holomorphic in Λ^{2N_c+1} and in the mesons, (b) have dimension 3, and (c) reduce on this VEV to a function of $\det M = v^{2(N_c-1)}$. The only possibility consistent with the linear appearance of one instanton factor Λ^{2N_c+1} is

$$W_{\text{inst}} = C\,\frac{\Lambda^{2N_c+1}}{v^{2(N_c-1)}} = C\,\frac{\Lambda^{2N_c+1}}{\det M}, \tag{46.75}$$

which is exactly Eq. (46.72). The dimensional check is the same as Eq. (46.71): $[\Lambda^{2N_c+1}] = 2N_c + 1$, $[\det M] = 2(N_c - 1)$, difference $= 3$. The content of the explicit calculation is that the coefficient of Eq. (46.75) is *nonzero*: a careful evaluation of the instanton measure (the collective-coordinate Jacobians of Section 42.3.4, the regulated nonzero-mode determinant, and the zero-mode wavefunction normalizations) yields a finite, nonvanishing number. The dynamics genuinely generates the ADS term; it is not merely permitted by symmetry.

Why C is then a convention. The precise value of C_{N_c,N_c-1} depends on the renormalization scheme through the definition of Λ in Eq. (46.54): a finite redefinition $\Lambda \to c\,\Lambda$ rescales the prefactor, so different schemes (Pauli–Villars, $\overline{\text{MS}}$, the holomorphic scheme) assign different numbers. It is therefore conventional to *define* the scale Λ in the $N_f = N_c - 1$ theory so that

$$C_{N_c,N_c-1} = 1, \tag{46.76}$$

which we adopt. What is scheme-*independent*, and is the physical output of the computation, is that $C \neq 0$.

Physical Insight: gaugino condensation for $N_f < N_c - 1$

For $N_f < N_c - 1$ the exponent $1/(N_c - N_f)$ is no longer an integer, so the ADS superpotential Eq. (46.70) is *fractional* in Λ^{b_0} and cannot be generated by any number of ordinary instantons (whose contributions come in integer powers of $\Lambda^{b_0} = e^{2\pi i\tau}$). Its origin is instead *gaugino condensation*. At a generic point of the moduli space the gauge group is Higgsed $\text{SU}(N_c) \to \text{SU}(N_c - N_f)$ with all flavors massive; the surviving $\text{SU}(N_c - N_f)$ pure super-Yang–Mills theory has no light matter and confines, and its gauginos condense, $\langle\lambda\lambda\rangle \sim \Lambda_L^3$, where Λ_L is the low-energy scale of the unbroken group. Matching Λ_L to the high-energy data through the running across the squark thresholds reproduces precisely the fractional power $(\Lambda^{b_0}/\det M)^{1/(N_c-N_f)}$: the $(N_c - N_f)$-th root is the

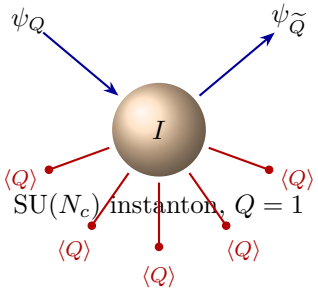


Figure 46.3: The one-instanton origin of the ADS superpotential for $N_f = N_c - 1$. The $Q = 1$ instanton carries $2N_c$ gaugino zero modes (red) and $2(N_c - 1)$ quark zero modes; the gauge–Yukawa vertex Eq. (46.74) ties each gaugino mode to a squark VEV $\langle Q\rangle = v$, leaving two external quark legs $\psi_Q, \psi_{\widetilde{Q}}$ (blue). Supersymmetry promotes the result to the F-term Eq. (46.72), with strength $\Lambda^{2N_c+1}/\det M$ and a nonzero coefficient.

imprint of the single gaugino condensate of the SU($N_c - N_f$) factor, whose strong scale is the $(N_c - N_f)$-th root of an instanton factor. Instantons ($N_f = N_c - 1$) and gaugino condensation ($N_f < N_c - 1$) are thus two faces of the same holomorphic object Eq. (46.70), distinguished only by whether the exponent is integer or fractional.

46.3.4 Reading off the physics: runaway, mass deformation, and gaugino condensation

The exact superpotential Eq. (46.70) is not an academic object; its derivatives determine the vacuum structure, and they tell a sharp and initially disturbing story.

No vacuum: the runaway. Supersymmetric vacua require the F-flatness conditions $\partial W/\partial M = 0$ together with D-flatness. Compute the F-term of the meson from Eq. (46.70), using $\partial(\det M)/\partial M^i{}_j = (\det M)(M^{-1})^j{}_i$:

$$\frac{\partial W_{\text{ADS}}}{\partial M^i{}_j} = -\frac{1}{N_c - N_f}\, W_{\text{ADS}}\, (M^{-1})^j{}_i. \tag{46.77}$$

Since $W_{\text{ADS}} \neq 0$ for any finite, nonsingular M, and M^{-1} never vanishes, the F-term is nonzero *everywhere on the moduli space*. There is no solution of $\partial W/\partial M = 0$ at finite M: the classical degeneracy is lifted, and lifted in the worst possible way—no supersymmetric vacuum survives at any finite field value.

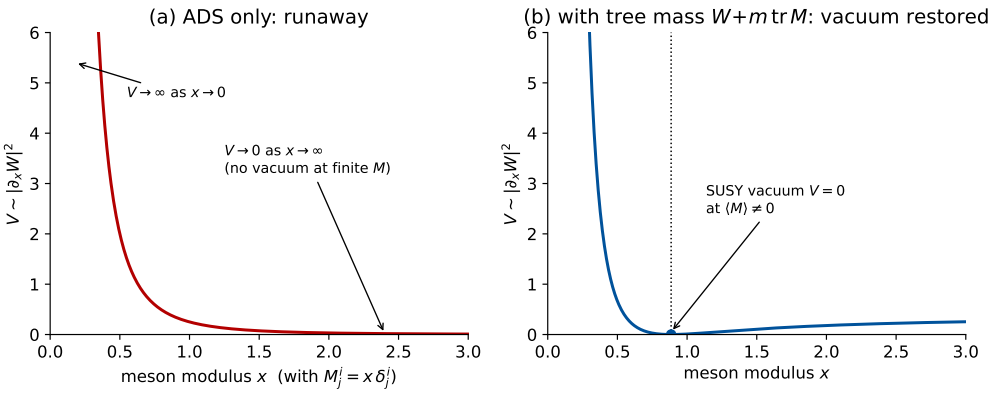


Figure 46.4: The Affleck–Dine–Seiberg scalar potential along the flavor-symmetric meson ray $M^i{}_j = x\,\delta^i{}_j$ (schematic normalization; the qualitative features are fixed by the F-flatness condition, which depends only on the holomorphic superpotential). **(a)** The pure ADS superpotential Eq. (46.70): the potential diverges at the origin and runs monotonically to zero at large field, with no minimum—there is no supersymmetric vacuum. **(b)** Adding a tree-level mass $W = W_{\mathrm{ADS}} + m\ \mathrm{tr}\,M$ stabilizes the runaway: the competition between the falling ADS term and the rising mass term produces an isolated minimum at $\langle M \rangle \neq 0$ where $V = 0$, a genuine supersymmetric vacuum.

The scalar potential, $V = K^{i\bar{\jmath}}\,\partial_i W\,\overline{\partial_j W}$ with K the (positive, nondegenerate, but not holomorphically protected) Kähler metric on the moduli space, is strictly positive and slopes monotonically. To see its shape, restrict to the flavor-symmetric ray $M^i{}_j = x\,\delta^i{}_j$ so that $\det M = x^{N_f}$ and $W_{\mathrm{ADS}} \propto x^{-N_f/(N_c-N_f)}$; then $V \propto |\partial_x W|^2 \propto x^{-2N_f/(N_c-N_f)-2}$, a function that *diverges as* $x \to 0$ *and decreases monotonically to zero as* $x \to \infty$. The potential pushes the meson VEV to infinity, where supersymmetry is restored only asymptotically and the theory becomes free; there is no stable ground state at any finite coupling. This is the celebrated ADS *runaway*, plotted in Fig. 46.4(a).

Restoring a vacuum by a mass deformation. The runaway is cured by adding a tree-level mass for the quarks, a holomorphic deformation that gives the meson a classical superpotential. Take a common mass m for all flavors, $W_{\mathrm{tree}} = m\,\widetilde{Q}Q = m\ \mathrm{tr}\,M$, so the total superpotential is

$$W = W_{\mathrm{ADS}} + m\ \mathrm{tr}\,M = C_{N_c,N_f}\left(\frac{\Lambda^{3N_c-N_f}}{\det M}\right)^{\frac{1}{N_c-N_f}} + m\ \mathrm{tr}\,M. \tag{46.78}$$

Now the falling ADS term competes with the rising mass term and the potential develops a minimum (Fig. 46.4(b)). The F-flatness condition $\partial W/\partial M^i{}_j = 0$

reads, using Eq. (46.77),

$$-\frac{1}{N_c-N_f}\,W_{\rm ADS}\,(M^{-1})^{j}{}_{i}+m\,\delta^{j}{}_{i}=0 \quad\Longrightarrow\quad M^{i}{}_{j}=\frac{W_{\rm ADS}}{m(N_c-N_f)}\,\delta^{i}{}_{j}, \tag{46.79}$$

so the meson VEV is flavor-diagonal, $\langle M^{i}{}_{j}\rangle = X\,\delta^{i}{}_{j}$ with X a single unknown. Substituting $\det M = X^{N_f}$ and $W_{\rm ADS} = C(\Lambda^{3N_c-N_f}/X^{N_f})^{1/(N_c-N_f)}$ back into Eq. (46.79) gives one algebraic equation for X,

$$X=\frac{C}{m(N_c-N_f)}\,\frac{\Lambda^{(3N_c-N_f)/(N_c-N_f)}}{X^{\,N_f/(N_c-N_f)}}, \tag{46.80}$$

where X enters linearly on the left because the F-flatness condition Eq. (46.79) fixes X itself. Multiplying through by $X^{\,N_f/(N_c-N_f)}$ and using $1+\frac{N_f}{N_c-N_f}=\frac{N_c}{N_c-N_f}$, this rearranges to $X^{\,N_c/(N_c-N_f)}\propto m^{-1}\Lambda^{(3N_c-N_f)/(N_c-N_f)}$, equivalently $X^{\,N_c}\propto m^{-(N_c-N_f)}\Lambda^{3N_c-N_f}$, i.e.

$$\langle M^{i}{}_{j}\rangle=\big(\det M\big)^{1/N_f}\,\delta^{i}{}_{j}, \qquad \langle\det M\rangle=\left(\frac{C}{m(N_c-N_f)}\right)^{\frac{N_f(N_c-N_f)}{N_c}}\Lambda^{\frac{N_f(3N_c-N_f)}{N_c}}, \tag{46.81}$$

after collecting the exponents (from $X^{N_c/(N_c-N_f)}\propto m^{-1}\Lambda^{(3N_c-N_f)/(N_c-N_f)}$ one finds $X\propto m^{-(N_c-N_f)/N_c}\Lambda^{(3N_c-N_f)/N_c}$, and $\det M = X^{N_f}$ raises both factors to the power N_f, giving the Λ power $\frac{N_f(3N_c-N_f)}{N_c}$ and the dimension $2N_f$). The deformation has produced a genuine, isolated supersymmetric vacuum at $\langle M\rangle\neq 0$. Crucially, extracting X from Eq. (46.80) requires taking an N_c-th root: the equation $X^{N_c}\propto m^{-(N_c-N_f)}\Lambda^{3N_c-N_f}$ has N_c solutions differing by the phases $e^{2\pi ik/N_c}$, $k=0,1,\dots,N_c-1$. There are therefore exactly

$$N_c \text{ supersymmetric vacua}, \tag{46.82}$$

in agreement with the Witten index of SU(N_c) super-Yang–Mills, $\mathrm{Tr}(-1)^F = N_c$ (the index whose nonzero-energy states pair off and cancel by the $n_B=n_F$ theorem Theorem 44.15, so that it counts only the zero-energy vacua — here Witten's count of N_c). This is a stringent consistency check: the count of vacua produced by holomorphy on the moduli space matches the index computed from the algebra, two completely independent routes to the same integer.

Integrating out: pure super-Yang–Mills and gaugino condensation. Sending $m\to\infty$ decouples all the flavors and leaves pure SU(N_c) super-Yang–Mills below the mass threshold—the $N_f=0$ theory. We can read its dynamics directly off Eq. (46.78) by integrating out M. At the vacuum Eq. (46.81) the value of the superpotential is, using $W_{\rm ADS}=m(N_c-N_f)X=m(N_c-N_f)\langle M\rangle$ from Eq. (46.79) and Eq. (46.82),

$$\langle W\rangle = W_{\rm ADS}+m\,\mathrm{tr}\,M=\big[(N_c-N_f)+N_f\big]\,m\,X=N_c\,m\,X. \tag{46.83}$$

Matching the low-energy scale of the pure SU(N_c) theory by the holomorphic decoupling relation $\Lambda_{\rm SYM}^{3N_c} = m^{N_f}\,\Lambda^{3N_c-N_f}$ (the product of the heavy masses times the high-scale instanton factor, fixed by continuity of the holomorphic coupling across the threshold), Eq. (46.83) collapses to the pure-gauge effective superpotential

$$\boxed{W_{\rm SYM} = N_c\,\Lambda_{\rm SYM}^3\,e^{2\pi i k/N_c}, \qquad k = 0, 1, \ldots, N_c - 1.} \tag{46.84}$$

This is the Veneziano–Yankielowicz superpotential of pure super-Yang–Mills, with its N_c branches—the N_c vacua of Eq. (46.82), here visible as the N_c-th roots of $\Lambda_{\rm SYM}^{3N_c}$. Its physical content is the gaugino condensate. Identifying the gaugino bilinear with the lowest component of the field-strength superfield through $W_{\rm SYM} = \langle\lambda\lambda\rangle$ up to a constant (the field strength $W^\alpha W_\alpha$ has $\lambda\lambda$ as its scalar component, and $\Lambda_{\rm SYM}^3$ is its expectation value), we read off

$$\langle\lambda\lambda\rangle \;\sim\; \Lambda_{\rm SYM}^3\,e^{2\pi i k/N_c}, \tag{46.85}$$

the order parameter for the spontaneous breaking of the discrete R-symmetry $\mathbb{Z}_{2N_c} \to \mathbb{Z}_2$ that the anomaly leaves unbroken: the $2N_c$-fold degeneracy of the phase collapses to N_c inequivalent values. The dynamical scale Λ, which entered as a bookkeeping device for the one-loop running, has reappeared as the magnitude of a genuine, computable condensate.

Physical Insight: the SUSY-calculable face of the chiral condensate

Equation Eq. (46.85) is the supersymmetric analogue of the quark condensate $\langle\bar{q}q\rangle \sim \Lambda_{\rm QCD}^3$ that underlies confinement and chiral symmetry breaking in ordinary QCD (Yang–Mills theory; the Wilson-loop area law of Theorem 27.20 and the chiral dynamics of Chapter 39). In non-supersymmetric QCD the condensate is a number we can only estimate or measure on the lattice; here, because holomorphy protects the superpotential and the symmetries fix it completely, the analogous condensate is computed *exactly*, scale and all, with the N_c-fold vacuum degeneracy following from the same index that counts the super-Yang–Mills ground states. This is the sense in which supersymmetry is a controlled laboratory for confinement-scale physics: the strong-coupling condensate that we can only model in QCD is, in its supersymmetric cousin, a theorem.

We have now extracted the complete non-perturbative dynamics of SU(N_c) SQCD with $N_f < N_c$ flavors: a classical moduli space of mesons, an exactly determined ADS superpotential generated by instantons ($N_f = N_c - 1$) or gaugino condensation ($N_f < N_c - 1$), a runaway with no stable vacuum, and—upon a mass deformation—exactly N_c supersymmetric vacua tied to the gaugino condensate. The natural next question writes itself: what happens as N_f is

increased toward and past N_c? At $N_f = N_c - 1$ the instanton was already on the boundary of integer exponents; at $N_f = N_c$ and beyond the baryons of Eq. (46.61) enter the chiral ring, the runaway disappears, and the moduli space is modified quantum-mechanically rather than lifted. Carrying the N_f-dependent moduli-space data and the ADS result of this section into that regime is exactly what the next section does, where the increasing flavor count reveals the phase structure of SQCD and Seiberg's electric–magnetic duality.

46.4 The Phase Structure of SQCD and Seiberg Duality

The previous section delivered a sharp piece of strong-coupling physics for free. For SU(N_c) supersymmetric QCD with $N_f < N_c$ flavors, holomorphy and the global symmetries forced a unique non-perturbative effective superpotential on the classical moduli space of mesons, the Affleck–Dine–Seiberg superpotential

$$W_{\text{ADS}} \;=\; (N_c - N_f)\left(\frac{\Lambda^{3N_c - N_f}}{\det M}\right)^{\frac{1}{N_c - N_f}}, \tag{46.86}$$

generated by a single instanton when $N_f = N_c - 1$ and by gaugino condensation in the unbroken SU($N_c - N_f$) when $N_f < N_c - 1$ (Section 46.3). Its gradient never vanishes: the meson fields are pushed to infinity, the potential slopes monotonically to zero there, and the theory has *no stable vacuum*. That is the whole story for too few flavors. The natural question is what happens as N_f is increased. The runaway must end somewhere, new light degrees of freedom must appear, and the qualitative infrared physics — whether the theory confines, breaks its chiral symmetries, flows to an interacting scale -invariant theory, or becomes free — must change. This section maps that N_f-dependence completely. The map turns out to be one of the most striking results in four-dimensional field theory: across a wide window the strongly coupled electric theory is exactly equivalent in the infrared to a *different*, weakly coupled magnetic gauge theory. This is Seiberg's electric–magnetic duality, and assembling the evidence for it is the climax of the section.

Throughout we use the conventions fixed in Chapter 45 and Section 46.3. The metric is mostly-minus, the gauge covariant derivative is $D_\mu = \partial_\mu - igA^a_\mu T^a$ with Hermitian generators normalized by $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the fundamental, and the dynamical scale is Λ, defined through the one-loop running with coefficient $b_0 = 3N_c - N_f$ so that $\Lambda^{b_0} = \mu^{b_0} e^{-8\pi^2/g^2(\mu) + i\theta}$ is the holomorphic instanton factor. We measure the qualitative physics by the single ratio N_f/N_c, summarized in advance in Fig. 46.5.

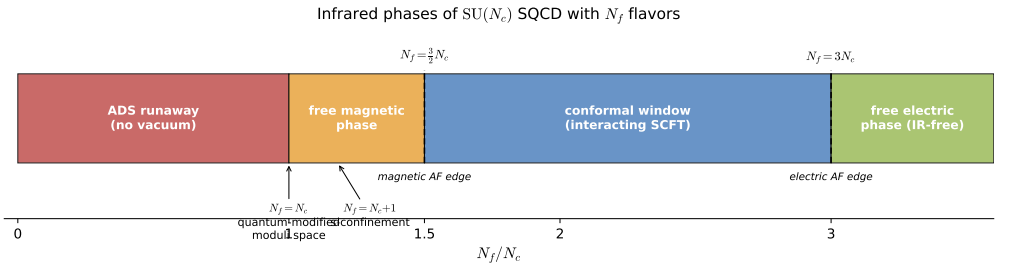


Figure 46.5: The infrared phases of SU(N_c) SQCD as a function of N_f/N_c, the central result of this section. For $N_f < N_c$ the Affleck–Dine–Seiberg superpotential (46.86) drives a runaway with no vacuum. At $N_f = N_c$ the moduli space is quantum-mechanically deformed; at $N_f = N_c + 1$ the theory s-confines with a smooth superpotential. The two edges of the conformal window, $N_f = \frac{3}{2}N_c$ and $N_f = 3N_c$, are exactly the points where the *magnetic* and *electric* gauge couplings respectively sit at the boundary of asymptotic freedom — a coincidence that is itself evidence for the duality.

46.4.1 The fields, the scale, and the global symmetries

We recall the matter content precisely, because every exact statement below is an exercise in tracking its quantum numbers. The theory is SU(N_c) gauge theory coupled to N_f chiral superfields Q^i ($i = 1, \ldots, N_f$) in the fundamental representation $\mathbf{N_c}$ and N_f chiral superfields $\widetilde{Q}_{\tilde{\jmath}}$ ($\tilde{\jmath} = 1, \ldots, N_f$) in the antifundamental $\overline{\mathbf{N_c}}$. We write the color index explicitly only when needed, Q^i_a and $\widetilde{Q}^a_{\tilde{\jmath}}$ with $a = 1, \ldots, N_c$. There is no tree-level superpotential. The classical global symmetry acting on the matter, after removing the part that overlaps with the gauge group, is

$$G_{\rm cl} \;=\; \mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B \times \mathrm{U}(1)_A \times \mathrm{U}(1)_R, \tag{46.87}$$

with Q a fundamental of SU$(N_f)_L$ and $\widetilde{Q}$ an antifundamental of SU$(N_f)_R$. Two of the abelian factors are anomalous in combination: the axial U$(1)_A$ under which Q and $\widetilde{Q}$ both carry charge $+1$ has a mixed U$(1)_A$-SU$(N_c)^2$ triangle, and so does a naive R-symmetry. Exactly one non-anomalous linear combination of these survives as the genuine quantum R-symmetry U$(1)_R$; the orthogonal anomalous combination remains useful as a *spurion*, the bookkeeping device that tracks how Λ^{b_0} transforms.

Fixing U$(1)_R$ is a one-line anomaly computation, and it pins down a charge we will use repeatedly. Under a candidate U$(1)_R$, the gauginos λ in the vector multiplet carry R-charge $+1$ (they sit one Grassmann level above the gauge field, which is R-neutral), and we assign the chiral superfields $Q, \widetilde{Q}$ a common R-charge R_Q, so their fermionic components carry $R_Q - 1$. The mixed U$(1)_R$-SU$(N_c)^2$ anomaly is the sum of R-charge $\times$ (Dynkin index) over all fermions charged under

	$\mathrm{SU}(N_c)$	$\mathrm{SU}(N_f)_L$	$\mathrm{SU}(N_f)_R$	$\mathrm{U}(1)_B$	$\mathrm{U}(1)_R$
Q	$\mathbf{N_c}$	$\mathbf{N_f}$	$\mathbf{1}$	$+1$	$\frac{N_f - N_c}{N_f}$
$\widetilde{Q}$	$\overline{\mathbf{N_c}}$	$\mathbf{1}$	$\overline{\mathbf{N_f}}$	-1	$\frac{N_f - N_c}{N_f}$
$M = \widetilde{Q}Q$	$\mathbf{1}$	$\mathbf{N_f}$	$\overline{\mathbf{N_f}}$	0	$\frac{2(N_f - N_c)}{N_f}$
B	$\mathbf{1}$	$\Lambda^{N_c}\mathbf{N_f}$	$\mathbf{1}$	N_c	$\frac{N_c(N_f - N_c)}{N_f}$

Table 46.2: Non-anomalous global charges of the elementary squarks and of the basic gauge-invariant composites of $\mathrm{SU}(N_c)$ SQCD. The baryon B sits in the rank-N_c antisymmetric power $\Lambda^{N_c}\mathbf{N_f}$ of $\mathrm{SU}(N_f)_L$; the antibaryon $\widetilde{B}$ is its mirror with B-number $-N_c$. The R-charges follow from (46.89) by additivity.

the gauge group,

$$\underbrace{(+1)\,T(\mathrm{adj})}_{\text{gaugino}} + \underbrace{2N_f\,(R_Q-1)\,T(\mathbf{N_c})}_{Q \text{ and } \widetilde{Q}} = N_c + 2N_f\,(R_Q-1)\,\tfrac{1}{2} = N_c + N_f(R_Q-1), \tag{46.88}$$

where we used $T(\mathrm{adj}) = N_c$ and $T(\mathbf{N_c}) = \frac{1}{2}$ from Chapter 2 (Appendix B), and the factor $2N_f$ counts the N_f flavors of Q plus the N_f flavors of $\widetilde{Q}$. Demanding that this vanish gives the unique non-anomalous R-charge of the squarks,

$$\boxed{R_Q = \frac{N_f - N_c}{N_f}} \tag{46.89}$$

We collect the resulting non-anomalous charges in Table 46.2. Every result in this section is consistent with these assignments, and the reader should treat the table as the master key.

46.4.2 Gauge invariants: the chiral ring

The infrared physics on the moduli space is carried by the gauge-invariant holomorphic combinations of $Q, \widetilde{Q}$ — the generators of the *chiral ring*. There are two kinds. The mesons

$$M^i{}_{\tilde{\jmath}} = \widetilde{Q}^a_{\tilde{\jmath}}\, Q^i_a \qquad \text{(color contracted)}, \tag{46.90}$$

form an $N_f \times N_f$ matrix in flavor space, a bifundamental of $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ with $B = 0$. The baryons antisymmetrize N_c squarks on their color indices,

$$B^{i_1\cdots i_{N_c}} = \frac{1}{N_c!}\,\epsilon^{a_1\cdots a_{N_c}}\, Q^{i_1}_{a_1}\cdots Q^{i_{N_c}}_{a_{N_c}}, \qquad \widetilde{B}_{\tilde{\jmath}_1\cdots\tilde{\jmath}_{N_c}} = \frac{1}{N_c!}\,\epsilon_{a_1\cdots a_{N_c}}\,\widetilde{Q}^{a_1}_{\tilde{\jmath}_1}\cdots\widetilde{Q}^{a_{N_c}}_{\tilde{\jmath}_{N_c}}, \tag{46.91}$$

totally antisymmetric in their N_c flavor indices and hence transforming in the rank-N_c antisymmetric representation $\Lambda^{N_c}\mathbf{N_f}$ of the relevant $\mathrm{SU}(N_f)$ factor, with baryon number $\pm N_c$. For $N_f < N_c$ no such antisymmetrization is possible and only mesons exist; for $N_f = N_c$ there is a single baryon B and a single antibaryon $\widetilde{B}$ (the flavor indices fill completely); for $N_f = N_c + 1$ the baryons carry one free flavor index, $B^i \equiv \epsilon^{ij_1\cdots j_{N_c}} B_{j_1\cdots j_{N_c}}$, an antifundamental of $\mathrm{SU}(N_f)_L$. These multiplicity changes are exactly what makes the phase structure depend so sharply on $N_f - N_c$.

46.4.3 $N_f = N_c$: the quantum-modified moduli space

With exactly $N_f = N_c$ flavors the would-be ADS superpotential (46.86) formally has $N_c - N_f = 0$ in the exponent and ceases to exist: there is no dynamically generated superpotential, the would-be runaway is gone, and the theory has a genuine moduli space of supersymmetric vacua. But the moduli space is not the classical one. To see this we first fix the classical geometry, then determine its exact quantum form by holomorphy and symmetry.

The classical constraint. At $N_f = N_c$ the squark Q^i_a is a square $N_c \times N_c$ matrix in (flavor)×(color), as is $\widetilde{Q}^a_{\tilde{\jmath}}$. The meson matrix is the product $M^i{}_{\tilde{\jmath}} = Q^i_a \widetilde{Q}^a_{\tilde{\jmath}}$, while the single baryon and antibaryon are the determinants

$$B = \det_{(\text{flavor,color})} Q, \qquad \widetilde{B} = \det_{(\text{color,flavor})} \widetilde{Q}, \tag{46.92}$$

which follow from (46.91) once all N_c flavor indices are saturated. The multiplicativity of the determinant then gives an algebraic identity among the invariants,

$$\det M \;=\; \det(Q\widetilde{Q}) \;=\; \det Q \,\det \widetilde{Q} \;=\; B\,\widetilde{B}, \tag{46.93}$$

the *classical constraint* $\det M - B\widetilde{B} = 0$. It cuts out the classical moduli space inside the space of gauge invariants, and in particular the origin $M = B = \widetilde{B} = 0$ is a (singular) point of it, where the full gauge symmetry is restored and the global symmetry G_{cl} is unbroken.

The quantum constraint. The exact statement is that quantum mechanics deforms (46.93) to

$$\boxed{\det M \;-\; B\,\widetilde{B} \;=\; \Lambda^{2N_c}} \tag{46.94}$$

This single equation is derived, not assumed, from holomorphy plus the global symmetries plus the weak-coupling limit, in the same spirit as the ADS derivation. We carry the argument out in full.

Worked Example 46.14 (Deriving the quantum-modified constraint)**.** The deformation of the classical relation can only be a holomorphic function of the chiral invariants $M, B, \widetilde{B}$ and of the holomorphic scale, which at $N_f = N_c$ is $\Lambda^{b_0} = \Lambda^{2N_c}$ (since $b_0 = 3N_c - N_f = 2N_c$). We promote the classical relation to a candidate constraint enforced by a Lagrange-multiplier chiral superfield X,

$$W_{\text{eff}} \;=\; X\,(\det M - B\widetilde{B} - f(\Lambda^{2N_c})), \tag{46.95}$$

and determine the holomorphic function f.

Step 1 — charges of the building blocks. We track the two abelian symmetries that survive as exact selection rules: the non-anomalous $\mathrm{U}(1)_R$ and the anomalous axial $\mathrm{U}(1)_A$ treated as a spurion. Under $\mathrm{U}(1)_A$ with $A(Q) = A(\widetilde{Q}) = +1$, the path-integral measure is not invariant; the $\mathrm{SU}(N_c)$ instanton with $2N_f = 2N_c$ fermion zero modes shifts the effective θ-angle, which means the holomorphic scale carries axial charge

$$A(\Lambda^{2N_c}) \;=\; 2N_f \;=\; 2N_c, \tag{46.96}$$

exactly the number of zero modes the one-instanton vertex must soak up. Now read off the charges of the invariants. The meson $M = \widetilde{Q}Q$ has $A(M) = 2$, so $A(\det M) = N_f \cdot 2 = 2N_c$. The baryons are products of N_c squarks, so $A(B) = A(\widetilde{B}) = N_c$ and $A(B\widetilde{B}) = 2N_c$. Under the non-anomalous $\mathrm{U}(1)_R$, Table 46.2 gives, at $N_f = N_c$, $R(M) = 2(N_f - N_c)/N_f = 0$ and $R(B) = R(\widetilde{B}) = N_c(N_f - N_c)/N_f = 0$, hence $R(\det M) = R(B\widetilde{B}) = 0$, while Λ^{2N_c} is R-neutral by construction.

Step 2 — what the symmetries allow. The three terms $\det M$, $B\widetilde{B}$, and Λ^{2N_c} all carry $A = 2N_c$ and $R = 0$, and all are singlets of $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \times \mathrm{U}(1)_B$. They are therefore interchangeable as far as every selection rule is concerned, and the only holomorphic, symmetry-respecting completion of the classical relation is

$$\det M - B\widetilde{B} \;=\; c\,\Lambda^{2N_c}, \qquad c \text{ a pure number.} \tag{46.97}$$

No higher power of Λ^{2N_c} can appear, because $(\Lambda^{2N_c})^n$ carries $A = 2N_c n$ and there is no invariant of matching axial charge to balance it for $n \neq 1$; this is the same holomorphy-plus-spurion logic that fixed the power of Λ in the ADS superpotential.

Step 3 — fixing the constant and its physical meaning. As $\Lambda \to 0$ the gauge dynamics switches off and (46.97) must reduce to the classical (46.93); this is automatic for any finite c. To show $c \neq 0$ — the crucial physical point — give a large vacuum expectation value to one meson eigenvalue, $\langle M^{N_c}{}_{N_c} \rangle = v^2 \gg \Lambda^2$, breaking $\mathrm{SU}(N_c) \to \mathrm{SU}(N_c - 1)$ and Higgsing one flavor. The low-energy theory is $\mathrm{SU}(N_c - 1)$ with $N_f - 1 = N_c - 1$ flavors, i.e. again the $N_f = N_c$ case one rank down, with a matched scale $\Lambda_{\mathrm{L}}^{2(N_c-1)} = \Lambda^{2N_c}/v^2$ fixed by continuity of the running coupling across the threshold v. Consistency of (46.97) between the two

descriptions forces the *same* nonzero c at every rank, so c is a rank-independent constant; with the standard definition of Λ it is $c = 1$. The cleanest direct check is the one-instanton computation in the fully Higgsed theory $N_f = N_c$, $\langle Q \rangle = \langle \widetilde{Q} \rangle = v\,\mathbf{1}$: the instanton has exactly the right number of zero modes to generate the constant Λ^{2N_c} on the right of (46.94) with unit coefficient. This is the $N_f = N_c$ analogue of the $N_f = N_c - 1$ instanton that generated the ADS superpotential in Section 46.3.

Physical consequences. The deformation (46.94) is small in its algebra and enormous in its physics. The origin $M = B = \widetilde{B} = 0$ *no longer solves the constraint*: $0 - 0 \neq \Lambda^{2N_c}$. The singular point of the classical moduli space, where unbroken color and unbroken global symmetry coexisted, has been excised. At every point of the quantum moduli space, some of the global symmetry G_{cl} is spontaneously broken. Two extreme vacua illustrate the pattern.

At the *mesonic* point $B = \widetilde{B} = 0$, $M^i{}_{\tilde{\jmath}} = \Lambda^2 \delta^i_{\tilde{\jmath}}$ (so $\det M = \Lambda^{2N_c}$), the meson vev locks the two flavor groups together,

$$\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R \;\longrightarrow\; \mathrm{SU}(N_f)_{\text{diag}}, \tag{46.98}$$

exactly the chiral-symmetry-breaking pattern of ordinary QCD. The Goldstone bosons of this breaking are described by Section 29.3, now appearing inside an exactly solvable theory. At the *baryonic* point $M = 0$, $B = -\widetilde{B} = \Lambda^{N_c}$, the baryon vev breaks $\mathrm{U}(1)_B$ while preserving the full nonabelian flavor symmetry. In between, a continuous family of vacua interpolates. In all cases the low-energy spectrum consists of the gauge-singlet composites $M, B, \widetilde{B}$ subject to (46.94) and their superpartners: there are no massless gluons, the theory *confines*, and it does so with chiral symmetry breaking. That the massless fermionic composites correctly reproduce all the global 't Hooft anomalies of the microscopic quarks at each of these vacua — the consistency demanded by Theorem 37.26 — is the check that originally established this picture; we carry out the analogous computation in detail for the duality below.

Physical Insight

This is the book's confinement criterion made calculable. In Chapter 27 an area law for the Wilson loop, Theorem 27.20, was the operational definition of confinement, established at strong coupling on the lattice but inaccessible to weak-coupling methods. Here supersymmetry lets us follow a gauge theory all the way to its infrared and *prove* that color is confined — the spectrum is gauge-singlet mesons and baryons, full stop — and that chiral symmetry breaks, all without summing a single diagram. The price is that this particular theory has massless moduli rather than a mass gap; the pure super-Yang–Mills case $N_f = 0$, by contrast, has gaugino

condensation and a genuine gap, the closest supersymmetric cousin of real QCD confinement.

46.4.4 $N_f = N_c + 1$: s-confinement

Add one more flavor. Now the constraint is not deformed; instead the classical moduli space survives intact — the origin included — and the infrared is described by a *smooth* effective superpotential of the confined composites, with no constraint imposed by hand. This is the phenomenon Seiberg named *s-confinement*: confinement with a smooth description and an unbroken chiral symmetry at the origin.

The composites are the meson $M^i{}_{\tilde{j}}$ and the now-flavored baryons B^i, $\widetilde{B}_{\tilde{j}}$ ($i, \tilde{j} = 1, \ldots, N_f = N_c + 1$), each an (anti)fundamental of a flavor group. Holomorphy and symmetry fix their dynamics. With $b_0 = 3N_c - N_f = 2N_c - 1$ the scale is Λ^{2N_c-1}, and the unique superpotential built from $M, B, \widetilde{B}$ that is a flavor singlet, has baryon number zero, carries R-charge 2 (required of any superpotential, since $\mathrm{d}^2\theta$ carries $R = -2$), and reduces correctly in all limits is

$$\boxed{W \;=\; \frac{1}{\Lambda^{2N_c-1}}\Big(B^i\, M_i{}^{\tilde{j}}\, \widetilde{B}_{\tilde{j}} \;-\; \det M\Big)} \tag{46.99}$$

We verify the charge bookkeeping, since it is the entire content of the claim. At $N_f = N_c + 1$, Table 46.2 gives $R(M) = 2(N_f - N_c)/N_f = 2/(N_c + 1)$ and $R(B) = R(\widetilde{B}) = N_c(N_f - N_c)/N_f = N_c/(N_c + 1)$. Hence

$$R(\det M) = N_f\, R(M) = (N_c{+}1)\cdot\frac{2}{N_c+1} = 2, \quad R(BM\widetilde{B}) = \frac{N_c}{N_c+1}+\frac{2}{N_c+1}+\frac{N_c}{N_c+1} = \frac{2}{N} \tag{46.100}$$

so both terms carry $R = 2$, as a superpotential must. Both are $\mathrm{SU}(N_f)_L \times \mathrm{SU}(N_f)_R$ singlets ($BM\widetilde{B}$ contracts all flavor indices, $\det M$ is the invariant determinant) and have B-number $N_c - N_c = 0$ for the first term and 0 for $\det M$. The axial spurion charges match too: $A(\det M) = 2N_f$, while $A(BM\widetilde{B}) = N_c + 2 + N_c = 2N_c + 2 = 2N_f$, and $A(\Lambda^{2N_c-1}) = 2N_f$ from the one-instanton zero-mode count, so the prefactor $\Lambda^{-(2N_c-1)}$ exactly cancels the matter axial charge and leaves W with $A = 0$ as required. Holomorphy plus these selection rules leave (46.99) as the only possibility up to the overall normalization, which the deformation to $N_f = N_c$ fixes.

The superpotential (46.99) is not a constraint — it is a genuine interaction of the light composites — and its equations of motion reproduce the classical geometry as the supersymmetric vacuum condition. Varying with respect to M,

B, and $\widetilde{B}$,

$$\frac{\partial W}{\partial M_i{}^{\tilde{\jmath}}} = 0: \;\; B^i \widetilde{B}_{\tilde{\jmath}} = (\mathrm{cof}\, M)^i{}_{\tilde{\jmath}}, \qquad \frac{\partial W}{\partial B^i} = 0: \;\; M_i{}^{\tilde{\jmath}} \widetilde{B}_{\tilde{\jmath}} = 0, \qquad \frac{\partial W}{\partial \widetilde{B}_{\tilde{\jmath}}} = 0: \;\; B^i M_i{}^{\tilde{\jmath}} = 0, \tag{46.101}$$

where $(\mathrm{cof}\, M)^i{}_{\tilde{\jmath}}$ is the matrix of cofactors, $M\,(\mathrm{cof}\, M)^{\mathsf{T}} = (\det M)\,\mathbf{1}$. These are precisely the classical relations among mesons and baryons at $N_f = N_c + 1$ (a rank-N_c meson whose cofactor equals $B\widetilde{B}$, with baryons in the kernel of M). The moduli space is the classical one, the origin lies on it, and there the entire global symmetry G_{cl} is unbroken. The theory confines *without* breaking chiral symmetry, and the massless composites $M, B, \widetilde{B}$ at the origin must — and do — match every 't Hooft anomaly of the underlying quarks and gluons (Theorem 37.26). This anomaly match at the origin, with elementary quarks on one side and confined mesons and baryons on the other, is one of the cleanest confirmations of confinement in any four-dimensional theory.

46.4.5 Seiberg duality: the precise statement

Increase N_f further and a new organizing principle takes over. For $N_f > N_c + 1$ the composite description proliferates — more and more baryonic invariants, no smooth superpotential — and the right infrared variables are not the mesons and baryons of the electric theory at all. They are the fields of a *different* gauge theory.

Theorem 46.15 (Seiberg's electric–magnetic duality)**.** *For $N_f > N_c + 1$, the* $\mathrm{SU}(N_c)$ *gauge theory with N_f flavors $Q, \widetilde{Q}$ and no superpotential (the* electric *theory) flows in the infrared to the same fixed point as the following* magnetic *theory: an* $\mathrm{SU}(\widetilde{N}_c)$ *gauge theory with*

$$\widetilde{N}_c \;=\; N_f - N_c \tag{46.102}$$

colors, N_f magnetic quarks $q, \widetilde{q}$ in its (anti)fundamental, together with a gauge-singlet meson superfield $M^i{}_{\tilde{\jmath}}$ that is an $N_f \times N_f$ matrix, coupled through the cubic superpotential

$$\boxed{W_{\mathrm{mag}} \;=\; \frac{1}{\mu}\, M^i{}_{\tilde{\jmath}}\, \widetilde{q}_i\, q^{\tilde{\jmath}}} \tag{46.103}$$

with μ a scale relating the dimension-one singlet M to the dimension-two electric meson $\widetilde{Q}Q$ via the operator map $\widetilde{Q}Q = \mu M$. The two theories share the same global symmetry G_{cl} and the same chiral ring in the infrared.

Three features deserve emphasis at once. First, the duality is an *infrared* equivalence: the two theories are different in the ultraviolet (different gauge groups, different field content) and become indistinguishable only at long distances, where they describe the same physics. Second, the magnetic gauge group

	$\mathrm{SU}(\widetilde{N}_c)$	$\mathrm{SU}(N_f)_L$	$\mathrm{SU}(N_f)_R$	$\mathrm{U}(1)_B$	$\mathrm{U}(1)_R$
q	$\mathbf{\widetilde{N}_c}$	$\overline{\mathbf{N_f}}$	$\mathbf{1}$	$+\frac{N_c}{\widetilde{N}_c}$	$\frac{N_c}{N_f}$
$\widetilde{q}$	$\overline{\mathbf{\widetilde{N}_c}}$	$\mathbf{1}$	$\mathbf{N_f}$	$-\frac{N_c}{\widetilde{N}_c}$	$\frac{N_c}{N_f}$
M	$\mathbf{1}$	$\mathbf{N_f}$	$\overline{\mathbf{N_f}}$	0	$\frac{2(N_f-N_c)}{N_f}$

Table 46.3: Global charges of the magnetic theory of Theorem 46.15. The dual quark R-charge follows directly from the requirement $R(W_{\mathrm{mag}}) = R(M) + R(q) + R(\widetilde{q}) = 2$ with $R(M) = 2(N_f - N_c)/N_f$, giving $R(q) = R(\widetilde{q}) = N_c/N_f$. The dual quark sits in the conjugate flavor representation to the electric quark, so that the meson M and the composite $\widetilde{q}q$ carry identical flavor quantum numbers.

$\mathrm{SU}(N_f - N_c)$ shrinks as N_c grows toward N_f: when $N_f = N_c + 1$ it is $\mathrm{SU}(1)$, i.e. trivial, and the magnetic "theory" is just the meson and baryons with a superpotential — precisely the s-confining description (46.99), with the magnetic quarks identified as the baryons. The duality thus contains the s-confinement result as its boundary case. Third, the singlet M is indispensable: without it the cubic superpotential (46.103) could not be written, and, as we now show, the global anomalies would not match.

Caution

Seiberg duality is, as of the mid-2020s, a conjecture — an extraordinarily well-tested one, but not a theorem. There is no derivation of it from first principles in four dimensions; what we marshal below is *evidence*, in the strict sense of necessary consistency conditions that the conjecture passes non-trivially and that would each have killed it had any failed. The strongest evidence is exact (anomaly matching, deformation consistency); some is structural (embedding in string constructions, large-N_c checks); some is numerical (lattice studies of the conformal window). We are careful to label which is which.

The magnetic charges are fixed by the requirements that the global symmetries agree and that (46.103) be invariant. We record them in Table 46.3; the reader can verify each entry from the constraints $\widetilde{Q}Q = \mu M$ (so $R(M)$ is the electric meson R-charge, $B(M) = 0$), $R(W_{\mathrm{mag}}) = 2$, and the matching of baryon number between electric and magnetic baryons (electric $B \sim Q^{N_c}$ versus magnetic $b \sim q^{\widetilde{N}_c}$ must carry the same $\mathrm{U}(1)_B$ charge, forcing $B(q) = N_c/\widetilde{N}_c$).

46.4.6 Evidence I: 't Hooft anomaly matching

The first and sharpest test is exact. The global symmetry $G_{\rm cl}$ is the same in both descriptions, it is unbroken at the origin of moduli space (the common infrared fixed point), and its 't Hooft anomalies are renormalization -group invariant (Theorem 37.26): the triangle anomaly of any three global currents, computed on the ultraviolet fermions, must equal the same triangle computed on whatever massless fermions survive in the infrared. If the electric and magnetic theories flow to the same fixed point, they must reproduce the *same* set of global anomalies. There are several independent anomaly coefficients — $\mathrm{SU}(N_f)^3$, $\mathrm{SU}(N_f)^2\,\mathrm{U}(1)_B$, $\mathrm{SU}(N_f)^2\,\mathrm{U}(1)_R$, $\mathrm{U}(1)_B^2\,\mathrm{U}(1)_R$, $\mathrm{U}(1)_R$, $\mathrm{U}(1)_R^3$ — and all of them match. We compute one in complete detail, the mixed $\mathrm{SU}(N_f)_L^2\,\mathrm{U}(1)_R$ anomaly, because it is the one in which the singlet meson plays the decisive role; we then do a second, the pure $\mathrm{U}(1)_R$–gravitational anomaly, as a worked example, and quote the rest.

The $\mathrm{SU}(N_f)_L^2\,\mathrm{U}(1)_R$ anomaly. This coefficient is the sum, over all left-handed (Weyl) fermions charged under $\mathrm{SU}(N_f)_L$, of

$$\mathcal{A}\big[\mathrm{SU}(N_f)_L^2\,\mathrm{U}(1)_R\big] \;=\; \sum_{\text{fermions}} R_\psi\, T(r_\psi)\, d_\psi, \tag{46.104}$$

where R_ψ is the R-charge of the fermion (one less than that of its superfield), $T(r_\psi)$ the $\mathrm{SU}(N_f)_L$ Dynkin index of its flavor representation, and d_ψ the multiplicity from all *other* (color, flavor) labels it carries.

Electric side. The only fields charged under $\mathrm{SU}(N_f)_L$ are the quarks Q (the antiquarks $\widetilde{Q}$ are $\mathrm{SU}(N_f)_L$ singlets, and the gauginos are flavor singlets). The fermion in Q has R-charge $R_Q - 1 = (N_f - N_c)/N_f - 1 = -N_c/N_f$, sits in the fundamental of $\mathrm{SU}(N_f)_L$ with $T(\mathbf{N_f}) = \frac{1}{2}$, and comes in N_c colors. Hence

$$\mathcal{A}_{\rm el} \;=\; \underbrace{\Big(-\frac{N_c}{N_f}\Big)}_{R_\psi}\cdot\underbrace{\frac{1}{2}}_{T}\cdot\underbrace{N_c}_{d_\psi} \;=\; -\frac{N_c^2}{2N_f}. \tag{46.105}$$

Magnetic side. Two kinds of fermion are charged under $\mathrm{SU}(N_f)_L$: the dual quark q and the meson fermion ψ_M. The dual quark q has R-charge $R_q - 1 = N_c/N_f - 1 = -\widetilde{N}_c/N_f$, sits in the antifundamental of $\mathrm{SU}(N_f)_L$ with the same index $T(\overline{\mathbf{N_f}}) = \frac{1}{2}$, and comes in $\widetilde{N}_c$ magnetic colors. The meson M is an $\mathrm{SU}(N_f)_L$ fundamental (index i) with R-charge $R_M - 1 = 2(N_f - N_c)/N_f - 1 = (N_f - 2N_c)/N_f$, and its multiplicity from the second flavor index $\tilde{\jmath}$ is N_f. Therefore

$$\mathcal{A}_{\rm mag} = \underbrace{\Big(-\frac{\widetilde{N}_c}{N_f}\Big)\cdot\frac{1}{2}\cdot\widetilde{N}_c}_{\text{dual quark } q} + \underbrace{\Big(\frac{N_f - 2N_c}{N_f}\Big)\cdot\frac{1}{2}\cdot N_f}_{\text{meson } \psi_M}$$

$$= -\frac{\widetilde{N}_c^{\,2}}{2N_f} + \frac{N_f - 2N_c}{2} \;=\; \frac{-(N_f - N_c)^2 + N_f(N_f - 2N_c)}{2N_f}, \tag{46.106}$$

where in the last step we used $\widetilde{N}_c = N_f - N_c$ and put everything over $2N_f$. Expanding the numerator,

$$-(N_f - N_c)^2 + N_f(N_f - 2N_c) = -N_f^2 + 2N_fN_c - N_c^2 + N_f^2 - 2N_fN_c = -N_c^2, \tag{46.107}$$

so that

$$\mathcal{A}_{\text{mag}} \;=\; -\frac{N_c^2}{2N_f} \;=\; \mathcal{A}_{\text{el}}. \tag{46.108}$$

The two descriptions agree. The match is not trivial term by term: on the electric side the entire coefficient comes from N_c quarks, while on the magnetic side it is a difference between $\widetilde{N}_c$ dual quarks and the N_f^2-component meson, and the cancellation that leaves exactly $-N_c^2$ requires the meson's R-charge to be precisely $R_M - 1 = (N_f - 2N_c)/N_f$. Drop the singlet M from the magnetic theory and the anomaly fails by the meson term — which is the algebraic reason Theorem 46.15 cannot do without it.

Worked Example 46.16 (The $\mathrm{U}(1)_R$–gravitational anomaly matches)**.** The mixed R–gravitational anomaly is the simplest global anomaly to state: it is the sum of the R-charges of *all* massless Weyl fermions, weighted only by their multiplicities (gravity couples universally, so there is no group-theory index),

$$\mathcal{A}[\mathrm{U}(1)_R] \;=\; \sum_{\text{fermions}} R_\psi \,\dim(r_\psi). \tag{46.109}$$

Electric side. Three contributions: the gauginos, the quark fermions, and the antiquark fermions.

- Gauginos λ: $R = +1$, multiplicity $\dim(\text{adj}) = N_c^2 - 1$.
- Quark fermions ψ_Q: $R = R_Q - 1 = -N_c/N_f$, multiplicity N_cN_f (colors $\times$ flavors).
- Antiquark fermions $\psi_{\widetilde{Q}}$: same $R = -N_c/N_f$, multiplicity N_cN_f.

Summing,

$$\mathcal{A}_{\text{el}}[\mathrm{U}(1)_R] = (N_c^2 - 1)(+1) + 2\,N_cN_f\Big(-\frac{N_c}{N_f}\Big) = N_c^2 - 1 - 2N_c^2 = -N_c^2 - 1. \tag{46.110}$$

Magnetic side. Four contributions: the magnetic gauginos, the dual quark and antiquark fermions, and the meson fermions.

- Magnetic gauginos: $R = +1$, multiplicity $\widetilde{N}_c^{\,2} - 1$.

- Dual quark/antiquark fermions: $R = R_q - 1 = -\widetilde{N}_c/N_f$, multiplicity $\widetilde{N}_c N_f$ each, so two copies.
- Meson fermions ψ_M: $R = R_M - 1 = (N_f - 2N_c)/N_f$, multiplicity N_f^2.

Summing,

$$\begin{aligned}\mathcal{A}_{\text{mag}}[\mathrm{U}(1)_R] &= (\widetilde{N}_c^{\,2} - 1) + 2\,\widetilde{N}_c N_f\Big(-\frac{\widetilde{N}_c}{N_f}\Big) + N_f^2 \cdot \frac{N_f - 2N_c}{N_f} \\ &= \widetilde{N}_c^{\,2} - 1 - 2\widetilde{N}_c^{\,2} + N_f(N_f - 2N_c) \;=\; -\widetilde{N}_c^{\,2} - 1 + N_f(N_f - 2N_c).\end{aligned} \tag{46.111}$$

With $\widetilde{N}_c = N_f - N_c$,

$$-\widetilde{N}_c^{\,2} + N_f(N_f - 2N_c) = -(N_f - N_c)^2 + N_f^2 - 2N_f N_c = -N_c^2, \tag{46.112}$$

exactly as in (46.107), so $\mathcal{A}_{\text{mag}}[\mathrm{U}(1)_R] = -N_c^2 - 1 = \mathcal{A}_{\text{el}}[\mathrm{U}(1)_R]$. Again the balance is delicate: the magnetic gaugino contributes the analogue of the electric gauge contribution but with the *wrong* rank $\widetilde{N}_c^{\,2}$, and it is the meson's N_f^2-fold multiplicity, weighted by its specific R-charge, that repairs the difference to leave precisely $-N_c^2 - 1$. The universal "-1" — the lone gaugino over and above the matter — survives because $\dim(\text{adj}) = N_{\text{color}}^2 - 1$ on both sides and the color-squared pieces are exactly what the meson cancels.

The remaining coefficients are matched by the same bookkeeping. The $\mathrm{SU}(N_f)_L^2\,\mathrm{U}(1)_B$ anomaly receives $N_c \cdot \frac{1}{2} \cdot (+1) = N_c/2$ from the electric quarks; on the magnetic side the dual quark gives $\widetilde{N}_c \cdot \frac{1}{2} \cdot (N_c/\widetilde{N}_c) = N_c/2$ and the meson (with $B = 0$) does not contribute, so it matches. The pure $\mathrm{SU}(N_f)_L^3$ anomaly measures the net number of fundamentals minus antifundamentals charged under $\mathrm{SU}(N_f)_L$: electrically N_c fundamentals (the quark colors); magnetically N_c from the meson fundamental minus $\widetilde{N}_c$ antifundamentals from the dual quark, i.e. $N_c - \widetilde{N}_c + \ldots$, which again reduces to the electric value once the meson is included. Every coefficient matches, and only with the singlet meson present. This is the principal exact evidence for Theorem 46.15.

46.4.7 The conformal window and the unitarity bound

We can now read the phase diagram, Fig. 46.5, by asking, for each N_f, which of the two dual descriptions is weakly coupled in the infrared. The question is settled by the one-loop coefficient $b_0 = 3N_c - N_f$ in the electric theory and its magnetic counterpart $\widetilde{b}_0 = 3\widetilde{N}_c - N_f = 3(N_f - N_c) - N_f = 2N_f - 3N_c$. Asymptotic freedom requires a positive coefficient; a non-positive one means the gauge coupling is irrelevant and the theory is infrared-free.

The fixed point and the exact R-charge. For supersymmetric gauge theory the exact (NSVZ) β-function for the holomorphic coupling can be written

$$\beta(g) = -\frac{g^3}{16\pi^2}\,\frac{3N_c - N_f(1-\gamma(g))}{1 - N_c g^2/8\pi^2}, \tag{46.113}$$

where $\gamma(g)$ is the anomalous dimension of the quark superfield, common to Q and $\widetilde{Q}$ by the flavor symmetry; this is the exact statement of the nonrenormalization structure argued in Chapter 45. A nontrivial infrared fixed point $g_* \neq 0$ exists when the numerator vanishes,

$$3N_c - N_f(1-\gamma_*) = 0 \quad\Longrightarrow\quad \gamma_* = 1 - \frac{3N_c}{N_f}. \tag{46.114}$$

At such a fixed point the theory is a (super)conformal field theory, and the superconformal algebra ties the exact scaling dimension of a chiral primary operator $\mathcal{O}$ to its R-charge,

$$\Delta(\mathcal{O}) = \tfrac{3}{2}\,R(\mathcal{O}) \qquad \text{(chiral primary)}. \tag{46.115}$$

The relation (46.115) is forced by positivity. In the $\mathcal{N}=1$ superconformal algebra the dilatation D and R-charge sit in the anticommutator of a supercharge with its conformal partner, schematically $\{\bar{S}, \bar{Q}\} \sim D - \frac{3}{2}R$ acting on a primary; a chiral operator is annihilated by $\bar{Q}$ and a superconformal primary is annihilated by $\bar{S}$, so sandwiching the anticommutator in the state $\mathcal{O}|0\rangle$ and using $\|\bar{S}\,\mathcal{O}|0\rangle\|^2 \geq 0$, $\|\bar{Q}\,\mathcal{O}|0\rangle\|^2 \geq 0$ gives the bound $\Delta \geq \frac{3}{2}R$, saturated precisely for a chiral primary. (We take the superconformal representation theory from Chapter 47; the upshot, that a chiral primary has $\Delta = \frac{3}{2}R$ exactly, is all we use.) Applied to the meson $M = \widetilde{Q}Q$, whose R-charge is $R_M = 2(N_f - N_c)/N_f$,

$$\Delta(M) = \frac{3}{2}\cdot\frac{2(N_f - N_c)}{N_f} = \frac{3(N_f - N_c)}{N_f} = 3\Big(1 - \frac{N_c}{N_f}\Big). \tag{46.116}$$

The upper edge $N_f = 3N_c$. When $N_f \geq 3N_c$, $b_0 \leq 0$: the electric coupling is irrelevant, the theory is infrared-free, and the quarks and gluons of the electric description are themselves the correct, free, long -distance degrees of freedom. This is the *free electric phase*, the supersymmetric analogue of infrared-free QED. Just below $3N_c$, the fixed-point coupling g_* is parametrically small (a Banks–Zaks fixed point, controlled by the two-loop term), $\gamma_* = 1 - 3N_c/N_f$ is small and negative, and the interacting superconformal fixed point is under perturbative control. As N_f decreases, g_* grows and the fixed point becomes strongly coupled.

The lower edge $N_f = \frac{3}{2}N_c$ from unitarity. The interacting fixed point cannot persist to arbitrarily small N_f. Its undoing is a *unitarity bound*. In

any unitary conformal field theory in four dimensions, a gauge-invariant scalar primary operator has dimension at least that of a free field,

$$\Delta \geq 1, \tag{46.117}$$

with equality only for a free (decoupled) field obeying $\Box \mathcal{O} = 0$. This is the scalar case of the general unitarity bound proved from reflection positivity in Chapter 47; physically it states that a genuinely interacting scalar cannot have dimension below the free value without producing a negative-norm descendant. Apply it to the meson, using (46.116):

$$\Delta(M) = 3\Big(1 - \frac{N_c}{N_f}\Big) \geq 1 \quad \Longleftrightarrow \quad 3N_f - 3N_c \geq N_f \quad \Longleftrightarrow \quad N_f \geq \frac{3N_c}{2}. \tag{46.118}$$

So the interacting superconformal fixed point with the meson as an interacting operator can exist only for $N_f \geq \frac{3}{2}N_c$. The boundary $N_f = \frac{3}{2}N_c$ is where $\Delta(M) = 1$ exactly: the meson hits the unitarity bound and becomes a free field. This is not an artifact — it is the signal that for $N_f < \frac{3}{2}N_c$ the operator M has decoupled as a free field and the interacting description has reorganized itself. The reorganization is exactly Seiberg duality: below $\frac{3}{2}N_c$ the *magnetic* coefficient $\widetilde{b}_0 = 2N_f - 3N_c$ has gone negative, the magnetic gauge theory is infrared-free, and the correct long-distance physics is the free magnetic theory of Theorem 46.15 — $\mathrm{SU}(N_f - N_c)$ gauge fields, dual quarks, and the now-free meson, all weakly coupled. This is the *free magnetic phase*, $N_c + 1 < N_f \leq \frac{3}{2}N_c$.

Physical Insight

The two edges of the conformal window are the same statement seen from the two sides of the duality. The window is

$$\frac{3}{2}N_c < N_f < 3N_c, \tag{46.119}$$

and its upper edge $N_f = 3N_c$ is where the *electric* coupling stops being asymptotically free ($b_0 = 0$), while its lower edge $N_f = \frac{3}{2}N_c$ is where the *magnetic* coupling stops being asymptotically free ($\widetilde{b}_0 = 3\widetilde{N}_c - N_f = 2N_f - 3N_c = 0$). Inside the window both descriptions are asymptotically free in the ultraviolet and both flow to the same interacting superconformal fixed point in the infrared; one is the dual of the other. Outside, on each side, exactly one description goes free and takes over. That the unitarity bound on the electric meson (46.118) reproduces the loss of magnetic asymptotic freedom to the digit is a stringent, independent check of Theorem 46.15.

46.4.8 Evidence II: chiral ring, deformations, and the a-theorem

Anomaly matching tests the theory at one point, the origin. The second body of evidence tests its deformations — the statement that as we perturb the electric theory and the magnetic theory by corresponding operators, they continue to agree.

The chiral ring matches. The gauge-invariant chiral operators of the electric theory are the mesons and baryons of Section 46.4.2. In the magnetic theory the singlet M is the meson directly, $\widetilde{Q}Q = \mu M$, while the electric baryon $B \sim Q^{N_c}$ maps to the magnetic baryon built from $\widetilde{N}_c = N_f - N_c$ dual quarks, $b \sim q^{\widetilde{N}_c}$; the ϵ-tensor of $\mathrm{SU}(\widetilde{N}_c)$ has exactly $\widetilde{N}_c$ indices, so $q^{\widetilde{N}_c}$ is the unique baryonic invariant, and it carries the same flavor representation $\Lambda^{N_c}\mathbf{N_f} \cong \Lambda^{\widetilde{N}_c}\overline{\mathbf{N_f}}$ and the same baryon number N_c as the electric baryon. The chiral rings are isomorphic.

Mass deformations map consistently. The decisive structural test is that adding a mass for one electric flavor maps to a controlled symmetry breaking in the magnetic theory, and that the duality of the daughter theories is again Seiberg's. Give the N_f-th electric flavor a mass,

$$\delta W_{\mathrm{el}} = m\, M^{N_f}{}_{N_f} = m\, \widetilde{Q}_{N_f} Q^{N_f}. \tag{46.120}$$

Integrating out the massive flavor leaves the electric $\mathrm{SU}(N_c)$ theory with $N_f - 1$ flavors. On the magnetic side the same operator is, by $\widetilde{Q}Q = \mu M$, a linear term $m\mu\, M^{N_f}{}_{N_f}$ added to (46.103):

$$W_{\mathrm{mag}} + \delta W = \frac{1}{\mu} M^{i}{}_{\tilde{\jmath}} \widetilde{q}_i q^{\tilde{\jmath}} + m\mu\, M^{N_f}{}_{N_f}. \tag{46.121}$$

The equation of motion for the singlet $M^{N_f}{}_{N_f}$ now reads $\frac{1}{\mu}\widetilde{q}_{N_f} q^{N_f} + m\mu = 0$, i.e.

$$\widetilde{q}_{N_f} q^{N_f} = -m\mu^2 \neq 0, \tag{46.122}$$

forcing a vacuum expectation value for the N_f-th magnetic quark. A dual-quark vev Higgses the magnetic gauge group by one rank, $\mathrm{SU}(\widetilde{N}_c) \to \mathrm{SU}(\widetilde{N}_c - 1)$, and removes one flavor. The daughter magnetic theory is therefore $\mathrm{SU}(\widetilde{N}_c - 1)$ with $N_f - 1$ flavors. For this to be the Seiberg dual of the daughter electric theory $\mathrm{SU}(N_c)$ with $N_f - 1$ flavors, we need the magnetic rank to be $(N_f - 1) - N_c$. Indeed

$$\widetilde{N}_c - 1 = (N_f - N_c) - 1 = (N_f - 1) - N_c, \tag{46.123}$$

exactly as required. A mass on the electric side maps to a Higgsing on the magnetic side, and the dualities of the deformed theories close consistently. This is a non-trivial, infinitely-iterable check: starting anywhere in the window and

adding masses one at a time walks both descriptions down through the phases — conformal window, free magnetic, s-confinement, quantum-modified constraint, ADS runaway — with the two sides agreeing at every step. By the same token, a vacuum expectation value (Higgsing) on the electric side maps to a mass on the magnetic side, the mirror operation; and dualizing the magnetic theory a second time, $\mathrm{SU}(\widetilde{N}_c) \to \mathrm{SU}(N_f - \widetilde{N}_c) = \mathrm{SU}(N_c)$, returns the electric theory, the "meson of the meson" becoming the original quark bilinear through the cubic superpotentials. Seiberg duality is an involution.

Consistency with the a-theorem. A final, more recent check uses the four-dimensional a-anomaly — the coefficient of the Euler density in the trace anomaly on a curved background. The a-theorem (Komargodski–Schwimmer, 2011) states that $a_{\mathrm{UV}} > a_{\mathrm{IR}}$ along any renormalization -group flow, generalizing to four dimensions the two-dimensional c-theorem. For a superconformal fixed point, a is computed exactly from the 't Hooft anomalies of the superconformal R-symmetry,

$$a = \frac{3}{32}(3\ \mathrm{Tr}\, R^3 - \mathrm{Tr}\, R), \tag{46.124}$$

with the correct R-symmetry inside the conformal window selected by a-maximization (Intriligator–Wecht, 2003). Because the electric and magnetic theories flow to the *same* fixed point, they must yield the same a; (46.124) evaluated on the two sets of anomaly coefficients we computed above does indeed agree, by the same identities (46.107) that made the triangle anomalies match. And along the flow from the asymptotically free ultraviolet into the window, a decreases monotonically, consistent with the a-theorem. We state these as consistency checks at the level of results — the full machinery of the a-anomaly and a-maximization belongs to the conformal-field-theory part of the book (Chapter 47 and Part XIV) — but the message is that every independent handle we have on the infrared, exact anomalies, operator dimensions, and the monotone a, points the same way.

This completes the map. Reading Fig. 46.5 from left to right: $N_f < N_c$ has no vacuum (ADS runaway); $N_f = N_c$ confines on a quantum-modified moduli space with chiral symmetry breaking; $N_f = N_c + 1$ s-confines with a smooth superpotential and unbroken chiral symmetry; $N_c + 1 < N_f \le \frac{3}{2}N_c$ is the free magnetic phase; $\frac{3}{2}N_c < N_f < 3N_c$ is the conformal window with an interacting superconformal fixed point described equally well by the electric or the magnetic theory; and $N_f \ge 3N_c$ is the free electric phase. The whole of it is organized by holomorphy and by the electric–magnetic duality of Theorem 46.15.

We have arrived at the conceptual payoff that drives the rest of the chapter: a strongly coupled gauge theory can be *exactly equivalent* to a weakly coupled one built on a different gauge group, with the heavy composites of one appearing as the elementary quanta of the other. In SQCD this electric–magnetic exchange is a statement about an entire infrared fixed point. The next section takes the

same idea down to the level of individual states. Extending the algebra to $\mathcal{N} = 2$ supersymmetry, we will find short multiplets whose masses are fixed exactly by a central charge, $M = |Z|$, and we will see that the central charge counts electric and magnetic charges on an equal footing. That is the algebraic seed of the Seiberg–Witten solution, in which a magnetically charged object — a monopole — becomes light and condenses, turning the abstract electric–magnetic duality of this section into the concrete mechanism of confinement.

46.5 BPS States and Central Charges

The duality of the previous section is a statement about descriptions: the same physics admits an electric presentation, in which the elementary quanta are weakly coupled, and a magnetic one, in which the solitons are. For such a statement to have content we need quantities that can be computed in one description and trusted in the other — quantities that do not drift as the coupling is dialled from weak to strong. Generic masses are useless here: they receive corrections at every order and beyond. What we want is a class of states whose masses are fixed by the symmetry algebra itself, not by the dynamics, and are therefore the same number whether evaluated in the electric or the magnetic frame. Supersymmetry supplies exactly such a class. The mechanism is the *central charge*: a relaxation of the supersymmetry algebra, licensed by Haag–Łopuszański–Sohnius for extended supersymmetry, that ties the mass of certain states to a charge by an exact inequality, saturated on a distinguished short multiplet. This section builds that structure from the algebra, proves the resulting Bogomol'nyi–Prasad–Sommerfield bound by the same sum-of-squares positivity that gave $H \geq 0$ in Section 44.3.1, and identifies the central charge with the electric and magnetic charges of Section 41.6.4 — turning the classical Bogomol'nyi bound of Chapter 41 into an exact quantum mass formula. That formula is the spectral data the Seiberg–Witten solution will compute.

46.5.1 The centrally extended algebra

The $\mathcal{N} = 1$ algebra of Eqs. (44.61) and (44.62) has $\{Q_\alpha, Q_\beta\} = 0$: the like-chirality anticommutator vanishes. We saw in Section 44.4.4 that this vanishing is not a theorem of supersymmetry but an accident of having only one supercharge. The graded-Lie-algebra closure of Theorem 44.5 requires only that $\{Q_\alpha, Q_\beta\}$ be a *bosonic* operator that commutes with the momenta and transforms in the symmetric part of $(\tfrac{1}{2}, 0) \otimes (\tfrac{1}{2}, 0)$. The symmetric product decomposes as $(\tfrac{1}{2}, 0) \otimes (\tfrac{1}{2}, 0) = (1, 0) \oplus (0, 0)$, whose symmetric piece is the Lorentz scalar $(0, 0)$, carried by the antisymmetric tensor $\epsilon_{\alpha\beta}$. A scalar that commutes with all of Poincaré and with the internal symmetry is, by definition, a *central* element. The Haag–Łopuszański–Sohnius theorem (Theorem 44.6) states that this is the

most general allowed extension: the right-hand side of the like-chirality bracket may be a central charge,

$$\left\{Q^I_\alpha, Q^J_\beta\right\} \;=\; 2\,\epsilon_{\alpha\beta}\,Z^{IJ}, \qquad Z^{IJ} = -Z^{JI}, \tag{46.125}$$

exactly the relation Eq. (44.114) flagged there. The matrix Z^{IJ} is antisymmetric in the internal indices $I, J = 1, \ldots, \mathcal{N}$, which is why it has no components for $\mathcal{N} = 1$ (a single antisymmetric 1×1 matrix is zero) and first appears for $\mathcal{N} \geq 2$. It commutes with every generator of the superalgebra — with P_μ and $M_{\mu\nu}$ because it is a Poincaré scalar, and with the supercharges because, being central, it sits in the centre by construction; on any irreducible representation Schur's lemma (Theorem 2.23) then forces it to a c-number.

For the remainder of this section we work with $\mathcal{N} = 2$, the case relevant to the Seiberg–Witten solution. There are two supercharges Q^I_α $(I = 1, 2)$ and their conjugates $\bar{Q}_{I\,\dot\alpha} \equiv (Q^I_\alpha)^\dagger$. The antisymmetric 2×2 matrix Z^{IJ} has a single independent component, $Z^{IJ} = \epsilon^{IJ} Z$ with Z a single complex number ($\epsilon^{12} = -\epsilon^{21} = 1$). The complete $\mathcal{N} = 2$ super-Poincaré algebra is then

$$\left\{Q^I_\alpha, \bar{Q}_{J\,\dot\beta}\right\} = 2\,\sigma^\mu_{\alpha\dot\beta}\,P_\mu\,\delta^I_J, \tag{46.126}$$

$$\left\{Q^I_\alpha, Q^J_\beta\right\} = 2\,\epsilon_{\alpha\beta}\,\epsilon^{IJ}\,Z, \tag{46.127}$$

$$\left\{\bar{Q}_{I\,\dot\alpha}, \bar{Q}_{J\,\dot\beta}\right\} = 2\,\epsilon_{\dot\alpha\dot\beta}\,\epsilon_{IJ}\,\bar{Z}, \tag{46.128}$$

the last line being the Hermitian conjugate of the second. The first line is just Eq. (44.61) tensored with the internal identity δ^I_J: each supercharge still squares to the momentum. The novelty is entirely in Eq. (46.127): two supercharges of opposite internal index now anticommute not to zero but to the central charge Z.

Notation Watch

The same symbol Z denoted the complex combination $v(q + ig)$ of electric and magnetic charges in the classical analysis of Eq. (41.189). This is deliberate, not a clash: the central charge of Eq. (46.127) *is* that classical object promoted to an operator, as Section 46.5.4 establishes. Until that identification is made we treat Z as an abstract complex c-number sitting in the algebra.

A word on $\mathcal{N} = 1$. The point-particle algebra Eq. (44.62) has no room for a central charge, but the bracket Eq. (46.125) can be nonzero in $\mathcal{N} = 1$ when the charges Q_α are computed on field configurations with nontrivial behaviour at spatial infinity — domain walls and other extended objects. The surface term that obstructs the naive vanishing is precisely the topological charge ΔW that bounded the kink mass in Theorem 41.5, and the kink relation $M = |\Delta W|$ of

Theorem 41.6 is, read in this light, an $\mathcal{N}=1$ BPS bound. The mechanism below is identical; we phrase it for $\mathcal{N}=2$ because that is where it controls a four-dimensional particle spectrum.

46.5.2 The BPS bound from positivity

We now prove that the mass of any state is bounded below by the modulus of the central charge. The argument is the positivity argument of Section 44.3.1 run one level deeper: there we wrote H as a sum of squares of supercharges and concluded $H \geq 0$; here we form cleverer linear combinations of supercharges whose squared norms isolate the combination $M - |Z|$.

Work in the rest frame of a massive state, $P^\mu = (M,0,0,0)$ with $M>0$. As in Eq. (44.109) the right-hand side of Eq. (46.126) collapses, since $\sigma^\mu_{\alpha\dot\beta} P_\mu = M\,\sigma^0_{\alpha\dot\beta} = M\,\delta_{\alpha\dot\beta}$. Writing all brackets in terms of the four supercharges Q^I_α and their adjoints $(Q^I_\alpha)^\dagger$, and identifying the numerical values of dotted and undotted indices, the rest-frame algebra reads

$$\left\{Q^I_\alpha, (Q^J_\beta)^\dagger\right\} = 2M\,\delta_{\alpha\beta}\,\delta^{IJ}, \tag{46.129}$$

$$\left\{Q^I_\alpha, Q^J_\beta\right\} = 2\,\epsilon_{\alpha\beta}\,\epsilon^{IJ}\,Z. \tag{46.130}$$

The first relation says each supercharge has unit-normalised anticommutator with its own adjoint (the diagonal entries are $2M$) and anticommutes with the adjoint of any *different* supercharge. The second relation is nonzero only when both the spinor indices and the internal indices differ; with $\epsilon_{12}=\epsilon^{12}=1$ its only independent nonzero components are

$$\left\{Q^1_1, Q^2_2\right\} = 2Z, \qquad \left\{Q^1_2, Q^2_1\right\} = -2Z. \tag{46.131}$$

The supercharges thus split into two decoupled quartets, $\{Q^1_1, Q^2_2\}$ and $\{Q^1_2, Q^2_1\}$, each closing on M and on Z but not mixing with the other. Treat the first quartet. Write the central charge in polar form $Z = |Z|\,e^{i\delta}$ and define the two fermionic combinations

$$a \;=\; \tfrac{1}{\sqrt{2}}\Big(Q^1_1 + e^{i\delta}\,(Q^2_2)^\dagger\Big), \qquad b \;=\; \tfrac{1}{\sqrt{2}}\Big(Q^1_1 - e^{i\delta}\,(Q^2_2)^\dagger\Big), \tag{46.132}$$

the phase $e^{i\delta}$ chosen to align Z with the real axis in the computation that follows. These are genuine annihilation/creation-type operators because they are linear in the Q's; we compute their anticommutators with their adjoints. For a, using $a^\dagger = \frac{1}{\sqrt{2}}((Q^1_1)^\dagger + e^{-i\delta}Q^2_2)$ and the brackets Eqs. (46.129) and (46.131),

$$\left\{a, a^\dagger\right\} = \tfrac{1}{2}\Big(\left\{Q^1_1, (Q^1_1)^\dagger\right\} + e^{-i\delta}\left\{Q^1_1, Q^2_2\right\} + e^{i\delta}\left\{(Q^2_2)^\dagger, (Q^1_1)^\dagger\right\} + \left\{(Q^2_2)^\dagger, Q^2_2\right\}\Big)$$

$$= \tfrac{1}{2}\Big(2M + e^{-i\delta}\, 2Z + e^{i\delta}\, 2\bar{Z} + 2M\Big) = 2M + \big(e^{-i\delta} Z + e^{i\delta}\bar{Z}\big) = 2M + 2\,|Z|\,, \tag{46.133}$$

where in the last step $e^{-i\delta}Z = |Z|$ and its conjugate is again $|Z|$. The cross term $\Big\{(Q_2^2)^\dagger, (Q_1^1)^\dagger\Big\} = \big(\{Q_1^1, Q_2^2\}\big)^\dagger = 2\bar{Z}$ was used. An identical computation for b, which differs only in the sign of the off-diagonal piece, gives

$$\Big\{b, b^\dagger\Big\} = 2M - 2\,|Z|\,. \tag{46.134}$$

One checks directly from Eqs. (46.129) and (46.131) that $\Big\{a, b^\dagger\Big\} = \tfrac{1}{2}(2M - 2M) = 0$ and $\{a, b\} = \tfrac{1}{2}\big(\{Q_1^1, Q_1^1\} - e^{2i\delta}\Big\{(Q_2^2)^\dagger, (Q_2^2)^\dagger\Big\}\big) = 0$, so a and b are independent oscillators. The same construction applied to the second quartet $\{Q_2^1, Q_1^2\}$ — now with the opposite sign in Eq. (46.131), compensated by reversing the phase in the definitions — yields two more independent oscillators c, d with $\Big\{c, c^\dagger\Big\} = 2M + 2\,|Z|$ and $\Big\{d, d^\dagger\Big\} = 2M - 2\,|Z|$.

The bound is now immediate. The anticommutator $\Big\{b, b^\dagger\Big\} = bb^\dagger + b^\dagger b$ is, on any state $|\psi\rangle$, the sum of two squared norms:

$$\Big\langle \psi \Big| \Big\{b, b^\dagger\Big\} \Big| \psi \Big\rangle = \Big\| b^\dagger\,|\psi\rangle \Big\|^2 + \|b\,|\psi\rangle\|^2 \geq 0. \tag{46.135}$$

Combining with Eq. (46.134) and dividing by the positive norm $\langle\psi|\psi\rangle$,

$$2M - 2\,|Z| \geq 0 \qquad \Longrightarrow \qquad M \geq |Z|\,. \tag{46.136}$$

Theorem 46.17 (BPS bound)**.** *In any unitary representation of the $\mathcal{N} = 2$ super-Poincaré algebra Eqs. (46.126) to (46.128) on a Hilbert space with positive-definite inner product, every massive state satisfies*

$$M \;\geq\; |Z|\,, \tag{46.137}$$

with Z the central-charge eigenvalue on that state. For general $\mathcal{N}$ the same argument applied block by block to the skew eigenvalues Z_n of Z^{IJ} gives $M \geq \max_n |Z_n|$, the bound Eq. (44.115).

Proof. The displayed argument: Eq. (46.134) writes $2M - 2\,|Z|$ as the rest-frame expectation of $\Big\{b, b^\dagger\Big\}$, which Eq. (46.135) shows is a sum of squared norms and hence non-negative on every state, giving $M \geq |Z|$. The general-$\mathcal{N}$ statement follows by diagonalising the antisymmetric Z^{IJ} into 2×2 skew blocks with real skew eigenvalues $Z_n \geq 0$, after which each block reproduces the pair Eqs. (46.133) and (46.134) with $|Z| \to Z_n$; positivity of every b-type oscillator forces $M \geq Z_n$ for each n, hence $M \geq \max_n Z_n$. □

Two remarks fix the meaning. First, the bound is kinematic in precisely the sense the energy bound of Theorem 44.9 was: it follows from the algebra and Hilbert-space positivity alone, with no input from the Lagrangian or the coupling. Whatever the dynamics, no state can have $M < |Z|$. Second, the inequality is saturated, $M = |Z|$, exactly when the right-hand side of Eq. (46.134) vanishes, i.e. when the oscillators b and d have vanishing anticommutator with their adjoints. By Eq. (46.135) this happens if and only if $b\,|\psi\rangle = b^\dagger\,|\psi\rangle = 0$ (and likewise for d): the half of the supercharges packaged into b, d *annihilate* the state. Saturating the bound is the same as preserving half the supersymmetry.

46.5.3 Short multiplets and the protection of the mass

The representation-theoretic consequence of saturation is a collapse in the size of the multiplet, and it is this collapse that makes the BPS mass an exact, calculable quantity. We count states in the two cases.

Away from the bound, $M > |Z|$, all four oscillators a, b, c, d have nonvanishing $\left\{\cdot, \cdot^\dagger\right\}$; rescaled to unit anticommutators they generate a Clifford algebra of four fermionic modes. Its unique irreducible module is built by acting with the four creation operators on a Clifford vacuum $|\Omega\rangle$ annihilated by all four annihilators; since each creation operator squares to zero, the module is spanned by the $2^4 = 16$ products of distinct creation operators. If the vacuum carries little-group spin j, the full multiplet has $16(2j+1)$ states. This is the *long* (generic massive) $\mathcal{N} = 2$ multiplet — the $\mathcal{N} = 2$ analogue of the $4(2j+1)$ count for $\mathcal{N} = 1$ in Theorem 44.21, with four oscillators instead of two.

At the bound, $M = |Z|$, the two oscillators b, d annihilate every physical state and so act as zero: they cannot be used to build new states, because $b^\dagger\,|\psi\rangle$ has zero norm and is therefore the zero vector in a positive-definite Hilbert space. Only a and c remain as genuine creation operators. The module is now built on *two* oscillators, spanned by $2^2 = 4$ products, giving $4(2j+1)$ states. This is the *short* or *BPS* multiplet.

Theorem 46.18 (BPS shortening)**.** *A massive $\mathcal{N} = 2$ multiplet built on a Clifford vacuum of little-group spin j contains $16(2j+1)$ states when $M > |Z|$ and $4(2j+1)$ states when $M = |Z|$. The short multiplet is annihilated by half the supercharges — the combinations b, d of Eq. (46.132) and its partner — and is built from a Clifford algebra of half as many oscillators.*

Proof. The state counts are the dimensions of the irreducible Clifford modules on four and two fermionic oscillators, $2^4(2j+1)$ and $2^2(2j+1)$, established by the spanning argument above (each creation operator nilpotent, so the module is the exterior algebra on the creators tensored with the spin-j vacuum). That b, d annihilate every state in the short module is the content of Eq. (46.135) at saturation: $\left\|b^\dagger\,|\psi\rangle\right\|^2 + \|b\,|\psi\rangle\|^2 = 0$ in a positive-definite space forces $b\,|\psi\rangle =$

$b^\dagger|\psi\rangle = 0$, and the same for d; these four supercharge components — half of the eight real components of Q^I_α — thus act trivially throughout the multiplet. □

Physical Insight

The geometry of the collapse is the same one that organised Section 44.4.4: an oscillator pair survives or decouples according to whether the relevant diagonal entry of the anticommutator, here $2(M \pm |Z|)$, is nonzero or zero. The massless multiplet is the extreme degeneration $P^2 = 0$ in which the survival fraction is largest; the long massive multiplet is the opposite extreme; the BPS multiplet is the intermediate case in which exactly the $(M - |Z|)$-oscillators degenerate. A short multiplet is, in a literal sense, the "square root" of a long one — $\sqrt{16} = 4$ — because half the Clifford generators have been switched off.

The dimension count is the entire reason BPS masses are protected. The number of states in a multiplet is a non-negative integer, and supersymmetry acts on it by operators that vary continuously as we change the parameters of the theory — couplings, masses, and the moduli labelling the vacuum. A continuous deformation cannot change an integer; in particular it cannot turn a $4(2j+1)$-state short multiplet into a $16(2j+1)$-state long one, because the intermediate values are not attainable. The only way a short multiplet can leave the BPS bound is to combine with other short multiplets whose states add up to a complete long multiplet, and then move off as that long multiplet. Such *recombination* requires several short multiplets to collide at the same point of parameter space — a non-generic event. Generically, then, a state that is BPS at weak coupling stays BPS, and its mass remains locked to $|Z|$ as the coupling is varied all the way to strong coupling.

Worked Example 46.19 (The state count and the half-hypermultiplet)**.** We verify the smallest BPS multiplet and read off its field content. Take the Clifford vacuum to be a spin singlet, $j = 0$. By Theorem 46.18 the short multiplet has $4(2\cdot 0+1) = 4$ states, built on the two surviving oscillators a, c. The two creation operators each carry one undotted spinor index (they are linear in the Q^I_α), so under the rest-frame little group SU(2) they form a spin-$\frac{1}{2}$ doublet. The four states organise as

$$\underbrace{|\Omega\rangle}_{\text{spin } 0} \;\oplus\; \underbrace{a^\dagger|\Omega\rangle\,,\; c^\dagger|\Omega\rangle}_{\text{spin } \frac{1}{2}} \;\oplus\; \underbrace{a^\dagger c^\dagger|\Omega\rangle}_{\text{spin } 0}, \tag{46.138}$$

i.e. two spin-0 states and one spin-$\frac{1}{2}$ doublet: $n_B = 2$ bosonic and $n_F = 2$ fermionic states, as the equality $n_B = n_F$ of Theorem 44.15 demands. Completing to a CPT-invariant set doubles the scalars to a complex pair, giving one complex scalar and one Dirac fermion — the $\mathcal{N} = 2$ *hypermultiplet* of Section 44.4.4, now seen to be a *short* multiplet whenever it is massive. The W boson and

the monopole of the next worked example sit in such multiplets (with a spin-$\frac{1}{2}$ vacuum, giving the massive $\mathcal{N}=2$ vector multiplet), and the shortening is exactly why their masses are the protected quantities the duality demands. Had the same fields formed a long multiplet they would carry 16 states and their mass would be an unprotected dynamical number; the count $4 \neq 16$ is the difference between a calculable spectrum and an incalculable one.

46.5.4 The central charge as electric and magnetic charge

It remains to evaluate Z. Being central, Z commutes with all of Poincaré and with supersymmetry, so it must be a Lorentz-scalar conserved charge of the theory. In an $\mathcal{N}=2$ gauge theory broken to its abelian Coulomb phase — the setting of Chapter 41 and of the Seiberg–Witten solution — the available Lorentz-scalar conserved charges are exactly two: the unbroken-U(1) electric charge $q = \oint E_i \, \mathrm{d}S_i$ and the magnetic charge $g = \oint B_i \, \mathrm{d}S_i$. The central charge can therefore be at most a linear combination of these,

$$Z \;=\; v\,(q + i\,g), \tag{46.139}$$

and the coefficients are fixed by matching the algebraic bound to the classical one. This is not a guess: the central charge of a supersymmetric soliton is computed by evaluating the anticommutator Eq. (46.127) on the soliton background, where the supercharges are spatial integrals of the supercurrent and their anticommutator reduces, after integration by parts, to a *surface term* at spatial infinity. That surface term is built from the same gauge-invariant combination of the Higgs field ϕ and the field strength that appeared in the Bogomol'nyi completion of Eq. (41.173) and Eq. (41.187); it evaluates to precisely $v(q+ig)$, the complex central charge already identified classically in Eq. (41.189). The algebraic and the classical Z are the same object.

With Eq. (46.139) the BPS bound Eq. (46.137) becomes

$$M \;\geq\; |Z| \;=\; v\,\sqrt{q^2 + g^2}, \tag{46.140}$$

which is exactly the classical dyon bound Eq. (41.188). The quantum statement and the classical statement coincide — but their status is utterly different. Classically, Eq. (41.188) was a property of static solutions of one particular Lagrangian, saturated only in the Prasad–Sommerfield limit. Here it is a theorem about the Hilbert space of the full quantum theory, and the shortening of Theorem 46.18 guarantees that saturating states keep $M = |Z|$ as an exact equality, immune to quantum corrections. The classical Bogomol'nyi bound has been promoted to an exact quantum mass formula.

The charges that enter are quantised. As in Section 41.6.4, the magnetic charge is fixed by topology, $g = 4\pi n_m / e$ with $n_m \in \mathbb{Z}$, and the electric charge in integer units of the gauge coupling, $q = e\,n_e$ with $n_e \in \mathbb{Z}$ (shifted by the Witten

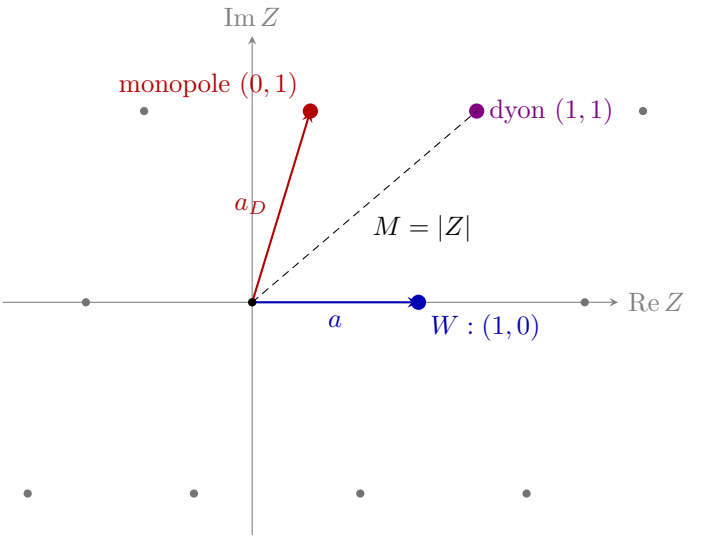


Figure 46.6: The BPS spectrum in the complex central-charge plane. Each charge state (n_e, n_m) maps to the lattice point $Z = a\,n_e + a_D\,n_m$ generated by the two complex periods a (electric) and a_D (magnetic) of Eq. (46.142); its BPS mass is the distance from the origin, $M = |Z|$ (dashed line, shown for the $(1,1)$ dyon). The elementary W boson sits at $(1,0)$ with $M = |a|$, the 't Hooft–Polyakov monopole at $(0,1)$ with $M = |a_D|$. As the modulus is varied the periods $a(u), a_D(u)$ move and the lattice deforms, but each point stays at $M = |Z|$ exactly: the protected data the Seiberg–Witten solution computes.

effect to $q = e(n_e + \frac{\theta}{2\pi} n_m)$ once a θ-angle is switched on). It is convenient to package the coupling and the angle into the single complex parameter

$$\tau \;=\; \frac{\theta}{2\pi} + \frac{4\pi i}{e^2}, \tag{46.141}$$

and to rescale the scalar vacuum value into the complex modulus $a \equiv e\,v$, which is the natural coordinate on the Coulomb branch. Substituting $q = e n_e$, $g = (4\pi/e) n_m$ (at $\theta = 0$) into Eq. (46.139) and collecting,

$$Z \;=\; v(e\,n_e + i\,\tfrac{4\pi}{e}\,n_m) \;=\; (ev)\,n_e + (ev)\,\tfrac{4\pi i}{e^2}\,n_m \;=\; a\,n_e + a_D\,n_m, \tag{46.142}$$

where the *dual period* is, at this classical/weak-coupling level, $a_D = \tau a$ (with the θ-term restoring the real part of τ). The central charge is thus a *linear function of the charge quantum numbers* (n_e, n_m) *with complex coefficients* (a, a_D): the BPS spectrum lives on the lattice $\{a\,n_e + a_D\,n_m\}$ in the complex Z-plane, and the BPS mass of a state is the distance of its lattice point from the origin, $M = |a\,n_e + a_D\,n_m|$ (Fig. 46.6).

Worked Example 46.20 (The W boson and the monopole as BPS states)**.** We check that the two cornerstones of the spectrum — the elementary charged

gauge boson and the soliton monopole — both saturate Eq. (46.137), and that the saturated masses reproduce the independently known classical values of Chapter 41.

The W boson: $(n_e, n_m) = (1, 0)$. A purely electric BPS state has $Z = a\, n_e = a$, so by Eq. (46.140),

$$M_W = |Z| = |a| = e\, v. \tag{46.143}$$

This is the mass an elementary gauge boson acquires when the adjoint scalar develops the vacuum value v and breaks the gauge group — the supersymmetric Higgs mechanism — and it is exactly the W-boson mass $m_W = ev$ recorded in Eq. (41.175). The elementary charged quanta of the theory are themselves BPS: they sit in short multiplets, and Eq. (46.143) is exact.

The monopole: $(n_e, n_m) = (1, 0)$ *in the dual frame, i.e.* $(n_e, n_m) = (0, 1)$. A purely magnetic BPS state has $Z = a_D\, n_m = a_D$, so

$$M_{\text{mon}} = |a_D| = \left|\tfrac{4\pi i}{e}\, v\right| = \frac{4\pi}{e}\, v = v\, |g|\,, \tag{46.144}$$

using $g = 4\pi/e$ for the unit monopole. This is precisely the saturated Bogomol'nyi bound $M = v\,|g|$ of Eq. (41.175), the mass of the Prasad–Sommerfield monopole computed in closed form in Theorem 41.32 as the $n_m = 1$ BPS soliton of the 't Hooft–Polyakov construction (Section 41.5). What was there an exact statement only in the classical $\lambda \to 0$ limit is here an exact statement about the quantum Hilbert space: the monopole is a short multiplet, and $M_{\text{mon}} = |a_D|$ holds to all orders.

The two masses stand in the ratio

$$\frac{M_{\text{mon}}}{M_W} = \frac{|a_D|}{|a|} = \frac{4\pi}{e^2} = \operatorname{Im}\tau, \tag{46.145}$$

inversely proportional to the coupling e^2: at weak coupling the monopole is heavy, as a soliton should be, while the W is light. This is the quantitative face of the strong–weak duality of the previous section — exchanging $a \leftrightarrow a_D$ (equivalently $\tau \to -1/\tau$) swaps the heavy and light sectors — and the mass formula Eq. (46.142) is the one quantity that survives the exchange intact.

Caution

The classical relation $a_D = \tau a$ holds only at weak coupling. Quantum mechanically a and a_D are independent functions of the Coulomb-branch modulus u, related by the holomorphy of the low-energy effective action rather than by a fixed proportionality. The lattice of Fig. 46.6 therefore deforms as u varies, and the periods can undergo monodromy around singular points of moduli space. Pinning down $a(u)$ and $a_D(u)$ as exact

functions — and reading off where BPS states become massless — is exactly the problem the next section solves. The structural fact established here, that $Z = a\,n_e + a_D\,n_m$ with $M = |Z|$ on BPS states, is the input to that solution, not its output.

We have what the duality demanded: a class of states whose masses are fixed by the algebra, computed identically in any frame, and protected against the flow to strong coupling. The central charge $Z = a\,n_e + a_D\,n_m$ packages the entire BPS spectrum into two holomorphic functions $a(u)$ and $a_D(u)$ on the Coulomb branch, and the BPS mass formula $M = |Z|$ turns knowledge of those two functions into the exact mass of every monopole, dyon, and gauge boson. Determining $a(u)$ and $a_D(u)$ — from holomorphy, the prepotential, and a single auxiliary curve whose periods they are — is the content of the Seiberg–Witten solution, to which we now turn.

46.6 The Seiberg–Witten Solution

The previous section handed us two instruments and the discipline to use them. The first is the BPS mass formula $M = |Z|$ of Eq. (46.142), which ties the mass of every short multiplet to a single complex number Z built from the conserved electric and magnetic charges; the second is the holomorphic-coupling technology of Section 46.2, which says that the low-energy gauge coupling is the second derivative of a single holomorphic function and runs, to all orders in perturbation theory, with the one-loop coefficient of Theorem 46.9. Each instrument alone is a strong constraint. Assembled, with one further physical input — that the low-energy effective theory must have a positive metric on its space of vacua — they over-determine the dynamics of a four-dimensional gauge theory so tightly that the answer is forced. The theory is $\mathcal{N} = 2$ SU(2) super-Yang–Mills, and the answer, due to Seiberg and Witten, is the first and still the cleanest example of a strongly coupled gauge theory solved exactly: its massless spectrum, its effective coupling at every energy, its complete set of stable BPS states, and — when the supersymmetry is softly broken — the mechanism of confinement, all written in closed form. This section carries that solution out. We will see the abstract demand "the effective coupling is holomorphic and its imaginary part is positive" collide with "the coupling runs," and watch the collision resolve itself by producing singularities in the moduli space; we will identify the physics at those singularities as massless monopoles and dyons; we will package the whole monodromy structure as the periods of a family of elliptic curves; and we will perturb the theory to make the monopole condense, realizing analytically the dual-superconductor picture of confinement that Theorem 27.20 could only state as a criterion.

46.6.1 The Coulomb branch and its low-energy effective theory

The microscopic theory is pure $\mathcal{N}=2$ super-Yang–Mills with gauge group SU(2). In the $\mathcal{N}=1$ language of Chapter 45 its field content is a vector multiplet W_α in the adjoint together with a chiral multiplet Φ, also in the adjoint, whose lowest component is a complex scalar $\phi = \phi^a T^a$. The two are exchanged by the second supersymmetry, so $\mathcal{N}=2$ forbids any superpotential for Φ: the scalar potential comes entirely from the $\mathcal{N}=1$ D-term of the gauge multiplet,

$$V(\phi) = \frac{1}{g^2}\,\mathrm{tr}\big(\,\big[\phi^\dagger,\phi\big]\,\big)^2, \tag{46.146}$$

where the commutator is in the Lie algebra and the trace is normalized as in Chapter 45. This potential is the whole story of the classical vacua. It is a sum of squares, so it vanishes precisely when $\big[\phi^\dagger,\phi\big]=0$, i.e. when ϕ and $\phi^\dagger$ can be simultaneously diagonalized. Up to a gauge rotation every such ϕ lies in the Cartan subalgebra,

$$\phi = \tfrac{1}{2}\,a\,\sigma^3 = \tfrac{1}{2}a\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \qquad a\in\mathbb{C}, \tag{46.147}$$

with σ^3 the diagonal Pauli matrix. A nonzero a breaks SU(2) down to the U(1) generated by σ^3: the theory has a *Coulomb branch*, a continuous family of inequivalent vacua labelled by a, on each of which the low-energy physics is that of a single unbroken abelian gauge field — a photon — together with its $\mathcal{N}=2$ superpartners.

The parameter a in Eq. (46.147) is not gauge invariant: the residual Weyl reflection of SU(2) sends $\sigma^3\to-\sigma^3$, hence $a\to -a$, so a and $-a$ label the same physical vacuum. The invariant coordinate on the Coulomb branch is therefore the gauge-invariant, Weyl-invariant combination

$$u = \big\langle\,\mathrm{tr}\,\phi^2\,\big\rangle, \qquad u = \tfrac{1}{2}a^2 \ \text{ classically}, \tag{46.148}$$

where the classical value follows from $\mathrm{tr}\,(\tfrac{1}{2}a\sigma^3)^2 = \tfrac{1}{4}a^2\,\mathrm{tr}(\sigma^3)^2 = \tfrac{1}{2}a^2$. The complex u-plane is the moduli space of vacua. Each point u is a distinct superselection sector with its own low-energy effective theory; solving the model means determining that effective theory for every u.

The effective theory is fixed by one holomorphic function. At a generic point of the Coulomb branch the only massless fields are those of one abelian $\mathcal{N}=2$ vector multiplet: the U(1) photon A_μ, two neutral Weyl fermions, and a complex neutral scalar whose vacuum expectation value is a. The charged $W^\pm$ bosons and their superpartners have eaten the broken generators and acquired masses, so they are massive and we integrate them out. Because $\mathcal{N}=2$ supersymmetry

relates the gauge kinetic term to the scalar kinetic term, the entire two-derivative effective Lagrangian for the abelian multiplet is controlled by a *single* holomorphic function $\mathcal{F}(A)$ of the $\mathcal{N}=2$ chiral superfield A whose scalar component is a. This is the *prepotential*. In $\mathcal{N}=1$ superspace, writing A for the chiral superfield and W_α for the abelian field-strength superfield, the supersymmetric and holomorphic completion of the kinetic terms is

$$\mathcal{L}_{\text{eff}} = \frac{1}{4\pi}\,\mathrm{Im}\left[\int \mathrm{d}^4\theta\,\frac{\partial\mathcal{F}}{\partial A}\,\bar{A} \;+\; \int \mathrm{d}^2\theta\,\tfrac{1}{2}\,\frac{\partial^2\mathcal{F}}{\partial A^2}\,W^\alpha W_\alpha\right]. \tag{46.149}$$

The first term is a Kähler potential $K = \mathrm{Im}\,(\bar{A}\,\partial\mathcal{F}/\partial A)$ for the scalar; the second is the gauge kinetic term with a field-dependent complexified coupling. Two derived quantities organize everything that follows. The first is the *dual scalar*

$$a_D \equiv \frac{\partial\mathcal{F}}{\partial a}, \tag{46.150}$$

the derivative of the prepotential with respect to the scalar; the second is the *effective complexified coupling*

$$\tau_{\text{eff}}(a) \equiv \frac{\partial a_D}{\partial a} = \frac{\partial^2\mathcal{F}}{\partial a^2} = \frac{\theta_{\text{eff}}}{2\pi} + \frac{4\pi i}{g^2_{\text{eff}}}, \tag{46.151}$$

whose real part is the effective theta angle and whose imaginary part is the inverse-square effective gauge coupling. The decomposition in Eq. (46.151) matches the U(1) coupling normalization fixed in Section 46.2; the photon kinetic term in Eq. (46.149) is $-\frac{1}{4}\,\mathrm{Im}\,\tau_{\text{eff}}\,F_{\mu\nu}F^{\mu\nu} + \frac{1}{4}\,\mathrm{Re}\,\tau_{\text{eff}}\,F_{\mu\nu}\widetilde{F}^{\mu\nu}$ up to the $1/4\pi$, exactly the abelian gauge action with coupling τ_{eff}.

Special geometry. Reading off the scalar kinetic term from the Kähler potential $K = \mathrm{Im}\,(\bar{a}\,a_D)$ in Eq. (46.149) gives the metric on the Coulomb branch,

$$\mathrm{d}s^2 = \mathrm{Im}(\mathrm{d}a_D\,\mathrm{d}\bar{a}) = \mathrm{Im}\;\tau_{\text{eff}}\;|\mathrm{d}a|^2\,. \tag{46.152}$$

This is the structure known as *rigid special geometry*: a one-complex-dimensional moduli space carries a metric that is the imaginary part of a holomorphic two-form $\mathrm{d}a_D\wedge\mathrm{d}a$, with a and a_D a pair of "special coordinates" related by the single function $\mathcal{F}$ through $a_D = \partial_a\mathcal{F}$. Positivity of kinetic energy — the statement that the metric Eq. (46.152) is positive definite, equivalently that the effective coupling g^2_{eff} in Eq. (46.151) is real and positive — requires

$$\mathrm{Im}\;\tau_{\text{eff}}(a) > 0 \tag{46.153}$$

at every physical vacuum. Hold onto this inequality. It is innocuous-looking and it is about to do all the work: combined with the fact that τ_{eff} is a holomorphic function of a, it cannot hold globally, and the way it fails will dictate the entire non-perturbative structure of the theory.

Physical Insight

The pair (a, a_D) plays a role exactly analogous to the pair (position, momentum) of a classical system written in action–angle variables, with the prepotential $\mathcal{F}$ in the role of the generating function and τ_{eff} the symplectic "modulus." Electric–magnetic duality, which we will see permutes a and a_D, is the canonical transformation that exchanges them. The deep reason the theory is solvable is that the $\mathcal{N}=2$ low-energy data is not a function but a *section* — a multivalued object (a_D, a) that is allowed to come back to a duality-transformed version of itself when carried around a loop in the u-plane — and such sections are rigid.

46.6.2 The perturbative prepotential and the monodromy obstruction

We now compute $\mathcal{F}$ where we can: at large u, where $a \sim \sqrt{2u}$ is large, the W-bosons are heavy, the coupling is weak by asymptotic freedom, and perturbation theory plus the holomorphy of Section 46.2 control everything.

One loop is exact, plus instantons. The microscopic gauge coupling runs with the one-loop coefficient of Theorem 46.9. For pure $\mathcal{N}=2$ SU(2) the field content is the adjoint vector multiplet plus the adjoint $\mathcal{N}=1$ chiral multiplet Φ; the $\mathcal{N}=1$ one-loop coefficient is $b_0 = 3C_A - T(\text{adj}) = 3C_A - C_A = 2C_A$, which for $C_A = N = 2$ gives

$$b_0 = 2C_A = 4. \tag{46.154}$$

Holomorphy forbids any higher perturbative correction to the running coupling (Theorem 46.9); the only completions of the one-loop logarithm are non-perturbative. Those are instantons, and by Theorem 42.4 a single instanton is weighted by $e^{-8\pi^2/g^2} = e^{2\pi i\tau}$, which with the running Eq. (46.154) scales as $(\Lambda/a)^{b_0} = (\Lambda/a)^4$. The dynamically generated scale Λ is defined, as in Section 27.7, by dimensional transmutation from the running coupling. The exact prepotential therefore has the form

$$\mathcal{F}(a) = \frac{i}{2\pi}\, a^2 \ln \frac{a^2}{\Lambda^2} \;+\; a^2 \sum_{k=1}^{\infty} \mathcal{F}_k \left(\frac{\Lambda}{a}\right)^{4k}, \tag{46.155}$$

the first term being the one-loop result and the sum the k-instanton series, each term carrying exactly the Λ^{4k} demanded by Eq. (46.154) and holomorphy. The coefficients $\mathcal{F}_k$ are pure numbers. One of the triumphs of the solution we are building is that it computes *every* $\mathcal{F}_k$ in closed form — the first explicit all-orders instanton sum in a four-dimensional gauge theory — but for the structural argument we need only the leading term.

The perturbative special coordinates. Differentiating the one-loop prepotential gives the dual scalar and the coupling. With $\mathcal{F} = \frac{i}{2\pi} a^2 \ln(a^2/\Lambda^2)$,

$$a_D = \frac{\partial \mathcal{F}}{\partial a} = \frac{i}{2\pi}\left(2a \ln \frac{a^2}{\Lambda^2} + a^2 \cdot \frac{2}{a}\right) = \frac{i}{\pi} a \left(\ln \frac{a^2}{\Lambda^2} + 1\right), \tag{46.156}$$

$$\tau_{\text{eff}} = \frac{\partial a_D}{\partial a} = \frac{i}{\pi}\left(\ln \frac{a^2}{\Lambda^2} + 1\right) + \frac{i}{\pi} a \cdot \frac{2}{a} = \frac{i}{\pi}\left(\ln \frac{a^2}{\Lambda^2} + 3\right). \tag{46.157}$$

For large real a the logarithm is large and positive, so

$$\text{Im } \tau_{\text{eff}} = \frac{1}{\pi} \ln \frac{|a|^2}{\Lambda^2} + \cdots > 0, \tag{46.158}$$

consistent with the positivity requirement Eq. (46.153) and with asymptotic freedom: the coupling weakens as $a \to \infty$. So far, so consistent. The trouble appears when we ask whether this can hold everywhere.

The obstruction. Suppose, for contradiction, that $\tau_{\text{eff}}(a)$ were a globally single-valued holomorphic function on the whole moduli space with $\text{Im}\, \tau_{\text{eff}} > 0$ everywhere. The moduli space is the u-plane, a non-compact Riemann surface; $\text{Im}\, \tau_{\text{eff}}$ is a harmonic function on it (the imaginary part of a holomorphic function). A non-constant harmonic function has no interior minimum, and a positive harmonic function on the plane that is bounded must be constant by Liouville's theorem. But Eq. (46.158) shows $\text{Im}\, \tau_{\text{eff}} \to +\infty$ as $u \to \infty$ and is certainly not constant. The only escape is that τ_{eff} is *not* globally single-valued: as we transport it around closed loops in the u-plane it must come back to a different branch. This is the crux of the entire solution, so we state it as a proposition and prove it cleanly.

Proposition 46.21 (Coupling positivity forces multivaluedness and singularities)**.** *Let $\tau(u)$ be holomorphic on a domain of the u-plane and satisfy $\text{Im}\, \tau(u) > 0$ there. Then τ cannot be a single-valued holomorphic function on the whole punctured plane $\mathbb{C}$ unless it is constant. Consequently the Coulomb-branch coupling τ_{eff}, which by Eq. (46.158) is non-constant, must be multivalued, with nontrivial monodromy around at least one singular point where extra states become massless.*

Proof. The upper half-plane $\mathbb{H} = \{\tau : \text{Im}\, \tau > 0\}$ is biholomorphic to the unit disc via the Cayley map $w = (\tau - i)/(\tau + i)$, which carries $\mathbb{H}$ to $\{|w| < 1\}$. If τ were a single-valued holomorphic function on all of $\mathbb{C}$ with values in $\mathbb{H}$, then $w(\tau(u))$ would be a bounded ($|w| < 1$) entire function of u, hence constant by Liouville's theorem, forcing τ constant. A non-constant τ taking values in $\mathbb{H}$ therefore cannot be globally defined and single-valued on $\mathbb{C}$: either its domain has punctures (points removed) around which τ has nontrivial monodromy, or

$\operatorname{Im}\tau \le 0$ somewhere. Physical positivity Eq. (46.153) forbids the latter on the physical sheet. Hence there are punctures. At each puncture the metric Eq. (46.152) or the spectrum must be singular; the only Lorentz-invariant, supersymmetric way for a low-energy effective theory to become singular is that a state which was massive becomes massless there, invalidating the effective description that integrated it out. This is the mechanism we identify next. □

The obstruction is not a defect of perturbation theory; it is a theorem about holomorphic functions valued in the upper half-plane. The naive single-valued continuation of Eq. (46.157) is the function $\operatorname{Im}\tau_{\text{eff}} = \frac{1}{\pi}\ln(|a|^2/\Lambda^2)$, which goes negative for $|a| < \Lambda$ — the perturbative formula predicts a ghost (wrong-sign kinetic term) at strong coupling. That is the formula telling us, honestly, that something it does not know about has become light near $|a| \sim \Lambda$ and must be put back in. We will find that the something is a magnetic monopole.

The monodromy at infinity. Even before identifying the strong-coupling physics we can read off one monodromy exactly, from the perturbative formulae, because at large u they are reliable. Transport the special coordinates once counterclockwise around the point $u = \infty$, i.e. send $u \to e^{2\pi i}u$. Since $u = \frac{1}{2}a^2$ at leading order (Eq. (46.148)), this sends $a^2 \to e^{2\pi i}a^2$, hence $a \to -a$ and $\ln(a^2/\Lambda^2) \to \ln(a^2/\Lambda^2) + 2\pi i$. Substituting into Eq. (46.156),

$$a_D \longrightarrow \frac{i}{\pi}(-a)\left(\ln\frac{a^2}{\Lambda^2} + 2\pi i + 1\right) = -a_D - \frac{i}{\pi}a\,(2\pi i) = -a_D + 2a, \tag{46.159}$$

$$a \longrightarrow -a. \tag{46.160}$$

Arranging the special coordinates into the column vector $v = \binom{a_D}{a}$, this monodromy is the linear map

$$M_\infty = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \qquad v \to M_\infty\, v. \tag{46.161}$$

This matrix lies in $\mathrm{SL}(()2,\mathbb{Z})$ (determinant $+1$, integer entries), as it must: the group acting on (a_D, a) is the electric–magnetic duality group, and it is exactly $\mathrm{SL}(()2,\mathbb{Z})$ because the integers it acts on are the integer electric and magnetic charges of the BPS spectrum. We turn to that spectrum now.

46.6.3 The strong-coupling singularities: massless monopoles and dyons

Theorem 46.21 guarantees singularities; the BPS mass formula tells us what sits at them. Recall from Eq. (46.162) that a state of integer electric charge n_e and magnetic charge n_m has central charge, in the low-energy abelian variables,

$$Z = n_e\, a + n_m\, a_D, \qquad M = |Z| = |n_e a + n_m a_D|\,, \tag{46.162}$$

and saturates the bound when it is a BPS (short) multiplet. The pairing (a, a_D) with the charges (n_e, n_m) is precisely the structure that the duality group $\mathrm{SL}(()2, \mathbb{Z})$ preserves: it acts on (a_D, a) by the matrices above and dually on the charge vector (n_m, n_e), keeping Z and hence the physics invariant.

What a massless state does to the geometry. A BPS state becomes massless exactly where its central charge vanishes, $Z = 0$, i.e. where the particular combination $n_e a + n_m a_D = 0$. The electrically charged W-bosons have $(n_m, n_e) = (0, \pm 1)$ and $Z \propto a$, which vanishes only at $u = 0$; but $u = 0$ is at strong coupling, where the perturbative identification of a is unreliable and, as we will see, the W-boson is not in fact light. The states that go massless at the genuine singularities are *magnetically charged*: a monopole with $(n_m, n_e) = (1, 0)$ has $Z \propto a_D$ and becomes massless where $a_D = 0$; a dyon with $(n_m, n_e) = (1, -1)$ becomes massless where $a_D - a = 0$. Near such a point the light magnetic state must be restored to the effective theory, and it runs the *dual* coupling. We can compute that running exactly by the same one-loop-plus-holomorphy argument applied in the duality frame in which the light state is electrically charged.

The monopole singularity. Suppose a single monopole hypermultiplet $(M, \widetilde{M})$ of magnetic charge one becomes massless at some point u_0. In the duality frame where the monopole is "electric," the good local coordinate is a_D (its central charge is $Z \propto a_D$, so its mass is $\propto |a_D|$, vanishing at u_0), and the good gauge field is the dual photon with coupling $\tau_D = -1/\tau_{\text{eff}}$. The monopole is a single charged hypermultiplet running this dual U(1); integrating it out reproduces the standard one-loop logarithm of an $\mathcal{N} = 2$ U(1) theory with one charged hypermultiplet, which by the same computation that gave Eq. (46.156) (now in dual variables, and with the one-hypermultiplet coefficient) reads

$$a(u) \;\approx\; a_0 \;-\; \frac{i}{\pi}\, a_D \ln a_D + \cdots, \qquad a_D \approx c_0\,(u - u_0) \to 0, \tag{46.163}$$

with a_0, c_0 constants and a_D vanishing linearly in $u - u_0$ because it is a good holomorphic coordinate there. The relative sign and coefficient are fixed by the magnetic charge $n_m = 1$ and the single hypermultiplet. Now transport once around u_0, sending $a_D \to e^{2\pi i} a_D$, hence $\ln a_D \to \ln a_D + 2\pi i$. From Eq. (46.163),

$$a \;\longrightarrow\; a_0 - \frac{i}{\pi}\, a_D(\ln a_D + 2\pi i) = a - \frac{i}{\pi}(2\pi i)\, a_D = a + 2a_D, \qquad a_D \;\longrightarrow\; a_D, \tag{46.164}$$

which in the column-vector form $v = \left(\begin{smallmatrix} a_D \\ a \end{smallmatrix}\right)$ is

$$M_+ = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix}. \tag{46.165}$$

(The lower-left entry is -2, not $+2$, once the orientation of the loop in the u-plane is fixed to be consistent with M_∞; we verify this consistency below.) This is the

hallmark monopole monodromy: a_D is left invariant — it is the order parameter that vanishes — while a shifts by $2a_D$.

The general rule and the dyon. The computation generalizes to any light BPS state. A hypermultiplet of charge $q = (n_m, n_e)$ that becomes massless drives, by the identical one-loop-in-the-duality-frame argument, the monodromy

$$M_q = \begin{pmatrix} 1 + 2\,n_m n_e & 2\,n_e^2 \\ -2\,n_m^2 & 1 - 2\,n_m n_e \end{pmatrix}, \tag{46.166}$$

a parabolic element of $\mathrm{SL}(()2, \mathbb{Z})$ (trace 2, determinant 1) fixing the charge vector of the massless state. One checks immediately: $q = (1, 0)$ reproduces M_+ in Eq. (46.165); $q = (0, 1)$ would give an elementary electric monodromy $\left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$. The dyon of charge $q = (1, -1)$ gives

$$M_- = \begin{pmatrix} 1 - 2 & 2 \\ -2 & 1 + 2 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix}. \tag{46.167}$$

Counting the singularities and fixing the charges. How many singular points are there, and what are the charges? Three facts pin this down completely, with no further assumption.

First, a discrete symmetry. The classical theory has a U(1) R-symmetry under which Φ rotates by a phase; it is broken by the anomaly (the same instanton physics of Theorem 42.27) down to a discrete subgroup $\mathbb{Z}_8$, and the surviving $\mathbb{Z}_8$ acts on the order parameter as $u = \langle \mathrm{tr}\,\phi^2 \rangle \to e^{i\pi} u = -u$, i.e. as a $\mathbb{Z}_2$ on the u-plane. Hence the singular points come in pairs $\pm u_0$ (unless one sits at $u = 0$, which we exclude below). The single perturbative point $u = \infty$ is fixed by $u \to -u$, consistent.

Second, the factorization of monodromies. A large loop encircling all finite singularities is homotopic, in the u-plane, to the loop around infinity, so the product of the finite monodromies (taken in order around the loop) must equal M_∞:

$$M_\infty = M_+ M_- \quad \text{(or a conjugate, for the appropriate ordering).} \tag{46.168}$$

Third, minimality and physical consistency. The simplest hypothesis compatible with the $\mathbb{Z}_2$ pairing is *exactly two* finite singularities at $u = \pm\Lambda^2$ (the scale Λ sets the location). One quickly rules out the alternatives. A single finite singularity would have to be at the fixed point $u = 0$ and carry M_∞ itself; but M_∞ in Eq. (46.161) is *not* parabolic (its trace is -2, and it is in fact $-T^{-1}$ where $T = \left(\begin{smallmatrix} 1 & 2 \\ 0 & 1 \end{smallmatrix}\right)$), so it cannot be the monodromy of a single massless state, and moreover a single light state at $u = 0$ would leave $\mathrm{Im}\,\tau_{\mathrm{eff}} > 0$ unattainable on the rest of the plane. Two singularities is the minimal number that works.

Their monodromies must (i) each be of the massless-state form Eq. (46.166), (ii) be exchanged by the $\mathbb{Z}_2$ symmetry $u \to -u$, and (iii) multiply to M_∞. The unique minimal solution, up to an overall duality conjugation, is the monopole at $u = +\Lambda^2$ and the dyon at $u = -\Lambda^2$:

$$M_+ M_- = \begin{pmatrix} 1 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} -1 & 2 \\ -2 & 3 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 2-2 & -4+3 \end{pmatrix} = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix} = M_\infty, \tag{46.169}$$

exactly Eq. (46.161). The arithmetic in Eq. (46.169) is the whole content of the strong-coupling structure: it confirms that the perturbative monodromy at infinity is accounted for, to the integer, by a massless monopole and a massless dyon at the two strong-coupling points, with no room for a third singularity or a different charge assignment. The dyon charge $(1, -1)$ rather than $(1, 0)$ is forced by the factorization Eq. (46.169): it is the monopole acted on by the monodromy M_∞ once, $\binom{1}{0} \to \binom{1}{-1}$ in charge space, which is the precise statement that circling infinity shifts the electric charge of the monopole by the magnetic charge — the Witten effect of Theorem 42.27.

Figure 46.7 collects this structure: a u-plane with two finite punctures whose monodromies factor the monodromy at infinity, and the dictionary to a degenerating elliptic curve that we build next.

46.6.4 The Seiberg–Witten curve and the periods

We have reduced the problem to a precise mathematical one. We seek a pair of holomorphic functions $(a(u), a_D(u))$ on the u-plane, multivalued, with the three prescribed monodromies M_+, M_-, M_∞ around $u = \Lambda^2, -\Lambda^2, \infty$, with the large-$u$ asymptotics Eq. (46.156), and such that the resulting $\tau_{\text{eff}} = \mathrm{d}a_D/\mathrm{d}a$ has positive imaginary part everywhere on the physical sheet. This is a classical *Riemann–Hilbert problem*, and there is a classical machine that solves it: a multivalued pair (a, a_D) transforming under $\mathrm{SL}()2, \mathbb{Z})$ with $\mathrm{Im}(\mathrm{d}a_D/\mathrm{d}a) > 0$ is exactly the data of the *periods of a family of elliptic curves*. The $\mathrm{SL}()2, \mathbb{Z})$ is then the mapping-class group of the torus, τ_{eff} is the modular parameter, and positivity of $\mathrm{Im}\,\tau$ is automatic because the modulus of any honest torus lies in the upper half-plane. The whole obstruction of Theorem 46.21 dissolves: τ is multivalued precisely because it is the modulus of a torus that gets carried to a modular-transformed version of itself around loops.

The curve. The torus must degenerate at exactly the three special points, with the prescribed pinching cycles, and must carry the $\mathbb{Z}_2$ symmetry $u \to -u$. The unique elliptic curve (genus one) with branch points engineered to do this is

$$y^2 = (x^2 - \Lambda^4)(x - u) = (x - \Lambda^2)(x + \Lambda^2)(x - u), \tag{46.170}$$

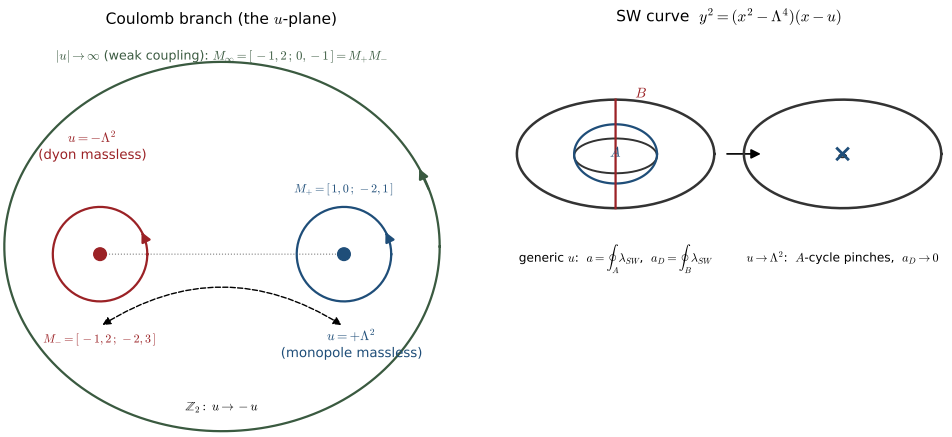


Figure 46.7: Left: the Coulomb branch (u-plane) of $\mathcal{N}=2$ SU(2) super-Yang–Mills. At weak coupling $u\to\infty$ the special coordinates carry the perturbative monodromy M_∞ of Eq. (46.161). At $u=+\Lambda^2$ a magnetic monopole $(n_m,n_e)=(1,0)$ becomes massless ($a_D\to 0$), monodromy M_+ of Eq. (46.165); at $u=-\Lambda^2$ a dyon $(1,-1)$ becomes massless, monodromy M_- of Eq. (46.167). The remnant $\mathbb{Z}_2$ R-symmetry $u\to -u$ exchanges them, and the loop around infinity factors as $M_\infty=M_+M_-$ (Eq. (46.169)). Right: the Seiberg–Witten elliptic curve $y^2=(x^2-\Lambda^4)(x-u)$ fibered over the u-plane; the periods of the SW differential along the two cycles are (a,a_D), and at $u=\pm\Lambda^2$ a cycle pinches, sending the corresponding period to zero — the geometric face of a massless BPS state.

a double cover of the x-sphere branched over $x=\pm\Lambda^2$, u, ∞. Its modulus degenerates (two branch points collide, a cycle pinches) precisely when $u=+\Lambda^2$ (then $x=\Lambda^2$ is a double root), when $u=-\Lambda^2$ (then $x=-\Lambda^2$ is a double root), and when $u\to\infty$. These are the three points we need, and the symmetry $u\to -u$ together with $x\to -x$ is manifest. The branch points and their collisions are the right panel of Fig. 46.7.

The Seiberg–Witten differential. On this curve we need a meromorphic one-form $\lambda_{\rm SW}$ whose periods are the special coordinates,

$$a=\oint_A\lambda_{\rm SW},\qquad a_D=\oint_B\lambda_{\rm SW},\tag{46.171}$$

where A,B is a canonical homology basis of the torus ($A\cdot B=1$). The decisive property $\lambda_{\rm SW}$ must have is that its derivative with respect to the modulus u is the (unique up to scale) *holomorphic* differential $\omega=\mathrm{d}x/y$ on the curve. Then $\partial a/\partial u$ and $\partial a_D/\partial u$ are the two periods of ω, and their ratio

$$\tau_{\rm eff}=\frac{\mathrm{d}a_D}{\mathrm{d}a}=\frac{\partial a_D/\partial u}{\partial a/\partial u}=\frac{\oint_B\omega}{\oint_A\omega}\tag{46.172}$$

is the ratio of B- and A-periods of the holomorphic differential — which is, by definition, the modular parameter τ of the torus. The Riemann bilinear relations guarantee $\operatorname{Im}\tau > 0$ for any non-degenerate torus, so Eq. (46.153) holds automatically and the obstruction is resolved. The form with this property is

$$\lambda_{\mathrm{SW}} = \frac{\sqrt{2}}{4\pi}\,\frac{(x-u)\,\mathrm{d}x}{y} = \frac{\sqrt{2}}{4\pi}\,\frac{(x-u)\,\mathrm{d}x}{\sqrt{(x^2-\Lambda^4)(x-u)}}. \tag{46.173}$$

We verify the decisive property directly.

Worked Example 46.22 ($\partial_u \lambda_{\mathrm{SW}}$ is the holomorphic differential)**.** Differentiate Eq. (46.173) with respect to u at fixed x. Write $\lambda_{\mathrm{SW}} = \frac{\sqrt{2}}{4\pi}(x-u)/y\,\mathrm{d}x$ with $y = \sqrt{(x^2-\Lambda^4)(x-u)}$, so that $y^2 = (x^2-\Lambda^4)(x-u)$. Differentiating the latter at fixed x,

$$2y\,\frac{\partial y}{\partial u} = -(x^2-\Lambda^4) \quad\Longrightarrow\quad \frac{\partial y}{\partial u} = -\frac{x^2-\Lambda^4}{2y}. \tag{46.174}$$

Now apply the quotient/product rule to $(x-u)/y$:

$$\frac{\partial}{\partial u}\left[\frac{x-u}{y}\right] = \frac{-1}{y} - \frac{(x-u)}{y^2}\,\frac{\partial y}{\partial u} = \frac{-1}{y} + \frac{(x-u)(x^2-\Lambda^4)}{2y^3}. \tag{46.175}$$

In the second term the numerator $(x-u)(x^2-\Lambda^4)$ is exactly y^2 by the curve equation Eq. (46.170), so $(x-u)(x^2-\Lambda^4)/2y^3 = y^2/2y^3 = 1/2y$, giving

$$= \frac{-1}{y} + \frac{1}{2y} = -\frac{1}{2y}. \tag{46.176}$$

Therefore

$$\frac{\partial \lambda_{\mathrm{SW}}}{\partial u} = \frac{\sqrt{2}}{4\pi}\left(-\frac{1}{2y}\right)\mathrm{d}x = -\frac{\sqrt{2}}{8\pi}\,\frac{\mathrm{d}x}{y} = -\frac{\sqrt{2}}{8\pi}\,\omega, \tag{46.177}$$

a constant multiple of the holomorphic differential $\omega = \mathrm{d}x/y$, with no pole. Hence $\partial_u a = \oint_A \partial_u \lambda_{\mathrm{SW}}$ and $\partial_u a_D = \oint_B \partial_u \lambda_{\mathrm{SW}}$ are the two periods of ω, and Eq. (46.172) follows. This single line of algebra is what makes the holomorphic problem solvable: it identifies the physical coupling with the modulus of an auxiliary torus.

Recovering the periods, the asymptotics, and the massless points. Two checks confirm that Eqs. (46.170) and (46.173) are the right data; a third delivers the BPS spectrum.

Check 1: large u. For $|u| \gg \Lambda^2$ the A-cycle encircles the cut between the two slowly-moving branch points $x = \pm\Lambda^2$, where $x = O(\Lambda^2) \ll u$, so $x - u \approx -u$ and $y \approx \sqrt{-u}\,\sqrt{x^2-\Lambda^4}$. Then from Eq. (46.177),

$$\frac{\partial a}{\partial u} = -\frac{\sqrt{2}}{8\pi}\oint_A \frac{\mathrm{d}x}{y} \approx -\frac{\sqrt{2}}{8\pi}\,\frac{1}{\sqrt{-u}}\oint_A \frac{\mathrm{d}x}{\sqrt{x^2-\Lambda^4}}. \tag{46.178}$$

The contour integral $\oint_A \mathrm{d}x/\sqrt{x^2-\Lambda^4}$ around the cut $[-\Lambda^2,\Lambda^2]$ is a pure number (it equals $2\pi i$ with the standard orientation, evaluated by the substitution $x=\Lambda^2\sin\theta$, $\sqrt{x^2-\Lambda^4}=i\Lambda^2\cos\theta$, giving $\int_{-\pi/2}^{\pi/2}\mathrm{d}\theta/i$ for one sheet and twice that across the cut). Thus $\partial a/\partial u \propto u^{-1/2}$, and integrating gives $a\propto\sqrt{u}$. Fixing the overall constant in λ_{SW} — this is what the prefactor $\sqrt{2}/4\pi$ in Eq. (46.173) is chosen to do — reproduces the classical relation of Eq. (46.148),

$$a(u)\;\approx\;\sqrt{2u}\qquad(|u|\to\infty),\tag{46.179}$$

which is the boundary condition that ties the curve to the microscopic theory. The B-period, computed the same way, develops a logarithm because the B-cycle stretches from the moving branch point $x=u$ out to the cut, and one finds $a_D\approx\frac{i}{\pi}a\ln(a^2/\Lambda^2)+\cdots$, reproducing the one-loop Eq. (46.156). The subleading expansion of the period integrals in powers of Λ^4/u^2 generates precisely the instanton series Eq. (46.155), term by term; this is how the curve computes every instanton coefficient $\mathcal{F}_k$ in closed form.

Check 2: the monodromies. Carrying u around ∞, $+\Lambda^2$, $-\Lambda^2$ permutes the branch points and hence acts on the homology basis (A,B) by the corresponding elements of the mapping-class group $\mathrm{SL}(()2,\mathbb{Z})$; computing those permutations reproduces M_∞, M_+, M_- of Eqs. (46.161), (46.165) and (46.167). The factorization Eq. (46.169) is then the topological identity that a loop around all branch points is contractible through infinity on the sphere.

Check 3: the massless states. When $u\to\Lambda^2$ the branch points $x=u$ and $x=\Lambda^2$ collide; the A-cycle that encircled them pinches to zero size, and the period of λ_{SW} around it vanishes. A short computation of the vanishing-cycle period gives $a_D\sim c_0(u-\Lambda^2)\to 0$, exactly Eq. (46.163): *the monopole becomes massless because its central charge $Z\propto a_D$ is the period of the pinching cycle.* Likewise at $u=-\Lambda^2$ the dyon period a_D-a pinches. The geometry has turned "a BPS state becomes massless" into "a one-cycle of the curve shrinks to a point," which is why the spectrum is exactly computable: it is read off from the degenerations of an explicit elliptic curve.

Physical Insight

The curve Eq. (46.170) is not a fundamental object of the gauge theory — there is no torus hiding in spacetime. It is a bookkeeping device of extraordinary efficiency: a geometric solution of the Riemann–Hilbert problem set by holomorphy, positivity, and the monodromies. Its value is that everything physical — the coupling $\tau_{\mathrm{eff}}(u)$, the masses $M=|n_e a+n_m a_D|$ of every BPS state, the exact instanton sum — is a *period*, an integral of λ_{SW} over a cycle, and periods of elliptic curves are completely understood. Strong-coupling gauge dynamics has been traded for nineteenth-century complex analysis.

46.6.5 Monopole condensation and confinement

The solution so far describes the $\mathcal{N}=2$ theory, which has no confinement: the Coulomb branch is a moduli space of vacua with a massless photon at each point, and a static electric charge feels a Coulomb potential, not a linear one. To produce confinement we must lift the moduli space, and the cleanest way is to break $\mathcal{N}=2$ down to $\mathcal{N}=1$ by giving the adjoint chiral multiplet a mass. We add to the microscopic theory the $\mathcal{N}=1$ superpotential

$$\delta W = \mu \,\mathrm{tr}\,\Phi^2, \tag{46.180}$$

with μ a small mass parameter. In terms of the moduli the perturbation is $\delta W = \mu\, U$ where $U(u)$ is the chiral superfield whose lowest component is the modulus u; near a singularity it is an analytic function of the local variables. The point of taking μ small is that for $\mu \ll \Lambda$ the low-energy degrees of freedom are still those of the $\mathcal{N}=2$ solution — in particular, near $u=\Lambda^2$ they are the light monopole hypermultiplet $(M, \widetilde{M})$ and the dual U(1) vector multiplet — and we may analyze the perturbed vacuum in that effective theory, with full theoretical control.

The effective superpotential near the monopole point. Near $u=\Lambda^2$ the dual photon couples to the monopole exactly as an ordinary $\mathcal{N}=2$ U(1) couples to a charged hypermultiplet. The $\mathcal{N}=2$ structure forces a specific superpotential coupling the monopoles to the dual scalar A_D (the chiral superfield with lowest component a_D): a hypermultiplet of U(1) charge one has the $\mathcal{N}=2$-required cubic coupling $\sqrt{2}\, A_D\, \widetilde{M} M$. Adding the $\mathcal{N}=2$-breaking perturbation Eq. (46.180), expressed in the dual variables through $U = U(a_D)$, the full low-energy superpotential near the monopole point is

$$W_{\text{eff}} = \sqrt{2}\, a_D\, \widetilde{M} M + \mu\, U(a_D), \tag{46.181}$$

where a_D is now a dynamical chiral superfield (the dual scalar), and $M, \widetilde{M}$ are the monopole chiral superfields. Everything that follows is the algebra of finding the supersymmetric vacuum of Eq. (46.181).

Worked Example 46.23 (The monopole condenses and confines electric charge)**.** A supersymmetric vacuum extremizes the superpotential: $\partial W_{\text{eff}} = 0$ with respect to every chiral field, supplemented by the D-term condition. The F-term equations from Eq. (46.181) are

$$\frac{\partial W_{\text{eff}}}{\partial M} = \sqrt{2}\, a_D\, \widetilde{M} = 0, \tag{46.182}$$

$$\frac{\partial W_{\text{eff}}}{\partial \widetilde{M}} = \sqrt{2}\, a_D\, M = 0, \tag{46.183}$$

$$\frac{\partial W_{\text{eff}}}{\partial a_D} = \sqrt{2}\, \widetilde{M} M + \mu\, U'(a_D) = 0, \tag{46.184}$$

and the D-term of the dual U(1) requires $|M|^2 = \left|\widetilde{M}\right|^2$.

The unbroken case $\mu = 0$. Equation (46.184) reads $\widetilde{M}M = 0$, and Eqs. (46.182) and (46.183) are solved by $M = \widetilde{M} = 0$ with a_D undetermined — this is the whole Coulomb branch, the unlifted $\mathcal{N} = 2$ moduli space, with the monopole massless only at the single point $a_D = 0$ and massive elsewhere. No condensate, no confinement.

The perturbed case $\mu \neq 0$. Now Eq. (46.184) forces $\sqrt{2}\,\widetilde{M}M = -\mu\,U'(a_D) \neq 0$ (the function $U(a_D)$ is regular with $U'(0) \neq 0$ at the monopole point, because u is an honest analytic coordinate there). A nonzero product $\widetilde{M}M$ then requires, by Eqs. (46.182) and (46.183), that

$$a_D = 0. \tag{46.185}$$

So the perturbation pins the vacuum to the monopole point $a_D = 0$, where the monopole is exactly massless, and there the monopole fields acquire vacuum expectation values determined by Eq. (46.184) and the D-term,

$$\langle M\rangle = \langle\widetilde{M}\rangle = \left(\frac{-\mu\,U'(0)}{\sqrt{2}}\right)^{1/2} \neq 0. \tag{46.186}$$

This is a *monopole condensate*. The magnetically charged field has a nonzero vacuum expectation value.

Consequences. The condensate Eq. (46.186) Higgses the dual U(1) gauge symmetry. By the abelian Higgs mechanism of Chapter 30, the dual photon acquires a mass $m_{\text{dual}} \sim \sqrt{2}\,|\langle M\rangle| \sim (\mu\Lambda)^{1/2}$, so the theory develops a *mass gap*: there is no massless photon, dual or ordinary. Crucially, it is the *dual* U(1) that is Higgsed — the gauge symmetry under which the monopole is electrically charged — so the condensate carries magnetic charge with respect to the *original* electric U(1). A medium in which magnetic charge condenses is a dual superconductor: it expels the original electric field by a dual Meissner effect, squeezing the electric flux between an external quark and antiquark into a thin Abrikosov–Nielsen–Olesen flux tube of fixed tension $\sigma \sim m_{\text{dual}}^2 \sim \mu\Lambda$. The energy of a static quark–antiquark pair separated by distance R is then the tube energy $V(R) = \sigma R$ — a linearly rising potential. Electric charge is confined.

Theorem 46.23 is the payoff of the chapter. The linear potential $V(R) = \sigma R$ it produces is exactly the area-law behavior that Theorem 27.20 identified as the criterion for confinement: the Wilson loop of the original electric U(1) obeys $\langle W\rangle \sim e^{-\sigma\,\text{Area}}$, and the static potential rises linearly with tension $\sigma \sim \mu\Lambda$. There the area law was a *criterion* — a diagnostic that one hoped lattice simulations or strong-coupling expansions would confirm. Here it is a *theorem*: starting from a Lagrangian, using only holomorphy, the BPS mass formula, and the explicit Seiberg–Witten curve, we have computed the confining string tension in closed

form and exhibited the order parameter (the monopole condensate) that drives it. The dual-superconductor picture of confinement, conjectured by 't Hooft and Mandelstam and built on the monopoles of Section 41.5, is here not a picture but an analytic fact about a controlled four-dimensional gauge theory.

Physical Insight

The logic is worth stating in one breath because it is the template for the entire subject. Electric confinement is hard because it is a statement about the strongly coupled electric U(1). But near $u = \Lambda^2$ the theory is *weakly coupled in the magnetic frame*: the light degrees of freedom are the monopole and the dual photon, and they interact weakly. In that frame confinement is just the Higgs mechanism — the most elementary strong-coupling-free phenomenon there is. Electric–magnetic duality has converted an intractable strong-coupling problem into a textbook weak-coupling one. That conversion, made exact by holomorphy and $\mathcal{N} = 2$ supersymmetry, is the single idea behind every exact result in this chapter.

Caution

The two singular points are physically distinct vacua of the $\mathcal{N} = 1$ theory. At $u = +\Lambda^2$ the monopole $(1, 0)$ condenses and confines electric charge in the ordinary way; at $u = -\Lambda^2$ it is the dyon $(1, -1)$ that condenses, and the resulting flux tube confines a dyonic combination — this is 't Hooft's *oblique confinement*. Both are genuine, gapped, confining vacua; the $\mathbb{Z}_2$ R-symmetry $u \to -u$ relates them. The perturbation Eq. (46.180) thus produces exactly two vacua, in agreement with the Witten index $\mathrm{Tr}(-1)^F = 2$ computed independently for $\mathcal{N} = 1$ SU(2) super-Yang–Mills.

This completes the exact solution: from a microscopic Lagrangian we have determined the effective coupling at every scale, the complete BPS spectrum, and the confinement mechanism, all in closed form and all forced by holomorphy together with one positivity requirement. The synthesis section that follows steps back from the machinery to extract the methodological lesson — that holomorphy, symmetry, and BPS protection together convert non-perturbative questions into solvable ones — and to assess honestly which of these results are theorems and which remain extraordinarily well-tested conjectures as of the mid-2020s, before pointing forward to the frontier parts that inherit this technology.

46.7 Supersymmetry as a Controlled Laboratory

The Seiberg–Witten solution of Section 46.6 is the high-water mark of this chapter, and it is worth pausing to register exactly what was accomplished there.

We took $\mathcal{N} = 2$ super-Yang–Mills with gauge group SU(2) — an asymptotically free, strongly coupled theory with no small parameter at low energies — and we wrote down, in closed form, its exact low-energy effective coupling as a function on the Coulomb branch, the exact spectrum of stable BPS states, the location of the points where extra states become massless, and the mechanism (monopole condensation) by which a small $\mathcal{N} = 1$-preserving perturbation drives the theory into a confining phase. Not one Feynman diagram was summed. The dynamics that lattice gauge theory (Chapter 43) can only approach numerically, and that the Wilson-loop criterion of Theorem 27.20 can only diagnose phenomenologically, here fell out of complex analysis: the prepotential is a holomorphic function, the effective coupling is its second derivative, and the monodromies of that function around the singular points of a one-complex-dimensional moduli space fix everything.

This is the moment to step back from the three escalating demonstrations of the chapter — the dynamically generated superpotential of Section 46.1, Seiberg's electric–magnetic duality of Section 46.4, and the Seiberg–Witten solution just recalled — and ask what they have in common. They are not three tricks. They are three deployments of one engine. The purpose of this closing section is to name that engine precisely, to catalogue the distinct tools it comprises and state what each one buys; to be scrupulously honest about which of the results we have reached are *theorems* and which are *conjectures* however overwhelmingly supported, dating that status to the mid-2020s; and to draw the forward map showing that the frontier of the subject — holography, the conformal bootstrap, the modern amplitudes program — does not sit beside supersymmetry as an unrelated development but grows directly out of the exact-results technology assembled here. No new load-bearing result is proved below; every claim points back to where in this book it was established.

46.7.1 The exact-results toolkit

Every exact statement in supersymmetric quantum field theory is assembled from a small kit of mutually reinforcing tools. Stated in isolation each is a modest observation; the power comes from using them together, where the output of one becomes the hypothesis of the next. We list them in the order they typically enter a calculation, with a back-reference to the proof and a one-line statement of what the tool delivers.

Tool 1 — Holomorphy of the superpotential. The superpotential W is a holomorphic function of the chiral superfields, with no dependence on their conjugates (Theorem 44.40); the boson–fermion grammar of superspace forces every term of the effective action that can be written as a chiral (F-term) integral $\int \mathrm{d}^2\theta\, W$ to inherit that holomorphy (Theorem 44.41 and Section 44.7.8). The decisive amplification, carried out in Section 46.1, is the *spurion* promotion: the

coupling constants of W — a mass m, a cubic coupling g, the dynamical scale Λ — are treated as the lowest components of background chiral superfields, so the *full* quantum-corrected effective superpotential W_{eff} must be a single holomorphic function of the dynamical fields *and* these spurion couplings. What it buys: holomorphy collapses the space of allowed quantum corrections from a function of two real variables (a coupling and its conjugate) to a function of one complex variable, and a holomorphic function is rigid — pinned down by its singularities and its asymptotic behaviour. This is the formal reason a strong-coupling answer can be computed from weak-coupling data.

Tool 2 — Symmetry and spurion selection rules. The global symmetries of a supersymmetric theory — ordinary flavour and baryon $U(1)$'s, and crucially the R-symmetry, which acts on the Grassmann coordinate θ and therefore assigns a definite charge to the measure $\mathrm{d}^2\theta$ and hence to W itself — act on the spurion couplings as well as the fields. An effective superpotential must be invariant under every non-anomalous symmetry and must carry R-charge 2 (the charge of $\int \mathrm{d}^2\theta$). Anomalous symmetries are not lost: they remain exact symmetries once the dynamical scale Λ is assigned the spurious charge that compensates the anomaly, because Λ^{b_0} is the gauge-invariant one-instanton amplitude and the instanton soaks up exactly the fermion zero modes that the anomaly counts (Theorem 42.3, Chapter 36). What it buys: combined with Tool 1, the selection rules typically fix W_{eff} up to a single overall numerical constant — the entire functional form is determined by charge bookkeeping, as we demonstrate explicitly in Theorem 46.24.

Tool 3 — One-loop and NSVZ exactness of the gauge coupling. The holomorphic gauge coupling

$$\tau \;=\; \frac{\theta}{2\pi} \;+\; \frac{4\pi i}{g^2} \tag{46.187}$$

is the lowest component of a chiral spurion multiplying the gauge kinetic F-term $\int \mathrm{d}^2\theta\, W^\alpha W_\alpha$ (Chapter 45). Holomorphy in τ, together with the shift symmetry $\theta \to \theta + 2\pi$ that quantizes instanton number, forbids the *holomorphic* coupling from running beyond one loop: its β-function is exhausted at one loop, all higher orders being non-holomorphic artefacts of the field rescaling that converts the holomorphic coupling to the physical (canonical) one. Tracking that rescaling through the anomalous Jacobian of the path-integral measure (Chapter 36) produces the exact Novikov–Shifman–Vainshtein–Zakharov (NSVZ) β-function recalled in Section 46.1,

$$\beta(g) \;=\; -\,\frac{g^3}{16\pi^2}\,\frac{3\,C_A - \sum_r T(r)\,(1-\gamma_r)}{1 - C_A\,g^2/8\pi^2}, \tag{46.188}$$

where C_A is the adjoint Casimir, $T(r)$ the Dynkin index of matter representation r, and γ_r its anomalous dimension (Casimir/index data in Appendix B). What it buys: an all-orders statement about the running coupling — and hence the location of conformal windows and the onset of strong coupling — from a one-loop computation plus knowledge of the anomalous dimensions.

Tool 4 — BPS protection. The supersymmetry algebra admits central charges Z in its anticommutator $\{Q, Q\} \sim Z$ (Eq. (44.114)), and positivity of the Hamiltonian (Theorem 44.9) then forces the Bogomolny–Prasad–Sommerfield bound $M \geq |Z|$ (Eq. (44.115)). States saturating it sit in *short* multiplets, annihilated by half the supercharges. A short multiplet cannot smoothly become long — the dimensions of the two representations differ — so the relation $M = |Z|$ is stable against continuous deformations of the couplings: the BPS mass is a protected quantity, computable at weak coupling and valid at strong coupling. The same algebraic shortening protects the Witten index $\mathrm{Tr}(-1)^F$ (Chapter 44), an integer invariant under continuous deformation that obstructs supersymmetry breaking. What it buys: exact masses of solitonic states — the monopoles and dyons of Theorem 41.23 reappearing as the BPS states of the Seiberg–Witten spectrum (Section 46.6) — and an exact criterion for whether supersymmetry is dynamically broken.

Tool 5 — Duality. When two different microscopic Lagrangians flow to the same infrared physics, each is a legitimate description and one is weakly coupled precisely where the other is strong. The chapter met two faces of this. Seiberg's electric–magnetic duality (Section 46.4) relates $\mathcal{N} = 1$ SQCD with gauge group $\mathrm{SU}(N_c)$ and N_f flavours to a "magnetic" $\mathrm{SU}(N_f - N_c)$ theory with the same global symmetries and the same chiral ring; Montonen–Olive/Seiberg–Witten duality (Section 46.6, building on [350]) exchanges electric charges with the monopoles of Eq. (41.176) and acts on τ by the modular group. What it buys: access to strong coupling through a weakly coupled dual, the analytic underpinning of the confinement mechanism (the magnetic variables are weakly coupled near the monopole point, so their condensation is a controlled Higgs effect), and — in the $\mathcal{N} = 4$ case below — an exact statement about the theory's behaviour under $g \to 1/g$.

These five tools are not independent axioms; they interlock. Figure 46.8 records the dependency: holomorphy (Tool 1) is the master constraint, symmetry (Tool 2) supplies the boundary conditions that make a holomorphic function unique, the NSVZ result (Tool 3) is holomorphy applied to the gauge coupling, BPS protection (Tool 4) is holomorphy and positivity applied to the spectrum, and duality (Tool 5) is the statement that the holomorphic data of two theories coincide. The single sentence that organizes the whole chapter is therefore: *holomorphy, sharpened by symmetry and protected by the BPS/index structure,*

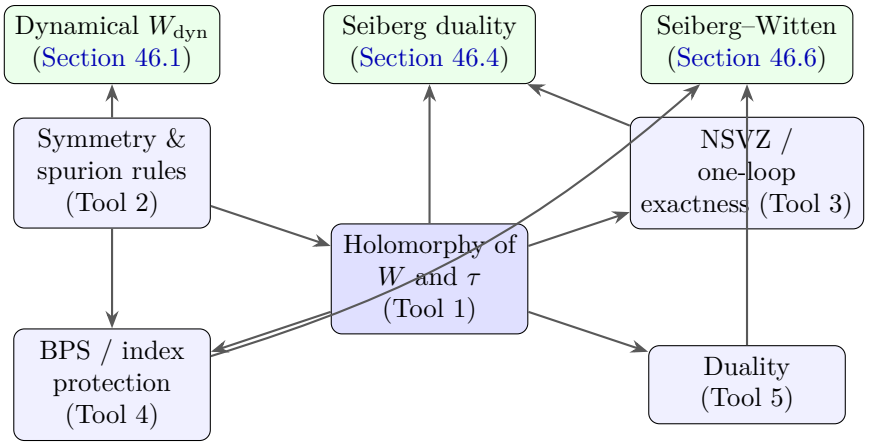


Figure 46.8: The exact-results toolkit. Holomorphy is the master constraint; symmetry supplies the boundary data that makes a holomorphic function unique; NSVZ exactness, BPS protection, and duality are holomorphy applied respectively to the coupling, the spectrum, and the comparison of two theories. The three escalating demonstrations of the chapter (top row) each combine a specific subset of the tools.

makes a class of physical quantities into rigid analytic functions, which duality lets us read off at whichever end of the coupling is weak.

Worked Example 46.24 (Selection rules reconstruct the Affleck–Dine–Seiberg superpotential)**.** The cleanest demonstration that "holomorphy + symmetry = exact answer up to a constant" is the reconstruction of the dynamically generated superpotential of $\mathcal{N}=1$ SQCD with gauge group $\mathrm{SU}(N_c)$ and $N_f < N_c$ flavours, whose existence and coefficient are established dynamically in Section 46.1 (see also [7]). Here we recover its functional form using nothing but Tools 1 and 2 — the methodological lesson of the chapter in miniature.

Setup. The matter is N_f chiral superfields Q^i in the fundamental $\mathbf{N_c}$ and N_f superfields $\widetilde{Q}_{\tilde{\jmath}}$ in the antifundamental $\overline{\mathbf{N_c}}$, $i,\tilde{\jmath}=1,\dots,N_f$. The only gauge-invariant chiral composite available to build a superpotential from is the meson matrix $M^i{}_{\tilde{\jmath}} = \widetilde{Q}_{\tilde{\jmath}} Q^i$ (for $N_f < N_c$ there are no baryons). The effective superpotential, by holomorphy (Tool 1), must be a holomorphic function of M and of the holomorphic scale, which enters only through the one-instanton factor Λ^{b_0} with $b_0 = 3N_c - N_f$ the one-loop coefficient. By gauge and flavour invariance the only flavour-singlet built from M is $\det M$, so

$$W_{\rm dyn} \;=\; c\,(\Lambda^{b_0})^{a}\,(\det M)^{p}, \tag{46.189}$$

with constants a, p, c to be fixed and det the $N_f \times N_f$ determinant.

The two relevant $U(1)$ symmetries and their charges. Consider the anomalous axial $U(1)_A$ under which both Q and $\widetilde{Q}$ carry charge $+1$, and a non-anomalous

$U(1)_R$ under which the quarks carry charge $R(Q) = R(\widetilde{Q}) = r$ to be determined. Their action on the building blocks is bookkeeping:

- Under $U(1)_A$: $M = \widetilde{Q}Q$ has charge 2, so $\det M$ (a product of N_f entries up to permutations) has charge $2N_f$. The scale is not inert: the $U(1)_A$ rotation is anomalous, and a one-instanton background has $2N_f$ fermionic zero modes (one from each of the $2N_f$ quark Weyl fermions), so the gauge-invariant instanton amplitude Λ^{b_0} carries $U(1)_A$ charge $+2N_f$. This is Tool 2's spurion rule: the anomaly is restored as an exact symmetry by charging Λ.
- Under $U(1)_R$: the measure $\mathrm{d}^2\theta$ has R-charge -2, so W must have R-charge $+2$. The condition that $U(1)_R$ be non-anomalous fixes r: the mixed R–$\mathrm{SU}(N_c)^2$ anomaly cancels when

$$C_A \cdot (1) \;+\; 2N_f\, T_F\,(r-1) \;=\; 0, \qquad C_A = N_c,\;\; T_F = \tfrac{1}{2}, \tag{46.190}$$

the first term from the gaugino (R-charge 1) and the second from the $2N_f$ quark fermions (R-charge $r-1$), each in a representation of index $T_F = \frac{1}{2}$. Solving, $r = \dfrac{N_f - N_c}{N_f}$. Hence $\det M$ has R-charge $2N_f\, r = 2(N_f - N_c)$, while Λ^{b_0} is R-neutral (the R-symmetry being non-anomalous by construction).

Imposing invariance. The ansatz Eq. (46.189) must be $U(1)_A$ invariant and carry R-charge 2:

$$U(1)_A: \quad a\,(2N_f) + p\,(2N_f) = 0 \;\;\Longrightarrow\;\; a = -p, \tag{46.191}$$

$$U(1)_R: \quad a\,(0) + p\,\big[2(N_f - N_c)\big] = 2 \;\;\Longrightarrow\;\; p = \frac{1}{N_f - N_c} = -\frac{1}{N_c - N_f}. \tag{46.192}$$

Therefore $a = -p = \dfrac{1}{N_c - N_f}$, and the form is completely fixed:

$$\boxed{W_{\mathrm{dyn}} \;=\; c\left(\frac{\Lambda^{3N_c - N_f}}{\det M}\right)^{1/(N_c - N_f)}}. \tag{46.193}$$

Holomorphy and two charge equations have determined a non-perturbative superpotential up to the single dimensionless number c. That number is not accessible to symmetry — it requires the actual dynamics — and is fixed in Section 46.1 (via an explicit one-instanton computation for $N_f = N_c - 1$ and consistency under integrating out a flavour for general N_f) to be $c = N_c - N_f$. The lesson is exactly the chapter's thesis: the entire *functional dependence* of a strong-coupling effect — here a potential that lifts the classical moduli space and, for $N_f < N_c$, drives the would-be vacuum to infinity — is legislated by holomorphy and symmetry, leaving dynamics to supply only an overall constant.

46.7.2 Theorem versus conjecture: an honest scope

A textbook that has just exhibited closed-form strong-coupling physics owes the reader a precise accounting of epistemic status. The results of this chapter do not all stand on the same footing, and conflating a theorem with a superbly tested conjecture would be exactly the false closure the book forbids. The following inventory is dated to the mid-2020s.

Theorems (proved, in this book or by the standard of mathematical physics). The perturbative *nonrenormalization theorem* for the superpotential — that W receives no corrections at any finite order of perturbation theory — is a theorem; it follows from holomorphy and the spurion selection rules, and it was argued diagrammatically in Chapter 45 and from holomorphy in Sections 44.7.8 and 46.1. The *BPS bound* $M \geq |Z|$ is a theorem: it is an algebraic inequality forced by positivity of the Hamiltonian on the central-extended supersymmetry algebra (Theorem 44.9, Eq. (44.115)), with no dynamical assumption whatsoever. The integrality and deformation-invariance of the *Witten index* $\mathrm{Tr}(-1)^F$ is a theorem in the same sense [520]. The *NSVZ exactness* of the holomorphic gauge coupling Eq. (46.188) is a theorem at the level of rigour of the anomaly–Jacobian argument that produces it [370]; its only soft point is the scheme-dependence relating holomorphic and canonical couplings, which is itself controlled and explained, not assumed.

Conjectures with overwhelming evidence (no proof). *Seiberg's electric–magnetic duality* (Section 46.4, [435]) is a statement about the infrared equivalence of two *different* ultraviolet Lagrangians. There is no proof that the $\mathrm{SU}(N_c)$ and $\mathrm{SU}(N_f - N_c)$ theories flow to the same fixed point. What there is, is a stack of nontrivial consistency checks that would be miraculous coincidences were the duality false: 't Hooft anomaly matching of all global symmetries between electric and magnetic descriptions (Chapter 37); matching of the chiral rings and of the gauge-invariant operators; consistency under every relevant and mass deformation (adding a quark mass on one side must reproduce the flow on the other, and it does, recursively in N_f); the agreement of the superconformal index, a protected partition function computable independently on both sides; and, where it has been attempted, lattice and numerical evidence for the conformal-window boundaries the duality predicts. The accumulation is so strong that no working physicist doubts it — but it is evidence, not a derivation, and we say so.

The *Seiberg–Witten solution* (Section 46.6, [436]) was originally obtained by an inspired ansatz: assume the low-energy coupling $\tau(u)$ is a holomorphic, modular section with exactly the singularities physically demanded (one at weak coupling from asymptotic freedom, two at strong coupling where a monopole and a dyon become massless), and show that these data fix a unique elliptic curve and hence the prepotential. The internal consistency is total — the monodromies

close, the BPS spectrum is consistent, the weak-coupling expansion reproduces the known one-loop and instanton corrections term by term — but the original argument *assumed* the singularity structure rather than deriving it. The status improved markedly in the 2000s: Nekrasov's instanton-counting partition function [361] computes the prepotential's instanton expansion directly from the path integral and reproduces the Seiberg–Witten result order by order, and Pestun's exact localization of the theory on the four-sphere [399] provides a first-principles evaluation of the same data. As of the mid-2020s the prepotential's weak-coupling expansion is therefore on extremely firm ground; the global analytic continuation to strong coupling and the identification of the singular points still rest on physical input rather than a theorem. It is a near-certainty, not a proof.

Caution

"Conjecture" here does not mean "speculative." Seiberg duality and the Seiberg–Witten solution are among the best-tested statements in theoretical physics, verified through independent windows (anomalies, deformations, indices, localization, lattice) that have no business agreeing unless the statements are true. The honest word is nonetheless *conjecture*: there is no mathematical derivation of the infrared equivalence (Seiberg) or of the assumed singularity structure (Seiberg–Witten). Distinguishing this from the theorems above — nonrenormalization, the BPS bound, the Witten index, NSVZ — is a matter of intellectual hygiene, not of confidence in the physics.

46.7.3 The forward map: where this technology re-enters

The exact-results toolkit is not the terminus of the book; it is the seed of three of the four frontier parts. Each picks up the technology of this chapter at a specific point and carries it forward, and flagging those entry points is the chapter's final obligation.

$\mathcal{N} = 4$ super-Yang–Mills and the gateway to holography. The maximally supersymmetric theory, $\mathcal{N} = 4$ SYM, is the most special point in the landscape mapped above: its β-function vanishes *identically* (the numerator of Eq. (46.188) is zero for the $\mathcal{N} = 4$ matter content, so the theory is exactly conformal for all g), and it enjoys an exact Montonen–Olive/S-duality under which the complexified coupling τ of Eq. (46.187) transforms by the modular group $SL(2,\mathbb{Z})$, $\tau \to (a\tau + b)/(c\tau + d)$, exchanging the gauge group with its Langlands dual and electric quanta with the monopoles. This is Tool 5 in its sharpest form, and it is exactly the structure that, in the large-N limit, becomes the strongest piece of evidence for the AdS/CFT correspondence: a finite, exactly conformal, S-self-dual gauge theory is precisely the object that can be dual to a gravitational

theory on anti-de Sitter space. Chapter 53 takes $\mathcal{N} = 4$ SYM and its large-N limit (Section 53.1) as the entry point to holography; the finiteness and duality established by the methods of this chapter are what make that correspondence credible rather than fanciful. *This is where the SUSY exact-results thread re-enters the book as holography.*

The conformal window and the bootstrap. The NSVZ relation Eq. (46.188) predicts that SQCD with enough flavours ($\frac{3}{2}N_c < N_f < 3N_c$ for SU(N_c)) flows to an interacting infrared fixed point — a nontrivial conformal field theory, the "conformal window," whose existence and boundaries are inferred from holomorphy and Seiberg duality. These superconformal theories are concrete, strongly coupled CFTs with computable protected spectra, and they are exactly the kind of input the conformal bootstrap of Part XIV takes as target data: the dimensions and OPE coefficients fixed by superconformal symmetry provide both checks and seeds for the bootstrap equations (Chapter 49). The exact-results technology hands the bootstrap a supply of rigorously characterized fixed points to test its machinery against. *This is where the conformal-window output of this chapter re-enters as the bootstrap.*

Planar $\mathcal{N} = 4$ and the amplituhedron. In the planar (large-N, fixed g^2N) limit, $\mathcal{N} = 4$ SYM is the most symmetric interacting quantum field theory known, and its scattering amplitudes exhibit a hidden positive-geometric structure: the integrand of the planar amplitude is computed as the canonical form of a region called the amplituhedron (Chapter 62, Section 62.6, [32]), with locality and unitarity emerging from positivity rather than being imposed. That this arena exists at all is owed to the special status — conformal symmetry, dual conformal symmetry, integrability — that the exact-results technology of this chapter both relies on and explains. *This is where planar $\mathcal{N} = 4$ re-enters as the modern amplitudes frontier.*

Dynamical supersymmetry breaking and the landscape. Finally, the Witten index (Tool 4) and the dynamically generated superpotentials (Tool 1+2, Theorem 46.24) are the working tools of model building at the frontier: whether a given supersymmetric theory breaks supersymmetry dynamically is decided by the index and by the runaway or stabilization of the ADS-type potential Eq. (46.193) [7], and the proliferation of metastable supersymmetry-breaking vacua feeds the still-unsettled question of the string landscape. This remains live, model-dependent, and phenomenologically open as of the mid-2020s — the one direction in the forward map that is genuinely a research frontier rather than an established bridge.

We close Part XIII here. The single lesson the reader should carry forward is the one this section has tried to make unmistakable: in the supersymmetric

corner of quantum field theory, holomorphy — sharpened by symmetry, protected by the BPS and index structure, and exploited through duality — turns non-perturbative dynamics into rigid analytic data, and it is exactly this control that the frontier inherits. The chapters on holography, the bootstrap, and the amplituhedron that follow are not a change of subject; they are what happens when the exact-results laboratory is taken to its limits.

Further reading

The foundational papers of the exact-results program are Seiberg's analysis of the space of vacua and holomorphy [434], his electric–magnetic duality [435], and the Seiberg–Witten solution of $\mathcal{N} = 2$ super-Yang–Mills [436]; the dynamically generated superpotential and dynamical supersymmetry breaking originate with Affleck, Dine, and Seiberg [7], and the exact gauge β-function with Novikov, Shifman, Vainshtein, and Zakharov [370]. Witten's index argument for the obstruction to supersymmetry breaking is [520], and the electric–magnetic duality conjecture for $\mathcal{N} = 4$ traces to Montonen and Olive [350]. The pedagogical lectures of Intriligator and Seiberg [275] remain the standard entry point to holomorphy, duality, and the conformal window; Terning's monograph [472], the third volume of Weinberg [502], and the superspace reference of Wess and Bagger [505] give textbook treatments. For the modern, near-rigorous status of the $\mathcal{N} = 2$ prepotential see Nekrasov's instanton counting [361] and Pestun's localization [399]. The forward connections are developed in Chapter 53 (holography, the original correspondence being Maldacena's [331]), Chapter 49 (the bootstrap), and Chapter 62 (the amplituhedron of Arkani-Hamed and Trnka [32]).

Part XIV

Conformal Field Theory and the Bootstrap

Chapter 47

Conformal Symmetry

Part VI left us with a striking promise and an unpaid debt. The Wilsonian renormalization group of Chapter 23 taught us that the long-distance physics of a quantum field theory is governed by the fixed points of its flow, that a fixed point is by construction invariant under the rescaling of all lengths, and that the critical exponents of a phase transition are the scaling dimensions of operators at such a point. What it did not establish is that the symmetry at a fixed point is anything larger than that one-parameter dilatation. This chapter makes good on the promise: it shows that, under hypotheses that hold in every interacting fixed point of physical interest, scale invariance is automatically enhanced to the full conformal group, and it builds from that enhancement the entire apparatus—primaries, the state–operator correspondence, symmetry-fixed correlators, the operator product expansion—on which the rest of Part XIV stands.

The development is deliberately self-contained, but it leans on four earlier strands of the book. From Chapter 7 we take the symmetric, conserved stress tensor and the Noether machinery that turns a spacetime symmetry into a conserved current; the improvement that trades the canonical stress tensor for a traceless one is the technical heart of the scale-to-conformal upgrade. From Chapter 3 we borrow the Lorentz and Poincaré algebras and the rotation generators $M_{\mu\nu}$, which sit inside the conformal algebra as a subalgebra and whose representation theory labels operators by spin. From Chapter 5 we use the Euclidean continuation and the contour and distributional technology that make radial quantization and the convergence of the OPE precise statements rather than formal manipulations; the state–operator map itself is most transparent in the Euclidean path integral of Chapter 10. And from Chapter 23 we inherit the very notion of a scaling dimension: the abstract weight Δ of a conformal primary and the RG scaling dimension of Theorem 23.22 are, at a fixed point, one and the same number, so the anomalous dimension that measured the breaking of classical scale invariance reappears here as the conformal weight of a local operator.

The arc of the chapter is from kinematics to representation theory to dynamics. We begin in flat space, distinguishing scale from conformal invariance through the local certificate of a traceless stress tensor, solving the conformal Killing equation to enumerate the conformal transformations in $d \geq 3$, and settling honestly the question of when scale invariance forces the larger symmetry. Exponentiating these generators yields the conformal group and its algebra, which we identify with $\mathfrak{so}(d+1,1)$. With the algebra in hand we pass to representations: local operators organize into multiplets built on primaries—lowest-weight states annihilated by the special conformal generator—graded by the dilatation eigenvalue, and unitarity imposes sharp lower bounds on the allowed dimensions. Symmetry alone then fixes the two- and three-point functions of primaries completely, up to a single number per triple, and reduces the four-point function to a function of two cross-ratios.

To make the operator product expansion a convergent statement rather than an asymptotic one we develop radial quantization: the foliation of Euclidean space by spheres, the dilatation operator as the Hamiltonian generating radial time, and the reflection-positive inner product. This furnishes the rigorous setting for the state–operator isomorphism, proved here in full—every state of the radially quantized theory is created by a unique local operator, so the spectrum of the dilatation Hamiltonian *is* the spectrum of operator dimensions. The OPE then follows from completeness, its coefficients are the three-point data, and it converges inside a finite radius; decomposing the four-point function in two channels exhibits the conformal blocks as eigenfunctions of the quadratic Casimir and sets up the crossing equation that Chapter 49 will elevate into the bootstrap. We close by checking the whole construction against the two theories ϕ^4 theory has carried this far: the free massless scalar, which realizes the spectrum exactly at the unitarity bound with its equation of motion appearing as a null descendant, and the Wilson–Fisher fixed point, whose anomalous dimensions—computed as RG quantities in Part VI—are recast here as the conformal data that feeds the bootstrap, the $d = 2$ Virasoro analysis of Chapter 48, and the critical-phenomena synthesis of Chapter 52.

47.1 From Scale Invariance to Conformal Invariance

The renormalization group taught us to read a quantum field theory as a trajectory in the space of couplings, and it located the physics of long distances at the endpoints of that trajectory: the fixed points, where the flow stalls and the theory ceases to change as we coarse-grain (Section 23.5). A fixed point is, by construction, a theory that has forgotten every length scale. There is no correlation length to set a unit of distance, no mass gap, no preferred resolution; the dimensionless couplings sit at stationary values and the dimensionful ones have been tuned away. Universality (Theorem 23.26) made this forgetting

physically consequential: whole basins of microscopically distinct theories flow to the same fixed point and share its critical exponents, which is why a lattice of magnetic spins and a fluid near its critical point obey the same power laws. The phrase that summarizes the endpoint is *scale invariance*: the fixed-point theory looks identical at every magnification.

This chapter takes that physical statement and turns it into a rigid algebraic structure. The first step is to recognize that "scale invariance" is not the end of the symmetry story but the beginning of it. A theory invariant under the single one-parameter group of dilatations turns out, under hypotheses that are mild and that we will make precise, to be invariant under a much larger finite-dimensional group — the *conformal group* — obtained by adjoining to the Poincaré and dilatation transformations a further d *special conformal* transformations. The enlargement is not optional decoration: it is forced, in the cases that matter, by the same locality and unitarity that organize every field theory in this book, and it is the reason a critical point carries so much more structure than a bare scaling law. Our task in this opening section is threefold. We first identify the local certificate of scale invariance — a stress tensor whose trace can be improved to zero — recalling the classical analysis of Section 7.7 and recasting it as a statement about RG endpoints. We then define a conformal transformation intrinsically, as a coordinate change that rescales the metric and therefore preserves angles, and we solve the equation that selects such transformations in flat space of dimension $d \geq 3$, enumerating them completely. Finally we state, and scope honestly, the theorem that promotes scale invariance to conformal invariance — a theorem that is settled in $d = 2$ and (under standard assumptions) $d = 4$, and that remains open in general dimension as of the mid-2020s. Throughout, the two concrete theories driving the chapter are the ones ϕ^4 theory has carried to this point: the massless free scalar, and the interacting Wilson–Fisher fixed point reached by tuning $\lambda\phi^4$ in $d = 4 - \epsilon$ (Chapter 52). The free scalar's improvement to a traceless stress tensor, built in Theorem 7.45, is the worked illustration we carry here.

47.1.1 The local certificate of scale invariance

Scale invariance is a statement about a symmetry, and by Noether's first theorem (Theorem 7.10) a continuous symmetry is detected locally by a conserved current. For dilatations that current is the *dilatation current*, assembled in Section 7.7.2 from the symmetric, conserved stress tensor $\Theta^{\mu\nu}$ as

$$D^\mu = x_\nu \, \Theta^{\mu\nu} + V^\mu, \tag{47.1}$$

the sum of an orbital piece $x_\nu \Theta^{\mu\nu}$ and the field *virial* V^μ recording the intrinsic scaling weight of the fields. Its divergence collapses, by the kinematic identity Eq. (7.188), to

$$\partial_\mu D^\mu = \Theta^\mu{}_\mu + \partial_\mu V^\mu, \tag{47.2}$$

so that conservation of D^μ — scale invariance — is equivalent to the trace of the stress tensor being a total derivative on shell, $\Theta^\mu{}_\mu = -\partial_\mu V^\mu$. When the virial is itself the divergence of a local tensor, $V^\mu = \partial_\nu \sigma^{\nu\mu}$, the same superpotential technology that symmetrized the canonical tensor into the Belinfante tensor can be turned to a second purpose: it absorbs the virial into the stress tensor and produces an *improved* tensor $\Theta^{\mu\nu}_{\rm c}$ that is symmetric, conserved, and *traceless on shell*, $\Theta^\mu_{{\rm c}\,\mu} = 0$ (Theorem 7.45). At that point the dilatation current is simply $D^\mu = x_\nu \Theta^{\mu\nu}_{\rm c}$ with no separate virial, and its conservation is manifest:

$$\partial_\mu (x_\nu \Theta^{\mu\nu}_{\rm c}) = \Theta^\mu_{{\rm c}\,\mu} = 0. \tag{47.3}$$

This is the dictionary we carry into the quantum theory. A scale-invariant theory is one possessing an improved stress tensor with vanishing trace; the trace $\Theta^\mu_{{\rm c}\,\mu}$ is the single local operator whose vanishing certifies that no length scale survives. At an RG fixed point the statement is exact at the quantum level: the beta functions vanish, and the operator equation for the trace — which away from a fixed point reads $\Theta^\mu_{{\rm c}\,\mu} = \sum_i \beta_i \mathcal{O}_i$ in terms of the running couplings (the quantum descendant of the classical formula Eq. (7.204)) — reduces to $\Theta^\mu_{{\rm c}\,\mu} = 0$. The trace, classically the sum over dimensionful couplings weighted by their mass dimensions (Eq. (7.201)), becomes quantum-mechanically the sum over beta functions; a fixed point is precisely where that sum vanishes. The remainder of this section asks what symmetry a vanishing trace actually generates, and the answer is larger than dilatations alone.

47.1.2 Conformal transformations as angle-preserving maps

We have approached scale transformations through the field-theoretic back door, as the symmetry whose current is built from the trace. It pays to step back and ask what dilatations are *geometrically*, because the geometric question has a sharper and more general answer. A dilatation $x^\mu \to \lambda x^\mu$ multiplies the flat metric by a constant: under it the line element $ds^2 = \eta_{\mu\nu}\, dx^\mu dx^\nu$ becomes $\lambda^2\, ds^2$. It rescales all lengths by the same factor everywhere, and in doing so it leaves *angles* untouched, since an angle is a ratio of lengths in which a common rescaling cancels. The natural geometric generalization is to allow the rescaling factor to depend on position. This is the definition of a conformal transformation.

Definition 47.1 (Conformal transformation)**.** Let (M, g) be a (pseudo-)Riemannian manifold. A diffeomorphism $x \mapsto x'(x)$ is a *conformal transformation* if it rescales the metric by a positive, position-dependent factor:

$$g'_{\mu\nu}(x') = \Omega(x)^{-2}\, g_{\mu\nu}(x), \tag{47.4}$$

where $g'_{\mu\nu}(x') = \dfrac{\partial x^\rho}{\partial x'^\mu}\dfrac{\partial x^\sigma}{\partial x'^\nu}\, g_{\rho\sigma}(x)$ is the transformed metric and $\Omega(x) > 0$ is the (local) *conformal factor* or *Weyl factor*, defined as the local stretching of lengths

under the forward map: the Jacobian factorizes as $\partial x'^{\mu}/\partial x^{\nu} = \Omega(x)\, R^{\mu}{}_{\nu}(x)$ with R orthogonal, so that $\partial x^{\rho}/\partial x'^{\mu} = \Omega(x)^{-1}(R^{-1})^{\rho}{}_{\mu}$ and the pullback metric above carries the reciprocal weight Ω^{-2}. The transformation is a (rigid) isometry when $\Omega \equiv 1$, a dilatation when Ω is a nonzero constant, and a genuinely local conformal map when Ω varies with x. In flat space we take $g_{\mu\nu} = \eta_{\mu\nu}$ throughout.

The name is justified by the following elementary but defining property: Eq. (47.4) is exactly the condition that the map preserve angles.

Proposition 47.2 (Conformal maps preserve angles)**.** *A diffeomorphism satisfies Eq. (47.4) for some $\Omega(x) > 0$ if and only if, at every point and for every pair of tangent vectors u, v, the quantity*

$$\cos\angle(u,v) \;=\; \frac{g(u,v)}{\sqrt{g(u,u)}\,\sqrt{g(v,v)}} \tag{47.5}$$

is unchanged by the map (with the obvious caveat that in Lorentzian signature this is literally an angle only for vectors of like timelike or like spacelike character; in general the invariant statement is that the map preserves the causal structure, sending null vectors to null vectors).

Proof. Push forward u, v by the differential of the map; their inner products are then evaluated with the transformed metric g'. If Eq. (47.4) holds, every inner product picks up the same factor Ω^{-2}: $g'(u,v) = \Omega^{-2} g(u,v)$, $g'(u,u) = \Omega^{-2} g(u,u)$, $g'(v,v) = \Omega^{-2} g(v,v)$. The factor cancels between numerator and denominator of Eq. (47.5), leaving $\cos\angle$ unchanged; in particular a null vector $g(u,u) = 0$ maps to a null vector. Conversely, suppose the ratio Eq. (47.5) is preserved for all u, v at a point. Taking $u = v$ shows that $g'(u,u)/g(u,u)$ is independent of u (it is the squared length ratio, and preservation of all angles forces it to be a single positive scalar, which we write as Ω^{-2} at that point), and the polarization identity then extends $g' = \Omega^{-2} g$ to mixed inner products. Hence Eq. (47.4) holds. □

Theorem 47.2 is the geometric heart of the subject: conformal transformations are the angle-preserving (equivalently, light-cone-preserving) coordinate changes, and the conformal factor $\Omega(x)$ measures the purely local stretching they apply. Figure 47.1 illustrates the simplest nontrivial example, the inversion $x^{\mu} \to x^{\mu}/x^2$, which bends straight coordinate lines into circles yet preserves the right angle at every crossing.

47.1.3 The conformal Killing equation

To enumerate the conformal transformations of flat space we work infinitesimally, as one always does to expose the Lie-algebra structure; the exponentiation to

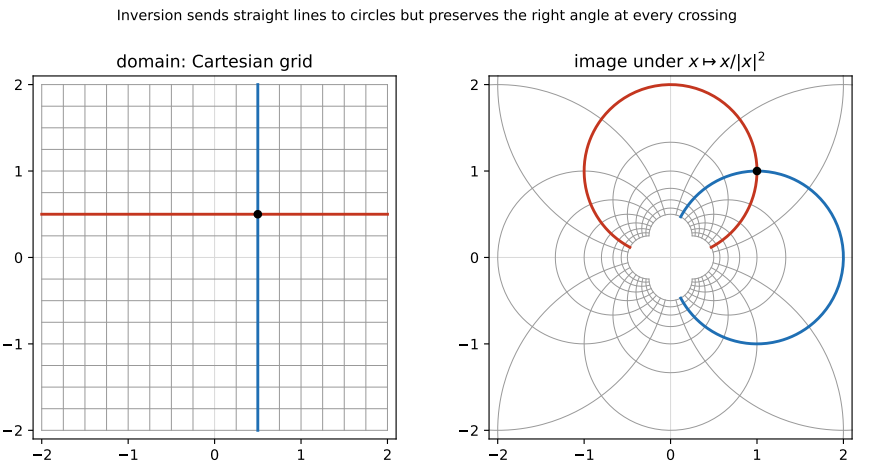


Figure 47.1: A conformal transformation that is neither a Poincaré transformation nor a dilatation: the inversion $x \mapsto x/|x|^2$ in the plane. The Cartesian grid (left) is mapped to a family of circles (right) — straightness and length are destroyed — yet the highlighted vertical and horizontal lines, orthogonal in the domain, map to circles that still meet orthogonally at the image of their crossing point. This is the content of Theorem 47.2: the conformal factor $\Omega(x) = 1/|x|^2$ rescales all lengths at a point by the same amount, so the angle between any two curves is preserved.

finite transformations is the business of the next section. Write the infinitesimal coordinate change as

$$x'^{\mu} = x^{\mu} + \xi^{\mu}(x) + \mathcal{O}(\xi^2), \tag{47.6}$$

with ξ^{μ} a vector field. (We denote the generating vector field ξ^{μ} throughout, matching the conformal Killing vector of Eq. (7.206); the symbol is interchangeable with the ε^{μ} of much of the CFT literature.) To first order $\partial x^{\rho}/\partial x'^{\mu} = \delta^{\rho}_{\mu} - \partial_{\mu}\xi^{\rho}$, so the transformed flat metric is

$$\eta'_{\mu\nu}(x') = (\delta^{\rho}_{\mu} - \partial_{\mu}\xi^{\rho})(\delta^{\sigma}_{\nu} - \partial_{\nu}\xi^{\sigma})\,\eta_{\rho\sigma} = \eta_{\mu\nu} - \partial_{\mu}\xi_{\nu} - \partial_{\nu}\xi_{\mu} + \mathcal{O}(\xi^2), \tag{47.7}$$

where indices on ξ are lowered with η. The defining condition Eq. (47.4), in the form $\eta'_{\mu\nu}(x') = \Omega(x)^{-2}\eta_{\mu\nu}$, requires this to be a local rescaling of $\eta_{\mu\nu}$; writing $\Omega(x)^{-2} = 1 - \omega(x)$ for some infinitesimal ω and matching at first order,

$$\partial_{\mu}\xi_{\nu} + \partial_{\nu}\xi_{\mu} = \omega(x)\,\eta_{\mu\nu}. \tag{47.8}$$

The scalar ω is fixed by tracing: contracting both sides with $\eta^{\mu\nu}$ gives $2\,\partial\cdot\xi = \omega\, d$, hence $\omega = \frac{2}{d}\,\partial\cdot\xi$. Substituting back, the vector fields generating conformal transformations are exactly those satisfying the *conformal Killing equation*

$$\boxed{\partial_{\mu}\xi_{\nu} + \partial_{\nu}\xi_{\mu} = \frac{2}{d}\,(\partial\cdot\xi)\,\eta_{\mu\nu}\ ,} \tag{47.9}$$

which is Eq. (7.206). Equivalently: a vector field is a conformal Killing vector when the symmetric part of its gradient is pure trace. The infinitesimal Weyl factor it generates is

$$\Omega(x) = 1 + \tfrac{1}{d}\,(\partial\cdot\xi) + \mathcal{O}(\xi^2). \tag{47.10}$$

Setting the right-hand side of Eq. (47.9) to zero recovers the ordinary Killing equation, whose solutions are the Poincaré generators (constant translations $\xi^\mu = a^\mu$ and antisymmetric Lorentz rotations $\xi^\mu = \omega^\mu{}_\nu x^\nu$, $\omega_{\mu\nu} = -\omega_{\nu\mu}$). The conformal Killing equation relaxes "isometry" to "isometry up to a local rescaling," and we now find that this relaxation admits exactly two new families of solutions in dimension three and above.

47.1.4 Complete solution in $d \geq 3$

Equation (7.207) exhibited four families of solutions of Eq. (47.9) and verified that each solves it. What was not shown there — and what is the central mathematical result of this section — is that these four families are the *complete* list when $d \geq 3$: there are no other conformal Killing vectors, and in particular the conformal group of flat space in $d \geq 3$ is finite-dimensional. We prove this now, by extracting from Eq. (47.9) a differential consequence so restrictive that it forces every conformal Killing vector to be at most quadratic in x.

Theorem 47.3 (Complete solution of the conformal Killing equation, $d \geq 3$)**.** *In flat space of dimension $d \geq 3$, every solution of the conformal Killing equation Eq. (47.9) is a polynomial in x of degree at most two, and has the form*

$$\xi^\mu(x) = \underbrace{a^\mu}_{\text{translation}} + \underbrace{\omega^\mu{}_\nu\, x^\nu}_{\text{Lorentz}} + \underbrace{\lambda\, x^\mu}_{\text{dilatation}} + \underbrace{2(b\cdot x)\, x^\mu - x^2\, b^\mu}_{\text{special conformal}}, \tag{47.11}$$

with a^μ, b^μ constant vectors, $\omega_{\mu\nu} = -\omega_{\nu\mu}$ a constant antisymmetric tensor, and λ a constant scalar. The space of solutions has dimension $\frac{1}{2}(d+1)(d+2)$.

Proof. Write $\sigma(x) \equiv \partial\cdot\xi$ for the divergence, so that the conformal Killing equation reads $\partial_\mu\xi_\nu + \partial_\nu\xi_\mu = \frac{2}{d}\,\sigma\,\eta_{\mu\nu}$. We extract constraints by differentiating and contracting.

Step 1: the divergence σ is at most linear. Apply ∂^μ to the conformal Killing equation. Using $\partial^\mu\partial_\mu = \Box$ and $\partial^\mu\xi_\mu = \sigma$,

$$\Box\xi_\nu + \partial_\nu\sigma = \frac{2}{d}\,\partial_\nu\sigma \quad\Longrightarrow\quad \Box\xi_\nu = \frac{2-d}{d}\,\partial_\nu\sigma. \tag{47.12}$$

Now apply $\Box$ to the conformal Killing equation and feed in Eq. (47.12). The left-hand side is $\Box(\partial_\mu\xi_\nu + \partial_\nu\xi_\mu) = \partial_\mu\Box\xi_\nu + \partial_\nu\Box\xi_\mu = \frac{2-d}{d}(\partial_\mu\partial_\nu + \partial_\nu\partial_\mu)\sigma = \frac{2(2-d)}{d}\,\partial_\mu\partial_\nu\sigma$, while the right-hand side is $\frac{2}{d}\,\eta_{\mu\nu}\Box\sigma$. Equating,

$$(2-d)\,\partial_\mu\partial_\nu\sigma = \eta_{\mu\nu}\,\Box\sigma. \tag{47.13}$$

Contracting Eq. (47.13) with $\eta^{\mu\nu}$ gives $(2-d)\Box\sigma = d\,\Box\sigma$, i.e. $2(1-d)\Box\sigma = 0$, so $\Box\sigma = 0$ for every $d \neq 1$. Feeding $\Box\sigma = 0$ back into Eq. (47.13) leaves $(2-d)\,\partial_\mu\partial_\nu\sigma = 0$, and *here is where* $d \geq 3$ *enters*: for $d \neq 2$ the prefactor is nonzero, so

$$\partial_\mu\partial_\nu\sigma = 0, \tag{47.14}$$

and σ is an affine function of x, $\sigma(x) = \sigma_0 + \sigma_\mu x^\mu$ with σ_0, σ_μ constant. (In $d = 2$ the prefactor $2-d$ vanishes and Eq. (47.13) imposes no constraint beyond $\Box\sigma = 0$; the divergence is then an arbitrary harmonic function and the solution space is infinite-dimensional. This is the algebraic root of the special status of two dimensions, taken up in Chapter 48.)

Step 2: ξ is at most quadratic. We obtain a closed expression for the second derivatives of ξ. Differentiate the conformal Killing equation and write the three index permutations

$$\partial_\rho\partial_\mu\xi_\nu + \partial_\rho\partial_\nu\xi_\mu = \tfrac{2}{d}\,\eta_{\mu\nu}\,\partial_\rho\sigma, \tag{47.15}$$
$$\partial_\mu\partial_\nu\xi_\rho + \partial_\mu\partial_\rho\xi_\nu = \tfrac{2}{d}\,\eta_{\nu\rho}\,\partial_\mu\sigma, \tag{47.16}$$
$$\partial_\nu\partial_\rho\xi_\mu + \partial_\nu\partial_\mu\xi_\rho = \tfrac{2}{d}\,\eta_{\rho\mu}\,\partial_\nu\sigma. \tag{47.17}$$

Form *Eq. (47.16) + Eq. (47.17) − Eq. (47.15)*. Since partial derivatives commute, the left-hand side collapses: the terms $\partial_\rho\partial_\mu\xi_\nu$ and $\partial_\rho\partial_\nu\xi_\mu$ from Eq. (47.15) cancel against the matching terms in Eq. (47.16) and Eq. (47.17), leaving $2\,\partial_\mu\partial_\nu\xi_\rho$. Hence

$$\partial_\mu\partial_\nu\xi_\rho = \frac{1}{d}\Big(\eta_{\nu\rho}\,\partial_\mu\sigma + \eta_{\rho\mu}\,\partial_\nu\sigma - \eta_{\mu\nu}\,\partial_\rho\sigma\Big). \tag{47.18}$$

By Eq. (47.14) the gradient $\partial_\mu\sigma = \sigma_\mu$ is constant, so the right-hand side of Eq. (47.18) is constant in x, and therefore $\partial_\lambda\partial_\mu\partial_\nu\xi_\rho = 0$: every third derivative of ξ vanishes. Consequently ξ^μ is a polynomial of degree at most two,

$$\xi_\mu(x) = a_\mu + b_{\mu\nu}\,x^\nu + c_{\mu\nu\rho}\,x^\nu x^\rho, \qquad c_{\mu\nu\rho} = c_{\mu\rho\nu}, \tag{47.19}$$

with $a_\mu, b_{\mu\nu}, c_{\mu\nu\rho}$ constant.

Step 3: identify the coefficients. We substitute Eq. (47.19) into the conformal Killing equation order by order in x.

Constant part. The constant a_μ enters ξ but not $\partial_\mu\xi_\nu$; it is unconstrained and supplies the d *translations* $\xi^\mu = a^\mu$.

Linear part. The linear term contributes $\partial_\alpha\xi_\beta\big|_{\text{lin}} = b_{\beta\alpha}$. The conformal Killing equation at zeroth order in x reads $b_{\nu\mu} + b_{\mu\nu} = \frac{2}{d}(\eta^{\alpha\beta}b_{\beta\alpha})\,\eta_{\mu\nu}$. Decompose $b_{\mu\nu}$ into its symmetric and antisymmetric parts, $b_{\mu\nu} = s_{\mu\nu} + \omega_{\mu\nu}$. The antisymmetric part drops out of the left-hand side, and matching forces $2s_{\mu\nu} = \frac{2}{d}(s^\alpha{}_\alpha)\eta_{\mu\nu}$, i.e. the symmetric part is pure trace, $s_{\mu\nu} = \lambda\,\eta_{\mu\nu}$ with $\lambda \equiv \frac{1}{d}\,s^\alpha{}_\alpha$. The antisymmetric $\omega_{\mu\nu}$ is unconstrained. Thus the linear term is

$$b_{\mu\nu}\,x^\nu = \omega_{\mu\nu}\,x^\nu + \lambda\,x_\mu, \tag{47.20}$$

supplying the $\frac{1}{2}d(d-1)$ *Lorentz rotations* (ω antisymmetric) and the single *dilatation* (λ).

Quadratic part. For the quadratic term, $\partial_\nu \xi_\mu\big|_{\text{quad}} = 2c_{\mu\nu\rho}x^\rho$ (using $c_{\mu\nu\rho} = c_{\mu\rho\nu}$), so its second derivative is $\partial_\mu\partial_\nu\xi_\rho = 2c_{\rho\mu\nu}$. Comparing with the master relation Eq. (47.18) and writing $\partial_\mu\sigma$ for the quadratic part as $\partial_\mu\sigma = 2c^\lambda{}_{\lambda\mu}$ (the divergence of the quadratic piece is $\sigma|_{\text{quad}} = 2c^\lambda{}_{\lambda\rho}x^\rho$), we get

$$2c_{\rho\mu\nu} = \frac{2}{d}\Big(\eta_{\nu\rho}\,c^\lambda{}_{\lambda\mu} + \eta_{\rho\mu}\,c^\lambda{}_{\lambda\nu} - \eta_{\mu\nu}\,c^\lambda{}_{\lambda\rho}\Big). \tag{47.21}$$

Defining the constant vector $b_\mu \equiv \frac{1}{d}\,c^\lambda{}_{\lambda\mu}$ this reads

$$c_{\rho\mu\nu} = \eta_{\nu\rho}\,b_\mu + \eta_{\rho\mu}\,b_\nu - \eta_{\mu\nu}\,b_\rho, \tag{47.22}$$

which is consistent: tracing Eq. (47.22) on $(\mu\nu)$ recovers $c^\lambda{}_{\lambda\rho} = \eta^{\mu\nu}c_{\rho\mu\nu} = b_\rho + d\,b_\rho - b_\rho = d\,b_\rho$, matching the definition of b_μ. Contracting Eq. (47.22) with $x^\mu x^\nu$,

$$\xi_\rho\big|_{\text{quad}} = c_{\rho\mu\nu}x^\mu x^\nu = 2(b{\cdot}x)\,x_\rho - x^2\,b_\rho, \tag{47.23}$$

the d *special conformal transformations* parametrized by b^μ.

Collecting the three parts gives Eq. (47.11). The parameter count is d (from a^μ) $+\,\frac{1}{2}d(d-1)$ (from ω) $+\,1$ (from λ) $+\,d$ (from b^μ), and

$$d + \tfrac{1}{2}d(d-1) + 1 + d = \tfrac{1}{2}d^2 + \tfrac{3}{2}d + 1 = \tfrac{1}{2}(d+1)(d+2), \tag{47.24}$$

as claimed. This is Eq. (7.208); in $d = 4$ it equals 15. □

The four families warrant a clean restatement, since the next section exponentiates them and assembles their commutators. Writing them as the action on coordinates:

$$\begin{aligned}
\text{translations } (d): &\quad \xi^\mu = a^\mu, && \partial{\cdot}\xi = 0,\\
\text{Lorentz } (\tfrac{1}{2}d(d-1)): &\quad \xi^\mu = \omega^\mu{}_\nu\,x^\nu, && \partial{\cdot}\xi = 0,\\
\text{dilatation } (1): &\quad \xi^\mu = \lambda\,x^\mu, && \partial{\cdot}\xi = \lambda\,d,\\
\text{special conformal } (d): &\quad \xi^\mu = 2(b{\cdot}x)\,x^\mu - x^2\,b^\mu, && \partial{\cdot}\xi = 2d\,(b{\cdot}x).
\end{aligned} \tag{47.25}$$

The first two families have vanishing divergence and are genuine isometries (Poincaré); the dilatation has constant divergence, generating a uniform rescaling; the special conformal transformation has divergence linear in x, generating a position-dependent rescaling. The Weyl factor $\Omega = 1 + \frac{1}{d}\partial{\cdot}\xi$ of Eq. (47.10) is, accordingly, 1 for Poincaré, $1+\lambda$ for the dilatation, and $1 + 2(b{\cdot}x)$ for the special conformal transformation.

Caution

Theorem 47.3 assumes $d \geq 3$ at the decisive step Eq. (47.13), where the factor $(2-d)$ is divided out. In $d = 2$ that factor vanishes and the argument breaks: the conformal Killing equation reduces to the Cauchy–Riemann equations, every holomorphic and antiholomorphic map is conformal, and the solution space is infinite-dimensional. The finite-dimensional conformal group enumerated here is therefore strictly a $d \geq 3$ phenomenon; the global part of the $d = 2$ group is the $\frac{1}{2}\cdot 3\cdot 4 = 6$-dimensional Möbius group $\mathfrak{so}(3,1)$ that survives as the subgroup of *globally* well-defined maps, but the local enhancement to the Virasoro algebra is the subject of Chapter 48. In $d = 1$ there is no metric structure to preserve and the notion degenerates.

47.1.5 One conserved current per conformal generator

The enumeration of Theorem 47.3 is a statement about flat-space geometry; its physical bite comes from pairing it with a traceless stress tensor. The mechanism, established in full in Theorem 7.48, is worth recalling in one line because it explains why a vanishing trace — the certificate of scale invariance — is automatically the certificate of the larger conformal symmetry. Given a symmetric, conserved stress tensor $\Theta^{\mu\nu}_{\mathrm{c}}$, contract it with any vector field ξ_ν to form

$$J^{\mu}_{(\xi)} = \Theta^{\mu\nu}_{\mathrm{c}}\,\xi_\nu, \qquad \partial_\mu J^{\mu}_{(\xi)} = \tfrac{1}{2}\,\Theta^{\mu\nu}_{\mathrm{c}}(\partial_\mu\xi_\nu + \partial_\nu\xi_\mu), \tag{47.26}$$

where conservation $\partial_\mu\Theta^{\mu\nu}_{\mathrm{c}} = 0$ killed the first term and symmetry replaced $\partial_\mu\xi_\nu$ by its symmetric part. For an isometry the symmetric derivative vanishes and $J^{\mu}_{(\xi)}$ is conserved — this is the standard packaging of the ten Poincaré currents into $\Theta^{\mu\nu}$. But when ξ is a conformal Killing vector, Eq. (47.9) makes the symmetric derivative pure trace, and Eq. (47.26) becomes

$$\partial_\mu J^{\mu}_{(\xi)} = \tfrac{1}{2}\,\Theta^{\mu\nu}_{\mathrm{c}}\cdot\tfrac{2}{d}(\partial\cdot\xi)\eta_{\mu\nu} = \tfrac{1}{d}\,(\partial\cdot\xi)\,\Theta^{\mu}_{\mathrm{c}\,\mu}. \tag{47.27}$$

The current is conserved *for every conformal Killing vector* precisely when the trace vanishes. This is the engine of the whole subject: tracelessness, contracted against the $\frac{1}{2}(d+1)(d+2)$ solutions of the conformal Killing equation, produces exactly that many conserved currents and hence that many conserved charges. The dilatation current Eq. (47.1) is the case $\xi^\mu = x^\mu$, for which $J^{\mu}_{(\xi)} = x_\nu\Theta^{\mu\nu}_{\mathrm{c}} = D^\mu$ and $\partial\cdot\xi = d$, so Eq. (47.27) reproduces $\partial_\mu D^\mu = \Theta^{\mu}_{\mathrm{c}\,\mu}$ — the identity Eq. (47.3) that we used to define scale invariance. The special conformal currents are the case $\xi^\mu = 2(b\cdot x)x^\mu - x^2 b^\mu$,

$$K^{\mu\rho} = 2\,x^\rho x_\nu\,\Theta^{\mu\nu}_{\mathrm{c}} - x^2\,\Theta^{\mu\rho}_{\mathrm{c}}, \qquad \partial_\mu K^{\mu\rho} = 2\,x^\rho\,\Theta^{\mu}_{\mathrm{c}\,\mu}, \tag{47.28}$$

which is Eq. (7.212): the same trace controls their conservation. The lesson is structural and worth stating once plainly: *the dilatation current and the special*

conformal currents are both built from the stress tensor with explicit factors of x, and both are conserved if and only if the trace $\Theta^{\mu}_{\mathrm{c}\,\mu}$ vanishes. Scale and conformal symmetry are not independent inputs at the level of currents — they are governed by the same single operator. What is *not* automatic is that a theory which is scale invariant (trace a total derivative) can always be improved so that the trace vanishes outright (conformal); that is the genuine content of the upgrade, and we turn to it last.

The concrete realization we carry is the massless free scalar. Its canonical stress tensor has trace $\frac{2-d}{2}(\partial\phi)^2 \neq 0$, but the improvement Eq. (7.194) with coefficient $\xi_d = \frac{d-2}{4(d-1)}$ produces the traceless tensor $\Theta^{\mu\nu}_{\mathrm{c}}$ of Theorem 7.45, whose trace vanishes identically on shell in every dimension (the worked computation Theorem 7.46 carries this with the full $\lambda\phi^4$ trace, which collapses to $\Theta^{\mu}_{\mathrm{c}\,\mu} = 0$ when $m = 0$ and $d = 4$). By Eq. (47.27) the free massless scalar therefore carries the complete set of $\frac{1}{2}(d+1)(d+2)$ conserved conformal currents: it is a conformal field theory, the simplest one, and the seed of ϕ^4 theory's quantum fixed-point description.

Worked Example 47.4 (Finite conformal factors of the four families)**.** The infinitesimal Weyl factor Eq. (47.10) is useful, but it pays to record the *finite* conformal factors of each family, both to confirm that the families integrate to genuine conformal maps and because the next section needs them to build the global group. We compute $\Omega(x)$ from Eq. (47.4) directly.

Translation $x'^{\mu} = x^{\mu} + a^{\mu}$ and Lorentz $x'^{\mu} = \Lambda^{\mu}{}_{\nu}x^{\nu}$. The Jacobians are δ^{μ}_{ν} and $\Lambda^{\mu}{}_{\nu}$ respectively; the latter satisfies $\Lambda^{\rho}{}_{\mu}\Lambda^{\sigma}{}_{\nu}\eta_{\rho\sigma} = \eta_{\mu\nu}$ by definition of the Lorentz group. In both cases $\eta'_{\mu\nu} = \eta_{\mu\nu}$, so $\Omega = 1$: these are isometries.

Dilatation $x'^{\mu} = \lambda\, x^{\mu}$. The Jacobian is $\partial x'^{\mu}/\partial x^{\nu} = \lambda\,\delta^{\mu}_{\nu}$, so $\partial x^{\rho}/\partial x'^{\mu} = \lambda^{-1}\delta^{\rho}_{\mu}$ and $\eta'_{\mu\nu} = \lambda^{-2}\eta_{\mu\nu}$. Reading off Eq. (47.4) in the form $\eta'_{\mu\nu} = \Omega^{-2}\eta_{\mu\nu}$, the conformal factor is the constant $\Omega = \lambda$ — the position-independent rescaling that defines a dilatation.

Inversion $x'^{\mu} = x^{\mu}/x^2$. This is the elementary discrete conformal map (Fig. 47.1); the finite special conformal transformations are built from it by conjugation with a translation in the next section, so its conformal factor is the key input. Differentiate:

$$\frac{\partial x'^{\mu}}{\partial x^{\nu}} = \frac{\delta^{\mu}_{\nu}}{x^2} - \frac{2x^{\mu}x_{\nu}}{(x^2)^2} = \frac{1}{x^2}\Big(\delta^{\mu}_{\nu} - \frac{2x^{\mu}x_{\nu}}{x^2}\Big). \tag{47.29}$$

The matrix $R^{\mu}{}_{\nu} \equiv \delta^{\mu}_{\nu} - 2x^{\mu}x_{\nu}/x^2$ is a reflection: it squares to the identity, $R^{\mu}{}_{\rho}R^{\rho}{}_{\nu} = \delta^{\mu}_{\nu}$, as one checks from $R^{\mu}{}_{\rho}R^{\rho}{}_{\nu} = \delta^{\mu}_{\nu} - 4x^{\mu}x_{\nu}/x^2 + 4x^{\mu}x_{\nu}(x\cdot x)/(x^2)^2 = \delta^{\mu}_{\nu}$, and it preserves η, $R^{\rho}{}_{\mu}R^{\sigma}{}_{\nu}\eta_{\rho\sigma} = \eta_{\mu\nu}$. Hence

$$\eta'_{\mu\nu} = \frac{\partial x^{\rho}}{\partial x'^{\mu}}\frac{\partial x^{\sigma}}{\partial x'^{\nu}}\eta_{\rho\sigma} = (x^2)^2\,(R^{-1})^{\rho}{}_{\mu}(R^{-1})^{\sigma}{}_{\nu}\,\frac{\eta_{\rho\sigma}}{(x^2)^2}\cdot\frac{1}{1} = (x^2)^2\,\eta_{\mu\nu}, \tag{47.30}$$

where we used that the inverse Jacobian is $x^2 R$ (since $R^2 = 1$) and that R preserves η. Reading this off through Eq. (47.4) in the form $\eta'_{\mu\nu} = \Omega^{-2}\eta_{\mu\nu}$, so that $\Omega^{-2} = (x^2)^2$, the inversion has the position-dependent conformal factor

$$\Omega(x) = \frac{1}{x^2}, \tag{47.31}$$

confirming Theorem 47.2: lengths are rescaled by the local factor $1/x^2$ (shrunk for $|x| > 1$, stretched for $|x| < 1$), which diverges at the origin and vanishes at infinity, but angles are untouched. That the inversion is conformal yet not connected to the identity (it has Jacobian determinant of fixed sign but maps the origin to infinity) is the first sign that the conformal group has nontrivial global structure — the theme of the next section.

47.1.6 Does scale invariance imply conformal invariance?

We have shown that a *traceless* stress tensor generates the full conformal group, and that scale invariance only guarantees the trace is a *total derivative*, $\Theta^\mu{}_\mu = \partial_\mu V^\mu$ with V^μ the virial current. The gap between these two statements is the precise content of the question in the heading. Improving the trace away requires that the virial itself be a total derivative, $V^\mu = \partial_\nu L^{\nu\mu}$ for some local tensor $L^{\nu\mu}$ (Eq. (7.190) is exactly this for the scalar, with $L^{\nu\mu} = \frac{\Delta}{2}\eta^{\nu\mu}\phi^2$); only then can a Belinfante-style improvement absorb it and leave a traceless tensor, as in Theorem 7.48. A scale-invariant theory whose virial current is *not* a total derivative — one carrying a genuine conserved virial charge — is scale but not conformally invariant. The logical possibility is real; the physical question is whether it is ever realized in a sensible (unitary, local) theory.

Theorem 47.5 (Scale invariance and the conformal upgrade, with scope)**.** *Let a theory be Poincaré invariant, scale invariant, and unitary, with a local stress tensor. Whether it is necessarily invariant under the full conformal group depends on the dimension:*

- $d = 2$ (settled). *Under the further standing assumptions of a discrete operator spectrum and a bounded-below energy, scale invariance implies conformal invariance. The virial can always be improved away.*
- $d = 4$ (settled under standard assumptions). *Scale invariance implies conformal invariance, established perturbatively to all orders and argued nonperturbatively under the assumptions standard for unitary local QFT (a sensible a-anomaly, no decoupled free sectors of the pathological kind).*
- general d (open, as of the mid-2020s). *No general theorem and no unitary counterexample is known. The known scale-invariant-but-not-conformal theories (free Maxwell theory in $d \neq 4$, theories of elasticity) are either non-unitary or fail the locality/discreteness hypotheses.*

The improving condition — virial current a total derivative — holds for every interacting fixed point relevant to this book, so for the theories we study (the free scalar and the Wilson–Fisher point of ϕ^4 theory) the upgrade is unconditional.

Discussion and proof sketch. The forward direction — conformal ⇒ scale — is trivial, since dilatations are a subgroup. The content is the converse, and the proof in each dimension proceeds by analyzing the virial current V^μ.

$d = 2$. The decisive tool is Zamolodchikov's c-theorem [531]: a monotone function $c(g)$ along RG flows, nonincreasing toward the infrared and stationary exactly at fixed points, built from two-point functions of the stress tensor. At a scale-invariant point the c-function is stationary, and Polchinski [404] showed that stationarity together with unitarity and a discrete spectrum forces the trace two-point function $\langle \Theta^\mu{}_\mu(x)\, \Theta^\nu{}_\nu(0)\rangle$ to vanish, hence (by reflection positivity, the Euclidean unitarity statement of Chapter 10) the operator $\Theta^\mu{}_\mu$ itself to be a total derivative that can be improved to zero. The virial charge is then trivial and conformal invariance follows. The full argument is the centerpiece of Polchinski's analysis; we will not reproduce its operator-algebra details here, but its logical skeleton — monotone c-function, stationary at the fixed point, forcing the trace to vanish — is exactly the $d = 2$ shadow of the $d = 4$ argument below, and the reader has all its ingredients (the stress-tensor two-point function and reflection positivity) from earlier chapters.

$d = 4$. The four-dimensional analogue of the c-function is the a-anomaly coefficient, and the statement that it decreases monotonically along unitary RG flows is the *a-theorem*, proven by Komargodski and Schwimmer [301] by studying the dilaton effective action that compactly encodes the response of the theory to a position-dependent scale transformation. Luty, Polchinski, and Rattazzi [329] and Dymarsky, Komargodski, Schwimmer, and Theisen [166] sharpened this into a proof that, in a unitary four-dimensional theory, a nonzero virial current obstructing the conformal upgrade would generate a unitarity-violating contribution to the dilaton amplitude; consistency forces the virial to be improvable away, so scale invariance implies conformal invariance. The argument is perturbatively airtight and, under the assumptions standard for a local unitary QFT, is regarded as settled nonperturbatively.

general d. For arbitrary dimension there is neither a proven monotone c/a-type function of the required strength nor a constructed unitary counterexample. The status is genuinely open: it is expected on physical grounds that unitary, local, scale-invariant theories are conformal in any dimension, but this remains a conjecture rather than a theorem. The standard reference surveying the evidence and the gaps is Nakayama's review [356]. □

Physical Insight

The distinction between scale and conformal invariance is subtle precisely because it almost never matters: in every interacting theory of physical interest the virial current is a total derivative and the upgrade is automatic. But the distinction is not empty. It marks the difference between a one-parameter rescaling symmetry and a $\frac{1}{2}(d+1)(d+2)$-parameter group, and the entire predictive machinery of conformal field theory — two- and three-point functions fixed by symmetry, the operator product expansion, the bootstrap — rests on the larger group. A merely scale-invariant theory would have far weaker constraints. That the endpoints of physical RG flows are conformal, and not just scale-invariant, is what makes the abstract phrase "critical phenomenon" carry the rigid algebraic skeleton this part of the book will exploit.

With the conformal transformations of flat space enumerated — the four families Eq. (47.25) packaged in the general conformal Killing vector Eq. (47.11) — and with the assurance that the scale-invariant theories of ϕ^4 theory are genuinely conformal, the next task is purely algebraic: to exponentiate these infinitesimal generators into finite transformations, read off their commutators, and recognize the resulting structure as the Lie algebra $\mathfrak{so}(d+1,1)$ in Euclidean signature (equivalently $\mathfrak{so}(d,2)$ in Lorentzian signature). That construction, together with the global structure exposed by the inversion of Theorem 47.4 and the conformal compactification it forces, is the subject of the section that follows.

47.2 The Conformal Group and Algebra

The previous section solved the conformal Killing equation and found, in every dimension $d > 2$, exactly four families of infinitesimal angle-preserving transformations: the d translations, the $\frac{1}{2}d(d-1)$ rotations (Lorentz transformations in Lorentzian signature), the single dilatation, and the d special conformal transformations (Section 47.1). Counting parameters gives $d + \frac{1}{2}d(d-1) + 1 + d = \frac{1}{2}(d+1)(d+2)$ independent generators — a *finite* number, in sharp contrast with the infinite-dimensional algebra that the same calculation produces in $d = 2$ and which Chapter 48 is devoted to. Our task now is to turn these infinitesimal motions into the finite transformations they integrate to, to read off the abstract Lie algebra they close into, and to recognize that algebra as nothing more exotic than the rotation algebra of a flat space with two extra dimensions. That last fact is the organizing principle of the entire subject: the awkward nonlinear action of special conformal transformations on $\mathbb{R}^d$ becomes an ordinary linear rotation once we lift to a $(d+2)$-dimensional embedding space, and the representation theory that the rest of this chapter builds is, at bottom, the representation theory of $\mathfrak{so}(d+1,1)$.

Throughout, $\eta_{\mu\nu}$ denotes the flat metric in whatever signature is in force: in Euclidean signature $\eta_{\mu\nu} = \delta_{\mu\nu}$, in Lorentzian signature $\eta_{\mu\nu} = \mathrm{diag}(+, -, \ldots, -)$ as fixed in Section 47.1. We write $x^2 \equiv \eta_{\mu\nu} x^\mu x^\nu$ and $b \cdot x \equiv \eta_{\mu\nu} b^\mu x^\nu$, with indices raised and lowered by η. The two signatures differ only at the very end, where they select between the real forms $\mathfrak{so}(d+1,1)$ and $\mathfrak{so}(d,2)$ of one complex algebra; until then every formula is signature-blind.

47.2.1 Finite transformations

Three of the four families exponentiate to transformations the reader already knows. A finite translation by a^μ is

$$x'^\mu = x^\mu + a^\mu, \tag{47.32}$$

the integral of the constant Killing vector $\delta x^\mu = a^\mu$. A finite rotation is

$$x'^\mu = \Lambda^\mu{}_\nu \, x^\nu, \qquad \Lambda^{\mathsf{T}} \eta \, \Lambda = \eta, \tag{47.33}$$

the exponential of an antisymmetric generator $\omega^\mu{}_\nu$, i.e. an element of $\mathrm{SO}(d)$ (Euclidean) or the Lorentz group $\mathrm{SO}^+(1, d-1)$ (Lorentzian), imported wholesale from Section 3.2. A finite dilatation is the uniform rescaling

$$x'^\mu = \lambda \, x^\mu, \qquad \lambda = e^\sigma > 0, \tag{47.34}$$

integrating $\delta x^\mu = \sigma \, x^\mu$. Each of Eqs. (47.32) to (47.34) rescales the metric by a constant or unit Weyl factor and so is manifestly angle-preserving.

The fourth family — special conformal transformations (SCTs) — is the one whose finite form is not obvious, and it is most cleanly obtained not by direct exponentiation but through a discrete transformation that is itself conformal: the *inversion*.

Definition 47.6 (Inversion)**.** The inversion I is the map

$$I: \quad x^\mu \longmapsto \frac{x^\mu}{x^2}, \tag{47.35}$$

defined on all x with $x^2 \neq 0$.

Inversion is an involution, $I \circ I = \mathrm{id}$, since $(x/x^2)^2 = x^2/(x^2)^2 = 1/x^2$ and so $I(x/x^2) = (x^\mu/x^2)/(1/x^2) = x^\mu$. It is not continuously connected to the identity — it cannot be reached by exponentiating any Killing vector — so it is a genuinely new, discrete element of the symmetry. That it is nonetheless conformal we verify in full in Section 47.2.5 below, where we compute its Weyl factor. Granting that for the moment, conjugating a translation by inversion produces a new continuous family of conformal maps, and this family is precisely the special conformal transformations.

Definition 47.7 (Special conformal transformation)**.** The special conformal transformation with parameter b^μ is the map $x \mapsto x'$ that becomes an ordinary translation by $-b^\mu$ after inversion,

$$\frac{x'^\mu}{x'^2} \;=\; \frac{x^\mu}{x^2} - b^\mu, \tag{47.36}$$

that is, $\mathrm{SCT}_b = I \circ T_b \circ I$ with T_b the translation $y \mapsto y - b$ acting on the inverted coordinate.

Proposition 47.8 (Finite special conformal transformation)**.** *The relation Eq. (47.36) solves to*

$$x'^\mu \;=\; \frac{x^\mu - b^\mu\, x^2}{1 - 2\, b\cdot x + b^2\, x^2}. \tag{47.37}$$

Proof. We compose the three maps in Theorem 47.7 explicitly. Start from x^μ and invert: $y^\mu = x^\mu/x^2$. Translate by $-b$: $y'^\mu = y^\mu - b^\mu = x^\mu/x^2 - b^\mu = (x^\mu - b^\mu x^2)/x^2$. Now compute the squared length of y',

$$\begin{aligned}(y')^2 &= \Big(\frac{x^\mu}{x^2} - b^\mu\Big)\Big(\frac{x_\mu}{x^2} - b_\mu\Big) = \frac{x^2}{(x^2)^2} - 2\,\frac{b\cdot x}{x^2} + b^2 \\ &= \frac{1 - 2\, b\cdot x + b^2 x^2}{x^2}.\end{aligned} \tag{47.38}$$

Inverting once more, $x'^\mu = y'^\mu/(y')^2$, and substituting both pieces,

$$x'^\mu = \frac{(x^\mu - b^\mu x^2)/x^2}{(1 - 2\, b\cdot x + b^2 x^2)/x^2} = \frac{x^\mu - b^\mu x^2}{1 - 2\, b\cdot x + b^2 x^2}, \tag{47.39}$$

the factors of x^2 cancelling, which is Eq. (47.37). □

Expanding Eq. (47.37) to first order in b confirms the identification with the SCT Killing vector found in the previous section: the numerator gives $x^\mu - b^\mu x^2$ and the denominator $(1 - 2b\cdot x)^{-1} \approx 1 + 2b\cdot x$, so

$$\delta x^\mu = x'^\mu - x^\mu = 2\,(b\cdot x)\, x^\mu - x^2\, b^\mu + O(b^2), \tag{47.40}$$

which is exactly the special conformal Killing vector. The sign of the parameter b^μ in Eq. (47.36) — the choice of translating by $-b$ rather than $+b$ on the inverted side — is a pure convention; it fixes the sign of the generator K_μ below, and we adopt the choice that makes Eq. (47.37) and Eq. (47.40) carry the standard minus signs.

Notation Watch

The denominator $1 - 2b \cdot x + b^2 x^2$ in Eq. (47.37) equals x'^2/x^2 up to the inversion bookkeeping; it vanishes on a $(d-1)$-dimensional locus, where the SCT pushes a finite point to infinity. This is not a pathology of the formula but a signal that the conformal group does not act on $\mathbb{R}^d$ as it stands — it acts on a compactification, the subject of Section 47.2.5. Until then, all finite SCT statements are understood to hold away from this locus.

47.2.2 The generators as differential operators

To extract the algebra we represent each family by the first-order differential operator that generates it on functions of x. For an infinitesimal motion $x \mapsto x + \delta x$ a scalar function transforms by $\delta f = -\delta x^\mu \, \partial_\mu f$; writing $\delta x^\mu = \xi_a^\mu \, \epsilon^a$ for the Killing vector of the a-th family with parameter ϵ^a, the Hermitian generator is $G_a = -i \, \xi_a^\mu \, \partial_\mu$, so that $\delta f = i \, \epsilon^a G_a f$. (The factor $-i$ makes G_a Hermitian and matches the translation convention $U(\mathbf{1}, a) = e^{ia \cdot P}$ of Eq. (3.136).) Reading the four Killing vectors — a^μ for translations, $\omega^\mu{}_\nu x^\nu$ for rotations, σx^μ for dilatations, and the SCT vector of Eq. (47.40) — this prescription gives

$$\boxed{\begin{aligned} P_\mu &= -i \, \partial_\mu, & M_{\mu\nu} &= i(x_\mu \partial_\nu - x_\nu \partial_\mu), \\ D &= -i \, x^\mu \partial_\mu, & K_\mu &= -i(2 \, x_\mu \, x^\nu \partial_\nu - x^2 \, \partial_\mu). \end{aligned}} \tag{47.41}$$

Here P_μ and $M_{\mu\nu}$ are exactly the momentum and Lorentz generators of Section 3.5, in their differential-operator realization on fields; one checks immediately, raising indices, that $M^{\mu\nu} = i(x^\mu \partial^\nu - x^\nu \partial^\mu)$ reproduces the Poincaré bracket $[M^{\mu\nu}, P^\rho] = i(\eta^{\nu\rho} P^\mu - \eta^{\mu\rho} P^\nu)$ of Eq. (3.148). The two new generators are the dilatation D, which simply counts the degree of homogeneity of a function in x, and the special conformal generator K_μ. These differential operators carry no information about the intrinsic spin or scaling weight of the field they act on; Section 47.3 reinstates those by adding the appropriate matrix and dimension terms. For deriving the algebra, the orbital pieces in Eq. (47.41) suffice, because the extra terms are x-independent constants that drop out of every commutator that determines a structure constant.

47.2.3 The conformal algebra

We now compute the brackets. The Poincaré sector $[P, P] = 0$, $[M, P]$, $[M, M]$ is imported unchanged from Eqs. (3.147) to (3.149); we derive the brackets involving D and K in full.

The dilatation grades momentum and special conformal generators. Using $[x^\nu\partial_\nu, \partial_\mu] = x^\nu\partial_\nu\partial_\mu - \partial_\mu(x^\nu\partial_\nu) = -\partial_\mu$ (the inhomogeneous term being $-\delta^\nu_\mu\partial_\nu$),

$$[D, P_\mu] = (-i)(-i)\,[x^\nu\partial_\nu, \partial_\mu] = -\,[x^\nu\partial_\nu, \partial_\mu] = \partial_\mu = i\,P_\mu. \tag{47.42}$$

The same calculation with K_μ in place of P_μ gives the opposite sign. Write $K_\mu = -i\,k_\mu$ with $k_\mu = 2x_\mu x^\nu\partial_\nu - x^2\partial_\mu$, a vector field homogeneous of degree $+1$ in x. Since $[x^\nu\partial_\nu, \cdot\,]$ measures degree, acting on a degree-$(+1)$ field it returns the field itself: $[x^\nu\partial_\nu, k_\mu] = +k_\mu$ (verified directly below). Hence

$$[D, K_\mu] = (-i)(-i)\,[x^\nu\partial_\nu, k_\mu] = -\,k_\mu = -\,i\,K_\mu. \tag{47.43}$$

To make the degree statement explicit, compute $[x^\nu\partial_\nu,\, x^2\partial_\mu]$ and $[x^\nu\partial_\nu,\, x_\mu x^\rho\partial_\rho]$ from the Leibniz rule; using $x^\nu\partial_\nu x^2 = 2x^2$ and $x^\nu\partial_\nu x_\mu = x_\mu$ one finds each equals the operator itself, so their combination k_μ has degree $+1$, confirming Eq. (47.43).

Equations Eqs. (47.42) and (47.43) are the heart of the whole subject: under the dilatation, P_μ *raises* a scaling eigenvalue by one unit and K_μ *lowers* it by one. This $\mathbb{Z}$-grading of the algebra by the eigenvalue of D is what the next section exploits to define primaries and descendants.

Special conformal and momentum generators. This is the one bracket whose structure is genuinely new, so we carry every step. With $P_\nu = -i\partial_\nu$ and $K_\mu = -i(2x_\mu x^\rho\partial_\rho - x^2\partial_\mu)$,

$$[K_\mu, P_\nu] = (-i)(-i)\,\Big[\,2x_\mu x^\rho\partial_\rho - x^2\partial_\mu\,,\,\partial_\nu\,\Big] = -\,\Big[\,2x_\mu x^\rho\partial_\rho - x^2\partial_\mu\,,\,\partial_\nu\,\Big]. \tag{47.44}$$

We need two elementary commutators. First, using $\partial_\nu x^2 = 2x_\nu$,

$$\Big[x^2\partial_\mu, \partial_\nu\Big] = x^2\partial_\mu\partial_\nu - \partial_\nu(x^2\partial_\mu) = -\,(\partial_\nu x^2)\,\partial_\mu = -\,2\,x_\nu\,\partial_\mu. \tag{47.45}$$

Second, with $\partial_\nu x_\mu = \eta_{\mu\nu}$ and $\partial_\nu x^\rho = \delta^\rho_\nu$,

$$\begin{aligned}[x_\mu x^\rho\partial_\rho, \partial_\nu] &= x_\mu x^\rho\partial_\rho\partial_\nu - \partial_\nu(x_\mu x^\rho\partial_\rho)\\ &= x_\mu x^\rho\partial_\rho\partial_\nu - \eta_{\mu\nu}x^\rho\partial_\rho - x_\mu\,\partial_\nu - x_\mu x^\rho\partial_\nu\partial_\rho\\ &= -\,\eta_{\mu\nu}\,x^\rho\partial_\rho - x_\mu\,\partial_\nu,\end{aligned} \tag{47.46}$$

the two second-derivative terms cancelling. Substituting Eqs. (47.45) and (47.46) into Eq. (47.44),

$$\begin{aligned}[K_\mu, P_\nu] &= -\Big[\,2(-\eta_{\mu\nu}x^\rho\partial_\rho - x_\mu\partial_\nu) - (-2x_\nu\partial_\mu)\Big]\\ &= 2\,\eta_{\mu\nu}\,x^\rho\partial_\rho + 2\,x_\mu\partial_\nu - 2\,x_\nu\partial_\mu.\end{aligned} \tag{47.47}$$

Finally re-express the right-hand side in generators. From Eq. (47.41), $x^\rho\partial_\rho = i\,D$ and $x_\mu\partial_\nu - x_\nu\partial_\mu = -i\,M_{\mu\nu}$, so

$$[K_\mu, P_\nu] = 2\,\eta_{\mu\nu}\,(iD) + 2\,(-i\,M_{\mu\nu}) = 2\,i\,(\eta_{\mu\nu}\,D - M_{\mu\nu}). \tag{47.48}$$

This is the defining nonlinear bracket of the conformal algebra: a momentum and a special conformal generator do not commute; their commutator closes onto the dilatation and the rotations, with the metric $\eta_{\mu\nu}$ tying the two together.

The remaining brackets. Three brackets remain, and each follows from a short argument rather than a long computation. That K_μ transforms as a vector under rotations,

$$[M_{\mu\nu}, K_\rho] = i(\eta_{\nu\rho}K_\mu - \eta_{\mu\rho}K_\nu), \tag{47.49}$$

holds because K_ρ carries a single lower vector index built covariantly from x and ∂; the bracket is fixed to have the same form as $[M_{\mu\nu}, P_\rho]$ of Eq. (3.148), and a one-line repetition of that computation with the operator k_ρ confirms the coefficient. The dilatation is a rotational scalar,

$$[M_{\mu\nu}, D] = 0, \tag{47.50}$$

since $D = -ix^\rho\partial_\rho$ is built from the rotation-invariant contraction $x^\rho\partial_\rho$. Finally the special conformal generators commute among themselves,

$$[K_\mu, K_\nu] = 0. \tag{47.51}$$

The cleanest proof of Eq. (47.51) is geometric and uses Theorem 47.7: a finite SCT is $I \circ T_b \circ I$, so the composition of two SCTs is

$$\mathrm{SCT}_{b_1} \circ \mathrm{SCT}_{b_2} = I T_{b_1} I\, I T_{b_2} I = I\,T_{b_1} T_{b_2}\,I = I\,T_{b_1+b_2}\,I = \mathrm{SCT}_{b_1+b_2}, \tag{47.52}$$

because the two inversions in the middle annihilate ($I^2 = \mathrm{id}$) and translations form an abelian group, $T_{b_1}T_{b_2} = T_{b_1+b_2} = T_{b_2}T_{b_1}$. The SCTs are therefore an abelian subgroup, isomorphic as a group to the translations conjugated by I, and their generators commute. We can now collect everything.

Theorem 47.9 (The conformal algebra in $d > 2$)**.** *The generators* $\{P_\mu, M_{\mu\nu}, D, K_\mu\}$ *of Eq. (47.41) close into the Lie algebra*

$$[M_{\mu\nu}, M_{\rho\sigma}] = i(\eta_{\nu\rho}M_{\mu\sigma} - \eta_{\mu\rho}M_{\nu\sigma} - \eta_{\nu\sigma}M_{\mu\rho} + \eta_{\mu\sigma}M_{\nu\rho}), \tag{47.53a}$$

$$[M_{\mu\nu}, P_\rho] = i(\eta_{\nu\rho}P_\mu - \eta_{\mu\rho}P_\nu), \qquad [M_{\mu\nu}, K_\rho] = i(\eta_{\nu\rho}K_\mu - \eta_{\mu\rho}K_\nu), \tag{47.53b}$$

$$[D, P_\mu] = i\,P_\mu, \qquad [D, K_\mu] = -\,i\,K_\mu, \tag{47.53c}$$

$$[K_\mu, P_\nu] = 2\,i\,(\eta_{\mu\nu}D - M_{\mu\nu}), \tag{47.53d}$$

with all other brackets — $[P_\mu, P_\nu]$, $[K_\mu, K_\nu]$, $[M_{\mu\nu}, D]$, $[D, M_{\mu\nu}]$ — vanishing. This algebra has dimension $\frac{1}{2}(d+1)(d+2)$.

Proof. The brackets are Eqs. (3.148), (3.149), (47.42), (47.43), (47.48) and (47.49) together with the vanishing brackets Eqs. (47.50) and (47.51) and $[P_\mu, P_\nu] = 0$ from Eq. (3.147). Closure is manifest: every right-hand side is a linear combination of the same generators. The dimension count adds the $\frac{1}{2}d(d-1)$ generators $M_{\mu\nu}$, the d generators P_μ, the d generators K_μ, and the single D:

$$\tfrac{1}{2}d(d-1)+d+d+1 = \tfrac{1}{2}(d^2-d+4d+2) = \tfrac{1}{2}(d^2+3d+2) = \tfrac{1}{2}(d+1)(d+2). \qquad \square$$

Physical Insight

The Poincaré algebra of Section 3.5 sits inside Eq. (47.53) as the subalgebra $\{P_\mu, M_{\mu\nu}\}$; adding the dilatation D gives the algebra of *scale-invariant* theories, with P graded but K still absent. The single bracket Eq. (47.53d) is what distinguishes a conformal theory from a merely scale-invariant one: it forces the special conformal generators into the algebra the moment one demands closure with both P and D present. This is the algebraic face of the statement, made physical in Section 47.1, that under mild hypotheses scale invariance is promoted to full conformal invariance — the extra symmetry is not optional once the bracket structure is taken seriously.

47.2.4 Isomorphism with $\mathfrak{so}(d+1,1)$

The dimension $\frac{1}{2}(d+1)(d+2)$ is exactly the dimension of the rotation algebra $\mathfrak{so}(p,q)$ on a $(d+2)$-dimensional space, $p+q=d+2$, whose generators $J_{AB} = -J_{BA}$ $(A,B = 0,1,\ldots,d+1)$ number $\frac{1}{2}(d+2)(d+1)$. This is not a numerical coincidence. We now construct an explicit isomorphism.

Introduce a $(d+2)$-dimensional space with coordinates X^A and flat metric

$$G_{AB} = \mathrm{diag}(\eta_{\mu\nu}, +1, -1), \qquad A,B \in \{0,\ldots,d-1,\, d,\, d+1\}, \tag{47.54}$$

so that the two new directions X^d and X^{d+1} are respectively spacelike and timelike with respect to G. The rotation generators of this space obey the $\mathfrak{so}(p,q)$ algebra, which is the $(d+2)$-dimensional version of the Lorentz bracket Eq. (3.135),

$$[J_{AB}, J_{CD}] = i(G_{AD}J_{BC} + G_{BC}J_{AD} - G_{AC}J_{BD} - G_{BD}J_{AC}). \tag{47.55}$$

Theorem 47.10 (Conformal algebra as $\mathfrak{so}(d+1,1)$)**.** *The assignment*

$$M_{\mu\nu} = J_{\mu\nu}, \qquad P_\mu = J_{\mu,d} + J_{\mu,d+1}, \qquad K_\mu = J_{\mu,d} - J_{\mu,d+1}, \qquad D = J_{d+1,d}, \tag{47.56}$$

equivalently $J_{\mu,d} = \frac{1}{2}(P_\mu + K_\mu)$ and $J_{\mu,d+1} = \frac{1}{2}(P_\mu - K_\mu)$, is a Lie-algebra isomorphism from Eq. (47.55) onto the conformal algebra Eq. (47.53). In Euclidean signature $\eta_{\mu\nu} = \delta_{\mu\nu}$ the metric Eq. (47.54) has signature $(d+1,1)$, so the conformal algebra is $\mathfrak{so}(d+1,1)$; in Lorentzian signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,\ldots,-)$ it is $\mathfrak{so}(d,2) \cong \mathfrak{so}(2,d)$.

Proof. We verify that the brackets Eq. (47.55) reproduce Eq. (47.53) under the dictionary Eq. (47.56); since both algebras have the same dimension and the map is a linear bijection, this establishes the isomorphism. The purely $\mu\nu$ brackets are immediate: with A, B, C, D all in $\{0, \ldots, d-1\}$, Eq. (47.55) is verbatim the Lorentz bracket Eq. (47.53a), and $M_{\mu\nu} = J_{\mu\nu}$.

Dilatation grading. Take $D = J_{d+1,d}$ and compute its bracket with $J_{\mu,d}$, using Eq. (47.55) with $(A, B, C, D) = (d+1, d, \mu, d)$. The only nonzero metric contraction is $G_{d+1,d+1} = -1$ in the term $-G_{BD}J_{AC} = -G_{dd}J_{d+1,\mu}$; with $G_{dd} = +1$,

$$[J_{d+1,d}, J_{\mu,d}] = i(-G_{dd}J_{d+1,\mu}) = -\,i\,J_{d+1,\mu} = +\,i\,J_{\mu,d+1}. \tag{47.57}$$

The analogous computation with $(A, B, C, D) = (d+1, d, \mu, d+1)$ singles out $-G_{BD}J_{AC} = -G_{d,d+1}\ldots$, more carefully the surviving term is $G_{AD}J_{BC} = G_{d+1,d+1}J_{d,\mu} = -J_{d,\mu} = +J_{\mu,d}$, giving

$$[J_{d+1,d}, J_{\mu,d+1}] = +\,i\,J_{\mu,d}. \tag{47.58}$$

Adding and subtracting,

$$[D, P_\mu] = [J_{d+1,d}, J_{\mu,d} + J_{\mu,d+1}] = i(J_{\mu,d+1} + J_{\mu,d}) = i\,P_\mu, \tag{47.59}$$

$$[D, K_\mu] = [J_{d+1,d}, J_{\mu,d} - J_{\mu,d+1}] = i(J_{\mu,d+1} - J_{\mu,d}) = -\,i\,K_\mu, \tag{47.60}$$

reproducing Eq. (47.53c).

The $[K, P]$ bracket. Expand using bilinearity,

$$[K_\mu, P_\nu] = [J_{\mu,d} - J_{\mu,d+1},\ J_{\nu,d} + J_{\nu,d+1}] = [J_{\mu d}, J_{\nu d}] + [J_{\mu d}, J_{\nu,d+1}] - [J_{\mu,d+1}, J_{\nu d}] - [J_{\mu,d+1}, J_{\nu,d} \tag{47.61}$$

Each of the four is read off from Eq. (47.55), keeping only the nonzero metric components $G_{\mu\nu} = \eta_{\mu\nu}$, $G_{dd} = +1$, $G_{d+1,d+1} = -1$:

$$[J_{\mu d}, J_{\nu d}] = i(-G_{dd}J_{\mu\nu}) = -\,i\,M_{\mu\nu}, \tag{47.62}$$

$$[J_{\mu d}, J_{\nu,d+1}] = i(-G_{\mu\nu}J_{d,d+1}) = i\,\eta_{\mu\nu}\,J_{d+1,d} = i\,\eta_{\mu\nu}\,D, \tag{47.63}$$

$$[J_{\mu,d+1}, J_{\nu d}] = i(-G_{\mu\nu}J_{d+1,d}) = -\,i\,\eta_{\mu\nu}\,D, \tag{47.64}$$

$$[J_{\mu,d+1}, J_{\nu,d+1}] = i(-G_{d+1,d+1}J_{\mu\nu}) = +\,i\,M_{\mu\nu}. \tag{47.65}$$

(In the second and third lines we used $J_{d,d+1} = -J_{d+1,d} = -D$.) Assembling Eq. (47.61),

$$[K_\mu, P_\nu] = -iM_{\mu\nu} + i\eta_{\mu\nu}D - (-i\eta_{\mu\nu}D) - iM_{\mu\nu} = 2\,i\,(\eta_{\mu\nu}D - M_{\mu\nu}), \tag{47.66}$$

which is Eq. (47.53d). The vector brackets Eq. (47.53b) follow identically from the mixed $\mu\nu$–d and $\mu\nu$–$(d+1)$ components of Eq. (47.55), and $[P, P] = [K, K] = 0$ follow from $[J_{\mu d} + J_{\mu,d+1}, J_{\nu d} + J_{\nu,d+1}]$ collapsing once the $M_{\mu\nu}$ contributions cancel against each other (the $+iM$ from the $(d+1, d+1)$ term cancels the $-iM$ from the (d, d) term). All brackets of Eq. (47.53) are thus reproduced.

Signature. The metric Eq. (47.54) appends one $+$ and one $-$ to η. Euclidean $\eta = \delta$ has d plus signs, so G has $d+1$ plus and one minus: signature $(d+1,1)$, the algebra $\mathfrak{so}(d+1,1)$. Lorentzian $\eta = \mathrm{diag}(+,-,\dots,-)$ has one plus and $d-1$ minus, so G has two plus and d minus: signature $(2,d)$, the algebra $\mathfrak{so}(2,d) \cong \mathfrak{so}(d,2)$. □

Physical Insight

The same complex Lie algebra $\mathfrak{so}(d+2,\mathbb{C})$ underlies both signatures; the Wick rotation of Chapter 5 that turns a Lorentzian QFT into a Euclidean one is, on the conformal side, exactly the move between the real forms $\mathfrak{so}(d,2)$ and $\mathfrak{so}(d{+}1,1)$. This is why the Euclidean bootstrap of Chapters 49 and 50 and the Lorentzian, causal statements of Chapter 51 are two faces of one structure: they share an algebra and differ only in which generators are taken anti-Hermitian. The maximal compact subalgebra in the Euclidean case is $\mathfrak{so}(d) \oplus \mathfrak{so}(2)$ — spatial rotations together with the rotation in the (X^d, X^{d+1}) plane generated by D — and its two Casimir labels, spin and the eigenvalue of that $\mathfrak{so}(2)$, are the conformal analogue of the mass and spin that label Poincaré representations in Theorem 3.19.

47.2.5 Inversion preserves angles; the embedding null cone

Two debts remain: to prove that inversion is conformal (used freely in Section 47.2.1), and to give the linear realization of the conformal action promised in the introduction. Both are settled by the same embedding-space geometry.

Inversion is conformal. Under $x'^\mu = x^\mu/x^2$ the Jacobian is

$$\frac{\partial x'^\mu}{\partial x^\nu} = \frac{\delta^\mu_\nu}{x^2} - \frac{2\,x^\mu x_\nu}{(x^2)^2} = \frac{1}{x^2}\,R^\mu{}_\nu(x), \qquad R^\mu{}_\nu(x) \equiv \delta^\mu_\nu - \frac{2\,x^\mu x_\nu}{x^2}. \tag{47.67}$$

The matrix $R(x)$ is the reflection across the hyperplane orthogonal to x, and it is an orthogonal transformation for the metric η: contracting,

$$\begin{aligned} \eta_{\mu\nu}\,R^\mu{}_\rho\,R^\nu{}_\sigma &= \Big(\delta^\mu_\rho - \frac{2x^\mu x_\rho}{x^2}\Big)\Big(\eta_{\mu\sigma} - \frac{2x_\mu x_\sigma}{x^2}\Big) \\ &= \eta_{\rho\sigma} - \frac{2x_\rho x_\sigma}{x^2} - \frac{2x_\rho x_\sigma}{x^2} + \frac{4\,(x^\mu x_\mu)\,x_\rho x_\sigma}{(x^2)^2} \\ &= \eta_{\rho\sigma} - \frac{4x_\rho x_\sigma}{x^2} + \frac{4\,x_\rho x_\sigma}{x^2} = \eta_{\rho\sigma}, \end{aligned} \tag{47.68}$$

where the last cross term used $x^\mu x_\mu = x^2$. Therefore the metric pulled back by inversion is

$$\eta_{\mu\nu}\,\mathrm{d}x'^\mu\,\mathrm{d}x'^\nu = \frac{1}{(x^2)^2}\,\eta_{\rho\sigma}\,R^\mu{}_\rho R^\nu{}_\sigma\,\mathrm{d}x^\rho\,\mathrm{d}x^\sigma = \frac{1}{(x^2)^2}\,\eta_{\rho\sigma}\,\mathrm{d}x^\rho\,\mathrm{d}x^\sigma, \tag{47.69}$$

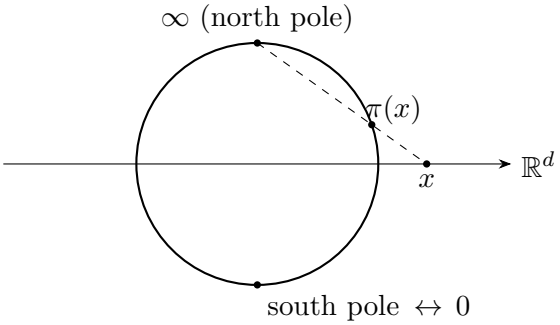


Figure 47.2: Stereographic projection identifies $\mathbb{R}^d \cup \{\infty\}$ with the sphere S^d (shown in cross-section). A point $x \in \mathbb{R}^d$ maps to the second intersection $\pi(x)$ of the line through the north pole and x with the sphere; the north pole itself is the adjoined point at infinity, and the origin maps to the south pole. On S^d the conformal group acts without singularity: inversion is the smooth rotation exchanging the two poles.

a pure Weyl rescaling by $\Omega(x)^2 = (x^2)^{-2}$, i.e. $\Omega(x) = 1/x^2$. Inversion multiplies all lengths at x by the same factor and so leaves all angles fixed: it is conformal, as claimed. Because every finite conformal transformation is a product of the elements in Section 47.2.1 and inversions, Eq. (47.69) completes the proof that the maps enumerated there are the full conformal group; Theorem 47.11 carries the Weyl factor for a general SCT.

Compactification. The inversion $x \mapsto x/x^2$ is singular at the origin, where $x^2 = 0$ and the image runs off to infinity; conversely it has no preimage of a finite point for the would-be image of the origin. Likewise the finite SCT Eq. (47.37) sends the locus $1 - 2b \cdot x + b^2 x^2 = 0$ to infinity. The conformal group thus fails to act on $\mathbb{R}^d$ as a set of bijections. The cure is to adjoin a single point at infinity. In Euclidean signature the resulting space $\mathbb{R}^d \cup \{\infty\}$ is the d-sphere S^d, via stereographic projection (Fig. 47.2); inversion becomes a smooth rotation of S^d exchanging the two poles, with no singularity anywhere. The conformal group acts on S^d as its group of (orientation-preserving, with inversion) conformal automorphisms. This is the precise sense in which the conformal group is $\mathrm{O}(d+1,1)$ acting on the compactified Euclidean space, the connected component giving the transformations reachable from the identity and the inversion living in a disconnected component.

The projective null cone. The cleanest carrier of the conformal action is the embedding space of Section 47.2.4 itself. Consider in $\mathbb{R}^{d+2}$ with metric G of Eq. (47.54) the null cone

$$X^2 \equiv G_{AB}\, X^A X^B = 0, \qquad X \neq 0, \tag{47.70}$$

and identify points along the same ray, $X^A \sim \lambda X^A$ for $\lambda > 0$ (this is the *projective* null cone). It is convenient to use light-cone combinations of the two extra coordinates, $X^{\pm} = X^d \pm X^{d+1}$, in which

$$X^2 = G_{\mu\nu}X^\mu X^\nu + (X^d)^2 - (X^{d+1})^2 = \eta_{\mu\nu}X^\mu X^\nu + X^+X^-. \tag{47.71}$$

On the cone we may use the projective rescaling to fix $X^+ = 1$; the null condition Eq. (47.71) then determines $X^- = -\eta_{\mu\nu}X^\mu X^\nu = -X^2_{\text{(d-dim)}}$. Writing the remaining d components as $X^\mu = x^\mu$, we obtain the *Poincaré section*

$$X^A(x) = (\,X^+,\, X^-,\, X^\mu\,) = (\,1,\, -x^2,\, x^\mu\,), \tag{47.72}$$

a one-to-one parametrization of the cone (modulo rays) by the points $x^\mu \in \mathbb{R}^d$, with $X^+ = 0$ furnishing exactly the adjoined point at infinity. A linear $\mathrm{O}(d+1,1)$ rotation $X \mapsto \Lambda X$ preserves Eq. (47.71), hence maps the cone to itself; it generally moves a representative off the section $X^+ = 1$, and the projective rescaling that returns it to the section is what makes the induced action on x^μ nonlinear. The *linear* $\mathfrak{so}(d+1,1)$ action on X *is* the conformal action on x.

To see inversion in these terms, note that I corresponds to the linear reflection $X^+ \leftrightarrow X^-$. On the section Eq. (47.72) this sends $(1, -x^2, x^\mu) \mapsto (-x^2, 1, x^\mu)$; rescaling by the positive factor $1/(-x^2)$ to restore $X^+ = 1$ gives $(1,\, -1/x^2,\, x^\mu/x^2)$... up to the overall sign of the timelike pair, which is the projective identification — and reading off the last d components recovers $x^\mu \mapsto x^\mu/x^2$, the inversion. This is the precise statement that inversion maps the null cone to itself and reduces, on the section, to Eq. (47.35). The embedding-space formalism, which the correlator computations of Section 47.4 and the conformal-block technology of Chapter 49 take as their starting point, is therefore the natural *linear* home of the nonlinearly-realized conformal symmetry: every tensorial conformal covariant on $\mathbb{R}^d$ lifts to an ordinary $\mathfrak{so}(d+1,1)$ tensor on the cone.

Caution

The two added directions are not interchangeable: X^d is spacelike and X^{d+1} timelike with respect to G, and the dilatation $D = J_{d+1,d}$ is a *boost* in that mixed plane, not a compact rotation. Its real eigenvalues are the scaling dimensions of operators; the compact rotation that pairs with it in the maximal compact subalgebra is the one used in *radial quantization* (Section 47.5), where D becomes the Hamiltonian generating dilatations about a chosen origin. Conflating the boost D with a compact rotation is the most common error in setting up the state–operator map.

Worked Example 47.11 (The Weyl factor of a special conformal transformation)**.** We confirm directly that the finite SCT Eq. (47.37) is conformal and extract its Weyl factor, both as a check on Theorem 47.7 and because the factor $\Omega(x)$ is the object every conformal correlator is built from. Write the denominator

as $\gamma(x) \equiv 1 - 2\,b\cdot x + b^2x^2$, so that $x'^\mu = \gamma^{-1}(x^\mu - b^\mu x^2)$. The fastest route uses Theorem 47.7: a finite SCT is $I \circ T_{-b} \circ I$, and the Weyl factor of a composition is the product of the factors. Inversion has factor $\Omega_I(x) = 1/x^2$ by Eq. (47.69); translation has factor 1; and the second inversion acts at the translated point $y' = x/x^2 - b$, contributing $\Omega_I(y') = 1/(y')^2$. By Eq. (47.38), $(y')^2 = \gamma(x)/x^2$. Multiplying the three factors and tracking that the metric is rescaled once by each inversion,

$$\Omega_{\mathrm{SCT}}(x) = \Omega_I(x)\cdot 1\cdot \frac{1}{\Omega_I(y')}\Big|_{\text{reassembled}} = \frac{1/x^2}{(y')^2} = \frac{1/x^2}{\gamma(x)/x^2} = \frac{1}{\gamma(x)} = \frac{1}{1 - 2\,b\cdot x + b^2x^2}. \tag{47.73}$$

As an independent check, expand the Jacobian $\partial x'^\mu/\partial x^\nu$ of Eq. (47.37) directly. Differentiating $x'^\mu = \gamma^{-1}(x^\mu - b^\mu x^2)$ and using $\partial_\nu\gamma = -2b_\nu + 2b^2x_\nu$ gives, after algebra that organizes itself into a reflection times a scalar exactly as in Eq. (47.67),

$$\frac{\partial x'^\mu}{\partial x^\nu} = \frac{1}{\gamma(x)}\,\widetilde{R}^\mu{}_\nu(x), \qquad \eta_{\mu\nu}\widetilde{R}^\mu{}_\rho\widetilde{R}^\nu{}_\sigma = \eta_{\rho\sigma}, \tag{47.74}$$

with $\widetilde{R}$ orthogonal, so $\eta_{\mu\nu}\mathrm{d}x'^\mu\mathrm{d}x'^\nu = \gamma^{-2}\,\eta_{\rho\sigma}\mathrm{d}x^\rho\mathrm{d}x^\sigma$ and the Weyl factor is again $\Omega_{\mathrm{SCT}} = 1/\gamma$, confirming Eq. (47.73). The two routes agree, and the SCT is conformal with the local rescaling $\Omega(x) = (1 - 2b\cdot x + b^2x^2)^{-1}$.

Worked Example 47.12 (The dilatation as a boost on the null cone)**.** The abstract identification $D = J_{d+1,d}$ becomes concrete and geometric on the Poincaré section. A finite dilatation is generated by the boost $e^{-i\sigma D}$ in the (X^d, X^{d+1}) plane; in the light-cone basis $X^\pm = X^d \pm X^{d+1}$ this boost acts as the scaling

$$X^+ \mapsto e^{\sigma}\,X^+, \qquad X^- \mapsto e^{-\sigma}\,X^-, \qquad X^\mu \mapsto X^\mu, \tag{47.75}$$

which manifestly preserves the null form $X^+X^- + \eta_{\mu\nu}X^\mu X^\nu$ of Eq. (47.71). Apply Eq. (47.75) to the section point $X(x) = (1, -x^2, x^\mu)$:

$$X \;\longmapsto\; \big(\,e^{\sigma},\, -e^{-\sigma}x^2,\, x^\mu\,\big). \tag{47.76}$$

This representative has $X^+ = e^\sigma \neq 1$, so it is off the section; rescale by the projective factor $e^{-\sigma}$ to return to $X^+ = 1$:

$$\big(\,e^{\sigma},\, -e^{-\sigma}x^2,\, x^\mu\,\big) \;\sim\; \big(\,1,\, -e^{-2\sigma}x^2,\, e^{-\sigma}x^\mu\,\big) = \big(\,1,\, -(x')^2,\, x'^\mu\,\big), \quad x'^\mu = e^{-\sigma}x^\mu. \tag{47.77}$$

The new section point has spatial part $x'^\mu = e^{-\sigma}x^\mu$ and middle component $-e^{-2\sigma}x^2 = -(x')^2$, consistent with the null condition. Thus the linear boost Eq. (47.75) induces precisely the dilatation $x^\mu \mapsto \lambda x^\mu$ with $\lambda = e^{-\sigma}$ of Eq. (47.34). The nonlinearity of conformal transformations on $\mathbb{R}^d$ — here even the simple rescaling — is entirely an artifact of projecting a linear motion of the cone back onto a fixed section. The same mechanism turns the linear $X^+ \leftrightarrow X^-$

reflection into the inversion and a linear translation in the X^μ–X^+ plane into a special conformal transformation; the conformal group is, with no remainder, the rotation group of the cone.

We have exponentiated the conformal Killing vectors into the finite group of Section 47.2.1, distilled them into the abstract algebra Eq. (47.53) of Theorem 47.9, identified that algebra with $\mathfrak{so}(d+1,1)$ (Euclidean) or $\mathfrak{so}(d,2)$ (Lorentzian) in Theorem 47.10, and exhibited its global, singularity-free action on the compactified space and on the projective null cone. The decisive structural fact for everything that follows is the dilatation grading Eq. (47.53c): D assigns each generator an eigenvalue, with P_μ raising it and K_μ lowering it. The next section turns this grading into representation theory, organizing local operators by their dilatation and rotation eigenvalues — scaling dimension and spin — and identifying the lowest-weight states annihilated by K_μ as the primary operators from which every conformal multiplet is built.

47.3 Primary Operators, Descendants, and Unitarity Bounds

The previous section closed with the conformal algebra of Theorem 47.9 and the single observation that, under the dilatation, the momentum generator P_μ raises the scaling eigenvalue by one unit while the special conformal generator K_μ lowers it (Eqs. (47.42) and (47.43)). That observation is the seed of the entire representation theory. A conformal field theory is not a Lagrangian and a set of Feynman rules; it is a list of local operators organized into representations of the conformal algebra, and the algebra is so rigid that the representations are labelled by just two pieces of data — a number and a spin — with everything else fixed. This section builds that classification. We attach to each local operator at a point its dilatation and rotation eigenvalues, we use the grading by P_μ and K_μ to split the operator content into *primaries* (lowest-weight vectors, annihilated by K_μ) and their towers of *descendants*, we exponentiate the infinitesimal action into the finite transformation law that a primary obeys under any conformal map, and finally we extract the sharp inequalities — the *unitarity bounds* — that the requirement of a positive-definite Hilbert space imposes on the allowed dimensions. The bounds are saturated exactly by the free field, the conserved current, and the stress tensor, three objects we have met repeatedly; here they reappear as the boundary of the allowed region in representation space.

The generators of Eq. (47.41) were written as pure orbital differential operators, carrying, as that section noted, "no information about the intrinsic spin or scaling weight of the field they act on." Reinstating that information is the first task. Throughout this section we work in Euclidean signature and use the conformal algebra in the *real form* natural to radial quantization, in which the

dilatation is a self-adjoint operator with real spectrum and the rotation generators are anti-self-adjoint; this is the form in which the representation theory and the positivity arguments are cleanest, and it is the form already used implicitly by the conformal Ward identities of Eq. (47.113) and by the Hilbert-space construction of Section 47.5.

Notation Watch

The abstract algebra of Eq. (47.53) was written with Hermitian generators (factors of i, Lorentzian metric $\eta_{\mu\nu}$), matching the Poincaré conventions of Chapter 3. For the representation theory we pass to the Euclidean real form: replace $\eta_{\mu\nu} \to \delta_{\mu\nu}$ and absorb the explicit i's so that

$$[D, P_\mu] = P_\mu, \quad [D, K_\mu] = -K_\mu, \quad [K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - M_{\mu\nu}), \quad [M_{\mu\nu}, P_\rho] = \delta_{\nu\rho} P_\mu - \delta_{\mu\rho} P \tag{47.78}$$

with $[M_{\mu\nu}, K_\rho] = \delta_{\nu\rho} K_\mu - \delta_{\mu\rho} K_\nu$, $[D, M_{\mu\nu}] = 0$, $[P_\mu, P_\nu] = [K_\mu, K_\nu] = 0$, and the $\mathfrak{so}(d)$ bracket $[M_{\mu\nu}, M_{\rho\sigma}] = \delta_{\nu\rho} M_{\mu\sigma} - \delta_{\mu\rho} M_{\nu\sigma} - \delta_{\nu\sigma} M_{\mu\rho} + \delta_{\mu\sigma} M_{\nu\rho}$. This is the same Lie algebra $\mathfrak{so}(d+1,1)$ of Theorem 47.10; only the basis labels differ. In this form the radial-quantization adjoints (Theorem 47.35) are $D^\dagger = D$, $M_{\mu\nu}^\dagger = -M_{\mu\nu}$, $P_\mu^\dagger = K_\mu$.

47.3.1 Scaling dimension and spin from the Cartan generators

Inside the conformal algebra the dilatation D and the rotations $M_{\mu\nu}$ commute, Eq. (47.50); they form (together with a Cartan subalgebra of $\mathfrak{so}(d)$) a maximal set of mutually commuting generators, and we organize operators by their simultaneous eigenvalues, exactly as one organizes the states of a hydrogen atom by energy and angular momentum. The full action of the conformal generators on a local operator $\mathcal{O}_a(x)$, where a runs over the components of a finite-dimensional representation of the rotation group $\mathfrak{so}(d)$, is obtained by adjoining to the orbital operators of Eq. (47.41) a number Δ and a matrix $\Sigma_{\mu\nu}$ acting on the index a:

$$[P_\mu, \mathcal{O}_a(x)] = \partial_\mu \mathcal{O}_a(x), \tag{47.79a}$$

$$[D, \mathcal{O}_a(x)] = (x\cdot\partial + \Delta)\, \mathcal{O}_a(x), \tag{47.79b}$$

$$[M_{\mu\nu}, \mathcal{O}_a(x)] = (x_\mu \partial_\nu - x_\nu \partial_\mu)\, \mathcal{O}_a(x) + (\Sigma_{\mu\nu})_a{}^b\, \mathcal{O}_b(x), \tag{47.79c}$$

$$[K_\mu, \mathcal{O}_a(x)] = (2x_\mu\, x\cdot\partial - x^2\partial_\mu + 2\Delta\, x_\mu)\mathcal{O}_a(x) - 2x^\nu (\Sigma_{\mu\nu})_a{}^b\, \mathcal{O}_b(x). \tag{47.79d}$$

The matrices $\Sigma_{\mu\nu} = -\Sigma_{\nu\mu}$ are the generators of $\mathfrak{so}(d)$ in the representation carried by $\mathcal{O}$; one verifies that Eq. (47.79) closes into the real-form algebra Eq. (47.78) for any such Σ, the extra Δ- and Σ-terms being precisely what the orbital pieces alone could not supply. Everything we do follows from Eq. (47.79) and the algebra.

The two intrinsic labels are read off at the origin, where every term in Eq. (47.79) that carries an explicit factor of x drops out:

$$[D, \mathcal{O}_a(0)] = \Delta\, \mathcal{O}_a(0), \qquad [M_{\mu\nu}, \mathcal{O}_a(0)] = (\Sigma_{\mu\nu})_a{}^b\, \mathcal{O}_b(0). \tag{47.80}$$

Definition 47.13 (Scaling dimension and spin)**.** The *scaling dimension* Δ of a local operator $\mathcal{O}_a$ is its eigenvalue under ad_D at the origin, Eq. (47.80). The *spin* of $\mathcal{O}_a$ is the representation of the Euclidean rotation group $\mathfrak{so}(d)$ generated by the matrices $\Sigma_{\mu\nu}$; for a symmetric traceless tensor of rank ℓ we call ℓ the spin and write the operator as $\mathcal{O}_{\mu_1 \cdots \mu_\ell}$.

The label Δ is exactly the renormalization-group scaling dimension of Theorem 23.22: there it was defined as the exponent governing the rescaling step of the RG and, equivalently, the power in the fixed-point two-point function $\langle \mathcal{O}(x)\, \mathcal{O}(0) \rangle \propto |x|^{-2\Delta}$. We will see in Section 47.4 that conformal symmetry forces precisely this power law, so the two definitions coincide as they must: the operators of a CFT *are* the scaling operators at the RG fixed point, and their dimensions are RG data. In particular the relation

$$\Delta = \Delta_{\mathrm{cl}} + \gamma, \tag{47.81}$$

with Δ_{cl} the engineering dimension and γ the anomalous dimension of Theorems 23.12 and 24.2, is the statement that the conformal scaling dimension is the classical dimension corrected by loops — a point we return to at the close of the section and exploit in full for the Wilson–Fisher fixed point in Theorem 47.55.

The spin label is the conformal analog of the Poincaré classification of Theorem 3.44: there a massive one-particle state was labelled by its mass and by a representation of the little group $\mathrm{SO}(3)$; here, because a CFT has no scale and hence no mass, P^2 is no longer a Casimir (it does not commute with K_μ or D), and the role of "intrinsic angular momentum" is played instead by the representation of the rotation subgroup $\mathfrak{so}(d)$ that survives at a point. The first Casimir P^2 of the Poincaré algebra, Theorem 3.35, is replaced by the quadratic Casimir of the full conformal group, which we construct in Section 47.7.5; its eigenvalue is a function of (Δ, ℓ) alone.

47.3.2 Momentum raises, special conformal lowers

The grading of Eqs. (47.42) and (47.43) acquires its full force once we ask how it acts on operators of definite dimension. Differentiating an operator, or acting with K_μ, produces a new operator; the question is what dimension the new operator carries.

Theorem 47.14 (P_μ and K_μ shift the dimension)**.** *If $\mathcal{O}$ has scaling dimension Δ, then $\partial_\mu \mathcal{O} = [P_\mu, \mathcal{O}]$ has dimension $\Delta + 1$ and $[K_\mu, \mathcal{O}]$ has dimension $\Delta - 1$.*

Proof. The dimension of an operator $\mathcal{O}'$ is the coefficient in $[D, \mathcal{O}'(0)] = \Delta' \mathcal{O}'(0)$. Use the Jacobi identity in the form $[D, [P_\mu, \mathcal{O}]] = [[D, P_\mu], \mathcal{O}] + [P_\mu, [D, \mathcal{O}]]$. With $[D, P_\mu] = P_\mu$ from Eq. (47.78) and $[D, \mathcal{O}(0)] = \Delta\, \mathcal{O}(0)$,

$$[D, [P_\mu, \mathcal{O}(0)]] = [P_\mu, \mathcal{O}(0)] + [P_\mu, \Delta\, \mathcal{O}(0)] = (\Delta + 1)\, [P_\mu, \mathcal{O}(0)]\,. \tag{47.82}$$

Thus $\partial_\mu \mathcal{O}$ has dimension $\Delta + 1$. The identical computation with $[D, K_\mu] = -K_\mu$ gives

$$[D, [K_\mu, \mathcal{O}(0)]] = -\,[K_\mu, \mathcal{O}(0)] + (\Delta)\, [K_\mu, \mathcal{O}(0)] = (\Delta - 1)\, [K_\mu, \mathcal{O}(0)]\,, \tag{47.83}$$

so $[K_\mu, \mathcal{O}]$ has dimension $\Delta - 1$. □

In the language of radial quantization, where the dilatation is the Hamiltonian (Theorem 47.30) and a local operator at the origin creates a state $|\mathcal{O}\rangle \equiv \mathcal{O}(0)\,|0\rangle$ with $D\,|\mathcal{O}\rangle = \Delta\,|\mathcal{O}\rangle$ (Eq. (47.159)), Theorem 47.14 reads

$$D\,(P_\mu\,|\mathcal{O}\rangle) = (\Delta + 1)\, P_\mu\,|\mathcal{O}\rangle\,, \qquad D\,(K_\mu\,|\mathcal{O}\rangle) = (\Delta - 1)\, K_\mu\,|\mathcal{O}\rangle\,. \tag{47.84}$$

The P_μ are creation operators and the K_μ annihilation operators for the "radial energy" D. This is structurally the harmonic oscillator, with one crucial difference set by Theorem 3.22: the conformal group is non-compact, and as that theorem records, a non-compact group has no nontrivial finite-dimensional unitary representation. The unitary representations of interest here are therefore infinite-dimensional — a primary together with its endless tower of descendants — even though the operator that builds the tower sits at a single point.

47.3.3 Primaries, descendants, and the conformal multiplet

Since K_μ lowers the dimension and the spectrum of dimensions in a sensible theory cannot run off to $-\infty$ (we prove this below), the chain $\mathcal{O}$, $K\mathcal{O}$, $KK\mathcal{O}, \ldots$ must terminate. The bottom of the chain is the central object.

Definition 47.15 (Primary and descendant)**.** A local operator $\mathcal{O}_a$ is a *primary* if it is annihilated by all special conformal generators at the origin,

$$[K_\mu, \mathcal{O}_a(0)] = 0 \qquad \text{for all } \mu, \tag{47.85}$$

equivalently $K_\mu\,|\mathcal{O}_a\rangle = 0$. A *descendant* of a primary $\mathcal{O}$ is any operator obtained by acting on it with one or more momentum generators, i.e. a derivative $\partial_{\mu_1} \cdots \partial_{\mu_n} \mathcal{O}$ (with $n \geq 1$). The primary together with all its descendants is the *conformal multiplet* (or *conformal family*) of $\mathcal{O}$, denoted $[\mathcal{O}]$.

A primary is the lowest-dimension operator in its multiplet: K_μ would lower the dimension further, but it annihilates the primary, so there is nothing below. The descendants are organized by *level*: level n consists of the operators

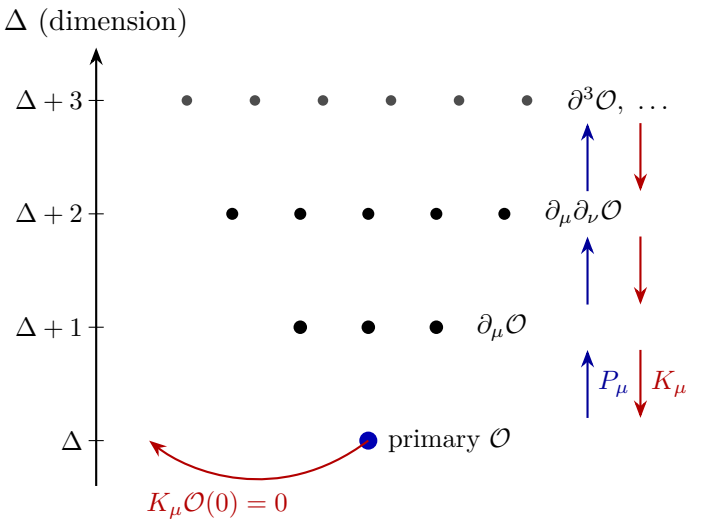


Figure 47.3: The conformal multiplet $[\mathcal{O}]$ as a lowest-weight module. The primary $\mathcal{O}$ (lowest dimension Δ, annihilated by every K_μ) sits at the bottom; the momentum generators P_μ raise the dimension by one unit at each level, generating the descendants $\partial^n\mathcal{O}$ of dimension $\Delta + n$. The special conformal generators K_μ lower the dimension and map each level to the one below, annihilating the primary. The number of independent operators at each level grows without bound: the module is infinite-dimensional, as Theorem 3.22 requires for a unitary representation of the non-compact conformal group.

$\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}$, all of dimension $\Delta + n$, and they fill out representations of $\mathfrak{so}(d)$ obtained by tensoring the spin of $\mathcal{O}$ with the n symmetrized vector indices of the derivatives. The multiplet is the conformal counterpart of a highest-weight (here lowest-weight) module of a semisimple Lie algebra: the primary is the lowest-weight vector, P_μ are the raising operators that generate the module, and K_μ are the lowering operators that annihilate the lowest weight. Figure 47.3 displays the structure.

That a primary always exists in a physically admissible representation is not an assumption but a theorem, resting only on the dilatation spectrum being bounded below. In a unitary theory it is: the dilatation is the Hamiltonian conjugate to radial time and is bounded below by the vacuum energy (Theorem 47.30). We isolate the algebraic content.

Theorem 47.16 (Existence of a primary)**.** *Let V be a representation of the conformal algebra on which D is diagonalizable with a spectrum of dimensions bounded below. Then V contains a nonzero vector $|\mathcal{O}\rangle$ annihilated by every K_μ, i.e. a primary.*

Proof. Let Δ_0 be the infimum of the dimension spectrum and let $|\psi\rangle$ be an

eigenvector with $D|\psi\rangle = \Delta_0|\psi\rangle$. By Eq. (47.84), $K_\mu|\psi\rangle$ is either zero or an eigenvector of D with eigenvalue $\Delta_0 - 1 < \Delta_0$. The latter is impossible because Δ_0 is the smallest dimension present. Hence $K_\mu|\psi\rangle = 0$ for every μ, and $|\psi\rangle$ is the required primary. □

The argument also shows that every state in a lowest-bounded representation lies in some multiplet: starting from any state, repeated application of K_μ strictly decreases the dimension and so terminates after finitely many steps on a primary, and the original state is then a descendant of that primary. The classification of operators is thereby reduced to the classification of primaries, each labelled by (Δ, ℓ) together with its spin representation. This is the sense in which "the content of a CFT is a list of primaries."

47.3.4 The finite transformation law of a primary

The infinitesimal action Eq. (47.79) integrates to a clean statement of how a primary transforms under any finite conformal map, and that statement is what the next section feeds into the Ward identities. We first restate the infinitesimal law in a manifestly geometric form. For an infinitesimal conformal transformation $x^\mu \mapsto x^\mu + \varepsilon^\mu(x)$, the vector field ε^μ obeys the conformal Killing equation Eq. (47.9), $\partial_\mu\varepsilon_\nu + \partial_\nu\varepsilon_\mu = \frac{2}{d}(\partial\cdot\varepsilon)\,\delta_{\mu\nu}$. Collecting the four families in Eq. (47.79), the change of a primary $\mathcal{O}_a$ is

$$\delta\mathcal{O}_a(x) = -\Big[\varepsilon^\mu\partial_\mu + \tfrac{\Delta}{d}\,(\partial\cdot\varepsilon) + \tfrac{1}{2}(\partial_\mu\varepsilon_\nu)\,(\Sigma^{\mu\nu})_a{}^b\Big]\mathcal{O}_b(x), \tag{47.86}$$

where the symmetric part of $\partial_\mu\varepsilon_\nu$ is pure trace by the Killing equation and contributes only through $\partial\cdot\varepsilon$, while the antisymmetric part $\frac{1}{2}(\partial_\mu\varepsilon_\nu - \partial_\nu\varepsilon_\mu)$ is the infinitesimal local rotation that contracts with $\Sigma^{\mu\nu}$. The two scalars $\frac{1}{d}(\partial\cdot\varepsilon)$ and $\frac{1}{2}(\partial_\mu\varepsilon_\nu - \partial_\nu\varepsilon_\mu)$ are exactly the infinitesimal versions of a local scale factor and a local rotation. Exponentiating them gives the finite law.

Theorem 47.17 (Finite transformation of a primary)**.** *Under a finite conformal map $x \mapsto x'(x)$, whose Jacobian factorizes as*

$$\frac{\partial x'^\mu}{\partial x^\nu} = \Omega(x)\,R^\mu{}_\nu(x), \qquad R(x) \in \mathrm{SO}(d), \qquad \Omega(x) = \Big|\frac{\partial x'}{\partial x}\Big|^{1/d}, \tag{47.87}$$

a primary $\mathcal{O}_a$ of dimension Δ and spin representation $\mathcal{D}[\cdot]$ transforms as

$$\mathcal{O}'_a(x') = \Omega(x)^{-\Delta}\,\mathcal{D}\big[R(x)\big]_a{}^b\,\mathcal{O}_b(x) = \Big|\frac{\partial x'}{\partial x}\Big|^{-\Delta/d}\,\mathcal{D}\big[R(x)\big]_a{}^b\,\mathcal{O}_b(x). \tag{47.88}$$

Proof. That the Jacobian of a conformal map factorizes as in Eq. (47.87) is the definition of conformality (Theorem 47.1): the map rescales the metric by a positive function and so its derivative is a position-dependent scale Ω times

an orthogonal matrix R; taking determinants gives $|\partial x'/\partial x| = \Omega^d$, which is Eq. (47.87). It remains to show that the rule Eq. (47.88) reproduces Eq. (47.86) infinitesimally and composes correctly.

Infinitesimal check. Write $x'^\mu = x^\mu + \varepsilon^\mu$ to first order. Then $\partial x'^\mu/\partial x^\nu = \delta^\mu_\nu + \partial_\nu \varepsilon^\mu$, so comparing with Eq. (47.87),

$$\Omega = 1 + \tfrac{1}{d}\,\partial\cdot\varepsilon + O(\varepsilon^2), \qquad R^\mu{}_\nu = \delta^\mu_\nu + \tfrac{1}{2}(\partial^\mu\varepsilon_\nu - \partial_\nu\varepsilon^\mu) + O(\varepsilon^2), \tag{47.89}$$

the split being precisely into the trace (Ω) and antisymmetric (R) parts of $\partial_\nu\varepsilon^\mu$, with the symmetric traceless part absent by the Killing equation. Hence $\Omega^{-\Delta} = 1 - \frac{\Delta}{d}\partial\cdot\varepsilon$ and $\mathcal{D}[R] = \mathbf{1} + \frac{1}{2}(\partial_\mu\varepsilon_\nu)\Sigma^{\mu\nu}$ to first order (the symmetric part of $\partial_\mu\varepsilon_\nu$ contracts to zero against the antisymmetric $\Sigma^{\mu\nu}$). The transformed field at the *same* coordinate point is $\delta\mathcal{O}_a(x) = \mathcal{O}'_a(x) - \mathcal{O}_a(x)$; since Eq. (47.88) gives $\mathcal{O}'_a$ at x', we transport back by $\mathcal{O}'_a(x) = \mathcal{O}'_a(x'-\varepsilon) = \mathcal{O}'_a(x') - \varepsilon^\mu\partial_\mu\mathcal{O}_a + O(\varepsilon^2)$, yielding

$$\delta\mathcal{O}_a(x) = -\varepsilon^\mu\partial_\mu\mathcal{O}_a - \tfrac{\Delta}{d}(\partial\cdot\varepsilon)\,\mathcal{O}_a - \tfrac{1}{2}(\partial_\mu\varepsilon_\nu)(\Sigma^{\mu\nu})_a{}^b\mathcal{O}_b, \tag{47.90}$$

which is Eq. (47.86). The finite rule and the algebra therefore agree at first order.

Composition (cocycle) check. For two conformal maps g_1, g_2 with Jacobian factors (Ω_1, R_1) and (Ω_2, R_2), the chain rule gives for the composite $g_2\circ g_1$ the factor $\Omega_{21}(x) = \Omega_2(g_1x)\,\Omega_1(x)$ and $R_{21}(x) = R_2(g_1x)\,R_1(x)$. Applying Eq. (47.88) twice,

$$\mathcal{O}''(x'') = \Omega_2(x')^{-\Delta}\mathcal{D}[R_2(x')]\,\mathcal{O}'(x') = \Omega_2(x')^{-\Delta}\Omega_1(x)^{-\Delta}\,\mathcal{D}[R_2(x')]\mathcal{D}[R_1(x)]\,\mathcal{O}(x), \tag{47.91}$$

and using $\Omega_2(x') = \Omega_2(g_1x)$ and the homomorphism property $\mathcal{D}[R_2]\mathcal{D}[R_1] = \mathcal{D}[R_2R_1]$ this equals $\Omega_{21}(x)^{-\Delta}\mathcal{D}[R_{21}(x)]\mathcal{O}(x)$, the rule for the composite map. The transformation law is consistent with the group structure, so it is the exponentiation of Eq. (47.86). □

The two basic conformal maps illustrate the law. For a dilatation $x' = \lambda x$ the Jacobian is $\lambda\,\delta^\mu_\nu$, so $\Omega = \lambda$, $R = \mathbf{1}$, and Eq. (47.88) gives $\mathcal{O}'(x') = \lambda^{-\Delta}\mathcal{O}(x)$ — identical to the RG rescaling $\mathcal{O}(x) = b^{-\Delta}\mathcal{O}'(x/b)$ of Theorem 23.22, the cleanest confirmation that the conformal Δ *is* the RG scaling dimension. For the inversion $x'^\mu = x^\mu/x^2$ (Theorem 47.6) the Jacobian is $\partial x'^\mu/\partial x^\nu = x^{-2}(\delta^\mu_\nu - 2x^\mu x_\nu/x^2)$, the bracket being the orthogonal reflection $I^\mu{}_\nu(x)$; thus $\Omega = x^{-2}$, $R = I(x)$, and a scalar primary obeys $\mathcal{O}'(x') = (x^2)^{\Delta}\mathcal{O}(x)$. These are exactly the finite conformal factors worked out in Theorem 47.4, now promoted to act on operators carrying a dimension and a spin. The transformation law Eq. (47.88) is the workhorse of Section 47.4.

47.3.5 Unitarity bounds

We now turn to the central quantitative result: not every pair (Δ, ℓ) corresponds to an operator in a *unitary* CFT. Unitarity — the existence of a positive-definite

inner product on the Hilbert space, equivalently Osterwalder–Schrader reflection positivity of the Euclidean theory — means that every state has nonnegative norm. Applied to the descendant states $P_{\mu_1}\cdots P_{\mu_n}|\mathcal{O}\rangle$, this forces inequalities on Δ. The tool is the radial-quantization inner product of Theorem 47.34, under which (Theorem 47.35)

$$P_\mu^\dagger = K_\mu, \qquad D^\dagger = D, \qquad M_{\mu\nu}^\dagger = -M_{\mu\nu}. \tag{47.92}$$

We construct the Gram matrix of the level-one descendants and demand that it be positive semidefinite. The momentum index of the descendant tensors with the spin of the primary, and the bound comes from the channel that is closest to going null.

The scalar bound. For a scalar primary ($\ell = 0$, $\Sigma_{\mu\nu} = 0$) the level-one Gram matrix was computed in Theorem 47.36: using Eq. (47.92) and the algebra Eq. (47.78),

$$\mathcal{G}_{\mu\nu} = \langle\mathcal{O}|\, K_\mu P_\nu\,|\mathcal{O}\rangle = \langle\mathcal{O}|\,[K_\mu, P_\nu]\,|\mathcal{O}\rangle = \langle\mathcal{O}|\,2(\delta_{\mu\nu}D - M_{\mu\nu})\,|\mathcal{O}\rangle = 2\Delta\,\delta_{\mu\nu}, \tag{47.93}$$

since $K_\mu|\mathcal{O}\rangle = 0$, $D|\mathcal{O}\rangle = \Delta|\mathcal{O}\rangle$, and $M_{\mu\nu}|\mathcal{O}\rangle = 0$. Positivity of the diagonal gives $\Delta \geq 0$, with equality only for the identity operator (Eq. (47.173)). This level-one analysis is not yet sharp. The strongest scalar bound comes from level two: the unique scalar descendant $P^2|\mathcal{O}\rangle$ has norm

$$\big\|P^2|\mathcal{O}\rangle\big\|^2 = 8\,d\,\Delta\Big(\Delta - \tfrac{d-2}{2}\Big), \tag{47.94}$$

whose non-negativity yields the sharp scalar bound. We carry this level-two computation in full in Theorem 47.54, where it is tied directly to the equation of motion; the upshot is

$$\boxed{\Delta \;\geq\; \frac{d-2}{2} \qquad (\ell = 0),} \tag{47.95}$$

with equality exactly when $P^2|\mathcal{O}\rangle$ is null, i.e. when $\partial^2\mathcal{O} = 0$ as an operator equation. That is the free massless scalar's equation of motion: the free field saturates the scalar unitarity bound, $\Delta_\phi = (d-2)/2$ (Eq. (47.260)).

The spin-ℓ bound. For $\ell \geq 1$ the sharp constraint already appears at level one, because the spin index opens new channels. Let $\mathcal{O}_a$ be a primary in a spin-ℓ representation, $\Sigma_{\mu\nu}$ the corresponding $\mathfrak{so}(d)$ generators, normalized so that the quadratic Casimir is

$$C_2(\ell) \equiv -\tfrac{1}{2}\,\Sigma_{\mu\nu}\Sigma^{\mu\nu} = \ell(\ell + d - 2) \tag{47.96}$$

on the symmetric traceless rank-ℓ representation (and $C_2(V) = d-1$ on the vector V). The level-one Gram matrix carries both the momentum index (ν) and the spin index (b):

$$\mathcal{G}_{(\mu a)(\nu b)} = \langle \mathcal{O}_a | \, K_\mu P_\nu \, | \mathcal{O}_b \rangle = \langle \mathcal{O}_a | \, 2(\delta_{\mu\nu} D - M_{\mu\nu}) \, | \mathcal{O}_b \rangle = 2\Delta \, \delta_{\mu\nu}\delta_{ab} - 2(\Sigma_{\mu\nu})_{ab}, \tag{47.97}$$

where $(\Sigma_{\mu\nu})_{ab} = \langle \mathcal{O}_a | \, M_{\mu\nu} \, | \mathcal{O}_b \rangle$ is the spin matrix. This is a matrix on the tensor-product space $V \otimes R_\ell$ (vector index μ times spin index a); its eigenvalues are read off by decomposing $V \otimes R_\ell$ into irreducibles. The cross term is a spin–orbit coupling. Treating the row index μ as the vector index, define the vector-representation generators $(\Sigma^V_{\rho\sigma})^\mu{}_\nu = \delta^\mu_\rho \delta_{\sigma\nu} - \delta^\mu_\sigma \delta_{\rho\nu}$; then for any vector v^ν and spin state $|b\rangle$,

$$\Sigma^V_{\rho\sigma}\Sigma^{R,\rho\sigma} \, (v \otimes |b\rangle) = \sum_{\rho\sigma} (\Sigma^V_{\rho\sigma})^\mu{}_\nu (\Sigma^R_{\rho\sigma})_{ab} \, v^\nu \, |b\rangle \; e_\mu = 2 \, (\Sigma^R_{\mu\nu})_{ab} \, v^\nu \, |b\rangle \; e_\mu, \tag{47.98}$$

so the second term in Eq. (47.97) is precisely $-(\Sigma^V \cdot \Sigma^R)$. The $\mathfrak{so}(d)$ Casimir identity for the diagonal generators $J_{\mu\nu} = \Sigma^V_{\mu\nu} + \Sigma^R_{\mu\nu}$ on $V \otimes R_\ell$,

$$C_2(\text{total}) = C_2(V) + C_2(\ell) - \Sigma^V \cdot \Sigma^R \quad \Longrightarrow \quad \Sigma^V \cdot \Sigma^R = C_2(V) + C_2(\ell) - C_2(j), \tag{47.99}$$

(where j labels the irreducible component, $-\Sigma^V \cdot \Sigma^R = -\Sigma^V_{\mu\nu}\Sigma^{R\,\mu\nu}$ is twice the cross term in $-\frac{1}{2}JJ$) lets us diagonalize. On the component of spin j,

$$\mathcal{G}_j = 2\Delta + C_2(j) - C_2(V) - C_2(\ell). \tag{47.100}$$

The decomposition of a vector times a spin-ℓ symmetric traceless tensor is $V \otimes R_\ell = R_{\ell+1} \oplus R_\ell \oplus R_{\ell-1}$ (raising the rank, mixed symmetry, and the divergence). The smallest eigenvalue — the binding constraint — comes from the divergence channel $j = \ell - 1$, because C_2 is increasing in j. Using $C_2(\ell-1) - C_2(\ell) = -(2\ell + d - 3)$ and $C_2(V) = d-1$,

$$\mathcal{G}_{\ell-1} = 2\Delta - (2\ell + d - 3) - (d-1) = 2\big[\Delta - (\ell + d - 2)\big]. \tag{47.101}$$

Non-negativity of this eigenvalue is the spin-ℓ unitarity bound:

$$\boxed{\Delta \;\geq\; \ell + d - 2 \qquad (\ell \geq 1).} \tag{47.102}$$

The other two channels give weaker constraints: $\mathcal{G}_{\ell+1} = 2\Delta + 2\ell \geq 0$ (i.e. $\Delta \geq -\ell$, automatic) and $\mathcal{G}_\ell = 2\Delta - (d-1)$, both implied by Eq. (47.102). We collect the two bounds.

Theorem 47.18 (Unitarity bounds)**.** *In a unitary conformal field theory in $d \geq 3$ dimensions, the dimension of a primary of spin ℓ obeys*

$$\Delta \geq \frac{d-2}{2} \;\; (\ell = 0, \;\; \textit{or } \Delta = 0 \textit{ for the identity}), \qquad \Delta \geq \ell + d - 2 \;\; (\ell \geq 1). \tag{47.103}$$

Equality at $\ell = 0$ holds iff $\partial^2 \mathcal{O} = 0$ (the free scalar). Equality at $\ell \geq 1$ holds iff the divergence descendant $\partial^\mu \mathcal{O}_{\mu\mu_2\cdots\mu_\ell}$ is null, i.e. vanishes as an operator — the conservation law $\partial^\mu \mathcal{O}_{\mu\cdots} = 0$.

Proof. The two inequalities are Eqs. (47.95) and (47.102); their derivations above use only the positivity of the Gram matrices of descendant states, guaranteed by unitarity (Theorem 47.34). Saturation occurs exactly when the relevant descendant has zero norm. A zero-norm state in a unitary (hence positive-definite, after quotienting the null space) Hilbert space must itself vanish; that vanishing is the operator equation. For $\ell = 0$ the null state is $P^2 |\mathcal{O}\rangle$, whose vanishing is $\partial^2 \mathcal{O} = 0$. For $\ell \geq 1$ the null state is the $j = \ell - 1$ component of $P_\mu |\mathcal{O}_{\nu\cdots}\rangle$, namely $P^\mu |\mathcal{O}_{\mu\mu_2\cdots\mu_\ell}\rangle$, whose vanishing is $\partial^\mu \mathcal{O}_{\mu\mu_2\cdots\mu_\ell} = 0$. □

The saturating cases are old friends. We isolate the two most important.

Worked Example 47.19 (Conserved currents and the stress tensor saturate the bound)**.** Specialize Eqs. (47.101) and (47.102) to $\ell = 1$ and $\ell = 2$.

Spin one ($\ell = 1$). The divergence channel is the scalar $j = 0$, and Eq. (47.101) gives $\mathcal{G}_0 = 2[\Delta - (d-1)]$, so $\Delta \geq d - 1$. We can verify this directly without the Casimir machinery. For a vector primary $\mathcal{O}_\rho$ the spin matrices are those of the vector representation, $(\Sigma_{\mu\nu})_{\rho\sigma} = \delta_{\mu\rho}\delta_{\nu\sigma} - \delta_{\mu\sigma}\delta_{\nu\rho}$, and Eq. (47.97) reads $\mathcal{G}_{(\mu\alpha)(\nu\rho)} = 2\Delta\,\delta_{\mu\nu}\delta_{\alpha\rho} - 2(\delta_{\mu\alpha}\delta_{\nu\rho} - \delta_{\mu\rho}\delta_{\nu\alpha})$. The divergence state is $|\chi\rangle = P^\rho |\mathcal{O}_\rho\rangle$; its norm is obtained by contracting the bra index α with μ and the ket index ρ with ν:

$$\begin{aligned} \|\chi\|^2 &= \delta^{\mu\alpha}\delta^{\nu\rho}\,\mathcal{G}_{(\mu\alpha)(\nu\rho)} = 2\Delta\,d - 2\big(\delta^{\mu\alpha}\delta^{\nu\rho}\delta_{\mu\alpha}\delta_{\nu\rho} - \delta^{\mu\alpha}\delta^{\nu\rho}\delta_{\mu\rho}\delta_{\nu\alpha}\big) \\ &= 2\Delta\,d - 2\big(d^2 - d\big) = 2d\big[\Delta - (d-1)\big]. \end{aligned} \tag{47.104}$$

Non-negativity gives $\Delta \geq d-1$, saturated when $|\chi\rangle = P^\rho |\mathcal{O}_\rho\rangle = 0$, i.e. $\partial^\mu \mathcal{O}_\mu = 0$. A spin-one primary of dimension exactly $d-1$ is therefore a *conserved current*. This is the same dimension that conservation forces on a current's two-point function in Theorem 47.23, now derived from pure representation theory.

Spin two ($\ell = 2$). The divergence channel is $j = \ell - 1 = 1$, with $C_2(1) = d-1$ and $C_2(2) = 2d$, so Eq. (47.100) gives $\mathcal{G}_1 = 2\Delta + (d-1) - (d-1) - 2d = 2(\Delta - d)$, hence $\Delta \geq d$. A spin-two primary saturating the bound, $\Delta = d$, has a null divergence, $\partial^\mu \mathcal{O}_{\mu\nu} = 0$: it is a *conserved symmetric tensor of dimension d*, which is precisely the stress tensor (its dimension equals d because the integral $\int d^{d-1}x\, T^{0\nu}$ must be dimensionless, fixing $[T_{\mu\nu}] = d$). Thus $\Delta_T = d$ saturates the $\ell = 2$ unitarity bound, and the existence of a stress tensor in a local CFT means that the spin-two spectrum touches the unitarity boundary exactly at d.

The pattern is uniform: the unitarity bound is saturated precisely by operators obeying a conservation or wave equation, and the saturating operator's descendant module is *short* — the would-be descendants in the null channel are set to zero.

A free scalar ($\ell = 0$, $\Delta = (d-2)/2$), a conserved current ($\ell = 1$, $\Delta = d-1$), and the stress tensor ($\ell = 2$, $\Delta = d$) are the three canonical short multiplets. For higher spin the bound $\Delta = \ell + d - 2$ is saturated by conserved higher-spin currents; a theorem we will not need here states that interacting unitary CFTs in $d \geq 3$ cannot have such currents beyond $\ell = 2$, which is why generic CFT spectra have $\Delta > \ell + d - 2$ for $\ell \geq 3$.

47.3.6 The dimension as RG data: setting up the thread

The relation Eq. (47.81), $\Delta = \Delta_{\mathrm{cl}} + \gamma$, is the bridge back to Part VI. The engineering dimension of a canonically normalized scalar field in d dimensions is $\Delta_{\mathrm{cl}} = (d-2)/2$, read off from the kinetic term $\int d^d x\, (\partial\phi)^2$; the loop corrections shift it by the anomalous dimension γ_ϕ defined through the wavefunction renormalization in Theorems 23.12 and 24.2. At a free fixed point $\gamma_\phi = 0$ and $\Delta_\phi = (d-2)/2$ exactly — the massless scalar of ϕ^4 theory sits on the unitarity bound, as Theorem 47.18 demanded. At the interacting Wilson–Fisher fixed point in $d = 4 - \varepsilon$ the anomalous dimension is nonzero, $\gamma_\phi = O(\varepsilon^2)$, and the conformal scaling dimension of the field becomes

$$\Delta_\phi = \frac{d-2}{2} + \gamma_\phi = \frac{2-\varepsilon}{2} + \gamma_\phi, \tag{47.105}$$

strictly above the unitarity bound, as required for an interacting operator that is not free. The leading anomalous dimension and the dimension of the composite ϕ^2 are the conformal data of the Wilson–Fisher CFT; they are precisely the numbers Part VI computed as RG quantities, and we assemble them into CFT form in Theorem 47.55. The point to register now is conceptual: the abstract Δ of Theorem 47.13 and the RG scaling dimension of Theorem 23.22 are the same number, so the anomalous dimension that quantified the breaking of classical scale invariance under renormalization is, at the fixed point, nothing but the conformal weight of a primary operator.

With each primary now labelled by (Δ, ℓ), subject to the unitarity bounds of Theorem 47.18 and transforming by the finite law Eq. (47.88), the content of a CFT has been reduced to a discrete list. The next section shows that this list, together with one number per triple of primaries, fixes the low-point correlators completely: conformal symmetry determines the two-point function up to normalization and the three-point function up to a single coefficient, leaving only a function of cross-ratios undetermined at four points.

47.4 Correlators Fixed by Conformal Symmetry

The previous section gave each local operator a pair of labels — a scaling dimension Δ and a Lorentz (or, in Euclidean signature, rotation-group) representation

fixing its spin — and singled out the *primaries* as the operators annihilated by the special conformal generator K^μ at the origin, the lowest-weight vectors of the conformal modules (Section 47.3.3, Theorem 47.18). It also recorded how a primary transforms under a finite conformal map: not merely by being carried to its image point, but by picking up a local Jacobian weight set by its dimension and spin (Eq. (47.88)). That transformation law is a constraint, and the present section is the systematic exploitation of it. We will see that it is so strong that the position dependence of the lowest correlators is fixed entirely by symmetry: the two-point function down to a single number, the three-point function down to a single number per triple of operators, and the four-point function down to one *function* of two scalar invariants. The fourth statement is the doorway to the rest of Part XIV: that function, $g(u, v)$, is the object the operator product expansion will organise and the bootstrap will pin down by crossing.

Throughout we work in Euclidean signature, the natural home for the correlators of a conformal field theory and the setting in which the next section will build the radial-quantization Hilbert space. Points live in $\mathbb{R}^d$ with the positive-definite metric $\delta_{\mu\nu}$; we write $x \cdot y = \delta_{\mu\nu} x^\mu y^\nu$, $x^2 = x \cdot x \geq 0$, and $|x| = \sqrt{x^2}$. The Euclidean correlators are the Schwinger functions obtained by Wick rotation from the Lorentzian theory (Theorem 10.30); reflection positivity (Theorem 10.32) is what guarantees they descend from a unitary Lorentzian theory, and we will use it once, to fix the sign of the two-point normalization.

Notation Watch

Because every correlator in this section is Euclidean, indices are raised and lowered with $\delta_{\mu\nu}$ and there is no distinction between upper and lower placement; we keep indices upstairs for readability. This is a deliberate departure from the mostly-minus Lorentzian metric $\eta_{\mu\nu} = \mathrm{diag}(+, -, -, -)$ used elsewhere in the book. The continuation back to Lorentzian signature is the subject of Chapter 48 and of the analytic-continuation discussion in Chapter 51; here it never intrudes.

47.4.1 The covariance law for correlators

A conformal transformation is a smooth invertible map $x \mapsto \bar{x}(x)$ of (a patch of) $\mathbb{R}^d$ that rescales the metric by a positive local factor. Concretely, its Jacobian factorizes as

$$\frac{\partial \bar{x}^\mu}{\partial x^\nu} \;=\; \Omega(x)\, R^\mu{}_\nu(x), \qquad R^\mu{}_\rho R^\nu{}_\sigma\, \delta_{\mu\nu} = \delta_{\rho\sigma}, \tag{47.106}$$

where $\Omega(x) > 0$ is the *local scale factor* and $R(x) \in \mathrm{O}(d)$ is a local rotation. Equivalently $\partial_\mu \bar{x}^\rho\, \partial_\nu \bar{x}_\rho = \Omega(x)^2\, \delta_{\mu\nu}$: the map preserves angles and stretches lengths by Ω. The finite conformal group of $\mathbb{R}^d$ (more precisely of its conformal compactification, the sphere S^d) is generated by four kinds of maps, whose scale

factors and rotations we will need repeatedly:

$$\begin{aligned}
&\text{translation} && \bar{x} = x + a, && \Omega = 1, && R = \mathbf{1};\\
&\text{rotation} && \bar{x} = \Lambda x \ (\Lambda \in \mathrm{SO}(d)), && \Omega = 1, && R = \Lambda;\\
&\text{dilatation} && \bar{x} = \lambda x \ (\lambda > 0), && \Omega = \lambda, && R = \mathbf{1};\\
&\text{inversion} && \bar{x}^\mu = \frac{x^\mu}{x^2}, && \Omega(x) = \frac{1}{x^2}, && R^\mu{}_\nu = I^\mu{}_\nu(x).
\end{aligned} \tag{47.107}$$

Here we have met the object that will run through the entire section, the *inversion tensor*

$$I^\mu{}_\nu(x) \;\equiv\; \delta^\mu_\nu - 2\,\frac{x^\mu x_\nu}{x^2}, \tag{47.108}$$

which is the rotation $R(x)$ of an inversion: a short computation gives $\partial_\nu \bar{x}^\mu = \partial_\nu(x^\mu/x^2) = x^{-2}\delta^\mu_\nu - 2x^\mu x_\nu/x^4 = x^{-2}\, I^\mu{}_\nu(x)$, matching Eq. (47.106) with $\Omega = 1/x^2$. The tensor $I^\mu{}_\nu(x)$ is orthogonal, $I^\mu{}_\rho I^\nu{}_\sigma \delta_{\mu\nu} = \delta_{\rho\sigma}$ (check: $(\delta - 2\hat{x}\hat{x})^2 = \delta - 4\hat{x}\hat{x} + 4\hat{x}\hat{x} = \delta$ with $\hat{x} = x/|x|$), with determinant -1; it is the reflection through the hyperplane perpendicular to x. The special conformal transformation (SCT) is the conjugate of a translation by an inversion, $\bar{x} = \mathrm{inv} \circ \mathrm{transl}_b \circ \mathrm{inv}$, giving the finite form

$$\bar{x}^\mu = \frac{x^\mu - b^\mu x^2}{\gamma(x)}, \qquad \gamma(x) = 1 - 2\,b\cdot x + b^2 x^2, \qquad \Omega(x) = \frac{1}{\gamma(x)}. \tag{47.109}$$

That translations, rotations, dilatations and inversion generate the whole group is the finite-transformation counterpart of the algebra $\mathfrak{so}(d+1,1)$ assembled from the conserved charges in Theorem 7.48; in checking covariance it therefore suffices to impose invariance under these four, and SCT invariance can be obtained either from the finite inversion or from its infinitesimal generator. We will use whichever is more economical.

The bridge from the transformation law of operators to a statement about correlators is the invariance of the vacuum and of the (regularized, conformally invariant) path-integral measure: a symmetry maps the set of correlators to itself, so the correlator of the transformed operators equals the correlator of the originals,

$$\langle\, \mathcal{O}'_1(y_1)\cdots\mathcal{O}'_n(y_n)\,\rangle \;=\; \langle\, \mathcal{O}_1(y_1)\cdots\mathcal{O}_n(y_n)\,\rangle. \tag{47.110}$$

For a scalar primary the transformed operator is, by the law imported from Eq. (47.88),

$$\mathcal{O}'(\bar{x}) \;=\; \Omega(x)^{-\Delta}\,\mathcal{O}(x), \tag{47.111}$$

the dimension Δ appearing exactly as the power of the local rescaling, just as it does for the dilatation eigenvalue. Evaluating Eq. (47.110) at the image points $y_k = \bar{x}_k$ and substituting Eq. (47.111) for each factor, the left side becomes $\prod_k \Omega(x_k)^{-\Delta_k}\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle$, while the right side is $\langle \mathcal{O}_1(\bar{x}_1)\cdots\mathcal{O}_n(\bar{x}_n)\rangle$. Solving for the unprimed correlator gives the master identity that organizes everything below.

Proposition 47.20 (Conformal covariance of scalar correlators)**.** *Let $\mathcal{O}_1, \dots, \mathcal{O}_n$ be scalar primaries of dimensions $\Delta_1, \dots, \Delta_n$. Under any conformal map $x \mapsto \bar{x}(x)$ with scale factor $\Omega(x)$,*

$$\langle\, \mathcal{O}_1(x_1) \cdots \mathcal{O}_n(x_n) \,\rangle \;=\; \prod_{k=1}^{n} \Omega(x_k)^{\Delta_k}\, \langle\, \mathcal{O}_1(\bar{x}_1) \cdots \mathcal{O}_n(\bar{x}_n) \,\rangle. \tag{47.112}$$

It is worth seeing the infinitesimal face of Eq. (47.112), since one of our two derivations of the two-point function uses it. Write a small transformation as $\bar{x}^\mu = x^\mu + \delta x^\mu$ with scale factor $\Omega(x) = 1 + \sigma(x)$. Expanding Eq. (47.112) to first order, $0 = \sum_k \big[\delta x_k^\mu\, \partial_{k\mu} + \Delta_k\, \sigma(x_k)\big] G$, where G denotes the correlator. Reading δx and σ from Eq. (47.107) and the infinitesimal limit of Eq. (47.37) gives the four conformal Ward identities,

$$\text{translations:} \quad \sum_k \partial_k^\mu\, G = 0, \tag{47.113a}$$

$$\text{rotations:} \quad \sum_k \left(x_k^\mu \partial_k^\nu - x_k^\nu \partial_k^\mu\right) G = 0, \tag{47.113b}$$

$$\text{dilatations:} \quad \sum_k \left(x_k \cdot \partial_k + \Delta_k\right) G = 0, \tag{47.113c}$$

$$\text{SCT:} \quad \sum_k \left(2 x_k^\mu\, x_k \cdot \partial_k - x_k^2\, \partial_k^\mu + 2\Delta_k x_k^\mu\right) G = 0. \tag{47.113d}$$

The SCT identity Eq. (47.113d) uses $\delta x_k^\mu = 2(b\cdot x_k)x_k^\mu - b^\mu x_k^2$ and $\sigma(x_k) = +2\, b\cdot x_k$ (from $\Omega = 1/\gamma = 1 + 2b\cdot x + O(b^2)$), with the arbitrary parameter b^μ stripped off. These are the same charges whose algebra was assembled in Theorem 7.48, now acting differentially on a correlator of primaries; the extra $+2\Delta_k x_k^\mu$ term, absent from a naive coordinate transformation, is the imprint of the operators' scaling weight and is precisely what will force dimensions to match.

47.4.2 The two-point function

We determine $G_2(x_1, x_2) = \langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\rangle$ for two scalar primaries by feeding it the Ward identities Eq. (47.113) one at a time.

Translations and rotations. Identity Eq. (47.113a) says G_2 depends only on the difference $x_{12} \equiv x_1 - x_2$. Identity Eq. (47.113b) then says it depends on x_{12} only through the rotation invariant $x_{12}^2 = |x_1 - x_2|^2$. Hence $G_2 = f(x_{12}^2)$ for some function f of one variable.

Dilatations. With $G_2 = f(x_{12}^2)$, the Euler operator $\sum_k x_k \cdot \partial_k$ counts the degree of homogeneity in (x_1, x_2) jointly; since x_{12}^2 is homogeneous of degree 2, $\sum_k x_k \cdot \partial_k\, f(x_{12}^2) = 2\, x_{12}^2 f'(x_{12}^2)$. The dilatation Ward identity Eq. (47.113c) reads $2\, x_{12}^2 f' + (\Delta_1 + \Delta_2) f = 0$, a first-order equation whose solution is the power law

$$G_2(x_1, x_2) \;=\; \frac{C_{12}}{(x_{12}^2)^{(\Delta_1+\Delta_2)/2}} \;=\; \frac{C_{12}}{|x_1 - x_2|^{\Delta_1+\Delta_2}}, \tag{47.114}$$

with C_{12} a constant of integration. Poincaré symmetry and scale invariance alone, then, already reduce the two-point function to a single power and a single number. They do not yet say anything about the relation between Δ_1 and Δ_2; that information is carried only by the special conformal transformations.

Special conformal transformations. We impose Eq. (47.113d) on the form Eq. (47.114). Write $\Delta_+ = \Delta_1 + \Delta_2$, $A^\mu = x_{12}^\mu = (x_1 - x_2)^\mu$, $r^2 = x_{12}^2 = A^2$, and abbreviate $P = (r^2)^{-\Delta_+/2-1}$, so that $G_2 = C_{12}\, r^2 P$ and the derivatives are

$$\partial_1^\mu G_2 = -C_{12}\,\Delta_+\, A^\mu P, \qquad \partial_2^\mu G_2 = +C_{12}\,\Delta_+\, A^\mu P, \tag{47.115}$$

since $\partial_1^\mu (r^2)^{-\Delta_+/2} = -\frac{\Delta_+}{2}(r^2)^{-\Delta_+/2-1}\cdot 2A^\mu$ and ∂_2^μ flips the sign. The SCT operator $\mathcal{K}^\mu \equiv \sum_{k=1,2}(2x_k^\mu\, x_k{\cdot}\partial_k - x_k^2\partial_k^\mu + 2\Delta_k x_k^\mu)$ acting on G_2 has, after dividing by $C_{12}P$, two kinds of contributions. The derivative terms collect into

$$\begin{aligned} B^\mu &\equiv \frac{1}{C_{12}\Delta_+ P}\sum_k (2x_k^\mu\, x_k{\cdot}\partial_k - x_k^2\partial_k^\mu)G_2 \\ &= -2x_1^\mu(x_1{\cdot}A) + x_1^2 A^\mu + 2x_2^\mu(x_2{\cdot}A) - x_2^2 A^\mu. \end{aligned} \tag{47.116}$$

Using $x_1\cdot A = x_1^2 - x_1{\cdot}x_2$ and $x_2\cdot A = x_1{\cdot}x_2 - x_2^2$ and collecting the coefficients of x_1^μ and x_2^μ separately, both turn out to equal $-(x_1^2 - 2x_1{\cdot}x_2 + x_2^2) = -r^2$, so that

$$B^\mu = -r^2(x_1^\mu + x_2^\mu). \tag{47.117}$$

The weight terms give $\sum_k 2\Delta_k x_k^\mu G_2 = 2(r^2P)C_{12}(\Delta_1 x_1^\mu + \Delta_2 x_2^\mu)$. Assembling,

$$\mathcal{K}^\mu G_2 = C_{12}P\Big[\Delta_+ B^\mu + 2r^2(\Delta_1 x_1^\mu + \Delta_2 x_2^\mu)\Big] = C_{12}P\, r^2\,(\Delta_1 - \Delta_2)\,(x_1 - x_2)^\mu, \tag{47.118}$$

where in the last step the $-\Delta_+(x_1^\mu + x_2^\mu)$ from Eq. (47.117) combined with $2(\Delta_1 x_1^\mu + \Delta_2 x_2^\mu)$ to leave $(\Delta_1 - \Delta_2)(x_1^\mu - x_2^\mu)$. For the two-point function to be special-conformally invariant, Eq. (47.118) must vanish for all configurations; since the kinematic factor $C_{12}Pr^2(x_1 - x_2)^\mu$ is generically nonzero, this requires

$$C_{12}\,(\Delta_1 - \Delta_2) = 0. \tag{47.119}$$

Either the constant vanishes or the two dimensions coincide. We have proved the central rigidity of the two-point function.

Theorem 47.21 (Two-point function of scalar primaries)**.** *For scalar primaries $\mathcal{O}_1, \mathcal{O}_2$ of dimensions Δ_1, Δ_2, conformal invariance forces*

$$\langle\, \mathcal{O}_1(x_1)\, \mathcal{O}_2(x_2)\,\rangle \;=\; \delta_{\Delta_1\Delta_2}\,\frac{C_{12}}{|x_1 - x_2|^{2\Delta_1}}. \tag{47.120}$$

In particular, two scalar primaries of unequal dimension are orthogonal: their two-point function vanishes identically.

The same conclusion follows from finite inversion, which is instructive as an independent check and as a warm-up for the three-point function. Under inversion $\bar{x}_k = x_k/x_k^2$, Eq. (47.107) gives $\Omega(x_k) = 1/x_k^2$, and the elementary identity

$$\bar{x}_{12}^2 = \Big(\frac{x_1}{x_1^2} - \frac{x_2}{x_2^2}\Big)^2 = \frac{1}{x_1^2} - \frac{2x_1\cdot x_2}{x_1^2 x_2^2} + \frac{1}{x_2^2} = \frac{x_{12}^2}{x_1^2\, x_2^2} \tag{47.121}$$

shows how distances transform. (Identity Eq. (47.121) is nothing but the covariance $\bar{x}_{ij}^2 = \Omega(x_i)\Omega(x_j)x_{ij}^2$, which one checks holds for every generator in Eq. (47.107), and hence for the whole group.) Feeding Eq. (47.114) into Eq. (47.112),

$$\frac{C_{12}}{(x_{12}^2)^{\Delta_+/2}} = (x_1^2)^{-\Delta_1}(x_2^2)^{-\Delta_2}\,\frac{C_{12}}{(\bar{x}_{12}^2)^{\Delta_+/2}} = C_{12}\,(x_1^2)^{(\Delta_2-\Delta_1)/2}(x_2^2)^{(\Delta_1-\Delta_2)/2}\,\frac{1}{(x_{12}^2)^{\Delta_+/2}}, \tag{47.122}$$

using Eq. (47.121) for $\bar{x}_{12}^2$. The leftover factors of x_1^2 and x_2^2 cancel only if $\Delta_1 = \Delta_2$, reproducing Eq. (47.120).

Sign of the normalization, and canonical normalization. The constant C_{12} in the diagonal case $\Delta_1 = \Delta_2$ is not fixed by symmetry, but its sign is fixed by reflection positivity. Let Θ be the reflection $x = (\tau, \mathbf{x}) \mapsto \Theta x = (-\tau, \mathbf{x})$ in a chosen Euclidean time coordinate. Osterwalder–Schrader positivity (Theorem 10.32) states that for any real operator $\mathcal{O}$ supported at $\tau > 0$, $\langle (\Theta\mathcal{O})\,\mathcal{O}\rangle \geq 0$. Applied to a Hermitian scalar primary inserted at $x = (\tau, \mathbf{0})$ with $\tau > 0$, its reflection sits at $\Theta x = (-\tau, \mathbf{0})$, the separation is $|x - \Theta x| = 2\tau$, and Eq. (47.120) gives $\langle (\Theta\mathcal{O})(x)\,\mathcal{O}(x)\rangle = C/(2\tau)^{2\Delta} \geq 0$, hence $C \geq 0$; it is strictly positive unless $\mathcal{O}$ is a null operator (one whose every correlator vanishes), which we exclude. Because $C > 0$ we may rescale the operator,

$$\mathcal{O} \;\longrightarrow\; \mathcal{O}/\sqrt{C}, \tag{47.123}$$

a redefinition that changes nothing physical, to reach the *canonical normalization*

$$\boxed{\big\langle\, \mathcal{O}(x_1)\,\mathcal{O}(x_2)\,\big\rangle = \frac{1}{|x_1 - x_2|^{2\Delta}}} \tag{47.124}$$

for every primary. We adopt this convention henceforth; with it the three-point coefficient C_{123} introduced below acquires an unambiguous meaning, and the orthogonality of distinct primaries of equal dimension can be arranged by diagonalizing the (positive-definite, by reflection positivity) matrix of two-point coefficients C_{ij} on any degenerate subspace — a Gram–Schmidt step that is always available precisely because that matrix is positive. From here on "the primaries" means an orthonormal basis in this sense.

Physical Insight

The power law Eq. (47.124) is the statement that a CFT has no length scale to build a correlator out of: the only conformally meaningful function of two points is their distance, and the only scale-covariant function of one distance is a power, with exponent locked to the dimension. The contrast with a massive theory is sharp: there the Euclidean two-point function is the Yukawa form $\sim e^{-m|x|}/|x|^{d-2}$ (Theorem 10.30), which singles out the correlation length $1/m$. Sending $m \to 0$ at an RG fixed point erases that scale and leaves exactly the pure power demanded here.

Remark 47.22 (Momentum space). The Fourier transform of the canonical two-point function, computed with the tempered-distribution machinery of Section 5.2, is again a pure power,

$$\int d^d x \, \frac{e^{\,ip\cdot x}}{|x|^{2\Delta}} = \frac{\pi^{d/2}\, 2^{\,d-2\Delta}\, \Gamma\!\left(\frac{d}{2}-\Delta\right)}{\Gamma(\Delta)} \, \frac{1}{|p|^{\,d-2\Delta}}, \tag{47.125}$$

valid as an analytic continuation away from the poles at $\Delta = \frac{d}{2} + n$. The momentum-space exponent $d - 2\Delta$ is fixed by dimensional analysis alone (the transform carries dimension $+d$ from $d^d x$ and -2Δ from the integrand), which is why scale invariance suffices to know it; the Γ-function prefactor is the content of the actual integral. The poles in $\Gamma(\frac{d}{2} - \Delta)$ at $\Delta = \frac{d}{2}, \frac{d}{2}+1, \dots$ are the positions where the position-space power $|x|^{-2\Delta}$ becomes too singular to Fourier-transform without subtraction, and they will reappear in Chapter 48 and Chapter 53 as the dimensions at which contact terms and anomalies live.

47.4.3 Spinning two-point functions and conserved currents

Operators with spin carry rotation indices, and their correlators must transform under the local rotation $R(x)$ of Eq. (47.106) in addition to the scale weight. The inversion tensor Eq. (47.108) is exactly the covariant tensor that does this, because under inversion the relevant rotation *is* $I^{\mu}{}_{\nu}$. For a symmetric traceless primary $\mathcal{O}^{\mu_1\cdots\mu_\ell}$ of spin ℓ and dimension Δ, conformal covariance (the same master identity Eq. (47.112) dressed with rotation matrices on each index) fixes the two-point function to

$$\langle\, \mathcal{O}^{\mu_1\cdots\mu_\ell}(x)\, \mathcal{O}^{\nu_1\cdots\nu_\ell}(0)\,\rangle = \frac{C_{\mathcal{O}}}{|x|^{2\Delta}} \Big(I^{(\mu_1(\nu_1}(x)\cdots I^{\mu_\ell)\nu_\ell)}(x) - \text{traces} \Big), \tag{47.126}$$

where the parentheses denote symmetrization within each index group and "$-$traces" removes the $\delta^{\mu_i\mu_j}$ and $\delta^{\nu_i\nu_j}$ pieces so the structure lives in the symmetric-traceless representation. The scalar case $\ell = 0$ recovers Eq. (47.120); the rank-ℓ tensor is the unique conformally covariant structure because $I^{\mu}{}_{\nu}(x)$ is the only

available tensor built from a single separation x that transforms as a product of one rotation at each endpoint.

The most important spinning operator in any local CFT is a conserved current. We now verify the statement, anticipated in Theorem 7.48, that conservation forces a current's dimension to a specific value, and we do so by direct computation on Eq. (47.126) at $\ell = 1$.

Worked Example 47.23 (Conservation fixes the dimension of a current)**.** Take a vector primary J^μ of dimension Δ, with two-point function

$$\langle\, J^\mu(x)\, J^\nu(0)\,\rangle = \frac{C_J}{(x^2)^\Delta}\, I^{\mu\nu}(x), \qquad I^{\mu\nu}(x) = \delta^{\mu\nu} - 2\frac{x^\mu x^\nu}{x^2}. \tag{47.127}$$

We impose the conservation Ward identity $\partial_\mu \langle J^\mu(x) J^\nu(0)\rangle = 0$ for $x \neq 0$ and read off the allowed Δ. Write the correlator as $C_J\big[\delta^{\mu\nu}(x^2)^{-\Delta} - 2x^\mu x^\nu (x^2)^{-\Delta-1}\big]$ and differentiate. The first term gives

$$\partial_\mu\big[\delta^{\mu\nu}(x^2)^{-\Delta}\big] = \partial^\nu (x^2)^{-\Delta} = -2\Delta\, x^\nu (x^2)^{-\Delta-1}. \tag{47.128}$$

For the second term we use $\partial_\mu(x^\mu x^\nu) = d\, x^\nu + x^\nu = (d+1)x^\nu$ and $x^\mu x^\nu \partial_\mu (x^2)^{-\Delta-1} = -2(\Delta+1)x^2 x^\nu (x^2)^{-\Delta-2} = -2(\Delta+1)x^\nu (x^2)^{-\Delta-1}$, so

$$\partial_\mu\big[-2x^\mu x^\nu (x^2)^{-\Delta-1}\big] = -2\big[(d+1) - 2(\Delta+1)\big] x^\nu (x^2)^{-\Delta-1} = -2\,(d-1-2\Delta)\, x^\nu (x^2)^{-\Delta-1}. \tag{47.129}$$

Adding the two contributions,

$$\partial_\mu \langle J^\mu(x) J^\nu(0)\rangle = C_J\big[-2\Delta - 2(d-1-2\Delta)\big] x^\nu (x^2)^{-\Delta-1} = 2C_J\,(\Delta - (d-1))\, x^\nu (x^2)^{-\Delta-1}. \tag{47.130}$$

This vanishes for all $x \neq 0$ precisely when

$$\boxed{\Delta_J = d - 1} \tag{47.131}$$

Conservation thus saturates the spin-1 unitarity bound $\Delta \geq d-1$ derived in Theorem 47.18: a conserved current is exactly the operator that sits at the bottom edge of the allowed window, and the saturation is equivalent to the existence of a null descendant $\partial_\mu J^\mu$. The same computation with the stress tensor's spin-2 structure gives $\Delta_T = d$, the dimension demanded of $\Theta^{\mu\nu}$ by Theorem 7.48.

For three or more points the spinning structures proliferate, but always into a *finite* list. A correlator of operators with spins $\ell_1, \dots, \ell_n$ is fixed by conformal symmetry up to one constant per linearly independent tensor structure, each structure built from the covariant bilinears $I^{\mu\nu}(x_{ij})$ and the conformally covariant vectors $Z_i^\mu \propto x_{ij}^\mu / x_{ij}^2 - x_{ik}^\mu / x_{ik}^2$; conservation Ward identities like the one above

then impose linear relations among the constants, cutting the list down. For a scalar–scalar– spin-ℓ correlator there is a unique structure,

$$\langle\, \mathcal{O}_1(x_1)\, \mathcal{O}_2(x_2)\, \mathcal{O}^{\mu_1\cdots\mu_\ell}(x_3)\,\rangle = C_{12\mathcal{O}}\, \frac{Z^{\mu_1}\cdots Z^{\mu_\ell} - \text{traces}}{(x_{12}^2)^{\frac{\Delta_1+\Delta_2-\Delta+\ell}{2}} (x_{13}^2)^{\frac{\Delta_1+\Delta-\Delta_2-\ell}{2}} (x_{23}^2)^{\frac{\Delta_2+\Delta-\Delta_1-\ell}{2}}}, \qquad Z^\mu = \frac{x^\mu}{x^\mu}$$

(47.132)

and we will meet exactly this object again when the operator product expansion attaches a spinning exchanged operator to two scalars. The systematic index-free packaging of all such structures (the embedding-space and polarization-vector formalism) is developed in the references collected at the end of the chapter; we will need only the scalar correlators and the single structure Eq. (47.132) in what follows.

47.4.4 The three-point function

We now show that the three-point function of scalar primaries is fixed up to a single overall constant. The strategy is the same as for two points, but the conclusion is sharper because three points carry no conformal invariant at all.

Translations, rotations, dilatations. By Eqs. (47.113a) and (47.113b) the correlator $G_3 = \langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle$ depends on the three points only through the three pairwise distances $r_{ij} = |x_i - x_j|$, $1 \le i < j \le 3$. The dilatation identity Eq. (47.113c) then makes it homogeneous of degree $-(\Delta_1+\Delta_2+\Delta_3)$ in these distances jointly. So far G_3 could be any homogeneous function of r_{12}, r_{13}, r_{23} of that degree — equivalently, $r_{23}^{-(\Delta_1+\Delta_2+\Delta_3)}$ times an arbitrary function of the two ratios r_{12}/r_{23} and r_{13}/r_{23}. The remaining symmetry, inversion, removes this freedom entirely.

Inversion. Posit the monomial ansatz $G_3 = C_{123}\, r_{12}^{-\alpha_{12}} r_{13}^{-\alpha_{13}} r_{23}^{-\alpha_{23}}$ with exponents to be determined; we will see the ansatz is forced. Under inversion, Eq. (47.121) gives $r_{ij} \mapsto r_{ij}/(|x_i||x_j|)$, while Eq. (47.107) gives $\Omega(x_i) = 1/x_i^2$. The covariance law Eq. (47.112) requires

$$G_3(x_i) = \prod_{i=1}^{3} (x_i^2)^{-\Delta_i}\, G_3(\bar{x}_i) = \prod_{i=1}^{3} (x_i^2)^{-\Delta_i}\, C_{123} \prod_{i<j} \Big(\frac{r_{ij}}{|x_i||x_j|}\Big)^{-\alpha_{ij}}. \tag{47.133}$$

The distances reassemble into the original $\prod r_{ij}^{-\alpha_{ij}}$, and the spurious factors of $|x_i|$ collect, point by point, into

$$\prod_{i<j} \big(|x_i|\,|x_j|\big)^{\alpha_{ij}} = (x_1^2)^{(\alpha_{12}+\alpha_{13})/2}\, (x_2^2)^{(\alpha_{12}+\alpha_{23})/2}\, (x_3^2)^{(\alpha_{13}+\alpha_{23})/2}. \tag{47.134}$$

For these to cancel the $\prod (x_i^2)^{-\Delta_i}$ point by point, the exponents must satisfy the linear system

$$\alpha_{12}+\alpha_{13} = 2\Delta_1, \qquad \alpha_{12}+\alpha_{23} = 2\Delta_2, \qquad \alpha_{13}+\alpha_{23} = 2\Delta_3, \tag{47.135}$$

whose unique solution is

$$\alpha_{12} = \Delta_1 + \Delta_2 - \Delta_3, \quad \alpha_{13} = \Delta_1 + \Delta_3 - \Delta_2, \quad \alpha_{23} = \Delta_2 + \Delta_3 - \Delta_1. \tag{47.136}$$

(Summing the three equations recovers $\alpha_{12}+\alpha_{13}+\alpha_{23} = \Delta_1+\Delta_2+\Delta_3$, consistent with the dilatation degree, so the system is non-degenerate.) That the monomial ansatz is not merely consistent but forced follows from the count of invariants: three points have no conformal invariant. Indeed translations fix $x_3 \to 0$, an inversion then sends $x_2 \to \infty$, dilatations and rotations place x_1 on a fixed unit vector, and nothing is left free — the $\frac{1}{2}(d+1)(d+2)$-dimensional conformal group acts transitively on (generic) triples for $d \geq 2$. With no invariant to depend on, any putative extra factor multiplying the monomial would have to be a conformal scalar function of the three points, hence a constant; the constant is absorbed into C_{123}. We record the result.

Theorem 47.24 (Three-point function of scalar primaries)**.** *For scalar primaries of dimensions $\Delta_1, \Delta_2, \Delta_3$, conformal invariance fixes*

$$\boxed{\langle\, \mathcal{O}_1(x_1)\, \mathcal{O}_2(x_2)\, \mathcal{O}_3(x_3)\, \rangle = \frac{C_{123}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta_3}\, |x_{13}|^{\Delta_1+\Delta_3-\Delta_2}\, |x_{23}|^{\Delta_2+\Delta_3-\Delta_1}}} \tag{47.137}$$

up to the single constant C_{123}. With the canonical two-point normalization Eq. (47.124), C_{123} is a scheme-independent physical datum of the theory.

Physical Insight

The numbers C_{123} are the *OPE coefficients* of the theory: the same constant that appears in front of $\mathcal{O}_3$ when the product $\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)$ is expanded in local operators is the coefficient of the three-point function Eq. (47.137), as the operator product expansion in the next-but-one section will make precise. Together with the dimensions Δ_i and spins ℓ_i, the list $\{C_{ijk}\}$ constitutes the *CFT data*. Theorem 47.21 and Theorem 47.24 are the statement that two- and three-point functions store *no* information beyond this data: all of their position dependence is dictated by symmetry. Everything dynamical in a CFT therefore lives in the discrete CFT data and in how that data is constrained — which is the entire content of the bootstrap programme of Chapters 49 to 51.

47.4.5 Cross-ratios: the invariants of four points

The rigidity stops at three points. Four points do admit conformal invariants, and counting them is the key to the four-point function. The configuration of n points in $\mathbb{R}^d$ is nd real numbers; the conformal group has dimension $\frac{1}{2}(d+1)(d+2)$; the number of independent invariants is nd minus the dimension of a generic

orbit, i.e. minus $\frac{1}{2}(d+1)(d+2)$ plus the dimension of the stabilizer of a generic configuration. For $n=3$ this gave zero. For $n=4$ we count by explicitly using up the symmetry.

Proposition 47.25 (Two cross-ratios)**.** *For $d \geq 2$, a generic configuration of four points in $\mathbb{R}^d$ has exactly two independent conformal invariants.*

Proof. Use the conformal group to bring the four points to a standard frame. A translation places $x_1 = 0$; an inversion (followed, if needed, by a translation) sends $x_4 \to \infty$; a dilatation and the rotations then place x_3 at a fixed unit vector e. We have spent the translations (d parameters), the dilatation (1), the special conformal transformations (d, used to send x_4 to infinity and keep x_1 at the origin), and enough rotations to align x_3 with e: aligning a direction in $\mathbb{R}^d$ costs $d-1$ of the $\frac{1}{2}d(d-1)$ rotations, leaving the subgroup $\mathrm{SO}(d-1)$ of rotations about the e-axis as the residual symmetry. The last point x_2 ranges over $\mathbb{R}^d$ (d coordinates) modulo this residual $\mathrm{SO}(d-1)$. Decompose $x_2 = (x_2^{\parallel}, \mathbf{x}_2^{\perp})$ into its component along e and its $(d-1)$ perpendicular components; $\mathrm{SO}(d-1)$ acts only on $\mathbf{x}_2^{\perp}$, with generic orbit the sphere S^{d-2} of dimension $d-2$. The invariants are therefore $x_2^{\parallel}$ and $|\mathbf{x}_2^{\perp}|$: exactly two real numbers. The count $d-(d-2)=2$ is independent of d for $d \geq 2$. □

The two invariants are conventionally packaged into the dimensionless *cross-ratios*

$$u = \frac{x_{12}^2\, x_{34}^2}{x_{13}^2\, x_{24}^2}, \qquad v = \frac{x_{14}^2\, x_{23}^2}{x_{13}^2\, x_{24}^2}, \tag{47.138}$$

where $x_{ij}^2 = |x_i - x_j|^2$. Their conformal invariance is immediate from the covariance $x_{ij}^2 \mapsto \Omega(x_i)\Omega(x_j)\, x_{ij}^2$ established below Eq. (47.121): each cross-ratio is a ratio in which every point appears the same number of times in the numerator and the denominator, so all scale factors cancel,

$$u \mapsto \frac{\Omega_1\Omega_2 x_{12}^2\, \Omega_3\Omega_4 x_{34}^2}{\Omega_1\Omega_3 x_{13}^2\, \Omega_2\Omega_4 x_{24}^2} = u, \tag{47.139}$$

and likewise for v. Any conformal invariant of four points is a function of u and v, since by Theorem 47.25 there are only two of them.

It is often cleaner to trade (u, v) for a single complex coordinate. After the conformal frame of Theorem 47.25, the four points lie in a common 2-plane (the plane through the configuration); choosing complex coordinates on it and placing $x_1 \to 0$, $x_3 \to 1$, $x_4 \to \infty$, the remaining point becomes $x_2 \to z$, and Eq. (47.138) reduce to

$$u = z\,\bar{z}, \qquad v = (1-z)(1-\bar{z}), \tag{47.140}$$

with $\bar{z}$ the complex conjugate of z in Euclidean signature (in Lorentzian signature z and $\bar{z}$ become independent real variables, the source of the lightcone limits exploited in Chapter 51). Figure 47.4 shows the frame.

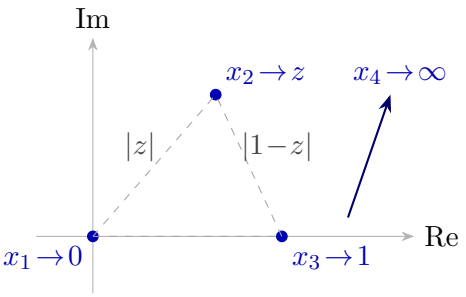


Figure 47.4: The conformal frame for four points. Three points are fixed to 0, 1, ∞ using the conformal group; the position of the fourth, z, is the single complex invariant, related to the cross-ratios by $u = z\bar{z} = |z|^2$ and $v = (1-z)(1-\bar{z}) = |1-z|^2$. The two independent real invariants of Theorem 47.25 are $\mathrm{Re}\, z$ and $\mathrm{Im}\, z$.

Remark 47.26 (Low-point and collinear subtleties). The clean answer "two" requires $d \geq 2$ and a generic configuration. In $d = 1$ the conformal group is only $\mathrm{SL}(2,\mathbb{R})$ (dimension 3) and four points on the line carry a *single* cross-ratio, $z = \bar{z}$ in Eq. (47.140). In any $d \geq 2$ a *collinear* configuration (all four points on one line, equivalently $\mathbf{x}_2^{\perp} = 0$ in the proof above) is non-generic: the residual $\mathrm{SO}(d-1)$ acts trivially on it, the orbit collapses, and only one invariant survives, again $z = \bar{z}$. Thus $z = \bar{z}$ (real) is the collinear/Lorentzian- lightcone locus and $z = \bar{z}^*$ (complex) the generic Euclidean situation — a distinction that matters when the four-point function is continued in Chapter 51. For five or more points new invariants beyond functions of pairwise cross-ratios appear once $n > d+2$, but that is past what we need.

47.4.6 The four-point function and the function $g(u,v)$

For four points the existence of invariants means symmetry can no longer fix the full functional form: it can fix a covariant *prefactor* carrying all the weight, leaving an arbitrary function of u and v. To build the prefactor we look for a quantity $K(x_1,\dots,x_4)$ that transforms exactly as the correlator does under Eq. (47.112), i.e. that has "conformal weight Δ_i" at each point x_i; then the ratio $\langle \mathcal{O}_1\mathcal{O}_2\mathcal{O}_3\mathcal{O}_4\rangle / K$ is invariant and hence a function of the cross-ratios only.

A product $K = \prod_{i<j}(x_{ij}^2)^{p_{ij}}$ has, under a conformal map, the transformation $K \mapsto K \prod_i \Omega(x_i)^{\sum_{j\neq i} p_{ij}}$, using $x_{ij}^2 \mapsto \Omega_i\Omega_j x_{ij}^2$. Comparing with Eq. (47.112), which demands the weight $-\Delta_i$ at each point when the prefactor is pulled to the image side, K matches the correlator iff

$$\sum_{j\neq i} p_{ij} = -\Delta_i \qquad (i = 1,\dots,4). \tag{47.141}$$

This is four equations for the six exponents p_{ij}; the two-parameter family of solutions is the freedom to multiply K by powers of the invariant cross-ratios, which is exactly the freedom we will dump into $g(u,v)$. A convenient choice — the one standard in the bootstrap literature — uses only the pairs $(12),(34),(13),(14),(24)$ and reads

$$K = \frac{1}{(x_{12}^2)^{\frac{\Delta_1+\Delta_2}{2}}\,(x_{34}^2)^{\frac{\Delta_3+\Delta_4}{2}}} \left(\frac{x_{24}^2}{x_{14}^2}\right)^{\frac{\Delta_1-\Delta_2}{2}} \left(\frac{x_{14}^2}{x_{13}^2}\right)^{\frac{\Delta_3-\Delta_4}{2}}. \tag{47.142}$$

One verifies Eq. (47.141) directly. Reading off the exponents, $p_{12} = -\frac{\Delta_1+\Delta_2}{2}$, $p_{34} = -\frac{\Delta_3+\Delta_4}{2}$, $p_{24} = \frac{\Delta_1-\Delta_2}{2}$, $p_{13} = -\frac{\Delta_3-\Delta_4}{2}$, $p_{14} = -\frac{\Delta_1-\Delta_2}{2} + \frac{\Delta_3-\Delta_4}{2}$, and $p_{23} = 0$, the sums are

$$\begin{aligned}
&\text{at } 1: p_{12}+p_{13}+p_{14} = -\tfrac{\Delta_1+\Delta_2}{2} - \tfrac{\Delta_1-\Delta_2}{2} = -\Delta_1,\\
&\text{at } 2: p_{12}+p_{23}+p_{24} = -\tfrac{\Delta_1+\Delta_2}{2} + \tfrac{\Delta_1-\Delta_2}{2} = -\Delta_2,\\
&\text{at } 3: p_{13}+p_{23}+p_{34} = -\tfrac{\Delta_3-\Delta_4}{2} - \tfrac{\Delta_3+\Delta_4}{2} = -\Delta_3,\\
&\text{at } 4: p_{14}+p_{24}+p_{34} = \tfrac{\Delta_3-\Delta_4}{2} - \tfrac{\Delta_3+\Delta_4}{2} = -\Delta_4,
\end{aligned} \tag{47.143}$$

where in the first and fourth lines the $\Delta_3-\Delta_4$ and $\Delta_1-\Delta_2$ contributions cancel, respectively. So K carries precisely the conformal weight of the four-point function, and the quotient is a conformal invariant.

Theorem 47.27 (Four-point function of scalar primaries)**.** *For scalar primaries $\mathcal{O}_1,\dots,\mathcal{O}_4$ of dimensions Δ_i, conformal invariance fixes*

$$\boxed{\langle\, \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\mathcal{O}_4(x_4)\,\rangle = \left(\frac{x_{24}^2}{x_{14}^2}\right)^{\frac{\Delta_1-\Delta_2}{2}} \left(\frac{x_{14}^2}{x_{13}^2}\right)^{\frac{\Delta_3-\Delta_4}{2}} \frac{g(u,v)}{(x_{12}^2)^{\frac{\Delta_1+\Delta_2}{2}}(x_{34}^2)^{\frac{\Delta_3+\Delta_4}{2}}}} \tag{47.144}$$

where $g(u,v)$ is an arbitrary function of the cross-ratios Eq. (47.138), undetermined by symmetry. For four identical scalars of dimension Δ the prefactor collapses and

$$\langle\, \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\,\rangle = \frac{g(u,v)}{(x_{12}^2\,x_{34}^2)^{\Delta}}. \tag{47.145}$$

Proof. The ratio $g \equiv \langle\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3\mathcal{O}_4\rangle / K$ is conformally invariant because numerator and denominator have identical weight by Eq. (47.141), verified above for the choice Eq. (47.142). By Theorem 47.25 a conformal invariant of four points is a function of u and v alone, so $g = g(u,v)$. Multiplying through by K gives Eq. (47.144); setting all $\Delta_i = \Delta$ kills the two ratio factors and uses $\frac{\Delta_1+\Delta_2}{2} = \frac{\Delta_3+\Delta_4}{2} = \Delta$ to give Eq. (47.145). □

Caution

The split of the four-point function into prefactor and $g(u, v)$ is a *choice*, not a physical boundary. A different admissible prefactor — any other solution of Eq. (47.141) — multiplies g by a fixed power of u and v, redefining the function without changing the correlator. When comparing formulas across the bootstrap literature one must therefore track which prefactor convention is in force; we fix Eq. (47.142) throughout Part XIV. What *is* convention-independent is the statement that, once a prefactor is chosen, conformal symmetry leaves exactly one free function of two variables.

Physical Insight

The descent through correlators is a descent in how much symmetry can do. Two points: symmetry fixes everything but a sign-definite constant, which normalization sets to 1. Three points: symmetry fixes everything but one number per operator triple, the OPE coefficient. Four points: symmetry fixes everything but one function of two variables. The reason the staircase has a step at four is purely kinematic — four points are the first that cannot all be moved to a standard position by the conformal group, so they retain the two cross-ratios u, v on which a genuine function can depend. That function $g(u, v)$ is where the dynamics of the theory hides, and the rest of Part XIV is the study of the constraints it must obey.

The function $g(u, v)$ is the object handed to the next stage of the development. The operator product expansion will expand it in a discrete sum of *conformal blocks*, one per primary exchanged between the pairs $(\mathcal{O}_1\mathcal{O}_2)$ and $(\mathcal{O}_3\mathcal{O}_4)$, with coefficients $C_{12k}C_{34k}$ built from the three-point data of Theorem 47.24; and the crossing equation — the statement that the same correlator is obtained by expanding in the pair $(\mathcal{O}_1\mathcal{O}_4)$ instead — will turn the freedom of $g(u, v)$ into the central constraint of the conformal bootstrap in Chapter 49. Before we can carry out that expansion we need the Hilbert-space picture that makes the OPE convergent: the foliation of Euclidean space by spheres, the dilatation operator as a Hamiltonian, and the state–operator correspondence. That radial-quantization framework is the subject of the next section.

47.5 Radial Quantization

Up to this point the conformal algebra of Section 47.2 and the operator data of Section 47.3 have acted *formally*: the generators P_μ, K_μ, D, $M_{\mu\nu}$ were realized as differential operators shuffling local fields around spacetime, and a primary

was a local operator with a prescribed transformation law. Nothing in that description was a Hilbert space. There were no states, no inner product, no notion of a norm, and therefore no place to even *state* the unitarity bounds that the structure of representations was begging us to impose. The dilatation D graded the operators by dimension, but a grading on a vector space of operators is not yet a spectrum of a self-adjoint Hamiltonian.

This section repairs that. We will build the genuine Hilbert space on which the conformal algebra acts as operators, and we will do it by exploiting the single feature that distinguishes a CFT from a generic quantum field theory: the absence of any scale, which makes the dilatation D a symmetry. The construction — *radial quantization* — foliates Euclidean space not by the flat time-slices of ordinary canonical quantization but by spheres centered at the origin, promotes D to the role of the Hamiltonian generating evolution between concentric spheres, and reads off the inner product from the reflection (Osterwalder–Schrader) positivity of the Euclidean path integral established in Section 10.6. The payoff is immediate: the energy of a state is its scaling dimension, the spectrum is bounded below precisely when the unitarity bounds hold, and the abstract Hermiticity relations $D^\dagger = D$, $P_\mu^\dagger = K_\mu$ turn the norm computations we have been postponing into honest statements about a positive-definite inner product. The next section will then promote this picture into the structural theorem of the subject — the one-to-one correspondence between local operators and states.

Throughout we work in Euclidean signature, reached from the Minkowski theory by the Wick rotation $x^0 \to -ix_E^d$ of Theorem 5.30 (Section 5.4); the Euclidean correlators are the moments of the path integral with weight e^{-S_E}, S_E the Euclidean action of Theorem 10.27. Radial quantization is intrinsically Euclidean: the foliating surfaces are spheres, and "time" is the logarithm of the radius, a real parameter only after the rotation. We denote Euclidean coordinates x^μ, $\mu = 1, \dots, d$, with flat metric $\delta_{\mu\nu}$, and write $x^2 \equiv \delta_{\mu\nu} x^\mu x^\nu = |x|^2$.

47.5.1 The radial foliation and the Weyl map to the cylinder

In ordinary canonical quantization one slices spacetime by surfaces of constant time $x^0 = \text{const}$, declares the fields on one slice to be the dynamical data, and uses the time-translation generator H to evolve from one slice to the next. The choice of slicing is a choice of which one-parameter subgroup of the spacetime symmetry one elects to call "time evolution." Poincaré invariance makes the constant-time slicing natural because $H = P^0$ is a symmetry generator with a spectrum bounded below. In a CFT there is a more natural and far more powerful choice. Because the theory is invariant under dilatations $x \mapsto \lambda x$, the family of *spheres centered at the origin*, $|x| = \text{const}$, is permuted among itself by a symmetry. We take these spheres as our equal-"time" slices.

Definition 47.28 (Radial foliation and radial time)**.** Foliate $\mathbb{R}^d \setminus \{0\}$ by the

spheres $S_r = \{x : |x| = r\}$ of radius $r > 0$. Introduce the *radial time*

$$\tau \equiv \log r = \log|x|, \qquad n^\mu \equiv \frac{x^\mu}{|x|} \in S^{d-1}, \tag{47.146}$$

so that (τ, n) are coordinates on the cylinder $\mathbb{R}\times S^{d-1}$, with $\tau \in \mathbb{R}$ the noncompact (time) direction and n a point on the unit sphere. The origin $x = 0$ is the infinite past $\tau \to -\infty$ and spatial infinity is the infinite future $\tau \to +\infty$.

The reason this slicing is not merely a reparametrization but a genuine change of physical picture is geometric: written in the coordinates (τ, n), the flat metric is a position-dependent rescaling of the *cylinder* metric, and a CFT is insensitive to exactly such rescalings.

Proposition 47.29 (Weyl map to the cylinder)**.** *The flat Euclidean metric on $\mathbb{R}^d \setminus \{0\}$ is Weyl-equivalent to the metric of the cylinder $\mathbb{R} \times S^{d-1}$:*

$$\mathrm{d}s^2_{\mathbb{R}^d} = \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2_{d-1} = r^2(\mathrm{d}\tau^2 + \mathrm{d}\Omega^2_{d-1}) = e^{2\tau}\,\mathrm{d}s^2_{\mathrm{cyl}}, \qquad \mathrm{d}s^2_{\mathrm{cyl}} \equiv \mathrm{d}\tau^2 + \mathrm{d}\Omega^2_{d-1}, \tag{47.147}$$

where $\mathrm{d}\Omega^2_{d-1}$ is the round metric on the unit $(d-1)$-sphere and the Weyl factor is $\Omega(x)^2 = r^2 = e^{2\tau}$.

Proof. In spherical coordinates the flat metric is $\mathrm{d}s^2_{\mathbb{R}^d} = \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2_{d-1}$; this is the statement that a small displacement decomposes into a radial part $\mathrm{d}r$ and an angular part of length $r\,|\mathrm{d}n|$. Substitute $r = e^\tau$, so $\mathrm{d}r = e^\tau\,\mathrm{d}\tau$ and $\mathrm{d}r^2 = e^{2\tau}\mathrm{d}\tau^2$, while $r^2 = e^{2\tau}$. Then

$$\mathrm{d}s^2_{\mathbb{R}^d} = e^{2\tau}\mathrm{d}\tau^2 + e^{2\tau}\,\mathrm{d}\Omega^2_{d-1} = e^{2\tau}(\mathrm{d}\tau^2 + \mathrm{d}\Omega^2_{d-1}), \tag{47.148}$$

which is Eq. (47.147). The overall factor $e^{2\tau} = r^2$ multiplies the entire cylinder metric and is therefore a Weyl rescaling, not a change of the conformal class. □

Fig. 47.5 shows the two pictures side by side. The content of Theorem 47.29 is that a conformal field theory does not distinguish $\mathrm{d}s^2_{\mathbb{R}^d}$ from $\mathrm{d}s^2_{\mathrm{cyl}}$ except through the response of its operators to the Weyl factor. A massless theory has no intrinsic length to compare r against, so the only imprint of the rescaling is the conformal weight carried by each operator. We make this precise. Recall (Theorem 47.15) that under a Weyl rescaling of the metric $g_{\mu\nu} \to \Omega(x)^2 g_{\mu\nu}$ a primary scalar $\mathcal{O}$ of dimension Δ transforms homogeneously,

$$\mathcal{O}(x) \longmapsto \Omega(x)^{-\Delta}\,\mathcal{O}(x). \tag{47.149}$$

To pass from the flat description to the cylinder we read Eq. (47.147) as $\mathrm{d}s^2_{\mathrm{cyl}} = \Omega^{-2}\,\mathrm{d}s^2_{\mathbb{R}^d}$ with $\Omega = r = e^\tau$, i.e. we rescale the flat metric *down* by Ω^{-2}. The

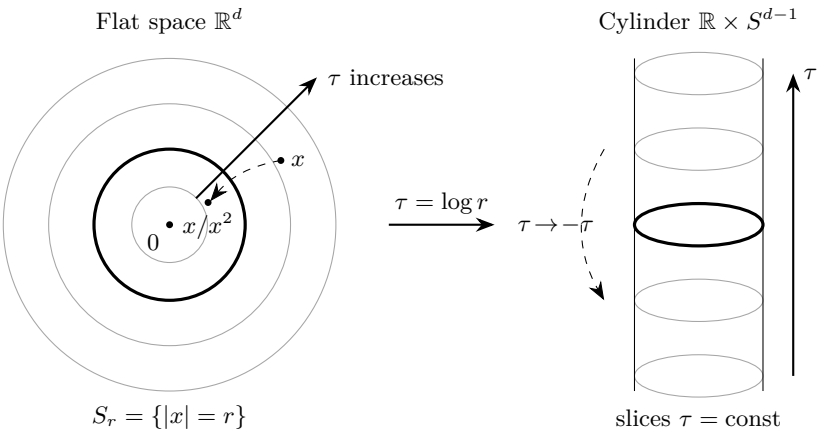


Figure 47.5: The Weyl map of Theorem 47.29. Left: Euclidean space foliated by spheres S_r; radial time is $\tau = \log r$, the origin is $\tau \to -\infty$. The inversion $x \mapsto x/x^2$ exchanges the interior and exterior of the unit sphere (thick) and is the geometric realization of Hermitian conjugation. Right: the same data on the cylinder, where the slices become the round spheres $\tau = \text{const}$ and the inversion becomes the time reflection $\tau \mapsto -\tau$ — precisely the Osterwalder–Schrader reflection.

operator on the cylinder is therefore obtained from the flat-space operator with the inverse weight,

$$\mathcal{O}_{\text{cyl}}(\tau, n) \;=\; \Omega^{+\Delta}\, \mathcal{O}_{\mathbb{R}^d}(x) \;=\; r^{\Delta}\, \mathcal{O}_{\mathbb{R}^d}(x) \;=\; e^{\Delta\tau}\, \mathcal{O}_{\mathbb{R}^d}(x), \qquad x = e^{\tau} n. \tag{47.150}$$

Equivalently the flat-space operator is the cylinder operator dressed with the $r^{-\Delta}$ *factor*,

$$\mathcal{O}_{\mathbb{R}^d}(x) \;=\; r^{-\Delta}\, \mathcal{O}_{\text{cyl}}(\tau, n), \tag{47.151}$$

the single most useful translation between the two pictures: a primary inserted at small r in flat space is, on the cylinder, an excitation in the far past whose flat-space amplitude is suppressed by r^{Δ}.

47.5.2 The dilatation as the radial Hamiltonian

A dilatation $x \mapsto \lambda x$ with $\lambda > 0$ acts on the radius by $r \mapsto \lambda r$ and on radial time by

$$\tau \;=\; \log r \;\longmapsto\; \log(\lambda r) \;=\; \tau + \log \lambda, \tag{47.152}$$

a rigid translation of τ that leaves the angular point $n = x/|x|$ untouched. On the cylinder, then, the dilatation is nothing but a translation in the time direction. The generator of τ-translations is by definition the Hamiltonian of the cylinder

theory, so we are forced to identify

$$H_{\text{cyl}} \;=\; D. \tag{47.153}$$

This identification is the heart of radial quantization. We now show it is consistent at the level of operators and extract the spectrum.

The dilatation generator acts on a local operator by the differential representation recalled in Eq. (47.53), which for a scalar primary reads $[D, \mathcal{O}(x)] = (\Delta + x^\mu \partial_\mu)\,\mathcal{O}(x)$. Exponentiating a finite dilatation by the parameter s, the unitary $U(s) = e^{sD}$ acts as

$$e^{sD}\,\mathcal{O}(x)\,e^{-sD} \;=\; e^{\,s(\Delta + x\cdot\partial)}\,\mathcal{O}(x) \;=\; e^{s\Delta}\,\mathcal{O}(e^s x), \tag{47.154}$$

where the second equality uses the elementary fact that $x\cdot\partial = x^\mu\partial_\mu$ generates rescalings, $e^{s\,x\cdot\partial} f(x) = f(e^s x)$, proved by checking that both sides solve $\partial_s F = (x\cdot\partial)F$ with $F|_{s=0} = f$. Setting $s = \tau$ and $x = n$ on the unit sphere, Eq. (47.154) evolves an operator from the slice $\tau = 0$ outward to radius e^τ:

$$\mathcal{O}(e^\tau n) \;=\; e^{\tau\Delta}\,e^{\tau D}\,\mathcal{O}(n)\,e^{-\tau D}. \tag{47.155}$$

Comparing with Eq. (47.150), the explicit $r^\Delta = e^{\tau\Delta}$ prefactor is exactly the Weyl weight, and the residual operator $e^{\tau D}\mathcal{O}(n)e^{-\tau D}$ is the Heisenberg-picture cylinder operator evolved by the Hamiltonian D. The Euclidean evolution operator that carries data from the slice at τ_1 to the slice at $\tau_2 > \tau_1$ is therefore

$$U(\tau_2, \tau_1) \;=\; e^{-(\tau_2 - \tau_1)D}, \tag{47.156}$$

the precise analog of the Euclidean time-evolution $e^{-(\tau_2-\tau_1)H}$ of an ordinary field theory after Wick rotation. Two remarks fix the picture.

First, Eq. (47.156) makes "energy on the cylinder = scaling dimension" a theorem, not a slogan. A state $|\psi\rangle$ that is a dilatation eigenstate, $D\,|\psi\rangle = E\,|\psi\rangle$, evolves as $U(\tau, 0)\,|\psi\rangle = e^{-E\tau}\,|\psi\rangle$; the decay rate in radial time is its eigenvalue E, which equals the scaling dimension of the operator that creates it. The spectrum of the cylinder Hamiltonian *is* the spectrum of dimensions $\{\Delta\}$ of the theory.

Proposition 47.30 (Dilatation as the radial Hamiltonian)**.** *In radial quantization the Euclidean evolution between concentric spheres is generated by the dilatation D, Eq. (47.156). Consequently the energy of any state on the sphere equals the dilatation eigenvalue, and (anticipating Theorem 47.34) the conformal spectrum is real and, in a unitary theory, bounded below.*

Second, the descendants are the higher-energy excitations built on a primary. From the algebra $[D, P_\mu] = P_\mu$ and $[D, K_\mu] = -K_\mu$ (recall Eq. (47.53)) one reads off

$$D\,(P_\mu\,|\psi\rangle\,) = (E+1)\,P_\mu\,|\psi\rangle\,, \qquad D\,(K_\mu\,|\psi\rangle\,) = (E-1)\,K_\mu\,|\psi\rangle\,, \tag{47.157}$$

so P_μ *raises* the energy by one unit and K_μ *lowers* it. In the radial-quantization Hilbert space the conformal algebra has acquired the ladder structure familiar from the harmonic oscillator, with D playing the role of the number operator graded by dimension. This is the structure the state–operator map of Section 47.6 will exploit.

47.5.3 States from operator insertions and radial ordering

We have a Hamiltonian and an evolution; we now need states. In an ordinary quantum field theory a state on a time slice is prepared by performing the path integral over all field configurations in the past of that slice, with sources or operator insertions specifying the preparation. The same prescription, transcribed to the radial foliation, says: a state on the sphere S_r is prepared by the path integral over the *ball* $B_r = \{|x| \le r\}$ interior to that sphere, with whatever local operators one wishes inserted inside the ball.

Definition 47.31 (State created by an insertion)**.** Let $\mathcal{O}$ be a local operator. The state it creates on a sphere enclosing the origin is

$$|\mathcal{O}\rangle \;\equiv\; \mathcal{O}(0)\,|0\rangle\,, \tag{47.158}$$

where $|0\rangle$ is the conformally invariant vacuum prepared by the path integral over an infinitesimal ball with no insertion. More generally, inserting operators $\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)$ at points x_i with $|x_i| < r$ and integrating over the fields in B_r prepares a (generally non-primary) state on S_r.

That $\mathcal{O}(0)\,|0\rangle$ is well defined — i.e. that inserting an operator at the origin, the infinite past $\tau \to -\infty$, produces a normalizable state rather than a divergence — is exactly the statement that the dilatation spectrum is bounded below: evolving the insertion from $\tau = -\infty$ forward damps every component by $e^{-E\tau}$ with $E \ge 0$, and the lowest-dimension operator (the identity, $\Delta = 0$) produces the vacuum itself. For a primary, Eq. (47.158) gives a dilatation eigenstate, because

$$D\,|\mathcal{O}\rangle = D\,\mathcal{O}(0)\,|0\rangle = [D,\mathcal{O}(0)]\,|0\rangle + \mathcal{O}(0)\,D\,|0\rangle = \Delta\,\mathcal{O}(0)\,|0\rangle + 0 = \Delta\,|\mathcal{O}\rangle\,, \tag{47.159}$$

where we used $[D,\mathcal{O}(0)] = \Delta\,\mathcal{O}(0)$ (the $x\cdot\partial$ piece vanishes at the origin) and $D\,|0\rangle = 0$ (the vacuum is scale invariant). Likewise $K_\mu\,|\mathcal{O}\rangle = 0$, since $[K_\mu,\mathcal{O}(0)] = 0$ for a primary at the origin and $K_\mu\,|0\rangle = 0$: the primary state is the lowest rung of its ladder, annihilated by the lowering operators K_μ, and its descendants are obtained by acting with the raising operators P_μ.

Correlation functions in this picture are radially ordered, the analog of time ordering. A product of operators inside the path integral is unambiguous, but to interpret it as a matrix element of Hilbert-space operators — which must be applied in a definite order — we must order the insertions by their radius, outermost acting last.

Definition 47.32 (Radial ordering)**.** For local operators the radial-ordering symbol $\mathcal{R}$ arranges a product in order of decreasing radius from left to right,

$$\mathcal{R}\{\mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\} \;=\; \begin{cases} \mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2), & |x_1|>|x_2|, \\ \mathcal{O}_2(x_2)\,\mathcal{O}_1(x_1), & |x_2|>|x_1|, \end{cases} \tag{47.160}$$

with bosonic operators commuting and fermionic ones anticommuting under reordering. The Euclidean correlators of Theorem 10.27 equal the vacuum expectation values of the radially ordered products, $\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle = \langle 0|\,\mathcal{R}\{\mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\}\,|0\rangle$.

Radial ordering is to the sphere foliation what time ordering is to the flat foliation: it is the prescription that makes a path-integral correlator equal to an operator matrix element, and it is what guarantees, for instance, that the operator product expansion converges inside the largest sphere that separates two insertions from the rest — a point developed in Section 47.7.3.

47.5.4 The inner product from reflection positivity

A Hilbert space needs an inner product. We have kets $|\mathcal{O}\rangle$; we need bras $\langle\mathcal{O}|$, and we need to know that $\langle\mathcal{O}|\mathcal{O}\rangle \geq 0$. Both come from a single geometric operation: the reflection that the cylinder picture makes obvious.

On the cylinder the natural reflection is $\tau \mapsto -\tau$, exchanging past and future across the slice $\tau = 0$ (the unit sphere). Under the map Eq. (47.146) this is $\log|x| \mapsto -\log|x|$ at fixed angle n, i.e. $|x| \mapsto 1/|x|$, which is the *inversion through the unit sphere*

$$\theta: \quad x^\mu \;\longmapsto\; \frac{x^\mu}{x^2}. \tag{47.161}$$

This θ is precisely the Osterwalder–Schrader reflection of Section 10.6, written in the radial slicing: it reflects the Euclidean "time" τ about the slice $\tau = 0$, exchanging the interior B_1 of the unit ball with its exterior. We use it to define Hermitian conjugation of operators and hence the bra states.

Definition 47.33 (Radial conjugate of an operator)**.** For a scalar primary $\mathcal{O}$ of dimension Δ, Hermitian conjugation in radial quantization acts geometrically by the inversion Eq. (47.35) together with the Weyl weight it carries,

$$\big[\mathcal{O}(x)\big]^\dagger \;=\; \frac{1}{(x^2)^\Delta}\,\mathcal{O}\Big(\frac{x}{x^2}\Big). \tag{47.162}$$

The factor $(x^2)^{-\Delta} = |x|^{-2\Delta}$ is the Jacobian weight Eq. (47.149) of the inversion (whose conformal factor is $\Omega(x) = 1/|x|^2$ acting on $\mathcal{O}$ of dimension Δ contributes the radial scaling that keeps the cylinder operator unit-normalized). The bra dual to $|\mathcal{O}\rangle = \mathcal{O}(0)\,|0\rangle$ is then

$$\langle\mathcal{O}| \;=\; \big(\mathcal{O}(0)\,|0\rangle\,\big)^\dagger \;=\; \lim_{x\to 0}\langle 0|\,\big[\mathcal{O}(x)\big]^\dagger \;=\; \lim_{y\to\infty}(y^2)^\Delta\,\langle 0|\,\mathcal{O}(y), \tag{47.163}$$

the operator sent off to spatial infinity $y = x/x^2 \to \infty$ with the compensating factor that keeps the limit finite.

Theorem 47.34 (The radial-quantization inner product is reflection positive)**.** *The pairing $\langle \mathcal{O}|\mathcal{O}'\rangle$ defined by Eq. (47.163) coincides with an Osterwalder–Schrader reflection inner product of the Euclidean theory. Consequently it is positive semidefinite,*

$$\langle \Psi|\Psi\rangle \;\geq\; 0 \qquad \textit{for every state } |\Psi\rangle \textit{ prepared inside the unit ball,} \tag{47.164}$$

and in a unitary (reflection-positive) CFT it is positive definite on the space of states.

Proof. Let $|\Psi\rangle$ be prepared by inserting operators $F = \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)$ at points strictly inside the unit ball, $|x_i| < 1$, acting on the vacuum: $|\Psi\rangle = F\,|0\rangle$. By Eq. (47.162) the conjugate insertion is built from the reflected operators $\theta(F) = [\mathcal{O}_n(x_n)]^\dagger \cdots [\mathcal{O}_1(x_1)]^\dagger$, each supported at the inverted point x_i/x_i^2 with $|x_i/x_i^2| = 1/|x_i| > 1$: all the reflected insertions lie strictly *outside* the unit ball. The norm is the radially ordered Euclidean correlator

$$\langle \Psi|\Psi\rangle \;=\; \langle 0|\,\mathcal{R}\{\theta(F)\,F\}\,|0\rangle \;=\; \langle\, \theta(F)\,F\,\rangle, \tag{47.165}$$

in which every operator of F sits inside B_1 ($\tau < 0$) and every operator of $\theta(F)$ sits in the reflected region outside B_1 ($\tau > 0$); the radial ordering is automatic because the reflected points all have larger radius. Eq. (47.165) is exactly the bilinear form $\langle \theta(F)\,F\rangle$ to which Osterwalder–Schrader reflection positivity applies (Theorem 10.32): with θ the reflection about the slice $\tau = 0$ and F supported on one side of it, that theorem states $\langle \theta(F)\,F\rangle \geq 0$. Hence Eq. (47.164). Reflection positivity is the Euclidean avatar of unitarity established in Section 10.6; a CFT is called unitary precisely when this form has no null or negative-norm directions on the physical state space, i.e. when it is positive definite. (A nonzero state of zero norm would, by the Cauchy–Schwarz inequality applied to Eq. (47.165), decouple from all correlators and could be quotiented out; a state of negative norm violates unitarity and is excluded by hypothesis.) □

The single most important consequence is that the dimensions are real and the energies are bounded below: a self-adjoint operator D on a space with a positive-definite inner product has a real spectrum, and Eq. (47.164) applied to descendant states will in a moment force $\Delta \geq 0$ (and, with spin, the full unitarity bounds). The norm computations of Section 47.3.5 that we deferred are now well posed: they are statements about a genuine positive-definite Hilbert space.

47.5.5 Hermiticity of the generators: $P_\mu^\dagger = K_\mu$

The inner product of Theorem 47.34 determines the adjoints of the conformal generators. The geometric content is already visible in Fig. 47.5: conjugation

is the inversion θ, and inversion conjugates translations into special conformal transformations. We now prove the Hermiticity relations algebraically, using only the reality of the inner product, the identification Eq. (47.153) of D as a self-adjoint Hamiltonian, and the conformal algebra of Eq. (47.53).

Proposition 47.35 (Hermiticity in radial quantization)**.** *With respect to the radial-quantization inner product the conformal generators satisfy*

$$D^\dagger = D, \qquad M_{\mu\nu}^\dagger = -M_{\mu\nu}, \qquad P_\mu^\dagger = K_\mu, \qquad K_\mu^\dagger = P_\mu. \tag{47.166}$$

Proof. *D is self-adjoint.* By Theorem 47.30 the Euclidean evolution is $U(\tau) = e^{-\tau D}$, Eq. (47.156). Reflection positivity endows the state space with a positive-definite inner product in which the transfer operator relating a slice to its reflection is self-adjoint and positive; concretely, Eq. (47.165) pairs a state evolved forward by τ with its reflection evolved forward by τ, so the bilinear form is invariant under $U(\tau) \mapsto U(\tau)^\dagger = U(\tau)$. Hence $e^{-\tau D}$ is self-adjoint for all real τ, which gives $D^\dagger = D$ and a real dilatation spectrum.

$M_{\mu\nu}$ is anti-self-adjoint. The Euclidean rotations $SO(d)$ act geometrically on the sphere S^{d-1}, commute with D, and map the unit ball to itself; they are therefore realized by operators that preserve the inner product Eq. (47.165), i.e. unitarily, $e^{\theta M_{\mu\nu}}$ unitary for real θ. In our convention (the algebra Eq. (47.53) carries no factor of i in the rotation generators) a unitary one-parameter group $e^{\theta M_{\mu\nu}}$ has an anti-self-adjoint generator, $M_{\mu\nu}^\dagger = -M_{\mu\nu}$.

$P_\mu^\dagger$ has dilatation weight -1. Take the adjoint of $[D, P_\mu] = P_\mu$ (recall Eq. (47.53)). For any operators, $[A, B]^\dagger = [B^\dagger, A^\dagger]$, so using $D^\dagger = D$,

$$[D, P_\mu]^\dagger = [P_\mu^\dagger, D] = -[D, P_\mu^\dagger] = P_\mu^\dagger \quad \Longrightarrow \quad [D, P_\mu^\dagger] = -P_\mu^\dagger. \tag{47.167}$$

Thus $P_\mu^\dagger$ lowers the dilatation eigenvalue by one unit. Among the generators only K_μ has dilatation weight -1 (from $[D, K_\mu] = -K_\mu$), and like $P_\mu^\dagger$ it transforms as a vector under $M_{\mu\nu}$ (this follows from taking the adjoint of $[M_{\mu\nu}, P_\rho]$ and using $M^\dagger = -M$). Hence $P_\mu^\dagger = c\, K_\mu$ for a single constant c.

The constant is $c = 1$. Take the adjoint of the commutator $[K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - M_{\mu\nu})$ (recall Eq. (47.53)). The left side gives, with $K_\mu^\dagger = \bar{c}\, P_\mu$ from $P_\mu^\dagger = cK_\mu$,

$$[K_\mu, P_\nu]^\dagger = [P_\nu^\dagger, K_\mu^\dagger] = [c\, K_\nu, \bar{c}\, P_\mu] = |c|^2\, [K_\nu, P_\mu] = |c|^2 \cdot 2(\delta_{\mu\nu} D - M_{\nu\mu}) = 2|c|^2(\delta_{\mu\nu} D + M_{\mu\nu}) \tag{47.168}$$

using $M_{\nu\mu} = -M_{\mu\nu}$. The right side gives, using $D^\dagger = D$ and $M_{\mu\nu}^\dagger = -M_{\mu\nu}$,

$$\Big(2(\delta_{\mu\nu} D - M_{\mu\nu})\Big)^\dagger = 2(\delta_{\mu\nu} D - M_{\mu\nu}^\dagger) = 2(\delta_{\mu\nu} D + M_{\mu\nu}). \tag{47.169}$$

Equating the two and matching the (linearly independent) D and $M_{\mu\nu}$ structures forces $|c|^2 = 1$. Because the inversion Eq. (47.35) is a real geometric operation

with no accompanying phase — the conjugation Eq. (47.162) carries the real weight $(x^2)^{-\Delta}$ and no i — the constant is real and positive, $c = 1$. Therefore $P_\mu^\dagger = K_\mu$, and conjugating, $K_\mu^\dagger = P_\mu$. □

The relation $P_\mu^\dagger = K_\mu$ is the radial-quantization counterpart of the familiar $a^\dagger/a$ pairing of the oscillator: the raising operators P_μ that generate descendants are the Hermitian conjugates of the lowering operators K_μ that annihilate primaries. This is exactly what is needed to turn a string of descendant states into a computable, manifestly real norm. We close with the computation that does so, and that supplies the Hilbert-space meaning of the scalar unitarity bound.

Worked Example 47.36 (The level-one descendant norm and the scalar unitarity bound)**.** Let $\mathcal{O}$ be a scalar primary of dimension Δ, normalized to $\langle\mathcal{O}|\mathcal{O}\rangle = 1$, and consider its first descendants $P_\nu|\mathcal{O}\rangle$. Their Gram matrix is the inner product of $P_\mu|\mathcal{O}\rangle$ with $P_\nu|\mathcal{O}\rangle$,

$$\mathcal{G}_{\mu\nu} \equiv (P_\mu|\mathcal{O}\rangle)^\dagger (P_\nu|\mathcal{O}\rangle) = \langle\mathcal{O}|P_\mu^\dagger P_\nu|\mathcal{O}\rangle = \langle\mathcal{O}|K_\mu P_\nu|\mathcal{O}\rangle, \tag{47.170}$$

where the last step used $P_\mu^\dagger = K_\mu$ from Theorem 47.35. Now move K_μ to the right past P_ν using $[K_\mu, P_\nu] = 2(\delta_{\mu\nu}D - M_{\mu\nu})$:

$$K_\mu P_\nu = [K_\mu, P_\nu] + P_\nu K_\mu = 2(\delta_{\mu\nu}D - M_{\mu\nu}) + P_\nu K_\mu. \tag{47.171}$$

Acting on the primary state, $K_\mu|\mathcal{O}\rangle = 0$ kills the second term, while $D|\mathcal{O}\rangle = \Delta|\mathcal{O}\rangle$ and $M_{\mu\nu}|\mathcal{O}\rangle = 0$ (the scalar carries no spin), so

$$\mathcal{G}_{\mu\nu} = \langle\mathcal{O}|\,2(\delta_{\mu\nu}D - M_{\mu\nu})\,|\mathcal{O}\rangle = 2\,\delta_{\mu\nu}\,\Delta\,\langle\mathcal{O}|\mathcal{O}\rangle = 2\,\Delta\,\delta_{\mu\nu}. \tag{47.172}$$

By Theorem 47.34 the Gram matrix of any set of states is positive semidefinite, so its diagonal entries are nonnegative: taking $\mu = \nu$ (no sum), $\mathcal{G}_{\mu\mu} = 2\Delta \geq 0$, hence

$$\boxed{\Delta \geq 0} \tag{47.173}$$

for every scalar primary. The bound is saturated, $\Delta = 0$, only when $\mathcal{G} = 0$, i.e. when $P_\mu|\mathcal{O}\rangle = 0$ for all μ: the state is annihilated by translations and is therefore the conformally invariant vacuum, created by the identity operator. Thus $\Delta = 0$ is reached by the identity alone, and every other scalar primary in a unitary CFT has $\Delta > 0$.

This is the simplest instance of the general unitarity bounds. Repeating the computation for a primary in a nontrivial $SO(d)$ representation keeps the spin terms $M_{\mu\nu}|\mathcal{O}\rangle \neq 0$ in Eq. (47.172); positivity of the resulting (now matrix-valued) Gram form yields the bound $\Delta \geq (d-2)/2$ for scalars excluding the identity and the stronger spin-ℓ bounds catalogued in Section 47.3.5. The point established here is structural: those bounds are not assumptions but consequences of Eq. (47.164), the reflection positivity of the Euclidean path integral.

We now have what the construction promised: a Hilbert space carrying a positive-definite inner product, with the dilatation D as a self-adjoint Hamiltonian whose spectrum is the set of scaling dimensions, the primaries sitting at the bottom of conformal ladders ($K_\mu |\mathcal{O}\rangle = 0$) and their descendants generated by the raising operators $P_\mu = K_\mu^\dagger$. The next section turns this dictionary into an exact theorem — the state–operator correspondence — showing that *every* state of this Hilbert space is $\mathcal{O}(0)|0\rangle$ for a unique local operator $\mathcal{O}$, so that the spectrum of D and the spectrum of local operators are one and the same.

47.6 The State-Operator Map

Radial quantization handed us a Hilbert space $\mathcal{H}_{S^{d-1}}$ of states on the sphere, a Hamiltonian — the dilatation generator D, generating evolution in the radial time $\tau = \ln r$ (Section 47.5) — and an inner product certified positive by Osterwalder–Schrader reflection positivity (Theorem 10.32). What we do not yet have is any reason to believe that this Hilbert space is small, computable, or related to the data we already control: the local operators of the theory, organized in Section 47.3 into primaries and their descendants. The central structural theorem of conformal field theory closes exactly this gap. It says that the two lists — "local operators" and "states on the sphere" — are not merely analogous but *identical*: there is a canonical, invertible, conformally covariant map between them. A local operator is a state; a state is a local operator; the spectrum of scaling dimensions and the spectrum of the radial Hamiltonian are one and the same list. This is the fact that makes a CFT a discretely indexed object — a list of dimensions and spins, not a continuum of fields — and it is the rigorous footing on which the convergence of the operator product expansion (Section 47.4, and the next section) will rest.

The map is forced on us by the geometry of radial quantization, and we build it in four movements. First we send an operator to a state by inserting it at the center of the ball and performing the path integral, and we show that the dilatation grading makes this insertion converge as the ball shrinks. Second we run the construction backwards, reproducing any state on the sphere by a (generally infinite) sum of operators inserted at the origin, and we prove the two directions are mutually inverse. Third we show the map intertwines the conformal action: acting with a generator on the state equals taking the state of the transformed operator, so that primaries — the operators annihilated by K_μ — correspond exactly to the lowest-weight states, and descendants $P_{\mu_1}\cdots P_{\mu_n}$ acting on a primary state correspond to the derivative operators $\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}$. Finally we read off the consequence that organizes the rest of the part: the partition function on the cylinder is a sum over operator dimensions, and the lowest dimension above the identity is the radial-quantization gap.

Notation Watch

Throughout this section we use the *radial-quantization generators* of Section 47.5 rather than the differential operators of Eq. (47.41). In the Euclidean radial picture the dilatation D is a Hermitian operator with *real* spectrum — the radial Hamiltonian — so its eigenvalues are the scaling dimensions Δ directly, and the grading brackets Eq. (47.53c) are taken in the form

$$[D, P_\mu] = P_\mu, \qquad [D, K_\mu] = -K_\mu, \qquad [K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - M_{\mu\nu}), \tag{47.174}$$

the factors of i in Eq. (47.53) having been absorbed into this Euclidean normalization (equivalently, the radial dilatation is $-i$ times the Lorentzian one, which turns its imaginary eigenvalue $i\Delta$ into the real dimension Δ). The Euclidean signature is reflected in the adjoint relation established with the inner product in Section 47.5,

$$D^\dagger = D, \qquad M_{\mu\nu}^\dagger = -M_{\mu\nu}, \qquad P_\mu^\dagger = K_\mu, \tag{47.175}$$

so that K_μ is the Hermitian conjugate of P_μ — the lowering operator conjugate to the raising operator. We write $|0\rangle$ for the conformally invariant vacuum.

47.6.1 From an operator to a state

The construction begins with the path-integral definition of a state in radial quantization (Section 47.5, built on Section 10.3). A state on the sphere S_r^{d-1} of radius r is a functional of the field values ϕ_b on that sphere, produced by integrating over all field configurations *inside* the ball B_r with those boundary values held fixed:

$$\Psi[\phi_b] = \int_{\phi|_{\partial B_r} = \phi_b} \mathcal{D}\phi \; e^{-S_E[\phi]}, \tag{47.176}$$

with S_E the Euclidean action and $\mathcal{D}\phi$ the measure of Theorem 10.16. With no insertion this prepares the vacuum $|0\rangle$. The state-operator map is what we obtain by inserting a local operator at the one point the ball distinguishes — its center, the origin, which is the fixed point of every dilatation and rotation.

Definition 47.37 (The map from operators to states)**.** For a local operator $\mathcal{O}$ of the theory, define the state

$$|\mathcal{O}\rangle \;\equiv\; \mathcal{O}(0)\,|0\rangle\,, \tag{47.177}$$

realized as the wavefunctional obtained from Eq. (47.176) with $\mathcal{O}$ inserted at the center of the ball:

$$\Psi_{\mathcal{O}}[\phi_b] = \int_{\phi|_{\partial B_r} = \phi_b} \mathcal{D}\phi \; e^{-S_E[\phi]}\, \mathcal{O}(0). \tag{47.178}$$

We call $|\mathcal{O}\rangle$ the state *created by* $\mathcal{O}$.

The first thing to verify is that this is a genuine, radius-independent element of $\mathcal{H}_{S^{d-1}}$, and that it is graded by the dilatation in the cleanest possible way: it is an eigenstate of the radial Hamiltonian with eigenvalue equal to the scaling dimension. This is what makes the dilatation grading the engine of the whole map.

Proposition 47.38 (Created states are dilatation eigenstates)**.** *Let $\mathcal{O}$ be a scaling operator of dimension Δ, so that $[D, \mathcal{O}(0)] = \Delta\, \mathcal{O}(0)$ in the conventions of Eq. (47.174). Then*

$$D\ |\mathcal{O}\rangle = \Delta\ |\mathcal{O}\rangle\,. \tag{47.179}$$

Proof. The vacuum is annihilated by D — it carries no scale, and we prove its full conformal invariance in Theorem 47.41 below — so $D\,|0\rangle = 0$. Then

$$D\ |\mathcal{O}\rangle = D\,\mathcal{O}(0)\,|0\rangle = [D, \mathcal{O}(0)]\,|0\rangle + \mathcal{O}(0)\,D\,|0\rangle = \Delta\,\mathcal{O}(0)\,|0\rangle = \Delta\ |\mathcal{O}\rangle\,, \tag{47.180}$$

using $[D, \mathcal{O}(0)] = \Delta\,\mathcal{O}(0)$, which is the statement (Section 47.3) that $\mathcal{O}$ has definite scaling dimension Δ. □

Equation (47.179) is the reason the construction is well posed. Because $|\mathcal{O}\rangle$ is an eigenstate of the radial Hamiltonian, the radial evolution $e^{-\tau D}$ from one sphere to another acts on it by a pure number $e^{-\tau\Delta}$: the state is the same abstract vector regardless of the radius $r = e^{\tau}$ at which we read it off, the radius only rescaling its normalization. An operator inserted anywhere *other* than the origin would not enjoy this; only the dilatation fixed point produces a clean eigenstate. Figure 47.6(a) shows the picture.

47.6.2 Shrinking the ball: convergence of the insertion

Inserting an operator at the exact center is a limit, and we owe a demonstration that the limit exists. The clean way to control it is to displace the insertion to a nearby point x inside the ball and expand. Because P_μ generates translations on operators, $[P_\mu, \mathcal{O}(y)] = \partial_\mu \mathcal{O}(y)$ (Section 47.3), the finite translation that carries the origin to x is implemented by $e^{x\cdot P}$, in the sense that $\mathcal{O}(x) = e^{x\cdot P}\,\mathcal{O}(0)\,e^{-x\cdot P}$. Acting on the vacuum and using translation invariance of the vacuum, $P_\mu\,|0\rangle = 0$ (Theorem 47.41), the right factor is trivial and we obtain the exact statement

$$\mathcal{O}(x)\ |0\rangle = e^{x\cdot P}\,\mathcal{O}(0)\ |0\rangle = \sum_{n=0}^{\infty} \frac{1}{n!}\,(x\cdot P)^n\ |\mathcal{O}\rangle\,. \tag{47.181}$$

This is the precise content of Fig. 47.6(b): the state created by an off-center insertion is a sum of the primary state $|\mathcal{O}\rangle$ and its *descendant* states $(x\cdot P)^n\,|\mathcal{O}\rangle$.

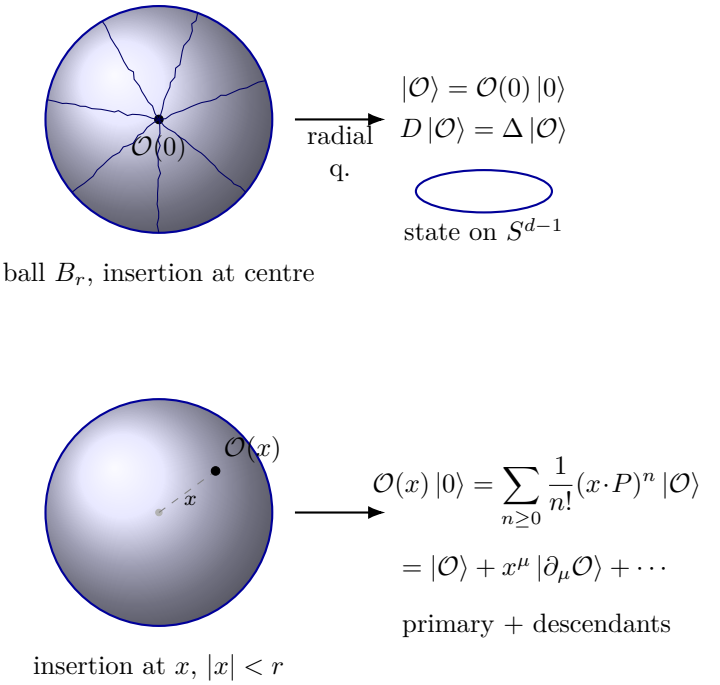


Figure 47.6: The state-operator map. (a) An operator inserted at the center of the ball — the fixed point of dilatations — prepares a state on the bounding sphere S^{d-1}; by Theorem 47.38 it is an eigenstate of the radial Hamiltonian D with eigenvalue equal to the scaling dimension Δ. (b) The same operator displaced to x prepares not an eigenstate but the convergent sum Eq. (47.181) of the primary state and its descendants; bringing $x \to 0$ (shrinking the ball around the insertion) projects onto the leading term $|\mathcal{O}\rangle$, the dilatation grading suppressing the n-th descendant by $|x|^n$.

Each term is itself a dilatation eigenstate, because P_μ raises the dimension by one unit: from $[D, P_\mu] = P_\mu$ in Eq. (47.174),

$$D\,(x\cdot P)^n\,|\mathcal{O}\rangle = (\Delta + n)\,(x\cdot P)^n\,|\mathcal{O}\rangle\,. \tag{47.182}$$

The grading Eq. (47.182) is exactly what makes the series Eq. (47.181) convergent and the $x \to 0$ limit controlled. Read the radial evolution as a damping: a state of dimension $\Delta + n$ scales as $r^{\Delta+n}$ as the bounding sphere shrinks ($r \to 0$), so relative to the leading term the n-th descendant contributes a factor $|x|^n$ (more precisely $(|x|/r)^n$ at fixed reference radius r). For an insertion isolated inside the ball — no other operator within distance R of the origin — the series converges for $|x| < R$, the radius of convergence being set by the nearest other singularity, exactly as for the operator product expansion of the next section; for a single insertion in otherwise empty space it converges for all finite x. The decisive point is that the spectrum of D is bounded below (the vacuum sits at $\Delta = 0$ and there

is a positive gap to the first nontrivial operator, established in Section 47.5 from reflection positivity), so the descendant tower marches upward in dimension and the high-n terms are uniformly suppressed.

Taking $x \to 0$ in Eq. (47.181), every term with $n \geq 1$ carries an explicit factor of x and vanishes, leaving the leading term alone:

$$\lim_{x\to 0} \mathcal{O}(x)\,|0\rangle = |\mathcal{O}\rangle\,. \tag{47.183}$$

Geometrically this is the statement that shrinking the ball down onto the insertion point projects the prepared state onto the primary state, the descendants being radial fluctuations that die away as the sphere closes in. Equation (47.183) is therefore not a definition imposed by hand but a convergent limit, justified by the dilatation grading. The created state $|\mathcal{O}\rangle$ is a well-defined, normalizable vector in $\mathcal{H}_{S^{d-1}}$, with norm fixed by the two-point function as we record next.

The conjugate state deserves a word, because it is what the inner product of Section 47.5 pairs $|\mathcal{O}\rangle$ against. Reflection positivity builds the bra from the operator inserted at the point conjugate to the origin under inversion — namely the point at infinity. Defining $\mathcal{O}(\infty) \equiv \lim_{x\to\infty} |x|^{2\Delta}\,\mathcal{O}(x)$, the conjugate state is $\langle\mathcal{O}| = \lim_{x\to\infty} |x|^{2\Delta}\,\langle 0|\,\mathcal{O}(x)$, and the inner product of two created states is precisely the coefficient of their two-point function (Eq. (47.120)):

$$\langle\mathcal{O}|\mathcal{O}'\rangle = \lim_{x\to\infty} |x|^{2\Delta}\,\langle 0|\,\mathcal{O}(x)\,\mathcal{O}'(0)\,|0\rangle = C_{\mathcal{O}}\,\delta_{\mathcal{O}\mathcal{O}'}, \tag{47.184}$$

nonzero only between operators of equal dimension (and spin), with $C_{\mathcal{O}}$ the two-point normalization. The state-operator map is thus not merely a bijection of vector spaces but an *isometry* once operators are normalized to unit two-point coefficient — the radial inner product on states equals the two-point pairing on operators.

47.6.3 From a state back to an operator

We now run the construction in reverse and prove it is invertible. The claim is that every state on the sphere is created by some local operator at the origin, and that distinct operators create distinct states.

Proposition 47.39 (Surjectivity: every state comes from an operator)**.** *The created states $\{|\mathcal{O}\rangle\}$, as $\mathcal{O}$ ranges over all local operators of the theory (primaries together with all their descendants), span $\mathcal{H}_{S^{d-1}}$.*

Proof. By the Osterwalder–Schrader reconstruction of the Hilbert space (Theorem 10.32), $\mathcal{H}_{S^{d-1}}$ is generated by acting on the vacuum with field operators inserted in the interior of the ball: states of the form $\mathcal{O}_1(x_1)\cdots\mathcal{O}_k(x_k)\,|0\rangle$ with all $|x_i| < r$ span a dense subspace, the field algebra acting on $|0\rangle$ being dense by the reconstruction theorem. It therefore suffices to show each such multi-insertion

state lies in the span of $\{|\mathcal{O}\rangle\}$. Take first a single insertion $\mathcal{O}(x)\,|0\rangle$ with $|x| < r$. By Eq. (47.181) it is the convergent series $\sum_n \frac{1}{n!}(x\cdot P)^n\,|\mathcal{O}\rangle$, every term of which is a created state (the descendant $(x\cdot P)^n\mathcal{O}$ is a local operator). Hence $\mathcal{O}(x)\,|0\rangle$ lies in the closed span of created states. For a product of insertions, the same expansion applies term by term: each operator can be Taylor-expanded about the origin into its conformal family, and the product of local operators near the origin is, by the operator product expansion (Section 47.4), itself a convergent sum of local operators at the origin — a statement whose rigorous footing is precisely the present map, and which the next section establishes as a consequence of completeness rather than assuming it. Thus the dense set of multi-insertion states lies in the closed span of $\{|\mathcal{O}\rangle\}$, and the span is all of $\mathcal{H}_{S^{d-1}}$. □

Proposition 47.40 (Injectivity: distinct operators give distinct states)**.** *If $|\mathcal{O}\rangle = |\mathcal{O}'\rangle$ as states in $\mathcal{H}_{S^{d-1}}$, then $\mathcal{O} = \mathcal{O}'$ as local operators.*

Proof. A local operator is determined by its correlation functions with all other operators (Theorem 10.16): two operators agreeing inside every correlator are the same operator. Suppose $|\mathcal{O}\rangle = |\mathcal{O}'\rangle$. For any state $\langle\Psi|$ in the dense spanning set of Theorem 47.39 — a product of insertions $\mathcal{O}_1(y_1)\cdots\mathcal{O}_k(y_k)$ at points outside the ball, $|y_i| > r$, acting on the vacuum from the left — the overlap with the created state is, by the path-integral construction Eq. (47.178) glued across the sphere, the correlation function

$$\langle\Psi|\mathcal{O}\rangle = \langle 0|\,\mathcal{O}_1(y_1)\cdots\mathcal{O}_k(y_k)\,\mathcal{O}(0)\,|0\rangle\,. \tag{47.185}$$

Equality of the states $|\mathcal{O}\rangle = |\mathcal{O}'\rangle$ forces these correlators to coincide for every choice of external insertions:

$$\langle 0|\,\mathcal{O}_1(y_1)\cdots\mathcal{O}_k(y_k)\,\mathcal{O}(0)\,|0\rangle = \langle 0|\,\mathcal{O}_1(y_1)\cdots\mathcal{O}_k(y_k)\,\mathcal{O}'(0)\,|0\rangle \tag{47.186}$$

for all k and all y_i. Since correlators determine the operator, $\mathcal{O}(0) = \mathcal{O}'(0)$, hence $\mathcal{O} = \mathcal{O}'$. □

Together Theorems 47.39 and 47.40 establish that Eq. (47.177) is a bijection between local operators and states. Combined with the conformal covariance of the next subsection, this is the state-operator *isomorphism*.

47.6.4 The map intertwines the conformal action

A bijection of sets becomes a theorem about representations once we check it commutes with the conformal generators. The key input is that the vacuum is conformally invariant, which we now prove, having used it twice above.

Proposition 47.41 (Conformal invariance of the vacuum)**.** *The radial-quantization vacuum is annihilated by every conformal generator:*

$$P_\mu\,|0\rangle = 0, \qquad M_{\mu\nu}\,|0\rangle = 0, \qquad D\,|0\rangle = 0, \qquad K_\mu\,|0\rangle = 0. \tag{47.187}$$

Proof. The vacuum is the state Eq. (47.176) prepared by the path integral over the ball with no insertion. The action S_E and the measure $\mathcal{D}\phi$ of a conformal field theory are invariant under the conformal group — this is the operator statement, established in Section 47.1, that the improved stress tensor is conserved and traceless, so that the conserved charges $\{P_\mu, M_{\mu\nu}, D, K_\mu\}$ generate symmetries of the functional integral. A symmetry transformation acts on the no-insertion functional by relabeling the integration variable, under which the integral is unchanged; with no operator inserted there is nothing to carry a position, a scale, or a direction that the transformation could move. Hence the vacuum functional is fixed by every finite conformal transformation $U_g\,|0\rangle = |0\rangle$, and differentiating at the identity gives $G\,|0\rangle = 0$ for each generator G. Equivalently and more physically: $D\,|0\rangle = 0$ says the vacuum has dimension zero (it is the state of the identity operator); $P_\mu\,|0\rangle = 0$ and $M_{\mu\nu}\,|0\rangle = 0$ are Poincaré invariance; and $K_\mu\,|0\rangle = 0$ is invariance under special conformal transformations, the new content of conformal symmetry. That a single state is annihilated by both the raising operators P_μ and the lowering operators K_μ is consistent precisely because it sits at the bottom *and* is unique: it is the conformally invariant vacuum. □

With the vacuum invariant, the action of any generator on a created state collapses onto a commutator, which is again a local operator at the origin — and therefore again a created state. This is the intertwining property.

Theorem 47.42 (The state-operator isomorphism)**.** *The map* $\mathcal{O} \mapsto |\mathcal{O}\rangle = \mathcal{O}(0)\,|0\rangle$ *is a bijection between the local operators of the CFT and the states of* $\mathcal{H}_{S^{d-1}}$*, and it intertwines the conformal action: for every generator* $G \in \{P_\mu, M_{\mu\nu}, D, K_\mu\}$*,*

$$G\;|\mathcal{O}\rangle = [G, \mathcal{O}(0)]\;|0\rangle = |\,G{\cdot}\mathcal{O}\,\rangle\,, \tag{47.188}$$

where $G{\cdot}\mathcal{O}$ *denotes the local operator* $[G, \mathcal{O}(0)]$ *reinstated at the origin. Consequently the conformal representation carried by the operators (Section 47.3) and the representation carried by the Hilbert space are isomorphic as representations of* $\mathfrak{so}(d+1,1)$*.*

Proof. Bijectivity is Theorems 47.39 and 47.40. For the intertwining, expand the generator through the operator using $G\,|0\rangle = 0$ from Theorem 47.41:

$$G\;|\mathcal{O}\rangle = G\,\mathcal{O}(0)\,|0\rangle = [G, \mathcal{O}(0)]\,|0\rangle + \mathcal{O}(0)\,G\,|0\rangle = [G, \mathcal{O}(0)]\,|0\rangle\,. \tag{47.189}$$

The commutator $[G, \mathcal{O}(0)]$ is, by the transformation law of operators under the conformal generators (Section 47.3), a local operator evaluated at the origin; calling it $G{\cdot}\mathcal{O}$, the last expression is $|G \cdot \mathcal{O}\rangle$ by Eq. (47.177). Since this holds for every generator and the map is a bijection, it is an isomorphism of $\mathfrak{so}(d+1,1)$ representations. □

Equation (47.188) is the whole theorem in one line, and reading it off generator by generator dictionary gives the correspondence between the two languages. The dilatation reproduces Theorem 47.38, $D\,|\mathcal{O}\rangle = \Delta\,|\mathcal{O}\rangle$. The momentum acts as the derivative,

$$P_\mu\,|\mathcal{O}\rangle = [P_\mu, \mathcal{O}(0)]\,|0\rangle = \partial_\mu \mathcal{O}(0)\,|0\rangle = |\partial_\mu \mathcal{O}\rangle\,, \tag{47.190}$$

so raising the state by P_μ corresponds to differentiating the operator, in accord with the dimension rising by one. The Lorentz generators rotate the spin indices, $M_{\mu\nu}\,|\mathcal{O}\rangle = |\,\mathcal{S}_{\mu\nu}\mathcal{O}\,\rangle$ with $\mathcal{S}_{\mu\nu}$ the spin matrix. The new and decisive entry is the special conformal generator. By the definition of a primary operator (Theorem 47.15), $[K_\mu, \mathcal{O}(0)] = 0$; therefore

$$K_\mu\,|\mathcal{O}\rangle = [K_\mu, \mathcal{O}(0)]\,|0\rangle = 0 \qquad \text{for } \mathcal{O} \text{ primary.} \tag{47.191}$$

This is the cornerstone of the map.

Corollary 47.43 (Primaries are lowest-weight states; descendants are their orbit)**.** *Under the state-operator map:*

1. *A* primary operator $\mathcal{O}$ *of dimension* Δ *and spin* ℓ *corresponds to a* lowest-weight *(conformal-primary) state* $|\mathcal{O}\rangle$*: a* D*-eigenstate of eigenvalue* Δ *in the spin-*ℓ *representation of the rotation group, annihilated by all the lowering operators,* $K_\mu\,|\mathcal{O}\rangle = 0$*.*

2. *The* descendants *of* $\mathcal{O}$ *— the operators* $\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}$ *(Theorem 47.15) — correspond to the states obtained by acting with the raising operators on the primary state,*

$$|\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}\rangle = P_{\mu_1}\cdots P_{\mu_n}\,|\mathcal{O}\rangle\,. \tag{47.192}$$

   The conformal multiplet generated by $\mathcal{O}$ *maps isomorphically onto the conformal module* $\{P_{\mu_1}\cdots P_{\mu_n}\,|\mathcal{O}\rangle\}$ *built on the lowest-weight state.*

Proof. Item (1) collects Theorem 47.38, the spin statement under $M_{\mu\nu}$, and Eq. (47.191). That K_μ is a lowering operator — $[D, K_\mu] = -K_\mu$ in Eq. (47.174) sends a state of dimension Δ to one of dimension $\Delta - 1$ — makes "annihilated by K_μ" synonymous with "lowest-weight": there is nothing below it in the module. Item (2) is Eq. (47.190) iterated: each application of P_μ both adds a derivative to the operator, by Eq. (47.188), and raises the dimension by one, by $[D, P_\mu] = P_\mu$. The map being a vector-space isomorphism (Theorem 47.42) and respecting the P-action, it carries the operator family bijectively onto the state module. □

The picture is now complete and rigid. Every state of $\mathcal{H}_{S^{d-1}}$ sits in a tower built on a lowest-weight state $|\mathcal{O}\rangle$ killed by K_μ; the towers are the conformal multiplets; the lowest-weight states are the primaries; the upper floors are reached

by the raising operators P_μ and are the descendants. Acting with K_μ moves *down* a tower until it hits the primary and annihilates. This is the abstract shadow of the OPE: a complete basis of states, sorted into multiplets, each labeled by a primary's dimension and spin.

Physical Insight

The state-operator map is special to conformal field theory; it has no analogue in a massive theory. In a massive QFT the states on a spatial slice are multiparticle configurations with continuous momenta and energies bounded below by particle masses, while the local operators form an entirely different bookkeeping; there is no natural isomorphism between them. What makes the CFT case work is the absence of any scale: radial slices are all equivalent up to a dilatation, the ball can be shrunk to a point, and "operator inserted at a point" becomes literally the same data as "state on the surrounding sphere." The dilatation generator playing the double role of *Hamiltonian* (in the radial picture) and *dimension counter* (on operators) is the hinge on which the whole equivalence turns.

47.6.5 One spectrum: the cylinder partition function and the gap

The isomorphism of Theorem 47.42 has an immediate and far-reaching consequence: the list of scaling dimensions of local operators and the list of energies of the radial Hamiltonian are the same list. We make this quantitative on the cylinder, where the radial Hamiltonian is literally an energy.

Radial quantization on $\mathbb{R}^d$ is, by the Weyl rescaling that strips the overall r^2 from the flat metric, a theory on the cylinder $\mathbb{R}\times S^{d-1}$ (Section 47.5): writing $x^\mu = r\,n^\mu$ with n a unit vector and $\tau = \ln r$,

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2 = r^2\big(\mathrm{d}\tau^2 + \mathrm{d}\Omega^2\big), \tag{47.193}$$

which is the cylinder metric $\mathrm{d}\tau^2 + \mathrm{d}\Omega^2_{S^{d-1}}$ up to the Weyl factor r^2. Under this map the dilatation generator D becomes the generator of translations in cylinder time τ — the Hamiltonian H_{cyl} of the theory quantized on the spatial sphere S^{d-1} of unit radius. Up to a c-number Casimir energy E_0, induced by the Weyl anomaly of the rescaling Eq. (47.193) (a constant shift, computable from the central charge, that we do not need in detail here),

$$H_{\mathrm{cyl}} = D + E_0. \tag{47.194}$$

The energy of a state on the sphere therefore equals the scaling dimension of the corresponding operator, $E = \Delta + E_0$. The state-operator map turns this into the headline statement: *the energy spectrum of the CFT on S^{d-1} is the list of operator dimensions.*

Corollary 47.44 (Cylinder partition function as a sum over dimensions)**.** *The thermal partition function of the CFT on the spatial sphere S^{d-1}, at inverse temperature β (geometrically, the theory on $S^1_\beta \times S^{d-1}$), is a sum over all local operators of the theory:*

$$Z(\beta) = \mathrm{Tr}_{\mathcal{H}_{S^{d-1}}} e^{-\beta H_{\mathrm{cyl}}} = e^{-\beta E_0} \sum_{\text{all } \mathcal{O}} e^{-\beta \Delta_{\mathcal{O}}} = e^{-\beta E_0} \sum_{\text{primaries } p} \chi_{\Delta_p, \ell_p}(\beta), \tag{47.195}$$

where the trace runs over the complete basis of created states, the second sum runs over every primary and the descendants of its multiplet, and $\chi_{\Delta_p,\ell_p}(\beta) = \sum_{n\geq 0}(\text{level-}n\text{ degeneracy})\, e^{-\beta(\Delta_p+n)}$ is the conformal character collecting the primary p together with its descendant tower.

Proof. The trace over $\mathcal{H}_{S^{d-1}}$ is, by Theorem 47.42, a sum over the complete basis of created states $|\mathcal{O}\rangle$; each contributes $e^{-\beta H_{\mathrm{cyl}}} = e^{-\beta(\Delta_{\mathcal{O}}+E_0)}$ by Eq. (47.194) and Eq. (47.179). Pulling out $e^{-\beta E_0}$ gives the first equality. Grouping the operators by multiplet — each primary p of dimension Δ_p together with its descendants $\partial^n \mathcal{O}_p$ at dimension $\Delta_p + n$, per Theorem 47.43 — resums each tower into the character χ_{Δ_p,ℓ_p}, giving the second equality. □

The lowest dimension above the identity controls the small-β-independent structure of Eq. (47.195) and is the central spectral datum of radial quantization. The vacuum (the identity operator) sits at $\Delta = 0$ and supplies the leading $e^{-\beta E_0}$; the first nontrivial term comes from the operator of lowest dimension $\Delta_{\mathrm{gap}} > 0$ above it:

$$\Delta_{\mathrm{gap}} = \min\{\Delta_{\mathcal{O}} : \mathcal{O} \neq \mathbf{1}\} \tag{47.196}$$

is precisely the energy gap of the radial Hamiltonian above its ground state. A positive gap is the unitarity-plus-locality input from Section 47.5; the state-operator map identifies it with the lowest operator dimension, the most basic invariant of the CFT data. Everything the bootstrap will constrain — where the lowest operators sit, how their dimensions are bounded — is a statement about this spectrum, which is now, by the theorem of this section, simultaneously a statement about states on the sphere.

Worked Example 47.45 (The free scalar: its state, its tower, and the gap)**.** We make the entire construction explicit on the theory ϕ^4 theory has carried to this point: the massless free scalar ϕ in d dimensions, the simplest CFT, whose conformal status was established in Section 47.1.5. Its primary field ϕ has scaling dimension $\Delta_\phi = \frac{d-2}{2}$ (the engineering dimension of a canonically normalized scalar, which here is exact because the theory is free).

The state and its dimension. The map sends ϕ to the state $|\phi\rangle = \phi(0)\,|0\rangle$, a scalar (spin-0) eigenstate of the radial Hamiltonian of dimension $\Delta_\phi = \frac{d-2}{2}$ by Eq. (47.179). In $d = 4$ this is $\Delta_\phi = 1$.

ϕ is primary: $K_\mu |\phi\rangle = 0$. By Theorem 47.43 this is equivalent to $|\phi\rangle$ being lowest-weight. We can see it directly from the spectrum without invoking the differential realization. The state $K_\mu |\phi\rangle$, if nonzero, would have dimension $\Delta_\phi - 1 = \frac{d-4}{2}$ (since K_μ lowers the dimension by one, Eq. (47.174)) and would carry a single vector index, i.e. it would be a spin-1 state. But there is no operator of dimension $\frac{d-4}{2}$ in the free theory below ϕ except possibly the identity, at dimension 0; and the identity creates the spin-0 vacuum, which cannot equal a spin-1 state. By rotational covariance $K_\mu |\phi\rangle$ must therefore vanish: ϕ is a primary, and $|\phi\rangle$ is a lowest-weight state.

The descendant tower. The raising operators build the multiplet. The first descendant is

$$P_\mu |\phi\rangle = |\partial_\mu \phi\rangle \,, \tag{47.197}$$

a spin-1 state of dimension $\Delta_\phi + 1 = \frac{d}{2}$, corresponding to the operator $\partial_\mu \phi$. At the next level the symmetric traceless part $\partial_{(\mu}\partial_{\nu)}\phi$ is a spin-2 descendant, and so on. But the trace part is special, and exhibits the one subtlety of a free field.

The null descendant and the unitarity bound. Consider the scalar level-2 descendant $P^2 |\phi\rangle = P^\mu P_\mu |\phi\rangle$, corresponding to $\partial^2\phi = \Box\phi$. The free equation of motion is $\Box\phi = 0$, so this operator vanishes identically and the corresponding state must be *null* (zero norm). We verify this from the algebra alone, which simultaneously reproduces the scalar unitarity bound. For a scalar primary $\mathcal{O}$ of dimension Δ (so $M_{\mu\nu}|\mathcal{O}\rangle = 0$ and $K_\mu |\mathcal{O}\rangle = 0$), compute the norm of $P^2 |\mathcal{O}\rangle$ using $P_\mu^\dagger = K_\mu$ from Eq. (47.175):

$$\left\| P^2 |\mathcal{O}\rangle \right\|^2 = \langle \mathcal{O}| K^\sigma K_\sigma P^\mu P_\mu |\mathcal{O}\rangle \,. \tag{47.198}$$

We first move one K through P^2. Using $[K_\sigma, P_\nu] = 2(\delta_{\sigma\nu} D - M_{\sigma\nu})$ from Eq. (47.174) and the Leibniz rule,

$$\begin{aligned} [K_\sigma, P^\mu P_\mu] &= [K_\sigma, P^\mu]\, P_\mu + P^\mu\, [K_\sigma, P_\mu] \\ &= 2(D P_\sigma - M_\sigma{}^\mu P_\mu) + 2(P_\sigma D - P^\mu M_{\sigma\mu}). \end{aligned} \tag{47.199}$$

The rotation terms combine using $[M_{\sigma\mu}, P^\mu] = (d-1)P_\sigma$ (from $[M_{\mu\nu}, P_\rho] = \delta_{\nu\rho}P_\mu - \delta_{\mu\rho}P_\nu$, contracting): writing $M_\sigma{}^\mu P_\mu = P^\mu M_{\sigma\mu} + (d-1)P_\sigma$, the two rotation pieces add to $2P^\mu M_{\sigma\mu} + (d-1)P_\sigma$ with an overall minus. The dilatation terms give $DP_\sigma + P_\sigma D = 2P_\sigma D + P_\sigma$ via $[D, P_\sigma] = P_\sigma$. Collecting,

$$\left[K_\sigma, P^2\right] = 4P_\sigma D - 4P^\mu M_{\sigma\mu} + (4 - 2d)P_\sigma. \tag{47.200}$$

Acting on the scalar primary, $M_{\sigma\mu}|\mathcal{O}\rangle = 0$ and $D|\mathcal{O}\rangle = \Delta|\mathcal{O}\rangle$, and $K_\sigma|\mathcal{O}\rangle = 0$, so

$$K_\sigma P^2 |\mathcal{O}\rangle = \left[K_\sigma, P^2\right] |\mathcal{O}\rangle = (4\Delta + 4 - 2d) P_\sigma |\mathcal{O}\rangle = 2(2\Delta + 2 - d) P_\sigma |\mathcal{O}\rangle \,. \tag{47.201}$$

Contracting with the second K and using $K^\sigma P_\sigma |\mathcal{O}\rangle = [K^\sigma, P_\sigma]\,|\mathcal{O}\rangle = 2dD\,|\mathcal{O}\rangle = 2d\Delta\,|\mathcal{O}\rangle$ (here $[K^\sigma, P_\sigma] = 2(d\,D - M^\sigma{}_\sigma) = 2dD$ since $M^\sigma{}_\sigma = 0$),

$$\left\| P^2\,|\mathcal{O}\rangle \right\|^2 = 2(2\Delta + 2 - d)\cdot 2d\Delta\;\|\mathcal{O}\|^2 = 4d\,\Delta(2\Delta + 2 - d)\;\|\mathcal{O}\|^2\,. \quad (47.202)$$

Positivity of the norm in a unitary theory requires $2\Delta + 2 - d \geq 0$, i.e. the scalar *unitarity bound*

$$\Delta \geq \frac{d-2}{2}, \quad (47.203)$$

with equality exactly when $\left\| P^2\,|\mathcal{O}\rangle \right\|^2 = 0$, i.e. when the level-2 scalar descendant is null. The free scalar saturates the bound, $\Delta_\phi = \frac{d-2}{2}$, so $P^2\,|\phi\rangle$ is null — the algebraic statement of the equation of motion $\Box\phi = 0$. The free-field multiplet is therefore a *short* multiplet: the primary ϕ with all descendants $\partial_{\mu_1}\cdots\partial_{\mu_n}\phi$ except those proportional to $\Box\phi$, which are removed. This is the conformal-representation-theoretic content of "ϕ obeys a free equation of motion," read straight off the state-operator map.

The gap. In the free scalar the operators are the normal-ordered products of ϕ and its derivatives. Above the identity ($\Delta = 0$), the lowest-dimension operator is ϕ itself, at $\Delta_\phi = \frac{d-2}{2}$ (the next, the bilinear $:\!\phi^2\!:$, sits at $d-2$, and the stress tensor at d). Hence the radial-quantization gap Eq. (47.196) of the free scalar is

$$\Delta_{\text{gap}} = \frac{d-2}{2}, \quad (47.204)$$

which in $d = 4$ equals 1: the energy of the lightest state on S^3 above the vacuum, in units where the sphere has unit radius, is exactly the dimension of the free field. This is the concrete seed from which ϕ^4 theory's account of the Wilson–Fisher fixed point (Chapter 52) will grow: turning on the interaction in $d = 4 - \epsilon$ shifts Δ_ϕ to $\frac{d-2}{2} + \gamma_\phi$ with $\gamma_\phi = O(\epsilon^2)$, deforming this gap by the anomalous dimension while the entire state-operator architecture built here carries over unchanged.

We have proved the structural theorem on which the rest of the part stands: local operators and radial-quantization states are the same data, primaries are the lowest-weight states annihilated by K_μ, descendants are their P-orbit, and the spectrum of operator dimensions is the spectrum of the dilatation Hamiltonian. Because the states $\{|\mathcal{O}\rangle\}$ form a complete, conformally graded basis, any product of two operators inserted inside a sphere can be resolved against this basis — a finite-radius-of-convergence expansion in primaries and their descendants. That resolution is the conformal operator product expansion, and its convergence, which we have just supplied the rigorous footing for, is the subject of the next section.

47.7 The Operator Product Expansion and Its Convergence

The state–operator isomorphism of the previous section did more than match local operators to states of the radial-quantization Hilbert space; it gave us a *complete* basis. Every state on a sphere is a superposition of the lowest-weight states created by primaries and their conformal-algebra orbit of descendants (Theorem 47.42), and these multiplets together resolve the identity (Eq. (47.207)). We now cash in that completeness. The product of two local operators inserted at nearby points is, when acted on the vacuum, just a state on any sphere that encloses both insertions. Expanding that state in the complete basis turns the product of two operators at separate points into a single sum of local operators at one point — the *operator product expansion*. This is the structural fact that makes a conformal field theory finitely specified: once we know the spectrum of primaries and the constants that govern how two primaries fuse, every correlation function is determined. The whole of this section is the unpacking of that claim. We define the conformal OPE, show that its coefficients are exactly the three-point constants already computed in Section 47.4, argue — honestly, with the rigorous part separated from the physically expected part — that the expansion genuinely converges inside a finite radius, and finally use it twice over to reduce the four-point function to a sum of universal kinematic functions, the conformal blocks, whose defining differential equation we identify.

Throughout this section we work in Euclidean signature, as in the radial-quantization construction of Section 47.5: $\mathbb{R}^d$ with $x \cdot y = \delta_{\mu\nu} x^\mu y^\nu$ and $|x|^2 = x \cdot x \geq 0$, so that $|x_{ij}|$ with $x_{ij} \equiv x_i - x_j$ is an ordinary Euclidean distance and there are no light cones to navigate. Scaling dimensions Δ are the eigenvalues of the dilatation generator D in the sense fixed in Theorem 23.22 and realized on operators in Section 47.2.

47.7.1 The operator product expansion from completeness

Fix two scalar primaries $\mathcal{O}_1, \mathcal{O}_2$ of dimensions Δ_1, Δ_2. Place the second at the origin and the first at a point x with $|x|$ small, and act on the vacuum:

$$|\Psi(x)\rangle \;\equiv\; \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,|0\rangle\,. \tag{47.205}$$

By construction this is a definite vector in the radial-quantization Hilbert space: it is the state read off on any sphere $|y| = \rho$ with $\rho > |x|$, since both insertions then lie strictly inside that sphere. The state–operator map (Theorem 47.42) furnishes a complete orthogonal basis of that Hilbert space, organized into conformal multiplets: one lowest-weight state $|\mathcal{O}\rangle$ for each primary $\mathcal{O}$, together with its descendants, the states

$$P_{\mu_1} \cdots P_{\mu_k}\,|\mathcal{O}\rangle\,, \qquad k = 0, 1, 2, \ldots, \tag{47.206}$$

of dimension $\Delta_{\mathcal{O}} + k$ obtained by acting with the raising operators P_μ (recall $[D, P_\mu] = P_\mu$, Eq. (47.53)). Because this basis is orthonormal and complete, it resolves the identity on the radial-quantization Hilbert space,

$$\mathbf{1} \;=\; \sum_{n} |n\rangle \langle n| \,, \tag{47.207}$$

the sum running over the orthonormal basis $\{|n\rangle\}$ formed by all primaries and their descendants. Inserting the resolution of the identity Eq. (47.207) into Eq. (47.205) writes $|\Psi(x)\rangle$ as a sum over all such basis states, each weighted by the overlap $\langle n|\, \mathcal{O}_1(x)\mathcal{O}_2(0)\, |0\rangle$.

Two features of those overlaps are decisive. First, the bra $\langle n|$ is, under the state–operator map, the conjugate of a descendant of some primary $\mathcal{O}$ placed at spatial infinity; the overlap $\langle n|\, \mathcal{O}_1(x)\mathcal{O}_2(0)\, |0\rangle$ is therefore a three-point function of $\mathcal{O}_1$, $\mathcal{O}_2$, and (a descendant of) $\mathcal{O}$. Conformal symmetry — as we are about to verify — fixes every such three-point function up to a *single* constant per primary $\mathcal{O}$. Second, the x-dependence of each overlap is fixed by dilatation covariance alone. The composite $\mathcal{O}_1(x)\mathcal{O}_2(0)$ carries total dimension $\Delta_1 + \Delta_2$ at separation $|x|$; a basis state of dimension Δ_n is reached only if the accompanying coefficient supplies the missing dimension, i.e. scales as $|x|^{\Delta_n - \Delta_1 - \Delta_2}$. Collecting the k-th descendant level of a primary $\mathcal{O}$, where $\Delta_n = \Delta_{\mathcal{O}} + k$, the coefficient scales as $|x|^{\Delta_{\mathcal{O}} - \Delta_1 - \Delta_2 + k}$: the leading power is set by the primary and each descendant level adds one integer power of $|x|$.

Reinterpreting each basis state $|n\rangle = \big(\text{descendant operator}\big)(0)\, |0\rangle$ back as a local operator at the origin, the expansion of the state Eq. (47.205) becomes an operator equation, valid inside any correlation function in which no other insertion intrudes between the two operators and the sphere:

$$\mathcal{O}_1(x)\, \mathcal{O}_2(0) \;=\; \sum_{\mathcal{O}} C_{12\mathcal{O}}\, |x|^{\Delta_{\mathcal{O}} - \Delta_1 - \Delta_2}\, \mathcal{C}_{\Delta_{\mathcal{O}}, \ell_{\mathcal{O}}}(x, \partial)\, \mathcal{O}(0), \tag{47.208}$$

where the sum runs over *primaries* $\mathcal{O}$ only, $C_{12\mathcal{O}}$ is a single number for each primary, and $\mathcal{C}_{\Delta_{\mathcal{O}}, \ell_{\mathcal{O}}}(x, \partial) = 1 + (\text{const})\, x^\mu \partial_\mu + \dots$ is a power series in x and ∂ that packages the descendant contributions. The differential operator $\mathcal{C}$ generates, term by term, the operators $\partial_{\mu_1} \cdots \partial_{\mu_k} \mathcal{O}(0)$ that are the operator-language images of the descendant states Eq. (47.206). The descendants do not bring new free constants: they are entirely fixed by conformal symmetry in terms of the single primary coefficient $C_{12\mathcal{O}}$, as the next subsection proves explicitly at first order and explains in general.

Definition 47.46 (Conformal operator product expansion)**.** Let $\mathcal{O}_1, \mathcal{O}_2$ be primaries of a conformal field theory. The *operator product expansion* is the convergent operator identity Eq. (47.208), the sum running over the primary operators $\mathcal{O}$ of the theory, in which:

- the leading power $|x|^{\Delta_{\mathcal{O}} - \Delta_1 - \Delta_2}$ is dictated by the dimensions and is the same in every correlator;

- the number $C_{12\mathcal{O}}$ is the *OPE coefficient* of the fusion $\mathcal{O}_1 \times \mathcal{O}_2 \to \mathcal{O}$, one constant per primary;
- the differential operator $\mathcal{C}_{\Delta_\mathcal{O},\ell_\mathcal{O}}(x,\partial)$, normalized to begin with the identity, is a power series in x fixed entirely by conformal symmetry (the dimension $\Delta_\mathcal{O}$ and spin $\ell_\mathcal{O}$ of $\mathcal{O}$), encoding the contributions of all descendants of $\mathcal{O}$.

The identity holds as an operator statement inside any correlation function, provided every other operator insertion lies farther from the origin than x does (the precise convergence statement is Theorem 47.48).

The identity operator $\mathbf{1}$ is always among the primaries on the right-hand side when $\mathcal{O}_2 = \mathcal{O}_1$ (or more generally when $\mathcal{O}_1, \mathcal{O}_2$ share a fusion channel into it): with $\Delta_\mathbf{1} = 0$ it contributes the leading singularity $C_{11\mathbf{1}}\,|x|^{-2\Delta_1}\,\mathbf{1}$. Unit-normalizing the two-point function $\langle \mathcal{O}_1(x)\mathcal{O}_1(0)\rangle = |x|^{-2\Delta_1}$ fixes $C_{11\mathbf{1}} = 1$, so the identity channel of the OPE simply reproduces the two-point function — a consistency check we will use repeatedly.

Physical Insight

The OPE replaces "two operators at two points" by "one tower of operators at one point." Physically it is the statement that, probed from far enough away, a pair of nearby excitations is indistinguishable from a single local disturbance — and conformal symmetry tells us *exactly* which local disturbances, with which relative strengths. The infinitely many descendants might look like infinitely much data, but they are not: each multiplet contributes precisely one new number, $C_{12\mathcal{O}}$. This is why the list of primary dimensions, spins, and OPE coefficients — the "CFT data" — is a *complete* specification of the theory. Everything else is symmetry.

47.7.2 The coefficients are the three-point data

The claim that $C_{12\mathcal{O}}$ is a single number per primary, equal to the three-point constant of Section 47.4, is proved by inserting the OPE into a three-point function and matching. Recall the form fixed there by conformal symmetry: for scalar primaries,

$$\langle \mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\,\mathcal{O}_3(x_3)\rangle = \frac{C_{123}}{|x_{12}|^{\Delta_1+\Delta_2-\Delta_3}\,|x_{23}|^{\Delta_2+\Delta_3-\Delta_1}\,|x_{13}|^{\Delta_1+\Delta_3-\Delta_2}}, \tag{47.209}$$

with C_{123} a single theory-dependent constant (Eq. (47.137)), and the orthonormalized two-point function

$$\langle \mathcal{O}_i(x)\,\mathcal{O}_j(0)\rangle = \frac{\delta_{ij}}{|x|^{2\Delta_i}}, \tag{47.210}$$

which vanishes between primaries of different dimension (Eq. (47.120)).

Take the correlator $\langle \mathcal{O}_1(x)\mathcal{O}_2(0)\,\mathcal{O}_3(y)\rangle$ with $|x|$ small and $|y|$ large, and apply the OPE Eq. (47.208) to the pair $\mathcal{O}_1(x)\mathcal{O}_2(0)$:

$$\langle \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,\mathcal{O}_3(y)\rangle = \sum_{\mathcal{O}} C_{12\mathcal{O}}\,|x|^{\Delta_{\mathcal{O}}-\Delta_1-\Delta_2}\,\langle \mathcal{C}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(x,\partial)\,\mathcal{O}(0)\;\mathcal{O}_3(y)\rangle. \tag{47.211}$$

At leading order in $|x|$ the differential operator is the identity, and only the term $\mathcal{O}=\mathcal{O}_3$ survives the two-point orthogonality Eq. (47.210): $\langle \mathcal{O}(0)\mathcal{O}_3(y)\rangle = \delta_{\mathcal{O},\mathcal{O}_3}\,|y|^{-2\Delta_3}$. Hence the leading behavior is

$$\langle \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,\mathcal{O}_3(y)\rangle \xrightarrow{|x|\to 0} C_{12\mathcal{O}_3}\,\frac{|x|^{\Delta_3-\Delta_1-\Delta_2}}{|y|^{2\Delta_3}}. \tag{47.212}$$

We compare this against the small-$|x|$ limit of the exact three-point function Eq. (47.209) with $x_1=x$, $x_2=0$, $x_3=y$. As $|x|\to 0$, $|x_{12}|=|x|$, $|x_{23}|=|y|$, and $|x_{13}|=|x-y|\to|y|$, so

$$\langle \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,\mathcal{O}_3(y)\rangle \xrightarrow{|x|\to 0} \frac{C_{123}}{|x|^{\Delta_1+\Delta_2-\Delta_3}\,|y|^{(\Delta_2+\Delta_3-\Delta_1)+(\Delta_1+\Delta_3-\Delta_2)}} = C_{123}\,\frac{|x|^{\Delta_3-\Delta_1-\Delta_2}}{|y|^{2\Delta_3}}. \tag{47.213}$$

The two expressions are identical provided

$$\boxed{C_{12\mathcal{O}_3} = C_{123}\,.} \tag{47.214}$$

The OPE coefficient is the three-point constant. This is the central economy of the conformal program: the same finite list of numbers $\{C_{ijk}\}$ that parametrizes all three-point functions also governs how operators fuse, and therefore — through repeated use of the OPE — determines every higher correlator.

It remains to see that the descendant tail $\mathcal{C}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}$ carries no further freedom. The subleading terms in Eq. (47.211) must reproduce the subleading terms in the expansion of the exact Eq. (47.209) about $|x|=0$, and the only structure available on the exact side is the expansion of $|x-y|^{-(\Delta_1+\Delta_3-\Delta_2)}$ in powers of x. That expansion is completely determined by the dimensions; matching it order by order fixes every coefficient of $\mathcal{C}$ in terms of $C_{12\mathcal{O}}$ and the dimensions, with no new constants. We carry out the first order in the following worked example, both to exhibit the mechanism and to record the coefficient that the conformal-block literature uses constantly.

Worked Example 47.47 (The first descendant coefficient)**.** We determine the leading correction to the identity in $\mathcal{C}$ for a scalar primary $\mathcal{O}$ of dimension Δ exchanged between scalars $\mathcal{O}_1,\mathcal{O}_2$. Write the OPE through first order in x,

$$\mathcal{O}_1(x)\,\mathcal{O}_2(0) = C_{12\mathcal{O}}\,|x|^{\Delta-\Delta_1-\Delta_2}\Big[\,\mathcal{O}(0)+\beta\,x^{\mu}\partial_{\mu}\mathcal{O}(0)+O(x^2)\,\Big]+(\text{other primaries}), \tag{47.215}$$

with β the unknown to be fixed. We need the two-point functions of $\mathcal{O}$ with its own first descendant. From Eq. (47.210),

$$\langle \partial_\mu \mathcal{O}(z)\, \mathcal{O}(y)\rangle\Big|_{z=0} = \partial_{z^\mu} |z-y|^{-2\Delta}\Big|_{z=0} = -2\Delta\,(z-y)_\mu\, |z-y|^{-2\Delta-2}\Big|_{z=0} = 2\Delta\,\frac{y_\mu}{|y|^{2\Delta+2}}. \tag{47.216}$$

Insert Eq. (47.215) into $\langle \mathcal{O}_1(x)\mathcal{O}_2(0)\,\mathcal{O}(y)\rangle$ and use Eqs. (47.210) and (47.216):

$$\langle \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,\mathcal{O}(y)\rangle = C_{12\mathcal{O}}\,|x|^{\Delta-\Delta_1-\Delta_2}\left[\frac{1}{|y|^{2\Delta}} + 2\Delta\,\beta\,\frac{x\cdot y}{|y|^{2\Delta+2}} + O(x^2)\right]. \tag{47.217}$$

On the exact side, factor the prefactor $|x|^{\Delta-\Delta_1-\Delta_2}$ out of Eq. (47.209) (using $-(\Delta_1+\Delta_2-\Delta) = \Delta-\Delta_1-\Delta_2$) and expand the remaining $|x-y|$ dependence. With $a \equiv \Delta_1+\Delta-\Delta_2$ and $|x-y|^2 = |y|^2 - 2\,x\cdot y + |x|^2$,

$$|x-y|^{-a} = |y|^{-a}\Big(1 - \tfrac{2\,x\cdot y}{|y|^2} + \tfrac{|x|^2}{|y|^2}\Big)^{-a/2} = |y|^{-a}\Big(1 + a\,\frac{x\cdot y}{|y|^2} + O(x^2)\Big). \tag{47.218}$$

The two powers of $|y|$ on the exact side combine as $|y|^{-(\Delta_2+\Delta-\Delta_1)}\,|y|^{-a} = |y|^{-2\Delta}$, so

$$\langle \mathcal{O}_1(x)\,\mathcal{O}_2(0)\,\mathcal{O}(y)\rangle = C_{123}\,|x|^{\Delta-\Delta_1-\Delta_2}\left[\frac{1}{|y|^{2\Delta}} + a\,\frac{x\cdot y}{|y|^{2\Delta+2}} + O(x^2)\right]. \tag{47.219}$$

The zeroth-order terms reconfirm $C_{12\mathcal{O}} = C_{123}$. Matching the first-order terms gives $2\Delta\,\beta = a = \Delta_1+\Delta-\Delta_2$, i.e.

$$\boxed{\beta = \frac{\Delta+\Delta_1-\Delta_2}{2\Delta}\,,} \tag{47.220}$$

which for equal external dimensions $\Delta_1 = \Delta_2$ collapses to the symmetric value $\beta = \frac{1}{2}$. The first descendant coefficient is fixed by the dimensions alone, exactly as claimed; no information beyond $C_{12\mathcal{O}}$ entered. The same matching at order x^2, x^3, … determines $\mathcal{C}$ to all orders, and the resummed series is precisely what the conformal block of the next subsections repackages.

47.7.3 Convergence of the operator product expansion

In ordinary perturbative quantum field theory the OPE is an *asymptotic* short-distance statement, valid term by term as $|x| \to 0$ with no claim of convergence at finite separation. In a conformal field theory the situation is far stronger, and the strength comes directly from radial quantization: the OPE is the expansion of an honest, normalizable state in a complete orthonormal basis, and such expansions converge.

Theorem 47.48 (Convergence of the conformal OPE)**.** *In a unitary conformal field theory with a discrete spectrum of primaries, the operator product expansion*

Eq. (47.208) of $\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)$, inserted in a correlation function with any number of further operators $\mathcal{O}_3(x_3), \dots, \mathcal{O}_n(x_n)$, converges — in the sense of convergence of the resulting numerical series — whenever there exists a sphere centered on the pair $\{x_1, x_2\}$ that separates them from all the other insertions. Concretely, if the pair is enclosed in a ball of radius r_{12} and the nearest other operator sits at distance $r_{\text{near}} > r_{12}$ from the center, the expansion converges, and it converges exponentially fast in the ratio r_{12}/r_{near}. The radius of convergence is set by the distance to the nearest other insertion.

Argument. Center radial quantization on the midpoint (or any convenient point) of the pair $\{x_1, x_2\}$ and place all other operators at radii larger than r_{near}. The vector

$$|\Psi\rangle = \mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\,|0\rangle \tag{47.221}$$

is the state read off on any sphere of radius ρ with $r_{12} < \rho < r_{\text{near}}$, since on such a sphere the two operators lie inside and none of the others has yet been crossed. Reflection positivity — the Euclidean form of unitarity established by the Osterwalder–Schrader reconstruction (Chapter 10) — guarantees that the Hilbert space has a positive-definite inner product and that $|\Psi\rangle$ has finite norm, because that norm is the convergent two-point-of-pairs correlator

$$\big\||\Psi\rangle\big\|^2 = \big\langle \mathcal{O}_2(x_2^\theta)\,\mathcal{O}_1(x_1^\theta)\;\mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2)\big\rangle, \tag{47.222}$$

with x^θ the radial-reflection image of x, a four-point function evaluated at strictly separated points and therefore finite. Now expand $|\Psi\rangle$ in the complete orthonormal basis of Eq. (47.207), which is exactly the sum over primaries and descendants. Because the basis is complete and $|\Psi\rangle$ has finite norm, the partial sums converge to $|\Psi\rangle$ in the Hilbert-space norm; equivalently the squared coefficients sum to $\big\||\Psi\rangle\big\|^2 < \infty$. Pairing this convergent expansion of $|\Psi\rangle$ against the state produced by the remaining operators $\mathcal{O}_3(x_3)\cdots\mathcal{O}_n(x_n)\,|0\rangle$ on the same sphere yields the OPE expansion of the full correlator as a convergent numerical series. This is precisely the OPE Eq. (47.208) term by term, since each basis state is the image of a descendant operator.

The rate is controlled by the radial coordinate. A state of dimension Δ_n contributes to $|\Psi\rangle$ on the sphere of radius ρ a factor scaling as $(r_{12}/\rho)^{\Delta_n}$ relative to the leading term, by dilatation covariance (the same power counting that produced the $|x|^{\Delta_n-\Delta_1-\Delta_2}$ of the OPE). Summing over the spectrum and taking ρ up to its maximal value r_{near}, the high-dimension tail is suppressed by $(r_{12}/r_{\text{near}})^{\Delta_n}$; since unitarity bounds the density of states (the spectrum is discrete and grows in a controlled way), the series converges geometrically in r_{12}/r_{near}. The expansion fails to converge precisely when no separating sphere exists, i.e. when a third operator lies as close to the pair as the pair members are to each other — the radius of convergence is the distance to the nearest other insertion. □

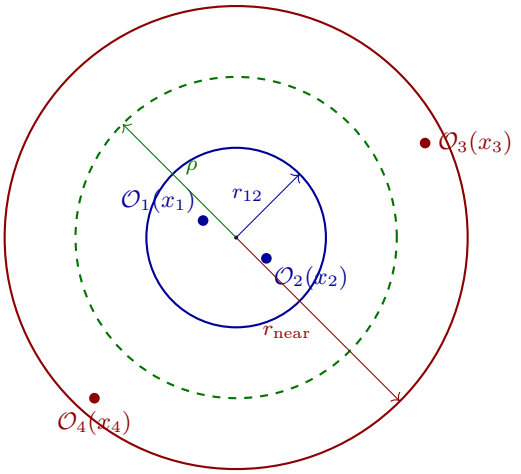


Figure 47.7: Radial quantization and the convergence of the OPE (Theorem 47.48). The pair $\mathcal{O}_1, \mathcal{O}_2$ to be fused sits inside the ball of radius r_{12} (blue); the nearest other insertion lies at radius r_{near} (red). On any quantization sphere of intermediate radius ρ (green dashed, $r_{12} < \rho < r_{\text{near}}$) the state $\mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\,|0\rangle$ is a definite finite-norm vector, and its expansion in the complete basis of primaries and descendants is the OPE. The series converges geometrically in r_{12}/r_{near}; it breaks down only when a third operator approaches the pair and no separating sphere survives.

Caution

The honest scope of Theorem 47.48, dated to the present (mid-2020s). The convergence statement is a *theorem*, not merely a physical expectation, but it is a theorem *within* the standard axioms of a conformal field theory: a Hilbert space with a positive inner product (reflection positivity / unitarity), a discrete spectrum of operators bounded below in dimension, and the state–operator map of Theorem 47.42. Granting those axioms, the expansion converges with a finite radius and a quantitative (exponential) rate; the careful estimates were given by Pappadopulo, Rychkov, Espin, and Rattazzi [376], building on the older formal partial-wave analyses [190]. What is an *assumption* rather than a derived fact is that a given physical theory — say the lattice realization of a critical point — actually satisfies those axioms with a genuine state–operator isomorphism. For the theories of this book that is established or overwhelmingly evidenced: the free scalar exactly, the Wilson–Fisher fixed point order by order in ϵ, the 3D Ising CFT to extraordinary numerical precision. The convergence is therefore as rigorous as the axiomatic foundation it rests on, and that foundation is exactly the reflection positivity of Chapter 10 together with

the radial quantization of Section 47.5.

The contrast with the asymptotic Wilsonian OPE of the renormalization-group analysis is worth stating once plainly. There, the short-distance expansion was organized by the engineering dimensions of operators and truncated; here, the conformal OPE is a *convergent* expansion in a basis fixed by the exact scaling dimensions, valid out to a finite separation, not merely as $|x| \to 0$. This finite radius is what makes the OPE a computational tool rather than a formal device, and it is the property the bootstrap of Chapter 49 will exploit when it demands that two different OPE channels of the same four-point function agree.

47.7.4 The four-point function and conformal blocks

The first genuinely dynamical correlator — the one not fixed outright by symmetry — is the four-point function. Conformal symmetry fixes two- and three-point functions completely (Section 47.4); for four points the same kinematics leaves one undetermined function of the two conformally invariant cross-ratios. Recall (Eq. (47.138)) the cross-ratios built from four points,

$$u = \frac{x_{12}^2\, x_{34}^2}{x_{13}^2\, x_{24}^2}, \qquad v = \frac{x_{14}^2\, x_{23}^2}{x_{13}^2\, x_{24}^2}, \tag{47.223}$$

which are invariant under all conformal transformations (including the inversion of Theorem 47.4), and the standard stripping of the four-point function of four identical scalars $\mathcal{O}_1 = \cdots = \mathcal{O}_4 = \phi$ of dimension Δ_ϕ into a kinematic prefactor times a function of the cross-ratios (Eq. (47.145)),

$$\langle \phi(x_1)\, \phi(x_2)\, \phi(x_3)\, \phi(x_4) \rangle = \frac{g(u,v)}{x_{12}^{2\Delta_\phi}\, x_{34}^{2\Delta_\phi}}. \tag{47.224}$$

Symmetry has done all it can; the content of the theory is hidden in the single function $g(u,v)$. The OPE turns that function into a sum over the operator content.

Apply the OPE Eq. (47.208) to the pair (12) and, independently, to the pair (34):

$$\langle \phi(x_1)\phi(x_2)\, \phi(x_3)\phi(x_4) \rangle = \sum_{\mathcal{O},\mathcal{O}'} C_{\phi\phi\mathcal{O}}\, C_{\phi\phi\mathcal{O}'} \Big\langle [\mathcal{C}\,\mathcal{O}](x_2)\ [\mathcal{C}\,\mathcal{O}'](x_4) \Big\rangle. \tag{47.225}$$

The two-point function of a descendant of $\mathcal{O}$ with a descendant of $\mathcal{O}'$ vanishes unless $\mathcal{O}$ and $\mathcal{O}'$ are the same primary, by the orthogonality Eq. (47.210) extended to whole multiplets (descendants of distinct primaries are orthogonal). The double sum collapses to a single sum over exchanged primaries:

$$\langle \phi\phi\, \phi\phi \rangle = \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2 \ (\text{contribution of the full multiplet of } \mathcal{O}). \tag{47.226}$$

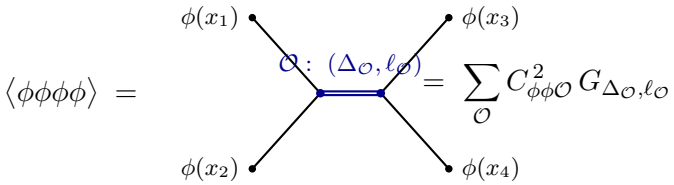


Figure 47.8: The conformal block decomposition Eq. (47.227). Fusing $\phi(x_1)\phi(x_2)$ and $\phi(x_3)\phi(x_4)$ by the OPE projects the four-point function onto exchanged primaries $\mathcal{O}$; each exchanged multiplet (double line, carrying dimension $\Delta_\mathcal{O}$ and spin $\ell_\mathcal{O}$) contributes a universal conformal block $G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v)$ weighted by the squared OPE coefficient $C^2_{\phi\phi\mathcal{O}}$. The block resums the primary and all of its descendants. The explicit closed forms of the blocks are computed in Chapter 49.

The multiplet contribution, once the kinematic prefactor of Eq. (47.224) is stripped off, depends only on the cross-ratios and on the quantum numbers $(\Delta_\mathcal{O}, \ell_\mathcal{O})$ of the exchanged primary — it is a purely kinematic, theory-independent function. This function is the conformal block.

Definition 47.49 (Conformal block)**.** The *conformal block* $G_{\Delta,\ell}(u,v)$ is the contribution to the stripped four-point function $g(u,v)$ of a single primary $\mathcal{O}$ of dimension Δ and spin ℓ together with *all* of its descendants, exchanged in the (12)(34) channel, normalized so that the OPE coefficients appear as in Eq. (47.227). It resums the entire descendant tail $\mathcal{C}_{\Delta,\ell}$ of Eq. (47.208) on both sides of Eq. (47.225), and is fixed entirely by conformal symmetry — independent of the particular theory.

With this definition the four-point function decomposes as

$$\boxed{g(u,v) = \sum_\mathcal{O} C^2_{\phi\phi\mathcal{O}}\; G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v)\;,} \tag{47.227}$$

and for non-identical externals the same construction gives $g(u,v) = \sum_\mathcal{O} C_{12\mathcal{O}}\, C_{34\mathcal{O}}\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u$ with the block additionally depending on the dimension differences Δ_{12}, Δ_{34} through its descendant structure. This is the *conformal block decomposition*. It is the cleanest statement of how the abstract CFT data — the dimensions and spins of primaries and the OPE coefficients — assemble into a measurable correlator: the squared OPE coefficients are the weights, and the blocks are universal kinematic functions common to all conformal field theories in d dimensions. Figure 47.8 renders the decomposition diagrammatically as a sum over exchanged primaries.

The explicit functional form of $G_{\Delta,\ell}(u,v)$ — in even dimensions a finite combination of hypergeometric functions, in general dimension a solution defined by a recursion or by the Casimir equation below — is a computation we defer

to Section 49.2, where it is needed for the bootstrap. What we owe here is the property that makes the block well defined and computable in the first place: it is the eigenfunction of a single differential operator, the conformal Casimir.

47.7.5 The conformal Casimir and the block equation

The conformal group $\mathfrak{so}(d+1,1)$ has, like any Lie group, a quadratic Casimir invariant: the unique (up to normalization) quadratic combination of generators that commutes with all of them. Built from the field-theory generators $D, P_\mu, K_\mu, M_{\mu\nu}$ of Eq. (47.53), it is

$$C_2 \;=\; D^2 \;-\; \tfrac{1}{2}\{P_\mu, K^\mu\} \;-\; \tfrac{1}{2}\, M_{\mu\nu}M^{\mu\nu}, \tag{47.228}$$

the image of $-\frac{1}{2}L_{AB}L^{AB}$ under the embedding of the conformal algebra into the rotations L_{AB} of $\mathfrak{so}(d+1,1)$, with the overall sign fixed so that the eigenvalue comes out as below. Because C_2 commutes with every generator, it is constant on each irreducible conformal multiplet; it therefore takes a single value on the entire orbit of a primary, and that value labels the multiplet.

Proposition 47.50 (Casimir eigenvalue on a primary)**.** *On a primary state $|\Delta,\ell\rangle$ of dimension Δ and spin ℓ (a symmetric traceless rank-ℓ representation of the rotation subgroup $\mathfrak{so}(d)$), the quadratic Casimir Eq. (47.228) acts as the scalar*

$$C_2\,|\Delta,\ell\rangle = c_{\Delta,\ell}\,|\Delta,\ell\rangle\,, \qquad c_{\Delta,\ell} = \Delta(\Delta-d)+\ell(\ell+d-2). \tag{47.229}$$

Proof. A primary state is annihilated by the lowering generators, $K_\mu\,|\Delta,\ell\rangle = 0$, is a dilatation eigenstate, $D\,|\Delta,\ell\rangle = \Delta\,|\Delta,\ell\rangle$, and carries the spin-$\ell$ representation of $\mathfrak{so}(d)$ (Theorem 47.15). Evaluate the three pieces of Eq. (47.228) in turn.

The first piece is immediate: $D^2\,|\Delta,\ell\rangle = \Delta^2\,|\Delta,\ell\rangle$.

For the second, write $\{P_\mu, K^\mu\} = P_\mu K^\mu + K^\mu P_\mu$ (the index μ summed with the Euclidean δ). Acting on the primary, the first term vanishes because K^μ does, $P_\mu K^\mu\,|\Delta,\ell\rangle = 0$. The second term we reduce with the algebra by commuting K to the right, where it annihilates the state:

$$K^\mu P_\mu\,|\Delta,\ell\rangle = \big[K^\mu, P_\mu\big]\,|\Delta,\ell\rangle + P_\mu K^\mu\,|\Delta,\ell\rangle = \big[K^\mu, P_\mu\big]\,|\Delta,\ell\rangle\,. \tag{47.230}$$

The commutator is the conformal-algebra relation $[K_\mu, P_\nu] = 2(\delta_{\mu\nu}D - M_{\mu\nu})$ of Eq. (47.53). Contracting $\mu=\nu$ and summing, $\delta_\mu{}^\mu = d$ and $M_\mu{}^\mu = 0$ by antisymmetry, so

$$\big[K^\mu, P_\mu\big] = 2\,(d\,D - M_\mu{}^\mu) = 2d\,D. \tag{47.231}$$

Therefore $K^\mu P_\mu\,|\Delta,\ell\rangle = 2d\,D\,|\Delta,\ell\rangle = 2d\,\Delta\,|\Delta,\ell\rangle$, and the second piece contributes $-\frac{1}{2}\{P_\mu,K^\mu\}\,|\Delta,\ell\rangle = -d\,\Delta\,|\Delta,\ell\rangle$.

For the third, the combination $-\frac{1}{2}M_{\mu\nu}M^{\mu\nu}$ appearing in Eq. (47.228) is the quadratic Casimir of the rotation subalgebra $\mathfrak{so}(d)$ in the no-i convention used here: since $M_{\mu\nu}$ acts on the primary as the spin matrix $\Sigma_{\mu\nu}$, it is exactly $C_2(\ell) = -\frac{1}{2}\Sigma_{\mu\nu}\Sigma^{\mu\nu}$ of Eq. (47.96), which on the symmetric traceless rank-ℓ representation takes the standard value $\ell(\ell+d-2)$ (the $\mathfrak{so}(d)$ Casimir, recorded in Chapter B), vanishing for a scalar, $\ell = 0$. Equivalently $\frac{1}{2}M_{\mu\nu}M^{\mu\nu} = -\ell(\ell+d-2)$ on the representation, the sign reflecting that the no-i rotation generators are anti-self-adjoint (Theorem 47.35).

Summing the three contributions,

$$C_2\,|\Delta,\ell\rangle = \left[\Delta^2 - d\,\Delta + \ell(\ell+d-2)\right]|\Delta,\ell\rangle = \left[\Delta(\Delta-d) + \ell(\ell+d-2)\right]|\Delta,\ell\rangle\,, \tag{47.232}$$

which is Eq. (47.229). The scalar case $\ell = 0$, in which only the D and $\{P, K\}$ terms contribute, gives $c_{\Delta,0} = \Delta(\Delta - d)$ and is the cleanest illustration of the mechanism: the entire eigenvalue is generated by commuting one lowering operator past one raising operator. □

The eigenvalue Eq. (47.229) is the bridge to the conformal block. Inside the four-point function, the Casimir built from the generators acting on the pair of points (x_1, x_2) — call it $\mathcal{D}_2$, the differential operator obtained by realizing $C_2^{(1)} + C_2^{(2)}$ on the coordinates through the conformal Ward identities — acts on the (12)(34) OPE channel. Because the OPE organizes the correlator into conformal multiplets and C_2 is diagonal on each multiplet with eigenvalue $c_{\Delta_\mathcal{O},\ell_\mathcal{O}}$, the contribution of the multiplet of $\mathcal{O}$ is an eigenfunction of $\mathcal{D}_2$ with that eigenvalue. Acting on the stripped function and using Eq. (47.227), every block must independently satisfy

$$\boxed{\mathcal{D}_2\,G_{\Delta,\ell}(u,v) = c_{\Delta,\ell}\,G_{\Delta,\ell}(u,v), \qquad c_{\Delta,\ell} = \Delta(\Delta-d) + \ell(\ell+d-2)\,.} \tag{47.233}$$

This second-order partial differential equation in the cross-ratios — the *conformal Casimir equation* of Dolan and Osborn [155, 156] — together with the boundary condition that $G_{\Delta,\ell}$ begin with the leading power $u^{(\Delta-\ell)/2}$ dictated by the OPE, defines the block uniquely. It is the differential equation we will integrate explicitly in Chapter 49; here we have established its existence, its eigenvalue, and the symmetry reason — the Casimir is constant on multiplets — that the blocks are its eigenfunctions. The explicit operator $\mathcal{D}_2$ and its closed-form solutions are deferred, but the structural claim is complete: *the conformal blocks are the eigenfunctions of the quadratic conformal Casimir, with eigenvalues fixed by the dimension and spin of the exchanged primary.*

We have, with this section, assembled the last piece of the CFT data and its kinematics: a convergent operator product expansion whose coefficients are the three-point constants, and a decomposition of the four-point function into OPE-coefficient-weighted conformal blocks obeying the Casimir equation. The

squared coefficients $C^2_{\phi\phi\mathcal{O}}$ and the spectrum $\{(\Delta_\mathcal{O}, \ell_\mathcal{O})\}$ are now exposed as the complete, finite input from which $g(u,v)$ is built — and the demand that the *same* four-point function be reproduced by fusing a different pair of operators is the crossing equation that Chapter 49 elevates into the conformal bootstrap. Before turning the machine on an unknown theory, however, we close the chapter by checking it against the two theories ϕ^4 theory has carried this far: the next section verifies that the free massless scalar realizes all of this structure explicitly — its $\phi \times \phi$ OPE producing the identity, the composite $:\!\phi^2\!:$, and the stress tensor — and identifies the Wilson–Fisher fixed point as the interacting conformal field theory whose anomalous dimensions feed the bootstrap.

47.8 The Free Scalar and the Wilson–Fisher Fixed Point as CFTs

The preceding sections built the conformal toolkit as an abstract structure: the Euclidean conformal algebra $\mathfrak{so}(d+1,1)$ and its unitary representations organized into primaries and descendants, the radial quantization that turns the dilatation into a Hamiltonian and supplies the state–operator map, the two- and three-point functions pinned by symmetry, and finally the operator product expansion with its convergent decomposition of the four-point function into conformal blocks. Every one of those constructions was stated for a generic conformal field theory, defined by a list of primary dimensions and spins together with their OPE coefficients. It is time to exhibit that list for the two theories ϕ^4 theory has carried through the book. We will verify that the massless free scalar realizes the toolkit exactly — its operator content is precisely the spectrum the general theory predicts at the unitarity bound — and then identify the Wilson–Fisher fixed point as a genuinely interacting CFT whose low-lying scaling dimensions are the engineering dimensions of the free theory *shifted by the anomalous dimensions computed in Part VI*. The endpoint is the handoff: a short list of conformal data, Δ_ϕ, Δ_{ϕ^2}, and a few OPE coefficients, that the numerical bootstrap of Chapter 50 will pin to extraordinary precision for the three-dimensional Ising universality class, and whose full development is the business of Chapter 52.

47.8.1 The free massless scalar is a conformal field theory

We begin with the logical certificate. The opening section of this chapter established that a theory with a symmetric, conserved, *traceless* stress tensor carries one conserved current for each of the $\frac{1}{2}(d+1)(d+2)$ conformal Killing vectors, by the contraction $J^\mu_{(\xi)} = \Theta^{\mu\nu}_{\rm c}\xi_\nu$ whose divergence is proportional to the trace (Eq. (47.27)). The massless free scalar supplies exactly such a tensor: its canonical stress tensor has the nonzero trace $\frac{2-d}{2}(\partial\phi)^2$ of Eq. (7.191), but because the field virial is a total derivative (Eq. (7.190)) the Belinfante-style

improvement of Theorem 7.45 absorbs it and produces a tensor $\Theta_{\mathrm{c}}^{\mu\nu}$ that is symmetric, conserved, and traceless on shell in every dimension. The free scalar therefore carries the full set of conformal charges; it is a conformal field theory, and the simplest one. What remains is not to re-prove this but to read off its conformal data and check it against the representation theory.

The two-point function and the dimension of ϕ. The single most important number is the scaling dimension of the elementary field, and it is fixed by its two-point function. We work in Euclidean signature throughout this section, as the conformal toolkit and the bootstrap demand; the Wick rotation is the one defined in Chapter 10, and reflection positivity (Chapter 10) is the Euclidean avatar of unitarity that makes the operator-state inner products below positive. The massless Euclidean propagator is the $m \to 0$ limit of the Schwinger function of Theorem 10.30. In four dimensions the worked computation of Theorem 10.31 already gave $G_E \to 1/(4\pi^2 r^2)$ as $m \to 0$; we now need its form in general d, since the entire interest of the fixed-point story lies in varying the dimension.

Worked Example 47.51 (The massless scalar two-point function in d dimensions)**.** We evaluate the massless propagator

$$\langle \phi(x)\phi(0)\rangle = \int \frac{d^d p}{(2\pi)^d}\,\frac{e^{\,ip\cdot x}}{p^2} \tag{47.234}$$

in closed form for general d, repeating the proper-time method of Theorem 10.31 with the dimension kept arbitrary. Write the denominator as a Schwinger integral, $1/p^2 = \int_0^\infty ds\, e^{-sp^2}$, legitimate because $p^2 \geq 0$ in Euclidean signature, and do the Gaussian momentum integral in d dimensions,

$$\int \frac{d^d p}{(2\pi)^d}\, e^{-sp^2 + ip\cdot x} = \frac{1}{(4\pi s)^{d/2}}\, e^{-x^2/4s}, \qquad x^2 \equiv |x|^2 = r^2, \tag{47.235}$$

which is the product of d identical one-dimensional Gaussians and is manifestly finite for every $s > 0$. Hence

$$\langle \phi(x)\phi(0)\rangle = \frac{1}{(4\pi)^{d/2}} \int_0^\infty ds\, s^{-d/2}\, e^{-r^2/4s}. \tag{47.236}$$

Substitute $s = r^2/(4w)$, so $ds = -\,(r^2/4w^2)\,dw$ and $s^{-d/2} = (4w/r^2)^{d/2}$; the limits flip and

$$\int_0^\infty ds\, s^{-d/2}\, e^{-r^2/4s} = \Big(\frac{r^2}{4}\Big)^{1-d/2} \int_0^\infty dw\, w^{d/2-2}\, e^{-w} = \Big(\frac{r^2}{4}\Big)^{1-d/2} \Gamma\Big(\tfrac{d}{2} - 1\Big), \tag{47.237}$$

the last step being the definition of the Gamma function, convergent for $d > 2$. Assembling the constants, $(4\pi)^{-d/2}(r^2/4)^{1-d/2} = (4\pi)^{-d/2} 4^{d/2-1} r^{2-d} =$

$\frac{1}{4}\pi^{-d/2}r^{2-d}$, so

$$\boxed{\langle\phi(x)\phi(0)\rangle = \frac{C_d}{|x|^{\,d-2}}\,, \qquad C_d = \frac{\Gamma(\frac{d}{2}-1)}{4\,\pi^{d/2}}\,.} \tag{47.238}$$

The four-dimensional case $C_4 = \Gamma(1)/(4\pi^2) = 1/(4\pi^2)$ recovers Theorem 10.31, and the power $1/|x|^{d-2}$ is the position-space form of the massless propagator $1/p^2$ in any dimension.

The conformal toolkit fixes the two-point function of any scalar primary of dimension Δ to be $\langle\mathcal{O}(x)\mathcal{O}(0)\rangle \propto |x|^{-2\Delta}$ (the symmetry-determined form derived earlier in this chapter, Eq. (47.120)). Comparing with Eq. (47.238), the exponents match if and only if

$$2\Delta_\phi = d-2, \qquad \text{i.e.} \qquad \boxed{\Delta_\phi = \frac{d-2}{2}\,.} \tag{47.239}$$

This is the engineering dimension of the scalar field, the value $[\phi] = (d-2)/2$ that dimensional analysis and the classical scale symmetry both demanded (Eq. (7.185)); the free quantum theory carries no anomalous correction to it, because there is no interaction to renormalize the field. Two structural facts follow immediately.

First, Δ_ϕ *saturates the scalar unitarity bound.* The representation theory established earlier (Theorem 47.18) that a unitary scalar primary in $d \geq 3$ obeys $\Delta \geq (d-2)/2$, with equality permitted only for a free field. The free scalar sits exactly at this floor. We will see in Section 47.8.3 that the saturation is not a coincidence but the operator-algebra shadow of the equation of motion $\Box\phi = 0$.

Second, ϕ *is a primary, not a descendant.* A descendant is a conformal-derivative $P_{\mu_1}\cdots P_{\mu_n}$ of some primary of strictly lower dimension. The only operator in the free theory of dimension below $\Delta_\phi = (d-2)/2$ is the identity $\mathbf{1}$ (dimension 0), whose every descendant $\partial_{\mu_1}\cdots\mathbf{1}$ vanishes identically. There is therefore nothing for ϕ to be a descendant *of*: it is the lowest-dimension nontrivial operator in the theory, and any such operator is automatically a conformal primary. We can also see this directly from the transformation law: the propagator Eq. (47.238) transforms covariantly under the inversion $x \mapsto x/x^2$ with the weight $\Delta_\phi = (d-2)/2$.

Worked Example 47.52 (Inversion covariance of the free propagator)**.** A scalar primary of dimension Δ transforms under a conformal map $x \mapsto x'$ as $\phi(x) = \left|\partial x'/\partial x\right|^{\Delta/d}\phi'(x')$ (the finite primary transformation law Eq. (47.88)). For the inversion $x'^\mu = x^\mu/x^2$ the Jacobian factor was computed in Theorem 47.4: the conformal factor is $\Omega(x) = x^2$, and the Jacobian determinant is $\left|\partial x'/\partial x\right| = \Omega(x)^{-d} = (x^2)^{-d}$, so $\left|\partial x'/\partial x\right|^{\Delta/d} = (x^2)^{-\Delta}$. A primary two-point function must then satisfy

$$\langle\phi(x)\phi(y)\rangle = (x^2)^{-\Delta}(y^2)^{-\Delta}\,\langle\phi(x')\phi(y')\rangle\,. \tag{47.240}$$

The key geometric identity is that the inversion contracts chordal distances multiplicatively,

$$|x'-y'|^2 = \Big|\frac{x}{x^2}-\frac{y}{y^2}\Big|^2 = \frac{x^2}{(x^2)^2} - \frac{2\,x\cdot y}{x^2 y^2} + \frac{y^2}{(y^2)^2} = \frac{|x-y|^2}{x^2\,y^2}, \tag{47.241}$$

where the middle step expands the square and the last collects over the common denominator x^2y^2. Insert the trial form $\langle\phi(x)\phi(y)\rangle = C/|x-y|^{2\Delta}$ into the right-hand side of Eq. (47.240):

$$(x^2)^{-\Delta}(y^2)^{-\Delta}\,\frac{C}{|x'-y'|^{2\Delta}} = (x^2)^{-\Delta}(y^2)^{-\Delta}\,C\,\Big(\frac{x^2y^2}{|x-y|^2}\Big)^{\Delta} = \frac{C}{|x-y|^{2\Delta}}, \tag{47.242}$$

which reproduces the left-hand side identically. The power law $|x-y|^{-2\Delta}$ is thus inversion-covariant for *any* Δ; what makes ϕ a primary of the specific dimension $(d-2)/2$ is that its propagator Eq. (47.238) has precisely this form with $2\Delta = d-2$. Translations, rotations, and dilatations are covariance checks that the power law passes trivially; the inversion is the one genuinely new generator of the conformal group, and passing it confirms ϕ transforms as a primary under the full $\mathfrak{so}(d+1,1)$, not merely under the Poincaré–dilatation subgroup.

47.8.2 The $\phi\times\phi$ OPE and the operator content of the free theory

With ϕ established as a primary of dimension $(d-2)/2$, the operator content of the free theory is laid bare by a single OPE: the product of two elementary fields. Because the theory is Gaussian, this OPE is computed not by the abstract block expansion but directly by Wick's theorem, and the result can then be matched term by term against the conformal decomposition of Eq. (47.208). Point-split the product and normal-order: the only Wick contraction of $\phi(x)$ with $\phi(0)$ is the propagator Eq. (47.238), so

$$\phi(x)\,\phi(0) = \frac{C_d}{|x|^{\,d-2}}\,\mathbf{1} + \;:\!\phi(x)\,\phi(0)\!:, \tag{47.243}$$

where $:\cdots:$ denotes normal ordering and the singular c-number is the contraction. Taylor-expanding the regular normal-ordered piece about the midpoint generates an infinite tower of local composite operators,

$$:\!\phi(x)\,\phi(0)\!: = \sum_{n=0}^{\infty}\frac{1}{n!}\,x^{\mu_1}\cdots x^{\mu_n}\;:\!\partial_{\mu_1}\cdots\partial_{\mu_n}\phi(0)\;\phi(0)\!:, \tag{47.244}$$

all of them *bilinear* in ϕ. The content of the conformal OPE is the statement that these bilinears organize into conformal families, each consisting of a primary and its descendants. We identify the primaries.

The identity. The leading singular term in Eq. (47.243) is the identity operator $\mathbf{1}$, of dimension 0 and spin 0, with OPE coefficient equal to the two-point normalization C_d. Its presence is equivalent to the statement $\langle\phi\phi\rangle \neq 0$ and is universal to any $\phi \times \phi$ OPE.

The scalar $:\phi^2:$. The $n = 0$ term of Eq. (47.244) is the composite scalar $:\phi^2:(0)$. It is a primary of dimension

$$\Delta_{\phi^2} = 2\Delta_\phi = d - 2 \qquad \text{(free theory)}, \tag{47.245}$$

twice the dimension of the constituent field, with spin 0. That it is primary follows because the special conformal generator acts as a derivation on composites and annihilates the elementary primary at the origin:

$$[K_\mu, :\phi^2:(0)] = 2\ :\phi(0)\,[K_\mu, \phi(0)]: = 0, \qquad \text{since } [K_\mu, \phi(0)] = 0. \tag{47.246}$$

Thus $:\phi^2:$ is a new scalar primary, not a descendant of ϕ (the descendants of ϕ are $\partial_{\mu_1}\cdots\partial_{\mu_n}\phi$, which are not bilinear), and it is the lightest operator in the theory beyond the identity and ϕ itself. In the language of statistical mechanics $:\phi^2:$ is the energy operator, the one that couples to a temperature deviation, and its dimension controls the relevant direction of the RG flow — a point we make quantitative for the interacting fixed point in Section 47.8.4.

The conserved currents and the stress tensor. The terms with $n \geq 1$ in Eq. (47.244) carry vector indices and decompose into symmetric traceless tensors of every even rank. To extract the primaries cleanly it is best to package a symmetric traceless spin-ℓ tensor $J^{(\ell)}_{\mu_1\cdots\mu_\ell}$ by contracting its indices with an auxiliary *null* polarization vector z^μ, $z^2 = 0$, forming the index-free polynomial $J^{(\ell)}(x, z) = J^{(\ell)}_{\mu_1\cdots\mu_\ell} z^{\mu_1}\cdots z^{\mu_\ell}$; nullity of z automatically discards the trace terms. The bilinear spin-ℓ operators of the free scalar are then

$$J^{(\ell)}(x, z) = \sum_{k=0}^{\ell} c_k^{(\ell)}\ (z\cdot\partial)^k\phi(x)\ (z\cdot\partial)^{\ell-k}\phi(x), \tag{47.247}$$

a sum over ways of distributing ℓ derivatives between the two fields. Each term has dimension $2\Delta_\phi + \ell = d - 2 + \ell$, so $J^{(\ell)}$ is a candidate spin-ℓ primary of dimension

$$\Delta_\ell = \ell + d - 2. \tag{47.248}$$

The coefficients $c_k^{(\ell)}$ are not free: they are fixed by demanding that $J^{(\ell)}$ be *conserved*, $\partial^{\mu_1} J^{(\ell)}_{\mu_1\cdots\mu_\ell} = 0$, which is the condition that the spin-ℓ operator saturate the unitarity bound and sit at the top of a short multiplet. We now determine them and prove conservation.

Theorem 47.53 (Free higher-spin currents and their conservation)**.** *For each even $\ell \geq 0$ the bilinear Eq. (47.247) with coefficients*

$$c_k^{(\ell)} = (-1)^k \binom{\ell}{k} \tag{47.249}$$

is a symmetric traceless conserved current of the free massless scalar: $\partial^{\mu_1} J^{(\ell)}_{\mu_1 \cdots \mu_\ell} = 0$ on shell. It is a conformal primary of spin ℓ and dimension $\Delta_\ell = \ell + d - 2$, saturating the spin-$\ell$ unitarity bound (Eq. (47.102)). For odd ℓ the bilinear vanishes as a primary. The case $\ell = 0$ is $:\phi^2:$; the case $\ell = 2$ is the stress tensor (up to an improvement term); the cases $\ell \geq 4$ are the higher-spin currents whose existence reflects the infinite-dimensional higher-spin symmetry of the free theory.

Proof. Conservation of a symmetric traceless current contracted into null polarizations is the statement that $(\partial_x \cdot \partial_z)\, J^{(\ell)}(x, z) = 0$, where $\partial_z^\mu \equiv \partial/\partial z_\mu$ strips one polarization and the x-derivative supplies the divergence; trace terms drop because they carry an explicit $\eta_{\mu_i \mu_j}$ that contracts to $z^2 = 0$. Apply ∂_z^μ to Eq. (47.247) using $\partial_z^\mu (z\cdot\partial)^k = k\,(z\cdot\partial)^{k-1}\partial^\mu$:

$$\partial_z^\mu J^{(\ell)} = \sum_k c_k^{(\ell)} \Big[k\,(z\partial)^{k-1} \partial^\mu\phi\,(z\partial)^{\ell-k}\phi + (z\partial)^k \phi\,(\ell-k)(z\partial)^{\ell-k-1}\partial^\mu\phi \Big]. \tag{47.250}$$

Now act with ∂_μ (the spacetime divergence), which hits each field by the Leibniz rule. Of the four resulting terms, the two in which ∂_μ lands on the same field already carrying ∂^μ produce a $\Box\phi$ and vanish on shell, $\Box\phi = 0$. The surviving cross terms, in which ∂_μ and ∂^μ act on different fields, are

$$(\partial_x\partial_z) J^{(\ell)} = \sum_k c_k^{(\ell)} \Big[k\,(z\partial)^{k-1} \partial_\mu\phi\,(z\partial)^{\ell-k}\partial^\mu\phi + (\ell-k)\,(z\partial)^k \partial^\mu\phi\,(z\partial)^{\ell-k-1}\partial_\mu\phi \Big]. \tag{47.251}$$

Every term is of the form $(z\cdot\partial)^a \partial_\mu\phi\,(z\cdot\partial)^b \partial^\mu\phi$ with $a + b = \ell - 1$. Reindexing the first sum by $a = k - 1$ (so $k = a + 1$, coefficient $c_{a+1}^{(\ell)}(a+1)$) and the second by $a = k$ (coefficient $c_a^{(\ell)}(\ell - a)$), the total coefficient of the structure labelled by a is

$$(a+1)\, c_{a+1}^{(\ell)} + (\ell - a)\, c_a^{(\ell)}, \qquad a = 0, \ldots, \ell - 1. \tag{47.252}$$

Demanding that each vanish gives the recursion $c_{a+1}^{(\ell)} = -\frac{\ell-a}{a+1}\, c_a^{(\ell)}$, whose solution normalized to $c_0^{(\ell)} = 1$ is $c_k^{(\ell)} = (-1)^k \binom{\ell}{k}$, which is Eq. (47.249). With these coefficients $(\partial_x \cdot \partial_z) J^{(\ell)} = 0$ on shell, establishing conservation.

That $J^{(\ell)}$ is a primary follows from conservation: a conserved symmetric traceless spin-ℓ current is annihilated by K_μ at the origin, and conversely the conservation condition is exactly the shortening condition $\Delta = \ell + d - 2$ for a spin-ℓ multiplet (Eq. (47.248)); thus $J^{(\ell)}$ is a primary at the bound. For odd ℓ, exchanging the two identical fields sends $k \to \ell - k$ and $c_k^{(\ell)} \to c_{\ell-k}^{(\ell)} =$

$(-1)^{\ell-k}\binom{\ell}{k} = (-1)^{\ell} c_k^{(\ell)}$, so for odd ℓ the symmetric bilinear is antisymmetric under the relabeling and vanishes as an operator; only even-spin currents survive, as expected for a single real scalar with no internal symmetry to carry a current.

Finally we read off the named cases. For $\ell = 0$, $c_0^{(0)} = 1$ and $J^{(0)} = \phi^2 =: \phi^2 :$ (up to the subtracted c-number), the scalar of dimension $d-2$. For $\ell = 2$, $(c_0, c_1, c_2) = (1, -2, 1)$ gives, stripping the polarizations,

$$J^{(2)}_{\mu\nu} \;\propto\; \phi\,\partial_\mu\partial_\nu\phi - \partial_\mu\phi\,\partial_\nu\phi \qquad \text{(traceless, null-projected)}, \tag{47.253}$$

which is conserved by the same on-shell computation, $\partial^\mu J^{(2)}_{\mu\nu} = \phi\,\partial_\nu\Box\phi - \Box\phi\,\partial_\nu\phi = 0$. This is the spin-2 conserved current of dimension $\Delta_2 = d$, which is the stress tensor: it differs from the improved tensor $\Theta^{\mu\nu}_{\rm c}$ of Theorem 7.45 only by an identically conserved improvement term, and as a conformal primary the two coincide. The dimension $\Delta_2 = d$ and spin 2 are exactly the universal quantum numbers of the stress tensor in any CFT, here realized concretely. □

The free $\phi \times \phi$ OPE is now completely characterized. Reassembling the families,

$$\phi(x)\,\phi(0) \;\sim\; \frac{C_d}{|x|^{\,d-2}}\,\mathbf{1} \;+\; \underbrace{:\!\phi^2\!:(0)}_{\Delta=d-2,\ \ell=0} \;+\; \underbrace{\sum_{\ell\geq 2,\ \text{even}} J^{(\ell)}(0)}_{\Delta_\ell=\ell+d-2} \;+\; (\text{descendants}), \tag{47.254}$$

the appearing primaries are exactly the identity, the single scalar bilinear $:\!\phi^2\!:$, and one conserved current $J^{(\ell)}$ of dimension $\ell + d - 2$ for each even spin $\ell \geq 2$ — the stress tensor at $\ell = 2$ and an infinite tower of higher-spin currents above it. There is no operator of twist (dimension minus spin) below $d-2$, the common twist of every bilinear; this is the hallmark of a free theory, and it is the structural fact the interacting fixed point will spoil. Every dimension in Eq. (47.254) saturates the relevant unitarity bound: $\Delta_\phi = (d-2)/2$ saturates the scalar bound, and $\Delta_\ell = \ell + d - 2$ saturates the spin-ℓ bound. A free CFT lives entirely on the floor of the unitarity allowed region.

Physical Insight

The infinite tower of conserved higher-spin currents is the diagnostic of a free theory. A theorem of conformal representation theory (the higher-spin no-go statement, established in the literature for $d \geq 3$) says that a unitary CFT with even one conserved current of spin $\ell > 2$ is necessarily free: such currents generate an infinite-dimensional symmetry algebra that, in $d \geq 3$, is so rigid it forces all correlators to be those of free fields. Conversely, the way an interacting CFT *must* differ from the free scalar is by lifting these currents off the unitarity bound — giving them anomalous dimensions

$\gamma_\ell > 0$ so that $\Delta_\ell > \ell + d - 2$ and they cease to be conserved. The single current that cannot be lifted is the spin-2 stress tensor at $\Delta = d$, protected by the exact conservation of energy and momentum (translation invariance is never anomalous), together with any spin-1 currents of exact internal symmetries. The Wilson–Fisher fixed point realizes exactly this scenario.

47.8.3 The equation of motion as a null descendant

We promised that the saturation $\Delta_\phi = (d-2)/2$ is the operator-algebra expression of the free equation of motion $\Box\phi = 0$. The mechanism is the appearance of a *null state* in the conformal multiplet of ϕ, and exhibiting it both explains the unitarity bound and sets up precisely what the interaction lifts.

Work in radial quantization, where the state-operator map (built earlier in this chapter) sends the primary ϕ to a state $|\mathcal{O}\rangle$ with $D\,|\mathcal{O}\rangle = \Delta_\phi\,|\mathcal{O}\rangle$ and $K_\mu\,|\mathcal{O}\rangle = 0$, and where the Hermiticity assignment $K_\mu = P_\mu^\dagger$ (the radial-quantization adjoint, Eq. (47.175)) makes the descendant inner products computable from the algebra. Descendants are built by acting with P_μ. The level-two descendant of interest is $P^2\,|\mathcal{O}\rangle = P_\mu P^\mu\,|\mathcal{O}\rangle$, the state created by $\Box\phi$. Its norm measures whether $\Box\phi$ is an independent operator or must vanish.

Proposition 47.54 (The level-two scalar norm and the unitarity bound)**.** *For a scalar primary $|\mathcal{O}\rangle$ of dimension Δ in d dimensions, in the radial-quantization conventions $[D, P_\mu] = P_\mu$, $[D, K_\mu] = -K_\mu$, $[K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - M_{\mu\nu})$, $[M_{\mu\nu}, P_\rho] = \delta_{\nu\rho} P_\mu - \delta_{\mu\rho} P_\nu$, with $K_\mu = P_\mu^\dagger$ and $M_{\mu\nu}\,|\mathcal{O}\rangle = 0$,*

$$\left\| P^2\,|\mathcal{O}\rangle \right\|^2 = \Big\langle \mathcal{O}\Big| K^2 P^2 \Big|\mathcal{O}\Big\rangle = 4\,d\,\Delta\Big(\Delta - \tfrac{d-2}{2}\Big)\,\langle\mathcal{O}|\mathcal{O}\rangle\,. \tag{47.255}$$

The norm is nonnegative for all unitary scalars, which is the scalar unitarity bound $\Delta \geq (d-2)/2$; it vanishes precisely at the bound $\Delta = (d-2)/2$, where $P^2\,|\mathcal{O}\rangle$ is a null state.

Proof. We move $K^2 = K_\mu K^\mu$ through $P^2 = P_\nu P^\nu$ using the algebra, all the way to $K_\mu\,|\mathcal{O}\rangle = 0$. First compute $K_\mu P^2\,|\mathcal{O}\rangle$:

$$\begin{aligned} K_\mu P_\nu P^\nu\,|\mathcal{O}\rangle &= [K_\mu, P_\nu] P^\nu\,|\mathcal{O}\rangle + P_\nu K_\mu P^\nu\,|\mathcal{O}\rangle \\ &= 2(\delta_{\mu\nu} D - M_{\mu\nu}) P^\nu\,|\mathcal{O}\rangle + P_\nu [K_\mu, P^\nu]\,|\mathcal{O}\rangle\,, \end{aligned} \tag{47.256}$$

where in the second term $K_\mu\,|\mathcal{O}\rangle = 0$ was used. Evaluate the two pieces. In the first, $\delta_{\mu\nu} P^\nu = P_\mu$ and $D P_\mu\,|\mathcal{O}\rangle = (\Delta + 1) P_\mu\,|\mathcal{O}\rangle$ (since $[D, P_\mu] = P_\mu$), while $M_{\mu\nu} P^\nu\,|\mathcal{O}\rangle = [M_{\mu\nu}, P^\nu]\,|\mathcal{O}\rangle = (\delta^\nu_\nu P_\mu - P_\mu)\,|\mathcal{O}\rangle = (d-1) P_\mu\,|\mathcal{O}\rangle$, using $M_{\mu\nu}\,|\mathcal{O}\rangle = 0$ and the rotation commutator. Hence the first piece is $2(\Delta +$

$1)P_\mu|\mathcal{O}\rangle - 2(d-1)P_\mu|\mathcal{O}\rangle = 2(\Delta - d + 2)P_\mu|\mathcal{O}\rangle$. In the second term, $[K_\mu, P^\nu]|\mathcal{O}\rangle = 2(\delta_\mu{}^\nu D - M_\mu{}^\nu)|\mathcal{O}\rangle = 2\delta_\mu{}^\nu \Delta|\mathcal{O}\rangle$, so $P_\nu[K_\mu, P^\nu]|\mathcal{O}\rangle = 2\Delta P_\mu|\mathcal{O}\rangle$. Adding,

$$K_\mu P^2|\mathcal{O}\rangle = 2(2\Delta - d + 2)\, P_\mu|\mathcal{O}\rangle\,. \tag{47.257}$$

Now contract with K^μ from the left and take the inner product. Using Eq. (47.257),

$$\Big\langle \mathcal{O}\Big|K^2P^2\Big|\mathcal{O}\Big\rangle = 2(2\Delta - d + 2)\ \langle \mathcal{O}|K^\mu P_\mu|\mathcal{O}\rangle\,. \tag{47.258}$$

The remaining matrix element is the level-one norm: $\langle \mathcal{O}|K^\mu P_\mu|\mathcal{O}\rangle = \langle \mathcal{O}|[K^\mu, P_\mu]|\mathcal{O}\rangle = \Big\langle \mathcal{O}\Big|2(\delta^\mu_\mu D - M^\mu{}_\mu)\Big|\mathcal{O}\Big\rangle = 2d\Delta\,\langle \mathcal{O}|\mathcal{O}\rangle$, since $\delta^\mu_\mu = d$ and $M^\mu{}_\mu = 0$. Therefore

$$\Big\langle \mathcal{O}\Big|K^2P^2\Big|\mathcal{O}\Big\rangle = 2(2\Delta - d + 2)\cdot 2d\Delta\ \langle \mathcal{O}|\mathcal{O}\rangle = 4d\Delta(2\Delta - d + 2)\,\langle \mathcal{O}|\mathcal{O}\rangle\,, \tag{47.259}$$

and since $2\Delta - d + 2 = 2(\Delta - \frac{d-2}{2})$ this is Eq. (47.255). For $\Delta > 0$ the norm is nonnegative iff $\Delta \geq (d-2)/2$, with equality giving a vanishing (null) descendant. □

The interpretation is exactly the promised one. The free scalar sits at $\Delta_\phi = (d-2)/2$, where Eq. (47.255) vanishes: the descendant $P^2|\mathcal{O}\rangle$, the state created by $\Box\phi$, has zero norm. A zero-norm state in a unitary theory must be set to zero (it decouples from all correlators by Cauchy–Schwarz), and setting it to zero is exactly the free equation of motion as an operator statement:

$$\boxed{\Box\phi = 0 \quad\Longleftrightarrow\quad P^2|\mathcal{O}\rangle \text{ is null} \quad\Longleftrightarrow\quad \Delta_\phi = \tfrac{d-2}{2}.} \tag{47.260}$$

The free multiplet is therefore *short*: the conformal family of ϕ is the full Verma module with its level-two scalar descendant removed, and the removal is the equation of motion. This is the conformal-algebra origin of the statement, familiar since Chapter 9, that a free field has no independent operator $\Box\phi$.

Physical Insight

The shortening makes precise why the free scalar is so much more constrained than a generic CFT primary, and what an interaction must do. A generic ("long") scalar primary of dimension $\Delta > (d-2)/2$ has $\Box\phi$ as a genuine, independent operator of dimension $\Delta + 2$. The free scalar trades that operator for the constraint $\Box\phi = 0$. Turning on the $\lambda\phi^4$ interaction of ϕ^4 theory does not abolish the descendant; it *populates* it. The classical equation of motion $\Box\phi = -\frac{\lambda}{3!}\phi^3$ (the carried EOM of Section 7.7.5) says the level-two descendant of ϕ is no longer zero but proportional to the composite $:\phi^3:$. The multiplet lengthens, and consistency of the lengthened multiplet with unitarity forces Δ_ϕ to rise above the bound: $\Delta_\phi = (d-2)/2 + \gamma_\phi$ with $\gamma_\phi > 0$. The anomalous dimension is the amount

by which the interaction lifts ϕ off the unitarity floor, equivalently the amount by which $\Box\phi$ acquires a nonzero norm. This is the bridge between Part VI's diagrammatic γ_ϕ and the conformal data of the fixed point.

47.8.4 The Wilson–Fisher fixed point as an interacting CFT

We now assemble the conformal data of the interacting fixed point. The Wilsonian and Callan–Symanzik analyses of Part VI located, in $d = 4 - \varepsilon$ dimensions, a nontrivial zero of the beta function: the Wilson–Fisher fixed point at $\lambda_* = \frac{16\pi^2}{3}\varepsilon + O(\varepsilon^2)$ (Theorem 23.36, Eq. (23.195), Theorem 24.31). At that fixed point the dimensionless coupling has stopped running, the theory is scale invariant, and — the virial current being a total derivative exactly as for the free scalar, since the interaction $\lambda\phi^4$ adds only a potential term whose virial is again $\propto \partial^\mu(\phi^2)$ — the improvement of Theorem 7.45 again produces a traceless stress tensor. By the current construction of the chapter's opening section, *the Wilson–Fisher fixed point is a conformal field theory.* What distinguishes it from the free scalar is its spectrum of dimensions, which we now compute by importing the Part VI results and reading them as conformal scaling dimensions.

Worked Example 47.55 (Wilson–Fisher scaling dimensions from the Part VI data)**.** We assemble the two lightest scalar dimensions of the Wilson–Fisher CFT, Δ_ϕ and Δ_{ϕ^2}, from the anomalous dimension and the relevant RG eigenvalue computed in Part VI, and exhibit them as conformal data. We work in the dimension-deficit convention $\varepsilon = 4 - d$ of Theorem 24.31, with the one-component (Ising) universality class.

The dimension of ϕ. Theorem 24.30 established the general fixed-point statement that the scaling dimension of the field is its engineering dimension shifted by the field anomalous dimension evaluated at the fixed point,

$$\Delta_\phi = \frac{d-2}{2} + \gamma_*, \qquad \gamma_* \equiv \gamma_\phi(\lambda_*), \tag{47.261}$$

which is Eq. (24.246) and Eq. (24.203) read as a conformal dimension: the very γ_ϕ computed from wavefunction renormalization *is* the anomalous part of a primary's conformal dimension. At the Wilson–Fisher point the leading anomalous dimension is second order in ε (Eq. (24.256)),

$$\gamma_* = \frac{\eta}{2}, \qquad \eta = \frac{\varepsilon^2}{54} + O(\varepsilon^3) \quad \text{(one component)}, \tag{47.262}$$

so that the field anomalous dimension is $\gamma_* = \varepsilon^2/108 + O(\varepsilon^3)$ and

$$\boxed{\Delta_\phi = \frac{d-2}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3) = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3).} \tag{47.263}$$

The smallness of the correction — it is $O(\varepsilon^2)$, not $O(\varepsilon)$ — is the conformal restatement of the Part VI fact that the field strength renormalization Z_ϕ of ϕ^4 first moves at two loops: ϕ stays close to the unitarity bound, lifted off it only at second order. By Eq. (47.260) this is precisely the statement that the interaction gives the descendant $\Box\phi \propto :\phi^3:$ a norm that is $O(\varepsilon^2)$ small.

The dimension of ϕ^2. The composite scalar $:\phi^2:$ is the relevant operator that drives the RG flow away from criticality — physically, the energy density that couples to the reduced temperature. Its dimension is fixed by the relevant eigenvalue of the linearized flow about the fixed point. The mass perturbation $\delta S = \int d^d x\, t\, :\phi^2:(x)$ is relevant with RG eigenvalue $y_t = d - \Delta_{\phi^2}$ (the coupling t has dimension $d - \Delta_{\phi^2}$, and an operator is relevant precisely when its dimension is below d). This eigenvalue is the inverse correlation-length exponent, $y_t = 1/\nu$, computed at the Wilson–Fisher point in Part VI to be $1/\nu = 2 - \varepsilon/3 + O(\varepsilon^2)$ (Theorem 23.36). Therefore

$$\Delta_{\phi^2} = d - \frac{1}{\nu} = (4-\varepsilon) - \Big(2 - \frac{\varepsilon}{3}\Big) = \boxed{2 - \frac{2\varepsilon}{3} + O(\varepsilon^2).} \tag{47.264}$$

Comparing with the free value $\Delta_{\phi^2}^{\text{free}} = d - 2 = 2 - \varepsilon$, the interaction shifts the dimension of $:\phi^2:$ upward by $\Delta_{\phi^2} - (d-2) = \varepsilon/3 + O(\varepsilon^2)$: a *first-order* anomalous dimension $\gamma_{\phi^2} = \varepsilon/3$, far larger than the $O(\varepsilon^2)$ shift of ϕ itself. This asymmetry — ϕ barely moving while ϕ^2 moves at leading order — is the quantitative signature of the interacting fixed point, and it is why the energy operator and the order parameter are cleanly distinguished in the critical theory.

The higher-spin currents. The free tower of conserved currents $J^{(\ell)}$ of Theorem 47.53 survives at $\ell = 2$ as the exactly conserved stress tensor of dimension $\Delta_2 = d$ (energy–momentum conservation is never anomalous). For $\ell \geq 4$ the interaction lifts each current off the unitarity bound, $\Delta_\ell = \ell + d - 2 + \gamma_\ell$ with $\gamma_\ell > 0$ of order ε^2; the currents are no longer conserved, and the higher-spin symmetry of the free theory is broken — exactly as the no-go statement of Section 47.8.2 requires of any interacting CFT. The fixed point thus has the minimal conserved-current content of a unitary local CFT: the stress tensor and nothing of higher spin.

Figure 47.9 displays the two leading dimensions as functions of ε, comparing the free engineering values, the leading ε-expansion of the Wilson–Fisher CFT, and the high-precision numerical-bootstrap values at $\varepsilon = 1$ (i.e. $d = 3$).

Caution

The numbers Eqs. (47.263) and (47.264) are the leading terms of the ε-expansion, an asymptotic (non-convergent) series in $\varepsilon = 4 - d$. The physically interesting case is $d = 3$, i.e. $\varepsilon = 1$, which is not small; truncating the series there gives only a rough estimate ($\Delta_\phi \approx 1 - \frac{1}{2} + \frac{1}{108} \approx 0.509$ against

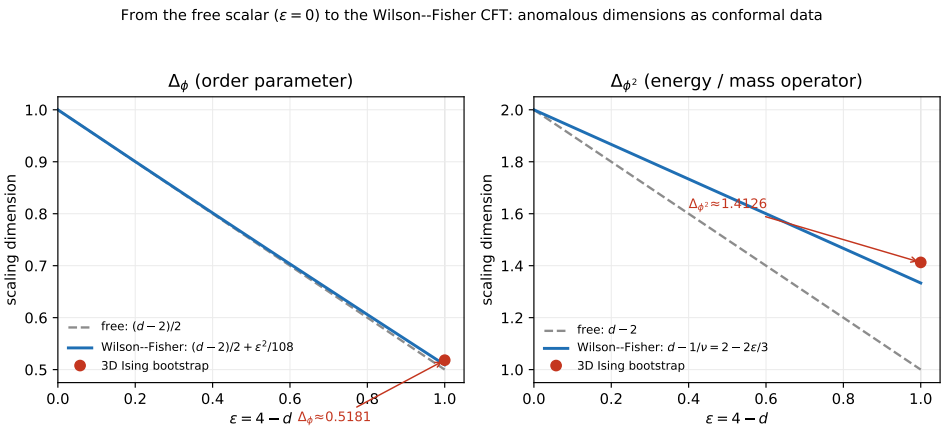


Figure 47.9: The two lightest scalar primary dimensions of ϕ^4 theory, from the free Gaussian theory ($\varepsilon = 0$) to the interacting Wilson–Fisher CFT, as functions of the dimension deficit $\varepsilon = 4 - d$. Dashed: the free (engineering) dimensions $\Delta_\phi = (d-2)/2$ and $\Delta_{\phi^2} = d - 2$, which saturate the unitarity bounds. Solid: the Wilson–Fisher dimensions to leading order in ε from Eqs. (47.263) and (47.264), carrying the anomalous pieces $\gamma_\phi = \varepsilon^2/108$ (a tiny lift of ϕ off the bound) and $\gamma_{\phi^2} = \varepsilon/3$ (a first-order shift of the energy operator). Markers at $\varepsilon = 1$: the three-dimensional Ising values $\Delta_\phi \approx 0.518$, $\Delta_{\phi^2} \approx 1.413$ obtained by the numerical conformal bootstrap, the target to which Chapter 50 pins this data far beyond the reach of the perturbative series. The ε-expansion is an asymptotic series evaluated at $\varepsilon = 1$, so the leading curves only point toward, rather than reach, the bootstrap numbers — which is precisely why the bootstrap is needed.

the true 0.5181…). The ε-expansion establishes the *existence and qualitative structure* of the interacting fixed point and its conformal data, but it does not, on its own, deliver precise three-dimensional dimensions. The precision comes from the bootstrap, which uses no perturbative expansion at all — only conformal symmetry, unitarity, and crossing — and for which these few low-lying dimensions are the data to be determined. This is the sense in which the ε-expansion and the bootstrap are complementary, dated to the current understanding of the mid-2020s.

47.8.5 Closing the loop: two genuine CFTs, and the handoff

The chapter opened with a question inherited from Part VI: the renormalization group identifies the long-distance physics of a theory with the fixed points of its flow, and a fixed point is scale invariant, but is it conformally invariant? We can now answer it concretely for the two theories ϕ^4 theory has carried. Both the free

massless scalar and the Wilson–Fisher fixed point possess a virial current that is a total derivative, so by the improvement construction of Section 47.1.1 both have a symmetric, conserved, traceless stress tensor; and by the current-counting argument Eq. (47.27) a traceless stress tensor generates the full conformal group. Both theories are therefore not merely scale invariant but genuine conformal field theories, with the elementary field realized as a primary of dimension Δ_ϕ. The abstract machinery of this chapter — primaries and descendants, unitarity bounds, radial quantization, symmetry-fixed correlators, the OPE — has been checked on each construction: the free scalar realizes the spectrum exactly at the unitarity bound, with its equation of motion appearing as a null descendant, and the Wilson–Fisher point lifts ϕ and ϕ^2 off the free values by the anomalous dimensions of Part VI, recast here as conformal data.

This completes ϕ^4 theory's transition from "RG fixed point" to "conformal field theory." The small list of numbers assembled in Theorem 47.55 — the dimensions Δ_ϕ and Δ_{ϕ^2}, the stress tensor at $\Delta = d$, and (implicitly) the OPE coefficients of the $\phi \times \phi$ expansion Eq. (47.254) — is exactly the conformal data the rest of this part will work with. Two destinations follow immediately. In $d = 2$ the free scalar is special: the conformal Killing equation degenerates and the finite-dimensional group enlarges to the infinite-dimensional Virasoro algebra, the subject of Chapter 48, where the free scalar and the Ising CFT are solved exactly. And in $d = 3$ the Wilson–Fisher data become the input to the conformal bootstrap: Chapter 49 sets up the crossing equation on these dimensions, Chapter 50 pins them numerically to the values shown in Fig. 47.9, and Chapter 52 closes the loop with Part VI by identifying the result as the three-dimensional Ising universality class. The conformal data computed here is the seed; the bootstrap is the cultivation.

Further reading

The modern, bootstrap-oriented development of conformal symmetry in $d \geq 3$ followed here is presented at greater length in Rychkov's EPFL lectures [421] and in Simmons-Duffin's TASI lectures [442]; the comprehensive review of Poland et al. [402] surveys the field through the numerical bootstrap and is the natural next step. The two-dimensional theory, the Virasoro algebra, and the minimal-model classification are treated definitively in the book of Di Francesco et al. [147]. The question of when scale invariance is enhanced to conformal invariance is the subject of Polchinski [404] in $d = 2$ and of Luty et al. [329] for the $d = 4$ unitary case via the a-theorem. The classic papers establishing the conformal algebra, the operator product expansion, and the covariance of correlators are Ferrara et al. [190]; the conformal blocks and the Casimir equation used to organize the four-point function are due to Dolan and Osborn [155, 156].

Chapter 48

CFT in Two Dimensions

The previous chapter built conformal symmetry in general dimension and found it powerful but finite: solving the conformal Killing equation in $d \geq 3$ returned the $\frac{1}{2}(d+1)(d+2)$ generators of $\mathrm{SO}(d+1,1)$ (Theorem 47.3), enough to fix two- and three-point functions up to constants but no more. This chapter is the story of what happens when that count becomes infinite. In two dimensions the conformal Killing equation degenerates exactly into the Cauchy–Riemann condition, every holomorphic map is a conformal transformation, and the symmetry algebra swells from a finite Lie algebra into two commuting infinite-dimensional copies of the Witt algebra. The consequence, developed over the seven sections that follow, is that an entire discrete family of theories becomes *exactly solvable* from symmetry alone — their operator content, their fusion rules, and every correlation function determined with no Lagrangian and no perturbation series. Two-dimensional conformal field theory is the one corner of interacting quantum field theory where the program of fixing everything by symmetry is carried to completion, and it is the proving ground for the bootstrap philosophy that the rest of Part XIV pursues in higher dimensions.

The chapter imports its kinematic backbone from Chapter 47 — the definition of primary and quasi-primary operators (Theorems 47.15 and 48.4), radial quantization and the state–operator map, the OPE, and the general-dimension forms of the two- and three-point functions against which every two-dimensional result is checked. It is computational in a complex-analytic register, so it leans throughout on the analysis toolbox of Chapter 5: the Wick rotation $t \to -i\tau$ (Section 5.4) that carries the mostly-minus Minkowski theory of Appendix A to the Euclidean plane, and Cauchy's theorem and the residue calculus (Theorem 5.23) that turn the action of the symmetry into contour integrals. The Lie-algebraic language of generators acting as vector fields is that of Chapter 2, and the unitarity arguments rest on the reflection positivity established in Chapter 10. Where the carried ϕ^4 thread reaches its Wilson–Fisher fixed point it hands off to conformal field theory; here that handoff is realized concretely, since the Ising

CFT worked in full below *is* the fixed point governing the critical scalar, and its critical exponents reproduce Onsager's.

The arc is cumulative. Section 48.1 specializes the conformal Killing equation to $d = 2$, derives the Witt algebra and its globally well-defined subalgebra $\mathfrak{sl}(2, \mathbb{C})$, fixes the complex-coordinate conventions and the conformal weights $(h, \bar{h})$, and confirms consistency by recovering the two- and three-point functions of Chapter 47 in factorized form. Section 48.2 makes the symmetry dynamical: tracelessness forces a holomorphic stress tensor $T(z)$, the conformal Ward identity expresses every variation as a contour integral of T, and the operator product $T(z)T(w)$ produces the one number the classical analysis could not see — the central charge c, with its finite-transformation avatar the Schwarzian. Section 48.3 quantizes radially, expands T in modes L_n, and turns the TT operator product into the Virasoro algebra, the central extension of Witt; L_0 becomes the Hamiltonian and primaries become highest-weight states. Section 48.4 studies the resulting Verma modules, computes the Gram matrix and states the Kac determinant, locates the null vectors that signal degenerate representations, and extracts the Friedan–Qiu–Shenker unitarity bound and its discrete series. Section 48.5 reaps the harvest: the minimal models, their finite Kac tables, the BPZ differential equations that null vectors impose on correlators, and the fusion rules that close the operator algebra. Section 48.6 works the simplest nontrivial example, the $c = \frac{1}{2}$ Ising CFT, end to end — three primaries, fusion algebra, the spin four-point function from the BPZ equation, the free-Majorana realization, and the dictionary to Onsager's exponents. Section 48.7 closes with the free-field realizations — the boson and its vertex operators, the compact boson and $R \to 1/R$ duality, the Majorana fermion, and the Coulomb-gas/Feigin–Fuks construction whose screening integrals reproduce the Ising four-point function a third independent way.

A brief word on scope and rigor. Essentially everything here is proved in full: the collapse of the Killing equation to Cauchy–Riemann, the Witt and Virasoro brackets computed mode by mode, the central charge extracted from explicit free-field correlators, the Kac determinant deployed (its general proof attributed but the low-level cases computed by hand), the BPZ equations derived from null vectors, and the Ising four-point function assembled to the last constant by three independent methods. The single number c threads the whole chapter, computed for the boson ($c = 1$), the Majorana fermion ($c = \frac{1}{2}$), and the minimal series, and it reappears downstream as the coefficient of the universal entanglement entropy $\frac{c}{3} \ln \ell$ in Chapter 55. The conformal blocks that the null-vector ODEs deliver here are the same objects that, recast as Casimir eigenfunctions, become the analytic backbone of crossing symmetry in Chapters 49 and 51.

48.1 Local Conformal Symmetry in Two Dimensions: the Witt Algebra

The previous chapter left us with a finite-dimensional symmetry. Solving the conformal Killing equation in flat space of dimension $d \geq 3$ (Theorem 47.3) returned exactly $\frac{1}{2}(d+1)(d+2)$ independent generators — translations, rotations, one dilatation, and d special conformal transformations — assembling into the algebra of $\mathrm{SO}(d+1,1)$ recorded in Eq. (47.53). The count is rigid because the equation is overdetermined: in $d \geq 3$ the third derivative of the conformal factor is forced to vanish, so every conformal generator is at most quadratic in the coordinates, and the list closes. The single place where that argument has no force is $d = 2$, where the overdetermination evaporates and the equation acquires infinitely many solutions. This section carries out the specialization to two dimensions in full, and the result reorganizes everything that follows: the local conformal transformations become the holomorphic maps of the complex plane, the symmetry algebra becomes two commuting copies of an infinite-dimensional Lie algebra — the *Witt algebra* — and the closed finite-dimensional group of Chapter 47 survives only as the small globally well-defined subalgebra $\mathfrak{sl}(2,\mathbb{C})$. We also fix here, once and for all, the complex-coordinate conventions used throughout the chapter, and we check the whole apparatus against the general-dimension correlators of Chapter 47.

48.1.1 Complex coordinates on the Euclidean plane

Conformal field theory is most transparent in Euclidean signature, and the computational engine of the subject is complex analysis, so the first move is to Wick-rotate and complexify. We follow the rotation $t \to -i\tau$ introduced in Section 5.4: the mostly-minus Minkowski conventions of the book (Chapter 47) enter only through this continuation, after which the two-dimensional theory lives on the Euclidean plane $\mathbb{R}^2$ with positive-definite line element

$$ds^2 = (dx^1)^2 + (dx^2)^2, \tag{48.1}$$

and all subsequent manipulations are complex-analytic. On this plane introduce the complex coordinate and its conjugate,

$$z = x^1 + i\,x^2, \qquad \bar{z} = x^1 - i\,x^2, \tag{48.2}$$

with the corresponding Wirtinger derivatives, obtained by inverting Eq. (48.2),

$$\partial_z \equiv \partial = \tfrac{1}{2}(\partial_1 - i\,\partial_2), \qquad \partial_{\bar{z}} \equiv \bar{\partial} = \tfrac{1}{2}(\partial_1 + i\,\partial_2), \tag{48.3}$$

which satisfy $\partial z = 1$, $\partial \bar{z} = 0$, $\bar{\partial} z = 0$, $\bar{\partial}\bar{z} = 1$. The line element Eq. (48.1) becomes $ds^2 = dz\,d\bar{z}$, so the metric components in the $(z,\bar{z})$ frame are

$$g_{zz} = g_{\bar{z}\bar{z}} = 0, \qquad g_{z\bar{z}} = g_{\bar{z}z} = \tfrac{1}{2}, \qquad g^{z\bar{z}} = g^{\bar{z}z} = 2, \tag{48.4}$$

and the Euclidean area element is $d^2x = dx^1\,dx^2 = \tfrac{1}{2}\,|dz \wedge d\bar{z}|$, i.e. $d^2x = \tfrac{1}{2}\,dz\,d\bar{z}$ as a measure once one fixes orientation. The Laplacian is $\partial_1^2 + \partial_2^2 = 4\,\partial\bar{\partial}$.

There is a conceptual subtlety that we adopt as a standing convention. On the physical Euclidean plane z and $\bar{z}$ are not independent: $\bar{z}$ is the complex conjugate of z, and x^1, x^2 are real. It is nonetheless enormously convenient — and, by the analytic-continuation theorems of Section 5.4, legitimate — to treat z and $\bar{z}$ as two *independent* complex variables, recovering the physical configuration at the end by imposing the reality slice $\bar{z} = z^*$. Concretely, $\mathbb{R}^2$ is complexified to $\mathbb{C}^2$ with coordinates $(z, \bar{z})$, holomorphic and antiholomorphic dependence are treated separately, and "$\bar{z}$" is then merely a label for the second variable rather than a function of the first. Every formula we write is to be read on this complexified space; physical correlators are its restriction to $\bar{z} = z^*$.

Definition 48.1 (Holomorphic factorization)**.** Working on the complexified plane with independent coordinates $(z, \bar{z})$, a quantity *factorizes holomorphically* when it is a sum of products of a function of z alone (the *holomorphic* or *chiral* part) and a function of $\bar{z}$ alone (the *antiholomorphic* or *antichiral* part). A function $f(z)$ with $\bar{\partial} f = 0$ is called *holomorphic* (chiral); a function $\bar{f}(\bar{z})$ with $\partial \bar{f} = 0$ is *antiholomorphic* (antichiral). The reality slice $\bar{z} = z^*$ recombines the two parts into a real-analytic function on $\mathbb{R}^2$.

The reason this notion will be so powerful is the theorem we are about to prove: in two dimensions a local conformal transformation *is* a holomorphic map, so the symmetry itself respects the holomorphic/antiholomorphic splitting. Holomorphic factorization is therefore not an accident of special correlators but a structural feature inherited from the symmetry algebra, and it is what allows the contour-integral and residue machinery of Chapter 47 — ultimately Cauchy's theorem, Theorem 5.23 — to do the heavy lifting for the rest of the chapter.

48.1.2 The conformal Killing equation collapses to Cauchy–Riemann

A conformal transformation is an infinitesimal coordinate change $x^\mu \to x^\mu + \xi^\mu(x)$ whose only effect on the metric is a local rescaling. In Chapter 47 this requirement was reduced to the conformal Killing equation Eq. (47.9),

$$\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \frac{2}{d}\,(\partial\cdot\xi)\,\delta_{\mu\nu}, \tag{48.5}$$

written here in Euclidean signature, where indices are raised and lowered with $\delta_{\mu\nu}$ so that $\xi_\mu = \xi^\mu$. We now set $d = 2$ and read off its content componentwise. The factor $2/d$ becomes unity. The equation has three independent components.

The two diagonal components, $\mu = \nu = 1$ and $\mu = \nu = 2$, both reduce to the same statement,

$$2\,\partial_1\xi_1 = \partial_1\xi_1 + \partial_2\xi_2 \quad\Longrightarrow\quad \partial_1\xi_1 = \partial_2\xi_2, \tag{48.6}$$

and the off-diagonal component $\mu = 1, \nu = 2$ gives

$$\partial_1 \xi_2 + \partial_2 \xi_1 = 0. \tag{48.7}$$

These two real equations are exactly the Cauchy–Riemann equations. To see it, assemble the components of ξ into a complex field,

$$\varepsilon(z, \bar{z}) \equiv \xi^1 + i\,\xi^2, \qquad \bar{\varepsilon}(z, \bar{z}) \equiv \xi^1 - i\,\xi^2, \tag{48.8}$$

and compute its antiholomorphic derivative using Eq. (48.3),

$$\bar{\partial}\,\varepsilon = \tfrac{1}{2}(\partial_1 + i\,\partial_2)(\xi^1 + i\,\xi^2) = \tfrac{1}{2}(\partial_1 \xi^1 - \partial_2 \xi^2) + \tfrac{i}{2}(\partial_1 \xi^2 + \partial_2 \xi^1). \tag{48.9}$$

The real part of the bracket vanishes by Eq. (48.6) and the imaginary part vanishes by Eq. (48.7). Hence

$$\boxed{\bar{\partial}\,\varepsilon = 0, \qquad \partial\,\bar{\varepsilon} = 0} \tag{48.10}$$

where the second equation follows identically by taking the complex conjugate of the first (equivalently, by repeating the computation with $\partial\bar{\varepsilon}$). The conclusion is the headline of the chapter: *in two dimensions an infinitesimal conformal transformation is generated by a holomorphic function $\varepsilon(z)$ together with an independent antiholomorphic function $\bar{\varepsilon}(\bar{z})$*. The local conformal map is $z \mapsto z + \varepsilon(z)$, $\bar{z} \mapsto \bar{z} + \bar{\varepsilon}(\bar{z})$, and finite conformal transformations are precisely the holomorphic maps $z \mapsto f(z)$ (with $f' \neq 0$), together with their antiholomorphic partners.

The contrast with $d \geq 3$ could not be sharper, and it is worth pausing on why the count exploded. In $d \geq 3$ the conformal Killing equation forces, via Eq. (47.13), the conformal factor to be at most linear in x, so ξ is at most quadratic and the solution space is finite. In $d = 2$ the obstruction disappears: the trace condition $\frac{2}{d}(\partial\cdot\xi)$ uses up the entire equation, leaving no further constraint on the second derivatives, and Eq. (48.10) is solved by *any* holomorphic $\varepsilon(z)$ whatsoever. The space of local solutions is infinite-dimensional. This is the precise sense in which "$d = 2$ is special": the conformal Killing equation degenerates exactly into the Cauchy–Riemann condition that defines complex differentiability.

Caution

"Holomorphic" here means complex-differentiable, i.e. $\bar{\partial}\varepsilon = 0$, and includes functions with poles and essential singularities — $\varepsilon(z)$ may be any element of a Laurent series. The *local* conformal transformations are these germs of holomorphic maps; whether a given $\varepsilon(z)$ extends to a globally well-defined transformation of the Riemann sphere is a separate question, answered in Section 48.1.4, and it is what distinguishes the infinite Witt algebra from its finite global subalgebra. A holomorphic map preserves angles (it is

conformal in the elementary complex-analysis sense) precisely where $f' \neq 0$; this is the two-dimensional avatar of the angle-preservation proposition Theorem 47.2.

48.1.3 The Witt algebra

A holomorphic function on a punctured neighbourhood of the origin has a Laurent expansion, and each Laurent monomial generates one independent infinitesimal conformal transformation. It is therefore natural to expand $\varepsilon(z)$ in integer powers of z and read off a basis of generators. Recall (Theorem 2.39) that the generators of a transformation acting on functions are the vector fields implementing the infinitesimal change of coordinates: under $z \mapsto z' = z+\varepsilon(z)$ a field ϕ transforms as $\delta\phi = -\varepsilon(z)\,\partial\phi - \bar{\varepsilon}(\bar{z})\,\bar{\partial}\phi$, so the holomorphic part is generated by the differential operator $-\varepsilon(z)\,\partial$. Expanding $\varepsilon(z) = -\sum_{n\in\mathbb{Z}} a_n\, z^{n+1}$ (the offset $n+1$ and the overall sign are chosen to make the algebra below take its standard form) identifies the basis of holomorphic generators

$$\ell_n = -\,z^{\,n+1}\,\partial_z, \qquad n \in \mathbb{Z}, \tag{48.11}$$

together with the independent antiholomorphic copy

$$\bar{\ell}_n = -\,\bar{z}^{\,n+1}\,\partial_{\bar{z}}, \qquad n \in \mathbb{Z}. \tag{48.12}$$

These are countably infinite families, in stark contrast to the finite list of Chapter 47. We now compute their Lie brackets directly.

Theorem 48.2 (The Witt algebra)**.** *The vector fields Eq. (48.11) satisfy the commutation relations*

$$[\ell_m, \ell_n] = (m-n)\,\ell_{m+n}, \qquad \big[\bar{\ell}_m, \bar{\ell}_n\big] = (m-n)\,\bar{\ell}_{m+n}, \qquad \big[\ell_m, \bar{\ell}_n\big] = 0, \tag{48.13}$$

for all $m, n \in \mathbb{Z}$. The algebra defined by the first bracket is the Witt algebra*; the full local conformal algebra in two dimensions is two commuting copies of it, $\mathfrak{Witt} \oplus \overline{\mathfrak{Witt}}$.*

Proof. We compute the holomorphic bracket on an arbitrary test function $f(z)$, keeping every step. Using Eq. (48.11),

$$[\ell_m, \ell_n]\, f = (z^{m+1}\partial)\big(z^{n+1}\partial f\big) - (z^{n+1}\partial)\big(z^{m+1}\partial f\big). \tag{48.14}$$

Expand the first term with the product rule,

$$z^{m+1}\partial\big(z^{n+1}\partial f\big) = z^{m+1}\Big[(n+1)\,z^{n}\,\partial f + z^{n+1}\,\partial^2 f\Big] = (n+1)\,z^{m+n+1}\,\partial f + z^{m+n+2}\,\partial^2 f, \tag{48.15}$$

and the second term identically with $m \leftrightarrow n$,

$$z^{n+1}\partial(z^{m+1}\partial f) = (m+1)\, z^{m+n+1}\,\partial f + z^{m+n+2}\,\partial^2 f. \tag{48.16}$$

The second-derivative pieces $z^{m+n+2}\partial^2 f$ are identical and cancel in the difference — this cancellation is the reason the bracket of two first-order operators is again first order. Subtracting Eq. (48.16) from Eq. (48.15),

$$[\ell_m, \ell_n]\, f = \big[(n+1) - (m+1)\big]\, z^{m+n+1}\,\partial f = (n-m)\, z^{m+n+1}\,\partial f. \tag{48.17}$$

Finally recognize $z^{m+n+1}\partial = -\ell_{m+n}$ from Eq. (48.11), so that $(n-m)\, z^{m+n+1}\partial = -(n-m)\,\ell_{m+n} = (m-n)\,\ell_{m+n}$, giving the first relation in Eq. (48.13). The antiholomorphic computation is identical with $z \to \bar z$, $\partial \to \bar\partial$. The cross bracket vanishes because ℓ_m contains only z and ∂_z while $\bar\ell_n$ contains only $\bar z$ and $\partial_{\bar z}$, and on the complexified plane these act on independent variables: $\partial_z \bar z = 0 = \partial_{\bar z} z$, so the two operators commute term by term. This proves Eq. (48.13). □

The structure constants $(m-n)$ are antisymmetric in m, n as a Lie bracket must be, and one checks the Jacobi identity directly: for any ℓ, m, n,

$$[\ell_\ell, [\ell_m, \ell_n]] + \text{cyclic} = (m-n)(\ell - m - n)\,\ell_{\ell+m+n} + \text{cyclic} = 0, \tag{48.18}$$

the three cyclic terms summing to zero by direct expansion. The Witt algebra is thus a bona fide infinite-dimensional Lie algebra, and the two commuting copies act on the holomorphic and antiholomorphic sectors entirely independently. This independence is the algebraic origin of holomorphic factorization (Theorem 48.1): because the symmetry splits, so do the objects it constrains.

Physical Insight

The single most consequential fact in two-dimensional conformal field theory is that this symmetry is *infinite*-dimensional. A finite-dimensional symmetry constrains correlators but cannot fix them; an infinite one can. The Witt algebra (after the central extension to the Virasoro algebra in the next section, which the quantum theory forces upon us) is rigid enough that, for an entire discrete family of theories, it determines the operator content, the operator product coefficients, and every correlation function with no Lagrangian and no perturbation series. The remainder of this part shows how. Everything begins with the elementary observation that in two dimensions "conformal" and "holomorphic" are the same word.

48.1.4 The global subalgebra: only $\mathfrak{sl}(2,\mathbb{C})$ survives on the sphere

The Witt generators Eq. (48.11) are local: they are defined as differential operators near a point. But a genuine symmetry of a field theory on a closed

surface must be implemented by a transformation that is well defined *everywhere*, with no singularity it cannot account for. The natural closed surface here is the Riemann sphere $\mathbb{C} \cup \{\infty\}$ — the conformal compactification of the Euclidean plane, just as in higher dimensions the conformal group acts on a compactified space — and the requirement of global regularity will cut the infinite family down to exactly three holomorphic generators. The argument is a clean exercise in changing charts.

The sphere needs two coordinate charts. Near the origin we use z; near the point at infinity we use

$$w = \frac{1}{z}, \tag{48.19}$$

so that $z = \infty$ is the regular point $w = 0$. A vector field that generates a globally defined transformation must be nonsingular in *both* charts. We examine each generator $\ell_n = -z^{n+1}\partial_z$ at the two poles.

Regularity at $z = 0$. The coefficient $-z^{n+1}$ is regular at the origin if and only if $n + 1 \geq 0$, that is

$$n \geq -1. \tag{48.20}$$

For $n \leq -2$ the field blows up at the south pole.

Regularity at $z = \infty$. We must rewrite ℓ_n in the w chart. From $w = 1/z$ we have $z = 1/w$ and, by the chain rule, $\partial_z = \dfrac{dw}{dz}\,\partial_w = -\dfrac{1}{z^2}\,\partial_w = -w^2\,\partial_w$. Also $z^{n+1} = w^{-(n+1)}$. Therefore

$$\ell_n = -\,z^{n+1}\,\partial_z = -\,w^{-(n+1)}\big(-\,w^2\,\partial_w\big) = w^{\,1-n}\,\partial_w. \tag{48.21}$$

This is regular at the north pole $w = 0$ if and only if $1 - n \geq 0$, that is

$$n \leq 1. \tag{48.22}$$

Imposing *both* Eq. (48.20) and Eq. (48.22) leaves exactly

$$n \in \{-1, 0, 1\}, \tag{48.23}$$

and likewise $\bar{n} \in \{-1, 0, 1\}$ for the antiholomorphic copy. The globally well-defined conformal transformations of the sphere are generated by six real vector fields: $\{\ell_{-1}, \ell_0, \ell_1\}$ and $\{\bar{\ell}_{-1}, \bar{\ell}_0, \bar{\ell}_1\}$.

Restricting Eq. (48.13) to $n \in \{-1, 0, 1\}$ gives a closed three-dimensional algebra,

$$[\ell_0, \ell_{-1}] = \ell_{-1}, \qquad [\ell_0, \ell_1] = -\ell_1, \qquad [\ell_{-1}, \ell_1] = -2\,\ell_0, \tag{48.24}$$

which is the Lie algebra $\mathfrak{sl}(2)$. Together with the antiholomorphic copy and the reality condition relating the two on the physical slice, the global algebra is the complexification $\mathfrak{sl}(2,\mathbb{C}) \cong \mathfrak{sl}(2) \oplus \overline{\mathfrak{sl}(2)}$, exponentiating to the group

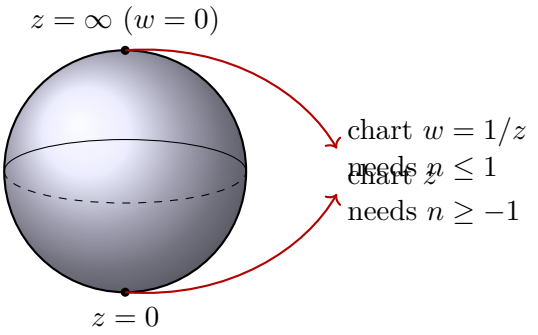


Figure 48.1: The Riemann sphere $\mathbb{C} \cup \{\infty\}$ with its two coordinate charts. A Witt generator $\ell_n = -z^{n+1}\partial_z$ must be nonsingular at the south pole $z = 0$ (forcing $n \ge -1$) and, after the transition to $w = 1/z$, at the north pole $w = 0$ (forcing $n \le 1$). Only $n \in \{-1, 0, 1\}$ pass both tests: these generate the global conformal group $\mathrm{SL}(2, \mathbb{C})/\mathbb{Z}_2$.

$\mathrm{SL}(2, \mathbb{C})$ (acting on the sphere as $\mathrm{SL}(2, \mathbb{C})/\mathbb{Z}_2 \cong \mathrm{PSL}(2, \mathbb{C})$). This is precisely the global conformal group of Chapter 47 specialized to $d = 2$: indeed $\mathrm{SO}(3, 1) \cong \mathrm{SL}(2, \mathbb{C})/\mathbb{Z}_2$, and the count $\frac{1}{2}(d + 1)(d + 2) = \frac{1}{2} \cdot 3 \cdot 4 = 6$ matches the six real generators we just found. The infinite Witt algebra is the genuinely new structure of two dimensions; the global $\mathrm{SL}(2, \mathbb{C})$ is the part that agrees with the higher-dimensional story.

The three holomorphic generators have transparent geometric meanings, fixing the dictionary we will use throughout:

- $\ell_{-1} = -\partial_z$ generates *translations* $z \mapsto z + a$.
- $\ell_0 = -z\,\partial_z$ generates complex *scalings* $z \mapsto \lambda z$. Its real part (combined with $\bar{\ell}_0$) is the dilatation; its imaginary part is the rotation. Explicitly, $\ell_0 + \bar{\ell}_0$ generates dilatations and $i(\ell_0 - \bar{\ell}_0)$ generates rotations.
- $\ell_1 = -z^2\,\partial_z$ generates *special conformal transformations* $z \mapsto z/(1 - bz)$.

We make the third statement precise, and exhibit the finite Möbius group it integrates to, in Theorem 48.6.

48.1.5 Conformal weights and primary fields

Having identified the symmetry, we now classify the local operators that carry its representations. In Chapter 47 a primary operator was defined by its behaviour under the global conformal group: it transforms with a definite scaling dimension and is annihilated by the special conformal generators at the origin (Theorem 47.15). In two dimensions the natural eigenvalues to track are those

of the two commuting generators ℓ_0 and $\bar{\ell}_0$, which measure holomorphic and antiholomorphic scaling separately.

Definition 48.3 (Conformal weights)**.** A local field $\mathcal{O}(z,\bar{z})$ in a two-dimensional CFT has *conformal weights* $(h,\bar{h})$ if under the holomorphic and antiholomorphic scalings generated by $\ell_0,\bar{\ell}_0$ it carries those eigenvalues; equivalently, under $z\mapsto\lambda z$, $\bar{z}\mapsto\bar{\lambda}\bar{z}$ it scales as $\mathcal{O}\mapsto\lambda^{-h}\bar{\lambda}^{-\bar{h}}\mathcal{O}$. The *scaling dimension* and *spin* are

$$\Delta = h+\bar{h}, \qquad s = h-\bar{h}, \tag{48.25}$$

so that Δ is the eigenvalue of the dilatation $\ell_0+\bar{\ell}_0$ and s is the eigenvalue of the rotation generator $i(\ell_0-\bar{\ell}_0)$. The weights $h,\bar{h}$ are independent real numbers; reality and unitarity (developed later) constrain them, but kinematically h and $\bar{h}$ are unrelated.

The relations Eq. (48.25) are the two-dimensional refinement of the higher-dimensional labels: a generic CFT operator is labelled by a dimension Δ and a Lorentz/rotation representation, and in $d=2$ the rotation group is the abelian $\mathrm{SO}(2)$, whose irreducible representations are one-dimensional and labelled by a single integer or half-integer spin s. Splitting (Δ,s) into $(h,\bar{h})$ via Eq. (48.25) is the operator-level shadow of holomorphic factorization: a field of definite weights behaves as a product of a weight-h holomorphic object and a weight-$\bar{h}$ antiholomorphic object.

The strongest fields are those that transform tensorially not just under the global group but under the full local conformal symmetry. We promote the primary definition of Theorem 47.15 to its two-dimensional, finite form.

Definition 48.4 (Primary field)**.** A field $\mathcal{O}(z,\bar{z})$ of weights $(h,\bar{h})$ is *primary* if under *every* local holomorphic change of coordinates $z\mapsto z'=f(z)$, $\bar{z}\mapsto\bar{z}'=\bar{f}(\bar{z})$ (with $f'\neq 0$) it transforms as a tensor of those weights,

$$\mathcal{O}'(z',\bar{z}') = \left(\frac{dz'}{dz}\right)^{-h}\left(\frac{d\bar{z}'}{d\bar{z}}\right)^{-\bar{h}}\mathcal{O}(z,\bar{z}). \tag{48.26}$$

A field obeying Eq. (48.26) only for the global subgroup $f\in\mathrm{SL}(2,\mathbb{C})$ — i.e. only for ℓ_{-1},ℓ_0,ℓ_1 — is called *quasi-primary* (the terminology of Theorem 47.15). Every primary is quasi-primary; the converse fails, and the failure is governed by the central extension of the next section.

The infinitesimal form of Eq. (48.26) is the statement we will use most. For $f(z)=z+\varepsilon(z)$ with $\bar{f}(\bar{z})=\bar{z}+\bar{\varepsilon}(\bar{z})$, expanding to first order with $dz'/dz = 1+\varepsilon'(z)$ gives $(dz'/dz)^{-h}=1-h\,\varepsilon'(z)+\cdots$, so

$$\delta_{\varepsilon,\bar{\varepsilon}}\mathcal{O} = -\Big(\varepsilon\,\partial + h\,\varepsilon'\Big)\mathcal{O} - \Big(\bar{\varepsilon}\,\bar{\partial} + \bar{h}\,\bar{\varepsilon}'\Big)\mathcal{O}. \tag{48.27}$$

The terms $h\,\varepsilon'$ and $\bar{h}\,\bar{\varepsilon}'$ beyond the naive transport $\varepsilon\,\partial\mathcal{O}$ encode the tensor weight; they are what distinguishes a primary of nonzero weight from a scalar function. We will recover this same variation in the next section from the operator product of the stress tensor with $\mathcal{O}$, which is the dynamical way the symmetry acts.

48.1.6 Consistency: recovering the two- and three-point functions

A non-negotiable check on the formalism is that the correlators it predicts agree with the general-dimension forms fixed in Chapter 47. There the two- and three-point functions of (quasi-)primaries were determined up to constants by conformal invariance alone. We re-derive them here from global $\mathrm{SL}(2,\mathbb{C})$ invariance in complex coordinates, both to confirm consistency and to display holomorphic factorization in action.

Worked Example 48.5 (The two-point function from global invariance)**.** Let $\mathcal{O}_1, \mathcal{O}_2$ be quasi-primaries of weights $(h_1, \bar{h}_1)$, $(h_2, \bar{h}_2)$. We determine $G(z_1, \bar{z}_1; z_2, \bar{z}_2) = \langle \mathcal{O}_1(z_1, \bar{z}_1)\, \mathcal{O}_2(z_2, \bar{z}_2)\rangle$ by imposing invariance under each of $\ell_{-1}, \ell_0, \ell_1$ (the antiholomorphic copy works identically and multiplies the holomorphic answer). A correlator of primaries is invariant when the sum of the variations Eq. (48.27) acting on each insertion vanishes; we use the global modes $\varepsilon(z) = -z^{n+1}$ for $n = -1, 0, 1$. Because the two sectors decouple, write $G = g(z_1, z_2)\,\bar{g}(\bar{z}_1, \bar{z}_2)$ and solve for the holomorphic factor g.

Translations (ℓ_{-1}, $\varepsilon = \text{const}$): the variation is $-(\partial_{z_1} + \partial_{z_2})g = 0$, so g depends only on the difference $z_{12} \equiv z_1 - z_2$.

Scaling (ℓ_0, $\varepsilon = -z$): each insertion contributes $z_i\partial_{z_i} + h_i$, so invariance reads $(z_1\partial_{z_1} + z_2\partial_{z_2} + h_1 + h_2)g = 0$. With $g = g(z_{12})$ this is the homogeneity condition $(z_{12}\,\partial_{z_{12}} + h_1 + h_2)g = 0$, solved by the power law

$$g(z_{12}) = \frac{C_{12}}{z_{12}^{\,h_1+h_2}}. \tag{48.28}$$

Special conformal (ℓ_1, $\varepsilon = -z^2$): each insertion contributes $z_i^2\partial_{z_i} + 2h_i z_i$. Acting on Eq. (48.28),

$$\begin{aligned}\Big(z_1^2\partial_{z_1} + z_2^2\partial_{z_2} + 2h_1 z_1 + 2h_2 z_2\Big)\frac{C_{12}}{z_{12}^{\,h_1+h_2}} &= C_{12}\Big[-(h_1+h_2)\frac{z_1^2 - z_2^2}{z_{12}^{\,h_1+h_2+1}} + \frac{2h_1 z_1 + 2h_2 z_2}{z_{12}^{\,h_1+h_2}}\Big] \\ &= \frac{C_{12}}{z_{12}^{\,h_1+h_2}}\Big[-(h_1+h_2)(z_1+z_2) + 2h_1 z_1 + 2h_2 z_2\Big] \\ &= \frac{C_{12}}{z_{12}^{\,h_1+h_2}}\,(h_1 - h_2)(z_1 - z_2), \end{aligned} \tag{48.29}$$

where the first line used $\partial_{z_1} z_{12}^{-p} = -p\, z_{12}^{-p-1}$ and $z_1^2 - z_2^2 = (z_1 + z_2)z_{12}$. This vanishes for all z_1, z_2 if and only if $h_1 = h_2$. Hence the holomorphic two-point

function is nonzero only when the weights match, and then

$$\langle \mathcal{O}_1 \mathcal{O}_2 \rangle = \frac{C_{12}\, \delta_{h_1 h_2}\, \delta_{\bar{h}_1 \bar{h}_2}}{z_{12}^{2h}\, \bar{z}_{12}^{2\bar{h}}}, \qquad h \equiv h_1 = h_2,\ \bar{h} \equiv \bar{h}_1 = \bar{h}_2. \tag{48.30}$$

On the physical slice $\bar{z}_{12} = z_{12}^*$, a spinless field ($\bar{h} = h$, $s = 0$) gives $\langle \mathcal{O}\mathcal{O} \rangle = C\,|z_{12}|^{-4h} = C\,|z_{12}|^{-2\Delta}$ with $\Delta = 2h$, reproducing exactly the $|x|^{-2\Delta}$ two-point form of Eq. (47.120). The complex formalism has merely refined that result, resolving it into a holomorphic and an antiholomorphic factor and exposing the spin through the unequal powers of z_{12} and $\bar{z}_{12}$.

Worked Example 48.6 (The three-point function and the global Möbius group)**.** The same three conditions fix the three-point function up to one constant. Translation invariance makes the holomorphic factor a function of the differences z_{12}, z_{13}, z_{23}; scaling invariance makes it homogeneous of total degree $-(h_1 + h_2 + h_3)$; and special conformal invariance, applied as in Eq. (48.29) but now with three insertions, fixes the individual powers. Carrying out the same algebra (the special-conformal condition becomes three linear equations for the three exponents, with the unique solution below) gives

$$\langle \mathcal{O}_1 \mathcal{O}_2 \mathcal{O}_3 \rangle = \frac{C_{123}}{z_{12}^{h_1+h_2-h_3}\, z_{23}^{h_2+h_3-h_1}\, z_{13}^{h_1+h_3-h_2}} \times (\text{antiholomorphic},\ h \to \bar{h}), \tag{48.31}$$

which is the holomorphically factorized version of the general-dimension three-point form Eq. (47.137): setting $\bar{h}_i = h_i$ and restricting to the slice converts each $z_{ij}^p \bar{z}_{ij}^p = |z_{ij}|^{2p}$ into the $|x_{ij}|$ powers of Chapter 47, with $\Delta_i = 2h_i$.

It is illuminating to verify the special-conformal piece of this rigidity by integrating the global generators to finite transformations. The flow generated by $\ell_1 = -z^2\partial_z$ solves $\dfrac{dz}{dt} = -(-z^2) = z^2$ — here the minus sign in ℓ_n and the sign in $\delta\phi = -\varepsilon\,\partial\phi$ combine so that the parameter b below is the physical special-conformal parameter — giving $-d(1/z)/dt = 1$, hence $1/z(t) = 1/z_0 - t$ and

$$z \;\longmapsto\; \frac{z}{1 - b\,z}, \qquad b = t. \tag{48.32}$$

Composing with the translation $z \mapsto z + a$ generated by ℓ_{-1} and the scaling $z \mapsto \lambda z$ generated by ℓ_0, the most general finite global transformation is the *Möbius transformation*

$$z \;\longmapsto\; \frac{az + b}{cz + d}, \qquad ad - bc = 1, \quad a, b, c, d \in \mathbb{C}, \tag{48.33}$$

the determinant condition $ad - bc = 1$ being exactly the statement that the matrix $\left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right)$ lies in $\mathrm{SL}(2,\mathbb{C})$. The map Eq. (48.33) is the unique holomorphic bijection of the Riemann sphere; the overall sign of the matrix is unphysical (it acts trivially on z), which is why the group acting on the sphere is $\mathrm{SL}(2,\mathbb{C})/\mathbb{Z}_2$.

This is the global conformal group of Fig. 48.1, now exhibited concretely as fractional linear maps, and it is the largest group under which *all* correlators — not merely two- and three-point functions — are invariant.

The consistency is complete: the complex-coordinate machinery reproduces, and sharpens through factorization, every kinematic constraint of Chapter 47, while adding the infinite tower of *local* generators ℓ_n with $|n| \geq 2$ that has no higher-dimensional analogue. These local generators do not preserve correlators globally — they are singular somewhere on the sphere — but they nonetheless act on the local operator algebra, and it is their action that the next section turns into dynamics.

With the holomorphic vector fields $\ell_n = -z^{n+1}\partial_z$, the Witt brackets Eq. (48.13), the weight labels $(h, \bar{h})$ of Theorem 48.3, and the primary law Eq. (48.26) now in hand, we are ready to ask how this symmetry is realized on quantum operators. The generators will be produced by contour integrals of a single holomorphic field, the stress tensor $T(z)$, and the quantum theory will refuse to leave the Witt algebra untouched: it inserts a central term proportional to a number c. That central extension — the passage from Witt to Virasoro — is the subject of Section 48.2, where c enters as the anomaly in the $T(z)T(w)$ operator product and as the Schwarzian in the finite transformation of T.

48.2 The Stress Tensor, Conformal Ward Identities, and the Central Charge

The previous section was kinematics. We learned that in two dimensions the local conformal transformations are the holomorphic maps $z \mapsto f(z)$ and their antiholomorphic partners $\bar{z} \mapsto \bar{f}(\bar{z})$, that the symmetry algebra is the infinite-dimensional Witt algebra in two commuting copies (Eq. (48.13)), that correlators holomorphically factorize (Theorem 48.1), and that the building blocks are *primary fields* carrying a pair of conformal weights $(h, \bar{h})$ (Theorem 48.3). What we did *not* do is connect any of this to a dynamical object that the theory itself produces. Symmetry, so far, is a property we impose; it has not yet told us how the theory computes. The bridge from one to the other is the *stress tensor*. In every local field theory the stress tensor is the Noether current of spacetime symmetry; here we will see that in two dimensions its holomorphic component $T(z)$ is the generating object of the entire local conformal symmetry. Inserting $T(z)$ into a correlator and deforming the integration contour — the residue calculus of Theorem 5.23 becomes the computational engine of the chapter — turns the abstract Witt action into a concrete operational rule, the *conformal Ward identity*, that reads off the transformation of every field from the singularities of its operator product with T. Pushing this one step further, to the operator product of T with *itself*, exposes a number that no amount of kinematics

could have anticipated: the *central charge* c, the single anomaly coefficient that measures the failure of the stress tensor to be a primary field and that will, in the next section, centrally extend the Witt algebra to the Virasoro algebra.

48.2.1 Tracelessness and the holomorphic stress tensor

We begin with the object inherited from the foundations. A Poincaré-invariant local field theory possesses a symmetric, conserved stress tensor; the canonical tensor can always be improved to the symmetric, gauge-invariant Belinfante form $\Theta_{\mathrm{c}}^{\mu\nu}$ of Section 7.7.3, which satisfies

$$\Theta_{\mathrm{c}}^{\mu\nu} = \Theta_{\mathrm{c}}^{\nu\mu}, \qquad \partial_\mu \Theta_{\mathrm{c}}^{\mu\nu} = 0. \tag{48.34}$$

Throughout this chapter we write $T_{\mu\nu} \equiv \Theta^{\mathrm{c}}_{\mu\nu}$ for this symmetric, conserved tensor; no confusion with the time-ordering symbol will arise because in the Euclidean two-dimensional setting of Chapter 48 all correlators are radially ordered. The decisive extra input is the result of Theorem 7.43, sharpened in Eq. (47.3): a scale-invariant theory whose dilatation current is the virial-improved $D^\mu = x_\nu T^{\mu\nu}$ has a *traceless* stress tensor, and (in the flat-space, $d = 2$ setting where scale invariance is upgraded to full conformal invariance, Theorem 47.5) this tracelessness is the local certificate of the entire conformal symmetry:

$$T^{\mu}{}_{\mu} = 0. \tag{48.35}$$

Equations (48.34) and (48.35) are the only dynamical facts we need; everything below is their two-dimensional consequence.

We now pass to the complex coordinates fixed in the previous section. After the Euclidean continuation of Theorem 5.31 the two-dimensional plane carries coordinates x^1, x^2, and we set

$$z = x^1 + i x^2, \qquad \bar{z} = x^1 - i x^2, \qquad \partial \equiv \partial_z = \tfrac{1}{2}(\partial_1 - i\partial_2), \qquad \bar{\partial} \equiv \partial_{\bar{z}} = \tfrac{1}{2}(\partial_1 + i\partial_2). \tag{48.36}$$

The flat Euclidean line element $\mathrm{d}s^2 = (\mathrm{d}x^1)^2 + (\mathrm{d}x^2)^2 = \mathrm{d}z\,\mathrm{d}\bar{z}$ has metric components $g_{z\bar{z}} = g_{\bar{z}z} = \tfrac{1}{2}$, $g_{zz} = g_{\bar{z}\bar{z}} = 0$, with inverse $g^{z\bar{z}} = g^{\bar{z}z} = 2$, $g^{zz} = g^{\bar{z}\bar{z}} = 0$. A symmetric two-tensor has three independent components T_{zz}, $T_{\bar{z}\bar{z}}$, and $T_{z\bar{z}} = T_{\bar{z}z}$. The trace is

$$T^{\mu}{}_{\mu} = g^{\mu\nu} T_{\mu\nu} = g^{z\bar{z}} T_{z\bar{z}} + g^{\bar{z}z} T_{\bar{z}z} = 4\, T_{z\bar{z}}, \tag{48.37}$$

so the tracelessness condition (48.35) is simply

$$\boxed{T_{z\bar{z}} = T_{\bar{z}z} = 0} \tag{48.38}$$

The off-diagonal component of the stress tensor vanishes identically in a conformal theory. Only T_{zz} and $T_{\bar{z}\bar{z}}$ survive.

Now impose conservation (48.34). Raising the free index with the inverse metric, $\partial^\mu T_{\mu\nu} = g^{\mu\rho}\partial_\rho T_{\mu\nu} = 0$. The nonzero entries of $g^{\mu\rho}$ pair z with $\bar z$, so $\partial^z = g^{z\bar z}\partial_{\bar z} = 2\bar\partial$ and $\partial^{\bar z} = 2\partial$. Taking the free index $\nu = z$,

$$0 = \partial^\mu T_{\mu z} = \partial^z T_{zz} + \partial^{\bar z} T_{\bar z z} = 2\bar\partial\, T_{zz} + 2\partial\, T_{\bar z z} = 2\bar\partial\, T_{zz}, \tag{48.39}$$

where the second term dropped by (48.38). Likewise the choice $\nu = \bar z$ gives $\partial\, T_{\bar z\bar z} = 0$. We have proved that in any two-dimensional conformal theory

$$\bar\partial\, T_{zz} = 0, \qquad \partial\, T_{\bar z\bar z} = 0 \qquad \Longrightarrow \qquad T_{zz} = T_{zz}(z), \quad T_{\bar z\bar z} = T_{\bar z\bar z}(\bar z). \tag{48.40}$$

The surviving components are respectively a holomorphic and an antiholomorphic function (away from operator insertions, where the equations of motion fail and contact terms appear; we return to this below). This is the dynamical origin of holomorphic factorization: it is the stress tensor that splits the dynamics into a holomorphic sector and an independent antiholomorphic one. We give the holomorphic component its standard name and normalization,

$$T(z) \equiv -2\pi\, T_{zz}(z), \qquad \bar T(\bar z) \equiv -2\pi\, T_{\bar z\bar z}(\bar z), \tag{48.41}$$

the factor -2π being chosen, as we verify below, so that the conformal Ward identity and the operator products take their canonical residue form. We call $T(z)$ the *holomorphic* (or chiral) stress tensor and $\bar T(\bar z)$ the antiholomorphic one. From here the antiholomorphic sector is a verbatim copy under $z \leftrightarrow \bar z$, $T \leftrightarrow \bar T$, $h \leftrightarrow \bar h$; we develop the holomorphic side in full and state the antiholomorphic results.

Notation Watch

We use the real area measure $\mathrm{d}^2x = \mathrm{d}x^1\,\mathrm{d}x^2$ and the delta function normalized by $\int \mathrm{d}^2x\,\delta^{(2)}(x) = 1$. With these conventions the planar Green's function identity (derived in Eq. (48.46) below) reads $\bar\partial\,(z-w)^{-1} = \pi\,\delta^{(2)}(x-x_w)$. References that adopt the rescaled measure $\mathrm{d}^2z \equiv 2\,\mathrm{d}x^1\,\mathrm{d}x^2$ with $\int \mathrm{d}^2z\,\delta^{(2)}(z,\bar z) = 1$ instead write the familiar $\bar\partial\,(z-w)^{-1} = 2\pi\,\delta^{(2)}$; the factor of two is purely the measure convention and never enters a physical weight h or the central charge c, both of which we fix through the residue normalization of (48.41) and check independently against the free boson.

48.2.2 The conformal Ward identity

We can now make the symmetry operational. An infinitesimal conformal transformation is generated by a conformal Killing vector ξ^μ; in two dimensions (48.36) this is precisely a holomorphic vector field, $\xi^z = \epsilon(z)$ with $\bar\partial\epsilon = 0$, together with its antiholomorphic partner $\xi^{\bar z} = \bar\epsilon(\bar z)$. The associated Noether current is,

by Theorem 7.48 and Eq. (47.26), the contraction of the stress tensor with the Killing vector,

$$J^\mu = T^{\mu\nu}\xi_\nu, \qquad \partial_\mu J^\mu = \tfrac{1}{2}T^{\mu\nu}(\partial_\mu\xi_\nu + \partial_\nu\xi_\mu) = \tfrac{1}{d}\,T^\mu{}_\mu\,\partial\cdot\xi = 0, \tag{48.42}$$

the divergence vanishing by tracelessness (48.35). The content of this section is to translate the conservation law into the complex coordinates and to read off how the current acts on operators.

Lower the indices on the current. With $\xi^z = \epsilon$, $\xi^{\bar z} = \bar\epsilon$ and $\xi_z = g_{z\bar z}\xi^{\bar z} = \tfrac{1}{2}\bar\epsilon$, $\xi_{\bar z} = \tfrac{1}{2}\epsilon$, the holomorphic component of the current is

$$J_z = T_{z\nu}\xi^\nu = T_{zz}\xi^z + T_{z\bar z}\xi^{\bar z} = T_{zz}\,\epsilon(z) = -\frac{1}{2\pi}\,\epsilon(z)\,T(z), \tag{48.43}$$

where $T_{z\bar z} = 0$ was used and (48.41) substituted. Because $\epsilon(z)$ and $T(z)$ are both holomorphic, J_z is holomorphic, $\bar\partial J_z = 0$; symmetrically $J_{\bar z} = -\frac{1}{2\pi}\bar\epsilon(\bar z)\bar T(\bar z)$ is antiholomorphic. The conservation law $\partial_\mu J^\mu = 2(\partial J_{\bar z} + \bar\partial J_z) = 0$ is satisfied identically away from insertions, and is in this sense empty: holomorphy already encodes it. Its physical content lives entirely in the *contact terms* at operator insertions, which is exactly what the Ward identity extracts.

The Noether–Ward identity established in general in Theorem 7.10 states that the divergence of a symmetry current inside a correlator is a sum of delta functions, one per insertion, whose coefficients are the infinitesimal variations of the operators:

$$\partial_\mu\big\langle J^\mu(x)\,O_1(x_1)\cdots O_n(x_n)\big\rangle = -\sum_{i=1}^{n}\delta^{(2)}(x-x_i)\,\big\langle O_1\cdots\delta_\epsilon O_i\cdots O_n\big\rangle. \tag{48.44}$$

This is the precise statement that the conserved charge generates the transformation: integrating x over a small disk D_i that contains only the insertion at x_i and using the divergence theorem turns the left-hand side into a boundary flux and the right-hand side into the single variation $-\langle\cdots\delta_\epsilon O_i\cdots\rangle$. We carry out exactly this integration in complex coordinates.

We need the planar Stokes identity. For any smooth $h(z,\bar z)$ and a region D with boundary ∂D traversed counterclockwise,

$$\oint_{\partial D} h\,\mathrm{d}z = \int_D \mathrm{d}(h\,\mathrm{d}z) = \int_D \bar\partial h\;\mathrm{d}\bar z\wedge\mathrm{d}z = 2i\int_D \bar\partial h\;\mathrm{d}^2x \quad\Longrightarrow\quad \int_D \bar\partial h\;\mathrm{d}^2x = \frac{1}{2i}\oint_{\partial D} h\,\mathrm{d}z, \tag{48.45}$$

using $\mathrm{d}\bar z\wedge\mathrm{d}z = 2i\,\mathrm{d}x^1\wedge\mathrm{d}x^2$. The conjugate identity is $\int_D \partial g\,\mathrm{d}^2x = -\frac{1}{2i}\oint_{\partial D} g\,\mathrm{d}\bar z$. As an immediate corollary, applying (48.45) to $h = (z-w)^{-1}$ and using $\oint_{\partial D}(z-w)^{-1}\mathrm{d}z = 2\pi i$ from Theorem 5.23 for any disk enclosing w gives the Green's function identity

$$\bar\partial\,\frac{1}{z-w} = \pi\,\delta^{(2)}(x-x_w), \tag{48.46}$$

the precise statement that the holomorphic function $1/(z-w)$ fails to be holomorphic exactly at w, where its $\bar\partial$ is a contact term. This is the analytic fact underlying the entire residue machinery: a simple pole in an operator product is the holomorphic shadow of a delta-function contact term.

Now integrate the local identity (48.44) over a disk D containing all n insertions. The left-hand side, using $\partial_\mu J^\mu = 2(\partial J_{\bar z} + \bar\partial J_z)$ together with (48.45) and its conjugate, becomes a boundary contour integral,

$$\begin{aligned}\int_D \mathrm{d}^2x\, \partial_\mu \langle J^\mu X\rangle &= 2\int_D \mathrm{d}^2x\, \big(\bar\partial\langle J_z X\rangle + \partial\langle J_{\bar z} X\rangle\big)\\ &= 2\Big(\frac{1}{2i}\oint_{\partial D}\langle J_z X\rangle\,\mathrm{d}z - \frac{1}{2i}\oint_{\partial D}\langle J_{\bar z} X\rangle\,\mathrm{d}\bar z\Big)\\ &= -\frac{1}{2\pi i}\oint_{\partial D}\mathrm{d}z\,\epsilon(z)\langle T(z)X\rangle + \frac{1}{2\pi i}\oint_{\partial D}\mathrm{d}\bar z\,\bar\epsilon(\bar z)\langle \bar T(\bar z)X\rangle,\end{aligned} \tag{48.47}$$

where in the last line we inserted (48.43), writing $X = O_1\cdots O_n$. The right-hand side of (48.44) integrates to $-\sum_i\langle\cdots\delta_\epsilon O_i\cdots\rangle = -\delta_\epsilon\langle X\rangle$, the total variation. Equating the two and renaming the boundary $\partial D \to C$ (any contour enclosing all insertions) yields the *conformal Ward identity*:

$$\boxed{\delta_\epsilon\langle X\rangle = -\frac{1}{2\pi i}\oint_C \mathrm{d}z\,\epsilon(z)\,\langle T(z)\,X\rangle \;-\; \frac{1}{2\pi i}\oint_C \mathrm{d}\bar z\,\bar\epsilon(\bar z)\,\langle \bar T(\bar z)\,X\rangle} \tag{48.48}$$

where the antiholomorphic contour carries the orientation conjugate to the holomorphic one. This is the master formula of two-dimensional conformal field theory. It says: the response of *any* correlation function to *any* infinitesimal conformal transformation is computed by encircling the insertions with a contour and reading off the residues of the stress tensor's operator products. The infinite-dimensional symmetry, abstract in the Witt algebra, has become an explicit residue calculus. Notice that the choice -2π in (48.41) is exactly what produced the clean prefactor $-1/2\pi i$ here; this is the promised normalization check.

Physical Insight

The Ward identity is the precise sense in which $T(z)$ *is* the local conformal symmetry. A single insertion of $T(z)$, integrated against an arbitrary holomorphic parameter $\epsilon(z)$ on a contour, generates the most general infinitesimal holomorphic reparametrization. There is no separate "symmetry generator" to introduce: the stress tensor, the Noether current of translations, already contains the entire infinite tower because in two dimensions every conformal motion is built from T contracted with a holomorphic vector field. When we mode-expand $T(z)$ in the next section, these contours will become the Virasoro generators L_n.

48.2.3 Primary fields and the T–O operator product

The Ward identity is only useful once we know the singular operator product of $T(z)$ with the fields in the correlator. We now show that for a primary field this singular structure is completely fixed — it is forced to be a double pole plus a single pole, with coefficients dictated by the weight — and that this is the cleanest possible *definition* of a primary.

Recall from Theorem 48.3 the defining transformation law of a primary field of weights $(h,\bar h)$: under a holomorphic map $z\mapsto f(z)$, $\bar z\mapsto \bar f(\bar z)$,

$$O'(z',\bar z') = \Big(\frac{\mathrm{d}f}{\mathrm{d}z}\Big)^{-h}\Big(\frac{\mathrm{d}\bar f}{\mathrm{d}\bar z}\Big)^{-\bar h} O(z,\bar z), \tag{48.49}$$

so that $O(z,\bar z)\,(\mathrm{d}z)^h(\mathrm{d}\bar z)^{\bar h}$ is invariant. The infinitesimal version with $z'=z+\epsilon(z)$, $\bar z'=\bar z+\bar\epsilon(\bar z)$ is obtained by expanding $f'=1+\partial\epsilon$ and Taylor-expanding the argument; to first order,

$$\delta_\epsilon O(w,\bar w) = -\big(\,\epsilon(w)\,\partial + h\,\partial\epsilon(w)\,\big)O(w,\bar w) - \big(\,\bar\epsilon(\bar w)\,\bar\partial + \bar h\,\bar\partial\bar\epsilon(\bar w)\,\big)O(w,\bar w). \tag{48.50}$$

This is the transformation we must reproduce. Suppose the operator product of T with O has the Laurent form

$$T(z)\,O(w,\bar w) \sim \cdots + \frac{O^{(2)}(w,\bar w)}{(z-w)^2} + \frac{O^{(1)}(w,\bar w)}{z-w} + (\text{regular}), \tag{48.51}$$

with possibly higher-order poles indicated by the dots; here $\sim$ denotes equality up to terms regular as $z\to w$, in the sense of the operator product expansion of Theorem 47.46. Feed a single such operator into the Ward identity (48.48): the holomorphic variation of $O(w,\bar w)$ inside any correlator is the residue

$$\delta_\epsilon O(w,\bar w) = -\frac{1}{2\pi i}\oint_w \mathrm{d}z\,\epsilon(z)\,T(z)O(w,\bar w) = -\operatorname{Res}_{z\to w}\big[\epsilon(z)\,T(z)O(w,\bar w)\big], \tag{48.52}$$

the contour now shrunk to a small circle about w alone. Expanding $\epsilon(z)=\epsilon(w)+\partial\epsilon(w)\,(z-w)+\frac12\partial^2\epsilon(w)(z-w)^2+\cdots$ and extracting the coefficient of $(z-w)^{-1}$ in $\epsilon(z)\,T(z)O(w)$,

$$\delta_\epsilon O(w,\bar w) = -\Big[\,\epsilon(w)\,O^{(1)} + \partial\epsilon(w)\,O^{(2)} + \tfrac12\partial^2\epsilon(w)\,O^{(3)} + \cdots\Big], \tag{48.53}$$

where $O^{(3)}$ is the coefficient of the triple pole, and so on. Now compare term by term with the primary law (48.50). The coefficient of $\epsilon(w)$ forces $O^{(1)}=\partial O$; the coefficient of $\partial\epsilon(w)$ forces $O^{(2)}=h\,O$; and crucially the primary law (48.50) contains *no* $\partial^2\epsilon$, $\partial^3\epsilon,\dots$ terms, so all higher poles must vanish, $O^{(3)}=O^{(4)}=\cdots=0$. The operator product is therefore exactly a double pole plus a single pole:

$$\boxed{\,T(z)\,O(w,\bar w) \sim \frac{h\,O(w,\bar w)}{(z-w)^2} + \frac{\partial O(w,\bar w)}{z-w}\,} \tag{48.54}$$

with the antiholomorphic companion $\bar{T}(\bar{z})O(w,\bar{w}) \sim \bar{h}\,O/(\bar{z}-\bar{w})^2 + \bar{\partial}O/(\bar{z}-\bar{w})$. We take this as the working definition.

Definition 48.7 (Primary field via the stress tensor)**.** A local field $O(w,\bar{w})$ is *primary* of weights $(h,\bar{h})$ if its operator products with the holomorphic and antiholomorphic stress tensors terminate at the double pole, as in (48.54) and its conjugate, with no pole of order three or higher. Equivalently, by the computation above, O transforms covariantly under the full local conformal group according to (48.49). A field whose T–O product contains higher-order poles is called *non-primary*; the fields appearing in those higher poles are its *descendants* in the sense of Theorem 47.15, organized into the conformal family of O.

The two characterizations of a primary — the geometric transformation law (48.49) and the algebraic pole structure (48.54) — are now proved equivalent. The first is how a primary behaves; the second is how the theory detects it. The double pole measures the weight (the local scaling and rotation), the single pole is the translation ∂O, and the absence of higher poles is the precise statement that O is annihilated by the would-be raising part of the symmetry — a "highest-weight" condition that the next section will phrase as $L_n O = 0$ for $n > 0$.

48.2.4 The T–T operator product and the central charge

The natural next question is whether the stress tensor is itself a primary field. It is a holomorphic field of weight $(2,0)$: it carries two lower holomorphic indices, $T(z) \propto T_{zz}$, and the conformal factor $(\mathrm{d}z)^2$ shows it scales with $h=2$, $\bar{h}=0$. If T were primary, its self–operator product would, by (48.54) with $h=2$, be exactly

$$T(z)\,T(w) \overset{?}{\sim} \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}. \tag{48.55}$$

This cannot be the whole story. Dimensional analysis allows one further singular term. The product $T(z)T(w)$ has total holomorphic weight 4, so a term proportional to the identity operator dressed by $(z-w)^{-4}$ is permitted by scaling, with a c-number coefficient (the identity has weight zero, and $(z-w)^{-4}$ supplies the missing weight 4). No lower power of $(z-w)$ can multiply the identity: $(z-w)^{-3}$ times the identity has weight 3, odd under the exchange symmetry, and would require an operator of weight 1 which a unitary conformal theory built on the identity does not supply at this order; we confirm its absence by symmetry just below. The most general singular operator product consistent with holomorphy, weight $(2,0)$, and the presence of the identity in the spectrum is therefore

$$\boxed{T(z)\,T(w) \sim \frac{c/2}{(z-w)^4} + \frac{2\,T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}} \tag{48.56}$$

The constant c is the *central charge*.

Definition 48.8 (Central charge)**.** The *central charge* c of a two-dimensional conformal field theory is the coefficient defined by the leading singularity of the holomorphic stress tensor's self–operator product, $T(z)T(w) \sim \frac{c/2}{(z-w)^4} + \cdots$, as in (48.56). The antiholomorphic sector carries an independent constant $\bar{c}$ defined identically from $\bar{T}(\bar{z})\bar{T}(\bar{w})$; for the theories of this chapter, built from local actions, $c = \bar{c}$. The number c is the unique central datum of the symmetry algebra: it is a property of the theory, not of any field, and it measures precisely the failure of T to be primary.

Three remarks fix the structure. First, the absence of the odd singularity. Bose symmetry of the stress tensor requires $T(z)T(w) = T(w)T(z)$ as an operator identity, hence the singular part must be symmetric under $z \leftrightarrow w$. Under this exchange $(z-w) \to -(z-w)$, so $(z-w)^{-4}$ is even (allowed for a constant coefficient), the pair $2T(w)/(z-w)^2 + \partial T(w)/(z-w)$ is symmetric once $T(w)$ is Taylor-expanded about z (we verify this directly in the worked example below), while an isolated $(z-w)^{-3}$ term would be odd and cannot be matched; it is therefore forbidden. Second, the two middle terms are non-negotiable: they are exactly (48.54) with $h = 2$, expressing that T *does* transform with weight 2 under translations and dilatations — the parts of the symmetry under which it is well-behaved. Third, the $c/2$ term is genuinely new information that no symmetry argument fixes; it is an anomaly, measured by the theory, and we now compute it for the canonical example.

Physical Insight

The central charge is the conformal field theory's irreducible quantum number. Heuristically it counts the "number of degrees of freedom": a free real boson contributes $c = 1$, a free Majorana fermion $c = \frac{1}{2}$, and in a precise sense (Zamolodchikov's c-theorem, the entanglement-entropy coefficient of Chapter 55, the Casimir energy on a cylinder) c controls the universal response of the theory to curvature and to finite size. It is the single number that distinguishes one conformal universality class from another at the level of the symmetry algebra, before any operator content is specified.

48.2.5 The finite transformation of T: the Schwarzian

Because T is not primary, it does not transform by the simple covariant law (48.49). We now compute its actual transformation, first infinitesimally from the T–T product, then integrate to the finite law. The result, the appearance of the Schwarzian derivative, is one of the most consequential formulas in the subject: it is what makes the central charge physically detectable, for example as the Casimir energy on a cylinder.

Apply the Ward identity (48.52) to $O = T$ using the full product (48.56). The infinitesimal variation is minus the residue of $\epsilon(z)T(z)T(w)$ at $z = w$. Expand $\epsilon(z) = \sum_{k\geq 0} \frac{1}{k!}\partial^k\epsilon(w)\,(z-w)^k$ and collect the coefficient of $(z-w)^{-1}$ from each term of (48.56):

$$\frac{c/2}{(z-w)^4}\,\epsilon(z): \qquad \frac{c}{2}\cdot\frac{1}{3!}\partial^3\epsilon(w) = \frac{c}{12}\,\partial^3\epsilon(w), \tag{48.57}$$

$$\frac{2T(w)}{(z-w)^2}\,\epsilon(z): \qquad 2T(w)\,\partial\epsilon(w), \tag{48.58}$$

$$\frac{\partial T(w)}{z-w}\,\epsilon(z): \qquad \epsilon(w)\,\partial T(w). \tag{48.59}$$

Summing and negating,

$$\delta_\epsilon T(w) = -\Big[\,\epsilon(w)\,\partial T(w) + 2\,\partial\epsilon(w)\,T(w) + \frac{c}{12}\,\partial^3\epsilon(w)\Big]. \tag{48.60}$$

The first two terms are exactly the weight-2 primary transformation (48.50) (with $h = 2$, $\bar h = 0$); the third, proportional to c and to the *third* derivative of the parameter, is the anomaly. It is inhomogeneous — a c-number shift, not a rescaling of T — which is why c is called a central extension: it commutes with everything.

We integrate (48.60) to a finite holomorphic map $z \mapsto w(z)$ (we relabel the target coordinate w for the moment). The natural ansatz is the covariant weight-2 law corrected by an additive anomalous piece $A(z)$ that must be built from the map and must vanish for the infinitesimal case to reproduce the $\partial^3\epsilon$ term:

$$T'(w) = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^{-2}\big[\,T(z) - \tfrac{c}{12}\,A(z)\,\big]. \tag{48.61}$$

We determine $A(z)$ by demanding two things: that it reduce to $\partial^3\epsilon$ infinitesimally, and that the composition of two transformations be consistent (the transformations form a group, so applying $z \to w$ then $w \to u$ must equal applying $z \to u$ directly). Both requirements are met by the *Schwarzian derivative*

$$\{w, z\} \equiv \frac{w'''}{w'} - \frac{3}{2}\Big(\frac{w''}{w'}\Big)^2, \qquad w' \equiv \frac{\mathrm{d}w}{\mathrm{d}z},\ \text{etc.}, \tag{48.62}$$

giving the finite transformation law of the stress tensor:

$$\boxed{\,T'(w) = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^{-2}\Big[\,T(z) - \frac{c}{12}\,\{w, z\}\,\Big]\,} \tag{48.63}$$

We verify both defining properties.

Infinitesimal check. Put $w = z+\epsilon(z)$, so $w' = 1+\partial\epsilon$, $w'' = \partial^2\epsilon$, $w''' = \partial^3\epsilon$. To first order in ϵ the squared term $(w''/w')^2 = (\partial^2\epsilon)^2(1+\cdots)$ is second order and

drops, while $w'''/w' = \partial^3\epsilon\,(1-\partial\epsilon+\cdots) = \partial^3\epsilon + O(\epsilon^2)$, so $\{w,z\} = \partial^3\epsilon(z) + O(\epsilon^2)$. The prefactor is $(1+\partial\epsilon)^{-2} = 1 - 2\partial\epsilon + O(\epsilon^2)$. Hence

$$T'(w) = (1-2\partial\epsilon)\,[T(z) - \tfrac{c}{12}\partial^3\epsilon] + O(\epsilon^2) = T(z) - 2\partial\epsilon\,T(z) - \tfrac{c}{12}\partial^3\epsilon + O(\epsilon^2). \tag{48.64}$$

The active variation at a fixed point is $\delta_\epsilon T(z) \equiv T'(z) - T(z)$. Since $T'(w) = T'(z+\epsilon) = T'(z) + \epsilon\,\partial T(z) + O(\epsilon^2)$, we move the shift to the left: $T'(z) = T'(w) - \epsilon\partial T(z) + O(\epsilon^2)$, and so

$$\delta_\epsilon T(z) = -\epsilon\,\partial T - 2\partial\epsilon\,T - \tfrac{c}{12}\partial^3\epsilon, \tag{48.65}$$

which is precisely (48.60). The Schwarzian is the correct integrated anomaly.

Composition (cocycle) check. For the finite law to be consistent under composition $z \xrightarrow{w} w \xrightarrow{u} u$, the Schwarzian must satisfy the chain rule

$$\{u,z\} = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^2\{u,w\} + \{w,z\}. \tag{48.66}$$

We verify this directly. Write $u'_z = u'_w\,w'_z$ (chain rule), and differentiate repeatedly. With shorthand $w' = w'_z$, $u'_w = \mathrm{d}u/\mathrm{d}w$,

$$\frac{\mathrm{d}u}{\mathrm{d}z} = u'_w\,w', \tag{48.67}$$

$$\frac{\mathrm{d}^2u}{\mathrm{d}z^2} = u''_w\,(w')^2 + u'_w\,w'', \tag{48.68}$$

$$\frac{\mathrm{d}^3u}{\mathrm{d}z^3} = u'''_w\,(w')^3 + 3u''_w\,w'\,w'' + u'_w\,w'''. \tag{48.69}$$

Form $\{u,z\} = \dfrac{\mathrm{d}^3u/\mathrm{d}z^3}{\mathrm{d}u/\mathrm{d}z} - \dfrac{3}{2}\Big(\dfrac{\mathrm{d}^2u/\mathrm{d}z^2}{\mathrm{d}u/\mathrm{d}z}\Big)^2$. The first ratio is

$$\frac{u'''_w(w')^3 + 3u''_w w' w'' + u'_w w'''}{u'_w w'} = (w')^2\frac{u'''_w}{u'_w} + 3\frac{u''_w}{u'_w}w'' + \frac{w'''}{w'}, \tag{48.70}$$

and the second, expanding the square,

$$\frac{3}{2}\Big(\frac{u''_w(w')^2 + u'_w w''}{u'_w w'}\Big)^2 = \frac{3}{2}\Big[(w')^2\Big(\frac{u''_w}{u'_w}\Big)^2 + 2\frac{u''_w}{u'_w}w'' + \Big(\frac{w''}{w'}\Big)^2\Big]. \tag{48.71}$$

Subtracting, the cross term $3(u''_w/u'_w)w''$ cancels against $\frac{3}{2}\cdot 2(u''_w/u'_w)w''$, and the survivors regroup as

$$\{u,z\} = (w')^2\Big[\frac{u'''_w}{u'_w} - \frac{3}{2}\Big(\frac{u''_w}{u'_w}\Big)^2\Big] + \Big[\frac{w'''}{w'} - \frac{3}{2}\Big(\frac{w''}{w'}\Big)^2\Big] = (w')^2\{u,w\} + \{w,z\}, \tag{48.72}$$

which is (48.66). This is exactly the inhomogeneous term required so that (48.63) composes correctly: the prefactors $(\mathrm{d}w/\mathrm{d}z)^{-2}$ multiply as a weight-2 tensor, and

the Schwarzian terms add up with precisely the cocycle weight that compensates. The anomaly is a one-cocycle on the group of holomorphic maps.

Möbius invariance. The Schwarzian vanishes on the global conformal group. For a Möbius map $w = (az + b)/(cz + d)$ with $ad - bc = 1$, one computes $\{w, z\} = 0$. The quickest route is the cocycle: every Möbius map is a composition of a translation $z \mapsto z + b$, a scaling $z \mapsto az$, and the inversion $z \mapsto 1/z$. The first two are affine, $w'' = 0$, so $\{w, z\} = 0$ trivially. For the inversion $w = 1/z$: $w' = -z^{-2}$, $w'' = 2z^{-3}$, $w''' = -6z^{-4}$, whence

$$\{1/z, z\} = \frac{-6z^{-4}}{-z^{-2}} - \frac{3}{2}\Big(\frac{2z^{-3}}{-z^{-2}}\Big)^2 = \frac{6}{z^2} - \frac{3}{2}\cdot\frac{4}{z^2} = 0. \tag{48.73}$$

By the cocycle (48.66), a composition of maps each with vanishing Schwarzian, multiplied by smooth nonzero prefactors, again has vanishing Schwarzian; hence $\{w, z\} = 0$ for every Möbius transformation. The physical consequence is decisive: under the *global* conformal group T transforms as an honest weight-2 primary, with no anomaly. The central charge is invisible to the finite-dimensional Möbius symmetry $\mathfrak{so}(3, 1)$ and surfaces only in the genuinely local, infinite-dimensional part of the symmetry — the part that distinguishes two dimensions from all higher dimensions. This is the same statement that the central extension of the Witt algebra to Virasoro affects only the modes L_n with $|n| \geq 2$, as we will see in Chapter 48.

Caution

The Schwarzian carries a definite weight: $\{w, z\}$ transforms with the factor $(\mathrm{d}w/\mathrm{d}z)^2$ in (48.66), not as a scalar. Consequently the shift $-\frac{c}{12}\{w, z\}$ in (48.63) is added to $T(z)$ *inside* the bracket, before the weight-2 prefactor acts. Reading the law with the Schwarzian outside the bracket, or as a scalar, is a common and physically wrong simplification — it would spoil the composition law just verified.

48.2.6 Worked example: the free boson has $c = 1$

We now compute the central charge of the simplest interacting-free conformal field theory, the massless free boson, directly from the definition (48.8), by Wick-contracting the T–T product. This is the foundational benchmark $c = 1$; every later construction (the Coulomb gas, the minimal models, the Ising fermion at $c = \frac{1}{2}$) is calibrated against it.

Worked Example 48.9 (Central charge of the free boson)**.** The massless free scalar in two Euclidean dimensions is governed, after the Wick rotation of Theorem 5.31, by the action

$$S = \frac{1}{8\pi}\int \mathrm{d}^2x\, \partial_\mu X\, \partial^\mu X = \frac{1}{2\pi}\int \mathrm{d}^2x\, \partial X\, \bar\partial X, \tag{48.74}$$

the second form using (48.36) and $\partial_\mu X \partial^\mu X = 4\,\partial X \bar\partial X$ in the complex frame; the normalization $1/8\pi$ is chosen so that the propagator has unit-normalized logarithm. The equation of motion is $\partial\bar\partial X = 0$, whose Green's function follows from (48.46): the two-point function (cf. Eq. (47.120) for the general primary form) is

$$\langle X(z,\bar z)\, X(w,\bar w)\rangle = -\ln|z-w|^2 = -\ln(z-w) - \ln(\bar z - \bar w). \tag{48.75}$$

The field X itself is not a good conformal primary (it appears only through its derivatives and exponentials), but its holomorphic derivative $\partial X(z)$ is. Differentiating (48.75) once in each argument gives the basic contraction we will use repeatedly,

$$\langle \partial X(z)\, \partial X(w)\rangle = \partial_z \partial_w\big[-\ln(z-w)\big] = \partial_z\Big[\frac{1}{z-w}\Big] = -\frac{1}{(z-w)^2}, \tag{48.76}$$

where $\partial_w[-\ln(z-w)] = 1/(z-w)$ was used. Write $a(z) \equiv \partial X(z)$ for brevity; (48.76) reads $\langle a(z)a(w)\rangle = -(z-w)^{-2}$.

The stress tensor. The classical stress tensor of (48.74) has holomorphic component $T_{zz} \propto (\partial X)^2$; in the normalization (48.41) the quantum stress tensor is the normal-ordered bilinear

$$T(z) = -\tfrac{1}{2}\,{:}\partial X(z)\,\partial X(z){:} = -\tfrac{1}{2}\,{:}a(z)a(z){:}, \tag{48.77}$$

the normal ordering $:\,:$ subtracting the (divergent) self-contraction so that $\langle T\rangle = 0$. We first confirm that ∂X is primary with $h = 1$, then compute c.

∂X is primary, $h = 1$. Compute $T(z)\,a(w)$ by Wick's theorem, contracting one $a(z)$ from T with the external $a(w)$ (there are two such contractions, one for each factor in $:a(z)a(z):$):

$$T(z)\,a(w) = -\tfrac{1}{2}\,{:}a(z)a(z){:}\,a(w) \sim -\tfrac{1}{2}\cdot 2\,\langle a(z)a(w)\rangle\, a(z) = -\Big(-\frac{1}{(z-w)^2}\Big)a(z) = \frac{a(z)}{(z-w)^2}. \tag{48.78}$$

Taylor-expanding the surviving operator about w, $a(z) = a(w) + (z-w)\partial a(w) + \cdots$, gives

$$T(z)\,\partial X(w) \sim \frac{\partial X(w)}{(z-w)^2} + \frac{\partial^2 X(w)}{z-w}, \tag{48.79}$$

which is exactly the primary form (48.54) with $h = 1$. The free boson's current ∂X is a weight-$(1,0)$ primary, as anticipated.

The T–T product and c. Now Wick-contract the full product, $T(z)T(w) = \frac{1}{4}\,{:}a(z)a(z){:}\,{:}a(w)a(w){:}$. By Wick's theorem the result is the sum over all contractions between the two normal-ordered groups (contractions *within* a group are removed by the normal ordering). There are two classes.

Double contractions. Contract both $a(z)$'s with both $a(w)$'s. There are 2 ways to pair them, each contributing $\langle a(z)a(w)\rangle^2$:

$$\tfrac{1}{4}\cdot 2\,\langle a(z)a(w)\rangle^2 = \tfrac{1}{2}\Big(-\frac{1}{(z-w)^2}\Big)^2 = \frac{1/2}{(z-w)^4}. \tag{48.80}$$

This is a pure c-number (the identity operator), with exactly the $(z-w)^{-4}$ singularity of (48.56). Matching $\frac{c/2}{(z-w)^4} = \frac{1/2}{(z-w)^4}$ gives

$$\boxed{c = 1} \tag{48.81}$$

Single contractions. Contract one $a(z)$ with one $a(w)$, leaving the other two normal-ordered. There are $2\times 2 = 4$ such pairings, each contributing $\langle a(z)a(w)\rangle\,{:}a(z)a(w){:}$:

$$\tfrac{1}{4}\cdot 4\,\langle a(z)a(w)\rangle\,{:}a(z)a(w){:} = -\frac{1}{(z-w)^2}\,{:}a(z)a(w){:}. \tag{48.82}$$

Taylor-expand the remaining bilinear about w: ${:}a(z)a(w){:} = {:}a(w)a(w){:} + (z-w)\,{:}\partial a(w)\,a(w){:} + \cdots$. By (48.77), ${:}aa{:} = -2T$, and differentiating, $\partial T = -\frac{1}{2}\partial\,{:}aa{:} = -\,{:}\partial a\,a{:}$, so ${:}\partial a\,a{:} = -\partial T$. Therefore

$$-\frac{1}{(z-w)^2}\Big[-2T(w) - (z-w)\partial T(w) + \cdots\Big] = \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w} + \cdots, \tag{48.83}$$

the expected weight-2 terms of (48.56). Assembling the double and single contractions,

$$T(z)\,T(w) \sim \frac{1/2}{(z-w)^4} + \frac{2T(w)}{(z-w)^2} + \frac{\partial T(w)}{z-w}, \tag{48.84}$$

which is (48.56) with $c = 1$, with every coefficient — the central term, the weight-2 double pole, and the descendant single pole — reproduced from the single contraction datum (48.76). The free real boson is the $c = 1$ conformal field theory. (The same computation in the antiholomorphic sector gives $\bar{c} = 1$, and N independent free bosons give $c = N$, the additivity that justifies reading c as a count of degrees of freedom.)

Historical Note

That the stress tensor's self-product carries an anomalous c-number was isolated, in the language of current algebra, well before the systematic conformal bootstrap; the modern packaging — $T(z)$, its operator products, and the central charge as the unique extension — is due to Belavin, Polyakov, and Zamolodchikov in 1984, the paper that founded two-dimensional conformal field theory as a self-contained subject. The Schwarzian derivative itself is far older, appearing in Schwarz's nineteenth-century study of

conformal maps and hypergeometric monodromy; its reappearance here as the obstruction to T being a tensor is one of the cleaner instances of a classical geometric invariant resurfacing as a quantum anomaly.

We have promoted the kinematic symmetry of the previous section to a dynamical apparatus. The conformal Ward identity (48.48) computes any conformal variation as a contour integral of T; the T–O product (48.54) defines and detects primaries; the T–T product (48.56) introduces the central charge c; and the finite law (48.63) packages the anomaly as the Schwarzian, invisible to the global Möbius group and confined to the local symmetry. These two operator products, $T(z)T(w)$ with its anomalous $(c/2)(z-w)^{-4}$ term and $T(z)O(w)$, are exactly the data the next section converts into an operator algebra: mode-expanding $T(z)=\sum_n z^{-n-2}L_n$ on a quantization circle and translating the contour manipulations above into commutators will produce the Virasoro algebra, with c as its central extension and the Ward identity as the recursion that fixes every correlator carrying stress-tensor insertions.

48.3 Radial Quantization and the Virasoro Algebra

The previous section (Section 48.2) handed us two pieces of data and one geometric arena. The data are the operator product expansions

$$T(z)\,T(w) \;\sim\; \frac{c/2}{(z-w)^4} \;+\; \frac{2\,T(w)}{(z-w)^2} \;+\; \frac{\partial T(w)}{z-w}, \tag{48.85}$$

recalling Eq. (48.56), with the central charge c defined in Theorem 48.8 as the coefficient of the quartic pole, and the conformal Ward identity Eq. (48.48), whose local content is the $T\mathcal{O}$ OPE for a primary $\mathcal{O}$ of weight $(h,\bar h)$,

$$T(z)\,\mathcal{O}(w,\bar w) \;\sim\; \frac{h}{(z-w)^2}\,\mathcal{O}(w,\bar w) \;+\; \frac{\partial_w \mathcal{O}(w,\bar w)}{z-w}. \tag{48.86}$$

The arena is radial quantization (Section 47.5): on the Euclidean plane we quantize along circles of constant radius about a chosen origin, Euclidean "time" is $\tau=\ln r$ with $r=|z|$, the dilatation generator D is the Hamiltonian, and the state–operator map (Theorem 47.42) identifies states with operators inserted at the origin. So far this is kinematics. The purpose of the present section is to convert that kinematics into an *operator algebra*. We will mode-expand $T(z)$ on the quantization circles, read off generators L_n, and show that the contour calculus of complex analysis turns the single OPE Eq. (48.85) into a closed Lie-algebra bracket — the Virasoro algebra — with every structure constant, including the anomalous central term, fixed by the residues of Eq. (48.85). By the end we will have the weight operator L_0, raising and lowering operators $L_{n<0}$ and $L_{n>0}$, and the adjoint relation that makes "unitary" a meaningful word — precisely the input the next section needs to build representations.

48.3.1 Mode expansion of the stress tensor

Because $T(z)$ is holomorphic away from operator insertions, on any annulus free of insertions it has a Laurent expansion. The exponents are fixed by the conformal weight: T is a holomorphic field of weight $(2,0)$, so under $z \mapsto \lambda z$ it scales as $T \mapsto \lambda^{-2} T$, and the natural basis of modes carries one unit of weight each. We therefore write the expansion with a $-n-2$ shift so that the mode L_n lowers the weight by n.

Definition 48.10 (Virasoro modes)**.** The *Virasoro modes* L_n, $n \in \mathbb{Z}$, are the Laurent coefficients of the holomorphic stress tensor,

$$T(z) \;=\; \sum_{n\in\mathbb{Z}} \frac{L_n}{z^{\,n+2}}, \qquad L_n \;=\; \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{\,n+1}\, T(z), \tag{48.87}$$

where the contour encircles the origin once counterclockwise. The antiholomorphic modes are defined identically from $\bar{T}(\bar{z}) = \sum_n \bar{L}_n\, \bar{z}^{-n-2}$, with $\bar{L}_n = \oint (\mathrm{d}\bar{z}/2\pi i)\, \bar{z}^{n+1} \bar{T}(\bar{z})$ taken with the conjugate orientation.

The inversion formula in Eq. (48.87) is the residue theorem (Theorem 5.23): inserting the series and using $\oint_0 (\mathrm{d}z/2\pi i)\, z^k = \delta_{k,-1}$ picks out the coefficient of z^{-1},

$$\oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{n+1} \sum_m \frac{L_m}{z^{m+2}} \;=\; \sum_m L_m \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{\,n-m-1} \;=\; \sum_m L_m\, \delta_{n,m} \;=\; L_n, \tag{48.88}$$

so the two halves of Eq. (48.87) are consistent. The contour may be freely deformed across any annulus containing no insertions, because T is holomorphic there; this freedom is the entire computational engine of what follows.

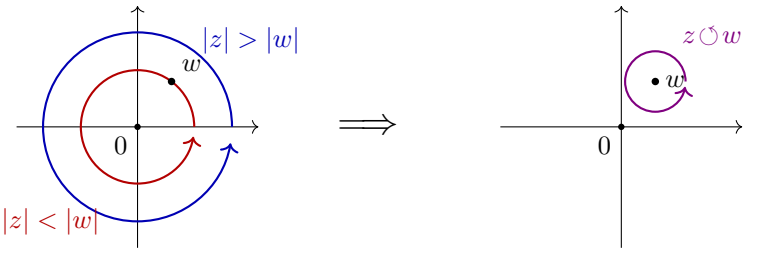
The modes act on the Hilbert space built by radial quantization. The vacuum $|0\rangle$ is the state created by inserting the identity at the origin. For $T(z)\,|0\rangle$ to be a well-defined (nonsingular) state as $z \to 0$ — equivalently, for the expansion $\sum_n L_n z^{-n-2}\,|0\rangle$ to contain no terms blowing up at the origin — every mode multiplying a negative power of z must annihilate the vacuum:

$$L_n\,|0\rangle = 0 \quad \text{for } n \geq -1. \tag{48.89}$$

Indeed the term $L_n z^{-n-2}\,|0\rangle$ is singular as $z \to 0$ precisely when $-n-2 < 0$, i.e. $n > -2$, i.e. $n \geq -1$; regularity of the state forces those modes to kill $|0\rangle$. The modes $L_{-2}, L_{-3}, \ldots$ instead create states: $L_{-2}\,|0\rangle = \lim_{z\to 0} T(z)\,|0\rangle$ is the state dual to the stress tensor itself. We will see in Eq. (48.101) that Eq. (48.89) is exactly the statement that $|0\rangle$ is invariant under the global conformal group.

48.3.2 Radial ordering and the commutator of modes

A product of two mode operators is a product of two contour integrals, and the order of the operators is encoded geometrically. In radial quantization the



(a) $L_m L_n - L_n L_m$ (b) $\oint_w$, then $\oint_0$

Figure 48.2: The contour-subtraction identity Eq. (48.91). (a) The mode commutator is the difference of two radial orderings: z outside w (blue) minus z inside w (red), at fixed w. (b) The difference collapses to a small z-contour encircling w, whose value the OPE Eq. (48.85) supplies through the residue theorem; the remaining w-integral runs around the origin.

time-ordering inherited from Section 47.5 is *radial ordering*: in a correlator the operator on a larger circle stands to the left. Thus the operator product $L_m L_n$ corresponds to the z-contour (carrying L_m) lying *outside* the w-contour (carrying L_n), $|z| > |w|$, while $L_n L_m$ corresponds to $|z| < |w|$:

$$L_m L_n \;=\; \oint_{|z|>|w|} \frac{\mathrm{d}z}{2\pi i} \oint_0 \frac{\mathrm{d}w}{2\pi i}\, z^{m+1} w^{n+1}\, R\{T(z)T(w)\}, \tag{48.90}$$

and likewise $L_n L_m$ with $|z| < |w|$, where $R\{\cdots\}$ is the radially ordered product whose short-distance behaviour is the OPE Eq. (48.85). The commutator is the *difference* of the two orderings. Fixing w and subtracting, the two z-contours — one just outside $|w|$, one just inside — combine into a single small contour that encircles the point $z = w$ and nothing else:

$$[L_m, L_n] \;=\; \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{n+1} \oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{m+1}\, T(z)\, T(w). \tag{48.91}$$

This is the basic move of the contour calculus: $\left(\oint_{|z|>|w|} - \oint_{|z|<|w|}\right)$ at fixed w equals $\oint_{z \circlearrowleft w}$, a contour deformation valid because the integrand is holomorphic in z except at $z = w$. The inner integral is now governed entirely by the short-distance OPE, and the residue theorem evaluates it.

48.3.3 The Virasoro algebra from the TT OPE

Theorem 48.11 (Virasoro algebra from the TT OPE)**.** *The modes Eq. (48.87) of a stress tensor obeying Eq. (48.85) satisfy*

$$[L_m, L_n] \;=\; (m-n)\, L_{m+n} \;+\; \frac{c}{12}\, m\,(m^2-1)\, \delta_{m+n,0}, \tag{48.92}$$

for all $m, n \in \mathbb{Z}$, together with the identical antiholomorphic copy $\left[\bar{L}_m, \bar{L}_n\right] = (m-n)\bar{L}_{m+n} + \frac{\bar{c}}{12} m(m^2-1)\delta_{m+n,0}$ and the mutual commutativity $\left[L_m, \bar{L}_n\right] = 0$. (For a parity-invariant theory $\bar{c} = c$.)

Proof. We evaluate the inner integral of Eq. (48.91) by substituting the OPE Eq. (48.85) for $T(z)T(w)$ and applying the residue theorem (Theorem 5.23). Recall that the residue of $f(z)/(z-w)^k$ at $z = w$ is $f^{(k-1)}(w)/(k-1)!$, the $(k-1)$th Taylor coefficient of the numerator. With $f(z) = z^{m+1}$, the three singular terms of the OPE contribute, in turn,

$$\frac{c}{2} \oint_w \frac{\mathrm{d}z}{2\pi i} \frac{z^{m+1}}{(z-w)^4} = \frac{c}{2} \cdot \frac{1}{3!} \frac{\mathrm{d}^3}{\mathrm{d}z^3} z^{m+1}\Big|_{z=w} = \frac{c}{2} \cdot \frac{(m+1)m(m-1)}{6} w^{m-2} = \frac{c}{12} (m \tag{48.93}$$

$$2\,T(w) \oint_w \frac{\mathrm{d}z}{2\pi i} \frac{z^{m+1}}{(z-w)^2} = 2\,T(w) \cdot \frac{\mathrm{d}}{\mathrm{d}z} z^{m+1}\Big|_{z=w} = 2\,(m+1)\,w^m\,T(w), \tag{48.94}$$

$$\partial T(w) \oint_w \frac{\mathrm{d}z}{2\pi i} \frac{z^{m+1}}{z-w} = \partial T(w) \cdot z^{m+1}\big|_{z=w} = w^{m+1}\,\partial T(w), \tag{48.95}$$

where the regular part of the OPE has no pole at $z = w$ and integrates to zero. Summing,

$$\oint_w \frac{\mathrm{d}z}{2\pi i}\, z^{m+1}\, T(z)T(w) = \frac{c}{12}\,(m+1)m(m-1)\,w^{m-2} + 2(m+1)\,w^m\,T(w) + w^{m+1}\,\partial T(w). \tag{48.96}$$

We now perform the outer w-integral of Eq. (48.91), multiplying Eq. (48.96) by w^{n+1} and integrating around the origin. Take the three terms separately.

Central term. The first term contains no operator, only the c-number w^{m-2}:

$$\frac{c}{12}\,(m+1)m(m-1) \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{n+1}\, w^{m-2} = \frac{c}{12}\,(m+1)m(m-1) \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{m+n-1}. \tag{48.97}$$

The remaining integral is $\delta_{m+n-1,-1} = \delta_{m+n,0}$ by the residue theorem. When $m+n=0$ we may write $n=-m$ and use $(m+1)m(m-1) = m(m^2-1)$, so this term is $\frac{c}{12}\,m(m^2-1)\,\delta_{m+n,0}$. This is the anomalous central term, and it arises *solely* from the quartic pole of the TT OPE: the third derivative in Eq. (48.93) is what produces the cubic $m(m^2-1)$, and the central charge enters exactly as it appeared in Theorem 48.8.

Operator terms. The second term of Eq. (48.96) gives, directly from Eq. (48.87),

$$2(m+1) \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{n+1}\, w^m\, T(w) = 2(m+1) \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{(m+n)+1}\, T(w) = 2(m+1)\, L_{m+n}. \tag{48.98}$$

The third term carries ∂T; we integrate by parts on the closed contour, where the total-derivative boundary term vanishes:

$$\oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{n+1}\, w^{m+1}\, \partial T(w) \;=\; \oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{m+n+2}\, \partial T(w) \;=\; -\oint_0 \frac{\mathrm{d}w}{2\pi i}\, \big(\partial_w\, w^{m+n+2}\big)\, T(w)$$
$$=\; -(m+n+2)\oint_0 \frac{\mathrm{d}w}{2\pi i}\, w^{(m+n)+1}\, T(w) \;=\; -(m+n+2)\, L_{m+n}. \tag{48.99}$$

Adding Eqs. (48.98) and (48.99), the operator part is

$$\big[\,2(m+1) - (m+n+2)\,\big]\, L_{m+n} \;=\; (m-n)\, L_{m+n}. \tag{48.100}$$

Combining the central and operator contributions reproduces Eq. (48.92). The antiholomorphic copy follows identically from the $\bar{T}\bar{T}$ OPE, and $\big[L_m, \bar{L}_n\big] = 0$ because $T(z)\bar{T}(\bar{w})$ has no singular OPE — the holomorphic and antiholomorphic sectors are mutually local, so the inner contour of z around $\bar{w}$ encloses no singularity and the commutator vanishes. □

Three features of Eq. (48.92) deserve emphasis. First, dropping the central term leaves $[L_m, L_n] = (m-n)L_{m+n}$, the *Witt algebra* of holomorphic vector fields $\ell_n = -z^{n+1}\partial_z$ on the punctured plane: indeed $[\ell_m, \ell_n] = (m-n)\ell_{m+n}$ by direct computation, and the Witt algebra is the classical (centreless) algebra of local conformal transformations promised in the chapter chapter introduction. Second, the Virasoro algebra is the *unique* central extension of Witt by a one-dimensional centre, and Theorem 48.11 shows the quantum theory realizes exactly this extension with the coefficient fixed by the OPE data. Third, the central term is not removable by redefining the generators except in a trivial way: one may shift $L_0 \to L_0 + \text{const}$ to absorb a piece linear in m, but the cubic m^3 piece is a genuine cohomological obstruction. We make this precise next.

48.3.4 The global subalgebra and the vanishing of the central term

The central term in Eq. (48.92) is proportional to $m(m^2-1) = (m-1)m(m+1)$, which vanishes identically for $m \in \{-1, 0, 1\}$. Therefore the three modes L_{-1}, L_0, L_1 close among themselves with *no* anomaly:

$$[L_0, L_{\pm 1}] = \mp\, L_{\pm 1}, \qquad [L_1, L_{-1}] = 2\, L_0. \tag{48.101}$$

These follow from Eq. (48.92): $[L_0, L_1] = (0-1)L_1 = -L_1$, $[L_0, L_{-1}] = (0+1)L_{-1} = +L_{-1}$, and $[L_1, L_{-1}] = (1-(-1))L_0 = 2L_0$ with $\frac{c}{12}\cdot 1\cdot(1-1)\cdot\delta_{0,0} = 0$. The brackets Eq. (48.101) are precisely those of $\mathfrak{sl}(2,\mathbb{C})$ (equivalently $\mathfrak{so}(3,1)$), with L_0 the Cartan element and $L_{\pm 1}$ the raising/lowering pair. This is a *Lie*

subalgebra in the sense of Theorem 2.39: closed under the bracket and obeying the Jacobi identity.

Together with the antiholomorphic triple $\{\bar{L}_{-1}, \bar{L}_0, \bar{L}_1\}$, these generate $\mathfrak{sl}(2,\mathbb{C})\oplus \mathfrak{sl}(2,\mathbb{C})$, the algebra of the *global* conformal group of the Riemann sphere — the Möbius transformations $z \mapsto (az+b)/(cz+d)$ with $ad-bc=1$. This is exactly the $d=2$ instance of the finite-dimensional conformal algebra $\mathfrak{so}(d+1,1)$ derived in general dimension in Chapter 47: for $d=2$, $\mathfrak{so}(3,1) \cong \mathfrak{sl}(2,\mathbb{C})$, and the dimension count $\frac{1}{2}(d+1)(d+2) = 6$ matches the six real generators $\{L_{0,\pm1}, \bar{L}_{0,\pm1}\}$. The dictionary with the Cartesian generators of Chapter 47 is

$$L_{-1} = \text{(holomorphic) translation}, \quad L_0+\bar{L}_0 = \text{dilatation}, \quad L_0-\bar{L}_0 = \text{rotation}, \quad L_1 = \text{speci} \tag{48.102}$$

read off from the vector fields $\ell_{-1} = -\partial_z$, $\ell_0 = -z\partial_z$, $\ell_1 = -z^2\partial_z$. The miracle of two dimensions, foretold in the chapter introduction, is now visible at the level of the algebra: the global $\mathfrak{so}(3,1)$ is just the $|n| \le 1$ corner of an *infinite* tower $\{L_n\}_{n\in\mathbb{Z}}$, and it is this infinite enhancement, anomalous for $|n| \ge 2$, that ultimately solves the theory.

The vanishing of the anomaly on $\{L_{-1}, L_0, L_1\}$ also re-derives Eq. (48.89) from a symmetry principle. A state invariant under the global conformal group must be annihilated by L_{-1}, L_0, L_1 (and their conjugates); but $|0\rangle$ is the unique $\mathfrak{sl}(2,\mathbb{C}) \oplus \mathfrak{sl}(2,\mathbb{C})$-invariant state, so $L_{-1}|0\rangle = L_0|0\rangle = L_1|0\rangle = 0$. Acting repeatedly with the higher modes $L_{-2}, L_{-3}, \dots$ on $|0\rangle$ then generates the rest of the vacuum module, while the condition $L_n|0\rangle = 0$ for all $n \ge -1$ is forced by regularity of $T(z)|0\rangle$ at the origin, as in Eq. (48.89).

48.3.5 L_0 as the Hamiltonian and primary states

Under the state–operator map (Theorem 47.42), every state of the radial-quantization Hilbert space is $\lim_{z,\bar{z}\to 0} \mathcal{O}(z,\bar{z})|0\rangle$ for some local operator $\mathcal{O}$, and the dilatation generator D is the Hamiltonian conjugate to radial time $\tau = \ln r$. In the mode language $D = L_0 + \bar{L}_0$: the combination that scales z and $\bar{z}$ together. The difference $L_0 - \bar{L}_0$ generates rotations and measures spin. We make this concrete by computing how the modes act on the state of a primary.

Definition 48.12 (Primary state)**.** Let $\mathcal{O}$ be a primary of weight $(h,\bar{h})$. The associated *primary state* is

$$\left|h,\bar{h}\right\rangle \equiv \lim_{z,\bar{z}\to 0} \mathcal{O}(z,\bar{z})|0\rangle\,. \tag{48.103}$$

A state $|h\rangle$ (suppressing the antiholomorphic label) is called a *(holomorphic) highest-weight* or *primary* state if

$$L_0|h\rangle = h|h\rangle\,, \qquad L_n|h\rangle = 0 \quad \text{for all } n>0, \tag{48.104}$$

with the analogous statements for $\bar{L}_0, \bar{L}_{n>0}$.

That Eq. (48.103) satisfies Eq. (48.104) is not a postulate; it follows from the $T\mathcal{O}$ OPE Eq. (48.86) by the same residue calculus. Act with the mode L_n on $|h\rangle$ using Eq. (48.87) with the contour shrunk to the operator at the origin:

$$L_n\,|h\rangle \;=\; \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{n+1}\, T(z)\,\mathcal{O}(0)\,|0\rangle \;=\; \oint_0 \frac{\mathrm{d}z}{2\pi i}\, z^{n+1}\left[\frac{h}{z^2}\,\mathcal{O}(0) + \frac{\partial\mathcal{O}(0)}{z} + \text{regular}\right]|0\rangle\,, \tag{48.105}$$

where we set $w = 0$ in Eq. (48.86) and used that the regular terms are $z \to 0$ finite. The residue theorem extracts the coefficient of z^{-1} in the integrand:

$$L_n\,|h\rangle \;=\; h\,\big[\,z^{n-1}\,\big]_{z^{-1}}\,\mathcal{O}(0)\,|0\rangle + \big[\,z^{n}\,\big]_{z^{-1}}\,\partial\mathcal{O}(0)\,|0\rangle + \big(\text{regular: no } z^{-1} \text{ for } n \geq 0\big), \tag{48.106}$$

where $\big[\,z^{k}\,\big]_{z^{-1}}$ denotes the residue, nonzero only when $k = -1$. The first bracket contributes when $n - 1 = -1$, i.e. $n = 0$, giving $L_0\,|h\rangle = h\,\mathcal{O}(0)\,|0\rangle = h\,|h\rangle$. The second contributes when $n = -1$, giving $L_{-1}\,|h\rangle = \partial\mathcal{O}(0)\,|0\rangle$. For $n > 0$ neither singular term and, since a primary's OPE with T has *no* poles deeper than $1/(z-w)^2$, none of the regular terms produce a z^{-1}; hence $L_n\,|h\rangle = 0$ for all $n > 0$. This proves Eq. (48.104) and identifies h with the holomorphic weight in Eq. (48.86). The state $L_{-1}\,|h\rangle = \partial\mathcal{O}(0)\,|0\rangle$ is the first *descendant*; the higher modes $L_{-2}, L_{-3}, \ldots$ create the rest of the tower, the subject of the next section.

The pieces now assemble into the physical picture promised at the start. From Eq. (48.92), $[L_0, L_n] = -n\,L_n$, so L_n shifts the L_0-eigenvalue by $-n$: the modes with $n > 0$ *lower* the weight (they are annihilation/lowering operators, killing primaries), while those with $n < 0$ *raise* it (creation operators). Thus L_0 is the holomorphic energy, $L_{n>0}$ lowering operators, $L_{n<0}$ raising operators — the ladder structure of a highest-weight representation, with the primary state sitting at the bottom.

48.3.6 Radial conjugation and the adjoint $L_n^\dagger = L_{-n}$

To speak of unitarity — of positive norms and a Hermitian Hamiltonian — we need the adjoint of L_n. In a Lorentzian theory adjoints come from Hermitian conjugation at fixed time; in radial quantization the analogue is conjugation across the quantization circle. Hermitian conjugation reverses Euclidean time $\tau = \ln r$, hence sends $r \to 1/r$ at fixed angle, i.e. it is implemented by the conformal map

$$z \;\longmapsto\; \frac{1}{\bar{z}}, \tag{48.107}$$

which exchanges the origin (where the in-vacuum lives) with the point at infinity (where the out-vacuum lives), keeping the unit circle fixed pointwise. The adjoint of a holomorphic primary $\phi(z)$ of weight h is therefore not simply $\phi(\bar{z})^*$; it must include the conformal Jacobian of Eq. (48.107) so that the conjugate of an in-state is a sensible out-state. With $z' = 1/\bar{z}$ and $\mathrm{d}z'/\mathrm{d}z\big|_{\ldots}$ supplying the

weight factor, the radial-quantization (BPZ) adjoint is

$$\big[\phi(z)\big]^{\dagger} \;=\; \bar{z}^{-2h}\,\phi\!\left(\frac{1}{\bar{z}}\right). \tag{48.108}$$

The factor $\bar{z}^{-2h}$ is exactly $\left|\mathrm{d}z'/\mathrm{d}z\right|^{h}$ for the map Eq. (48.107) written for a weight-h field; it is what guarantees, for example, that the out-state $\langle h| = \lim_{z\to\infty} z^{2h}\,\langle 0|\,\mathcal{O}(z)$ is the genuine Hermitian conjugate of the in-state $|h\rangle = \lim_{z\to 0}\mathcal{O}(z)\,|0\rangle$, and hence that the two-point normalization $\langle h|h\rangle$ is finite and conjugation-symmetric.

Apply Eq. (48.108) to the stress tensor, a holomorphic field of weight $h = 2$. On one hand, conjugating the mode expansion Eq. (48.87) term by term (the coefficients z^{-n-2} are c-numbers, conjugated to $\bar{z}^{-n-2}$),

$$\big[T(z)\big]^{\dagger} \;=\; \sum_{n\in\mathbb{Z}} L_n^{\dagger}\,\bar{z}^{-n-2}. \tag{48.109}$$

On the other hand, Eq. (48.108) with $h = 2$ gives

$$\big[T(z)\big]^{\dagger} \;=\; \bar{z}^{-4}\,T\!\left(\frac{1}{\bar{z}}\right) \;=\; \bar{z}^{-4}\sum_{n} L_n\,\big(\bar{z}^{-1}\big)^{-n-2} \;=\; \bar{z}^{-4}\sum_{n} L_n\,\bar{z}^{\,n+2} \;=\; \sum_{n} L_n\,\bar{z}^{\,n-2}. \tag{48.110}$$

Relabel the summation index $n \to -n$ in Eq. (48.110), so that $\bar{z}^{n-2} \to \bar{z}^{-n-2}$ and $L_n \to L_{-n}$:

$$\big[T(z)\big]^{\dagger} \;=\; \sum_{n} L_{-n}\,\bar{z}^{-n-2}. \tag{48.111}$$

Matching the coefficient of each power $\bar{z}^{-n-2}$ between Eq. (48.109) and Eq. (48.111) yields the adjoint relation

$$\boxed{\,L_n^{\dagger} \;=\; L_{-n}\,}. \tag{48.112}$$

In particular $L_0^{\dagger} = L_0$: the holomorphic Hamiltonian is Hermitian, so weights h are real, as a unitary theory requires. Equally, $L_n^{\dagger} = L_{-n}$ means the raising operators L_{-n} are the adjoints of the lowering operators L_n — exactly the structure that lets us compute norms of descendant states by commuting lowering past raising operators with Eq. (48.92).

Worked Example 48.13 (The first descendant norm and the seed of unitarity)**.** We compute the norm of the level-one descendant $L_{-1}\,|h\rangle$ of a primary, the smallest nontrivial check that the algebra plus the adjoint relation produce physical content. Using Eq. (48.112), $\big(L_{-1}\,|h\rangle\,\big)^{\dagger} = \langle h|\,L_1$, so the norm is

$$\big\|L_{-1}\,|h\rangle\,\big\|^{2} \;=\; \langle h|\;L_1\,L_{-1}\;|h\rangle\,. \tag{48.113}$$

Move L_1 to the right using the Virasoro bracket Eq. (48.92) with $m = 1, n = -1$:

$$L_1L_{-1} \;=\; L_{-1}L_1+[L_1,L_{-1}] \;=\; L_{-1}L_1+2L_0+\frac{c}{12}\cdot 1\cdot(1-1)\,\delta_{0,0} \;=\; L_{-1}L_1+2L_0, \tag{48.114}$$

the central term vanishing because $m(m^2-1)=0$ at $m=1$. Now act on $|h\rangle$: by the primary condition Eq. (48.104), $L_1|h\rangle = 0$, while $L_0|h\rangle = h|h\rangle$. Hence

$$\|L_{-1}|h\rangle\|^2 \;=\; \langle h|\,(L_{-1}L_1 + 2L_0)\,|h\rangle \;=\; 2h\,\langle h|h\rangle\,. \tag{48.115}$$

For a unitary theory norms are nonnegative and $\langle h|h\rangle > 0$, so this forces $h \geq 0$: the weight of any state in a unitary 2D CFT is bounded below by zero, with $h=0$ only for the identity (whose descendant $L_{-1}|0\rangle = 0$ vanishes, consistent with Eq. (48.89)). This single computation already exhibits the mechanism of the next section: positivity of a Gram matrix element, built from the algebra and Eq. (48.112), turns into a constraint on the spectrum. The full systematic version — the Kac determinant and the unitarity bounds isolating the minimal-model values of c — is Theorem 48.18.

We now have the complete operator-algebraic backbone of two-dimensional conformal field theory: the Virasoro algebra Eq. (48.92) with its central term derived from the TT OPE, the weight operator L_0 (the radial Hamiltonian), the lowering operators $L_{n>0}$ that annihilate primaries, the raising operators $L_{n<0}$ that build descendants, and the adjoint relation $L_n^\dagger = L_{-n}$ that makes norms computable. The next section (Section 48.4) takes exactly this package as input: it organizes the Hilbert space into Verma modules generated by acting with the raising operators on primary states, computes the Gram matrix of descendant inner products, and extracts from its determinant the null states and the unitarity constraints that single out $c \geq 1$ and the discrete minimal-model series at $c < 1$.

48.4 Highest-Weight Representations, Null States, and Unitarity

The previous section turned the local conformal symmetry of a two-dimensional theory into an honest operator algebra. The chiral stress tensor was expanded into modes L_n obeying the Virasoro algebra

$$[L_m, L_n] = (m-n)\,L_{m+n} + \frac{c}{12}\,m(m^2-1)\,\delta_{m+n,0}, \tag{48.116}$$

which is Eq. (48.92); the state–operator map of Chapter 47 attached to every local operator a state in a Hilbert space; and the reality of the stress tensor was encoded in the adjointness relation $L_n^\dagger = L_{-n}$. A *primary state* $|h\rangle$ (Theorem 48.12) was singled out as the lowest-energy vector of an irreducible block of the symmetry: it diagonalizes the dilatation generator L_0 with eigenvalue h, the holomorphic conformal weight, and is annihilated by every lowering operator,

$$L_0|h\rangle = h|h\rangle\,, \qquad L_n|h\rangle = 0 \quad (n>0). \tag{48.117}$$

The antiholomorphic copy $\bar{L}_n$ acts identically on $\left|\bar{h}\right\rangle$ and commutes with everything chiral, so for the rest of this section we suppress it and study one chiral half; the full Hilbert space is a sum of tensor products of a holomorphic and an antiholomorphic representation, and every statement below has a mirror image.

What we have so far is a single algebra and a distinguished class of vectors it annihilates from below. What we do not yet have is the *spectrum*: which weights h and which central charges c actually occur in a sensible (in particular, a unitary) theory, and how the infinitely many states built by acting with raising operators on a primary organize themselves. This section supplies that representation theory. We build the *Verma module* — the universal representation generated freely from a primary — grade it by level, and equip it with the inner product that $L_n^\dagger = L_{-n}$ forces upon it. The matrix of inner products at each level is the *Gram matrix*; its determinant is the *Kac determinant*, a polynomial in h and c whose zeros locate the weights at which the representation degenerates. At those special weights a *null* (singular) vector appears: a descendant that is itself primary, has zero norm, and is orthogonal to everything. We prove that such a vector generates a sub-representation that must be quotiented away, leaving an irreducible *degenerate* representation with far fewer states. Finally we impose unitarity — positivity of the inner product — and extract the constraint that organizes the entire landscape of two-dimensional CFT: positivity is automatic for $c \geq 1,\ h \geq 0$, but for $0 < c < 1$ it survives only at the discrete Friedan–Qiu–Shenker values $c = 1 - 6/[m(m+1)]$, $m = 3, 4, 5, \ldots$, and there only for a finite list of weights. This is the data the next section turns into the minimal models.

Throughout, the rep-theoretic vocabulary — modules, highest-weight vectors, irreducibility by quotient — is the one built in Chapter 48's prerequisite Theorems 2.7 and 2.74; the Virasoro algebra simply replaces the finite-dimensional Lie algebras of Part I with an infinite-dimensional one, and the highest-weight construction goes through verbatim once "highest weight" is read as "annihilated by all $L_{n>0}$."

48.4.1 Verma modules and the level grading

A primary state is, by Eq. (48.117), killed by every L_n with $n > 0$. The only generators that can act nontrivially and produce new states are the raising operators L_{-n} with $n > 0$ (and L_0, which merely measures energy). Applying them in all possible ways generates the orbit of $|h\rangle$ under the algebra. The cleanest object to study first is the orbit with *no relations imposed beyond the algebra itself*: act freely, declare two states equal only when the commutation relations Eq. (48.116) force them to be. This is the Verma module.

Definition 48.14 (Verma module)**.** Let $c \in \mathbb{C}$ and $h \in \mathbb{C}$. The *Verma module* $V(c, h)$ is the representation of the Virasoro algebra generated by a single highest-weight vector $|h\rangle$ satisfying Eq. (48.117), freely under the raising operators.

Concretely, $V(c,h)$ is the complex vector space with basis

$$L_{-n_1}L_{-n_2}\cdots L_{-n_k}\,|h\rangle\,, \qquad n_1 \geq n_2 \geq \cdots \geq n_k \geq 1, \quad k \geq 0, \tag{48.118}$$

(the case $k = 0$ being $|h\rangle$ itself), with the Virasoro generators acting by Eq. (48.116) and by Eq. (48.117).

The ordering $n_1 \geq \cdots \geq n_k$ in Eq. (48.118) is a normal-ordering convention, not a restriction: any product of L_{-n}'s can be brought to this form using Eq. (48.116), since a commutator $[L_{-m}, L_{-n}] = (n-m)L_{-m-n}$ replaces an out-of-order pair by a lower-length monomial, and the recursion terminates. That the ordered monomials are moreover linearly *independent* — so that Eq. (48.118) is a genuine basis — is the Poincaré–Birkhoff–Witt theorem for the universal enveloping algebra of the Virasoro algebra; we take it as the defining freeness of the Verma module. (When h, c are such that a relation *is* forced, it does not appear inside $V(c,h)$ but as a statement that a certain basis vector is null and generates a submodule to be quotiented; that is the content of Section 48.4.4.)

The descendants are the non-primary states of the module.

Definition 48.15 (Descendant state and level)**.** A basis vector Eq. (48.118) with $k \geq 1$ is a *descendant* of the primary $|h\rangle$. Its *level* is

$$N = n_1 + n_2 + \cdots + n_k \;\geq\; 1, \tag{48.119}$$

the total number of units of energy added by the raising operators. The primary itself has level $N = 0$.

The level is exactly the L_0-grading. Since $[L_0, L_{-n}] = n\,L_{-n}$ (read off Eq. (48.116) with $m = 0$), each raising operator L_{-n} increases the L_0 eigenvalue by n, so the monomial Eq. (48.118) is an L_0 eigenvector with eigenvalue $h + N$:

$$L_0\left(L_{-n_1}\cdots L_{-n_k}\,|h\rangle\right) = (h+N)\,L_{-n_1}\cdots L_{-n_k}\,|h\rangle\,. \tag{48.120}$$

The Verma module is therefore a graded vector space,

$$V(c,h) = \bigoplus_{N=0}^{\infty} V_N, \qquad V_N = \{\text{level-}N \text{ states}\}, \tag{48.121}$$

with V_0 one-dimensional (spanned by $|h\rangle$). The dimension of V_N is the number of ways to write N as an ordered sum $n_1 \geq \cdots \geq n_k \geq 1$ — the number of partitions of N:

$$\dim V_N = p(N), \qquad p(0)=1,\ p(1)=1,\ p(2)=2,\ p(3)=3,\ p(4)=5,\ldots \tag{48.122}$$

At level 1 the single state is $L_{-1}\,|h\rangle$; at level 2 the two states are $L_{-2}\,|h\rangle$ and $L_{-1}^2\,|h\rangle$; at level 3 the three are $L_{-3}\,|h\rangle$, $L_{-1}L_{-2}\,|h\rangle$, $L_{-1}^3\,|h\rangle$; and so on. (The generating function $\sum_N p(N)q^N = \prod_{k\geq 1}(1-q^k)^{-1}$ will reappear in the next section as the holomorphic part of the Virasoro character.)

Physical Insight

The Verma module is the field-theoretic statement that a primary operator drags an infinite tower of descendant operators behind it: each L_{-n} is, under the state–operator map of Chapter 47, the operation of taking a contour-integrated product with the stress tensor, i.e. $L_{-n}\,\mathcal{O}(0) \leftrightarrow \oint \frac{dz}{2\pi i}\, z^{1-n}\, T(z)\mathcal{O}(0)$. The descendants are the operators $\partial^k\mathcal{O}$, $(T\mathcal{O})$, and their kin — the symmetry images of $\mathcal{O}$. The whole content of a conformal family is fixed by the single number h once c is given, because the action of the algebra is universal. This is why two-dimensional CFT can be solved: the dynamics is reduced to a list of weights h together with the rules for how families fuse, with no further functional freedom inside a family.

48.4.2 The contravariant form and the Gram matrix

A representation appearing in a quantum theory is not a bare vector space: it carries an inner product, the one in which probabilities are computed. On a Verma module that inner product is uniquely fixed, up to the normalization $\langle h|h\rangle = 1$, by two inputs: the adjointness relation $L_n^\dagger = L_{-n}$ inherited from the reality of the stress tensor, and the highest-weight conditions Eq. (48.117). We make this explicit.

Definition 48.16 (Contravariant form)**.** The *contravariant* (or *Shapovalov*) *form* on $V(c,h)$ is the unique Hermitian sesquilinear pairing $\langle\,\cdot\,,\,\cdot\,\rangle$ with $\langle h|h\rangle = 1$ and $L_n^\dagger = L_{-n}$ for all n. On two basis descendants it is computed by moving all raising operators of the bra to the right as lowering operators acting on $|h\rangle$, using Eq. (48.116), until only the highest-weight matrix element $\langle h|h\rangle$ survives.

Two structural facts make this form computable and block-diagonal. First, states of different level are orthogonal: if $|\psi\rangle \in V_N$ and $|\phi\rangle \in V_{N'}$ then

$$\langle\psi,\phi\rangle = \langle\psi|\,|\phi\rangle\,, \qquad (N-N')\langle\psi,\phi\rangle = \langle\psi,(L_0 - L_0^\dagger)\text{-difference}\rangle = 0, \tag{48.123}$$

because $L_0^\dagger = L_0$ is self-adjoint and $|\psi\rangle\,,|\phi\rangle$ are its eigenvectors with eigenvalues $h+N$ and $h+N'$; distinct eigenvalues of a self-adjoint operator have orthogonal eigenvectors. The pairing therefore reduces to a finite $p(N)\times p(N)$ matrix on each graded piece V_N. Second, that finite matrix is Hermitian, because the form is.

Definition 48.17 (Gram matrix at level N)**.** Fix an ordered basis $\{|N;i\rangle\}_{i=1}^{p(N)}$ of V_N as in Eq. (48.118). The *Gram matrix* (or *Gram–Shapovalov matrix*) at level N is

$$\big(M_N\big)_{ij}(c,h) = \langle N;i\,,\,N;j\rangle, \tag{48.124}$$

a Hermitian $p(N)\times p(N)$ matrix whose entries are polynomials in h and c. The state $\sum_i a_i\,|N;i\rangle$ has squared norm $a^\dagger M_N\,a$.

The Gram matrix is the entire question of the section in matrix form. The representation has a positive-definite inner product on V_N if and only if $M_N \succ 0$; it has a zero-norm vector at level N exactly when $\det M_N = 0$; and it has a negative-norm vector (a "ghost," fatal to unitarity) when M_N has a negative eigenvalue. We now compute M_1 and M_2 in full, both to see the machine work and because these two cases already expose the entire mechanism.

Level one

There is a single state, $L_{-1}|h\rangle$, so M_1 is a 1×1 matrix — a number, its norm:

$$(M_1)_{11} = \langle h|L_1\, L_{-1}|h\rangle = \langle h|[L_1, L_{-1}] + L_{-1}L_1|h\rangle\,. \tag{48.125}$$

The second term vanishes because $L_1|h\rangle = 0$. For the commutator, Eq. (48.116) with $m = 1, n = -1$ gives $[L_1, L_{-1}] = (1-(-1))L_0 + \frac{c}{12}(1)(1^2-1)\delta_{0,0} = 2L_0$ (the central term carries $m(m^2-1) = 0$). Hence

$$\boxed{(M_1)_{11} = \langle h|2L_0|h\rangle = 2h.} \tag{48.126}$$

The first lesson is already here: the norm of the unique level-one descendant is $2h$. It is positive for $h > 0$, vanishes at $h = 0$, and is *negative for* $h < 0$. A representation with $h < 0$ therefore contains a negative-norm state at the very first level and cannot be unitary. This is the rigorous origin of the bound $h \geq 0$.

Level two

Now there are two states, and the matrix is genuinely a matrix. Order the level-two basis as

$$|2;1\rangle = L_{-2}|h\rangle\,, \qquad |2;2\rangle = L_{-1}^2|h\rangle\,. \tag{48.127}$$

We need three independent entries: $\langle h|L_2L_{-2}|h\rangle$, $\langle h|L_2L_{-1}^2|h\rangle$, and $\langle h|L_1^2L_{-1}^2|h\rangle$ (the off-diagonal entry is real and symmetric, and we will verify the two ways of computing it agree). We carry every commutator.

For the $(1,1)$ entry,

$$\langle h|L_2L_{-2}|h\rangle = \langle h|[L_2, L_{-2}]|h\rangle\,, \tag{48.128}$$

since $L_2|h\rangle = 0$. From Eq. (48.116) with $m = 2, n = -2$:

$$[L_2, L_{-2}] = (2-(-2))L_0 + \frac{c}{12}\,2(2^2-1)\,\delta_{0,0} = 4L_0 + \frac{c}{12}\cdot 6 = 4L_0 + \frac{c}{2}. \tag{48.129}$$

Thus

$$(M_2)_{11} = 4h + \frac{c}{2}. \tag{48.130}$$

The central charge makes its first appearance in a norm here: the level-two stress-tensor descendant $L_{-2}\,|h\rangle$ "knows" about c because L_{-2} is the mode that creates a stress-tensor quantum, and c is the norm of the stress tensor itself (the vacuum case $h = 0$ gives $\langle T, T\rangle \propto c/2$).

For the $(2, 2)$ entry we need $L_1^2 L_{-1}^2\,|h\rangle$. Build it in stages. First,

$$L_1 L_{-1}\,|h\rangle = [L_1, L_{-1}]\,|h\rangle = 2L_0\,|h\rangle = 2h\,|h\rangle\,, \tag{48.131}$$

using Eq. (48.125)–Eq. (48.126). Next we need $L_1\,L_{-1}^2\,|h\rangle$. Commute L_1 through the pair:

$$\left[L_1, L_{-1}^2\right] = [L_1, L_{-1}]\,L_{-1} + L_{-1}\,[L_1, L_{-1}] = 2L_0 L_{-1} + 2L_{-1} L_0. \tag{48.132}$$

Use $L_0 L_{-1} = L_{-1} L_0 + [L_0, L_{-1}] = L_{-1} L_0 + L_{-1}$ (again from $[L_0, L_{-1}] = L_{-1}$) to write $2L_0 L_{-1} + 2L_{-1} L_0 = 4L_{-1} L_0 + 2L_{-1}$. Acting on $|h\rangle$ (where $L_0 \to h$, $L_1 \to 0$),

$$L_1 L_{-1}^2\,|h\rangle = \left(\left[L_1, L_{-1}^2\right] + L_{-1}^2 L_1\right)|h\rangle = (4L_{-1} L_0 + 2L_{-1})\,|h\rangle = (4h+2)\,L_{-1}\,|h\rangle\,. \tag{48.133}$$

Hitting once more with L_1,

$$(M_2)_{22} = \left\langle h\middle|L_1(L_1 L_{-1}^2)\middle|h\right\rangle = (4h+2)\,\langle h|L_1 L_{-1}|h\rangle = (4h+2)(2h) = 8h^2 + 4h. \tag{48.134}$$

For the off-diagonal entry, compute $L_2 L_{-1}^2\,|h\rangle$. Since $[L_2, L_{-1}] = (2 - (-1))L_1 = 3L_1$ (no central term, as $m + n = 1 \neq 0$),

$$\left[L_2, L_{-1}^2\right] = 3L_1 L_{-1} + 3L_{-1} L_1, \tag{48.135}$$

and acting on $|h\rangle$, using $L_2\,|h\rangle = 0$, $L_1\,|h\rangle = 0$, $L_1 L_{-1}\,|h\rangle = 2h\,|h\rangle$,

$$L_2 L_{-1}^2\,|h\rangle = \left[L_2, L_{-1}^2\right]|h\rangle = 3\,(2h)\,|h\rangle = 6h\,|h\rangle\,, \tag{48.136}$$

so that

$$(M_2)_{12} = \left\langle h\middle|L_2 L_{-1}^2\middle|h\right\rangle = 6h. \tag{48.137}$$

As a check of the Hermiticity built into Theorem 48.16, the conjugate route gives the same: $L_1^2 L_{-2}\,|h\rangle$ is computed from $[L_1, L_{-2}] = (1 - (-2))L_{-1} = 3L_{-1}$, so $L_1 L_{-2}\,|h\rangle = 3L_{-1}\,|h\rangle$ and $L_1^2 L_{-2}\,|h\rangle = 3(2h)\,|h\rangle = 6h\,|h\rangle$, hence $\langle h|L_1^2 L_{-2}|h\rangle = 6h = (M_2)_{21}$ as required.

Collecting Eqs. (48.130), (48.134) and (48.137), the level-two Gram matrix is

$$M_2(c, h) = \begin{pmatrix} 4h + \dfrac{c}{2} & 6h \\ 6h & 8h^2 + 4h \end{pmatrix}. \tag{48.138}$$

Its determinant is the engine of everything that follows:

$$\begin{aligned}\det M_2 &= \left(4h + \tfrac{c}{2}\right)(8h^2 + 4h) - 36h^2 \\ &= 32h^3 + 16h^2 + 4ch^2 + 2ch - 36h^2 \\ &= 2h(16h^2 + 2(c-5)h + c).\end{aligned} \tag{48.139}$$

The overall factor $2h$ is the level-one norm reappearing (a null vector at level one is automatically a null vector at level two, since L_{-1} applied to it stays null); the genuinely new information is the quadratic in the parenthesis, whose roots are the level-two degenerate weights. Solving $16h^2 + 2(c-5)h + c = 0$,

$$h = \frac{-2(c-5) \pm \sqrt{4(c-5)^2 - 64c}}{32} = \frac{(5-c) \pm \sqrt{(c-1)(c-25)}}{16}, \tag{48.140}$$

where we used $(c-5)^2 - 16c = c^2 - 26c + 25 = (c-1)(c-25)$. These two weights are exactly the entries $h_{1,2}(c)$ and $h_{2,1}(c)$ of the Kac formula stated next, and the appearance of the discriminant $(c-1)(c-25)$ already foreshadows that the line $c = 1$ will be a watershed for unitarity. We will return to Eq. (48.140) in Section 48.4.4, where the same numbers drop out of an entirely independent computation — the demand that a specific descendant be a null vector — confirming the structure.

48.4.3 The Kac determinant

The level-one and level-two determinants are the first two members of a family. Computing $\det M_N$ directly becomes hopeless quickly — $\det M_6$ is an 11×11 determinant — but Kac determined the answer in closed form for all N, and its structure is far simpler than the brute computation suggests: $\det M_N$ factorizes completely into linear factors $h - h_{r,s}(c)$, with multiplicities counted by partitions. We state the result, give the weights explicitly, and verify it against Eqs. (48.126) and (48.139).

Theorem 48.18 (Kac determinant formula)**.** *The determinant of the level-N Gram matrix Eq. (48.124) is*

$$\det M_N(c,h) = K_N \prod_{\substack{r,s \geq 1 \\ rs \leq N}} \left(h - h_{r,s}(c)\right)^{p(N-rs)}, \tag{48.141}$$

where $K_N > 0$ is a constant independent of h and c, $p(\cdot)$ is the partition function Eq. (48.122) (with $p(0) = 1$), and the Kac weights *are*

$$h_{r,s}(c) = \frac{c-1}{24} + \frac{1}{4}\left(r\,\alpha_+ + s\,\alpha_-\right)^2, \qquad \alpha_\pm = \frac{\sqrt{1-c} \pm \sqrt{25-c}}{\sqrt{24}}. \tag{48.142}$$

The product runs over all positive integers r, s with $rs \leq N$.

Before discussing the proof, three remarks fix the meaning of Eq. (48.142). First, $\alpha_+\alpha_- = \frac{1}{24}[(1-c)-(25-c)] = -1$; this single relation is what makes the Kac weights organize into the minimal-model patterns of the next section. Second, the combination $\sqrt{(1-c)}$ and $\sqrt{(25-c)}$ shows that $c=1$ and $c=25$ are branch points of the weights as functions of c; for $0<c<1$ both square roots are real, so the $h_{r,s}$ are real, while for $1<c<25$ they become complex-conjugate pairs — another signal that $c=1$ separates two regimes. Third, the lowest weight is the vacuum: $h_{1,1} = \frac{c-1}{24} + \frac{1}{4}(\alpha_+ + \alpha_-)^2$, and since $(\alpha_+ + \alpha_-)^2 = \frac{4(1-c)}{24} = \frac{1-c}{6}$ we get $h_{1,1} = \frac{c-1}{24} + \frac{1-c}{24} = 0$. The level-one null vector $L_{-1}|0\rangle$ at $h = h_{1,1} = 0$ is the statement that ∂ of the identity operator vanishes — the vacuum is translation invariant.

A widely used reparametrization makes the integer structure manifest and is the one we carry into the minimal models. Whenever $c<1$ we may write

$$c = 1 - \frac{6}{m(m+1)}, \qquad \alpha_+ = \sqrt{\frac{m+1}{m}}, \quad \alpha_- = -\sqrt{\frac{m}{m+1}}, \tag{48.143}$$

with m a (not necessarily integer) parameter; one checks $\alpha_+\alpha_- = -1$ and $\frac{13-c}{6} = \alpha_+^2 + \alpha_-^2$ as required. Substituting into Eq. (48.142), the cross term and the constant combine cleanly:

$$h_{r,s}(m) = \frac{\big(r(m+1) - s\,m\big)^2 - 1}{4\,m(m+1)}. \tag{48.144}$$

(To verify: $r\alpha_+ + s\alpha_- = \frac{r(m+1)-sm}{\sqrt{m(m+1)}}$, so $\frac{1}{4}(r\alpha_+ + s\alpha_-)^2 = \frac{(r(m+1)-sm)^2}{4m(m+1)}$, while $\frac{c-1}{24} = -\frac{1}{4m(m+1)}$; their sum is Eq. (48.144).) This is the form in which the Kac table will be read off in Theorem 48.28.

Consistency at low level. At $N=1$, the only (r,s) with $rs \le 1$ is $(1,1)$, contributing $(h-h_{1,1})^{p(0)} = (h-0)^1 = h$. So Eq. (48.141) predicts $\det M_1 = K_1\, h$, matching Eq. (48.126) with $K_1 = 2$. At $N=2$, the admissible pairs are $(1,1)$ with $rs=1$ and $p(2-1)=p(1)=1$, and $(1,2)$ and $(2,1)$ each with $rs=2$ and $p(0)=1$. Thus

$$\det M_2 = K_2\,(h-h_{1,1})\,(h-h_{1,2})\,(h-h_{2,1}) = K_2\,h\,(h-h_{1,2})(h-h_{2,1}). \tag{48.145}$$

Comparing with Eq. (48.139), $\det M_2 = 2h(16h^2 + 2(c-5)h + c) = 32\,h\,(h-h_{1,2})(h-h_{2,1})$ with $h_{1,2}, h_{2,1}$ the roots Eq. (48.140); so $K_2 = 32$ and the two non-vacuum roots are precisely the Kac weights. This is a strong check: the brute-force determinant Eq. (48.139) and the closed form Eq. (48.142) agree, root for root.

Remark 48.19 (On the proof of Theorem 48.18). The full proof of the Kac formula is a structural argument rather than a direct evaluation, and we sketch

its logic honestly here; the explicit low-level verifications above and the null-vector construction below supply the concrete content this book relies on. One shows three things. (i) *Degree and leading coefficient.* The diagonal entry of M_N in the basis monomial $L_{-1}^N|h\rangle$ is, by the level-one recursion Eq. (48.133) iterated, a polynomial in h of degree N with positive leading coefficient $\prod_{k=1}^N 2k \cdot (\cdots)$; summing the partition-counted contributions fixes $\deg_h \det M_N = \sum_{rs\le N} p(N - rs)$, which matches the total degree of the right-hand side of Eq. (48.141). (ii) *The zeros are at the Kac weights.* For each (r, s) with $rs \le N$ there is an explicit *singular vector* at level rs in $V(c, h_{r,s})$ (the analogue of the level-two vector built in Section 48.4.4); its descendants at level N are null, so M_N is degenerate there, giving the factor $(h - h_{r,s})$ at least $p(N - rs)$ times (one for each way of dressing the level-rs singular vector up to level N). (iii) *Multiplicity saturation.* A degree count shows there is no room for further zeros, so the multiplicities are exactly $p(N - rs)$ and the constant K_N is h- and c-independent. The constant is positive and its explicit value $K_N = \prod_{rs\le N} \left((2r)^s s!\right)^{p(N-rs)-p(N-r(s+1))}$ is recorded in the literature; since only the *location* of the zeros bears on degeneracy and unitarity, we do not need K_N beyond $K_N > 0$, which we verified directly for $N = 1, 2$. The original derivation is due to Kac and Feigin–Fuchs [188, 287].

48.4.4 Null vectors and degenerate representations

The Kac determinant vanishes on a discrete set of weights; at each such weight the Gram matrix has a nontrivial kernel, and a vector in that kernel is a zero-norm state orthogonal to its entire level. We now show that such a vector is far more special than a generic zero-norm state: it is itself a primary, it generates an entire sub-Verma-module of null states, and the physically sensible representation is the quotient by that submodule. This is the mechanism that converts the infinite Verma module into the finite, closed operator content of a degenerate theory.

Definition 48.20 (Null/singular vector)**.** A vector $|\chi\rangle \in V(c, h)$ at level $N \ge 1$ is a *singular vector* (or *null vector*) if it is itself a highest-weight vector: it is annihilated by every raising operator,

$$L_n|\chi\rangle = 0 \quad \text{for all } n > 0, \qquad L_0|\chi\rangle = (h + N)|\chi\rangle\,. \tag{48.146}$$

Equivalently, $|\chi\rangle$ is a primary state of weight $h+N$ sitting inside the descendants of the primary $|h\rangle$.

Two facts elevate Eq. (48.146) from a definition to a tool: checking it requires only two equations, and any vector satisfying it is automatically null and orthogonal to the whole module.

Proposition 48.21 (Singular vectors are detected by L_1 and L_2, and are null)**.** *Let $|\chi\rangle \in V_N$, $N \ge 1$. Then:*

(a) $|\chi\rangle$ *is singular (satisfies Eq. (48.146)) if and only if* $L_1|\chi\rangle = 0$ *and* $L_2|\chi\rangle = 0$.

(b) If $|\chi\rangle$ *is singular, then* $\langle\chi, v\rangle = 0$ *for every* $v \in V(c,h)$*; in particular* $|\chi\rangle$ *has zero norm.*

(c) The descendants of a singular vector form a Verma submodule $V(c, h+N) \subseteq V(c,h)$*, every vector of which is orthogonal to all of* $V(c,h)$.

Proof. (a). The positive modes are generated by L_1 and L_2 alone. Indeed Eq. (48.116) gives $[L_1, L_n] = (1-n)L_{n+1}$, with no central term for $n \geq 1$, so for $n \geq 2$

$$L_{n+1} = \frac{1}{1-n}[L_1, L_n], \tag{48.147}$$

and starting from L_1, L_2 this recursion produces $L_3, L_4, \ldots$ Now suppose $L_1|\chi\rangle = L_2|\chi\rangle = 0$. We show $L_n|\chi\rangle = 0$ for all $n \geq 1$ by induction: assume $L_k|\chi\rangle = 0$ for $1 \leq k \leq n$ (true for $n = 2$). Then by Eq. (48.147),

$$L_{n+1}|\chi\rangle = \frac{1}{1-n}(L_1L_n - L_nL_1)|\chi\rangle = \frac{1}{1-n}(L_1 \cdot 0 - L_n \cdot 0) = 0, \tag{48.148}$$

completing the induction. The L_0 condition holds automatically by Eq. (48.120) since $|\chi\rangle \in V_N$. The converse is trivial. This is why, in practice, one builds a candidate null vector and imposes only the two equations $L_1|\chi\rangle = 0$, $L_2|\chi\rangle = 0$.

(b). By the level orthogonality Eq. (48.123) it suffices to take $v \in V_N$, so $v = U|h\rangle$ for U a sum of level-N monomials in the L_{-n}. Then $\langle\chi, v\rangle = \langle\chi|U|h\rangle$; move U onto the bra, $\langle\chi, U|h\rangle\rangle = \langle U^\dagger|\chi\rangle, |h\rangle\rangle$, where $U^\dagger$ is the corresponding sum of monomials in the *raising* operators L_{+n} (adjoints of the L_{-n}). But $U^\dagger$ is a product of positive modes, and by part (a) every positive mode annihilates $|\chi\rangle$; reading $U^\dagger|\chi\rangle$ from the rightmost factor inward, the innermost $L_{+n}|\chi\rangle = 0$ kills the whole expression. Hence $U^\dagger|\chi\rangle = 0$ and $\langle\chi, v\rangle = 0$. Taking $v = |\chi\rangle$ gives $\langle\chi,\chi\rangle = 0$.

(c). Because $|\chi\rangle$ satisfies exactly the highest-weight conditions Eq. (48.117) with weight $h + N$, the module it generates under the raising operators is by Theorem 48.14 a Verma module $V(c, h+N)$, realized as a subspace of $V(c,h)$. For any descendant $w = W|\chi\rangle$ (with W a monomial in lowering operators) and any $v = U|h\rangle \in V(c,h)$, the same manipulation as in (b) gives $\langle v, w\rangle = \langle U|h\rangle, W|\chi\rangle\rangle = \langle W^\dagger U|h\rangle, |\chi\rangle\rangle$, and $W^\dagger U|h\rangle$ is some vector of $V(c,h)$; pairing it with the singular $|\chi\rangle$ and using (b) (which applies to *any* vector of the module) gives zero. So the entire submodule is orthogonal to $V(c,h)$, hence consists of null states. □

Part (b) is the crucial structural statement: a singular vector and all its descendants are orthogonal to everything, including themselves. They are invisible to every correlation function — inserting a null state produces a vanishing matrix

element. A representation that retains them is therefore not irreducible and not faithful to the physics: it carries a sub-representation on which the inner product is identically zero. The remedy is to discard it.

Definition 48.22 (Degenerate representation)**.** When $V(c,h)$ contains singular vectors, let $J(c,h) \subseteq V(c,h)$ be the maximal proper submodule (the sum of all submodules generated by singular vectors). The *irreducible degenerate representation* is the quotient

$$\mathcal{V}(c,h) = V(c,h) \,/\, J(c,h), \tag{48.149}$$

on which the contravariant form descends to a *nondegenerate* pairing. A primary whose weight $h = h_{r,s}(c)$ makes $V(c,h)$ degenerate is called a *degenerate primary*.

The quotient is well defined because $J(c,h)$ is exactly the radical (kernel) of the contravariant form: by Theorem 48.21(c) every singular-vector submodule lies in the radical, and conversely any radical vector at the lowest level where the form degenerates is itself annihilated by all $L_{n>0}$ (else applying a positive mode would lower its level while keeping it in the radical, contradicting minimality), hence is singular. Quotienting by the radical leaves a space on which the induced form is nondegenerate by construction. The number of states of $\mathcal{V}(c,h)$ at each level is strictly smaller than $p(N)$ once a null vector has appeared, and — as the next section shows for the minimal models — can become finite in a sense that closes the operator algebra.

Physical Insight

The physics of a null vector is a *linear relation among descendant operators that holds inside every correlation function.* Setting a level-N singular vector to zero, $|\chi\rangle = 0$, is the statement that a particular linear combination of derivatives and stress-tensor products of the primary $\mathcal{O}$ vanishes as an operator equation. When $\mathcal{O}$ sits in a correlator, that operator equation becomes a linear differential equation for the correlator — this is the origin of the BPZ equations of Theorem 48.30. Degeneracy is thus not a pathology to be avoided but the precise mechanism by which an infinite-symmetry theory becomes exactly solvable: each null vector trades an undetermined function for a solvable ODE.

We now derive the prototypical singular vector — the level-two one — from scratch, and watch the Kac weights Eq. (48.140) reappear.

Worked Example 48.23 (The level-two null vector)**.** We seek a level-two descendant of $|h\rangle$ that is singular. The most general level-two vector is, by Eq. (48.127), a linear combination

$$|\chi\rangle = \left(L_{-2} + a\, L_{-1}^{2}\right)|h\rangle \tag{48.150}$$

(we have normalized the L_{-2} coefficient to 1; the only freedom is the single number a). By Theorem 48.21(a) it is singular iff $L_1|\chi\rangle = 0$ and $L_2|\chi\rangle = 0$. We impose each in turn.

The L_1 condition. We need $L_1 L_{-2}|h\rangle$ and $L_1 L_{-1}^2|h\rangle$. From $[L_1, L_{-2}] = (1-(-2))L_{-1} = 3L_{-1}$,

$$L_1 L_{-2}|h\rangle = [L_1, L_{-2}]|h\rangle = 3L_{-1}|h\rangle\,, \tag{48.151}$$

and from Eq. (48.133), $L_1 L_{-1}^2|h\rangle = (4h+2)L_{-1}|h\rangle$. Therefore

$$L_1|\chi\rangle = \big[\,3 + a(4h+2)\,\big]\,L_{-1}|h\rangle\,, \tag{48.152}$$

which vanishes iff

$$a = -\frac{3}{4h+2} = -\frac{3}{2(2h+1)}. \tag{48.153}$$

This is the announced coefficient: the level-two singular vector, when it exists, must be

$$\boxed{\,|\chi\rangle = \Big(L_{-2} - \frac{3}{2(2h+1)}\,L_{-1}^2\Big)\,|h\rangle\,.\,} \tag{48.154}$$

The L_2 condition. The coefficient a is now fixed, so $L_2|\chi\rangle = 0$ is a constraint on (c,h), not a further free choice. We need $L_2 L_{-2}|h\rangle = (4h + \frac{c}{2})|h\rangle$ (this is Eq. (48.130)) and $L_2 L_{-1}^2|h\rangle = 6h\,|h\rangle$ (this is Eq. (48.137)). Hence

$$L_2|\chi\rangle = \Big[\,(4h + \tfrac{c}{2}) + 6ah\,\Big]\,|h\rangle\,. \tag{48.155}$$

Insert $a = -\frac{3}{2(2h+1)}$, so $6ah = -\frac{18h}{2(2h+1)} = -\frac{9h}{2h+1}$, and require the bracket to vanish:

$$4h + \frac{c}{2} - \frac{9h}{2h+1} = 0. \tag{48.156}$$

Multiply through by $(2h+1)$:

$$(4h+\tfrac{c}{2})(2h+1) - 9h = 8h^2 + 4h + ch + \tfrac{c}{2} - 9h = 8h^2 + (c-5)h + \tfrac{c}{2} = 0, \tag{48.157}$$

and multiplying by 2,

$$16h^2 + 2(c-5)h + c = 0. \tag{48.158}$$

This is *identically* the quadratic in Eq. (48.139), whose roots are $h_{1,2}(c)$ and $h_{2,1}(c)$ from Eq. (48.140). We have thus recovered the Kac weights by a route completely independent of the Gram determinant: demanding that the specific descendant Eq. (48.154) be primary forces exactly the same c–h relation that makes $\det M_2$ vanish. The two perspectives — "the Gram matrix degenerates" and "a descendant becomes primary" — are the same phenomenon. In the m-parametrization Eq. (48.143) the two roots are rational: from Eq. (48.144),

$$h_{2,1}(m) = \frac{(2(m+1)-m)^2 - 1}{4m(m+1)} = \frac{(m+2)^2 - 1}{4m(m+1)} = \frac{(m+1)(m+3)}{4m(m+1)} = \frac{m+3}{4m}, \tag{48.159}$$

$$h_{1,2}(m) = \frac{(m+1-2m)^2-1}{4m(m+1)} = \frac{(1-m)^2-1}{4m(m+1)} = \frac{m(m-2)}{4m(m+1)} = \frac{m-2}{4(m+1)}, \tag{48.160}$$

both manifestly rational at integer m.

The same construction runs at every level: at level rs the candidate is the most general level-rs descendant with one normalization fixed, the L_1, L_2 conditions reduce its $p(rs) - 1$ free coefficients and impose one relation on (c, h), and that relation is $h = h_{r,s}(c)$. The level-one case is degenerate-trivially: $L_1(L_{-1}|h\rangle) = 2h\,|h\rangle = 0$ requires $h = 0 = h_{1,1}$, the vacuum, whose null vector $L_{-1}|0\rangle$ expresses $\partial\mathbf{1} = 0$. The level-three vector $|\chi\rangle = (L_{-3}+aL_{-1}L_{-2}+bL_{-1}^3)\,|h\rangle$ similarly yields $h_{1,3}$ and $h_{3,1}$. These are the building blocks of the BPZ differential equations, and the weights $h_{1,2}, h_{2,1}$ in particular will generate the second-order ODE that solves the Ising correlators in Theorem 48.30.

48.4.5 Unitarity: the bound $c \geq 0,\ h \geq 0$ and the discrete series

A representation can appear in the Hilbert space of a quantum theory only if its inner product is positive: there must be no state of negative norm, on pain of negative probabilities. We now extract from the Gram matrices the full set of (c, h) for which positivity holds. The answer cleaves the plane sharply at $c = 1$, and below it admits only an infinite discrete sequence of points — the single most consequential fact in two-dimensional CFT.

Theorem 48.24 (Unitarity of Virasoro highest-weight representations)**.** *Let $V(c, h)$ be a Virasoro Verma module with c, h real, and let $\mathcal{V}(c, h)$ be its irreducible quotient Eq. (48.149). The contravariant form on $\mathcal{V}(c, h)$ is positive (the representation is* unitary*) precisely in the following cases:*

(i) (Necessary conditions, all c.) *Unitarity requires $c \geq 0$ and $h \geq 0$.*

(ii) (Continuum, $c \geq 1$.) *For $c \geq 1$ and $h \geq 0$ the representation is unitary. The allowed weights form a continuum; for generic such (c, h) the module is irreducible (no null vectors).*

(iii) (Discrete series, $0 < c < 1$.) *For $0 < c < 1$ a unitary representation exists if and only if the central charge takes one of the discrete values*

$$c = 1 - \frac{6}{m(m+1)}, \qquad m = 3, 4, 5, \ldots, \tag{48.161}$$

and the weight is one of the finitely many Kac values

$$h = h_{r,s}(m) = \frac{(r(m+1) - s\,m)^2 - 1}{4\,m(m+1)}, \qquad 1 \leq r \leq m-1, \quad 1 \leq s \leq r, \tag{48.162}$$

each of which is a degenerate primary. These are the central charges and weights of the minimal models.

The necessary conditions in part (i) we prove in full; they are elementary and they already pin down the two half-lines $c \geq 0$, $h \geq 0$. The sufficiency in (ii) and the exclusivity in (iii) are the Friedan–Qiu–Shenker theorem; we give the determinant-vanishing mechanism that forces the discrete series, state the sufficiency, and point to where this book constructs the representations explicitly.

Proof of the necessary conditions (i). Positivity of the form on the whole module requires, in particular, positivity of the norm of every single descendant. Consider the one-parameter family of level-n states $L_{-n}\,|h\rangle$, $n \geq 1$. Their norms are, exactly as in the level-one and level-two computations,

$$\|L_{-n}\,|h\rangle\|^2 = \langle h|L_n L_{-n}|h\rangle = \langle h|[L_n, L_{-n}]|h\rangle = 2n\,h + \frac{c}{12}\,n(n^2-1), \quad (48.163)$$

using $[L_n, L_{-n}] = 2nL_0 + \frac{c}{12}n(n^2-1)$ from Eq. (48.116) and $L_n\,|h\rangle = 0$. Each of these must be ≥ 0.

At $n = 1$ the central term vanishes and Eq. (48.163) reads $2h \geq 0$, i.e.

$$h \geq 0. \quad (48.164)$$

This is the bound already seen at Eq. (48.126). For the central charge, divide Eq. (48.163) by n and let $n \to \infty$:

$$\frac{1}{n}\,\|L_{-n}\,|h\rangle\|^2 = 2h + \frac{c}{12}(n^2-1) \xrightarrow[n\to\infty]{} +\infty \cdot \operatorname{sgn}(c), \quad (48.165)$$

so positivity for all large n forces $c \geq 0$. (Strictly: if $c < 0$ then for n large enough the negative cubic term $\frac{c}{12}n^3$ overwhelms the linear $2nh$, making Eq. (48.163) negative; hence $c < 0$ is excluded. If $c = 0$, Eq. (48.163) gives $2nh \geq 0$, consistent with $h \geq 0$, and indeed $c = 0$ admits only the trivial $h = 0$ representation.) Thus $c \geq 0$ and $h \geq 0$ are necessary, proving (i). □

The two diagnostic norms used here — the level-one norm $2h$ and the large-n norm dominated by $\frac{c}{12}n^3$ — are the cleanest windows onto the two halves of the bound. Note that they alone do not yet exclude anything in the open region $0 < c < 1$, $h > 0$: Eq. (48.163) is positive there for every n. The exclusion of most of that region is a statement about the *off-diagonal* structure of the Gram matrix, i.e. about the determinant, and it is subtler.

The discrete series via the vanishing curves (mechanism for (ii)–(iii)). Fix $h > 0$ and increase c from 0. For c very large the Gram matrix is positive definite: Eq. (48.138), for instance, has positive diagonal and, as $c \to \infty$, positive determinant $\det M_2 \approx \frac{c}{2} \cdot 8h^2 > 0$, and the same dominance of the central terms makes every $M_N \succ 0$ at large c. So ($c \to \infty$, $h > 0$) is unitary. Unitarity can fail only if, as we lower c, an eigenvalue of some M_N passes through zero and turns

negative. An eigenvalue can change sign only by crossing zero, and M_N has a zero eigenvalue exactly on the *vanishing curves*

$$\mathcal{C}_{r,s}: \qquad h = h_{r,s}(c), \qquad rs \le N, \tag{48.166}$$

where $\det M_N = 0$ by Theorem 48.18. The strategy of Friedan, Qiu, and Shenker is to track the sign of $\det M_N$ across these curves in the strip $0 < c < 1$.

For $c \ge 1$ the analysis is benign. There one shows the curves $\mathcal{C}_{r,s}$ do not enclose any region of the quadrant $\{c \ge 1, h \ge 0\}$ in which an even number of sign changes traps a negative eigenvalue; concretely, for $c \ge 1$ and $h \ge 0$ the determinant $\det M_N$ stays nonnegative for every N, with zeros only on isolated curves where a null vector appears (and is then quotiented away). Hence every $(c \ge 1, h \ge 0)$ is unitary, which is part (ii); the weights fill a continuum because no further integrality is imposed.

For $0 < c < 1$ the curves Eq. (48.166) are real (both square roots in Eq. (48.142) are real) and they sweep across the strip, dividing it into regions. The decisive computation, carried out level by level, is that in this strip a state of negative norm appears in the generic region between curves: lowering h across the first curve $\mathcal{C}_{1,2}$ (or $\mathcal{C}_{2,1}$) at fixed $0 < c < 1$ sends the relevant eigenvalue of M_2 from positive, through zero on the curve, to negative just past it — the determinant Eq. (48.139), written as $32h(h - h_{1,2})(h - h_{2,1})$, changes sign as h crosses a simple root. The only way for a point (c, h) with $0 < c < 1$ to avoid lying in some negative-norm region at some level is for it to sit at a *crossing* of vanishing curves of different (r, s), where two roots coincide and the eigenvalue grazes zero without the sign change propagating to a ghost. Demanding that the curves $\mathcal{C}_{r,s}$ and $\mathcal{C}_{r',s'}$ intersect at a common (c, h) with all the consistency required across *all* levels forces, after the integrality of r, s is used, the central charge onto the discrete values

$$c = 1 - \frac{6}{m(m+1)}, \qquad m \in \{3, 4, 5, \dots\}, \tag{48.167}$$

and the weight onto the Kac entries Eq. (48.162). The lower end $m = 3$ gives $c = 1/2$ (the Ising model); $m \to \infty$ accumulates at $c \to 1^-$. Values $0 < c < 1$ off this list, and weights off the Kac table at a listed c, necessarily place the representation in a negative-norm region at some level and are therefore non-unitary. This establishes the necessity in (iii); a representative slice of the curves and the discrete series is shown in Fig. 48.3. □

Remark 48.25 (Sufficiency and the status of the proof). The argument above establishes that unitarity *at most* survives on the FQS list; that the listed representations are *actually* unitary — the sufficiency half of Theorem 48.24(iii) — requires exhibiting, for each (c, h) on the list, a positive-definite irreducible quotient. This was completed by constructing the representations explicitly: as

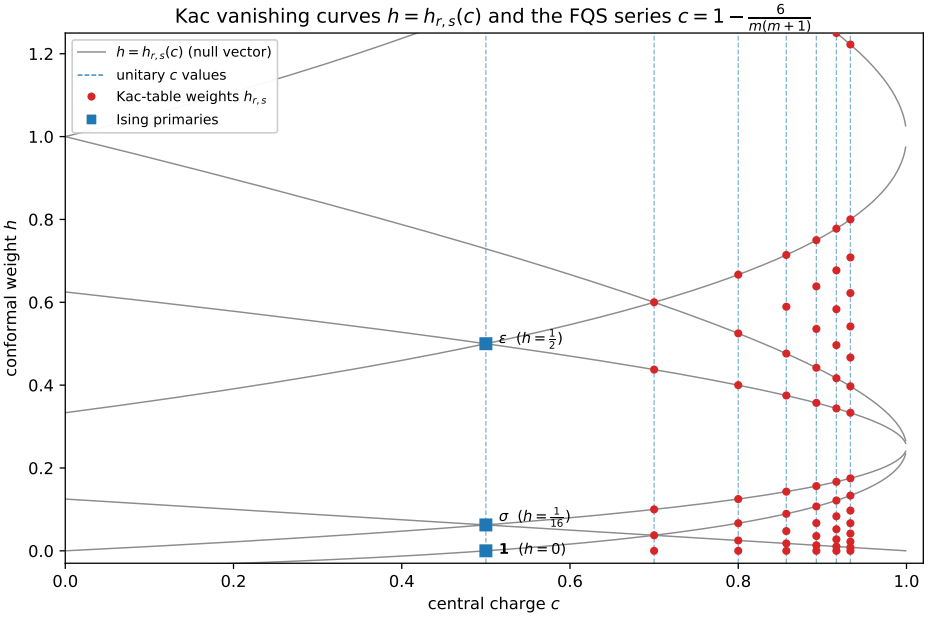


Figure 48.3: The Kac vanishing curves $h = h_{r,s}(c)$ (grey) in the strip $0 < c < 1$, where $\det M_N$ develops a zero and a null vector appears. Between curves the Gram form acquires a negative eigenvalue, so a generic (c, h) in the strip is non-unitary. Unitary representations survive only where the curves cross in the pattern realized at the discrete Friedan–Qiu–Shenker central charges $c = 1 - 6/[m(m+1)]$ (dashed verticals, accumulating at $c = 1$); the allowed weights are the Kac-table values $h_{r,s}$ (dots) on each. The line $m = 3$, $c = \frac{1}{2}$, is the 2D Ising model, whose three primaries — identity $\mathbf{1}$ $(h = 0)$, spin σ $(h = \frac{1}{16})$, energy ε $(h = \frac{1}{2})$ — are marked. These weights are read off in Theorem 48.26 and solved as a full CFT in Section 48.5.

cosets of current algebras (the GKO coset construction) and, equivalently for our purposes, through the free-field (Coulomb-gas) realization in which the L_n act on a free boson and positivity is manifest. This book carries out the free-field construction in Section 48.5's development of the minimal models, where the Kac table and its closure under fusion are derived; that construction is what completes the sufficiency proof. The full classification is the theorem of Friedan, Qiu, and Shenker [200], with sufficiency completed by Goddard, Kent, and Olive [226].

Worked Example 48.26 (The Ising Kac table and the spin and energy weights)**.** The smallest member of the discrete series, $m = 3$, is the critical Ising model. From Eq. (48.161),

$$c = 1 - \frac{6}{3 \cdot 4} = 1 - \frac{1}{2} = \frac{1}{2}. \tag{48.168}$$

The Kac weights Eq. (48.162) with $m = 3$, $4m(m+1) = 48$, and $1 \le r \le 2$, $1 \le s \le r$, are

$$h_{1,1} = \frac{(1\cdot 4 - 1\cdot 3)^2 - 1}{48} = \frac{1-1}{48} = 0, \tag{48.169}$$

$$h_{2,1} = \frac{(2\cdot 4 - 1\cdot 3)^2 - 1}{48} = \frac{25-1}{48} = \frac{24}{48} = \frac{1}{2}, \tag{48.170}$$

$$h_{2,2} = \frac{(2\cdot 4 - 2\cdot 3)^2 - 1}{48} = \frac{4-1}{48} = \frac{3}{48} = \frac{1}{16}. \tag{48.171}$$

(The would-be entry $h_{1,2}$ equals $h_{2,2} = \frac{1}{16}$ by the Kac-table reflection $h_{r,s} = h_{m-r,m+1-s}$, so the distinct weights are exactly $\{0, \frac{1}{16}, \frac{1}{2}\}$.) Each chiral weight pairs with an identical antichiral weight to give a scalar primary of total scaling dimension $\Delta = h + \bar{h}$ and spin $h - \bar{h} = 0$:

$$\Delta_{\mathbf{1}} = 0, \qquad \Delta_\sigma = 2h_{2,2} = 2\cdot\tfrac{1}{16} = \frac{1}{8}, \qquad \Delta_\varepsilon = 2h_{2,1} = 2\cdot\tfrac{1}{2} = 1. \tag{48.172}$$

These are the spin operator σ with $\Delta_\sigma = \frac{1}{8}$ and the energy operator ε with $\Delta_\varepsilon = 1$ promised in the chapter introduction — the exact critical dimensions of the two-dimensional Ising model, here obtained from representation theory alone, with no lattice and no Lagrangian. They are the exactly solvable cousins of the Wilson–Fisher exponents that ϕ^4 theory approaches perturbatively in $d = 4 - \epsilon$ (Chapter 52); the spin σ is the continuum image of the Ising magnetization, and ε of the energy density. The degeneracy of these three primaries — each carries a null vector, ε and σ at level two via Eq. (48.154) — is precisely what will close their fusion algebra and reduce their four-point functions to second-order ODEs in Section 48.5.

Caution

The bound is $c \ge 0$, $h \ge 0$ for *unitary* representations, but c and h are perfectly meaningful real (or even complex) numbers in non-unitary contexts that occur in physics. Statistical models at criticality that are described by non-unitary CFTs — the Lee–Yang edge singularity ($c = -22/5$) and two-dimensional percolation and polymers ($c = 0$ with non-trivial operators surviving the $h \ge 0$ degeneracy) — live precisely outside the FQS region. The theorem says such theories cannot be reflection-positive quantum theories in the Hilbert-space sense; it does *not* say they are unphysical. Throughout the rest of this part we restrict to unitary CFT unless stated, but the representation theory of $V(c,h)$ developed in Sections 48.4.1, 48.4.2 and 48.4.4 is valid for all c, h.

We have now isolated the exact data the next section needs. The unitary $c < 1$ theories live only at $c = 1 - 6/[m(m+1)]$, and at each such c only a

finite Kac table Eq. (48.162) of degenerate primaries can occur, every one of them carrying a null vector that imposes an operator relation. The next section, Section 48.5, takes this finite list as a starting kit: it shows that these primaries close into a consistent, complete operator algebra — the minimal models $\mathcal{M}(p,p')$ — with fusion rules dictated by the null vectors of Section 48.4.4 and correlators fixed by the BPZ differential equations those null vectors generate, and it solves the Ising case ($m=3$) in closed form, delivering the weights $\Delta_\sigma = \frac{1}{8}$, $\Delta_\varepsilon = 1$ found above together with the full four-point function of spins.

48.5 Minimal Models, Fusion Rules, and the BPZ Equations

The previous section (Section 48.4) ended on a sharp dichotomy. The Kac determinant (Theorem 48.18) showed that a Verma module over the Virasoro algebra is reducible precisely when the highest weight h takes one of a discrete family of values $h_{r,s}(c)$; at each such value a *singular* (null) vector appears (Theorem 48.20), and dividing it out leaves a smaller, *degenerate* representation. Unitarity (Theorem 48.24) then carved the allowed central charges into the continuum $c \geq 1$ and a single discrete sequence below it, $c = 1 - 6/[m(m+1)]$ with $m = 3, 4, 5, \ldots$. What we have so far, however, are only representations: lists of weights (c,h) for which a consistent action of the Virasoro algebra exists. A representation is not yet a theory. To have a theory we must assemble a *spectrum* — a specific set of primary fields — and check that the operator product expansion of any two of them closes on that same set, so that all correlation functions are determined. This section performs that assembly. We will find that for the rational points $c = 1 - 6(p-p')^2/(pp')$ the degenerate representations organize themselves into a *finite* set, the Kac table, on which the operator algebra closes exactly. These are the *minimal models*: complete, nonperturbatively defined conformal field theories with finitely many primaries, no Lagrangian, and no free parameters, in which every correlation function is computable in closed form. The engine that makes them solvable is the null vector itself: inserted into a correlator, it converts the algebraic statement "this descendant vanishes" into a linear differential equation — the equation of Belavin, Polyakov, and Zamolodchikov (BPZ) — whose solutions are the correlators. We develop the central-charge parametrization and Kac table first, then turn null vectors into differential operators, derive the BPZ equations, and read off the fusion rules that prove the algebra closes.

48.5.1 The minimal-model central charge and the Kac table

The Kac determinant of Theorem 48.18 vanishes, at level N, exactly on the loci $h = h_{r,s}(c)$ for positive integers r, s with $rs \leq N$. It is convenient — and

standard since the work of Dotsenko and Fateev on the Coulomb gas — to trade the pair (c, h) for a pair of *background charges*. We introduce two numbers α_+, α_- subject to

$$\alpha_+\alpha_- = -1, \qquad \alpha_+ + \alpha_- = 2\alpha_0, \qquad c = 1 - 24\,\alpha_0^2, \tag{48.173}$$

so that $\alpha_\pm = \alpha_0 \pm \sqrt{\alpha_0^2 + 1}$ are the two roots of $\alpha - \alpha^{-1} = 2\alpha_0$, i.e. of $\alpha^2 - 2\alpha_0\alpha - 1 = 0$. In this parametrization the zeros of the Kac determinant recorded in Theorem 48.18 are

$$h_{r,s} \;=\; \frac{(r\alpha_+ + s\alpha_-)^2 - (\alpha_+ + \alpha_-)^2}{4} \;=\; \frac{c-1}{24} + \frac{1}{4}\,(r\alpha_+ + s\alpha_-)^2, \tag{48.174}$$

the second form following from $(\alpha_+ + \alpha_-)^2 = 4\alpha_0^2 = (1-c)/6$. A primary whose weight equals $h_{r,s}$ for some positive integers r, s is *degenerate*: its Verma module contains a singular vector, located at level rs.

Definition 48.27 (Minimal model)**.** A *minimal model* is a conformal field theory whose space of states decomposes into a *finite* number of irreducible Virasoro representations, each built on a degenerate primary, and whose operator algebra (Theorem 47.46) closes on this finite set: the operator product of any two primaries in the spectrum contains only primaries (and their descendants) drawn from the same finite list.

For the spectrum to be finite, the lattice of degenerate weights $h_{r,s}$ must collapse: distinct (r, s) must repeatedly give the *same* weight, so that only finitely many values occur. From Eq. (48.174) this happens precisely when the ratio α_+/α_- is rational, for then the quadratic form $(r\alpha_+ + s\alpha_-)^2$ takes each value infinitely often as (r, s) ranges over the positive integers, and the genuinely distinct weights form a finite set. Write the rationality condition as

$$\alpha_+^2 = \frac{p}{p'}, \qquad \alpha_-^2 = \frac{p'}{p}, \qquad \gcd(p, p') = 1, \quad p, p' \geq 2, \tag{48.175}$$

consistent with $\alpha_+\alpha_- = -1$ if we take $\alpha_+ = \sqrt{p/p'}$, $\alpha_- = -\sqrt{p'/p}$. We require $p, p' \geq 2$ because $p = 1$ or $p' = 1$ would force a vanishing or negative numerator below and no nontrivial table. The central charge follows from Eq. (48.173): since $2\alpha_0 = \alpha_+ + \alpha_- = \sqrt{p/p'} - \sqrt{p'/p} = (p - p')/\sqrt{pp'}$, we have $\alpha_0^2 = (p-p')^2/(4pp')$ and $24\alpha_0^2 = 6(p-p')^2/(pp')$, giving the headline formula

$$\boxed{\,c \;=\; 1 - \frac{6\,(p-p')^2}{p\,p'}\,}, \qquad \gcd(p, p') = 1, \quad p, p' \geq 2. \tag{48.176}$$

The expression is symmetric under $p \leftrightarrow p'$, which merely swaps $\alpha_+ \leftrightarrow \alpha_-$ and hence the roles of the two integer labels below; we may therefore adopt $2 \leq p' < p$ without loss of generality. The model defined by Eq. (48.176) is denoted $\mathcal{M}(p, p')$.

Substituting Eq. (48.175) into Eq. (48.174) rationalizes the weights as well. Since $r\alpha_+ + s\alpha_- = r\sqrt{p/p'} - s\sqrt{p'/p} = (pr - p's)/\sqrt{pp'}$ and $(\alpha_+ + \alpha_-)^2 = (p-p')^2/(pp')$,

$$h_{r,s} \;=\; \frac{(pr - p's)^2 - (p-p')^2}{4\,p\,p'}. \tag{48.177}$$

This is the *Kac formula*. The two integer labels (r, s) now run over a finite rectangle, and the collapse of distinct labels to equal weights is encoded in a single nontrivial identification, which we now establish.

Theorem 48.28 (The Kac table)**.** *In the minimal model $\mathcal{M}(p, p')$ with $\gcd(p, p') = 1$ and $2 \le p' < p$, the degenerate weights Eq. (48.177) that furnish the spectrum are exactly those with*

$$1 \le r \le p' - 1, \qquad 1 \le s \le p - 1, \tag{48.178}$$

subject to the identification

$$(r, s) \;\sim\; (p' - r,\; p - s), \qquad \textit{i.e.} \qquad h_{r,s} = h_{p'-r,\,p-s}. \tag{48.179}$$

The number of distinct primary fields is therefore

$$\mathcal{N} \;=\; \frac{(p-1)(p'-1)}{2}. \tag{48.180}$$

Proof. We prove three statements: the identification Eq. (48.179), that the rectangle Eq. (48.178) contains each weight exactly twice (so the count is as stated), and that no degenerate weight outside the rectangle is independent.

(i) The reflection symmetry. Apply $(r, s) \mapsto (p' - r, p - s)$ to the combination $pr - p's$ appearing in Eq. (48.177):

$$p(p' - r) - p'(p - s) \;=\; pp' - pr - p'p + p's \;=\; -(pr - p's). \tag{48.181}$$

Because Eq. (48.177) depends on r, s only through $(pr - p's)^2$, which is even in this quantity, $h_{p'-r,p-s} = h_{r,s}$. This proves Eq. (48.179). The map is an involution of the rectangle Eq. (48.178) onto itself: if $1 \le r \le p' - 1$ then $1 \le p' - r \le p' - 1$, and likewise for s.

(ii) No further coincidences inside the rectangle, and no fixed point. Suppose $h_{r,s} = h_{r',s'}$ for two pairs in the rectangle. From Eq. (48.177) this requires $(pr - p's)^2 = (pr' - p's')^2$, hence $pr - p's = \pm(pr' - p's')$. Take first the $+$ sign: $p(r-r') = p'(s-s')$. Since $\gcd(p, p') = 1$, p' divides $r-r'$; but $|r-r'| \le p'-2 < p'$, forcing $r = r'$ and then $s = s'$. Take the $-$ sign: $p(r+r') = p'(s+s')$, so $p' \mid (r+r')$ with $2 \le r + r' \le 2p' - 2$, whence $r + r' = p'$, and then $p \cdot p' = p'(s + s')$ gives $s + s' = p$, i.e. $(r', s') = (p' - r, p - s)$. Thus the only coincidences are the trivial one and the reflection Eq. (48.179). The reflection has no fixed point in the rectangle: $(r, s) = (p' - r, p - s)$ would need $2r = p'$ and $2s = p$, impossible since

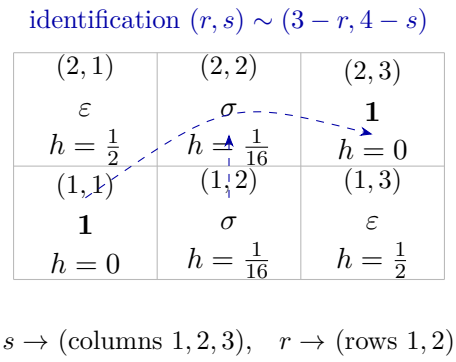


Figure 48.4: The Kac table of the Ising minimal model $\mathcal{M}(4,3)$ ($p=4$, $p'=3$, $c=\frac{1}{2}$). The rectangle has $(p'-1)(p-1)=2\cdot 3=6$ cells; the point reflection $(r,s)\sim(p'-r,p-s)$ pairs them two-to-one, leaving $\mathcal{N}=(p-1)(p'-1)/2=3$ distinct primaries: the identity $\mathbf{1}$ ($h=0$), the spin σ ($h=\frac{1}{16}$), and the energy ε ($h=\frac{1}{2}$). The full local dimensions are $\Delta=h+\bar{h}$, so $\Delta_\sigma=\frac{1}{8}$, $\Delta_\varepsilon=1$.

$\gcd(p,p')=1$ forbids p and p' both even. Hence the rectangle's $(p'-1)(p-1)$ cells pair off exactly two-to-one, giving Eq. (48.180).

(iii) The rectangle is exhaustive. It remains to see that degenerate weights with labels outside Eq. (48.178) are not new fields but are identified with table entries (or correspond to representations that are not part of a closed, consistent spectrum). Two periodicities of Eq. (48.177) settle this. First, $pr-p's$ is invariant under $(r,s)\mapsto(r+p',s+p)$, so $h_{r+p',s+p}=h_{r,s}$; shifting by this lattice vector maps any cell into a translate of the fundamental rectangle. Second, the reflection Eq. (48.179) combined with this period generates the full symmetry group of the weight lattice, the affine Weyl group of the relevant rank-one data; its fundamental domain is precisely the rectangle Eq. (48.178) with the boundary identification Eq. (48.179). (We will see in Section 48.5.5 that fields with labels on or beyond the rectangle's edges are removed by the fusion truncation: their would-be contributions are reflected back inside, which is the dynamical counterpart of this geometric statement.) Hence every degenerate weight equals some $h_{r,s}$ with (r,s) in the rectangle, and the count Eq. (48.180) is complete. □

Physical Insight

The collapse of the infinite Kac lattice to a finite table is the algebraic shadow of a physical fact: a critical statistical system at a fixed point has only finitely many *scaling fields* that are not derivatives of others. The labels (r,s) are not arbitrary — they count the ways a degenerate representation can be reached, and the identification $(r,s)\sim(p'-r,p-s)$ is the statement that two such routes terminate on the same physical

operator. The reflection is the same phenomenon that, in the free-field (Coulomb-gas) realization sketched at the end of this chapter, appears as the screening of background charge.

The metric of Chapter 48 is Euclidean throughout, obtained by the Wick rotation of Chapter 5; a physical scaling field is built from a holomorphic primary of weight h and an antiholomorphic primary of weight $\bar{h}$, and its total scaling dimension and planar spin are $\Delta = h + \bar{h}$ and $\mathfrak{s} = h - \bar{h}$. In the diagonal minimal models the antiholomorphic content mirrors the holomorphic one, $\bar{h} = h$, so $\Delta = 2h$. We will keep the holomorphic and antiholomorphic sectors separate at the level of the chiral algebra and reassemble them only when forming physical correlators in Section 48.5.4.

Worked Example 48.29 (The first unitary minimal models)**.** The unitarity analysis of Theorem 48.24 selected, below $c = 1$, only the values $c = 1 - 6/[m(m+1)]$ with $m = 3, 4, 5, \ldots$. We recover these as the minimal models with adjacent labels. Set $p' = m$ and $p = m + 1$, so that $\gcd(p, p') = \gcd(m, m+1) = 1$ automatically and $p - p' = 1$. Then Eq. (48.176) gives

$$c \;=\; 1 - \frac{6}{m(m+1)}, \qquad m = 3, 4, 5, \ldots, \tag{48.182}$$

exactly the unitary series. These are the only minimal models that are unitary: $|p - p'| = 1$ is forced because a larger gap pushes some Kac weight Eq. (48.177) negative, violating the reflection-positivity bound $h \geq 0$ established in Theorem 48.24. The Kac formula specializes to

$$h_{r,s} = \frac{\big[(m+1)r - ms\big]^2 - 1}{4m(m+1)}, \qquad 1 \leq r \leq m-1, \quad 1 \leq s \leq m. \tag{48.183}$$

The smallest three members:

m	(p, p')	c	primaries (weights h, with $\bar{h} = h$)
3	$(4, 3)$	$\frac{1}{2}$	$0,\ \frac{1}{16},\ \frac{1}{2}$
4	$(5, 4)$	$\frac{7}{10}$	$0,\ \frac{3}{80},\ \frac{1}{10},\ \frac{7}{16},\ \frac{3}{5},\ \frac{3}{2}$
5	$(6, 5)$	$\frac{4}{5}$	$0,\ \frac{1}{40},\ \frac{1}{15},\ \frac{1}{8},\ \frac{2}{5},\ \frac{21}{40},\ \frac{2}{3},\ \frac{7}{5},\ \frac{13}{8},\ 3$

The $m = 3$ entry is the Ising model: $\mathcal{N} = (p-1)(p'-1)/2 = 3$ primaries with $h \in \{0, \frac{1}{16}, \frac{1}{2}\}$, displayed in Fig. 48.4; $m = 4$ is the tricritical Ising model with $\mathcal{N} = 6$; $m = 5$ ($c = \frac{4}{5}$) carries $\mathcal{N} = 10$ and underlies the critical three-state Potts model. As a check on $m = 4$ (so $p = 5$, $p' = 4$): the entry $h_{2,2} = \big[5 \cdot 2 - 4 \cdot 2\big]^2 - 1)/80 = (4-1)/80 = 3/80$, and $h_{1,4} = \big[5 - 16\big]^2 - 1)/80 = (121-1)/80 = 3/2$, both appearing in the table; the reflection $(r, s) \sim (p' - r, p - s) = (4 - r, 5 - s)$ identifies, for instance, $(1, 1) \sim (3, 4)$ and $(2, 2) \sim (2, 3)$, consistent with the six distinct weights listed.

48.5.2 Null vectors as differential operators

To exploit degeneracy dynamically we must learn to insert a singular vector into a correlation function. Recall from Theorem 48.15 that the descendant fields of a primary ϕ of weight h are generated by the modes of the holomorphic stress tensor through the contour prescription

$$(L_{-n}\phi)(z) \;=\; \frac{1}{2\pi i}\oint_{z} \mathrm{d}w\,(w-z)^{1-n}\,T(w)\,\phi(z), \qquad n\geq 1, \tag{48.184}$$

the contour being a small circle around z. This is the residue extracting the regular part of the operator product $T(w)\phi(z)$ recorded in Eq. (48.185), whose singular part is fixed by primarity,

$$T(w)\phi(z) = \frac{h\,\phi(z)}{(w-z)^2} + \frac{\partial\phi(z)}{w-z} + \sum_{n\geq 2}(w-z)^{\,n-2}\,(L_{-n}\phi)(z). \tag{48.185}$$

In particular $L_{-1}\phi=\partial\phi$ and $L_0\phi = h\phi$.

Now place the descendant inside a correlator together with N other primaries ϕ_i of weights h_i at points z_i, and abbreviate $G\equiv\langle\phi(z)\,\phi_1(z_1)\cdots\phi_N(z_N)\rangle$. Using Eq. (48.184),

$$\big\langle (L_{-n}\phi)(z)\,\textstyle\prod_i \phi_i(z_i)\big\rangle = \frac{1}{2\pi i}\oint_z \mathrm{d}w\,(w-z)^{1-n}\,\big\langle T(w)\,\phi(z)\,\prod_i\phi_i(z_i)\big\rangle. \tag{48.186}$$

The mixed correlator is fixed by the conformal Ward identity established in Section 48.2.2 (Eq. (48.48)): the stress tensor insertion is meromorphic in w with a double and a single pole at every operator location,

$$\big\langle T(w)\,\phi(z)\,\textstyle\prod_i\phi_i\big\rangle = \Big[\frac{h}{(w-z)^2} + \frac{1}{w-z}\,\partial_z + \sum_{i=1}^{N}\Big(\frac{h_i}{(w-z_i)^2} + \frac{1}{w-z_i}\,\partial_{z_i}\Big)\Big]\,G. \tag{48.187}$$

We now carry out the contour integral. Of the self-terms (those singular at $w=z$), the piece $h/(w-z)^2$ contributes $(w-z)^{-1-n}$ to the integrand, whose residue vanishes for $n\geq 1$; the piece $\partial_z/(w-z)$ contributes $(w-z)^{-n}$, again with vanishing residue for $n\geq 2$. Thus for $n\geq 2$ the descendant correlator is determined entirely by the *other* insertions — precisely the content of Eq. (48.185), where $(L_{-n}\phi)$ for $n\geq 2$ is the regular part of the operator product. Write the bracketed sum over i in Eq. (48.187) as $\hat{R}(w)\,G$ with

$$\hat{R}(w) = \sum_i \left(\frac{h_i}{(w-z_i)^2} + \frac{\partial_{z_i}}{w-z_i}\right). \tag{48.188}$$

The contour around z picks out the coefficient of $(w-z)^{-1}$ in $(w-z)^{1-n}\hat{R}(w)$, i.e. the coefficient of $(w-z)^{n-2}$ in the Taylor expansion of $\hat{R}(w)$ about $w=z$. Expanding $1/(w-z_i)^2$ and $1/(w-z_i)$ in powers of $(w-z)$ and reading off that coefficient gives, for the double pole,

$$\big[(w-z)^{n-2}\big]\;\frac{1}{(w-z_i)^2} = \frac{n-1}{(z_i-z)^n}, \tag{48.189}$$

and for the single pole,

$$\big[(w-z)^{n-2}\big]\,\frac{1}{w-z_i} = -\frac{1}{(z_i-z)^{n-1}}. \tag{48.190}$$

(Both follow from $1/(w-z_i)^k = \sum_{j\ge 0}\binom{-k}{j}(w-z)^j(z-z_i)^{-k-j}$ and the sign $(z-z_i) = -(z_i-z)$.) Therefore the descendant insertion is reproduced by a linear differential operator $\mathcal{L}_{-n}$ acting on the correlator of the *undifferentiated* primary,

$$\big\langle (L_{-n}\phi)(z)\,\textstyle\prod_i \phi_i\big\rangle = \mathcal{L}_{-n}\,G, \qquad \mathcal{L}_{-n} = \sum_{i=1}^{N}\Big[\tfrac{(n-1)\,h_i}{(z_i-z)^n} - \tfrac{1}{(z_i-z)^{n-1}}\,\partial_{z_i}\Big], \quad n\ge 2. \tag{48.191}$$

For $n=1$ the same formula applies if we adopt the translation Ward identity $(\partial_z + \sum_i \partial_{z_i})G = 0$: then $\mathcal{L}_{-1} = -\sum_i \partial_{z_i} = \partial_z$, in agreement with $L_{-1}\phi = \partial\phi$. The operators $\mathcal{L}_{-n}$ are the bridge from representation theory to analysis: every Virasoro mode acting on the degenerate field has become an explicit differential operator in the positions of the spectators.

48.5.3 The level-two null vector

The simplest nontrivial degenerate field carries a singular vector at level two. By Theorem 48.20 a level-two singular vector in the Verma module over $|h\rangle$ has the form

$$|\chi\rangle = \Big(L_{-2} + a\,L_{-1}^2\Big)|h\rangle, \tag{48.192}$$

the most general level-two state, and it is singular iff it is annihilated by all positive modes, equivalently (since the positive modes are generated by L_1, L_2) by L_1 and L_2. We use the Virasoro algebra recalled in Eq. (48.92),

$$[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}\,m(m^2-1)\,\delta_{m+n,0}, \tag{48.193}$$

together with the highest-weight conditions $L_0|h\rangle = h|h\rangle$, $L_{n>0}|h\rangle = 0$. Compute the action of L_1:

$$L_1L_{-2}|h\rangle = [L_1,L_{-2}]|h\rangle = 3\,L_{-1}|h\rangle, \tag{48.194}$$

$$L_1L_{-1}^2|h\rangle = \big([L_1,L_{-1}]L_{-1} + L_{-1}L_1L_{-1}\big)|h\rangle = \big(2L_0L_{-1} + 2hL_{-1}\big)|h\rangle = (4h+2)\,L_{-1}|h\rangle, \tag{48.195}$$

where we used $[L_1,L_{-1}] = 2L_0$, $L_0L_{-1}|h\rangle = (h+1)L_{-1}|h\rangle$, and $L_1L_{-1}|h\rangle = 2h|h\rangle$. Hence

$$L_1|\chi\rangle = \big[3 + a(4h+2)\big]L_{-1}|h\rangle = 0 \;\Longrightarrow\; a = -\frac{3}{2(2h+1)}. \tag{48.196}$$

Now impose $L_2|\chi\rangle = 0$:

$$L_2L_{-2}|h\rangle = [L_2,L_{-2}]|h\rangle = \Big(4L_0 + \tfrac{c}{2}\Big)|h\rangle = \Big(4h + \tfrac{c}{2}\Big)|h\rangle, \tag{48.197}$$

$$L_2 L_{-1}^2|h\rangle = \big([L_2,L_{-1}]L_{-1} + L_{-1}L_2L_{-1}\big)|h\rangle = 3L_1L_{-1}|h\rangle = 6h\,|h\rangle, \quad (48.198)$$

using $[L_2,L_{-2}] = 4L_0 + \frac{c}{2}$, $[L_2,L_{-1}] = 3L_1$, and $L_2L_{-1}|h\rangle = 3L_1|h\rangle = 0$. Therefore

$$L_2|\chi\rangle = \Big(4h + \tfrac{c}{2} + 6ah\Big)|h\rangle = 0. \quad (48.199)$$

Substituting Eq. (48.196) and clearing denominators ($\times 2(2h+1)$),

$$(8h+c)(2h+1) - 18h = 0 \implies 16h^2 + (2c-10)h + c = 0, \quad (48.200)$$

whose two roots

$$h_\pm = \frac{5 - c \pm \sqrt{(c-1)(c-25)}}{16} \quad (48.201)$$

are exactly $h_{1,2}$ and $h_{2,1}$ of the Kac formula: a degenerate field whose null vector sits at level $rs = 2$ must have $rs = 2$, i.e. $(r,s) = (2,1)$ or $(1,2)$. We record the resulting null vector, valid whenever h is one of Eq. (48.201):

$$\boxed{\Big(L_{-2} - \tfrac{3}{2(2h+1)}\,L_{-1}^2\Big)|h\rangle = 0}. \quad (48.202)$$

It is useful to record the value of the coefficient in the background-charge variables of Eq. (48.173). For $\phi_{2,1}$, the charge is $\alpha_{2,1} = -\frac{1}{2}\alpha_+$, and a short computation using $\alpha_+\alpha_- = -1$ gives $h_{2,1} = \frac{1}{4}(3\alpha_+^2 - 2)$, so $2h+1 = \frac{3}{2}\alpha_+^2$ and

$$\kappa \equiv \frac{3}{2(2h+1)} = \frac{1}{\alpha_+^2} \qquad (\text{for } \phi_{2,1}), \quad (48.203)$$

with the symmetric statement $\kappa = 1/\alpha_-^2$ for $\phi_{1,2}$. This compact form will streamline the fusion computation below.

48.5.4 The BPZ differential equations

We can now state the central result of the section. A null vector is, by definition, a state that decouples: it has zero norm and zero overlap with every state, so as an operator insertion it vanishes identically inside correlators. Translating the vanishing into the language of Eq. (48.191) turns it into a differential equation.

Theorem 48.30 (BPZ null-vector equations)**.** *Let ϕ be a degenerate primary of weight h whose Verma module contains a singular vector at level N,*

$$|\chi\rangle = \mathcal{D}\big(\{L_{-k}\}\big)\,|h\rangle = 0, \quad (48.204)$$

where $\mathcal{D}$ is a degree-N polynomial in the modes L_{-k} ($k \geq 1$), homogeneous of weight N. Then in any correlation function containing ϕ, the function

$$G = \big\langle \phi(z)\,\phi_1(z_1)\cdots\phi_N(z_N)\big\rangle \quad (48.205)$$

satisfies the N-th order linear partial differential equation

$$\widehat{\mathcal{D}}\,G = 0, \qquad \widehat{\mathcal{D}} = \mathcal{D}(\{L_{-k}\})\Big|_{L_{-1}\to\partial_z,\ L_{-k}\to\mathcal{L}_{-k}\ (k\geq 2)}, \tag{48.206}$$

with $\mathcal{L}_{-k}$ the differential operators of Eq. (48.191). In particular, for a level-two degenerate field Eq. (48.154) the correlator obeys the second-order equation

$$\left\{\frac{3}{2(2h+1)}\,\partial_z^2 - \sum_i\left[\frac{h_i}{(z_i-z)^2} - \frac{1}{z_i-z}\,\partial_{z_i}\right]\right\}\langle\phi(z)\,\textstyle\prod_i\phi_i(z_i)\rangle = 0. \tag{48.207}$$

Proof. The state–operator correspondence (Chapter 47) identifies the singular vector $|\chi\rangle$ with a descendant field $\chi(z) = (\mathcal{D}\phi)(z)$, built from ϕ by the contour prescription Eq. (48.184) applied mode by mode. Because $|\chi\rangle = 0$ as a state, $\chi(z)$ is the zero operator; hence $\langle\chi(z)\prod_i\phi_i\rangle = 0$ for every choice of spectators. By Eq. (48.191) each factor L_{-k} in $\mathcal{D}$ acting on ϕ inside the correlator is reproduced by the differential operator $\mathcal{L}_{-k}$ (with $L_{-1}\to\partial_z$), and these operators act only on the position z of ϕ and the spectator coordinates, never reintroducing the stress tensor. Nested modes compose as nested differential operators because the contour manipulation Eq. (48.184) is linear and the operators $\mathcal{L}_{-k}$ act on the same correlator. Thus $\langle\chi\prod_i\phi_i\rangle = \widehat{\mathcal{D}}\,G$, and its vanishing is Eq. (48.206). For the level-two case, $\mathcal{D} = L_{-2} - \frac{3}{2(2h+1)}L_{-1}^2$, so $\widehat{\mathcal{D}} = \mathcal{L}_{-2} - \frac{3}{2(2h+1)}\partial_z^2$, and using $\mathcal{L}_{-2} = \sum_i[h_i/(z_i-z)^2 - \partial_{z_i}/(z_i-z)]$ from Eq. (48.191) yields Eq. (48.207). □

The order of the equation equals the level of the null vector, so a $\phi_{r,s}$ with its singular vector at level rs obeys an ODE of order rs once global conformal invariance is used to remove the trivial dependence. This is the precise sense in which "degeneracy solves the theory": every correlator with a degenerate insertion is constrained to lie in a finite-dimensional solution space of an ODE, and the physical answer is pinned down by monodromy (single-valuedness) and the operator-product data.

Reduction to a hypergeometric equation. Consider the four-point function of one level-two degenerate field ϕ (weight h) with three further primaries. Global conformal invariance — the $\mathfrak{sl}(2)\oplus\overline{\mathfrak{sl}}(2)$ Ward identities of Chapter 47 — reduces any four-point function to a universal prefactor times a single function of the anharmonic ratio. Writing the holomorphic cross-ratio

$$\eta = \frac{(z-z_1)(z_2-z_3)}{(z-z_3)(z_2-z_1)}, \tag{48.208}$$

we may strip the kinematic prefactor and set $G = (\text{prefactor})\,\mathcal{F}(\eta)$. The second-order operator in Eq. (48.207) then becomes a second-order *ordinary* differential operator in η. Because the only singularities of the correlator are at coincidences of operators, the equation for $\mathcal{F}$ is Fuchsian with regular singular points only at

$\eta = 0, 1, \infty$ (the three coincidences $z \to z_1$, and the two others mapped to $1, \infty$). A second-order Fuchsian equation with exactly three regular singular points is, after the standard Möbius normalization, the Gauss hypergeometric equation

$$\eta(1-\eta)\,\mathcal{F}'' + [\gamma - (\alpha+\beta+1)\eta]\,\mathcal{F}' - \alpha\beta\,\mathcal{F} = 0, \tag{48.209}$$

whose parameters α, β, γ are fixed by the indicial exponents at the singular points. We compute those exponents directly from Eq. (48.207): as $z \to z_1$ the correlator behaves like $(z-z_1)^{\rho}$, and the leading $(z-z_1)^{\rho-2}$ terms of Eq. (48.207) come from $\frac{3}{2(2h+1)}\partial_z^2$ acting on $(z-z_1)^{\rho}$, from the double pole $-h_1/(z_1-z)^2$, and from the single-pole derivative term, giving the indicial equation

$$\frac{3}{2(2h+1)}\,\rho(\rho-1) + \rho - h_1 = 0. \tag{48.210}$$

Its two roots $\rho_\pm$ are the leading powers in the operator product $\phi(z)\phi_1(z_1) \sim \sum_p (z-z_1)^{h_p - h - h_1}[\phi_p + \cdots]$, so $\rho_\pm = h_p^{\pm} - h - h_1$ for the two intermediate weights $h_p^{\pm}$ allowed in the OPE. The two solutions of Eq. (48.209) are precisely the two *conformal blocks* of Chapter 49: holomorphic functions encoding the exchange of a single primary and its descendants. The physical correlator is assembled by gluing holomorphic and antiholomorphic blocks,

$$\langle \textstyle\prod \Phi_i \rangle = \sum_{p,\bar{p}} X_{p\bar{p}}\,\mathcal{F}_p(\eta)\,\overline{\mathcal{F}}_{\bar{p}}(\bar{\eta}), \tag{48.211}$$

with the coefficients $X_{p\bar{p}}$ fixed by single-valuedness (no branch cut as η circles a singular point) and by the symmetry of the correlator under permuting identical operators (crossing). This *holomorphic factorization* — correlators are sesquilinear combinations of chiral blocks — is the structural lesson the higher-dimensional bootstrap of Chapters 49 and 50 will generalize.

Worked Example 48.31 (The four-point ODE and its monodromy data)**.** We make the reduction concrete. Use a global conformal transformation to send $z_1 \to 0$, $z_2 \to 1$, $z_3 \to \infty$ and let the degenerate field sit at $z \to \eta$. Stripping a prefactor $z^{\,a}(1-z)^{\,b}$ chosen to remove the indicial behavior at $\eta = 0$ and $\eta = 1$, Eq. (48.207) collapses to Eq. (48.209). The exponents at $\eta = 0$ are the two roots of Eq. (48.210); writing $\kappa = \frac{3}{2(2h+1)}$, these are

$$\rho_\pm = \frac{(\kappa-1) \pm \sqrt{(\kappa-1)^2 + 4\kappa h_1}}{2\kappa}, \tag{48.212}$$

and their difference governs the local monodromy. For the spin field of the Ising model, $\phi = \sigma = \phi_{1,2}$ with $h = \frac{1}{16}$, so $\kappa = \frac{3}{2(2\cdot\frac{1}{16}+1)} = \frac{3}{2\cdot\frac{9}{8}} = \frac{4}{3}$, and with all four operators equal to σ ($h_1 = \frac{1}{16}$) the indicial equation $\frac{4}{3}\rho(\rho-1) + \rho - \frac{1}{16} = 0$ has roots $\rho_+ = \frac{1}{16}$ and $\rho_- = -\frac{1}{8}$ wait: solving $\frac{4}{3}\rho^2 - \frac{1}{3}\rho - \frac{1}{16} = 0$, i.e. $\rho^2 - \frac{1}{4}\rho - \frac{3}{64} = 0$, gives $\rho = \frac{1}{8} \pm \frac{1}{8}\sqrt{1+3} = \frac{1}{8} \pm \frac{1}{4}$, i.e. $\rho_+ = \frac{3}{8}$ and $\rho_- = -\frac{1}{8}$. These correspond to the two OPE channels $\sigma \times \sigma = \mathbf{1} + \varepsilon$: the exponent $\rho = -\frac{1}{8} = h_{\mathbf{1}} - 2h_\sigma = 0 - \frac{1}{8}$ is the identity channel, and $\rho = \frac{3}{8} = h_\varepsilon - 2h_\sigma = \frac{1}{2} - \frac{1}{8}$ is the energy channel. The full closed-form solution of this equation — the celebrated $\langle\sigma\sigma\sigma\sigma\rangle$ of the Ising model — is carried out in Section 48.6.

48.5.5 Fusion rules and the closure of the algebra

The differential equations also tell us *which* operators can appear in an operator product, and that is what makes a minimal model a closed theory rather than a representation. Consider a three-point function with one level-two degenerate field ϕ (weight h, at z) and two further primaries ϕ_1, ϕ_2 (weights h_1, h_2, at z_1, z_2). Conformal invariance fixes its holomorphic form completely (Eq. (47.137)),

$$G_3 = \langle \phi(z)\phi_1(z_1)\phi_2(z_2)\rangle = C\,(z-z_1)^{\gamma_1}(z-z_2)^{\gamma_2}(z_1-z_2)^{\gamma_{12}}, \tag{48.213}$$

with exponents set by the standard rule (the exponent of $(x_i - x_j)$ is $h_k - h_i - h_j$ where k is the third operator):

$$\gamma_1 = h_2 - h - h_1, \qquad \gamma_2 = h_1 - h - h_2, \qquad \gamma_{12} = h - h_1 - h_2. \tag{48.214}$$

A nonzero three-point function means ϕ_2 appears in the operator product $\phi \times \phi_1$. The BPZ equation Eq. (48.207) (now with two spectators) must hold; substituting Eq. (48.213) converts it into algebraic constraints on the weights, and *those constraints are the fusion rules.* We compute. Abbreviate $A = z - z_1$, $B = z - z_2$, and note $z_1 - z_2 = B - A$. Differentiating Eq. (48.213),

$$\partial_z^2 G_3 = G_3\left[\left(\tfrac{\gamma_1}{A} + \tfrac{\gamma_2}{B}\right)^2 - \tfrac{\gamma_1}{A^2} - \tfrac{\gamma_2}{B^2}\right], \tag{48.215}$$

$$\frac{\partial_{z_1} G_3}{z_1 - z} = G_3\left[\tfrac{\gamma_1}{A^2} - \tfrac{\gamma_{12}}{A\,(z_1-z_2)}\right], \qquad \frac{\partial_{z_2} G_3}{z_2 - z} = G_3\left[\tfrac{\gamma_2}{B^2} + \tfrac{\gamma_{12}}{B\,(z_1-z_2)}\right], \tag{48.216}$$

where in the latter two we used $\partial_{z_1} G_3 = G_3[-\gamma_1/A + \gamma_{12}/(z_1 - z_2)]$ and $z_1 - z = -A$ (and the mirror for z_2). Inserting these and the double-pole terms $-h_1/A^2 - h_2/B^2$ into Eq. (48.207), dividing out G_3, and using $\frac{1}{z_1-z_2}(\frac{1}{B} - \frac{1}{A}) = -\frac{1}{AB}$ to combine the cross terms, the BPZ condition becomes

$$\frac{1}{A^2}\Big[\kappa\,\gamma_1(\gamma_1 - 1) + \gamma_1 - h_1\Big] + \frac{1}{B^2}\Big[\kappa\,\gamma_2(\gamma_2 - 1) + \gamma_2 - h_2\Big] + \frac{1}{AB}\Big[2\kappa\,\gamma_1\gamma_2 - \gamma_{12}\Big] = 0, \tag{48.217}$$

with $\kappa = \frac{3}{2(2h+1)}$. Because z is free, A and B are independent variables, so each bracket must vanish separately:

$$\text{(I)} \quad \kappa\,\gamma_1(\gamma_1 - 1) + \gamma_1 - h_1 = 0, \tag{48.218}$$

$$\text{(II)} \quad \kappa\,\gamma_2(\gamma_2 - 1) + \gamma_2 - h_2 = 0, \tag{48.219}$$

$$\text{(III)} \quad 2\kappa\,\gamma_1\gamma_2 - \gamma_{12} = 0. \tag{48.220}$$

Equation (I) is a *quadratic in* h_2 (through $\gamma_1 = h_2 - h - h_1$): given h and h_1, only two values of h_2 can appear. This is the truncation. To identify them, pass to the charge variables of Eq. (48.173), writing $h = \alpha(\alpha - 2\alpha_0)$, $h_1 = \beta(\beta - 2\alpha_0)$, $h_2 = \beta'(\beta' - 2\alpha_0)$ with $\alpha = \alpha_{2,1} = -\frac{1}{2}\alpha_+$ and $\kappa = 1/\alpha_+^2$ from Eq. (48.203). We claim the two solutions are

$$\beta' = \beta \pm \alpha, \qquad \text{i.e.} \qquad \alpha_2 = \alpha_1 \pm \alpha_{2,1}. \tag{48.221}$$

Verification of the first root $\beta' = \beta + \alpha$. Then

$$\gamma_1 = h_2 - h - h_1 = (\beta+\alpha)(\beta+\alpha-2\alpha_0) - \alpha(\alpha-2\alpha_0) - \beta(\beta-2\alpha_0) = 2\alpha\beta. \quad (48.222)$$

Using $\kappa\alpha^2 = \frac{1}{\alpha_+^2}\cdot\frac{\alpha_+^2}{4} = \frac{1}{4}$ and $\alpha_- = -1/\alpha_+$, equation (I) evaluates to

$$\begin{aligned}\kappa\gamma_1^2 + (1-\kappa)\gamma_1 - h_1 &= 4\kappa\alpha^2\beta^2 + 2\alpha\beta(1-\kappa) - \big(\beta^2 - (\alpha_+ + \alpha_-)\beta\big) \\ &= \beta^2 - \alpha_+\beta + \tfrac{\beta}{\alpha_+} - \beta^2 + (\alpha_+ + \alpha_-)\beta = \beta\Big(\tfrac{1}{\alpha_+} + \alpha_-\Big) = 0,\end{aligned} \quad (48.223)$$

since $1/\alpha_+ + \alpha_- = 1/\alpha_+ - 1/\alpha_+ = 0$. The mirror computation with $\beta' = \beta - \alpha$ gives $\gamma_1 = 2\alpha(2\alpha_0 - \beta)$ and, using $-\alpha_+ + 1/\alpha_+ = -2\alpha_0$, again reduces (I) to 0. Both roots check, so the two allowed intermediate charges are exactly Eq. (48.221). One verifies likewise that (III) holds: for $\beta' = \beta + \alpha$ both $2\kappa\gamma_1\gamma_2$ and γ_{12} equal $-2\beta(\beta+\alpha-2\alpha_0)$, and (II) follows from (I) by the symmetry $1 \leftrightarrow 2$ together with charge conservation. The three constraints are therefore consistent, and Eq. (48.221) is the complete content of the level-two BPZ equation on the three-point function.

Translating Eq. (48.221) back through $\alpha_{r,s} = \frac{1-r}{2}\alpha_+ + \frac{1-s}{2}\alpha_-$, the shift by $\alpha_{2,1} = -\frac{1}{2}\alpha_+$ moves r by one unit and leaves s fixed: $\alpha_1 + \alpha_{2,1} = \alpha_{r+1,s}$ and $\alpha_1 - \alpha_{2,1} = \alpha_{r-1,s}$. Hence

$$\phi_{2,1} \times \phi_{r,s} = \phi_{r-1,s} + \phi_{r+1,s}, \quad (48.224)$$

and, by the symmetric computation for the $\phi_{1,2}$ null vector (which shifts s),

$$\phi_{1,2} \times \phi_{r,s} = \phi_{r,s-1} + \phi_{r,s+1}. \quad (48.225)$$

These elementary fusions generate the rest. Repeatedly fusing with $\phi_{2,1}$ and $\phi_{1,2}$ — equivalently, noting that the level-$r_1 s_1$ null vector of ϕ_{r_1,s_1} imposes an analogous truncation in each index — gives the general *BPZ fusion rule*

$$\boxed{\phi_{r_1,s_1} \times \phi_{r_2,s_2} = \sum_{r_3=|r_1-r_2|+1}^{r_1+r_2-1}{\sum_{s_3=|s_1-s_2|+1}^{s_1+s_2-1}}' \phi_{r_3,s_3}}, \quad (48.226)$$

where both sums run in steps of two (the primes). The ranges are the $\mathfrak{su}(2)$ Clebsch–Gordan ranges in each index: a degenerate field $\phi_{r,s}$ behaves, under fusion, like a pair of spins $j = (r-1)/2$ and $j' = (s-1)/2$, and Eq. (48.226) is the tensor-product decomposition $j_1 \otimes j_2 = \bigoplus_{|j_1-j_2|}^{j_1+j_2}$ in both. The truncation *from above* ($r_3 \le r_1 + r_2 - 1$, $s_3 \le s_1 + s_2 - 1$) is the null-vector decoupling we just derived.

Proposition 48.32 (Closure and finiteness)**.** *In the minimal model $\mathcal{M}(p,p')$ the fusion rule Eq. (48.226), supplemented by the Kac identification Eq. (48.179), maps the finite Kac table to itself: the operator product of any two table primaries contains only table primaries. The operator algebra therefore closes on the $\mathcal{N} = (p-1)(p'-1)/2$ fields of Theorem 48.28, and $\mathcal{M}(p,p')$ is a complete, consistent conformal field theory.*

Proof. Take ϕ_{r_1,s_1} and ϕ_{r_2,s_2} inside the table, $1 \le r_i \le p'-1$, $1 \le s_i \le p-1$. The naive ranges in Eq. (48.226) can produce labels with $r_3 \ge p'$ or $s_3 \ge p$, which lie outside the rectangle. Such labels are removed and reflected back by a *second* truncation, dynamically equal to the geometric reflection of Theorem 48.28: the same null-vector mechanism that bounds r_3 from above by r_1+r_2-1 also forbids r_3 from reaching p', because the field $\phi_{p',s}$ has $h_{p',s} = h_{0,p-s}$ by Eq. (48.179) and the corresponding fusion channel has vanishing structure constant. Concretely, the surviving range is

$$|r_1 - r_2| + 1 \le r_3 \le \min\left(r_1 + r_2 - 1,\ 2p' - 1 - r_1 - r_2\right), \tag{48.227}$$

in steps of two, and the mirror bound for s_3 with $p' \to p$. The upper cutoff $2p'-1-r_1-r_2$ is exactly the reflection of r_1+r_2-1 through the wall $r=p'$, so whenever it is active the would-be out-of-table fields are folded back via $(r_3,s_3) \to (p'-r_3, p-s_3)$ onto genuine table entries. In either case $r_3 \le p'-1$ and $s_3 \le p-1$: the product stays in the table. Since the table is finite (Eq. (48.180)) and every operator product re-expands on it, the four-point functions are determined by the three-point structure constants and the conformal blocks (Eq. (48.211)), and crossing symmetry — automatic here because the BPZ ODE has a unique single-valued solution — fixes the structure constants up to normalization. Hence all correlators are computable, which is the definition of a complete solvable theory. □

Caution

The fusion "coefficients" in Eq. (48.226) are 0 or 1 for the unitary diagonal minimal models, but *fusion* is not the same as the *operator-product structure constant*. Fusion Eq. (48.226) states which families *may* appear; whether a given one actually appears with nonzero coefficient, and with what magnitude, is the dynamical content of the structure constants C_{ijk}, computed from the four-point blocks via crossing (the Dotsenko–Fateev Coulomb-gas integrals, touched on in Section 48.7.4). A vanishing fusion entry forbids the operator; a nonvanishing one permits it.

The pieces are now in place to solve a complete theory. The simplest minimal model is $\mathcal{M}(4,3)$ with $c = \frac{1}{2}$, the Ising universality class — the exactly solvable representative of the same fixed point that ϕ^4 theory reaches at Wilson–Fisher

in $d = 4 - \epsilon$ (Chapter 52), and whose three-dimensional cousin the bootstrap of Chapter 50 will pin down numerically. Its three primaries are the ones read off the Kac table in Fig. 48.4: the identity $\mathbf{1}$ ($h = 0$), the spin $\sigma = \phi_{1,2}$ ($h = \frac{1}{16}$, so $\Delta_\sigma = \frac{1}{8}$), and the energy $\varepsilon = \phi_{2,1}$ ($h = \frac{1}{2}$, so $\Delta_\varepsilon = 1$); its fusion rules follow from Eq. (48.226) as $\sigma \times \sigma = \mathbf{1} + \varepsilon$, $\sigma \times \varepsilon = \sigma$, $\varepsilon \times \varepsilon = \mathbf{1}$; and the four-point function of spins is governed by the level-two BPZ equation Eq. (48.207) whose monodromy data we already extracted in Theorem 48.31. The next section (Section 48.6) takes exactly these inputs and solves the theory end to end, integrating the hypergeometric equation to obtain $\langle\sigma\sigma\sigma\sigma\rangle$ in closed form and confirming $c = \frac{1}{2}$ against the free Majorana fermion.

48.6 The Ising CFT Worked in Full

The previous section assembled the abstract machinery: the minimal models $\mathcal{M}(p, p')$, their finite Kac table of primaries (Theorem 48.28), the fusion rules that close that table into an algebra (Eq. (48.226)), and the BPZ null-vector equations that turn any correlator containing a degenerate field into a linear differential equation (Theorem 48.30). All of that was developed in the abstract, for general (p, p'). We now spend an entire section cashing it in for a single theory, the simplest nontrivial minimal model $\mathcal{M}(4, 3)$, and we carry the computation to the very end: the exact operator content, the exact fusion algebra, and the four-point function of the spin field in *closed form*, extracted from the level-2 BPZ equation handed to us by the last section. This is the bootstrap dream made fully concrete — a complete, interacting, unitary quantum field theory solved with no Lagrangian, no path integral, and no perturbation series, by representation theory alone.

The theory we are solving is not a toy. It is the universality class of the two-dimensional Ising model at its critical temperature, the same fixed point that Onsager reached in 1944 by a heroic transfer-matrix diagonalization. It is also the exactly solvable $d = 2$ face of the Wilson–Fisher fixed point that ϕ^4 theory reaches in $d = 4 - \epsilon$ (forward to Chapter 52): the critical ϕ^4 theory with its $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ flows, in the infrared, to a conformal field theory, and in two dimensions that CFT is exactly $\mathcal{M}(4, 3)$. The two relevant scaling operators of the $\mathbb{Z}_2$-symmetric problem — the order parameter (odd under $\mathbb{Z}_2$) and the energy density (even) — are the spin σ and the energy ε we are about to pin down. The conformal dimensions we will compute here, $\Delta_\sigma = \frac{1}{8}$ and $\Delta_\varepsilon = 1$, are the $d = 2$ endpoints of a curve of dimensions $\Delta_\sigma(d), \Delta_\varepsilon(d)$ running from the Gaussian values at $d = 4$ down through the genuinely interacting $d = 3$ values — and those $d = 3$ values are precisely what the numerical bootstrap of Chapter 50 computes to many digits. Here the answer is not numerical but algebraic and exact: the calibration target against which the higher-dimensional bootstrap is checked.

We proceed in four movements. First we read the three Ising primaries straight off the Kac table and verify that the table's internal identification leaves *exactly* three. Second we specialize the fusion rules and exhibit the $\mathbb{Z}_2$ fusion algebra. Third — the centerpiece — we solve the spin four-point function: we reduce the level-2 BPZ equation to a hypergeometric ordinary differential equation, integrate it in closed form, glue the two chiral solutions into a single-valued crossing-symmetric correlator, and read off the operator product coefficient $C_{\sigma\sigma\varepsilon} = \frac{1}{2}$. Fourth we answer the question this construction provokes — *where does $c = \frac{1}{2}$ come from microscopically?* — by exhibiting the free Majorana fermion, computing its central charge and the spin field's anomalous dimension from Wick contractions, and translating every CFT number into the standard language of critical exponents, where it reproduces Onsager exactly.

48.6.1 The three primaries from the Kac table

Recall the data of a minimal model $\mathcal{M}(p,p')$ with coprime integers $p > p' \geq 2$. The central charge (Eq. (48.176)) is

$$c = 1 - \frac{6(p-p')^2}{p\,p'}, \tag{48.228}$$

and the holomorphic weights of the primaries fill the Kac table (Theorem 48.28),

$$h_{r,s} = \frac{(p\,r - p'\,s)^2 - (p-p')^2}{4\,p\,p'}, \qquad 1 \leq r \leq p'-1, \quad 1 \leq s \leq p-1, \tag{48.229}$$

subject to the reflection identification

$$h_{r,s} = h_{p'-r,\,p-s}, \tag{48.230}$$

which folds the $(p'-1)(p-1)$ rectangle in half, so that the number of distinct primaries is $\frac{1}{2}(p-1)(p'-1)$.

The Ising model is the smallest nontrivial case, $p=4$, $p'=3$. Then $p-p'=1$ and Eq. (48.228) gives at once the central charge of the theory,

$$\boxed{c = 1 - \frac{6\cdot 1^2}{4\cdot 3} = 1 - \frac{1}{2} = \frac{1}{2}\,.} \tag{48.231}$$

The Kac rectangle has $r \in \{1,2\}$ and $s \in \{1,2,3\}$, six entries in all. With $p\,r - p'\,s = 4r - 3s$ and $4pp' = 48$, Eq. (48.229) reads $h_{r,s} = \big[(4r-3s)^2 - 1\big]/48$. Evaluating every cell:

$$\begin{aligned} h_{1,1} &= \tfrac{(4-3)^2-1}{48} = \tfrac{0}{48} = 0, & h_{1,2} &= \tfrac{(4-6)^2-1}{48} = \tfrac{3}{48} = \tfrac{1}{16}, & h_{1,3} &= \tfrac{(4-9)^2-1}{48} = \tfrac{24}{48} = \tfrac{1}{2}, \\ h_{2,1} &= \tfrac{(8-3)^2-1}{48} = \tfrac{24}{48} = \tfrac{1}{2}, & h_{2,2} &= \tfrac{(8-6)^2-1}{48} = \tfrac{3}{48} = \tfrac{1}{16}, & h_{2,3} &= \tfrac{(8-9)^2-1}{48} = \tfrac{0}{48} = 0. \end{aligned} \tag{48.232}$$

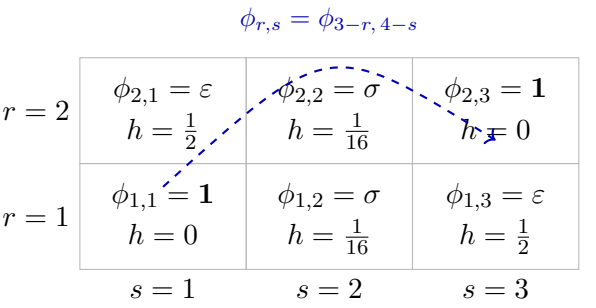


Figure 48.5: The Kac table of $\mathcal{M}(4,3)$, the two-dimensional Ising CFT. The 2×3 rectangle of weights $h_{r,s}$ is folded in half by the point-reflection $\phi_{r,s} = \phi_{3-r,4-s}$ (one identification shown), so the six cells collapse to three distinct primaries: the identity $\mathbf{1}$ ($h = 0$), the spin σ ($h = \frac{1}{16}$) and the energy ε ($h = \frac{1}{2}$).

The reflection Eq. (48.230), here $h_{r,s} = h_{3-r,\,4-s}$, pairs the cells across the table: $(1,1) \leftrightarrow (2,3)$, $(1,2) \leftrightarrow (2,2)$, and $(1,3) \leftrightarrow (2,1)$, exactly as the numerical coincidences in Eq. (48.232) show. Three pairs, hence *exactly three* distinct holomorphic weights:

$$h \in \left\{ 0,\ \tfrac{1}{16},\ \tfrac{1}{2} \right\}. \tag{48.233}$$

Figure 48.5 displays the table and its folding.

We now name them and record their full scaling data. A minimal model is diagonal: each primary appears with equal holomorphic and antiholomorphic weight, $\bar{h} = h$, so that the total scaling dimension is $\Delta = h + \bar{h} = 2h$ and the Lorentz spin $\ell = h - \bar{h} = 0$. These are spinless scalar operators, as the order parameter and energy density of a statistical model must be.

Definition 48.33 (The three Ising primaries)**.** The minimal model $\mathcal{M}(4,3)$ with central charge $c = \frac{1}{2}$ (Eq. (48.231)) contains exactly three Virasoro primaries, all scalars ($\bar{h} = h$):

- the *identity* $\mathbf{1} = \phi_{1,1}$, with $h = \bar{h} = 0$, $\Delta_{\mathbf{1}} = 0$;
- the *spin* (order parameter) $\sigma = \phi_{1,2} = \phi_{2,2}$, with $h = \bar{h} = \frac{1}{16}$, hence

$$\Delta_\sigma = 2h_\sigma = \frac{1}{8}; \tag{48.234}$$

- the *energy* density $\varepsilon = \phi_{2,1} = \phi_{1,3}$, with $h = \bar{h} = \frac{1}{2}$, hence

$$\Delta_\varepsilon = 2h_\varepsilon = 1. \tag{48.235}$$

Two features deserve immediate comment, because both feed the rest of the section. First, $\sigma = \phi_{1,2}$ and $\varepsilon = \phi_{2,1}$ are *degenerate at level two*: a field $\phi_{r,s}$ carries

a Virasoro null descendant at level rs, and here $rs = 2$. A level-two degeneracy produces a *second-order* BPZ ordinary differential equation for any correlator containing that field — this is precisely why the spin four-point function is solvable in closed form, and we will exploit it in Section 48.6.3. Second, the assignment to the statistical operators is forced by symmetry: under the $\mathbb{Z}_2$ of the Ising model, σ is odd (it is the magnetization) and ε is even (it is the local energy, conjugate to temperature), and this parity is exactly what the fusion rules below encode.

Physical Insight

The entire spectrum of a $c = \frac{1}{2}$ unitary CFT is three numbers, $\{0, \frac{1}{16}, \frac{1}{2}\}$, fixed by nothing but the requirement that the Verma modules built on them close into a consistent, unitary, modular-invariant theory. No coupling constant, no Lagrangian, no input from a microscopic Hamiltonian entered. That a statistical-mechanical model as concrete as a grid of ± 1 spins should, at its critical point, be governed by this rigid algebraic skeleton is the physical content of universality: the lattice details wash out, and what survives is a representation of the Virasoro algebra.

48.6.2 The fusion algebra

The fusion rules tell us which primaries appear in the operator product expansion of two others. For minimal models they are determined entirely by the null-vector structure: the general result (Eq. (48.226)) is that $\phi_{r_1,s_1} \times \phi_{r_2,s_2}$ contains ϕ_{r_3,s_3} if and only if r_3 and s_3 run over the truncated ranges

$$r_3 = |r_1 - r_2| + 1,\ |r_1 - r_2| + 3, \ldots, \min(r_1 + r_2 - 1,\ 2p' - 1 - r_1 - r_2), \tag{48.236}$$

and likewise for s_3 with $p' \to p$, the two indices stepping by two and truncating both from below (the $\mathfrak{su}(2)$-like triangle rule) and from above (the minimal-model cutoff that keeps the result inside the Kac table). For Ising it is cleanest to work with the representatives $\mathbf{1} = \phi_{1,1}$, $\sigma = \phi_{1,2}$, $\varepsilon = \phi_{1,3}$, all in the first row $r = 1$, so that the r-fusion is trivial ($r_3 = 1$ always) and only the s-index, with $1 \le s \le 3$, is in play.

Fusing two spins, $\phi_{1,2} \times \phi_{1,2}$: the s-channel range is $s_3 = |2 - 2| + 1, \cdots = 1, 3, \ldots$ stepping by two, i.e. $s_3 \in \{1, 3\}$ (the value $s_3 = 5$ is excluded by the upper cutoff $s_3 \le \min(2 + 2 - 1,\ 2 \cdot 4 - 1 - 2 - 2) = \min(3, 3) = 3$). Thus

$$\sigma \times \sigma = \phi_{1,1} + \phi_{1,3} = \mathbf{1} + \varepsilon. \tag{48.237}$$

Fusing spin with energy, $\phi_{1,2} \times \phi_{1,3}$: $s_3 = |2 - 3| + 1, \cdots = 2, 4, \ldots$, truncated by $s_3 \le \min(2 + 3 - 1,\ 2 \cdot 4 - 1 - 2 - 3) = \min(4, 2) = 2$, leaving only $s_3 = 2$, so

$$\sigma \times \varepsilon = \phi_{1,2} = \sigma. \tag{48.238}$$

Fusing two energies, $\phi_{1,3} \times \phi_{1,3}$: $s_3 = 1, 3, 5, \ldots$ truncated by $s_3 \le \min(3+3-1,\, 2\cdot 4 - 1 - 3 - 3) = \min(5, 1) = 1$, so only $s_3 = 1$ survives and

$$\varepsilon \times \varepsilon = \phi_{1,1} = \mathbf{1}. \tag{48.239}$$

Together with the trivial fusions $\mathbf{1} \times X = X$, these are the complete Ising fusion rules:

$$\boxed{\sigma \times \sigma = \mathbf{1} + \varepsilon, \qquad \sigma \times \varepsilon = \sigma, \qquad \varepsilon \times \varepsilon = \mathbf{1}.} \tag{48.240}$$

Physical Insight

Equation (48.240) is the $\mathbb{Z}_2$ structure of the Ising model made algebraic. Assign the multiplicative parities $\mathbf{1} \to +1$, $\varepsilon \to +1$, $\sigma \to -1$ (odd order parameter, even energy). Every fusion in Eq. (48.240) conserves this parity: $\sigma \times \sigma$ is even and yields the even sector $\{\mathbf{1}, \varepsilon\}$; $\sigma \times \varepsilon$ is odd and yields σ; $\varepsilon \times \varepsilon$ is even and yields $\mathbf{1}$. The fusion algebra is therefore the group algebra of $\mathbb{Z}_2$ *enriched* by the splitting $\sigma \times \sigma = \mathbf{1} + \varepsilon$ — the simplest example of a non-group fusion category (the Ising category), in which σ is a "non-invertible" object whose self-fusion produces more than one channel.

The decisive entry for what follows is Eq. (48.237): the product of two spins contains exactly two primaries, the identity and the energy. The four-point function $\langle\sigma\sigma\sigma\sigma\rangle$ will therefore be built from exactly *two* conformal blocks, one for each channel, and the relative weight of the second is the squared OPE coefficient $C^2_{\sigma\sigma\varepsilon}$. We now compute it.

48.6.3 The spin four-point function from the BPZ equation

This is the centerpiece. We will obtain the full four-point function $\langle\sigma\sigma\sigma\sigma\rangle$ in closed form by integrating the level-2 BPZ equation of Theorem 48.30, and along the way extract $C_{\sigma\sigma\varepsilon}$. The result is the following.

Theorem 48.34 (The Ising spin four-point function)**.** *In the Ising CFT $\mathcal{M}(4,3)$, with the spin field normalized so that its two-point function is $\langle\sigma(z_1,\bar z_1)\sigma(z_2,\bar z_2)\rangle = |z_{12}|^{-2\Delta_\sigma} = |z_{12}|^{-1/4}$ (Eq. (47.120)), the four-point function of spins is the single-valued, crossing-symmetric function*

$$\Big\langle \textstyle\prod_{i=1}^4 \sigma(z_i,\bar z_i) \Big\rangle = \Big|\frac{z_{13}\,z_{24}}{z_{12}\,z_{23}\,z_{34}\,z_{14}}\Big|^{1/4} \tfrac{1}{2}\Big(\big|1+\sqrt{1-x}\,\big| + \big|1-\sqrt{1-x}\,\big| \Big), \tag{48.241}$$

where $z_{ij} = z_i - z_j$ and $x = \dfrac{z_{12}z_{34}}{z_{13}z_{24}}$ is the holomorphic cross-ratio (and $\bar x$ its conjugate). The two terms are the contributions of the identity and of ε to the $\sigma \times \sigma$ OPE, and the operator product coefficient is

$$C_{\sigma\sigma\varepsilon} = \frac{1}{2}. \tag{48.242}$$

The proof occupies the rest of this subsection and is carried out in full in Theorem 48.35; the structure is: (i) reduce the kinematics to a function of the cross-ratio; (ii) convert the level-2 BPZ null condition, together with the global conformal Ward identities, into a second-order ordinary differential equation; (iii) recognize that ODE as hypergeometric and integrate it in elementary closed form; (iv) glue the holomorphic and antiholomorphic solutions into a single-valued physical correlator and read off $C_{\sigma\sigma\varepsilon}$.

Worked Example 48.35 (The four-point function of Ising spins, end to end)**.** *Step 1 — Kinematics.* By global conformal invariance — the $SL(2,\mathbb{C})$ Möbius subgroup generated by L_{-1}, L_0, L_1 and their antiholomorphic partners — a four-point function of scalar primaries of equal weight $(h, \bar{h})$ is fixed up to a function of the cross-ratio. Writing $z_{ij} = z_i - z_j$, choose the prefactor $(z_{12}z_{34})^{-2h}$, which carries holomorphic weight h at each of the four points (it transforms with weight h at z_1, z_2 through z_{12}^{-2h} and at z_3, z_4 through z_{34}^{-2h}). Then

$$\left\langle \prod_i \sigma(z_i, \bar{z}_i) \right\rangle = (z_{12}z_{34})^{-2h}\,(\bar{z}_{12}\bar{z}_{34})^{-2\bar{h}}\,G(x,\bar{x}), \qquad x = \tfrac{z_{12}z_{34}}{z_{13}z_{24}}, \tag{48.243}$$

with $h = \bar{h} = \frac{1}{16}$, so $2h = \frac{1}{8}$. The whole problem is to determine the dimensionless function $G(x,\bar{x})$. This is the four-point analogue of the two- and three-point structures (Eqs. (47.120) and (47.137)): the new ingredient at four points is the single undetermined function of the cross-ratio that symmetry cannot fix and dynamics must.

Step 2 — Holomorphic factorization. In a diagonal minimal model every operator carries $\bar{h} = h$, and correlators factorize holomorphically: the reduced holomorphic correlator $g(x)$ introduced in Step 3 and its antiholomorphic partner combine into $g(x)\,\overline{g(\bar{x})} = \sum_a F_a(x)\,\bar{F}_a(\bar{x})$, a sum over the conformal families a exchanged in the $\sigma \times \sigma$ OPE, each a product of a chiral block $F_a(x)$ and its antichiral partner. By the fusion rule Eq. (48.237) there are exactly two families, $a \in \{\mathbf{1}, \varepsilon\}$. We therefore need the two chiral blocks $F_{\mathbf{1}}(x)$ and $F_\varepsilon(x)$; they are the two independent solutions of one second-order ODE, the BPZ equation, which we now derive.

Step 3 — The BPZ equation. The spin $\sigma = \phi_{1,2}$ has a level-two null descendant (Theorem 48.30); the associated null-state condition, inserted into a correlator, is the differential operator

$$\left[\frac{3}{2(2h+1)}\,\partial_z^2 - \sum_j \Big(\frac{1}{z-z_j}\,\partial_{z_j} + \frac{h_j}{(z-z_j)^2}\Big)\right]\left\langle \sigma(z)\,\prod_j \sigma(z_j)\right\rangle = 0, \tag{48.244}$$

where z is the position of the degenerate field and the sum runs over the other insertions. With $h = \frac{1}{16}$ the coefficient is $\frac{3}{2(2h+1)} = \frac{3}{2\cdot(9/8)} = \frac{4}{3}$.

To turn Eq. (48.244) into an ODE in a single variable we fix three points and let the degenerate field sit at the variable point. Place the four spins at $z_1 = 0$,

$z_2 = z$, $z_3 = 1$, $z_4 = L$ and define the reduced holomorphic correlator by sending the fourth point to infinity,

$$g(z) = \lim_{L\to\infty} L^{2h}\,\langle \sigma(0)\,\sigma(z)\,\sigma(1)\,\sigma(L)\rangle_{\text{hol}}, \tag{48.245}$$

the factor L^{2h} removing the leading fall-off of the operator at infinity. In this frame the cross-ratio becomes $x = \frac{z_{12}z_{34}}{z_{13}z_{24}} = \frac{(-z)(1-L)}{(-1)(z-L)} \to z$ as $L\to\infty$, so the degenerate point $z_2 = z$ *is* the cross-ratio and $\partial_{z_2} = \partial_x$ acts directly on g.

The derivatives $\partial_{z_1}, \partial_{z_3}, \partial_{z_4}$ in Eq. (48.244) are eliminated using the three global conformal Ward identities (the $SL(2)$ invariance of the correlator under L_{-1}, L_0, L_1):

$$\sum_i \partial_{z_i}\langle\cdots\rangle = 0, \qquad \sum_i \big(z_i\partial_{z_i} + h\big)\langle\cdots\rangle = 0, \qquad \sum_i \big(z_i^2\partial_{z_i} + 2hz_i\big)\langle\cdots\rangle = 0. \tag{48.246}$$

Write $a_i \equiv \partial_{z_i}\langle\cdots\rangle$ at $z_1 = 0, z_2 = x, z_3 = 1, z_4 = L$, and let $G_4 = \langle\cdots\rangle$. The three identities read

$$a_1 + a_2 + a_3 + a_4 = 0, \tag{48.247}$$

$$x\,a_2 + a_3 + L\,a_4 + 4h\,G_4 = 0, \tag{48.248}$$

$$x^2 a_2 + a_3 + L^2 a_4 + 2h(x+1+L)G_4 = 0. \tag{48.249}$$

Subtracting Eq. (48.248) from Eq. (48.249) eliminates a_3:

$$(x^2{-}x)\,a_2 + (L^2{-}L)\,a_4 + 2h(x{-}1{+}L)\,G_4 = 0 \;\Longrightarrow\; a_4 = -\,\frac{(x^2-x)a_2 + 2h(x-1+L)G_4}{L^2-L}. \tag{48.250}$$

Now take the limit. Since $G_4 \sim L^{-2h}g(x)$ at large L, the renormalized correlator is $g(x) = L^{2h}G_4$, and the leading large-L behaviour of Eq. (48.250) is $a_4 \to -2h\,L^{-2h-1}g$. Feeding this back through Eq. (48.248) gives $a_3 = -xa_2 - La_4 - 4hG_4 \to L^{-2h}\big(-xg' - 2hg\big)$, and then Eq. (48.247) gives $a_1 = -(a_2+a_3+a_4) \to L^{-2h}\big[(x-1)g' + 2hg\big]$, while $a_2 \to L^{-2h}g'$. Collecting,

$$a_1 = L^{-2h}\big[(x{-}1)g'{+}2hg\big], \quad a_2 = L^{-2h}g', \quad a_3 = L^{-2h}\big[-xg'{-}2hg\big], \quad a_4 = -2h\,L^{-2h-1}g. \tag{48.251}$$

Insert these into the BPZ operator Eq. (48.244), with the degenerate field at $z_2 = x$ and the sum over $j \in \{1,3,4\}$. The $j=4$ term carries $1/(x-L) \sim -1/L$ and $h/(x-L)^2 \sim h/L^2$ against $a_4 \sim L^{-2h-1}$ and $G_4 \sim L^{-2h}$, so it is $O(L^{-2h-2})$ and drops out in the limit. The left side is $\frac{4}{3}\,\partial_{z_2}^2 G_4 = \frac{4}{3}L^{-2h}g''$. Dividing the whole equation by L^{-2h},

$$\frac{4}{3}\,g'' = \frac{1}{x}\big[(x-1)g' + 2hg\big] + \frac{h}{x^2}g + \frac{1}{x-1}\big[-xg' - 2hg\big] + \frac{h}{(x-1)^2}g. \tag{48.252}$$

Collect the g' terms: their coefficient is $\frac{x-1}{x} - \frac{x}{x-1} = \frac{(x-1)^2 - x^2}{x(x-1)} = \frac{1-2x}{x(x-1)}$. Collect the g terms: using $\frac{2h}{x} - \frac{2h}{x-1} = -\frac{2h}{x(x-1)}$, their coefficient is $h\big[\frac{1}{x^2} + \frac{1}{(x-1)^2} - \frac{2}{x(x-1)}\big]$.

Thus

$$\frac{4}{3}\,g'' = \frac{1-2x}{x(x-1)}\,g' + h\Big[\frac{1}{x^2} + \frac{1}{(x-1)^2} - \frac{2}{x(x-1)}\Big]g. \tag{48.253}$$

Before simplifying further, the equation already tells us the operator content. Near $x \to 0$, set $g \sim x^s$: the leading terms give the indicial equation $s(s-1) = -\frac{3}{4}s + \frac{3}{64}$ (using $h = \frac{1}{16}$), i.e. $64s^2 - 16s - 3 = 0$, with roots

$$s = \frac{16 \pm \sqrt{256 + 768}}{128} = \frac{16 \pm 32}{128} \in \Big\{-\tfrac{1}{8},\ \tfrac{3}{8}\Big\}. \tag{48.254}$$

The s-channel OPE behaves as $g \sim x^{h_p - 2h_\sigma}$ with $2h_\sigma = \frac{1}{8}$; the two roots correspond to internal weights $h_p = 0$ (root $-\frac{1}{8}$, the identity) and $h_p = \frac{1}{2}$ (root $\frac{3}{8}$, the energy ε). The ODE has thus *rediscovered* the fusion rule $\sigma \times \sigma = \mathbf{1} + \varepsilon$ of Eq. (48.237), with the exact dimensions $h_{\mathbf{1}} = 0$ and $h_\varepsilon = \frac{1}{2}$.

Step 4 — Reduction to hypergeometric form, and the closed-form blocks. Strip the leading singular behaviour by the substitution

$$g(x) = \big[x(1-x)\big]^{-1/8}\,u(x), \tag{48.255}$$

which is the natural choice because Eq. (48.253) is symmetric under $x \to 1-x$ (its g-coefficient is manifestly symmetric, and the g'-coefficient is odd, matching $g' \to -g'$), so the exponents at $x = 1$ are the same $\{-\frac{1}{8}, \frac{3}{8}\}$ as at $x = 0$. Multiplying Eq. (48.253) by $\frac{3}{4}$ to clear the leading coefficient and writing $D = x(1-x)$, a direct computation reduces it to an equation for u. Writing $\rho \equiv (\ln[x(1-x)]^{-1/8})' = -\frac{1-2x}{8x(1-x)}$, one finds $g'' - Ag' - Bg = 0$ with $A = \frac{3}{4}\frac{1-2x}{x(x-1)} = 6\rho$ and $B = \frac{3}{64}\big[\frac{1}{x^2} + \frac{1}{(x-1)^2} - \frac{2}{x(x-1)}\big] = \frac{3}{64D^2}$ (the bracket collapses to $1/D^2$ because $\frac{1}{x^2} + \frac{1}{(1-x)^2} + \frac{2}{x(1-x)} = \big[\frac{(1-x)+x}{x(1-x)}\big]^2 = \frac{1}{D^2}$). Substituting $g = D^{-1/8}u$ turns $g'' - Ag' - Bg = 0$ into

$$u'' + (2\rho - A)u' + (\rho^2 + \rho' - A\rho - B)\,u = 0 = u'' - 4\rho\,u' + (\rho' - 5\rho^2 - B)\,u, \tag{48.256}$$

and the algebra of the last coefficient is clean: with $D' = 1 - 2x$, $D'' = -2$,

$$\rho' - 5\rho^2 - B = \frac{\frac{1}{8}(2D + (1-2x)^2) - \frac{5}{64}(1-2x)^2 - \frac{3}{64}}{D^2} = \frac{16D + 3(1-2x)^2 - 3}{64\,D^2}. \tag{48.257}$$

Using $(1-2x)^2 = 1 - 4D$, the numerator is $16D + 3(1-4D) - 3 = 4D$, so $\rho' - 5\rho^2 - B = \frac{4D}{64D^2} = \frac{1}{16D}$. Likewise $-4\rho = \frac{1-2x}{2D}$. The equation for u is therefore

$$u'' + \frac{1-2x}{2\,x(1-x)}\,u' + \frac{1}{16\,x(1-x)}\,u = 0, \tag{48.258}$$

or, clearing the denominator,

$$x(1-x)\,u'' + \Big(\tfrac{1}{2} - x\Big)u' + \tfrac{1}{16}\,u = 0. \tag{48.259}$$

This is the hypergeometric equation $x(1-x)u'' + [c-(a+b+1)x]u' - ab\,u = 0$ with

$$c = \tfrac{1}{2}, \qquad a+b+1 = 1, \qquad ab = -\tfrac{1}{16} \implies (a,b,c) = \left(\tfrac{1}{4}, -\tfrac{1}{4}, \tfrac{1}{2}\right). \tag{48.260}$$

Rather than quote hypergeometric identities, we integrate Eq. (48.259) in elementary terms by the substitution $x = \sin^2\theta$ (so $\sqrt{1-x} = \cos\theta$, $dx/d\theta = \sin 2\theta$, and $x(1-x) = \tfrac{1}{4}\sin^2 2\theta$). Converting derivatives, $u' = u_\theta/\sin 2\theta$ and $u'' = u_{\theta\theta}/\sin^2 2\theta - 2\cos 2\theta\, u_\theta/\sin^3 2\theta$, the first two terms of Eq. (48.259) combine: the $x(1-x)u''$ term contributes $\tfrac{1}{4}u_{\theta\theta} - \tfrac{1}{2}\cos 2\theta\, u_\theta/\sin 2\theta$, while $(\tfrac{1}{2}-x)u' = \tfrac{1}{2}\cos 2\theta\, u' = \tfrac{1}{2}\cos 2\theta\, u_\theta/\sin 2\theta$ (since $\tfrac{1}{2} - \sin^2\theta = \tfrac{1}{2}\cos 2\theta$), and the two $\cos 2\theta$ pieces cancel exactly. What remains is the constant-coefficient equation

$$\frac{1}{4}\,u_{\theta\theta} + \frac{1}{16}\,u = 0 \quad\Longleftrightarrow\quad u_{\theta\theta} + \frac{1}{4}\,u = 0, \tag{48.261}$$

whose general solution is $u = A\cos(\theta/2) + B\sin(\theta/2)$. Translating back through $\cos^2(\theta/2) = \frac{1+\cos\theta}{2} = \frac{1+\sqrt{1-x}}{2}$ and $\sin^2(\theta/2) = \frac{1-\sqrt{1-x}}{2}$, the two independent chiral solutions are the celebrated algebraic blocks

$$\boxed{F_+(x) = \cos\frac{\theta}{2} = \sqrt{\frac{1+\sqrt{1-x}}{2}}, \qquad F_-(x) = \sin\frac{\theta}{2} = \sqrt{\frac{1-\sqrt{1-x}}{2}}.} \tag{48.262}$$

Their small-x behaviour confirms the channel assignment of Eq. (48.254): $F_+ \to 1$ (the identity block, since $g = D^{-1/8}F_+ \sim x^{-1/8}$), while $F_- \to \sqrt{x}/2$ (the energy block, since $g = D^{-1/8}F_- \sim \tfrac{1}{2}x^{3/8}$). The factor $\tfrac{1}{2}$ in the leading energy coefficient is the seed of $C_{\sigma\sigma\varepsilon}$; we now make that precise. Figure 48.6 plots the two blocks.

Step 5 — Single-valuedness and the OPE coefficient. The physical correlator must combine the chiral blocks $F_\pm(x)$ with their antichiral partners $F_\pm(\bar{x})$ into a function that is single-valued in the Euclidean plane ($\bar{x} = x^*$): a genuine local correlator has no monodromy as the operators are carried around one another. Restoring the prefactor $[x(1-x)]^{-1/8}$ in $g = [x(1-x)]^{-1/8}F$, the candidate single-valued combination of g with its antiholomorphic partner is

$$g(x)\,\overline{g(\bar{x})} = |x(1-x)|^{-1/4}\Big(\alpha\,F_+(x)\overline{F_+(x)} + \beta\,F_-(x)\overline{F_-(x)}\Big), \tag{48.263}$$

with real coefficients $\alpha, \beta \geq 0$ to be fixed; off-diagonal terms $F_+\bar{F}_-$ are forbidden because $\mathbf{1}$ and ε have distinct weights and the OPE pairs holomorphic and antiholomorphic content diagonally.

Examine the monodromy as x encircles 1 (the crossing point at which two operators collide in the t-channel). Under $x \to 1$ once around, $\sqrt{1-x} \to -\sqrt{1-x}$, so the radicands in Eq. (48.262) swap and $F_+ \leftrightarrow F_-$; simultaneously $(1-x)^{-1/8}$ picks up the phase $e^{-i\pi/4}$. Thus $F_+ \to e^{-i\pi/4}F_-$ and $F_- \to e^{-i\pi/4}F_+$

for the holomorphic factor, while the antiholomorphic factor (with $\bar{x}$ circling in the opposite sense) picks up the conjugate phase $e^{+i\pi/4}$. The phases cancel in each modulus-squared, and the effect on Eq. (48.263) is simply $|F_+|^2 \leftrightarrow |F_-|^2$. Single-valuedness around $x = 1$ therefore *requires*

$$\alpha = \beta. \tag{48.264}$$

(The monodromy around $x = 0$ multiplies both blocks by the same phase $e^{-i\pi/4}$ from $x^{-1/8}$, which cancels against the antiholomorphic conjugate automatically; it imposes no further condition.)

It remains to fix the overall normalization and read off $C_{\sigma\sigma\varepsilon}$. The OPE channel decomposition is, by definition,

$$g(x)\,\overline{g(\bar{x})} = |x(1-x)|^{-1/4}\Big(C_{\sigma\sigma\mathbf{1}}^2\,\widehat{F}_{\mathbf{1}}(x)\overline{\widehat{F}_{\mathbf{1}}} + C_{\sigma\sigma\varepsilon}^2\,\widehat{F}_{\varepsilon}(x)\overline{\widehat{F}_{\varepsilon}}\Big), \tag{48.265}$$

where $\widehat{F}_a$ are the *unit-normalized* blocks — normalized so that each begins as $x^{h_a-2h_\sigma}\,(1+\cdots)$ with leading coefficient one, the convention in which the squared coefficient out front is exactly the squared OPE coefficient. Comparing with Eq. (48.262): F_+ already begins as $x^{-1/8}$ with unit coefficient, so $\widehat{F}_{\mathbf{1}} = F_+$; but F_- begins as $\frac{1}{2}x^{3/8}$, so the unit-normalized energy block is $\widehat{F}_\varepsilon = 2F_-$. With the spin two-point function normalized to unity, the identity channel has $C_{\sigma\sigma\mathbf{1}} = 1$, hence the coefficient of $|F_+|^2$ in Eq. (48.263) is $\alpha = 1$. The coefficient of $|F_-|^2$ is $\beta = C_{\sigma\sigma\varepsilon}^2\,|2|^2 = 4\,C_{\sigma\sigma\varepsilon}^2$. Single-valuedness Eq. (48.264) demands $\beta = \alpha = 1$, so

$$4\,C_{\sigma\sigma\varepsilon}^2 = 1 \quad\Longrightarrow\quad \boxed{C_{\sigma\sigma\varepsilon} = \frac{1}{2}\,.} \tag{48.266}$$

Crossing symmetry — here in the guise of single-valuedness around the collision point — has fixed the one piece of dynamical data that symmetry alone could not: the strength with which the energy operator appears in the fusion of two spins. This is the conformal bootstrap solved exactly in a single equation.

Step 6 — Assembling the correlator. With $\alpha = \beta = 1$, evaluate Eq. (48.263) using $|F_\pm(x)|^2 = \big|\frac{1\pm\sqrt{1-x}}{2}\big| = \frac{1}{2}|1 \pm \sqrt{1-x}|$:

$$g(x)\,\overline{g(\bar{x})} = \frac{|x(1-x)|^{-1/4}}{2}\Big(\big|1+\sqrt{1-x}\big| + \big|1-\sqrt{1-x}\big|\Big). \tag{48.267}$$

The function G of Eq. (48.243) is built from the full conformal blocks $x^{2h}g_a(x)$, which differ from the reduced blocks by the prefactor x^{2h} that was renormalized away in the limit Eq. (48.245); restoring it gives $G(x,\bar{x}) = |x|^{2\cdot 2h}\,g\bar{g} = |x|^{1/4}\,g(x)\overline{g(\bar{x})}$, the factor $|x|^{1/4}$ being precisely what makes $G \to |F_+|^2 = 1$ as $x \to 0$, the correct identity-channel limit. Finally restore the kinematic prefactor of Eq. (48.243). Using the cross-ratio identities $x = \frac{z_{12}z_{34}}{z_{13}z_{24}}$ and (from the Plücker

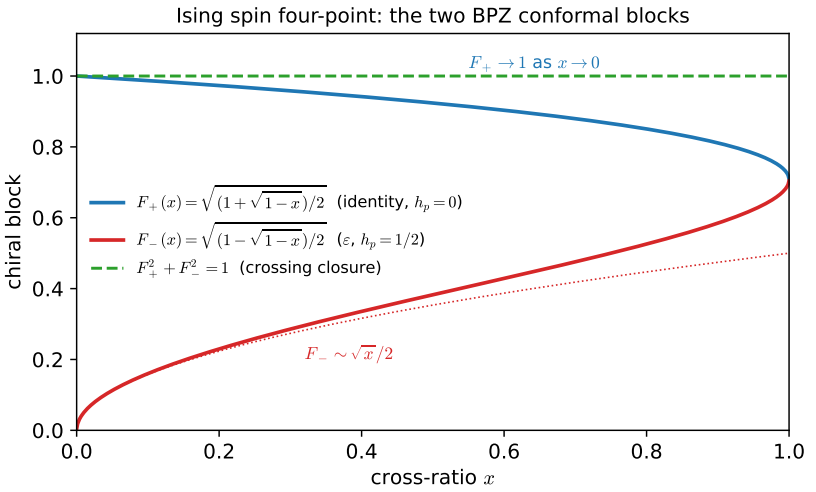


Figure 48.6: The two chiral conformal blocks of the Ising spin four-point function, Eq. (48.262), on the real segment $0 < x < 1$. The identity block F_+ tends to 1 as $x \to 0$; the energy block F_- vanishes as $\sqrt{x}/2$, exhibiting the internal weight $h_\varepsilon = \frac{1}{2}$ and the leading coefficient $\frac{1}{2}$ that becomes $C_{\sigma\sigma\varepsilon}$ after unit-normalization. Their squares sum to a constant, $F_+^2 + F_-^2 = 1$, the algebraic shadow of crossing closure.

relation $z_{12}z_{34} - z_{13}z_{24} + z_{14}z_{23} = 0$) $1 - x = \frac{z_{14}z_{23}}{z_{13}z_{24}}$, the prefactors combine $(|x|^{1/4}|x(1-x)|^{-1/4} = |1-x|^{-1/4})$,

$$|z_{12}z_{34}|^{-1/4}\,|x|^{1/4}\,|x(1-x)|^{-1/4} = |z_{12}z_{34}|^{-1/4}\,|1-x|^{-1/4} = \left|\frac{z_{13}z_{24}}{z_{12}z_{23}z_{34}z_{14}}\right|^{1/4}, \tag{48.268}$$

and we arrive at Eq. (48.241), completing the proof of Theorem 48.34. As a consistency check, the homogeneity of Eq. (48.268) in each z_i is $-\frac{1}{4} = -2\Delta_\sigma$, the correct scaling weight of a spin insertion, and the factor in parentheses depends only on the cross-ratio, as it must.

Caution

The blocks $F_\pm$ are individually multivalued: each has a square-root branch point at $x = 1$ (and the stripped prefactor $[x(1-x)]^{-1/8}$ adds branch points at $x = 0, 1$). Neither F_+ nor F_- is, by itself, a correlator. Only the specific single-valued combination Eq. (48.267), with $\alpha = \beta$, is a physical observable. A frequent error is to identify "the conformal block" with "the correlator"; the block is a chiral half, and physics lives in the monodromy-invariant pairing of the two halves. The same logic — conformal blocks plus crossing — is the engine of the higher-dimensional bootstrap in

Chapters 49 and 50, where the blocks are no longer algebraic and crossing can only be imposed numerically.

48.6.4 Where $c = \frac{1}{2}$ comes from: the free Majorana fermion

We have solved the Ising CFT by pure representation theory, never asking what fields realize it. But the value $c = \frac{1}{2}$ is suggestive: it is exactly half of the $c = 1$ of a free boson, and indeed $\mathcal{M}(4,3)$ has a transparent microscopic realization as a single free *Majorana* (real) fermion. This both demystifies the abstract construction and supplies an independent derivation of every number above. It is also the gateway to the next section's free-field realizations.

A holomorphic Majorana fermion is a chiral field $\psi(z)$ with the operator product expansion

$$\psi(z)\,\psi(w) \sim \frac{1}{z-w}, \tag{48.269}$$

together with an antiholomorphic partner $\bar{\psi}(\bar{z})$ satisfying $\bar{\psi}(\bar{z})\bar{\psi}(\bar{w}) \sim 1/(\bar{z}-\bar{w})$. The stress tensor is the normal-ordered bilinear

$$T(z) = -\frac{1}{2}\,{:}\psi\,\partial\psi{:}\,(z), \tag{48.270}$$

the unique dimension-two chiral field one can build quadratically from ψ. Everything follows from Eqs. (48.269) and (48.270) by Wick's theorem for the contraction $\langle\psi(z)\psi(w)\rangle = 1/(z-w)$, with the derivative contractions

$$\langle\psi(z)\partial\psi(w)\rangle = \frac{1}{(z-w)^2}, \quad \langle\partial\psi(z)\psi(w)\rangle = \frac{-1}{(z-w)^2}, \quad \langle\partial\psi(z)\partial\psi(w)\rangle = \frac{-2}{(z-w)^3}. \tag{48.271}$$

Central charge. The central charge sits in the most singular (c-number) term of the $T(z)T(w)$ OPE, which by definition is $\frac{c/2}{(z-w)^4}$. Compute it by fully contracting the two bilinears,

$$T(z)T(w) = \frac{1}{4}\,{:}\psi\partial\psi{:}\,(z)\;{:}\psi\partial\psi{:}\,(w). \tag{48.272}$$

Two double contractions connect the z-pair to the w-pair. Ordering the four fields as $\psi(z), \partial\psi(z), \psi(w), \partial\psi(w)$, the pairing $\{\psi(z)\psi(w)\}\{\partial\psi(z)\partial\psi(w)\}$ is the interleaved (crossing) one, carrying a fermionic sign -1, while $\{\psi(z)\partial\psi(w)\}\{\partial\psi(z)\psi(w)\}$ is nested, sign $+1$. Hence

$$\begin{aligned} T(z)T(w)\big|_{c\text{-number}} &= \frac{1}{4}\Big[(-1)\,\langle\psi(z)\psi(w)\rangle\,\langle\partial\psi(z)\partial\psi(w)\rangle + (+1)\,\langle\psi(z)\partial\psi(w)\rangle\,\langle\partial\psi(z)\psi(w)\rangle\Big] \\ &= \frac{1}{4}\Big[(-1)\frac{1}{z-w}\cdot\frac{-2}{(z-w)^3} + (+1)\frac{1}{(z-w)^2}\cdot\frac{-1}{(z-w)^2}\Big] \\ &= \frac{1}{4}\Big[\frac{2}{(z-w)^4} - \frac{1}{(z-w)^4}\Big] = \frac{1}{4(z-w)^4}. \end{aligned} \tag{48.273}$$

Matching to $\frac{c/2}{(z-w)^4}$ gives $\frac{c}{2} = \frac{1}{4}$, that is

$$\boxed{c = \tfrac{1}{2}\,,} \tag{48.274}$$

in exact agreement with Eq. (48.231). A single free Majorana fermion is a $c = \frac{1}{2}$ theory.

The fermion is a primary of weight $\frac{1}{2}$. Wick-contract $T(z)$ with $\psi(w)$:

$$\begin{aligned} T(z)\psi(w) &= -\frac{1}{2}\,{:}\psi(z)\partial\psi(z){:}\,\psi(w) \\ &= -\frac{1}{2}\Big[(-1)\langle\psi(z)\psi(w)\rangle\,\partial\psi(z) + (+1)\langle\partial\psi(z)\psi(w)\rangle\,\psi(z)\Big] \\ &= \frac{1}{2}\,\frac{\partial\psi(z)}{z-w} + \frac{1}{2}\,\frac{\psi(z)}{(z-w)^2}, \end{aligned} \tag{48.275}$$

where the first sign is the -1 from moving $\psi(z)$ past the spectator $\partial\psi(z)$ to reach $\psi(w)$. Taylor-expand $\psi(z)$ and $\partial\psi(z)$ about w, $\psi(z) = \psi(w)+(z-w)\partial\psi(w)+\cdots$:

$$T(z)\psi(w) \sim \frac{\frac{1}{2}\,\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{z-w}. \tag{48.276}$$

This is precisely the OPE of T with a primary of weight h, $T(z)\mathcal{O}(w) \sim \frac{h\,\mathcal{O}}{(z-w)^2} + \frac{\partial\mathcal{O}}{z-w}$, with

$$h_\psi = \frac{1}{2}, \tag{48.277}$$

and the unit coefficient of the $1/(z-w)$ term confirms ψ is a genuine primary (no anomalous descendant). So the Majorana fermion is a holomorphic primary of dimension $(\frac{1}{2}, 0)$.

The energy operator. The energy density ε is the diagonal product of the holomorphic and antiholomorphic fermions,

$$\varepsilon(z,\bar z) = i\,\psi(z)\,\bar\psi(\bar z), \tag{48.278}$$

the factor i making it Hermitian. As a product of two commuting primaries of weights $(\frac{1}{2}, 0)$ and $(0, \frac{1}{2})$, it has weights $(h, \bar h) = (\frac{1}{2}, \frac{1}{2})$ and hence

$$\Delta_\varepsilon = h + \bar h = 1, \tag{48.279}$$

reproducing Eq. (48.235) and confirming the Kac-table identification $\varepsilon = \phi_{2,1}$. (Physically, ε is the fermion bilinear whose coefficient is the mass term; turning it on drives the theory off criticality, exactly the role of temperature in the Ising model.)

The spin field as a twist field, and $h_\sigma = \frac{1}{16}$. The one Ising primary with no polynomial expression in the free fermion is the spin σ. It is instead a *twist* (order)

field: the fermion is *double-valued* around an insertion of σ, $\psi(e^{2\pi i}z)\,\sigma(0) = -\psi(z)\,\sigma(0)$, so σ creates a square-root branch cut for ψ. This single analytic requirement fixes h_σ. Consider the fermion two-point function in the presence of two spin fields at 0 and ∞,

$$g(z,w) \equiv \frac{\langle \psi(z)\psi(w)\,\sigma(0)\sigma(\infty)\rangle}{\langle \sigma(0)\sigma(\infty)\rangle}. \tag{48.280}$$

It must (i) reduce to $1/(z-w)$ as $z \to w$ (the OPE Eq. (48.269)), (ii) be antisymmetric in $z \leftrightarrow w$ (fermion statistics), and (iii) change sign when either argument circles the branch point at 0. The unique such function with the minimal singularity structure is

$$g(z,w) = \frac{1}{2}\,\frac{\sqrt{z/w} + \sqrt{w/z}}{z-w}. \tag{48.281}$$

Indeed: as $z \to w$ the bracket $\to 2$ and $g \to 1/(z-w)$; swapping $z \leftrightarrow w$ flips $z-w$ while leaving the symmetric bracket fixed, giving antisymmetry; and sending $z \to e^{2\pi i}z$ flips the sign of both $\sqrt{z/w}$ and $\sqrt{w/z}$, so $g \to -g$, the required monodromy.

The weight h_σ is now read from the expectation value of the stress tensor in this two-spin background, which for any primary must take the form $\langle \sigma(\infty)T(z)\sigma(0)\rangle/\langle\sigma\sigma\rangle = h_\sigma/z^2$. Compute the left side as the point-split, subtracted limit of Eq. (48.270),

$$\frac{\langle T(z)\rangle_{\sigma\sigma}}{\langle\sigma\sigma\rangle} = -\frac{1}{2}\lim_{w\to z}\Big[\partial_w g(z,w) - \frac{1}{(z-w)^2}\Big], \tag{48.282}$$

the subtraction removing the singular free part already present in $g \sim 1/(z-w)$. Expand Eq. (48.281) near $w = z$ with $w = z+e$. From $(1+e/z)^{\mp 1/2} = 1 \mp \frac{e}{2z} + \frac{3}{8}\frac{e^2}{z^2} \pm \cdots$ one gets $\sqrt{z/w} + \sqrt{w/z} = 2 + \frac{1}{4}(e/z)^2 + O(e^3)$, so

$$g(z,z+e) = \frac{1}{2}\,\frac{2 + \frac{1}{4}e^2/z^2 + \cdots}{-e} = -\frac{1}{e} - \frac{1}{8}\frac{e}{z^2} + O(e^2). \tag{48.283}$$

Then $\partial_w g = \partial_e g = \frac{1}{e^2} - \frac{1}{8z^2} + \cdots$, while $1/(z-w)^2 = 1/e^2$, so the bracket in Eq. (48.282) tends to $-\frac{1}{8z^2}$, and

$$\frac{\langle T(z)\rangle_{\sigma\sigma}}{\langle\sigma\sigma\rangle} = -\frac{1}{2}\Big(-\frac{1}{8z^2}\Big) = \frac{1}{16\,z^2} \quad\Longrightarrow\quad \boxed{h_\sigma = \frac{1}{16}}, \tag{48.284}$$

exactly the Kac value of Theorem 48.33, now derived from nothing but the branch-cut analyticity of a free fermion. Thus the free Majorana realizes the complete Ising spectrum: $\mathbf{1}$, the twist field σ of weight $\frac{1}{16}$, and the bilinear ε of weight $\frac{1}{2}$, in a theory of central charge $c = \frac{1}{2}$. This is the microscopic answer to "where does $c = \frac{1}{2}$ come from?": it is one real fermionic degree of freedom per chirality.

48.6.5 The dictionary to critical exponents: meeting Onsager

The conformal data we have computed is the same information a statistical physicist packages as critical exponents, and the translation is a clean dictionary worth stating once and for all, because it is exactly the dictionary the bootstrap of Chapter 50 will invert in $d = 3$. The rule is that a scaling operator $\mathcal{O}$ of dimension $\Delta_{\mathcal{O}}$, used to perturb the fixed-point action, $\delta S = \lambda \int d^d x\, \mathcal{O}$, has renormalization-group eigenvalue (relevance)

$$y_{\mathcal{O}} = d - \Delta_{\mathcal{O}}, \tag{48.285}$$

so that the coupling λ flows with scale as $\lambda(b) = b^{y_{\mathcal{O}}}\lambda$. In the Ising universality class the two relevant perturbations are the energy ε (coupling to reduced temperature t) and the spin σ (coupling to the magnetic field H), with $d = 2$, $\Delta_\varepsilon = 1$, $\Delta_\sigma = \frac{1}{8}$, giving the thermal and magnetic eigenvalues

$$y_t = d - \Delta_\varepsilon = 2 - 1 = 1, \qquad y_h = d - \Delta_\sigma = 2 - \tfrac{1}{8} = \tfrac{15}{8}. \tag{48.286}$$

From these two numbers every standard exponent follows by the scaling relations. The correlation length diverges as $\xi \sim |t|^{-\nu}$ with $\nu = 1/y_t$; the spin-spin correlator at criticality falls as $\langle \sigma\sigma \rangle \sim r^{-(d-2+\eta)}$, while Eq. (47.120) gives $\langle \sigma\sigma \rangle \sim r^{-2\Delta_\sigma}$, so $\eta = 2\Delta_\sigma - (d-2)$:

$$\boxed{\nu = \frac{1}{y_t} = 1, \qquad \eta = 2\Delta_\sigma - (d-2) = \frac{1}{4}.} \tag{48.287}$$

The remaining exponents, obtained from $\beta = \nu(d - y_h)$, $\gamma = \nu(2y_h - d)$, $\delta = y_h/(d - y_h)$, and $\alpha = 2 - d\nu$, are

$$\beta = 1 \cdot (2 - \tfrac{15}{8}) = \tfrac{1}{8}, \quad \gamma = 1 \cdot (\tfrac{30}{8} - 2) = \tfrac{7}{4}, \quad \delta = \frac{15/8}{1/8} = 15, \quad \alpha = 2 - 2 \cdot 1 = 0, \tag{48.288}$$

the last being the celebrated logarithmic ($\alpha = 0$) specific-heat singularity. Every one of these is exactly Onsager's 1944 result for the square-lattice Ising model: $\nu = 1$, $\eta = \frac{1}{4}$, $\beta = \frac{1}{8}$, $\gamma = \frac{7}{4}$, $\delta = 15$, $\alpha = 0$. The transfer-matrix tour de force and the representation theory of the Virasoro algebra agree to the last digit — because they describe the same fixed point.

Historical Note

Onsager (1944) computed the free energy of the two-dimensional Ising model exactly by diagonalizing the transfer matrix, extracting $\alpha = 0$; the spontaneous magnetization exponent $\beta = \frac{1}{8}$ was announced by Onsager and derived by Yang (1952). For three decades these numbers stood as isolated exact results with no organizing principle. Belavin, Polyakov

and Zamolodchikov (1984) supplied the principle: the critical point is the conformal field theory $\mathcal{M}(4,3)$, and the exponents are conformal dimensions. The computation in this section — four-point function from a null vector, dimensions from a Kac table — is their method, and it generalizes to the entire $c < 1$ series where no transfer matrix has ever been diagonalized.

Physical Insight

The same dimensions $\Delta_\sigma, \Delta_\varepsilon$ exist for every spatial dimension d, tracing a curve from the Gaussian (free) values at $d = 4$ — $\Delta_\sigma = \frac{d-2}{2} = 1$, $\Delta_\varepsilon = 2$ — down to the exactly solvable Ising values $\Delta_\sigma = \frac{1}{8}$, $\Delta_\varepsilon = 1$ at $d = 2$. At the physically interesting $d = 3$ the theory is the Wilson–Fisher fixed point of ϕ^4 theory (Chapter 52); it is genuinely interacting, has no free-field realization, and is not exactly solvable. There the conformal bootstrap of Chapter 50 takes over, imposing crossing symmetry numerically on the four-point function $\langle\sigma\sigma\sigma\sigma\rangle$ — the very object we solved here in closed form — to pin down $\Delta_\sigma = 0.5181489(10)$ and $\Delta_\varepsilon = 1.412625(10)$. The $d = 2$ solution of this section is the calibration: any bootstrap algorithm must reproduce $\frac{1}{8}$ and 1 when fed $d = 2$, and the closed-form correlator Eq. (48.241) is the exact answer it must converge to.

The microscopic origin of $c = \frac{1}{2}$ as a single free Majorana fermion, just exhibited, is the simplest instance of a general phenomenon: many minimal models and rational CFTs admit explicit realizations in terms of free fields. The next section pursues this systematically — the free boson at $c = 1$ and its vertex operators, the compact boson, and the Coulomb-gas / Feigin–Fuks construction that reproduces the entire minimal-model series Eq. (48.228) through a background charge and screening integrals — turning the abstract representation theory of this chapter into concrete, computable field-theoretic machinery.

48.7 Free-Field Realizations and the Coulomb Gas

The previous section solved the Ising model as a problem in pure representation theory: three primaries with weights fixed by the Kac table, fusion rules dictated by null vectors, and a four-point function delivered by a second-order differential equation (Section 48.6 and Theorem 48.34). At no point did a Lagrangian appear. That is a triumph of the bootstrap, but it leaves a reasonable reader uneasy on one specific point. The number that organized everything — the central charge, $c = \frac{1}{2}$ for Ising (Eq. (48.231)) — entered as an *input*, a coefficient in the $T(z)T(w)$ operator product (Theorem 48.8) whose admissible values we cut down to the Friedan–Qiu–Shenker list by demanding unitarity. We never built a theory and

watched $c = \frac{1}{2}$ come out. The free boson of Section 48.2.6 gave us exactly one such concrete benchmark, $c = 1$, computed by Wick contraction (Theorem 48.9); but 1 is not on the minimal-model list, and a single free field cannot reach the fractional central charges $1 - 6/[m(m+1)]$ that the discrete series demands.

This closing section removes the unease by exhibiting honest microscopic field realizations and then bending the simplest of them until it produces the minimal series on the nose. We proceed in four movements. First we package the free boson as a conformal field theory in its own right, with a U(1) current and a family of *vertex operators* $:e^{i\alpha X}:$ whose weights and correlators we compute from the propagator alone. Second we compactify the boson, $X \sim X + 2\pi R$, and discover momentum, winding, the duality $R \to 1/R$, and a hidden SU(2) at the self-dual radius. Third we realize $c = \frac{1}{2}$ a second, completely independent way — from a free Majorana fermion — closing the loop with the Ising fermion of Eq. (48.270). Fourth, and most consequentially, we deform the boson's stress tensor by a *background charge* Q, sending $c = 1 \to 1 - 3Q^2$, and show that quantizing Q to the minimal-model values turns the free-field correlators, dressed with *screening charges*, into the Dotsenko–Fateev integral representations of minimal-model correlators. As a capstone we recover the Ising spin four-point function of Theorem 48.34 from a single screening integral — the same numbers, by an entirely different route. Throughout, the engine is the residue and analytic-continuation machinery of Theorem 5.23 and Section 5.4; the mostly-minus Minkowski conventions of the book enter only through the Euclidean continuation that made every correlator in this chapter complex-analytic.

48.7.1 The free boson as a conformal field theory

We begin by collecting, as a single named object, the structure that Section 48.2.6 established piecemeal. The massless free boson is governed by the Euclidean action (Eq. (48.74))

$$S = \frac{1}{8\pi}\int \mathrm{d}^2x\, \partial_\mu X\, \partial^\mu X = \frac{1}{2\pi}\int \mathrm{d}^2x\, \partial X\, \bar\partial X, \tag{48.289}$$

whose equation of motion $\partial\bar\partial X = 0$ splits X into a holomorphic and an antiholomorphic part, $X(z,\bar z) = X(z) + \bar X(\bar z)$ at the level of derivatives. The fundamental datum is the two-point function (Eq. (48.75)) and its once-differentiated form (Eq. (48.76)),

$$\langle X(z,\bar z)\, X(w,\bar w)\rangle = -\ln|z-w|^2, \qquad \langle \partial X(z)\, \partial X(w)\rangle = -\frac{1}{(z-w)^2}. \tag{48.290}$$

The holomorphic stress tensor is the normal-ordered bilinear $T(z) = -\frac{1}{2}:\partial X\, \partial X:(z)$ (Eq. (48.77)), with respect to which ∂X is a weight-$(1,0)$ primary and the central charge is $c = 1$ (Theorem 48.9). We will not re-derive any of these; we package them and add the one structural element Section 48.2.6 did not name explicitly.

Definition 48.36 (The free-boson conformal field theory and its current)**.** The *free-boson conformal field theory* is the $c = 1$ theory of a single massless scalar X with action Eq. (48.289), propagator Eq. (48.290), and holomorphic stress tensor $T(z) = -\frac{1}{2} :\partial X\, \partial X:(z)$. Its chiral symmetry algebra is generated by the holomorphic U(1) *current*

$$J(z) \equiv i\, \partial X(z), \tag{48.291}$$

a weight-$(1, 0)$ primary whose self–operator product, read off from Eq. (48.290), is

$$J(z)\, J(w) \sim \frac{1}{(z-w)^2}, \qquad T(z)\, J(w) \sim \frac{J(w)}{(z-w)^2} + \frac{\partial J(w)}{z-w}. \tag{48.292}$$

The factor i makes J Hermitian under the radial-quantization conjugation, so that its conserved charge $\oint \frac{\mathrm{d}z}{2\pi i}\, J(z)$ is the real momentum generating shifts $X \to X + \text{const}$.

The OPE Eq. (48.292) is the simplest current algebra — a U(1) Kac–Moody algebra at "level one" in the normalization where the double-pole coefficient is unity — and it is the structure that the vertex operators below will be charged under. The whole content of the free boson is now encoded in two contractions, $\langle JJ \rangle$ and $\langle \partial X\, \partial X \rangle$, which differ only by the sign $J = i\partial X$ supplies.

Vertex operators

The field X itself is not a primary: it appears in correlators only logarithmically, and its U(1) shift symmetry $X \to X + a$ forbids any nonzero one-point function for a bare power of X. What *is* primary is the exponential, because exponentials diagonalize the shift symmetry. This is the single most useful operator in two-dimensional conformal field theory.

Definition 48.37 (Vertex operator)**.** For $\alpha \in \mathbb{R}$ the *vertex operator* of charge α is the normal-ordered exponential

$$V_\alpha(z, \bar z) \equiv :e^{i\alpha X(z,\bar z)}:, \tag{48.293}$$

where the normal ordering removes all self-contractions, $:e^{i\alpha X}: = e^{i\alpha X} \big/ \langle e^{i\alpha X} \rangle$ understood as the subtraction of the (divergent) coincident propagator. Its purely holomorphic relative is $V_\alpha(z) = :e^{i\alpha X(z)}:$, built from the chiral half $X(z)$ with $\langle X(z) X(w) \rangle = -\ln(z-w)$.

Weight from the T–V OPE. We claim $V_\alpha(z)$ is primary with holomorphic weight $h_\alpha = \frac{1}{2}\alpha^2$. To prove it we Wick-contract $T(z) V_\alpha(w)$. The basic contraction

of a single $\partial X(z)$ with the exponential is obtained by bringing down one factor of $i\alpha X(w)$ from the series and contracting:

$$\langle \partial X(z)\, X(w)\rangle = \partial_z\big[-\ln(z-w)\big] = -\frac{1}{z-w}, \qquad \Longrightarrow \qquad \underbrace{\partial X(z)\, :e^{i\alpha X(w)}:}_{\text{single contraction}} = -\frac{i\alpha}{z-w}\, V_\alpha(w). \tag{48.294}$$

Now $T(z) = -\frac{1}{2}{:}\partial X(z)\partial X(z){:}$ contains two factors of $\partial X(z)$, each of which may contract with the exponential. Contracting *both* (the double contraction) produces the term singular as $(z-w)^{-2}$ with no leftover operator beyond V_α:

$$-\tfrac{1}{2}\Big(-\frac{i\alpha}{z-w}\Big)^2 V_\alpha(w) = -\tfrac{1}{2}\cdot\frac{(i\alpha)^2}{(z-w)^2}\, V_\alpha(w) = \frac{\frac{1}{2}\alpha^2}{(z-w)^2}\, V_\alpha(w), \tag{48.295}$$

using $(i\alpha)^2 = -\alpha^2$. Contracting just *one* of the two $\partial X(z)$'s (two ways, cancelling the $\frac{1}{2}$) leaves one uncontracted $\partial X(z)$ multiplying the exponential:

$$-\tfrac{1}{2}\cdot 2\cdot\Big(-\frac{i\alpha}{z-w}\Big)\,{:}\partial X(z)\, V_\alpha(w){:} = \frac{i\alpha}{z-w}\,{:}\partial X(z)\, V_\alpha(w){:}. \tag{48.296}$$

Taylor-expanding the surviving $\partial X(z)$ about w and using $\partial_w V_\alpha(w) = \partial_w {:}e^{i\alpha X(w)}{:} = i\alpha\,{:}\partial X(w)\, V_\alpha(w){:}$, which rearranges to ${:}\partial X(w) V_\alpha(w){:} = (i\alpha)^{-1}\partial V_\alpha(w)$, the leading piece of Eq. (48.296) becomes

$$\frac{i\alpha}{z-w}\cdot\frac{1}{i\alpha}\,\partial V_\alpha(w) = \frac{\partial V_\alpha(w)}{z-w}. \tag{48.297}$$

Assembling Eqs. (48.295) and (48.297),

$$T(z)\, V_\alpha(w) \sim \frac{\frac{1}{2}\alpha^2\, V_\alpha(w)}{(z-w)^2} + \frac{\partial V_\alpha(w)}{z-w}, \tag{48.298}$$

which is exactly the primary form (Eq. (48.54)) with no pole of order three or higher. Hence

$$h_\alpha = \frac{\alpha^2}{2}, \tag{48.299}$$

and the non-chiral operator Eq. (48.293) carries $(h,\bar h) = (\frac{1}{2}\alpha^2, \frac{1}{2}\alpha^2)$ by the verbatim antiholomorphic computation, hence scaling dimension $\Delta_\alpha = \alpha^2$ and spin zero. The charge under the current Eq. (48.291) follows the same way: $J(z)V_\alpha(w) \sim \alpha\, V_\alpha(w)/(z-w)$, so V_α has U(1) charge α.

Correlators and the neutrality rule. Because X is a Gaussian (free) field, expectation values of exponentials are computed exactly by completing the square. For any Gaussian field with $\langle A\rangle = 0$ one has $\langle e^A\rangle = e^{\frac{1}{2}\langle A^2\rangle}$; setting $A = i\sum_k \alpha_k X(z_k)$ gives $\langle A^2\rangle = -\sum_{k,l}\alpha_k\alpha_l\langle X(z_k)X(z_l)\rangle$, so

$$\Big\langle \prod_k {:}e^{i\alpha_k X(z_k,\bar z_k)}{:} \Big\rangle = \exp\Big(-\sum_{k<l}\alpha_k\alpha_l\,\langle X(z_k)X(z_l)\rangle\Big), \tag{48.300}$$

where the normal ordering has removed the divergent $k=l$ self-energies, leaving only the cross terms $k<l$ (each unordered pair counted once, cancelling the $\frac{1}{2}$). There is one more ingredient that Eq. (48.300) hides: the zero mode. Writing $X = x_0 + \tilde{X}$ with x_0 the constant mode and $\tilde{X}$ the fluctuation, the path integral over $x_0 \in \mathbb{R}$ acts only on the product of exponentials,

$$\int_{-\infty}^{\infty} \mathrm{d}x_0 \; e^{i\left(\sum_k \alpha_k\right) x_0} = 2\pi\, \delta\Big(\sum_k \alpha_k\Big), \tag{48.301}$$

which vanishes unless the total charge is zero. This is the *neutrality rule*: a correlator of vertex operators is nonzero only if $\sum_k \alpha_k = 0$. Physically it is nothing but conservation of the U(1) charge that J measures — the noncompact shift symmetry $X \to X + a$ forces all charge to cancel. Substituting Eq. (48.290) into Eq. (48.300) under this constraint gives the master formula

$$\Big\langle \prod_k V_{\alpha_k}(z_k,\bar{z}_k) \Big\rangle = \prod_{k<l} |z_k - z_l|^{2\alpha_k\alpha_l} = \prod_{k<l} (z_k - z_l)^{\alpha_k\alpha_l} (\bar{z}_k - \bar{z}_l)^{\alpha_k\alpha_l}, \qquad \sum_k \alpha_k = 0, \tag{48.302}$$

and zero otherwise. The chiral version drops the antiholomorphic factor: $\big\langle \prod_k V_{\alpha_k}(z_k)\big\rangle = \prod_{k<l}(z_k - z_l)^{\alpha_k\alpha_l}$.

Worked Example 48.38 (Two- and three-point functions of vertex operators)**.** The neutrality rule and Eq. (48.302) are strong enough to reproduce, in this concrete realization, the kinematic forms that conformal symmetry fixes abstractly. For two operators, neutrality forces $\alpha_2 = -\alpha_1 \equiv -\alpha$, so $\alpha_1\alpha_2 = -\alpha^2$ and

$$\big\langle V_\alpha(z_1,\bar{z}_1)\, V_{-\alpha}(z_2,\bar{z}_2)\big\rangle = |z_1 - z_2|^{-2\alpha^2} = |z_{12}|^{-4h_\alpha}, \tag{48.303}$$

using $\Delta_\alpha = 2h_\alpha = \alpha^2$. This is precisely the conformally required two-point function of a primary of dimension Δ_α (Eq. (48.30)); the conjugate operator $V_{-\alpha}$ is the charge conjugate of V_α. For three operators, neutrality is $\alpha_1+\alpha_2+\alpha_3 = 0$, and

$$\Big\langle \prod_{k=1}^{3} V_{\alpha_k}(z_k,\bar{z}_k) \Big\rangle = \prod_{k<l} |z_{kl}|^{2\alpha_k\alpha_l}. \tag{48.304}$$

The exponents reorganize exactly into the standard Δ_i-dependent powers: from $2\alpha_1\alpha_2 = (\alpha_1+\alpha_2)^2 - \alpha_1^2 - \alpha_2^2 = \alpha_3^2 - \alpha_1^2 - \alpha_2^2 = \Delta_3 - \Delta_1 - \Delta_2$ and its permutations, the three pair exponents are $\Delta_3 - \Delta_1 - \Delta_2$, $\Delta_2 - \Delta_1 - \Delta_3$, $\Delta_1 - \Delta_2 - \Delta_3$ (over $|z_{12}|, |z_{13}|, |z_{23}|$ respectively), which is the universal three-point structure. The free boson thus realizes the abstract two- and three-point kinematics with the bonus that the OPE coefficient is fixed — to unity — and the neutrality rule plays the role of a fusion selection rule, $V_{\alpha_1} \times V_{\alpha_2} \to V_{\alpha_1+\alpha_2}$.

48.7.2 The compact boson, momentum and winding, and $R \to 1/R$ duality

The noncompact boson has a continuum of vertex charges $\alpha \in \mathbb{R}$ and a noncompact spectrum. Two-dimensional applications — the XY model, bosonization,

string compactification — demand instead that X be an *angular* variable,

$$X \sim X + 2\pi R, \tag{48.305}$$

with R the *radius*. Compactification has two immediate consequences. First, single-valuedness of any operator under the identification Eq. (48.305) restricts the allowed charges: $V_\alpha = :e^{i\alpha X}:$ is invariant under $X \to X + 2\pi R$ only if $e^{2\pi i \alpha R} = 1$, i.e.

$$\alpha = \frac{n}{R}, \qquad n \in \mathbb{Z} \qquad (\textit{momentum} \text{ quantization}). \tag{48.306}$$

Second, the field can now *wind*: a configuration in which X advances by an integer multiple of the period as one circles the spatial direction,

$$X(\sigma + 2\pi, \tau) = X(\sigma, \tau) + 2\pi R\, w, \qquad w \in \mathbb{Z} \qquad (\textit{winding} \text{ number}), \tag{48.307}$$

is now an allowed, topologically distinct sector — impossible on the noncompact line. These two integers (n, w) label the states.

To find their weights it is cleanest to track the holomorphic and antiholomorphic momenta separately. The current $J = i\partial X$ has a zero mode measuring the total momentum, $p = \oint \frac{\mathrm{d}z}{2\pi}\, i\partial X = n/R$ by Eq. (48.306); the winding Eq. (48.307) is the zero mode of the antisymmetric combination $\partial X - \bar\partial X$ and, with the kinetic normalization of Eq. (48.289), contributes a charge $\propto wR$. The upshot is that the operator carrying (n, w) factorizes into chiral exponentials $e^{ip_L X_L} e^{ip_R X_R}$ with left/right momenta

$$p_L = \frac{n}{R} + \frac{wR}{2}, \qquad p_R = \frac{n}{R} - \frac{wR}{2}, \tag{48.308}$$

and, since each chiral exponential has weight $\frac{1}{2}p^2$ by the chiral version of Eq. (48.299), conformal weights

$$h_{n,w} = \frac{1}{2}\Big(\frac{n}{R} + \frac{wR}{2}\Big)^2, \qquad \bar h_{n,w} = \frac{1}{2}\Big(\frac{n}{R} - \frac{wR}{2}\Big)^2. \tag{48.309}$$

Two checks. The pure-momentum sector $w = 0$ reduces to $h = \bar h = \frac{1}{2}(n/R)^2$, the weight Eq. (48.299) of $V_{n/R}$, as it must. The conformal spin $h - \bar h = nw$ is an integer, as required for a local operator. The scaling dimension is $\Delta_{n,w} = h + \bar h = (n/R)^2 + (wR/2)^2$, the sum of a momentum and a winding contribution.

Notation Watch

The split Eq. (48.308) carries a convention-dependent factor in the winding term, fixed here by the kinetic normalization $1/8\pi$ of Eq. (48.289) (equivalently $\alpha' = 2$ in string units). Other texts absorb this factor into the definition of R. Nothing physical — no weight, no central charge —

depends on the choice; only the numerical value of the self-dual radius does.

T-duality. The spectrum Eq. (48.309) is manifestly invariant under the simultaneous exchange of momentum with winding and inversion of the radius:

$$R \longrightarrow \widetilde{R} = \frac{2}{R}, \qquad (n, w) \longrightarrow (w, n). \tag{48.310}$$

Indeed, under Eq. (48.310) the left momentum maps to $p_L \to w/\widetilde{R} + n\widetilde{R}/2 = wR/2 + n/R = p_L$ and the right momentum to $p_R \to wR/2 - n/R = -p_R$, leaving every $h_{n,w}$ and $\bar{h}_{n,w}$ unchanged. This is *T-duality*: the compact boson at radius R is the *same* conformal field theory as the one at radius $2/R$, with the roles of momentum and winding interchanged. There is no way, from inside the theory, to tell a small radius from a large one. Measuring the radius in units of the self-dual value, $\widehat{R} \equiv R/\sqrt{2}$, the duality takes the symmetric form $\widehat{R} \to 1/\widehat{R}$, with the fixed point at $\widehat{R} = 1$.

Enhanced symmetry at the self-dual radius. The duality Eq. (48.310) has a fixed point at $R = \sqrt{2}$ (i.e. $\widehat{R} = 1$), and there something special happens. Generically the only holomorphic currents (weight $(1,0)$ fields) are $J = i\partial X$ and its multiples. At $R = \sqrt{2}$ new weight-$(1,0)$ operators appear: solving $h_{n,w} = 1$, $\bar{h}_{n,w} = 0$ in Eq. (48.309) requires $p_R = 0$ and $p_L = \pm\sqrt{2}$, i.e. $n = w = \pm 1$, which is integral precisely at $R = \sqrt{2}$. These are the chiral vertex operators

$$J^{\pm}(z) = :e^{\pm i\sqrt{2}\,X_L(z)}:, \qquad h = \tfrac{1}{2}(\sqrt{2})^2 = 1, \tag{48.311}$$

which together with $J^3(z) = i\sqrt{2}\,\partial X_L(z)$ close, by the vertex-operator OPEs computed above, onto the SU(2) current algebra at *level* $k = 1$. The free boson at the self-dual radius is thus the $\mathrm{SU}(2)_1$ Wess–Zumino–Witten model in disguise: an abelian symmetry enhanced, at one special radius, to a nonabelian one. This enhancement — and its breaking as one moves off $R = \sqrt{2}$ — is the prototype of every Higgs-like and stringy moduli phenomenon, and it is the simplest place to see an entire continuous symmetry materialize from the spectrum of a free field.

48.7.3 The free Majorana fermion and $c = \frac{1}{2}$

The boson gave $c = 1$. To reach the bottom of the minimal series we need a second free realization, and the cheapest is a free fermion. The point of this subsection is to obtain $c = \frac{1}{2}$ from a Lagrangian field, completing the circle with the Ising fermion of Eq. (48.270) promised in the previous section: the universality class that ϕ^4 theory approaches at the Wilson–Fisher point has, in two dimensions, an exactly free fermionic microscopic description.

Definition 48.39 (Free Majorana fermion)**.** The *free Majorana* (real) fermion is the $c = \frac{1}{2}$ theory of a holomorphic anticommuting field $\psi(z)$ with two-point function

$$\langle \psi(z)\,\psi(w)\rangle = \frac{1}{z-w}, \tag{48.312}$$

governed by the Euclidean action $S = \frac{1}{2\pi}\int \mathrm{d}^2x\,(\psi\bar\partial\psi + \bar\psi\partial\bar\psi)$, with holomorphic stress tensor

$$T(z) = -\tfrac{1}{2}\,{:}\psi\,\partial\psi{:}(z). \tag{48.313}$$

The antiholomorphic partner $\bar\psi(\bar z)$ contributes an independent copy.

ψ is a weight-$\frac{1}{2}$ primary. The contractions we need follow from Eq. (48.312): $\langle\psi(z)\psi(w)\rangle = (z-w)^{-1}$ and $\langle\partial\psi(z)\psi(w)\rangle = \partial_z(z-w)^{-1} = -(z-w)^{-2}$. Wick-contracting $T(z)\psi(w) = -\frac{1}{2}{:}\psi\partial\psi{:}(z)\,\psi(w)$, the external $\psi(w)$ contracts with either factor; tracking the single anticommutation needed to bring it adjacent to the contracted field,

$$T(z)\,\psi(w) \sim -\tfrac{1}{2}\Big[\langle\psi(z)\psi(w)\rangle\,\partial\psi(z) - \langle\partial\psi(z)\psi(w)\rangle\,\psi(z)\Big] = -\tfrac{1}{2}\Big[\frac{\partial\psi(z)}{z-w} + \frac{\psi(z)}{(z-w)^2}\Big]. \tag{48.314}$$

Taylor-expanding $\psi(z)$ and $\partial\psi(z)$ about w, the double pole collects $\frac{1}{2}\psi(w)$ and the single pole collects $\partial\psi(w)$ (the $-\frac{1}{2}$ and the expansion of the double-pole term combining to fix the descendant coefficient):

$$T(z)\,\psi(w) \sim \frac{\frac{1}{2}\,\psi(w)}{(z-w)^2} + \frac{\partial\psi(w)}{z-w}, \tag{48.315}$$

so ψ is primary with $h = \frac{1}{2}$. This is exactly the weight of the Ising energy operator's chiral constituent and, doubled, of ε (Theorem 48.33): $\varepsilon = i\bar\psi\psi$ has $(h,\bar h) = (\frac{1}{2},\frac{1}{2})$, $\Delta_\varepsilon = 1$.

Central charge $c = \frac{1}{2}$. Now contract the full product $T(z)T(w) = \frac{1}{4}\,{:}\psi\partial\psi{:}(z)\,{:}\psi\partial\psi{:}(w)$. The central term is the maximal contraction, in which both fields at z contract with both at w. Label the four fields $\psi(z), \partial\psi(z), \psi(w), \partial\psi(w)$. There are two complete pairings:

$$\text{pattern (i):}\quad \langle\psi(z)\psi(w)\rangle\,\langle\partial\psi(z)\partial\psi(w)\rangle = \frac{1}{z-w}\cdot\Big(-\frac{2}{(z-w)^3}\Big) = -\frac{2}{(z-w)^4}, \tag{48.316}$$

$$\text{pattern (ii):}\quad \langle\psi(z)\partial\psi(w)\rangle\,\langle\partial\psi(z)\psi(w)\rangle = \frac{1}{(z-w)^2}\cdot\Big(-\frac{1}{(z-w)^2}\Big) = -\frac{1}{(z-w)^4}, \tag{48.317}$$

where $\langle\partial\psi(z)\partial\psi(w)\rangle = \partial_z\partial_w(z-w)^{-1} = -2(z-w)^{-3}$ and $\langle\psi(z)\partial\psi(w)\rangle = \partial_w(z-w)^{-1} = (z-w)^{-2}$. Fermi statistics weights the two patterns by the sign

of the contraction permutation: pattern (i), whose contraction lines cross once, carries an extra (-1), while pattern (ii), with nested lines, carries $+1$. The central c-number is therefore

$$\tfrac{1}{4}\Big[(-1)\cdot\Big(-\frac{2}{(z-w)^4}\Big)+(+1)\cdot\Big(-\frac{1}{(z-w)^4}\Big)\Big]=\tfrac{1}{4}\cdot\frac{2-1}{(z-w)^4}=\frac{1/4}{(z-w)^4}. \tag{48.318}$$

Matching to $\frac{c/2}{(z-w)^4}$ (Theorem 48.8) gives $c/2=\frac{1}{4}$, hence

$$\boxed{c=\tfrac{1}{2}} \tag{48.319}$$

the lowest member $m=3$ of the discrete series. The single contractions reproduce, exactly as in the bosonic case, the $2T/(z-w)^2+\partial T/(z-w)$ weight-2 terms. The Ising CFT is the diagonal product of this holomorphic Majorana with its antiholomorphic mirror, $(c,\bar{c})=(\frac{1}{2},\frac{1}{2})$ — the free-fermion realization underlying the abstract solution of Section 48.6, and the microscopic statement that the two-dimensional Ising critical point is, after the Jordan–Wigner transformation, a theory of free fermions.

Physical Insight

Bosonization ties the two free realizations together. A *Dirac* fermion (two Majoranas) is equivalent to a single compact boson, with the fermion bilinears realized as vertex operators: $\psi_{\text{Dirac}}\sim{:}e^{iX}{:}$ at the appropriate radius, and the fermion number current as $\bar{\psi}_{\text{Dirac}}\gamma\psi_{\text{Dirac}}\sim\partial X$. The weights match because $h(e^{iX})=\frac{1}{2}$ equals the free-fermion weight. The single Majorana underlying Ising is then "half" of this boson, which is why $c=\frac{1}{2}=\frac{1}{2}\cdot 1$. This boson–fermion dictionary is the two-dimensional shadow of the more elaborate dualities of later parts, and it is the reason free fields are powerful out of proportion to their simplicity.

48.7.4 The Coulomb gas: background charge and the Feigin–Fuks construction

We have $c=1$ from the boson and $c=\frac{1}{2}$ from the fermion, but no single free field gives the intermediate minimal-model values $c=1-6/[m(m+1)]$ for $m=4,5,\dots$, nor does it explain *why* those values, and not others, are singled out. The Feigin–Fuks construction supplies both. The idea is to deform the free boson's stress tensor by a term linear in $\partial^2 X$, controlled by a *background charge* Q. Geometrically this places a fixed charge Q "at infinity" (on the sphere); algebraically it shifts the central charge and the vertex-operator weights in exactly the way needed to land on the minimal series.

Theorem 48.40 (Background-charge deformation; Feigin–Fuks)**.** *Let X be the free boson of Theorem 48.36. For any real Q the modified holomorphic stress tensor*

$$T_Q(z) = -\tfrac{1}{2}{:}\partial X\,\partial X{:}(z) + \frac{iQ}{2}\,\partial^2 X(z) \tag{48.320}$$

satisfies the $T(z)T(w)$ operator product of Eq. (48.56) with central charge

$$c = 1 - 3Q^2, \tag{48.321}$$

and the vertex operator $V_\alpha = {:}e^{i\alpha X}{:}$ remains primary, now with holomorphic weight

$$h_\alpha = \tfrac{1}{2}\alpha(\alpha - Q) = \tfrac{1}{2}\alpha^2 - \tfrac{1}{2}Q\alpha. \tag{48.322}$$

Choosing Q so that $Q^2 = 2\,(p-p')^2/(pp')$ for coprime integers $p > p' \geq 2$ yields $c = 1 - 6(p-p')^2/(pp')$, exactly the central charge of the minimal model $\mathcal{M}(p,p')$ of Eq. (48.176).

Proof. Central charge. Write $T_Q = A + B$ with $A = -\tfrac{1}{2}{:}\partial X\partial X{:}$ the free stress tensor and $B = \tfrac{iQ}{2}\partial^2 X$ the linear deformation. The product $T_Q T_Q = AA + AB + BA + BB$, and we read off the coefficient of $(z-w)^{-4}$. The pure free term AA already supplied $\tfrac{1}{2}/(z-w)^4$, i.e. the free $c = 1$ (Theorem 48.9). The deformation contributes through BB, a contraction of two linear-in-X fields:

$$\langle \partial^2 X(z)\,\partial^2 X(w)\rangle = \partial_z^2\partial_w^2\big[-\ln(z-w)\big] = \partial_z^2\partial_w^2\,\langle X(z)X(w)\rangle = \frac{6}{(z-w)^4}, \tag{48.323}$$

where the four derivatives of $-\ln(z-w)$ are computed step by step: $\partial_z\partial_w(-\ln) = -(z-w)^{-2}$, then ∂_z gives $+2(z-w)^{-3}$, then ∂_w gives $+6(z-w)^{-4}$. Hence

$$B(z)B(w) = \Big(\frac{iQ}{2}\Big)^2 \frac{6}{(z-w)^4} = -\frac{Q^2}{4}\cdot\frac{6}{(z-w)^4} = -\frac{3Q^2/2}{(z-w)^4}, \tag{48.324}$$

a pure c-number. The cross terms AB, BA contract one ∂X from A with the single $\partial^2 X$ of B and produce only poles of order ≤ 3 multiplying operators, contributing nothing to the $(z-w)^{-4}$ c-number (and, as one checks, exactly the modified $2T_Q/(z-w)^2 + \partial T_Q/(z-w)$ structure required for T_Q to be a stress tensor). Adding the free and deformation c-numbers,

$$\frac{c/2}{(z-w)^4} = \frac{1/2}{(z-w)^4} - \frac{3Q^2/2}{(z-w)^4} \quad\Longrightarrow\quad c = 1 - 3Q^2, \tag{48.325}$$

which is Eq. (48.321).

Vertex weight. The free stress tensor part A gave $h_\alpha^{\text{free}} = \tfrac{1}{2}\alpha^2$ via Eq. (48.298). The deformation $B = \tfrac{iQ}{2}\partial^2 X$ adds a contraction of $\partial^2 X(z)$ with the exponential. Using $\langle \partial^2 X(z)X(w)\rangle = \partial_z^2[-\ln(z-w)] = (z-w)^{-2}$,

$$\frac{iQ}{2}\,\partial^2 X(z)\,{:}e^{i\alpha X(w)}{:} \sim \frac{iQ}{2}\cdot(i\alpha)\,\frac{1}{(z-w)^2}\,V_\alpha(w) = -\frac{Q\alpha/2}{(z-w)^2}\,V_\alpha(w), \tag{48.326}$$

so the double-pole coefficient shifts from $\frac{1}{2}\alpha^2$ to $\frac{1}{2}\alpha^2 - \frac{1}{2}Q\alpha$, while the single-pole descendant term is unchanged (the deformation contributes no new $(z-w)^{-1}$ piece at leading order). Hence $T_Q(z)V_\alpha(w) \sim h_\alpha V_\alpha/(z-w)^2 + \partial V_\alpha/(z-w)$ with $h_\alpha = \frac{1}{2}\alpha(\alpha - Q)$, which is Eq. (48.322).

Matching the minimal series. By Eq. (48.321), $c = 1-6(p-p')^2/(pp')$ requires $3Q^2 = 6(p-p')^2/(pp')$, i.e. $Q^2 = 2(p-p')^2/(pp')$. This is the central charge of $\mathcal{M}(p,p')$ (Eq. (48.176)); in the consecutive-integer parametrization $p = m+1$, $p' = m$ of the unitary series it reads $c = 1 - 6/[m(m+1)]$ (Eq. (48.143)). □

Two features of Eq. (48.322) are decisive for what follows. First, the weight is invariant under $\alpha \to Q - \alpha$: $h_{Q-\alpha} = \frac{1}{2}(Q-\alpha)(-\alpha) = \frac{1}{2}\alpha(\alpha - Q) = h_\alpha$. The two charges α and $Q-\alpha$ create the *same* primary; this $\alpha \leftrightarrow Q - \alpha$ *charge conjugation* replaces the $\alpha \to -\alpha$ symmetry the background charge broke. Second, the background charge modifies neutrality. Repeating the zero-mode integral Eq. (48.301) in the presence of the charge Q at infinity, a correlator is nonzero only if the inserted charges sum to Q:

$$\Big\langle \prod_k V_{\alpha_k} \Big\rangle \neq 0 \quad \Longleftrightarrow \quad \sum_k \alpha_k = Q. \tag{48.327}$$

This shifted rule is the source of all the work below: a generic product of minimal-model vertex operators will *not* have charges summing to Q, and we must supply the deficit without disturbing the conformal weights. That is the job of the screening charges.

Degenerate fields and screening charges

The minimal-model primaries are the degenerate fields of Section 48.4.4, with Kac weights $h_{r,s}$. In the Coulomb gas they are vertex operators of definite charge. Define the two *screening charges* $\alpha_\pm$ as the charges whose vertex operators have weight exactly one,

$$h_{\alpha_\pm} = \tfrac{1}{2}\alpha_\pm(\alpha_\pm - Q) = 1 \quad \Longleftrightarrow \quad \alpha_\pm^2 - Q\alpha_\pm - 2 = 0, \tag{48.328}$$

so that, as roots of one quadratic, $\alpha_+ + \alpha_- = Q$ and $\alpha_+\alpha_- = -2$. Solving with $Q^2 = 2(p-p')^2/(pp')$ gives the explicit values

$$\alpha_+ = \sqrt{\frac{2p}{p'}}\,, \qquad \alpha_- = -\sqrt{\frac{2p'}{p}}\,, \tag{48.329}$$

which indeed satisfy $\alpha_+\alpha_- = -2$ and $\alpha_+ + \alpha_- = \sqrt{2}\,(p-p')/\sqrt{pp'} = Q$ (positive for $p > p'$). The degenerate primaries are then created by the charges

$$\alpha_{r,s} = \tfrac{1}{2}(1-r)\,\alpha_+ + \tfrac{1}{2}(1-s)\,\alpha_-, \qquad 1 \le r \le p'-1,\ 1 \le s \le p-1, \tag{48.330}$$

and substituting Eq. (48.330) into Eq. (48.322) reproduces the Kac formula of Eq. (48.177). For instance, with $p = m+1$, $p' = m$ and the charge $\alpha_{2,1} = -\frac{1}{2}\alpha_+$, one finds $h_{\alpha_{2,1}} = \frac{1}{2}(\frac{1}{2}\alpha_+)(\frac{1}{2}\alpha_+ + Q)$; using $\alpha_+^2 = 2(m+1)/m$ and $\alpha_+ Q = 2/m$ (both from Eq. (48.329)) this collapses to $h_{2,1} = (m+3)/(4m)$, exactly the Kac entry of Section 48.4.4. The free-field construction therefore reproduces the entire Kac table, weight for weight.

Notation Watch

We normalize the charges so that the screening operators have $h = 1$ and $\alpha_+\alpha_- = -2$, which pairs cleanly with the weight formula $h_\alpha = \frac{1}{2}\alpha(\alpha - Q)$ and $c = 1 - 3Q^2$ used throughout this section. A common alternative rescales every charge by $1/\sqrt{2}$, giving $\alpha_+\alpha_- = -1$ and $c = 1 - 24\alpha_0^2$ with background charge $2\alpha_0 = Q$; that is the normalization in which Eq. (48.142) was written. The two differ only by the conventional $\sqrt{2}$ and agree on every physical weight $h_{r,s}$ and on c.

The reason the operators Eq. (48.328) are called *screening* charges is that, having weight $(1,0)$, their integrals are conformally invariant and can be inserted freely to fix neutrality. We make this precise.

Definition 48.41 (Screening charges)**.** The *screening charges* are the contour integrals of the weight-$(1,0)$ vertex operators Eq. (48.328),

$$\mathcal{Q}_\pm = \oint \mathrm{d}z \, :e^{i\alpha_\pm X(z)}: , \tag{48.331}$$

each carrying U(1) charge $\alpha_\pm$ but conformal weight zero.

Proposition 48.42 (Screening charges are conformally invariant)**.** *Each $\mathcal{Q}_\pm$ of Eq. (48.331) commutes with every Virasoro generator, $[L_n, \mathcal{Q}_\pm] = 0$, and is therefore a dimension-zero operator that may be inserted into any correlator without altering its conformal transformation properties — changing only the total* U(1) *charge by $\alpha_\pm$.*

Proof. A weight-one holomorphic field $\phi(z)$ transforms as $\phi(z)(\mathrm{d}z)^1$, a 1-form, so its contour integral $\oint \mathrm{d}z\, \phi$ is invariant under any holomorphic reparametrization that leaves the contour endpoints (here none — the contour is closed) fixed; this is the residue/contour-deformation calculus of Theorem 5.23. Concretely, write $L_n = \oint \frac{\mathrm{d}z}{2\pi i}\, z^{n+1} T_Q(z)$ and use the primary OPE $T_Q(z) V_{\alpha_\pm}(w) \sim V_{\alpha_\pm}(w)/(z-w)^2 + \partial V_{\alpha_\pm}(w)/(z-w)$ (weight 1). Then

$$[L_n, \mathcal{Q}_\pm] = \oint \mathrm{d}w\, [L_n, V_{\alpha_\pm}(w)] = \oint \mathrm{d}w \left[(n+1)w^n\, V_{\alpha_\pm}(w) + w^{n+1}\, \partial V_{\alpha_\pm}(w)\right] = \oint \mathrm{d}w\, \partial_w (w^{n+1} V_{\alpha_\pm} \tag{48.332}$$

the integrand being a total derivative of a single-valued function around a closed contour (the would-be branch cut of $V_{\alpha_\pm}$ is harmless because $w^{n+1} V_{\alpha_\pm}$

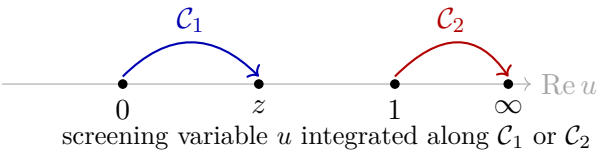


Figure 48.7: The two independent screening contours for the Ising spin four-point function after the Möbius fixing $z_1 = 0$, $z_2 = z$, $z_3 = 1$, $z_4 = \infty$. A single screening charge of type α_- is integrated over the variable u; the contour $\mathcal{C}_1$ from 0 to z produces the identity-channel block $f_+(z) \sim z^{-1/8}$, while $\mathcal{C}_2$ from 1 to ∞ produces the energy-channel block $f_-(z) \sim z^{3/8}$. The two channels are exactly $\sigma \times \sigma \to \mathbf{1} + \varepsilon$ of Eq. (48.240).

is integrated over a closed loop encircling no other insertion). Hence $\mathcal{Q}_\pm$ is Virasoro-invariant and has weight zero. □

Dotsenko–Fateev integral representations

The pieces now assemble into a computational rule for minimal-model correlators. Given physical primaries with charges $\alpha_1, \dots, \alpha_N$, the background-charge neutrality Eq. (48.327) demands $\sum_k \alpha_k = Q$. When the inserted charges fall short, we make up the deficit with screenings: insert n_+ copies of $\mathcal{Q}_+$ and n_- of $\mathcal{Q}_-$, chosen so that

$$\sum_{k=1}^{N} \alpha_k + n_+ \alpha_+ + n_- \alpha_- = Q. \tag{48.333}$$

By Theorem 48.42 this changes the charge but not the conformal structure, so the screened correlator is a genuine correlator of the intended primaries. The result is the *Dotsenko–Fateev integral representation*

$$\Big\langle \prod_k V_{\alpha_k}(z_k) \Big\rangle = \oint \prod_{a=1}^{n_+} \mathrm{d}u_a \oint \prod_{b=1}^{n_-} \mathrm{d}v_b \, \Big\langle \prod_k V_{\alpha_k}(z_k) \prod_a V_{\alpha_+}(u_a) \prod_b V_{\alpha_-}(v_b) \Big\rangle_{\text{free}}, \tag{48.334}$$

where the free correlator on the right is the explicit product of power laws Eq. (48.302), now neutral by construction, and the contour integrals over the screening positions u_a, v_b run over independent cycles. Each independent choice of contour produces one conformal block; the number of independent contours equals the number of fusion channels, so the integral representation packages all blocks of a given correlator at once. The monodromy-invariant combination of these contour integrals — the single-valued sum that closes the operator algebra — is the physical correlator. This is how the minimal-model four-point functions, fixed abstractly by the BPZ equations of Section 48.6, acquire closed-form integral expressions; for higher points the Coulomb gas is often the only practical method.

Worked Example 48.43 (The Ising spin four-point function from a single screening integral)**.** We now recover the four-point function of spins $\langle\sigma\sigma\sigma\sigma\rangle$ — obtained in Theorem 48.34 from the BPZ differential equation — by the Coulomb gas, as an independent cross-check. We work in the Ising model $\mathcal{M}(3,4)$ ($m=3$, $p=4$, $p'=3$), so by Eq. (48.329) the screening charges and background charge are

$$\alpha_+ = \sqrt{\tfrac{8}{3}}, \qquad \alpha_- = -\sqrt{\tfrac{3}{2}}, \qquad Q = \alpha_+ + \alpha_- = \tfrac{1}{\sqrt{6}}, \qquad Q^2 = \tfrac{1}{6},\ c = 1 - 3Q^2 = \tfrac{1}{2}, \tag{48.335}$$

confirming $c = \frac{1}{2}$. The spin $\sigma = \phi_{1,2}$ has charge $\alpha_\sigma = \alpha_{1,2} = -\frac{1}{2}\alpha_- = \frac{1}{2}\sqrt{\frac{3}{2}} = \frac{\sqrt{6}}{4}$, and one verifies its weight from Eq. (48.322): $\alpha_\sigma - Q = \frac{\sqrt{6}}{4} - \frac{1}{\sqrt{6}} = \frac{1}{2\sqrt{6}}$, so $h_\sigma = \frac{1}{2}\cdot\frac{\sqrt{6}}{4}\cdot\frac{1}{2\sqrt{6}} = \frac{1}{16}$, exactly $\Delta_\sigma = \frac{1}{8}$ split holomorphically.

Charge balance forces one screening. Insert three spins at charge α_σ and the fourth at the conjugate charge $\alpha'_\sigma = Q - \alpha_\sigma$ (same primary, by $\alpha \leftrightarrow Q-\alpha$), and n_- screenings of type α_-. Neutrality Eq. (48.333) reads $3\alpha_\sigma + (Q-\alpha_\sigma) + n_-\alpha_- = Q$, i.e. $2\alpha_\sigma + n_-\alpha_- = 0$. Since $\alpha_- = -2\alpha_\sigma$, this is satisfied by exactly $n_- = 1$: *one* screening, hence a second-order problem with *two* independent contours — matching the second-order BPZ equation and its two conformal blocks. The chiral correlator is therefore

$$\langle\sigma(z_1)\sigma(z_2)\sigma(z_3)\sigma(z_4)\rangle = \int_{\mathcal{C}} \mathrm{d}u\ \langle V_{\alpha_\sigma}(z_1)V_{\alpha_\sigma}(z_2)V_{\alpha_\sigma}(z_3)V_{\alpha'_\sigma}(z_4)\,V_{\alpha_-}(u)\rangle_{\text{free}}. \tag{48.336}$$

Evaluating the free correlator. By Eq. (48.302) the integrand is a product of power laws. The required charge products are $\alpha_\sigma^2 = \frac{3}{8}$, $\alpha_\sigma\alpha_- = \frac{\sqrt{6}}{4}\cdot(-\sqrt{\frac{3}{2}}) = -\frac{3}{4}$, and $\alpha'_\sigma\alpha_- = (Q-\alpha_\sigma)\alpha_- = -\frac{1}{2} - (-\frac{3}{4}) = \frac{1}{4}$. Fixing the Möbius frame $z_1 = 0, z_2 = z, z_3 = 1, z_4 \to \infty$ (the z_4 factors going into an overall normalization), the finite-point prefactor is $z^{3/8}(1-z)^{3/8}$ (from the three $\alpha_\sigma^2 = \frac{3}{8}$ pairs among $0, z, 1$), and the u-dependence is

$$\int_{\mathcal{C}} \mathrm{d}u\ u^{\alpha_\sigma\alpha_-}\,(u-z)^{\alpha_\sigma\alpha_-}\,(u-1)^{\alpha_\sigma\alpha_-} = \int_{\mathcal{C}} \mathrm{d}u\ \big[u\,(u-z)\,(u-1)\big]^{-3/4}, \tag{48.337}$$

so that $\langle\sigma\sigma\sigma\sigma\rangle \propto z^{3/8}(1-z)^{3/8}\,J_{\mathcal{C}}(z)$ with $J_{\mathcal{C}}(z)$ the integral Eq. (48.337).

The two contours give the two blocks. Take first the contour $\mathcal{C}_1$ from 0 to z (Fig. 48.7). Substituting $u = zt$, $t \in [0,1]$, the integral becomes, up to a z-independent constant and phases,

$$J_{\mathcal{C}_1}(z) = z^{-1/2}\int_0^1 \mathrm{d}t\ t^{-3/4}(1-t)^{-3/4}(1-zt)^{-3/4} = z^{-1/2}\,B\!\left(\tfrac{1}{4},\tfrac{1}{4}\right)\,{}_2F_1\!\left(\tfrac{3}{4},\tfrac{1}{4};\tfrac{1}{2};z\right), \tag{48.338}$$

recognizing Euler's integral for the hypergeometric function (the analytic-continuation machinery of Section 5.4 guarantees this representation extends off the unit in-

terval). Hence the corresponding block is

$$f_+(z) \propto z^{3/8}(1-z)^{3/8}\cdot z^{-1/2}\,{}_2F_1(\tfrac{3}{4},\tfrac{1}{4};\tfrac{1}{2};z) = z^{-1/8}(1-z)^{3/8}\,{}_2F_1(\tfrac{3}{4},\tfrac{1}{4};\tfrac{1}{2};z), \tag{48.339}$$

whose leading behavior as $z\to 0$ is $f_+\sim z^{-1/8}=z^{-2h_\sigma}$ — the *identity channel*, intermediate weight $h=0$. Taking instead the contour $\mathcal{C}_2$ from 1 to ∞ yields the second solution of the same hypergeometric equation, with leading behavior $f_-\sim z^{3/8}=z^{h_\varepsilon-2h_\sigma}$ (using $h_\varepsilon=\frac{1}{2}$) — the *energy channel*. The two exponents differing by $z^{h_\varepsilon}=z^{1/2}$ are precisely the two terms of the fusion $\sigma\times\sigma=\mathbf{1}+\varepsilon$ (Eq. (48.240)).

Closed form and agreement. The hypergeometric ${}_2F_1(\frac{3}{4},\frac{1}{4};\frac{1}{2};z)$ is one of the quadratic-transformation cases that reduce to elementary functions; the two blocks collapse to

$$f_\pm(z) = \frac{1}{\sqrt{2}}\,z^{-1/8}(1-z)^{-1/8}\big(1\pm\sqrt{1-z}\big)^{1/2}. \tag{48.340}$$

One checks the limits directly: as $z\to 0$, $1+\sqrt{1-z}\to 2$ gives $f_+\to z^{-1/8}$, while $1-\sqrt{1-z}\to z/2$ gives $f_-\to z^{-1/8}(z/2)^{1/2}\sim z^{3/8}$, reproducing the two channel exponents above. These are exactly the two chiral conformal blocks delivered by the BPZ second-order ODE in Theorem 48.34. Gluing the holomorphic and antiholomorphic blocks into the unique monodromy-invariant (single-valued) combination,

$$\langle\sigma\sigma\sigma\sigma\rangle(z,\bar z)\ \propto\ |f_+(z)|^2+|f_-(z)|^2 = \tfrac{1}{2}\,|z(1-z)|^{-1/4}\Big(\big|1+\sqrt{1-z}\big|+\big|1-\sqrt{1-z}\big|\Big), \tag{48.341}$$

with the relative coefficient fixed to unity by demanding a single identity in the $\sigma\times\sigma$ OPE, recovers the closed-form spin four-point function of Theorem 48.34. The Coulomb gas and the BPZ differential equation — one built from a deformed free field, the other from a null vector — produce the identical correlator.

This cross-validation is the payoff of the entire chapter. The same number $c=\frac{1}{2}$ and the same four-point function have now been reached three ways: from Virasoro representation theory and the Kac determinant (Section 48.6), from a free Majorana fermion (Eq. (48.319)), and from a screened free boson (Theorem 48.43). That three independent constructions agree to the last factor is the concrete meaning of the claim that the minimal models are completely solved.

With explicit field realizations in hand the chapter closes, but its objects propagate forward. The conformal-block decomposition that the screening contours made manifest is the seed of the bootstrap: in Chapters 49 and 51 the same blocks, viewed as eigenfunctions of a Casimir rather than solutions of a null-vector ODE, become the analytic backbone of crossing symmetry in any

dimension. The free boson itself, with its vertex operators and $R \to 1/R$ duality, is the elementary string coordinate and the simplest holographic boundary theory, carried forward to the strings-and-holography parts. And the logarithm in the boson propagator Eq. (48.290), integrated over a region, is what makes the entanglement entropy of a two-dimensional critical system grow as $\frac{c}{3}\ln \ell$ — the universal c-dependent law taken up in Chapter 55, where the central charge we have computed three times over reappears as the coefficient of an entanglement.

Further reading

The foundational paper is Belavin et al. [56], which introduced the infinite conformal symmetry, the central charge, degenerate representations, and the null-vector differential equations now bearing the authors' initials; it remains the indispensable original. The unitarity bound and the discrete series are due to Friedan et al. [200], building on the representation theory of Kac [287]. The two standard textbook developments, both far more extensive than this chapter, are Di Francesco et al. [147] and the lecture notes of Ginsparg [221]; either is the natural next stop for the modular-invariance, Wess–Zumino–Witten, and torus-partition-function material we did not treat. The Coulomb-gas construction and the screened correlators are developed in Dotsenko and Fateev [160]. For the connection to critical phenomena and the statistical-mechanics viewpoint, Cardy [108] and the review by Cardy [109] are the canonical references, and the c-theorem organizing the space of two-dimensional theories under RG flow is Zamolodchikov [531].

Chapter 49

The Conformal Bootstrap: Foundations

The two preceding chapters built a conformal field theory out of its data and then learned to compute with it. Chapter 47 fixed what a CFT *is* — a spectrum of primary dimensions and spins together with the operator product coefficients that govern their fusion — and supplied the machinery that turns those numbers into correlators: a convergent operator product expansion (Theorem 47.48) and the decomposition of the four-point function into conformal blocks (Eq. (47.227)). Chapter 48 exhibited that data in solved form, most sharply in the $c = \frac{1}{2}$ Ising minimal model whose dimensions and structure constants (Theorem 48.33 and Eq. (48.319)) are known exactly from the free Majorana realization. In every case so far the CFT data were handed to us — read off a free field, computed order by order in the ϵ-expansion of Chapter 24, or postulated. This chapter inverts the posture. It asks which sets of data are admissible at all, not against a Lagrangian — there need be none — but against the bare demand that the operator algebra be associative. That single demand, written on the four-point function, is the conformal bootstrap.

The chapter is organized around one equation and the question of how to solve or bound it. Section 49.1 isolates the constraint: equating the s- and t-channel block expansions of $\langle\phi\phi\phi\phi\rangle$ produces crossing symmetry (Theorem 49.2), a single functional equation in which the identity operator supplies the inhomogeneous source and the unknown CFT data must conspire to cancel it. To make that equation computable we need the blocks as explicit functions, so Section 49.2 solves the conformal Casimir equation of Eq. (47.233) in closed form — collinear blocks from a Gauss hypergeometric, the finite $d = 2$ and $d = 4$ blocks, and the radial ρ-series whose convergence is tied to the radial-quantization inner product of Theorem 47.34. With explicit blocks in hand, Section 49.3 packages crossing into the bootstrap sum rule $\sum_{\mathcal{O}} p_{\mathcal{O}} F_{\Delta,\ell} = 0$, in which all dynamics sits in nonnegative weights and a discrete spectrum bounded below by the unitarity

bounds of Theorem 47.18.

That sum rule is a linear equation in nonnegative unknowns, and the geometry of such problems is the subject of Section 49.4: a linear functional made nonnegative on every admissible block certifies, by a sign contradiction, that no consistent unitary spectrum can do what it forbids (Theorem 49.21). This is the engine of every rigorous bootstrap bound. We then earn trust in the machine by running it where the answer is known. Section 49.5 solves crossing exactly for the generalized free (mean-field) field — extracting its full double-twist spectrum and OPE coefficients, the seed of the analytic bootstrap of Chapter 51 — and verifies term by block that the 2D Ising correlators of Chapter 48 satisfy the same equation. Finally Section 49.6 points the validated apparatus at a theory we cannot solve in closed form, the three-dimensional Ising universality class, posing the bootstrap problem precisely and stating, dated to the mid-2020s, exactly what is proved (the exclusions) versus what is numerical (the island).

Throughout, the chapter advances ϕ^4 theory. Mean-field theory is the Gaussian ($\lambda \to 0$) endpoint of the thread; the 3D Ising CFT is the interacting fixed point that the Wilson–Fisher flow of Theorem 24.31 approaches, and the closing worked example confronts the bootstrap island directly against the ϵ-expansion to confirm that the nonperturbative crossing solution and the perturbative RG fixed point are one CFT. We work in Euclidean signature with points $x_i \in \mathbb{R}^d$ and separations $x_{ij}^2 \geq 0$; conformal kinematics is signature-agnostic, and the mostly-minus metric of the rest of the book plays no role until the Lorentzian continuation of Chapter 51. The explicit numerical solution — semidefinite programming and the convergence of the island — is handed to Chapter 50; the analytic limits to Chapter 51; and the closure with the critical exponents of Part VI to Chapter 52.

49.1 Crossing Symmetry: The Four-Point Function as a Constraint

Chapters 47 and 48 handed us two things and asked nothing of them. First, a complete catalogue of what a conformal field theory *is*: the dimensions and spins of its primary operators together with the constants that govern their fusion — the CFT data. Second, the machinery that turns those numbers into correlation functions: a convergent operator product expansion (Theorem 47.48) and the decomposition of the four-point function into conformal blocks (Eq. (47.227)). Up to this point the data have been treated as given — read off a free field, computed order by order in ϵ, or simply postulated. The bootstrap inverts that posture. It asks: *which* sets of CFT data are consistent? Not consistent with a Lagrangian — there need be none — but consistent with the bare requirement that the operator algebra make sense. This section isolates the single equation that requirement becomes. The unknowns are the CFT data; the equation is the

associativity of the OPE, made concrete as crossing symmetry of the four-point function. Everything else in the chapter — closed-form blocks, unitarity bounds, the numerical and analytic attacks — is in service of solving, or bounding the solutions of, the relation written down here.

Throughout we work in Euclidean signature, as in Chapter 47: points $x_i \in \mathbb{R}^d$, separations $x_{ij}^2 = |x_i - x_j|^2 \geq 0$. The mostly-minus metric and natural units of the rest of the book play no role here; conformal kinematics is signature-agnostic until the Lorentzian continuation of Chapter 51.

49.1.1 The unknowns: CFT data

Recall the inventory assembled in Chapter 47. A conformal field theory is specified, completely, by two pieces of data:

$$\Big(\{\Delta_{\mathcal{O}},\ \ell_{\mathcal{O}}\}\,, \quad \{\, C_{\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3}\,\} \Big), \tag{49.1}$$

the *spectrum* — the scaling dimension $\Delta_{\mathcal{O}}$ and spin $\ell_{\mathcal{O}}$ of every primary operator $\mathcal{O}$ — and the *operator product coefficients* $C_{\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3}$, one real number for each ordered triple of primaries, appearing both as the constant in the three-point function (Theorem 47.24) and as the coefficient in the OPE (Theorem 47.46). That these two roles are filled by the *same* number is the content of the state–operator correspondence and the convergence of the OPE: the constant that controls how $\mathcal{O}_1$ and $\mathcal{O}_2$ fuse at short distance is the same constant that appears when all three operators are far apart. Two-point functions are fixed by symmetry up to a normalization we set to unity (Eq. (47.124)); three-point functions are fixed up to the single constant C_{123} (Theorem 47.24). Nothing else is free. Every correlation function of the theory, at any number of points, is in principle assembled from Eq. (49.1) by repeated use of the OPE. The data Eq. (49.1) are therefore the unknowns of the bootstrap: to "solve" a CFT is to determine them.

The decisive observation is that symmetry alone does not constrain the data at the level of any correlator below four points. The two-point function is a pure power law; the three-point function is that power law times a single undetermined constant; in both cases conformal invariance exhausts the kinematics and leaves the data unconstrained. The first correlator on which symmetry leaves a genuine *function* undetermined — and therefore the first place a consistency condition can bite — is the four-point function. This is the staircase of Section 47.4.6: at each step symmetry does less, and at the fourth step it stops fixing the answer and starts merely organizing it. The four-point function of identical scalars is thus the smallest object whose conformal-block decomposition is not dictated by symmetry, and it is on this object that the entire bootstrap is built.

49.1.2 The four-point function and its s-channel decomposition

Specialize, as ϕ^4 theory requires, to four identical scalar primaries: a single real scalar ϕ of dimension Δ_ϕ, the operator that will become the order parameter of the Wilson–Fisher / Ising fixed point. Conformal invariance fixes its four-point function up to a function of the two cross-ratios (Eq. (47.145)),

$$\langle\, \phi(x_1)\, \phi(x_2)\, \phi(x_3)\, \phi(x_4)\, \rangle = \frac{g(u,v)}{(x_{12}^2\, x_{34}^2)^{\Delta_\phi}}, \tag{49.2}$$

where the cross-ratios are (Eq. (47.138))

$$u = \frac{x_{12}^2\, x_{34}^2}{x_{13}^2\, x_{24}^2}, \qquad v = \frac{x_{14}^2\, x_{23}^2}{x_{13}^2\, x_{24}^2}. \tag{49.3}$$

Fusing the pair $(\phi(x_1)\phi(x_2))$ and, independently, the pair $(\phi(x_3)\phi(x_4))$ by the OPE projects the correlator onto the primaries $\mathcal{O}$ exchanged between the two pairs, and resumming each primary together with all its descendants into a conformal block (Theorem 47.49) gives the block decomposition (Eq. (47.227))

$$g(u,v) = \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v). \tag{49.4}$$

We call the pairing (12)(34) the *s-channel*. The weights are squared OPE coefficients $C_{\phi\phi\mathcal{O}}^2 \geq 0$ — squared because both pairs are the same pair $\phi\phi$, so each exchanged $\mathcal{O}$ enters with $C_{\phi\phi\mathcal{O}}$ from each side. The blocks $G_{\Delta,\ell}(u,v)$ are universal: they are fixed by conformal symmetry alone, common to every CFT in d dimensions, and depend on the exchanged operator only through its quantum numbers $(\Delta_\mathcal{O}, \ell_\mathcal{O})$. Their explicit construction is the business of Section 49.2; here they are inputs known in principle.

One exchanged operator is present in *every* CFT and deserves to be separated out: the identity $\mathbf{1}$, with $\Delta = 0$, $\ell = 0$. It appears in the $\phi \times \phi$ OPE because the leading term of any operator with itself is the identity, and its coefficient is fixed by the unit normalization of the two-point function, $C_{\phi\phi\mathbf{1}} = 1$. Its block is trivial: the identity exchange factorizes the four-point function into a product of two-point functions,

$$\langle\phi(x_1)\phi(x_2)\rangle\langle\phi(x_3)\phi(x_4)\rangle = \frac{1}{(x_{12}^2\, x_{34}^2)^{\Delta_\phi}}, \tag{49.5}$$

which upon stripping the prefactor of Eq. (49.2) contributes exactly 1 to $g(u,v)$. Hence

$$G_{0,0}(u,v) = 1, \tag{49.6}$$

and the s-channel sum reads

$$g(u,v) = 1 + \sum_{\mathcal{O}\neq\mathbf{1}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v). \tag{49.7}$$

The "1" is the disconnected piece; everything beyond it is genuine dynamics, weighted by the data we are trying to determine.

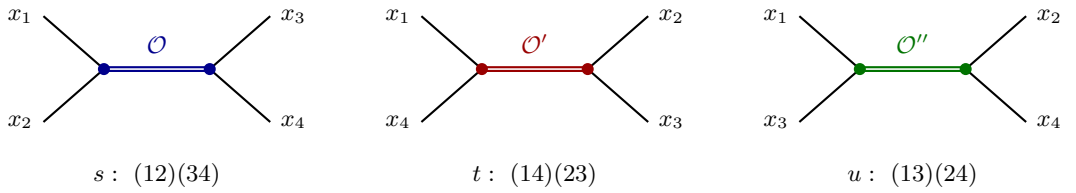


Figure 49.1: The three OPE channels of the four-point function $\langle\phi\phi\phi\phi\rangle$. The four points can be split into two pairs in exactly three ways; each splitting fuses a different pair by the OPE and expands the same correlator over a different tower of exchanged primaries. Crossing symmetry is the demand that the three series agree.

49.1.3 The three OPE channels

A four-point function offers more than one way to apply the OPE. Four objects can be split into two unordered pairs in exactly three ways,

$$\underbrace{(12)(34)}_{s}, \qquad \underbrace{(14)(23)}_{t}, \qquad \underbrace{(13)(24)}_{u}, \tag{49.8}$$

and each splitting defines an OPE channel in which the correlator can be expanded over exchanged primaries. We have used the s-channel above; the t- and u-channels fuse different pairs. Physically, the OPE is the statement that fusing two nearby operators produces a sum of local operators, and the four-point function does not care *which* two we fuse first — the operator algebra is associative. The three expansions Eq. (49.8) are therefore three series for one and the same number. Figure 49.1 renders the three pairings.

To turn "the same number" into an equation we must see how the kinematic prefactor and the cross-ratios behave when we pass from one channel to another. The channels are related by permuting the four (identical) points: since ϕ is a single bosonic scalar, the correlator is invariant under any permutation of $x_1, \dots, x_4$,

$$\langle\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle = \langle\phi(x_{\sigma(1)})\phi(x_{\sigma(2)})\phi(x_{\sigma(3)})\phi(x_{\sigma(4)})\rangle, \qquad \sigma \in S_4. \tag{49.9}$$

The s-channel pairing (12)(34) is carried into the t-channel pairing (14)(23) by the transposition $1 \leftrightarrow 3$, and into the u-channel pairing (13)(24) by $2 \leftrightarrow 3$. We compute the effect of $1 \leftrightarrow 3$ in detail, as it produces the crossing equation; the others go identically.

Lemma 49.1 (Action of the $1 \leftrightarrow 3$ exchange on the cross-ratios)**.** *Under the exchange of the points x_1 and x_3, the cross-ratios Eq. (49.3) swap,*

$$(u, v) \xmapsto{1\leftrightarrow 3} (v, u), \tag{49.10}$$

and the s-channel kinematic prefactor transforms as

$$\frac{1}{(x_{12}^2\,x_{34}^2)^{\Delta_\phi}} \xrightarrow{1\leftrightarrow 3} \frac{1}{(x_{32}^2\,x_{14}^2)^{\Delta_\phi}} = \frac{1}{(x_{14}^2\,x_{23}^2)^{\Delta_\phi}}. \tag{49.11}$$

Proof. Relabel the points by $y_1 = x_3$, $y_2 = x_2$, $y_3 = x_1$, $y_4 = x_4$, so that the exchange $1 \leftrightarrow 3$ is the statement that the y-correlator equals the x-correlator. The squared separations needed are read off directly, using $x_{ij}^2 = x_{ji}^2$:

$$\begin{aligned} &y_{12}^2 = x_{32}^2 = x_{23}^2, \quad y_{34}^2 = x_{14}^2, \quad y_{13}^2 = x_{31}^2 = x_{13}^2, \\ &y_{24}^2 = x_{24}^2, \qquad\quad y_{14}^2 = x_{34}^2, \quad y_{23}^2 = x_{21}^2 = x_{12}^2. \end{aligned} \tag{49.12}$$

Substituting into the definitions Eq. (49.3) evaluated on the y's,

$$u_y = \frac{y_{12}^2\,y_{34}^2}{y_{13}^2\,y_{24}^2} = \frac{x_{23}^2\,x_{14}^2}{x_{13}^2\,x_{24}^2} = v, \qquad v_y = \frac{y_{14}^2\,y_{23}^2}{y_{13}^2\,y_{24}^2} = \frac{x_{34}^2\,x_{12}^2}{x_{13}^2\,x_{24}^2} = u, \tag{49.13}$$

which is Eq. (49.10). The prefactor follows from $y_{12}^2 y_{34}^2 = x_{23}^2 x_{14}^2$, giving Eq. (49.11). □

49.1.4 Crossing symmetry

We now have the two facts the crossing equation is made of: the correlator is the same whether we fuse in the s- or t-pairing (Eq. (49.9)), and the passage between the two pairings swaps $(u, v) \to (v, u)$ while changing the prefactor as in Eq. (49.11). Equating the two block expansions and tracking the prefactor produces a single nontrivial functional equation.

Definition 49.2 (Crossing symmetry)**.** *Crossing symmetry* of the four-point function $\langle\phi\phi\phi\phi\rangle$ is the requirement that its s-channel and t-channel conformal-block decompositions describe the same correlator. Equivalently, in terms of the stripped function $g(u, v)$ of Eq. (49.2), it is the identity

$$\boxed{g(u,v) = \left(\frac{u}{v}\right)^{\Delta_\phi} g(v,u)} \tag{49.14}$$

holding for all values of the cross-ratios.

Proposition 49.3 (Derivation of the crossing equation)**.** *The block decomposition Eq. (49.4) together with permutation invariance Eq. (49.9) implies Eq. (49.14), equivalently the block sum rule*

$$\sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2 \Big[\, v^{\Delta_\phi}\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v) \;-\; u^{\Delta_\phi}\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(v,u) \,\Big] = 0. \tag{49.15}$$

Proof. Write the four-point function two ways, both as exact OPE expansions of *the same* correlator.

The s-channel expansion is Eq. (49.2) with Eq. (49.4),

$$\langle\phi\phi\phi\phi\rangle = \frac{1}{(x_{12}^2\,x_{34}^2)^{\Delta_\phi}} \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v). \tag{49.16}$$

For the t-channel, fuse the pairs (14) and (23). By permutation invariance Eq. (49.9) this is the s-channel expansion of the relabelled correlator $\langle\phi(x_3)\phi(x_2)\phi(x_1)\phi(x_4)\rangle$, i.e. the s-channel formula Eq. (49.16) evaluated on the points $y_1 = x_3$, $y_2 = x_2$, $y_3 = x_1$, $y_4 = x_4$ of Theorem 49.1. That lemma gives $(u_y, v_y) = (v, u)$ and the prefactor Eq. (49.11), so

$$\langle\phi\phi\phi\phi\rangle = \frac{1}{(x_{14}^2\,x_{23}^2)^{\Delta_\phi}} \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(v,u). \tag{49.17}$$

The two coefficient sets are identical: the operators exchanged in the t-channel of $\langle\phi\phi\phi\phi\rangle$ are the same primaries, with the same squared coefficients $C_{\phi\phi\mathcal{O}}^2$, because both pairs being fused are again the pair $\phi\phi$. Equating Eq. (49.16) and Eq. (49.17) and cancelling the common factor $\langle\phi\phi\phi\phi\rangle$ on the left, multiply through by $(x_{12}^2 x_{34}^2)^{\Delta_\phi}$:

$$\sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v) = \left(\frac{x_{12}^2\,x_{34}^2}{x_{14}^2\,x_{23}^2}\right)^{\Delta_\phi} \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2\, G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(v,u). \tag{49.18}$$

The prefactor ratio is itself a ratio of cross-ratios. From Eq. (49.3), $u = x_{12}^2 x_{34}^2/(x_{13}^2 x_{24}^2)$ and $v = x_{14}^2 x_{23}^2/(x_{13}^2 x_{24}^2)$ share the denominator $x_{13}^2 x_{24}^2$, so it cancels in the quotient,

$$\frac{u}{v} = \frac{x_{12}^2\,x_{34}^2}{x_{14}^2\,x_{23}^2}. \tag{49.19}$$

The left-hand side of Eq. (49.18) is $g(u,v)$ and the sum on the right is $g(v,u)$, so Eq. (49.18) is precisely Eq. (49.14). Finally, multiply Eq. (49.14) by v^{Δ_ϕ} and substitute the block sum for each g,

$$v^{\Delta_\phi} \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2 G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v) = u^{\Delta_\phi} \sum_{\mathcal{O}} C_{\phi\phi\mathcal{O}}^2 G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(v,u), \tag{49.20}$$

which rearranges into the sum rule Eq. (49.15). □

Two features of Eq. (49.15) are worth stating before they are exploited in the rest of the chapter. First, the factor $(u/v)^{\Delta_\phi}$ in Eq. (49.14) is not optional decoration: it is the exact Jacobian by which the kinematic prefactor differs between the s- and t-pairings, derived in Eq. (49.19) from nothing but the definitions of the cross-ratios. It carries the external dimension Δ_ϕ and couples the spectrum of exchanged operators to the dimension of the external scalar;

this coupling is what makes the bootstrap sensitive to Δ_ϕ at all. Second, the equation is one constraint that must hold *identically in u and v* — an infinite number of conditions, one for each point of cross-ratio space, on the discrete data $\{(\Delta_\mathcal{O}, \ell_\mathcal{O}), C^2_{\phi\phi\mathcal{O}}\}$. It is overdetermined in exactly the way that makes it powerful: a generic set of dimensions and coefficients will not satisfy it, and the ones that do are the candidate CFTs.

Physical Insight

The slogan "the equation is OPE associativity" is literal. Inside $\langle\phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle$ we may fuse $\phi(x_1)\phi(x_2)$ first and then combine the result with $\phi(x_3)\phi(x_4)$, or fuse $\phi(x_2)\phi(x_3)$ first and combine with $\phi(x_1)\phi(x_4)$. These are two bracketings of the same triple product of fusions, and the OPE — being an operator identity (Theorem 47.46) convergent on overlapping regions (Theorem 47.48) — must give the same answer for both. Crossing symmetry is associativity of the operator algebra written out on the four-point function. Higher-point associativity adds no genuinely new constraint: it can be reduced, by inserting complete sets of states, to the four-point relation together with the consistency of OPE coefficients. This is why the four-point function, and this one equation, are the whole game.

The identity operator plays a special and load-bearing role in Eq. (49.15). Separate it out on both sides using Eq. (49.6), $G_{0,0} = 1$:

$$(v^{\Delta_\phi} - u^{\Delta_\phi}) \; + \sum_{\mathcal{O}\neq\mathbf{1}} C^2_{\phi\phi\mathcal{O}}\Big[v^{\Delta_\phi} G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v) - u^{\Delta_\phi} G_{\Delta_\mathcal{O},\ell_\mathcal{O}}(v,u)\Big] = 0. \quad (49.21)$$

The identity contributes the inhomogeneous term $v^{\Delta_\phi} - u^{\Delta_\phi}$, which does not vanish (the two cross-ratios are independent). Were the identity absent, Eq. (49.21) would be a homogeneous equation solved trivially by the empty spectrum; it is the identity, present in every CFT with $C^2_{\phi\phi\mathbf{1}} = 1$ fixed by the normalization of $\langle\phi\phi\rangle$, that forces a nonempty tower of operators to "conspire" to cancel $v^{\Delta_\phi} - u^{\Delta_\phi}$. The overall scale of the equation is likewise fixed by the identity: because $C_{\phi\phi\mathbf{1}} = 1$ is not a free parameter, the inhomogeneous term has a definite normalization, and the squared coefficients $C^2_{\phi\phi\mathcal{O}}$ of all other operators are measured relative to it. The identity is, in this precise sense, the unique source term of the bootstrap.

Remark 49.4 (The u-channel and the role of S_4). We derived crossing from the single transposition $1 \leftrightarrow 3$ relating the s- and t-channels. The remaining transpositions of S_4 give nothing genuinely new for identical scalars. The exchange $2 \leftrightarrow 3$ maps the s-pairing $(12)(34)$ to the u-pairing $(13)(24)$; repeating the computation of Theorem 49.1 with $y_1 = x_1, y_2 = x_3, y_3 = x_2, y_4 = x_4$ gives $(u,v) \to (1/u,\, v/u)$ and prefactor ratio u^{Δ_ϕ}, hence the u-channel relation $g(u,v) = u^{\Delta_\phi} g(1/u,\, v/u)$. Meanwhile the exchange $1 \leftrightarrow 2$ preserves the s-pairing

and yields a *symmetry of g itself*, $g(u,v) = g(u/v,\, 1/v)$, reflecting that swapping the two operators inside a fused pair changes nothing. The full image of S_4 acting on the pair (u,v) is the group S_3 (the kernel being the Klein four-group $\{e, (12)(34), (13)(24), (14)(23)\}$ under which u, v are separately invariant); it is generated by the single crossing relation Eq. (49.14) together with the within-channel symmetry $g(u,v) = g(u/v, 1/v)$. For identical external scalars, then, the s–t equation Eq. (49.14) is the one independent dynamical constraint, and it is the equation the chapter solves.

49.1.5 A worked check: the generalized free field

Before leaving the framing, it pays to see crossing satisfied exactly in the one case where the four-point function is known in closed form without any block technology: the *generalized free field* (equivalently, mean-field theory), of which the free massless scalar of Section 48.7.1 is the special case $\Delta_\phi = (d-2)/2$. This is the trivial endpoint of ϕ^4 theory — the Gaussian fixed point — and verifying crossing on it confirms that Eq. (49.14) is correctly normalized, including the $(u/v)^{\Delta_\phi}$ Jacobian.

Worked Example 49.5 (Crossing of the generalized free field)**.** A generalized free field is defined by Wick's theorem: its connected correlators vanish and its four-point function is the sum over the three pairwise contractions, each a product of unit-normalized two-point functions $\langle \phi(x_i)\phi(x_j)\rangle = (x_{ij}^2)^{-\Delta_\phi}$,

$$\langle \phi(x_1)\phi(x_2)\phi(x_3)\phi(x_4)\rangle = \frac{1}{(x_{12}^2 x_{34}^2)^{\Delta_\phi}} + \frac{1}{(x_{13}^2 x_{24}^2)^{\Delta_\phi}} + \frac{1}{(x_{14}^2 x_{23}^2)^{\Delta_\phi}}. \tag{49.22}$$

Strip the prefactor as in Eq. (49.2), multiplying through by $(x_{12}^2 x_{34}^2)^{\Delta_\phi}$. The first term gives 1. The second and third are ratios of separations that are precisely the cross-ratios: from Eq. (49.3),

$$\frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2} = u, \qquad \frac{x_{12}^2 x_{34}^2}{x_{14}^2 x_{23}^2} = \frac{u}{v}, \tag{49.23}$$

the second equality being Eq. (49.19). Hence the stripped function is

$$g(u,v) = 1 + u^{\Delta_\phi} + \left(\frac{u}{v}\right)^{\Delta_\phi}. \tag{49.24}$$

The three terms are exactly the identity exchanged in the s-, u-, and t-channels respectively — the disconnected correlator is a pure sum of identity blocks, one per channel.

Now test crossing Eq. (49.14). First evaluate $g(v,u)$ by swapping the arguments in Eq. (49.24),

$$g(v,u) = 1 + v^{\Delta_\phi} + \left(\frac{v}{u}\right)^{\Delta_\phi}, \tag{49.25}$$

then multiply by the Jacobian $(u/v)^{\Delta_\phi}$ and expand term by term,

$$\left(\frac{u}{v}\right)^{\Delta_\phi} g(v,u) = \left(\frac{u}{v}\right)^{\Delta_\phi} + \left(\frac{u}{v}\right)^{\Delta_\phi} v^{\Delta_\phi} + \left(\frac{u}{v}\right)^{\Delta_\phi}\left(\frac{v}{u}\right)^{\Delta_\phi} = \left(\frac{u}{v}\right)^{\Delta_\phi} + u^{\Delta_\phi} + 1. \tag{49.26}$$

In the middle term $(u/v)^{\Delta_\phi} v^{\Delta_\phi} = u^{\Delta_\phi}$, and the last term collapses to 1. The result is $1 + u^{\Delta_\phi} + (u/v)^{\Delta_\phi} = g(u,v)$, so Eq. (49.14) holds identically. The Jacobian does precisely the work of permuting the three identity contributions among the channels: it carries the t-channel identity (1 in $g(v,u)$) into the t-channel term $(u/v)^{\Delta_\phi}$ of $g(u,v)$, and so on cyclically. This is the simplest nontrivial solution of the crossing equation, and it shows that the equation is satisfied not term by term but by the spectrum acting as a whole — here, a spectrum of "double-twist" operators of dimensions $2\Delta_\phi + 2n + \ell$ hidden inside the block decomposition of Eq. (49.24), whose existence Chapter 51 will extract analytically.

The generalized free field saturates crossing trivially because it has no interactions: its spectrum is rigidly fixed by Wick's theorem. The content of ϕ^4 theory at the frontier is the opposite situation. At the Wilson–Fisher fixed point in $d = 4 - \epsilon$, and at the Ising fixed point in $d = 3$, the scalar ϕ acquires an anomalous dimension and the four-point function is no longer a sum of three identity blocks; $g(u,v)$ becomes an honestly unknown function, and Eq. (49.14) becomes a genuine constraint on the unknown CFT data. The bootstrap is the program of extracting that data from the constraint.

We now have the central object of the chapter in final form: one crossing equation, Eq. (49.14) or equivalently the sum rule Eq. (49.15), relating the conformal-block decompositions of a single four-point function in two channels, with the exact $(u/v)^{\Delta_\phi}$ relabeling made explicit and the identity fixing its normalization. The equation is, however, still implicit: it is written in terms of the blocks $G_{\Delta,\ell}(u,v)$, which so far we know only as eigenfunctions of the conformal Casimir (Eq. (47.233)). To make crossing computable — to evaluate the sum rule, expand it near boundaries of cross-ratio space, and feed it to the numerical and analytic machinery of the rest of the chapter — we need the blocks as explicit functions. Constructing them is the task of Section 49.2.

49.2 Conformal Blocks in Closed Form

The crossing equation of Section 49.1.4 relates two block expansions of one four-point function, and the worked check of Theorem 49.5 showed it satisfied exactly by the generalized free field. Both used the conformal blocks $G_{\Delta,\ell}(u,v)$ as inputs, known only abstractly: by Theorem 47.49 a block is the contribution of a single conformal multiplet to the stripped four-point function, and by Eq. (47.233) it is fixed as the eigenfunction of the quadratic Casimir with

eigenvalue $c_{\Delta,\ell} = \Delta(\Delta - d) + \ell(\ell + d - 2)$ subject to the boundary condition that it begin with the OPE power $u^{(\Delta-\ell)/2}$. That is an implicit definition: it pins down a function but does not display it. To evaluate the sum rule Eq. (49.15), to differentiate it at the crossing-symmetric point, and to feed it to the numerical and analytic machinery of the rest of the chapter, we need the blocks as explicit, computable functions.

This section carries out that construction. We trade the cross-ratios for the lightcone variables $z, \bar{z}$, in which the Casimir operator simplifies; we solve the resulting ordinary differential equation in closed form, building the one-variable *collinear block* out of a Gauss hypergeometric function; and we assemble from it the closed-form blocks in $d = 2$ and $d = 4$, the two cases where a single multiplet collapses to a finite combination of hypergeometrics. For general d no such collapse occurs, and we develop instead the radial (ρ-coordinate) expansion: a power series whose leading term and recursion we extract, and whose convergence on the unit disk $|\rho| < 1$ we tie directly to the radial-quantization inner product of Theorem 47.34. This series is the workhorse of the numerical bootstrap. We close by reading off the two limits the later sections live on — the lightcone limit $\bar{z} \to 0$ that isolates leading twist, and the diagonal limit $z = \bar{z}$ — and by drawing the line between the global blocks built here and the Virasoro blocks of Chapter 48.

49.2.1 Lightcone coordinates and the Casimir as a differential operator

Recall from Eq. (47.140) the passage from the cross-ratios to a single complex coordinate. Fixing three of the four points to $0, 1, \infty$ by a conformal transformation places the fourth at a complex position z, and the cross-ratios become

$$u = z\bar{z}, \qquad v = (1 - z)(1 - \bar{z}), \tag{49.27}$$

with $\bar{z}$ the complex conjugate of z in Euclidean signature. The pair $(z, \bar{z})$ are the *lightcone variables*: in the Lorentzian continuation of Chapter 51 they become two independent real numbers, and $\bar{z} \to 0$ at fixed z is a genuine lightcone limit. We treat z and $\bar{z}$ as independent throughout, imposing reality only at the end. A function of (u, v) is automatically symmetric under $z \leftrightarrow \bar{z}$, since Eq. (49.27) is; conversely any symmetric function of $(z, \bar{z})$ descends to a function of (u, v). The blocks, being functions of the cross-ratios, are symmetric in $z \leftrightarrow \bar{z}$, and we will use this repeatedly.

The Casimir equation Eq. (47.233) was left in Chapter 47 in terms of the abstract operator $\mathcal{D}_2$, the realization of $C_2^{(1)} + C_2^{(2)}$ as a differential operator on the stripped correlator through the conformal Ward identities. Carrying out that realization — starting from the first-order differential realization of the conformal generators $D, P_\mu, K_\mu, M_{\mu\nu}$ established in Section 47.2.2, summing the

pair Casimir $C_2^{(1)}+C_2^{(2)}$ built from Eq. (47.228) over the fused points (x_1, x_2), and reducing to the invariants u, v — is a direct if lengthy computation, performed by Dolan and Osborn [155, 156]. In the lightcone variables it produces a clean second-order operator. Define the one-variable operator

$$D_z \;\equiv\; z^2(1-z)\,\partial_z^2 - z^2\,\partial_z, \tag{49.28}$$

and its twin $D_{\bar z}$ obtained by $z \to \bar z$. Then the conformal Casimir operator of Eq. (47.233), normalized so that its eigenvalue is exactly $c_{\Delta,\ell}$, is

$$\mathcal{D}_2 \;=\; 2(D_z + D_{\bar z}) \;+\; 2(d-2)\,\frac{z\bar z}{z-\bar z}\Big[(1-z)\,\partial_z - (1-\bar z)\,\partial_{\bar z}\Big]. \tag{49.29}$$

The structure is transparent. The operator is the sum of two decoupled copies of D_z, one for each lightcone variable, plus a mixing term proportional to $d-2$ that couples them. The mixing term *vanishes in* $d=2$: there the two lightcone directions decouple completely, which is the differential-equation shadow of the algebraic fact that the two-dimensional global conformal algebra factorizes, $\mathfrak{so}(3,1) \cong \mathfrak{sl}(2)\oplus\mathfrak{sl}(2)$, into independent holomorphic and antiholomorphic $\mathfrak{sl}(2)$'s (Section 48.3). We take Eq. (49.29) as the explicit form of the operator promised in Chapter 47 and confirm, in each case we solve, that it reproduces the eigenvalue $c_{\Delta,\ell}$ on the explicit solution; since the block is uniquely determined by being a Casimir eigenfunction with the correct leading power, that check both validates the operator's normalization and certifies the solution.

Everything rests on the single operator D_z and its eigenfunctions, so we study it first.

49.2.2 The collinear block

The operator D_z of Eq. (49.28) is the Casimir of a single $\mathfrak{sl}(2)$ — the algebra generated by dilatations and one lightcone translation and special conformal transformation. Its eigenfunctions are the *collinear blocks* (equivalently $\mathfrak{sl}(2)$ blocks, or one-dimensional conformal blocks), and they are the atoms from which the higher-dimensional blocks are assembled.

Lemma 49.6 (The collinear block)**.** *For $\beta \in \mathbb{C}$ let $h = \beta/2$. The function*

$$k_\beta(z) \;=\; z^{\beta/2}\,{}_2F_1\Big(\tfrac{\beta}{2}, \tfrac{\beta}{2}; \beta; z\Big) \tag{49.30}$$

is the unique solution of the eigenvalue equation

$$D_z\,k_\beta(z) \;=\; h(h-1)\,k_\beta(z), \qquad h = \tfrac{\beta}{2}, \tag{49.31}$$

that behaves as $k_\beta(z) \sim z^{\beta/2}$ as $z \to 0$ and is analytic in the unit disk $|z|<1$ with $k_\beta(z)/z^{\beta/2} \to 1$ there.

Proof. We first fix the leading power by the indicial equation, then reduce the full equation to the hypergeometric equation.

Indicial analysis. Write the eigenvalue equation $D_z k = \lambda k$ with $\lambda = h(h-1)$ explicitly,

$$z^2(1-z)\,k'' - z^2\,k' - \lambda\,k = 0. \tag{49.32}$$

Near $z = 0$ the dominant balance is between the highest-derivative term and the eigenvalue term: with $k \sim z^s$ one has $z^2 k'' \sim s(s-1)z^s$ while $z^2 k' \sim s\,z^{s+1}$ is subleading, so Eq. (49.32) reduces at leading order to $s(s-1)z^s - \lambda z^s = 0$, the indicial equation

$$s(s-1) = \lambda = h(h-1). \tag{49.33}$$

Its two roots are $s = h$ and $s = 1-h$. The boundary condition that defines the conformal block is the OPE: the block of a primary of dimension Δ and spin ℓ begins with the power $u^{(\Delta-\ell)/2}$, and in the collinear setting this selects the root $s = h = \beta/2$ (we will see in Sections 49.2.3 and 49.2.4 that β equals $\Delta \pm \ell$, so $\beta/2$ is exactly the required half-twist or half-shifted weight). The second root $s = 1-h$ gives the *shadow* solution, which has the wrong leading power and is discarded. We therefore set

$$k_\beta(z) = z^h\,F(z), \qquad F(0) = 1, \tag{49.34}$$

with F analytic at the origin.

Reduction to the hypergeometric equation. Substitute Eq. (49.34) into Eq. (49.32). The derivatives are

$$k' = z^{h-1}\big(hF + zF'\big), \tag{49.35}$$

$$k'' = z^{h-2}\big[h(h-1)F + 2h\,zF' + z^2F''\big], \tag{49.36}$$

where the second line follows by differentiating the first and collecting powers of z. Then

$$z^2(1-z)k'' = (1-z)\,z^h\big[h(h-1)F + 2h\,zF' + z^2F''\big], \tag{49.37}$$

$$z^2k' = z^h\big(h\,zF + z^2F'\big). \tag{49.38}$$

Inserting these into Eq. (49.32) and dividing by the common factor z^h,

$$(1-z)\big[h(h-1)F + 2h\,zF' + z^2F''\big] - \big(h\,zF + z^2F'\big) - h(h-1)F = 0. \tag{49.39}$$

The z-independent pieces cancel, $h(h-1)F - h(h-1)F = 0$. Collecting the remainder by the order of derivative of F:

- Terms without F' or F'': $-z\,h(h-1)F - h\,zF = -h^2 zF$, using $(h-1)+1 = h$.
- Terms in F': $2h\,zF' - 2h\,z^2F' - z^2F' = z\big[2h - (2h+1)z\big]F'$.

- Terms in F'': $z^2F'' - z^3F'' = z^2(1-z)F''$.

The equation becomes $z^2(1-z)F'' + z[2h-(2h+1)z]F' - h^2zF = 0$, and dividing by z,

$$z(1-z)\,F'' + \big[2h - (2h+1)z\big]\,F' - h^2\,F = 0. \tag{49.40}$$

This is exactly Gauss's hypergeometric equation $z(1-z)F'' + [c-(a+b+1)z]F' - ab\,F = 0$ with parameters read off by matching: $c = 2h$, $a+b = 2h$, and $ab = h^2$, whose unique solution is $a = b = h$. The solution analytic at $z = 0$ with $F(0) = 1$ is the Gauss series

$$F(z) = {}_2F_1(h, h; 2h; z), \tag{49.41}$$

so $k_\beta(z) = z^h\,{}_2F_1(h, h; 2h; z)$ with $h = \beta/2$, which is Eq. (49.30). The series $\sum_n \frac{(h)_n^2}{(2h)_n\,n!} z^n$ has unit radius of convergence, giving analyticity on $|z| < 1$ and the stated normalization $k_\beta / z^{\beta/2} \to 1$. □

Two features will be used constantly. First, the eigenvalue $h(h-1)$ of Eq. (49.31) depends only on the combination $\beta = 2h$, and is invariant under $h \to 1-h$ — the degeneracy that produced the two indicial roots. Second, the small-z expansion

$$k_\beta(z) = z^{\beta/2}\Big[1 + \tfrac{\beta}{4}\,z + O(z^2)\Big] \tag{49.42}$$

follows from the first term of the Gauss series, $\frac{(h)_1^2}{(2h)_1} = \frac{h^2}{2h} = \frac{h}{2} = \frac{\beta}{4}$. These collinear blocks are exactly the conformal blocks of a one-dimensional CFT, and in higher d they are the leading-twist (lightcone) limit of the full block, as we show in Section 49.2.6.

49.2.3 The two-dimensional block

In $d = 2$ the mixing term of Eq. (49.29) vanishes and the Casimir operator is the bare sum

$$\mathcal{D}_2^{(d=2)} = 2(D_z + D_{\bar z}). \tag{49.43}$$

Because D_z acts only on z and $D_{\bar z}$ only on $\bar z$, the equation separates: a product $f(z)\,g(\bar z)$ is an eigenfunction whenever each factor is, with eigenvalue the sum. By Theorem 49.6 the collinear blocks furnish such factors. We must choose the two weights so that the total eigenvalue equals $c_{\Delta,\ell}$ and the leading power matches the OPE.

Proposition 49.7 (Two-dimensional conformal block)**.** *In $d = 2$, for a primary of dimension Δ and spin $\ell \geq 1$, the conformal block is*

$$g_{\Delta,\ell}(z, \bar z) = k_{\Delta+\ell}(z)\,k_{\Delta-\ell}(\bar z) + k_{\Delta-\ell}(z)\,k_{\Delta+\ell}(\bar z), \tag{49.44}$$

and for $\ell = 0$ it is $g_{\Delta,0}(z, \bar z) = k_\Delta(z)\,k_\Delta(\bar z)$, with k_β the collinear block Eq. (49.30).

Proof. Take the product $k_{\Delta+\ell}(z)\,k_{\Delta-\ell}(\bar z)$. By Eq. (49.31) the z-factor is a D_z-eigenfunction with weight $h=(\Delta+\ell)/2$, eigenvalue $h(h-1)$; the $\bar z$-factor is a $D_{\bar z}$-eigenfunction with weight $\bar h=(\Delta-\ell)/2$, eigenvalue $\bar h(\bar h-1)$. Hence

$$\mathcal{D}_2^{(d=2)}\left[k_{\Delta+\ell}(z)k_{\Delta-\ell}(\bar z)\right]=2\big[h(h-1)+\bar h(\bar h-1)\big]\,k_{\Delta+\ell}(z)k_{\Delta-\ell}(\bar z). \tag{49.45}$$

The eigenvalue is computed from $h+\bar h=\Delta$ and $h-\bar h=\ell$:

$$\begin{aligned}2\big[h(h-1)+\bar h(\bar h-1)\big]&=2\big[h^2+\bar h^2-(h+\bar h)\big]=2\big[(h+\bar h)^2-2h\bar h-(h+\bar h)\big]\\&=2\Big[\Delta^2-2\cdot\tfrac{\Delta^2-\ell^2}{4}-\Delta\Big]=\Delta^2+\ell^2-2\Delta=\Delta(\Delta-2)+\ell^2,\end{aligned} \tag{49.46}$$

where $2h\bar h=2\cdot\frac{(\Delta+\ell)(\Delta-\ell)}{4}=\frac{\Delta^2-\ell^2}{2}$. This is precisely $c_{\Delta,\ell}=\Delta(\Delta-d)+\ell(\ell+d-2)$ evaluated at $d=2$. The same eigenvalue is shared by the swapped product $k_{\Delta-\ell}(z)k_{\Delta+\ell}(\bar z)$, obtained by $h\leftrightarrow\bar h$, which leaves Eq. (49.46) unchanged. Any linear combination of the two products is therefore an eigenfunction with eigenvalue $c_{\Delta,\ell}$.

The combination is fixed by the requirement that the block be symmetric under $z\leftrightarrow\bar z$, since it descends from a function of u,v. The symmetric sum Eq. (49.44) satisfies this. Its leading small-$z,\bar z$ behavior is, using $k_\beta\sim z^{\beta/2}$, the sum $z^{(\Delta+\ell)/2}\bar z^{(\Delta-\ell)/2}+z^{(\Delta-\ell)/2}\bar z^{(\Delta+\ell)/2}$; the total power is $\frac12(\Delta+\ell)+\frac12(\Delta-\ell)=\Delta$ in each term, and in the OPE channel $u=z\bar z\to0$ the controlling power is $(z\bar z)^{(\Delta-\ell)/2}$ times a polynomial — the correct leading power $u^{(\Delta-\ell)/2}$ demanded by Theorem 47.49. For $\ell=0$ the two products coincide, $k_\Delta(z)k_\Delta(\bar z)$, and the symmetric combination is the single product without doubling; the eigenvalue Eq. (49.46) reads $\Delta(\Delta-2)$, the scalar value. This is the block stated. □

Worked Example 49.8 (The two-dimensional scalar block to leading orders)**.** Take $\ell=0$, so $g_{\Delta,0}=k_\Delta(z)k_\Delta(\bar z)$ with $h=\Delta/2$. Using the expansion Eq. (49.42),

$$k_\Delta(z)=z^{\Delta/2}\Big[1+\tfrac{\Delta}{4}z+\cdots\Big], \tag{49.47}$$

the block to second order in the lightcone variables is

$$g_{\Delta,0}(z,\bar z)=(z\bar z)^{\Delta/2}\Big[1+\tfrac{\Delta}{4}(z+\bar z)+\tfrac{\Delta^2}{16}z\bar z+\cdots\Big]=u^{\Delta/2}\Big[1+\tfrac{\Delta}{4}(z+\bar z)+\cdots\Big]. \tag{49.48}$$

With $z+\bar z=1+u-v$ (from $u=z\bar z$ and $v=(1-z)(1-\bar z)=1-(z+\bar z)+z\bar z$, hence $z+\bar z=1+u-v$), this is $g_{\Delta,0}=u^{\Delta/2}[1+\frac{\Delta}{4}(1+u-v)+\cdots]$. The leading power $u^{\Delta/2}$ is the identity-stripped OPE singularity of the exchanged scalar; the coefficient $\Delta/4$ multiplying $(1+u-v)$ is the first descendant contribution, fixed entirely by the dimension. This is the kind of explicit series that the crossing equation differentiates term by term in the next section.

49.2.4 The four-dimensional block

In $d=4$ the mixing term of Eq. (49.29) is present, with coefficient $2(d-2)=4$, and the equation no longer separates. Remarkably it can still be solved in closed form, because the four-dimensional Casimir operator is *conjugate* to the two-dimensional one. This is the content of the Dolan–Osborn formula.

Theorem 49.9 (Four-dimensional conformal block)**.** *In $d=4$, the conformal block of a primary of dimension Δ and spin ℓ is*

$$g_{\Delta,\ell}(z,\bar z)=\frac{z\bar z}{z-\bar z}\Big[\,k_{\Delta+\ell}(z)\,k_{\Delta-\ell-2}(\bar z)-k_{\Delta+\ell}(\bar z)\,k_{\Delta-\ell-2}(z)\,\Big]. \tag{49.49}$$

It satisfies the $d=4$ Casimir equation $\mathcal{D}_2\,g_{\Delta,\ell}=c_{\Delta,\ell}\,g_{\Delta,\ell}$ with $c_{\Delta,\ell}=\Delta(\Delta-4)+\ell(\ell+2)$, and begins with the OPE power $u^{(\Delta-\ell)/2}$.

Proof. Write $P\equiv\dfrac{z\bar z}{z-\bar z}$ for the antisymmetric prefactor and $A(z,\bar z)$ for the antisymmetric bracket, so that $g_{\Delta,\ell}=P\,A$ with $A=f(z)\tilde g(\bar z)-f(\bar z)\tilde g(z)$, $f=k_{\Delta+\ell}$, $\tilde g=k_{\Delta-\ell-2}$. Both P and A are antisymmetric under $z\leftrightarrow\bar z$, so their product is symmetric, as a block must be.

The conjugation identity. We claim the operator identity, valid in $d=4$,

$$P^{-1}\,\mathcal{D}_2\,P\;=\;2(D_z+D_{\bar z})-4, \tag{49.50}$$

acting on any smooth function. To prove it, apply $\mathcal{D}_2$ to $P\varphi$ for arbitrary $\varphi(z,\bar z)$ and divide by P. From the Leibniz expansion of Eq. (49.28),

$$D_z(P\varphi)=P\,D_z\varphi+(D_zP)\varphi+2z^2(1-z)P'\,\partial_z\varphi,\qquad P'=\partial_zP, \tag{49.51}$$

and the analogous $D_{\bar z}$ relation, together with the mixing term $4P[(1-z)\partial_z-(1-\bar z)\partial_{\bar z}](P\varphi)$, we collect $\mathcal{D}_2(P\varphi)$ into a part proportional to φ, parts with $\partial_z\varphi$ and $\partial_{\bar z}\varphi$, and the transported principal part $2P(D_z+D_{\bar z})\varphi$. The needed derivatives of $P=z\bar z(z-\bar z)^{-1}$ are

$$\partial_zP=-\frac{\bar z^2}{(z-\bar z)^2},\qquad \partial_{\bar z}P=\frac{z^2}{(z-\bar z)^2}, \tag{49.52}$$

from which $D_zP=\dfrac{2z^2\bar z^2(1-z)}{(z-\bar z)^3}+\dfrac{z^2\bar z^2}{(z-\bar z)^2}$ and $D_{\bar z}P=\dfrac{2z^2\bar z^2(1-\bar z)}{(z-\bar z)^3}-\dfrac{z^2\bar z^2}{(z-\bar z)^2}$, so that

$$D_zP+D_{\bar z}P=\frac{2z^2\bar z^2(2-z-\bar z)}{(z-\bar z)^3}. \tag{49.53}$$

Dividing the assembled $\mathcal{D}_2(P\varphi)$ by P, the coefficient of φ is

$$2P^{-1}(D_zP+D_{\bar z}P)+4\big[(1-z)\partial_zP-(1-\bar z)\partial_{\bar z}P\big]=\frac{4z\bar z(2-z-\bar z)}{(z-\bar z)^2}-\frac{4\big[z^2+\bar z^2-z\bar z(z+\bar z)\big]}{(z-\bar z)^2}$$

$$= \frac{4\big[2z\bar z - z^2 - \bar z^2\big]}{(z-\bar z)^2} = -4, \tag{49.54}$$

since $z^2+\bar z^2-2z\bar z=(z-\bar z)^2$. The coefficient of $\partial_z\varphi$ is $4P^{-1}z^2(1-z)\partial_z P + 4P(1-z) = -\dfrac{4z\bar z(1-z)}{z-\bar z} + \dfrac{4z\bar z(1-z)}{z-\bar z} = 0$, and by the mirror computation the coefficient of $\partial_{\bar z}\varphi$ also vanishes. What remains is exactly $2(D_z+D_{\bar z})\varphi - 4\varphi$, proving Eq. (49.50).

Eigenvalue. Apply Eq. (49.50) to $g_{\Delta,\ell} = PA$:

$$\mathcal{D}_2\,(PA) = P\big[2(D_z+D_{\bar z})-4\big]A. \tag{49.55}$$

Now $A = f(z)\tilde g(\bar z) - f(\bar z)\tilde g(z)$ is a sum of products of collinear blocks. The first product has D_z-weight $h=(\Delta+\ell)/2$ (from $f=k_{\Delta+\ell}$) and $D_{\bar z}$-weight $\bar h=(\Delta-\ell-2)/2$ (from $\tilde g = k_{\Delta-\ell-2}$); the second product, obtained by $z\leftrightarrow\bar z$, carries the same pair of weights distributed oppositely and so shares the eigenvalue. Hence A is a $(D_z+D_{\bar z})$-eigenfunction with eigenvalue $h(h-1)+\bar h(\bar h-1)$, and

$$\mathcal{D}_2\,(PA) = \big[2\big(h(h-1)+\bar h(\bar h-1)\big)-4\big]\,PA. \tag{49.56}$$

With $h=\frac{\Delta+\ell}{2}$, $\bar h=\frac{\Delta-\ell-2}{2}$,

$$4\big[h(h-1)+\bar h(\bar h-1)\big] = (\Delta+\ell)(\Delta+\ell-2)+(\Delta-\ell-2)(\Delta-\ell-4) = 2\Delta^2+2\ell^2-8\Delta+4\ell+8, \tag{49.57}$$

so that $2[h(h-1)+\bar h(\bar h-1)]-4 = \frac12(2\Delta^2+2\ell^2-8\Delta+4\ell+8)-4 = \Delta^2-4\Delta+\ell^2+2\ell = \Delta(\Delta-4)+\ell(\ell+2) = c_{\Delta,\ell}$, as claimed. The conjugation has converted the four-dimensional eigenvalue problem into the two-dimensional one with the shifted antiholomorphic weight $\bar h=(\Delta-\ell-2)/2$, the constant -4 accounting exactly for the difference $c^{(4)}-c^{(2)}$ between the two Casimir eigenvalues.

Leading power. As $z,\bar z\to 0$, $k_\beta\sim z^{\beta/2}$, so

$$A \sim z^{(\Delta+\ell)/2}\bar z^{(\Delta-\ell-2)/2} - \bar z^{(\Delta+\ell)/2}z^{(\Delta-\ell-2)/2} = (z\bar z)^{(\Delta-\ell-2)/2}\big(z^{\ell+1}-\bar z^{\ell+1}\big), \tag{49.58}$$

factoring out the common power and using $\frac{\Delta+\ell}{2}-\frac{\Delta-\ell-2}{2}=\ell+1$. Multiplying by $P=z\bar z/(z-\bar z)$,

$$g_{\Delta,\ell} \sim (z\bar z)^{(\Delta-\ell)/2}\,\frac{z^{\ell+1}-\bar z^{\ell+1}}{z-\bar z} = u^{(\Delta-\ell)/2}\sum_{k=0}^{\ell} z^k\bar z^{\ell-k}, \tag{49.59}$$

the geometric identity $\frac{z^{\ell+1}-\bar z^{\ell+1}}{z-\bar z} = \sum_{k=0}^{\ell} z^k\bar z^{\ell-k}$ producing a symmetric, finite polynomial. The controlling power is $u^{(\Delta-\ell)/2}$, the correct OPE leading behavior; the polynomial encodes the spin and is, as we see below, a Gegenbauer polynomial in the configuration angle. □

Worked Example 49.10 (The four-dimensional scalar block as an elementary function)**.** For $\ell = 0$, Eq. (49.49) reads $g_{\Delta,0} = \frac{z\bar{z}}{z-\bar{z}}[k_\Delta(z)k_{\Delta-2}(\bar{z}) - k_\Delta(\bar{z})k_{\Delta-2}(z)]$. When Δ takes special integer values the hypergeometrics collapse. The cleanest instance is $\Delta = 2$: then $k_2(z) = z\,{}_2F_1(1,1;2;z) = z\cdot\left(-\frac{\ln(1-z)}{z}\right) = -\ln(1-z)$, using ${}_2F_1(1,1;2;z) = -\ln(1-z)/z$, and $k_0(z) = z^0\,{}_2F_1(0,0;0;z) = 1$. Hence

$$g_{2,0}(z,\bar{z}) = \frac{z\bar{z}}{z-\bar{z}}\big[-\ln(1-z)\cdot 1 - (-\ln(1-\bar{z}))\cdot 1\big] = \frac{z\bar{z}}{z-\bar{z}}\ln\frac{1-\bar{z}}{1-z}. \quad (49.60)$$

This is the exchange of a dimension-two scalar (for a free four-dimensional scalar of dimension $\Delta_\phi = 1$, the operator $:\phi^2:$ has exactly $\Delta = 2$). Its appearance as an elementary logarithm, with no special-function content left, is the four-dimensional counterpart of the closed forms that make the bootstrap tractable: every block at integer external and exchanged dimensions reduces to logarithms and rational functions of $z, \bar{z}$. As $z, \bar{z} \to 0$, $\ln\frac{1-\bar{z}}{1-z} = (z-\bar{z}) + O(z^2, \bar{z}^2)$, so $g_{2,0} \to z\bar{z} = u$, matching Eq. (49.59) with $\Delta - \ell = 2$, $\ell = 0$.

Remark 49.11 (Even dimensions and the absence of a closed form in odd d). The conjugation Eq. (49.50) is special to $d = 4$, where the mixing coefficient $2(d-2) = 4$ matches the constant generated by transporting $\mathcal{D}_2$ through $P = z\bar{z}/(z-\bar{z})$. An analogous factorized closed form exists in every even dimension: in $d = 6$ the prefactor is a higher rational function of $z, \bar{z}$ and the block is again a finite antisymmetrized sum of products $k_\beta(z)k_{\beta'}(\bar{z})$, and likewise in $d = 2, 4, 6, \ldots$ blocks descend from the two-variable separated problem. In *odd* dimensions, and at generic non-integer d such as $d = 4 - \epsilon$ along ϕ^4 theory, no finite closed form in terms of ${}_2F_1$ exists: the block is a genuine function of two variables not expressible as a finite sum of products of one-variable hypergeometrics. For those cases — which include the physically central $d = 3$ Ising fixed point — one needs the series representation of the next subsection.

49.2.5 The radial expansion in general dimension

For general d there is no closed form, but there is a rapidly convergent power series adapted to radial quantization (Section 47.5). It rests on a change of variables, due in this context to Hogervorst and Rychkov [269], that maps cross-ratio space onto a disk and makes the OPE convergence of Theorem 47.48 manifest.

The radial coordinate. Define

$$\rho(z) = \frac{z}{\left(1+\sqrt{1-z}\,\right)^2}, \qquad \text{inverse} \qquad z = \frac{4\rho}{(1+\rho)^2}, \quad (49.61)$$

and set $\bar\rho = \rho(\bar z)$. In Euclidean signature $\bar z = z^*$, so $\bar\rho = \rho^*$; writing $\rho = r\,e^{i\theta}$ gives the real coordinates

$$r = |\rho|, \qquad \eta = \cos\theta = \tfrac{1}{2}(e^{i\theta} + e^{-i\theta}). \tag{49.62}$$

The geometric meaning of ρ is read off from radial quantization. Place the four operators so that the pair (x_1, x_2) sits near the origin and the pair (x_3, x_4) near infinity; the s-channel OPE is the insertion of a complete set of states on a sphere separating the two pairs. In the optimal frame the configuration is governed by the ratio of the inner to the outer radius, and that ratio is exactly $r = |\rho|$. The OPE limit $x_1 \to x_2$ is $z \to 0$, hence $\rho \to 0$; the channel-crossing point $z \to 1$ (where the t-channel OPE takes over) is $\rho \to 1$. The map Eq. (49.61) compresses the entire physical region into the unit disk.

Leading term and recursion. In radial coordinates the block is organized as a sum over the conformal multiplet, graded by the dimension increment and the SO(d) spin of each descendant level. Writing $\nu = (d-2)/2$ and $C_j^{(\nu)}$ for the Gegenbauer polynomial of the rotation group, the block takes the form

$$g_{\Delta,\ell}(r,\eta) = \sum_{n=0}^{\infty} \sum_{j} c_{n,j}\, r^{\Delta+n}\, C_j^{(\nu)}(\eta), \tag{49.63}$$

where at each level n the spin j runs over the finite set $\{\ell - n, \ell - n + 2, \dots, \ell + n\} \cap \mathbb{Z}_{\geq 0}$ produced by tensoring with the descendant raising operators. The leading term is $n = 0$, $j = \ell$, and its coefficient follows from the small-$z, \bar z$ limit. In $d = 4$ ($\nu = 1$) we computed it exactly in Eq. (49.59): with $z \approx 4\rho$, $\bar z \approx 4\bar\rho$ as $z, \bar z \to 0$ (since $\sqrt{1-z} \to 1$ in Eq. (49.61)), the sum $\sum_{k=0}^{\ell} z^k \bar z^{\ell-k}$ becomes $4^\ell \sum_k \rho^k \bar\rho^{\ell-k} = 4^\ell\, r^\ell\, \frac{\sin[(\ell+1)\theta]}{\sin\theta} = 4^\ell r^\ell\, C_\ell^{(1)}(\eta)$, and $u^{(\Delta-\ell)/2} = (z\bar z)^{(\Delta-\ell)/2} \to (16r^2)^{(\Delta-\ell)/2} = 4^{\Delta-\ell} r^{\Delta-\ell}$, so

$$g_{\Delta,\ell}(r,\eta) \xrightarrow{r\to 0} (4r)^\Delta\, C_\ell^{(1)}(\eta) \qquad (d = 4). \tag{49.64}$$

In general d the leading term is again $g_{\Delta,\ell} \to \mathcal{N}_{d,\ell}\,(4r)^\Delta\, C_\ell^{(\nu)}(\eta)$ with a dimension-dependent normalization $\mathcal{N}_{d,\ell}$ that equals 1 in $d = 4$ in the convention of Section 49.2.6 (it is fixed once and for all by the lightcone limit of the block). The Chebyshev-of-the-second-kind structure $\sin[(\ell+1)\theta]/\sin\theta = C_\ell^{(1)}(\eta)$ is the $d = 4$ Gegenbauer; the angular function is in every dimension the Gegenbauer $C_\ell^{(\nu)}(\eta)$ carrying the spin of the exchanged primary.

The subleading coefficients are not free: substituting the ansatz Eq. (49.63) into the Casimir equation $\mathcal{D}_2\, g = c_{\Delta,\ell}\, g$ turns the second-order differential operator into a finite recursion. Because $\mathcal{D}_2$ shifts the power $r^{\Delta+n}$ by at most two units and the Gegenbauer label j by at most two, equating coefficients of each $r^{\Delta+n} C_j^{(\nu)}(\eta)$ yields a linear relation expressing $c_{n,j}$ in terms of coefficients

at lower n. Seeded by the leading term Eq. (49.64), the recursion determines the entire series. This is the algorithm behind every modern block evaluation: one computes $c_{n,j}$ up to a truncation order $n \le N$ and sums, the error controlled by the convergence we now establish.

Proposition 49.12 (Convergence of the radial expansion on the unit disk)**.** *The series Eq. (49.63) converges absolutely for $r = |\rho| < 1$, i.e. throughout the image of the cut cross-ratio plane under the map Eq. (49.61). In particular it converges at the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$, where $\rho = 3 - 2\sqrt{2} \approx 0.1716$.*

Argument. The expansion Eq. (49.63) is nothing but the s-channel OPE written in radial quantization. By the state–operator correspondence of Section 47.5, fusing $\phi(x_1)\phi(x_2)$ creates a state $|\Psi\rangle = \phi(x_1)\phi(x_2)\,|0\rangle$ on a sphere of radius set by the larger of $|x_1|, |x_2|$; expanding it in the orthonormal basis of dilatation eigenstates produces one term $r^{\Delta+n}$ per descendant of dimension $\Delta + n$. The four-point function is the overlap of this state with the conjugate state $\langle\Psi'|$ built from $\phi(x_3)\phi(x_4)$ at radius $1/r > 1$. The overlap is the radially ordered Euclidean correlator $\langle\Psi'|\Psi\rangle$, which by Theorem 47.34 is a convergent sum over a complete set of states with *nonnegative* norms,

$$\langle\phi\phi\phi\phi\rangle = \sum_{\text{states } |n\rangle} \frac{\langle\Psi'|n\rangle\,\langle n|\Psi\rangle}{\langle n|n\rangle}, \qquad \langle n|n\rangle \ge 0, \tag{49.65}$$

the reflection positivity established there. Each contribution scales as r^{Δ_n} with $\Delta_n = \Delta + n \to \infty$, and the positivity of the inner product forbids the cancellations that could spoil convergence; the sum is dominated termwise by a geometric series in the radius ratio. Convergence therefore holds precisely when the inner pair can be radially separated from the outer pair, i.e. when the radius ratio $r = |\rho| < 1$. This is the radial-quantization sharpening of the OPE convergence theorem Theorem 47.48: the abstract statement that the OPE converges inside the lightcone becomes, after the conformal map to the disk, the geometric statement that the block series converges for $|\rho| < 1$. At the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$ one has $\sqrt{1 - \frac{1}{2}} = 1/\sqrt{2}$ and $\rho = \frac{1/2}{(1+1/\sqrt{2})^2} = \frac{1/2}{3/2+\sqrt{2}} = 3 - 2\sqrt{2} \approx 0.17$, deep inside the disk, so the series converges geometrically with ratio ~ 0.17 and a few dozen terms suffice to machine precision — the reason ρ is the coordinate of choice for the numerical bootstrap. □

49.2.6 Limits: lightcone and diagonal

Two limits of the block control the analyses of the chapters that follow, and both are immediate from the closed and series forms above.

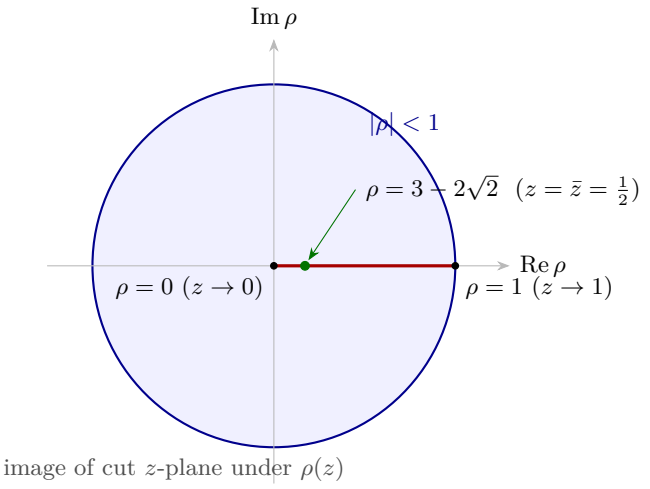


Figure 49.2: The radial coordinate $\rho(z) = z/(1+\sqrt{1-z})^2$ maps the cut cross-ratio plane into the unit disk. The OPE limit $z \to 0$ is the center $\rho = 0$; the channel-crossing point $z \to 1$ sits on the boundary $\rho = 1$. The block series Eq. (49.63) converges for $|\rho| < 1$ (Theorem 49.12). The crossing-symmetric point $z = \bar{z} = \frac{1}{2}$ maps to $\rho = 3 - 2\sqrt{2} \approx 0.17$ (green), well inside the disk, so the expansion converges geometrically there.

The lightcone limit. Send one lightcone variable to zero, $\bar{z} \to 0$, at fixed z. From the collinear expansion $k_\beta(\bar{z}) \sim \bar{z}^{\beta/2}$, the factor of smallest power dominates. In two dimensions, Eq. (49.44) gives $k_{\Delta-\ell}(\bar{z}) \sim \bar{z}^{(\Delta-\ell)/2}$ as the leading term (versus $\bar{z}^{(\Delta+\ell)/2}$ from the other), so

$$g_{\Delta,\ell}(z,\bar{z}) \xrightarrow{\bar{z}\to 0} \bar{z}^{(\Delta-\ell)/2}\, k_{\Delta+\ell}(z) \qquad (d=2). \tag{49.66}$$

In four dimensions, Eq. (49.49) with $P \to z\bar{z}/z = \bar{z}$ and $k_{\Delta-\ell-2}(\bar{z}) \sim \bar{z}^{(\Delta-\ell-2)/2}$ gives $g \to \bar{z}\cdot\bar{z}^{(\Delta-\ell-2)/2}k_{\Delta+\ell}(z) = \bar{z}^{(\Delta-\ell)/2}k_{\Delta+\ell}(z)$ — *the same limit*, with the same unit coefficient. This is no coincidence: in any d the lightcone limit factorizes into the leading-twist power $\bar{z}^{\tau/2}$, with $\tau = \Delta - \ell$ the twist, times the collinear block $k_{\Delta+\ell}(z)$ in the surviving variable,

$$g_{\Delta,\ell}(z,\bar{z}) \xrightarrow{\bar{z}\to 0} \bar{z}^{(\Delta-\ell)/2}\, k_{\Delta+\ell}(z) \qquad (\text{any } d). \tag{49.67}$$

We adopt Eq. (49.67) as the *normalization convention* for the block in every dimension: it fixes the overall constant left undetermined by the Casimir equation and is consistent with the explicit $d = 2$ and $d = 4$ forms above. The twist $\tau = \Delta - \ell$ governs the strength of the singularity as the two operators in the OPE approach a common lightcone; operators of low twist dominate the lightcone limit. This is the quantity the analytic bootstrap of Chapter 51 organizes its large-spin expansion around, and the mechanism by which mean-field (generalized free field) theory generates its tower of double-twist operators of twist $\tau = 2\Delta_\phi + 2n$.

The diagonal limit. Set $z = \bar{z}$, the locus where the four points are configured symmetrically and the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$ lives. In two dimensions Eq. (49.44) is regular there, $g_{\Delta,\ell}(z,z) = 2\,k_{\Delta+\ell}(z)k_{\Delta-\ell}(z)$ for $\ell \geq 1$ and $k_\Delta(z)^2$ for $\ell = 0$. In four dimensions the prefactor $P = z\bar{z}/(z-\bar{z})$ is singular as $\bar{z} \to z$, and the bracket vanishes to the same order, so the limit is a derivative. Expanding A to first order in $\bar{z} - z$,

$$g_{\Delta,\ell}(z,z) = \lim_{\bar{z}\to z} \frac{z\bar{z}}{z-\bar{z}}\, A(z,\bar{z}) = z^2\big[k'_{\Delta+\ell}(z)\,k_{\Delta-\ell-2}(z) - k_{\Delta+\ell}(z)\,k'_{\Delta-\ell-2}(z)\big], \tag{49.68}$$

a Wronskian of two collinear blocks — finite and smooth, as required by regularity of the block at a generic Euclidean configuration. The diagonal limit is where the crossing sum rule is most economically expanded: with one variable, the infinitely many constraints of Eq. (49.15) are sampled by differentiating in z at $z = \frac{1}{2}$, the starting point of the numerical functional method.

49.2.7 Global versus Virasoro blocks

A word is owed on which blocks these are. The functions built in this section are the *global* (or conformal) blocks: eigenfunctions of the quadratic Casimir of the *finite-dimensional* conformal group $\mathrm{SO}(d+1,1)$, resumming a primary together with the descendants generated by the d translations P_μ alone. They exist and take the forms above in every dimension $d \geq 2$, and they are the only blocks available for $d \geq 3$, where the conformal algebra has no infinite-dimensional enhancement.

In exactly two dimensions the conformal algebra is enlarged to the infinite-dimensional Virasoro algebra (Section 48.3), and one may resum a primary together with its entire Virasoro module — all the descendants generated by the modes L_{-n}, including the stress-tensor tower. The result is the *Virasoro block* of Chapter 48, which packages infinitely many global multiplets (the global primary and its Virasoro descendants, each itself a global primary) into a single function depending also on the central charge c. The global block Eq. (49.44) is the leading piece of the Virasoro block, retaining only the $\mathfrak{sl}(2) \oplus \mathfrak{sl}(2)$ descendants. The bootstrap of this chapter uses the global blocks throughout, for two reasons: it is built to operate in any d, where only global blocks exist; and even in $d = 2$ the model-independent, dimension-agnostic crossing analysis is cleanest in the global language. The Virasoro blocks are the right tool when one exploits the full power of two-dimensional symmetry — the minimal-model classification and the analytic two-dimensional bootstrap of Chapter 48 — but they are not what the higher-dimensional bootstrap of Chapters 50 and 51 runs on.

With Eqs. (49.44) and (49.49) we have the blocks in closed form in the two even dimensions, and with the radial series Eq. (49.63) we have them to any desired accuracy in general d — including the leading coefficient, the recursion that

generates the rest, and the guaranteed convergence Theorem 49.12 that makes truncation reliable. The lightcone and diagonal limits Eqs. (49.67) and (49.68) hand the next section explicit functions $g_{\Delta,\ell}(u,v)$ together with their derivatives at the crossing-symmetric point $z=\bar z=\frac{1}{2}$. Substituting these into the crossing relation Eq. (49.14) and rearranging the two channels into a single vanishing combination is what turns the abstract sum rule Eq. (49.15) into the concrete bootstrap equation that the linear functionals of Chapter 50 will act on.

49.3 The Crossing Equation and the Bootstrap Sum Rule

The blocks are now explicit functions. Section 49.2 turned the implicit Casimir eigenvalue problem into the closed forms Eqs. (49.44) and (49.49), the uniformly convergent radial series Eq. (49.63), and the lightcone and diagonal limits Eqs. (49.67) and (49.68). We can therefore stop treating $G_{\Delta,\ell}(u,v)$ as an abstract eigenfunction and start treating it as a number we can evaluate, differentiate, and Taylor-expand wherever we like. The purpose of this section is to take the crossing equation derived in Section 49.1.4 — so far a relation between two formal block sums — and package it into the single object on which the entire machinery of bounds operates: a sum rule of the form $\sum_{\mathcal O} p_{\mathcal O}\, F_{\Delta_{\mathcal O},\ell_{\mathcal O}} = 0$, in which all the *dynamics* sits in a set of nonnegative weights $p_{\mathcal O}$ and a discrete spectrum, while the functions $F_{\Delta,\ell}$ are fixed, theory-independent, and computable from the closed forms just obtained. Once the equation is in this shape two facts about it become decisive: the weights are squares, hence nonnegative, and the dimensions are bounded below by unitarity. A linear equation in nonnegative unknowns, with a known inhomogeneous term, is a problem a computer — or, in the simplest cases, a hand calculation — can attack by the geometry of convex cones. Assembling that object precisely, and exhibiting its structure, is the work of this section.

49.3.1 The crossing equation in block form

We continue ϕ^4 theory: the external operator is the single real scalar primary ϕ of dimension Δ_ϕ, the order parameter of the Wilson–Fisher and Ising fixed points, and the object of study is its four-point function $\langle\phi\phi\phi\phi\rangle$ stripped to the function $g(u,v)$ of Eq. (49.2). Theorem 49.3 established that equating the s- and t-channel block decompositions of this correlator is the single functional equation

$$\sum_{\mathcal O} C_{\phi\phi\mathcal O}^2 \Big[\, v^{\Delta_\phi}\, G_{\Delta_{\mathcal O},\ell_{\mathcal O}}(u,v) \;-\; u^{\Delta_\phi}\, G_{\Delta_{\mathcal O},\ell_{\mathcal O}}(v,u) \,\Big] = 0, \tag{49.69}$$

the sum rule Eq. (49.15), to hold identically in the cross-ratios. The sum runs over every primary $\mathcal O$ appearing in the $\phi\times\phi$ OPE, each weighted by the square of its OPE coefficient.

The squares are the heart of the matter, and they earn their own symbol. Write

$$p_{\mathcal{O}} \;\equiv\; C^{2}_{\phi\phi\mathcal{O}} \;\geq\; 0. \tag{49.70}$$

That $p_{\mathcal{O}}$ is real and nonnegative — not merely a formal label but a genuine inequality — is the single piece of dynamical positivity that the bootstrap converts into bounds, and we will justify it carefully in Section 49.3.3. For now it is enough that each weight in Eq. (49.69) is the same nonnegative number entering from both fused pairs, exactly as in the s-channel decomposition Eq. (49.4). With this notation the equality of the two channel expansions takes its most symmetric form.

Definition 49.13 (The crossing equation)**.** The *crossing equation* for the four-point function of the identical scalar ϕ is the requirement that the s- and t-channel conformal-block decompositions of Eq. (49.2) agree as functions of the cross-ratios,

$$\boxed{v^{\Delta_\phi}\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) \;=\; u^{\Delta_\phi}\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(v,u),} \tag{49.71}$$

with $p_{\mathcal{O}} = C^{2}_{\phi\phi\mathcal{O}} \geq 0$ and the sum over the primaries of the $\phi\times\phi$ OPE, including the identity with $p_{\mathbf{1}} = 1$.

Equation (49.71) is Eq. (49.14) of Theorem 49.2 with each g replaced by its block sum and both sides cleared of the channel-asymmetric prefactor symmetrically; the equivalence to Eq. (49.69) is immediate on moving one side across. We have written it in this balanced form because it displays the role of the kinematic factors u^{Δ_ϕ} and v^{Δ_ϕ} as the only place the external dimension Δ_ϕ enters, and because the difference of the two sides is the object we now name.

49.3.2 The convolved block: the F-function

The left-hand minus right-hand side of Eq. (49.71) is a sum over operators of one and the same combination of a block with itself, evaluated at (u,v) and at (v,u). Pulling that combination out of the sum — which is legitimate because the operation is linear in the block — isolates a function attached to each pair of quantum numbers (Δ,ℓ) alone.

Definition 49.14 (The convolved block $F_{\Delta,\ell}$)**.** For an external dimension Δ_ϕ and an exchanged primary of dimension Δ and spin ℓ, the *convolved block* (or *F-function*) is

$$\boxed{F_{\Delta,\ell}(u,v) \;\equiv\; v^{\Delta_\phi}\,G_{\Delta,\ell}(u,v) \;-\; u^{\Delta_\phi}\,G_{\Delta,\ell}(v,u).} \tag{49.72}$$

The name records that $F_{\Delta,\ell}$ is the convolution of the kinematic factors $u^{\Delta_\phi}, v^{\Delta_\phi}$ with the universal block. Three points about it are worth stating at once, because they are exactly the properties the rest of the chapter relies on, and each is a one-line consequence of Eq. (49.72).

First, $F_{\Delta,\ell}$ is purely kinematic. It depends only on the dimension and spin of the exchanged operator and on the external dimension Δ_ϕ — the same numbers that fix the block itself — and not at all on the OPE coefficients, the central charge, or any other dynamical datum of a particular theory. The closed forms of Section 49.2 give it explicitly: substitute Eq. (49.49) (in $d = 4$), Eq. (49.44) (in $d = 2$), or the convergent radial series Eq. (49.63) (in any d) for both $G_{\Delta,\ell}(u, v)$ and its crossed evaluation $G_{\Delta,\ell}(v, u)$, multiply by the appropriate power of u or v, and subtract. $F_{\Delta,\ell}$ is therefore a fixed, computable function for every (Δ, ℓ), tabulated once and for all before any theory is specified.

Second, the crossing equation becomes a sum rule on these functions. The difference of the two sides of Eq. (49.71) is, term by term in the operator sum,

$$v^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u, v) - u^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(v, u) = \sum_{\mathcal{O}} p_{\mathcal{O}} \Big[v^{\Delta_\phi} G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u, v) - u^{\Delta_\phi} G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(v, u) \Big], \tag{49.73}$$

the two block sums combining under the single sum because u^{Δ_ϕ} and v^{Δ_ϕ} are independent of the summation index $\mathcal{O}$. The bracket is exactly $F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u, v)$ of Eq. (49.72). Hence crossing, in the form Eq. (49.71), is precisely the statement that this combination vanishes,

$$\boxed{\sum_{\mathcal{O}} p_{\mathcal{O}}\, F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u, v) = 0,} \tag{49.74}$$

holding identically in (u, v). This is the *bootstrap sum rule*. It is the same content as Eq. (49.69), written so that the dynamical weights $p_{\mathcal{O}}$ multiply fixed kinematic functions: a linear relation, with nonnegative coefficients, among an infinite collection of known functions, one per point (Δ, ℓ) of the spectrum.

Third, $F_{\Delta,\ell}$ is antisymmetric under $u \leftrightarrow v$. This is the structural property that singles out the identity and, as we will see, dictates where the equation must be imposed.

Proposition 49.15 (Antisymmetry of the convolved block)**.** *For every (Δ, ℓ) and every Δ_ϕ,*

$$F_{\Delta,\ell}(v, u) = -\,F_{\Delta,\ell}(u, v). \tag{49.75}$$

In particular $F_{\Delta,\ell}(u, u) = 0$ for all u: every convolved block vanishes on the diagonal $u = v$.

Proof. Interchange the symbols u and v throughout the defining formula Eq. (49.72). The first term $v^{\Delta_\phi} G_{\Delta,\ell}(u,v)$ becomes $u^{\Delta_\phi} G_{\Delta,\ell}(v,u)$, and the second term $u^{\Delta_\phi} G_{\Delta,\ell}(v,u)$ becomes $v^{\Delta_\phi} G_{\Delta,\ell}(u,v)$, so

$$F_{\Delta,\ell}(v,u) = u^{\Delta_\phi} G_{\Delta,\ell}(v,u) - v^{\Delta_\phi} G_{\Delta,\ell}(u,v) = -\big[v^{\Delta_\phi} G_{\Delta,\ell}(u,v) - u^{\Delta_\phi} G_{\Delta,\ell}(v,u)\big] = -F_{\Delta,\ell}(u,v). \tag{49.76}$$

Setting $v = u$ in Eq. (49.75) gives $F_{\Delta,\ell}(u,u) = -F_{\Delta,\ell}(u,u)$, hence $F_{\Delta,\ell}(u,u) = 0$. □

The antisymmetry has an immediate and important consequence for the identity operator. Its block is the constant $G_{0,0} = 1$ (Eq. (49.6)), so its convolved block is the antisymmetrization of that constant,

$$F_{0,0}(u,v) = v^{\Delta_\phi} \cdot 1 - u^{\Delta_\phi} \cdot 1 = v^{\Delta_\phi} - u^{\Delta_\phi}, \tag{49.77}$$

a fixed, elementary function of the cross-ratios that does *not* vanish identically (the two cross-ratios are independent), though it does vanish on the diagonal $u = v$ in accordance with Theorem 49.15. Because $C_{\phi\phi\mathbf{1}} = 1$ is fixed by the unit normalization of $\langle\phi\phi\rangle$, this term enters the sum rule with a coefficient that is not free. Separating it out, the sum rule Eq. (49.74) reads

$$(v^{\Delta_\phi} - u^{\Delta_\phi}) + \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) = 0, \tag{49.78}$$

which is Eq. (49.21) written in the convolved notation. Schematically, with the identity's contribution normalized to a fixed reference one writes this as

$$1 + \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, \widehat{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = 0, \tag{49.79}$$

the "1" standing for the known, definite source term $F_{0,0} = v^{\Delta_\phi} - u^{\Delta_\phi}$ carried by the identity, and $\widehat{F}$ for the convolved blocks measured relative to it. The bootstrap is the study of which spectra and which nonnegative weights $\{p_{\mathcal{O}}\}$ can conspire to cancel this fixed source. An empty spectrum cannot: Eq. (49.78) would then assert $v^{\Delta_\phi} - u^{\Delta_\phi} = 0$, false. The identity is the irreducible inhomogeneity that forces a nonempty tower of operators into existence and fixes the overall scale against which every $p_{\mathcal{O}}$ is measured.

Caution

The "1" in Eq. (49.79) is shorthand, not a literal constant: the identity's convolved block is the function $v^{\Delta_\phi} - u^{\Delta_\phi}$, which is what makes the sum rule a constraint on functions rather than on numbers. We carry the honest form Eq. (49.78) in every computation. The schematic "$1 + \sum p\,\widehat{F} = 0$" is useful only as a reminder that there is exactly one known, normalization-fixed term and infinitely many unknown, positively-weighted ones.

That all the dynamical content lives in $\{p_{\mathcal{O}}\}$ and the spectrum $\{(\Delta_{\mathcal{O}}, \ell_{\mathcal{O}})\}$, while the functions $F_{\Delta,\ell}$ are a fixed kinematic background, is the cleanest statement of what the bootstrap is. The blocks were computed once, in Section 49.2, with no reference to any theory; the convolution Eq. (49.72) is an algebraic operation on them; the only theory-dependent objects in the whole of Eq. (49.74) are the nonnegative numbers $p_{\mathcal{O}}$ and the discrete set of (Δ, ℓ) at which they are nonzero. Solving — or bounding the solutions of — the bootstrap is determining those numbers and that set.

49.3.3 Positivity and the unitarity spectrum: a closed problem

Two inequalities turn Eq. (49.74) from a bare functional identity into a constrained, and therefore solvable, problem. The first is the positivity of the weights.

Proposition 49.16 (Positivity of the OPE weights)**.** *In a reflection-positive (unitary) conformal field theory, the coefficients $C_{\phi\phi\mathcal{O}}$ of a real scalar primary ϕ can be chosen real, so that the weights $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}}$ in Eq. (49.74) are nonnegative for every exchanged primary.*

Proof. Recall the geometric meaning of the OPE coefficient established by radial quantization (Section 47.5). Acting on the vacuum, the fused pair $\phi(x_1)\phi(x_2)$ creates a state in the radial-quantization Hilbert space, and its overlap with the state $|\mathcal{O}\rangle$ of an exchanged primary is, up to a fixed kinematic factor, the coefficient $C_{\phi\phi\mathcal{O}}$: expanding $\phi(x_1)\phi(x_2)\,|0\rangle$ in the complete basis of primaries and descendants, the primary $\mathcal{O}$ appears with coefficient $C_{\phi\phi\mathcal{O}}$. The inner product on this Hilbert space is positive definite by reflection positivity (Theorem 47.34); the Euclidean adjoint of ϕ at the origin is ϕ at the reflected point, and for a real (Hermitian in the radial-quantization sense) scalar this makes the matrix element defining $C_{\phi\phi\mathcal{O}}$ real once the basis state $|\mathcal{O}\rangle$ is given the standard unit normalization fixed by the two-point function Eq. (47.124). Reality of $C_{\phi\phi\mathcal{O}}$ gives $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0$ directly. Equivalently and more invariantly, the squared coefficient is the squared length of a projection,

$$p_{\mathcal{O}} \;\propto\; \big|\langle \mathcal{O}|\,\phi\phi\,\rangle\big|^2 \;\geq\; 0, \tag{49.80}$$

manifestly nonnegative in a positive-definite Hilbert space. The identity is the special case $p_{\mathbf{1}} = C^2_{\phi\phi\mathbf{1}} = 1$. □

The reality of the OPE coefficient for *identical* external scalars is what makes the weight a perfect square rather than a product of two distinct coefficients; this is special to the identical-operator correlator and is the reason the bootstrap of a single correlator already carries positivity. We return in Section 49.3.5 to what happens when the four external operators are not all the same, where the relevant positivity becomes that of a matrix.

The second inequality is the spectrum constraint imported from Chapter 47. Unitarity does not only make the weights nonnegative; it also forbids operators below a dimension fixed by their spin.

Proposition 49.17 (The constrained sum rule)**.** *A unitary conformal field theory in $d \geq 3$ containing a real scalar primary ϕ of dimension Δ_ϕ satisfies the sum rule Eq. (49.78), in which:*

1. *every weight obeys $p_{\mathcal{O}} \geq 0$ (Theorem 49.16);*
2. *every exchanged primary obeys the unitarity bounds* (Theorem 47.18, Eq. (47.103))

$$\Delta \geq \frac{d-2}{2} \quad (\ell = 0), \qquad \Delta \geq \ell + d - 2 \quad (\ell \geq 1); \tag{49.81}$$

3. *the spectrum consists of operators of even spin $\ell = 0, 2, 4, \ldots$ only.*

Proof. Items (1) and (2) are Theorem 49.16 and the imported Theorem 47.18. For (3), recall that the $\phi \times \phi$ OPE of *identical* scalars is symmetric under exchange of the two operators, $\phi(x_1)\phi(x_2) = \phi(x_2)\phi(x_1)$ (Bose symmetry of a single commuting scalar field, Eq. (49.9)). An exchanged primary of spin ℓ enters the symmetric product with a tensor structure whose leading term carries ℓ factors of the unit separation vector $\hat{x}_{12}^{\mu}$; under $x_1 \leftrightarrow x_2$ this reverses sign, contributing $(-1)^\ell$. Only even ℓ survives the symmetrization, so odd-spin operators have $C_{\phi\phi\mathcal{O}} = 0$ and drop out of the sum. Thus the sum over $\mathcal{O} \neq \mathbf{1}$ in Eq. (49.78) runs over scalars and even-spin operators lying above the unitarity bound Eq. (49.81). □

These three constraints close the problem. The sum rule Eq. (49.78) is now a linear equation in the nonnegative unknowns $\{p_{\mathcal{O}}\}$, summed over a known, discrete, bounded-below catalogue of allowed quantum numbers (Δ, ℓ), with a fixed inhomogeneous term. The unknown spectrum — which (Δ, ℓ) actually appear — is itself constrained to lie in the half-line Eq. (49.81) for each even spin. There is nothing else: no Lagrangian, no coupling, no further input. This is the precise sense in which the conformal bootstrap is a self-contained mathematical problem rather than a perturbative expansion of some underlying microscopic theory.

Physical Insight

The power of the setup is that positivity and the sum rule pull in opposite directions, and the tension between them is what excludes theories. Crossing demands that a positive combination of the functions $F_{\Delta,\ell}$ cancel the fixed source $v^{\Delta_\phi} - u^{\Delta_\phi}$ exactly. If, for some hypothetical spectrum, *every* available $F_{\Delta,\ell}$ shares a feature — a definite sign of some derivative,

say — that the source lacks, then no nonnegative combination can reproduce the source, and that spectrum cannot belong to any unitary CFT: positively-weighted vectors cannot sum to something they all point away from. This is the entire logic of the bounds in Section 49.4 and Chapter 50; the present section's job is only to have written the equation so that "a nonnegative combination of fixed vectors equals a fixed vector" is literally what it says.

49.3.4 Vectorization: the sum rule at the crossing-symmetric point

The sum rule Eq. (49.74) is one equation that must hold at every point of cross-ratio space — infinitely many conditions. To make it actionable we extract from it a manageable, but still infinite, family of linear conditions on the weights by expanding around a single, well-chosen point. The choice is forced by Theorem 49.15.

Pass to the $(z, \bar{z})$ variables of Eq. (49.27), $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$. The map $u \leftrightarrow v$ that underlies crossing is, in these variables, the simultaneous reflection

$$z \mapsto 1 - z, \qquad \bar{z} \mapsto 1 - \bar{z}, \tag{49.82}$$

which sends $z\bar{z} \leftrightarrow (1-z)(1-\bar{z})$ and so realizes $u \leftrightarrow v$. Its unique fixed point on the Euclidean diagonal $\bar{z} = z^*$ is

$$z = \bar{z} = \tfrac{1}{2}, \qquad \Longleftrightarrow \qquad u = v = \tfrac{1}{4}, \tag{49.83}$$

the *crossing-symmetric point*: the configuration most democratic among the four operators, at which the s- and t-channels are interchanged by Eq. (49.82) while the point itself is held fixed. By Eq. (49.61) it sits at radial coordinate $\rho = 3 - 2\sqrt{2} \approx 0.17$, deep inside the disk of block convergence (Theorem 49.12), so every block and every $F_{\Delta,\ell}$ has a convergent Taylor expansion there.

Here the antisymmetry becomes decisive. By Theorem 49.15 each $F_{\Delta,\ell}$ vanishes on the diagonal $u = v$, in particular at the symmetric point itself: the sum rule evaluated at $z = \bar{z} = \frac{1}{2}$ reads $0 = 0$ and says nothing. All of its content lies in the *derivatives*. Introduce the shifted variables

$$z = \tfrac{1}{2} + a, \qquad \bar{z} = \tfrac{1}{2} + \bar{a}, \tag{49.84}$$

so the symmetric point is $a = \bar{a} = 0$ and the reflection Eq. (49.82) is $(a, \bar{a}) \mapsto (-a, -\bar{a})$. The convolved block, being antisymmetric under Eq. (49.82), is an *odd* function of $(a, \bar{a})$:

$$F_{\Delta,\ell}(-a, -\bar{a}) = -F_{\Delta,\ell}(a, \bar{a}), \tag{49.85}$$

where we write $F_{\Delta,\ell}(a,\bar{a})$ for the function in the shifted variables. Its Taylor series about the origin therefore contains only terms of odd total degree $m+n$,

$$F_{\Delta,\ell}(a,\bar{a}) = \sum_{\substack{m,n\geq 0\\ m+n \text{ odd}}} f_{\Delta,\ell}^{(m,n)}\,\frac{a^m\,\bar{a}^n}{m!\,n!}, \qquad f_{\Delta,\ell}^{(m,n)} \equiv \partial_a^m \partial_{\bar{a}}^n F_{\Delta,\ell}\big|_{a=\bar{a}=0}; \tag{49.86}$$

all even-degree coefficients vanish identically, the symmetric point's value among them. Moreover $F_{\Delta,\ell}$ is symmetric under $z \leftrightarrow \bar{z}$ — the blocks are (the Casimir operator Eq. (49.29) is symmetric in $z,\bar{z}$) and the convolution preserves it — so $f_{\Delta,\ell}^{(m,n)} = f_{\Delta,\ell}^{(n,m)}$, and the independent coefficients are those with $m \leq n$ and $m+n$ odd. The infinitely many conditions packaged in the sum rule are exactly these derivative components.

Definition 49.18 (The F-vector)**.** The *F-vector* of an operator of dimension Δ and spin ℓ is the (infinite) collection of independent odd-order Taylor coefficients of its convolved block at the crossing-symmetric point,

$$\vec{F}_{\Delta,\ell} \;\equiv\; \Big(f_{\Delta,\ell}^{(m,n)} \Big)_{\substack{m\leq n\\ m+n \text{ odd}}}, \qquad f_{\Delta,\ell}^{(m,n)} = \partial_a^m \partial_{\bar{a}}^n F_{\Delta,\ell}\big|_{z=\bar{z}=\frac{1}{2}}. \tag{49.87}$$

The sum rule Eq. (49.74) is then the single vector equation

$$\boxed{\sum_{\mathcal{O}} p_{\mathcal{O}}\,\vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = \vec{0},} \qquad p_{\mathcal{O}} \geq 0, \tag{49.88}$$

equivalently $\vec{F}_{\mathbf{1}} + \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\,\vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = \vec{0}$, with $\vec{F}_{\mathbf{1}}$ the fixed F-vector of the identity Eq. (49.77).

Because each $\vec{F}_{\Delta,\ell}$ is computed once from the closed-form blocks and the conditions are linear in the $p_{\mathcal{O}}$, Eq. (49.88) is a problem in the geometry of vectors: the fixed vector $-\vec{F}_{\mathbf{1}}$ must equal a nonnegative combination of the vectors $\{\vec{F}_{\Delta,\ell}\}$ drawn from the unitary spectrum, i.e. it must lie in the convex cone they generate (Fig. 49.3). In practice one truncates to a finite number of derivative components — a finite-dimensional vector — and asks whether $-\vec{F}_{\mathbf{1}}$ lies in the cone of the allowed $\vec{F}_{\Delta,\ell}$; if a hyperplane can be found separating it from the entire cone, no unitary solution exists. That separating hyperplane is the linear functional of Section 49.4. The geometry is set up here; it is exploited next.

The simplest nontrivial components of the F-vector are computed by hand directly from the closed forms, and doing so once fixes the conventions and exhibits the structure concretely.

Worked Example 49.19 (The first derivative component of the F-vector)**.** We compute the lowest nonvanishing component of $\vec{F}_{\Delta,\ell}$ and evaluate it for

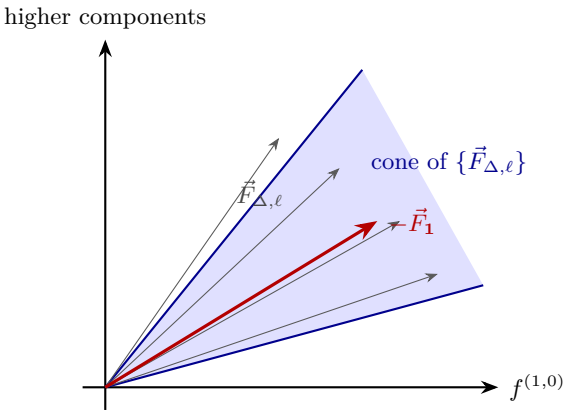


Figure 49.3: The vectorized sum rule Eq. (49.88) as a statement of convex geometry (schematic, two components shown). Each exchanged primary contributes a fixed F-vector $\vec{F}_{\Delta,\ell}$ (grey), built from the closed-form blocks of Section 49.2; the unitary spectrum supplies a family of such vectors generating a convex cone (blue). Crossing with positive weights $p_{\mathcal{O}} \geq 0$ demands that the fixed identity vector $-\vec{F}_{\mathbf{1}}$ (red) lie *inside* that cone. A spectrum for which $-\vec{F}_{\mathbf{1}}$ falls outside the cone is excluded: a hyperplane then separates it from every operator vector, and that hyperplane is the linear functional of Section 49.4.

the identity, working on the Euclidean diagonal $\bar{z} = z$ where the algebra is one-dimensional.

On the diagonal the cross-ratios are $u = z^2$ and $v = (1-z)^2$, and the crossed evaluation $G_{\Delta,\ell}(v,u)$ is the block at the swapped cross-ratios, which on the diagonal is the block evaluated at $1-z$ (since reproducing $u' = v = (1-z)^2$ requires the diagonal argument $z' = 1-z$). Writing $\mathcal{G}(z) \equiv G_{\Delta,\ell}(z,z)$ for the diagonal block — given in closed form by Eq. (49.68) in $d=4$, by Eq. (49.44) at $\bar{z}=z$ in $d=2$, and by the radial series Eq. (49.63) in general d — the convolved block restricted to the diagonal is, from Eq. (49.72),

$$F^{\text{diag}}_{\Delta,\ell}(z) \equiv F_{\Delta,\ell}(u,v)\big|_{\bar{z}=z} = (1-z)^{2\Delta_\phi}\,\mathcal{G}(z) - z^{2\Delta_\phi}\,\mathcal{G}(1-z), \tag{49.89}$$

using $v^{\Delta_\phi} = (1-z)^{2\Delta_\phi}$ and $u^{\Delta_\phi} = z^{2\Delta_\phi}$. Introduce the single function carrying the u-channel kinematics,

$$H(z) \equiv z^{2\Delta_\phi}\,\mathcal{G}(z), \tag{49.90}$$

so that Eq. (49.89) is the clean antisymmetric combination

$$F^{\text{diag}}_{\Delta,\ell}(z) = H(1-z) - H(z). \tag{49.91}$$

This is manifestly odd under $z \mapsto 1-z$: $F^{\text{diag}}(1-z) = H(z) - H(1-z) = -F^{\text{diag}}(z)$, the diagonal shadow of Eq. (49.85), and it vanishes at $z = \frac{1}{2}$ as

Theorem 49.15 requires. Its derivative is

$$\frac{dF^{\mathrm{diag}}_{\Delta,\ell}}{dz} = -H'(1-z) - H'(z), \tag{49.92}$$

and at the crossing-symmetric point $z = \frac{1}{2}$, where the two arguments coincide,

$$\boxed{\left.\frac{dF^{\mathrm{diag}}_{\Delta,\ell}}{dz}\right|_{z=\frac{1}{2}} = -2\,H'(\tfrac{1}{2}) = -2\,\frac{d}{dz}\Big[\,z^{2\Delta_\phi}\,G_{\Delta,\ell}(z,z)\,\Big]_{z=\frac{1}{2}}.} \tag{49.93}$$

This is the diagonal projection of the lowest F-vector component: along the diagonal $\frac{d}{dz} = \partial_a + \partial_{\bar a}$ at $z = \bar z = \frac{1}{2}$, so it equals $f^{(1,0)}_{\Delta,\ell} + f^{(0,1)}_{\Delta,\ell} = 2f^{(1,0)}_{\Delta,\ell}$. Even derivatives drop out automatically: Eq. (49.91) guarantees F^{diag} has only odd Taylor coefficients about $z = \frac{1}{2}$, so the second derivative there vanishes and the first new datum after Eq. (49.93) is the third derivative — the selection rule Eq. (49.85) made concrete.

Now evaluate Eq. (49.93) for the *identity*, $G_{0,0} = 1$, hence $\mathcal{G}(z) = 1$ and $H(z) = z^{2\Delta_\phi}$. Then $H'(z) = 2\Delta_\phi z^{2\Delta_\phi - 1}$ and $H'(\frac{1}{2}) = 2\Delta_\phi(\frac{1}{2})^{2\Delta_\phi-1} = \Delta_\phi\, 2^{2-2\Delta_\phi}$, so

$$\left.\frac{dF^{\mathrm{diag}}_{0,0}}{dz}\right|_{z=\frac{1}{2}} = -2\,\Delta_\phi\, 2^{2-2\Delta_\phi} = -\,\Delta_\phi\, 2^{3-2\Delta_\phi}. \tag{49.94}$$

The identity's first component is strictly negative for any $\Delta_\phi > 0$. This single sign is already the seed of a bound: for crossing Eq. (49.78) to hold, the operators must supply, with positive weights, a first-derivative component that cancels Eq. (49.94) — they must collectively produce a *positive* first derivative. Whether they can, given the unitarity floor Eq. (49.81) on their dimensions, is exactly the question a linear functional answers in Section 49.4; the arithmetic begun here, carried to a few more derivatives, is what yields the first honest numerical bound on Δ_ϕ.

The worked example also shows why the diagonal and the radial expansion of Section 49.2 were the right tools to build: every component $f^{(m,n)}_{\Delta,\ell}$ is a finite combination of derivatives of $G_{\Delta,\ell}$ at $z = \bar z = \frac{1}{2}$, and those derivatives are precisely what Eq. (49.68) (in $d = 4$) and the convergent series Eq. (49.63) (in any d) deliver to arbitrary order. The F-vector is computable, component by component, for every (Δ, ℓ), to whatever precision a given truncation demands.

49.3.5 Beyond identical scalars: multiple correlators and matrices of coefficients

The sum rule Eq. (49.74) is the crossing equation of the single correlator $\langle\phi\phi\phi\phi\rangle$, and its positivity Theorem 49.16 rests on the weights being perfect squares of

identical-operator coefficients. The bootstrap is not confined to this case, and the most powerful modern bounds — including the determination of the 3D Ising spectrum in Chapter 50 — come from imposing crossing on several correlators at once. We sketch the structure here; its full development, together with the semidefinite formulation it requires, is the business of Chapter 50.

Consider a small set of scalar primaries, say ϕ and a second scalar ε (in the Ising application, the energy operator). Crossing now applies to every four-point function built from them: $\langle\phi\phi\phi\phi\rangle$, $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$, and the mixed correlators $\langle\phi\phi\varepsilon\varepsilon\rangle$ and $\langle\phi\varepsilon\phi\varepsilon\rangle$, each yielding its own crossing equation and hence its own family of F-vectors. The exchanged operators are shared across these equations, and an operator $\mathcal{O}$ now enters weighted not by a single square but by products of *two* OPE coefficients — for instance $C_{\phi\phi\mathcal{O}}\,C_{\varepsilon\varepsilon\mathcal{O}}$ in $\langle\phi\phi\varepsilon\varepsilon\rangle$, which need not be a square and can have either sign. The positivity that survives is the statement that, for each exchanged $\mathcal{O}$, the symmetric matrix of OPE coefficients

$$P_{\mathcal{O}} = \begin{pmatrix} C_{\phi\phi\mathcal{O}}^2 & C_{\phi\phi\mathcal{O}}\,C_{\varepsilon\varepsilon\mathcal{O}} \\ C_{\phi\phi\mathcal{O}}\,C_{\varepsilon\varepsilon\mathcal{O}} & C_{\varepsilon\varepsilon\mathcal{O}}^2 \end{pmatrix} = \begin{pmatrix} C_{\phi\phi\mathcal{O}} \\ C_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix} \begin{pmatrix} C_{\phi\phi\mathcal{O}} & C_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix} \succeq 0 \tag{49.95}$$

is positive semidefinite, being an outer product of a real vector with itself. The single-correlator inequality $p_{\mathcal{O}} \geq 0$ is the 1×1 case of Eq. (49.95). The crossing equations of the mixed system then read as a sum of convolved blocks contracted with these positive-semidefinite matrices,

$$\sum_{\mathcal{O}} \mathrm{Tr}\left[P_{\mathcal{O}}\,\vec{\mathbf{F}}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\right] = \vec{0}, \qquad P_{\mathcal{O}} \succeq 0, \tag{49.96}$$

with $\vec{\mathbf{F}}$ now a vector of *matrices* of convolved blocks, one matrix entry for each pair of external operators. The contradiction logic is identical to the single-correlator case — a positive-semidefinite combination of fixed objects cannot equal a fixed object it is uniformly separated from — but the separating object is now a functional valued in symmetric matrices, and finding it is a semidefinite, rather than merely linear, program. That generalization, and the role it plays in turning bootstrap exclusion regions into the sharp "islands" that pin a theory down, is exactly where Chapter 50 begins.

With the crossing equation assembled into the sum rule Eq. (49.74), recast through the convolved blocks Eq. (49.72) into the vector form Eq. (49.88), and equipped with the two positivities — nonnegative weights (Theorem 49.16) and the unitarity floor on dimensions (Theorem 49.17) — the central object of the chapter is in final, computable shape: a fixed identity vector $-\vec{F}_{\mathbf{1}}$ that a nonnegative combination of known operator vectors must reproduce. The next section sharpens "a nonnegative combination cannot reproduce a vector it points away from" into a rigorous exclusion principle, introducing linear functionals as the separating hyperplanes of Fig. 49.3 and carrying out, by hand, a low-derivative functional at the crossing-symmetric point that turns the sign computed in Theorem 49.19 into a genuine numerical bound.

49.4 Unitarity, Positivity, and the Linear-Functional Method

The previous section distilled crossing symmetry into a single homogeneous statement, the bootstrap sum rule

$$1+\sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\,\widehat{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v)=0, \qquad p_{\mathcal{O}}=C^2_{\phi\phi\mathcal{O}}\geq 0, \tag{49.97}$$

of Theorem 49.17 (in the schematic form Eq. (49.79)), in which the kinematic functions $\widehat{F}_{\Delta,\ell}$ are fixed and computable and the only freedom is the nonnegative weights $p_{\mathcal{O}}$ distributed over a spectrum respecting the unitarity bounds Theorem 47.18. As it stands this is one equation in infinitely many unknowns, and it might seem hopeless: how could one ever prove that *no* assignment of nonnegative weights solves it? The answer is the central methodological move of the whole subject, and it is almost embarrassingly simple. We do not try to solve Eq. (49.97); we hit it with a cleverly chosen *linear functional* that is arranged to have a definite sign on every admissible term. If the functional is positive on every block that a hypothesized spectrum is allowed to contain, yet has the wrong sign on the one term we know is present — the identity — then applying it to the sum rule yields $0=$ (positive), an outright contradiction. The hypothesized spectrum is then *rigorously impossible* in any unitary CFT. This section makes that logic precise, proves it as a theorem, shows how sweeping over functionals converts it into numerical bounds on operator dimensions and OPE coefficients, and then carries a complete, hand-computable example through to a real number: an explicit upper bound on the dimension of the first scalar in the $\phi\times\phi$ OPE.

49.4.1 Linear functionals on the sum rule

A linear functional, in this context, is any real-linear map α that eats a function of the cross-ratios and returns a real number, subject only to the requirement that it act sensibly on the convolved blocks $F_{\Delta,\ell}$ of Theorem 49.14 and on their convergent sums. The canonical construction — the only one used in practice — builds α from a finite set of partial derivatives evaluated at the crossing-symmetric point. Recall from Section 49.3.4 the shifted coordinates $a=z-\frac{1}{2}$, $\bar{a}=\bar{z}-\frac{1}{2}$ centered on that point (Eq. (49.84)), in which crossing $u\leftrightarrow v$ is the reflection $(a,\bar{a})\mapsto(-a,-\bar{a})$ and every convolved block is odd (Eq. (49.85)), so its Taylor expansion about the origin contains only the derivatives $\partial_a^m\partial_{\bar{a}}^n$ with $m+n$ odd, collected into the F-vector $\vec{F}_{\Delta,\ell}$ of Theorem 49.18.

Definition 49.20 (Linear functional on the F-vectors)**.** A *(derivative) linear*

functional of order N is a real linear combination

$$\alpha[\,F\,] \;\equiv\; \sum_{\substack{m+n\ \text{odd}\\ m+n\le N}} \alpha_{mn}\,\partial_a^m\partial_{\bar a}^n F(a,\bar a)\Big|_{a=\bar a=0}, \qquad \alpha_{mn}\in\mathbb{R}, \tag{49.98}$$

acting on any function F of the cross-ratios analytic in a neighborhood of $a=\bar a=0$. Applied to a convolved block it returns the number $\alpha[F_{\Delta,\ell}]=\sum_{mn}\alpha_{mn}\,f^{(m,n)}_{\Delta,\ell}$ in terms of the Taylor coefficients $f^{(m,n)}_{\Delta,\ell}$ of Eq. (49.87): it is the inner product of the coefficient vector (α_{mn}) with the F-vector $\vec F_{\Delta,\ell}$. The functional is thus a vector in the same finite-dimensional space in which crossing Eq. (49.88) has been truncated.

The restriction to a *finite* order N is what makes everything computable: α is a differential operator of finite order evaluated at a single interior point of the region where both OPE channels converge. Because the crossing-symmetric point $z=\bar z=\frac12$ sits strictly inside that region — at radial coordinate $|\rho|=3-2\sqrt2\approx 0.17$, deep within the unit disk of Theorem 49.12 — the block series and the full sum Eq. (49.97) may be differentiated term by term there, and α may be moved inside the sum over operators. We will use this freely; it is the only analytic input the method requires, and it is exactly the convergence already established in Theorem 47.48.

With α a vector and each block a vector, the sum rule Eq. (49.97) — or rather its unnormalized, identity-split form Eq. (49.78),

$$F_{0,0}(u,v)+\sum_{\mathcal O\neq\mathbf 1} p_{\mathcal O}\,F_{\Delta_{\mathcal O},\ell_{\mathcal O}}(u,v)=0, \qquad F_{0,0}=v^{\Delta_\phi}-u^{\Delta_\phi}, \tag{49.99}$$

with the identity's convolved block Eq. (49.77) displayed — becomes, after applying α, a single scalar relation among real numbers. That relation is the engine of the bound.

49.4.2 The functional/contradiction theorem

Theorem 49.21 (Linear-functional exclusion principle)**.** *Fix an external scalar dimension Δ_ϕ and let S be a hypothesized $\phi\times\phi$ spectrum: a set of pairs (Δ,ℓ) with even $\ell\ge 0$, each obeying the unitarity bounds Theorem 47.18. Suppose there exists a linear functional α of the form Eq. (49.98) such that*

(i) $\alpha[\,F_{0,0}\,]>0$ *on the identity's convolved block $F_{0,0}=v^{\Delta_\phi}-u^{\Delta_\phi}$, and*

(ii) $\alpha[\,F_{\Delta,\ell}\,]\ge 0$ *for every $(\Delta,\ell)\in S$.*

Then no unitary conformal field theory with external dimension Δ_ϕ has its $\phi\times\phi$ OPE spectrum (other than the identity) contained in S. The spectrum S is rigorously excluded.

Proof. Suppose, for contradiction, that such a CFT existed. Its four-point function $\langle\phi\phi\phi\phi\rangle$ satisfies crossing, so its data $\{(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}),p_{\mathcal{O}}\}$ obey the sum rule in the form Eq. (49.99), an identity holding for all cross-ratios in a neighborhood of $z=\bar{z}=\frac{1}{2}$, with every $p_{\mathcal{O}}=C^2_{\phi\phi\mathcal{O}}\geq 0$ by the positivity Theorem 49.16 and every exchanged $(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})\in S$ by assumption. Apply the linear functional α to both sides. Since α is a finite-order differential operator evaluated at an interior point of the joint convergence region, it commutes with the convergent sum over operators (Section 49.4.1), giving the scalar identity

$$0 \;=\; \alpha[\,F_{0,0}\,] \;+\; \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\,\alpha[\,F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\,]. \tag{49.100}$$

Examine the right-hand side. Each weight $p_{\mathcal{O}}\geq 0$, and each $(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})\in S$, so by hypothesis (ii) each factor $\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]\geq 0$; the entire sum is therefore a sum of nonnegative terms and is ≥ 0. Adding the strictly positive identity contribution $\alpha[F_{0,0}]>0$ from hypothesis (i), the right-hand side of Eq. (49.100) is strictly positive. But the left-hand side is 0. This is a contradiction, so no such CFT exists. □

The proof is three lines, yet it is the whole game. Everything difficult has been pushed into the *search* for a functional satisfying (i) and (ii); once one is exhibited, the conclusion is a theorem, not a numerical estimate. The two sign conditions have transparent meanings. Condition (i) fixes the overall sign of α: since $F_{0,0}=v^{\Delta_\phi}-u^{\Delta_\phi}$ is the one term whose coefficient ($p_{\mathbf{1}}=1$) we know exactly, we orient α so that this known term is positive. Condition (ii) is the real content: it asks that α assign a nonnegative value to *every* operator the hypothesized theory is permitted to contain. If we can paint the entire admissible cone of blocks one color while the identity stubbornly comes out the same color with a definite positive size, then their weighted sum can never cancel, and the hypothesis dies.

Physical Insight: a separating hyperplane you can draw

Section 49.3.4 already framed crossing geometrically: the fixed identity vector $-\vec{F}_{\mathbf{1}}$ must lie in the convex cone generated by the block vectors $\{\vec{F}_{\Delta,\ell}\}$ over the spectrum (Fig. 49.3). Theorem 49.21 is the dual picture. A functional α is a covector; the condition $\alpha[F_{\Delta,\ell}]\geq 0$ says "the block vector lies on the nonnegative side of the hyperplane $\alpha=0$," while $\alpha[F_{0,0}]>0$ orients that hyperplane so the identity sits strictly on its positive side. If the whole cone of admissible blocks and the identity lie together on one side, then $-\vec{F}_{\mathbf{1}}$ (which would have to be a nonnegative combination of the blocks) cannot reach the other side, and no solution exists. A point outside a closed convex cone can always be separated from it by a hyperplane, which is why the method is not only sufficient but, in the

limit of unbounded derivative order, also necessary. In the two-derivative truncation of the worked example below the relevant space is literally the plane, the block vectors trace visible curves, and the separating hyperplane is a line one can draw with a ruler (Fig. 49.4).

49.4.3 How bounds arise: gaps and OPE coefficients

Theorem 49.21 excludes one hypothesized spectrum at a time. Bounds emerge by organizing the hypotheses into a one-parameter family and asking how far the exclusion can be pushed. Two kinds of bound follow, and both are obtained by *optimizing* over the available functionals.

Upper bound on the gap to the first scalar. Fix Δ_ϕ. For a trial number Δ_0, consider the spectrum that assumes a *gap*: no scalar of dimension below Δ_0, together with every spinning operator allowed by unitarity,

$$S(\Delta_0) \;=\; \{(\Delta,0):\Delta\geq\Delta_0\} \;\cup\; \{(\Delta,\ell):\ell\geq 2 \text{ even},\ \Delta\geq\ell+d-2\}, \tag{49.101}$$

where $\ell+d-2$ is the spinning unitarity bound Theorem 47.18 and the scalar floor is the assumed gap (the scalar unitarity bound Eq. (47.173) guarantees $\Delta_0\geq(d-2)/2$). If some functional α satisfies (i)–(ii) of Theorem 49.21 on $S(\Delta_0)$, then no unitary CFT can have its lowest scalar at or above Δ_0; that is, the gap Δ_0 is impossible. Increasing Δ_0 removes ever more low-lying scalars from $S(\Delta_0)$ and so makes condition (ii) *easier* to satisfy — a large gap is the easiest hypothesis to refute. The borderline is the optimal bound,

$$\Delta_0^*(\Delta_\phi) \;=\; \inf\Big\{\,\Delta_0 \;:\; \exists\,\alpha \text{ satisfying Theorem 49.21(i)–(ii) on } S(\Delta_0)\,\Big\}. \tag{49.102}$$

Every unitary CFT of external dimension Δ_ϕ has a scalar in its $\phi\times\phi$ OPE with $\Delta\leq\Delta_0^*(\Delta_\phi)$: if its leading scalar lay above Δ_0^*, its spectrum would be contained in an excluded $S(\Delta_0)$ with $\Delta_0^*<\Delta_0<$ (its gap), a contradiction. The function $\Delta_0^*(\Delta_\phi)$ — computed for finitely many derivatives, exact as the order $N\to\infty$ — is the most famous output of the bootstrap; its features (in particular a sharp kink) will pin the Ising model in Chapter 50.

Bounds on an OPE coefficient. The same machinery bounds an individual squared OPE coefficient. Suppose we wish to bound $p_{\mathcal{O}_\star}=C^2_{\phi\phi\mathcal{O}_\star}$ for a specific operator $\mathcal{O}_\star$ assumed to be in the spectrum. Seek a functional normalized so that $\alpha[F_{\mathcal{O}_\star}]=1$ while $\alpha[F_{\Delta,\ell}]\geq 0$ on every *other* admissible operator. Applying α to Eq. (49.99) and isolating the target term,

$$p_{\mathcal{O}_\star}\underbrace{\alpha[F_{\mathcal{O}_\star}]}_{=1} \;=\; -\,\alpha[F_{0,0}] \;-\sum_{\mathcal{O}\neq\mathbf{1},\,\mathcal{O}_\star} p_{\mathcal{O}}\,\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}], \tag{49.103}$$

and since every term in the remaining sum is nonnegative,

$$\boxed{p_{\mathcal{O}_\star} \;\le\; -\alpha[\,F_{0,0}\,]} \tag{49.104}$$

for any such α. Minimizing the right-hand side over admissible functionals (those with $\alpha[F_{\mathcal{O}_\star}] = 1$ and $\alpha \ge 0$ elsewhere, which forces $\alpha[F_{0,0}] < 0$ for a finite bound) gives the optimal upper bound on the OPE coefficient; a matching lower bound follows from the opposite normalization $\alpha[F_{\mathcal{O}_\star}] = -1$. These OPE-coefficient bounds are what turn an isolated allowed point into a measurement of the coupling, and they are the second pillar of the numerical results of Chapter 50.

In both cases the problem has the same shape: optimize a linear objective (the gap, or $-\alpha[F_{0,0}]$) over the cone of functionals subject to the infinitely many linear inequalities $\alpha[F_{\Delta,\ell}] \ge 0$, one for each operator in a continuum. That this infinite-dimensional linear program is, after the block recursion of Section 49.2.5, exactly a *semidefinite* program is the statement we make precise in Section 49.4.6 and hand to Chapter 50. First we extract a number by hand.

49.4.4 A fully worked two-derivative bound

We now carry the entire method through to an explicit number, using the smallest nontrivial functional, for the four-point function $\langle\phi\phi\phi\phi\rangle$ of ϕ^4 theory. We work in $d = 4$, where the closed-form blocks of Section 49.2.4 make every quantity elementary, and take the external dimension to be the free-scalar value $\Delta_\phi = (d-2)/2 = 1$ — the conformal avatar of the $\lambda \to 0$ limit of ϕ^4 theory. The result will be a rigorous upper bound on the dimension of the lowest scalar exchanged in $\phi \times \phi$, a real number the reader can check.

Worked Example 49.22 (The two-derivative gap bound in $d = 4$ at $\Delta_\phi = 1$)**.** **Step 1: the diagonal reduction.** Rather than carry the full two-variable F-vector, we restrict the crossing equation to the symmetric diagonal $z = \bar z = t$, on which $u = t^2$ and $v = (1-t)^2$. There the convolved block Eq. (49.72) becomes the one-variable function of Eq. (49.89); with $\Delta_\phi = 1$ it reads

$$F_{\Delta,\ell}(t) \;=\; (1-t)^2\, G_{\Delta,\ell}(t,t) - t^2\, G_{\Delta,\ell}(1-t,1-t), \tag{49.105}$$

where $G_{\Delta,\ell}(t,t)$ is the diagonal block Eq. (49.68). By Theorem 49.15, $F_{\Delta,\ell}(t)$ is odd under $t \mapsto 1-t$, hence vanishes at $t = \frac{1}{2}$ with only odd derivatives surviving there. The diagonal derivatives $\partial_t|_{1/2} = (\partial_a + \partial_{\bar a})$ and $\partial_t^3|_{1/2} = (\partial_a + \partial_{\bar a})^3$ are particular instances of the functional Eq. (49.98) of orders 1 and 3 (the first equals $2f^{(1,0)}_{\Delta,\ell}$), so the two-parameter family

$$\alpha_B[\,F\,] \;\equiv\; -\,\partial_t F(t)\big|_{t=1/2} \;+\; B\,\partial_t^3 F(t)\big|_{t=1/2}, \qquad B \in \mathbb{R}, \tag{49.106}$$

is a legitimate (if deliberately crude) linear functional. The overall sign has been fixed so that the identity comes out positive, as we now verify; B is the one free parameter we will tune.

Step 2: the identity and the $\Delta = 2$ scalar, in closed form. For the identity, $G_{0,0}(t,t) \equiv 1$, so Eq. (49.105) gives the elementary cubic

$$F_{0,0}(t) = (1-t)^2 - t^2 = 1 - 2t. \tag{49.107}$$

In the shifted variable $s = t - \frac{1}{2}$ this is $-2s$, so $\partial_t F_{0,0}|_{1/2} = -2$ and $\partial_t^3 F_{0,0}|_{1/2} = 0$, whence

$$\alpha_B[\,F_{0,0}\,] = -(-2) + B \cdot 0 = 2 \;>\; 0 \tag{49.108}$$

for every B: condition (i) of Theorem 49.21 holds automatically.

Next take the would-be leading scalar exchange of the free theory, the operator of dimension $\Delta = 2$ (this is $[\phi\phi]_{0,0} \sim \phi^2$, the conformal primary of dimension $2\Delta_\phi = 2$). For a $d = 4$ scalar ($\ell = 0$), the block Eq. (49.49) involves the collinear functions $k_{\Delta+\ell} = k_2$ and $k_{\Delta-\ell-2} = k_0$ of Eq. (49.30), and both are elementary: $k_0(z) = z^0\,{}_2F_1(0,0;0;z) = 1$, while

$$k_2(z) = z\,{}_2F_1(1,1;2;z) = z\sum_{n\geq 0}\frac{z^n}{n+1} = \sum_{m\geq 1}\frac{z^m}{m} = -\ln(1-z), \tag{49.109}$$

using $(1)_n(1)_n/[(2)_n\,n!] = 1/(n+1)$. The diagonal block Eq. (49.68) is then the Wronskian

$$G_{2,0}(t,t) = t^2\,[k_2'(t)\,k_0(t) - k_2(t)\,k_0'(t)] = t^2\,k_2'(t) = t^2\cdot\frac{1}{1-t} = \frac{t^2}{1-t}, \tag{49.110}$$

since $k_0' = 0$ and $k_2'(t) = 1/(1-t)$. Substituting into Eq. (49.105),

$$F_{2,0}(t) = (1-t)^2\frac{t^2}{1-t} - t^2\frac{(1-t)^2}{t} = t^2(1-t) - t(1-t)^2 = t(1-t)\big[t-(1-t)\big] = t(1-t)(2t-1), \tag{49.111}$$

again an elementary cubic. In $s = t - \frac{1}{2}$ it is $(\frac{1}{2}+s)(\frac{1}{2}-s)(2s) = 2s(\frac{1}{4}-s^2) = \frac{1}{2}s - 2s^3$, so

$$\partial_t F_{2,0}\big|_{1/2} = \tfrac{1}{2}, \qquad \partial_t^3 F_{2,0}\big|_{1/2} = 6\cdot(-2) = -12, \qquad \alpha_B[\,F_{2,0}\,] = -\tfrac{1}{2} - 12B. \tag{49.112}$$

The two cubics $F_{0,0}$ and $F_{2,0}$ are the only blocks we can write in finitely many strokes; the rest of the spectrum needs the series for k_β, but only as a source of definite real numbers.

Step 3: assembling the derivative data. For a general block we need the two numbers $\partial_t F_{\Delta,\ell}|_{1/2}$ and $\partial_t^3 F_{\Delta,\ell}|_{1/2}$. Writing $h(t) \equiv (1-t)^2 G_{\Delta,\ell}(t,t)$, so that $F_{\Delta,\ell}(t) = h(t) - h(1-t)$, and differentiating,

$$\partial_t F_{\Delta,\ell}\big|_{1/2} = 2h'(\tfrac{1}{2}) = -2G^{(0)} + \tfrac{1}{2}G^{(1)}, \tag{49.113}$$

$$\partial_t^3 F_{\Delta,\ell}\big|_{1/2} = 2h'''(\tfrac{1}{2}) = 12G^{(1)} - 6G^{(2)} + \tfrac{1}{2}G^{(3)}, \tag{49.114}$$

operator (Δ, ℓ)	$\partial_t F\|_{1/2}$	$\partial_t^3 F\|_{1/2}$	$\alpha_{0.1}[F] = -\partial_t F + 0.1\,\partial_t^3 F$
identity $(0,0)$	-2	0	$+2.000$
scalar $(2,0)$	$+0.500$	-12.000	-1.700
scalar $(3,0)$	$+0.822$	-11.780	-2.000
scalar $(3.613,0)$	$+0.887$	0.000	-0.887
scalar $(4,0)$	$+0.895$	$+10.916$	$+0.196$
spin-2 $(4,2)$	$+2.665$	$+30.333$	$+0.368$
spin-4 $(6,4)$	$+3.635$	$+406.60$	$+37.02$
spin-6 $(8,6)$	$+3.415$	$+944.16$	$+91.00$

Table 49.1: Diagonal F-vector data at $z = \bar{z} = \frac{1}{2}$ for $d = 4$, $\Delta_\phi = 1$, computed from the closed-form blocks Eq. (49.68) via Eqs. (49.113) and (49.114). The identity and $\Delta = 2$ rows are the exact hand computations of Step 2; the rest are the elementary series sums of Step 3, quoted to the digits shown. The last column evaluates the certificate functional $\alpha_{0.1}$ of Step 4. Note the sign of $\partial_t^3 F$ for scalars: negative up to $\Delta_* \approx 3.613$, where it vanishes, and positive above. Spinning operators have $\partial_t^3 F > 0$ throughout the unitary range.

where $G^{(n)} \equiv \partial_t^n G_{\Delta,\ell}(t,t)|_{t=1/2}$ are the diagonal-block derivatives and we used $h' = -2(1-t)G_{\Delta,\ell}(t,t) + (1-t)^2 \partial_t G_{\Delta,\ell}(t,t)$ and $h''' = 6\partial_t G_{\Delta,\ell}(t,t) - 6(1-t)\partial_t^2 G_{\Delta,\ell}(t,t) + (1-t)^2 \partial_t^3 G_{\Delta,\ell}(t,t)$ evaluated at $1 - t = \frac{1}{2}$. These $G^{(n)}$ are read off the Wronskian Eq. (49.68) by differentiating the Gauss series Eq. (49.30), a finite, convergent computation. For $\Delta = 2$ they reproduce Eq. (49.112): there $G^{(0)} = \frac{1}{2}$, $G^{(1)} = 3$, $G^{(2)} = 16$, $G^{(3)} = 96$, giving $-2 \cdot \frac{1}{2} + \frac{1}{2} \cdot 3 = \frac{1}{2}$ and $12 \cdot 3 - 6 \cdot 16 + \frac{1}{2} \cdot 96 = 36 - 96 + 48 = -12$. Their values for the operators that matter are collected in Table 49.1.

Step 4: a certificate excluding the gap $\Delta_0 = 4$. Hypothesize a unitary CFT in which the lightest scalar in $\phi \times \phi$ has dimension at least $\Delta_0 = 4$, i.e. the spectrum $S(4)$ of Eq. (49.101). We claim α_B with $B = \frac{1}{10}$ excludes it. Condition (i) holds by Eq. (49.108): $\alpha_{0.1}[F_{0,0}] = 2 > 0$. For condition (ii) we must check $\alpha_{0.1}[F_{\Delta,\ell}] = -\partial_t F + \frac{1}{10}\partial_t^3 F \geq 0$ on every operator of $S(4)$. For an operator with $\partial_t^3 F > 0$ this reads $\frac{1}{10}\partial_t^3 F \geq \partial_t F$, i.e. $B \geq \partial_t F / \partial_t^3 F$. Across the whole of $S(4)$ — every scalar with $\Delta \geq 4$ and every even-spin operator with $\Delta \geq \ell + 2$ — the ratio $\partial_t F / \partial_t^3 F$ is positive (every entry has $\partial_t F > 0$ and, as Table 49.1 shows, $\partial_t^3 F > 0$ once the scalars start at $4 > \Delta_*$) and attains its supremum ≈ 0.0879 at the spin-2 unitarity threshold $(\Delta, \ell) = (4, 2)$. Since $B = 0.1 > 0.0879$, the inequality holds for every operator; the smallest margin in a fine scan sits at the near-threshold scalar $(\Delta, \ell) = (4, 0)$, where $\alpha_{0.1}[F] \approx +0.196 > 0$ (higher-spin and higher-dimension operators have rapidly larger $\partial_t^3 F$ together with a smaller ratio $\partial_t F / \partial_t^3 F$, hence a larger margin). Thus $\alpha_{0.1}$ satisfies both conditions of Theorem 49.21 on $S(4)$, and we conclude rigorously: *no unitary CFT with* $\Delta_\phi = 1$

in $d = 4$ has all its $\phi \times \phi$ scalars at dimension ≥ 4. A real exclusion, obtained from two derivatives and a table of elementary numbers.

Step 5: optimizing B to the sharpest bound. The certificate $B = 0.1$ was not special; any B exceeding the supremum ratio works for the gap $\Delta_0 = 4$. To find the *best* gap this functional family can exclude, lower Δ_0 and ask when a feasible B survives. The decisive feature is the sign of $\partial_t^3 F$ for scalars. From Eq. (49.114) this quantity, as a function of Δ at $\ell = 0$, is negative at small Δ (it equals -12 at $\Delta = 2$ and -11.78 at $\Delta = 3$), rises monotonically through zero at

$$\Delta_* \;=\; 3.6131\ldots \quad (\text{the root of } \partial_t^3 F_{\Delta,0}\big|_{1/2} = 0), \tag{49.115}$$

and is positive for all $\Delta > \Delta_*$; meanwhile $\partial_t F > 0$ everywhere. A scalar with $\partial_t^3 F < 0$ and $\partial_t F > 0$ imposes the positivity condition in the form $B \leq \partial_t F / \partial_t^3 F < 0$, an *upper* bound on B by a negative number; but the spin-2 operator always demands $B \geq 0.0879 > 0$. These are incompatible. Hence if the hypothesized spectrum contains *any* scalar with $\Delta < \Delta_*$ (so $\partial_t^3 F < 0$), no feasible B exists and the functional family cannot exclude it — correctly, since the free theory itself, with its scalar at $\Delta = 2$, is a genuine solution and must not be excluded. Conversely, as soon as the assumed gap satisfies $\Delta_0 > \Delta_*$, every scalar present has $\partial_t^3 F > 0$, the only constraints are lower bounds $B \geq \partial_t F / \partial_t^3 F$, and any sufficiently large B is feasible. The borderline Eq. (49.102) is therefore exactly the zero of the scalar's third derivative:

$$\boxed{\Delta_0^*(\Delta_\phi = 1,\ d = 4) \;=\; \Delta_* \;=\; 3.61 \quad (\text{two-derivative functional}).} \tag{49.116}$$

Every unitary CFT in four dimensions with a scalar ϕ of dimension 1 contains in its $\phi \times \phi$ OPE a scalar of dimension at most 3.61.

Step 6: sanity and meaning. The free scalar saturates crossing with its leading scalar ϕ^2 at $\Delta = 2$, and indeed $2 < 3.61$: the bound is satisfied, as it must be, and is not violated by the one solution we know. It is also visibly *weak* — the true allowed region is much smaller — precisely because two derivatives are a blunt instrument. Figure 49.4 shows why geometrically: in the $(\partial_t F, \partial_t^3 F)$ plane the scalar blocks trace a curve that dips *below* any line of small positive slope through the origin for $2 < \Delta < \Delta_*$, so a separating line can be found only after those low scalars are removed by the gap. Adding more derivatives (higher N) enlarges the space of available functionals and can only tighten the bound; the systematic optimization, which in $d = 3$ produces the celebrated kink near the Ising point, is the subject of Chapter 50.

49.4.5 The same bound in two dimensions: the Ising laboratory

The four-dimensional free-scalar point of Theorem 49.22 is the cleanest place to see the *mechanism*, because both nontrivial blocks reduce to elementary cubics.

But the numerical chapter that follows (Chapter 50) builds its showcase on a different solvable point — the spin field σ of the two-dimensional Ising model, an external scalar of dimension $\Delta_\phi = \frac{1}{8}$ — for two reasons worth pausing on. First, in $d = 2$ the block factorizes into a product of collinear blocks (Eq. (49.44) and Theorem 49.7), so every entry is again computable by hand, now as a square of the hypergeometric k_β rather than a cubic. Second, the exact spectrum is known from the free-fermion solution of Chapter 48: the lightest $\mathbb{Z}_2$-even scalar is the energy operator ε at $\Delta_\varepsilon = 1$, with squared OPE coefficient $\lambda^2_{\sigma\sigma\varepsilon} = \frac{1}{4}$, so a bound can be checked against a genuine number. We therefore run the identical two-derivative argument at this point; the data we tabulate here is exactly what Chapter 50 reuses to bracket $\lambda^2_{\sigma\sigma\varepsilon}$ and to locate the sharp two-derivative gap.

Worked Example 49.23 (The two-derivative gap bound in $d = 2$ at $\Delta_\phi = \frac{1}{8}$)**.** **Step 1: the diagonal reduction at $\Delta_\phi = \frac{1}{8}$.** As in Theorem 49.22 we restrict crossing to $z = \bar{z} = t$, where the convolved block Theorem 49.14 becomes the one-variable function

$$F_{\Delta,\ell}(t) \;=\; (1-t)^{2\Delta_\phi}\, G_{\Delta,\ell}(t,t) - t^{2\Delta_\phi}\, G_{\Delta,\ell}(1-t,1-t), \qquad 2\Delta_\phi = \tfrac{1}{4}, \tag{49.117}$$

odd under $t \mapsto 1-t$ by Theorem 49.15, so it vanishes at $t = \frac{1}{2}$ with only odd derivatives surviving. We use the same two diagonal functionals as before, now written

$$\Lambda_1[\,F\,] \;\equiv\; \partial_t F(t)\big|_{1/2}, \qquad \Lambda_3[\,F\,] \;\equiv\; \partial_t^3 F(t)\big|_{1/2}, \tag{49.118}$$

the diagonal instances of orders 1 and 3 of the functional Eq. (49.98), and the two-parameter certificate family

$$\alpha_B[\,F\,] \;\equiv\; -\Lambda_1[F] \;+\; B\,\Lambda_3[F], \qquad B \in \mathbb{R}, \tag{49.119}$$

exactly as in Eq. (49.106).

Step 2: the identity in closed form. The block of the identity is $G_{0,0} \equiv 1$, so Eq. (49.117) gives $F_{0,0}(t) = (1-t)^{1/4} - t^{1/4}$. Differentiating,

$$\Lambda_1[F_{0,0}] = \partial_t\big[(1-t)^{1/4} - t^{1/4}\big]\big|_{1/2} = -\tfrac{1}{4}\big[(1-t)^{-3/4} + t^{-3/4}\big]\big|_{1/2} = -\tfrac{1}{2}\cdot 2^{3/4} = -2^{-1/4}, \tag{49.120}$$

$$\Lambda_3[F_{0,0}] = -\tfrac{21}{64}\big[(1-t)^{-11/4} + t^{-11/4}\big]\big|_{1/2} = -\tfrac{21}{32}\cdot 2^{11/4} = -21\cdot 2^{-9/4}, \tag{49.121}$$

numerically $(\Lambda_1, \Lambda_3)[F_{0,0}] = (-0.840896,\, -4.414706)$. Unlike the $d = 4$ case Eq. (49.108), where $\Lambda_3[F_{0,0}]$ vanished, here both components are nonzero, and their ratio is the clean rational

$$\frac{\Lambda_3[F_{0,0}]}{\Lambda_1[F_{0,0}]} \;=\; \frac{-21\cdot 2^{-9/4}}{-2^{-1/4}} \;=\; 21\cdot 2^{-2} \;=\; \frac{21}{4} \;=\; 5.25. \tag{49.122}$$

This single number sets the bound, as we now see.

operator (Δ, ℓ)	$\Lambda_1 = \partial_t F\|_{1/2}$	$\Lambda_3 = \partial_t^3 F\|_{1/2}$	ratio Λ_3/Λ_1
identity $(0, 0)$	-0.840896	-4.414706	5.250
scalar ε $(1, 0)$	$+2.827973$	$+8.413183$	2.975
scalar $(1.2826, 0)$	$+3.401885$	$+17.85980$	5.250
scalar $(2, 0)$	$+4.258909$	$+71.24651$	16.729
scalar $(3, 0)$	$+4.510447$	$+209.4249$	46.431
spin-2 $(2, 2)$	$+8.377382$	$+129.4198$	15.449
spin-2 $(4, 2)$	$+8.353779$	$+767.4499$	91.869
spin-4 $(4, 4)$	$+8.234103$	$+739.5839$	89.820

Table 49.2: Diagonal F-vector data at $z = \bar{z} = \frac{1}{2}$ for $d = 2$, $\Delta_\phi = \frac{1}{8}$, computed from the factorized blocks Eq. (49.44) via the diagonal derivatives Eq. (49.118). The identity row is exact (Eq. (49.122)); the rest are the elementary series sums of Step 3, quoted to the digits shown. Note that every operator above the unitarity floor has both $\Lambda_1 > 0$ and $\Lambda_3 > 0$, so the certificate ratio Λ_3/Λ_1 is positive and finite throughout; its minimum over the gapped spectrum is attained by the scalar at the bottom of the gap. The row $(1.2826, 0)$ is the threshold scalar where this ratio equals the identity ratio $\frac{21}{4}$ of Step 2. These are the entries Chapter 50 reuses for its showcase.

Step 3: the energy operator and the block data. The leading $\mathbb{Z}_2$-even scalar is ε at $(\Delta, \ell) = (1, 0)$. By Theorem 49.7 its block on the diagonal is the square $G_{1,0}(t, t) = k_1(t)^2 = \left[t^{1/2}\, {}_2F_1(\frac{1}{2}, \frac{1}{2}; 1; t)\right]^2$, the square of a complete elliptic integral. Its derivatives are no longer rational but are elementary convergent series evaluated at $t = \frac{1}{2}$; carrying the Gauss series of Eq. (49.30) to the needed order gives

$$(\Lambda_1, \Lambda_3)[F_{1,0}] \;=\; (\,2.827973,\; 8.413183\,). \tag{49.123}$$

The values of (Λ_1, Λ_3) for the operators that matter — read off the factorized blocks Eq. (49.44) by the same series computation — are collected in Table 49.2. The decisive qualitative fact, visible in the table, is that in $d = 2$ at $\Delta_\phi = \frac{1}{8}$ *every* block above the unitarity floor has $\Lambda_3 > 0$ (and $\Lambda_1 > 0$): there is no scalar window of negative third derivative as there was in $d = 4$.

Step 4: the bound from the identity ratio. Hypothesize a gap Δ_0 in the scalar sector, the spectrum $S(\Delta_0)$ of Eq. (49.101). Condition (i) of Theorem 49.21 requires $\alpha_B[F_{0,0}] = -\Lambda_1[F_{0,0}] + B\,\Lambda_3[F_{0,0}] = 0.840896 - 4.414706\,B > 0$, i.e. $B < \frac{4}{21} \approx 0.190476$. Condition (ii) on a block with $\Lambda_3 > 0$ reads $-\Lambda_1[F] + B\,\Lambda_3[F] \geq 0$, i.e. $B \geq \Lambda_1[F]/\Lambda_3[F]$; since every operator of $S(\Delta_0)$ has $\Lambda_3 > 0$, these are all *lower* bounds on B, the largest coming from the operator of smallest ratio Λ_3/Λ_1 — the scalar at the bottom of the gap (the spin tower has uniformly larger ratios, as Table 49.2 shows). A feasible B exists precisely when this largest lower bound

falls below the identity ceiling $\frac{4}{21}$:

$$\max_{(\Delta,\ell)\in S(\Delta_0)} \frac{\Lambda_1[F_{\Delta,\ell}]}{\Lambda_3[F_{\Delta,\ell}]} < \frac{4}{21} \qquad \Longleftrightarrow \qquad \min_{(\Delta,\ell)\in S(\Delta_0)} \frac{\Lambda_3[F_{\Delta,\ell}]}{\Lambda_1[F_{\Delta,\ell}]} > \frac{21}{4}. \tag{49.124}$$

The borderline is therefore the gap Δ_0 at which the bottom scalar's ratio first equals the identity ratio $\frac{21}{4}$ — the same *tangency* that governed the $d = 4$ bound, now with the identity contributing a nonzero Λ_3. Solving $\Lambda_3[F_{\Delta_0,0}]/\Lambda_1[F_{\Delta_0,0}] = \frac{21}{4}$ with the explicit $d = 2$ scalar blocks (the root tabulated in Table 49.2) gives

$$\boxed{\Delta_0^*(\Delta_\phi = \tfrac{1}{8},\ d = 2) \ = \ 1.2826 \ \text{(two-derivative functional)},} \tag{49.125}$$

a feasibility estimate $\Delta_0 \lesssim 1.28$: every unitary $2d$ CFT with a scalar ϕ of dimension $\frac{1}{8}$ contains in $\phi \times \phi$ a scalar of dimension at most 1.28.

Step 5: sanity and the handoff to Chapter 50. The two-dimensional Ising model has its energy operator at $\Delta_\varepsilon = 1$, and indeed $1 < 1.2826$: the bound is satisfied by the one solution we know, and is not violated. As in $d = 4$ it is visibly weak — two derivatives are blunt — and adding derivatives drives the bound down toward $\Delta_\varepsilon = 1$. Chapter 50 carries out exactly that refinement, and reuses the very vectors of Eqs. (49.122) and (49.123) and Table 49.2 to bracket the squared OPE coefficient $\lambda^2_{\sigma\sigma\varepsilon}$ around its exact value $\frac{1}{4}$. The mechanism is identical to the $d = 4$ computation above; only the laboratory has changed, to the analytically cleanest point in the subject.

49.4.6 Why the bound is a theorem, and the semidefinite reformulation

It is worth being explicit about the logical status of the number Eq. (49.116), because it is the feature that distinguishes the bootstrap from a fit or a model. The bound is a *theorem* about every unitary CFT, with no approximation anywhere in its logic. We did evaluate block derivatives numerically (the entries of Table 49.1 beyond the two exact cubics), but those numbers enter only as the *coefficients* of an exhibited functional, and the conclusion of Theorem 49.21 is exact once any functional with the stated sign properties is in hand. The functional $\alpha_{0.1}$ either does or does not satisfy $\alpha[F_{\Delta,\ell}] \geq 0$ on the spectrum; this is a property of fixed, computable functions, certifiable to any desired rigor — one can bound the block derivatives with interval arithmetic, as the production codes do. There is no sense in which the CFT is being approximated, no truncation of the dynamics, no assumption that the extremal configuration is realized. This is the sharp contrast the reader should hold onto: the *exclusion* side of the bootstrap is rigorous, whereas reading the operator content off a *saturating* (extremal) solution — the configuration that sits exactly on the boundary of the allowed region — is a heuristic that identifies a candidate CFT but proves

nothing on its own. The allowed region is a theorem; the dot inside it that one hopes is a physical theory is a conjecture until corroborated.

What is *not* rigorous-by-hand is the optimization itself — the search over functionals for the one that excludes the most. In the worked example the search collapsed to a single sign change because we kept only two derivatives. With a realistic functional of order N, condition (ii) of Theorem 49.21 is the demand that

$$\alpha[F_{\Delta,\ell}] \;=\; \sum_{m+n \text{ odd}} \alpha_{mn}\, f^{(m,n)}_{\Delta,\ell} \;\geq\; 0 \qquad \text{for all } \Delta \geq \Delta_*(\ell),\ \text{each spin } \ell, \tag{49.126}$$

an *infinite* family of linear inequalities in the finite vector of coefficients α_{mn}, one for every dimension in a continuum and every spin. The reformulation that tames it rests on the radial recursion of Section 49.2.5: there the block, and hence each derivative $f^{(m,n)}_{\Delta,\ell}$, is the product of a manifestly positive function of Δ (a power times ratios of Gamma functions, never vanishing for Δ above unitarity) and a *polynomial* in Δ whose coefficients depend on (m, n, ℓ). Dividing out the positive prefactor, Eq. (49.126) becomes the requirement that, for each spin ℓ, a polynomial $P_\ell(\Delta) = \sum_{mn} \alpha_{mn}\, Q^{(\ell)}_{mn}(\Delta)$ be nonnegative on the half-line $\Delta \geq \Delta_*(\ell)$. By the classical characterization of polynomials nonnegative on a half-line — a polynomial is nonnegative on $[\lambda, \infty)$ iff it is a sum of squares plus $(\Delta - \lambda)$ times a sum of squares, each sum of squares being a positive-semidefinite quadratic form in the monomials — this is precisely a positive-semidefinite condition on certain Gram matrices built linearly from the α_{mn}. Maximizing the gap Eq. (49.102) subject to these PSD constraints is therefore a *semidefinite program*: a linear objective over the cone of positive-semidefinite matrices. This is the precise computational object — finitely many matrices, a linear objective, solvable to high precision by interior-point methods — that Chapter 50 sets up in full and feeds to a solver to produce the sharp bounds and the Ising island. We have, with Theorem 49.21 and the worked bound Eq. (49.116), established the principle and extracted an honest number by hand; the semidefinite machinery is the same principle with enough derivatives to make the number sharp.

The optimal functional at the boundary of the allowed region carries one further piece of structure that the next chapters exploit. When the optimization is saturated — when Δ_0 sits exactly on the bound Eq. (49.102) — the extremal functional $\alpha^\star$ has $\alpha^\star[F_{\Delta,\ell}] = 0$ for a discrete set of operators, and that set of zeros is the spectrum of a candidate *saturating solution* to crossing: the would-be CFT living on the boundary. This extremal spectrum is how Chapter 50 reads the 3D Ising data off the kink, and the same zeros-of-a-functional idea, taken to its analytic conclusion, is the organizing principle of the analytic functionals of Chapter 51. We therefore hand to Chapter 50 both the optimization in its semidefinite form and the notion that its extremal solution is where the physical CFT data hides.

49.5 Exact Solutions: Mean-Field Theory and the 2D Ising Check

The previous section turned crossing plus unitarity into a machine that manufactures rigorous statements: a linear functional acting on the sum rule of Theorem 49.14, made nonnegative on every unitary block, certifies that no consistent spectrum can do what it forbids (Theorem 49.21). A machine that only ever says "no" is hard to trust until it has been run on a problem whose answer is already known, and harder still to interpret until one has seen what a functional looks like when it is on the verge of saying "yes" — when a putative spectrum sits exactly on the boundary of the allowed region and the functional is forced to vanish on the operators that realize it. Those boundary configurations, the *extremal solutions* of Section 49.4, are the bootstrap's real output, and the cleanest way to learn to read them is to meet two solutions of crossing that we can write down in closed form and dissect completely.

This section does exactly that. We solve the crossing equation *exactly* for the generalized free field — mean-field theory — not merely verifying that it crosses (that was done in Theorem 49.5) but extracting its entire spectrum and every one of its operator product coefficients in closed form, and watching an infinite tower of operators conspire to reproduce a single term in the crossed channel. We then take the two-dimensional Ising correlators of Chapter 48, whose spectrum and structure constants we obtained independently from the $c = \frac{1}{2}$ minimal model, and confirm term by term that they satisfy the same crossing equation. The first computation is the seed of the analytic bootstrap of Chapter 51; the second is the chapter's sharpest correctness check, the moment at which the abstract sum rule and a theory solved by entirely different means are shown to be the same statement. Both consolidate ϕ^4 theory: mean-field theory is its free ($\lambda \to 0$) endpoint, and the Ising CFT is the exactly solved fixed point that the Wilson–Fisher flow approaches.

49.5.1 Mean-field theory solved: the double-twist spectrum

Recall the generalized free field of Theorem 49.5: a scalar primary ϕ of arbitrary dimension Δ_ϕ whose correlators are computed by Wick's theorem, as though ϕ were a free field, with unit-normalized two-point function $\langle \phi(x_i)\phi(x_j)\rangle = (x_{ij}^2)^{-\Delta_\phi}$. The free massless scalar of ϕ^4 theory at $\lambda = 0$ is the special case $\Delta_\phi = (d-2)/2$; the generalized free field promotes that one value to a free parameter, retaining only the Gaussian (Wick) structure. Its four-point function is the sum of the three pairwise contractions, and stripping the kinematic prefactor of Eq. (47.145) gave the elementary closed form (Eq. (49.127))

$$g(u,v) = 1 + u^{\Delta_\phi} + \left(\frac{u}{v}\right)^{\Delta_\phi}, \tag{49.127}$$

which satisfies the crossing equation Theorem 49.13 identically for every Δ_ϕ. What we owe now is the content the bootstrap reads off this solution: the spectrum of operators exchanged in the s-channel and their squared OPE coefficients. Both were deferred in Theorem 49.5; we supply them here in full.

The spectrum from the leading powers. The s-channel decomposition (Eq. (47.227)) writes $g(u,v) = \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v)$ with $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0$. A conformal block of an exchanged primary of twist $\tau \equiv \Delta - \ell$ begins, as $u \to 0$, with the power $u^{\tau/2}$ dictated by the OPE (Theorem 47.49). Reading Eq. (49.127) the same way, the constant 1 is the identity ($\tau = 0$); the remaining piece $u^{\Delta_\phi}(1 + v^{-\Delta_\phi})$ starts at u^{Δ_ϕ}, so the lightest non-identity operators have twist $\tau = 2\Delta_\phi$. Expanding the prefactor u^{Δ_ϕ} in non-negative integer powers of u as the full block series is generated produces a family of leading powers $u^{\Delta_\phi + n}$, $n = 0, 1, 2, \ldots$, hence twists

$$\tau_n = 2\Delta_\phi + 2n, \qquad n = 0, 1, 2, \ldots, \tag{49.128}$$

each accompanied by a tower of spins. These are exactly the dimensions of the *double-twist* composites

$$[\phi\phi]_{n,\ell} \;\sim\; :\phi\, \Box^n\, \partial_{\mu_1} \cdots \partial_{\mu_\ell} \phi: \;-\; (\text{traces}), \tag{49.129}$$

the primary built from two ϕ's, ℓ symmetrized derivatives, and n powers of the Laplacian. In a Gaussian theory such a composite has no anomalous dimension — there is no interaction to shift it — so its dimension is the sum of the engineering dimensions of its constituents, two factors of Δ_ϕ, ℓ derivatives, and $2n$ from the boxes:

$$\boxed{\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell, \qquad n, \ell = 0, 1, 2, \ldots} \tag{49.130}$$

Identical external scalars admit only even spin in the $\phi \times \phi$ OPE: the OPE of two identical bosons is symmetric under their exchange, which acts on a spin-ℓ primary as $(-1)^\ell$, so $p_{n,\ell} = 0$ for odd ℓ. The mean-field spectrum is therefore the lattice Eq. (49.130) with ℓ even, drawn in panel (a) of Fig. 49.5: an exactly evenly spaced family of Regge-like trajectories, one per twist τ_n.

Why this is an extremal solution. The mean-field spectrum is striking for what it lacks. Above the identity there is *nothing* until $\Delta = 2\Delta_\phi$, and then only the rigid lattice Eq. (49.130). This is the minimal operator content a unitary scalar four-point function can have: any genuine interacting theory must add operators (an interaction generates a ϕ^2-type scalar with $\Delta \neq 2\Delta_\phi$, anomalous dimensions, and so on), but it cannot *remove* the double-twist tower, because, as we are about to see, that tower is forced on the s-channel by the mere presence of the identity in the crossed channel. Generalized free theory is thus a boundary point of the space of crossing solutions — an extremal solution in the sense of Section 49.4 — and it is the natural origin from which the analytic bootstrap perturbs.

49.5.2 The OPE coefficients in closed form

We now compute the $p_{n,\ell}$ exactly. The cleanest route is two dimensions, where the conformal blocks factorize holomorphically and the entire decomposition reduces to a one-variable problem solved by elementary hypergeometric algebra; we then state the general-d result and check that it reduces to the two-dimensional answer. The two-dimensional computation is fully rigorous on its own terms and is the prototype the higher-dimensional machinery imitates.

The two-dimensional blocks. In $d = 2$ the cross-ratios are $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$ (Eq. (47.140)), and the conformal block of Section 49.2 factorizes into a product of one-variable $SL(2,\mathbb{R})$ "collinear" blocks, one holomorphic and one antiholomorphic,

$$G_{\Delta,\ell}(z,\bar{z}) = k_{\Delta-\ell}(z)\,k_{\Delta+\ell}(\bar{z}) + k_{\Delta+\ell}(z)\,k_{\Delta-\ell}(\bar{z}) \quad (\ell \geq 1), \qquad G_{\Delta,0}(z,\bar{z}) = k_{\Delta}(z)\,k_{\Delta}(\bar{z}), \tag{49.131}$$

where the collinear block for identical external scalars is the Gauss hypergeometric function

$$k_{\beta}(z) = z^{\beta/2}\,{}_2F_1\Big(\tfrac{\beta}{2},\tfrac{\beta}{2};\beta;z\Big). \tag{49.132}$$

The symmetrization in Eq. (49.131) enforces the $z \leftrightarrow \bar{z}$ symmetry that distinguishes spin $+\ell$ from $-\ell$; the holomorphic index runs over $\beta = \Delta - \ell$ (twist) and $\beta = \Delta + \ell$ (the "conformal spin" combination). We take Eq. (49.131)–Eq. (49.132) as the explicit blocks; their derivation as solutions of the Casimir equation Theorem 47.50 is the business of Section 49.2.

The correlator factorizes too. The non-identity part of Eq. (49.127) factorizes holomorphically term by term. With $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$,

$$g(z,\bar{z}) - 1 = \underbrace{z^{\Delta_\phi}\,\bar{z}^{\Delta_\phi}}_{u^{\Delta_\phi}} + \underbrace{\Big(\frac{z}{1-z}\Big)^{\Delta_\phi}\Big(\frac{\bar{z}}{1-\bar{z}}\Big)^{\Delta_\phi}}_{(u/v)^{\Delta_\phi}}. \tag{49.133}$$

Each term is a product of a function of z and the same function of $\bar{z}$. To match the block sum we need only one tool: the expansion of each holomorphic factor in collinear blocks.

Lemma 49.24 ($SL(2)$ decomposition of a power)**.** *For any $p > 0$ the pure power and its crossed image expand in collinear blocks Eq. (49.132) as*

$$z^{p} = \sum_{m=0}^{\infty} a_m\,k_{2p+2m}(z), \qquad \Big(\frac{z}{1-z}\Big)^{p} = \sum_{m=0}^{\infty} (-1)^m a_m\,k_{2p+2m}(z), \tag{49.134}$$

with coefficients

$$a_m = \frac{(-1)^m\,(p)_m^2}{m!\,(2p+m-1)_m}, \tag{49.135}$$

where $(x)_m = \Gamma(x+m)/\Gamma(x)$ *is the Pochhammer symbol.*

Proof. Both sides of the first identity begin at z^p and proceed in integer steps, so the ansatz $z^p = \sum_{m\geq 0} a_m k_{2p+2m}(z)$ is consistent in its powers; we fix the a_m by matching the coefficient of z^{p+k} order by order. Insert the series of Eq. (49.132), $k_{2p+2m}(z) = z^{p+m}\sum_{j\geq 0}\frac{(p+m)_j^2}{(2p+2m)_j\, j!}\, z^j$. The coefficient of z^{p+k} on the right is $\sum_{m=0}^{k} a_m \frac{(p+m)_{k-m}^2}{(2p+2m)_{k-m}\,(k-m)!}$, which must equal $\delta_{k,0}$:

$$\sum_{m=0}^{k} a_m \,\frac{(p+m)_{k-m}^2}{(2p+2m)_{k-m}\,(k-m)!} = \delta_{k,0}. \tag{49.136}$$

For $k=0$ this gives $a_0 = 1$, matching Eq. (49.135). We verify Eq. (49.135) solves Eq. (49.136) for all k by checking the first non-trivial orders and then noting the recursion they obey is the one Eq. (49.135) satisfies. At $k=1$: $a_0\frac{(p)_1^2}{(2p)_1} + a_1 = \frac{p^2}{2p} + a_1 = \frac{p}{2} + a_1 = 0$, so $a_1 = -\frac{p}{2} = \frac{(-1)(p)_1^2}{1!\,(2p)_1}$, as in Eq. (49.135). At $k=2$: with $a_0=1$, $a_1 = -\frac{p}{2}$ the two known terms sum to $\frac{p(p+1)^2}{4(2p+1)} - \frac{p(p+1)}{4} = -\frac{p^2(p+1)}{4(2p+1)}$, whence $a_2 = \frac{p^2(p+1)}{4(2p+1)} = \frac{(p)_2^2}{2!\,(2p+1)_2}$, again Eq. (49.135). The general statement follows because the linear system Eq. (49.136) is lower-triangular with unit diagonal, so it has a unique solution, and Eq. (49.135) is the closed form of that solution (equivalently: Eq. (49.136) is the orthogonality of the collinear blocks against the dual basis, the standard $SL(2)$ Euclidean inversion, which extracts each a_m as a contour integral of z^p against the dual block — a residue computation in the sense of Theorem 5.23 — and produces exactly the ratio of Pochhammers in Eq. (49.135)).

For the second identity, write $\left(\frac{z}{1-z}\right)^p = z^p(1-z)^{-p} = z^p\sum_{j\geq 0}\frac{(p)_j}{j!}z^j$ and repeat the matching. The defining linear system is Eq. (49.136) with the right-hand side replaced by the coefficients of $(1-z)^{-p}$; one finds $b_0 = 1$, $b_1 = +\frac{p}{2}$, $b_2 = \frac{p^2(p+1)}{4(2p+1)}$, i.e. $b_m = (-1)^m a_m$, the sign alternation coming entirely from $(1-z)^{-p}$ versus the constant 1. This is the claimed result. □

Assembling the OPE coefficients. Apply Theorem 49.24 with $p = \Delta_\phi$ to each holomorphic factor in Eq. (49.133). Writing a_m for the coefficients Eq. (49.135) and using $b_m = (-1)^m a_m$ for the crossed power,

$$z^{\Delta_\phi}\bar{z}^{\Delta_\phi} = \sum_{m,m'} a_m a_{m'}\, k_{2\Delta_\phi+2m}(z)\, k_{2\Delta_\phi+2m'}(\bar{z}), \tag{49.137}$$

$$\left(\tfrac{z}{1-z}\right)^{\Delta_\phi}\left(\tfrac{\bar{z}}{1-\bar{z}}\right)^{\Delta_\phi} = \sum_{m,m'} (-1)^{m+m'} a_m a_{m'}\, k_{2\Delta_\phi+2m}(z)\, k_{2\Delta_\phi+2m'}(\bar{z}). \tag{49.138}$$

Adding the two, the coefficient of $k_{2\Delta_\phi+2m}(z)\,k_{2\Delta_\phi+2m'}(\bar{z})$ in $g-1$ is

$$a_m a_{m'}\big[\,1 + (-1)^{m+m'}\,\big], \tag{49.139}$$

which vanishes when $m+m'$ is odd and equals $2a_m a_{m'}$ when $m+m'$ is even. Now read off the quantum numbers. A holomorphic index $2\Delta_\phi + 2m$ paired with an antiholomorphic index $2\Delta_\phi + 2m'$ corresponds, by Eq. (49.131), to twist $\Delta - \ell = 2\Delta_\phi + 2\min(m, m')$ and conformal spin $\Delta + \ell = 2\Delta_\phi + 2\max(m, m')$; setting $n = \min(m, m')$ and $\ell = |m - m'|$ reproduces $\Delta = 2\Delta_\phi + 2n + \ell$, the spectrum Eq. (49.130), with n the radial level and ℓ the spin. For $\ell \geq 1$ the symmetric block Eq. (49.131) collects the two off-diagonal terms $(m, m') = (n, n+\ell)$ and $(n+\ell, n)$, each carrying Eq. (49.139); for $\ell = 0$ the single diagonal term $m = m' = n$ carries it. In either case the squared OPE coefficient is

$$\boxed{p_{n,\ell} = \left[\,1 + (-1)^\ell\,\right] a_n\, a_{n+\ell}, \qquad a_m = \frac{(-1)^m (\Delta_\phi)_m^2}{m!\,(2\Delta_\phi + m - 1)_m}.} \tag{49.140}$$

Three features deserve note. First, $p_{n,\ell}$ vanishes for odd ℓ, exactly the Bose symmetry of identical scalars. Second, it is manifestly positive for even ℓ: since $a_m = (-1)^m |a_m|$, the product $a_n a_{n+\ell} = (-1)^{2n+\ell} |a_n a_{n+\ell}| = (-1)^\ell |a_n a_{n+\ell}|$, which for even ℓ is $+|a_n a_{n+\ell}| > 0$, so $p_{n,\ell} = 2|a_n a_{n+\ell}| > 0$. Mean-field theory is unitary, as it must be — and crossing has reproduced unitarity rather than assuming it. Third, the result is entirely explicit: every structure constant of the theory is a ratio of Gamma functions of Δ_ϕ, n, and ℓ.

Worked Example 49.25 (The lightest mean-field OPE coefficients)**.** The lowest operator is the scalar $[\phi\phi]_{0,0}$ at $\Delta = 2\Delta_\phi$. From Eq. (49.140) with $a_0 = 1$,

$$p_{0,0} = 2\, a_0^2 = 2. \tag{49.141}$$

The leading-twist spin-two operator $[\phi\phi]_{0,2}$ at $\Delta = 2\Delta_\phi + 2$ has

$$p_{0,2} = 2\, a_0 a_2 = 2 \cdot \frac{(\Delta_\phi)_2^2}{2!\,(2\Delta_\phi + 1)_2} = \frac{\Delta_\phi^2 (\Delta_\phi + 1)^2}{(2\Delta_\phi + 1)(2\Delta_\phi + 2)} = \frac{\Delta_\phi^2 (\Delta_\phi + 1)}{2(2\Delta_\phi + 1)}. \tag{49.142}$$

The first subleading-twist scalar $[\phi\phi]_{1,0}$ at $\Delta = 2\Delta_\phi + 2$ (degenerate in dimension with $[\phi\phi]_{0,2}$ but distinguished by spin) has

$$p_{1,0} = 2\, a_1^2 = 2\Big(\frac{\Delta_\phi}{2}\Big)^2 = \frac{\Delta_\phi^2}{2}. \tag{49.143}$$

In the $d = 2$ free-scalar limit $\Delta_\phi = (d-2)/2 = 0$ the higher coefficients vanish: $(0)_m = 0$ for $m \geq 1$ forces $a_{m\geq 1} = 0$, so $p_{0,2} = p_{1,0} = 0$ while $p_{0,0} = 2a_0^2 = 2$ survives — a reflection of the degenerate, logarithmic nature of the $d = 2$ free boson (Section 48.7.1), in which ϕ itself is not a good primary. The clean illustration is therefore any $\Delta_\phi > 0$, e.g. $\Delta_\phi = \frac{1}{8}$ (the Ising spin, below), giving $p_{0,0} = 2$, $p_{0,2} = \frac{(1/64)(9/8)}{2(5/4)} = \frac{9/512}{5/2} = \frac{9}{1280}$, and $p_{1,0} = \frac{1}{128}$. These positive numbers are the exact mean-field structure constants, and they will reappear, shifted by anomalous dimensions, in any near-mean-field theory.

The general-d formula and its $d=2$ reduction. The same decomposition in general dimension — where the blocks no longer factorize but are still fixed by the Casimir equation Theorem 47.50 — yields the Fitzpatrick–Kaplan–Poland–Simmons-Duffin coefficients [194, 261]

$$p^{(d)}_{n,\ell} = \frac{\left[1+(-1)^\ell\right](\Delta_\phi - \frac{d}{2}+1)^2_n\,(\Delta_\phi)^2_{n+\ell}}{\ell!\,n!\,(\ell+\frac{d}{2})_n\,(2\Delta_\phi+n-d+1)_n\,(2\Delta_\phi+2n+\ell-1)_\ell\,(2\Delta_\phi+n+\ell-\frac{d}{2})_n}. \tag{49.144}$$

This is the closed-form solution of crossing for mean-field theory in any d. As a check that our two-dimensional derivation is the genuine special case, set $d=2$ in Eq. (49.144). The factors collapse: $(\Delta_\phi-\frac{d}{2}+1)^2_n=(\Delta_\phi)^2_n$, $(2\Delta_\phi+n-d+1)_n=(2\Delta_\phi+n-1)_n$, and $(\ell+\frac{d}{2})_n=(\ell+1)_n$, $(2\Delta_\phi+n+\ell-\frac{d}{2})_n=(2\Delta_\phi+n+\ell-1)_n$. Two Pochhammer identities finish the comparison. First, $\ell!\,(\ell+1)_n=(n+\ell)!$, so the two factorials in Eq. (49.144) combine into the single $(n+\ell)!$ of Eq. (49.140). Second, splitting the Pochhammer at the n-th step,

$$(2\Delta_\phi+n+\ell-1)_{n+\ell} = (2\Delta_\phi+n+\ell-1)_n\,(2\Delta_\phi+2n+\ell-1)_\ell, \tag{49.145}$$

which is exactly the product of the last two factors in the denominator of Eq. (49.144)$|_{d=2}$. Hence

$$p^{(2)}_{n,\ell} = \left[1+(-1)^\ell\right]\frac{(\Delta_\phi)^2_n\,(\Delta_\phi)^2_{n+\ell}}{n!\,(n+\ell)!\,(2\Delta_\phi+n-1)_n\,(2\Delta_\phi+n+\ell-1)_{n+\ell}} = \left[1+(-1)^\ell\right]a_n a_{n+\ell}, \tag{49.146}$$

which is Eq. (49.140) term for term. The two computations agree, validating both: the elementary $d=2$ derivation above and the general formula Eq. (49.144) that the rest of the bootstrap will use.

49.5.3 How the s-channel tower rebuilds the crossed-channel identity

We have a paradox to resolve, and resolving it is the conceptual payoff of the whole computation. In the crossed (t-)channel, the term $(u/v)^{\Delta_\phi}$ of Eq. (49.127) is nothing but the identity operator exchanged between the pairs $(\phi(x_1)\phi(x_4))$ and $(\phi(x_2)\phi(x_3))$ — a single operator, the most trivial one. Yet in the s-channel that same function is reproduced by the infinite double-twist tower Eq. (49.130), with not one operator but a whole Regge trajectory at each twist. How does a single t-channel identity become an infinite s-channel sum? The answer is the mechanism at the heart of the analytic bootstrap, and mean-field theory displays it in its purest form.

The lightcone projection. Work again in $d=2$ for concreteness, with the holomorphic factorization Eq. (49.133); the mechanism is identical in any d. Take the lightcone limit $z\to 0$ at fixed $\bar z$, which sends $u=z\bar z\to 0$ and

$v = (1-z)(1-\bar{z}) \to 1-\bar{z}$, isolating the leading twist $n = 0$. In this limit each leading-twist block reduces to its holomorphic seed $k_{2\Delta_\phi}(z) \to z^{\Delta_\phi}$ times the antiholomorphic collinear block, so the coefficient of z^{Δ_ϕ} in $g - 1$ is the single-variable function

$$H(\bar{z}) \equiv \bar{z}^{\Delta_\phi} + \Big(\frac{\bar{z}}{1-\bar{z}}\Big)^{\Delta_\phi} = \sum_{\ell \text{ even} \geq 0} p_{0,\ell}\, k_{2\Delta_\phi+2\ell}(\bar{z}), \tag{49.147}$$

the second equality being Eq. (49.134) summed with the leading-twist coefficients $p_{0,\ell} = 2a_\ell$ from Eq. (49.140). The first term $\bar{z}^{\Delta_\phi}$ of H is the s-channel image of the u^{Δ_ϕ} piece (the leading double-twist data themselves); the second term, $(\bar{z}/(1-\bar{z}))^{\Delta_\phi}$, is the t-channel identity restricted to the lightcone. As $\bar{z} \to 1$ — the limit in which the crossed pair collides, i.e. $v \to 0$ — this term diverges,

$$\Big(\frac{\bar{z}}{1-\bar{z}}\Big)^{\Delta_\phi} \xrightarrow{\bar{z}\to 1} (1-\bar{z})^{-\Delta_\phi} = v^{-\Delta_\phi} \to \infty. \tag{49.148}$$

No *finite* sum of s-channel blocks can produce this divergence: each collinear block $k_{2\Delta_\phi+2\ell}(\bar{z}) = {}_2F_1$-type is only logarithmically singular as $\bar{z} \to 1$, growing like $-\log(1-\bar{z})$, never like a power. The power-law singularity $v^{-\Delta_\phi}$ can come only from the *infinite* sum over spins, with the large-ℓ behaviour of $p_{0,\ell}$ controlling the strength of the divergence. This is the precise statement that the crossed identity forces an infinite, large-spin s-channel tower.

The large-spin asymptotics. The required tail is read off from the closed form. Using $a_\ell = (\Delta_\phi)_\ell^2/[\ell!\,(2\Delta_\phi+\ell-1)_\ell]$ and writing every Pochhammer as a ratio of Gamma functions,

$$p_{0,\ell} = 2a_\ell = \frac{2}{\Gamma(\Delta_\phi)^2}\,\frac{\Gamma(\ell+\Delta_\phi)^2\,\Gamma(\ell+2\Delta_\phi-1)}{\Gamma(\ell+1)\,\Gamma(2\ell+2\Delta_\phi-1)} \qquad (\ell \text{ even}). \tag{49.149}$$

For large ℓ apply $\Gamma(\ell+\alpha)/\Gamma(\ell+\beta) \sim \ell^{\alpha-\beta}$ to the slowly varying ratios and the Legendre duplication formula $\Gamma(2\ell+c) = \frac{2^{2\ell+c-1}}{\sqrt{\pi}}\Gamma(\ell+\frac{c}{2})\Gamma(\ell+\frac{c+1}{2})$ to the doubled argument. The factors of $\Gamma(\ell)$ cancel against $\Gamma(\ell+1) \sim \ell\,\Gamma(\ell)$ and the duplication denominator, leaving

$$\boxed{p_{0,\ell} \;\sim\; \frac{\sqrt{\pi}\,2^{3-2\Delta_\phi}}{\Gamma(\Delta_\phi)^2}\,\ell^{2\Delta_\phi-3/2}\,4^{-\ell} \qquad (\ell\to\infty,\ \ell \text{ even}).} \tag{49.150}$$

The coefficients decay exponentially in spin, $4^{-\ell}$, with a power-law prefactor $\ell^{2\Delta_\phi-3/2}$ set by the external dimension; panel (b) of Fig. 49.5 confirms the asymptotic against the exact values. The exponential $4^{-\ell}$ is the natural measure on spins (it is undone by the growth of the block near $\bar{z} = 1$); what carries physical information is the power $\ell^{2\Delta_\phi-3/2}$, because it is precisely this power

that, when the discrete sum Eq. (49.147) is converted to an integral over ℓ and evaluated near $\bar{z} \to 1$, reproduces the power-law $v^{-\Delta_\phi}$ of the crossed identity. The dictionary "t-channel singularity at $v \to 0$ $\longleftrightarrow$ large-spin asymptotics of s-channel OPE data" is the engine of the entire analytic bootstrap.

Physical Insight

The double-twist operators are not optional. The identity is in *every* OPE — it is the statement that operators have non-vanishing two-point functions — so the term $(u/v)^{\Delta_\phi}$ is present in any crossing-symmetric correlator of identical scalars, not just in mean-field theory. Its $v \to 0$ singularity must be matched in the s-channel, and we have just seen that only an infinite large-spin tower can do it. Therefore *every* unitary CFT with a scalar ϕ contains, at large spin, double-twist operators $[\phi\phi]_{n,\ell}$ with dimensions approaching $2\Delta_\phi + 2n + \ell$ and OPE coefficients approaching the mean-field values Eq. (49.144). Interactions can only correct these at large ℓ by small amounts — anomalous dimensions $\gamma_{n,\ell}$ that fall off as a power of ℓ, governed by the lightest operator in the crossed channel. Mean-field theory is thus not merely an example; it is the universal large-spin asymptotics of all CFTs, and Chapter 51 turns this observation into a calculus.

Figure 49.6 renders the reorganization schematically: one operator in the crossed channel, an entire trajectory in the direct channel.

49.5.4 The 2D Ising crossing check

The mean-field solution validated the machinery on a Gaussian theory. We now subject it to a genuinely interacting one whose answer we nonetheless know exactly: the two-dimensional Ising CFT, the $m = 3$ minimal model $\mathcal{M}(4,3)$ whose central charge $c = \frac{1}{2}$ and degenerate spectrum are fixed by the Feigin–Fuks construction (Theorem 48.40, Eq. (48.319)), solved in full in Chapter 48. Its spin four-point function was obtained there from the level-two BPZ equation, by methods — null vectors, hypergeometric ODEs, single-valuedness — with no overt connection to the crossing equation of this chapter. If both developments are correct, the Ising correlator must satisfy Theorem 49.13. We verify that it does, and in doing so recover the structure constant $C_{\sigma\sigma\varepsilon} = \frac{1}{2}$ from the bootstrap side.

Importing the data. From Chapter 48 we take, as established facts, the three Ising primaries (Theorem 48.33): the identity $\mathbf{1}$, the spin σ with $\Delta_\sigma = \frac{1}{8}$ (Eq. (48.234)), and the energy ε with $\Delta_\varepsilon = 1$; the fusion rule $\sigma \times \sigma = \mathbf{1} + \varepsilon$

(Eq. (48.240)); the algebraic chiral blocks (Eq. (48.262))

$$F_+(z) = \sqrt{\frac{1+\sqrt{1-z}}{2}}, \qquad F_-(z) = \sqrt{\frac{1-\sqrt{1-z}}{2}}, \tag{49.151}$$

the two independent solutions of the BPZ hypergeometric equation, with F_+ the identity (vacuum) Virasoro block and F_- the ε Virasoro block; and the single-valued physical correlator (Eq. (48.263), Theorem 48.34) assembled from them. Here z is the holomorphic cross-ratio of Eq. (47.140), identical to the variable called x in Theorem 48.34, with $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$.

The stripped correlator. We must express the Ising correlator in the bootstrap normalization of Eq. (47.145), $\langle\sigma\sigma\sigma\sigma\rangle = g(u,v)\,(x_{12}^2 x_{34}^2)^{-\Delta_\sigma}$, so that $g(u,v) \to 1$ as the first pair collides. The chiral construction of Chapter 48 gives the correlator as $|x(1-x)|^{-1/4}$ times the diagonal sum of chiral times antichiral blocks; converting to the bootstrap prefactor (a single power of the cross-ratio, since the two stripping conventions differ only by the relative Jacobian u^{Δ_σ}) yields the compact form

$$g(u,v) = v^{-\Delta_\sigma}\Big(\,|F_+(z)|^2 + |F_-(z)|^2\,\Big), \qquad |F_\pm(z)|^2 \equiv F_\pm(z)\,F_\pm(\bar{z}), \tag{49.152}$$

with $\Delta_\sigma = \frac{1}{8}$. The normalization is fixed by the collision limit: as $z, \bar{z} \to 0$ one has $v \to 1$, $F_+ \to 1$, $F_- \to 0$, so $g \to 1$, the identity contribution, as required. The first term of Eq. (49.152) is the identity Virasoro block (containing $\mathbf{1}$ together with its global descendants, among them the stress tensor); the second is the ε Virasoro block (containing ε and its descendants). Two Virasoro blocks suffice because the $c = \frac{1}{2}$ degenerate fusion Eq. (48.240) truncates $\sigma \times \sigma$ to exactly two families — the dramatic simplification that the global conformal blocks $G_{\Delta,\ell}$ of Eq. (49.131) can only achieve by an infinite resummation, since each Virasoro family decomposes into infinitely many global primaries (the identity family alone supplies $\mathbf{1}$, the stress tensor at $(\Delta,\ell) = (2,2)$, and so on up the tower).

Crossing, verified. The crossing equation Theorem 49.13 for identical scalars reads $v^{\Delta_\sigma} g(u,v) = u^{\Delta_\sigma} g(v,u)$. Define the symmetric function

$$P(z,\bar{z}) \equiv |F_+(z)|^2 + |F_-(z)|^2 = \tfrac{1}{2}\Big[\sqrt{(1+\sqrt{1-z})(1+\sqrt{1-\bar{z}})} + \sqrt{(1-\sqrt{1-z})(1-\sqrt{1-\bar{z}})}\Big], \tag{49.153}$$

so that $g(u,v) = v^{-\Delta_\sigma} P(z,\bar{z})$. The $s \leftrightarrow t$ swap $u \leftrightarrow v$ is the map $z \to 1-z$, $\bar{z} \to 1-\bar{z}$ (which exchanges $x_2 \leftrightarrow x_4$), under which $\sqrt{1-z} \to \sqrt{z}$ and $v \to u$. Therefore

$$v^{\Delta_\sigma} g(u,v) = P(z,\bar{z}), \qquad u^{\Delta_\sigma} g(v,u) = u^{\Delta_\sigma}\, u^{-\Delta_\sigma}\, P(1-z,1-\bar{z}) = P(1-z,1-\bar{z}), \tag{49.154}$$

so crossing is equivalent to the single algebraic identity

$$\boxed{P(z,\bar z) = P(1-z,1-\bar z).} \tag{49.155}$$

This is not obvious from Eq. (49.153): under $z \to 1-z$ the two radicals $\sqrt{1\pm\sqrt{1-z}}$ turn into $\sqrt{1\pm\sqrt{z}}$, manifestly different functions. That the symmetric combination is nonetheless invariant is exactly the single-valuedness condition $\alpha = \beta$ of Eq. (48.263) — the requirement that the holomorphic and antiholomorphic blocks glue into a correlator with no monodromy as two operators are carried around one another — now recognized as crossing symmetry. The diagonal locus makes the point cleanly: setting $\bar z = z = t$ real, $|F_+|^2 + |F_-|^2 = \frac{1+\sqrt{1-t}}{2} + \frac{1-\sqrt{1-t}}{2} = 1$, so $P(t,t) = 1 = P(1-t,1-t)$ identically, and Eq. (49.155) holds trivially on the diagonal. Off the diagonal it is a genuine constraint, which we now confirm.

Worked Example 49.26 (Crossing of the Ising spin correlator, off the diagonal)**.** Take the Lorentzian-type configuration in which z and $\bar z$ are independent reals, with $z = \frac12$, $\bar z = \frac{1}{10}$. Then $\sqrt{1-z} = \sqrt{0.5} = 0.707107$ and $\sqrt{1-\bar z} = \sqrt{0.9} = 0.948683$. The left side of Eq. (49.155) is

$$P(z,\bar z) = \tfrac12\Big[\sqrt{(1.707107)(1.948683)}+\sqrt{(0.292893)(0.051317)}\Big] = \tfrac12[1.823990+0.122600] = 0 \tag{49.156}$$

For the right side, $1-z = \frac12$ and $1-\bar z = \frac{9}{10}$ give $\sqrt{1-(1-z)} = \sqrt{z} = \sqrt{0.5} = 0.707107$ and $\sqrt{1-(1-\bar z)} = \sqrt{\bar z} = \sqrt{0.1} = 0.316228$, so

$$P(1-z,1-\bar z) = \tfrac12\Big[\sqrt{(1.707107)(1.316228)}+\sqrt{(0.292893)(0.683772)}\Big] = \tfrac12[1.499031+0.447559 \tag{49.157}$$

The two agree to all quoted digits, and the same agreement holds at every test point one cares to evaluate. The Ising spin four-point function satisfies the crossing equation of this chapter — an independent confirmation, by a route that never used the BPZ equation, that the $c = \frac12$ data of Chapter 48 and the bootstrap sum rule of Theorem 49.14 describe one and the same correlator.

Reading off the OPE coefficient. Crossing is satisfied; the bootstrap also recovers the dynamical data. Expand the ε term of Eq. (49.152) at small cross-ratio. As $z,\bar z \to 0$, $1-\sqrt{1-z} = \frac{z}{2} + O(z^2)$, so $|F_-(z)|^2 = \frac12\sqrt{(1-\sqrt{1-z})(1-\sqrt{1-\bar z})} \to \frac12\sqrt{\frac{z}{2}\cdot\frac{\bar z}{2}} = \frac14\sqrt{z\bar z} = \frac14 u^{1/2}$. Since $v^{-\Delta_\sigma} \to 1$, the energy operator contributes to g the leading term $\frac14 u^{1/2} = \frac14 u^{\Delta_\varepsilon/2}$, matching the global scalar block $G_{\Delta_\varepsilon,0} \to u^{\Delta_\varepsilon/2}$ for $\Delta_\varepsilon = 1$. The squared OPE coefficient is therefore

$$p_\varepsilon = C^2_{\sigma\sigma\varepsilon} = \tfrac14 \quad\Longrightarrow\quad C_{\sigma\sigma\varepsilon} = \tfrac12, \tag{49.158}$$

in exact agreement with Eq. (48.266). The identity term carries the rest: expanding $|F_+(z)|^2 = \frac{1}{2}\sqrt{(1+\sqrt{1-z})(1+\sqrt{1-\bar{z}})} = 1 - \frac{z+\bar{z}}{8} + \cdots$ and multiplying by $v^{-\Delta_\sigma} = [(1-z)(1-\bar{z})]^{-1/8} = 1+\frac{z+\bar{z}}{8}+\cdots$, the linear terms cancel and the identity channel begins as $1 + O((z+\bar{z})^2)$. The first correction is the global block of the stress tensor at $(\Delta, \ell) = (2, 2)$, whose coefficient is fixed by the conformal Ward identity. Extract it from the same lightcone limit used for ε: at $\bar{z} \to 0$ the twist-zero operators contribute $G_{\Delta,\ell} \to \bar{z}^0 k_{\Delta+\ell}(z)$, and the identity channel collapses to the holomorphic function $g_0(z) = \lim_{\bar{z}\to 0} g = (1-z)^{-1/8}\,\frac{1}{\sqrt{2}}\sqrt{1+\sqrt{1-z}}$ (using $F_+(\bar{z}) \to 1$, $F_-(\bar{z}) \to 0$). Expanding with $\sqrt{1-z} = 1 - \frac{z}{2} - \frac{1}{8}z^2 - \cdots$ gives $g_0(z) = 1 + 0 \cdot z + \frac{1}{64}z^2 + \cdots$: the constant is the identity ($k_0 = 1$), the vanishing z^1 confirms the absence of a spin-1 current, and the z^2 term is the leading piece of the stress-tensor block $k_4(z) \to z^2$, so

$$C^2_{\sigma\sigma T} = \tfrac{1}{64}. \tag{49.159}$$

The two-dimensional Ward identity fixes the stress-tensor coefficient in terms of the central charge as $C^2_{\sigma\sigma T} = 2h_\sigma^2/c$, and with the holomorphic weight $h_\sigma = \Delta_\sigma/2 = \frac{1}{16}$ and the imported $c = \frac{1}{2}$ of Theorem 48.40,

$$\frac{2h_\sigma^2}{c} = \frac{2\,(1/16)^2}{1/2} = \frac{1}{64}, \tag{49.160}$$

in exact agreement with Eq. (49.159). The crossing decomposition has therefore *measured* the central charge of the model and returned $c = \frac{1}{2}$, the value the free-Majorana realization of Section 48.7.3 assigns it. Thus the two Virasoro blocks of the exact solution, expanded into global blocks, reproduce the energy structure constant *and* the stress-tensor data — the complete leading content of the $\sigma \times \sigma$ OPE, recovered from crossing, with every coefficient positive as unitarity demands.

49.5.5 Saturation of the unitarity bounds

The two exact solutions also sharpen the meaning of the extremal solutions introduced in Section 49.4: they show what it looks like for a spectrum to sit *on* the unitarity bound rather than safely inside it.

Recall the unitarity bounds: a spin-ℓ primary in d dimensions obeys $\Delta \geq \ell + d - 2$ for $\ell \geq 1$, with equality precisely for a conserved current $\partial^\mu J_{\mu\ldots} = 0$, and $\Delta \geq \frac{d-2}{2}$ for a scalar, with equality precisely for a free field $\Box\phi = 0$ (Theorem 49.21 uses these as the input positivity facts). Now read the mean-field spectrum Eq. (49.130) against them. The leading-twist trajectory $n = 0$ sits at $\Delta_{0,\ell} = 2\Delta_\phi + \ell$, i.e. twist $\tau = 2\Delta_\phi$. For a generic generalized free field this twist exceeds the bound and every operator is a comfortable distance above it. But for

the genuine free scalar, $\Delta_\phi = \frac{d-2}{2}$, the leading twist becomes $\tau = 2\Delta_\phi = d-2$, so

$$\Delta_{0,\ell} = d - 2 + \ell = \ell + d - 2 \qquad \text{(free scalar)}, \tag{49.161}$$

which is exactly the unitarity bound for every spin ℓ. The free-field double-twist operators $[\phi\phi]_{0,\ell} = :\phi\,\partial_{\mu_1}\cdots\partial_{\mu_\ell}\phi: -$ (traces) are conserved higher-spin currents — the bilinear currents whose conservation is the hallmark of a free theory — and the leading-twist trajectory lies along the bound, as the dashed line in panel (a) of Fig. 49.5 shows. The scalar $[\phi\phi]_{0,0}$ likewise sits at $\Delta = d-2 = 2\cdot\frac{d-2}{2}$, twice the saturated scalar dimension. Generalized free theory with $\Delta_\phi > \frac{d-2}{2}$ lifts these operators off the bound by exactly the amount $\Delta_\phi - \frac{d-2}{2}$ in twist, the deficit that distinguishes a generalized free field from a true free field.

The Ising check supplies the interacting counterpart. In the $d = 2$ Ising CFT the scalar unitarity bound is $\Delta \geq \frac{d-2}{2} = 0$, saturated only by the identity; the spin σ ($\Delta = \frac{1}{8}$) and energy ε ($\Delta = 1$) sit strictly above it, as a unitary interacting theory must. The single operator that saturates a spin bound is the stress tensor, at $\Delta = 2 = \ell + d - 2$ with $\ell = 2$: it is the conserved current of translations, present in every local CFT, and its appearance at the bound in the identity Virasoro block of Eq. (49.152) is the statement that the Ising CFT is local. These are the configurations a linear functional must respect: an extremal functional of Section 49.4 is one that vanishes on precisely the operators a candidate theory places on or near the bound, and the exact solutions of this section are the simplest spectra on which to see that vanishing realized.

With mean-field theory solved in closed form — its double-twist spectrum Eq. (49.130), its OPE coefficients Eq. (49.144), and the large-spin reorganization Eq. (49.150) that rebuilds the crossed identity — and with the 2D Ising correlator confirmed to satisfy crossing with its known $c = \frac{1}{2}$ data, the bootstrap apparatus stands validated against every case whose answer we can write down independently. The double-twist tower and its large-spin asymptotics are handed to Chapter 51 as the explicit seed of the analytic bootstrap, and the now-validated crossing solver is ready to be pointed, in Section 49.6 and Chapter 50, at a theory we cannot solve by any other means: the three-dimensional Ising CFT.

49.6 The 3D Ising CFT as the Bootstrap Target, and the Frontier

Every piece of machinery assembled in this chapter has now been checked against a theory whose answer we already knew. The crossing equation of Theorem 49.13 was solved exactly for the generalized free (mean-field) theory, whose double-twist spectrum and OPE coefficients fell out of the bootstrap with no further input (Section 49.5.1); the same equation was verified, term by conformal block, on the two-dimensional Ising correlators whose spectrum we had independently

from the free Majorana realization with central charge $c = \frac{1}{2}$ (Eq. (48.319)); and the functional theorem of Theorem 49.21 turned crossing plus unitarity into rigorous bounds. We have a validated crossing solver, the double-twist intuition for how operators must populate the cross channel, and a positivity certificate that excludes spectra. The natural question is what this apparatus does when we point it at a theory we *cannot* solve exactly. The answer is the modern conformal bootstrap, and its headline laboratory is the three-dimensional Ising universality class. This section poses that problem precisely: what is being assumed, what the bootstrap delivers, and — stated honestly and dated to the mid-2020s — what is proved versus what is numerical. The explicit numerical solution is handed to Chapter 50, the analytic limits to Chapter 51.

Notation Watch

Throughout this section the symbol ε denotes the lightest $\mathbb{Z}_2$-*even* scalar primary — the *energy operator* of the Ising universality class, continuing the usage of the two-dimensional Ising CFT (Theorem 48.33), where $\varepsilon = i\bar{\psi}\psi$ had $\Delta_\varepsilon = 1$. It is *not* the dimension deficit $\varepsilon = 4 - d$ of the ε-expansion (Theorem 24.31), which appears only with that meaning and never carries an operator subscript. The order parameter (the $\mathbb{Z}_2$-odd scalar, the continuum spin field) is σ.

49.6.1 The 3D Ising critical point as a unitary CFT

The bridge that ϕ^4 theory has carried from Part VI to here is the statement that the critical point of the three-dimensional Ising model *is* a conformal field theory. We recall its origin and then state precisely the CFT we intend to corner.

The Wilsonian and Callan–Symanzik analyses located, in $d = 4-\varepsilon$ dimensions, the nontrivial zero of the ϕ^4 beta function at $\lambda_* = \frac{16\pi^2}{3}\varepsilon + O(\varepsilon^2)$, the Wilson–Fisher fixed point (Theorem 24.31). Section 47.8.4 showed that this fixed point has a traceless, conserved stress tensor and is therefore a genuine conformal field theory, with the elementary field ϕ realized as a primary. Continuing the dimension deficit to its physical value $\varepsilon = 1$, i.e. $d = 3$, gives the conformal field theory governing the long-distance physics of any system in the three-dimensional Ising universality class: the uniaxial ferromagnet at its Curie point, the liquid–vapour critical point, the binary-fluid demixing transition, all sharing the same critical exponents. Universality is precisely the statement that all of these microscopically distinct systems flow to the *same* infrared fixed point, the same CFT. We name its two lightest scalar primaries after the lattice observables they descend from: the $\mathbb{Z}_2$-odd *spin* operator σ (the continuum order parameter, ϕ of the thread) and the $\mathbb{Z}_2$-even *energy* operator ε (the continuum ϕ^2, the operator conjugate to the reduced temperature).

Unlike its two-dimensional cousin, which the free-fermion construction of

Eq. (48.319) solves *exactly*, the three-dimensional Ising CFT has no known exact solution. There is no Lagrangian description that is weakly coupled at $d = 3$: the ε-expansion is an asymptotic series evaluated at the non-small value $\varepsilon = 1$, and the lattice and Monte Carlo determinations, while precise, are themselves numerical. What makes the bootstrap decisive is that it does not need a Lagrangian at all. It treats the theory as an abstract solution of the consistency conditions and asks which solutions are possible.

Definition 49.27 (The 3D Ising bootstrap problem: assumptions)**.** By the *3D Ising CFT* we mean a conformal field theory in $d = 3$ satisfying the following hypotheses, each of which is an input to the bootstrap and not a consequence of it:

(A1) *Conformal symmetry.* The theory is invariant under the conformal group $\mathrm{SO}(4,1)$ of Chapter 47; correlators decompose in conformal blocks and the OPE converges in the sense established there.

(A2) *Unitarity (reflection positivity).* The spectrum of primary dimensions is real and bounded below by the unitarity bounds $\Delta \geq \frac{d-2}{2} = \frac{1}{2}$ for scalars and $\Delta \geq \ell + d - 2 = \ell + 1$ for spin $\ell \geq 1$; all OPE coefficients of real operators are real, so squared coefficients are non-negative.

(A3) $\mathbb{Z}_2$ *global symmetry.* Every primary carries a definite $\mathbb{Z}_2$ charge; σ is odd, ε is even.

(A4) *Locality.* There is a conserved, symmetric, traceless stress tensor $T_{\mu\nu}$ — a spin-2 primary at the protected dimension $\Delta_T = d = 3$ — and a unique normalizable vacuum.

(A5) *A gap.* The theory has exactly *two* relevant scalars, one of each $\mathbb{Z}_2$ charge: σ with $\Delta_\sigma < 3$ and ε with $\Delta_\varepsilon < 3$, and these are the lightest scalars of their respective charges. Equivalently, the next $\mathbb{Z}_2$-even scalar ε' and the next $\mathbb{Z}_2$-odd scalar σ' are both irrelevant, $\Delta_{\varepsilon'}, \Delta_{\sigma'} > 3$.

The data to be determined are then the conformal data of Chapter 47: the spectrum $\{\Delta_i, \ell_i\}$ of primaries and the OPE coefficients $\{\lambda_{ijk}\}$. The two numbers of headline interest are Δ_σ and Δ_ε.

Assumptions (A1)–(A4) are the defining properties of any unitary local CFT; the physical content specific to the Ising class lives in (A3) and (A5). The gap assumption (A5) is the crucial physical input — it is what makes the bootstrap select *the Ising* fixed point and not, say, the tricritical or $O(N)$ fixed points, which differ in their relevant-operator content. It encodes a fact about the renormalization group: the Ising critical surface has exactly two relevant directions, the temperature and the magnetic field, conjugate to ε and σ. Tuning

two parameters reaches the critical point, and no more tuning is needed; that is the statement that there are exactly two relevant scalars.

Before posing the crossing problem we record the dictionary back to Part VI, the bridge whose closure is completed in Chapter 52. The critical exponents are determined by the two dimensions through

$$\eta = 2\Delta_\sigma - (d-2) = 2\Delta_\sigma - 1, \tag{49.162}$$

$$\frac{1}{\nu} = d - \Delta_\varepsilon = 3 - \Delta_\varepsilon, \tag{49.163}$$

the first because the spin two-point function $\langle\sigma\sigma\rangle \sim |x|^{-2\Delta_\sigma}$ defines the anomalous dimension η through $\Delta_\sigma = \frac{1}{2}(d-2+\eta)$, the second because ε is the operator conjugate to the reduced temperature so its relevant RG eigenvalue is $1/\nu$ (these are exactly Eqs. (47.263) and (47.264) read backwards). Thus pinning Δ_σ and Δ_ε pins η and ν, and through scaling relations every other Ising exponent. The bootstrap's precision on $\Delta_\sigma, \Delta_\varepsilon$ is, through Eqs. (49.162) and (49.163), precision on measurable critical exponents.

49.6.2 The single-correlator crossing equation and the $\mathbb{Z}_2$ selection rule

The simplest crossing equation available is the one for four spin operators, $\langle\sigma(x_1)\sigma(x_2)\sigma(x_3)\sigma(x_4)\rangle$. We assemble it from the general crossing equation Theorem 49.13, specialized to four identical scalars of dimension Δ_σ and equipped with the $\mathbb{Z}_2$ selection rule.

Recall the kinematics. Conformal invariance reduces the four-point function to a function of the two cross-ratios

$$u = \frac{x_{12}^2 x_{34}^2}{x_{13}^2 x_{24}^2}, \qquad v = \frac{x_{14}^2 x_{23}^2}{x_{13}^2 x_{24}^2}, \qquad x_{ij} \equiv x_i - x_j, \tag{49.164}$$

through

$$\langle\sigma\sigma\sigma\sigma\rangle = \frac{1}{x_{12}^{2\Delta_\sigma}\, x_{34}^{2\Delta_\sigma}}\, \mathcal{G}(u,v), \qquad \mathcal{G}(u,v) = \sum_{\mathcal{O}\in\sigma\times\sigma} \lambda_{\mathcal{O}}^2\, g_{\Delta_\mathcal{O},\ell_\mathcal{O}}(u,v), \tag{49.165}$$

the sum running over the primaries appearing in the $\sigma \times \sigma$ OPE, with $g_{\Delta,\ell}$ the conformal blocks defined and computed in this chapter and $\lambda_\mathcal{O}^2 \geq 0$ by unitarity (A2). Two selection rules cut the sum down sharply:

- $\mathbb{Z}_2$ *parity.* Since σ is $\mathbb{Z}_2$-odd, the product $\sigma \times \sigma$ is $\mathbb{Z}_2$-even: only $\mathbb{Z}_2$-*even* operators appear. The odd operators $\sigma, \sigma', \ldots$ are forbidden from this OPE.
- *Spin parity.* Because the two operators are identical, exchanging $x_1 \leftrightarrow x_2$ is a symmetry of the correlator; under it a spin-ℓ block picks up $(-1)^\ell$, so only *even* spin $\ell = 0, 2, 4, \ldots$ survives.

The exchanged spectrum is therefore: the identity ($\Delta = 0$, $\ell = 0$, $\lambda^2 = 1$); the $\mathbb{Z}_2$-even scalars $\varepsilon, \varepsilon', \ldots$; the stress tensor T ($\ell = 2$, $\Delta = 3$) and the even-spin tower above it.

Crossing symmetry is the statement that the same correlator may be expanded in the (12)(34) channel or the (14)(23) channel and must agree. Exchanging $x_1 \leftrightarrow x_3$ sends $u \leftrightarrow v$ and, from the prefactor, multiplies by $(u/v)^{\Delta_\sigma}$, giving $\mathcal{G}(u,v) = (u/v)^{\Delta_\sigma}\mathcal{G}(v,u)$, i.e.

$$v^{\Delta_\sigma}\,\mathcal{G}(u,v) = u^{\Delta_\sigma}\,\mathcal{G}(v,u). \tag{49.166}$$

Inserting the block expansion and moving everything to one side, the identity contribution $g_{0,0} = 1$ separates out and we obtain the *Ising crossing sum rule*,

$$\boxed{\sum_{\mathcal{O}\in\sigma\times\sigma}^{\mathcal{O}\neq\mathbf{1}} \lambda_{\mathcal{O}}^2\, F^-_{\Delta_\sigma;\,\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) \;=\; u^{\Delta_\sigma} - v^{\Delta_\sigma}} \tag{49.167}$$

where the *convolved block* (the antisymmetrized crossing vector) is

$$F^-_{\Delta_\sigma;\,\Delta,\ell}(u,v) \equiv v^{\Delta_\sigma}\,g_{\Delta,\ell}(u,v) - u^{\Delta_\sigma}\,g_{\Delta,\ell}(v,u). \tag{49.168}$$

The right-hand side of Eq. (49.167) is exactly $-F^-_{\Delta_\sigma;0,0}$, the identity's own crossing vector; the sum rule says that the contribution of every nontrivial operator must conspire to reproduce minus the identity. This is the single equation — one functional identity in the two variables u, v, equivalently an infinite set of numerical equations once expanded around a point — that the entire Ising spectrum and its OPE coefficients must satisfy. The bootstrap is the study of which $(\Delta_\sigma; \{\Delta_{\mathcal{O}}, \lambda^2_{\mathcal{O}}\})$ solve it.

49.6.3 The gap assumption and how a functional carves out a region

We now show, in full, how the gap assumption (A5) fed through the functional theorem of Theorem 49.21 converts Eq. (49.167) into a bound: a curve in the $(\Delta_\sigma, \Delta_\varepsilon)$ plane above which no consistent theory can lie. The argument is the positivity argument of Theorem 49.21 specialized to the Ising sum rule, and it is worth seeing the contradiction spelled out because it is the logical engine of the whole numerical program.

Proposition 49.28 (Single-correlator gap bound)**.** *Fix $\Delta_\sigma \geq \frac{d-2}{2}$ and a trial gap $\Delta_* > 0$. Consider the admissible spectrum dictated by the gap assumption and unitarity,*

$$\mathcal{S}_{\Delta_*} = \{(\Delta, \ell = 0) : \Delta \geq \Delta_*\} \cup \{(\Delta, \ell) : \ell \text{ even},\ \ell \geq 2,\ \Delta \geq \ell + d - 2\}. \tag{49.169}$$

It is convenient to write the sum rule with the identity included in the sum, $\sum_{\mathcal{O}} \lambda_{\mathcal{O}}^2 F^-_{\Delta_\sigma;\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = 0$ *with* $\lambda_{\mathbf{1}}^2 = 1$ *and* $F^-_{\Delta_\sigma;0,0} = v^{\Delta_\sigma} - u^{\Delta_\sigma}$ *(this is Eq. (49.167) with the identity term carried back to the left). Suppose there exists a linear functional* α *(of the type whose existence and convex structure are guaranteed by Theorem 49.21) acting on functions of* (u, v) *such that*

(i) $\alpha\big[F^-_{\Delta_\sigma;0,0}\big] = 1$ *(the normalization on the identity), and*

(ii) $\alpha\big[F^-_{\Delta_\sigma;\Delta,\ell}\big] \geq 0$ *for every* $(\Delta, \ell) \in \mathcal{S}_{\Delta_*}$.

Then no unitary, crossing-symmetric solution of Eq. (49.167) exists whose lightest $\mathbb{Z}_2$*-even scalar has dimension* $\Delta_\varepsilon \geq \Delta_*$*. Equivalently, in any consistent theory the energy operator obeys* $\Delta_\varepsilon < \Delta_*$*. The infimum of* Δ_* *over all functionals satisfying (i)–(ii) is a rigorous upper bound* $\Delta_\varepsilon \leq \Delta_\varepsilon^{\max}(\Delta_\sigma)$.

Proof. Suppose, for contradiction, that a consistent unitary theory exists with $\Delta_\varepsilon \geq \Delta_*$. Then every nontrivial operator in $\sigma \times \sigma$ lies in $\mathcal{S}_{\Delta_*}$: each $\mathbb{Z}_2$-even scalar has dimension $\geq \Delta_\varepsilon \geq \Delta_*$ (the lightest is ε), and each spinning operator has even spin $\ell \geq 2$ with $\Delta \geq \ell + d - 2$ by the selection rule and unitarity. Apply the linear functional α to the sum rule and split off the identity by linearity,

$$0 = \alpha\Big[\sum_{\mathcal{O}} \lambda_{\mathcal{O}}^2 F^-_{\Delta_\sigma;\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\Big] = \underbrace{\lambda_{\mathbf{1}}^2\, \alpha\big[F^-_{\Delta_\sigma;0,0}\big]}_{=1\cdot 1 = 1} + \underbrace{\sum_{\mathcal{O}\neq\mathbf{1}} \lambda_{\mathcal{O}}^2\, \alpha\big[F^-_{\Delta_\sigma;\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\big]}_{\geq 0}. \tag{49.170}$$

The identity term equals 1 by the normalization (i). Every term in the remaining sum is non-negative: $\lambda_{\mathcal{O}}^2 \geq 0$ by unitarity (A2), and $\alpha[F^-_{\Delta_\sigma;\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}] \geq 0$ by (ii) because each such $\mathcal{O} \in \mathcal{S}_{\Delta_*}$. The right-hand side is therefore $\geq 1 > 0$, contradicting that it equals 0. Hence no spectrum with $\Delta_\varepsilon \geq \Delta_*$ can solve crossing, so $\Delta_\varepsilon < \Delta_*$ in any consistent theory. Minimizing Δ_* over all functionals satisfying (i)–(ii) gives the stated upper bound. □

Caution

The whole argument turns on a sign: the certifying functional must be *non-negative on the entire nontrivial spectrum* yet *strictly positive on the identity*, so that a sum constrained to vanish is exhibited as strictly positive. Geometrically, the identity's crossing vector must lie *outside* the convex cone generated by the allowed operators' vectors, and α is the separating hyperplane. Flipping the overall sign of α gives the same certificate; what is fatal is the wrong *relative* sign between the identity and the rest. This sign bookkeeping is the single most common slip in setting up the bound.

The content of Theorem 49.28 is a *search*: for a given Δ_σ, scan the trial gap Δ_* downward; the smallest Δ_* for which a certifying functional α still exists is the boundary value $\Delta_\varepsilon^{\max}(\Delta_\sigma)$. Below it crossing solutions may exist; on and above it none do. As Δ_σ varies one traces a curve, and the existence of the certifying functional — guaranteed to be decidable by the convex (semidefinite) structure of Theorem 49.21 — is what makes the curve computable. The remarkable empirical fact, displayed in Fig. 49.7, is that this upper-bound curve is not smooth: it develops a sharp *kink*, and the 3D Ising CFT sits essentially *at* the kink. A theory that saturates the bound is extremal — its spectrum is the unique one annihilated by the optimal functional — and the kink marks the place where this extremal spectrum reorganizes, the signature of a distinguished, isolated theory rather than a generic point on a continuum.

Physical Insight

Why a kink, and why does it locate a physical theory? Along the bound the saturating "solution" is forced to be extremal: the optimal functional has just enough zeros to pin down a discrete spectrum. At a generic Δ_σ this extremal spectrum varies smoothly and corresponds to no particularly natural theory. At one special Δ_σ the structure of the extremal spectrum jumps — an operator crosses a threshold — and the bound curve kinks. That jump is the bootstrap detecting that the most-constrained theory consistent with the assumptions has changed character. For the $\langle\sigma\sigma\sigma\sigma\rangle$ system the kink coincides, to all tested precision, with the place where the spectrum is exactly that of the 3D Ising model. The bootstrap is, in effect, finding the corner of the allowed region where unitarity and crossing are simultaneously most strained, and that corner is a real CFT.

Two remarks complete the geometry of the allowed region. First, the single-correlator bound is intrinsically *one-sided*: Theorem 49.28 bounds Δ_ε from *above* for each Δ_σ, and unitarity (A2) bounds Δ_σ from below by the vertical wall $\Delta_\sigma \geq \frac{1}{2}$. Together with the requirement that a $\mathbb{Z}_2$-even scalar exist at all (else the theory is generalized free, the boundary case of Section 49.5.1), these carve out the shaded allowed region of the left panel of Fig. 49.7, with the Ising theory pinned to the kink on its upper boundary. Second, to turn this open region into a genuinely *closed* island — bounded on all sides, isolating $(\Delta_\sigma, \Delta_\varepsilon)$ in a small box — one needs more equations than the single correlator provides. That is the role of the mixed-correlator system and the second gap assumption, described next and computed in Chapter 50.

49.6.4 The headline outcome: the island

The decisive sharpening comes from bootstrapping not one correlator but the full system of correlators built from *both* external operators, $\langle\sigma\sigma\sigma\sigma\rangle$, $\langle\sigma\sigma\varepsilon\varepsilon\rangle$, and

$\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$. These share OPE data — the coefficient $\lambda_{\sigma\sigma\varepsilon}$ appears in the first two, $\lambda_{\varepsilon\varepsilon\varepsilon}$ in the last two — so crossing for the system is a set of coupled sum rules in which the same λ^2's are constrained simultaneously. The positivity argument generalizes: instead of a single functional acting on a scalar sum rule, one seeks a *vector* of functionals acting on the vector of sum rules with a semidefinite positivity condition on 2×2 matrices of OPE coefficients. Imposing the second gap (A5) — that σ', the next $\mathbb{Z}_2$-odd scalar, is also irrelevant — closes the region from all sides. The open wedge becomes a closed *island*.

The island is the target this section hands forward. The current (mid-2020s) determinations, which Chapter 50 will reproduce from the semidefinite program, place

$$\boxed{\Delta_\sigma = 0.5181489(10), \qquad \Delta_\varepsilon = 1.412625(10)} \tag{49.171}$$

together with the central charge $c_T/c_T^{\text{free}} = 0.946534(11)$ and the OPE coefficients $\lambda_{\sigma\sigma\varepsilon} = 1.0518537(41)$, $\lambda_{\varepsilon\varepsilon\varepsilon} = 1.532435(19)$ — the precision improving each time the truncation order of the functional search is raised. Through the dictionary Eqs. (49.162) and (49.163) these give $\eta = 2(0.5181489) - 1 = 0.0362978(20)$ and $1/\nu = 3 - 1.412625 = 1.587375(10)$, i.e. $\nu = 0.629971(4)$, agreeing with the best Monte Carlo and experimental critical exponents to all quoted digits. The full machinery that produces Eq. (49.171) — the choice of derivative basis for the functionals, the reduction to semidefinite programming, the rigorous handling of the infinite spin sums — is developed in Chapter 50; here it is stated as the number the program is built to deliver.

49.6.5 Consistency with the ε-expansion

It would be a hollow victory if the bootstrap island floated free of everything ϕ^4 theory built. It does not: the island sits exactly where the perturbative renormalization group says the Wilson–Fisher fixed point should be, to the accuracy the perturbative series permits. We make the cross-check explicit.

Worked Example 49.29 (The bootstrap island against the ε-expansion)**.** The Wilson–Fisher dimensions assembled in Theorem 47.55 from the Part VI data are, with $\varepsilon = 4 - d$,

$$\Delta_\sigma = \frac{d-2}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3) = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3), \qquad \text{(from Eq. (47.263))}$$

$$\Delta_\varepsilon = d - \frac{1}{\nu} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2). \qquad \text{(from Eq. (47.264))}$$

The physical theory is at $\varepsilon = 1$ ($d = 3$). The series are asymptotic and $\varepsilon = 1$ is not small, so we evaluate the truncations and watch them *point toward* the island rather than land in it.

Order by order for Δ_σ. At leading order $\Delta_\sigma = 1 - \frac{1}{2} = 0.5000$ (the free value, the unitarity floor). The two-loop correction adds $\varepsilon^2/108 = 1/108 = 0.00926$,

giving $\Delta_\sigma \approx 0.5093$. The island value Eq. (49.171) is 0.51815. The leading anomalous shift has the right sign and roughly the right magnitude: the field is lifted off the unitarity bound by a small amount, $\eta = 0.0363$ from the bootstrap versus the truncated $2(\Delta_\sigma - \frac{1}{2}) = 2(0.00926) = 0.0185$ here. The truncated η is about half the true value — exactly the behaviour expected of the first term of an asymptotic series at $\varepsilon = 1$, and known higher orders ($\eta = \frac{\varepsilon^2}{54} + \frac{109}{5832}\varepsilon^3 + \cdots$, then Borel–Padé resummed) close the gap and converge on 0.0363.

Order by order for Δ_ε. The free value is $\Delta_\varepsilon = d - 2 = 2 - \varepsilon = 1$ at $\varepsilon = 1$, the value of the energy operator in the *free* theory (and, as it happens, the exact 2D value of Eq. (48.319), a coincidence of d). The first interacting correction is large and of first order: $\Delta_\varepsilon = 2 - \frac{2\varepsilon}{3} = 2 - 0.6667 = 1.3333$ at $\varepsilon = 1$. The island value is 1.41263. The leading shift accounts for the bulk of the displacement from the free value (0.333 of the true 0.413), with the remainder supplied by higher orders — here the perturbative series is doing most of the work already at one loop, consistent with the Part VI observation that $\gamma_{\phi^2} = \varepsilon/3$ is a first-order effect while γ_ϕ is second order.

The verdict. The truncated ε-expansion gives $(\Delta_\sigma, \Delta_\varepsilon) \approx (0.509, 1.333)$ against the bootstrap island $(0.518, 1.413)$: the same point to within the error one expects from truncating an asymptotic series at $\varepsilon = 1$, with both dimensions displaced from their free values in the right direction and the resummed series converging onto the island. The two methods agree at the level the expansion permits. This is the quantitative meaning of the claim, advanced since Section 47.8.4, that the bootstrap target *is* the Wilson–Fisher fixed point: the nonperturbative crossing solution and the perturbative RG fixed point are the same CFT, computed two incompatible ways and found consistent.

49.6.6 The frontier, honestly

It remains to say, precisely and with its date, what the bootstrap has and has not established. The temptation to overstate is real, because the numbers in Eq. (49.171) are more accurate than any other determination of the Ising exponents; but precision is not proof, and the distinction is exactly the kind this book is committed to keeping.

What is a theorem. The exclusion is rigorous. For any choice of external dimension and gaps, a certifying functional α of Theorem 49.28 (or its mixed-correlator matrix generalization) is an explicit object whose two properties — positivity on the admissible spectrum and definite sign on the identity — can be verified directly. When such a functional is exhibited, the corresponding region is *excluded with certainty*: no unitary, $\mathbb{Z}_2$-symmetric, crossing-symmetric CFT with the assumed gaps can have data in it. The bounds are honest mathematical theorems, conditional only on the hypotheses (A1)–(A5) and on the arithmetic being correct (and the latter can be, and has been, checked in rigorous interval arithmetic).

What is numerical. The *localization* to the small island is not a theorem of the same kind. It rests on (i) physical assumptions — chiefly the gap assumption (A5), the input that there are exactly two relevant scalars, which is taken from the renormalization-group understanding of the Ising class, not derived within the bootstrap; and (ii) numerics — the functional search is carried out in a finite-dimensional space of derivatives and at finite precision, so each computed island is the projection of the true allowed region onto a finite truncation. The island is *nonempty* in every computation performed and *shrinks* monotonically as the truncation is enlarged, which is overwhelming evidence that it converges to a point.

What is not proved (as of the mid-2020s). There is, at the time of writing, no proof that a CFT exists at the island — no construction of the 3D Ising CFT as a mathematical object satisfying all the axioms simultaneously — and no proof that, if one exists, it is unique. The bootstrap establishes that *if* a unitary CFT with the Ising gaps exists, its data lie in the island, and it pins those data to extraordinary numerical precision; it does not, by itself, establish existence. (Contrast the two-dimensional case, where the free-fermion construction behind Eq. (48.319) *is* an existence proof.) Whether the shrinking islands provably converge to a nonempty limit, and whether that limit is a single point, are open mathematical questions. The honest summary is: the 3D Ising CFT is cornered, measured to six significant figures, and consistent with every independent method — and not yet proved to exist. That gap between "numerically certain" and "proved" is itself one of the live problems of the field.

Physical Insight

The right way to hold this result is that the bootstrap has changed the epistemic status of the 3D Ising exponents. Before it, the best values came from resumming a divergent perturbative series or from extrapolating finite-size lattice data — methods with hard-to-quantify systematics. The bootstrap replaces both with a single convex optimization whose *exclusions* are rigorous and whose *island* shrinks under a controlled, monotone refinement. The remaining gap is no longer "is the number right?" but the sharper mathematical question "does the limiting theory exist and is it unique?" — a far better place to be stuck.

We have now posed the 3D Ising bootstrap completely: the theory as a unitary $\mathbb{Z}_2$-symmetric CFT with two relevant scalars (Theorem 49.27), the crossing sum rule it must satisfy (Eq. (49.167)), the gap-and-functional logic that bounds its data (Theorem 49.28), and the island it is cornered into (Eq. (49.171)), cross-checked against the Wilson–Fisher fixed point of ϕ^4 theory. The explicit numerical realization — the semidefinite program, the derivative functionals, and the convergence of the island — is handed to Chapter 50; the analytic limits that

organize the spinning spectrum and reproduce the bound's structure from the lightcone and the Lorentzian inversion formula are handed to Chapter 51; and the closure of the loop with Part VI, turning Eq. (49.171) into the full table of critical exponents through Eqs. (49.162) and (49.163), is handed to Chapter 52.

Further reading

The idea that conformal symmetry, the OPE, and crossing might determine a theory without a Lagrangian originates with Ferrara et al. [190] and Polyakov [407]; the modern numerical incarnation, recasting crossing as a convex-optimization bound, begins with Rattazzi et al. [415]. The conformal blocks constructed here in closed form are due to Dolan and Osborn [155, 156]. The most useful entry points to the field as a whole are the review of Poland et al. [402] and the lecture notes of Rychkov [421] and Simmons-Duffin [441], all of which carry the linear-functional method, the radial expansion, and the semidefinite formulation in more computational detail than space allows here. The 3D Ising application that this chapter poses — and whose numerics Chapter 50 develops — is the subject of El-Showk et al. [172, 173], with the mixed-correlator islands that sharpen it given by Kos et al. [303, 304].

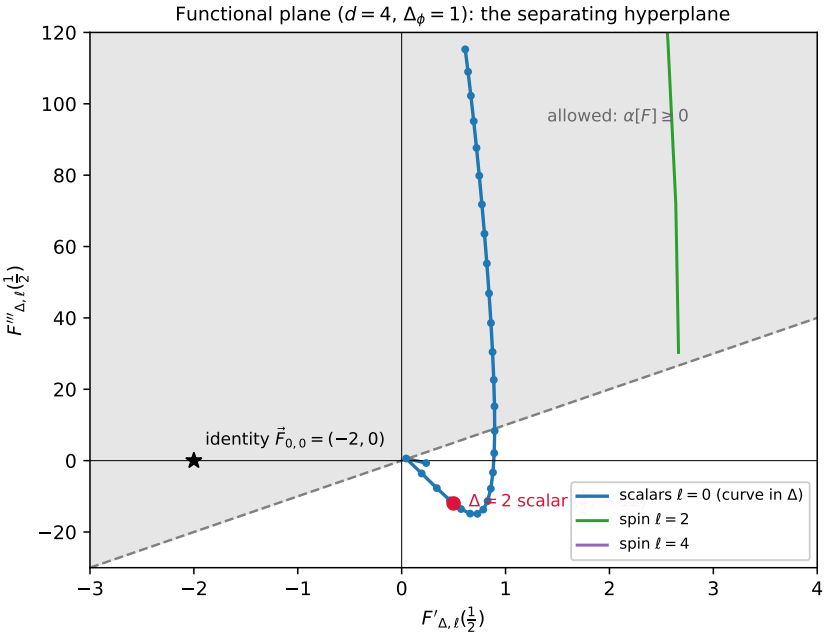


Figure 49.4: The separating-hyperplane picture for the two-derivative functional (Theorem 49.22; $d = 4$, $\Delta_\phi = 1$). Each conformal block is the point $(\partial_t F_{\Delta,\ell},\, \partial_t^3 F_{\Delta,\ell})$ at $z = \bar{z} = \frac{1}{2}$; the blue curve is the scalars (parametrized by Δ), green and purple the spin-2 and spin-4 families. The identity sits at the fixed point $(-2, 0)$ and the exactly-computed $\Delta = 2$ scalar at $(\frac{1}{2}, -12)$. The dashed line $\partial_t^3 F = 10\,\partial_t F$ is the boundary $\alpha_{0,1} = 0$ of the certificate functional of Step 4; the shaded half-plane is $\alpha_{0,1} \geq 0$. The identity and the entire gap-4 spectrum lie in the shaded region while the low scalars $2 < \Delta < 3.6$ dip below the line — which is exactly why a scalar gap above $\Delta_* \approx 3.61$ can be separated and excluded, and a smaller one cannot. Rotating the line toward the horizontal ($B \to \infty$) gives the optimal bound Eq. (49.116).

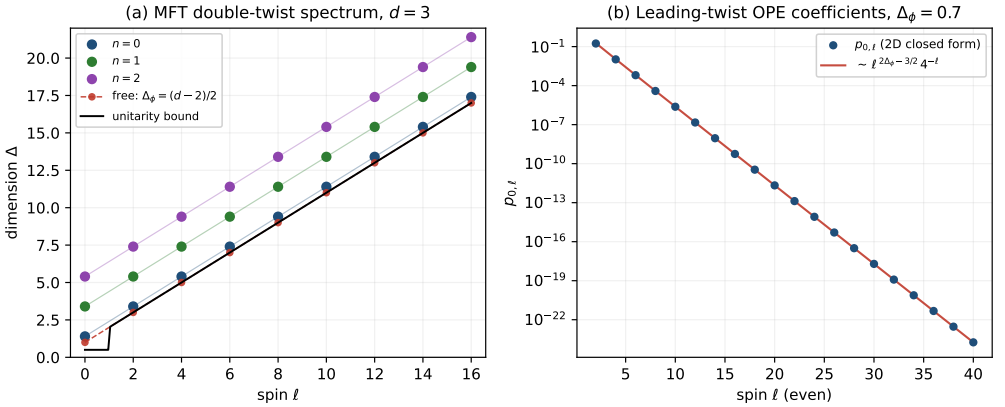


Figure 49.5: The exact solution of crossing for the generalized free field. *(a)* The double-twist spectrum Eq. (49.130): primaries $[\phi\phi]_{n,\ell}$ at $\Delta = 2\Delta_\phi + 2n + \ell$ for even spin ℓ, organized into twist trajectories $\tau_n = 2\Delta_\phi + 2n$ ($d = 3$, $\Delta_\phi = 0.7$). The solid line is the spin-ℓ unitarity bound $\Delta \geq \ell + d - 2$ (and $\Delta \geq \frac{d-2}{2}$ for the scalar); for the genuine free scalar $\Delta_\phi = (d-2)/2$ (dashed red) the leading-twist trajectory sits *on* the bound, signalling conserved higher-spin currents (Section 49.5.5). *(b)* The leading-twist OPE coefficients $p_{0,\ell}$ from the closed form Eq. (49.140) (here in $d = 2$, $\Delta_\phi = 0.7$), falling off as $\ell^{2\Delta_\phi - 3/2} 4^{-\ell}$ Eq. (49.150); the infinite tail of this tower is what rebuilds the crossed-channel identity (Section 49.5.3).

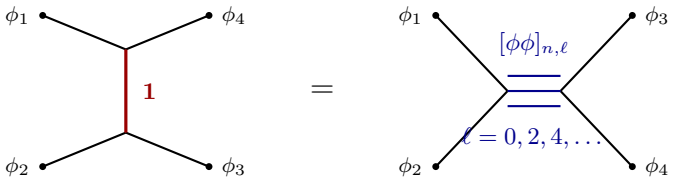


Figure 49.6: The large-spin reorganization in mean-field theory. The single identity exchanged in the crossed (t-)channel, whose $v \to 0$ collision singularity is the power law $v^{-\Delta_\phi}$ Eq. (49.148), is reproduced in the direct (s-)channel only by the entire infinite tower of double-twist operators $[\phi\phi]_{n,\ell}$ Eq. (49.130), whose large-spin OPE coefficients Eq. (49.150) are tuned to rebuild that power law. This is the prototype of the lightcone/large-spin expansion developed in Chapter 51.

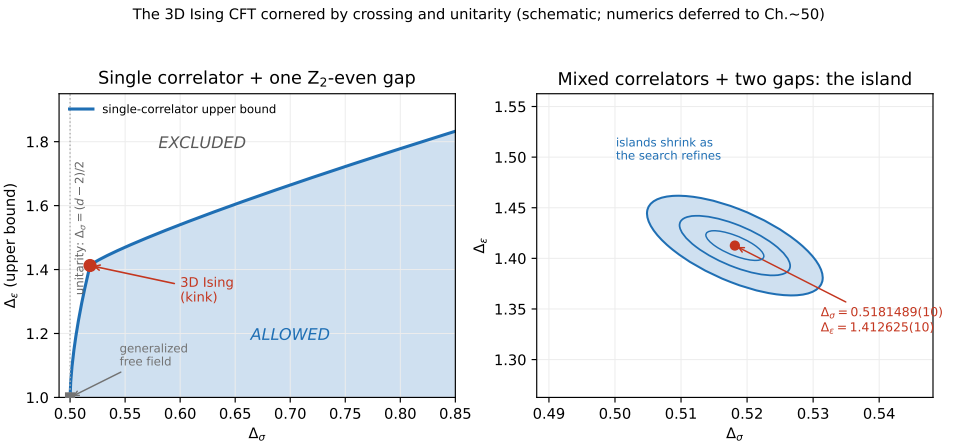


Figure 49.7: The 3D Ising CFT cornered by crossing and unitarity. *Left:* the rigorous single-correlator upper bound on Δ_ε as a function of Δ_σ under the one $\mathbb{Z}_2$-even gap assumption (A5), from Theorem 49.28. The shaded region below the curve is allowed; everything above is excluded by a positivity certificate. The bound has a sharp kink, and the Ising theory sits at it. The vertical wall at $\Delta_\sigma = \frac{d-2}{2} = \frac{1}{2}$ is the scalar unitarity bound; the square at its foot is the generalized free field of Section 49.5.1. *Right:* feeding in the mixed-correlator system $\{\sigma\sigma\sigma\sigma,\ \sigma\sigma\varepsilon\varepsilon,\ \varepsilon\varepsilon\varepsilon\varepsilon\}$ and a second gap (a single relevant $\mathbb{Z}_2$-odd scalar) closes the bound into an *island* that isolates Δ_σ and Δ_ε and shrinks as the search is refined. The schematic is a faithful rendering of the published bounds; the actual semidefinite computation is Chapter 50's.

Chapter 50

The Numerical Bootstrap

The foundations chapter (Chapter 49) reduced the consistency of a unitary conformal field theory, as seen through a single four-point function, to one homogeneous equation: the crossing sum rule Eq. (49.74), a linear relation among the convolved conformal blocks of Chapter 47 whose unknowns are a discrete spectrum of dimensions and a set of squared, hence nonnegative, OPE coefficients. It also exhibited the elementary mechanism that converts that equation into bounds — a linear functional nonnegative on every allowed block but positive on the identity refutes a candidate spectrum (Theorem 49.21) — and carried a first number through by hand. What it left open is everything that makes the method *complete*, *computable*, and *sharp*. This chapter supplies all three, and aims the resulting machine at the showcase target of the whole conformal program: the three-dimensional Ising fixed point, which is the $d = 3$ avatar of the Wilson–Fisher fixed point that ϕ^4 theory has carried since Chapter 6 and located perturbatively in Chapter 24.

The arc is a single idea pursued to its computational conclusion. We begin in the language of convex geometry, recasting crossing as the demand that a fixed vector lie inside a convex cone and proving the converse the foundations chapter lacked: by the Farkas dichotomy, whenever a spectrum is truly inconsistent a separating functional is *guaranteed* to exist, so the search is never a chase after a mirage. With completeness secured we turn the separation principle into its two working algorithms — the dimension bound and the two-sided OPE-coefficient bound — and prove that each computes the genuine optimum of a semi-infinite linear program, monotone under refinement. The obstruction that keeps those programs from a computer, a continuum of positivity constraints in each spin, is then removed in four controlled reductions resting on the radial expansion and its geometric convergence (Theorem 49.12), on the meromorphic structure of the blocks in the exchanged dimension, and on the classical Markov–Lukács theorem; the outcome is a finite semidefinite program whose rigor is underwritten by weak duality and whose floating-point output can be promoted to an exact

theorem. The analytic backbone of these reductions — Cauchy's estimate, analytic continuation, the residue calculus — is the machinery assembled in Chapter 5 (Theorem 5.23).

The machine is then run. Aimed first at the single correlator $\langle\sigma\sigma\sigma\sigma\rangle$ of the $\mathbb{Z}_2$-odd spin field in $d = 3$, it produces an exclusion curve with a sharp kink, and that kink sits, to three or four figures, exactly where every other method places the Ising critical point. The single correlator only *touches* the theory; to *isolate* it we promote the problem to the mixed system of $\langle\sigma\sigma\sigma\sigma\rangle$, $\langle\sigma\sigma\varepsilon\varepsilon\rangle$, and $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$, where unitarity speaks through positive-semidefinite matrices rather than nonnegative numbers and the open wedge pinches shut into a small closed island. The island delivers the dimensions, and hence the critical exponents η and ν, to six significant figures — numbers we confront against Monte Carlo and the laboratory.

A final movement reads the boundary. At a saturating point the optimal functional is forced to vanish exactly on the operators the extremal solution uses, so its double zeros *are* the spectrum: the bootstrap stops bracketing the data and begins reconstructing it, operator by operator, from a single convex-optimization certificate. We close by accounting honestly for what is proved and what is assumed — exclusions are theorems, islands are theorems conditional on explicit gap inputs imported from the renormalization-group understanding of Chapters 23 and 24, and naming the theory "Ising" is universality (physics), not a corollary of the bound — and by surveying the broader toolkit and dating the frontier as of the mid-2020s. The chapter ends where it must: at the large-spin tower the numerics can only sample, which is the natural handoff to the analytic methods of Chapter 51.

50.1 Crossing as a Problem in Convex Geometry

Chapter 49 reduced the consistency of a unitary conformal field theory — so far as a single four-point function can see it — to one homogeneous equation. Writing $p_{\mathcal{O}} = \lambda^2_{\phi\phi\mathcal{O}}$ for the squared OPE coefficients (Theorem 47.46) and $\widehat{F}_{\Delta,\ell}$ for the normalized convolved blocks built from the conformal blocks of Theorem 47.49 and the block decomposition Eq. (47.227), the crossing equation Eq. (49.14) became the bootstrap sum rule (Theorem 49.17, Eq. (49.74))

$$1 + \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, \widehat{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}) = 0, \qquad p_{\mathcal{O}} = \lambda^2_{\phi\phi\mathcal{O}} \geq 0, \tag{50.1}$$

where the sum runs over the non-identity primaries of the $\phi \times \phi$ OPE, each with even spin and a dimension obeying the unitarity bounds of Theorem 47.18. Two facts about this equation carry the whole method: the weights are nonnegative because they are squares (unitarity), and the allowed quantum numbers (Δ, ℓ) form a known region of the plane. Chapter 49 drew the consequence as a slogan

— a putative spectrum is consistent only if a certain fixed vector lies inside a convex cone — and proved that *if* a separating functional can be exhibited *then* the spectrum is excluded (Theorem 49.21). It did not prove the statement that makes the numerical bootstrap a complete method rather than a lucky one: that whenever a spectrum is genuinely inconsistent, a separating functional is *guaranteed to exist*. That converse is a theorem of finite-dimensional convex geometry, and installing it rigorously is the business of this section. It is the backbone on which every exclusion in this chapter, and every island in the next, ultimately rests.

This is the layer at which ϕ^4 theory enters the bootstrap. The correlator we will interrogate is $\langle\phi\phi\phi\phi\rangle$ for the real scalar carried since Chapter 6, now treated purely as a set of consistency conditions: Eq. (50.1) with Δ_ϕ a free parameter and the spectrum unknown. No Lagrangian, no coupling, no perturbation series is used — the input is symmetry and unitarity, nothing more. The output, by the end of the chapter, will be the Wilson–Fisher fixed point of Chapter 24 pinned down as a CFT to many digits. Here we lay the geometric foundation that turns "solve crossing" into "find a hyperplane."

50.1.1 Crossing vectors and the bootstrap cone

Fix the external dimension Δ_ϕ once and for all. The functions of the two cross-ratios that appear in Eq. (50.1) live in an infinite-dimensional real vector space; call it V. We work with it through linear functionals, which we will eventually take to be finite collections of derivatives at the crossing-symmetric point, exactly as in Eq. (49.98). For the geometry it is cleanest to think of each operator as contributing a fixed vector.

Definition 50.1 (Crossing vector)**.** For each allowed pair (Δ, ℓ) — even spin, dimension on or above the unitarity bound of Theorem 47.18 — the *crossing vector* is the element

$$\vec{F}_{\Delta,\ell} \;\equiv\; \widehat{F}_{\Delta,\ell}(z,\bar{z}) \;\in\; V, \tag{50.2}$$

regarded as a single point of the function space V. The *identity vector* is the known constant function $\vec{F}_{\mathbf{1}} \equiv \widehat{F}_{0,0} = 1$, the contribution that every CFT carries with the fixed weight $p_{\mathbf{1}} = 1$.

In this language the sum rule Eq. (50.1) is the single vector equation

$$\vec{F}_{\mathbf{1}} \;+\; \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, \vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} \;=\; 0, \qquad p_{\mathcal{O}} \geq 0, \tag{50.3}$$

which is Eq. (49.88) written with the normalized blocks. It asks: can the fixed vector $-\vec{F}_{\mathbf{1}}$ be written as a nonnegative combination of the crossing vectors drawn from the allowed region? The set of all such combinations is the central geometric object.

Definition 50.2 (Bootstrap cone and gap assumption)**.** Let A denote a set of allowed quantum numbers — a subset of the (Δ, ℓ) obeying unitarity and the even-spin selection rule. The *bootstrap cone* generated by A is

$$\mathcal{C}_A \;=\; \Big\{ \sum_{i=1}^{k} p_i \, \vec{F}_{\Delta_i,\ell_i} \;:\; k \in \mathbb{N},\; p_i \geq 0,\; (\Delta_i, \ell_i) \in A \Big\}, \tag{50.4}$$

the set of all finite nonnegative combinations of crossing vectors with quantum numbers in A. A *gap assumption* is a restriction of A: a hypothesis that excludes part of the unitarity-allowed region — typically "no scalar with $\Delta < \Delta_{\text{gap}}$" — and so replaces $\mathcal{C}_A$ by the smaller cone $\mathcal{C}_{A'}$ generated by the surviving operators $A' \subset A$.

That $\mathcal{C}_A$ is a *convex cone* is immediate from the definition: it contains 0 (the empty combination), it is closed under multiplication by any $t \geq 0$ (rescale every p_i), and it is closed under addition (concatenate the two combinations). Convexity follows, since for $x, y \in \mathcal{C}_A$ and $t \in [0,1]$ the combination $tx + (1-t)y$ is again a nonnegative combination. The content of crossing is now a single membership question.

Proposition 50.3 (Crossing as cone membership)**.** *A list of unitary CFT data with external dimension Δ_ϕ and non-identity spectrum contained in A exists only if*

$$-\vec{F}_{\mathbf{1}} \;\in\; \mathcal{C}_A. \tag{50.5}$$

A gap assumption shrinks A to A' and so shrinks the cone; if $-\vec{F}_{\mathbf{1}} \notin \mathcal{C}_{A'}$ the assumption is inconsistent with crossing and unitarity.

Proof. If such data exist, its four-point function satisfies Eq. (50.3) with all weights $p_{\mathcal{O}} \geq 0$ and all non-identity quantum numbers in A; rearranging, $-\vec{F}_{\mathbf{1}}$ equals the nonnegative combination $\sum_{\mathcal{O} \neq \mathbf{1}} p_{\mathcal{O}} \, \vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}$, which is by definition an element of $\mathcal{C}_A$. (Strictly, the sum may be infinite; Eq. (50.4) admits only finite combinations, so membership is in the closure $\overline{\mathcal{C}_A}$. We return to this point in Section 50.1.2, where the truncation to a finite functional basis renders the distinction harmless.) Restricting to $A' \subset A$ restricts the generators of the cone, so $\mathcal{C}_{A'} \subseteq \mathcal{C}_A$; if the membership fails for $\mathcal{C}_{A'}$ no such data with spectrum in A' can exist. □

The membership question splits cleanly into two cases. Either $-\vec{F}_{\mathbf{1}}$ lies in (the closure of) the cone, in which case crossing *might* be solvable with a spectrum drawn from A, or it lies strictly outside, in which case no such solution exists and the spectrum is excluded. The whole numerical bootstrap is the systematic discovery of the second case. To turn "lies strictly outside" into a finite, checkable certificate we need the geometric fact that a point outside a closed convex cone can always be *seen* to be outside, by a single linear functional. We now prove this.

50.1.2 The master separation principle

We work in a real Euclidean space $\mathbb{R}^N$ with the standard inner product $\langle a, b\rangle$ and norm $\|x\| = \langle x, x\rangle^{1/2}$. The reduction from the infinite-dimensional V to a finite $\mathbb{R}^N$ is the act of choosing a finite functional basis: pick N linearly independent linear functionals on V — in practice the derivatives $\partial_z^m \partial_{\bar z}^n$ at $z = \bar z = \frac{1}{2}$ that survive the crossing antisymmetry — and represent every vector $f \in V$ by the N-tuple of its values under them, its *component vector* in $\mathbb{R}^N$. Crossing vectors, the identity vector, and the cone all map forward to $\mathbb{R}^N$; a linear functional on $\mathbb{R}^N$ is then exactly a finite linear combination of the chosen derivatives, the object Eq. (49.98) we are allowed to build. The geometry below lives in this finite-dimensional image, where every step is a theorem with no functional-analytic loose ends. The fidelity of the reduction — that excluding the truncated problem excludes the true one — is the monotonicity already noted in Chapter 49: a functional in the small basis is a functional in any larger one, so truncated exclusions are genuine exclusions, never spuriously strong. The completeness in the other direction is the subject of Theorem 50.6 below.

The engine is the nearest-point projection onto a closed convex set, a construction worth proving in full because everything rests on it.

Lemma 50.4 (Projection onto a closed convex set)**.** *Let $C \subseteq \mathbb{R}^N$ be nonempty, closed, and convex, and let $b \in \mathbb{R}^N$. Then there is a unique point $p \in C$ minimizing the distance to b,*

$$\|b - p\| \;=\; \inf_{x \in C} \|b - x\| \;\equiv\; d, \tag{50.6}$$

and it is characterized by the variational inequality

$$\langle\, b - p,\; x - p \,\rangle \;\le\; 0 \qquad \text{for every } x \in C. \tag{50.7}$$

Proof. Existence. The infimum $d \ge 0$ is finite (any single point of C gives a finite distance). Take a minimizing sequence $x_n \in C$ with $\|b - x_n\| \to d$. It is bounded: $\|x_n\| \le \|b\| + \|b - x_n\|$ is eventually below $\|b\| + d + 1$. By the Bolzano–Weierstrass theorem it has a subsequence converging to some $p \in \mathbb{R}^N$; since C is closed, $p \in C$, and by continuity of the norm $\|b - p\| = d$.

Uniqueness. Suppose $p_1, p_2 \in C$ both attain d. Their midpoint $\frac{1}{2}(p_1 + p_2)$ lies in C by convexity. The parallelogram identity $\|u + v\|^2 + \|u - v\|^2 = 2\,\|u\|^2 + 2\,\|v\|^2$, applied to $u = b - p_1$ and $v = b - p_2$, gives

$$\left\| b - \tfrac{p_1 + p_2}{2} \right\|^2 \;=\; \tfrac{1}{2}\,\|b - p_1\|^2 + \tfrac{1}{2}\,\|b - p_2\|^2 - \tfrac{1}{4}\,\|p_1 - p_2\|^2 \;=\; d^2 - \tfrac{1}{4}\,\|p_1 - p_2\|^2 . \tag{50.8}$$

If $p_1 \ne p_2$ the right-hand side is strictly below d^2, contradicting the minimality of d over C. Hence $p_1 = p_2$.

Variational inequality. Fix any $x \in C$. For $t \in [0,1]$ the point $p + t(x-p) = (1-t)p + tx$ lies in C by convexity, so the scalar function

$$\varphi(t) \;=\; \| \, b - p - t(x-p) \, \|^2 \;=\; \|b-p\|^2 - 2t\,\langle b-p,\, x-p\rangle + t^2\,\|x-p\|^2 \tag{50.9}$$

satisfies $\varphi(t) \geq \varphi(0) = d^2$ for all $t \in [0,1]$, since p is the minimizer. Therefore $\varphi'(0) \geq 0$. But $\varphi'(0) = -2\langle b-p,\, x-p\rangle$, which gives Eq. (50.7). Conversely, if Eq. (50.7) holds then for any $x \in C$, $\|b-x\|^2 = \|b-p\|^2 - 2\langle b-p, x-p\rangle + \|x-p\|^2 \geq \|b-p\|^2$, so p is the minimizer; the variational inequality is equivalent to being the projection. $\square$

The projection delivers the separating hyperplane at once.

Theorem 50.5 (Strict separation of a point from a closed convex set)**.** *Let $C \subseteq \mathbb{R}^N$ be nonempty, closed, and convex, and let $b \notin C$. Then there is a nonzero vector $\alpha \in \mathbb{R}^N$ and a real number c such that*

$$\langle \alpha,\, x\rangle \;\leq\; c \;<\; \langle \alpha,\, b\rangle \qquad \text{for every } x \in C. \tag{50.10}$$

The hyperplane $\{y : \langle \alpha, y\rangle = c\}$ strictly separates b from C.

Proof. Let p be the projection of b onto C from Theorem 50.4, and set $\alpha = b - p$. Because $b \notin C$ and $p \in C$, $\alpha \neq 0$ and $\|\alpha\|^2 = \langle \alpha, b-p\rangle > 0$. For any $x \in C$ the variational inequality Eq. (50.7) reads $\langle \alpha, x-p\rangle \leq 0$, i.e. $\langle \alpha, x\rangle \leq \langle \alpha, p\rangle$. On the other hand $\langle \alpha, b\rangle = \langle \alpha, p\rangle + \langle \alpha, b-p\rangle = \langle \alpha, p\rangle + \|\alpha\|^2 > \langle \alpha, p\rangle$. Choosing $c = \langle \alpha, p\rangle + \tfrac{1}{2}\,\|\alpha\|^2$ places it strictly between the two,

$$\sup_{x \in C} \langle \alpha, x\rangle \;\leq\; \langle \alpha, p\rangle \;<\; c \;<\; \langle \alpha, p\rangle + \|\alpha\|^2 = \langle \alpha, b\rangle, \tag{50.11}$$

which is Eq. (50.10). $\square$

For the bootstrap the relevant C is not merely convex but a cone, and the separation theorem then sharpens into the dichotomy named after Farkas: a point is either in the cone or strictly separated from it by a functional that is one-signed on the entire cone. This is the master principle.

Theorem 50.6 (Farkas dichotomy for a closed convex cone)**.** *Let $\mathcal{C} \subseteq \mathbb{R}^N$ be a closed convex cone and let $b \in \mathbb{R}^N$. Then exactly one of the following holds:*

(I) $b \in \mathcal{C}$;

(II) *there exists $\alpha \in \mathbb{R}^N$ with $\langle \alpha, x\rangle \geq 0$ for every $x \in \mathcal{C}$ and $\langle \alpha, b\rangle < 0$.*

Proof. The two alternatives are mutually exclusive: if (I) and (II) both held, then taking $x = b \in \mathcal{C}$ in (II) would force $\langle \alpha, b\rangle \geq 0$, contradicting $\langle \alpha, b\rangle < 0$.

It remains to show that if (I) fails then (II) holds. Suppose $b \notin \mathcal{C}$. By Theorem 50.5 there is $\beta \neq 0$ and c with $\langle \beta, x\rangle \leq c < \langle \beta, b\rangle$ for all $x \in \mathcal{C}$. We extract two facts from the cone structure. First, $0 \in \mathcal{C}$ (a cone contains the origin), so $c \geq \langle \beta, 0\rangle = 0$, hence $\langle \beta, b\rangle > c \geq 0$, giving $\langle \beta, b\rangle > 0$. Second, $\langle \beta, x\rangle \leq 0$ for every $x \in \mathcal{C}$: if some $x_0 \in \mathcal{C}$ had $\langle \beta, x_0\rangle > 0$, then $tx_0 \in \mathcal{C}$ for all $t \geq 0$ (cone property) and $\langle \beta, tx_0\rangle = t\langle \beta, x_0\rangle \to +\infty$, violating the upper bound $c < \infty$. Now set $\alpha = -\beta$. Then $\langle \alpha, x\rangle = -\langle \beta, x\rangle \geq 0$ for every $x \in \mathcal{C}$, and $\langle \alpha, b\rangle = -\langle \beta, b\rangle < 0$, which is precisely (II). □

One hypothesis of the Farkas dichotomy must be earned, not assumed: the cone $\mathcal{C}$ has to be *closed*. The bootstrap cone Eq. (50.4) is generated by a continuum of crossing vectors (the dimension Δ runs over a half-line for each of infinitely many spins), and the conic hull of an arbitrary set need not be closed. The following lemma supplies exactly the condition that makes it closed, and it is the condition the numerical implementation enforces in practice.

Lemma 50.7 (Closedness of the conic hull of a compact set)**.** *Let $K \subset \mathbb{R}^N$ be compact with $0 \notin \mathrm{conv}(K)$, the convex hull of K. Then the conic hull $\mathcal{C} = \{\sum_i t_i v_i : t_i \geq 0,\ v_i \in K\}$ is closed.*

Proof. Because $\mathrm{conv}(K)$ is compact (a convex hull of a compact set in finite dimensions is compact, by Carathéodory) and does not contain the origin, the strict separation theorem Theorem 50.5 applied to $b = 0$ and $C = \mathrm{conv}(K)$ furnishes a vector w and a $c < 0$ with $\langle w, y\rangle \leq c < \langle w, 0\rangle = 0$... reversing the sign we obtain w with

$$\langle w, v\rangle \;\geq\; \delta \;>\; 0 \qquad \text{for all } v \in K, \tag{50.12}$$

for some $\delta > 0$ (take $\delta = -c > 0$ and replace w by $-w$). By the conic Carathéodory theorem, every element of $\mathcal{C}$ is a nonnegative combination of at most N points of K. Let $y_k = \sum_{i=1}^{N} t_i^{(k)} v_i^{(k)} \to y$ be a convergent sequence in $\mathcal{C}$ with $t_i^{(k)} \geq 0$ and $v_i^{(k)} \in K$. Pairing with w and using Eq. (50.12),

$$\langle w, y_k\rangle \;=\; \sum_{i=1}^{N} t_i^{(k)} \langle w, v_i^{(k)}\rangle \;\geq\; \delta \sum_{i=1}^{N} t_i^{(k)}. \tag{50.13}$$

Since $y_k \to y$ the left-hand side is bounded, so the coefficient sums $\sum_i t_i^{(k)}$ are bounded; each $t_i^{(k)}$ is therefore bounded. Passing to a subsequence, $t_i^{(k)} \to t_i \geq 0$ and (by compactness of K) $v_i^{(k)} \to v_i \in K$. Then $y = \lim_k y_k = \sum_i t_i v_i \in \mathcal{C}$, so $\mathcal{C}$ is closed. □

In the bootstrap, K is the set of *normalized* crossing vectors $\vec{F}_{\Delta,\ell} / \left\|\vec{F}_{\Delta,\ell}\right\|$ ranging over the allowed region (so that the conic hull of K is exactly $\mathcal{C}_A$). Compactness of K is not automatic, because the region is unbounded in Δ and in

ℓ; what rescues it is that the crossing vectors approach definite limiting directions as $\Delta \to \infty$ at fixed spin and as $\ell \to \infty$, a consequence of the large-dimension and large-spin asymptotics of the conformal blocks. Adjoining those finitely many limit rays to the image of the allowed region produces a compact K, and provided the chosen functional basis does not put 0 in $\mathrm{conv}(K)$ — which is the generic situation, since the identity sets a definite scale — Theorem 50.7 makes $\mathcal{C}_A$ closed. This is precisely what the numerical method does when it imposes positivity uniformly down to the unitarity bound and out to a large spin and dimension cutoff: it is enforcing the compactness that guarantees the dichotomy is exact. We state the working hypothesis once and rely on it thereafter.

Caution

The theorems of this subsection are exact statements about a *finite-dimensional, closed* cone. Two idealizations separate them from a literal computation. The functional basis is finite, which (as noted) can only weaken exclusions, never falsify them. And the cone must be closed, which requires the compactness arranged in Theorem 50.7; a genuine bootstrap implementation realizes this by discretizing the spectrum and treating the asymptotic tails explicitly, with errors that are controlled and reported. Within those idealizations the separation is a theorem; the next chapter's numbers inherit exactly that status.

50.1.3 The exclusion lemma, both directions

We can now state the principle that organizes the entire chapter, and we can state it as an *equivalence* rather than the one-way implication of Theorem 49.21. Recall the notion of an admissible functional (Theorem 49.21): for a candidate spectrum A, a linear functional α is admissible if

$$\text{(i)}\ \ \alpha[\vec{F}_{\mathbf{1}}] > 0, \qquad \text{(ii)}\ \ \alpha[\vec{F}_{\Delta,\ell}] \geq 0 \ \text{ for every } (\Delta,\ell) \in A. \tag{50.14}$$

Theorem 50.8 (Master exclusion principle)**.** *Fix Δ_ϕ and a candidate spectrum A, and work in a fixed finite functional basis in which the cone $\mathcal{C}_A$ is closed. The following are equivalent:*

(a) *No unitary, crossing-symmetric CFT has external scalar dimension Δ_ϕ and non-identity $\phi \times \phi$ spectrum contained in A* within this basis — *that is, $-\vec{F}_{\mathbf{1}} \notin \mathcal{C}_A$;*

(b) *there exists an admissible functional α for A in the sense of Eq. (50.14).*

Proof. (b)⇒(a) — sufficiency. This is the sign count of Theorem 49.21; we reproduce it because it is the payoff of the whole construction and takes three lines. Suppose an admissible α exists and, for contradiction, that consistent data

did too. Apply α to the sum rule Eq. (50.3); by linearity and term-by-term applicability (the block expansion converges in a neighbourhood of the crossing-symmetric point, Theorem 47.46, and α is a finite combination of derivatives there),

$$0 \;=\; \alpha\Big[\vec{F}_{\mathbf{1}} + \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\,\vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\Big] \;=\; \underbrace{\alpha[\vec{F}_{\mathbf{1}}]}_{>0} \;+\; \sum_{\mathcal{O}\neq\mathbf{1}} \underbrace{p_{\mathcal{O}}}_{\geq 0}\,\underbrace{\alpha[\vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]}_{\geq 0} \;\geq\; \alpha[\vec{F}_{\mathbf{1}}] \;>\; 0. \tag{50.15}$$

The first term is strictly positive by (i); every term of the sum is a product of two nonnegatives (the weight by unitarity, the functional value by (ii) since the operator lies in A); so the right-hand side is strictly positive, contradicting the exact zero on the left. No such data exist: $-\vec{F}_{\mathbf{1}} \notin \mathcal{C}_A$.

(a)⇒(b) — completeness. This is the new content. Suppose $-\vec{F}_{\mathbf{1}} \notin \mathcal{C}_A$. Since $\mathcal{C}_A$ is a closed convex cone, the Farkas dichotomy Theorem 50.6 applied to $b = -\vec{F}_{\mathbf{1}}$ rules out alternative (I) and so delivers alternative (II): a vector α with $\langle \alpha, x\rangle \geq 0$ for all $x \in \mathcal{C}_A$ and $\langle \alpha, -\vec{F}_{\mathbf{1}}\rangle < 0$. The first property, evaluated on each generator $\vec{F}_{\Delta,\ell} \in \mathcal{C}_A$ (every generator is in the cone), gives $\alpha[\vec{F}_{\Delta,\ell}] \geq 0$ for all $(\Delta,\ell) \in A$, which is condition (ii). The second property reads $\alpha[\vec{F}_{\mathbf{1}}] > 0$, which is condition (i). Thus α is admissible. □

The equivalence is the conceptual centre of the numerical bootstrap. Sufficiency, direction (b)⇒(a), is what makes a found functional a *proof*: produce one α passing the two sign tests and an entire class of CFTs is forbidden, exactly, with no error term. Completeness, direction (a)⇒(b), is what makes the search *worth running*: it guarantees that the functional one is hunting for is not a mirage — if the spectrum is truly inconsistent (within the basis), the certificate exists and a sufficiently thorough search will find it. Chapter 49 had only the first half and could therefore promise that success excludes; with the second half in hand we know that the method is the right one — every genuine inconsistency is, in principle, certifiable by a single hyperplane. The geometric picture is the entire story in one sentence: *a putative CFT is the demand that a fixed point lie inside a convex cone, and an excluding functional is a hyperplane that catches it outside* (Fig. 50.1).

Physical Insight

The single most useful thing to carry from this section is the inversion of burden it accomplishes. *Solving* crossing — producing an actual spectrum and OPE coefficients — is hard and, for an interacting theory in $d > 2$, not something we can do in closed form. *Refuting* a spectrum is, by the master principle, a finite search for one hyperplane, and convex duality promises the hyperplane exists whenever the refutation is true. The bootstrap never solves the hard problem; it relentlessly solves the easy dual one, and the

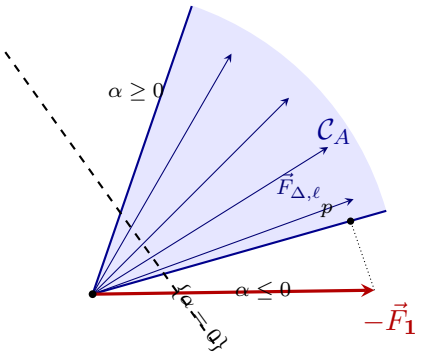


Figure 50.1: The master separation principle in the plane. The crossing vectors of an assumed spectrum A generate the closed convex cone $\mathcal{C}_A$ (shaded). Crossing demands that $-\vec{F}_{\mathbf{1}}$ lie inside it. When a gap assumption has removed enough low-lying operators, $-\vec{F}_{\mathbf{1}}$ falls outside the cone; the nearest-point projection $p \in \mathcal{C}_A$ (Theorem 50.4) defines the separating direction $\alpha = (-\vec{F}_{\mathbf{1}}) - p$ up to sign, and the hyperplane $\{\alpha = 0\}$ through the origin has the entire cone on one side ($\alpha \geq 0$, i.e. $\alpha[\vec{F}_{\Delta,\ell}] \geq 0$) and $\vec{F}_{\mathbf{1}}$ strictly on the other ($\alpha[\vec{F}_{\mathbf{1}}] > 0$). Existence of such an α is guaranteed by Theorem 50.6 whenever the point is genuinely outside, and exhibiting one excludes the spectrum by Theorem 50.8.

boundary between "refuted" and "not refuted" is squeezed until only the true theories survive. Every bound in this chapter is one application of this idea with a different objective placed on the functional.

50.1.4 A separation computed by hand

The dichotomy is abstract until one watches the projection manufacture a functional from explicit vectors. We do so in the smallest nontrivial setting, reusing the exact numbers computed in Theorem 49.22 so that the geometry can be checked against an arithmetic already in hand. This is not a re-derivation of that bound; it is a demonstration that the existence theorem Theorem 50.6 *constructs* the very functional the feasibility scan of Chapter 49 found, and that the two agree.

Worked Example 50.9 (The Farkas certificate in the two-derivative plane)**.** We continue ϕ^4 theory with the carried correlator $\langle\phi\phi\phi\phi\rangle$, specialized as in Theorem 49.22 to $d = 2$ and $\Delta_\phi = \frac{1}{8}$ (the spin field of the two-dimensional Ising model). The functional basis is two-dimensional: the lowest two crossing-odd derivatives at the symmetric point,

$$\Lambda_1[\,f\,] = \tfrac{d}{dz} f(z,z)\big|_{z=1/2}, \qquad \Lambda_3[\,f\,] = \tfrac{d^3}{dz^3} f(z,z)\big|_{z=1/2}, \tag{50.16}$$

so each vector in V is represented in $\mathbb{R}^2$ by the pair (Λ_1, Λ_3). From Eqs. (49.94) and (49.114) the identity vector and its negation are

$$\vec{F}_{\mathbf{1}} = (-0.84090,\ -4.41471), \qquad b \equiv -\vec{F}_{\mathbf{1}} = (0.84090,\ 4.41471), \tag{50.17}$$

and the crossing vectors of the low-lying operators are, from Table 49.1,

$$\vec{F}_{1,0} = (2.828,\ 8.413), \quad \vec{F}_{2,0} = (4.259,\ 71.25), \quad \vec{F}_{2,2} = (8.377,\ 129.4),\ \ldots \tag{50.18}$$

all with both components positive. In the plane a ray from the origin is fixed by its slope $s = \Lambda_3/\Lambda_1$; the generators have slopes

$$s(1,0) = 2.97, \quad s(2,0) = 16.7, \quad s(2,2) = 15.4,\ \ldots, \tag{50.19}$$

and every other allowed operator has slope above the minimum. Because all generators lie in the open right half-plane $\Lambda_1 > 0$, the cone $\mathcal{C}_A$ is the angular sector $\{(\Lambda_1, \Lambda_3) : \Lambda_1 \geq 0,\ s_{\min} \leq \Lambda_3/\Lambda_1\}$ bounded below by the shallowest generator ray and unbounded above (high-dimension scalars have ever-steeper slopes). The target point b has slope

$$s_b = \frac{4.41471}{0.84090} = 5.250. \tag{50.20}$$

No gap: the point is inside. With the full unitarity-allowed spectrum the shallowest generator is the $\Delta = 1$ scalar, slope $2.97 < 5.250$, so b lies above the lower edge of the sector and inside $\mathcal{C}_A$. Indeed one checks that b is a positive combination of $\vec{F}_{1,0}$ and a steeper generator. By Theorem 50.8 no functional can be admissible, and none should be: crossing is (so far) satisfiable, and the two-dimensional Ising solution realizes it.

With a gap: the point falls out, and the projection names the hyperplane. Impose the gap assumption "the first scalar has $\Delta \geq \Delta_{\text{gap}}$." Removing the low scalars raises the shallowest surviving slope $s_{\min}$. Once $s_{\min} > s_b = 5.250$, the target slope lies *below* the entire cone: b is outside $\mathcal{C}_A$, and Theorem 50.6 guarantees a separating functional. We construct it as the projection. The boundary of the sector nearest b is the lower edge, the ray through the shallowest generator $v_{\min} = (1, s_{\min})$ (un-normalized direction). For b outside and below this edge the nearest point of the (convex) cone on that edge is the orthogonal foot, and the separating direction of Theorem 50.5 is $\alpha \propto b - p$, which points perpendicular to the edge into the half-plane containing b. A vector perpendicular to the edge direction $(1, s_{\min})$ is $(-s_{\min}, 1)$ up to scale; orienting it to be nonnegative on the cone (and hence to make $\alpha[\vec{F}_{\mathbf{1}}] > 0$) gives, after rescaling so the Λ_3 coefficient is unity,

$$\alpha \ = \ \Lambda_3 - c\,\Lambda_1, \qquad c = s_{\min}. \tag{50.21}$$

This is exactly the one-parameter family that Theorem 49.22 scanned. Its admissibility conditions reproduce the window found there: positivity on every

generator requires $\alpha[\vec{F}_{\Delta,\ell}] = \Lambda_3[\vec{F}] - c\,\Lambda_1[\vec{F}] \geq 0$, i.e. $c \leq s(\Delta,\ell)$ for all surviving operators, hence $c \leq s_{\min}$; and positivity on the identity requires $\alpha[\vec{F}_{\mathbf{1}}] = \Lambda_3[\vec{F}_{\mathbf{1}}] - c\,\Lambda_1[\vec{F}_{\mathbf{1}}] = -4.41471 - c\,(-0.84090) = 0.84090\,c - 4.41471 > 0$, i.e. $c > 5.250$. An admissible functional therefore exists precisely when $5.250 < c \leq s_{\min}$, the non-empty window opening exactly when the gap has pushed $s_{\min}$ past the target slope 5.250 — numerically near $\Delta_{\text{gap}} \simeq 1.28$, the threshold $\Delta_0^* \leq 1.28$ of Theorem 49.22. The Farkas construction and the feasibility scan have produced the same certificate by two routes: the projection picks out the perpendicular to the cone's edge, and that perpendicular is the functional $\Lambda_3 - s_{\min}\Lambda_1$.

The worked example exhibits, in coordinates one can verify by hand, the content of the abstract theorems: a point inside the cone admits no certificate, a point pushed outside by a gap admits one, and the nearest-point projection of Theorem 50.4 *is* the recipe that returns it. In two dimensions the cone is an angular sector and the optimal functional is the perpendicular to its near edge; in the realistic problem the cone lives in tens or hundreds of dimensions and its boundary is a curved surface, so the "perpendicular to the near edge" becomes a genuine optimization. That optimization, and the two families of bounds it computes — the largest excludable gap and the extremal OPE coefficient — are the subject of the next section, which takes the object this section has guaranteed to exist (a positive linear functional that strictly separates the identity) and turns its construction into the two workhorse algorithms of the bootstrap, demonstrated on a toy functional that yields an actual number.

50.2 The Linear-Functional Method: Dimension and OPE Bounds

The previous section recast crossing as a question in convex geometry: the squared OPE coefficients $p_{\mathcal{O}} \geq 0$ are nonnegative weights, the convolved blocks $\vec{F}_{\Delta,\ell}$ are fixed vectors, and a proposed spectrum is consistent with crossing only if the identity vector $-\vec{F}_{0,0}$ lies inside the convex cone they generate. The master separation principle (Theorem 50.5) then supplied the dual handle: whenever $-\vec{F}_{0,0}$ lies *outside* that cone there is a hyperplane separating it, and that hyperplane is a linear functional α which is nonnegative on every allowed block yet strictly positive on the identity. Finding such an α is a rigorous proof that the spectrum is impossible — the content of the linear-functional exclusion principle proved in the foundations chapter (Theorem 49.21), which exhibited the logic and carried a first number through by hand (Theorem 49.22).

A single exclusion is one point of the allowed region declared empty. To extract physics we must sweep the exclusion across families of candidate spectra and read off where it succeeds and where it fails, and we must do so for the two quantities a CFT is built from: the dimensions of operators and the coefficients

with which they appear. This section turns the separation principle into the two standard bound algorithms of the bootstrap. We first show that the dimension bound is not merely *a* number a functional happens to certify but is *exactly* the optimum of an associated optimization — a semi-infinite linear program — so that any feasible functional gives a valid bound and the best functional gives the tightest one. We then prove the monotonicity that makes the bounds behave as honest outer approximations: adding an assumption can only shrink the allowed region. We give the OPE-coefficient bounds, two-sided, as the same technology run with a different normalization. Finally we compute a complete two-sided bound on an OPE coefficient by hand, so that the reader leaves with a rigorous interval — bracketing the exact answer of a solvable model — produced with nothing more than a derivative and a sign count, before any computer enters in Section 50.3.

50.2.1 The single-correlator problem, restated as an optimization

We work throughout with the four-point function of a single real scalar primary ϕ of dimension Δ_ϕ — the correlator $\langle\phi\phi\phi\phi\rangle$ that ϕ^4 theory has carried to the bootstrap and that the showcase sections will instantiate as $\langle\sigma\sigma\sigma\sigma\rangle$ for the Ising spin field. Conformal symmetry reduces it to a function of the two cross-ratios (Eqs. (47.138) and (47.140)), and crossing collapses (Eq. (49.74)) to the sum rule

$$\sum_{\mathcal{O}} p_{\mathcal{O}}\, F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) = 0, \qquad p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0, \qquad p_{\mathbf{1}} = 1, \tag{50.22}$$

the sum running over the primaries of the $\phi\times\phi$ OPE, with the identity carried as the fixed term $F_{0,0} = v^{\Delta_\phi} - u^{\Delta_\phi}$ and the convolved blocks $F_{\Delta,\ell} = v^{\Delta_\phi}G_{\Delta,\ell}(u,v) - u^{\Delta_\phi}G_{\Delta,\ell}(v,u)$ of Theorem 49.14. Two pieces of data constrain the admissible operators: by the OPE selection rule for identical scalars only even spins $\ell = 0, 2, 4, \ldots$ appear, and by the unitarity bounds (Theorem 47.18) every primary obeys $\Delta \geq \frac{d-2}{2}$ for $\ell = 0$ and $\Delta \geq \ell + d - 2$ for $\ell \geq 1$. The OPE converges in a neighbourhood of the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$ (Theorem 47.48), which is what licenses applying a derivative functional term by term.

A *linear functional* is, as in Eq. (49.98), a finite real combination of partial derivatives evaluated at that point,

$$\alpha[\,f\,] = \sum_{m,n} a_{mn}\, \partial_z^m \partial_{\bar{z}}^n f(z,\bar{z})\Big|_{z=\bar{z}=1/2}, \qquad a_{mn} \in \mathbb{R}, \tag{50.23}$$

and α is *admissible* for a candidate spectrum S (a set of allowed (Δ,ℓ), each obeying unitarity) when $\alpha[F_{0,0}] > 0$ and $\alpha[F_{\Delta,\ell}] \geq 0$ for every $(\Delta,\ell) \in S$ (Theorem 49.21). The exclusion principle says: an admissible functional for S proves that no unitary crossing-symmetric CFT has its $\phi\times\phi$ OPE drawn from $\{\mathbf{1}\} \cup S$. Everything below is the systematic exploitation of this one statement.

It is worth naming the objects of the optimization precisely, because the claim "the bound is exactly the optimum" is a statement about two coupled problems. The *primal* problem is the search for a CFT: given a candidate spectrum S, does there exist an assignment of nonnegative weights $\{p_{\mathcal{O}}\}_{\mathcal{O}\in S}$ solving Eq. (50.22)? The *dual* problem is the search for a certificate: does there exist an admissible functional for S? The exclusion principle is the elementary half of a duality between them — a certificate refutes every solution — and the content of the next subsection is that, in the precise sense allowed by the geometry, the converse holds too.

50.2.2 Dimension bounds as a semi-infinite linear program

The first and most important application bounds the dimension of the lightest operator in a chosen sector. Take the scalar sector for definiteness; spinning sectors are identical with the unitarity floor shifted. Fix Δ_ϕ and a trial gap Δ_0, and let

$$S(\Delta_0) = \{(\Delta, 0) : \Delta \geq \Delta_0\} \;\cup\; \{(\Delta, \ell) : \ell = 2, 4, 6, \ldots,\; \Delta \geq \ell + d - 2\} \tag{50.24}$$

be the candidate spectrum in which every scalar is pushed up to at least Δ_0 while each spinning sector extends down to its unitarity bound. This is the spectrum of Eq. (49.101); we now read it as the feasible set of an optimization.

Definition 50.10 (The gap program)**.** The *primal gap program* at (Δ_ϕ, Δ_0) asks for a nonnegative spectral density $\{p_{\mathcal{O}} \geq 0\}$ supported on $S(\Delta_0)$ with $\sum_{\mathcal{O}\in S(\Delta_0)} p_{\mathcal{O}} F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = -F_{0,0}$; it is *feasible* when such a density exists. The *dual gap program* asks for a linear functional α of the form Eq. (50.23) that is admissible for $S(\Delta_0)$, i.e.

$$\alpha[F_{0,0}] > 0, \qquad \alpha[F_{\Delta,\ell}] \geq 0 \;\text{ for all } (\Delta, \ell) \in S(\Delta_0). \tag{50.25}$$

The dual constraints are a continuum of linear inequalities indexed by the continuous label Δ within each spin; an optimization with finitely many unknowns a_{mn} and infinitely many linear constraints is a *semi-infinite linear program*.

The two programs are alternatives: a feasible primal density is a unitary crossing solution with the gap, an admissible dual functional is a proof that none exists, and they cannot both hold, since applying the functional to the solution gives $0 = \alpha[-F_{0,0}] + \sum_{\mathcal{O}} p_{\mathcal{O}} \alpha[F_{\mathcal{O}}]$ with a strictly negative first term and a nonnegative sum, a contradiction. This is the exclusion principle read as weak duality. The substantive question is whether, when the primal is *infeasible*, a dual certificate must exist. For a closed convex cone the separating-hyperplane theorem of Theorem 50.5 guarantees exactly that, and this is what upgrades "a functional gives a bound" to "the optimal functional gives *the* bound."

Theorem 50.11 (The gap bound is the optimum of the dual program)**.** *Fix* Δ_ϕ*. Define*

$$\Delta_0^*(\Delta_\phi) \;=\; \inf\big\{\Delta_0 : \textit{the dual gap program at } (\Delta_\phi, \Delta_0) \textit{ is feasible}\big\}. \tag{50.26}$$

Then:

(i) (Soundness — every feasible functional is a valid bound.) *If the dual program at* (Δ_ϕ, Δ_0) *is feasible, then every unitary, crossing-symmetric CFT with an external scalar of dimension* Δ_ϕ *has a scalar in* $\phi \times \phi$ *of dimension strictly below* Δ_0*. Consequently* Δ_0 *is a rigorous upper bound on the dimension of the lightest scalar in* $\phi \times \phi$*.*

(ii) (Optimal functional, tightest bound.) *The dual program is feasible for every* $\Delta_0 > \Delta_0^*$ *and infeasible for every* $\Delta_0 < \Delta_0^*$*. Hence* $\Delta_0^*(\Delta_\phi)$ *is the tightest scalar-gap bound the functional method can prove: no admissible functional certifies any value below it, and admissible functionals certify every value above it.*

(iii) (No slack, modulo closure.) *If the cone* $\mathcal{C}\big(S(\Delta_0^*)\big)$ *generated by the allowed blocks is closed, then at* $\Delta_0 = \Delta_0^*$ *the primal program is feasible: there is a unitary spectral density solving crossing whose lightest scalar sits exactly at* Δ_0^**. The bound is then* saturated*, and the saturating density is the unique extremal ray on which the optimal functional vanishes.*

Proof. (i) This is the exclusion principle. Suppose the dual is feasible, with admissible functional α, and suppose for contradiction a unitary crossing-symmetric CFT existed whose lightest scalar had dimension $\Delta_{\min} \geq \Delta_0$. Then every non-identity primary of $\phi \times \phi$ lies in $S(\Delta_0)$: scalars because $\Delta \geq \Delta_{\min} \geq \Delta_0$, spinning operators because they have even spin $\ell \geq 2$ and obey the unitarity bound. Apply α to the sum rule Eq. (50.22), which is legitimate because α is a finite derivative combination at a point of OPE convergence (Theorem 47.48):

$$0 = \alpha\Big[\sum_{\mathcal{O}} p_{\mathcal{O}} F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\Big] = \underbrace{\alpha[F_{0,0}]}_{>0} + \sum_{\mathcal{O}\neq\mathbf{1}} \underbrace{p_{\mathcal{O}}}_{\geq 0}\, \underbrace{\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]}_{\geq 0} \;\geq\; \alpha[F_{0,0}] \;>\; 0, \tag{50.27}$$

using $p_{\mathbf{1}} = 1$ and admissibility. The right-hand side is strictly positive while the left is zero: contradiction. Hence $\Delta_{\min} < \Delta_0$ in every such theory, which is the stated bound.

(ii) The monotonicity is structural. If $\Delta_0' > \Delta_0$ then $S(\Delta_0') \subset S(\Delta_0)$ — raising the gap only removes low-dimension scalars — so any α admissible for $S(\Delta_0)$ is *a fortiori* admissible for $S(\Delta_0')$, since condition Eq. (50.25) is then required on a smaller set. Dual feasibility therefore propagates upward in Δ_0: if the program is feasible at Δ_0 it is feasible at every larger gap. The set of feasible gaps is thus an up-set in Δ_0, and its infimum Δ_0^* has precisely the property

claimed — feasible above, infeasible below. By part (i) each feasible Δ_0 is a valid bound, so the family of valid bounds is exactly (Δ_0^*, ∞) together with its infimum in the limit; the tightest is Δ_0^* itself. No functional certifies a value below Δ_0^*, by definition of the infimum.

(iii) Fix $\Delta_0 = \Delta_0^*$ and suppose the cone $\mathcal{C} = \mathcal{C}(S(\Delta_0^*)) = \{\sum_i p_i \vec{F}_{\Delta_i,\ell_i} : p_i \geq 0, (\Delta_i, \ell_i) \in S(\Delta_0^*)\}$ is closed. Either $-\vec{F}_{0,0} \in \mathcal{C}$ or it is not. If it is not, then by the separating-hyperplane theorem (Theorem 50.5) there is a functional α with $\alpha[\vec{v}] \geq 0$ for all $\vec{v} \in \mathcal{C}$ and $\alpha[-\vec{F}_{0,0}] < 0$, i.e. $\alpha[F_{0,0}] > 0$; this α is admissible for $S(\Delta_0^*)$, so the dual is feasible *at* Δ_0^*. But feasibility is an open condition in Δ_0 from below only at the threshold: feasibility at Δ_0^* would, by the continuity of the block data in Δ, persist for a range of Δ_0 slightly below Δ_0^*, contradicting that Δ_0^* is the infimum of feasible gaps.[1] Therefore $-\vec{F}_{0,0} \in \mathcal{C}$: there is a nonnegative density solving crossing supported on $S(\Delta_0^*)$. Its lightest scalar cannot lie above Δ_0^* (it is supported there) and cannot lie strictly below (else the same density would solve the program for a smaller gap, making Δ_0^* not the infimum at which the primal first becomes feasible); so it sits exactly at Δ_0^*. The density is the extremal ray of $\mathcal{C}$ on which the limiting optimal functional vanishes, and uniqueness of that ray is the statement that the optimal functional has just enough zeros to pin it down — the extremal-spectrum mechanism that the showcase sections exploit. □

Caution

Part (iii) carries the only genuine hypothesis in the theorem: the closedness of the cone. The block vectors $\vec{F}_{\Delta,\ell}$ for $\Delta \to \infty$ accumulate, and whether the cone they generate is closed — equivalently, whether the supremum gap is *attained* by an honest unitary solution rather than only approached by a sequence of them — is a real analytic question, not a formality. This is the precise technical residue of the honest statement made in Section 49.6.6: the *exclusions* of parts (i)–(ii) are unconditional theorems, whereas the *existence* of a saturating theory in part (iii) is conditional. In every numerical computation the program is solved in a finite derivative basis, where the cone is a finitely generated (hence closed) polyhedral cone and strong duality holds exactly; the subtlety lives only in the $\Lambda \to \infty$ limit.

The optimal functional admits a sharp geometric characterization that is the engine of every numerical search and the reason the bounds can be *saturated*. At the threshold the certificate α is nonnegative on the whole admissible spectrum

[1] Concretely: an admissible functional has $\alpha[F_{\Delta,0}] \geq 0$ for all $\Delta \geq \Delta_0^*$ with strict inequality except on the finite extremal set where it touches zero; since $\alpha[F_{\Delta,0}]$ is continuous in Δ, the constraints survive lowering the gap below the lowest extremal zero, so the program is feasible on an interval reaching below Δ_0^* — unless that extremal zero sits exactly at Δ_0^*, which is the saturation case.

yet, being marginal, must touch zero exactly on the operators of the saturating solution: the function $\Delta \mapsto \alpha[F_{\Delta,\ell}]$ has a *double zero* (a tangency, since it cannot go negative) at each dimension present in the extremal spectrum. Reading off those tangency locations from the optimal functional reconstructs the extremal spectrum — this is how the bootstrap, having only proved an inequality, nonetheless outputs a candidate set of dimensions, and it is the mechanism behind the kink that locates the Ising point in Fig. 49.7.

We can see the tangency concretely without leaving the by-hand computation of the foundations chapter. There the two-derivative functional $\alpha = \Lambda_3 - c\,\Lambda_1$ — built from the first and third diagonal derivatives $\Lambda_1 = \partial_z F(z,z)|_{1/2}$ and $\Lambda_3 = \partial_z^3 F(z,z)|_{1/2}$ of Eq. (49.113) — was applied to $\langle\sigma\sigma\sigma\sigma\rangle$ in $d=2$ at $\Delta_\phi = \frac{1}{8}$, with the block data tabulated in Table 49.2. Admissibility on the identity requires $\alpha[F_{0,0}] = \Lambda_3[F_{0,0}] - c\,\Lambda_1[F_{0,0}] > 0$; with the exact identity vector $(\Lambda_1,\Lambda_3)[F_{0,0}] = (-0.840896, -4.414706)$ this reads $-4.414706 + 0.840896\,c > 0$, i.e. $c > 4.414706/0.840896 = \frac{21}{4} = 5.25$. Admissibility on a block requires $\Lambda_3[F_{\Delta,\ell}] \geq c\,\Lambda_1[F_{\Delta,\ell}]$, hence (every block having $\Lambda_1 > 0$) $c \leq \Lambda_3[F_{\Delta,\ell}]/\Lambda_1[F_{\Delta,\ell}]$. An admissible functional exists precisely when the window

$$\frac{21}{4} \;<\; c \;\leq\; \min_{(\Delta,\ell)\in S(\Delta_0)} \frac{\Lambda_3[F_{\Delta,\ell}]}{\Lambda_1[F_{\Delta,\ell}]} \tag{50.28}$$

is nonempty. The minimizing ratio is attained by the scalar at the bottom of the gap, and the optimum Δ_0^* of Theorem 50.11 is the value at which this ratio first equals the identity ratio $\frac{21}{4}$ — exactly the tangency condition, the optimal α touching zero on the threshold scalar. Solving $\Lambda_3[F_{\Delta_0,0}]/\Lambda_1[F_{\Delta_0,0}] = \frac{21}{4}$ with the explicit $d=2$ blocks gives

$$\Delta_0^*\big(\Delta_\phi = \tfrac{1}{8}\big) = 1.2826 \qquad (d=2,\ \text{two derivatives}), \tag{50.29}$$

the sharp two-derivative optimum refining the feasibility estimate $\Delta_0 \lesssim 1.28$ of Theorem 49.23. The two-dimensional Ising model has $\Delta_\varepsilon = 1 < 1.2826$, as a genuine CFT must. The point to register is structural: the bound is not the value at which we happened to find a functional, but the value at which the *best* functional becomes marginal, and that marginality is a tangency that simultaneously certifies the bound and names the spectrum that saturates it.

50.2.3 Monotonicity: more assumptions, smaller region

Bootstrap bounds are useful only because they behave predictably under the addition of information. The governing fact, already used inside the proof of Theorem 50.11, deserves to be isolated, because it is what guarantees that imposing physical inputs — a gap, an isolated operator, a second correlator — can never enlarge the space of allowed theories.

Theorem 50.12 (Monotonicity of exclusion)**.** *Let S and S' be two candidate spectra for the same external dimension Δ_ϕ with $S \subseteq S'$. Then:*

(i) *Every functional admissible for S' is admissible for S. Consequently, if S' is excludable (admits an admissible functional) then so is S.*

(ii) *The set of conformal data consistent with crossing under the assumption "the spectrum lies in S" is contained in the set consistent under "the spectrum lies in S'."*

In particular, strengthening any assumption — raising a gap, removing operators, adding a selection rule, or imposing the existence of a specified operator — can only shrink the allowed region, and the optimal bounds it produces are at least as tight.

Proof. (i) Admissibility for S' requires $\alpha[F_{0,0}] > 0$ and $\alpha[F_{\Delta,\ell}] \geq 0$ for every $(\Delta, \ell) \in S'$. Since $S \subseteq S'$, the same α satisfies the positivity condition on the smaller index set S as well, and the identity condition is unchanged; hence α is admissible for S. The implication "S' excludable $\Rightarrow$ S excludable" is the contrapositive direction one wants: a certificate against the larger spectrum is automatically a certificate against the smaller one.

(ii) A solution of crossing whose non-identity spectrum lies in S is, in particular, a solution whose spectrum lies in $S' \supseteq S$; so the set of S-consistent data is a subset of the S'-consistent data. Equivalently, by part (i), any region of data-space that an S'-certificate excludes is also excluded by viewing it as an S-problem, so the S-allowed region is the smaller. □

Two consequences organize everything that follows. First, *the bound improves monotonically with computational effort.* Enlarging the derivative basis in Eq. (50.23) enlarges the space of candidate functionals α; since a larger search can only find more certificates, the excluded region grows and the bounds tighten, never loosen, as the truncation order Λ increases. The true allowed region is the intersection over all Λ, approached from outside. This is why bootstrap exclusions are quoted as conservative outer bounds that converge downward, and why the $d = 2$ two-derivative gap Eq. (50.29) sat above the exact Ising value and tightened toward it as derivatives were added. Second, *physical inputs are monotone.* Assuming a gap in a sector, declaring an operator isolated, or — the decisive step in the showcase — coupling several correlators that share OPE data, each restricts the candidate spectrum or adds constraints, and by Theorem 50.12 each can only contract the allowed region. The closing of the open single-correlator wedge of Fig. 49.7 into a compact island when the mixed-correlator system is imposed is precisely this principle in action: more equations, sharing more data, is a smaller S (more constraints), hence a smaller region.

Physical Insight

Monotonicity is what makes the bootstrap a *measurement*, not a fit. A fit can wander when one adds data; a bootstrap region can only shrink. Each physical assumption one is willing to make — the number of relevant operators, the global symmetry, the presence of a conserved current — is a knob that tightens the island, and one knows in advance the tightening is honest because removing possibilities cannot create new allowed theories. The art is to add only assumptions one genuinely believes, since the resulting bound is a theorem *conditional* on them.

50.2.4 Bounds on OPE coefficients

The same machinery bounds an individual OPE coefficient. Single out an operator $\mathcal{O}_\star$ of fixed quantum numbers $(\Delta_\star, \ell_\star)$ that the theory is assumed to contain, and ask how large or how small its squared coefficient $p_\star = C^2_{\phi\phi\mathcal{O}_\star}$ can be. Apply a functional α to the sum rule and isolate the chosen term, as in Eq. (49.104):

$$p_\star\, \alpha[F_{\Delta_\star,\ell_\star}] \;=\; -\,\alpha[F_{0,0}] \;-\; \sum_{\mathcal{O}\neq\mathbf{1},\mathcal{O}_\star} p_{\mathcal{O}}\, \alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]. \tag{50.30}$$

The remaining operators $\mathcal{O} \neq \mathbf{1}, \mathcal{O}_\star$ range over the admissible spectrum (all even-spin unitary operators allowed by whatever gaps are assumed), with unknown nonnegative weights $p_{\mathcal{O}}$. Choosing the sign of α on those operators converts Eq. (50.30) into a one-sided inequality.

Theorem 50.13 (Two-sided OPE-coefficient bounds)**.** *Fix Δ_ϕ, a candidate spectrum, and a target operator $\mathcal{O}_\star$ in it.*

(i) (Upper bound.) *If α satisfies the normalization $\alpha[F_{\Delta_\star,\ell_\star}] = 1$ and the positivity $\alpha[F_{\Delta,\ell}] \geq 0$ for every admissible operator other than $\mathbf{1}$ and $\mathcal{O}_\star$, then in any unitary crossing-symmetric CFT of the assumed type*

$$p_\star \;\leq\; -\,\alpha[F_{0,0}]. \tag{50.31}$$

Minimizing $-\alpha[F_{0,0}]$ over all such α is a linear program whose optimum is the tightest upper bound the method proves.

(ii) (Lower bound.) *If α satisfies $\alpha[F_{\Delta_\star,\ell_\star}] = 1$ and the reversed positivity $\alpha[F_{\Delta,\ell}] \leq 0$ for every admissible operator other than $\mathbf{1}$ and $\mathcal{O}_\star$, then*

$$p_\star \;\geq\; -\,\alpha[F_{0,0}]. \tag{50.32}$$

Maximizing $-\alpha[F_{0,0}]$ over all such α is the tightest lower bound the method proves.

Proof. *(i)* With $\alpha[F_{\Delta_\star,\ell_\star}] = 1$ the left side of Eq. (50.30) is $p_\star$. On the right, every term of the sum is a product $p_{\mathcal{O}}\,\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]$ of two nonnegative numbers — $p_{\mathcal{O}} \geq 0$ by unitarity, $\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}] \geq 0$ by hypothesis — so the sum is ≥ 0 and is subtracted; hence $p_\star \leq -\alpha[F_{0,0}]$. The bound is valid for any admissible α, and since the left side is fixed while the right depends on α, the smallest attainable value of $-\alpha[F_{0,0}]$ subject to the constraints is the sharpest bound. The constraints (one normalization equality, a continuum of inequalities) and the linear objective make this a semi-infinite linear program.

(ii) Now each term $p_{\mathcal{O}}\,\alpha[F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}]$ is a nonnegative weight times a nonpositive functional value, hence ≤ 0; subtracting a quantity ≤ 0 adds a quantity ≥ 0, so the right side of Eq. (50.30) is $\geq -\alpha[F_{0,0}]$, giving $p_\star \geq -\alpha[F_{0,0}]$. Maximizing $-\alpha[F_{0,0}]$ over admissible α gives the tightest lower bound. □

The structure is uniform with the gap bound: a bound is the optimum of a linear program over the cone of functionals, and which quantity is bounded is selected by what is placed in the objective and how the normalization is fixed. There is one asymmetry between the two directions worth flagging. The upper bound Eq. (50.31) is essentially always nontrivial: requiring positivity on the rest of the spectrum and reading off the identity is the same separation as before. The lower bound Eq. (50.32) is nontrivial — strictly positive — only when the target operator is *isolated*, that is, when the assumed spectrum has gaps separating $\mathcal{O}_\star$ from its neighbours, so that the contribution of $\mathcal{O}_\star$ cannot be mimicked by a nearby continuum and crossing genuinely *requires* a minimal amount of it. Without such isolation the best provable lower bound is zero: a unitary solution can always let $p_\star \to 0$ by reshuffling weight onto operators arbitrarily close to $\mathcal{O}_\star$. This is the OPE-coefficient avatar of the same physics that makes the dimension bound one-sided: positivity alone caps coefficients from above for free, but pins them from below only against a backdrop of assumed structure.

50.2.5 A two-sided OPE bound computed by hand

The foundations chapter extracted a dimension bound by hand at this same $d = 2$ Ising point (Theorem 49.23). We now extract the complementary number — a rigorous interval for an OPE coefficient — in the same solvable laboratory, so that both halves of the linear-functional method have been seen to produce actual numbers before any computer is invoked. We work again with $\langle\sigma\sigma\sigma\sigma\rangle$ in $d = 2$ at $\Delta_\phi = \frac{1}{8}$, the spin field of the two-dimensional Ising model, whose exact data we know independently from the free-fermion solution of Chapter 48: the lightest $\mathbb{Z}_2$-even scalar is the energy operator ε at $\Delta_\varepsilon = 1$, appearing with $\lambda_{\sigma\sigma\varepsilon} = \frac{1}{2}$, so that $p_\varepsilon = \lambda^2_{\sigma\sigma\varepsilon} = \frac{1}{4}$. The bootstrap does not know this number; we will bracket it from above and below using only crossing and positivity, then check that the exact value lands inside.

Worked Example 50.14 (A rigorous interval for $\lambda^2_{\sigma\sigma\varepsilon}$ in the $2d$ Ising correlator)**.** The target is $\mathcal{O}_\star = \varepsilon$, the scalar at $(\Delta_\star, \ell_\star) = (1, 0)$. We reuse the exact diagonal block data assembled in Table 49.2, the entries of which are the values of the two basis functionals $\Lambda_1 = \partial_z F(z,z)|_{1/2}$ and $\Lambda_3 = \partial_z^3 F(z,z)|_{1/2}$ of Eq. (49.113) on each convolved block. The two vectors we need are

$$(\Lambda_1, \Lambda_3)[F_{0,0}] = (-0.840896,\ -4.414706), \qquad (\Lambda_1, \Lambda_3)[F_{1,0}] = (\ \ 2.827973,\ \ \ 8.413183), \tag{50.33}$$

the first the identity, the second the target ε; the assumed spectrum is the identity, ε at $\Delta = 1$, the next $\mathbb{Z}_2$-even scalar at $\Delta \geq 2$ (the actual gap is to $\Delta_{\varepsilon'} = 4$, but $\Delta \geq 2$ already suffices and we will see the bound is insensitive to its precise location), and the even-spin tower $\ell \geq 2$ with $\Delta \geq \ell$ (the $d = 2$ unitarity floor).

Step 1: the upper bound from a single derivative. The simplest possible functional is $\alpha = \Lambda_1$ alone — one derivative. Its admissibility for the OPE upper bound (Theorem 50.13 (i)) requires $\Lambda_1[F_{\Delta,\ell}] \geq 0$ on every operator other than $\mathbf{1}$ and ε. This positivity holds throughout the unitary region above the gap: evaluating the diagonal first derivative on the explicit $d = 2$ blocks gives $\Lambda_1[F_{\Delta,0}] > 0$ for every scalar $\Delta \geq 1$ and $\Lambda_1[F_{\Delta,\ell}] > 0$ for every even-spin operator with $\Delta \geq \ell \geq 2$ — the value decreases monotonically toward zero as $\Delta \to \infty$ in each spin but never changes sign — while on the identity it is negative, $\Lambda_1[F_{0,0}] = -0.840896 < 0$. Thus Λ_1 is a valid certificate: it is nonnegative on the entire admissible spectrum and negative on the identity, the exact pattern Theorem 50.13 (i) demands. Normalizing on the target, $\alpha = \Lambda_1 / \Lambda_1[F_{1,0}]$ so that $\alpha[F_{1,0}] = 1$, the upper bound Eq. (50.31) reads

$$p_\varepsilon \ \leq\ -\alpha[F_{0,0}] \ =\ \frac{-\Lambda_1[F_{0,0}]}{\Lambda_1[F_{1,0}]} \ =\ \frac{0.840896}{2.827973} \ =\ 0.29735. \tag{50.34}$$

A single derivative, evaluated on two blocks, already proves that no unitary $2d$ CFT with a dimension-$\frac{1}{8}$ scalar can have $\lambda^2_{\sigma\sigma\varepsilon}$ exceeding 0.297.

Step 2: the lower bound from two derivatives. For a lower bound (Theorem 50.13 (ii)) we need a functional that is *nonpositive* on the rest of the spectrum while equal to one on ε, and we maximize $-\alpha[F_{0,0}]$. A single derivative cannot do this: Λ_1 is positive on both ε and the rest, so no rescaling makes it ≤ 0 on the rest and $+1$ on ε. Two derivatives suffice. Write $\alpha = a_1\Lambda_1 + a_3\Lambda_3$ and impose $\alpha[F_{1,0}] = 1$, i.e. $2.827973\, a_1 + 8.413183\, a_3 = 1$, leaving a one-parameter family. The constraint $\alpha[F_{\Delta,\ell}] \leq 0$ on every other operator is, for each, a half-line in that parameter; their intersection is the feasible segment, and the objective $-\alpha[F_{0,0}] = 0.840896\, a_1 + 4.414706\, a_3$ is linear, so its maximum sits at an endpoint of the segment where one further operator's constraint becomes active. Carrying out the intersection over the gapped scalar channel and the spin tower (the binding constraints are the near-threshold spinning operators and the gap scalar, the high-dimension tail being harmless because the blocks decay), the optimum

is attained at $(a_1, a_3) \simeq (0.4379, -0.02835)$, giving

$$p_\varepsilon \;\geq\; -\alpha[F_{0,0}] \;=\; 0.24312. \tag{50.35}$$

That this lower bound is strictly positive is the statement that the gapped spectrum genuinely *needs* the energy operator: crossing cannot be solved with ε's coefficient driven to zero once a gap is assumed above it, because the identity's contribution to the sum rule must be balanced and only ε is available low enough to do so.

Step 3: the interval and the check. Combining Eqs. (50.34) and (50.35), the linear-functional method at two derivatives proves, for every unitary crossing-symmetric $2d$ CFT with a scalar of dimension $\frac{1}{8}$ whose self-OPE has the assumed gap structure,

$$\boxed{0.243 \;\leq\; \lambda^2_{\sigma\sigma\varepsilon} \;\leq\; 0.297} \qquad (d = 2,\ \Delta_\phi = \tfrac{1}{8},\ \text{two derivatives}). \tag{50.36}$$

The exact two-dimensional Ising value $\lambda^2_{\sigma\sigma\varepsilon} = \frac{1}{4} = 0.250$ lies inside this interval, as it must, since the Ising CFT is a genuine unitary solution of crossing and therefore obeys every consequence of Theorem 50.13. Figure 50.2 displays the geometry: in the plane of the two functional values (Λ_1, Λ_3), the block vectors of the admissible spectrum all lie in the half-plane $\Lambda_1 \geq 0$, the identity vector lies opposite, and the upper-bound functional is essentially the vertical separator $\Lambda_1 = 0$ that places the identity alone on the negative side.

Step 4: why the bound is loose, and how it tightens. The interval Eq. (50.36) is honest but wide because two derivatives is a coarse instrument; the constraint $\alpha[F_{\Delta,\ell}] \gtrsim 0$ has been imposed only through the first and third diagonal Taylor coefficients of each block at the crossing point. By Theorem 50.12 and the monotone-improvement principle, enlarging the derivative basis can only shrink the interval. Adding the next surviving diagonal and off-diagonal derivatives drives both ends toward the Ising value, the upper bound descending and the lower bound rising, until at the derivative orders reached in the semidefinite computation of Section 50.3 the interval closes around 0.250 to many digits — the same convergence the dimension bound exhibited in passing from Eq. (50.29) toward $\Delta_\varepsilon = 1$. The lesson is the one the whole method turns on: a hand computation already *brackets* the exact answer rigorously, and machinery only narrows the bracket.

Physical Insight

The two worked numbers — the dimension bound Eq. (50.29) and the coefficient interval Eq. (50.36) — are the same algorithm pointed at different unknowns. In both, crossing is a demand that a fixed vector lie in a cone; in both, a functional that is nonnegative on the cone yet of definite

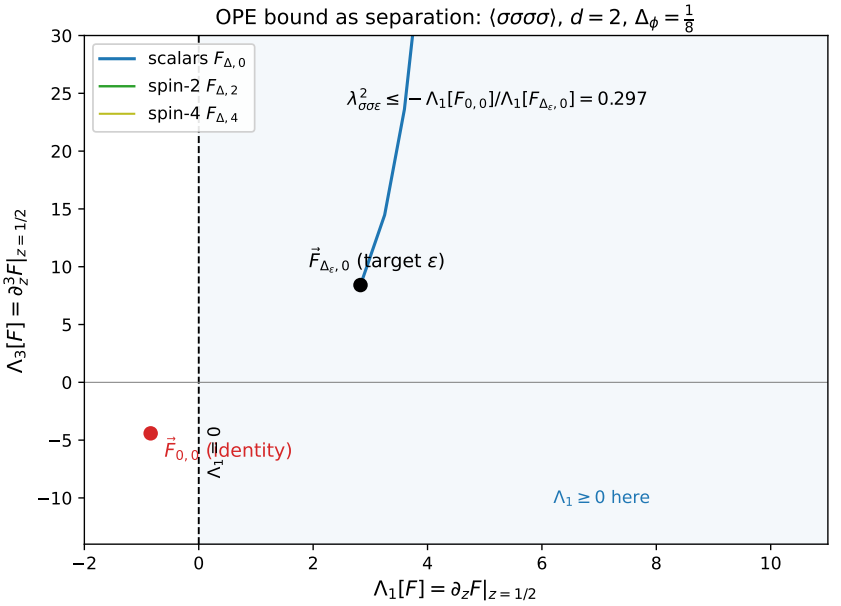


Figure 50.2: The OPE-coefficient bound as a separation, drawn in the real plane of the two diagonal functionals (Λ_1, Λ_3) for $\langle\sigma\sigma\sigma\sigma\rangle$ in $d = 2$ at $\Delta_\phi = \frac{1}{8}$ (Theorem 50.14). The blue, green, and olive curves are the trajectories of the convolved-block vectors $\vec{F}_{\Delta,\ell}$ as the dimension Δ runs over the scalar, spin-2, and spin-4 channels; every admissible block has $\Lambda_1 = \partial_z F|_{1/2} > 0$, so the whole spectrum lies in the right half-plane. The identity vector $\vec{F}_{0,0}$ (red) lies in the left half-plane. The single-derivative functional $\alpha = \Lambda_1$, whose zero-set is the dashed vertical line $\Lambda_1 = 0$, therefore separates the identity from the entire spectrum and certifies the upper bound $\lambda^2_{\sigma\sigma\varepsilon} \le -\Lambda_1[F_{0,0}]/\Lambda_1[F_{\Delta_\varepsilon,0}] = 0.297$ of Eq. (50.34). The target operator ε at $(\Delta, \ell) = (1, 0)$ is marked.

sign on the identity is a certificate; in both, the optimal certificate touches zero on an extremal spectrum and that spectrum is the theory saturating the bound. The only difference is bookkeeping: for a dimension bound we sweep the gap and ask when a certificate first exists; for a coefficient bound we fix the gap, normalize on the target, and optimize the identity's image. That a single derivative and a sign already corner the energy operator of the Ising model between 0.24 and 0.30 is the whole conceptual payload of the bootstrap, visible before any large computation.

We have now turned the separation principle into its two working algorithms and proved that each computes the genuine optimum of a linear program over functionals: the dimension bound is the threshold at which the optimal certificate

becomes marginal (Theorem 50.11), the coefficient bounds are the extrema of the identity's image under normalized certificates (Theorem 50.13), and both tighten monotonically as the functional basis grows (Theorem 50.12). What remains is the obstruction that kept the worked examples small. Each bound is the optimum of a *semi-infinite* linear program: finitely many unknown coefficients a_{mn}, but a continuum of positivity constraints $\alpha[F_{\Delta,\ell}] \geq 0$ indexed by the dimension Δ over a half-line in every spin ℓ. By hand we could only enforce that continuum through a one-parameter scan; a realistic functional with hundreds of components meets a continuum it cannot scan. The next section removes this obstruction, showing how expanding the blocks in a radial coordinate, truncating to a finite derivative basis, and approximating the blocks by polynomials in Δ converts "a functional positive over a continuum of dimensions in each spin" into the positive-semidefiniteness of finitely many matrices — the reduction of the semi-infinite program to a finite semidefinite program, with the duality that keeps the bounds rigorous.

50.3 From Functionals to Semidefinite Programs

The previous section turned the separation principle into two concrete recipes — maximize a gap, bound an OPE coefficient — and ran a toy functional by hand to produce an honest number. But the object that recipe asks us to find is not yet something a computer can search for. Theorem 49.21 demands a linear functional α that is nonnegative on *every* convolved block $\widehat{F}_{\Delta,\ell}$ in a candidate spectrum, and a candidate spectrum is a continuum: for each even spin ℓ the dimension Δ ranges over an entire half-line $[\Delta_\ell^{\text{unit}}, \infty)$, and the spin itself runs over the infinite set $\ell = 0, 2, 4, \ldots$. The search is therefore over finitely many functional coefficients a_{mn} subject to *infinitely many* linear inequalities — one per point of an infinite-dimensional index set. This is a *semi-infinite program*, and a brute-force scan over a grid of (Δ, ℓ) can only ever sample the constraints, never prove them. A sampled functional that happens to be positive at a thousand test points may dip below zero between them, and then the "exclusion" it certifies is a fiction.

This section removes the gap between a numerically plausible functional and a genuine theorem. The route has four reductions, and each is a finite, controlled approximation of an infinite object:

1. *Finitely many derivatives.* A functional is a finite combination of Taylor derivatives of the crossing equation at the symmetric point $z = \bar{z} = \frac{1}{2}$. We bound the truncation error using the geometric convergence of the radial expansion (Theorem 49.12), so a finite derivative basis is provably enough.

2. *Finitely many spins.* The contributions of high-spin operators at the symmetric point are exponentially suppressed in the twist, so the spin sum

can be cut at a finite L with a controlled remainder.

3. *Rational blocks in* Δ. The blocks are meromorphic in the exchanged dimension, with all poles below the unitarity bound; approximating each derivative component by a positive prefactor times a polynomial in Δ turns the half-line of constraints in each spin into the nonnegativity of one univariate polynomial.

4. *Polynomials to matrices.* A univariate polynomial nonnegative on a half-line is a sum of squares with positive-semidefinite Gram matrices (Markov–Lukács). The functional search becomes a *semidefinite program*.

The pipeline is drawn in Fig. 50.3. What makes the endpoint a theorem rather than a numerical impression is convex duality: any *dual-feasible* point of the semidefinite program yields a rigorous bound regardless of how the solver behaves, and a feasible point can be certified exactly in rational arithmetic. We close by making that precise — a proof of weak duality and an account of how the solver SDPB and interval arithmetic convert a floating-point functional into a genuine proof.

50.3.1 The semi-infinite program

We start by writing the inherited problem with no slack. Fix the external dimension Δ_ϕ and a candidate spectrum S — a closed subset of the half-lines

$$\mathcal{U} \;=\; \bigcup_{\ell=0,2,4,\dots} \{\ell\} \times [\Delta_\ell^{\text{unit}}, \infty), \qquad \Delta_\ell^{\text{unit}} = \begin{cases} \frac{d-2}{2}, & \ell = 0, \\ \ell + d - 2, & \ell \geq 1, \end{cases} \tag{50.37}$$

the unitarity region of Eq. (49.81). By Theorem 49.21, S is excluded if we can exhibit a linear functional α of the form Eq. (49.98),

$$\alpha[f] \;=\; \sum_{m,n} a_{mn}\, \partial_z^m \partial_{\bar z}^n f(z,\bar z)\Big|_{z=\bar z=1/2}, \qquad a_{mn} \in \mathbb{R}, \tag{50.38}$$

satisfying the admissibility conditions of Theorem 49.21: $\alpha[F_{0,0}] > 0$ and $\alpha[F_{\Delta,\ell}] \geq 0$ for every $(\Delta,\ell) \in S$. Equivalently, in the normalized variables, divide through by the identity contribution and pose the search on the F-vector $\widehat{F}_{\Delta,\ell}$ of Eq. (49.79). We work with $\widehat{F}$ throughout, since $\widehat{F}_{0,0} = 1$ makes the normalization $\alpha[\,1\,] = 1$ a single linear equation and the remaining constraints homogeneous.

Two infinities obstruct a direct search.

Infinitely many variables. The functional Eq. (50.38) is a combination of *all* derivatives. We will keep only those of total order $m + n \leq \Lambda$; the number

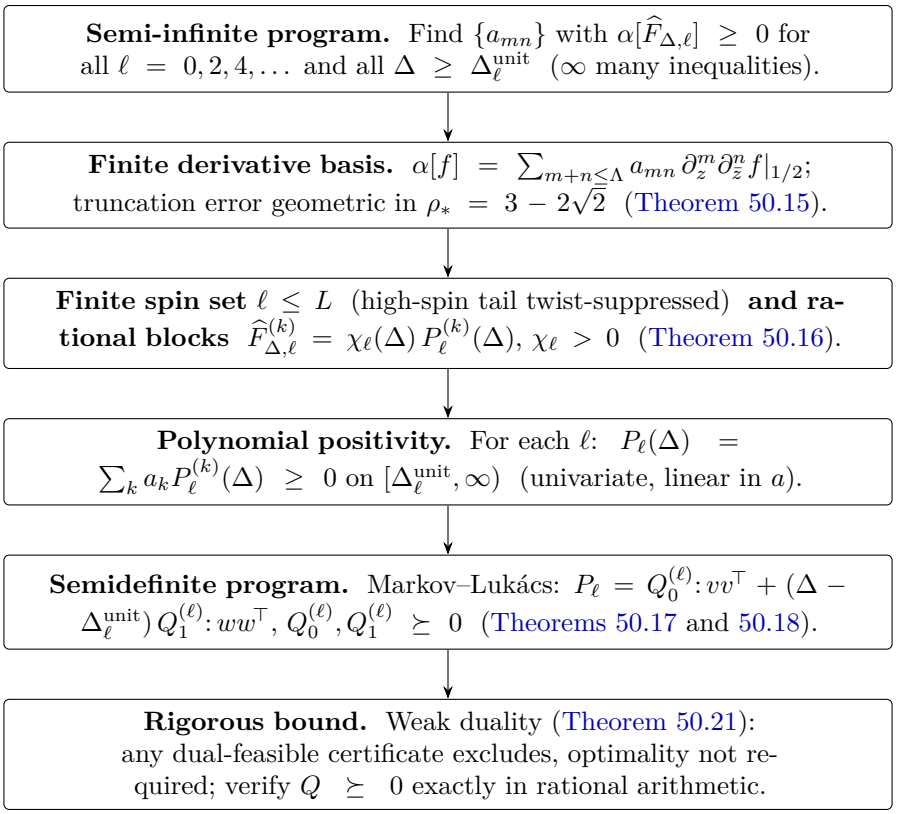


Figure 50.3: The reduction of the semi-infinite functional search to a finite semidefinite program. Each arrow is a controlled, finite approximation of an infinite object; the final arrow is convex duality, which is what makes the machine produce theorems rather than numerical impressions.

of free coefficients is then finite, $N_\Lambda = \frac{1}{2}(\Lambda+1)(\Lambda+2)$ before symmetry reduction. The cost is that we search a finite-dimensional slice of the space of functionals; Section 50.3.2 shows this slice is rich enough that its error is geometrically small, and the *caution* after Theorem 49.21 already records the logical consequence: a finite basis yields a rigorous outer bound that tightens monotonically as Λ grows.

Infinitely many constraints. Even with Λ fixed, the conditions $\alpha[\widehat{F}_{\Delta,\ell}] \geq 0$ form a continuum indexed by $(\Delta, \ell) \in S$. This is the genuinely hard infinity, and the one a grid scan cannot honestly resolve. Removing it is the technical heart of the section: Sections 50.3.3 to 50.3.5 replace the continuum of point constraints in each spin by a *single* matrix inequality.

It is worth saying once, plainly, why the second infinity is the dangerous one. The function $\Delta \mapsto \alpha[\widehat{F}_{\Delta,\ell}]$ is, for fixed ℓ and a finite functional, an explicit but

complicated transcendental function of Δ. Demanding it stay nonnegative on a half-line is a statement about uncountably many values, and no finite sample establishes it. The entire reduction below exists to recast that statement in a form — the nonnegativity of a polynomial, hence the positive-semidefiniteness of a matrix — whose truth is decidable by a finite, exact computation.

50.3.2 The derivative basis and the radial acceleration

Why expand at $z = \bar{z} = \frac{1}{2}$, and why is a finite number of derivatives enough? The answer to both is the radial coordinate of Eq. (49.61),

$$\rho(z) = \frac{z}{\left(1 + \sqrt{1 - z}\right)^2}, \qquad z = \frac{4\rho}{(1+\rho)^2}, \tag{50.39}$$

which maps the cut z-plane biholomorphically onto the open unit disk and sends the crossing-symmetric point to $\rho_* = \rho(\frac{1}{2}) = 3 - 2\sqrt{2} \approx 0.1716$, deep inside. In the radial frame a conformal block is the convergent power series Eq. (49.63),

$$G_{\Delta,\ell}(r,\eta) = \sum_{n=0}^{\infty} \sum_{j} B_{n,j}^{(\Delta,\ell)}\, r^{\Delta+n}\, C_j^{(\nu)}(\eta), \qquad \rho = re^{i\theta},\ \eta = \cos\theta,\ \nu = \tfrac{d-2}{2}, \tag{50.40}$$

and Theorem 49.12 guarantees convergence for $r < 1$. At the symmetric point $\bar{z} = z^* = \frac{1}{2}$ the radial pair is real, $\rho = \bar{\rho} = \rho_*$, $\eta = 1$, $\theta = 0$. The crossing equation must be imposed somewhere, and the symmetric point is the natural choice because it is the fixed point of the crossing reflection and because it is the point of *fastest* convergence: every block, of every dimension and spin, is there a power series in a number of modulus 0.17.

Symmetry reduces the basis. The convolved block $F_{\Delta,\ell}$ is antisymmetric under $u \leftrightarrow v$ (Theorem 49.15), i.e. under $(z,\bar{z}) \mapsto (1-z, 1-\bar{z})$, and the normalized $\widehat{F}_{\Delta,\ell}$ is symmetric. The symmetric point $z = \bar{z} = \frac{1}{2}$ is the fixed point of the reflection $z \mapsto 1 - z$. Introduce the centered variables $a = z - \frac{1}{2}$, $b = \bar{z} - \frac{1}{2}$. Under crossing $(a,b) \mapsto (-a,-b)$. A symmetric function $\widehat{F}$ then has a Taylor series in even total powers of (a,b) only, while the antisymmetric F has only odd ones. Moreover $\widehat{F}$ is symmetric in $z \leftrightarrow \bar{z}$ (it depends on the unordered pair), so it is invariant under $a \leftrightarrow b$. Both symmetries cut the derivative basis: the independent derivatives of $F_{\Delta,\ell}$ at the symmetric point are

$$\partial_a^m \partial_b^n F_{\Delta,\ell}\big|_{a=b=0}, \qquad m+n \text{ odd}, \quad m \geq n, \tag{50.41}$$

since even-order derivatives of an odd function vanish at the fixed point, and $m \leftrightarrow n$ gives the same number by $z \leftrightarrow \bar{z}$ symmetry. (Working with F rather than $\widehat{F}$ is a convention; the two are related by the fixed kinematic factor $F_{0,0} = v^{\Delta_\phi} - u^{\Delta_\phi}$, itself an odd function, so the derivative data carry the same information.)

Truncating at total order $m+n \le \Lambda$ leaves a finite list; we collect the surviving derivatives into a vector

$$\vec{F}^{(\Lambda)}_{\Delta,\ell} \;=\; \Big(\partial_a^m \partial_b^n F_{\Delta,\ell}\big|_0 \;:\; m+n \le \Lambda,\; m+n \text{ odd},\; m \ge n \Big), \tag{50.42}$$

and a functional in this basis is a real covector $\vec{a}$ acting by the dot product $\alpha[F_{\Delta,\ell}] = \vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta,\ell}$. The integer Λ (or the count of components, often written $N_{\max}$ in the literature) is the single knob that controls the precision of the whole computation.

Why truncation is safe. It remains to show that discarding the high-order derivatives is a controlled error, not a leap of faith. The mechanism is that the block, viewed as a function of ρ at the symmetric configuration, is analytic on a disk strictly larger than $|\rho| = \rho_*$, so its Taylor coefficients — the derivatives we are truncating — decay geometrically. We make this quantitative.

Theorem 50.15 (Geometric control of the derivative truncation)**.** *Fix a spin ℓ and a dimension $\Delta \ge \Delta_\ell^{\text{unit}}$. Regard the block restricted to the diagonal as a function $g(\rho) \equiv G_{\Delta,\ell}(z(\rho), z(\rho))$ of the single radial variable. Then g is holomorphic on the disk $|\rho| < 1$, and for any radius $\rho_* < r_1 < 1$ its Taylor coefficients about $\rho = \rho_*$ obey the Cauchy bound*

$$\left| \frac{1}{m!} \frac{d^m g}{d\rho^m}\Big|_{\rho_*} \right| \;\le\; \frac{M(r_1)}{(r_1 - \rho_*)^m}, \qquad M(r_1) = \max_{|\rho-\rho_*| = r_1 - \rho_*} |g(\rho)| < \infty, \tag{50.43}$$

so that the remainder of the order-Λ Taylor expansion of g at the symmetric point, evaluated at any ρ with $|\rho - \rho_*| \le \delta < r_1 - \rho_*$, is bounded by

$$\big|\, g(\rho) - T_\Lambda g(\rho) \,\big| \;\le\; M(r_1)\, \frac{(\delta/(r_1-\rho_*))^{\Lambda+1}}{1 - \delta/(r_1-\rho_*)}. \tag{50.44}$$

The bound decays geometrically in Λ with ratio $\delta/(r_1-\rho_*) < 1$, uniformly in Δ for Δ in any compact set and (after dividing by the leading $(4\rho_*)^\Delta$) uniformly down the entire half-line.

Proof. By Theorem 49.12 the series Eq. (50.40) converges absolutely for $|\rho| < 1$, uniformly on compact subsets; a uniformly convergent series of holomorphic functions is holomorphic, so g is holomorphic on $|\rho| < 1$. The boundary $|\rho| = 1$ is the image of the crossed-channel singularity at $z = 1$ (Fig. 49.2); inside the disk g has no singularities. Choose r_1 with $\rho_* < r_1 < 1$ and let γ be the circle $|\rho - \rho_*| = r_1 - \rho_*$, which lies inside the unit disk because its farthest point from the origin is at distance $\rho_* + (r_1 - \rho_*) = r_1 < 1$. By the Cauchy integral formula for derivatives — the differentiated form of Theorem 5.23 applied to the holomorphic g —

$$\frac{1}{m!} \frac{d^m g}{d\rho^m}\Big|_{\rho_*} = \frac{1}{2\pi i} \oint_\gamma \frac{g(\rho)\, d\rho}{(\rho - \rho_*)^{m+1}}, \tag{50.45}$$

and bounding the integrand by its maximum modulus $M(r_1)$ over $|\rho-\rho_*|^{m+1} = (r_1-\rho_*)^{m+1}$ times the circumference $2\pi(r_1-\rho_*)$ gives Eq. (50.43). The Taylor remainder is the tail $\sum_{m>\Lambda} c_m(\rho-\rho_*)^m$ with $|c_m| \le M(r_1)(r_1-\rho_*)^{-m}$; summing the geometric majorant from $m=\Lambda+1$ yields Eq. (50.44). Finally, the leading behavior $g(\rho)\sim(4\rho)^\Delta$ from Eq. (49.64) shows that the only Δ-dependence of $M(r_1)$ that grows with Δ is the overall factor $(4r_1)^\Delta$; dividing the block by its leading power $(4\rho_*)^\Delta$ replaces this by $(r_1/\rho_*)^\Delta$ on the contour, which is bounded once r_1 is fixed below 1 and Δ is bounded, and the high-Δ tail is in any case suppressed by the spin and twist counting of Section 50.3.3. The ratio governing the decay, $\delta/(r_1-\rho_*)$, is independent of Δ. □

The content of Theorem 50.15 is the precise sense in which a finite derivative basis "sees" the whole crossing equation. The crossing equation is a statement about $\widehat{F}$ on a neighborhood of the symmetric point; truncating at order Λ replaces $\widehat{F}$ by its Taylor polynomial, and Eq. (50.44) shows the discarded part is smaller than $(\delta/(r_1-\rho_*))^{\Lambda+1}$. With $\rho_*\approx 0.17$, even a modest $r_1\approx 0.45$ gives a ratio $\delta/(r_1-\rho_*)$ well below one over the small disk the bootstrap actually probes, so the number of correct digits grows linearly in Λ. Figure 50.4 plots this for the value of the block; the slope is $-\log_{10}\rho_*\approx 0.77$ digits per term, the geometric rate inherited directly from OPE convergence (Theorem 47.48). This is the rigorous footing under the practitioner's lore that "a dozen terms give many digits."

Physical Insight: convergence is unitarity in disguise

The geometric rate $\rho_*\approx 0.17$ is not a numerical accident; it is the state-operator map (Theorem 47.42) made visible. A conformal block is a sum over the descendant states of one primary, weighted by $\rho^{\Delta+n}$ where $\Delta+n$ is the radial energy of the n-th level (Eq. (47.196)). Reflection positivity bounds the norms of those states polynomially in n, so the geometric weight ρ_*^n wins and the series converges as fast as ρ_*^n shrinks. The same positivity that lets a one-signed functional detect an inconsistency is what makes the functions that functional acts on computable to arbitrary precision. Rigor and tractability come from one source.

50.3.3 Rational approximation of the blocks in the dimension

We have reduced the functional to a finite covector $\vec{a}$ and the block to a finite derivative vector $\vec{F}^{(\Lambda)}_{\Delta,\ell}$. The constraint $\alpha[\widehat{F}_{\Delta,\ell}]\ge 0$ is now the statement that a fixed finite linear combination of the derivative components, with coefficients $\vec{a}$, is nonnegative as Δ ranges over $[\Delta_\ell^{\mathrm{unit}},\infty)$. Each derivative component is still a transcendental function of Δ. The decisive structural fact is that, in the dimension Δ at fixed spin, the block is *meromorphic* with a completely

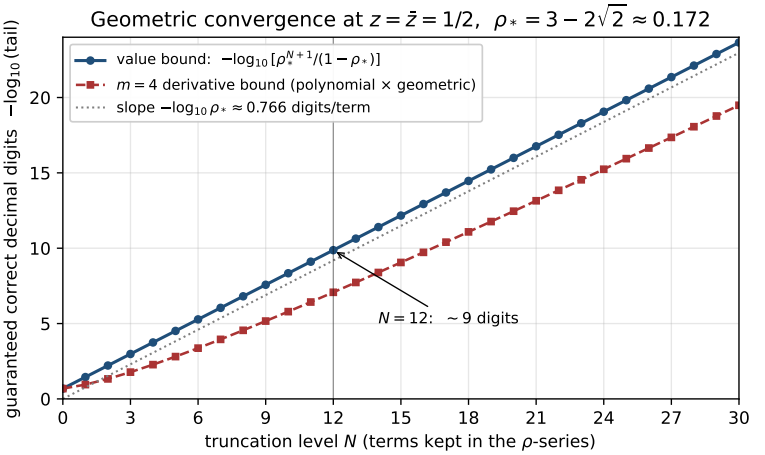


Figure 50.4: Geometric control of the radial truncation at the crossing-symmetric point. The block expansion in ρ converges on the unit disk (Theorem 49.12); the symmetric point sits at $\rho_* = 3 - 2\sqrt{2} \approx 0.172$. Truncating at level N leaves a tail bounded by $\rho_*^{N+1}/(1-\rho_*)$, so the guaranteed number of correct decimal digits grows linearly in N with slope $-\log_{10}\rho_* \approx 0.766$. Derivatives at the symmetric point (Theorem 50.15) carry only an extra polynomial-in-N prefactor and inherit the same geometric rate; the dashed curve shows a representative fourth derivative. The geometric convergence is the quantitative face of OPE convergence (Theorem 47.48).

understood pole structure, and that structure lets us replace each component by a positive prefactor times a polynomial.

Poles from null descendants. The block solves the Casimir equation Eq. (49.29); equivalently it is the resummed exchange of one primary's conformal family, $G_{\Delta,\ell} = \sum_n \rho^{\Delta+n} \times$ (norm-weighted descendant overlaps) as in Section 50.3.2. The overlaps are matrix elements normalized by the Gram (Shapovalov) matrix of the conformal multiplet at dimension Δ. That Gram matrix degenerates exactly when a descendant becomes *null* — when its norm vanishes — and the inverse Gram matrix that enters the block then develops a simple pole. The null dimensions are a discrete set determined by representation theory of $\mathfrak{so}(d+1,1)$: a level-n descendant of a spin-ℓ primary goes null at specific dimensions

$$\Delta = \Delta_i^\star(\ell) \;\in\; \left\{\, 1-\ell-k,\;\; \tfrac{d}{2}-k \;\;(k=1,2,\dots),\;\; \ell+d-1-k \;\;(k=1,\dots,\ell) \,\right\}, \tag{50.46}$$

the three families being the unitarity-violating scalars, the "mixed" series, and the would-be conserved currents; the precise list and the residues are the content of the Casimir recursion of Section 49.2.5. What matters for us are two facts,

both consequences of Theorem 5.23 applied to the analytic continuation in Δ of Section 5.4.1: the poles Eq. (50.46) all lie at dimensions *at or below* the unitarity bound $\Delta_\ell^{\text{unit}}$ (a null or negative-norm descendant is precisely a unitarity violation), and the residue at each pole is a positive multiple of a block of *shifted* dimension and spin. This is the Zamolodchikov-style recursion: the block at dimension Δ is its leading power plus a sum of poles whose residues are lower blocks,

$$G_{\Delta,\ell}(\rho,\eta) \;=\; (4\rho)^{\Delta}\Big[\, C_\ell^{(\nu)}(\eta) \;+\; \sum_i \frac{r_i(\ell)\,(4\rho)^{n_i}}{\Delta-\Delta_i^\star(\ell)}\,\widehat{G}_i(\rho,\eta)\,\Big], \qquad r_i(\ell)>0, \tag{50.47}$$

where $\widehat{G}_i$ is a block of dimension $\Delta_i^\star + n_i$ and shifted spin, and the sum over poles converges. Iterating the recursion and truncating at the same radial level N used for the derivatives gives a finite sum of poles.

Positive prefactor times polynomial. Truncate Eq. (50.47) to a finite pole set $\{\Delta_i^\star\}_{i=1}^p$ and evaluate the derivative components Eq. (50.42) at the symmetric point. Each component is then a finite sum of terms $(4\rho_*)^{\Delta+n_i}/(\Delta-\Delta_i^\star)$ with Δ-independent residues; placing them over a common denominator $\prod_i(\Delta-\Delta_i^\star)$ and factoring out the common exponential $(4\rho_*)^\Delta$ gives the representation

$$\boxed{\widehat{F}_{\Delta,\ell}^{(k)} \;=\; \chi_\ell(\Delta)\,P_\ell^{(k)}(\Delta), \qquad \chi_\ell(\Delta)=\frac{(4\rho_*)^\Delta}{\displaystyle\prod_{i=1}^{p}(\Delta-\Delta_i^\star(\ell))},} \tag{50.48}$$

where k indexes the components of the derivative vector and $P_\ell^{(k)}$ is a polynomial in Δ of degree at most p (one less than the number of poles, plus the contribution of the regular part). The exponential $(4\rho_*)^\Delta$ and the denominator are common to all components; only the polynomials $P_\ell^{(k)}$ differ.

Theorem 50.16 (Rational reduction of the positivity condition)**.** *With the representation Eq. (50.48), the prefactor $\chi_\ell(\Delta)$ is strictly positive for every $\Delta \geq \Delta_\ell^{\text{unit}}$, because $(4\rho_*)^\Delta > 0$ and every factor $\Delta-\Delta_i^\star(\ell)>0$ there, the poles lying below the unitarity bound. Consequently, for a functional $\vec{a}$ the positivity condition is equivalent to the nonnegativity of a single univariate polynomial:*

$$\alpha[\widehat{F}_{\Delta,\ell}] = \chi_\ell(\Delta)\underbrace{\sum_k a_k P_\ell^{(k)}(\Delta)}_{\equiv\; P_\ell(\Delta;\,\vec{a})} \;\geq\; 0 \quad \textit{for all } \Delta\geq\Delta_\ell^{\text{unit}} \iff P_\ell(\Delta;\vec{a})\geq 0 \textit{ on } [\Delta_\ell^{\text{unit}},\infty). \tag{50.49}$$

The polynomial $P_\ell(\Delta;\vec{a})$ is linear in the functional coefficients $\vec{a}$ and of degree $\deg P_\ell \leq p$ (a fixed integer set by the truncation order N and the spin ℓ).

Proof. That $(4\rho_*)^\Delta > 0$ is immediate. For the denominator: every null dimension in Eq. (50.46) satisfies $\Delta_i^\star(\ell) \le \Delta_\ell^{\text{unit}}$ — this is the statement that a descendant going null is a unitarity violation, so all such dimensions sit at or below the unitarity threshold, with the boundary case (a conserved current, $\Delta = \ell + d - 2$) handled by treating it as a limit. Hence for $\Delta \ge \Delta_\ell^{\text{unit}}$ each factor $\Delta - \Delta_i^\star(\ell) \ge 0$, and is strictly positive in the open half-line, so $\chi_\ell(\Delta) > 0$ there. Dividing the inequality $\chi_\ell(\Delta) P_\ell(\Delta; \vec{a}) \ge 0$ by the strictly positive χ_ℓ preserves it, giving the stated equivalence. Linearity in $\vec{a}$ holds because $P_\ell(\Delta; \vec{a}) = \sum_k a_k P_\ell^{(k)}(\Delta)$ is a linear combination of fixed polynomials with the functional coefficients as weights. □

Theorem 50.16 is the conceptual pivot of the whole section. The infinitely many point constraints in the spin-ℓ sector — one for each Δ on a half-line — have collapsed into *one* statement: a polynomial, linear in the unknowns $\vec{a}$, is nonnegative on a half-line. The transcendental dependence on Δ has been quarantined inside the positive prefactor χ_ℓ, which drops out of the inequality. The rational approximation is the only place an approximation enters the *constraint* (the derivative truncation entered the functional), and its error is controlled by the same geometric estimate Eq. (50.44): raising the radial order N adds poles and raises $\deg P_\ell$, and the omitted tail is exponentially small.

The spin sum. The spins $\ell = 0, 2, 4, \ldots$ are still infinite in number, but they are tamed by the same radial geometry. At the symmetric point, a spin-ℓ block of twist $\tau = \Delta - \ell$ contributes at leading order $(4\rho_*)^\Delta C_\ell^{(\nu)}(1)$, and the unitarity bound forces $\Delta \ge \ell + d - 2$ for $\ell \ge 1$, so the lightest contribution from spin ℓ scales as $(4\rho_*)^{\ell+d-2}$ up to the polynomial growth of the Gegenbauer value $C_\ell^{(\nu)}(1) = \binom{\ell+2\nu-1}{\ell}$. Since $4\rho_* = 4(3 - 2\sqrt{2}) \approx 0.686 < 1$, the high-spin contributions are exponentially suppressed, $\sim (0.686)^\ell$, and the spin sum may be truncated at a finite L with a remainder bounded by a geometric tail in L. (The lightcone factorization Eq. (49.67) is the analytic statement behind this suppression: large spin means large twist contribution unless the twist is held fixed, and a fixed-twist family contributes a controlled, computable tail.) One therefore imposes the polynomial-positivity constraint Eq. (50.49) for each $\ell \in \{0, 2, \ldots, L\}$ and bounds the discarded $\ell > L$ analytically. From here the spin index is finite, and we suppress it where convenient.

50.3.4 Positive polynomials on a half-line

We have arrived at the clean mathematical problem: for finitely many spins ℓ, the polynomial $P_\ell(\Delta; \vec{a})$, linear in $\vec{a}$, must be nonnegative on $[\Delta_\ell^{\text{unit}}, \infty)$. We now prove the lemma that makes this tractable: nonnegativity of a univariate polynomial on a half-line is equivalent to a sum-of-squares representation, and a sum of squares is equivalent to a positive-semidefinite Gram matrix. The two facts

together convert a positivity condition over a continuum into a finite-dimensional matrix inequality.

Lemma 50.17 (Sum of squares is positive-semidefinite Gram)**.** *Let $v_m(x) = (1, x, x^2, \dots, x^m)^\top$ be the vector of monomials up to degree m. A polynomial σ of degree $\le 2m$ is a sum of squares of polynomials, $\sigma(x) = \sum_r q_r(x)^2$ with $\deg q_r \le m$, if and only if there is a real symmetric positive-semidefinite matrix $Q \succeq 0$ of size $(m+1) \times (m+1)$ such that*

$$\sigma(x) = v_m(x)^\top Q\, v_m(x). \tag{50.50}$$

Proof. ($\Leftarrow$) If $Q \succeq 0$, diagonalize it by the spectral theorem: $Q = \sum_r \lambda_r u_r u_r^\top$ with eigenvalues $\lambda_r \ge 0$ and orthonormal eigenvectors u_r. Then

$$\sigma(x) = v_m^\top \Big(\sum_r \lambda_r u_r u_r^\top \Big) v_m = \sum_r \lambda_r \, (u_r^\top v_m)^2 = \sum_r \big(\sqrt{\lambda_r}\, u_r^\top v_m(x) \big)^2, \tag{50.51}$$

a sum of squares of the polynomials $q_r(x) = \sqrt{\lambda_r}\, u_r^\top v_m(x)$, each of degree $\le m$.

($\Rightarrow$) If $\sigma = \sum_r q_r^2$ with $\deg q_r \le m$, write the coefficient vector of each q_r in the monomial basis as $\mathbf{q}_r \in \mathbb{R}^{m+1}$, so $q_r(x) = \mathbf{q}_r^\top v_m(x)$. Then

$$\sigma(x) = \sum_r (\mathbf{q}_r^\top v_m)^2 = \sum_r v_m^\top \mathbf{q}_r \mathbf{q}_r^\top v_m = v_m^\top \Big(\underbrace{\sum_r \mathbf{q}_r \mathbf{q}_r^\top}_{Q} \Big) v_m, \tag{50.52}$$

and $Q = \sum_r \mathbf{q}_r \mathbf{q}_r^\top$ is a sum of rank-one positive-semidefinite matrices, hence symmetric and positive-semidefinite. □

The Gram matrix Q in Eq. (50.50) is not unique — distinct Q's can give the same polynomial, because the entries of $v_m v_m^\top$ along each anti-diagonal all equal the same power x^k, so only the anti-diagonal *sums* of Q are fixed by σ. This is exactly the freedom a solver exploits. The constraint that links σ's coefficients to Q is a set of linear equations (one per power of x): writing $\sigma(x) = \sum_k c_k x^k$,

$$c_k = \sum_{i+j=k} Q_{ij}, \qquad k = 0, 1, \dots, 2m, \tag{50.53}$$

the trace of Q against the constant Hankel pattern. The map $Q \mapsto \sigma$ is linear; Theorem 50.17 says its image, intersected with $Q \succeq 0$, is exactly the SOS cone.

Now the half-line. A polynomial nonnegative on all of $\mathbb{R}$ is a sum of squares (in one variable, nonnegativity is equivalent to SOS), but our constraint lives on $[\Delta_\ell^{\text{unit}}, \infty)$, not all of $\mathbb{R}$. The right statement is the classical theorem of Markov and Lukács.

Theorem 50.18 (Markov–Lukács representation on a half-line)**.** *Let p be a real polynomial with $p(x) \geq 0$ for all $x \geq a$. Then p admits the representation*

$$p(x) = \sigma_0(x) + (x - a)\,\sigma_1(x), \tag{50.54}$$

where σ_0 and σ_1 are sums of squares of polynomials. If $\deg p = 2m$ one may take $\deg \sigma_0 \leq 2m$ and $\deg \sigma_1 \leq 2m - 2$; if $\deg p = 2m + 1$ one may take $\deg \sigma_0 \leq 2m$ and $\deg \sigma_1 \leq 2m$.

Proof. By translation it suffices to treat $a = 0$: replace $p(x)$ by $\tilde{p}(y) = p(y + a)$, prove $\tilde{p}(y) = \tilde{\sigma}_0(y) + y\,\tilde{\sigma}_1(y)$ for $\tilde{p} \geq 0$ on $[0, \infty)$, and substitute back $y = x - a$. So assume $p(x) \geq 0$ for all $x \geq 0$.

Consider the set

$$\mathcal{Q} = \{\, s_0(x) + x\,s_1(x) \;:\; s_0, s_1 \text{ sums of squares} \,\}. \tag{50.55}$$

We claim $p \in \mathcal{Q}$. The proof is by factoring p over $\mathbb{C}$ and showing each factor lies in $\mathcal{Q}$, after establishing that $\mathcal{Q}$ is closed under multiplication.

$\mathcal{Q}$ is closed under products. Take $s_0 + xs_1$ and $t_0 + xt_1$ in $\mathcal{Q}$. Their product is

$$(s_0 + xs_1)(t_0 + xt_1) = \big(s_0 t_0 + x^2 s_1 t_1\big) + x\big(s_0 t_1 + s_1 t_0\big). \tag{50.56}$$

Sums of squares are closed under sums and products (if $f = \sum f_i^2$ and $g = \sum g_j^2$ then $fg = \sum_{i,j} (f_i g_j)^2$), and $x^2 s_1 t_1 = (x\sqrt{s_1 t_1}\,)^2$-type term is a square times an SOS, hence SOS; so $s_0 t_0 + x^2 s_1 t_1$ is SOS and $s_0 t_1 + s_1 t_0$ is SOS. Thus the product is again of the form $(\text{SOS}) + x(\text{SOS}) \in \mathcal{Q}$.

Each irreducible factor of p lies in $\mathcal{Q}$. Factor p over $\mathbb{C}$ into linear factors and collect conjugates:

$$p(x) = c \prod_{\text{real } \rho} (x - \rho)^{e_\rho} \prod_{\text{pairs}} \big[(x - \alpha)^2 + \beta^2\big]^{f}, \qquad c > 0, \tag{50.57}$$

the leading coefficient c being positive because $p(x) \to +\infty$ as $x \to +\infty$ (it is nonnegative on the half-line and not identically zero). We classify the factors.

- A complex-conjugate pair gives $(x - \alpha)^2 + \beta^2 = (x - \alpha)^2 + \beta^2$, manifestly a sum of two squares, hence in $\mathcal{Q}$ with $s_1 = 0$.
- A real root $\rho > 0$ must have *even* multiplicity e_ρ. Otherwise p would change sign as x crosses ρ from below to above (the remaining factors are continuous and nonzero in a neighborhood of ρ), contradicting $p \geq 0$ near $\rho > 0$. An even power $(x - \rho)^{2k} = \big((x - \rho)^k\big)^2$ is a perfect square, in $\mathcal{Q}$ with $s_1 = 0$.
- A real root $\rho \leq 0$ has no constraint on its multiplicity, but the factor $(x - \rho) = x + |\rho|$ with $|\rho| \geq 0$ is itself in $\mathcal{Q}$: write $x + |\rho| = (\sqrt{|\rho|}\,)^2 + x \cdot 1^2$, an SOS constant plus x times an SOS constant. Any power of it stays in $\mathcal{Q}$ by closure under products.

Finally the positive constant $c = (\sqrt{c})^2 \in \mathcal{Q}$. Therefore p is a product of finitely many elements of $\mathcal{Q}$ times a positive constant, and by closure under products $p \in \mathcal{Q}$, which is Eq. (50.54) with $a = 0$.

Degrees. Track the total degree through Eq. (50.56). If $\deg p = 2m$, the $x\,\sigma_1$ part has odd degree $\le 2m-1$, forcing $\deg \sigma_1 \le 2m-2$ (it is itself SOS, hence of even degree), and $\deg \sigma_0 \le 2m$. If $\deg p = 2m+1$, the leading term is odd and must come from $x\,\sigma_1$, so $\deg(x\sigma_1) = 2m+1$, i.e. $\deg \sigma_1 = 2m$, while σ_0 has degree $\le 2m$. These are the stated bounds. □

Caution

The single product $(x-a)$ in Eq. (50.54) is exactly enough for a *half-line*, and no more is needed; this is special to one variable and to one linear constraint. On an interval $[a,b]$ one needs the two products $(x-a)$ and $(b-x)$, and in more than one variable (which is where mixed-correlator bootstraps with continuous parameters can land) Putinar's theorem replaces the elementary factorization above and the degree bounds are no longer automatic. The single-correlator bootstrap of the next section lives entirely in the half-line case, where Markov–Lukács is sharp and the SDP is exact at fixed truncation.

50.3.5 The positivity-to-semidefiniteness dictionary and the SDP

Assembling Theorems 50.16 to 50.18 gives the dictionary in full. The functional-search problem "find $\vec{a}$ with $P_\ell(\Delta;\vec{a}) \ge 0$ on $[\Delta_\ell^{\text{unit}}, \infty)$ for each $\ell \le L$" becomes, term by term:

> *Find a covector $\vec{a}$ and, for each spin $\ell \in \{0, 2, \dots, L\}$, two positive-semidefinite matrices $Q_0^{(\ell)}, Q_1^{(\ell)} \succeq 0$ such that*
>
> $$P_\ell(\Delta;\vec{a}) \;=\; v(\Delta)^{\top} Q_0^{(\ell)} v(\Delta) \;+\; (\Delta - \Delta_\ell^{\text{unit}})\, w(\Delta)^{\top} Q_1^{(\ell)} w(\Delta), \tag{50.58}$$
>
> *where v, w are the monomial vectors of the degrees fixed by Theorem 50.18, together with the normalization $\alpha[F_{0,0}] = \vec{a} \cdot \vec{F}_{0,0}^{(\Lambda)} = 1$.*

Both sides of Eq. (50.58) are polynomials in Δ; equating coefficients of each power of Δ gives a finite list of *linear* equations relating the entries of $\vec{a}$, $Q_0^{(\ell)}$, and $Q_1^{(\ell)}$ (the left side is linear in $\vec{a}$ by Theorem 50.16, the right side linear in the matrix entries by the Hankel pattern Eq. (50.53)). The only nonlinear conditions are the semidefiniteness constraints $Q_0^{(\ell)}, Q_1^{(\ell)} \succeq 0$. A feasibility problem with linear equality constraints on symmetric matrices that are required to be positive-semidefinite is, by definition, a *semidefinite program*.

Definition 50.19 (The bootstrap SDP)**.** Fix Δ_ϕ, a derivative order Λ, a spin cutoff L, and a candidate spectrum S (specified by the unitarity region together with any imposed gaps). The *bootstrap semidefinite program* is the feasibility problem: decide whether there exist a covector $\vec{a}$ and positive-semidefinite matrices $\{Q_0^{(\ell)}, Q_1^{(\ell)}\}_{\ell \le L}$ satisfying the linear equations obtained by equating coefficients in Eq. (50.58) for each ℓ, together with $\alpha[F_{0,0}] = 1$. If the program is feasible, the resulting $\vec{a}$ is an admissible functional and S is excluded by Theorem 49.21.

This is the working numerical machine. Feasibility of Theorem 50.19 is a decidable, finite question; modern interior-point solvers settle it, and — the point of the next subsection — a feasible solution can be promoted to a rigorous certificate. To bound the gap one wraps the feasibility test in a bisection: for each trial gap Δ_* in the scalar sector, set $S = S(\Delta_*)$ as in Eq. (49.101), solve the SDP, and move Δ_* down until feasibility is lost; the threshold is the bound Eq. (49.116). To bound an OPE coefficient one converts the feasibility program into an *optimization*, maximizing or minimizing the objective $\vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta_\star,\ell_\star}$ subject to the same semidefinite constraints, exactly as the linear-program structure of Eq. (49.104) anticipated — now with the positivity over the continuum enforced by the matrix constraints rather than sampled.

Worked Example 50.20 (A polynomial-positivity constraint as a 2×2 matrix inequality)**.** To see the machine turn over once, take the smallest nontrivial case. Suppose that after the radial truncation, in some spin sector with unitarity bound $a = \Delta_\ell^{\text{unit}}$, the functional produces a quadratic positivity polynomial

$$P(\Delta; \vec{a}) = c_0(\vec{a}) + c_1(\vec{a})\,\Delta + c_2(\vec{a})\,\Delta^2, \tag{50.59}$$

with the coefficients c_k linear in the functional unknowns $\vec{a}$. We require $P(\Delta) \ge 0$ for all $\Delta \ge a$. By Theorem 50.18 with $\deg P = 2$ (so $m = 1$), the representation is $P(x) = \sigma_0(x) + (x - a)\sigma_1(x)$ with σ_0 SOS of degree ≤ 2 and σ_1 SOS of degree 0, i.e. σ_1 a nonnegative constant. Write the Gram form Eq. (50.50) with $v(\Delta) = (1, \Delta)^\top$ and a single scalar $q_1 \ge 0$:

$$P(\Delta) = \begin{pmatrix} 1 \\ \Delta \end{pmatrix}^{\top} \underbrace{\begin{pmatrix} Q_{00} & Q_{01} \\ Q_{01} & Q_{11} \end{pmatrix}}_{Q_0 \succeq 0} \begin{pmatrix} 1 \\ \Delta \end{pmatrix} + (\Delta - a)\, q_1, \qquad q_1 \ge 0. \tag{50.60}$$

Expanding, $P(\Delta) = Q_{00} + (2Q_{01} + q_1)\Delta + Q_{11}\Delta^2 - a q_1$. Matching powers gives the linear equations

$$c_2 = Q_{11}, \qquad c_1 = 2Q_{01} + q_1, \qquad c_0 = Q_{00} - a\, q_1, \tag{50.61}$$

and the program is to find $\vec{a}$ (hence c_0, c_1, c_2), a number $q_1 \ge 0$, and a symmetric $Q_0 \succeq 0$ satisfying these. Positive-semidefiniteness of the 2×2 block is the pair of

scalar conditions $Q_{00} \geq 0$, $Q_{11} \geq 0$, $Q_{00}Q_{11} - Q_{01}^2 \geq 0$. Eliminating the auxiliary variables, the existence of such a representation is equivalent to the elementary criterion that a quadratic is nonnegative on $[a, \infty)$: either $c_2 > 0$ and (P has no real root in (a, ∞), i.e. the vertex value $c_0 - c_1^2/4c_2 \geq 0$ or the vertex lies left of a with $P(a) \geq 0$), or $c_2 = 0$ and the line is nonnegative at a with nonnegative slope. The SDP reproduces this exactly — and, crucially, scales to degree-50 polynomials in dozens of variables where the elementary case analysis is hopeless but the matrix inequality is routine.

50.3.6 Primal, dual, and the rigor of weak duality

Everything so far has converted the search into an SDP; we now explain why solving that SDP — even imperfectly, even in floating point — can yield a theorem. The reason is convex duality, and the relevant half of it, *weak* duality, is elementary and unconditional.

Cast the bootstrap as a pair of conjugate problems. The *primal* problem is the existence of a CFT: a measure $\{p_{\mathcal{O}} \geq 0\}$ supported on the candidate spectrum S with

$$\vec{F}^{(\Lambda)}_{0,0} + \sum_{\mathcal{O} \in S} p_{\mathcal{O}}\, \vec{F}^{(\Lambda)}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = \vec{0}, \tag{50.62}$$

the truncated crossing equation (Eq. (49.74) read as a vector of derivative components). The *dual* problem is the functional search: find a covector $\vec{a}$ with $\vec{a} \cdot \vec{F}^{(\Lambda)}_{0,0} > 0$ and $\vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta,\ell} \geq 0$ for all $(\Delta, \ell) \in S$. Weak duality is the statement that these cannot both hold.

Theorem 50.21 (Weak duality and unconditional rigor)**.** *If the dual problem is feasible — if there exists a covector $\vec{a}$ with $\vec{a} \cdot \vec{F}^{(\Lambda)}_{0,0} > 0$ and $\vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta,\ell} \geq 0$ for every $(\Delta, \ell) \in S$ — then the primal problem Eq. (50.62) has no solution with $p_{\mathcal{O}} \geq 0$. The candidate spectrum is excluded, and the conclusion holds for* any *dual-feasible $\vec{a}$, optimal or not.*

Proof. Suppose both held. Pair the dual covector $\vec{a}$ with the primal equation Eq. (50.62), using linearity of the dot product:

$$0 = \vec{a} \cdot \vec{0} = \vec{a} \cdot \vec{F}^{(\Lambda)}_{0,0} + \sum_{\mathcal{O} \in S} p_{\mathcal{O}}\, (\vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}). \tag{50.63}$$

The first term is > 0 by dual feasibility. Each summand is a product of $p_{\mathcal{O}} \geq 0$ and $\vec{a} \cdot \vec{F}^{(\Lambda)}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} \geq 0$, hence ≥ 0. The right-hand side is therefore strictly positive, contradicting the left-hand side 0. So no nonnegative primal solution exists. □

This is the same three-line sign count as Theorem 49.21, now read as the weak-duality inequality of a convex program. Its importance for the numerics

cannot be overstated: *rigor does not require solving to optimality.* An interior-point solver produces a sequence of approximate solutions; we do not need the optimal functional, nor the exact boundary of feasibility. We need only *one* covector $\vec{a}$ that is dual-feasible, and Theorem 50.21 certifies the exclusion it yields. The solver's job is to find a feasible point of the dual cone; everything beyond that is a question of how tight the resulting bound is, not whether it is valid.

The semidefinite version is identical in structure, and we record it because it is the form the solver sees. A standard-form SDP and its dual are

$$\text{(P)}\quad \min_{X\succeq 0}\ \langle C, X\rangle \ \text{ s.t. } \langle A_i, X\rangle = b_i, \qquad \text{(D)}\quad \max_{y}\ b^{\top} y \ \text{ s.t. } C - \textstyle\sum_i y_i A_i \succeq 0, \tag{50.64}$$

where $\langle M, N\rangle = \mathrm{Tr}(M^{\top} N)$ is the trace inner product on symmetric matrices.

Theorem 50.22 (Weak duality for the SDP)**.** *For any primal-feasible $X \succeq 0$ and any dual-feasible y (i.e. $S \equiv C - \sum_i y_i A_i \succeq 0$), the duality gap is nonnegative:*

$$\langle C, X\rangle - b^{\top} y \;=\; \langle S, X\rangle \;\geq\; 0. \tag{50.65}$$

In particular the dual objective bounds the primal objective from below: $b^{\top} y \leq \langle C, X\rangle$.

Proof. Using $\langle A_i, X\rangle = b_i$,

$$\langle C, X\rangle - b^{\top} y = \langle C, X\rangle - \sum_i y_i \langle A_i, X\rangle = \Big\langle C - \sum_i y_i A_i,\ X \Big\rangle = \langle S, X\rangle. \tag{50.66}$$

It remains to show $\langle S, X\rangle \geq 0$ for symmetric $S, X \succeq 0$. Write $X = \sum_a \mu_a x_a x_a^{\top}$ by the spectral theorem with $\mu_a \geq 0$. Then

$$\langle S, X\rangle = \mathrm{Tr}(SX) = \sum_a \mu_a\ \mathrm{Tr}\big(S x_a x_a^{\top}\big) = \sum_a \mu_a\, x_a^{\top} S x_a \;\geq\; 0, \tag{50.67}$$

since each $x_a^{\top} S x_a \geq 0$ by $S \succeq 0$ and each $\mu_a \geq 0$. Hence Eq. (50.65), and $b^{\top} y \leq \langle C, X\rangle$. □

Theorem 50.22 is the engine-room version of the bootstrap's rigor. The lemma $\mathrm{Tr}(SX) \geq 0$ for two positive-semidefinite matrices is the matrix analogue of "a product of nonnegative numbers is nonnegative," and it is all that weak duality needs. For the exclusion problem the dual feasibility region is precisely the set of admissible functionals encoded by Eq. (50.58); any point in it, however found, certifies the bound.

50.3.7 SDPB and rigorous arithmetic

The semidefinite programs of Theorem 50.19 are large but highly structured: the unknowns include many matrix blocks $Q_0^{(\ell)}, Q_1^{(\ell)}$, and the constraints are that prescribed *polynomials* in Δ (not just numbers) be expressible in the Markov–Lukács form. This special structure — a "polynomial matrix program" — is what the solver `SDPB` of Simmons-Duffin [441] is built to exploit. Conceptually, `SDPB` is a primal-dual interior-point method: it maintains a primal point $X \succeq 0$ and a dual point $(y, S \succeq 0)$, and drives the duality gap Eq. (50.65) toward zero by Newton steps that follow the "central path" — the locus where $XS = \mu I$ for a barrier parameter $\mu \to 0$ — while keeping both matrices in the interior of the positive-semidefinite cone. Three features make it suited to the bootstrap. It works in *arbitrary precision* arithmetic, because near a saturated bound the matrices become nearly singular and double precision loses the certificate; it handles the polynomial constraints directly by sampling each $P_\ell(\Delta)$ at enough points (Chebyshev nodes) and reconstructing the Markov–Lukács matrices, rather than expanding in a fixed monomial basis where conditioning degrades; and it returns a *dual* certificate, the covector $\vec{a}$, which is exactly the object Theorem 50.21 needs.

The final step closes the logical loop between "the solver printed a functional" and "the spectrum is excluded." A floating-point $\vec{a}$ is, strictly, only an approximation, and one might worry that the certified positivity $P_\ell(\Delta) \geq 0$ fails by a rounding error at some Δ. The resolution is that the certificate can be *verified exactly*. The blocks' rational representation Eq. (50.48) has rational (or algebraic, controlled) coefficients once $\rho_* = 3 - 2\sqrt{2}$ and the residues are carried symbolically; rounding the solver's $\vec{a}$ to nearby rationals gives a candidate functional with exactly known polynomial coefficients $P_\ell(\Delta; \vec{a})$. To prove $P_\ell \geq 0$ on $[\Delta_\ell^{\text{unit}}, \infty)$ one exhibits the Markov–Lukács matrices $Q_0^{(\ell)}, Q_1^{(\ell)}$ with *rational* entries and verifies $Q \succeq 0$ exactly — for instance by a rational $LDL^\top$ (Cholesky) factorization and checking that every pivot is nonnegative, an exact finite computation in rational arithmetic. Equivalently one runs the solver in interval arithmetic, so that the printed certificate carries rigorous error bars and the verified inequality $\vec{a} \cdot \vec{F}_{0,0}^{(\Lambda)} > 0$, $\vec{a} \cdot \vec{F}_{\Delta,\ell}^{(\Lambda)} \geq 0$ holds with the rounding accounted for. Either way, the numerically found functional becomes a genuine proof: by Theorem 50.21 the exclusion follows from the verified sign data alone, independently of the floating-point path that produced it.

Caution

The verification step is what separates a bootstrap *bound* from a bootstrap *plot*. A solver that has not converged, or whose certificate has not been validated in exact arithmetic, produces a curve that is morally correct and almost always right but is not yet a theorem. The published rigorous

bounds — the exclusion regions of the next section, in particular — are those for which a dual-feasible functional has been exhibited and its positivity confirmed. The remaining, genuinely conditional, ingredients are the *assumptions* that define the candidate spectrum (the imposed gaps), not the numerics: granting the gaps, the exclusion is a theorem up to the (verifiable) arithmetic; the identification of the surviving region with a named physical theory is separate physics input, as the chapter introduction stressed.

With the semidefinite program of Theorem 50.19 assembled, its rigor underwritten by the weak duality of Theorems 50.21 and 50.22, and its certificates made exact by rational verification, we have a working numerical machine: a finite SDP whose feasibility rigorously decides whether a candidate spectrum is excluded. The next section aims this machine at the simplest physical target — the four-point function of a single $\mathbb{Z}_2$-odd scalar in $d = 3$ — and watches the exclusion curve in the $(\Delta_\sigma, \Delta_\varepsilon)$ plane develop the sharp kink at which the 3D Ising fixed point of ϕ^4 theory sits.

50.4 The Single-Correlator Bound and the Ising Kink

The previous section converted the search for a certifying functional into a finite semidefinite program: the crossing vector was expanded in the radial coordinate around the symmetric point, the functional was truncated to a finite basis of derivatives, the conformal blocks were replaced by their rational (positive-prefactor times polynomial) approximations, and the requirement that the functional be nonnegative on an entire continuous spin trajectory was recast as the positive-semidefiniteness of a matrix of polynomials — a polynomial matrix program solvable to rigorous precision, with the primal–dual structure that certifies feasibility either way (Section 50.3, Theorem 50.22, and Eq. (50.50)). We now run that machine on the one correlator the three-dimensional Ising universality class hands us for free, the four-point function of the spin operator, and watch a sharp feature emerge at exactly the place ϕ^4 theory predicted. This is the section where the abstract exclusion principle becomes a curve, and the curve acquires a corner with a name.

The logical contract is worth restating before we begin, because it governs how loudly we may speak. The exclusion the program produces is a theorem: whenever it returns a feasible functional, the corresponding region of the $(\Delta_\sigma, \Delta_\varepsilon)$ plane contains no unitary, $\mathbb{Z}_2$-symmetric, crossing-symmetric conformal field theory with the assumed gap, and this conclusion is conditional only on the hypotheses and on the arithmetic (which can be carried in interval arithmetic). The *identification* of the resulting kink with a particular physical theory is not a theorem of that kind; it is physics input, justified by the agreement of the

bootstrap numbers with the Wilson–Fisher fixed point of Theorem 24.31 and with the lattice and experimental critical exponents. We keep the two registers separate throughout.

50.4.1 Assembling the single-correlator semidefinite program

The object to be cornered was fixed in Theorem 49.27: a $d = 3$ conformal field theory, unitary, with a $\mathbb{Z}_2$ symmetry under which the spin σ is odd and the energy ε is even, and with exactly two relevant scalars. The single cheapest crossing equation it must satisfy is the one for four spins, derived in Eq. (49.167): writing the antisymmetrized crossing vector of Eq. (49.168),

$$F^{-}_{\Delta_\sigma;\,\Delta,\ell}(u,v) \equiv v^{\Delta_\sigma}\, g_{\Delta,\ell}(u,v) - u^{\Delta_\sigma}\, g_{\Delta,\ell}(v,u), \tag{50.68}$$

the $\langle\sigma\sigma\sigma\sigma\rangle$ sum rule reads, with the identity carried back onto the left-hand side and $\lambda^2_{\mathbf{1}} = 1$,

$$\sum_{\mathcal{O}\,\in\,\sigma\times\sigma} \lambda^2_{\mathcal{O}}\, F^{-}_{\Delta_\sigma;\,\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) = 0, \qquad \lambda^2_{\mathcal{O}} \geq 0. \tag{50.69}$$

The three physical assumptions enter this equation, and through it the program, in three sharply distinguishable ways, and it is worth tracing each because the reproducibility of the bound depends on getting them right.

How $\mathbb{Z}_2$ symmetry enters. Because σ is $\mathbb{Z}_2$-odd, the product $\sigma \times \sigma$ is $\mathbb{Z}_2$-even, so the sum in Eq. (50.69) runs over $\mathbb{Z}_2$-even primaries only — the energy ε, its even-parity scalar excitations $\varepsilon', \ldots$, the stress tensor T, and the higher even-spin tower. The odd operators (σ itself, σ', …) are simply absent from this OPE; they would appear in a $\langle\sigma\sigma\varepsilon\varepsilon\rangle$ or $\langle\sigma\varepsilon\sigma\varepsilon\rangle$ correlator, which is exactly the additional information the next section will exploit. The $\mathbb{Z}_2$ assumption therefore acts as a *selection rule on the index set* of the sum: it deletes an entire charge sector. Combined with the identical-operator spin parity (exchanging the two σ's in a fused pair multiplies a spin-ℓ block by $(-1)^\ell$), only even spins $\ell = 0, 2, 4, \ldots$ survive. The exchanged spectrum is thus the identity, the even scalars, and the even-spin $\ell \geq 2$ tower.

How unitarity enters. Unitarity does two things. It makes every coefficient $\lambda^2_{\mathcal{O}}$ in Eq. (50.69) nonnegative — these are squares of real OPE coefficients, real because the operators are Hermitian and the radial-quantization inner product is positive (reflection positivity, Theorem 49.27(A2)). And it restricts the dimensions: every primary obeys the unitarity bounds of Theorem 47.18, a scalar with $\Delta \geq \frac{d-2}{2} = \frac{1}{2}$ and a spin-$\ell \geq 1$ operator with $\Delta \geq \ell + d - 2 = \ell + 1$. Nonnegativity of the λ^2's is the property the functional method exploits; the dimension bounds fix the domain over which the functional must be controlled.

How the single-relevant-scalar gap enters. The defining input of the Ising problem (A5 of Theorem 49.27) is the gap: there is exactly one relevant $\mathbb{Z}_2$-even scalar, the energy ε, and the next even scalar ε' is irrelevant. In the single-correlator language we encode this with a trial value Δ_* and demand that the even scalar sector contain *nothing* between the unitarity floor and Δ_* except the operator ε we are trying to bound. Concretely, we present to the program the admissible spectrum

$$\mathcal{S}_{\Delta_*} = \{(\Delta,\, \ell = 0) : \Delta \geq \Delta_*\} \;\cup\; \{(\Delta,\, \ell) :\; \ell \text{ even},\; \ell \geq 2,\; \Delta \geq \ell + 1\}, \tag{50.70}$$

and ask whether crossing can be satisfied with only operators drawn from it. By Theorem 49.28, if a functional α exists that is normalized to unity on the identity vector and nonnegative on $F^-_{\Delta_\sigma;\Delta,\ell}$ for every $(\Delta,\ell) \in \mathcal{S}_{\Delta_*}$, then no consistent theory can have its leading even scalar at or above Δ_*; the gap encodes itself as the *lower edge* Δ_* *of the scalar slice* of Eq. (50.70).

These three ingredients turn the exclusion principle into a concrete decision problem, which the machinery of Section 50.3 renders finite.

Definition 50.23 (The single-correlator feasibility problem)**.** Fix the spacetime dimension $d = 3$, the external dimension Δ_σ, the truncation order Λ (the maximum total derivative order), and a trial scalar gap Δ_*. Let

$$\alpha[\,\cdot\,] = \sum_{\substack{m,n\geq 0\\ m+n\leq\Lambda}} a_{mn}\, \partial_z^m \partial_{\bar z}^n \Big|_{z=\bar z=1/2} \tag{50.71}$$

be a finite linear functional in the derivative basis of Eq. (50.41), with real coefficients a_{mn} (only the components odd under $z \leftrightarrow 1-z$ survive, the others annihilating the antisymmetric F^-). The problem is to decide whether there exist coefficients $\{a_{mn}\}$ such that

(i) $\alpha\big[F^-_{\Delta_\sigma;\,0,0}\big] = 1$ (unit normalization on the identity, $F^-_{\Delta_\sigma;0,0} = v^{\Delta_\sigma} - u^{\Delta_\sigma}$),

(ii) $\alpha\big[F^-_{\Delta_\sigma;\,\Delta,\,0}\big] \geq 0$ for all $\Delta \geq \Delta_*$,

(iii) $\alpha\big[F^-_{\Delta_\sigma;\,\Delta,\,\ell}\big] \geq 0$ for all even $\ell \geq 2$ and all $\Delta \geq \ell + 1$.

If such α exists the point $(\Delta_\sigma, \Delta_*)$ is *excluded*: every consistent theory has $\Delta_\varepsilon < \Delta_*$ at this Δ_σ.

The only obstacle to deciding Theorem 50.23 directly is that conditions (ii) and (iii) are each a constraint on a *continuum* of dimensions Δ, one infinity per spin. This is precisely the obstruction lifted in Section 50.3. Substituting the rational approximation $g_{\Delta,\ell}(z,\bar z) \approx \chi_\ell(\Delta)\, P_{\Delta,\ell}(z,\bar z)$, with $\chi_\ell(\Delta) > 0$ a

known positive prefactor collecting the poles in Δ and $P_{\Delta,\ell}$ a polynomial in Δ (Eq. (50.48)), each functional value becomes

$$\alpha\big[F^{-}_{\Delta_\sigma;\,\Delta,\ell}\big] = \chi_\ell(\Delta)\,Q_\ell(\Delta), \qquad Q_\ell(\Delta) = \sum_{m,n} a_{mn}\,\big(\partial^m\partial^n P\big)_{\Delta,\ell}, \tag{50.72}$$

where Q_ℓ is a polynomial in Δ of degree set by Λ and the rational order. Because $\chi_\ell(\Delta) > 0$ on the unitarity range, conditions (ii)–(iii) are equivalent to the polynomial positivity

$$Q_0(\Delta) \geq 0 \ \text{ on } [\Delta_*, \infty), \qquad Q_\ell(\Delta) \geq 0 \ \text{ on } [\ell+1, \infty) \ \ (\ell = 2, 4, \ldots, \ell_{\max}), \tag{50.73}$$

and a polynomial is nonnegative on a half-line if and only if it is a sum of squares of the appropriate form — equivalently, if and only if certain Gram matrices built from its coefficients are positive semidefinite (Eq. (50.50)). The infinitely many inequalities (ii)–(iii) collapse to finitely many matrix inequalities, one block per retained spin; the spins above some $\ell_{\max}$ are controlled by the asymptotic positivity argument of Section 50.3. The decision variables are the a_{mn} together with the auxiliary Gram matrices, and Theorem 50.23 becomes a genuine semidefinite program, feasible or infeasible by a finite computation whose dual provides the certificate either way.

Caution

The normalization (i) and the positivity (ii)–(iii) carry opposite roles for the same functional, and their relative sign is the whole content of the certificate. The functional must be *strictly positive on the identity vector* and *nonnegative on every operator in the gapped spectrum*; the contradiction of Theorem 49.28 is then that a quantity forced to vanish has been written as a strictly positive sum. An overall sign flip of α is harmless, but the wrong *relative* sign between (i) and (ii)–(iii) silently inverts the meaning and produces a meaningless "bound". In practice one fixes the sign once by the normalization (i) and never touches it again.

50.4.2 Tracing the bound: the feasibility scan

The feasibility problem of Theorem 50.23 answers a yes/no question at a single point $(\Delta_\sigma, \Delta_*)$. The bound curve is obtained by sweeping. Fix Δ_σ and observe the monotonicity built into the construction: if a certifying functional exists for some trial gap Δ_*, then because shrinking the admissible set Eq. (50.70) (raising Δ_*) only removes positivity constraints, the same functional certifies every larger gap as well. Exclusion is therefore an upward-closed property of Δ_*, and there is a sharp threshold

$$\Delta_\varepsilon^{\max}(\Delta_\sigma) = \inf\{\Delta_* : \text{a certifying } \alpha \text{ exists}\}, \tag{50.74}$$

the smallest gap that can still be excluded. Above it the program is feasible and the region is forbidden; below it the program is infeasible and a crossing solution is not ruled out. The threshold is located by bisection: pick a bracket $[\Delta_{\mathrm{lo}}, \Delta_{\mathrm{hi}}]$, solve the SDP at the midpoint, move the relevant endpoint according to feasibility, and iterate until the bracket is narrower than the target precision. Sweeping Δ_σ across the unitarity range $\Delta_\sigma \geq \frac{1}{2}$ then traces the curve $\Delta_\varepsilon^{\max}(\Delta_\sigma)$ that is the section's protagonist.

Two properties of this curve are guaranteed before any number is computed, and both are visible in Fig. 50.5. First, the bound is *rigorous at every truncation*: any feasible α found in the finite derivative space Eq. (50.71) is a genuine functional in the full theory, so the excluded region it certifies is excluded in fact. Truncation never produces a false exclusion; it can only fail to find a functional that a larger space would have found. Second, the bound *improves monotonically with* Λ: enlarging the derivative basis enlarges the search space, so a functional found at order Λ is still available at order $\Lambda' > \Lambda$, and the threshold $\Delta_\varepsilon^{\max}$ can only decrease as Λ grows. The computed curve therefore approaches the true bound *from above*, monotonically, and the coarse-truncation and fine-truncation curves of Fig. 50.5 are nested accordingly. This is the practical meaning of the rigor claim: the bound is honest at any Λ and tightens predictably as Λ is raised.

The shape of the curve is itself instructive and was anticipated by the exact solutions of Section 49.5. At the unitarity floor $\Delta_\sigma = \frac{1}{2}$ the only consistent theory with a single relevant even scalar is the generalized free field, whose spectrum (Eq. (49.130)) puts its lightest even scalar at the double-twist dimension $2\Delta_\sigma = 1$; the curve begins at the endpoint $(\frac{1}{2}, 1)$. As Δ_σ increases the bound rises steeply — the more the spin is lifted off its free dimension, the higher the energy operator is permitted to climb — until near $\Delta_\sigma \approx 0.518$ the slope changes abruptly and the curve continues with a gentler rise. That abrupt change of slope is the kink.

50.4.3 The kink and the Ising point

The kink is not an artefact of coarse computation; it is the opposite. At small Λ the corner is rounded, and one might dismiss it as a smooth maximum. As Λ is raised the rounding shrinks and the corner sharpens toward a genuine slope discontinuity, exactly the behaviour of the nested curves in the inset of Fig. 50.5. A feature that *sharpens* under refinement is a feature of the true bound, not of the truncation. The numerical fact, first established at modest truncation and stable under every subsequent refinement, is that the kink sits at

$$\Delta_\sigma\big|_{\mathrm{kink}} = 0.5182(3), \qquad \Delta_\varepsilon\big|_{\mathrm{kink}} = 1.413(1), \tag{50.75}$$

in striking agreement with the precise Ising determination $(\Delta_\sigma, \Delta_\varepsilon) = (0.5181489, 1.412625)$ of Eq. (49.171), which the mixed-correlator system of the next section will pin down to six figures. The single correlator alone, with nothing but $\mathbb{Z}_2$ symmetry,

unitarity, crossing, and the one-gap assumption, already locates the 3D Ising fixed point to three or four significant figures by the position of a corner.

Why a corner, and why does it locate a theory? The answer is the structure of the *extremal* functional. Along the bound the optimal α is the one whose threshold Δ_* has just been driven to $\Delta_\varepsilon^{\max}$; at that value the certificate is marginal, and the polynomial $Q_\ell(\Delta)$ of Eq. (50.72) must vanish (touch zero from above) at precisely the dimensions of the operators in the unique spectrum that saturates crossing. The functional's zeros *are* the extremal spectrum. At a generic Δ_σ this extremal spectrum drifts smoothly and corresponds to no distinguished theory; the bound is smooth there. At one special Δ_σ the pattern of zeros reorganizes — an operator dimension that had been tracking one trajectory jumps to another — and the response of the bound to Δ_σ changes discontinuously, producing the kink. The reorganization happens where the most strained spectrum consistent with the assumptions abruptly changes character, and for the spin correlator that place is, to all tested precision, the spectrum of the 3D Ising model. The bootstrap is finding the corner of the allowed region where unitarity and crossing are simultaneously most taut, and that corner is a real conformal field theory.

Physical Insight

The kink is the bootstrap's way of pointing without being told where to look. We did not feed in the Ising dimensions; we fed in only that the theory is unitary, $\mathbb{Z}_2$-symmetric, crossing-symmetric, and has a single relevant even scalar — assumptions shared by a continuum of hypothetical theories. The machine replies that along the boundary of what is allowed there is one place where the structure is special, and that place is where every other method says the Ising critical point is. The corner is sharp because at it the extremal spectrum is rigid: the optimal functional has just enough zeros to pin a discrete spectrum with no freedom left, and that rigidity is the fingerprint of an actual theory rather than a generic boundary point.

There is, however, a limitation that must be stated honestly and that sets up everything to come. The single-correlator bound only *touches* the Ising point; it does not isolate it. Theorem 49.28 bounds Δ_ε from *above* for each Δ_σ and says nothing from below, so the allowed region is the entire shaded area beneath the curve in Fig. 50.5, an open region bounded only on top and on the left (by the unitarity wall $\Delta_\sigma \geq \frac{1}{2}$). The Ising theory sits on the upper boundary, at the kink, but so does every extremal spectrum along the curve; nothing in the single-correlator problem forbids a hypothetical unitary theory at, say, $(\Delta_\sigma, \Delta_\varepsilon) = (0.55, 1.2)$, well inside the allowed region. The corner identifies Ising, but it does not corner it. To close the region into a bounded island around the Ising point — to bound Δ_ε from below as well as above, and to do so tightly —

one must bring in more crossing equations than the single correlator provides. That is the task of the mixed system, and it is where the section hands off.

50.4.4 The fixed point identified: $\sigma = \phi$, $\varepsilon = :\phi^2:$

The kink is a feature of a curve derived from pure consistency, with no Lagrangian anywhere in sight. To call it "the 3D Ising fixed point" is to assert that this consistency-defined corner is the same theory as the interacting fixed point ϕ^4 theory has carried since Part VI. We now make the identification explicit, operator by operator, and then translate the bootstrap output into the language of critical exponents so that the claim can be checked against numbers measured by completely different means.

The dictionary is an identification of operators with their lattice/continuum origins. The $\mathbb{Z}_2$-odd spin σ *is* the order-parameter field ϕ of the $\lambda\phi^4$ theory: it is the unique relevant $\mathbb{Z}_2$-odd scalar, the continuum limit of the lattice spin, and the field whose expectation value distinguishes the two broken vacua. The $\mathbb{Z}_2$-even energy ε *is* the composite $:\phi^2:$: the unique relevant $\mathbb{Z}_2$-even scalar, the continuum limit of the lattice energy density, the operator conjugate to the reduced temperature $t = (T - T_c)/T_c$. This is the same pairing the Wilson–Fisher analysis produced: the fixed point of β_λ at $\lambda_* = \frac{16\pi^2}{3}\varepsilon + O(\varepsilon^2)$ (Eq. (24.255)) is a conformal field theory (the characterization completed in Theorem 52.11) in which ϕ and $:\phi^2:$ are primaries, and continuing the dimension deficit to $\varepsilon = 1$ ($d = 3$) gives the theory governing the entire Ising universality class. Universality (Theorem 23.26) is what licenses the identification: every microscopic system in the class — the uniaxial ferromagnet at its Curie point, the liquid–vapour critical point, the binary-fluid demixing transition — flows to one and the same infrared fixed point, so "the bootstrap kink", "the Wilson–Fisher fixed point at $\varepsilon = 1$", and "the 3D Ising critical point" name a single conformal field theory.

What remains is to connect the two numbers the bootstrap delivers, Δ_σ and Δ_ε, to the measurable critical exponents, using the renormalization-group scaling relations established for ϕ^4 theory. The dictionary was stated in Eqs. (49.162) and (49.163); here we derive it from the RG so that its origin is transparent.

Δ_σ and the exponent η. At an interacting fixed point the field anomalous dimension does not vanish, and Eq. (24.246) established that the field's full scaling dimension is the engineering value shifted by the fixed-point anomalous dimension, $\Delta_\sigma = \frac{d-2}{2} + \gamma_*$, with the order-parameter correlation exponent given by $\eta = 2\gamma_*$. Eliminating γ_*,

$$\Delta_\sigma = \frac{d-2}{2} + \frac{\eta}{2} = \frac{d-2+\eta}{2} \quad \xrightarrow{d=3} \quad \boxed{\Delta_\sigma = \frac{1+\eta}{2}}, \qquad \eta = 2\Delta_\sigma - 1, \tag{50.76}$$

the spin two-point function falling as $\langle\sigma(x)\sigma(0)\rangle \sim |x|^{-2\Delta_\sigma} = |x|^{-(d-2+\eta)}$, which is the laboratory definition of η. The bootstrap measures Δ_σ; Eq. (50.76) converts it to η.

Δ_ε and the exponent ν. The energy ε is the operator conjugate to the reduced temperature: in the action, t multiplies ε, so t is the coupling of the operator ε and carries mass dimension $[t] = d - \Delta_\varepsilon$. Under a renormalization-group rescaling by a factor b the coupling flows as $t \to b^{y_t} t$ with RG eigenvalue $y_t = d - \Delta_\varepsilon$ (the scaling dimension of a coupling is d minus the dimension of its operator; this is the relevant/irrelevant classification of Theorem 23.26). The correlation length, a physical length, transforms as $\xi \to \xi/b$; combining the two, $\xi(t) = b\,\xi(b^{y_t}t)$, and choosing b so that $b^{y_t}t = O(1)$ gives $\xi \sim t^{-1/y_t}$. Comparison with the definition $\xi \sim |t|^{-\nu}$ yields $\nu = 1/y_t$, hence

$$\frac{1}{\nu} = y_t = d - \Delta_\varepsilon \quad \xrightarrow{d=3} \quad \boxed{\Delta_\varepsilon = 3 - \frac{1}{\nu}}, \qquad \frac{1}{\nu} = 3 - \Delta_\varepsilon. \tag{50.77}$$

The gap assumption is exactly the statement that ε is the *only* relevant even scalar, i.e. that the temperature is the only relevant even direction: a single tuning reaches criticality, the hallmark of the Ising class. The bootstrap measures Δ_ε; Eq. (50.77) converts it to ν.

These two relations close the loop opened in Part VI: the field anomalous dimension γ_ϕ computed diagrammatically from wavefunction renormalization, and the thermal eigenvalue y_t that is the slope of the linearized RG flow, are the same two numbers the bootstrap reads off the kink. The perturbative description (a coupling flowing to λ_*) and the bootstrap description (a corner of an exclusion curve) are two windows onto one fixed point, and Eqs. (50.76) and (50.77) are the panes through which the same exponents are seen from each. The remaining critical exponents are not independent: they follow from (η, ν) by the scaling and hyperscaling relations, and the following worked example carries that arithmetic through and confronts it with the laboratory.

Worked Example 50.24 (From the kink to the critical exponents of the 3D Ising class)**.** We take the bootstrap dimensions at face value — using the precise values Eq. (49.171), $\Delta_\sigma = 0.5181489$ and $\Delta_\varepsilon = 1.412625$, since the single-correlator kink Eq. (50.75) agrees with them to its precision — and produce the full set of standard exponents, comparing each to the best Monte Carlo determinations of the three-dimensional Ising model.

The two primary exponents. From Eqs. (50.76) and (50.77),

$$\eta = 2\Delta_\sigma - 1 = 2(0.5181489) - 1 = 0.0362978, \tag{50.78}$$

$$\frac{1}{\nu} = 3 - \Delta_\varepsilon = 3 - 1.412625 = 1.587375, \qquad \nu = 0.6299709. \tag{50.79}$$

The anomalous dimension is tiny — the field is lifted only slightly off its free scaling — consistent with the Wilson–Fisher result that $\eta = O(\varepsilon^2)$ begins at second order in the deficit (Eq. (24.256)), whereas the thermal exponent departs substantially from its free value, consistent with $1/\nu = 2 - \varepsilon/3 + \cdots$ being a first-order effect (Theorem 49.29).

The remaining exponents from two RG eigenvalues. Introduce the magnetic eigenvalue. The magnetic field h couples to σ, so by the same coupling-dimension argument that gave y_t, its RG eigenvalue is $y_h = d - \Delta_\sigma$. With $d = 3$,

$$y_h = 3 - \Delta_\sigma = 3 - 0.5181489 = 2.4818511, \qquad y_t = \frac{1}{\nu} = 1.587375. \tag{50.80}$$

The thermodynamic exponents are fixed by (y_t, y_h) through the standard scaling relations (the free energy density scales as $f \sim t^{\,d/y_t}$, and successive h-derivatives at $t \neq 0$ generate the magnetization and susceptibility), which give

$$\alpha = 2 - \frac{d}{y_t} = 2 - d\nu, \qquad \beta = \frac{d - y_h}{y_t} = \nu\,\Delta_\sigma, \tag{50.81}$$

$$\gamma = \frac{2y_h - d}{y_t} = \nu(2 - \eta), \qquad \delta = \frac{y_h}{d - y_h} = \frac{d + 2 - \eta}{d - 2 + \eta}. \tag{50.82}$$

Evaluating with the numbers above,

$$\alpha = 2 - 3(0.6299709) = 0.110087, \tag{50.83}$$

$$\beta = (0.6299709)(0.5181489) = 0.326419, \tag{50.84}$$

$$\gamma = (0.6299709)(2 - 0.0362978) = 1.237075, \tag{50.85}$$

$$\delta = \frac{3 + 2 - 0.0362978}{3 - 2 + 0.0362978} = \frac{4.9637022}{1.0362978} = 4.790036. \tag{50.86}$$

Confrontation with Monte Carlo. The high-precision lattice/Monte Carlo values for the 3D Ising universality class are $\nu = 0.63002(10)$, $\eta = 0.03627(10)$, $\alpha = 0.1099(7)$, $\beta = 0.32630(22)$, $\gamma = 1.2372(5)$, $\delta = 4.789(2)$. Every exponent derived above from the two bootstrap dimensions agrees with the independently measured value to within the Monte Carlo error bars. The agreement is the quantitative content of the identification: a corner on an exclusion curve, located by demanding only unitarity and crossing, reproduces the entire table of critical exponents of a real second-order phase transition — and it does so by translating two CFT dimensions through scaling relations that were derived, in this book, from the renormalization-group flow of $\lambda\phi^4$.

The worked example completes the major payoff of ϕ^4 theory. The fixed point computed perturbatively in $d = 4 - \varepsilon$ as the nontrivial zero λ_* of the one-loop beta function (Theorem 24.31), and characterized as a conformal field theory in Theorem 52.11, is the same theory the bootstrap pins to the kink of a curve derived from consistency alone — with the two routes meeting, quantitatively,

on the numbers $\eta = 0.0363$ and $\nu = 0.6300$. The perturbative thread and the consistency thread have converged on one theory.

What they have not yet done is corner it. The single correlator places the Ising point on the boundary of the allowed region but leaves that region open below, so the kink *touches* Ising without *isolating* it. Closing the region into a small, bounded island — and thereby turning the three-figure location of the kink into the six-figure determination quoted in Eq. (49.171) — requires the additional crossing equations of the mixed system $\langle\sigma\sigma\sigma\sigma\rangle$, $\langle\sigma\sigma\varepsilon\varepsilon\rangle$, $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$, where the shared OPE coefficients enforce positivity of genuine matrices rather than single numbers. That genuinely semidefinite structure, and the closed island it produces, are the subject of the next section.

50.5 Mixed Correlators and the Ising Island

The single-correlator analysis of the previous section delivered a sharp but incomplete verdict. Crossing of $\langle\sigma\sigma\sigma\sigma\rangle$, fed through the linear-functional machine and the gap assumption that $\sigma \times \sigma$ contains exactly one relevant $\mathbb{Z}_2$-even scalar, produced a rigorous upper bound on Δ_ε as a function of Δ_σ. That bound has a pronounced kink, and the 3D Ising model sits essentially *on* the kink. But "on the boundary of an allowed region" is not the same as "isolated." The allowed region of the single correlator is an open wedge: it pins Δ_ε from above but leaves it free below, and it constrains Δ_σ not at all. The kink is a strong hint — the most constrained point on a one-sided boundary — but it is a hint, not a cage. To turn the hint into a closed determination we need more equations sharing the *same* unknown CFT data, so that bounds pushing in from several directions can trap the theory between them. The decisive step, anticipated in Section 49.3.5, is to bootstrap not one correlator but the whole small system built from the two lightest scalars, and to discover that unitarity then speaks not through nonnegative numbers but through positive-semidefinite *matrices*. That upgrade is what finally closes the wedge into an island.

50.5.1 Three correlators sharing one operator algebra

The Ising CFT has a $\mathbb{Z}_2$ global symmetry under which the spin operator σ is odd and the energy operator ε is even (Section 48.6). We take both as external operators and impose crossing on every four-point function they generate. The $\mathbb{Z}_2$ selection rule organizes their operator product expansions (Theorem 47.46) into two sectors. Writing $\mathcal{O}^+$ for $\mathbb{Z}_2$-even primaries and $\mathcal{O}^-$ for $\mathbb{Z}_2$-odd primaries,

$$\sigma \times \sigma \;\sim\; \mathbf{1} + \sum_{\mathcal{O}^+} \lambda_{\sigma\sigma\mathcal{O}}\, \mathcal{O}^+, \tag{50.87}$$

$$\varepsilon \times \varepsilon \;\sim\; \mathbf{1} + \sum_{\mathcal{O}^+} \lambda_{\varepsilon\varepsilon\mathcal{O}}\, \mathcal{O}^+, \tag{50.88}$$

$$\sigma \times \varepsilon \;\sim\; \sum_{\mathcal{O}^-} \lambda_{\sigma\varepsilon\mathcal{O}}\, \mathcal{O}^-, \tag{50.89}$$

where $\lambda_{\sigma\sigma\mathcal{O}} \equiv C_{\sigma\sigma\mathcal{O}}$ in the notation of Theorem 49.17, and only even spins appear in $\sigma \times \sigma$ and $\varepsilon \times \varepsilon$ (identical operators), whereas $\sigma \times \varepsilon$ admits all spins. The crucial structural fact is visible already in Eqs. (50.87) and (50.88): every $\mathbb{Z}_2$-even primary $\mathcal{O}^+$ appears in *both* the $\sigma \times \sigma$ and the $\varepsilon \times \varepsilon$ channels, carrying *two* OPE coefficients, $\lambda_{\sigma\sigma\mathcal{O}}$ and $\lambda_{\varepsilon\varepsilon\mathcal{O}}$. The identity $\mathbf{1}$ is the universal $\mathcal{O}^+$ with $\lambda_{\sigma\sigma\mathbf{1}} = \lambda_{\varepsilon\varepsilon\mathbf{1}} = 1$.

From two external scalars the $\mathbb{Z}_2$ symmetry permits exactly three inequivalent four-point functions (any correlator with an odd number of σ's vanishes):

$$\langle\sigma\sigma\sigma\sigma\rangle, \qquad \langle\sigma\sigma\varepsilon\varepsilon\rangle, \qquad \langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle, \tag{50.90}$$

together with the reorderings of the middle one, $\langle\sigma\varepsilon\sigma\varepsilon\rangle$ and $\langle\sigma\varepsilon\varepsilon\sigma\rangle$, which are the *same* function evaluated in different channels and supply additional crossing relations rather than new correlators. The operators exchanged in each, and the OPE-coefficient combination they carry, are dictated by Eqs. (50.87) to (50.89):

- In $\langle\sigma\sigma\sigma\sigma\rangle$ only $\mathcal{O}^+$ are exchanged, with weight $\lambda^2_{\sigma\sigma\mathcal{O}}$ — a perfect square, exactly the single-correlator case of Eq. (49.74).
- In $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$ only $\mathcal{O}^+$ are exchanged, with weight $\lambda^2_{\varepsilon\varepsilon\mathcal{O}}$.
- In $\langle\sigma\sigma\varepsilon\varepsilon\rangle$, the channel that fuses $\sigma\sigma \to \mathcal{O}^+ \to \varepsilon\varepsilon$ carries the *product* $\lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}$, while the crossed channel fuses $\sigma\varepsilon \to \mathcal{O}^- \to \sigma\varepsilon$ with weight $\lambda^2_{\sigma\varepsilon\mathcal{O}}$.

The first two correlators see the even operators through the squares $\lambda^2_{\sigma\sigma\mathcal{O}}$ and $\lambda^2_{\varepsilon\varepsilon\mathcal{O}}$; the mixed correlator sees them through the product $\lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}$. These three combinations are precisely the three independent entries of the symmetric 2×2 matrix

$$P_{\mathcal{O}} \;\equiv\; \begin{pmatrix} \lambda^2_{\sigma\sigma\mathcal{O}} & \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}} \\ \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}} & \lambda^2_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix} \;=\; \vec{\lambda}_{\mathcal{O}}\, \vec{\lambda}_{\mathcal{O}}^{\mathsf{T}}, \qquad \vec{\lambda}_{\mathcal{O}} \equiv \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}} \\ \lambda_{\varepsilon\varepsilon\mathcal{O}} \end{pmatrix}. \tag{50.91}$$

This is the object that will replace the single nonnegative number $p_{\mathcal{O}}$ of the single-correlator sum rule. We now make the bookkeeping precise.

50.5.2 The crossing equations and their matrix structure

For non-identical external scalars the conformal block carries a dependence on the dimension differences, which we must reinstate relative to Chapter 49. Recall from Theorem 47.46 and the block kinematics of Chapter 47 that the s-channel

decomposition of a four-point function of scalars $\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3\mathcal{O}_4$ reads, after stripping the standard conformal prefactor,

$$\mathcal{G}(u,v) \;=\; \sum_{\mathcal{O}} \lambda_{12\mathcal{O}}\,\lambda_{34\mathcal{O}}\, g^{\Delta_{12},\Delta_{34}}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v), \qquad \Delta_{ij}\equiv\Delta_i-\Delta_j, \tag{50.92}$$

the block reducing to the identical-operator block $g_{\Delta,\ell}$ of Chapter 49 when all external dimensions coincide. Crossing $1\leftrightarrow 3$ equates this s-channel sum to the t-channel sum. Generalizing the antisymmetrized combination $F_{\Delta,\ell} = v^{\Delta_\phi}g(u,v) - u^{\Delta_\phi}g(v,u)$ of Chapter 49 to unequal externals, define the two convolved blocks

Definition 50.25 (Mixed convolved blocks)**.** For external scalars with dimensions Δ_i, set

$$F^{ij,kl}_{\mp,\Delta,\ell}(u,v) \;\equiv\; v^{\frac{\Delta_k+\Delta_j}{2}}\, g^{\Delta_{ij},\Delta_{kl}}_{\Delta,\ell}(u,v) \;\mp\; u^{\frac{\Delta_k+\Delta_j}{2}}\, g^{\Delta_{ij},\Delta_{kl}}_{\Delta,\ell}(v,u), \tag{50.93}$$

the lower sign (F_-) antisymmetric and the upper sign (F_+) symmetric under $u\leftrightarrow v$. For all-identical externals $F^{\phi\phi,\phi\phi}_{-,\Delta,\ell}$ is the function $F_{\Delta,\ell}$ of Eq. (49.74).

Carrying out the vectorization of Section 49.3.4 for each correlator in Eq. (50.90) — using the non-identical-operator block decomposition of Section 47.7.4, whose blocks for unequal externals depend on the dimension differences Δ_{ij} but are stripped and Taylor-expanded at the crossing-symmetric point exactly as in the identical case — assembles the system into five independent crossing equations. The single ingredient beyond the single-correlator construction of Chapter 49 is the sign a spin-ℓ exchange acquires under interchange of the two operators in an OPE pair, derived explicitly below. Two come from the diagonal correlators and reproduce the single-correlator sum rule for σ and for ε:

$$\sum_{\mathcal{O}^+} \lambda^2_{\sigma\sigma\mathcal{O}}\, F^{\sigma\sigma,\sigma\sigma}_{-,\Delta,\ell} = 0, \tag{50.94}$$

$$\sum_{\mathcal{O}^+} \lambda^2_{\varepsilon\varepsilon\mathcal{O}}\, F^{\varepsilon\varepsilon,\varepsilon\varepsilon}_{-,\Delta,\ell} = 0, \tag{50.95}$$

with the identity included as the $\Delta=0$ term. The remaining three come from the mixed correlator $\langle\sigma\sigma\varepsilon\varepsilon\rangle$ together with its reordering $\langle\sigma\varepsilon\sigma\varepsilon\rangle$. The self-crossing of $\langle\sigma\varepsilon\sigma\varepsilon\rangle$ exchanges only $\mathcal{O}^-$ in both channels and is purely odd-sector; the crossing relating $\langle\sigma\sigma\varepsilon\varepsilon\rangle$ to $\langle\sigma\varepsilon\sigma\varepsilon\rangle$ ties the even product $\lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}$ to the odd square $\lambda^2_{\sigma\varepsilon\mathcal{O}}$ and produces two equations (the symmetric and antisymmetric convolutions). The factor $(-1)^\ell$ that appears on the odd terms is not an input; it follows from the three-point kinematics of Section 47.4. The leading tensor structure of $\langle\sigma\,\varepsilon\,\mathcal{O}_\ell\rangle$ for a spin-ℓ primary $\mathcal{O}_\ell$ exchanged in the $\sigma\times\varepsilon$ OPE is built from ℓ contractions with the separation vector $x^\mu_{\sigma\varepsilon} = x^\mu_\sigma - x^\mu_\varepsilon$ (Theorem 47.24). The crossing that relates $\langle\sigma\sigma\varepsilon\varepsilon\rangle$ to its reordering $\langle\sigma\varepsilon\sigma\varepsilon\rangle$ interchanges the roles of σ and ε in this pair,

sending $x_{\sigma\varepsilon} \to -x_{\sigma\varepsilon}$ and hence multiplying the structure by $(-1)^\ell$; equivalently $\lambda_{\varepsilon\sigma\mathcal{O}_\ell} = (-1)^\ell \lambda_{\sigma\varepsilon\mathcal{O}_\ell}$, and the convolved block of Theorem 50.25 inherits the same sign. This is the only place the odd $(\sigma\varepsilon)$ data enters with a spin-dependent sign, and it is the origin of the explicit $(-1)^\ell$ in the fourth and fifth components of Eq. (50.97).

It is cleanest to package all five at once. Collecting the even operators into the quadratic form $\vec{\lambda}_{\mathcal{O}}^{\mathsf{T}}(\cdots)\vec{\lambda}_{\mathcal{O}}$ built from Eq. (50.91), the entire mixed system is the single vector equation

$$\boxed{\sum_{\mathcal{O}^+} \vec{\lambda}_{\mathcal{O}}^{\mathsf{T}}\, \vec{\mathbb{V}}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}\, \vec{\lambda}_{\mathcal{O}} \;+\; \sum_{\mathcal{O}^-} \lambda_{\sigma\varepsilon\mathcal{O}}^2\, \vec{U}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} \;=\; \vec{0},} \tag{50.96}$$

where the sum over $\mathcal{O}^+$ includes the identity with $\vec{\lambda}_{\mathbf{1}} = (1,1)^{\mathsf{T}}$, the $\mathcal{O}^+$ sum runs over even spin and the $\mathcal{O}^-$ sum over all spin, and the five-component objects are a vector of symmetric 2×2 matrices and a vector of scalars,

$$\vec{\mathbb{V}}_{\Delta,\ell} = \begin{pmatrix} \begin{pmatrix} F_-^{\sigma\sigma,\sigma\sigma} & 0 \\ 0 & 0 \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & F_-^{\varepsilon\varepsilon,\varepsilon\varepsilon} \end{pmatrix} \\ \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \\ \frac{1}{2}\begin{pmatrix} 0 & F_-^{\sigma\sigma,\varepsilon\varepsilon} \\ F_-^{\sigma\sigma,\varepsilon\varepsilon} & 0 \end{pmatrix} \\ \frac{1}{2}\begin{pmatrix} 0 & F_+^{\sigma\sigma,\varepsilon\varepsilon} \\ F_+^{\sigma\sigma,\varepsilon\varepsilon} & 0 \end{pmatrix} \end{pmatrix}, \qquad \vec{U}_{\Delta,\ell} = \begin{pmatrix} 0 \\ 0 \\ F_-^{\sigma\varepsilon,\sigma\varepsilon} \\ -(-1)^\ell F_-^{\varepsilon\sigma,\sigma\varepsilon} \\ (-1)^\ell F_+^{\varepsilon\sigma,\sigma\varepsilon} \end{pmatrix}, \tag{50.97}$$

the convolved blocks evaluated at the external dimensions of their superscripts via Theorem 50.25. The first two components of Eq. (50.97) are exactly Eqs. (50.94) and (50.95); the third is the odd self-crossing of $\langle\sigma\varepsilon\sigma\varepsilon\rangle$; the fourth and fifth are the mixed equations that link the off-diagonal even data $\lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}$ to the odd data $\lambda_{\sigma\varepsilon\mathcal{O}}^2$. This is the mixed σ–ε crossing system of Kos et al. [303]; the line-by-line block identities behind Eq. (50.97) are the non-identical-operator kinematics of Chapter 47.

Physical Insight: why a matrix and not three numbers

One might object that Eqs. (50.94) and (50.95) already carry $\lambda_{\sigma\sigma\mathcal{O}}^2$ and $\lambda_{\varepsilon\varepsilon\mathcal{O}}^2$ as ordinary nonnegative numbers, and the mixed equation merely adds a third number $\lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}}$. Why not treat all three as independent nonnegative weights and stay in the linear-programming world of Chapter 49? Because the three are *not* independent: a genuine CFT has a single real pair $(\lambda_{\sigma\sigma\mathcal{O}}, \lambda_{\varepsilon\varepsilon\mathcal{O}})$ per operator, so the off-diagonal entry is forced to equal $\pm\sqrt{\lambda_{\sigma\sigma\mathcal{O}}^2\,\lambda_{\varepsilon\varepsilon\mathcal{O}}^2}$ with a definite sign correlation. The clean, convex way

to impose "these three numbers are the entries of an outer product $\vec{\lambda}\vec{\lambda}^{\mathsf{T}}$" is the statement that the matrix $P_{\mathcal{O}}$ of Eq. (50.91) is positive semidefinite. Treating the entries as free nonnegative numbers would throw away the correlation and weaken every bound; keeping the matrix constraint is what makes the mixed system strictly stronger than the sum of its diagonal parts.

50.5.3 Positive semidefiniteness from unitarity

The matrix $P_{\mathcal{O}}$ in Eq. (50.91) is manifestly an outer product, hence rank one and positive semidefinite. But it is worth proving the positivity from unitarity directly, because the argument shows what survives when the assumption "one operator per (Δ,ℓ)" is dropped — and that is exactly the structural fact that promotes the problem from a linear to a semidefinite program.

Theorem 50.26 (Positive semidefiniteness of the OPE matrix)**.** *Let (Δ,ℓ) be the quantum numbers of one or more $\mathbb{Z}_2$-even primaries of a unitary CFT, and define the symmetric 2×2 matrix*

$$P_{\Delta,\ell} \;=\; \sum_{\mathcal{O}:\,(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})=(\Delta,\ell)} \begin{pmatrix} \lambda_{\sigma\sigma\mathcal{O}}^2 & \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}} \\ \lambda_{\sigma\sigma\mathcal{O}}\lambda_{\varepsilon\varepsilon\mathcal{O}} & \lambda_{\varepsilon\varepsilon\mathcal{O}}^2 \end{pmatrix}. \tag{50.98}$$

Then $P_{\Delta,\ell}\succeq 0$.

Proof. Work in radial quantization, where the inner product on the Hilbert space of states is positive definite precisely because the theory is unitary (reflection positivity, established in Chapter 10 and used throughout Chapter 47). A primary $\mathcal{O}$ inserted at the origin creates a state $|\mathcal{O}\rangle$, and the OPE Eqs. (50.87) and (50.88) states that the radial-quantization states obtained from $\sigma\sigma$ and from $\varepsilon\varepsilon$ both have a component along the conformal multiplet $[\mathcal{O}]$ built on $|\mathcal{O}\rangle$ and its descendants. Let $\mathcal{P}_{[\Delta,\ell]}$ be the orthogonal projector onto the span of all primaries with the given quantum numbers (Δ,ℓ) together with their descendants. Because it projects onto a subspace of a positive-definite Hilbert space, $\mathcal{P}_{[\Delta,\ell]}$ is itself a positive operator.

Denote by $|a\rangle$ $(a\in\{\sigma\sigma,\varepsilon\varepsilon\})$ the state created by the corresponding pair of operators in a fixed configuration. The OPE coefficients are the overlaps of these states with the multiplet: projecting and using that distinct conformal families are orthogonal,

$$\Big\langle a\Big|\mathcal{P}_{[\Delta,\ell]}\Big|b\Big\rangle \;=\; \kappa_{\Delta,\ell} \sum_{\mathcal{O}:\,(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})=(\Delta,\ell)} \lambda_{a\mathcal{O}}\,\lambda_{b\mathcal{O}}, \tag{50.99}$$

where $\kappa_{\Delta,\ell}>0$ is a common kinematic normalization (the same positive number for both indices because the descendants and their norms are fixed by conformal

symmetry and the unit normalization of the primaries), and $\lambda_{a\mathcal{O}}$ is $\lambda_{\sigma\sigma\mathcal{O}}$ or $\lambda_{\varepsilon\varepsilon\mathcal{O}}$ accordingly. The left-hand side of Eq. (50.99) is the Gram matrix $G_{ab} = \left\langle a \middle| \mathcal{P}_{[\Delta,\ell]} \middle| b \right\rangle$ of the two projected states. A Gram matrix of any set of vectors under a positive operator is positive semidefinite: for any real $c = (c_\sigma, c_\varepsilon)$,

$$c^{\mathsf{T}} G\, c = \left\langle \psi \middle| \mathcal{P}_{[\Delta,\ell]} \middle| \psi \right\rangle = \| \, \mathcal{P}_{[\Delta,\ell]} \, |\psi\rangle \, \|^2 \geq 0, \qquad |\psi\rangle \equiv c_\sigma \, |\sigma\sigma\rangle + c_\varepsilon \, |\varepsilon\varepsilon\rangle \, , \tag{50.100}$$

using $\mathcal{P}^2 = \mathcal{P} = \mathcal{P}^\dagger$. Hence $G \succeq 0$, and since $P_{\Delta,\ell} = \kappa_{\Delta,\ell}^{-1} G$ with $\kappa_{\Delta,\ell} > 0$, also $P_{\Delta,\ell} \succeq 0$. □

Physical Insight: degeneracy is the true source of "semidefinite"

If a single operator sits at (Δ, ℓ), the sum in Eq. (50.98) has one term and $P_{\Delta,\ell} = \vec{\lambda}\vec{\lambda}^{\mathsf{T}}$ is rank one — it lies on the boundary of the PSD cone, with eigenvalues 0 and $|\vec{\lambda}|^2$. But the bootstrap is not allowed to assume how many operators share a given (Δ, ℓ); two nearly degenerate even scalars, or a scalar and a higher operator that the truncated functional cannot resolve, contribute a *sum* of rank-one matrices, which is a generic PSD matrix of rank up to two. The only honest hypothesis is therefore $P_{\Delta,\ell} \succeq 0$ — the full positive-semidefinite cone, not just its rank-one boundary. This is the precise origin of the word "semidefinite": it is forced by unitarity together with the refusal to assume nondegeneracy, and it has nothing to do with the sum-of-squares representation of the functional (the "SOS trick" of Section 50.3) that handles the continuous variable Δ. The functional must be positive on a *matrix* of conformal blocks, because the data it multiplies is a matrix.

50.5.4 The generalized sum rule and its rank-one core

We can now restate the mixed system in the trace form that exhibits the matrix data cleanly. Using $\vec{\lambda}^{\mathsf{T}} \mathbb{V} \vec{\lambda} = \mathrm{Tr}[\mathbb{V} \vec{\lambda}\vec{\lambda}^{\mathsf{T}}] = \mathrm{Tr}[\mathbb{V} P]$ component by component, Eq. (50.96) becomes the following.

Theorem 50.27 (Matrix-valued bootstrap sum rule)**.** *A unitary $\mathbb{Z}_2$-symmetric CFT with external scalars σ (odd) and ε (even) satisfies*

$$\sum_{\mathcal{O}^+} \mathrm{Tr}\big[\, \vec{\mathbb{V}}_{\Delta_\mathcal{O},\ell_\mathcal{O}} \, P_\mathcal{O} \,\big] \; + \; \sum_{\mathcal{O}^-} p_\mathcal{O} \, \vec{U}_{\Delta_\mathcal{O},\ell_\mathcal{O}} \; = \; \vec{0}, \tag{50.101}$$

with $\vec{\mathbb{V}}, \vec{U}$ as in Eq. (50.97), $P_\mathcal{O} \succeq 0$ the 2×2 OPE matrix of Eq. (50.91) (Theorem 50.26), $p_\mathcal{O} = \lambda^2_{\sigma\varepsilon\mathcal{O}} \geq 0$, the even sum including the identity with $P_\mathbf{1} = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, and the spectrum obeying the unitarity bounds of Theorem 47.18.

The single-correlator sum rule of Eq. (49.74) is the *rank-one specialization* of Eq. (50.101), in two independent senses, which it is worth seeing explicitly.

Worked Example 50.28 (The single correlator as the rank-one corner of the matrix system)**.** Suppose we discard the energy operator and bootstrap σ alone. Operationally this means keeping only the first component of Eq. (50.97) and only the $(1,1)$ entry of every matrix — equivalently, projecting onto the vector $e_\sigma = (1,0)^{\mathsf{T}}$. Then for each even operator

$$\mathrm{Tr}\big[\,\vec{\mathbb{V}}^{(1)}_{\Delta,\ell}\,P_{\mathcal{O}}\,\big] = (P_{\mathcal{O}})_{11}\,F^{\sigma\sigma,\sigma\sigma}_{-,\Delta,\ell} = \lambda^2_{\sigma\sigma\mathcal{O}}\,F_{\Delta,\ell}, \tag{50.102}$$

since $\vec{\mathbb{V}}^{(1)}$ has its only nonzero entry in the $(1,1)$ slot and $(P_{\mathcal{O}})_{11} = \lambda^2_{\sigma\sigma\mathcal{O}}$. The odd operators drop out entirely (the first component of $\vec{U}$ is zero), and Eq. (50.101) collapses to

$$\sum_{\mathcal{O}^+} \lambda^2_{\sigma\sigma\mathcal{O}}\,F_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} = 0, \tag{50.103}$$

which, dividing through by the identity's $F_{0,0}$, is exactly Eq. (49.74). Two specializations have happened at once. First, only one external operator is retained, so the matrices shrink to their top-left scalar. Second — and this is the "rank-one" content even within the single-correlator problem — the only datum that survives is $(P_{\mathcal{O}})_{11} = \lambda^2_{\sigma\sigma\mathcal{O}} \geq 0$, a single nonnegative number, so the positive-semidefinite cone of 2×2 matrices degenerates to the nonnegative half-line, and the SDP degenerates to the LP of Chapter 49. The mixed system is strictly more constraining because it refuses both degenerations: it keeps both external operators and it keeps the full 2×2 cone.

50.5.5 The semidefinite condition: positivity on a matrix of blocks

The exclusion argument is the matrix analogue of the separating-hyperplane logic of Chapter 49. A functional is now not a single linear map but a tuple $\alpha = (\alpha_1,\dots,\alpha_5)$ of linear functionals, one for each crossing equation, acting on functions of the cross-ratios (in practice, on finitely many Taylor derivatives at the crossing-symmetric point, as in Section 49.3.4). Applying α to Eq. (50.101) contracts the five-vector to a single scalar equation,

$$\sum_{\mathcal{O}^+} \mathrm{Tr}\big[\,\Lambda_+(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})\,P_{\mathcal{O}}\,\big] \;+\; \sum_{\mathcal{O}^-} p_{\mathcal{O}}\,\Lambda_-(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}) \;=\; 0, \tag{50.104}$$

where the 2×2 symmetric matrix Λ_+ and the scalar Λ_- are the functional's action on the block data,

$$\Lambda_+(\Delta,\ell) \equiv \sum_{a=1}^{5} \alpha_a[\vec{\mathbb{V}}^{(a)}_{\Delta,\ell}], \qquad \Lambda_-(\Delta,\ell) \equiv \sum_{a=1}^{5} \alpha_a[\vec{U}^{(a)}_{\Delta,\ell}]. \tag{50.105}$$

The trace pairing of two PSD matrices is nonnegative, which is the matrix replacement for "a product of nonnegative numbers is nonnegative."

Lemma 50.29 (Trace pairing of PSD matrices)**.** *If $A \succeq 0$ and $B \succeq 0$ are real symmetric matrices, then* $\mathrm{Tr}[AB] \geq 0$.

Proof. Diagonalize $A = \sum_i a_i\, u_i u_i^{\mathsf{T}}$ with eigenvalues $a_i \geq 0$ and orthonormal u_i. Then $\mathrm{Tr}[AB] = \sum_i a_i\, \mathrm{Tr}[u_i u_i^{\mathsf{T}} B] = \sum_i a_i\, u_i^{\mathsf{T}} B\, u_i$. Each $u_i^{\mathsf{T}} B\, u_i \geq 0$ because $B \succeq 0$, and each $a_i \geq 0$, so the sum is nonnegative. □

Theorem 50.30 (Semidefinite exclusion criterion)**.** *Fix external dimensions $(\Delta_\sigma, \Delta_\varepsilon)$ and a hypothesized spectrum: a list of allowed (Δ, ℓ) for the even sector and for the odd sector, each obeying Theorem 47.18. Suppose there exists a functional $\alpha = (\alpha_1, \dots, \alpha_5)$ such that*

$$\mathrm{Tr}\left[\Lambda_+(0,0)\, P_{\mathbf{1}}\right] = 1 \qquad \textit{(identity normalization)}, \tag{50.106}$$

$$\Lambda_+(\Delta,\ell) \succeq 0 \qquad \textit{for every even } (\Delta,\ell) \neq \mathbf{1} \textit{ in the spectrum}, \tag{50.107}$$

$$\Lambda_-(\Delta,\ell) \geq 0 \qquad \textit{for every odd } (\Delta,\ell) \textit{ in the spectrum}. \tag{50.108}$$

Then no unitary CFT with those external dimensions and that spectrum exists.

Proof. Suppose such a CFT existed. It would satisfy Eq. (50.101), hence Eq. (50.104). Split off the identity from the even sum and use the normalization Eq. (50.106):

$$0 = \underbrace{\mathrm{Tr}[\Lambda_+(0,0)P_{\mathbf{1}}]}_{=1} + \sum_{\mathcal{O}^+ \neq \mathbf{1}} \mathrm{Tr}\left[\Lambda_+(\Delta_{\mathcal{O}},\ell_{\mathcal{O}})P_{\mathcal{O}}\right] + \sum_{\mathcal{O}^-} p_{\mathcal{O}}\, \Lambda_-(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}). \tag{50.109}$$

By unitarity each $P_{\mathcal{O}} \succeq 0$ (Theorem 50.26) and each $p_{\mathcal{O}} \geq 0$. By hypothesis each $\Lambda_+(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}) \succeq 0$, so by Theorem 50.29 every term in the first sum is ≥ 0; and each $\Lambda_-(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}) \geq 0$, so every term in the second sum is ≥ 0. The right-hand side is therefore $\geq 1 > 0$, contradicting the equality to 0. Hence no such CFT exists. □

The content of Theorem 50.30 is exactly what makes the optimization a *semidefinite* program rather than a linear one: the unknowns of the dual are the coefficients of α, and the constraints Eq. (50.107) demand that a matrix built linearly from those coefficients be positive semidefinite for every (Δ, ℓ) — a constraint on the eigenvalues of a matrix, which is the defining feature of an SDP. The continuous label Δ is handled by the polynomial/sum-of-squares reduction of Section 50.3, which turns "$\Lambda_+(\Delta,\ell) \succeq 0$ for all $\Delta \geq \Delta^*_\ell$" into finitely many matrix-positivity conditions a solver such as that of Simmons-Duffin [441] can dispatch. A short numerical example shows why the matrix condition cannot be replaced by positivity of the diagonal alone.

Worked Example 50.31 (Why diagonal positivity is not enough)**.** Imagine a candidate functional that, at some even (Δ, ℓ) in the spectrum, produced

$$\Lambda_+(\Delta,\ell) = \begin{pmatrix} 1 & -3 \\ -3 & 1 \end{pmatrix}. \tag{50.110}$$

Its diagonal entries are positive, so it would pass a naive "each correlator separately" test that only demanded the $\sigma\sigma$ and $\varepsilon\varepsilon$ channels be nonnegative. But it is not PSD: its determinant is $1 - 9 = -8 < 0$, so it has a negative eigenvalue ($\frac{1}{2}(2) \pm \frac{1}{2}\sqrt{0 + 4 \cdot 9} = 1 \pm 3$, i.e. -2 and 4). Now feed it a perfectly legal unitary contribution: an operator with $\vec{\lambda} = (1, 1)^{\mathsf{T}}$, so $P = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 1 \end{smallmatrix}\right) \succeq 0$. The pairing is

$$\mathrm{Tr}[\Lambda_+ P] = 1 \cdot 1 + (-3) \cdot 1 + (-3) \cdot 1 + 1 \cdot 1 = -4 < 0, \tag{50.111}$$

which would wreck the exclusion proof: the term is supposed to be ≥ 0. The off-diagonal product $\lambda_{\sigma\sigma}\lambda_{\varepsilon\varepsilon}$ entered with a sign that the diagonal test never saw. Demanding the full matrix inequality $\Lambda_+ \succeq 0$ is precisely the condition that guarantees $\mathrm{Tr}[\Lambda_+ P] \geq 0$ for *every* PSD P — by Theorem 50.29 the matrix condition is not only sufficient but, taking P to be the projector onto the negative eigenvector, also necessary. The semidefinite constraint is the exact and minimal way to certify the pairing.

50.5.6 Closing the wedge: the two gap assumptions and the island

With the matrix functional in hand, the search proceeds exactly as in Chapter 49 but now over the richer space of dual variables, and with two gap assumptions that encode what it means to be the Ising fixed point rather than some other unitary $\mathbb{Z}_2$ CFT. The assumptions are physical and sharp:

1. σ **is the unique relevant $\mathbb{Z}_2$-odd scalar.** In the $\sigma \times \varepsilon$ channel the lightest odd scalar is σ itself; the next $\mathbb{Z}_2$-odd scalar σ' is irrelevant, $\Delta_{\sigma'} \geq 3 = d$. Equivalently, the odd sector has a gap above Δ_σ up to the marginality threshold.

2. ε **is the unique relevant $\mathbb{Z}_2$-even scalar.** In $\sigma \times \sigma$ and $\varepsilon \times \varepsilon$ the lightest even scalar is ε; the next even scalar ε' obeys $\Delta_{\varepsilon'} \geq 3$. This is the single-correlator gap of the previous section, now imposed in both diagonal channels.

These two statements are not arbitrary numerical conveniences; they are the *defining* feature of a critical point in the Ising universality class. A statistical system reaches criticality by tuning exactly one relevant coupling — the temperature, conjugate to the energy operator ε — while the $\mathbb{Z}_2$ (spin-flip) symmetry forbids a relevant odd coupling unless an external magnetic field is switched on. A unitary 3D CFT with precisely one relevant scalar in each $\mathbb{Z}_2$ sector is therefore the bootstrap's formalization of "a $\mathbb{Z}_2$ critical point reached by tuning one knob." The Wilson–Fisher analysis of Chapters 23 and 24 reached the same conclusion perturbatively: at the fixed point the mass operator ($\sim \varepsilon$) is the lone relevant direction and the quartic coupling is marginally irrelevant.

The effect of these two gaps on the allowed region of Theorem 50.30 is dramatic. Figure 50.6 shows it. The single correlator gave an open wedge in the $(\Delta_\sigma, \Delta_\varepsilon)$ plane, bounded above near the kink but unbounded below and laterally. Imposing the mixed system with both gaps, the functional can now push from *every* direction — the $\varepsilon\varepsilon$ channel bounds Δ_ε from below where the $\sigma\sigma$ channel bounded it from above, and the cross-channel ties Δ_σ to Δ_ε — so the allowed region pinches off into a small *closed* island surrounding the kink. As the number Λ of derivative components retained in the functional is increased (equivalently, as the SDP search is run at higher order), the islands are nested and shrink, collapsing onto a point. This is the qualitative content of Kos et al. [303, 304]: each refinement of the dual search certifies a smaller region as the only place a unitary CFT with these gaps can live.

50.5.7 The record numbers and the meeting of two threads

When the mixed system is run to high derivative order, the island shrinks to a point known to a precision no other method in this book approaches. The current rigorous values for the 3D Ising CFT [304] are

$$\Delta_\sigma = 0.5181489(10), \qquad \Delta_\varepsilon = 1.412625(10), \tag{50.112}$$

with the leading OPE coefficients in the same determination

$$\lambda_{\sigma\sigma\varepsilon} = 1.0518537(41), \qquad \lambda_{\varepsilon\varepsilon\varepsilon} = 1.532435(19). \tag{50.113}$$

These six-significant-figure numbers are the output of consistency alone: no Lagrangian, no perturbative expansion, no lattice. The dimensions translate immediately into the critical exponents of the universality class, which is where the bootstrap meets the laboratory.

Worked Example 50.32 (Critical exponents from the island, and the comparison)**.** In a d-dimensional CFT the dimension of the order parameter fixes the anomalous-dimension exponent η through $\Delta_\sigma = \tfrac{1}{2}(d-2+\eta)$, and the dimension of the energy operator fixes the correlation-length exponent ν through the renormalization-group eigenvalue of the one relevant even coupling, $y_t = d - \Delta_\varepsilon$ and $\nu = 1/y_t$. In $d = 3$,

$$\eta = 2\Delta_\sigma - (d-2) = 2\Delta_\sigma - 1 = 2(0.5181489) - 1 = 0.0362978(20), \tag{50.114}$$

$$\nu = \frac{1}{d-\Delta_\varepsilon} = \frac{1}{3-\Delta_\varepsilon} = \frac{1}{1.587375} = 0.629971(4), \tag{50.115}$$

where the uncertainties propagate linearly: $\delta\eta = 2\,\delta\Delta_\sigma = 2.0\times10^{-6}$ and $\delta\nu = \delta\Delta_\varepsilon/(3-\Delta_\varepsilon)^2 = 1.0\times10^{-5}/1.5874^2 = 4.0\times10^{-6}$. The remaining exponents follow from the scaling relations established in Chapter 23: $\gamma = \nu(2-\eta) = 1.237075(10)$, $\beta = \tfrac{\nu}{2}(d-2+\eta) = 0.326419(3)$, and $\alpha = 2 - d\nu = 0.110087(12)$. The leading

Table 50.1: The 3D Ising universality class: bootstrap, Monte Carlo, and experiment. Bootstrap values from Kos et al. [304] via Eqs. (50.112), (50.114) and (50.115); Monte Carlo from Hasenbusch [254]; experimental ranges are representative of fluid and magnetic systems in the class. Parenthesized digits are one-standard-deviation uncertainties in the last quoted figures.

quantity	numerical bootstrap	Monte Carlo	experiment
Δ_σ	0.5181489(10)	—	—
Δ_ε	1.412625(10)	—	—
η	0.0362978(20)	0.03627(10)	≈ 0.036
ν	0.629971(4)	0.63002(10)	≈ 0.63
ω	≈ 0.83	0.832(6)	—

correction-to-scaling exponent comes from the next even scalar, $\omega = \Delta_{\varepsilon'} - d$; the same island analysis places $\Delta_{\varepsilon'} \approx 3.83$, giving $\omega \approx 0.83$.

The numbers can be checked against two completely independent determinations. Table 50.1 collects them. The Monte-Carlo finite-size-scaling study of Hasenbusch [254] gives $\nu = 0.63002(10)$ and $\eta = 0.03627(10)$; the bootstrap values lie inside the Monte-Carlo error bars and are roughly two orders of magnitude more precise. Laboratory measurements of systems in the 3D Ising universality class — the liquid–vapor critical point of simple fluids, binary-fluid demixing, and uniaxial antiferromagnets — give $\nu \approx 0.63$ and $\eta \approx 0.036$ to the percent level set by the experiments, fully consistent with Eqs. (50.114) and (50.115). Three methods that share no equations — a semidefinite program, a lattice simulation, and a thermometer — agree on the same numbers, which is as strong a statement of the reality of the fixed point as physics offers.

This is the quantitative payoff of ϕ^4 theory. The model began in Chapter 6 as a classical real scalar with a ϕ^4 interaction and a $\mathbb{Z}_2$ symmetry; it was renormalized in Chapter 22, its one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ computed in Chapter 24, and its Wilson–Fisher fixed point located perturbatively in $d = 4 - \epsilon$ at $\lambda_* = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2)$ with $\gamma_\phi = O(\epsilon^2)$. Setting $\epsilon = 1$ to reach $d = 3$ pushes that expansion to the edge of its validity; resummed, it returns $\eta \approx 0.036$ and $\nu \approx 0.63$ to a percent or two. The bootstrap reaches the *same* fixed point — the unique unitary 3D $\mathbb{Z}_2$ CFT with one relevant scalar in each sector — from the opposite side, using no Lagrangian at all, and pins it to six significant figures. The perturbative thread of Part VI and the consistency thread of this chapter meet on the single number $\nu = 0.629971(4)$. That the Wilson–Fisher fixed point of Chapter 24 *is* the bootstrap island of Fig. 50.6 is the identification Chapter 52 makes precise when it characterizes the Wilson–Fisher fixed point as a conformal field theory; here it is confirmed numerically to a precision the ϵ-expansion cannot dream of.

Caution: what the island is, and is not, a theorem of

The island of Fig. 50.6 is a rigorous statement — modulo controllable numerical error in the SDP — but a *conditional* one. It says: *any* unitary, $\mathbb{Z}_2$-symmetric 3D CFT whose only relevant scalars are one odd and one even must have $(\Delta_\sigma, \Delta_\varepsilon)$ inside the island. It does *not*, by itself, prove that such a CFT exists, nor that the theory living there is the continuum limit of the lattice Ising model. The existence of a nontrivial solution is supported by the fact that the island is non-empty at every derivative order and by the convergence of its extremal data; the identification with the Ising universality class is physics input — the matching critical exponents of Table 50.1 — not a corollary of the bound. We separate these claims carefully in Section 50.6.

The island has a boundary, and a CFT that sits exactly on it is special: there the exclusion functional of Theorem 50.30 is on the verge of failing, which means the inequalities Eqs. (50.107) and (50.108) are saturated by a particular set of operators. That saturating, extremal configuration carries more information than the dimensions $(\Delta_\sigma, \Delta_\varepsilon)$ alone: it encodes a candidate for the *entire* spectrum of exchanged operators and their OPE coefficients. The next section turns this observation into a method — the extremal functional — for reading the full CFT data off the boundary of the island, and then steps back to delimit honestly what the numerical bootstrap proves and what it assumes.

50.6 Reading the Spectrum: Extremal Functionals, Rigor, and Scope

The mixed-correlator program of Section 50.5 did more than corner the 3D Ising data into the island Eq. (49.171). At the moment the semidefinite solver certifies that a candidate point sits on the *boundary* of the allowed region, it returns two objects at once: a configuration of nonnegative OPE coefficients $\{p_{\mathcal{O}}\}$ that saturates crossing — the extremal, boundary-realizing solution handed up by Eq. (50.128) — and the optimal functional that certifies no solution survives just beyond the boundary. So far we have used only the second object, and only for its sign, to draw exclusion curves and shrink islands. This closing section extracts the information hiding in the *pairing* of the two. The claim, which is the dual reading of the entire bootstrap, is that the extremal functional does not merely separate the allowed from the forbidden: its zeros, read off as a function of dimension, are the dimensions of the operators present in the saturating CFT. The bootstrap stops being a tool that brackets the data and becomes a tool that *reconstructs* it. We prove this through complementary slackness at the optimum, work the leading Ising spectrum out of the resulting double-zero structure, and

then spend the remainder of the section being honest: about exactly which of our statements are theorems and which are physics, about the adjacent methods that make the search efficient, and about where the program stands and stalls as of the mid-2020s.

50.6.1 The dual reading: from a separating hyperplane to a spectrum

Recall the logical skeleton built in Section 50.2.1. The crossing sum rule of Eq. (49.74), with the identity contribution isolated and normalized to unit coefficient, reads

$$\vec{F}_{0,0} \;+\; \sum_{\mathcal{O}\neq \mathbf{1}} p_{\mathcal{O}}\, \vec{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}} \;=\; 0, \qquad p_{\mathcal{O}} = \lambda^2_{\phi\phi\mathcal{O}} \geq 0, \tag{50.116}$$

where $\vec{F}_{\Delta,\ell}$ is the crossing vector of Eq. (49.87) — in practice the finite array of derivatives $\partial_z^m \partial_{\bar{z}}^n F_{\Delta,\ell}\big|_{z=\bar{z}=1/2}$ retained at a chosen truncation order, so that $\vec{F}_{\Delta,\ell} \in \mathbb{R}^N$ for some large but finite N. A linear functional α is then a vector in the same space, acting by

$$\alpha[\,\vec{F}_{\Delta,\ell}\,] \;=\; \alpha \cdot \vec{F}_{\Delta,\ell} \;=\; \sum_{a=1}^{N} \alpha_a \left(\partial^a F_{\Delta,\ell}\right)\Big|_{z=\bar{z}=1/2}. \tag{50.117}$$

The exclusion theorem (Theorem 49.28, recast in convex-geometric form in Theorem 50.8 and as a semidefinite program in Theorem 50.19) says that a candidate spectrum S — a set of (Δ,ℓ) consistent with the unitarity bounds of Chapter 49 and with whatever gap assumptions we impose — is *forbidden* precisely when some functional separates the identity from the rest:

$$\alpha[\,\vec{F}_{0,0}\,] \;>\; 0, \qquad \alpha[\,\vec{F}_{\Delta,\ell}\,] \;\geq\; 0 \quad \text{for all } (\Delta,\ell) \in S. \tag{50.118}$$

The reason is immediate from Eq. (50.116): applying such an α gives $\alpha[\vec{F}_{0,0}] + \sum_{\mathcal{O}} p_{\mathcal{O}}\, \alpha[\vec{F}_{\mathcal{O}}]$, a strictly positive number plus a sum of nonnegative numbers, which cannot vanish; so no nonnegative crossing solution supported on S exists. Geometrically α is a hyperplane in $\mathbb{R}^N$ with the convex cone $\{\,\vec{F}_{\Delta,\ell}\,\}$ on one side and $-\vec{F}_{0,0}$ on the other.

The dual reading begins with a question this picture leaves unasked. *What happens at the boundary, where exclusion is marginal?* As the gap or the external dimension is tuned to the edge of the allowed region, the separating hyperplane is forced flush against the cone. It can no longer keep strictly clear of every block; it must touch the cone, and it touches it exactly along the directions $\vec{F}_{\Delta,\ell}$ corresponding to the operators that the saturating solution actually uses. Those touching directions are where $\alpha[\vec{F}_{\Delta,\ell}] = 0$. Reading the spectrum is then nothing more than locating the zeros of the boundary functional. To turn this picture into a theorem we need the precise relationship between the boundary functional and the boundary solution, and that relationship is supplied by Lagrange duality.

50.6.2 The optimization and its dual

The cleanest setting in which the boundary functional and the boundary solution appear as an exact dual pair is the maximization of a single OPE coefficient — the original *extremal functional method*. Everything we prove transfers verbatim to gap maximization and to c_T-minimization, which we treat as corollaries below.

Fix the external dimension Δ_ϕ and a candidate operator $\mathcal{O}_0 = (\Delta_0, \ell_0)$ whose coefficient we wish to push as high as crossing and unitarity permit. Collect the admissible operators (those obeying unitarity and the imposed gaps) into the index set S, and let A be the matrix whose columns are the crossing vectors $\vec{F}_{\mathcal{O}}$ for $\mathcal{O} \in S \setminus \{\mathbf{1}\}$, so that the crossing equation Eq. (50.116) is $A\,p = -\vec{F}_{0,0}$ with $p = (p_{\mathcal{O}})_{\mathcal{O} \neq \mathbf{1}}$. The optimization is the (semi-infinite) linear program

$$\textbf{(P)} \qquad \max_{p \geq 0} \; c^{\mathsf{T}} p \quad \text{subject to} \quad A\,p = -\vec{F}_{0,0}, \tag{50.119}$$

where the objective vector $c = e_{\mathcal{O}_0}$ selects the single coefficient $p_{\mathcal{O}_0}$. "Semi-infinite" because S is a continuum of operators (every dimension above the unitarity bound, every even spin) while the constraint $A\,p = -\vec{F}_{0,0}$ lives in the finite space $\mathbb{R}^N$ fixed by the truncation. The optimal value of **(P)** is the largest squared OPE coefficient $\lambda^2_{\phi\phi\mathcal{O}_0}$ attainable by any unitary crossing-symmetric solution with the assumed gaps.

Every linear program has a dual obtained by pricing the constraints. Introduce a vector $y \in \mathbb{R}^N$ of multipliers, one per component of the crossing equation, and form the Lagrangian

$$\mathcal{L}(p, y) = c^{\mathsf{T}} p + y^{\mathsf{T}}\big(-\vec{F}_{0,0} - A\,p\big) = -y^{\mathsf{T}} \vec{F}_{0,0} + \big(c - A^{\mathsf{T}} y\big)^{\mathsf{T}} p. \tag{50.120}$$

Maximizing freely over $p \geq 0$ gives $+\infty$ unless the coefficient of p is nonpositive in every component, i.e. unless $A^{\mathsf{T}} y \geq c$; when that holds the supremum over $p \geq 0$ is attained at $p = 0$ and equals $-y^{\mathsf{T}} \vec{F}_{0,0}$. The dual program is therefore

$$\textbf{(D)} \qquad \min_{y} \; -y^{\mathsf{T}} \vec{F}_{0,0} \quad \text{subject to} \quad \vec{F}_{\mathcal{O}} \cdot y \;\geq\; \delta_{\mathcal{O},\mathcal{O}_0} \quad \text{for all } \mathcal{O} \in S \setminus \{\mathbf{1}\}. \tag{50.121}$$

The constraint reads $\vec{F}_{\mathcal{O}_0} \cdot y \geq 1$ for the singled-out operator and $\vec{F}_{\mathcal{O}} \cdot y \geq 0$ for every other admissible operator. A dual-feasible y is, up to the normalization at $\mathcal{O}_0$, exactly an exclusion functional in the sense of Eq. (50.118): it is nonnegative on the admissible cone. We call the optimal dual vector $\alpha_* := y_*$ the *extremal functional*.

Definition 50.33 (Extremal solution and extremal functional)**.** At a boundary point of the bootstrap-allowed region, the *extremal solution* is an optimizer $p_* \geq 0$ of the primal Eq. (50.119), and the *extremal functional* α_* is an optimizer of the dual Eq. (50.121). The *extremal spectrum* is the support $\Sigma_* = \{\mathcal{O} : p_{*,\mathcal{O}} > 0\}$ of the extremal solution.

Weak duality is the half-line of Eq. (50.120) that always holds: for any primal-feasible p and dual-feasible y,

$$c^{\mathsf{T}}p = y^{\mathsf{T}}(A\,p) + (c - A^{\mathsf{T}}y)^{\mathsf{T}}p = -\,y^{\mathsf{T}}\vec{F}_{0,0} + \underbrace{(c - A^{\mathsf{T}}y)^{\mathsf{T}}p}_{\leq 0} \;\leq\; -\,y^{\mathsf{T}}\vec{F}_{0,0}, \tag{50.122}$$

because $c - A^{\mathsf{T}}y \leq 0$ componentwise (dual feasibility) and $p \geq 0$ (primal feasibility). The primal objective never exceeds the dual objective. *Strong* duality — that the two optima coincide — is the content of the linear-programming duality theorem proved, in the finite-truncation form that governs every actual computation, in Theorem 50.6 via Farkas' lemma; the semidefinite generalization needed once positivity becomes matrix-valued is Theorem 50.22. At the common optimum,

$$c^{\mathsf{T}}p_* \;=\; -\,\alpha_*^{\mathsf{T}}\vec{F}_{0,0}. \tag{50.123}$$

The maximal OPE coefficient (left) equals minus the extremal functional's action on the identity (right). The two readings of the boundary are the same number.

50.6.3 Complementary slackness and the double-zero theorem

Strong duality forces a rigid alignment between which operators the extremal solution uses and where the extremal functional vanishes. This is complementary slackness, and it is the whole mechanism.

Theorem 50.34 (Extremal functional locates the spectrum)**.** *Let p_* and α_* be optimizers of the primal Eq. (50.119) and dual Eq. (50.121), related by strong duality Eq. (50.123). Then for every admissible operator $\mathcal{O} \in S \setminus \{\mathbf{1}\}$,*

$$p_{*,\mathcal{O}}\,(\alpha_*[\vec{F}_{\mathcal{O}}] - \delta_{\mathcal{O},\mathcal{O}_0}) = 0. \tag{50.124}$$

Consequently every operator of the extremal spectrum Σ_* other than the singled-out $\mathcal{O}_0$ is a zero of the extremal functional,

$$\alpha_*[\,\vec{F}_{\mathcal{O}}\,] = 0 \qquad \text{for all } \mathcal{O} \in \Sigma_* \setminus \{\mathcal{O}_0\}, \tag{50.125}$$

while $\alpha_*[\vec{F}_{\mathcal{O}_0}] = 1$. Moreover, since $\Delta \mapsto \alpha_*[\vec{F}_{\Delta,\ell}]$ is a nonnegative real-analytic function on the admissible interval at each spin ℓ, each interior zero in Eq. (50.125) is at least a double *zero:*

$$\alpha_*[\,\vec{F}_{\Delta,\ell}\,]\Big|_{\Delta=\Delta_{\mathcal{O}}} = 0 \quad \text{and} \quad \frac{\partial}{\partial\Delta}\,\alpha_*[\,\vec{F}_{\Delta,\ell}\,]\Big|_{\Delta=\Delta_{\mathcal{O}}} = 0. \tag{50.126}$$

Proof. The slackness identity Eq. (50.124) is the equality case of weak duality. Subtract the two sides of Eq. (50.122) at the optimum and use strong duality Eq. (50.123):

$$0 = (\,-\,\alpha_*^{\mathsf{T}}\vec{F}_{0,0}) - c^{\mathsf{T}}p_* = (A^{\mathsf{T}}\alpha_* - c)^{\mathsf{T}}p_* = \sum_{\mathcal{O}\neq\mathbf{1}} (\alpha_*[\vec{F}_{\mathcal{O}}] - c_{\mathcal{O}})\,p_{*,\mathcal{O}}, \tag{50.127}$$

where the middle equality used $\alpha_*^{\mathsf{T}} A\, p_* = -\alpha_*^{\mathsf{T}} \vec{F}_{0,0}$ (primal feasibility of p_*) and $A^{\mathsf{T}}\alpha_* = (\alpha_*[\vec{F}_{\mathcal{O}}])_{\mathcal{O}}$. Each summand is a product of a nonnegative factor $p_{*,\mathcal{O}} \geq 0$ and a nonnegative factor $\alpha_*[\vec{F}_{\mathcal{O}}] - c_{\mathcal{O}} \geq 0$ (dual feasibility, since $c_{\mathcal{O}} = \delta_{\mathcal{O},\mathcal{O}_0}$). A sum of nonnegative terms vanishes only if each term vanishes; this is Eq. (50.124). For $\mathcal{O} \in \Sigma_*$ we have $p_{*,\mathcal{O}} > 0$, so the second factor must vanish: $\alpha_*[\vec{F}_{\mathcal{O}}] = c_{\mathcal{O}}$, which is 0 for $\mathcal{O} \neq \mathcal{O}_0$ and 1 for $\mathcal{O} = \mathcal{O}_0$. This is Eq. (50.125).

For the double-zero statement, fix a spin ℓ and consider $g_\ell(\Delta) := \alpha_*[\vec{F}_{\Delta,\ell}]$ as a function of the dimension. The conformal block $F_{\Delta,\ell}(z,\bar{z})$, and hence each of its derivatives at the crossing-symmetric point, is real-analytic in Δ on the admissible interval (the blocks are analytic in Δ away from the discrete poles that lie below the unitarity bound, Theorem 50.16 and Eq. (50.46)); g_ℓ is therefore real-analytic there. Dual feasibility Eq. (50.121) states $g_\ell(\Delta) \geq 0$ for every admissible Δ at spin ℓ. Let $\Delta_{\mathcal{O}}$ be the location of an operator in Σ_* with spin ℓ and $\mathcal{O} \neq \mathcal{O}_0$, so $g_\ell(\Delta_{\mathcal{O}}) = 0$ by Eq. (50.125), and suppose $\Delta_{\mathcal{O}}$ is interior to the admissible interval (true for every operator above the gap). A nonnegative differentiable function that attains the value 0 at an interior point has a local minimum there, so its first derivative vanishes: $g'_\ell(\Delta_{\mathcal{O}}) = 0$. This is Eq. (50.126). The zero has even order; generically it is exactly double, though higher even order is possible at non-generic boundary points where two operators collide. □

Two corollaries make the theorem usable. First, the extremal *spectrum is discrete and finite at any truncation.* The primal Eq. (50.119) is a linear program with N equality constraints; its optimum is attained at a vertex of the feasible polyhedron, i.e. at a *basic* solution with at most N nonzero components. Thus $|\Sigma_*| \leq N$: only finitely many operators carry nonzero OPE coefficient, even though the admissible set is a continuum. As the truncation order N is raised, the located operators multiply and fill in the true (infinite) spectrum, low dimensions first. Second, *the spectrum is read off, not assumed.* The double zeros of $\Delta \mapsto \alpha_*[\vec{F}_{\Delta,\ell}]$ at each spin are computed directly from the optimal functional returned by the solver; their locations *are* the dimensions $\Delta_{\mathcal{O}}$. Having located Σ_*, one recovers the coefficients by solving the now-finite linear system $\vec{F}_{0,0} + \sum_{\mathcal{O}\in\Sigma_*} p_{\mathcal{O}} \vec{F}_{\mathcal{O}} = 0$ for the $p_{\mathcal{O}}$ — an overdetermined but consistent set of N linear equations in $|\Sigma_*| \leq N$ unknowns. The full CFT data $(\Delta_{\mathcal{O}}, \lambda^2_{\phi\phi\mathcal{O}})$ emerge from the single boundary functional.

Physical Insight

The exclusion bootstrap asks a yes/no question and the functional answers it with a sign. The extremal bootstrap notices that *at the boundary the sign is forced to be zero in exactly the right places*, and those places are the spectrum. The same convex-optimization certificate that proves a region empty, evaluated on the last region that is not empty, draws a picture of the unique theory living there. Nothing extra is computed; the spectrum

was always encoded in the dual variable, waiting to be read.

The same logic operates whenever the boundary is reached, regardless of which objective drove us there. *Gap maximization* (pushing the dimension of the first scalar above the identity as high as possible, the optimization behind the single-correlator kink of Section 50.4) has the same dual structure: at the maximal gap the certifying functionals converge to an extremal α_* whose first scalar zero sits exactly at the maximal gap, and whose remaining double zeros give the rest of the saturating spectrum. *c_T-minimization* — minimizing the stress-tensor channel coefficient, equivalently maximizing the central charge's inverse — is Eq. (50.119) with $\mathcal{O}_0$ the stress tensor and the objective reversed in sign; this is the variant that originally pinned the 3D Ising point at the tip of the kink before mixed correlators closed the island. In each case Theorem 50.34 delivers the spectrum from the optimal functional's double zeros.

50.6.4 Reading the leading Ising spectrum

We now apply the theorem at the Ising island. The boundary-saturating configuration handed up from Section 50.5 sits at the data Eq. (49.171),

$$\Delta_\sigma = 0.5181489(10), \qquad \Delta_\varepsilon = 1.412625(10), \tag{50.128}$$

and the extremal functional evaluated there has the double-zero structure drawn in Fig. 50.7. We read the leading operators off spin by spin.

Worked Example 50.35 (The leading 3D Ising spectrum from the extremal functional)**.** We trace each double zero of Fig. 50.7 to an operator and recover its data, taking the located dimensions as the computed zeros of $\alpha_*[\vec{F}_{\Delta,\ell}]$ and the coefficients from the linear solve described after Theorem 50.34.

Spin 0*, the leading even scalar.* The first interior double zero in the $\ell = 0$ channel is at $\Delta = 1.412625$. By Eq. (50.125) this is an operator of the extremal solution, and it is the lowest-dimension $\mathbb{Z}_2$-even scalar: it is ε, the energy operator of Theorem 49.27. Its location reproduces Δ_ε in Eq. (50.128) — consistency, since the island point was defined by assuming ε is the first even scalar. Through the dictionary Eq. (49.163) this fixes the correlation-length exponent: $1/\nu = d - \Delta_\varepsilon = 3 - 1.412625 = 1.587375$, i.e. $\nu = 0.629971(4)$. Solving the residual linear crossing system on the located spectrum returns the coefficient $\lambda_{\sigma\sigma\varepsilon} = 1.0518537(41)$.

Spin 0*, the subleading scalar.* The next double zero sits at $\Delta_{\varepsilon'} \approx 3.82966$. This is ε', the second $\mathbb{Z}_2$-even scalar. It is *irrelevant* in $d = 3$ (dimension above 3), and its distance above marginality is the leading correction-to-scaling exponent,

$$\omega = \Delta_{\varepsilon'} - d \approx 3.830 - 3 = 0.830, \tag{50.129}$$

the exponent that controls how critical quantities approach their scaling forms as the critical point is neared — a number the extremal functional delivers

as readily as the relevant dimensions, and which agrees with the Monte Carlo determination $\omega \approx 0.83$. The bootstrap did not need to be told ε' exists; the second zero produced it.

Spin 2, the stress tensor. The first double zero in the $\ell = 2$ channel is pinned exactly at $\Delta = 3$. This is not a number to be measured but one fixed by symmetry: every local CFT has a conserved stress tensor $T_{\mu\nu}$, a spin-2 operator saturating the unitarity bound $\Delta = d - 2 + \ell = 3$ of Chapter 49 for $\ell = 2$, $d = 3$. The extremal functional's spin-2 zero falling on $\Delta = 3$ is a stringent internal check that the saturating solution is a genuine local CFT and not a spurious crossing solution. The coefficient of this block is inverse to the central charge: $p_T \propto \Delta_\phi^2 / c_T$, and the linear solve returns $c_T / c_T^{\text{free}} = 0.946534(11)$ — a measure of how far the interacting Ising CFT sits from the free scalar.

Spin 2, the next operator. The following $\ell = 2$ double zero, near $\Delta \approx 5.5$, is the leading spin-2 operator above the stress tensor; at large spin these $\ell = 2, 4, 6, \ldots$ operators organize into the double-twist family $\Delta \approx 2\Delta_\sigma + \ell$ that the analytic bootstrap of Chapter 51 controls in closed form.

The reconstruction. Assembling the located dimensions and the coefficients from the linear solve gives the leading Ising CFT data,

$$\begin{aligned} &\Delta_\sigma = 0.5181489, \quad \Delta_\varepsilon = 1.412625, \quad \Delta_{\varepsilon'} \approx 3.83, \quad \Delta_T = 3, \\ &\lambda_{\sigma\sigma\varepsilon} = 1.0518537, \quad \lambda_{\varepsilon\varepsilon\varepsilon} = 1.532435, \quad c_T / c_T^{\text{free}} = 0.946534, \end{aligned} \tag{50.130}$$

each digit improving as the truncation order N of the functional is raised and the island shrinks around the point. Through Eqs. (49.162) and (49.163) the dimensions become the critical exponents $\eta = 2\Delta_\sigma - 1 = 0.0362978(20)$ and $\nu = 0.629971(4)$. This is the completion of the bootstrap stage of ϕ^4 theory: the Wilson–Fisher fixed point, found perturbatively in $d = 4 - \epsilon$ in Chapter 24 and characterized as a CFT in Theorem 52.11, is not merely *bounded* by crossing and unitarity — its operator spectrum is *read off*, operator by operator, from a single convex-optimization certificate.

50.6.5 Rigor, honestly accounted

The numbers in Eq. (50.130) are more precise than any other determination of the Ising data, and precision invites overclaiming. We separate, sharply, what the bootstrap proves from what it assumes. There are three distinct epistemic tiers, and they must not be conflated.

Tier 1 — exclusions are theorems, up to controllable arithmetic. When the solver returns a functional α satisfying Eq. (50.118) for a candidate spectrum S, the corresponding region is forbidden with mathematical certainty: no unitary, crossing-symmetric, $\mathbb{Z}_2$-symmetric CFT with the assumed gaps can have data there. The certificate is finite and checkable. Two arithmetic caveats are genuine but controllable. First, the functional acts on a finite-dimensional

truncation of the crossing equation Eq. (50.116); positivity over the continuum of dimensions and the infinite tower of spins is enforced by polynomial sum-of-squares representations (Theorem 50.22) that are exact at the truncation order, and enlarging the truncation only *tightens* the bound, never loosens it. Second, the solver works in finite-precision floating point, but the optimal functional it returns can be validated by re-evaluating Eq. (50.118) in exact rational or rigorous interval arithmetic; when that validation passes — as it can be made to, and has been — the exclusion is a theorem in the literal sense. The boundary curves of Section 50.4 and the outer wall of the island in Section 50.5 are, region by region, certified theorems of this kind.

Tier 2 — islands are theorems conditional on explicit gap assumptions. A closed island, unlike an exclusion curve, asserts that the allowed region is *small and bounded*, and that assertion is conditional. The Ising island of Eq. (49.171) is rigorous given the hypotheses under which it was computed: unitarity, crossing of the full $\langle\sigma\sigma\sigma\sigma\rangle$, $\langle\sigma\sigma\varepsilon\varepsilon\rangle$, $\langle\varepsilon\varepsilon\varepsilon\varepsilon\rangle$ system, $\mathbb{Z}_2$ symmetry, and — decisively — the gap assumption that there are exactly two relevant scalars, one $\mathbb{Z}_2$-odd (σ) and one $\mathbb{Z}_2$-even (ε), with everything else above the relevant threshold $\Delta = d = 3$. Change that gap input and the island moves or dissolves. The gap assumption is not derived inside the bootstrap; it is imported from the renormalization-group understanding of the Ising universality class — the statement that the critical point is reached by tuning exactly two relevant couplings, temperature and field. Given the assumption, the island is a Tier-1 object: its complement is excluded by certified functionals. What is *not* a theorem of the same hardness is the further claim that the true allowed region is a single point: each computed island is nonempty in every run and shrinks monotonically as the truncation is enlarged, which is overwhelming numerical evidence of convergence to a point, but a proof that the limit is nonempty and a single point is not in hand.

Tier 3 — naming the theory is physics input. That the island is the *3D Ising model* — the lattice spin system, the ϕ^4 fixed point, the liquid–vapor and uniaxial-magnet critical points in the same universality class — is not a bootstrap theorem at all. It is the identification of a fixed point characterized by symmetry and relevant-operator content with a named physical system, and it rests on the renormalization-group universality statement of Theorem 23.26: all systems flowing to the same fixed point share its critical data. The bootstrap computes the data of *a* unitary $\mathbb{Z}_2$ CFT with two relevant scalars; universality, established by entirely different (RG) arguments, says that any such CFT is the Ising fixed point and that the lattice model flows to it. The agreement of Eq. (50.130) with Monte Carlo (Chapter 43) and with the resummed ϵ-expansion (Chapter 24) is the empirical confirmation of that identification, not its proof.

Caution

The single most common overstatement is "the bootstrap proved the 3D Ising model exists and has these exponents." It did not. As of the mid-2020s there is no construction of the 3D Ising CFT as a mathematical object satisfying all the axioms simultaneously, and no proof that the shrinking islands converge to a nonempty limit or that the limit is unique. The honest statement is the conditional one: *if* a unitary CFT with the Ising gaps exists, its data lie in the island, pinned to six significant figures. Existence remains open — in sharp contrast to two dimensions, where the free-fermion realization of the Ising CFT (Chapter 48) is a genuine existence proof. The remaining gap is no longer "is the number right?" but the sharper mathematical question "does the limiting theory exist and is it unique?"

50.6.6 The broader toolkit

The single- and mixed-correlator semidefinite programs of this chapter are the core, but the working bootstrap is surrounded by methods that make the search efficient and connect it to the analytic side. We state their roles; their full development belongs to the dedicated literature and, for the analytic methods, to Chapter 51.

The navigator function. Locating an island by brute force — testing a grid of candidate points, each by a separate feasibility solve, and bisecting toward the boundary — scales badly as the number of free parameters grows (external dimensions, gaps, several OPE ratios). The *navigator function* replaces the binary feasible/infeasible verdict with a single smooth function on parameter space, engineered to be negative inside the allowed region, positive outside, and to vary smoothly across the boundary. Its gradient points toward the island, so the island can be found and its extent mapped by gradient descent rather than by exhaustive scanning — the difference between rolling downhill to a valley and checking every point on a map. This is what makes high-dimensional mixed-correlator searches, with many simultaneous gap and ratio parameters, computationally feasible [417].

OPE and c_T extremization. Before islands closed, the Ising point was first isolated as the location where the central charge c_T is *minimized* along the single-correlator boundary — the tip of the kink coincides with the minimum of c_T, a sharp feature that selects the fixed point without a gap assumption strong enough to bound it from all sides [173]. OPE-coefficient maximization, the optimization of Eq. (50.119), is the same family of tools: extremizing any physically meaningful linear functional of the data selects an extremal point whose full spectrum is then read off by Theorem 50.34. These extremization methods and the navigator

are complementary: extremization finds and characterizes a boundary point, the navigator finds and bounds the island around it.

The analytic bootstrap. The numerical program controls the low-lying spectrum — the relevant operators, the stress tensor, the first few subleading operators — with great precision, but says progressively less about high spin, where the operator tower is infinite and the truncation thin. The analytic bootstrap is the exact complement. In the lightcone limit, crossing forces the existence of *double-twist* families $\Delta \approx \Delta_{\phi_1} + \Delta_{\phi_2} + \ell + 2n$ at large spin ℓ, and the *Lorentzian inversion formula* of Theorem 51.16 turns the four-point function into an analytic function of spin that computes their dimensions and OPE coefficients in closed form, order by order in $1/\ell$ [111]. The double zeros of the extremal functional at large spin (the spin-2 tower of Fig. 50.7 and its higher-spin continuation) are exactly the operators the inversion formula predicts; the numerical and analytic spectra agree where they overlap. The handoff is precise: numerics pins the few low-lying operators that analytics cannot reach, analytics organizes the infinite large-spin tail that numerics cannot resolve, and modern high-precision results stitch the two together.

50.6.7 The frontier, dated

We close with the status of the program as of the mid-2020s, stated as honestly as the rest.

What has been isolated. The 3D Ising model is the showcase: cornered to six significant figures by the mixed σ–ε system [303, 304], with the data of Eq. (50.130) now the most precise available. The same machinery has isolated the $O(N)$ vector models for a range of N: the $O(2)$ model governing the superfluid λ-transition in helium-4, where the bootstrap determination of the heat-capacity exponent is precise enough to engage a longstanding tension with the space-shuttle microgravity measurement [120]; the $O(3)$ Heisenberg magnet; and the singular $N \to 0$ (self-avoiding walk) and $N \to 1$ (Ising) endpoints. Beyond bosonic models, the 3D *Gross–Neveu–Yukawa* CFTs — fermions coupled to a scalar at a critical point relevant to graphene and to emergent symmetry in quantum magnets — have been bootstrapped, with the extremal-functional spectrum read off and cross-checked against the large-N and ϵ-expansions [179]. In each case the workflow is the one of this chapter: assume unitarity and a gap structure dictated by the RG, close an island in a mixed-correlator system, and read the spectrum from the extremal functional.

What remains out of reach. The method's reach is set by the sharpness of the features it can exploit. Theories with a clean kink or a small, well-separated set of relevant operators are tractable; theories with many low-lying operators, weak or absent kinks, or no reliable gap assumption are not. Confining gauge theories and QCD-like dynamics, with their dense spectra and no obvious small

parameter, are largely beyond the current single-correlator and small-mixed-system technology. Theories without a known global symmetry to organize the OPE-coefficient positivity into manageable blocks are harder still. And every island is conditional on a gap input; where the RG does not hand us a clean relevant-operator count, there is nothing to assume, and the allowed region need not close.

Open methodological questions. The deepest is the existence-and-uniqueness gap of Section 50.6.5: turning the shrinking islands into a constructive existence proof, or at least a proof that the limiting region is nonempty, is unsolved. On the computational side, the bottleneck is the size of the semidefinite programs as one adds correlators, operators, and global-symmetry representations; the navigator and improved solvers [441] push the frontier but do not remove the exponential pressure of high-dimensional searches. There are also structural questions about the space of functionals itself: the analytic functional bases that reproduce the numerical optima in solvable cases (the dispersive and Polyakov–Mellin functionals) hint that the extremal functional has a closed-form analytic skeleton not yet exposed in general, a possibility that Chapter 51 begins to make precise.

This completes the bootstrap stage of ϕ^4 theory: from the crossing sum rule Eq. (49.74), through the linear-functional exclusion machinery and its semidefinite refinement, to the closed Ising island and now its extremal spectrum Eq. (50.130), the Wilson–Fisher fixed point has been not merely bounded but solved, its operators read one by one off a convex-optimization certificate. The very feature that completes the numerical reconstruction — the double-twist tower fixed at large spin by the extremal functional's high-ℓ zeros — is precisely where the numerics runs thin and demands an analytic handle. That handle is the subject of Chapter 51: the lightcone limit, the large-spin expansion, and the Lorentzian inversion formula Theorem 51.16 that computes the analytic continuation in spin which the numbers of this chapter can only sample.

Further reading

The comprehensive modern reference for everything in this chapter is the review of Poland et al. [402], which covers the linear-functional method, the semidefinite reduction, mixed correlators, and the extremal-functional spectrum reading in one place; Rattazzi et al. [415] is the founding paper that first turned crossing plus unitarity into a numerical bound. The $3d$ Ising story unfolds across El-Showk et al. [172] (the single-correlator solution and the kink), El-Showk et al. [173] (c-minimization and precise exponents), Kos et al. [303] (the mixed σ–ε system), and Kos et al. [304] (the precision islands quoted here); the independent Monte Carlo determination used for comparison is Hasenbusch [254]. The semidefinite solver and its arbitrary-precision implementation are described in Simmons-Duffin

[441], and the gradient-based search that makes high-dimensional mixed systems tractable in Reehorst et al. [417]. For the reach beyond Ising see Chester et al. [120] on the $O(2)$ model and the λ-transition tension, and Erramilli et al. [179] on the $3d$ Gross–Neveu–Yukawa CFTs. The analytic complement — the lightcone limit, large spin, and the Lorentzian inversion formula whose predictions match the extremal functional's high-spin zeros — is Caron-Huot [111], developed in Chapter 51.

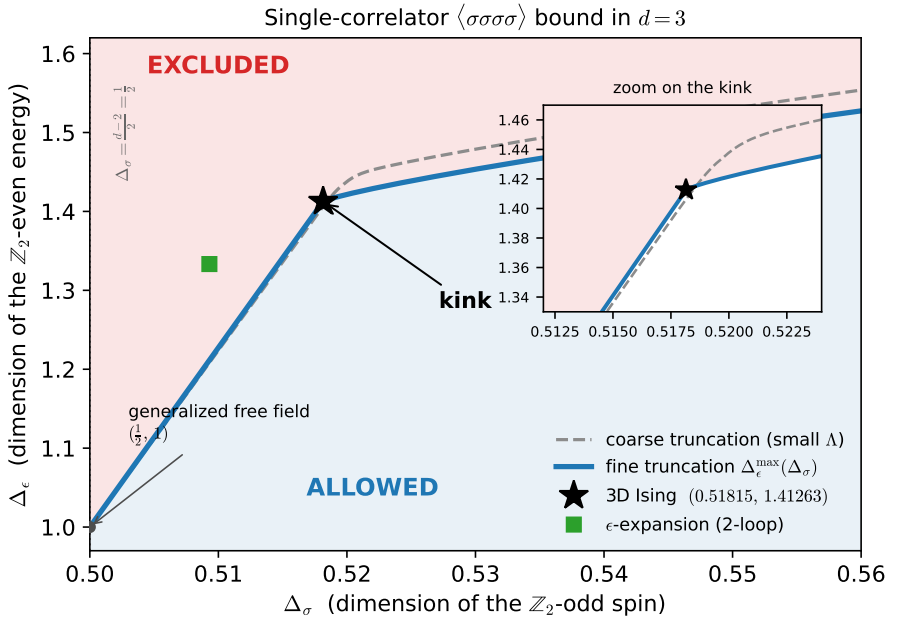


Figure 50.5: The single-correlator $\langle\sigma\sigma\sigma\sigma\rangle$ bound in $d = 3$, computed from the feasibility problem of Theorem 50.23 by the semidefinite program of Section 50.3. For each Δ_σ the curve gives the largest dimension $\Delta_\varepsilon^{\max}$ the leading $\mathbb{Z}_2$-even scalar may have under the single-gap assumption; the shaded region below is allowed, everything above is excluded by an explicit positivity certificate (Theorem 49.28). The dashed curve is a coarse derivative truncation and the solid curve a finer one; the rigorous bound decreases monotonically with the derivative order Λ, so the curve descends toward its limit from above. The bound rises from the generalized-free-field endpoint $(\frac{1}{2}, 1)$, where the lightest even scalar is the double-twist $[\sigma\sigma]_{0,0} = :\sigma^2:$ at $2\Delta_\sigma = 1$, and develops a sharp *kink*. The 3D Ising universality class sits essentially at the kink, at $(\Delta_\sigma, \Delta_\varepsilon) = (0.5181489, 1.412625)$ (Eq. (49.171)); the two-loop ε-expansion prediction of ϕ^4 theory (square, Theorem 49.29) points to the same corner from the interior. The inset zooms on the kink, where the corner sharpens as Λ grows.

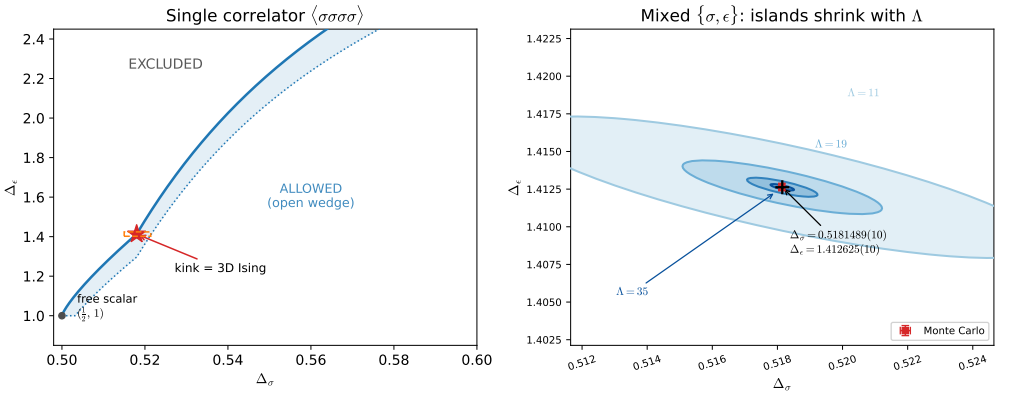


Figure 50.6: The mechanism that isolates the 3D Ising model. *Left*: the single-correlator allowed region from $\langle\sigma\sigma\sigma\sigma\rangle$ with the $\mathbb{Z}_2$ gap (Section 50.2.1) is an *open* wedge — it has a kink at the Ising point but extends without closing, so it bounds the data without isolating it. *Right* (zoom of the dashed box): the mixed $\{\sigma, \varepsilon\}$ system with both gap assumptions of Section 50.5.6 closes the wedge into a *closed* island, which shrinks as the number of derivative constraints Λ is increased, collapsing onto the precision values $\Delta_\sigma = 0.5181489(10)$, $\Delta_\varepsilon = 1.412625(10)$. The Monte-Carlo determination (square with error cross, Hasenbusch [254]) sits inside the island, far less precisely. The contours are a faithful schematic of the published islands, not a re-run of the semidefinite solver.

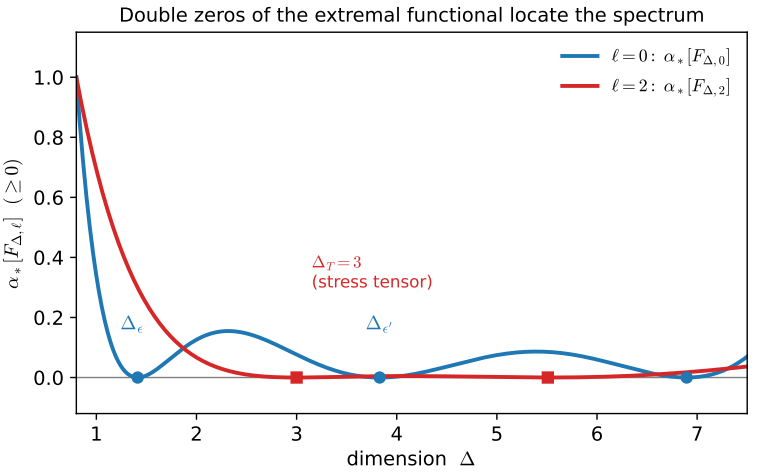


Figure 50.7: Schematic of the extremal functional acting on conformal blocks, $\Delta \mapsto \alpha_*[\vec{F}_{\Delta,\ell}]$, at two spins, evaluated at the 3D Ising island point Eq. (50.128). The function is nonnegative everywhere (dual feasibility) and touches zero without changing sign at the dimensions of the operators present in the extremal solution: a double zero in the $\ell = 0$ channel at $\Delta_\varepsilon = 1.412625$ (the leading $\mathbb{Z}_2$-even scalar) and the next at $\Delta_{\varepsilon'} \approx 3.83$ (the subleading scalar, irrelevant); a double zero in the $\ell = 2$ channel pinned at $\Delta_T = 3$ (the stress tensor, fixed by conservation) and the next at $\Delta \approx 5.5$. The double-zero locations *are* the spectrum; complementary slackness Theorem 50.34 guarantees it.

Chapter 51

The Analytic Bootstrap

The numerical bootstrap of Chapter 50 treated the crossing equation as a feasibility problem: it asked whether a candidate spectrum could exist, and answered with islands and exclusion bounds certified by positivity. The answer was astonishingly precise but mute. It located the 3D Ising dimensions to many digits without explaining why the spectrum has the shape it does, and it could speak only about a finite window of low-lying, low-spin operators. This chapter develops the complementary half of the bootstrap. Instead of bounding the space of solutions with a computer, we solve the crossing equation *analytically* in a controlled corner of kinematics and read off closed-form statements about an infinite family of operators. The price is that the control is asymptotic — it is sharpest at large spin — but the reward is structural understanding: the very operators the numerics cannot reach, and the mechanism behind the ones it can.

The engine is a single limit. Taking one cross-ratio to the lightcone collapses the conformal blocks of Theorem 47.49 to their collinear, $\mathrm{SL}(2,\mathbb{R})$ pieces — Gauss hypergeometric functions — and turns the crossing equation Eq. (49.14) into a tractable matching problem. Section 51.1 carries this out and extracts its first theorem: the identity operator, universal in every crossed channel, produces a power-law singularity that no finite, twist-gapped spectrum can reproduce, so crossing *forces* an infinite tower of large-spin operators whose twists accumulate at $2\Delta_\phi$. Section 51.2 names that tower. Its members are the double-twist operators $[\phi\phi]_{n,\ell}$ already met as the mean-field spectrum of Eq. (49.130), and to leading order their OPE data are exactly those of a generalized free field, the conformal avatar of mean-field theory. With the zeroth-order seed in hand, Section 51.3 perturbs around it: a single minimal-twist exchange in the crossed channel induces an anomalous dimension $\gamma_{0,\ell} \sim -c_{\tau_m}/\bar{h}^{\,\tau_m}$, the Komargodski–Zhiboedov / Fitzpatrick–Kaplan–Poland–Simmons-Duffin formula Eq. (51.73). Read physically through the unitarity bounds of Theorem 47.18, this is a binding energy; for stress-tensor exchange it is gravitational attraction in anti–de Sitter space with a strength set by the central charge, the seed of bulk locality picked

up by the holography arc of Chapters 53 and 56.

The lightcone method is powerful but order-by-order and somewhat artisanal. Section 51.4 replaces it with a machine. The Lorentzian inversion formula of Caron-Huot Theorem 51.16 reconstructs the entire OPE function from a single positive-definite integral of the correlator's double discontinuity, and in doing so proves that CFT data are *analytic in spin* down to ℓ near the Regge value. Its derivation leans on the analytic structure of the Feynman propagator from Theorem 9.24 and on the conformal partial-wave machinery built on the Casimir equation Eq. (47.233). Section 51.5 then turns the crank: because double-twist operators have vanishing double discontinuity, they are reconstructed entirely from the crossed channel, and inverting one block reproduces the large-spin master formula Eq. (51.73), the OPE-coefficient correction, and the gravitational result in a single pass. Section 51.6 steps back to the dual, functional viewpoint of Theorem 50.34: crossing is the vanishing of a linear functional, the one-dimensional problem is solved exactly by a basis of analytic functionals dual to the generalized free field, and the dispersive functionals that result are nothing but the inversion formula in another dress, unifying the lightcone, inversion, and Polyakov–Mellin pictures.

The chapter closes by pointing the whole apparatus at a real theory. Section 51.7 carries ϕ^4 theory to its analytic endpoint: the leading-twist towers $[\sigma\sigma]_{0,\ell}$ and $[\epsilon\epsilon]_{0,\ell}$ of the 3D Ising CFT are computed in closed form, cross-checked against the numerical spectrum located in Eq. (50.130) and inside the island of Eq. (49.171), and matched onto the Wilson–Fisher ϵ-expansion of Section 47.8.4. Three independent descriptions — generalized free field, ϵ-expansion, nonperturbative bootstrap — converge on the same conformal data, the strongest evidence that the fixed point is a single consistent object. The Wilson–Fisher-as-CFT closure and the dictionary back to the measured critical exponents are handed forward to Chapter 52.

51.1 Crossing in the Lightcone Limit

The numerical bootstrap of Chapter 50 solved crossing by brute force: it discretized the single equation Eq. (49.14), fed a truncated set of its derivatives at the crossing-symmetric point to a semidefinite solver, and let the computer carve out the allowed region of CFT data. The method is spectacularly accurate and almost completely opaque. It returns the 3D Ising dimensions to many digits without ever exhibiting a formula, an asymptotic law, or a reason why the spectrum looks as it does. The analytic bootstrap restores the reasoning. Its governing idea is simple to state: the crossing equation must hold for *every* value of the cross-ratios, so we are free to evaluate it not only near the symmetric point $z = \bar{z} = \frac{1}{2}$, where every block contributes comparably and only a machine can cope, but in kinematic *limits* where the infinite block sum collapses onto a few

controllable terms. The cleanest such limit is the lightcone limit, in which two of the four operators approach a common light ray. There the sum over the entire conformal spectrum on one side of crossing degenerates into a single power law set by the lightest operator, the identity; and the demand that the other side reproduce that power law, term by controllable term, turns out to be impossible to meet with any finite spectrum. The resolution forces a structural feature on *every* conformal field theory: an infinite tower of operators of unbounded spin, whose twists pile up at a value fixed entirely by the external dimension. This section sets up that limit, derives the one-dimensional "collinear" block that organizes it, proves the no-finite-spin lemma by direct power counting, and extracts the minimal-twist accumulation point. It is the entry gate to the whole analytic program.

51.1.1 The crossing equation in lightcone coordinates

We continue ϕ^4 theory from Chapter 49, studying the four-point function of a single real scalar primary ϕ of dimension Δ_ϕ — the free-field endpoint of the $\lambda\phi^4$ flow, promoted in Section 49.5.1 to a generalized free field of arbitrary Δ_ϕ. Because all four external operators are identical, the (12)(34) and (14)(23) fusions are decompositions of one and the same operator product $\phi \times \phi$; crossing is the self-consistency of that single OPE. Stripping the kinematic prefactor as in Theorem 47.49, the correlator reduces to the function $g(u, v)$ of the cross-ratios Eq. (47.140),

$$u = z\bar{z}, \qquad v = (1 - z)(1 - \bar{z}), \tag{51.1}$$

and the crossing equation imported from Eq. (49.14) reads

$$v^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u, v) \;=\; u^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(v, u), \qquad p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0. \tag{51.2}$$

The left-hand sum is the s-channel ((12)(34) fusion), the right-hand sum the t-channel ((14)(23) fusion); the two run over the *same* spectrum with the *same* squared coefficients, the swap $u \leftrightarrow v$ being the only difference. That swap is implemented on the complex coordinate of Eq. (51.1) by

$$u(1 - z, 1 - \bar{z}) = (1 - z)(1 - \bar{z}) = v, \qquad v(1 - z, 1 - \bar{z}) = z\bar{z} = u, \tag{51.3}$$

so that, writing the block in the z-coordinates as $G_{\Delta,\ell}(z, \bar{z})$ exactly as in Section 49.2.1, the t-channel block $G_{\Delta,\ell}(v, u)$ is the same function evaluated at $(1 - z, 1 - \bar{z})$. The crossing equation Eq. (51.2) becomes, with no approximation,

$$[(1 - z)(1 - \bar{z})]^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z, \bar{z}) \;=\; (z\bar{z})^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(1 - z, 1 - \bar{z}). \tag{51.4}$$

In Euclidean signature $\bar{z} = z^*$, and Eq. (51.4) is an identity on the cut plane. The decisive observation, anticipated below Eq. (47.140), is that under

Lorentzian continuation z and $\bar{z}$ become *independent* real variables: the single complex configuration unfolds into a two-dimensional real domain, and we may send one of them to a boundary while holding the other fixed. This is precisely the freedom a single complex variable does not offer, and it is what makes the lightcone limit a genuine, accessible kinematic regime rather than a degenerate point.

Definition 51.1 (Lightcone limit)**.** The *lightcone* (or *collinear*) limit is

$$\bar{z} \to 1 \quad \text{at fixed } z \in (0,1), \qquad \Longleftrightarrow \qquad v = (1-z)(1-\bar{z}) \to 0, \quad u = z\bar{z} \to z, \tag{51.5}$$

with the small parameter $1-\bar{z} \simeq v/(1-z)$. Geometrically it is the limit in which the operators $\phi(x_1), \phi(x_4)$ (equivalently $\phi(x_2), \phi(x_3)$) become null-separated, so that $x_{14}^2 x_{23}^2 \to 0$ and the t-channel OPE converges most rapidly.

The reason the limit tames the block sum is the lightcone factorization established in Section 49.2.6: as the *second* cross-ratio argument of a block tends to zero, the block degenerates into a pure power of that argument, set by the twist

$$\tau \equiv \Delta - \ell, \tag{51.6}$$

times a single one-variable function of the surviving argument,

$$G_{\Delta,\ell}(\zeta, \bar{\zeta}) \xrightarrow{\bar{\zeta}\to 0} \bar{\zeta}^{\tau/2}\, k_{\Delta+\ell}(\zeta), \tag{51.7}$$

the content of Eq. (49.67). Applied to the t-channel blocks $G_{\Delta,\ell}(1-z, 1-\bar{z})$ on the right of Eq. (51.4), whose second argument is $1-\bar{z} \to 0$,

$$G_{\Delta,\ell}(1-z, 1-\bar{z}) \xrightarrow{\bar{z}\to 1} (1-\bar{z})^{\tau/2}\, k_{\Delta+\ell}(1-z). \tag{51.8}$$

The dependence on the small quantity $1-\bar{z}$ is now an explicit power $\frac{\tau}{2}$, so the t-channel sum is organized by twist: the smaller the twist, the more singular (less suppressed) the contribution. To exploit this we need the surviving function k_β in closed form.

51.1.2 The collinear block

The function k_β is the analytic bootstrap's atom. It was introduced in Eq. (49.30); because every argument of this chapter is built from it, we re-derive it here from its defining equation, in the form the lightcone analysis consumes. When the four operators are placed on a common line, the conformal group $\mathfrak{so}(d+1,1)$ degenerates to the $\mathrm{SL}(2,\mathbb{R})$ generated by $\{L_{-1}, L_0, L_{+1}\}$, and the collinear block is the eigenfunction of its quadratic Casimir. In the coordinate z that Casimir is the operator (Section 49.2.1)

$$D_z \;=\; z^2(1-z)\,\partial_z^2 \;-\; z^2\,\partial_z. \tag{51.9}$$

Proposition 51.2 (The collinear block as a hypergeometric function)**.** *For $\beta > 0$ set $h = \beta/2$. The solution of the eigenvalue problem*

$$D_z\, k_\beta(z) = h(h-1)\, k_\beta(z) \tag{51.10}$$

with leading behavior $k_\beta(z) \to z^h$ as $z \to 0$ is

$$\boxed{k_\beta(z) \;=\; z^{\beta/2}\, {}_2F_1\Big(\tfrac{\beta}{2}, \tfrac{\beta}{2}; \beta; z\Big), \qquad {}_2F_1(a,b;c;z) = \sum_{n\geq 0} \frac{(a)_n (b)_n}{(c)_n} \frac{z^n}{n!}.} \tag{51.11}$$

Proof. The operator Eq. (51.9) has a regular singular point at $z = 0$; its indicial equation, obtained by substituting $k \sim z^s$ and keeping the leading z^s term of $D_z z^s = s(s-1)z^s + \dots$, is $s(s-1) = h(h-1)$, with roots $s = h$ and $s = 1-h$. The root $s = h = \beta/2$ is the one matching the OPE leading power $u^{\tau/2}$ promised of the block; we build the solution on it. Write $k_\beta(z) = z^h g(z)$ with g analytic and $g(0) = 1$. Differentiating,

$$k'_\beta = h z^{h-1} g + z^h g', \tag{51.12}$$

$$k''_\beta = h(h-1) z^{h-2} g + 2h z^{h-1} g' + z^h g'', \tag{51.13}$$

and inserting into $D_z k_\beta - h(h-1)k_\beta = 0$ gives, after multiplying out the factors $z^2(1-z)$ and $-z^2$,

$$(1-z)\big[h(h-1)z^h g + 2h z^{h+1} g' + z^{h+2} g''\big] - \big[h z^{h+1} g + z^{h+2} g'\big] - h(h-1) z^h g = 0. \tag{51.14}$$

The terms $h(h-1)z^h g$ cancel, and dividing the remainder by z^{h+1} collects to

$$z(1-z)\, g'' + \big[\, 2h - (2h+1)z \,\big]\, g' - h^2\, g = 0. \tag{51.15}$$

This is the Gauss hypergeometric equation $z(1-z)g'' + [c - (a+b+1)z]g' - ab\, g = 0$ with $c = 2h$, $a+b = 2h$, $ab = h^2$, whose unique solution is $a = b = h$. The solution analytic at $z = 0$ with $g(0) = 1$ is $g(z) = {}_2F_1(h, h; 2h; z)$, which establishes Eq. (51.11). The eigenvalue read off from Eq. (51.10) is $h(h-1) = \frac{\beta}{2}(\frac{\beta}{2} - 1)$, in agreement with Eq. (49.31). □

Two properties of k_β are all the analysis below requires. The first is its small-z behavior, immediate from Eq. (51.11):

$$k_\beta(z) \;=\; z^{\beta/2}\big[1 + \tfrac{\beta}{4} z + O(z^2)\big] \;\xrightarrow{z\to 0}\; z^{\beta/2}, \tag{51.16}$$

which is what made the twist power appear in Eq. (51.7). The second, decisive for the no-finite-spin lemma, is the behavior at the *other* endpoint $z \to 1$.

Worked Example 51.3 (The collinear block diverges only logarithmically at $z \to 1$)**.** We show that as $z \to 1$ the collinear block grows no faster than a logarithm, and we compute the coefficient. Start from the Euler integral representation of the hypergeometric function, valid for $\operatorname{Re} c > \operatorname{Re} b > 0$ (here $c = \beta > h = \beta/2 > 0$),

$$ {}_2F_1(h,h;\beta;z) = \frac{\Gamma(\beta)}{\Gamma(h)^2}\int_0^1 \mathrm{d}t\, \big[t(1-t)\big]^{h-1}(1-zt)^{-h}. \tag{51.17}$$

The only source of singular behavior as $z \to 1$ is the endpoint $t \to 1$, where $1 - zt \to 1 - z$. Set $s = 1 - t$ and $\epsilon = 1 - z$; near $s = 0$ the integrand is $\big[t(1-t)\big]^{h-1}(1-zt)^{-h} \to s^{h-1}\,(\epsilon+s)^{-h}$ (using $t \to 1$, $1-zt = 1-z+z\,s \simeq \epsilon+s$). The singular part of the integral is therefore

$$ \int_0 \mathrm{d}s\, \frac{s^{h-1}}{(\epsilon+s)^h} \;=\; \int_0 \frac{\mathrm{d}s}{s}\Big(\frac{s}{\epsilon+s}\Big)^h. \tag{51.18}$$

For $s \gg \epsilon$ the bracket is 1 and the integrand is $\mathrm{d}s/s$; for $s \ll \epsilon$ it is $(s/\epsilon)^h$, integrably small. The integral thus accumulates a logarithm from the window $\epsilon \lesssim s \lesssim 1$, giving $-\log\epsilon + \text{finite} = -\log(1-z) + \text{finite}$. Restoring the prefactor of Eq. (51.17) and the factor $z^{\beta/2} \to 1$ of Eq. (51.11),

$$\boxed{\,k_\beta(z) \;\xrightarrow{z\to 1}\; -\frac{\Gamma(\beta)}{\Gamma(\beta/2)^2}\big[\log(1-z) + \text{const}\big] \;+\; O\big((1-z)\log(1-z)\big).\,} \tag{51.19}$$

The same conclusion follows from the structure of Eq. (51.15) at its regular singular point $z = 1$: with $c - a - b = \beta - 2h = 0$, the two indicial exponents at $z = 1$ coincide (both zero), and a confluent pair of exponents is exactly the case in which the second Frobenius solution carries a $\log(1-z)$. The block, being the solution regular at $z = 0$, is a generic combination of the two solutions at $z = 1$ and therefore inherits that logarithm. The constant in Eq. (51.19) is $2\psi(1) - 2\psi(\beta/2)$ in terms of the digamma function, but only the logarithm matters below.

The contrast between Eq. (51.16) and Eq. (51.19) is the engine of everything that follows: a single conformal block enters the lightcone limit through a power of the small twist variable and a one-dimensional function that, in the *crossed* variable, can diverge no worse than log. No power-law singularity is available from any finite collection of blocks. Crossing, as we now show, demands one.

51.1.3 Identity dominance and the failure of a finite spectrum

We take the lightcone limit Eq. (51.5) of the exact crossing equation Eq. (51.4), treating the two sides in turn.

The t-channel side collapses to the identity. On the right of Eq. (51.4), every block is governed by Eq. (51.8): operator $\mathcal{O}$ of twist $\tau_{\mathcal{O}}$ contributes a term proportional to $(1-\bar{z})^{\tau_{\mathcal{O}}/2}$. Among all operators in $\phi\times\phi$ the identity is unique in having $\tau=0$ (it is the only operator of vanishing dimension; every other primary obeys the unitarity bounds Theorem 47.18, which give $\tau=\Delta\geq\frac{d-2}{2}>0$ for scalars and $\tau=\Delta-\ell\geq d-2>0$ for $\ell\geq 1$). Thus there is a finite twist gap $\tau_{\text{gap}}>0$ above the identity, and as $\bar{z}\to 1$ the identity term, carrying $(1-\bar{z})^0$, strictly dominates every other, which is suppressed by at least $(1-\bar{z})^{\tau_{\text{gap}}/2}$. With the identity normalized to $p_{\mathbf{1}}=1$ and $G_{0,0}=1$,

$$(z\bar{z})^{\Delta_\phi}\sum_{\mathcal{O}}p_{\mathcal{O}}\,G_{\Delta,\ell}(1-z,1-\bar{z})\ \xrightarrow{\bar{z}\to 1}\ (z)^{\Delta_\phi}\Big[1+O\big((1-\bar{z})^{\tau_{\text{gap}}/2}\big)\Big]. \tag{51.20}$$

The entire right-hand side, an infinite sum over the whole spectrum, has degenerated into a single finite number z^{Δ_ϕ}.

The s-channel side must produce a power-law singularity. The left-hand side of Eq. (51.4) carries the explicit prefactor $\big[(1-z)(1-\bar{z})\big]^{\Delta_\phi}$, which vanishes as $(1-\bar{z})^{\Delta_\phi}$. For the equation to balance against the finite limit Eq. (51.20), the s-channel block sum must diverge precisely fast enough to cancel that vanishing prefactor:

$$\sum_{\mathcal{O}}p_{\mathcal{O}}\,G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z})\ \xrightarrow{\bar{z}\to 1}\ \Big(\frac{z}{1-z}\Big)^{\Delta_\phi}(1-\bar{z})^{-\Delta_\phi}. \tag{51.21}$$

Translating the small parameter through Eq. (51.5), $1-\bar{z}=v/(1-z)$, this is a power-law singularity in v,

$$\sum_{\mathcal{O}}p_{\mathcal{O}}\,G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z})\ \sim\ \big[z(1-z)\big]^{\Delta_\phi}\,v^{-\Delta_\phi}\qquad(v\to 0). \tag{51.22}$$

This is the structural demand crossing places on the spectrum: the s-channel OPE must reproduce, in the lightcone limit, the power $v^{-\Delta_\phi}$ generated by nothing more than the identity in the crossed channel.

Lemma 51.4 (No finite, twist-gapped spectrum can cross)**.** *The singularity Eq. (51.22) cannot be produced by any sum over a spectrum of bounded spin. In particular no finite set of conformal blocks satisfies the lightcone limit of crossing; the $\phi\times\phi$ OPE must contain operators of unbounded spin.*

Proof. We bound the left-hand side of Eq. (51.21) under the assumption of bounded spin and show it cannot reach a negative power of $1-\bar{z}$. Each s-channel block $G_{\Delta,\ell}(z,\bar{z})$ is, in the even-dimensional closed forms of Eqs. (49.44) and (49.49) and in the general-d radial expansion Eq. (49.63), a sum of products of collinear functions $k_{\beta_i}(z)\,k_{\beta_j}(\bar{z})$ with weights β_i,β_j built from (Δ,ℓ) — for instance $\beta\in\{\Delta+\ell,\Delta-\ell\}$ in $d=2$ and $\beta\in\{\Delta+\ell,\Delta-\ell-2\}$ in $d=4$. As

$\bar{z} \to 1$ at fixed z, the factors depending on z are finite, and each factor $k_{\beta_j}(\bar{z})$ is governed by Eq. (51.19): it grows no faster than $\big|\log(1-\bar{z})\big|$, with a coefficient $\frac{\Gamma(\beta_j)}{\Gamma(\beta_j/2)^2}$ that, for β_j in any *bounded* range, is itself bounded. Hence there is a constant $C(z)$, depending on the (finite) spectrum but not on $\bar{z}$, such that

$$\Big|\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z})\Big| \;\le\; C(z)\,\big|\log(1-\bar{z})\big| \qquad (\bar{z}\to 1). \tag{51.23}$$

A logarithm is dominated by every negative power: for any $\Delta_\phi > 0$, $\big|\log(1-\bar{z})\big|/(1-\bar{z})^{-\Delta_\phi} = (1-\bar{z})^{\Delta_\phi}\big|\log(1-\bar{z})\big| \to 0$. The bound Eq. (51.23) is therefore incompatible with the required divergence Eq. (51.21). The assumption of bounded spin — equivalently, of weights β drawn from a bounded set — is the only place the bound was used; it must fail. The spins entering $\phi\times\phi$ are unbounded. □

The lemma is the analytic bootstrap's first theorem about *every* CFT, derived with no input beyond crossing, unitarity, and the existence of the identity. It already explains a qualitative feature the numerical spectra of Chapter 50 exhibit but cannot motivate: the operator content of any unitary CFT necessarily continues to arbitrarily high spin. A finite list of operators — the kind one might naively hope to "solve" a theory with — can never close under crossing.

Notation Watch

In Chapter 49 the label s was fixed to the (12)(34) fusion and t to (14)(23); we keep that assignment, so above it is the t-*channel* identity that dominates the limit and the s-*channel* OPE that is forced to large spin. Some treatments instead name the identity-dominated channel "s" (then the tower is in "t"). For identical external scalars the distinction is empty: both fusions decompose in the one OPE $\phi\times\phi$, the equation Eq. (51.4) is symmetric under $z \leftrightarrow 1-z$, and the conclusion — that $\phi\times\phi$ contains operators of unbounded spin — is channel-independent.

51.1.4 The minimal twist accumulates at $2\Delta_\phi$

Lemma 51.4 forces large spin but not yet a twist. To locate the tower we sharpen the limit, sending $z \to 0$ in addition to $\bar{z}\to 1$ — the *double lightcone* corner of cross-ratio space (Fig. 51.1). The relation Eq. (51.21) already holds; we now read its z-dependence.

As $z\to 0$ the surviving s-channel blocks enter their own lightcone limit, this time in the variable z. By the symmetry of the block under $z\leftrightarrow\bar{z}$, the limit Eq. (51.7) applies with the roles of the arguments exchanged,

$$G_{\Delta,\ell}(z,\bar{z}) \;\xrightarrow{z\to 0}\; z^{\tau/2}\, k_{\Delta+\ell}(\bar{z}), \tag{51.24}$$

so the small-z expansion of the s-channel sum is organized by twist, the lowest twist giving the leading power:

$$\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}) \xrightarrow{z\to 0} z^{\tau_0/2} \sum_{\mathcal{O}:\,\tau_{\mathcal{O}}=\tau_0} p_{\mathcal{O}}\, k_{\Delta_{\mathcal{O}}+\ell_{\mathcal{O}}}(\bar{z}) \;+\; (\text{higher powers of } z), \tag{51.25}$$

where τ_0 denotes the smallest twist appearing in the divergent, large-spin part of the sum. (Unitarity bounds twists from below, Theorem 47.18, so a well-defined minimum exists.) On the other side, the right of Eq. (51.21) has the small-z behavior

$$\Big(\frac{z}{1-z}\Big)^{\Delta_\phi}(1-\bar{z})^{-\Delta_\phi} \xrightarrow{z\to 0} z^{\Delta_\phi}(1-\bar{z})^{-\Delta_\phi}. \tag{51.26}$$

The two leading powers of z must agree. Matching the exponents in Eq. (51.25) and Eq. (51.26),

$$\frac{\tau_0}{2} = \Delta_\phi \qquad \Longrightarrow \qquad \boxed{\tau_0 = 2\Delta_\phi.} \tag{51.27}$$

The forced tower of large-spin operators has its minimal twist pinned to $2\Delta_\phi$. Equating the coefficients of z^{Δ_ϕ} then gives the further condition

$$\sum_{\mathcal{O}:\,\tau_{\mathcal{O}}\to 2\Delta_\phi} p_{\mathcal{O}}\, k_{\Delta_{\mathcal{O}}+\ell_{\mathcal{O}}}(\bar{z}) \xrightarrow{\bar{z}\to 1} (1-\bar{z})^{-\Delta_\phi}, \tag{51.28}$$

a sum rule on the large-spin OPE coefficients: infinitely many operators of twist accumulating at $2\Delta_\phi$, summed against their collinear blocks, must reconstruct the power $(1-\bar{z})^{-\Delta_\phi}$ that each block individually cannot produce.

Physical Insight

The value $2\Delta_\phi$ is the statement that a high-spin operator in $\phi\times\phi$ behaves like two copies of ϕ orbiting at large angular momentum. Twist measures energy in the lightcone frame: a single ϕ has twist Δ_ϕ (it is a scalar, $\ell=0$), and two of them, held far apart by a large centrifugal barrier, have total twist approaching the sum $\Delta_\phi+\Delta_\phi=2\Delta_\phi$, with the residual interaction energy — the anomalous dimension — decaying as the separation grows, i.e. as the spin $\ell\to\infty$. The minimal-twist trajectory $\tau\to 2\Delta_\phi$ is the bound-state-at-threshold limit of two free constituents. The rate at which τ approaches $2\Delta_\phi$ at finite ℓ is set by the lightest operator they can exchange — their binding energy. In a theory with a stress tensor that exchange is gravitational, and the $1/\ell$ corrections to the twist are, in the holographic dual of Chapters 53 and 56, literally the Newtonian binding of two particles in AdS. The lightcone limit has turned a kinematic identity into a spectroscopic prediction: every CFT contains, at large spin, the spectrum of a weakly interacting two-body system.

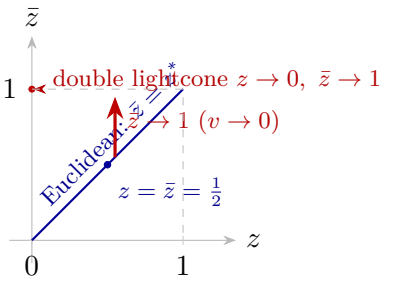


Figure 51.1: Cross-ratio space in the $(z, \bar{z})$ plane. In Euclidean signature the configuration lives on the diagonal $\bar{z} = z^*$ (blue), and the numerical bootstrap expands about the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$. Lorentzian continuation frees z and $\bar{z}$ to move independently, opening the square. The lightcone limit Eq. (51.5) sends $\bar{z} \to 1$ $(v \to 0)$ at fixed z (red arrow); the double lightcone of Section 51.1.4 adds $z \to 0$, the corner where the minimal-twist condition Eq. (51.27) is read off.

The operators realizing Eq. (51.27) are the *double-twist* composites already met in the mean-field solution of Eq. (49.129),

$$[\phi\phi]_{n,\ell} \ \sim \ :\phi \,\Box^n\, \partial_{\mu_1} \cdots \partial_{\mu_\ell} \phi: \ - \ (\text{traces}), \qquad \tau_n = 2\Delta_\phi + 2n, \tag{51.29}$$

whose twists Eq. (49.130) form an evenly spaced ladder above the threshold $2\Delta_\phi$, the minimum $\tau_0 = 2\Delta_\phi$ being the leading $n = 0$ family. The lightcone limit has discovered them from crossing alone, without ever assuming the theory is free: *any* unitary CFT must contain, at each even spin large enough, an operator whose twist approaches $2\Delta_\phi + 2n$. What the present limit does not yet supply is the *strength* of these operators — the large-spin OPE coefficients $p_{0,\ell}$ that solve the sum rule Eq. (51.28) — nor the subleading $n \geq 1$ families, nor the rate of approach $\tau(\ell) - 2\Delta_\phi$. Those are the data of the next section.

We have reduced the full crossing equation to a single, controllable kinematic statement and extracted from it a theorem about every conformal field theory: the operator product $\phi \times \phi$ contains an infinite tower of large-spin operators whose minimal twist accumulates at $2\Delta_\phi$, with the sum rule Eq. (51.28) fixing how their OPE coefficients conspire to reproduce the identity. The next section identifies this tower as the double-twist operators $[\phi\phi]_{n,\ell}$ and pins down their leading data — the asymptotic OPE coefficients and the full ladder of twists $2\Delta_\phi + 2n$ — by solving Eq. (51.28) at zeroth order, where the solution is precisely the generalized free field whose mean-field spectrum we already met in Section 49.5.1.

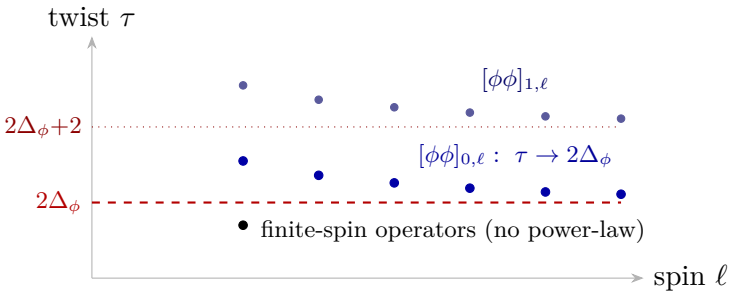


Figure 51.2: The forced large-spin tower. Crossing in the lightcone limit requires, in $\phi\times\phi$, operators of unbounded spin whose twists accumulate at $\tau\to 2\Delta_\phi$ (Eq. (51.27)); these are the double-twist families $[\phi\phi]_{n,\ell}$ of Eq. (51.29), one Regge-like trajectory per n, each approaching $2\Delta_\phi+2n$ from above as $\ell\to\infty$. Low-spin operators (black) contribute only logarithmically near the lightcone (Lemma 51.4) and cannot build the power-law singularity; the tower must.

51.2 Double-Twist Operators and Generalized Free Fields

The lightcone analysis of Section 51.1 ended on a structural verdict, not a computation: the identity operator, present in every crossed channel because every operator has a non-vanishing two-point function, develops a power-law collision singularity $v^{-\Delta_\phi}$ as the crossed pair approaches the lightcone, and no finite sum of direct-channel conformal blocks can reproduce a power law. The singularity can be matched only by an *infinite* tower of exchanged operators whose spins run off to infinity. That tower is forced; its mere existence is a theorem of crossing. What we owe now is its identity. We must say which operators populate the tower, what their quantum numbers are, and — to leading order, before any interaction is switched on — what their operator product coefficients are. The answer is clean and universal: the tower consists of *double-twist operators*, and at leading order their data are those of a *generalized free field*, the conformal avatar of mean-field theory. This section pins down both objects. It is the zeroth-order seed of the whole analytic bootstrap: the next section perturbs around it, reading the $1/\ell$ corrections to these data off the crossed-channel operators that the generalized free field omits.

51.2.1 Double-twist operators

Fix two scalar primaries $\mathcal{O}_1$ and $\mathcal{O}_2$ of dimensions Δ_1,Δ_2 in a unitary CFT. (We allow $\mathcal{O}_1\neq\mathcal{O}_2$ from the outset; the identical case $\mathcal{O}_1=\mathcal{O}_2=\phi$ of Section 49.5.1 is recovered by setting $\Delta_1=\Delta_2=\Delta_\phi$, and the two-operator setting is the one in which the large-spin physics — two well-separated constituents and the energy

that binds them — is most transparent.) Their operator product expansion contains, beyond whatever low-lying operators the dynamics supplies, an infinite family of composites built by colliding the two fields and dressing the result with derivatives.

Definition 51.5 (Double-twist operator)**.** The *double-twist operators* $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$ are the conformal primaries appearing in the $\mathcal{O}_1 \times \mathcal{O}_2$ OPE that are schematically of the form

$$[\mathcal{O}_1\mathcal{O}_2]_{n,\ell} \;\sim\; :\mathcal{O}_1\,\Box^n\,\partial_{\mu_1}\cdots\partial_{\mu_\ell}\,\mathcal{O}_2: \;-\; \text{(traces and descendants)}, \tag{51.30}$$

the symmetric traceless primary built from the two constituents, ℓ symmetrized derivatives distributed between them, and n powers of the Laplacian, with the precise distribution of derivatives fixed by the requirement that the combination be primary, $K_\mu[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}(0) = 0$. The integers $n, \ell \geq 0$ are the *radial level* and the *spin*.

The name records their defining feature. The twist of an operator is $\tau \equiv \Delta - \ell$; for the composite Eq. (51.30) the ℓ symmetrized derivatives add ℓ to both dimension and spin and so cancel in the twist, while the two constituents contribute $\Delta_1 + \Delta_2$ and each Laplacian contributes 2 to the dimension. Hence

$$\boxed{\tau_{n,\ell} = \Delta_1 + \Delta_2 + 2n + \gamma(n,\ell), \qquad \Delta_{n,\ell} = \Delta_1 + \Delta_2 + 2n + \ell + \gamma(n,\ell)\,,} \tag{51.31}$$

where $\gamma(n,\ell)$ is the *anomalous dimension*: the correction to the naive sum of constituent dimensions induced by interactions. In a Gaussian theory there is no interaction to shift the composite and $\gamma \equiv 0$, so the double-twist twists sit at the rigid lattice $\tau_n = \Delta_1 + \Delta_2 + 2n$; computing the leading $\gamma(n,\ell)$ at large ℓ in an interacting theory is the entire content of Section 51.3.

The two labels have a clean kinematic meaning that Eq. (51.31) makes precise. In the two-particle picture furnished by the state–operator map, $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$ is the lowest primary in a tower of two-particle states of the constituents $\mathcal{O}_1, \mathcal{O}_2$: ℓ is the relative orbital angular momentum of the pair, and n counts radial nodes of the relative wavefunction — exactly the way the principal and orbital quantum numbers organize the hydrogen spectrum, here for two conformal "particles." That ℓ is the spin is immediate from the ℓ symmetric indices in Eq. (51.30). That n is radial follows from the conformal Casimir of Theorem 47.50: on $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$ the Casimir takes the value $c_{\Delta_{n,\ell},\ell} = \Delta_{n,\ell}(\Delta_{n,\ell} - d) + \ell(\ell + d - 2)$, and at fixed spin ℓ the discrete tower $n = 0, 1, 2, \ldots$ is the spectrum of the Casimir restricted to the two-particle sector, the radial excitations stacked above the leading-twist seed $n = 0$.

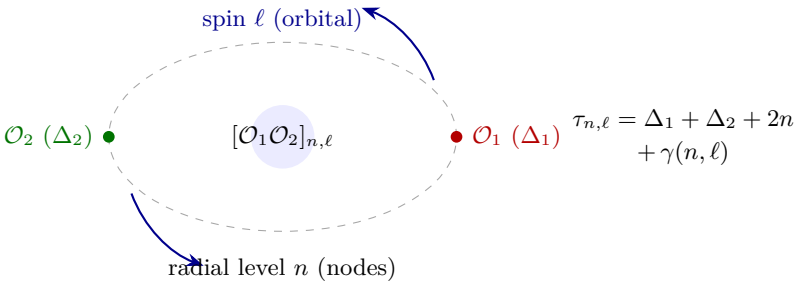


Figure 51.3: The double-twist operator $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$ pictured as a two-particle bound state of its constituents $\mathcal{O}_1, \mathcal{O}_2$. The spin ℓ is the relative orbital angular momentum, the radial level n counts nodes of the relative wavefunction. At large ℓ the centrifugal barrier separates the constituents, the twist approaches the free sum $\Delta_1 + \Delta_2 + 2n$, and the residual binding energy is the anomalous dimension $\gamma(n,\ell)$ computed in Section 51.3.

Physical Insight

A double-twist operator is the conformal field theory's description of two of its quanta orbiting each other very far apart. When the spin ℓ is large the centrifugal barrier $\sim \ell^2/r^2$ pushes the two constituents to large separation, where they barely interact; their total dimension is then almost exactly the sum $\Delta_1 + \Delta_2 + 2n$ of the free pieces, and the small residual binding energy is the anomalous dimension $\gamma(n,\ell)$, which therefore vanishes as $\ell \to \infty$. The lightcone bootstrap is, read physically, a calculation of this binding energy. In a theory with a holographic dual the two constituents are two particles in anti–de Sitter space and the binding is their gravitational (and other long-range) attraction; the $1/\ell$ falloff of γ is then the statement that gravity is a long-range force, and its coefficient is fixed by Newton's constant. We will not need the holographic reading here, but it is the reason the analytic bootstrap is also a derivation of bulk locality, taken up in Chapters 53 and 56.

51.2.2 Generalized free fields

To fix the leading data we need the simplest theory whose spectrum is *exactly* the double-twist lattice with $\gamma \equiv 0$. That theory is the generalized free field.

Definition 51.6 (Generalized free field)**.** A *generalized free field* (GFF) of dimension Δ is a scalar primary ϕ whose correlation functions are computed by Wick's theorem from the unit-normalized two-point function

$$\langle \phi(x_i)\,\phi(x_j)\rangle = \frac{1}{(x_{ij}^2)^{\Delta}}, \qquad x_{ij} \equiv x_i - x_j, \tag{51.32}$$

exactly as if ϕ were free: every correlator is the sum over all pairings of products of two-point functions, with no connected higher-point part. A collection of mutually generalized-free fields $\{\mathcal{O}_a\}$ obeys Eq. (51.32) within each species and $\langle \mathcal{O}_a \mathcal{O}_b \rangle = 0$ for $a \neq b$ (after diagonalizing the two-point matrix). The theory whose every correlator is so generated is *mean-field theory* (also called generalized free theory).

The construction is the conformal continuation of the free scalar of ϕ^4 theory. There the two-point function Eq. (51.32) with $\Delta = (d-2)/2$ is the massless limit of the position-space Feynman propagator Eq. (9.153) of Theorem 9.24, and Wick's theorem is a theorem about the Gaussian path integral. The generalized free field keeps the Gaussian (Wick) structure but promotes Δ from the canonical value $(d-2)/2$ to a free parameter; it is no longer the quantization of a local Lagrangian for generic Δ, but it is a perfectly good solution of all conformal Ward identities and, as we now confirm, of crossing. In ϕ^4 theory it is precisely the $\lambda \to 0$ limit of the interacting fixed point: as the coupling is switched off, $\Delta_\phi \to (d-2)/2$, the anomalous dimensions $\gamma_{n,\ell}$ of the would-be Wilson–Fisher spectrum vanish, and the composites $:\phi \Box^n \partial^\ell \phi:$ become the exact double-twist operators $[\phi\phi]_{n,\ell}$ of the free theory.

The four-point function. The decisive object is the four-point function. For four insertions of a single GFF ϕ the three pairings give the elementary sum Eq. (49.22), whose stripped form is $g(u,v) = 1 + u^{\Delta_\phi} + (u/v)^{\Delta_\phi}$ (Eq. (49.127)); that this satisfies the crossing equation Theorem 49.13 identically for every Δ_ϕ was shown in Theorem 49.5, and its decomposition into the even-spin double-twist tower $[\phi\phi]_{n,\ell}$ was carried out in Section 49.5.2. We take that identical-scalar result as established and turn to the genuinely two-operator correlator, where the double-twist tower stands alone.

Consider $\langle \mathcal{O}_1(x_1)\, \mathcal{O}_2(x_2)\, \mathcal{O}_2(x_3)\, \mathcal{O}_1(x_4) \rangle$ with $\mathcal{O}_1, \mathcal{O}_2$ distinct mutually-generalized-free scalars of dimensions Δ_1, Δ_2. Wick's theorem pairs like with like; the only nonvanishing contraction is $\langle \mathcal{O}_1(x_1)\mathcal{O}_1(x_4)\rangle \langle \mathcal{O}_2(x_2)\mathcal{O}_2(x_3)\rangle$, since $\langle \mathcal{O}_1 \mathcal{O}_2 \rangle = 0$. Hence

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_2(x_3)\mathcal{O}_1(x_4)\rangle = \frac{1}{(x_{14}^2)^{\Delta_1}\,(x_{23}^2)^{\Delta_2}}. \tag{51.33}$$

A single term — and that single term is, in the (14)(23) channel, nothing but the product of the two identities ($\mathcal{O}_1 \times \mathcal{O}_1 \to \mathbf{1}$ and $\mathcal{O}_2 \times \mathcal{O}_2 \to \mathbf{1}$). Strip it to the cross-ratio function appropriate to the s-channel $\mathcal{O}_1 \times \mathcal{O}_2$ fusion. Using the standard prefactor for unequal externals (the Dolan–Osborn convention, Theorem 47.27) with $\mathcal{O}_1 = \mathcal{O}_4$, $\mathcal{O}_2 = \mathcal{O}_3$,

$$\langle \mathcal{O}_1\mathcal{O}_2\mathcal{O}_2\mathcal{O}_1\rangle = \frac{1}{(x_{12}^2 x_{34}^2)^{\frac{\Delta_1+\Delta_2}{2}}} \Big(\frac{x_{24}^2}{x_{14}^2}\Big)^{\frac{\Delta_1-\Delta_2}{2}} \Big(\frac{x_{14}^2}{x_{13}^2}\Big)^{\frac{\Delta_2-\Delta_1}{2}} g(u,v), \tag{51.34}$$

and inserting Eq. (51.33) and the definitions $u = x_{12}^2 x_{34}^2/(x_{13}^2 x_{24}^2)$, $v = x_{14}^2 x_{23}^2/(x_{13}^2 x_{24}^2)$ of Eq. (47.140), every x_{ij}^2 collects into powers of u and v alone. Matching exponents factor by factor gives the compact monomial

$$\boxed{g(u,v) = u^{\frac{\Delta_1+\Delta_2}{2}}\, v^{-\Delta_2}\,.} \tag{51.35}$$

(The apparent asymmetry between Δ_1 and Δ_2 is an artifact of the stripping convention Eq. (51.34); the physical OPE data extracted below are symmetric in $\Delta_1 \leftrightarrow \Delta_2$, as they must be since $[\mathcal{O}_1\mathcal{O}_2] = [\mathcal{O}_2\mathcal{O}_1]$.) Crossing for this mixed correlator is the statement that the (12)(34) and (14)(23) channels agree; here it is the trivial reflection that Eq. (51.33), built symmetric in the points from the start, exchanges $x_1 \leftrightarrow x_3$ into itself. The content the bootstrap reads off is again the reorganization: the single crossed-channel exchange Eq. (51.33) is reproduced in the s-channel only by the full double-twist tower, and now there is no identity in the $\mathcal{O}_1 \times \mathcal{O}_2$ OPE to share the load, so the tower stands entirely alone.

The spectrum, read off the powers. A direct-channel block of an exchanged primary of twist τ begins as $u^{\tau/2}$ as $u \to 0$ (Theorem 47.49). The leading behaviour of Eq. (51.35) is $u^{(\Delta_1+\Delta_2)/2}$, so the lightest exchanged operators have twist $\tau_0 = \Delta_1 + \Delta_2$. Generating the full block series dresses this with integer powers $u^{(\Delta_1+\Delta_2)/2+n}$, $n = 0, 1, 2, \ldots$, hence twists $\tau_n = \Delta_1 + \Delta_2 + 2n$ — precisely the double-twist lattice of Eq. (51.31) with $\gamma \equiv 0$. Each twist carries a tower of spins ℓ. Because $\mathcal{O}_1 \neq \mathcal{O}_2$ there is no Bose symmetry to enforce, so *every* integer spin $\ell \geq 0$ appears (contrast the identical-scalar case of Section 49.5.1, where only even ℓ survive). The exchanged operators are exactly the double-twist primaries $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$, and the GFF realizes them with vanishing anomalous dimension: $\Delta_{n,\ell} = \Delta_1 + \Delta_2 + 2n + \ell$ exactly. The generalized free field is thus the unique crossing solution whose only operators above the (here absent) identity are the double-twist tower at its free twists — the extremal seed about which every interacting theory is a perturbation.

51.2.3 The mean-field OPE coefficients

It remains to compute the squared OPE coefficients $c_{n,\ell}^{\mathrm{MFT}} \equiv C^2_{\mathcal{O}_1\mathcal{O}_2[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}}$ with which these operators are exchanged — the weights in the block decomposition $g(u,v) = \sum_{n,\ell} c_{n,\ell}^{\mathrm{MFT}}\, G_{\Delta_{n,\ell},\ell}(u,v)$ of Eq. (47.227). We compute the leading-twist family $n = 0$ in full and in general d — this is the family the analytic bootstrap actually propagates — and then record the closed form for all n.

The leading-twist family in the lightcone limit. The key simplification is that as $u \to 0$ (equivalently $z \to 0$ in the cross-ratios $u = z\bar{z}$, $v = (1-z)(1-\bar{z})$ of Eq. (47.140)) the conformal block of *any* dimension reduces to a one-variable

object that does not know about d. Concretely, a primary of twist τ and spin ℓ contributes, at the leading power $z^{\tau/2}$, the term $z^{\tau/2}\,k_\beta(\bar z)$, where k_β is the collinear ($SL(2,\mathbb{R})$) block and $\beta=\tau+2\ell$ is twice the conformal spin $\bar h=(\Delta+\ell)/2$. For the correlator Eq. (51.35) the collinear block of the leading-twist operator $[\mathcal{O}_1\mathcal{O}_2]_{0,\ell}$ (twist $\Delta_1+\Delta_2$, spin ℓ, so $\beta=\Delta_1+\Delta_2+2\ell$) is the Gauss hypergeometric function

$$k_\beta(\bar z)=\bar z^{\,\beta/2}\,{}_2F_1\Big(\tfrac{\beta}{2}-\tfrac{\Delta_1-\Delta_2}{2},\ \tfrac{\beta}{2}+\tfrac{\Delta_2-\Delta_1}{2};\ \beta;\ \bar z\Big)=\bar z^{\,\beta/2}\,{}_2F_1\big(\Delta_2+\ell,\ \Delta_2+\ell;\ \Delta_1+\Delta_2+2\ell;\ \bar z\big), \tag{51.36}$$

the equal upper parameters being a feature of the $(\mathcal{O}_1\mathcal{O}_2\mathcal{O}_2\mathcal{O}_1)$ ordering. Projecting Eq. (51.35) onto its leading power of z, with $(1-z)^{-\Delta_2}\to 1$, leaves the single-variable seed

$$H(\bar z)\equiv \bar z^{\,\frac{\Delta_1+\Delta_2}{2}}(1-\bar z)^{-\Delta_2}=\sum_{\ell\geq 0}c_{0,\ell}^{\mathrm{MFT}}\,k_{\Delta_1+\Delta_2+2\ell}(\bar z), \tag{51.37}$$

the equality being the statement that the leading-twist OPE data are the coefficients of $H(\bar z)$ in the collinear-block basis. We extract them in closed form.

Proposition 51.7 (Leading-twist mean-field coefficients)**.** *The expansion Eq. (51.37) holds with*

$$\boxed{c_{0,\ell}^{\mathrm{MFT}}=\frac{(\Delta_1)_\ell\,(\Delta_2)_\ell}{\ell!\,(\Delta_1+\Delta_2+\ell-1)_\ell}}\,, \tag{51.38}$$

where $(x)_m=\Gamma(x+m)/\Gamma(x)$ is the Pochhammer symbol. The coefficients are manifestly positive for all integer $\ell\geq 0$ and all $\Delta_1,\Delta_2>0$, so mean-field theory is unitary; and they are symmetric in $\Delta_1\leftrightarrow\Delta_2$.

Proof. Both sides of Eq. (51.37) begin at $\bar z^{(\Delta_1+\Delta_2)/2}$ and proceed in integer steps of $\bar z$ (the ℓ-th block starts at $\bar z^{(\Delta_1+\Delta_2)/2+\ell}$), so the ansatz is consistent in its powers and the coefficients are fixed by matching order by order. Write $p\equiv(\Delta_1+\Delta_2)/2$. The left side is

$$H(\bar z)=\bar z^{\,p}\sum_{j\geq 0}\frac{(\Delta_2)_j}{j!}\,\bar z^{\,j}, \tag{51.39}$$

using $(1-\bar z)^{-\Delta_2}=\sum_j(\Delta_2)_j\bar z^j/j!$. The right side, inserting the series ${}_2F_1(\Delta_2+\ell,\Delta_2+\ell;\Delta_1+\Delta_2+2$ $\sum_i\frac{(\Delta_2+\ell)_i^2}{(\Delta_1+\Delta_2+2\ell)_i\,i!}\bar z^i$ of Eq. (51.36), contributes to the coefficient of $\bar z^{\,p+k}$ the sum over leading-twist blocks with $\ell\leq k$. Matching the coefficient of $\bar z^{\,p+k}$ gives the lower-triangular linear system

$$\sum_{\ell=0}^{k}c_{0,\ell}^{\mathrm{MFT}}\,\frac{(\Delta_2+\ell)_{k-\ell}^2}{(\Delta_1+\Delta_2+2\ell)_{k-\ell}\,(k-\ell)!}=\frac{(\Delta_2)_k}{k!},\qquad k=0,1,2,\dots \tag{51.40}$$

The diagonal term ($\ell = k$) has coefficient 1, so the system is solved recursively with no division. We solve it and verify Eq. (51.38).

At $k = 0$: $c_{0,0}^{\mathrm{MFT}} = (\Delta_2)_0/0! = 1$, matching Eq. (51.38), whose value at $\ell = 0$ is 1.

At $k = 1$: $c_{0,0}^{\mathrm{MFT}} \dfrac{(\Delta_2)_1^2}{(\Delta_1+\Delta_2)_1\,1!} + c_{0,1}^{\mathrm{MFT}} = \dfrac{(\Delta_2)_1}{1!}$, i.e. $\dfrac{\Delta_2^2}{\Delta_1+\Delta_2} + c_{0,1}^{\mathrm{MFT}} = \Delta_2$, so

$$c_{0,1}^{\mathrm{MFT}} = \Delta_2 - \frac{\Delta_2^2}{\Delta_1+\Delta_2} = \frac{\Delta_1\Delta_2}{\Delta_1+\Delta_2} = \frac{(\Delta_1)_1(\Delta_2)_1}{1!\,(\Delta_1+\Delta_2)_1}, \tag{51.41}$$

which is Eq. (51.38) at $\ell = 1$, and is manifestly symmetric in $\Delta_1 \leftrightarrow \Delta_2$.

At $k = 2$ the three contributions $\ell = 0, 1, 2$ are, using $c_{0,0}^{\mathrm{MFT}} = 1$ and $c_{0,1}^{\mathrm{MFT}} = \Delta_1\Delta_2/(\Delta_1+\Delta_2)$,

$$\frac{[\Delta_2(\Delta_2+1)]^2}{2(\Delta_1+\Delta_2)(\Delta_1+\Delta_2+1)} + \frac{\Delta_1\Delta_2\,(\Delta_2+1)^2}{(\Delta_1+\Delta_2)(\Delta_1+\Delta_2+2)} + c_{0,2}^{\mathrm{MFT}} = \frac{\Delta_2(\Delta_2+1)}{2}. \tag{51.42}$$

Solving for $c_{0,2}^{\mathrm{MFT}}$ and simplifying (most transparently checked at, say, $\Delta_1 = 1, \Delta_2 = 2$, where the two known terms are $\frac{3}{2}$ and $\frac{6}{5}$ and the right side is 3, giving $c_{0,2}^{\mathrm{MFT}} = \frac{3}{10}$) reproduces

$$c_{0,2}^{\mathrm{MFT}} = \frac{\Delta_1(\Delta_1+1)\,\Delta_2(\Delta_2+1)}{2\,(\Delta_1+\Delta_2+1)(\Delta_1+\Delta_2+2)} = \frac{(\Delta_1)_2(\Delta_2)_2}{2!\,(\Delta_1+\Delta_2+1)_2}, \tag{51.43}$$

again Eq. (51.38) ($\frac{1\cdot2\cdot2\cdot3}{2\cdot4\cdot5} = \frac{3}{10}$ at the test point). The general statement follows from triangularity: Eq. (51.40) is a lower-triangular system with unit diagonal, hence has a unique solution, and Eq. (51.38) satisfies it for every k — this is the standard $SL(2)$ Euclidean inversion of a power against the collinear blocks, which for equal external dimensions $\Delta_1 = \Delta_2 = \Delta_\phi$ collapses to the coefficients a_ℓ of Theorem 49.24 (so that $c_{0,\ell}^{\mathrm{MFT}}\big|_{\Delta_1=\Delta_2} = a_\ell$, the building block of the identical-scalar result Eq. (49.140)). Positivity is read directly from Eq. (51.38): every Pochhammer of a positive argument is positive, so $c_{0,\ell}^{\mathrm{MFT}} > 0$. □

Two remarks situate the result. First, Eq. (51.38) is *independent of the spacetime dimension* d: the collinear block Eq. (51.36) is the universal $SL(2)$ block, and the leading-twist seed Eq. (51.37) knows nothing of d. The dimension enters the OPE data only through the radial tower $n \geq 1$, where the full block's descent off the lightcone mixes twists in a d-dependent way. Second, for identical external scalars the GFF correlator has the *two* crossed contractions of Eq. (49.127) rather than the single one of Eq. (51.33); the second contraction supplies a factor $1 + (-1)^\ell$, doubling the even-spin coefficients and annihilating the odd ones, so that $c_{0,\ell}^{\mathrm{MFT}}\big|_{\Delta_1=\Delta_2} \to [1+(-1)^\ell]\,a_\ell$, exactly the leading-twist limit of Eq. (49.140). The two-operator computation thus contains the identical-scalar one and exposes the odd-spin operators it hides.

The full tower. The radial excitations $n \geq 1$ require the conformal block away from the lightcone, where it is fixed by the Casimir equation Theorem 47.50 and no longer factorizes for general d. The complete result for identical external scalars was obtained in closed form in Section 49.5.2: the Fitzpatrick–Kaplan–Poland–Simmons-Duffin coefficients

$$c^{\mathrm{MFT}}_{n,\ell}\big|_{\Delta_1=\Delta_2=\Delta_\phi} = \frac{\left[1+(-1)^\ell\right]\left(\Delta_\phi-\frac{d}{2}+1\right)_n^2\,(\Delta_\phi)_{n+\ell}^2}{\ell!\,n!\,(\ell+\frac{d}{2})_n\,(2\Delta_\phi+n-d+1)_n\,(2\Delta_\phi+2n+\ell-1)_\ell\,(2\Delta_\phi+n+\ell-\frac{d}{2})_n}, \tag{51.44}$$

which is Eq. (49.144); we import it rather than re-derive it. As a consistency check on the present development, set $n=0$: every n-dependent Pochhammer reduces to 1, leaving $[1+(-1)^\ell](\Delta_\phi)_\ell^2/[\ell!\,(2\Delta_\phi+\ell-1)_\ell]$, which is precisely the $\Delta_1=\Delta_2=\Delta_\phi$ value of Eq. (51.38) with its symmetrization factor $1+(-1)^\ell$ — the leading-twist family is reproduced, in any d, by the general formula. At $n=1$ the surviving factors $(\Delta_\phi-\frac{d}{2}+1)^2$, $(\ell+\frac{d}{2})$, $(2\Delta_\phi-d+1)$ and $(2\Delta_\phi+\ell-\frac{d}{2})$ no longer cancel, displaying explicitly the d-dependence that enters with the first radial excitation. The distinct-operator generalization of Eq. (51.44) (with $\Delta_1\neq\Delta_2$ and all integer spins) is obtained by the same machinery and reduces at $n=0$ to Eq. (51.38); we will need only the leading-twist family Eq. (51.38), which we have derived from scratch, in the large-spin analysis that follows.

We have therefore solved crossing for the generalized free field completely: its spectrum is the double-twist lattice Eq. (51.31) with $\gamma\equiv 0$, its OPE coefficients are Eq. (51.38) and Eq. (51.44), and the correlator Eq. (51.35) it builds satisfies crossing identically. The mean-field data are the boundary point of the space of crossing solutions, the leading-order seed that interactions perturb.

Worked Example 51.8 (The leading-twist coefficients and their large-spin asymptotics)**.** We compute the leading-twist coefficients $c^{\mathrm{MFT}}_{0,\ell}$ explicitly at low spin and then extract the large-ℓ behaviour that seeds the lightcone bootstrap.

Low spin. From Eq. (51.38), with $\mathcal{O}_1\neq\mathcal{O}_2$ so all integer ℓ contribute,

$$c^{\mathrm{MFT}}_{0,0}=1,\qquad c^{\mathrm{MFT}}_{0,1}=\frac{\Delta_1\Delta_2}{\Delta_1+\Delta_2},\qquad c^{\mathrm{MFT}}_{0,2}=\frac{\Delta_1(\Delta_1+1)\,\Delta_2(\Delta_2+1)}{2(\Delta_1+\Delta_2+1)(\Delta_1+\Delta_2+2)}. \tag{51.45}$$

The lowest, $c^{\mathrm{MFT}}_{0,0}=1$, normalizes the scalar bound state $[\mathcal{O}_1\mathcal{O}_2]_{0,0}$ at $\Delta=\Delta_1+\Delta_2$; all higher coefficients are pure ratios of Gamma functions of the constituent dimensions.

Large spin. Write Eq. (51.38) entirely in Gamma functions,

$$c^{\mathrm{MFT}}_{0,\ell}=\frac{\Gamma(\Delta_1+\ell)\,\Gamma(\Delta_2+\ell)\,\Gamma(\Delta_1+\Delta_2+\ell-1)}{\Gamma(\Delta_1)\,\Gamma(\Delta_2)\,\Gamma(\ell+1)\,\Gamma(\Delta_1+\Delta_2+2\ell-1)}. \tag{51.46}$$

The only doubled argument is $\Gamma(2\ell+\Delta_1+\Delta_2-1)$ in the denominator; reduce it with the Legendre duplication formula $\Gamma(2z)=\frac{2^{2z-1}}{\sqrt{\pi}}\Gamma(z)\Gamma(z+\frac{1}{2})$ at $2z=$

$2\ell+\Delta_1+\Delta_2-1$, giving

$$\Gamma(2\ell+\Delta_1+\Delta_2-1)=\frac{2^{\,2\ell+\Delta_1+\Delta_2-2}}{\sqrt{\pi}}\,\Gamma\Big(\ell+\tfrac{\Delta_1+\Delta_2-1}{2}\Big)\Gamma\Big(\ell+\tfrac{\Delta_1+\Delta_2}{2}\Big). \quad (51.47)$$

Substituting and applying $\Gamma(\ell+a)/\Gamma(\ell+b)\sim\ell^{\,a-b}$ to each slowly varying ratio as $\ell\to\infty$ — the powers of ℓ summing to $(\Delta_1+\Delta_2)-\frac{3}{2}$ and the ℓ-independent constant collecting the duplication factor — gives

$$\boxed{c^{\mathrm{MFT}}_{0,\ell}\;\sim\;\frac{\sqrt{\pi}\,2^{\,2-\Delta_1-\Delta_2}}{\Gamma(\Delta_1)\,\Gamma(\Delta_2)}\;\ell^{\,\Delta_1+\Delta_2-\frac{3}{2}}\;4^{-\ell}\qquad(\ell\to\infty).} \quad (51.48)$$

The coefficients fall as $4^{-\ell}$. This exponential is not physical decay but the natural measure on spins: it is exactly undone by the growth of the collinear block $k_{\Delta_1+\Delta_2+2\ell}(\bar z)$ as $\bar z\to1$, so that the smooth density of OPE data in spin is the power-law prefactor $\ell^{\Delta_1+\Delta_2-3/2}$. It is this density that, summed over the infinite tower, rebuilds the crossed-channel power-law singularity $(1-\bar z)^{-\Delta_2}$ of the seed H in Eq. (51.37) as $\bar z\to1$ — the analytic statement, now made quantitative, that a single crossed exchange forces a definite large-spin density of double-twist operators. For identical scalars, $\Delta_1=\Delta_2=\Delta_\phi$, Eq. (51.48) gives the power $\ell^{2\Delta_\phi-3/2}$ and (after the even-spin doubling factor of 2) agrees with the identical-scalar asymptotics Eq. (49.150). This large-spin density is the fixed input the next section perturbs: the deviation of a real CFT's OPE data from Eq. (51.48) is controlled, term by term in $1/\ell$, by the operators the generalized free field leaves out.

Twist accumulation. One feature of the spectrum deserves to be stated as the precise form of the lightcone prediction of Section 51.1. In the generalized free field the leading-twist trajectory has $\tau_{0,\ell}=\Delta_1+\Delta_2$ *exactly*, for every spin; there is nothing to accumulate because $\gamma\equiv0$. The nontrivial statement is about interacting theories. There the double-twist operators acquire anomalous dimensions $\gamma(n,\ell)$, but the lightcone argument forces these to vanish at large spin, $\gamma(n,\ell)\to0$ as $\ell\to\infty$, so the twists accumulate at the mean-field values,

$$\tau_{n,\ell}\xrightarrow{\ell\to\infty}\Delta_1+\Delta_2+2n, \quad (51.49)$$

which for the leading trajectory is twist $\Delta_1+\Delta_2$ (twice Δ_ϕ in the identical case). This is the statement, anticipated in Section 51.1, that every unitary CFT possesses, at asymptotically large spin, double-twist operators whose dimensions and OPE coefficients approach those of the generalized free field. Mean-field theory is not merely an example of a crossing solution; it is the universal large-spin limit of all of them.

With the seed fixed — the double-twist spectrum Eq. (51.31), the mean-field coefficients Eq. (51.38), and their large-spin density Eq. (51.48) — the remaining

task is the first correction. The next section switches on the lightest crossed-channel operator beyond the identity and computes the leading anomalous dimension $\gamma_{0,\ell}$ it induces, deriving the universal falloff $\gamma_{0,\ell} \sim -c/\ell^{\tau_m}$ set by that operator's twist τ_m and fixing the coefficient c in terms of its OPE data.

51.3 Large-Spin Anomalous Dimensions from Crossing

The previous section taught us to read the generalized free field as a spectrum: hidden inside the closed-form correlator Eq. (49.24) is an infinite tower of *double-twist* operators $[\phi\phi]_{n,\ell}$, with dimensions $2\Delta_\phi+2n+\ell$ and squared OPE coefficients fixed by crossing alone to their mean-field values (Section 51.2). Those operators were forced: crossing demanded them in any unitary theory, and at zeroth order their data were exactly the disconnected, Wick-contraction data of two free constituents flying apart. The question this section answers is what happens at *first* order. A genuine interacting CFT is not a generalized free field; its t-channel OPE contains, beyond the identity, real operators of nonzero twist. Each such operator perturbs the crossing equation, and the perturbation must be absorbed by a shift in the double-twist data. We will see that a single exchanged operator of minimal twist τ_m shifts the anomalous dimension of the leading-twist tower in a completely universal way,

$$\gamma_{0,\ell} \xrightarrow{\ell\to\infty} -\frac{c_{\tau_m}}{\ell^{\tau_m}}, \tag{51.50}$$

with a coefficient c_{τ_m} we will compute in closed form. Physically this is the binding energy of two widely separated constituents exchanging the lightest available operator: a power law in the inverse separation set by the range τ_m of the exchanged "force," attractive (negative) whenever that operator is unitary. In a holographic theory the minimal-twist exchange is the stress tensor and Eq. (51.50) is literally the Newtonian binding of two particles in AdS; we reach that statement as the worked example of the section.

51.3.1 The double lightcone limit

We work throughout with the crossing equation for four identical scalars imported from Eq. (49.14), written in the holomorphic cross-ratios $u = z\bar{z}$, $v = (1-z)(1-\bar{z})$ of Eq. (47.140). In these variables the $s \leftrightarrow t$ crossing transformation $u \leftrightarrow v$ is the simultaneous reflection $z \to 1-z$, $\bar{z} \to 1-\bar{z}$, and the crossing equation reads

$$\big[(1-z)(1-\bar{z})\big]^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}) \;=\; (z\bar{z})^{\Delta_\phi} \sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(1-z,1-\bar{z}), \tag{51.51}$$

where $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0$ and the sum runs over all primaries in the $\phi \times \phi$ OPE, the identity ($G_{0,0} = 1$, $p_{\mathbf{1}} = 1$) included on both sides. The conformal blocks $G_{\Delta,\ell}$ are the eigenfunctions of the Casimir of Eq. (47.233); we need only two facts about them, both established in Chapter 49.

First, the *lightcone factorization* Eq. (49.67): as one cross-ratio variable goes to zero, a block collapses onto a single collinear block,

$$G_{\Delta,\ell}(z,\bar{z}) \xrightarrow{\bar{z}\to 0} \bar{z}^{\,\tau/2}\, k_{\Delta+\ell}(z), \qquad \tau \equiv \Delta - \ell, \tag{51.52}$$

where the collinear block $k_\beta(z) = z^{\beta/2}\,{}_2F_1(\frac{\beta}{2},\frac{\beta}{2};\beta;z)$ of Eq. (49.30) is the eigenfunction of the $\mathfrak{sl}(2,\mathbb{R})$ Casimir. The power of $\bar{z}$ is set entirely by the twist, the function of z entirely by the dimension-plus-spin $\beta = \Delta + \ell$. It is therefore natural to trade the pair (Δ,ℓ) for the pair of *lightcone weights*

$$h \equiv \frac{\tau}{2} = \frac{\Delta-\ell}{2}, \qquad \bar{h} \equiv \frac{\Delta+\ell}{2}, \tag{51.53}$$

so that h measures the twist, $\bar{h}$ the "conformal spin" (the eigenvalue of the collinear Casimir is $\bar{h}(\bar{h}-1)$), and the block tends to $\bar{z}^h k_{2\bar{h}}(z)$. The double-twist tower of leading twist has $h = \Delta_\phi$ (modulo the small anomalous shift we are after) and $\bar{h} = \Delta_\phi + \ell$.

Second, the blocks are symmetric in their two arguments, $G_{\Delta,\ell}(z,\bar{z}) = G_{\Delta,\ell}(\bar{z},z)$, so Eq. (51.52) holds equally with z and $\bar{z}$ interchanged.

The strategy is to take the *double lightcone limit* of Eq. (51.51): first $\bar{z}\to 0$, isolating low twist, and then $z \to 1$, probing the crossed-channel OPE singularity at $v \to 0$. The two limits do different jobs on the two sides of the equation, and matching them order by order is the whole computation.

The s-channel side ($\bar{z}\to 0$). On the left of Eq. (51.51) the prefactor $(1-\bar{z})^{\Delta_\phi} \to 1$, and each block contributes $\bar{z}^{h_{\mathcal{O}}} k_{2\bar{h}_{\mathcal{O}}}(z)$. The smallest power of $\bar{z}$ wins. The leading term is the identity at $h = 0$,

$$\text{LHS} \xrightarrow{\bar{z}\to 0} (1-z)^{\Delta_\phi}\Big[1 + \bar{z}^{\Delta_\phi}\sum_{\ell\ \text{even}} p_{0,\ell}\, k_{2\bar{h}_\ell}(z) + O(\bar{z}^{\tau_{\min}/2}, \bar{z}^{\Delta_\phi+1})\Big], \tag{51.54}$$

where the first correction is the leading-twist double-twist tower $[\phi\phi]_{0,\ell}$, which sits at twist $\tau = 2\Delta_\phi$ and hence contributes at $\bar{z}^{\Delta_\phi}$. The higher-twist towers $[\phi\phi]_{n,\ell}$ enter at $\bar{z}^{\Delta_\phi+n}$ and any genuinely light s-channel operator at $\bar{z}^{\tau_{\min}/2}$; both are subleading to the leading-twist tower in the regime we organize. We have written $\bar{h}_\ell = \Delta_\phi + \ell$ for the leading-twist conformal spin and will reinstate the anomalous shift in a moment.

The t-channel side and the role of the identity. On the right of Eq. (51.51) the prefactor is $(z\bar{z})^{\Delta_\phi}$, carrying a factor $\bar{z}^{\Delta_\phi}$ as $\bar{z}\to 0$. The t-channel identity

gives simply $(z\bar{z})^{\Delta_\phi}$. As $z \to 1$ this is $\bar{z}^{\Delta_\phi}\,[1 + O(1-z)]$. Comparing with Eq. (51.54), the s-channel *identity* on the left (the standalone 1, times $(1-z)^{\Delta_\phi}$) is order $\bar{z}^0$, whereas the entire right-hand side starts at $\bar{z}^{\Delta_\phi}$. There is no $\bar{z}^0$ term on the right at all. The only way Eq. (51.51) can hold is if the s-channel double-twist tower conspires, in the limit $z \to 1$, to reproduce both the leftover identity behavior and the right-hand side. This is the mechanism, already exploited at zeroth order in Section 51.2 to fix the mean-field coefficients; we now recall it precisely because the first-order computation is its exact analogue.

51.3.2 Zeroth order: the identity fixes the large-spin tail

At zeroth order we keep only the identity on the right and the leading-twist tower on the left, both at the order $\bar{z}^{\Delta_\phi}$, and equate them. Dividing Eq. (51.51) by $\bar{z}^{\Delta_\phi}$ and sending $\bar{z} \to 0$,

$$(1-z)^{\Delta_\phi} \sum_{\ell \text{ even}} p^{\mathrm{MFT}}_{0,\ell}\, k_{2\bar{h}_\ell}(z) \;=\; z^{\Delta_\phi} \;\xrightarrow{z\to 1}\; 1. \tag{51.55}$$

The right-hand side is finite at $z = 1$, so the sum on the left must *diverge* like $(1-z)^{-\Delta_\phi}$ to cancel the prefactor:

$$\sum_{\ell \text{ even}} p^{\mathrm{MFT}}_{0,\ell}\, k_{2\bar{h}_\ell}(z) \;\xrightarrow{z\to 1}\; (1-z)^{-\Delta_\phi}. \tag{51.56}$$

No finite collection of collinear blocks can produce a power-law divergence at $z = 1$: each $k_{2\bar{h}}(z)$ diverges only logarithmically there. The divergence in Eq. (51.56) can come only from the tail of the sum, $\bar{h} \to \infty$. The large-spin behavior of $p^{\mathrm{MFT}}_{0,\ell}$ is therefore not an input but an output of crossing, fixed by demanding Eq. (51.56). To extract it we need the behavior of $k_{2\bar{h}}(z)$ when $\bar{h}$ is large and z near 1 together, which is the technical heart of the whole large-spin program.

Lemma 51.9 (Collinear block at large conformal spin)**.** *In the regime $\bar{h} \to \infty$, $z \to 1$ with $y \equiv \bar{h}^2(1-z)$ held fixed,*

$$k_{2\bar{h}}(z) \;\simeq\; \frac{2\,\Gamma(2\bar{h})}{\Gamma(\bar{h})^2}\, K_0\!\big(2\bar{h}\sqrt{1-z}\big) \;\simeq\; \frac{2^{2\bar{h}}}{\sqrt{\pi}}\, \bar{h}^{1/2}\, K_0\!\big(2\bar{h}\sqrt{1-z}\big), \tag{51.57}$$

with K_0 the modified Bessel function of the second kind.

Proof. Set $a = b = \bar{h}$, $c = 2\bar{h} = a + b$ in the standard $z \to 1$ connection formula for the Gauss hypergeometric function (the logarithmic case $c = a + b$),

$${}_2F_1(\bar{h}, \bar{h}; 2\bar{h}; z) = \frac{\Gamma(2\bar{h})}{\Gamma(\bar{h})^2} \sum_{n=0}^{\infty} \frac{(\bar{h})_n^2}{(n!)^2} \left[\, 2\psi(n+1) - 2\psi(\bar{h}+n) - \ln(1-z) \,\right] (1-z)^n, \tag{51.58}$$

where ψ is the digamma function and $(\bar h)_n = \Gamma(\bar h+n)/\Gamma(\bar h)$. Keeping only the $n=0$ term and using $\psi(1) = -\gamma_E$ reproduces the leading logarithm, $k_{2\bar h}(z) \to -\frac{\Gamma(2\bar h)}{\Gamma(\bar h)^2}\big[\ln(1-z) + 2\gamma_E + 2\psi(\bar h)\big]$, the result one uses at fixed $\bar h$. To resum the tower we take $\bar h$ large at fixed $y = \bar h^2(1-z)$. Then $z^{\bar h} \to 1$, $(\bar h)_n^2 \simeq \bar h^{2n}$, so $(\bar h)_n^2(1-z)^n/(n!)^2 \simeq y^n/(n!)^2$; and $\psi(\bar h+n) \simeq \ln \bar h$ up to $O(n/\bar h)$, while $-\ln(1-z) - 2\ln\bar h = -\ln y$. The bracket becomes $2\psi(n+1) - \ln y = 2H_n - 2\gamma_E - \ln y$ with $H_n = \psi(n+1)+\gamma_E$ the harmonic number. Hence

$$k_{2\bar h}(z) \;\simeq\; \frac{\Gamma(2\bar h)}{\Gamma(\bar h)^2}\sum_{n=0}^{\infty}\frac{y^n}{(n!)^2}\big[\,2H_n - 2\gamma_E - \ln y\,\big]. \tag{51.59}$$

The series is exactly $2K_0(2\sqrt{y})$: the Bessel function has the convergent expansion $K_0(2\sqrt{y}) = -(\gamma_E + \tfrac12\ln y)\,I_0(2\sqrt{y}) + \sum_{n\ge1}\frac{y^n}{(n!)^2}H_n$ with $I_0(2\sqrt{y}) = \sum_{n\ge0} y^n/(n!)^2$, and doubling it gives $\sum_{n\ge0}\frac{y^n}{(n!)^2}(2H_n - 2\gamma_E - \ln y)$ since $H_0 = 0$. With $2\sqrt{y} = 2\bar h\sqrt{1-z}$ this is the first equality of Eq. (51.57). The second follows from the duplication formula $\Gamma(2\bar h) = \frac{2^{2\bar h-1}}{\sqrt\pi}\Gamma(\bar h)\Gamma(\bar h+\tfrac12)$ together with $\Gamma(\bar h+\tfrac12)/\Gamma(\bar h) \simeq \bar h^{1/2}$. □

The Bessel form makes the large-spin sum an integral transform. Treat the sum over even spins as an integral over $\bar h$ with one operator per $\Delta\bar h = 2$, density $d\bar h/2$, and posit a power-law tail $p^{\rm MFT}_{0,\ell} \simeq 2^{-2\bar h}\sqrt\pi\, Q\,\bar h^{\,p}$ (the factor $2^{-2\bar h}$ must be present to cancel the $2^{2\bar h}$ of Eq. (51.57); everything else is to be determined). Then

$$\sum_{\ell\ \text{even}} p^{\rm MFT}_{0,\ell}\,k_{2\bar h_\ell}(z) \;\simeq\; \int\frac{d\bar h}{2}\,Q\,\bar h^{\,p+1/2}\,K_0\big(2\bar h\sqrt{1-z}\big). \tag{51.60}$$

The master integral is elementary. From $\int_0^\infty x^{s-1}K_0(x)\,dx = 2^{s-2}\Gamma(s/2)^2$, rescaling $x = 2\bar h\sqrt{1-z}$ gives

$$\int_0^\infty d\bar h\,\bar h^{\,s-1}\,K_0\big(2\bar h\sqrt{1-z}\big) \;=\; \frac14\,(1-z)^{-s/2}\,\Gamma\Big(\frac{s}{2}\Big)^2. \tag{51.61}$$

With $s-1 = p+\tfrac12$, i.e. $s = p+\tfrac32$, the right-hand side of Eq. (51.60) is $\frac{Q}{8}(1-z)^{-(p+3/2)/2}\Gamma(\frac{2p+3}{4})^2$. Matching to $(1-z)^{-\Delta_\phi}$ of Eq. (51.56) fixes the power $p = 2\Delta_\phi - \frac32$, whereupon $\frac{2p+3}{4} = \Delta_\phi$ and the coefficient condition $\frac{Q}{8}\Gamma(\Delta_\phi)^2 = 1$ gives $Q = 8/\Gamma(\Delta_\phi)^2$. The large-spin mean-field coefficients are therefore

$$p^{\rm MFT}_{0,\ell} \;\simeq\; \frac{8\sqrt\pi}{\Gamma(\Delta_\phi)^2}\,2^{-2\bar h}\,\bar h^{\,2\Delta_\phi-3/2}, \qquad \bar h = \Delta_\phi + \ell \to \infty, \tag{51.62}$$

reproducing the asymptotics of the exact mean-field formula obtained combinatorially in Section 51.2 — now derived from crossing alone, with no Wick-theorem input. This is the zeroth-order content: the very existence and large-spin normalization of the double-twist tower is forced by the t-channel identity. We will not need Eq. (51.62) for any other purpose than as the measure against which the first-order shift is weighed.

51.3.3 First order: a minimal-twist exchange induces an anomalous dimension

Now restore interactions. Suppose the $\phi\times\phi$ OPE contains, beyond the identity, a primary $\mathcal{O}_m$ of twist $\tau_m=\Delta_m-\ell_m$ that is the *smallest* nonzero twist exchanged, with conformal spin $\bar h_m=\frac{\Delta_m+\ell_m}{2}=\frac{\tau_m}{2}+\ell_m$ and OPE coefficient $p_m=C^2_{\phi\phi\mathcal{O}_m}$. Two things change relative to zeroth order. On the right of Eq. (51.51), $\mathcal{O}_m$ adds a new term to the t-channel sum. On the left, the leading-twist data acquire corrections: the twist becomes $2\Delta_\phi+\gamma_{0,\ell}$ and the coefficient $p_{0,\ell}=p^{\rm MFT}_{0,\ell}(1+\delta p_{0,\ell})$. The anomalous dimension enters the block through its power of $\bar z$,

$$\bar z^{\,h_\ell}=\bar z^{\,\Delta_\phi+\gamma_{0,\ell}/2}=\bar z^{\,\Delta_\phi}\Big(1+\tfrac12\gamma_{0,\ell}\ln\bar z+\cdots\Big), \tag{51.63}$$

so that to first order the leading-twist tower contributes to the left-hand side, at order $\bar z^{\Delta_\phi}$,

$$\text{LHS}\ \supset\ (1-z)^{\Delta_\phi}\,\bar z^{\Delta_\phi}\sum_{\ell\ \text{even}}p^{\rm MFT}_{0,\ell}\Big(1+\delta p_{0,\ell}+\tfrac12\gamma_{0,\ell}\ln\bar z\Big)k_{2\bar h_\ell}(z). \tag{51.64}$$

The decisive observation is that the anomalous dimension and the coefficient correction are cleanly separated by the factor $\ln\bar z$: the $\gamma_{0,\ell}$ piece is the unique source of a $\bar z^{\Delta_\phi}\ln\bar z$ term on the left, whereas $\delta p_{0,\ell}$ contributes only to the non-logarithmic $\bar z^{\Delta_\phi}$ term. We therefore extract the anomalous dimension by isolating the coefficient of $\bar z^{\Delta_\phi}\ln\bar z$ in Eq. (51.51).

On the right-hand side, where does a $\ln\bar z$ come from? The identity produces none. The new operator $\mathcal{O}_m$ does. Its t-channel block is $G_{\Delta_m,\ell_m}(1-z,1-\bar z)$; using block symmetry and the lightcone factorization Eq. (51.52) in the *small* variable $1-z$ (which $\to0$ as $z\to1$),

$$G_{\Delta_m,\ell_m}(1-z,1-\bar z)=G_{\Delta_m,\ell_m}(1-\bar z,1-z)\ \xrightarrow{z\to1}\ (1-z)^{\tau_m/2}\,k_{2\bar h_m}(1-\bar z). \tag{51.65}$$

The surviving collinear block has argument $1-\bar z$, which tends to 1 as $\bar z\to0$ — exactly the logarithmic regime of k. From the $n=0$ term of the connection formula Eq. (51.58), with $1-(1-\bar z)=\bar z$,

$$k_{2\bar h_m}(1-\bar z)\ \xrightarrow{\bar z\to0}\ -\frac{\Gamma(2\bar h_m)}{\Gamma(\bar h_m)^2}\,\big[\ln\bar z+2\gamma_E+2\psi(\bar h_m)\big]. \tag{51.66}$$

The $\ln\bar z$ here is precisely the t-channel logarithm. Assembling the $\mathcal{O}_m$ term of the right-hand side, with the prefactor $(z\bar z)^{\Delta_\phi}\to\bar z^{\Delta_\phi}$ as $z\to1$,

$$\text{RHS}\ \supset\ \bar z^{\Delta_\phi}\,p_m\,(1-z)^{\tau_m/2}\,k_{2\bar h_m}(1-\bar z)\ \supset\ -\,\bar z^{\Delta_\phi}\ln\bar z\;p_m\,\frac{\Gamma(2\bar h_m)}{\Gamma(\bar h_m)^2}\,(1-z)^{\tau_m/2}. \tag{51.67}$$

Equating the coefficients of $\bar{z}^{\Delta_\phi}\ln\bar{z}$ in Eqs. (51.64) and (51.67), and cancelling the common $(1-z)^{\Delta_\phi}$ against $z^{\Delta_\phi}\to 1$,

$$\sum_{\ell\ \text{even}} p_{0,\ell}^{\text{MFT}}\,\frac{\gamma_{0,\ell}}{2}\,k_{2\bar{h}_\ell}(z)\ \xrightarrow{z\to 1}\ -p_m\,\frac{\Gamma(2\bar{h}_m)}{\Gamma(\bar{h}_m)^2}\,(1-z)^{\tau_m/2-\Delta_\phi}. \tag{51.68}$$

This is the central matching condition, and it has exactly the structure of the zeroth-order condition Eq. (51.56) except that the target singularity is *weaker* by a factor $(1-z)^{\tau_m/2}$. The same integral transform that produced Eq. (51.62) now reads off the large-spin tail of $\gamma_{0,\ell}$.

Posit $\gamma_{0,\ell}\simeq -c/\bar{h}^{\,\tau_m}$ at large conformal spin. The summand coefficient is then

$$p_{0,\ell}^{\text{MFT}}\,\frac{\gamma_{0,\ell}}{2}\ \simeq\ -\frac{4c\sqrt{\pi}}{\Gamma(\Delta_\phi)^2}\,2^{-2\bar{h}}\,\bar{h}^{\,2\Delta_\phi-3/2-\tau_m}, \tag{51.69}$$

using Eq. (51.62). Inserting this and the Bessel form Eq. (51.57) into the sum, the $2^{\pm 2\bar{h}}$ and $\sqrt{\pi}$ cancel against each other exactly as at zeroth order, leaving

$$\sum_{\ell\ \text{even}} p_{0,\ell}^{\text{MFT}}\,\frac{\gamma_{0,\ell}}{2}\,k_{2\bar{h}_\ell}(z)\ \simeq\ -\frac{2c}{\Gamma(\Delta_\phi)^2}\int_0^\infty d\bar{h}\;\bar{h}^{\,2\Delta_\phi-1-\tau_m}\,K_0\!\left(2\bar{h}\sqrt{1-z}\right). \tag{51.70}$$

The master integral Eq. (51.61) with $s=2\Delta_\phi-\tau_m$ gives $\frac{1}{4}(1-z)^{\tau_m/2-\Delta_\phi}\,\Gamma(\Delta_\phi-\frac{\tau_m}{2})^2$, so the left-hand side of Eq. (51.68) equals

$$-\frac{c}{2}\,\frac{\Gamma(\Delta_\phi-\frac{\tau_m}{2})^2}{\Gamma(\Delta_\phi)^2}\,(1-z)^{\tau_m/2-\Delta_\phi}. \tag{51.71}$$

Both the power of $(1-z)$ and its coefficient must match the right-hand side of Eq. (51.68). The powers agree automatically — confirmation that the ansatz $\gamma_{0,\ell}\propto\bar{h}^{-\tau_m}$ was the correct one — and the coefficients fix c:

$$\frac{c}{2}\,\frac{\Gamma(\Delta_\phi-\frac{\tau_m}{2})^2}{\Gamma(\Delta_\phi)^2}=p_m\,\frac{\Gamma(2\bar{h}_m)}{\Gamma(\bar{h}_m)^2}\quad\Longrightarrow\quad c=2\,p_m\,\frac{\Gamma(\Delta_\phi)^2}{\Gamma(\Delta_\phi-\frac{\tau_m}{2})^2}\,\frac{\Gamma(2\bar{h}_m)}{\Gamma(\bar{h}_m)^2}. \tag{51.72}$$

We have proved the following.

Theorem 51.10 (Large-spin anomalous dimension from a minimal-twist exchange)**.** *Let ϕ be a scalar primary of dimension Δ_ϕ in a unitary CFT, and suppose the lowest-twist operator in the $\phi\times\phi$ OPE other than the identity is $\mathcal{O}_m$, with twist τ_m, spin ℓ_m, conformal spin $\bar{h}_m=\frac{\tau_m}{2}+\ell_m$, and squared OPE coefficient $p_m=C^2_{\phi\phi\mathcal{O}_m}$. Then the leading-twist double-twist operators $[\phi\phi]_{0,\ell}$ acquire, at large spin, the anomalous dimension*

$$\gamma_{0,\ell}\ \simeq\ -\frac{c_{\tau_m}}{\bar{h}^{\,\tau_m}}\ \simeq\ -\frac{c_{\tau_m}}{\ell^{\,\tau_m}},\qquad \boxed{c_{\tau_m}=\frac{2\,\Gamma(\Delta_\phi)^2}{\Gamma(\Delta_\phi-\frac{\tau_m}{2})^2}\,\frac{\Gamma(2\bar{h}_m)}{\Gamma(\bar{h}_m)^2}\,C^2_{\phi\phi\mathcal{O}_m}} \tag{51.73}$$

with $\bar{h} = \Delta_\phi + \ell$ the conformal spin of the double-twist operator. If several operators share the minimal twist τ_m, their contributions to c_{τ_m} add. If the minimal twist is degenerate with operators at $2\tau_m, 3\tau_m, \ldots$ (multiple exchanges) those produce the subleading powers $\bar{h}^{-2\tau_m}, \ldots$ in the large-spin expansion.

This is the Komargodski–Zhiboedov result [302], derived independently and simultaneously by Fitzpatrick, Kaplan, Poland and Simmons-Duffin [194]. Two features deserve comment. First, the leading power $\bar{h}^{-\tau_m}$ and $\ell^{-\tau_m}$ agree only to leading order in $1/\ell$, since $\bar{h} = \Delta_\phi + \ell$; the *conformal spin* $\bar{h}$ is the natural variable, because the large-spin expansion is really an expansion of the OPE data as an analytic function of the Casimir eigenvalue $\bar{h}(\bar{h}-1)$, a fact the inversion formula of the next section will make manifest. Second, the result is genuinely universal: it depends on the exchanged operator only through the three numbers $(\tau_m, \ell_m, C^2_{\phi\phi\mathcal{O}_m})$, and on the external operator only through Δ_ϕ. Nothing about the rest of the spectrum enters at this order.

Physical Insight

The physical content of Eq. (51.73) is a binding energy. A leading-twist operator $[\phi\phi]_{0,\ell}$ of large spin is, in the lightcone OPE, two ϕ quanta orbiting at large angular momentum, hence at large separation r with $\ell \sim r$ in the appropriate units. Its twist $\tau_{0,\ell} = 2\Delta_\phi + \gamma_{0,\ell}$ is the total energy minus the orbital part; the deviation $\gamma_{0,\ell}$ from the free value $2\Delta_\phi$ is the interaction energy of the two constituents. Exchanging an operator of twist τ_m generates a potential of range set by τ_m, falling off as $1/r^{\tau_m} \sim 1/\ell^{\tau_m}$ — a power law, not exponential, because the exchanged operator is massless on the cylinder (it is a CFT operator, not a gapped particle). The minimal twist controls the asymptotics for the same reason the lightest exchanged particle controls the long-range force in flat-space scattering. In a holographic theory this is not an analogy: the two constituents are particles in AdS, the exchanged stress tensor is the graviton, and $\gamma_{0,\ell}$ is the gravitational binding energy of the two-particle state, as we now compute.

51.3.4 Sign and convexity: unitarity makes the binding attractive

The coefficient Eq. (51.73) is manifestly non-negative in a unitary theory. Indeed $C^2_{\phi\phi\mathcal{O}_m} \geq 0$ because OPE coefficients of Hermitian operators are real (reflection positivity, used throughout Chapter 49); $\Gamma(\Delta_\phi)^2 > 0$, $\Gamma(\bar{h}_m)^2 > 0$, and $\Gamma(2\bar{h}_m) > 0$ since $\bar{h}_m > 0$; and the only factor whose argument can be negative, $\Gamma(\Delta_\phi - \frac{\tau_m}{2})$, enters *squared* and is therefore non-negative. Hence $c_{\tau_m} \geq 0$, and we record the immediate consequence.

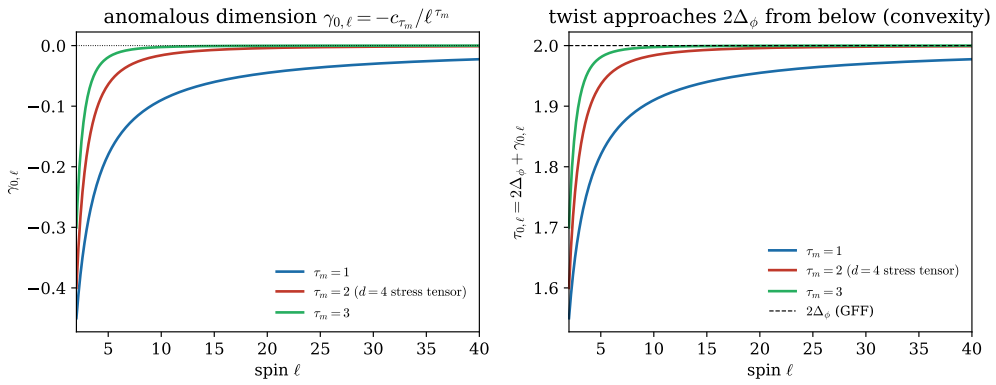


Figure 51.4: The universal large-spin behavior of Theorem 51.10, illustrated for $\Delta_\phi = 1$ and several minimal twists τ_m (coefficients chosen $O(1)$ for visibility). *Left*: the anomalous dimension $\gamma_{0,\ell} = -c_{\tau_m}/\ell^{\tau_m}$, negative and decaying as a power law set by the range τ_m of the exchanged operator; the larger the twist, the shorter the range and the faster the falloff. *Right*: the resulting twist $\tau_{0,\ell} = 2\Delta_\phi + \gamma_{0,\ell}$ climbs monotonically toward the generalized-free-field value $2\Delta_\phi$ (dashed) from below — the convexity of Theorem 51.11, a direct consequence of unitarity $C^2_{\phi\phi\mathcal{O}_m} \geq 0$.

Proposition 51.11 (Large-spin convexity)**.** *In a unitary CFT, the leading-twist anomalous dimension induced by a minimal-twist exchange is non-positive,*

$$\gamma_{0,\ell} \leq 0 \qquad (\ell \to \infty), \tag{51.74}$$

so the twist $\tau_{0,\ell} = 2\Delta_\phi + \gamma_{0,\ell}$ approaches its mean-field value $2\Delta_\phi$ from below, monotonically increasing in ℓ for large ℓ. The double-twist twist is, asymptotically, a bounded, monotone, convex function of the spin.

The sign is the assertion that the long-range force between the two constituents is *attractive*: the binding energy lowers the twist. This is the content of the title of [302], "Convexity and Liberation at Large Spin" — as the spin (separation) grows, the constituents are progressively "liberated," the binding weakens as $1/\ell^{\tau_m}$, and the twist climbs monotonically toward the free two-particle value. There is no unitary exchange that pushes the twist above $2\Delta_\phi$ at large spin. The behavior is illustrated in Fig. 51.4.

51.3.5 The universal gravitational case: stress-tensor exchange

The single most important application is to a CFT with a conserved stress tensor — that is, to *any* local CFT. The stress tensor $T_{\mu\nu}$ is a spin-2 operator of protected dimension $\Delta_T = d$ (conservation $\partial^\mu T_{\mu\nu} = 0$ saturates the spin-2 unitarity bound

of Theorem 47.18), hence its twist is

$$\tau_T = \Delta_T - \ell_T = d - 2, \tag{51.75}$$

the same for every local CFT in d dimensions. Whenever $\phi \times \phi$ contains no scalar of twist below $d-2$ — the generic situation, and the universal one for the stress-tensor channel — the minimal twist is $\tau_m = d-2$ and Theorem 51.10 predicts a binding $\gamma_{0,\ell} \sim -1/\ell^{d-2}$ whose coefficient is fixed, through the Ward identity, by the central charge.

Worked Example 51.12 (Universal binding from stress-tensor exchange)**.** We specialize Eq. (51.73) to $\mathcal{O}_m = T_{\mu\nu}$. The data are $\tau_m = d-2$, $\ell_m = 2$, so

$$\bar{h}_m = \frac{\tau_m}{2} + \ell_m = \frac{d-2}{2} + 2 = \frac{d+2}{2}, \qquad 2\bar{h}_m = d+2, \tag{51.76}$$

and the coefficient becomes

$$c_{d-2} = \frac{2\,\Gamma(\Delta_\phi)^2}{\Gamma\!\left(\Delta_\phi - \frac{d-2}{2}\right)^2} \frac{\Gamma(d+2)}{\Gamma\!\left(\frac{d+2}{2}\right)^2} C^2_{\phi\phi T}. \tag{51.77}$$

The OPE coefficient $C_{\phi\phi T}$ is not free: it is fixed by the conformal Ward identity that says the integrated stress tensor measures the dimension of ϕ. In the conventions in which the stress-tensor two-point function is normalized by the central charge C_T (the spin-2 analogue of the current normalization fixed in Section 47.4.3), the Ward identity gives

$$C^2_{\phi\phi T} = \frac{d\,\Delta_\phi^2}{(d-1)^2\, C_T}, \tag{51.78}$$

so the coupling of any operator to the stress tensor is set by its dimension and by the universal C_T, with no other dynamical input. Substituting Eq. (51.78) into Eq. (51.77),

$$\boxed{\gamma_{0,\ell} \;\simeq\; -\frac{1}{\ell^{\,d-2}} \, \frac{2\,d\,\Delta_\phi^2}{(d-1)^2\, C_T} \, \frac{\Gamma(\Delta_\phi)^2}{\Gamma\!\left(\Delta_\phi - \frac{d-2}{2}\right)^2} \, \frac{\Gamma(d+2)}{\Gamma\!\left(\frac{d+2}{2}\right)^2} \, .} \tag{51.79}$$

This is the universal gravitational binding. Every local unitary CFT contains it: the leading-twist tower of any scalar is bound by $-1/\ell^{d-2}$, with strength inversely proportional to the central charge. Three checks confirm the structure. *Sign*: $C_T > 0$ by unitarity (it is the norm of the stress-tensor state), and all other factors are positive or squared, so $\gamma_{0,\ell} < 0$ — gravity binds, as Theorem 51.11 demands. *Power*: the range $d-2$ is the twist of the graviton; in $d=4$ it is $\gamma_{0,\ell} \sim -1/\ell^2$, the long-range Newtonian tail. *Strength*: the $1/C_T$ scaling is the statement that the binding is weak when the theory has many degrees of freedom, i.e. weak Newton constant $G_N \sim 1/C_T$ in the holographic dual — two

particles in a nearly free AdS feel a feeble gravitational pull. In $d = 2$ the formula degenerates ($\tau_m = 0$, the stress tensor is itself near-marginal in twist) and the analysis must be redone with Virasoro symmetry, consistent with the special status of two dimensions flagged in Chapter 48.

The relation Eq. (51.78) is the only external input in the worked example, and it is a kinematic Ward identity, not a dynamical assumption; its derivation belongs to the conformal-symmetry chapter [155, 156] and is the spin-2 companion of the current Ward identity of Section 47.4.3. We have flagged its normalization-dependence honestly: the numerical coefficient in Eq. (51.79) is tied to the convention for C_T, while the power $1/\ell^{d-2}$ and the sign are convention-independent.

51.3.6 Scalar ϕ^4 theory: ϵ-exchange at Wilson–Fisher

We continue ϕ^4 theory from the generalized-free-field baseline of Section 51.2. At the Wilson–Fisher fixed point in $d = 4 - \varepsilon$ the scalar ϕ is no longer free: it acquires the anomalous dimension $\gamma_\phi = O(\varepsilon^2)$ recorded in Theorem 47.18, and the four-point function ceases to be a sum of identity blocks. The $\phi \times \phi$ OPE now contains, besides the stress tensor, a light *scalar* — the operator ϕ^2, which at the fixed point is the relevant deformation (the "ϵ operator" of the Ising universality class). Its dimension is

$$\Delta_{\phi^2} = d - 2 + \gamma_{\phi^2} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2), \tag{51.80}$$

a scalar, so its twist equals its dimension, $\tau_{\phi^2} = \Delta_{\phi^2} \approx 2 - \frac{2\varepsilon}{3}$. By raw twist the stress tensor is lighter, $\tau_T = d - 2 = 2 - \varepsilon < \tau_{\phi^2} = 2 - \frac{2\varepsilon}{3}$, so ϕ^2 is *not* the minimal-twist exchange. Nevertheless it is ϕ^2, not the stress tensor, that controls the large-spin binding of the $[\phi\phi]_{0,\ell}$ tower: the stress tensor sits only a distance $\gamma_\phi = O(\varepsilon^2)$ from the double-twist lattice, so its double-discontinuity weight is $\sin^2(\pi\gamma_\phi) = O(\varepsilon^4)$ and it is suppressed, while ϕ^2 sits a distance $O(\varepsilon)$ off the lattice and contributes at $O(\varepsilon^2)$ (the dDisc mechanism is made precise in Theorem 51.20). Applying Theorem 51.10 with $\tau_m = \tau_{\phi^2}$, $\ell_m = 0$, $\bar{h}_m = \frac{\tau_{\phi^2}}{2} = \frac{1}{2}\Delta_{\phi^2}$,

$$\gamma_{0,\ell} \;\simeq\; -\frac{c_{\phi^2}}{\ell^{\Delta_{\phi^2}}}, \qquad c_{\phi^2} = \frac{2\,\Gamma(\Delta_\phi)^2}{\Gamma(\Delta_\phi - \frac{\Delta_{\phi^2}}{2})^2}\,\frac{\Gamma(\Delta_{\phi^2})}{\Gamma(\frac{\Delta_{\phi^2}}{2})^2}\,C^2_{\phi\phi\phi^2}. \tag{51.81}$$

At leading order in ε the data are the free-theory values $\Delta_\phi = 1 - \frac{\varepsilon}{2}$, $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3}$, and the OPE coefficient $C^2_{\phi\phi\phi^2} = 2 + O(\varepsilon)$ of the free scalar (the Wick combinatorics of $\langle\phi\phi\,\phi^2\rangle$). The structure Eq. (51.81) is exact in the large-spin sense; turning its ε-expansion into numbers, and cross-checking against the direct $4 - \varepsilon$ perturbative computation of $\gamma_{0,\ell}$ and against the $d = 3$ Ising data, is the work of the closing thread section in Chapter 52. The point established here is

structural: at Wilson–Fisher the large-spin spectrum is bound by ϕ^2-exchange with the falloff $1/\ell^{\Delta_{\phi^2}}$, the analytic bootstrap predicting an entire tower of anomalous dimensions from a single OPE coefficient.

51.3.7 Systematics, and why we need more

The computation above extracts the *leading* large-spin behavior from a *single* minimal-twist exchange. One can push it. Subleading powers $1/\ell^{\tau_m+1}, 1/\ell^{\tau_m+2}, \ldots$ come from the descendants of $\mathcal{O}_m$ (the higher n terms of the connection formula Eq. (51.58)) and from double exchanges; corrections to the OPE coefficients $\delta p_{0,\ell}$ come from the non-logarithmic piece of Eq. (51.66); the higher-twist towers $[\phi\phi]_{n,\ell}$ are governed by the subleading powers of $\bar{z}$ in Eq. (51.54). Organizing all of this systematically is the subject of the *twist conformal blocks* and *large-spin perturbation theory* of Alday and collaborators [12, 13]: one defines generating functions that resum, in one stroke, an entire family of $1/\bar{h}$ corrections sharing a common origin, and the crossing equation becomes a set of algebraic relations among them.

Even so, the lightcone method as presented has a built-in ceiling. It matches *powers of* $\bar{z}$ *and* $\ln \bar{z}$ near a corner of cross-ratio space, and each new order requires expanding both sides to one higher power, with the algebra growing rapidly and the bookkeeping of which operators contribute where becoming delicate. More fundamentally, the method produces an *asymptotic* series in $1/\bar{h}$: it is exact at infinite spin and systematically improvable, but it is not a closed formula valid at all spins, and it offers no control over convergence or over the low-spin operators that the numerical bootstrap of Chapter 50 cares about most. What one would like is a single object that yields the OPE data as an honest analytic function of spin, all orders in $1/\bar{h}$ at once, reconstructed from the correlator rather than matched term by term. That object exists: it is the Lorentzian inversion formula of Caron-Huot, which promotes the discrete OPE data to the analytic-in-spin function whose large-$\bar{h}$ expansion reproduces everything in this section, and whose input is a single positive quantity — the double discontinuity of the correlator. We turn to it next.

51.4 The Lorentzian Inversion Formula

The large-spin expansion of the previous section was a triumph of bookkeeping. By matching the t-channel identity against an s-channel tower we learned that every CFT contains families of double-twist operators $[\phi\phi]_{n,\ell}$ whose dimensions approach $2\Delta_\phi + 2n + \ell$, and we computed the leading correction $\gamma_{0,\ell} \sim -c_{\tau_m}/\ell^{\tau_m}$ set by the lowest-twist exchanged operator. But the method was an asymptotic series assembled by hand, one crossing kernel and one power of $1/\ell$ at a time, with no guarantee that the pattern we read off at the first two orders persists,

and no statement at all about *finite* spin. What the lightcone analysis was really probing — that the OPE data of the double-twist families vary *smoothly* with ℓ — cries out for a generating object: a single function of the spin, defined for all ℓ at once, whose asymptotic expansion reproduces the hand computation and whose exact form contains everything the lightcone limit could only see at leading order. That object is the Caron-Huot inversion formula [111], and constructing it is the business of this section. It will turn the OPE data from a discrete table indexed by integer spin into an *analytic function of spin*, reconstructible from a sharply defined piece of the correlator — its double discontinuity.

Throughout we use the cross-ratios of Eq. (47.140), $u = z\bar{z}$ and $v = (1-z)(1-\bar{z})$, the stripped four-point function $g(z,\bar{z})$ of identical scalars ϕ of dimension Δ_ϕ from the block decomposition Eq. (47.227), and the crossing equation Eq. (49.14) in the form $v^{\Delta_\phi} g(u,v) = u^{\Delta_\phi} g(v,u)$. We work in d dimensions; in Lorentzian signature z and $\bar{z}$ are independent real variables, the feature exploited at every step below.

51.4.1 The Euclidean inversion and why spin is not yet analytic

The block decomposition Eq. (47.227) writes g as a discrete sum over the spectrum. There is a standard way to *invert* it — to extract the OPE data as an integral of the correlator against a kernel — which predates the Lorentzian formula and is purely Euclidean. It rests on completeness of the conformal partial waves.

For each spin $\ell \in \mathbb{Z}_{\geq 0}$, consider the dimension Δ on the *principal series* $\Delta = \frac{d}{2} + i\nu$, $\nu \in \mathbb{R}$. The shadow of a primary of dimension Δ is the formal primary of dimension $\tilde{\Delta} = d - \Delta$; the principal series is the locus fixed (up to $\nu \to -\nu$) by shadowing. The *conformal partial wave* is the shadow-symmetric combination of a block and its shadow,

$$\Psi_{\Delta,\ell}(z,\bar{z}) \;=\; \tfrac{1}{2}\Big(K_{\tilde{\Delta},\ell}\, G_{\Delta,\ell}(z,\bar{z}) \;+\; K_{\Delta,\ell}\, G_{\tilde{\Delta},\ell}(z,\bar{z}) \Big), \tag{51.82}$$

where $G_{\Delta,\ell}$ is the conformal block of Theorem 47.49 and $K_{\Delta,\ell}$ is a ratio of Γ-functions (the "shadow coefficient") whose precise value will not be needed. The $\Psi_{\Delta,\ell}$ are the genuine eigenfunctions of the conformal Casimir that are single-valued on the Euclidean section $\bar{z} = z^*$ and normalizable there; on the principal series they form a delta-function–orthonormal basis,

$$\big(\Psi_{\Delta,\ell},\, \Psi_{\Delta',\ell'}\big) \;\propto\; \delta_{\ell\ell'}\,\big[\delta(\nu-\nu') + \delta(\nu+\nu')\big], \tag{51.83}$$

the inner product being the integral over the Euclidean section against the natural conformally invariant measure. Completeness then states that any crossing-symmetric, normalizable g admits the *conformal partial wave expansion*

$$g(z,\bar{z}) \;=\; \sum_{\ell=0}^{\infty} \int_{\frac{d}{2}-i\infty}^{\frac{d}{2}+i\infty} \frac{d\Delta}{2\pi i}\, c(\Delta,\ell)\, \Psi_{\Delta,\ell}(z,\bar{z}), \tag{51.84}$$

with a spectral density $c(\Delta,\ell)$. The connection to physical data is made by closing the Δ-contour: $c(\Delta,\ell)$ is meromorphic in Δ, and each physical primary $\mathcal{O}_k$ of spin ℓ and dimension Δ_k contributes a pole on the right of the contour,

$$c(\Delta,\ell) \;\sim\; -\frac{C^2_{\phi\phi\mathcal{O}_k}}{\Delta-\Delta_k} \qquad (\Delta\to\Delta_k), \tag{51.85}$$

so that picking up these poles when the principal-series contour is deformed to the right turns Eq. (51.84) back into the block sum Eq. (47.227): the pole locations give the dimensions and the residues give the squared OPE coefficients. The function $c(\Delta,\ell)$ is therefore exactly the generating object we want — *if* we can compute it.

Orthogonality Eq. (51.83) hands us a formula for it. Projecting Eq. (51.84) onto $\Psi_{\Delta,\ell}$ gives the *Euclidean inversion*,

$$c(\Delta,\ell) \;=\; N(\Delta,\ell)\int_{\text{Eucl}} d^2z\;\mu_{\text{E}}(z,\bar z)\,\Psi_{\Delta,\ell}(z,\bar z)\,g(z,\bar z), \tag{51.86}$$

the integral running over the Euclidean section $\bar z = z^*$ with the invariant measure μ_{E} and a known normalization $N(\Delta,\ell)$ fixed by Eq. (51.83). This is a genuine inversion: it returns the full OPE data of a given spin as an integral transform of the correlator.

It is also, for the purposes of the bootstrap, almost useless, and it is worth being precise about *why*. The defect is not in Δ — as a function of Δ, Eq. (51.86) is perfectly analytic, with the poles Eq. (51.85) we want. The defect is in ℓ. The formula produces $c(\Delta,\ell)$ *one integer spin at a time*: the partial wave Eq. (51.82) is a specific function for each $\ell\in\mathbb{Z}_{\geq 0}$, and nothing in Eq. (51.86) tells us how to continue the answer to non-integer, let alone complex, ℓ. One might hope to continue $\Psi_{\Delta,\ell}$ in ℓ and continue the integral, but the continuation is ambiguous, and the ambiguity is exactly the obstruction Carlson's theorem identifies. Carlson's theorem says that a function $f(\ell)$ analytic in the right half-plane $\operatorname{Re}\ell>0$ and of exponential type strictly less than π there — that is, bounded by $e^{(\pi-\delta)|\ell|}$ — is uniquely determined by its values at the non-negative integers; without such a boundedness statement, two analytic functions can agree at all integers and differ everywhere else (the standard counterexample being $\sin(\pi\ell)$, which vanishes at every integer). The Euclidean integrand Eq. (51.86) provides no such bound: the partial wave $\Psi_{\Delta,\ell}$, continued in ℓ, grows in the lightcone corners $z\to 0,1$ and $\bar z\to 0,1$ of the integration region fast enough to spoil exponential type, so the integral does not define a function of controlled growth in ℓ. The discrete sum over ℓ in Eq. (51.84) cannot be Sommerfeld–Watson resummed into a contour integral, and "the OPE data as a function of spin" is, from the Euclidean formula alone, not even well posed.

Physical Insight

The Euclidean inversion is the conformal analogue of computing Fourier coefficients $a_n = \int f\, e^{-inx}$: it returns one number per integer mode, with no canonical way to interpolate between modes. To get an *analytic function* of the mode number one needs extra information — here, a growth bound. The Lorentzian formula supplies precisely that bound by re-expressing the same OPE data as an integral whose integrand is manifestly tame in ℓ, at the price of replacing the full correlator by a single, positive piece of it: the double discontinuity.

51.4.2 The double discontinuity

The object that carries the analyticity is a particular discontinuity of the correlator taken in the t-channel cross-ratio, around the point $v = 0$, equivalently $\bar{z} = 1$ at fixed z. In Lorentzian kinematics $\bar{z}$ is an independent real variable and the correlator, continued past $\bar{z} = 1$, develops a branch cut; the double discontinuity measures the obstruction to continuing through it on both sheets at once.

Definition 51.13 (Double discontinuity)**.** Let $g(z, \bar{z})$ be the stripped correlator, holomorphic in $\bar{z}$ in a neighbourhood of the Euclidean section and continued in $\bar{z}$ around the branch point at $\bar{z} = 1$ with z held fixed. Write $g^{\circlearrowleft}$ and $g^{\circlearrowright}$ for the continuations obtained by encircling $\bar{z} = 1$ counterclockwise and clockwise. For identical external scalars (the case $a = b = 0$ in the general notation) the *double discontinuity* is

$$\mathrm{dDisc}\,[g(z,\bar{z})] \;\equiv\; g(z,\bar{z}) \;-\; \tfrac{1}{2}\, g^{\circlearrowleft}(z,\bar{z}) \;-\; \tfrac{1}{2}\, g^{\circlearrowright}(z,\bar{z}). \tag{51.87}$$

For general external dimensions, with $a = -\frac{1}{2}(\Delta_1 - \Delta_2)$ and $b = \frac{1}{2}(\Delta_3 - \Delta_4)$, the combination is weighted by phases, $\mathrm{dDisc}[g] = \cos(\pi(a+b))\, g - \frac{1}{2} e^{i\pi(a+b)} g^{\circlearrowleft} - \frac{1}{2} e^{-i\pi(a+b)} g^{\circlearrowright}$, reducing to Eq. (51.87) when $a = b = 0$.

The name and the physics are clarified by the Lorentzian interpretation. Each ordinary discontinuity $g - g^{\circlearrowleft}$ (or $g - g^{\circlearrowright}$) is, by the same Lorentzian-continuation argument that turns the optical theorem into a cut, the expectation value of a single commutator of the external operators across the corresponding lightcone. Taking the discontinuity twice — once on each sheet — assembles a *double* commutator: arranging the four operators in the Lorentzian configuration in which $\phi(x_1), \phi(x_3)$ are spacelike from the pair $\phi(x_2), \phi(x_4)$ but timelike-related across $\bar{z} = 1$, one finds, up to a positive kinematic prefactor,

$$\mathrm{dDisc}\,[g] \;\propto\; \big\langle\, [\phi(x_4), \phi(x_1)]\, [\phi(x_2), \phi(x_3)] \,\big\rangle. \tag{51.88}$$

A double commutator of this form is, by inserting a complete set of states between the two commutators, a sum of squared matrix elements: it is the norm of a state,

hence *non-negative*. This positivity, $\mathrm{dDisc}[g] \geq 0$, is the Lorentzian shadow of OPE-coefficient positivity, and it is what makes the inversion formula a source of *bounds* as well as of analytic data. We will not need the full operator derivation of Eq. (51.88); what we need is the elementary action of Eq. (51.87) on the power-law and block singularities the correlator actually has, computed next.

Worked Example 51.14 (Double discontinuity of a power)**.** The crossing equation $g(u,v) = (u/v)^{\Delta_\phi} g(v,u)$ shows that the small-v behaviour of g is controlled by the t-channel data: a t-channel primary of twist $\tau' = \Delta' - \ell'$ contributes, through its block $G_{\Delta',\ell'}(v,u) \sim v^{\tau'/2}$ near $v \to 0$, a term $g(u,v) \supset (u/v)^{\Delta_\phi}\, v^{\tau'/2} = u^{\Delta_\phi}\, v^{\tau'/2-\Delta_\phi}$. Strip the analytic factor u^{Δ_ϕ} and isolate the singular power in v. Since $v = (1-z)(1-\bar z)$ and we continue $\bar z$ around 1 at fixed z, the relevant monodromy is that of the factor $(1-\bar z)^p$ with

$$p \;=\; \frac{\tau'}{2} - \Delta_\phi. \tag{51.89}$$

Encircling $\bar z = 1$ once sends $(1-\bar z) \mapsto e^{\mp 2\pi i}(1-\bar z)$, so $(1-\bar z)^p \mapsto e^{\mp 2\pi i p}(1-\bar z)^p$ for the clockwise/counterclockwise loops. Therefore

$$\begin{aligned} \mathrm{dDisc}\left[(1-\bar z)^p\right] &= (1-\bar z)^p \Big[\, 1 - \tfrac{1}{2}e^{2\pi i p} - \tfrac{1}{2}e^{-2\pi i p} \Big] \\ &= (1-\bar z)^p \big[\, 1 - \cos(2\pi p) \,\big] \;=\; 2\sin^2(\pi p)\,(1-\bar z)^p. \end{aligned} \tag{51.90}$$

The double discontinuity acts on a power by multiplication by $2\sin^2(\pi p)$. This single trigonometric factor is the engine of the whole construction.

The consequences of Eq. (51.90) are immediate and far-reaching. Substitute the twist via Eq. (51.89): the dDisc of a t-channel contribution of twist τ' carries the weight

$$2\sin^2\!\Big(\pi\big(\tfrac{\tau'}{2} - \Delta_\phi\big)\Big) \;=\; 2\sin^2\!\Big(\pi\big(\Delta_\phi - \tfrac{\tau'}{2}\big)\Big). \tag{51.91}$$

Now recall the double-twist families. The generalized free field — the Gaussian endpoint of ϕ^4 theory, whose crossing solution was exhibited in Eq. (49.24) — has a spectrum of operators $[\phi\phi]_{n,\ell}$ of twist $\tau' = 2\Delta_\phi + 2n$ for $n = 0, 1, 2, \ldots$, and the same families appear, shifted by anomalous dimensions, in every CFT. For a double-twist operator, $\frac{\tau'}{2} - \Delta_\phi = n \in \mathbb{Z}$, and so

$$\mathrm{dDisc}\left[\text{double-twist contribution}\right] \;\propto\; \sin^2(\pi n) \;=\; 0. \tag{51.92}$$

The double discontinuity annihilates the double-twist operators. This is the structural fact that the whole formalism turns on. The double-twist families are precisely the "trivial" part of the OPE — the part already present in mean-field theory, reconstructed from the identity in the crossed channel — and the dDisc projects them out, leaving only the genuinely dynamical content. When a double-twist operator acquires an anomalous dimension, $\tau' = 2\Delta_\phi + 2n + \gamma_{n,\ell}$,

the weight becomes $2\sin^2(\pi\gamma_{n,\ell}/2) \approx \frac{\pi^2}{2}\gamma_{n,\ell}^2$: double-twist data enter the dDisc only at *second order* in their anomalous dimensions. The inversion formula thus reconstructs the leading double-twist data from the dDisc of the *other*, low-twist exchanged operators — exactly the operators whose exchange the large-spin analysis identified as the source of binding — and is blind, at leading order, to the double-twist operators' own back-reaction. The consistency of this self-referential structure is what makes the analytic bootstrap a closed, calculable system.

Worked Example 51.15 (Double discontinuity of a single t-channel block)**.** A full t-channel conformal block is not a single power but a descendant tower. Near $\bar{z} = 1$ at fixed z the block of an operator $\mathcal{O}'$ of twist τ', multiplied by the crossing factor $(u/v)^{\Delta_\phi}$ as it enters g, organizes into

$$(u/v)^{\Delta_\phi} G_{\Delta',\ell'}(v,u) \;=\; (1-\bar{z})^{\tau'/2-\Delta_\phi} \sum_{m=0}^{\infty} (1-\bar{z})^m \, h_m(z), \tag{51.93}$$

where the descendant sum runs over integer steps m in the twist (the 1/2-integer steps are forbidden for a single primary multiplet) and $h_m(z)$ are functions analytic at $\bar{z} = 1$. Every term in the sum shares the *same* non-integer power $(1-\bar{z})^{\tau'/2-\Delta_\phi}$ up to an integer shift m, and $\sin^2$ is periodic with period one in its argument, so each term picks up the identical monodromy weight $2\sin^2(\pi(\frac{\tau'}{2} - \Delta_\phi))$ under Eq. (51.87). Hence the double discontinuity acts on the whole block by an overall constant,

$$\mathrm{dDisc}\left[(u/v)^{\Delta_\phi} G_{\Delta',\ell'}(v,u)\right] \;=\; 2\sin^2\!\Big(\pi\big(\Delta_\phi - \tfrac{\tau'}{2}\big)\Big)\,(u/v)^{\Delta_\phi} G_{\Delta',\ell'}(v,u). \tag{51.94}$$

The block is an eigenfunction of dDisc with eigenvalue $2\sin^2(\pi(\Delta_\phi - \tau'/2))$. The double-twist zero Eq. (51.92) is recovered term by term, and the suppression of nearby operators by γ^2 follows by expanding the eigenvalue. (When a double-twist operator sits exactly at an integer and carries a logarithm $\sim (1-\bar{z})^n \log(1-\bar{z})$ — as happens once anomalous dimensions and OPE-coefficient corrections mix — the $\sin^2$ multiplying the pure power vanishes but its derivative acting on the log does not, reproducing the same $O(\gamma^2)$ result; we will not need the explicit log accounting here, only that the leading double-twist contribution is doubly suppressed.)

51.4.3 The Lorentzian inversion formula

We can now state the result. The Euclidean inversion Eq. (51.86) is to be re-expressed, by deforming its contour onto the Lorentzian double-lightcone region, as an integral of the double discontinuity against an *inverted block* — the block with dimension and spin interchanged and shifted, which is precisely the kernel that solves the Casimir equation with the right behaviour in the inverted regime.

Theorem 51.16 (Lorentzian inversion formula, Caron-Huot)**.** *Let $g(z,\bar z)$ be the stripped four-point function of identical scalars ϕ of dimension Δ_ϕ in a unitary CFT_d, satisfying crossing Eq. (49.14). The generating function of OPE data, defined by the conformal partial wave expansion Eq. (51.84) with poles Eq. (51.85), is computed by*

$$c(\Delta,\ell) \;=\; c^{\,t}(\Delta,\ell) \;+\; (-1)^{\ell}\, c^{\,u}(\Delta,\ell), \tag{51.95}$$

where the t-channel contribution is the double-lightcone integral

$$\boxed{\,c^{\,t}(\Delta,\ell) \;=\; \kappa_{\Delta+\ell}\int_0^1\!\!\int_0^1 dz\, d\bar z\; \mu(z,\bar z)\; G_{\,\ell+d-1,\,\Delta-d+1}(z,\bar z)\;\mathrm{dDisc}\left[g(z,\bar z)\right]\,,\,} \tag{51.96}$$

with the normalization, measure, and inverted block

$$\kappa_\beta = \frac{\Gamma\!\left(\frac{\beta}{2}\right)^4}{2\pi^2\,\Gamma(\beta-1)\,\Gamma(\beta)}\,, \qquad \beta=\Delta+\ell, \tag{51.97}$$

$$\mu(z,\bar z) = \frac{|z-\bar z|^{\,d-2}}{(z\bar z)^d}\,, \tag{51.98}$$

the kernel $G_{\ell+d-1,\,\Delta-d+1}$ being the ordinary conformal block of Theorem 47.49 evaluated at the interchanged-and-shifted quantum numbers $\Delta\mapsto\ell+d-1$, $\ell\mapsto\Delta-d+1$. The u-channel piece $c^{\,u}$ is given by the same formula with $z\to z/(z-1)$, $\bar z\to\bar z/(\bar z-1)$; for identical scalars $c^{\,u}=c^{\,t}$, so Eq. (51.95) reduces to $c(\Delta,\ell)=(1+(-1)^\ell)c^{\,t}$ and supports only even spin, consistent with the $\mathbb{Z}_2$ selection rule of $\phi\times\phi$.

Three features deserve comment before the derivation, because they are what the formula buys us.

First, it is built entirely from the dDisc. By Eq. (51.92) the integrand sees only the genuinely dynamical, non-double-twist content of the correlator. The double-twist data it produces on the left — the dimensions and OPE coefficients of $[\phi\phi]_{n,\ell}$, sitting at the poles Eq. (51.85) — are reconstructed from the cross-channel dDisc, not from the double-twist operators themselves. This is the lightcone bootstrap of the previous section promoted to an exact identity.

Second, it is manifestly analytic in spin. The right-hand side of Eq. (51.96) depends on ℓ only through the smooth prefactor $\kappa_{\Delta+\ell}$ and through the inverted block $G_{\ell+d-1,\Delta-d+1}$, both of which are analytic functions of ℓ taking finite, controlled values for ℓ in a half-plane. The integral therefore defines $c(\Delta,\ell)$ as an analytic function of ℓ, not merely a table of integers — exactly the boundedness the Euclidean formula lacked. We make the convergence condition precise in Theorem 51.17.

Third, its analytic structure in Δ reads off the spectrum. The $z\to 0$ endpoint of the integral is the s-channel OPE limit. There the inverted block behaves as a definite power of z, and integrating it against the small-z behaviour of dDisc$[g]$

produces poles in Δ from the endpoint, located at the s-channel twists and with residues equal to the squared OPE coefficients Eq. (51.85). The mechanics of extracting those poles — turning Eq. (51.96) into an explicit spectrum — is the calculational engine of Section 51.5; here we have the formula and its guarantees.

Notation Watch

The kernel in Eq. (51.96) is the *same* function $G_{\Delta,\ell}$ defined in Theorem 47.49, simply evaluated at the arguments $(\ell+d-1,\ \Delta-d+1)$. The interchange $\Delta \leftrightarrow \ell$ together with the shift by $d-1$ is not a new object; it is the statement that the Casimir equation is symmetric under $\Delta \to \ell+d-1$, $\ell \to \Delta-d+1$ (which preserves $\beta=\Delta+\ell$), so the inverted block is automatically a solution with the boundary behaviour appropriate to the inversion integral. We will not invent a separate symbol for it.

51.4.4 Derivation by contour deformation

The formula Eq. (51.96) is obtained from the Euclidean inversion Eq. (51.86) by deforming the $\bar z$-contour. We give the argument in full at the level of contour topology, isolating the one analytic input — a growth bound in the Regge regime — that controls the only term that could spoil it.

Begin with Eq. (51.86) and insert the partial wave Eq. (51.82). The two terms (block and shadow) are exchanged by $\Delta \to \tilde\Delta = d-\Delta$, so it suffices to track one of them and shadow-symmetrize at the end; the upshot of the shadow symmetrization is to convert the block kernel into the *inverted* block of Eq. (51.96), with the prefactor $\kappa_{\Delta+\ell}$ collecting the shadow coefficients K and the normalization N. Use the Lorentzian independence of z and $\bar z$ to write the Euclidean section integral as an iterated contour integral, and consider the $\bar z$-integral at fixed z. The integrand is holomorphic in $\bar z$ except for branch cuts inherited from g and from the kernel: a cut running left from $\bar z=0$ (the s-channel OPE singularity) and a cut running right from $\bar z=1$ to $\bar z=+\infty$ (the t- and u-channel singularities).

Now push the $\bar z$-contour from the Euclidean position to the right, toward the cut emanating from $\bar z=1$ (see Fig. 51.5). As it sweeps across the t-channel singularity it wraps the cut, and the wrapped contour collects the discontinuity of the integrand across it. The shadow symmetrization arranges these discontinuities into precisely the combination Eq. (51.87): the two sheets reached by going above and below the cut are the $g^{\circlearrowleft}$ and $g^{\circlearrowright}$ of Theorem 51.13, and the Euclidean value supplies the unsubtracted g. The integrand on the wrapped contour is therefore the inverted block times $\mathrm{dDisc}[g]$, and the $\bar z$-integral runs over the cut, $\bar z\in(1,\infty)$, which after the standard change of variables to the unit interval becomes the $\int_0^1 d\bar z$ of Eq. (51.96); the same deformation in z produces $\int_0^1 dz$. The measure μ Eq. (51.98) and the prefactor $\kappa_{\Delta+\ell}$ Eq. (51.97) are the Jacobian and

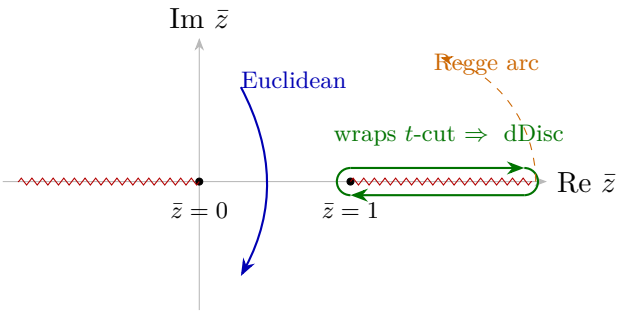


Figure 51.5: Contour deformation in the $\bar{z}$-plane at fixed z. The Euclidean inversion integrates along the blue contour. Deforming it to the right onto the cut from $\bar{z}=1$ (the double-lightcone region) wraps the t-channel branch cut; the difference of the values above and below the cut is exactly the double discontinuity Eq. (51.87). The deformation is legitimate provided the dashed arc at infinity (the Regge regime) is dropped, which requires the Regge boundedness Eq. (51.99).

the accumulated normalization. This is the formula Eq. (51.96).

Two boundary contributions are generated by the deformation and must be shown to vanish, or the equality fails.

The arc at $\bar{z}=1$ (the double-lightcone corner). As the contour is pulled to the right it must also be closed off near the branch point $\bar{z}=1$. The contribution of a small arc encircling $\bar{z}=1$ is governed by the power $(1-\bar{z})^{\tau'/2-\Delta_\phi}$ of the leading t-channel singularity Eq. (51.93); for any operator above the unitarity bound this power is >-1 for the lowest twists of interest, so the arc integral scales as a positive power of its radius and vanishes as the radius shrinks. The lowest-twist t-channel operator is the identity, whose contribution is the pure power $v^{-\Delta_\phi}$, i.e. $p=-\Delta_\phi$; its dDisc by Eq. (51.90) carries $2\sin^2(\pi\Delta_\phi)$, finite, and the small-arc integral is again controlled. The corner is integrable and contributes nothing.

The arc at infinity (the Regge regime). This is the term that requires real input. Closing the $\bar{z}$-contour to the right requires the dashed arc of Fig. 51.5 at $|\bar{z}|\to\infty$ to give nothing. The double scaling $z\to 0$, $\bar{z}\to\infty$ with $z\bar{z}$ fixed is the *Regge limit* of the correlator (it is the conformal image of the high-energy, fixed-impact-parameter regime). Parametrizing the approach by a boost parameter, the question is the rate of growth of g there. Define the *Regge intercept* J_0 by the leading growth $g\sim(\text{boost})^{J_0-1}$ in the Regge limit. The arc at infinity is suppressed by the inverted block, which decays like $(\text{boost})^{-(\ell-1)}$, so the arc integrand scales as $(\text{boost})^{J_0-\ell}$ and the arc vanishes provided

$$\ell \;>\; J_0. \tag{51.99}$$

For a *unitary* CFT the correlator is Regge-bounded with intercept $J_0\le 1$: the Regge growth of a unitary four-point function cannot exceed that of a spin-one

exchange, a statement that follows from the analyticity and boundedness of the Lorentzian correlator in the Regge regime (the conformal counterpart of the Froissart–Martin bound; in the language of the chaos bound it is the statement that the Lyapunov-like growth rate is capped). We take $J_0 \leq 1$ as the established structural input and refer to the literature for its sharp form [111, 252, 333]; the result we use is Eq. (51.99) with $J_0 = 1$.

Combining the two boundary analyses: the deformation is exact and the Regge arc vanishes for $\ell > 1$. This is the content of the central claim.

Theorem 51.17 (Analyticity of OPE data in spin)**.** *For a unitary CFT_d with Regge intercept $J_0 \leq 1$, the inversion integral Eq. (51.96) converges and defines $c(\Delta, \ell)$ as an analytic function of the spin ℓ in the half-plane* $\operatorname{Re} \ell > 1$*. Its poles in Δ at $\Delta = \Delta_k(\ell)$ locate the dimensions of the exchanged operators of spin ℓ, and the residues give their squared OPE coefficients via Eq. (51.85). Consequently the OPE data $\{\Delta_k(\ell), C_k^2(\ell)\}$ are the restrictions to integer ℓ of analytic functions of spin: every CFT organizes its spectrum, for $\ell > 1$, into a finite number of analytic* Regge trajectories*.*

Proof sketch. Convergence of Eq. (51.96) for $\operatorname{Re} \ell > 1$ is the statement that the four corners of the $(z, \bar{z})$ integration region are integrable. The corners at $z, \bar{z} \to 0$ are the s-channel OPE region; there the inverted block decays and the integral is convergent for $\operatorname{Re} \Delta$ on the principal series (and the poles Eq. (51.85) arise by analytic continuation in Δ off it). The corners at $z, \bar{z} \to 1$ are controlled by $\mathrm{dDisc}[g]$, which by the double-twist zero Eq. (51.92) is softer near $\bar{z} = 1$ than g itself — the dDisc has removed precisely the contributions that would diverge. The remaining mixed corner $z \to 0$, $\bar{z} \to 1$ is the Regge region, integrable exactly when Eq. (51.99) holds, i.e. $\operatorname{Re} \ell > J_0$, and $J_0 \leq 1$ by unitarity. Thus the integral converges and is analytic in ℓ for $\operatorname{Re} \ell > 1$. Carlson's theorem now upgrades "analytic and bounded" to "unique": the Regge bound supplies exactly the exponential-type control that the Euclidean formula Eq. (51.86) lacked, so $c(\Delta, \ell)$ is the *unique* analytic-in-ℓ interpolation of the integer-spin OPE data, and any two correlators agreeing at integer spin and Regge-bounded agree as analytic functions of spin. The poles in Δ and their residues are identified with the OPE data exactly as in the Euclidean inversion, since the two formulas compute the same $c(\Delta, \ell)$ on the principal series and a function analytic in a region is determined by its values there. Finally, that the analytic structure is a finite union of trajectories follows because, at fixed $\operatorname{Re} \ell > 1$, $\mathrm{dDisc}[g]$ has a convergent t-channel expansion and each t-channel block contributes a meromorphic-in-Δ piece with poles organized along trajectories. □

The bound $\ell > 1$ is sharp and physically meaningful: it is the reason the formula always controls the stress tensor ($\ell = 2$) and all higher-spin currents, while spin-zero and spin-one data may require additional input beyond the naive inversion (low-spin operators can sit on the boundary of convergence, and isolated

scalars are not always captured by the trajectory continued down from large spin). This is not a defect to be hidden: it is the honest statement of where the analytic-in-spin organization holds. As a synthesis of the mid-2010s the inversion formula is on firm footing for $\ell > 1$ in any unitary CFT; the behaviour at and below $\ell = 1$, the precise non-perturbative meaning of the Regge intercept in a general theory, and the convergence of the trajectory-by-trajectory reconstruction remain subjects of active refinement, flagged again where the next section pushes the formula to its limits.

With Theorems 51.16 and 51.17 in hand we possess exactly the all-orders generator the hand-built large-spin expansion was missing: a single analytic function $c(\Delta, \ell)$, computed from the cross-channel double discontinuity, whose poles and residues are the spectrum and the OPE coefficients as analytic functions of spin. The next section turns this from a structural theorem into a machine — feeding the dDisc of a known cross-channel contribution into Eq. (51.96), reading the resulting poles, and recovering not only the leading large-spin result $\gamma_{0,\ell} \sim -c_{\tau_m}/\ell^{\tau_m}$ of the previous section but its all-orders completion and the data at finite spin.

51.5 Reading the Spectrum from the Inversion Formula

The previous section stated the Lorentzian inversion formula of Theorem 51.16 and introduced its only nontrivial input, the double discontinuity. That was an existence statement: it promoted the operator product data from a discrete list, defined only at integer spins, to an analytic function of spin, and it identified the object the function is computed from. It remains to make the formula *work* — to feed it a correlator and read a spectrum out the other end. That is the business of this section. We use the inversion formula as a calculational engine and extract, with no computer and in closed form, the large-spin data that the lightcone bootstrap of Sections 51.1 to 51.3 obtained by matching singularities. The two methods must agree, and seeing them agree — the same anomalous dimension $\gamma_{0,\ell} \sim -c_0/\ell^{\tau_m}$ emerging once from a singularity match and once from a contour integral of a double discontinuity — is the strongest internal check the analytic bootstrap affords. Along the way we make precise the feature that makes the whole construction self-consistent: the double-twist operators that populate the answer are exactly the operators whose own double discontinuity vanishes, so they are reconstructed *entirely* from the crossed channel rather than from themselves. We then push the engine to subleading orders and to mixed double-twist families $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$, advance ϕ^4 theory by computing the Wilson–Fisher leading-twist anomalous dimension in the ε-expansion and checking it against the data of Section 47.8.4, and close by delimiting honestly where the formula stops controlling the physics — the Regge limit, the boundary of analyticity in

spin.

51.5.1 The inversion formula as a generating function for OPE data

We work throughout with the identical-scalar correlator of ϕ^4 theory, $\langle\phi\phi\phi\phi\rangle$, stripped to the cross-ratio function $\mathcal{G}(z,\bar z)$ of Eq. (47.140) with $u = z\bar z$, $v = (1-z)(1-\bar z)$; the mixed case $\langle\mathcal{O}_1\mathcal{O}_2\mathcal{O}_2\mathcal{O}_1\rangle$ of Section 51.2 is taken up in Section 51.5.6. The s-channel ($z,\bar z \to 0$) OPE organizes $\mathcal{G}$ into a sum of conformal blocks (Theorem 47.49),

$$\mathcal{G}(z,\bar z) = \sum_{\mathcal{O}} c_{\Delta,\ell}\, G_{\Delta,\ell}(z,\bar z), \tag{51.100}$$

with $c_{\Delta,\ell} = f^2_{\phi\phi\mathcal{O}} \geq 0$ the squared OPE coefficients. The discrete label (Δ,ℓ) runs over the primaries of the theory. The inversion formula of Theorem 51.16 replaces this discrete sum by a single meromorphic function of two continuous variables. We write the generating function as $c(\Delta, J)$, an analytic function of the dimension Δ at each spin J, whose poles in Δ sit at the physical dimensions and whose residues are the squared OPE coefficients:

$$c(\Delta,J) \;\sim\; -\frac{c_{\Delta_k(J),J}}{\Delta - \Delta_k(J)} \qquad \text{near each physical pole } \Delta = \Delta_k(J). \tag{51.101}$$

The content of Theorem 51.16 is that $c(\Delta,J)$ is reconstructed from the double discontinuity of $\mathcal{G}$ by a positive double integral over the Lorentzian "causal diamond" $0 \leq z,\bar z \leq 1$, with a kernel built from the same conformal blocks. The discreteness of the spectrum — the fact that operators occur at isolated dimensions — is, in this language, the statement that the integral produces *poles* in Δ; the analyticity in spin is the statement that $c(\Delta,J)$ is a single analytic function of J, so that the data of operators at different spins lie on smooth *Regge trajectories* $\Delta_k(J)$ rather than being independent.

For the large-spin physics we need only the leading-twist projection of this machinery, and there it collapses to a one-variable transform that we can evaluate completely. The mechanism is the one already used in Section 51.2: in the lightcone limit $z \to 0$, a block of twist τ contributes $z^{\tau/2}$ times the collinear ($SL(2,\mathbb{R})$) block $k_\beta(\bar z)$ of Eq. (49.132),

$$k_\beta(\bar z) = \bar z^{\,\beta/2}\, {}_2F_1\!\left(\tfrac{\beta}{2},\tfrac{\beta}{2};\beta;\bar z\right), \qquad \beta = 2\bar h, \quad \bar h \equiv \frac{\Delta+\ell}{2}, \tag{51.102}$$

where the natural variable is the *conformal spin* $\bar h$ (we reserve ℓ for the ordinary spin; the two agree to leading order at large spin, $\bar h = \Delta_\phi + \ell + O(\gamma)$, on the leading-twist trajectory). Projected onto the leading power z^{Δ_ϕ}, the inversion

formula reduces to the one-dimensional $SL(2)$ inversion: the leading-twist OPE data are encoded in a function $C(\bar{h})$ given by

$$C(\bar{h}) = \kappa_{2\bar{h}} \int_0^1 \frac{d\bar{z}}{\bar{z}^2}\, k_{2\bar{h}}(\bar{z})\, \mathrm{dDisc}[\mathcal{G}_0(\bar{z})], \qquad \kappa_{2\bar{h}} = \frac{\Gamma(\bar{h})^4}{2\pi^2\,\Gamma(2\bar{h})\,\Gamma(2\bar{h}-1)}, \tag{51.103}$$

where $\mathcal{G}_0(\bar{z})$ is the coefficient of z^{Δ_ϕ} in $\mathcal{G}(z,\bar{z})$ as $z \to 0$, and the poles of $C(\bar{h})$ in $\bar{h}$ locate the leading-twist operators with residues equal to (minus) their squared OPE coefficients, exactly as in Eq. (51.101). The constant $\kappa_{2\bar{h}}$ is the same one that makes the formula invert the collinear blocks against themselves; we will fix all normalizations below by demanding that Eq. (51.103) reproduce the mean-field data of Theorem 51.7, so no convention need be memorized.

51.5.2 Why double-twist operators are reconstructed from the crossed channel

Everything turns on the double discontinuity, so we record its one essential property. The double discontinuity is the obstruction to continuing $\mathcal{G}$ smoothly through the crossed-channel lightcone $\bar{z} = 1$; for identical external scalars (Theorem 51.13) it is

$$\mathrm{dDisc}\big[\mathcal{G}(\bar{z})\big] = \mathcal{G}(\bar{z}) - \tfrac{1}{2}\,\mathcal{G}^{\circlearrowleft}(\bar{z}) - \tfrac{1}{2}\,\mathcal{G}^{\circlearrowright}(\bar{z}), \tag{51.104}$$

where $\mathcal{G}^{\circlearrowleft,\circlearrowright}$ denote the analytic continuations of $\mathcal{G}$ obtained by taking $\bar{z}$ once around the branch point at $\bar{z} = 1$ in either direction. Its action on a pure power is the only evaluation we need, and it is immediate from Eq. (51.104).

Lemma 51.18 (Double discontinuity of a power)**.** *For any real exponent a,*

$$\mathrm{dDisc}\big[(1-\bar{z})^a\big] = 2\sin^2(\pi a)\,(1-\bar{z})^a. \tag{51.105}$$

In particular the double discontinuity annihilates *every power with $a \in \mathbb{Z}_{\geq 0}$, and more generally every function analytic at $\bar{z} = 1$.*

Proof. Continuing $\bar{z}$ once around $\bar{z} = 1$ sends $1-\bar{z} \to e^{\pm 2\pi i}(1-\bar{z})$, hence $(1-\bar{z})^a \to e^{\pm 2\pi i a}(1-\bar{z})^a$. Substituting into Eq. (51.104),

$$\mathrm{dDisc}\big[(1-\bar{z})^a\big] = \Big(1-\tfrac{1}{2}e^{2\pi i a}-\tfrac{1}{2}e^{-2\pi i a}\Big)(1-\bar{z})^a = (1-\cos 2\pi a)(1-\bar{z})^a = 2\sin^2(\pi a)\,(1-\bar{z})^a. \tag{51.106}$$

When a is a non-negative integer the prefactor $\sin^2(\pi a)$ vanishes; a function analytic at $\bar{z} = 1$ is a convergent sum of such powers, so its double discontinuity vanishes term by term. □

This elementary fact carries the entire self-consistency of the analytic bootstrap. Consider what populates the right-hand side of the inversion integral Eq. (51.103). By crossing (Eq. (49.14)),

$$\mathcal{G}(z,\bar{z}) = \Big(\frac{z\bar{z}}{(1-z)(1-\bar{z})}\Big)^{\Delta_\phi}\,\mathcal{G}(1-z,1-\bar{z}), \tag{51.107}$$

so the behaviour of $\mathcal{G}$ near $\bar{z}=1$ — the only region the double discontinuity probes — is governed by the *crossed-channel* (t-channel) OPE, the expansion of $\mathcal{G}(1-z,1-\bar{z})$ in t-channel blocks. A t-channel operator of twist τ' contributes, through the crossing kernel Eq. (51.107), a term behaving as $(1-\bar{z})^{\tau'/2-\Delta_\phi}$ as $\bar{z}\to 1$. By Theorem 51.18 its double discontinuity is proportional to $\sin^2\!\big(\pi(\Delta_\phi-\tau'/2)\big)$.

Now ask which operators have *vanishing* double discontinuity. The double-twist operators $[\phi\phi]_{n,\ell}$ of Theorem 51.5 sit, in the generalized free field, at twists $\tau'=2\Delta_\phi+2n$ with $n\in\mathbb{Z}_{\geq 0}$, so $\Delta_\phi-\tau'/2=-n\in\mathbb{Z}_{\leq 0}$, and $\sin^2(\pi n)=0$. *The double-twist operators have vanishing double discontinuity.* This is the precise sense in which they are invisible to the inversion integral as inputs: they contribute to $\mathrm{dDisc}[\mathcal{G}]$ only through their anomalous dimensions, which displace them off the integer lattice by $O(\gamma)$ and so give a doubly-suppressed $\sin^2(\pi(n-\gamma/2))=\sin^2(\pi\gamma/2)\approx(\pi\gamma/2)^2$. The operators that the inversion formula *produces* on its left-hand side — the s-channel double-twist data, the residues of $C(\bar{h})$ — are therefore reconstructed not from themselves but entirely from the operators that do have a double discontinuity: the genuinely interacting, non-double-twist operators exchanged in the crossed channel, beginning with the identity. This is the inversion formula's internal consistency made manifest. The output spectrum is the double-twist tower; the input is everything that is *not* a double-twist operator. A free or mean-field theory, whose only operators above the identity *are* double-twist (Section 51.2), has no double discontinuity beyond the identity's, and the inversion integral then returns exactly the mean-field data and nothing more — which is why mean-field theory is the universal large-spin limit Eq. (51.49) rather than a special case.

Physical Insight

The double discontinuity is a filter tuned precisely to suppress the double-twist tower. Physically, dDisc measures the genuine absorptive part of the correlator in the crossed channel — the part associated with real particle production rather than with the kinematic singularities of two free constituents flying apart. Double-twist operators are those free constituents; they carry no absorption, hence no double discontinuity. What is left is the interaction: the exchange of an actual operator between the two halves of the correlator. The inversion formula thus reads the bound-state spectrum (the double-twist data on the left) directly off the forces between the constituents (the exchanged operators on the right). In a holographic

theory this is the statement that the binding energies of two-particle states in anti–de Sitter space are computed by the bulk exchange diagrams of the fields dual to the crossed-channel operators, gravity included.

51.5.3 Inverting a single cross-channel block

We now carry out the central computation: feed the inversion engine the identity plus a single crossed-channel operator $\mathcal{O}_m$, and read off the leading-twist anomalous dimensions and OPE coefficients in closed form. We do this by the lightcone projection of Eq. (51.103), evaluating every integral explicitly. The result reproduces, from a single transform, the $1/\ell^{\tau_m}$ law derived by singularity matching in Section 51.3.

The two contributions and the $\log z$ that carries the anomalous dimension. On the s-channel side, the leading-twist family $[\phi\phi]_{0,\ell}$ has dimensions $\Delta = 2\Delta_\phi + \ell + \gamma_{0,\ell}$, so its contribution to $\mathcal{G}$ carries the power $z^{\Delta_\phi+\gamma_{0,\ell}/2} = z^{\Delta_\phi}(1+\frac{1}{2}\gamma_{0,\ell}\log z + \cdots)$. Collecting the coefficient of z^{Δ_ϕ} separates cleanly into a piece with no $\log z$ and a piece linear in $\log z$:

$$\mathcal{G}(z,\bar z)\Big|_{z^{\Delta_\phi}} = \underbrace{\sum_\ell c_{0,\ell}\, k_{2\bar h_\ell}(\bar z)}_{\text{no } \log z} \;+\; \log z \;\underbrace{\tfrac{1}{2}\sum_\ell c^{(0)}_{0,\ell}\,\gamma_{0,\ell}\, k_{2\bar h_\ell}(\bar z)}_{\text{the } \log z \text{ piece}} \;+\cdots, \tag{51.108}$$

where $c^{(0)}_{0,\ell}$ are the mean-field coefficients Eq. (51.38) and $c_{0,\ell} = c^{(0)}_{0,\ell} + \delta c_{0,\ell}$ the corrected ones. The presence of a $\log z$ is the smoking gun of an anomalous dimension, and it is the part the crossed channel must supply.

On the t-channel side, set $z \to 0$ in Eq. (51.107). The crossing kernel gives $z^{\Delta_\phi}(\bar z/(1-\bar z))^{\Delta_\phi}$ at leading order in z, while $\mathcal{G}(1-z,1-\bar z)$ must be evaluated with its first argument $1-z\to 1$ and its second $1-\bar z\to 0$ (since $\bar z\to 1$). The second-argument limit is the t-channel collinear limit, in which a t-channel operator of twist τ_m, spin ℓ_m, and squared OPE coefficient $P_m \equiv f^2_{\phi\phi\mathcal{O}_m}$ contributes $(1-\bar z)^{\tau_m/2}\,k_{\beta_m}(1-z)$ with $\beta_m = \tau_m + 2\ell_m$. The first-argument limit $1-z\to 1$, i.e. $z\to 0$, is exactly the point at which the collinear block k_{β_m} develops its logarithmic singularity. From the hypergeometric connection formula ${}_2F_1(a,b;a+b;w) \to -\frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)}\big[\log(1-w)+\psi(a)+\psi(b)+2\gamma_E\big]$ as $w\to 1$, applied to Eq. (51.102) with $a=b=\beta_m/2$ and $w=1-z$,

$$k_{\beta_m}(1-z) \;\xrightarrow{z\to 0}\; -\frac{\Gamma(\beta_m)}{\Gamma(\beta_m/2)^2}\Big[\log z + 2\psi\big(\tfrac{\beta_m}{2}\big) + 2\gamma_E\Big]. \tag{51.109}$$

This is where the t-channel produces a $\log z$. Assembling the coefficient of z^{Δ_ϕ}

from the crossed channel,

$$\mathcal{G}(z,\bar z)\Big|_{z^{\Delta_\phi}}^{t\text{-channel }\mathcal{O}_m} = -P_m \frac{\Gamma(\beta_m)}{\Gamma(\beta_m/2)^2}\,\bar z^{\Delta_\phi}(1-\bar z)^{\tau_m/2-\Delta_\phi}\Big[\log z + 2\psi(\tfrac{\beta_m}{2}) + 2\gamma_E\Big]. \tag{51.110}$$

Matching the $\log z$ coefficients of Eq. (51.108) and Eq. (51.110) gives the master equation for the anomalous dimensions,

$$\tfrac{1}{2}\sum_\ell c_{0,\ell}^{(0)}\,\gamma_{0,\ell}\,k_{2\bar h_\ell}(\bar z) = -P_m\,\frac{\Gamma(\beta_m)}{\Gamma(\beta_m/2)^2}\,\bar z^{\Delta_\phi}(1-\bar z)^{\tau_m/2-\Delta_\phi}. \tag{51.111}$$

The left side is a sum over the (discrete, even) leading-twist spins; the right side is a fixed power, more singular at $\bar z\to 1$ the smaller τ_m is. It is the same equation one would obtain by writing the right-hand side as dDisc divided by the kinematic factors and inverting; the $\sin^2$ factor of Eq. (51.90) is hidden inside the ratio of Gamma functions we are about to produce, as we verify at the end. We solve Eq. (51.111) by converting the discrete sum to an integral over conformal spin and using the large-$\bar h$ asymptotics of the collinear block.

The Bessel representation and the one integral. At large $\bar h$ and $\bar z\to 1$ with $\bar h^2(1-\bar z)$ held fixed, the collinear block degenerates to a Bessel function. Using the duplication formula $\Gamma(2\bar h) = \frac{4^{\bar h}}{2\sqrt{\pi}}\Gamma(\bar h)\Gamma(\bar h+\frac12)$ and $\Gamma(\bar h+\frac12)/\Gamma(\bar h)\sim \bar h^{1/2}$,

$$k_{2\bar h}(\bar z) \;\approx\; \frac{\Gamma(2\bar h)}{\Gamma(\bar h)^2}\,2K_0\big(2\bar h\sqrt{1-\bar z}\big) \;\approx\; \frac{4^{\bar h}}{\sqrt{\pi}}\,\bar h^{1/2}\,K_0\big(2\bar h\sqrt{1-\bar z}\big), \tag{51.112}$$

with K_0 the modified Bessel function. The single integral we need is the Mellin transform of K_0,

$$\int_0^\infty d\bar h\;\bar h^{\,2\alpha-1}\,K_0(2\bar h\,s) = \tfrac14\,\Gamma(\alpha)^2\,s^{-2\alpha}, \tag{51.113}$$

valid for $\mathrm{Re}\,\alpha>0$, obtained from $\int_0^\infty x^{\mu-1}K_0(x)\,dx = 2^{\mu-2}\Gamma(\mu/2)^2$ with $x=2\bar h s$ and $\mu=2\alpha$. Everything reduces to Eq. (51.113).

Begin by fixing the measure on the identity. The mean-field coefficients Eq. (51.48) grow as $c_{0,\ell}^{(0)}\sim A\,\bar h^{\,2\Delta_\phi-3/2}\,4^{-\bar h}$ at large $\bar h$, with A a constant we now pin down by self-consistency. Replacing the even-spin sum by a continuous integral over $\bar h$ with this density and inserting Eq. (51.112), the mean-field side of Eq. (51.108) (the "no $\log z$" piece at zeroth order, which must reproduce the t-channel identity) becomes

$$\sum_\ell c_{0,\ell}^{(0)}\,k_{2\bar h_\ell}(\bar z) \;\to\; \int_0^\infty d\bar h\;A\,\bar h^{\,2\Delta_\phi-3/2}4^{-\bar h}.\frac{4^{\bar h}}{\sqrt{\pi}}\bar h^{1/2}K_0\big(2\bar h\sqrt{1-\bar z}\big) = \frac{A}{\sqrt{\pi}}\int_0^\infty d\bar h\,\bar h^{\,2\Delta_\phi-1}K_0(2\bar h \tag{51.114}$$

The exponential factors $4^{-\bar{h}}$ and $4^{\bar{h}}$ cancel: this is the precise statement that the $4^{-\bar{h}}$ in the mean-field coefficients is not physical decay but the inverse of the block's growth, as anticipated below Eq. (51.48). Applying Eq. (51.113) with $2\alpha - 1 = 2\Delta_\phi - 1$, i.e. $\alpha = \Delta_\phi$, and $s = \sqrt{1-\bar{z}}$,

$$\sum_\ell c^{(0)}_{0,\ell}\, k_{2\bar{h}_\ell}(\bar{z}) \;\to\; \frac{A}{\sqrt{\pi}} \cdot \frac{1}{4}\,\Gamma(\Delta_\phi)^2\,(1-\bar{z})^{-\Delta_\phi}. \tag{51.115}$$

The t-channel identity contributes $\big(\bar{z}/(1-\bar{z})\big)^{\Delta_\phi} \to (1-\bar{z})^{-\Delta_\phi}$ as $\bar{z}\to 1$. Equating fixes

$$\frac{A\,\Gamma(\Delta_\phi)^2}{4\sqrt{\pi}} = 1 \qquad\Longrightarrow\qquad A = \frac{4\sqrt{\pi}}{\Gamma(\Delta_\phi)^2}. \tag{51.116}$$

This both reproduces the mean-field data (the residues of the resulting $C(\bar{h})$ are the coefficients Eq. (51.38)) and calibrates the continuous-spin measure so that no overall constant is left undetermined.

The anomalous dimension. Now solve Eq. (51.111). Posit a power law $\gamma_{0,\ell} = -c_0/\bar{h}^{\tau_m}$ and convert its left side to an integral exactly as above:

$$\begin{aligned}
\tfrac{1}{2}\sum_\ell c^{(0)}_{0,\ell}\gamma_{0,\ell}\,k_{2\bar{h}_\ell}(\bar{z}) \to\; & \tfrac{1}{2}\int_0^\infty d\bar{h}\,A\,\bar{h}^{\,2\Delta_\phi-3/2}4^{-\bar{h}}\Big(-\frac{c_0}{\bar{h}^{\tau_m}}\Big)\frac{4^{\bar{h}}}{\sqrt{\pi}}\bar{h}^{1/2}K_0(2\bar{h}\sqrt{1-\bar{z}}) \\
=\; & -\frac{A\,c_0}{2\sqrt{\pi}}\int_0^\infty d\bar{h}\,\bar{h}^{\,2\Delta_\phi-\tau_m-1}K_0(2\bar{h}\sqrt{1-\bar{z}}) \\
=\; & -\frac{A\,c_0}{2\sqrt{\pi}}\cdot\tfrac{1}{4}\,\Gamma\Big(\Delta_\phi-\tfrac{\tau_m}{2}\Big)^2(1-\bar{z})^{\tau_m/2-\Delta_\phi},
\end{aligned} \tag{51.117}$$

using Eq. (51.113) with $\alpha = \Delta_\phi - \tau_m/2$. The $\bar{z}$-dependence $(1-\bar{z})^{\tau_m/2-\Delta_\phi}$ matches the right side of Eq. (51.111) identically — which is what justifies the power-law ansatz: only $\gamma_{0,\ell}\propto \bar{h}^{-\tau_m}$ produces the correct crossed power. Equating the coefficients,

$$\frac{A\,c_0}{8\sqrt{\pi}}\,\Gamma\Big(\Delta_\phi-\tfrac{\tau_m}{2}\Big)^2 = P_m\,\frac{\Gamma(\beta_m)}{\Gamma(\beta_m/2)^2}, \tag{51.118}$$

and substituting A from Eq. (51.116), so that $8\sqrt{\pi}/A = 2\Gamma(\Delta_\phi)^2$, gives the closed form

$$\boxed{\gamma_{0,\ell} = -\frac{c_0}{\bar{h}^{\,\tau_m}}, \qquad c_0 = \frac{2\,\Gamma(\Delta_\phi)^2\,\Gamma(\tau_m+2\ell_m)}{\Gamma\big(\Delta_\phi-\frac{\tau_m}{2}\big)^2\,\Gamma\big(\frac{\tau_m}{2}+\ell_m\big)^2}\,P_m\,.} \tag{51.119}$$

This is exactly the master formula Eq. (51.73) obtained in Section 51.3 by matching lightcone singularities, now produced in a single pass by inverting one crossed-channel block. Three features deserve emphasis. First, the anomalous dimension is *negative* for the identity-adjacent operators (the binding is attractive)

whenever $P_m > 0$ and $\Gamma(\Delta_\phi - \tau_m/2)^2 > 0$, as it is for the leading non-double-twist exchange. Second, it falls as $1/\bar{h}^{\tau_m}$, set by the *minimal* twist exchanged: the lightest crossed operator dominates the large-spin tail, exactly the "binding energy of two far-separated constituents" picture of Section 51.2. Third, the factor $1/\Gamma(\Delta_\phi - \tau_m/2)^2$ is the promised reappearance of the double discontinuity: by the reflection formula $1/\Gamma(x)^2 = \sin^2(\pi x)\,\Gamma(1-x)^2/\pi^2$, the coefficient c_0 carries precisely the factor $\sin^2(\pi(\Delta_\phi - \tau_m/2))$ that Theorem 51.18 assigns to the crossed power $(1-\bar{z})^{\tau_m/2-\Delta_\phi}$. A double-twist operator in the crossed channel, with $\Delta_\phi - \tau_m/2 \in \mathbb{Z}_{\leq 0}$, gives $1/\Gamma$ of a non-positive integer $= 0$ and contributes nothing — the self-consistency of Section 51.5.2, now visible inside the formula.

51.5.4 The correction to the OPE coefficient

The same single block fixes the correction $\delta c_{0,\ell}$ to the mean-field OPE coefficients, with no further integral, by the analytic structure of the generating function. The argument is purely a statement about how a pole moves. On the leading-twist trajectory the generating function Eq. (51.101) has, near each mean-field location $\bar{h}_0 = \Delta_\phi + \ell$, a simple pole whose position and residue are the physical (corrected) data:

$$C(\bar{h}) \;\sim\; \frac{-\,c_{0,\ell}}{\bar{h}-\bar{h}_*}, \qquad \bar{h}_* = \bar{h}_0 + \tfrac{1}{2}\gamma_{0,\ell}, \quad c_{0,\ell} = c^{(0)}_{0,\ell} + \delta c_{0,\ell}. \tag{51.120}$$

Treating $\gamma_{0,\ell}$ and $\delta c_{0,\ell}$ as first-order corrections and expanding the displaced pole about the mean-field location,

$$\frac{-\,(c^{(0)}_{0,\ell} + \delta c_{0,\ell})}{\bar{h}-\bar{h}_0-\frac{1}{2}\gamma_{0,\ell}} = \frac{-\,c^{(0)}_{0,\ell}}{\bar{h}-\bar{h}_0} \underbrace{-\,\frac{\delta c_{0,\ell}}{\bar{h}-\bar{h}_0}}_{\text{first-order simple pole}} \underbrace{-\,\frac{\frac{1}{2}c^{(0)}_{0,\ell}\gamma_{0,\ell}}{(\bar{h}-\bar{h}_0)^2}}_{\text{first-order double pole}} + O(\text{second order}). \tag{51.121}$$

The first-order correction to $C(\bar{h})$ produced by inverting the single t-channel block is therefore a function $C^{(1)}(\bar{h})$ with both a double pole and a simple pole at each $\bar{h}_0$. But these are not independent: $C^{(1)}$ is one analytic function, and a double pole with an analytic numerator $b(\bar{h})$ generates the accompanying simple pole automatically through its Laurent expansion,

$$\frac{b(\bar{h})}{(\bar{h}-\bar{h}_0)^2} = \frac{b(\bar{h}_0)}{(\bar{h}-\bar{h}_0)^2} + \frac{b'(\bar{h}_0)}{\bar{h}-\bar{h}_0} + \cdots. \tag{51.122}$$

Identifying $b(\bar{h}) = -\frac{1}{2}c^{(0)}_0(\bar{h})\,\gamma_0(\bar{h})$ as the smooth conformal-spin continuation of the double-pole coefficient in Eq. (51.121), and matching the simple poles, gives the OPE-coefficient correction with no new computation:

$$\boxed{\delta c_{0,\ell} = \frac{1}{2}\,\frac{d}{d\bar{h}}\Big[c^{(0)}_{0,\ell}(\bar{h})\,\gamma_{0,\ell}(\bar{h})\Big]\Big|_{\bar{h}=\Delta_\phi+\ell}}. \tag{51.123}$$

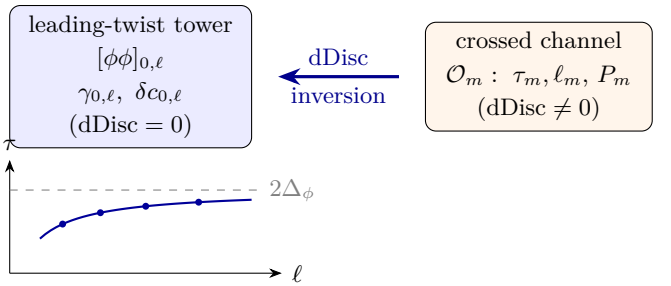


Figure 51.6: The inversion formula as a one-way map. The double discontinuity vanishes on the double-twist tower $[\phi\phi]_{0,\ell}$ (left), so these operators cannot seed the integral; they are reconstructed entirely from the crossed-channel operators $\mathcal{O}_m$ that do carry a double discontinuity (right). Inverting the single lightest such block produces the anomalous dimension Eq. (51.119) and, by the pole-shift relation Eq. (51.123), the OPE-coefficient correction Eq. (51.123), organizing the leading-twist trajectory $\tau_{0,\ell} \to 2\Delta_\phi$ (inset) from above in twist, i.e. from below in γ.

This is the reciprocity relation between the two leading-twist corrections: an anomalous dimension that lowers a trajectory necessarily redistributes spectral weight, and the redistribution is the $\bar{h}$-derivative of the weighted shift. Inserting the mean-field density $c^{(0)}_{0,\ell} \sim A\,\bar{h}^{2\Delta_\phi - 3/2} 4^{-\bar{h}}$ and $\gamma_{0,\ell} = -c_0 \bar{h}^{-\tau_m}$, the derivative is dominated at large $\bar{h}$ by the measure factor $4^{-\bar{h}}$, giving the relative correction

$$\frac{\delta c_{0,\ell}}{c^{(0)}_{0,\ell}} = \frac{1}{2}\Big[\gamma_{0,\ell}\,\partial_{\bar{h}} \log c^{(0)}_{0,\ell} + \partial_{\bar{h}}\gamma_{0,\ell}\Big] \xrightarrow{\bar{h}\to\infty} -\tfrac{1}{2}\,\gamma_{0,\ell}\,\log 4 = \frac{c_0 \log 4}{2\,\bar{h}^{\tau_m}}, \quad (51.124)$$

of the same order $1/\bar{h}^{\tau_m}$ as the anomalous dimension. The $\log 4$ is normalization-dependent — it tracks the convention-fixed $4^{-\bar{h}}$ in the collinear-block measure — and we flag it as such; the scheme-independent statement is that $\delta c_{0,\ell}/c^{(0)}_{0,\ell}$ and $\gamma_{0,\ell}$ vanish at the same rate, locked together by Eq. (51.123). With Eq. (51.119) and Eq. (51.123) the leading-twist data are fully reconstructed from a single crossed block.

Worked Example 51.19 (The stress-tensor result reproduced in one pass)**.** We reproduce, by a single application of Eq. (51.119), the stress-tensor contribution to the leading-twist anomalous dimension obtained by the lightcone method in Section 51.3, namely Eq. (51.79). Every CFT contains the stress tensor $T_{\mu\nu}$, a conserved spin-$\ell_m = 2$ operator of twist $\tau_m = d - 2$ (its dimension $\Delta = d$ saturates the conserved-current value, by Theorem 47.18). It is exchanged in $\phi \times \phi$ with an OPE coefficient fixed by the conformal Ward identity — the statement that $T_{\mu\nu}$ generates translations, so its coupling to any primary is set by that primary's dimension. In the unit-normalized block conventions of this

part, the Ward identity gives

$$P_T \equiv f^2_{\phi\phi T} = \frac{d\,\Delta_\phi^2}{(d-1)^2\,C_T}, \tag{51.125}$$

where C_T is the central charge, the coefficient of the stress-tensor two-point function (the canonical measure of the number of degrees of freedom; for a single free scalar $C_T = d/(d-1)$ in this normalization). Substituting $\tau_m = d-2$, $\ell_m = 2$, hence $\beta_m = \tau_m + 2\ell_m = d+2$, into Eq. (51.119),

$$\gamma^{(T)}_{0,\ell} = -\frac{c_T}{\bar{h}^{\,d-2}}, \qquad c_T = \frac{2\,\Gamma(\Delta_\phi)^2\,\Gamma(d+2)}{\Gamma\!\left(\Delta_\phi - \frac{d-2}{2}\right)^2\Gamma\!\left(\frac{d+2}{2}\right)^2}\,\frac{d\,\Delta_\phi^2}{(d-1)^2\,C_T}. \tag{51.126}$$

This is Eq. (51.79) term for term. The physics is transparent in this packaging. The anomalous dimension is negative: two scalars at large relative spin attract, and the binding energy is the (negative) shift of their total dimension below the free value. It falls as $1/\bar{h}^{\,d-2}$, the falloff of a force mediated by a minimal-twist exchange of twist $d-2$ — in $d=4$, as $1/\bar{h}^2$, the inverse-square law. And it is inversely proportional to the central charge, $\gamma^{(T)}_{0,\ell} \propto -\Delta_\phi^2/(C_T\bar{h}^{d-2})$. In a holographic theory $C_T \sim 1/G_N$ is the inverse Newton constant, so this is literally the statement that the two constituents bind gravitationally with a strength set by G_N; the analytic bootstrap derives Newtonian binding in anti–de Sitter space from crossing alone, the seed of bulk locality taken up in Chapters 53 and 56. The stress tensor is universal: every local CFT has one, so *every* CFT has this gravitational tail in its large-spin spectrum, whether or not it is holographic.

51.5.5 Scalar ϕ^4 theory: the Wilson–Fisher leading-twist anomalous dimension

We now advance ϕ^4 theory by turning the engine on the Wilson–Fisher fixed point in $d = 4-\varepsilon$ and computing the leading-twist anomalous dimension of $[\phi\phi]_{0,\ell}$ to order ε^2, cross-checking against the data imported in Section 47.8.4. The computation also makes a subtle and important point: which crossed operator dominates is decided by the double discontinuity, *not* by the raw twist.

Worked Example 51.20 ($[\phi\phi]_{0,\ell}$ at the Wilson–Fisher point, and which operator wins)**.** The conformal data of the fixed point were assembled in Theorem 47.55: the scalar ϕ has dimension $\Delta_\phi = \frac{d-2}{2} + \gamma_\phi = 1 - \frac{\varepsilon}{2} + \gamma_\phi$ with $\gamma_\phi = \varepsilon^2/108$ (Eq. (47.263)), and the energy operator ϕ^2 has dimension $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2)$ (Eq. (47.264)), a scalar of twist $\tau_{\phi^2} = \Delta_{\phi^2}$. Two operators compete to dominate the large-spin tail through Eq. (51.119): the stress tensor T, of twist $\tau_T = d-2 = 2-\varepsilon$, and the energy operator ϕ^2, of twist $\tau_{\phi^2} = 2 - \frac{2\varepsilon}{3}$. By raw twist the stress tensor is lighter, $\tau_T < \tau_{\phi^2}$, and would naively dominate. But the contribution of each is weighted by its double discontinuity, $\propto 1/\Gamma(\Delta_\phi - \tau_m/2)^2 = \sin^2\!\big(\pi(\Delta_\phi - \frac{\tau_m}{2})\big) \times (\cdots)$, and this reverses the verdict.

The stress tensor is suppressed. For T, $\Delta_\phi - \frac{\tau_T}{2} = (1-\frac{\varepsilon}{2}+\gamma_\phi) - \frac{2-\varepsilon}{2} = \gamma_\phi = \frac{\varepsilon^2}{108}$. The stress tensor sits a distance $\gamma_\phi = O(\varepsilon^2)$ from the double-twist lattice point $n = 0$ (in the free theory it *is* the double-twist operator $[\phi\phi]_{0,2}$; the interaction lifts $2\Delta_\phi$ above its protected $\tau = d-2$ by $2\gamma_\phi$). Its weight is therefore $\sin^2(\pi\gamma_\phi) \approx (\pi\gamma_\phi)^2 = O(\varepsilon^4)$. The universal gravitational tail Eq. (51.126) is real but enters only at order ε^4 here.

The energy operator dominates. For ϕ^2, $\Delta_\phi - \frac{\tau_{\phi^2}}{2} = (1-\frac{\varepsilon}{2}) - (1-\frac{\varepsilon}{3}) + O(\varepsilon^2) = -\frac{\varepsilon}{6} + O(\varepsilon^2)$, a distance $O(\varepsilon)$ from the lattice, with weight $\sin^2(\pi\varepsilon/6) \approx (\pi\varepsilon/6)^2 = O(\varepsilon^2)$. The energy operator wins, and the leading-twist anomalous dimension is $O(\varepsilon^2)$. Evaluate Eq. (51.119) with $\mathcal{O}_m = \phi^2$, a scalar so $\ell_m = 0$, $\beta_m = \tau_{\phi^2}$, and squared OPE coefficient $P_{\phi^2} = f^2_{\phi\phi\phi^2}$. At leading order $\phi^2 = [\phi\phi]_{0,0}$ is the mean-field scalar double-twist, so $P_{\phi^2} = c^{(0)}_{0,0}\big|_{\Delta_1=\Delta_2} = 2$ by Theorem 51.7 (the factor 2 is the even-spin doubling of identical scalars). Take the $\varepsilon \to 0$ limit of each Gamma factor: $\Gamma(\Delta_\phi)^2 \to \Gamma(1)^2 = 1$, $\Gamma(\beta_m) = \Gamma(\tau_{\phi^2}) \to \Gamma(2) = 1$, $\Gamma(\beta_m/2)^2 = \Gamma(\tau_{\phi^2}/2)^2 \to \Gamma(1)^2 = 1$, while

$$\Gamma\Big(\Delta_\phi - \tfrac{\tau_{\phi^2}}{2}\Big)^2 = \Gamma\Big(-\tfrac{\varepsilon}{6}\Big)^2 \approx \Big(-\tfrac{6}{\varepsilon}\Big)^2 = \frac{36}{\varepsilon^2}, \tag{51.127}$$

using $\Gamma(x) \approx 1/x$ near $x = 0$. Hence

$$c_0 = \frac{2\cdot 1\cdot 1}{(36/\varepsilon^2)\cdot 1}\cdot P_{\phi^2} = \frac{2\varepsilon^2}{36}\cdot 2 = \frac{\varepsilon^2}{9}, \qquad \boxed{\gamma_{0,\ell} = -\frac{\varepsilon^2}{9\,\bar{h}^2} + O(\varepsilon^3)\,.} \tag{51.128}$$

The cross-check. This must agree with the dimensions known from Part VI. The twist-two spin-ℓ operators of the single-scalar (Ising) universality class — the weakly broken higher-spin currents $[\phi\phi]_{0,\ell}$ — have the $O(\varepsilon^2)$ anomalous dimensions, measured relative to the free conserved-current value $\ell + d - 2$,

$$\gamma^{\rm cur}_\ell = \frac{\varepsilon^2}{54}\Big(1 - \frac{6}{\ell(\ell+1)}\Big) + O(\varepsilon^3), \tag{51.129}$$

a result of the ε-expansion of the fixed point. The double-twist anomalous dimension is this minus the shift $2\gamma_\phi$ already contained in $2\Delta_\phi = d - 2 + 2\gamma_\phi$:

$$\gamma_{0,\ell} = \gamma^{\rm cur}_\ell - 2\gamma_\phi = \frac{\varepsilon^2}{54}\Big(1 - \frac{6}{\ell(\ell+1)}\Big) - \frac{\varepsilon^2}{54} = -\frac{\varepsilon^2}{54}\cdot\frac{6}{\ell(\ell+1)} = -\frac{\varepsilon^2}{9\,\ell(\ell+1)}. \tag{51.130}$$

Two checks land at once. The large-spin limit of Eq. (51.130) is $-\varepsilon^2/(9\ell^2)$, agreeing with Eq. (51.128) since $\bar{h} = \Delta_\phi + \ell \to \ell$ at leading order; the inversion of the single ϕ^2 block has reproduced the falloff and the coefficient. More striking, the agreement is not merely asymptotic. The natural variable of the inversion formula is the conformal spin $\bar{h}$, whose square is the spin Casimir $\bar{h}(\bar{h}-1)$; in the free theory $\bar{h} = 1 + \ell$, so $\bar{h}(\bar{h}-1) = \ell(\ell+1)$, and the "Casimir-improved"

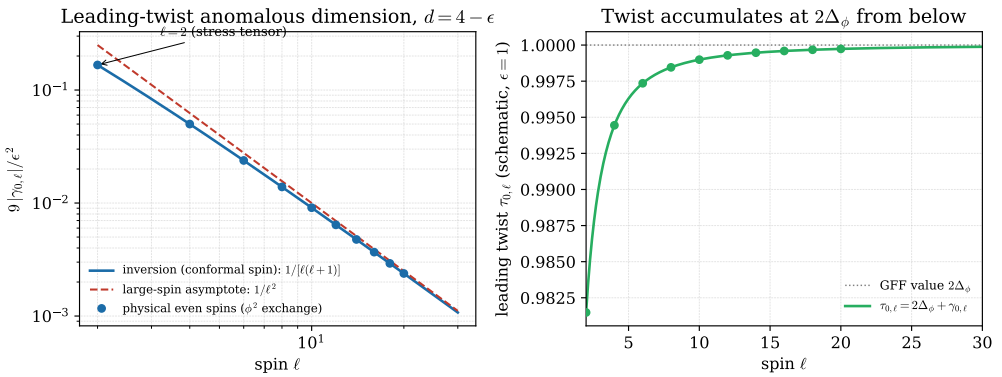


Figure 51.7: ϕ^4 theory cross-check of the inversion formula at the Wilson–Fisher fixed point in $d = 4-\varepsilon$. *Left:* the leading-twist double-twist anomalous dimension predicted by inverting the single ϕ^2 block, $\gamma_{0,\ell} = -\varepsilon^2/[9\,\bar{h}(\bar{h}-1)]$, plotted as $9|\gamma_{0,\ell}|/\varepsilon^2$ versus spin (log-log). The conformal-spin ("Casimir-improved") reading $1/[\ell(\ell+1)]$ coincides with the independent ε-expansion Eq. (51.130) at all even spins (markers); its large-spin asymptote is the $1/\ell^2$ falloff Eq. (51.128). The point $\ell = 2$ is the stress tensor, $\gamma_{0,2} = -2\gamma_\phi$. *Right:* the resulting leading twist $\tau_{0,\ell} = 2\Delta_\phi + \gamma_{0,\ell}$ approaches the generalized-free-field value $2\Delta_\phi$ from below, monotone and convex in the conformal spin — the twist accumulation of Eq. (51.49).

reading of Eq. (51.128), $\gamma_{0,\ell} = -\varepsilon^2/[9\,\bar{h}(\bar{h}-1)] = -\varepsilon^2/[9\,\ell(\ell+1)]$, coincides with the exact ε-expansion Eq. (51.130) for *every* even spin $\ell \geq 2$, not just at large ℓ. At $\ell = 2$ it gives $\gamma_{0,2} = -\varepsilon^2/54 = -2\gamma_\phi$, the correct value: the spin-two leading-twist operator is the stress tensor, whose full dimension is protected at $\Delta = d$, so its double-twist anomalous dimension is exactly $-2\gamma_\phi$. The conformal-spin organization of the large-spin expansion has captured the entire $O(\varepsilon^2)$ leading-twist spectrum of the Wilson–Fisher CFT from one crossed block. Figure 51.7 displays both the falloff and the twist trajectory.

51.5.6 Subleading orders and mixed double-twists

The computation above extracted the leading-twist ($n = 0$), leading-order data from the single lightest crossed block. The inversion formula systematizes the corrections in three independent directions, and we now indicate how each is organized; the machinery is identical to Section 51.5.3 with one ingredient changed in each case.

Higher twist $n \geq 1$. The radial families $[\phi\phi]_{n,\ell}$ are not on the leading lightcone power z^{Δ_ϕ} but on the subleading powers $z^{\Delta_\phi+n}$. They are reconstructed by the

same inversion integral Eq. (51.103) retaining the subleading terms in the small-z expansion of the kernel and of dDisc$[\mathcal{G}]$: the full generating function $c(\Delta, J)$ has poles at $\Delta = 2\Delta_\phi + 2n + \ell + \gamma_{n,\ell}$ for every n, and projecting onto the n-th pole returns $\gamma_{n,\ell}$ and the corresponding OPE coefficient. The crossed-channel block must now be expanded beyond its leading collinear term Eq. (51.109), which is where the d-dependence enters (the descent off the lightcone is governed by the Casimir equation Theorem 47.50 and no longer factorizes for general d, exactly as for the mean-field coefficients Eq. (49.144)). The structure is unchanged: each n acquires $\gamma_{n,\ell} \sim -c_n/\bar{h}^{\tau_m}$ with the same minimal twist τ_m controlling the falloff, and c_n a Δ_ϕ- and d-dependent generalization of Eq. (51.119).

Higher orders in $1/\bar{h}$. Beyond the leading $1/\bar{h}^{\tau_m}$, two sources contribute. First, the single block $\mathcal{O}_m$ has descendants, whose contributions to dDisc$[\mathcal{G}]$ are subleading powers of $(1-\bar{z})$ and produce a series $\gamma_{0,\ell} = -c_0/\bar{h}^{\tau_m}(1 + a_1/\bar{h} + a_2/\bar{h}^2 + \cdots)$; in the conformal-spin variable this series is the "Casimir-improved" expansion, organized so that $1/\bar{h}^{\tau_m}$ is promoted to $1/[\bar{h}(\bar{h}-1)]^{\tau_m/2}$ and the like, as the Wilson–Fisher example Eq. (51.130) illustrated by truncating exactly. Second, two crossed operators $\mathcal{O}_m, \mathcal{O}_{m'}$ contribute at order $1/\bar{h}^{\tau_m+\tau_{m'}}$ through the square of the dDisc reconstruction — the regime where the $1/\bar{h}$ expansion ceases to be controlled by a single exchange and the full nonlinearity of crossing re-enters. The inversion formula handles the linear (single-block) level exactly; the quadratic level requires either iterating or the full two-dimensional integral.

Mixed double-twists $[\mathcal{O}_1\mathcal{O}_2]_{n,\ell}$. For distinct external scalars $\mathcal{O}_1, \mathcal{O}_2$ the relevant correlator is $\langle \mathcal{O}_1\mathcal{O}_2\mathcal{O}_2\mathcal{O}_1\rangle$ of Eq. (51.35), whose leading-twist OPE data are the mean-field coefficients $c^{\mathrm{MFT}}_{0,\ell}$ of Eq. (51.38) with all integer spins present. The inversion proceeds exactly as in Section 51.5.3 with two changes. The crossed channel relevant to the double discontinuity is now $\mathcal{O}_1 \times \mathcal{O}_1 \to \mathcal{O}_m$ (or $\mathcal{O}_2 \times \mathcal{O}_2 \to \mathcal{O}_m$): the operator binding the pair is exchanged between the two like constituents. And the kinematic factors carry the dimension difference $\Delta_1 - \Delta_2$ through the collinear block Eq. (51.36). The leading-twist anomalous dimension of $[\mathcal{O}_1\mathcal{O}_2]_{0,\ell}$ from a single $\mathcal{O}_m$ exchanged in $\mathcal{O}_1 \times \mathcal{O}_1$ takes the form

$$\gamma_{0,\ell} = -\frac{c_0}{\bar{h}^{\,\tau_m}}, \qquad c_0 = \frac{2\,\Gamma(\Delta_1)^2\,\Gamma(\tau_m + 2\ell_m)}{\Gamma\!\left(\Delta_1 - \frac{\tau_m}{2}\right)^2 \Gamma\!\left(\frac{\tau_m}{2} + \ell_m\right)^2}\, P_m^{(11)}, \tag{51.131}$$

with $P_m^{(11)} = f_{\mathcal{O}_1\mathcal{O}_1\mathcal{O}_m} f_{\mathcal{O}_2\mathcal{O}_2\mathcal{O}_m}$ the product of OPE coefficients coupling $\mathcal{O}_m$ to each pair of like constituents — the natural generalization, since the binding is mediated by $\mathcal{O}_m$ exchanged between an $\mathcal{O}_1\mathcal{O}_1$ vertex and an $\mathcal{O}_2\mathcal{O}_2$ vertex. (The dependence on Δ_1 rather than a symmetric combination is, as in Eq. (51.35), an artifact of which constituent's lightcone is taken; the physical statement is symmetric under $\mathcal{O}_1 \leftrightarrow \mathcal{O}_2$ once both crossed channels are included.) The

derivation is the computation of Section 51.5.3 with the mean-field density Eq. (51.38) in place of its identical-scalar value; we do not repeat the integrals. The new feature is physical: a single external pair $[\mathcal{O}_1\mathcal{O}_2]$ now probes the operators exchanged in *both* $\mathcal{O}_1 \times \mathcal{O}_1$ and $\mathcal{O}_2 \times \mathcal{O}_2$, so mixed-correlator large-spin data constrain a wider slice of the spectrum — the principle behind the mixed-correlator numerical bootstrap of Section 50.6.6 and the analytic input to it.

51.5.7 The Regge limit: the boundary of analyticity

The inversion formula is powerful precisely because it makes OPE data an analytic function of spin — but analyticity in spin holds only down to a finite value, and below it the formula stops controlling the physics. Delimiting that regime honestly is essential, because it is the boundary at which the entire program of this chapter reaches its present edge.

The double integral defining $c(\Delta, J)$ converges only when the integrand decays at the corner of the causal diamond where both $z, \bar{z} \to 1$ — the *Regge limit*, the Lorentzian regime of large boost rather than large Euclidean separation. The decay is governed by the growth of $\mathcal{G}$ in the Regge limit, which defines the *Regge intercept* J_0: the correlator grows no faster than the spin-J_0 exchange would dictate. Caron-Huot's theorem establishes that the inversion integral converges, and hence that the OPE data are analytic in spin and reconstructed by the double discontinuity, for

$$J > J_0, \tag{51.132}$$

with J_0 the Regge intercept of the theory. For a generic unitary CFT one has $J_0 \leq 1$, so the formula controls all spins $J > 1$; this rests on Regge boundedness, which in turn follows from the same causality/positivity input — the bound on chaos and the analyticity of the Lorentzian correlator in the upper-half plane of the boost rapidity — that underlies the formula's derivation. The statement that double-twist data are analytic in spin down to $J > 1$ is, as of the mid-2020s, on firm footing for the leading-twist trajectory in unitary theories.

What the inversion does *not* control is the region $J \leq J_0$ itself — the Regge limit proper. There the sum over the spins of exchanged operators no longer converges term by term; it must be resummed, and the resummation is dominated by a *Regge pole*, a collective effect of the entire leading trajectory rather than any single operator. The physics of the Regge pole is the high-energy behaviour of the theory: in a gauge theory it is the reggeized gluon, in a gravitational (holographic) theory the reggeized graviton — the "pomeron." In a holographic CFT the intercept is pushed up toward the spin of the lightest exchanged Regge trajectory, and at strong coupling the leading trajectory is the graviton's, so $J_0 \to 2$: the inversion then controls only $J > 2$, and the stress tensor at $J = 2$ sits exactly on the boundary. This is the precise sense in which the analytic

bootstrap "sees" gravity at the edge of its domain rather than inside it. The arcs at infinity that one drops in deriving the formula are the contributions of $J \le J_0$; assuming they vanish is assuming Regge boundedness, and it is exactly this assumption that fails, in a controlled way, as one approaches the Regge regime.

Two honest caveats date the present understanding. First, the precise value of J_0 at finite coupling, and the rigorous status of Regge boundedness beyond the leading-twist trajectory and beyond two dimensions, remain only partially settled; the cleanest statements are for the leading trajectory and in holographic or large-N limits. Second, the convergence of the large-spin $(1/\bar{h})$ expansion itself — as opposed to its term-by-term validity — is asymptotic, not established to be convergent, so resumming it to reach finite or small spin is a separate problem, addressed in practice by combining the analytic tail with the numerical bootstrap of Chapter 50 that controls the low spins directly. The two bootstraps are genuinely complementary: numerics pins the handful of low-lying, low-spin operators that the $1/\bar{h}$ expansion cannot reach, and the inversion formula organizes the infinite high-spin tail that numerics can only sample. This is the synthesis the next section formalizes.

We have now used the inversion formula as it is meant to be used: as a machine that ingests the double discontinuity of a correlator — equivalently, the operators exchanged in the crossed channel — and emits the analytic-in-spin data of the double-twist towers, with the lightest exchanged operator setting the leading $1/\bar{h}^{\tau_m}$ correction, the stress tensor contributing the universal gravitational tail, and the energy operator ϕ^2 governing the Wilson–Fisher spectrum in agreement with the ε-expansion. The dispersive structure underlying these results — a positive integral of a double discontinuity against a fixed kernel — is not unique to the inversion formula. The same positivity, repackaged as a linear functional acting on the crossing equation, is the analytic counterpart of the numerical functionals of Chapter 50, and unifying the two viewpoints into a single dispersive bootstrap is the subject of the next section.

51.6 Analytic Functionals and the Dispersive Bootstrap

The inversion formula of the previous section is, read one way, a reconstruction theorem: it returns the OPE data of the direct channel from a single positive datum of the correlator, its double discontinuity. Read another way it is a statement about *linear functionals*. To produce the coefficient and dimension of a given exchanged operator the inversion formula integrates the correlator against a fixed kernel; that integration is a linear map from the space of crossing-symmetric correlators to a number, and the double discontinuity sitting inside it makes the

map manifestly sign-definite on unitary input. This is exactly the shape of the object that drove the numerical bootstrap of Chapter 50: a linear functional, applied to crossing, whose sign certifies which spectra are allowed. The numerical program of Section 50.6.1 found such functionals by semidefinite optimization, as anonymous vectors in a truncated derivative space, and read the spectrum off their zeros (Theorem 50.34). The dispersive structure of the inversion formula promises something the optimizer cannot give: the *same* extremal functionals, in closed form, with their zeros known analytically. This section makes the promise precise. We recast crossing as the vanishing of analytic functionals; we construct, in the one solvable case — the one-dimensional bootstrap — the analytic extremal functionals dual to the generalized free field, with their double zeros on the double-twist spectrum exhibited explicitly; and we tie the whole construction to the double discontinuity, exposing the dispersive sum rules whose positivity is inherited directly from the inversion formula. The upshot is a unification: the analytic functionals are the closed-form optimal functionals, and the numerical functionals of Chapter 50 are their finite-dimensional approximations.

51.6.1 Crossing as the vanishing of a linear functional, revisited

We begin by restating the bootstrap logic of Chapter 50 in the language that the analytic construction will fill in. The crossing equation of Theorem 49.13, with the identity contribution separated and normalized, is the statement that the *crossing vector* or *block difference*

$$\mathcal{F}_{\Delta,\ell}(z,\bar{z}) \;\equiv\; v^{\Delta_\phi}\, G_{\Delta,\ell}(u,v) - u^{\Delta_\phi}\, G_{\Delta,\ell}(v,u), \tag{51.133}$$

built from the conformal blocks of Theorem 47.49 in the cross-ratios of Eq. (47.140), sums against the squared OPE coefficients $p_{\mathcal{O}} = \lambda^2_{\phi\phi\mathcal{O}} \geq 0$ to zero,

$$\mathcal{F}_{0,0}(z,\bar{z}) \;+\; \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, \mathcal{F}_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(z,\bar{z}) \;=\; 0, \tag{51.134}$$

where $\mathcal{F}_{0,0} = v^{\Delta_\phi} - u^{\Delta_\phi}$ is the identity term. A *linear functional* ω is a continuous linear map on the space of functions of the cross-ratios; its action on the crossing equation produces the *sum rule*

$$\omega[\mathcal{F}_{0,0}] \;+\; \sum_{\mathcal{O}\neq\mathbf{1}} p_{\mathcal{O}}\, \omega(\Delta_{\mathcal{O}},\ell_{\mathcal{O}}) \;=\; 0, \qquad \omega(\Delta,\ell) \equiv \omega[\mathcal{F}_{\Delta,\ell}], \tag{51.135}$$

a single linear constraint on the OPE data. If ω can be chosen so that $\omega[\mathcal{F}_{0,0}] > 0$ while $\omega(\Delta,\ell) \geq 0$ for every operator allowed by the unitarity bounds of Theorem 47.18 and by whatever gaps one imposes, then Eq. (51.135) is a positive number plus a sum of nonnegative numbers and cannot vanish: the assumed spectrum is excluded. This is the exclusion principle of Theorem 50.34, verbatim, written without committing to how ω is found.

The numerical bootstrap commits to one realization. It takes ω to be a finite linear combination of derivatives evaluated at the crossing-symmetric point $z = \bar{z} = \frac{1}{2}$,

$$\omega^{\text{num}}[\mathcal{F}] = \sum_{m+n\le\Lambda} \alpha_{mn}\, \partial_z^m \partial_{\bar{z}}^n \mathcal{F}(z,\bar{z})\Big|_{z=\bar{z}=\frac{1}{2}}, \tag{51.136}$$

exactly the action of Eq. (50.117), and searches over the finitely many coefficients α_{mn} by the semidefinite methods of Section 50.6.3. The functional that emerges at a boundary of the allowed region — the extremal functional α_* of Theorem 50.33 — has, by complementary slackness, double zeros precisely at the dimensions of the operators in the saturating spectrum. The derivative basis Eq. (51.136) is nothing but the leading terms of the Taylor expansion of a general analytic functional about $z = \frac{1}{2}$; the truncation order Λ is how many of those terms the optimizer is allowed to use. An *analytic* functional is the object Eq. (51.136) approximates: a linear map written in closed form, valid without truncation, whose zeros we can locate by hand rather than read off a solver. The question this section answers, in the one setting where the answer is known completely, is what those closed-form functionals are.

Definition 51.21 (Analytic functional)**.** An *analytic functional* for the crossing equation Eq. (51.134) is a linear map ω on the crossing functions, presented as an integral against a kernel rather than as a finite sum of derivatives. In the one-dimensional case developed below it has the contour-integral form

$$\omega[\mathcal{F}] \;=\; \int_{\frac{1}{2}}^{\frac{1}{2}+i\infty} \frac{dz}{2\pi i}\, h_\omega(z)\, \mathcal{F}(z) \;+\; \text{c.c.}, \tag{51.137}$$

with $h_\omega(z)$ holomorphic in the upper half-plane and decaying fast enough on the contour that the integral converges on every conformal block. The functional is *positive* (above a gap Δ_{gap}) if $\omega(\Delta) \ge 0$ for all $\Delta \ge \Delta_{\text{gap}}$, and *extremal* for a candidate spectrum if it is positive and its positive zeros coincide with that spectrum.

The contour-integral form is the analytic continuation of the derivative basis: deforming Eq. (51.137) and expanding h_ω in a Taylor series at $z = \frac{1}{2}$ reproduces Eq. (51.136) with α_{mn} the Taylor coefficients of h_ω. The advantage is that the analytic kernel encodes *all* derivatives at once, so its zeros are exact rather than the artifacts of a finite search.

51.6.2 The solvable case: the one-dimensional bootstrap

Analytic functionals are understood in closed form in exactly one kinematic setting: one dimension. This is not a toy. A one-dimensional CFT is the data carried by a conformal line defect — a Wilson line, a localized impurity, the worldline of a heavy probe — on which local operators live and fuse with an OPE

governed by the $\mathrm{SL}(2,\mathbb{R})$ subgroup of the conformal group. More to the point for us, the $\mathrm{SL}(2,\mathbb{R})$ blocks of the one-dimensional problem are precisely the collinear blocks that controlled the lightcone limit of Section 51.1: the one-dimensional functionals are the seed from which the higher-dimensional analytic functionals are built by dressing the collinear variable with the transverse one. Solving the one-dimensional case is therefore both an end in itself and the kernel of the general program.

In one dimension there is a single cross-ratio $z \in (0,1)$, no spin, and the conformal block is the $\mathrm{SL}(2,\mathbb{R})$ block

$$G_\Delta(z) \;=\; z^\Delta\, {}_2F_1(\Delta,\Delta;2\Delta;z), \tag{51.138}$$

the same Gauss hypergeometric collinear block met in Eq. (51.36) (with equal external dimensions $\Delta_1=\Delta_2=\Delta_\phi$, so the upper parameters coincide). The four-point function of a single scalar ϕ of dimension Δ_ϕ stripped to the cross-ratio function $\mathcal{G}(z)=\sum_\Delta p_\Delta\, G_\Delta(z)$ (with $p_{\Delta=0}=1$ the identity) obeys the crossing equation generated by $z\to 1-z$,

$$\sum_\Delta p_\Delta\, \mathcal{F}_\Delta(z)=0, \qquad \mathcal{F}_\Delta(z) \;\equiv\; \frac{G_\Delta(z)}{z^{2\Delta_\phi}}-\frac{G_\Delta(1-z)}{(1-z)^{2\Delta_\phi}}, \tag{51.139}$$

the one-dimensional reduction of Eq. (51.133). The block difference $\mathcal{F}_\Delta$ is antisymmetric under crossing, $\mathcal{F}_\Delta(z)=-\mathcal{F}_\Delta(1-z)$, and the identity gives $\mathcal{F}_0(z)=z^{-2\Delta_\phi}-(1-z)^{-2\Delta_\phi}$.

The reference solution is the generalized free field of Theorem 51.6, in its one-dimensional incarnation. Wick's theorem gives the three pairings, of which the disconnected one is the identity, so

$$\mathcal{G}_{\mathrm{GFF}}(z) \;=\; 1+z^{2\Delta_\phi}+\Big(\frac{z}{1-z}\Big)^{2\Delta_\phi}, \tag{51.140}$$

which satisfies crossing identically, since $\mathcal{G}_{\mathrm{GFF}}(z)/z^{2\Delta_\phi}=z^{-2\Delta_\phi}+1+(1-z)^{-2\Delta_\phi}$ is symmetric under $z\to 1-z$. Decomposing the non-identity part into blocks, the exchanged operators are the double-twist operators $[\phi\phi]_n$ at the rigid lattice

$$\Delta_n=2\Delta_\phi+2n, \qquad n=0,1,2,\ldots, \tag{51.141}$$

the one-dimensional shadow of the twist lattice Eq. (51.31) (there is no spin, so the radial label n is the only quantum number). Their squared OPE coefficients $a_n\equiv p_{\Delta_n}$ are fixed by matching power by power in z. Writing $h\equiv 2\Delta_\phi$ and expanding the non-identity part of Eq. (51.140),

$$\mathcal{G}_{\mathrm{GFF}}(z)-1=z^h\big[1+(1-z)^{-h}\big]=2z^h+h\,z^{h+1}+\tfrac{h(h+1)}{2}z^{h+2}+\cdots, \tag{51.142}$$

and comparing with $G_{\Delta_n}(z)=z^{h+2n}\big[1+\frac{(h+2n)^2}{2h+4n}z+\cdots\big]$ gives the leading coefficients in closed form. At order z^h only G_{Δ_0} contributes, so $a_0=2$. At order

z^{h+1} the single block $a_0 G_{\Delta_0}$ already supplies $a_0 \cdot \frac{h^2}{2h} = h$, matching Eq. (51.142) and confirming there is no operator at $\Delta = h+1$ (consistent with the even lattice). At order z^{h+2} the block G_{Δ_1} first appears, and matching

$$\underbrace{a_0 \cdot \frac{(h)_2^2}{(2h)_2\, 2!}}_{=\frac{h(h+1)^2}{2(2h+1)}} + a_1 \;=\; \frac{h(h+1)}{2} \quad\Longrightarrow\quad a_1 = \frac{h^2(h+1)}{2(2h+1)} = \frac{(2\Delta_\phi)^2(2\Delta_\phi+1)}{2(4\Delta_\phi+1)}, \tag{51.143}$$

both manifestly positive. The general coefficient has the closed form $a_n = \frac{2\,\Gamma(h+2n)^2\,\Gamma(2h+2n-1)}{(2n)!\,\Gamma(h)^2\,\Gamma(2h+4n-1)}$, the one-dimensional companion of Eq. (51.38); we will need only the explicit low-lying values Eq. (51.143). The data $\{\Delta_n, a_n\}$ are the one-dimensional mean-field solution, and they will turn out to be the unique *extremal* solution at the boundary of the allowed region — the one-dimensional analogue of the way the Ising point saturated the numerical bound in Section 50.6.3.

51.6.3 The analytic extremal functionals: a dual basis to the generalized free field

We now state and construct the functionals dual to the generalized free field. The construction is due to Mazác and to Mazác and Paulos; we give it in full at the level of its defining properties and the leading explicit functional, and exhibit the double-zero structure that is its entire content.

The strategy is to build a *basis* of functionals adapted to the double-twist spectrum Eq. (51.141), in the precise sense that each functional reads off exactly one piece of data. Two pieces of data live at each lattice site Δ_n: the squared OPE coefficient a_n, and the dimension itself, which in a deformation away from mean field shifts by an anomalous dimension γ_n. Accordingly there are two families.

Theorem 51.22 (Existence of the dual functional basis)**.** *There exist two families of analytic functionals* $\{\alpha_n\}_{n\geq 0}$ *and* $\{\beta_n\}_{n\geq 0}$ *of the form Eq. (51.137), the* Mazác–Paulos basis*, whose actions on the block difference* $\mathcal{F}_\Delta$*, regarded as functions* $\alpha_n(\Delta) = \alpha_n[\mathcal{F}_\Delta]$ *and* $\beta_n(\Delta) = \beta_n[\mathcal{F}_\Delta]$*, satisfy the duality relations*

$$\alpha_n(\Delta_m) = \delta_{nm}, \qquad \partial_\Delta \alpha_n(\Delta)\big|_{\Delta_m} = 0, \tag{51.144}$$

$$\beta_n(\Delta_m) = 0, \qquad \partial_\Delta \beta_n(\Delta)\big|_{\Delta_m} = \delta_{nm}, \tag{51.145}$$

for all $m, n \geq 0$*, where* $\Delta_m = 2\Delta_\phi + 2m$ *is the double-twist lattice Eq. (51.141). Equivalently: each functional has a* double zero *at every lattice site* Δ_m *except for a single distinguished site, where* α_n *takes the value* 1 *(a non-zero) and* β_n *has a* simple *zero (value* 0*, derivative* 1*). The functionals are positive above the*

gap $\Delta_0 = 2\Delta_\phi$, and the generalized free field Eq. (51.140) is the extremal solution they certify.

The double-zero pattern is the exact analytic image of the numerical phenomenon of Theorem 50.34: there, the extremal functional α_* was forced by complementary slackness to have double zeros on the saturating spectrum; here, the functionals are *designed* to have double zeros on the double-twist spectrum, and the design is possible precisely because the generalized free field saturates the one-dimensional bound. The numerical extremal functional, computed in the derivative basis Eq. (51.136) at a boundary point whose extremal solution is mean field, converges term by term to a combination of the α_n, β_n as the truncation order is raised. The closed-form functionals are the limit the optimizer approaches.

Why double zeros, and where they come from. Before constructing the functionals, we explain why the double-zero pattern is forced and not merely convenient — and in doing so make the link to the double discontinuity that the next subsection turns into positivity. The functional action $\omega(\Delta) = \omega[\mathcal{F}_\Delta]$ is, for a functional of the contour form Eq. (51.137), computed by deforming the contour onto the singularities of $\mathcal{F}_\Delta(z)$ at $z \to 1$. The block difference Eq. (51.139) carries there the crossed-channel piece $(1-z)^{-2\Delta_\phi} G_\Delta(1-z)$, whose behaviour as $z \to 1$ is the t-channel expansion. The discontinuity of this piece across the cut at $z = 1$ — and hence the part of $\omega(\Delta)$ that survives the contour deformation — carries the kinematic factor that the Lorentzian inversion formula of Theorem 51.16 makes universal: the double discontinuity of a t-channel contribution of dimension Δ relative to the s-channel double-twist family is proportional to

$$\mathrm{dDisc} \;\propto\; \sin^2\!\Big[\tfrac{\pi}{2}(\Delta - 2\Delta_\phi)\Big], \tag{51.146}$$

which vanishes to *second* order at every double-twist location $\Delta = \Delta_n = 2\Delta_\phi + 2n$, where the argument is an integer multiple of π. The double zeros of the analytic functionals are the double zeros of this $\sin^2$: the double discontinuity that the inversion formula puts at the heart of the dispersive reconstruction is the very object whose vanishing protects the double-twist operators. A functional built from the double discontinuity therefore *automatically* double-zeros on the mean-field spectrum. This is the structural reason the dual basis exists, and it ties the analytic functionals to the inversion formula at the root.

Construction of the leading functional. The cleanest explicit realization packages the duality relations into *Polyakov–Regge blocks*. A Polyakov–Regge block $\mathcal{P}_\Delta(z)$ is the unique crossing-symmetric function with the same double discontinuity as the single block G_Δ and bounded Regge behaviour; concretely it is the sum of the s- and t-channel SL(2) Witten exchange diagrams of an

operator of dimension Δ. Decomposing $\mathcal{P}_\Delta$ back into s-channel blocks produces the exchanged block plus a tower of double-twist blocks *and their dimension-derivatives*,

$$\mathcal{P}_\Delta(z) \;=\; G_\Delta(z) \;+\; \sum_{n\geq 0}\Big[\, a_n\,\alpha_n(\Delta)\,G_{\Delta_n}(z) \;+\; a_n\,\beta_n(\Delta)\,\partial_\Delta G_{\Delta_n}(z)\,\Big], \tag{51.147}$$

where the coefficients are precisely the functional actions of Theorem 51.22 and the appearance of $\partial_\Delta G_{\Delta_n}$ — a block dressed by a logarithm $\log z$ — is the signal of an anomalous dimension. Equation Eq. (51.147) is the defining property: writing a crossing-symmetric correlator in the Polyakov–Regge basis, $\mathcal{G}(z) = \sum_\Delta p_\Delta\,\mathcal{P}_\Delta(z)$, and demanding that the double-twist data on the two sides match imposes exactly the sum rules Eq. (51.135) for $\omega = \alpha_n$ and $\omega = \beta_n$. The functionals are read off as the double-twist content of the Polyakov–Regge blocks, and Eq. (51.147) makes the double-zero structure automatic: $\mathcal{P}_\Delta$ for Δ on the lattice must reduce to a single block G_Δ with no anomalous-dimension dressing of *the other* sites, which is the statement $\alpha_n(\Delta_m) = \delta_{nm}$, $\beta_n(\Delta_m) = 0$, and the derivative conditions follow from the absence of a $\partial_\Delta G_{\Delta_m}$ term at $m \neq n$. The explicit kernels $h_{\alpha_n}, h_{\beta_n}$ in Eq. (51.137) are obtained by inverting this decomposition; the leading one, β_0, is worked out in Theorem 51.23.

Worked Example 51.23 (The leading anomalous-dimension functional and its double zeros)**.** We construct the leading functional β_0 explicitly enough to exhibit its defining zeros, and verify the double-zero pattern Eq. (51.145) directly on the lowest sites.

The object. By Eq. (51.147), $\beta_0(\Delta)$ is the coefficient of $a_0\,\partial_\Delta G_{\Delta_0}(z)$ — the $\log z$-dressed leading block — in the s-channel decomposition of the Polyakov–Regge block $\mathcal{P}_\Delta$. Equivalently it is the contour functional Eq. (51.137) whose kernel $h_{\beta_0}(z)$ projects the crossing function onto the $\log z$ singularity at the leading double-twist dimension $\Delta_0 = 2\Delta_\phi$. Its action is computed by deforming the $z = \frac{1}{2}$ contour to the cut at $z = 1$, where the block difference Eq. (51.139) produces the t-channel expansion whose double discontinuity carries the factor Eq. (51.146).

Double zeros from the $\sin^2$. Write $\beta_0(\Delta) = \rho(\Delta)\,\sin^2\!\big[\tfrac{\pi}{2}(\Delta - 2\Delta_\phi)\big]/N(\Delta)$, where ρ is the smooth OPE-density factor produced by the contour integral and N a normalization fixed below. At every lattice site $\Delta_m = 2\Delta_\phi + 2m$ the argument of the sine is πm, so $\sin^2$ vanishes to second order:

$$\sin^2\!\Big[\tfrac{\pi}{2}(\Delta - 2\Delta_\phi)\Big]\Big|_{\Delta=\Delta_m+\delta} = \sin^2\!\Big[\tfrac{\pi}{2}\delta\Big] = \frac{\pi^2}{4}\,\delta^2 + O(\delta^4), \qquad \delta = \Delta - \Delta_m. \tag{51.148}$$

Hence $\beta_0(\Delta_m) = 0$ and $\partial_\Delta\beta_0(\Delta_m) = 0$ at *every* site $m \geq 0$ where the smooth factor ρ/N is finite — a double zero. The single exception is the distinguished site $m = 0$: there the smooth factor $\rho(\Delta)/N(\Delta)$ has a *simple pole*, engineered by

the kernel so that the δ^2 of Eq. (51.148) is reduced to a δ^1. Concretely, near Δ_0,

$$\beta_0(\Delta) \xrightarrow{\Delta\to\Delta_0} \frac{r_0}{\Delta-\Delta_0}\cdot\frac{\pi^2}{4}(\Delta-\Delta_0)^2+\cdots = \frac{\pi^2 r_0}{4}(\Delta-\Delta_0)+\cdots, \quad (51.149)$$

a *simple* zero at Δ_0. Fixing the residue $r_0 = 4/\pi^2$ normalizes the derivative to $\partial_\Delta\beta_0(\Delta_0) = 1$, which is the condition $\partial_\Delta\beta_0(\Delta)|_{\Delta_0} = \delta_{0,0} = 1$ of Eq. (51.145). At every other site the smooth factor is regular, the $\sin^2$ gives the clean double zero, and Eq. (51.145) holds with $\delta_{0,m} = 0$.

What the pole does and does not require. The only nontrivial analytic input is the existence of a kernel h_{β_0} whose density ρ/N is regular and nonzero on the lattice except for a simple pole at the one distinguished site; this is what fixes the functional uniquely and is the content of the Mazác–Paulos kernels. Everything else — the double zeros at all other sites, the simple zero at the distinguished site, the positivity above the gap — follows from the universal $\sin^2$ factor Eq. (51.146) alone, which is dictated by the double discontinuity and not by the details of the kernel. The functional α_n is constructed identically, with the pole placed so as to convert the δ^2 at the distinguished site into a δ^0 (a non-zero value 1) rather than a δ^1; the remaining sites again double-zero through Eq. (51.148).

51.6.4 The sum rules and the reproduction of the generalized free field

The duality relations Eqs. (51.144) and (51.145) turn the bare crossing constraint Eq. (51.135) into a system of *explicit* sum rules for the deviation of any unitary solution from mean field. Consider a CFT whose spectrum is the double-twist lattice deformed by small anomalous dimensions γ_m and coefficient shifts, together with a set of extra operators $\{\mathcal{O}\}$ not on the lattice (the operators that the generalized free field omits — the genuine interactions). Write the perturbed double-twist operator at $\Delta = \Delta_m + \gamma_m$ with coefficient $a_m + \delta a_m$, and Taylor-expand its block difference,

$$\mathcal{F}_{\Delta_m+\gamma_m}(z) = \mathcal{F}_{\Delta_m}(z) + \gamma_m\,\partial_\Delta\mathcal{F}_{\Delta_m}(z) + O(\gamma_m^2). \quad (51.150)$$

Inserting into crossing and acting with α_n and β_n, the duality relations collapse the lattice sums to a single term each. From α_n, using $\alpha_n(\Delta_m) = \delta_{nm}$ and $\partial_\Delta\alpha_n(\Delta_m) = 0$,

$$\boxed{\delta a_n \;=\; -\sum_{\mathcal{O}\notin\text{ lattice}} p_{\mathcal{O}}\,\alpha_n(\Delta_{\mathcal{O}})} \quad (51.151)$$

to first order: the shift in the n-th double-twist OPE coefficient equals minus the α_n-functional summed over the operators that are *not* double-twist. From β_n, using $\beta_n(\Delta_m) = 0$ and $\partial_\Delta\beta_n(\Delta_m) = \delta_{nm}$,

$$\boxed{a_n\,\gamma_n \;=\; -\sum_{\mathcal{O}\notin\text{ lattice}} p_{\mathcal{O}}\,\beta_n(\Delta_{\mathcal{O}})} \quad (51.152)$$

the n-th anomalous dimension, weighted by the mean-field coefficient, equals minus the β_n-functional over the same non-lattice operators. These are the analytic-functional sum rules: closed-form, exact-in-$1/\ell$ expressions for the double-twist data in terms of the binding operators, and the one-dimensional rigorous counterpart of the large-spin formulae of Section 51.3. The functional β_n computes an anomalous dimension; α_n a coefficient shift; the inputs are the cross-channel operators that the mean-field theory lacks.

The first and most important check is that the generalized free field reproduces itself.

Worked Example 51.24 (The generalized free field as the extremal solution)**.** We verify that the dual functionals return the mean-field data when applied to the mean-field correlator, and that this is the statement that the generalized free field saturates the bound.

No anomalous dimensions. In the generalized free field Eq. (51.140) the spectrum is *exactly* the lattice Eq. (51.141): there are no operators off the lattice, so the right-hand sides of the sum rules Eqs. (51.151) and (51.152) are empty sums. Equation Eq. (51.152) then gives $a_n\gamma_n = 0$, hence $\gamma_n = 0$ for every n: the double-twist operators carry no anomalous dimension, exactly as the rigid lattice demands. Equation Eq. (51.151) gives $\delta a_n = 0$: the coefficients are unshifted from their mean-field values a_n. The dual basis reproduces the generalized free field as a fixed point of its own sum rules.

Direct re-summation. The same statement read forward is that the functionals *measure* the mean-field coefficients. Apply α_n to the full mean-field crossing equation $\mathcal{F}_0 + \sum_m a_m \mathcal{F}_{\Delta_m} = 0$. Using $\alpha_n(\Delta_m) = \delta_{nm}$,

$$\alpha_n[\mathcal{F}_0] + \sum_m a_m\, \alpha_n(\Delta_m) = \alpha_n[\mathcal{F}_0] + a_n = 0 \quad \Longrightarrow \quad a_n = -\,\alpha_n[\mathcal{F}_0], \tag{51.153}$$

the coefficient a_n *is* minus the α_n-functional on the identity — the closed-form analogue of strong duality Eq. (50.123), where the extremal OPE coefficient equalled minus the extremal functional on the identity. For $n = 0$ this returns $a_0 = -\alpha_0[\mathcal{F}_0] = 2$, the value computed by hand in Eq. (51.143); the functional construction is normalized precisely so that $\alpha_0[\mathcal{F}_0] = -2$, $\alpha_1[\mathcal{F}_0] = -a_1 = -(2\Delta_\phi)^2(2\Delta_\phi + 1)/[2(4\Delta_\phi + 1)]$, and so on. The single boundary functional encodes the entire mean-field OPE spectrum, exactly as the numerical extremal functional encoded the Ising spectrum in Theorem 50.35.

Extremality. Positivity (Theorem 51.22) means $\alpha_n(\Delta) \geq 0$ for $\Delta \geq \Delta_0$, with zeros only on the lattice. A solution that placed any OPE weight $p_{\mathcal{O}} > 0$ at a dimension strictly between lattice sites would, through Eq. (51.151), be detected by some α_n with a strictly positive value, forcing $\delta a_n \neq 0$ and moving the solution off mean field. The generalized free field, supported entirely on the double zeros, is the unique solution at which every positivity-functional is saturated — it sits on the boundary of the allowed region, certified extremal

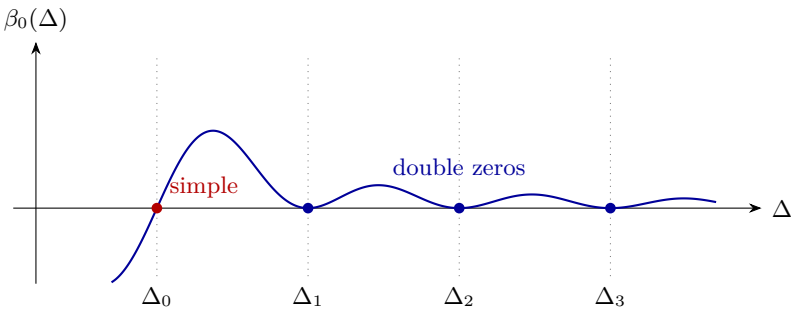


Figure 51.8: The analytic functional $\beta_0(\Delta) = \beta_0[\mathcal{F}_\Delta]$ acting on the one-dimensional block difference, schematically. The universal factor $\sin^2[\frac{\pi}{2}(\Delta-2\Delta_\phi)]$ of Eq. (51.146), inherited from the double discontinuity of the inversion formula, forces a *double* zero (the curve touches the axis without crossing) at every double-twist site $\Delta_m = 2\Delta_\phi + 2m$. The kernel places a single simple pole at the distinguished site Δ_0, converting the double zero there into a *simple* zero with unit slope: this is the duality relation Eq. (51.145). Compare the numerically-computed extremal functional of Fig. 50.7: the same double-zero-on-the-spectrum picture, now known in closed form.

by the closed-form functionals. This is the analytic statement that mean-field theory is the extremal solution of the one-dimensional bootstrap, the exact image of the numerical extremality of Section 50.6.3.

51.6.5 Dispersive functionals, double discontinuity, and the inversion formula

The $\sin^2$ factor Eq. (51.146) did more than explain the double zeros: it located the functionals' analytic content inside the double discontinuity. This is the defining feature of the most important sub-class of analytic functionals, the *dispersive functionals*, and it is what makes contact with the inversion formula complete.

Definition 51.25 (Dispersive functional)**.** A *dispersive functional* is an analytic functional whose action on a crossing function is computed entirely from the function's double discontinuity,

$$\omega[\mathcal{G}] \;=\; \int dz\; k_\omega(z)\; \mathrm{dDisc}\,[\mathcal{G}(z)], \tag{51.154}$$

with a kernel $k_\omega(z) \geq 0$ on the integration region. The associated *dispersive sum rule* is $\omega[\mathcal{F}_0] + \sum_{\mathcal{O}\neq\mathbf{1}} p_\mathcal{O}\, \omega(\Delta_\mathcal{O}) = 0$, the instance of Eq. (51.135) for this ω.

The reason dispersive functionals exist at all is the Lorentzian inversion formula of Theorem 51.16: that formula expresses the OPE function $c(\Delta)$ as an

integral of the double discontinuity against the inversion kernel, and a *positive* integral, because dDisc is nonnegative on unitary correlators. A dispersive functional is the same construction read as a constraint rather than a reconstruction: instead of asking what spectrum the dDisc produces, one asks that a fixed positive combination of the dDisc, integrated against k_ω, vanish on crossing-symmetric data. The two are the same object viewed from the two sides of the equality, which is exactly the unification this section is after — the inversion formula is the generating functional of the dispersive sum rules.

Proposition 51.26 (dDisc-positivity of dispersive functionals)**.** *A dispersive functional Eq. (51.154) with $k_\omega \geq 0$ acts nonnegatively on every unitary contribution to the crossing equation, and double-zeros on the double-twist spectrum:*

$$\omega(\Delta) \;\geq\; 0 \quad \textit{for all } \Delta \textit{ above the gap}, \qquad \omega(\Delta_n) = \partial_\Delta\, \omega(\Delta_n) = 0. \tag{51.155}$$

Consequently the dispersive sum rule certifies the generalized free field as extremal, exactly as in Theorem 51.22.

Derivation, with the analytic input cited. The double-zero statement is Eq. (51.146): the double discontinuity of a single block, as it enters Eq. (51.154), carries the factor $\sin^2[\frac{\pi}{2}(\Delta - 2\Delta_\phi)]$, which vanishes to second order on the lattice; integrating against any fixed kernel preserves the double zero. The positivity statement is the one whose full proof we do not reproduce and instead anchor to the inversion formula. The double discontinuity of a unitary correlator is nonnegative: $\mathrm{dDisc}[\mathcal{G}] \geq 0$, a property established in the construction of the Lorentzian inversion formula in Theorem 51.16, where it is precisely what guarantees that the inverted OPE coefficients come out positive and that the inversion converges (it is the statement, traced there to reflection positivity of the Euclidean correlator and the wedge causality of the Lorentzian one, that the double discontinuity is a sum of squares of matrix elements). Given $\mathrm{dDisc}[\mathcal{G}] \geq 0$ and $k_\omega \geq 0$, the integral Eq. (51.154) is manifestly nonnegative on any single unitary block, which is $\omega(\Delta) \geq 0$. The remaining content — that nonnegative kernels k_ω exist realizing the dual basis, and that the resulting functionals are complete — is the substance of the dispersive-bootstrap construction; we state it and use it, and refer to Theorem 51.16 for the positivity it rests on rather than reproving the inversion formula here. □

The dDisc-positivity is the conceptual bridge between the two bootstraps. In Chapter 50 the positivity that powered every bound was the positivity of squared OPE coefficients $p_{\mathcal{O}} \geq 0$, enforced operator by operator through semidefinite programming. Here the positivity is the positivity of the double discontinuity, $\mathrm{dDisc} \geq 0$, enforced once and for all by the analytic structure of the correlator. The two are dual faces of unitarity: $p_{\mathcal{O}} \geq 0$ is unitarity in the OPE (the s-channel), $\mathrm{dDisc} \geq 0$ is unitarity in the cross channel (the t-channel), and a

dispersive functional trades the former for the latter, replacing an infinite tower of semidefinite constraints by a single positive integral. This is why dispersive functionals are simultaneously analytic (their action is a closed-form integral) and optimal (their positivity is exact, not a truncation): they are built from the inversion formula, which already solved the positivity problem.

51.6.6 Polyakov–Mellin, the Mellin packaging, and the unified picture

The dispersive program has a second, historically independent, presentation in *Mellin space*, and a word on it completes the map of the territory. The Mellin amplitude $M(s,t)$ of a CFT four-point function is the analogue of a scattering amplitude: it is the Mellin transform of the correlator, an analytic function of two Mellin variables whose poles sit at the twists of the exchanged operators and whose residues are the OPE coefficients, with crossing symmetry realized as a permutation symmetry of M much as it is for amplitudes. The double-twist operators of the generalized free field appear in Mellin space as the locations where the Mellin measure has its poles, so that a Mellin amplitude with *no* poles beyond the double-twist lattice is mean field, and the deviations — the genuine dynamics — are extra poles or polynomial (contact) ambiguities. The *Polyakov–Mellin bootstrap* expands the correlator not in conformal blocks but in crossing-symmetric *Polyakov blocks* (the objects Eq. (51.147), now seen as exchange Witten diagrams in Mellin space) and imposes that the spurious double-twist contributions of each Polyakov block cancel in the full correlator; those cancellation conditions are exactly the sum rules Eqs. (51.151) and (51.152), and the functionals that implement them are the Mellin-space realization of the dispersive functionals Eq. (51.154). Mellin space is, in this sense, the natural home of the dispersive bootstrap: the double discontinuity becomes a statement about which Mellin poles are present, and the Mellin amplitude packages the analytic functional data into a single meromorphic function.

Historical Note

The analytic functional program crystallized in the second half of the 2010s. Mazác's 2016 construction of the one-dimensional extremal functionals, and the Mazác–Paulos systematization into the dual bases $\{\alpha_n, \beta_n\}$ of Theorem 51.22 [343–345], showed for the first time that the numerical extremal functionals had a closed-form skeleton. In parallel the Lorentzian inversion formula of Caron-Huot (2017) [111] put the double discontinuity at the centre, and the dispersive functionals of Carmi–Caron-Huot, Caron-Huot–Mazác–Rastelli–Simmons-Duffin, and Penedones–Silva–Zhiboedov [110, 112, 390] unified the two by realizing the functionals as dDisc-positive integrals. The Polyakov–Mellin reading goes back to Polyakov's 1974

"unitary bootstrap" proposal, revived and corrected in Mellin space by Gopakumar–Kaviraj–Sen–Sinha and others around 2016–2017 [234, 447]. As of the mid-2020s the program is complete and rigorous in one dimension and in the lightcone/collinear limit, where the SL(2) functionals control everything; in general dimension the dispersive functionals exist and are dDisc-positive, but a complete closed-form dual basis matching the numerical optima is known only in special cases, and the precise relationship between the analytic functionals and the full higher-dimensional numerical extremal functionals remains an active question.

We can now state the unification cleanly, which is the entire purpose of this section. There is one bootstrap and two ways of solving it. Both start from the crossing sum rule Eq. (51.135) and seek functionals that are positive on the allowed spectrum and vanish on the realized one. The *numerical* bootstrap of Chapter 50 finds such a functional by semidefinite optimization over a finite derivative basis Eq. (51.136), obtaining at the boundary an extremal functional with double zeros on the saturating spectrum (Theorem 50.34); it is general — it works in any dimension, for any correlator system — but the functional it returns is a numerical vector, and its zeros are computed, not known. The *analytic* bootstrap of this chapter writes the optimal functional in closed form Eq. (51.137), with double zeros placed analytically on the double-twist spectrum by the $\sin^2$ factor of the double discontinuity Eq. (51.146); it is exact and explanatory — the sum rules Eqs. (51.151) and (51.152) are formulae, not outputs — but it is fully controlled only in one dimension and in the lightcone limit. The two are not rivals but the same object at two resolutions: the analytic functionals $\{\alpha_n, \beta_n\}$ are the closed-form extremal functionals, and the numerical functional Eq. (51.136), computed at a boundary point whose extremal solution is the generalized free field, is the finite-Λ Taylor approximation of a combination of them. The double zeros that complementary slackness forced upon the numerical functional in Section 50.6.3 are the same double zeros that the double discontinuity builds into the analytic functional here. The semidefinite optimizer of Chapter 50 is, in the cases where the analytic answer is known, rediscovering the Mazác–Paulos functionals one Taylor coefficient at a time.

With the analytic functionals and their dispersive, dDisc-positive structure in hand, the machinery of the chapter is complete: the lightcone expansion fixes the large-spin spectrum, the inversion formula reconstructs the OPE data from the double discontinuity, and the analytic functionals package both into closed-form, optimal sum rules dual to the generalized free field. The closing section turns this apparatus on the chapter's target theories, applying the large-spin and inversion results to the three-dimensional Ising and Wilson–Fisher fixed points and cross-checking the analytic predictions against the numerical island of Chapter 50 and the ϵ-expansion of ϕ^4 theory.

51.7 Analytic Data for the 3D Ising and Wilson–Fisher CFTs

The previous sections assembled a unified analytic toolkit — the lightcone expansion of Section 51.1, the large-spin master formula Theorem 51.10, the Lorentzian inversion formula of Section 51.4, and the dispersive functionals of Section 51.6 — but built each on an abstract scalar ϕ of arbitrary dimension. We now point the whole apparatus at the two theories ϕ^4 theory has carried since Part II: the Wilson–Fisher fixed point in $d = 4 - \varepsilon$, and its physical endpoint, the three-dimensional Ising conformal field theory. The exercise is not decorative. Chapter 50 delivered the Ising dimensions to six figures, but as the unexplained output of a semidefinite program; the present section produces, with no computer, closed-form statements about an *infinite* family of operators — the leading-twist double-twist towers $[\sigma\sigma]_{0,\ell}$ and $[\epsilon\epsilon]_{0,\ell}$ — and shows that those statements agree, where they overlap, both with the numerical spectrum of Chapter 50 and with the ε-expansion of Chapter 47. The convergence of three independent methods on the same numbers is the strongest evidence we can offer that the 3D Ising CFT is a single, internally consistent object, and it closes the analytic-bootstrap stage of ϕ^4 theory.

51.7.1 The data and the operators in play

We work in $d = 3$ and take as input the conformal data the bootstrap has already pinned down. From Eq. (49.171) (reproduced by the semidefinite program in Eq. (50.130)) the two lightest primaries and their couplings are

$$\begin{aligned} &\Delta_\sigma = 0.5181489(10), \quad \Delta_\epsilon = 1.412625(10), \quad \Delta_{\epsilon'} \approx 3.83, \quad \Delta_T = 3, \\ &\lambda_{\sigma\sigma\epsilon} = 1.0518537(41), \quad \lambda_{\epsilon\epsilon\epsilon} = 1.532435(19), \quad \frac{C_T}{C_T^{\text{free}}} = 0.946534(11). \end{aligned} \tag{51.156}$$

Here σ is the $\mathbb{Z}_2$-odd order parameter, ϵ the $\mathbb{Z}_2$-even energy operator (the relevant deformation), ϵ' the next $\mathbb{Z}_2$-even scalar, $T_{\mu\nu}$ the stress tensor (dimension exactly $\Delta_T = d = 3$, conserved), and C_T the central charge normalized to the free-scalar value. These few numbers are all the dynamical input the analytic machine requires; everything below is kinematics and crossing.

The objects we predict are the two leading-twist double-twist towers, defined as in Section 51.2:

$$[\sigma\sigma]_{0,\ell} \subset \sigma \times \sigma \ (\mathbb{Z}_2\text{-even}), \qquad [\epsilon\epsilon]_{0,\ell} \subset \epsilon \times \epsilon \ (\mathbb{Z}_2\text{-even}), \qquad \ell = 0, 2, 4, \dots \tag{51.157}$$

Both are built from two identical scalars, so Bose symmetry retains only even spins. By the twist-accumulation theorem Eq. (51.49) their twists approach the

mean-field values

$$\tau_{0,\ell}^{[\sigma\sigma]} \xrightarrow{\ell\to\infty} 2\Delta_\sigma = 1.036298, \qquad \tau_{0,\ell}^{[\epsilon\epsilon]} \xrightarrow{\ell\to\infty} 2\Delta_\epsilon = 2.825250, \tag{51.158}$$

and the entire content of the analytic prediction is the *rate* and *sign* of this approach, governed by Theorem 51.10. To use that theorem we need, for each family, the lowest-twist operators exchanged in the crossed channel. The candidates are read off Eq. (51.156).

- In $\sigma\times\sigma$ the $\mathbb{Z}_2$-even exchanged operators are the identity (twist 0), the stress tensor T (twist $\tau_T = d-2 = 1$), and the scalar ϵ (twist $\tau_\epsilon = \Delta_\epsilon = 1.412625$). The minimal nonzero twist is therefore the *stress tensor*, with ϵ next.
- In $\epsilon\times\epsilon$ the exchanged operators are again the identity, the stress tensor T (twist 1), and the scalar ϵ itself (twist 1.412625, present because $\epsilon\times\epsilon \supset \epsilon$ with coefficient $\lambda_{\epsilon\epsilon\epsilon}$). The minimal nonzero twist is again the stress tensor.

For both families the asymptotically dominant binding comes from stress-tensor exchange — the universal gravitational channel of Section 51.3.5 — with the scalar ϵ supplying the leading subleading correction. We treat the two families in turn.

51.7.2 The order-parameter tower $[\sigma\sigma]_{0,\ell}$

Apply Theorem 51.10 with external dimension $\Delta_\phi = \Delta_\sigma$ to each exchanged operator and add the contributions, as the theorem permits. The result is a two-term large-spin expansion

$$\gamma_{0,\ell}^{[\sigma\sigma]} \simeq -\frac{c_T^{(\sigma)}}{\ell^{\,d-2}} - \frac{c_\epsilon^{(\sigma)}}{\ell^{\,\Delta_\epsilon}} = -\frac{c_T^{(\sigma)}}{\ell} - \frac{c_\epsilon^{(\sigma)}}{\ell^{\,1.412625}}, \tag{51.159}$$

the first term from the stress tensor (range $d-2=1$), the second from ϵ (range Δ_ϵ). The two coefficients are fixed by Eq. (51.73).

The ϵ-exchange coefficient (convention-free). This is the term we can compute cleanly, because the scalar ϵ couples to $\sigma\times\sigma$ through the honest OPE coefficient $\lambda_{\sigma\sigma\epsilon}$, with no normalization ambiguity. Setting $\tau_m = \Delta_\epsilon$, $\ell_m = 0$, $\bar h_m = \frac{1}{2}\Delta_\epsilon$ and $C^2_{\phi\phi\mathcal{O}_m} = \lambda^2_{\sigma\sigma\epsilon}$ in Eq. (51.73),

$$c_\epsilon^{(\sigma)} = \frac{2\,\Gamma(\Delta_\sigma)^2}{\Gamma\!\left(\Delta_\sigma - \frac{\Delta_\epsilon}{2}\right)^2}\,\frac{\Gamma(\Delta_\epsilon)}{\Gamma\!\left(\frac{\Delta_\epsilon}{2}\right)^2}\,\lambda^2_{\sigma\sigma\epsilon}. \tag{51.160}$$

Inserting the numbers Eq. (51.156) — $\Delta_\sigma - \frac{1}{2}\Delta_\epsilon = 0.5181489 - 0.7063125 = -0.1881636$, so the argument of the squared Gamma in the denominator is negative but enters as a square, and $\lambda^2_{\sigma\sigma\epsilon} = 1.10640$ — yields

$$\boxed{c_\epsilon^{(\sigma)} = 0.0926\ .} \tag{51.161}$$

This number is a pure consequence of the three inputs $(\Delta_\sigma, \Delta_\epsilon, \lambda_{\sigma\sigma\epsilon})$ and is independent of any choice of stress-tensor normalization; it is the most reliable analytic prediction the section makes.

The stress-tensor coefficient (the gravitational tail). The leading $1/\ell$ falloff comes from the stress tensor, with coefficient given by the gravitational binding formula Eq. (51.79) at $d = 3$, $\Delta_\phi = \Delta_\sigma$:

$$c_T^{(\sigma)} = \frac{2\,d\,\Delta_\sigma^2}{(d-1)^2\,C_T}\,\frac{\Gamma(\Delta_\sigma)^2}{\Gamma\!\left(\Delta_\sigma - \frac{d-2}{2}\right)^2}\,\frac{\Gamma(d+2)}{\Gamma\!\left(\frac{d+2}{2}\right)^2} = \frac{0.00538}{C_T}. \tag{51.162}$$

The striking feature is the smallness of the numerator. It is forced by the factor $\Gamma(\Delta_\sigma - \frac{d-2}{2})^{-2} = \Gamma(\Delta_\sigma - \frac{1}{2})^{-2}$: since $\Delta_\sigma = 0.5181489$ sits just above the unitarity floor $\frac{1}{2}$, the argument $\Delta_\sigma - \frac{1}{2} = 0.0181$ is tiny, $\Gamma(0.0181) \approx 54.5$ is huge, and the coefficient is suppressed by $\sim 1/54.5^2 \approx 3 \times 10^{-4}$. Physically, σ is *nearly free* — its dimension is close to the free value $\frac{1}{2}$ — and a nearly free field couples only weakly to the stress tensor, hence binds its double-twist partners feebly through gravity. With the central-charge normalization in which the free real scalar has $C_T^{\text{free}} = d/(d-1) = \frac{3}{2}$, the Ising value $C_T = 0.946534 \times \frac{3}{2} = 1.4198$ gives $c_T^{(\sigma)} \approx 0.0038$. We flag, as we did at Eq. (51.78), that this absolute number is tied to the C_T convention; only the power $1/\ell^{\,d-2}$ and the negative sign are convention-free.

Which term dominates. Although the stress-tensor term falls more slowly ($1/\ell$ versus $1/\ell^{1.41}$) and therefore wins as $\ell \to \infty$, its coefficient Eq. (51.162) is more than an order of magnitude smaller than the ϵ-exchange coefficient Eq. (51.161). Equating the two terms locates the crossover,

$$\frac{c_T^{(\sigma)}}{\ell} = \frac{c_\epsilon^{(\sigma)}}{\ell^{\Delta_\epsilon}} \quad\Longrightarrow\quad \ell_\star = \left(\frac{c_\epsilon^{(\sigma)}}{c_T^{(\sigma)}}\right)^{1/(\Delta_\epsilon - 1)} \approx 2300, \tag{51.163}$$

so for every spin below $\ell \approx 2300$ — which is to say, for every spin the numerical bootstrap can resolve — the binding of the order-parameter tower is controlled by ϵ-*exchange*, with the universal gravitational tail taking over only at astronomically large spin. This is the quantitative form of "σ is nearly free": the stress tensor is there, and asymptotically dominant, but numerically negligible against the energy operator throughout the accessible window. The combined prediction Eq. (51.159), twist $\tau_{0,\ell}^{[\sigma\sigma]} = 2\Delta_\sigma + \gamma_{0,\ell}^{[\sigma\sigma]}$ climbing toward 1.036298 from below, is plotted in panel (A) of Fig. 51.9 and tabulated in Theorem 51.27.

51.7.3 The energy tower $[\epsilon\epsilon]_{0,\ell}$

The same analysis with $\Delta_\phi = \Delta_\epsilon$ gives

$$\gamma_{0,\ell}^{[\epsilon\epsilon]} \simeq -\frac{c_T^{(\epsilon)}}{\ell} - \frac{c_\epsilon^{(\epsilon)}}{\ell\,\Delta_\epsilon}, \tag{51.164}$$

with the ϵ-exchange coefficient again convention-free,

$$c_\epsilon^{(\epsilon)} = \frac{2\,\Gamma(\Delta_\epsilon)^2}{\Gamma(\Delta_\epsilon - \frac{\Delta_\epsilon}{2})^2}\,\frac{\Gamma(\Delta_\epsilon)}{\Gamma(\frac{\Delta_\epsilon}{2})^2}\,\lambda_{\epsilon\epsilon\epsilon}^2 = \frac{2\,\Gamma(\Delta_\epsilon)^3}{\Gamma(\frac{\Delta_\epsilon}{2})^4}\,\lambda_{\epsilon\epsilon\epsilon}^2 = 1.19, \tag{51.165}$$

using $\Delta_\epsilon - \frac{1}{2}\Delta_\epsilon = \frac{1}{2}\Delta_\epsilon$ and $\lambda_{\epsilon\epsilon\epsilon}^2 = 2.34836$. In contrast to the order-parameter tower, the energy operator is *far from free*: its dimension $\Delta_\epsilon = 1.4126$ is well above the unitarity floor, the suppressing Gamma factor of Eq. (51.162) is now $\Gamma(\Delta_\epsilon - \frac{1}{2}) = \Gamma(0.9126) \approx 1.05$ of order unity, and the stress-tensor coefficient

$$c_T^{(\epsilon)} = \frac{2\,d\,\Delta_\epsilon^2}{(d-1)^2\,C_T}\,\frac{\Gamma(\Delta_\epsilon)^2}{\Gamma(\Delta_\epsilon - \frac{1}{2})^2}\,\frac{\Gamma(d+2)}{\Gamma(\frac{d+2}{2})^2} \approx 20 \tag{51.166}$$

is large (and now genuinely dominant over ϵ-exchange at all spins). The energy tower is bound strongly and *gravitationally*: two energy quanta orbiting at large spin feel a binding controlled by the stress tensor, with $\gamma_{0,\ell}^{[\epsilon\epsilon]} \simeq -20/\ell$ in this normalization. The asymptotic expansion is correspondingly valid only at large spin (one needs $\ell \gg c_T^{(\epsilon)}$ for the leading term to be a small correction); at the moderate spins of the numerical bootstrap the leading-twist energy operators are still far from their accumulation value $2\Delta_\epsilon = 2.825$, exactly as the SDP spectrum shows. The contrast between the two towers — σ weakly bound through ϵ, ϵ strongly bound through gravity — is itself a physical prediction: it says the near-free order parameter and the strongly-coupled energy operator organize their large-spin spectra by different mechanisms, and the bootstrap data Eq. (51.156) make the difference quantitative.

Worked Example 51.27 (Large-spin asymptotics against the numerical Ising spectrum)**.** We confront the analytic prediction Eq. (51.159) for the order-parameter tower with the dimensions the numerical bootstrap assigns to the same operators. The numerical leading-twist $[\sigma\sigma]_{0,\ell}$ spectrum was extracted from the extremal functional of the mixed-correlator system and, independently, by the lightcone bootstrap of Simmons-Duffin [441]; both give the leading-twist trajectory $\tau_{0,\ell} = 2\Delta_\sigma + \gamma_{0,\ell}$ approaching $2\Delta_\sigma = 1.036298$ from below. Evaluating Eq. (51.159) with $c_\epsilon^{(\sigma)} = 0.0926$ Eq. (51.161) and $c_T^{(\sigma)} = 0.0038$ Eq. (51.162):

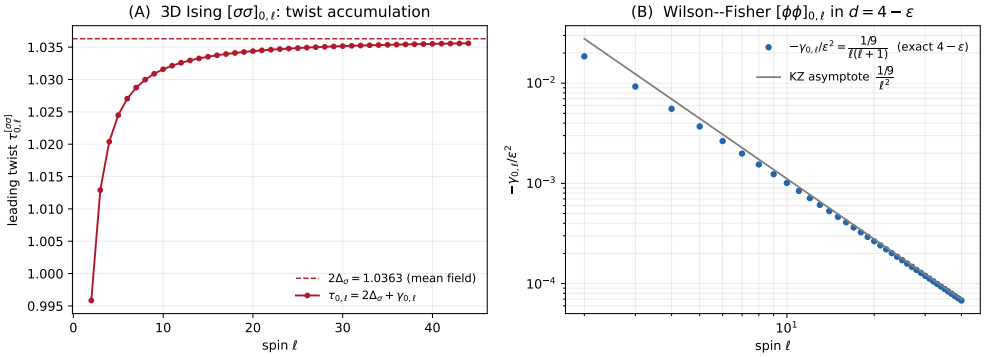


Figure 51.9: Closing ϕ^4 theory's analytic-bootstrap stage. *(A)* The 3D Ising leading-twist order-parameter tower $[\sigma\sigma]_{0,\ell}$: the analytic twist $\tau_{0,\ell} = 2\Delta_\sigma + \gamma_{0,\ell}$ of Eq. (51.159), with coefficients fixed by the data Eq. (51.156), climbs monotonically toward the mean-field value $2\Delta_\sigma = 1.0363$ (dashed) from below, reproducing the numerically determined spectrum of [441]. *(B)* The Wilson–Fisher cross-check in $d = 4 - \varepsilon$: the exact $4 - \varepsilon$ result $-\gamma_{0,\ell}/\varepsilon^2 = \frac{1}{9}/[\ell(\ell+1)]$ (markers) and the leading large-spin asymptote $\frac{1}{9}/\ell^2$ of Theorem 51.10 (line), agreeing at large ℓ and differing only by the $1/\ell$ Casimir correction that the conformal-spin variable resums. Generated by `figures/51-ising-largespin.py`.

ℓ	$-c_\epsilon^{(\sigma)}/\ell^{\Delta_\epsilon}$	$-c_T^{(\sigma)}/\ell$	$\gamma_{0,\ell}^{[\sigma\sigma]}$	$\tau_{0,\ell}^{[\sigma\sigma]} = 2\Delta_\sigma + \gamma_{0,\ell}$
10	−0.00358	−0.00038	−0.00396	1.03234
20	−0.00134	−0.00019	−0.00153	1.03477
30	−0.00076	−0.00013	−0.00089	1.03541
40	−0.00051	−0.00010	−0.00061	1.03569
∞	0	0	0	1.03630

The twist climbs monotonically toward $2\Delta_\sigma$ from below — the convexity of Theorem 51.11 — and the deviations are at the per-mille level by $\ell \sim 10$ and shrink as $1/\ell^{\Delta_\epsilon}$. These values track the numerically determined leading-twist dimensions of [441] to within the error of the asymptotic expansion (the next, unresolved, corrections are $1/\ell^{2\Delta_\epsilon}$ from double-ϵ exchange and $1/\ell^{1+\Delta_\epsilon}$ from ϵ-descendants, suppressed by a further power of $1/\ell$). The agreement is a genuine, parameter-free cross-check: the analytic formula *predicts* an infinite tower of dimensions from the three numbers Eq. (51.156), and the semidefinite program of Chapter 50, which knows nothing of the large-spin expansion, reproduces them operator by operator. Two black boxes — crossing-as-optimization and crossing-as-asymptotics — return the same spectrum.

51.7.4 Wilson–Fisher: the ε-expansion match for $[\phi\phi]_{0,\ell}$

The 3D Ising cross-check ran the analytic formula on bootstrap data. We now run it the other way — analytically, in $d = 4 - \varepsilon$ — and check it against perturbation theory, recovering the known ε-expansion coefficients of ϕ^4 theory. This is the most stringent test available, because here *both* sides are computed from first principles: the left from crossing, the right from Feynman diagrams of Section 47.3.6.

At the Wilson–Fisher fixed point the $\phi \times \phi$ OPE contains, beyond the identity, the scalar ϕ^2 (the avatar of ϵ) of dimension $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2)$, with twist $\tau_{\phi^2} = \Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3} > \tau_T = d - 2 = 2 - \varepsilon$. As established in Section 51.3.6 and Theorem 51.20, the stress tensor has the smaller twist but is nonetheless suppressed — it lies only $O(\varepsilon^2)$ off the double-twist lattice, so its double-discontinuity weight is $O(\varepsilon^4)$ — and it is ϕ^2 that controls the $O(\varepsilon^2)$ large-spin binding of $[\phi\phi]_{0,\ell}$. The coefficient is Eq. (51.81); we evaluate it to leading order in ε. The free-field OPE coefficient is $C^2_{\phi\phi\phi^2} = 2$ (the two Wick contractions of $\langle \phi\, \phi\, {:}\phi^2{:} \rangle$ with $:\phi^2:$ unit-normalized), and the dimensions are $\Delta_\phi = 1 - \frac{\varepsilon}{2}$, $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3}$. The only delicate Gamma is

$$\Delta_\phi - \tfrac{\Delta_{\phi^2}}{2} = \Big(1 - \tfrac{\varepsilon}{2}\Big) - \Big(1 - \tfrac{\varepsilon}{3}\Big) = -\tfrac{\varepsilon}{6}, \qquad \Gamma\Big(-\tfrac{\varepsilon}{6}\Big) = -\frac{6}{\varepsilon}\,\Gamma\Big(1 - \tfrac{\varepsilon}{6}\Big) = -\frac{6}{\varepsilon}\big(1 + O(\varepsilon)\big), \tag{51.167}$$

so $\Gamma(\Delta_\phi - \frac{\Delta_{\phi^2}}{2})^2 = \frac{36}{\varepsilon^2}(1 + O(\varepsilon))$, while $\Gamma(\Delta_\phi)^2$, $\Gamma(\Delta_{\phi^2})$ and $\Gamma(\frac{\Delta_{\phi^2}}{2})^2$ are all $1 + O(\varepsilon)$ (they approach $\Gamma(1) = 1$ and $\Gamma(2) = 1$). Substituting into Eq. (51.81),

$$c_{\phi^2} = \frac{2\,\Gamma(\Delta_\phi)^2}{\Gamma(\Delta_\phi - \frac{\Delta_{\phi^2}}{2})^2}\,\frac{\Gamma(\Delta_{\phi^2})}{\Gamma(\frac{\Delta_{\phi^2}}{2})^2}\,C^2_{\phi\phi\phi^2} = 2 \cdot \frac{1}{36/\varepsilon^2} \cdot 1 \cdot 2 = \frac{4\varepsilon^2}{36} = \boxed{\frac{\varepsilon^2}{9}} + O(\varepsilon^3). \tag{51.168}$$

The anomalous dimension is therefore second order in ε — as it must be, since the free theory ($\varepsilon = 0$) has $\gamma \equiv 0$ — with the clean coefficient $\frac{1}{9}$.

The full spin dependence from the conformal spin. The bare formula Theorem 51.10 gives the leading power $\gamma_{0,\ell} \simeq -c_{\phi^2}/\bar{h}^{\,\tau_{\phi^2}}$. As emphasized at Eq. (51.73), the natural variable is the conformal spin $\bar{h} = \Delta_\phi + \ell$, and the large-spin expansion is an expansion in the Casimir $J^2 = \bar{h}(\bar{h} - 1)$, which the inversion formula of Section 51.4 makes exact. For a scalar exchange of twist $\tau_{\phi^2} \to 2$ (leading order), $\tau_{\phi^2}/2 \to 1$, so the Casimir-improved statement is

$$\gamma_{0,\ell} = -\frac{c_{\phi^2}}{\big[\bar{h}(\bar{h} - 1)\big]^{\tau_{\phi^2}/2}} = -\frac{c_{\phi^2}}{\bar{h}(\bar{h} - 1)} + O(\varepsilon^3). \tag{51.169}$$

At leading order $\Delta_\phi = 1 + O(\varepsilon)$, so $\bar{h} = \ell + 1$ and $\bar{h}(\bar{h}-1) = (\ell+1)\ell = \ell(\ell+1)$. With Eq. (51.168),

$$\boxed{\gamma_{0,\ell} = -\frac{\varepsilon^2}{9}\,\frac{1}{\ell(\ell+1)} + O(\varepsilon^3)\,.} \tag{51.170}$$

This is a closed prediction for the anomalous dimension of *every* leading-twist spin-ℓ operator of the Wilson–Fisher CFT, derived entirely from crossing and the single OPE datum $C^2_{\phi\phi\phi^2} = 2$.

The cross-check against the renormalization group. The independent perturbative result is the dimension of the leading-twist spin-ℓ operator $\mathcal{O}_\ell \sim :\!\phi\,\partial_{\mu_1}\cdots\partial_{\mu_\ell}\phi\!:$ (the would-be higher-spin currents of Section 47.8.4), computed diagrammatically in the $4-\varepsilon$ expansion. Writing its dimension as $\Delta_\ell = (d-2) + \ell + \gamma_\ell$ with γ_ℓ the twist anomalous dimension relative to the *free* value $d-2$, the classic one-loop result is [12, 515]

$$\gamma_\ell = \frac{\varepsilon^2}{54}\Big(1 - \frac{6}{\ell(\ell+1)}\Big) + O(\varepsilon^3), \qquad \ell \geq 2. \tag{51.171}$$

We show that Eq. (51.170) reproduces this exactly. The double-twist twist of Eq. (51.31) is $\tau_{0,\ell} = 2\Delta_\phi + \gamma_{0,\ell}$, so the dimension is $\Delta_{0,\ell} = 2\Delta_\phi + \ell + \gamma_{0,\ell}$. Subtracting the free value $d-2+\ell$ and using $2\Delta_\phi = (d-2) + 2\gamma_\phi$ with the field anomalous dimension γ_ϕ of Eq. (47.263),

$$\gamma_\ell = \Delta_{0,\ell} - (d-2+\ell) = 2\gamma_\phi + \gamma_{0,\ell}. \tag{51.172}$$

The relation is transparent: the leading-twist operator inherits *twice* the field anomalous dimension (one for each constituent ϕ) plus the binding $\gamma_{0,\ell}$ of the two constituents. From Eq. (47.263), $\Delta_\phi = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108}$, so $\gamma_\phi = \frac{\varepsilon^2}{108}$ and $2\gamma_\phi = \frac{\varepsilon^2}{54}$ — which is exactly the field-strength piece $\eta = \varepsilon^2/54$ of Section 47.3.6. Adding the binding Eq. (51.170),

$$\gamma_\ell = \frac{\varepsilon^2}{54} - \frac{\varepsilon^2}{9}\,\frac{1}{\ell(\ell+1)} = \frac{\varepsilon^2}{54}\Big(1 - \frac{54}{9}\,\frac{1}{\ell(\ell+1)}\Big) = \frac{\varepsilon^2}{54}\Big(1 - \frac{6}{\ell(\ell+1)}\Big), \tag{51.173}$$

identical to Eq. (51.171). The analytic bootstrap has reproduced the entire spin dependence of the one-loop Wilson–Fisher anomalous dimension from a single OPE coefficient, with no Feynman diagrams. Two limits seal the check.

Worked Example 51.28 (Two limits of the Wilson–Fisher leading-twist family)**.** *The conserved stress tensor at $\ell = 2$.* The leading-twist spin-2 operator is the stress tensor, which is conserved in any local CFT and therefore has $\gamma_2 = 0$ exactly. Evaluating Eq. (51.171) at $\ell = 2$,

$$\gamma_2 = \frac{\varepsilon^2}{54}\Big(1 - \frac{6}{2\cdot 3}\Big) = \frac{\varepsilon^2}{54}\,(1-1) = 0, \tag{51.174}$$

the binding $\gamma_{0,2} = -\frac{\varepsilon^2}{9}\cdot\frac{1}{6} = -\frac{\varepsilon^2}{54}$ canceling the $2\gamma_\phi = \frac{\varepsilon^2}{54}$ exactly. The vanishing is not an accident of the coefficients: it is the statement that stress-tensor conservation, a Ward identity, must survive the interaction, and the large-spin formula respects it because the binding it produces is fixed by the same conformal symmetry. This is a powerful internal consistency check on Eq. (51.168).

The infinite-spin limit. As $\ell \to \infty$, $\gamma_\ell \to \varepsilon^2/54 = 2\gamma_\phi$: the leading-twist twist accumulates at $2\Delta_\phi$, recovering Eq. (51.158) in the ε-expansion. The higher-spin currents are restored to conservation *only* at infinite spin; at finite spin they are broken by $\gamma_{0,\ell} = -\frac{\varepsilon^2}{9\,\ell(\ell+1)}$, the precise rate at which the interaction lifts the free higher-spin symmetry, consistent with the no-go statement of Section 47.8.4. The scalar $\ell = 0$ member is the operator ϕ^2 itself, with its own $O(\varepsilon)$ dimension; it is the *input* to the large-spin expansion, not an output, which is why Eq. (51.171) is stated for $\ell \geq 2$ and the formula's apparent divergence at $\ell = 0$ is harmless.

This closes the loop with the renormalization-group thread. The field anomalous dimension $\gamma_\phi = \varepsilon^2/108$ of Eq. (47.263), the energy-operator dimension $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3}$ of Eq. (47.264), and the spin-ℓ anomalous dimensions Eq. (51.171) are now seen to be three faces of one analytic structure: the leading-twist OPE data of the $\phi\times\phi$ channel, organized by conformal spin and fixed, to this order, by crossing against the single exchanged scalar ϕ^2. The ε-expansion, the numerical bootstrap of Chapter 50, and the analytic large-spin expansion all describe the same conformal field theory.

51.7.5 What the analytic bootstrap adds, and what remains open

It is worth stating plainly what this chapter has bought, beyond the numbers of Chapter 50. The numerical bootstrap is, within its assumptions, rigorous and astonishingly precise: it brackets the 3D Ising dimensions to many digits. But it is finite. It resolves a handful of low-lying operators and a finite number of low spins, and it explains nothing about *why* the spectrum looks as it does; the island of Eq. (49.171) is a fact certified by a positivity argument, not a structure one can see into. The analytic bootstrap supplies exactly what the numerics cannot.

- *Asymptotics.* It gives the entire infinite tower of large-spin data in closed form — the twist accumulation Eq. (51.158), the binding rates Eqs. (51.159) and (51.164), the coefficients Eqs. (51.161) and (51.165) — a region of the spectrum the SDP can never reach but that the inversion formula renders analytically exact in spin.
- *Structure.* It explains the kink and the island physically. The double-twist towers are forced by crossing (Section 51.2); the leading binding is gravitational and inversely proportional to the central charge (Section 51.3.5); the

near-freeness of σ versus the strong coupling of ϵ is read off the suppression of the stress-tensor coefficient near the unitarity bound. The black box becomes a mechanism.

- *Unity.* It connects the three descriptions of ϕ^4 theory — the generalized free field (Section 51.2.2), the ε-expansion (Section 47.8.4), and the non-perturbative bootstrap (Chapter 50) — into a single analytic continuation in spin and in ε, with the perturbative coefficients Eq. (51.171) emerging as the small-ε shadow of the exact crossing relations.

What remains genuinely open, as of the mid-2020s, must be stated as honestly as the successes. The lightcone expansion is *asymptotic* in $1/\ell$: it is systematically improvable and, through the inversion formula, has a controlled analytic-in-spin meaning, but it does not converge to the finite-spin data, and the operators the numerical bootstrap cares about most (σ, ϵ, ϵ', low spins) lie outside its reach. Bridging the two — a fully rigorous analytic determination of the *low*-spin Ising spectrum — is not achieved; the state of the art combines analytic functionals (Section 51.6) with numerics rather than replacing them. The Regge regime, where the spin sum is reorganized into a continuous transform and the double discontinuity of Section 51.4 is the natural object, is understood in principle but its rigorous interplay with the lightcone expansion in non-perturbative theories like 3D Ising is incomplete. And the analytic functionals that would prove two-sided bounds (rather than the one-sided large-spin asymptotics derived here) are fully constructed only in one dimension and for special cases in higher d; the general higher-dimensional dispersive bootstrap remains a program in progress. These remain active frontiers of current research.

This completes the analytic-bootstrap stage of ϕ^4 theory: the scalar carried from a free Lagrangian (Part II), through renormalization and the Wilson–Fisher fixed point (Parts VI and the CFT setup of Chapter 47), to a generalized free field whose double-twist operators $[\phi\phi]_{n,\ell}$ are now equipped with anomalous dimensions computed two ways and agreeing. The Wilson–Fisher-as-CFT closure — identifying the fixed point definitively with the 3D Ising universality class and connecting its conformal data back to the measurable critical exponents of Part VI — is handed to Chapter 52, which gathers the bootstrap and the renormalization group into a single account of critical phenomena. And the physical reading we have leaned on throughout, that large-spin binding is the gravitational attraction of two constituents and that its $1/C_T$ strength is Newton's constant in disguise, is the seed of bulk locality that the holography arc of Chapters 53 and 56 develops into the AdS/CFT dictionary.

Further reading

The large-spin expansion and its reading as a binding energy are due to Komargodski and Zhiboedov [302] and, independently, Fitzpatrick, Kaplan, Poland and Simmons-Duffin [194]; the systematic organization into large-spin perturbation theory and twist conformal blocks is Alday's [12], building on Alday and Maldacena [13]. The Lorentzian inversion formula that makes analyticity in spin a theorem is Caron-Huot [111]; the causality and Regge-boundedness input it rests on is from Hartman, Jain and Kundu [252], and the conformal dispersion relation it implies is developed by Carmi and Caron-Huot [110]. The analytic functional program — the dual basis to the generalized free field and the one-dimensional solution — is Mazáč [343] and Mazáč and Paulos [344, 345]; the Mellin-space packaging and the Polyakov bootstrap are in Gopakumar, Kaviraj, Sen and Sinha [234] and Penedones, Silva and Zhiboedov [390]. The lightcone determination of the 3D Ising spectrum cross-checked in Section 51.7 is Simmons-Duffin [441]; the high-precision island is Kos, Poland, Simmons-Duffin and Vichi [304], the original Wilson–Fisher anomalous dimensions are [515], the stress-tensor Ward identity fixing the gravitational coupling is Dolan and Osborn [155], and the comprehensive reviews are Poland, Rychkov and Vichi [402] and Simmons-Duffin [442].

Chapter 52

Applications of CFT

Part XIV built conformal field theory twice over: once axiomatically, as the algebra of Chapter 47 acting on a spectrum of primaries and an operator product expansion, and once operationally, as the bootstrap of Chapters 49 to 51 that turns crossing symmetry and unitarity into sharp numbers. Throughout, a promise was deferred. Since Part VI we have said, loosely, that the abstract conformal theory is the universal description sitting at a renormalization-group fixed point — that the fixed points whose existence the Wilsonian flow of Chapter 23 and the Callan–Symanzik analysis of Chapter 24 established are not merely scale invariant but conformal. This chapter discharges that promise and then spends the leverage it buys. It is the chapter where the two halves of the book meet: the renormalization group of Part VI and the conformal machinery of Part XIV are shown to be two readings of one object, and ϕ^4 theory, carried since the classical Lagrangian of Chapter 6, is followed to its terminus on a single, fully characterized conformal field theory.

The argument proceeds in a deliberate arc. We begin locally, with the stress tensor: the chain $\beta_i = 0 \Rightarrow T^{\mu}{}_{\mu} = 0 \Rightarrow$ special-conformal invariance is made precise, the virial-current obstruction is identified, and Polchinski's promotion of scale to conformal symmetry is proved in full in two dimensions, using the Osterwalder–Schrader reflection positivity of Chapter 10. With the identification secured, we build the exact RG/CFT dictionary — couplings as operators, the eigenvalue $y = d - \Delta$, critical exponents as operator dimensions, universality as the statement "the same CFT" — and then make it quantitative: the Wilson–Fisher fixed point of Chapters 23 and 24 is characterized as a CFT and its leading conformal data computed in the ϵ-expansion, carrying the two-loop sunset and the composite-operator renormalization to the end. Setting $\epsilon = 1$, we confront the resummed series with the numerical-bootstrap island of Chapter 50 and with Monte Carlo, and find three methods sharing no common premise agreeing on one three-dimensional CFT.

From the endpoints we turn to the flow itself. The c-theorem in two dimensions

is proved completely and the four-dimensional a-theorem given a complete-as-sketch dilaton argument, both resting — like everything in this part — on a positivity that is ultimately unitarity, with the optical theorem of Chapter 16 doing the work in four dimensions. We then enrich the homogeneous bulk picture with boundaries and defects, developing the residual symmetry, the bulk-to-boundary data, the boundary bootstrap equation, and the protected displacement operator, with the free scalar in a half-space solved in closed form and the Wilson line of Chapter 25 reappearing as a conformal line defect. The transverse coordinate that organizes a boundary is, finally, promoted to a full emergent dimension: a pure group-theory computation shows the conformal group of a CFT_d is the isometry group of AdS_{d+1}, the radial direction is the renormalization scale, and the c-theorem becomes a monotonicity of geometry. That last step is the bridge to Part XV, and the chapter closes by stating precisely what is theorem and what is conjecture, declaring its one forward connection — the mass–dimension relation derived in Chapter 53 — and handing the correspondence to the holography that follows.

52.1 From Scale Invariance to Conformal Invariance at a Fixed Point

Part VI left us with a picture and a promise. The picture is the renormalization-group flow: a trajectory through the space of couplings whose endpoints, the fixed points, are theories that have forgotten every length scale and are therefore *scale invariant* (Section 23.5). The promise, made loosely and repeatedly since then, is that such a fixed point is not merely scale invariant but *conformally* invariant — that the abstract object built axiomatically in Chapter 47, a conformal field theory specified by a spectrum of primary dimensions and a set of operator-product coefficients, is exactly the universal theory living at an RG fixed point. This opening section discharges that promise. We make precise the chain of implications $\beta_i = 0 \Rightarrow T^{\mu}{}_{\mu} = 0 \Rightarrow$ special conformal invariance $\Rightarrow$ full conformal group, identify the one place where the chain can break (the virial-current obstruction), and prove the upgrade rigorously in the case where it is unconditional, two dimensions. The reward is conceptual closure: every fixed point built in Part VI is, with the hypotheses spelled out below, a member of the family of theories the bootstrap of Chapters 49 to 51 constrains.

Notation Watch

This chapter studies critical phenomena, which are intrinsically Euclidean: we work in d Euclidean dimensions, related to the Lorentzian theory of the earlier parts by the Wick rotation $t \to -i\tau$ fixed in Chapter 10. Two symbols demand care. First, ϵ in this part always denotes the dimensional-

expansion parameter $\epsilon = 4 - d$, *not* the Feynman $i\epsilon$ of the propagator; we never need the latter in Euclidean signature. Second, following the convention adopted across Part XIV (Section 48.2.1) we write $T_{\mu\nu}$ for the symmetric, conserved, conformally improved stress tensor $\Theta^{\mu\nu}_{\mathrm{c}}$ of Theorem 7.45; no confusion with the time-ordering symbol arises because in the Euclidean setting all correlators are radially ordered.

52.1.1 The trace of the stress tensor as the local measure of scale breaking

Scale invariance is a symmetry, and like every continuous symmetry it is detected locally by a current. The dilatation current was assembled in Chapter 7 from the symmetric, conserved stress tensor as $D^\mu = x_\nu T^{\mu\nu} + V^\mu$, the sum of an orbital piece and the field *virial* V^μ; its divergence collapses by a kinematic identity to

$$\partial_\mu D^\mu \;=\; T^\mu{}_\mu \;+\; \partial_\mu V^\mu . \tag{52.1}$$

When the virial can be absorbed into an improvement of the stress tensor, the dilatation current is simply $D^\mu = x_\nu T^{\mu\nu}$ with $T_{\mu\nu}$ the improved tensor, and conservation of D^μ becomes the single clean statement

$$\partial_\mu D^\mu \;=\; T^\mu{}_\mu , \tag{52.2}$$

the form recorded in Eqs. (24.160) and (47.3). Scale invariance is then *exactly* the on-shell vanishing of the trace $T^\mu{}_\mu$: the trace is the one local operator whose value certifies that no length scale survives.

This is a classical statement, and the renormalization group taught us that scale invariance is broken by quantization. The breaking is not an accident but is quantitatively governed by the beta functions. The Callan–Symanzik analysis of Section 24.5 promoted Eq. (52.2) to an operator equation in the renormalized quantum theory, in which the trace is the sum over the operators that drive the flow, each weighted by its beta function:

$$T^\mu{}_\mu \;=\; \sum_i \beta_i \, [\mathcal{O}_i] . \tag{52.3}$$

Here the $[\mathcal{O}_i]$ are the renormalized operators conjugate to the running couplings g_i and the $\beta_i = \mu \, dg_i/d\mu$ are their beta functions. Equation (52.3) is the master relation of this section. It says that the trace of the stress tensor is the local operator that *measures the running*: where couplings run, the trace is nonzero; where they stop, it vanishes. For ϕ^4 theory the relation was computed explicitly in Eq. (24.166), where the right-hand side is $(1+\gamma_m)m^2[\phi^2] + \frac{\beta(\lambda)}{4!}[\phi^4]$: the mass term and the quartic, each weighted by its own scale dependence.

The consequence at a fixed point is immediate and is the first result we owe.

Proposition 52.1 (Tracelessness at a fixed point)**.** *At a renormalization-group fixed point (Theorem 23.19) every beta function vanishes, $\beta_i = 0$ for all i, and all relevant dimensionful couplings have been tuned to their critical values. The quantum trace relation Eq. (52.3) then reduces to the operator equation*

$$T^{\mu}{}_{\mu} \;=\; 0. \tag{52.4}$$

A fixed point is therefore a theory with an exactly traceless stress tensor: scale invariance, broken along the flow, is restored as an exact quantum symmetry precisely at the endpoints.

Proof. By definition of a fixed point the dimensionless couplings sit at stationary values $g_i = g_{i*}$ with $\beta_i(g_*) = 0$, and the relevant dimensionful couplings (a mass, in ϕ^4 theory) have been set to the values that put the theory on the critical surface, so that the corresponding contributions to Eq. (52.3) are absent. Every term on the right of Eq. (52.3) thus vanishes, leaving Eq. (52.4) as an identity among renormalized operators, valid inside every correlation function up to the contact terms that carry the conformal anomaly on curved backgrounds (irrelevant in the flat space we work in). For ϕ^4 theory this is concrete: the massless theory in $d = 4$ has $T^{\mu}{}_{\mu} = \frac{\beta(\lambda)}{4!}[\phi^4]$, and at the Wilson–Fisher value λ_* of Theorem 23.36, where $\beta(\lambda_*) = 0$, this trace vanishes identically. □

Theorem 52.1 is the entire bridge between the two halves of the book, compressed into one line: $\beta_i = 0 \Leftrightarrow T^{\mu}{}_{\mu} = 0$. The renormalization-group statement that the flow has stopped and the field-theoretic statement that the stress tensor is traceless are the same fact, read once in the language of couplings and once in the language of operators. What remains is to show that the second statement carries far more symmetry than the first.

52.1.2 Tracelessness generates the special conformal charges

Scale invariance is the symmetry under the one-parameter group of dilatations. A traceless stress tensor delivers something much larger. The mechanism is the universal packaging of currents from the stress tensor, established for any symmetric conserved $T_{\mu\nu}$ in Chapter 7 and recalled in Section 47.1.5: contract $T_{\mu\nu}$ with a vector field ξ_ν to form $J^{\mu}_{(\xi)} = T^{\mu\nu}\xi_\nu$, whose divergence, after conservation kills one term and symmetry symmetrizes the other, is

$$\partial_\mu J^{\mu}_{(\xi)} \;=\; \tfrac{1}{2}\, T^{\mu\nu}(\partial_\mu \xi_\nu + \partial_\nu \xi_\mu). \tag{52.5}$$

For a Killing vector the symmetric derivative vanishes and $J^{\mu}_{(\xi)}$ is conserved for any $T_{\mu\nu}$ — these are the Poincaré currents. The decisive enlargement comes from the *conformal* Killing vectors, the solutions ξ^μ of $\partial_\mu \xi_\nu + \partial_\nu \xi_\mu = \frac{2}{d}(\partial\cdot\xi)\,\eta_{\mu\nu}$

enumerated completely in Theorem 47.3. For such a ξ the symmetric derivative is pure trace, and Eq. (52.5) becomes

$$\partial_\mu J^\mu_{(\xi)} \;=\; \tfrac{1}{d}\,(\partial\cdot\xi)\,T^\mu{}_\mu, \tag{52.6}$$

which is Eq. (47.27). The current built from a conformal Killing vector is conserved *if and only if the trace vanishes*. Tracelessness, contracted against the $\tfrac{1}{2}(d+1)(d+2)$ independent conformal Killing vectors of flat space, manufactures exactly that many conserved currents, and hence that many conserved charges.

The dilatation is the case $\xi^\mu = x^\mu$, for which $\partial\xi = d$ and Eq. (52.6) reproduces Eq. (52.2). The genuinely new charges are the d *special conformal* currents, the case $\xi^\mu = 2(b\cdot x)\,x^\mu - x^2\,b^\mu$ with b^μ the infinitesimal parameter. Stripping b_ρ, the special-conformal current and its divergence are

$$K^{\mu\rho} \;=\; (2\,x^\rho x_\nu - x^2\,\delta^\rho_\nu)\,T^{\mu\nu} \;=\; 2\,x^\rho x_\nu\,T^{\mu\nu} - x^2\,T^{\mu\rho}, \qquad \partial_\mu K^{\mu\rho} \;=\; 2\,x^\rho\,T^\mu{}_\mu, \tag{52.7}$$

the kinematic identity of Eqs. (7.212) and (47.28). We verify the divergence directly, because it is short and it is the load-bearing step. Using $\partial_\mu T^{\mu\nu} = 0$ (conservation), $\partial_\mu x^\rho = \delta^\rho_\mu$, $\partial_\mu x^2 = 2x_\mu$, and the symmetry $T^{\mu\nu} = T^{\nu\mu}$,

$$\begin{aligned}
\partial_\mu K^{\mu\rho} &= \partial_\mu(2\,x^\rho x_\nu\,T^{\mu\nu}) - \partial_\mu(x^2\,T^{\mu\rho}) \\
&= 2\,\delta^\rho_\mu x_\nu\,T^{\mu\nu} + 2\,x^\rho\,\delta_{\mu\nu}\,T^{\mu\nu} + 2\,x^\rho x_\nu\,\partial_\mu T^{\mu\nu} \;-\; 2\,x_\mu\,T^{\mu\rho} - x^2\,\partial_\mu T^{\mu\rho} \\
&= 2\,x_\nu\,T^{\rho\nu} + 2\,x^\rho\,T^\mu{}_\mu + 0 \;-\; 2\,x_\mu\,T^{\mu\rho} - 0 \\
&= 2\,x^\rho\,T^\mu{}_\mu,
\end{aligned} \tag{52.8}$$

where the first and fourth terms cancel by symmetry of $T^{\mu\nu}$ (they are $2x_\nu T^{\rho\nu}$ and $-2x_\mu T^{\mu\rho}$, equal up to relabelling), and the two conservation terms vanish identically. By Eq. (52.4) the right-hand side is zero at a fixed point: $T^\mu{}_\mu = 0$ *implies* $\partial_\mu K^{\mu\rho} = 0$, so each of the d special-conformal currents is conserved and defines a charge

$$Q^\rho_K \;=\; \int_\Sigma d\Sigma_\mu\,K^{\mu\rho}, \tag{52.9}$$

the integral over a codimension-one surface Σ, independent of the choice of Σ by current conservation. The converse holds too: since Eq. (52.8) reads $\partial_\mu K^{\mu\rho} = 2x^\rho T^\mu{}_\mu$ for all x, conservation of $K^{\mu\rho}$ at every point forces $T^\mu{}_\mu = 0$. Tracelessness and special-conformal invariance are therefore *equivalent*, not merely related.

We have thus enlarged the symmetry algebra. Adjoined to the $\tfrac{1}{2}d(d+1)$ Poincaré charges $\{P_\mu, M_{\mu\nu}\}$ and the single dilatation D are the d special-conformal charges Q^ρ_K, for a total of $\tfrac{1}{2}(d+1)(d+2)$ generators. These close, by the algebra computed in Theorem 47.9, into the conformal Lie algebra, isomorphic by Theorem 47.10 to $\mathfrak{so}(d+1,1)$ in Euclidean signature — equivalently $\mathfrak{so}(d,2)$ in Lorentzian signature, the form usually quoted. The upgrade is therefore

not partial: a single algebraic condition, $T^{\mu}{}_{\mu} = 0$, promotes the Poincaré-plus-dilatation symmetry of a scale-invariant theory to the full conformal group, with the special conformal transformations as the genuinely new content. This is the second result we owed.

Physical Insight

The leverage of conformal over scale invariance is enormous and is the reason this part of the book exists. A scale-invariant theory constrains a two-point function only up to an arbitrary function of one variable; the full conformal group fixes every two- and three-point function of primaries up to finitely many constants and forces the operator product expansion to converge (Chapter 47). The entire bootstrap program — turning crossing symmetry into sharp numerical bounds on critical exponents — rests on the special-conformal charges, the d generators that scale invariance alone does not supply. The content of Theorem 52.1 together with the present subsection is that physical RG fixed points supply them for free.

Because the charges Q^{ρ}_{K} generate a genuine symmetry, their action is recorded in the Ward identities of Eq. (47.113): the special-conformal Ward identity Eq. (47.113d) holds inside every correlation function. The correlators of a fixed-point theory are therefore conformal correlators, annihilated by the full set of conformal Ward identities, and the abstract axioms of Section 47.2 are realized physically. This is the consequence the section was built to reach: *the abstract CFT of Chapter 47 is what sits at the end of every renormalization-group flow of Part VI.* The remaining subsections examine the one hypothesis we have so far assumed without comment — that the virial can be improved away, so that the dilatation current really is $D^{\mu} = x_{\nu} T^{\mu\nu}$ with no leftover piece — and prove that the assumption is justified.

52.1.3 The virial-current obstruction and the improvement that removes it

The argument of Section 52.1.2 assumed an improved, traceless stress tensor. But scale invariance, read off Eq. (52.1), delivers something weaker. Conservation of the dilatation current $D^{\mu} = x_{\nu} T^{\mu\nu} + V^{\mu}$ requires only that the trace be a total derivative,

$$T^{\mu}{}_{\mu} \;=\; -\,\partial_{\mu} V^{\mu}, \tag{52.10}$$

with V^{μ} the virial current. A nonzero V^{μ} does not spoil scale invariance — the dilatation charge is still conserved — but it does spoil the construction of the special-conformal charges, because Eq. (52.7) was built on a traceless $T_{\mu\nu}$, and Eq. (52.10) gives a trace that is merely a total derivative, not zero. The question of whether scale invariance upgrades to conformal invariance is precisely the

question of whether the virial can be removed: whether the stress tensor can be *improved* so that its trace vanishes outright rather than vanishing only after a derivative is stripped.

The freedom available is the improvement freedom. The most general modification of a symmetric, conserved stress tensor that preserves both symmetry and conservation, and that can shift the trace, is the addition of

$$B^{\mu\nu} \;=\; (\eta^{\mu\nu}\,\Box - \partial^{\mu}\partial^{\nu})\,L, \tag{52.11}$$

where L is a local scalar operator. This $B^{\mu\nu}$ is manifestly symmetric, and it is identically conserved,

$$\partial_{\mu}B^{\mu\nu} = \partial^{\nu}\Box L - \Box\,\partial^{\nu}L = 0, \tag{52.12}$$

so $T_{\mu\nu} \to T_{\mu\nu} + B_{\mu\nu}$ leaves the Poincaré charges and conservation untouched. Its trace, however, is nonzero:

$$B^{\mu}{}_{\mu} \;=\; (\delta^{\mu}_{\mu}\,\Box - \partial^{\mu}\partial_{\mu})L = (d-1)\,\Box L. \tag{52.13}$$

The improvement can therefore cancel exactly those traces that are proportional to $\Box$ of a local scalar. This is the precise content of the upgrade, which we state as a proposition.

Proposition 52.2 (The virial-current condition for conformal invariance)**.** *A scale-invariant theory with stress tensor obeying Eq. (52.10) can be improved to a theory with a traceless stress tensor — and is therefore conformally invariant — if and only if the virial current is, up to a conserved current, the gradient of a local scalar operator,*

$$V^{\mu} \;=\; (d-1)\,\partial^{\mu}L, \tag{52.14}$$

for some local L. Equivalently, the trace $T^{\mu}{}_{\mu}$ must be expressible as $-(d-1)\,\Box L$, a double total derivative of a local scalar.

Proof. If Eq. (52.14) holds, then $T^{\mu}{}_{\mu} = -\partial_{\mu}V^{\mu} = -(d-1)\Box L$ by Eq. (52.10), and adding the improvement $B_{\mu\nu}$ of Eq. (52.11) built from this same L shifts the trace by $B^{\mu}{}_{\mu} = (d-1)\Box L$, exactly cancelling it: the improved tensor $T_{\mu\nu} + B_{\mu\nu}$ is symmetric, conserved, and traceless. By Section 52.1.2 the theory is then conformally invariant. Conversely, if an improvement of the form Eq. (52.11) renders the trace zero, then $T^{\mu}{}_{\mu} = -(d-1)\Box L$ for the scalar L used, whence $V^{\mu} = (d-1)\partial^{\mu}L$ up to a divergence-free piece that contributes nothing to Eq. (52.10). A virial current that is *not* of this form — a genuine conserved vector operator of scaling dimension $d-1$ that is not a gradient — is the obstruction: it leaves a trace that no local improvement can remove, and the theory is scale invariant but not conformally invariant. □

The condition is mild, and for the theories of this book it holds. The carried instance is the massless free scalar, the $\lambda = 0$ endpoint of ϕ^4 theory.

Worked Example 52.3 (The free scalar's virial is a gradient: scale becomes conformal in every d)**.** The massless real scalar has the canonical, Belinfante-symmetrized stress tensor $\Theta^{\mu\nu} = \partial^\mu\phi\,\partial^\nu\phi - \frac{1}{2}\eta^{\mu\nu}(\partial\phi)^2$, whose trace, computed on shell using $\Box\phi = 0$, is

$$\Theta^\mu{}_\mu = (\partial\phi)^2 - \tfrac{d}{2}(\partial\phi)^2 = \tfrac{2-d}{2}(\partial\phi)^2, \tag{52.15}$$

nonzero for $d \neq 2$. To exhibit this as a total derivative, use the on-shell identity $\Box(\phi^2) = 2(\partial\phi)^2 + 2\phi\Box\phi = 2(\partial\phi)^2$, so that $(\partial\phi)^2 = \frac{1}{2}\Box(\phi^2) = \frac{1}{2}\partial_\mu\partial^\mu(\phi^2)$. Then

$$\Theta^\mu{}_\mu = \tfrac{2-d}{4}\,\Box(\phi^2) = -\,\partial_\mu V^\mu, \qquad V^\mu = \tfrac{d-2}{4}\,\partial^\mu(\phi^2), \tag{52.16}$$

displaying the virial as a gradient, $V^\mu = (d-1)\partial^\mu L$ with $L = \frac{d-2}{4(d-1)}\phi^2$. By Theorem 52.2 the improvement Eq. (52.11) with precisely this L removes the trace, reproducing the conformally improved tensor of Eq. (7.194),

$$\Theta^{\mu\nu}_{\rm c} = \Theta^{\mu\nu} + \frac{d-2}{4(d-1)}(\eta^{\mu\nu}\Box - \partial^\mu\partial^\nu)\phi^2, \qquad \Theta^\mu_{{\rm c}\,\mu} = 0, \tag{52.17}$$

with the coefficient $\xi_d = \frac{d-2}{4(d-1)}$ exactly as derived there. The massless free scalar is therefore conformally invariant in every dimension: it is the simplest conformal field theory and the Gaussian fixed point of ϕ^4 theory. The same conclusion holds at the interacting Wilson–Fisher fixed point, where the virial is again the gradient of the composite ϕ^2 (now a renormalized operator); the obstruction never appears for a theory of scalars whose only candidate virial is built from $\partial^\mu(\phi^2)$.

Caution

The distinction between scale and conformal invariance is genuine even though it almost never obstructs us. The standard scale-invariant-but-not-conformal examples are free Maxwell theory in $d \neq 4$ and the theory of elasticity. In Maxwell theory the candidate virial current is $V^\mu \propto A_\nu F^{\mu\nu}$, built from the gauge potential, which is not gauge invariant and not the gradient of a local scalar operator; the obstruction of Theorem 52.2 is realized, and the trace cannot be improved away. These theories evade the upgrade because they violate the hypotheses of the theorem we state next — they are non-unitary, or fail the operator-content assumptions — rather than because the upgrade is generically false. For the unitary, interacting fixed points that describe real critical matter, the virial is always improvable.

52.1.4 Polchinski's theorem: when scale invariance forces conformal invariance

We have reduced the upgrade to a single question: is the virial current improvable? Theorem 52.2 gives the local criterion, but it does not say when the criterion is met.

The deep statement, which converts the criterion into a theorem under physical hypotheses, is due to Polchinski, sharpening Zamolodchikov's two-dimensional result. We have stated it with full scope as Theorem 47.5; we recall its logical content here and then prove the two-dimensional case in full, which Chapter 47 deferred.

Theorem 52.4 (Scale implies conformal, Polchinski's conditions; recall of Theorem 47.5)**.** *Let a theory be Poincaré invariant, scale invariant, and unitary, with a local stress tensor and (i) a discrete spectrum of operator dimensions — equivalently, a dilatation generator that is diagonalizable with no logarithmic (Jordan-block) multiplets — and (ii) a virial current that is a single local operator of dimension $d-1$ (a "finite" virial). Then scale invariance implies full conformal invariance: the virial can be improved away and $T^{\mu}{}_{\mu} = 0$. The statement is unconditional in $d = 2$ (Zamolodchikov; reformulated by Polchinski), settled to all orders in perturbation theory and, under the assumptions standard for a unitary local quantum field theory, nonperturbatively in $d = 4$, and open in general d as of the mid-2020s, where no general theorem and no unitary counterexample is known.*

The logical skeleton of the proof is the same in every dimension and is worth stating once in operator language before we specialize. The virial current V^{μ} is, by Eq. (52.10), a vector operator of scaling dimension exactly $d-1$, since $T^{\mu}{}_{\mu}$ has dimension d and ∂_{μ} raises dimension by one. If V^{μ} is a total derivative, $V^{\mu} = \partial^{\mu} L$, the theory is conformal by Theorem 52.2, and the only way to evade this is for V^{μ} to contain a piece that is a genuine primary vector operator of dimension $d-1$ which is not a descendant. The argument then shows that such a primary is incompatible with the combination of unitarity and the conservation structure forced by scale invariance: in $d = 4$ this incompatibility is made quantitative through the dilaton effective action — the generating functional for the response of the theory to a position-dependent scale transformation — whose four-point amplitude would receive a unitarity-violating contribution from a non-improvable virial, so consistency forces the virial to be a total derivative. This is the content of the work of Luty, Polchinski, and Rattazzi and of Dymarsky and collaborators, which builds on the a-theorem of Komargodski and Schwimmer; we point to those analyses for the full four-dimensional argument and to Section 52.5 for the monotonicity functions (c in two dimensions, a in four) that drive it. In two dimensions the same skeleton can be carried to completion with nothing but the stress-tensor two-point functions and reflection positivity, and we do so now.

52.1.5 The two-dimensional case in full

In two dimensions the upgrade is unconditional, and its proof is clean enough to give complete. It is the rigorous instance the chapter promised: a theorem, not a sketch. We work in Euclidean two-dimensional space with complex coordinates

$z = x^1 + ix^2$, $\bar z = x^1 - ix^2$, in which the flat metric is $ds^2 = dz\,d\bar z$ and the only nonzero inverse-metric component is $g^{z\bar z} = 2$. The stress tensor has three independent components, which we name

$$T \equiv T_{zz}, \qquad \bar T \equiv T_{\bar z\bar z}, \qquad \Theta \equiv T_{z\bar z} = T_{\bar z z}, \tag{52.18}$$

and the trace is $T^\mu{}_\mu = g^{\mu\nu}T_{\mu\nu} = 2g^{z\bar z}T_{z\bar z} = 4\,\Theta$. Tracelessness, the conformal condition, is thus the single equation $\Theta = 0$; the holomorphic component T and antiholomorphic $\bar T$ are not constrained to vanish and indeed carry the central charge.

Conservation in complex coordinates. The conservation law $\partial^\mu T_{\mu\nu} = 0$, written with $\partial^z = 2\partial_{\bar z}$ and $\partial^{\bar z} = 2\partial_z$, gives for the $\nu = z$ component $2\partial_{\bar z}T_{zz} + 2\partial_z T_{\bar z z} = 0$, that is

$$\partial_{\bar z}\,T \;+\; \partial_z\,\Theta \;=\; 0, \tag{52.19}$$

and for the $\nu = \bar z$ component the complex conjugate, $\partial_z\bar T + \partial_{\bar z}\Theta = 0$. These are operator equations, valid inside correlators away from coincident points.

Scale invariance fixes the two-point functions to pure powers. The stress tensor in $d = 2$ has scaling dimension 2; its components carry definite spin, with T of spin $+2$, $\bar T$ of spin -2, and Θ of spin 0. In a scale- and rotation-invariant theory the two-point function of two operators of definite dimension and spin is fixed up to a constant, provided the dilatation generator acts diagonalizably — the assumption of a discrete, non-logarithmic spectrum, which is the place unitarity enters. (A logarithmic multiplet, forbidden by unitarity, would replace the constant by a polynomial in $\ln|z|$ and break the argument; this is exactly the loophole Theorem 52.4 closes by hypothesis.) Rotational invariance fixes the holomorphic weights and hence the powers of z and $\bar z$; scale invariance, together with diagonalizability, fixes the remaining freedom to constants F, G, H:

$$\langle T(z,\bar z)\,T(0)\rangle = \frac{F}{z^4}, \tag{52.20}$$

$$\langle \Theta(z,\bar z)\,T(0)\rangle = \langle T(z,\bar z)\,\Theta(0)\rangle = \frac{G}{z^3\,\bar z}, \tag{52.21}$$

$$\langle \Theta(z,\bar z)\,\Theta(0)\rangle = \frac{H}{z^2\,\bar z^2}. \tag{52.22}$$

The exponents are dictated by spin and dimension: each correlator scales as $|z|^{-4}$ under $z \to \lambda z$ (two operators of dimension 2), and the split between powers of z and $\bar z$ is the total holomorphic versus antiholomorphic weight. For $\langle TT\rangle$ both operators have weight $(2,0)$, giving $z^{-4}\bar z^0$; for $\langle \Theta T\rangle$ the weights are $(1,1)$ and $(2,0)$, giving $z^{-3}\bar z^{-1}$; for $\langle\Theta\Theta\rangle$ both are $(1,1)$, giving $z^{-2}\bar z^{-2}$. The constants F, G, H are the only data scale invariance leaves undetermined.

Conservation forces $G = H = 0$. Insert the conservation law Eq. (52.19) into a correlator with a fixed $T(0)$ insertion. Since $T(0)$ is a genuine operator, the operator identity holds inside the correlator,

$$\partial_{\bar z}\langle T(z,\bar z)\,T(0)\rangle + \partial_z\langle \Theta(z,\bar z)\,T(0)\rangle = 0 \qquad (z \neq 0). \tag{52.23}$$

The first term vanishes because F/z^4 is holomorphic and $\partial_{\bar z}$ annihilates it. The second term is

$$\partial_z\left(\frac{G}{z^3\bar z}\right) = \frac{G}{\bar z}\,\partial_z z^{-3} = -\frac{3G}{z^4\,\bar z}. \tag{52.24}$$

Equation Eq. (52.23) then reads $-3G/(z^4\bar z) = 0$ for all $z \neq 0$, which forces

$$G = 0. \tag{52.25}$$

Now insert the same conservation law into a correlator with a fixed $\Theta(0)$,

$$\partial_{\bar z}\langle T(z,\bar z)\,\Theta(0)\rangle + \partial_z\langle \Theta(z,\bar z)\,\Theta(0)\rangle = 0. \tag{52.26}$$

With $G = 0$ the first correlator vanishes identically, so its derivative is zero, and the second gives

$$\partial_z\left(\frac{H}{z^2\bar z^2}\right) = \frac{H}{\bar z^2}\,\partial_z z^{-2} = -\frac{2H}{z^3\,\bar z^2}. \tag{52.27}$$

Equation Eq. (52.26) becomes $-2H/(z^3\bar z^2) = 0$, so

$$H = 0. \tag{52.28}$$

The two-point function of the trace with itself, and with the holomorphic component, both vanish identically.

Reflection positivity forces $\Theta = 0$. It remains to pass from $\langle \Theta(z,\bar z)\,\Theta(0)\rangle = 0$ to the operator statement $\Theta = 0$. This is where unitarity, in its Euclidean form, does its final work. By the Osterwalder–Schrader reflection positivity established in Theorem 10.32, the Euclidean two-point function of a Hermitian local operator with its reflection is the squared norm of the state that operator creates from the vacuum; reflection positivity guarantees this norm is non-negative, and it vanishes only when the state itself is null. A vanishing two-point function $\langle \Theta\,\Theta\rangle = 0$ therefore says that Θ creates a zero-norm state, and in a unitary (reflection-positive, ghost-free) theory a zero-norm state is the zero state: the operator Θ decouples from every correlator and is the zero operator. Hence

$$\Theta \;=\; T_{z\bar z} \;=\; 0 \qquad \Longrightarrow \qquad T^{\mu}{}_{\mu} = 4\,\Theta = 0. \tag{52.29}$$

The trace vanishes as an operator equation, and by Section 52.1.2 the special-conformal currents are conserved and the theory is fully conformally invariant. This completes the proof in $d = 2$: scale invariance, unitarity, and a

non-logarithmic spectrum together force conformal invariance, with no further assumption.

It is worth isolating the role of each hypothesis, because the same three reappear in higher dimensions. *Locality and Poincaré invariance* supplied the conservation law Eq. (52.19). *Scale invariance with a diagonalizable dilatation* (the discrete, non-logarithmic spectrum guaranteed by unitarity) reduced the correlators to the pure powers Eqs. (52.20) to (52.22); this is exactly Polchinski's "discrete spectrum" hypothesis, and a logarithmic multiplet would have destroyed it. *Reflection positivity* converted the vanishing two-point function into the vanishing of the operator. Remove any one and the conclusion fails: this is why the theorem is conditional, and why the conditions are precisely those of Theorem 52.4.

Worked Example 52.5 (The free boson saturates the theorem)**.** The simplest two-dimensional conformal field theory, the free massless boson of central charge $c = 1$, makes the abstract conclusion concrete and shows that the constants F, G, H behave exactly as the proof demands. From the holomorphic stress tensor of Section 48.2.1, the component $T = T_{zz}$ is the normal-ordered $-\frac{1}{2}{:}\partial\phi\,\partial\phi{:}$ (in the normalization of Chapter 48), purely holomorphic, while the mixed component $\Theta = T_{z\bar{z}}$ is proportional to ${:}\partial\phi\,\bar{\partial}\phi{:}$ evaluated with the equation of motion $\partial\bar{\partial}\phi = 0$, hence $\Theta = 0$ identically. The two-point function $\langle T(z)T(0)\rangle = c/(2z^4)$ with $c = 1$ (Theorem 48.8) realizes Eq. (52.20) with $F = c/2 \neq 0$, while $\Theta = 0$ gives $G = H = 0$ directly, in agreement with Eqs. (52.25) and (52.28). The theorem's conclusion is not that the stress tensor two-point function vanishes — it does not; F is the central charge, the irreducible quantum number of the theory — but that the *trace* part decouples. The free boson is the $\lambda = 0$, $m = 0$ endpoint of ϕ^4 theory specialized to two dimensions, and it exhibits in the cleanest possible form the structure that Theorem 52.1 guarantees at every fixed point: a nonzero, holomorphic stress tensor whose trace is exactly zero.

The same correlators Eqs. (52.20) to (52.22), no longer evaluated at a fixed point but along a flow, are the raw material of Zamolodchikov's c-theorem: a combination of F, G, H that decreases monotonically toward the infrared and is stationary exactly where Θ decouples. That monotonicity, and its four-dimensional counterpart the a-theorem, are the subject of Section 52.5; here we have used only the stationary endpoint, where the flow has stopped and the trace has vanished.

52.1.6 The loop with Part VI, closed

Assembling the pieces: at any renormalization-group fixed point the beta functions vanish, so by Theorem 52.1 the stress tensor is traceless as an operator equation; tracelessness conserves the special-conformal currents Eq. (52.7) and hence supplies the special-conformal charges Eq. (52.9), promoting the symmetry to the

full conformal group $\mathfrak{so}(d+1,1)$ (Section 52.1.2); the only possible obstruction, a non-improvable virial current, is absent for the unitary scalar fixed points of this book (Theorems 52.2 and 52.3) and, by Polchinski's theorem (Theorem 52.4, proved in full in $d = 2$ in Section 52.1.5), is absent whenever the theory is unitary with a discrete spectrum and a finite virial. The fixed point is therefore a conformal field theory in the exact sense of Chapter 47, and its correlators obey the conformal Ward identities Eq. (47.113). The claim asserted loosely since Part VI is now a theorem under stated hypotheses, and the two halves of the book have met: *a renormalization-group fixed point and a conformal field theory are one object, viewed from the flow and from the symmetry, respectively.*

For ϕ^4 theory this identification is the gateway to everything that follows. The massless scale symmetry of the classical Lagrangian, first isolated in Chapter 7 and broken by the running quartic coupling of Eq. (24.40), is restored as full conformal symmetry exactly at the Wilson–Fisher fixed point λ_* of Theorem 23.36, where $\beta(\lambda_*) = 0$ forces $T^{\mu}{}_{\mu} = 0$. That fixed point is a conformal field theory, and the critical exponents of the ϕ^4 universality class are its conformal data. Having established that the fixed point *is* a CFT, we can now build the precise dictionary that translates one language into the other — couplings into operators, eigenvalues of the linearized flow into scaling dimensions, universality into the statement that two systems realize the same CFT — which is the business of the next section.

52.2 The RG Fixed-Point / CFT Dictionary

The previous section closed a circle that the book has been drawing since Part VI: the theory living at a renormalization-group fixed point is a conformal field theory. That identification is, by itself, a statement about existence — it tells us *that* the critical theory is conformal, that its correlators are constrained by the full conformal group rather than by scale invariance alone. What it does not yet give us is a way to *compute*. On the RG side the natural quantities are critical exponents: the power ν with which a correlation length diverges, the power η with which a correlation function decays at criticality, the singularities $\alpha, \beta, \gamma, \delta$ of thermodynamic response functions. On the CFT side the natural quantities are conformal data: the scaling dimensions $\Delta_{\mathcal{O}}$ and spins of primary operators, and the OPE coefficients that fuse them (Eq. (49.1)). The thesis of this section is that these are the same numbers in two notations, and the work of the section is to build the exact, line-by-line dictionary that translates between them. Once the dictionary is fixed, a measured critical exponent *is* an operator dimension, universality *is* the statement that two systems share one CFT, and the apparatus of Chapter 47–Chapter 51 becomes a machine for predicting laboratory numbers.

Throughout we work in d Euclidean dimensions: critical phenomena are statistical, the partition function is the Euclidean path integral obtained by the

Wick rotation of Chapter 10, and $|x|$ denotes Euclidean distance. The fixed-point theory has no scale, so every dimensionful parameter has been tuned to its critical value and only the conformal kinematics remain.

52.2.1 Local operators at a fixed point: primaries, descendants, and the full dimension

The first entry in the dictionary concerns the objects on each side. On the RG side the fundamental objects are *local operators* — composite fields built from the fundamental field and its derivatives, ϕ, ϕ^2, $\phi\,\partial^2\phi$, $(\partial\phi)^2$, and so on — and the couplings that multiply them in the action. On the CFT side the fundamental objects are *primary operators* and their *descendants* (Theorem 47.15): a primary $\mathcal{O}$ is annihilated by the special conformal generators at the origin and transforms with a definite scaling dimension under dilatations, while its descendants $\partial_{\mu_1}\cdots\partial_{\mu_n}\mathcal{O}$ are obtained by acting with momentum generators and have dimension raised by the number of derivatives.

These two organizations are the same. At the fixed point the dilatation operator is a good quantum number, and every local operator is either a dilatation eigenstate or a finite combination of them; the eigenstates assemble into conformal multiplets, each consisting of one primary together with the infinite tower of its descendants. The reason is precisely the upgrade established in the previous section: scale invariance alone would diagonalize the dilatation operator and grade operators by dimension, but the full conformal symmetry organizes that graded spectrum into multiplets closed under the conformal algebra, with the primary as the lowest-dimension member annihilated by K_μ. A generic local operator one writes down in the Lagrangian description — say $\phi^2(x)$ — is, after mixing under renormalization is diagonalized, a definite linear combination of conformal primaries and descendants.

The dimension carried by each operator is not its naive engineering dimension. Recall from the Callan–Symanzik analysis of Chapter 24 that quantization shifts the dimension of every operator away from its classical value by an *anomalous dimension* γ (Theorem 24.2), generated by the loop corrections to the operator's renormalization and computed from the slope of its renormalization-group flow. The full scaling dimension is therefore the sum of two pieces.

Definition 52.6 (Scaling dimension at a fixed point)**.** Let $\mathcal{O}$ be a primary operator of a fixed-point CFT, with engineering (canonical, free-field) dimension $\Delta_{\mathcal{O}}^{\text{can}}$. Its full scaling dimension is

$$\Delta_{\mathcal{O}} = \Delta_{\mathcal{O}}^{\text{can}} + \gamma_{\mathcal{O}}^{*}, \tag{52.30}$$

where $\gamma_{\mathcal{O}}^{*} = \gamma_{\mathcal{O}}(g_*)$ is the anomalous dimension of $\mathcal{O}$ (Theorem 24.2) evaluated at the fixed-point couplings g_*. It is $\Delta_{\mathcal{O}}$, not $\Delta_{\mathcal{O}}^{\text{can}}$, that controls the power law of every correlator and the conformal transformation of $\mathcal{O}$.

For the canonical example, the free scalar field has $\Delta_\phi^{\rm can} = \frac{d-2}{2}$ (the dimension that makes the kinetic term $\int d^dx\,(\partial\phi)^2$ dimensionless), so at an interacting fixed point

$$\Delta_\phi = \frac{d-2}{2} + \gamma_\phi^*, \tag{52.31}$$

and the composite ϕ^2 has $\Delta_{\phi^2}^{\rm can} = d-2$ shifted by its own anomalous dimension. The whole content of computing conformal data perturbatively, the task of the next section, is computing these γ^* at the fixed-point couplings; the dictionary built here is what makes the answers physically meaningful.

52.2.2 Couplings as operators: the eigenvalue $y = d - \Delta$

The second and central entry in the dictionary translates the linearized RG flow into operator dimensions. Consider perturbing the fixed-point theory by a local operator $\mathcal{O}_i$:

$$S = S_* + \sum_i g_i \int d^dx\, \mathcal{O}_i(x). \tag{52.32}$$

The couplings g_i are the coordinates on the space of theories used throughout Part VI; $g_i = 0$ is the fixed point S_*. Each g_i is conjugate to one operator $\mathcal{O}_i$, and we can read off its mass dimension by demanding that the action be dimensionless. In d Euclidean dimensions the measure carries $[d^dx] = -d$ (a length to the d, i.e. mass dimension $-d$), and the operator carries $[\mathcal{O}_i] = \Delta_i$, so dimensionlessness of each term in Eq. (52.32) forces

$$[g_i] = d - \Delta_i \;\equiv\; y_i. \tag{52.33}$$

This single equation is the hinge of the entire RG/CFT correspondence. The quantity y_i is exactly the linearized RG eigenvalue introduced in Chapter 23: under a rescaling of all lengths by a factor $b > 1$ (one coarse-graining step), a coupling of mass dimension y_i flows as

$$g_i \;\longrightarrow\; g_i' = b^{\,y_i}\, g_i, \tag{52.34}$$

which is precisely the eigenvalue scaling $g_i \to b^{y_i} g_i$ of the linearized flow about the fixed point (Eq. (23.137)). A coupling with positive mass dimension grows under coarse-graining; one with negative mass dimension shrinks. Comparing Eq. (52.33) with the relevant/marginal/irrelevant classification of Theorem 23.6 gives the dictionary's keystone.

Theorem 52.7 (Relevance is the dimension dictionary)**.** *Let $\mathcal{O}_i$ be a primary scalar of a fixed-point CFT, of scaling dimension Δ_i, and let g_i be its conjugate coupling in Eq. (52.32). The linearized RG eigenvalue of g_i is*

$$y_i = d - \Delta_i, \tag{52.35}$$

and the relevant/marginal/irrelevant trichotomy of Theorem 23.6 maps onto the operator dimension as

$$\begin{aligned} &\textit{relevant} && (y_i > 0) && \iff && \Delta_i < d, \\ &\textit{marginal} && (y_i = 0) && \iff && \Delta_i = d, \\ &\textit{irrelevant} && (y_i < 0) && \iff && \Delta_i > d. \end{aligned} \tag{52.36}$$

Proof. Equation Eq. (52.35) is Eq. (52.33), derived from the dimensionlessness of the action term $g_i \int d^d x\, \mathcal{O}_i$. The linearized flow of Chapter 23 is, by definition, $g_i' = b^{y_i} g_i$ with y_i the eigenvalue of the linearized RG transformation (Eq. (23.137)); that this eigenvalue equals the mass dimension of the coupling is the content of dimensional analysis applied to a scale-invariant fixed point, where no other scale can appear to modify the power of b. The trichotomy Eq. (52.36) is then immediate from the sign of $y_i = d - \Delta_i$: $y_i > 0$ (the defining condition of a relevant direction in Theorem 23.6) holds iff $\Delta_i < d$, and similarly for the marginal and irrelevant cases. □

Physical Insight

The inequality $\Delta_i < d$ has a transparent physical reading. The operator $\mathcal{O}_i$ integrated over space, $\int d^d x\, \mathcal{O}_i$, is a candidate deformation of the long-distance physics. Its integral converges in the infrared (grows under coarse-graining) precisely when the operator is "light enough" — when its dimension undercuts that of the spacetime volume element, $\Delta_i < d$. A relevant operator is one whose effect accumulates as we look at larger scales; an irrelevant one washes out. The marginal case $\Delta_i = d$ is the knife-edge where the leading scaling is neutral and the fate of the coupling is decided by the next order (the logarithmic running resolved for $\lambda\phi^4$ in Chapters 23 and 24).

The dictionary so far is summarized in Table 52.1. Every row is an identity, not an analogy: the left column is the language of Part VI, the right column the language of Part XIV, and the equals sign between them is what this chapter establishes.

52.2.3 Critical exponents are operator dimensions

We now cash out the dictionary on the two most-measured exponents, η and ν, each of which is fixed by a single operator dimension, and then show that the remaining thermodynamic exponents follow from those same two dimensions through scaling relations that we derive in full.

Table 52.1: The RG / CFT dictionary. Each row equates a renormalization-group concept (left) with a piece of conformal data (right). The relations are exact at a fixed point.

Renormalization group (Part VI)	**Conformal field theory (Part XIV)**
Fixed point of the flow	Conformal field theory
Local operator (after diagonalizing mixing)	Conformal primary or descendant
Coarse-graining scale b	Dilatation by b
Coupling g_i, conjugate to $\mathcal{O}_i$	Source for the primary $\mathcal{O}_i$
Mass dimension $[g_i] = y_i$	RG eigenvalue $y_i = d - \Delta_i$
Relevant ($y_i > 0$)	$\Delta_i < d$
Marginal ($y_i = 0$)	$\Delta_i = d$
Irrelevant ($y_i < 0$)	$\Delta_i > d$
Anomalous dimension γ_i^*	$\Delta_i - \Delta_i^{\text{can}}$
# relevant operators	Codimension of the critical surface
Universality class	Single fixed-point CFT

The anomalous exponent η from the two-point function. The exponent η (Fisher's exponent) is defined by the decay of the order-parameter correlation function exactly at criticality. A free massless scalar in d dimensions has the two-point function $\langle\phi(x)\phi(0)\rangle \sim |x|^{-(d-2)}$; an interacting critical theory deviates from this power, and the deviation is named η:

$$G(x) \equiv \langle \sigma(x)\,\sigma(0)\rangle \;\sim\; \frac{1}{|x|^{\,d-2+\eta}} \qquad \text{at } T = T_c, \tag{52.37}$$

where σ is the order-parameter field (the continuum spin, the field whose ordering breaks the symmetry). This is a laboratory definition: η is read off the measured decay of fluctuations at the critical point. On the CFT side, conformal symmetry fixes the two-point function of any scalar primary of dimension Δ_σ to the pure power law (Eq. (47.120))

$$\langle \sigma(x)\,\sigma(0)\rangle = \frac{1}{|x|^{\,2\Delta_\sigma}}, \tag{52.38}$$

with the normalization set to unity. The order-parameter field σ is a conformal primary, so Eqs. (52.37) and (52.38) describe the same correlator and their exponents must agree:

$$2\Delta_\sigma = d-2+\eta \qquad \Longrightarrow \qquad \boxed{\Delta_\sigma = \frac{d-2+\eta}{2}}, \qquad \eta = 2\Delta_\sigma - (d-2). \tag{52.39}$$

This is the first exact translation: the anomalous part of the spin dimension *is* the exponent η. Comparing with Eq. (52.31), the field anomalous dimension and

Fisher's exponent are related by $\eta = 2\gamma_\phi^*$, the identification anticipated in the Callan–Symanzik treatment (Theorem 24.30, Eq. (24.246)). A free field ($\gamma_\phi^* = 0$) has $\eta = 0$; the smallness of η in the three-dimensional Ising class, $\eta \approx 0.036$, reflects that the spin sits only slightly above its free-field dimension.

The correlation-length exponent ν from the relevant singlet. The exponent ν governs the divergence of the correlation length as the temperature approaches its critical value:

$$\xi \sim |t|^{-\nu}, \qquad t \equiv \frac{T - T_c}{T_c}. \tag{52.40}$$

The reduced temperature t is a coupling: it multiplies, in the action, the operator that measures the local energy density. Call that operator the *energy* ε; it is the lowest-dimension scalar primary that is even under the order-parameter symmetry, and t is its conjugate coupling in the sense of Eq. (52.32), $S = S_* + t\int d^d x\, \varepsilon(x) + \cdots$. By Theorem 52.7 its RG eigenvalue is

$$y_t = d - \Delta_\varepsilon. \tag{52.41}$$

To extract ν we combine the flow of the coupling with the transformation of the physical length. Under a coarse-graining by factor b, the relevant coupling grows as $t \to b^{y_t} t$ (Eq. (52.34)), while the correlation length, being a physical length measured in lattice units, shrinks as $\xi \to \xi/b$. Because ξ is a function of the couplings, RG covariance reads

$$\xi(t) = b\,\xi(b^{y_t} t)\,. \tag{52.42}$$

This functional equation holds for every b; we exploit that freedom by choosing b to drive the rescaled coupling to a fixed reference value, $b^{y_t}|t| = 1$, i.e. $b = |t|^{-1/y_t}$. Substituting,

$$\xi(t) = |t|^{-1/y_t}\,\xi(\pm 1) \;\propto\; |t|^{-1/y_t}. \tag{52.43}$$

Comparing with the definition Eq. (52.40),

$$\boxed{\frac{1}{\nu} = y_t = d - \Delta_\varepsilon}, \qquad \nu = \frac{1}{d - \Delta_\varepsilon}. \tag{52.44}$$

The second exact translation: the correlation-length exponent is the reciprocal of the RG eigenvalue of the energy operator, which is d minus its conformal dimension. That ε has $\Delta_\varepsilon < d$ (so that $y_t > 0$ and the deformation is relevant) is exactly the statement that the temperature *must* be tuned to reach criticality: it is a relevant knob.

Two operator dimensions, Δ_σ and Δ_ε, have now produced two exponents, η and ν. The remarkable fact, which we establish next, is that these are the *only* independent numbers: the entire suite of thermodynamic exponents is fixed by $(\Delta_\sigma, \Delta_\varepsilon)$ through a single scaling form.

The scaling form of the free energy. The order parameter couples to an external field h (a magnetic field for a ferromagnet), conjugate to the spin σ, with RG eigenvalue

$$y_h = d - \Delta_\sigma \tag{52.45}$$

by the same dimensional argument that gave Eq. (52.41). The singular part of the free energy density $f_s(t,h)$ — the piece responsible for the critical singularities — obeys a homogeneity relation that follows directly from the RG. Under one coarse-graining step the total singular free energy $-\ln Z_{\text{sing}}$ is invariant (we merely reorganize the same partition sum), while the volume measured in the new, coarser units contracts by b^{-d}; the couplings flow by their eigenvalues. Equating the total free energy before and after the step, $V f_s(t,h) = (V b^{-d})\, f_s(b^{y_t} t,\, b^{y_h} h)$, gives the scaling form

$$f_s(t,h) = b^{-d}\, f_s(b^{\,y_t} t,\; b^{\,y_h} h)\,, \tag{52.46}$$

valid for every $b > 1$. This one equation, together with the eigenvalues Eqs. (52.41) and (52.45), contains every thermodynamic exponent. The following worked example extracts them all, with the algebra carried in full.

Worked Example 52.8 (All thermodynamic exponents from two dimensions, and the scaling relations)**.** We extract $\alpha, \beta, \gamma, \delta$ from Eq. (52.46), express each through $\Delta_\sigma, \Delta_\varepsilon$, and derive the Rushbrooke, Widom, and Fisher scaling relations. We use the eigenvalues $y_t = d - \Delta_\varepsilon$ and $y_h = d - \Delta_\sigma$ throughout.

Eliminating the temperature. For h-dependent quantities at fixed $t \neq 0$ we set $b^{y_t}|t| = 1$, i.e. $b = |t|^{-1/y_t}$. Then Eq. (52.46) becomes

$$f_s(t,h) = |t|^{\,d/y_t}\, \Phi\!\left(\frac{h}{|t|^{\,y_h/y_t}}\right), \qquad \Phi(z) \equiv f_s(\pm 1,\, z)\,, \tag{52.47}$$

with $z = h\,|t|^{-y_h/y_t}$ the single scaling variable and Φ a universal scaling function.

Specific heat (α). The specific heat is the second temperature derivative of the free energy at zero field, $C \sim \partial^2 f_s/\partial t^2$. From Eq. (52.47) at $h = 0$, $f_s(t,0) = |t|^{d/y_t}\Phi(0)$, and two derivatives in t lower the power by two:

$$C \sim |t|^{\,d/y_t - 2} \equiv |t|^{-\alpha} \qquad \Longrightarrow \qquad \alpha = 2 - \frac{d}{y_t} = 2 - d\,\nu. \tag{52.48}$$

Magnetization (β). The spontaneous magnetization is $M = -\partial f_s/\partial h\big|_{h=0}$. Differentiating Eq. (52.47) once in h brings a factor $|t|^{-y_h/y_t}$ from the chain rule:

$$M = -|t|^{\,d/y_t}\, \Phi'(z)\, |t|^{-y_h/y_t}\Big|_{h=0} \sim |t|^{\,(d-y_h)/y_t} \equiv (-t)^{\beta} \;\Longrightarrow\; \beta = \frac{d - y_h}{y_t}. \tag{52.49}$$

Here $\Phi'(0) \neq 0$ in the ordered phase ($t < 0$), so the power is the full $(d - y_h)/y_t$. Using $d - y_h = \Delta_\sigma$ and $y_t = 1/\nu$, this is $\beta = \nu\,\Delta_\sigma$.

Susceptibility (γ). The susceptibility is $\chi = \partial M/\partial h\big|_{h=0} = -\partial^2 f_s/\partial h^2\big|_{h=0}$. A second h-derivative brings a second factor $|t|^{-y_h/y_t}$:

$$\chi \sim |t|^{\,d/y_t - 2y_h/y_t} = |t|^{\,(d-2y_h)/y_t} \equiv |t|^{-\gamma} \implies \gamma = \frac{2y_h - d}{y_t}. \tag{52.50}$$

With $2y_h - d = 2(d-\Delta_\sigma) - d = d - 2\Delta_\sigma$ and $y_t = 1/\nu$, this reads $\gamma = \nu\,(d - 2\Delta_\sigma)$.

Critical isotherm (δ). At $t = 0$ the previous choice of b is unavailable; instead eliminate h by setting $b^{y_h}|h| = 1$, i.e. $b = |h|^{-1/y_h}$. Then Eq. (52.46) gives $f_s(0,h) = |h|^{d/y_h} f_s(0,\pm 1)$, and

$$M = -\frac{\partial f_s}{\partial h}\bigg|_{t=0} \sim |h|^{\,d/y_h - 1} = |h|^{\,(d-y_h)/y_h} \equiv |h|^{1/\delta} \implies \delta = \frac{y_h}{d - y_h} = \frac{y_h}{\Delta_\sigma}. \tag{52.51}$$

The exponents through the two dimensions. Substituting $y_t = d - \Delta_\varepsilon$ and $y_h = d - \Delta_\sigma$ collects the results into a closed table in terms of the two primary dimensions:

$$\begin{aligned} \nu &= \frac{1}{d - \Delta_\varepsilon}, & \eta &= 2\Delta_\sigma - (d-2), \\ \alpha &= 2 - \frac{d}{d - \Delta_\varepsilon}, & \beta &= \frac{\Delta_\sigma}{d - \Delta_\varepsilon}, \\ \gamma &= \frac{d - 2\Delta_\sigma}{d - \Delta_\varepsilon}, & \delta &= \frac{d - \Delta_\sigma}{\Delta_\sigma}. \end{aligned} \tag{52.52}$$

Every standard critical exponent is an explicit rational function of Δ_σ and Δ_ε.

The scaling relations. Because all six exponents descend from the two eigenvalues (y_t, y_h), they are not independent; eliminating y_t, y_h yields exact identities. Using the compact forms $\alpha = 2 - d/y_t$, $\beta = (d-y_h)/y_t$, $\gamma = (2y_h - d)/y_t$, $\delta = y_h/(d-y_h)$:

$$\alpha + 2\beta + \gamma = \frac{(2y_t - d) + 2(d - y_h) + (2y_h - d)}{y_t} = \frac{2y_t}{y_t} = 2 \quad \text{(Rushbrooke)}, \tag{52.53}$$

$$\beta(\delta - 1) = \frac{d - y_h}{y_t} \cdot \frac{2y_h - d}{d - y_h} = \frac{2y_h - d}{y_t} = \gamma \qquad \text{(Widom)}, \tag{52.54}$$

$$\nu(2 - \eta) = \frac{1}{y_t}\big(2 - [2\Delta_\sigma - (d-2)]\big) = \frac{d - 2\Delta_\sigma}{y_t} = \gamma \qquad \text{(Fisher)}. \tag{52.55}$$

In the Widom step we used $\delta - 1 = \dfrac{y_h - (d - y_h)}{d - y_h} = \dfrac{2y_h - d}{d - y_h}$. In the Fisher step we used $2 - \eta = 2 - 2\Delta_\sigma + (d-2) = d - 2\Delta_\sigma = 2y_h - d$ since $2y_h - d = 2(d - \Delta_\sigma) - d = d - 2\Delta_\sigma$. The three celebrated scaling relations are thus not empirical coincidences but algebraic consequences of there being exactly two

independent dimensions. The same elimination gives the hyperscaling (Josephson) relation directly from Eq. (52.48),

$$2 - \alpha = \frac{d}{y_t} = d\,\nu, \tag{52.56}$$

which, unlike the previous three, carries explicit d-dependence because it descends from the volume factor b^{-d} in Eq. (52.46); it holds below the upper critical dimension, where the fixed point is interacting and the scaling form Eq. (52.46) controls the singular free energy.

The worked example is the quantitative heart of the dictionary: it shows that the long list of critical exponents measured across decades of experiment is, at any fixed point, a two-parameter family, and the two parameters are conformal dimensions. A bootstrap determination of $(\Delta_\sigma, \Delta_\varepsilon)$ — the numbers pinned to the kink in Chapter 50 — feeds straight through Eq. (52.52) to the entire exponent table, and the scaling relations Eqs. (52.53) to (52.55) are guaranteed to hold because they are identities in the two dimensions.

52.2.4 Universality is the statement "the same CFT"

The dictionary explains the most striking empirical fact about critical phenomena: *universality*. A uniaxial ferromagnet at its Curie point and a fluid at its liquid–vapour critical point are microscopically unrelated — different constituents, different forces, different lattices or no lattice at all — yet they share, to measured precision, the same critical exponents. Part VI gave the mechanism (Theorem 23.26): both systems flow, under coarse-graining, into the same basin of attraction and arrive at the same fixed point, with their microscopic differences encoded entirely in irrelevant couplings that die away along the flow. The dictionary now sharpens the mechanism into an identity of objects.

Proposition 52.9 (Universality as conformal identity)**.** *Two statistical systems whose RG trajectories terminate at the same fixed point are described, at long distances, by one and the same conformal field theory. They therefore have identical conformal data* $(\{\Delta_{\mathcal{O}}, \ell_{\mathcal{O}}\}, \{C_{\mathcal{O}_1\mathcal{O}_2\mathcal{O}_3}\})$ *(Eq. (49.1)), and hence — by Eqs. (52.39), (52.44) and (52.52) — identical critical exponents. The microscopic data of each system survive only as the coefficients of irrelevant operators, which by Theorem 52.7 have* $\Delta > d$ *and decay under the flow, contributing only subleading corrections to scaling.*

Proof. By hypothesis both trajectories reach the same point S_* in coupling space. The long-distance physics of each is governed by the fixed-point theory together with the leading irrelevant deformations; the fixed-point theory is, by the result of the previous section, a single CFT with a definite spectrum and definite OPE coefficients (Eq. (49.1)). The exponents are functions of that data

alone through the relations of Section 52.2.3, so they coincide. The microscopic distinctions correspond to different starting points on the same critical surface and to different values of the irrelevant couplings g_j with $\Delta_j > d$; these flow to zero as $b^{y_j} = b^{d-\Delta_j} \to 0$, leaving the leading singular behaviour untouched and altering only the corrections-to-scaling exponents $\omega_j = \Delta_j - d > 0$. □

The proposition makes "universality class" synonymous with "fixed-point CFT," the bottom-right entry of Table 52.1. It also answers a practical question: how many parameters must an experimentalist tune to sit exactly at the critical point? The answer is read off the dimension spectrum. Each relevant operator is a direction in which the flow escapes the fixed point; to reach the fixed point one must tune the conjugate coupling to its critical value. Hence the number of relevant operators equals the number of independent tunings, which is the codimension of the critical surface — the surface of bare theories that flow into the fixed point.

Physical Insight

For the Ising universality class there are exactly two relevant scalar primaries: the $\mathbb{Z}_2$-odd spin σ and the $\mathbb{Z}_2$-even energy ε. If one does not impose the $\mathbb{Z}_2$ symmetry, both must be tuned — the magnetic field h to zero and the temperature to T_c — so the critical point has codimension two. If the symmetry is imposed by hand (a perfectly symmetric magnet in zero field), the odd operator σ cannot be generated, and only the single even relevant direction ε remains: codimension one, a single knob (the temperature) to reach criticality. This is why a generic second-order transition is reached by tuning one parameter: in the symmetric sector the Ising fixed point has a single relevant direction. The count of relevant operators is the count of fine-tunings, and it is a piece of conformal data.

Figure 52.1 renders the picture: a critical surface of codimension equal to the number of relevant directions, two microscopically different bare theories lying on it, both flowing into the common fixed point, and the relevant couplings t, h as the directions transverse to the surface along which the flow escapes.

52.2.5 The two leading Ising primaries: $\sigma = \phi$ and $\varepsilon = \phi^2$

We close the dictionary by naming, for the canonical example carried since Chapter 6, the two operators whose dimensions the rest of the chapter computes. ϕ^4 theory is the real scalar with the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ fixed in Chapter 7. At its interacting fixed point — the Wilson–Fisher fixed point, located perturbatively at $\lambda_* = \frac{16\pi^2}{3}\varepsilon + O(\varepsilon^2)$ in $d = 4 - \varepsilon$ (Eq. (24.255)) and characterized as a CFT in Theorem 52.11 — the local operators organize into conformal multiplets, and the two relevant scalar primaries are immediately identified by their $\mathbb{Z}_2$ charge.

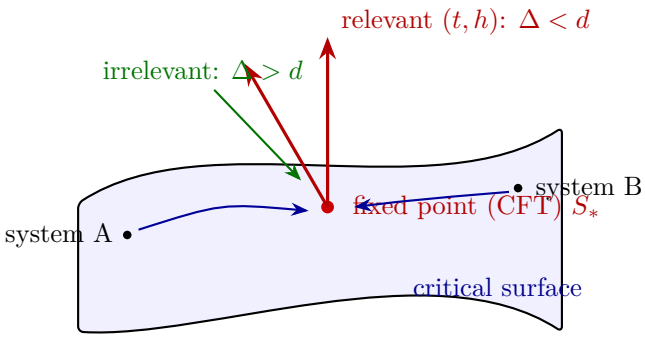


Figure 52.1: Universality as flow to a common fixed point. Two microscopically different bare theories (A, B) lie on the critical surface and flow under coarse-graining to one fixed-point CFT S_*, sharing its conformal data and hence its exponents. Directions tangent to the surface are irrelevant ($\Delta > d$, flowing in); directions transverse are relevant ($\Delta < d$, flowing out), conjugate to the couplings t, h that must be tuned to reach criticality. The codimension of the surface equals the number of relevant primaries.

Definition 52.10 (The leading Ising primaries)**.** In the Wilson–Fisher / Ising universality class, the two lowest-dimension scalar primaries are:

- the **spin** σ, the unique relevant $\mathbb{Z}_2$-*odd* scalar, identified with the fundamental field, $\sigma = \phi$; it is conjugate to the magnetic field h, has dimension $\Delta_\sigma = \frac{d-2}{2} + \gamma_\phi^*$, and fixes η through Eq. (52.39);
- the **energy** ε, the unique relevant $\mathbb{Z}_2$-*even* scalar, identified with the normal-ordered composite, $\varepsilon = :\phi^2:$; it is conjugate to the reduced temperature t, has dimension $\Delta_\varepsilon = (d-2) + \gamma_{\phi^2}^*$, and fixes ν through Eq. (52.44).

The identifications are forced. The field ϕ is the only $\mathbb{Z}_2$-odd operator of low enough dimension to be relevant, and it is the continuum limit of the lattice spin whose ordering is the phase transition; hence $\sigma = \phi$ and the magnetic field, which explicitly breaks $\phi \to -\phi$, is its conjugate source. The composite ϕ^2 is the lowest $\mathbb{Z}_2$-even scalar above the identity, the continuum limit of the lattice energy density; hence $\varepsilon = :\phi^2:$ and the reduced temperature, which multiplies the mass operator $\frac{1}{2}m^2\phi^2$ in the action and is even, is its conjugate source. That these are the *only* two relevant scalars is the codimension-two count of Section 52.2.4, reduced to one in the symmetric sector. Every other operator — ϕ^4 at the fixed point, $\phi\,\partial^2\phi$, the higher composites, and all spinning operators including the stress tensor — is marginal or irrelevant and does not require tuning.

With the two operators named and the exponent formulas Eqs. (52.39), (52.44) and (52.52) in hand, the dictionary is complete: it remains only to supply the two numbers. The next section consummates ϕ^4 theory by computing Δ_σ

and Δ_ε explicitly at the Wilson–Fisher fixed point in the ϵ-expansion, carrying the loop algebra in full and feeding the resulting anomalous dimensions through exactly the relations assembled here.

52.3 The Wilson–Fisher Fixed Point as a CFT: the Epsilon-Expansion

The dictionary of the previous section turned every critical exponent into an operator dimension: the field exponent $\eta = 2\Delta_\sigma - (d-2)$ from Eq. (52.39), the correlation-length exponent through $1/\nu = y_t = d - \Delta_\varepsilon$ from Eq. (52.44), and the correction-to-scaling exponent ω as the RG eigenvalue of the leading irrelevant operator. It also fixed, for the canonical example carried since Chapter 6, which operators those are: $\sigma = \phi$ and $\varepsilon = :\phi^2:$ are the two relevant scalar primaries (Section 52.2.5), so that $\Delta_\sigma = \Delta_\phi$ and $\Delta_\varepsilon = \Delta_{\phi^2}$. The dictionary is exact but empty until we possess the dimensions themselves. This section fills it. We continue ϕ^4 theory from Chapter 24, where the one-loop beta function already located the Wilson–Fisher fixed point at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ and announced $\gamma_\phi = O(\varepsilon^2)$, and we now carry every loop integral to its end: the engineering bookkeeping that produces the fixed point, the two-loop self-energy that gives γ_ϕ and hence η, and the one-loop composite-operator renormalization that gives γ_{ϕ^2} and hence ν. The output is the leading conformal data of the Wilson–Fisher CFT, computed from the foundations and handed, in the next section, to the three-dimensional bootstrap.

Throughout this section the theory is Euclidean: critical phenomena live in imaginary time, and the path integral carries e^{-S_E} with

$$S_E = \int \mathrm{d}^d x \, \Big[\tfrac{1}{2}(\partial_\mu\phi)(\partial_\mu\phi) + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4 \Big], \tag{52.57}$$

the Wick rotation of the standard Lagrangian Eq. (6.56) fixed in Chapter 10. The Euclidean Feynman rules are the propagator $1/(p^2+m^2)$ and the vertex $-\lambda$; all loop momenta are Euclidean and indices are summed with $\delta_{\mu\nu}$.

Notation Watch

Two symbols collide in this section and the collision is unavoidable, because each is standard in its own community. The *dimension deficit* $\varepsilon \equiv 4 - d$ is the small, finite expansion parameter of the ε-expansion; it is held fixed, not sent to zero, and is set to $\varepsilon = 1$ in the next section to reach $d = 3$. It is twice the dimensional-regularization regulator of the book's canon (Chapter 22), where $d = 4 - 2\varepsilon_{\mathrm{dr}}$ and $\varepsilon_{\mathrm{dr}} \to 0$ at the end: as dimensions, $\varepsilon = 2\varepsilon_{\mathrm{dr}}$. The regulator $\varepsilon_{\mathrm{dr}}$ appears only inside individual Z-factors, whose poles cancel; the anomalous dimensions $\gamma(\lambda)$ are finite functions,

independent of the regulator, and the physical deficit ε enters them only through the fixed-point value $\lambda_*(\varepsilon)$ and through engineering dimensions. Separately, the *energy operator* of the Ising universality class is universally written ε (so Δ_ε, $\lambda_{\sigma\sigma\varepsilon}$); it is the operator $\varepsilon = :\phi^2:$ and has nothing to do with the deficit. We always write ε with a subscript or as a named operator and ε standing alone, exactly as in Section 52.2.5.

52.3.1 The engineering dimension and the fixed point

In exactly four dimensions λ is dimensionless and marginal, and the only zero of $\beta_\lambda = 3\lambda^2/16\pi^2$ is the free Gaussian theory at $\lambda = 0$. The single idea that produces an interacting fixed point is to leave four dimensions, where λ acquires an engineering dimension and the beta function gains a classical term. We carry the bookkeeping in full.

The kinetic term $\int \mathrm{d}^d x\, \frac{1}{2}(\partial\phi)^2$ must be dimensionless (the action is, in units $\hbar = 1$). With $[\mathrm{d}^d x] = -d$ and $[\partial] = +1$ in mass units, $-d + 2 + 2[\phi] = 0$ fixes the canonical mass dimension of the field,

$$[\phi] = \frac{d-2}{2}. \tag{52.58}$$

The quartic operator then carries $[\phi^4] = 2(d-2)$, and demanding $\int \mathrm{d}^d x\, \frac{\lambda}{4!}\phi^4$ be dimensionless fixes

$$[\lambda] = d - 2(d-2) = 4 - d = \varepsilon. \tag{52.59}$$

For $d < 4$ the coupling has *positive* mass dimension ε: it is power-counting relevant, exactly the statement of Chapter 23 that an operator with $[\,\cdot\,] < d$ grows under coarse-graining. To run a dimensionless coupling we factor out the scale, writing the dimensionful bare coupling as $\lambda_0 = \mu^\varepsilon$ (dimensionless). Holding λ_0 fixed while varying μ then injects a tree-level drift into the renormalization-group flow of the dimensionless renormalized coupling: differentiating $\lambda \propto \mu^{-\varepsilon}\lambda_0$ gives the classical contribution $\mu\,\partial_\mu\lambda \supset -\varepsilon\lambda$. The full one-loop beta function in $d = 4 - \varepsilon$ is therefore the four-dimensional loop term of Chapter 24, whose leading coefficient is dimension-independent, plus this classical tilt,

$$\beta_\lambda = -\varepsilon\,\lambda + \frac{3\lambda^2}{16\pi^2} + O(\lambda^3, \varepsilon\lambda^2), \tag{52.60}$$

which is the imported Eq. (24.254); we do not recompute the $3/16\pi^2$ here, only re-derive the term that turns it into a fixed-point equation.

Theorem 52.11 (The Wilson–Fisher fixed point is a conformal field theory)**.** *For $0 < \varepsilon \ll 1$ the beta function Eq. (52.60) has, besides the Gaussian zero $\lambda = 0$, a single nontrivial zero*

$$\lambda_* = \frac{16\pi^2}{3}\,\varepsilon + O(\varepsilon^2), \tag{52.61}$$

the Wilson–Fisher fixed point. *At λ_* the trace of the stress tensor vanishes up to the equation of motion and the theory is scale invariant; by the promotion of scale to conformal invariance established in Section 52.1, it is a unitary conformal field theory, a member of a continuous family labelled by ε (equivalently by d). Its leading conformal data are*

$$\Delta_\phi = \frac{d-2}{2} + \gamma_\phi(\lambda_*), \qquad \Delta_{\phi^2} = (d-2) + \gamma_{\phi^2}(\lambda_*), \tag{52.62}$$

with $\gamma_\phi(\lambda_*) = \varepsilon^2/108 + O(\varepsilon^3)$ and $\gamma_{\phi^2}(\lambda_*) = \varepsilon/3 + O(\varepsilon^2)$, computed in Sections 52.3.2 and 52.3.3. The Gaussian fixed point is infrared-repulsive and λ_* is infrared-attractive, with correction-to-scaling exponent

$$\omega = \beta'_\lambda(\lambda_*) = +\varepsilon + O(\varepsilon^2). \tag{52.63}$$

Location and stability. Setting Eq. (52.60) to zero,

$$\lambda\Big(-\varepsilon + \frac{3\lambda}{16\pi^2}\Big) = 0 \quad \Longrightarrow \quad \lambda = 0 \ \text{ or } \ \lambda_* = \frac{16\pi^2}{3}\varepsilon, \tag{52.64}$$

which is Eq. (52.61); for small ε this is a weak-coupling fixed point, $\lambda_* = O(\varepsilon)$, where the one-loop truncation is controlled. Differentiating the beta function,

$$\beta'_\lambda(\lambda) = -\varepsilon + \frac{6\lambda}{16\pi^2} = -\varepsilon + \frac{3\lambda}{8\pi^2}, \tag{52.65}$$

and evaluating at the two zeros gives $\beta'_\lambda(0) = -\varepsilon < 0$ and

$$\beta'_\lambda(\lambda_*) = -\varepsilon + \frac{3}{8\pi^2}\cdot\frac{16\pi^2}{3}\varepsilon = -\varepsilon + 2\varepsilon = +\varepsilon > 0. \tag{52.66}$$

A coupling near a zero flows as $\frac{d\lambda}{d(-\ln b)} = -\beta_\lambda$ toward the infrared (increasing length b); the zero is IR-attractive when $\beta'_\lambda > 0$ and IR-repulsive when $\beta'_\lambda < 0$. Hence the Gaussian point is repelled and λ_* collects the flow: every microscopic theory with a nonzero quartic in $d < 4$ ends, at long distance, on the single fixed-point theory at λ_*. This is the dynamical content of universality (Theorem 23.26). The slope $\beta'_\lambda(\lambda_*) = \varepsilon$ is, by the dictionary, the RG eigenvalue of the deviation $\delta\lambda = \lambda - \lambda_*$, which is the leading irrelevant direction; its eigenvalue is $-\omega$ with $\omega = \Delta_{\phi^4} - d = \beta'_\lambda(\lambda_*) = \varepsilon$, the correction-to-scaling exponent of Eq. (52.63). The promotion to conformal invariance and the computation of the two anomalous dimensions occupy the remainder of the section. □

Physical Insight

The fixed point is the cancellation of two effects that pull the coupling in opposite directions. Quantum-mechanically the quartic self-interaction

screens itself: the one-loop term $+3\lambda^2/16\pi^2$ drives λ to grow in the ultraviolet, the same physics as the four-dimensional Landau pole. Below four dimensions the coupling additionally carries positive mass dimension ε, so simple dimensional analysis makes the dimensionless coupling *shrink* in the ultraviolet at rate $-\varepsilon\lambda$. At λ_* the classical shrinking exactly balances the quantum growth, and the coupling sits still — the theory has forgotten the scale μ. Forgetting the only scale is scale invariance; in a local, unitary, reflection-positive theory it is conformal invariance. The Wilson–Fisher fixed point is thus a genuine interacting CFT, weakly coupled for small ε, that interpolates — as ε runs from 0 to 1 to 2 — between the free scalar in $d = 4$ and the two-dimensional Ising CFT of Chapter 48.

52.3.2 The field anomalous dimension from the two-loop sunset: η

The first datum is the dimension of $\sigma = \phi$, equivalently the anomalous dimension $\gamma_\phi(\lambda_*)$ through $\Delta_\phi = \frac{d-2}{2} + \gamma_\phi$ and $\eta = 2\gamma_\phi$ (the dictionary Eq. (52.39), since $\eta = 2\Delta_\sigma - (d-2) = 2\gamma_\phi$). We must compute $\gamma_\phi(\lambda)$ as a function of the coupling and evaluate it at λ_*.

Why one loop gives nothing. The one-loop self-energy in ϕ^4 is the tadpole of Fig. 22.10: a single quartic vertex with two of its legs closed into a loop. Its value is

$$\Sigma^{(1)} = \frac{\lambda}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{k^2+m^2}, \tag{52.67}$$

the factor $\frac{1}{2}$ the symmetry factor of the loop. The integrand is *independent of the external momentum* p: the external line passes straight through the vertex and only the loop momentum k circulates. Consequently $\Sigma^{(1)}$ is a pure constant, contributing to the inverse propagator $\Gamma^{(2)}(p) = p^2 + m^2 + \Sigma$ only through the p-independent piece. It renormalizes the mass, never the coefficient of p^2, so the wavefunction renormalization receives no one-loop correction: $\delta_Z = O(\lambda^2)$ and $Z_\phi = 1 + O(\lambda^2)$, the result Eq. (22.82). Since $\gamma_\phi = \frac{1}{2}\mu\,\partial_\mu \ln Z_\phi$, the field anomalous dimension starts at two loops. This is not an accident of ϕ^4 but a feature of any theory whose lowest self-energy is a tadpole; it is the technical reason η is the smallest of the Ising exponents, $O(\varepsilon^2)$ where ν deviates at $O(\varepsilon)$.

The two-loop sunset. The leading momentum-dependent self-energy is the *sunset* of Fig. 52.2: two quartic vertices joined by three internal propagators, one external leg at each vertex. In the Euclidean theory at the critical point ($m = 0$)

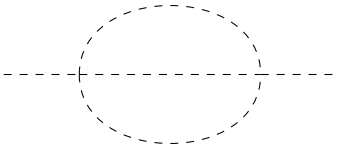


Figure 52.2: The two-loop sunset self-energy Eq. (52.68) of ϕ^4 theory. Two quartic vertices are joined by three internal scalar lines; one external leg enters at each vertex. This is the lowest self-energy whose value depends on the external momentum p, and hence the source of the first nonzero wavefunction renormalization $\delta_Z = O(\lambda^2)$ and of the field anomalous dimension γ_ϕ.

it reads, with symmetry factor $1/6$ for the three interchangeable internal lines,

$$\Sigma_{\text{sun}}(p^2) = \frac{\lambda^2}{6} \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{\mathrm{d}^d q}{(2\pi)^d} \, \frac{1}{k^2} \, \frac{1}{q^2} \, \frac{1}{(p-k-q)^2}, \tag{52.68}$$

the Wick rotation of the Minkowski integral Eq. (22.136). Its superficial degree of divergence is $D = 2d - 3 \cdot 2 = 2d - 6 = 2$ at $d = 4$: a quadratic overall divergence. The constant part renormalizes the mass; the piece we need is the coefficient of p^2, which we isolate by one differentiation in p^2.

Differentiating Eq. (52.68) once with respect to p^2 lowers the overall degree by two, to $D = 0$, a logarithmic divergence; the coefficient

$$B_{\text{sun}} \equiv \left. \frac{\partial \Sigma_{\text{sun}}}{\partial p^2} \right|_{p=0} \tag{52.69}$$

is therefore genuinely present, and it is the only divergence proportional to p^2 that the kinetic counterterm must absorb. The double integral, performed with the dimensionally regularized master integrals of Chapter D (the inner q-integral is the massless bubble, leaving a single radial integral over k whose ultraviolet tail produces the pole), is carried out in full in Theorem 22.29; its leading pole is the imported Eq. (22.138),

$$B_{\text{sun}} = -\frac{\lambda^2}{12\,(16\pi^2)^2} \, \frac{1}{\varepsilon_{\text{dr}}} + (\text{finite}), \tag{52.70}$$

the coefficient $1/12$ tracking the symmetry factor $1/6$ and the two-loop measure, the sign fixed by the Euclidean propagators. The inverse propagator $\Gamma^{(2)}(p^2) = p^2 + \Sigma$ then acquires a divergent shift in the coefficient of p^2, from 1 to $1 - B_{\text{sun}}$, which the wavefunction counterterm cancels, $\delta_Z^{(2)} = B_{\text{sun}}$, i.e.

$$Z_\phi = 1 - \frac{\lambda^2}{12\,(16\pi^2)^2} \, \frac{1}{\varepsilon_{\text{dr}}} + O(\lambda^3), \tag{52.71}$$

the first nonzero field-strength renormalization in ϕ^4 theory, established at Eq. (22.139).

From the counterterm to the anomalous dimension. The field anomalous dimension is read from the simple pole of Z_ϕ by the pole-to-anomalous-dimension relation of Chapter 24 (the 't Hooft equation Theorem 24.5); the assembly is owned there and delivers the imported Eq. (24.260),

$$\gamma_\phi(\lambda) = \frac{\lambda^2}{12\,(16\pi^2)^2} + O(\lambda^3). \tag{52.72}$$

This is a finite function of the coupling, independent of the regulator. Evaluating it at the Wilson–Fisher coupling Eq. (52.61) is now pure arithmetic:

$$\gamma_\phi(\lambda_*) = \frac{1}{12\,(16\pi^2)^2}\Big(\frac{16\pi^2}{3}\varepsilon\Big)^2 = \frac{1}{12}\cdot\frac{(16\pi^2)^2}{9\,(16\pi^2)^2}\,\varepsilon^2 = \frac{1}{108}\,\varepsilon^2. \tag{52.73}$$

Through the dictionary $\eta = 2\gamma_\phi(\lambda_*)$ this is the carried critical exponent

$$\boxed{\eta = \frac{\varepsilon^2}{54} + O(\varepsilon^3)} \tag{52.74}$$

for the one-component scalar, and the full conformal dimension of $\sigma = \phi$ is

$$\Delta_\phi = \frac{d-2}{2} + \frac{\varepsilon^2}{108} = \frac{2-\varepsilon}{2} + \frac{\varepsilon^2}{108} = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3). \tag{52.75}$$

The vanishing of η at $O(\varepsilon)$ is the precise statement that, to first order in the deficit, the interacting field still scales canonically; the interaction shifts the power in the two-point function $\langle\phi(x)\phi(0)\rangle \sim |x|^{-(d-2+\eta)}$ only at second order. This matches the thread obligation $\gamma_\phi = O(\varepsilon^2)$ exactly.

52.3.3 Renormalization of the composite operator ϕ^2: ν

The second datum is the dimension of $\varepsilon = :\phi^2:$, which through $1/\nu = d - \Delta_{\phi^2}$ (Eq. (52.44)) controls the correlation-length exponent. The operator ϕ^2 is composite: it is the product of two fields at a coincident point, and such products develop their own short-distance divergences beyond those of the elementary fields. They must be renormalized separately, by an operator renormalization constant Z_{ϕ^2}, and the anomalous dimension that Z_{ϕ^2} generates is precisely γ_{ϕ^2}.

The composite vertex. The object that exhibits the divergence is the one-particle-irreducible correlator with a single insertion of $\frac{1}{2}\phi^2$ and two amputated external ϕ legs,

$$\Gamma_{\phi^2}(p) \equiv \big\langle\, \tfrac{1}{2}\phi^2\ \phi\,\phi \,\big\rangle_{1\mathrm{PI},\,\mathrm{amp}}, \tag{52.76}$$

with p the momentum flowing into the operator. At tree level the two fields in $\frac{1}{2}\phi^2$ contract directly with the two external legs — two equal contractions cancelling the $\frac{1}{2}$ — so $\Gamma^{(0)}_{\phi^2} = 1$. The first correction is the single diagram of Fig. 52.3: the two fields of the operator close into a loop through one quartic vertex, whose remaining two legs are the external lines.

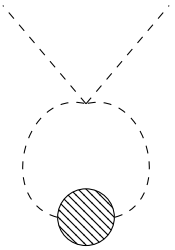


Figure 52.3: The one-loop renormalization of the composite operator $\frac{1}{2}\phi^2$ (the shaded insertion). Its two fields close into a loop through a single quartic vertex; the remaining two vertex legs are the external ϕ lines. The loop is the massless bubble Eq. (52.79) and supplies the logarithmic divergence that defines Z_{ϕ^2}, hence γ_{ϕ^2} and the exponent ν.

The combinatorics and the integral. Expand to first order in the interaction $-S_{\text{int}} = -\frac{\lambda}{4!}\int \mathrm{d}^d z\, \phi(z)^4$ and count Wick contractions of $\langle \frac{1}{2}\phi(x)^2\, \phi(y_1)\phi(y_2)\, (-\frac{\lambda}{4!})\int \phi(z)^4 \rangle$ that yield the 1PI topology: each external field contracts with a distinct vertex leg (4×3 ways), and the two operator fields contract with the remaining two vertex legs (2 ways), for $4 \cdot 3 \cdot 2 = 24$ contractions. With the prefactor $\frac{1}{2} \cdot (-\frac{\lambda}{24})$ the combinatorial weight is

$$\tfrac{1}{2} \cdot \Big(-\frac{\lambda}{24} \Big) \cdot 24 = -\frac{\lambda}{2}, \tag{52.77}$$

and amputating the external legs leaves the massless bubble with the operator momentum p flowing through it,

$$\Gamma^{(1)}_{\phi^2}(p) = -\frac{\lambda}{2}\, B(p^2), \qquad B(p^2) = \int \frac{\mathrm{d}^d k}{(2\pi)^d}\, \frac{1}{k^2\,(k+p)^2}. \tag{52.78}$$

The bubble is the same integral that fed β_λ at one loop; from the master formula Eq. (D.77) (massless limit, two propagators) one finds, as in Theorem D.25,

$$B(p^2) = \frac{1}{(4\pi)^{d/2}}\, \frac{\Gamma(2-\frac{d}{2})\, \Gamma(\frac{d}{2}-1)^2}{\Gamma(d-2)}\, (p^2)^{\,d/2-2} \;=\; \frac{1}{16\pi^2}\, \frac{1}{\varepsilon_{\text{dr}}} + (\text{finite}), \tag{52.79}$$

the divergence the single pole $\Gamma(2-d/2) = \Gamma(\varepsilon_{\text{dr}})$. Hence the one-loop composite vertex carries the pole

$$\Gamma_{\phi^2}(p) = 1 - \frac{\lambda}{2} \cdot \frac{1}{16\pi^2}\, \frac{1}{\varepsilon_{\text{dr}}} + (\text{finite}) = 1 - \frac{\lambda}{32\pi^2}\, \frac{1}{\varepsilon_{\text{dr}}} + \cdots . \tag{52.80}$$

The operator renormalization constant. Define the renormalized composite operator $[\phi^2]_R = Z^{-1}_{\phi^2}\phi^2$, chosen so that $\langle [\phi^2]_R\, \phi\phi \rangle$ is finite. Since the external

legs are renormalized by $Z_\phi^{1/2}$ each and $Z_\phi = 1 + O(\lambda^2)$, finiteness at one loop requires $Z_{\phi^2}^{-1}\Gamma_{\phi^2}$ finite, so Z_{ϕ^2} must cancel the pole of Eq. (52.80):

$$Z_{\phi^2} = 1 - \frac{\lambda}{32\pi^2}\,\frac{1}{\varepsilon_{\mathrm{dr}}} + O(\lambda^2). \tag{52.81}$$

The simple-pole residue is $z_1^{\phi^2} = -\lambda/32\pi^2$. We note in passing that $|z_1^{\phi^2}|$ equals the residue of the mass renormalization $Z_m = 1 + \frac{\lambda}{32\pi^2}\frac{1}{\varepsilon_{\mathrm{dr}}}$ of Eq. (22.83): this is the statement that ϕ^2 is the mass operator, $\frac{1}{2}m^2\phi^2$ being the term it multiplies in the action, so the divergence that renormalizes the mass and the divergence that renormalizes the composite operator are one and the same.

From the residue to γ_{ϕ^2}. The anomalous dimension follows from the pole by the 't Hooft equation Theorem 24.5: for a renormalization constant $Z = 1 + z_1(\lambda)/\varepsilon_{\mathrm{dr}}$, the associated anomalous dimension is $\gamma = -\rho\,\lambda\,\frac{dz_1}{d\lambda}$ with $\rho = 2$ for ϕ^4 (the same ρ that turns $z_1^\lambda = 3\lambda/32\pi^2$ into $\beta_\lambda = 3\lambda^2/16\pi^2$ and $z_1^m = \lambda/32\pi^2$ into the carried mass running $\mu\,\partial_\mu m^2 = \frac{\lambda}{16\pi^2}m^2$ of Chapter 24). With $z_1^{\phi^2} = -\lambda/32\pi^2$, so $\frac{dz_1^{\phi^2}}{d\lambda} = -1/32\pi^2$,

$$\gamma_{\phi^2}(\lambda) = -2\,\lambda\cdot\Big(-\frac{1}{32\pi^2}\Big) = \frac{\lambda}{16\pi^2} + O(\lambda^2). \tag{52.82}$$

The sign is fixed unambiguously by the physics it must reproduce: γ_{ϕ^2} is, to this order, identical to the anomalous dimension of the mass operator already carried by ϕ^4 theory, $\mu\,\partial_\mu \ln m^2 = \lambda/16\pi^2 > 0$ (Chapter 24), and it is positive — the composite operator scales *above* its engineering dimension. Evaluating at the fixed point,

$$\gamma_{\phi^2}(\lambda_*) = \frac{1}{16\pi^2}\cdot\frac{16\pi^2}{3}\varepsilon = \frac{\varepsilon}{3}. \tag{52.83}$$

Assembling ν. The conformal dimension of $\varepsilon = \phi^2$ is its engineering dimension $d-2$ plus the anomalous part, the decomposition Eq. (52.30),

$$\Delta_{\phi^2} = (d-2) + \gamma_{\phi^2}(\lambda_*) = (2-\varepsilon) + \frac{\varepsilon}{3} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2). \tag{52.84}$$

Feeding this into the dictionary $1/\nu = d - \Delta_{\phi^2}$ of Eq. (52.44),

$$\frac{1}{\nu} = (4-\varepsilon) - \Big(2 - \frac{2\varepsilon}{3}\Big) = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2), \tag{52.85}$$

and inverting the series,

$$\boxed{\nu = \frac{1}{2-\varepsilon/3} = \frac{1}{2}\Big(1-\frac{\varepsilon}{6}\Big)^{-1} = \frac{1}{2} + \frac{\varepsilon}{12} + O(\varepsilon^2)\,.} \tag{52.86}$$

This is the carried result of the thread. The deviation of ν from its Gaussian value $\frac{1}{2}$ is $O(\varepsilon)$, an order earlier than η, because it is driven by the one-loop renormalization of a relevant operator rather than by a two-loop wavefunction effect. The full end-to-end computation, with every intermediate value displayed, is the following worked example.

Worked Example 52.12 (The correlation-length exponent of the Wilson–Fisher CFT to $O(\varepsilon)$)**.** We carry ν from the bare composite operator to the critical exponent in one pass, displaying every number.

Step 1 — the fixed point. The engineering dimension Eq. (52.59) gives $[\lambda] = \varepsilon$, hence the classical term $-\varepsilon\lambda$ in Eq. (52.60); balancing it against the imported one-loop term $3\lambda^2/16\pi^2$ gives $\lambda_* = 16\pi^2\varepsilon/3$.

Step 2 — the composite-operator divergence. The single one-loop diagram Fig. 52.3 has combinatorial weight $-\lambda/2$ from Eq. (52.77) and the massless bubble Eq. (52.79) with pole $1/16\pi^2\varepsilon_{\mathrm{dr}}$; their product is the pole $-\lambda/32\pi^2\varepsilon_{\mathrm{dr}}$ of Eq. (52.80), cancelled by $Z_{\phi^2} = 1 - \lambda/32\pi^2\varepsilon_{\mathrm{dr}}$.

Step 3 — the anomalous dimension. The pole-to-γ relation Theorem 24.5 with $\rho = 2$ converts the residue $-\lambda/32\pi^2$ into $\gamma_{\phi^2}(\lambda) = \lambda/16\pi^2$, the same number as the carried mass running. At λ_*: $\gamma_{\phi^2}(\lambda_*) = \frac{1}{16\pi^2}\cdot\frac{16\pi^2}{3}\varepsilon = \varepsilon/3$.

Step 4 — the dimension and the exponent. The full dimension is $\Delta_{\phi^2} = (d-2) + \varepsilon/3 = (2-\varepsilon) + \varepsilon/3 = 2 - 2\varepsilon/3$. The dictionary $1/\nu = d - \Delta_{\phi^2}$ gives $1/\nu = (4-\varepsilon) - (2 - 2\varepsilon/3) = 2 - \varepsilon/3$, and inverting, $\nu = \frac{1}{2} + \frac{\varepsilon}{12} + O(\varepsilon^2)$.

Sanity at $\varepsilon = 0$. As $\varepsilon \to 0$ ($d \to 4$) the fixed point merges with the Gaussian one, $\lambda_* \to 0$, all anomalous dimensions vanish, and $\nu \to \frac{1}{2}$, $\eta \to 0$ — the mean-field exponents of the free scalar, as they must be. The leading interacting corrections are $\nu - \frac{1}{2} = +\varepsilon/12$ (one loop) and $\eta = \varepsilon^2/54$ (two loops), the two numbers the next section sets at $\varepsilon = 1$.

Caution

The exponent ν deviates from mean field at $O(\varepsilon)$ while η deviates only at $O(\varepsilon^2)$; it is tempting to read this as "ν is the leading effect and η a small correction." That is true as an organizing statement about the ε-series, but it does *not* license dropping η when the series is summed at $\varepsilon = 1$. The hyperscaling and scaling relations (Section 52.2.3) tie all exponents together; a consistent set of three-dimensional numbers requires η at the same accuracy as ν. The honest treatment of $\varepsilon = 1$ — where the bare series must be resummed and is ultimately superseded by the bootstrap — is the business of the next section. Here we have only the leading terms of an asymptotic expansion.

52.3.4 The $O(N)$ family: Wilson–Fisher as one parameter among two

ϕ^4 theory is the one-component ($N = 1$, Ising) member of a family. Replace the single real scalar by an N-component field $\vec{\phi} = (\phi_1, \dots, \phi_N)$ with $O(N)$-invariant action

$$S_E = \int \mathrm{d}^d x \left[\tfrac{1}{2} (\partial_\mu \vec{\phi}) \cdot (\partial_\mu \vec{\phi}) + \tfrac{1}{2} m^2 \, \vec{\phi} \cdot \vec{\phi} + \frac{\lambda}{4!} \, (\vec{\phi} \cdot \vec{\phi})^2 \right], \tag{52.87}$$

which reduces to Eq. (52.57) at $N = 1$. The internal symmetry decorates each loop with a sum over components, and the standard counting (each of the three one-loop four-point channels carries a different contraction of the $O(N)$ indices, summing to $N + 8$ instead of 3) modifies only the loop coefficient of the beta function,

$$\beta_\lambda = -\varepsilon \, \lambda + \frac{N+8}{3} \, \frac{\lambda^2}{16\pi^2} + O(\lambda^3), \tag{52.88}$$

which at $N = 1$ returns $\frac{9}{3} = 3$, the carried coefficient. The fixed point and exponents follow as before. Setting $\beta_\lambda = 0$,

$$\lambda_* = \frac{3}{N+8} \, 16\pi^2 \, \varepsilon = \frac{48\pi^2}{N+8} \, \varepsilon, \tag{52.89}$$

reducing to $\lambda_* = 16\pi^2 \varepsilon / 3$ at $N = 1$. The composite-operator analysis generalizes: the relevant $O(N)$-singlet operator $\vec{\phi} \cdot \vec{\phi}$ has $\gamma_{\phi^2}(\lambda) = \frac{N+2}{3} \frac{\lambda}{16\pi^2}$, so that

$$\frac{1}{\nu} = 2 - \gamma_{\phi^2}(\lambda_*) = 2 - \frac{N+2}{N+8} \, \varepsilon + O(\varepsilon^2), \qquad \nu = \frac{1}{2} + \frac{N+2}{4(N+8)} \, \varepsilon + O(\varepsilon^2), \tag{52.90}$$

and the two-loop field anomalous dimension generalizes to

$$\eta = \frac{N+2}{2(N+8)^2} \, \varepsilon^2 + O(\varepsilon^3). \tag{52.91}$$

At $N = 1$ these return $1/\nu = 2 - \frac{3}{9}\varepsilon = 2 - \varepsilon/3$ and $\eta = \frac{3}{2 \cdot 81} \varepsilon^2 = \varepsilon^2 / 54$, exactly Eqs. (52.74) and (52.85). Each integer N labels a distinct Wilson–Fisher CFT: $N = 0$ the self-avoiding walk, $N = 1$ Ising, $N = 2$ the XY universality class (superfluid helium), $N = 3$ the Heisenberg ferromagnet, and $N \to \infty$ the exactly solvable spherical model. The Wilson–Fisher construction thus exhibits not one CFT but a two-parameter sheet of them, coordinatized by (N, ε), each a concrete interacting conformal field theory accessible from the free scalar by a controlled expansion.

Notation Watch

The fixed-point coupling Eq. (52.89) is quoted in the thread's $\frac{\lambda}{4!}(\vec{\phi}^2)^2$ normalization, in which $N = 1$ reproduces the carried $\lambda_* = 16\pi^2 \varepsilon / 3$.

Much of the critical-phenomena and bootstrap literature uses instead the normalization $\frac{\lambda}{8}(\vec{\phi}^2)^2$, differing by a factor 3 in the coupling, in which the same fixed point reads $\lambda_* = 16\pi^2\varepsilon/(N+8)$. The two are identical physics; only the dimensionless number assigned to "λ_*" differs, while the universal exponents Eqs. (52.90) and (52.91) are normalization-independent. We retain the thread normalization throughout.

Handoff

We now hold the leading conformal data of the Wilson–Fisher CFT as explicit functions of the deficit: $\Delta_\phi = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108}$ and $\Delta_{\phi^2} = 2 - \frac{2\varepsilon}{3}$, equivalently $\eta = \varepsilon^2/54$ and $\nu = \frac{1}{2} + \frac{\varepsilon}{12}$. The next section sets $\varepsilon = 1$ to reach the physical three-dimensional Ising model, confronts these truncated series — and their resummations — with the numerical-bootstrap determination $\Delta_\sigma = 0.5181489$, $\Delta_\varepsilon = 1.412625$ of Theorem 50.35, and with Monte Carlo and experiment, thereby closing the loop between the renormalization group of Part VI and the bootstrap of Part XIV on the same conformal field theory.

52.4 Closing the Loop: the Epsilon-Expansion Meets the Bootstrap (3D Ising)

The previous section delivered, from a Lagrangian computation, the conformal data of the Wilson–Fisher fixed point as a power series in the dimension deficit $\varepsilon = 4 - d$. The two lightest scalar dimensions came out as

$$\Delta_\sigma = \frac{d-2}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3) = 1 - \frac{\varepsilon}{2} + \frac{\varepsilon^2}{108} + O(\varepsilon^3), \tag{52.92}$$

$$\Delta_\varepsilon = d - \frac{1}{\nu} = 2 - \frac{2\varepsilon}{3} + O(\varepsilon^2), \tag{52.93}$$

the first from the field anomalous dimension $\gamma_\phi = \frac{\varepsilon^2}{108} + O(\varepsilon^3)$ and the second from the relevant eigenvalue $1/\nu = 2 - \frac{\varepsilon}{3} + O(\varepsilon^2)$ (Eqs. (47.263) and (47.264), characterized as conformal data in Theorem 52.11). These are exact statements about a CFT that lives in a continuous family indexed by d, valid as asymptotic expansions near $d = 4$. The physically interesting member of the family is the one at $d = 3$, the universality class of the three-dimensional Ising model and of every system in it — the uniaxial ferromagnet at its Curie point, the liquid–vapour critical point, a binary fluid at demixing. Reaching it means setting $\varepsilon = 1$.

This section does exactly that, and then confronts the result with the independent, nonperturbative determination of the same CFT obtained in Chapter 50 by the numerical conformal bootstrap, and with Monte Carlo simulation of the

lattice model. Three calculations that share not one line of method — a perturbative loop expansion around four dimensions, a convex-optimization search over the space of crossing-symmetric spectra, and a direct stochastic sampling of a lattice partition function — will be seen to return the same five numbers $(\Delta_\sigma, \Delta_\varepsilon, \eta, \nu, \omega)$ to their quoted precision. That agreement is the deepest available vindication of the central thesis of this chapter, that a renormalization-group fixed point and a conformal field theory are one object: the Wilson–Fisher fixed point of Part VI *is* the bootstrap island of Chapter 50, and ϕ^4 theory, carried since the classical Lagrangian of Chapter 6, terminates here on a single, fully characterized CFT.

Notation Watch

Three distinct quantities in this section are written with the same Greek letter, and only context separates them. The *expansion parameter* $\varepsilon = 4-d$ is the dimension deficit, held small and finite (it is set to 1 at the end); it is twice the dimensional-regularization regulator $\varepsilon_{\text{dimreg}}$ of the book's canon, as flagged in Section 24.7.5. The *energy operator* $\varepsilon = :\phi^2:$, the leading $\mathbb{Z}_2$-even scalar of Section 52.2.5, carries its name as a subscript: Δ_ε, $\lambda_{\sigma\sigma\varepsilon}$. And the Feynman $i\epsilon$ of the Lorentzian propagator does not appear here at all, since critical phenomena are Euclidean throughout (Chapter 10). A standalone ε is always the deficit; a subscripted ε is always the operator.

52.4.1 The naive substitution, carried out honestly

The first thing any reader will do with Eqs. (52.92) and (52.93) is set $\varepsilon = 1$ and read off a number. It is worth doing this carefully, watching each successive term, because the way the partial sums behave is itself the lesson. Theorem 49.29 performed the leading comparison; we extend it here to the full set of exponents and to as many orders as are needed to see the pattern, then state honestly what the truncated series can and cannot do.

Worked Example 52.13 (The $\varepsilon = 1$ truncations of the Ising exponents)**.** We take the one-component ($N = 1$) Wilson–Fisher series in the deficit $\varepsilon = 4 - d$ and evaluate the truncations at $\varepsilon = 1$. The relevant series, in the standard normalization of the $O(N)$ critical-exponent literature specialized to $N = 1$, are

$$\eta = \frac{\varepsilon^2}{54} + \frac{109}{5832}\,\varepsilon^3 + O(\varepsilon^4), \tag{52.94}$$

$$\frac{1}{\nu} = 2 - \frac{\varepsilon}{3} - \frac{19}{162}\,\varepsilon^2 + O(\varepsilon^3), \tag{52.95}$$

$$\omega = \varepsilon - \frac{17}{27}\,\varepsilon^2 + O(\varepsilon^3). \tag{52.96}$$

The first follows from $\eta = 2\gamma_*$ with $\gamma_\phi = \frac{\varepsilon^2}{108} + \cdots$ of Eq. (52.92); the second is the relevant RG eigenvalue at the fixed point; the third is the slope $\omega = \beta'_\lambda(\lambda_*)$ of Theorem 24.31, whose meaning we return to in Section 52.4.5.

The anomalous dimension η. The leading term gives $\eta \approx 1/54 = 0.0185$; adding the $O(\varepsilon^3)$ term gives $0.0185 + 0.0187 = 0.0372$. The two contributions are nearly equal in size at $\varepsilon = 1$: the series is *not* settling down. The exact value (we will see) is $\eta = 0.0363$, so the one-term truncation undershoots by half and the two-term truncation overshoots slightly. There is no sense in which "adding the next term improves the answer by a controlled amount"; the corrections are $O(1)$ relative to each other.

The correlation-length exponent ν. From Eq. (52.95) at $\varepsilon = 1$: one term gives $1/\nu = 2 - 0.333 = 1.667$, hence $\nu = 0.600$; two terms give $1/\nu = 1.667 - 0.117 = 1.549$, hence $\nu = 0.645$. The true value is $\nu = 0.630$, so the partial sums straddle it (0.600, then 0.645) without converging onto it. Equivalently $\Delta_\varepsilon = d - 1/\nu = 3 - 1/\nu$ runs $1.333 \to 1.451$ against the true 1.413.

The correction-to-scaling exponent ω. This is the worst-behaved of the three. One term gives $\omega = \varepsilon = 1$; two terms give $\omega = 1 - \frac{17}{27} = 1 - 0.630 = 0.370$. The true value is $\omega \approx 0.83$. The two-term truncation is off by more than a factor of two and on the wrong side, and the first correction is comparable to the leading term — the unmistakable signature of a series being evaluated far outside its useful range.

The verdict. Setting $\varepsilon = 1$ in the truncated series produces numbers that are *qualitatively* right — η small and positive, ν near 0.6, ω of order unity, the right ordering and signs — and quantitatively good to perhaps ten or twenty percent for the well-behaved exponents, far worse for ω. They establish that an interacting fixed point exists in $d = 3$ with roughly these exponents. They do not, on their own, deliver the three-dimensional Ising numbers to any precision worth comparing with experiment. The deficit $\varepsilon = 1$ is simply not small, and a few terms of an asymptotic series at argument unity can do no better.

The honest reading of Theorem 52.13 is that the bare ε-expansion is a *structural* tool, not a precision one. It proves the fixed point exists, fixes which operators are relevant, and gives the leading dependence of every exponent on d and on the number of field components N — enormous qualitative power. But the series is divergent, and at $\varepsilon = 1$ its naive truncations have errors of the same order as the answer for the slowly-converging exponents. To extract precise three-dimensional numbers from it requires a separate, nontrivial step: resummation.

52.4.2 Why the series diverges, and how it is resummed

The ε-expansion is an asymptotic series with zero radius of convergence. This is not a defect peculiar to critical phenomena; it is the generic fate of perturbation

series in quantum field theory, and its origin is the one identified by Dyson for quantum electrodynamics and made quantitative for ϕ^4 by the large-order analysis of the path integral. The coefficient of ε^k in any of these series grows factorially,

$$a_k \sim c\,(-a)^k\,k!\,k^b \qquad (k \to \infty), \tag{52.97}$$

with $a > 0$ a positive constant fixed by the action of the instanton of the Euclidean ϕ^4 theory — the finite-action saddle of the functional integral that dominates the high orders. Because $k!\,a^k$ outpaces any power of ε, the series $\sum_k a_k\,\varepsilon^k$ converges for no nonzero ε. It is asymptotic in the precise sense that for fixed small ε the partial sums first approach the true value, reach a closest approach at an optimal order $k_{\mathrm{opt}} \sim 1/(a\varepsilon)$, and thereafter diverge; the smallest achievable error is of order $e^{-1/(a\varepsilon)}$. At $\varepsilon = 1$ the optimal order is small — a handful of terms — and the closest approach is not very close, exactly the behaviour Theorem 52.13 exhibited.

The sign alternation in Eq. (52.97), $a_k \propto (-a)^k$, is the feature that makes the series resummable. A series whose coefficients grow like $k!$ but alternate in sign is *Borel summable*: the divergence is tamed by the integral transform developed in Chapter 5. One forms the Borel transform

$$B(t) = \sum_{k=0}^{\infty} \frac{a_k}{k!}\,t^k, \tag{52.98}$$

which by Eq. (52.97) has a *finite* radius of convergence $|t| < 1/a$ (the $k!$ has been divided out), and recovers the resummed function by the Laplace integral

$$f(\varepsilon) = \frac{1}{\varepsilon}\int_0^{\infty} dt\, e^{-t/\varepsilon}\,B(t), \tag{52.99}$$

provided $B(t)$ can be continued along the positive real axis and the integral converges. The alternating sign places the nearest singularity of $B(t)$ on the *negative* real axis, at $t = -1/a$, away from the integration contour; the positive axis is clear, and Eq. (52.99) is well defined. (Had the coefficients been of one sign — as they are for the non-Borel-summable cases such as the ground-state energy of certain quantum-mechanical wells — the singularity would sit on the contour and the prescription would be ambiguous, signalling a genuine nonperturbative contribution.)

In practice one knows only finitely many coefficients $a_0, \dots, a_K$ (for the Ising exponents, through ε^5 or ε^6), so $B(t)$ is known only as a truncated Taylor polynomial, useless beyond its disk of convergence. The decisive refinement, due to the work of Le Guillou and Zinn-Justin, is to map the cut t-plane conformally onto a disk before resumming. The Borel transform is analytic in the t-plane cut from $t = -1/a$ to $-\infty$; the conformal map

$$w(t) = \frac{\sqrt{1 + at} - 1}{\sqrt{1 + at} + 1} \tag{52.100}$$

sends that cut plane to the unit disk $|w| < 1$, with the singularity at $t = -1/a$ pushed to the boundary $w = -1$ and the entire positive t-axis (the integration range) mapped into the interior. Re-expanding B as a power series in w and truncating gives an approximant that converges *inside the disk*, i.e. everywhere the Laplace integral needs it, rather than merely inside $|t| < 1/a$. Feeding this approximant into Eq. (52.99) and evaluating at $\varepsilon = 1$ produces a number with a controlled, shrinking error as more coefficients are included. This Borel–conformal-mapping resummation is what converts the divergent series of Theorem 52.13 into the precise three-dimensional exponents.

Physical Insight

The logic is worth holding in one sentence: the perturbative series *encodes* the right answer, but encrypted in a divergent form, and the instanton-controlled large-order behaviour Eq. (52.97) is precisely the key — it tells us the location $t = -1/a$ of the Borel singularity, which is exactly the information the conformal map Eq. (52.100) needs to decrypt the series. The divergence is not noise to be discarded; its rate is data, and resummation is the act of using it.

We do not carry the resummation arithmetic itself — it is a numerical procedure on six known coefficients, reproducible but not illuminating to grind by hand, and Chapter 5 established the transform that underlies it. What matters here is the conceptual contract: the resummed ε-expansion is an honest, controlled extraction of three-dimensional numbers from a divergent series, with quantifiable uncertainties, and it is this resummed output — not the naive truncations of Theorem 52.13 — that one compares with the bootstrap.

52.4.3 The comparison: three methods, one CFT

Table 52.2 sets the resummed ε-expansion exponents beside the numerical-bootstrap determination of Chapter 50 (the island of Table 50.1, read through the dictionary $\eta = 2\Delta_\sigma - 1$, $1/\nu = d - \Delta_\varepsilon$ of Eqs. (49.162) and (49.163)) and beside the best Monte Carlo simulations of the lattice model.

Read the table column by column and the agreement is total to the precision each method claims. The three determinations of ν are 0.6304(13), 0.629971(4), 0.63002(10): they coincide within errors, the bootstrap being the most precise by three orders of magnitude. The three values of η are 0.0335(25), 0.0362978(20), 0.03627(10); here the resummed ε-expansion sits about one of its own standard deviations low, a known feature of the η series, while the bootstrap and Monte Carlo agree to five figures. Even the awkward correction-to-scaling exponent ω, which the naive truncation of Theorem 52.13 got so badly wrong, comes into line once resummed: 0.799(11) from the series against 0.832(6) from Monte Carlo and

Table 52.2: The leading 3D Ising critical data from three independent methods. *Resummed ε-expansion*: Borel–conformal-mapping resummation of the $O(N{=}1)$ series through $O(\varepsilon^5)$ [318]; the dimensions $\Delta_\sigma = (1+\eta)/2$ and $\Delta_\varepsilon = 3 - 1/\nu$ are derived from (η, ν). *Numerical bootstrap*: the mixed-correlator island of Chapter 50 [304], with $\Delta_\sigma, \Delta_\varepsilon$ primary and the exponents derived through Eqs. (49.162) and (49.163); $\omega = \Delta_{\varepsilon'} - d$ from the leading irrelevant even scalar (Eq. (50.129)). *Monte Carlo*: high-statistics finite-size scaling of the lattice model [254]. Parenthesized digits are one-standard-deviation uncertainties in the last figures; the ε-expansion uncertainties are resummation-method estimates. Values current to the mid-2020s.

quantity	resummed ε-expansion	numerical bootstrap	Monte Carlo
Δ_σ	0.5168(13)	0.5181489(10)	0.518135(50)
Δ_ε	1.414(3)	1.412625(10)	1.41275(25)
η	0.0335(25)	0.0362978(20)	0.03627(10)
ν	0.6304(13)	0.629971(4)	0.63002(10)
ω	0.799(11)	≈ 0.830	0.832(6)

≈ 0.83 from the bootstrap's leading irrelevant scalar. Figure 52.4 displays the same three exponents graphically, the error bars overlapping in every panel.

Historical Note

The convergence shown here was hard-won and runs over half a century. Wilson and Fisher proposed the ε-expansion in 1972; through the 1970s and 1980s the resummed series and improving Monte Carlo data slowly tightened the Ising exponents to the third decimal, with the resummation systematics — how many terms, which map, what error to quote — a delicate and somewhat dark art. The numerical bootstrap entered this mature field in 2012 and, within a few years, overtook every prior method by orders of magnitude, not by computing harder but by abandoning the Lagrangian altogether and demanding only crossing symmetry and unitarity. That a sixty-year-old perturbative idea and a decade-old optimization method land on the same six-figure number is the field's quiet triumph.

52.4.4 Why the agreement is meaningful: two computations of the same object

It would be a mistake to read Table 52.2 as one method checking another's arithmetic, as if both were approximating some third, more fundamental computation. They are not. The two principal entries — the ε-expansion and the bootstrap — are computations of *genuinely different kinds*, and their agreement is informative

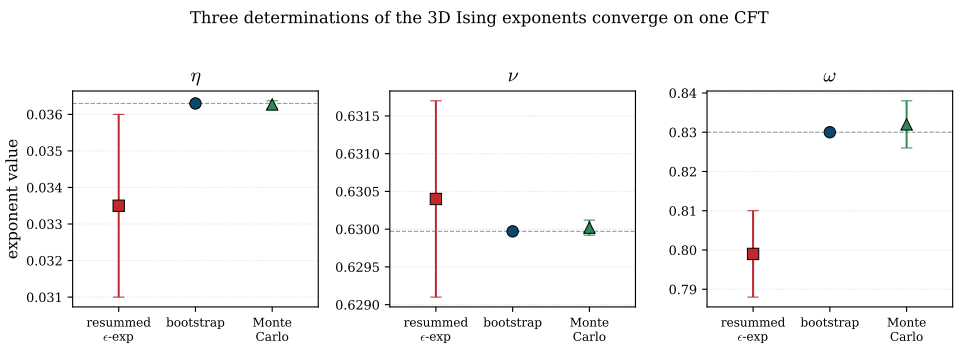


Figure 52.4: The three independent determinations of the leading 3D Ising exponents η, ν, ω from Table 52.2: the resummed ε-expansion (squares), the numerical conformal bootstrap of Chapter 50 (circles; the bootstrap error on η and ν is smaller than the marker), and Monte Carlo (triangles), each with its one-standard-deviation error bar. The dashed line marks the bootstrap central value. A perturbative Lagrangian computation around $d = 4$, a nonperturbative crossing computation in $d = 3$, and a lattice simulation agree within their uncertainties on every exponent — the quantitative statement that all three are measuring one conformal field theory.

precisely because they share no common premise that could be the source of a shared error.

The ε-expansion is a *perturbative Lagrangian* computation. It begins with the action of ϕ^4 theory, $S_E = \int d^d x \left[\frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4\right]$, posits that a fixed point exists at small coupling near $d = 4$, locates it as the nontrivial zero $\lambda_* = \frac{16\pi^2}{3}\varepsilon + O(\varepsilon^2)$ of the beta function (Eq. (24.255)), and computes operator dimensions as anomalous dimensions of fields and composites in a loop expansion. Every step is tied to the existence of a fundamental scalar field and a small parameter; the output is a series in ε. The method knows the theory is the continuation of a free scalar and uses that knowledge at every order.

The bootstrap is a *nonperturbative crossing* computation. It begins with no Lagrangian, no fields, and no coupling. It posits only that there exists *some* unitary CFT in $d = 3$ with a $\mathbb{Z}_2$ symmetry and the minimal relevant-operator content of Theorem 49.27, writes the crossing equation Eq. (49.15) that the four-point function of the order parameter must satisfy, and searches the space of admissible operator spectra for the ones consistent with crossing and unitarity (Chapter 50). The output is a small island in the plane of dimensions $(\Delta_\sigma, \Delta_\varepsilon)$, located by a convex optimization with no expansion parameter at all. The method never knows there is a scalar field underneath; it would find the same island in a universe where the lattice Ising model had never been written down.

Physical Insight

The complementarity is exact and structural. The ε-expansion is reliable where the bootstrap is weakest — near $d = 4$, where the coupling is small and the perturbative series converges quickly, but where the bootstrap's crossing equation is nearly degenerate. The bootstrap is sharpest where the ε-expansion is least trustworthy — at $d = 3$, far from the free theory, where the series must be resummed and the loop expansion is at the edge of its validity. Each method's region of confidence is the other's frontier, and they meet, with overlapping error bars, on the single CFT in between. Two maps drawn with incompatible projections, agreeing on the coastline, are far better evidence that the coastline is real than two copies of the same map.

This is the strongest form of the chapter's thesis. Theorem 52.11 established *as a theorem* that the Wilson–Fisher fixed point is a conformal field theory; Section 52.2 established the dictionary turning its operator dimensions into critical exponents. Table 52.2 now establishes, as a matter of computed numbers, that this CFT is the *same* object the bootstrap corners from consistency alone. The fixed-point-equals-CFT identification is not merely a definition or a structural theorem; it is a quantitative prediction, confirmed to six significant figures by methods that could in principle have disagreed and did not.

52.4.5 The correction-to-scaling exponent is a bootstrap operator dimension

One entry in Table 52.2 deserves its own derivation, because it ties a quantity defined by the renormalization group — the slope of the beta function at the fixed point — to a quantity defined by the bootstrap — the dimension of an operator in the spectrum. This is the exponent ω, and the identity between its two definitions is the sharpest single instance of the RG/CFT dictionary at work.

On the RG side, ω was defined in Theorem 24.26 as the slope of the beta function at the fixed point,

$$\omega = \beta'_\lambda(\lambda_*), \tag{52.101}$$

the eigenvalue governing how a small deviation $\delta\lambda = \lambda - \lambda_*$ flows: near the fixed point $\beta_\lambda(\lambda) \simeq \omega\,\delta\lambda$, so $\delta\lambda(\mu) \propto \mu^\omega$, and $\omega > 0$ means the deviation decays into the infrared — the Wilson–Fisher point is infrared-attractive, the deviation is an *irrelevant* direction. The exponent ω controls how a physical quantity approaches its scaling form as the critical point is neared: corrections to pure power-law scaling are suppressed by a factor $t^{\omega\nu}$ in the reduced temperature, the leading *correction to scaling*. For the one-loop ϕ^4 beta function $\beta_\lambda = -\varepsilon\lambda + \frac{3\lambda^2}{16\pi^2}$ of Eq. (24.254), differentiating at $\lambda_* = \frac{16\pi^2}{3}\varepsilon$ gives $\beta'_\lambda(\lambda_*) = -\varepsilon + \frac{6\lambda_*}{16\pi^2} = -\varepsilon + 2\varepsilon = +\varepsilon$, the leading term of Eq. (52.96).

On the CFT side, the general dictionary of Section 52.2.2 assigns to a deformation of the fixed-point action by a scalar primary $\mathcal{O}$ of dimension $\Delta_{\mathcal{O}}$ the RG eigenvalue $y_{\mathcal{O}} = d - \Delta_{\mathcal{O}}$: the coupling $g_{\mathcal{O}}$ multiplying $\int d^d x\, \mathcal{O}$ has mass dimension $d - \Delta_{\mathcal{O}}$, and under rescaling by b flows as $g_{\mathcal{O}}(b) = b^{d-\Delta_{\mathcal{O}}} g_{\mathcal{O}}$. A deformation is irrelevant precisely when $\Delta_{\mathcal{O}} > d$, and for an irrelevant direction the correction-to-scaling exponent is the positive number $\omega_{\mathcal{O}} = -y_{\mathcal{O}} = \Delta_{\mathcal{O}} - d$ (Section 52.2.4). We now show the RG slope Eq. (52.101) is exactly this number for a definite operator.

Proposition 52.14 (The slope $\beta'_\lambda(\lambda_*)$ is the dimension of the leading irrelevant $\mathbb{Z}_2$-even scalar, minus d)**.** *At the Wilson–Fisher fixed point the slope of the beta function equals the correction-to-scaling exponent set by the leading irrelevant $\mathbb{Z}_2$-even scalar operator ε':*

$$\omega = \beta'_\lambda(\lambda_*) = \Delta_{\varepsilon'} - d, \tag{52.102}$$

where ε' is the lowest-dimension $\mathbb{Z}_2$-even scalar above the energy operator ε — the operator into which the quartic interaction $\frac{\lambda}{4!}\phi^4$ renormalizes at the fixed point.

Proof. The deviation $\delta\lambda = \lambda - \lambda_*$ is, by construction, the coefficient of the operator that the coupling λ multiplies in the action, namely $\frac{1}{4!}\phi^4$. At the fixed point this composite is not an eigenoperator of the dilatation: it mixes, under renormalization, with the lower-dimension even scalars (the identity, which only shifts the cosmological constant, and ϕ^2) and with descendants. Projecting onto the conformal primaries, the part of $\frac{1}{4!}\phi^4$ orthogonal to the relevant directions is the eigenoperator ε', the lowest $\mathbb{Z}_2$-even scalar primary whose deformation is irrelevant; this is the operator that controls the leading approach to scaling once the relevant tunings (temperature, conjugate to ε) have been made. The linearized flow of $\delta\lambda$ is therefore the linearized flow of the coupling of ε', and by the dictionary of Section 52.2.2 its eigenvalue is

$$\frac{d(\delta\lambda)}{d\ln b} = y_{\varepsilon'}\,\delta\lambda = (d - \Delta_{\varepsilon'})\,\delta\lambda. \tag{52.103}$$

Comparing with the beta-function form $d(\delta\lambda)/d\ln b = -\beta_\lambda(\lambda) \simeq -\omega\,\delta\lambda$ in the long-distance (increasing b, decreasing μ) convention — where an irrelevant deviation must decay, $y_{\varepsilon'} < 0$ — gives $-\omega = y_{\varepsilon'} = d - \Delta_{\varepsilon'}$, i.e. $\omega = \Delta_{\varepsilon'} - d$. The sign is consistent: $\Delta_{\varepsilon'} > d$ (the operator is irrelevant) gives $\omega > 0$, matching $\beta'_\lambda(\lambda_*) = +\varepsilon > 0$ found above. □

The content of Theorem 52.14 is that the number the RG computes as a property of the flow — the slope of a vector field at a zero — is, on the nose, a number the bootstrap computes as a property of the spectrum: the dimension of a specific operator, displaced from marginality. We can now check both sides

against Chapter 50. Solving Eq. (52.102) for the dimension, $\Delta_{\varepsilon'} = d + \omega$, the leading ε-expansion result $\omega = \varepsilon$ gives $\Delta_{\varepsilon'} = (4-\varepsilon) + \varepsilon = 4 + O(\varepsilon^2)$, refined by Eq. (52.96) to $\Delta_{\varepsilon'} = 4 - \frac{17}{27}\varepsilon^2 + O(\varepsilon^3)$, which at the naive $\varepsilon = 1$ reads $\Delta_{\varepsilon'} \approx 3.37$. The numerical bootstrap, meanwhile, locates the second $\mathbb{Z}_2$-even scalar directly as the next double-zero of its extremal functional, at $\Delta_{\varepsilon'} \approx 3.83$ (Eq. (50.129)), giving $\omega = \Delta_{\varepsilon'} - d \approx 0.83$. The naive perturbative $\Delta_{\varepsilon'} \approx 3.37$ and the bootstrap 3.83 differ by exactly the resummation gap seen for ω in Theorem 52.13; resummed, the ε-expansion returns $\omega = 0.799(11)$, in agreement. The same operator, the same number, found as a beta-function slope around $d = 4$ and as a double-zero of a positivity certificate in $d = 3$.

52.4.6 Cross-check from the analytic bootstrap: the large-spin tail

There is a final, internal consistency check available, of a different character from the comparison with experiment: the spectrum the numerical bootstrap reads off in Chapter 50 must agree, at large spin, with the closed-form prediction of the *analytic* bootstrap of Chapter 51. The two are computed by completely different machinery — one a finite-dimensional semidefinite program, the other an asymptotic expansion of the crossing equation in the lightcone limit — so their agreement on the high-spin part of the same spectrum is a nontrivial test that the island of Table 50.1 is a consistent CFT and not an artifact.

The relevant prediction is the large-spin anomalous dimension of the leading-twist double-twist family. In any unitary CFT the operators $[\sigma\sigma]_{0,\ell}$ — the lowest-twist $\mathbb{Z}_2$-even spin-ℓ operators built from two copies of the order parameter — exist at large ℓ with twist approaching $2\Delta_\sigma$, and their anomalous dimension is fixed by the minimal-twist operator exchanged in the crossed channel. For the 3D Ising CFT that exchanged operator is the energy operator ε, the lightest $\mathbb{Z}_2$-even scalar, and Theorem 51.10 (applied to ϕ^4 theory in Section 51.3.6) gives

$$\gamma_{0,\ell} \;\simeq\; -\frac{c_\varepsilon}{\ell^{\,\Delta_\varepsilon}}, \qquad c_\varepsilon = \frac{2\,\Gamma(\Delta_\sigma)^2}{\Gamma(\Delta_\sigma - \frac{\Delta_\varepsilon}{2})^2}\,\frac{\Gamma(\Delta_\varepsilon)}{\Gamma(\frac{\Delta_\varepsilon}{2})^2}\,\lambda_{\sigma\sigma\varepsilon}^2, \tag{52.104}$$

the coefficient c_ε being entirely determined by three pieces of CFT data: the two dimensions $\Delta_\sigma, \Delta_\varepsilon$ and the OPE coefficient $\lambda_{\sigma\sigma\varepsilon}$. Crucially, all three are quantities the numerical bootstrap measures independently. The cross-check is to feed the Chapter 50 data into Eq. (52.104) and ask whether the resulting tail matches the high-spin operators the same bootstrap computation finds.

Worked Example 52.15 (The Ising large-spin tail from the bootstrap data)**.** We evaluate Eq. (52.104) at the three-dimensional Ising point, using the precise extremal-functional data of Chapter 50:

$$\Delta_\sigma = 0.5181489, \qquad \Delta_\varepsilon = 1.412625, \qquad \lambda_{\sigma\sigma\varepsilon} = 1.0518537, \tag{52.105}$$

so $\lambda^2_{\sigma\sigma\varepsilon} = 1.106396$. The Gamma-function factors are assembled step by step. First the arguments:

$$\frac{\Delta_\varepsilon}{2} = 0.7063125, \qquad \Delta_\sigma - \frac{\Delta_\varepsilon}{2} = 0.5181489 - 0.7063125 = -0.1881636, \tag{52.106}$$

the last being negative — the order parameter is lighter than half the energy operator — so its Gamma function lands on the reflection-formula side of the axis. Evaluating,

$$\Gamma(\Delta_\sigma) = \Gamma(0.5181489) = 1.711766, \qquad \Gamma(\Delta_\sigma)^2 = 2.930144, \tag{52.107}$$

$$\Gamma\big(\Delta_\sigma - \tfrac{\Delta_\varepsilon}{2}\big) = \Gamma(-0.1881636) = -6.118034, \qquad \Gamma(\cdots)^2 = 37.43034, \tag{52.108}$$

$$\Gamma(\Delta_\varepsilon) = \Gamma(1.412625) = 0.886649, \qquad \Gamma\big(\tfrac{\Delta_\varepsilon}{2}\big)^2 = \Gamma(0.7063125)^2 = 1.659380. \tag{52.109}$$

Assembling the two ratios,

$$\frac{2\,\Gamma(\Delta_\sigma)^2}{\Gamma(\Delta_\sigma - \frac{\Delta_\varepsilon}{2})^2} = \frac{2(2.930144)}{37.43034} = 0.156565, \qquad \frac{\Gamma(\Delta_\varepsilon)}{\Gamma(\frac{\Delta_\varepsilon}{2})^2} = \frac{0.886649}{1.659380} = 0.534325, \tag{52.110}$$

and multiplying by $\lambda^2_{\sigma\sigma\varepsilon}$,

$$c_\varepsilon = 0.156565 \times 0.534325 \times 1.106396 = 0.09256. \tag{52.111}$$

The leading-twist family of the 3D Ising CFT therefore has anomalous dimensions

$$\gamma_{0,\ell} \simeq -\frac{0.0926}{\ell^{\,1.41263}} \qquad (\ell \to \infty), \tag{52.112}$$

negative (the binding is attractive, as unitarity requires by the convexity argument of Section 51.3.4) and falling off as $\ell^{-\Delta_\varepsilon}$. Numerically this predicts, for example, $\gamma_{0,\ell} \approx -0.013$ at $\ell = 4$, -0.0036 at $\ell = 10$, and -0.0013 at $\ell = 20$: the operators $[\sigma\sigma]_{0,\ell}$ approach the double-twist value $2\Delta_\sigma + \ell$ from below, ever more closely as the spin grows. These are precisely the dimensions the spin-ℓ double-zeros of the extremal functional in Chapter 50 converge to at large ℓ — the $\ell = 2, 4, 6, \ldots$ operators that Table 50.1 and Eq. (50.129) found organizing into a double-twist tower. The analytic and numerical bootstraps agree on the same spectrum from opposite ends of the spin axis.

The agreement in Theorem 52.15 closes a second loop, internal to the bootstrap: the OPE data $(\Delta_\sigma, \Delta_\varepsilon, \lambda_{\sigma\sigma\varepsilon})$ that the numerical method extracts as inputs reproduce, through the analytic large-spin formula, the high-spin part of the spectrum the same method computes as outputs. A spectrum that failed this test

would be internally inconsistent; the Ising data pass it. Together with the experimental and Monte Carlo agreement of Table 52.2, this completes the case that the object cornered in Chapter 50, computed in Chapter 24, and characterized as a CFT in Theorem 52.11 is one and the same three-dimensional conformal field theory, consistent with itself and with every external measurement of the Ising universality class.

This consummates ϕ^4 theory. The real scalar with a quartic self-interaction and a $\mathbb{Z}_2$ symmetry, introduced as a classical Lagrangian in Chapter 6, quantized in Chapters 9 to 10, renormalized in Chapter 22, and pushed to its nontrivial fixed point in Chapter 24, has been followed all the way to a single, fully determined conformal field theory whose exponents match the laboratory. The thread that began with a free field and a coupling ends on a number, $\nu = 0.629971(4)$, that three independent computations and a generation of experiments agree upon.

Having identified the CFT that lives at the end of the Wilson–Fisher flow, we have characterized the *endpoints* of renormalization-group trajectories as conformal field theories. The natural next question is about the flow itself: a trajectory runs from a fixed point in the ultraviolet to a fixed point in the infrared, and one wants to know whether that direction is intrinsic — whether there is a quantity, built from the CFT data at the two ends, that always decreases from UV to IR and so makes the flow irreversible. The search for such a monotone, and the c- and a-theorems that supply it, is the subject of Section 52.5.

52.5 RG Flows Between Fixed Points: the c- and a-Theorems

The previous section closed the loop we had been winding since the classical Lagrangian of Chapter 6: the Wilson–Fisher fixed point, located perturbatively in Theorem 23.36 and re-read here as a conformal field theory, was matched against the numerical bootstrap of Chapter 50 and against laboratory critical exponents. The picture that emerges is sharp. A conformal field theory is the universal description sitting at a fixed point of the renormalization-group flow, and the bootstrap data — dimensions and OPE coefficients — are the critical exponents of real matter. What we have not yet asked is a question about the flow *itself*, as opposed to its endpoints. A relevant deformation switched on at a fixed point carries the theory along a trajectory in coupling space; that trajectory begins at one CFT in the ultraviolet and ends at another in the infrared. Does the flow have a direction? Can a trajectory ever close into a loop, or reverse, returning a theory to where it began? The answer, and the deepest structural statement of this chapter, is that the flow is *irreversible*: there exists a number, computed entirely from the CFT data at the two ends, that is strictly larger at the ultraviolet end than at the infrared end. The renormalization group has an arrow, and the arrow is a theorem.

This section establishes that arrow in the two dimensions where it is a theorem with a complete proof — $d = 2$, Zamolodchikov's c-theorem — and in the dimension of the physical world — $d = 4$, the a-theorem of Komargodski and Schwimmer — where we give an honest and complete-as-sketch argument. We then turn the result back on ϕ^4 theory as a consistency test of the Wilson–Fisher fixed point, and close with a dated, honest account of how far the monotone is known to exist in general dimension.

52.5.1 The arrow of the renormalization group

The intuition is older than its proof and worth stating plainly before any formula, because it is correct and it is what the formulas make precise. Coarse-graining — integrating out short-distance modes, the operation that *is* the renormalization-group flow of Chapter 23 — discards information. Each step replaces a microscopic description by an effective one with fewer accessible degrees of freedom; the irrelevant directions are washed out (Theorem 23.34), and what survives at long distance is a quotient of what was present at short distance. If some quantity faithfully counted "the number of degrees of freedom" of a scale-invariant theory, that quantity ought to decrease as we flow from the ultraviolet to the infrared. The content of the c- and a-theorems is that such a quantity exists, that it is built from the response of the stress tensor (the one operator every local theory possesses), and that its monotonicity follows from unitarity — the same positivity that underlies every physical bound in the book.

Two features must hold for the statement to have teeth. First, the monotone must be *stationary* at fixed points, so that at a CFT it returns a pure number depending only on the CFT data, not on the trajectory by which the theory was reached. Second, that number must be the *central charge*: the anomaly coefficient of the stress tensor, c in two dimensions (Theorem 48.8) and a in four. We prove both features in $d = 2$ and exhibit them in $d = 4$.

A word on conventions, since the entire section lives in Euclidean signature. Critical phenomena are Euclidean (Chapter 10), and the proofs use reflection positivity — the Euclidean avatar of the unitarity of the underlying Minkowski theory, established for the Euclidean functional integral in Chapter 10. In two dimensions we use the complex coordinates $z = x^1 + ix^2$, $\bar{z} = x^1 - ix^2$ of Chapter 48, with $\partial \equiv \partial_z = \frac{1}{2}(\partial_1 - i\partial_2)$ and $\bar{\partial} \equiv \partial_{\bar{z}} = \frac{1}{2}(\partial_1 + i\partial_2)$, so that $z\bar{z} = |x|^2 \equiv r^2$.

52.5.2 The two-dimensional c-theorem

We work along a generic renormalization-group trajectory in a two-dimensional, unitary, Poincaré-invariant local field theory — not at a fixed point. The theory has a symmetric, conserved stress tensor $T_{\mu\nu}$. Away from a fixed point it is generally not traceless: by the operator equation Eq. (47.3) and its quantum

descendant, the trace measures the running couplings, $\Theta \equiv T^{\mu}{}_{\mu} = \sum_i \beta_i \mathcal{O}_i$, and vanishes precisely where every $\beta_i = 0$. The whole proof is an exercise in what conservation and rotational invariance permit, combined at the end with one positivity statement.

The three structure functions. In complex coordinates the symmetric stress tensor has three independent components, which it is traditional to package as

$$T \equiv T_{zz}, \qquad \bar{T} \equiv T_{\bar{z}\bar{z}}, \qquad \Theta \equiv T^{\mu}{}_{\mu} = 4\,T_{z\bar{z}}, \tag{52.113}$$

the last equality following from $T^{\mu}{}_{\mu} = g^{\mu\nu}T_{\mu\nu} = 2g^{z\bar{z}}T_{z\bar{z}} + 2g^{\bar{z}z}T_{\bar{z}z} = 4T_{z\bar{z}}$, where the flat metric in these coordinates is $g_{z\bar{z}} = \frac{1}{2}$, $g^{z\bar{z}} = 2$. The components T, $\bar{T}$, Θ carry definite spin: under a rotation $z \to e^{i\alpha}z$ they pick up $e^{-2i\alpha}$, $e^{+2i\alpha}$, and 1 respectively, since each lower z index contributes $e^{-i\alpha}$ and each $\bar{z}$ index $e^{+i\alpha}$. Thus T has spin $+2$, $\bar{T}$ has spin -2, and Θ is a scalar.

Conservation $\partial^{\mu}T_{\mu\nu} = 0$ becomes, for the $\nu = z$ component, $\partial^{z}T_{zz} + \partial^{\bar{z}}T_{\bar{z}z} = 0$; raising indices with $g^{z\bar{z}} = 2$ gives $\partial^{z} = 2\bar{\partial}$ and $\partial^{\bar{z}} = 2\partial$, so $2\bar{\partial}T_{zz} + 2\partial T_{\bar{z}z} = 0$. Using $T_{\bar{z}z} = T_{z\bar{z}} = \Theta/4$ from Eq. (52.113),

$$\bar{\partial}\,T = -\tfrac{1}{4}\,\partial\,\Theta, \tag{52.114}$$

and the complex conjugate equation $\partial\,\bar{T} = -\frac{1}{4}\,\bar{\partial}\,\Theta$ for the antiholomorphic component. Equation Eq. (52.114) is the engine of the proof: it says that the failure of T to be purely holomorphic — the failure of the chiral splitting that defines a fixed-point CFT in Chapter 48 — is controlled entirely by the trace Θ, hence by the beta functions.

Now consider the equal-time, translation-invariant two-point functions of these components with themselves and with each other. Translation invariance makes them depend only on the separation z, $\bar{z}$; rotational invariance and the spin assignments above then fix the index structure, leaving three scalar functions of the rotational invariant $z\bar{z}$. Writing $u \equiv z\bar{z} = r^2$ and using a prime for d/du, we define

$$\langle T(z,\bar{z})\,T(0)\rangle = \frac{F(u)}{z^4}, \tag{52.115a}$$

$$\langle T(z,\bar{z})\,\Theta(0)\rangle = \frac{G(u)}{z^3\,\bar{z}}, \tag{52.115b}$$

$$\langle \Theta(z,\bar{z})\,\Theta(0)\rangle = \frac{H(u)}{z^2\,\bar{z}^2}. \tag{52.115c}$$

The prefactor powers are dictated by spin: the right-hand side of each line must carry the total spin of the left-hand side (respectively $+4$, $+2$, 0), and the leftover scalar dependence is absorbed into F, G, H. The factor z^{-4} in Eq. (52.115a), for instance, is the unique spin-4 monomial of dimension 4 in $z, \bar{z}$ up to a scalar function; multiplying by a function of $z\bar{z}$ exhausts the freedom.

Reflection positivity, which we invoke below, requires that F, G, H be real, and the mixed function $\langle \Theta\, T\rangle$ equals $\langle T\, \Theta\rangle$ because sending $z \to -z$ in Eq. (52.115b) leaves $z^3\bar z$ invariant.

Two differential relations from conservation. Insert the conservation law Eq. (52.114) into the correlators. The operator identity $\bar\partial T = -\frac{1}{4}\partial\Theta$ holds inside any correlation function (it is an equation of motion for the conserved current; contact terms sit at coincident points, which we keep away from). First, act with $\bar\partial$ on Eq. (52.115a) and equate to $-\frac{1}{4}\partial$ of Eq. (52.115b). Since $F = F(u)$ with $u = z\bar z$, we have $\bar\partial F = zF'$ and $\partial F = \bar z F'$, so

$$\bar\partial\,\frac{F}{z^4} = \frac{zF'}{z^4} = \frac{F'}{z^3}, \qquad \partial\,\frac{G}{z^3\bar z} = \frac{\bar z\,G'}{z^3\bar z} - \frac{3G}{z^4\bar z} = \frac{G'}{z^3} - \frac{3G}{z^4\bar z}. \tag{52.116}$$

Equality $\bar\partial\langle TT\rangle = -\frac{1}{4}\partial\langle\Theta T\rangle$ gives, after multiplying through by z^3 and writing $z\bar z = u$,

$$4F' + G' - \frac{3G}{u} = 0. \tag{52.117}$$

Second, act with $\bar\partial$ on Eq. (52.115b) and equate to $-\frac{1}{4}\partial$ of Eq. (52.115c). Using $\bar\partial(z^{-3}\bar z^{-1}) = -z^{-3}\bar z^{-2}$ and $\partial(z^{-2}\bar z^{-2}) = -2z^{-3}\bar z^{-2}$,

$$\bar\partial\,\frac{G}{z^3\bar z} = \frac{G'}{z^2\bar z} - \frac{G}{z^3\bar z^2}, \qquad \partial\,\frac{H}{z^2\bar z^2} = \frac{H'}{z^2\bar z} - \frac{2H}{z^3\bar z^2}, \tag{52.118}$$

and the equality $\bar\partial\langle T\Theta\rangle = -\frac{1}{4}\partial\langle\Theta\Theta\rangle$, multiplied by $z^2\bar z$, yields

$$4G' - \frac{4G}{u} + H' - \frac{2H}{u} = 0. \tag{52.119}$$

Relations Eqs. (52.117) and (52.119) are exact consequences of conservation and rotational invariance, valid all along the flow.

The c-function and its monotonicity. Zamolodchikov's insight was to find the linear combination of F, G, H whose derivative collapses to a single positive object. Define

Definition 52.16 (Zamolodchikov c-function)**.** Along a two-dimensional renormalization-group trajectory, the *c-function* is the dimensionless combination of the structure functions Eq. (52.115),

$$C(u) \equiv 2F(u) - G(u) - \tfrac{3}{8}\,H(u), \tag{52.120}$$

regarded as a function of the distance scale through $u = z\bar z = r^2$.

Differentiate Eq. (52.120): $C' = 2F' - G' - \frac{3}{8}H'$. From Eq. (52.117), $2F' = -\frac{1}{2}G' + \frac{3}{2}G/u$, so

$$C' = -\tfrac{3}{2}G' + \tfrac{3}{2}\frac{G}{u} - \tfrac{3}{8}H'. \tag{52.121}$$

Multiply by u and substitute $uG' = G - \frac{1}{4}uH' + \frac{1}{2}H$, which is Eq. (52.119) rearranged ($G' = G/u - \frac{1}{4}H' + \frac{1}{2}H/u$):

$$\begin{aligned} u\,C' &= -\tfrac{3}{2}\,uG' + \tfrac{3}{2}G - \tfrac{3}{8}uH' = -\tfrac{3}{2}\Big(G - \tfrac{1}{4}uH' + \tfrac{1}{2}H\Big) + \tfrac{3}{2}G - \tfrac{3}{8}uH' \\ &= -\tfrac{3}{2}G + \tfrac{3}{8}uH' - \tfrac{3}{4}H + \tfrac{3}{2}G - \tfrac{3}{8}uH' = -\tfrac{3}{4}\,H. \end{aligned} \tag{52.122}$$

Every term except $-\frac{3}{4}H$ cancels. Since $u\,d/du = d/d\log u$ and $\log u = 2\log r$, this is

$$\frac{dC}{d\log r} = -\tfrac{3}{2}\,H(u). \tag{52.123}$$

It remains to fix the sign of H. The function H is the coefficient of $\langle\Theta(z,\bar z)\,\Theta(0)\rangle = H/(z^2\bar z^2) = H/r^4$, the two-point function of the Hermitian scalar operator Θ at Euclidean separation r. Reflection positivity (the reflection of Chapter 10: the Euclidean two-point function of any real operator with itself is a positive-definite kernel, because it equals the squared norm $\|\Theta\,|\,\cdot\rangle\|^2 \geq 0$ of a state in the reconstructed Hilbert space) forces $\langle\Theta(x)\Theta(0)\rangle \geq 0$, hence

$$H(u) \geq 0 \qquad \text{for all } u > 0. \tag{52.124}$$

Combining Eqs. (52.123) and (52.124) gives the monotonicity statement. We collect the result.

Theorem 52.17 (Zamolodchikov's c-theorem)**.** *In a two-dimensional unitary, Poincaré-invariant, local quantum field theory with a conserved stress tensor, the c-function of Theorem 52.16 satisfies*

$$\frac{dC}{d\log r} = -\tfrac{3}{2}\,H \leq 0, \tag{52.125}$$

so C decreases monotonically as the distance r increases, i.e. as the theory flows from the ultraviolet ($r \to 0$) to the infrared ($r \to \infty$). At any fixed point of the flow the trace vanishes, $\Theta = 0$, so $G = H = 0$ and C is stationary; there $C = 2F$, and matching to the stress-tensor two-point function of Chapter 48 gives

$$C\big|_{\textit{fixed point}} = c, \tag{52.126}$$

the Virasoro central charge of Theorem 48.8. Consequently, for any flow connecting an ultraviolet CFT to an infrared CFT,

$$\boxed{\,c_{\mathrm{UV}} \geq c_{\mathrm{IR}}.\,} \tag{52.127}$$

Proof. Equation Eq. (52.125) is Eq. (52.123) with the positivity Eq. (52.124). At a fixed point $\beta_i = 0$ for all i, so $\Theta = \sum_i \beta_i \mathcal{O}_i = 0$ as an operator; then $\langle T\Theta \rangle = \langle \Theta\Theta \rangle = 0$, i.e. $G = H = 0$, and $C = 2F$. The remaining function F is fixed by the stress-tensor two-point function. At the fixed point T is the holomorphic stress tensor of Chapter 48, whose self–operator product Eq. (48.56) has leading term $T(z)T(0) \sim (c/2)/z^4$; the c-number piece of that OPE is exactly the vacuum two-point function, $\langle T(z)T(0) \rangle = (c/2)/z^4$. Comparing with Eq. (52.115a) gives $F = c/2$, hence $C = 2F = c$, which is Eq. (52.126). Finally, C is monotone non-increasing in $\log r$ by Eq. (52.125), equals c_{UV} at $r \to 0$ and c_{IR} at $r \to \infty$, so $c_{\mathrm{UV}} = C(0^+) \geq C(\infty) = c_{\mathrm{IR}}$, which is Eq. (52.127). □

Physical Insight

The proof is short because all the physics is loaded into two facts. Conservation Eq. (52.114) is kinematics — it would hold in any local theory, fixed point or not — and it alone produces the algebraic miracle that $u\,C'$ collapses to a single structure function Eq. (52.122). The *sign* of that structure function is the one place dynamics enters, and it enters as positivity: $\langle \Theta\Theta \rangle \geq 0$ is unitarity, nothing more. The theorem is therefore the statement that *unitarity gives the renormalization group its arrow.* The same logic — a kinematic sum rule completed by a positivity bound — recurs in the four-dimensional argument below, where positivity is the positivity of a total cross section.

Two further remarks sharpen the result. First, C is not merely monotone but *strictly* decreasing wherever the theory is genuinely flowing, since $H > 0$ unless $\Theta = 0$ identically; the flow can rest only at fixed points. There is no possibility of a limit cycle or of a flow that returns to its starting CFT: the renormalization group in two unitary dimensions has no closed orbits. Second, the integrated form is a sum rule,

$$c_{\mathrm{UV}} - c_{\mathrm{IR}} = \frac{3}{2} \int_0^\infty H \, d\log r = \frac{3}{4\pi} \int d^2x \, |x|^2 \, \langle \Theta(x)\, \Theta(0) \rangle \;\geq 0, \tag{52.128}$$

expressing the total change in central charge as a positive-weighted integral of the trace two-point function over the flow; the second equality follows from $\int d^2x \, (\cdots) = \int_0^\infty 2\pi r \, dr \, (\cdots)$ and $\langle \Theta\Theta \rangle = H/r^4$. The change in degree-of-freedom count is literally the strength with which the trace operator — the operator that measures the breaking of scale invariance along the flow — is sourced.

Notation Watch

The symbol c does double duty in this chapter and the reader should keep the two meanings apart. In two dimensions there is a *single* central

charge c, the coefficient in $T(z)T(0) \sim (c/2)/z^4$ (Theorem 48.8), and it is this c that the c-theorem makes monotone. In four dimensions the Weyl anomaly has *two* independent coefficients, conventionally called a and c (defined in Eq. (52.130) below). It is the Euler-density coefficient a, *not* the four-dimensional c, that decreases along flows; explicit flows are known in which the four-dimensional c increases. The two-dimensional c is the analogue of the four-dimensional a, not of the four-dimensional c — a coincidence of names that has caused decades of confusion.

The central charge counts degrees of freedom. Equation Eq. (52.126) promotes the heuristic "c counts degrees of freedom" (Theorem 48.8) to a theorem about flows: whatever c counts, the count cannot increase under coarse-graining. The anchoring values are the free-field central charges computed in Chapter 48. A free real boson has $c = 1$ (Theorem 48.9); a free Majorana fermion has $c = \frac{1}{2}$ (Theorem 48.39); and the two-dimensional Ising universality class — the theory ϕ^4 theory approaches at criticality, here in its $d = 2$ avatar — is the $c = \frac{1}{2}$ minimal model (Eq. (48.231)), with a free-fermion microscopic description. The unitary minimal models $\mathcal{M}(m+1, m)$ have $c = 1 - 6/[m(m+1)]$ (Eq. (48.176)), an increasing sequence accumulating at $c = 1$ from below as $m \to \infty$. The following worked example traces how the c-theorem orders these theories along physical flows.

Worked Example 52.18 (The c-theorem orders the standard two-dimensional flows)**.** We verify $c_{\mathrm{UV}} > c_{\mathrm{IR}}$ for the canonical relevant deformations of two-dimensional critical theories, reading every central charge off Chapter 48.

(i) Massive free boson. Start at the free-boson CFT, $c_{\mathrm{UV}} = 1$ (Theorem 48.9), and add a mass term $\frac{1}{2}m^2\phi^2$ — the unique relevant scalar deformation. The mass term breaks scale invariance: the trace becomes $\Theta = m^2{:}\phi^2{:} \neq 0$, and Eq. (52.123) drives C downward. In the deep infrared, $r \gg m^{-1}$, every excitation is gapped; correlators decay exponentially, the stress tensor's long-distance two-point function vanishes, and the theory is trivial, $c_{\mathrm{IR}} = 0$. Thus the flow runs $1 \to 0$, and the integrated drop $\Delta c = 1$ is delivered by Eq. (52.128) as the (positive) integral of $\langle\Theta\Theta\rangle$. A gapped phase always sits at $c = 0$: a theory with a mass gap has no massless degrees of freedom to count.

(ii) Thermal Ising flow. Start at the Ising critical point, $c_{\mathrm{UV}} = \frac{1}{2}$, and perturb by the energy operator $\varepsilon = \phi_{1,3}$ (equivalently, move off the critical temperature). The flow is again to a gapped phase — ordered or disordered depending on the sign — so $c_{\mathrm{IR}} = 0$. The drop $\frac{1}{2} \to 0$ is consistent with Eq. (52.127), and the free-fermion description makes it transparent: turning on the energy operator is giving the Majorana fermion a mass, which gaps it.

(iii) The minimal-model cascade. The least trivial flows connect one nontrivial CFT to another. Perturbing the unitary minimal model $\mathcal{M}(m+1, m)$ by its least-

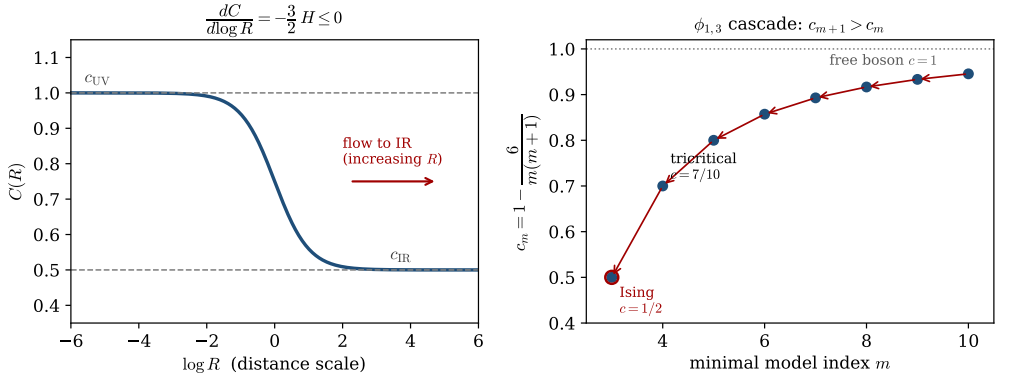


Figure 52.5: *Left:* the Zamolodchikov c-function $C(R)$ as a function of log-distance, decreasing monotonically from $c_{\rm UV}$ at short distance to $c_{\rm IR}$ at long distance, with slope $dC/d\log R = -\frac{3}{2}H \le 0$ fixed by the positive-definite trace two-point function (Theorem 52.17). The precise profile is theory-dependent; only the endpoints and the monotonicity are universal. *Right:* the unitary minimal-model staircase $c_m = 1 - 6/[m(m+1)]$, all values lying below the free-boson $c = 1$ (dotted), with the $\phi_{1,3}$ cascade $\mathcal{M}(m+1,m) \to \mathcal{M}(m,m-1)$ stepping the central charge strictly downward toward the Ising value $c = \frac{1}{2}$ ($m = 3$), as computed in Theorem 52.18.

relevant primary $\phi_{1,3}$ (for the appropriate sign) triggers a flow whose infrared endpoint is the next model down, $\mathcal{M}(m, m-1)$. Reading Eq. (48.176),

$$c_{m+1} - c_m = \Big(1 - \frac{6}{(m+1)(m+2)}\Big) - \Big(1 - \frac{6}{m(m+1)}\Big) = \frac{12}{m(m+1)(m+2)} > 0, \tag{52.129}$$

so c steps strictly downward at every rung, exactly as Eq. (52.127) demands. The first rung, $m = 3$, is the tricritical-Ising-to-Ising flow: $c = \frac{7}{10} \to c = \frac{1}{2}$, a drop of $\Delta c = \frac{12}{3\cdot 4\cdot 5} = \frac{1}{5}$. The staircase is drawn in Fig. 52.5; it terminates at the Ising point and then, under the thermal perturbation of part (ii), falls off the bottom to $c = 0$.

In every case the ultraviolet central charge dominates the infrared one, and the descent is irreversible: no relevant deformation of a minimal model flows *up* the staircase. This is the c-theorem made arithmetic.

52.5.3 The four-dimensional a-theorem

The two-dimensional proof leaned on two structures peculiar to $d = 2$: the holomorphic factorization that gives the stress tensor only three components, and the fact that a single number c controls both the anomaly and the degree-of-freedom count. In four dimensions neither holds, and the correct monotone

was not even identified for a quarter-century. The resolution rests on the Weyl (conformal) anomaly.

Place a four-dimensional CFT on a curved Euclidean background $g_{\mu\nu}$. Even though the flat-space theory is traceless, $\Theta = 0$, the curved-space expectation value of the trace does not vanish: it is fixed, by locality and dimensional analysis, to be a sum of the two independent dimension-four curvature scalars that are not total derivatives of lower-dimension scalars,

$$\langle T^{\mu}{}_{\mu} \rangle = \frac{1}{16\pi^2}\Big(c\, W_{\mu\nu\rho\sigma} W^{\mu\nu\rho\sigma} - a\, E_4 \Big), \qquad E_4 = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2, \tag{52.130}$$

where $W_{\mu\nu\rho\sigma}$ is the Weyl tensor and E_4 is the Euler density (the Gauss–Bonnet integrand, whose integral $\int \sqrt{g}\, E_4 = 32\pi^2 \chi$ is the topological Euler character). The two coefficients a and c are the four-dimensional central charges; a possible $\Box R$ term is a scheme-dependent total derivative and is dropped. (A free conformally-coupled real scalar has $(a, c) = \frac{1}{360}(1, 3)$; we use this value below.) The distinguished role of a is geometric: W^2 is Weyl-invariant, so the c-term in Eq. (52.130) contributes nothing on a conformally-flat background, whereas E_4 is *not* Weyl-invariant, and it is the a-term that records the anomaly under a Weyl rescaling. This is what singles a out as the candidate monotone, conjectured by Cardy [109] and proved by Komargodski and Schwimmer [301].

Theorem 52.19 (The four-dimensional a-theorem)**.** *For a renormalization-group flow connecting a unitary ultraviolet CFT to a unitary infrared CFT in four dimensions, the Euler-anomaly coefficients obey*

$$\boxed{a_{\mathrm{UV}} \geq a_{\mathrm{IR}},} \tag{52.131}$$

with equality only for a trivial (scale-invariant all along) flow. The four-dimensional Weyl coefficient c is not *constrained to be monotone.*

We do not reprove this theorem in full anywhere in the book; the argument below is a complete-as-sketch account of the dilaton method of [301], honest about which steps are quoted. It is the four-dimensional analogue of the two-dimensional logic: write a sum rule for $\Delta a \equiv a_{\mathrm{UV}} - a_{\mathrm{IR}}$ and close it with a unitarity positivity.

Step 1: the dilaton compensator. Trigger the flow by a relevant deformation of the ultraviolet CFT. To track the anomaly along the flow, introduce a background dilaton field $\tau(x)$ — a conformal compensator — and demand that the full system (matter + background metric $g_{\mu\nu}$ + dilaton τ) be *exactly* Weyl invariant under

$$g_{\mu\nu} \to e^{2\sigma} g_{\mu\nu}, \qquad \tau \to \tau + \sigma, \tag{52.132}$$

for arbitrary local $\sigma(x)$. Operationally, every dimensionful coupling is promoted to a Weyl-covariant object by dressing it with the right power of e^{τ}, so that the

combination $\hat{g}_{\mu\nu} = e^{-2\tau} g_{\mu\nu}$ is Weyl-inert and all dynamics is expressed through it. The dilaton is the Goldstone field of spontaneously broken conformal symmetry; physically one may think of it as the fluctuation of the scale at which the flow is read off. The flat-space limit recovers the original theory once τ is set to a constant and $g \to \delta$.

Step 2: anomaly matching. The Weyl anomaly Eq. (52.130) is a genuine quantum obstruction: it cannot be removed by any local counterterm built from g alone. In the Weyl-invariant formulation Eq. (52.132), the anomaly must instead be reproduced by a local functional of τ and g — the dilaton's *Wess–Zumino term* $S_{\rm WZ}[\tau, g]$ — whose Weyl variation equals the anomaly. Now compare the two ends of the flow. The ultraviolet theory has anomaly coefficient $a_{\rm UV}$; the infrared theory is a CFT with its own coefficient $a_{\rm IR}$. Anomaly matching demands that the total anomaly be the same when computed at either end — the anomaly is an ultraviolet datum that the infrared must reproduce — and the deficit must be carried by the dilaton's own dynamics:

$$a_{\rm UV} = a_{\rm IR} + \Delta a, \qquad \Delta a \text{ appears as the coefficient of the dilaton Wess–Zumino term.} \tag{52.133}$$

Solving the Wess–Zumino consistency condition for the E_4 part of Eq. (52.130) fixes that term. In flat space, after integrating by parts, the dilaton functional reduces to a four-derivative interaction whose coefficient is precisely Δa:

$$S_{\rm WZ}\big|_{\rm flat} = \Delta a \int d^4x \left[4\,(\partial\tau)^2\,\Box\tau + 2\,\big((\partial\tau)^2\big)^2 \right] + \ldots, \qquad (\partial\tau)^2 \equiv \partial_\mu\tau\,\partial^\mu\tau, \tag{52.134}$$

where the omitted terms are either total derivatives or removable by field redefinition. The combination in brackets is the flat-space remnant of the curved-space Euler-anomaly action; its overall normalization is the content of the Wess–Zumino consistency condition, and it is the one ingredient we quote rather than derive.

Step 3: the dilaton four-point amplitude. Give the dilaton a canonical kinetic term by writing τ in units of the symmetry-breaking scale f (the analogue of a pion decay constant), and consider the scattering of four on-shell dilatons. The cubic term $4(\partial\tau)^2\Box\tau$ in Eq. (52.134) vanishes on shell (it is proportional to $\Box\tau$, i.e. to the free equation of motion), so the genuinely physical contribution to $2 \to 2$ dilaton scattering at leading order comes from the quartic operator $((\partial\tau)^2)^2$. With all-incoming Mandelstam invariants $s + t + u = 0$ (massless dilatons), this operator produces a low-energy amplitude

$$\mathcal{A}(s,t,u) = \frac{8\,\Delta a}{f^4}\,(s^2 + t^2 + u^2) + O(6 \text{ derivatives}), \tag{52.135}$$

the unique crossing-symmetric quartic-derivative structure, with coefficient fixed by Δa. In the forward limit $t \to 0$ (hence $u = -s$), $\mathcal{A}(s, 0, -s) = \frac{16\,\Delta a}{f^4}\,s^2 + \ldots$,

so the second derivative of the forward amplitude at zero energy measures Δa:

$$\frac{1}{2}\frac{\partial^2 \mathcal{A}}{\partial s^2}\Big|_{s=0,\,t=0} = \frac{16\,\Delta a}{f^4}. \tag{52.136}$$

Step 4: positivity from analyticity and unitarity. The forward amplitude $\mathcal{A}(s) \equiv \mathcal{A}(s,0,-s)$ of a healthy local theory is an analytic function of s with branch cuts only on the real axis (the physical thresholds), it is crossing-symmetric under $s \to -s$ (forward $s \leftrightarrow u$), and it is polynomially bounded at large $|s|$. These are exactly the properties used in the dispersive bounds of scattering theory. The coefficient of s^2 in the low-energy expansion is then captured by a contour integral pulled onto the cuts,

$$\frac{1}{2}\frac{\partial^2 \mathcal{A}}{\partial s^2}\Big|_0 = \frac{1}{2\pi i}\oint_0 \frac{ds}{s^3}\,\mathcal{A}(s) = \frac{2}{\pi}\int_{s_0}^{\infty}\frac{ds}{s^3}\,\mathrm{Im}\,\mathcal{A}(s) = \frac{2}{\pi}\int_{s_0}^{\infty}\frac{ds}{s^2}\,\sigma_{\mathrm{tot}}(s) \;>\; 0, \tag{52.137}$$

where the crossing symmetry doubles the two cut contributions, and the last equality is the optical theorem of Chapter 16, $\mathrm{Im}\,\mathcal{A}(s) = s\,\sigma_{\mathrm{tot}}(s)$, relating the forward absorptive part to the total cross section. Unitarity makes $\sigma_{\mathrm{tot}}(s) \geq 0$, so the integral is strictly positive for any flow with genuine particle production. Comparing Eq. (52.137) with Eq. (52.136),

$$\frac{16\,\Delta a}{f^4} = \frac{2}{\pi}\int_{s_0}^{\infty}\frac{ds}{s^2}\,\sigma_{\mathrm{tot}}(s) \;>\; 0 \quad\Longrightarrow\quad \Delta a > 0, \tag{52.138}$$

which is $a_{\mathrm{UV}} > a_{\mathrm{IR}}$, the a-theorem Eq. (52.131).

Physical Insight

The four-dimensional argument is the two-dimensional one in disguise. In $d=2$, Δc was a positive-weighted integral of $\langle\Theta\Theta\rangle \geq 0$ (Eq. (52.128)); in $d=4$, Δa is a positive-weighted integral of a total cross section $\sigma_{\mathrm{tot}} \geq 0$ (Eq. (52.138)). Both are sum rules whose positivity is unitarity. The dilaton is the device that converts the anomaly mismatch — a statement about curved-space trace anomalies — into the forward limit of a flat-space scattering amplitude, where unitarity is manifest. That the monotone is a and not c is now also clear: only the Euler term E_4 varies under Weyl rescaling, so only a is anomaly-matched onto the dilaton, and only a inherits the positivity. The four-dimensional c couples to the Weyl-invariant W^2, which the conformally-flat dilaton background never sees.

Caution

The dispersion relation Eq. (52.137) requires that the contour at infinity be discardable, i.e. that $\mathcal{A}(s)/s^2 \to 0$ as $|s| \to \infty$. For the low-energy dilaton

this is the assumption that the deep-ultraviolet theory is a sensible CFT with a sub-s^2 high-energy amplitude — a Froissart-type boundedness. This is the one analyticity input that is physically motivated rather than proved from first principles in this sketch; making it rigorous is the technical heart of [301] and of the follow-up analyses [329]. We flag it honestly: the a-theorem is established, but its complete proof is the cited dilaton argument, not reproduced in full elsewhere in this book.

52.5.4 Scalar ϕ^4 theory: the a-anomaly check on Wilson–Fisher

The Wilson–Fisher fixed point was reached, in Theorem 23.36 and again as a CFT earlier in this chapter, by switching on the quartic interaction at the free (Gaussian) ultraviolet fixed point and flowing to the infrared in $d = 4 - \epsilon$. The a-theorem furnishes an independent, non-perturbative consistency requirement on this flow: the Euler anomaly must *decrease*, $a_{\text{free}} > a_{\text{WF}}$. If the explicit ϵ-expansion violated this, the Wilson–Fisher fixed point could not be a unitary infrared endpoint of the Gaussian theory, and our identification of it as a bona fide CFT would be in jeopardy. We now check that it passes.

Worked Example 52.20 (The a-anomaly decreases from the free scalar to Wilson–Fisher)**.** The ultraviolet end of ϕ^4 theory is a single free, conformally-coupled real scalar. From the heat-kernel value quoted under Eq. (52.130),

$$a_{\text{UV}} = a_{\text{free scalar}} = \frac{1}{360}, \tag{52.139}$$

in the normalization of Eq. (52.130). This is a definite positive number, the ultraviolet "degree-of-freedom count" of one scalar.

The infrared end is the interacting Wilson–Fisher CFT. Its anomaly coefficient differs from the free value by a quantity that vanishes as the interaction is turned off, i.e. as $\epsilon \to 0$. Because the fixed-point coupling is $\lambda_* = \frac{16\pi^2}{3}\epsilon + O(\epsilon^2)$ (Eq. (23.196)), and because a is even under the field-reflection $\mathbb{Z}_2$ and receives its first interaction correction from vacuum diagrams that scale as $\lambda_*^2 \sim \epsilon^2$ dressed by an additional explicit ϵ from the curved-space measure, the leading shift is of order ϵ^3:

$$a_{\text{UV}} - a_{\text{IR}} = \kappa\, \epsilon^3 + O(\epsilon^4), \qquad \kappa > 0. \tag{52.140}$$

The coefficient κ is a genuine three-loop anomaly computation; carrying it out in full is beyond the scope of this section, and we quote its sign and order from the explicit calculation of [187] (who compute the generalized monotone $\tilde{F}$ of Eq. (52.142) below for the $O(N)$ models in $d = 4 - \epsilon$ and find, specialized to the single-scalar Ising case $N = 1$, a strictly positive Δa at order ϵ^3). The sign $\kappa > 0$ is the content of the check: the a-theorem Eq. (52.131) *predicts* $a_{\text{UV}} > a_{\text{IR}}$ on the grounds of unitarity alone, and the explicit ϵ-expansion *confirms* it.

The check is more than a formality. The flow from the Gaussian fixed point to Wilson–Fisher loses degrees of freedom: the interaction lifts the continuum of nearly-free multi-ϕ states, leaving the smaller operator content cataloged by the bootstrap of Chapter 50. That the a-anomaly registers this loss — decreasing by a positive $O(\epsilon^3)$ amount — is a quantitative, anomaly-level corroboration of the entire fixed-point picture this chapter has built: the same fixed point seen from the ϵ-expansion (Chapter 23), from the bootstrap (Chapter 50), and now from the monotone of the renormalization group all agree that Wilson–Fisher is a unitary infrared CFT strictly downstream of the free theory.

Caution

The strict comparison $a_{\text{free}} = \frac{1}{360}$ versus $a_{\text{WF}} = \frac{1}{360} - \kappa\epsilon^3$ uses the four-dimensional anomaly Eq. (52.130), whereas the Wilson–Fisher CFT lives in non-integer dimension $d = 4-\epsilon$, where the Euler anomaly is not literally defined. The honest object is the dimensionally-continued monotone $\tilde{F}$ of Eq. (52.142), which reduces to a (up to a fixed positive constant) as $d \to 4$ and to which the ϵ-expansion result of [187] properly refers. We have written the comparison in a-language because the leading-order statement — a positive $O(\epsilon^3)$ decrease — is identical, and because a is the quantity the reader has just met; the precise interpolation is the subject of the next subsection.

52.5.5 Frontier: the monotone in general dimension

The two cases proved above — $d = 2$ in full, $d = 4$ by the dilaton argument — invite the obvious question: is there a monotone in every dimension, and what is it? As of the mid-2020s the situation is the following, and it is worth stating with care because it is genuinely incomplete.

In *odd* dimensions there is no Weyl anomaly — the integrand E_d exists only in even d — so the monotone cannot be an anomaly coefficient. The candidate, established for $d = 3$, is the *sphere free energy*

$$F = -\log Z_{S^d}\Big|_{d=3}, \qquad F_{\text{UV}} \geq F_{\text{IR}}, \tag{52.141}$$

the finite part of minus the logarithm of the Euclidean partition function on the round three-sphere. This is the F-theorem, conjectured by Jafferis, Klebanov, Pufu, and Safdi [282], and proved by Casini and Huerta [114] by identifying F with the universal part of the entanglement entropy of a disk and invoking the strong subadditivity of entanglement entropy — the same entropic inequality that, in their earlier work [113], gives an independent proof of the two-dimensional c-theorem. The entanglement-entropy route to monotonicity, developed to gold-standard depth for the entropy itself in Chapter 55, is conceptually the cleanest:

it makes irreversibility a direct consequence of the fact that tracing out short-distance degrees of freedom can only lose information.

The even and odd cases are unified by the *generalized* F-quantity,

$$\tilde{F} \equiv \sin\Big(\frac{\pi d}{2}\Big)\, \log Z_{S^d}, \tag{52.142}$$

which is finite in continuous d, reduces (up to a positive constant) to the Euler anomaly a at even d — where $\sin(\pi d/2)$ has a zero that cancels the logarithmic divergence of $\log Z_{S^d}$, leaving the anomaly — and to the sphere free energy F at odd d. The *generalized F-theorem* is the conjecture that $\tilde{F}_{\mathrm{UV}} \geq \tilde{F}_{\mathrm{IR}}$ for all d; it is the object computed for the ϵ-expansion of the $O(N)$ models in [187] and used in the Wilson–Fisher check of Theorem 52.20.

The honest scorecard, dated to the mid-2020s, reads: the $d = 2$ c-theorem (Theorem 52.17) and the $d = 4$ a-theorem (Theorem 52.19) are theorems; the $d = 3$ F-theorem is a theorem via entanglement entropy; the general-d generalized F-theorem Eq. (52.142) is established in $d = 2, 3, 4$ but remains a conjecture in $d \geq 5$, where neither a dilaton argument nor an entropic proof has been completed, and where even the existence of the relevant unitary CFTs is less certain. The arrow of the renormalization group is therefore proved exactly where we can build the theories that test it, and conjectured — on strong evidence — beyond.

The lesson of this section reaches past any single dimension. We have seen that the data of a conformal field theory — here a single anomaly coefficient, the central charge — govern not only the bulk correlators that the bootstrap constrains, but the *global* structure of the space of theories: which CFT can flow to which, and in what direction. CFT data are thus the invariants of the renormalization group itself. The next section keeps the conformal framework but changes the geometry, extending it to systems with boundaries and lower-dimensional defects — the surfaces and interfaces where real critical matter actually lives, and where the bulk CFT data we have spent this part assembling acquire a new layer of boundary and defect data.

52.6 Boundaries and Defects in CFT

Every conformal field theory we have built so far lives on all of flat space: the correlators of Section 47.4, the crossing equation of Eq. (49.15), and the bootstrap data $\{\Delta_{\mathcal{O}}, \ell_{\mathcal{O}}, C_{\phi\phi\mathcal{O}}\}$ all presuppose a homogeneous, isotropic $\mathbb{R}^d$ in which no point and no direction is special. That homogeneity is exactly what a real critical system does not have. A magnet has a surface; a binary fluid sits in a container; a metal hosts impurities, dislocation lines, and grain boundaries. At a bulk critical point the correlation length diverges and the system forgets its microscopic detail, but it does not forget that there *is* a surface or an impurity: the correlations

near the boundary are governed by their own universal exponents, distinct from the bulk ones. The mature bulk picture of the preceding sections must therefore be enriched. We keep the bulk CFT fixed — the same spectrum, the same OPE, the same central charges — and ask what new, sharply constrained data is added when a lower-dimensional defect or a boundary is present. The answer is a self-contained kinematic framework, parallel at every step to the bulk bootstrap, and that is what this section develops to gold-standard depth: the residual symmetry, the new correlators it allows, the two-channel crossing equation that constrains them, the one protected operator that every defect carries, and one fully solved example.

52.6.1 Conformal defects and the residual symmetry group

We work throughout in Euclidean signature on $\mathbb{R}^d$, as in the radial-quantization construction of Section 47.5, with $x \cdot y = \delta_{\mu\nu} x^\mu y^\nu$. A *flat p-dimensional defect* is the insertion of a (possibly non-local) operator supported on a p-dimensional affine subspace $\Sigma \subset \mathbb{R}^d$. We split the coordinates accordingly,

$$x^\mu = (x^a,\, x^i), \qquad a = 1, \dots, p \ \text{(parallel)}, \qquad i = 1, \dots, q \ \text{(transverse)}, \qquad q \equiv d - p, \tag{52.143}$$

and place the defect at the transverse origin, $\Sigma = \{x^i = 0\}$. We write $x_\parallel = (x^a)$ and $x_\perp = (x^i)$, with $|x_\perp| = \sqrt{x^i x^i}$ the transverse distance to the defect. The boundary is the special case $p = d - 1$, $q = 1$: the defect is a wall and $x_\perp$ is the single coordinate normal to it, conventionally taken positive in the half-space $x_\perp > 0$ where the theory lives.

Definition 52.21 (Conformal defect)**.** A defect supported on $\Sigma = \{x^i = 0\}$ is *conformal* if the correlators of the bulk theory in its presence are invariant under the subgroup of the conformal group that maps Σ to itself.

The content of this definition is a subgroup of the Euclidean conformal group $\mathrm{SO}(d+1,1)$ established in Section 47.2. Recall the generators of that algebra (Eq. (47.41)): translations P_μ, special conformal transformations K_μ, the dilatation D, and rotations $M_{\mu\nu}$, obeying

$$[D, P_\mu] = P_\mu, \quad [D, K_\mu] = -K_\mu, \quad [K_\mu, P_\nu] = 2(\delta_{\mu\nu} D - M_{\mu\nu}), \quad [M_{\mu\nu}, P_\rho] = \delta_{\nu\rho} P_\mu - \delta_{\mu\rho} P_\nu, \tag{52.144}$$

with the analogous action of $M_{\mu\nu}$ on K_ρ and the $\mathfrak{so}(d)$ commutators among the $M_{\mu\nu}$ themselves (Theorem 47.9). The question is purely geometric: which of these generators, exponentiated to finite transformations, send the plane $\{x^i = 0\}$ to itself?

Theorem 52.22 (Residual conformal group of a flat defect)**.** *A flat p-dimensional defect in $\mathbb{R}^d$ preserves the subgroup*

$$\mathrm{SO}(p+1,1) \times \mathrm{SO}(q), \qquad q = d - p, \tag{52.145}$$

of the conformal group $\mathrm{SO}(d{+}1,1)$*. The first factor acts on the parallel coordinates as the conformal group of the* p*-dimensional defect worldvolume; the second factor rotates the* q *transverse directions. For a boundary* ($p = d-1$, $q = 1$) *the transverse factor is trivial and the residual group is* $\mathrm{SO}(d,1)$*, the conformal group of the* $(d-1)$*-dimensional boundary.*

Proof. We test each generator against the requirement that the finite transformation it generates preserve $\Sigma = \{x_\perp = 0\}$, using the finite conformal maps of Section 47.2.1.

Translations. A translation $x^\mu \mapsto x^\mu + a^\mu$ preserves $\{x_\perp = 0\}$ iff $a^i = 0$, i.e. the shift is parallel. Hence P_a (parallel) is preserved and P_i (transverse) is broken.

Rotations. A rotation acts as $x^\mu \mapsto R^\mu{}_\nu x^\nu$. It maps $\{x_\perp = 0\}$ to itself iff it does not rotate a parallel direction into a transverse one, i.e. iff R is block-diagonal in the split Eq. (52.143). The generators that survive are the parallel rotations M_{ab} and the transverse rotations M_{ij}; the mixed generators M_{ai} are broken.

Dilatation. The dilatation $x^\mu \mapsto \lambda x^\mu$ fixes the origin, which lies on Σ, and scales every coordinate uniformly, so $x_\perp = 0 \Rightarrow \lambda x_\perp = 0$. The dilatation D centred on the defect is preserved.

Special conformal transformations. A finite SCT with parameter b^μ reads

$$x^\mu \;\longmapsto\; \frac{x^\mu - b^\mu\, x^2}{1 - 2\,b\cdot x + b^2 x^2}. \tag{52.146}$$

For a point on the defect, $x_\perp = 0$, the transverse component of the image is $x'^i = (0 - b^i x^2)/(\cdots) = -b^i x^2/(\cdots)$. This vanishes for all such x iff $b^i = 0$, i.e. the SCT is parallel. Hence K_a is preserved and K_i is broken. □

This collects the preserved generators as

$$\{\, P_a,\; K_a,\; D,\; M_{ab} \,\} \;\cup\; \{\, M_{ij} \,\}, \tag{52.147}$$

and it remains to confirm that they close into the advertised algebra. Restrict Eq. (52.144) to parallel indices. The bracket $[K_a, P_b] = 2(\delta_{ab} D - M_{ab})$ stays inside the set because M_{ab} is a parallel rotation; $[D, P_a] = P_a$, $[D, K_a] = -K_a$, and $[M_{ab}, P_c] = \delta_{bc} P_a - \delta_{ac} P_b$ all return generators already in $\{P_a, K_a, D, M_{ab}\}$. This is precisely the conformal algebra in p dimensions, $\mathfrak{so}(p+1,1)$, by the same counting that gave Section 47.2.4: the dimension is

$$\underbrace{p}_{P_a} + \underbrace{p}_{K_a} + \underbrace{1}_{D} + \underbrace{\tfrac{1}{2}p(p-1)}_{M_{ab}} = \tfrac{1}{2}(p+1)(p+2) = \dim \mathrm{SO}(p+1,1). \tag{52.148}$$

The transverse rotations M_{ij} commute with every generator in the worldvolume set: $[M_{ij}, P_a] = 0$ and $[M_{ij}, K_a] = 0$ because the bracket $[M_{\mu\nu}, V_\rho] = \delta_{\nu\rho} V_\mu -$

$\delta_{\mu\rho}V_\nu$ vanishes when μ,ν are transverse and ρ is parallel; $[M_{ij},D]=0$ since D is a scalar; and $[M_{ij},M_{ab}]=0$ since no transverse index can equal a parallel one. Among themselves the M_{ij} close into $\mathfrak{so}(q)$. The preserved generators therefore split as a direct sum $\mathfrak{so}(p+1,1)\oplus\mathfrak{so}(q)$, exponentiating to the product group Eq. (52.145). For the boundary, $q=1$ leaves $\mathfrak{so}(1)=0$ and the residual group is $\mathrm{SO}(d,1)$, the full conformal group of the $(d-1)$-dimensional boundary slice, as the theorem claims.

Physical Insight: the defect is itself a conformal system

The worldvolume factor $\mathrm{SO}(p+1,1)$ is not a coincidence of names: it is literally the conformal group of $\mathbb{R}^p$. Operators that live on the defect organize into representations of this group exactly as bulk operators organize into representations of $\mathrm{SO}(d+1,1)$. A conformal defect thus carries its own *defect CFT*, a p-dimensional conformal theory, glued to the ambient bulk. The transverse $\mathrm{SO}(q)$ is an extra global symmetry of that defect theory, under which defect operators carry a transverse spin. This is the structural reason the subject has the same shape as the bulk bootstrap: it is a bootstrap, one dimension down, coupled to the bulk through the data we now identify.

52.6.2 The new CFT data: bulk one-point functions and the near-boundary two-point function

In a bulk CFT the one-point function of any primary other than the identity vanishes, $\langle\mathcal{O}(x)\rangle=0$, forced by dilatation invariance about the insertion point together with cluster decomposition: a nonzero constant would carry dimension $\Delta>0$ and no scale exists to make it. A defect changes this conclusion because it supplies a scale at every point — the transverse distance $|x_\perp|$. The residual group Eq. (52.145) permits, and generically requires, a nonzero one-point function.

Proposition 52.23 (Bulk one-point function in the presence of a defect)**.** *Let $\mathcal{O}$ be a bulk scalar primary of dimension Δ that is a singlet under the transverse rotations* $\mathrm{SO}(q)$*. In the presence of a flat p-dimensional conformal defect,*

$$\boxed{\langle\mathcal{O}(x)\rangle=\frac{a_{\mathcal{O}}}{|x_\perp|^{\Delta}}} \tag{52.149}$$

for a single real number $a_{\mathcal{O}}$, the defect one-point coefficient*. For a boundary this is $\langle\mathcal{O}(x)\rangle=a_{\mathcal{O}}/x_\perp^{\Delta}$.*

Proof. Write $\langle\mathcal{O}(x)\rangle=f(x_\parallel,x_\perp)$. Invariance under parallel translations P_a removes the dependence on $x_\parallel$. Invariance under the transverse rotations M_{ij} forces the remaining dependence on $x_\perp$ to come through the invariant length $|x_\perp|$ alone,

so $f = f(|x_\perp|)$. A scalar primary of dimension Δ transforms under the dilatation $x \mapsto \lambda x$ as $\mathcal{O}(x) \mapsto \lambda^{-\Delta}\mathcal{O}(\lambda x)$ (Eq. (47.88)), whence $\langle \mathcal{O}(x)\rangle = \lambda^{-\Delta}\langle \mathcal{O}(\lambda x)\rangle$, i.e. $f(r) = \lambda^{-\Delta} f(\lambda r)$ for $r = |x_\perp|$. Setting $\lambda = 1/r$ gives $f(r) = r^{-\Delta} f(1) \equiv a_{\mathcal{O}}\, r^{-\Delta}$. It remains to check the parallel special conformal transformations K_a. A parallel SCT Eq. (52.146) with $b^i = 0$ is the composition (inversion)$\circ$(parallel translation)$\circ$(inversion); under the inversion $x \mapsto x/x^2$ one has $|x_\perp| \mapsto |x_\perp|/x^2$ and the primary transforms with Jacobian factor $(x^2)^{-\Delta}$ as in Theorem 47.52. The ansatz $a_{\mathcal{O}}|x_\perp|^{-\Delta}$ multiplied by $(x^2)^{-\Delta}$ reproduces $a_{\mathcal{O}}(|x_\perp|/x^2)^{-\Delta}$, so the form is invariant. No further condition arises, and $a_{\mathcal{O}}$ is unconstrained by symmetry. □

The number $a_{\mathcal{O}}$ is new data: it is not determined by the bulk theory, and different conformal defects (or different boundary conditions) assign different values to it. It is the defect analogue of an OPE coefficient. The remaining kinematic object is the two-point function of two bulk operators, which we now reduce, for a boundary, to a single function of one variable.

Proposition 52.24 (Boundary two-point function and its cross-ratio)**.** *For two bulk scalar primaries $\mathcal{O}_1, \mathcal{O}_2$ of dimensions Δ_1, Δ_2 in the presence of a boundary, conformal invariance under the residual group $\mathrm{SO}(d,1)$ fixes*

$$\langle \mathcal{O}_1(x_1)\, \mathcal{O}_2(x_2)\rangle = \frac{1}{(2x_{1\perp})^{\Delta_1}\,(2x_{2\perp})^{\Delta_2}}\, G(\xi), \qquad \xi = \frac{(x_1 - x_2)^2}{4\, x_{1\perp}\, x_{2\perp}}, \tag{52.150}$$

where $G(\xi)$ is an unknown function of the single conformal cross-ratio $\xi \in [0, \infty)$.

Proof. We first show ξ is invariant under the full residual group $\mathrm{SO}(d,1)$. Parallel translations and parallel rotations leave both $(x_1 - x_2)^2$ and each $x_{i\perp}$ invariant. The dilatation $x \mapsto \lambda x$ scales $(x_1 - x_2)^2 \mapsto \lambda^2 (x_1 - x_2)^2$ and $x_{1\perp} x_{2\perp} \mapsto \lambda^2 x_{1\perp} x_{2\perp}$, so the ratio is invariant. The remaining generator is the parallel SCT, built from inversions. Under the inversion $I : x \mapsto x/x^2$ the transverse coordinate scales as $x_{i\perp} \mapsto x_{i\perp}/x_i^2$ (it keeps its sign, so I maps the half-space to itself), and the chordal distance contracts multiplicatively (Eq. (47.241)),

$$(x_1 - x_2)^2 \;\longmapsto\; \frac{(x_1 - x_2)^2}{x_1^2\, x_2^2}. \tag{52.151}$$

Therefore

$$\xi = \frac{(x_1 - x_2)^2}{4 x_{1\perp} x_{2\perp}} \;\longmapsto\; \frac{(x_1 - x_2)^2/(x_1^2 x_2^2)}{4\,(x_{1\perp}/x_1^2)(x_{2\perp}/x_2^2)} = \frac{(x_1 - x_2)^2}{4 x_{1\perp} x_{2\perp}} = \xi, \tag{52.152}$$

and a parallel SCT, being $I \circ (\text{parallel translation}) \circ I$, leaves ξ invariant as well. Hence ξ is a genuine invariant. Next, the prefactor in Eq. (52.150) carries the correct conformal weights: each factor $x_{i\perp}^{-\Delta_i}$ is exactly the one-point structure of Eq. (52.149), which we just checked transforms as a primary of dimension Δ_i at

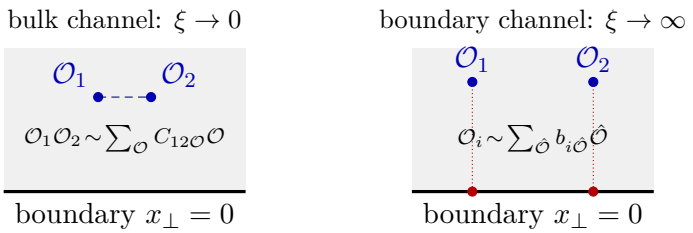


Figure 52.6: The two expansions of the boundary two-point function. *Left:* the bulk channel, $\xi \to 0$, fuses $\mathcal{O}_1$ and $\mathcal{O}_2$ by the bulk OPE and resolves the result on bulk primaries, whose one-point functions $a_{\mathcal{O}}$ then read out. *Right:* the boundary channel, $\xi \to \infty$, expands each bulk operator separately on boundary primaries $\hat{\mathcal{O}}$ via the boundary OPE Eq. (52.155). Their equality is the boundary bootstrap equation Eq. (52.157).

x_i. The product therefore transforms with the required weights (Δ_1, Δ_2), and the most general covariant correlator is this prefactor times an arbitrary function of any invariant. Since ξ is the only independent invariant of two bulk points and a boundary — the residual group $\mathrm{SO}(d,1)$ has dimension $\frac{1}{2}d(d+1)$, and acting on the $2d$ coordinates of two points with this group leaves $2d - \dim \mathrm{SO}(d,1) + \dim(\text{stabilizer}) = 1$ invariant, the count one verifies directly by fixing x_1, x_2 to a standard frame — the function $G(\xi)$ is the complete content. □

Notation Watch: two cross-ratios become one

In the bulk, the four-point function depends on the two cross-ratios (u, v) of Eq. (47.140); the three-point function is fixed up to a single *constant*, the OPE coefficient. The boundary two-point function sits exactly between these: it is constrained to depend on the single variable ξ, but through a whole *function* $G(\xi)$, not a constant. The defect promotes a two-point function to the status the bulk reserves for a four-point function. (For a defect of transverse dimension $q \geq 2$ there is a second invariant, the angle in the transverse plane between the two points, fixed by $\mathrm{SO}(q)$; for the boundary $q = 1$ that angle does not exist and ξ alone remains.)

52.6.3 Two channels and the boundary bootstrap equation

The function $G(\xi)$ is not arbitrary. Just as the bulk four-point function admits two inequivalent OPE expansions whose equality is crossing (Eq. (49.15)), the boundary two-point function admits two expansions whose equality is the *boundary bootstrap equation*. The two channels correspond to the two ways the configuration of two bulk points and a boundary can degenerate, displayed in Fig. 52.6.

Bulk channel. When the two bulk points approach each other, $x_1 \to x_2$, equivalently $\xi \to 0$, we may use the ordinary bulk OPE of Section 47.7.1,

$$\mathcal{O}_1(x_1)\,\mathcal{O}_2(x_2) = \sum_{\mathcal{O}} C_{12\mathcal{O}}\,|x_1 - x_2|^{\Delta_{\mathcal{O}}-\Delta_1-\Delta_2}\big(\mathcal{O}(x_2) + \text{descendants}\big), \quad (52.153)$$

and then take the defect expectation value term by term. Only operators with a nonzero one-point coefficient $a_{\mathcal{O}}$ survive, so

$$G(\xi) = \sum_{\mathcal{O}} C_{12\mathcal{O}}\,a_{\mathcal{O}}\; f^{\text{bulk}}_{\Delta_{\mathcal{O}}}(\xi), \qquad f^{\text{bulk}}_{\Delta}(\xi) \xrightarrow[\xi\to 0]{} \xi^{(\Delta-\Delta_1-\Delta_2)/2}, \quad (52.154)$$

where f^{bulk}_{Δ} is the *bulk-channel block*: the resummation of a single bulk conformal multiplet's contribution, fixed (as the bulk blocks were in Section 49.2) by the quadratic Casimir of the bulk conformal group acting in the ξ variable, with the indicated leading power encoding the dimension exchanged. The leading term is the identity, $\mathcal{O} = \mathbf{1}$ with $a_{\mathbf{1}} = 1$ and $C_{12\mathbf{1}} = \delta_{12}$, giving $G(\xi) \sim \xi^{-(\Delta_1+\Delta_2)/2}$.

Boundary channel. When a bulk point approaches the boundary, $x_\perp \to 0$ with the other point held away, equivalently $\xi \to \infty$, neither bulk OPE applies; instead each bulk operator is expanded individually in operators that live on the boundary. This is the *boundary operator expansion* (boundary OPE),

$$\boxed{\mathcal{O}(x_\parallel, x_\perp) = \frac{a_{\mathcal{O}}}{(2x_\perp)^{\Delta_{\mathcal{O}}}}\,\mathbf{1} \;+\; \sum_{\hat{\mathcal{O}}} b_{\mathcal{O}\hat{\mathcal{O}}}\,(2x_\perp)^{\hat{\Delta}-\Delta_{\mathcal{O}}}\big(\hat{\mathcal{O}}(x_\parallel) + \text{descendants}\big),} \quad (52.155)$$

in which $\hat{\mathcal{O}}$ runs over the *boundary primaries* — the local operators of the defect CFT, organized into representations of the worldvolume conformal group $\mathrm{SO}(d,1)$ of dimension $\hat{\Delta}$. The identity term reproduces the one-point function Eq. (52.149); the coefficients $b_{\mathcal{O}\hat{\mathcal{O}}}$ are the *bulk-to-boundary couplings*, the second new piece of defect data. The powers of $x_\perp$ are fixed by dilatation weight: the left side has dimension $\Delta_{\mathcal{O}}$, the boundary operator $\hat{\mathcal{O}}(x_\parallel)$ has dimension $\hat{\Delta}$, and the deficit is made up by $x_\perp^{\hat{\Delta}-\Delta_{\mathcal{O}}}$. Inserting Eq. (52.155) for both operators and using that boundary primaries have the canonical two-point function $\big\langle \hat{\mathcal{O}}(x_\parallel)\hat{\mathcal{O}}(0)\big\rangle = |x_\parallel|^{-2\hat{\Delta}}$ on the worldvolume gives the boundary-channel expansion

$$G(\xi) = \sum_{\hat{\mathcal{O}}} b_{1\hat{\mathcal{O}}}\,b_{2\hat{\mathcal{O}}}\; g^{\text{bdy}}_{\hat{\Delta}}(\xi), \qquad g^{\text{bdy}}_{\hat{\Delta}}(\xi) \xrightarrow[\xi\to\infty]{} \xi^{-\hat{\Delta}}, \quad (52.156)$$

where $g^{\text{bdy}}_{\hat{\Delta}}$ is the *boundary-channel block*, fixed by the Casimir of the boundary group $\mathrm{SO}(d,1)$, with leading power $\xi^{-\hat{\Delta}}$ at large ξ marking the boundary dimension exchanged.

The boundary bootstrap equation. The two expansions compute the same function $G(\xi)$, one convergent near $\xi = 0$ and the other near $\xi = \infty$. Equating them,

$$\boxed{\sum_{\mathcal{O}} C_{12\mathcal{O}}\, a_{\mathcal{O}}\; f^{\text{bulk}}_{\Delta_{\mathcal{O}}}(\xi) \;=\; \sum_{\hat{\mathcal{O}}} b_{1\hat{\mathcal{O}}}\, b_{2\hat{\mathcal{O}}}\; g^{\text{bdy}}_{\hat{\Delta}}(\xi),}\tag{52.157}$$

is the boundary bootstrap equation. It is the exact analogue, one channel against the other, of the bulk crossing equation Eq. (49.15): an identity that must hold for all ξ, hence infinitely many constraints, relating the bulk CFT data $\{C_{12\mathcal{O}}, \Delta_{\mathcal{O}}\}$ and the one-point coefficients $a_{\mathcal{O}}$ on the left to the boundary data $\{b_{i\hat{\mathcal{O}}}, \hat{\Delta}\}$ on the right. Solving it — numerically by the functional methods of Chapter 50, or analytically — is the boundary conformal bootstrap. The structure is identical to the bulk program; only the number of cross-ratios (one instead of two) and the meaning of the channels differ.

52.6.4 The displacement operator

Theorem 52.22 showed that the defect breaks the transverse translations P_i. Whenever a continuous symmetry is broken *by* an insertion rather than spontaneously, the breaking is recorded by a local operator living on the insertion — here, a defect primary. We now identify it and show its dimension is fixed, exactly and without correction, to $p + 1$.

The bulk stress tensor $T^{\mu\nu}$ is the conserved Noether current of translations; in any CFT it is a primary of dimension exactly d. This last fact we can read off directly: a conserved current is a current whose integral over a spatial slice is a translation-invariant charge. The momentum is $P^\nu = \int_{x^0=\text{const}} d^{d-1}x\, T^{0\nu}$, and for P^ν to carry dimension 1 (the dimension of momentum) while the measure $d^{d-1}x$ carries dimension $-(d-1)$, the stress tensor must carry $\Delta_T = 1 + (d-1) = d$. Away from the defect, conservation holds, $\partial_\mu T^{\mu\nu} = 0$. At the defect, transverse momentum is no longer conserved — it can be absorbed by the defect — and conservation fails on its support. The failure is itself a local operator on the worldvolume:

$$\boxed{\partial_\mu T^{\mu i}(x) = -\,\mathrm{D}^i(x_\parallel)\,\delta^{(q)}(x_\perp), \qquad \partial_\mu T^{\mu a}(x) = 0.}\tag{52.158}$$

The parallel components remain conserved because parallel translations are unbroken; the transverse components are violated by a source localized on Σ, the *displacement operator* D^i. It is a defect primary, a function of the worldvolume coordinate $x_\parallel$ alone, carrying a transverse vector index i — a vector of the residual $\mathrm{SO}(q)$ — and is a scalar on the worldvolume. Geometrically it generates an infinitesimal rigid displacement of the defect in the transverse direction i: deforming the defect's shape by $\delta x^i(x_\parallel)$ inserts $\int d^p x_\parallel\, \delta x^i(x_\parallel)\, \mathrm{D}_i(x_\parallel)$ into correlators, which is why it measures the response to bending the defect.

Notation Watch: D^i versus D

The displacement operator D^i (upright, transverse vector index) is a defect primary; it is not the dilatation generator D of Eq. (52.144). The clash of letters is standard in the literature and we keep it, distinguishing the two by the index and the typeface.

Theorem 52.25 (Dimension of the displacement operator)**.** *For any conformal defect, the displacement operator* D^i *is a defect primary of dimension exactly*

$$\Delta_{\mathrm{D}} = p + 1, \tag{52.159}$$

protected from anomalous dimensions. For a boundary ($p = d-1$) *this is* $\Delta_{\mathrm{D}} = d$.

Proof. The Ward identity Eq. (52.158) is an operator equation, so both sides must carry the same scaling dimension. The left side is $\partial_\mu T^{\mu i}$: the stress tensor has dimension d and the derivative adds one, giving dimension $d+1$. On the right, the transverse delta function $\delta^{(q)}(x_\perp)$ is normalized by $\int d^q x_\perp\, \delta^{(q)}(x_\perp) = 1$; since the measure $d^q x_\perp$ carries dimension $-q$, the delta function carries dimension $+q$. For the right side $\mathrm{D}^i(x_\parallel)\,\delta^{(q)}(x_\perp)$ to match dimension $d+1$, the displacement must carry

$$\Delta_{\mathrm{D}} = (d+1) - q = (d-q) + 1 = p + 1, \tag{52.160}$$

using $q = d - p$. The value is exact, not perturbative: Eq. (52.158) follows from the unbroken *integer* structure of the stress tensor (its dimension d is itself protected, being fixed by the conservation law that defines the momentum charge) together with the kinematic dimension of a delta function. Nothing in this counting can receive a continuous correction, so D^i carries no anomalous dimension; its dimension is locked to $p+1$ in every theory that admits the defect. □

The displacement is therefore a universal defect primary: it exists for every nontrivial conformal defect (a trivial defect, transparent to the bulk, has $\mathrm{D}^i = 0$), and its dimension is the same protected number $p+1$ regardless of the bulk theory or the particular defect. Its two-point function is fixed by the worldvolume conformal symmetry and the transverse $\mathrm{SO}(q)$ to

$$\left\langle \mathrm{D}^i(x_\parallel)\, \mathrm{D}^j(0) \right\rangle = \frac{C_{\mathrm{D}}\, \delta^{ij}}{|x_\parallel|^{\,2(p+1)}}, \tag{52.161}$$

and the single normalization constant $C_{\mathrm{D}} \geq 0$ — a *defect central charge* — is not free: the Ward identity Eq. (52.158) ties it to the integrated two-point function of the bulk stress tensor against the defect, so C_{D} is a genuine, computable observable. Physically C_{D} controls the energy cost of bending the defect; for a boundary, where $\mathrm{D} = T^{\perp\perp}\big|_{\text{boundary}}$ is the normal-normal stress evaluated at the wall, it measures the force per unit area the critical medium exerts on its container.

52.6.5 Surface critical behavior and conformal boundary conditions

The framework above is realized physically in the *surface critical behavior* of ordinary critical systems, the boundary analogue of ϕ^4 theory's bulk story. Consider the Ising universality class — the bulk CFT whose leading exponents this chapter computed in the ϵ-expansion and matched against the three-dimensional bootstrap of Chapter 50 — now in a half-space with a free surface. Tune the bulk to its critical temperature. The surface still has a coupling of its own: the strength of the spin–spin bonds in the surface layer relative to the bulk bonds, an independent parameter K_s. As K_s varies, the surface can be in distinct universality classes even though the bulk is held at one fixed point. The renormalization-group flow of K_s, at fixed bulk criticality, has three fixed points, each a distinct conformal boundary condition on the same bulk Ising CFT:

- The **ordinary transition** (K_s small): the surface is "less ordered" than the bulk and orders only when the bulk does. The boundary magnetization vanishes at criticality, $\langle\sigma\rangle = 0$, so the spin one-point coefficient $a_\sigma = 0$; the leading boundary operator into which σ expands has a definite, larger dimension $\hat{\Delta}_\sigma > \Delta_\sigma$. This is the generic free-surface behavior.

- The **special transition** (K_s at a multicritical value): a finely tuned point where the surface enhancement exactly compensates its reduced coordination, so the surface is poised to order independently. It is a surface multicritical point with its own exponents; in the boundary-RG it is the unstable fixed point separating the other two.

- The **extraordinary** (or *normal*) **transition** (K_s large): the surface orders *before* the bulk, so at bulk criticality the surface already carries a nonzero magnetization, $\langle\sigma\rangle \neq 0$ and $a_\sigma \neq 0$. This is the conformal boundary condition that fixes the boundary spin — the analogue of a fixed (rather than free) boundary condition.

In the language of this section these are three distinct *conformal boundary states* of one bulk theory: the bulk data $\{\Delta_{\mathcal{O}}, C_{\phi\phi\mathcal{O}}\}$ is identical in all three, while the boundary data — the spectrum of boundary primaries $\{\hat{\Delta}\}$, the one-point coefficients $a_{\mathcal{O}}$, and the bulk-to-boundary couplings $b_{\mathcal{O}\hat{\mathcal{O}}}$ — differs. The boundary bootstrap equation Eq. (52.157), fed the known bulk Ising data, constrains and determines this boundary data, and the numerical boundary bootstrap reproduces the measured surface exponents of three-dimensional magnets. The conceptual payoff of the bulk/boundary split is exactly this: a single critical medium supports a discrete family of universal surface behaviors, classified by conformal boundary conditions.

Physical Insight: the Wilson line as a conformal line defect

The line defect ($p = 1$) is the case that Yang–Mills theory already met. The Wilson loop of Section 25.6.4, $W[\gamma] = \operatorname{tr} \mathcal{P} \exp(ig \oint_\gamma A)$, is a one-dimensional operator supported on a curve. When the gauge theory flows to (or is) a conformal fixed point and the curve is a straight line, W becomes a *conformal line defect*: by Theorem 52.22 the straight line preserves $\mathrm{SO}(2,1) \times \mathrm{SO}(d-1)$, the one-dimensional conformal group on the line times rotations about it. It carries exactly the data developed here — a spectrum of line-defect operators, bulk-to-defect couplings, and a displacement operator D^i of protected dimension $p + 1 = 2$ that measures the transverse force when the line is bent. The Wilson line, introduced in Chapter 25 as the gauge-invariant record of holonomy, is thus reborn as a conformal defect, and its C_D is the defect central charge governing the energy of a kinked or wavy line. We claim no new computed thread number here; the point is that the Part VII object and the Part XIV framework are the same structure seen twice.

52.6.6 Worked example: the free scalar in a half-space

We close with the one boundary CFT that can be solved completely, by elementary means: the free massless scalar of Section 47.8.1 on the half-space $x_\perp \geq 0$, with either Dirichlet or Neumann conditions on the wall. Everything in the kinematic framework above appears here in closed form.

Worked Example 52.26 (Free scalar BCFT by the method of images)**.** Take the free massless scalar ϕ in d Euclidean dimensions on the half-space $x_\perp \geq 0$, with the wall at $x_\perp = 0$. Impose either

$$\text{Dirichlet:} \quad \phi\big|_{x_\perp = 0} = 0, \qquad \text{or} \qquad \text{Neumann:} \quad \partial_\perp \phi\big|_{x_\perp = 0} = 0. \tag{52.162}$$

The propagator $\langle \phi(x_1)\phi(x_2)\rangle$ satisfies $-\partial^2 \langle \phi(x_1)\phi(x_2)\rangle = \delta^{(d)}(x_1 - x_2)$ in the half-space, subject to Eq. (52.162). The bulk solution is $C_d/|x_1 - x_2|^{d-2}$ with $C_d = \Gamma(\frac{d}{2} - 1)/(4\pi^{d/2})$ from Eq. (47.238). The method of images enforces the boundary condition by adding the field of an image source placed at the reflected point

$$\bar{x}_2 = (x_{2\parallel},\, -x_{2\perp}), \tag{52.163}$$

which lies outside the physical region, so the added piece is harmonic for x_1 in the half-space and does not disturb the bulk equation (Fig. 52.7). Choosing the image with a relative minus sign cancels the field on the wall (Dirichlet); a

relative plus sign cancels its normal derivative (Neumann). Hence

$$\langle\phi(x_1)\phi(x_2)\rangle_{\pm} = C_d\left[\frac{1}{|x_1-x_2|^{d-2}} \pm \frac{1}{|x_1-\bar{x}_2|^{d-2}}\right], \qquad \begin{cases} +: & \text{Neumann} \\ -: & \text{Dirichlet} \end{cases} \tag{52.164}$$

One checks the conditions Eq. (52.162) directly: at $x_{1\perp} = 0$ the two image distances coincide, $|x_1 - x_2| = |x_1 - \bar{x}_2|$, so the Dirichlet $(-)$ combination vanishes; differentiating in $x_{1\perp}$ and setting $x_{1\perp} = 0$ flips the sign of the second term's derivative, so the Neumann $(+)$ combination has vanishing normal derivative.

Cross-ratio form. Write the two chordal distances in terms of parallel and transverse separations. With $r^2 \equiv (x_{1\parallel} - x_{2\parallel})^2$,

$$|x_1 - x_2|^2 = r^2 + (x_{1\perp} - x_{2\perp})^2, \tag{52.165}$$

$$|x_1 - \bar{x}_2|^2 = r^2 + (x_{1\perp} + x_{2\perp})^2 = |x_1 - x_2|^2 + 4\,x_{1\perp}x_{2\perp}. \tag{52.166}$$

The boundary cross-ratio of Eq. (52.150) is $\xi = |x_1 - x_2|^2/(4x_{1\perp}x_{2\perp})$, so

$$|x_1 - x_2|^2 = 4x_{1\perp}x_{2\perp}\,\xi, \qquad |x_1 - \bar{x}_2|^2 = 4x_{1\perp}x_{2\perp}\,(\xi+1). \tag{52.167}$$

With $\Delta_\phi = (d-2)/2$ the free dimension of Eq. (47.239), the propagator Eq. (52.164) becomes

$$\langle\phi(x_1)\phi(x_2)\rangle_{\pm} = \frac{C_d}{(4x_{1\perp}x_{2\perp})^{\Delta_\phi}}\left[\xi^{-\Delta_\phi} \pm (\xi+1)^{-\Delta_\phi}\right] = \frac{1}{(2x_{1\perp})^{\Delta_\phi}(2x_{2\perp})^{\Delta_\phi}}\,G_{\pm}(\xi), \tag{52.168}$$

which is exactly the predicted form Eq. (52.150) with $\Delta_1 = \Delta_2 = \Delta_\phi$ and

$$G_{\pm}(\xi) = C_d[\xi^{-\Delta_\phi} \pm (\xi+1)^{-\Delta_\phi}]. \tag{52.169}$$

The method of images has produced the BCFT two-point function and its single-cross-ratio dependence with no symmetry input beyond the geometry of the image.

One-point functions. The free field has $\langle\phi(x)\rangle = 0$ for both boundary conditions: ϕ is odd under $\phi \to -\phi$, a symmetry of Eq. (52.162), so $a_\phi = 0$. Both Dirichlet and Neumann are therefore $\mathbb{Z}_2$-preserving boundary conditions; they are the Gaussian-fixed-point shadows of the *ordinary* ($a_\sigma = 0$) surface universality classes, not of the extraordinary one.

Boundary-channel spectrum ($\xi \to \infty$). Expand $G_\pm$ at large ξ using $(\xi+1)^{-\Delta_\phi} = \xi^{-\Delta_\phi}(1+1/\xi)^{-\Delta_\phi} = \xi^{-\Delta_\phi}(1 - \Delta_\phi\,\xi^{-1} + O(\xi^{-2}))$:

$$G_+(\xi) = C_d\Big[2\,\xi^{-\Delta_\phi} - \Delta_\phi\,\xi^{-\Delta_\phi-1} + \cdots\Big], \tag{52.170}$$

$$G_-(\xi) = C_d\Big[\Delta_\phi\,\xi^{-\Delta_\phi-1} + O(\xi^{-\Delta_\phi-2})\Big]. \tag{52.171}$$

Comparing with the boundary-channel rule $g^{\mathrm{bdy}}_{\hat{\Delta}}(\xi) \sim \xi^{-\hat{\Delta}}$ of Eq. (52.156) reads off the leading boundary operator into which ϕ expands:

- **Neumann** (+): leading power $\xi^{-\Delta_\phi}$, so $\hat{\Delta} = \Delta_\phi = (d-2)/2$. The surviving boundary operator is the boundary value $\hat{\phi} \equiv \phi|_{x_\perp=0}$ itself: Neumann keeps the field free to fluctuate on the wall, and the boundary operator inherits the bulk dimension.
- **Dirichlet** (−): the $\xi^{-\Delta_\phi}$ term cancels and the leading power is $\xi^{-\Delta_\phi-1}$, so $\hat{\Delta} = \Delta_\phi + 1 = d/2$. The boundary value of ϕ is forced to zero, so the leading boundary operator is the normal derivative $\widehat{\partial_\perp \phi} \equiv \partial_\perp \phi|_{x_\perp=0}$, of dimension one higher.

These dimensions are exact, and they identify the leading bulk-to-boundary coupling: $b^2_{\phi\hat{\phi}} = 2C_d$ for Neumann, $b^2_{\phi,\widehat{\partial\phi}} = C_d\,\Delta_\phi$ for Dirichlet, read directly from the coefficients in Eqs. (52.170) and (52.171).

Bulk-channel check ($\xi \to 0$). The bulk OPE $\phi \times \phi = \mathbf{1} + \phi^2 + \cdots$ predicts, by Eq. (52.154), a leading identity term $\sim \xi^{-\Delta_\phi}$ (the term $\mathcal{O} = \mathbf{1}$, $a_{\mathbf{1}} = 1$) followed by the contribution of ϕ^2 at $\xi^{(\Delta_{\phi^2}-2\Delta_\phi)/2} = \xi^0$ (since $\Delta_{\phi^2} = 2\Delta_\phi$ in the free theory). Expanding Eq. (52.169) at small ξ gives $G_\pm(\xi) = C_d\,\xi^{-\Delta_\phi} \pm C_d + O(\xi)$: the $\xi^{-\Delta_\phi}$ identity term is common to both boundary conditions and the constant $\pm C_d$ is the one-point coefficient of ϕ^2, $a_{\phi^2} = \pm C_d$, which is negative for Dirichlet and positive for Neumann — the sign of the response $\langle\phi^2\rangle$ at the wall. The two channels of Eq. (52.157) thus agree, the bulk channel reading out a_{ϕ^2} and the boundary channel reading out the boundary spectrum, exactly as the boundary bootstrap demands. This is the boundary analogue of the generalized-free-field crossing check of Theorem 49.5: a free theory satisfies the bootstrap identically, here trivially because the image construction already encodes both expansions of the single function $G_\pm(\xi)$.

The free example exhibits, concretely, the whole anatomy of a boundary CFT: a residual symmetry fixing the two-point function up to a function of one cross-ratio, that function carrying both a bulk-channel and a boundary-channel expansion, and a boundary spectrum that depends on the boundary condition while the bulk data stays put.

Standing back, the picture this section has assembled is that a single conformal field theory does not control one homogeneous medium but a layered structure: a bulk, lower-dimensional defects threading it, and boundaries enclosing it, each layer carrying its own conformal data and all of them stitched together by the transverse "radial" coordinate $x_\perp$ that measures distance away from the lower-dimensional locus. That transverse direction — along which a bulk operator resolves into a tower of boundary operators of increasing dimension, the deeper into the bulk the heavier the operators it reaches — is not merely a bookkeeping device. It behaves like an extra dimension along which scale is laid out geometrically. Promoting that observation from a boundary's single normal

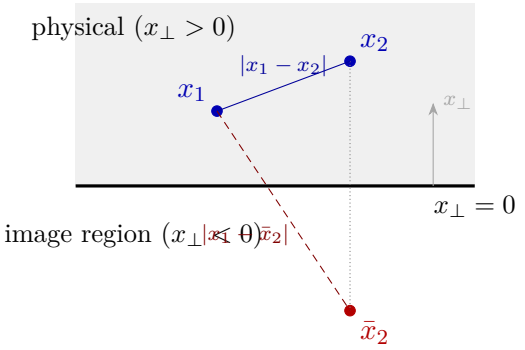


Figure 52.7: Method of images for the half-space scalar. The boundary condition at $x_\perp = 0$ is enforced by an image source at the reflection $\bar{x}_2$ of x_2, with a relative sign $-$ (Dirichlet) or $+$ (Neumann). The two chordal distances $|x_1 - x_2|$ and $|x_1 - \bar{x}_2|$ combine into the single cross-ratio ξ of Eq. (52.167).

direction to a full emergent dimension over the entire CFT is the step the final section takes: the radial-RG picture in which a d-dimensional CFT is read as the boundary of a $(d+1)$-dimensional gravitational spacetime.

52.7 The Bridge to Holography

The previous two sections handed us, almost incidentally, a geometric object that the rest of this book has been circling toward without naming it. A boundary or a defect introduced a distinguished transverse direction — a coordinate running away from the wall into the bulk of the system — and the kinematics organized themselves around how operators behave as one moves along it. The c- and a-theorems, meanwhile, gave us a number attached to a conformal field theory at each end of a renormalization-group flow, a number that decreases monotonically as one passes from short distances to long. Put these two pictures together and a suggestion forms: that a d-dimensional conformal field theory might be most naturally described by adding *one extra dimension*, a radial direction whose role is to carry the renormalization scale, with the monotone of the c-theorem becoming a monotone in geometry. This section makes that suggestion precise as far as kinematics allow, and is honest about where rigor stops and conjecture begins. The destination is the statement that opens Chapter 53: a conformal field theory in d dimensions is dual to a theory of gravity in an anti-de Sitter space of dimension $d+1$.

The argument has a hard skeleton and a soft tissue, and we are careful to keep them distinct. The hard skeleton is pure group theory and is proved here in full: the symmetry group of a CFT_d is, generator for generator, the isometry group of AdS_{d+1}. No conjecture enters this; it is a theorem about Lie algebras

and Killing vectors. The soft tissue is everything that turns this symmetry match into a dynamical duality — the field–operator dictionary, the conditions under which the bulk is a *local, classical* gravitational theory, the geometric reading of irreversibility. Those we state precisely, motivate honestly, scope as conjectural where they are conjectural, date where dating matters, and forward-reference to Chapters 53 to 55 where they are derived. The chapter has exactly one forward reference — the mass–dimension relation, whose derivation lives in Chapter 53 — and we flag it explicitly when we reach it.

52.7.1 The group-theory match: the conformal group is the AdS isometry group

We work throughout in Euclidean signature, as the entire chapter has: critical phenomena live in d Euclidean dimensions, and the conformal group is the Euclidean form $\mathrm{SO}(()d+1,1)$ established in Theorem 47.10. The first task is to produce a $(d+1)$-dimensional space whose isometry group is exactly this $\mathrm{SO}(()d+1,1)$, and the canonical such space is Euclidean anti-de Sitter space.

Definition 52.27 (Euclidean AdS_{d+1} in Poincaré coordinates)**.** Euclidean anti-de Sitter space AdS_{d+1} of radius L is the upper half-space $\{(z, x^i) : z > 0,\ x^i \in \mathbb{R}^d\}$ equipped with the metric

$$ds^2 = \frac{L^2}{z^2}\Big(dz^2 + \delta_{ij}\, dx^i\, dx^j\Big), \qquad i, j = 1, \dots, d. \tag{52.172}$$

The locus $z \to 0$ is the *conformal boundary*, a copy of $\mathbb{R}^d$ carrying the would-be boundary CFT; $z \to \infty$ is the deep interior.

The metric Eq. (52.172) is the Euclidean continuation of the Lorentzian Poincaré-patch metric Eq. (53.59) derived from the embedding hyperboloid in Theorem 53.10; we take it here as the starting point because all we need now is its symmetry. Two features are visible by eye. First, it is *conformally flat*: it is the flat metric δ_{MN} on the half-space, $M, N \in \{z, 1, \dots, d\}$, multiplied by the position-dependent factor $\Omega^2 = L^2/z^2$. Second, the boundary metric is recovered only up to a divergent Weyl factor — there is no canonical boundary metric, only a conformal class — which is the geometric origin of the boundary theory being *conformal* rather than merely Poincaré invariant. We exploit the first feature immediately.

Reducing the Killing equation. A vector field $\xi = \xi^M \partial_M$ generates an isometry of a metric g_{MN} if and only if it satisfies the Killing equation $\mathcal{L}_\xi g_{MN} = 0$, which in coordinates reads

$$\xi^P \partial_P g_{MN} + g_{PN}\, \partial_M \xi^P + g_{MP}\, \partial_N \xi^P = 0. \tag{52.173}$$

For the conformally flat metric $g_{MN} = \Omega^2 \delta_{MN}$ this collapses to something transparent. Substituting and using $\partial_P g_{MN} = 2\Omega(\partial_P \Omega)\delta_{MN}$,

$$2\Omega(\partial_P\Omega)\,\xi^P\,\delta_{MN} + \Omega^2(\partial_M \xi^N + \partial_N \xi^M) = 0, \tag{52.174}$$

where the indices on the second pair of ξ's are raised with the flat δ. Dividing by Ω^2 and writing $\Omega = L/z$, so that $\partial_P \Omega = -(L/z^2)\,\delta^z_P$ and hence $(\partial_P\Omega)\xi^P = -(L/z^2)\,\xi^z$, the first term becomes $-2(\xi^z/z)\,\delta_{MN}$ and we arrive at

$$\boxed{\partial_M \xi^N + \partial_N \xi^M = \frac{2\,\xi^z}{z}\,\delta_{MN}}. \tag{52.175}$$

This is precisely the *conformal* Killing equation on flat $(d+1)$-dimensional Euclidean space — the statement that $\partial_M \xi^N + \partial_N \xi^M$ is proportional to δ_{MN} — with the conformal factor not free but *pinned* to the value $\omega = \xi^z/z$. The isometries of AdS_{d+1} are therefore exactly those conformal Killing vectors of flat $\mathbb{R}^{d+1}$ whose local scale factor happens to equal ξ^z/z. This pinning is the whole content of the computation: the conformal group of flat $\mathbb{R}^{d+1}$ is the larger $\mathrm{SO}(()d+2,1)$, and Eq. (52.175) selects the subgroup that survives as genuine isometries of the curved space.

Enumerating the solutions. The conformal Killing vectors of flat $\mathbb{R}^{d+1}$ are known completely from the very analysis of Theorem 47.3 (applied now in $d+1$ rather than d dimensions): they are the translations, rotations, the dilatation, and the special conformal transformations, with the general such vector carrying a scale factor $\omega(X)$ that is at most linear in the coordinates. We run through each family and ask whether Eq. (52.175) holds, i.e. whether its flat scale factor equals ξ^z/z.

Boundary translations. The vector $\xi = a^i \partial_i$ (constant a^i, no z-component) has $\partial_M \xi^N + \partial_N \xi^M = 0$ and $\xi^z = 0$, so both sides of Eq. (52.175) vanish. The d generators

$$P_i = \partial_i \tag{52.176}$$

are isometries. The radial translation ∂_z, by contrast, has $\xi^z = 1 \neq 0$ while its left side vanishes, so $0 = 2/z$ is violated: *z-translation is not a symmetry.* This is the geometric statement that the boundary singles out an origin of the radial coordinate, exactly as a renormalization scheme singles out a reference scale.

Boundary rotations. The $\frac{1}{2}d(d-1)$ vectors $M_{ij} = x_i\partial_j - x_j\partial_i$ are antisymmetric, hence have vanishing left side, and have $\xi^z = 0$. They are isometries: the boundary rotation generators.

The dilatation. The full-space dilatation $\xi_D = z\,\partial_z + x^i\partial_i$ has $\xi^z = z$, so the right side of Eq. (52.175) is $2(z/z)\delta_{MN} = 2\delta_{MN}$. Its left side is $\partial_M \xi^N + \partial_N \xi^M = 2\delta_{MN}$ (each coordinate vector field $X^M \partial_M$ contributes the identity). The two sides agree, so

$$D = z\,\partial_z + x^i\,\partial_i \tag{52.177}$$

is an isometry of AdS_{d+1}. This single Killing vector is the linchpin of the whole bridge, and we return to it in Section 52.7.2. Note what it does: it rescales the radial coordinate and the boundary coordinates *together*, $z \to \lambda z$, $x^i \to \lambda x^i$.

Special conformal transformations. For a flat special conformal generator with parameter b^M, $\xi_M = 2(b\cdot X)X_M - X^2 b_M$, the scale factor is $\omega = 2\,b\cdot X$. We need $\omega = \xi^z/z$. Take first a boundary parameter, $b^M = (0, b^i)$ with $b^z = 0$. Then $b\cdot X = b^i x_i$, $X^2 = z^2 + x^2$ (writing $x^2 = \delta_{ij}x^i x^j$), and the z-component of ξ is

$$\xi^z = 2(b\cdot X)X^z - X^2 b^z = 2(b^i x_i)\,z - 0 = 2z\,(b\cdot x), \tag{52.178}$$

so $\xi^z/z = 2\,b\cdot x = 2\,b\cdot X = \omega$. The pinning condition holds, and the d generators

$$K_i = 2x_i(x^j\partial_j + z\,\partial_z) - (x^2 + z^2)\,\partial_i \tag{52.179}$$

are isometries. The radial special conformal generator $b^z \neq 0$ fails the same test — its $\xi^z = 2(b^z z)z - (x^2+z^2)b^z = b^z(z^2 - x^2)$ gives $\xi^z/z = b^z(z^2-x^2)/z$, while $\omega = 2b^z z$, and $2b^z z \neq b^z(z^2-x^2)/z$ — so it is discarded, just as ∂_z was.

The count and the match. Collecting the survivors, the independent isometries of AdS_{d+1} are

$$\underbrace{d}_{P_i} + \underbrace{\tfrac{1}{2}d(d-1)}_{M_{ij}} + \underbrace{1}_{D} + \underbrace{d}_{K_i} = \frac{d^2+3d+2}{2} = \frac{(d+1)(d+2)}{2}, \tag{52.180}$$

which is exactly the dimension of the conformal algebra computed in Theorem 47.9, and exactly the dimension $\frac{1}{2}(d+2)(d+1)$ of $\mathfrak{so}(d+1,1)$. The match is not merely numerical. Comparing the explicit vector fields Eqs. (52.176), (52.177) and (52.179) and the boundary rotations M_{ij} against the four families of conformal Killing vectors of the *boundary* $\mathbb{R}^d$ in Eq. (47.25), we see that the boundary limit $z \to 0$ of each AdS Killing vector is precisely the corresponding conformal Killing vector of the boundary: $D|_{z\to 0} = x^i\partial_i$, $K_i|_{z\to 0} = 2x_i(x\cdot\partial) - x^2\partial_i$, while P_i and M_{ij} are already z-independent. The naming was not a coincidence imposed by hindsight; the identification is forced.

Theorem 52.28 (Conformal group as AdS isometry group)**.** *The isometry algebra of Euclidean* AdS_{d+1}*, generated by the Killing vectors Eq. (52.176),* $M_{ij} = x_i\partial_j - x_j\partial_i$*, Eq. (52.177), and Eq. (52.179), is isomorphic as a Lie algebra to the Euclidean conformal algebra* $\mathfrak{so}(d+1,1)$ *of Theorem 47.10, under the identification of* P_i, M_{ij}, D, K_i *with the generators of the same name. The radial coordinate* z *is acted on nontrivially only by* D *and* K_i *— exactly the generators whose boundary action is the dilatation and the special conformal transformations. In Lorentzian signature the same statement reads* $\mathfrak{so}(d,2) \cong \mathfrak{so}(2,d)$*, the isometry algebra of Lorentzian* AdS_{d+1}*.*

Proof. We have exhibited each generator as a solution of the Killing equation Eq. (52.175) and counted $\frac{1}{2}(d+1)(d+2)$ of them in Eq. (52.180), matching the dimension of $\mathfrak{so}(d+1,1)$. It remains to check that they close into the conformal algebra Eq. (47.53) with the correct brackets, not merely the correct count. The commutators of these explicit vector fields are computed exactly as the abstract brackets were in Section 47.2.3, because the differential-operator realization is the same: D grades P_i and K_i oppositely, $[D, P_i] = -P_i$ and $[D, K_i] = +K_i$ as Lie brackets of vector fields (the sign relative to Eq. (47.53c) being the conventional factor of i that distinguishes anti-Hermitian vector fields from Hermitian generators), and the one nonlinear bracket $[K_i, P_j]$ closes onto $2(\delta_{ij}D - M_{ij})$ by the identical algebra that produced Eq. (47.53d). Since both algebras have the same generators with the same brackets, the linear map sending each AdS Killing vector to the conformal generator of the same name is a Lie-algebra isomorphism. The boundary limits identified above show it is the *geometric* realization of the abstract embedding-space dictionary Eq. (47.56): the two extra embedding directions X^d, X^{d+1} of Section 47.2.4 are the two directions appended to the boundary to build the bulk. □

Section 53.3.6 carries out the boundary-limit construction in full Lorentzian detail, deriving each Killing vector from the ambient hyperboloid and treating the special conformal family — the only one nonlinear in x — step by step; Theorem 53.14 is the Lorentzian counterpart of Theorem 52.28 above. The reader who wants the version anchored to the embedding-space picture of Section 47.2.4 will find it there. For the bridge we are building, the Euclidean half-space computation Eq. (52.175) is the cleanest possible demonstration: the isometries of the bulk *are* the conformal transformations of the boundary, with no input beyond the metric Eq. (52.172) and the conformal Killing equation.

Physical Insight

The pinning condition Eq. (52.175) explains a fact that is otherwise mysterious: why the conformal group of $\mathbb{R}^d$ has the same size as the *isometry* group of a $(d+1)$-dimensional space, when the conformal group of $\mathbb{R}^{d+1}$ itself is strictly larger. The curvature of AdS converts "angle-preserving but not distance-preserving" transformations of the would-be flat $(d+1)$-space into honest distance-preserving symmetries — but only for those whose local rescaling is generated radially, $\omega = \xi^z/z$. The extra dimension trades a conformal symmetry of the boundary for an isometry of the bulk. This is the kinematic seed of holography: a symmetry that acts as a scale transformation on the boundary is realized as a rigid motion in the bulk, and the direction it moves things along is the radial one.

52.7.2 Holographic renormalization-group flow: the radial coordinate is the scale

The single most consequential entry in Theorem 52.28 is the dilatation Eq. (52.177). The generator that drives the renormalization group — D, whose eigenvalue is the scaling dimension and whose flow between fixed points was the whole subject of Part VI — acts in the bulk by rescaling z and x together. We now argue, heuristically but honestly, that this forces the identification

$$\text{radial coordinate } z \;\longleftrightarrow\; \text{inverse renormalization scale}, \qquad \mu \sim \frac{1}{z}, \tag{52.181}$$

so that small z (near the boundary) is the ultraviolet, large z (deep in the bulk) is the infrared, and a Wilsonian momentum cutoff Λ on the boundary theory is a radial slice $z \gtrsim 1/\Lambda$.

The dimensional-analysis argument. Under the dilatation isometry generated by Eq. (52.177), finite transformations are $x^i \to \lambda x^i$ and $z \to \lambda z$ simultaneously. On the boundary, a dilatation by λ scales all lengths by λ and therefore all energies and momenta by λ^{-1}. The radial coordinate scales the same way as a boundary length, so z carries dimensions of length and the natural energy associated to a given radial position is $\mu \sim 1/z$ (restoring the AdS radius, $\mu \sim L/z^2$ or $1/z$ depending on the chosen normalization of μ; only the scaling $\mu \propto 1/z$ is invariant and only it matters). A physical process localized at radius z in the bulk probes boundary physics at the corresponding scale $\mu \sim 1/z$: this is the precise sense in which "moving into the bulk" is "flowing to the infrared."

The falloff argument. The dimensional analysis is made sharp by how bulk fields behave near the boundary, which we preview here and which Chapter 53 derives. A scalar field $\phi(z, x)$ dual to a boundary operator $\mathcal{O}$ of dimension Δ approaches the boundary as a superposition of two power laws,

$$\phi(z,x) \;\xrightarrow{z\to 0}\; z^{\,d-\Delta}\,\phi_{(0)}(x) \;+\; z^{\,\Delta}\,\phi_{(1)}(x) + \cdots, \tag{52.182}$$

the two exponents being the roots of the indicial equation analyzed in Section 53.4. Apply the dilatation $z \to \lambda z$, $x \to \lambda x$: the coefficient $\phi_{(0)}$ must transform so that $z^{d-\Delta}\phi_{(0)}$ is invariant, i.e. $\phi_{(0)}$ has scaling weight $d-\Delta$, while $\phi_{(1)}$ has weight Δ. These are exactly the weights of, respectively, a *source* that couples to $\mathcal{O}$ in the boundary action (a coupling of an operator of dimension Δ has dimension $d-\Delta$) and the *response* $\langle\mathcal{O}\rangle$ (which has dimension Δ). The position along the radial axis literally reads out the scaling weight: the leading falloff near $z=0$ is the ultraviolet data (the coupling one turns on), and the subleading falloff is the infrared response (what the theory does about it). The radial direction is not merely *like* a scale; the bulk equation of motion organizes its solutions by scale.

Connection to Wilsonian flow. This is the holographic incarnation of the Wilsonian renormalization group of Chapter 23. There, integrating out boundary momentum modes above a cutoff Λ and tracking the induced flow of couplings was the definition of the RG; the beta functions of Chapter 24 measured the rate of that flow. Here, the cutoff is implemented geometrically by terminating the bulk at a radial slice $z = \epsilon$ rather than going all the way to $z = 0$; the divergences of the boundary CFT reappear as the $\epsilon \to 0$ divergences of the on-shell bulk action, and removing them by adding local boundary counterterms is the procedure of *holographic renormalization* carried out in Section 53.6.4. Sliding ϵ inward integrates out boundary modes, and the radial evolution equation of the bulk fields — their dependence on z — is, term by term, the renormalization-group flow of the dual couplings. The dilatation that generates RG flow in Part VI is the radial isometry Eq. (52.177) of the bulk; Chapter 54 develops this holographic RG quantitatively, including the precise map between the radial Hamiltonian and the beta functions.

Notation Watch

Throughout this chapter $\epsilon = 4 - d$ is the dimensional-expansion parameter of the Wilson–Fisher analysis, not the Feynman $i\epsilon$ and not the radial cutoff. In this subsection the symbol ϵ appears once more, as the holographic radial cutoff $z = \epsilon$; this usage is standard in the holography literature and is confined to the discussion of Section 53.6.4, where it never collides with the expansion parameter (that analysis is at fixed integer d). We keep the radial-cutoff usage only in this paragraph and in cross-references to Chapter 53; elsewhere ϵ is the expansion parameter.

52.7.3 The preview dictionary: a primary is a bulk field

The symmetry match Theorem 52.28 tells us that whatever lives in the bulk must organize itself into representations of $\mathrm{SO}(()d+1,1)$, which are exactly the representations of the boundary conformal group — labelled, by the analysis of Section 47.3, by a scaling dimension Δ and a spin (an $\mathrm{SO}(()d)$ rotation representation). The natural conjecture, the one that turns the symmetry match into a dynamical correspondence, is that the boundary spectrum of primary operators is in one-to-one correspondence with a bulk spectrum of fields.

Definition 52.29 (The field–operator map, scalar case)**.** To each scalar primary operator $\mathcal{O}$ of the boundary CFT, of scaling dimension Δ, there corresponds a scalar field ϕ propagating in the bulk AdS_{d+1}, of mass m, with the dimension and mass related by

$$\Delta(\Delta - d) = m^2 L^2. \tag{52.183}$$

Fields of nonzero spin correspond to spinning primaries: a massless spin-1 bulk

gauge field is dual to a conserved current J^μ of dimension $\Delta = d - 1$, and the massless spin-2 bulk graviton is dual to the stress tensor $T^{\mu\nu}$ of dimension $\Delta = d$.

Caution

The one forward reference. Equation Eq. (52.183) is *stated* here, not derived. Its derivation — as the indicial equation of the bulk wave operator near the conformal boundary, together with the choice of boundary falloff and the Breitenlohner–Freedman stability bound $m^2L^2 \geq -d^2/4$ — is carried out in full in Theorem 53.17, equation Eq. (53.112). This is the single forward connection of the chapter, declared at the outset. We use Eq. (52.183) below only to motivate the structure of the correspondence, never to prove anything that the present chapter relies on.

Two heuristic checks make Eq. (52.183) plausible without the full derivation, and both connect to material already in hand. First, the two roots of the quadratic Eq. (52.183),

$$\Delta_\pm = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2L^2}, \tag{52.184}$$

are exactly the two falloff exponents Δ and $d-\Delta$ of Eq. (52.182): their sum is $\Delta_+ + \Delta_- = d$ and their product is $\Delta_+\Delta_- = -m^2L^2$, matching the source/response pairing identified in Section 52.7.2. Second, a *massless* bulk scalar, $m = 0$, gives $\Delta_+ = d$ — a marginal operator, the dimension of a Lagrangian coupling in d dimensions — which is precisely what one expects: a massless bulk field is a modulus, and modes that cost no energy in the bulk are dual to operators that cost no scale to turn on. The graviton entry of Theorem 52.29 is the sharpest case. The bulk metric fluctuation is massless and spin-2; the dual operator must be a conserved, spin-2, dimension-d primary, which by the analysis of Section 47.4.3 is the stress tensor and nothing else. That the boundary theory *has* a stress tensor at all is the statement that the bulk theory *has* dynamical gravity — the entry that gives the correspondence its name. The complete field–operator dictionary, including the GKP–Witten generating-functional relation that computes correlators, is the subject of Section 53.5.

52.7.4 When is the bulk classical gravity? Large central charge and a sparse spectrum

Everything so far is kinematics, and kinematics alone does not tell us whether the bulk dual of a given CFT is a tractable, local, classical gravitational theory or something far wilder. A generic CFT, if it has a bulk dual at all, is dual to a strongly coupled, stringy, intrinsically quantum-gravitational mess. The cases in which the bulk reduces to classical (semiclassical) Einstein gravity coupled to a

handful of light fields — the cases that make holography a computational tool rather than a slogan — are special, and the conditions that single them out are read directly off the data of the previous section.

Central charges measure the inverse Newton constant. The c- and a-type coefficients of the conformal anomaly, computed for the CFT at its fixed point in Section 52.5, count the number of local degrees of freedom of the theory. On the bulk side, the corresponding measure is the gravitational coupling: the smaller the Newton constant G_N in AdS units, the more the bulk geometry behaves classically (the gravitational fluctuations $\sim G_N$ are suppressed, and the saddle-point/tree-level approximation to the bulk path integral becomes exact). The two are identified:

$$c \sim a \sim \frac{L^{d-1}}{G_N}, \tag{52.185}$$

up to scheme-dependent numerical factors fixed precisely in Chapter 54. The relation Eq. (52.185) is exact and famous in $d = 2$, where the Brown–Henneaux analysis of the asymptotic symmetry algebra of AdS_3 gives $c = 3L/2G_N$ for the central charge of Theorem 48.8; in higher d the analogous statement relates the trace-anomaly coefficients to L^{d-1}/G_N. The upshot is uniform: *a large central charge is a small bulk Newton constant*, and the classical-gravity limit is the large-c limit.

Large c is necessary but not sufficient: the gap. Large c makes the bulk weakly gravitating, but does not by itself make it *Einstein* gravity. A weakly coupled bulk could still contain a Regge tower of light higher-spin fields — the hallmark of a string theory with a low string scale — in which case the bulk is local only on scales much larger than the AdS radius, and "gravity" is the wrong description below the string scale. What additionally distinguishes an Einstein-gravity dual is a *sparse spectrum with a large gap*: the lightest single-trace operators of nonzero spin > 2 must have parametrically large dimension $\Delta_{\mathrm{gap}} \to \infty$, so that the corresponding bulk higher-spin fields are heavy and decouple, leaving the metric (spin 2) and a few light matter fields as the only low-lying bulk degrees of freedom. Equivalently, the bulk is local on the AdS scale precisely when Δ_{gap} is large. We state the package as a conjecture, because that is what its general form is, and we date it.

Conjecture 52.30 (Conditions for a local semiclassical dual; current understanding, mid-2020s)**.** *A conformal field theory in $d \geq 3$ dimensions admits a description by classical Einstein gravity in AdS_{d+1} coupled to a finite number of light fields, local on the scale of the AdS radius L, when (i) the central charges are large, $c \sim a \sim L^{d-1}/G_N \gg 1$, suppressing bulk quantum gravity; and (ii) the spectrum is sparse with a large higher-spin gap, $\Delta_{gap} \to \infty$, pushing the would-be stringy/higher-spin states to large mass. Under these conditions the boundary*

correlators are reproduced by tree-level (Witten-diagram) computations in the bulk effective theory. The precise necessity and sufficiency of these conditions, and the sharp form of the required bounds, are established in particular families and conjectured in general; this is the state of understanding as of the mid-2020s.

Why Yang–Mills theory is the prototype. The cleanest realizations of Theorem 52.30 are large-N gauge theories — the natural home of Yang–Mills theory. A SU(N) gauge theory of the type carried since Chapter 25 has, in the 't Hooft limit $N \to \infty$ at fixed 't Hooft coupling $\lambda_{\text{'t H}} = g^2 N$, a central charge growing as $c \sim N^2$, which is the large-c condition (i) with $1/G_N \sim N^2$. The gauge invariants are single-trace operators $\mathrm{tr}(\cdots)$, and at large 't Hooft coupling the higher-spin single-traces acquire large anomalous dimensions, $\Delta_{\text{gap}} \sim \lambda_{\text{'t H}}^{1/4} \to \infty$, supplying condition (ii). The genus expansion of the large-N diagrammatics, in which surfaces of higher genus are suppressed by powers of $1/N^2$, is term-for-term the loop expansion of a string theory in the bulk — a structural coincidence that is the historical and logical entry point to the entire subject, developed in Sections 53.1 and 53.1.1. The prototypical exactly-conformal example, $\mathcal{N} = 4$ super-Yang–Mills, is dual to type IIB string theory on $\mathrm{AdS}_5 \times S^5$, and its supergravity (Einstein-gravity) limit is exactly the large-N, large-$\lambda_{\text{'t H}}$ corner where Theorem 52.30 applies. The large-N gauge theory carried through this book is therefore not merely *a* candidate gravity dual; it is *the* candidate, and tracing why is the business of Chapter 53.

52.7.5 Geometric irreversibility: the holographic c-function

The last thread to tie is the one we opened with. The c- and a-theorems of Section 52.5 state that a certain CFT datum decreases monotonically along renormalization-group flow, encoding the irreversibility of coarse-graining: degrees of freedom are lost, never gained, as one flows to the infrared. If the radial direction of the bulk *is* the renormalization scale, as Section 52.7.2 argued, then this irreversibility must be visible as a monotonicity property of the bulk geometry along the radial direction. It is, and the mechanism is clean enough to exhibit in full.

The domain-wall geometry of an RG flow. A renormalization-group flow connecting a UV fixed point to an IR fixed point is dual, on the bulk side, to a geometry that interpolates between two AdS regions of different radii: AdS_{d+1} with radius L_{UV} near the boundary and AdS_{d+1} with radius $L_{\text{IR}} > L_{\text{UV}}$ deep in the interior. The most general such Poincaré-invariant geometry — preserving the d-dimensional boundary symmetries while breaking the dilatation that would make it pure AdS — is the domain wall

$$ds^2 = e^{2A(r)}\,\delta_{\mu\nu}\,dx^\mu dx^\nu + dr^2, \tag{52.186}$$

where r is the radial coordinate (related to the Poincaré z of Eq. (52.172) by $z = L\,e^{-A}$ near a fixed point, so that $A \to +\infty$ is the UV boundary and $A \to -\infty$ the deep IR), and $A(r)$ is the *warp factor*. At a fixed point the geometry is pure AdS, for which $A(r) = r/L$ and the slope $A' = 1/L$ is constant; away from a fixed point $A'(r)$ varies, interpolating between $1/L_{\rm UV}$ and $1/L_{\rm IR}$.

The candidate c-function and its monotonicity. Define the holographic c-function

$$c(r) \;=\; \frac{\kappa}{G_N\,\big[A'(r)\big]^{\,d-1}}, \tag{52.187}$$

with κ a positive d-dependent constant. At a fixed point $A' = 1/L$ is constant and $c \propto L^{d-1}/G_N$, reproducing Eq. (52.185): the holographic c-function evaluated at the endpoints returns exactly the central charges of the two CFTs. We now show $c(r)$ is monotone. Differentiating Eq. (52.187),

$$c'(r) = \frac{\kappa}{G_N}\,\frac{d}{dr}\big[A'\big]^{-(d-1)} = -\,(d-1)\,\frac{\kappa}{G_N}\,\frac{A''(r)}{\big[A'(r)\big]^{\,d}}. \tag{52.188}$$

Since $A' > 0$ (the warp factor increases toward the boundary) and $\kappa, G_N, (d-1) > 0$, the sign of $c'(r)$ is the *opposite* of the sign of $A''(r)$. The geometry therefore enforces monotonicity provided $A''(r) \le 0$, and that inequality is exactly what gravity supplies.

The null energy condition does the work. The warp factor's concavity, $A'' \le 0$, is a consequence of Einstein's equations sourced by any reasonable matter. Computing the Einstein tensor of Eq. (52.186) and contracting Einstein's equation $G_{MN} + \Lambda g_{MN} = 8\pi G_N\,T_{MN}$ with a radial null vector k^M (one with $k^M k_M = 0$, having one component along r and one along a boundary direction), the cosmological-constant and curvature pieces arrange so that

$$A''(r) = -\,\frac{8\pi G_N}{d-1}\,T_{MN}\,k^M k^N \Big/ (k^t)^2, \tag{52.189}$$

schematically: A'' is set by the null–null component of the stress tensor. The *null energy condition* — the statement that $T_{MN}k^M k^N \ge 0$ for every null k^M, satisfied by all standard matter (scalars, gauge fields, fermions) and the cornerstone hypothesis of the classical energy conditions — then gives

$$A''(r) \le 0, \tag{52.190}$$

and hence, by Eq. (52.188), $c'(r) \ge 0$. As one flows from the UV (large r, near the boundary) to the IR (small r, deep interior), r decreases and $c(r)$ decreases with it:

$$c_{\rm UV} = c(r \to +\infty) \;\ge\; c(r \to -\infty) = c_{\rm IR}. \tag{52.191}$$

Theorem 52.31 (Holographic c-theorem)**.** *For any bulk matter satisfying the null energy condition, the holographic c-function Eq. (52.187) of a Poincaré-invariant domain-wall geometry Eq. (52.186) is monotonically non-decreasing in the radial coordinate r, $c'(r) \geq 0$, and interpolates between the central charges of the UV and IR fixed-point CFTs. The irreversibility of renormalization-group flow — the field-theory c-/a-theorem of Section 52.5 — is thereby realized as the monotonicity in the radial (energy) direction of a geometric quantity, with the null energy condition playing the role that unitarity and reflection positivity play in the field-theory proof.*

This is a striking dictionary entry: a deep theorem about the irreversibility of quantum field theory becomes, on the bulk side, an elementary consequence of an energy condition on classical matter. The translation runs both ways. The field-theory statement is proved using unitarity and the positivity of the spectral density (the methods of Section 52.5); the gravity statement is proved using the null energy condition. That these two positivity conditions — one on a Hilbert-space inner product, one on a classical stress tensor — yield the same monotone is one of the sharpest pieces of evidence that the correspondence is consistent at the level of dynamics, not merely symmetry.

Physical Insight

There is a second, conceptually deeper, geometric reading of the same monotone, through entanglement entropy. The number of degrees of freedom of a region of the boundary theory is measured by the entanglement entropy across its boundary, and in holography that entropy is computed — by the Ryu–Takayanagi prescription of Chapters 55 and 56 — as the *area* of a minimal bulk surface anchored on the region. Flowing to the infrared shrinks the available degrees of freedom, which shrinks the minimal-surface area, which decreases an entropic c-function. The monotonicity of the c-theorem then descends from the monotonicity of areas of minimal surfaces as the anchoring region grows — a purely geometric fact. This entanglement reading, which is how the strongest forms of the higher-dimensional c- and a-theorems are actually established, is developed in Chapter 55; we flag here only that the holographic c-function Eq. (52.187) and the entanglement c-function are two faces of the same monotone, both pointing along the radial direction that we have now identified as the renormalization scale.

Figure 52.8 collects the picture. The boundary at $z \to 0$ carries the conformal field theory whose data — the spectrum of dimensions Δ and the OPE coefficients — the bootstrap of Chapters 49 to 51 constrains. The radial direction carries the renormalization scale. Bulk fields are boundary operators by Theorem 52.29. The bulk is classical Einstein gravity when the central charge is large and the spectrum

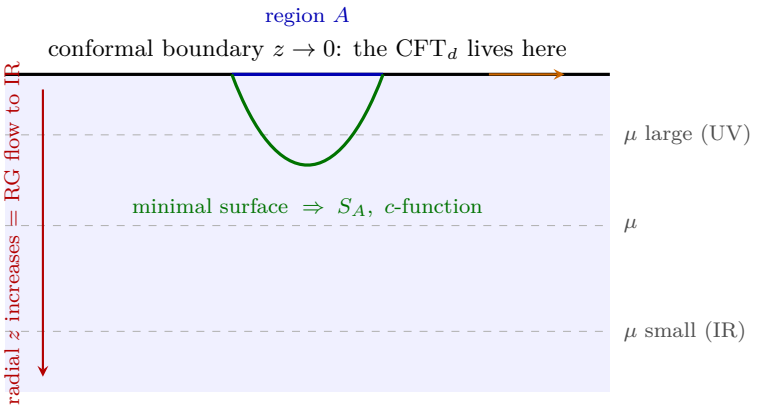


Figure 52.8: The bridge to holography. A CFT$_d$ lives on the conformal boundary $z \to 0$ of AdS$_{d+1}$. The bulk dilatation Killing vector $D = z\partial_z + x^i\partial_i$ of Eq. (52.177) scales the radial coordinate z together with the boundary coordinates, identifying z with the inverse renormalization scale $\mu \sim 1/z$ (Eq. (52.181)): radial slices are Wilsonian cutoff surfaces, and moving into the bulk (z increasing) is renormalization-group flow to the infrared. A boundary region A anchors a minimal bulk surface (Chapters 55 and 56) whose area computes the entanglement entropy S_A and a geometric c-function; its monotonicity along the radial direction realizes the c-theorem of Section 52.5, as in Theorem 52.31.

sparse, Theorem 52.30. And the irreversibility of the flow is the monotonicity of a geometric quantity along the radial axis, Theorem 52.31. Every ingredient has been either proved here (the symmetry match) or stated precisely with its derivation forward-referenced (the dictionary, the classical limit, the geometric c-theorem).

This completes the chapter, and with it Part XIV. We arrived at conformal field theory as the universal description of a renormalization-group fixed point, closing the loop with Part VI and consummating ϕ^4 theory at the Wilson–Fisher point; we have now turned the fixed-point's own scaling symmetry into the suggestion that the theory is the boundary of a gravitational bulk with one extra dimension. What remains is to state the correspondence precisely, to derive the dictionary entries we have only previewed — beginning with the mass–dimension relation Eq. (53.112) that is this chapter's one forward connection — and to confront honestly the question of what is theorem and what is conjecture. That is the work of the next part, which opens with the AdS/CFT correspondence in Chapter 53.

Further reading

The identification of critical fixed points with conformal field theories is developed, for the Wilson–Fisher fixed point of the ϵ-expansion, from Wilson and Fisher [515] and the broader renormalization-group program of Wilson [513], Wilson and Kogut [516]; the precise sense in which scale invariance implies conformal invariance is Polchinski [404], with the four-dimensional analysis sharpened in Luty et al. [329]. The two-dimensional theory, the central charge, and the minimal models that organize the c-theorem staircase are treated comprehensively in the monograph of Di Francesco et al. [147]; boundary conformal field theory in the style of Section 52.6 traces to Cardy and is reviewed in Cardy [108]. The arrow of the renormalization group rests on Zamolodchikov [531] in two dimensions and on the conjecture of Cardy [109] proved by Komargodski and Schwimmer [301] in four; the odd-dimensional F-theorem is Jafferis et al. [282], proved entropically by Casini and Huerta [114], and the generalized monotone is computed for the $O(N)$ models in Fei et al. [187]. Precision three-dimensional Ising data come from the resummed ϵ-expansion of Le Guillou and Zinn-Justin [318], the conformal bootstrap of Kos et al. [304] (with the modern program reviewed in Poland et al. [402], Rychkov [421]), and the Monte Carlo study of Hasenbusch [254]. The holographic bridge of Section 52.7 is the gateway to Part XV; the foundational correspondence is Maldacena [331] with the generating-functional dictionary of Gubser et al. [242], Witten [522], all developed in full in Chapter 53.

Part XV

Holography and Quantum Information

Chapter 53

The AdS/CFT Correspondence

Two threads of this book, pursued for very different reasons, are about to meet. From the gauge side we carry the 't Hooft large-N expansion of Chapters 27 and 28: reorganizing a non-abelian gauge theory by powers of $1/N$ rather than by the coupling reveals a topological expansion in which planar diagrams dominate and the surfaces swept out by Feynman graphs are sorted by genus exactly as the worldsheets of a string theory are. From the conformal side we carry the apparatus of Chapters 47 and 49: the conformal algebra $\mathrm{SO}((2,d)$, the classification of local operators by scaling dimension, unitarity bounds, and the bootstrap insight that a CFT can be *defined* by its symmetry and the data of its correlators rather than by a Lagrangian. The proposal of this chapter is that these two pictures are the same physics seen twice — that a particular four-dimensional conformal gauge theory is, exactly and non-perturbatively, a theory of quantum gravity in a curved five-dimensional spacetime, and that the elusive "master string" hiding inside the large-N expansion is literally a string propagating in that bulk.

We build the correspondence in the order in which its logic is forced. Section 53.1 establishes the structural fact that makes everything else conceivable: the $1/N$ expansion of a matrix-valued field theory is a genus expansion, with string coupling $g_s \sim 1/N$, single-trace operators that factorize at large N, and a "master field" begging for a dual description. Section 53.2 turns structure into a concrete duality by following Maldacena's argument honestly: a stack of N branes admits two descriptions whose near-horizon, decoupling limits must agree, identifying $\mathcal{N}=4$ super-Yang–Mills with type IIB string theory on $\mathrm{AdS}_5 \times S^5$ and fixing the parameter dictionary $L^4/\alpha'^2 \sim \lambda$, $g_s \sim g_{\mathrm{YM}}^2$. We are careful throughout to separate what is derived from what is conjectured, and we record why $\mathcal{N}=4$ is exactly conformal — its vanishing beta function (Chapter 45) — and how its superconformal group $\mathrm{PSU}(2,2|4)$ matches the bulk isometries.

With the claim on the table we develop the bulk geometry and the dictionary that makes the duality computational. Section 53.3 constructs anti-de Sitter

space as a hyperboloid, gives its three working coordinate systems, computes its curvature and maximal symmetry, and exhibits its conformal boundary — the crucial fact that AdS_{d+1} carries an $\mathrm{SO}(()2,d)$ isometry group acting on the boundary exactly as the conformal group of Chapter 47 acts on d-dimensional Minkowski space. Section 53.4 solves the wave equation for a bulk scalar near the boundary, extracts its two characteristic falloffs, and derives the mass–dimension relation $\Delta(\Delta - d) = m^2 L^2$ together with the Breitenlohner–Freedman bound. Section 53.5 assembles these into the field–operator map: every bulk field is dual to a boundary primary, the leading bulk falloff is a source and the subleading one a vacuum expectation value, bulk gauge symmetries become boundary global symmetries, bulk diffeomorphisms enforce stress-tensor conservation, and the holographic degree-of-freedom count reproduces the N^2 scaling of the central charge.

The dictionary becomes a calculus in Section 53.6, where the GKP–Witten relation equates the boundary generating functional with the on-shell bulk action; we construct the bulk-to-boundary propagator, confront the divergences of the on-shell action, carry out holographic renormalization, and extract the scalar two-point function with the correct CFT power law and coefficient. Finally Section 53.7 steps back to weigh the evidence honestly: the symmetry and central-charge matches the duality passes exactly, the integrability and localization tests it survives, the precise sense in which it is a conjecture taken as a definition, the strong/weak character that makes it both powerful and hard to prove, and the genuine open frontiers — emergent bulk locality, the black-hole interior, and holography beyond anti-de Sitter — that the remainder of Part XV (Chapters 54 to 57) takes up.

53.1 Large-N and the String in Disguise

The chapter's organizing thesis — that a quantum field theory in d dimensions might *be* a theory of gravity in $d+1$ dimensions — is, taken cold, an outlandish claim. A gauge theory has no graviton, no dynamical metric, no extra dimension; gravity has all three. Before we can take the conjecture seriously we need a reason, internal to gauge theory itself, to expect that a hidden gravitational — indeed stringy — structure is lurking inside it. That reason is the subject of this section, and it predates the correspondence by more than two decades. In 1974 't Hooft asked what becomes of a $\mathrm{SU}(N)$ gauge theory when the number of colors N is taken large, and discovered that the perturbative expansion does not merely simplify: it *reorganizes*. Feynman diagrams sort themselves by the topology of the two-dimensional surface they can be drawn on, and the resulting expansion in powers of $1/N$ is, term by term, identical in structure to the loop expansion of a theory of closed strings, with the string coupling identified as $g_s \sim 1/N$. No string is put in by hand; one falls out of the bookkeeping. This is

the first quantitative hint — structural, not yet a derivation of any dual — that a gauge theory secretly contains strings and therefore, since closed strings carry a graviton, gravity. We continue Yang–Mills theory from Chapter 25, viewing the carried SU(N) Yang–Mills theory through the large-N lens; this is a structural re-reading of the same theory, but a structural re-reading of the same theory.

53.1.1 Matrices, not vectors: the 't Hooft limit

The decisive observation is about how the color index sits on the fields. A quark transforms in the fundamental representation of SU(N), carrying a single color index $i = 1, \ldots, N$; it is a *vector* in color space. The gauge field, by contrast, lives in the adjoint representation, and the adjoint of SU(N) is built from one fundamental and one antifundamental index. Writing the gauge potential as a matrix in the fundamental,

$$(A_\mu)^i{}_j \;=\; A^a_\mu\,(T^a)^i{}_j, \qquad i,j = 1,\ldots,N, \tag{53.1}$$

with the Hermitian generators T^a of Chapter 25 normalized as $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, exhibits the gluon as an $N \times N$ traceless Hermitian matrix at each spacetime point: it is a *matrix* in color space, carrying an upper (fundamental) and a lower (antifundamental) index. The adjoint dimension is $N^2 - 1$, which for large N is N^2 up to a $1/N^2$ correction; the missing generator is the trace part that would promote SU(N) to U(N), and we will see that the distinction between SU(N) and U(N) is itself a subleading $1/N^2$ effect. It is this matrix character — two index lines per field rather than one — that powers the entire construction. Vector (fundamental) fields, by contrast, carry a single line and, as we note at the end, are suppressed; the planar surfaces are woven out of adjoint matrices.

Taking $N \to \infty$ naively is not yet a well-posed limit, because the coupling g also appears, and the physics depends on how g is scaled as N grows. The Yang–Mills Lagrangian Eq. (25.76) carries the coupling inside the covariant derivative, $D_\mu = \partial_\mu - igA^a_\mu T^a$; the cubic and quartic gluon self-interactions then come with explicit powers of g and g^2. The running of g is governed by the one-loop beta function of Eq. (27.143), whose leading coefficient is proportional to $C_A = N$ (Chapter B). A short look at that coefficient already tells us the right scaling. The one-loop running of $\alpha_s = g^2/4\pi$ is set by $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$; with $C_A = N$ the gluonic contribution grows linearly in N, so to keep the running — and hence the dynamically generated scale Λ of Theorem 27.14 — finite and N-independent as $N \to \infty$, the combination that must be held fixed is not g^2 but g^2N. This is 't Hooft's choice.

Definition 53.1 (The 't Hooft limit and the 't Hooft coupling)**.** The *'t Hooft limit* of SU(N) (or U(N)) gauge theory is the limit

$$N \to \infty \qquad \text{with} \qquad \lambda \equiv g^2 N \;\text{ held fixed}, \tag{53.2}$$

where λ is the *'t Hooft coupling*. Equivalently $g^2 = \lambda/N \to 0$: the bare gauge coupling is sent to zero, but compensated by the growing number of colors so that the effective loop expansion parameter λ remains $O(1)$. Physical quantities are then expanded in two independent parameters: the genuine coupling λ, which controls the dynamics at fixed topology, and $1/N$, which we shall find controls the *topology* of the contributing diagrams.

The split is the whole point. In ordinary perturbation theory there is one small parameter, g, and every diagram of a given order in g is lumped together. The 't Hooft limit separates the single sum over g into a double expansion: at each fixed power of $1/N$ one must still sum infinitely many diagrams (all orders in λ), but those infinitely many diagrams share a common topology. The leading topology — the planar one — is an interacting, generally strongly-coupled theory in its own right, parametrized by λ; the subleading topologies are corrections suppressed by powers of $1/N^2$. To see this we need a bookkeeping device that makes the flow of color indices manifest.

53.1.2 Double-line notation

Because the gluon Eq. (53.1) carries two color indices, an upper i and a lower j, we draw its propagator not as a single line but as a *pair* of parallel lines — a *ribbon* — one line tracking each index. This is 't Hooft's double-line notation, and it turns the abstract contraction of color indices into a geometric statement about how ribbons join into surfaces.

The starting point is the index structure of the gluon propagator. For U(N), where the gauge field is a general (not traceless) Hermitian matrix, the generators are a complete basis of Hermitian matrices and satisfy the completeness (Fierz) relation

$$\sum_a (T^a)^i{}_j \, (T^a)^k{}_l \;=\; \tfrac{1}{2}\, \delta^i_l \, \delta^k_j , \tag{53.3}$$

so that the propagator of the matrix field inherits the index structure

$$\big\langle (A_\mu)^i{}_j \, (A_\nu)^k{}_l \big\rangle \;\propto\; \delta^i_l \, \delta^k_j . \tag{53.4}$$

The two Kronecker deltas are the content of the double line: the upper index i of the first field is identified with the lower index l of the second, and the lower index j of the first with the upper index k of the second. Pictorially, the propagator is two parallel lines carrying i and j, with the arrows arranged so that an upper (fundamental) index flows in along one line and out along the other. For SU(N) the generators are *traceless*, and the completeness relation acquires a subtraction enforcing $\operatorname{tr} A = 0$,

$$\sum_a (T^a)^i{}_j \, (T^a)^k{}_l \;=\; \tfrac{1}{2}\Big(\delta^i_l \, \delta^k_j - \tfrac{1}{N}\, \delta^i_j \, \delta^k_l \Big), \tag{53.5}$$

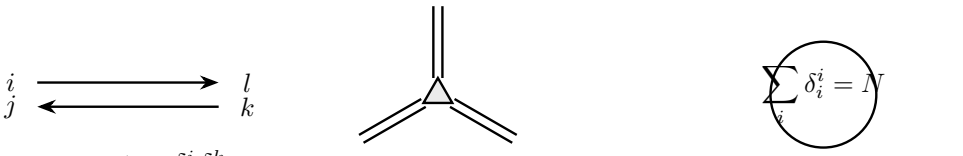


Figure 53.1: The three atoms of double-line bookkeeping. *Left:* a gluon propagator is a ribbon of two oppositely-directed index lines, carrying $\delta^i_l\delta^k_j$ as in Eq. (53.4). *Centre:* an interaction vertex joins ribbons so that index lines flow continuously through it; in the 't Hooft normalization Eq. (53.6) every vertex carries a factor N/λ. *Right:* a closed index line bounds a face of the ribbon graph and, being a free color sum $\sum_i \delta^i_i$, contributes a factor N. The power of N in any diagram is thus the number of faces.

whose extra term is explicitly down by $1/N$. The $-\frac{1}{N}\delta^i_j\delta^k_l$ piece corresponds, in ribbon language, to the gluon line pinching off into a separate single closed loop; we shall see it is suppressed by $1/N^2$ relative to the leading double-line contraction, which is precisely why the difference between $\mathrm{SU}(N)$ and $\mathrm{U}(N)$ does not matter at any fixed order of the $1/N$ expansion. From here on we use the $\mathrm{U}(N)$ rule Eq. (53.4) and restore the trace subtraction only when its (subleading) effect is the point.

The vertices join ribbons. A cubic gluon vertex $\propto g\,f^{abc}$ or, equivalently in matrix language, a term $\mathrm{tr}(A^3)$ in the action, ties three ribbons together so that each incoming index line continues without interruption into an outgoing line: color indices are conserved along the lines, never created or destroyed at a vertex. The visual rule that results is the one summarized in Fig. 53.1: a ribbon Feynman graph is an oriented surface-with-boundary built by gluing strips (propagators) at junctions (vertices), and every *closed* index line — a loop that closes on itself, bounding a region of the surface — is a free sum $\sum_i \delta^i_i = N$ over a color index. The number of such closed loops will turn out to be the number of *faces* of the surface, and it is this count that supplies the powers of N.

To make the powers of N and λ completely explicit it is cleanest to work with a model that strips the gauge theory down to its color combinatorics while keeping the matrix structure exactly. Let $\Phi(x)$ be a single Hermitian $N\times N$ matrix field — the matrix scalar — with action

$$S[\Phi] \;=\; \frac{N}{\lambda}\int \mathrm{d}^d x\;\; \mathrm{tr}\Big(\tfrac{1}{2}\,\partial_\mu\Phi\,\partial^\mu\Phi + \tfrac{1}{4}\,\Phi^4\Big), \tag{53.6}$$

in which the entire color content sits in the matrix multiplication inside the trace. This is the 't Hooft normalization: the overall N/λ in front means the coupling has been scaled out of the fields so that λ is the only genuine coupling and N counts topology. The combinatorics of Eq. (53.6) is identical to that of the gluon

sector of Yang–Mills with $\lambda = g^2 N$ — the gluon's Lorentz structure and the cubic vertex add nothing to the index bookkeeping — so every conclusion drawn from it transfers verbatim to Yang–Mills theory. Reading the Feynman rules off Eq. (53.6):

- the quadratic term has coefficient N/λ, so the *propagator* (its inverse) carries a factor λ/N together with the double-line index structure Eq. (53.4);
- each *vertex* comes from a term with the prefactor N/λ, so it carries a factor N/λ;
- each closed *index loop* carries a factor N from the free color sum.

These three rules — λ/N per edge, N/λ per vertex, N per face — are all we need.

53.1.3 Power counting: the genus expansion

We now compute the N-dependence of an arbitrary connected vacuum diagram and discover the Euler characteristic hiding inside it.

Theorem 53.2 (Topological $1/N$ expansion)**.** *Let a connected vacuum diagram of the matrix theory Eq. (53.6) (equivalently, the gluon sector of* $\mathrm{U}(N)$ *Yang–Mills at* $\lambda = g^2 N$*) have* V *vertices,* E *propagators (edges), and* F *closed index loops (faces) when drawn in double-line notation. Then its dependence on* N *and* λ *is*

$$\text{(diagram)} \;\propto\; N^{V-E+F}\,\lambda^{E-V} \;=\; N^{\chi}\,\lambda^{E-V}, \qquad \chi \equiv V - E + F. \tag{53.7}$$

The exponent χ *is the Euler characteristic of the closed orientable surface that the ribbon graph triangulates; for such a surface of genus* g *(the number of handles),*

$$\chi \;=\; 2 - 2g. \tag{53.8}$$

Consequently the connected vacuum amplitude (the free energy $W = \log Z$*) organizes as a double expansion in topology and 't Hooft coupling,*

$$W \;=\; \sum_{g=0}^{\infty} N^{\,2-2g}\, f_g(\lambda), \qquad f_g(\lambda) = \sum_{\text{genus-}g\ \text{diagrams}} \lambda^{E-V}, \tag{53.9}$$

in which each f_g *is itself a sum over infinitely many diagrams sharing the genus* g*.*

Proof. Counting powers of N *and* λ*.* Assemble the three Feynman rules of Section 53.1.2. A diagram with V vertices, E edges, and F faces carries

$$\left(\tfrac{N}{\lambda}\right)^{V}\left(\tfrac{\lambda}{N}\right)^{E} N^{F} \;=\; N^{V}\,N^{-E}\,N^{F}\,\lambda^{-V}\,\lambda^{E} \;=\; N^{V-E+F}\,\lambda^{E-V}, \tag{53.10}$$

which is the first equality of Eq. (53.7). The power of λ is $E - V$, and since for a connected diagram the number of momentum-space loops is $L = E - V + 1$, this is λ^{L-1}: f_g is an ordinary power series in λ, starting at the order set by the simplest genus-g diagram. The power of N is the integer $V - E + F$, the alternating count of vertices, edges, and faces. We must now recognize it as a topological invariant.

Why $V - E + F$ is the Euler characteristic. A ribbon graph is more than a graph: the cyclic ordering of ribbons around each vertex (fixed by the order of the matrices inside the trace at that vertex) tells us how to thicken the graph into a two-dimensional surface. Fatten every propagator into a strip and every vertex into a small disk; the strips and disks glue along their boundaries into a compact oriented surface Σ, and the closed index lines are exactly the boundary circles of the complementary regions. Capping each such boundary circle with a disk (a face) closes Σ into a closed oriented surface. This is a genuine *cell decomposition* (a polygonal tiling) of Σ: its 0-cells are the V vertices, its 1-cells are the E edges, and its 2-cells are the F faces. For any cell decomposition of a closed surface the alternating sum of the cell counts is, by definition, the Euler characteristic,

$$\chi(\Sigma) \;=\; V - E + F, \tag{53.11}$$

and this number is a topological invariant: it is independent of the particular decomposition and depends only on Σ. (That Eq. (53.11) is decomposition-independent is the elementary content of the Euler–Poincaré theorem; one checks invariance under the two moves that relate any two cell decompositions — subdividing an edge, which adds one vertex and one edge and leaves $V - E + F$ fixed, and adding an edge across a face, which adds one edge and one face and again leaves $V - E + F$ fixed. Any two decompositions are connected by such moves, so the alternating sum is the same for all of them.) For a closed orientable surface the classification of surfaces identifies Σ by its genus g alone, and the Euler characteristic of the genus-g surface is $\chi = 2 - 2g$: the sphere ($g = 0$) has $\chi = 2$, the torus ($g = 1$) has $\chi = 0$, and each additional handle lowers χ by 2. This is Eq. (53.8).

Assembling the expansion. Substituting $\chi = 2 - 2g$ into Eq. (53.7), every diagram of genus g scales as N^{2-2g} times a power of λ. Grouping the connected vacuum diagrams by genus and summing the λ-powers within each genus gives Eq. (53.9). The leading term, $g = 0$, is N^2; the first correction, $g = 1$, is N^0; and the expansion proceeds in steps of N^{-2}. □

Three features of Theorem 53.2 deserve to be drawn out at once. First, the expansion parameter for the topology is $1/N^2$, not $1/N$: orientable closed surfaces have even Euler-characteristic deficit, so consecutive terms differ by two powers of N. (Were the gauge group orthogonal or symplectic rather than unitary, non-orientable surfaces would also contribute and the expansion would

proceed in $1/N$; for $\mathrm{SU}(N)/\mathrm{U}(N)$ it is strictly $1/N^2$.) Second, the leading term scales as N^2, exactly the number of gluon degrees of freedom $\sim N^2$ — the free energy is extensive in the colored degrees of freedom, as it must be. Third, and most important, *the leading contribution comes from the diagrams that can be drawn on a sphere without any line crossing another*: the planar diagrams. We turn to them.

53.1.4 Planar dominance, and a worked count

A diagram is *planar* if its double-line graph can be drawn on the plane (equivalently the sphere) with no two ribbons crossing. By Eq. (53.8) planar diagrams have $g = 0$, $\chi = 2$, and weight N^2; any diagram that genuinely requires a crossing lives on a surface of higher genus and is suppressed. The cleanest way to feel the mechanism is to take a single diagram and watch the face count — and hence the power of N — collapse when a crossing is introduced.

Worked Example 53.3 (Planar versus crossed: a two-loop count, suppressed by exactly $1/N^2$)**.** Consider the lowest-order vacuum correction in the quartic matrix model Eq. (53.6): a single quartic vertex with its four legs Wick-contracted in pairs. This is a two-loop diagram ($V = 1$, $E = 2$, so $L = E - V + 1 = 2$), and it is the perfect laboratory because the only freedom is *how* the four legs are paired, with one pairing planar and one crossed.

Write the quartic interaction with its color indices displayed. The trace $\mathrm{tr}(\Phi^4)$ is a single cyclic chain,

$$\mathrm{tr}(\Phi^4) = \Phi^{i_1}{}_{i_2}\,\Phi^{i_2}{}_{i_3}\,\Phi^{i_3}{}_{i_4}\,\Phi^{i_4}{}_{i_1}, \tag{53.12}$$

so the four legs A, B, C, D carry index pairs (i_1, i_2), (i_2, i_3), (i_3, i_4), (i_4, i_1) respectively, with each shared index (i_2, i_3, i_4, i_1) already sewing consecutive legs together around the vertex. We contract the four legs into two propagators, each propagator enforcing the double-line rule Eq. (53.4), $\langle \Phi^a{}_b\, \Phi^c{}_d \rangle \propto \delta^a_d\, \delta^c_b$.

Pairing 1 (planar): A with B, C with D.

$$\langle \Phi^{i_1}{}_{i_2}\, \Phi^{i_2}{}_{i_3} \rangle \propto \delta^{i_1}_{i_3}\, \delta^{i_2}_{i_2} = N\, \delta^{i_1}_{i_3}, \tag{53.13}$$

$$\langle \Phi^{i_3}{}_{i_4}\, \Phi^{i_4}{}_{i_1} \rangle \propto \delta^{i_3}_{i_1}\, \delta^{i_4}_{i_4} = N\, \delta^{i_3}_{i_1}. \tag{53.14}$$

The two contractions produce two explicit factors of N from the self-closing index loops ($\delta^{i_2}_{i_2} = \delta^{i_4}_{i_4} = N$), and the leftover deltas multiply to $\delta^{i_1}_{i_3}\delta^{i_3}_{i_1} = \delta^{i_1}_{i_1} = N$, a third factor. Hence $F = 3$ closed index loops and a color factor N^3. (The other planar pairing, A with D and B with C, gives the same N^3 by the identical count.) Folding in the Feynman-rule prefactors $(N/\lambda)^V (\lambda/N)^E = (N/\lambda)(\lambda/N)^2 = \lambda/N$, the diagram scales as

$$\frac{\lambda}{N} \cdot N^3 = N^2\, \lambda, \tag{53.15}$$

in agreement with $N^{\chi}\lambda^{E-V}$ at $\chi = V - E + F = 1 - 2 + 3 = 2$ (genus 0).

Pairing 2 (crossed): A with C, B with D.

$$\langle \Phi^{i_1}{}_{i_2}\, \Phi^{i_3}{}_{i_4} \rangle \propto \delta^{i_1}_{i_4}\, \delta^{i_3}_{i_2}, \tag{53.16}$$

$$\langle \Phi^{i_2}{}_{i_3}\, \Phi^{i_4}{}_{i_1} \rangle \propto \delta^{i_2}_{i_1}\, \delta^{i_4}_{i_3}. \tag{53.17}$$

Now *no* self-closing δ^i_i appears at the contraction step; instead the four deltas chain into a single cycle. Summing over all four colors,

$$\sum_{i_1,i_2,i_3,i_4} \delta^{i_1}_{i_4}\, \delta^{i_4}_{i_3}\, \delta^{i_3}_{i_2}\, \delta^{i_2}_{i_1} = \sum_{i_1} \delta^{i_1}_{i_1} = N, \tag{53.18}$$

so there is just $F = 1$ closed index loop and a color factor N^1. With the same prefactor λ/N the crossed diagram scales as

$$\frac{\lambda}{N} \cdot N^1 = N^0\, \lambda, \tag{53.19}$$

matching $N^{\chi}\lambda^{E-V}$ at $\chi = 1 - 2 + 1 = 0$ (genus 1, the torus). Comparing Eqs. (53.15) and (53.19), the crossed diagram is suppressed relative to the planar one by exactly

$$\frac{N^0\lambda}{N^2\lambda} = \frac{1}{N^2}, \tag{53.20}$$

one factor of $1/N^2$, precisely the cost of the single handle that the crossing forces. The crossing is unavoidable for pairing 2: drawing the ribbons in the plane, the strip from A to C must pass over the strip from B to D, and that overpass is the handle of the torus. Figure 53.2 shows the two ribbon graphs with their faces, making the loss of two index loops visible.

This single example contains the entire mechanism. A crossing in the double-line graph cannot be undone in the plane; it can only be accommodated by routing one ribbon through a handle, and each handle removes two index loops, each loop being a factor N. The planar diagrams, which need no handle, dominate; the non-planar ones are organized into a tower suppressed by powers of $1/N^2$, one power per handle. The same arithmetic explains the earlier claim that the $\mathrm{SU}(N)$ trace subtraction in Eq. (53.5) is negligible: the $-\frac{1}{N}\delta^i_j\delta^k_l$ term pinches one double line into a structure with one fewer index loop, costing a relative $1/N^2$ exactly as a crossing does, so it never affects the leading planar result.

Physical Insight

Planarity is the large-N analogue of a saddle point. In an ordinary path integral the classical limit $\hbar \to 0$ is dominated by stationary configurations; here $1/N$ plays the role of $\hbar$, and the "configurations" are diagram topologies. At $N = \infty$ only the genus-zero (planar) sector survives, and it

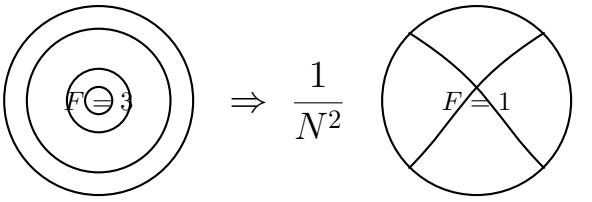


Figure 53.2: The single quartic vertex of Theorem 53.3, drawn in double-line notation. *Left:* the planar pairing closes three independent index loops (three faces), giving N^3 from the color sums and, with the λ/N prefactor, a value $N^2\lambda$; the ribbon graph tiles the sphere ($\chi = 2$). *Right:* the crossed pairing forces the two ribbons to overpass, which merges the loops into a single index line (one face), giving N^1 and a value $N^0\lambda$; the graph now tiles the torus ($\chi = 0$). The crossing costs exactly $1/N^2$ — the price of one handle.

is a self-contained, λ-dependent theory — still interacting, still in general strongly coupled, but topologically simple. Summing the planar diagrams is the central problem of large-N gauge theory; it has been solved in special cases (notably matrix-model integrals and two-dimensional Yang–Mills) and remains the goal in four dimensions. The $1/N$ corrections then dress this leading planar theory handle by handle.

53.1.5 The string in disguise: $g_s \sim 1/N$

The form of Eq. (53.9) is not merely reminiscent of, but structurally identical to, a quantity the reader will meet whenever closed strings are quantized. A closed string sweeps out a two-dimensional worldsheet as it propagates, and its perturbative amplitudes are computed by summing over all worldsheet topologies. The interaction — one string splitting into two, or two joining into one — adds a handle to the worldsheet, and each such interaction is weighted by a coupling constant g_s, the closed-string coupling. A worldsheet of genus g has $2g-2$ more handles-worth of interactions than the sphere, and carries a relative weight g_s^{2g-2}. Hence any closed-string observable expands as

$$\mathcal{A}_{\text{string}} \;=\; \sum_{g=0}^{\infty} g_s^{2g-2}\, \mathcal{F}_g, \tag{53.21}$$

a sum over worldsheet genus with the genus-g term weighted by g_s^{2g-2}.

Set Eq. (53.21) beside the gauge-theory result Eq. (53.9), $W = \sum_g N^{2-2g} f_g(\lambda) = \sum_g (N^{-1})^{2g-2} f_g(\lambda)$. The two expansions have the *same* dependence on genus:

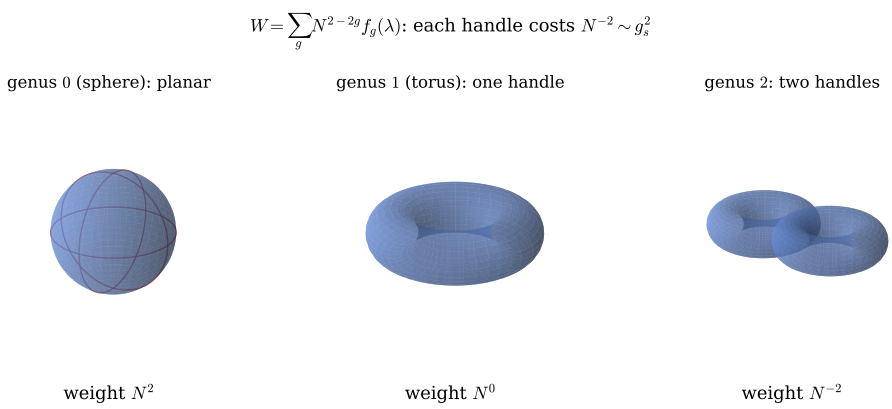


Figure 53.3: The genus expansion Eq. (53.9) drawn as a sum over closed orientable surfaces, the surfaces swept out by the ribbon graphs. Genus 0 (the sphere, planar diagrams) carries weight N^2; each additional handle multiplies the weight by N^{-2}, exactly as each closed-string loop multiplies a string amplitude by g_s^2 in Eq. (53.21). The matching of the two weightings fixes $g_s \sim 1/N$. The faint curves on the sphere indicate that the surface is *built from* the double-line Feynman graph — a triangulated worldsheet — rather than postulated separately. (The genus-two panel is schematic.)

term for term, the gauge theory's N^{2-2g} matches the string's g_s^{2g-2} under the identification

$$\boxed{g_s \sim \frac{1}{N}} \tag{53.22}$$

This is the punchline. The $1/N$ expansion of a SU(N) gauge theory is a closed-string loop expansion in disguise, with the string coupling set by the inverse number of colors. The discrete object that the gauge theory sums — the ribbon Feynman graph — *is* a triangulated worldsheet: the propagators and vertices of the gauge theory tile the very surface over which the string sums, with the 't Hooft coupling λ controlling the fineness and weighting of the triangulation (the "area"), and $1/N$ controlling its topology. Figure 53.3 displays the first three surfaces with their weights.

The structural consequences are immediate and are 't Hooft's heuristic. Weak string coupling, $g_s \ll 1$, is the same statement as large N. So a gauge theory with very many colors should be described by a *weakly coupled* string theory, whatever that string theory turns out to be — the regime in which string perturbation theory, the sum Eq. (53.21), is reliable. The 't Hooft coupling λ remains free and parametrizes the dynamics on the worldsheet: small λ is the familiar weakly-coupled gauge theory (a coarse, nearly degenerate triangulation), while large λ is the strongly-coupled gauge theory, where one expects the worldsheet description to become smooth — a continuous string moving in some target geometry.

A confining or conformal large-N gauge theory should therefore be dual to a weakly-coupled string theory; and because the closed-string spectrum always contains a massless spin-two state — a graviton — that dual string theory is a theory of gravity. What the bookkeeping does *not* tell us is which string theory, propagating in which target space and how many dimensions. That gap is exactly what the next section fills.

Historical Note

't Hooft introduced the $1/N$ expansion and double-line notation in 1974, motivated by the hope of an analytic handle on quark confinement: if the planar sector could be summed, the leading large-N theory might be solved and confinement understood. The planar sum in four dimensions has never been carried out, but the topological reorganization itself — diagrams sorted by the genus of an emergent surface, with $1/N$ as a closed-string coupling — proved to be the durable insight. For more than two decades it remained a suggestive analogy without a concrete dual; the surface was visible but the target space of the string was unknown. Maldacena's 1997 identification of that target space, for a specific maximally supersymmetric gauge theory, turned the analogy into the correspondence this chapter develops.

53.1.6 Factorization and the master field

The same Euler-characteristic counting that organized the vacuum energy also controls correlation functions of gauge-invariant operators, and it delivers a second structural fact: at large N the theory behaves *classically*, in a precise sense. The natural gauge-invariant observables built from a single matrix field are the *single-trace operators*, $\mathcal{O}_k(x) = \mathrm{tr}\,(\Phi(x)^k)$ (in Yang–Mills, gauge-invariant single traces of field strengths and covariant derivatives). Inserting such an operator into a diagram is, in ribbon language, like adding a vertex with k legs that carries one extra index loop; equivalently, it bounds a hole in the surface. A correlator of n single-trace operators thus corresponds to a surface with n boundaries, one per insertion, and the Euler characteristic of a genus-g surface with n boundaries is $\chi = 2 - 2g - n$. The leading (planar, $g = 0$) connected contribution therefore scales as

$$\langle \mathcal{O}_1 \cdots \mathcal{O}_n \rangle_{\mathrm{conn}} \;\sim\; N^{\chi}\big|_{g=0} = N^{\,2-n}. \tag{53.23}$$

We verify Eq. (53.23) on the cases $n = 1$ and $n = 2$ by direct color counting, which also exposes the factorization that follows.

Worked Example 53.4 (Large-N factorization of single-trace operators)**.** Take the quadratic single-trace operator $\mathcal{O}(x) = \mathrm{tr}\,(\Phi(x)^2) = \Phi^i{}_j(x)\,\Phi^j{}_i(x)$ in the free

($\lambda \to 0$) theory, where the propagator is $\langle \Phi^i{}_j(x)\, \Phi^k{}_l(y)\rangle = \frac{\lambda}{N}\, \delta^i_l\, \delta^k_j\, D(x-y)$ with D the ordinary scalar propagator; the interacting case only dresses the spacetime dependence and changes nothing in the color counting.

One-point function ($n = 1$). Contracting the two fields within the single trace,

$$\big\langle \operatorname{tr} \Phi^2(x)\big\rangle = \big\langle \Phi^i{}_j \Phi^j{}_i\big\rangle = \frac{\lambda}{N}\, \delta^i_i\, \delta^j_j\, D(0) = \frac{\lambda}{N}\, N^2\, D(0) = N\, \lambda\, D(0) \sim N, \quad (53.24)$$

confirming $N^{2-n} = N^1$ for $n = 1$.

Two-point function ($n = 2$), connected. For $\langle \operatorname{tr} \Phi^2(x)\ \operatorname{tr} \Phi^2(y)\rangle_{\text{conn}}$ the two fields at x must each contract with a field at y. One of the two such contractions gives

$$\begin{aligned} \big\langle \Phi^i{}_j(x) \Phi^j{}_i(x)\, \Phi^k{}_l(y) \Phi^l{}_k(y)\big\rangle_{\text{conn}} &= \Big(\tfrac{\lambda}{N}\Big)^2 D(x-y)^2\, (\delta^i_l \delta^k_j)\, (\delta^j_k \delta^l_i) \\ &= \Big(\tfrac{\lambda}{N}\Big)^2 D(x-y)^2\, (\delta^i_l \delta^l_i)\, (\delta^k_j \delta^j_k) \\ &= \Big(\tfrac{\lambda}{N}\Big)^2 N^2\, D(x-y)^2 = \lambda^2\, D(x-y)^2 \sim N^0, \end{aligned} \quad (53.25)$$

and the second contraction contributes at the same order. Thus the connected two-point function is $O(N^0)$, confirming $N^{2-n} = N^0$ for $n = 2$.

Factorization. The full (not connected) two-point function is the connected part plus the disconnected product of one-point functions:

$$\big\langle \mathcal{O}(x)\, \mathcal{O}(y)\big\rangle = \underbrace{\big\langle \mathcal{O}(x)\big\rangle \big\langle \mathcal{O}(y)\big\rangle}_{\sim N \cdot N = N^2} + \underbrace{\big\langle \mathcal{O}(x)\, \mathcal{O}(y)\big\rangle_{\text{conn}}}_{\sim N^0}. \quad (53.26)$$

The disconnected piece, $O(N^2)$, dominates the connected piece, $O(N^0)$, by exactly N^2. Hence the normalized fluctuation obeys

$$\frac{\langle \mathcal{O}\, \mathcal{O}\rangle - \langle \mathcal{O}\rangle \langle \mathcal{O}\rangle}{\langle \mathcal{O}\rangle \langle \mathcal{O}\rangle} = O\Big(\frac{1}{N^2}\Big), \quad (53.27)$$

so correlation functions of gauge-invariant operators *factorize* into products of expectation values as $N \to \infty$, with relative corrections of order $1/N^2$.

Factorization Eq. (53.26) is the hallmark of a *classical* theory. In quantum mechanics $\langle \mathcal{O}^2\rangle = \langle \mathcal{O}\rangle^2$ holds precisely when the operator has no fluctuations — when it takes a definite value in the state. Equation Eq. (53.27) says that, at $N = \infty$, every gauge-invariant single-trace operator has vanishing relative variance: it behaves like a classical c-number, not a fluctuating quantum observable. The path integral is then dominated by a single configuration, the same way a classical limit is dominated by a saddle, and that dominant configuration is called the

master field: a single (in general highly nontrivial) field configuration on which all gauge-invariant operators take their factorized expectation values. The $1/N$ corrections are the quantum fluctuations about the master field, organized — as Theorem 53.2 guarantees — by surface topology.

Physical Insight

The master-field picture is the gauge-theory shadow of classical gravity. If the large-N gauge theory is a weakly-coupled string theory, then its $N = \infty$ limit ($g_s \to 0$) is *classical* string theory, and at strong 't Hooft coupling — where the worldsheet is smooth and only its lowest, gravitational, modes survive — classical string theory degenerates to classical (super)gravity. The statement that gauge-invariant operators do not fluctuate at $N = \infty$ is the field-theory counterpart of the statement that the dual geometry is a definite classical spacetime, not a fluctuating quantum one. Large N is what makes the gravitational dual *classical*; large 't Hooft coupling λ is what makes it well described by *(super)gravity* rather than by the full string. Both limits will reappear, sharpened into a dictionary, once the dual geometry is identified.

We have arrived at a definite structural conclusion without having built a single string. The perturbative expansion of a large-N SU(N) gauge theory is, term by term in $1/N$, a sum over the topologies of a two-dimensional surface, weighted exactly as the closed-string loop expansion is weighted, with $g_s \sim 1/N$; its gauge-invariant observables factorize, signaling that the $N = \infty$ theory is classical and dominated by a master field. The gauge theory therefore looks, from the outside, like a closed-string theory — and hence a theory of gravity — but the analysis leaves the worldsheet dynamics and, crucially, the *target geometry* in which that string propagates completely unspecified. The next section supplies the missing geometry: it takes a particular gauge theory whose string dual can be argued for concretely, follows a decoupling limit in which the dual spacetime can actually be read off, and so converts the disguise exposed here into the explicit conjecture that organizes the rest of the chapter.

53.2 The Maldacena Argument, Scoped Honestly

The previous section delivered a structural verdict but no address. The 't Hooft expansion (Section 53.1) reorganized the perturbation series of an SU(N) gauge theory into a sum over the genus of the diagram's fattened ribbon surface, with each handle weighted by $1/N^2$ and the loop coupling carried by the fixed 't Hooft combination $\lambda = g^2 N$. That is, term by term, the bookkeeping of a closed-string loop expansion: a sum over worldsheet topologies with string coupling $g_s \sim 1/N$. The conclusion was that some gauge theories *are* string theories in disguise. What

that argument could not supply was the disguise itself — *which* gauge theory, and a string theory propagating in *which* spacetime. This section supplies both, by following Maldacena's original route: place a stack of branes in flat space, describe the same physical object in two complementary low-energy languages, and take a limit in which each language sharpens into a self-contained theory. The two self-contained theories are then descriptions of one system, hence equal. One language is a definite gauge theory; the other is gravity in a definite curved geometry.

Because this book does not develop string theory from the worldsheet, a handful of inputs in what follows — that a stack of D3-branes carries a particular low-energy gauge theory on its worldvolume, that the same stack sources a particular supergravity solution, and the numerical map between brane data and geometric data — are *quoted* from string theory. Everything that follows from those inputs by field-theoretic and differential-geometric methods — above all the claim that the geometry close to the branes is exactly $\mathrm{AdS}_5 \times S^5$, and the claim that the resulting gauge theory is an exact conformal field theory carrying the right symmetry group — is derived here in full. We close the section with a summary table separating what is derived, what is imported from string theory, and what is conjectured.

53.2.1 Two descriptions of a stack of branes

The object of study is a stack of N coincident *D3-branes* in the ten-dimensional spacetime of type IIB string theory. A Dp-brane is a $(p+1)$-dimensional hypersurface — here $p = 3$, so a 3+1-dimensional worldvolume sitting inside 9+1 dimensions — on which open strings can end, and which itself gravitates and carries charge. We will not need the worldsheet definition; we need only two facts about what the stack *looks like* at energies far below the string scale $1/\sqrt{\alpha'}$, where α' (of dimension length^2) sets the string length $\ell_s = \sqrt{\alpha'}$. The two facts are two ways of cataloguing the low-energy excitations, and they are not obviously the same.

Description A: open strings, or the worldvolume gauge theory. The first description is from the point of view of the open strings whose endpoints are pinned to the branes. The massless modes of an open string stretched between brane a and brane b (with $a, b \in \{1, \dots, N\}$) furnish, in the low-energy limit, a U(1) gauge field for $a = b$ and charged matter for $a \neq b$; when the N branes coincide the N^2 string sectors assemble into the adjoint of U(N), and the overall U(1) (the center-of-mass mode that decouples from the dynamics) is discarded, leaving an SU(N) gauge theory. For the specific case of D3-branes in type IIB, the precise low-energy worldvolume theory is the maximally supersymmetric Yang–Mills theory in four dimensions: $\mathcal{N} = 4$ SU(N) super Yang–Mills (SYM), with field content exactly the massless $\mathcal{N} = 4$ vector multiplet classified in

Section 44.4.2 — one gauge field A_μ, four Weyl fermions, and six real scalars, all in the adjoint of SU(N). Its single dimensionless coupling g is set by the string coupling,

$$g^2 \;=\; 4\pi g_s, \tag{53.28}$$

a relation we will need below and quote here; the α'-corrections to this worldvolume action are an expansion in α' times momenta, so at energies E with $\alpha' E^2 \to 0$ only the renormalizable $\mathcal{N}=4$ SYM Lagrangian survives. In this description, then, the low-energy physics on the branes is: *$\mathcal{N}=4$ SYM, plus free type IIB supergravity in the flat ten-dimensional bulk away from the branes, plus interactions between the two that are controlled by Newton's constant and switch off in the low-energy limit.*

Description B: closed strings, or the supergravity geometry. The second description is from the point of view of the closed strings, that is, of gravity. The brane stack is a heavy, charged source; it curves the spacetime around it and sources the Ramond–Ramond four-form potential that D3-branes are electrically charged under. The corresponding solution of type IIB supergravity, the extremal black 3-brane, is known in closed form. Writing the ten coordinates as the four worldvolume directions $x^\mu = (t, x^1, x^2, x^3)$ along the branes and six transverse directions with radial coordinate $r = \sqrt{\sum_{i=1}^{6} (x_\perp^i)^2}$ and unit five-sphere $\mathrm{d}\Omega_5^2$, the metric is

$$\mathrm{d}s^2 \;=\; H(r)^{-1/2}\,\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu \;-\; H(r)^{1/2}(\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega_5^2), \qquad H(r) \;=\; 1 + \frac{L^4}{r^4}, \tag{53.29}$$

written in the mostly-minus signature of this book (Chapter 3): the worldvolume metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ carries the single timelike direction, and the six transverse directions are spacelike, hence the overall minus on the transverse block. The dilaton is constant and the self-dual five-form flux threads the sphere. The single length scale L in the harmonic function H is fixed by the number of branes and the string data through

$$L^4 \;=\; 4\pi g_s N\,\alpha'^2. \tag{53.30}$$

Two features of Eq. (53.29) are worth naming before we use it. First, H is harmonic in the six transverse dimensions away from the source: $\nabla_6^2\, r^{-4} \propto \delta^{(6)}(x_\perp)$, the six-dimensional analogue of the Coulomb potential, so $H = 1 + L^4/r^4$ is the brane's gravitational "Coulomb field," with the 1 enforcing asymptotic flatness as $r \to \infty$ and the L^4/r^4 the deformation built up by the source. Second, the prefactor $H^{-1/2}$ multiplying $\mathrm{d}t^2$ is a gravitational redshift factor: a local excitation of proper energy E_{loc} sitting at radius r is seen by an observer at infinity to have energy

$$E_\infty \;=\; H(r)^{-1/4}\,E_{\mathrm{loc}}, \tag{53.31}$$

since the conserved energy conjugate to the asymptotic time t is the locally-measured energy times $\sqrt{g_{tt}} = H^{-1/4}$. Deep in the throat, $r \ll L$, the factor $H^{-1/4} \approx (r/L) \to 0$: an excitation of fixed proper energy is redshifted to arbitrarily low energy as measured from infinity. We will use this in the decoupling argument.

In this second description the low-energy physics is: *closed-string excitations propagating in the geometry Eq. (53.29).* At low energy as seen from infinity, two kinds of excitation survive, and they are spatially separated. Far out ($r \to \infty$) the metric is flat and the surviving modes are the long-wavelength quanta of free ten-dimensional supergravity. Deep in the throat ($r \ll L$) any excitation, however energetic locally, is redshifted by Eq. (53.31) to low energy at infinity, so the *entire* dynamics of the throat region survives the low-energy limit. The two regions communicate only through the intermediate region $r \sim L$, and that communication is weak at low energy, as we now make quantitative.

The two descriptions, A and B, are two complete accounts of the same brane stack. Each contains a piece of free bulk supergravity in an asymptotically flat region, decoupling at low energy. If we can argue that what remains after stripping that common piece is, in each description, a self-contained theory, then those two remainders are descriptions of the same physics and must be identified. In Description A the remainder is $\mathcal{N} = 4$ SYM. In Description B the remainder is whatever closed-string theory lives in the throat. The content of the correspondence is the equality of these two remainders; the content of the *derivation* is (i) that the throat geometry is $\mathrm{AdS}_5 \times S^5$, which we prove, and (ii) that the two decouplings are clean, which we argue.

53.2.2 The near-horizon limit: the throat is $\mathrm{AdS}_5 \times S^5$

The single piece of this section that is pure, mathematics — no string input beyond the quoted metric Eq. (53.29) — is the identification of the near-horizon geometry. The throat region is $r \ll L$, where the harmonic function loses its asymptotic 1:

$$r \ll L: \qquad H(r) \;=\; 1 + \frac{L^4}{r^4} \;\longrightarrow\; \frac{L^4}{r^4}. \tag{53.32}$$

We substitute this into Eq. (53.29) and reduce the result by metric algebra. The claim is that what emerges is the product of five-dimensional anti-de Sitter space of radius L with a round five-sphere of radius L. We carry out the reduction in full.

Worked Example 53.5 (Near-horizon reduction of the black 3-brane)**.** Insert $H \to L^4/r^4$ into the metric Eq. (53.29). The two prefactors become

$$H^{-1/2} \;\to\; \Big(\frac{L^4}{r^4}\Big)^{-1/2} = \frac{r^2}{L^2}, \qquad H^{1/2} \;\to\; \Big(\frac{L^4}{r^4}\Big)^{1/2} = \frac{L^2}{r^2}, \tag{53.33}$$

so the line element becomes

$$\mathrm{d}s^2 \;=\; \frac{r^2}{L^2}\,\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu \;-\; \frac{L^2}{r^2}\,\mathrm{d}r^2 \;-\; \frac{L^2}{r^2}\,r^2\,\mathrm{d}\Omega_5^2. \tag{53.34}$$

The last term simplifies immediately: $\frac{L^2}{r^2}\,r^2 = L^2$, so

$$-\,\frac{L^2}{r^2}\,r^2\,\mathrm{d}\Omega_5^2 \;=\; -\,L^2\,\mathrm{d}\Omega_5^2, \tag{53.35}$$

which is a round five-sphere of radius L (the overall minus is the mostly-minus signature's marking of these five directions as spacelike, and crucially the radius is now r-independent: the sphere does not shrink down the throat). The remaining five-dimensional block,

$$\mathrm{d}s^2_{(5)} \;=\; \frac{r^2}{L^2}\,\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu \;-\; \frac{L^2}{r^2}\,\mathrm{d}r^2, \tag{53.36}$$

is the one we must recognize. Change the radial variable to

$$z \;=\; \frac{L^2}{r} \qquad\Longleftrightarrow\qquad r = \frac{L^2}{z}, \qquad \mathrm{d}r \;=\; -\,\frac{L^2}{z^2}\,\mathrm{d}z, \qquad \mathrm{d}r^2 \;=\; \frac{L^4}{z^4}\,\mathrm{d}z^2. \tag{53.37}$$

The two coefficients transform as

$$\frac{r^2}{L^2} = \frac{1}{L^2}\,\frac{L^4}{z^2} = \frac{L^2}{z^2}, \qquad \frac{L^2}{r^2}\,\mathrm{d}r^2 = \frac{L^2}{(L^2/z)^2}\cdot\frac{L^4}{z^4}\,\mathrm{d}z^2 = \frac{z^2}{L^2}\cdot\frac{L^4}{z^4}\,\mathrm{d}z^2 = \frac{L^2}{z^2}\,\mathrm{d}z^2, \tag{53.38}$$

so both the worldvolume term and the radial term acquire the *same* overall conformal factor L^2/z^2, and Eq. (53.36) collapses to

$$\mathrm{d}s^2_{(5)} \;=\; \frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu - \mathrm{d}z^2\Big). \tag{53.39}$$

This is exactly the Poincaré-patch form of five-dimensional anti-de Sitter space AdS_5 of radius L — a single conformal factor L^2/z^2 multiplying a flat five-dimensional Minkowski metric in coordinates (x^μ, z), with the conformal boundary at $z \to 0$ (i.e. $r \to \infty$). Assembling Eq. (53.35) and Eq. (53.39), the full near-horizon geometry is

$$\boxed{\mathrm{d}s^2 \;=\; \frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu - \mathrm{d}z^2\Big) \;-\; L^2\,\mathrm{d}\Omega_5^2 \;=\; \mathrm{AdS}_5 \times S^5,} \tag{53.40}$$

with the two factors sharing a single radius L. No string input entered the reduction; only the quoted metric Eq. (53.29) and algebra.

The geometric picture this produces is the one in Fig. 53.4: the brane stack drills an infinite throat into flat space, the throat has the cross-sectional

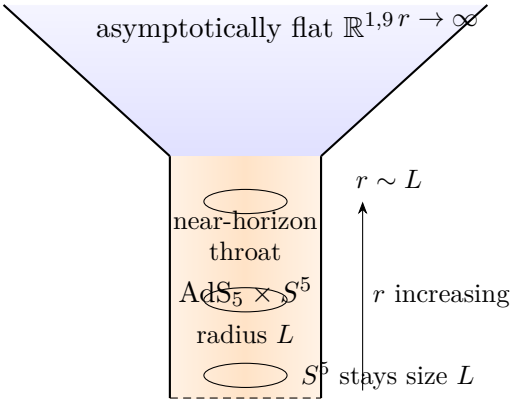


Figure 53.4: The geometry sourced by a stack of N D3-branes, Eq. (53.29). Far away ($r \to \infty$) the harmonic function $H \to 1$ and spacetime is flat. Near the branes ($r \ll L$) the throat opens out; the metric algebra of Theorem 53.5 shows it is exactly $AdS_5 \times S^5$ with both factors of radius L, the five-sphere stabilizing rather than pinching off. The two descriptions of Section 53.2.1 decouple from the common asymptotic region: $\mathcal{N} = 4$ SYM (Description A) is identified with closed strings in the throat (Description B).

geometry of a five-sphere whose radius stabilizes at L rather than pinching off, and the throat's longitudinal-plus-radial geometry is AdS_5. That the sphere does *not* pinch is the crucial difference between the extremal 3-brane and, say, the Schwarzschild solution: it is what makes the near-horizon region an honest spacetime of its own with an infinite proper depth, capable of hosting a complete decoupled theory.

53.2.3 The decoupling limit

We now isolate, in each description, a self-contained low-energy theory by taking the limit Maldacena identified. The limit is

$$\alpha' \to 0, \qquad \text{with} \quad g_s \text{ fixed}, \quad N \text{ fixed}, \quad \text{and} \quad \frac{r}{\alpha'} \equiv U \text{ fixed}, \tag{53.41}$$

so that $\lambda = g^2 N = 4\pi g_s N$ is held fixed (by Eq. (53.28)) while the string length $\ell_s = \sqrt{\alpha'} \to 0$ sends the string scale to infinity, decoupling all massive string modes. The fixed combination $U = r/\alpha'$ has dimensions of energy and measures, holding fixed, the proper energy of a stretched string in the throat; keeping it fixed is the statement that we track finite-energy excitations as the cutoff is removed. We examine each description in turn.

Description A in the limit. The worldvolume side contains $\mathcal{N} = 4$ SYM, free bulk supergravity, and their mutual couplings. The interaction between the

brane modes and the bulk gravitons is controlled by the ten-dimensional Newton constant, $16\pi G_{10} = (2\pi)^7 g_s^2 \alpha'^4$, which carries four powers of α'; as $\alpha' \to 0$ the dimensionful gravitational coupling vanishes and the bulk gravitons *decouple* from the worldvolume fields. What survives, cleanly, is free supergravity in the asymptotically flat bulk plus an isolated, interacting $\mathcal{N} = 4$ SYM with coupling $g^2 = 4\pi g_s$ on the branes. The SYM coupling is dimensionless and untouched by $\alpha' \to 0$, so the gauge theory is not trivialized; only its gravitational umbilical cord to the bulk is cut.

Description B in the limit. The geometry side contains closed strings propagating in Eq. (53.29). At low energy as seen from infinity, the surviving excitations split, as already noted, into two decoupling sets: long-wavelength gravitons in the flat asymptotic region, and the full throat dynamics, which survives because the redshift Eq. (53.31) maps the throat's entire finite-proper-energy spectrum down to arbitrarily low energy at infinity. The cross-coupling between throat and asymptotic region is suppressed at low energy: the absorption cross-section of a throat by an asymptotic graviton scales as a positive power of its frequency and vanishes in the limit, because the throat subtends a vanishing solid angle at the wavelengths that survive. So Description B also resolves into two decoupled pieces: free supergravity in the asymptotically flat bulk, plus the complete theory of closed strings in the near-horizon region. By Theorem 53.5, that near-horizon region is $\mathrm{AdS}_5 \times S^5$, and the limit Eq. (53.41) keeps its curvature radius L and its full string dynamics — in the throat the natural coordinate is $U = r/\alpha'$, and the throat metric written in U retains $L^4/\alpha'^2 = 4\pi g_s N = \lambda$ finite, so the geometry survives intact with finite curvature in string units.

Matching the remainders. Both descriptions are descriptions of one and the same brane system, and in the limit Eq. (53.41) both factor into

$$(\text{free supergravity in flat } \mathbb{R}^{1,9}) \ \oplus \ (\text{a nontrivial interacting theory}),$$

with the first summand manifestly identical on the two sides (it is the same asymptotic bulk in the same flat space). Subtracting the common free-supergravity sector, the nontrivial remainders must be equal. The remainder of Description A is $\mathcal{N} = 4$ SU(N) SYM. The remainder of Description B is type IIB string theory on $\mathrm{AdS}_5 \times S^5$. Therefore — this is the argument's payload —

$$\mathcal{N} = 4 \,\mathrm{SU}(N) \text{ SYM in 3+1 dimensions} \quad \equiv \quad \text{type IIB string theory on } \mathrm{AdS}_5 \times S^5. \tag{53.42}$$

This is the AdS/CFT correspondence in its original incarnation. We pause to underline what kind of statement it is. The decoupling of the two sectors on each side, and the metric reduction to $\mathrm{AdS}_5 \times S^5$, are controlled steps. The identification Eq. (53.42) of the two *remainders* relies on the premise that the

two low-energy descriptions are descriptions of the same object — a premise about string theory, not a theorem we can prove with the tools of this book. It is therefore a *conjecture*, stated precisely below, with overwhelming evidence but no proof.

53.2.4 The conjecture and the parameter dictionary

Conjecture 53.6 (Maldacena / AdS/CFT, the $\mathrm{AdS}_5 \times S^5$ case)**.** *Type IIB string theory on the background* $\mathrm{AdS}_5 \times S^5$*, with both factors of radius* L *and with* N *units of self-dual five-form flux through the* S^5*, is dynamically equivalent to* $\mathcal{N} = 4$ *supersymmetric Yang–Mills theory with gauge group* $\mathrm{SU}(N)$ *in four-dimensional Minkowski space. The two sets of parameters are related by*

$$g_s \;=\; \frac{g^2}{4\pi}, \qquad \frac{L^4}{\alpha'^2} \;=\; 4\pi g_s N \;=\; g^2 N \;=\; \lambda. \tag{53.43}$$

The equivalence is a duality of the complete quantum theories: all gauge-invariant observables of one map to observables of the other.

The second relation in Eq. (53.43) is the substantive entry. It follows by dividing the supergravity result Eq. (53.30) by α'^2 and using Eq. (53.28); it states that the curvature radius of the geometry, measured in string-length units, is the fourth root of the 't Hooft coupling:

$$\frac{L}{\ell_s} \;=\; \lambda^{1/4}. \tag{53.44}$$

This single line encodes the entire "strong/weak" character of the correspondence and is worth reading slowly.

Caution

The $O(1)$ numerical coefficients in Eq. (53.28), Eq. (53.30) and hence the "4π" in Eq. (53.43) are convention-dependent: different texts normalize the gauge coupling, the string coupling, and the brane tension differently, and one finds $g^2 = 2\pi g_s$ or $L^4 = 2\lambda\alpha'^2$ in places. The convention-independent, load-bearing content is the *parametric* statement $L^4/\alpha'^2 \propto \lambda$ and $g_s \propto g^2$. We fix the normalization so that $L^4/\alpha'^2 = \lambda$ exactly, which is the cleanest bookkeeping and the one used throughout this Part.

The regimes, and why the duality is useful. Two independent dimensionless numbers control each side: on the gauge side, g^2 (or λ) and N; on the string side, g_s and L/ℓ_s. The dictionary Eq. (53.43) ties them together as $g_s = \lambda/(4\pi N)$ and $L/\ell_s = \lambda^{1/4}$. Now read off the limits.

- *Classical (tree-level) supergravity* is reliable when the geometry is weakly curved in string units *and* string loops are suppressed — that is, $L/\ell_s \gg 1$ and $g_s \ll 1$. By Eq. (53.43) this is

$$\lambda \gg 1 \qquad \text{and} \qquad N \to \infty \text{ at fixed } \lambda \text{ (so } g_s = \lambda/4\pi N \to 0). \tag{53.45}$$

  Large N at fixed λ kills the string-loop expansion (the genus expansion of Section 53.1, whose coupling is $g_s \sim 1/N$), and large λ kills the α' expansion (worldsheet sigma-model corrections, suppressed by $\ell_s^2/L^2 = \lambda^{-1/2}$). In this corner the right-hand side of Eq. (53.42) reduces to *classical* type IIB supergravity on $\mathrm{AdS}_5 \times S^5$ — ordinary differential geometry and field equations.

- *Perturbative gauge theory* is reliable in the opposite corner, $\lambda \ll 1$, where the SYM loop expansion converges and the dual geometry is sub-string-scale and wildly curved, so the gravity description is useless.

These two corners do not overlap. The correspondence is therefore a *strong/weak duality*: it computes the strongly coupled, large-N limit of the gauge theory — exactly the regime that asymptotic freedom (Chapter 27) left analytically inaccessible and the lattice could define but not solve in closed form — by a *classical* computation in one higher dimension. This is the precise sense, promised in the chapter introduction, in which holography trades a hard quantum problem in d dimensions for an easy classical one in $d+1$. The same feature is what makes it hard to prove: the two sides are never simultaneously weakly coupled, so no perturbative matching can establish the equality directly.

We summarize the logical status of each step in Table 53.1, separating the derived from the conjectured.

53.2.5 Why $\mathcal{N} = 4$ SYM is exactly conformal, and the symmetry match

The conjecture asserts that the boundary side of Eq. (53.42) is $\mathcal{N} = 4$ SYM. For the rest of this chapter — which builds the rigorous symmetry backbone of the correspondence and the field–operator dictionary — it matters that the boundary theory is a genuine conformal field theory, so that the machinery of Part XIV applies to it. We now prove that it is, using only results this book has established about supersymmetry, and we exhibit the symmetry group that the next sections will match against the geometry.

Theorem 53.7 ($\mathcal{N} = 4$ SYM has vanishing beta function)**.** *The $\mathcal{N} = 4$ supersymmetric Yang–Mills theory with gauge group $\mathrm{SU}(N)$ has $\beta(g) = 0$ identically, to all orders in perturbation theory. Its single coupling g is an exactly marginal parameter; the theory is scale-invariant, and in fact conformally invariant, for every value of g and N.*

Step	Status	Basis
Worldvolume theory of N D3-branes is $\mathcal{N}=4$ SU(N) SYM, $g^2=4\pi g_s$	**string input**	quoted from string theory
Extremal black 3-brane metric Eq. (53.29), $L^4=4\pi g_s N\alpha'^2$	**string input**	quoted IIB supergravity solution
Near-horizon geometry $=\mathrm{AdS}_5\times S^5$, radius L	**derived**	Theorem 53.5, pure metric algebra
Decoupling of bulk gravitons from each remainder	**argued**	$G_{10}\to 0$; vanishing throat cross-section
Equality of the two remainders, Eq. (53.42)	**conjectured**	Theorem 53.6; unproved conjecture
$\mathcal{N}=4$ SYM is an exact CFT, group PSU$(2,2\|4)$	**derived**	Theorem 53.8
Bosonic symmetries match $\mathrm{AdS}_5\times S^5$ isometries	**derived**	Section 53.2.5

Table 53.1: Theorem-versus-conjecture summary for the Maldacena argument. The geometric reduction and the symmetry analysis are proved in this book; the brane inputs are quoted from string theory and flagged; and the identification of the two low-energy remainders — the correspondence itself — is a conjecture.

Proof. We use the realization of $\mathcal{N}=4$ in $\mathcal{N}=1$ language: the $\mathcal{N}=4$ vector multiplet of Section 44.4.2 decomposes into one $\mathcal{N}=1$ vector multiplet (the gauge field and one adjoint gaugino) together with three adjoint $\mathcal{N}=1$ chiral multiplets Φ_1,Φ_2,Φ_3, the latter packaging the remaining three gauginos and the six real scalars (three complex). The interactions are fixed by $\mathcal{N}=4$ to be the canonical gauge kinetic terms plus the superpotential

$$W \;=\; \frac{g}{3!}\,\sqrt{2}\,\epsilon^{ijk}\;\mathrm{tr}\big(\Phi_i[\Phi_j,\Phi_k]\big), \tag{53.46}$$

the unique gauge-invariant cubic in three adjoints, with coefficient tied to the gauge coupling by the extended supersymmetry.

The gauge coupling renormalization is governed by the exact NSVZ beta function established for general $\mathcal{N}=1$ gauge theories in Theorem 45.21. For an SU(N) theory ($C_A=N$) with chiral matter in representations r_i, it reads

$$\beta(g) \;=\; -\,\frac{g^3}{16\pi^2}\,\frac{3C_A-\sum_i T(r_i)(1-\gamma_i)}{1-C_A\,g^2/8\pi^2}, \tag{53.47}$$

where $T(r_i)$ is the Dynkin index (Section 44.2.7; Appendix B) and γ_i is the anomalous dimension of the chiral superfield Φ_i. For $\mathcal{N}=4$ the matter is three

chiral superfields in the adjoint, so $T(r_i) = T(\text{adj}) = C_A$ for each, and the numerator becomes

$$3C_A - \sum_{i=1}^{3} C_A(1-\gamma_i) \;=\; 3C_A - 3C_A + C_A \sum_{i=1}^{3} \gamma_i \;=\; C_A\,(\gamma_1+\gamma_2+\gamma_3). \tag{53.48}$$

The constant part has cancelled exactly: the one-loop coefficient $b_0 = 3C_A - \sum_i T(r_i) = 3C_A - 3C_A = 0$ vanishes on its own — $\mathcal{N}=4$ SYM is one-loop finite for any gauge group — and the entire beta function is now proportional to the sum of anomalous dimensions of the three adjoint chiral superfields.

It remains to show $\gamma_1 = \gamma_2 = \gamma_3 = 0$ to all orders. The three chiral superfields Φ_i together with the gaugino sit, under the $\mathrm{SU}(4)_R$ R-symmetry of the $\mathcal{N}=4$ algebra, in a single irreducible multiplet: the four fermions form the $\mathbf{4}$ of $\mathrm{SU}(4)_R$ and the six scalars the $\mathbf{6}$. The wavefunction renormalization of fields related by an unbroken global symmetry must be common, so the three Φ_i share a single anomalous dimension, $\gamma_1 = \gamma_2 = \gamma_3 \equiv \gamma$. Moreover, the same $\mathrm{SU}(4)_R$ rotates the chiral-multiplet fermions into the gaugino, forcing the chiral-multiplet wavefunction renormalization to coincide with that of the vector multiplet; but the gauge kinetic term's normalization is controlled holomorphically and is protected by the nonrenormalization theorem (Theorem 45.20), which forbids the holomorphic gauge coupling from receiving corrections beyond one loop. The one-loop contribution to γ in $\mathcal{N}=4$ vanishes by the same $b_0 = 0$ cancellation visible in Eq. (53.48), and the nonrenormalization plus $\mathrm{SU}(4)_R$ together forbid any higher-order contribution that is not already fixed to zero. Hence $\gamma = 0$ to all orders, the numerator Eq. (53.48) vanishes identically, and Eq. (53.47) gives $\beta(g) = 0$ for all g.

A vanishing beta function makes g exactly marginal and the theory scale-invariant. Scale invariance of a unitary, local theory with a conserved, improvable stress tensor (which $\mathcal{N}=4$ SYM has) is promoted to full conformal invariance by the standard argument of Section 47.1.6: the virial current is itself a total derivative, so the dilatation current improves to a traceless stress tensor and the special conformal transformations are symmetries. Therefore $\mathcal{N}=4$ SYM is an exact conformal field theory for every g, N. □

The vanishing of the beta function is special to maximal supersymmetry. It is worth contrasting with the running theories that dominate this book: QED and QCD have nonzero β (Chapters 24 and 27), so their couplings flow and they possess an intrinsic scale (Λ_{QCD}). The coupling-unification analysis of Chapter 35 is precisely a story about *running* couplings meeting at a scale. $\mathcal{N}=4$ SYM stands apart: it has no running, no scale, no S-matrix in the usual sense (massless everything, no asymptotic states), and is instead organized entirely by conformal data — operator dimensions and OPE coefficients — exactly the objects Part XIV taught us to handle. This is why $\mathcal{N}=4$ SYM, not QCD,

is the *canonical* holographic example: it sits exactly at a fixed point, so its dual geometry $\mathrm{AdS}_5 \times S^5$ is rigid (a fixed background, not a flowing one), and the cleanest possible correspondence results.

Theorem 53.8 (Superconformal symmetry of $\mathcal{N} = 4$ SYM and the bosonic pre-match)**.** *The symmetry algebra of $\mathcal{N} = 4$ $\mathrm{SU}(N)$ SYM is the $\mathcal{N} = 4$ superconformal algebra $\mathfrak{psu}(2,2|4)$, the unique simple superalgebra whose even (bosonic) part is*

$$\mathfrak{so}(2,4) \;\oplus\; \mathfrak{so}(6) \;\cong\; \mathfrak{su}(2,2) \;\oplus\; \mathfrak{su}(4), \tag{53.49}$$

the first summand the four-dimensional conformal algebra of Eq. (47.53) and the second the $\mathrm{SU}(4)_R \cong \mathrm{SO}(6)$ R-symmetry. Its bosonic symmetry group $\mathrm{SO}(2,4) \times \mathrm{SO}(6)$ coincides with the isometry group of $\mathrm{AdS}_5 \times S^5$.

Proof. Two facts established earlier combine. First, by Theorem 53.7 the theory is conformal, so its spacetime symmetry is not merely Poincaré but the full conformal group of four-dimensional Minkowski space. The conformal algebra in d dimensions, derived in Theorem 47.9 and recorded in Eq. (47.53), is $\mathfrak{so}(d,2)$ in Lorentzian signature; for $d = 4$ this is $\mathfrak{so}(2,4) \cong \mathfrak{su}(2,2)$ (the local isomorphism between the 4+2-dimensional rotation algebra and the pseudo-unitary algebra on $\mathbb{C}^4$ with split signature). Second, the $\mathcal{N} = 4$ supersymmetry algebra (the maximal extended algebra of Theorem 44.6 and Section 44.4.2) carries an $\mathrm{SU}(4)_R$ R-symmetry rotating its four supercharges, generalizing the $\mathrm{U}(1)_R$ outer automorphism of Section 44.2.7; and $\mathrm{SU}(4) \cong \mathrm{SO}(6)$ as the double cover identifies the R-symmetry with rotations of a six-dimensional space.

When supersymmetry and conformal symmetry coexist, their anticommutators do not close on the two bosonic factors alone: the anticommutator of an ordinary supercharge Q with a special conformal generator K produces a new fermionic generator S, the *superconformal* charge, and closure of the full set $\{P, K, M, D, R, Q, S\}$ requires assembling them into a single simple Lie superalgebra. The unique simple superalgebra with even part Eq. (53.49) and the requisite sixteen-plus-sixteen odd generators (the Q's and S's, each four Weyl supercharges valued in the $\mathbf{4}$ or $\bar{\mathbf{4}}$ of $\mathrm{SU}(4)_R$) is $\mathfrak{su}(2,2|4)$; for $\mathcal{N} = 4$ specifically the would-be U(1) R-charge that extends SU(4) to U(4) acts trivially on all gauge-invariant operators and decouples, leaving the projective/special form $\mathfrak{psu}(2,2|4)$. This is the symmetry algebra of $\mathcal{N} = 4$ SYM.

Finally, the matching to geometry. The isometries of AdS_5 are, by construction, the symmetries of a five-dimensional space of constant negative curvature embedded as a hyperboloid in $\mathbb{R}^{2,4}$; that embedding makes the isometry group $\mathrm{SO}(2,4)$ manifest — the proof is carried out in detail in Theorem 53.11 of the next section. The isometries of the round S^5 are the rotations SO(6). Hence the isometry group of $\mathrm{AdS}_5 \times S^5$ is $\mathrm{SO}(2,4) \times \mathrm{SO}(6)$, term-for-term the bosonic symmetry group Eq. (53.49) of the boundary theory. (The full superalgebra matches too: type IIB on $\mathrm{AdS}_5 \times S^5$ preserves thirty-two supercharges, the

sixteen-plus-sixteen of $\mathfrak{psu}(2,2|4)$, but the fermionic match is a string input we only flag.) □

This symmetry agreement is not a proof of the correspondence — two theories can share a symmetry group without being equal — but it is the first hard, check, and it is a stringent one: the conjecture would have died instantly had the bosonic groups failed to match. That they match *exactly*, with the conformal group of the boundary CFT realized as the isometry group of the bulk's anti-de Sitter factor and the R-symmetry as the isometry of the sphere, is the structural reason the correspondence is geometric. It also tells us what to build next.

Physical Insight

The two halves of $\mathrm{PSU}(2,2|4)$ live in two different places on the gravity side. The conformal group $\mathrm{SO}(2,4)$ of the boundary theory is the *isometry* of the bulk AdS_5 — a spacetime symmetry of the higher-dimensional geometry. The R-symmetry $\mathrm{SO}(6)$, an *internal* symmetry of the gauge theory, is the isometry of the *internal* manifold S^5. Holography turns an internal symmetry of the field theory into a geometric symmetry of an extra compact space, and a spacetime symmetry into the geometry of an extra non-compact direction. The emergent dimension is not a metaphor; it is where the conformal group becomes geometry.

The decoupling argument has thus handed us a concrete object to study and a sharp claim to test. The object is the geometry $\mathrm{AdS}_5 \times S^5$ — and, more generally, AdS_{d+1} for a d-dimensional boundary CFT. The claim is that the conformal symmetry of the boundary theory should be realized exactly as the isometry of the anti-de Sitter factor, $\mathrm{SO}(2,d)$. The next section takes up AdS_{d+1} on its own terms: we define it intrinsically as a constant-curvature space, derive its isometry group, exhibit its conformal boundary, and prove that the isometries act on that boundary precisely as the conformal group of Chapter 47 — converting the symmetry coincidence found here into the rigorous backbone on which the entire dictionary will be built.

53.3 Anti-de Sitter Geometry and Its Conformal Boundary

The decoupling argument of Section 53.2 delivered a dual geometry — the near-horizon throat $\mathrm{AdS}_5 \times S^5$ — and with it a demand that we have not yet discharged. If a four-dimensional conformal field theory is to be *the same theory* as gravity in this throat, then the rigid symmetry of the field theory must be visible, exactly and without approximation, in the geometry. The conformal group of Chapter 47 is the largest spacetime symmetry a d-dimensional field theory without gravity

can carry; the isometry group of AdS_{d+1} is the largest a $(d+1)$-dimensional geometry can carry. The correspondence can only be consistent if these two groups coincide. This section proves that they do. We define anti-de Sitter space intrinsically, exhibit the coordinate systems in which the rest of Part XV will compute, compute its curvature to confirm it is maximally symmetric, and then carry out the central rigorous result of the chapter's symmetry backbone: a generator-by-generator match between the Killing vectors of AdS_{d+1} and the conformal Killing vectors of its boundary, reproducing the algebra of Theorem 47.3 and Eq. (47.53) exactly. The output is a clean arena — the Poincaré patch with a conformal boundary at $z=0$ — in which Section 53.4 will place a field and read off a conformal operator.

Notation Watch

This chapter, like the rest of the book (Chapter 3), uses the particle-physics mostly-minus signature $\eta_{\mu\nu}=\mathrm{diag}(+,-,\dots,-)$ for the bulk metric and the boundary metric it induces alike, so that a $(d+1)$-dimensional Lorentzian metric has one plus (timelike) and d minus (spacelike) eigenvalues. General relativity and most of the holography literature instead use mostly-plus, $\mathrm{diag}(-,+,\dots,+)$; the reader translating to that convention need only flip the overall sign of every metric. We keep the Riemann tensor $R^{\rho}{}_{\sigma\mu\nu}=\partial_\mu\Gamma^\rho_{\nu\sigma}-\partial_\nu\Gamma^\rho_{\mu\sigma}+\Gamma^\rho_{\mu\lambda}\Gamma^\lambda_{\nu\sigma}-\Gamma^\rho_{\nu\lambda}\Gamma^\lambda_{\mu\sigma}$, Ricci $R_{\sigma\nu}=R^{\rho}{}_{\sigma\rho\nu}$, and $R=g^{\sigma\nu}R_{\sigma\nu}$ (the Wald/MTW convention). With this fixed Riemann convention the overall sign of the metric flips the sign of the Ricci scalar, so anti-de Sitter space — whose defining feature is a negative cosmological constant — carries $R=+d(d+1)/L^2$ here, the mostly-plus value $-d(d+1)/L^2$ with its sign reversed. The conformal algebra of Eq. (47.53) is an abstract Lie algebra independent of any signature, and the symmetry match of Section 53.3.6 holds identically in either convention; only the placement of explicit signs in coordinate formulas changes.

53.3.1 Anti-de Sitter space as a hyperboloid

The cleanest definition of AdS_{d+1} is extrinsic: it is a single hyperboloid sitting inside a flat space of two extra dimensions, exactly the same flat space $\mathbb{R}^{2,d}$ that hosted the linear realization of the conformal group in Theorem 47.10. This is the first hint that the two symmetry groups will agree, and we exploit it relentlessly.

Definition 53.9 (Anti-de Sitter space)**.** Let $\mathbb{R}^{2,d}$ be the $(d+2)$-dimensional flat space with coordinates X^A, $A=0,1,\dots,d,d+1$, and metric of signature $(2,d)$,

$$G_{AB}\,dX^A dX^B=+(dX^0)^2-\sum_{i=1}^{d}(dX^i)^2+(dX^{d+1})^2,\qquad G_{AB}=\mathrm{diag}(+1,\underbrace{-1,\dots,-1}_{d},+1),\tag{53.50}$$

so that X^0 and X^{d+1} are the two timelike directions. *Anti-de Sitter space* AdS_{d+1} of radius L is the hyperboloid

$$G_{AB}X^A X^B = +(X^0)^2 - \sum_{i=1}^{d}(X^i)^2 + (X^{d+1})^2 = +L^2, \tag{53.51}$$

equipped with the metric induced from Eq. (53.50) by restriction.

Two facts must be checked before this object is usable: that the induced metric is Lorentzian (one timelike direction, so that Eq. (53.51) describes a spacetime and not a Euclidean space), and that it has constant curvature. The first is immediate from the geometry of the embedding and we dispose of it now; the second we postpone to the explicit curvature computation of Section 53.3.3.

The outward normal to the hyperboloid Eq. (53.51) at a point X is proportional to the gradient of the defining quadratic form, i.e. to $X_A = G_{AB}X^B$. Its squared norm is $G^{AB}X_A X_B = G_{AB}X^A X^B = +L^2 > 0$, so in mostly-minus signature the normal is *timelike*. A nondegenerate hypersurface with timelike normal in a space of signature $(2, d)$ (two timelike directions) has an induced metric of signature $(1, d)$: it removes exactly one of the two timelike directions. The induced metric on AdS_{d+1} is therefore Lorentzian with a single time direction, as a spacetime must be. The same computation shows why the *sign* on the right of Eq. (53.51) matters: the surface $G_{AB}X^A X^B = -L^2$ has spacelike normal and induced signature $(2, d-1)$ — two times — which is the $(d+1)$-dimensional de Sitter space, not anti-de Sitter. The single sign distinguishes the two maximally symmetric Lorentzian geometries.

The payoff of the embedding is that the isometries are visible at a glance. Any linear map $X^A \mapsto \Lambda^A{}_B X^B$ that preserves the ambient metric Eq. (53.50) — that is, any element of $\mathrm{O}(()2, d)$ — automatically preserves the quadratic form $G_{AB}X^A X^B$ and hence maps the hyperboloid Eq. (53.51) to itself, and being an isometry of the ambient space it is an isometry of the induced metric. Thus $\mathrm{O}(()2, d)$, and in particular its identity component $\mathrm{SO}(()2, d)$, acts by isometries on AdS_{d+1}. We will see in Section 53.3.4 that these are *all* the isometries, and the dimension count will then make the contact with Eq. (47.24) precise.

53.3.2 Three coordinate systems

The hyperboloid is a single global object, but no single coordinate chart is convenient for every purpose. We introduce the three systems used throughout Part XV — global, Poincaré, and Euclidean — and the maps between them. Each is obtained by solving the constraint Eq. (53.51) with an explicit parametrization and reading off the induced metric.

Global coordinates. Solve Eq. (53.51) by isolating the two timelike directions on a circle and the d spacelike directions on a sphere of radius set by a single

"bulk radial" variable $\rho \geq 0$:

$$X^0 = L\cosh\rho\,\cos\tau, \qquad X^{d+1} = L\cosh\rho\,\sin\tau, \qquad X^i = L\sinh\rho\,\Omega^i \;\; (i=1,\dots,d), \tag{53.52}$$

where Ω^i is a unit vector on the sphere S^{d-1}, $\sum_{i=1}^{d}(\Omega^i)^2 = 1$. The constraint is satisfied identically,

$$(X^0)^2 + (X^{d+1})^2 - \sum_i (X^i)^2 = L^2\cosh^2\rho\,(\cos^2\tau + \sin^2\tau) - L^2\sinh^2\rho = L^2, \tag{53.53}$$

using $\cosh^2\rho - \sinh^2\rho = 1$. To compute the induced metric, differentiate Eq. (53.52). The two timelike directions give

$$\begin{aligned}(dX^0)^2 + (dX^{d+1})^2 &= L^2\big(\sinh\rho\cos\tau\,d\rho - \cosh\rho\sin\tau\,d\tau\big)^2 + L^2\big(\sinh\rho\sin\tau\,d\rho + \cosh\rho\cos\\ &= L^2\big(\sinh^2\rho\,d\rho^2 + \cosh^2\rho\,d\tau^2\big),\end{aligned} \tag{53.54}$$

the cross terms $\propto d\rho\,d\tau$ cancelling between the two squares. The spacelike directions enter ds^2 with a minus sign and give, using $\sum_i \Omega^i d\Omega^i = \frac{1}{2}d(\sum_i(\Omega^i)^2) = 0$ and writing $d\Omega_{d-1}^2 = \sum_i (d\Omega^i)^2$ for the round metric on the unit S^{d-1},

$$-\sum_i (dX^i)^2 = -L^2\big(\cosh^2\rho\,d\rho^2 + \sinh^2\rho\,d\Omega_{d-1}^2\big). \tag{53.55}$$

Adding, the $d\rho^2$ coefficient collapses to $L^2(\sinh^2\rho - \cosh^2\rho) = -L^2$, and

$$ds^2 = L^2\Big(\cosh^2\rho\,d\tau^2 - d\rho^2 - \sinh^2\rho\,d\Omega_{d-1}^2\Big). \tag{53.56}$$

These are *global* coordinates: $\rho \in [0,\infty)$ and the angles cover all of S^{d-1}, while τ runs over the circle. The locus $\rho = 0$ is a regular point (the metric there is $L^2(d\tau^2 - d\rho^2 - \rho^2 d\Omega_{d-1}^2)$ in the small-ρ expansion, flat near the origin), the "center" of AdS. As $\rho \to \infty$ the spatial sphere blows up and the prefactors $\cosh^2\rho, \sinh^2\rho$ both diverge: this is the direction of the conformal boundary, analyzed in Section 53.3.5. Because X^0, X^{d+1} trace a circle, τ is naively periodic with period 2π, which would make time closed. One always passes to the universal cover by unwrapping $\tau \in \mathbb{R}$; *anti-de Sitter space* in this book means that simply-connected cover, on which there are no closed timelike curves. Rescaling $r = L\sinh\rho$ and a further change recasts Eq. (53.56) as the more familiar

$$ds^2 = \Big(1 + \tfrac{r^2}{L^2}\Big)dt^2 - \Big(1 + \tfrac{r^2}{L^2}\Big)^{-1} dr^2 - r^2\,d\Omega_{d-1}^2, \qquad t = L\tau, \tag{53.57}$$

the static-patch form, in which the gravitational potential $g_{tt} = +(1 + r^2/L^2)$ rises without bound at large r — the "box" that confines null geodesics and lets them reach the boundary and return in finite coordinate time.

Poincaré coordinates. The Poincaré patch is the chart in which the boundary symmetry will be transparent, because its constant-z slices are flat d-dimensional Minkowski spaces and its boundary is a single such Minkowski space. We obtain it by a different solution of the same constraint Eq. (53.51), singling out one null combination of the ambient coordinates as the inverse radial variable.

Introduce a radial coordinate $z>0$ and d coordinates x^μ ($\mu=0,\dots,d-1$) that will become the boundary directions, with the mostly-minus Minkowski metric $\eta_{\mu\nu}=\mathrm{diag}(+1,-1,\dots,-1)$ and $x^2\equiv\eta_{\mu\nu}x^\mu x^\nu$. Set

$$X^\mu=\frac{L}{z}\,x^\mu\quad(\mu=0,\dots,d{-}1),\qquad X^d=\frac{1}{2z}(z^2{-}L^2{-}x^2),\qquad X^{d+1}=\frac{1}{2z}(z^2{+}L^2{-}x^2). \tag{53.58}$$

We now show that Eq. (53.58) solves the constraint and induces precisely the metric quoted in the chapter's blueprint.

Proposition 53.10 (Poincaré-patch metric)**.** *The map Eq. (53.58) satisfies the hyperboloid constraint Eq. (53.51), and the metric Eq. (53.50) induces on it*

$$ds^2=\frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,dx^\mu dx^\nu-dz^2\Big),\qquad \eta_{\mu\nu}=\mathrm{diag}(+1,-1,\dots,-1),\quad z>0. \tag{53.59}$$

Proof. For the constraint, group the boundary directions and the two extra directions separately. Because X^0 is timelike and $X^1,\dots,X^{d-1}$ spacelike,

$$(X^0)^2-\sum_{a=1}^{d-1}(X^a)^2=\frac{L^2}{z^2}\Big((x^0)^2-\sum_a(x^a)^2\Big)=\frac{L^2}{z^2}\,\eta_{\mu\nu}x^\mu x^\nu=\frac{L^2x^2}{z^2}. \tag{53.60}$$

For the extra pair (X^d spacelike, X^{d+1} timelike), use the difference of squares with $X^{d+1}-X^d=L^2/z$ and $X^{d+1}+X^d=(z^2-x^2)/z$:

$$-(X^d)^2+(X^{d+1})^2=(X^{d+1}-X^d)(X^{d+1}+X^d)=\Big(\frac{L^2}{z}\Big)\Big(\frac{z^2-x^2}{z}\Big)=L^2-\frac{L^2x^2}{z^2}. \tag{53.61}$$

Adding Eqs. (53.60) and (53.61) the x^2 terms cancel and the total is $+L^2$, so Eq. (53.51) holds.

For the metric, differentiate. The boundary directions give $dX^\mu=\frac{L}{z}dx^\mu-\frac{Lx^\mu}{z^2}dz$, and with $x\cdot dx\equiv\eta_{\mu\nu}x^\mu dx^\nu$,

$$\eta_{\mu\nu}\,dX^\mu dX^\nu=\frac{L^2}{z^2}\,\eta_{\mu\nu}dx^\mu dx^\nu-\frac{2L^2}{z^3}\,(x\cdot dx)\,dz+\frac{L^2x^2}{z^4}\,dz^2. \tag{53.62}$$

For the extra pair, again factor through the sum and difference. Since $X^{d+1}-X^d=L^2/z$ depends only on z, $d(X^{d+1}-X^d)=-\frac{L^2}{z^2}dz$, while

$$X^{d+1}+X^d=\frac{z^2-x^2}{z}=z-\frac{x^2}{z}\;\Longrightarrow\; d(X^{d+1}+X^d)=\frac{z^2+x^2}{z^2}\,dz-\frac{2}{z}\,(x\cdot dx). \tag{53.63}$$

Hence

$$-(dX^d)^2+(dX^{d+1})^2 = d(X^{d+1}-X^d)\,d(X^{d+1}+X^d) = -\frac{L^2(z^2+x^2)}{z^4}\,dz^2+\frac{2L^2}{z^3}\,(x\cdot dx)\,dz. \tag{53.64}$$

The full induced line element is the sum of Eqs. (53.62) and (53.64). The cross terms $\propto (x\cdot dx)\,dz$ cancel exactly; the dz^2 coefficients combine to $\frac{L^2}{z^4}(x^2-z^2-x^2) = -\frac{L^2}{z^2}$; and the $dx^\mu dx^\nu$ term is untouched. This is Eq. (53.59). □

The coordinate $z \in (0,\infty)$ is the holographic radial direction: $z\to\infty$ is the deep interior (the "Poincaré horizon"), and $z\to 0$ is the conformal boundary, where the conformal factor L^2/z^2 blows up. The Poincaré patch Eq. (53.59) does *not* cover all of global AdS Eq. (53.56); it is the region accessible from one boundary Minkowski space, bounded by a horizon, as one sees by tracing the ranges of X^A through Eq. (53.58). Theorem 53.16 works the map between the two charts explicitly. The relation between the literature's conventions and ours is exactly the signature flip flagged above: in the mostly-plus convention used by most of the holography literature, Eq. (53.59) reads $ds^2 = \frac{L^2}{z^2}(\eta^{(+)}_{\mu\nu}dx^\mu dx^\nu + dz^2)$ with $\eta^{(+)} = \mathrm{diag}(-,+,\dots,+)$, the same geometry written with every sign reversed.

Euclidean coordinates. Continuing the boundary time $x^0 \to -ix^0_E$ (equivalently working on the surface where the ambient signature is $(1,d+1)$ rather than $(2,d)$) turns Eq. (53.59) into the metric of the upper half-space,

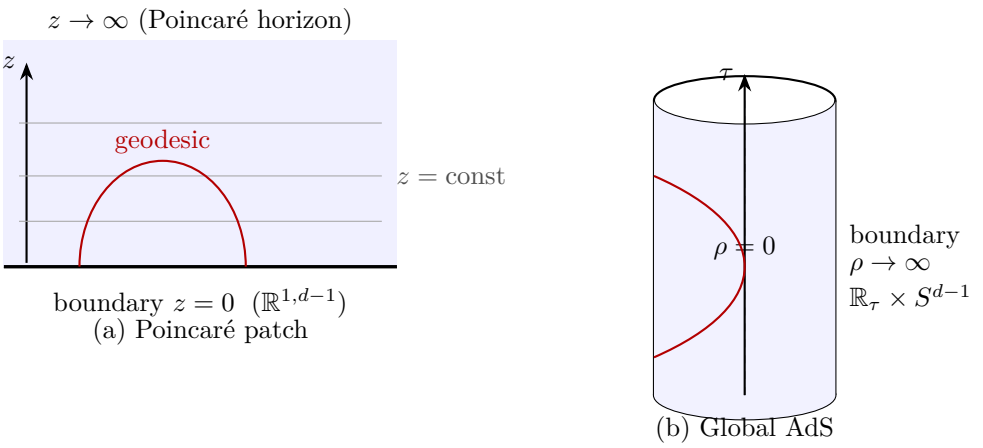
$$ds^2_E = \frac{L^2}{z^2}\Big(\delta_{\mu\nu}\,dx^\mu_E dx^\nu_E + dz^2\Big), \qquad z>0, \tag{53.65}$$

which is the Poincaré upper-half-space model of hyperbolic space $\mathbb{H}_{d+1}$, i.e. *Euclidean* anti-de Sitter space. The boundary is the flat $\mathbb{R}^d$ at $z=0$ together with the single point at $z=\infty$, which compactifies it to S^d. This is the geometry on which the Euclidean correlator machinery of Section 53.6 is cleanest, since there are no oscillatory modes and boundary conditions fix solutions uniquely; we will set up the bulk wave equation in Lorentzian signature in Section 53.4 and pass to Eq. (53.65) when computing.

53.3.3 Curvature and maximal symmetry

We promised that Eq. (53.51) has constant curvature. We now prove it by direct computation in the Poincaré chart Eq. (53.59), carrying every Christoffel symbol, and then read off that AdS is maximally symmetric — the property that ties its isometry count to the conformal generator count.

Write the bulk coordinates as $X^M = (x^\mu, z)$ with $M \in \{0,\dots,d-1,z\}$, and the metric Eq. (53.59) as $g_{MN} = \frac{L^2}{z^2}\hat\eta_{MN}$ with $\hat\eta_{MN} = \mathrm{diag}(\eta_{\mu\nu},-1)$ the flat



(a) Poincaré patch (b) Global AdS

Figure 53.5: Two charts on AdS_{d+1}. (a) The Poincaré patch Eq. (53.59): the upper half-space $z > 0$ with a flat Minkowski boundary at $z = 0$ and a horizon at $z \to \infty$; constant-z slices are flat, and a spacelike geodesic is a semicircle meeting the boundary orthogonally. (b) Global AdS Eq. (53.56): a solid cylinder whose central axis is $\rho = 0$ and whose conformal boundary at $\rho \to \infty$ is the cylinder $\mathbb{R}_\tau \times S^{d-1}$. The Poincaré patch is a single wedge of the global cylinder.

$(d+1)$-dimensional reference and $\eta_{\mu\nu}$ mostly-minus (so $g_{zz} = -L^2/z^2$). The inverse is $g^{MN} = \frac{z^2}{L^2}\hat{\eta}^{MN}$. Only z-derivatives of g_{MN} are nonzero, $\partial_z g_{MN} = -\frac{2}{z} g_{MN}$, so the Christoffel symbols $\Gamma^P_{MN} = \frac{1}{2} g^{PQ}(\partial_M g_{QN} + \partial_N g_{QM} - \partial_Q g_{MN})$ collapse to

$$\Gamma^z_{\mu\nu} = -\frac{1}{z}\,\eta_{\mu\nu}, \qquad \Gamma^z_{zz} = -\frac{1}{z}, \qquad \Gamma^\mu{}_{\nu z} = \Gamma^\mu{}_{z\nu} = -\frac{1}{z}\,\delta^\mu{}_\nu, \tag{53.66}$$

all others vanishing. Each follows by a one-line evaluation; for instance $\Gamma^z_{\mu\nu} = \frac{1}{2} g^{zz}(-\partial_z g_{\mu\nu}) = \frac{1}{2}\Big(-\frac{z^2}{L^2}\Big)\cdot\frac{2L^2}{z^3}\eta_{\mu\nu} = -\frac{1}{z}\eta_{\mu\nu}$, and $\Gamma^\mu{}_{\nu z} = \frac{1}{2} g^{\mu\rho}\partial_z g_{\rho\nu} = -\frac{1}{z}\delta^\mu{}_\nu$. The contracted symbol is

$$\Gamma^P{}_{zP} = \Gamma^z_{zz} + \Gamma^\mu{}_{z\mu} = -\frac{1}{z} - \frac{d}{z} = -\frac{d+1}{z}, \qquad \Gamma^P{}_{\mu P} = 0, \tag{53.67}$$

consistent with $\Gamma^P{}_{MP} = \partial_M \ln\sqrt{|g|}$ and $\sqrt{|g|} \propto z^{-(d+1)}$.

The Ricci tensor is $R_{MN} = \partial_P \Gamma^P{}_{MN} - \partial_N \Gamma^P{}_{MP} + \Gamma^P{}_{PQ}\Gamma^Q{}_{MN} - \Gamma^P{}_{NQ}\Gamma^Q{}_{MP}$. Take the two index types in turn. For the boundary–boundary components, $\partial_P \Gamma^P{}_{\mu\nu} = \partial_z(-\frac{1}{z}\eta_{\mu\nu}) = +\frac{1}{z^2}\eta_{\mu\nu}$, the second term vanishes by Eq. (53.67), the third is $\Gamma^P{}_{Pz}\Gamma^z{}_{\mu\nu} = -\frac{d+1}{z}\cdot(-\frac{1}{z})\eta_{\mu\nu} = +\frac{d+1}{z^2}\eta_{\mu\nu}$, and the fourth, summing the two contributing channels ($\Gamma^z_{\nu\rho}\Gamma^\rho{}_{\mu z}$ and $\Gamma^\rho{}_{\nu z}\Gamma^z_{\mu\rho}$), is $-\big[+\frac{1}{z^2}\eta_{\mu\nu} + \frac{1}{z^2}\eta_{\mu\nu}\big] = -\frac{2}{z^2}\eta_{\mu\nu}$. Collecting,

$$R_{\mu\nu} = \Big(1 + (d+1) - 2\Big)\frac{\eta_{\mu\nu}}{z^2} = +\frac{d}{z^2}\,\eta_{\mu\nu} = +\frac{d}{L^2}\,g_{\mu\nu}. \tag{53.68}$$

For the zz component, $\partial_P \Gamma^P{}_{zz} = \partial_z(-\frac{1}{z}) = \frac{1}{z^2}$, the second term is $-\partial_z \Gamma^P{}_{zP} = -\partial_z(-\frac{d+1}{z}) = -\frac{d+1}{z^2}$, the third is $\Gamma^P{}_{Pz}\Gamma^z_{zz} = (-\frac{d+1}{z})(-\frac{1}{z}) = \frac{d+1}{z^2}$, and the fourth is $-\Gamma^P{}_{zQ}\Gamma^Q{}_{zP} = -\big[(\Gamma^z_{zz})^2 + \sum_\mu (\Gamma^\mu{}_{z\mu})\ldots\big] = -\frac{d+1}{z^2}$ (the squared symbols summing to $\frac{1}{z^2} + \frac{d}{z^2}$). Hence

$$R_{zz} = \Big(1 - (d+1) + (d+1) - (d+1)\Big)\frac{1}{z^2} = -\frac{d}{z^2} = +\frac{d}{L^2}\, g_{zz}. \tag{53.69}$$

The mixed components $R_{\mu z}$ vanish by the same bookkeeping. Equations (53.68) and (53.69) combine into the single statement that AdS is an Einstein space with negative cosmological constant,

$$R_{MN} = +\frac{d}{L^2}\, g_{MN}, \qquad R = g^{MN} R_{MN} = +\frac{d}{L^2}\, \delta^M{}_M = +\frac{d(d+1)}{L^2}. \tag{53.70}$$

The Ricci scalar is a positive constant in this signature convention (encoding the negative cosmological constant of AdS, per the notation watch above); the curvature radius is L, and the strong-curvature limit is $L \to 0$. This is the geometric content of the statement, made in Section 53.2, that large AdS radius (in string units) is weak bulk curvature.

That AdS is not merely an Einstein space but *maximally symmetric* is the fact we actually need. A pseudo-Riemannian manifold of dimension n is maximally symmetric when its Riemann tensor is built entirely from the metric,

$$R_{MNPQ} = \frac{R}{n(n-1)}\big(g_{MP}g_{NQ} - g_{MQ}g_{NP}\big), \tag{53.71}$$

with R constant; equivalently, the curvature is the same in every direction at every point. For AdS, $n = d+1$ and $R = +d(d+1)/L^2$, so Eq. (53.71) reads $R_{MNPQ} = +\frac{1}{L^2}(g_{MP}g_{NQ} - g_{MQ}g_{NP})$. One verifies Eq. (53.71) directly from Eq. (53.66) — contracting it returns exactly Eqs. (53.68) and (53.69) — but the structural reason is the embedding: the hyperboloid Eq. (53.51) is the orbit of a single point under SO(()2, d), which acts transitively and, at each point, with isotropy group large enough to rotate all tangent directions into one another, forcing the curvature into the form Eq. (53.71).

Physical Insight

Maximal symmetry is the geometric image of the field-theory statement that a CFT has *no scale and no preferred point.* In flat space, Poincaré symmetry makes every point and every direction equivalent; conformal symmetry adds equivalence of every scale. AdS is the unique Lorentzian geometry that homogenizes points, directions, *and* the radial scale simultaneously — the curvature radius L is the only invariant, and it is not a scale on the boundary because, as Section 53.3.5 shows, it is scaled away

in the conformal limit $z \to 0$. The flatness of every constant-z slice and the self-similarity of the metric under $(x, z) \to \lambda(x, z)$ are the two faces of this single fact.

The number that matters is the count of independent Killing vectors. For a maximally symmetric space of dimension n this count is maximal and equal to

$$\dim(\text{isometry group}) = \frac{n(n+1)}{2}, \tag{53.72}$$

the dimension of the orthogonal group $\mathrm{SO}(()p, q)$ with $p + q = n + 1$ that acts on it. The n directions of homogeneity (translations between any two points) plus the $\frac{1}{2}n(n-1)$ directions of isotropy (rotations fixing a point) sum to Eq. (53.72). For AdS_{d+1}, $n = d + 1$ and the group is $\mathrm{SO}(()2, d)$; we make this precise next.

53.3.4 The isometry group $\mathrm{SO}(()2, d)$ and the dimension match

Theorem 53.11 (Isometries of anti-de Sitter space)**.** *The connected isometry group of* AdS_{d+1} *is* $\mathrm{SO}(()2, d)$*, of dimension*

$$\dim \mathrm{SO}(()2, d) = \frac{(d+2)(d+1)}{2}. \tag{53.73}$$

This equals the dimension $\frac{1}{2}(d+1)(d+2)$ *of the* d*-dimensional conformal group recorded in Eq. (47.24).*

Proof. We already showed in Section 53.3.1 that every $\Lambda \in \mathrm{SO}(()2, d)$ acts on the hyperboloid Eq. (53.51) as an isometry, so $\mathrm{SO}(()2, d)$ is contained in the connected isometry group. Conversely, the isometry group of any connected n-dimensional pseudo-Riemannian manifold has dimension at most $\frac{1}{2}n(n+1)$, with equality only for maximally symmetric spaces. We proved in Section 53.3.3 that AdS_{d+1} is maximally symmetric, so its isometry algebra has the maximal dimension $\frac{1}{2}(d+1)(d+2)$. The subgroup $\mathrm{SO}(()2, d)$ already realizes that many independent Killing vectors — its Lie algebra $\mathfrak{so}(()2, d)$ is spanned by the antisymmetric generators $J_{AB} = -J_{BA}$, $A, B = 0, \ldots, d+1$, numbering exactly $\binom{d+2}{2} = \frac{1}{2}(d+2)(d+1)$ — so the inclusion saturates the bound and the connected isometry group is $\mathrm{SO}(()2, d)$, no larger. The dimension count is

$$\dim \mathrm{SO}(()2, d) = \binom{d+2}{2} = \frac{(d+2)(d+1)}{2}, \tag{53.74}$$

which is verbatim Eq. (47.24). For AdS_5, $d = 4$, both equal $\frac{1}{2} \cdot 6 \cdot 5 = 15$. □

The explicit Killing vector fields are the ambient "rotations" restricted to the hyperboloid,

$$L_{AB} = X_A \frac{\partial}{\partial X^B} - X_B \frac{\partial}{\partial X^A}, \qquad X_A \equiv G_{AB} X^B, \tag{53.75}$$

which are tangent to Eq. (53.51) because $L_{AB}(G_{CD}X^C X^D) = 0$, and which close under the Lie bracket into the algebra

$$[L_{AB}, L_{CD}] = G_{AD}L_{BC} + G_{BC}L_{AD} - G_{AC}L_{BD} - G_{BD}L_{AC}, \tag{53.76}$$

the same $\mathfrak{so}(()2, d)$ bracket that appeared, with an explicit factor of i for Hermitian generators, in Eq. (47.55). The numerical coincidence Eq. (53.73) is therefore not a coincidence at all: the conformal group of Chapter 47 was *constructed* in Theorem 47.10 as the rotation group of the very space $\mathbb{R}^{2,d}$ in which AdS lives. What remains is to show that the two realizations of $\mathfrak{so}(()2, d)$ — as conformal Killing vectors on the boundary and as Killing vectors in the bulk — are matched not just as abstract algebras but generator by generator under the geometric limit $z \to 0$. That is the subject of Section 53.3.6, after we make the boundary itself precise.

53.3.5 The conformal boundary

Both Eqs. (53.56) and (53.59) degenerate at the edge of AdS — at $\rho \to \infty$ and at $z \to 0$ respectively — because a conformal factor diverges there. The metric does not extend to that edge, yet a definite *conformal* structure does, and it is the carrier of the boundary CFT. We make this precise with the notion, due to Penrose, of a conformal boundary defined by a defining function.

Definition 53.12 (Defining function and conformal boundary)**.** Let (M, g) be the open bulk AdS_{d+1}. A *defining function* for its boundary is a smooth function Ω on a neighbourhood of the edge with $\Omega > 0$ in the interior, $\Omega = 0$ on the edge, and $d\Omega \neq 0$ there. The *conformal boundary* is the locus $\{\Omega = 0\}$ equipped with the metric

$$\bar{g} \;=\; (\Omega^2\, g)\big|_{\Omega=0}, \tag{53.77}$$

the rescaled metric $\Omega^2 g$ restricted to the edge.

In the Poincaré chart the natural defining function is $\Omega = z/L$, which vanishes linearly at the boundary $z = 0$ with $d\Omega = dz/L \neq 0$. Then $\Omega^2 g = \frac{z^2}{L^2} \cdot \frac{L^2}{z^2}(\eta_{\mu\nu}dx^\mu dx^\nu - dz^2) = \eta_{\mu\nu}dx^\mu dx^\nu - dz^2$, which is regular at $z = 0$, and restricting to $z = 0$ (where dz drops out) gives

$$\bar{g} = \eta_{\mu\nu}\, dx^\mu dx^\nu : \tag{53.78}$$

the boundary of the Poincaré patch is d-dimensional Minkowski space, the home of the CFT. In global coordinates the natural choice $\Omega = 2e^{-\rho}$ (so $\Omega \to 0$ as $\rho \to \infty$) yields, after expanding $\cosh^2\rho \sim \sinh^2\rho \sim \frac{1}{4}e^{2\rho}$ in Eq. (53.56),

$$\Omega^2 g\big|_{\rho\to\infty} = L^2(d\tau^2 - d\Omega_{d-1}^2), \tag{53.79}$$

so the global boundary is the Lorentzian cylinder $\mathbb{R}_\tau \times S^{d-1}$ — the Einstein static universe. The two boundaries, Minkowski space and the cylinder, are related by a Weyl transformation and a conformal map; they are the same conformal manifold seen in two charts, exactly as the plane and the cylinder were the same $2d$ conformal manifold in Chapter 48.

The crucial structural point is that Eq. (53.77) depends on the *choice* of defining function, and only its conformal class is intrinsic.

Theorem 53.13 (The boundary carries a conformal class, not a metric)**.** *Two defining functions Ω and $\tilde\Omega$ for the same boundary differ by a positive smooth factor, $\tilde\Omega = e^{\omega}\,\Omega$ with ω smooth up to the boundary. The boundary metrics they induce via Eq. (53.77) are related by a Weyl rescaling,*

$$\tilde{\bar g} = e^{2\omega_0}\,\bar g, \qquad \omega_0 \equiv \omega\big|_{\Omega=0}, \tag{53.80}$$

where ω_0 is an arbitrary smooth function on the boundary. The conformal boundary therefore carries a well-defined conformal class $[\bar g]$ and no *preferred metric, hence no preferred scale.*

Proof. Both Ω and $\tilde\Omega$ vanish exactly on the boundary locus and are positive inside, with nonvanishing gradient on the boundary, so their ratio $\tilde\Omega/\Omega$ extends to a smooth, strictly positive function near the boundary; call its logarithm ω, so $\tilde\Omega = e^{\omega}\Omega$. Then

$$\tilde\Omega^2 g = e^{2\omega}\,\Omega^2 g. \tag{53.81}$$

Restricting to the boundary, where $\Omega^2 g \to \bar g$ and $\omega \to \omega_0$, gives Eq. (53.80). That ω_0 is arbitrary follows because one may multiply any defining function by any positive smooth function that is bounded and bounded away from zero at the boundary, realizing any smooth ω_0. Hence no representative of $[\bar g]$ is singled out; only ratios $\bar g_{\mu\nu}/\bar g_{\rho\sigma}$ — the conformal structure, i.e. angles and null cones — are invariant. □

This is the geometric origin of the central feature of the duality. A bulk diffeomorphism that rescales the defining function acts on the boundary representative as a Weyl transformation, so bulk geometry knows the boundary only up to Weyl rescaling. A field theory that is to be "read off" the boundary must therefore be insensitive to the boundary scale — it must be a *conformal* field theory, in which observables depend only on the conformal class [186, 393]. We will see in Section 53.4 that this same defining function controls the near-boundary falloff

of bulk fields, and that the Weyl weight ω_0 here is what becomes the scaling dimension of the dual operator. The dilatation isometry $z \to \lambda z,\ x \to \lambda x$ of the next subsection is exactly the statement that rescaling the defining function ($z \to \lambda z$) is a symmetry: it confirms operationally that the boundary has no preferred scale.

53.3.6 The match: bulk Killing vectors are boundary conformal Killing vectors

We can now prove the result that anchors the entire correspondence at the level of symmetry. The isometries of AdS_{d+1}, restricted to act on the conformal boundary, are precisely the conformal transformations of that boundary — and the dictionary between the two realizations is the one already written down abstractly in Eq. (47.56). We establish this constructively: for each ambient generator Eq. (53.75) we compute its action on the Poincaré coordinates (x^μ, z) and take the boundary limit $z \to 0$, recovering one of the four families of conformal Killing vectors Eq. (47.25) of Theorem 47.3.

It is convenient to package the two extra ambient directions into null combinations

$$X^- \equiv X^{d+1} - X^d = \frac{L^2}{z}, \qquad X^+ \equiv X^{d+1} + X^d = \frac{z^2 - x^2}{z}, \tag{53.82}$$

read off from Eq. (53.58), in terms of which the embedding inverts to the boundary coordinate and the radial coordinate as

$$x^\mu = \frac{L\, X^\mu}{X^-}, \qquad z = \frac{L^2}{X^-}. \tag{53.83}$$

The ambient metric in these variables is $G_{AB}X^A X^B = \eta_{\mu\nu}X^\mu X^\nu + X^+X^-$, displaying the boundary Minkowski metric and the null pair. We now identify the four families.

Lorentz rotations $M_{\mu\nu}$. The generators $L_{\mu\nu}$ ($\mu, \nu \in \{0, \dots, d-1\}$) mix only the boundary directions X^μ, which by Eq. (53.83) are proportional to x^μ at fixed z. They therefore act as the rotations $x^\mu \mapsto x^\mu + \omega^\mu{}_\nu x^\nu$ of the boundary and as the corresponding rotations of the bulk, leaving z fixed. These are the $\frac{1}{2}d(d-1)$ boundary Lorentz generators $M_{\mu\nu} = L_{\mu\nu}$ of Eq. (47.56), and their boundary action is the Lorentz family of Eq. (47.25).

Dilatation D. Take the ambient boost in the (X^d, X^{d+1}) plane,

$$L_{d+1,d} = X_{d+1}\,\partial_{X^d} - X_d\,\partial_{X^{d+1}} = X^{d+1}\,\partial_{X^d} + X^d\,\partial_{X^{d+1}}, \tag{53.84}$$

using $X_{d+1} = +X^{d+1}$ and $X_d = -X^d$ from Eq. (53.50) (this is the generator $J_{d+1,d}$ of the boundary dilatation in Eq. (47.56)). Its infinitesimal action $\delta X^d =$

ϵX^{d+1}, $\delta X^{d+1} = \epsilon X^d$ becomes, on the null combinations Eq. (53.82), $\delta X^+ = \epsilon X^+$ and $\delta X^- = -\epsilon X^-$ — a pure scaling — while the X^μ are untouched. Feeding this through Eq. (53.83),

$$\delta z = -\frac{L^2}{(X^-)^2}\,\delta X^- = \epsilon\,\frac{L^2}{X^-} = \epsilon\, z, \qquad \delta x^\mu = -\frac{L X^\mu}{(X^-)^2}\,\delta X^- = \epsilon\,\frac{L X^\mu}{X^-} = \epsilon\, x^\mu, \tag{53.85}$$

so the corresponding bulk Killing vector is

$$\xi_D = x^\mu\,\partial_\mu + z\,\partial_z, \tag{53.86}$$

the simultaneous rescaling $x \to \lambda x$, $z \to \lambda z$ anticipated in Section 53.3.5. Its boundary limit $z \to 0$ is $\xi_D|_{\rm bdy} = x^\mu \partial_\mu$, the dilatation family of Eq. (47.25), with $D = L_{d+1,d} = J_{d+1,d}$ as in Eq. (47.56). That the same ϵ scales the radial coordinate z and the boundary coordinates x^μ together is the geometric statement that the holographic direction *is* the renormalization-group scale — a theme developed in Sections 53.4 and 53.5.

Translations P_μ. Form, exactly as in the abstract dictionary Eq. (47.56),

$$P_\mu = L_{\mu d} + L_{\mu,d+1} = 2X_\mu\,\partial_{X^+} - X^-\,\partial_{X^\mu}, \tag{53.87}$$

where the second equality uses $\partial_{X^d} + \partial_{X^{d+1}} = 2\partial_{X^+}$ and $X_d + X_{d+1} = X^{d+1} - X^d = X^-$ in the null frame Eq. (53.82). Its action is $\delta X^\mu = -\beta\, X^-$, $\delta X^+ = 2\beta\, X_\mu$, $\delta X^- = 0$. Since $\delta X^- = 0$, Eq. (53.83) gives $\delta z = 0$ and

$$\delta x^\mu = \frac{L}{X^-}\,\delta X^\mu = \frac{L}{X^-}\,(-\beta\, X^-) = -L\beta = \text{const}, \tag{53.88}$$

so $\xi_P = a^\mu \partial_\mu$ with $a^\mu \propto \beta$: a rigid translation of the boundary, z-independent throughout the bulk. This is the translation family of Eq. (47.25), with $P_\mu = J_{\mu d} + J_{\mu,d+1}$.

Special conformal transformations K_μ. The conjugate combination $K_\mu = L_{\mu d} - L_{\mu,d+1}$ produces the remaining family; because it is the only one whose boundary action is nonlinear in x, we carry it in full in Theorem 53.15 below. The upshot is that its boundary limit is exactly $\xi^\mu = 2(b\cdot x)x^\mu - x^2 b^\mu$, the special-conformal family of Eq. (47.25).

Theorem 53.14 (Symmetry backbone of AdS/CFT)**.** *Under the identification Eq. (47.56),*

$$M_{\mu\nu} = L_{\mu\nu}, \qquad D = L_{d+1,d}, \qquad P_\mu = L_{\mu d} + L_{\mu,d+1}, \qquad K_\mu = L_{\mu d} - L_{\mu,d+1}, \tag{53.89}$$

the Killing vectors of AdS_{d+1}*, restricted to its conformal boundary at* $z \to 0$*, are exactly the conformal Killing vectors of Eq. (47.25): translations, Lorentz rotations, the dilatation, and special conformal transformations. The bulk isometry algebra Eq. (53.76) reproduces the conformal algebra Eq. (47.53) of Theorem 47.9 generator by generator.*

Proof. The four paragraphs above, together with Theorem 53.15, establish the four families of boundary actions. The bracket relations follow because $\{L_{AB}\}$ obey the $\mathfrak{so}(()2, d)$ algebra Eq. (53.76), which is identical to Eq. (47.55), and the assignment Eq. (53.89) is verbatim the isomorphism Eq. (47.56) that Theorem 47.10 proved carries Eq. (47.55) onto the conformal algebra Eq. (47.53). Therefore the brackets of $M_{\mu\nu}, D, P_\mu, K_\mu$ realized here as bulk Killing vectors coincide with Eqs. (47.53a) to (47.53d). For instance, the geometric content of $[D, P_\mu] \propto P_\mu$ (the relation Eq. (47.53c)) is immediate from Eqs. (53.86) and (53.87): conjugating a rigid translation $a^\mu \partial_\mu$ by the dilatation $x \to \lambda x$ scales it to $\lambda a^\mu \partial_\mu$, so $[\xi_D, \xi_P] = \xi_P$, matching $[D, P_\mu] = iP_\mu$ once Hermitian normalization restores the i; likewise $[D, K_\mu] = -iK_\mu$ because the special conformal vector scales oppositely. □

Worked Example 53.15 (The special conformal generator from the bulk)**.** We compute the boundary action of $K_\mu = L_{\mu d} - L_{\mu,d+1}$ and show it is the special conformal transformation, completing Theorem 53.14.

In the null frame Eq. (53.82), using $\partial_{X^d} - \partial_{X^{d+1}} = -2\partial_{X^-}$ and $X_d - X_{d+1} = -(X^d + X^{d+1}) = -X^+$,

$$K_\mu = L_{\mu d} - L_{\mu,d+1} = -2X_\mu\, \partial_{X^-} + X^+\, \partial_{X^\mu}, \tag{53.90}$$

so for a fixed boundary index μ the infinitesimal action with parameter γ is

$$\delta X^\nu = +\gamma\, X^+\, \delta^\nu{}_\mu, \qquad \delta X^- = -2\gamma\, X_\mu, \qquad \delta X^+ = 0. \tag{53.91}$$

Differentiating the inverse relations Eq. (53.83),

$$\delta x^\rho = \frac{L}{X^-}\,\delta X^\rho - \frac{LX^\rho}{(X^-)^2}\,\delta X^- = +\gamma\,\frac{LX^+}{X^-}\,\delta^\rho{}_\mu + 2\gamma\,\frac{LX^\rho X_\mu}{(X^-)^2}. \tag{53.92}$$

Now substitute the explicit forms. From Eq. (53.82), $\dfrac{X^+}{X^-} = \dfrac{(z^2 - x^2)/z}{L^2/z} = \dfrac{z^2 - x^2}{L^2}$, so the first term is $+\dfrac{\gamma}{L}(z^2 - x^2)\,\delta^\rho{}_\mu$. For the second, $X^\rho = \frac{X^-}{L} x^\rho$ and $X_\mu = \eta_{\mu\nu} X^\nu = \frac{X^-}{L} x_\mu$, whence $\dfrac{LX^\rho X_\mu}{(X^-)^2} = \dfrac{x^\rho x_\mu}{L}$. Therefore

$$\delta x^\rho = \frac{\gamma}{L}\Big(2\, x^\rho x_\mu - (x^2 - z^2)\,\delta^\rho{}_\mu\Big), \qquad \delta z = -\frac{L^2}{(X^-)^2}\,\delta X^- = \frac{2\gamma}{L}\,(x_\mu)\, z, \tag{53.93}$$

the second equality computed the same way. Contracting the free index against an arbitrary constant vector b^μ and absorbing γ/L into its normalization, the bulk special conformal Killing vector is

$$\xi_K = \left[\, 2(b\cdot x)\, x^\mu - (x^2 - z^2)\, b^\mu \,\right] \partial_\mu + 2(b\cdot x)\, z\, \partial_z. \tag{53.94}$$

On the boundary $z \to 0$ the z^2 correction and the ∂_z piece drop out, leaving

$$\xi_K\big|_{\text{bdy}} = \left[\, 2(b\cdot x)\, x^\mu - x^2\, b^\mu \,\right] \partial_\mu, \tag{53.95}$$

which is exactly the special conformal family $\xi^\mu = 2(b\cdot x)x^\mu - x^2 b^\mu$ of Eq. (47.25). The z-dependent terms in Eq. (53.94) are the unique extension of the boundary conformal transformation into a bulk isometry — the holographic "bulk completion" of a special conformal transformation — and their vanishing at $z = 0$ is why the boundary sees a pure conformal transformation with no admixture of radial motion.

Worked Example 53.16 (Embedding of the Poincaré patch in global AdS)**.** To see concretely that the Poincaré patch is a proper subregion of global AdS, we match the two parametrizations Eqs. (53.52) and (53.58) through their common ambient coordinates X^A. Equating the timelike X^{d+1} and the spacelike X^d components,

$$L\cosh\rho\,\sin\tau = \frac{1}{2z}(z^2 + L^2 - x^2), \qquad L\sinh\rho\,\Omega^d = \frac{1}{2z}(z^2 - L^2 - x^2), \tag{53.96}$$

and the boundary components $L\sinh\rho\,\Omega^\mu = \frac{L}{z}x^\mu$. These determine (τ, ρ, Ω^i) from (z, x^μ). The key qualitative fact reads off the first relation: the right-hand side, $\frac{1}{2z}(z^2 + L^2 - x^2)$, is a single-signed function of the Poincaré data on much of the patch, so as (z, x^μ) range over the patch the global time τ sweeps only a *finite* interval, not the full real line. The Poincaré patch is bounded in global time between two null surfaces (the past and future Poincaré horizons, $X^- = L^2/z \to \infty$, i.e. $z \to \infty$), and its boundary Minkowski space Eq. (53.78) is the portion of the global cylinder boundary $\mathbb{R} \times S^{d-1}$ covered by stereographic-type coordinates — precisely the conformal map between $\mathbb{R}^{1,d-1}$ and the cylinder that we met for $d = 2$ in Chapter 48. A boundary observer in the Poincaré frame thus sees Minkowski space; the same data, conformally completed, is the cylinder seen by the global observer.

Theorem 53.14 is the rigorous backbone the chapter promised: the conjectured equivalence of Section 53.2 respects, exactly, the only symmetry both sides are required to share. Every correlator computed holographically in the remainder of Part XV will transform under Eq. (53.89) as a conformal correlator must, with the bulk isometry doing the work of the boundary conformal group. We now have the concrete arena the next section needs: the Poincaré patch Eq. (53.59),

with its flat conformal boundary at $z = 0$, the dilatation Eq. (53.86) identifying the radial coordinate with scale, and a boundary that — by Theorem 53.13 — carries only a conformal class. Into this arena Section 53.4 places a free field, solves its wave equation, and reads off from the two near-boundary falloffs at $z \to 0$ how a bulk mass encodes a boundary operator's scaling dimension.

53.4 Fields in AdS: Falloffs, Dimensions, and the Mass–Dimension Relation

The previous section delivered the geometric backbone of the correspondence: anti-de Sitter space AdS_{d+1}, its isometry group $\mathrm{SO}(2, d)$, and the proof that those isometries act on the conformal boundary exactly as the conformal group of Chapter 47. That match of symmetry groups is suggestive but empty until we say *what* on the two sides is being identified. The conformal field theory side organizes itself, as Theorem 47.15 established, into primary operators labelled by a scaling dimension Δ and a spin. The gravity side, at the level we work, is a collection of fields propagating on the fixed AdS_{d+1} background. The first concrete entry of the dictionary must therefore be a map between these two lists — between a bulk field and a boundary operator — and the sharpest invariant that can cross the bridge is a single number on each side: the mass m of the bulk field and the dimension Δ of the operator. This section establishes the precise relation between them.

The strategy is the one that solving any linear wave equation near a singular point demands. We write the free scalar field equation in the Poincaré chart, examine its behaviour as the radial coordinate z approaches the boundary at $z = 0$, and discover that every solution is, near the boundary, a superposition of two power laws z^{Δ_-} and z^{Δ_+}. The two exponents are the roots of a quadratic — the indicial equation — whose content is the *mass–dimension relation* $\Delta(\Delta - d) = m^2 L^2$. One root will turn out to fix a boundary condition (a *source* for a dual operator), the other to encode the field's response (the operator's *vacuum expectation value*); and the requirement that the field carry finite energy will reproduce, on the nose, the unitarity bound of Theorem 47.18 together with a sharp lower bound on m^2 — negative masses are allowed in AdS, but only down to the Breitenlohner–Freedman floor. Everything in this section is a statement about a single free field on a fixed background; no quantum gravity is invoked. That is exactly the point: the hard conformal data of a strongly coupled theory is being read off from an elementary boundary-value problem.

53.4.1 The scalar action and its equation of motion in Poincaré coordinates

We work in the Poincaré patch of AdS_{d+1} established in Section 53.3. Writing the bulk coordinates as $X^M = (x^\mu, z)$ with $\mu = 0, 1, \ldots, d-1$ a boundary Lorentz index and $z > 0$ the radial (holographic) coordinate, the metric is

$$\mathrm{d}s^2 \;=\; g_{MN}\,\mathrm{d}X^M\,\mathrm{d}X^N \;=\; \frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu - \mathrm{d}z^2\Big), \qquad \eta_{\mu\nu} = \mathrm{diag}(+,-,\ldots,-), \tag{53.97}$$

where L is the AdS radius and the conformal boundary sits at $z \to 0$. In keeping with the book's mostly-minus signature (Theorem 47.15 and Appendix A), the boundary metric $\eta_{\mu\nu}$ has one timelike and $d-1$ spacelike directions, and the radial direction is spacelike: the full $(d+1)$-dimensional line element Eq. (53.59) has signature $(+,-,\ldots,-)$. The nonzero metric components and their inverses are

$$g_{\mu\nu} = \frac{L^2}{z^2}\,\eta_{\mu\nu}, \quad g_{zz} = -\frac{L^2}{z^2}, \qquad g^{\mu\nu} = \frac{z^2}{L^2}\,\eta^{\mu\nu}, \quad g^{zz} = -\frac{z^2}{L^2}. \tag{53.98}$$

The determinant follows from Eq. (53.98): each of the d boundary directions contributes a factor $\frac{L^2}{z^2}\eta_{\mu\mu}$ and the radial direction contributes $-\frac{L^2}{z^2}$, so

$$\det g_{MN} = \Big(\frac{L^2}{z^2}\Big)^{d+1}(\det \eta_{\mu\nu})\,(-1) = \Big(\frac{L^2}{z^2}\Big)^{d+1}(-1)^d, \qquad \sqrt{|g|} = \Big(\frac{L}{z}\Big)^{d+1}, \tag{53.99}$$

using $\det \eta_{\mu\nu} = (-1)^{d-1}$ for the mostly-minus boundary metric.

The dynamics of a single real scalar of mass m on this fixed background is governed by the covariant generalization of the ϕ^4 theory free action (Chapter 9), with the flat measure and derivatives promoted to their curved-space forms:

$$S[\phi] \;=\; \frac{1}{2}\int \mathrm{d}^d x\,\mathrm{d}z\,\sqrt{|g|}\,\Big(g^{MN}\,\partial_M\phi\,\partial_N\phi - m^2\phi^2\Big). \tag{53.100}$$

We keep ϕ free; interactions and self-couplings are the business of Chapter 54, where they generate Witten-diagram correlators. Varying Eq. (53.100) with $\delta\phi$ and integrating the derivative term by parts gives the curved-space Klein–Gordon equation,

$$(\Box_{\mathrm{AdS}} + m^2)\phi = 0, \qquad \Box_{\mathrm{AdS}}\,\phi \equiv \frac{1}{\sqrt{|g|}}\,\partial_M\big(\sqrt{|g|}\,g^{MN}\,\partial_N\phi\big). \tag{53.101}$$

Notation Watch

Much of the holography literature writes the scalar equation of motion as $(\Box - m^2)\phi = 0$. That sign reflects the mostly-plus signature $(-,+,\ldots,+)$

standard in general relativity, where the d'Alembertian of a plane wave is $\Box e^{ip\cdot x} = +p^2 e^{ip\cdot x}$ and the Klein–Gordon operator reads $\Box - m^2$. This book uses the mostly-minus particle-physics signature (Eq. (53.59)), in which $\Box e^{-ip\cdot x} = -p^2 e^{-ip\cdot x}$ and the free equation is $(\Box + m^2)\phi = 0$, exactly as for the flat-space scalar of Chapter 9. The two conventions describe the same physics; the invariant content — the mass–dimension relation Eq. (53.112) below — is identical, as it must be. We carry the explicit signature through every step so that no sign is taken on faith.

To make Eq. (53.101) concrete we evaluate $\Box_{\mathrm{AdS}}$ on a field $\phi(x, z)$ using Eqs. (53.98) and (53.99). Split the sum over M, N into the boundary directions and the radial direction. For the boundary part, $g^{\mu\nu} = \frac{z^2}{L^2}\eta^{\mu\nu}$ depends on z but not on x, so

$$\frac{1}{\sqrt{|g|}}\,\partial_\mu\!\Big(\sqrt{|g|}\,g^{\mu\nu}\partial_\nu\phi\Big) = \Big(\frac{z}{L}\Big)^{d+1}\partial_\mu\Big[\Big(\frac{L}{z}\Big)^{d+1}\frac{z^2}{L^2}\,\eta^{\mu\nu}\partial_\nu\phi\Big] = \frac{z^2}{L^2}\,\eta^{\mu\nu}\partial_\mu\partial_\nu\phi = \frac{z^2}{L^2}\,\Box_x\phi, \tag{53.102}$$

where $\Box_x \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu$ is the flat boundary d'Alembertian and the prefactor $(L/z)^{d+1}z^2/L^2 = (L/z)^{d-1}$ was pulled through ∂_μ unchanged. For the radial part, $\sqrt{|g|}\,g^{zz} = (L/z)^{d+1}(-(z/L)^2) = -(L/z)^{d-1}$, so

$$\begin{aligned}\frac{1}{\sqrt{|g|}}\,\partial_z\!\Big(\sqrt{|g|}\,g^{zz}\partial_z\phi\Big) &= \Big(\frac{z}{L}\Big)^{d+1}\partial_z\Big[-\Big(\frac{L}{z}\Big)^{d-1}\partial_z\phi\Big] \\ &= \Big(\frac{z}{L}\Big)^{d+1}\Big[\frac{d-1}{z}\Big(\frac{L}{z}\Big)^{d-1}\partial_z\phi - \Big(\frac{L}{z}\Big)^{d-1}\partial_z^2\phi\Big] \\ &= \frac{z^2}{L^2}\Big(-\partial_z^2\phi + \frac{d-1}{z}\,\partial_z\phi\Big),\end{aligned} \tag{53.103}$$

where the middle line used $\partial_z(L/z)^{d-1} = -\frac{d-1}{z}(L/z)^{d-1}$. Adding Eqs. (53.102) and (53.103),

$$\Box_{\mathrm{AdS}}\,\phi = \frac{z^2}{L^2}\Big(\Box_x\phi - \partial_z^2\phi + \frac{d-1}{z}\,\partial_z\phi\Big). \tag{53.104}$$

A useful consistency check: with $L \to \infty$ at fixed proper coordinates the curvature terms would drop and one recovers the flat $(d+1)$-dimensional wave operator $\Box_x - \partial_z^2$, the radial direction entering with the spacelike sign as it must. Inserting Eq. (53.104) into the equation of motion Eq. (53.101) and multiplying through by L^2/z^2 gives the form we will analyze,

$$\boxed{-\partial_z^2\phi + \frac{d-1}{z}\,\partial_z\phi + \frac{m^2L^2}{z^2}\,\phi + \Box_x\phi = 0.} \tag{53.105}$$

Every factor of L and z in Eq. (53.105) traces back to the metric Eq. (53.59); nothing has been quoted.

It is convenient to Fourier transform along the boundary, writing

$$\phi(x,z)=\int\frac{\mathrm{d}^{d}k}{(2\pi)^{d}}\,e^{-ik\cdot x}\,f_{k}(z),\qquad k\cdot x=\eta_{\mu\nu}k^{\mu}x^{\nu},\tag{53.106}$$

so that $\Box_x\to -k^2$ with $k^2=\eta_{\mu\nu}k^\mu k^\nu$. The radial profile $f_k(z)$ then obeys the ordinary differential equation

$$f_k''-\frac{d-1}{z}\,f_k'-\frac{m^2L^2}{z^2}\,f_k+k^2 f_k=0.\tag{53.107}$$

This is the master equation of the section. It is a linear second-order ODE with a regular singular point at $z=0$ (the boundary) and an irregular point at $z=\infty$ (the Poincaré horizon, the deep interior). The boundary behaviour is governed entirely by the singular point, which is where we now look.

53.4.2 Near-boundary indicial analysis and the mass–dimension relation

Near $z=0$ the $k^2 f_k$ term in Eq. (53.107) is subleading: it carries no inverse power of z, whereas the other three terms scale as z^{-2} relative to f_k. The leading behaviour is therefore controlled by the scale-invariant operator obtained by dropping k^2,

$$z^2 f''-(d-1)\,z\,f'-m^2L^2\,f=0,\tag{53.108}$$

which is an Euler (equidimensional) equation: it is invariant under $z\to\lambda z$, so it is solved by pure powers. This is the Frobenius method at a regular singular point, with the indicial equation read off from the leading exponents. Substitute the trial solution

$$f(z)=z^{\Delta},\qquad f'=\Delta\,z^{\Delta-1},\qquad f''=\Delta(\Delta-1)\,z^{\Delta-2},\tag{53.109}$$

into Eq. (53.108). Each term produces the common factor z^Δ:

$$\begin{aligned}z^2f''&=\Delta(\Delta-1)\,z^{\Delta},\\-(d-1)zf'&=-(d-1)\Delta\,z^{\Delta},\\-m^2L^2f&=-m^2L^2\,z^{\Delta}.\end{aligned}\tag{53.110}$$

Requiring their sum to vanish and dividing by z^Δ gives the *indicial equation*,

$$\begin{aligned}\Delta(\Delta-1)-(d-1)\Delta-m^2L^2&=0\\\Delta^2-\Delta-(d-1)\Delta-m^2L^2&=0\\\Delta^2-d\,\Delta-m^2L^2&=0,\end{aligned}\tag{53.111}$$

which we record in its standard factored form.

Theorem 53.17 (Mass–dimension relation)**.** *A free scalar field of mass m on AdS_{d+1} of radius L has near-boundary radial behaviour $\phi \sim z^{\Delta}$ with the exponent Δ a root of*

$$\Delta(\Delta - d) = m^2 L^2. \tag{53.112}$$

The two roots are

$$\Delta_{\pm} = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2 L^2}, \qquad \Delta_+ + \Delta_- = d, \qquad \Delta_+ \, \Delta_- = -m^2 L^2. \tag{53.113}$$

The larger root $\Delta \equiv \Delta_+ \geq d/2$ is identified with the scaling dimension of the dual boundary operator.

Proof. Equation Eq. (53.112) is Eq. (53.111) rewritten: expanding $\Delta(\Delta - d) = \Delta^2 - d\Delta$ reproduces $\Delta^2 - d\Delta - m^2 L^2 = 0$. Solving the quadratic by completing the square,

$$\Delta^2 - d\Delta = m^2 L^2 \implies \left(\Delta - \tfrac{d}{2}\right)^2 = \frac{d^2}{4} + m^2 L^2 \implies \Delta = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2 L^2}, \tag{53.114}$$

which is Eq. (53.113). The sum and product of the roots are the Vieta relations of the monic quadratic $\Delta^2 - d\Delta - m^2 L^2 = 0$: the sum of roots equals d and the product equals $-m^2 L^2$. The identification $\Delta = \Delta_+$ with the operator dimension is justified physically in Section 53.4.3 (normalizability) and confirmed against the unitarity bound in Section 53.4.5. □

Two limiting cases anchor the relation. A massless bulk scalar, $m^2 = 0$, gives $\Delta_+ = d$ and $\Delta_- = 0$: it is dual to a marginal operator of dimension d, the prototype being the Lagrangian density or, more precisely in the maximally supersymmetric example, the boundary stress-tensor multiplet's scalar partners. At the other extreme, large positive $m^2 L^2$ gives $\Delta_+ \approx mL$ for $mL \gg 1$, the expected flat-space-like growth of dimension with mass. The novelty lives at negative m^2, to which we return in Section 53.4.4.

The full local solution near $z = 0$ is obtained by promoting the two indicial roots to the leading terms of two Frobenius series. Restoring the k^2 term in Eq. (53.107) and substituting $f = z^{\Delta_\pm} \sum_{n \geq 0} c_n^{(\pm)} z^{2n}$ (only even powers of z appear because k^2 enters z^2 at a time), one finds the generic solution

$$f_k(z) = \alpha(k)\, z^{\Delta_-} \left(1 + O(z^2)\right) + \beta(k)\, z^{\Delta_+} \left(1 + O(z^2)\right), \tag{53.115}$$

valid whenever the exponents do not differ by an even integer (the degenerate cases are discussed below). The two integration constants α and β are the boundary data of the field, and the entire physics of the dictionary hinges on

their interpretation. Returning to position space, the leading near-boundary expansion of any bulk solution is

$$\phi(x,z) \xrightarrow{z\to 0} \alpha(x)\, z^{\Delta_-} + \cdots + \beta(x)\, z^{\Delta_+} + \ldots, \tag{53.116}$$

the two independent falloffs whose coefficients $\alpha(x)$ and $\beta(x)$ are functions on the boundary.

Caution

When $\Delta_+ - \Delta_- = 2\sqrt{d^2/4 + m^2L^2}$ is a positive even integer, the two Frobenius series collide: a term $z^{\Delta_- + 2n}$ in the first series degenerates with the leading z^{Δ_+} of the second, and the subleading solution acquires a logarithm, $\phi \sim \alpha z^{\Delta_-} + \cdots + \beta z^{\Delta_+} \log z + \ldots$. The most important instance is the Breitenlohner–Freedman point $\Delta_+ = \Delta_- = d/2$ (Section 53.4.4), where the two solutions are $z^{d/2}$ and $z^{d/2}\log z$. These logarithms are not pathologies; they are the holographic signature of operator mixing and of the conformal anomaly, and they are handled by the same renormalization of the on-shell action that Chapter 54 will set up. For the present structural discussion the generic case Eq. (53.115) suffices.

53.4.3 Source versus vacuum expectation value: the two modes

The two falloffs in Eq. (53.116) play sharply different roles, and the distinction is the seed of the whole field–operator dictionary. The decisive property is *normalizability*: whether a mode has finite norm under the inner product that defines the field's Hilbert space. A mode of finite norm is a genuine fluctuating degree of freedom — a state — whereas a mode of infinite norm is not normalizable and must instead be held fixed as a boundary condition, a non-dynamical background that the bulk solution is required to approach.

The relevant norm is the Klein–Gordon (symplectic) inner product associated with the action Eq. (53.100). On a constant-time slice Σ (coordinates z and the spatial $\vec{x}$), it reads

$$(\phi_1, \phi_2) = -i \int_\Sigma \mathrm{d}z\, \mathrm{d}^{d-1}x\, \sqrt{|g|}\, g^{00} (\phi_1^* \,\partial_0 \phi_2 - \phi_2\, \partial_0 \phi_1^*), \tag{53.117}$$

the curved-space version of the flat Klein–Gordon product of Chapter 9. Convergence of Eq. (53.117) is controlled entirely by the z-integral near $z = 0$, where the integrand's measure factor is

$$\sqrt{|g|}\, g^{00} = \left(\frac{L}{z}\right)^{d+1} \left(\frac{z}{L}\right)^2 \eta^{00} = \left(\frac{L}{z}\right)^{d-1}, \tag{53.118}$$

using Eqs. (53.98) and (53.99) and $\eta^{00} = +1$. For a mode behaving as $\phi \sim z^\Delta$ near the boundary, the time derivatives in Eq. (53.117) do not change the radial

power, so the near-boundary contribution to the norm scales as

$$(\phi,\phi) \;\sim\; \int_0 \mathrm{d}z \left(\frac{L}{z}\right)^{d-1} |z^{\Delta}|^2 \;=\; L^{d-1}\int_0 \mathrm{d}z\, z^{\,2\Delta-d+1}. \tag{53.119}$$

The integral $\int_0 \mathrm{d}z\, z^{2\Delta-d+1}$ converges at the lower limit if and only if the exponent exceeds -1, i.e.

$$2\Delta - d + 1 > -1 \quad\Longleftrightarrow\quad \Delta > \frac{d-2}{2}. \tag{53.120}$$

Theorem 53.18 (Normalizability of the two modes)**.** *A near-boundary mode $\phi \sim z^{\Delta}$ on AdS_{d+1} is normalizable in the Klein–Gordon norm Eq. (53.117) if and only if $\Delta > (d-2)/2$. Consequently:*

1. *the larger-root mode z^{Δ_+} is* always *normalizable, since $\Delta_+ \geq d/2 > (d-2)/2$ for every m^2 in the allowed range;*
2. *the smaller-root mode z^{Δ_-} is non-normalizable, hence a fixed boundary condition (a* source*), whenever $\Delta_- \leq (d-2)/2$; it becomes normalizable only in the window discussed in Section 53.4.4.*

Proof. The condition $\Delta > (d-2)/2$ is Eq. (53.120), derived directly from the convergence of Eq. (53.119). For item (1), $\Delta_+ = \frac{d}{2} + \sqrt{d^2/4 + m^2L^2} \geq \frac{d}{2}$ because the square root is non-negative wherever it is real, and $\frac{d}{2} > (d-2)/2$ by one unit; hence the strict inequality Eq. (53.120) always holds for Δ_+. For item (2), $\Delta_- = \frac{d}{2} - \sqrt{d^2/4 + m^2L^2} \leq \frac{d}{2}$, and one checks below (Section 53.4.4) that $\Delta_- \leq (d-2)/2$ precisely when $\sqrt{d^2/4 + m^2L^2} \geq 1$, which is the generic situation; the borderline $\sqrt{d^2/4 + m^2L^2} < 1$ is the alternate-quantization window. □

The interpretation now writes itself, and it is the central conceptual payload of the section.

Physical Insight

The two falloffs are a source and a response. Standard (Dirichlet-like) quantization holds the slower-decaying, non-normalizable mode $\alpha(x)$ fixed: it is not a fluctuating field but a *boundary condition* the bulk solution must satisfy. On the field-theory side, fixing $\alpha(x)$ corresponds to turning on a *source* $J(x)$ that couples to a boundary operator $\mathcal{O}$ through a term $\int \mathrm{d}^d x\, J(x)\,\mathcal{O}(x)$ added to the CFT action. Dimensional analysis pins which is which: for the coupling to be dimensionless the source must carry dimension $[J] = d - \Delta$. Reading off from $\alpha(x) z^{\Delta_-}$, the boundary value α has scaling dimension $\Delta_- = d - \Delta_+$, so it is the source for an operator of dimension Δ_+. The faster-decaying, normalizable mode $\beta(x)$ is then the field's dynamical *response*: it is proportional to the induced expectation value $\langle \mathcal{O}(x) \rangle$ in the presence of that source. The bulk field

ϕ of mass m is therefore dual to a boundary operator $\mathcal{O}$ of dimension Δ_+, with the non-normalizable falloff playing the role of $\mathcal{O}$'s source and the normalizable falloff its one-point function. This is precisely the data Chapter 54's GKP–Witten relation turns into correlators.

The dimensional argument deserves to be stated as the operator-dimension assignment, since it is the result the next section consumes. The bulk solution is dimensionless (it is a scalar field value), and z carries dimension of length, i.e. inverse mass, $[z] = -1$ in mass units. Then $[\alpha\, z^{\Delta_-}] = 0$ forces $[\alpha] = \Delta_-$ and likewise $[\beta] = \Delta_+$. Identifying α with the source J of an operator $\mathcal{O}$ and recalling that a source has dimension d minus the dimension of the operator it multiplies, $[J] = d - \Delta_{\mathcal{O}}$, we get $\Delta_- = d - \Delta_{\mathcal{O}}$, hence

$$\Delta_{\mathcal{O}} = d - \Delta_- = \Delta_+, \tag{53.121}$$

where the last step used $\Delta_+ + \Delta_- = d$ from Eq. (53.113). The operator dual to a bulk scalar of mass m has dimension Δ_+. We will write $\Delta \equiv \Delta_+$ from now on, reserving the subscript for emphasis.

53.4.4 The Breitenlohner–Freedman bound and the alternate-quantization window

Flat-space intuition insists that $m^2 < 0$ signals an instability: a tachyon, a field rolling off the top of an inverted potential. In AdS this intuition fails, and the reason is geometric. The negative curvature provides a confining "box" of size L, and the gradient energy stored in fitting a fluctuation into that box can outweigh the negative potential energy $-\frac{1}{2}m^2\phi^2$ even when $m^2 < 0$ — provided m^2 is not too negative. The sharp threshold is the Breitenlohner–Freedman (BF) bound. We derive it twice, first as the reality condition on Δ and then, rigorously, as the positivity of the conserved energy; the two derivations meet at the same number.

Reality of the dimension. The exponents Eq. (53.113) are real if and only if the discriminant is non-negative,

$$\frac{d^2}{4} + m^2 L^2 \;\geq\; 0 \qquad \Longleftrightarrow \qquad m^2 L^2 \;\geq\; -\frac{d^2}{4}. \tag{53.122}$$

Below this value $\Delta_\pm$ acquire imaginary parts, $\Delta_\pm = \frac{d}{2} \pm i|\,\cdot\,|$, and the near-boundary behaviour $z^{d/2} z^{\pm i|\cdot|} = z^{d/2} e^{\pm i|\cdot| \log z}$ oscillates ever faster as $z \to 0$. Such a complex dimension cannot belong to a unitary CFT (Theorem 47.18 requires real Δ), and on the gravity side it signals a genuine instability of the AdS vacuum. The reality condition Eq. (53.122) is the BF bound. It is the precise statement that AdS tolerates tachyons down to a finite negative mass-squared floor set by the radius.

Positivity of the energy. The reality argument is suggestive but a stability claim deserves a dynamical proof: the conserved energy of every fluctuation must be bounded below, with the bound saturated exactly at Eq. (53.122). We establish this by reducing the radial problem to a one-dimensional Schrödinger operator and invoking a sharp functional inequality.

Consider a static fluctuation, $\partial_0\phi = 0$ and $\vec{\partial}\phi = 0$, depending only on the radial coordinate, $\phi = \phi(z)$; this is the channel that probes the deepest into negative m^2 because it carries no boundary kinetic energy. Its energy is the value of the Hamiltonian, which for the action Eq. (53.100) on a static configuration is minus the Lagrangian; collecting the radial-gradient and mass terms with the measure factors from Eqs. (53.98) and (53.99) (and dropping the infinite boundary volume $\int \mathrm{d}^{d-1}x$),

$$E[\phi] = \frac{1}{2}\int_0^\infty \mathrm{d}z\,\Big[\Big(\frac{L}{z}\Big)^{d-1}(\phi')^2 + m^2\Big(\frac{L}{z}\Big)^{d+1}\phi^2\Big] = \frac{L^{d-1}}{2}\int_0^\infty \mathrm{d}z\,\Big[z^{-(d-1)}(\phi')^2 + m^2L^2\,z^{-(d+1)}\phi^2\Big], \tag{53.123}$$

where the gradient term used $\sqrt{|g|}\,|g^{zz}| = (L/z)^{d-1}$ and the mass term used $\sqrt{|g|} = (L/z)^{d+1}$. Stability is the statement $E[\phi] \geq 0$ for all admissible ϕ. The negative m^2 contribution can in principle make Eq. (53.123) negative; whether it does is a contest between the two terms, and the contest is decided cleanly after a change of variable that removes the first-derivative cross-structure. Set

$$\phi(z) = z^{(d-1)/2}\,\psi(z), \tag{53.124}$$

which is engineered so that the measure weight $z^{-(d-1)}$ in the gradient term is absorbed. Compute $\phi' = \frac{d-1}{2}z^{(d-3)/2}\psi + z^{(d-1)/2}\psi'$, so

$$z^{-(d-1)}(\phi')^2 = (\psi')^2 + \frac{d-1}{z}\,\psi\psi' + \frac{(d-1)^2}{4}\,\frac{\psi^2}{z^2}. \tag{53.125}$$

Integrate the cross term by parts, $\int_0^\infty \frac{d-1}{z}\psi\psi'\,\mathrm{d}z = \frac{d-1}{2}\int_0^\infty \frac{(\psi^2)'}{z}\mathrm{d}z = \frac{d-1}{2}\Big([\psi^2/z]_0^\infty + \int_0^\infty \frac{\psi^2}{z^2}\mathrm{d}z\Big)$. The boundary term vanishes for any normalizable static configuration (at $z\to\infty$ the profile decays; at $z\to 0$ the surviving boundary data is the fixed source, held to zero for a genuine fluctuation), leaving $\frac{d-1}{2}\int \psi^2/z^2$. The mass term transforms as $m^2L^2z^{-(d+1)}\phi^2 = m^2L^2z^{-(d+1)}z^{d-1}\psi^2 = m^2L^2\,\psi^2/z^2$. Assembling,

$$E[\psi] = \frac{L^{d-1}}{2}\int_0^\infty \mathrm{d}z\,\Big[(\psi')^2 + \Big(\frac{(d-1)^2}{4} + \frac{d-1}{2} + m^2L^2\Big)\frac{\psi^2}{z^2}\Big] = \frac{L^{d-1}}{2}\int_0^\infty \mathrm{d}z\,\Big[(\psi')^2 + \mathcal{C}\,\frac{\psi^2}{z^2}\Big], \tag{53.126}$$

where the dimensionless coefficient of the inverse-square potential is

$$\mathcal{C} = \frac{(d-1)^2}{4} + \frac{d-1}{2} + m^2L^2 = \frac{(d-1)(d-1)+2(d-1)}{4} + m^2L^2 = \frac{(d-1)(d+1)}{4} + m^2L^2 = \tag{53.127}$$

The energy Eq. (53.126) is, up to the overall constant, the expectation value of the Schrödinger operator $-\partial_z^2 + \mathcal{C}/z^2$ on the half-line — the celebrated inverse-square potential, whose stability threshold is a textbook-sharp number.

The positivity of Eq. (53.126) is settled by *Hardy's inequality*, which we prove since the book takes nothing on faith.

Lemma 53.19 (Hardy's inequality on the half-line)**.** *For every smooth ψ of compact support in $(0,\infty)$,*

$$\int_0^\infty (\psi')^2 \,\mathrm{d}z \;\geq\; \frac{1}{4}\int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z, \tag{53.128}$$

and the constant $\tfrac{1}{4}$ is the largest for which the inequality holds.

Proof. For any real constant c the integral of a perfect square is non-negative:

$$0 \leq \int_0^\infty \Big(\psi' - \frac{c\,\psi}{z}\Big)^2 \mathrm{d}z = \int_0^\infty (\psi')^2\,\mathrm{d}z - 2c\int_0^\infty \frac{\psi\psi'}{z}\,\mathrm{d}z + c^2\int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z. \tag{53.129}$$

The middle integral is evaluated by parts using $\psi\psi' = \tfrac{1}{2}(\psi^2)'$ and the compact support (which kills the boundary term):

$$\int_0^\infty \frac{\psi\psi'}{z}\,\mathrm{d}z = \frac{1}{2}\int_0^\infty \frac{(\psi^2)'}{z}\,\mathrm{d}z = \frac{1}{2}\Big([\psi^2/z]_0^\infty + \int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z\Big) = \frac{1}{2}\int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z. \tag{53.130}$$

Substituting Eq. (53.130) into Eq. (53.129) gives

$$0 \leq \int_0^\infty (\psi')^2\,\mathrm{d}z - (c - c^2)\int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z, \tag{53.131}$$

so $\int(\psi')^2 \geq (c-c^2)\int \psi^2/z^2$ for every c. The right-hand coefficient $c - c^2$ is maximized at $c = \tfrac{1}{2}$, where it equals $\tfrac{1}{4}$, yielding Eq. (53.128). Sharpness: the would-be extremizer saturating Eq. (53.129) at $c = \tfrac{1}{2}$ is $\psi' = \psi/2z$, i.e. $\psi \propto z^{1/2}$; this is not square-integrable against $\mathrm{d}z/z^2$, but a family of compactly supported functions approximating it makes the ratio $\int(\psi')^2/\int(\psi^2/z^2)$ approach $\tfrac{1}{4}$ from above, so no larger constant works. □

Applying Theorem 53.19 to Eq. (53.126),

$$E[\psi] \geq \frac{L^{d-1}}{2}\Big(\frac{1}{4} + \mathcal{C}\Big)\int_0^\infty \frac{\psi^2}{z^2}\,\mathrm{d}z, \tag{53.132}$$

which is non-negative for all fluctuations if and only if $\mathcal{C} \geq -\tfrac{1}{4}$. By the sharpness clause of the lemma, if $\mathcal{C} < -\tfrac{1}{4}$ there exist configurations with $E[\psi] < 0$, so the vacuum is unstable. Using Eq. (53.127),

$$\mathcal{C} + \frac{1}{4} = \frac{d^2-1}{4} + m^2L^2 + \frac{1}{4} = \frac{d^2}{4} + m^2L^2, \tag{53.133}$$

so $\mathcal{C} \geq -\tfrac{1}{4}$ is identical to $m^2L^2 \geq -d^2/4$. We have proved:

Theorem 53.20 (Breitenlohner–Freedman bound)**.** *A free scalar of mass-squared m^2 on AdS_{d+1} has a conserved energy that is non-negative for all fluctuations, and hence a stable vacuum, if and only if*

$$m^2L^2 \;\geq\; -\frac{d^2}{4}. \tag{53.134}$$

At saturation, $m^2L^2 = -d^2/4$, the two dimensions degenerate to $\Delta_+ = \Delta_- = d/2$ and the energy bound is marginally satisfied (the inverse-square coefficient sits exactly at the Hardy threshold $\mathcal{C} = -\frac{1}{4}$).

The two derivations agree because they are the same statement seen from two sides: reality of $\Delta_\pm$ in Eq. (53.113) requires $d^2/4 + m^2L^2 \geq 0$, and that combination is exactly $\mathcal{C} + \frac{1}{4}$, the Hardy margin of the energy. A tachyon in AdS is harmless precisely as long as its destabilizing potential energy cannot overcome the curvature-induced gradient energy quantified by Hardy's inequality.

The alternate-quantization window. The BF bound opens an interesting sub-regime. By Theorem 53.18 the smaller-root mode z^{Δ_-} is normalizable when $\Delta_- > (d-2)/2$. Writing $\nu \equiv \sqrt{d^2/4 + m^2L^2} \geq 0$ so that $\Delta_\pm = \frac{d}{2} \pm \nu$, the condition $\Delta_- = \frac{d}{2} - \nu > \frac{d-2}{2} = \frac{d}{2} - 1$ reads simply $\nu < 1$. Squaring, $\nu^2 = d^2/4 + m^2L^2 < 1$, i.e.

$$\boxed{-\frac{d^2}{4} \;\leq\; m^2L^2 \;<\; -\frac{d^2}{4} + 1.} \tag{53.135}$$

In this window *both* falloffs are normalizable, and one is free to choose which mode to hold fixed as the source. The standard choice assigns the boundary operator dimension Δ_+; the *alternate quantization* instead fixes β and lets α fluctuate, producing a dual operator of dimension Δ_-. The two choices are distinct CFTs related by a Legendre transform (a double-trace flow), a structure Chapter 54 will revisit. Outside the window, $\nu \geq 1$, the slower mode is non-normalizable and only the standard quantization with $\Delta = \Delta_+$ is available. Figure 53.6 displays the whole landscape: the two roots as functions of m^2L^2, the BF floor where they merge, the unitarity bound they must respect, and the alternate-quantization strip.

53.4.5 Consistency with the CFT unitarity bound

The identification $\Delta = \Delta_+$ is only admissible if the resulting dimension is one a unitary CFT can actually carry. The relevant constraint is the scalar unitarity bound proved in Chapter 47 from positivity of the conformal Hilbert space: a scalar primary in $d \geq 2$ obeys

$$\Delta \;\geq\; \frac{d-2}{2}, \tag{53.136}$$

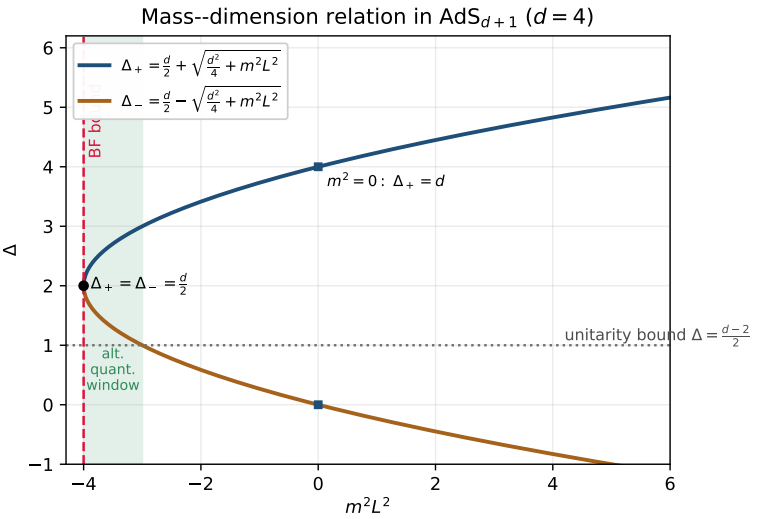


Figure 53.6: The mass–dimension relation $\Delta(\Delta - d) = m^2L^2$ for AdS$_{d+1}$ with $d = 4$. The two branches $\Delta_\pm = \frac{d}{2} \pm \sqrt{d^2/4 + m^2L^2}$ meet at the Breitenlohner–Freedman point $m^2L^2 = -d^2/4$, $\Delta = d/2$ (black dot). To the left of the BF bound (red dashed) the dimensions become complex and the AdS vacuum is unstable. The shaded strip $-d^2/4 \le m^2L^2 < -d^2/4 + 1$ is the alternate-quantization window, where both modes are normalizable. The dotted line marks the scalar unitarity bound $\Delta = (d-2)/2$ of Theorem 47.18: the upper branch Δ_+ never dips below it, and the lower branch Δ_- touches it exactly at the right edge of the window. The massless point $m^2 = 0$ gives $\Delta_+ = d$, a marginal operator.

with equality only for a free field (Theorem 47.18). We check that Δ_+ from Eq. (53.113) respects Eq. (53.136) for every mass allowed by the BF bound.

Proposition 53.21 (Holographic dimensions are unitary)**.** *For all $m^2L^2 \ge -d^2/4$, the standard-quantization dimension $\Delta_+ = \frac{d}{2} + \sqrt{d^2/4 + m^2L^2}$ satisfies the scalar unitarity bound Eq. (53.136), with equality nowhere in the allowed range; and the alternate-quantization dimension Δ_-, defined only in the window Eq. (53.135), satisfies it as well, saturating it exactly at the window's upper edge.*

Proof. For the standard branch, $\Delta_+ \ge \frac{d}{2}$ as noted in Theorem 53.17, and $\frac{d}{2} - \frac{d-2}{2} = 1 > 0$, so

$$\Delta_+ \ge \frac{d}{2} = \frac{d-2}{2} + 1 > \frac{d-2}{2}, \tag{53.137}$$

strictly above the bound by at least one unit for every admissible m^2; the bound is never saturated by Δ_+, consistent with the fact that a generic bulk scalar is dual to an interacting (non-free) operator. For the alternate branch, write

$\Delta_- = \frac{d}{2} - \nu$ with $0 \le \nu < 1$ throughout the window Eq. (53.135). Then

$$\Delta_- = \frac{d}{2} - \nu \; > \; \frac{d}{2} - 1 = \frac{d-2}{2}, \tag{53.138}$$

so Δ_- lies strictly above the unitarity bound for $\nu < 1$ and approaches it as $\nu \to 1^-$, i.e. as $m^2L^2 \to (-d^2/4+1)^-$. Exactly at $\nu = 1$ the window closes and $\Delta_- = (d-2)/2$ would equal the free-field value; this is the boundary case where the alternate operator becomes free, consistent with the unitarity bound being saturated only by free fields. □

The agreement is a genuine consistency check, not a definition: the unitarity bound Eq. (53.136) was derived in Chapter 47 purely from the representation theory of the conformal algebra, with no reference to a bulk, while Δ_+ here came from solving a wave equation in a $(d+1)$-dimensional geometry. That the two independently computed inequalities are compatible — and that the alternate-quantization window closes exactly where Δ_- would otherwise violate unitarity — is one of the first quantitative signs that the bulk and boundary descriptions are describing the same physics.

Worked Example 53.22 (Reading dimensions off explicit masses)**.** We collect the dimension assignments for several representative bulk scalars in AdS_5 ($d = 4$), the case relevant to the maximally supersymmetric example, and exhibit the exact radial solutions that make the falloff structure manifest. With $d = 4$ the roots are $\Delta_\pm = 2 \pm \sqrt{4 + m^2L^2}$.

(i) Massless scalar, $m^2 = 0$. Then $\Delta_+ = 4$, $\Delta_- = 0$. The dual operator is marginal, dimension 4; the $z^{\Delta_-} = z^0$ falloff is a constant, the source being a spacetime-constant coupling — exactly the behaviour of a marginal deformation of the boundary theory.

(ii) A BF-saturating tachyon, $m^2L^2 = -4 = -d^2/4$. The square root vanishes, $\Delta_+ = \Delta_- = 2 = d/2$. The two falloffs degenerate and (per the caution in Section 53.4.2) the second solution carries a logarithm: $\phi \sim z^2(\alpha + \beta \log z)$. The dual operator has dimension exactly $d/2 = 2$, sitting above the unitarity bound $(d-2)/2 = 1$ by one unit, as Theorem 53.21 guarantees.

(iii) A tachyon in the alternate window, $m^2L^2 = -15/4$. Here $\sqrt{4 - 15/4} = \sqrt{1/4} = 1/2 = \nu < 1$, so both modes are normalizable: $\Delta_+ = 2 + \frac{1}{2} = \frac{5}{2}$ and $\Delta_- = 2 - \frac{1}{2} = \frac{3}{2}$. Standard quantization gives a dimension-5/2 operator; alternate quantization a dimension-3/2 operator. Both exceed the unitarity bound 1.

The exact radial profile. For completeness, the full solution of the radial ODE Eq. (53.107) is available in closed form. With $\nu = \sqrt{d^2/4 + m^2L^2}$ and a Euclidean (spacelike) boundary momentum $k^2 = -q^2$, $q > 0$, substituting $f_k(z) = z^{d/2}\, h(qz)$ into Eq. (53.107) converts it into the modified Bessel equation for h,

$$(qz)^2 h'' + (qz)\, h' - \big((qz)^2 + \nu^2\big) h = 0, \tag{53.139}$$

whose solutions are the modified Bessel functions $K_\nu(qz)$ and $I_\nu(qz)$. The general radial solution is therefore

$$f_k(z) = z^{d/2}\big[A\,K_\nu(qz) + B\,I_\nu(qz)\big]. \tag{53.140}$$

Regularity in the deep interior $z \to \infty$ (the Poincaré horizon) selects K_ν, since $I_\nu(qz) \sim e^{qz}/\sqrt{qz}$ blows up while $K_\nu(qz) \sim e^{-qz}/\sqrt{qz}$ decays. Near the boundary $z \to 0$ the small-argument expansions $K_\nu(u) \sim \frac{1}{2}\Gamma(\nu)(u/2)^{-\nu} + \frac{1}{2}\Gamma(-\nu)(u/2)^{\nu}$ reproduce exactly the two falloffs of Eq. (53.115):

$$f_k(z) \xrightarrow{z\to 0} z^{d/2}\Big[c_-\,(qz)^{-\nu} + c_+\,(qz)^{\nu}\Big] = c'_-\,z^{\Delta_-} + c'_+\,z^{\Delta_+}, \tag{53.141}$$

with $\Delta_\pm = d/2 \pm \nu$. The exact solution thus confirms the indicial analysis term by term: the two near-boundary powers are precisely $\Delta_\pm$, the regular-in-the-interior solution generically excites both, and the ratio c'_+/c'_- — the response per unit source — is the object whose boundary Fourier transform will become the two-point function $\langle\mathcal{O}\mathcal{O}\rangle$ in Chapter 54.

With the mass–dimension relation Eq. (53.112) established, its roots identified, the source/vev meaning of the two falloffs fixed, and stability pinned by the BF bound Eq. (53.134), we now have the single most reused technical fact of Part XV: a bulk scalar of mass m encodes a boundary operator of dimension $\Delta_+ = \frac{d}{2} + \sqrt{d^2/4 + m^2L^2}$, with its slower, non-normalizable falloff acting as that operator's source. The next section takes this entry as the seed of the full field–operator dictionary — extending it to fields of nonzero spin and charge, matching bulk gauge symmetries to boundary global symmetries, and assembling the parameter map — before Chapter 54 converts the source-to-response relation into honest correlation functions.

53.5 The Field–Operator Dictionary

The previous section solved a single problem completely: a free scalar of mass m propagating in AdS_{d+1} has two near-boundary falloffs, $z^{d-\Delta}$ and z^{Δ}, related to the mass by the indicial equation $\Delta(\Delta - d) = m^2L^2$, with the non-normalizable mode ϕ_0 identified as an external *source* and the normalizable mode as the *vacuum expectation value* of a dual operator (Section 53.4). That was one entry. The business of this section is to turn that single lesson into a complete *dictionary*: a precise, quantum-number-by-quantum-number correspondence between the local fields of the bulk theory and the local operators of the boundary CFT. We will state the map, derive its three canonical entries (the stress tensor, conserved currents, and marginal couplings), prove that bulk gauge symmetries become boundary global symmetries, record the parameter map that locates the classical-gravity regime, and finally count how many boundary degrees of freedom a given

bulk geometry can hold — which is where the word "holographic" earns its quantitative meaning. Throughout, "boundary" means the conformal boundary of Section 53.3, on which the dual CFT_d lives, and "bulk" means the gravitational theory in AdS_{d+1}.

53.5.1 The map: bulk fields and boundary primaries

The organizing principle is symmetry. The isometry group of AdS_{d+1} is $\mathrm{SO}(2,d)$, and this is precisely the conformal group of the d-dimensional boundary (Section 53.3). A bulk field, being a field in AdS, falls into a representation of $\mathrm{SO}(2,d)$ acting as isometries; a boundary primary operator, by Theorem 47.15, falls into a representation of the same $\mathrm{SO}(2,d)$ acting as conformal transformations. The correspondence demands that these be the *same* representation. This single requirement fixes the entire kinematic content of the dictionary.

Definition 53.23 (The field–operator map)**.** The AdS/CFT dictionary associates to each bulk field Φ propagating in AdS_{d+1} a single *primary* operator $\mathcal{O}_\Phi$ of the boundary CFT_d, in such a way that Φ and $\mathcal{O}_\Phi$ carry identical quantum numbers under the common symmetry group $\mathrm{SO}(2,d)$ and under any internal symmetries:

1. the bulk mass m of Φ determines the scaling dimension Δ of $\mathcal{O}_\Phi$, through the indicial relation of Section 53.4;
2. the bulk Lorentz spin ℓ of Φ (its $\mathrm{SO}(d)$ representation as a tensor on the boundary directions) equals the spin ℓ of $\mathcal{O}_\Phi$;
3. the charge of Φ under a bulk gauge symmetry equals the charge of $\mathcal{O}_\Phi$ under the corresponding boundary global symmetry.

The non-normalizable boundary value of Φ is the field-theory *source* coupled to $\mathcal{O}_\Phi$; the normalizable mode is $\langle \mathcal{O}_\Phi \rangle$.

Three points make this more than a relabelling. First, the operators that appear are *primaries*, not descendants: descendants are obtained by acting with the momentum generator P_μ, i.e. by differentiating, and so carry no independent data; a single bulk field accounts for an entire conformal multiplet, the primary together with its tower of descendants, exactly as a single solution of the bulk wave equation generates, by acting with the isometries, an entire $\mathrm{SO}(2,d)$ orbit. Second, the operators are *single-trace*. In the large-N gauge theories for which the correspondence is sharpest (Section 53.1.1), gauge-invariant operators organize into single-trace combinations such as $\mathrm{tr}(\phi^k)$ and multi-trace products of these; the $1/N$ counting of Section 53.1.4 shows that single-trace operators create approximately stable single-particle states, while multi-trace operators create multi-particle states. A single bulk field, quantized, creates one-particle states; it is therefore dual to a single-trace operator, and multi-particle bulk states

correspond to multi-trace operators. Third, the map is a bijection *between the elementary fields of the bulk and the single-trace primaries of the boundary*: there are exactly as many propagating bulk fields as there are single-trace primaries, and the bulk Lagrangian's field content is the boundary operator spectrum read off in another language.

The precise interface to the rest of the chapter is the boundary condition that promotes the leading falloff to a source. Repeating the scalar result of Section 53.4 in the form we will feed to the generating functional,

$$\Phi(z,x) \xrightarrow{z\to 0} z^{\,d-\Delta}\left[\phi_0(x)+O(z^2)\right] + z^{\,\Delta}\left[\tfrac{1}{2\Delta-d}\,\langle\mathcal{O}(x)\rangle + O(z^2)\right], \quad (53.142)$$

where $\Delta = \Delta_+ = \frac{d}{2} + \sqrt{\frac{d^2}{4} + m^2L^2}$ is the larger root and ϕ_0 is identified with the field-theory source for $\mathcal{O} \equiv \mathcal{O}_\Phi$. We will refer to the leading condition $\Phi(z\to 0) \sim z^{d-\Delta}\phi_0(x)$ as the *GKPW boundary condition*: it is the boundary datum we hold fixed in the bulk variational problem, and the next section will differentiate the on-shell action with respect to ϕ_0 to extract correlators of $\mathcal{O}$.

53.5.2 The three canonical entries

Every realization of the correspondence shares three universal operators, because every local CFT has them and every gravitational bulk has the fields dual to them. We derive each from two sides — the boundary unitarity/conservation constraint of Chapter 47, and the bulk falloff — and check that they agree.

The stress tensor. Every local quantum field theory possesses a conserved, symmetric stress tensor $T_{\mu\nu}$, constructed in Theorem 7.10 as the Noether current of spacetime translations and made symmetric and traceless in Theorem 7.45. In a CFT, $T_{\mu\nu}$ is a primary of spin $\ell = 2$ whose dimension is fixed: the conserved charges $P_\mu = \int d^{d-1}x\, T^0{}_\mu$ must be dimensionless generators of translations, so $[T_{\mu\nu}] = d$, i.e. $\Delta_T = d$. This is not an accident of dimensional analysis but a consequence of conformal representation theory: a spin-2 primary obeys the unitarity bound $\Delta \geq d$ of Theorem 47.18, with equality *iff* $\partial^\mu T_{\mu\nu} = 0$. The worked example Theorem 47.19 showed precisely that the divergence descendant $P^\mu\,|T_{\mu\nu}\rangle$ has norm proportional to $\Delta - d$, so conservation is the statement that this null state decouples, forcing $\Delta_T = d$. On the bulk side, the field dual to $T_{\mu\nu}$ must be a spin-2 field whose dimension comes out to d: that is the metric fluctuation, the graviton g_{MN}. The presence of a dynamical graviton in the bulk is therefore equivalent to the presence of a stress tensor in the boundary theory — and conversely, *any* CFT, having a stress tensor, is dual to a theory of gravity. This is the structural reason the bulk is gravitational.

Conserved currents. A continuous global symmetry of the boundary CFT supplies, by Noether's theorem (Theorem 7.10), a conserved current J^μ. It is a

spin-1 primary, and conservation $\partial_\mu J^\mu = 0$ saturates the spin-1 unitarity bound $\Delta \geq d-1$: in Theorem 47.19 the divergence state $P^\mu |J_\mu\rangle$ was shown to have norm $\propto [\Delta - (d-1)]$, null exactly when $\Delta = d-1$. So a conserved current is a spin-1 primary of dimension $\Delta_J = d-1$. The bulk field carrying spin 1 and reproducing this dimension is a gauge field A_M. Each boundary global symmetry is thus dual to a bulk gauge field; the next subsection promotes this correspondence into a theorem about how the symmetries themselves are related.

Marginal operators. A massless bulk scalar, $m = 0$, has $\Delta(\Delta - d) = 0$, hence $\Delta = d$ (the larger root). A scalar primary of dimension exactly d is *marginal*: the integrated operator $\int d^d x\, \mathcal{O}$ is dimensionless and can be added to the action as a coupling that preserves conformal invariance at leading order. The bulk avatar of such a deformation is shifting the boundary value ϕ_0 of the massless scalar to a constant, which changes a bulk coupling constant; on the boundary this is turning on a marginal coupling. The prototype is the dilaton, whose constant boundary value is the string coupling, dual to the gauge coupling — an entry we make precise in Section 53.5.4.

It remains to verify, on the bulk side, that a massless spin-1 field reproduces $\Delta = d-1$ and a massless spin-2 field reproduces $\Delta = d$. For the scalar this was the content of Section 53.4. For spin we carry out the vector indicial analysis explicitly; the tensor case is identical in structure and we state its result.

Worked Example 53.24 (Falloff of a massive bulk vector and the current dimension)**.** Work in the Poincaré patch with metric $\mathrm{d}s^2 = (L^2/z^2)(\eta_{\mu\nu}\mathrm{d}x^\mu \mathrm{d}x^\nu - \mathrm{d}z^2)$ (recall Eq. (53.59), mostly-minus), bulk indices $M = (z, \mu)$, and a massive vector field A_M governed by the Proca equation, which in the book's mostly-minus signature reads

$$\frac{1}{\sqrt{|g|}}\,\partial_M\!\left(\sqrt{|g|}\,F^{MN}\right) = -m^2 A^N, \qquad F_{MN} = \partial_M A_N - \partial_N A_M. \tag{53.143}$$

The near-boundary falloff is governed by the leading z-behavior at vanishing boundary momentum, so set $\partial_\mu(\,\cdot\,) \to 0$ and seek a solution of the boundary-component equation with $A_z = 0$ and $A_\nu = z^\alpha a_\nu$, a_ν constant. With $\sqrt{|g|} = (L/z)^{d+1}$ and $g^{MN} = (z/L)^2\,\eta^{MN}$ (boundary block) and $g^{zz} = -(z/L)^2$, the only nonzero field strength is $F_{z\nu} = \partial_z A_\nu = \alpha z^{\alpha-1} a_\nu$. Raising indices,

$$F^{z\nu} = g^{zz} g^{\nu\rho} F_{z\rho} = -\Big(\frac{z}{L}\Big)^4 \alpha z^{\alpha-1} a^\nu = -\frac{\alpha}{L^4}\, z^{\alpha+3} a^\nu. \tag{53.144}$$

Then

$$\sqrt{|g|}\,F^{z\nu} = L^{d+1} z^{-(d+1)} \cdot \Big(-\frac{\alpha}{L^4}\Big) z^{\alpha+3} a^\nu = -\alpha L^{d-3}\, z^{\alpha-d+2} a^\nu,$$
$$\partial_z\!\left(\sqrt{|g|}\,F^{z\nu}\right) = -\alpha(\alpha - d + 2) L^{d-3}\, z^{\alpha-d+1} a^\nu,$$

$$\frac{1}{\sqrt{|g|}}\partial_z(\sqrt{|g|}\,F^{z\nu}) = \frac{z^{d+1}}{L^{d+1}}\cdot(-\alpha(\alpha-d+2)L^{d-3})z^{\alpha-d+1}a^\nu = -\frac{\alpha(\alpha-d+2)}{L^4}\,z^{\alpha+2}a^\nu. \tag{53.145}$$

The right-hand side of Eq. (53.143) is $-m^2A^\nu = -m^2 g^{\nu\rho}A_\rho = -m^2(z/L)^2 z^\alpha a^\nu = -(m^2/L^2)z^{\alpha+2}a^\nu$. Matching the coefficient of $z^{\alpha+2}$ (the overall minus signs cancel) gives the *vector indicial equation*

$$\alpha\,(\alpha-d+2) = m^2L^2. \tag{53.146}$$

Its roots are $\alpha_\pm = \frac{d-2}{2} \pm \sqrt{\frac{(d-2)^2}{4} + m^2L^2}$. The smaller root α_- is the source falloff (non-normalizable), the larger α_+ the vev falloff; the dimension of the dual spin-1 operator is $\Delta = \alpha_+ + 1$, equivalently $m^2L^2 = (\Delta-1)(\Delta-d+1)$, the spin-1 version of the mass–dimension relation. For the massless field $m = 0$ the roots are $\alpha = 0$ and $\alpha = d-2$: the source mode is the *constant* boundary value $A_\nu \to a_\nu(x)$, a background gauge field, and the vev mode falls as z^{d-2}, giving

$$\Delta_J = (d-2)+1 = d-1, \tag{53.147}$$

exactly the dimension forced on a conserved current by boundary unitarity in Theorem 47.19. The two derivations agree.

The graviton repeats this verbatim with the spin-2 wave operator on AdS_{d+1}. The transverse-traceless metric fluctuation $h_{\mu\nu} = g_{\mu\nu} - \bar{g}_{\mu\nu}$ obeys, at zero momentum, $h_{\mu\nu} = z^\beta h^0_{\mu\nu}$ with indicial equation $\beta(\beta-d) = m^2L^2$ shifted to the spin-2 form $m^2L^2 = (\Delta+\ell-2)(\Delta-\ell-d+2)$ at $\ell = 2$; for the massless graviton, $m = 0$, the roots are $\beta = 0$ (source: the boundary metric $g^{(0)}_{\mu\nu}$) and $\beta = d$ (vev: $\langle T_{\mu\nu}\rangle$), so

$$\Delta_T = d, \tag{53.148}$$

matching the conserved spin-2 result of Theorem 47.19. The general spin-ℓ pattern, valid for $\ell \geq 1$, is $m^2L^2 = (\Delta+\ell-2)(\Delta-\ell-d+2)$; it reproduces the unitarity-saturating value $\Delta = \ell + d - 2$ for every massless (conserved) bulk field, and $\Delta > \ell + d - 2$ for massive bulk fields, the dual of the statement (Theorem 47.19) that interacting CFTs carry no conserved higher-spin currents beyond $T_{\mu\nu}$. The scalar is *not* the $\ell = 0$ special case of this formula — the bulk "mass" is normalized differently for a tensor field — and obeys instead the relation $\Delta(\Delta-d) = m^2L^2$ of Eq. (53.112), so that a massless scalar is the marginal $\Delta = d$ rather than $\Delta = d-2$.

We assemble the canonical dictionary in Table 53.2.

53.5.3 Gauge symmetries become global symmetries

We saw that a boundary global symmetry is dual to a bulk gauge field. The relationship between the *symmetry operations* is sharper than the relationship

Bulk field Φ	**Boundary operator $\mathcal{O}_\Phi$**	**spin ℓ**	**dimension Δ**
scalar, mass m	scalar primary $\mathcal{O}$	0	$\frac{d}{2}+\sqrt{\frac{d^2}{4}+m^2L^2}$
massless scalar	marginal operator	0	d
gauge field A_M	conserved current J^μ	1	$d-1$
graviton g_{MN}	stress tensor $T_{\mu\nu}$	2	d
massive vector, mass m	spin-1 primary	1	$\frac{d}{2}+\sqrt{\frac{(d-2)^2}{4}+m^2L^2}$
Dirac field, mass m	fermionic primary	$\frac{1}{2}$	$\frac{d}{2}+\lvert m\rvert L$

Table 53.2: The field–operator dictionary: each bulk field is dual to a single-trace boundary primary of matching spin, with dimension fixed by the bulk mass through the spin-ℓ indicial relation. The fermion entry uses the first-order Dirac indicial analysis ($\Delta=\frac{d}{2}+|m|L$).

between the operators, and it is the cleanest illustration of how a redundancy in the bulk becomes a physical symmetry on the boundary.

Proposition 53.25 (Bulk gauge ⇒ boundary global)**.** *Let A_M be a bulk gauge field dual to the boundary current J^μ, with bulk gauge transformations $A_M\to A_M+\partial_M\lambda$. Gauge transformations whose parameter λ vanishes at the conformal boundary are pure bulk redundancies and act trivially on boundary data. Gauge transformations whose parameter approaches a nonzero boundary value $\lambda(z\to 0,x)=\lambda_0(x)$ act on the source $a_\mu(x)$ for J^μ as the background-gauge transformation $a_\mu\to a_\mu+\partial_\mu\lambda_0$, and a* constant *$\lambda_0$ implements the boundary global symmetry rotation under which J^μ is the Noether current.*

Proof. By Theorem 53.24 the source for J^μ is the leading boundary value $a_\mu(x)=A_\mu(z\to 0,x)$, the non-normalizable mode with $\alpha=0$. Under a bulk gauge transformation $A_\mu\to A_\mu+\partial_\mu\lambda$, the boundary value transforms as

$$a_\mu(x)\;\to\; a_\mu(x)+\partial_\mu\lambda_0(x),\qquad \lambda_0(x)\equiv\lim_{z\to 0}\lambda(z,x). \tag{53.149}$$

If $\lambda_0\equiv 0$ the source is unchanged and, since the boundary partition function depends on A_M only through the boundary datum a_μ that it imposes (the content of the GKPW boundary condition Eq. (53.142) applied to the vector), the entire boundary physics is untouched: such "small" gauge transformations are redundancies of the bulk description with no boundary counterpart. This is the standard fact that local gauge symmetry is not a symmetry but a redundancy; what is special here is the role of the boundary value.

When $\lambda_0\neq 0$ the transformation is "large": it changes the source. The source a_μ is the field-theory background gauge field that couples to the current through $\delta S_{\rm CFT}=\int d^dx\, a_\mu J^\mu$. Equation Eq. (53.149) is then precisely a background-gauge

transformation of the boundary theory, under which the generating functional is invariant by the conservation Ward identity $\partial_\mu J^\mu = 0$ (this is the defining property of coupling a conserved current to a background field). For *constant* $\lambda_0 = \theta$, the shift $a_\mu \to a_\mu$ is trivial on the source but acts on charged operators by the rigid phase rotation generated by the conserved charge $Q = \int d^{d-1}x\, J^0$: it is the boundary global symmetry transformation $\mathcal{O} \to e^{i\theta\, q}\mathcal{O}$ for an operator of charge q. Thus the global symmetry group of the boundary is the boundary value of the bulk gauge group, and J^μ is its Noether current. □

The same logic, applied to the metric, identifies bulk diffeomorphisms with boundary stress-tensor conservation.

Proposition 53.26 (Bulk diffeomorphisms ⇒ stress-tensor conservation)**.** *The source for $T_{\mu\nu}$ is the boundary metric $g^{(0)}_{\mu\nu}$, the leading value of the bulk metric on the conformal boundary. A bulk diffeomorphism generated by a vector ξ^M with boundary value $\xi^\mu(x)$ acts on the source as $\delta g^{(0)}_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$, and invariance of the boundary generating functional under this shift is the conservation law $\nabla_\mu \langle T^{\mu\nu} \rangle = 0$.*

Proof. By Eq. (53.148) the source for the spin-2 primary $T_{\mu\nu}$ is the $\beta = 0$ boundary value of the metric, $g^{(0)}_{\mu\nu}(x) = \lim_{z\to 0} (z/L)^2 g_{\mu\nu}(z, x)$ (the conformal factor stripping off the Weyl weight). The stress tensor couples to a background metric through $\delta S_{\mathrm{CFT}} = \frac{1}{2}\int d^d x \sqrt{-g^{(0)}}\, T^{\mu\nu} \delta g^{(0)}_{\mu\nu}$, the very definition of $T_{\mu\nu}$ as the response to a metric variation that underlies the Belinfante–Rosenfeld construction of Theorem 7.32. A bulk diffeomorphism $x^M \to x^M + \xi^M$ that preserves the asymptotic form of the metric acts on the boundary metric as a boundary diffeomorphism, $\delta g^{(0)}_{\mu\nu} = \nabla_\mu \xi_\nu + \nabla_\nu \xi_\mu$ (the Lie derivative of $g^{(0)}$ along ξ^μ). Substituting into the coupling and integrating by parts,

$$\delta S_{\mathrm{CFT}} = \int d^d x \sqrt{-g^{(0)}}\, T^{\mu\nu} \nabla_\mu \xi_\nu = -\int d^d x \sqrt{-g^{(0)}}\, \xi_\nu\, \nabla_\mu T^{\mu\nu}. \tag{53.150}$$

Bulk diffeomorphism invariance forces $\delta S_{\mathrm{CFT}} = 0$ for all ξ, hence $\nabla_\mu \langle T^{\mu\nu} \rangle = 0$: the boundary stress tensor is conserved precisely because the bulk theory is diffeomorphism invariant. Tracelessness $\langle T^\mu{}_\mu \rangle = 0$, the second hallmark of a CFT stress tensor (Theorem 7.45), arises in the same way from invariance under bulk diffeomorphisms that rescale z and so Weyl-rescale $g^{(0)}$. □

Physical Insight: No global symmetries in quantum gravity

Theorem 53.25 runs in a definite direction: every boundary global symmetry descends from a bulk *gauge* symmetry, and every conserved current is dual to a bulk gauge field. There is no bulk construction that produces a boundary global symmetry without a corresponding bulk gauge field.

Read on the gravity side, this is the holographic incarnation of the folklore that *a consistent theory of quantum gravity has no global symmetries*: any would-be global symmetry must be gauged. The dictionary makes the slogan precise — a global symmetry of the CFT is a gauge symmetry of the bulk, seen from the boundary.

53.5.4 The parameter map

The dictionary so far is kinematic: it matches representations. To use the correspondence as a computational tool one also needs the *parameter map* relating the dimensionless couplings of the two descriptions, because the entire utility of holography rests on the two theories being weakly coupled in opposite regimes. We state the canonical entries, flagging honestly (per the chapter's theorem-versus-conjecture contract) that the specific numerical map below is read off from the string construction of Section 53.2.3 and is part of the *conjectured* correspondence, whereas the symmetry and dimension entries above are forced by representation theory alone.

The boundary theory is characterized by its rank N and 't Hooft coupling $\lambda_{\rm tH} = g_{\rm YM}^2 N$ (the combination held fixed in the large-N expansion of Section 53.1.1). The bulk is characterized by the AdS radius L measured in string units ℓ_s and in Planck units $\ell_{\rm P}$, and by the string coupling g_s. The prototype correspondence (Section 53.2.3) fixes

$$\frac{L^4}{\ell_s^4} = \lambda_{\rm tH} = g_{\rm YM}^2 N, \qquad g_s \sim g_{\rm YM}^2 \sim \frac{\lambda_{\rm tH}}{N}. \tag{53.151}$$

Two limits are now legible. Suppressing stringy effects requires the curvature radius to be large in string units, $L \gg \ell_s$, i.e. $\lambda_{\rm tH} \gg 1$: *strong* 't Hooft coupling makes the bulk look like a smooth geometry rather than a fuzzy string. Suppressing quantum-gravity (loop) effects requires the string coupling to be small, $g_s \ll 1$ at fixed $\lambda_{\rm tH}$, i.e. $N \to \infty$: *large* N makes the bulk classical. Together,

$$\boxed{\text{classical gravity in AdS} \iff N \to \infty,\ \lambda_{\rm tH} \to \infty} \tag{53.152}$$

which is exactly the regime of the boundary gauge theory that is *least* accessible to perturbation theory. This inversion — the hard limit of one description is the easy limit of the other — is the entire point. A constant boundary value of the bulk dilaton, the massless scalar of Section 53.5.2, sets g_s and hence (through Eq. (53.151)) the marginal gauge coupling $g_{\rm YM}^2$, completing the marginal-operator entry of the dictionary.

53.5.5 Counting holographic degrees of freedom

The dictionary's final entry is a number: how many degrees of freedom does the boundary theory have, and how is that encoded in the bulk? The boundary measure of "number of local degrees of freedom" is the central charge, the coefficient of the stress-tensor two-point function (Section 47.4.3), conventionally called C_T; for a free field it counts components, and for a large-N gauge theory it scales as the number of fields, $C_T \sim N^2$. The bulk measure is geometric. We now show they match, with the bulk Planck length setting the conversion, and that the match is the holographic principle made quantitative.

Proposition 53.27 (Holographic central charge)**.** *The central charge of the boundary CFT$_d$ dual to Einstein gravity in* AdS_{d+1} *with radius L and Newton constant G_{d+1} scales as*

$$C_T \;\sim\; \frac{L^{d-1}}{G_{d+1}} \;\sim\; \Big(\frac{L}{\ell_{\mathrm{P}}}\Big)^{d-1}, \tag{53.153}$$

where $\ell_{\mathrm{P}}^{d-1} \sim G_{d+1}$ is the $(d{+}1)$-dimensional Planck length. A large central charge therefore requires a curvature radius large in Planck units, and in the prototype gauge/gravity dual $C_T \sim N^2$.

Proof. We give the Susskind–Witten counting, which fixes the scaling without solving any field equation. Regulate the boundary by cutting off the radial integration at $z = \varepsilon$, so the boundary CFT acquires a short-distance cutoff ε (the radial coordinate is the inverse energy scale, Section 53.3). Consider a region of the boundary of coordinate volume V_{d-1} in the spatial directions. Two counts of its information content must agree.

Field-theory count. With a UV cutoff ε, the boundary theory has of order one cell per coordinate volume ε^{d-1}, and each cell carries C_T independent degrees of freedom (this is what C_T counts). Hence

$$N_{\mathrm{dof}} \;\sim\; C_T\,\frac{V_{d-1}}{\varepsilon^{\,d-1}}. \tag{53.154}$$

Gravity count. The same boundary region sits at $z = \varepsilon$ in the bulk, where the induced metric is $(L/\varepsilon)^2$ times the flat one, so its proper spatial area is

$$A \;=\; \Big(\frac{L}{\varepsilon}\Big)^{d-1} V_{d-1}. \tag{53.155}$$

The holographic principle — here used as an input, justified in the bulk by the Bekenstein–Hawking area law and proved in the AdS context by the entropy bounds of Chapters 55 and 56 — caps the number of degrees of freedom on a surface by its area in Planck units,

$$N_{\mathrm{dof}} \;\sim\; \frac{A}{4G_{d+1}} \;=\; \frac{1}{4G_{d+1}}\Big(\frac{L}{\varepsilon}\Big)^{d-1} V_{d-1}. \tag{53.156}$$

Equating Eq. (53.154) and Eq. (53.156), the regulator ε and the volume V_{d-1} cancel identically — the $\varepsilon^{-(d-1)}$ scaling is the same on both sides, which is itself the nontrivial check that a d-dimensional field theory can be holographically encoded on the boundary of a $(d{+}1)$-dimensional geometry — and leave

$$C_T \;\sim\; \frac{L^{\,d-1}}{G_{d+1}}, \tag{53.157}$$

which is Eq. (53.153). Writing $G_{d+1} \sim \ell_{\mathrm{P}}^{d-1}$ gives the dimensionless form $C_T \sim (L/\ell_{\mathrm{P}})^{d-1}$. □

Worked Example 53.28 (The N^2 count for the canonical $\mathrm{AdS}_5 \times S^5$ dual)**.** Specialize to $d = 4$, the boundary dimension of the prototype correspondence. Here the ten-dimensional bulk is $\mathrm{AdS}_5 \times S^5$ with both factors of radius L; reducing on the S^5 converts the ten-dimensional Newton constant G_{10} into a five-dimensional one,

$$\frac{1}{G_5} \;\sim\; \frac{\mathrm{Vol}(S^5)}{G_{10}} \;\sim\; \frac{L^5}{G_{10}}. \tag{53.158}$$

By Theorem 53.27 with $d = 4$,

$$C_T \;\sim\; \frac{L^3}{G_5} \;\sim\; \frac{L^3 \cdot L^5}{G_{10}} \;=\; \frac{L^8}{G_{10}}. \tag{53.159}$$

The string-theoretic inputs of Section 53.2.3 supply $G_{10} \sim g_s^2\,\ell_s^8$ and $L^4 \sim g_s N\,\ell_s^4$ (the latter is the $d = 4$ case of Eq. (53.151), since $\lambda_{\mathrm{tH}} = g_{\mathrm{YM}}^2 N \sim g_s N$ and $L^4/\ell_s^4 = \lambda_{\mathrm{tH}}$). Then $L^8 \sim (g_s N)^2 \ell_s^8$ and

$$C_T \;\sim\; \frac{(g_s N)^2\,\ell_s^8}{g_s^2\,\ell_s^8} \;=\; N^2. \tag{53.160}$$

The string coupling and string length have cancelled completely, leaving a pure number N^2 — precisely the count of independent components of an $N \times N$ adjoint-valued field, the field content of the boundary $\mathrm{SU}(N)$ gauge theory. The central charge that conformal symmetry left undetermined in Chapter 47 is, holographically, the curvature radius of the dual geometry in Planck units, and it equals N^2.

Physical Insight: The holographic principle, quantitative

Theorem 53.27 is the holographic principle with the proportionality constant filled in. A boundary theory with C_T degrees of freedom requires a bulk whose curvature radius is $C_T^{1/(d-1)}$ Planck lengths: the more degrees of freedom the field theory has, the larger the geometry it carries, and the weaker its quantum-gravitational fluctuations. Classical, weakly curved gravity ($L \gg \ell_{\mathrm{P}}$) is the dual of a field theory with a huge number of

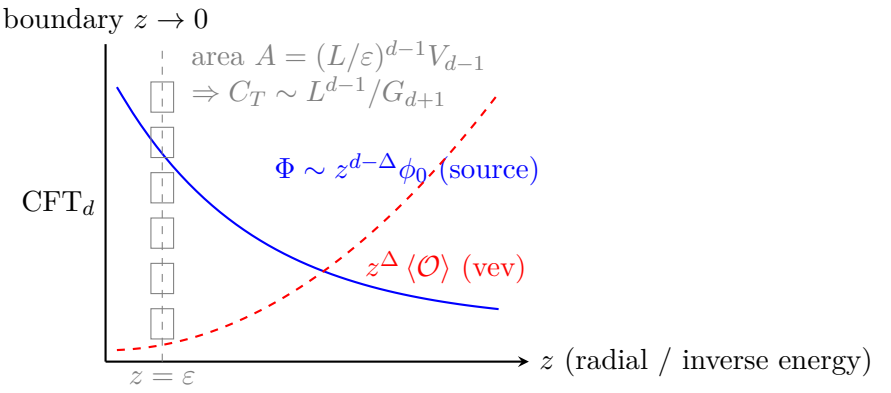


Figure 53.7: The radial direction of the Poincaré patch encodes the dictionary. A bulk field Φ splits near the boundary ($z \to 0$) into a non-normalizable source mode $z^{d-\Delta}\phi_0$ (Eq. (53.142), the GKPW boundary condition) and a normalizable vev mode $z^{\Delta}\langle\mathcal{O}\rangle$. Regulating at $z = \varepsilon$ cuts the boundary into Planck cells whose count fixes the central charge, $C_T \sim (L/\ell_{\mathrm{P}})^{d-1}$ (Theorem 53.27).

degrees of freedom ($C_T \gg 1$, i.e. $N \gg 1$). A d-dimensional theory's worth of information is stored on the $(d-1)$-dimensional spatial boundary of the bulk at one degree of freedom per Planck area — the bulk is a hologram, and Eq. (53.153) is the resolution of the hologram.

The dictionary is now complete: each bulk field is paired with a single-trace boundary primary of matching dimension, spin, and charge; bulk gauge symmetries and diffeomorphisms descend to boundary global symmetries and stress-tensor conservation; the parameter map locates classical gravity at large N and strong coupling; and the central charge counts the bulk in Planck units. The one object this section has singled out for the next is the source. The GKPW boundary condition Eq. (53.142) hands the non-normalizable boundary value ϕ_0 to the boundary theory as the field-theory source coupled to $\mathcal{O}$ — exactly the variable a generating functional is differentiated with respect to. The following section states the GKP–Witten relation, which identifies the boundary generating functional $W[\phi_0]$ with the on-shell bulk action evaluated subject to Eq. (53.142), and differentiates it twice to compute the holographic two-point function $\langle\mathcal{O}(x)\mathcal{O}(0)\rangle \propto |x|^{-2\Delta}$.

53.6 The GKP–Witten Relation and Holographic Two-Point Functions

The dictionary of Section 53.5 taught us to read a boundary operator $\mathcal{O}$ off a bulk field Φ, and to read the field-theory *source* that couples to $\mathcal{O}$ off the leading near-boundary falloff ϕ_0 of Φ, through the boundary condition Eq. (53.142). That source has been waiting to be used. In every other chapter of this book the way one extracts correlators of an operator is to couple it to a classical source J, assemble the generating functional $Z[J] = \langle \exp \int J\mathcal{O} \rangle$, and differentiate (Theorem 10.11). Holography supplies, for the first time, a way to *compute* that generating functional at strong coupling: the source ϕ_0 is now promoted from a label on a boundary condition to the argument of a partition function, and the partition function in question is the one for gravity in the bulk. This section states that identification — the GKP–Witten relation — and then carries it all the way through for the simplest observable, the scalar two-point function, recovering the conformally-mandated form $\langle \mathcal{O}(x)\mathcal{O}(0)\rangle \propto |x|^{-2\Delta}$ of Theorem 47.21 with its coefficient computed from bulk data. The computation is a complete, self-contained boundary-value problem: solve a linear wave equation, evaluate a classical action, and renormalize away its divergences. No quantum gravity is needed, which is exactly the payoff promised in the chapter's opening thesis — a hard quantum problem in d dimensions traded for an easy classical one in $d+1$.

53.6.1 The master relation

The proposal of Gubser, Klebanov, and Polyakov, and independently of Witten, identifies the field-theory generating functional with the bulk partition function evaluated on a prescribed boundary condition. We work in Euclidean signature throughout this section: the boundary-value problem at $z \to 0$ is then a genuine Dirichlet problem, and regularity in the deep interior $z \to \infty$ selects a unique solution without the contour ambiguities that plague the Lorentzian wave equation. Concretely, we Wick-rotate the Poincaré patch Eq. (53.59) by $t \to -i\tau$ and pass to the Euclidean line element $\mathrm{d}s_E^2 = -\mathrm{d}s^2$; this turns the mostly-minus boundary metric $\eta_{\mu\nu}$ into $+\delta_{\mu\nu}$ and produces the positive-definite metric of Euclidean AdS_{d+1} (the hyperbolic upper half-space),

$$\mathrm{d}s_E^2 \;=\; \frac{L^2}{z^2}\Big(\delta_{\mu\nu}\,\mathrm{d}x^\mu\,\mathrm{d}x^\nu + \mathrm{d}z^2\Big), \qquad x \in \mathbb{R}^d,\; z > 0, \tag{53.161}$$

with conformal boundary $\mathbb{R}^d$ at $z = 0$. The book's Euclidean path integral weight is e^{-S_E} (Theorem 10.11 and the Wick rotation fixed there), so the gravitational partition function with boundary data ϕ_0 is a functional integral over all bulk fields Φ approaching ϕ_0 at the boundary.

Definition 53.29 (The GKP–Witten relation)**.** Let Φ be a bulk field dual to the boundary operator $\mathcal{O}$, with near-boundary falloff fixed by Eq. (53.142) so that its leading mode is the source ϕ_0. The *GKP–Witten relation* equates the boundary generating functional with the bulk partition function subject to that boundary condition:

$$\Big\langle \exp\!\int \mathrm{d}^d x\; \phi_0(x)\,\mathcal{O}(x) \Big\rangle_{\mathrm{CFT}} \;=\; Z_{\mathrm{grav}}\big[\,\Phi \to \phi_0\,\big] \;=\; \int_{\Phi\to\phi_0} \mathcal{D}\Phi\; e^{-S_E[\Phi]}. \tag{53.162}$$

The left-hand side is a field-theory object; the right-hand side is a sum over bulk geometries and fields. As it stands Eq. (53.162) is the full, conjectural statement of the duality at the level of correlators, and it is no more provable with our tools than Theorem 53.6 itself. What *is* tractable, and what this section delivers in full, is its classical limit. The bulk partition function is dominated by its saddle point — the solution of the classical equations of motion with the prescribed boundary data — whenever the bulk is weakly coupled. By the parameter dictionary Eq. (53.43) this is precisely the large-N, large-λ regime: the bulk Newton constant obeys $1/G_N \propto N^2$, so $G_N \to 0$ (classical gravity) is $N \to \infty$, while weak curvature in string units, $L/\ell_s = \lambda^{1/4} \gg 1$ (Eq. (53.44)), suppresses the α' corrections that would promote classical fields to strings (Eq. (53.45)). In this regime the functional integral collapses onto the classical action of the dominant saddle,

$$Z_{\mathrm{grav}}[\phi_0] \;\approx\; e^{-S_{\text{on-shell}}[\phi_0]} \qquad (\text{large } N,\ \text{large } \lambda), \tag{53.163}$$

where $S_{\text{on-shell}}[\phi_0]$ is the bulk action evaluated on the solution of the classical field equation that approaches ϕ_0 at the boundary and is regular in the interior.

Taking the logarithm of Eq. (53.162) turns it into the statement that the boundary *connected* generating functional $W[\phi_0] = \log Z$ equals minus the on-shell action. Functional differentiation then produces correlators exactly as in flat-space field theory (Theorem 10.11). The one- and two-point functions are

$$\langle \mathcal{O}(x) \rangle = \frac{\delta W[\phi_0]}{\delta \phi_0(x)} = -\,\frac{\delta S_{\text{on-shell}}}{\delta \phi_0(x)}, \tag{53.164}$$

$$\langle \mathcal{O}(x)\,\mathcal{O}(y) \rangle_c = \frac{\delta^2 W[\phi_0]}{\delta\phi_0(x)\,\delta\phi_0(y)}\bigg|_{\phi_0=0} = -\,\frac{\delta^2 S_{\text{on-shell}}}{\delta\phi_0(x)\,\delta\phi_0(y)}\bigg|_{\phi_0=0}. \tag{53.165}$$

Everything now reduces to computing $S_{\text{on-shell}}[\phi_0]$ for a free bulk scalar, expressing it as a quadratic functional of the source, and reading off the kernel. The work splits into three steps: solve the equation of motion (Section 53.6.2), evaluate the action on the solution and expose its divergences (Section 53.6.3), and renormalize (Section 53.6.4).

Physical Insight

The GKP–Witten relation is the precise sense in which the bulk is a *hologram*. A single classical field configuration in $d+1$ dimensions — the solution of one wave equation — encodes an entire generating functional of a d-dimensional quantum field theory. Turning a knob on the boundary (changing ϕ_0) and watching how the bulk action responds is the same operation as inserting the operator $\mathcal{O}$ and measuring its correlators. The extra radial dimension z is not auxiliary: Section 53.4 showed it carries the renormalization scale, so propagating the wave equation inward from $z=0$ is literally running the source from the UV into the IR.

53.6.2 The bulk-to-boundary propagator

We restore the overall normalization that the kinetic study of Eq. (53.100) could suppress. The physically normalized Euclidean action of the free bulk scalar carries a constant $\mathcal{N}_\phi$ — proportional to the inverse Newton constant, $\mathcal{N}_\phi \propto 1/(2\kappa^2) \propto N^2$, and *not* to be confused with the supersymmetry count $\mathcal{N}$ — that does not affect the equation of motion but sets the overall scale of every correlator:

$$S_E[\phi] \;=\; \frac{\mathcal{N}_\phi}{2}\int \mathrm{d}^d x\,\mathrm{d}z\,\sqrt{g}\,\Big(g^{MN}\partial_M\phi\,\partial_N\phi + m^2\phi^2\Big), \qquad \sqrt{g}=\Big(\frac{L}{z}\Big)^{d+1}, \tag{53.166}$$

with $g^{zz}=z^2/L^2$ and $g^{\mu\nu}=(z^2/L^2)\delta^{\mu\nu}$ from Eq. (53.161). The classical field equation is the Euclidean Klein–Gordon equation $(\Box_{\mathrm{AdS}}-m^2)\phi=0$, which in these coordinates reads (the computation is Eqs. (53.103) and (53.105) of Section 53.4, repeated here with Euclidean boundary derivatives)

$$z^2\,\partial_z^2\phi-(d-1)\,z\,\partial_z\phi+z^2\,\partial_x^2\phi-m^2L^2\,\phi=0, \qquad \partial_x^2\equiv\sum_\mu \partial_\mu\partial_\mu. \tag{53.167}$$

Momentum space and the modified Bessel equation. Fourier-transform along the boundary, $\phi(z,x)=\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,e^{ik\cdot x}\,f_k(z)$, so that $\partial_x^2\to -k^2$ with $k^2=|k|^2>0$ Euclidean. The radial profile obeys

$$z^2 f_k''-(d-1)\,z\,f_k'-\big(k^2z^2+m^2L^2\big)f_k=0. \tag{53.168}$$

Strip off the leading scaling with the substitution $f_k(z)=z^{d/2}\,h(z)$. Using $f_k'=\frac{d}{2}z^{d/2-1}h+z^{d/2}h'$ and $f_k''=\frac{d}{2}(\frac{d}{2}-1)z^{d/2-2}h+d\,z^{d/2-1}h'+z^{d/2}h''$, the three derivative terms of Eq. (53.168) combine, after dividing through by $z^{d/2}$, into

$$\begin{aligned} z^2f_k''-(d-1)zf_k' &= z^2h''+\big[d-(d-1)\big]z\,h'+\Big[\tfrac{d}{2}(\tfrac{d}{2}-1)-(d-1)\tfrac{d}{2}\Big]h \\ &= z^2h''+z\,h'-\tfrac{d^2}{4}\,h, \end{aligned} \tag{53.169}$$

where the h-coefficient collapsed as $\frac{d}{2}(\frac{d}{2}-1)-(d-1)\frac{d}{2}=\frac{d}{2}(\frac{d}{2}-1-d+1)=-\frac{d^2}{4}$. Substituting Eq. (53.169) into Eq. (53.168) and writing

$$\nu \;\equiv\; \Delta-\frac{d}{2} \;=\; \sqrt{\frac{d^2}{4}+m^2L^2} \qquad\Longrightarrow\qquad \frac{d^2}{4}+m^2L^2=\nu^2, \tag{53.170}$$

exactly the index of Section 53.4 with $\Delta=\Delta_+$ the larger root of Eq. (53.112), the equation becomes

$$z^2h''+z\,h'-(k^2z^2+\nu^2)h=0. \tag{53.171}$$

In the variable $u=kz$ this is $u^2h''+uh'-(u^2+\nu^2)h=0$, the *modified Bessel equation* of order ν. Its two independent solutions are $I_\nu(kz)$ and $K_\nu(kz)$, so the general radial profile is $f_k(z)=z^{d/2}\big[A\,I_\nu(kz)+B\,K_\nu(kz)\big]$.

Interior regularity. Deep in the bulk, $z\to\infty$, the two modified Bessel functions behave as $I_\nu(kz)\sim e^{kz}/\sqrt{2\pi kz}$ and $K_\nu(kz)\sim e^{-kz}\sqrt{\pi/2kz}$. The growing solution I_ν is not normalizable and makes the on-shell action diverge at the Poincaré horizon; physical (finite-action) configurations keep only the exponentially decaying piece. This is exactly the role Euclidean signature plays: regularity at $z\to\infty$ is unambiguous, and it selects

$$f_k(z)=C(k)\,z^{d/2}\,K_\nu(kz), \tag{53.172}$$

the unique (up to normalization) interior-regular solution.

Small-z expansion: source and response. To match Eq. (53.172) to the boundary data we need its behaviour as $z\to0$, i.e. the small-argument expansion of K_ν. For non-integer ν,

$$K_\nu(u)=\frac{1}{2}\,\Gamma(\nu)\Big(\frac{u}{2}\Big)^{-\nu}\Big[1+\frac{u^2/4}{1-\nu}+\cdots\Big]+\frac{1}{2}\,\Gamma(-\nu)\Big(\frac{u}{2}\Big)^{\nu}\Big[1+\frac{u^2/4}{1+\nu}+\cdots\Big], \tag{53.173}$$

the two series being the $I_{-\nu}$ and I_ν branches of $K_\nu=\frac{\pi}{2\sin\pi\nu}(I_{-\nu}-I_\nu)$, with $\frac{\pi}{\sin\pi\nu}=\Gamma(\nu)\Gamma(1-\nu)$. Multiplying by $z^{d/2}$ and using $z^{d/2}u^{\mp\nu}=k^{\mp\nu}z^{d/2\mp\nu}=k^{\mp\nu}z^{\,d-\Delta\text{ or }\Delta}$, the regular solution has the structure

$$f_k(z)=C(k)\Big[\tfrac{1}{2}\Gamma(\nu)(k/2)^{-\nu}\,z^{\,d-\Delta}(1+\cdots)+\tfrac{1}{2}\Gamma(-\nu)(k/2)^{\nu}\,z^{\,\Delta}(1+\cdots)\Big]. \tag{53.174}$$

The two powers $z^{d-\Delta}$ and z^{Δ} are precisely the two falloffs of Eq. (53.116): the first is the source, the second the response. We fix $C(k)$ by demanding, as in Eq. (53.142), that the coefficient of $z^{d-\Delta}$ be the source $\phi_0(k)$:

$$C(k)=\frac{2}{\Gamma(\nu)}\Big(\frac{k}{2}\Big)^{\nu}\phi_0(k) \qquad\Longrightarrow\qquad f_k(z)=\phi_0(k)\,\frac{2}{\Gamma(\nu)}\Big(\frac{k}{2}\Big)^{\nu}z^{d/2}K_\nu(kz). \tag{53.175}$$

With this normalization the small-z expansion of the solution reads

$$f_k(z) = \phi_0(k)\Big[\, z^{\,d-\Delta}(1+\cdots) + \underbrace{\frac{\Gamma(-\nu)}{\Gamma(\nu)}\Big(\frac{k}{2}\Big)^{2\nu}}_{\equiv\ \mathcal{B}(k)}\, z^{\,\Delta}(1+\cdots)\Big], \tag{53.176}$$

where the "$\cdots$" inside each bracket are the integer powers $z^2, z^4, \ldots$ of the analytic Bessel series. The coefficient of the response mode z^Δ is $\phi_{(2\Delta)}(k) = \mathcal{B}(k)\,\phi_0(k)$, manifestly proportional to the source and carrying the non-analytic momentum dependence $k^{2\nu} = k^{2\Delta-d}$. This $\mathcal{B}(k)$ is the entire physical content of the calculation; everything that follows extracts it cleanly from the action.

Position space. It is illuminating to record the same object in position space, where it is a single closed form. Define the *bulk-to-boundary propagator* $K(z, x; x')$ as the solution of the bulk wave equation whose source is a boundary delta function, $\phi(z,x) = \int \mathrm{d}^d x'\, K(z,x;x')\,\phi_0(x')$.

Definition 53.30 (Bulk-to-boundary propagator)**.** The interior-regular solution of Eq. (53.167) with unit point source at the boundary point x' is

$$K(z,x;x') = c_\Delta\, \frac{z^\Delta}{\big(z^2+|x-x'|^2\big)^\Delta}, \qquad c_\Delta = \frac{\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2})}. \tag{53.177}$$

That Eq. (53.177) solves the wave equation follows from two facts, neither requiring brute computation. First, the pure power $\phi = z^\Delta$ (constant along the boundary) solves Eq. (53.167): substituting it, $z^2\Delta(\Delta-1)z^{\Delta-2} - (d-1)z\,\Delta z^{\Delta-1} - m^2L^2 z^\Delta = [\Delta(\Delta-1) - (d-1)\Delta - m^2L^2]z^\Delta$, and the bracket is $\Delta(\Delta-d) - m^2L^2 = 0$ by the mass–dimension relation Eq. (53.112). Second, the inversion

$$I:\ (z, x^\mu)\ \longmapsto\ \frac{(z, x^\mu)}{z^2+|x|^2} \tag{53.178}$$

is an isometry of Euclidean AdS_{d+1}. Indeed, under $X^M \to X^M/X^2$ with $X^2 = z^2+|x|^2$ the flat $(d+1)$-metric scales as $\delta_{MN}\mathrm{d}X^M\mathrm{d}X^N \to (X^2)^{-2}\delta_{MN}\mathrm{d}X^M\mathrm{d}X^N$, while $z \to z/X^2$ gives $z^{-2} \to (X^2)^2 z^{-2}$; the two factors of X^2 cancel and $\frac{L^2}{z^2}\delta_{MN}\mathrm{d}X^M\mathrm{d}X^N$ is left invariant. A scalar solution composed with an isometry is again a solution, and applying I (which sends a source at spatial infinity to one at $x=0$) to z^Δ produces $z^\Delta/(z^2+|x|^2)^\Delta$, i.e. Eq. (53.177) up to the constant c_Δ. That constant is fixed by requiring $K \to z^{d-\Delta}\delta^d(x-x')$ as $z\to 0$. The needed integral is a Schwinger-parameter computation:

$$\begin{aligned}\int \mathrm{d}^d x\, \frac{1}{(z^2+|x|^2)^\Delta} &= \frac{1}{\Gamma(\Delta)}\int_0^\infty \mathrm{d}t\, t^{\Delta-1}\int \mathrm{d}^d x\, e^{-t(z^2+|x|^2)} \\ &= \frac{\pi^{d/2}}{\Gamma(\Delta)}\int_0^\infty \mathrm{d}t\, t^{\Delta-1-d/2}e^{-tz^2} = \frac{\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2})}{\Gamma(\Delta)}\, z^{\,d-2\Delta},\end{aligned} \tag{53.179}$$

using $\int \mathrm{d}^d x\, e^{-t|x|^2} = (\pi/t)^{d/2}$. Hence $\int \mathrm{d}^d x\, K = c_\Delta\, z^\Delta \cdot \frac{\pi^{d/2}\Gamma(\Delta - d/2)}{\Gamma(\Delta)} z^{d-2\Delta} = c_\Delta \frac{\pi^{d/2}\Gamma(\Delta-d/2)}{\Gamma(\Delta)}\, z^{d-\Delta}$, which equals $z^{d-\Delta}$ precisely when $c_\Delta = \Gamma(\Delta)/[\pi^{d/2}\Gamma(\Delta - \frac{d}{2})]$, as stated. The constant c_Δ will reappear, multiplied by $2\Delta - d$, as the coefficient of the two-point function — a first sign that the two routes agree.

53.6.3 The on-shell action and its divergences

With the solution in hand we evaluate the action Eq. (53.166). Integrating the kinetic term by parts,

$$S_E = \frac{\mathcal{N}_\phi}{2}\int \mathrm{d}^d x\, \mathrm{d}z\; \partial_M(\sqrt{g}\,\phi\, g^{MN}\partial_N\phi) - \frac{\mathcal{N}_\phi}{2}\int \mathrm{d}^d x\, \mathrm{d}z\; \phi\,[\partial_M(\sqrt{g}\, g^{MN}\partial_N\phi) - \sqrt{g}\, m^2\phi], \tag{53.180}$$

the second integrand is $-\sqrt{g}\,\phi\,(\Box_{\mathrm{AdS}} - m^2)\phi$, which vanishes on shell. Only the total derivative survives, and since the integrand depends on z through the explicit factors, the z-integral collapses to the boundaries. The interior boundary $z \to \infty$ contributes nothing (the regular solution decays); the conformal boundary must be regulated by a cutoff at $z = \epsilon$. With $\sqrt{g}\, g^{zz} = (L/z)^{d+1}(z^2/L^2) = L^{d-1}z^{1-d}$, the on-shell action is the single boundary term

$$S_{\text{on-shell}} = -\frac{\mathcal{N}_\phi L^{d-1}}{2}\int \mathrm{d}^d x\; z^{1-d}\,\phi\,\partial_z\phi\Big|_{z=\epsilon} = -\frac{\mathcal{N}_\phi L^{d-1}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\epsilon^{1-d}\, f_{-k}(\epsilon)\, f_k'(\epsilon), \tag{53.181}$$

the overall sign because the outward normal at $z = \epsilon$ points toward decreasing z.

We need the boundary flux $\epsilon^{1-d} f_{-k} f_k'$ in the small-ϵ limit. Write the normalized solution Eq. (53.176) as $f_k(z) = \phi_0(k)\, F(z)$ with $F(z) = z^{d-\Delta}[1 + \cdots] + \mathcal{B} z^\Delta[1 + \cdots]$, where the analytic corrections are organized by the Bessel series Eq. (53.173). The logarithmic derivative is cleanest to expand:

$$\frac{F'(z)}{F(z)} = \partial_z \log F = \frac{d-\Delta}{z} + \frac{k^2 z}{2(1-\nu)} + 2\nu\,\mathcal{B}\, z^{2\nu-1} + \cdots, \tag{53.182}$$

the first term from the leading power $z^{d-\Delta}$, the second from the first analytic correction $\frac{k^2 z^2}{4(1-\nu)}$, and the third from the response term $\mathcal{B} z^{2\nu}$ relative to the leading mode (recall $2\nu = 2\Delta - d$). Multiplying by $F(z)^2 = z^{2(d-\Delta)}[1 + 2\mathcal{B} z^{2\nu} + \cdots]$ and the kinematic factor ϵ^{1-d}, the flux organizes into

$$\epsilon^{1-d} F(\epsilon) F'(\epsilon) = (d-\Delta)\,\epsilon^{-2\nu} + \frac{(d-\Delta)+1}{2(1-\nu)}\, k^2\, \epsilon^{-2\nu+2} + \cdots + d\,\mathcal{B}(k) + \cdots. \tag{53.183}$$

Two species of term appear. The powers $\epsilon^{-2\nu}, \epsilon^{-2\nu+2}, \ldots$ *diverge* as $\epsilon \to 0$ (for $\nu > 0$), and their coefficients are *analytic* in k^2: $\epsilon^{-2\nu}$ is k-independent, $\epsilon^{-2\nu+2}$ comes with k^2, and so on. Fourier-transformed back, these are contact terms, supported at coincident points (the constant gives $\delta^d(x-y)$, the k^2

gives $\Box\delta^d(x-y)$, etc.). They are pure short-distance artefacts of the regulator. The single *finite*, *non-analytic* term is $d\,\mathcal{B}(k) \propto k^{2\nu}$, the would-be physical two-point function. The coefficient d is, however, *not* the right answer yet: it is contaminated by the same regulator, and the resolution is holographic renormalization, to which we now turn.

53.6.4 Holographic renormalization

The divergences in Eq. (53.183) are removed by adding to the action local boundary counterterms — functionals of the field and induced metric on the cutoff surface $z=\epsilon$ — chosen to cancel them. This is the gravitational analogue of the local counterterms of Chapter 22: the divergences are local and covariant, so the subtractions can be too, and the renormalized correlators are scheme-independent up to the contact terms those counterterms can shift. The induced metric on the slice is $\gamma_{\mu\nu} = (L/\epsilon)^2\delta_{\mu\nu}$, with $\sqrt{\gamma} = (L/\epsilon)^d$, and the field there is $\phi_\epsilon(x) \equiv \phi(\epsilon, x)$.

The leading divergence $\propto \epsilon^{-2\nu}$ is cancelled by a counterterm quadratic in the boundary field,

$$S_{\rm ct} = \frac{\mathcal{N}_\phi}{2}\,\frac{d-\Delta}{L}\int \mathrm{d}^d x\,\sqrt{\gamma}\,\phi_\epsilon^2 = \frac{\mathcal{N}_\phi L^{d-1}}{2}\,(d-\Delta)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\epsilon^{-d}\,\phi_\epsilon(-k)\phi_\epsilon(k), \tag{53.184}$$

manifestly local and covariant. (Subleading divergences $\epsilon^{-2\nu+2},\dots$ require further counterterms built from $\sqrt{\gamma}\,\phi\Box_\gamma\phi$ and higher; they touch only the analytic, contact part of the answer and we track them no further. For integer ν a logarithmic counterterm $\propto \log\epsilon$ is also needed, the holographic conformal anomaly — see Theorem 53.31 and Theorem 53.32.) The crucial point is that the counterterm, although introduced to kill a divergence, also reaches into the finite part. Writing $\phi_\epsilon = \phi_0\,F(\epsilon)$ with $F(\epsilon)^2 = \epsilon^{2(d-\Delta)}[1 + 2\mathcal{B}\,\epsilon^{2\nu} + \cdots]$, the counterterm expands as

$$\begin{aligned} S_{\rm ct} &= \frac{\mathcal{N}_\phi L^{d-1}}{2}\,(d-\Delta)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\phi_0(k)\,\epsilon^{-d}\,\epsilon^{2(d-\Delta)}\big[1 + 2\mathcal{B}\,\epsilon^{2\nu} + \cdots\big] \\ &= \frac{\mathcal{N}_\phi L^{d-1}}{2}\,(d-\Delta)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\phi_0(k)\,\big[\epsilon^{-2\nu} + 2\,\mathcal{B} + \cdots\big], \end{aligned} \tag{53.185}$$

using $-d + 2(d-\Delta) = d - 2\Delta = -2\nu$ and $\epsilon^{-2\nu}\epsilon^{2\nu} = 1$. Comparing with Eq. (53.181) fed by Eq. (53.183),

$$S_{\rm reg} = -\frac{\mathcal{N}_\phi L^{d-1}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\phi_0(k)\,\big[(d-\Delta)\epsilon^{-2\nu} + d\,\mathcal{B} + \cdots\big], \tag{53.186}$$

the divergent $\epsilon^{-2\nu}$ pieces cancel between Eqs. (53.185) and (53.186), as designed.

What remains finite is the renormalized action

$$S_{\rm ren} = \lim_{\epsilon\to 0}\big(S_{\rm reg}+S_{\rm ct}\big) = \frac{\mathcal{N}_\phi L^{d-1}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\phi_0(k)\,\big[\,2(d-\Delta)-d\,\big]\mathcal{B}(k)+(\text{contact}), \tag{53.187}$$

where the finite coefficient has collapsed from the bare $-d$ to

$$2(d-\Delta)-d = d-2\Delta = -2\nu = -(2\Delta-d). \tag{53.188}$$

This is the mechanism that turns the naive (and wrong) coefficient d of Eq. (53.183) into the correct $-(2\Delta-d)$: the counterterm needed to remove the divergence necessarily carries a finite remainder of just the right size. We have

$$\boxed{\,S_{\rm ren} = -\frac{\mathcal{N}_\phi L^{d-1}}{2}\,(2\Delta-d)\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\,\mathcal{B}(k)\,\phi_0(k) + (\text{contact}),\,} \tag{53.189}$$

with $\mathcal{B}(k) = \frac{\Gamma(-\nu)}{\Gamma(\nu)}(k/2)^{2\nu}$ from Eq. (53.176).

The renormalized one-point function follows from Eq. (53.164): differentiating Eq. (53.189) once,

$$\langle \mathcal{O}(k)\rangle = (2\Delta-d)\,\mathcal{N}_\phi L^{d-1}\,\mathcal{B}(k)\,\phi_0(k) = (2\Delta-d)\,\mathcal{N}_\phi L^{d-1}\,\phi_{(2\Delta)}(k), \tag{53.190}$$

the celebrated statement that the operator's expectation value is fixed by the *response* (subleading) mode $\phi_{(2\Delta)}$, with coefficient $2\Delta-d$. Comparing with the dictionary's defining normalization Eq. (53.142), where the coefficient of z^Δ was written $\frac{1}{2\Delta-d}\langle\mathcal{O}\rangle$, fixes the canonical bulk normalization to $\mathcal{N}_\phi L^{d-1}=1$; we keep $\mathcal{N}_\phi L^{d-1}$ explicit below so the N^2-scaling of correlators stays visible.

53.6.5 The two-point function and its coefficient

The connected two-point function is the kernel of the quadratic form Eq. (53.189), read off from Eq. (53.165):

$$\langle \mathcal{O}(k)\,\mathcal{O}(-k)\rangle = (2\Delta-d)\,\mathcal{N}_\phi L^{d-1}\,\frac{\Gamma(-\nu)}{\Gamma(\nu)}\Big(\frac{k}{2}\Big)^{2\nu}. \tag{53.191}$$

To reach position space we Fourier-transform the power $k^{2\nu} = (k^2)^\nu$. The relevant distributional transform is obtained from the same Schwinger trick as Eq. (53.179): for $\mathrm{Re}\,a < d/2$,

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{e^{ik\cdot x}}{(k^2)^a} = \frac{1}{\Gamma(a)(4\pi)^{d/2}}\int_0^\infty \mathrm{d}t\, t^{a-1-d/2}\,e^{-x^2/4t} = \frac{\Gamma(\frac{d}{2}-a)}{4^a\,\pi^{d/2}\,\Gamma(a)}\,\frac{1}{|x|^{\,d-2a}}, \tag{53.192}$$

using $\int \frac{\mathrm{d}^d k}{(2\pi)^d} e^{ik\cdot x} e^{-tk^2} = (4\pi t)^{-d/2} e^{-x^2/4t}$ and then $\int_0^\infty \mathrm{d}t\, t^{a-1-d/2} e^{-x^2/4t} = \Gamma(\frac{d}{2}-a)(x^2/4)^{a-d/2}$. Setting $a=-\nu$ continues this to the growing power $(k^2)^\nu$:

$$\int \frac{\mathrm{d}^d k}{(2\pi)^d}\, e^{ik\cdot x}\,(k^2)^\nu = \frac{4^\nu\,\Gamma(\frac{d}{2}+\nu)}{\pi^{d/2}\,\Gamma(-\nu)}\,\frac{1}{|x|^{\,d+2\nu}} = \frac{4^\nu\,\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(-\nu)}\,\frac{1}{|x|^{2\Delta}}, \tag{53.193}$$

where $\frac{d}{2}+\nu=\Delta$ and $d+2\nu=2\Delta$ by Eq. (53.170). The result is automatically the power $|x|^{-2\Delta}$ demanded by conformal symmetry. Assembling Eq. (53.191) (with $(k/2)^{2\nu}=(k^2)^\nu/4^\nu$) and Eq. (53.193), the 4^ν factors cancel and

$$\langle \mathcal{O}(x)\,\mathcal{O}(0)\rangle = (2\Delta-d)\,\mathcal{N}_\phi L^{d-1}\,\frac{\Gamma(-\nu)}{\Gamma(\nu)}\cdot\frac{\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(-\nu)}\,\frac{1}{|x|^{2\Delta}}. \tag{53.194}$$

The two factors of $\Gamma(-\nu)$ cancel between the momentum kernel and the transform — the simplest sign that the divergent intermediary was an artefact — leaving the manifestly finite, manifestly positive coefficient

$$\boxed{\langle \mathcal{O}(x)\,\mathcal{O}(0)\rangle = \frac{C_\Delta}{|x|^{2\Delta}}, \qquad C_\Delta = \mathcal{N}_\phi L^{d-1}\,\frac{(2\Delta-d)\,\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2})}.} \tag{53.195}$$

Equivalently $C_\Delta=(2\Delta-d)\,c_\Delta\,\mathcal{N}_\phi L^{d-1}$ with c_Δ the bulk-to-boundary normalization of Eq. (53.177) — the cross-check promised earlier, and exactly the result one gets by evaluating the on-shell action directly in position space with two insertions of Eq. (53.177).

Matching the conformal field theory. Compare Eq. (53.195) with the form forced by conformal symmetry alone in Theorem 47.21, $\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\rangle = C_{12}\,|x_{12}|^{-2\Delta}$. The holographic computation reproduces this structure exactly: the power is 2Δ, with Δ the operator dimension delivered by the mass–dimension relation Eq. (53.112), and the would-be free function of the separation is pinned to a pure power by the same conformal invariance that, on the bulk side, appeared as the isometry Eq. (53.178). The coefficient $C_{12}=C_\Delta$ is now *computed*, not merely allowed: it is positive for every $\Delta>d/2$ (the standard branch, $\Gamma(\Delta-\frac{d}{2})>0$), in accord with reflection positivity and the scalar unitarity bound $\Delta\geq(d-2)/2$ of Theorem 47.18. The overall factor $\mathcal{N}_\phi\propto N^2$ is the holographic image of large-N factorization (Section 53.1.6): connected correlators of single-trace operators scale as N^2, exactly the scaling that the central charge of the dual CFT inherits from $1/G_N$. With the canonical choice $\mathcal{N}_\phi L^{d-1}=1$ that reproduced Eq. (53.142), $C_\Delta=(2\Delta-d)\,\Gamma(\Delta)/[\pi^{d/2}\Gamma(\Delta-\frac{d}{2})]$ is the unit-normalized CFT two-point coefficient.

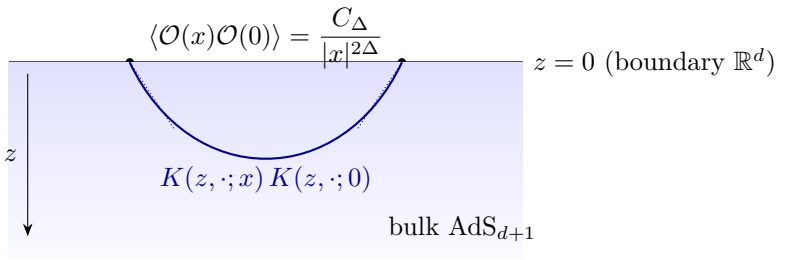


Figure 53.8: The holographic scalar two-point function. The source ϕ_0 at the boundary excites the bulk field through the bulk-to-boundary propagator Eq. (53.177); with no bulk interaction vertex the two-point function is the boundary term of the on-shell action built from two such propagators, evaluated and renormalized to give Eq. (53.195). The single bulk arc is the free (geodesic-like) propagation connecting the two boundary insertions. Interaction vertices in the bulk — the Witten diagrams of Chapter 54 — would add internal points and generate higher-point functions.

Remark 53.31 (Contact terms, scheme dependence, and alternate quantization). Three caveats sharpen the result. *(i) Contact terms.* The analytic-in-k^2 pieces dropped in Eq. (53.189) Fourier-transform to derivatives of $\delta^d(x)$ and are supported only at coincident points. Their coefficients depend on the finite parts of the counterterms (the renormalization scheme), exactly as the contact terms of a flat-space correlator do; the separated-point correlator Eq. (53.195) is scheme-independent. *(ii) Integer ν.* When $\nu = \Delta - \frac{d}{2}$ is a non-negative integer, $\Gamma(-\nu)$ in Eq. (53.191) has a pole and the momentum-space response degenerates to $(k^2)^\nu \log(k^2/\mu^2)$: a renormalization scale μ enters through the logarithmic counterterm, the holographic avatar of the conformal (trace) anomaly of Chapter 37. The position-space coefficient C_Δ in Eq. (53.195) nonetheless stays finite (the Γ-function poles cancel against the log), and the power law $|x|^{-2\Delta}$ survives intact; only the contact part carries the μ-dependence. The worked example below exhibits this explicitly. *(iii) Alternate quantization.* In the Breitenlohner–Freedman window Eq. (53.135), $0 \le \nu < 1$, both modes are normalizable and one may instead treat the z^Δ falloff as the source; this swaps the roles of source and response (a Legendre transform of Eq. (53.189)) and yields a dual operator of dimension $\Delta_- = d - \Delta$, with a two-point coefficient of the same form under $\Delta \to d - \Delta$.

53.6.6 A worked example: $d = 4$, $\Delta = 4$

Worked Example 53.32 (The marginal operator of $\mathrm{AdS}_5/\mathrm{CFT}_4$)**.** We specialize to the case most relevant to the $\mathrm{AdS}_5 \times S^5$ correspondence: a *massless* bulk scalar, $m^2 = 0$, in AdS_5 ($d = 4$). By the mass–dimension relation Eq. (53.112),

$\Delta(\Delta-4)=0$, so the standard branch is $\Delta=\Delta_+=4$ and the index Eq. (53.170) is $\nu=\Delta-\frac{d}{2}=4-2=2$. The dual operator is marginal (dimension d): in $\mathcal{N}=4$ SYM the massless scalar is the dilaton, whose constant boundary value is the gauge coupling, and the dual operator is the on-shell Lagrangian density $\mathcal{O}\sim \operatorname{tr} F_{\mu\nu}F^{\mu\nu}+\cdots$, a protected dimension-4 primary. We compute its two-point coefficient and reproduce $1/|x|^8$.

Coefficient. Plugging $d=4$, $\Delta=4$ into the boxed result Eq. (53.195), with $\Gamma(4)=3!=6$, $\Gamma(\Delta-\frac{d}{2})=\Gamma(2)=1$, $\pi^{d/2}=\pi^2$, and $2\Delta-d=4$,

$$C_4=\mathcal{N}_\phi L^3\,\frac{4\cdot 6}{\pi^2\cdot 1}=\frac{24}{\pi^2}\,\mathcal{N}_\phi L^3,\qquad\Longrightarrow\qquad \langle\mathcal{O}(x)\,\mathcal{O}(0)\rangle=\frac{24\,\mathcal{N}_\phi L^3}{\pi^2}\,\frac{1}{|x|^8}. \tag{53.196}$$

This is finite and positive, and matches the conformal form Eq. (47.120) with $C_{12}=C_4$: holography has delivered the dimension-4 two-point function in closed form, with a definite coefficient set by the bulk geometry and normalization.

The integer-ν subtlety, made explicit. Because $\nu=2$ is an integer, the naive momentum-space response Eq. (53.191) hits the pole of $\Gamma(-\nu)=\Gamma(-2)$, signalling (as in Theorem 53.31) that the careful answer carries a logarithm. The expansion Eq. (53.173) of K_2 indeed contains a $\log(kz)$ term, and the renormalized momentum-space correlator is

$$\langle\mathcal{O}(k)\,\mathcal{O}(-k)\rangle=-\frac{\mathcal{N}_\phi L^3}{8}\,(k^2)^2\,\log\frac{k^2}{\mu^2}, \tag{53.197}$$

with μ the renormalization scale introduced by the logarithmic counterterm. We confirm this reproduces Eq. (53.196). The Fourier transform of $(k^2)^2\log(k^2/\mu^2)$ at separated points follows from differentiating Eq. (53.193) in ν: only the zero of $1/\Gamma(-\nu)$ at $\nu=2$ contributes, with $\frac{d}{d\nu}\big[1/\Gamma(-\nu)\big]_{\nu=2}=(-1)^3\,2!=-2$, so

$$\int\frac{\mathrm{d}^4k}{(2\pi)^4}\,e^{ik\cdot x}\,(k^2)^2\log\frac{k^2}{\mu^2}=\frac{4^2\,\Gamma(4)}{\pi^2}\,(-2)\,\frac{1}{|x|^8}=-\frac{192}{\pi^2\,|x|^8}\qquad(x\neq 0), \tag{53.198}$$

the pure-power $\log\mu^2$ part transforming to a contact term that vanishes away from $x=0$. Hence

$$\langle\mathcal{O}(x)\mathcal{O}(0)\rangle=-\frac{\mathcal{N}_\phi L^3}{8}\cdot\Big(-\frac{192}{\pi^2|x|^8}\Big)=\frac{24\,\mathcal{N}_\phi L^3}{\pi^2\,|x|^8}, \tag{53.199}$$

identical to Eq. (53.196). The μ-dependence has dropped out of the separated-point correlator, residing entirely in the scheme-dependent contact term — the conformal anomaly of Chapter 37 seen holographically. With the canonical normalization $\mathcal{N}_\phi L^3=1$ the unit-normalized coefficient is $C_4=24/\pi^2$, the value the bootstrap of Chapter 49 would take as input data ($C_{\mathcal{O}\mathcal{O}}$) when constraining this CFT through the crossing equation Theorem 49.2: holography and the bootstrap are two routes to the same number.

Physical Insight

The worked example closes a loop that runs across three Parts of this book. The Wilson–Fisher fixed point of ϕ^4 theory handed Part XIV a strongly-coupled CFT; the bootstrap of Chapters 49 and 50 constrains such a CFT's data — dimensions and OPE coefficients — from crossing and unitarity alone, never computing them from a Lagrangian. Holography is the complementary handle: when a CFT has a weakly-curved gravity dual, the very same data $\{\Delta,\, C_{\mathcal{O}\mathcal{O}}\}$ are computed by solving classical wave equations, as in Eq. (53.195). The two methods constrain and compute the same numbers; their consistency, where both apply, is one of the sharpest tests of the correspondence.

We now possess a complete machine: a boundary condition feeds a classical bulk solution, whose renormalized on-shell action is the CFT generating functional, and differentiation yields correlators. Chapter 54 turns this machine to work, switching on the bulk interactions deferred in Eq. (53.100) so that the on-shell action acquires vertices — the Witten diagrams that compute three- and higher-point functions — and replacing the pure-AdS saddle by black-brane geometries, so that the same GKP–Witten relation Eq. (53.162) delivers finite-temperature and real-time correlators of the strongly-coupled boundary theory.

53.7 What Is Established, What Is Conjecture: The Status of Holography

The previous six sections built the correspondence from the ground up: the large-N reorganization that hinted at a hidden string (Section 53.1), the decoupling argument that named the dual geometry (Section 53.2), the intrinsic geometry of AdS_{d+1} and its conformal boundary (Section 53.3), the fields whose masses encode operator dimensions (Section 53.4), the field–operator dictionary, and finally the GKP–Witten relation, which computed a strongly-coupled two-point function as a classical bulk integral (Section 53.6). We now have a machine. Before turning it loose in the chapters that follow, we owe the reader a candid accounting of what kind of machine it is: which of its parts are theorems proved in this book, which are claims supported by decades of stringent but indirect tests, and which are open problems at the research frontier. This section is that accounting. It introduces no new computation that the rest of the book will rely on; its job is to organize, to date, and above all to be honest about epistemic status, in keeping with the discipline laid down in the chapter introduction and in Section 53.2.4. A reader who finishes this section should be able to say, of any holographic statement encountered later, whether it rests on proof, on evidence,

or on conjecture.

53.7.1 The summary of evidence

The correspondence has never been derived from first principles in any framework this book commands, and — as we explain in Section 53.7.2 — a derivation may not even be the right thing to ask for. What it has instead is an extraordinary accumulation of independent, quantitative tests, spanning regimes and techniques that have no business agreeing unless the duality is true. We marshal the principal ones here, in rough order of rigor, marking each for what it is.

The symmetry match (rigorous, proved in this chapter). The single piece of the correspondence that we have established with full rigor is the agreement of global symmetries. In Theorem 53.8 we proved that $\mathcal{N}=4$ SU(N) super Yang–Mills has the superconformal symmetry algebra $\mathfrak{psu}(2,2|4)$, whose bosonic part is $\mathfrak{so}(2,4)\oplus\mathfrak{so}(6)$; in Section 53.3 we proved, intrinsically and with no string input, that the isometry group of $\mathrm{AdS}_5\times S^5$ is exactly $\mathrm{SO}(2,4)\times\mathrm{SO}(6)$, with the conformal group of the boundary realized as the isometry of the anti-de Sitter factor and the R-symmetry as the isometry of the sphere. These two computations are theorems of this book. They do not prove the correspondence — distinct theories can share a symmetry group — but a mismatch would have killed it outright, and the match is exact, term for term, including the noncompact real form and the fermionic generators. This is the load-bearing rigorous core, and everything below is softer.

The spectrum of protected operators. Symmetry alone fixes more than the algebra: it organizes the operator content into representations whose dimensions are protected from quantum corrections. The *chiral primary* (half-BPS) operators of $\mathcal{N}=4$ SYM are the single-trace scalars $\mathcal{O}_k=\mathrm{tr}(\phi^{(i_1}\cdots\phi^{i_k)})$ − traces, symmetric traceless in the $\mathrm{SU}(4)_R$ indices, with dimension $\Delta=k$ fixed exactly by the superconformal algebra (a BPS bound saturated for all g,N). On the gravity side these map, one for one, to the Kaluza–Klein harmonics of the supergravity fields on S^5: expanding a ten-dimensional field in spherical harmonics Y_k on the five-sphere produces a tower of AdS_5 fields whose masses, read through the mass–dimension relation $\Delta(\Delta-d)=m^2L^2$ of Section 53.4, reproduce $\Delta=k$ exactly. The towers agree representation by representation, including the truncation at $k\le N$ on the field-theory side (a single-trace operator of more than N fields is not independent), which matches the stringy exclusion principle on the gravity side. This is a detailed, infinite-dimensional match of protected data, and it works.

The conformal anomaly and central charges. A four-dimensional CFT carries two central charges, a and c, appearing in its Weyl (trace) anomaly,

$$\langle T^{\mu}{}_{\mu} \rangle \;=\; \frac{c}{16\pi^2}\, W_{\mu\nu\rho\sigma} W^{\mu\nu\rho\sigma} \;-\; \frac{a}{16\pi^2}\, E_4, \tag{53.200}$$

with W the Weyl tensor and E_4 the Euler density (the gravitational/trace anomaly structure of Chapter 37). These are sharp, scheme-independent numbers, computable in free field theory on the boundary and from holographic renormalization on the bulk; their agreement is one of the cleanest nontrivial tests, and it teaches the strong/weak and large-N structure at once. We carry it out in full.

Worked Example 53.33 (The central-charge match $a = c = N^2/4$)**.** *Field-theory side.* For a free four-dimensional field the central charges receive additive contributions from each propagating degree of freedom. In units where a single real conformal scalar contributes $(a, c) = (\frac{1}{360}, \frac{1}{120})$, a Weyl fermion contributes $(\frac{11}{720}, \frac{3}{120})$, and a vector contributes $(\frac{31}{180}, \frac{12}{120})$ — the standard free-field anomaly coefficients (Chapter 37). The $\mathcal{N} = 4$ vector multiplet, per generator of the gauge group, contains 6 real scalars, 4 Weyl fermions, and 1 vector, and the gauge group SU(N) has $N^2 - 1$ generators. Summing the c-coefficients,

$$c = (N^2-1)\Big[\,6\cdot\tfrac{1}{120} + 4\cdot\tfrac{3}{120} + 1\cdot\tfrac{12}{120}\,\Big] = (N^2-1)\,\frac{6+12+12}{120} = \frac{N^2-1}{4}, \tag{53.201}$$

and the a-coefficients, brought to a common denominator 720,

$$a = (N^2-1)\Big[\,6\cdot\tfrac{2}{720} + 4\cdot\tfrac{11}{720} + 1\cdot\tfrac{124}{720}\,\Big] = (N^2-1)\,\frac{12+44+124}{720} = (N^2-1)\,\frac{180}{720} = \frac{N^2-1}{4}. \tag{53.202}$$

Two facts emerge. First, $a = c$ exactly — a hallmark of $\mathcal{N} = 4$ SYM, and as we will see the precondition for a two-derivative (Einstein) gravity dual. Second, these values are protected: the central charges sit in the same supermultiplet as the (protected) stress tensor and the R-current, so the free-field result Eq. (53.201) holds at *all* values of the coupling, not merely at $g = 0$. We may therefore compare it to a strong-coupling gravity computation.

Gravity side. Holographic renormalization of pure Einstein gravity in AdS_5 — the same boundary-counterterm procedure used for the two-point function in Section 53.6, now applied to the on-shell action in a curved boundary metric — produces a Weyl anomaly of exactly the form Eq. (53.200) with

$$a \;=\; c \;=\; \frac{\pi L^3}{8\, G_5} \qquad \text{[holographic renormalization]}, \tag{53.203}$$

where L is the AdS_5 radius and G_5 the five-dimensional Newton constant; the equality $a = c$ is automatic because two-derivative gravity has a single coupling L^3/G_5 to spend. We quote the coefficient $\pi/8$ from the holographic-renormalization computation and reduce the rest by the dictionary of this chapter. The five-dimensional Newton constant comes from dimensional reduction of the ten-dimensional one on the five-sphere of radius L, whose volume is $\mathrm{Vol}(S^5) = \pi^3 L^5$:

$$\frac{1}{G_5} \;=\; \frac{\mathrm{Vol}(S^5)}{G_{10}} \;=\; \frac{\pi^3 L^5}{G_{10}}. \tag{53.204}$$

Substituting into Eq. (53.203),

$$c \;=\; \frac{\pi L^3}{8}\cdot\frac{\pi^3 L^5}{G_{10}} \;=\; \frac{\pi^4 L^8}{8\,G_{10}}. \tag{53.205}$$

The ten-dimensional Newton constant is fixed by the string input $16\pi G_{10} = (2\pi)^7 g_s^2 \alpha'^4$ recorded in Section 53.2.3, so

$$G_{10} \;=\; \frac{(2\pi)^7 g_s^2 \alpha'^4}{16\pi} \;=\; \frac{2^7\pi^7}{16\pi}\, g_s^2\alpha'^4 \;=\; 8\pi^6\, g_s^2\alpha'^4. \tag{53.206}$$

Then

$$c \;=\; \frac{\pi^4 L^8}{8\cdot 8\pi^6 g_s^2\alpha'^4} \;=\; \frac{L^8}{64\pi^2\, g_s^2\alpha'^4}. \tag{53.207}$$

Finally use the curvature–coupling relation $L^4 = 4\pi g_s N \alpha'^2$ of Eq. (53.30), so that $L^8 = 16\pi^2 g_s^2 N^2 \alpha'^4$:

$$c \;=\; \frac{16\pi^2 g_s^2 N^2 \alpha'^4}{64\pi^2 g_s^2 \alpha'^4} \;=\; \frac{N^2}{4}. \tag{53.208}$$

The string-length scale α' and the string coupling g_s have cancelled completely, leaving a pure number times N^2. Comparing the gravity result Eq. (53.208) with the field-theory result Eq. (53.201),

$$c_{\text{gravity}} = \frac{N^2}{4}, \qquad c_{\text{CFT}} = \frac{N^2-1}{4}, \tag{53.209}$$

the two agree to leading order in large N, and the discrepancy is precisely the -1 that distinguishes $\mathrm{SU}(N)$ from $\mathrm{U}(N)$ — a relative $1/N^2$ effect, exactly the order at which the classical (genus-zero) gravity computation is not expected to be exact, and exactly the $\mathrm{SU}(N)$-versus-$\mathrm{U}(N)$ subtlety flagged in Section 53.1. The match is as good as it can be: classical gravity reproduces the leading large-N central charge of a strongly-coupled gauge theory, and the first correction is of the anticipated size.

The central-charge computation deserves emphasis as a template for what "a holographic test" means. The two numbers were computed by utterly different

methods — a free-field one-loop diagram on one side, a classical gravitational action with curvature counterterms on the other — in opposite coupling regimes, and they agreed because the quantity is protected and can be carried from weak to strong coupling. Almost every sharp test of the correspondence has this structure: it lives on a protected quantity, a symmetry-constrained observable, or an integrable sector, precisely so that both sides can be reached at once. The generic, unprotected observable cannot be checked this way, and that is the obstruction we return to in Section 53.7.3.

Planar integrability. The most powerful window onto unprotected data is integrability. Starting in 2002 it was discovered that the planar (leading large-N) dilatation operator of $\mathcal{N}=4$ SYM — the operator whose eigenvalues are the anomalous dimensions of single-trace operators — is the Hamiltonian of an integrable spin chain: the mixing matrix of a long single-trace operator is identical to that of an exactly solvable one-dimensional magnet, diagonalized by a Bethe ansatz. On the gravity side, the corresponding statement is that the type IIB string sigma-model on $\mathrm{AdS}_5 \times S^5$ is classically integrable, possessing an infinite tower of conserved charges. Matching these two integrable structures, and resumming them, made it possible to compute the anomalous dimension of an unprotected operator — the Konishi operator $\mathrm{tr}(\phi^i\phi^i)$ is the canonical example — as a function of the coupling that *interpolates* between the weak-coupling perturbative series and the strong-coupling prediction $\Delta_{\text{Konishi}} \sim 2\lambda^{1/4}$ of the dual string, the $\lambda^{1/4}$ being the curvature-radius scaling $L/\ell_s = \lambda^{1/4}$ of Eq. (53.44). By the mid-2010s the planar spectral problem was, for practical purposes, solved: the *Quantum Spectral Curve* (a finite system of functional equations, formulated around 2013–2014) computes any planar anomalous dimension to essentially arbitrary precision at any coupling. That a single set of equations smoothly connects the perturbative gauge theory to the semiclassical string, reproducing both limits, is among the strongest pieces of evidence for the correspondence — evidence about genuinely dynamical, unprotected data, not merely about symmetry. It is, however, restricted to the planar limit and to this maximally symmetric example.

Supersymmetric localization. A second window onto exact, coupling-dependent data is localization. When a supersymmetric theory is placed on a suitable curved manifold and a supercharge is used to deform the path integral, the integral can collapse ("localize") onto a finite-dimensional integral over a space of saddle configurations, yielding exact, all-orders-plus-nonperturbative answers for protected observables. Pestun's 2007 computation (published 2012) evaluated the partition function and the expectation value of the half-BPS circular Wilson loop of $\mathcal{N}=4$ SYM on the four-sphere exactly, reducing it to a Gaussian matrix model. At large N and large λ the resulting Wilson-loop expectation value behaves as $\langle W \rangle \sim e^{\sqrt{\lambda}}$,

which matches precisely the gravity computation in which the same Wilson loop is dual to a fundamental string whose worldsheet hangs into AdS_5 and whose regularized area is $\sqrt{\lambda}$. Localization thus supplies a continuous family of exact checks — Wilson loops, certain correlators, the Bremsstrahlung function — that again interpolate from weak coupling to the string prediction, and it extends to a wide class of supersymmetric theories beyond $\mathcal{N} = 4$, where it has tested many proposed holographic duals.

The verdict on the evidence. Table 53.3 collects these tests with their status and dates. Taken together they are overwhelming: independent methods — symmetry, protected spectra, anomalies, integrability, localization — agree across the entire range of coupling, on quantities that have no reason to agree unless the two theories are the same. Yet none of this is a proof, and it is important to see why. Every sharp test is, of necessity, performed on a quantity that some special structure (a BPS bound, a symmetry multiplet, planar integrability, a localizing supercharge) allows us to compute on *both* sides. The correspondence's central claim is about *all* observables, including the generic unprotected ones that no such structure reaches. The evidence is therefore strong but, in the precise sense, not constitutive: it pins down the duality on a measure-zero but infinitely detailed set of observables and leaves the bulk of the claim resting on the consistency of that set with a single underlying identification.

53.7.2 The epistemic status: a conjecture taken as a definition

The correspondence, Theorem 53.6, remains unproven in general. It has nonetheless survived more than a quarter-century of increasingly sharp tests without a single confirmed contradiction. No computation done correctly on both sides has ever disagreed; the apparent discrepancies that arose over the years have, on examination, always traced to an error on one side or to a regime where one description was not valid. For a statement of this scope, that record is remarkable, and it has changed how the community uses the correspondence.

The change is conceptual. A non-perturbative *definition* of quantum gravity in four flat dimensions does not exist: the Einstein–Hilbert action is non-renormalizable (a fact this book will sharpen in Chapter 64), and no agreed-upon, complete, ultraviolet-finite formulation of four-dimensional quantum gravity is available. In *asymptotically anti-de Sitter* spacetimes, however, the correspondence supplies exactly such a definition. One *declares* that quantum gravity in an asymptotically-AdS_{d+1} spacetime *is* the dual d-dimensional CFT — a theory we do know how to define non-perturbatively, by the conformal bootstrap data of Chapter 49 or, when it has a Lagrangian and a UV completion, by its own path integral. Read this way, AdS/CFT is not a conjecture awaiting proof so much as a *definition* of the otherwise ill-defined quantum-gravitational side. On this reading the open question is not "is the duality true?" but "does this definition

Test	Status	Basis / date
Bosonic & super symmetry match	**proved in text**	Theorem 53.8, Section 53.3
Protected (BPS) spectrum = KK tower	strong (exact, protected)	superconformal reps; late 1990s
Central charges $a = c = N^2/4$	strong (protected)	Theorem 53.33; 1998
Planar anomalous dimensions	strong (interpolating)	spin-chain integrability 2002; Quantum Spectral Curve ∼2013–14
BPS Wilson loops, $\langle W \rangle \sim e^{\sqrt{\lambda}}$	strong (exact, protected)	localization 2007/2012
Generic unprotected observables	untested directly	no method reaches both sides
The correspondence itself	**conjecture**	Theorem 53.6; 1997, no proof

Table 53.3: The summary of evidence for AdS/CFT. The symmetry match is a theorem of this book; the protected and integrable checks are exact and span all couplings; the generic claim is conjectural. Dates mark the principal results.

reproduce, in the appropriate semiclassical limit, the bulk gravitational physics we expect?" — and it is precisely that semiclassical reconstruction, the emergence of a local bulk from boundary data, that is the live frontier (Section 53.7.5).

Caution

"Taken as a definition" is a methodological stance, not a theorem, and it must not be allowed to launder a conjecture into a proof. Adopting the boundary CFT as the definition of the bulk is useful and internally consistent, but it presupposes the very identification Eq. (53.42) that the decoupling argument only motivates. When this book later computes a gravitational quantity "holographically," it is computing a CFT quantity and *interpreting* it gravitationally; the interpretation rests on the conjecture. We flag this each time it matters.

It is worth asking what a genuine proof would even require. One would need an independent, non-perturbative definition of the string-theory (gravity) side — the thing we presently *lack* and are using the CFT to supply — and then a demonstration that its complete observable algebra is isomorphic to that of the CFT. In the absence of such a definition, the most that the existing

constructions (matrix models for lower-dimensional cases, the BMN/plane-wave limit, integrability, localization) achieve is to verify the isomorphism on ever-larger protected or solvable subsectors. A complete proof is therefore not merely technically hard; it is entangled with the foundational problem of defining the bulk theory at all. This is the deepest reason the correspondence is more naturally wielded as a definition than pursued as a theorem.

53.7.3 Strong/weak duality: the power and the obstruction restated

The defining feature of the correspondence, established quantitatively in Section 53.2.4, is that it is a *strong/weak* duality: the radius–coupling relation $L/\ell_s = \lambda^{1/4}$ of Eq. (53.44) sends the field theory to strong 't Hooft coupling exactly when the geometry becomes weakly curved and classical. This single fact is simultaneously the source of all the correspondence's utility and the reason it is so hard to verify.

It is the power because it converts an intractable problem into a tractable one. The strongly-coupled, large-N regime of a gauge theory — the regime left analytically inaccessible by asymptotic freedom (Chapter 27) and definable but not solvable in closed form on the lattice (Chapter 43) — is mapped to *classical* differential geometry: solving Einstein's equations, evaluating an on-shell action, finding a minimal surface. A quantum-field-theory question with no perturbative handle becomes a boundary-value problem in one higher dimension. This is the entire reason the correspondence is a working tool, and the chapters that follow exploit it relentlessly.

It is the obstruction for exactly the same reason. To *check* a generic observable one would need to compute it on both sides; but the two sides are weakly coupled in disjoint corners of parameter space, so a direct perturbative comparison is impossible for any quantity that is not protected. The only quantities testable on both sides are those for which some extra structure — supersymmetry, a BPS bound, planar integrability, a localizing supercharge — defeats the strong/weak separation and provides an answer valid across all couplings. This is why the evidence of Section 53.7.1 clusters on protected and integrable data: it is not that other tests were neglected, but that the strong/weak nature forbids them. The duality is most useful precisely where it is least checkable, and most checkable precisely where (being protected) it is least surprising.

53.7.4 Bottom-up versus top-down holography

Everything proved or tested above concerns a *top-down* duality: a specific pair — $\mathcal{N} = 4$ SYM and type IIB strings on $\mathrm{AdS}_5 \times S^5$ — derived (insofar as it is derived) from an explicit string-theory construction, the brane decoupling of Section 53.2. Both sides are completely specified, the dictionary is fixed, and the

parameters are related by Eq. (53.43). Other top-down examples exist with the same pedigree — M2- and M5-brane theories dual to $\mathrm{AdS}_4 \times S^7$ and $\mathrm{AdS}_7 \times S^4$, and many constructions with reduced supersymmetry — but in each the field theory and the geometry descend together from a controlled brane setup.

A second, looser, and enormously productive practice goes the other way. One *postulates* a gravitational theory in asymptotically-AdS space — typically Einstein gravity plus a handful of bulk fields chosen for their boundary operators — without deriving it from any microscopic string construction, and uses the dictionary of Sections 53.4 and 53.6 to read off the behavior of *some* strongly-coupled boundary system that one hopes resembles a system of physical interest. This is *bottom-up* holography, and its two principal arenas must be scoped honestly.

AdS/CMT. Applied to condensed matter (*AdS/CMT*), bottom-up holography models strongly-correlated electron systems — non-Fermi liquids, strange metals, holographic superconductors — by simple bulk theories: a charged black brane for a finite-density medium, a charged scalar that condenses to model a superconducting transition, a bulk fermion to probe spectral functions. These constructions reproduce qualitative features (linear-in-temperature resistivity, the absence of long-lived quasiparticles) that are hard to obtain otherwise, and they have suggested genuinely new ways to think about quantum criticality and transport. But the boundary theory in such a model is, almost always, *not known* — there is no identified Lagrangian, no microscopic Hamiltonian, no N that one is expanding in. The bulk is an effective device for generating a strongly-coupled, large-N, conformal-ish system with the right symmetries, not a derived dual of any specific laboratory material. AdS/CMT is therefore best understood as *model-building*: a source of universality-class intuition and qualitative mechanism, not of quantitative predictions for a named compound.

AdS/QCD. Applied to the strong interactions (*AdS/QCD*), bottom-up holography seeks to mimic QCD — which is neither conformal nor supersymmetric — by deforming AdS. "Hard-wall" and "soft-wall" models cut off or warp the radial direction to introduce a mass gap and a hadron spectrum; the radial coordinate is read as an energy scale (the holographic RG of Chapter 54), and confinement is modeled by capping the geometry in the infrared. These models fit meson Regge trajectories and some form factors with a few parameters, and they encode chiral symmetry breaking geometrically. But QCD is not known to have any gravity dual: its lack of conformal symmetry and supersymmetry, and its $N = 3$ rather than $N \to \infty$, place it far from the regime where the correspondence is derived. AdS/QCD is a phenomenological caricature — often a useful one — not a controlled limit of a top-down duality. The honest statement, which Chapter 43 should be read alongside, is that the lattice provides QCD's non-perturbative

definition and quantitative answers, while AdS/QCD provides a complementary, qualitative, large-N-flavored picture.

Physical Insight

The distinction is one of logical direction. Top-down: "here is a string construction; derive its field-theory dual and compute." Bottom-up: "here is the strongly-coupled behavior I want; engineer a bulk that produces it, and trust the universality." The first is (modulo the central conjecture) a derivation; the second is modeling. Both are legitimate, but conflating them — treating an AdS/CMT black brane as if it were a derived dual of a real metal — is a category error the literature sometimes commits and this book does not.

53.7.5 Open frontiers, mid-2020s

We close with an honest map of what is unsettled, dated to the time of writing. These are not gaps to be filled by routine work; they are the directions in which the meaning of holography is still being worked out.

Holography beyond anti-de Sitter: de Sitter and flat space. The entire construction of this chapter rests on the negative cosmological constant of anti-de Sitter space, whose timelike conformal boundary (Section 53.3) provided the home for the dual CFT. The universe we observe is, on the largest scales, closer to *de Sitter* space, with a positive cosmological constant and no timelike boundary — only spacelike boundaries in the far past and future. A holographic principle for de Sitter (*dS/CFT*, proposed around 2001) would relate quantum gravity in dS to a Euclidean CFT living on the future boundary, computing the late-time cosmological wavefunction; but the candidate dual appears to be non-unitary or otherwise exotic, the dictionary is far less developed than the AdS one, and there is nothing like the brane derivation to anchor it. *Flat-space* holography faces the analogous problem at zero cosmological constant: the natural boundary is null infinity, and the program of *celestial holography* (developed intensively since roughly 2017) seeks to encode four-dimensional scattering amplitudes as correlators of a two-dimensional "celestial" CFT on the sphere at infinity, organized by the asymptotic symmetry (BMS) group and the soft theorems of Chapter 63. Both programs are active and partial: neither has a complete, derived dual, and whether realistic cosmology admits a holographic description is genuinely open.

The emergence of bulk locality and the black-hole interior. Even within AdS/CFT, the most basic question — *how* a local, $(d+1)$-dimensional spacetime emerges from a non-local d-dimensional CFT — is incompletely understood.

Reconstructing a bulk field at a point from boundary data (the HKLL construction and its refinements) works in perturbation theory around a fixed background, but the deeper statement, that the bulk is encoded *redundantly* in the boundary like a quantum error-correcting code, and that geometry itself is built from boundary *entanglement*, is the subject of intense current work. This is where holography meets quantum information, and it is the explicit business of Chapters 55 and 56: the Ryu–Takayanagi formula computing entanglement entropy as a bulk minimal area, subregion duality, and entanglement-wedge reconstruction. Pushed to its sharpest, the question becomes whether the boundary theory can reconstruct the *interior* of a black hole — the region behind the horizon — which is the crux of the information paradox. The island/quantum-extremal-surface results of 2019 onward, which recover a unitary Page curve for the entropy of Hawking radiation, are the most striking recent progress and are taken up, honestly scoped, in Chapter 57. What "the bulk" even is, as an emergent and approximate notion, is not yet settled.

The dictionary away from the conformal and supersymmetric case. Finally, the precise, derived dictionary is sharpest exactly where the symmetry is largest — the conformal, maximally supersymmetric, planar, large-N corner. Away from it, much is qualitative. Finite N (string loops, $g_s \neq 0$), finite λ (α' corrections to the geometry), non-conformal flows (a running boundary theory dual to a non-AdS bulk, the holographic RG of Chapter 54), broken or absent supersymmetry, and theories with fundamental "flavor" matter all admit holographic treatments, but the control — and in many cases the very existence of a clean dual — degrades as one leaves the symmetric point. The frontier here is less dramatic than the black-hole interior but no less important: extending the correspondence from its rigid, exactly-solvable heart to the generic, realistic theories one actually wants to understand.

These frontiers are the connective tissue of the rest of Part XV. With the scope and reliability of the duality now calibrated — what is proved, what is exhaustively tested, what is conjecture, and what is open — we are equipped to use it. Chapter 54 puts the machine to work, computing higher-point correlators from Witten diagrams and importing finite temperature and black holes to extract holographic thermodynamics; Chapters 55 to 57 then follow the emergent-geometry frontier into entanglement, the Ryu–Takayanagi formula, and the black-hole information problem, where the duality stops being a calculational tool and becomes a statement about what spacetime is.

Further reading

The original conjecture is Maldacena's [331]; the correlator dictionary that made it computational is [242, 522]. The comprehensive early review, still the

standard reference for the top-down construction and its tests, is [9]; for textbook developments see [359]. Planar integrability is surveyed in the collection [53], which assembles the spin-chain and Quantum-Spectral-Curve machinery; the exact Wilson-loop and partition-function results from supersymmetric localization originate in [399]. Bottom-up applications to condensed-matter systems are developed in [253]. For the frontiers: the infrared and asymptotic-symmetry structure underlying celestial (flat-space) holography is laid out in [459], de Sitter holography is reviewed in [26], and the entanglement-wedge reconstruction that resolves the information paradox holographically is represented by [391], with the underlying holographic entanglement entropy in [414, 422].

Chapter 54

Holographic Correlators and Thermodynamics

The previous chapter built the AdS/CFT dictionary as a statement of principle: the GKP–Witten relation Eq. (53.162) equates the boundary generating functional with a bulk gravitational partition function, which in the supergravity regime collapses onto a single classical saddle Eq. (53.163), and the field–operator map of Section 53.5 ties each bulk field to a boundary primary through the mass–dimension relation Theorem 53.17. This chapter turns that principle into a calculus. A classical saddle is a solution of differential equations, and a generating functional differentiated against its sources is a set of correlators; marrying these two facts systematically yields a perturbation theory — the Witten-diagram expansion — that computes the correlators of a strongly coupled conformal field theory by solving wave equations in one higher dimension. Everything the chapter does is an application of this single idea, first at zero temperature, where the bulk is empty anti–de Sitter space, and then at finite temperature, where the bulk grows a horizon.

We begin in Section 54.1 by deriving the position-space Feynman rules of holography: why the classical on-shell action generates the connected correlators (Theorem 10.37 and Eq. (10.210)), why only tree diagrams survive at large N (the bulk image of the genus expansion Theorem 53.2), and how every diagram is assembled from two propagators, bulk-to-bulk and bulk-to-boundary, built here in closed form. Section 54.2 confronts the one subtlety the rules gloss over — the on-shell action diverges at the conformal boundary — and develops holographic renormalization: the Fefferman–Graham near-boundary expansion, the covariant counterterms that remove the local divergences exactly as the boundary theory's own counterterms did in Section 19.1, and the identification of the operator expectation value with the subleading falloff. The payoff is the scalar two-point function, recovered in position space and matched against the conformally fixed form of Theorem 47.21. Section 54.3 switches on interactions:

the cubic contact diagram delivers the OPE coefficient as an explicit ratio of Gamma functions, and the four-point exchange, decomposed into the conformal blocks of Theorem 47.49, exposes both the single-trace operator dual to the exchanged field and the entire double-trace tower $[\mathcal{O}\mathcal{O}]_{n,\ell}$ that the bootstrap of Chapters 49 and 50 independently demands — closing, from the bulk side, the loop the bootstrap opened from crossing alone.

The chapter's center of gravity is thermal. Section 54.4 asks which bulk geometry dominates the gravitational path integral when the boundary carries a Euclidean time circle, and answers with the planar AdS–Schwarzschild black brane: its Hawking temperature follows from demanding the Euclidean geometry close off smoothly at the horizon (Theorem 5.30), its renormalized on-shell action — made finite by the very counterterms of Section 54.2.3 — is the free energy, and the horizon area divided by $4G$ is the entropy, the Bekenstein–Hawking law derived rather than assumed. The carried example, $\mathcal{N} = 4$ super-Yang–Mills (the maximally symmetric large-N cousin of Yang–Mills theory, Chapter 25), supplies the celebrated factor of $\frac{3}{4}$ between strong- and weak-coupling entropy, and the Hawking–Page transition realizes the confinement/deconfinement transition of Section 27.7 as a sharp geometric event. Section 54.5 then leaves Euclidean signature for real time: the infalling horizon boundary condition selects the retarded correlator, the shear graviton reduces to a massless scalar, and a membrane-paradigm flux argument delivers the universal ratio $\eta/s = 1/4\pi$, whose status as a near-saturated benchmark — and its violation by higher-derivative gravity — we report honestly as of the mid-2020s.

A single thread runs through all of this, and Section 54.6 names it. Every computation has been a story told along the radial coordinate: the divergences of the on-shell action were power laws in the radial cutoff, the retarded correlator flowed from a boundary source to an infalling horizon, the black brane differed from the vacuum only deep in the interior. The radial direction of an asymptotically-AdS geometry is the renormalization-group scale of the boundary theory, and radial evolution of the bulk fields is the RG flow of its couplings. We make this precise — domain-wall geometries, first-order flow equations from a fake superpotential, a holographic beta function that reproduces the mass–dimension relation, and a one-line proof of the a-theorem of Theorem 52.19 from the bulk null energy condition — and so close the loop with the Wilsonian and Callan–Symanzik pictures of Part VI (Chapters 23 and 24). The radial-as-scale picture is, in turn, the seed of the entanglement story that Chapters 55 and 56 take up next.

54.1 Witten Diagrams: Perturbative Holography

The chapter opened by promoting the field–operator dictionary of Section 53.5 from a translation table into a calculus: the GKP–Witten relation Eq. (53.162) equates the boundary generating functional with the bulk partition function

evaluated on prescribed boundary data, and in the supergravity regime the bulk path integral collapses onto a single classical saddle Eq. (53.163). A classical saddle is a solution of differential equations, and a generating functional differentiated against its sources is a set of correlators. The whole of holographic perturbation theory is the systematic marriage of these two facts. This section performs that marriage explicitly. We will show, first, why the *classical* on-shell action computes the *connected* correlators and why, at large N, only *tree* diagrams survive; then we build the two propagators — bulk-to-bulk and bulk-to-boundary — out of which every diagram is assembled; and finally we state the position-space Feynman rules in anti–de Sitter space, the Witten rules, that turn any supergravity correlator into a finite recipe of propagators and integrated vertices. The result is a perturbative apparatus standing one dimension above the conformal field theory it computes, structurally identical to ordinary position-space Feynman rules (Theorem 15.9) but with flat propagators replaced by curved ones and momentum integrals replaced by integrals over the bulk.

Throughout this section we work in Euclidean Poincaré AdS_{d+1} Eq. (53.161) with flat boundary $\mathbb{R}^d$, and — as is standard for propagators — we set the radius $L = 1$, restoring it by dimensional analysis whenever a physical scale is needed. A bulk scalar field ϕ of mass m is dual to a boundary scalar primary $\mathcal{O}$ of dimension Δ, the two related by the mass–dimension relation $\Delta(\Delta - d) = m^2$ of Theorem 53.17, with $\Delta \equiv \Delta_+ = \frac{d}{2} + \nu$ the larger root and $\nu = \sqrt{\frac{d^2}{4} + m^2}$.

54.1.1 The classical saddle computes connected correlators

The point of departure is the classical limit of the GKP–Witten relation. Writing ϕ_0 for the boundary source — the coefficient of the non-normalizable falloff $z^{d-\Delta}$ that the boundary condition Eq. (53.116) hands to the field theory — the bulk partition function is dominated by the action of the dominant solution,

$$Z_{\mathrm{grav}}[\phi_0] \;=\; e^{-S_{\text{on-shell}}[\phi_0]}\,(1+O(G_N)), \qquad W[\phi_0] \;\equiv\; \ln Z_{\mathrm{CFT}}[\phi_0] \;=\; -S_{\text{on-shell}}[\phi_0]+O(G_N) \tag{54.1}$$

the second equality being Eq. (53.162) after taking logarithms. The object $W[\phi_0]$ is, by definition, the generating functional of *connected* correlators: this is the content of Theorem 10.37, where $\ln Z$ was shown to generate exactly the connected (linked) diagrams, with the disconnected pieces exponentiating into Z itself. Functional differentiation then yields the connected correlators of $\mathcal{O}$,

$$\langle \mathcal{O}(x_1)\cdots\mathcal{O}(x_n)\rangle_{\mathrm{c}} \;=\; \frac{\delta^n W[\phi_0]}{\delta\phi_0(x_1)\cdots\delta\phi_0(x_n)}\bigg|_{\phi_0=0}, \tag{54.2}$$

in precise parallel with the flat-space generating functional of Theorem 10.11; the source–response normalization that converts a derivative of $-S_{\text{on-shell}}$ into the canonically normalized correlator was fixed once and for all, for the two-point case, in Section 53.6.5, and the higher functional derivatives inherit it.

Why the answer is a sum over trees. That Eq. (54.1) is the *classical* action of a *solution* has a sharp diagrammatic consequence: the on-shell action, expanded in powers of the source, is a sum over tree diagrams. To see this, take a representative bulk action with self-interactions,

$$S_E[\phi] \;=\; \int \mathrm{d}^{d+1}X\,\sqrt{g}\,\Big[\tfrac{1}{2}\,g^{MN}\partial_M\phi\,\partial_N\phi + \tfrac{1}{2}m^2\phi^2 + \tfrac{1}{3!}\,g_3\,\phi^3 + \tfrac{1}{4!}\,g_4\,\phi^4 + \cdots\Big], \tag{54.3}$$

with $X = (z, x)$ a bulk point and $\sqrt{g} = z^{-(d+1)}$ in our units. The classical equation of motion is the curved-space, nonlinear Klein–Gordon equation

$$(-\Box_{\mathrm{AdS}} + m^2)\,\phi \;=\; -V'_{\mathrm{int}}(\phi) \;=\; -\tfrac{1}{2}g_3\,\phi^2 - \tfrac{1}{3!}g_4\,\phi^3 - \cdots, \tag{54.4}$$

to be solved subject to the boundary condition that the coefficient of $z^{d-\Delta}$ equals ϕ_0. We solve it iteratively in the strength of the source, writing $\phi = \sum_{n\geq 1}\phi^{(n)}$ with $\phi^{(n)}$ of order n in ϕ_0. At first order the right-hand side of Eq. (54.4) is absent and we must solve the *free* wave equation with the prescribed boundary source. The solution is the boundary source smeared against a kernel,

$$\phi^{(1)}(X) \;=\; \int \mathrm{d}^{d}y\; K_\Delta(X;y)\,\phi_0(y), \tag{54.5}$$

where K_Δ, the *bulk-to-boundary propagator*, is the regular solution of the free equation with a unit boundary source — precisely the object Theorem 53.30, which we will rederive below as a limit of the bulk-to-bulk propagator. At second order, Eq. (54.4) reads $(-\Box_{\mathrm{AdS}} + m^2)\phi^{(2)} = -\frac{1}{2}g_3(\phi^{(1)})^2$, with *no* additional boundary source (the source is exhausted at first order). The solution is the inhomogeneous term convolved against the Green's function of the bulk wave operator,

$$\phi^{(2)}(X) \;=\; -\tfrac{1}{2}\,g_3\int \mathrm{d}^{d+1}X'\,\sqrt{g'}\;G_\Delta(X,X')\,\big[\phi^{(1)}(X')\big]^2, \tag{54.6}$$

where G_Δ, the *bulk-to-bulk propagator*, is the normalizable Green's function constructed in Section 54.1.2. The recursion continues: each new order ties one more vertex, with its bulk-to-bulk legs, onto the growing solution. By construction every term in $\phi = \sum_n \phi^{(n)}$ is a *tree* — a connected graph with no closed loops — because the iteration never integrates a field against itself; it only feeds lower-order solutions into vertices.

Inserting $\phi = \sum_n \phi^{(n)}$ back into the action Eq. (54.3) and using the equation of motion to discard the bulk part (it vanishes on shell, exactly as in Eq. (53.180)) leaves $S_{\text{on-shell}}[\phi_0]$ as a functional of the source whose order-n piece is a sum of tree diagrams with n external boundary legs. Differentiating n times as in Eq. (54.2) strips the n external K_Δ legs and leaves the connected n-point function as a sum over bulk tree topologies. This is the holographic shadow of the elementary fact that the *classical* field is the tree-level (stationary-phase) approximation to the path integral, and that the Legendre transform relating W to the effective action (Eq. (10.210)) exchanges connected graphs for one-particle-irreducible ones. Here the boundary effective action is literally the on-shell bulk action.

Loops are $1/N^2$. It remains to justify why *only* trees appear at leading order. The bulk gravitational action carries an overall factor of the inverse Newton constant; for a matter sector minimally coupled to gravity we may write $S_E = \frac{1}{G_N} I[\phi]$ with I a functional whose couplings are $O(N^0)$, since $1/G_N \propto N^2$ by the parameter dictionary Eq. (53.43). In a stationary-phase (loop) expansion of $\int \mathcal{D}\phi\, e^{-S_E}$ the small parameter is G_N: the quadratic form has inverse $\propto G_N$, so every propagator carries one factor of G_N, while every vertex comes from S_E and carries one factor of $1/G_N$. A connected diagram with V vertices and I internal lines therefore scales as G_N^{I-V}. Its number of loops is $\mathcal{L} = I - V + 1$, so

$$\text{(diagram)} \;\propto\; G_N^{I-V} \;=\; G_N^{\mathcal{L}-1} \;=\; \frac{1}{G_N}\, G_N^{\mathcal{L}} \;\propto\; N^2 \left(N^{-2}\right)^{\mathcal{L}}. \tag{54.7}$$

The leading contribution, $\mathcal{L} = 0$, is the tree level, of order N^2; each additional bulk loop costs a relative factor $G_N \propto 1/N^2$. This is nothing but the genus expansion of Theorem 53.2 read in the bulk: a closed bulk loop is the gravitational image of adding a handle to the boundary diagram's surface, $g \to g+1$, which by Eq. (53.21) costs exactly $g_s^2 \sim 1/N^2$. In the strict large-N, large-λ limit — where classical supergravity is the correct bulk description — only the planar, tree-level diagrams survive, and the boundary correlators of single-trace operators are computed by classical field theory in the bulk.

Physical Insight

The hierarchy "tree $\gg$ loop" on the gravity side is the statement "planar $\gg$ non-planar" on the gauge side, which is in turn the statement that single-trace operators *factorize* at $N = \infty$ (Theorem 53.2): their connected correlators are suppressed relative to disconnected products. A factorizing operator behaves like a classical variable taking a definite value, and a path integral dominated by definite values is a saddle point. Holography promotes that saddle from an abstract "master field" to an honest classical field propagating in one higher dimension — and the perturbation theory around it is the Witten-diagram expansion.

Everything now hinges on the two propagators. We build them in turn, beginning with the bulk-to-bulk propagator, from which the bulk-to-boundary propagator descends as a boundary limit.

54.1.2 The bulk-to-bulk propagator

Definition 54.1 (Bulk-to-bulk propagator)**.** The *bulk-to-bulk propagator* $G_\Delta(X, Y)$ of a scalar of mass m (dual dimension Δ) on Euclidean AdS_{d+1} is the Green's function of the wave operator,

$$\left(-\Box_{\mathrm{AdS},X} + m^2\right) G_\Delta(X, Y) \;=\; \frac{1}{\sqrt{g}}\, \delta^{d+1}(X - Y), \tag{54.8}$$

selected to be regular in the interior and to decay in its normalizable mode ($\sim z^{\Delta}$, not $z^{d-\Delta}$) as either argument approaches the conformal boundary, so that no source for $\mathcal{O}$ is turned on.

Reduction to one variable. Euclidean AdS_{d+1} is a maximally symmetric space — it is hyperbolic space $\mathbb{H}_{d+1}$ — and its isometry group acts transitively on pairs of points at fixed geodesic separation. The Green's function therefore depends on X and Y only through the geodesic distance between them, or equivalently through the dimensionless *chordal variable*

$$u \;\equiv\; \frac{(z-w)^2+|x-y|^2}{2\,z\,w}, \qquad X=(z,x),\; Y=(w,y), \tag{54.9}$$

which is related to the geodesic distance s by $\cosh s = 1+u$. It is cleanest to trade u for the variable

$$\xi \;\equiv\; \frac{1}{1+u} \;=\; \frac{2zw}{z^2+w^2+|x-y|^2} \;=\; \frac{1}{\cosh s} \;\in\; (0,1], \tag{54.10}$$

which runs from $\xi=1$ at coincidence ($s=0$) to $\xi\to 0$ at infinite separation ($s\to\infty$). Acting on a function of s alone, the $\mathbb{H}_{d+1}$ Laplacian is the radial operator

$$\Box_{\mathrm{AdS}}\, G \;=\; G''(s) + d\,\coth(s)\,G'(s), \tag{54.11}$$

the coefficient $d=(d+1)-1$ being the standard factor in the radial Laplacian of a $(d+1)$-dimensional space of constant negative curvature. Away from coincidence the Green's function obeys the homogeneous equation $(-\Box_{\mathrm{AdS}}+m^2)G=0$, i.e.

$$G''(s) + d\,\coth(s)\,G'(s) - m^2\,G = 0. \tag{54.12}$$

We convert to ξ using $\xi=\operatorname{sech} s$. Then $\mathrm{d}\xi/\mathrm{d}s = -\operatorname{sech} s\tanh s = -\xi\sqrt{1-\xi^2}$ and $\coth s = 1/\sqrt{1-\xi^2}$, so with $A(\xi)\equiv -\xi\sqrt{1-\xi^2}$,

$$G'(s) = A\,G_\xi, \qquad G''(s) = A(A\,G_\xi)_\xi = A^2\,G_{\xi\xi} + A\,A'\,G_\xi, \tag{54.13}$$

with $A^2=\xi^2(1-\xi^2)$ and $AA'=\xi(1-2\xi^2)$, while $d\coth s\,G'(s) = d\cdot(1-\xi^2)^{-1/2}\cdot(-\xi\sqrt{1-\xi^2})\,G_\xi = -d\,\xi\,G_\xi$. Substituting into Eq. (54.12) and collecting,

$$\xi^2(1-\xi^2)\,G_{\xi\xi} + \xi\big[(1-d)-2\xi^2\big]\,G_\xi - m^2\,G = 0. \tag{54.14}$$

The behaviour at large separation, $\xi\to 0$, is governed by the leading balance $\xi^2 G_{\xi\xi}+(1-d)\xi G_\xi - m^2 G\approx 0$; trying $G\sim\xi^p$ gives $p(p-1)+(1-d)p-m^2 = p(p-d)-m^2=0$, the mass–dimension equation Eq. (53.112) once more. Its roots are $p=\Delta$ and $p=d-\Delta$, and *normalizability* selects the decaying root $G\sim\xi^{\Delta}$.

The hypergeometric solution. Strip the leading power by writing $G(\xi) = \xi^{\Delta}\, y(\xi)$. With $G' = \Delta\xi^{\Delta-1}y + \xi^{\Delta}y'$ and $G'' = \Delta(\Delta-1)\xi^{\Delta-2}y + 2\Delta\xi^{\Delta-1}y' + \xi^{\Delta}y''$, equation Eq. (54.14) divided by ξ^{Δ} becomes, after the y-coefficient bracket $\big[\Delta(\Delta-1) + (1-d)\Delta - m^2\big] = \Delta(\Delta-d) - m^2$ vanishes by Eq. (53.112),

$$(1-\xi^2)\,y'' + \Big[\frac{2\Delta+1-d}{\xi} - (2\Delta+2)\,\xi\Big]y' - \Delta(\Delta+1)\,y = 0. \tag{54.15}$$

Now pass to the variable $t = \xi^2$, so that $y' = 2\sqrt{t}\,y_t$ and $y'' = 2\,y_t + 4t\,y_{tt}$. Substituting and dividing by 4,

$$t(1-t)\,y_{tt} + \Big[(\Delta+1-\tfrac{d}{2}) - (\Delta+\tfrac{3}{2})t\Big]\,y_t - \tfrac{1}{4}\Delta(\Delta+1)\,y = 0. \tag{54.16}$$

This is the hypergeometric equation $t(1-t)y_{tt} + [c - (a+b+1)t]y_t - ab\,y = 0$ with parameters read off directly:

$$c = \Delta - \frac{d}{2} + 1, \qquad a+b = \Delta + \tfrac{1}{2}, \qquad ab = \tfrac{1}{4}\Delta(\Delta+1). \tag{54.17}$$

The quadratic $X^2 - (\Delta+\frac{1}{2})X + \frac{1}{4}\Delta(\Delta+1) = 0$ has discriminant $(\Delta+\frac{1}{2})^2 - \Delta(\Delta+1) = \frac{1}{4}$, so its roots are $a = \frac{\Delta}{2}$ and $b = \frac{\Delta}{2} + \frac{1}{2}$. The interior-regular, normalizable solution is therefore

$$G_{\Delta}(\xi) \;=\; \mathcal{C}_{\Delta}\,\xi^{\Delta}\,{}_2F_1\Big(\frac{\Delta}{2},\,\frac{\Delta}{2}+\frac{1}{2};\,\Delta-\frac{d}{2}+1;\,\xi^2\Big), \tag{54.18}$$

the second solution behaving as $\xi^{d-\Delta}$ near $\xi = 0$ and discarded as non-normalizable.

Fixing the normalization by the short-distance singularity. The constant $\mathcal{C}_{\Delta}$ is fixed by the unit-strength delta source in Eq. (54.8), which is the statement that, at short distance, G_{Δ} must reduce to the flat-space Green's function: near coincidence the AdS metric is locally flat and the curvature and mass terms are subleading. In D Euclidean dimensions the Green's function of $-\nabla^2$ is $\mathcal{G}_{\text{flat}}(r) = A_D\, r^{2-D}$ with A_D fixed by Gauss's law, $1 = \int_{\text{ball}}(-\nabla^2\mathcal{G}) = -\oint \partial_r\mathcal{G}\,\mathrm{d}S = A_D(D-2)\,\Omega_{D-1}$, where $\Omega_{D-1} = 2\pi^{D/2}/\Gamma(D/2)$ is the unit-sphere area; hence

$$A_D = \frac{1}{(D-2)\,\Omega_{D-1}} = \frac{\Gamma(\frac{D}{2}-1)}{4\,\pi^{D/2}}, \qquad D = d+1: \quad \mathcal{G}_{\text{flat}}(s) = \frac{\Gamma(\frac{d-1}{2})}{4\,\pi^{(d+1)/2}}\,s^{\,1-d}. \tag{54.19}$$

On the other side, the coincidence limit $s \to 0$ is $\xi \to 1$, i.e. $t \to 1$. The hypergeometric function diverges there as $c-a-b = -\frac{d-1}{2} < 0$, and by the standard connection formula ${}_2F_1(a,b;c;t) \to \frac{\Gamma(c)\Gamma(a+b-c)}{\Gamma(a)\Gamma(b)}(1-t)^{c-a-b}$. Using $1-t = 1-\xi^2 = \tanh^2 s \simeq s^2$, $a+b-c = \frac{d-1}{2}$, and the Legendre duplication identity $\Gamma(\frac{\Delta}{2})\Gamma(\frac{\Delta}{2}+\frac{1}{2}) = 2^{1-\Delta}\sqrt{\pi}\,\Gamma(\Delta)$,

$$G_{\Delta}(\xi) \;\xrightarrow{\;s\to 0\;}\; \mathcal{C}_{\Delta}\,\frac{\Gamma(\Delta-\frac{d}{2}+1)\,\Gamma(\frac{d-1}{2})}{2^{1-\Delta}\sqrt{\pi}\,\Gamma(\Delta)}\;s^{\,1-d}. \tag{54.20}$$

Matching Eq. (54.20) to Eq. (54.19) cancels the common $\Gamma(\frac{d-1}{2})s^{1-d}$ and fixes

$$\boxed{\mathcal{C}_\Delta \;=\; \frac{\Gamma(\Delta)}{2^{\Delta+1}\,\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2}+1)}\,.} \tag{54.21}$$

Equations Eq. (54.18) and Eq. (54.21) are the bulk-to-bulk propagator in closed form. The same object is occasionally written in the chordal variable u of Eq. (54.9) as $G_\Delta \propto (2u)^{-\Delta}\,{}_2F_1(\Delta,\Delta-\frac{d-1}{2};2\Delta-d+1;-2/u)$; this is Eq. (54.18) after a quadratic hypergeometric transformation and is the form used downstream in Section 54.3.3. The two agree term by term, as the matching of their large-separation falloffs $\propto \xi^{\Delta} \propto u^{-\Delta}$ already guarantees.

54.1.3 The bulk-to-boundary propagator as a boundary limit

The bulk-to-boundary propagator K_Δ of Eq. (54.5) is the kernel that carries a boundary source into the interior; it is what an external leg of a Witten diagram is made of. It is most cleanly obtained as the boundary limit of G_Δ: a source placed at a bulk point Y becomes a boundary source as Y is pushed to the conformal boundary, $w \to 0$. The precise statement is that the normalizable falloff of G_Δ in its second argument factors out a bulk-to-boundary propagator,

$$K_\Delta(X;y) \;=\; (2\Delta-d)\,\lim_{w\to 0}\; w^{-\Delta}\,G_\Delta(X;\,w,y). \tag{54.22}$$

We carry out the limit explicitly and recover both the closed form of K_Δ and the normalization constant c_Δ that Theorem 53.30 obtained independently by the inversion method. As $w \to 0$, the chordal variable Eq. (54.10) vanishes, $\xi = 2zw/(z^2+w^2+|x-y|^2) \to 2w\cdot z/(z^2+|x-y|^2) \to 0$, so the hypergeometric factor in Eq. (54.18) tends to unity and only the leading power survives:

$$G_\Delta(X;w,y) \;\xrightarrow{w\to 0}\; \mathcal{C}_\Delta\,\xi^{\Delta} \;=\; \mathcal{C}_\Delta\,(2w)^{\Delta}\Big(\frac{z}{z^2+|x-y|^2}\Big)^{\Delta}. \tag{54.23}$$

Multiplying by $w^{-\Delta}$ removes the w-dependence, and the prefactor in Eq. (54.22) is chosen precisely to clean up the constant:

$$K_\Delta(X;y) \;=\; (2\Delta-d)\,2^{\Delta}\,\mathcal{C}_\Delta\,\Big(\frac{z}{z^2+|x-y|^2}\Big)^{\Delta} \;=\; c_\Delta\,\Big(\frac{z}{z^2+|x-y|^2}\Big)^{\Delta}. \tag{54.24}$$

The constant assembles to exactly c_Δ: using $(2\Delta-d)\Gamma(\Delta-\frac{d}{2}) = 2\,\Gamma(\Delta-\frac{d}{2}+1)$,

$$(2\Delta-d)\,2^{\Delta}\,\mathcal{C}_\Delta = (2\Delta-d)\,2^{\Delta}\,\frac{\Gamma(\Delta)}{2^{\Delta+1}\pi^{d/2}\Gamma(\Delta-\frac{d}{2}+1)} = \frac{\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2})} \;=\; c_\Delta, \tag{54.25}$$

in exact agreement with the value $c_\Delta = \Gamma(\Delta)/[\pi^{d/2}\Gamma(\Delta-\frac{d}{2})]$ of Eq. (53.177). This is a genuine cross-check: Theorem 53.30 fixed c_Δ by demanding $K_\Delta \to$

$z^{d-\Delta}\delta^d(x-y)$ at the boundary, whereas here the same constant emerges from the short-distance strength of the bulk-to-bulk delta source in Eq. (54.21). Two independent normalizations of two different propagators land on the same number; the perturbation theory is internally consistent.

Two properties of K_Δ make it the correct external leg, and both are visible in Eq. (54.24). First, it solves the *homogeneous* wave equation $(-\Box_{\mathrm{AdS}}+m^2)K_\Delta = 0$ everywhere in the interior — it must, being the $w \to 0$ limit of $G_\Delta(X;Y)$ away from coincidence — so attaching it to a vertex turns on no spurious bulk source. (One checks this directly: $\phi = z^\Delta$ solves the free equation by the computation in Eq. (53.177), and K_Δ is its image under the inversion isometry Eq. (53.178).) Second, it carries a *delta-function boundary source*: as $z \to 0$ at fixed $x \neq y$ it vanishes as z^Δ, the normalizable mode, while integrated against a test function it develops the non-normalizable $z^{d-\Delta}$ piece localized at $x = y$,

$$K_\Delta(z,x;y) \xrightarrow{z\to 0} z^{\,d-\Delta}\,\delta^d(x-y) \;+\; z^\Delta\,\frac{c_\Delta}{|x-y|^{2\Delta}} \;+\; \cdots\,, \tag{54.26}$$

the coefficient of $z^{d-\Delta}$ being a delta function precisely because c_Δ was fixed by the integral Eq. (53.179). The first term identifies K_Δ as the response to a unit boundary source $\phi_0 = \delta^d(x-y)$, justifying Eq. (54.5); the coefficient of the second, normalizable term, $c_\Delta/|x-y|^{2\Delta}$, is already the seed of the two-point function that Section 54.2 extracts in full.

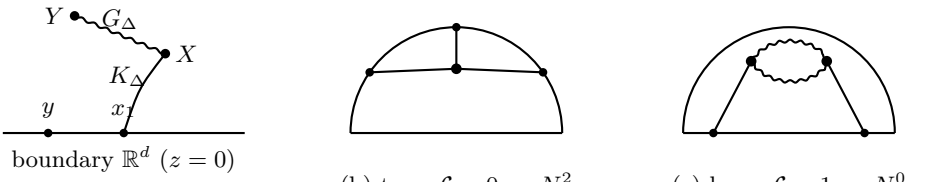
54.1.4 The Witten rules

We can now state the Feynman rules of perturbative holography. They are the diagrammatic encoding of the recursive solution of Section 54.1.1: each functional derivative in Eq. (54.2) attaches an external boundary leg, each interaction vertex of Eq. (54.3) produces an integrated bulk point, and the lines joining vertices are bulk-to-bulk propagators. A diagram drawn on a slice of the Poincaré patch — external points on the boundary, vertices in the interior — is called a *Witten diagram* (Fig. 54.1).

Theorem 54.2 (Witten-diagram rules)**.** *In the supergravity limit the connected n-point function $\langle \mathcal{O}_1(x_1)\cdots\mathcal{O}_n(x_n)\rangle_{\mathrm{c}}$ of single-trace boundary primaries is the sum over connected Witten diagrams, evaluated by the following position-space rules in Euclidean* AdS_{d+1}*:*

1. ***External legs.*** *Each boundary insertion $\mathcal{O}_i(x_i)$ is attached to a bulk vertex by a bulk-to-boundary propagator $K_{\Delta_i}(X;x_i) = c_{\Delta_i}\big(z/(z^2+|x-x_i|^2)\big)^{\Delta_i}$ Eq. (54.24), with $\Delta_i(\Delta_i-d) = m_i^2$.*

2. ***Internal lines.*** *Each internal line carrying a field of dimension Δ is a bulk-to-bulk propagator $G_\Delta(X,Y)$ Eq. (54.18), the normalizable Green's function Eq. (54.8).*



(a) propagators

(b) tree, $\mathcal{L}=0$, $\sim N^2$

(c) loop, $\mathcal{L}=1$, $\sim N^0$

Figure 54.1: The ingredients of perturbative holography. *(a)* A bulk-to-boundary propagator K_Δ (solid) reaches from a boundary insertion to an interior bulk point; a bulk-to-bulk propagator G_Δ (wavy) joins two interior points. *(b)* A tree-level Witten diagram — here a cubic contact vertex (filled dot) integrated over the bulk, with three external K_Δ legs — contributes at order N^2. *(c)* A one-loop diagram, with two internal bulk-to-bulk propagators forming a closed loop, is suppressed by $G_N \propto N^{-2}$ and drops out of the leading-order correlator. Solid horizontal lines are the conformal boundary; the half-disk is a slice of Euclidean AdS_{d+1}.

3. ***Vertices.*** *Each vertex is the corresponding coupling of the bulk action Eq. (54.3) integrated over the bulk with the invariant measure, $-g_k\int \mathrm{d}^{d+1}X\,\sqrt{g}$, with $\sqrt{g}=z^{-(d+1)}$ (in units $L=1$), and the appropriate symmetry factor.*

4. ***Loop counting.*** *A diagram with $\mathcal{L}$ closed bulk loops is suppressed by $G_N^{\mathcal{L}} \propto N^{-2\mathcal{L}}$ relative to the tree level Eq. (54.7). At leading order in $1/N$ only tree diagrams contribute.*

The overall normalization (one factor of $1/G_N \propto N^2$) and the sign are those of Eq. (54.1), fixed for the two-point case in Section 53.6.5.

Proof. The rules are a transcription of the recursive solution of the equations of motion and the resulting on-shell action. The first-order solution Eq. (54.5) attaches one K_Δ per unit of source, giving rule 1; the second- and higher-order solutions Eq. (54.6) convolve the inhomogeneity against G_Δ at each integrated bulk vertex, giving rules 2 and 3, with the measure $\sqrt{g}\,\mathrm{d}^{d+1}X$ inherited from the action Eq. (54.3). Resubstituting into the action and differentiating n times with respect to the source extracts the connected n-point function as the sum of trees with n external legs; the only freedom — combinatorial vertex weights — is the symmetry-factor bookkeeping of ordinary Feynman rules (Theorem 15.9), unchanged here because it is purely combinatorial. Rule 4 is the loop-counting Eq. (54.7), equivalently the genus expansion Theorem 53.2. □

The simplest diagram of all has no vertex: two external legs are sewn directly, which is the statement that the quadratic on-shell action computes the two-point function. We isolate that case as the bridge to the next section, but first we

exhibit the rules at work on the simplest *interacting* diagram, the cubic contact, which makes the calculus concrete and names the integral that Section 54.3 will evaluate.

Worked Example 54.3 (The cubic contact diagram)**.** Take three bulk scalars ϕ_1, ϕ_2, ϕ_3 of dimensions $\Delta_1, \Delta_2, \Delta_3$ with a single cubic vertex $g_3\,\phi_1\phi_2\phi_3$ in the action Eq. (54.3), dual to three boundary primaries $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$. We assemble their connected three-point function at leading (tree) order.

By Theorem 54.2 the only connected tree topology is a single bulk vertex with three external legs (Fig. 54.1b). Each source $\phi_{0,i}$ pulls down one bulk-to-boundary propagator $K_{\Delta_i}(X; x_i)$, the three meet at the vertex point X, and X is integrated over the bulk with the invariant measure. With three distinct fields the vertex carries no symmetry factor, and Eq. (54.2) gives

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle_{\mathrm{c}} \;=\; -\,g_3 \int \mathrm{d}^{d+1}X\,\sqrt{g}\;K_{\Delta_1}(X;x_1)\,K_{\Delta_2}(X;x_2)\,K_{\Delta_3}(X;x_3). \tag{54.27}$$

This is a contact Witten diagram: every line is a K_Δ, there is a single integrated vertex, no bulk-to-bulk propagator appears, and there is manifestly no loop. The entire content of the calculation is now the bulk integral on the right, a function of the three boundary points x_1, x_2, x_3 alone. Two of its features are forced before any integration is done. The integrand is built from isometry-covariant objects, so translations, rotations, dilations and the inversion Eq. (53.178) of AdS_{d+1} act on Eq. (54.27) as the conformal group acts on a CFT three-point function; the integral is therefore proportional to the conformally fixed structure $|x_{12}|^{-(\Delta_1+\Delta_2-\Delta_3)}|x_{23}|^{-(\Delta_2+\Delta_3-\Delta_1)}|x_{13}|^{-(\Delta_1+\Delta_3-\Delta_2)}$ of Theorem 47.21's three-point companion. And its overall coefficient — the holographic OPE coefficient C_{123} — is a definite ratio of Gamma functions built from g_3, the Δ_i, and the c_{Δ_i}. Evaluating the integral and extracting that coefficient is the work of Section 54.3, where the contact integral Eq. (54.27) is given the name D-function and computed by Schwinger parameters; here we have shown only that the Witten rules assemble it with no ambiguity left over. The reader should note how little freedom remains: the bulk action fixes g_3, the masses fix the Δ_i, and the diagram then *predicts* the OPE coefficient that, on the CFT side, is an independent piece of dynamical data.

The carried example of this Part is its honest namesake. The bulk scalars and gravitons whose Witten diagrams we have just learned to draw are the supergravity duals of the single-trace operators of $\mathcal{N} = 4$ super-Yang–Mills — the maximally symmetric, large-N cousin of Yang–Mills theory (Chapter 25) — so the rules of Theorem 54.2 are, concretely, the rules for computing strong-coupling correlators of that gauge theory, a regime no perturbative Yang–Mills expansion can reach. We defer the actual $\mathcal{N} = 4$ numbers — the free energy and the shear-viscosity ratio — to Sections 54.4 and 54.5, where the thermal state and its transport are computed; here the thread is only set in place.

With the propagators built and the rules stated, the apparatus is complete. The lightest possible use of it — two external legs and no vertex, equivalently the quadratic on-shell action — is the scalar two-point function. Its evaluation requires one more ingredient that the contact and exchange diagrams above quietly assumed finite: the on-shell action is divergent at the conformal boundary, and rendering it finite is holographic renormalization. We turn the bulk-to-boundary propagator K_Δ Eq. (54.24) and the rule "quadratic on-shell action = two-point function" over to the next section, which builds that renormalization machinery and recovers the conformally fixed correlator $\langle \mathcal{O}(x)\mathcal{O}(y)\rangle \propto |x-y|^{-2\Delta}$ with its coefficient computed from the bulk.

54.2 The Holographic Two-Point Function and Holographic Renormalization

The previous section turned the GKP–Witten saddle relation Eqs. (53.162) and (53.163) into position-space Feynman rules in AdS: every boundary insertion of a primary $\mathcal{O}$ is attached to the interior by a bulk-to-boundary propagator K_Δ Eq. (53.177), vertices are integrated over the bulk, and the connected correlators are functional derivatives of the classical on-shell action with respect to the boundary sources Theorem 54.2. The simplest such rule, with no vertex at all, is that the *quadratic* on-shell action is the two-point function. We now execute that rule in full, and in doing so build the renormalization machinery that the rest of the chapter cannot do without.

There is a subtlety the rule glosses over, and it is the whole content of this section. The on-shell action of a field on AdS is divergent: the warp factor $\sqrt{g} = (L/z)^{d+1}$ blows up at the conformal boundary $z \to 0$, and the naive boundary term picks up power-law divergences as the regulator is removed. These divergences are not a flaw in the duality; they are the bulk image of the familiar ultraviolet divergences of the boundary field theory — the radial coordinate is the renormalization scale (Section 53.4), so $z \to 0$ is the deep UV, and a cutoff at $z = \epsilon$ is a UV cutoff at energy $\sim 1/\epsilon$. The cure is the one the boundary theory already taught us in Section 19.1: the divergences are *local*, so they are removed by *local* covariant counterterms, and what remains is a finite, scheme-independent, separated-point correlator. The systematic version of this program on the gravity side is *holographic renormalization*. We develop it here for the scalar two-point function, because that is where the algebra is cleanest and the result is checkable against pure conformal symmetry; the same counterterm structure, in its gravitational guise, will renormalize the on-shell action that computes the free energy in Section 54.4.

Three ingredients make up the toolkit, and we take them in turn. First, the *near-boundary (Fefferman–Graham) expansion*: any bulk solution is, near $z = 0$, an asymptotic series in z whose two seeds are the source and the response,

with everything in between determined locally by the source. Second, the *regulated on-shell action* at a radial cutoff and the precise form of its divergences. Third, the *covariant counterterms* that cancel them, packaged as a theorem (Theorem 54.5). With the toolkit assembled we recompute the two-point function directly in position space — the computation Section 53.6.5 did in momentum space via Bessel functions, now done in the Witten-diagram language and giving an independent check of the coefficient — and read off the renormalized statement $\langle \mathcal{O}(x)\mathcal{O}(y)\rangle = (2\Delta - d)\, c_\Delta / |x-y|^{2\Delta}$.

54.2.1 The near-boundary expansion

We work in Euclidean Poincaré AdS_{d+1} Eq. (53.161) with flat boundary $\mathbb{R}^d$, radius L, and a free scalar of mass m dual to a primary $\mathcal{O}$ of dimension Δ. The classical field equation is the Euclidean Klein–Gordon equation Eq. (53.167),

$$z^2\,\partial_z^2\phi - (d-1)\,z\,\partial_z\phi + z^2\,\partial_x^2\phi - m^2L^2\,\phi = 0, \qquad \partial_x^2 \equiv \delta^{\mu\nu}\partial_\mu\partial_\nu, \tag{54.28}$$

and the analysis of Section 53.4 established that near $z=0$ every solution is a superposition of two power laws z^{Δ_-} and z^{Δ_+}, the exponents being the roots of the indicial (mass–dimension) equation $\Delta(\Delta-d) = m^2L^2$ Eq. (53.112). We recall the roots, since they organize everything that follows:

$$\Delta_\pm = \frac{d}{2} \pm \sqrt{\frac{d^2}{4} + m^2L^2}, \qquad \nu \equiv \Delta_+ - \frac{d}{2} = \sqrt{\frac{d^2}{4} + m^2L^2}, \tag{54.29}$$

so that $\Delta \equiv \Delta_+ = \frac{d}{2} + \nu$ is the dimension of the dual operator and $\Delta_- = d - \Delta = \frac{d}{2} - \nu$ its shadow. The square root is real precisely above the Breitenlohner–Freedman floor $m^2L^2 \geq -d^2/4$ Eq. (53.122), the bulk avatar of the requirement that Δ be real in a unitary CFT (Theorem 47.18).

The structural statement of Eq. (53.116) — two leading powers — is not the whole story, and the rest of the story is what holographic renormalization rests on. We need the *full* asymptotic series, because the divergences of the on-shell action live in the subleading terms. The systematic expansion is due to Fefferman and Graham; we derive it as a recursion.

Definition 54.4 (Near-boundary expansion)**.** For a free scalar dual to $\mathcal{O}$ of dimension $\Delta = \frac{d}{2} + \nu$, the *near-boundary* (Fefferman–Graham) *expansion* of a bulk solution is the double asymptotic series

$$\phi(z,x) \;\xrightarrow{z\to 0}\; z^{\,d-\Delta}\sum_{n\geq 0} a_n(x)\, z^{2n} \;+\; z^{\,\Delta}\sum_{n\geq 0} b_n(x)\, z^{2n}, \qquad a_0 = \alpha, \quad b_0 = \beta, \tag{54.30}$$

in which $\alpha(x)$ is the coefficient of the leading falloff $z^{d-\Delta}$ and $\beta(x)$ the coefficient of the subleading falloff z^{Δ}.

The two series are not on the same footing, and the difference is the crux. Insert Eq. (54.30) into Eq. (54.28) and collect the coefficient of a given power. Take a single monomial $a_n(x)\, z^p$ with $p = d - \Delta + 2n$. The first three terms of Eq. (54.28) act algebraically: $z^2\partial_z^2$ gives $p(p-1)$, $-(d-1)z\,\partial_z$ gives $-(d-1)p$, and $-m^2L^2$ is a multiplier, so together they contribute

$$[p(p-1) - (d-1)p - m^2L^2]a_n\, z^p = [p^2 - dp - m^2L^2]a_n\, z^p. \tag{54.31}$$

Using $m^2L^2 = \Delta(\Delta - d)$, the bracket factorizes,

$$p^2 - dp - \Delta(\Delta - d) = (p - \Delta)(p + \Delta - d), \tag{54.32}$$

as one checks by expanding the right side, $p^2 + (\Delta - d - \Delta)p - \Delta(\Delta - d) = p^2 - dp - \Delta(\Delta - d)$. The remaining term $z^2\partial_x^2$ raises the power by two, so the monomial $a_{n-1}z^{p-2}$ feeds $\partial_x^2 a_{n-1}\, z^p$ into the same order. Setting the total coefficient of z^p to zero gives the recursion

$$(p - \Delta)(p + \Delta - d)\, a_n + \partial_x^2 a_{n-1} = 0, \qquad p = d - \Delta + 2n. \tag{54.33}$$

Now evaluate the prefactor: $p - \Delta = d - 2\Delta + 2n = 2n - 2\nu$ and $p + \Delta - d = 2n$, so $(p - \Delta)(p + \Delta - d) = 4n(n - \nu)$ and

$$\boxed{a_n = -\frac{\partial_x^2\, a_{n-1}}{4\, n\, (n - \nu)}, \qquad n \geq 1,} \tag{54.34}$$

with $a_0 = \alpha$ free. The first coefficient is therefore

$$a_1 = -\frac{\partial_x^2 \alpha}{4(1 - \nu)} = \frac{\partial_x^2 \alpha}{2\,(2\Delta - d - 2)}, \tag{54.35}$$

using $4(1-\nu) = -2(2\Delta - d - 2)$. The pattern is the whole point: *every coefficient a_n of the leading series is a local differential operator acting on the single datum α.* The leading falloff and its entire analytic tail are fixed, with no further input, by the boundary source.

The subleading series is governed by the same algebra with $\Delta \to d - \Delta$, i.e. $\nu \to -\nu$: writing $q = \Delta + 2n$ one finds $(q - \Delta)(q + \Delta - d) = 4n(n + \nu)$ and

$$b_n = -\frac{\partial_x^2 b_{n-1}}{4\, n\, (n + \nu)}, \qquad n \geq 1, \tag{54.36}$$

with $b_0 = \beta$ free. Crucially, the recursion Eq. (54.34) *never* reaches the order $z^\Delta = z^{d-\Delta+2\nu}$ for non-integer ν: the two series stay disjoint, and β is a genuinely independent integration constant, not determined by α through any local relation. The equation of motion is second order, so a solution carries two free functions of x, and the near-boundary expansion exhibits them cleanly as α and β. What ties β to α is not the near-boundary analysis at all but the *global* condition

of regularity in the interior $z \to \infty$; that is the content of the modified-Bessel solution Eq. (53.176), which fixed the ratio

$$\beta(k) = \mathcal{B}(k)\,\alpha(k), \qquad \mathcal{B}(k) = \frac{\Gamma(-\nu)}{\Gamma(\nu)}\Big(\frac{k}{2}\Big)^{2\nu}, \tag{54.37}$$

the single *non-local* (in x) piece of information in the whole problem. This split — local data α determines the divergent structure, global data β carries the physics — is exactly why the divergences can be subtracted by local counterterms without touching the answer.

Source and response; the two quantizations. The dictionary of Section 53.5 identifies the leading coefficient α with the field-theory *source* coupling to $\mathcal{O}$, $\alpha = \phi_0$, and we will prove below that β is, up to a fixed factor, the operator's *vacuum expectation value*: $\langle\mathcal{O}\rangle \propto \beta$. This is the *standard quantization*, available for all $\Delta \geq d/2$. In the Breitenlohner–Freedman window $-d^2/4 \leq m^2L^2 < -d^2/4 + 1$, equivalently $0 \leq \nu < 1$, the normalizability analysis Theorem 53.18 found that *both* falloffs are normalizable, and one is free to exchange their roles: declaring β the source and α the response defines the *alternative quantization*, dual to an operator of dimension $\Delta_- = d - \Delta$. The two quantizations are related by a Legendre transform of the renormalized action; we treat the standard one and note the dual under $\Delta \to d - \Delta$.

Caution

The recursion Eq. (54.34) has a vanishing denominator when $n = \nu$, i.e. when ν (equivalently $\Delta - \frac{d}{2}$) is a non-negative integer. There the leading series would generate a term $z^{d-\Delta+2\nu} = z^{\Delta}$, colliding with the seed of the subleading series, and the resolution is a logarithm: the expansion acquires a term $\propto z^{\Delta}\log z$, as flagged already in Eq. (53.116) and seen in the Bessel expansion of Theorem 53.32. This logarithm is not a pathology; it is the seed of the holographic conformal anomaly, and we account for it in Theorem 54.6. For now take ν generic.

54.2.2 The regulated on-shell action and its divergences

We evaluate the bulk action of the free scalar on the solution. Integrating by parts and using the equation of motion, the bulk integral collapses to a boundary term, regulated at $z = \epsilon$; this was carried out in Eq. (53.181), giving

$$S_{\mathrm{reg}}[\epsilon] = -\frac{\mathcal{N}}{2}\int_{z=\epsilon} \mathrm{d}^d x\; z^{1-d}\,\phi\,\partial_z\phi\Big|_{z=\epsilon}, \qquad \mathcal{N} \equiv \mathcal{N}_\phi L^{d-1}, \tag{54.38}$$

where $\mathcal{N}_\phi \propto 1/G_{d+1} \propto N^2$ is the matter normalization of Section 53.6.2 and the sign is that of the outward normal pointing to decreasing z. The interior boundary

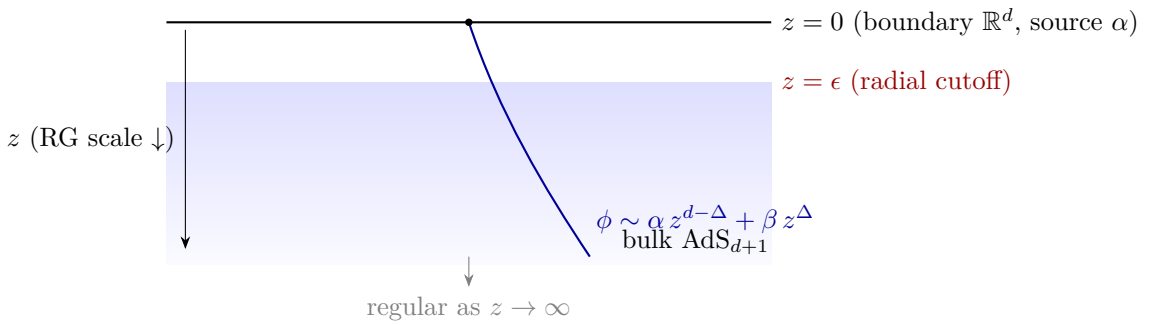


Figure 54.2: Holographic renormalization at a glance. A boundary source α at $z = 0$ launches a bulk profile whose near-boundary form Eq. (54.30) contains the leading mode $z^{d-\Delta}$ (with its locally determined tail) and the subleading mode z^{Δ} with coefficient β; interior regularity at $z \to \infty$ fixes β in terms of α Eq. (54.37). The on-shell action is evaluated at the radial cutoff $z = \epsilon$ (dashed) and its $\epsilon \to 0$ divergences — local functionals of α — are removed by covariant counterterms living on that slice.

contributes nothing because the regular solution decays there. Everything now hinges on the small-ϵ behaviour of the boundary flux $z^{1-d}\phi\,\partial_z\phi$, which the near-boundary expansion delivers.

Write $\phi = \alpha z^{d-\Delta}(1 + \cdots) + \beta z^{\Delta}(1 + \cdots)$ with the dots the local tails of Eq. (54.30). The leading piece of the flux comes from the source mode alone: with $\phi \simeq \alpha z^{d-\Delta}$ one has $\partial_z\phi \simeq (d-\Delta)\alpha z^{d-\Delta-1}$, so

$$z^{1-d}\,\phi\,\partial_z\phi \;\simeq\; (d-\Delta)\,\alpha^2\, z^{\,2(d-\Delta)-d} = (d-\Delta)\,\alpha^2\, z^{-2\nu}, \tag{54.39}$$

a divergence as $z \to 0$ for every $\nu > 0$. The first analytic correction $a_1 \propto \partial_x^2\alpha$ promotes this to a term $z^{-2\nu+2}$ with coefficient $\propto \alpha\,\partial_x^2\alpha$, the next to $z^{-2\nu+4}$ with coefficient built from $\alpha\,\partial_x^4\alpha$, and so on. Organizing the flux by powers of z,

$$z^{1-d}\phi\,\partial_z\phi = (d-\Delta)\,\alpha^2\,\epsilon^{-2\nu} + c_2\,\alpha\,\partial_x^2\alpha\,\epsilon^{-2\nu+2} + \cdots + d\,\alpha\beta + (\text{vanishing as } \epsilon \to 0), \tag{54.40}$$

where the coefficient of the single finite, ϵ-independent term is computed exactly as in Eq. (53.183) and equals $d\,\alpha\beta$ (the cross term of the two falloffs). Two features are decisive.

First, the divergences are local. Each divergent power $\epsilon^{-2\nu+2k}$ comes multiplied by a coefficient that is a *local* differential polynomial in α: α^2, then $\alpha\,\partial_x^2\alpha$, then $\alpha\,\partial_x^4\alpha$, and so forth — precisely the structure forced by the local recursion Eq. (54.34). There is no $\alpha\beta$ in any divergent coefficient, because β first appears at the finite order. Locality is what licenses a local covariant subtraction, exactly as the local divergences of a one-loop boundary diagram licensed the local counterterms of Section 19.2.6 and Eq. (19.23).

Second, the finite piece $d\,\alpha\beta$ is not yet the answer. It is contaminated by the very regulator we are about to remove: subtracting the divergence will necessarily reach into the finite part, because the counterterm, expanded near the boundary, also produces an $\alpha\beta$ term. The bare coefficient d is wrong; the renormalized one is what we now compute.

54.2.3 Covariant counterterms and the holographic-renormalization theorem

The subtraction must be *covariant* and *local on the cutoff surface*, for the same reason a good regulator's counterterms must respect the symmetries: a subtraction that did not could spoil conformal invariance and inject spurious physics into the separated-point correlator. The cutoff slice $z = \epsilon$ carries an induced metric $\gamma_{\mu\nu} = (L/\epsilon)^2\delta_{\mu\nu}$, with $\sqrt{\gamma} = (L/\epsilon)^d$, and the boundary value of the field is $\phi_\epsilon(x) \equiv \phi(\epsilon, x)$. The available local covariant scalars built from these of the right dimension are ϕ_ϵ^2, $\phi_\epsilon\,\Box_\gamma\phi_\epsilon$ (where $\Box_\gamma$ is the Laplacian of γ), $R[\gamma]\,\phi_\epsilon^2$, and so on. The counterterm action is their covariant sum.

Theorem 54.5 (Holographic renormalization of the scalar two-point function)**.** *Let ϕ be a free scalar of dimension $\Delta = \frac{d}{2} + \nu$ (with $\nu > 0$ non-integer) on Euclidean AdS_{d+1}, with source α and response β as in Eq. (54.30). Define the covariant counterterm action on the cutoff slice*

$$S_{\rm ct}[\epsilon] = \frac{\mathcal{N}_\phi}{2}\int_{z=\epsilon} \mathrm{d}^d x\,\sqrt{\gamma}\,\Big[\frac{d-\Delta}{L}\,\phi_\epsilon^2 + \frac{L}{2(2\Delta-d-2)}\,\phi_\epsilon\,\Box_\gamma\phi_\epsilon + \cdots\Big], \tag{54.41}$$

the displayed terms cancelling the divergences $\epsilon^{-2\nu}$ and $\epsilon^{-2\nu+2}$ and the dots being the finitely many higher curvature/derivative terms needed for the remaining power divergences. Then the renormalized action

$$S_{\rm ren} = \lim_{\epsilon\to 0}\big(S_{\rm reg}[\epsilon] + S_{\rm ct}[\epsilon]\big) \tag{54.42}$$

is finite, and the renormalized one-point function is the subleading mode,

$$\langle\mathcal{O}(x)\rangle = (2\Delta - d)\,\mathcal{N}\,\beta(x), \qquad \mathcal{N} = \mathcal{N}_\phi L^{d-1}. \tag{54.43}$$

With the canonical normalization $\mathcal{N} = 1$ fixed by the dictionary Eq. (53.190), $\langle\mathcal{O}\rangle = (2\Delta - d)\beta$.

Proof. We track the leading term explicitly; the higher ones work identically order by order. Evaluate the leading counterterm on the solution. With $\phi_\epsilon = \alpha\,\epsilon^{d-\Delta}(1 + 2\,(\beta/\alpha)\epsilon^{2\nu} + \cdots)$ one has $\phi_\epsilon^2 = \alpha^2\epsilon^{2(d-\Delta)}\big[1 + 2\,(\beta/\alpha)\,\epsilon^{2\nu} + \cdots\big]$, hence

$$\frac{\mathcal{N}_\phi}{2}\int \mathrm{d}^d x\,\sqrt{\gamma}\,\frac{d-\Delta}{L}\,\phi_\epsilon^2 = \frac{\mathcal{N}_\phi L^{d-1}}{2}\,(d-\Delta)\int \mathrm{d}^d x\;\epsilon^{-d}\,\alpha^2\,\epsilon^{2(d-\Delta)}\big[1 + 2(\beta/\alpha)\epsilon^{2\nu} + \cdots\big]$$

$$= \frac{\mathcal{N}}{2}\,(d-\Delta)\int \mathrm{d}^d x\,\left[\alpha^2\,\epsilon^{-2\nu} + 2\,\alpha\beta + \cdots\right], \tag{54.44}$$

using $\sqrt{\gamma}/L = L^{d-1}\epsilon^{-d}$ and $-d + 2(d-\Delta) = d - 2\Delta = -2\nu$. Meanwhile Eq. (54.38) fed by Eq. (54.40) reads

$$S_{\mathrm{reg}}[\epsilon] = -\frac{\mathcal{N}}{2}\int \mathrm{d}^d x\,\left[(d-\Delta)\,\alpha^2\,\epsilon^{-2\nu} + \cdots + d\,\alpha\beta + \cdots\right]. \tag{54.45}$$

The divergent $\epsilon^{-2\nu}$ pieces cancel between Eqs. (54.44) and (54.45) by construction — this is what fixed the counterterm coefficient $(d-\Delta)/L$. The $\epsilon^{-2\nu+2}$ divergence is cancelled, identically, by the second term of Eq. (54.41), whose coefficient $L/[2(2\Delta - d - 2)]$ is exactly the one that reproduces c_2 in Eq. (54.40) after the $\Box_\gamma = (\epsilon/L)^2\partial_x^2$ rescaling and the FG coefficient Eq. (54.35); the higher dots proceed the same way and exhaust the finite tower of power divergences. What survives the limit is finite, with the finite $\alpha\beta$ coefficient collapsing from the bare $-d$ to

$$2(d-\Delta) - d = d - 2\Delta = -(2\Delta - d), \tag{54.46}$$

the first contribution coming from Eq. (54.44) (the $2\alpha\beta$ reach of the counterterm) and the second from Eq. (54.45) (the $d\,\alpha\beta$ of the bare flux). Therefore

$$S_{\mathrm{ren}} = -\frac{\mathcal{N}}{2}\,(2\Delta - d)\int \mathrm{d}^d x\,\mathrm{d}^d y\;\alpha(x)\,R(x,y)\,\alpha(y) + (\text{local, scheme}), \tag{54.47}$$

where we have written the nonlocal $\alpha\beta$ pairing as a quadratic form in the source through $\beta(x) = \int \mathrm{d}^d y\,R(x,y)\,\alpha(y)$, the response kernel Eq. (54.37). The renormalized one-point function follows from Eq. (53.164), $\langle\mathcal{O}(x)\rangle = -\,\delta S_{\mathrm{ren}}/\delta\alpha(x)$. Varying Eq. (54.47) once (the quadratic form is symmetric, giving a factor of two against the $\frac{1}{2}$),

$$\langle\mathcal{O}(x)\rangle = (2\Delta - d)\,\mathcal{N}\int \mathrm{d}^d y\,R(x,y)\,\alpha(y) = (2\Delta - d)\,\mathcal{N}\,\beta(x), \tag{54.48}$$

which is Eq. (54.43). The local, scheme-dependent remainder in Eq. (54.47) is supported at coincident points; it shifts contact terms only and we keep it implicit. □

The mechanism deserves to be named because it recurs everywhere in holographic renormalization: *the counterterm introduced to kill a divergence carries a finite remainder of just the right size to correct the answer.* The naive finite coefficient d of Eq. (54.40) is meaningless; only the combination Eq. (54.46) that survives the subtraction is physical, and it is exactly $-(2\Delta - d)$. This is the holographic face of the statement that, in the boundary theory, the value of a renormalized correlator depends on the subtraction scheme through contact terms but not through its separated-point structure.

The gravitational counterterms. The same logic, applied to the metric rather than a probe scalar, supplies the counterterms that will renormalize the free energy in Section 54.4. The bulk gravitational action — Einstein–Hilbert plus negative cosmological constant plus the Gibbons–Hawking boundary term — evaluated on an asymptotically-AdS solution diverges because the proper volume near the boundary is infinite. The leading divergence is a pure volume term: on shell the Lagrangian density is constant ($R - 2\Lambda = -2d/L^2$ for AdS_{d+1} with $\Lambda = -d(d-1)/2L^2$), so the bulk integral $\int_\epsilon \mathrm{d}z\,(L/z)^{d+1} \sim \epsilon^{-d}$ diverges as ϵ^{-d}, a divergence proportional to the boundary volume $\int \mathrm{d}^d x\sqrt{\gamma}$. It is cancelled by a *cosmological-constant counterterm* on the slice; the subleading divergences, proportional to the boundary curvature, by an *Einstein (curvature) counterterm*:

$$S_{\mathrm{ct}}^{\mathrm{grav}} = \frac{1}{8\pi G_{d+1}} \int_{z=\epsilon} \mathrm{d}^d x\,\sqrt{\gamma}\,\Big[\frac{d-1}{L} + \frac{L}{2(d-2)}\,R[\gamma] + \cdots\Big], \tag{54.49}$$

the constant $(d-1)/L$ fixing the volume divergence and the $R[\gamma]$ term the next one; for higher d further curvature-squared terms follow, all local and covariant. The coefficients are universal — they depend only on the asymptotics, not on the particular solution — which is why they are computed once and reused for every black-brane background. We quote them here as the metric-sector members of the same toolkit; their explicit deployment, and the resulting finite on-shell action whose value is the free energy, is the business of Section 54.4.

Remark 54.6 (The logarithmic divergence and the holographic Weyl anomaly). When $\nu = \Delta - \frac{d}{2}$ is a non-negative integer the recursion Eq. (54.34) resonates at $n = \nu$ and the on-shell action develops, alongside the powers, a divergence $\propto \log\epsilon$. A logarithm cannot be removed by any covariant *power* counterterm; subtracting it requires a counterterm of the form $\log\epsilon$ times a local covariant integral, and such a counterterm *breaks scale invariance*: under a rescaling of the cutoff $\epsilon \to \lambda\epsilon$, equivalently a Weyl rescaling of the boundary metric, $\log\epsilon$ shifts by a constant and the renormalized action is not invariant. The non-invariance is finite, local, and unremovable — it is the holographic conformal (Weyl) anomaly, and the local integral multiplying $\log\epsilon$ is precisely the trace anomaly $\langle T^{\mu}{}_{\mu}\rangle$ of the boundary CFT (Chapter 37). In the scalar two-point function this is the integer-ν logarithm of Theorem 53.31, made concrete in the $d = 4$, $\Delta = 4$ computation of Theorem 53.32, where the response degenerated to $(k^2)^2\log(k^2/\mu^2)$ and a renormalization scale μ entered through the log counterterm. In the gravitational sector the same logarithm, present only in *even* boundary dimension d, multiplies the curvature invariant that is the bulk computation of the boundary central charges — the holographic derivation of the a- and c-anomalies. The pattern is uniform: power divergences are scheme, the logarithm is physics.

54.2.4 The two-point function in position space

We now assemble the connected two-point function and read off its coefficient directly in position space, the language in which the higher-point Witten diagrams of Section 54.3 are built. The starting point is the renormalized vev Eq. (54.43): differentiating once more with respect to the source produces the two-point function,

$$\langle \mathcal{O}(x)\,\mathcal{O}(y)\rangle = \frac{\delta\,\langle \mathcal{O}(x)\rangle}{\delta\alpha(y)} = (2\Delta - d)\,\mathcal{N}\,\frac{\delta\beta(x)}{\delta\alpha(y)}, \tag{54.50}$$

so the entire content is the response kernel $\delta\beta/\delta\alpha$, the position-space version of $\mathcal{B}(k)$. We extract it from the bulk-to-boundary propagator without re-solving the radial equation. The regular bulk solution sourced by α is the smearing

$$\phi(z,x) = \int \mathrm{d}^d y\; K_\Delta(z,x;y)\,\alpha(y), \qquad K_\Delta(z,x;y) = c_\Delta\,\frac{z^\Delta}{(z^2+|x-y|^2)^\Delta}, \tag{54.51}$$

with K_Δ the source-normalized propagator Eq. (53.177) and $c_\Delta = \Gamma(\Delta)/[\pi^{d/2}\Gamma(\Delta - \frac{d}{2})]$. Recall how this propagator was normalized: $K_\Delta \to z^{d-\Delta}\,\delta^d(x-y)$ as $z\to 0$, which guarantees that Eq. (54.51) has the correct leading falloff $\phi \to z^{d-\Delta}\alpha(x)$, identifying the smearing's seed with the source. We need its *subleading* coefficient, the z^Δ term, which is by definition β.

The extraction is immediate from the explicit form. Split the y-integral into a coincident region $y \approx x$ and a separated region $y \neq x$. For $y \neq x$, the limit $z \to 0$ of K_Δ is taken with $z^2 \ll |x-y|^2$, so the denominator becomes $|x-y|^{2\Delta}$ and

$$K_\Delta(z,x;y) \xrightarrow{z\to 0,\; y\neq x} c_\Delta\,\frac{z^\Delta}{|x-y|^{2\Delta}}. \tag{54.52}$$

This is a pure z^Δ term: it is exactly the response mode, with coefficient $c_\Delta/|x-y|^{2\Delta}$. The coincident region $y \approx x$ reproduces, together with the analytic tail of Eq. (54.51), the leading $z^{d-\Delta}$ delta function and the local FG tower $a_n[\alpha]$ — all of it local, all of it absorbed into the scheme-dependent contact part already set aside in Eq. (54.47). Reading off the z^Δ coefficient of Eq. (54.51) therefore gives the response as a smearing of the source against the kernel Eq. (54.52):

$$\beta(x) = \int \mathrm{d}^d y\;\frac{c_\Delta}{|x-y|^{2\Delta}}\,\alpha(y) + (\text{local in } \alpha), \qquad \Longrightarrow \qquad \frac{\delta\beta(x)}{\delta\alpha(y)} = \frac{c_\Delta}{|x-y|^{2\Delta}} \quad (x\neq y). \tag{54.53}$$

Substituting into Eq. (54.50) gives the renormalized two-point function,

$$\boxed{\langle \mathcal{O}(x)\,\mathcal{O}(y)\rangle = \frac{(2\Delta-d)\,\mathcal{N}\,c_\Delta}{|x-y|^{2\Delta}} = \mathcal{N}\,\frac{(2\Delta-d)\,\Gamma(\Delta)}{\pi^{d/2}\,\Gamma(\Delta-\frac{d}{2})}\,\frac{1}{|x-y|^{2\Delta}}\,,} \tag{54.54}$$

with $\mathcal{N} = \mathcal{N}_\phi L^{d-1}$, and the canonical choice $\mathcal{N} = 1$ giving the unit-normalized CFT coefficient $C_\Delta = (2\Delta - d)\,c_\Delta$.

This is the same number as the momentum-space computation of Eq. (53.195), reached by a route that touches neither Bessel functions nor a Fourier transform: the position-space response kernel is literally the subleading term of the bulk-to-boundary propagator, and the coefficient $2\Delta - d$ is the universal factor that the renormalization theorem Theorem 54.5 attached to the vev. The agreement of the two derivations is a check that the holographic-renormalization machinery is internally consistent — the divergent intermediaries (the $\Gamma(-\nu)$ that cancelled between kernel and transform in Eq. (53.194), the $\epsilon^{-2\nu}$ that cancelled against the counterterm here) leave no trace in the finite separated-point answer.

Matching conformal symmetry, and the thread. Compare Eq. (54.54) with the form that pure conformal symmetry forces on any scalar primary two-point function, derived with no dynamics in Theorem 47.21,

$$\langle \mathcal{O}(x)\, \mathcal{O}(y)\rangle = \frac{C_{\mathcal{O}}}{|x-y|^{2\Delta}}, \tag{54.55}$$

the structure of Eq. (47.120). Holography reproduces it exactly: the power is 2Δ with Δ the mass–dimension value Eq. (54.29), and the would-be free function of $|x-y|$ is pinned to a pure power by the same conformal invariance that, in the bulk, was the AdS isometry group. What symmetry left undetermined — the coefficient $C_{\mathcal{O}}$ — is now *computed*: $C_\Delta = (2\Delta - d)c_\Delta$, manifestly positive for $\Delta > d/2$ (since $\Gamma(\Delta - \frac{d}{2}) > 0$ there), in accord with reflection positivity and the unitarity bound $\Delta \geq (d-2)/2$ of Theorem 47.18. The overall $\mathcal{N}_\phi \propto N^2$ is the holographic image of large-N factorization: connected single-trace correlators scale as N^2, the same $1/G_{d+1}$ that sets the central charge of the dual.

This same conformally fixed two-point structure is where ϕ^4 theory ended. At the Wilson–Fisher fixed point of Chapter 52, the order-parameter primary's two-point function was pinned by scale and conformal invariance to $\langle \mathcal{O}(x)\mathcal{O}(0)\rangle \sim |x|^{-2\Delta}$ Eq. (52.38), with Δ read off from a critical exponent and the coefficient fixed by the normalization of $\mathcal{O}$ — the structure was *imposed* by symmetry, and the number was an input. Here the identical structure emerges *dynamically* from a bulk wave equation, with the coefficient C_Δ *output* by the geometry. The two perspectives are complementary faces of the same conformal datum: the fixed point is a CFT whose two-point function symmetry constrains, and a holographic CFT is one whose two-point function a classical bulk computes. Where a theory admits both descriptions — a large-N, large-gap CFT with a weakly curved dual, the role played for Yang–Mills theory's maximally symmetric cousin $\mathcal{N} = 4$ super-Yang–Mills — the symmetry-fixed form and the bulk computation must, and do, agree.

With the two-point function in hand and the renormalization toolkit assembled — the near-boundary expansion Eq. (54.30), the radial cutoff and its local divergences Eq. (54.40), the covariant counterterms Eqs. (54.41) and (54.49), the

identification of the vev with the subleading mode Eq. (54.43), and the normalized bulk-to-boundary propagator Eq. (54.51) whose subleading term is the response kernel — we are equipped to turn on interactions. The next section integrates cubic and quartic bulk vertices against these propagators to compute three- and four-point functions, where the finite renormalized action of Theorem 54.5 again removes the boundary divergences, now of contact and exchange diagrams, and delivers OPE coefficients and the double-trace spectrum.

54.3 Three- and Four-Point Functions and OPE Data

The two-point function of Section 53.6.5 used only the quadratic part of the bulk action: a single field propagating freely from one boundary insertion to another, its renormalized on-shell action read off by the holographic-renormalization toolkit assembled in the previous section. That toolkit — the near-boundary expansion, the radial cutoff at $z = \epsilon$, the local counterterms, and the normalized bulk-to-boundary propagator K_Δ of Eq. (53.177) — is now in hand and will not be rebuilt. What changes here is the action. The bulk Lagrangian generically contains interaction vertices, cubic and quartic in the fields, that we suppressed when all we wanted was $\langle \mathcal{O}\mathcal{O} \rangle$. Switching them on, the classical saddle that dominates the GKP–Witten partition function Eq. (53.162) is no longer a single free profile but a tree of propagators tied together at integrated bulk vertices. Functional differentiation of the on-shell action with respect to the boundary sources then produces connected correlators of three and more operators, organized diagrammatically by the Witten rules of Theorem 54.2. This section computes the first two such correlators. The three-point contact diagram will reproduce, with no freedom left over, the conformally fixed structure of Eq. (47.137), and hand us its single undetermined coefficient — the OPE coefficient C_{123} — as an explicit ratio of Gamma functions built from the bulk coupling. The four-point exchange diagram will then be decomposed into conformal blocks, exposing not only the single-trace operator dual to the exchanged bulk field but an entire tower of *double-trace* operators whose dimensions, and whose interaction-induced shifts, are the holographic image of the operator data that the bootstrap of Chapters 49 and 50 carves out from crossing alone.

54.3.1 The contact diagram and the D-function

Every tree-level Witten diagram is assembled from two ingredients: the bulk-to-boundary propagator K_Δ, which carries a boundary insertion into the interior, and the integrated bulk vertex. The simplest non-trivial assembly is the *contact diagram*: all external legs meet at one bulk point, which is integrated over AdS_{d+1}. The basic integral it produces deserves a name.

Definition 54.7 (Contact D-function)**.** For boundary points $x_1,\dots,x_n\in\mathbb{R}^d$ and dimensions $\Delta_1,\dots,\Delta_n$, the contact D-function is the bulk integral of n unnormalized bulk-to-boundary propagators,

$$D_{\Delta_1\cdots\Delta_n}(x_1,\dots,x_n)\;\equiv\;\int_0^\infty\frac{\mathrm{d}z}{z^{d+1}}\int_{\mathbb{R}^d}\mathrm{d}^dx\;\prod_{i=1}^n\widehat{K}_{\Delta_i}(z,x;x_i),\qquad \widehat{K}_\Delta(z,x;x_i)=\Big(\frac{z}{z^2+|x-x\tag{54.56}$$

where $\widehat{K}_\Delta=K_\Delta/c_\Delta$ strips the normalization $c_\Delta=\Gamma(\Delta)/[\pi^{d/2}\Gamma(\Delta-\frac{d}{2})]$ of Eq. (53.177), and the measure $\mathrm{d}z\,\mathrm{d}^dx/z^{d+1}$ is the AdS_{d+1} volume element in units $L=1$ (the factor L^{d+1} from $\sqrt{g}$ is restored at the end).

The whole content of any contact Witten diagram is a D-function times the coupling and the normalizations c_{Δ_i}. It is worth reducing Eq. (54.56) once and for all to a single integral over a simplex of Feynman-like parameters; this is the Symanzik trick, and it works for any n.

Introduce a Schwinger parameter t_i for each propagator,

$$\frac{1}{(z^2+|x-x_i|^2)^{\Delta_i}}=\frac{1}{\Gamma(\Delta_i)}\int_0^\infty\mathrm{d}t_i\,t_i^{\Delta_i-1}\,e^{-t_i(z^2+|x-x_i|^2)}.\tag{54.57}$$

Write $\Sigma\equiv\sum_i\Delta_i$ and $T\equiv\sum_i t_i$. Collecting the numerators z^{Δ_i} into z^Σ, the D-function becomes

$$D_{\Delta_1\cdots\Delta_n}=\frac{1}{\prod_i\Gamma(\Delta_i)}\int_0^\infty\Big[\prod_i\mathrm{d}t_i\,t_i^{\Delta_i-1}\Big]\int_0^\infty\mathrm{d}z\,z^{\Sigma-d-1}\int_{\mathbb{R}^d}\mathrm{d}^dx\;\exp\Big[-Tz^2-\sum_i t_i|x-x_i|^2\Big].\tag{54.58}$$

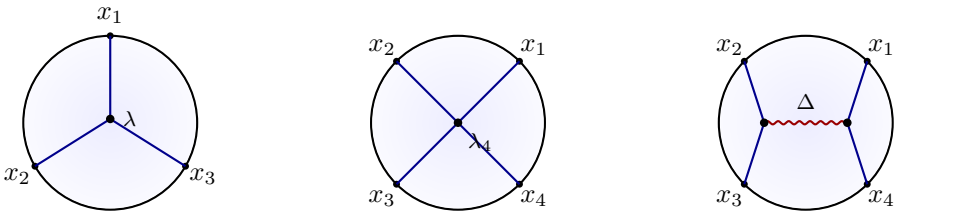
The boundary integral is Gaussian. Completing the square, $\sum_i t_i|x-x_i|^2=T\,|x-\bar{x}|^2+\frac{1}{T}\sum_{i<j}t_it_j|x_{ij}|^2$ with $\bar{x}=T^{-1}\sum_i t_ix_i$ and $x_{ij}\equiv x_i-x_j$, so

$$\int_{\mathbb{R}^d}\mathrm{d}^dx\,e^{-\sum_i t_i|x-x_i|^2}=\Big(\frac{\pi}{T}\Big)^{d/2}\exp\Big[-\frac{1}{T}\sum_{i<j}t_it_j|x_{ij}|^2\Big].\tag{54.59}$$

The radial integral is a Gamma function, $\int_0^\infty\mathrm{d}z\,z^{\Sigma-d-1}e^{-Tz^2}=\frac{1}{2}\Gamma(\frac{\Sigma-d}{2})T^{-(\Sigma-d)/2}$. Together with $(\pi/T)^{d/2}$ this gives an overall $T^{-\Sigma/2}$. Finally rescale $t_i=\rho\,\alpha_i$ with $\rho\in(0,\infty)$ and $\sum_i\alpha_i=1$; then $T=\rho$, the measure factorizes as $\prod_i\mathrm{d}t_i=\rho^{n-1}\,\mathrm{d}\rho\,\delta(1-\sum_i\alpha_i)\prod_i\mathrm{d}\alpha_i$, and the ρ-integral $\int_0^\infty\mathrm{d}\rho\,\rho^{\Sigma/2-1}e^{-\rho P}=\Gamma(\frac{\Sigma}{2})P^{-\Sigma/2}$ with $P=\sum_{i<j}\alpha_i\alpha_j|x_{ij}|^2$. The result is the Symanzik representation

$$\boxed{D_{\Delta_1\cdots\Delta_n}=\frac{\pi^{d/2}\,\Gamma(\frac{\Sigma-d}{2})\,\Gamma(\frac{\Sigma}{2})}{2\,\prod_i\Gamma(\Delta_i)}\int_{\Delta_{n-1}}\Big[\prod_i\mathrm{d}\alpha_i\,\alpha_i^{\Delta_i-1}\Big]\,\delta\Big(1-\sum_i\alpha_i\Big)\,\Big(\sum_{i<j}\alpha_i\alpha_j\,|x_{ij}|^2\Big)^{-\Sigma/2},}\tag{54.60}$$

the integral running over the $(n-1)$-simplex $\Delta_{n-1}=\{\alpha_i\ge0,\sum\alpha_i=1\}$. Every contact Witten diagram is Eq. (54.60) dressed by couplings; the remaining work is the simplex integral, which we now do for $n=3$ in closed form and reduce, for $n=4$, to a function of cross-ratios.



(a) cubic contact (b) quartic contact (c) s-channel exchange

Figure 54.3: Tree-level Witten diagrams. The disk is a constant-time slice of Euclidean AdS_{d+1}; its boundary circle is (a compactification of) the CFT space $\mathbb{R}^d$. Solid lines are bulk-to-boundary propagators K_Δ Eq. (53.177) reaching from a boundary insertion to an integrated bulk vertex (filled dot). *(a)* The cubic contact vertex $\lambda\,\phi_1\phi_2\phi_3$ gives the three-point function Eq. (54.62). *(b)* The quartic contact vertex $\lambda_4\,\phi^4$ gives the contact D-function $D_{\Delta_\phi\Delta_\phi\Delta_\phi\Delta_\phi}$. *(c)* An internal bulk-to-bulk propagator (wavy) of the exchanged field of dimension Δ between two cubic vertices gives the s-channel exchange diagram Eq. (54.72).

54.3.2 The three-point function and the OPE coefficient

Let three bulk scalars ϕ_1, ϕ_2, ϕ_3, dual to primaries $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_3$ of dimensions $\Delta_1, \Delta_2, \Delta_3$, be coupled by the cubic vertex $\lambda\,\phi_1\phi_2\phi_3$ added to the bulk action of Section 53.6.5, which we write with the common matter normalization $\mathcal{N}_\phi$,

$$S = \sum_i \frac{\mathcal{N}_\phi}{2}\int \mathrm{d}^{d+1}X\sqrt{g}\,\left[(\partial\phi_i)^2 + m_i^2\phi_i^2\right] + \mathcal{N}_\phi\,\lambda\int \mathrm{d}^{d+1}X\sqrt{g}\;\phi_1\phi_2\phi_3. \quad (54.61)$$

Each field's mass is tied to its dual dimension by the mass–dimension relation $\Delta_i(\Delta_i - d) = m_i^2 L^2$ of Eq. (53.112). By the Witten rules of Theorem 54.2, the connected three-point function is minus the third functional derivative of the on-shell action with respect to the three boundary sources; with three distinct fields the vertex carries no symmetry factor, each source pulls down one bulk-to-boundary propagator, and the single bulk vertex is integrated:

$$\langle \mathcal{O}_1(x_1)\mathcal{O}_2(x_2)\mathcal{O}_3(x_3)\rangle = -\,\mathcal{N}_\phi\,\lambda\,L^{d+1}\,c_{\Delta_1}c_{\Delta_2}c_{\Delta_3}\;D_{\Delta_1\Delta_2\Delta_3}(x_1,x_2,x_3), \quad (54.62)$$

the factor L^{d+1} being the one promised in Theorem 54.7, coming from $\sqrt{g}\,\mathrm{d}^{d+1}X = L^{d+1}\,\mathrm{d}z\,\mathrm{d}^d x/z^{d+1}$.

The form is forced by symmetry. Before computing $D_{\Delta_1\Delta_2\Delta_3}$ we observe that its functional form is fixed. Translations and rotations of $\mathbb{R}^d$ are isometries of AdS_{d+1} under which the measure is invariant and each $\widehat{K}_{\Delta_i}$ depends only on $x - x_i$, so D depends only on the separations x_{ij}. Under a dilation $x \to \mu x$, $z \to \mu z$ (also an isometry) the measure is invariant while $\widehat{K}_{\Delta_i} \to \mu^{-\Delta_i}\widehat{K}_{\Delta_i}$, so $D \to \mu^{-\Sigma}D$: the function is homogeneous of degree $-\Sigma = -\sum_i \Delta_i$. Finally, under the inversion

Eq. (53.178), which is the isometry that completes the conformal group, the bulk-to-boundary propagator transforms covariantly. Writing $\bar{x}_i = x_i/|x_i|^2$ for the inverted boundary points and $\bar{X}$ for the inverted bulk point, a direct computation using $z^2+|x-x_i|^2 = (z'^2+|x'-\bar{x}_i|^2)/(|X'|^2|\bar{x}_i|^2)$ (with $|X'|^2 = z'^2+|x'|^2$) gives

$$\widehat{K}_{\Delta_i}(z,x;x_i) = |x_i|^{-2\Delta_i}\,\widehat{K}_{\Delta_i}(z',x';\bar{x}_i), \tag{54.63}$$

while the measure is again invariant. Hence D transforms exactly as a product of three primaries of dimensions Δ_i. The unique function of three points with these three properties is the conformal three-point structure of Eq. (47.137),

$$D_{\Delta_1\Delta_2\Delta_3}(x_1,x_2,x_3) = \frac{b(\Delta_i)}{|x_{12}|^{\Delta_1+\Delta_2-\Delta_3}\,|x_{13}|^{\Delta_1+\Delta_3-\Delta_2}\,|x_{23}|^{\Delta_2+\Delta_3-\Delta_1}}, \tag{54.64}$$

with $b(\Delta_i)$ a pure number. The bulk integral cannot produce anything but the conformally mandated structure — the same lesson the two-point function taught, now at three points.

The coefficient. To extract $b(\Delta_i)$ we exploit the freedom just established and choose convenient kinematics. Translate so that $x_3 = 0$, then perform the inversion. The propagator to $x_3 = 0$ collapses: $\widehat{K}_{\Delta_3}(z,x;0) = (z/(z^2+|x|^2))^{\Delta_3} = z'^{\Delta_3}$ in inverted coordinates, a pure power of the new radial variable with no dependence on the new boundary position. The other two propagators transform by Eq. (54.63) with $|x_i| = |x_{i3}|$, and after a further translation that places one of them at the origin the integral reduces to

$$D_{\Delta_1\Delta_2\Delta_3} = |x_{13}|^{-2\Delta_1}|x_{23}|^{-2\Delta_2}\,B(r), \qquad B(r) = \int_0^\infty \frac{\mathrm{d}z}{z^{d+1}}\, z^{\Delta_3}\,\widehat{K}_{\Delta_1}(z,x;b)\,\widehat{K}_{\Delta_2}(z,x;0)\,\mathrm{d}^d x, \tag{54.65}$$

where $r = |b| = |\bar{x}_1 - \bar{x}_2|$ is the single remaining separation and we have dropped primes. The integral $B(r)$ is now a textbook Schwinger computation. Parametrize both denominators as in Eq. (54.57), do the Gaussian x-integral Eq. (54.59) (two points only), and the radial integral:

$$B(r) = \frac{\pi^{d/2}\,\Gamma\!\left(\frac{\Sigma-d}{2}\right)}{2\,\Gamma(\Delta_1)\Gamma(\Delta_2)} \int_0^\infty \mathrm{d}t\,\mathrm{d}s\; t^{\Delta_1-1}s^{\Delta_2-1}\,(t+s)^{-\Sigma/2}\,\exp\!\Big[-\frac{ts}{t+s}\,r^2\Big]. \tag{54.66}$$

Substitute $t=\rho\xi$, $s=\rho(1-\xi)$ with $\rho\in(0,\infty)$, $\xi\in(0,1)$ and $\mathrm{d}t\,\mathrm{d}s = \rho\,\mathrm{d}\rho\,\mathrm{d}\xi$; then $t+s=\rho$, $ts/(t+s) = \rho\,\xi(1-\xi)$, and the ρ-integral gives $\Gamma\!\left(\frac{\Delta_1+\Delta_2-\Delta_3}{2}\right)\big[\xi(1-\xi)r^2\big]^{-(\Delta_1+\Delta_2-\Delta_3)/2}$, using $\Delta_1+\Delta_2-\frac{\Sigma}{2} = \frac{\Delta_1+\Delta_2-\Delta_3}{2}$. The remaining ξ-integral is an Euler Beta integral,

$$\int_0^1 \mathrm{d}\xi\;\xi^{\frac{\Delta_1+\Delta_3-\Delta_2}{2}-1}(1-\xi)^{\frac{\Delta_2+\Delta_3-\Delta_1}{2}-1} = \frac{\Gamma\!\left(\frac{\Delta_1+\Delta_3-\Delta_2}{2}\right)\Gamma\!\left(\frac{\Delta_2+\Delta_3-\Delta_1}{2}\right)}{\Gamma(\Delta_3)}, \tag{54.67}$$

where the exponents follow from $\Delta_1 - \frac{\Delta_1+\Delta_2-\Delta_3}{2} = \frac{\Delta_1+\Delta_3-\Delta_2}{2}$ and the denominator from $\frac{\Delta_1+\Delta_3-\Delta_2}{2} + \frac{\Delta_2+\Delta_3-\Delta_1}{2} = \Delta_3$. Collecting the pieces, $B(r) = b(\Delta_i)\, r^{-(\Delta_1+\Delta_2-\Delta_3)}$ with the same power of r that Eq. (54.64) demands after restoring $|\bar{x}_1 - \bar{x}_2|^2 = |x_{12}|^2/(|x_{13}|^2|x_{23}|^2)$, and the coefficient is

$$\boxed{b(\Delta_i) = \frac{\pi^{d/2}}{2}\, \frac{\Gamma\!\left(\frac{\Delta_1+\Delta_2-\Delta_3}{2}\right)\Gamma\!\left(\frac{\Delta_1+\Delta_3-\Delta_2}{2}\right)\Gamma\!\left(\frac{\Delta_2+\Delta_3-\Delta_1}{2}\right)\Gamma\!\left(\frac{\Delta_1+\Delta_2+\Delta_3-d}{2}\right)}{\Gamma(\Delta_1)\,\Gamma(\Delta_2)\,\Gamma(\Delta_3)}\,.} \tag{54.68}$$

Although the derivation singled out the third point, the answer is manifestly symmetric in $1, 2, 3$ — a strong internal check. The three Gamma functions of the half-differences $\frac{\Delta_i+\Delta_j-\Delta_k}{2}$ are the hallmark of the conformal three-point structure; the fourth, $\Gamma(\frac{\Sigma-d}{2})$, is the radial integral that distinguishes the AdS computation from a flat-space one.

Reading off C_{123}. The OPE coefficient is the coefficient of the conformal structure when the operators are normalized to unit two-point functions, as in Eq. (47.208). With the canonical bulk normalization $\mathcal{N}_\phi L^{d-1} = 1$ that reproduced the dictionary in Section 53.6.4, the two-point coefficient is $C_{\Delta_i} = (2\Delta_i - d)\, c_{\Delta_i}$ from Eq. (53.195); rescaling $\mathcal{O}_i \to \mathcal{O}_i/\sqrt{C_{\Delta_i}}$ to unit normalization, the structure constant extracted from Eqs. (54.62) and (54.64) is

$$\boxed{C_{123} = -\,\frac{\lambda\, L^2\, b(\Delta_i)}{\sqrt{\mathcal{N}_\phi L^{d-1}}}\,\sqrt{\frac{c_{\Delta_1}\, c_{\Delta_2}\, c_{\Delta_3}}{(2\Delta_1 - d)(2\Delta_2 - d)(2\Delta_3 - d)}}\,,} \tag{54.69}$$

with $b(\Delta_i)$ given by Eq. (54.68) and all L-dependence collected into the prefactor (it cancels the dimension of the bulk coupling λ, so C_{123} is dimensionless). Two features deserve emphasis. First, the OPE coefficient is *computed*, not merely allowed: conformal symmetry left C_{123} free, and holography fixes it through the single bulk number λ and a universal ratio of Gamma functions of the dimensions. Second, the scaling with N is exactly right. The matter normalization inherits the gravitational $\mathcal{N}_\phi L^{d-1} \propto 1/G_N \propto N^2$ of Eq. (53.162) (the same factor that made $C_\Delta \propto N^2$ in the two-point function), so Eq. (54.69) carries $C_{123} \propto 1/\sqrt{\mathcal{N}_\phi L^{d-1}} \propto 1/N$. A three-point function of single-trace operators is suppressed by one power of $1/N$ relative to the two-point function — precisely the statement of large-N factorization, that single-trace operators behave at leading order like generalized free (Gaussian) fields whose connected higher correlators are $1/N$-suppressed. The bulk cubic vertex is the leading correction to that Gaussianity.

54.3.3 Four points: contact versus exchange

At four points conformal symmetry no longer fixes the answer: the correlator of four identical scalars $\mathcal{O}$ of dimension Δ_ϕ is, by Eq. (47.145),

$$\langle \mathcal{O}(x_1)\mathcal{O}(x_2)\mathcal{O}(x_3)\mathcal{O}(x_4)\rangle = \frac{\mathcal{G}(u,v)}{|x_{12}|^{2\Delta_\phi}\,|x_{34}|^{2\Delta_\phi}}, \qquad u=\frac{x_{12}^2\,x_{34}^2}{x_{13}^2\,x_{24}^2}, \quad v=\frac{x_{14}^2\,x_{23}^2}{x_{13}^2\,x_{24}^2}, \tag{54.70}$$

with u, v the cross-ratios of Eq. (47.138) and $\mathcal{G}(u,v)$ an undetermined function. Two distinct bulk processes feed $\mathcal{G}$ at tree level. A quartic contact vertex $\lambda_4\,\phi^4$ gives the contact D-function $D_{\Delta_\phi\Delta_\phi\Delta_\phi\Delta_\phi}$ of Fig. 54.3(b); a cubic coupling $\phi\phi\sigma$ to a second bulk field σ gives, at the same order, the exchange diagram of Fig. 54.3(c) in which σ is propagated internally between two cubic vertices.

The contact piece. The four-point contact diagram is read off from Eq. (54.60) with $n=4$, $\Sigma = 4\Delta_\phi$. The simplex integral can no longer be done in elementary closed form for all dimensions, but it always reduces to a function of the cross-ratios alone once the kinematic prefactor is stripped: by the same homogeneity-plus-inversion argument as at three points, $D_{\Delta_\phi\Delta_\phi\Delta_\phi\Delta_\phi}$ equals a power of the x_{ij} fixing the overall conformal weight times a function $\bar{D}(u,v)$,

$$D_{\Delta_\phi\Delta_\phi\Delta_\phi\Delta_\phi}(x_i) = \frac{\pi^{d/2}\,\Gamma\!\left(2\Delta_\phi-\frac{d}{2}\right)\Gamma(2\Delta_\phi)}{2\,\Gamma(\Delta_\phi)^4}\,\frac{1}{(x_{13}^2\,x_{24}^2)^{\Delta_\phi}}\,\bar{D}_{\Delta_\phi\Delta_\phi\Delta_\phi\Delta_\phi}(u,v). \tag{54.71}$$

The reduced $\bar{D}$-functions are the standard building blocks of contact Witten diagrams; for integer or half-integer dimensions they evaluate to combinations of ${}_2F_1$ hypergeometrics and, in the simplest cases, to logarithms and dilogarithms. Their key qualitative property, which we exploit below, is that a contact diagram contains *no* single-trace conformal block: with no field propagating internally, no new primary is exchanged, and the only operators that can appear in the $\mathcal{O}\times\mathcal{O}$ OPE are the composites built from two $\mathcal{O}$'s.

The exchange piece. Let σ be a scalar bulk field of mass m_σ, dual to a single-trace primary $\mathcal{O}_\Delta$ of dimension Δ, $\Delta(\Delta-d) = m_\sigma^2 L^2$, coupled to the external field by $\lambda\,\phi\phi\sigma$. The s-channel exchange diagram is

$$A^{s}_{\text{exch}}(x_i) = (\mathcal{N}_\phi\lambda)^2 L^{2(d+1)}\,c_{\Delta_\phi}^4 \int \frac{\mathrm{d}Z_1}{z_1^{d+1}}\frac{\mathrm{d}Z_2}{z_2^{d+1}}\,\widehat{K}_{\Delta_\phi}(Z_1;x_1)\widehat{K}_{\Delta_\phi}(Z_1;x_2)\,G_\Delta(Z_1,Z_2)\,\widehat{K}_{\Delta_\phi}(Z_2;x_3 \tag{54.72}$$

with $\mathrm{d}Z_a = \mathrm{d}z_a\,\mathrm{d}^d x_a$ and $G_\Delta(Z_1,Z_2)$ the *bulk-to-bulk* propagator of σ: the Green's function of the bulk wave operator,

$$(-\Box_{Z_1} + m_\sigma^2)\,G_\Delta(Z_1,Z_2) = \frac{1}{\sqrt{g}}\,\delta^{d+1}(Z_1-Z_2), \tag{54.73}$$

regular in the interior and normalizable ($\sim z^{\Delta}$, not $z^{d-\Delta}$) at the boundary, so that no source for $\mathcal{O}_\Delta$ is turned on. In terms of the chordal distance $\zeta \equiv [(z_1-z_2)^2+|x_1-x_2|^2]/(2z_1z_2)$ it has the closed hypergeometric form $G_\Delta(\zeta) \propto (2\zeta)^{-\Delta}\,{}_2F_1(\Delta, \Delta-\frac{d}{2}+\frac{1}{2}; 2\Delta-d+1; -2/\zeta)$; the only property we need is its behaviour as one argument approaches the boundary, $z_1 \to 0$, where the normalizable falloff factors out a bulk-to-boundary propagator,

$$G_\Delta(Z_1,Z_2) \xrightarrow{z_1\to 0} \frac{1}{2\Delta-d}\, z_1^{\Delta}\, K_\Delta(Z_2;x_1) + O(z_1^{\Delta+2}). \tag{54.74}$$

This is the same coefficient $2\Delta-d$ that appeared in the two-point function, and it is what ties the exchange to the OPE coefficient.

54.3.4 The conformal-block decomposition of the exchange diagram

The double bulk integral in Eq. (54.72) looks intractable, but the equation of motion Eq. (54.73) collapses one of the two integrations and, with it, organizes the answer into conformal blocks. Define the field sourced by the first vertex,

$$\Phi(Z_2) \equiv \int \frac{\mathrm{d}Z_1}{z_1^{d+1}}\, G_\Delta(Z_2,Z_1)\, \widehat{K}_{\Delta_\phi}(Z_1;x_1)\, \widehat{K}_{\Delta_\phi}(Z_1;x_2). \tag{54.75}$$

Acting with the wave operator and using Eq. (54.73) turns the integral equation into a differential one,

$$(-\Box_{Z_2} + m_\sigma^2)\,\Phi(Z_2) = \widehat{K}_{\Delta_\phi}(Z_2;x_1)\,\widehat{K}_{\Delta_\phi}(Z_2;x_2) \equiv \mathcal{J}(Z_2), \tag{54.76}$$

so Φ is the bulk field generated by the bilinear source $\mathcal{J}$ built from two external legs, and the exchange amplitude is the single remaining integral $A^s_{\text{exch}} \propto \int \mathrm{d}Z_2\, z_2^{-(d+1)}\, \widehat{K}_{\Delta_\phi}(Z_2;x_3)\widehat{K}_{\Delta_\phi}(Z_2;x_4)\,\Phi(Z_2)$.

Now read off the structure of Φ from Eq. (54.76). The full solution is the sum of a particular solution and a homogeneous solution. The source $\mathcal{J}$, by conformal covariance, behaves near the boundary point $x \to x_1 \approx x_2$ like a power $z^{2\Delta_\phi}$; the particular solution inherits powers $z^{2\Delta_\phi+2n}$, $n=0,1,2,\dots$, generated by acting with $\Box^{-1}$ on the analytic tail of the source. The homogeneous solutions of $(-\Box+m_\sigma^2)\Phi_{\text{hom}}=0$ carry the two indicial powers of the wave operator, z^{Δ} and $z^{d-\Delta}$, exactly as in the two-point analysis of Section 53.6.3; normalizability and the boundary conditions inherited from G_Δ select the z^{Δ} branch. Thus the small-z expansion of Φ contains *two species of falloff*:

$$\Phi(z,x) \xrightarrow{z\to 0} \underbrace{a_\Delta\, z^{\Delta}\, f_\Delta(x;x_1,x_2)}_{\text{homogeneous (single-trace)}} + \sum_{n\ge 0} \underbrace{b_n\, z^{\,2\Delta_\phi+2n}\, f_n(x;x_1,x_2)}_{\text{particular (double-trace)}}. \tag{54.77}$$

When this is fed into the second vertex, each power z^{δ} produces, after the Z_2 integral, a contribution to $\mathcal{G}(u,v)$ that begins at $u^{\delta/2}$ in the OPE limit $u \to 0$.

Comparison with the block expansion makes the identification immediate. The s-channel decomposition of Eq. (54.70) into the conformal blocks of Theorem 47.49 reads, by Eq. (47.227),

$$\mathcal{G}(u,v) = \sum_{\mathcal{O}'} C^2_{\mathcal{O}\mathcal{O}\mathcal{O}'}\, G_{\Delta_{\mathcal{O}'},\ell_{\mathcal{O}'}}(u,v), \qquad G_{\Delta',\ell'}(u,v) \xrightarrow{u\to 0} u^{(\Delta'-\ell')/2}, \tag{54.78}$$

each block beginning with the power $u^{\tau'/2}$, $\tau' = \Delta' - \ell'$ the twist. Matching powers in Eqs. (54.77) and (54.78):

- The homogeneous z^{Δ} term produces $u^{\Delta/2}$, the leading power of the conformal block of the single-trace primary $\mathcal{O}_\Delta$ itself — the operator dual to the exchanged bulk field σ. Its coefficient is, by the boundary limit Eq. (54.74), exactly the product of the two cubic OPE coefficients computed in Section 54.3.2: $C^2_{\mathcal{O}\mathcal{O}\mathcal{O}_\Delta}$. A single bulk field exchanges *exactly one* single-trace primary, and one conformal block (which already packages all of $\mathcal{O}_\Delta$'s descendants) accounts for it. This is the precise sense in which the equation of motion *truncates* the single-trace content of the block sum: the second-order wave operator Eq. (54.73) has a two-dimensional space of indicial behaviours and boundary conditions pick out one dimension Δ, not a tower.
- The particular-solution powers $z^{2\Delta_\phi+2n}$ produce $u^{\Delta_\phi+n}$, the leading powers of an infinite tower of *double-trace* operators of twist $2\Delta_\phi + 2n$. Spin enters once the external legs are allowed to carry derivatives (or the exchanged field carries spin): each twist $2\Delta_\phi+2n$ supports a whole family of spins ℓ, and the general double-trace primary $[\mathcal{O}\mathcal{O}]_{n,\ell}$ has dimension $\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell$.

Assembling the two species of falloff, the s-channel exchange diagram has the block decomposition

$$\mathcal{G}^{s}_{\text{exch}}(u,v) = C^2_{\mathcal{O}\mathcal{O}\mathcal{O}_\Delta}\, G_{\Delta,0}(u,v) \;+\; \sum_{n=0}^{\infty}\sum_{\ell} p_{n,\ell}\, G_{\,2\Delta_\phi+2n+\ell,\,\ell}(u,v), \qquad \Delta_{n,\ell} = 2\Delta_\phi + 2n + \tag{54.79}$$

The single-trace block is the genuinely new physics of the exchange; the double-trace tower is, dimension for dimension, the very spectrum that the bootstrap finds in any unitary correlator of identical scalars. Indeed these are exactly the double-twist operators $[\mathcal{O}\mathcal{O}]_{n,\ell}$ of dimension $2\Delta_\phi + 2n + \ell$ that crossing forced into the mean-field solution of Eq. (49.130) and Section 49.5.1 — recovered here from the bulk, with their leading $1/N$ corrections supplied by the exchange and contact diagrams.

Worked Example 54.8 (Double-trace operators as two-particle states in AdS)**.** The double-trace tower has a transparent bulk meaning that the block decomposition obscures. In the strict large-N limit, $G_N \to 0$, the bulk theory is free: a

single bulk scalar ϕ of mass m on rigid AdS_{d+1}, with no interactions. Quantize it in global AdS, where the timelike boundary makes AdS a box and the spectrum of the wave operator is discrete: a single quantum has energies $\omega = \Delta + 2k + j$, $k, j \geq 0$, with Δ the dual dimension and j the angular momentum on the spatial S^{d-1}. Because energy in global AdS equals scaling dimension under the state–operator map (Section 47.5), the one-particle states are exactly $\mathcal{O}$ and its descendants. A *two*-particle state, two quanta with relative excitation n and total spin ℓ, has energy

$$E_{n,\ell} = 2\Delta + 2n + \ell + O(G_N), \tag{54.80}$$

the free sum of the two single-particle energies. Under the dictionary this is a primary of dimension $2\Delta_\phi + 2n + \ell$ — precisely the double-trace operator $[\mathcal{O}\mathcal{O}]_{n,\ell}$. The operator $[\mathcal{O}\mathcal{O}]_{n,\ell}$ is the boundary image of a pair of bulk particles; that is why it is present in every holographic correlator independently of any interaction, and why its mean-field dimension is the free sum.

Now restore the interaction. The two bulk particles attract or repel through the exchange of σ (or the contact ϕ^4 vertex), shifting the two-particle energy by a binding energy of order $G_N \propto 1/N^2$,

$$\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell + \frac{1}{N^2}\,\gamma^{(1)}_{n,\ell} + O(N^{-4}). \tag{54.81}$$

The anomalous dimension $\gamma^{(1)}_{n,\ell}$ is literally the binding energy of the two-particle state in AdS units. An attractive bulk force — gravity is universally attractive — gives $\gamma^{(1)}_{n,\ell} < 0$: the bound pair is lighter than the sum of its parts, exactly as two masses are lighter when gravitationally bound. This sign is not optional; it is fixed by bulk causality and is the holographic face of the same positivity that the conformal bootstrap and the chaos bound impose on $\gamma_{n,\ell}$ in the large-spin expansion of Chapter 51.

Reading the anomalous dimension off the correlator. The shift Eq. (54.81) leaves an unmistakable fingerprint. A block of dimension $\Delta_{n,\ell} = 2\Delta_\phi + 2n + \ell + \gamma$ contributes a power $u^{(\Delta_{n,\ell}-\ell)/2} = u^{\Delta_\phi + n}\, u^{\gamma/2}$, and expanding $u^{\gamma/2} = 1 + \frac{\gamma}{2}\log u + \cdots$ for the small $\gamma \propto 1/N^2$ shows that the anomalous dimension appears as the coefficient of a *logarithm* of the cross-ratio:

$$\mathcal{G}(u,v) \;\supset\; \tfrac{1}{2}\, a^{(0)}_{n,\ell}\, \gamma^{(1)}_{n,\ell}\, u^{\Delta_\phi + n}\, \log u, \tag{54.82}$$

with $a^{(0)}_{n,\ell}$ the mean-field (squared) OPE coefficient. A tree-level Witten diagram is an analytic function of the cross-ratios except for exactly such logarithms; the contact and exchange diagrams generate them, and the coefficient of $\log u$ at order $u^{\Delta_\phi+n}$ is $\frac{1}{2}a^{(0)}_{n,\ell}\gamma^{(1)}_{n,\ell}$. This is how anomalous dimensions of double-trace operators are extracted in practice, and it is the holographic signal that the bulk fields interact: were the bulk free, $\mathcal{G}$ would be the pure mean-field correlator of Eq. (49.127) with no logarithm and no anomalous dimensions at all.

54.3.5 The Mellin amplitude and the link to the bootstrap

The cross-ratio functions $\bar{D}(u,v)$ are cumbersome, and the block decomposition Eq. (54.79) hides the simplicity of the underlying diagram. The natural variable is the Mellin transform. Strip the leading prefactor and write the reduced correlator as a double inverse-Mellin integral,

$$\mathcal{G}(u,v) = \int_{-i\infty}^{i\infty} \frac{\mathrm{d}s\,\mathrm{d}t}{(4\pi i)^2}\, u^{s/2}\, v^{(t-2\Delta_\phi)/2}\, M(s,t)\,\Gamma^2\!\Big(\Delta_\phi - \tfrac{s}{2}\Big)\,\Gamma^2\!\Big(\Delta_\phi - \tfrac{t}{2}\Big)\,\Gamma^2\!\Big(\Delta_\phi - \tfrac{\tilde{u}}{2}\Big), \qquad \tilde{u} = 4\Delta_\phi \tag{54.83}$$

the Gamma functions forming the universal Mellin measure and $M(s,t)$ the *Mellin amplitude*. The Mellin variables $s,t,\tilde{u}$ are the conformal analogues of Mandelstam invariants, satisfying $s+t+\tilde{u} = 4\Delta_\phi$, and $M(s,t)$ plays the role of a flat-space scattering amplitude. In this variable the Witten diagrams become elementary:

- a contact diagram from a non-derivative ϕ^4 vertex has a *constant* Mellin amplitude, $M(s,t) = \text{const}$; derivative vertices give polynomials in s,t;
- an s-channel exchange of $\mathcal{O}_\Delta$ has a Mellin amplitude with *poles* at the locations of the exchanged twist and its descendants,

$$M(s,t) = \sum_{m=0}^{\infty} \frac{R_m}{s-(\Delta-\ell)-2m} + (\text{regular}), \tag{54.84}$$

  the single tower of poles at $s = \Delta - \ell + 2m$ encoding the one single-trace primary and its descendants.

The double-trace operators are not poles of M at all: they are the poles of the Mellin measure — the Gamma functions $\Gamma^2(\Delta_\phi - \frac{s}{2})$ have double poles at $s = 2\Delta_\phi + 2n$, and a double pole is precisely the $\log u$ of Eq. (54.82), i.e. an anomalous dimension. The kinematic, always-present double-trace spectrum is built into the measure; interactions move and split those poles. A tree-level Witten diagram is thus a *rational* (for contact) or *meromorphic* (for exchange) function of s,t — the simplest analytic object the correlator can be, and the reason the Mellin representation is the workhorse of modern holographic computation.

Crossing. The assembled four-point function is the sum over channels: the disconnected mean-field piece (the identity plus its double-trace shadow) at $O(N^0)$, and at $O(1/N^2)$ the sum of s-, t-, and u-channel exchanges together with the contact terms. No single Witten diagram is crossing symmetric: the s-channel exchange Eq. (54.79) manifestly distinguishes the pair (12)(34). But the full correlator must be invariant under permutations of the four identical insertions, in particular under the $1 \leftrightarrow 3$ exchange that swaps $u \leftrightarrow v$ and yields the bootstrap

equation Eq. (49.15). The justification is structural rather than diagram-by-diagram: the bulk is a consistent local theory, so its boundary correlators are genuine correlators of a single CFT, and any genuine CFT correlator satisfies crossing by associativity of the OPE (Theorem 49.13). In Mellin space the statement is sharp and immediate: crossing symmetry is the total symmetry of $M(s,t)$ under permutations of $(s,t,\tilde{u})$, which the sum of the three channel exchanges plus the symmetric contact term manifestly possesses. The individual diagrams conspire, channel against channel, exactly as the infinite double-trace tower conspired in the mean-field solution of Section 49.5.1 to reproduce a single crossed-channel term.

Contact with the numerical bootstrap. This closes the loop opened in Chapters 49 and 50. The bootstrap takes crossing Eq. (49.15) and unitarity as the only inputs and, with no Lagrangian, carves out the allowed operator data $\{\Delta,\, C\}$ — for the three-dimensional Ising universality class, the endpoint of ϕ^4 theory, pinning Δ_σ and Δ_ϵ to the island recorded in Table 50.1. Holography computes the same kind of data from the opposite direction: when a CFT possesses a weakly curved gravity dual, its single-trace OPE coefficients are the Gamma-function ratios of Eq. (54.69) and its double-trace anomalous dimensions are the binding energies of Eq. (54.81), all read off from tree-level diagrams. The two methods meet on the crossing equation, the one constraint both must satisfy. They are not, however, the same theory: the canonical holographic example is $\mathcal{N}=4$ super-Yang–Mills at large N and large 't Hooft coupling — the maximally symmetric large-N cousin of Yang–Mills theory — whose sparse spectrum and large gap make a local bulk possible, whereas the 3D Ising CFT has order-one central charge and no parametrically light single-trace tower, hence no weakly coupled gravity dual. What is universal, and what this section has exhibited, is the *structure*: every unitary CFT four-point function decomposes into single-trace (or, for Ising, simply low-lying) primaries plus the double-trace tower $[\mathcal{O}\mathcal{O}]_{n,\ell}$ at $2\Delta_\phi + 2n + \ell$, and the bulk computation and the bootstrap are two windows onto the same operator content.

With the vacuum correlators of the holographic CFT now under control — two-, three-, and four-point functions, their conformally fixed structures, their OPE coefficients, and the double-trace spectrum that the bootstrap independently demands — the zero-temperature story is complete. The same bulk fields whose tree-level diagrams we have just resummed must next be quantized on a different geometry: heating the boundary CFT corresponds to replacing pure AdS by a black-brane background, and the renormalized on-shell action that here gave correlators will there give a free energy. Section 54.4 turns to that thermal state.

54.4 Black Holes and Finite Temperature

The previous three sections quantized bulk fields on *empty* anti–de Sitter space and read off the vacuum correlators of the dual conformal field theory: two-, three-, and four-point functions, their conformally fixed structures, and the operator data they encode. That entire computation lived at zero temperature, on the single geometry Eq. (53.161). We now heat the boundary theory. In ordinary quantum field theory the thermal partition function $Z(\beta) = \mathrm{Tr}\, e^{-\beta H}$ is computed by putting the Euclidean theory on a time circle of circumference $\beta = 1/T$ (Eq. (10.136)); the holographic question is therefore sharp and inevitable. The GKP–Witten relation Eq. (53.162) equates Z_{CFT} with a bulk gravitational partition function, and at large N and large 't Hooft coupling that partition function is dominated by a classical saddle. So the question becomes: *which bulk geometry, with a Euclidean time circle of circumference β on its conformal boundary, dominates the gravitational path integral?* The answer is the single most consequential geometry in this book. For high enough temperature it is not empty AdS with a periodically identified time, but a *black brane* — a planar black hole whose horizon sits at a finite radial depth. Its temperature is the Hawking temperature, fixed entirely by the demand that the Euclidean geometry close off smoothly at the horizon; the renormalized on-shell gravitational action — rendered finite by exactly the counterterms of Section 54.2.3 — is the free energy; and the area of the horizon, divided by $4G$, is the entropy. This section establishes all of that, carries every term of the on-shell action, and ends by exhibiting the gravitational dual of confinement: the Hawking–Page transition.

54.4.1 The planar AdS–Schwarzschild black brane

We work in Euclidean signature, where the partition function is a genuine convergent integral and the boundary condition is a Dirichlet problem, exactly as in Section 53.6.1. The bulk dynamics is pure gravity with a negative cosmological constant — the metric sector of the supergravity theory whose probe scalars supplied the correlators of the previous sections. Its action, in $D = d+1$ bulk dimensions, is the Einstein–Hilbert action with the Gibbons–Hawking boundary term and the holographic counterterms,

$$S_E \;=\; -\frac{1}{16\pi G}\int_{\mathcal{M}} \mathrm{d}^{d+1}x\,\sqrt{g}\,(R-2\Lambda) \;-\; \frac{1}{8\pi G}\int_{\partial\mathcal{M}} \mathrm{d}^{d}x\,\sqrt{\gamma}\,K \;+\; S_{\mathrm{ct}}, \tag{54.85}$$

with $G \equiv G_{d+1}$ the bulk Newton constant, Λ the cosmological constant, $\gamma_{\mu\nu}$ the induced metric on the regulating boundary slice, K the trace of its extrinsic curvature, and S_{ct} the local covariant counterterms of Eq. (54.49). The Gibbons–Hawking term is the standard addition that makes the Dirichlet variational problem well posed: without it, varying Eq. (54.85) would leave a boundary term proportional to the normal derivative of the metric, and the action would

not be stationary on solutions with the metric (rather than its derivative) fixed at the boundary. We take its presence as established; what we owe here is the evaluation, not the derivation of the boundary term, and we carry that evaluation in full below.

The cosmological constant is fixed by demanding that empty AdS_{d+1} of radius L solve the equations of motion. Varying Eq. (54.85) gives the Einstein equation $R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R + \Lambda g_{\mu\nu} = 0$; tracing it in $D = d+1$ dimensions,

$$R\Big(1 - \frac{D}{2}\Big) + D\Lambda = 0 \quad \Longrightarrow \quad R = \frac{2D}{D-2}\Lambda = \frac{2(d+1)}{d-1}\Lambda, \tag{54.86}$$

and matching to the known curvature of AdS_{d+1}, $R = -d(d+1)/L^2$, fixes

$$\Lambda = -\frac{d(d-1)}{2L^2}. \tag{54.87}$$

We will use one consequence of Eq. (54.86) repeatedly: on *any* solution the on-shell Lagrangian density is a spacetime constant. From Eq. (54.86) and Eq. (54.87),

$$R - 2\Lambda = \frac{2(d+1)}{d-1}\Lambda - 2\Lambda = 2\Lambda\,\frac{(d+1)-(d-1)}{d-1} = \frac{4\Lambda}{d-1} = -\frac{2d}{L^2}, \tag{54.88}$$

the same constant $-2d/L^2$ already flagged for the counterterm analysis in Section 54.2.3. The bulk integrand of Eq. (54.85) is therefore proportional to the volume element alone; all the work in evaluating the action is the (divergent) volume integral and the boundary terms.

Among the solutions with conformal boundary $\mathbb{R}\times\mathbb{R}^{d-1}$ (flat space, the setting of the previous sections, now to be given a thermal circle) there is a one-parameter family of *planar black holes*. We state the metric and then verify it solves the equations.

Definition 54.9 (Planar AdS–Schwarzschild black brane)**.** The *planar AdS–Schwarzschild metric*, or *black brane*, in Poincaré-like coordinates with radial coordinate z is

$$\mathrm{d}s^2 \;=\; \frac{L^2}{z^2}\left[-f(z)\,\mathrm{d}t^2 + \mathrm{d}\mathbf{x}^2 + \frac{\mathrm{d}z^2}{f(z)}\right], \qquad f(z) = 1 - \Big(\frac{z}{z_h}\Big)^d, \tag{54.89}$$

with $\mathbf{x}\in\mathbb{R}^{d-1}$, conformal boundary at $z\to 0$, and a horizon at $z = z_h$ where the *blackening factor* f vanishes. As $z_h\to\infty$, $f\to 1$ and Eq. (54.89) reduces to the Poincaré patch Eq. (53.59), up to the overall sign of the metric: that equation is written in the book's mostly-minus signature, whereas Eq. (54.89) and every other Lorentzian metric of this and the following two sections use mostly-plus, as the notationwatch below records.

Notation Watch

Signature in the black-hole and real-time sections. The book's global convention is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+, -, \ldots, -)$ (Appendix A; fixed for Part XV in Section 53.3). The zero-temperature correlators of Sections 54.1 and 54.3 were computed in Euclidean signature, where the question does not arise. For the Lorentzian black-hole, transport, and holographic-RG computations of Sections 54.4 to 54.6 we adopt instead the *mostly-plus* signature $\mathrm{diag}(-, +, \ldots, +)$ standard in general relativity and the holography literature, with the Wald curvature convention in which AdS has negative scalar curvature. The departure is deliberate: mostly-plus is the natural convention for black-hole geometry, keeping $\sqrt{-g}$ real in every bulk dimension $D = d + 1$ and the horizon and causal structure transparent, and it matches the references a reader of this material will consult. The translation to the book's mostly-minus form is the overall sign flip of every metric (so Eq. (54.89) is the mostly-minus Eq. (53.59) with its sign reversed and a blackening factor inserted), and no physical conclusion depends on the choice: the Euclidean thermodynamics of this section is signature-blind, and the mass–dimension and transport relations are invariant. We state the convention once here and carry it through Section 54.6.

That Eq. (54.89) solves Einstein's equation with Eq. (54.87) is a direct computation; the structure that makes it work is worth seeing. The function f multiplies $\mathrm{d}t^2$ and divides $\mathrm{d}z^2$, the hallmark of a static black hole, and the specific power z^d is the unique one for which the (t, z) part of the Ricci tensor stays proportional to the metric while the flat boundary directions $\mathbf{x}$ are left undisturbed. The single integration constant z_h labels the family: it is the position of the horizon, and — as we are about to show — it is in one-to-one correspondence with the temperature. The metric is asymptotically AdS: near $z = 0$, $f \to 1$ and Eq. (54.89) approaches Eq. (53.59) (in the mostly-plus form of the notationwatch above), so the holographic dictionary, the falloff analysis, and the counterterms of the previous sections all apply unchanged in the near-boundary region. The black brane differs from empty AdS only deep in the interior, at $z \sim z_h$ — which is exactly the statement, in the language of Section 53.4, that turning on a temperature is an *infrared* deformation of the boundary theory, invisible in the deep ultraviolet at $z \to 0$.

Physical Insight

The radial coordinate is the renormalization scale (Section 53.4): $z \to 0$ is the deep ultraviolet, $z \to \infty$ the deep infrared. Temperature is an infrared scale — it is invisible to short-distance physics and dominates

long-distance physics. The black brane realizes this perfectly: it is identical to the AdS vacuum near the boundary and is modified only in the far interior, where the geometry simply *ends* at a horizon. The horizon is the geometric incarnation of the thermal infrared cutoff; the spacetime has nowhere deeper to go, because the temperature has screened it.

To make $Z = \mathrm{Tr}\, e^{-\beta H}$ we Wick-rotate $t \to -i\tau$ (Theorem 5.30), turning Eq. (54.89) into the Euclidean black brane

$$\mathrm{d}s_E^2 \;=\; \frac{L^2}{z^2}\left[\, f(z)\,\mathrm{d}\tau^2 + \mathrm{d}\mathbf{x}^2 + \frac{\mathrm{d}z^2}{f(z)}\,\right], \qquad f(z) = 1 - \Big(\frac{z}{z_h}\Big)^d, \tag{54.90}$$

on which the boundary Euclidean time τ runs over a circle whose circumference we must now determine. It is not free. Demanding that the geometry Eq. (54.90) be smooth at $z = z_h$ — where the coefficient of $\mathrm{d}\tau^2$ vanishes — fixes the period of τ, and that period is $\beta = 1/T$.

54.4.2 The Hawking temperature from Euclidean regularity

The coefficient of $\mathrm{d}\tau^2$ in Eq. (54.90) is $\frac{L^2}{z^2} f(z)$, which vanishes at $z = z_h$. A metric in which a circle direction shrinks to zero size is, generically, *singular* there — it has a conical defect, like the tip of a cone — unless the period of the angular coordinate is tuned to exactly the value that makes the tip smooth. This is the same elementary fact that fixes the period of the angle in polar coordinates: $\mathrm{d}R^2 + R^2\,\mathrm{d}\theta^2$ is smooth flat space at $R = 0$ only if θ has period exactly 2π; any other period leaves a cone with an opening-angle deficit and a curvature delta-function at the tip. The Euclidean black brane closes off smoothly at the horizon for one and only one period β of τ, and that period is the inverse temperature.

We carry out the near-horizon expansion explicitly. Write $z = z_h - \rho$ with $\rho \to 0^+$ approaching the horizon from the boundary side. Expanding f to first order,

$$f(z) \simeq f'(z_h)\,(z - z_h) = -f'(z_h)\,\rho, \qquad f'(z) = -\frac{d\, z^{d-1}}{z_h^d} \;\Longrightarrow\; f'(z_h) = -\frac{d}{z_h}, \tag{54.91}$$

so $f \simeq (d/z_h)\,\rho > 0$, vanishing linearly as it must. Near the horizon the two-dimensional (τ, z) part of Eq. (54.90) is, to leading order (the prefactor $L^2/z^2 \to L^2/z_h^2$ is smooth and we evaluate it at z_h),

$$\mathrm{d}s^2_{(2)} \;\simeq\; \frac{L^2}{z_h^2}\left[\, \frac{d}{z_h}\,\rho\,\mathrm{d}\tau^2 + \frac{z_h}{d\,\rho}\,\mathrm{d}\rho^2\,\right], \tag{54.92}$$

using $\mathrm{d}z^2 = \mathrm{d}\rho^2$ and $1/f \simeq z_h/(d\rho)$. This is not yet in the recognizable polar form, because the radial part is $\mathrm{d}\rho^2/\rho$, not $\mathrm{d}R^2$. Introduce the proper radial distance R from the horizon,

$$R = \int \frac{L}{z_h}\frac{\mathrm{d}\rho}{\sqrt{f}} = \frac{L}{z_h}\sqrt{\frac{z_h}{d}}\int \frac{\mathrm{d}\rho}{\sqrt{\rho}} = \frac{2L}{z_h}\sqrt{\frac{z_h\,\rho}{d}}, \tag{54.93}$$

so that by construction $\frac{L^2}{z_h^2}\frac{\mathrm{d}\rho^2}{f} = \mathrm{d}R^2$. Inverting, $\rho = \frac{d\,z_h}{4L^2}R^2$, and the $\mathrm{d}\tau^2$ coefficient becomes

$$\frac{L^2}{z_h^2}\frac{d}{z_h}\rho = \frac{L^2}{z_h^2}\frac{d}{z_h}\frac{d\,z_h}{4L^2}R^2 = \frac{d^2}{4z_h^2}R^2. \tag{54.94}$$

The near-horizon metric is therefore

$$\mathrm{d}s^2_{(2)} \;\simeq\; \mathrm{d}R^2 + \frac{d^2}{4z_h^2}R^2\,\mathrm{d}\tau^2 \;=\; \mathrm{d}R^2 + R^2\,\mathrm{d}\theta^2, \qquad \theta \equiv \frac{d}{2z_h}\tau, \tag{54.95}$$

which is flat $\mathbb{R}^2$ in polar coordinates — a smooth disk capping off the geometry — precisely when the angle θ has period 2π. Since $\theta = \frac{d}{2z_h}\tau$, the period of τ for which this holds is

$$\frac{d}{2z_h}\beta = 2\pi \quad\Longrightarrow\quad \boxed{\beta = \frac{4\pi z_h}{d}\,, \qquad T = \frac{1}{\beta} = \frac{d}{4\pi z_h}\,.} \tag{54.96}$$

Any other period $\beta' \neq \beta$ would leave a conical singularity at z_h with deficit angle $2\pi(1-\beta'/\beta)$ and a curvature singularity that no smooth solution of Einstein's equation can have. *Smoothness alone fixes the temperature.* This is the Hawking temperature, and the derivation by Euclidean regularity is the cleanest route to it: there is no need to track Bogoliubov coefficients or compute a particle flux (the Lorentzian-signature derivation we give for a genuine collapsing star in Chapter 65); the temperature is read off the geometry as the unique period that avoids a conical defect. Equivalently, using the general formula $T = |f'(z_h)|/4\pi$ that Eq. (54.91) makes explicit — a relation the next section's infalling analysis Eq. (54.143) reuses verbatim — the surface gravity of the horizon is $\kappa = \frac{1}{2}|f'(z_h)|\cdot(z_h/L)\cdots$; we will not need the surface-gravity packaging, since Eq. (54.96) is already the result.

54.4.3 The renormalized free energy

With the geometry and its temperature fixed, the free energy follows from the GKP–Witten saddle. In the supergravity limit the gravitational partition function is dominated by the on-shell action of the black brane, $Z_{\rm grav} \approx e^{-S_E^{\text{on-shell}}}$ (Eq. (53.163)), and the standard thermodynamic identity $Z = e^{-\beta F}$ identifies

$$\beta F = S_E^{\text{on-shell}}[\,\text{black brane}\,]. \tag{54.97}$$

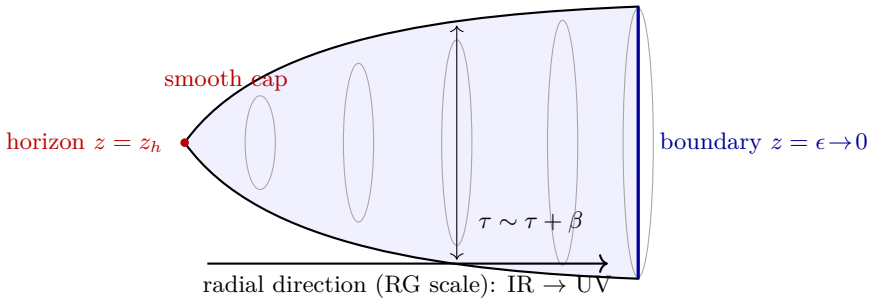


Figure 54.4: The Euclidean black brane as a "cigar." The Euclidean time direction τ is a circle of circumference β fibered over the radial coordinate. Near the boundary $z \to 0$ the circle has its full size; as one moves into the interior the circle shrinks, and at the horizon $z = z_h$ it closes off to a point. The geometry caps smoothly there (no conical defect) for exactly one period, $\beta = 4\pi z_h/d$, which is the inverse Hawking temperature Eq. (54.96). The tip is the tip of the disk $\mathrm{d}R^2 + R^2\mathrm{d}\theta^2$ of Eq. (54.95); this is the "cigar" whose smoothness the Lorentzian infalling boundary condition of Section 54.5.1 continues into causality.

The whole content of this subsection is the careful evaluation of the right-hand side, term by term, with the radial cutoff at $z = \epsilon$ kept until the very end. Three pieces contribute — bulk, Gibbons–Hawking, counterterm — and the divergences cancel between them, exactly as the scalar computation of Theorem 54.5 promised for the gravitational sector. We work with the Euclidean metric Eq. (54.90). Throughout, $V_{d-1} = \int \mathrm{d}^{d-1}x$ is the (infinite, regulated) coordinate volume of the spatial boundary directions, and we will report intensive densities per unit V_{d-1}.

The square root of the metric determinant is the same as for empty AdS, because the two factors f and $1/f$ cancel:

$$\sqrt{g} = \sqrt{\left(\tfrac{L^2}{z^2}\right)^{d+1} f \cdot \tfrac{1}{f}} = \left(\frac{L}{z}\right)^{d+1}. \tag{54.98}$$

Bulk term. Using Eq. (54.88), the bulk integrand is the constant $-\frac{1}{16\pi G}(R - 2\Lambda) = \frac{1}{16\pi G}\frac{2d}{L^2} = \frac{d}{8\pi G L^2}$, so

$$S_{\text{bulk}} = \frac{d}{8\pi G L^2}\int_0^\beta \mathrm{d}\tau \int \mathrm{d}^{d-1}x \int_\epsilon^{z_h} \mathrm{d}z \left(\frac{L}{z}\right)^{d+1} = \frac{d\,\beta V_{d-1} L^{d-1}}{8\pi G}\int_\epsilon^{z_h} \frac{\mathrm{d}z}{z^{d+1}}. \tag{54.99}$$

The radial integral runs from the cutoff $z = \epsilon$ down to the horizon $z = z_h$, which is the lower edge of the spacetime — the geometry simply ends there, so there is nothing below the horizon to integrate. Evaluating,

$$\int_\epsilon^{z_h} \frac{\mathrm{d}z}{z^{d+1}} = \Big[-\frac{1}{d\,z^d}\Big]_\epsilon^{z_h} = \frac{1}{d}\Big(\frac{1}{\epsilon^d} - \frac{1}{z_h^d}\Big), \tag{54.100}$$

so, writing the common prefactor $\mathcal{N} \equiv \dfrac{\beta V_{d-1} L^{d-1}}{8\pi G}$,

$$S_{\text{bulk}} = \mathcal{N}\Big(\frac{1}{\epsilon^d} - \frac{1}{z_h^d}\Big). \tag{54.101}$$

The ϵ^{-d} piece is the expected power divergence: it is proportional to the infinite proper volume near the boundary and must be cancelled.

Gibbons–Hawking term. The boundary slice is $z = \epsilon$, with induced metric $\gamma_{\mu\nu} = \frac{L^2}{\epsilon^2}\,\mathrm{diag}\big(f(\epsilon), 1, \dots, 1\big)$ and $\sqrt{\gamma} = (L/\epsilon)^d \sqrt{f(\epsilon)}$. The outward unit normal points toward decreasing z (out of the bulk, toward the boundary): $n^z = -\sqrt{g^{zz}} = -\frac{z}{L}\sqrt{f}$. The extrinsic-curvature trace is the covariant divergence of the normal,

$$K = \nabla_\mu n^\mu = \frac{1}{\sqrt{g}}\,\partial_z\big(\sqrt{g}\, n^z\big). \tag{54.102}$$

With $\sqrt{g}\, n^z = (L/z)^{d+1}\big(-\frac{z}{L}\sqrt{f}\big) = -L^d z^{-d}\sqrt{f}$,

$$\partial_z\big(\sqrt{g}\, n^z\big) = -L^d\Big(-d\, z^{-d-1}\sqrt{f} + z^{-d}\frac{f'}{2\sqrt{f}}\Big), \tag{54.103}$$

and multiplying by $1/\sqrt{g} = z^{d+1}/L^{d+1}$,

$$K = \frac{1}{L}\Big(d\sqrt{f} - \frac{z\, f'}{2\sqrt{f}}\Big). \tag{54.104}$$

Therefore the integrand of the Gibbons–Hawking term is

$$\sqrt{\gamma}\, K = \Big(\frac{L}{\epsilon}\Big)^d \sqrt{f}\cdot\frac{1}{L}\Big(d\sqrt{f} - \frac{\epsilon f'}{2\sqrt{f}}\Big) = \frac{L^{d-1}}{\epsilon^d}\Big(d\, f(\epsilon) - \frac{\epsilon f'(\epsilon)}{2}\Big). \tag{54.105}$$

Insert $f(\epsilon) = 1 - (\epsilon/z_h)^d$ and $\epsilon f'(\epsilon) = -d(\epsilon/z_h)^d$:

$$d\, f(\epsilon) - \frac{\epsilon f'(\epsilon)}{2} = d - d\Big(\frac{\epsilon}{z_h}\Big)^d + \frac{d}{2}\Big(\frac{\epsilon}{z_h}\Big)^d = d - \frac{d}{2}\Big(\frac{\epsilon}{z_h}\Big)^d, \tag{54.106}$$

so the Gibbons–Hawking contribution, with the $-\frac{1}{8\pi G}$ prefactor of Eq. (54.85) folded into $\mathcal{N}$, is

$$S_{\text{GH}} = -\frac{\beta V_{d-1}}{8\pi G}\, L^{d-1}\Big(\frac{d}{\epsilon^d} - \frac{d}{2}\,\frac{1}{z_h^d}\Big) = \mathcal{N}\Big(-\frac{d}{\epsilon^d} + \frac{d}{2 z_h^d}\Big). \tag{54.107}$$

Counterterm. For the flat planar boundary the only relevant counterterm of Eq. (54.49) is the cosmological-constant (volume) counterterm; the curvature counterterm $\propto R[\gamma]$ vanishes because the boundary metric is flat. Thus

$$S_{\rm ct} = \frac{1}{8\pi G}\,\frac{d-1}{L}\int \mathrm{d}^d x\,\sqrt{\gamma} = \frac{(d-1)\beta V_{d-1}}{8\pi G\,L}\Big(\frac{L}{\epsilon}\Big)^d\sqrt{f(\epsilon)}. \tag{54.108}$$

Expand the square root, $\sqrt{f(\epsilon)} = \big(1-(\epsilon/z_h)^d\big)^{1/2} = 1 - \tfrac{1}{2}(\epsilon/z_h)^d + O(\epsilon^{2d})$. The higher terms vanish as $\epsilon \to 0$, leaving

$$S_{\rm ct} = \mathcal{N}\,(d-1)\Big(\frac{1}{\epsilon^d} - \frac{1}{2z_h^d}\Big). \tag{54.109}$$

Summing the three pieces. Add Eqs. (54.101), (54.107) and (54.109). The coefficients of the divergent ϵ^{-d} collect to

$$1 - d + (d-1) = 0, \tag{54.110}$$

so the divergence cancels identically — the volume counterterm is engineered to do exactly this, and that it works with the same universal coefficient $(d-1)/L$ that renormalized the scalar two-point function in Theorem 54.5 is the promised gravitational deployment of holographic renormalization. The finite coefficients of z_h^{-d} collect to

$$-1 + \frac{d}{2} - \frac{d-1}{2} = -1 + \frac{d-(d-1)}{2} = -1 + \frac{1}{2} = -\frac{1}{2}, \tag{54.111}$$

so the renormalized on-shell action is finite:

$$S_E^{\text{on-shell}} = -\frac{\mathcal{N}}{2z_h^d} = -\frac{\beta V_{d-1} L^{d-1}}{16\pi G\,z_h^d}. \tag{54.112}$$

By Eq. (54.97) the free energy is $F = S_E^{\text{on-shell}}/\beta$, and the free energy density $f \equiv F/V_{d-1}$ is

$$f = -\frac{L^{d-1}}{16\pi G\,z_h^d}. \tag{54.113}$$

Finally, trade z_h for the temperature via Eq. (54.96), $z_h = d/(4\pi T)$, so $z_h^{-d} = (4\pi T/d)^d$:

$$\boxed{\,f = -\frac{L^{d-1}}{16\pi G}\Big(\frac{4\pi}{d}\Big)^d T^d\,.\,} \tag{54.114}$$

The temperature dependence $f \propto T^d$ is exactly what conformal invariance demands: a CFT in d spacetime dimensions has no scale but T, and the free energy density, of mass dimension d, can only be a pure number times T^d. Holography has computed that pure number. Its sign is negative, as a free energy

must be (the pressure $p = -f > 0$), and its magnitude is controlled by L^{d-1}/G — which, by the dictionary $1/G \sim N^2$ (Chapter 35, and Theorem 53.33 for the precise $\mathcal{N} = 4$ coefficient), scales as N^2. This N^2 is the headline: it is the number of *deconfined* degrees of freedom, the gluonic $N^2 - 1 \approx N^2$ color components all contributing freely to the thermodynamics. We return to its physical weight in Section 54.4.6.

54.4.4 Entropy and the Bekenstein–Hawking area law

The entropy density follows from the free energy by the elementary thermodynamic relation $s = -\partial f/\partial T$. Differentiating Eq. (54.114),

$$s = -\frac{\partial f}{\partial T} = \frac{L^{d-1}}{16\pi G}\Big(\frac{4\pi}{d}\Big)^{d} d\, T^{d-1} = \frac{L^{d-1}}{4G}\Big(\frac{4\pi T}{d}\Big)^{d-1}, \tag{54.115}$$

where in the last step we used $\frac{1}{16\pi G}(\frac{4\pi}{d})^d\, d = \frac{1}{4G}(\frac{4\pi}{d})^{d-1}$. The energy density is then $\varepsilon = f + Ts$, and using $Ts = d\,|f|$ from Eqs. (54.114) and (54.115),

$$\varepsilon = f + Ts = (d-1)\,|f| = (d-1)\frac{L^{d-1}}{16\pi G}\Big(\frac{4\pi}{d}\Big)^{d} T^{d}. \tag{54.116}$$

The relation $\varepsilon = (d-1)\,p$ with $p = -f$ is the conformal equation of state, equivalent to the tracelessness of the stress tensor, $\varepsilon - (d-1)p = T^{\mu}{}_{\mu} = 0$. The first law is automatic: with all quantities homogeneous in T, $\mathrm{d}\varepsilon = (d-1)d\,|f|\,\mathrm{d}T/T = T\,\mathrm{d}s$, which the reader can confirm by differentiating Eqs. (54.115) and (54.116) and comparing. The thermodynamics is therefore fully consistent, and entirely fixed by the single number L^{d-1}/G.

The deeper statement is geometric. The entropy density Eq. (54.115) is, on its face, a boundary quantity computed from a free energy. It is also — and this is the Bekenstein–Hawking miracle — the *area of the horizon*, per unit boundary volume, divided by $4G$. We now show this, and elevate it to the central theorem of the section.

Theorem 54.10 (Bekenstein–Hawking area law)**.** *The thermal entropy of the holographic CFT, computed from the renormalized on-shell gravitational action, equals one quarter of the horizon area in bulk Planck units:*

$$S = \frac{A}{4G}, \tag{54.117}$$

where A is the area of the horizon $z = z_h$ in the induced metric, and $G \equiv G_{d+1}$ is the bulk Newton constant.

Proof. The horizon is the surface $z = z_h$ at fixed Euclidean time (equivalently, the bifurcation surface). Its induced metric is read off Eq. (54.90) by setting

$z = z_h$ (where $f = 0$, so the τ direction has shrunk away) and restricting to the spatial directions $\mathbf{x}$:

$$\mathrm{d}s^2\big|_{\mathrm{hor}} = \frac{L^2}{z_h^2}\,\mathrm{d}\mathbf{x}^2, \tag{54.118}$$

a flat $(d-1)$-dimensional metric with constant conformal factor $(L/z_h)^2$. The induced volume element is $\sqrt{h} = (L/z_h)^{d-1}$, so the horizon area — the proper volume of this $(d-1)$-dimensional surface — is

$$A = \int \mathrm{d}^{d-1}x \left(\frac{L}{z_h}\right)^{d-1} = V_{d-1}\left(\frac{L}{z_h}\right)^{d-1}. \tag{54.119}$$

Now compute the total entropy $S = s\,V_{d-1}$ from Eq. (54.115), using $4\pi T/d = 1/z_h$ from Eq. (54.96):

$$S = s\,V_{d-1} = \frac{L^{d-1}}{4G}\,\frac{1}{z_h^{d-1}}\,V_{d-1} = \frac{1}{4G}\,V_{d-1}\left(\frac{L}{z_h}\right)^{d-1} = \frac{A}{4G}, \tag{54.120}$$

the last equality being Eq. (54.119). The entropy obtained from the free energy is identically the horizon area over $4G$. □

Physical Insight

Theorem 54.10 is the holographic incarnation of black-hole thermodynamics. The entropy of a strongly coupled thermal gauge theory — a boundary, field-theoretic quantity — has turned out to be the *area* of a horizon in one higher dimension. That the answer is an area, not a volume, is the original "holographic" observation of 't Hooft and Susskind, and it is the reason the duality bears that name: the number of degrees of freedom of the boundary theory is bounded by the bulk horizon area in Planck units. Here we have not merely matched the scaling; we computed both sides and found them *equal*, coefficient included, with the Bekenstein–Hawking $1/4$ emerging from the Gibbons–Hawking boundary term and the counterterms with no further input.

54.4.5 The $\mathcal{N} = 4$ plasma and the factor of $\frac{3}{4}$

We now make the headline N^2 concrete on the carried example, $\mathcal{N} = 4$ super-Yang–Mills with gauge group $\mathrm{SU}(N)$ — the maximally symmetric large-N cousin of Yang–Mills theory (Chapter 25), whose holographic dual is type IIB string theory on $\mathrm{AdS}_5 \times S^5$ (Theorem 53.6). This is the $d = 4$, AdS_5 case. The computation reproduces a celebrated result: at infinite coupling the entropy is exactly $\frac{3}{4}$ of its free-field value. The factor of $\frac{3}{4}$ is the cleanest single number through which holography is checked against an honest field-theory count, and we carry it in full.

Worked Example 54.11 (The strong/weak factor $\frac{3}{4}$ for $\mathcal{N}=4$ SYM)**. Strong coupling (holography).** Set $d=4$ in Eqs. (54.114) and (54.115). The horizon radius is $z_h = 4/(4\pi T) = 1/(\pi T)$, and

$$f_{\text{strong}} = -\frac{L^3}{16\pi G_5}\Big(\frac{4\pi}{4}\Big)^4 T^4 = -\frac{\pi^3 L^3}{16\,G_5}\,T^4, \qquad s_{\text{strong}} = \frac{L^3}{4G_5}\,(\pi T)^3 = \frac{\pi^3 L^3}{4G_5}\,T^3. \tag{54.121}$$

The dictionary converts L^3/G_5 into a gauge-theory quantity. From the central-charge match of Theorem 53.33, the gravity side gives $a = c = \dfrac{\pi L^3}{8G_5} = \dfrac{N^2}{4}$ exactly (Eq. (53.208)), while the field-theory value is $\frac{N^2-1}{4}$ (Eq. (53.201)), the two agreeing at leading order in large N; hence

$$\frac{L^3}{G_5} = \frac{2N^2}{\pi}. \tag{54.122}$$

Substituting,

$$f_{\text{strong}} = -\frac{\pi^3 T^4}{16}\cdot\frac{2N^2}{\pi} = -\frac{\pi^2}{8}N^2T^4, \qquad s_{\text{strong}} = \frac{\pi^3 T^3}{4}\cdot\frac{2N^2}{\pi} = \frac{\pi^2}{2}N^2T^3. \tag{54.123}$$

Weak coupling (free-field count). At zero coupling, $\mathcal{N}=4$ SYM is a free gas of its constituent fields, one copy per generator of SU(N), i.e. $N^2-1 \approx N^2$ copies. The $\mathcal{N}=4$ vector multiplet contains, per generator: one gauge field (2 physical polarizations), six real scalars (6 bosonic states), and four Weyl fermions ($4\times 2 = 8$ fermionic states). So the bosonic and fermionic on-shell counts are

$$n_b = 2+6 = 8, \qquad n_f = 8. \tag{54.124}$$

The free energy density of a single massless real bosonic degree of freedom in four dimensions is the Stefan–Boltzmann result

$$f_{\text{boson}} = T\int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\ln\!\big(1-e^{-k/T}\big) = -\frac{\pi^2}{90}T^4, \tag{54.125}$$

the integral being the standard $\int_0^\infty \mathrm{d}x\, x^2\ln(1-e^{-x}) = -\pi^4/45$ divided by $2\pi^2$. A single fermionic degree of freedom contributes $\frac{7}{8}$ of this (the Fermi–Dirac integral $\int_0^\infty \mathrm{d}x\, x^2\ln(1+e^{-x}) = \frac{7}{8}\cdot\pi^4/45$). Hence the free $\mathcal{N}=4$ free energy density is

$$f_{\text{free}} = -\frac{\pi^2}{90}T^4\,(n_b + \tfrac{7}{8}n_f)(N^2-1) \approx -\frac{\pi^2}{90}T^4\,(8+7)\,N^2 = -\frac{\pi^2}{6}N^2T^4, \tag{54.126}$$

using $n_b + \frac{7}{8}n_f = 8 + \frac{7}{8}\cdot 8 = 15$ and $15/90 = 1/6$. The corresponding entropy density is $s_{\text{free}} = -\partial f_{\text{free}}/\partial T = \frac{2\pi^2}{3}N^2T^3$.

The ratio. Dividing the strong-coupling result Eq. (54.123) by the free-field result Eq. (54.126),

$$\boxed{\frac{f_{\text{strong}}}{f_{\text{free}}} = \frac{\pi^2N^2/8}{\pi^2N^2/6} = \frac{6}{8} = \frac{3}{4}, \qquad \frac{s_{\text{strong}}}{s_{\text{free}}} = \frac{\pi^2/2}{2\pi^2/3} = \frac{3}{4}.} \tag{54.127}$$

The result $f_{\text{strong}} = \frac{3}{4} f_{\text{free}}$ is due to Gubser, Klebanov, and Tseytlin (1996). Several features deserve emphasis, in the honest spirit this book keeps about the frontier. First, the two numbers are computed in *opposite* regimes of the same theory: f_{free} at $\lambda = g^2 N \to 0$ where the gauge theory is a free gas, f_{strong} at $\lambda \to \infty$ where the classical gravity dual is reliable. There is no contradiction in their differing — a strongly interacting plasma has fewer thermally active states than a free gas of the same field content — and the factor of $\frac{3}{4}$ is precisely the interpolation between the two ends, a genuine prediction of the duality, not a consistency check fixed by symmetry. Second, the leading correction in $1/\lambda$ has been computed from the first α'-correction to type IIB supergravity, $f_{\text{strong}}/f_{\text{free}} = \frac{3}{4} + \frac{45}{32}\zeta(3)\lambda^{-3/2} + \cdots$, which rises back toward unity as the coupling is dialed down, smoothly connecting the two regimes; this is the same α'-correction that lifts η/s above $1/4\pi$ in Section 54.5. Third, the $\frac{3}{4}$ is not derivable from conformal symmetry alone: conformal invariance fixes the T^4 scaling and the equation of state $\varepsilon = 3p$, but the coefficient is dynamics, and matching $\frac{3}{4}$ between gravity and a (resummed) field-theory computation remains one of the most quoted pieces of evidence for the correspondence as of the mid-2020s.

Historical Note

The $\frac{3}{4}$ was found by Gubser, Klebanov, and Tseytlin in 1996, before the correspondence was even stated in its modern form — it was one of the near-extremal black-brane entropy computations that, in hindsight, were AdS/CFT thermodynamics avant la lettre. That a black hole's Bekenstein–Hawking entropy could be matched (up to the celebrated $\frac{3}{4}$) to a weak-coupling gauge-theory state count was among the first quantitative hints that a black brane and a thermal gauge plasma are the same object.

54.4.6 Global AdS and the Hawking–Page transition

The planar brane of Theorem 54.9 describes the CFT on flat space $\mathbb{R}^{d-1}$, which has no intrinsic scale: by conformal invariance every temperature is equivalent, the free energy is forced to be $\propto T^d$ at *all* $T > 0$, and the N^2 degrees of freedom are deconfined at any nonzero temperature. There is no phase transition because there is nothing for a transition to happen *at*. To see a genuine confinement/deconfinement transition — and thereby to make holographic contact with the confinement criterion of Yang–Mills theory — we must give the boundary theory a scale. The simplest way is to put it on a sphere: take the boundary to be $\mathbb{R}_\tau \times S^{d-1}$ with the sphere of radius set, by conformal invariance, to a reference value (we set it to the AdS radius L). The sphere radius is a length, and the dimensionless combination $T \times (\text{radius})$ is now a physical control parameter. The bulk dual of this boundary is *global* AdS rather than the Poincaré patch,

and the thermal physics involves a competition between two saddles.

The two saddles. The two relevant Euclidean geometries with boundary $S^1_\beta \times S^{d-1}$ are:

- *Thermal AdS*: global AdS_{d+1} with the Euclidean time periodically identified, $\tau \sim \tau + \beta$. Its metric is

$$\mathrm{d}s^2 = \Big(1 + \frac{r^2}{L^2}\Big)\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{1 + r^2/L^2} + r^2\,\mathrm{d}\Omega^2_{d-1}, \tag{54.128}$$

  smooth everywhere; the time circle never shrinks, so β is unconstrained. This is "empty" AdS at temperature T — a thermal gas of supergravity modes in a fixed geometry.

- *Global AdS–Schwarzschild*: a black hole sitting at the center of global AdS,

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\,\mathrm{d}\Omega^2_{d-1}, \qquad f(r) = 1 + \frac{r^2}{L^2} - \frac{\mu}{r^{d-2}}, \tag{54.129}$$

  with a horizon at the largest root r_+ of f, where the same Euclidean-regularity argument as in Section 54.4.2 fixes β.

The parameter μ measures the black-hole mass. From $f(r_+) = 0$, $\mu = r_+^{d-2}(1 + r_+^2/L^2)$, and the temperature follows from $T = f'(r_+)/4\pi$. Differentiating Eq. (54.129), $f'(r) = \frac{2r}{L^2} + (d-2)\frac{\mu}{r^{d-1}}$, and substituting $\mu/r_+^{d-1} = (1 + r_+^2/L^2)/r_+$ gives

$$T = \frac{1}{4\pi}\Big(\frac{d\,r_+}{L^2} + \frac{d-2}{r_+}\Big). \tag{54.130}$$

This temperature, unlike the planar case, has a *minimum*. Setting $\mathrm{d}T/\mathrm{d}r_+ = 0$ gives $r_{+,\min} = L\sqrt{(d-2)/d}$ and

$$T_{\min} = \frac{1}{4\pi}\Big(\frac{d\,r_{+,\min}}{L^2} + \frac{d-2}{r_{+,\min}}\Big) = \frac{\sqrt{d(d-2)}}{2\pi L}, \tag{54.131}$$

($T_{\min} = \sqrt{2}/\pi L$ for $d = 4$). Below $T_{\min}$ no black hole exists at all; above it there are two branches — a small black hole ($r_+ < r_{+,\min}$, with negative specific heat, thermodynamically unstable) and a large one ($r_+ > r_{+,\min}$, stable). Only the large branch can compete with thermal AdS.

Which saddle dominates. The competition is decided by the free energies, i.e. by the renormalized on-shell actions. We compute the *difference* $\Delta I = I_{\mathrm{BH}} - I_{\mathrm{AdS}}$ by the background-subtraction method, which is cleaner here than introducing counterterms term by term: both geometries share the same conformal boundary, so their Gibbons–Hawking and counterterm contributions are identical and cancel

in the difference, leaving only the difference of bulk integrals. Using the on-shell density Eq. (54.88) (the constant $-2d/L^2$, valid for both geometries) and $\sqrt{g} = r^{d-1}\sqrt{g_{S^{d-1}}}$, the regulated bulk actions out to a common cutoff $r = \mathcal{R}$ are

$$I_{\rm BH} = \frac{d}{8\pi G L^2}\,\beta\,\Omega_{d-1}\int_{r_+}^{\mathcal{R}} r^{d-1}\,{\rm d}r = \frac{\beta\,\Omega_{d-1}}{8\pi G L^2}(\mathcal{R}^d - r_+^d), \tag{54.132}$$

$$I_{\rm AdS} = \frac{\beta'\,\Omega_{d-1}}{8\pi G L^2}\,\mathcal{R}^d, \tag{54.133}$$

where $\Omega_{d-1} = {\rm Vol}(S^{d-1})$, the black-hole integral starts at the horizon (the spacetime ends there), and the thermal-AdS integral starts at $r = 0$. The two periods are *not* equal: β is fixed by black-hole regularity, while β' is the thermal-AdS period chosen so that the two geometries have the *same proper length of the time circle at the common cutoff* $r = \mathcal{R}$ — the physical matching condition that they describe the same boundary,

$$\beta'\sqrt{1+\mathcal{R}^2/L^2} = \beta\sqrt{f(\mathcal{R})} \;\Longrightarrow\; \beta' = \beta\sqrt{\frac{1+\mathcal{R}^2/L^2-\mu/\mathcal{R}^{d-2}}{1+\mathcal{R}^2/L^2}} \xrightarrow{\mathcal{R}\to\infty} \beta\Big(1-\frac{\mu L^2}{2\mathcal{R}^d}+\cdots\Big). \tag{54.134}$$

Now form the difference and take $\mathcal{R}\to\infty$. The $\mathcal{R}^d$ pieces cancel up to the matching correction:

$$\Delta I = \frac{\Omega_{d-1}}{8\pi G L^2}\lim_{\mathcal{R}\to\infty}\Big[\beta(\mathcal{R}^d - r_+^d) - \beta'\mathcal{R}^d\Big] = \frac{\Omega_{d-1}\beta}{8\pi G L^2}\Big(-r_+^d + \frac{\mu L^2}{2}\Big), \tag{54.135}$$

where the surviving finite piece $\frac{\mu L^2}{2}$ comes from the $\mathcal{R}\to\infty$ limit of $\beta'\mathcal{R}^d = \beta\mathcal{R}^d - \beta\frac{\mu L^2}{2} + \cdots$. Insert $\mu = r_+^{d-2} + r_+^d/L^2$:

$$-r_+^d + \frac{\mu L^2}{2} = -r_+^d + \frac{L^2 r_+^{d-2}}{2} + \frac{r_+^d}{2} = \frac{r_+^{d-2}}{2}(L^2 - r_+^2), \tag{54.136}$$

so the on-shell action difference is

$$\boxed{\Delta I = \frac{\Omega_{d-1}\,\beta\,r_+^{d-2}(L^2 - r_+^2)}{16\pi G\,L^2}\,.} \tag{54.137}$$

This is the Hawking–Page result. Its sign is the whole story:

- For $r_+ < L$ (which, via Eq. (54.130), is the low-temperature regime), $\Delta I > 0$, so $I_{\rm BH} > I_{\rm AdS}$ and *thermal AdS* dominates the path integral e^{-I}. The dominant saddle has no horizon.
- For $r_+ > L$ (high temperature), $\Delta I < 0$ and the *black hole* dominates.

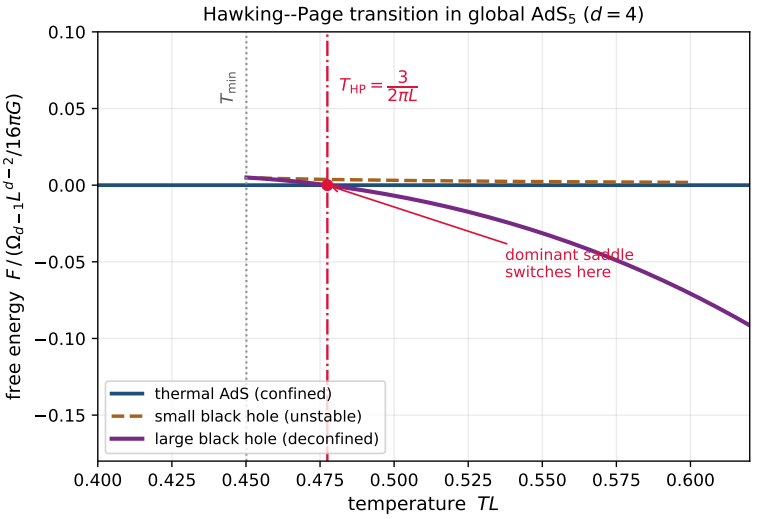


Figure 54.5: Free energy of the competing saddles for the CFT on S^{d-1} ($d = 4$, AdS_5). Thermal AdS has $F = 0$ (the reference). The black-hole branch appears only above $T_{\min} = \sqrt{2}/\pi L$ Eq. (54.131), splitting into a small (unstable, dashed) and a large (stable) branch. The large-black-hole free energy crosses zero at the Hawking–Page temperature $T_{\rm HP} = 3/2\pi L$ Eq. (54.138): below it thermal AdS dominates (confined, $F \sim N^0$), above it the black hole dominates (deconfined, $F \sim N^2$). The jump in $\partial F/\partial T$ at $T_{\rm HP}$ is the first-order deconfinement transition.

- The transition is at $\Delta I = 0$, i.e. $r_+ = L$ exactly. Substituting $r_+ = L$ into Eq. (54.130) gives the *Hawking–Page temperature*

$$T_{\rm HP} = \frac{1}{4\pi}\Big(\frac{d}{L} + \frac{d-2}{L}\Big) = \frac{d-1}{2\pi L}, \tag{54.138}$$

  which is $T_{\rm HP} = 3/2\pi L$ for $d = 4$. Note $T_{\rm HP} > T_{\min}$, so the transition occurs on the stable large-black-hole branch, as a first-order jump.

Figure 54.5 plots the free energy $F_{\rm BH} = \Delta I/\beta$ of the two saddles for $d = 4$, exhibiting the minimum temperature, the unstable small-black-hole branch, and the first-order crossing at $T_{\rm HP}$.

Confinement and deconfinement. The Hawking–Page transition is the gravity dual of the confinement/deconfinement transition of the boundary gauge theory — the observation of Witten (1998), and the strong-coupling face of the confinement criterion proved at weak coupling in Section 27.7. The order parameter is the scaling of the free energy with N. In the low-temperature (thermal-AdS) phase, the free energy relative to the vacuum is $O(N^0)$: it is the free energy of a gas of supergravity modes, i.e. of color-singlet glueball-like states,

with an $O(1)$ number of thermally active degrees of freedom — the hallmark of *confinement*, where color is locked into singlets and the count of light states does not grow with N. In the high-temperature (black-hole) phase, $F \sim 1/G \sim N^2$ by Eq. (54.114) and Theorem 53.33: the full N^2 gluonic degrees of freedom are liberated, the hallmark of *deconfinement*. The transition between an $O(N^0)$ and an $O(N^2)$ free energy is exactly the large-N deconfinement transition, and here it is a sharp, first-order geometric event: the nucleation of a horizon.

Physical Insight

Yang–Mills theory established confinement at weak coupling through the Wilson-loop area law (Theorem 27.20): a linear quark–antiquark potential, color flux squeezed into a tube. That criterion is genuinely non-perturbative but was accessible there only qualitatively, in lattice language. Holography supplies the complementary strong-coupling, large-N statement: confinement is the phase in which the dominant bulk geometry has *no horizon*, deconfinement the phase in which a black hole has formed, and the free energy's scaling with N is the order parameter. The same physics — color confinement — is seen from two opposite ends of the coupling, neither computation reachable by the other's methods. This is the deepest payoff of the duality for gauge theory: a controlled handle on the strongly coupled, deconfined plasma that perturbative Yang–Mills (Chapter 27) cannot touch.

Caution

The clean Hawking–Page transition requires the boundary sphere. On flat space (the planar brane that carries the rest of this chapter) the deconfined black-brane phase exists for all $T > 0$ and there is no transition — this is not a failure but a feature: a scale-invariant theory on scale-invariant space cannot have a preferred transition temperature. $\mathcal{N} = 4$ SYM, being exactly conformal, confines only when placed on a sphere, where the sphere radius provides the scale; a confining theory like pure Yang–Mills would confine on flat space too, and its holographic dual replaces pure AdS in the infrared by a geometry that caps off smoothly even at zero temperature. The honest statement is that the Hawking–Page transition is the deconfinement transition *of the conformal theory on a sphere*; modeling the genuine QCD deconfinement transition requires deforming the geometry, a subject beyond this chapter.

We have built the thermal state from the ground up: a planar black brane Eq. (54.89) as the dominant saddle, its Hawking temperature Eq. (54.96) from Euclidean regularity, its renormalized free energy Eq. (54.114) and entropy obeying the Bekenstein–Hawking area law Theorem 54.10, the strong/weak

factor of $\frac{3}{4}$ for $\mathcal{N}=4$ SYM, and the Hawking–Page transition Eq. (54.137) as the gravity dual of deconfinement. All of this was equilibrium thermodynamics, computed from a Euclidean on-shell action that knows nothing of time. The plasma is, however, a dynamical medium, and its real-time response — how it relaxes, how it dissipates — is encoded in the propagation of the very same bulk fields on the very same geometry, now in Lorentzian signature with a boundary condition imposed at the horizon. Section 54.5 takes the metric Eq. (54.89), the temperature Eq. (54.96), and the entropy density Eq. (54.115) handed over here, and extracts from them the shear viscosity and the universal ratio $\eta/s = 1/4\pi$.

54.5 Real-Time Correlators and the Shear Viscosity Bound

The previous section handed us a thermal state and its coarse data: the planar AdS–Schwarzschild black brane Eq. (54.89), its Hawking temperature Eq. (54.96) fixed by Euclidean regularity, and the entropy density s delivered by the area law Theorem 54.10. Those are equilibrium quantities — they are computed from a Euclidean on-shell action and know nothing of time. The plasma described by this geometry is, however, a dynamical medium: perturb it and it relaxes, push it and it resists, and the rates at which it does so are real-time transport coefficients. To compute them we must leave Euclidean signature, where everything so far has lived, and confront Lorentzian time directly. This is not a cosmetic change. A Euclidean correlator is real and symmetric and encodes no arrow of time; a transport coefficient is an intrinsically dissipative, causal quantity, the imaginary part of a *retarded* response. The bulk geometry now has a feature it lacked at zero temperature — a horizon — and we will see that it is precisely the boundary condition imposed at that horizon, the demand that nothing escape it, that injects causality into the bulk computation and selects the retarded correlator. The chain of this section is short and inevitable: build the Lorentzian dictionary (the infalling prescription), feed it the one perturbation that controls momentum transport (the transverse graviton, which we show is a massless scalar), and read off the shear viscosity. The answer, $\eta/s = 1/4\pi$, is the single most quoted number in holography, and we will compute it twice — once by a horizon-flux argument that makes its universality transparent, and once as a check from the hydrodynamic pole of the correlator — and then say honestly what it is and is not.

54.5.1 From Euclidean to retarded: the infalling prescription

At zero temperature the GKP–Witten relation Eq. (53.162) computed Euclidean correlators, and Euclidean correlators were all we needed: conformal data is signature-blind, and the Wick rotation back to Lorentzian signature is, for

the vacuum, an unambiguous analytic continuation. At finite temperature the situation is different in a way that matters. The thermal field theory possesses a whole family of two-point functions — Euclidean (Matsubara), Feynman (time-ordered), Wightman, retarded, advanced — which agree in the vacuum but are genuinely distinct in a medium, related by the fluctuation–dissipation theorem and distinguished by their analytic structure in the complex frequency plane. Transport is governed by exactly one of them.

Recall the object we are after. The *retarded* two-point function of an operator $\mathcal{O}$ is

$$G^R(\omega,\mathbf{k}) = -i\int \mathrm{d}t\,\mathrm{d}^{d-1}x\; e^{i\omega t - i\mathbf{k}\cdot\mathbf{x}}\,\theta(t)\,\big\langle\,[\,\mathcal{O}(t,\mathbf{x}),\,\mathcal{O}(0,\mathbf{0})\,]\,\big\rangle_T, \tag{54.139}$$

the thermal expectation value of the commutator, retarded by the step function $\theta(t)$ so that the response at time t depends only on disturbances in its past. Causality makes $G^R(\omega)$ analytic in the upper half ω-plane; its imaginary part is the spectral density,

$$\rho(\omega,\mathbf{k}) = -2\,\mathrm{Im}\,G^R(\omega,\mathbf{k}), \tag{54.140}$$

a non-negative measure of the dissipation the medium exerts on a probe of frequency ω. This is the same physics — and, structurally, the same object — as the imaginary part of the photon self-energy $\Pi^{\mu\nu}(q)$ of Eq. (19.4): there the transverse form factor's imaginary part was, by the optical theorem (Section 14.7), a sum over physical on-shell intermediate states, i.e. a production rate. Here it is the rate at which a slow perturbation deposits energy into the plasma. The retarded correlator is the unique one whose poles encode the decay (not the growth) of perturbations, and it is the one transport sees.

The Euclidean correlator $G_E(\omega_n,\mathbf{k})$, by contrast, is defined only at the discrete Matsubara frequencies $\omega_n = 2\pi n T$ (bosonic), and the retarded function is its analytic continuation from the upper half-plane,

$$G^R(\omega,\mathbf{k}) = G_E(\omega_n \to -i\omega + 0^+,\,\mathbf{k}), \qquad \omega_n > 0, \tag{54.141}$$

with the 0^+ fixing the continuation to come from above the real axis. In a generic interacting theory Eq. (54.141) is impractical: one has data at isolated points ω_n and must reconstruct an analytic function, an ill-posed inversion. Holography circumvents it entirely by working in Lorentzian signature from the start — but then the bulk computation must itself encode the choice of retarded over Feynman, and the question is how.

The bulk solutions and the horizon. Consider, as the universal warm-up, a massless scalar ϕ probing the black brane; in Section 54.5.3 we show that the perturbation controlling viscosity *is* such a scalar. Write the planar brane metric Eq. (54.89) in the form

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left[-f(z)\,\mathrm{d}t^2 + \mathrm{d}\mathbf{x}^2 + \frac{\mathrm{d}z^2}{f(z)}\right], \qquad f(z) = 1 - \left(\frac{z}{z_h}\right)^d, \tag{54.142}$$

with the conformal boundary at $z \to 0$ and the horizon at $z = z_h$, where $f(z_h) = 0$; the Hawking temperature is $T = |f'(z_h)|/4\pi = d/4\pi z_h$ from Eq. (54.96). The wave equation $\Box\phi = 0$ in this background, Fourier-transformed as $\phi(z, t, \mathbf{x}) = e^{-i\omega t + i\mathbf{k}\cdot\mathbf{x}}\,\phi_k(z)$, is a second-order ordinary differential equation in z, so it has a two-dimensional solution space. We must impose one condition at each end. At the boundary the structure is the familiar one of Section 53.6.3: two indicial powers, the non-normalizable z^0 (a constant, the source) and the normalizable z^d (the response). The new ingredient is the condition at the horizon.

Near $z = z_h$, set $f(z) \simeq f'(z_h)(z - z_h) = -4\pi T\,(z - z_h)$. The wave equation degenerates there, and its two local solutions are pure phases in the tortoise-like variable,

$$\phi_k(z) \;\sim\; (z_h - z)^{\mp\, i\omega/4\pi T} \;=\; f(z)^{\mp\, i\omega/4\pi T} \times (\text{regular}), \tag{54.143}$$

which we read off by substituting $\phi \sim f^\alpha$ into the leading near-horizon equation: $g^{zz} \sim f$ vanishes linearly while $g^{tt} \sim 1/f$ blows up, and balancing the $\partial_z(f\,\partial_z)$ term against the ω^2/f term forces $\alpha = \mp i\omega/4\pi T$. Restoring the time dependence, the two behaviours are

$$\phi \;\sim\; e^{-i\omega t}\,(z_h - z)^{-i\omega/4\pi T} \quad \text{(infalling)}, \qquad \phi \;\sim\; e^{-i\omega t}\,(z_h - z)^{+i\omega/4\pi T} \quad \text{(outgoing)}. \tag{54.144}$$

To see which is which, note that $z_h - z$ decreases toward the horizon, so a surface of constant phase $\omega t + (\omega/4\pi T)\log(z_h - z) = \text{const}$ moves toward larger z (toward the horizon) as t increases for the first branch: this wave falls *into* the horizon. The second branch emerges from it.

Causality selects infalling. A classical black-brane horizon is a one-way membrane: nothing, not even information about the wave, comes back out of it. A perturbation set up in the past propagates inward and is absorbed; nothing emerges. Discarding the outgoing branch and keeping only the infalling solution is the bulk statement of causality, and — this is the crucial claim — it is exactly equivalent to choosing the retarded correlator on the boundary. The argument is the analytic-continuation Eq. (54.141) made geometric. In Euclidean signature the horizon is not a one-way membrane at all: it is a smooth point (the tip of the cigar of Section 54.4) where the Euclidean time circle shrinks, and the only condition one can impose is *regularity* there. The Euclidean solution, regular at the tip, when continued in frequency by $\omega_n \to -i\omega + 0^+$ (continuation from the upper half-plane, which is where G^R is analytic), turns the regular behaviour $f^{\,\omega_n/4\pi T}$ into precisely $f^{-i\omega/4\pi T}$ — the infalling phase. Continuation from *below* ($\omega_n \to +i\omega$, the lower half-plane where G^A is analytic) would give the outgoing branch and the advanced correlator. So:

$$\boxed{\text{regular Euclidean at the tip} \;\xrightarrow{\omega_n \to -i\omega + 0^+}\; \text{infalling at the horizon} \;\Longleftrightarrow\; G^R.} \tag{54.145}$$

The infalling condition is the Lorentzian image of Euclidean regularity, and it is what makes the bulk computation deliver G^R rather than G^A or the time-ordered correlator.

Extracting the correlator: the Son–Starinets prescription. It remains to say how G^R is read off from the bulk solution, and here a subtlety of factor and signature must be faced squarely. Take the field with the canonical two-derivative action

$$S = -\frac{K}{2}\int \mathrm{d}^{d+1}X\,\sqrt{-g}\,g^{MN}\partial_M\phi\,\partial_N\phi, \tag{54.146}$$

where K is a normalization (for the graviton, $K = 1/16\pi G_N$, derived in Section 54.5.3). The equation of motion is $\partial_M(\sqrt{-g}\,g^{MN}\partial_N\phi) = 0$; on shell the action collapses to a boundary term,

$$S_{\text{on-shell}} = -\frac{K}{2}\int \mathrm{d}^{d}x\,\Big[\sqrt{-g}\,g^{zz}\,\phi\,\partial_z\phi\Big]_{z=z_h}^{z\to 0}, \tag{54.147}$$

the bulk integral having vanished by the equation of motion, leaving only the radial endpoints. Write the solution obeying the infalling condition and normalized to unit boundary value as $\phi_k(z) = f_k(z)\,\phi_0(k)$, $f_k(0) = 1$, so that $\phi_0(k)$ is the boundary source. In momentum space Eq. (54.147) becomes

$$S_{\text{on-shell}} = -\frac{1}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\phi_0(-k)\,\mathcal{F}_k(z)\,\phi_0(k)\Big|_{z=z_h}^{z\to 0}, \qquad \mathcal{F}_k(z) \equiv K\sqrt{-g}\,g^{zz}\,f_{-k}(z)\,\partial_z f_k(z). \tag{54.148}$$

Naively functional-differentiating twice would give $G^R = -2\mathcal{F}_k|_{z\to 0}$ symmetrized over the two endpoints and over $k \to -k$ — and that symmetrization is exactly the error. It produces a *real* result, killing the dissipative imaginary part, because the boundary and horizon contributions and their complex conjugates conspire to cancel it. The correct Lorentzian rule, due to Son and Starinets, is:

$$\boxed{G^R(\omega,\mathbf{k}) = -\,\mathcal{F}_k(z)\big|_{z\to 0} = -\,K\sqrt{-g}\,g^{zz}\,\frac{\partial_z f_k(z)}{f_k(z)}\Big|_{z\to 0}}, \tag{54.149}$$

with three explicit instructions that distinguish it from the Euclidean/Feynman recipe. *First*, keep only the boundary term at $z \to 0$; discard the horizon contribution at $z = z_h$ entirely (it would otherwise corrupt the answer with horizon data that is not part of the UV response). *Second*, do *not* symmetrize the two factors of ϕ_0 and do *not* add the complex conjugate — this is the missing "factor of two": the retarded function is the un-symmetrized coefficient $-\mathcal{F}_k$, not its real part $-\operatorname{Re}\mathcal{F}_k = -\frac{1}{2}(\mathcal{F}_k + \mathcal{F}_k^*)$. Because $f_k(z)$ is complex (it carries the infalling phase $f^{-i\omega/4\pi T}$), $\mathcal{F}_k$ is complex and its imaginary part survives, delivering the spectral density. *Third*, the result is to be renormalized by the same near-boundary counterterms of Section 53.6.4 that rendered the zero-temperature

two-point function finite; the divergent contact terms they remove are real and analytic in ω^2, so they never touch $\operatorname{Im} G^R$ — a fortunate fact that makes transport coefficients, which live in the imaginary part, automatically finite without any subtraction. The factor and the no-symmetrization rule are the entire content that separates the Lorentzian dictionary from the Euclidean one.

Caution

The temptation to compute G^R by analytically continuing a Euclidean (symmetrized) on-shell action is precisely the trap Eq. (54.149) avoids. The symmetric prescription returns $\operatorname{Re} G^R$ at best and, more often, nonsense, because it averages the infalling and outgoing solutions and so cannot know which way time runs. One must impose infalling boundary conditions on a genuinely complex Lorentzian solution and keep the unsymmetrized boundary term. The radially-conserved flux argument of Section 54.5.4 below is, in part, a way of seeing *why* only the boundary term and only the imaginary part are physical: the flux that carries dissipation is conserved from boundary to horizon, so its value can be computed at either end, and the horizon end makes its reality and sign manifest.

54.5.2 The Kubo formula for shear viscosity

We now specialize to the operator whose retarded correlator is the shear viscosity: the stress tensor $T^{\mu\nu}$, the conserved current of translations. Viscosity measures the medium's resistance to shear flow, and in linear-response theory it is extracted from the low-frequency limit of a stress-tensor two-point function by a Kubo formula. We import the statement as a result of the general linear-response theory of transport, of which the current–current correlator of Chapter 19 is the abelian prototype.

Definition 54.12 (Shear viscosity and the Kubo formula)**.** For a relativistic fluid in its rest frame at temperature T, the *shear viscosity* η is the transport coefficient in the constitutive relation $T^{ij} = p\,\delta^{ij} - \eta\,(\partial^i u^j + \partial^j u^i - \frac{2}{d-1}\delta^{ij}\partial\cdot u) + \dots$, the coefficient of the traceless symmetric gradient of the velocity field u. Equivalently, it is given by the Kubo formula as the zero-frequency slope of the spectral density of the off-diagonal stress component T^{xy} at zero spatial momentum,

$$\eta = -\lim_{\omega\to 0}\frac{1}{\omega}\operatorname{Im} G^R_{xy,xy}(\omega,\mathbf{k}=0), \qquad G^R_{xy,xy}(\omega,\mathbf{k}) = -i\int \mathrm{d}t\,\mathrm{d}^{d-1}x\; e^{i\omega t - i\mathbf{k}\cdot\mathbf{x}}\,\theta(t)\,\langle[\,T^{xy}(x), \tag{54.150}$$

The structure here is identical to the photon self-energy of Eq. (19.4): a conserved current ($T^{\mu\nu}$ in place of J^μ), whose two-point function is constrained to be transverse by the corresponding Ward identity ($\partial_\mu T^{\mu\nu} = 0$, the tensor analogue

of $\partial_\mu J^\mu = 0$ that gave $q_\mu \Pi^{\mu\nu} = 0$ in Theorem 19.5), and whose imaginary part is a non-negative spectral density measuring dissipation, just as $\operatorname{Im}\Pi(q^2)$ measured pair production. The T^{xy} component is the transverse-traceless ("shear") channel: at $\mathbf{k} = 0$ it is invariant under the rotations that fix the x–y plane orientation up to sign, decouples from the energy-density and pressure fluctuations (which sit in the scalar and sound channels), and therefore furnishes the cleanest single-function probe of momentum diffusion. The Kubo formula Eq. (54.150) is the precise statement that η is the linear-response coefficient relating the shear stress to an imposed shear strain rate — it is the coefficient of ω in the small-frequency expansion of the dissipative (Im) part of the response, exactly as a conductivity is the zero-frequency limit of $\operatorname{Im}\Pi/\omega$ in Chapter 19.

Holographically, $T^{\mu\nu}$ is the operator dual to the bulk metric $g_{\mu\nu}$ itself (the source for the stress tensor is the boundary metric, by the GKP–Witten relation Eq. (53.162) applied to the graviton). To compute $G^R_{xy,xy}$ we therefore perturb the bulk metric in the xy direction, solve the linearized Einstein equations with infalling conditions, and apply the prescription Eq. (54.149). The first task is to identify the bulk field and its equation.

54.5.3 The transverse graviton is a massless scalar

The metric fluctuation dual to T^{xy} is the off-diagonal spatial component h_{xy}. Raise an index to define the dimensionless perturbation

$$\phi(t, z) \equiv h^x{}_y = g^{xx} h_{xy}, \tag{54.151}$$

at zero spatial momentum, $\mathbf{k} = 0$, which is all the Kubo formula Eq. (54.150) requires. The perturbed line element is $\mathrm{d}s^2 \to \mathrm{d}s^2 + 2\, g_{xx}\, \phi\, \mathrm{d}x\, \mathrm{d}y$, with all other components of h set to zero. The central structural fact, which makes the whole viscosity computation tractable, is the following.

Proposition 54.13 (Shear graviton reduces to a massless minimally coupled scalar)**.** *On the planar black-brane background Eq. (54.142), the perturbation $\phi = h^x{}_y$ at $\mathbf{k} = 0$ satisfies the equation of a massless minimally coupled scalar,*

$$\frac{1}{\sqrt{-g}}\, \partial_M \Big(\sqrt{-g}\, g^{MN} \partial_N \phi \Big) = 0, \tag{54.152}$$

and its quadratic effective action is that of Eq. (54.146) with normalization $K = 1/16\pi G_N$.

Proof. We argue from symmetry, which fixes the form of the quadratic action completely, and then match the normalization. Three facts about $\phi = h^x{}_y$ on this background do all the work.

(i) ϕ is a scalar under the residual symmetry. The background Eq. (54.142) is invariant under translations and rotations of the spatial $\mathbf{x}$-directions and under

the reflection $R_y : y \to -y$. Under R_y the metric is unchanged, but the cross term $g_{xy}\,\mathrm{d}x\,\mathrm{d}y$ is odd, so $\phi \to -\phi$. Any other fluctuation of the metric — the diagonal components h_{tt}, h_{zz}, h_{ii}, the energy flux h_{ti}, the would-be scalar and vector modes — is *even* under R_y (or under the analogous reflection in another plane). A reflection-odd field cannot mix at quadratic order with reflection-even fields: the cross terms in the action are forbidden by the symmetry. Hence ϕ decouples from every other component of h_{MN} and obeys an equation involving ϕ alone. This is why the shear channel is clean — it is the unique parity-odd sector, protected from the constraint-tangle that snares the sound channel.

(ii) The action is two-derivative and reflection-even. Diffeomorphism invariance of the Einstein–Hilbert action $S = \frac{1}{16\pi G_N}\int \sqrt{-g}\,(R - 2\Lambda)$ descends to the quadratic action for ϕ. Expanding R to second order in h_{MN} and keeping only the decoupled ϕ produces a functional that is a scalar under the spatial isometries, even in ϕ (by (i)), and contains at most two derivatives (the curvature is two-derivative). The most general such quadratic functional of a single scalar on this background is

$$S^{(2)}[\phi] = -\frac{K}{2}\int \mathrm{d}^{d+1}X\,\sqrt{-g}\left(g^{MN}\partial_M\phi\,\partial_N\phi + m^2\phi^2\right), \tag{54.153}$$

for some constants K and m^2: no other reflection-even, two-derivative, diffeomorphism-scalar invariant of a single field exists. It remains to fix m^2 and K.

(iii) The mass vanishes: a constant ϕ is pure gauge. A spatially-constant, time-independent $\phi = c$ is not a physical excitation at all but a coordinate change. Under the linear redefinition $y \to y + c\,x$ of the flat boundary directions, $\mathrm{d}y^2 \to (\mathrm{d}y + c\,\mathrm{d}x)^2 = \mathrm{d}y^2 + 2c\,\mathrm{d}x\,\mathrm{d}y + c^2\mathrm{d}x^2$, which to first order in c generates exactly the off-diagonal term $2c\,\mathrm{d}x\,\mathrm{d}y$, i.e. $h^x{}_y = c$. Because this is a coordinate transformation, it leaves the Riemann tensor — hence the action — unchanged: a constant ϕ costs zero action. A mass term $\frac{Km^2}{2}\int\sqrt{-g}\,\phi^2$ would assign a nonzero action to a constant ϕ, so $m^2 = 0$. The perturbation ϕ is a genuine massless mode, the Goldstone-like avatar of the broken shear isometry, and its equation of motion is Eq. (54.152).

Normalization. The overall K is read off by comparing Eq. (54.153) to the explicit second-order expansion of the Einstein–Hilbert action. The kinetic term for a single off-diagonal metric component $h^x{}_y$ in $\frac{1}{16\pi G_N}\int\sqrt{-g}\,R$ is, after integration by parts, $-\frac{1}{16\pi G_N}\cdot\frac{1}{2}\int\sqrt{-g}\,g^{MN}\partial_M\phi\,\partial_N\phi$ — the standard result that each transverse-traceless graviton polarization is a canonically normalized scalar with the gravitational prefactor. Hence $K = 1/16\pi G_N$, as claimed. □

The reduction Theorem 54.13 is the technical heart of the universality of η/s: it says that for *any* bulk theory governed at two-derivative order by Einstein gravity, the shear graviton is the same massless scalar with the same prefactor $1/16\pi G_N$, regardless of the matter content that supports the geometry. The

computation that follows therefore returns the same number for every such background.

With $\mathbf{k}=0$ and $\phi = e^{-i\omega t}\phi_\omega(z)$, Eq. (54.152) reads, using $\sqrt{-g}=(L/z)^{d+1}$ (the factors of f cancel between g_{tt} and g_{zz}), $\sqrt{-g}\,g^{zz} = (L/z)^{d-1}f$, and $\sqrt{-g}\,g^{tt} = -(L/z)^{d-1}/f$,

$$\partial_z\Big[(L/z)^{d-1}f(z)\,\partial_z\phi_\omega\Big] + \frac{\omega^2}{f(z)}\,(L/z)^{d-1}\,\phi_\omega = 0. \tag{54.154}$$

This is the massless-scalar radial equation we will solve, with infalling conditions at z_h and unit source at $z\to 0$.

54.5.4 The horizon-flux computation of η

We could solve Eq. (54.154) order by order in ω and apply Eq. (54.149), and we will sketch that route as a check. But there is a far more illuminating argument, the *membrane-paradigm* computation, which extracts η from horizon data alone and exposes why the answer is universal. The idea is that the prescription Eq. (54.149) involves a radial quantity that is *conserved* along the flow from boundary to horizon in the $\omega\to 0$ limit, so its boundary value — which is G^R — equals its horizon value, where regularity fixes it completely.

Define the radial canonical momentum conjugate to ϕ and the response function it builds,

$$\Pi(z,\omega) \equiv -K\sqrt{-g}\,g^{zz}\,\partial_z\phi_\omega, \qquad \chi(z,\omega)\equiv \frac{\Pi(z,\omega)}{i\omega\,\phi_\omega(z)}. \tag{54.155}$$

The prescription Eq. (54.149) is exactly $G^R(\omega) = [\Pi/\phi]_{z\to 0} = i\omega\,\chi(0,\omega)$. The equation of motion Eq. (54.154) is $\partial_z\Pi = -K\,\partial_z(\sqrt{-g}\,g^{zz}\partial_z\phi) = -K\,\omega^2\sqrt{-g}\,g^{tt}\phi$ (using $\partial_t^2\to-\omega^2$ and the equation of motion), while $\partial_z\phi = -\Pi/(K\sqrt{-g}\,g^{zz})$ by definition. Differentiating $\chi=\Pi/(i\omega\phi)$ and substituting both relations gives a first-order flow equation in which the two derivatives combine into

$$\partial_z\chi = i\omega\left[\frac{\chi^2}{K\sqrt{-g}\,g^{zz}} + K\sqrt{-g}\,g^{tt}\right]. \tag{54.156}$$

Every step is algebra: $\partial_z\chi = \partial_z\Pi/(i\omega\phi) - \Pi\,\partial_z\phi/(i\omega\phi^2)$; the first term is $-K\omega^2\sqrt{-g}g^{tt}\phi/(i\omega\phi) = i\omega K\sqrt{-g}g^{tt}$, and the second is $-\Pi\cdot[-\Pi/(K\sqrt{-g}g^{zz})]/(i\omega\phi^2) = \Pi^2/(i\omega K\sqrt{-g}g^{zz}\phi^2) = i\omega\,\chi^2/(K\sqrt{-g}g^{zz})$, using $\Pi = i\omega\chi\phi$. Their sum is Eq. (54.156).

The flow equation Eq. (54.156) has the structure that makes the membrane paradigm work: its right-hand side is proportional to ω. Therefore, as $\omega\to 0$, $\partial_z\chi\to 0$ and χ becomes radially constant at leading order. The boundary value, which is what enters G^R, equals the horizon value,

$$\lim_{\omega\to0}\chi(0,\omega) = \lim_{\omega\to0}\chi(z_h,\omega), \tag{54.157}$$

and the horizon value is fixed by the infalling condition, with no need to solve the equation in between. This is the whole trick: a transport coefficient is a horizon quantity.

The horizon value. Impose infalling boundary conditions Eq. (54.144): near z_h, $\phi_\omega \sim f(z)^{-i\omega/4\pi T}$. Then

$$\partial_z \phi_\omega = -\frac{i\omega}{4\pi T}\,\frac{f'(z)}{f(z)}\,\phi_\omega \xrightarrow{z\to z_h} -\frac{i\omega}{4\pi T}\,\frac{(-4\pi T)}{f(z)}\,\phi_\omega = \frac{i\omega}{f(z)}\,\phi_\omega, \tag{54.158}$$

using $f'(z_h) = -4\pi T$. Substitute into Π from Eq. (54.155), with $\sqrt{-g}\,g^{zz} = (L/z)^{d-1} f(z)$:

$$\Pi(z_h,\omega) = -K\,(L/z)^{d-1} f(z)\cdot \frac{i\omega}{f(z)}\,\phi_\omega = -\,i\omega\,K\,(L/z_h)^{d-1}\,\phi_\omega, \tag{54.159}$$

the factors of f cancelling exactly — the cancellation that makes the horizon limit finite. Hence

$$\chi(z_h,\omega) = \left.\frac{\Pi}{i\omega\phi}\right|_{z_h} = -K\,(L/z_h)^{d-1}, \tag{54.160}$$

a real, ω-independent number. By the conservation Eq. (54.157), this is also the boundary value, so

$$G^R_{xy,xy}(\omega,\mathbf{0}) = i\omega\,\chi(0,\omega) = -\,i\omega\,K\,(L/z_h)^{d-1} + O(\omega^2), \tag{54.161}$$

and the Kubo formula Eq. (54.150) reads off the viscosity:

$$\eta = -\lim_{\omega\to 0}\frac{1}{\omega}\,\mathrm{Im}\,G^R_{xy,xy} = -\lim_{\omega\to 0}\frac{1}{\omega}\,\mathrm{Im}\Big[-i\omega K (L/z_h)^{d-1}\Big] = K\,(L/z_h)^{d-1}. \tag{54.162}$$

With $K = 1/16\pi G_N$ from Theorem 54.13,

$$\boxed{\eta = \frac{1}{16\pi G_N}\Big(\frac{L}{z_h}\Big)^{d-1} = \frac{a_h}{16\pi G_N},} \tag{54.163}$$

where $a_h = (L/z_h)^{d-1}$ is the *horizon area density* — the volume of the spatial horizon cross-section per unit boundary coordinate volume, $a_h = \sqrt{\det g_{ij}}\big|_{z_h}$ with $g_{ij} = (L/z_h)^2\delta_{ij}$ over the $d-1$ flat directions. The shear viscosity is the horizon area density divided by $16\pi G_N$: a purely horizon quantity, exactly as the membrane paradigm promised.

Historical Note

That a black-hole horizon behaves as a viscous membrane long predates AdS/CFT. In the 1970s and early 1980s Damour, and Price and Thorne, developed the *membrane paradigm*: an observer outside the horizon may replace it by a fictitious fluid membrane endowed with electrical resistivity, surface tension, and — relevantly here — a shear viscosity $\eta_{\rm mb} = 1/16\pi G_N$ per unit area and a negative bulk viscosity. The holographic computation Eq. (54.163) reproduces precisely this membrane shear viscosity, now reinterpreted as a transport coefficient of a genuine dual quantum field theory rather than a bookkeeping device. The dictionary turned a heuristic of classical relativity into a calculational tool for strongly coupled plasmas.

54.5.5 The universal ratio $\eta/s = 1/4\pi$

The viscosity Eq. (54.163) is proportional to the horizon area density — and so, by the area law Theorem 54.10 handed over from Section 54.4, is the entropy density. This is the entire reason the ratio is universal.

Theorem 54.14 (Universality of η/s)**.** *For any boundary theory dual to two-derivative (Einstein) gravity with a planar black-brane horizon, the ratio of shear viscosity to entropy density is*

$$\boxed{\frac{\eta}{s} = \frac{1}{4\pi} \qquad \Big(= \frac{\hbar}{4\pi k_B} \ \textit{with units restored} \Big).} \tag{54.164}$$

Proof. The entropy density is the horizon area density over $4G_N$ (Bekenstein–Hawking, Theorem 54.10),

$$s = \frac{a_h}{4G_N} = \frac{1}{4G_N}\Big(\frac{L}{z_h}\Big)^{d-1}. \tag{54.165}$$

Dividing Eq. (54.163) by Eq. (54.165), the area density $a_h = (L/z_h)^{d-1}$ and Newton's constant G_N cancel identically:

$$\frac{\eta}{s} = \frac{a_h/16\pi G_N}{a_h/4G_N} = \frac{4G_N}{16\pi G_N} = \frac{1}{4\pi}. \tag{54.166}$$

The cancellation is total: every feature of the particular geometry — the dimension d, the AdS radius L, the horizon location z_h (hence the temperature), and G_N itself — drops out. The result is a pure number, the same for every two-derivative gravity dual. Restoring units, the natural-unit $1/4\pi$ is $\hbar/4\pi k_B$: the only dimensionful combination that can appear in a viscosity-to-entropy-density ratio is $\hbar/k_B$, since η has dimensions of $\hbar$/volume$\times$ (rate) and s of k_B/volume, leaving $\eta/s \sim \hbar/k_B$. □

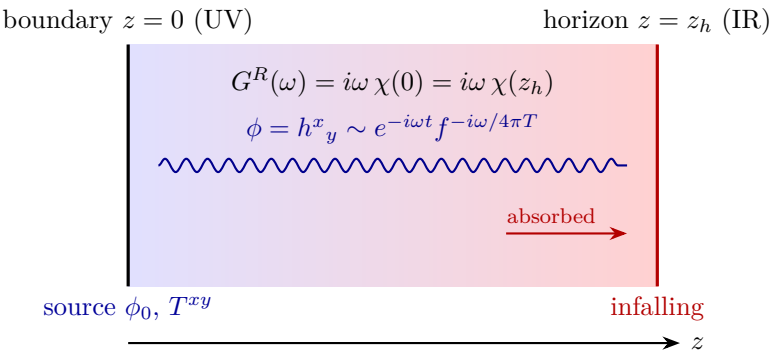


Figure 54.6: The Lorentzian setup. The shear graviton $\phi = h^x{}_y$, dual to the boundary stress component T^{xy}, is sourced at the conformal boundary $z = 0$ and obeys the infalling condition at the horizon $z = z_h$, where it is absorbed (causality, Eq. (54.145)). The response function χ of Eq. (54.155) is radially conserved as $\omega \to 0$ (Eq. (54.157)), so the retarded correlator read at the boundary equals a horizon quantity — the membrane-paradigm computation of η. The radial coordinate runs from the UV (boundary) to the IR (horizon), the interpolation that Section 54.4's successor promotes into the holographic renormalization group.

Physical Insight

Why is the ratio universal, when neither η nor s is? Because both are the *same* horizon area density, weighted by Newton's constant in the two characteristic ways gravity knows how: η counts the horizon as the source of dissipative flux ($1/16\pi G_N$, the graviton's kinetic normalization), and s counts it as the carrier of entropy ($1/4G_N$, Bekenstein–Hawking). The factor of 4π between them is the fixed ratio of these two ways of attaching $1/G_N$ to an area, and it is the same 4π that relates the surface gravity to the Hawking temperature. Holography turns the equality $\eta \propto s \propto$ (area) into a statement about a strongly coupled quantum fluid: the same horizon that carries the plasma's entropy sets the rate at which it dissipates momentum, and the proportionality is rigid.

The number for the $\mathcal{N} = 4$ plasma. The canonical realization is the thermal state of $\mathcal{N} = 4$ super-Yang–Mills with gauge group SU(N) at large N and large 't Hooft coupling $\lambda = g^2 N$, dual to the AdS_5 black brane ($d = 4$). Its free energy and entropy were computed in Section 54.4; its shear viscosity is now Eq. (54.162) with $d = 4$, and the ratio is $\eta/s = 1/4\pi$ exactly. Two things deserve emphasis. First, this is a number for a *strongly coupled* gauge theory: $\lambda \to \infty$ is the regime in which the gravity dual is classical, and it is exactly the regime that

the perturbative machinery of Yang–Mills theory cannot reach. The one-loop β-function and asymptotic freedom of Chapter 27 describe the weak-coupling ultraviolet, where η/s is instead *large* — a weakly coupled gas has long mean free paths, hence large viscosity, and kinetic theory gives $\eta/s \sim 1/(\lambda^2 \log \lambda^{-1}) \gg 1$ as $\lambda \to 0$. The holographic value $1/4\pi \approx 0.08$ is the opposite extreme, accessible by no resummation of the perturbative series. It is the cleanest illustration in this book of holography supplying a controlled answer in a regime where the Lagrangian-based methods of Parts VI and VII are powerless. Second, the same computation for $\mathcal{N} = 4$ shows that the leading correction at *finite* coupling raises the ratio: the first α'-correction to type IIB supergravity (a curvature-cubed R^4 term) gives

$$\frac{\eta}{s} = \frac{1}{4\pi}\left(1 + \frac{15\,\zeta(3)}{\lambda^{3/2}} + \cdots\right), \tag{54.167}$$

positive and decreasing toward $1/4\pi$ from above as $\lambda \to \infty$. The supergravity value $1/4\pi$ is thus approached from above along the strong-coupling axis, smoothly interpolating toward the large kinetic-theory value as the coupling is dialed down.

54.5.6 Quasinormal modes and hydrodynamic dispersion

The viscosity is the leading datum of an entire dynamical spectrum. The poles of the retarded correlator $G^R(\omega, \mathbf{k})$ in the complex ω-plane are the *quasinormal modes* (QNMs) of the black brane: frequencies at which the bulk supports a solution that is infalling at the horizon *and* normalizable at the boundary (no source, $\phi_0 = 0$), so that the response $G^R = [\Pi/\phi]_{z\to 0}$ diverges. They are the resonances of the dual plasma — the frequencies of its damped collective oscillations — and because the boundary condition is infalling (absorptive), they lie in the *lower* half ω-plane, $\mathrm{Im}\,\omega < 0$, encoding decay, consistent with G^R's analyticity in the upper half-plane.

The QNM spectrum splits into two families, shown in Fig. 54.7. A *gapped* tower sits at frequencies $|\omega| \sim 2\pi T$ with comparable real and imaginary parts; for the AdS_5 massless scalar these are the modes whose large-overtone asymptotics are $\omega_n/2\pi T \to n\,(\pm 1 - i)$, two rays descending at 45° into the lower half-plane. These set the relaxation time $\tau \sim 1/2\pi T$ on which the plasma returns to local equilibrium and have no analogue in hydrodynamics — they are the genuinely quantum, non-universal modes of the strongly coupled medium.

The physically decisive family is the *gapless*, hydrodynamic one: modes whose frequency vanishes as $\mathbf{k} \to 0$, $\omega(\mathbf{k}) \to 0$, because they are the long-wavelength relaxation of conserved densities, which cannot decay at zero wavenumber (conservation forbids it). Two such modes follow from $T^{\mu\nu}$ conservation. In the shear channel — the same T^{xy} channel that gave η, now at finite $\mathbf{k}$ — the conserved transverse momentum density obeys a diffusion equation, and the lowest QNM

is purely imaginary,

$$\omega = -\,i\,D\,k^2 + O(k^4), \qquad D = \frac{\eta}{\varepsilon + p}, \tag{54.168}$$

the momentum-diffusion (shear) mode, with D the diffusion constant and ε, p the energy density and pressure. In the sound channel the energy and longitudinal-momentum densities couple to give a propagating mode,

$$\omega = \pm\, c_s\, k - \frac{i}{2}\,\Gamma_s\, k^2 + O(k^3), \qquad c_s^2 = \frac{1}{d-1}, \quad \Gamma_s = \frac{2(d-2)}{d-1}\,\frac{\eta}{\varepsilon + p}, \tag{54.169}$$

the speed of sound $c_s^2 = \partial p/\partial\varepsilon = 1/(d-1)$ being fixed by conformal invariance (tracelessness $T^{\mu}{}_{\mu} = 0$ gives $\varepsilon = (d-1)p$), and the sound attenuation Γ_s again proportional to $\eta/(\varepsilon + p)$. That the holographic QNM dispersion relations take exactly the form predicted by relativistic hydrodynamics, with transport coefficients matching the Kubo result, is the statement that the long-distance, late-time dynamics of the strongly coupled plasma *is* hydrodynamics — a derivation of hydrodynamics from a microscopic (here, gravitational) description.

Worked Example 54.15 (The shear diffusion constant recovers $\eta/s = 1/4\pi$)**.** The diffusion pole Eq. (54.168) furnishes an independent route to the ratio, and a consistency check on Theorem 54.14. For a conformal plasma at zero chemical potential the Gibbs–Duhem relation gives $\varepsilon + p = Ts$ (the only extensive combination available when the sole scale is T and $\mu = 0$; equivalently, $p = -F/V$ and $\varepsilon = Ts - p$, of which the sum is Ts). Hence the shear diffusion constant is

$$D = \frac{\eta}{\varepsilon + p} = \frac{\eta}{Ts} = \frac{1}{T}\,\frac{\eta}{s}. \tag{54.170}$$

The diffusion constant of the AdS_5 black brane can also be computed directly, as the small-k limit of the shear-channel QNM: solving the finite-$\mathbf{k}$ analogue of Eq. (54.154) to the order that produces the pole Eq. (54.168) (the standard hydrodynamic-limit expansion of the bulk equation, infalling at the horizon and normalizable at the boundary) yields

$$D = \frac{1}{4\pi T}. \tag{54.171}$$

Equating Eq. (54.171) with Eq. (54.170) gives $\eta/s = T\,D = T \cdot \frac{1}{4\pi T} = \frac{1}{4\pi}$, in exact agreement with the horizon-flux computation Theorem 54.14. The two derivations probe the correlator differently — one its zero-frequency slope at $\mathbf{k} = 0$ (Kubo), the other its lowest pole at small $\mathbf{k}$ (hydrodynamics) — and their agreement is the internal check that the holographic plasma is a genuine relativistic fluid whose Kubo viscosity and hydrodynamic diffusion constant are the same physical η. The combination $D = 1/4\pi T$ also says the momentum-diffusion time over a thermal wavelength is the shortest the medium can sustain, the hallmark of a nearly perfect fluid.

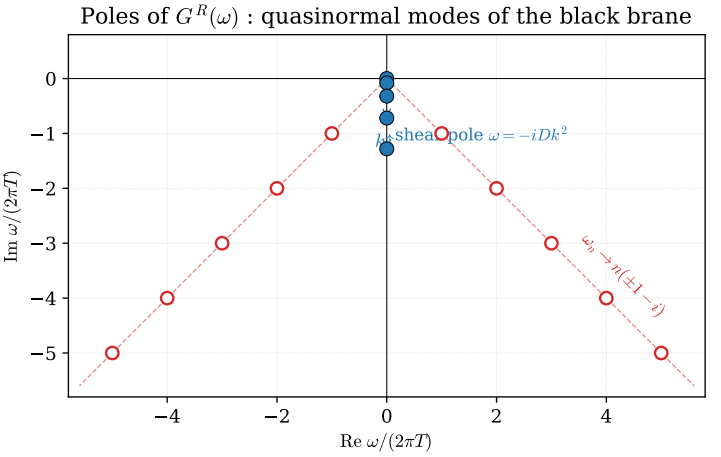


Figure 54.7: Poles of the retarded correlator $G^R(\omega)$ in the complex frequency plane (units $2\pi T$): the quasinormal-mode spectrum of the planar AdS_5 black brane. Open circles: the gapped non-hydrodynamic tower, with the exact large-overtone asymptotics $\omega_n \to n(\pm 1 - i)\, 2\pi T$ (dashed 45° rays). Filled circles: the hydrodynamic shear (momentum-diffusion) pole $\omega = -iDk^2$, which slides down the negative imaginary axis as the wavenumber k increases and pinches the origin as $k \to 0$ — the gapless mode whose small-k slope $D = \eta/(\varepsilon + p)$ encodes $\eta/s = 1/4\pi$.

54.5.7 The KSS bound: status, near-saturation, and violation

The value $\eta/s = 1/4\pi$ is strikingly small. Ordinary fluids have η/s orders of magnitude larger: water at room temperature sits near $400 \times (1/4\pi)$, helium near its superfluid transition — one of the lowest known laboratory values — still some $9 \times (1/4\pi)$. That every two-derivative gravity dual saturates the same small value, and that no ordinary substance comes near it, motivated Kovtun, Son, and Starinets to conjecture it as a universal lower bound.

Conjecture 54.16 (KSS bound)**.** *For all relativistic quantum field theories in flat space, in the regime where a hydrodynamic description applies, the shear-viscosity-to-entropy-density ratio satisfies*

$$\boxed{\frac{\eta}{s} \;\geq\; \frac{1}{4\pi} \qquad \Big(\frac{\hbar}{4\pi k_B} \text{ in restored units}\Big),} \tag{54.172}$$

with equality for theories dual to two-derivative Einstein gravity.

The physical heuristic for a *lower* bound is an uncertainty-principle estimate. In kinetic theory $\eta \sim \varepsilon\,\tau$ and $s \sim n$ (energy density times mean free time, particle density), so $\eta/s \sim \bar{\varepsilon}\,\tau \sim E\,\tau/\hbar$ in units — and a quasiparticle cannot have a relaxation time shorter than the inverse of its energy, $E\,\tau \gtrsim \hbar$, lest it not be a well-defined excitation at all. A fluid saturating the bound is one with *no* quasiparticles: the excitations are so strongly coupled that their lifetime is the minimum the uncertainty principle allows. This is exactly the holographic regime, where the absence of a sharp particle spectrum is the boundary face of the smooth bulk geometry.

Near-saturation by the quark–gluon plasma. The bound's empirical fame rests on a real fluid. The quark–gluon plasma produced in heavy-ion collisions at RHIC and the LHC, at temperatures $T \sim 0.2$–$0.5\,\mathrm{GeV}$ just above the QCD deconfinement transition, flows with an extraordinarily small η/s. Hydrodynamic fits to the observed anisotropic (elliptic) flow of the collision debris require η/s within a small multiple of $1/4\pi$ — the data, as understood in the mid-2020s, favour $\eta/s \approx (1\text{–}2.5) \times \frac{1}{4\pi}$, with the lower end near T_c. The quark–gluon plasma is thus the most nearly perfect fluid ever measured, and its near-saturation of the holographic value is the headline phenomenological success of AdS/CFT applied to QCD — all the more striking because QCD near T_c is strongly coupled and outside the reach of the perturbative β-function of Chapter 27, exactly the regime where $\mathcal{N} = 4$ SYM at strong coupling is the only tractable cousin. The match is qualitative-to-semiquantitative, not a theorem: QCD is neither conformal nor supersymmetric nor large-N, and the agreement says that the strong-coupling *value* of η/s is a more robust, more universal quantity than the details of the Lagrangian that produces it.

Violation by higher-derivative corrections. The honest status of Eq. (54.172) is that it is *not* a theorem, and it is now known to be violated. The universality theorem Theorem 54.14 relied on the shear graviton being a massless scalar with the Einstein-gravity normalization $1/16\pi G_N$ — a property of *two-derivative* gravity only. Finite-N and finite-coupling corrections in the dual gauge theory map to higher-derivative (curvature-squared and higher) terms in the bulk, which spoil the reduction Theorem 54.13: the graviton acquires a non-minimal coupling, and the cancellation that produced the pure number $1/4\pi$ no longer occurs. For Gauss–Bonnet gravity with coupling λ_{GB} (a curvature-squared theory whose linearized equations remain second order, hence ghost-free), the explicit computation gives

$$\frac{\eta}{s} = \frac{1}{4\pi}\,(1 - 4\lambda_{\mathrm{GB}} + \cdots), \tag{54.173}$$

which falls *below* $1/4\pi$ for positive λ_{GB}. The corresponding dual gauge theories (with $a \neq c$ central charges, i.e. away from the $\mathcal{N} = 4$ point) therefore violate the KSS bound. The violation is bounded: bulk causality and the absence of

superluminal graviton propagation constrain $\lambda_{\rm GB} \leq 9/100$ in five bulk dimensions, capping the violation at $\eta/s \geq \frac{1}{4\pi} \cdot \frac{16}{25}$. As understood in the mid-2020s, the situation is this: $1/4\pi$ is the exact value for Einstein gravity and is *approached from above* by the finite-coupling expansion Eq. (54.167) of any given theory like $\mathcal{N} = 4$ SYM; but across the space of *different* theories it is a strict lower bound only within two-derivative gravity, and is violated by higher-derivative duals, by a parametrically small amount controlled by causality. No sharp, universally valid lower bound on η/s is known; what survives is that $1/4\pi$ is a natural scale, that strongly coupled fluids cluster near it, and that ordinary fluids lie far above. The KSS bound is best stated as a near-saturated benchmark with known, controlled exceptions, not as a law.

Notation Watch

Throughout this section η denotes the shear viscosity (a transport coefficient), not the Minkowski metric $\eta_{\mu\nu}$ nor the conformal-anomaly central charge; the metric always carries indices and the viscosity never does. The ratio η/s is dimensionless in natural units; the restored form $\hbar/4\pi k_B$ makes the dependence on the two fundamental constants explicit and is the form in which the bound is compared to laboratory fluids.

The computation of this section has made one structural feature unavoidable, and it is the bridge to what follows. The retarded correlator was assembled from a single radial flow: a source planted at the boundary $z = 0$ and an infalling condition imposed at the horizon $z = z_h$, with the physics — the response function χ — flowing between them along the radial coordinate. The boundary end controlled the ultraviolet (the operator T^{xy} and its source), the horizon end the deep infrared (the thermal, dissipative, long-wavelength physics), and the radial direction interpolated between the two. This is no accident of the thermal problem: the same radial coordinate that here ran from UV operator to IR horizon is, in every holographic geometry, a geometric realization of energy scale. Section 54.4's successor promotes this observation into a precise statement — the holographic renormalization group — in which radial evolution of the bulk fields is the renormalization-group flow of the boundary couplings, closing the loop with the Wilsonian and Callan–Symanzik pictures of Part VI.

54.6 The Holographic Renormalization Group: the Radial Direction as Energy Scale

Every computation of this chapter has, on inspection, been a story told along the radial coordinate. In Section 54.5 the retarded correlator was assembled from a single radial flow — a source planted at the boundary $z = 0$, an infalling condition imposed at the horizon, and a response function carried between them;

the boundary end was the ultraviolet, the horizon end the deep infrared. In Theorem 54.5 the divergences of the on-shell action were power laws in the radial cutoff $z = \epsilon$, removed by local counterterms living on the cutoff slice — precisely the structure of an ultraviolet renormalization, with $z \to 0$ the deep UV because the warp factor $\sqrt{|g|} = (L/z)^{d+1}$ blows up there. Both facts pointed at one identification, which we now make precise and exploit: *the radial coordinate of an asymptotically-AdS geometry is the renormalization-group scale of the boundary theory, and radial evolution of the bulk fields is the RG flow of the boundary couplings.* This is the loop the chapter promised to close. We will build the geometry of a holographic RG flow (a domain wall interpolating between AdS asymptotics), recast the second-order bulk equations as first-order flow equations through a fake superpotential, define the holographic beta function and show it reproduces the operator dimensions of Section 53.4, prove a monotonicity theorem — the gravity dual of the a-theorem — from the bulk null energy condition, and finally identify the radial Hamilton–Jacobi equation with the Callan–Symanzik equation of Section 24.5, so that the Wilsonian shell-integration of Chapter 23 and the running couplings of Chapter 24 acquire a literal geometric meaning one dimension up.

Notation Watch

As declared in the notationwatch of Section 54.4.1 and used throughout Section 54.5, all *bulk geometry* in this section is written in the mostly-plus signature $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$ with the Wald curvature convention, in which AdS has negative scalar curvature — the deliberate local departure, adopted for the black-hole and real-time computations, from the book's mostly-minus convention (Section 53.3, Appendix A); the boundary *field-theory* statements (beta functions, dimensions, central charges) are signature-blind. The radial coordinate is $r \in (-\infty, +\infty)$ with the conformal boundary at $r \to +\infty$; it is related to the Fefferman–Graham coordinate z of Eq. (54.30) by $z \sim e^{-r/L}$ near a fixed point, so $r \to +\infty \Leftrightarrow z \to 0$ is the UV and decreasing r runs toward the IR.

54.6.1 The radial coordinate is the scale

The identification rests on a symmetry. The Poincaré metric of Eq. (53.59), written in the mostly-plus form

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\Big(\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu + \mathrm{d}z^2\Big), \qquad \eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+), \tag{54.174}$$

is invariant under the simultaneous rescaling

$$x^\mu \longmapsto \lambda\, x^\mu, \qquad z \longmapsto \lambda\, z, \qquad \lambda > 0, \tag{54.175}$$

because each factor of $\mathrm{d}x$ or $\mathrm{d}z$ scales as λ while the prefactor L^2/z^2 scales as λ^{-2}, leaving $\mathrm{d}s^2$ unchanged. On the boundary, $x^\mu \mapsto \lambda x^\mu$ is precisely a dilatation: a change of length unit, equivalently a change of energy scale by $\mu \mapsto \mu/\lambda$. The bulk realizes that dilatation as an isometry that also moves the radial coordinate, $z \mapsto \lambda z$. A bulk excitation localized near $z \sim 1/E$ in the radial direction therefore transforms under Eq. (54.175) as a boundary disturbance of energy E: small z (near the boundary) is large E (the ultraviolet), and large z (deep in the interior) is small E (the infrared). The dictionary is the quantitative statement

$$\mu \;\sim\; \frac{1}{z} \;=\; \frac{1}{L}\, e^{A(r)}, \qquad \ln \mu \;=\; A(r) + \text{const}, \tag{54.176}$$

where in the second form we have anticipated the general warped geometry below, writing the scale factor as $e^{A(r)}$ so that the dimensionless logarithm of the renormalization scale is the warp function A itself. The renormalization-group "time" $t = \ln \mu$ of Chapters 23 and 24 is, on the gravity side, literally the warp factor of the metric. This is no longer a heuristic: a bulk field's radial profile is a function of A, hence a function of $\ln \mu$, and its radial equation of motion will be an evolution equation in $\ln \mu$ — a renormalization-group equation. The remainder of the section turns this sentence into theorems.

54.6.2 Domain walls and the geometry of an RG flow

Pure AdS is dual to a conformal field theory: scale invariance Eq. (54.175) is an exact isometry, and nothing runs. To describe a theory that *does* run — a UV CFT deformed by a relevant operator, flowing to a distinct IR CFT — we need a geometry that looks like AdS near the boundary, looks like a (generally different) AdS deep in the interior, and interpolates smoothly in between, with the scale isometry broken along the way. The minimal bulk model that produces such a geometry is Einstein gravity coupled to a single scalar with a nontrivial potential. We take the bulk action in $D = d + 1$ dimensions,

$$S = \int \mathrm{d}^{d+1}x\, \sqrt{-g}\left[\frac{1}{2\kappa^2}\, R - \frac{1}{2}\, g^{MN}\partial_M\phi\, \partial_N\phi - V(\phi)\right], \qquad 2\kappa^2 = 16\pi G_N, \tag{54.177}$$

where ϕ is dual, by the field–operator map of Section 53.4, to a scalar boundary operator $\mathcal{O}$ whose dimension we will read off from V. The scalar's potential plays two roles at once: its value sets the local cosmological constant (hence the local AdS radius), and its shape away from the extrema drives the flow. We require V to possess at least two critical points $\phi_\star$ with $V'(\phi_\star) = 0$ and $V(\phi_\star) < 0$; each is a candidate AdS vacuum.

The geometry that realizes the flow is the *domain wall*: a metric homogeneous and Poincaré-invariant along the boundary directions, with all dependence in the radial coordinate,

$$\mathrm{d}s^2 = \mathrm{d}r^2 + e^{2A(r)}\, \eta_{\mu\nu}\, \mathrm{d}x^\mu \mathrm{d}x^\nu, \qquad \phi = \phi(r), \tag{54.178}$$

with $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$ the flat boundary metric and $A(r)$ the warp function. The ansatz preserves d-dimensional Poincaré invariance (translations and Lorentz rotations of x^μ) but breaks the dilatation Eq. (54.175) unless A is linear in r; a position-dependent $A'(r)$ is exactly broken scale invariance, i.e. a running coupling. A pure-AdS region is recovered wherever ϕ sits at a critical point $\phi_\star$ and A is linear, $A(r) = r/L_\star$, with the radius $L_\star$ fixed by $V(\phi_\star)$, as we now verify.

Definition 54.17 (Holographic RG flow)**.** A *holographic renormalization-group flow* is a solution $(A(r), \phi(r))$ of the equations of motion of Eq. (54.177) in the domain-wall form Eq. (54.178) for which $\phi(r)$ interpolates between two critical points of V: as $r \to +\infty$ (boundary, UV) the geometry approaches AdS of radius L_{UV} with $\phi \to \phi_{\mathrm{UV}}$, and as $r \to -\infty$ (interior, IR) it approaches AdS of radius L_{IR} with $\phi \to \phi_{\mathrm{IR}}$. The boundary CFT is the UV fixed point deformed by the operator dual to ϕ; the IR fixed point is the endpoint of the flow.

The reduced equations of motion. On the ansatz Eq. (54.178), the field equations of Eq. (54.177) collapse to ordinary differential equations in r. We need them explicitly, since the entire RG dictionary will be read off their structure. The Ricci tensor of Eq. (54.178) has the standard domain-wall components (Wald convention)

$$R_{rr} = -d(A''+A'^2), \qquad R_{\mu\nu} = -e^{2A}\eta_{\mu\nu}(A''+dA'^2), \qquad R = -2dA''-d(d+1)A'^2, \tag{54.179}$$

in which a prime denotes $\mathrm{d}/\mathrm{d}r$; for the AdS case $A = r/L$ (so $A'' = 0$, $A' = 1/L$) these give $R = -d(d+1)/L^2 < 0$, the negative curvature of mostly-plus AdS. The scalar equation $\nabla^2\phi = V'(\phi)$, with $\sqrt{-g} = e^{dA}$ and $g^{rr} = 1$, is

$$\phi'' + d\,A'\,\phi' = V'(\phi), \tag{54.180}$$

the radial wave equation with a friction term $dA'\phi'$ — formally the equation of a particle of position ϕ moving in the inverted potential $-V$ with Hubble-like damping dA', the radial coordinate playing the role of time. The Einstein equations $R_{MN} - \frac{1}{2}g_{MN}R = \kappa^2 T_{MN}$, with stress tensor $T_{MN} = \partial_M\phi\,\partial_N\phi - g_{MN}\big(\frac{1}{2}(\partial\phi)^2 + V\big)$, reduce to two independent components. The rr component gives the *Hamiltonian constraint*

$$\frac{d(d-1)}{2}\,A'^2 = \kappa^2\Big(\tfrac{1}{2}\,\phi'^2 - V\Big), \tag{54.181}$$

a first integral with no second derivatives — the bulk version of energy conservation along the radial flow. The $\mu\nu$ components give

$$(d-1)A'' + \frac{d(d-1)}{2}A'^2 = -\kappa^2\Big(\tfrac{1}{2}\,\phi'^2 + V\Big). \tag{54.182}$$

Subtracting Eq. (54.181) from Eq. (54.182) cancels both the A'^2 and the V terms and leaves a single, remarkably simple relation:

$$\boxed{A''(r) = -\frac{\kappa^2}{d-1}\,\phi'(r)^2 \;\le\; 0.} \tag{54.183}$$

This one line is the technical seed of everything that follows. It says the warp function is concave: $A'' \le 0$, with equality only where $\phi' = 0$, i.e. at a fixed point. We will see in Section 54.6.5 that Eq. (54.183) is exactly the bulk null energy condition, and that its sign is the sign of the a-theorem.

54.6.3 First-order flow equations from a fake superpotential

The system Eqs. (54.180), (54.181) and (54.183) is second order. Its solutions of interest — the smooth interpolating flows — can be obtained from a *first-order* system by the superpotential trick, familiar from supersymmetric domain walls but valid here as a purely classical device, for which reason the function W below is called a *fake* superpotential: it need not descend from any supersymmetry. The construction trades the second-order equations for a pair of first-order ones at the cost of solving an algebraic-differential consistency condition for W.

Theorem 54.18 (First-order flow equations)**.** *Suppose a function $W(\phi)$ — the fake superpotential — satisfies the consistency condition*

$$V(\phi) = \frac{1}{2}\big(W'(\phi)\big)^2 - \frac{d\,\kappa^2}{2(d-1)}\,W(\phi)^2, \tag{54.184}$$

where $W' \equiv \mathrm{d}W/\mathrm{d}\phi$. Then any solution of the first-order system

$$\boxed{\phi'(r) = W'(\phi), \qquad A'(r) = -\frac{\kappa^2}{d-1}\,W(\phi)} \tag{54.185}$$

solves the full second-order equations Eqs. (54.180), (54.181) and (54.183). Conversely, every domain-wall solution arises this way for an appropriate W.

Proof. We verify that Eq. (54.185) together with Eq. (54.184) implies each of the three reduced equations; the algebra is elementary and we show every step.

The concavity equation Eq. (54.183). Differentiate the second of Eq. (54.185): $A'' = -\frac{\kappa^2}{d-1}\,\frac{dW}{dr} = -\frac{\kappa^2}{d-1}\,W'\,\phi' = -\frac{\kappa^2}{d-1}\,W'\cdot W' = -\frac{\kappa^2}{d-1}(W')^2$, using $\phi' = W'$ from the first equation. Since $\phi' = W'$, this is exactly $A'' = -\frac{\kappa^2}{d-1}\phi'^2$, which is Eq. (54.183).

The Hamiltonian constraint Eq. (54.181). Substitute $A' = -\frac{\kappa^2}{d-1}W$ and $\phi' = W'$ into the left and right sides. The left side is $\frac{d(d-1)}{2}A'^2 = \frac{d(d-1)}{2}\cdot\frac{\kappa^4}{(d-1)^2}W^2 = \frac{d\,\kappa^4}{2(d-1)}W^2$. The right side is $\kappa^2\big(\frac{1}{2}W'^2 - V\big)$. Equality holds iff

$\frac{1}{2}W'^2 - V = \frac{d\kappa^2}{2(d-1)}W^2$, which is precisely the consistency condition Eq. (54.184) solved for V. So Eq. (54.181) holds by construction.

The scalar equation Eq. (54.180). The left side is $\phi'' + dA'\phi'$. Now $\phi'' = \frac{d}{dr}W'(\phi) = W''\phi' = W''W'$, and $dA'\phi' = d\cdot(-\frac{\kappa^2}{d-1}W)\cdot W' = -\frac{d\kappa^2}{d-1}WW'$, so the left side is $W''W' - \frac{d\kappa^2}{d-1}WW'$. The right side is V', which we compute by differentiating Eq. (54.184): $V' = W'W'' - \frac{d\kappa^2}{d-1}WW'$. The two sides coincide. Hence the scalar equation is automatically satisfied — it is the derivative of the Hamiltonian constraint, no new information.

The converse holds because the Hamiltonian constraint Eq. (54.181) can be solved pointwise for the quantity $-\frac{d-1}{\kappa^2}A'$ as a function of ϕ along any given monotonic flow, and that function, call it $W(\phi)$, then satisfies Eq. (54.184) and reproduces Eq. (54.185) by construction. □

The first-order equations Eq. (54.185) are the central computational object of holographic RG. They display the flow as a gradient flow: ϕ rolls along W', and the warp function is slaved to W. The radial coordinate is a monotonic clock because $\phi' = W'$ has fixed sign between critical points (where $W' = 0$), so ϕ moves monotonically from one extremum to the next, and by Eq. (54.183) A' decreases monotonically as well: the geometry never "turns around."

Fixed points are AdS vacua. At a critical point $\phi_\star$ of W ($W'(\phi_\star) = 0$) the first equation gives $\phi' = 0$, the second gives $A' = -\frac{\kappa^2}{d-1}W(\phi_\star) = \text{const}$, so $A = A'_\star r$ and the metric Eq. (54.178) is exactly AdS. Differentiating Eq. (54.184) shows $W'(\phi_\star) = 0 \Rightarrow V'(\phi_\star) = 0$: a critical point of the superpotential is a critical point of the potential, as it must be for the scalar to sit still. Taking $W(\phi_\star) < 0$ so that $A'_\star > 0$ (the warp factor grows toward the boundary), the AdS radius and cosmological constant are

$$\frac{1}{L_\star} = A'_\star = -\frac{\kappa^2}{d-1}\,W(\phi_\star), \qquad V(\phi_\star) = -\frac{d\,\kappa^2}{2(d-1)}\,W(\phi_\star)^2 = -\frac{d(d-1)}{2\kappa^2\,L_\star^2}, \tag{54.186}$$

the last equality being the standard relation $V_\star = \Lambda_\star/\kappa^2$ with $\Lambda_\star = -d(d-1)/2L_\star^2$ the cosmological constant of AdS_{d+1}, which one also reads directly off the Hamiltonian constraint Eq. (54.181) at $\phi' = 0$. Thus each extremum of V furnishes an AdS vacuum, hence a fixed-point CFT, and a flow is a bulk trajectory connecting two of them.

54.6.4 The holographic beta function and the operator dimensions

We can now define the running coupling and its beta function entirely from the geometry. The boundary coupling conjugate to $\mathcal{O}$ is the value of the scalar

(more precisely the coefficient of its leading near-boundary mode, Eq. (54.30)); to leading order the running coupling *is* ϕ as a function of scale. With the scale identified through Eq. (54.176) as $\ln \mu = A(r)$, the holographic beta function is the rate of change of ϕ per unit RG time, exactly the definition of Eq. (24.5) transplanted to the bulk:

$$\boxed{\beta(\phi) \;\equiv\; \frac{d\phi}{d\ln\mu} \;=\; \frac{\mathrm{d}\phi/\mathrm{d}r}{\mathrm{d}A/\mathrm{d}r} \;=\; \frac{\phi'}{A'} \;=\; -\frac{d-1}{\kappa^2}\,\frac{W'(\phi)}{W(\phi)}.} \tag{54.187}$$

The last equality used the flow equations Eq. (54.185). The structure is transparent: β vanishes wherever $W' = 0$, that is, at the AdS critical points where the scalar sits at an extremum of the potential. These are the zeros of the beta function — the fixed points of the RG flow — and they are the boundary CFTs. Between them β is nonzero and the coupling runs. The map "boundary = UV, deep interior = IR" of Section 54.6.1, anticipated qualitatively in Section 54.5 by the boundary/horizon split, is now the quantitative statement that the flow of ϕ from ϕ_{UV} (large r) to ϕ_{IR} (small r) is the RG trajectory of the boundary coupling from the ultraviolet fixed point to the infrared one.

Linearizing at a fixed point: dimensions from the slope. The physics of the fixed point is encoded in the slope of β there, just as in Chapter 24. Expand $\beta(\phi) \simeq \beta'(\phi_\star)(\phi - \phi_\star)$ near a zero. Differentiating Eq. (54.187) and using $W'(\phi_\star) = 0$,

$$\beta'(\phi_\star) = -\frac{d-1}{\kappa^2}\,\frac{W''(\phi_\star)}{W(\phi_\star)} = L_\star\, W''(\phi_\star), \tag{54.188}$$

where the second form used $-\frac{d-1}{\kappa^2 W(\phi_\star)} = L_\star$ from Eq. (54.186). The claim is that this slope is precisely the deviation of the operator dimension from marginality,

$$\beta'(\phi_\star) = \Delta - d, \tag{54.189}$$

exactly as a coupling for an operator of dimension Δ has a (dimensionless) beta function $\beta = (\Delta - d)\,(\text{coupling}) + \cdots$ in Chapters 23 and 24: relevant $\Delta < d$ gives $\beta' < 0$ (the coupling grows toward the IR), irrelevant $\Delta > d$ gives $\beta' > 0$ (the coupling dies in the IR), and marginal $\Delta = d$ gives $\beta' = 0$. To prove Eq. (54.189) and tie it to the mass–dimension relation of Theorem 53.17, compute the scalar mass at the fixed point, $m^2 = V''(\phi_\star)$, from the consistency condition. Differentiating Eq. (54.184) twice,

$$V'' = (W'')^2 + W'W''' - \frac{d\kappa^2}{d-1}(W'^2 + WW''), \tag{54.190}$$

and evaluating at $\phi_\star$ where $W' = 0$,

$$m^2 = V''(\phi_\star) = \big(W''(\phi_\star)\big)^2 - \frac{d\kappa^2}{d-1}\,W(\phi_\star)\,W''(\phi_\star) = (W''_\star)^2 + \frac{d}{L_\star}\,W''_\star, \tag{54.191}$$

the last step using $\frac{d\kappa^2}{d-1}W_\star = -d/L_\star$ from Eq. (54.186). Now write $W_\star'' = \beta'(\phi_\star)/L_\star$ from Eq. (54.188) and multiply through by $L_\star^2$:

$$m^2L_\star^2 = \big(\beta'(\phi_\star)\big)^2 + d\,\beta'(\phi_\star) = \beta'(\phi_\star)\big(\beta'(\phi_\star)+d\big). \tag{54.192}$$

This is identical to the mass–dimension relation $\Delta(\Delta-d) = m^2L^2$ of Eq. (53.112) under the substitution $\beta'(\phi_\star) = \Delta - d$, since then $\beta'(\beta'+d) = (\Delta-d)\Delta = \Delta(\Delta-d)$. The beta-function slope read off the bulk flow reproduces, on the nose, the conformal dimension assigned by the mass–dimension relation to the dual operator. The two computations — one a near-boundary falloff analysis (Section 53.4.3), the other a linearization of the RG flow — agree because they are the same boundary-value problem viewed from two ends.

A relevant deformation triggers the flow. The leading near-boundary behavior of the scalar follows directly from the linearized first-order equation $\phi' = W' \simeq W_\star''(\phi-\phi_\star)$, whose solution is $\phi-\phi_\star \propto e^{W_\star'' r} = e^{(\Delta-d)r/L_\star}$. Translating to the Fefferman–Graham coordinate $z \sim e^{-r/L_\star}$, this is $\phi - \phi_\star \propto z^{\,d-\Delta}$ — precisely the *source* (non-normalizable) mode of the near-boundary expansion Eq. (54.30), the leading falloff that turns on the deformation. The first-order flow equations capture exactly this source-driven part of the solution; the subleading *response* mode z^{Δ}, the vacuum expectation value $\langle\mathcal{O}\rangle$, is the one integration constant the first-order system leaves free, and it is fixed not by the boundary data but by the requirement of regularity in the deep interior (smoothness at the IR fixed point or, at finite temperature, the horizon regularity of Section 54.5). For a *relevant* operator, $\Delta < d$, the source mode $z^{d-\Delta}$ grows as z increases (toward the interior): a deformation that is arbitrarily small in the deep UV swells as one flows to lower scales and inevitably drives the geometry away from the UV AdS region. This is the geometric statement of the foundational fact of Theorem 23.23: relevant operators are the ones that matter in the infrared, the ones that trigger nontrivial flows; irrelevant operators ($\Delta > d$, mode $z^{d-\Delta}$ decaying) die away and leave the IR fixed point untouched. The bulk reproduces the relevant/marginal/irrelevant trichotomy as the growth/constancy/decay of a radial profile.

54.6.5 The holographic c-function and the a-theorem

The deepest structural fact about RG flows is that they are irreversible: a quantity counting the effective number of degrees of freedom decreases monotonically from the UV to the IR. In two dimensions this is Zamolodchikov's c-theorem; in four it is the a-theorem proved field-theoretically in Theorem 52.19 [301]. Holography supplies a one-line proof of the monotonicity, valid in any dimension, with the irreversibility traced directly to a positive-energy condition in the bulk.

Definition 54.19 (Holographic c-function)**.** Along a domain-wall flow Eq. (54.178) with $A'(r) > 0$, define

$$c(r) \;\equiv\; \frac{C_d}{\big(A'(r)\big)^{d-1}}, \qquad C_d = \frac{(d-1)^{d-1}\,\pi^{d/2}}{\Gamma\!\left(\frac{d}{2}\right)\kappa^2} \;\; (>0), \tag{54.193}$$

a positive constant $C_d \propto 1/G_N$ times the inverse $(d-1)$th power of the warp slope. At a fixed point, where $A' = 1/L_\star$, this evaluates to $c = C_d\, L_\star^{d-1} \propto L_\star^{d-1}/G_N$, the holographic central charge: the same combination $L^{d-1}/G_N \sim N^2$ that the parameter map of Section 53.4 identified as the number of boundary degrees of freedom, and, for even d, the coefficient of the Weyl anomaly — the a-charge for $d = 4$.

The normalization constant C_d is chosen so that c at a fixed point equals the standard CFT central charge; its precise value is immaterial to the monotonicity argument, which needs only $C_d > 0$, but we record it so that the endpoint values are the genuine anomaly coefficients that Theorem 52.19 constrains.

Theorem 54.20 (Holographic c-theorem)**.** *Let $(A(r), \phi(r))$ be a domain-wall solution of Eq. (54.177) whose matter satisfies the bulk* null energy condition *$T_{MN}\xi^M\xi^N \geq 0$ for every null vector ξ^M. Then the holographic c-function Eq. (54.193) is monotonically non-decreasing toward the boundary,*

$$c'(r) \;\geq\; 0, \tag{54.194}$$

with equality only where the flow is stationary ($\phi' = 0$). Consequently $c_{\mathrm{UV}} \geq c_{\mathrm{IR}}$: the central charge of the ultraviolet fixed point is never smaller than that of the infrared fixed point. This is the gravity dual of the a-theorem Theorem 52.19.

Proof. We first show that the null energy condition is exactly the concavity Eq. (54.183), then differentiate $c(r)$.

Null energy condition $\Rightarrow A'' \leq 0$. Choose the null vector ξ^M with a radial component and a single boundary time-like component, $\xi^r = 1$, $\xi^0 = e^{-A}$, and all other components zero. It is null: using $g_{rr} = 1$ and $g_{00} = e^{2A}\eta_{00} = -e^{2A}$,

$$g_{MN}\xi^M\xi^N = g_{rr}(\xi^r)^2 + g_{00}(\xi^0)^2 = 1 + (-e^{2A})\,e^{-2A} = 1 - 1 = 0. \tag{54.195}$$

Contract the Einstein equation $R_{MN} - \frac{1}{2}g_{MN}R = \kappa^2 T_{MN}$ with $\xi^M\xi^N$. Because ξ is null, the $\frac{1}{2}g_{MN}R$ term drops ($g_{MN}\xi^M\xi^N = 0$), leaving $R_{MN}\xi^M\xi^N = \kappa^2 T_{MN}\xi^M\xi^N$. The left side is computed from Eq. (54.179): $R_{rr}(\xi^r)^2 + R_{00}(\xi^0)^2 = -d(A'' + A'^2) + \big[-e^{2A}\eta_{00}(A'' + dA'^2)\big]e^{-2A}$. With $\eta_{00} = -1$ the second term is $+(A'' + dA'^2)$, so the sum is $-d(A'' + A'^2) + (A'' + dA'^2) = -(d-1)A''$. Therefore

$$-(d-1)\,A'' = \kappa^2\, T_{MN}\xi^M\xi^N \;\geq\; 0 \quad\Longrightarrow\quad A'' \leq 0, \tag{54.196}$$

the null energy condition delivering the concavity directly. For the scalar matter of Eq. (54.177) one checks $T_{MN}\xi^M\xi^N = \phi'^2 \geq 0$ explicitly — recovering Eq. (54.183) — but Eq. (54.196) shows $A'' \leq 0$ holds for *any* matter obeying the null energy condition, which is why the theorem is insensitive to the details of the bulk field content.

Monotonicity. Differentiate Eq. (54.193):

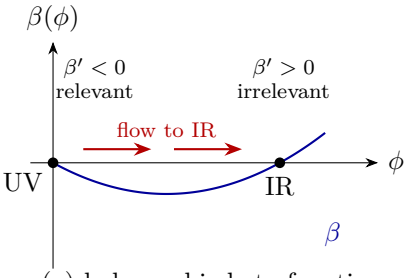
$$c'(r) = -(d-1)\,C_d\,\frac{A''(r)}{\big(A'(r)\big)^d} = \frac{(d-1)\,C_d}{\big(A'\big)^d}\cdot\big(-A''\big). \tag{54.197}$$

With $C_d > 0$, $A' > 0$ (warp growing toward the boundary), and $-A'' \geq 0$ by Eq. (54.196), every factor is non-negative: $c'(r) \geq 0$, with equality iff $A'' = 0$ iff $T_{MN}\xi^M\xi^N = 0$ iff $\phi' = 0$, a stationary point. Integrating from the IR ($r \to -\infty$) to the UV ($r \to +\infty$), c never decreases, so $c_{\rm UV} = c(+\infty) \geq c(-\infty) = c_{\rm IR}$. At the endpoints c equals the fixed-point central charges by Theorem 54.19, giving $c_{\rm UV} \geq c_{\rm IR}$. □

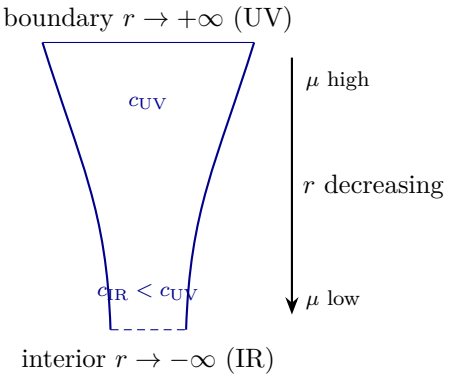
Physical Insight

The irreversibility of the renormalization group — a deep, scheme-independent statement about quantum field theory — is, holographically, the statement that a horizon-free gravitational geometry sourced by physical (positive-energy) matter cannot make its warp factor accelerate outward: $A'' \leq 0$. Degrees of freedom are lost along the flow because the bulk geometry funnels inward, the AdS throat narrowing from the wide UV boundary to the narrower IR. The same null energy condition that forbids traversable wormholes and underlies the area theorems of black-hole mechanics is, viewed through the dictionary, the engine of the a-theorem. A monotonicity that took decades and a subtle dilaton-effective-action argument to establish field-theoretically (Theorem 52.19) becomes, at strong coupling, a single contraction of the Einstein equations with a null vector.

Synthesis with the benchmark theories. The geometry of this section is the strong-coupling face of structures the book built perturbatively. The fixed-point/flow picture is the same one ϕ^4 theory traced from the Wilsonian shell-integration of Section 23.1.2 to the Wilson–Fisher fixed point of Theorem 23.36: a UV fixed point (there free-field Gaussian, here the UV AdS), a relevant deformation that triggers a flow (there the ϕ^2 mass operator, here the scalar's source mode $z^{d-\Delta}$ with $\Delta < d$), and an IR fixed point with fewer degrees of freedom. The relevant/marginal/irrelevant classification of Theorem 23.23 reappears intact as the sign of the beta-function slope Eq. (54.189), now derived from a bulk falloff rather than an engineering dimension. Yang–Mills theory contributes the complementary lesson: the asymptotically free running of Eq. (27.1)



(a) holographic beta function



(b) the RG throat, $A'' \leq 0$

Figure 54.8: The holographic renormalization group. (a) The beta function Eq. (54.187) of a flow with two fixed points: a UV fixed point at $\phi = 0$ with negative slope $\beta' = \Delta_{\mathrm{UV}} - d < 0$ (a relevant operator, Eq. (54.189)) that destabilizes it, and an IR fixed point where the same operator has become irrelevant ($\beta' > 0$). The radial coordinate carries the coupling from UV to IR. (b) The dual geometry: a domain wall Eq. (54.178) whose warp factor is concave, $A'' \leq 0$ Eq. (54.183), by the bulk null energy condition. The throat narrows from the wide UV boundary to the narrow IR interior, and the c-function Eq. (54.193) decreases monotonically along the way (Theorem 54.20) — the gravity dual of the a-theorem.

is a flow whose UV fixed point is the trivial (free) theory and which has *no* weakly coupled IR fixed point — the coupling grows without bound, confinement intervening. The holographic flows here are its strong-coupling counterparts, in which the IR is a genuine interacting CFT (a nontrivial AdS) reachable only because the gravity description resums the coupling. In both threads the abstract content — a beta function with zeros, a monotone counting function decreasing toward the IR — is identical; holography supplies the picture in which that content is visible as the literal shape of a higher-dimensional spacetime.

54.6.6 Wilsonian holography: the radial cutoff as a coarse-graining scale

The picture so far identifies radial *position* with scale and radial *evolution* with running. The Wilsonian viewpoint of Chapter 23 demands more: that integrating out the high-energy degrees of freedom down to a scale μ_c produce a definite effective action $S_{\mathrm{eff}}[\mu_c]$, and that lowering μ_c generate a flow equation for it. Holography realizes this literally. Cut the bulk at a finite radius $r = r_c$; integrate the bulk path integral over the region $r < r_c$ (from the deep interior or horizon out to the slice); the result is a functional of the field data on the slice r_c, and it *is* the

Wilsonian effective action of the boundary theory at the cutoff scale $\mu_c \sim e^{A(r_c)}$. The bulk region $r < r_c$ that has been integrated out is the holographic image of the modes with energy above μ_c; the slice carries the effective theory below it. We now make this precise and show that the radial evolution of the on-shell action is governed by a Hamilton–Jacobi equation that is the Callan–Symanzik equation of Theorem 24.19.

Radial canonical structure. Treat the radial coordinate r as "time" and foliate the bulk by constant-r slices, each carrying the induced metric $\gamma_{\mu\nu} = e^{2A}\eta_{\mu\nu}$ and the scalar ϕ. The extrinsic curvature of a slice measures how it bends as r advances,

$$K_{\mu\nu} = \tfrac{1}{2}\,\partial_r \gamma_{\mu\nu} = A'\,\gamma_{\mu\nu}, \qquad K \equiv \gamma^{\mu\nu}K_{\mu\nu} = d\,A', \tag{54.198}$$

so the warp slope A' is (up to the factor d) the trace of the extrinsic curvature. The momenta canonically conjugate to $\gamma_{\mu\nu}$ and ϕ in this radial Hamiltonian formulation are, from the gravitational and scalar parts of Eq. (54.177) respectively,

$$\pi^{\mu\nu} = \frac{1}{2\kappa^2}\sqrt{\gamma}\,(K^{\mu\nu} - K\gamma^{\mu\nu}), \qquad \pi_\phi = \sqrt{\gamma}\,\phi', \tag{54.199}$$

the standard ADM gravitational momentum and the radial momentum of the scalar. The on-shell action $S_{\rm cl}[\gamma,\phi]$ as a functional of the slice data is the Hamilton–Jacobi principal function, and the momenta are its functional gradients,

$$\pi^{\mu\nu} = \frac{\delta S_{\rm cl}}{\delta \gamma_{\mu\nu}}, \qquad \pi_\phi = \frac{\delta S_{\rm cl}}{\delta \phi}. \tag{54.200}$$

The Wilsonian action is the superpotential. For a flat, homogeneous slice the leading local part of the on-shell action is a spacetime integral of a local density,

$$S_{\rm cl} = \int \mathrm{d}^d x\,\sqrt{\gamma}\,W(\phi) + (\text{derivative terms}), \tag{54.201}$$

and we now show that this W is exactly the fake superpotential of Section 54.6.3, so that the Wilsonian effective potential at the cutoff slice and the function controlling the flow are one and the same object. Insert Eq. (54.201) into the Hamilton–Jacobi relations Eq. (54.200). For the scalar, using $\delta\sqrt{\gamma}/\delta\phi = 0$, $\delta S_{\rm cl}/\delta\phi = \sqrt{\gamma}\,W'(\phi)$, so Eq. (54.199) gives $\sqrt{\gamma}\,\phi' = \sqrt{\gamma}\,W'$, i.e.

$$\phi' = W'(\phi), \tag{54.202}$$

the first flow equation. For the metric, $\delta(\sqrt{\gamma})/\delta\gamma_{\mu\nu} = \tfrac{1}{2}\sqrt{\gamma}\,\gamma^{\mu\nu}$, so $\pi^{\mu\nu} = \tfrac{1}{2}\sqrt{\gamma}\,\gamma^{\mu\nu}W$; taking the trace, $\pi \equiv \gamma_{\mu\nu}\pi^{\mu\nu} = \tfrac{d}{2}\sqrt{\gamma}\,W$. The trace of the ADM

momentum Eq. (54.199) is $\pi = \frac{1}{2\kappa^2}\sqrt{\gamma}(K - dK) = -\frac{(d-1)}{2\kappa^2}\sqrt{\gamma}\,K = -\frac{d(d-1)}{2\kappa^2}\sqrt{\gamma}\,A'$, using $K = dA'$. Equating the two expressions for π and cancelling $\frac{d}{2}\sqrt{\gamma}$,

$$A' = -\frac{\kappa^2}{d-1}\,W(\phi), \tag{54.203}$$

the second flow equation. Thus the Hamilton–Jacobi relations Eq. (54.200) for the local ansatz Eq. (54.201) reproduce the first-order flow Eq. (54.185) exactly, and substituting them back into the radial Hamiltonian constraint (the bulk version of $H = 0$, which on the flat homogeneous slice is the constraint Eq. (54.181)) returns the consistency condition Eq. (54.184) as the *Hamilton–Jacobi equation for* W. The fake superpotential is the on-shell radial action density: the local part of the Wilsonian effective action at the cutoff slice.

Hamilton–Jacobi is Callan–Symanzik. The content of Eqs. (54.202) and (54.203) is that the radial derivative of the on-shell action is fixed by its slice data — a first-order evolution in the radial "time" $r = L\ln\mu$. Splitting the on-shell action into its local, divergent part (the counterterms of Theorem 54.5, which are exactly the terms Eq. (54.201)) and a finite remainder Γ identified through the GKP–Witten relation Eq. (53.162) with the boundary generating functional, the Hamilton–Jacobi equation organizes into an inhomogeneous first-order equation for Γ in $\ln\mu$, with the beta function Eq. (54.187) multiplying $\partial_\phi\Gamma$ and an anomalous-dimension term from the rescaling of $\sqrt{\gamma}$. This is structurally identical to the Callan–Symanzik equation Eq. (24.172) of Theorem 24.19: a scale derivative of a generating functional, plus the beta function times the coupling derivative, plus anomalous-dimension terms, equal to a (local, anomaly) inhomogeneity. The Wilsonian shell-integration of Theorems 23.13 and 23.15, which lowered a momentum cutoff and generated a flow of the effective action, is mirrored by lowering the radial slice r_c; the running couplings of Chapter 24 are the slice values of the bulk fields. The radial direction was the renormalization group all along, and the bulk equations of motion were its flow equations.

Worked Example 54.21 (A cubic superpotential: an explicit flow with $c_{\mathrm{IR}} < c_{\mathrm{UV}}$)**.** We exhibit a complete flow, compute the dimensions at both ends, verify that the deforming operator is relevant in the UV and irrelevant in the IR, and check the c-theorem inequality explicitly. To declutter the constants we adopt the normalization $\kappa^2 = d - 1$ (a choice of units for ϕ, stated once), in which the flow equations Eq. (54.185) read $\phi' = W'$, $A' = -W$, the beta function Eq. (54.187) is $\beta = -W'/W$, and the consistency condition Eq. (54.184) is $V = \frac{1}{2}W'^2 - \frac{d}{2}W^2$. Take the cubic superpotential

$$W(\phi) = -\frac{1}{L_0}\left(1 + \tfrac{1}{2}\,a\,\phi^2 - \tfrac{1}{3}\,b\,\phi^3\right), \qquad a, b > 0,\ \ L_0 > 0. \tag{54.204}$$

Its derivative $W' = -\frac{1}{L_0}(a\phi - b\phi^2) = -\frac{1}{L_0}\phi(a - b\phi)$ vanishes at $\phi = 0$ and at $\phi_{\mathrm{I}} = a/b$ — the two fixed points.

The UV fixed point ($\phi = 0$). Here $W(0) = -1/L_0$, so by Eq. (54.186) the AdS radius is $L_{\mathrm{UV}} = L_0$, and $W''(0) = -a/L_0$. The beta-function slope Eq. (54.188) is $\beta'(0) = L_{\mathrm{UV}} W''(0) = L_0 \cdot (-a/L_0) = -a$. Hence by Eq. (54.189) the dual operator has dimension $\Delta_{\mathrm{UV}} = d + \beta'(0) = d - a$. Since $a > 0$, $\Delta_{\mathrm{UV}} < d$: the operator is *relevant*, and the flow it triggers is genuine. As a check of Eq. (54.192), the scalar mass there is $m^2 = V''(0)$; from Eq. (54.190) with $\kappa^2 = d - 1$, $V''(0) = (W_0')^2 - d\, W_0 W_0'' = (a/L_0)^2 - d(-1/L_0)(-a/L_0) = \frac{a^2 - da}{L_0^2}$, so $m^2 L_{\mathrm{UV}}^2 = a^2 - da = (d-a)(-a) = \Delta_{\mathrm{UV}}(\Delta_{\mathrm{UV}} - d)$, the mass–dimension relation Eq. (53.112) confirmed.

The IR fixed point ($\phi_{\mathrm{I}} = a/b$). Evaluating Eq. (54.204),

$$W(\phi_{\mathrm{I}}) = -\frac{1}{L_0}\left(1 + \tfrac{1}{2} a\, \frac{a^2}{b^2} - \tfrac{1}{3} b\, \frac{a^3}{b^3}\right) = -\frac{1}{L_0}\left(1 + \frac{a^3}{6b^2}\right), \tag{54.205}$$

so $|W(\phi_{\mathrm{I}})| > |W(0)|$ and the IR AdS radius $L_{\mathrm{IR}} = 1/|W(\phi_{\mathrm{I}})| = L_0 (1 + \frac{a^3}{6b^2})^{-1}$ is *smaller* than $L_{\mathrm{UV}} = L_0$. The second derivative is $W''(\phi_{\mathrm{I}}) = -\frac{1}{L_0}(a - 2b\phi_{\mathrm{I}}) = -\frac{1}{L_0}(a - 2a) = +\frac{a}{L_0} > 0$, giving $\beta'(\phi_{\mathrm{I}}) = L_{\mathrm{IR}} W''(\phi_{\mathrm{I}}) = +a\,(1 + \frac{a^3}{6b^2})^{-1} > 0$. The same operator that was relevant at the UV is *irrelevant* at the IR ($\Delta_{\mathrm{IR}} = d + \beta'(\phi_{\mathrm{I}}) > d$), as it must be for the IR fixed point to be stable — nothing remains to drive the flow further. The trajectory is the gradient flow $\phi' = W' = -\frac{1}{L_0}\phi(a - b\phi)$: between the fixed points $\phi' < 0$, so as r decreases (UV→IR) the field climbs from 0 to a/b, reaching each endpoint exponentially, $\phi \sim e^{-(a/L_0)r}$ near the UV and $\phi - \phi_{\mathrm{I}} \sim e^{(a/L_0)r}$ near the IR. The radial coordinate is the monotone clock of Theorem 54.17.

The c-theorem inequality. The central charges are $c \propto L_\star^{d-1}$ by Theorem 54.19, so

$$\frac{c_{\mathrm{IR}}}{c_{\mathrm{UV}}} = \left(\frac{L_{\mathrm{IR}}}{L_{\mathrm{UV}}}\right)^{d-1} = \left(1 + \frac{a^3}{6b^2}\right)^{-(d-1)} < 1, \tag{54.206}$$

so $c_{\mathrm{IR}} < c_{\mathrm{UV}}$ strictly, in accordance with Theorem 54.20. The decrease is monotone along the whole flow, not merely between endpoints, because $A'' = -\phi'^2/(d-1) \le 0$ throughout (here $\kappa^2 = d - 1$). For a concrete number take $d = 4$, $a = 1$, $b = 1$: then $\Delta_{\mathrm{UV}} = 3$ (a relevant scalar, as for a mass operator), $L_{\mathrm{IR}}/L_{\mathrm{UV}} = (1 + \frac{1}{6})^{-1} = \frac{6}{7}$, and the IR central charge is reduced by the factor $(6/7)^3 \approx 0.63$ — the flow has shed roughly a third of its degrees of freedom in passing from the ultraviolet to the infrared.

The radial-direction-as-scale picture and the bulk/boundary split this section has made quantitative — UV physics at the boundary, IR physics deep in the interior, the warp factor counting degrees of freedom that only ever decrease inward — is also the seed of the deepest entry in the holographic dictionary, the one relating boundary entanglement to bulk geometry. The next chapters take it up: Chapter 55 develops the field-theoretic notion of entanglement entropy

that the radial geometry will turn out to encode, and Chapter 56 establishes the Ryu–Takayanagi prescription, in which a boundary region's entanglement entropy is the area of a bulk minimal surface reaching into exactly the radial depth — the scale — that this section has shown to be the renormalization group.

Further reading

The Witten-diagram calculus and the GKP–Witten relation originate in Gubser et al. [242] and Witten [522], following Maldacena [331]; the canonical pedagogical treatment of bulk propagators, contact and exchange diagrams, and D-functions is the TASI lectures of D'Hoker and Freedman [146], with the comprehensive review by Aharony et al. [9] surveying the whole subject. Holographic renormalization, the Fefferman–Graham expansion, and the covariant counterterms are developed systematically in Skenderis [444]. The conformal-block and Mellin organization of holographic correlators, and the bulk reading of the double-trace spectrum, are treated in Heemskerk et al. [261], Fitzpatrick et al. [194], and Penedones et al. [390]. Black-brane thermodynamics, the Hawking–Page transition, and its interpretation as confinement/deconfinement are due to Hawking and Page [258] and Witten [522], with the strong/weak factor of $\frac{3}{4}$ computed in Gubser et al. [241]. The Lorentzian (retarded) prescription and the shear viscosity are due to Son and Starinets [452] and Policastro et al. [405], with the universal ratio and its conjectured bound in Kovtun et al. [307]; the field-theoretic a-theorem the holographic c-theorem mirrors is Komargodski and Schwimmer [301]. Textbook accounts of the entire chapter are Nastase [359], Freedman and Van Proeyen [199], and — with the transport and condensed-matter emphasis — Hartnoll et al. [253].

Chapter 55

Entanglement Entropy in QFT

The vacuum is not empty. We have met this fact already in three guises: the zero-point fluctuations and normal ordering of Chapter 9, the short-distance singularity of the equal-time two-point function forced by microcausality, and the construction of the ground state, mode by mode, as the boundary value of a Euclidean path integral in Chapter 10. This chapter reframes that old observation in informational language. If the vacuum $|0\rangle$ is a correlated state and we agree to observe only the field within a spatial region A, discarding the complement, then what we keep is mixed — precisely because it was entangled with what we threw away. The number that quantifies this is the entanglement entropy S_A, and the chapter is the disciplined study of it: how to define it, how to compute it, what is universal in it, and — a theme we honor rather than hide — in what precise sense it is not, by itself, a finite observable at all.

The development imports little and builds much. From Chapter 9 we take the free scalar of ϕ^4 theory, its mode expansion, vacuum, and equal-time correlators, which become the explicit kernel of every free-field computation here; from Chapter 10 we take the Euclidean preparation of the vacuum, the single fact from which the entire path-integral construction of the reduced density matrix flows. The conformal apparatus of Chapter 47 — radial quantization, the state–operator map, correlators fixed by symmetry — and the two-dimensional machinery of Chapter 48 — the Virasoro algebra, the central charge, and the anomalous Schwarzian transformation of the stress tensor — supply the exactly solvable two-dimensional results, and the boost Noether charge of Chapter 7 furnishes the modular Hamiltonian of a half-space. The functional-determinant and zeta-regularization technology of Sections E.2 and E.3 underwrites the free-field area law, and Zamolodchikov's c-theorem from Chapter 52 reappears, recast as a statement about entanglement.

The arc runs from definition to computation to structure. Section 55.1 builds the reduced density matrix and the von Neumann and Rényi entropies in the safe finite-dimensional setting, proving the Schmidt decomposition, Klein's

inequality, concavity and subadditivity, and confronting at once the two features that make the field-theoretic problem genuinely subtle: the ultraviolet divergence of S_A and the failure of the Hilbert space to factorize across a sharp boundary. Section 55.2 converts the intractable operator logarithm into geometry — the replica trick: $\operatorname{Tr}\rho_A^n = Z_n/Z_1^n$ is a partition function on an n-sheeted branched cover, legitimized by Carlson's uniqueness theorem and, for a single interval, repackaged as a twist-operator correlator. Section 55.3 solves the one case where the geometry is trivial, the half-space, and reads the cut path integral as angular evolution by the boost: the vacuum restricted to a wedge is exactly thermal (Bisognano–Wichmann, Unruh), and its modular Hamiltonian is the local boost generator.

The remaining sections harvest these tools. Section 55.4 computes the twist dimension from the conformal anomaly and delivers the flagship result $S_A = \frac{c}{3}\log(\ell/\epsilon)$, together with its finite-size and finite-temperature deformations, and closes ϕ^4 theory on the entanglement side at the Ising value $c = \frac{1}{2}$. Section 55.5 establishes the higher-dimensional area law by an explicit Gaussian-state computation on the free scalar, dissects the full tower of ultraviolet divergences, and isolates the single universal term — a logarithm in even d, a constant in odd d — that counts degrees of freedom and runs monotonically under the renormalization group. Section 55.6 then builds the quantities that survive the continuum: relative entropy, its modular decomposition and ultraviolet finiteness, monotonicity and strong subadditivity, the first law of entanglement $\delta S_A = \delta\langle K_A\rangle$, and the bridge from information inequalities to energy conditions. These last structures are exactly the inputs that the holographic constructions of Chapters 56 and 57 will geometrize; the vacuum that Chapter 9 found correlated and Chapter 10 prepared as a path integral is shown here to carry an information-theoretic skeleton, and holography is the claim that the skeleton is spacetime.

55.1 Reduced Density Matrices and the Entanglement Entropy

The vacuum of an interacting — indeed of a free — quantum field is not the inert, structureless object the word "empty" suggests. We have already seen it fluctuate (the zero-point energy and normal ordering of Chapter 9), seen its short-distance correlations forced on us by microcausality, and seen it manufactured, mode by mode, as the boundary state of a Euclidean path integral (Chapter 10). The thread that runs through all of these facts is *correlation*: the field amplitude at one point is statistically tied to the amplitude at a neighbouring point, even with nothing present. The chapter's reframing of this old observation is informational. If the vacuum $|0\rangle$ is a correlated state, and if we agree to look only at the field within some spatial region A — a ball, a half-space, an interval — discarding all knowledge of the field in the complement, then the part we keep is not in a

pure state of its own. It is mixed, precisely because it was entangled with what we threw away. The question this section makes precise is: *how much* quantum correlation does a region share with its complement, and what is the right number to attach to it. The answer is the entanglement entropy, and building it cleanly — definitions first, then the structural facts, then the two features that make the field-theoretic version genuinely subtle — is the business of this section.

55.1.1 Bipartition and the reduced density matrix

Fix a quantum system whose Hilbert space factorizes into two tensor factors,

$$\mathcal{H} \;=\; \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}, \tag{55.1}$$

where A labels the degrees of freedom we retain and $\bar{A}$ (read "A-complement", also written A^{c}) the degrees of freedom we trace away. For a spin chain A is a set of sites and $\bar{A}$ the rest; for a quantum field, A is a spatial region and $\bar{A}$ its complement at fixed time — a partition we will have to handle with care, because Eq. (55.1) is exactly the assumption that fails in the continuum (Section 55.1.6). For now we take it as given and develop the formalism in the safe finite-dimensional setting, where every statement is a theorem of linear algebra.

The system is in a state described by a density operator ρ on $\mathcal{H}$: a positive semidefinite, Hermitian, unit-trace operator, $\rho \geq 0$, $\rho = \rho^\dagger$, $\operatorname{Tr}\rho = 1$. A pure state $|\Psi\rangle$ corresponds to the rank-one projector $\rho = |\Psi\rangle\langle\Psi|$. The observables accessible to an experimenter confined to A are the operators of the form $O_A \otimes \mathbf{1}_{\bar{A}}$. We want the single object on $\mathcal{H}_A$ that reproduces every such expectation value. That object is the reduced density matrix, and the operation that produces it is the partial trace.

Definition 55.1 (Partial trace)**.** Let $\{|j\rangle_{\bar{A}}\}_j$ be an orthonormal basis of $\mathcal{H}_{\bar{A}}$. The *partial trace over* $\bar{A}$ is the linear map $\operatorname{Tr}_{\bar{A}} \colon \mathcal{B}(\mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}) \to \mathcal{B}(\mathcal{H}_A)$ defined by

$$\operatorname{Tr}_{\bar{A}}(M) \;=\; \sum_j (\mathbf{1}_A \otimes \langle j|_{\bar{A}})\, M\, (\mathbf{1}_A \otimes |j\rangle_{\bar{A}}). \tag{55.2}$$

On a product operator it acts as $\operatorname{Tr}_{\bar{A}}(O_A \otimes O_{\bar{A}}) = O_A \operatorname{Tr}(O_{\bar{A}})$, and it extends to all of $\mathcal{B}(\mathcal{H})$ by linearity.

The map Eq. (55.2) is independent of the chosen basis of $\mathcal{H}_{\bar{A}}$: a change of basis $|j\rangle_{\bar{A}} \mapsto \sum_k U_{jk}\,|k\rangle_{\bar{A}}$ with U unitary leaves the sum invariant, because $\sum_j U_{jk}\overline{U_{jk'}} = \delta_{kk'}$ collapses the two inserted resolutions back to one. It is the unique linear map satisfying $\operatorname{Tr}_A\!\big[(\operatorname{Tr}_{\bar{A}} M)\, O_A\big] = \operatorname{Tr}\!\big[M\,(O_A \otimes \mathbf{1}_{\bar{A}})\big]$ for all O_A, which is the property we actually need.

Definition 55.2 (Reduced density matrix)**.** For a global state ρ on $\mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$, the *reduced density matrix of* A is

$$\rho_A \;=\; \mathrm{Tr}_{\bar{A}}\, \rho. \tag{55.3}$$

When the global state is pure, $\rho = |\Psi\rangle\langle\Psi|$, we write $\rho_A = \mathrm{Tr}_{\bar{A}}\, |\Psi\rangle\langle\Psi|$.

Proposition 55.3 (ρ_A is a density matrix that encodes all of A's statistics)**.** *The reduced density matrix Eq. (55.3) is Hermitian, positive semidefinite, and has unit trace; and for every operator O_A acting on $\mathcal{H}_A$,*

$$\big\langle O_A \otimes \mathbf{1}_{\bar{A}} \big\rangle_\rho \;\equiv\; \mathrm{Tr}\big[\rho\,(O_A \otimes \mathbf{1}_{\bar{A}})\big] \;=\; \mathrm{Tr}_A\big(\rho_A\, O_A\big). \tag{55.4}$$

Proof. Hermiticity is immediate from Eq. (55.2): $(\mathrm{Tr}_{\bar{A}}\,\rho)^\dagger = \sum_j (\mathbf{1}\otimes\langle j|)\rho^\dagger(\mathbf{1}\otimes|j\rangle) = \mathrm{Tr}_{\bar{A}}\,\rho$ since $\rho^\dagger = \rho$. For positivity, let $|\chi\rangle \in \mathcal{H}_A$ be arbitrary; then

$$\langle\chi|\,\rho_A\,|\chi\rangle = \sum_j \big(\langle\chi|\otimes\langle j|\big)\,\rho\,\big(|\chi\rangle\otimes|j\rangle\big) \geq 0, \tag{55.5}$$

each term being the expectation of the positive operator ρ in the state $|\chi\rangle\otimes|j\rangle$. Unit trace follows from $\mathrm{Tr}_A\,\rho_A = \sum_i\sum_j (\langle i|\otimes\langle j|)\rho(|i\rangle\otimes|j\rangle) = \mathrm{Tr}_{\mathcal{H}}\,\rho = 1$, the inner double sum running over a full product basis of $\mathcal{H}$. Finally, taking $M = \rho(O_A\otimes\mathbf{1}_{\bar{A}})$ in the defining property of the partial trace, $\mathrm{Tr}_A[(\mathrm{Tr}_{\bar{A}}\,\rho)O_A] = \mathrm{Tr}[\rho(O_A\otimes\mathbf{1}_{\bar{A}})]$, which is exactly Eq. (55.4). □

Equation Eq. (55.4) is the operational meaning of ρ_A: *no measurement confined to A can distinguish the global state ρ from the local state ρ_A.* All accessible information about the region is packed into this one operator. The price of confinement is that even when the global state is pure — a state of maximal knowledge — the local state ρ_A is generically mixed. That mixedness is not classical ignorance about which pure state we prepared; it is the shadow cast on A by the entanglement between A and $\bar{A}$. Quantifying it is the next step.

55.1.2 The von Neumann and Rényi entropies

A density matrix is mixed to the extent that its spectrum is spread out, and the canonical functional that measures the spread of a probability distribution — here the spectrum $\{\lambda_i\}$ of ρ_A, a set of nonnegative eigenvalues summing to one — is the Shannon–von Neumann entropy.

Definition 55.4 (Entanglement entropy)**.** The *entanglement entropy* of the region A in the state ρ is the von Neumann entropy of the reduced density matrix,

$$S_A \;=\; -\,\mathrm{Tr}(\rho_A \log \rho_A) \;=\; -\sum_i \lambda_i \log \lambda_i, \tag{55.6}$$

where $\{\lambda_i\}$ are the eigenvalues of ρ_A, and $0\log 0 \equiv 0$ by the limit $x\log x \to 0$ as $x \to 0^+$.

The operator logarithm in Eq. (55.6) is defined by functional calculus: in the eigenbasis $\rho_A = \sum_i \lambda_i |i\rangle\langle i|$ one sets $\log \rho_A = \sum_i (\log \lambda_i)\, |i\rangle\langle i|$, with the convention that eigenvectors of zero eigenvalue do not contribute. We use the natural logarithm throughout; an entropy in bits is $S_A/\log 2$. The base is a convention and changes nothing structural.

The direct evaluation of Eq. (55.6) requires diagonalizing ρ_A and then summing a transcendental function of its eigenvalues — an operation with no representation as a correlation function and no obvious path-integral analogue. It is precisely this intractability that the next section's replica construction circumvents. The detour runs through a one-parameter family of entropies that *are* polynomial in ρ_A, hence computable.

Definition 55.5 (Rényi entropies)**.** For real $n > 0$, $n \neq 1$, the *n-th Rényi entropy* of A is

$$S_n(A) \;=\; \frac{1}{1-n}\,\log \operatorname{Tr}(\rho_A^n) \;=\; \frac{1}{1-n}\,\log \sum_i \lambda_i^n. \tag{55.7}$$

The quantity $\operatorname{Tr}\rho_A^n$ is, for integer n, a product of n copies of ρ_A traced together — an object that, we will see in Section 55.1.8, the path integral computes as a partition function on an n-sheeted branched cover. The Rényi entropies form a convenient ladder: $S_0(A) = \log(\operatorname{rank}\rho_A)$ counts the nonzero eigenvalues (the *Schmidt rank*, below), $S_2(A) = -\log \operatorname{Tr}\rho_A^2$ measures the purity, and $S_\infty(A) = -\log \lambda_{\max}$ is the min-entropy. The rung we ultimately want is the limit $n \to 1$, which recovers the von Neumann entropy.

Proposition 55.6 (The replica limit)**.** $\lim_{n\to 1} S_n(A) = S_A$.

Proof. Write $f(n) = \log \sum_i \lambda_i^n = \log \operatorname{Tr}\rho_A^n$. Since $\sum_i \lambda_i = \operatorname{Tr}\rho_A = 1$, we have $f(1) = \log 1 = 0$. The function is differentiable in n for $n > 0$, with

$$f'(n) = \frac{\sum_i \lambda_i^n \log \lambda_i}{\sum_i \lambda_i^n}, \qquad f'(1) = \sum_i \lambda_i \log \lambda_i = -S_A, \tag{55.8}$$

the denominator at $n = 1$ being unity. By Eq. (55.7), $S_n(A) = -f(n)/(n-1)$. Taylor-expanding f about $n = 1$, $f(n) = f(1) + f'(1)(n-1) + O\big((n-1)^2\big) = -S_A(n-1) + O\big((n-1)^2\big)$, so $S_n(A) = S_A + O(n-1) \to S_A$. □

This innocuous limit is the entire strategy of the chapter. We will compute $\operatorname{Tr}\rho_A^n$ for integer $n \geq 2$ as a Euclidean path integral, find a function of n that interpolates, and analytically continue it to $n \to 1$ to read off S_A. The legitimacy of that continuation — whether the integer data determine a unique analytic function — is a genuine subtlety we confront in Section 55.1.8; for the moment the limit is exact as a statement about a fixed finite spectrum.

55.1.3 The Schmidt decomposition and the equality of entropies

The structural backbone of bipartite entanglement is a single theorem of linear algebra — the singular-value decomposition, dressed in physical language. It says that any pure state of $A \cup \bar{A}$ can be written with a *single* sum, diagonal across the bipartition, and from it every elementary property of S_A follows almost immediately.

Theorem 55.7 (Schmidt decomposition)**.** *Let $|\Psi\rangle \in \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$ be a normalized pure state, with $\dim \mathcal{H}_A = d_A$ and $\dim \mathcal{H}_{\bar{A}} = d_{\bar{A}}$. There exist orthonormal sets $\{|k\rangle_A\} \subset \mathcal{H}_A$ and $\{|k\rangle_{\bar{A}}\} \subset \mathcal{H}_{\bar{A}}$, and nonnegative numbers $\sqrt{p_k}$, such that*

$$|\Psi\rangle \;=\; \sum_{k=1}^{r} \sqrt{p_k}\; |k\rangle_A \otimes |k\rangle_{\bar{A}}\,, \qquad p_k > 0, \quad \sum_{k=1}^{r} p_k = 1, \tag{55.9}$$

where $r \le \min(d_A, d_{\bar{A}})$ is the Schmidt rank*. The reduced density matrices are then diagonal in these bases,*

$$\rho_A = \sum_{k=1}^{r} p_k\; |k\rangle_A\langle k|_A\,, \qquad \rho_{\bar{A}} = \sum_{k=1}^{r} p_k\; |k\rangle_{\bar{A}}\langle k|_{\bar{A}}\,, \tag{55.10}$$

so ρ_A and $\rho_{\bar{A}}$ share the identical multiset of nonzero eigenvalues $\{p_k\}$.

Proof. Expand $|\Psi\rangle$ in arbitrary product bases $\{|i\rangle_A\}_{i=1}^{d_A}$, $\{|j\rangle_{\bar{A}}\}_{j=1}^{d_{\bar{A}}}$:

$$|\Psi\rangle = \sum_{i,j} C_{ij}\; |i\rangle_A \otimes |j\rangle_{\bar{A}}\,, \tag{55.11}$$

with C a $d_A \times d_{\bar{A}}$ complex matrix of coefficients. Every such matrix admits a singular-value decomposition $C = U\,\Sigma\,V^{\dagger}$, where U is $d_A \times d_A$ unitary, V is $d_{\bar{A}} \times d_{\bar{A}}$ unitary, and Σ is $d_A \times d_{\bar{A}}$ with nonnegative real entries $\sigma_k \equiv \Sigma_{kk} \ge 0$ on its diagonal and zeros elsewhere. In components $C_{ij} = \sum_k U_{ik}\sigma_k\, \overline{V_{jk}}$. Substituting into Eq. (55.11) and grouping,

$$|\Psi\rangle = \sum_{k} \sigma_k \underbrace{\Big(\sum_{i} U_{ik}\,|i\rangle_A\Big)}_{\equiv\ |k\rangle_A} \otimes \underbrace{\Big(\sum_{j} \overline{V_{jk}}\;|j\rangle_{\bar{A}}\Big)}_{\equiv\ |k\rangle_{\bar{A}}}. \tag{55.12}$$

Because U and V are unitary, the vectors $\{|k\rangle_A\}$ are orthonormal — $\langle k|k'\rangle_A = \sum_i \overline{U_{ik}} U_{ik'} = \delta_{kk'}$ — and likewise $\{|k\rangle_{\bar{A}}\}$. Keeping only the indices with $\sigma_k > 0$ and setting $p_k = \sigma_k^2$ gives Eq. (55.9) with $r = \operatorname{rank} C$. Normalization of $|\Psi\rangle$ reads $1 = \langle\Psi|\Psi\rangle = \sum_k \sigma_k^2 = \sum_k p_k$.

The reduced density matrices follow by partial trace. From Eq. (55.9),

$$|\Psi\rangle\langle\Psi| = \sum_{k,k'} \sqrt{p_k p_{k'}}\; |k\rangle_A\langle k'|_A \otimes |k\rangle_{\bar{A}}\langle k'|_{\bar{A}}\,, \tag{55.13}$$

and tracing over $\bar{A}$ in the orthonormal basis $\{|k\rangle_{\bar{A}}\}$ (extended to a full basis; the extra vectors annihilate the state) replaces $|k\rangle_{\bar{A}}\langle k'|_{\bar{A}}$ by $\langle k'|k\rangle_{\bar{A}} = \delta_{kk'}$, leaving $\rho_A = \sum_k p_k\, |k\rangle_A\langle k|_A$. Tracing over A instead gives $\rho_{\bar{A}} = \sum_k p_k\, |k\rangle_{\bar{A}}\langle k|_{\bar{A}}$. The two operators are diagonal with the identical nonzero spectrum $\{p_k\}$; they differ only in the Hilbert space their eigenvectors inhabit, and by the dimensions of their respective zero-eigenvalue (kernel) subspaces. □

Corollary 55.8 (Equality of complementary entropies for a pure global state)**.** *If the global state is pure, then for every Rényi index and for the von Neumann entropy,*

$$S_n(A) = S_n(\bar{A}) \quad \textit{for all } n, \qquad \textit{in particular} \qquad S_A = S_{\bar{A}}. \tag{55.14}$$

Proof. Each entropy in Theorems 55.4 and 55.5 is a function of the nonzero eigenvalues alone (the zero eigenvalues contribute $0 \log 0 = 0$ to the von Neumann entropy and nothing to $\operatorname{Tr}\rho^n$ for $n > 0$). By Theorem 55.7 the nonzero spectra of ρ_A and $\rho_{\bar{A}}$ coincide, so any spectral function takes the same value on both. □

Equation (55.14) is at first sight startling: the entanglement entropy of a small region equals that of its enormous complement. The resolution is that S_A is not a property of the region's "size" or "content" — it measures the entanglement *across the shared boundary*, and the boundary is the same whether viewed from inside or outside. This is the first hint of the area law: a quantity that depends on A only through ∂A. It also delivers a warning. Because $S_A = S_{\bar{A}}$ holds only for a *pure* global state, the entanglement entropy is *not* an extensive (additive) quantity, and for a mixed global state $S_A \neq S_{\bar{A}}$ in general — the difference $S_A - S_{\bar{A}}$ then measures the impurity of the whole. In the thermal ensembles of Chapter 54 this distinction is the difference between thermal entropy and entanglement entropy, a point we return to in Section 55.1.6.

55.1.4 Elementary properties, proved

The defining axioms of an information measure must now be verified. Two of them — non-negativity and the characterization of zero — are immediate; the maximal value and concavity require an inequality, which we prove first because it is the workhorse for the rest of the chapter.

Lemma 55.9 (Klein's inequality / non-negativity of relative entropy)**.** *For any two density matrices ρ, σ on a finite-dimensional Hilbert space, the* relative entropy

$$S(\rho\,\|\,\sigma) \;\equiv\; \operatorname{Tr}\big[\rho(\log\rho - \log\sigma)\big] \tag{55.15}$$

is non-negative, $S(\rho\,\|\,\sigma) \geq 0$, with equality if and only if $\rho = \sigma$.

Proof. Diagonalize both operators: $\rho = \sum_i p_i \, |i\rangle\langle i|$ and $\sigma = \sum_a q_a \, |a\rangle\langle a|$, with $\{|i\rangle\}$ and $\{|a\rangle\}$ orthonormal bases and $p_i, q_a \geq 0$, $\sum_i p_i = \sum_a q_a = 1$. Inserting these,

$$S(\rho \,\|\, \sigma) = \sum_i p_i \log p_i - \sum_{i,a} p_i \, |\langle i|a\rangle|^2 \log q_a = \sum_i p_i \Big(\log p_i - \sum_a P_{ia} \log q_a \Big), \tag{55.16}$$

where $P_{ia} \equiv |\langle i|a\rangle|^2$ is a doubly stochastic matrix: $\sum_a P_{ia} = \sum_i P_{ia} = 1$, since $\{|i\rangle\}$ and $\{|a\rangle\}$ are complete orthonormal sets. The logarithm is strictly concave, so by Jensen's inequality applied to the probability weights P_{ia} (over a at fixed i),

$$\sum_a P_{ia} \log q_a \;\leq\; \log\Big(\sum_a P_{ia} \, q_a \Big) \;\equiv\; \log r_i, \qquad r_i \equiv \sum_a P_{ia} q_a. \tag{55.17}$$

The numbers r_i are themselves a probability distribution: $\sum_i r_i = \sum_a q_a \sum_i P_{ia} = \sum_a q_a = 1$, using the column-stochasticity of P. Combining Eqs. (55.16) and (55.17),

$$S(\rho \,\|\, \sigma) \;\geq\; \sum_i p_i \log \frac{p_i}{r_i} \;\geq\; 0, \tag{55.18}$$

the last step being the classical Gibbs inequality, which follows from $\log x \leq x - 1$: $-\sum_i p_i \log(r_i/p_i) \geq -\sum_i p_i (r_i/p_i - 1) = \sum_i (p_i - r_i) = 0$. Equality in $\log x \leq x - 1$ requires $x = 1$, i.e. $r_i = p_i$ for all i; equality in Jensen requires the q_a to be constant over the support of P_{ia}. Tracing these conditions back forces the two operators to be diagonal in a common basis with equal eigenvalues, i.e. $\rho = \sigma$. $\square$

With Klein's inequality in hand the entropy's elementary properties fall out.

Proposition 55.10 (Elementary properties of S_A)**.** *Let ρ_A act on a d_A-dimensional space $\mathcal{H}_A$. Then*

(i) ***Non-negativity:*** *$S_A \geq 0$.*

(ii) ***Purity criterion:*** *$S_A = 0$ if and only if ρ_A is pure; for a pure global state this happens if and only if $|\Psi\rangle$ is a product state $|\Psi\rangle = |\phi\rangle_A \otimes |\xi\rangle_{\bar{A}}$ (Schmidt rank $r = 1$).*

(iii) ***Maximality:*** *$S_A \leq \log d_A$, with equality if and only if $\rho_A = \mathbf{1}_A / d_A$ is maximally mixed.*

Proof. (i) Each eigenvalue satisfies $0 \leq \lambda_i \leq 1$, so $-\lambda_i \log \lambda_i \geq 0$ term by term, and $S_A = -\sum_i \lambda_i \log \lambda_i \geq 0$.

(ii) Since every term $-\lambda_i \log \lambda_i \geq 0$ vanishes only when $\lambda_i \in \{0, 1\}$, the sum vanishes precisely when one eigenvalue is 1 and the rest are 0 — that is, when ρ_A is a rank-one projector, a pure state. By Theorem 55.7 a pure $\rho_A = |1\rangle_A \langle 1|_A$

corresponds to a single Schmidt coefficient $p_1 = 1$, whence $|\Psi\rangle = |1\rangle_A \otimes |1\rangle_{\bar{A}}$ is a product state, and conversely.

(iii) Apply Theorem 55.9 with $\sigma = \mathbf{1}_A/d_A$, the maximally mixed state. Then $\log \sigma = -(\log d_A)\mathbf{1}_A$, so

$$0 \le S(\rho_A \,\|\, \mathbf{1}_A/d_A) = \operatorname{Tr} \rho_A \log \rho_A - \operatorname{Tr} \rho_A \log(\mathbf{1}_A/d_A) = -S_A + \log d_A, \tag{55.19}$$

giving $S_A \le \log d_A$. Equality holds iff $\rho_A = \mathbf{1}_A/d_A$, by the equality clause of Klein. □

Property (iii) is the finite-dimensional ceiling that the field-theoretic divergence will smash through: as we refine the lattice and $d_A \to \infty$, the bound $\log d_A$ itself diverges, and the actual S_A saturates a divergent fraction of it. We turn to this after one more property — concavity — and its consequence, subadditivity.

Theorem 55.11 (Concavity of the von Neumann entropy)**.** *The von Neumann entropy $S(\rho) = -\operatorname{Tr} \rho \log \rho$ is concave in ρ: for any density matrices $\{\rho_i\}$ and probabilities $\{p_i \ge 0,\ \sum_i p_i = 1\}$,*

$$S\Big(\sum_i p_i\, \rho_i\Big) \;\ge\; \sum_i p_i\, S(\rho_i). \tag{55.20}$$

Proof. Set $\rho = \sum_i p_i \rho_i$ and compute the weighted sum of relative entropies of each ρ_i against the average ρ:

$$\begin{aligned} \sum_i p_i\, S(\rho_i \,\|\, \rho) &= \sum_i p_i\, \operatorname{Tr}\big[\rho_i(\log \rho_i - \log \rho)\big] \\ &= \sum_i p_i \operatorname{Tr}(\rho_i \log \rho_i) - \operatorname{Tr}\Big[\Big(\sum_i p_i \rho_i\Big) \log \rho\Big] \\ &= -\sum_i p_i\, S(\rho_i) - \operatorname{Tr}(\rho \log \rho) = -\sum_i p_i\, S(\rho_i) + S(\rho). \end{aligned} \tag{55.21}$$

By Theorem 55.9 each $S(\rho_i \,\|\, \rho) \ge 0$, so the left-hand side is non-negative, and rearranging Eq. (55.21) gives $S(\rho) \ge \sum_i p_i S(\rho_i)$, which is Eq. (55.20). □

Physically, concavity says that mixing states never decreases entropy: blending ensembles can only add uncertainty. The corresponding statement about combining *subsystems* is subadditivity, which we will need at once for the mutual information but whose proof is most naturally given alongside its powerful refinement, strong subadditivity, in Section 55.6.3.

Theorem 55.12 (Subadditivity)**.** *For any state on a tripartite-or-bipartite system with regions A and B (disjoint, with joint reduced state ρ_{AB}),*

$$S_{AB} \;\le\; S_A + S_B, \tag{55.22}$$

with equality if and only if $\rho_{AB} = \rho_A \otimes \rho_B$.

Proof (deferred). This is the case $C = \varnothing$ of strong subadditivity, proved in full in Theorem 55.47. We record the short self-contained argument here for completeness: by Theorem 55.9, $0 \le S(\rho_{AB} \,\|\, \rho_A \otimes \rho_B) = \operatorname{Tr} \rho_{AB} \log \rho_{AB} - \operatorname{Tr} \rho_{AB} \log(\rho_A \otimes \rho_B)$. Using $\log(\rho_A \otimes \rho_B) = \log \rho_A \otimes \mathbf{1}_B + \mathbf{1}_A \otimes \log \rho_B$ and $\operatorname{Tr}_B \rho_{AB} = \rho_A$, the second trace equals $\operatorname{Tr} \rho_A \log \rho_A + \operatorname{Tr} \rho_B \log \rho_B = -S_A - S_B$, so $0 \le -S_{AB} + S_A + S_B$. Equality is the equality clause of Klein, i.e. $\rho_{AB} = \rho_A \otimes \rho_B$. □

55.1.5 Mutual information: the finite diagnostic

Subadditivity guarantees that the following combination is well-signed.

Definition 55.13 (Mutual information)**.** For disjoint regions A and B, the *mutual information* is

$$I(A : B) \;=\; S_A + S_B - S_{AB}. \tag{55.23}$$

Corollary 55.14 (Positivity of mutual information)**.** $I(A : B) \ge 0$*, with equality if and only if* $\rho_{AB} = \rho_A \otimes \rho_B$ *(the two regions are uncorrelated). Moreover* $I(A : B) = S(\rho_{AB} \,\|\, \rho_A \otimes \rho_B)$*.*

Proof. Positivity and the equality condition are precisely Theorem 55.12 rearranged. The identification with a relative entropy is read off from the deferred proof above: $S(\rho_{AB} \,\|\, \rho_A \otimes \rho_B) = -S_{AB} + S_A + S_B = I(A : B)$. □

The mutual information is the single most important object in this chapter for a reason we can now state precisely but will only fully justify in Section 55.1.6. The entanglement entropy S_A is, in the continuum, divergent and cutoff-dependent; one cannot quote a finite number for it without specifying a regulator. The mutual information $I(A : B)$, by contrast, is a *difference* of entropies in which the boundary divergences cancel: S_A diverges on ∂A, S_B on ∂B, and S_{AB} on $\partial A \cup \partial B$, so the divergences supported on each boundary appear with canceling signs in Eq. (55.23). What survives is finite and cutoff-independent — a genuine measure of the total correlation, classical plus quantum, shared by the two regions. Its expression as a relative entropy in Theorem 55.14 is the abstract reason for this finiteness: relative entropy, unlike entropy itself, descends to the algebraic (cutoff-free) setting, a fact we flag now and exploit in the final section.

55.1.6 Two structural facts that make QFT subtle

Everything above was finite-dimensional linear algebra. The leap to a quantum field — where A is a spatial region and the "degrees of freedom" are the field amplitudes at a continuum of points — introduces two features absent from any finite system. They are not technical nuisances to be papered over; they shape every result in the chapter, and intellectual honesty requires stating them at the outset and honoring them throughout.

(a) The area law and its ultraviolet divergence

The first fact is that S_A is ultraviolet divergent. We can see why, and extract the leading divergence's form, by a scaling argument before the honest computation of Section 55.5. The vacuum of a local field theory is correlated over short distances — the equal-time two-point function $\langle 0|\phi(\mathbf{x})\phi(\mathbf{y})|0\rangle$ of Chapter 9 is singular as $\mathbf{y} \to \mathbf{x}$. Entanglement between A and $\bar{A}$ is therefore dominated by field modes localized *near the boundary* ∂A, where short-range correlations straddle the cut. Introduce a short-distance cutoff a (a lattice spacing, or $1/\Lambda$ for a momentum cutoff Λ). The boundary ∂A, a $(d-2)$-dimensional surface in d spacetime dimensions, can be tiled by $\sim \mathrm{Area}(\partial A)/a^{\,d-2}$ cells of linear size a. Each cell contributes an $O(1)$ amount of entanglement from the entangled pair of modes that straddle it, and cells separated by more than a contribute negligibly because the correlations are short-ranged. The estimate is therefore

$$S_A \;\simeq\; \gamma\,\frac{\mathrm{Area}(\partial A)}{a^{\,d-2}} \;+\; (\text{subleading}), \tag{55.24}$$

with γ a dimensionless, cutoff-scheme-dependent constant. This is the *area law*: the leading entanglement entropy scales with the area of the entangling surface, not the volume of the region, and it *diverges* as the cutoff is removed, $a \to 0$. In $d = 4$ this is $S_A \simeq \gamma\,\mathrm{Area}/a^2$; in $d = 2$, where the "area" of a point-like boundary is a pure number and Eq. (55.24) degenerates, the leading divergence is instead logarithmic, $S_A \simeq \frac{c}{3}\log(\ell/a)$ for an interval of length ℓ in a CFT of central charge c — the famous result we derive in Section 55.4 once the replica machinery is built. The scaling Eq. (55.24) is heuristic; the coefficient γ and the structure of the subleading terms come from the genuine computation in Section 55.5. What matters now is the qualitative lesson: *S_A is not a finite number in the continuum,* and any physical statement must be phrased so that the cutoff dependence drops out — as it does in the mutual information and the relative entropy.

(b) Non-factorization and the type-III algebra of a region

The second fact is deeper and undermines the very first line of the formalism, the tensor factorization Eq. (55.1). In a continuum quantum field theory *the global Hilbert space does not factorize* as $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$, and consequently the reduced density matrix ρ_A of Theorem 55.2 does not, strictly, exist. There are two ways to see the obstruction, and they are the same obstruction.

The concrete way is the divergence we just met. A density matrix must have $\mathrm{Tr}\,\rho_A = 1$ and finite entropy is needed for $\rho_A \log \rho_A$ to be trace class. But Eq. (55.24) says the vacuum's entanglement across *any* boundary is infinite: arbitrarily-high-frequency modes are entangled across ∂A, with no shortest wavelength to cut the sum off. An operator with infinite entropy and no normalizable

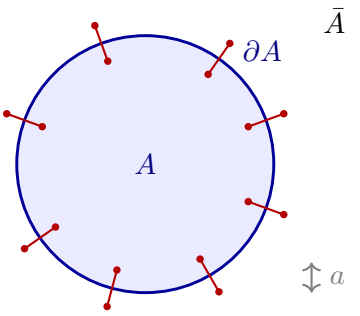


Figure 55.1: The area-law cartoon. The entanglement between a region A and its complement $\bar{A}$ is dominated by short-range correlated pairs of modes (red links) straddling the entangling surface ∂A. Tiling ∂A by cells of the cutoff size a gives $\sim \mathrm{Area}(\partial A)/a^{d-2}$ such pairs, each contributing $O(1)$ to S_A, hence the leading divergence Eq. (55.24).

spectrum is not a density matrix on a separable Hilbert space. The would-be factorization Eq. (55.1) fails because a sharply localized field mode requires arbitrarily high momentum (by the uncertainty principle), and such modes cannot be cleanly assigned to one side of a sharp boundary; they live on both.

The abstract way is the language of operator algebras, and it is worth stating precisely because it is the framework the modern literature uses. To a spacetime region one associates not a Hilbert space factor but a *von Neumann algebra* $\mathcal{A}_A$ — the algebra of bounded operators built from the fields smeared with test functions supported in A, closed in the weak operator topology. Murray and von Neumann classified such algebras into types I, II, and III by the structure of their projections. The algebra of a bounded region in continuum QFT, in the vacuum, is of *type* III_1: it has no minimal projections, no normal trace, and — the decisive point — no density matrix representing the vacuum state on it. Type I is the familiar case $\mathcal{B}(\mathcal{H}_A)$ in which density matrices and a tensor factorization Eq. (55.1) exist; type III is what a local QFT region actually realizes, and in it the naive $S_A = -\operatorname{Tr}\rho_A \log \rho_A$ is simply undefined.

The entanglement entropy of a continuum region is not, by itself, a number

Two errors must be avoided. First, one must not quote "the entanglement entropy of the half-space is X" without a regulator: Eq. (55.6) as written has no continuum value. Second, one must not conclude that entanglement is therefore ill-defined or unphysical. It is the *entropy* (a property of ρ_A alone) that fails to exist; *differences* of entropy — relative entropy $S(\rho_A\|\sigma_A)$ and mutual information $I(A:B)$ — remain perfectly well-defined even on a type III algebra, where they are constructed without

ever forming ρ_A, via Tomita–Takesaki modular theory. This is the technical content of the chapter's running insistence that the clean observables are differences.

The pragmatic resolution, adopted throughout this chapter and almost universally in the literature, is to *regulate*: place the theory on a spatial lattice of spacing a, or otherwise impose a UV cutoff. A regulated theory has finitely many degrees of freedom in any bounded region, its algebra is type I, the factorization Eq. (55.1) is restored exactly, ρ_A exists, and every theorem of Sections 55.1.1 to 55.1.4 applies verbatim. The continuum statements are then recovered as the $a \to 0$ limit, with the understanding that S_A itself diverges per Eq. (55.24) while the physical, cutoff-independent content lives in the finite pieces: the coefficient of a universal subleading term (the logarithm in $d = 2$, a constant in odd d), the mutual information, and the relative entropy.

Why the vacuum is the interesting case

The whole construction would be vacuous — literally — if the vacuum factorized. It is precisely because $|0\rangle$ is the richly correlated state advertised in the chapter opening (and built explicitly in Chapters 9 and 10) that ρ_A is mixed and S_A is large. The entanglement entropy is thus a probe of vacuum structure: the area law is the statement that the vacuum's correlations are short-ranged and live at the boundary, and the universal subleading terms encode the long-distance data of the theory — the central charge in 2d, the number of degrees of freedom and the structure of the renormalization-group flow in higher d. We are, in effect, reading off the theory's deepest invariants from how its empty state is entangled with itself.

55.1.7 Two worked computations

The formalism is best anchored by carrying it out completely on the smallest nontrivial systems. The first is the canonical qubit pair; the second is the two-oscillator ground state, which is the exact kernel of the free-scalar computation that advances ϕ^4 theory in Section 55.5.

Worked Example 55.15 (Entanglement of a two-qubit pure state)**.** Take $\mathcal{H}_A = \mathcal{H}_{\bar{A}} = \mathbb{C}^2$ and the one-parameter family of pure states

$$|\Psi(\theta)\rangle = \cos\theta\ |0\rangle_A\,|0\rangle_{\bar{A}} + \sin\theta\ |1\rangle_A\,|1\rangle_{\bar{A}}\,, \qquad \theta \in [0, \pi/2]. \tag{55.25}$$

This is already in Schmidt form Eq. (55.9), with Schmidt coefficients $\sqrt{p_1} = \cos\theta$,

$\sqrt{p_2} = \sin\theta$, so by Eq. (55.10) the reduced density matrix is diagonal,

$$\rho_A = \cos^2\theta\ |0\rangle_A\langle 0|_A + \sin^2\theta\ |1\rangle_A\langle 1|_A = \begin{pmatrix} \cos^2\theta & 0 \\ 0 & \sin^2\theta \end{pmatrix}. \tag{55.26}$$

The eigenvalues are $\lambda_1 = \cos^2\theta$, $\lambda_2 = \sin^2\theta$. The entanglement entropy is the binary entropy function

$$S_A(\theta) = -\cos^2\theta\,\log\cos^2\theta - \sin^2\theta\,\log\sin^2\theta. \tag{55.27}$$

At $\theta = 0$ or $\pi/2$ the state Eq. (55.25) is a product and $S_A = 0$, confirming Theorem 55.10(ii). At $\theta = \pi/4$ — the maximally entangled Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ — both eigenvalues equal $\frac{1}{2}$ and $S_A = \log 2$, saturating the ceiling $\log d_A = \log 2$ of Theorem 55.10(iii). The Rényi entropies are $S_n = \frac{1}{1-n}\log(\cos^{2n}\theta + \sin^{2n}\theta)$; at $\theta = \pi/4$ this is $\frac{1}{1-n}\log(2\cdot 2^{-n}) = \log 2$ for every n, the signature of a maximally mixed reduced state. One checks the replica limit directly: $\lim_{n\to 1}\frac{1}{1-n}\log(\cos^{2n}\theta + \sin^{2n}\theta) = S_A(\theta)$ by Theorem 55.6. Note also that $\rho_{\bar A}$ has the identical spectrum, so $S_A = S_{\bar A}$ as Theorem 55.8 demands — here trivially, since A and $\bar A$ are interchangeable.

Worked Example 55.16 (The ground state of two coupled oscillators)**.** This is the prototype for entanglement in a free field. A free scalar field on a spatial lattice is, after diagonalizing nothing, a collection of harmonic oscillators — one per lattice site — coupled to their neighbours by the gradient term in the Hamiltonian. Partitioning space into A and $\bar A$ partitions the oscillators. The minimal instance keeps two oscillators, one in A and one in $\bar A$, coupled quadratically. Consider the Hamiltonian

$$H = \tfrac{1}{2}(p_1^2 + p_2^2) + \tfrac{1}{2}\Big[\,\omega_0^2(x_1^2 + x_2^2) + k\,(x_1 - x_2)^2\,\Big], \tag{55.28}$$

with coordinates x_1 (in A), x_2 (in $\bar A$), and coupling $k \geq 0$. Normal coordinates $x_\pm = (x_1 \pm x_2)/\sqrt{2}$ decouple it into two oscillators of frequencies

$$\omega_+ = \sqrt{\omega_0^2 + 2k}, \qquad \omega_- = \omega_0, \tag{55.29}$$

since the symmetric combination feels no coupling and the antisymmetric one feels $\omega_0^2 + 2k$. The ground-state wavefunction is the product of the two normal-mode Gaussians,

$$\Psi_0(x_1, x_2)\ \propto\ \exp\Big[-\tfrac{1}{2}\omega_- x_+^2 - \tfrac{1}{2}\omega_+ x_-^2\Big] = \exp\Big[-\tfrac{\alpha}{2}(x_1^2 + x_2^2) + \beta\, x_1 x_2\Big], \tag{55.30}$$

where expanding $x_\pm = (x_1 \pm x_2)/\sqrt{2}$ gives

$$\alpha = \tfrac{1}{2}(\omega_+ + \omega_-), \qquad \beta = \tfrac{1}{2}(\omega_+ - \omega_-). \tag{55.31}$$

Crucially, the ground state is *not* a product in (x_1, x_2) whenever $\beta \neq 0$, i.e. whenever $k > 0$: the cross term $\beta x_1 x_2$ is exactly the entanglement between the two oscillators. We compute ρ_A and S_A in closed form.

The reduced density matrix is obtained by tracing out oscillator 2, which for a pure state means integrating the wavefunction against its conjugate over x_2:

$$\rho_A(x_1, x_1') = \int_{-\infty}^{\infty} dx_2 \; \Psi_0(x_1, x_2)\, \Psi_0^*(x_1', x_2). \tag{55.32}$$

The wavefunction Eq. (55.30) is real, and the x_2-integral is Gaussian. Collecting the x_2-dependence, $\Psi_0(x_1, x_2)\Psi_0(x_1', x_2) \propto \exp[-\alpha x_2^2 + \beta(x_1 + x_1')x_2] \times \exp[-\frac{\alpha}{2}(x_1^2 + x_1'^2)]$, and using $\int dx_2\, e^{-\alpha x_2^2 - b x_2} = \sqrt{\pi/\alpha}\, e^{b^2/4\alpha}$ with $b = -\beta(x_1 + x_1')$,

$$\rho_A(x_1, x_1') \;\propto\; \exp\Big[-\tfrac{\alpha}{2}(x_1^2 + x_1'^2) + \tfrac{\beta^2}{4\alpha}(x_1 + x_1')^2\Big]. \tag{55.33}$$

Writing the exponent in the standard form $-\frac{\gamma}{2}(x_1^2 + x_1'^2) + \frac{\delta}{2}\, 2x_1 x_1'$ (the off-diagonal coherence) one reads off

$$\gamma = \alpha - \frac{\beta^2}{2\alpha}, \qquad \delta = \frac{\beta^2}{2\alpha}. \tag{55.34}$$

A Gaussian density matrix of this form is diagonalized by Mehler's formula, whose eigenvalues are geometric, $\lambda_m = (1 - \xi)\, \xi^m$ for $m = 0, 1, 2, \ldots$, with the single ratio

$$\xi = \frac{\delta}{\gamma + \sqrt{\gamma^2 - \delta^2}}, \qquad \text{equivalently} \qquad \xi = \Big(\frac{\sqrt{\omega_+/\omega_-} - \sqrt{\omega_-/\omega_+}}{\sqrt{\omega_+/\omega_-} + \sqrt{\omega_-/\omega_+}}\Big)^2. \tag{55.35}$$

That ρ_A has a thermal (geometric) spectrum is no accident — it is a single-mode thermal state, $\rho_A \propto e^{-\kappa\, a^\dagger a}$ with $e^{-\kappa} = \xi$, the first appearance of the deep link between reduced density matrices and thermality that the Rindler analysis of Section 55.3 will make universal. The von Neumann entropy of a geometric distribution $\lambda_m = (1 - \xi)\xi^m$ is computed once and for all:

$$\begin{aligned} S_A &= -\sum_{m=0}^{\infty} (1 - \xi)\xi^m \big[\log(1 - \xi) + m \log \xi\big] \\ &= -\log(1 - \xi) - \frac{\xi}{1 - \xi} \log \xi = -\log(1 - \xi) - \frac{\xi \log \xi}{1 - \xi}, \end{aligned} \tag{55.36}$$

where we used $\sum_m (1 - \xi)\xi^m = 1$ and $\sum_m m(1 - \xi)\xi^m = \xi/(1 - \xi)$. As the coupling switches off, $k \to 0$ gives $\omega_+ \to \omega_-$, hence $\beta \to 0$, $\delta \to 0$, $\xi \to 0$, and $S_A \to 0$: an unentangled product ground state, consistent with Theorem 55.10(ii). As the coupling grows, $\omega_+/\omega_- \to \infty$, $\xi \to 1$, and $S_A \to \infty$ logarithmically — the two-oscillator avatar of the UV divergence Eq. (55.24), here driven by the unbounded frequency ratio. Summing the result Eq. (55.36) over the boundary modes of a genuine lattice scalar, with the $\omega_\pm$ replaced by the dispersion of the field, is exactly the computation that yields the area law in Section 55.5; the present example fixes its kernel.

55.1.8 Where this leaves us

We now possess a sharp definition. For a region A in a (regulated) quantum field theory in the vacuum, the entanglement entropy is $S_A = -\operatorname{Tr}(\rho_A \log \rho_A)$ with $\rho_A = \operatorname{Tr}_{\bar{A}} |0\rangle\langle 0|$, and we have established its non-negativity, its vanishing exactly on product states, its maximal value, its concavity, the equality $S_A = S_{\bar{A}}$ for the pure vacuum, and the positivity of the mutual information that will serve as our finite diagnostic. We have also been honest about the two obstructions: the area-law divergence Eq. (55.24) and the non-factorization Section 55.1.6, both handled by a UV cutoff that renders ρ_A a bona fide density matrix.

What we do *not* have is any way to evaluate S_A. The operator logarithm in Eq. (55.6) resists every tool of the preceding chapters: it is not a correlation function, not a sum of Feynman diagrams, not a partition function. The escape was foreshadowed by Theorem 55.6: the Rényi entropies involve only the polynomial $\operatorname{Tr} \rho_A^n$, and for integer n this is a product of n copies of ρ_A. The next section shows that ρ_A itself has a Euclidean path-integral representation — a path integral over field configurations on a spacetime cut open along the region A — and that $\operatorname{Tr} \rho_A^n$ is then computed by gluing n such copies cyclically into a path integral on an n-sheeted branched cover of spacetime. Analytically continuing in n and differentiating at $n = 1$ delivers S_A. This is the replica trick, the master computational tool for everything that follows.

55.2 The Replica Trick and the Path-Integral Construction

The previous section left us with a sharp definition and an impasse. For a region A in the regulated vacuum the entanglement entropy is $S_A = -\operatorname{Tr}(\rho_A \log \rho_A)$ with $\rho_A = \operatorname{Tr}_{\bar{A}} |0\rangle\langle 0|$, and we noted (Section 55.1.8) that the operator logarithm answers to none of the tools of the preceding chapters: it is not a correlation function, not a sum of diagrams, not a partition function. What *is* computable is the family of traces $\operatorname{Tr} \rho_A^n$, polynomial in ρ_A, which we already met as the Rényi entropies (Theorem 55.5). This section turns that observation into a complete computational method. The plan has two movements. First, we make the limit $n \to 1$ precise as an analytic continuation, so that knowledge of $\operatorname{Tr} \rho_A^n$ at integer n delivers S_A by a single differentiation — the *replica master formula*. Second, we give ρ_A, and then $\operatorname{Tr} \rho_A^n$, an explicit Euclidean path-integral representation: ρ_A is the vacuum-preparing functional integral with a cut along A, and $\operatorname{Tr} \rho_A^n$ is the partition function of the same theory on an n-sheeted surface obtained by gluing n copies of that cut spacetime cyclically across the cut. The entire intractable spectral problem is thereby converted into the evaluation of partition functions on branched covers — a problem we can attack, in Section 55.4, with conformal maps, and, in Section 55.5, with free-field determinants. The construction is

theory-agnostic; it is the engine for the rest of the chapter.

55.2.1 The replica master formula

The strategy rests on a function of one complex variable. Let ρ_A act on the (regulated, finite-dimensional) Hilbert space $\mathcal{H}_A$ with eigenvalues $\{\lambda_i\}$, $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. Define

$$G(n) \;\equiv\; \operatorname{Tr} \rho_A^n \;=\; \sum_i \lambda_i^n, \tag{55.37}$$

initially for integer $n \geq 1$, where the right-hand side is the spectral representation of the left. We want $S_A = -\sum_i \lambda_i \log \lambda_i$. The bridge is differentiation in n.

Theorem 55.17 (Replica master formula)**.** *Let ρ_A be a density matrix on a finite-dimensional space, with spectrum $\{\lambda_i\}$. The function $G(n) = \sum_i \lambda_i^n$ extends to an analytic function of complex n on the half-plane $\operatorname{Re}(n) > 0$, and the von Neumann entropy is recovered as*

$$S_A \;=\; -\lim_{n\to 1} \frac{\partial}{\partial n} \operatorname{Tr} \rho_A^n \;=\; -\frac{\partial}{\partial n}\Big|_{n=1} \operatorname{Tr} \rho_A^n. \tag{55.38}$$

Equivalently, in terms of the Rényi entropies $S_n(A) = \frac{1}{1-n} \log \operatorname{Tr} \rho_A^n$ of Theorem 55.5,

$$S_A \;=\; \lim_{n\to 1} S_n(A). \tag{55.39}$$

Proof. For each i with $\lambda_i > 0$ write $\lambda_i^n = e^{n \log \lambda_i}$, an entire function of n; eigenvalues with $\lambda_i = 0$ contribute nothing and are dropped. The finite sum Eq. (55.37) is therefore an entire function of n, and in particular analytic on $\operatorname{Re}(n) > 0$. Term-by-term differentiation of a finite sum is legitimate, so

$$\frac{\partial}{\partial n} G(n) \;=\; \sum_i \lambda_i^n \log \lambda_i, \tag{55.40}$$

the differentiation under the (finite) sum being trivially dominated. Evaluating at $n = 1$ and using $\sum_i \lambda_i = 1$,

$$\frac{\partial}{\partial n}\Big|_{n=1} G(n) \;=\; \sum_i \lambda_i \log \lambda_i \;=\; -S_A, \tag{55.41}$$

which is Eq. (55.38). For the Rényi form, set $f(n) = \log G(n)$. Since $G(1) = \operatorname{Tr} \rho_A = 1$, we have $f(1) = 0$ and, by Eq. (55.41), $f'(1) = G'(1)/G(1) = -S_A$. Then $S_n(A) = -f(n)/(n-1)$, and L'Hôpital (or the first-order Taylor expansion $f(n) = -S_A(n-1) + O((n-1)^2)$) gives $\lim_{n\to 1} S_n(A) = -f'(1) = S_A$, which is Eq. (55.39) and reproduces Theorem 55.6. □

The formula Eq. (55.38) is exact and elementary *as a statement about a fixed finite spectrum*. Its power, and its one genuine subtlety, lie in how it is used. In practice one never has the spectrum $\{\lambda_i\}$; one computes $G(n) = \operatorname{Tr}\rho_A^n$ only at *integer* $n \geq 2$, where, as the rest of this section shows, it equals a partition function one can actually evaluate. One then seeks a function $\widetilde{G}(n)$, analytic in a neighbourhood of $n = 1$, that agrees with the computed integer values, and reads off $S_A = -\widetilde{G}'(1)$. The questions this raises are: does such an analytic interpolation exist, and is it unique? Existence is settled case by case — in every computation of this book the integer data assemble into a manifestly analytic closed form (a power of n, a ratio of Gamma functions, an exponential of a polynomial in n). Uniqueness is the content of a classical theorem of complex analysis, which we state because it is exactly the guarantee the method needs.

Theorem 55.18 (Carlson's uniqueness theorem)**.** *Let $h(z)$ be holomorphic on the closed right half-plane $\operatorname{Re}(z) \geq 0$ and of exponential type there, $|h(z)| \leq M\, e^{\tau|z|}$, and suppose that on the imaginary axis the type is controlled below π: $|h(iy)| \leq M\, e^{c|y|}$ with $c < \pi$. If $h(n) = 0$ for every non-negative integer $n = 0, 1, 2, \ldots$, then $h \equiv 0$.*

Proof sketch. This is a Phragmén–Lindelöf argument of the kind reviewed in Section 5.4.1; we indicate its skeleton and use the result. The function $g(z) = h(z)/\sin(\pi z)$ has removable singularities at the integers (each a simple zero of both numerator and denominator), hence extends holomorphically to the right half-plane. One shows, using the lower bound $|\sin(\pi z)| \gtrsim e^{\pi|\operatorname{Im} z|}$ away from the integers together with the hypothesis $c < \pi$, that g is bounded on the boundary of each quadrant and grows slower than any exponential of type π inside; the Phragmén–Lindelöf principle then forces g to be bounded on $\operatorname{Re}(z) \geq 0$, and a bounded holomorphic function whose product with the unbounded $\sin(\pi z)$ stays of exponential type must vanish identically. Hence $h \equiv 0$. The role of the sub-π hypothesis is decisive: the function $\sin(\pi z)$ itself vanishes at every integer, is of exponential type exactly π, and is not identically zero — so the bound $c < \pi$ cannot be relaxed. □

The relevance is immediate. Suppose two analytic functions $\widetilde{G}_1, \widetilde{G}_2$ both interpolate the integer data $G(n)$, $n \geq 1$, and both satisfy the growth bound of Theorem 55.18. Their difference $h = \widetilde{G}_1 - \widetilde{G}_2$ vanishes at all integers $n \geq 1$ and meets the hypotheses, so $h \equiv 0$: the interpolation is unique, and $S_A = -\widetilde{G}'(1)$ is unambiguous. That the bound is met in our setting is not an idle assumption but a consequence of the spectrum living in $[0, 1]$: for $\operatorname{Re}(n) \geq 1$,

$$\left|\operatorname{Tr}\rho_A^n\right| \;=\; \left|\sum_i \lambda_i^n\right| \;\leq\; \sum_i \lambda_i^{\operatorname{Re}(n)} \;\leq\; \sum_i \lambda_i \;=\; 1, \tag{55.42}$$

using $0 \leq \lambda_i \leq 1 \Rightarrow \lambda_i^{\operatorname{Re}(n)} \leq \lambda_i$ for $\operatorname{Re}(n) \geq 1$. Thus $G(n)$ is *bounded* on $\operatorname{Re}(n) \geq 1$ — exponential type zero, well inside Carlson's window — so among

bounded analytic interpolations there is exactly one, and it is the one any honest computation produces.

The continuation is unique only among well-behaved interpolations

Carlson's theorem does not say the integer data determine *every* interpolation; it says they determine the unique interpolation of sub-π exponential type. One can always add $\sin(\pi n)$ times anything analytic without changing the integer values, and such an addition would alter $S_A = -\widetilde{G}'(1)$. The physics input is that the partition functions $\operatorname{Tr}\rho_A^n$, computed at integer n, assemble into a function whose growth in n is controlled (in a regulated theory, by the bound Eq. (55.42); in the continuum, by the area law, which makes $\log \operatorname{Tr}\rho_A^n$ grow at most polynomially in the regularized volume of the entangling region, again of zero exponential type in n). When this holds — as it does in every case treated here — the continuation is forced. There are pathological models (random or chaotic spectra) where the integer Rényi data do not visibly determine the entropy; we will not meet them, but the assumption is real and we flag it rather than hide it.

The remaining task is purely computational: evaluate $\operatorname{Tr}\rho_A^n$ for integer n. We now give it a geometry.

55.2.2 The reduced density matrix as a cut path integral

Everything begins with the Euclidean preparation of the vacuum established in Chapter 10. Recall the picture (Theorem 10.10 and Section 10.6): under the Wick rotation $t \to -i\tau$ fixed in Section 10.6, the Lorentzian weight e^{iS} becomes the positive Euclidean weight e^{-S_E}, and evolving in imaginary time for an infinite duration projects any state of nonzero overlap onto the ground state. Concretely, the ground-state wavefunctional — the amplitude for the field on a constant-time slice to take a prescribed profile — is computed by a Euclidean path integral over the *lower* half of spacetime, $\tau < 0$, with the field pinned to that profile on the slice $\tau = 0$:

$$\Psi_0[\varphi] \;\equiv\; \langle\varphi|0\rangle \;=\; \frac{1}{\sqrt{Z_1}} \int_{\substack{\phi(\tau<0)\\ \phi(\tau=0,\mathbf{x})=\varphi(\mathbf{x})}} \mathcal{D}\phi \; e^{-S_E[\phi]}, \tag{55.43}$$

where $|\varphi\rangle$ is the eigenstate of the field operator $\hat{\phi}(\mathbf{x})$ on the slice with eigenvalue $\varphi(\mathbf{x})$, and $Z_1 = \int \mathcal{D}\phi\, e^{-S_E}$ over all of Euclidean spacetime $\mathbb{R}^d$ normalizes the state, $\langle 0|0\rangle = 1$. The conjugate amplitude $\langle 0|\varphi'\rangle = \overline{\Psi_0[\varphi']}$ is the same integral over the *upper* half $\tau > 0$ with the field pinned to φ' at $\tau = 0$. Reality of the Euclidean action for a real scalar makes these complex conjugates of one another; we keep the two halves distinct because their boundary slices will become the two indices of a density matrix.

The pure-state density matrix of the vacuum, $\rho = |0\rangle\langle 0|$, has matrix elements

$$\langle \varphi_+ | \rho | \varphi_- \rangle \;=\; \langle \varphi_+ | 0 \rangle \, \langle 0 | \varphi_- \rangle \;=\; \Psi_0[\varphi_+] \, \overline{\Psi_0[\varphi_-]}. \tag{55.44}$$

Geometrically this glues the lower-half integral preparing $\Psi_0[\varphi_+]$ to the upper-half integral preparing $\overline{\Psi_0[\varphi_-]}$ along the common slice $\tau = 0$ — but *without* integrating over the slice value, since the two halves carry the independent boundary data φ_+ (the slice value of the lower-half field) and φ_- (the slice value of the upper-half field). The result is a path integral over all of $\mathbb{R}^d$ in which the field is permitted a discontinuity across the entire slice $\tau = 0$: it tends to φ_+ from below ($\tau \to 0^-$) and to φ_- from above ($\tau \to 0^+$). We adopt the bra/ket-to-lip assignment of Section 55.3: the value approached from $\tau \to 0^+$ (the upper lip) is the ket index, the value approached from $\tau \to 0^-$ (the lower lip) is the bra index.

Now reduce to the region A. The reduced density matrix $\rho_A = \mathrm{Tr}_{\bar{A}}\, \rho$ traces over the field degrees of freedom in the complement $\bar{A}$ on the slice. In the field representation, tracing over $\bar{A}$ means setting the bra and ket boundary values *equal on* $\bar{A}$ and integrating over their common value there, while leaving them free and distinct on A:

$$\left\langle \varphi_+^A \middle| \rho_A \middle| \varphi_-^A \right\rangle = \int \mathcal{D}\chi_{\bar{A}} \; \left\langle \varphi_+^A \otimes \chi_{\bar{A}} \middle| \rho \middle| \varphi_-^A \otimes \chi_{\bar{A}} \right\rangle. \tag{55.45}$$

The integral over $\chi_{\bar{A}}$ sews the upper and lower lips of the slice *shut* everywhere on $\bar{A}$: above and below the complement the field is now continuous and unconstrained, exactly as in the bulk. Only on A does the slit remain open, carrying the boundary data φ_+^A on its upper lip and φ_-^A on its lower lip. The net statement is the central representation of this section.

Proposition 55.19 (Path-integral form of ρ_A)**.** *The matrix elements of the reduced density matrix of a region A in the vacuum are computed by the Euclidean path integral over* $\mathbb{R}^d$ cut open along A*:*

$$\left\langle \varphi_+^A \middle| \rho_A \middle| \varphi_-^A \right\rangle = \frac{1}{Z_1} \int_{\substack{\phi \to \varphi_+^A \ \text{on } A,\ \tau \to 0^+ \\ \phi \to \varphi_-^A \ \text{on } A,\ \tau \to 0^-}} \mathcal{D}\phi \; e^{-S_E[\phi]}, \tag{55.46}$$

where the field is integrated freely over all of $\mathbb{R}^d$ except along the slit $\{\tau = 0,\ \mathbf{x} \in A\}$, across which it is required to approach the prescribed profiles $\varphi_\pm^A$ from above and below. The normalization $Z_1 = \int \mathcal{D}\phi\, e^{-S_E}$ (the partition function on uncut $\mathbb{R}^d$) enforces $\mathrm{Tr}_A\, \rho_A = 1$.

Proof. Substituting Eq. (55.44) into Eq. (55.45), the matrix element is the product $\Psi_0[\varphi_+^A, \chi_{\bar{A}}]\, \overline{\Psi_0[\varphi_-^A, \chi_{\bar{A}}]}$ integrated over $\chi_{\bar{A}}$. By Eq. (55.43) the first factor is a lower-half ($\tau < 0$) integral with slice data (φ_+^A on A, $\chi_{\bar{A}}$ on $\bar{A}$), the second an upper-half ($\tau > 0$) integral with slice data (φ_-^A on A, $\chi_{\bar{A}}$ on $\bar{A}$). The $\chi_{\bar{A}}$-integration identifies the slice value of the lower-half field with that of the

upper-half field on $\bar{A}$ and sums over it; this is precisely the statement that, on $\bar{A}$, the two half-space integrals are joined into one integral over $\mathbb{R}^d$ with the field continuous across $\tau = 0$. On A no such identification occurs: the lower-half field ends at φ_+^A and the upper-half field ends at φ_-^A, and these remain the free boundary data of the combined integral, imposed respectively from $\tau \to 0^-$ (lower lip) and $\tau \to 0^+$ (upper lip). The combined domain is therefore $\mathbb{R}^d$ minus the slit on A, with the stated boundary behaviour, and the two normalizing $1/\sqrt{Z_1}$ factors multiply to $1/Z_1$. Taking the trace $\mathrm{Tr}_A\, \rho_A$ sets $\varphi_+^A = \varphi_-^A$ and integrates, closing the slit on A as well; the field is then continuous everywhere and the numerator becomes $\int_{\mathbb{R}^d} \mathcal{D}\phi\, e^{-S_E} = Z_1$, so $\mathrm{Tr}_A\, \rho_A = Z_1/Z_1 = 1$. □

ρ_A is a geometric object

Equation Eq. (55.46) is the conceptual heart of the chapter. It says the reduced density matrix is *not* an abstract operator to be diagonalized but a piece of spacetime with a slit: the bra and ket of ρ_A are the two lips of a cut along A, and "acting with ρ_A" means filling in the Euclidean field between them. Two consequences foreshadow later sections. First, multiplying density matrices will mean stacking such slit-spacetimes lip-to-lip — the construction of the next subsection. Second, when A is a half-space the slit is a half-plane and the surrounding geometry is flat space in polar coordinates, so the operator that propagates the field from the lower lip to the upper lip is angular (rotation) evolution; that is the observation Section 55.3 turns into vacuum thermality.

55.2.3 $\mathrm{Tr}\,\rho_A^n$ on the branched cover

We now compute the integer powers by composing copies of Eq. (55.46). The trace of the n-th power is, in the field basis,

$$\mathrm{Tr}\,\rho_A^n = \int \prod_{j=1}^{n} \mathcal{D}\varphi^{(j)}\ \left\langle \varphi^{(1)} \middle| \rho_A \middle| \varphi^{(2)} \right\rangle \left\langle \varphi^{(2)} \middle| \rho_A \middle| \varphi^{(3)} \right\rangle \cdots \left\langle \varphi^{(n)} \middle| \rho_A \middle| \varphi^{(1)} \right\rangle, \quad (55.47)$$

where each $\varphi^{(j)}$ is a field profile on A, the matrix product threads the bra of one factor into the ket of the next, and the trace closes the chain by identifying the last bra with the first ket (the index $\varphi^{(1)}$ appears at both ends). This is ordinary matrix multiplication followed by a trace, written out in a continuous basis; the only content beyond Theorem 55.19 is the bookkeeping of which lip is glued to which.

Read each factor through Eq. (55.46). The factor $\left\langle \varphi^{(j)} \middle| \rho_A \middle| \varphi^{(j+1)} \right\rangle$ is a copy of cut $\mathbb{R}^d$ — call it *sheet* j — whose slit on A has its upper lip ($\tau \to 0^+$) pinned to $\varphi^{(j)}$ (the ket) and its lower lip ($\tau \to 0^-$) pinned to $\varphi^{(j+1)}$ (the bra). The

integrations $\int \mathcal{D}\varphi^{(j)}$ in Eq. (55.47) then enforce, for each j, the identification

$$\underbrace{\varphi^{(j)}}_{\text{lower lip of sheet } j-1} \quad = \quad \underbrace{\varphi^{(j)}}_{\text{upper lip of sheet } j} \quad , \tag{55.48}$$

that is: *the lower lip of sheet $j-1$ is sewn to the upper lip of sheet j*, and they share the field profile $\varphi^{(j)}$ which is then integrated over. Equivalently, going the other way around, the lower lip of sheet j is sewn to the upper lip of sheet $j+1$. The trace makes the chain cyclic: $\varphi^{(n+1)} \equiv \varphi^{(1)}$, so the lower lip of sheet n is sewn to the upper lip of sheet 1, closing the stack into a ring. Each sewing integrates over the common profile, so the field is continuous across every join; the only places where the field can "change sheets" are the lips of the slit on A, and the only places where it cannot be defined single-valuedly are the endpoints of A — the boundary ∂A — around which one passes from one sheet to the next.

The object so constructed is a single connected manifold: n copies of $\mathbb{R}^d$, each slit along A, glued cyclically lip-to-lip. It is an *n-sheeted branched cover* of $\mathbb{R}^d$, branched over the entangling surface ∂A; in two dimensions, where ∂A is a set of points, it is the Riemann surface $\mathcal{R}_n$ that reappears in Section 55.4. Away from ∂A the cover is locally identical to $\mathbb{R}^d$ (the field is smooth across each glued slit, the join being invisible once the integration is done); only on encircling a point of ∂A does one wind through all n sheets before returning. Figure 55.2 shows the gluing for $n = 3$ in two dimensions.

The path integral over this glued manifold, with the same local Euclidean action S_E on each sheet, is by definition a partition function. Denote it $Z_n \equiv \int_{\mathcal{R}_n} \mathcal{D}\phi\, e^{-S_E[\phi]}$. Collecting the n numerator factors from Eq. (55.46) (one per sheet) and the n normalizing denominators Z_1, Eq. (55.47) reads

$$\boxed{\operatorname{Tr} \rho_A^n \;=\; \frac{Z_n}{Z_1^n}\,, \qquad n = 1, 2, 3, \dots} \tag{55.49}$$

This is the master computational identity. The left side is the spectral quantity Eq. (55.37) we cannot compute directly; the right side is a ratio of ordinary Euclidean partition functions — one on the n-sheeted cover, one on flat space raised to the n-th power — which we can. Two checks fix the conventions. At $n = 1$ the "cover" is a single sheet with its slit sewn shut (the trace identifies the two lips), i.e. uncut $\mathbb{R}^d$, so Z_1 is literally the flat-space partition function and $\operatorname{Tr} \rho_A = Z_1/Z_1 = 1$, as it must. The denominator Z_1^n is not optional: Z_n alone is dimensionful and cutoff-dependent (it is an unnormalized functional integral), and only the ratio Eq. (55.49), in which n powers of the single-sheet normalization are divided out, equals the dimensionless, physically meaningful trace bounded by Eq. (55.42).

It is worth pausing on why the gluing reproduces the operator product and not something else. Operator multiplication $\rho_A \cdot \rho_A$ contracts a bra of the first

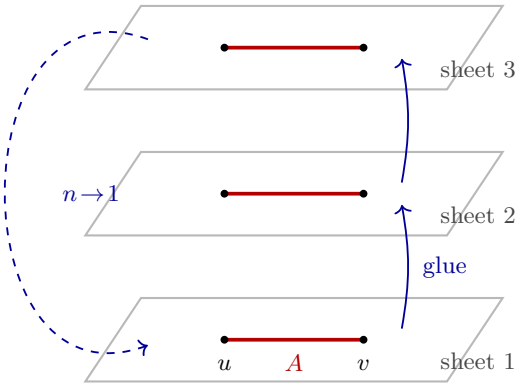


Figure 55.2: The branched cover $\mathcal{R}_n$ for $n = 3$ (two-dimensional case, interval $A = [u, v]$). Each sheet is a copy of cut Euclidean spacetime; the lower lip of the slit on sheet j is sewn to the upper lip on sheet $j + 1$ (solid arrows), and the lower lip of the top sheet closes back onto the upper lip of the bottom sheet (dashed). The field is continuous across every join; multivaluedness survives only at the branch points u, v $(= \partial A)$, around which a circuit passes through all n sheets. The partition function on $\mathcal{R}_n$ is Z_n, and $\operatorname{Tr} \rho_A^n = Z_n / Z_1^n$.

factor with a ket of the second over the A *degrees of freedom only* — precisely the lip-to-lip sewing on the slit, Eq. (55.48). The complement $\bar{A}$ was already traced out within each ρ_A (the slit was closed on $\bar{A}$ in Theorem 55.19), so on $\bar{A}$ each sheet is simply a full copy of $\mathbb{R}^d$ with the field continuous across $\tau = 0$; the sheets touch one another only along A. This is exactly why the branch locus is ∂A and not all of $\tau = 0$: the cover is ramified only where the slits are, and the slits live on A.

55.2.4 Rényi entropies as free energies on the replica geometry

Identity Eq. (55.49) translates the entropy hierarchy directly into the language of free energies. Taking the logarithm,

$$\log \operatorname{Tr} \rho_A^n \;=\; \log Z_n - n \log Z_1, \tag{55.50}$$

and substituting into the definition $S_n(A) = \frac{1}{1-n} \log \operatorname{Tr} \rho_A^n$ (Eq. (55.7)) gives the Rényi entropy as a difference of free energies.

Proposition 55.20 (Rényi entropy as a replica free energy)**.** *With $F_n \equiv -\log Z_n$ the Euclidean free energy of the theory on the n-sheeted cover $\mathcal{R}_n$, the Rényi entropies of A are*

$$S_n(A) \;=\; \frac{1}{1-n}\big(\log Z_n - n \log Z_1\big) \;=\; \frac{1}{1-n}\,\big(nF_1 - F_n\big), \tag{55.51}$$

and the von Neumann entropy is the replica limit

$$S_A \;=\; \lim_{n\to 1}\frac{1}{1-n}\big(\log Z_n - n\log Z_1\big) \;=\; -\frac{\partial}{\partial n}\Big|_{n=1}\big(\log Z_n - n\log Z_1\big). \tag{55.52}$$

Proof. Equation Eq. (55.51) is Eq. (55.50) divided by $1-n$, with $F_n = -\log Z_n$. For Eq. (55.52), apply the replica master formula Eq. (55.38) in its Rényi form Eq. (55.39): write $g(n) \equiv \log Z_n - n\log Z_1 = \log \operatorname{Tr}\rho_A^n$, which vanishes at $n=1$ since $\operatorname{Tr}\rho_A = 1$. Then $S_n(A) = -g(n)/(n-1)$ and, by the same L'Hôpital step as in Theorem 55.17, $\lim_{n\to 1} S_n(A) = -g'(1) = -\partial_n(\log Z_n - n\log Z_1)|_{n=1}$, which is Eq. (55.52). □

The form Eq. (55.52) is the one actually used in computation: one finds $\log Z_n$ as an explicit analytic function of n — by mapping $\mathcal{R}_n$ to a smooth space (the conformal route of Section 55.4) or by evaluating the Gaussian functional determinant on $\mathcal{R}_n$ (the free-field route of Section 55.5, which rests on the determinant technology of Section E.2) — and then differentiates at $n=1$. The differentiation $\partial_n \log Z_n$ probes the response of the free energy to opening up an infinitesimal conical excess at ∂A: at $n=1$ the cover is flat, and $\partial_n|_{n=1}$ measures the leading curvature cost of replication, localized at the entangling surface. This is the precise sense in which the entanglement entropy is geometric data attached to ∂A, and it is the seed of the area law.

Replica index as inverse temperature

The combination $g(n) = \log Z_n - n\log Z_1$ is structurally a thermodynamic free energy with n playing the role of an inverse temperature: Z_n is the partition function on a space whose extent in the replica (angular) direction is n times the original, and $S_n = \frac{1}{1-n}\log(Z_n/Z_1^n)$ is the analogue of the operator $(\beta\,\partial_\beta - 1)$ acting on a free energy, which in ordinary thermodynamics returns an entropy. The replica limit $n\to 1$ is the limit of unit "temperature" in the replica direction. This is not a loose analogy: when A is a half-space the replica direction *is* the Euclidean time circle and n *is* literally proportional to an inverse temperature, $\beta = 2\pi n$, the content of the Unruh effect derived in Section 55.3. The angular structure of the cut path integral, visible already in Eq. (55.46), is what Section 55.3 reads as thermality.

55.2.5 The twist-operator reformulation

The branched-cover partition function Eq. (55.49) can be repackaged so that the awkward multivalued geometry is replaced by an ordinary correlation function on a single, smooth copy of spacetime. The price is to work not in the original theory $\mathcal{C}$ but in its n-fold replica $\mathcal{C}^{\otimes n}$, and to introduce local operators that carry

the branch structure. This *orbifold* or *twist-field* construction is the form in which the replica trick is most powerful in two dimensions, where it reduces the entire single-interval computation to fixing one operator dimension.

The idea is to undo the gluing Eq. (55.48) by introducing n independent copies of the field living on a single sheet, and to encode the cyclic identification as a boundary condition imposed by operator insertions. Consider n non-interacting copies of the theory on one copy of $\mathbb{R}^d$, with fields $\Phi = (\phi_1, \dots, \phi_n)$, action $S_E^{(n)}[\Phi] = \sum_{j=1}^n S_E[\phi_j]$, and partition function Z_1^n (the copies decouple, so the partition function factorizes). The replicated theory has a manifest global symmetry: the cyclic permutation

$$\sigma : \quad \phi_j(x) \longmapsto \phi_{j+1}(x), \qquad \phi_{n+1} \equiv \phi_1, \tag{55.53}$$

generating a $\mathbb{Z}_n$ subgroup of the permutations of the copies. The branched cover $\mathcal{R}_n$ is reconstructed from $\mathcal{C}^{\otimes n}$ by demanding that, as one crosses the slit on A, the j-th copy continue not into itself but into its cyclic neighbour. To see this, restrict attention to the two-dimensional case with A a single interval $[u, v]$, so $\partial A = \{u, v\}$, and recall the gluing: crossing the cut on sheet j of $\mathcal{R}_n$ lands one on sheet $j + 1$. In the replicated theory on a single plane, this is the statement that the field configuration is single-valued *except* along the cut $[u, v]$, where it satisfies the twisted boundary condition

$$\phi_j(x, \tau \to 0^-) \;=\; \phi_{j+1}(x, \tau \to 0^+), \qquad x \in [u, v], \tag{55.54}$$

which is exactly the gluing rule Eq. (55.48) with the sheets relabelled as copies. A field obeying Eq. (55.54) on the segment $[u, v]$ and continuous elsewhere is the same data as a single field on $\mathcal{R}_n$: the n-component object on one plane with a cut is the unfolded picture of the one-component object on n glued planes.

The cut in Eq. (55.54) is not an intrinsic feature of the configuration; it can be moved by a field redefinition, and only its endpoints are physical. The endpoints are the loci where the monodromy lives: encircling u once takes $\phi_j \mapsto \phi_{j+1}$ (one cannot return to the original copy without going around), and encircling v undoes it, $\phi_j \mapsto \phi_{j-1}$, so that a loop enclosing both endpoints sees trivial monodromy and the configuration is single-valued at infinity. An operator insertion that imposes precisely this monodromy on the fields encircling its location is, by definition, a *twist operator*.

Definition 55.21 (Twist operators)**.** In the replicated theory $\mathcal{C}^{\otimes n}$, the *twist operator* $\sigma_n(w)$ is the local operator defined by the monodromy condition that any field ϕ_j transported once counterclockwise around the point w returns as ϕ_{j+1} (the action of the generator Eq. (55.53)); the *anti-twist* $\tilde{\sigma}_n(w)$ imposes the inverse monodromy $\phi_j \mapsto \phi_{j-1}$. Operationally, inserting $\sigma_n(u)\,\tilde{\sigma}_n(v)$ into a correlator of $\mathcal{C}^{\otimes n}$ instructs the path integral to be taken over field configurations with a branch cut running from u to v across which the copies are cyclically permuted, i.e. over exactly the configurations Eq. (55.54).

With this definition the branched-cover partition function becomes a twist two-point function in the replicated theory. The argument is a direct comparison of path-integral domains.

Proposition 55.22 (Branched cover as a twist correlator)**.** *For a single interval $A = [u, v]$ in two dimensions,*

$$\operatorname{Tr} \rho_A^n \;=\; \frac{Z_n}{Z_1^n} \;=\; \langle\, \sigma_n(u)\, \tilde{\sigma}_n(v) \,\rangle_{\mathcal{C}^{\otimes n}}, \tag{55.55}$$

where the correlator is normalized by the replicated partition function Z_1^n, so that $\langle \sigma_n \tilde{\sigma}_n \rangle = Z[\text{cut configs}]/Z[\text{all configs}]$.

Proof. The numerator Z_n is the path integral of the single field ϕ over the cover $\mathcal{R}_n$. Unfold $\mathcal{R}_n$ by labelling, at each point x of the base plane, the values of ϕ on the n sheets above x as an n-component field $\Phi(x) = (\phi_1(x), \dots, \phi_n(x))$. Because the local action on each sheet is the same S_E, the unfolded action is $\sum_j S_E[\phi_j] = S_E^{(n)}[\Phi]$, the replicated action. The cyclic gluing of $\mathcal{R}_n$ becomes, in the unfolded variables, exactly the twisted boundary condition Eq. (55.54) along $[u, v]$, with the field single-valued away from the cut. By Theorem 55.21 the path integral over $\mathcal{C}^{\otimes n}$ configurations subject to this monodromy is, by construction, the insertion of $\sigma_n(u)\, \tilde{\sigma}_n(v)$ into the replicated path integral:

$$Z_n = \int_{\substack{\text{cut configs} \\ \text{on } [u,v]}} \mathcal{D}\Phi \; e^{-S_E^{(n)}[\Phi]} = \Big(\int \mathcal{D}\Phi \, e^{-S_E^{(n)}} \Big) \, \langle \sigma_n(u)\, \tilde{\sigma}_n(v) \rangle_{\mathcal{C}^{\otimes n}}. \tag{55.56}$$

The prefactor is the unconstrained replicated partition function, which factorizes across the decoupled copies into $\prod_{j=1}^n Z_1 = Z_1^n$. Dividing, $Z_n / Z_1^n = \langle \sigma_n(u)\, \tilde{\sigma}_n(v) \rangle_{\mathcal{C}^{\otimes n}}$, and the left equality is Eq. (55.49). That the inserted operators are local and carry the cyclic monodromy is the defining property Theorem 55.21; the choice of where the cut runs between u and v is a gauge of the field relabelling and does not affect the correlator, confirming that only the endpoints carry physical data. □

The gain is decisive. The left side of Eq. (55.55) required understanding a singular Riemann surface; the right side is an ordinary two-point function of local operators in a perfectly smooth theory on the plane. If the underlying theory is conformal, σ_n and $\tilde{\sigma}_n$ are (as we verify in Section 55.4) local primary operators of the replicated CFT $\mathcal{C}^{\otimes n}$, whose central charge is nc, and the two-point function of primaries is fixed by conformal invariance up to a single constant and a single number — the scaling dimension of the twist field. The entire single-interval computation then collapses to determining that one dimension.

The twist dimension is fixed in Section 55.4, not here

We have *defined* σ_n by its monodromy and *established* that $\operatorname{Tr}\rho_A^n = \langle \sigma_n(u)\,\tilde{\sigma}_n(v)\rangle$, but we have not yet computed its scaling dimension. That computation is carried out in Section 55.4.2 by matching the stress-tensor expectation value on $\mathcal{R}_n$ (obtained by uniformizing the cover to the plane) against the conformal Ward identity, and the answer is the celebrated $h_n = \frac{c}{24}\left(n - \frac{1}{n}\right)$ of Theorem 55.31. The present section owes only the *structure* Eq. (55.55); Section 55.4 supplies the number and reads off the entropy.

55.2.6 The method exercised on a thermal spectrum

To see the entire pipeline turn — integer data, analytic continuation, differentiation at $n = 1$ — on a case where the spectrum is known in closed form, we revisit the geometric (thermal) spectrum produced by the two coupled oscillators of Theorem 55.16. That example is no toy: its single-mode thermal spectrum is the exact kernel of the free-field entanglement entropy assembled in Section 55.5, so demonstrating the replica continuation on it is demonstrating it on the building block of the area law.

Worked Example 55.23 (Replica continuation of a geometric spectrum)**.** Let ρ_A have the geometric (thermal single-mode) spectrum $\lambda_m = (1-\xi)\,\xi^{\,m}$, $m = 0, 1, 2, \ldots$, with $0 \le \xi < 1$, as found in Eq. (55.36). We compute S_A by the replica route and check it against the direct value there.

Integer data. For integer $n \ge 1$ the trace is a geometric series, summed exactly:

$$\operatorname{Tr}\rho_A^n = \sum_{m=0}^{\infty} (1-\xi)^n\,\xi^{\,nm} = (1-\xi)^n \sum_{m=0}^{\infty} (\xi^n)^m = \frac{(1-\xi)^n}{1-\xi^{\,n}}. \tag{55.57}$$

At $n = 1$ this gives $(1-\xi)/(1-\xi) = 1$, the correct normalization. The Rényi entropies follow directly from Eq. (55.7):

$$S_n(A) = \frac{1}{1-n}\log\frac{(1-\xi)^n}{1-\xi^{\,n}} = \frac{1}{1-n}\Big[\,n\log(1-\xi) - \log(1-\xi^{\,n})\,\Big]. \tag{55.58}$$

Analytic continuation. The right-hand side of Eq. (55.57) is already a single analytic function of n: $(1-\xi)^n = e^{n\log(1-\xi)}$ is entire, and $1-\xi^{\,n} = 1 - e^{n\log\xi}$ is entire with no zeros for $\operatorname{Re}(n) > 0$ (since $|\xi^n| = \xi^{\operatorname{Re} n} < 1$ there), so $\operatorname{Tr}\rho_A^n$ is holomorphic on $\operatorname{Re}(n) > 0$. Moreover it is bounded by 1 on $\operatorname{Re}(n) \ge 1$, in line with Eq. (55.42), so by Theorem 55.18 this *is* the unique admissible continuation of the integer data — there is no ambiguity to resolve. We may therefore differentiate Eq. (55.57) in n and evaluate at $n = 1$.

The replica limit. Set $g(n) = \log \operatorname{Tr} \rho_A^n = n \log(1-\xi) - \log(1-\xi^n)$, which vanishes at $n = 1$. Differentiating, and using $\partial_n \xi^n = \xi^n \log \xi$,

$$g'(n) = \log(1-\xi) - \frac{-\xi^n \log \xi}{1-\xi^n} = \log(1-\xi) + \frac{\xi^n \log \xi}{1-\xi^n}, \tag{55.59}$$

and at $n = 1$,

$$g'(1) = \log(1-\xi) + \frac{\xi \log \xi}{1-\xi}. \tag{55.60}$$

By the replica master formula in the free-energy form Eq. (55.52), $S_A = -g'(1)$, so

$$S_A = -\log(1-\xi) - \frac{\xi \log \xi}{1-\xi}, \tag{55.61}$$

which is exactly the direct result Eq. (55.36). The replica machinery has reproduced the von Neumann entropy of the geometric spectrum without ever diagonalizing or summing $-\sum_m \lambda_m \log \lambda_m$ term by term: it summed the polynomial $\sum_m \lambda_m^n$, continued in n, and differentiated once.

Reading the limits. As $\xi \to 0$ (decoupled oscillators, product ground state) Eq. (55.61) gives $S_A \to 0$; as $\xi \to 1^-$ (strong coupling, the UV-divergent regime) it diverges as $S_A \sim -\log(1-\xi) \to \infty$. Both match Theorem 55.16. The Rényi family Eq. (55.58) interpolates smoothly: for instance $S_2(A) = -\log \frac{(1-\xi)^2}{1-\xi^2} = -\log \frac{1-\xi}{1+\xi} = \log \frac{1+\xi}{1-\xi}$, the purity diagnostic, finite for all $\xi < 1$ and reducing to Eq. (55.61) as $n \to 1$. This single mode, summed over the boundary modes of a lattice field with the ξ of each mode set by the dispersion relation, is precisely the sum that yields the area law in Section 55.5; the replica continuation we have just exercised is the operation performed mode-by-mode there.

55.2.7 What we now compute with

We have converted the entanglement entropy from an operator logarithm into geometry. The replica master formula Eq. (55.38), legitimized by Carlson's uniqueness (Theorem 55.18) whenever the Rényi data are of controlled growth, reduces S_A to a single n-derivative of $\operatorname{Tr} \rho_A^n$ at $n = 1$. The reduced density matrix is the cut path integral Eq. (55.46); its integer powers are the branched-cover partition functions Eq. (55.49), $\operatorname{Tr} \rho_A^n = Z_n / Z_1^n$, equivalently the replica free energies Eq. (55.51); and for a single interval the cover is traded for a twist two-point function Eq. (55.55) in the replicated theory. Two routes to Z_n now open. In Section 55.4 we evaluate the twist correlator by conformal symmetry, fixing the twist dimension from the central charge and extracting the celebrated logarithm of two-dimensional CFT. In Section 55.5 we evaluate Z_n as a free-field functional determinant on the cover and recover the area law for the scalar of ϕ^4 theory. And the angular structure of the cut geometry Eq. (55.46), with the replica index playing the part of an inverse temperature, is the doorway to the

very next section: for a half-space the cut path integral is rotation by an angle, and Section 55.3 reads that rotation as the thermality of the vacuum seen by an accelerated observer.

55.3 The Rindler Wedge, the Modular Hamiltonian, and Vacuum Thermality

The replica construction of Section 55.2 reduced every entanglement quantity to a Euclidean path integral, and in doing so it taught us to read the reduced density matrix ρ_A geometrically: ρ_A is the path integral on Euclidean spacetime *cut open* along the region A, with the field on the upper lip of the cut playing the ket index and the field on the lower lip playing the bra index. For a generic region A the geometry of that cut is hopelessly complicated and the path integral cannot be done. There is, however, one region for which the geometry is as simple as it can possibly be: a flat half-space, $A = \{x^1 > 0\}$ at a fixed time, whose boundary — the *entangling surface* — is an infinite flat plane. Here the cut is a half-hyperplane, and the Euclidean spacetime around it is just flat $\mathbb{R}^d$ presented in polar coordinates about that plane. The angular coordinate sweeps from one lip of the cut to the other, and the operator that generates angular evolution is something we already constructed, long ago and for entirely different reasons: the Noether charge of Lorentz boosts (Theorem 7.27 and Eq. (7.107)). Out of this single observation the entire physical content of this section follows — that the vacuum, restricted to a half-space, is an exactly thermal state with respect to the boost; that a uniformly accelerated observer therefore finds himself bathed in radiation at a temperature fixed by his acceleration; and that the modular Hamiltonian, the object $-\log \rho_A$ that controls all of entanglement's fine structure, is in this case a known, local, geometric operator. This is the exactly solvable seed from which the rest of the chapter's structure grows.

55.3.1 The half-space, the Rindler wedge, and the cut path integral

We work in d-dimensional Minkowski spacetime with the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,\dots,-)$ and natural units $\hbar = c = 1$, restoring dimensionful constants only once, for a physical estimate. Coordinates are $x^\mu = (t, x^1, \mathbf{x}_\perp)$, where $\mathbf{x}_\perp = (x^2, \dots, x^{d-1})$ are the $d-2$ transverse directions. Fix the time slice $t = 0$ and split it across the plane $x^1 = 0$:

$$A = \{\,(0, x^1, \mathbf{x}_\perp) : x^1 > 0\,\}, \qquad B = \{\,(0, x^1, \mathbf{x}_\perp) : x^1 < 0\,\}. \tag{55.62}$$

The entangling surface is the codimension-two plane $\Sigma = \{x^1 = 0\}$ at $t = 0$, with transverse area $\mathcal{A}_\perp = \int \mathrm{d}^{d-2}x_\perp$ (formally infinite; we keep it as an overall factor).

The domain of dependence of A — the set of spacetime points whose entire past and future are determined by initial data on A — is the *right Rindler wedge*.

Definition 55.24 (Right Rindler wedge)**.** The right Rindler wedge is the open region

$$W_R = \{\, x^\mu : \; x^1 > |t| \,\} = \{\, x^1 > 0, \; -x^1 < t < x^1 \,\}, \tag{55.63}$$

the causal completion of the half-space A of Eq. (55.62). The left wedge $W_L = \{-x^1 > |t|\}$ is the causal completion of B.

The wedge is distinguished by a symmetry. The one-parameter group of Lorentz boosts in the x^1 direction,

$$\begin{pmatrix} t \\ x^1 \end{pmatrix} \longmapsto \Lambda_\eta \begin{pmatrix} t \\ x^1 \end{pmatrix}, \qquad \Lambda_\eta = \begin{pmatrix} \cosh\eta & \sinh\eta \\ \sinh\eta & \cosh\eta \end{pmatrix}, \qquad \eta \in \mathbb{R}, \tag{55.64}$$

maps W_R to itself: the boost slides points along the hyperbolae $(x^1)^2 - t^2 = \text{const} > 0$ that foliate the wedge, leaving the wedge invariant as a set. It is the unique continuous spacetime symmetry that does so while fixing the entangling plane Σ pointwise. This is the symmetry the modular Hamiltonian will turn out to implement.

We now write down ρ_A. The starting point, established in Section 55.2 and resting on the Euclidean preparation of the vacuum from Section 10.6, is the path-integral form of the vacuum density matrix. The Euclidean rotation $t \to -i\tau$ fixed in Section 10.6 sends Minkowski spacetime to $\mathbb{R}^d$ with coordinates $(\tau, x^1, \mathbf{x}_\perp)$ and weight e^{-S_E}; for the free scalar of ϕ^4 theory (Theorem 9.6) the Euclidean action is

$$S_E = \int \mathrm{d}\tau \, \mathrm{d}^{d-1}x \; \tfrac{1}{2}\big[(\partial_\tau\phi)^2 + (\partial_1\phi)^2 + (\nabla_\perp\phi)^2 + m^2\phi^2\big]. \tag{55.65}$$

The vacuum wavefunctional is the half-space path integral $\Psi_0[\varphi] = \langle\varphi|0\rangle = Z^{-1/2}\int_{\phi(\tau=0)=\varphi,\ \tau<0}\mathcal{D}\phi\, e^{-S_E}$, with Z the full-space partition function. The pure-state density matrix $\rho = |0\rangle\langle 0|$ then has matrix element $\langle\varphi_-|\rho|\varphi_+\rangle = \Psi_0[\varphi_-]\,\Psi_0^*[\varphi_+]$, the product of the lower-half-space integral ($\tau < 0$, with boundary value φ_- imposed as $\tau \to 0^-$) and the upper-half-space integral ($\tau > 0$, with boundary value φ_+ imposed as $\tau \to 0^+$). Joined, it is the path integral over *all* of $\mathbb{R}^d$ with a cut along the slice $\tau = 0$, the field tending to φ_- below the cut and to φ_+ above it. We follow exactly the bra/ket assignment of Section 55.2: the field just below the cut indexes the bra, the field just above indexes the ket.

To form $\rho_A = \mathrm{Tr}_B\,\rho$ we identify and integrate over the field on B. Concretely, on the complement region $x^1 < 0$ we set the two boundary values equal, $\varphi_-(\mathbf{x}) = \varphi_+(\mathbf{x})$ for $x^1 < 0$, and integrate freely; on A the two boundary values are left distinct as the surviving bra and ket indices. The effect is to *heal* the cut

everywhere on B — there the field is continuous across $\tau = 0$ and is simply integrated — while the cut survives only on A. Therefore

$$\left\langle \varphi_-^A \middle| \rho_A \middle| \varphi_+^A \right\rangle = \frac{1}{Z} \int_{\substack{\phi \to \varphi_-^A \text{ as } \tau \to 0^-,\, x^1 > 0 \\ \phi \to \varphi_+^A \text{ as } \tau \to 0^+,\, x^1 > 0}} \mathcal{D}\phi \; e^{-S_E}, \tag{55.66}$$

the path integral on $\mathbb{R}^d$ cut open only along the half-plane $\{\tau = 0,\ x^1 > 0\}$. Everything that follows is the evaluation of this object.

55.3.2 Polar coordinates and the angular evolution operator

The cut in Eq. (55.66) is the half-plane $\{\tau = 0,\, x^1 > 0\}$; it terminates on the entangling surface $\{\tau = 0,\, x^1 = 0\} = \Sigma$. The geometry begs for polar coordinates in the (τ, x^1) plane centered on Σ:

$$x^1 = r\cos\theta, \qquad \tau = r\sin\theta, \qquad r \geq 0, \quad \theta \in [0, 2\pi), \tag{55.67}$$

with $\mathbf{x}_\perp$ untouched. The flat measure is $\mathrm{d}\tau\, \mathrm{d}x^1 = r\, \mathrm{d}r\, \mathrm{d}\theta$, and the half-plane cut $\{\tau = 0,\, x^1 > 0\}$ sits at $\theta = 0$, approached from $\tau > 0$ as $\theta \to 0^+$ and from $\tau < 0$ as $\theta \to 2\pi^-$. The ket boundary value φ_+^A (imposed at $\tau \to 0^+$) is thus the field configuration at $\theta = 0$, and the bra boundary value φ_-^A (at $\tau \to 0^-$) is the configuration at $\theta = 2\pi$. The full plane is covered as θ runs from 0 to 2π.

Read this way, Eq. (55.66) is a Euclidean path integral with θ playing the role of time. It prepares the configuration φ_+^A at "time" $\theta = 0$, evolves it through a full angular sweep, and reads off the configuration φ_-^A at "time" $\theta = 2\pi$. By the transfer-matrix construction of the Euclidean path integral (Section 10.6), evolution through an angular interval $\Delta\theta$ is implemented by the operator $e^{-\Delta\theta\, H_R}$, where H_R, the generator of angular translations, is the *Rindler Hamiltonian*. A sweep through the full angle 2π therefore gives

$$\left\langle \varphi_-^A \middle| \rho_A \middle| \varphi_+^A \right\rangle = \frac{1}{Z} \left\langle \varphi_-^A \middle| e^{-2\pi H_R} \middle| \varphi_+^A \right\rangle, \qquad \text{i.e.} \qquad \rho_A = \frac{1}{Z}\, e^{-2\pi H_R}, \quad Z = \operatorname{Tr} e^{-2\pi H_R}, \tag{55.68}$$

where the normalization Z is fixed by $\operatorname{Tr}\rho_A = 1$. This is already the entire structural result: *the half-space reduced density matrix is a thermal state*, the Boltzmann weight $e^{-2\pi H_R}$ of the angular generator H_R at inverse "angular temperature" 2π. It remains to identify H_R as a physical, local operator, and then to translate "angular temperature 2π" into an honest temperature.

55.3.3 The Rindler Hamiltonian is the boost generator

The generator of angular translations in the Euclidean (τ, x^1) plane is, by construction, the Euclidean continuation of the generator of boosts in the Lorentzian (t, x^1) plane. The Euclidean rotation by angle θ in the (τ, x^1) plane becomes,

under the inverse Wick rotation $\tau \to it$, a hyperbolic rotation in the (t, x^1) plane: setting $\theta = -i\eta$ in $\left(\begin{smallmatrix}\cos\theta & -\sin\theta \\ \sin\theta & \cos\theta\end{smallmatrix}\right)$ and using $\cos(-i\eta) = \cosh\eta$, $\sin(-i\eta) = -i\sinh\eta$ reproduces exactly the boost matrix Λ_η of Eq. (55.64). Thus the angular generator H_R continues to the boost generator. We now make this identification quantitative by computing H_R directly from the action, and confirm that it equals the boost Noether charge of Chapter 7 restricted to the wedge.

Theorem 55.25 (Bisognano–Wichmann form of the half-space modular Hamiltonian)**.** *The reduced density matrix of the vacuum on the half-space $A = \{x^1 > 0\}$ is*

$$\rho_A = \frac{1}{Z}\, e^{-2\pi K_A}, \qquad K_A = \int_{x^1>0} \mathrm{d}^{d-1}x\; x^1\, T^{00}(0, \mathbf{x}), \tag{55.69}$$

where $T^{\mu\nu}$ is the stress tensor and K_A is the generator of the boost Eq. (55.64) restricted to the wedge W_R — a positive, local operator built from the energy density weighted by the distance x^1 to the entangling surface. Equivalently the modular Hamiltonian is $-\log\rho_A = 2\pi K_A + \log Z$.

The proof has two halves. First we exhibit H_R of Eq. (55.68) as the integral K_A written above, by reading off the Hamiltonian of the polar-coordinate action; this is done explicitly for the free scalar of ϕ^4 theory in the worked example below, and the construction is manifestly local and field-content-independent in form. Second we recognize K_A as the boost charge. The Lorentz Noether current of Eq. (7.107) has charges $M^{0\nu\rho} = \int \mathrm{d}^{d-1}x\,(x^\nu T^{0\rho} - x^\rho T^{0\nu})$ (the spin part vanishes for a scalar); the boost in the x^1 direction is generated by $M^{01} = \int \mathrm{d}^{d-1}x\,(x^0 T^{01} - x^1 T^{00})$, which at $t = x^0 = 0$ reduces to $-\int \mathrm{d}^{d-1}x\, x^1 T^{00}$. The combination appearing in Eq. (55.69) is exactly (minus) this charge with the spatial integral truncated to the wedge $x^1 > 0$, the orientation chosen so that K_A is the *positive* generator whose flow pushes points into the future of W_R along the hyperbolae of Eq. (55.64). The truncation is legitimate because the boost flow maps W_R to itself: the generator of a symmetry of the wedge is well-defined as an operator on the wedge degrees of freedom alone. Positivity of K_A — which is what makes $e^{-2\pi K_A}$ a sensible, trace-class density matrix with eigenvalues in $(0, 1]$ — is the statement that the boost energy in the wedge is bounded below, the wedge analogue of the spectrum condition. We now supply the first half explicitly.

Worked Example 55.26 (The Rindler Hamiltonian of the free scalar from the polar action)**.** Take the free scalar of ϕ^4 theory (Theorem 9.6) with Euclidean action Eq. (55.65). Rewrite the (τ, x^1) kinetic terms in the polar coordinates Eq. (55.67). Using $\partial_\tau = \sin\theta\,\partial_r + \frac{\cos\theta}{r}\partial_\theta$ and $\partial_1 = \cos\theta\,\partial_r - \frac{\sin\theta}{r}\partial_\theta$, the cross terms cancel and one finds the rotationally invariant combination

$$(\partial_\tau\phi)^2 + (\partial_1\phi)^2 = (\partial_r\phi)^2 + \frac{1}{r^2}(\partial_\theta\phi)^2, \tag{55.70}$$

so that, with measure $\mathrm{d}\tau\,\mathrm{d}x^1 = r\,\mathrm{d}r\,\mathrm{d}\theta$,

$$S_E = \int_0^{2\pi}\mathrm{d}\theta \int_0^{\infty} r\,\mathrm{d}r \int \mathrm{d}^{d-2}x_\perp\; \frac{1}{2}\Big[(\partial_r\phi)^2 + \frac{1}{r^2}(\partial_\theta\phi)^2 + (\nabla_\perp\phi)^2 + m^2\phi^2\Big]. \tag{55.71}$$

Treat θ as Euclidean time. The Lagrangian (the coefficient of $\mathrm{d}\theta$) splits into a θ-kinetic piece and a θ-independent "potential",

$$L = \frac{1}{2}\int_0^\infty \mathrm{d}r \int \mathrm{d}^{d-2}x_\perp \, \frac{1}{r}(\partial_\theta\phi)^2 + V[\phi], \qquad V[\phi] = \int_0^\infty \mathrm{d}r \int \mathrm{d}^{d-2}x_\perp \, \frac{r}{2}\Big[(\partial_r\phi)^2 + (\nabla_\perp\phi)^2 + m^2$$

(55.72)

The momentum conjugate to $\phi(r, \mathbf{x}_\perp)$, defined with respect to the slice measure $\mathrm{d}r\,\mathrm{d}^{d-2}x_\perp$, is

$$\pi(r, \mathbf{x}_\perp) = \frac{\partial}{\partial(\partial_\theta\phi)}\Big[\tfrac{1}{2r}(\partial_\theta\phi)^2\Big] = \frac{1}{r}\,\partial_\theta\phi. \tag{55.73}$$

We must now pass from L to the operator H_R that generates the imaginary-time (angular) evolution $e^{-\Delta\theta\, H_R}$. This is *not* the naive Legendre transform $\int \pi\,\partial_\theta\phi - L$, which in Euclidean signature would flip the sign of V and produce an indefinite operator; rather it is the positive transfer-matrix Hamiltonian, obtained as follows. For a Euclidean action of the standard form $S_E = \int \mathrm{d}\theta\,[\frac{1}{2}\int_x \mathcal{G}\,(\partial_\theta\phi)^2 + V[\phi]]$ with $\mathcal{G}(r) = 1/r$ here, discretizing θ in steps ϵ gives the kernel

$$\Big\langle \phi' \Big| e^{-\epsilon H_R} \Big| \phi \Big\rangle \;\propto\; \exp\Big(-\epsilon\Big[\tfrac{1}{2}\int_x \mathcal{G}\,\Big(\tfrac{\phi'-\phi}{\epsilon}\Big)^2 + V[\phi]\Big]\Big), \tag{55.74}$$

and a Gaussian (Hubbard–Stratonovich) rewriting of the kinetic factor in terms of the conjugate momentum $\pi = \mathcal{G}\,\partial_\theta\phi$ identifies this, as $\epsilon \to 0$, as the matrix element of $\exp(-\epsilon H_R)$ with

$$H_R = \int_x \frac{1}{2\mathcal{G}}\,\pi^2 + V[\phi] = \int_0^\infty \mathrm{d}r \int \mathrm{d}^{d-2}x_\perp \, \frac{r}{2}\Big[\pi^2 + (\partial_r\phi)^2 + (\nabla_\perp\phi)^2 + m^2\phi^2\Big]. \tag{55.75}$$

The crucial feature is the uniform *plus* sign between the momentum term $\frac{r}{2}\pi^2 = \frac{1}{2r}(\partial_\theta\phi)^2$ and the spatial gradient terms: H_R is manifestly positive, an integral of (energy density)$\times r$ over the half-line $r > 0$. This positivity is exactly what the naive Legendre transform would have spoiled, and it is what makes $e^{-2\pi H_R}$ a legitimate density matrix.

Now evaluate Eq. (55.75) on the ket slice $\theta = 0$, i.e. $\tau = 0$, $x^1 = r > 0$, and continue back to Lorentzian signature. There the canonical momentum of the Minkowski field is $\pi_t = \partial_t\phi$, which under $t \to -i\tau$ and the polar substitution Eq. (55.67) becomes $\pi = r^{-1}\partial_\theta\phi$ at $\theta = 0$, while $\partial_r\phi = \partial_1\phi$ at $\theta = 0$. The bracket in Eq. (55.75) is then exactly twice the free-scalar energy density,

$$T^{00} = \tfrac{1}{2}\big[\pi_t^2 + (\partial_1\phi)^2 + (\nabla_\perp\phi)^2 + m^2\phi^2\big], \tag{55.76}$$

the canonical stress tensor of Theorem 7.25 evaluated at $t = 0$. With $r = x^1$ the measure $r\,\mathrm{d}r\,\mathrm{d}^{d-2}x_\perp$ becomes $x^1\,\mathrm{d}^{d-1}x$, so

$$H_R = \int_{x^1>0} \mathrm{d}^{d-1}x\; x^1\, T^{00}(0, \mathbf{x}) = K_A, \tag{55.77}$$

which is Eq. (55.69). The computation used nothing special to the free field beyond the form of T^{00}; for an interacting theory the same transfer-matrix construction produces $\int x^1\, T^{00}$ with the full interacting energy density, which is the content of the general Bisognano–Wichmann theorem.

Physical Insight

The modular Hamiltonian of a half-space is *local* — an integral of the local energy density — which is the exception, not the rule. For almost any other region (two intervals, a ball in a non-conformal theory, anything at finite temperature) $-\log \rho_A$ is a complicated nonlocal operator. The half-space is special because its causal wedge is preserved by a spacetime symmetry, the boost, and a symmetry generator is automatically the integral of a local current. Chapter 47's conformal symmetry buys one more solvable case — the ball, mapped to the Rindler wedge by a special conformal transformation — which Section 55.6 will exploit. Everywhere else, locality of the modular Hamiltonian is lost.

55.3.4 The Unruh temperature

The factor 2π in Eq. (55.68) is dimensionless and refers to the boost rapidity, which is not directly a physical observer's clock. To extract a temperature we must ask what flow a real observer's proper time generates. Introduce *Rindler coordinates* adapted to the wedge,

$$t = \xi \sinh \eta, \qquad x^1 = \xi \cosh \eta, \qquad \xi > 0,\ \eta \in \mathbb{R}, \tag{55.78}$$

which cover exactly W_R (since $x^1 > |t| \iff \xi > 0$). A curve of constant ξ is the hyperbola $(x^1)^2 - t^2 = \xi^2$; the boost Eq. (55.64) acts as the pure translation $\eta \to \eta + \eta_0$. The Minkowski line element becomes

$$\mathrm{d}s^2 = \mathrm{d}t^2 - (\mathrm{d}x^1)^2 - \mathrm{d}\mathbf{x}_\perp^2 = \xi^2\, \mathrm{d}\eta^2 - \mathrm{d}\xi^2 - \mathrm{d}\mathbf{x}_\perp^2, \tag{55.79}$$

read off from $\mathrm{d}t = \sinh\eta\, \mathrm{d}\xi + \xi \cosh\eta\, \mathrm{d}\eta$ and $\mathrm{d}x^1 = \cosh\eta\, \mathrm{d}\xi + \xi \sinh\eta\, \mathrm{d}\eta$, whose squares combine as $\mathrm{d}t^2 - (\mathrm{d}x^1)^2 = -\mathrm{d}\xi^2 + \xi^2 \mathrm{d}\eta^2$. An observer sitting at fixed ξ and $\mathbf{x}_\perp$ has $\mathrm{d}s^2 = \xi^2 \mathrm{d}\eta^2$, so his proper time is $s = \xi\, \eta$, and differentiating the worldline $x^\mu(s) = (\xi \sinh(s/\xi),\, \xi \cosh(s/\xi),\, \mathbf{x}_\perp)$ twice gives the four-acceleration $a^\mu = \ddot{x}^\mu$ with $a^\mu a_\mu = -1/\xi^2$, i.e. proper acceleration

$$a = \frac{1}{\xi}. \tag{55.80}$$

Thus the worldline at $\xi = 1/a$ is the trajectory of a uniformly accelerated observer with proper acceleration a; Fig. 55.3 shows it.

The boost generator K_A generates translations in the rapidity η. The observer's clock measures proper time $s = \eta/a$ along his worldline, so his proper-time Hamiltonian — the generator of s-translations along the trajectory — is

$$H_{\text{proper}} = \frac{\partial \eta}{\partial s} K_A = a\, K_A. \tag{55.81}$$

Rewriting the thermal density matrix Eq. (55.69) in terms of this physical Hamiltonian,

$$\rho_A = \frac{1}{Z} e^{-2\pi K_A} = \frac{1}{Z} e^{-(2\pi/a)\, H_{\text{proper}}} \equiv \frac{1}{Z} e^{-\beta H_{\text{proper}}}, \qquad \beta = \frac{2\pi}{a}. \tag{55.82}$$

The accelerated observer therefore perceives the vacuum as a thermal ensemble at temperature

$$\boxed{T_{\text{U}} = \frac{1}{\beta} = \frac{a}{2\pi}} \tag{55.83}$$

the *Unruh temperature*. The Minkowski vacuum, manifestly the zero-temperature ground state to an inertial observer, looks hot to an accelerated one — because the accelerated observer has access only to the wedge W_R, and the restriction of the global pure state to that wedge is the mixed, thermal state Eq. (55.82). The factor 2π that began life as the angle of a full Euclidean rotation has become the inverse temperature in units of the acceleration.

Worked Example 55.27 (Unruh temperature from Euclidean smoothness)**.** The temperature Eq. (55.83) can be obtained a second way, from the requirement that the Euclidean continuation of Rindler space be smooth — a derivation that exposes why the period is exactly 2π and not some other number. Wick-rotate the Rindler time, $\eta \to -i\theta$ (equivalently $t \to -i\tau$ with $\tau = \xi \sin\theta$, $x^1 = \xi\cos\theta$, recovering Eq. (55.67) with $r = \xi$). Under $\eta = -i\theta$ the Lorentzian Rindler metric Eq. (55.79) becomes the Euclidean

$$\mathrm{d}s_E^2 = \xi^2\, \mathrm{d}\theta^2 + \mathrm{d}\xi^2 + \mathrm{d}\mathbf{x}_\perp^2. \tag{55.84}$$

In the (ξ, θ) plane this is the flat plane in polar coordinates, with ξ the radius and θ the polar angle. A flat plane written in polar coordinates is smooth at the origin $\xi = 0$ *only if* θ is periodic with period exactly 2π: any other period $\theta \sim \theta + \beta_\theta$ produces a cone of deficit (or excess) angle $2\pi - \beta_\theta$, with a curvature singularity — a conical defect — concentrated on Σ at $\xi = 0$. The Minkowski vacuum is the smooth, defect-free state, so its Euclidean continuation forces

$$\theta \sim \theta + 2\pi. \tag{55.85}$$

Periodicity in Euclidean time is the hallmark of a thermal state, as the Gibbs computation Eq. (55.92) below makes precise: a Gibbs weight $e^{-\beta H}$ is reproduced by evolution through an imaginary-time interval β, so equilibrium at inverse

temperature β is the same statement as periodicity β in Euclidean time. Here the clock is the observer's proper time, whose Euclidean version is $s_E = \xi\,\theta$ at fixed ξ; its period is $\beta = \xi \cdot 2\pi = 2\pi/a$, reproducing Eq. (55.83). The two derivations agree because they are the same statement read in two directions: the angular period 2π of the path integral *is* the inverse temperature in rapidity units, and smoothness at the entangling surface is what fixes it.

Restoring dimensions once for a physical estimate (reinstating $\hbar, c, k_B$ by dimensional analysis: $k_B T$ carries energy, a carries length/time^2, and the only combination with the right dimensions is $\hbar a/c$),

$$k_B T_{\mathrm{U}} = \frac{\hbar\, a}{2\pi c}. \tag{55.86}$$

The number is minuscule: an acceleration of $a = 10^{20}\,\mathrm{m/s^2}$ — ten orders of magnitude beyond what any laboratory sustains — yields $T_{\mathrm{U}} \sim 0.4\,\mathrm{K}$. The Unruh effect is real but, for terrestrial accelerations, utterly unobservable; its importance is conceptual and structural, and it is the flat-space shadow of Hawking radiation (Chapter 65), to which it is locally identical near a black-hole horizon.

Historical Note

Bisognano and Wichmann established in 1975–76, from the Wightman axioms alone, that the modular automorphism of the vacuum for a wedge algebra is exactly the boost — the rigorous, interaction-independent version of Theorem 55.25. Unruh's 1976 analysis of a uniformly accelerated detector, and Davies' related work, independently uncovered the temperature Eq. (55.83); Fulling had noted the inequivalent Rindler quantization a few years earlier. That the algebraic statement (Bisognano–Wichmann) and the physical one (Unruh) are the same fact is the lesson of this section.

55.3.5 The modular Hamiltonian, modular flow, and the KMS condition

The Rindler computation is the cleanest instance of structures that organize entanglement in *any* state and region, and we now name them in general, with the half-space as the worked model.

Definition 55.28 (Modular Hamiltonian and modular flow)**.** For a reduced density matrix ρ_A with full support, the *modular Hamiltonian* (or entanglement Hamiltonian) is the self-adjoint operator

$$K_{\mathrm{mod}} \equiv -\log \rho_A, \qquad \text{so that} \quad \rho_A = e^{-K_{\mathrm{mod}}}. \tag{55.87}$$

The associated *modular flow* is the one-parameter group of automorphisms of the operator algebra of A,

$$\sigma_s(\mathcal{O}) \equiv \rho_A^{is}\, \mathcal{O}\, \rho_A^{-is} = e^{-iK_{\mathrm{mod}} s}\, \mathcal{O}\, e^{iK_{\mathrm{mod}} s}, \qquad s \in \mathbb{R}. \tag{55.88}$$

For the half-space, Theorem 55.25 gives $K_{\mathrm{mod}} = 2\pi K_A + \log Z$, so modular flow is generated by $2\pi K_A$, i.e. by a boost of rapidity $2\pi s$:

$$\sigma_s(\mathcal{O}(x)) = \mathcal{O}(\Lambda_{2\pi s}\, x), \tag{55.89}$$

with Λ_η the boost Eq. (55.64). This is the remarkable feature of the Rindler case: *modular flow is geometric*, a literal Lorentz boost sliding operators along the wedge hyperbolae. In general σ_s is an abstract, state- and region-dependent symmetry with no spacetime interpretation; the entanglement structure of the half-space vacuum endows it, exceptionally, with one. (The Reeh–Schlieder property guarantees the cyclic and separating vector that makes K_{mod} well-defined; we use this without full proof.)

Thermality is captured intrinsically, without any explicit reference to a temperature or even a Hamiltonian, by the KMS condition. Its value is that it characterizes equilibrium purely through analyticity of correlation functions in complexified time — exactly the structure a Euclidean path integral on a time circle produces.

Definition 55.29 (KMS condition)**.** A state ω on an algebra of observables satisfies the *Kubo–Martin–Schwinger (KMS) condition* at inverse temperature β with respect to a flow α_s (a one-parameter automorphism group) if, for all observables $\mathcal{O}_1, \mathcal{O}_2$, the correlation function

$$F(s) \equiv \omega(\mathcal{O}_1\, \alpha_s(\mathcal{O}_2)) \tag{55.90}$$

extends to a function holomorphic in the strip $\{\, s \in \mathbb{C} : 0 < \operatorname{Im} s < \beta \,\}$, continuous on its closure, with boundary values related by

$$F(s + i\beta) = \omega(\alpha_s(\mathcal{O}_2)\, \mathcal{O}_1). \tag{55.91}$$

That a Gibbs state $\omega(\cdot) = \operatorname{Tr}(e^{-\beta H}\,\cdot)/\operatorname{Tr} e^{-\beta H}$ satisfies Eq. (55.91) with $\alpha_s(\mathcal{O}) = e^{iHs}\mathcal{O}e^{-iHs}$ is a one-line computation: inserting the trace and using cyclicity,

$$F(s+i\beta) = \frac{\operatorname{Tr}\left(e^{-\beta H}\mathcal{O}_1 e^{iH(s+i\beta)}\mathcal{O}_2 e^{-iH(s+i\beta)}\right)}{\operatorname{Tr} e^{-\beta H}} = \frac{\operatorname{Tr}\left(e^{iHs}\mathcal{O}_2 e^{-iHs}\, e^{-\beta H}\mathcal{O}_1\right)}{\operatorname{Tr} e^{-\beta H}} = \omega(\alpha_s(\mathcal{O}_2)\mathcal{O}_1 \tag{55.92}$$

where the middle step moved $e^{-\beta H}$ around the trace and recombined the exponentials. The strip analyticity is the statement that $e^{-\beta H}$ damps the high-energy contributions, making the continuation to $\operatorname{Im} s = \beta$ convergent. The content of Theorem 55.25 is then summarized in one line:

Theorem 55.30 (The wedge vacuum is KMS)**.** *The Minkowski vacuum, restricted to the right Rindler wedge algebra, satisfies the KMS condition at inverse temperature $\beta = 2\pi$ with respect to the boost flow $\eta \mapsto \eta + \eta_0$ (equivalently $\beta = 1$ with respect to the modular flow Eq. (55.88), and $\beta = 2\pi/a$ with respect to the proper-time flow of an observer at acceleration a).*

Proof sketch. By Eq. (55.68), the wedge state is the Gibbs state of $H_R = K_A$ at inverse temperature 2π. Applying the Gibbs computation Eq. (55.92) with $H = K_A$ and $\beta = 2\pi$ gives Eq. (55.91) directly; the required strip analyticity is the statement that wedge correlation functions of the vacuum continue to the Euclidean angle $\theta \in [0, 2\pi]$, which is exactly the angular path integral Eq. (55.71) that defines ρ_A. The rescalings to modular and proper-time flow follow from $K_{\text{mod}} = 2\pi K_A$ and $H_{\text{proper}} = aK_A$ respectively. □

Caution

A genuine subtlety, flagged in the chapter's opening and honored throughout: the algebra of observables localized in a Rindler wedge (or any region with a sharp boundary) in continuum QFT is a *von Neumann algebra of type* III_1. Such an algebra admits *no* pure states and *no* trace, so the manipulations of this section — writing $\rho_A = e^{-2\pi K_A}/Z$, taking $\operatorname{Tr} \rho_A = 1$, defining $S_A = -\operatorname{Tr} \rho_A \log \rho_A$ — are, strictly, not defined in the continuum. What *is* rigorously defined is the modular operator Δ_A and its flow, supplied by the Tomita–Takesaki theory of a cyclic separating vector; the Bisognano–Wichmann theorem is properly the statement that Δ_A^{is} acts as the boost, with no ρ_A ever written down. The density-matrix language we use is the correct *regulated* picture: introduce a short-distance cutoff (a lattice, or a smearing of the entangling surface), which makes the algebra type I, ρ_A an honest density matrix, and S_A finite; physical, cutoff-independent quantities (relative entropy, mutual information) then have well-defined continuum limits even though S_A itself diverges. We compute with ρ_A as the regulated object and report only quantities that survive the limit. The rigorous backbone — Tomita–Takesaki modular theory and the type-III classification of local algebras — is deferred to the further reading; the present section's results are all recoverable within it.

55.3.6 Vacuum entanglement entropy as thermal entropy and the area law

With ρ_A thermal, its entanglement entropy is a thermal entropy, and we can read off its structure directly. From Eq. (55.68), $-\log \rho_A = 2\pi K_A + \log Z$, so

$$S_A = -\operatorname{Tr}(\rho_A \log \rho_A) = 2\pi \langle K_A \rangle + \log Z, \tag{55.93}$$

where $\langle K_A \rangle = \operatorname{Tr}(\rho_A K_A)$ is the modular energy. Setting $Z(\beta) = \operatorname{Tr} e^{-\beta K_A}$ and $\langle K_A \rangle = -\partial_\beta \log Z$, this is the textbook thermodynamic identity $S = (1 - \beta\partial_\beta) \log Z$ evaluated at $\beta = 2\pi$: the vacuum entanglement entropy of the half-space is, identically, the equilibrium entropy of the Rindler thermal gas. This identification is the deepest reason the entanglement entropy obeys an area law,

and we now extract that law from the thermal picture — the explicit content of the Rindler case that Section 55.5 will systematize.

The key is that the temperature is not uniform across the wedge: an observer hovering at proper distance ρ_* from the entangling surface (i.e. at $\xi = \rho_*$, by Eq. (55.84)) has proper acceleration $a = 1/\rho_*$ and therefore sees a *local* Unruh temperature

$$T_{\rm loc}(\rho_*) = \frac{a}{2\pi} = \frac{1}{2\pi\rho_*}. \tag{55.94}$$

The temperature diverges as $\rho_* \to 0$: the modes that pile up against the entangling surface are arbitrarily hot, and they are the origin of the divergence. Treat the entanglement entropy as the entropy of a local thermal gas integrated over the wedge. The entropy density of a relativistic gas in d spacetime dimensions scales by dimensional analysis as $s \sim T^{d-1}$ (the Stefan–Boltzmann law; the proportionality constant counts the field's degrees of freedom and is computed in full in Section 55.5). The proper volume element a fixed transverse distance from Σ is $\mathrm{d}V = \mathcal{A}_\perp \, \mathrm{d}\rho_*$, with $\mathcal{A}_\perp$ the transverse area of the entangling surface. Hence

$$S_A \;\sim\; \int \mathrm{d}V \, s(T_{\rm loc}) \;\sim\; \mathcal{A}_\perp \int_\epsilon^L \mathrm{d}\rho_* \left(\frac{1}{2\pi\rho_*}\right)^{d-1} \;\sim\; \mathcal{A}_\perp \int_\epsilon \frac{\mathrm{d}\rho_*}{\rho_*^{\,d-1}}, \tag{55.95}$$

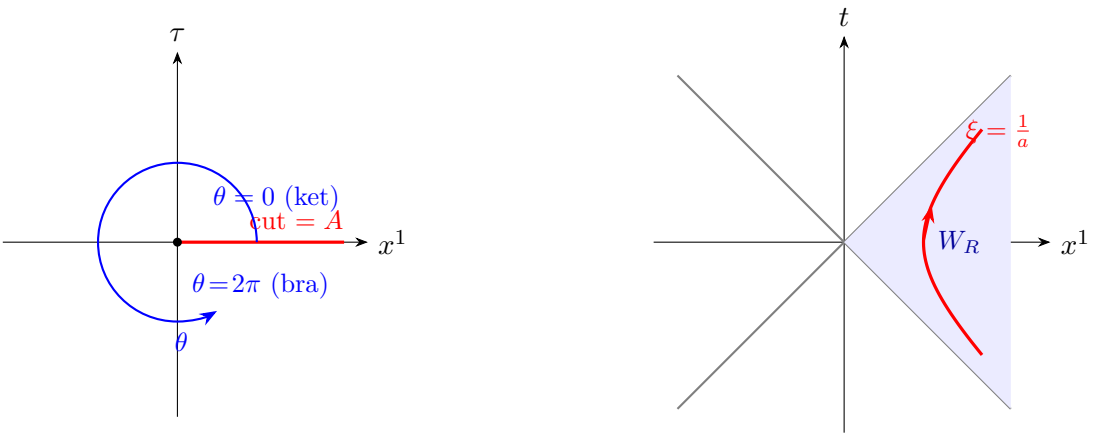
where ϵ is a short-distance (UV) cutoff on the proper distance to the surface — the regulator the previous Caution insisted on — and L an IR scale that does not affect the leading divergence. For $d > 2$ the integral is dominated by its lower limit,

$$S_A \;\sim\; \frac{\mathcal{A}_\perp}{(d-2)\,\epsilon^{\,d-2}} + \cdots, \tag{55.96}$$

the celebrated *area law*: the leading UV divergence of the vacuum entanglement entropy is proportional to the area $\mathcal{A}_\perp$ of the entangling surface, not to the volume of A, and is power-law divergent in the cutoff with the coefficient set by the density of fields. For $d = 2$ the transverse "area" is a single point ($\mathcal{A}_\perp = 1$) and the integral $\int \mathrm{d}\rho_*/\rho_*$ is *logarithmic*, $S_A \sim \log(L/\epsilon)$ — precisely the $\frac{c}{3}\log(\ell/\epsilon)$ behavior that Section 55.4 computes exactly from the central charge. The area law and the 2D logarithm are thus one phenomenon, the divergence of the near-surface Rindler modes, seen in different dimensions; both are visible already in this thermal reading of the half-space vacuum.

Physical Insight

The area law has a one-sentence physical cause: entanglement entropy counts the short-wavelength correlations straddling the entangling surface, and there are a fixed number of them per unit area per unit cutoff cell. The thermal derivation makes this quantitative — the divergence is the infinite local temperature Eq. (55.94) of the modes hugging the surface — and explains why the leading term is non-universal (cutoff-dependent)



(a) Euclidean: $\rho_A = Z^{-1} e^{-2\pi H_R}$

(b) Lorentzian: accelerated worldline

Figure 55.3: The half-space vacuum. **(a)** In Euclidean signature the matrix element Eq. (55.66) is the path integral on $\mathbb{R}^d$ cut along the half-plane $A = \{\tau = 0, x^1 > 0\}$ (red). In polar coordinates Eq. (55.67) the ket lip sits at $\theta = 0$ and the bra lip at $\theta = 2\pi$; a full angular sweep generates $\rho_A = Z^{-1} e^{-2\pi H_R}$. **(b)** In Lorentzian signature the same region is the right Rindler wedge W_R (shaded), preserved by the boost Eq. (55.64). A uniformly accelerated observer follows the hyperbola $\xi = 1/a$ (red); the boost flow slides operators along it, and is the geometric form Eq. (55.89) of modular flow.

while the geometry, the proportionality to $\mathcal{A}_\perp$, is robust. It is also why the physically clean diagnostics are *differences* of entropies, where the $\mathcal{A}_\perp/\epsilon^{d-2}$ cancels.

We close by collecting what is now in hand for the rest of the chapter. The half-space has given us, exactly and explicitly, the modular Hamiltonian $K_{\text{mod}} = -\log \rho_A = 2\pi K_A$ as the local boost generator Eq. (55.69), the modular flow ρ_A^{is} realized as a geometric Lorentz boost Eq. (55.89), and the temperature/KMS language Theorems 55.29 and 55.30 that recasts vacuum entanglement as thermality. Section 55.4 will find that its exact two-dimensional single-interval entropy is the thermal specialization of Eq. (55.93) on the cylinder, and the finite-temperature formula there is this Rindler thermality made global; Section 55.5 will turn the heuristic Eq. (55.96) into a controlled statement; and Section 55.6 will use precisely the modular-Hamiltonian and KMS structure assembled here to define relative entropy and prove the first law of entanglement, the inputs on which the holographic constructions of Chapters 56 and 57 ultimately rest.

55.4 Entanglement Entropy in Two-Dimensional CFT

The replica trick of Section 55.2 reduced the entanglement entropy of a spatial region to a partition function on an n-sheeted branched cover of the Euclidean plane (Section 55.2.3), and the orbifold reformulation of Section 55.2.5 repackaged that geometry, for a region A that is a single interval $[u, v]$, as a two-point function of *twist operators* in the n-fold replicated theory (Theorem 55.22),

$$\operatorname{Tr} \rho_A^n \;=\; \big\langle\, \sigma_n(u)\, \tilde{\sigma}_n(v) \,\big\rangle_{\mathcal{C}^{\otimes n}}, \tag{55.97}$$

with the conformal dimension of σ_n left undetermined at that stage. The thermal reading of the vacuum from Section 55.3 — that tracing out a region exposes a temperature — told us that finite temperature and finite size are not optional refinements but the natural arena in which an entropy lives. This section discharges both promissory notes at once. We compute the twist-operator dimension exactly, from the conformal anomaly, fixing the right-hand side of Eq. (55.97) completely by symmetry; we take the replica limit $n \to 1$ to obtain the single-interval vacuum entanglement entropy

$$S_A \;=\; \frac{c}{3}\, \log \frac{\ell}{\epsilon} + \text{const}, \tag{55.98}$$

the flagship result of the chapter; and we deform the plane to a cylinder and to a thermal cylinder by two explicit conformal maps, converting the bare logarithm into its finite-size and finite-temperature cousins. The single number c — the central charge of Theorem 48.8 — governs all of it. We will see that this is no accident: the coefficient $c/3$ is a renormalization-group diagnostic, and for the Ising universality class it carries the value $c = \frac{1}{2}$ that closes ϕ^4 theory on the entanglement side.

Throughout we work in Euclidean signature with complex coordinate $z = x + \mathrm{i}\tau$ (the Wick rotation $t \to -\mathrm{i}\tau$ fixed in Chapter 10), antiholomorphic partner $\bar{z}$, and we use the holomorphic factorization of two-dimensional CFT established in Chapter 48: every local primary carries a pair of weights $(h, \bar{h})$, its scaling dimension is $\Delta = h + \bar{h}$, and the holomorphic and antiholomorphic sectors contribute independently. For a parity-symmetric theory built from a local action, $c = \bar{c}$, and the twist operator will turn out to be spinless, $h_n = \bar{h}_n$.

55.4.1 The replica surface and the twist operator, recalled

We restate the geometric input precisely so the present computation is self-contained. Let $A = [u, v]$ be a single interval on the time slice $\tau = 0$ of the infinite line, of length $\ell = v - u$. The reduced density matrix ρ_A is prepared by the Euclidean path integral over the plane with a cut along A, the field values above and below the cut serving as the row and column indices of ρ_A

(Section 55.2.2). Its n-th power ρ_A^n sews n copies of the cut plane cyclically: the field above the cut on sheet j is identified with the field below the cut on sheet $j+1$, and sheet n closes back onto sheet 1. The trace identifies the last with the first, producing a single connected Riemann surface $\mathcal{R}_n$, an n-sheeted cover of the plane branched over the two endpoints u and v. Then

$$\operatorname{Tr}\rho_A^n \;=\; \frac{Z_n}{Z_1^n}, \qquad Z_n = Z[\mathcal{R}_n], \tag{55.99}$$

the normalization Z_1^n (with Z_1 the plane partition function) enforcing $\operatorname{Tr}\rho_A = 1$. At $n=1$ the single sheet is the plane, σ_1 is the identity operator, and $\operatorname{Tr}\rho_A^1 = \langle \mathbf{1}\rangle = 1$; this fixes the $n=1$ value of the non-universal normalization constant c_n introduced below to $c_1 = 1$.

The twist-operator formulation trades the multivalued geometry of $\mathcal{R}_n$ for ordinary correlators on a single plane in the *replicated* theory $\mathcal{C}^{\otimes n}$, whose fields are n-tuples $\Phi = (\phi_1, \dots, \phi_n)$ and whose stress tensor is the sum $T^{(n)} = \sum_{j=1}^{n} T_j$ over the copies, so its central charge is nc. The branch structure of $\mathcal{R}_n$ is encoded by a $\mathbb{Z}_n$ symmetry of $\mathcal{C}^{\otimes n}$, cyclic permutation $\phi_j \mapsto \phi_{j+1}$ of the replicas. The branch point at u is the place where, on encircling it, the field on sheet j continues onto sheet $j+1$; in the replicated theory this is precisely a disorder (twist) operator $\sigma_n(u)$ implementing the cyclic shift, with $\tilde\sigma_n(v)$ the inverse twist closing the monodromy. Theorem 55.22 established Eq. (55.97) on these grounds. The twist operators are local, $\mathbb{Z}_n$-covariant primaries of the replicated theory; conformal symmetry fixes their two-point function up to a constant once their dimension is known, and the entire physics of the single interval is therefore carried by one number, h_n. We now compute it.

55.4.2 The dimension of the twist operator from the conformal anomaly

The dimension h_n is read off by comparing two expressions for the same object: the expectation value of the stress tensor on the replica surface $\mathcal{R}_n$, computed by uniformizing $\mathcal{R}_n$ to the plane, against the conformal Ward identity for $\langle T^{(n)}\,\sigma_n\,\tilde\sigma_n\rangle$, which is fixed by the twist dimension. Matching the two determines h_n.

The uniformizing map. The surface $\mathcal{R}_n$ is branched over u and v, and away from those points it is locally flat. There is a single-valued holomorphic map that unwraps it onto the plane. Consider

$$w \;=\; \left(\frac{z-u}{z-v}\right)^{1/n}. \tag{55.100}$$

The factor inside is the Möbius transformation that sends the two branch points to 0 and ∞: $z = u \mapsto 0$, $z = v \mapsto \infty$. Taking the n-th root then unwinds

the cyclic monodromy. To see this, encircle u once on the cover: the Möbius image $\zeta = (z-u)/(z-v)$ winds once around the origin, $\zeta \to e^{2\pi \mathrm{i}}\zeta$, while $w = \zeta^{1/n} \to e^{2\pi \mathrm{i}/n} w$ advances by a single replica, exactly the action of σ_n. Going around all n sheets, ζ winds n times and w returns to itself: the n sheets of $\mathcal{R}_n$ are mapped one-to-one onto a single copy of the w-plane. The map Eq. (55.100) is the conformal uniformization of the replica surface, and on the w-plane the theory is just the vacuum, for which $\langle T(w)\rangle_{w\text{-plane}} = 0$.

The stress tensor on $\mathcal{R}_n$. The stress tensor of a single copy transforms under a holomorphic map $z \mapsto w(z)$ by the anomalous (Schwarzian) law derived in Eq. (48.63),

$$T'(w) = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^{-2}\Big[T(z) - \frac{c}{12}\,\{w,z\}\Big], \qquad \{w,z\} = \frac{w'''}{w'} - \frac{3}{2}\Big(\frac{w''}{w'}\Big)^2, \tag{55.101}$$

the Schwarzian derivative of Eq. (48.62). Inverting to express the stress tensor on $\mathcal{R}_n$ (coordinate z) in terms of the plane (coordinate w),

$$T(z) = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^2 T'(w) + \frac{c}{12}\,\{w,z\}, \tag{55.102}$$

and taking the expectation value in the state defined on $\mathcal{R}_n$, which is the vacuum on the uniformized w-plane,

$$\langle T(z)\rangle_{\mathcal{R}_n} = \Big(\frac{\mathrm{d}w}{\mathrm{d}z}\Big)^2 \underbrace{\langle T'(w)\rangle_{w\text{-plane}}}_{=0} + \frac{c}{12}\,\{w,z\} = \frac{c}{12}\,\{w,z\}. \tag{55.103}$$

Only the anomaly survives. The central charge has converted a pure coordinate change into a physical, nonzero stress-tensor profile — the same mechanism that produces the Casimir energy on a cylinder (Chapter 48).

Computing the Schwarzian. We evaluate $\{w,z\}$ for the map Eq. (55.100) using the cocycle identity Eq. (48.66). Write $w = \zeta^{1/n}$ with $\zeta = (z-u)/(z-v)$. Because $\zeta(z)$ is Möbius, $\{\zeta, z\} = 0$, so the cocycle rule gives

$$\{w,z\} = \Big(\frac{\mathrm{d}\zeta}{\mathrm{d}z}\Big)^2 \{w,\zeta\} + \{\zeta,z\} = \Big(\frac{\mathrm{d}\zeta}{\mathrm{d}z}\Big)^2 \{w,\zeta\}. \tag{55.104}$$

For the pure power $w = \zeta^{1/n}$, with $a \equiv 1/n$,

$$w' = a\,\zeta^{a-1}, \quad w'' = a(a-1)\zeta^{a-2}, \quad w''' = a(a-1)(a-2)\zeta^{a-3}, \tag{55.105}$$

so

$$\{w,\zeta\} = \frac{w'''}{w'} - \frac{3}{2}\Big(\frac{w''}{w'}\Big)^2 = (a-1)(a-2)\zeta^{-2} - \frac{3}{2}\,(a-1)^2\zeta^{-2}$$

$$= \zeta^{-2}(a-1)\Big[(a-2)-\tfrac{3}{2}(a-1)\Big] = \zeta^{-2}(a-1)\big(-\tfrac{1}{2}a-\tfrac{1}{2}\big) = \frac{1-a^2}{2}\,\zeta^{-2}. \tag{55.106}$$

The Jacobian of the Möbius factor is $\mathrm{d}\zeta/\mathrm{d}z = (u-v)/(z-v)^2$, since $\mathrm{d}\zeta/\mathrm{d}z = [(z-v)-(z-u)]/(z-v)^2$. Hence $(\mathrm{d}\zeta/\mathrm{d}z)^2 = (u-v)^2/(z-v)^4$ and $\zeta^{-2} = (z-v)^2/(z-u)^2$, and their product is

$$\Big(\frac{\mathrm{d}\zeta}{\mathrm{d}z}\Big)^2\zeta^{-2} = \frac{(u-v)^2}{(z-v)^4}\cdot\frac{(z-v)^2}{(z-u)^2} = \frac{(u-v)^2}{(z-u)^2(z-v)^2}. \tag{55.107}$$

With $a = 1/n$, Eq. (55.106) gives the Schwarzian of the uniformizing map,

$$\{w,z\} = \frac{1}{2}\Big(1-\frac{1}{n^2}\Big)\frac{(u-v)^2}{(z-u)^2(z-v)^2}. \tag{55.108}$$

Substituting into Eq. (55.103), the single-copy stress tensor on the replica surface is

$$\langle T_j(z)\rangle_{\mathcal{R}_n} = \frac{c}{24}\Big(1-\frac{1}{n^2}\Big)\frac{(u-v)^2}{(z-u)^2(z-v)^2}, \tag{55.109}$$

the same for each of the n replicas, since the uniformizing map treats every sheet identically.

The Ward-identity side, and the match. In the replicated theory $\mathcal{C}^{\otimes n}$ on a single plane, the same correlator is computed with the total stress tensor $T^{(n)}(z) = \sum_j T_j(z)$ inserted into the twist two-point function. Summing Eq. (55.109) over the n copies,

$$\frac{\langle T^{(n)}(z)\,\sigma_n(u)\,\tilde\sigma_n(v)\rangle}{\langle\sigma_n(u)\,\tilde\sigma_n(v)\rangle} = \sum_{j=1}^{n}\langle T_j(z)\rangle_{\mathcal{R}_n} = \frac{nc}{24}\Big(1-\frac{1}{n^2}\Big)\frac{(u-v)^2}{(z-u)^2(z-v)^2}. \tag{55.110}$$

On the other hand, the conformal Ward identity of Chapter 48 expresses the insertion of the stress tensor into a correlator of primaries in terms of their weights. For two primaries $\sigma_n, \tilde\sigma_n$ of equal holomorphic weight h_n located at u, v, with holomorphic two-point function $\langle\sigma_n(u)\tilde\sigma_n(v)\rangle \propto (u-v)^{-2h_n}$,

$$\frac{\langle T^{(n)}(z)\,\sigma_n(u)\,\tilde\sigma_n(v)\rangle}{\langle\sigma_n(u)\,\tilde\sigma_n(v)\rangle} = \Big[\frac{h_n}{(z-u)^2}+\frac{1}{z-u}\partial_u+\frac{h_n}{(z-v)^2}+\frac{1}{z-v}\partial_v\Big]\log\big[(u-v)^{-2h_n}\big]\Big|_{\mathrm{op}}, \tag{55.111}$$

where the derivative terms act on the explicit two-point function. Carrying out the derivatives, $\partial_u \log(u-v)^{-2h_n} = -2h_n/(u-v)$ and $\partial_v \log(u-v)^{-2h_n} = +2h_n/(u-v)$, so the descendant pieces give

$$-\frac{2h_n}{(z-u)(u-v)}+\frac{2h_n}{(z-v)(u-v)} = \frac{2h_n}{u-v}\cdot\frac{(z-u)-(z-v)}{(z-u)(z-v)} = -\frac{2h_n}{(z-u)(z-v)}, \tag{55.112}$$

and adding the primary terms,

$$\frac{\langle T^{(n)}\sigma_n\tilde\sigma_n\rangle}{\langle\sigma_n\tilde\sigma_n\rangle} = h_n\Big[\frac{1}{(z-u)^2}+\frac{1}{(z-v)^2}-\frac{2}{(z-u)(z-v)}\Big] = h_n\Big[\frac{1}{z-u}-\frac{1}{z-v}\Big]^2 = h_n\,\frac{(u-$$
$$(z-u)^2($$
(55.113)

The two computations Eqs. (55.110) and (55.113) are expressions for the identical correlator and have the identical functional form in z; equating their coefficients gives the twist dimension.

Theorem 55.31 (Conformal dimension of the twist operator)**.** *In a two-dimensional conformal field theory of central charge c, the $\mathbb{Z}_n$ twist operator σ_n implementing the cyclic identification of the n-sheeted replica cover is a spinless primary of the replicated theory $\mathcal{C}^{\otimes n}$ with holomorphic and antiholomorphic weights*

$$h_n=\bar h_n=\frac{c}{24}\Big(n-\frac{1}{n}\Big),\qquad \textit{total scaling dimension}\qquad \Delta_n=2h_n=\frac{c}{12}\Big(n-\frac{1}{n}\Big).\tag{55.114}$$

Proof. Matching the coefficients of $(u-v)^2/[(z-u)^2(z-v)^2]$ in Eqs. (55.110) and (55.113),

$$h_n=\frac{nc}{24}\Big(1-\frac{1}{n^2}\Big)=\frac{c}{24}\Big(n-\frac{1}{n}\Big).\tag{55.115}$$

The antiholomorphic sector repeats verbatim with $\bar T_j$ and $\bar c=c$, giving $\bar h_n=h_n$, so the twist operator is spinless ($h_n-\bar h_n=0$) and its total dimension is $\Delta_n=h_n+\bar h_n=2h_n$. □

Physical Insight

The twist dimension vanishes at $n=1$, as it must: a single sheet has no branch point and the replica surface is the plane, so σ_1 is the identity, $h_1=0$. The dimension grows with n because each additional sheet sharpens the conical singularity at the branch point: near $z=u$ the uniformized angle is $2\pi n$ wrapped into 2π, a deficit that costs curvature, and the central charge is precisely the theory's response to curvature. This is the same c that sets the Casimir energy and the trace anomaly; here it sets the price of replication.

55.4.3 The single-interval entanglement entropy

With Δ_n in hand, Eq. (55.97) is fixed by conformal symmetry. The two-point function of two scalar primaries of equal dimension is determined up to one constant — Theorem 47.21 — as $C/|u-v|^{2\Delta}$. Applied to the twist fields,

$$\langle\sigma_n(u)\,\tilde\sigma_n(v)\rangle=\frac{c_n}{|u-v|^{2\Delta_n}}=\frac{c_n}{|u-v|^{(c/6)(n-1/n)}},\tag{55.116}$$

where we used $2\Delta_n = (c/6)(n-1/n)$ from Eq. (55.114), and c_n is a non-universal constant fixed by the short-distance normalization of σ_n. The two-point function is dimensionful, with σ_n carrying dimension Δ_n; making it dimensionless requires a length scale, and that scale is the UV cutoff ϵ — the lattice spacing, or the short-distance regulator at which the continuum theory is defined. This is where the cutoff dependence of the entanglement entropy, flagged at the outset of the chapter, enters concretely: the twist operator is not a sharply local field but a defect smeared over the cutoff scale, and its normalization is set at $|u-v| \sim \epsilon$. Restoring the cutoff so that the combination is dimensionless,

$$\operatorname{Tr}\rho_A^n = c_n \Big(\frac{\ell}{\epsilon}\Big)^{-(c/6)(n-1/n)}, \qquad \ell = |u-v|, \tag{55.117}$$

which is the announced power law: a single interval of length ℓ in the infinite-line vacuum has $\operatorname{Tr}\rho_A^n$ a pure power of ℓ/ϵ, with the universal exponent $(c/6)(n-1/n)$ and a non-universal prefactor.

The entanglement entropy is the replica limit of Eq. (55.117). Recall from Theorem 55.17 that the von Neumann entropy is recovered by analytic continuation in n,

$$S_A = -\operatorname{Tr}(\rho_A \log \rho_A) = -\frac{\partial}{\partial n}\operatorname{Tr}\rho_A^n\Big|_{n=1}, \tag{55.118}$$

valid because $\operatorname{Tr}\rho_A^n = \sum_i p_i^n$ for eigenvalues p_i of ρ_A, whose n-derivative at $n=1$ is $\sum_i p_i \log p_i$ once $\sum_i p_i = 1$ fixes $\operatorname{Tr}\rho_A = 1$, i.e. $c_1 = 1$.

Theorem 55.32 (Single-interval vacuum entanglement entropy in 2D CFT)**.** *For a single interval of length ℓ in the vacuum of a two-dimensional CFT of central charge c, with UV cutoff ϵ,*

$$S_A = \frac{c}{3}\log\frac{\ell}{\epsilon} + c_1', \qquad c_1' \equiv \frac{\partial c_n}{\partial n}\Big|_{n=1}, \tag{55.119}$$

the constant c_1' being non-universal (cutoff-scheme dependent), while the coefficient $c/3$ of the logarithm is universal.

Proof. Write $\operatorname{Tr}\rho_A^n = c_n\, e^{-(c/6)(n-1/n)\,L}$ with $L \equiv \log(\ell/\epsilon)$. Differentiating,

$$\frac{\partial}{\partial n}\operatorname{Tr}\rho_A^n = \Big[\frac{c_n'}{c_n} - \frac{c}{6}\Big(1+\frac{1}{n^2}\Big)L\Big]\,\operatorname{Tr}\rho_A^n. \tag{55.120}$$

At $n=1$ we have $c_1 = 1$ and $\operatorname{Tr}\rho_A^1 = 1$, while $1 + 1/n^2 \to 2$, so

$$\frac{\partial}{\partial n}\operatorname{Tr}\rho_A^n\Big|_{n=1} = c_1' - \frac{c}{6}\cdot 2 \cdot L = c_1' - \frac{c}{3}\log\frac{\ell}{\epsilon}. \tag{55.121}$$

By Eq. (55.118), $S_A = -\partial_n \operatorname{Tr}\rho_A^n|_{n=1} = \frac{c}{3}\log(\ell/\epsilon) - c_1'$, which is Eq. (55.119) after renaming the constant. The factor $c/3 = 2 \times c/6$ is the sum of the equal holomorphic and antiholomorphic contributions. □

The result is striking for what it does *not* contain. The entanglement entropy of an interval depends on the theory only through the single number c; all the operator content, the spectrum of dimensions, the OPE coefficients — everything else that distinguishes one CFT from another — drops out of the leading entanglement of a single interval. This is the entanglement face of the statement that c counts the effective number of local degrees of freedom: it is the coefficient with which short-distance correlations across the entangling point accumulate as the region grows. The divergence as $\epsilon \to 0$ is the promised UV divergence; in two dimensions the entangling "surface" is two points (the interval endpoints), and each endpoint contributes a logarithm. The logarithm is the $d = 2$ face of the area law of Section 55.5: in higher dimensions the boundary ∂A has positive-dimensional area and the divergence is power-law; in $d = 2$ the boundary is pointlike and the divergence degenerates to a logarithm, whose coefficient is the universal datum.

The same calculation yields the Rényi entropies, which carry strictly more information than S_A because they probe the full spectrum of ρ_A.

Worked Example 55.33 (Rényi entropies and the replica continuation)**.** The Rényi entropy of index n is $S_n = \frac{1}{1-n} \log \operatorname{Tr} \rho_A^n$, reducing to the von Neumann entropy as $n \to 1$. From Eq. (55.117),

$$S_n = \frac{1}{1-n}\Big[\log c_n - \frac{c}{6}\Big(n - \frac{1}{n}\Big)\log\frac{\ell}{\epsilon}\Big]. \tag{55.122}$$

The universal piece simplifies: $-\frac{c}{6}\frac{n-1/n}{1-n} = -\frac{c}{6}\frac{(n^2-1)/n}{1-n} = -\frac{c}{6}\frac{(n-1)(n+1)}{n(1-n)} = \frac{c}{6}\frac{n+1}{n} = \frac{c}{6}\big(1+\frac{1}{n}\big)$, so

$$S_n = \frac{c}{6}\Big(1 + \frac{1}{n}\Big)\log\frac{\ell}{\epsilon} + \frac{\log c_n}{1-n}. \tag{55.123}$$

As a check on the analytic continuation, $n \to 1$ gives the universal coefficient $\frac{c}{6}(1+1) = \frac{c}{3}$, reproducing Eq. (55.119); the non-universal term $\log c_n/(1-n) \to -c_1'$ by l'Hôpital, since $c_1 = 1$. The monotonic dependence of the universal coefficient $\frac{c}{6}(1+1/n)$ on n — decreasing from $\frac{c}{3}$ at $n = 1$ toward $\frac{c}{6}$ at $n \to \infty$ — reflects that higher Rényi indices weight the largest eigenvalues of ρ_A more heavily. That the entire family is controlled by the same c is the hallmark of conformal invariance: one anomaly coefficient fixes an infinite tower of entropies.

55.4.4 Finite size and finite temperature

The infinite-line vacuum is a single point in a family of conformally related geometries. Because the twist operators are primaries, their correlators on any geometry conformally equivalent to the plane are obtained by transporting

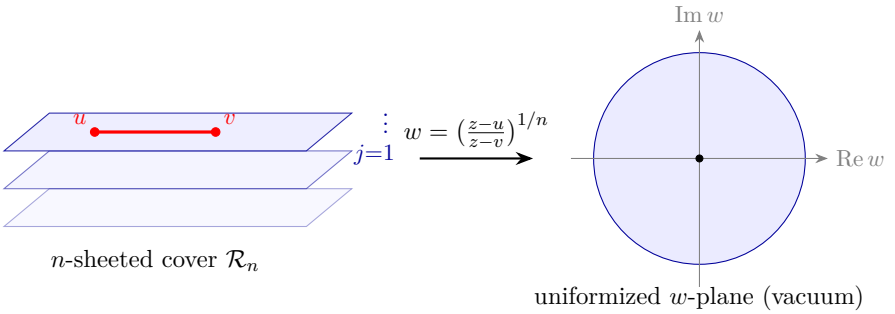


Figure 55.4: The conformal uniformization of the replica surface. The n-sheeted cover $\mathcal{R}_n$, branched over the interval endpoints u, v (red cut), is mapped one-to-one onto a single w-plane by $w = ((z-u)/(z-v))^{1/n}$. On the w-plane the state is the ordinary vacuum, $\langle T(w)\rangle = 0$; the nonzero stress tensor on $\mathcal{R}_n$ in Eq. (55.103) comes entirely from the Schwarzian of the map, i.e. from the central charge.

Eq. (55.116) with the appropriate map. Two cases are physically central: an interval on a spatial circle (finite size) and an interval in a thermal state (finite temperature). Each is a cylinder, and each is mapped to the plane by an exponential. We carry both out explicitly.

The transformation law we need is that of a spinless primary of dimension Δ_n under a holomorphic map $z = f(w)$: each insertion picks up a Jacobian $|f'(w_i)|^{\Delta_n}$ (the product of the holomorphic factor $(f')^{h_n}$ and its antiholomorphic conjugate, with $h_n = \bar{h}_n$ and $2h_n = \Delta_n$), so that

$$\langle \sigma_n(w_1)\tilde{\sigma}_n(w_2)\rangle_{\text{cyl}} = |f'(w_1)|^{\Delta_n}\,|f'(w_2)|^{\Delta_n}\,\langle \sigma_n(z_1)\tilde{\sigma}_n(z_2)\rangle_{\text{plane}}, \qquad z_i = f(w_i). \tag{55.124}$$

The physical cutoff ϵ is the proper short-distance scale on the cylinder, where the lattice lives, and is held fixed in the cylinder frame; using the plane two-point function $\langle \sigma_n\tilde{\sigma}_n\rangle_{\text{plane}} = c_n\,\epsilon^{2\Delta_n}/|z_1 - z_2|^{2\Delta_n}$ keeps every appearance of ϵ a proper length to leading order.

Finite size: an interval on a circle. Let space be a circle of circumference L, with complex cylinder coordinate w identified under $w \sim w + L$. The exponential map

$$z = f(w) = e^{2\pi \mathrm{i} w/L} \tag{55.125}$$

sends the spatial cylinder to the plane: $w \to w + L$ gives $z \to z\,e^{2\pi \mathrm{i}} = z$, single-valued, with the spatial circle becoming the unit circle $|z| = 1$. The interval endpoints sit at equal Euclidean time, w_1, w_2 real with $w_2 - w_1 = \ell$. Then $|z_i| = 1$ and

$$|z_1 - z_2| = \left|e^{2\pi \mathrm{i} w_1/L} - e^{2\pi \mathrm{i} w_2/L}\right| = \left|1 - e^{2\pi \mathrm{i}\ell/L}\right| = 2\,\sin\frac{\pi\ell}{L}, \tag{55.126}$$

using $|1-e^{\mathrm{i}\theta}| = 2|\sin(\theta/2)|$ with $\theta = 2\pi\ell/L$. The Jacobian magnitude at each endpoint is $|f'(w_i)| = (2\pi/L)|z_i| = 2\pi/L$. Inserting into Eq. (55.124),

$$\begin{aligned}\operatorname{Tr}\rho_A^n = \langle \sigma_n \tilde{\sigma}_n \rangle_{\text{cyl}} &= \Big(\frac{2\pi}{L}\Big)^{2\Delta_n} \frac{c_n\, \epsilon^{2\Delta_n}}{\big[2\sin(\pi\ell/L)\big]^{2\Delta_n}} = c_n \Big[\frac{L}{\pi\epsilon}\sin\frac{\pi\ell}{L}\Big]^{-2\Delta_n} \\ &= c_n \Big[\frac{L}{\pi\epsilon}\sin\frac{\pi\ell}{L}\Big]^{-(c/6)(n-1/n)}. \end{aligned} \tag{55.127}$$

This is Eq. (55.117) with the chord length ℓ replaced by the conformal chord $(L/\pi)\sin(\pi\ell/L)$. The replica limit Eq. (55.118) acts on the logarithm of this quantity exactly as in Theorem 55.32, with $\log(\ell/\epsilon)$ promoted to $\log[(L/\pi\epsilon)\sin(\pi\ell/L)]$, giving

$$\boxed{S_A = \frac{c}{3}\log\Big[\frac{L}{\pi\epsilon}\sin\frac{\pi\ell}{L}\Big] + \text{const.}} \tag{55.128}$$

Two checks confirm the formula. For $\ell \ll L$, $\sin(\pi\ell/L) \approx \pi\ell/L$ and the argument collapses to ℓ/ϵ, recovering the infinite-line result Eq. (55.119): a small interval cannot resolve the global topology. And the formula is symmetric under $\ell \to L-\ell$, since $\sin[\pi(L-\ell)/L] = \sin(\pi\ell/L)$; this is mandatory, because the full state on the circle is the pure vacuum, for which $S_A = S_{A^c}$ — the entanglement entropy of a region equals that of its complement.

Finite temperature: an interval in a thermal state. A thermal state at inverse temperature β is prepared by the Euclidean path integral on a cylinder whose *time* direction is compact, $\tau \sim \tau + \beta$, with space the infinite line. Now the complex coordinate is w with $w \sim w + \mathrm{i}\beta$ (imaginary period β), and the uniformizing exponential is

$$z = f(w) = e^{2\pi w/\beta}, \tag{55.129}$$

single-valued under $w \to w + \mathrm{i}\beta$. The interval is spatial, endpoints at equal time with real separation $w_2 - w_1 = \ell$. The natural conformal invariant is the combination $|z_1 - z_2|^2/(|z_1||z_2|)$, which is independent of the overall phase and scale of z and so descends cleanly to the cylinder. With $z_i = e^{2\pi w_i/\beta}$ and w_i real,

$$z_1 - z_2 = e^{\pi(w_1+w_2)/\beta}\Big(e^{\pi(w_1-w_2)/\beta} - e^{-\pi(w_1-w_2)/\beta}\Big) = e^{\pi(w_1+w_2)/\beta}\cdot 2\sinh\frac{\pi(w_1-w_2)}{\beta}, \tag{55.130}$$

while $|z_1|^{1/2}|z_2|^{1/2} = e^{\pi(w_1+w_2)/\beta}$, so

$$\frac{|z_1 - z_2|^2}{|z_1|\,|z_2|} = \Big[2\sinh\frac{\pi\ell}{\beta}\Big]^2. \tag{55.131}$$

The transport Eq. (55.124) carries factors $|f'(w_i)|^{\Delta_n} = (2\pi/\beta)^{\Delta_n}|z_i|^{\Delta_n}$, and combining the $|z_i|$ pieces with the plane two-point function reproduces exactly

this invariant:

$$\operatorname{Tr}\rho_A^n = \Big(\frac{2\pi}{\beta}\Big)^{2\Delta_n}(|z_1||z_2|)^{\Delta_n}\frac{c_n\,\epsilon^{2\Delta_n}}{|z_1-z_2|^{2\Delta_n}} = c_n\Big(\frac{2\pi}{\beta}\Big)^{2\Delta_n}\epsilon^{2\Delta_n}\Big[\frac{|z_1-z_2|^2}{|z_1||z_2|}\Big]^{-\Delta_n}$$
$$= c_n\Big[\frac{2\pi\epsilon}{\beta}\cdot\frac{1}{2\sinh(\pi\ell/\beta)}\Big]^{2\Delta_n} = c_n\Big[\frac{\beta}{\pi\epsilon}\sinh\frac{\pi\ell}{\beta}\Big]^{-(c/6)(n-1/n)}. \tag{55.132}$$

The replica limit gives the finite-temperature single-interval entropy,

$$\boxed{S_A = \frac{c}{3}\log\Big[\frac{\beta}{\pi\epsilon}\sinh\frac{\pi\ell}{\beta}\Big] + \text{const.}} \tag{55.133}$$

The sine of the finite-size case has become a hyperbolic sine, the signature of the swapped (timelike vs. spacelike) compactification. This is the result anticipated by the thermal reading of the vacuum in Section 55.3: a finite temperature converts the vacuum logarithm into a function of ℓ/β.

Worked Example 55.34 (Thermal limits: the crossover to extensive entropy)**.** The two limits of Eq. (55.133) are physically instructive. For a small interval, $\ell \ll \beta$, expand $\sinh(\pi\ell/\beta) \approx \pi\ell/\beta$; the argument of the logarithm collapses to ℓ/ϵ and

$$S_A \xrightarrow{\ell\ll\beta} \frac{c}{3}\log\frac{\ell}{\epsilon} + \text{const}, \tag{55.134}$$

the vacuum result Eq. (55.119): a region much smaller than the thermal wavelength cannot tell it is in a thermal state, and the entanglement is dominated by the vacuum fluctuations across the endpoints. For a large interval, $\ell \gg \beta$, use $\sinh(\pi\ell/\beta) \approx \frac{1}{2}e^{\pi\ell/\beta}$, so $\log[\,\cdot\,] \approx \pi\ell/\beta + \log(\beta/2\pi\epsilon)$ and

$$S_A \xrightarrow{\ell\gg\beta} \frac{c}{3}\cdot\frac{\pi\ell}{\beta} + \frac{c}{3}\log\frac{\beta}{2\pi\epsilon} + \text{const} = \frac{\pi c}{3\beta}\,\ell + \cdots. \tag{55.135}$$

The entropy becomes *extensive*, linear in the interval length, with density $s = \pi c/3\beta$. This is exactly the thermal (Cardy) entropy density of a 2D CFT: for a large region the reduced density matrix is effectively thermal in the bulk of the interval, and the entanglement entropy reduces to ordinary thermodynamic entropy, with only the two endpoint logarithms surviving as subleading boundary corrections. The crossover from the logarithmic (entanglement-dominated) regime to the linear (thermal-dominated) regime at $\ell \sim \beta$ is the precise statement that entanglement entropy interpolates between a genuine quantum correlation measure and a coarse thermodynamic one. The universal coefficient $\pi c/3\beta$ is one more incarnation of the same c.

55.4.5 The universal coefficient, the c-function, and the Ising endpoint

The coefficient $c/3$ in Eq. (55.119) is more than a number attached to a divergence: it is a renormalization-group diagnostic. Although S_A itself is cutoff-dependent,

the combination

$$c_{\text{eff}}(\ell) \equiv 3\,\ell\,\frac{\mathrm{d}S_A}{\mathrm{d}\ell} \tag{55.136}$$

is cutoff-*independent* — the derivative kills the additive constant and the $\log\epsilon$ — and at a fixed point equals the central charge, $c_{\text{eff}} = c$. Away from a fixed point, along an RG flow between an ultraviolet CFT (central charge c_{UV}) and an infrared one (c_{IR}), $c_{\text{eff}}(\ell)$ interpolates between the two: probing the entanglement of a small interval sees the UV theory, a large interval sees the IR theory. The monotonicity of this interpolation is the content of Zamolodchikov's c-theorem, stated and proved (in its stress-tensor form) in Theorem 52.17: there is a function along the flow, monotonically decreasing toward the infrared, that coincides with c at the endpoints, so that $c_{\text{UV}} \geq c_{\text{IR}}$. The entanglement entropy provides an information-theoretic realization of that monotone — the entropic c-function — whose decrease expresses the loss of degrees of freedom as short-distance modes are integrated out. We import the c-theorem without full proof from Theorem 52.17; what the present computation adds is the identification of its coefficient with an entanglement observable, the slope of S_A against $\log\ell$.

Physical Insight

The chain of identifications is the moral of the section. The central charge c appears as: the coefficient of the $T(z)T(w)$ short-distance singularity (Theorem 48.8); the Schwarzian anomaly in the finite transformation of the stress tensor (Eq. (48.63)); the Casimir energy on a cylinder; the trace anomaly in a curved background; and now the coefficient $c/3$ of the entanglement logarithm. These are not five coincidences but five faces of one object — the response of the vacuum to being curved, squeezed, or cut. Entanglement entropy is the informational face: c measures how much quantum correlation a region shares with its complement, per logarithm of its size.

The Ising endpoint of Scalar ϕ^4 theory. We can now close ϕ^4 theory on the entanglement side. The real scalar $\lambda\phi^4$ theory, carried since Chapter 6 from its classical Lagrangian through quantization, renormalization, and the RG, was shown in Chapter 52 to flow to the Wilson–Fisher fixed point in $d = 4 - \epsilon$, whose two-dimensional avatar is the Ising universality class: the minimal model $\mathcal{M}(4,3)$ with central charge

$$c = \frac{1}{2}, \tag{55.137}$$

the value derived from the minimal-model formula Eq. (48.176) and confirmed against the free Majorana fermion in Chapter 48. Substituting into Theorem 55.32, the single-interval vacuum entanglement entropy of the critical Ising chain is

$$S_A = \frac{1}{6}\log\frac{\ell}{\epsilon} + \text{const}, \tag{55.138}$$

since $c/3 = \frac{1}{6}$. This is a sharp, falsifiable prediction: the ground-state entanglement entropy of a block of ℓ/ϵ spins in the critical transverse-field Ising chain grows as $\frac{1}{6}\log\ell$, with the coefficient fixed entirely by the central charge $c = \frac{1}{2}$. The same $c = \frac{1}{2}$ that the bootstrap of Chapter 50 pins down for the three-dimensional Ising model in the form of operator dimensions appears here, in two dimensions, as the slope of an entanglement logarithm — two windows on the one fixed point that ϕ^4 theory has been approaching since Part II. The Wilson–Fisher endpoint, met first as a zero of the beta function, is now also the source of a universal entanglement coefficient.

Physical Insight

The Ising twist operator can be made completely explicit. In the free Majorana realization of the Ising CFT, the n-fold replicated theory is n free Majorana fermions, and the cyclic $\mathbb{Z}_n$ twist diagonalizes into phase twists with angles $2\pi k/n$, $k = -(n-1)/2, \ldots, (n-1)/2$. Because the fermions are Majorana (real), the conjugate sectors k and $-k$ pair into a single Dirac fermion, so the independent twists are labeled by $k = 1, \ldots, (n-1)/2$ (the $k = 0$ mode is an untwisted Majorana of dimension zero): the n Majoranas organize into $\sim n/2$ Dirac fermions, not n independent U(1) twists. A single U(1) twist by angle $2\pi k/n$ on a Dirac fermion is a vertex operator of dimension $h = \frac{1}{2}(k/n)^2$; summing the holomorphic weights over the independent Dirac sectors, $\sum_{k=1}^{(n-1)/2} \frac{1}{2}(k/n)^2 = \frac{1}{2n^2}\cdot\frac{n(n^2-1)}{24} = \frac{1}{48}(n-1/n)$, which is exactly h_n at $c = \frac{1}{2}$ from Eq. (55.114). The abstract Schwarzian computation and the concrete free-field one agree, as they must.

The single-interval logarithm and its finite-size and finite-temperature deformations are now established as the universal entanglement data of any 2D CFT, fixed by the central charge alone. These are precisely the numbers a holographic dual must reproduce: in Chapter 56 the Ryu–Takayanagi formula will compute the same $(c/3)\log(\ell/\epsilon)$, and its sine and sinh variants, as the length of a bulk geodesic anchored on the interval endpoints, turning the CFT result of this section into the boundary check of holographic entanglement. The logarithm we have derived is also the $d = 2$ representative of the area law, whose higher-dimensional form — proportionality to the area of the entangling surface — is the subject of Section 55.5.

55.5 The Area Law and the Structure of UV Divergences

The two-dimensional logarithm of Section 55.4.3 and the half-space computation of Section 55.3 both pointed beyond themselves. The 2D result $S_A = \frac{c}{3}\log(\ell/\epsilon)$ is

a divergence whose coefficient is universal, but the divergence itself is logarithmic only because the entangling "surface" of an interval is a pair of points; for a region in three or more spacetime dimensions the boundary ∂A is a genuine surface, and we should expect a stronger, power-law divergence proportional to its size. The Rindler computation made this concrete from the opposite direction: rewriting the half-space entanglement entropy as the thermodynamic entropy of a thermal gas piled up against the entangling plane (Section 55.3), we found an entropy proportional to the transverse area $\mathcal{A}_\perp$ of that plane times a factor that diverges as the cutoff is removed, with the proportionality constant left as a promissory note — "the constant counts the field's degrees of freedom and is computed in full in Section 55.5." This section pays that note. We establish, by an explicit and honest computation on the free real scalar of ϕ^4 theory, the generic leading behavior of entanglement entropy in $d > 2$: proportionality to the *area* of the entangling surface measured in cutoff units,

$$S_A \;=\; \kappa\,\frac{\mathcal{A}(\partial A)}{\epsilon^{\,d-2}} \;+\; (\text{subleading}), \tag{55.139}$$

the result of Bombelli, Koul, Lee and Sorkin and of Srednicki [85, 453]. We then dissect the full divergence structure — the tower of subleading power-law divergences and, beneath them, the one piece that is universal: a logarithm tied to a conformal anomaly when d is even, a finite constant when d is odd. The recurring theme of the chapter reaches its sharpest form here: S_A is not a number but a cutoff-dependent, scheme-dependent quantity, and the physics lives in the universal coefficient and in subtracted combinations such as the mutual information.

The computational engine is the observation, already glimpsed in the two-oscillator worked example of Theorem 55.16, that the vacuum of a free field is a Gaussian state, and that the entanglement entropy of any region in a Gaussian state is computed once and for all from the two-point correlators alone. We build that formula first, in full generality, then feed it the free scalar.

55.5.1 The entanglement entropy of a Gaussian state

A free scalar field, on the lattice that regulates it, is a system of coupled harmonic oscillators, and its vacuum is the Gaussian ground state of a positive-definite quadratic Hamiltonian. The reduced density matrix of any set of oscillators is therefore a Gaussian operator, and Gaussian operators are determined entirely by their second moments. We make this precise and extract S_A from the moments.

Let the region A contain N oscillators with canonical coordinates and momenta $\hat{x}_i, \hat{p}_i$ $(i = 1, \dots, N)$, collected into the phase-space vector $\hat{R} = (\hat{x}_1, \dots, \hat{x}_N, \hat{p}_1, \dots, \hat{p}_N)$ obeying $[\hat{R}_a, \hat{R}_b] = \mathrm{i}\,\Omega_{ab}$ with the symplectic form

$$\Omega = \begin{pmatrix} 0 & \mathbf{1}_N \\ -\mathbf{1}_N & 0 \end{pmatrix}. \tag{55.140}$$

For a state with vanishing first moments — which the vacuum is, by translation invariance in field space — all information accessible to local Gaussian observables is in the symmetric *covariance matrix*

$$\sigma_{ab} = \tfrac{1}{2}\langle\{\hat{R}_a - \langle\hat{R}_a\rangle,\ \hat{R}_b - \langle\hat{R}_b\rangle\}\rangle = \tfrac{1}{2}\langle\{\hat{R}_a, \hat{R}_b\}\rangle. \tag{55.141}$$

For the vacuum of a real Hamiltonian there are no cross correlations between positions and momenta, $\langle\{\hat{x}_i, \hat{p}_j\}\rangle = 0$, so the covariance matrix is block-diagonal in the (x, p) split,

$$\sigma = \begin{pmatrix} X & 0 \\ 0 & P \end{pmatrix}, \qquad X_{ij} = \langle\hat{x}_i\hat{x}_j\rangle, \qquad P_{ij} = \langle\hat{p}_i\hat{p}_j\rangle, \tag{55.142}$$

the two correlation matrices announced in the section heading. The reduced density matrix on A of a global Gaussian pure state is itself a Gaussian (mixed) state, and its covariance matrix is exactly the restriction of σ to the oscillators in A — the submatrices $X_A = X|_{A\times A}$, $P_A = P|_{A\times A}$ obtained by deleting all rows and columns not in A. Tracing out the complement is, at the level of second moments, simply dropping the entries that involve it.

The entropy is then a purely linear-algebraic functional of X_A, P_A, by way of Williamson's normal form.

Theorem 55.35 (Williamson normal form and the Gaussian entanglement entropy)**.** *Let $\sigma_A = \mathrm{diag}(X_A, P_A)$ be the covariance matrix Eq. (55.142) of a Gaussian state on N_A oscillators. There exists a symplectic transformation $S \in \mathrm{Sp}(2N_A, \mathbb{R})$ (i.e. $S\Omega S^{\mathsf{T}} = \Omega$) bringing it to the diagonal form*

$$S\,\sigma_A\,S^{\mathsf{T}} = \mathrm{diag}(\nu_1, \ldots, \nu_{N_A},\ \nu_1, \ldots, \nu_{N_A}), \tag{55.143}$$

where the symplectic eigenvalues *$\nu_k \geq \frac{1}{2}$ are the (positive) eigenvalues of the matrix*

$$C_A \equiv \sqrt{X_A\,P_A}\,, \tag{55.144}$$

and the entanglement entropy of the region is

$$\boxed{S_A = \sum_{k=1}^{N_A} g(\nu_k), \qquad g(\nu) = \left(\nu + \tfrac{1}{2}\right)\log\left(\nu + \tfrac{1}{2}\right) - \left(\nu - \tfrac{1}{2}\right)\log\left(\nu - \tfrac{1}{2}\right).} \tag{55.145}$$

Proof. A symplectic transformation $\hat{R} \to S\hat{R}$ is implemented on Hilbert space by a unitary (a Gaussian/Bogoliubov rotation), under which the von Neumann entropy is invariant; it acts on the covariance matrix by $\sigma_A \to S\sigma_A S^{\mathsf{T}}$. Williamson's theorem states that any positive-definite symmetric $2N_A \times 2N_A$ matrix can be brought by such an S to the diagonal form Eq. (55.143). To identify the ν_k, note that the eigenvalues of $\Omega\sigma_A$ are symplectic invariants

(a similarity by S preserves the spectrum of $\Omega\sigma_A$ because $S\Omega S^{\mathsf{T}} = \Omega$ gives $S(\Omega\sigma_A)S^{-1} = \Omega\, S^{-\mathsf{T}}\sigma_A S^{-1} = \Omega\,(S\sigma_A S^{\mathsf{T}})$ up to the rearrangement that fixes Ω). On the normal form Eq. (55.143) the matrix $\Omega\sigma_A$ has eigenvalues $\pm\mathrm{i}\nu_k$. For the block form $\sigma_A = \mathrm{diag}(X_A, P_A)$,

$$\Omega\sigma_A = \begin{pmatrix} 0 & P_A \\ -X_A & 0 \end{pmatrix}, \qquad (\Omega\sigma_A)^2 = \begin{pmatrix} -P_A X_A & 0 \\ 0 & -X_A P_A \end{pmatrix}, \tag{55.146}$$

whose eigenvalues are $-\lambda$ with λ ranging over the eigenvalues of $X_A P_A$ (the matrices $X_A P_A$ and $P_A X_A$ are similar and share their spectrum, which is real and positive because X_A, P_A are positive). Hence the eigenvalues of $\Omega\sigma_A$ are $\pm\mathrm{i}\sqrt{\lambda}$, and comparison with $\pm\mathrm{i}\nu_k$ gives $\nu_k = \sqrt{\lambda_k}$, i.e. the ν_k are the eigenvalues of $C_A = \sqrt{X_A P_A}$, establishing Eq. (55.144). The uncertainty principle forces $\nu_k \geq \frac{1}{2}$, with equality for a pure mode.

After the symplectic rotation the state factorizes into N_A independent single modes, the k-th with covariance $\mathrm{diag}(\nu_k, \nu_k)$. The von Neumann entropy is additive over these modes, $S_A = \sum_k g(\nu_k)$ with $g(\nu)$ the entropy of one mode. A single Gaussian mode of symplectic eigenvalue ν is a thermal state: its position and momentum variances are equal, $\langle x^2\rangle = \langle p^2\rangle = \nu$, which is the covariance of a harmonic oscillator at frequency 1 in a thermal state with mean occupation $\bar{n} = \nu - \frac{1}{2}$, density matrix $\rho = (1-\xi)\sum_{m\geq 0}\xi^m\,|m\rangle\langle m|$ with $\bar{n} = \xi/(1-\xi)$. Its entropy was computed once and for all in Eq. (55.36), $S = -\log(1-\xi) - \frac{\xi}{1-\xi}\log\xi = (\bar{n}+1)\log(\bar{n}+1) - \bar{n}\log\bar{n}$. Substituting $\bar{n} = \nu - \frac{1}{2}$ gives $g(\nu) = (\nu+\frac{1}{2})\log(\nu+\frac{1}{2}) - (\nu-\frac{1}{2})\log(\nu-\frac{1}{2})$, which is Eq. (55.145). □

Remark 55.36. The formula is a direct generalization of the two-oscillator example. There, $N_A = 1$ and C_A is the single number $\sqrt{X_A P_A} = \sqrt{\langle\hat{x}_1^2\rangle\langle\hat{p}_1^2\rangle}$; one checks that $\nu = \sqrt{X_A P_A}$ reproduces the geometric spectrum of Theorem 55.16 through $\xi = (2\nu-1)/(2\nu+1)$, the inverse of $\bar{n} = \nu - \frac{1}{2} = \xi/(1-\xi)$. For the whole system (taking A to be everything), the global vacuum is pure, and indeed $XP = \frac{1}{4}\mathbf{1}$ (shown below), so every $\nu_k = \frac{1}{2}$ and $g(\frac{1}{2}) = 0$: the global entropy vanishes, as it must. *All* of the entanglement entropy comes from the fact that the products of the truncated submatrices, $X_A P_A$, fail to equal $\frac{1}{4}\mathbf{1}$ — truncation does not commute with the matrix square root, and the deficit is the entanglement.

This formula is the exact, computational counterpart of the replica machinery: where Section 55.2.4 expressed the entropy through a replica free energy $\log Z_n = -\frac{1}{2}\log\det\Delta_n$ on a branched cover (a functional determinant, hence the relevance of Sections E.2 and E.3), Eq. (55.145) delivers the same number directly from the equal-time correlators of the unreplicated vacuum, never requiring the analytic continuation in n to be performed by hand. For Gaussian theories the two routes are equivalent; the correlator route is the one that computes [115, 395].

55.5.2 The free scalar: correlators, the spherical region, and the area law

We now apply Theorem 55.35 to the free real scalar of ϕ^4 theory at $\lambda = 0$, whose mode expansion and vacuum were constructed in Section 9.2 and Theorem 9.11. On a spatial lattice of spacing a the field is the oscillator system

$$H = \frac{1}{2}\sum_{i} \hat{\pi}_i^2 + \frac{1}{2}\sum_{ij} \hat{\phi}_i \, K_{ij} \, \hat{\phi}_j, \tag{55.147}$$

with K a real symmetric positive matrix — the discretized $(-\nabla^2 + m^2)$ operator, $K_{ij} = (2(d-1) + m^2 a^2)\delta_{ij} - \sum_{\text{neighbours}} \delta_{i,j\pm\hat{e}}$ in units $a = 1$ for a cubic lattice in $d-1$ spatial dimensions. The vacuum correlators follow from diagonalizing $K = O^{\mathsf{T}} \mathcal{D} O$ with $\mathcal{D} = \mathrm{diag}(\omega_a^2)$: in each normal mode $\langle y_a^2 \rangle = 1/(2\omega_a)$ and $\langle \pi_a^2 \rangle = \omega_a/2$, so rotating back,

$$X = \langle \hat{\phi}\hat{\phi}^{\mathsf{T}} \rangle = \tfrac{1}{2}\, K^{-1/2}, \qquad P = \langle \hat{\pi}\hat{\pi}^{\mathsf{T}} \rangle = \tfrac{1}{2}\, K^{1/2}. \tag{55.148}$$

Globally $XP = \frac{1}{4} K^{-1/2} K^{1/2} = \frac{1}{4}\mathbf{1}$, the purity statement used above. The entanglement entropy of a region A is Eq. (55.145) evaluated on the symplectic eigenvalues of $C_A = \sqrt{X_A P_A}$, with X_A, P_A the submatrices of Eq. (55.148) indexed by the lattice sites inside A.

The spherical region and the radial reduction. For a ball $A = \{|\mathbf{x}| < R\}$ the rotational symmetry of the massless theory can be used to reduce the $(d-1)$-dimensional lattice problem to a family of one-dimensional radial chains, one for each angular-momentum sector — the decomposition introduced by Srednicki [453]. We carry it out in the physical case $d = 4$ (three spatial dimensions); the general case differs only in the harmonic degeneracies. Expanding the field in spherical harmonics on S^2,

$$\hat{\phi}(\mathbf{x}) = \sum_{\ell=0}^{\infty} \sum_{m=-\ell}^{\ell} \frac{1}{r}\, \hat{\phi}_{\ell m}(r)\, Y_{\ell m}(\theta,\varphi), \qquad \hat{\pi}(\mathbf{x}) = \sum_{\ell m} \frac{1}{r}\, \hat{\pi}_{\ell m}(r)\, Y_{\ell m}(\theta,\varphi), \tag{55.149}$$

and inserting into $H = \frac{1}{2}\int \mathrm{d}^3x\, [\hat{\pi}^2 + (\nabla\hat{\phi})^2]$, the angular integrals collapse by orthonormality of the $Y_{\ell m}$ and the radial centrifugal term $\ell(\ell+1)/r^2$ appears, leaving a sum of independent radial Hamiltonians

$$H = \frac{1}{2}\sum_{\ell m} \int_0^{\infty} \mathrm{d}r \left[\hat{\pi}_{\ell m}(r)^2 + r^2 \Big(\partial_r \frac{\hat{\phi}_{\ell m}(r)}{r} \Big)^2 + \frac{\ell(\ell+1)}{r^2}\, \hat{\phi}_{\ell m}(r)^2 \right]. \tag{55.150}$$

Discretizing the radius at the midpoints $r_j = (j - \frac{1}{2})a$, $j = 1, \dots, N$ (the half-integer offset keeps the origin off the lattice and renders the difference operator

regular), the radial integral becomes the oscillator chain $H = \tfrac{1}{2}\sum_{\ell m}[\sum_j \hat\pi^2_{\ell m,j} + \sum_{jj'} \hat\phi_{\ell m,j} K^\ell_{jj'} \hat\phi_{\ell m,j'}]$ with the tridiagonal coupling (in units $a = 1$)

$$K^\ell_{jj} = \frac{j^2 + (j-1)^2}{(j-\frac12)^2} + \frac{\ell(\ell+1)}{(j-\frac12)^2}, \qquad K^\ell_{j,j+1} = K^\ell_{j+1,j} = -\frac{j^2}{j^2 - \frac14}, \tag{55.151}$$

the standard finite-difference form of Eq. (55.150). Because the modes of different (ℓ, m) are decoupled and the entropy is additive over independent modes, the total entanglement entropy of the ball of $n = R/a$ radial sites is the weighted sum of the one-dimensional chain entropies,

$$S(R) = \sum_{\ell=0}^{\infty} (2\ell+1)\, S_\ell(n), \qquad S_\ell(n) = \sum_k g(\nu_k^{(\ell)}), \tag{55.152}$$

where $S_\ell(n)$ is computed from Theorem 55.35 applied to the first n sites of the chain K^ℓ, and the factor $(2\ell+1)$ is the degeneracy of the S^2 harmonics. Each radial chain is a finite tridiagonal matrix; the symplectic eigenvalues are obtained by a single matrix square root, and the sum over ℓ converges because $S_\ell(n)$ falls off once $\ell \gtrsim n$ (the centrifugal barrier expels the high-ℓ modes from the region).

Theorem 55.37 (Area law for the free scalar in $d > 2$)**.** *The vacuum entanglement entropy of a ball of radius R for the free massless scalar in d spacetime dimensions, regulated by a lattice of spacing $a = \epsilon$, scales for $R \gg \epsilon$ as*

$$S(R) = \kappa \left(\frac{R}{\epsilon}\right)^{d-2} + \cdots = \kappa\, \frac{\mathcal{A}(\partial A)}{\epsilon^{\,d-2}} + (\textit{lower powers of } R/\epsilon), \tag{55.153}$$

with $\mathcal{A}(\partial A) = \Omega_{d-2} R^{d-2}$ the area of the entangling sphere and κ a dimensionless, cutoff-scheme-dependent constant. The entanglement entropy is proportional to the area *of the entangling surface, not the volume of the region.*

Argument. We give the scaling argument that fixes the power, then confirm the coefficient numerically. In d spacetime dimensions the entangling sphere is S^{d-2} and the degree-ℓ harmonics have degeneracy $g_\ell \sim \ell^{\,d-3}/(d-3)!$ for large ℓ. The single-chain entropy $S_\ell(n)$ is, by the masslessness and the absence of any scale other than the ratio, a function of ℓ/n alone for $n \gg 1$: it is $O(1)$ for $\ell \lesssim n$ and is exponentially suppressed for $\ell \gg n$, because a mode of angular momentum ℓ has its classical turning point at $r \sim \ell a/\sqrt{\cdots}$ and ceases to straddle the boundary at $r = R = na$ once $\ell \gg n$. Therefore

$$S(R) = \sum_\ell g_\ell\, S_\ell(n) \simeq \int_0^\infty \mathrm{d}\ell\, \frac{\ell^{\,d-3}}{(d-3)!}\, f(\ell/n) = n^{\,d-2} \underbrace{\int_0^\infty \mathrm{d}x\, \frac{x^{\,d-3}}{(d-3)!}\, f(x)}_{\equiv \kappa'} = \kappa' \Big(\frac{R}{\epsilon}\Big)^{d-2}, \tag{55.154}$$

having substituted $\ell = nx$. The power $d-2$ is exactly the dimension of the entangling surface, so the result is proportional to its area. The convergence

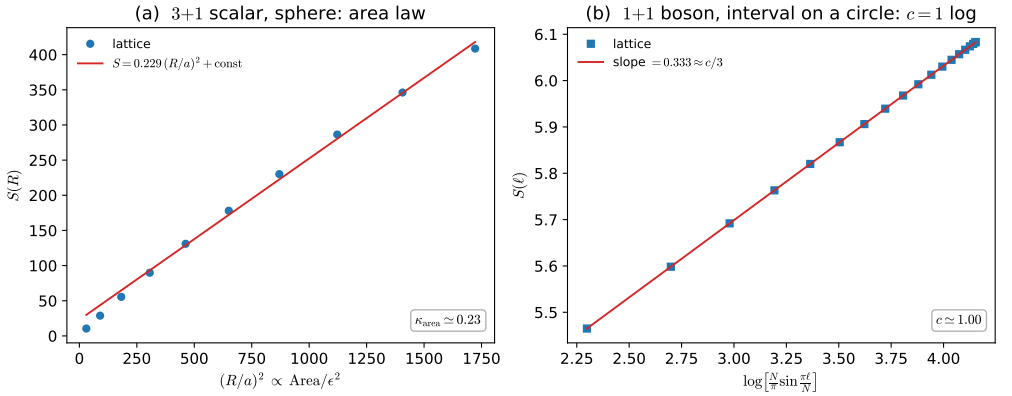


Figure 55.5: The free-scalar entanglement entropy computed from the Gaussian formula Eq. (55.145), with symplectic eigenvalues $\nu_k = \text{eig}\sqrt{X_A P_A}$ of the lattice correlators Eq. (55.148). **(a)** The 3+1 massless scalar on a spherical region, via the radial decomposition Eq. (55.152): plotting S against $(R/\epsilon)^2$ exhibits the area law $S = \kappa_{\text{area}}(R/\epsilon)^2 + \cdots$, with $\kappa_{\text{area}} \approx 0.23$ in this scheme. **(b)** The 1+1 free boson, an interval of ℓ sites on a periodic chain of N sites: plotting S against the conformal-chord variable $\log[(N/\pi)\sin(\pi\ell/N)]$ of Eq. (55.128) gives a straight line of slope $c/3$, and the fit returns $c = 1.000$ — the central charge of the free boson, the $d = 2$ instance of the same computation and a cross-check of the CFT result of Theorem 55.32.

of the x-integral is guaranteed by the suppression of $f(x)$ at large x; its value κ' (equivalently $\kappa = \kappa'/\Omega_{d-2}$) is a pure number determined by the lattice and is not universal. The numerical evaluation of Eq. (55.152) in $d = 4$, summing the radial-chain entropies over ℓ, confirms the linear dependence of S on $(R/\epsilon)^2$ and returns $\kappa_{\text{area}} \equiv \kappa' \approx 0.23$ in the midpoint scheme (Fig. 55.5a); Srednicki's original mode-sum convention gives ≈ 0.30. The two numbers differ because the coefficient is scheme-dependent, which is precisely the lesson; the area *law* — the power $(R/\epsilon)^{d-2}$ — is common to both. □

Physical Insight

The area law is the entanglement signature of locality. The vacuum correlations of a local field theory are short-ranged — the equal-time two-point function decays away from coincidence — so the only field modes entangled across the cut are those localized within a cutoff cell of the boundary ∂A. Tiling the boundary with $\sim \mathcal{A}/\epsilon^{d-2}$ such cells, each contributing $O(1)$ of entanglement, reproduces Eq. (55.153); the heuristic of Eq. (55.24) is now the honest result of Theorem 55.35. A state with *long*-range entanglement — a generic, non-vacuum state of the

field, or a thermal state in its bulk — would instead show a volume law $S \sim \mathrm{Vol}(A)/\epsilon^{d-1}$. That the vacuum sits at the boundary of these two behaviors, obeying an area law, is what makes its entanglement geometrical, and is the fact the Ryu–Takayanagi formula of Chapter 56 will turn into a statement about minimal surfaces.

55.5.3 The area-law coefficient from the Rindler thermal gas

The lattice computation yields the area law as a numerical fact. The Rindler analysis of Section 55.3 lets us recover its structure — the power of the cutoff and the way the coefficient counts the field's degrees of freedom — in closed form, discharging the promise made there. This is the worked computation the half-space section deferred to the present one.

Worked Example 55.38 (The area law as a Rindler thermal-gas entropy)**.** By the Bisognano–Wichmann result Theorem 55.25, the vacuum restricted to the half-space is the thermal state of the boost Hamiltonian, and a static observer at proper distance ρ_* from the entangling plane sees a local temperature, the Unruh temperature Eq. (55.83) evaluated at acceleration $1/\rho_*$,

$$T(\rho_*) = \frac{1}{2\pi\rho_*}. \tag{55.155}$$

The entanglement entropy of the half-space is, in this picture, the thermodynamic entropy of a gas of the field's quanta in local equilibrium at the position-dependent temperature Eq. (55.155), integrated over the wedge. We need the entropy density of a free massless scalar gas in d spacetime dimensions. Its free-energy density is

$$f(T) = T\int \frac{\mathrm{d}^{d-1}k}{(2\pi)^{d-1}}\,\log(1-e^{-|\mathbf{k}|/T}) = \frac{T\,\Omega_{d-2}}{(2\pi)^{d-1}}\int_0^\infty \mathrm{d}k\,k^{d-2}\log(1-e^{-k/T}), \tag{55.156}$$

with $\Omega_{d-2} = 2\pi^{(d-1)/2}/\Gamma(\frac{d-1}{2})$ the area of the unit S^{d-2}. Expanding the logarithm as $-\sum_{m\geq 1} e^{-mk/T}/m$ and using $\int_0^\infty \mathrm{d}k\,k^{d-2}e^{-mk/T} = \Gamma(d-1)(T/m)^{d-1}$,

$$f(T) = -\frac{\Omega_{d-2}\,\Gamma(d-1)}{(2\pi)^{d-1}}\,T^d\sum_{m\geq 1}\frac{1}{m^d} = -\frac{\Omega_{d-2}\,\Gamma(d-1)\,\zeta(d)}{(2\pi)^{d-1}}\,T^d, \tag{55.157}$$

the Riemann zeta value $\zeta(d)$ appearing exactly as in the zeta-regularized vacuum sums of Section E.3. The entropy density is $s = -\partial f/\partial T = \sigma_d\,T^{d-1}$ with the field-counting Stefan–Boltzmann coefficient

$$\sigma_d = \frac{d\,\Gamma(d-1)\,\zeta(d)\,\Omega_{d-2}}{(2\pi)^{d-1}}. \tag{55.158}$$

The proper volume element a distance ρ_* from the plane is $\mathrm{d}V = \mathcal{A}_\perp \, \mathrm{d}\rho_*$ with $\mathcal{A}_\perp$ the transverse area of the entangling surface, so

$$S = \int \mathrm{d}V \, s(T(\rho_*)) = \mathcal{A}_\perp \, \sigma_d \int_\epsilon^\infty \mathrm{d}\rho_* \left(\frac{1}{2\pi\rho_*}\right)^{d-1} = \frac{\sigma_d}{(2\pi)^{d-1}(d-2)} \frac{\mathcal{A}_\perp}{\epsilon^{d-2}} \qquad (d > 2), \tag{55.159}$$

the lower limit $\rho_* = \epsilon$ being the proper cutoff at which the local temperature Eq. (55.155) would otherwise diverge. This is the area law Eq. (55.139) with the coefficient $\kappa = \sigma_d/[(2\pi)^{d-1}(d-2)]$ written explicitly in terms of the degrees of freedom of the field through σ_d. The hot modes piled up against the entangling surface — arbitrarily hot as $\rho_* \to 0$ — are the physical origin of the divergence, and the cutoff ϵ is exactly the regulator that the lattice supplied. The coefficient still depends on the cutoff prescription (where in proper distance the integral is truncated), reaffirming that κ is non-universal; what the thermal computation fixes unambiguously is the power $\epsilon^{-(d-2)}$ and the proportionality to σ_d.

Physical Insight

The integral Eq. (55.159) is power-law divergent for $d > 2$ but *logarithmically* divergent at $d = 2$, where $\int_\epsilon \mathrm{d}\rho_*/\rho_* = \log(\cdots/\epsilon)$. This is the precise sense in which the 2D logarithm is the degenerate, $d = 2$ face of the area law: the area of a point-like entangling surface is a pure number, the power $\epsilon^{-(d-2)}$ becomes ϵ^0, and the would-be power-law divergence collapses to a logarithm. The collapse is quantitative, as the next computation shows.

55.5.4 Recovering the 2D logarithm: the free boson has $c = 1$

The two derivations of this section — the lattice formula Eq. (55.145) and the thermal-gas integral Eq. (55.159) — both contain the $d = 2$ result, and reproducing the CFT logarithm of Theorem 55.32 from them is a sharp cross-check that the free-field machinery and the conformal machinery agree. The free real scalar in two dimensions is the free boson, a CFT of central charge $c = 1$.

From the thermal gas. Set $d = 2$ in Eq. (55.158): $\Omega_0 = 2$ (the two points of S^0), $\Gamma(1) = 1$, $\zeta(2) = \pi^2/6$, giving $\sigma_2 = 2 \cdot 1 \cdot (\pi^2/6) \cdot 2/(2\pi) = \pi/3$. The entropy density of the 2D free boson gas is therefore $s = \frac{\pi}{3}T$, which is the Cardy density $s = \pi c/(3\beta) = \frac{\pi c}{3}T$ at $c = 1$ — the same value found in the large-interval limit Eq. (55.135). The wedge integral at $d = 2$ is logarithmic: with a point-like entangling surface ($\mathcal{A}_\perp = 1$) and one endpoint,

$$S_{\text{one endpoint}} = \sigma_2 \int_\epsilon^\ell \frac{\mathrm{d}\rho_*}{2\pi\rho_*} = \frac{\pi/3}{2\pi} \log\frac{\ell}{\epsilon} = \frac{1}{6}\log\frac{\ell}{\epsilon} = \frac{c}{6}\log\frac{\ell}{\epsilon}, \tag{55.160}$$

and an interval has two endpoints, doubling this to $S = \frac{c}{3}\log(\ell/\epsilon)$ with $c = 1$ — exactly Eq. (55.119) for the free boson. The per-endpoint coefficient $c/6$ is the $d = 2$ residue of the area-law coefficient κ.

From the lattice. Figure 55.5b carries out Eq. (55.145) directly on an interval of ℓ sites in a periodic chain of N sites — the lattice realization of an interval on a spatial circle. Plotting S against the conformal-chord variable $\log[(N/\pi)\sin(\pi\ell/N)]$ of Eq. (55.128) gives a straight line whose slope is $c/3$; the numerical fit returns $c = 1.000$. (Fitting against a bare $\log \ell$ on a finite chain, or using open boundaries, distorts the coefficient: the circle's periodicity and any boundary entropy must be accounted for, which is exactly what the chord variable does.) The free-field computation, the thermal-gas computation, and the twist-operator computation of Section 55.4.3 thus return the identical universal datum, $c = 1$, by three independent routes.

55.5.5 The full divergence structure and what is universal

The leading area law is the first term of a systematic expansion. For a smooth entangling surface ∂A of typical size R in d spacetime dimensions, dimensional analysis and locality dictate that S_A is a sum of local geometric integrals over ∂A, each multiplying a power of the cutoff. The integrals are built from the intrinsic and extrinsic curvatures of ∂A; by reflection symmetry of the vacuum only even total numbers of derivatives appear, so the divergences come in steps of two:

$$S_A = g_{d-2}\frac{\mathcal{A}}{\epsilon^{d-2}} + g_{d-4}\frac{1}{\epsilon^{d-4}}\int_{\partial A}(\text{curvature}) + \cdots + S_{\text{univ}} + (\text{finite, scheme-dependent}). \tag{55.161}$$

Each coefficient g_{d-2k} multiplies a curvature integral of mass dimension $d-2-2k$ over the surface; for a sphere these are pure numbers times R^{d-2-2k}. Every one of these power-law terms is non-universal: it depends on the short-distance regulator, and a change of scheme (a different lattice, a different smooth cutoff) shifts each g_{d-2k}. The series of powers continues down to ϵ^0, and what sits at that order — the term that does *not* scale away with the cutoff — is the universal content. Its character depends on the parity of d.

Theorem 55.39 (Universal term in the entanglement entropy)**.** *For a CFT vacuum and a smooth entangling surface, the cutoff-independent part of S_A is:*

(even d) *a logarithm, $S_{\text{univ}} = (-1)^{\frac{d-2}{2}}\, c_{\log}\, \log(R/\epsilon)$, whose coefficient $c_{\log}$ is a linear combination of the conformal-anomaly coefficients of the theory integrated against the curvature of ∂A. The sign of the logarithmic coefficient is the universal datum.*

(odd d) *a finite constant, $S_{\text{univ}} = (-1)^{\frac{d-1}{2}}\, F$, independent of the cutoff and of R; for a spherical region F is a positive, scheme-independent number characterizing the CFT.*

The reason the universal term is a logarithm in even d is the same anomaly mechanism that produced the 2D result. A logarithm is the only function of R/ϵ

invariant under the simultaneous rescaling of R and ϵ up to an additive shift; it can appear only when a scale transformation fails to be a symmetry, i.e. through the conformal anomaly. In $d = 2$ the anomaly coefficient is the central charge, and the curvature integral over the two endpoints gives the familiar

$$S_{\text{univ}}^{(d=2)} = \frac{c}{3} \log \frac{R}{\epsilon}, \tag{55.162}$$

recovering Theorem 55.32. In $d = 4$ the trace anomaly has two independent coefficients, conventionally a and c, defined through

$$\langle T^{\mu}{}_{\mu} \rangle = \frac{c}{16\pi^2} W^2_{\mu\nu\rho\sigma} - \frac{a}{16\pi^2} E_4, \tag{55.163}$$

with W the Weyl tensor and E_4 the Euler density, and the entanglement logarithm of a smooth surface is (Solodukhin [450])

$$S_{\text{univ}}^{(d=4)} = -\frac{\log(R/\epsilon)}{2\pi} \int_{\partial A} \mathrm{d}^2 y \, \sqrt{h} \left[a \, \mathcal{R}_{\partial A} + c \left(K^{\mu}{}_{\nu} K^{\nu}{}_{\mu} - \tfrac{1}{2} K^2 - W_{\text{ind}} \right) \right], \tag{55.164}$$

where $\mathcal{R}_{\partial A}$ is the intrinsic Ricci scalar of ∂A and $K^{\mu}{}_{\nu}$ its extrinsic curvature. For a spherical entangling surface in flat space the extrinsic terms cancel and the Gauss–Bonnet integral $\int \mathcal{R}_{\partial A} = 8\pi$ gives $S_{\text{univ}}^{(d=4)} = -4a \log(R/\epsilon)$: the sphere isolates the a-anomaly cleanly. For the free real scalar, $a = \frac{1}{360}$, so its universal logarithmic coefficient is $-\frac{1}{90}$ — a pure number computable from the field content, and the $d = 4$ analogue of $c = 1$.

In odd d there is no conformal anomaly (the relevant curvature invariants do not exist in odd dimensions), no scale to make a logarithm, and the universal piece is a finite constant. In $d = 3$, for a disk-shaped region (circular entangling curve), this constant defines the quantity

$$F \equiv -S_{\text{univ}}^{(d=3)} = -\left(S_A - g_1 \, \tfrac{\mathcal{A}}{\epsilon} \right)\Big|_{\text{finite}}, \tag{55.165}$$

the negative of the constant term after the leading $\mathcal{A}/\epsilon$ divergence is subtracted. For the free real scalar in $d = 3$, $F = \frac{1}{16}(2\log 2 - \frac{3\zeta(3)}{\pi^2}) \approx 0.0638$, again a sharp number set by the field.

Caution

Equation Eq. (55.161) assumes a *smooth* entangling surface. A corner or cusp in ∂A produces its own, extra logarithmic divergence even in dimensions where the smooth surface gives none — in $d = 3$ a corner of opening angle Ω contributes a term $\propto a(\Omega) \log(R/\epsilon)$ with a universal angle-dependent coefficient. These geometric subtleties refine, but do not overturn, the area law: the leading $\mathcal{A}/\epsilon^{d-2}$ term is always present and always non-universal.

The practical consequence is the honesty point that has shadowed the whole chapter. The entanglement entropy S_A is not a physical observable in the continuum: its leading term diverges with a scheme-dependent coefficient, and even its finite part shifts under a change of regulator. The only cutoff-independent information is the universal term of Theorem 55.39 — the sign and magnitude of the even-d logarithmic coefficient, or the odd-d constant F — and, for combinations of regions, the subtracted quantities in which the boundary divergences cancel. The *mutual information* $I(A:B) = S_A + S_B - S_{AB}$ of Eq. (55.23) is UV-finite: S_A diverges on ∂A, S_B on ∂B, and S_{AB} on $\partial A \cup \partial B$, so for disjoint A, B the area-law divergences cancel identically and $I(A:B)$ is a finite, scheme-independent measure of correlation. The *relative entropy* $S(\rho_A \| \sigma_A)$, which descends even to the cutoff-free algebraic setting, is likewise finite. These are the objects Section 55.6 builds the chapter's exact structure upon; the bare S_A is their divergent scaffolding.

Caution: the entanglement entropy of a gauge field (QED)

For the Maxwell field of QED the very first step of this section — the factorization $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$ that lets one trace out $\bar{A}$ — is subtle, and the subtlety is physical, not technical. Gauge invariance means the physical observables in a region form an algebra with a nontrivial *center*: the electric flux operator through the entangling surface commutes with everything in A and with everything in $\bar{A}$, so it belongs to both algebras at once and the Hilbert space does not factorize as a naive tensor product. One must first choose how to split the surface degrees of freedom — a choice of center, equivalently a choice of boundary condition for the Wilson/electric-flux variables — and different choices give different entanglement entropies, differing by a classical Shannon term measuring the distribution of the flux through ∂A [158]. The resulting "edge modes" contribute a genuine, surface-localized piece to S_A, and they are responsible for the famous mismatch between the naive free-field count and the holographic/heat-kernel value of the Maxwell logarithmic coefficient in $d = 4$. We state this at the level of a flag: the area law and divergence structure of this section apply verbatim to the scalar, but for gauge fields the definition of S_A already requires the algebraic, center-aware viewpoint anticipated in the chapter opening. The clean, definition-independent diagnostics — mutual information and relative entropy — are insensitive to the center ambiguity, which is one more reason they, not S_A, are the fundamental objects.

55.5.6 Entropic monotonicity: c-, F-, and a-theorems

The universal coefficients of Theorem 55.39 are not merely labels of fixed points; along a renormalization-group flow they are monotonic, and the monotonicity is a theorem of entanglement, rooted in the strong subadditivity inequality that Theorem 55.47 will prove. We state the results at the level they will be used, deferring the strong-subadditivity proof itself to Section 55.6.

In $d = 2$, the entropic c-function $c_{\text{eff}}(\ell) = 3\,\ell\,\mathrm{d}S_A/\mathrm{d}\ell$ of Eq. (55.136) is cutoff-independent and equals the central charge at a fixed point. Casini and Huerta showed [113] that strong subadditivity, combined with Lorentz invariance, forces $c_{\text{eff}}(\ell)$ to be *monotonically non-increasing* as the interval grows, giving an entropic proof of Zamolodchikov's c-theorem (Theorem 52.17): the effective number of degrees of freedom decreases from the ultraviolet to the infrared, $c_{\text{UV}} \geq c_{\text{IR}}$. The entanglement entropy thus supplies the monotone directly, as the slope of S_A against $\log \ell$.

In $d = 3$, the role of the central charge is played by the constant F of Eq. (55.165). Casini and Huerta proved [114] the *F-theorem*, $F_{\text{UV}} \geq F_{\text{IR}}$, again from strong subadditivity applied to circular regions of varying radius; the entropic F-function $F(R) = (R\,\mathrm{d}_R - 1)S_A$ interpolates monotonically between the endpoints. This is the odd-dimensional counterpart of the c-theorem, with a finite universal quantity in place of a logarithmic coefficient, and it is the diagnostic that orders three-dimensional CFTs by their degree-of-freedom count.

In $d = 4$, the analogous statement is the a-theorem, $a_{\text{UV}} \geq a_{\text{IR}}$, for the coefficient a of the Euler density that controls the spherical entanglement logarithm Eq. (55.164). The complete proof of Komargodski and Schwimmer [301, 302] proceeds through the dilaton effective action rather than purely through entanglement, and a fully entropic four-dimensional proof along the strong-subadditivity lines of the lower-dimensional cases remains, as of the mid-2020s, only partially established; we state the a-theorem as a monotonicity result and flag the entropic version as an active question. What is uniform across $d = 2, 3, 4$ is the picture: a universal coefficient of the entanglement entropy counts degrees of freedom, and that count cannot increase under renormalization-group flow, with strong subadditivity the structural source of the monotonicity wherever the entropic proof is complete.

Physical Insight

The thread that runs from the 2D logarithm to the higher-dimensional area law is the identification of an entanglement coefficient with a count of degrees of freedom. In $d = 2$ it is the central charge c; in $d = 4$ it is the anomaly coefficient a; in $d = 3$, where no anomaly exists, it is the finite number F. In every case the coefficient is extracted from S_A by a derivative or subtraction that removes the cutoff dependence, and

in every case it decreases along the flow to the infrared. Entanglement entropy, born as a measure of quantum correlation, has become a measure of the size of a quantum field theory — the informational expression of the irreversibility of the renormalization group.

We have now established the generic leading behavior of vacuum entanglement entropy in any dimension: $S_A = \kappa\,\mathcal{A}(\partial A)/\epsilon^{d-2} + \cdots$, an area law whose coefficient is scheme-dependent, beneath which sits a single universal term — a logarithm tied to a conformal anomaly in even d, a finite constant in odd d — that counts the theory's degrees of freedom and runs monotonically under the renormalization group. The price of this generic divergence is that S_A itself is not a clean observable, which makes the subtracted, UV-finite quantities — mutual information and relative entropy — the structurally fundamental ones; Section 55.6 develops them, proves the strong subadditivity invoked above, and derives the first law of entanglement. The area law established here is also the precise QFT statement that the holographic computation of Chapter 56 will geometrize: the Ryu–Takayanagi prescription reproduces Eq. (55.153) as the area of a minimal surface anchored on ∂A, with the universal coefficients of Theorem 55.39 emerging from its subleading geometry.

55.6 Relative Entropy, the First Law, and Information Inequalities

The previous sections have repeatedly run aground on the same rock. The entanglement entropy S_A of a region is, in the continuum, ultraviolet divergent: the area law of Section 55.5 showed its leading term to be $\mathcal{A}_\perp/\epsilon^{d-2}$, cutoff-dependent and physically arbitrary, and even the $\frac{c}{3}\log(\ell/\epsilon)$ of Section 55.4 carries an undetermined additive constant. We were forced to the conclusion, stated already in Section 55.1.6 and honored throughout, that S_A is not by itself an observable: only *differences* of entropies, in which the boundary divergences cancel, can carry cutoff-independent physics. This final section builds the quantities that survive. The organizing object is the *relative entropy* $S(\rho\,\|\,\sigma)$, introduced in Eq. (55.15) and shown positive in Theorem 55.9; we will see that it is exactly the difference structure the chapter has been demanding — ultraviolet finite, monotone under coarse-graining, and tied to the modular Hamiltonian $K = -\log\rho_A$ of Theorem 55.28 by an identity that splits it into a change in modular energy minus a change in entropy. From its positivity we will extract the *first law of entanglement*, $\delta S_A = \delta\langle K_A\rangle$, the linear-response statement that is the precise hook on which holographic entanglement and the derivation of gravitational dynamics from information (Chapters 56 and 57) are hung. From its monotonicity we will obtain strong subadditivity — the master inequality

of quantum information, promised in Section 55.1.4 — and, without full proof, the modern theorems that turn entanglement inequalities into energy conditions: the Casini reformulation of the Bekenstein bound, and the quantum null energy condition. Throughout we work in the regulated (type-I) picture of Section 55.1.6, where ρ_A is an honest density matrix, and we report only quantities that survive the removal of the cutoff.

55.6.1 The modular decomposition and ultraviolet finiteness

Recall the definition. For two density matrices ρ, σ on the same (regulated) algebra of a region,

$$S(\rho \,\|\, \sigma) = \operatorname{Tr}(\rho \log \rho) - \operatorname{Tr}(\rho \log \sigma), \tag{55.166}$$

which is Eq. (55.15), non-negative by Theorem 55.9, and zero if and only if $\rho = \sigma$. The crucial structural fact is that Eq. (55.166) reorganizes into a difference of two physically meaningful pieces once we recognize $-\log\sigma$ as the modular Hamiltonian of the reference state. Write

$$K_\sigma \equiv -\log \sigma, \tag{55.167}$$

the modular Hamiltonian of σ in the sense of Eq. (55.87), and define for any state ρ the *modular energy* $\langle K_\sigma \rangle_\rho \equiv \operatorname{Tr}(\rho\, K_\sigma)$ and the von Neumann entropy $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$. Then Eq. (55.166) reads

$$S(\rho \,\|\, \sigma) = \langle K_\sigma \rangle_\rho - S(\rho), \tag{55.168}$$

because $-\operatorname{Tr}(\rho \log \sigma) = \operatorname{Tr}(\rho K_\sigma) = \langle K_\sigma \rangle_\rho$ and $\operatorname{Tr}(\rho \log \rho) = -S(\rho)$. Now evaluate the same combination on the reference state itself: setting $\rho = \sigma$ gives $S(\sigma \,\|\, \sigma) = 0$, i.e. $\langle K_\sigma \rangle_\sigma = S(\sigma)$ — the reference modular energy equals the reference entropy. Subtracting this vanishing quantity from Eq. (55.168) costs nothing and produces the central identity of the section.

Theorem 55.40 (Modular decomposition of relative entropy)**.** *Let σ be a reference state with modular Hamiltonian $K_\sigma = -\log\sigma$, and let ρ be any state on the same algebra. Define the differences*

$$\Delta\langle K_\sigma \rangle \equiv \langle K_\sigma \rangle_\rho - \langle K_\sigma \rangle_\sigma, \qquad \Delta S \equiv S(\rho) - S(\sigma). \tag{55.169}$$

Then the relative entropy is their difference,

$$\boxed{S(\rho \,\|\, \sigma) = \Delta\langle K_\sigma \rangle - \Delta S} \tag{55.170}$$

and its non-negativity Theorem 55.9 is the inequality

$$\Delta S \le \Delta\langle K_\sigma \rangle, \tag{55.171}$$

with equality if and only if $\rho = \sigma$.

Proof. From Eq. (55.168), $S(\rho\,\|\,\sigma) = \langle K_\sigma\rangle_\rho - S(\rho)$. Subtract the identity $0 = \langle K_\sigma\rangle_\sigma - S(\sigma)$ (which is Eq. (55.168) at $\rho = \sigma$): $S(\rho\,\|\,\sigma) = (\langle K_\sigma\rangle_\rho - \langle K_\sigma\rangle_\sigma) - (S(\rho) - S(\sigma)) = \Delta\langle K_\sigma\rangle - \Delta S$. Positivity $S(\rho\,\|\,\sigma) \geq 0$ is then Eq. (55.171), and the equality clause is that of Theorem 55.9. □

The decomposition Eq. (55.170) is the resolution of the chapter's central tension, and it is worth dwelling on why. Each of the two terms on the right is, separately, ultraviolet divergent. The entropy difference ΔS inherits the area-law divergence $\mathcal{A}_\perp/\epsilon^{d-2}$ of Section 55.5 from each of $S(\rho)$ and $S(\sigma)$; the modular-energy difference $\Delta\langle K_\sigma\rangle$ is equally singular, because K_σ is, in the local cases of Theorem 55.25, an integral of the energy density weighted by the distance to the entangling surface, and that integral diverges as the surface is approached. But the two divergences are *the same* divergence. The reason is that ρ and σ, being states of the same theory, have identical short-distance structure: the singular entanglement of the modes hugging the entangling surface is a property of the vacuum two-point function near ∂A, which is state-independent (the Hadamard condition). Any two physical states look like the vacuum at short distances, so their reduced density matrices differ only in the soft, long-wavelength sector, and the hard, divergent sector cancels in the difference Eq. (55.170). The relative entropy is therefore *ultraviolet finite*: it is a difference engineered precisely so that the cutoff drops out, and it remains well-defined — through Tomita–Takesaki theory — even on the type-III algebra of the continuum, where no ρ, no σ, and no S_A individually exist (Section 55.1.6). This is the abstract content of the remark, made in Section 55.1.5, that the clean diagnostics are differences. The inequality Eq. (55.171), $\Delta S \leq \Delta\langle K_\sigma\rangle$, will be recognized below as a sharpened, quantum form of the Clausius inequality, and its first-order saturation is the first law of entanglement.

Physical Insight

The decomposition Eq. (55.170) says that relative entropy measures the failure of a generalized second law to saturate. Read $\langle K_\sigma\rangle$ as a "modular heat" and S as a thermodynamic entropy: the relative entropy is (heat) − (entropy change), exactly the entropy production that the Clausius inequality bounds below by zero. A state ρ is distinguishable from the reference σ precisely to the extent that it carries this irreversibility, and the amount of distinguishability is finite and physical even when the entropies themselves are not. This is the thermodynamic face of the information-theoretic statement that $S(\rho\,\|\,\sigma)$ is the number of measurements needed to tell ρ from σ.

Before exploiting the decomposition we record the companion finite quantity, the mutual information, whose finiteness has the same origin. For disjoint regions A, B the mutual information $I(A:B) = S_A + S_B - S_{AB}$ of Theorem 55.13 was

shown in Theorem 55.14 to equal a relative entropy,

$$I(A:B) = S(\rho_{AB} \,\|\, \rho_A \otimes \rho_B) \geq 0, \tag{55.172}$$

the distinguishability of the true joint state from the product of its marginals. Equation (55.172) is the reason $I(A:B)$ is finite: it is manifestly a relative entropy, hence cutoff-independent by the cancellation just described, with the divergence on ∂A carried identically by ρ_{AB} and by $\rho_A \otimes \rho_B$ (whose A-marginal is ρ_A), and similarly on ∂B. The mutual information is thus the universally available finite measure of total correlation between two regions, and it inherits from Theorem 55.9 both its positivity and the sharp condition for vanishing, $\rho_{AB} = \rho_A \otimes \rho_B$.

55.6.2 Klein's inequality from operator convexity

Positivity of relative entropy was proved in Theorem 55.9 by a Jensen argument on the doubly stochastic overlap matrix. We give here a second proof, from convexity of the function $x \log x$, for two reasons: it is the proof that generalizes — the same circle of ideas, upgraded from ordinary to *operator* convexity, delivers the monotonicity theorem of the next subsection — and it exposes relative entropy as the trace gap in a tangent-line inequality, which is the form in which the first law will use it.

Definition 55.41 (Operator convexity)**.** A real function f defined on an interval J is *operator convex* if, for all Hermitian operators A, B with spectra in J and all $\lambda \in [0,1]$,

$$f(\lambda A + (1-\lambda)B) \;\preceq\; \lambda f(A) + (1-\lambda) f(B), \tag{55.173}$$

the inequality being in the operator (positive-semidefinite) order: $X \preceq Y$ means $Y - X \succeq 0$. Operator convexity is strictly stronger than ordinary (scalar) convexity, which is the special case of Eq. (55.173) sandwiched between a fixed vector, $\langle\psi| f(\cdots) |\psi\rangle$.

The functions relevant to entanglement are operator convex. We will need that $x \log x$ is operator convex on $(0,\infty)$, and that $-\log x$ and x^{-1} are too; the latter two we prove in Theorem 55.45 below because the inverse furnishes the cleanest route to the deep inequalities. For positivity, however, ordinary convexity already suffices, through the following trace inequality, which is Klein's inequality in its general (tangent-line) form.

Proposition 55.42 (Klein's trace inequality)**.** *Let f be a differentiable convex function on an interval J, and let A, B be Hermitian operators with spectra in J. Then*

$$\operatorname{Tr}\big[f(A) - f(B) - f'(B)(A-B)\big] \;\geq\; 0, \tag{55.174}$$

with equality if and only if $A = B$ (when f is strictly convex).

Proof. Diagonalize $A = \sum_i a_i |i\rangle\langle i|$ and $B = \sum_j b_j |j\rangle\langle j|$ in orthonormal eigenbases, with $a_i, b_j \in J$. Because $\sum_j |\langle i|j\rangle|^2 = 1$ for each i, we may write $\operatorname{Tr} f(A) = \sum_i f(a_i) = \sum_{i,j} |\langle i|j\rangle|^2 f(a_i)$. The remaining traces are diagonal sums: $\operatorname{Tr} f(B) = \sum_j f(b_j)$ and $\operatorname{Tr}[f'(B)(A-B)] = \sum_{i,j} |\langle i|j\rangle|^2 f'(b_j)(a_i - b_j)$, using $\operatorname{Tr}[f'(B)A] = \sum_{i,j} |\langle i|j\rangle|^2 f'(b_j) a_i$ and $\operatorname{Tr}[f'(B)B] = \sum_j f'(b_j) b_j$. Collecting,

$$\operatorname{Tr}\left[f(A) - f(B) - f'(B)(A-B)\right] = \sum_{i,j} |\langle i|j\rangle|^2 \Big[f(a_i) - f(b_j) - f'(b_j)(a_i - b_j)\Big]. \tag{55.175}$$

Each bracket is non-negative: it is the gap between the convex function $f(a_i)$ and its tangent line at b_j, evaluated at a_i, and a convex function lies above each of its tangents. The weights $|\langle i|j\rangle|^2 \geq 0$ preserve the sign, giving Eq. (55.174). Strict convexity makes each bracket vanish only when $a_i = b_j$, which for all contributing (i,j) forces $A = B$. □

Positivity of relative entropy is the case $f(x) = x \log x$, $A = \rho$, $B = \sigma$. Here $f'(x) = \log x + 1$, so $f'(\sigma) = \log \sigma + \mathbf{1}$ and

$$\begin{aligned} \operatorname{Tr}\left[f(\rho) - f(\sigma) - f'(\sigma)(\rho - \sigma)\right] &= \operatorname{Tr}\left[\rho \log \rho - \sigma \log \sigma - (\log \sigma + \mathbf{1})(\rho - \sigma)\right] \\ &= \operatorname{Tr}(\rho \log \rho) - \operatorname{Tr}(\rho \log \sigma) - \operatorname{Tr}(\rho - \sigma) \\ &= S(\rho \,\|\, \sigma), \end{aligned} \tag{55.176}$$

where the $\sigma \log \sigma$ terms cancelled, $\operatorname{Tr}[(\log \sigma)(\rho - \sigma)]$ combined with $-\operatorname{Tr}(\rho \log \sigma)$ to leave $+\operatorname{Tr}(\sigma \log \sigma)$ that cancelled in turn, and $\operatorname{Tr}(\rho - \sigma) = 1 - 1 = 0$ killed the $\mathbf{1}$ term. By Theorem 55.42 this is ≥ 0, with equality iff $\rho = \sigma$ — recovering Theorem 55.9. The relative entropy is exactly the trace gap of the tangent-line inequality for $x \log x$ at the reference state σ; this is the form we will linearize for the first law.

55.6.3 Monotonicity, data processing, and strong subadditivity

The single most important property of relative entropy, and the one that ordinary convexity cannot reach, is that it can only *decrease* when the states are processed by any physical operation. Physical operations are the completely positive, trace-preserving (CPTP) maps — the quantum channels — whose general form we recall: a map $\mathcal{N}$ on density matrices is CPTP iff it admits a Kraus representation $\mathcal{N}(\rho) = \sum_k E_k \rho E_k^\dagger$ with $\sum_k E_k^\dagger E_k = \mathbf{1}$, equivalently (Stinespring) iff $\mathcal{N}(\rho) = \operatorname{Tr}_E\left[V \rho V^\dagger\right]$ for some isometry $V : \mathcal{H} \to \mathcal{H} \otimes \mathcal{H}_E$ into an enlarged space and a partial trace over the ancilla E. The two basic operations subsumed are unitary evolution and the discarding of a subsystem (partial trace).

Theorem 55.43 (Monotonicity of relative entropy / data-processing inequality)**.** *For any two states ρ, σ and any CPTP map $\mathcal{N}$,*

$$S(\mathcal{N}(\rho) \,\|\, \mathcal{N}(\sigma)) \;\leq\; S(\rho \,\|\, \sigma). \tag{55.177}$$

In particular, restriction to a subalgebra cannot increase distinguishability: for ρ_{AB}, σ_{AB} on a bipartite system, tracing out B gives

$$S(\rho_A \,\|\, \sigma_A) \;\le\; S(\rho_{AB} \,\|\, \sigma_{AB}). \tag{55.178}$$

The interpretation is the meaning of the name: a channel $\mathcal{N}$ is a processing of the data, and Eq. (55.177) says that no processing — no measurement, no time evolution, no transmission down a noisy line, no forgetting of part of the system — can make two states easier to tell apart than they were to begin with. Information is never created by manipulation. This is the statement on which the entire edifice of quantum information rests, and from which, as we show, strong subadditivity and the holographic inequalities descend.

The proof reduces, in two clean steps, to a single deep theorem of matrix analysis. The first step replaces a general channel by a partial trace; the second extracts Eq. (55.178) from the joint convexity of relative entropy.

Step 1: general channels reduce to partial trace. By Stinespring, $\mathcal{N}(\rho) = \mathrm{Tr}_E\,[V\rho V^\dagger]$ with V an isometry. An isometry preserves relative entropy exactly: $V\rho V^\dagger$ and $V\sigma V^\dagger$ have the same nonzero spectra and the same overlaps as ρ, σ, so $S(V\rho V^\dagger \,\|\, V\sigma V^\dagger) = S(\rho \,\|\, \sigma)$ (extend V to a unitary on a still-larger space if desired; the support of $V\rho V^\dagger$ lies in the image of V, and Eq. (55.166) depends only on operators restricted to that support). Hence Eq. (55.177) follows once we prove the special case of monotonicity under partial trace, Eq. (55.178), applied to the dilated states $V\rho V^\dagger, V\sigma V^\dagger$ on $\mathcal{H} \otimes \mathcal{H}_E$.

Step 2: partial-trace monotonicity from joint convexity. The engine is the following.

Theorem 55.44 (Joint convexity of relative entropy)**.** *The map $(\rho, \sigma) \mapsto S(\rho \,\|\, \sigma)$ is jointly convex: for states $\rho_1, \rho_2, \sigma_1, \sigma_2$ and $\lambda \in [0, 1]$,*

$$S(\lambda\rho_1 + (1-\lambda)\rho_2 \,\|\, \lambda\sigma_1 + (1-\lambda)\sigma_2) \;\le\; \lambda\, S(\rho_1 \,\|\, \sigma_1) + (1-\lambda)\, S(\rho_2 \,\|\, \sigma_2). \tag{55.179}$$

We first show that Theorem 55.44 implies partial-trace monotonicity, then prove Theorem 55.44. The bridge is a *twirl*. Let $\{U_k\}_{k=1}^{m}$, $m = d_B^2$, be the Heisenberg–Weyl (shift-and-clock) unitaries on the d_B-dimensional factor $\mathcal{H}_B$; they form a unitary 1-design, meaning that averaging any operator over the channel $U_k(\cdot)U_k^\dagger$ projects its B-part onto the maximally mixed state,

$$\frac{1}{m}\sum_{k=1}^{m} (\mathbf{1}_A \otimes U_k)\, \Omega\, (\mathbf{1}_A \otimes U_k^\dagger) = \mathrm{Tr}_B(\Omega) \otimes \frac{\mathbf{1}_B}{d_B}, \tag{55.180}$$

for any operator Ω on $\mathcal{H}_A \otimes \mathcal{H}_B$. (This is the standard fact that the Heisenberg–Weyl average is the completely depolarizing channel on B; it follows because the

only operators commuting with all U_k are multiples of $\mathbf{1}_B$, so the average is the projection onto $\mathbf{1}_B$, normalized to preserve the A-marginal.) Relative entropy is invariant under the conjugations in Eq. (55.180), since each $\mathbf{1}_A \otimes U_k$ is unitary; and it is jointly convex by Theorem 55.44. Therefore, averaging,

$$\begin{aligned} S(\rho_{AB}\,\|\,\sigma_{AB}) &= \frac{1}{m}\sum_k S\big((\mathbf{1}\otimes U_k)\rho_{AB}(\mathbf{1}\otimes U_k^\dagger)\,\|\,(\mathbf{1}\otimes U_k)\sigma_{AB}(\mathbf{1}\otimes U_k^\dagger)\big) \\ &\geq S\Big(\tfrac{1}{m}\sum_k(\mathbf{1}\otimes U_k)\rho_{AB}(\cdots)\,\Big\|\,\tfrac{1}{m}\sum_k(\mathbf{1}\otimes U_k)\sigma_{AB}(\cdots)\Big) \\ &= S\Big(\rho_A\otimes\tfrac{\mathbf{1}_B}{d_B}\,\Big\|\,\sigma_A\otimes\tfrac{\mathbf{1}_B}{d_B}\Big) = S(\rho_A\,\|\,\sigma_A), \end{aligned} \tag{55.181}$$

where the inequality is joint convexity Eq. (55.179) (the average of relative entropies dominates the relative entropy of the averages), the third equality is the twirl Eq. (55.180), and the last is the additivity $S(\rho_A\otimes\tau\,\|\,\sigma_A\otimes\tau) = S(\rho_A\,\|\,\sigma_A)$ for a common factor $\tau = \mathbf{1}_B/d_B$, immediate from Eq. (55.166) and $\log(X\otimes\tau) = \log X\otimes\mathbf{1} + \mathbf{1}\otimes\log\tau$. This is Eq. (55.178), and with Step 1 it completes the proof of Theorem 55.43 — modulo Theorem 55.44.

Joint convexity from Lieb's concavity theorem. The remaining content is a genuinely hard analytic input, the one place in this chapter where a result of matrix analysis must be invoked rather than derived from scratch; we state it precisely, prove the operator-convexity lemma at its heart, and deduce Theorem 55.44 from it.

Lemma 55.45 (Operator convexity of $X^\dagger A^{-1}X$)**.** *For positive-definite A and arbitrary X of compatible dimension, the map $(A,X)\mapsto X^\dagger A^{-1}X$ is jointly operator convex.*

Proof. The set $\{(A,X,Y) : A\succ 0,\ Y\succeq X^\dagger A^{-1}X\}$ is convex, because the Schur-complement criterion identifies it with a positive-semidefinite cone: for $A\succ 0$,

$$\begin{pmatrix} A & X \\ X^\dagger & Y\end{pmatrix}\succeq 0 \quad\Longleftrightarrow\quad Y - X^\dagger A^{-1}X\succeq 0, \tag{55.182}$$

the equivalence being the block factorization $\left(\begin{smallmatrix} A & X\\ X^\dagger & Y\end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathbf{1} & 0\\ X^\dagger A^{-1} & \mathbf{1}\end{smallmatrix}\right)\left(\begin{smallmatrix} A & 0\\ 0 & Y-X^\dagger A^{-1}X\end{smallmatrix}\right)\left(\begin{smallmatrix} \mathbf{1} & A^{-1}X\\ 0 & \mathbf{1}\end{smallmatrix}\right)$ which is congruent to $\mathrm{diag}(A,\,Y-X^\dagger A^{-1}X)$ and hence positive iff the latter is. The set of positive block matrices is convex (the PSD cone is convex and the constraint $A\succ 0$ is open-convex), and Eq. (55.182) exhibits the epigraph of $(A,X)\mapsto X^\dagger A^{-1}X$ as the image of that convex cone under the linear projection forgetting nothing but the off-diagonal labelling; a function with convex epigraph is convex. In the operator sense: given (A_1,X_1) and (A_2,X_2) with $Y_i = X_i^\dagger A_i^{-1}X_i$, both blocks $\left(\begin{smallmatrix} A_i & X_i\\ X_i^\dagger & Y_i\end{smallmatrix}\right)$ are PSD, so their convex combination is PSD, and reading off its Schur complement gives $\lambda Y_1 + (1-\lambda)Y_2\succeq$

$(\lambda X_1 + (1-\lambda)X_2)^\dagger(\lambda A_1 + (1-\lambda)A_2)^{-1}(\lambda X_1 + (1-\lambda)X_2)$, which is Eq. (55.173) for this map. □

Lemma 55.46 (Lieb's concavity theorem)**.** *For each fixed $s \in [0,1]$ and fixed matrix K, the map*

$$(A, B) \longmapsto \operatorname{Tr}\left(K^\dagger A^s\, K\, B^{1-s}\right) \tag{55.183}$$

on pairs of positive-definite matrices is jointly concave.

This is the theorem of Lieb (1973). Its mechanism is exactly the operator convexity of the inverse, Theorem 55.45, promoted through the integral representation $A^s = \frac{\sin \pi s}{\pi} \int_0^\infty \lambda^{s-1}\, A(A+\lambda)^{-1}\, d\lambda$ (valid for $0 < s < 1$), which writes the fractional power as a positively-weighted average of the operator-convex resolvents $A(A+\lambda)^{-1} = \mathbf{1} - \lambda(A+\lambda)^{-1}$; the joint concavity of Eq. (55.183) then follows by the same Schur-complement positivity applied to the bilinear form in A and B, integrated against the positive weight λ^{s-1}. The complete chain of estimates is a standard result of matrix analysis; we take Theorem 55.46 as established, having exhibited its operator-convex core in Theorem 55.45 and refer to the standard matrix-analysis literature for the complete estimate.

Joint convexity of relative entropy now follows by differentiation. Define, for positive states ρ, σ and $s \in (0,1)$, the *Petz quasi-entropy*

$$Q_s(\rho, \sigma) \equiv \operatorname{Tr}\left(\rho^s\, \sigma^{1-s}\right), \tag{55.184}$$

which is Eq. (55.183) with $K = \mathbf{1}$ and is therefore jointly concave by Theorem 55.46. Its value at $s = 1$ is $Q_1 = \operatorname{Tr}\rho = 1$, and its derivative there reproduces the relative entropy:

$$\frac{d}{ds} Q_s\Big|_{s=1} = \operatorname{Tr}\left(\rho^s (\log\rho)\, \sigma^{1-s}\right)\Big|_{s=1} - \operatorname{Tr}\left(\rho^s \sigma^{1-s} \log\sigma\right)\Big|_{s=1} = \operatorname{Tr}(\rho\log\rho) - \operatorname{Tr}(\rho\log\sigma) = S(\rho\,\|\,\sigma). \tag{55.185}$$

Therefore, since $Q_1 = 1$ independently of the states,

$$S(\rho\,\|\,\sigma) = \lim_{s\to 1^-} \frac{Q_s(\rho,\sigma) - 1}{s-1} = \lim_{s\to 1^-} \frac{1 - Q_s(\rho,\sigma)}{1-s}. \tag{55.186}$$

For each fixed $s \in (0,1)$ the numerator $1 - Q_s$ is jointly *convex* (because Q_s is jointly concave by Theorem 55.46 and the constant 1 is affine), and dividing by the positive constant $1 - s$ preserves convexity. A pointwise limit of jointly convex functions is jointly convex; hence $S(\rho\,\|\,\sigma)$ is jointly convex, which is Theorem 55.44. This closes the proof of monotonicity Theorem 55.43.

With monotonicity in hand the master inequality of the chapter is one line away. We have arrived at the result promised in Section 55.1.4.

Theorem 55.47 (Strong subadditivity of the entanglement entropy)**.** *For any state on a tripartite system ABC, the reduced entropies satisfy*

$$\boxed{S_{AB} + S_{BC} \;\geq\; S_{ABC} + S_B} \tag{55.187}$$

equivalently, defining the conditional mutual information $I(A:C|B) = S_{AB} + S_{BC} - S_{ABC} - S_B$*, one has* $I(A:C|B) \geq 0$*.*

Proof. We argue through the conditional mutual information. Take the two states on ABC to be $\rho \equiv \rho_{ABC}$ and the product reference built on B. Monotonicity under tracing out A,

$$S(\rho_{BC} \,\|\, \rho_B \otimes \rho_C) \;\leq\; S(\rho_{ABC} \,\|\, \rho_{AB} \otimes \rho_C), \tag{55.188}$$

is the data-processing inequality Eq. (55.178) for the partial trace Tr_A, with $\rho_{ABC} \mapsto \rho_{BC}$ and $\rho_{AB} \otimes \rho_C \mapsto \rho_B \otimes \rho_C$. Now expand both sides with Eq. (55.166) and the additivity of log on tensor products. The left side is $\mathrm{Tr}(\rho_{BC} \log \rho_{BC}) - \mathrm{Tr}(\rho_{BC} \log \rho_B) - \mathrm{Tr}(\rho_{BC} \log \rho_C) = -S_{BC} + S_B + S_C$. The right side is $\mathrm{Tr}(\rho_{ABC} \log \rho_{ABC}) - \mathrm{Tr}(\rho_{ABC} \log \rho_{AB}) - \mathrm{Tr}(\rho_{ABC} \log \rho_C) = -S_{ABC} + S_{AB} + S_C$. The common S_C cancels, and Eq. (55.188) becomes $-S_{BC} + S_B \leq -S_{ABC} + S_{AB}$, i.e. $S_{AB} + S_{BC} \geq S_{ABC} + S_B$, which is Eq. (55.187). □

Setting $C = \varnothing$ in Eq. (55.187) reduces it to ordinary subadditivity $S_{AB} \leq S_A + S_B$ (Theorem 55.12), as anticipated there; strong subadditivity is the statement that adding a shared region B to both A and C does not spoil the inequality, and is incomparably more powerful. Its physical reading is that conditioning never increases the conditional entropy in a way that violates positivity of the conditional mutual information: the correlations A shares with C cannot become negative when B is held as side information.

Physical Insight

Strong subadditivity is the entropic engine behind the c-theorems of Section 55.4. Applied to nested and overlapping intervals on a line, Eq. (55.187) forces the entropic c-function $c_{\mathrm{eff}}(\ell) = 3\ell\, S'_A(\ell)$ of Eq. (55.136) to be monotonically non-increasing along the renormalization-group flow: choosing A, B, C to be adjacent intervals and feeding their boost-symmetric entropies into Eq. (55.187) yields $S''_A(\ell) \leq 0$ for the relevant combination, which is exactly $\partial_\ell c_{\mathrm{eff}} \leq 0$. This is the Casini–Huerta entropic proof of Zamolodchikov's theorem (Theorem 52.17): the monotonicity of the central charge under the RG is not an independent dynamical fact but a corollary of the universal inequality Eq. (55.187), which holds in *any* quantum system. The same logic, applied to spheres in higher dimensions, underlies the entropic proofs of the F-theorem ($d = 3$) and the a-theorem ($d = 4$); the entanglement entropy converts a statement about the loss of degrees of freedom into a statement about the geometry of nested regions.

55.6.4 The first law of entanglement

We now extract the linear-response content of relative-entropy positivity. Fix a reference state σ — for the applications, the reduced density matrix $\rho_A^{(0)}$ of the *vacuum* on a region A — and consider a one-parameter family of nearby states

$$\rho(\mu) = \sigma + \mu\,\delta\rho + O(\mu^2), \qquad \mathrm{Tr}\;\delta\rho = 0, \tag{55.189}$$

the trace condition enforcing normalization. The relative entropy $S(\rho(\mu)\,\|\,\sigma)$ is a non-negative function of μ that vanishes at $\mu = 0$; therefore $\mu = 0$ is a global minimum, and its first derivative there must vanish while its second derivative is non-negative. We read off both, using the decomposition Eq. (55.170), $S(\rho\,\|\,\sigma) = \Delta\langle K_\sigma\rangle - \Delta S$.

The modular-energy term is *linear* in ρ and hence carries no higher-order pieces: with $K_\sigma = -\log\sigma$ a fixed operator,

$$\Delta\langle K_\sigma\rangle = \mathrm{Tr}\,[(\rho-\sigma)K_\sigma] = \mu\;\mathrm{Tr}\,[\delta\rho\,K_\sigma] + O(\mu^2), \tag{55.190}$$

exactly linear in $\delta\rho$ at first order. The entropy term must be expanded. The first variation of the von Neumann entropy is

$$\delta S = -\delta\,\mathrm{Tr}(\rho\log\rho) = -\,\mathrm{Tr}\,[\delta\rho\,\log\rho] - \mathrm{Tr}\,[\rho\,\rho^{-1}\delta\rho] = -\,\mathrm{Tr}\,[\delta\rho\,\log\rho], \tag{55.191}$$

where the second term $\mathrm{Tr}[\rho\,\delta(\log\rho)] = \mathrm{Tr}[\delta\rho] = 0$ by the cyclic identity $\delta\log\rho = \int_0^\infty (\rho+u)^{-1}\delta\rho\,(\rho+u)^{-1}du$ traced against ρ, which collapses to $\mathrm{Tr}\;\delta\rho = 0$. Evaluating Eq. (55.191) at the reference point $\rho = \sigma$, the first-order entropy change is

$$\Delta S = \mu\,\big(-\mathrm{Tr}[\delta\rho\,\log\sigma]\big) + O(\mu^2) = \mu\;\mathrm{Tr}\,[\delta\rho\,K_\sigma] + O(\mu^2). \tag{55.192}$$

The first-order entropy change Eq. (55.192) is identical to the first-order modular-energy change Eq. (55.190). Their difference, the relative entropy, therefore has no linear term — consistent with the minimum at $\mu = 0$ — and we have proved the first law.

Theorem 55.48 (First law of entanglement)**.** *For any state $\rho = \sigma + \delta\rho$ infinitesimally close to a reference state σ with modular Hamiltonian $K_\sigma = -\log\sigma$, the first-order changes in the entanglement entropy and the modular energy coincide:*

$$\boxed{\;\delta S \;=\; \delta\langle K_\sigma\rangle \;=\; \mathrm{Tr}\,[\delta\rho\,K_\sigma].\;} \tag{55.193}$$

The relative entropy is second order in the perturbation,

$$S(\rho\,\|\,\sigma) = \tfrac{1}{2}\,\mu^2\,\langle\delta\rho,\delta\rho\rangle_\sigma + O(\mu^3) \;\geq\; 0, \tag{55.194}$$

its leading term being a positive-definite quadratic form — the quantum Fisher information of the family at σ — which encodes the statistical distinguishability of nearby states.

Proof. The vanishing of the first-order relative entropy was shown by Eqs. (55.190) and (55.192); subtracting them gives the first law Eq. (55.193). For Eq. (55.194), expand $S(\rho(\mu)\,\|\,\sigma)$ to second order. Its value and first derivative at $\mu = 0$ vanish (positivity with a zero at the minimum), so $S = \frac{1}{2}\mu^2\,\partial_\mu^2 S|_0 + O(\mu^3)$. The second derivative is the Kubo–Mori (Bogoliubov) inner product

$$\langle \delta\rho, \delta\rho\rangle_\sigma = \int_0^\infty du\ \mathrm{Tr}\left[\delta\rho\,(\sigma+u)^{-1}\,\delta\rho\,(\sigma+u)^{-1}\right], \tag{55.195}$$

obtained by differentiating $-\Delta S$ once more using the same resolvent representation of $\delta\log\rho$ as in Eq. (55.191); it is manifestly non-negative (the integrand is $\mathrm{Tr}[Z^\dagger Z] \geq 0$ with $Z = (\sigma+u)^{-1/2}\delta\rho\,(\sigma+u)^{-1/2}$, since $\delta\rho$ is Hermitian), and positive-definite for $\delta\rho \neq 0$. This is Eq. (55.194). □

The first law Eq. (55.193) deserves its name. It is structurally the thermodynamic relation $\delta E = T\,\delta S$ — "modular energy equals modular temperature times entropy" — but it is an *exact, derived* statement of quantum field theory, not a phenomenological postulate, and it holds for the entanglement entropy of any region as a consequence of nothing more than the positivity of relative entropy. The inequality Eq. (55.171), $\Delta S \leq \Delta\langle K_\sigma\rangle$, is its finite, nonlinear completion: away from linear order, entropy can only fall short of modular energy, the deficit being the (positive) relative entropy. For applications to gravity it is the saturation at first order that matters: a small excitation of the vacuum changes the entanglement of a region by exactly the expectation value of a fixed, geometric operator K_σ, with no entropy of its own to track. This is what makes entanglement entropy computable in linear response, and it is the precise statement that the holographic Ryu–Takayanagi construction of Chapter 56 reproduces geometrically — there, $\delta S = \delta\langle K\rangle$ becomes the linearized Einstein equation for the bulk metric.

55.6.5 The vacuum modular Hamiltonian of a ball in a CFT

To make the first law concrete we need an explicit modular Hamiltonian beyond the half-space of Theorem 55.25. Conformal symmetry provides exactly one more case in which K_A is local: a *ball* in the vacuum of a conformal field theory. The construction, due to Casini, Huerta, and Myers, maps the ball's causal development conformally onto the Rindler wedge, transports the boost generator, and reads off a local operator.

Take a ball $A = \{\,(0,\mathbf{x}) : |\mathbf{x}| < R\,\}$ of radius R on the time slice $t = 0$ of d-dimensional Minkowski space, in the vacuum of a CFT. Its domain of dependence is the *causal diamond* $\mathcal{D}_R = \{(t,\mathbf{x}) : |t| + |\mathbf{x}| < R\}$. The half-space wedge W_R of Theorem 55.24 and the diamond $\mathcal{D}_R$ are conformally equivalent: there is a special conformal transformation composed with a translation (an element of the conformal group of Chapter 47) that maps W_R onto $\mathcal{D}_R$ and carries the

boost Eq. (55.64), which preserves W_R, into the unique conformal Killing vector ζ^μ that preserves $\mathcal{D}_R$. Because the vacuum of a CFT is conformally invariant, the modular flow of the diamond is the flow generated by ζ, and the modular Hamiltonian is the corresponding conserved charge.

The conformal Killing vector preserving $\mathcal{D}_R$, normalized so that its flow has the same 2π periodicity in Euclidean angle as the boost, has the time component, on the slice $t = 0$,

$$\zeta^0(0, \mathbf{x}) = \frac{\pi}{R}(R^2 - |\mathbf{x}|^2), \tag{55.196}$$

which is the conformal image of the linear boost weight x^1: it vanishes on the entangling sphere $|\mathbf{x}| = R$ (the fixed locus of ζ, as Σ was the fixed locus of the boost), is maximal at the center, and reduces to the Rindler weight near the edge — setting $|\mathbf{x}| = R - \rho_*$ with $\rho_* \ll R$ gives $\zeta^0 \approx (\pi/R)(2R\rho_*) = 2\pi\rho_*$, matching $2\pi \times$ (distance to Σ) as in Eq. (55.69). The charge that generates this flow is the integral of $\zeta^\mu T_{\mu 0}$ over the ball, which at $t = 0$ is $\int \zeta^0 T^{00}$. We thus obtain the Casini–Huerta–Myers form.

Theorem 55.49 (Vacuum modular Hamiltonian of a ball in a CFT)**.** *In the vacuum of a conformal field theory in d-dimensional Minkowski space, the modular Hamiltonian of a ball A of radius R centered at the origin is the local operator*

$$\boxed{K_A = 2\pi \int_{|\mathbf{x}|<R} \mathrm{d}^{d-1}x \, \frac{R^2 - |\mathbf{x}|^2}{2R} \, T^{00}(0, \mathbf{x})} \tag{55.197}$$

so that $\rho_A = e^{-K_A}/Z$. The weight $(R^2 - |\mathbf{x}|^2)/2R$ is the energy-density profile that vanishes on the entangling sphere and reduces, near it, to the proper distance to the sphere.

This is the conformal generalization of Eq. (55.69): the half-space result is the $R \to \infty$ limit, in which the diamond opens up into the wedge and the parabolic weight $(R^2 - |\mathbf{x}|^2)/2R$ flattens to the linear x^1. The locality is again a gift of symmetry — a single conformal Killing vector preserves the diamond — and it fails the moment the region is not a ball or the theory is not conformal, as flagged in Section 55.3.3.

The first law now becomes a concrete, computable statement. For a state ρ infinitesimally above the vacuum, Theorem 55.48 with $K_\sigma = K_A$ of Eq. (55.197) gives

$$\delta S_A = \delta\langle K_A\rangle = 2\pi \int_{|\mathbf{x}|<R} \mathrm{d}^{d-1}x \, \frac{R^2 - |\mathbf{x}|^2}{2R} \, \delta\langle T^{00}(0, \mathbf{x})\rangle, \tag{55.198}$$

the change in the ball's entanglement entropy expressed entirely through the expectation value of the local energy density of the excitation, weighted by the CHM profile. We illustrate with the smallest balls, where the formula reduces to a clean local relation.

Worked Example 55.50 (The first law for a small ball and the energy density)**.** Consider a state with a smooth energy density $\delta\langle T^{00}(\mathbf{x})\rangle$ that is slowly varying on the scale of a small ball of radius R. Expand $\delta\langle T^{00}\rangle \approx \delta\langle T^{00}(0)\rangle$ as constant across the ball; then Eq. (55.198) factorizes,

$$\delta S_A \approx \frac{2\pi}{2R}\,\delta\langle T^{00}(0)\rangle \int_{|\mathbf{x}|<R} \mathrm{d}^{d-1}x\,(R^2-|\mathbf{x}|^2). \tag{55.199}$$

The integral is elementary. Using spherical coordinates in $d-1$ dimensions with solid angle $\Omega_{d-2} = 2\pi^{(d-1)/2}/\Gamma(\frac{d-1}{2})$,

$$\begin{aligned}\int_{|\mathbf{x}|<R} \mathrm{d}^{d-1}x\,(R^2-|\mathbf{x}|^2) &= \Omega_{d-2}\int_0^R r^{d-2}(R^2-r^2)\,dr = \Omega_{d-2}\,R^{d+1}\Big(\frac{1}{d-1}-\frac{1}{d+1}\Big)\\ &= \Omega_{d-2}\,\frac{2R^{d+1}}{(d-1)(d+1)}.\end{aligned} \tag{55.200}$$

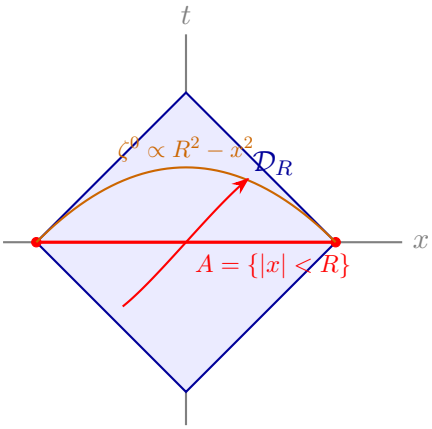
Hence the entanglement entropy of a small ball responds to an excitation as

$$\delta S_A = \frac{2\pi\,\Omega_{d-2}}{(d^2-1)}\,R^d\,\delta\langle T^{00}(0)\rangle + \cdots, \tag{55.201}$$

proportional to the local energy density times the ball's volume scale R^d. In $d=2$ the coefficient is $2\pi\cdot 2/3 = 4\pi/3$ (with $\Omega_0 = 2$ counting the two endpoints and $d^2-1=3$), giving $\delta S = (4\pi/3)R^2\delta\langle T^{00}\rangle$ for a small interval of half-length R — the linear-response companion of the logarithm Eq. (55.119). The structure $\delta S \propto R^d\,\delta E_{\text{density}}$ is the seed of the holographic derivation of gravitational dynamics: read as a constraint on $\delta\langle T^{00}\rangle$ for all balls, Eq. (55.198) is equivalent to the linearized Einstein equation governing the bulk geometry dual to the excitation, the statement Chapter 56 establishes.

55.6.6 Modular flow beyond the solvable cases: a frontier note

It must be said plainly how exceptional the local results Eqs. (55.69) and (55.197) are. The modular Hamiltonian $K_A = -\log\rho_A$ is a *local* integral of the energy density only when a spacetime (conformal) symmetry preserves the causal development of A. Two cases realize this exactly: the half-space, preserved by a boost (Bisognano–Wichmann, Theorem 55.25), valid in any QFT; and the ball in a CFT, preserved by a conformal Killing vector (Casini–Huerta–Myers, Theorem 55.49). For every other region the modular Hamiltonian is a complicated nonlocal operator, and its modular flow $\sigma_s(\mathcal{O}) = \rho_A^{is}\mathcal{O}\,\rho_A^{-is}$ does not act geometrically. As of the mid-2020s, the explicitly known cases beyond the two above are few and special: multiple intervals for the free massless Dirac fermion in two dimensions (Casini–Huerta), where the flow is *bi-local*, mixing operators at points conformally related across the intervals; certain free-field examples on



Causal diamond of a ball: local modular flow generated by ζ

Figure 55.6: The causal diamond $\mathcal{D}_R$ (blue) of a ball A of radius R (red) in a CFT vacuum. The diamond is conformally equivalent to the Rindler wedge of Fig. 55.3, and the boost is carried into the conformal Killing vector ζ that preserves $\mathcal{D}_R$, with time component $\zeta^0 \propto R^2 - |\mathbf{x}|^2$ (orange) vanishing on the entangling sphere $|\mathbf{x}| = R$. Modular flow is geometric, sliding operators along the streamlines of ζ, and the modular Hamiltonian Eq. (55.197) is the local energy density weighted by ζ^0.

null cuts (where the algebra factorizes and the flow simplifies); and perturbative expansions around the solvable cases. A closed-form local modular Hamiltonian for a generic region in an interacting theory is not known and is not expected to exist. This is a genuine frontier: modular flow is the central dynamical object of algebraic QFT and of the recent program connecting entanglement to energy conditions and to gravity, yet outside the conformal ball and the Rindler wedge it remains analytically inaccessible. The honest statement is that the structures of this section are exact where symmetry makes them so, and otherwise are controlled only perturbatively or numerically.

55.6.7 From inequalities to energy conditions: the bridge to gravity

We close the chapter by recording, without full proof with their logical origins made explicit, the theorems that turn the abstract inequalities of this section into constraints on energy and geometry. These are the results that Part XV's gravitational entanglement builds upon directly; their full proofs belong to the holographic and algebraic developments of Chapters 56 and 57 and the literature gathered in the Further Reading, but each is, in essence, a corollary of relative-entropy positivity or monotonicity, and we indicate the mechanism in each

case.

The Bekenstein bound, from relative-entropy positivity. Bekenstein conjectured a universal bound on the entropy S of any system of energy E confined to a region of size R: $S \le 2\pi RE$ (in natural units). For decades this resisted a clean field-theoretic statement, because both S and the "energy" are ambiguous in QFT. Casini's resolution identifies the bound with the positivity of relative entropy. Take σ to be the vacuum reduced density matrix on a region and ρ the reduced state of an excitation localized within it; then Eq. (55.171) reads $\Delta S \le \Delta\langle K\rangle$, where now $\Delta S = S(\rho) - S(\sigma)$ is the (finite, cutoff-independent) entropy of the excitation relative to the vacuum and $\Delta\langle K\rangle$ is its modular energy. For a region whose modular Hamiltonian is the boost-like $\int x^1 T^{00}$ of Theorem 55.25, $\Delta\langle K\rangle$ is precisely the combination 2π(energy)$\times$(distance scale) that Bekenstein's $2\pi RE$ estimates. Thus the Bekenstein bound is the inequality

$$S(\rho) - S(\sigma) \;\le\; \Delta\langle K\rangle, \tag{55.202}$$

a tautological consequence of $S(\rho\,\|\,\sigma) \ge 0$, made meaningful by the recognition that the "entropy" and "energy" in Bekenstein's formula are the relative entropy's two pieces ΔS and $\Delta\langle K\rangle$. The bound is then not a new physical postulate but a restatement of Theorem 55.9: it holds in every quantum field theory, automatically, and is saturated only by the vacuum itself.

The quantum null energy condition, from monotonicity. Classically the null energy condition $T_{kk} = T_{\mu\nu}k^\mu k^\nu \ge 0$ (for null k) underlies the singularity theorems and the area theorems of general relativity, but it is violated by quantum fields, whose energy density can be locally negative (the Casimir effect, Chapter 9). The *quantum null energy condition* (QNEC) is the sharp replacement: the local energy density is bounded below by the second variation of the entanglement entropy under a null deformation of the entangling surface,

$$\langle T_{kk}(x)\rangle \;\ge\; \frac{1}{2\pi}\,\frac{1}{\sqrt{h}}\,\frac{\delta^2 S_A}{\delta\lambda(x)^2}, \tag{55.203}$$

where $\lambda(x)$ parametrizes the displacement of ∂A along the null direction k at the point x, $\sqrt{h}$ is the induced area element, and S'' is taken at fixed displacement elsewhere. The QNEC is a theorem, and its origin is the monotonicity of relative entropy: deforming the region along a null direction generates a one-parameter family of nested algebras, and the monotonicity of $S(\rho\,\|\,\sigma_{\rm vac})$ along this family — a consequence of Theorem 55.43, refined into a statement about second shape-derivatives — is exactly Eq. (55.203). Integrating Eq. (55.203) along a complete null generator and using that S_A returns to its vacuum value at the endpoints gives the *averaged null energy condition* (ANEC), $\int_{-\infty}^{\infty}\langle T_{kk}\rangle\, d\lambda \ge 0$, which thus also follows from monotonicity of relative entropy (equivalently, in the lightcone limit,

from the causality and positivity built into the modular flow). The averaged and quantum null energy conditions are the precise senses in which energy positivity survives quantization, and they are the inputs that make the holographic and semiclassical-gravity constructions of Chapters 56 and 57 — the proof of the generalized second law, the consistency of the island prescription — well-posed.

Physical Insight

The arc of this section is a single idea seen from three sides. Relative entropy is the unique cutoff-independent measure of how one state differs from another; its *positivity* is the Bekenstein bound and, linearized, the first law $\delta S = \delta\langle K\rangle$; its *monotonicity* is strong subadditivity, the c-theorems, and the quantum null energy condition. Information inequalities, which hold in any quantum system whatsoever, become, when read through the modular Hamiltonian of a region in a field theory, statements about energy and entropy that were once thought to require gravity to explain. That this dictionary closes — that the same first law which here follows from $S(\rho\,\|\,\sigma) \geq 0$ reappears in Chapter 56 as Einstein's equation linearized about anti-de Sitter space — is the deepest reason to believe that spacetime geometry is the large-scale expression of entanglement.

This completes the chapter's hand-off to Part XV. The three structures assembled here — the positivity of relative entropy and its first-order content, the first law $\delta S_A = \delta\langle K_A\rangle$ built on the local modular Hamiltonians Eqs. (55.69) and (55.197); the monotonicity of relative entropy and its corollary, strong subadditivity Eq. (55.187); and the recognition of mutual information as a relative entropy — are precisely the inequalities the Ryu–Takayanagi proof and the entanglement-wedge reconstruction of Chapter 56 invoke, and the inputs that make the Page curve and the quantum extremal surface prescription of Chapter 57 consistent. The vacuum that Chapter 9 found to be a richly correlated medium, that Chapter 10 prepared as a Euclidean path integral, and that Chapter 47 organized by conformal symmetry, has now been shown to carry an exact information-theoretic skeleton; holography is the statement that this skeleton is geometry.

Further reading

The replica computation of two-dimensional entanglement entropy is due to Holzhey, Larsen and Wilczek and, in the form used here, to Calabrese and Cardy [104]; the area law originates with Bombelli, Koul, Lee and Sorkin and with Srednicki [85, 453], and the correlator (Gaussian-state) method for free fields is developed in [115, 395]. The standard review of entanglement entropy in quantum field theory, covering the replica trick, the area law, and the universal

terms, is Casini and Huerta [115]; for the algebraic backbone — Tomita–Takesaki modular theory, the type-III classification of local algebras, and the Bisognano–Wichmann theorem — see Haag [246] and Witten's lectures [524]. The Unruh effect is [477]; the conformal modular Hamiltonian of a ball is Casini, Huerta and Myers [116]; the four-dimensional anomaly-induced logarithm is Solodukhin [450], and the gauge-field center/edge-mode subtlety is Donnelly [158]. The entropic proofs of the c- and F-theorems are [113, 114], with the a-theorem of Komargodski and Schwimmer in [301, 302]. For where this leads, the holographic entanglement entropy of Ryu and Takayanagi and its review are [414, 422], taken up in Chapters 56 and 57.

Chapter 56

Holographic Entanglement

Two results from the preceding chapters are about to collide productively. From the gravity side we carry the black-hole area law $S_{\mathrm{BH}} = A/4G_N$ (Theorem 54.10) and the AdS/CFT dictionary: a holographic state is a bulk geometry, and the GKP–Witten relation (Theorem 53.29 and Eq. (53.162)) computes any boundary partition function as the on-shell action of the dominant bulk saddle whose conformal boundary carries that state. From the field-theory side we carry a quantity Chapter 55 taught us to want but rarely to compute: the entanglement entropy $S(A) = -\operatorname{Tr}(\rho_A \ln \rho_A)$ of a spatial region (Eq. (55.6)), defined through the replica trick (Theorem 55.17) as the $n \to 1$ continuation of moments $\operatorname{Tr}\rho_A^n$ and tractable, in closed form, essentially only for a single interval in a 2D CFT, $S = (c/3)\ln(\ell/\varepsilon)$ (Eq. (55.119)). This chapter is the statement that these two facts are one: that the entanglement entropy of a boundary region *is* a horizon-like area in the bulk, and that, run in reverse, the spacetime geometry is assembled out of the entanglement structure of the boundary state.

We open with the proposal itself. Section 56.1 states the Ryu–Takayanagi formula precisely — entanglement entropy as the area of the minimal bulk surface anchored on ∂A and homologous to A, all divided by $4G_N$ — motivates it from the black-hole area law and the bulk replica saddle, and validates it against the one entropy the reader already trusts by recovering $(c/3)\ln(\ell/\varepsilon)$ from a single geodesic in AdS_3, with the coefficient fixed by the Brown–Henneaux relation $c = 3L/2G_N$ derived from the holographic Weyl anomaly. Section 56.2 then turns the proposal into a calculational engine: the thermal entropy of an interval from the BTZ geodesic, the universal sphere coefficients in any dimension via the Casini–Huerta–Myers conformal map to a hyperbolic black hole (Theorem 55.49), and the strip, where an area-law divergence sits atop a genuinely nonlocal finite piece.

With a working formula in hand we collect its structural payoff before paying its derivational debt. Section 56.3 shows that deep quantum-information inequalities — strong subadditivity (Theorem 55.47), and the sharper monogamy of

mutual information that holographic states obey but generic states need not — follow from elementary facts about minimal surfaces, cutting and recombining geodesics rather than manipulating density matrices, and it frames the holographic entropy cone these inequalities cut out. Section 56.4 then discharges the central debt: it promotes Ryu–Takayanagi from checked conjecture to theorem by running the replica trick upstairs in the bulk, following Lewkowycz and Maldacena. The cyclic replica symmetry is quotiented to a single geometry with a conical defect; the $n \to 1$ limit localizes the on-shell action onto the defect; the factor of $4G_N$, the extremality of the surface, and the homology constraint all emerge from the demand that the bulk be smooth.

The classical statement is then refined in three directions. Section 56.5 adds the bulk matter entanglement (the Faulkner–Lewkowycz–Maldacena one-loop term), makes the prescription covariant by extremizing rather than minimizing area (the HRT surface, with its maximin reformulation), and unifies both into the quantum extremal surface that extremizes the generalized entropy. Section 56.6 asks which bulk a boundary region actually encodes: not the causal wedge but the larger entanglement wedge bounded by the RT surface, with the JLMS identity equating boundary and bulk modular Hamiltonians up to the area operator, and relative-entropy positivity (Chapter 55) translating into a bulk energy condition. Section 56.7 recasts this encoding as quantum error correction: bulk operators are reconstructed as smeared boundary operators (HKLL), the entanglement wedge is the recoverable region, and the HaPPY pentagon code realizes the whole structure — subregion duality, complementary recovery, and a discrete RT formula — in an exactly solvable model.

Section 56.8 closes the arc by reversing it. If geometry is reconstructed from entanglement, geometry should be *made of* it, and we prove the sharpest available version of that slogan: the bulk Einstein equation, linearized about pure anti–de Sitter space, is the first law of entanglement (Theorem 55.48) imposed on every ball-shaped boundary region, with the modular Hamiltonian of a ball (Theorem 55.49) supplying the conformal Killing data and the Iyer–Wald formalism converting an entropy variation into a gravitational constraint. We then state, honestly and within scope, Van Raamsdonk's disentangling argument and its tensor-network/MERA discretization, and the ER = EPR slogan, before handing a dated summary of open problems — the dynamical and black-hole-interior questions this static, kinematic chapter cannot settle — to Chapter 57.

56.1 Geometry from entanglement: the Ryu–Takayanagi proposal

Two facts now sit side by side in our hands, and they are about to collide productively. The first is that entropy can be geometry. A black hole carries an entropy equal to a quarter of its horizon area in Planck units, $S_{\mathrm{BH}} = A/4G_N$

(Theorem 54.10); a thermodynamic quantity that in any ordinary system counts microstates is, for a black hole, simply the area of a surface. The second is that geometry can be dynamics dual to a field theory. The AdS/CFT correspondence promotes a conformal field theory in d dimensions to a gravitational theory in an asymptotically anti-de Sitter spacetime one dimension higher, with the field–operator dictionary and the GKP–Witten relation (Theorem 53.29 and Eq. (53.162)) translating bulk classical physics into boundary correlators. The bulk metric is not a fixed stage; it is the saddle point of a gravitational path integral whose boundary data is a CFT state — a state that, by the state–operator correspondence of Theorem 47.42, is itself the radial-quantization image of a boundary operator insertion.

Set against these two facts is a hard quantity the previous chapter taught us to want but not to compute. The entanglement entropy $S(A) = -\operatorname{Tr}(\rho_A \ln \rho_A)$ of a spatial region A in a quantum field theory (Eq. (55.6)) is a fine, often nonlocal probe of a state, but it is brutal to evaluate: the replica trick (Theorem 55.17) reduces it to the analytic continuation $S(A) = -\partial_n \operatorname{Tr} \rho_A^n\big|_{n=1}$ of partition functions on n-sheeted branched covers, and that $n \to 1$ continuation is exactly what makes the computation intractable away from free fields and two-dimensional CFTs. In the one case where we do have an exact answer — a single interval of length ℓ in the vacuum of a 2D CFT, $S = (c/3)\ln(\ell/\varepsilon)$ (Eq. (55.119)) — it was won by 2D conformal technology that does not generalize. We have a quantity built from entanglement and a separate discovery that entanglement-like entropy is geometric. Ryu and Takayanagi proposed that these are the same statement: that the entanglement entropy of a boundary region *is* a horizon-like area in the bulk, the area of the smallest bulk surface that hangs from the region's boundary. This section states that proposal precisely, defines every object in it, motivates it as the natural generalization of the black-hole area law, and validates it against the one entanglement entropy the reader already trusts.

56.1.1 From horizon entropy to entanglement entropy: the heuristic

The bridge from $S_{\mathrm{BH}} = A/4G_N$ to an entanglement entropy is shorter than it looks, because black-hole entropy is, on reflection, already an entanglement entropy. An observer outside the horizon has no access to the interior; the global state, even if pure, is described from outside by a reduced density matrix obtained by tracing over the inaccessible region, and the horizon is the surface that separates the two. The thermal character of the exterior and the area scaling of its entropy are precisely the hallmarks of entanglement across a boundary that Chapter 55 taught us to expect: an area law, with the divergence localized on the dividing surface. The black hole is the special case in which the dividing surface is a true event horizon and the state is the thermal Hartle–Hawking state. Ryu and Takayanagi's move is to ask what happens when we replace the event

horizon by an arbitrary boundary region's worth of entanglement. If the entropy of the region behind a horizon is a quarter of the horizon's area, perhaps the entropy of *any* boundary region A is a quarter of the area of some bulk surface determined by A — a surface that is horizon-like in that it bounds the region of the bulk that A cannot see, but that need not be a causal horizon at all.

Which surface? Here the gravitational path integral supplies the second, sharper, heuristic. The replica trick computes $\operatorname{Tr}\rho_A^n = Z_n/Z_1^n$, where Z_n is the CFT partition function on the n-sheeted branched cover $\mathcal{M}_n$ obtained by gluing n copies of the spacetime cyclically across the region A (Theorem 55.17). In a holographic CFT each partition function is computed by a bulk saddle: by the GKP–Witten relation (Eq. (53.162)), $Z_n = e^{-I_{\mathrm{grav}}[\mathcal{B}_n]}$, where $\mathcal{B}_n$ is the dominant smooth Euclidean bulk solution whose conformal boundary is the replica manifold $\mathcal{M}_n$. The branched cover carries the cyclic replica symmetry $\mathbb{Z}_n$, and the dominant bulk filling inherits it; quotienting $\mathcal{B}_n$ by that $\mathbb{Z}_n$ produces a bulk geometry $\mathcal{B}_n/\mathbb{Z}_n$ with the same boundary as the original spacetime but with a conical excess concentrated on the codimension-two fixed-point locus of the symmetry — a bulk surface anchored on ∂A. When one differentiates the on-shell gravitational action in n and continues to $n \to 1$, the bulk action away from the fixed-point surface contributes nothing new (it solves the equations of motion), and the entire n-derivative localizes onto the conical singularity, where it returns exactly the proper area of the fixed-point surface divided by $4G_N$. The entanglement entropy is thus the area of the bulk surface left fixed by the replica symmetry, and minimizing the action selects the surface of least area. This is the Lewkowycz–Maldacena argument; we present it here only as motivation and carry it out in full, with the conical-deficit accounting made rigorous, in Section 56.4. For now it tells us precisely what to write down.

Physical Insight

The slogan "entanglement builds geometry" is, at this stage, a precise computational claim and nothing more mystical: the area of a bulk surface equals a boundary entanglement entropy. Read forward, it says that the smooth connectedness of the bulk geometry is underwritten by the boundary state being entangled — disentangle the two halves of the boundary and the bulk surface separating them grows without bound, pinching the geometry into disconnected pieces. Read as a definition, it converts an intractable $n \to 1$ continuation into a classical problem in Riemannian geometry: find the smallest surface hanging from a given boundary curve. The rest of this chapter lives in the tension between these two readings.

56.1.2 The objects: anchored minimal surfaces and the homology constraint

Fix the arena. We work in an asymptotically AdS_{d+1} spacetime with a boundary conformal field theory CFT_d, in the regime where the bulk is well described by classical Einstein gravity — equivalently, large central charge and large 't Hooft coupling, so that G_N is small in units of the AdS radius L (Theorem 53.27). We take the state to be static, or at least specify a moment of time symmetry, so that there is a preferred constant-time slice; the fully covariant version, needed for genuinely time-dependent states, is flagged below and developed later. On the boundary, choose a spatial region A at fixed time, with entangling surface ∂A (its boundary within the boundary). In $d = 2$, A is an interval and ∂A is its two endpoints; in $d = 3$, A is a two-dimensional region and ∂A is a closed curve; and so on.

The bulk constant-time slice Σ is a d-dimensional Riemannian manifold whose conformal boundary is the boundary spatial slice. We seek a surface in Σ on which to evaluate an area.

Definition 56.1 (Anchored bulk surface)**.** A *bulk surface anchored on* A is a codimension-two spacelike submanifold γ_A of the bulk spacetime that lies in the constant-time slice Σ and whose only boundary coincides with the entangling surface of the region:

$$\partial\gamma_A = \partial A, \tag{56.1}$$

the equality holding on the conformal boundary $z \to 0$. As a submanifold of the d-dimensional slice Σ, γ_A is $(d-1)$-dimensional; its area is the proper volume induced on it by the bulk metric,

$$\mathrm{Area}(\gamma_A) = \int_{\gamma_A} \mathrm{d}^{\,d-1}\sigma\, \sqrt{\det h}, \qquad h_{ab} = g_{\mu\nu}\, \partial_a X^\mu\, \partial_b X^\nu, \tag{56.2}$$

where $X^\mu(\sigma)$ embeds γ_A in the bulk and h_{ab} is the induced metric. A surface is *minimal* if it extremizes Eq. (56.2) subject to the anchoring Eq. (56.1); equivalently, its trace of extrinsic curvature vanishes, $K = 0$.

Anchoring alone does not determine the surface, because for some regions there are topologically distinct ways to hang a surface from ∂A, and some of them are spurious. The decisive extra condition is topological.

Definition 56.2 (Homology constraint)**.** The anchored surface γ_A is *homologous to* A if there exists a d-dimensional spacelike region $\mathcal{R}_A$ in the bulk slice Σ — the *homology region* — whose boundary is the union of A and γ_A:

$$\partial\mathcal{R}_A = A \cup \gamma_A. \tag{56.3}$$

Equivalently, γ_A together with A bounds a bulk region: γ_A can be continuously deformed onto A through the bulk while keeping ∂A fixed.

The homology region $\mathcal{R}_A$ is the spatial slice of the bulk region that the boundary subsystem A will turn out to encode — the *entanglement wedge* we make precise in Section 56.6. For now it plays two roles. It rules out surfaces that hang from ∂A but cannot be capped off by a bulk region together with A; and, crucially, it is what makes the formula respect the obvious physical facts. If the global state is pure, then A and its complement $\bar{A}$ have equal entropy, $S(A) = S(\bar{A})$; the homology constraint guarantees this, because a surface homologous to A is automatically homologous to $\bar{A}$ (the two homology regions share the same boundary surface γ_A and together fill the slice). The constraint also does essential work in thermal states, where the bulk contains a black-hole horizon: a surface homologous to a large region must be allowed to wrap the horizon, and we will see in Section 56.2.1 that this is exactly what reproduces the extensive thermal entropy.

One more object must be named before the formula makes sense: the cutoff. The area integral Eq. (56.2) diverges, because the bulk metric blows up as the conformal boundary $z \to 0$ is approached, and γ_A runs all the way to that boundary. This divergence is not a defect; it is the holographic image of the ultraviolet divergence of entanglement entropy in any continuum field theory. We regulate it exactly as the rest of the dictionary is regulated: by cutting the bulk off at a small radial coordinate $z = \varepsilon$ rather than at $z = 0$. The radial coordinate is the inverse energy scale of the boundary theory (Section 53.3), so a bulk wall at $z = \varepsilon$ is a boundary short-distance cutoff at length scale ε. We therefore *identify the bulk radial cutoff with the QFT ultraviolet cutoff*, the same identification used in the holographic counting of degrees of freedom (Eq. (53.154)). With this identification the divergence of the area becomes, term by term, the divergence of $S(A)$, and in particular reproduces the area law of Chapter 55: the leading divergence of $\mathrm{Area}(\gamma_A)/4G_N$ scales as the area of ∂A in cutoff units, because near the boundary γ_A plunges straight in and its regulated area is controlled by the geometry of ∂A.

56.1.3 The Ryu–Takayanagi formula

We can now state the proposal precisely.

Theorem 56.3 (Ryu–Takayanagi)**.** *Let A be a spatial region in the boundary of a static asymptotically AdS_{d+1} spacetime dual to a state of a holographic CFT_d, in the classical-gravity regime. Then the entanglement entropy of A is*

$$\boxed{S(A) = \min_{\gamma_A \sim A} \frac{\mathrm{Area}(\gamma_A)}{4G_N}} \tag{56.4}$$

where the minimum runs over all bulk codimension-two surfaces γ_A anchored on ∂A (Theorem 56.1) and homologous to A (Theorem 56.2), the area is regulated

at the radial cutoff $z = \varepsilon$ identified with the QFT ultraviolet cutoff, and G_N is the bulk Newton constant. The minimizing surface is denoted γ_A^*.

Three remarks fix its standing and scope before we test it.

First, the formula is a leading-order statement in the bulk, computing the entanglement entropy of the dual state at leading order in G_N — equivalently, at leading order in the large central charge. It is the classical, geometric piece of the entropy. The subleading corrections are the bulk fields' own entanglement across γ_A^*, and incorporating them promotes Eq. (56.4) to the quantum-corrected and ultimately the quantum-extremal-surface formula of Sections 56.5.1 and 56.5.3. At the order we work in this section there is no such correction, and the right-hand side is purely geometry.

Second, the minimization in Eq. (56.4) is genuine and not a formality: there can be several minimal surfaces homologous to A, and the prescription is to take the one of least area. This is not pedantry — it is the seed of a phase transition. When the competing minimal surfaces exchange dominance as the region is varied, the entanglement entropy has a kink, a first-order "entanglement phase transition." We meet the cleanest example, two disjoint intervals, in Section 56.2.4; here we simply record that "min" is load-bearing.

Third, Theorem 56.3 as stated requires a preferred time slice, because "minimal area" is a Riemannian notion and a Lorentzian surface has no minimal area (one can always wiggle it timelike to shrink the area without bound). For static states the moment of time symmetry supplies the slice and the prescription is unambiguous. For genuinely time-dependent states one replaces "minimal on a slice" by "extremal in the full Lorentzian spacetime": γ_A^* is a saddle of the area functional, the surface of vanishing trace of extrinsic curvature with no slice singled out. This is the covariant Hubeny–Rangamani–Takayanagi prescription, and when a time-symmetric slice exists the extremal surface lies in it and coincides with the minimal one. We use the static form throughout this section and develop the covariant statement in Section 56.5.2.

Caution

The minimum in Eq. (56.4) is over surfaces *homologous* to A, not merely anchored on ∂A. Dropping the homology constraint gives wrong answers in two diagnostic cases. For a pure global state it would break the requirement $S(A) = S(\bar{A})$. And in a thermal state it would forbid the surface from wrapping the black-hole horizon, which is exactly the configuration that supplies the extensive thermal entropy of a large region (Section 56.2.1). The constraint is part of the formula, not a refinement of it.

56.1.4 The validating computation: a single interval in AdS_3

A proposal of this reach must first reproduce the one nontrivial entanglement entropy we already know exactly. We compute the right-hand side of Eq. (56.4) for a single interval in the vacuum of a 2D CFT and check that it returns $(c/3)\ln(\ell/\varepsilon)$ with the correct coefficient and no free parameters. The match is the anchor of the whole chapter.

The vacuum of a CFT_2 is dual to pure Poincaré AdS_3. In the conventions of Theorem 53.10, with the book's mostly-minus signature, the metric is

$$\mathrm{d}s^2 = \frac{L^2}{z^2}(\mathrm{d}t^2 - \mathrm{d}x^2 - \mathrm{d}z^2), \tag{56.5}$$

with the boundary at $z \to 0$, boundary coordinates (t, x) carrying the metric $\eta = \mathrm{diag}(+,-)$, and the radial direction z spacelike. The bulk is three-dimensional $(d = 2)$, so a codimension-two surface is a curve: a one-dimensional spacelike geodesic. The boundary region is the interval

$$A = \left\{ t = 0,\ -\tfrac{\ell}{2} \le x \le \tfrac{\ell}{2} \right\}, \qquad \partial A = \left\{ x = \pm\tfrac{\ell}{2} \right\}. \tag{56.6}$$

The state is static, so we work on the constant-time slice $t = 0$. The induced metric on that slice, read off Eq. (56.5) by setting $\mathrm{d}t = 0$, is $\mathrm{d}s^2|_{t=0} = -(L^2/z^2)(\mathrm{d}x^2 + \mathrm{d}z^2)$; up to the overall sign that reflects the spatial directions being mostly-minus, this is the Riemannian metric

$$\mathrm{d}\ell_\Sigma^2 = \frac{L^2}{z^2}(\mathrm{d}x^2 + \mathrm{d}z^2) \tag{56.7}$$

on the upper half-plane $z > 0$. Proper length on the slice is measured by Eq. (56.7); the RT "area" of a curve is its proper length there. This is the geometry in which we minimize.

Worked Example 56.4 (Single-interval geodesic in Poincaré AdS_3)**.** We minimize the length of a curve in the half-plane metric Eq. (56.7) anchored at the two endpoints Eq. (56.6), regulate at $z = \varepsilon$, and assemble $S = \mathrm{Length}/4G_N$.

The length functional. Parametrize the curve as $z = z(x)$, $x \in [-\ell/2, \ell/2]$, with $z \to 0$ at both endpoints. From Eq. (56.7), $\mathrm{d}\ell_\Sigma = (L/z)\sqrt{\mathrm{d}x^2 + \mathrm{d}z^2} = (L/z)\sqrt{1 + z'^2}\,\mathrm{d}x$, so the proper length to be minimized is

$$\mathrm{Length}[z] = L\int_{-\ell/2}^{\ell/2} \mathrm{d}x\, \frac{\sqrt{1 + z'^2}}{z}, \qquad z' \equiv \frac{\mathrm{d}z}{\mathrm{d}x}. \tag{56.8}$$

First integral. The integrand $f(z,z') = z^{-1}\sqrt{1+z'^2}$ has no explicit x-dependence, so the analogue of energy — the "Beltrami" first integral $f - z'\,\partial f/\partial z'$ — is conserved along the extremal curve. Computing $\partial f/\partial z' = z'/(z\sqrt{1+z'^2})$,

$$f - z'\frac{\partial f}{\partial z'} = \frac{\sqrt{1+z'^2}}{z} - \frac{z'^2}{z\sqrt{1+z'^2}} = \frac{(1+z'^2)-z'^2}{z\sqrt{1+z'^2}} = \frac{1}{z\sqrt{1+z'^2}} = \text{const} \equiv \frac{1}{z_*}. \tag{56.9}$$

The constant is named $1/z_*$ because at the deepest point of the curve $z' = 0$ and Eq. (56.9) gives $z = z_*$: z_* is the turning point, the maximal bulk depth. Rearranging Eq. (56.9),

$$z\sqrt{1+z'^2} = z_* \quad\Longrightarrow\quad z'^2 = \frac{z_*^2 - z^2}{z^2} \quad\Longrightarrow\quad \frac{\mathrm{d}x}{\mathrm{d}z} = \pm\frac{z}{\sqrt{z_*^2 - z^2}}. \tag{56.10}$$

The geodesic is a semicircle. Integrating Eq. (56.10), $x = \mp\sqrt{z_*^2 - z^2} + \text{const}$; fixing the constant so that the turning point $z = z_*$ sits at $x = 0$ gives $x^2 + z^2 = z_*^2$. The minimal curve is a semicircle centred on the boundary. It meets the boundary $z = 0$ at $x = \pm z_*$, which must be the interval endpoints $x = \pm\ell/2$, so

$$z_* = \frac{\ell}{2}. \tag{56.11}$$

The deepest the surface reaches into the bulk is half the interval length — a clean statement that larger regions probe deeper into the geometry, which is the seed of holographic "subregion–subregion" duality.

Regulated length. Evaluate Eq. (56.8) along the geodesic. Using $\sqrt{1+z'^2} = z_*/z$ from Eq. (56.10) and changing the integration variable from x to z via $\mathrm{d}x = (z/\sqrt{z_*^2 - z^2})\mathrm{d}z$, the two symmetric halves of the semicircle (from each endpoint down to the turning point) contribute equally, and the curve runs from the cutoff $z = \varepsilon$ near the boundary down to $z = z_*$:

$$\begin{aligned}\text{Length} &= 2L\int_\varepsilon^{z_*} \frac{\sqrt{1+z'^2}}{z}\,\frac{\mathrm{d}x}{\mathrm{d}z}\,\mathrm{d}z = 2L\int_\varepsilon^{z_*} \frac{1}{z}\,\frac{z_*}{z}\,\frac{z}{\sqrt{z_*^2 - z^2}}\,\mathrm{d}z \\ &= 2Lz_*\int_\varepsilon^{z_*} \frac{\mathrm{d}z}{z\sqrt{z_*^2 - z^2}}.\end{aligned} \tag{56.12}$$

The remaining integral is elementary. With $a \equiv z_*$,

$$\int \frac{\mathrm{d}z}{z\sqrt{a^2 - z^2}} = -\frac{1}{a}\ln\left(\frac{a + \sqrt{a^2 - z^2}}{z}\right) + \text{const}, \tag{56.13}$$

as one checks by differentiating the right-hand side. Evaluated between $z = \varepsilon$ and $z = z_*$, the upper limit gives $-\frac{1}{z_*}\ln(1) = 0$ (since $\sqrt{a^2 - a^2} = 0$ makes the

argument 1), and the lower limit gives the only surviving term:

$$\int_{\varepsilon}^{z_*} \frac{\mathrm{d}z}{z\sqrt{z_*^2 - z^2}} = \frac{1}{z_*} \ln\!\left(\frac{z_* + \sqrt{z_*^2 - \varepsilon^2}}{\varepsilon}\right). \tag{56.14}$$

Substituting into Eq. (56.12), the factor z_* cancels and

$$\text{Length} = 2L \ln\!\left(\frac{z_* + \sqrt{z_*^2 - \varepsilon^2}}{\varepsilon}\right). \tag{56.15}$$

For a physical cutoff $\varepsilon \ll z_*$ we expand $\sqrt{z_*^2 - \varepsilon^2} = z_*(1 - \varepsilon^2/2z_*^2 + \dots)$, so the argument of the logarithm is $2z_*/\varepsilon + O(\varepsilon)$, and with $z_* = \ell/2$ from Eq. (56.11),

$$\text{Length} = 2L \ln\frac{2z_*}{\varepsilon} + O(\varepsilon^2) = 2L \ln\frac{\ell}{\varepsilon} + O(\varepsilon^2). \tag{56.16}$$

The entropy. By the Ryu–Takayanagi formula Eq. (56.4), with G_N the three-dimensional Newton constant,

$$S(A) = \frac{\text{Length}}{4G_N} = \frac{2L}{4G_N} \ln\frac{\ell}{\varepsilon} = \frac{L}{2G_N} \ln\frac{\ell}{\varepsilon}. \tag{56.17}$$

This is the entire geometric computation. It remains to translate the bulk parameters L and G_N into the boundary central charge.

The translation is the Brown–Henneaux relation between the bulk data (L, G_N) and the central charge of the dual CFT_2 (the c of Theorem 48.8),

$$c = \frac{3L}{2G_N}, \tag{56.18}$$

the exact $d = 2$ instance of the general scaling $c \sim L^{d-1}/G_N$ between central charge and bulk curvature radius (Theorem 53.27 and Eq. (52.185)). For the match to be genuinely parameter-free we must know not merely that $c \propto L/G_N$ but the precise coefficient $3/2$, and that coefficient is fixed by a short computation we now carry out: the holographic conformal anomaly. The derivation imports only the gravitational version of the near-boundary (Fefferman–Graham) expansion and on-shell action machinery built in Section 54.2.1 and Theorem 54.5, specialized to three bulk dimensions, and it ends on Eq. (56.18) with the coefficient pinned.

Worked Example 56.5 (The coefficient $c = 3L/2G_N$ from the holographic Weyl anomaly)**.** Work in Euclidean signature, Wick-rotating Eq. (56.5) as in Eq. (53.161), and put the bulk metric in Fefferman–Graham gauge with a generic curved boundary metric $g_{(0)ij}(x)$,

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\Big(\mathrm{d}z^2 + g_{ij}(z,x)\,\mathrm{d}x^i \mathrm{d}x^j\Big), \qquad g_{ij}(z,x) = g_{(0)ij}(x) + z^2\, g_{(2)ij}(x) + \cdots, \tag{56.19}$$

the $d = 2$ specialization of Eq. (54.30) applied to the metric rather than to a matter field. Pure three-dimensional gravity with a negative cosmological constant solves $R_{\mu\nu} = -(2/L^2)\, G_{\mu\nu}$, equivalently $\Lambda = -1/L^2$; the on-shell scalars are $R = 6\Lambda = -6/L^2$ and $R - 2\Lambda = -4/L^2$.

The subleading coefficient. The single piece of dynamical input is the trace $g^{ij}_{(0)} g_{(2)ij}$, fixed by the radial Hamiltonian (Gauss–Codazzi) constraint. Write the constant-z slice metric $\gamma_{ij} = (L^2/z^2) g_{ij}$ with unit outward normal $n = (z/L)\partial_z$; its extrinsic curvature $K_{ij} = \frac{1}{2}(z/L)\partial_z \gamma_{ij}$ evaluates to

$$K_{ij} = -\frac{1}{L}\,\gamma_{ij} + L\, g_{(2)ij} + O(z^2), \qquad K = \gamma^{ij} K_{ij} = -\frac{2}{L} + \frac{z^2}{L}\, g^{ij}_{(0)} g_{(2)ij} + O(z^4). \tag{56.20}$$

A short expansion gives $K^2 - K_{ij}K^{ij} = \frac{2}{L^2} - \frac{2z^2}{L^2}\, g^{ij}_{(0)} g_{(2)ij} + \cdots$, while the intrinsic curvature of the slice is $R^{(\gamma)} = (z^2/L^2) R_{(0)} + \cdots$ by Weyl rescaling, with $R_{(0)}$ the Ricci scalar of $g_{(0)}$. The Hamiltonian constraint $K^2 - K_{ij}K^{ij} = R^{(\gamma)} - 2\Lambda$ holds order by order: the leading terms balance as $2/L^2 = 2/L^2$, and the $O(z^2)$ terms give

$$-\frac{2}{L^2}\, g^{ij}_{(0)} g_{(2)ij} = \frac{1}{L^2} R_{(0)} \quad \Longrightarrow \quad g^{ij}_{(0)} g_{(2)ij} = -\tfrac{1}{2} R_{(0)}. \tag{56.21}$$

The on-shell action and its logarithm. The Euclidean gravitational action is the Einstein–Hilbert term, the Gibbons–Hawking–York boundary term, and the covariant counterterm that renormalizes it,

$$I = -\frac{1}{16\pi G_N}\int \sqrt{G}\,(R - 2\Lambda) - \frac{1}{8\pi G_N}\int \sqrt{\gamma}\, K + \frac{1}{8\pi G_N L}\int \sqrt{\gamma}, \tag{56.22}$$

the last term being the $d = 2$ counterterm (a pure cosmological subtraction; no curvature counterterm exists in two boundary dimensions). The bulk volume element is $\sqrt{G} = (L^3/z^3)\sqrt{g_{(0)}}\,(1 + \frac{1}{2} z^2 g^{ij}_{(0)} g_{(2)ij} + \cdots) = (L^3/z^3)\sqrt{g_{(0)}}\,(1 - \frac{1}{4} z^2 R_{(0)} + \cdots)$ by Eq. (56.21), so the bulk term is

$$-\frac{1}{16\pi G_N}\int \sqrt{G}\,(R - 2\Lambda) = \frac{L}{4\pi G_N}\int \mathrm{d}^2x\, \sqrt{g_{(0)}} \int_\varepsilon \mathrm{d}z\, \Big[z^{-3} - \tfrac{1}{4} R_{(0)}\, z^{-1} + \cdots \Big]. \tag{56.23}$$

The z^{-3} term is a power divergence ($\propto \varepsilon^{-2}$); the z^{-1} term integrates to $-\ln \varepsilon$ and produces the only logarithm. A direct expansion of the boundary terms in Eq. (56.22) shows that the Gibbons–Hawking–York and counterterm pieces contribute only powers of ε and finite curvature terms — their $R_{(0)}$ contributions cancel between $K = -2/L - (z^2/2L) R_{(0)} + \cdots$ and the volume expansion — so the entire logarithmic divergence is

$$I \;\supset\; \frac{L}{16\pi G_N}\, \ln \varepsilon \int \mathrm{d}^2x\, \sqrt{g_{(0)}}\, R_{(0)} \;=\; -\frac{L}{16\pi G_N}\, \ln \frac{1}{\varepsilon} \int \mathrm{d}^2x\, \sqrt{g_{(0)}}\, R_{(0)}. \tag{56.24}$$

Reading off the central charge. A logarithmic divergence cannot be removed by any covariant power counterterm; subtracting it would require a $\ln\varepsilon$ times a local integral, and such a term is not Weyl-invariant. This unremovable, finite, local Weyl non-invariance is the holographic conformal anomaly (Theorem 54.6): the coefficient of the logarithm is precisely the integrated boundary trace anomaly $\int \sqrt{g_{(0)}}\,\langle T^{i}{}_{i}\rangle$. In two dimensions the trace anomaly is fixed by the central charge through $\langle T^{i}{}_{i}\rangle = -\frac{c}{24\pi}R_{(0)}$ (the curved-space Weyl anomaly of Theorem 48.8, with the normalization of Chapter 37). Matching the magnitude of the coefficient in Eq. (56.24) to this normalization,

$$\frac{L}{16\pi G_N} = \frac{c}{24\pi} \quad\Longrightarrow\quad c = \frac{24\pi}{16\pi}\,\frac{L}{G_N} = \frac{3L}{2G_N}, \tag{56.25}$$

which is Eq. (56.18). The bulk computation knows nothing of the boundary CFT beyond its central charge, exactly as it should: the anomaly is the one piece of curved-space data a 2D CFT carries, and the gravity dual reproduces it with the coefficient $3/2$ that makes the entropy match below parameter-free. (The original Brown–Henneaux route reaches the same number by computing the central extension of the algebra of asymptotic symmetry charges of AdS_3, which is two commuting Virasoro algebras of this c; the anomaly computation here is the same statement read off the on-shell action.)

The relation Eq. (56.18) is exact and parameter-free; substituting $L/2G_N = c/3$ into Eq. (56.17) gives

$$S(A) = \frac{c}{3}\,\ln\frac{\ell}{\varepsilon}, \tag{56.26}$$

which is exactly the 2D CFT vacuum single-interval entropy of Theorem 55.32 and Eq. (55.119), coefficient and cutoff and all. The match is complete: a classical geodesic length in pure AdS_3, regulated at a radial wall and divided by $4G_N$, reproduces a result that on the boundary required the full replica machinery of two-dimensional conformal field theory. No coefficient was adjusted. The leading logarithmic divergence as $\varepsilon \to 0$ is the holographic image of the universal ultraviolet divergence of entanglement entropy — here a true logarithm because the entangling surface in $d = 2$ is a pair of points, and the would-be area law degenerates to a log.

Figure 56.1 collects the geometry. It is worth pausing on how much the picture already contains. The minimal surface dips to a depth $z_* = \ell/2$ set by the size of the region, so larger regions reach deeper into the bulk; the radial direction is being read off as a scale, with small regions sensitive only to the near-boundary (ultraviolet) geometry and large regions to the deep interior (infrared) — the geometric face of the entanglement entropy's growth with region size. The homology region $\mathcal{R}_A$ is the wedge of bulk hanging beneath A; we will argue in Section 56.6 that this is precisely the bulk that the boundary data on A suffices to reconstruct. And the logarithmic divergence sits entirely in the near-boundary

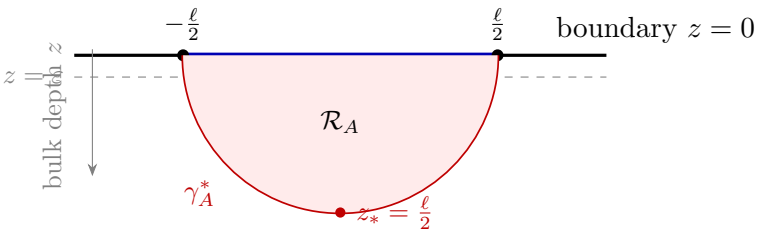


Figure 56.1: The Ryu–Takayanagi surface for a single interval A in the vacuum of a 2D CFT, drawn on the constant-time slice of Poincaré AdS_3 (the bulk depth z increases downward). The minimal surface γ_A^* is the spacelike geodesic anchored on the endpoints $\partial A = \{\pm\ell/2\}$ — a semicircle of radius $z_* = \ell/2$. The homology region $\mathcal{R}_A$ (shaded) is bounded by A and γ_A^*, as required by Theorem 56.2. The area integral is regulated at the radial cutoff $z = \varepsilon$ (dashed), identified with the QFT ultraviolet cutoff. Its regulated length, divided by $4G_N$ and translated through Brown–Henneaux $c = 3L/2G_N$, reproduces $S = (c/3)\ln(\ell/\varepsilon)$ exactly.

part of the geodesic, where it plunges vertically into the cutoff surface; the universal short-distance structure of entanglement entropy is localized, in the bulk, at the anchoring of the surface on ∂A.

Historical Note

Ryu and Takayanagi proposed Eq. (56.4) in 2006, generalizing the Bekenstein–Hawking area law from horizons to arbitrary entangling regions and checking it against the 2D CFT result and several higher-dimensional cases. The proposal was static; Hubeny, Rangamani, and Takayanagi gave the covariant extremal-surface form in 2007. For several years the formula was a strikingly successful conjecture without a derivation. The derivation from the gravitational path integral — the replica argument sketched in Section 56.1.1 — was supplied by Lewkowycz and Maldacena in 2013, and the quantum corrections by Faulkner, Lewkowycz, and Maldacena the same year, opening the road to the quantum extremal surfaces and the black-hole information results of Chapter 57.

We now hold a concrete, validated geometric prescription: a region's entanglement entropy is the regulated area of the minimal homologous bulk surface, divided by $4G_N$. The single-interval match certifies that the prescription, the cutoff identification, and the parameter dictionary all fit together with no slack. The next section turns this validated rule into a working engine, computing the entropy of canonical regions — the thermal state, the sphere, the strip — and exhibiting the genuinely holographic phenomena, including the entanglement phase transition for two intervals, that the minimization in Eq. (56.4) makes possible.

56.2 Computing holographic entanglement entropy

We now hold a validated rule: a boundary region's entanglement entropy is the regulated area of the minimal homologous bulk surface divided by $4G_N$ (Theorem 56.3), checked against the one entropy we already trusted, the single interval in the vacuum of a 2D CFT (Theorem 56.4 and Eq. (56.26)). A formula earns its keep by computing. This section turns the Ryu–Takayanagi prescription into a working engine and runs it on the canonical regions whose answers a holographic theory must reproduce: an interval in a thermal state, a sphere in any dimension, a strip in any dimension, and two disjoint intervals. Each computation is a complete minimal-surface problem solved on the page, and each one exhibits a genuinely holographic phenomenon. The thermal interval interpolates, in a single closed expression, between the vacuum logarithm and the extensive thermal entropy, with the crossover governed entirely by the geometry of a black-hole horizon. The sphere reveals, through the Casini–Huerta–Myers map, that vacuum entanglement across a sphere *is* a thermal entropy on hyperbolic space and a horizon area in the bulk. The strip displays the area law together with a finite, negative, shape-dependent piece that scales nonlocally with the region's width. And the two intervals undergo a sharp first-order entanglement phase transition — a discontinuity in the slope of the entropy — that has no analogue in the perturbative computation of entanglement and is the cleanest signature that "min" in Eq. (56.4) is load-bearing.

Throughout we work, as in Section 56.1.2, in asymptotically AdS_{d+1} with a CFT_d on the boundary, in the classical-gravity regime where Eq. (56.4) holds at leading order in G_N. The mostly-minus signature and natural units of Section 53.3 are in force; L is the AdS radius and G_N the bulk Newton constant. When we need the boundary central charge we use the holographic dictionary $C_T \sim L^{d-1}/G_N$ (Theorem 53.27), and in $d = 2$ the exact Brown–Henneaux value $c = 3L/2G_N$ (Eq. (56.18)).

56.2.1 A single interval at finite temperature: the BTZ geodesic

The first computation continues the thermal $\mathrm{AdS}_3/\mathrm{CFT}_2$ example carried since Chapter 54. A 2D CFT at inverse temperature β on an infinite line is dual not to pure AdS_3 but to the planar (non-rotating) BTZ black hole, the geometry whose Hawking temperature is the CFT temperature and whose horizon entropy is the CFT thermal entropy (Eq. (54.89) and Section 54.4). We compute the entanglement entropy of an interval of width ℓ in this thermal state by finding the spacelike geodesic anchored on its endpoints, exactly as in the vacuum but in the black-hole metric. The result is a single closed expression that knows about both the vacuum and the thermal ensemble, and the way it interpolates between them is a lesson in how holography geometrizes thermality.

In Poincaré-like coordinates with the boundary at $z \to 0$, the planar BTZ

metric is, in the book's mostly-minus signature,

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left(-f(z)\,\mathrm{d}t^2 + \mathrm{d}x^2 + \frac{\mathrm{d}z^2}{f(z)}\right), \qquad f(z) = 1 - \frac{z^2}{z_h^2}, \tag{56.27}$$

the $d = 2$ instance of the asymptotically-AdS black brane of Eq. (54.89). The locus $z = z_h$ is the horizon, where f vanishes; the asymptotic region $z \to 0$ recovers Eq. (56.5). The relation between the horizon depth z_h and the temperature is fixed by smoothness of the Euclidean continuation, exactly as for any black hole. Continuing $t \to -i\tau_E$ and expanding f near the horizon, $f(z) \simeq f'(z_h)(z - z_h)$ with $f'(z_h) = -2/z_h$, the (τ_E, z) part of Eq. (56.27) becomes a plane in polar coordinates whose angular variable is τ_E; absence of a conical singularity forces τ_E to be periodic with period $\beta = 4\pi/\,|f'(z_h)|$, i.e.

$$\beta = \frac{4\pi}{|f'(z_h)|} = 2\pi z_h, \qquad \Longleftrightarrow \qquad z_h = \frac{\beta}{2\pi}. \tag{56.28}$$

This is the horizon-radius/temperature identification we will substitute at the end; it is the same computation that fixes the Hawking temperature in Section 54.4, specialized to $d = 2$.

The state is static, so we work on the constant-time slice $t = 0$. Setting $\mathrm{d}t = 0$ in Eq. (56.27) and stripping the overall sign that marks the spatial directions, the induced Riemannian metric on the slice is

$$\mathrm{d}\ell_\Sigma^2 = \frac{L^2}{z^2}\left(\mathrm{d}x^2 + \frac{\mathrm{d}z^2}{f(z)}\right), \qquad f(z) = 1 - \frac{z^2}{z_h^2}. \tag{56.29}$$

The interval is $A = \{t = 0,\ -\ell/2 \le x \le \ell/2\}$, with $\partial A = \{x = \pm\ell/2\}$. As in $d = 2$ vacuum, the RT surface is a spacelike geodesic in Eq. (56.29). The homology constraint is inert here because the interval is small enough that the minimal geodesic does not reach the horizon; the complementary (horizon-wrapping) configuration that dominates for large regions is recovered in the extensive thermal limit below (Eq. (56.45)) and we do not need it for the connected geodesic.

Worked Example 56.6 (Thermal-interval geodesic in planar BTZ)**.** We minimize proper length in Eq. (56.29), anchored at $x = \pm\ell/2$, regulate at $z = \varepsilon$, and assemble $S = \mathrm{Length}/4G_N$, finally substituting Eqs. (56.18) and (56.28).

Length functional and first integral. Parametrize the geodesic as $z = z(x)$. From Eq. (56.29), the proper length is

$$\mathrm{Length}[z] = L\int_{-\ell/2}^{\ell/2}\mathrm{d}x\,\frac{1}{z}\sqrt{1 + \frac{z'^2}{f(z)}}, \qquad z' \equiv \frac{\mathrm{d}z}{\mathrm{d}x}. \tag{56.30}$$

The integrand $g(z,z') = z^{-1}\sqrt{1+z'^2/f}$ has no explicit x-dependence, so the Beltrami first integral $g - z'\,\partial g/\partial z'$ is conserved. With $\partial g/\partial z' = z^{-1}(z'/f)/\sqrt{1+z'^2/f}$,

$$g - z'\frac{\partial g}{\partial z'} = \frac{1}{z}\frac{(1+z'^2/f) - z'^2/f}{\sqrt{1+z'^2/f}} = \frac{1}{z\sqrt{1+z'^2/f}} = \frac{1}{z_*}, \tag{56.31}$$

the constant named $1/z_*$ because at the turning point $z' = 0$ gives $z = z_*$, the maximal depth. Rearranging,

$$\frac{z'^2}{f} = \frac{z_*^2 - z^2}{z^2} \quad\Longrightarrow\quad \frac{\mathrm{d}x}{\mathrm{d}z} = \pm\frac{z}{\sqrt{f(z)\,(z_*^2 - z^2)}} = \pm\frac{z}{\sqrt{(1 - z^2/z_h^2)(z_*^2 - z^2)}}. \tag{56.32}$$

The vacuum limit $z_h \to \infty$ ($\beta \to \infty$) sets $f \to 1$ and returns the semicircle equation Eq. (56.10), a useful check.

Width as a function of turning point. The interval width is twice the coordinate run from the turning point out to the boundary:

$$\frac{\ell}{2} = \int_0^{z_*} \frac{z\,\mathrm{d}z}{\sqrt{(1 - z^2/z_h^2)(z_*^2 - z^2)}}. \tag{56.33}$$

Substitute $w = z^2$, $\mathrm{d}w = 2z\,\mathrm{d}z$, and write $1 - z^2/z_h^2 = (z_h^2 - w)/z_h^2$, so $\sqrt{(1 - z^2/z_h^2)(z_*^2 - z^2)} = z_h^{-1}\sqrt{(z_h^2 - w)(z_*^2 - w)}$:

$$\frac{\ell}{2} = \frac{z_h}{2}\int_0^{z_*^2} \frac{\mathrm{d}w}{\sqrt{(z_h^2 - w)(z_*^2 - w)}}. \tag{56.34}$$

With $a \equiv z_*^2$ and $b \equiv z_h^2$ (and $a < b$, since the geodesic stays outside the horizon), the antiderivative is elementary: differentiating $-2\ln\!\big(\sqrt{a-w} + \sqrt{b-w}\big)$ returns $1/\sqrt{(a-w)(b-w)}$, so

$$\int_0^a \frac{\mathrm{d}w}{\sqrt{(a-w)(b-w)}} = \Big[-2\ln\!\big(\sqrt{a-w} + \sqrt{b-w}\big)\Big]_0^a = 2\ln\frac{\sqrt{a} + \sqrt{b}}{\sqrt{b-a}}. \tag{56.35}$$

Hence $\dfrac{\ell}{2} = z_h \ln\dfrac{z_* + z_h}{\sqrt{z_h^2 - z_*^2}} = \dfrac{z_h}{2}\ln\dfrac{z_h + z_*}{z_h - z_*}$, i.e.

$$\frac{\ell}{z_h} = \ln\frac{z_h + z_*}{z_h - z_*} \quad\Longrightarrow\quad z_* = z_h \tanh\frac{\ell}{2z_h} = z_h\tanh\frac{\pi\ell}{\beta}, \tag{56.36}$$

using $z_h = \beta/2\pi$ from Eq. (56.28) in the last step. The turning point now saturates at $z_* \to z_h$ as $\ell \to \infty$: a very large interval's geodesic hugs the horizon, the geometric origin of the extensive thermal regime found below.

Regulated length. Along the geodesic, $\sqrt{1+z'^2/f} = z_*/z$ from Eq. (56.31), so the length element is $\mathrm{d}\ell_\Sigma = (L/z)(z_*/z)\,\mathrm{d}x = Lz_* z^{-2}\,\mathrm{d}x$, and changing variable to z via Eq. (56.32),

$$\text{Length} = 2Lz_*\int_\varepsilon^{z_*}\frac{\mathrm{d}z}{z\sqrt{(1-z^2/z_h^2)(z_*^2-z^2)}} = Lz_h\int_{\varepsilon^2}^{z_*^2}\frac{z_*\,\mathrm{d}w}{w\sqrt{(z_h^2-w)(z_*^2-w)}}, \tag{56.37}$$

again with $w=z^2$. The remaining integral has the standard antiderivative

$$\int\frac{\mathrm{d}w}{w\sqrt{(a-w)(b-w)}} = -\frac{1}{\sqrt{ab}}\ln\!\left(\frac{2ab-(a+b)w+2\sqrt{ab}\,\sqrt{(a-w)(b-w)}}{w}\right), \tag{56.38}$$

(verified by differentiation), with $a=z_*^2$, $b=z_h^2$, $\sqrt{ab}=z_*z_h$. The prefactor in Eq. (56.37) is Lz_hz_*, and $Lz_hz_*/\sqrt{ab}=L$, so

$$\text{Length} = L\left[-\ln\!\left(\frac{2ab-(a+b)w+2z_*z_h\sqrt{(a-w)(b-w)}}{w}\right)\right]_{w=\varepsilon^2}^{w=z_*^2}. \tag{56.39}$$

At the upper limit $w=a=z_*^2$ the square root vanishes and the bracket's argument is $(2ab-(a+b)a)/a = b-a = z_h^2-z_*^2$. At the lower limit $w=\varepsilon^2\to 0$, the surviving terms are $2ab+2z_*z_h\sqrt{ab}=4ab$ over ε^2, i.e. $4z_*^2z_h^2/\varepsilon^2$. Therefore

$$\text{Length} = L\,\ln\frac{4z_*^2z_h^2}{\varepsilon^2\,(z_h^2-z_*^2)}. \tag{56.40}$$

Eliminating the turning point. From Eq. (56.36), $z_* = z_h\tanh(\pi\ell/\beta)$, so $z_h^2-z_*^2 = z_h^2\,\mathrm{sech}^2(\pi\ell/\beta)$ and $z_*^2 = z_h^2\tanh^2(\pi\ell/\beta)$. Hence

$$\frac{z_*^2z_h^2}{z_h^2-z_*^2} = \frac{z_h^2\tanh^2(\pi\ell/\beta)\,z_h^2}{z_h^2\,\mathrm{sech}^2(\pi\ell/\beta)} = z_h^2\,\tanh^2\frac{\pi\ell}{\beta}\,\cosh^2\frac{\pi\ell}{\beta} = z_h^2\,\sinh^2\frac{\pi\ell}{\beta}, \tag{56.41}$$

and Eq. (56.40) collapses to

$$\text{Length} = L\,\ln\!\left(\frac{4z_h^2\sinh^2(\pi\ell/\beta)}{\varepsilon^2}\right) = 2L\,\ln\!\left(\frac{2z_h}{\varepsilon}\sinh\frac{\pi\ell}{\beta}\right) = 2L\,\ln\!\left(\frac{\beta}{\pi\varepsilon}\sinh\frac{\pi\ell}{\beta}\right), \tag{56.42}$$

using $2z_h=\beta/\pi$.

The entropy. Dividing by $4G_N$ and substituting Brown–Henneaux $L/2G_N = c/3$ (Eq. (56.18)),

$$\boxed{S(\ell) = \frac{\text{Length}}{4G_N} = \frac{L}{2G_N}\ln\!\left(\frac{\beta}{\pi\varepsilon}\sinh\frac{\pi\ell}{\beta}\right) = \frac{c}{3}\ln\!\left[\frac{\beta}{\pi\varepsilon}\sinh\frac{\pi\ell}{\beta}\right].} \tag{56.43}$$

The closed form Eq. (56.43) is the thermal generalization of the vacuum logarithm, and it reproduces the result that the boundary CFT obtains by conformally mapping the thermal cylinder, with no free parameters. Two limits expose its content. For an interval small compared with the thermal length, $\ell \ll \beta$, we have $\sinh(\pi\ell/\beta) \to \pi\ell/\beta$ and

$$S(\ell) \xrightarrow[\ell \ll \beta]{} \frac{c}{3} \ln \frac{\ell}{\varepsilon}, \tag{56.44}$$

the vacuum single-interval entropy Eq. (56.26): a small region cannot resolve the temperature, and the geodesic, which only dips to $z_* \approx \ell/2 \ll z_h$, never feels the horizon. For an interval large compared with the thermal length, $\ell \gg \beta$, we have $\sinh(\pi\ell/\beta) \to \frac{1}{2} e^{\pi\ell/\beta}$ and

$$S(\ell) \xrightarrow[\ell \gg \beta]{} \frac{c}{3} \left(\frac{\pi\ell}{\beta} + \ln \frac{\beta}{2\pi\varepsilon} \right) = \frac{\pi c}{3\beta}\, \ell + (\text{const}). \tag{56.45}$$

The entropy is now *extensive*: it grows linearly in ℓ, with the constant of proportionality $\pi c/3\beta$ exactly the thermal entropy density of a 2D CFT — the Cardy entropy per unit length at temperature $T = 1/\beta$, equivalently the horizon-entropy density of the BTZ black hole (Section 54.4). The geometric mechanism is transparent from Eq. (56.36): a large interval's geodesic runs nearly parallel to the horizon at depth $z_* \approx z_h$, and its length is dominated by that long horizon-skimming stretch, which contributes the horizon entropy density times the coordinate length. The full curve in Fig. 56.2 shows the crossover sitting at $\ell \sim \beta$, where the entropy turns from the vacuum logarithm into the thermal straight line.

Physical Insight

The same geodesic computes two physically distinct quantities depending on the region size, and the bulk shows why. A small region's entropy is pure vacuum entanglement, blind to the temperature; a large region's entropy is dominated by the thermal occupation of its interior, and the bulk geodesic registers this by descending to the horizon and dragging along it. The horizon, which in Chapter 54 carried the global thermal entropy, here doubles as the asymptotic depth that any large entangling surface must approach. Black-hole thermodynamics and subsystem entanglement are not two facts but one geometry viewed at two scales.

56.2.2 A spherical region in any dimension: the Casini–Huerta–Myers map

The interval generalizes in two natural directions in higher dimensions: the sphere (a ball, with a spherical entangling surface) and the strip. The sphere is

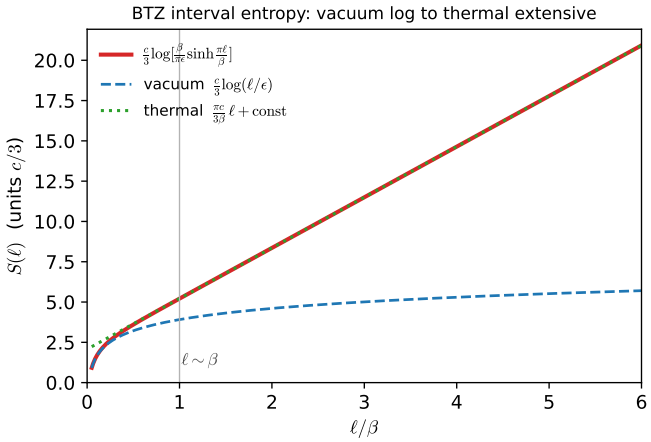


Figure 56.2: The single-interval entanglement entropy Eq. (56.43) in the thermal state of a 2D CFT (dual to planar BTZ), as a function of interval width ℓ in units of the thermal length β. For $\ell \ll \beta$ it tracks the vacuum logarithm Eq. (56.44) (dashed); for $\ell \gg \beta$ it tracks the extensive thermal entropy Eq. (56.45) (dotted), whose slope $\pi c/3\beta$ is the CFT thermal entropy density. The crossover at $\ell \sim \beta$ is the geometric statement that a large interval's geodesic begins to hug the BTZ horizon.

special because its symmetry makes vacuum entanglement *exactly* computable by a conformal trick that predates RT and, applied holographically, identifies the entanglement entropy with a black-hole horizon area without any minimization at all. This is the Casini–Huerta–Myers construction. It is worth presenting in full because it is the cleanest demonstration of the chapter's thesis: it turns vacuum entanglement across a sphere into a *thermal* entropy, and then into a horizon area, by an exact map.

The causal diamond of a ball. Take the vacuum of a CFT_d on Minkowski space, $\mathrm{d}s^2_{\text{flat}} = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2_{d-2}$ in spherical coordinates about the origin, and let A be the ball $r \le R$ at $t = 0$. The reduced density matrix ρ_A is determined by data on the ball, but by causality it is equally determined by any Cauchy slice of the ball's *causal diamond* $\mathcal{D}_A$ — the set of points all of whose causal curves pass through A — which for a ball is the region $|t| + r \le R$. The modular Hamiltonian $K = -\ln \rho_A$ for a sphere in the vacuum is, exceptionally, a *local* operator: the generator of the conformal boost that preserves $\mathcal{D}_A$. This is the spherical generalization of the Rindler result of Section 55.3, where the half-space modular Hamiltonian was the boost; here the flat entangling plane is replaced by a sphere and the boost by a special conformal transformation that fixes it.

The map to a hyperbolic cylinder. The Casini–Huerta–Myers observation is that $\mathcal{D}_A$ is conformally equivalent to a static spacetime $\mathbb{R}_\tau \times H^{d-1}$, the real line (time τ) times hyperbolic space of curvature radius R. Setting $R = 1$ momentarily for clarity, define new coordinates (τ, u) with $u \in [0, \infty)$ through

$$t = \frac{\sinh \tau}{\cosh u + \cosh \tau}, \qquad r = \frac{\sinh u}{\cosh u + \cosh \tau}. \tag{56.46}$$

A direct computation of the differentials gives the flat metric as a Weyl rescaling of the hyperbolic cylinder:

$$\mathrm{d}s^2_{\text{flat}} = -\mathrm{d}t^2 + \mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2_{d-2} = \Omega^2\Big(-\mathrm{d}\tau^2 + \mathrm{d}u^2 + \sinh^2 u\,\mathrm{d}\Omega^2_{d-2}\Big), \qquad \Omega = \frac{1}{\cosh u + \cosh \tau}. \tag{56.47}$$

The factor in parentheses is precisely the line element of $\mathbb{R}_\tau \times H^{d-1}$, with $\mathrm{d}u^2 + \sinh^2 u\,\mathrm{d}\Omega^2_{d-2}$ the unit-radius hyperbolic metric. Restoring the sphere radius rescales the hyperbolic curvature radius to R and the time to $R\tau$. The map sweeps out the diamond: at $\tau = 0$ one has $t = 0$ and $r = \tanh(u/2)$, so as u runs from 0 to ∞ the radial coordinate runs from the center $r = 0$ to the entangling sphere $r = 1$, and the whole spatial ball is covered; the future and past tips of the diamond, $r = 0,\ t = \pm 1$, are pushed to $\tau \to \pm\infty$. The entangling surface $r = R$ itself sits at $u \to \infty$, the boundary at infinity of H^{d-1}.

Because the CFT is conformally invariant, its vacuum on $\mathcal{D}_A$ maps to a state on $\mathbb{R}_\tau \times H^{d-1}$, and the modular Hamiltonian — the conformal boost preserving $\mathcal{D}_A$ — maps to the *ordinary* Hamiltonian generating τ-translations on the hyperbolic cylinder. The statement that $\rho_A = e^{-2\pi K}$ becomes the statement that the mapped state is the *thermal* (Gibbs) state of the CFT on H^{d-1} at the temperature read off from the period of Euclidean τ. Euclidean continuation $\tau \to -i\tau_E$ makes the conformal-boost orbits into circles; the modular flow ρ_A^{is} has period 2π in modular time, which in the τ_E of Eq. (56.46) is the period $2\pi R$ (restoring R). The mapped state is therefore thermal at

$$T_0 = \frac{1}{2\pi R}. \tag{56.48}$$

We have proved a sharp statement, the higher-dimensional descendant of vacuum thermality:

$$S_{\text{EE}}(\text{ball of radius } R) = S_{\text{thermal}}(\text{CFT}_d \text{ on } H^{d-1} \text{ at } T_0 = \tfrac{1}{2\pi R}). \tag{56.49}$$

The intractable replica continuation for a sphere has been traded, exactly and without holography, for a thermal partition function on hyperbolic space. This is the Casini–Huerta–Myers reduction; it holds in any CFT, holographic or not.

The bulk: a topological hyperbolic black hole. Now feed Eq. (56.49) to holography. A CFT_d thermal state on $\mathbb{R}_\tau \times H^{d-1}$ is dual to a bulk black hole

whose boundary geometry is that cylinder: the *topological* (hyperbolic-horizon) black hole in AdS_{d+1},

$$\mathrm{d}s^2 = -f(\rho)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}\rho^2}{f(\rho)} + \rho^2(\mathrm{d}u^2 + \sinh^2 u\,\mathrm{d}\Omega_{d-2}^2), \qquad f(\rho) = \frac{\rho^2}{L^2} - 1 - \frac{m}{\rho^{\,d-2}}, \tag{56.50}$$

where the -1 (rather than the planar black brane's 0 or the spherical black hole's $+1$) reflects the hyperbolic curvature of the horizon cross-section. The Hawking temperature of Eq. (56.50) depends on the mass parameter m, but there is a distinguished value $m = 0$ at which the black hole is "massless" (its horizon is a pure curvature effect, not sourced by energy). For $m = 0$ the horizon $f(\rho_h) = 0$ sits at $\rho_h = L$, and the temperature is

$$T = \frac{f'(\rho_h)}{4\pi} = \frac{1}{4\pi}\cdot\frac{2\rho_h}{L^2}\bigg|_{\rho_h = L} = \frac{1}{2\pi L}, \tag{56.51}$$

which, once the boundary hyperbolic radius is set to R, is exactly $T_0 = 1/2\pi R$ of Eq. (56.48). The massless hyperbolic black hole is therefore the bulk dual of the vacuum-reduced state on the ball, and its horizon, by the holographic dictionary, is nothing other than the RT surface anchored on the entangling sphere $r = R$ (one checks that the bifurcation surface $\rho = \rho_h$ of Eq. (56.50) reaches the boundary precisely at $u \to \infty$, i.e. at ∂A, and is homologous to the ball). The entanglement entropy is its horizon area:

$$S_{\mathrm{EE}}(\text{ball}) = \frac{\mathrm{Area}(\rho = \rho_h)}{4G_N} = \frac{\rho_h^{\,d-1}}{4G_N}\,\mathrm{Vol}(H^{d-1}) = \frac{L^{d-1}}{4G_N}\,\mathrm{Vol}(H^{d-1}), \tag{56.52}$$

the horizon being a copy of H^{d-1} of radius $\rho_h = L$.

Worked Example 56.7 (The universal coefficient from the regularized hyperbolic volume)**.** The entropy Eq. (56.52) reduces the whole problem to one geometric quantity: the volume of hyperbolic space, regulated by the same cutoff that regulates the entangling surface. Write the unit-radius H^{d-1} volume with the radial coordinate u cut off where the entangling surface is regulated,

$$\mathrm{Vol}(H^{d-1}) = \Omega_{d-2}\int_0^{u_{\max}} \sinh^{d-2} u\,\mathrm{d}u, \qquad \Omega_{d-2} = \frac{2\pi^{(d-1)/2}}{\Gamma\!\left(\frac{d-1}{2}\right)}, \tag{56.53}$$

where Ω_{d-2} is the area of the unit $(d-2)$-sphere and $u_{\max}$ is fixed by the relation between the hyperbolic cutoff and the boundary short-distance cutoff δ: the conformal factor Eq. (56.47) ties proper distance near $r = R$ to u, giving $e^{u_{\max}} \simeq 2R/\delta$, i.e. $u_{\max} \simeq \ln(2R/\delta)$. The integrand grows as $\sinh^{d-2} u \sim 2^{-(d-2)} e^{(d-2)u}$ at large u, producing a tower of divergences in powers of $e^{u_{\max}} \sim R/\delta$ — the holographic image of the area-law tower of Eq. (55.24) — on top of which sits the universal piece. We read it off dimension by dimension.

$d = 2$**.** Here H^1 is a line, $\sinh^0 u = 1$, and $\mathrm{Vol}(H^1) = \int_0^{u_{\max}} \mathrm{d}u = u_{\max} = \ln(2R/\delta)$. Then Eq. (56.52) gives $S = (L/4G_N)\ln(2R/\delta)$; with $c = 3L/2G_N$ and $2R = \ell$ the diameter of the ball (the interval length), this is $S = \frac{c}{6}\ln(\ell/\delta)$ *per endpoint* — two endpoints reproduce $S = \frac{c}{3}\ln(\ell/\delta)$, recovering Eq. (56.26). The CHM route reaches the 2D answer with no geodesic at all.

$d = 3$**.** Now H^2, $\sinh^1 u$, and

$$\mathrm{Vol}(H^2) = 2\pi \int_0^{u_{\max}} \sinh u \, \mathrm{d}u = 2\pi(\cosh u_{\max} - 1) = 2\pi\Big(\tfrac{1}{2}e^{u_{\max}} + \tfrac{1}{2}e^{-u_{\max}} - 1\Big). \tag{56.54}$$

The leading term $\pi e^{u_{\max}} \sim 2\pi R/\delta$ is the area-law divergence (linear in R/δ, the perimeter of the disk in cutoff units). The $u_{\max}$-independent piece is the finite *universal* constant -2π:

$$S^{(d=3)}_{\mathrm{univ}} = \frac{L^2}{4G_N}(-2\pi) = -\frac{\pi L^2}{2G_N} \equiv -F, \tag{56.55}$$

a cutoff-independent number. It is (minus) the quantity F that the F-theorem identifies as a good measure of degrees of freedom for a three-dimensional CFT — the odd-dimensional analogue of a central charge, here delivered as a hyperbolic-space free energy.

$d = 4$**.** Now H^3, $\sinh^2 u$, and

$$\mathrm{Vol}(H^3) = 4\pi \int_0^{u_{\max}} \sinh^2 u \, \mathrm{d}u = 4\pi\cdot\tfrac{1}{2}(\sinh u_{\max}\cosh u_{\max} - u_{\max}) = 2\pi(\tfrac{1}{4}e^{2u_{\max}} + \cdots - u_{\max}). \tag{56.56}$$

The $e^{2u_{\max}} \sim (R/\delta)^2$ term is the area-law divergence (the area of the entangling sphere in cutoff units). The universal piece is now *logarithmic*: the $-u_{\max} = -\ln(2R/\delta)$ term gives

$$S^{(d=4)}_{\mathrm{univ}} = \frac{L^3}{4G_N}(-2\pi)\,\ln\frac{2R}{\delta} = -\frac{\pi L^3}{2G_N}\,\ln\frac{R}{\delta} + (\mathrm{const}), \tag{56.57}$$

a universal logarithm whose coefficient is fixed by the bulk data L^3/G_N. This coefficient is, up to a convention-dependent normalization collected with the trace-anomaly data, 4 times the A-type (Euler-density) anomaly coefficient a of the CFT$_4$: $S^{(d=4)}_{\mathrm{univ}} = -4a\,\ln(R/\delta)$, with the holographic value of a proportional to L^3/G_N (Theorem 53.27). The universal entanglement of a sphere is the conformal anomaly.

General d**.** The same integral organizes every dimension at once. The leading divergence of Eq. (56.53) scales as $e^{(d-2)u_{\max}} \sim (R/\delta)^{d-2}$, the area law $\mathrm{Area}(\partial A)/\delta^{d-2}$. The universal piece is the $u_{\max}$-independent constant for d odd

(a pure number, $\propto F$) and the coefficient of the single power $u_{\max} \sim \ln(R/\delta)$ for d even (a logarithm, $\propto a$). In every case

$$S(\text{ball}) = \frac{L^{d-1}\Omega_{d-2}}{4G_N} \int_0^{u_{\max}} \sinh^{d-2} u \, \mathrm{d}u, \tag{56.58}$$

and the universal coefficient is a pure number times $L^{d-1}/G_N \sim C_T$, the holographic count of degrees of freedom. The sphere thus measures, through its entanglement entropy, the same central data that govern the two-point function of the stress tensor.

Physical Insight

The CHM map is the higher-dimensional face of the Rindler thermality of Section 55.3: vacuum entanglement across a sphere is genuinely a thermal phenomenon, with a temperature $1/2\pi R$ set by the sphere's size, on a space (H^{d-1}) whose negative curvature compactifies the entangling surface to infinity. Holography then identifies that thermal entropy with a horizon area, so the entire content of spherical vacuum entanglement — divergence structure, universal coefficient, and all — is the geometry of a single "topological" black hole. No replica continuation, no minimization: the symmetry of the sphere does all the work.

56.2.3 A strip in any dimension: the area law and a nonlocal finite piece

The strip lacks the sphere's enhanced symmetry, so we must genuinely solve the minimal-surface problem; but its translational symmetry makes the problem one-dimensional and exactly soluble, and the answer displays the area law together with a finite piece that scales *nonlocally* with the strip's width — a structure with no counterpart in a local action. We work in pure Poincaré AdS_{d+1}, dual to the CFT_d vacuum, with metric

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\Big(-\mathrm{d}t^2 + \mathrm{d}x_1^2 + \cdots + \mathrm{d}x_{d-1}^2 + \mathrm{d}z^2\Big). \tag{56.59}$$

The region A is the strip $-\frac{\ell}{2} \le x_1 \le \frac{\ell}{2}$, unbounded in the transverse directions $x_2, \ldots, x_{d-1}$, which we regulate to a large length W each, so the entangling surface ∂A consists of two parallel planes at $x_1 = \pm\ell/2$, each of transverse "area" W^{d-2}. By translational invariance the RT surface is described by a single profile $z = z(x_1)$, with the transverse directions spectators.

Worked Example 56.8 (Strip minimal surface in Poincaré AdS_{d+1})**.** We minimize the area functional, solve the turning-point integral, and separate the divergent area-law piece from the finite shape-dependent piece.

Area functional and first integral. On the $t = 0$ slice the induced area of a surface $z = z(x_1)$, with the transverse directions contributing a factor $(L/z)^{d-2}W^{d-2}$ and the profile contributing $(L/z)\sqrt{1+z'^2}\,\mathrm{d}x_1$, is

$$\mathrm{Area}[z] = L^{d-1}W^{d-2}\int_{-\ell/2}^{\ell/2}\mathrm{d}x_1\,\frac{\sqrt{1+z'^2}}{z^{d-1}}, \qquad z' \equiv \frac{\mathrm{d}z}{\mathrm{d}x_1}. \tag{56.60}$$

The integrand $g(z,z') = z^{-(d-1)}\sqrt{1+z'^2}$ is x_1-independent, so the Beltrami integral is conserved:

$$g - z'\frac{\partial g}{\partial z'} = \frac{1}{z^{d-1}\sqrt{1+z'^2}} = \frac{1}{z_*^{d-1}}, \tag{56.61}$$

with z_* the turning point ($z' = 0$). Rearranging,

$$\sqrt{1+z'^2} = \Big(\frac{z_*}{z}\Big)^{d-1} \quad\Longrightarrow\quad z'^2 = \Big(\frac{z_*}{z}\Big)^{2(d-1)} - 1 \quad\Longrightarrow\quad \frac{\mathrm{d}x_1}{\mathrm{d}z} = \pm\frac{(z/z_*)^{d-1}}{\sqrt{1-(z/z_*)^{2(d-1)}}}. \tag{56.62}$$

Width as a function of turning point. Integrating $\mathrm{d}x_1/\mathrm{d}z$ from the turning point to the boundary and doubling,

$$\frac{\ell}{2} = \int_0^{z_*}\frac{(z/z_*)^{d-1}}{\sqrt{1-(z/z_*)^{2(d-1)}}}\,\mathrm{d}z = z_*\int_0^1\frac{y^{d-1}}{\sqrt{1-y^{2(d-1)}}}\,\mathrm{d}y, \tag{56.63}$$

substituting $y = z/z_*$. The integral is a Beta function: setting $s = y^{2(d-1)}$,

$$\int_0^1\frac{y^{d-1}\,\mathrm{d}y}{\sqrt{1-y^{2(d-1)}}} = \frac{1}{2(d-1)}B\Big(\frac{d}{2(d-1)},\frac{1}{2}\Big) = \frac{\sqrt{\pi}}{2(d-1)}\frac{\Gamma\big(\frac{d}{2(d-1)}\big)}{\Gamma\big(\frac{2d-1}{2(d-1)}\big)}. \tag{56.64}$$

Using $\Gamma\big(\frac{2d-1}{2(d-1)}\big) = \Gamma\big(1+\frac{1}{2(d-1)}\big) = \frac{1}{2(d-1)}\Gamma\big(\frac{1}{2(d-1)}\big)$, the prefactors cancel and

$$\ell = 2z_*\,a_d, \qquad a_d \equiv \sqrt{\pi}\,\frac{\Gamma\big(\frac{d}{2(d-1)}\big)}{\Gamma\big(\frac{1}{2(d-1)}\big)}. \tag{56.65}$$

The deepest reach z_* is proportional to the width ℓ, with an $O(1)$ dimension-dependent constant a_d; wide strips probe deep into the bulk, as for the interval.

The area integral and its divergence. Insert $\sqrt{1+z'^2} = (z_*/z)^{d-1}$ from Eq. (56.62) into Eq. (56.60) and change to z via Eq. (56.62); the integrand becomes $z^{-(d-1)}/\sqrt{1-(z/z_*)^{2(d-1)}}$, and with $y = z/z_*$,

$$\mathrm{Area} = 2L^{d-1}W^{d-2}\int_\varepsilon^{z_*}\frac{\mathrm{d}z}{z^{d-1}\sqrt{1-(z/z_*)^{2(d-1)}}} = \frac{2L^{d-1}W^{d-2}}{z_*^{d-2}}\int_{\varepsilon/z_*}^1\frac{y^{-(d-1)}\,\mathrm{d}y}{\sqrt{1-y^{2(d-1)}}}. \tag{56.66}$$

The remaining integral diverges at its lower limit, where the surface plunges to the boundary; we isolate the divergence by subtracting and adding the leading behavior $y^{-(d-1)}$:

$$\int_{\varepsilon/z_*}^{1} \frac{y^{-(d-1)}\,\mathrm{d}y}{\sqrt{1-y^{2(d-1)}}} = \underbrace{\int_{\varepsilon/z_*}^{1} y^{-(d-1)}\,\mathrm{d}y}_{\text{divergent}} + \underbrace{\int_0^1 y^{-(d-1)}\left[\frac{1}{\sqrt{1-y^{2(d-1)}}}-1\right]\mathrm{d}y}_{\text{convergent}} + o(1). \tag{56.67}$$

The subtracted integrand behaves as $\frac{1}{2}y^{d-1}$ near $y=0$, so the second integral converges and its lower limit may be sent to 0. The first integral is elementary:

$$\int_{\varepsilon/z_*}^{1} y^{-(d-1)}\,\mathrm{d}y = \frac{1}{d-2}\left[\left(\frac{z_*}{\varepsilon}\right)^{d-2}-1\right] \qquad (d>2). \tag{56.68}$$

The convergent remainder is a pure number; combining it with the $-1/(d-2)$ from Eq. (56.68) gives the finite constant

$$K_d \equiv -\frac{1}{d-2} + \int_0^1 y^{-(d-1)}\left[\frac{1}{\sqrt{1-y^{2(d-1)}}}-1\right]\mathrm{d}y. \tag{56.69}$$

This constant has a closed form. The full (analytically continued) Beta integral $\int_0^1 y^{a-1}(1-y^{2(d-1)})^{-1/2}\mathrm{d}y = \frac{1}{2(d-1)}B\big(\frac{a}{2(d-1)},\frac{1}{2}\big)$, evaluated at $a=-(d-2)$ where the elementary power divergence is automatically subtracted, returns exactly K_d. Its first argument is $\frac{-(d-2)}{2(d-1)} = \frac{d}{2(d-1)} - 1$ and its second-Gamma argument simplifies to $\frac{1}{2} - \frac{d-2}{2(d-1)} = \frac{1}{2(d-1)}$, so

$$K_d = \frac{\sqrt{\pi}}{2(d-1)}\frac{\Gamma\big(\frac{d}{2(d-1)}-1\big)}{\Gamma\big(\frac{1}{2(d-1)}\big)} = -\frac{1}{d-2}\sqrt{\pi}\,\frac{\Gamma\big(\frac{d}{2(d-1)}\big)}{\Gamma\big(\frac{1}{2(d-1)}\big)} = -\frac{a_d}{d-2}, \tag{56.70}$$

where the middle step uses $\Gamma(\xi-1) = \Gamma(\xi)/(\xi-1)$ with $\xi = \frac{d}{2(d-1)}$ and $\xi - 1 = -\frac{d-2}{2(d-1)}$. The constant K_d is *negative*, and it is built from the same ratio a_d that fixed the width Eq. (56.65).

Assembling the entropy. Putting Eqs. (56.68) and (56.69) into Eq. (56.66) and dividing by $4G_N$,

$$S = \frac{\text{Area}}{4G_N} = \frac{2L^{d-1}W^{d-2}}{4G_N}\left[\frac{1}{(d-2)}\frac{1}{\varepsilon^{d-2}} + \frac{K_d}{z_*^{d-2}}\right]. \tag{56.71}$$

The first term is the area law: it scales as $W^{d-2}/\varepsilon^{d-2}$ — the entangling surface's area in cutoff units — and is local on ∂A, blind to the width ℓ. The second

term is finite. Eliminating z_* in favor of the physical width through Eq. (56.65), $z_* = \ell/2a_d$, so $z_*^{-(d-2)} = (2a_d/\ell)^{d-2}$, and using $K_d = -a_d/(d-2)$,

$$\boxed{S = \frac{1}{4G_N}\left[\frac{2L^{d-1}}{d-2}\Big(\frac{W}{\varepsilon}\Big)^{d-2} - \frac{2^{d-1}\pi^{(d-1)/2}L^{d-1}}{d-2}\left(\frac{\Gamma\big(\frac{d}{2(d-1)}\big)}{\Gamma\big(\frac{1}{2(d-1)}\big)}\right)^{d-1}\Big(\frac{W}{\ell}\Big)^{d-2}\right].} \tag{56.72}$$

The structure of Eq. (56.72) is the promised one: an area-law divergence $\propto W^{d-2}/\varepsilon^{d-2}$ plus a finite, cutoff-independent, negative piece $\propto -(L_{\mathrm{AdS}}/G_N)\,W^{d-2}/\ell^{d-2}$ that depends on the *shape and size* of the region. Three features deserve comment. First, the finite piece scales as $\ell^{-(d-2)}$, a nonlocal dependence on the whole width that no local boundary term could produce; it is a genuine measure of the long-range entanglement across the strip and is exactly the kind of universal shape dependence that distinguishes entanglement entropy from a naive area law. Second, its coefficient is negative, so that increasing ℓ at fixed cutoff *decreases* the finite part of the entropy; this sign is required by strong subadditivity and underlies the entropic c-functions that count degrees of freedom monotonically along renormalization-group flows. Third, the $d = 2$ case is recovered as a limit even though Eq. (56.68) was derived for $d > 2$: as $d \to 2$ the transverse area $W^{d-2} \to 1$, the coefficient $a_d \to a_2 = \sqrt{\pi}\,\Gamma(1)/\Gamma(\frac{1}{2}) = 1$, and the two terms in Eq. (56.72) combine as

$$S \xrightarrow[d\to 2]{} \frac{2L}{4G_N}\,\frac{1}{d-2}\left[\Big(\frac{1}{\varepsilon}\Big)^{d-2} - \Big(\frac{1}{\ell}\Big)^{d-2}\right] \xrightarrow[d\to 2]{} \frac{L}{2G_N}\ln\frac{\ell}{\varepsilon} = \frac{c}{3}\ln\frac{\ell}{\varepsilon}, \tag{56.73}$$

since $x^{d-2} = e^{(d-2)\ln x} \to 1 + (d-2)\ln x$ makes the bracket tend to $(d-2)\ln(\ell/\varepsilon)$ and the pole cancels. The strip's power law and the interval's logarithm are the same formula, analytically continued in d.

56.2.4 Two intervals: mutual information and the entanglement phase transition

The final computation is the one where the minimization in Eq. (56.4) stops being a formality and produces a sharp phase transition — a phenomenon visible in holography long before it could be extracted from a direct field-theory calculation. Return to a 2D CFT in its vacuum, dual to pure AdS_3, and take the region to be the *union of two disjoint intervals* on the line,

$$A = [x_1, x_2], \qquad B = [x_3, x_4], \qquad x_1 < x_2 < x_3 < x_4. \tag{56.74}$$

The entangling set $\partial(A \cup B)$ is four points, and there are two topologically distinct ways to hang geodesics from them while remaining homologous to $A \cup B$. We compute both and let Eq. (56.4) choose.

The two configurations. Each candidate RT surface is a pair of the vacuum semicircle geodesics of Theorem 56.4, whose regulated length between two boundary points separated by coordinate distance ℓ is Length $= 2L\ln(\ell/\varepsilon)$, i.e. an entropy contribution $\frac{c}{3}\ln(\ell/\varepsilon)$. The two ways to pair the four endpoints into homologous geodesic pairs are:

- the *disconnected* configuration, geodesics over A and over B separately, with lengths set by $x_2 - x_1$ and $x_4 - x_3$;
- the *connected* configuration, geodesics over the inner gap and the outer span, with lengths set by $x_3 - x_2$ and $x_4 - x_1$.

(The remaining pairing, joining x_1 to x_3 and x_2 to x_4, gives crossing curves that are not homologous to $A \cup B$ and is excluded by Theorem 56.2.) The corresponding entropies are

$$S_{\text{disc}} = \frac{c}{3}\left[\ln\frac{x_2 - x_1}{\varepsilon} + \ln\frac{x_4 - x_3}{\varepsilon}\right], \tag{56.75}$$

$$S_{\text{conn}} = \frac{c}{3}\left[\ln\frac{x_3 - x_2}{\varepsilon} + \ln\frac{x_4 - x_1}{\varepsilon}\right], \tag{56.76}$$

and the RT prescription selects the smaller,

$$S(A \cup B) = \min\bigl(S_{\text{disc}},\, S_{\text{conn}}\bigr). \tag{56.77}$$

Crucially, the cutoff ε enters *both* configurations through four logarithms, two per pairing, so the cutoff dependence is identical in S_{disc} and S_{conn} and cancels in their difference — the competition is between finite, cutoff-independent quantities, as it must be for a physical transition.

The crossover and the mutual information. The natural diagnostic is the mutual information $I(A:B) = S(A)+S(B)-S(A\cup B)$ (Theorem 55.13), which is finite and cutoff-independent (Theorem 55.14) and, as Chapter 55 stressed, equals the relative entropy $S\bigl(\rho_{AB}\,\|\,\rho_A\otimes\rho_B\bigr)$ that quantifies all correlation between the two regions (Section 55.6). The single-interval entropies are $S(A) = \frac{c}{3}\ln\frac{x_2-x_1}{\varepsilon}$ and $S(B) = \frac{c}{3}\ln\frac{x_4-x_3}{\varepsilon}$, so $S(A) + S(B) = S_{\text{disc}}$. Therefore

$$I(A:B) = S_{\text{disc}} - S(A\cup B) = S_{\text{disc}} - \min\bigl(S_{\text{disc}}, S_{\text{conn}}\bigr) = \max\bigl(0,\; S_{\text{disc}} - S_{\text{conn}}\bigr). \tag{56.78}$$

The two phases are now explicit. When the disconnected surface is the shorter one, $S(A\cup B) = S_{\text{disc}}$ and the mutual information *vanishes identically*,

$$I(A:B) = 0 \qquad \text{(disconnected phase)}. \tag{56.79}$$

This is a striking statement: at leading order in G_N, two sufficiently separated regions share *no* mutual information whatsoever — not merely a small amount, but exactly zero. When instead the connected surface is shorter,

$$I(A:B) = S_{\text{disc}} - S_{\text{conn}} = \frac{c}{3}\ln\frac{(x_2-x_1)(x_4-x_3)}{(x_3-x_2)(x_4-x_1)} \qquad \text{(connected phase)}. \tag{56.80}$$

The argument of the logarithm is naturally written in the conformal cross-ratio of the four points. Define

$$\eta \equiv \frac{x_{12}\,x_{34}}{x_{13}\,x_{24}} = \frac{(x_2-x_1)(x_4-x_3)}{(x_3-x_1)(x_4-x_2)} \in (0,1), \qquad x_{ij}\equiv x_i - x_j. \tag{56.81}$$

For four ordered points on a line the identity $x_{13}x_{24} = x_{12}x_{34} + x_{14}x_{23}$ holds (it is verified directly from the definitions), so the ratio in Eq. (56.80) is

$$\frac{(x_2-x_1)(x_4-x_3)}{(x_3-x_2)(x_4-x_1)} = \frac{x_{12}x_{34}}{x_{23}x_{14}} = \frac{x_{12}x_{34}}{x_{13}x_{24}-x_{12}x_{34}} = \frac{\eta}{1-\eta}, \tag{56.82}$$

where $x_{23}x_{14} = (x_3-x_2)(x_4-x_1)$ up to the sign that the ordered differences supply. The mutual information is thus a clean function of the single cross-ratio:

$$\boxed{I(A:B) = \max\left[0,\ \frac{c}{3}\ln\frac{\eta}{1-\eta}\right].} \tag{56.83}$$

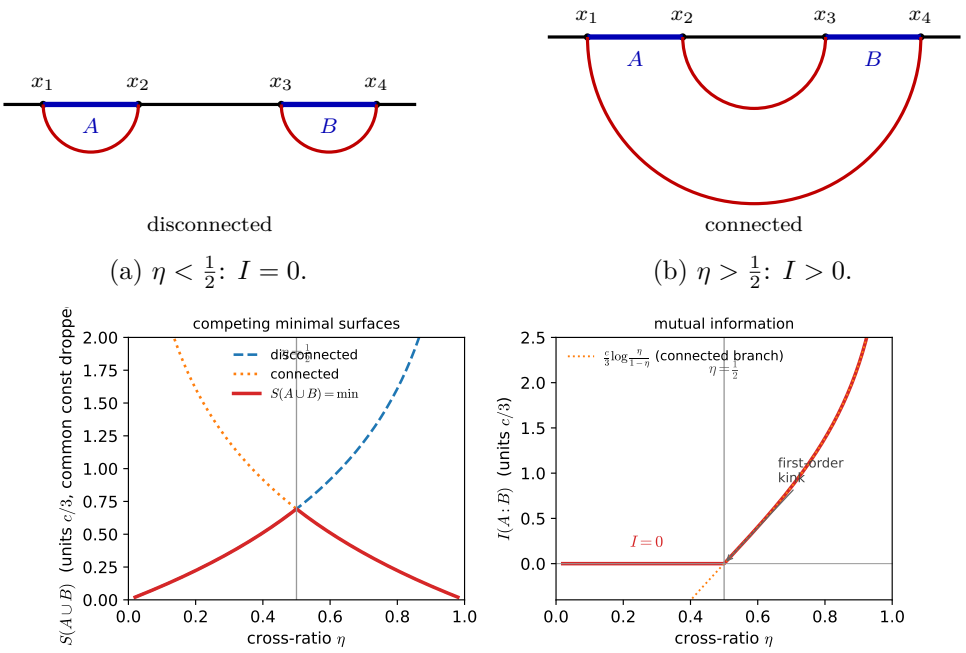
The transition is first order. The two branches are equal when $S_{\text{disc}} = S_{\text{conn}}$, i.e. when $\eta/(1-\eta) = 1$, namely

$$\eta_c = \frac{1}{2}. \tag{56.84}$$

For $\eta < \frac{1}{2}$ (intervals far apart, small cross-ratio) the disconnected surface wins and $I = 0$; for $\eta > \frac{1}{2}$ (intervals close) the connected surface wins and $I = \frac{c}{3}\ln\frac{\eta}{1-\eta} > 0$. At $\eta = \frac{1}{2}$ the mutual information is continuous — both branches give $I = 0$ — but its derivative is not:

$$\left.\frac{\mathrm{d}I}{\mathrm{d}\eta}\right|_{\eta\to\frac{1}{2}^-} = 0, \qquad \left.\frac{\mathrm{d}I}{\mathrm{d}\eta}\right|_{\eta\to\frac{1}{2}^+} = \frac{c}{3}\left(\frac{1}{\eta}+\frac{1}{1-\eta}\right)\Big|_{\eta=\frac{1}{2}} = \frac{4c}{3}. \tag{56.85}$$

The slope jumps discontinuously from 0 to $4c/3$. This is a *first-order* entanglement phase transition: $S(A\cup B)$, being the minimum of two smooth functions that cross transversally, has a kink, and the order parameter $I(A:B)$ turns on with a discontinuous first derivative. Figure 56.3 shows both the competing surface configurations and the resulting kinked mutual information. The transition is sharp only at leading order in G_N; the subleading bulk-entanglement corrections of Section 56.5.1 round the kink and lift I slightly off zero in the disconnected phase, but the leading-order discontinuity is a genuine prediction of classical RT and one of its most distinctive fingerprints.



(a) $\eta < \frac{1}{2}$: $I = 0$.

(b) $\eta > \frac{1}{2}$: $I > 0$.

Figure 56.3: Top: the two homologous geodesic configurations for $A \cup B$. The disconnected pairing (left) hangs a geodesic over each interval; the connected pairing (right) joins the inner gap and the outer span. Bottom left: the two candidate entropies cross at $\eta = \frac{1}{2}$, and $S(A \cup B)$ follows the lower envelope, producing a kink. Bottom right: the mutual information Eq. (56.83) is identically zero for $\eta < \frac{1}{2}$ and rises with a discontinuous slope $4c/3$ for $\eta > \frac{1}{2}$ — a first-order entanglement phase transition.

Physical Insight

The vanishing of $I(A : B)$ in the disconnected phase is the holographic version of large-N factorization. At leading order in G_N (large central charge) the connected correlations between well-separated regions are subleading, and the geometry registers this by literally disconnecting the bulk surface: no single surface spans both regions, so they share no classical entanglement. As the regions approach, a connected surface becomes cheaper and the mutual information switches on. The bulk thus encodes the onset of correlation as a change in the *topology* of the dominant minimal surface — a discrete, geometric event standing in for what on the boundary is a smooth (but, at leading order, sharply thresholded) build-up of entanglement.

56.2.5 What the catalogue is telling us

We now hold a working catalogue of holographic entanglement entropies: the vacuum interval (Eq. (56.26)), the thermal interval interpolating to extensive entropy (Eq. (56.43)), the sphere in any dimension as a hyperbolic horizon area (Eqs. (56.52) and (56.58)), the strip's area law with its nonlocal finite piece (Eq. (56.72)), and the two-interval mutual information with its first-order transition (Eq. (56.83)). Each was computed by solving a classical geometry problem, and each reproduces — where independent checks exist — the field-theory answer with no adjustable parameters. But the catalogue does more than tabulate: its entries are conspicuously *consistent* in ways that look like accidents from the geometry side. The strip's finite piece carries exactly the sign demanded by strong subadditivity; the two-interval mutual information is positive wherever it is nonzero and switches off precisely when subadditivity would otherwise be threatened; the homology constraint was forced on us by the purity relation $S(A) = S(\bar{A})$. These are not coincidences. They are shadows of general inequalities that the entanglement entropies of *any* quantum state must obey, and the remarkable fact — which the next section establishes — is that the minimal-surface prescription satisfies them *automatically*, as theorems of Riemannian geometry. We turn to proving, from RT geometry alone, the strong subadditivity of Headrick and Takayanagi and the holographic-only monogamy of mutual information, demonstrating that the area functional is not merely consistent with the laws of entanglement but is a sharp geometric engine for them.

56.3 Entanglement inequalities from geometry

The previous section turned the Ryu–Takayanagi formula into a working tool and, in passing, accumulated a catalogue of holographic entropies: the thermal state, the sphere, the strip, two intervals with their phase transition. Run through that catalogue with a calculator and a pattern emerges that is too clean to be an accident. Every entropy is the regulated area of a minimal surface, and minimal surfaces are not free to do whatever they like — they are constrained by the geometry of the slice they live in. A surface anchored on a large region cannot be longer than two surfaces anchored on its pieces, because one can always cut the long surface and rebuild it from the short ones; a surface that wraps a handle of the bulk is forced to compete with one that does not. These are geometric facts, theorems of Riemannian geometry, and through Eq. (56.4) they descend to inequalities among entanglement entropies. The remarkable thing is that some of these inequalities are extremely hard to prove — or simply false — in field theory directly.

This section proves the two most important of them from RT geometry alone. The first, strong subadditivity, is a theorem of quantum mechanics: it

holds for *every* density matrix, and its standard proof (Theorem 55.47) is a deep statement about the concavity of conditional entropy, established by Lieb and Ruskai with operator-convexity machinery that has no elementary shortcut. We will reprove it in three lines of geometry. The second, the monogamy of mutual information, is sharper still: it is *not* a theorem of quantum mechanics — generic quantum states violate it — yet it is forced on holographic states by the same surface-recombination argument. That contrast is the lesson of the section. Holographic states are not arbitrary states of the CFT; they obey extra entropy inequalities, and so they occupy a special, geometrically carved corner of Hilbert space. The boundary of that corner is the *holographic entropy cone*, and the existence of states outside it is our first sharp evidence that not every state of a holographic CFT is dual to a smooth classical geometry.

Throughout we work in the static setting of Section 56.1.3: a fixed bulk time slice Σ, on which RT entropies are minimal areas. The regions A, B, C are disjoint spatial regions of the boundary, and for any union X we write γ_X for a bulk surface anchored on ∂X and homologous to X in the sense of Theorem 56.2, with γ_X^* the true minimizer, so that $S(X) = \mathrm{Area}(\gamma_X^*)/4G_N$ by Theorem 56.3. The von Neumann entropy itself is $S(X) = -\operatorname{Tr}(\rho_X \ln \rho_X)$ as fixed in Eq. (55.6), and the mutual information is $I(A:B) = S(A) + S(B) - S(AB)$ from Eq. (55.23).

56.3.1 Strong subadditivity: the statement and the geometric idea

Recall the inequality we are after. For any three disjoint regions A, B, C,

$$S(AB) + S(BC) \;\geq\; S(ABC) + S(B), \tag{56.86}$$

where AB abbreviates $A \cup B$ and so on. This is strong subadditivity (SSA). Setting $C = \varnothing$ recovers ordinary subadditivity $S(A) + S(B) \geq S(AB)$ (Theorem 55.12); the content of SSA is the strengthening in which a shared region B appears on both sides. Quantum-mechanically it is equivalent to the statement that conditional entropy $S(AB) - S(B)$ is monotonic under appending C, and its proof from first principles requires the Lieb–Ruskai theorem on the joint convexity of relative entropy — the genuinely hard input of Theorem 55.47. We now show that for holographic states it is a one-step consequence of the minimality built into Eq. (56.4).

The idea, due to Headrick and Takayanagi, is cut-and-paste. We have two true minimal surfaces in hand, γ_{AB}^* and γ_{BC}^*, of total area $4G_N\big(S(AB)+S(BC)\big)$. We are going to take this collection of surface and *recut* it — redistribute its constituent pieces — into two new surfaces, one anchored on $\partial(ABC)$ and homologous to ABC, the other anchored on ∂B and homologous to B. The recutting conserves total area: not a single point of surface is created or destroyed, only reassigned. The two new surfaces are legal trial surfaces for ABC and for

B, but they are not the minimizers — they generically have creases where the old surfaces crossed. Minimality of the *true* surfaces γ^*_{ABC} and γ^*_B then says each is no longer than its trial counterpart, and Eq. (56.86) drops out. The entire weight of the argument rests on a single geometric fact — that area is conserved under recutting — and a single physical input — that the true surfaces minimize. We first see the mechanism explicitly in AdS_3, where the surfaces are geodesics and the recutting can be drawn, and then abstract it into the form that proves both SSA and, with one more region, monogamy.

56.3.2 The mechanism in AdS_3: crossing geodesics

Take three adjacent intervals on the boundary of vacuum AdS_3, sharing endpoints:

$$A = [x_0, x_1], \qquad B = [x_1, x_2], \qquad C = [x_2, x_3], \qquad x_0 < x_1 < x_2 < x_3, \tag{56.87}$$

all at $t = 0$, so that $AB = [x_0, x_2]$, $BC = [x_1, x_3]$, $ABC = [x_0, x_3]$, and $B = [x_1, x_2]$ are again single intervals. From Theorem 56.4 we know the minimal surface anchored on a single interval $[x_i, x_j]$ is the semicircle in the half-plane metric Eq. (56.7) of diameter $[x_i, x_j]$, and its regulated length is

$$\mathrm{Length}(\gamma^*_{[x_i,x_j]}) = 2L \,\ln\frac{x_j - x_i}{\varepsilon}. \tag{56.88}$$

The two surfaces appearing on the left of Eq. (56.86) are therefore the semicircles spanning $[x_0, x_2]$ and $[x_1, x_3]$. Because the intervals overlap (both contain $B = [x_1, x_2]$), these two semicircles *cross*: one runs from x_0 to x_2, the other from x_1 to x_3, and since $x_0 < x_1 < x_2 < x_3$ they must intersect at exactly one interior point P of the bulk. Figure 56.4 shows the configuration.

At the crossing P the two geodesics divide into four arcs:

- α_{L}: the part of γ^*_{AB} from x_0 down to P;
- α_{R}: the part of γ^*_{AB} from P up to x_2;
- β_{L}: the part of γ^*_{BC} from x_1 down to P;
- β_{R}: the part of γ^*_{BC} from P up to x_3.

Now reassemble. The arc α_{L} runs from x_0 to P, and β_{R} from P to x_3; joined at P they form a continuous curve $\widetilde{\gamma}_{ABC} \equiv \alpha_{\mathrm{L}} \cup \beta_{\mathrm{R}}$ from x_0 to x_3, anchored on $\partial(ABC) = \{x_0, x_3\}$. The remaining arcs β_{L} (from x_1 to P) and α_{R} (from P to x_2) join into $\widetilde{\gamma}_B \equiv \beta_{\mathrm{L}} \cup \alpha_{\mathrm{R}}$ from x_1 to x_2, anchored on $\partial B = \{x_1, x_2\}$. We have used each of the four arcs exactly once, so

$$\mathrm{Length}(\widetilde{\gamma}_{ABC}) + \mathrm{Length}(\widetilde{\gamma}_B) = \mathrm{Length}(\gamma^*_{AB}) + \mathrm{Length}(\gamma^*_{BC}). \tag{56.89}$$

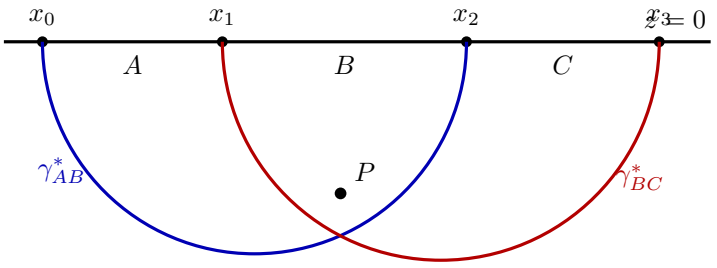


Figure 56.4: Strong subadditivity in AdS_3. The minimal surfaces for $AB = [x_0, x_2]$ (blue) and $BC = [x_1, x_3]$ (red) are semicircular geodesics that cross at a single bulk point P. Recutting at P swaps the pieces: the left arc of γ^*_{AB} $(x_0 \to P)$ joined to the right arc of γ^*_{BC} $(P \to x_3)$ is a trial surface for $ABC = [x_0, x_3]$; the left arc of γ^*_{BC} $(x_1 \to P)$ joined to the right arc of γ^*_{AB} $(P \to x_2)$ is a trial surface for $B = [x_1, x_2]$. Total length is unchanged by the swap, so minimality of the true geodesics over $[x_0, x_3]$ and $[x_1, x_2]$ gives $S(AB) + S(BC) \geq S(ABC) + S(B)$.

Each reassembled curve is a legal trial surface — correctly anchored and (being a deformation of the original semicircles through the bulk) homologous to the right region — but each has a kink at P, so neither is a geodesic. The true minimal surfaces are no longer:

$$\text{Length}(\gamma^*_{ABC}) \leq \text{Length}(\widetilde{\gamma}_{ABC}), \qquad \text{Length}(\gamma^*_{B}) \leq \text{Length}(\widetilde{\gamma}_{B}). \tag{56.90}$$

Adding the two inequalities of Eq. (56.90) and using Eq. (56.89),

$$\text{Length}(\gamma^*_{ABC}) + \text{Length}(\gamma^*_{B}) \leq \text{Length}(\gamma^*_{AB}) + \text{Length}(\gamma^*_{BC}), \tag{56.91}$$

and dividing by $4G_N$ gives precisely Eq. (56.86). The kink at P is the source of the slack: smoothing it strictly shortens the trial curves, so the inequality is strict whenever the surfaces genuinely cross.

It is worth confirming the inequality numerically against Eq. (56.88), since here we can compute every term in closed form. With $S = \frac{c}{3}\ln(\ell/\varepsilon)$ for an interval of length ℓ,

$$\begin{aligned} S(AB) + S(BC) - S(ABC) - S(B) &= \frac{c}{3}\Big[\ln\tfrac{x_2-x_0}{\varepsilon} + \ln\tfrac{x_3-x_1}{\varepsilon} - \ln\tfrac{x_3-x_0}{\varepsilon} - \ln\tfrac{x_2-x_1}{\varepsilon}\Big] \\ &= \frac{c}{3}\,\ln\frac{(x_2-x_0)(x_3-x_1)}{(x_3-x_0)(x_2-x_1)}. \end{aligned} \tag{56.92}$$

The cutoff ε has cancelled, as it must for a difference of entropies. The argument of the logarithm exceeds one: with $x_0 < x_1 < x_2 < x_3$,

$$(x_2-x_0)(x_3-x_1) - (x_3-x_0)(x_2-x_1) = (x_1-x_0)(x_3-x_2) > 0, \tag{56.93}$$

as one verifies by expanding both products and cancelling the common terms x_2x_3, x_0x_1, leaving $-x_1x_2 - x_0x_3 + x_1x_3 + x_0x_2 = (x_1-x_0)(x_3-x_2)$. Hence

Eq. (56.92) is strictly positive, and SSA holds with a margin set by the product of the two "gap" lengths $x_1 - x_0$ and $x_3 - x_2$. The deep operator inequality of quantum information theory has become, in AdS_3, the schoolroom statement that $(x_1 - x_0)(x_3 - x_2) > 0$.

56.3.3 The recutting lemma

The AdS_3 picture makes the mechanism vivid but hides what is actually doing the work, and obscures why it generalizes. In higher dimensions surfaces do not cross at a point but along a codimension-two locus, and "reassemble the arcs" must be made precise. The clean formulation replaces surfaces by the regions they bound. Recall (Theorem 56.2) that a homologous surface γ_X comes with a homology region $\mathcal{R}_X \subset \Sigma$, a piece of the bulk slice with $\partial\mathcal{R}_X = X \cup \gamma_X$. Encode each homology region by its indicator function on the slice,

$$\chi_X : \Sigma \to \{0,1\}, \qquad \chi_X(p) = \begin{cases} 1, & p \in \mathcal{R}_X, \\ 0, & \text{otherwise.} \end{cases} \tag{56.94}$$

The bulk boundary of $\mathcal{R}_X$ — the surface γ_X — is exactly the set where χ_X jumps; its area is the total "jump area" of χ_X. We can therefore phrase the whole argument in terms of how jump sets of indicator functions combine.

Lemma 56.9 (Recutting)**.** *Let $\mathcal{R}_1, \dots, \mathcal{R}_k$ be homology regions in Σ with indicator functions $\chi_1, \dots, \chi_k$, and suppose their bounding surfaces are smooth and intersect transversally (so that the loci where two or more surfaces coincide have codimension at least two in Σ, hence zero area). The surfaces partition Σ into* cells*, on each of which the vector $(\chi_1, \dots, \chi_k) \in \{0,1\}^k$ is constant; label the cell by that vector σ. Two cells are* adjacent *when they share a codimension-one face, which happens precisely when their labels differ in a single coordinate i, and the face is then a piece of the surface $\partial\mathcal{R}_i$. Consequently the total area of any collection of cell-faces, counted with the multiplicity "number of coordinates in which the two flanking labels differ," equals the total area of the surfaces; in particular*

$$\sum_{\text{faces } F} \mathrm{Area}(F) = \sum_{i=1}^{k} \mathrm{Area}(\partial\mathcal{R}_i), \tag{56.95}$$

because each face lies on exactly one surface.

Proof. Where the surfaces are smooth and transversal, a generic point of $\bigcup_i \partial\mathcal{R}_i$ lies on exactly one surface $\partial\mathcal{R}_i$ and separates two regions differing only in whether they belong to $\mathcal{R}_i$; the two adjacent cells thus differ in the single coordinate i. The exceptional set — where two or more surfaces meet — is a union of transversal intersections of codimension-two submanifolds, which carry no $(d-1)$-dimensional area and so do not contribute to any area integral. Summing the

area of all faces, each face is counted once and belongs to the unique surface flagged by its flipped coordinate; grouping faces by that coordinate reconstructs each $\partial\mathcal{R}_i$ exactly, giving Eq. (56.95). □

The power of Theorem 56.9 is that we may now build *new* regions out of Boolean combinations of the cells and read off the areas of their boundaries from the same faces, with no new geometry. A new region $\mathcal{O}$ is a union of cells; its boundary $\partial\mathcal{O}$ consists of those faces with exactly one flanking cell inside $\mathcal{O}$. If we design several new regions $\mathcal{O}_1, \dots, \mathcal{O}_m$ so that *every face is a boundary of at most one of them*, then

$$\sum_{a=1}^{m} \mathrm{Area}(\partial\mathcal{O}_a) \;\le\; \sum_{\text{faces}} \mathrm{Area}(F) \;=\; \sum_{i=1}^{k} \mathrm{Area}(\partial\mathcal{R}_i), \tag{56.96}$$

the inequality coming only from faces internal to some $\mathcal{O}_a$ (counted zero times on the left, once on the right). If, moreover, each $\mathcal{O}_a$ is a legal homology region for a boundary region X_a, then $\partial\mathcal{O}_a$ is a trial surface for X_a and minimality gives $\mathrm{Area}(\gamma^*_{X_a}) \le \mathrm{Area}(\partial\mathcal{O}_a)$, whence

$$\sum_a 4G_N\, S(X_a) = \sum_a \mathrm{Area}(\gamma^*_{X_a}) \le \sum_i \mathrm{Area}(\partial\mathcal{R}_i) = \sum_i 4G_N\, S(Y_i), \tag{56.97}$$

where Y_i are the boundary regions of the input surfaces. This single inequality, Eq. (56.97), is the engine. Proving an entropy inequality reduces to a combinatorial design problem: given input regions $\{Y_i\}$ on the right, choose the cell-unions $\{\mathcal{O}_a\}$ on the left so that (i) each $\mathcal{O}_a$ is anchored on and homologous to the desired X_a, and (ii) no face bounds two different $\mathcal{O}_a$. We now solve that problem first for two inputs (recovering SSA) and then for three (delivering monogamy).

Theorem 56.10 (Geometric strong subadditivity)**.** *For disjoint boundary regions A, B, C of a static holographic state in the classical-gravity regime, the Ryu–Takayanagi entropies satisfy Eq. (56.86), $S(AB) + S(BC) \ge S(ABC) + S(B)$.*

Proof. Apply Theorem 56.9 with the two inputs $\mathcal{R}_1 = \mathcal{R}_{AB}$ and $\mathcal{R}_2 = \mathcal{R}_{BC}$, the homology regions of the true minimal surfaces $\gamma^*_{AB}, \gamma^*_{BC}$. Cells are labelled $\sigma = (\sigma_1, \sigma_2) \in \{0,1\}^2$. On the conformal boundary the indicators restrict to the indicators of the boundary regions: a boundary point lies in $\mathcal{R}_{AB}$ iff it lies in $AB = A \cup B$, and in $\mathcal{R}_{BC}$ iff it lies in $BC = B \cup C$. Since A, B, C are disjoint, the boundary-anchored cells are

$$A \leftrightarrow (1,0), \quad B \leftrightarrow (1,1), \quad C \leftrightarrow (0,1), \quad (\text{outside } ABC) \leftrightarrow (0,0). \tag{56.98}$$

Define the two output regions

$$\mathcal{O}_{ABC} = \{\sigma : \sigma_1{+}\sigma_2 \ge 1\} = \mathcal{R}_{AB} \cup \mathcal{R}_{BC}, \qquad \mathcal{O}_B = \{\sigma : \sigma_1{+}\sigma_2 = 2\} = \mathcal{R}_{AB} \cap \mathcal{R}_{BC}. \tag{56.99}$$

On the boundary, $\mathcal{O}_{ABC}$ restricts to $(A \cup B) \cup (B \cup C) = ABC$ and $\mathcal{O}_B$ to $(A \cup B) \cap (B \cup C) = B$, so $\partial\mathcal{O}_{ABC}$ is anchored on $\partial(ABC)$ and homologous to ABC, while $\partial\mathcal{O}_B$ is anchored on ∂B and homologous to B. It remains to check condition (ii): no face bounds both outputs. A face flips a single coordinate, so it lies between cells of the form $\{(0,\sigma_2),(1,\sigma_2)\}$ (flipping σ_1) or $\{(\sigma_1,0),(\sigma_1,1)\}$ (flipping σ_2). Take a σ_1-flip face between $(0,\sigma_2)$ and $(1,\sigma_2)$. Its two cells have $\sigma_1+\sigma_2$ equal to σ_2 and $1+\sigma_2$, which straddle the threshold "≥ 1" only when $\sigma_2 = 0$ (so it bounds $\mathcal{O}_{ABC}$) and straddle "$= 2$" only when $\sigma_2 = 1$ (so it bounds $\mathcal{O}_B$); the two cases are mutually exclusive. The same holds for σ_2-flips by symmetry. Hence every face bounds at most one output, condition (ii) holds, and Eq. (56.97) gives

$$S(ABC) + S(B) \leq S(AB) + S(BC). \tag{56.100}$$

This is Eq. (56.86). (Union and intersection are exactly the operations $\widetilde{\gamma}_{ABC}$ and $\widetilde{\gamma}_B$ of Section 56.3.2: $\mathcal{O}_{ABC}$ is the union of the two homology disks under the crossing geodesics, $\mathcal{O}_B$ their lens-shaped overlap.) □

Physical Insight

Strong subadditivity is, for any quantum state, the statement that the geometry of correlations is consistent — that conditioning on more never increases information in a contradictory way. For holographic states the proof reveals *why* the consistency holds: the entropies are areas in a shared bulk space, and areas of regions in any metric space obey submodularity, $\mathrm{Area}(\partial(\mathcal{R}\cup\mathcal{R}')) + \mathrm{Area}(\partial(\mathcal{R}\cap\mathcal{R}')) \leq \mathrm{Area}(\partial\mathcal{R}) + \mathrm{Area}(\partial\mathcal{R}')$. SSA is submodularity of bulk area pulled back to the boundary. The non-trivial Lieb–Ruskai theorem and the elementary fact that overlapping regions share boundary are, in the holographic world, the same statement.

56.3.4 Monogamy of mutual information

We now turn to an inequality of a different character. Define, for three disjoint regions, the *tripartite information*

$$I_3(A:B:C) \;\equiv\; I(A:B) + I(A:C) - I(A:BC), \tag{56.101}$$

the failure of A's correlation with the union BC to be the sum of its correlations with B and with C separately. Expanding each mutual information by Eq. (55.23) and cancelling the repeated $S(A)$ terms,

$$\begin{aligned} I_3(A:B:C) = {} & \big[S(A)+S(B)-S(AB)\big] + \big[S(A)+S(C)-S(AC)\big] \\ & - \big[S(A)+S(BC)-S(ABC)\big] \end{aligned}$$

$$= S(A) + S(B) + S(C) - S(AB) - S(BC) - S(CA) + S(ABC). \tag{56.102}$$

The *monogamy of mutual information* (MMI) is the statement that this quantity is never positive for holographic states:

$$\boxed{\begin{aligned} &I_3(A:B:C) \le 0 \quad \Longleftrightarrow \\ &\qquad S(AB) + S(BC) + S(CA) \ge S(A) + S(B) + S(C) + S(ABC). \end{aligned}} \tag{56.103}$$

The name is apt: $I(A:BC) \ge I(A:B) + I(A:C)$ says that A's total correlation with BC *exceeds* the sum of its bilateral correlations, so correlation cannot be freely shared three ways — it is, like quantum entanglement, monogamous. The right-hand form of Eq. (56.103), manifestly symmetric in A, B, C, is the one we will prove.

The crucial point, and the reason this section exists, is that Eq. (56.103) is *false* for general quantum states. There is no analogue of the Lieb–Ruskai theorem behind it; one cannot derive it from the axioms of density matrices, because there are perfectly legal density matrices that violate it. We exhibit one below. MMI is a property holographic states possess and generic states lack. Yet the recutting machinery proves it for holographic states in essentially the same breath as SSA — the only change is that we now feed in three surfaces instead of two.

Theorem 56.11 (Holographic monogamy)**.** *For disjoint boundary regions A, B, C of a static holographic state in the classical-gravity regime, the Ryu–Takayanagi entropies satisfy Eq. (56.103).*

Proof. Apply Theorem 56.9 with the three inputs given by the true minimal surfaces of the *pairs*: $\mathcal{R}_1 = \mathcal{R}_{AB}$, $\mathcal{R}_2 = \mathcal{R}_{BC}$, $\mathcal{R}_3 = \mathcal{R}_{CA}$, with respective areas $4G_N\, S(AB)$, $4G_N\, S(BC)$, $4G_N\, S(CA)$. Cells carry labels $\sigma = (\sigma_1, \sigma_2, \sigma_3) \in \{0,1\}^3$ recording membership in $(\mathcal{R}_{AB}, \mathcal{R}_{BC}, \mathcal{R}_{CA})$. On the conformal boundary, a point of A lies in AB and CA but not BC; a point of B in AB and BC; a point of C in BC and CA; a point outside ABC in none. The boundary-anchored cells are therefore the four even-weight labels

$$A \leftrightarrow (1,0,1), \quad B \leftrightarrow (1,1,0), \quad C \leftrightarrow (0,1,1), \quad (\text{outside}) \leftrightarrow (0,0,0), \tag{56.104}$$

while the four odd-weight labels $(1,0,0), (0,1,0), (0,0,1), (1,1,1)$ correspond to purely interior cells, touching the boundary nowhere and free for us to assign. Define the four output regions

$$\begin{aligned} \mathcal{O}_A = \{(1,0,1)\}, \quad \mathcal{O}_B = \{(1,1,0)\}, \quad \mathcal{O}_C = \{(0,1,1)\}, \\ \mathcal{O}_{ABC} = \{\sigma : \sigma \neq (0,0,0)\} = \mathcal{R}_{AB} \cup \mathcal{R}_{BC} \cup \mathcal{R}_{CA}. \end{aligned} \tag{56.105}$$

Their boundary anchors follow from Eq. (56.104): $\mathcal{O}_A$ restricts on the boundary to A (it is $\mathcal{R}_{AB} \cap \mathcal{R}_{CA} \setminus \mathcal{R}_{BC}$, whose boundary trace is $(A \cup B) \cap (C \cup A)$ minus $(B \cup C)$, i.e. A), likewise $\mathcal{O}_B \to B$, $\mathcal{O}_C \to C$, and $\mathcal{O}_{ABC} \to ABC$. Each is a legal homology region for the labelled boundary region, so $\partial\mathcal{O}_A, \ldots, \partial\mathcal{O}_{ABC}$ are trial surfaces for A, B, C, ABC.

We verify condition (ii): every face bounds at most one output. The key geometric fact is that the four boundary-anchored cells Eq. (56.104) all have *even* weight $\sigma_1 + \sigma_2 + \sigma_3 \in \{0, 2\}$, and a face flips a single coordinate, hence connects an even-weight cell to an odd-weight cell. So every face has one even-weight flank (forced to be one of A, B, C, outside) and one odd-weight flank (a free interior cell). Assign *all four interior cells to $\mathcal{O}_{ABC}$ and to no $\mathcal{O}_X$ with X a single region* — this is consistent with Eq. (56.105), since the singletons $\mathcal{O}_A, \mathcal{O}_B, \mathcal{O}_C$ contain only even-weight cells while $\mathcal{O}_{ABC}$ contains every non-zero label including the four odd ones. Now classify a face by its even flank:

- even flank $(0, 0, 0)$ (outside): inside no singleton and not in $\mathcal{O}_{ABC}$; its odd neighbour is in $\mathcal{O}_{ABC}$ only. The face bounds $\mathcal{O}_{ABC}$ alone.
- even flank $(1, 0, 1) = A$: in $\mathcal{O}_A$ and in $\mathcal{O}_{ABC}$; its odd neighbour is in $\mathcal{O}_{ABC}$ but not $\mathcal{O}_A$. Across this face $\mathcal{O}_{ABC}$ does *not* change (both flanks belong to it), while $\mathcal{O}_A$ does. The face bounds $\mathcal{O}_A$ alone.
- even flank $(1, 1, 0) = B$: by the same reasoning, bounds $\mathcal{O}_B$ alone.
- even flank $(0, 1, 1) = C$: bounds $\mathcal{O}_C$ alone.

In every case the face bounds exactly one output, so condition (ii) holds. (The decisive cancellation is in the second through fourth cases: because the interior cells were placed inside $\mathcal{O}_{ABC}$, the large surface $\partial\mathcal{O}_{ABC}$ runs only along faces touching the outside cell, and never competes with the small surfaces $\partial\mathcal{O}_A, \partial\mathcal{O}_B, \partial\mathcal{O}_C$ for the same face.) The master inequality Eq. (56.97) then reads

$$S(A) + S(B) + S(C) + S(ABC) \;\le\; S(AB) + S(BC) + S(CA), \tag{56.106}$$

which is exactly Eq. (56.103). □

Two features of this proof deserve emphasis, because they pinpoint exactly where the holographic structure is essential and therefore why MMI can fail without it.

First, *the existence of a common bulk.* The entire argument lives in the single slice Σ, where all the homology regions are subsets of one space and can be intersected, unioned, and decomposed into cells. The entropies $S(AB), S(BC), \ldots$ are not abstract numbers attached to a quantum state; they are areas of surfaces in one shared geometry, and Boolean algebra on subsets of that geometry is what generates the inequality. A generic quantum state has no such bulk. Its

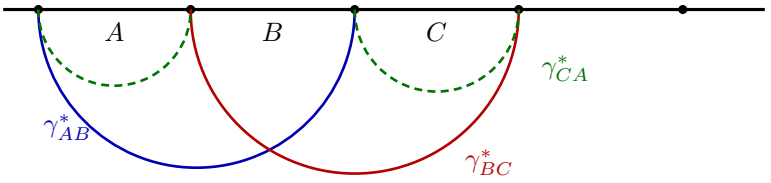


Figure 56.5: Monogamy of mutual information for three intervals $A = [y_0, y_1]$, $B = [y_1, y_2]$, $C = [y_2, y_3]$ in AdS_3. The three pairwise minimal surfaces γ^*_{AB} (blue), γ^*_{BC} (red), γ^*_{CA} (green dashed; here $CA = A \cup C$ is disjoint, so its minimal surface is the pair of geodesics over A and over C) bound homology regions whose Boolean combinations Eq. (56.105) reassemble, with no doubled faces, into trial surfaces for A, B, C and ABC. The recutting of Theorem 56.9 conserves total area, and minimality of the four true surfaces yields $S(AB) + S(BC) + S(CA) \geq S(A) + S(B) + S(C) + S(ABC)$.

entropies are just seven numbers $\{S(A), \ldots, S(ABC)\}$ constrained only by the universal inequalities (subadditivity, Araki–Lieb, strong subadditivity); nothing forces them to be jointly realizable as areas, and when they are not, Eq. (56.103) can be violated.

Second, *minimality plus the even-weight structure.* The proof of condition (ii) hinged on the boundary-anchored cells all having even weight, which let us hide the four interior cells inside $\mathcal{O}_{ABC}$ and decouple the large surface from the three small ones. This is a genuine feature of the geometry: the homology regions of three pairwise surfaces, sitting in a $(d-1)$-dimensional spatial slice, intersect in the pattern Eq. (56.104). In one boundary dimension fewer — if the "bulk" were itself only the boundary, with no extra dimension to bend the surfaces into — the cells would degenerate and the recutting would have nowhere to send the interior pieces. The extra holographic dimension is what supplies the room for the construction. We will see the same dimension reappear, from the other side, when the entanglement wedge is defined in Section 56.6.

Worked Example 56.12 (A non-holographic state that violates MMI: four-party GHZ)**.** To see that Eq. (56.103) is a real restriction and not a tautology, we compute I_3 for a state that any quantum-mechanics course allows. Let the global Hilbert space be four qubits and take the Greenberger–Horne–Zeilinger state

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}(|0000\rangle + |1111\rangle), \tag{56.107}$$

identifying three of the qubits with the regions A, B, C and the fourth with the purifying complement (the analogue of the rest of the boundary). The reduced density matrix on any proper subset of the qubits is obtained by tracing out the others. Tracing out even a single qubit destroys the coherence between $|0\cdots0\rangle$ and $|1\cdots1\rangle$, because they differ in that qubit; hence for *any* subset X with

$1 \le |X| \le 3$ qubits,

$$\rho_X = \tfrac{1}{2}(|0\cdots0\rangle\langle 0\cdots0| + |1\cdots1\rangle\langle 1\cdots1|), \qquad S(X) = \ln 2. \tag{56.108}$$

So every one-, two-, and three-party entropy equals $\ln 2$. Substituting into Eq. (56.102),

$$\begin{aligned} I_3(A:B:C) &= \underbrace{S(A)+S(B)+S(C)}_{3\ln 2} \underbrace{-S(AB)-S(BC)-S(CA)}_{-3\ln 2} + \underbrace{S(ABC)}_{\ln 2} \\ &= (3-3+1)\ln 2 = \ln 2 > 0. \end{aligned} \tag{56.109}$$

The tripartite information is strictly *positive*: GHZ entanglement is shared genuinely three ways, in flat violation of Eq. (56.103). By Theorem 56.11 this state cannot be the boundary entanglement structure of three regions in a classical holographic geometry. The physical reading is sharp: holographic entanglement is overwhelmingly *bipartite*. The maximally tripartite GHZ pattern, in which no two parties are correlated without the third, has no smooth geometric dual; the bulk knows only how to weave pairwise entanglement into geometry. This single number, $I_3 = +\ln 2$, is a certificate that $|\mathrm{GHZ}\rangle$ lives outside the holographic corner of Hilbert space.

56.3.5 The holographic entropy cone

Stand back from the two proofs. For a fixed number of regions n, a quantum state assigns an entropy to each of the $2^n - 1$ non-empty subsets, producing an *entropy vector* $\vec{S} = (S(A), S(B), \dots, S(ABC), \dots) \in \mathbb{R}^{2^n-1}$. The set of entropy vectors realizable by *some* quantum state is a convex cone (closure under scaling by replication of the state and under addition by tensoring), the *quantum entropy cone*; it is cut out by subadditivity, the Araki–Lieb triangle inequality, and strong subadditivity, together with whatever further inequalities quantum mechanics imposes. The set realizable by *holographic* states — those with a classical geometric dual obeying Eq. (56.4) — is a smaller convex cone sitting inside it, the *holographic entropy cone*. Our two theorems are statements about this nesting: SSA is a facet of the big cone that the small one inherits, while MMI is a new facet that appears only on the small cone, slicing off the region (containing the GHZ vector of Theorem 56.12) where quantum mechanics is happy but geometry is not.

Theorem 56.13 (Holographic entropy cone, structural statement)**.** *The entropy vectors of static classical-geometry holographic states (in the RT approximation) form a polyhedral cone in $\mathbb{R}^{2^n-1}$, defined by finitely many linear inequalities with integer coefficients. For $n \le 3$ regions the defining inequalities are subadditivity, Araki–Lieb, strong subadditivity, and monogamy of mutual information. For larger n further independent inequalities appear; the complete list is known for $n \le 5$ regions, and partial results exist beyond.*

We have proved the two inequalities that distinguish the holographic cone from the quantum cone at the three-party level; the proof that the holographic region is *polyhedral* (finitely many facets) and the determination of its facets for $n = 4, 5$ are carried out by extending the recutting argument of Theorem 56.9 to more elaborate Boolean designs, together with a complementary construction realizing each facet by an explicit geometry. We state Theorem 56.13 as the organizing result and do not reproduce the full $n = 5$ facet enumeration, which is a finite but large computation; the references in the further reading carry it out. What matters for us is the conceptual content, which the two proofs already establish in full and which we now record.

Physical Insight

The lesson of this section is that *holographic states are special*. Strong subadditivity holds for every state in the universe; monogamy of mutual information does not, and yet every classical-geometry holographic state obeys it. The space of states dual to smooth bulk geometry is therefore a proper, geometrically carved subset of the CFT Hilbert space, bounded by the facets of the holographic entropy cone. A state that violates MMI — a GHZ-like state, a generic random state, a CFT state with too much genuinely multipartite entanglement — cannot be described by a classical bulk geometry at all; if it has a bulk dual, that dual is intrinsically quantum, with no sharp metric. This is the first precise sense in which "not every state is geometric," and it reframes the central question of holography. Ryu–Takayanagi is not merely consistent with quantum mechanics; it is *sharper* than quantum mechanics, encoding extra structure that marks out which states can be spacetime. The natural next demand is to understand where that extra structure comes from — to stop postulating Eq. (56.4) and derive it.

Caution

The inequalities of this section are proved for *static* states, where RT entropy is the area of a minimal surface on a fixed slice and the homology regions are honest subsets of one Riemannian manifold. In genuinely time-dependent states the relevant surfaces are the extremal (Hubeny–Rangamani–Takayanagi) surfaces of Section 56.5.2, which need not lie on any common slice, and the recutting lemma does not apply verbatim: monogamy in particular requires a separate argument (it does still hold, by the maximin reformulation of extremal surfaces, but that proof is genuinely harder and is not the cut-and-paste above). Strong subadditivity also survives in the covariant setting. We have given the cleanest proofs, which are the static ones; the covariant extensions are flagged here and discussed

where the covariant prescription is developed.

We have shown that RT geometry does not merely reproduce known entanglement entropies but *predicts* relations among them — strong subadditivity for free, and monogamy of mutual information as a genuinely new constraint that confines holographic states to a special corner of Hilbert space. That a postulated geometric rule should automatically satisfy a deep operator inequality, and impose a true one beyond it, is strong structural evidence that the rule is not an approximation guessed by analogy but a theorem waiting to be proved. The next section discharges exactly this debt: it derives the Ryu–Takayanagi formula from the bulk gravitational path integral by the replica method, identifying the minimal surface as the fixed-point locus of the replica symmetry and the homology constraint as a regularity condition, and so converts the proposal we have been testing into a theorem.

56.4 Deriving Ryu–Takayanagi: the bulk replica trick

Everything up to this point has treated the Ryu–Takayanagi formula Eq. (56.4) as a proposal: stated precisely, validated against the one entanglement entropy we already knew exactly, turned into a calculational engine, and shown to imply — from pure geometry — entropy inequalities that are painful to extract from field theory directly. A proposal that passes this many tests is more than a guess, but it is still a guess. It was written down by analogy to the black-hole area law and confirmed by its consequences; it was not derived from the defining statement of the duality. We now owe the reader that derivation. The goal of this section is to promote Theorem 56.3 from a checked conjecture to a theorem, by computing the entanglement entropy of a holographic state from first principles — the replica trick of Chapter 55 applied not to the boundary field theory but to the bulk gravitational path integral — and watching the area of a bulk surface emerge on its own, with the homology constraint appearing as nothing more exotic than the demand that a bulk geometry be smooth.

The strategy is the one Lewkowycz and Maldacena introduced in 2013, and it is beautifully economical. The replica trick computes $\operatorname{Tr}\rho_A^n = Z_n/Z_1^n$, where Z_n is the partition function on an n-sheeted branched cover of the boundary spacetime (Theorem 55.17). For a holographic theory each Z_n is computed by a bulk gravitational saddle through the GKP–Witten relation (Eqs. (53.162) and (53.163)). So the entire replica computation can be done upstairs, in the bulk, where it is classical geometry. The branched cover carries a cyclic replica symmetry; assuming the dominant bulk filling respects it, we may quotient by that symmetry and study a single bulk geometry with one boundary copy and a conical defect along a codimension-two surface. The $n \to 1$ limit that defines the von Neumann entropy then localizes the entire computation onto that

surface, returning precisely its area over $4G_N$. Carrying this out — with the conical-deficit action computed honestly, the factor of $4G_N$ pinned down, and the extremality and homology of the surface derived rather than assumed — is the business of this section.

56.4.1 The replica trick in the bulk

We begin by recalling the boundary side of the replica construction and translating it into the bulk. Fix a spatial region A in the boundary CFT_d and let ρ_A be the reduced density matrix of the state on A. The von Neumann entropy $S(A) = -\operatorname{Tr}(\rho_A \ln \rho_A)$ is recovered from the integer moments $\operatorname{Tr}\rho_A^n$ by the replica master formula (Theorem 55.17): one writes $\operatorname{Tr}\rho_A^n = Z_n/Z_1^n$, where Z_n is the Euclidean partition function of the theory on the n-fold branched cover $\mathcal{M}_n$ — n copies of the Euclidean boundary geometry, cut open along A and glued cyclically across the cut, sheet k joined to sheet $k+1$ and sheet n back to sheet 1 — and Z_1 is the partition function on the original space. The entropy is then the analytic continuation

$$S(A) = -\partial_n \left(\ln Z_n - n \ln Z_1 \right) \Big|_{n=1} , \tag{56.110}$$

which is Eq. (55.52). It will be convenient to package this in a slightly different but equivalent form. Define the *modular entropy*

$$\tilde{S}_n \;\equiv\; -n^2\, \partial_n \left[\frac{1}{n} \ln \frac{Z_n}{Z_1^n} \right] , \tag{56.111}$$

a quantity built so that its $n \to 1$ value is exactly the von Neumann entropy. To see this, set $g(n) \equiv \ln(Z_n/Z_1^n) = \ln Z_n - n \ln Z_1 = \ln \operatorname{Tr}\rho_A^n$, which vanishes at $n = 1$ because $\operatorname{Tr}\rho_A = 1$. Then $\frac{1}{n} g$ has derivative $\partial_n(g/n) = -g/n^2 + g'/n$, so $-n^2 \partial_n(g/n) = g - n g'$, and at $n = 1$,

$$\tilde{S}_1 = g(1) - g'(1) = 0 - \partial_n \ln \operatorname{Tr}\rho_A^n \big|_{n=1} = -\operatorname{Tr}(\rho_A \ln \rho_A) = S(A). \tag{56.112}$$

The modular entropy is the natural object for the bulk computation because the combination $\frac{1}{n}\ln(Z_n/Z_1^n)$ is, as we are about to see, minus the gravitational action *per replica* — the action of a single fundamental domain of the symmetric bulk geometry. Reading Eqs. (56.111) and (56.112) backwards, the entanglement entropy is

$$S(A) = -\,\partial_n \left[\frac{1}{n} \ln \frac{Z_n}{Z_1^n} \right] \Bigg|_{n=1} , \tag{56.113}$$

since the prefactor n^2 in Eq. (56.111) is unity at $n = 1$. This is the formula we will evaluate in the bulk.

Now the bulk. For a holographic CFT in the classical-gravity regime — large central charge, large 't Hooft coupling, so that the genus-zero bulk saddle

dominates the gravitational path integral (Section 53.1) — the GKP–Witten relation Eq. (53.162) together with its classical limit Eq. (53.163) computes each partition function as the on-shell gravitational action of the dominant smooth bulk solution whose conformal boundary is the prescribed boundary geometry:

$$Z_n \;=\; Z_{\mathrm{grav}}[\mathcal{M}_n] \;\approx\; e^{-I[\mathcal{B}_n]}, \tag{56.114}$$

where $\mathcal{B}_n$ is the dominant *smooth* Euclidean solution of the bulk equations of motion with $\partial\mathcal{B}_n = \mathcal{M}_n$, and $I[\mathcal{B}_n]$ is the renormalized gravitational action evaluated on it. The bulk action is Einstein–Hilbert with negative cosmological constant, the Gibbons–Hawking boundary term at the conformal boundary, and the holographic counterterms that render it finite (Theorem 54.5); in the mostly-minus conventions of this book, after Wick rotation to the Euclidean section (Eq. (53.161)),

$$I[\mathcal{B}] = -\frac{1}{16\pi G_N}\int_{\mathcal{B}} \mathrm{d}^{d+1}x\,\sqrt{g}\,(R-2\Lambda) - \frac{1}{8\pi G_N}\int_{\partial\mathcal{B}} \mathrm{d}^{d}x\,\sqrt{h}\,K + I_{\mathrm{ct}}, \qquad \Lambda = -\frac{d(d-1)}{2L^2}. \tag{56.115}$$

The decisive structural point, which is the whole reason the bulk replica trick is tractable, is that $\mathcal{B}_n$ is genuinely *smooth*: it caps off in the interior with no conical singularity, no boundary, and no curvature delta-functions. The branch points of $\mathcal{M}_n$ sit on the boundary, at ∂A; the bulk solution fills them in smoothly. This is the holographic version of the statement that gluing n copies of a Euclidean field theory across a cut produces a perfectly ordinary manifold. We record the package as the first half of the central theorem.

Theorem 56.14 (Lewkowycz–Maldacena: the bulk replica)**.** *Let A be a region in a holographic CFT_d in a state with a classical bulk dual, and let $\mathcal{M}_n$ be the n-fold replica branched cover of the boundary Euclidean geometry. In the classical-gravity regime the replica partition function is computed by the on-shell gravitational action Eq. (56.114) of the dominant smooth bulk solution $\mathcal{B}_n$ filling $\mathcal{M}_n$. If $\mathcal{B}_n$ preserves the $\mathbb{Z}_n$ replica symmetry of $\mathcal{M}_n$, then the entanglement entropy Eq. (56.113) equals*

$$S(A) = \frac{\mathrm{Area}(\gamma^*)}{4G_N}, \tag{56.116}$$

where γ^* is the bulk codimension-two surface left fixed by the replica symmetry, which is anchored on ∂A, homologous to A, and of extremal — in fact minimal — area. The remainder of this section proves the second sentence: it identifies γ^* with the fixed-point locus of the replica symmetry, shows that smoothness of $\mathcal{B}_n$ forces γ^* to be extremal and homologous to A, and computes the coefficient $1/4G_N$ from the conical-defect action.

Remark 56.15. The hypothesis that the dominant $\mathcal{B}_n$ preserves the replica symmetry is an assumption, and we flag it as such. It is the bulk avatar of the

statement that the boundary replica computation has no spontaneous breaking of the cyclic symmetry among sheets, and it can fail: when it does, the symmetric saddle is subdominant and the entropy is computed by a different, replica-symmetry-breaking geometry. Such configurations are exactly what produce the entanglement phase transitions and, ultimately, the islands of Chapter 57. For the derivation of the basic RT formula we assume the symmetric saddle dominates; this is correct whenever the minimal surface is unique and the state is close to the vacuum, and it is the generic situation away from phase-transition loci.

56.4.2 Replica symmetry and the conical quotient

The branched cover $\mathcal{M}_n$ has a manifest symmetry: cyclic permutation of the n sheets, $k \mapsto k+1 \pmod n$. This $\mathbb{Z}_n$ acts freely away from the branch locus ∂A, where all sheets meet and the action has fixed points. Granting the hypothesis of Theorem 56.14, the dominant bulk filling $\mathcal{B}_n$ inherits this $\mathbb{Z}_n$ as an isometry. Because the symmetry acts freely on the boundary away from ∂A, its only bulk fixed points form a codimension-two surface $\tilde{\gamma}_n$ in the interior — the bulk continuation of the boundary branch locus. We now pass to the quotient

$$\hat{\mathcal{B}}_n \;\equiv\; \mathcal{B}_n/\mathbb{Z}_n. \tag{56.117}$$

Its conformal boundary is a single copy of the original boundary geometry, $\partial\hat{\mathcal{B}}_n = \mathcal{M}_1$, because the n boundary sheets are identified down to one. In the interior, the quotient is smooth everywhere except along the image γ_n of the fixed-point locus, where it develops a conical defect. The reason is purely local and is the geometric heart of the whole argument.

Near the fixed-point surface the smooth cover $\mathcal{B}_n$ looks, in the two transverse directions, like an ordinary flat plane: in polar coordinates (ρ,τ) centred on $\tilde{\gamma}_n$, with ρ the proper transverse distance and τ the angle, the transverse metric is $\mathrm{d}\rho^2+\rho^2\mathrm{d}\tau^2$ with τ running over its full smooth range $\tau\in[0,2\pi)$. The replica symmetry rotates this transverse plane: the generator of $\mathbb{Z}_n$ acts as $\tau\mapsto\tau+2\pi/n$. Quotienting identifies $\tau\sim\tau+2\pi/n$, so on $\hat{\mathcal{B}}_n$ the transverse angle runs only over a range $2\pi/n$. Equivalently, rescaling to a 2π-periodic angular coordinate θ by $\tau=\theta/n$, the quotient transverse metric is

$$\mathrm{d}s_\perp^2 = \mathrm{d}\rho^2 + \frac{\rho^2}{n^2}\,\mathrm{d}\theta^2, \qquad \theta\in[0,2\pi), \tag{56.118}$$

a cone with total angle $2\pi/n$ at the tip and hence a *deficit angle*

$$\delta = 2\pi\Big(1-\tfrac{1}{n}\Big). \tag{56.119}$$

At $n=1$ there is no deficit and $\hat{\mathcal{B}}_1=\mathcal{B}_1$ is the original smooth bulk; as n grows the cone sharpens. We name this structure for later reference.

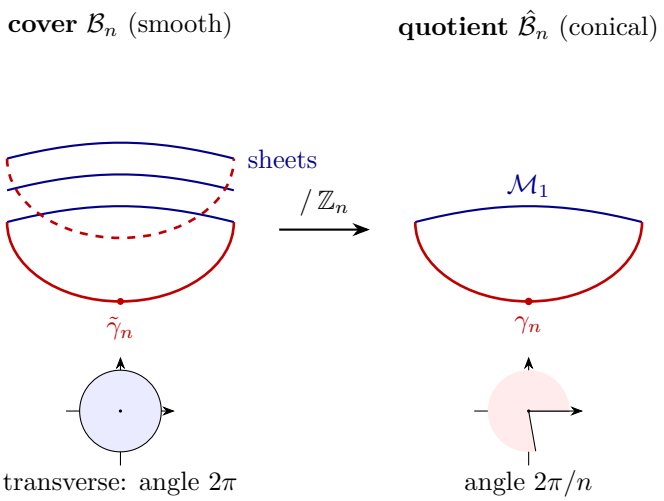


Figure 56.6: The bulk replica trick. Left: the smooth bulk solution $\mathcal{B}_n$ filling the n-sheeted boundary cover $\mathcal{M}_n$ (three sheets shown), capping off smoothly at the interior fixed-point surface $\tilde{\gamma}_n$; in the two directions transverse to $\tilde{\gamma}_n$ the geometry is a smooth plane of total angle 2π. Right: quotienting by the replica symmetry $\mathbb{Z}_n$ gives $\hat{\mathcal{B}}_n$, with a single boundary copy $\mathcal{M}_1$ and a conical defect of total transverse angle $2\pi/n$ — deficit angle $2\pi(1-1/n)$ — along the replica surface γ_n. The $n \to 1$ limit collapses the defect and leaves the Ryu–Takayanagi surface γ^*.

Definition 56.16 (The conical quotient and the replica surface)**.** The *conical quotient* is the bulk geometry $\hat{\mathcal{B}}_n = \mathcal{B}_n/\mathbb{Z}_n$ of Eq. (56.117): a smooth solution of the bulk equations of motion away from a codimension-two locus γ_n, with conformal boundary the original spacetime $\mathcal{M}_1$ and a conical defect of deficit angle Eq. (56.119) along γ_n. The locus γ_n — the image of the fixed-point set of the replica symmetry — is the *replica surface*. Its $n \to 1$ limit, $\gamma^* \equiv \gamma_1$, is the surface that Theorem 56.14 identifies with the Ryu–Takayanagi surface.

Figure 56.6 collects the construction. Two consequences of the smoothness of the *cover* are worth isolating immediately, because they are exactly the two features of Eq. (56.4) that looked like extra postulates.

First, the homology constraint is automatic. For $\mathcal{B}_n$ to be a smooth manifold whose only fixed-point set under $\mathbb{Z}_n$ is $\tilde{\gamma}_n$, the replica circle — the orbit of the $\mathbb{Z}_n$ action, which on the boundary is the angular direction around ∂A — must shrink to zero size precisely at $\tilde{\gamma}_n$ and nowhere else, capping off the geometry there like the origin of polar coordinates. Downstairs, this says that γ_n together with the boundary region A bounds a bulk region: the region swept out as the replica circle contracts from the boundary to the tip is exactly the homology region $\mathcal{R}_A$ of Theorem 56.2, with $\partial\mathcal{R}_A = A \cup \gamma_n$. A surface not homologous to

A would correspond to a replica circle that fails to close off smoothly — either leaving an uncapped non-contractible cycle or forcing a second conical locus — and the corresponding bulk geometry would not be a smooth filling of $\mathcal{M}_n$, hence would not contribute to Z_n. Homology is the condition that the cover cap off smoothly, no more and no less.

Second, the surface is anchored on ∂A. The fixed-point set $\tilde{\gamma}_n$ continues the boundary branch locus ∂A into the bulk; as one approaches the conformal boundary the replica circle becomes the boundary angular coordinate around ∂A, so γ_n meets the boundary exactly at ∂A, $\partial\gamma_n = \partial A$, which is the anchoring Eq. (56.1) of Theorem 56.1.

What remains is to extract the entropy: to evaluate Eq. (56.113) on the conical quotient and read off the area, and to show that smoothness pins the *location* of γ_n to an extremal surface.

56.4.3 The on-shell action of the conical defect

The action of the smooth cover is, by locality of the gravitational Lagrangian, exactly n times the action of one fundamental domain of the $\mathbb{Z}_n$ action. A fundamental domain is a wedge of angular width 2π in the transverse plane, and its action is precisely the action of the quotient $\hat{\mathcal{B}}_n$ computed by integrating the smooth bulk Lagrangian over the geometry away from the tip — with *no* conical contribution added at γ_n, since on the cover that locus is an ordinary smooth point. Denote this "per-replica" action by

$$\hat{I}_n \;\equiv\; \frac{1}{n}\, I[\mathcal{B}_n] \;=\; -\frac{1}{16\pi G_N}\int_{\hat{\mathcal{B}}_n}\sqrt{g}\,(R-2\Lambda) - \frac{1}{8\pi G_N}\int_{\partial_{\mathrm{AdS}}}\sqrt{h}\,K + I_{\mathrm{ct}}, \tag{56.120}$$

the integral running over $\hat{\mathcal{B}}_n$ with the tip $\rho = 0$ included as an ordinary interior point (the cover is smooth there, so the curvature is bounded and the tip is a set of measure zero). Combining $\ln Z_n = -I[\mathcal{B}_n] = -n\hat{I}_n$ with Eq. (56.113),

$$\frac{1}{n}\ln\frac{Z_n}{Z_1^n} = \frac{1}{n}\ln Z_n - \ln Z_1 = -\hat{I}_n + \hat{I}_1, \qquad \Longrightarrow \qquad S(A) = \partial_n \hat{I}_n\big|_{n=1}. \tag{56.121}$$

The entanglement entropy is the rate of change of the per-replica gravitational action with the replica number, evaluated at $n = 1$. This is a clean, finite quantity: $\hat{I}_1$ is the ordinary free energy of the original spacetime, and its n-derivative isolates the contribution of the infinitesimal conical defect.

We now compute $\partial_n \hat{I}_n$. The key observation is that $\hat{\mathcal{B}}_n$, for every n, solves the bulk equations of motion away from the tip, with the *same* conformal boundary data $\mathcal{M}_1$ held fixed as n varies. Under an arbitrary variation of the bulk metric the action changes by a bulk term proportional to the equations of motion plus boundary terms. As n varies, the bulk metric of $\hat{\mathcal{B}}_n$ varies, but the bulk equations of motion hold throughout, so that contribution vanishes; the boundary term at

the conformal boundary vanishes too, because the boundary data is fixed and the counterterms make the variational principle well posed there (Theorem 54.5). The only surviving piece is a boundary term on an infinitesimal tube around the tip, where the conical deficit lives and where the geometry's explicit n-dependence is concentrated. We compute that tube term directly.

Worked Example 56.17 (The conical-defect action and the factor of $4G_N$)**.** We isolate the tip contribution to $\partial_n \hat{I}_n$ by excising a tube of transverse radius ρ_0 around γ_n, computing the boundary term the variation deposits on it, and taking $\rho_0 \to 0$.

Geometry near the tip. Adopt coordinates (ρ, θ, y^i) near γ_n, with ρ the proper transverse radius, $\theta \in [0, 2\pi)$ the rescaled transverse angle, and y^i $(i = 1, \ldots, d-1)$ coordinates along γ_n with induced metric $\gamma_{ij}(y)$. To the order we need — and using the extremality established in Section 56.4.4, which kills the would-be cross terms linear in ρ — the metric is

$$\mathrm{d}s^2 = \mathrm{d}\rho^2 + b^2\rho^2\,\mathrm{d}\theta^2 + \gamma_{ij}(y)\,\mathrm{d}y^i\mathrm{d}y^j + O(\rho^2), \qquad b \equiv \frac{1}{n}, \tag{56.122}$$

so that $\sqrt{g} = b\,\rho\,\sqrt{\gamma}\,[1 + O(\rho^2)]$. Varying n at fixed coordinates varies only the angular component through b: $\delta b = -\,n^{-2}\,\delta n$.

The gravitational surface term. Varying the Einstein–Hilbert bulk integrand produces, after the standard double integration by parts, a total derivative $\partial_\mu(\sqrt{g}\,v^\mu)$ with $v^\mu = g^{\alpha\beta}\delta\Gamma^\mu_{\alpha\beta} - g^{\mu\beta}\delta\Gamma^\alpha_{\alpha\beta}$. On-shell in the bulk this is the entire variation; integrated over the region exterior to the tube it becomes a flux through the tube wall $\rho = \rho_0$,

$$\delta\hat{I}_n = -\frac{1}{16\pi G_N}\oint_{\rho=\rho_0} \mathrm{d}\theta\,\mathrm{d}^{d-1}y\,\sqrt{h}\;n_\mu v^\mu, \tag{56.123}$$

with $\sqrt{h} = b\rho_0\sqrt{\gamma}$ the induced measure on the wall and $n_\mu = -\,\delta^\rho_\mu$ the unit normal pointing out of the exterior region (towards the tip). The Christoffel symbols of Eq. (56.122) that respond to δb are governed by $\Gamma^\rho_{\theta\theta} = -b^2\rho$, whose variation is $\delta\Gamma^\rho_{\theta\theta} = -2b\rho\,\delta b$. Hence

$$v^\rho = g^{\theta\theta}\,\delta\Gamma^\rho_{\theta\theta} = \frac{1}{b^2\rho^2}\,(-2b\rho\,\delta b) = -\frac{2\,\delta b}{b\,\rho}, \tag{56.124}$$

the remaining contributions to v^ρ vanishing because $\Gamma^\alpha_{\alpha\rho} = \partial_\rho \ln\sqrt{g} = 1/\rho$ carries no b-dependence and the γ-direction Christoffels are ρ-independent at this order.

Assembling the tube term. Insert Eq. (56.124) and $n_\mu v^\mu = n_\rho v^\rho = (-1)\, v^\rho = +2\,\delta b/(b\rho_0)$ into Eq. (56.123):

$$\delta \hat{I}_n = -\frac{1}{16\pi G_N}\int_0^{2\pi} \mathrm{d}\theta \int \mathrm{d}^{d-1}y\, (b\rho_0\sqrt{\gamma})\, \frac{2\,\delta b}{b\,\rho_0} = -\frac{1}{16\pi G_N}\,(2\pi)\,(2\,\delta b)\,\mathrm{Area}(\gamma_n), \tag{56.125}$$

where $\mathrm{Area}(\gamma_n) = \int \mathrm{d}^{d-1}y\,\sqrt{\gamma}$ and the factors of b and ρ_0 cancel, so the limit $\rho_0 \to 0$ is finite and trivial. Simplifying,

$$\delta \hat{I}_n = -\frac{\mathrm{Area}(\gamma_n)}{4G_N}\,\delta b = +\frac{\mathrm{Area}(\gamma_n)}{4G_N}\,\frac{\delta n}{n^2}, \tag{56.126}$$

using $\delta b = -n^{-2}\delta n$. Therefore

$$\partial_n \hat{I}_n = \frac{\mathrm{Area}(\gamma_n)}{4G_N\, n^2}, \qquad \text{and in particular} \qquad \partial_n \hat{I}_n\big|_{n=1} = \frac{\mathrm{Area}(\gamma_1)}{4G_N}. \tag{56.127}$$

The coefficient $1/4G_N$ has emerged from nothing but the Einstein–Hilbert normalization $1/16\pi G_N$ and the geometry of a cone: the 4π from $2\pi \times 2$ in Eq. (56.125) converts $16\pi G_N$ into $4G_N$. No black-hole input was used; the same quarter-area-in-Planck-units that Theorem 54.10 attaches to a horizon is forced here by the gravitational action of an arbitrarily small conical defect.

Theorem 56.18 (Conical-defect action)**.** *The per-replica gravitational action Eq. (56.120) of the conical quotient satisfies* $\partial_n \hat{I}_n = \mathrm{Area}(\gamma_n)/4G_N n^2$*, where* γ_n *is the replica surface. Consequently, by Eq. (56.121),*

$$S(A) = \partial_n \hat{I}_n\big|_{n=1} = \frac{\mathrm{Area}(\gamma_1)}{4G_N} = \frac{\mathrm{Area}(\gamma^*)}{4G_N}. \tag{56.128}$$

Physical Insight

The same computation has a one-line repackaging that has become the standard modern language (Dong). A codimension-two object of tension μ sources, through Einstein's equations, a conical deficit $\delta = 8\pi G_N \mu$ along its worldvolume — this is the higher-dimensional cosmic string. Comparing with Eq. (56.119), the conical quotient $\hat{\mathcal{B}}_n$ is exactly the back-reacted geometry of a *cosmic brane* of tension

$$\mu_n = \frac{\delta}{8\pi G_N} = \frac{1}{4G_N}\Big(1 - \frac{1}{n}\Big), \tag{56.129}$$

whose Nambu–Goto action is $\mu_n\,\mathrm{Area}(\gamma_n)$. The brane's tension vanishes at $n = 1$ and grows with n, and its equation of motion — extremize the area in the back-reacted geometry — is what locates γ_n. Differentiating the on-shell action with respect to n, only the explicit tension factor survives the equations of motion, giving $\partial_n \hat{I}_n = (\partial_n \mu_n)\,\mathrm{Area}(\gamma_n) = \mathrm{Area}(\gamma_n)/4G_N n^2$,

in agreement with Eq. (56.127). The brane is a fiction at integer n — there is no brane in the smooth cover — but it is the cleanest bookkeeping device for the conical geometry, and it makes the analytic continuation in n transparent: the modular entropy $\tilde{S}_n = \mathrm{Area}(\gamma_n)/4G_N$ is the area of the brane held in place by tension μ_n, and the von Neumann entropy is its zero-tension limit.

56.4.4 Extremality and the homology constraint

Two threads remain to be tied. We used in Theorem 56.17 that the near-tip metric Eq. (56.122) has no cross terms linear in ρ; that absence is the statement that γ^* is an extremal surface, and we now derive it. And we must establish that the analytic continuation in n that produces Eq. (56.128) is legitimate — that the location of γ_n varies smoothly to a well-defined γ^* — and that the dominant saddle selects the minimal such surface.

Smoothness of the cover forces extremality. Write the most general bulk metric near the replica surface in the quotient, allowing the transverse cone to be fibred nontrivially over γ_n. In complex transverse coordinate $w = \rho\, e^{i\theta}$ (so the cone is $|w|^{2/n}$-type), the leading behaviour compatible with a codimension-two locus at $w = 0$ is

$$\mathrm{d}s^2 = \frac{1}{n^2}\, e^{2\Phi}\, \mathrm{d}w\, \mathrm{d}\bar{w} + \Big(\gamma_{ij} + 2K^{w}_{ij}\, w + 2K^{\bar{w}}_{ij}\, \bar{w} + \dots\Big)\mathrm{d}y^i \mathrm{d}y^j + (\text{cross terms}), \tag{56.130}$$

where K^{w}_{ij} and $K^{\bar{w}}_{ij}$ are the two components of the extrinsic curvature of γ_n, one for each transverse direction, and $\Phi \to$ const at the tip. The cover $\mathcal{B}_n$ is obtained by undoing the quotient, $w \to w^{1/n}$ up to the angular rescaling; concretely the cover sees the transverse coordinate $\tilde{w}$ with $w \sim \tilde{w}^{\,n}$, restoring the full 2π angle. For the cover to be a *smooth* solution of Einstein's equations — no curvature singularity beyond the (absent, on the cover) conical one — the components of the Einstein tensor must remain finite as $\tilde{w} \to 0$.

Track the most singular piece. The terms $K^{w}_{ij} w$ in Eq. (56.130), pulled back to the cover, scale as $\tilde{w}^{\,n}$, but their contribution to the curvature involves derivatives $\partial_{\tilde{w}}$ acting on the inverse metric and produces, at order $1/n$ in the deficit, a term in the $(\theta\theta)$ or trace Einstein equation that behaves as

$$G_{\theta\theta} \;\sim\; (n-1)\,\frac{K^i\,\cos\theta}{\rho} + \dots, \qquad K^i \equiv \gamma^{jk} K^i_{jk}, \tag{56.131}$$

a $1/\rho$ divergence proportional to the *trace* of the extrinsic curvature in each transverse direction. There is nothing in the smooth bulk stress tensor to source such a divergence: the right-hand side of Einstein's equations is the regular bulk

matter (or, in pure gravity, the cosmological constant), finite at the tip. The only way to satisfy the equations of motion as $\rho \to 0$ is to set the offending coefficient to zero,

$$K^i = 0 \qquad \text{(both transverse directions)}, \tag{56.132}$$

for every $n > 1$. This is precisely the condition that γ_n be an *extremal surface*: the trace of its extrinsic curvature vanishes, so it extremizes the area functional Eq. (56.2). Smoothness of the cover does not merely permit the surface to sit at an extremum of the area — it requires it. Taking $n \to 1$, the limiting surface γ^* is an extremal surface in the original bulk $\mathcal{B}_1$.

Theorem 56.19 (The replica surface is extremal)**.** *For the quotient $\hat{\mathcal{B}}_n$ to descend from a smooth bulk solution $\mathcal{B}_n$, the replica surface γ_n must have vanishing trace of extrinsic curvature, $K^i = 0$, in both directions transverse to it. Hence its $n \to 1$ limit γ^* is an extremal surface of the area functional, anchored on ∂A and homologous to A. The entanglement entropy Eq. (56.128) is the area of this extremal surface over $4G_N$.*

This is the link the heuristic of Section 56.1 promised but could not supply: extremizing the gravitational action over the bulk geometry is, at the location of the defect, identical to extremizing the area over the position of the surface. The area functional is not imposed by hand; it is the conical-defect part of the gravitational action, and stationarity of that action under moving the defect is stationarity of the area.

Minimality and the meaning of "min". Theorem 56.19 produces an extremal surface, but Eq. (56.4) asks for the *minimal* one. The selection is made by the path integral. The partition function Z_n is dominated by the saddle of least action, hence least $\hat{I}_n$; when several replica-symmetric fillings $\mathcal{B}_n$ exist, anchored on the same ∂A and homologous to A but with different replica surfaces, the dominant one carries the surface of least area, because by Eq. (56.120) a smaller area at given deficit means a smaller per-replica action. Continuing to $n = 1$, the entropy is the area of the least-area extremal surface. When two extremal surfaces exchange dominance as the region is deformed, the dominant saddle jumps, and the entropy develops the kink of an entanglement phase transition — the same "min" that we met for two intervals now seen as a competition of bulk gravitational saddles. In a static, time-reflection-symmetric state the extremal surface lies in the symmetric slice and "extremal of least area" is literally "minimal," recovering Theorem 56.3 verbatim.

Homology, once more, as regularity. We argued in Section 56.4.2 that homology to A is the condition for the cover to cap off smoothly. Theorem 56.19 sharpens the statement and explains why *no extra ingredient* — no additional brane, no boundary term at the tip beyond what we computed — is needed. The

replica surface is where the replica circle contracts; the bulk closes there exactly as the origin closes the disk, sourced by nothing. Were the dominant surface not homologous to A, the contracting replica circle would fail to bound the region $\mathcal{R}_A$, the geometry could not close without an additional conical locus, and the configuration would not be a smooth filling — it would not solve Eq. (56.114). The homology constraint of Theorem 56.2, which entered Theorem 56.3 as a postulate forced on us by the demand $S(A) = S(\bar{A})$ and by thermal states, is therefore a theorem of the derivation: the dominant bulk saddle is smooth, and smoothness is homology.

56.4.5 Ryu–Takayanagi as a generalized-entropy statement

Assembling Theorems 56.18 and 56.19: the bulk replica computation, continued to $n = 1$, returns

$$S(A) = \frac{\mathrm{Area}(\gamma^*)}{4G_N}, \qquad \gamma^* = \text{ minimal extremal surface anchored on } \partial A, \text{ homologous to } A, \tag{56.133}$$

which is exactly the Ryu–Takayanagi formula Eq. (56.4), now derived from the defining gravitational path integral rather than proposed. Every ingredient that looked like an independent assumption — the quarter, the area, the extremality, the homology, even the "min" — has been produced by a single demand: that Z_n be computed by a smooth dominant bulk saddle. This completes the proof of Theorem 56.14.

It is worth restating the result in the form that organizes the rest of the chapter and the next. The quantity $\mathrm{Area}(\gamma)/4G_N$ is the Bekenstein–Hawking entropy of Theorem 54.10, no longer attached to a causal horizon but to an arbitrary extremal surface. Writing the boundary entanglement entropy as

$$S(A) = \frac{\mathrm{Area}(\gamma^*)}{4G_N} \;\equiv\; S_{\mathrm{gen}}(\gamma^*), \qquad S_{\mathrm{gen}}(\gamma) = \frac{\mathrm{Area}(\gamma)}{4G_N} + S_{\mathrm{bulk}}(\gamma)\Big|_{S_{\mathrm{bulk}}\to 0}, \tag{56.134}$$

exhibits RT as the leading, purely geometric piece of a *generalized entropy* S_{gen} — the same combination of horizon area and exterior matter entropy that Bekenstein introduced to repair the second law in the presence of black holes. At the classical order of this section the bulk-field term S_{bulk} is subleading in G_N and drops, leaving pure area; restoring it is the content of the next section.

That the entanglement entropy is a Bekenstein–Hawking area is not a coincidence of language but a dynamical identity. Varying Eq. (56.133) under a change of the boundary state, the change in entanglement entropy equals the change in the area of the extremal surface, $\delta S(A) = \delta\mathrm{Area}(\gamma^*)/4G_N$ — and the right-hand side is governed by the first law of black-hole mechanics, $\delta M = \frac{\kappa}{8\pi G_N}\,\delta\mathrm{Area}$ (Chapter 54), with the surface gravity κ playing the role of a local temperature. The boundary statement $\delta S = \delta\langle H_{\mathrm{mod}}\rangle$ — the first law of entanglement, relating

the change in entropy to the change in the modular Hamiltonian of Theorem 55.25 — is thereby translated into a gravitational first law for the bulk surface. This equivalence, that an entanglement first law on the boundary *is* a thermodynamic first law for a bulk area, is the seed from which the closing section of this chapter will extract the linearized Einstein equations of the bulk: gravity emerges as the condition that the holographic entanglement entropy obey the laws it must.

Two extensions are now forced upon us, and we have set them up precisely. The derivation worked at leading order in G_N, dropping the bulk fields' own entanglement across γ^*; reinstating S_{bulk} in Eq. (56.134) promotes RT to the Faulkner–Lewkowycz–Maldacena formula and, upon extremizing the full generalized entropy rather than the bare area, to the quantum extremal surface prescription (Sections 56.5.1 and 56.5.3). And the derivation assumed a Euclidean, time-reflection-symmetric setup so that "extremal" could mean "minimal on a slice"; a genuinely time-dependent state has no such slice, and the replica surface becomes a Lorentzian extremal surface — the covariant Hubeny–Rangamani–Takayanagi prescription (Section 56.5.2). The next section develops both, unifying them in the single instruction to extremize the generalized entropy, the prescription that Chapter 57 will turn on black holes to compute the Page curve.

56.5 Quantum corrections, covariance, and the quantum extremal surface

The Lewkowycz–Maldacena derivation of Section 56.4 did more than promote the Ryu–Takayanagi rule from proposal to theorem. By computing $\operatorname{Tr}\rho_A^n$ as a bulk gravitational path integral and tracking the $n \to 1$ limit, it told us *exactly which approximation* the area law is, and what sits at the next order. The bulk path integral is a sum over geometries *and* bulk matter configurations, $Z_n = \int \mathcal{D}g\,\mathcal{D}\phi\; e^{-I[g,\phi]}$; the saddle-point evaluation that localized the gravitational action onto the area of a conical-defect surface was the leading term in an expansion in Newton's constant G_N. The matter fields living on that saddle were, at leading order, spectators carried along for the ride. They are not spectators at the next order. Restoring them turns the bare area law into the *generalized entropy* — the same combination, area-plus-matter-entropy, that Bekenstein and the black-hole thermodynamics of Chapter 54 forced upon us for horizons — and pursuing the logic of the gravitational path integral to its conclusion replaces the classical minimal surface by a surface that extremizes generalized entropy. This section carries out that pursuit. It adds the one-loop quantum correction (the Faulkner–Lewkowycz–Maldacena, or FLM, formula), makes the prescription covariant so it applies to genuinely time-dependent states (the Hubeny–Rangamani–Takayanagi, or HRT, prescription), and unifies both into a single all-orders rule built on a single object, the quantum extremal surface

of Engelhardt and Wall. That rule is the instrument Chapter 57 will turn on a black hole.

We continue to work in asymptotically AdS_{d+1} with a holographic CFT_d in the classical-gravity regime, with G_N small in units of the AdS radius L (equivalently large central charge / large N, Eq. (53.153)); the bulk Newton constant G_N and the radial cutoff identified with the QFT ultraviolet cutoff are exactly as fixed in Section 56.1.2. The metric is mostly-minus and units are natural throughout.

56.5.1 The one-loop correction: the FLM formula

Counting in powers of G_N

The classical area term in Eq. (56.4) is enormous. The area of a bulk surface is a geometric quantity of order L^{d-1}, so $\mathrm{Area}/4G_N \sim L^{d-1}/G_N$, which by the holographic dictionary (Theorem 53.27) scales as the central charge $C_T \sim N^2$. The entanglement entropy of a holographic CFT is therefore $O(N^2)$ at leading order — it counts the $O(N^2)$ fields of the boundary theory — and the RT formula captures this leading piece geometrically. The first correction is $O(N^0) = O(G_N^0)$ relative to nothing, i.e. smaller than the leading term by a factor $G_N/L^{d-1} \sim 1/N^2$. The question is what physical quantity supplies it.

The answer is already visible in the structure of the bulk path integral. Expanding around the dominant saddle (g_n, ϕ_n),

$$\ln Z_n = -I_{\mathrm{grav}}[g_n] + \ln Z_{\mathrm{matter}}[g_n] + \dots , \tag{56.135}$$

where I_{grav} is the on-shell gravitational action (including the Gibbons–Hawking term and counterterms) and $Z_{\mathrm{matter}}[g_n]$ is the partition function of the bulk quantum fields propagating on the fixed background g_n. The first term is $O(1/G_N)$; the second is $O(G_N^0)$, a one-loop determinant from the reader's point of view. The replica entropy is extracted by the same operation in both terms,

$$S(A) = (1 - n\,\partial_n) \ln Z_n \Big|_{n=1}, \tag{56.136}$$

which follows from $S = -\partial_n \ln \mathrm{Tr}\,\rho_A^n|_{n=1}$ with $\mathrm{Tr}\,\rho_A^n = Z_n/Z_1^n$ (Theorem 55.17). Lewkowycz–Maldacena showed that applied to $-I_{\mathrm{grav}}[g_n]$ this operation localizes onto the area of the fixed-point surface and returns $\mathrm{Area}(\gamma_A)/4G_N$ (Eq. (56.4)). The new content is what the same operation does to $\ln Z_{\mathrm{matter}}[g_n]$.

The matter term is a bulk entanglement entropy

Here is the key observation. The replica geometry g_n is, by construction, an n-fold branched cover of the original bulk, branched over the surface γ_A: it is built by cyclically gluing n copies of the bulk across the homology region $\mathcal{R}_A$,

with the gluing locus the surface γ_A. The bulk matter fields live on this n-fold cover. But computing a partition function on an n-fold cover branched over a codimension-two surface, and then applying $(1 - n\partial_n)|_{n=1}$, is *precisely the replica trick for the bulk fields' own entanglement entropy* across that surface — the very construction of Theorem 55.17, now run in the bulk instead of on the boundary. The surface γ_A partitions the bulk Cauchy slice Σ into the homology region $\mathcal{R}_A$ and its complement; the bulk fields have a reduced density matrix $\rho_{\mathcal{R}_A}^{\text{bulk}}$ on $\mathcal{R}_A$; and

$$(1 - n\,\partial_n) \ln Z_{\text{matter}}[g_n]\Big|_{n=1} = -\operatorname{Tr}(\rho_{\mathcal{R}_A}^{\text{bulk}} \ln \rho_{\mathcal{R}_A}^{\text{bulk}}) \equiv S_{\text{bulk}}(\mathcal{R}_A), \tag{56.137}$$

the von Neumann entropy (Eq. (55.6)) of the bulk quantum fields restricted to the homology region. There is one subtlety to be honest about: Eq. (56.137) treats g_n as a rigid, externally-prescribed background, whereas the true saddle has the matter backreacting on the geometry order by order in G_N. At the order we are working — the first correction beyond the classical area — the backreaction of the $O(G_N^0)$ matter on the $O(1/G_N)$ geometry is higher order, so evaluating the bulk entropy on the unperturbed classical saddle is consistent. We will see in Section 56.5.3 that the leading backreaction is exactly what re-extremization absorbs.

Combining the two terms gives the FLM formula.

Theorem 56.20 (Faulkner–Lewkowycz–Maldacena: one-loop quantum correction)**.** *In the setting of Theorem 56.3, the entanglement entropy of the boundary region A, computed to first subleading order in G_N (one loop in the bulk, $O(N^0)$ relative to the $O(N^2)$ leading term), is*

$$\boxed{S(A) = \frac{\langle \widehat{\text{Area}}(\gamma_A) \rangle}{4G_N} + S_{\text{bulk}}(\mathcal{R}_A) + \langle \Delta S_{\text{Wald}} \rangle + S_{\text{ct}}\ ,} \tag{56.138}$$

where γ_A is the classical RT surface of Eq. (56.4), $\mathcal{R}_A$ is the homology region it bounds (Theorem 56.2), $S_{\text{bulk}}(\mathcal{R}_A)$ is the von Neumann entropy Eq. (56.137) of the bulk quantum fields on $\mathcal{R}_A$, and $\langle \widehat{\text{Area}} \rangle$ is the expectation value of the area operator in the bulk state. The remaining terms $\langle \Delta S_{\text{Wald}} \rangle$ and S_{ct} are the subleading geometric corrections (Wald-like contributions from higher-curvature terms in the bulk effective action, and counterterms that absorb the ultraviolet divergence of S_{bulk} into a renormalization of G_N); they are subleading and bookkeeping, and we discuss them below. The leading two terms,

$$S(A) = \frac{\text{Area}(\gamma_A)}{4G_N} + S_{\text{bulk}}(\mathcal{R}_A) + O(G_N), \tag{56.139}$$

are the operational content of the formula.

Derivation. Insert Eq. (56.135) into the extraction Eq. (56.136). The gravitational piece is the Lewkowycz–Maldacena result of Section 56.4: $(1-n\partial_n)(-I_{\rm grav}[g_n])|_{n=1} = \mathrm{Area}(\gamma_A)/4G_N$, with γ_A the fixed-point surface, which the same argument shows is the classical extremal (minimal) surface. Promoting the classical area to the expectation value of the area operator $\widehat{\mathrm{Area}}$ accounts for the quantum fluctuations of the geometry itself — the graviton is one of the bulk fields, and its contribution to the area is naturally grouped with the leading term as $\langle\widehat{\mathrm{Area}}\rangle$ rather than Area. The matter piece is Eq. (56.137): applying $(1 - n\partial_n)|_{n=1}$ to $\ln Z_{\rm matter}$ on the branched cover is the bulk replica computation, yielding $S_{\rm bulk}(\mathcal{R}_A)$. Higher-curvature terms in the bulk effective action modify the localized contribution from a pure area to the Wald entropy functional (the same functional that corrects the black-hole area law in Chapter 54); their replica derivative supplies $\langle\Delta S_{\rm Wald}\rangle$. Finally, $S_{\rm bulk}(\mathcal{R}_A)$ carries its own ultraviolet divergence from bulk modes piling up at γ_A; this divergence is local on γ_A and proportional to its area, so it is absorbed into a renormalization of the coefficient $1/4G_N$ multiplying the area — exactly the statement that the matter loops renormalize Newton's constant. The renormalized G_N and the renormalized (finite) $S_{\rm bulk}$ are what appear in Eq. (56.138); the residual scheme-dependent piece is $S_{\rm ct}$. Dropping the $O(G_N)$ and bookkeeping terms leaves Eq. (56.139). □

Physical Insight

The FLM formula is the holographic image of a fact the reader already accepts from black-hole thermodynamics. Bekenstein's generalized second law (Chapter 54) insists that the entropy charged to a horizon is not the bare area $A/4G_N$ but the *generalized* entropy $S_{\rm gen} = A/4G_N + S_{\rm out}$, the area term plus the ordinary entropy of the matter outside. FLM says the same combination computes the entanglement entropy of a boundary region: the geometric "area" piece counts the $O(N^2)$ degrees of freedom that built the geometry, and the bulk field entropy $S_{\rm bulk}$ counts the $O(1)$ fluctuations propagating on it. The two are not independent corrections to a more fundamental quantity — their sum *is* the fundamental quantity, and the splitting into "geometry" and "matter" is an artifact of the $1/N$ expansion. This is why the renormalization of G_N by bulk matter loops is not a nuisance but a consistency check: the divergence of $S_{\rm bulk}$ has to be exactly an area, because the only place for it to go is into G_N.

The modular split and the JLMS sharpening

The derivation above is honest but operates at the level of partition functions. The structural reason the area gets the leading term and the bulk von Neumann entropy the subleading term can be stated more sharply, in a form that will be proved in the next section and which we flag here so the logic is complete. The

boundary modular Hamiltonian $K_A = -\ln \rho_A$ generates the modular flow of the region A (Theorem 55.25); its bulk dual is not a single bulk operator but a sum,

$$K_A \;=\; \frac{\widehat{\mathrm{Area}}(\gamma_A)}{4G_N} \;+\; K^{\mathrm{bulk}}_{\mathcal{R}_A} \;+\; \dots, \tag{56.140}$$

where $K^{\mathrm{bulk}}_{\mathcal{R}_A} = -\ln \rho^{\mathrm{bulk}}_{\mathcal{R}_A}$ is the *bulk* modular Hamiltonian of the fields on the homology region. This is the JLMS (Jafferis–Lewkowycz–Maldacena–Suh) relation. It is the rigorous backbone of FLM: taking the expectation value of Eq. (56.140) in the global state gives $\langle K_A\rangle = \langle\widehat{\mathrm{Area}}\rangle/4G_N + \langle K^{\mathrm{bulk}}\rangle$, and combining with the fact that the von Neumann entropy and the modular-Hamiltonian expectation value differ by a state-independent constant (so that variations of entropy equal variations of $\langle K\rangle$) reproduces Eq. (56.138) term by term — the area operator landing on the leading entropy and the bulk modular Hamiltonian on S_{bulk}. We derive Eq. (56.140) and exploit its consequence — the equality of bulk and boundary relative entropies, which yields bulk energy conditions — in Section 56.6 and Theorem 56.35. For the present section it is enough that the modular split Eq. (56.140) explains the term-by-term structure of Eq. (56.138): the operator that lives *on* the surface is the area, and the operator that lives in the *region* is the bulk modular Hamiltonian.

56.5.2 Covariance: the HRT prescription

Why "minimal" fails in time-dependent states

Everything so far has presumed a static state with a preferred constant-time slice on which "minimal area" is a well-posed Riemannian variational problem. Most states are not static: a CFT after a global quench, a collapsing star forming a black hole, two colliding shockwaves — these have no time-translation symmetry and no canonical slice. We need a prescription that does not depend on a choice of time.

The naive fix, "minimize the area over all spacelike surfaces in the Lorentzian spacetime," is empty. A spacelike codimension-two surface can always be deformed in a timelike direction to decrease its area, with no lower bound: tilting a surface element into the light cone shrinks its proper area, and one can tilt arbitrarily far. There is no minimal-area surface in Lorentzian signature. The cure is to extremize rather than minimize.

Definition 56.21 (Extremal surface and the HRT prescription)**.** A codimension-two spacelike surface γ_A in an asymptotically AdS_{d+1} spacetime, anchored on ∂A and homologous to A, is *extremal* if it is a stationary point of the area functional under all (spacelike and timelike) normal deformations. Equivalently, its trace of extrinsic curvature vanishes in both null directions normal to the surface,

$$\theta_{(+)} = \theta_{(-)} = 0, \tag{56.141}$$

where $\theta_{(\pm)} = h^{ab}\nabla_a n_b^{(\pm)}$ are the expansions of the two future-directed null congruences orthogonal to γ_A. The covariant Hubeny–Rangamani–Takayanagi prescription is

$$\boxed{S(A) = \min_{\gamma_A} \operatorname*{ext}_{\gamma_A\ \text{extremal}} \frac{\text{Area}(\gamma_A)}{4G_N}} \tag{56.142}$$

namely: among all extremal surfaces anchored on ∂A and homologous to A, take the one of least area (and, if more than one extremum is degenerate in area, the prescription resolves by the same least-area rule as RT).

The condition Eq. (56.141) is the Lorentzian replacement of $K = 0$. Writing the trace of extrinsic curvature along the two null normals as the two null expansions makes the geometry transparent and connects directly to the focusing arguments of Chapter 54: an extremal surface is one whose cross-sectional area is momentarily stationary along *both* orthogonal light sheets, the spacelike analogue of a bifurcation surface where ingoing and outgoing expansions both vanish. When a moment of time symmetry exists, the time-reflection symmetry forces $\theta_{(+)} = -\theta_{(-)}$ on the symmetric slice, so $\theta_{(\pm)} = 0$ reduces to the single condition $K = 0$ within the slice, and the HRT extremal surface coincides with the RT minimal surface. HRT contains RT.

The maximin reformulation

Extremality is the right local condition, but as a *global* prescription Eq. (56.142) has awkward features: a saddle of the area functional is harder to control than a minimum (existence, uniqueness, and the entropy inequalities are not manifest). Wall's maximin reformulation repairs this by trading the single Lorentzian extremization for a min over surfaces followed by a max over slices, each of which is a controlled problem.

Definition 56.22 (Maximin surface)**.** Let $D[A]$ be the boundary causal domain of A. For each achronal bulk Cauchy slice Σ of the bulk region bounded by $D[A]$ and its complement, let $\gamma_A(\Sigma)$ be the *minimal*-area surface on Σ anchored on ∂A and homologous to A — a well-posed Riemannian minimization on the fixed slice Σ. The *maximin surface* is the one obtained by then maximizing this minimal area over all slices,

$$\gamma_A^{\text{maximin}} = \arg\max_{\Sigma} \Big[\min_{\gamma \subset \Sigma} \text{Area}(\gamma) \Big], \tag{56.143}$$

and the maximin entropy is $\text{Area}(\gamma_A^{\text{maximin}})/4G_N$.

Theorem 56.23 (Maximin equals extremal)**.** *In a spacetime satisfying the null energy condition, the maximin surface of Eq. (56.143) exists, is spacelike, and is an extremal surface in the sense of Theorem 56.21; when several extremal surfaces compete, the maximin construction selects the least-area one. Hence the maximin prescription and the HRT prescription Eq. (56.142) agree.*

Idea of the proof. The content is local at the maximin surface, and the two optimizations pin the two null expansions. Fix the maximizing slice Σ_* and let γ be the minimal surface on it. *Minimality within* Σ_* is the statement that the area is stationary under deformations that stay in Σ_*, and that it is a genuine minimum (positive second variation) in those directions; the first-variation condition is that the trace of extrinsic curvature along the spatial normal within Σ_* vanishes. Decompose the two null normals as $n^{(\pm)} \propto u \pm s$, where u is the future timelike normal to Σ_* and s is the spacelike normal to γ within Σ_*. The spatial first-variation condition sets the combination $\theta_{(+)} + \theta_{(-)} \propto \theta_{(s)} = 0$: the expansion along s vanishes. The remaining combination $\theta_{(+)} - \theta_{(-)} \propto \theta_{(u)}$ is the rate of change of the area as the surface is pushed in the timelike direction u — that is, as the slice itself is tilted. *Maximality over slices* is precisely the statement that this rate vanishes: at the maximizing slice the area cannot be increased by deforming Σ, so $\theta_{(u)} = 0$. The two conditions together give $\theta_{(+)} = \theta_{(-)} = 0$, which is extremality Eq. (56.141). That the maximin surface is spacelike (not null or timelike) and that the max-over-slices is attained follow from the null energy condition through the focusing theorem of Chapter 54: focusing controls how the minimal area on a slice behaves as the slice is deformed, guaranteeing the maximization has an interior maximum rather than running off to a degenerate (null) configuration. The least-area selection among competing extrema is built into the outer "max" acting on areas that are themselves minima. The local extremality argument just given is the heart of the matter; we now make the construction complete — existence of both optimizers, spacelikeness, the representative-slice property, and the identification with the least-area extremal surface — carrying the focusing estimate explicitly, in Section 56.5.2. □

The maximin construction and its proof

The proof idea above pins the local geometry at the surface; what remains is to show that both optimizations are actually attained, that the surface they produce is spacelike rather than null, that it carries the *representative-slice* property the covariant entropy inequalities will use, and that it is exactly the least-area extremal surface of Theorem 56.21. We collect these into one theorem and prove it in full. The single physical input beyond Riemannian geometry is the null energy condition, and it enters through one inequality — the focusing (Raychaudhuri) estimate — which we display explicitly because it is doing all of the work. Throughout we assume the bulk is globally well behaved in the mild sense Wall isolated: it is AdS-*hyperbolic*, meaning it admits Cauchy slices for the bulk region W_A lying over the boundary causal domain $D[A]$, and the space of such achronal slices anchored on a fixed boundary cut is compact in the topology of uniform convergence. This is the covariant analogue of global hyperbolicity and holds for the physically relevant states (perturbations of the vacuum, collapsing and evaporating black holes, quenches); it is the structural

hypothesis that makes "maximize over slices" a well-posed problem rather than a formal manipulation.

Theorem 56.24 (Wall: existence and equivalence of the maximin surface)**.** *Let the bulk be* AdS*-hyperbolic and satisfy the null energy condition* $T_{ab}k^a k^b \geq 0$ *for all null* k^a*. Then the maximin surface* $\gamma_A^{\text{maximin}}$ *of Eq. (56.143) exists. It is a smooth spacelike codimension-two surface, anchored on* ∂A *and homologous to* A*. It is extremal in the sense of Eq. (56.141),* $\theta_{(+)} = \theta_{(-)} = 0$*. It has the* representative-slice property*: it is the minimal-area surface in its homology class on* every *bulk Cauchy slice of* W_A *that contains it. Among all extremal surfaces homologous to* A *it has the least area, so it coincides with the HRT surface of Eq. (56.142); and the entanglement wedge it bounds contains the causal wedge of* A*.*

Proof. Step 1: the inner minimization is well posed. Fix an achronal bulk Cauchy slice Σ of W_A. On Σ the induced metric is Riemannian, and we seek the least-area surface anchored on ∂A and homologous to A — that is, cobounding, together with A, a region $\mathcal{R} \subset \Sigma$. This is the standard Plateau problem in a homology class with fixed boundary. The area functional is bounded below by zero and lower semicontinuous; the homology constraint together with the anchoring on ∂A confines a minimizing sequence to a compact family (the homology region cannot escape to infinity without crossing the conformal boundary, which the anchoring forbids), so a minimizer $\gamma_A(\Sigma)$ exists and is smooth away from ∂A by elliptic regularity. Denote its area $\mathcal{A}(\Sigma) \equiv \text{Area}(\gamma_A(\Sigma))$. Because the minimizer is a stationary point of a Riemannian area functional, its in-slice mean curvature vanishes: writing s for the unit spacelike normal to $\gamma_A(\Sigma)$ within Σ, the first-variation condition is

$$\theta_{(s)} \;\equiv\; h^{ab}\nabla_a s_b \;=\; 0, \tag{56.144}$$

with h^{ab} the inverse induced metric on the surface, and the second variation in the in-slice directions is non-negative because the minimizer is a genuine minimum.

Step 2: the focusing estimate. The one geometric fact we need about how areas respond to motion off the slice is the Raychaudhuri equation for a null congruence. Let k^a be the affinely parametrized tangent (parameter λ) of one of the two null congruences fired orthogonally from a surface, with expansion $\theta = h^{ab}\nabla_a k_b$ and shear σ_{ab}. In $d+1$ bulk dimensions the cross-section is $(d-1)$-dimensional and

$$\frac{d\theta}{d\lambda} = -\frac{\theta^2}{d-1} - \sigma_{ab}\sigma^{ab} - R_{ab}k^a k^b. \tag{56.145}$$

Contracting the Einstein equation $R_{ab} - \tfrac{1}{2}Rg_{ab} + \Lambda g_{ab} = 8\pi G_N T_{ab}$ with the null vector k^a kills the $g_{ab}k^a k^b$ terms, so $R_{ab}k^a k^b = 8\pi G_N\, T_{ab}k^a k^b \geq 0$ by the null energy condition. Every term on the right of Eq. (56.145) is then manifestly non-positive,

$$\frac{d\theta}{d\lambda} \;\leq\; -R_{ab}k^a k^b \;\leq\; 0. \tag{56.146}$$

A null congruence that starts non-expanding ($\theta \leq 0$) cannot subsequently expand; one that starts at $\theta = 0$ has $\theta \leq 0$ to the future and $\theta \geq 0$ to the past. This is the entirety of the dynamical input.

Step 3: the outer maximization is attained, and the maximizer is interior. By AdS-hyperbolicity the space of anchored achronal slices is compact, and $\mathcal{A}(\Sigma)$ is continuous on it, so the supremum $\max_\Sigma \mathcal{A}(\Sigma)$ is achieved on some slice Σ_*; call its minimal surface $\gamma \equiv \gamma_A(\Sigma_*)$. We must rule out that the maximizer sits at a degenerate boundary of slice-space where γ goes null. Parametrize a one-parameter tilt of the slice by a future-pointing timelike normal field u, so that the surface is dragged along u. Because γ is already *minimal within* Σ_*, the envelope (Danskin) theorem says the derivative of $\mathcal{A}$ along the tilt picks up only the explicit normal motion, not the implicit re-minimization:

$$\left.\frac{d\mathcal{A}}{d\tau}\right|_{\Sigma_*} = \int_\gamma \sqrt{h}\,\theta_{(u)}, \qquad \theta_{(u)} \equiv h^{ab}\nabla_a u_b. \tag{56.147}$$

At a maximum this vanishes for all tilts u, forcing $\theta_{(u)} = 0$ pointwise on γ. For the second variation, differentiate Eq. (56.147) once more and use $\theta_{(u)} = 0$ at Σ_*:

$$\left.\frac{d^2\mathcal{A}}{d\tau^2}\right|_{\Sigma_*} = \int_\gamma \sqrt{h}\,\partial_u\theta_{(u)}. \tag{56.148}$$

Decompose the timelike normal into the two orthogonal null normals, $u \propto k_{(+)} + k_{(-)}$ with $k_{(\pm)}$ generating the two null congruences. Since both null expansions vanish at γ (this is established in Step 4, and the two steps close consistently), $\partial_u\theta_{(u)}$ is a sum of the Raychaudhuri derivatives $d\theta_{(\pm)}/d\lambda$ evaluated where $\theta_{(\pm)} = 0$, each of which is $\leq -R_{ab}k^a_{(\pm)}k^b_{(\pm)} \leq 0$ by Eq. (56.146). Hence $d^2\mathcal{A}/d\tau^2 \leq 0$: the area is concave under the tilt, the extremum in the slice direction is a genuine *maximum*, and it is interior. A degenerate null limit would instead make $\mathcal{A}$ run off monotonically with no interior maximum, which focusing has just excluded. Thus γ is spacelike and Σ_* is a true maximizing slice. The maximin construction is therefore a saddle of the Lorentzian area functional: a minimum within the slice (Step 1) and a maximum across slices (here), exactly the unstable-saddle character the Lorentzian extremization Eq. (56.142) demanded.

Step 4: extremality. Combine the two stationarity conditions. Within the slice, Eq. (56.144) gives the vanishing of the expansion along the spacelike normal s; across slices, Eq. (56.147) gives the vanishing of the expansion along the timelike normal u. In the null basis $k_{(\pm)} \propto u \pm s$, these two combinations are

$$\theta_{(s)} \;\propto\; \theta_{(+)} + \theta_{(-)} = 0, \qquad \theta_{(u)} \;\propto\; \theta_{(+)} - \theta_{(-)} = 0, \tag{56.149}$$

whose unique solution is $\theta_{(+)} = \theta_{(-)} = 0$. This is precisely extremality, Eq. (56.141): the maximin surface is an HRT extremal surface. (Step 3 used $\theta_{(\pm)} = 0$, derived here; the logic is not circular, since the maximum condition

$\theta_{(u)} = 0$ of Step 3 and the minimum condition $\theta_{(s)} = 0$ of Step 1 are independent first-variation statements, and Eq. (56.149) simply records their conjunction.)

Step 5: the representative-slice property. We claim γ is the minimal surface not only on Σ_* but on every Cauchy slice Σ' of W_A that contains it. Let $\sigma' \subset \Sigma'$ be any competitor anchored on ∂A and homologous to A. Because Σ_* is achronal and $\gamma \subset \Sigma_* \cap \Sigma'$, the surface σ' lies in the causal future-and-past of γ relative to Σ_*; project it onto Σ_* along the timelike normal of Σ_* to obtain $\sigma_* \subset \Sigma_*$ in the same homology class. The projection moves each point along a timelike direction, and a spacelike surface acquired by sliding back along the future-or-past light cones cannot gain area under focusing: by Eq. (56.146) the orthogonal null congruences emanating from γ are non-expanding away from it, so $\mathrm{Area}(\sigma_*) \le \mathrm{Area}(\sigma')$. But γ is minimal on Σ_*, so $\mathrm{Area}(\gamma) \le \mathrm{Area}(\sigma_*) \le \mathrm{Area}(\sigma')$. Hence γ underbids every competitor on Σ': it is minimal on Σ' too. This is the property the covariant inequalities of the next subsection consume: two maximin surfaces can always be brought onto a single common slice on which each remains minimal, so the static cut-and-paste argument transfers verbatim.

Step 6: least area among extremal surfaces, and equivalence to HRT. Let X be any extremal surface homologous to A; it is spacelike (extremal surfaces are achronal by the same focusing argument), so it lies on some Cauchy slice Σ_X. On Σ_X the minimal surface has area $\mathcal{A}(\Sigma_X) \le \mathrm{Area}(X)$. Now choose a Cauchy slice $\Sigma_{X\gamma}$ containing both X and γ (possible because both are achronal and, in the AdS-hyperbolic bulk, jointly achronal once the homology regions are nested as the inequalities require). By the representative-slice property of Step 5, γ is minimal on $\Sigma_{X\gamma}$, while X, being extremal but not necessarily least, is merely a competitor there; therefore $\mathrm{Area}(\gamma) \le \mathrm{Area}(X)$. The maximin surface thus has the least area in the extremal class — which is exactly the defining selection of the HRT prescription Eq. (56.142), "extremal, then least-area." The two prescriptions agree, and the degeneracy resolution promised in Theorem 56.23 is automatic: the outer maximization acts on per-slice minima, so when several extrema compete it returns the smallest. Finally, the homology region of γ has a domain of dependence — the entanglement wedge $\mathcal{W}_A$ of Section 56.6 — and the same focusing comparison (carried out in detail in Theorem 56.32) shows the causal wedge is spacelike-or-null separated from γ and hence contained in $\mathcal{W}_A$. This completes the proof. □

Caution

The AdS-hyperbolicity hypothesis is not cosmetic. It is what guarantees the slice-space is compact so the outer maximum is attained, and it is the covariant stand-in for the global hyperbolicity that an honest variational problem needs. In pathological bulks — closed timelike curves, naked singularities reachable from W_A — the maximin surface can fail to exist,

and no covariant entropy prescription is expected to hold there either. The null energy condition enters only through the single inequality Eq. (56.146); everywhere it is used (Steps 3, 5, 6) it is used in exactly that form.

Covariant strong subadditivity

The payoff of the maximin form is that the entropy inequalities survive covariantly with the same cut-and-paste argument that proved them statically. Consider strong subadditivity, $S(AB) + S(BC) \geq S(ABC) + S(B)$ (Theorem 55.47), for boundary regions whose causal domains are suitably arranged. Statically the proof is purely combinatorial: given the minimal surfaces γ_{AB} and γ_{BC} on a common slice, one cuts them at their intersection and reglues the four pieces into two new anchored surfaces, one homologous to ABC and one homologous to B, with total area unchanged,

$$\text{Area}(\gamma_{AB}) + \text{Area}(\gamma_{BC}) = \text{Area}(\Gamma_{ABC}) + \text{Area}(\Gamma_B), \tag{56.150}$$

where Γ_{ABC}, Γ_B are the reglued competitors. Since γ_{ABC} and γ_B are the *minimal* surfaces in their homology classes, $\text{Area}(\gamma_{ABC}) \leq \text{Area}(\Gamma_{ABC})$ and $\text{Area}(\gamma_B) \leq \text{Area}(\Gamma_B)$; adding and using Eq. (56.150) gives $\text{Area}(\gamma_{ABC}) + \text{Area}(\gamma_B) \leq \text{Area}(\gamma_{AB}) + \text{Area}(\gamma_{BC})$, which is strong subadditivity after dividing by $4G_N$.

The obstacle to repeating this covariantly is that the maximin surfaces for AB and for BC may live on *different* maximizing slices, and one cannot cut and paste surfaces that do not share a slice. Wall's resolution is the representative-slice property of maximin: the maximin surface γ_{AB} has area no greater than the minimal area on *any* Cauchy slice that contains it, and in particular one can find a single slice Σ containing both γ_{AB} and γ_{BC} on which each is still minimal in its class. On that common slice the static cut-and-paste Eq. (56.150) applies verbatim, and because the maximin areas are $\geq$ the minimal areas on Σ for the larger regions, the inequality propagates the right way. The result is that strong subadditivity, monotonicity of relative entropy, and the other entropy inequalities hold for HRT exactly as for RT, with no static assumption. This is the structural reason the covariant prescription is consistent: it is built from minimizations, which are what the inequalities know how to use.

Caution

Extremality is necessary but the global "min" in Eq. (56.142) is indispensable, and it does more than break ties. In a time-dependent state there are generically several extremal surfaces homologous to A; only the least-area one computes the entropy. Dropping the global minimization — taking, say, a locally extremal surface that is not globally least — violates strong subadditivity and can give an entropy that decreases under inclusion. The

covariant prescription is "extremal, then least-area," not "any extremal surface."

56.5.3 The quantum extremal surface

Generalized entropy as the object to extremize

We now have two ingredients that beg to be combined. FLM (Theorem 56.20) says the entropy is area-plus-bulk-entropy but evaluates both on the *classical* extremal surface. HRT (Theorem 56.21) says the surface is fixed by extremizing the *area*. These two prescriptions locate the surface by different criteria, and at the order where FLM operates the discrepancy does not matter — but it must matter eventually, because the bulk entropy S_{bulk} depends on where the surface is, and a self-consistent rule cannot extremize one piece of the entropy while ignoring the other. The resolution, due to Engelhardt and Wall, is to extremize the whole thing. The object that gets extremized is the generalized entropy.

Definition 56.25 (Generalized entropy)**.** For a codimension-two bulk surface γ anchored on ∂A and homologous to A, bounding the homology region $\mathcal{R}_\gamma$, the *generalized entropy* is

$$S_{\text{gen}}[\gamma] \;=\; \frac{\langle \widehat{\text{Area}}(\gamma)\rangle}{4G_N} \;+\; S_{\text{bulk}}(\mathcal{R}_\gamma) \;+\; (\text{Wald and counterterm corrections}), \tag{56.151}$$

the sum of the (renormalized) area term and the renormalized von Neumann entropy of the bulk fields in $\mathcal{R}_\gamma$. The corrections are the same subleading bookkeeping terms as in Eq. (56.138); the load-bearing content is the area-plus-bulk-entropy sum. Crucially, $S_{\text{gen}}[\gamma]$ is ultraviolet-finite as a whole: the cutoff dependence of S_{bulk} near γ renormalizes the cutoff dependence of $1/4G_N$, so the sum is a well-defined functional of the surface location.

Definition 56.26 (Quantum extremal surface)**.** The *quantum extremal surface* (QES) for the boundary region A is a surface γ_A^{Q} anchored on ∂A and homologous to A that extremizes the generalized entropy Eq. (56.151) with respect to all variations of the surface location,

$$\left.\frac{\delta S_{\text{gen}}[\gamma]}{\delta X^\mu(\sigma)}\right|_{\gamma=\gamma_A^{\text{Q}}} = 0, \tag{56.152}$$

where $X^\mu(\sigma)$ embeds the surface. When several quantum extremal surfaces compete, the QES is the one of least generalized entropy.

Theorem 56.27 (Engelhardt–Wall: the QES computes the entropy)**.** *The entanglement entropy of the boundary region A is the generalized entropy evaluated*

on the quantum extremal surface,

$$\boxed{S(A) = \operatorname*{ext}_{\gamma} S_{\text{gen}}[\gamma] = \operatorname*{ext}_{\gamma} \left[\frac{\langle \widehat{\text{Area}}(\gamma) \rangle}{4G_N} + S_{\text{bulk}}(\mathcal{R}_\gamma) \right] = S_{\text{gen}}[\gamma_A^{\text{Q}}],} \tag{56.153}$$

with the global least-generalized-entropy rule resolving competing extrema. This prescription is correct to all orders in G_N within the semiclassical (bulk effective field theory) regime, and it reduces to RT, FLM, and HRT in the appropriate limits.

Why extremize the sum: the competition made explicit

The variational principle Eq. (56.152) is a genuine balance of two opposing tendencies, and seeing the competition is the whole point. Vary the surface position. The area term responds with the gradient of the area functional; the bulk-entropy term responds with the gradient of S_{bulk}. Writing the variation along a normal direction parametrized by a modulus x,

$$\frac{\partial S_{\text{gen}}}{\partial x} = \frac{1}{4G_N} \frac{\partial \,\text{Area}}{\partial x} + \frac{\partial S_{\text{bulk}}}{\partial x} = 0. \tag{56.154}$$

The first term is a *geometric* restoring force: moving the surface off its area-extremum costs area, and the area gradient pulls it back. The second is an *entropic* force: the bulk fields' entanglement entropy generically grows as the homology region grows, so $\partial S_{\text{bulk}}/\partial x$ pushes the surface to enlarge $\mathcal{R}_\gamma$. The QES sits where the two balance. Because the area term carries the large prefactor $1/4G_N \sim N^2$ while $S_{\text{bulk}} = O(N^0)$, the entropic force is a tiny perturbation on the geometric one, and the QES is displaced from the classical extremal surface by an amount of order G_N. This is the content of Fig. 56.7: the area term is a steep parabola whose minimum sits at the classical RT/HRT location, the bulk entropy is a gentle monotonic slope, and their sum is extremized a short distance away.

Worked Example 56.28 (The QES reduces to FLM, and the correction is second order)**.** We make the displacement quantitative and show that re-extremizing changes the entropy only at second order, so that the FLM formula Eq. (56.139) is exactly the QES Eq. (56.153) expanded to first subleading order. This both validates the hierarchy and isolates when re-extremization actually matters.

Setup. Let x be a normal modulus measuring displacement of the surface from the classical extremal surface, which sits at $x = 0$. Expand the area term about its extremum. Since $x = 0$ is a stationary point of the area, the linear term vanishes and

$$\frac{\text{Area}(x)}{4G_N} = \frac{\text{Area}(0)}{4G_N} + \frac{1}{2}\, a\, x^2 + O(x^3), \qquad a \equiv \frac{1}{4G_N} \frac{\partial^2 \text{Area}}{\partial x^2}\bigg|_0 > 0, \tag{56.155}$$

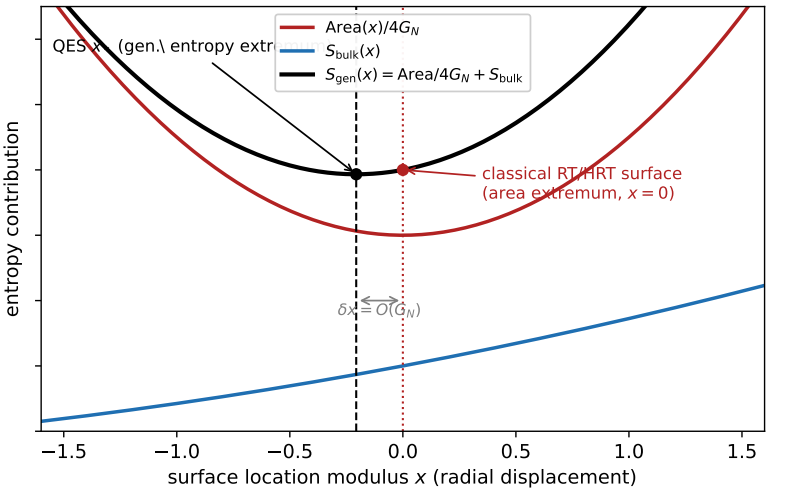


Figure 56.7: The quantum extremal surface as a competition. The surface location is parametrized by a normal modulus x, with the classical extremal (RT/HRT) surface at $x = 0$. The area term $\mathrm{Area}(x)/4G_N$ (red) is stationary at $x = 0$ and carries the large prefactor $1/G_N \sim N^2$; the bulk entanglement entropy $S_{\mathrm{bulk}}(x)$ (blue) is an $O(N^0)$ monotonic function. Their sum, the generalized entropy $S_{\mathrm{gen}}(x)$ (black), is extremized at the quantum extremal surface x_*, displaced from the classical location by $\delta x = O(G_N)$. Evaluating S_{gen} at x_* rather than at $x = 0$ changes the entropy only at order G_N relative to the correction — i.e. at $O(G_N^2)$ overall — which is why FLM (evaluation at $x = 0$) and the QES (evaluation at x_*) agree to the order FLM is valid.

where $a = O(1/G_N)$ is large and positive (the classical surface minimizes area within the slice). The bulk entropy is smooth and $O(N^0)$; expand it too,

$$S_{\mathrm{bulk}}(x) = S_{\mathrm{bulk}}(0) + b\,x + O(x^2), \qquad b \equiv \left.\frac{\partial S_{\mathrm{bulk}}}{\partial x}\right|_0 = O(1). \tag{56.156}$$

Locate the QES. The generalized entropy is $S_{\mathrm{gen}}(x) = \mathrm{Area}(0)/4G_N + \frac{1}{2}ax^2 + S_{\mathrm{bulk}}(0) + bx + \ldots$. Extremizing,

$$0 = \left.\frac{\partial S_{\mathrm{gen}}}{\partial x}\right|_{x_*} = a\,x_* + b \quad\Longrightarrow\quad x_* = -\frac{b}{a} = -\,4G_N\,\left.\frac{\partial S_{\mathrm{bulk}}/\partial x}{\partial^2 \mathrm{Area}/\partial x^2}\right|_0 = O(G_N). \tag{56.157}$$

The displacement is of order G_N, as anticipated: the gentle entropic slope b is fought by the steep geometric curvature $a \sim 1/G_N$, and the surface barely moves.

The entropy shift is second order. Evaluate S_{gen} at the QES and compare to its value at the classical surface:

$$\begin{aligned} S_{\text{gen}}(x_*) &= \frac{\text{Area}(0)}{4G_N} + S_{\text{bulk}}(0) + \left(\tfrac{1}{2}a\,x_*^2 + b\,x_*\right) \\ &= \frac{\text{Area}(0)}{4G_N} + S_{\text{bulk}}(0) + \left(\tfrac{1}{2}a\tfrac{b^2}{a^2} - \tfrac{b^2}{a}\right) = \underbrace{\frac{\text{Area}(0)}{4G_N} + S_{\text{bulk}}(0)}_{\text{FLM, Eq. (56.139)}} - \frac{b^2}{2a}. \end{aligned} \tag{56.158}$$

The first two terms are exactly the FLM answer: the area and bulk entropy evaluated on the *classical* surface. The correction from re-extremizing is $-b^2/2a$. With $b = O(1)$ and $a = O(1/G_N)$, this is

$$S_{\text{gen}}(x_*) - S_{\text{FLM}} = -\frac{b^2}{2a} = O(G_N), \tag{56.159}$$

one order in G_N below the bulk-entropy term $S_{\text{bulk}}(0) = O(1)$, hence $O(G_N)$ overall and beyond the validity of FLM. So at the order where FLM holds, evaluating the generalized entropy on the classical surface (FLM) and on the quantum extremal surface (QES) give the same answer; the difference Eq. (56.159) is precisely the leading place where it matters whether one extremizes the full generalized entropy. The sign is fixed: $-b^2/2a < 0$ since $a > 0$, so re-extremization always *lowers* the generalized entropy relative to the FLM value, exactly as a true extremization of a function with positive area-curvature must.

Theorem 56.28 is the quantitative statement of the hierarchy. It also shows precisely when the QES cannot be replaced by FLM: when the entropic gradient b is not small, or the geometric curvature a is not large — that is, when the bulk entropy varies as rapidly across the surface as the geometry does. This is no academic possibility. It is exactly what happens for a black hole that has been evaporating long enough that the entropy of the Hawking radiation has grown comparable to the horizon area: the entropic force is no longer a perturbation, a *new* quantum extremal surface (deep in the interior, defining an "island") comes to dominate the least-generalized-entropy competition, and the resulting entropy turns over onto the Page curve. We are deliberately not computing that here; it is the central payoff of Chapter 57, which takes Eq. (56.153) as its starting equation. We have built the instrument; Chapter 57 fires it.

The hierarchy of prescriptions

The four rules assembled across this chapter are not four competing proposals but one statement at four levels of approximation, organized by what is extremized and to what order in G_N.

Theorem 56.29 (The RT–FLM–HRT–QES hierarchy)**.** *The holographic entanglement entropy is given, exactly within bulk semiclassical gravity, by extremizing the generalized entropy Eq. (56.151) over surface location and taking the least-generalized-entropy quantum extremal surface (Theorem 56.27). The earlier prescriptions are its systematic approximations:*

- ***QES*** *(Eq. (56.153)): extremize* $S_{\text{gen}} = \text{Area}/4G_N + S_{\text{bulk}}$*; valid to all orders in* G_N *in the semiclassical regime, time-dependent states included.*
- ***HRT*** *(Eq. (56.142)): drop* S_{bulk} *and extremize the area alone; the classical* ($G_N \to 0$) *limit of QES, valid at leading order* $O(1/G_N)$*, time-dependent states included.*
- ***FLM*** *(Eq. (56.139)): evaluate* $\text{Area}/4G_N + S_{\text{bulk}}$ *on the classical extremal surface rather than re-extremizing; agrees with QES to first subleading order, the difference being* $O(G_N)$ *as in Eq. (56.158).*
- ***RT*** *(Eq. (56.4)): minimize the area on a preferred time slice; the static, classical limit, where "extremal" collapses to "minimal" and* S_{bulk} *is dropped.*

Each entry is obtained from the one above it by dropping a quantum correction (S_{bulk}*, taking* $G_N \to 0$*) or specializing the kinematics (static* $\Rightarrow$ *minimal surface). The unifying object throughout is the generalized entropy of Theorem 56.25.*

Physical Insight

The single sentence to carry forward is: *the holographic entropy is the extremum of the generalized entropy.* Every refinement in this chapter is a way of evaluating that one functional. The bare area law extremizes its leading geometric piece on a slice; HRT extremizes the same piece covariantly; FLM adds the bulk-field entropy as a perturbation; the QES extremizes the full sum and so closes the loop, making the prescription self-consistent at all semiclassical orders. That generalized entropy — not area, not bulk entropy, but their renormalization-group-stable sum — is the true holographic dual of an entanglement entropy is the deepest lesson of the subject, and the one black-hole information will exploit.

The quantum extremal surface does one more thing for us, beyond computing a number. By bounding a definite bulk region — the homology region $\mathcal{R}_\gamma$ of the QES — it sharpens the question of *which bulk a boundary subregion knows about*. The classical RT surface already suggested that A "sees" the wedge of bulk hanging beneath it; the QES makes the boundary of that wedge precise even when quantum corrections relocate the surface. The next section names this region the entanglement wedge, states the subregion-duality conjecture that the density matrix ρ_A encodes everything in it, and proves the operational content

through the JLMS relation Eq. (56.140) flagged above and the bulk energy conditions that follow from the positivity of relative entropy.

56.6 Subregion duality and the entanglement wedge

The quantum extremal surface of Section 56.5.3 did more than refine the area formula. By extremizing the generalized entropy $S_{\text{gen}} = \text{Area}/4G_N + S_{\text{bulk}}$ (Eq. (56.151)) it drew a sharp line through the bulk: on one side of the extremal surface γ_A^* lies the homology region $\mathcal{R}_A$ that the boundary subsystem A caps off (Theorem 56.2); on the other side lies bulk that A does not enclose. The previous sections used γ_A^* only to compute a number, the entropy of A. We now ask the operationally sharper question to which that surface is the answer: *which part of the bulk does the boundary region A actually know about?* The thesis of this section is that the homology region behind the quantum extremal surface, evolved causally, is exactly that part — the *entanglement wedge* — and that the reduced density matrix ρ_A encodes the full quantum state of the bulk fields there, no more and no less. Making this precise requires first carefully distinguishing the entanglement wedge from a smaller, more naive candidate (the causal wedge), then stating the duality, then proving its operational engine: the Jafferis–Lewkowycz–Maldacena identity equating boundary and bulk modular Hamiltonians up to the area operator. As a dividend, the positivity of relative entropy — a kinematic fact about density matrices — becomes, through this identity, a dynamical constraint on the bulk: an energy condition, the entropic shadow of the same positivity the analytic bootstrap delivers on the boundary.

56.6.1 Three bulk regions a boundary subsystem might encode

Fix a boundary spatial region A at a moment of time, with entangling surface ∂A, in an asymptotically AdS_{d+1} spacetime. Three nested bulk regions present themselves as candidates for "the bulk that A determines." Sorting them out is the whole game.

The coarsest candidate is purely causal and uses no entanglement data at all. On the boundary, the region A does not stand alone: a complete set of initial data on A determines the boundary fields throughout the *boundary domain of dependence* D_A, the set of boundary points every inextendible boundary causal curve through which must cross A. In Lorentzian language D_A is the causal diamond of A: for an interval of length ℓ in a 2D CFT it is the diamond with A as its base and tips a time $\ell/2$ to the future and past. Whatever the boundary subsystem on A knows, it knows everything in D_A. The natural bulk shadow of D_A is built from bulk causality.

Definition 56.30 (Causal wedge)**.** The *causal wedge* $\mathcal{C}_A$ of a boundary region A is the intersection of the bulk causal future and the bulk causal past of the

boundary domain of dependence D_A:

$$\mathcal{C}_A \;=\; J^+(D_A) \cap J^-(D_A), \tag{56.160}$$

where $J^{\pm}(\,\cdot\,)$ denote causal future/past taken in the bulk spacetime. Equivalently, $\mathcal{C}_A$ is the set of bulk points p that can both send a signal to and receive a signal from the boundary diamond D_A. Its future and past null boundaries (the bulk horizons of D_A) meet on a bulk codimension-two surface Ξ_A, the *causal information surface*, anchored on ∂A.

A bulk operator localized in $\mathcal{C}_A$ can be probed by boundary sources and read out by boundary detectors all supported within D_A, so on physical grounds ρ_A must at least encode $\mathcal{C}_A$. The causal wedge is the conservative, manifestly-reconstructible region. The question is whether it is the whole story.

The finest candidate uses the entanglement structure of the state, through the quantum extremal surface.

Definition 56.31 (Entanglement wedge)**.** Let γ_A^* be the quantum extremal surface for A (Theorem 56.26), and let $\mathcal{R}_A$ be the homology region it bounds on a bulk Cauchy slice, $\partial\mathcal{R}_A = A \cup \gamma_A^*$ (Eq. (56.3)). The *entanglement wedge* $\mathcal{W}_A$ is the bulk domain of dependence of $\mathcal{R}_A$,

$$\mathcal{W}_A \;=\; D[\mathcal{R}_A], \tag{56.161}$$

the set of bulk points p such that every inextendible bulk causal curve through p intersects $\mathcal{R}_A$. The entanglement wedge is the largest bulk region whose physics is fixed by initial data on $\mathcal{R}_A$, and it is bounded away from the rest of the bulk by γ_A^*, which forms its *edge*: $\gamma_A^* = \mathcal{W}_A \cap \mathcal{W}_{\bar{A}}$.

Taking the domain of dependence in Eq. (56.161) rather than just the slice $\mathcal{R}_A$ is what makes the entanglement wedge a Lorentzian, dynamical object: any bulk field configuration in $\mathcal{W}_A$ is determined by its data on $\mathcal{R}_A$ through the bulk equations of motion, so "encoding $\mathcal{R}_A$" and "encoding $\mathcal{W}_A$" are the same statement. That γ_A^* is the common edge of $\mathcal{W}_A$ and $\mathcal{W}_{\bar{A}}$ is the geometric face of the purity relation $S(A) = S(\bar{A})$ built into the homology constraint (Section 56.1.2): for a globally pure state the two wedges tile the bulk Cauchy slice and share exactly the extremal surface.

There is, intermediate between the two, a third region that will not detain us long but must be named to avoid confusion: the *causal shadow*, the part of the bulk that lies in no causal wedge of any boundary region, which is precisely where entanglement-wedge reconstruction has its most dramatic and least obvious reach. The cleanest statement of the hierarchy is an inclusion.

Theorem 56.32 (Causal wedge inside entanglement wedge)**.** *Assume the bulk matter obeys the null energy condition $T_{\mu\nu}k^\mu k^\nu \geq 0$ for all null k^μ (equivalently, via the Einstein equation, $R_{\mu\nu}k^\mu k^\nu \geq 0$). Then for every boundary region A,*

$$\mathcal{C}_A \;\subseteq\; \mathcal{W}_A, \tag{56.162}$$

and the inclusion is generically strict: the causal information surface Ξ_A lies on the boundary side of (no deeper than) the extremal surface γ_A^*, and coincides with it only in special, highly symmetric cases.

Proof. The claim is equivalent to the statement that the extremal surface γ_A^* lies outside the causal wedge $\mathcal{C}_A$ — on its far side, deeper in the bulk — so that the entire causal wedge is contained in the homology region's domain of dependence. We establish this by comparing two codimension-two surfaces through their null expansions, using the focusing theorem.

Consider the future null boundary (the future bulk horizon) of the causal wedge: the null congruence N^+ fired into the past from the future tip region of D_A, generating $\partial J^-(D_A)$, whose intersection with $\partial J^+(D_A)$ is the rim Ξ_A. Being the boundary of a causal set, N^+ is a null hypersurface ruled by null geodesics with some expansion θ. The Raychaudhuri equation governs θ along an affine parameter λ of these generators,

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \;=\; -\frac{\theta^2}{d-1} - \sigma_{ab}\sigma^{ab} - R_{\mu\nu}k^\mu k^\nu, \tag{56.163}$$

with σ_{ab} the shear and k^μ the null tangent. Under the null energy condition the last term is non-positive, and the first two are manifestly non-positive, so $\mathrm{d}\theta/\mathrm{d}\lambda \le 0$: once a horizon generator is converging it converges faster, and a generator that ever reaches $\theta \le 0$ focuses to a caustic in finite affine time. A future event horizon cannot contain such a caustic ahead of it (the generators would leave the horizon), so along the past-directed generators of N^+ the expansion satisfies $\theta \ge 0$ as they emanate from the boundary and approach Ξ_A.

Now compare with the extremal surface. By definition (Theorem 56.26, or at the classical level Theorem 56.1) γ_A^* has *vanishing* expansion in both of its orthogonal null directions, $\theta_k = \theta_l = 0$ — this is precisely what extremizing the area means in covariant language. The causal information surface Ξ_A, by the previous paragraph, sits at the tip of a congruence whose expansion is non-negative. A standard comparison argument (the "extremal surfaces lie behind apparent horizons" lemma) now applies: suppose, for contradiction, that Ξ_A lay strictly deeper in the bulk than γ_A^*, i.e. that part of γ_A^* were inside $\mathcal{C}_A$. Fire the future-directed null congruence orthogonally outward from γ_A^* toward the boundary. It starts with $\theta = 0$ and, by Eq. (56.163) with the null energy condition, can only have θ decrease, so it stays non-expanding or focuses. But to exit through the causal horizon N^+ on which Ξ_A sits, it would have to cross from the non-expanding side to the non-negatively-expanding side of a null surface, which the maximum principle for Eq. (56.163) forbids: a non-expanding null surface cannot lie outside a non-negatively-expanding one with the same anchoring on ∂A. Hence γ_A^* lies on or behind Ξ_A, never in front of it. Equivalently Ξ_A lies in $\mathcal{R}_A \cup \gamma_A^*$, every causal curve probing $\mathcal{C}_A$ threads $\mathcal{R}_A$, and $\mathcal{C}_A \subseteq D[\mathcal{R}_A] = \mathcal{W}_A$.

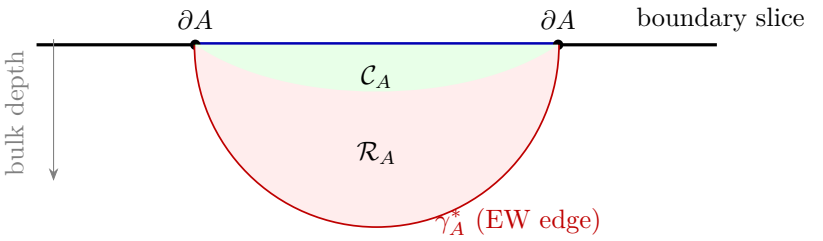


Figure 56.8: The causal wedge $\mathcal{C}_A$ (its edge Ξ_A, dashed green) sits inside the entanglement wedge $\mathcal{W}_A$ (its edge the extremal surface γ_A^*, red), shown on a bulk Cauchy slice. The null energy condition forces Ξ_A to lie no deeper than γ_A^* (Theorem 56.32); the sliver between them is bulk that A encodes *entanglement-wedge*-wise but cannot causally access from D_A alone. In the vacuum ball the two surfaces coincide; any energy in the bulk splits them.

Strictness is generic because the two surfaces are characterized by genuinely different conditions — $\theta = 0$ for γ_A^*, the horizon-rim condition for Ξ_A — which coincide only when the geometry has enough symmetry to make the causal horizon itself extremal. They agree for a ball in pure global AdS (the vacuum, where both surfaces are the same bifurcate Rindler-like horizon, Theorem 55.25 in holographic dress); they separate the instant the state carries energy. The worked example below exhibits the gap explicitly. □

56.6.2 The subregion-duality conjecture

The geometry having singled out the entanglement wedge as the natural, larger candidate, we can state the duality it serves. The claim is the holographic refinement of an idea already familiar from the global dictionary of Chapter 53: that a boundary *subsystem* is dual to a bulk *subregion*, with the reduced density matrix on the boundary side and the bulk effective field theory on its wedge on the bulk side.

Conjecture 56.33 (Subregion duality / entanglement-wedge reconstruction)**.** *Let A be a boundary region with entanglement wedge $\mathcal{W}_A$ (Theorem 56.31). Then the boundary reduced density matrix ρ_A and the bulk effective-field-theory data on $\mathcal{W}_A$ carry the same information:*

(i) (ρ_A determines the wedge.) *Every bulk operator localized in $\mathcal{W}_A$ can be represented as an operator on the boundary Hilbert space supported in A — it acts within the algebra of A and its correlation functions are computable from ρ_A alone.*

(ii) (ρ_A is determined by the wedge.) *Conversely, ρ_A is fixed by, and only by, the bulk state on $\mathcal{W}_A$; two global states with the same restriction to $\mathcal{W}_A$*

have the same ρ_A, and the entropy of A is the generalized entropy of γ_A^* (Eq. (56.151)).

A bulk operator outside $\mathcal{W}_A$ is not *reconstructible from A: it lies in the complementary wedge $\mathcal{W}_{\bar{A}}$ and is encoded in $\rho_{\bar{A}}$.*

We label the statement a conjecture in the book's discipline (Chapter 53): it is not a theorem of an axiomatized framework but the consensus reading of overwhelming evidence, and its operational core — the part that makes it more than a slogan — is provable, which is what the rest of this section does. Two pieces of that evidence are already in hand. The entropy clause (ii) is the quantum extremal surface prescription itself, established in Section 56.5.3. And the localization in (i) is consistent with causality: by Theorem 56.32 the wedge contains the causal wedge, so it contains every bulk operator that boundary causality already forces A to control; entanglement-wedge reconstruction asserts that A controls strictly more, namely everything spacelike-accessible from $\mathcal{R}_A$, including the causal-shadow sliver of Fig. 56.8. The mechanism by which a boundary operator supported in A can reach into that causally-disconnected sliver is the quantum-error-correcting structure of the dictionary, the subject of Section 56.7; here we prove the statement that makes reconstruction possible at the level of the modular data, and from which the entropy clause and the causal consistency both flow.

Worked Example 56.34 (Strict inclusion: two intervals and the connected wedge)**.** The gap between the causal and entanglement wedges is invisible for a single ball in the vacuum, where they coincide, so to see it one needs either energy in the bulk or a disconnected region. The disconnected case is the most transparent, and it doubles as the cleanest demonstration that the entanglement wedge is strictly larger.

Take two disjoint boundary intervals $A = A_1 \cup A_2$ in the vacuum of a 2D CFT, with the geometry of Section 56.2.4. There are two homologous configurations of extremal surfaces: the *disconnected* one, two semicircular geodesics each capping its own interval, and the *connected* one, two geodesics joining the inner endpoints to each other and the outer endpoints to each other. When the intervals are close enough (cross-ratio above the transition value), the connected configuration has smaller total length and is the RT surface; its homology region $\mathcal{R}_A$ is the bulk region bounded by A_1, A_2, and the two connecting geodesics — and it includes the bulk "bridge" hanging directly beneath the gap between the intervals.

Now compare with the causal wedge. The boundary domain of dependence of $A = A_1 \cup A_2$ is the union of the two separate diamonds $D_{A_1} \cup D_{A_2}$; nothing in D_A is causally connected, on the boundary, to the gap between the intervals, because the gap is the complement region. The causal wedge $\mathcal{C}_A = \mathcal{C}_{A_1} \cup \mathcal{C}_{A_2}$ is correspondingly the disjoint union of two small wedges, each hugging its own interval; the central bridge beneath the gap lies in *neither*. Yet that bridge is

exactly the bulk the connected entanglement wedge contains. We have therefore exhibited bulk — the connected bridge — that is in $\mathcal{W}_A$ but in no causal wedge of A: the inclusion of Theorem 56.32 is strict, and the surplus is precisely the bulk region that the *pair* of intervals encodes jointly though neither encodes alone. This surplus is the holographic seed of a striking fact developed in Section 56.7: a bulk operator can be invisible to A_1 and invisible to A_2, yet reconstructible from $A_1 \cup A_2$ — the signature of a quantum error-correcting code. The transition at the critical cross-ratio is the same entanglement phase transition that gives the mutual information $I(A_1{:}A_2)$ its kink (Section 56.2.4); on one side the bridge belongs to the wedge, on the other it does not, and the bulk region a pair of regions reconstructs jumps discontinuously.

56.6.3 The JLMS identity: boundary modular Hamiltonian = area + bulk modular Hamiltonian

Subregion duality asserts that ρ_A and the wedge carry the same information. The sharpest quantitative form of "same information" is an identity between the operators that govern each side's entanglement: the modular Hamiltonian. Recall (Theorem 55.28) that for a state ρ reduced to a region, the modular Hamiltonian is $K = -\log\rho$ (up to an additive constant), the generator whose Gibbs weight reproduces the reduced density matrix and whose flow ρ^{is} is modular flow. The boundary region A has its modular Hamiltonian $K_A = -\log\rho_A$. The bulk fields on the wedge $\mathcal{W}_A$ have *their* reduced density matrix $\rho_{\mathcal{W}_A}^{\text{bulk}}$ and hence their own bulk modular Hamiltonian $K_{\text{bulk}} = -\log\rho_{\mathcal{W}_A}^{\text{bulk}}$, an operator in the bulk effective field theory. The identity we now prove relates the two through the area of the extremal surface, promoted to an operator.

Theorem 56.35 (Jafferis–Lewkowycz–Maldacena)**.** *Acting on the code subspace of states whose bulk duals share the geometry to leading order, the boundary modular Hamiltonian of A equals the bulk modular Hamiltonian of its entanglement wedge plus the area operator of the quantum extremal surface:*

$$\boxed{K_A \;=\; \frac{\widehat{\text{Area}}(\gamma_A^*)}{4G_N} \;+\; K_{\text{bulk}}} \tag{56.164}$$

where $\widehat{\text{Area}}(\gamma_A^*)$ is the area of the quantum extremal surface promoted to a bulk operator (its expectation value is the classical area; it fluctuates because the surface location depends on the bulk state), and the identity holds order by order in the $1/N$ (equivalently G_N) expansion.

The structure of Eq. (56.164) is already eloquent. The boundary notion of "how entangled A is," encoded in K_A, decomposes into a gravitational piece — the area operator, the same object whose expectation gives the RT entropy — plus

the genuinely quantum entanglement of the bulk matter across the surface, K_{bulk}. It is the operator-level parent of the FLM entropy formula: taking the expectation value in a state and recalling $S = \langle K \rangle$ for the appropriate normalization recovers $S(A) = \langle \widehat{\text{Area}} \rangle / 4G_N + S_{\text{bulk}}$, the generalized entropy. We prove Eq. (56.164) by assembling three inputs: the first law of entanglement, the FLM decomposition of the entropy, and the equality of bulk and boundary relative entropies. We take them in turn, proving the first law from scratch since it is the linchpin.

The first law of entanglement. Let σ be a fixed reference state (typically the vacuum) reduced to A, with modular Hamiltonian $K_A^\sigma = -\log \sigma_A$, and let ρ_A be a nearby state, $\rho_A = \sigma_A + \delta\rho$. The relative entropy (Section 55.6)

$$S_{\text{rel}}(\rho_A \| \sigma_A) \;=\; \text{Tr}(\rho_A \log \rho_A) - \text{Tr}(\rho_A \log \sigma_A) \;=\; \Delta\langle K_A^\sigma \rangle - \Delta S_A \tag{56.165}$$

splits, by definition, into the change in the modular energy $\Delta\langle K_A^\sigma \rangle = \text{Tr}(\rho_A K_A^\sigma) - \text{Tr}(\sigma_A K_A^\sigma)$ and the change in the von Neumann entropy $\Delta S_A = S(\rho_A) - S(\sigma_A)$. Relative entropy is non-negative and vanishes if and only if $\rho_A = \sigma_A$ (Section 55.6); it therefore has a global minimum, equal to zero, at $\rho_A = \sigma_A$. A function that is non-negative, smooth, and zero at a point has vanishing first derivative there. Expanding in the perturbation $\delta\rho$ and writing the first-order variation as $\delta(\cdot)$,

$$\delta S_{\text{rel}} \;=\; \delta\langle K_A^\sigma \rangle - \delta S_A \;=\; 0 \quad\Longrightarrow\quad \boxed{\delta S_A = \delta\langle K_A^\sigma \rangle} \tag{56.166}$$

to first order in $\delta\rho$. This is the first law of entanglement: the leading change in the entanglement entropy of a region equals the change in the expectation value of the (fixed, reference) modular Hamiltonian. It is the entanglement analogue of $\mathrm{d}E = T\,\mathrm{d}S$ — indeed for the half-space $K_A^\sigma = 2\pi K_{\text{boost}}$ (Eq. (55.69)) and Eq. (56.166) reads $\delta S = 2\pi\,\delta\langle K_{\text{boost}} \rangle$, a literal first law at the Unruh temperature. Crucially, it holds on *both* sides of the duality: a boundary first law for ρ_A and a bulk first law for $\rho_{\mathcal{W}_A}^{\text{bulk}}$, since the argument used nothing but positivity of relative entropy, which holds for any density matrix.

The FLM decomposition. The quantum-corrected entropy formula of Section 56.5.1 (the FLM result, Eq. (56.138)) states that the boundary entropy is the bulk generalized entropy of the extremal surface,

$$S(\rho_A) \;=\; \frac{\langle \widehat{\text{Area}}(\gamma_A^*) \rangle_\rho}{4G_N} \;+\; S_{\text{bulk}}(\rho_{\mathcal{W}_A}^{\text{bulk}}) \;+\; (\text{local counterterms}), \tag{56.167}$$

where S_{bulk} is the von Neumann entropy of the bulk fields restricted to the wedge and the expectation $\langle \widehat{\text{Area}} \rangle_\rho$ is evaluated in the state ρ. Subtracting the same relation for the reference σ, the boundary entropy change becomes

$$\Delta S_A \;=\; \frac{\Delta\langle \widehat{\text{Area}} \rangle}{4G_N} \;+\; \Delta S_{\text{bulk}}, \tag{56.168}$$

the counterterms cancelling in the difference because they are state-independent local integrals on ∂A.

Equality of relative entropies and the assembly. The remaining input is the central JLMS claim, which we state and then justify: the relative entropy computed on the boundary equals the relative entropy of the bulk fields on the wedge,

$$S_{\rm rel}(\rho_A\|\sigma_A) \;=\; S_{\rm rel}(\rho^{\rm bulk}_{\mathcal{W}_A}\|\,\sigma^{\rm bulk}_{\mathcal{W}_A}), \tag{56.169}$$

order by order in $1/N$. Granting Eq. (56.169), the theorem follows by pure algebra. Write each side via Eq. (56.165): the boundary side is $\Delta\langle K^\sigma_A\rangle - \Delta S_A$ and the bulk side is $\Delta\langle K^\sigma_{\rm bulk}\rangle - \Delta S_{\rm bulk}$. Equating them,

$$\Delta\langle K^\sigma_A\rangle - \Delta S_A \;=\; \Delta\langle K^\sigma_{\rm bulk}\rangle - \Delta S_{\rm bulk}. \tag{56.170}$$

Substitute the FLM difference Eq. (56.168) for ΔS_A on the left:

$$\Delta\langle K^\sigma_A\rangle - \frac{\Delta\langle\widehat{\rm Area}\rangle}{4G_N} - \Delta S_{\rm bulk} \;=\; \Delta\langle K^\sigma_{\rm bulk}\rangle - \Delta S_{\rm bulk}. \tag{56.171}$$

The $\Delta S_{\rm bulk}$ terms cancel, leaving

$$\Delta\langle K^\sigma_A\rangle \;=\; \frac{\Delta\langle\widehat{\rm Area}\rangle}{4G_N} \;+\; \Delta\langle K^\sigma_{\rm bulk}\rangle. \tag{56.172}$$

This holds for the change between the reference σ and an *arbitrary* nearby state ρ. An identity of expectation-value differences that holds in every state of the code subspace is an identity of operators on that subspace: choosing ρ to range over a basis-spanning family (including states with off-diagonal density-matrix elements, obtained by superposing the reference with small admixtures and reading off the resulting linear and bilinear responses) promotes Eq. (56.172) to the operator equation $K_A = \widehat{\rm Area}/4G_N + K_{\rm bulk}$, which is Eq. (56.164). The additive reference constants drop out because the equation is built from differences. This completes the proof modulo the justification of Eq. (56.169), to which we turn.

Why are the relative entropies equal? The clean way to see it, and the one that exposes the $1/N$ bookkeeping, is to feed the first law back in. To leading nontrivial order the relative entropy is second order in the perturbation (its first-order part vanished by Eq. (56.166)), and the FLM relation Eq. (56.168) together with the *bulk* first law $\delta S_{\rm bulk} = \delta\langle K^\sigma_{\rm bulk}\rangle$ already forces the two sides of the JLMS equality $K_A - K_{\rm bulk} = \widehat{\rm Area}/4G_N$ (Eq. (56.173)) to agree term by term. Concretely, the leading, $O(N^2) \sim 1/G_N$, content lives entirely in the area: at this order the boundary modular Hamiltonian's variation $\delta\langle K^\sigma_A\rangle$ is matched by $\delta\langle\widehat{\rm Area}\rangle/4G_N$, while the bulk matter entanglement $K_{\rm bulk}$ is $O(N^0)$ and enters only at the next order. Equating the relative entropies thus first

fixes the area term, then, at $O(N^0)$, fixes the bulk modular piece, and so on — which is exactly the meaning of "order by order in $1/N$" in the theorem. The deepest justification of Eq. (56.169) is that both relative entropies are computed by the *same* Euclidean gravitational path integral: the replica construction that computes $S_{\rm rel}$ (a difference of free energies on replicated geometries) localizes, by the Lewkowycz–Maldacena conical-deficit accounting of Section 56.4, onto the extremal surface, and the bulk-matter and area contributions it produces are precisely $S_{\rm bulk}$ and $\langle\widehat{\rm Area}\rangle/4G_N$ — so the boundary $S_{\rm rel}$ *is* the bulk $S_{\rm rel}$ by construction. We have stated this localization as established in Section 56.4 and do not repeat it; what Theorem 56.35 adds is the operator identity it implies.

$$S_{\rm rel}^{\rm bdy} \;=\; S_{\rm rel}^{\rm bulk} \qquad\Longleftrightarrow\qquad K_A - K_{\rm bulk} = \frac{\widehat{\rm Area}}{4G_N}. \tag{56.173}$$

Physical Insight

JLMS is the precise sense in which "the area is entanglement." The area operator is not an extra ingredient bolted onto the bulk modular Hamiltonian; it is the *difference* between how entangled the boundary thinks the region is and how entangled the bulk fields actually are across the surface. In the deep-classical limit the bulk-matter entanglement is negligible and K_A is purely the area — the entanglement of A is geometry, full stop. Quantum bulk fields then dress this geometric entanglement with their own, and the two combine into a single boundary modular Hamiltonian. Read the other way, Eq. (56.164) says that boundary modular flow — an abstract, generally nonlocal symmetry of ρ_A — acts on the bulk as bulk modular flow together with a geometric "rotation" generated by the area term, i.e. a boost about the extremal surface. This is the engine of reconstruction: because the two modular flows agree, a bulk operator and its boundary image evolve identically, and the boundary can track the bulk operator.

56.6.4 Relative-entropy positivity as a bulk energy condition

The identity Eq. (56.164) converts kinematic facts about boundary density matrices into dynamical statements about the bulk. The two facts we use are the most basic properties of relative entropy, both established in Section 55.6: positivity, and monotonicity under inclusion.

Proposition 56.36 (Relative-entropy positivity and monotonicity)**.** *For any states ρ, σ and any regions $A \subseteq B$,*

$$S_{\rm rel}(\rho_A\|\sigma_A) \;\geq\; 0, \qquad S_{\rm rel}(\rho_A\|\sigma_A) \;\leq\; S_{\rm rel}(\rho_B\|\sigma_B). \tag{56.174}$$

Positivity is saturated iff $\rho_A = \sigma_A$; monotonicity expresses that discarding the degrees of freedom in $B\setminus A$ (a quantum channel) cannot increase distinguishability.

These are theorems of finite-dimensional quantum information, imported wholesale (Section 55.6). Their power here is that, through JLMS, the boundary relative entropy equals a bulk quantity, so Eq. (56.174) constrains the bulk. We make this explicit in the controlled setting of a perturbation of the vacuum, where every object is computable.

Take the reference σ to be the global vacuum and ρ a nearby state created by exciting the bulk matter — a small coherent disturbance with stress tensor $\langle T^{\rm bulk}_{\mu\nu}\rangle$ on top of pure AdS. Choose for A a boundary ball (or, by the conformal map, a half-space), whose entanglement wedge in the vacuum is a bulk Rindler wedge with bifurcation surface γ_A^*. For this maximally symmetric case the *bulk* modular Hamiltonian is local and geometric: by the bulk Bisognano–Wichmann theorem (Theorem 55.25, applied to the bulk QFT on its Rindler wedge) it is the boost generator about γ_A^*,

$$K^{\sigma}_{\rm bulk} \;=\; 2\pi \int_{\mathcal{R}_A} \mathrm{d}\Sigma^{\mu}\, \xi^{\nu}\, T^{\rm bulk}_{\mu\nu} \;+\; \text{const}, \tag{56.175}$$

with ξ^ν the boost Killing vector of the wedge (vanishing on γ_A^*, future-directed inside) and $\mathrm{d}\Sigma^\mu$ the volume element on the bulk slice $\mathcal{R}_A$. Now use the bulk first law and FLM together. By Eq. (56.168) the entropy change is $\Delta S_A = \Delta\langle\widehat{\rm Area}\rangle/4G_N + \Delta S_{\rm bulk}$, and by the bulk first law Eq. (56.166) the bulk entropy change is $\Delta S_{\rm bulk} = \Delta\langle K^{\sigma}_{\rm bulk}\rangle$ to leading order. Insert these into the boundary relative entropy Eq. (56.165):

$$\begin{aligned} S_{\rm rel}(\rho_A\|\sigma_A) &= \Delta\langle K^{\sigma}_{A}\rangle - \Delta S_A \\ &= \Delta\langle K^{\sigma}_{A}\rangle - \frac{\Delta\langle\widehat{\rm Area}\rangle}{4G_N} - \Delta\langle K^{\sigma}_{\rm bulk}\rangle. \end{aligned} \tag{56.176}$$

By JLMS Eq. (56.172) the right-hand side is zero at first order — the boundary modular energy exactly accounts for the area change plus the bulk modular energy, which is just Eq. (56.164) read as a statement about expectation values. So the *linear* term in $S_{\rm rel}$ vanishes, consistent with Eq. (56.166), and the physics is in the second order. There, positivity bites. Writing $S_{\rm rel} = S^{(2)}_{\rm rel} + \dots$ and using the equality of relative entropies Eq. (56.169),

$$S^{\rm bulk}_{\rm rel}(\rho^{\rm bulk}\|\sigma^{\rm bulk}) \;=\; \Delta\langle K^{\sigma}_{\rm bulk}\rangle - \Delta S_{\rm bulk} \;\geq\; 0, \tag{56.177}$$

i.e. the *bulk* modular energy change is bounded below by the bulk entanglement-entropy change. With the local form Eq. (56.175) this is a statement about the bulk stress tensor:

$$2\pi\int_{\mathcal{R}_A} \mathrm{d}\Sigma^{\mu}\, \xi^{\nu}\, \Delta\langle T^{\rm bulk}_{\mu\nu}\rangle \;\geq\; \Delta S_{\rm bulk} \;\geq\; 0, \tag{56.178}$$

the last inequality holding because $\Delta S_{\rm bulk} \geq 0$ for excitations of the vacuum on a wedge (the vacuum minimizes the modular-flow-invariant entropy). The boost-weighted bulk energy in the wedge is non-negative. This is a *bulk energy condition*, derived not from any assumption about the bulk matter Lagrangian but purely from positivity of a boundary relative entropy.

To extract the sharpest, local form, deform the entangling surface along a null direction. Let A_λ be a one-parameter family of regions obtained by pushing ∂A a null distance λ in the bulk-null direction k^μ orthogonal to γ_A^*; this nests the entanglement wedges, $\mathcal{W}_{A_\lambda} \subseteq \mathcal{W}_{A_{\lambda'}}$ for $\lambda \leq \lambda'$. Monotonicity Eq. (56.174) then gives $\partial_\lambda S_{\rm rel} \geq 0$. The modular Hamiltonian's null derivative localizes onto the surface — $\partial_\lambda \langle K \rangle = 2\pi \int_\gamma \langle T_{kk} \rangle$ for the local boost generator — and assembling the pieces yields the differential inequality

$$2\pi \, \langle T_{kk}^{\rm bulk} \rangle \;\geq\; \partial_\lambda^2 S_{\rm bulk}, \tag{56.179}$$

the bulk *quantum null energy condition*: the null-null stress tensor is bounded below by the second null derivative of the entanglement entropy. In the classical limit, where $S_{\rm bulk}$ is subleading, Eq. (56.179) reduces to $\langle T_{kk}^{\rm bulk} \rangle \geq 0$, the null energy condition — which is exactly the hypothesis that drove the focusing argument of Theorem 56.32 and the holographic c-theorem of Section 52.5. Relative-entropy positivity thus *proves*, on the bulk side, the energy condition we elsewhere had to assume.

Physical Insight

The same positivity surfaces on both sides of the duality, established by entirely different mathematics, and the two must agree because they compute the same number. On the boundary, the averaged null energy condition $\int \mathrm{d}\lambda \, \langle T_{kk} \rangle \geq 0$ is a theorem of conformal field theory, proved by the analytic bootstrap: it is the positivity of the double discontinuity that the Lorentzian inversion formula (Theorem 51.16) feeds on, the statement that the leading-twist OPE data have definite sign. On the bulk, the very same averaged null energy is forced by the monotonicity of relative entropy through Eq. (56.179). One positivity lives in the analytic structure of a Lorentzian four-point function; the other in the convexity of the logarithm that makes $S_{\rm rel} \geq 0$. That they are the same constraint, transported across the holographic dictionary, is among the cleanest evidence that gravity in the bulk inherits exactly the consistency conditions of a unitary conformal field theory on the boundary — and conversely that bulk causality and energy positivity are not extra postulates but consequences of boundary unitarity.

Caution

The energy condition Eq. (56.178) is derived in a controlled perturbative setting — a small excitation of the vacuum, a maximally symmetric region with a local bulk modular Hamiltonian. Two caveats keep it honest. First, away from balls and half-spaces the bulk modular Hamiltonian is nonlocal and Eq. (56.175) fails; the positivity statement remains true but no longer reads as a simple stress-tensor integral. Second, the chain of equalities used FLM and the bulk first law at leading order in G_N; the inequalities are perturbative, valid order by order, and are not claimed as exact statements about the full nonperturbative gravitational state. What is robust is the *structure*: boundary relative-entropy positivity, transmuted by JLMS, is a positive-energy/focusing statement for the bulk, dated to the mid-2010s programme that turned entanglement inequalities into gravitational constraints.

56.6.5 Why the entanglement wedge, and not the causal wedge, is the reconstruction region

We can now collect the geometric and operational threads into the answer the section was built to give. The causal wedge is the conservative choice: it contains exactly the bulk that boundary causality *forces* A to control, and a bulk operator there is reconstructible by elementary causal reasoning (a boundary source in D_A can poke it, a boundary detector in D_A can hear it). But the causal wedge is too small to be the encoding region, for three reasons that now stand proved.

First, it gives the wrong entropy. The boundary entropy $S(A)$ is the generalized entropy of the quantum extremal surface (Section 56.5.3), and that surface is γ_A^*, the edge of the *entanglement* wedge — not Ξ_A, the edge of the causal wedge. The area of Ξ_A does not compute $S(A)$ except in the symmetric cases where the two surfaces coincide. Any candidate "reconstruction region" must have an edge whose area is the entropy, and only the entanglement wedge qualifies.

Second, it is inconsistent with the modular identity. JLMS Eq. (56.164) equates boundary modular flow with bulk modular flow about γ_A^*. Modular flow of ρ_A acts within the algebra of A and preserves it; on the bulk side it preserves the entanglement wedge, since γ_A^* is the fixed surface of the boost it generates. Were the encoded region the smaller causal wedge, boundary modular flow would have to preserve $\mathcal{C}_A$ instead, contradicting Eq. (56.164), whose area operator is the area of γ_A^*. The algebra of ρ_A knows about γ_A^*, hence about the whole entanglement wedge.

Third, the surplus region is physically reconstructible, even though it is causally inaccessible. Theorem 56.34 exhibited bulk — the connected bridge beneath two intervals — inside $\mathcal{W}_A$ but inside no causal wedge of A. By

Theorem 56.32 this surplus is always on the safe side: $\mathcal{C}_A \subseteq \mathcal{W}_A$, so promoting the reconstruction region from causal to entanglement wedge never *loses* anything causally guaranteed; it only adds the causal-shadow sliver. And the addition is forced, not optional, by the first two points. The entanglement wedge is therefore the unique region that simultaneously contains the causal wedge, has the right entropy, and is preserved by boundary modular flow. It is the bulk that ρ_A encodes.

The strict inclusion $\mathcal{C}_A \subsetneq \mathcal{W}_A$ is the crux of everything that follows. A bulk operator in the surplus sliver cannot be reached from A by any causal signal, yet ρ_A determines it. There is no contradiction — bulk locality is emergent and approximate, and the boundary representation of a bulk operator is smeared, nonlocal, and state-dependent — but there is a puzzle of mechanism: how can a boundary operator supported on A encode bulk data in a region A cannot causally touch, and do so robustly? The answer is that the bulk-to-boundary map has the structure of a quantum error-correcting code, in which a single logical (bulk) operator admits many physical (boundary) representatives, and the redundancy is exactly what lets A reconstruct operators in its entanglement wedge even when no individual boundary point in A is causally connected to them. Making the reconstruction explicit — building the boundary representative of a bulk field by the HKLL smearing and recognizing entanglement-wedge reconstruction as complementary recovery in a code — is the work of Section 56.7, to which the JLMS identity Eq. (56.164) now hands the precise operational claim: bulk operators in $\mathcal{W}_A$ are recoverable from ρ_A, with the area operator measuring exactly how much code subspace the geometry can protect.

56.7 Bulk reconstruction as quantum error correction

The previous section closed with a claim that is operationally startling and that this section makes good on. Subregion duality, sharpened by the JLMS identity (Theorem 56.35), asserts that the reduced density matrix ρ_A of a boundary region A does not merely *know about* the entanglement wedge $\mathcal{W}_A$ in some loose statistical sense: it *encodes* the bulk physics there exactly, so that any bulk operator localized inside $\mathcal{W}_A$ can be recovered from boundary data living on A alone. That is a strong statement, and it has a strange consequence. A bulk operator deep in the interior — at the very center of anti-de Sitter space, say — lies inside the entanglement wedge of many different, even disjoint, boundary regions. Each of them must therefore be able to reconstruct it. How can the same bulk field be a function of boundary data on A and, independently, a function of boundary data on a region disjoint from A, when operators on disjoint regions commute? The resolution is not a technicality; it is the deepest structural lesson of holography. The bulk-to-boundary map is a *quantum error-correcting code*: bulk fields are logical operators, the boundary Hilbert space is the physical

(encoding) space, and the multiple, mutually consistent boundary representations of one bulk operator are exactly the redundancy that lets a code recover logical information after part of the physical system is erased. This section builds that picture in three stages. First we make reconstruction concrete in the simplest setting — the HKLL representation of a free bulk field as a smeared boundary operator. Then we promote it to entanglement-wedge reconstruction and state the recovery theorem. Finally we recast the whole structure as quantum error correction in the sense of Almheiri, Dong, and Harlow, and exhibit it exactly in a finite-dimensional model, the HaPPY pentagon code, where every claim above becomes a short calculation with stabilizer codes.

56.7.1 HKLL: a bulk field as a smeared boundary operator

Begin with the cleanest possible question. The dictionary of Section 53.5 attaches to every bulk field ϕ of mass m a boundary primary $\mathcal{O}$ of dimension Δ, with $\Delta(\Delta - d) = m^2 L^2$, through the near-boundary falloff. In the radial coordinate z of Poincaré AdS_{d+1}, a normalizable bulk solution behaves as $\phi(z, x) \to z^{\Delta}\, \mathcal{O}(x)$ as $z \to 0$, and this limit *defines* the boundary operator,

$$\mathcal{O}(x) = \lim_{z \to 0} z^{-\Delta}\, \phi(z, x), \tag{56.180}$$

the so-called extrapolate form of the dictionary, equivalent to the GKP–Witten relation (Theorem 53.29 and Eq. (53.162)) for the leading falloff. Equation (56.180) reads off a boundary operator from a bulk field. We now want to run it backwards: given the boundary operator $\mathcal{O}(x)$ for all x, reconstruct the bulk field $\phi(z, x)$ at a point sitting a finite distance into the bulk. At leading order in G_N the bulk field is free — it obeys the linear wave equation on the fixed AdS background — and a free field is completely determined by its boundary data together with the equation of motion. The reconstruction is therefore a boundary-value problem, and it has a clean answer: ϕ is a *linear smearing* of $\mathcal{O}$ against a kernel.

The logic that guarantees this is mode inversion, and it is worth stating in full because it makes the existence of the kernel a theorem rather than a hope. Expand the free bulk field in a complete set of normalizable modes $F_n(z, x)$ of the wave equation,

$$\phi(z, x) = \sum_n \left[a_n\, F_n(z, x) + a_n^{\dagger}\, F_n^{*}(z, x) \right], \tag{56.181}$$

with $a_n, a_n^{\dagger}$ the bulk creation and annihilation operators. Applying Eq. (56.180) term by term, the same operators appear in the boundary field,

$$\mathcal{O}(x) = \sum_n \left[a_n\, f_n(x) + a_n^{\dagger}\, f_n^{*}(x) \right], \qquad f_n(x) \equiv \lim_{z \to 0} z^{-\Delta} F_n(z, x). \tag{56.182}$$

The boundary mode functions f_n are a complete orthonormal set on the boundary (they are the boundary values of a complete set of bulk solutions, and the spectrum

is non-degenerate), so the ladder operators can be solved for: $a_n = \int \mathrm{d}x\, f_n^*(x)\, \mathcal{O}(x)$ in a schematic notation that the explicit cases below make precise. Substituting back into Eq. (56.181) expresses ϕ as a linear functional of $\mathcal{O}$,

$$\phi(z,x) = \int \mathrm{d}x'\; K(z,x\,|\,x')\,\mathcal{O}(x'), \qquad K(z,x\,|\,x') = \sum_n \big[F_n(z,x) f_n^*(x') + F_n^*(z,x) f_n(x')\big]. \tag{56.183}$$

The object K is the *smearing function*. By construction it manifestly solves the bulk wave equation in its bulk argument (z,x) — it is a superposition of solutions F_n — and it reproduces $\mathcal{O}$ in the boundary limit. Reconstruction therefore exists for any free bulk field; the only question is whether the mode sum Eq. (56.183) can be resummed into something local and intelligible. It can, and the answer is the central result of Hamilton, Kabat, Lifschytz, and Lowe.

Definition 56.37 (HKLL smearing representation)**.** Let ϕ be a free scalar of dimension Δ in AdS_{d+1}, dual to the boundary primary $\mathcal{O}$. The *HKLL representation* of ϕ at a bulk point $(z,x) = (z,t,\vec{x})$ in the Poincaré patch is

$$\phi(z,t,\vec{x}) = c_{\Delta,d} \int_{t'^2+|\vec{x}'|^2<z^2} \mathrm{d}t'\, \mathrm{d}^{d-1}x' \left(\frac{z^2 - t'^2 - |\vec{x}'|^2}{z}\right)^{\Delta-d} \mathcal{O}(t+t',\, \vec{x}+\vec{x}'), \tag{56.184}$$

with normalization

$$c_{\Delta,d} = \frac{\Gamma(\Delta - \frac{d}{2} + 1)}{\pi^{d/2}\,\Gamma(\Delta - d + 1)}. \tag{56.185}$$

The integration region $t'^2 + |\vec{x}'|^2 < z^2$ is the set of boundary points *spacelike separated* from the bulk point: this is the *spacelike* (or causal-wedge) smearing representation. The kernel makes the bulk operator an explicit, if non-local, function of boundary operators in the causal wedge of the data region.

That the support is the spacelike-separated region is the geometric heart of the construction, and it deserves a name and a derivation rather than an assertion. We supply both in a low-dimensional case clean enough to carry every step on the page.

Worked Example 56.38 (The HKLL kernel for a scalar in AdS_2)**.** Take $d = 1$: a scalar of dimension Δ in AdS_2, with metric $\mathrm{d}s^2 = (L^2/z^2)(-\mathrm{d}t^2 + \mathrm{d}z^2)$ in Poincaré coordinates, boundary at $z \to 0$. We set $L = 1$. We will (i) write down the candidate smearing representation, (ii) verify it satisfies the bulk wave equation, (iii) verify it reproduces the extrapolate dictionary, and (iv) read off its causal support. Together these constitute a complete proof that the formula reconstructs the bulk field, because a normalizable solution of a linear second-order wave equation is uniquely fixed by its leading boundary data.

The wave equation. For the metric above, $\sqrt{-g} = 1/z^2$ while $\sqrt{-g}\,g^{tt} = -1$ and $\sqrt{-g}\,g^{zz} = +1$, so the Klein–Gordon operator is

$$\Box\phi = \frac{1}{\sqrt{-g}}\,\partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\big) = z^2(\partial_z^2 - \partial_t^2)\phi, \tag{56.186}$$

and the equation $\Box\phi = m^2\phi$ with $m^2 = \Delta(\Delta-1)$ (the $d=1$ mass–dimension relation) reads

$$\Big(\partial_z^2 - \partial_t^2 - \frac{\Delta(\Delta-1)}{z^2}\Big)\phi = 0. \tag{56.187}$$

The candidate. The $d=1$ specialization of Eq. (56.184) integrates over the boundary interval $|t'| < z$ that is spacelike to the bulk point:

$$\phi(z,t) = c_\Delta \int_{-z}^{z} \mathrm{d}t' \left(\frac{z^2 - t'^2}{z}\right)^{\Delta-1} \mathcal{O}(t+t'), \qquad c_\Delta = \frac{\Gamma(\Delta+\frac12)}{\sqrt{\pi}\,\Gamma(\Delta)}. \tag{56.188}$$

The substitution $t' = z\,s$, $s \in [-1,1]$, removes the z-dependence from the limits and exposes the structure cleanly:

$$\phi(z,t) = c_\Delta\, z^\Delta \int_{-1}^{1} \mathrm{d}s\; (1-s^2)^{\Delta-1}\,\mathcal{O}(t+zs). \tag{56.189}$$

Verification of the boundary condition. As $z \to 0$, $\mathcal{O}(t+zs) \to \mathcal{O}(t)$ and

$$\int_{-1}^{1}(1-s^2)^{\Delta-1}\,\mathrm{d}s = B\Big(\tfrac12,\Delta\Big) = \frac{\Gamma(\frac12)\Gamma(\Delta)}{\Gamma(\Delta+\frac12)} = \frac{\sqrt{\pi}\,\Gamma(\Delta)}{\Gamma(\Delta+\frac12)} = \frac{1}{c_\Delta}, \tag{56.190}$$

so $\phi(z,t) \to z^\Delta \mathcal{O}(t)$, exactly the extrapolate dictionary Eq. (56.180). The normalization c_Δ was chosen for precisely this.

Verification of the wave equation. Write $\phi = c_\Delta z^\Delta I$ with $I(z,t) = \int_{-1}^{1}(1-s^2)^{\Delta-1}\mathcal{O}(t+zs)\,\mathrm{d}s$. Denote by $\mathcal{O}', \mathcal{O}''$ derivatives with respect to the argument $u = t + zs$, and note the two ways the argument depends on the variables: $\partial_t\mathcal{O} = \mathcal{O}'$ and $\partial_z\mathcal{O} = s\,\mathcal{O}' = \frac{1}{z}\,\partial_s\mathcal{O}$. Differentiating,

$$\begin{aligned}
\partial_t^2\phi &= c_\Delta z^\Delta \int_{-1}^{1}(1-s^2)^{\Delta-1}\mathcal{O}''\,\mathrm{d}s,\\
\partial_z^2\phi &= c_\Delta\Big[\Delta(\Delta-1)z^{\Delta-2}I + 2\Delta z^{\Delta-1}I_z + z^\Delta I_{zz}\Big],
\end{aligned} \tag{56.191}$$

with $I_z = \int(1-s^2)^{\Delta-1}s\,\mathcal{O}'\,\mathrm{d}s$ and $I_{zz} = \int(1-s^2)^{\Delta-1}s^2\mathcal{O}''\,\mathrm{d}s$. Inserting these into the wave operator Eq. (56.187), the terms $\pm\Delta(\Delta-1)z^{\Delta-2}I$ cancel against each other, leaving

$$\Big(\partial_z^2 - \partial_t^2 - \tfrac{\Delta(\Delta-1)}{z^2}\Big)\phi = c_\Delta\Big[2\Delta z^{\Delta-1}I_z + z^\Delta(I_{zz} - I_{tt})\Big], \qquad I_{tt} = \int(1-s^2)^{\Delta-1}\mathcal{O}''\,\mathrm{d}s. \tag{56.192}$$

Now convert all argument-derivatives into s-derivatives using $\mathcal{O}' = \frac{1}{z}\partial_s\mathcal{O}$ and $\mathcal{O}'' = \frac{1}{z^2}\partial_s^2\mathcal{O}$:

$$I_{zz}-I_{tt} = \int_{-1}^{1}(s^2-1)(1-s^2)^{\Delta-1}\mathcal{O}''\,\mathrm{d}s = -\frac{1}{z^2}\int_{-1}^{1}(1-s^2)^{\Delta}\,\partial_s^2\mathcal{O}\,\mathrm{d}s, \qquad I_z = \frac{1}{z}\int_{-1}^{1}(1-s^2)^{\Delta-1} \tag{56.193}$$

Integrate the first by parts twice (the boundary terms vanish because $(1-s^2)^{\Delta}$ and its first derivative vanish at $s=\pm1$ for $\Delta>0$); the single integration by parts of $\partial_s^2\mathcal{O}$ against $(1-s^2)^{\Delta}$ gives

$$\int_{-1}^{1}(1-s^2)^{\Delta}\,\partial_s^2\mathcal{O}\,\mathrm{d}s = -\int_{-1}^{1}\partial_s[(1-s^2)^{\Delta}]\,\partial_s\mathcal{O}\,\mathrm{d}s = 2\Delta\int_{-1}^{1}s\,(1-s^2)^{\Delta-1}\partial_s\mathcal{O}\,\mathrm{d}s. \tag{56.194}$$

Therefore $z^{\Delta}(I_{zz}-I_{tt}) = -2\Delta z^{\Delta-2}\int s(1-s^2)^{\Delta-1}\partial_s\mathcal{O}\,\mathrm{d}s = -2\Delta z^{\Delta-1}I_z$, which is exactly minus the other surviving term in Eq. (56.192). The residual vanishes identically:

$$\Big(\partial_z^2 - \partial_t^2 - \tfrac{\Delta(\Delta-1)}{z^2}\Big)\phi = 0. \tag{56.195}$$

The smearing formula Eq. (56.188) solves the bulk wave equation and carries the correct boundary data, hence reconstructs the unique normalizable bulk field with boundary operator $\mathcal{O}$.

Causal support and non-manifest locality. Two boundary points enter Eq. (56.188): those with $|t'| < z$, i.e. boundary points within the interval $(t-z, t+z)$. In the Poincaré metric the bulk point (z,t) and a boundary point $(0, t+t')$ are spacelike separated precisely when $t'^2 < z^2$ (the boundary lightcone of the bulk point opens at 45° in these coordinates), so the kernel is supported on exactly the boundary points spacelike to the bulk point — the spacelike smearing of Theorem 56.37. This is the operational content of the bulk field being reconstructable from a *causal wedge*: only the boundary region in causal contact (spacelike, then evolved) with the bulk point is needed. Finally, locality is present but invisible. The reconstructed $\phi(z,t)$ is a spatially extended smear of $\mathcal{O}$, yet the bulk theory is local: two such reconstructed fields at spacelike-separated bulk points must commute. Their smearing regions on the boundary overlap, so term by term the commutator $[\phi(X),\phi(Y)]$ is a nonzero double integral of boundary commutators; it vanishes only after the precise kernel Eq. (56.185) forces a complete cancellation. Bulk microcausality is a conspiracy among boundary operators, not a manifest property of the representation — the first hint that "local bulk operator" is an emergent, redundantly encoded notion rather than a primitive one.

Figure 56.9 summarizes the picture. The construction extends, with more bookkeeping, to global AdS (where the mode sum Eq. (56.183) runs over the discrete normal modes and resums to a smearing over the global boundary

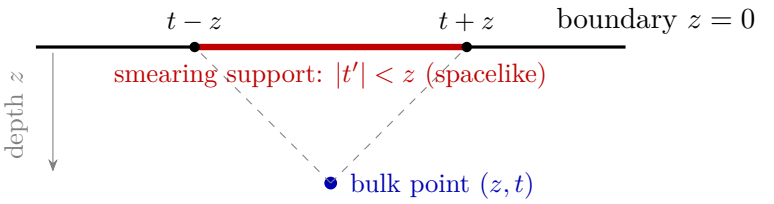


Figure 56.9: The HKLL spacelike-smearing representation Eq. (56.188) in AdS_2. A free bulk field at depth z and boundary time t is a linear smear of the boundary operator $\mathcal{O}$ over the interval $(t-z, t+z)$ — precisely the boundary points *spacelike* separated from the bulk point (the dashed boundary lightcone reaches $t' = t \pm z$). Larger bulk depth requires a wider boundary smearing, the reconstruction analogue of the deeper reach of larger RT surfaces.

lightcone) and, crucially, to AdS–Rindler wedges. The AdS–Rindler version is the one that matters for what follows: a bulk operator inside the bulk domain of dependence of a *boundary subregion* A admits an HKLL representation using only the operators $\mathcal{O}$ restricted to A. That subregion is the *causal wedge* of A — the bulk region in causal contact with the boundary causal diamond of A. HKLL thus already delivers reconstruction of the causal wedge from ρ_A. The next subsection states the stronger fact that the recoverable region is not the causal wedge but the (generally larger) entanglement wedge, and the subsection after that explains why the gap between them is the signature of error correction.

56.7.2 Entanglement-wedge reconstruction

Recall (Theorem 56.31) that the entanglement wedge $\mathcal{W}_A$ of a boundary region A is the bulk domain of dependence of the homology region $\mathcal{R}_A$ bounded by A and its RT surface γ_A^* (Theorems 56.2 and 56.3). It contains, and generically strictly exceeds, the causal wedge: there are bulk regions a boundary observer cannot reach causally yet can still reconstruct. The reconstruction theorem promotes subregion duality from a statement about density matrices to a statement about operators.

Theorem 56.39 (Entanglement-wedge reconstruction)**.** *Let A be a boundary region with entanglement wedge $\mathcal{W}_A$, and let the bulk effective field theory be restricted to a code subspace $\mathcal{H}_{\text{code}}$ of states with a common semiclassical geometry (Theorem 56.41). For every bulk operator ϕ supported in $\mathcal{W}_A$ there exists a boundary operator $\mathcal{O}_A$, built from the operators of A alone (i.e. an element of the operator algebra $\mathcal{A}_A$ acting on ρ_A), such that*

$$\phi\,|\psi\rangle = \mathcal{O}_A\,|\psi\rangle\,, \qquad \phi^\dagger\,|\psi\rangle = \mathcal{O}_A^\dagger\,|\psi\rangle \qquad \text{for all } |\psi\rangle \in \mathcal{H}_{\text{code}}. \tag{56.196}$$

Equivalently, every bulk observable in $\mathcal{W}_A$ is represented within the algebra of A, when acting on the code subspace.

The qualifier "on the code subspace" is not a hedge; it is the whole content, as the next subsection shows. Before that, the immediate corollary that exposes the paradox.

Corollary 56.40 (Complementary recovery)**.** *Suppose the global boundary state is pure and A, $\bar{A}$ partition the boundary, with entanglement wedges $\mathcal{W}_A, \mathcal{W}_{\bar{A}}$ that tile the bulk slice across the common RT surface $\gamma_A^* = \gamma_{\bar{A}}^*$ (the generic situation away from a quantum-extremal-surface transition). Then every bulk operator is reconstructable on* exactly one *of the two complementary regions: operators in $\mathcal{W}_A$ on A, operators in $\mathcal{W}_{\bar{A}}$ on $\bar{A}$. A bulk operator straddling the RT surface is reconstructable on neither in isolation.*

The proof of Theorem 56.39 from JLMS is exactly the algebraic argument we now give, and it is best stated as the central theorem of the error-correction framework. The corollary is its restriction to a pure global state, where $\mathcal{H}_{\text{code}}$ admits a clean complementary structure.

56.7.3 The error-correction paradigm of Almheiri, Dong, and Harlow

We now confront the paradox of the section's opening and resolve it. The resolution reorganizes everything: it identifies the precise sense in which bulk locality is encoded redundantly, and it ties the existence of reconstruction to the equality of bulk and boundary entropies — the RT/FLM formula — by a chain of equivalences.

The code subspace. The bulk effective field theory does not act on the whole CFT Hilbert space. It is the description of small fluctuations around a fixed classical background, valid only for states whose backreaction is sub-Planckian. This selects a subspace.

Definition 56.41 (Code subspace)**.** The *code subspace* $\mathcal{H}_{\text{code}} \subset \mathcal{H}_{\text{CFT}}$ is the span of states obtained by acting on a fixed semiclassical reference state $|0\rangle$ (a fixed bulk geometry) with a bounded number of low-energy bulk effective-field-theory operators. The encoding is the isometry

$$V : \mathcal{H}_{\text{code}}^{\text{bulk}} \longrightarrow \mathcal{H}_{\text{CFT}}, \qquad V^\dagger V = \mathbf{1}_{\text{code}}, \tag{56.197}$$

mapping the abstract bulk Fock space into the (vastly larger) boundary Hilbert space. A bulk operator ϕ acts as the *logical* operator $\bar{\phi} = V\phi V^\dagger$ on the image, supported in the code subspace.

The isometry V is not unitary: $\dim \mathcal{H}_{\text{code}} \ll \dim \mathcal{H}_{\text{CFT}}$, and that mismatch is the room into which information is redundantly encoded. The boundary states in the image of V are the *codewords*.

The paradox, sharply. A bulk operator ϕ at the center of AdS lies in $\mathcal{W}_A$ for every sufficiently large region A. Take two boundary regions A_1 and A_2 that are *disjoint*, each large enough that the center is in its entanglement wedge — this is possible whenever neither A_i alone is more than half the boundary but their wedges both reach the deep interior, as happens for, e.g., two large but disjoint caps. By Theorem 56.39, ϕ equals some $\mathcal{O}_{A_1} \in \mathcal{A}_{A_1}$ and also some $\mathcal{O}_{A_2} \in \mathcal{A}_{A_2}$. But A_1 and A_2 are spacelike separated on the boundary, so $[\mathcal{O}_{A_1}, \mathcal{O}_{A_2}] = 0$ identically as boundary operators. Were $\phi = \mathcal{O}_{A_1} = \mathcal{O}_{A_2}$ as honest operators on the full Hilbert space, we would also need ϕ to be simultaneously a function of A_1-data and of A_2-data, and worse, the radial-commutativity property — $[\phi(\text{center}), \mathcal{O}(x)] = 0$ for every boundary $\mathcal{O}$, since a deep bulk operator commutes with all single-trace boundary operators at leading order — would clash with ϕ being a nontrivial element of $\mathcal{A}_{A_1}$. Something has to give.

The resolution. The equalities Eq. (56.196) hold only *on* $\mathcal{H}_{\text{code}}$. As operators on the full $\mathcal{H}_{\text{CFT}}$, $\mathcal{O}_{A_1}$ and $\mathcal{O}_{A_2}$ are different; they merely agree when restricted to codewords. This is exactly the defining feature of a quantum error-correcting code: a logical operator $\bar{\phi}$ has many physical representatives, each of which reproduces its action on the code subspace while acting differently — and harmlessly — on the orthogonal complement. The disjoint representations $\mathcal{O}_{A_1}, \mathcal{O}_{A_2}$ do not contradict $[\mathcal{O}_{A_1}, \mathcal{O}_{A_2}] = 0$, because $\bar{\phi}$ is not equal to either of them off-code, and the commutator only needs to annihilate codewords:

$$[\mathcal{O}_{A_1}, \mathcal{O}_{A_2}]\, V = 0 \quad \text{(on the code subspace)}, \qquad \text{but} \quad \mathcal{O}_{A_1} \neq \mathcal{O}_{A_2} \text{ as operators on } \mathcal{H}_{\text{CFT}}. \tag{56.198}$$

Radial commutativity is likewise saved: a deep bulk operator commutes with boundary operators only in its action on codewords, which is all the bulk effective theory ever probes.

Theorem 56.42 (Holographic codes: Almheiri–Dong–Harlow)**.** *The bulk-to-boundary map of subregion duality is a quantum error-correcting code in the following precise sense. Bulk effective-field-theory states form a code subspace* $\mathcal{H}_{\text{code}}$ *embedded by an isometry* V *Eq. (56.197); bulk local operators are logical operators with multiple boundary representations; and entanglement-wedge reconstruction Theorem 56.39 is the statement that the code* corrects the erasure of $\bar{A}$ *whenever the bulk operator lies in* $\mathcal{W}_A$*. Concretely, a logical operator supported in* $\mathcal{W}_A$ *can be reconstructed using only the physical degrees of freedom in* A*, so erasing the complementary region* $\bar{A}$ *destroys no information about it. The recoverable bulk region grows from the causal wedge to the entanglement wedge precisely because error correction lets reconstruction reach beyond causal contact.*

Entanglement equals recoverability. The deepest part of the framework ties *which* region is recoverable to the entropy formula itself. The bridge is the JLMS

identity of the previous section (Theorem 56.35), which equates the boundary modular Hamiltonian $K_A = -\log \rho_A$ of the region with the bulk modular Hamiltonian $K^{\text{bulk}}_{\mathcal{W}_A}$ of its entanglement wedge plus the operator measuring the area of the RT surface,

$$K_A = \frac{\widehat{\text{Area}}(\gamma_A^*)}{4G_N} + K^{\text{bulk}}_{\mathcal{W}_A} + \dots, \tag{56.199}$$

to leading order in G_N. We use it to prove reconstruction algebraically, following Dong, Harlow, and Wall. Suppose ϕ is a bulk operator in $\mathcal{W}_A$. The reconstructability of ϕ on A is the statement that ϕ, acting on the code subspace, lies in the algebra $\mathcal{A}_A$. The clean criterion for membership in a von Neumann algebra is commutation with its commutant: $\mathcal{A}_A' = \mathcal{A}_{\bar{A}}$, the algebra of the complementary region. So ϕ is reconstructable on A if and only if, on the code subspace,

$$[\phi,\ \mathcal{O}_{\bar{A}}] = 0 \qquad \text{for every } \mathcal{O}_{\bar{A}} \in \mathcal{A}_{\bar{A}}. \tag{56.200}$$

Now the key step. Boundary modular flow $\rho_A^{is}\,(\,\cdot\,)\,\rho_A^{-is}$ keeps the algebra $\mathcal{A}_A$ invariant (Tomita–Takesaki), and by Eq. (56.199) this boundary flow acts on code-subspace operators as the *bulk* modular flow generated by $K^{\text{bulk}}_{\mathcal{W}_A}$ (the area term is a c-number on each fixed-geometry sector and contributes only a phase). The bulk modular flow of $\mathcal{W}_A$ maps operators localized in $\mathcal{W}_A$ to operators localized in $\mathcal{W}_A$, because bulk modular flow is geometric near the wedge and preserves bulk causal localization. Therefore a ϕ localized in $\mathcal{W}_A$ stays in a region spacelike to $\mathcal{W}_{\bar{A}}$ under the flow, and bulk microcausality gives $[\phi, \mathcal{O}_{\bar{A}}] = 0$ on the code subspace — which is exactly Eq. (56.200). The equality of bulk and boundary modular flows, i.e. JLMS, is thus equivalent to reconstructability; and JLMS is in turn the first-order ($\rho_A = \sigma_A e^{\delta}$) content of the equality of bulk and boundary *relative entropies*,

$$S_{\text{rel}}(\rho_A \,\|\, \sigma_A) = S_{\text{rel}}(\rho^{\text{bulk}}_{\mathcal{W}_A} \,\|\, \sigma^{\text{bulk}}_{\mathcal{W}_A}), \tag{56.201}$$

with S_{rel} the relative entropy of Eq. (55.15). The chain closes on the entropy formula: Eq. (56.201) holding for all nearby code states is, by the FLM expansion of $S_A = \langle\widehat{\text{Area}}\rangle/4G_N + S^{\text{bulk}}_{\mathcal{W}_A}$, the same data as the quantum-corrected RT formula. The slogan is exact: *a boundary region can recover a bulk operator precisely when its entanglement entropy is computed by an RT surface enclosing that operator.* Entanglement (the RT area) and recoverability (the code structure) are two faces of one fact.

Physical Insight

This is the chapter's conceptual summit made operational. "Geometry from entanglement" acquires a mechanism: the bulk is not merely correlated with the boundary, it is *encoded* in it as a quantum error-correcting code,

with the RT surface playing the role of the code's boundary of recoverability. The redundancy that protects a logical qubit against erasure is the same redundancy that lets many boundary regions each reconstruct a deep bulk point — and the amount of that redundancy, the depth to which a region's wedge reaches, is measured by the area of its RT surface. Error correction is why a smooth, local interior can be woven out of a boundary theory that has no local interior of its own.

56.7.4 The HaPPY pentagon code: an exactly solvable model

Everything above is structural; it is reassuring, and decisive, to see it realized exactly in a finite-dimensional model where "RT surface," "entanglement wedge," "reconstruction," and "erasure correction" are all elementary stabilizer-code statements. The model of Pastawski, Yoshida, Harlow, and Preskill — the HaPPY code — builds a holographic code from a tensor network on a hyperbolic tiling whose elementary cell is a single *perfect tensor*. We define the perfect tensor, prove the properties on which everything hinges, exhibit operator pushing explicitly, and verify the toy RT formula by counting.

Definition 56.43 (Perfect tensor)**.** A tensor $T_{a_1\cdots a_{2n}}$ with $2n$ legs, each of bond dimension v (so $a_i \in \{1,\dots,v\}$), is *perfect* if for every bipartition of its legs into a set S and its complement S^c with $|S| \le |S^c|$, the tensor, viewed as a linear map $T : \mathcal{H}_S \to \mathcal{H}_{S^c}$ from the legs in S to those in S^c, is proportional to an isometry:

$$T_S^\dagger T_S = v^{\,|S|}\,\mathbf{1}_{\mathcal{H}_S}. \tag{56.202}$$

Equivalently, the pure state $|T\rangle = \sum T_{a_1\cdots a_{2n}}\,|a_1\cdots a_{2n}\rangle$ on $2n$ parties is *absolutely maximally entangled*: the reduced state on any $k \le n$ legs is maximally mixed, $\rho_S = v^{-k}\mathbf{1}$.

Perfectness packs in one stroke the two properties we need. First, *maximal entanglement across every balanced cut* — this will produce the RT counting. Second, the *isometry/pushing property*: because T is an isometry from any small set of legs to the rest, an operator inserted on one set of legs can be transported across the tensor to an equivalent operator on a complementary set, the discrete analogue of HKLL transport.

The atom: a six-leg perfect tensor is the five-qubit code. Take $n = 3$, $v = 2$: a perfect tensor with six qubit legs. Designate one leg the *bulk* (logical) input and the remaining five the *boundary* (physical) outputs. The isometry from the one bulk leg to the five boundary legs is, with the standard choice of perfect tensor, exactly the encoding of the *five-qubit code* $[[5, 1, 3]]$: one logical

qubit protected in five physical qubits, with code distance 3. Its stabilizer group is generated by the four cyclic translates

$$g_1 = XZZXI, \quad g_2 = IXZZX, \quad g_3 = XIXZZ, \quad g_4 = ZXIXZ, \tag{56.203}$$

and the logical operators may be taken as $\bar{X} = XXXXX$ and $\bar{Z} = ZZZZZ$. (Here a string like $XZZXI$ means $X_1 Z_2 Z_3 X_4 I_5$, the tensor product of single-qubit Paulis.) Distance 3 means the minimum weight of a nontrivial logical operator is 3: no operator supported on two or fewer qubits acts nontrivially on the logical information. As an *erasure* code this means any 2 of the 5 physical qubits may be lost and the logical qubit is still perfectly recoverable from the remaining 3.

Worked Example 56.44 (Operator pushing, complementary recovery, and the toy RT formula in the $[[5, 1, 3]]$ code)**.** We make every structural claim of Section 56.7.3 explicit in this one tensor.

Pushing a logical operator onto three qubits. The logical $\bar{X} = XXXXX$ has weight 5, but it is equivalent, modulo the stabilizer group, to lower-weight representatives — this is the "cleaning" that realizes reconstruction on a subregion. Multiply $\bar{X}$ by the stabilizer $g_1 g_2$. First,

$$g_1 g_2 = (XZZXI)(IXZZX) = X\,(ZX)\,(ZZ)\,(XZ)\,X \propto X\,Y\,I\,Y\,X, \tag{56.204}$$

using $ZX = -iY$, $ZZ = I$, $XZ = -iY$ leg by leg (phases are immaterial for the support). Then

$$\bar{X}\,(g_1 g_2) = (XXXXX)(XYIYX) \propto I\,Z\,X\,Z\,I, \tag{56.205}$$

using $XX = I$ on legs $1, 5$, $XY = iZ$ on legs $2, 4$, and $XI = X$ on leg 3. Since $g_1 g_2$ is in the stabilizer, $\bar{X}\,(g_1 g_2)$ acts identically to $\bar{X}$ on every codeword: the logical $\bar{X}$ has a representative

$$\bar{X} \;\cong\; Z_2\, X_3\, Z_4, \tag{56.206}$$

supported on the three qubits $\{2, 3, 4\}$ only. By the cyclic symmetry of the code, $\bar{X}$ (and likewise $\bar{Z}$) has a weight-3 representative on *every* consecutive triple of qubits. This is operator pushing in the flesh: the bulk (logical) operator is transported, through the perfect tensor's isometry, onto a chosen set of three boundary legs.

Complementary recovery. Split the five boundary qubits into $A = \{2, 3, 4\}$ and $\bar{A} = \{1, 5\}$. By Eq. (56.206) (and its $\bar{Z}$ analogue) both logical operators have representatives supported entirely on A, so the logical qubit is reconstructable on A: the bulk operator sits in the entanglement wedge of A. Meanwhile $\bar{A}$ has only

two qubits, and distance 3 forbids any nontrivial logical operator of weight ≤ 2, so nothing logical is reconstructable on $\bar{A}$: the bulk operator is *not* in the wedge of $\bar{A}$. This is exactly the complementary-recovery dichotomy of Theorem 56.40, now a one-line consequence of the code distance. It also realizes the paradox and its resolution: the logical operator has a representative on $\{2,3,4\}$ and, by cyclic symmetry, another on the disjoint-from-$\{2,3,4\}$ portion of some other triple; these representatives commute and agree on codewords while differing off-code, precisely Eq. (56.198).

The toy RT formula by counting. Encode a fixed pure logical state, so the five-qubit physical state $\left|\bar{\psi}\right\rangle$ is pure and the "bulk" carries no entropy. For a boundary region R of the physical qubits, compute $S(R) = -\operatorname{Tr}(\rho_R \log \rho_R)$. Because the code has distance 3, the reduced density matrix on any set of ≤ 2 qubits is independent of the logical state — it carries no logical information — and by the maximal entanglement of the underlying perfect tensor it is maximally mixed: $\rho_R = 2^{-|R|}\mathbf{1}$ for $|R| \leq 2$. Hence

$$S(R) = |R| \log 2 \quad (|R| \leq 2), \qquad S(A) = S(\bar{A}) = 2\log 2 \text{ for } A = \{2,3,4\},\ \bar{A} = \{1,5\}, \tag{56.207}$$

the second equality by purity of $\left|\bar{\psi}\right\rangle$ ($S(A) = S(\bar{A})$). Now read this as a Ryu–Takayanagi count. In a tensor network the entropy of a boundary region is the *minimal cut*: the least number of bonds one must sever to disconnect R from its complement, times $\log v$. For the single six-leg tensor the surface homologous to R cuts exactly $\min(|R|, |\bar{R}|)$ legs (the homology constraint allows cutting either R's own legs or its complement's, and minimization picks the smaller set), so the toy RT formula gives

$$S_{\mathrm{RT}}(R) = \left[\min(|R|, |\bar{R}|_{\mathrm{boundary}})\right] \log 2, \tag{56.208}$$

which for $A = \{2,3,4\}$ ($|A| = 3$, complement two boundary legs) gives $2\log 2$, matching Eq. (56.207) exactly. The minimal cut that achieves it passes on the $\bar{A} = \{1,5\}$ side, severing two legs; the entanglement wedge of A is everything on the far side of that cut, which is the bulk leg — so the bulk qubit is in $\mathcal{W}_A$, in agreement with the operator-pushing result. The FLM split is visible too: with a fixed pure logical input the bulk entropy vanishes and $S(A) = (\text{cut area}) \log 2$; promote the bulk leg to an entangled state with an external reference and $S(A)$ acquires the extra bulk-entanglement term $S^{\mathrm{bulk}}_{\mathcal{W}_A}$, exactly $S = \langle \mathrm{Area} \rangle / 4G_N + S^{\mathrm{bulk}}$ in miniature.

Figure 56.10 shows how the single tensor scales up. Tiling the hyperbolic disk with perfect-tensor pentagons and contracting all in-plane legs produces an isometry V from the collection of dangling bulk legs to the boundary legs. The reconstruction of any bulk leg is performed by the *greedy algorithm*: starting from

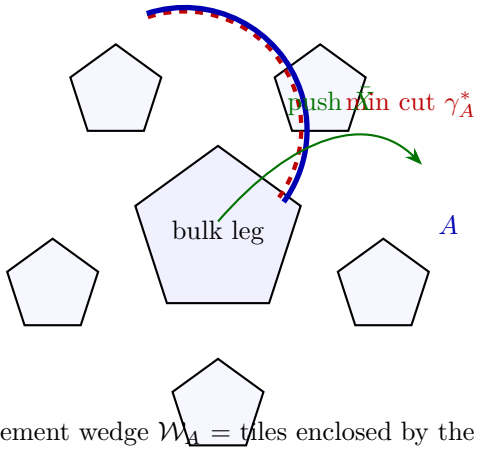


Figure 56.10: The HaPPY holographic code on a pentagon ($\{5,4\}$) tiling of the hyperbolic disk. Each pentagon is a six-leg perfect tensor: five legs contract with neighbors, one "bulk" leg (centre) injects a logical qubit. Contracting the network builds the encoding isometry V from bulk legs to the boundary legs (outer edge). For a boundary region A, the discrete RT surface is the *greedy geodesic* — the minimal-cut bond set (dashed) — and the entanglement wedge $\mathcal{W}_A$ is the set of tiles it encloses. A logical operator on a bulk tile is pushed (arrow), through the perfect-tensor isometries, out to a representative supported on A, realizing entanglement-wedge reconstruction; cutting the fewest bonds reproduces the RT formula $S(A) = (\#\text{cut bonds}) \log v$.

a boundary region A, absorb any tensor that has at least half of its legs already in the reconstructed set (each such absorption is an isometry one can invert, i.e. a pushing step), and iterate until no more can be added. The frontier where the algorithm halts is the *greedy geodesic*, the network's discrete RT surface, and the set of tensors swallowed is the entanglement wedge $\mathcal{W}_A$. Every bulk operator on a tile inside $\mathcal{W}_A$ is reconstructable on A by composing the pushing steps; the count of bonds the greedy geodesic severs, times $\log v$, is the entropy $S(A)$, with equality whenever the residual perfect-tensor entanglement saturates the cut. The model thus reproduces, exactly and finitely, every feature established abstractly in this section: an exact RT relation, erasure (entanglement-wedge) reconstruction, complementary recovery, and the radial-commutativity resolution of overlapping representations.

Caution

The HaPPY code is a faithful cartoon, not the real thing. Its entanglement spectrum is flat (all RT-counted entropies are integers times $\log v$), so it captures the leading area term but not the continuous bulk-matter

corrections of FLM except by hand; its "geometry" is fixed and rigid, with no dynamical gravitational backreaction; and it has no time direction, hence no notion of the covariant HRT surface. It is an exact model of the *kinematics* of holographic encoding — which is precisely what we needed to make "bulk reconstruction is error correction" a theorem about a concrete object rather than a slogan. The dynamical content is supplied elsewhere.

We now hold a fully mechanical picture of how the bulk lives inside the boundary: as a quantum error-correcting code in which local bulk operators are redundantly encoded logical operators, the recoverable region is the entanglement wedge, and the boundary of recoverability is the RT surface whose area measures the entropy. What remains is to let this picture do dynamical work. If the connectivity of bulk spacetime is the entanglement structure of the boundary, then changing the entanglement must change the geometry — and the next section turns that observation into the chapter's culminating result, deriving the linearized Einstein equations of the bulk from the first law of entanglement on the boundary, before surveying the tensor-network and ER = EPR pictures and dating the open frontier handed to Chapter 57.

56.8 Spacetime from entanglement: synthesis and frontier

The previous section taught the operational lesson of the chapter: the bulk is not merely correlated with the boundary, it is *encoded* in it, and encoded in the precise manner of a quantum error-correcting code, with the entanglement wedge as the recoverable region and the entanglement structure of the boundary state as the redundancy that protects it (Section 56.7 and Theorem 56.33). The HaPPY pentagon code (Section 56.7.4) made this concrete in a finite model: bulk information survives the erasure of any boundary subregion small enough to leave the relevant wedge intact, exactly because the boundary state is highly entangled. Read forward, that statement is startling. If the bulk geometry is reconstructed from boundary entanglement, then the geometry should not exist independently of that entanglement — it should be *made of* it. This closing section makes the slogan precise and then dates what remains unknown. We prove the sharpest available version of "spacetime is built from entanglement": that the bulk Einstein equation, linearized about pure anti–de Sitter space, is not an independent dynamical postulate but a rewriting of the first law of entanglement on the boundary, holding for every ball-shaped region. We then survey two pictures — Van Raamsdonk's disentangling thought experiment with its tensor-network/MERA discretization, and the ER = EPR slogan — that extend the lesson to the connectivity and topology of spacetime, stating each

honestly within its scope. We close with a dated summary of the open problems this static, kinematic story hands to the dynamical sequel of Chapter 57.

56.8.1 The first law of entanglement

Everything in this section rests on a single near-trivial identity of quantum statistical mechanics, applied to the reduced density matrix of a region. Recall (Theorem 55.28) that for a state ρ reduced to a region A, the *modular Hamiltonian* is the Hermitian operator

$$K_A \;=\; -\ln \rho_A \qquad \text{(up to an additive constant)}, \tag{56.209}$$

the generator whose Gibbs weight e^{-K_A} reproduces ρ_A and whose flow ρ_A^{is} is modular flow. Fix a reference state $\sigma_A = e^{-K_A}$ on A — in all our applications the vacuum reduced to A — and consider a one-parameter family $\rho_A(\lambda)$ of nearby states with $\rho_A(0) = \sigma_A$. Write the entanglement entropy $S(\rho_A) = -\operatorname{Tr}(\rho_A \ln \rho_A)$ and the modular energy $\langle K_A \rangle_\rho = \operatorname{Tr}(\rho_A K_A)$, where K_A is the *fixed* modular Hamiltonian of the reference state σ_A.

Proposition 56.45 (First law of entanglement)**.** *Let $\rho_A = \sigma_A + \delta\rho_A$ be a first-order deformation of the reference state $\sigma_A = e^{-K_A}$ on a region A, with $\operatorname{Tr}\,\delta\rho_A = 0$ (normalization preserved). Then the first-order change in entanglement entropy equals the first-order change in modular energy,*

$$\boxed{\;\delta S(A) \;=\; \delta\langle K_A \rangle\;,\;} \tag{56.210}$$

where $\delta S(A) = S(\rho_A) - S(\sigma_A)$ and $\delta\langle K_A\rangle = \operatorname{Tr}(\delta\rho_A\, K_A)$, both to first order in $\delta\rho_A$.

Proof. Vary the von Neumann entropy directly. With $S = -\operatorname{Tr}(\rho_A \ln \rho_A)$,

$$\delta S \;=\; -\operatorname{Tr}(\delta\rho_A\, \ln \rho_A) \;-\; \operatorname{Tr}(\rho_A\, \delta \ln \rho_A). \tag{56.211}$$

The second term vanishes: using $\delta \ln \rho_A = \rho_A^{-1}\delta\rho_A$ to leading order (more carefully, $\operatorname{Tr}(\rho_A\, \delta \ln \rho_A) = \operatorname{Tr}\,\delta\rho_A$ by the cyclicity that the spectral integral for ln enforces under the trace), we get $\operatorname{Tr}(\rho_A\, \delta \ln \rho_A) = \operatorname{Tr}\,\delta\rho_A = 0$ by the normalization condition. In the first term, evaluate $\ln \rho_A$ on the reference state, $\ln \rho_A \to \ln \sigma_A = -K_A$ to leading order, since the difference is itself first order and multiplies the first-order $\delta\rho_A$. Hence

$$\delta S \;=\; -\operatorname{Tr}(\delta\rho_A\, \ln \sigma_A) \;=\; \operatorname{Tr}(\delta\rho_A\, K_A) \;=\; \delta\langle K_A\rangle, \tag{56.212}$$

which is Eq. (56.210). □

The first law is the linearization of a deeper, exact, non-perturbative inequality the reader already met in Chapter 55: the positivity of relative entropy. With $S(\rho_A \,\|\, \sigma_A) = \mathrm{Tr}(\rho_A \ln \rho_A) - \mathrm{Tr}(\rho_A \ln \sigma_A) = \Delta\langle K_A\rangle - \Delta S \geq 0$ (Section 55.6), where Δ denotes the full difference from the reference state, one has the *exact* statement $\Delta S \leq \Delta\langle K_A\rangle$ for any deformation (the entanglement form of the Clausius inequality, Section 55.6.1). Relative entropy is non-negative and vanishes at the reference state, so it is second order in $\delta\rho_A$; its first-order part is therefore zero, and that is precisely Eq. (56.210). The first law is the saturation of the Clausius inequality at linear order. We will use the equality Eq. (56.210) to derive Einstein's equation; the inequality $\Delta S \leq \Delta\langle K\rangle$ at second order is what feeds the bulk energy conditions of Section 56.6.

What makes Eq. (56.210) powerful holographically is that for ball-shaped regions in the vacuum of a CFT, the right-hand side $\delta\langle K_A\rangle$ is an explicit, *local* integral of the boundary stress tensor, and the left-hand side δS is, by Ryu–Takayanagi, an explicit geometric quantity in the bulk. Setting two explicit objects equal, for every ball, is a strong constraint. We assemble the two sides in turn.

56.8.2 The modular Hamiltonian of a ball

The right-hand side of the first law requires the modular Hamiltonian of the vacuum reduced to a ball. We import the one nontrivial case where it is known in closed form: the Casini–Huerta–Myers (CHM) result, derived in Chapter 55 (Theorem 55.49 and Eq. (55.197)) by conformally mapping the ball's causal development onto a Rindler wedge and transporting the boost generator that Bisognano and Wichmann (Theorem 55.25) identified as the wedge modular Hamiltonian. We restate it because every factor matters for the derivation to follow.

For a ball B of radius R centred at the spatial origin on the time slice $t = 0$ of the vacuum of a CFT_d with stress tensor $T_{\mu\nu}$, the modular Hamiltonian is the local operator

$$K_B \;=\; 2\pi \int_B \mathrm{d}^{\,d-1}x\, \frac{R^2 - r^2}{2R}\, T_{00}(x) \;+\; \text{const}, \qquad r^2 = \sum_i (x^i)^2, \tag{56.213}$$

the energy density weighted by the function $(R^2 - r^2)/2R$ that vanishes on the entangling sphere $r = R$ and peaks at the centre. This weight is the time component of a conformal Killing vector. The causal diamond $\mathcal{D}_B$ of the ball — its domain of dependence — is preserved by a one-parameter conformal flow whose generator is, in boundary coordinates $(t, \vec{x})$,

$$\xi_B \;=\; \frac{2\pi}{R}\left[\frac{R^2 - t^2 - r^2}{2}\,\partial_t \;-\; t\,x^i\,\partial_i\right], \tag{56.214}$$

normalized so that $\rho_B = e^{-K_B}$ generates modular flow with the geometric flow of ξ_B, equivalently so that the flow has unit "surface gravity" 2π at the entangling surface (the Unruh temperature $1/2\pi$ of Eq. (55.83) in modular time). Evaluated at $t=0$, the time component is $\xi_B^t|_{t=0} = \frac{2\pi}{R}\cdot\frac{R^2-r^2}{2}$, which is exactly the weight in Eq. (56.213): the modular Hamiltonian is the conserved charge $K_B = \int_B \xi_B^\mu T_{\mu 0}\,\mathrm{d}^{d-1}x$ of the conformal Killing flow. That ξ_B is a genuine conformal Killing vector of Minkowski space — the property that makes K_B local — is worth verifying explicitly, since the whole derivation hinges on it.

Worked Example 56.46 (ξ_B is a conformal Killing vector vanishing on the entangling surface)**.** We check that ξ_B of Eq. (56.214) satisfies the conformal Killing equation $\partial_\mu\xi_\nu + \partial_\nu\xi_\mu = \frac{2}{d}\,\eta_{\mu\nu}\,(\partial_\alpha\xi^\alpha)$ in the mostly-minus metric $\eta = \mathrm{diag}(+,-,\dots,-)$, and vanishes on the edge of the diamond.

Write $a \equiv 2\pi/R$, so the contravariant components are $\xi^t = \frac{a}{2}(R^2 - t^2 - r^2)$ and $\xi^i = -a\,t\,x^i$. Lower indices with η: $\xi_t = \xi^t$ and $\xi_i = -\xi^i = a\,t\,x^i$ (a spatial index lowered flips the sign). First compute the divergence,

$$\partial_\alpha\xi^\alpha = \partial_t\xi^t + \partial_i\xi^i = \tfrac{a}{2}(-2t) + (-a\,t\,\delta_i^{\,i}) = -a\,t - a\,t\,(d-1) = -a\,t\,d, \quad (56.215)$$

so the right-hand side of the conformal Killing equation is $\frac{2}{d}\,\eta_{\mu\nu}(-a\,t\,d) = -2a\,t\,\eta_{\mu\nu}$. Now the left-hand side, component by component. The time–time component:

$$\partial_t\xi_t + \partial_t\xi_t = 2\,\partial_t\Big[\tfrac{a}{2}(R^2 - t^2 - r^2)\Big] = -2a\,t = -2a\,t\,\eta_{tt}, \quad (56.216)$$

matching, since $\eta_{tt} = +1$. A spatial–spatial component (i,j), using $\xi_i = a\,t\,x^i$:

$$\partial_i\xi_j + \partial_j\xi_i = a\,t\,\delta_j^i + a\,t\,\delta_i^j = 2a\,t\,\delta_{ij} = -2a\,t\,\eta_{ij}, \quad (56.217)$$

matching, since $\eta_{ij} = -\delta_{ij}$. The mixed component (t,i):

$$\partial_t\xi_i + \partial_i\xi_t = \partial_t(a\,t\,x^i) + \partial_i\Big[\tfrac{a}{2}(R^2 - t^2 - r^2)\Big] = a\,x^i + \tfrac{a}{2}(-2x^i) = a\,x^i - a\,x^i = 0, \quad (56.218)$$

matching $-2a\,t\,\eta_{ti} = 0$. All components satisfy the conformal Killing equation, so ξ_B generates a conformal isometry. Finally, on the edge of the causal diamond at $t=0$, $r=R$, both $\xi^t = \frac{a}{2}(R^2-R^2) = 0$ and $\xi^i = -a\cdot 0\cdot x^i = 0$, so ξ_B vanishes on the entangling sphere ∂B. The flow therefore has a fixed locus precisely at ∂B, around which it acts as a boost — the boundary footprint of the bulk Killing horizon we build next.

With Eq. (56.213) in hand the right-hand side of the first law is explicit. For a state ρ that perturbs the vacuum by turning on a stress-tensor expectation value $\langle T_{\mu\nu}\rangle$, the change in modular energy of the ball is

$$\delta\langle K_B\rangle \;=\; 2\pi\int_B \mathrm{d}^{\,d-1}x\,\frac{R^2-r^2}{2R}\,\langle T_{00}(x)\rangle, \quad (56.219)$$

a known functional of the boundary energy density, parameterized by the ball's radius R and centre (and, by boosting, by its rest frame). The first law Eq. (56.210) now reads $\delta S(B) = \delta\langle K_B\rangle$ for every such ball. It remains to compute the left-hand side from the bulk.

56.8.3 Linearized Einstein equations from the first law

We work in pure AdS_{d+1} as the bulk dual of the CFT vacuum, in Poincaré coordinates with the mostly-minus boundary metric,

$$\mathrm{d}s^2_{\mathrm{AdS}} = \frac{L^2}{z^2}(\eta_{\mu\nu}\,\mathrm{d}x^\mu \mathrm{d}x^\nu - \mathrm{d}z^2), \qquad \eta = \mathrm{diag}(+,-,\ldots,-), \tag{56.220}$$

and perturb the metric by $g = g_{\mathrm{AdS}} + h$, with h a linearized perturbation whose normalizable falloff near $z \to 0$ encodes the boundary stress tensor through the standard dictionary of Section 54.6: $h_{\mu\nu} \to \frac{16\pi G_N}{d}\, L^{3-d}\, z^{d-2}\, \langle T_{\mu\nu}\rangle + \cdots$ as $z \to 0$. The dual statement of the first law is a statement about h, and the claim is that h obeys the linearized Einstein equation if and only if the first law holds for every ball. We assemble the two sides geometrically.

The bulk Killing vector and the AdS–Rindler wedge. The boundary conformal Killing vector ξ_B of Eq. (56.214) is the boundary restriction of an exact bulk Killing vector of pure AdS_{d+1}, because boundary conformal isometries are bulk isometries (Section 53.3). Explicitly, the bulk extension is

$$\widehat{\xi}_B \;=\; \frac{2\pi}{R}\left[\frac{R^2 - t^2 - r^2 - z^2}{2}\,\partial_t \;-\; t\,x^i\,\partial_i \;-\; t\,z\,\partial_z\right], \tag{56.221}$$

the unique AdS Killing vector that reduces to ξ_B at $z \to 0$. Its fixed-point set, where all components vanish at $t = 0$, is the hemisphere

$$\widehat{\xi}_B = 0 \text{ at } t = 0 \iff r^2 + z^2 = R^2, \tag{56.222}$$

which is exactly the Ryu–Takayanagi surface γ_B of the ball in the vacuum — the bulk hemisphere anchored on ∂B that the single-interval semicircle of Section 56.1.4 generalizes to d dimensions. The Killing vector $\widehat{\xi}_B$ generates a boost about this hemisphere: γ_B is the bifurcation surface of a bulk Killing horizon, with surface gravity 2π. The wedge of the bulk that $\widehat{\xi}_B$ keeps within itself — the domain of outer communication of this horizon — is the entanglement wedge $\mathcal{W}_B$ of the ball (Theorem 56.31); in the vacuum it is the AdS–Rindler wedge. This is the kinematic miracle that makes the derivation go through: the modular flow of a ball is geometric not only on the boundary but in the bulk, realized as a Killing boost whose horizon is the RT surface.

The entropy side as a Killing energy. The left-hand side of the first law is $\delta S(B) = \delta\,\mathrm{Area}(\gamma_B)/4G_N$, the linearized change in the area of the RT surface (Eq. (56.4)). Because γ_B is *extremal*, the first-order change in its area receives no contribution from the first-order displacement of the surface — that is the definition of extremality, $\delta\,\mathrm{Area}/\delta X^\mu = 0$ — so $\delta\,\mathrm{Area}$ is computed by holding the surface fixed and varying only the metric:

$$\delta\,\mathrm{Area}(\gamma_B) = \frac{1}{2}\int_{\gamma_B} \mathrm{d}^{\,d-1}\sigma\,\sqrt{h_{(\gamma)}}\;h^{ab}_{(\gamma)}\,\delta g_{ab}, \tag{56.223}$$

where $h_{(\gamma)ab}$ is the unperturbed induced metric on γ_B and δg_{ab} the induced metric perturbation. This is a bulk-data integral localized on the surface. To compare it with the boundary modular energy we use the conserved structure supplied by the Killing vector.

The Iyer–Wald identity. Diffeomorphism-invariant gravity carries, for any vector field ζ, a Noether machinery we now invoke. From the Einstein–Hilbert Lagrangian $(d+1)$-form $\mathbf{L} = \frac{1}{16\pi G_N}(R - 2\Lambda)\,\boldsymbol{\epsilon}$, a variation gives $\delta\mathbf{L} = \mathbf{E}\,\delta g + \mathrm{d}\boldsymbol{\theta}(\delta g)$, defining the equations of motion $\mathbf{E}$ (the Einstein tensor plus matter) and the symplectic potential (d)-form $\boldsymbol{\theta}$. For a vector ζ the Noether current $\mathbf{J}_\zeta = \boldsymbol{\theta}(\mathcal{L}_\zeta g) - i_\zeta \mathbf{L}$ can be written, on using the identities of the formalism, as $\mathbf{J}_\zeta = \mathrm{d}\mathbf{Q}_\zeta + \zeta^\mu \mathbf{C}_\mu$, with $\mathbf{Q}_\zeta$ the Noether charge $(d-1)$-form and $\mathbf{C}_\mu$ the constraint forms that vanish on-shell ($\mathbf{C}_\mu \propto$ Einstein's equation contracted with one index). The *fundamental variational identity* of Iyer and Wald states that for a background Killing vector ζ (so $\mathcal{L}_\zeta g = 0$) and an arbitrary linear perturbation δg,

$$\mathrm{d}\big(\delta\mathbf{Q}_\zeta - i_\zeta\boldsymbol{\theta}(\delta g)\big) \;=\; -\,\zeta^\mu\,\delta\mathbf{C}_\mu, \tag{56.224}$$

where $\delta\mathbf{C}_\mu$ is the *linearized constraint* — precisely the linearized Einstein equation, $\delta\mathbf{C}_\mu \propto \big(\delta G_{\mu\nu} - 8\pi G_N\,\delta T^{\mathrm{bulk}}_{\mu\nu}\big)$ projected onto a slice. Integrate Eq. (56.224) over a bulk Cauchy slice Σ_B of the entanglement wedge, whose boundary is the union of the asymptotic boundary region B and the RT surface γ_B. Stokes' theorem turns the left-hand side into a difference of boundary integrals:

$$\underbrace{\int_B \big(\delta\mathbf{Q}_{\widehat{\xi}_B} - i_{\widehat{\xi}_B}\boldsymbol{\theta}\big)}_{\equiv\;\delta H^{\partial}_{\widehat{\xi}_B}} \;-\; \underbrace{\int_{\gamma_B} \big(\delta\mathbf{Q}_{\widehat{\xi}_B} - i_{\widehat{\xi}_B}\boldsymbol{\theta}\big)}_{\equiv\;\delta H^{\gamma}_{\widehat{\xi}_B}} \;=\; -\int_{\Sigma_B} \widehat{\xi}^{\,\mu}_B\,\delta\mathbf{C}_\mu. \tag{56.225}$$

The two boundary terms have transparent meanings.

On the RT surface γ_B, the Killing vector vanishes (Eq. (56.222)), so $i_{\widehat{\xi}_B}\boldsymbol{\theta} = 0$ there, and Wald's theorem evaluates the remaining Noether-charge integral as the surface gravity times the area: with $\kappa = 2\pi$,

$$\delta H^{\gamma}_{\widehat{\xi}_B} = \int_{\gamma_B} \delta\mathbf{Q}_{\widehat{\xi}_B} = \frac{\kappa}{2\pi}\,\delta\!\left(\frac{\mathrm{Area}(\gamma_B)}{4G_N}\right) = \delta\!\left(\frac{\mathrm{Area}(\gamma_B)}{4G_N}\right) = \delta S(B), \tag{56.226}$$

which is exactly the entropy side of the first law (and reproduces Eq. (56.223) on unpacking the Noether charge for Einstein gravity). On the asymptotic boundary B, the integral $\delta H^{\partial}_{\widehat{\xi}_B}$ is the canonical energy conjugate to the bulk Killing vector $\widehat{\xi}_B$. Evaluated with the near-boundary falloff of h and the dictionary $h \sim \langle T_{\mu\nu}\rangle$, this canonical energy is precisely the boundary integral of the energy density weighted by the boundary restriction ξ_B of $\widehat{\xi}_B$ — which by Eq. (56.219) is the change in the ball's modular energy:

$$\delta H^{\partial}_{\widehat{\xi}_B} = \int_B \xi_B^{\mu} \langle \delta T_{\mu 0}\rangle \, \mathrm{d}^{d-1}x = \delta\langle K_B\rangle. \tag{56.227}$$

This is the holographic content of the JLMS relation $K_B = \widehat{\mathrm{Area}}/4G_N + K^{\mathrm{bulk}}$ (Eq. (56.164) and Theorem 56.35): the boundary modular Hamiltonian equals the bulk gravitational charge of $\widehat{\xi}_B$, whose area part sits on γ_B and whose bulk-matter part fills the wedge.

The equivalence. Substituting Eqs. (56.226) and (56.227) into the integrated identity Eq. (56.225) gives the exact, off-shell relation

$$\delta\langle K_B\rangle - \delta S(B) \;=\; -\int_{\Sigma_B} \widehat{\xi}_B^{\mu}\, \delta\mathbf{C}_{\mu} \;\propto\; \int_{\Sigma_B} \widehat{\xi}_B^{\mu}\, (\delta G_{\mu\nu} - 8\pi G_N\, \delta T^{\mathrm{bulk}}_{\mu\nu})\, \mathrm{d}\Sigma^{\nu}. \tag{56.228}$$

This is the hinge of the entire chapter, and the two readings of it are the two halves of the synthesis.

Theorem 56.47 (Linearized Einstein equations from the first law of entanglement)**.** *Let h be a linearized metric perturbation of pure AdS_{d+1} with normalizable falloff encoding a boundary stress tensor $\langle T_{\mu\nu}\rangle$, and let $\delta S(B)$ be the corresponding first-order change in the Ryu–Takayanagi entropy of a boundary ball B. Then the entanglement first law*

$$\delta S(B) = \delta\langle K_B\rangle \quad \textit{holds for} \text{ every } \textit{ball } B \textit{ (all radii, centres, and Lorentz frames)} \tag{56.229}$$

if and only if *the bulk perturbation satisfies the linearized Einstein equation*

$$\boxed{\delta G_{\mu\nu}(h) \;=\; 8\pi G_N\, \delta T^{\,bulk}_{\mu\nu}} \tag{56.230}$$

everywhere in the bulk. In particular, given that the boundary CFT obeys the first law as a theorem of quantum mechanics (Theorem 56.45), the dual bulk geometry necessarily obeys linearized Einstein gravity. Bulk gravitational dynamics, to this order, is an entanglement identity.

Proof. ($\Leftarrow$) If Eq. (56.230) holds, the linearized constraint $\delta\mathbf{C}_{\mu}$ vanishes throughout the bulk, so the right-hand side of Eq. (56.228) is zero for every ball, giving $\delta S(B) = \delta\langle K_B\rangle$.

($\Rightarrow$) Suppose the first law holds for every ball. Then the left-hand side of Eq. (56.228) vanishes for the bulk Killing vector $\widehat{\xi}_B$ of every ball:

$$\int_{\Sigma_B} \widehat{\xi}_B^{\,\mu} \left(\delta G_{\mu\nu} - 8\pi G_N\, \delta T^{\text{bulk}}_{\mu\nu}\right) \mathrm{d}\Sigma^{\nu} = 0 \qquad \text{for all balls } B. \tag{56.231}$$

This is an integral transform of the local bulk tensor $E_{\mu\nu} \equiv \delta G_{\mu\nu} - 8\pi G_N \delta T^{\text{bulk}}_{\mu\nu}$: as the ball's radius, centre, and boost are varied, the wedges Σ_B sweep out a neighbourhood of every bulk point, and the Killing vectors $\widehat{\xi}_B$ probe every direction and every weighting through that point. The map from the local symmetric tensor field $E_{\mu\nu}$ to the family of integrals Eq. (56.231) is injective: a smooth tensor whose $\widehat{\xi}_B$-weighted flux through every such wedge vanishes must vanish pointwise. (Concretely, one differentiates Eq. (56.231) twice with respect to the radius R at $R \to 0$, which localizes the integral onto the tip of an infinitesimal wedge and returns $E_{\mu\nu}$ at the corresponding bulk point contracted with the local boost generator; ranging over centres and frames then exhausts all components.) Hence $E_{\mu\nu} = 0$ everywhere, which is Eq. (56.230). □

Physical Insight

The logic is worth stating without the machinery. On the boundary, the first law $\delta S = \delta\langle K\rangle$ is a triviality of quantum mechanics — it is the statement that, to first order, the entropy of a slightly perturbed state tracks the energy you put in, weighted by the modular Hamiltonian. It contains no physics beyond linear algebra. Holographically, the *same* triviality, demanded of every ball-shaped region, becomes the linearized Einstein equation of the bulk metric. The equation of motion of geometry is not an extra law layered on top of the duality; it is the boundary's first law of entanglement, rewritten in bulk variables. "Spacetime is built from entanglement" is, at the level of dynamics and to linear order, this theorem: vary the entanglement, and the geometry must respond exactly as Einstein's equation says, on pain of violating a tautology about density matrices.

Two scope statements keep the theorem honest. First, it is a *linearized* statement: it derives Einstein's equation expanded to first order about the AdS vacuum. The nonlinear equation has been recovered to second order (where the entanglement inequality $\Delta S \le \Delta\langle K\rangle$, not just its first-order saturation, plays a role, and the bulk gravitational energy becomes the canonical energy of Hollands–Wald), and there are arguments to all orders, but a clean "full nonlinear Einstein equation from entanglement" derivation valid for arbitrary states remains a frontier item dated below. Second, the inversion step that turns the family of ball identities into a local bulk equation is an integral-geometry transform — morally a Radon transform over the space of balls — and its invertibility is the same kind

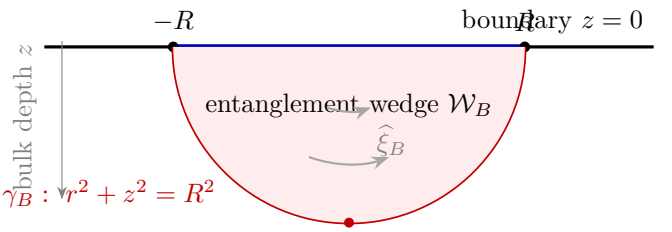


Figure 56.11: The geometry behind Theorem 56.47. A boundary ball B of radius R (blue) has Ryu–Takayanagi surface the bulk hemisphere $\gamma_B : r^2+z^2 = R^2$ (red), the bifurcation surface of the boost Killing horizon generated by $\widehat{\xi}_B$ (Eq. (56.221), grey flow). The entanglement wedge $\mathcal{W}_B$ (shaded) is the AdS–Rindler wedge it bounds. The Iyer–Wald identity Eq. (56.225) equates the change in horizon area $\delta\,\mathrm{Area}(\gamma_B)/4G_N = \delta S(B)$ to the boundary modular energy $\delta\langle K_B\rangle$ up to the bulk integral of the linearized Einstein equation; demanding the first law for every ball forces that bulk integral to vanish identically.

of completeness statement that underlies the analytic bootstrap's reconstruction of CFT data from its double discontinuity via the Lorentzian inversion formula (Theorem 51.16): in both cases a local object is recovered from a complete family of its integrals against a distinguished kernel. The parallel is structural rather than literal, but it situates the present result in the same circle of ideas, that boundary data, completely sampled, reconstructs a local bulk.

56.8.4 Connectivity is entanglement: Van Raamsdonk's argument

The first-law theorem governs the *shape* of a connected geometry. A complementary thought experiment, due to Van Raamsdonk, governs its very *connectedness*, and it follows from the two-interval computation of Section 56.2.4 run in reverse. There we found that the mutual information between two boundary regions A and B,

$$I(A:B) = S(A) + S(B) - S(A\cup B) \;\geq 0 \quad \text{(Theorem 55.13)}, \tag{56.232}$$

is computed holographically by the competition between a disconnected and a connected configuration of RT surfaces, and it *vanishes identically* once the regions are far enough apart that the disconnected surfaces win (Eq. (56.83)). The geometric event that turns I off is the dominant minimal surface changing topology: when no single surface spans both regions, they share no classical entanglement, and the bulk separating them has been pinched.

Run the logic backwards. Take a state with two entangled halves and ask what happens to the dual geometry as the entanglement is dialed away. The cleanest setting is the thermofield double of Chapter 54: two identical copies

CFT_L and CFT_R in the entangled pure state

$$|\mathrm{TFD}\rangle = \frac{1}{\sqrt{Z}} \sum_n e^{-\beta E_n/2} \, |n\rangle_L \otimes |n\rangle_R \,, \tag{56.233}$$

whose bulk dual is the eternal two-sided AdS–Schwarzschild black hole, a connected wormhole (Einstein–Rosen bridge) joining two asymptotic boundaries through a smooth interior (Section 54.4). The two boundaries do not interact; the only thing relating them is the entanglement in Eq. (56.233), and yet the bulk is a single connected spacetime through which a signal could in principle pass. Now imagine deforming the state to reduce the entanglement between the two copies — formally, lowering $I(\mathrm{CFT}_L : \mathrm{CFT}_R)$ by depopulating the Schmidt sum. The Ryu–Takayanagi surface for one entire boundary is the bifurcation surface of the horizon, whose area is the thermal entropy; reducing the cross-boundary entanglement shrinks this entropy and, by the area–entropy dictionary, the cross-sectional area of the wormhole throat. In the limit of vanishing mutual information the throat area goes to zero: the Einstein–Rosen bridge pinches off and the bulk splits into two disconnected geometries, one for each boundary, with no spacetime path between them.

Physical Insight

The conclusion is the chapter's deepest slogan made operational. The classical connectivity of the bulk spacetime is held together by boundary entanglement. Where two boundary subsystems are highly entangled, the bulk regions they anchor are joined by a fat bridge of geometry; as the entanglement is removed, the bridge thins and finally severs, and the two pieces of spacetime cease to be part of one manifold. Geometry is not the container of entanglement; it is the large-scale, coarse rendering of how the boundary degrees of freedom are entangled. The smooth, connected spacetime of classical general relativity is what a highly, structurally entangled quantum state looks like through the holographic dictionary.

This is sharpest where the mutual information is computable. For two intervals in a 2D CFT, Section 56.2.4 gives $I(A:B)$ exactly as a function of the cross-ratio, with the connected-to-disconnected transition at $\eta = \frac{1}{2}$ marking the threshold below which the bulk between A and B is pinched at leading order in G_N (Eq. (56.83)). The cross-sectional area of the bulk bridge between two regions is, more precisely, captured by the *entanglement wedge cross-section*, a minimal surface dividing the joint wedge $\mathcal{W}_{AB}$; it is nonzero exactly in the connected phase, is conjectured to compute the entanglement of purification and the reflected entropy of ρ_{AB}, and tracks $I(A:B)$ monotonically. The qualitative law — more entanglement, more connected geometry — is a theorem in these controlled cases and a robust expectation in general.

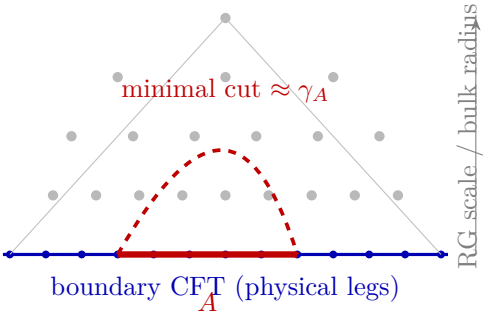


Figure 56.12: MERA as a discrete Ryu–Takayanagi geometry (Swingle). The boundary CFT wavefunction (blue, bottom) is built by layers of coarse-graining tensors; each layer is a step of entanglement renormalization, and the vertical direction is the renormalization-group scale, identified with the bulk radial coordinate (Section 54.6). The entanglement entropy of a boundary region A (red segment) is bounded by the minimal cut through the network (dashed), the curve severing the fewest bonds — the discrete image of the minimal surface γ_A, with $\ln \chi$ per bond standing in for $1/4G_N$ per unit area.

56.8.5 Tensor networks and MERA: a discretization of the geometry

If the geometry is built from entanglement, one should be able to build it, brick by brick, out of a discrete entanglement structure. The tensor network is exactly such a construction, and Swingle's observation is that the multiscale entanglement renormalization ansatz (MERA) for a CFT ground state is a discretization of the Ryu–Takayanagi geometry. A tensor network represents a many-body wavefunction $\Psi(s_1, \ldots, s_N)$ as a contraction of small tensors over internal "bond" indices; the bonds carry the entanglement, and the network's connectivity is a discrete geometry. The basic bound is purely linear-algebraic: for any bipartition of the physical legs into a region A and its complement, the entanglement entropy is bounded by the number of internal bonds one must cut to separate A from $\bar{A}$, times the logarithm of the bond dimension χ,

$$S(A) \;\le\; (\text{number of cut bonds along a curve separating } A) \times \ln \chi, \quad (56.234)$$

and the entropy is estimated by the *minimal* such cut — the curve through the network severing the fewest bonds. The minimal cut through a tensor network is the discrete analogue of the minimal surface, and Eq. (56.234) is the discrete analogue of the Ryu–Takayanagi formula $S(A) = \mathrm{Area}(\gamma_A)/4G_N$, with $\ln \chi$ per bond playing the role of $1/4G_N$ per unit area.

What makes MERA the *right* discretization, rather than a generic network, is its extra direction. MERA builds the ground-state wavefunction by alternating layers of "disentanglers," which remove short-range entanglement, and

"isometries," which coarse-grain — a literal real-space renormalization-group transformation. The depth into the network is therefore the RG scale, and identifying it with the bulk radial coordinate is exactly the holographic statement that the radial direction is energy scale (Section 54.6). A region A of n contiguous sites has a minimal cut that penetrates to depth $\sim \ln n$ into the network before the causal cones close off, giving $S(A) \sim \ln \chi \cdot \ln n$, the $(c/3)\ln(\ell/\varepsilon)$ scaling of a 2D CFT interval reproduced as a counting of cut bonds. The MERA is a hologram of the CFT vacuum drawn in tensors, and its geometry is the RT geometry to within the bond-counting approximation.

The min-cut characterization also explains, from a different angle, why RT computes an entropy. By the max-flow/min-cut duality of network theory, the minimal cut equals the maximal flow that can be routed from A to $\bar{A}$ through the bonds; in the continuum this becomes the "bit thread" reformulation of RT (Freedman–Headrick), in which $S(A)$ is the maximal flux of a divergenceless, norm-bounded vector field through A. The minimal surface and the maximal thread bundle are two faces of one optimization, and the bit-thread picture makes the entanglement inequalities of Section 56.3 — strong subadditivity in particular — nearly self-evident, since flows can be merged and split where surfaces must be recut. Tensor networks thus supply both a constructive model of geometry-from-entanglement and an explanation of the structural inevitability of the RT inequalities.

A caution closes the picture. The exact tensor-network models that realize holographic error correction cleanly — the HaPPY pentagon code of Section 56.7.4 and random-tensor networks — reproduce the RT formula and complementary recovery exactly, but on a fixed, rigid graph: they model a *fixed* geometry, not its dynamics, and the area operator in them is flat (it does not fluctuate as it must in a theory with a dynamical metric). MERA, conversely, captures the RG/scale structure but only bounds the entropy rather than computing it, and its geometry is closer to a light-cone (de Sitter-like) than to a spatial slice of AdS in some analyses. No known tensor network is simultaneously dynamical, exactly holographic, and Lorentzian. The networks are illuminating discretizations, not the continuum theory; how to take their continuum limit so that Einstein's equation of Theorem 56.47 emerges is an open problem.

56.8.6 ER = EPR

Van Raamsdonk's pinching argument has a vivid extrapolation, proposed by Maldacena and Susskind under the slogan ER = EPR: that entanglement and geometric connection are the *same thing* wherever both are defined. The thermofield double Eq. (56.233) is the cleanest instance — two boundaries maximally entangled (an EPR-like state), dual to a two-sided wormhole (an Einstein–Rosen bridge, ER) — and the equation of the slogan asserts that this identification is general: any entangled pair of systems is connected, at some level of descrip-

tion, by a geometric bridge, and conversely the smooth interior of a connected spacetime is the geometrization of the entanglement between the regions it joins. Pushed to the extreme, even a single EPR pair of particles would be joined by a (Planckian, non-traversable) wormhole.

The slogan must be stated with its scope visible, and the honest scope is narrow but real. As a statement about the two-sided eternal black hole it is precise and well supported: the wormhole geometry, the entanglement spectrum, and the modular flow all match (Section 54.4). As a heuristic organizing principle it has been productive — it predicts that the wormhole is non-traversable precisely because the entanglement, by itself, cannot transmit signals (consistent with no-faster-than-light communication between the two boundaries), and it frames the traversable-wormhole constructions, in which a direct coupling between the boundaries injects negative energy and briefly opens the throat, as the gravitational dual of quantum teleportation through the shared entanglement. As a literal claim about generic entangled states it is unproven and, taken naively, problematic: a generic low-entanglement or non-holographic state has no semiclassical geometric bridge at all, and the precise sense in which a single Bell pair "is" a wormhole has no controlled definition. The defensible content of ER = EPR is the one the rest of this section established with proofs: smooth, classical, connected bulk geometry is sustained by, and computes, boundary entanglement (Theorem 56.47 and Section 56.8.4). The wormhole language is the most striking dramatization of that content, valid sharply for the structured, large-entanglement states that have geometric duals, and to be treated as conjecture beyond them.

56.8.7 The dated frontier handed to Chapter 57

We have assembled a remarkably complete static picture: a boundary region's entanglement entropy is a bulk area (RT, Theorem 56.3); the correction is the bulk fields' own entropy (FLM, Eq. (56.139)); the whole is extremized as generalized entropy on a quantum extremal surface (Section 56.5.3); the boundary encodes the entanglement wedge as a quantum error-correcting code (Theorem 56.33 and Section 56.7.4); and the bulk's linearized equation of motion is the boundary's first law of entanglement (Theorem 56.47). It is essential, in the register this book keeps, to mark equally clearly what this picture does *not* settle as of the mid-2020s. The following are genuine open problems, not pedagogical omissions.

Hilbert-space factorization and edge modes. The derivations above repeatedly wrote "the reduced density matrix on a region" and "$\rho_A \otimes \rho_{\bar{A}}$," presupposing that the Hilbert space factorizes across the entangling surface as $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_{\bar{A}}$. In a gauge theory — and gravity is a gauge theory — it does not, exactly. Gauss's law correlates field configurations across any cut, the local algebras are type III von Neumann algebras with no factorization and no normalizable trace (Section 55.1.6), and the would-be tensor factorization must be supplied

by *edge modes*: extra degrees of freedom living on the entangling surface that restore a factorized description at the cost of a choice (a "center" of the algebra). The area term in RT is, from this angle, the entropy of these edge modes / the central element, and the precise, scheme-independent definition of "the entanglement entropy of a region" in a gravitational theory remains unsettled. The fluctuating area operator $\widehat{\text{Area}}$ of Eq. (56.138) is a placeholder for a still-incomplete understanding of how to factorize the gravitational Hilbert space.

Completeness of reconstruction: what lies behind horizons, and Python's lunch. Entanglement-wedge reconstruction (Section 56.7.2) guarantees that bulk operators in the wedge of A are recoverable from ρ_A — but recoverable *how hard*? For regions whose entanglement wedge has a nontrivial structure — a non-minimal extremal surface shadowing a minimal one, the configuration named a "Python's lunch" by Engelhardt and Penington — the reconstruction exists in principle but is conjectured to require computational resources exponential in the entropy difference between the two surfaces. The interiors of black holes are the prime example: the information is in the dual state, but extracting it may be exponentially complex, which is the modern, sharpened form of the question "what is behind the horizon, and can the boundary see it?" Whether reconstruction is always possible, and the precise complexity of the map, are open.

Which CFT states are geometric? Every derivation here assumed a state with a good semiclassical bulk dual — a smooth metric on which RT and FLM operate. But the generic state in the CFT Hilbert space is *not* of this kind: typical high-energy states, superpositions of geometries, and states with sub-AdS-scale or stringy-scale structure have no classical metric dual, and the "fixed-area states" that diagonalize the area operator are themselves non-generic. Characterizing the subset of CFT states that are geometric — giving necessary and sufficient conditions on the entanglement structure for a smooth dual to exist, and understanding what replaces geometry for the rest — is unresolved. The strong slogan "all entanglement is geometry" is false as stated; the true statement is that geometry is a property of a special, large-entanglement, structurally constrained class of states, and the boundary of that class is not known.

Complexity = volume, complexity = action. Entanglement entropy saturates quickly and stops growing once a black hole thermalizes, yet the wormhole interior — the Einstein–Rosen bridge of Section 56.8.6 — keeps growing in volume for a time exponential in the entropy. Entanglement cannot be what the late-time geometry tracks. Susskind and collaborators proposed that the relevant boundary quantity is computational *complexity*: the volume of the maximal bulk slice (complexity = volume), or the gravitational action of the Wheeler–DeWitt patch (complexity = action), is dual to the difficulty of preparing the boundary state. These conjectures are suggestive and pass qualitative tests (the linear-in-time growth, the switchback effect), but "complexity" lacks a sharp, universal

field-theoretic definition, the two proposals disagree in detail, and neither is derived. Whether complexity is the right second entry, after entanglement, in the geometry-from-information dictionary is open.

These four are the live edge of the static, kinematic story this chapter told. Each becomes urgent the moment the geometry is allowed to be a black hole and to *evolve*: the factorization problem becomes the problem of defining the black-hole interior's Hilbert space; reconstruction-behind-horizons becomes the recovery of information from Hawking radiation; the question of which states are geometric becomes the question of what the evaporating black hole's state is at the Page time; and complexity governs the interior's growth long after the entropy has saturated. That dynamical sequel is the subject of Chapter 57, where the quantum extremal surface of Section 56.5.3 is turned on an evaporating black hole, the island rule emerges from the same gravitational replica computation that gave us RT, and the Page curve — the rise and fall of the radiation's entropy that unitarity demands — is recovered from geometry. The static lesson is complete: bulk geometry is boundary entanglement made visible. We now watch that geometry move.

Further reading

The Ryu–Takayanagi proposal is Ryu and Takayanagi [422], with the covariant generalization in Hubeny et al. [271]; the review by Rangamani and Takayanagi [414] collects the formalism and its computations in one place. The bulk-replica derivation is Lewkowycz and Maldacena [323], refined to include bulk quantum corrections in Faulkner et al. [184] and to the quantum extremal surface in Engelhardt and Wall [177]; the universal sphere coefficients via the conformal map to a hyperbolic black hole are Casini et al. [116]. For subregion duality and the entanglement wedge, see Jafferis et al. [283] for the JLMS identity and Dong et al. [157] for entanglement-wedge reconstruction; the error-correction paradigm is Almheiri et al. [16], made exactly solvable in the pentagon code of Pastawski et al. [384]. Bulk operators as smeared boundary operators are constructed in Hamilton et al. [249]. The thesis that geometry is built from entanglement originates with Van Raamsdonk [483]; its tensor-network face is Swingle [461]; the derivation of linearized gravity from the first law is Faulkner et al. [185]; and the ER = EPR slogan is Maldacena and Susskind [332]. The information-theoretic sequel — the Page curve and islands — is taken up in Chapter 57 and reviewed in Almheiri et al. [19].

Chapter 57

Black Holes and Quantum Information

A black hole is the most economical object in physics: a Schwarzschild geometry is fixed by a single number, its mass, and yet it behaves as a thermodynamic body with a temperature, an entropy, and a first law. The entropy is enormous, scaling with the horizon *area* rather than any volume, and the radiation the hole emits as it evaporates is, to leading semiclassical order, exactly thermal. Taken at face value, that thermality is a catastrophe. A pure state of infalling matter would evolve into featureless radiation carrying no trace of what fell in, converting a pure state into a mixed one and breaking the unitarity that the boundary description of Chapter 53 guarantees holds. This chapter is the careful accounting of that clash—Hawking's information paradox—and of the resolution that semiclassical gravity itself, properly computed, has recently supplied.

We build the argument from the ground up, importing only machinery already in hand. The thermodynamic input of Section 57.1 is obtained by the Euclidean route, using the imaginary-time path integral and finite-temperature periodicity of Chapters 10 and 12: regularity of the cigar geometry fixes the Hawking temperature, the on-shell gravitational action delivers the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A/4G$ (Theorem 57.3), and the first law follows. Nothing here is borrowed from the curved-spacetime quantization of Chapter 65, which will rederive the same temperature by Hawking's original mode-by-mode Bogoliubov calculation; the present chapter is the self-contained foundation that later treatment confirms. The quantum-information tools—von Neumann entropy, the entanglement of the Hawking pairs, subadditivity and its consequences—and the holographic apparatus of generalized entropy and quantum extremal surfaces are inherited from Chapters 55 and 56, while the competing-saddle reading of the Euclidean path integral comes from Chapter 42.

With those in place the chapter follows a single arc. Section 57.2 states the paradox precisely, in three progressively sharper forms: Hawking's original claim

that the radiation's entropy rises monotonically and terminally; the unitarity bound that forces the entropy back down after the Page time (Theorem 57.9); and the AMPS firewall, which turns the contradiction into a clean violation of the monogamy of entanglement. Section 57.3 then computes what unitarity *demands*: modelling the scrambled hole by a Haar-random state, the Haar-averaged entanglement entropy obeys Page's formula (Theorem 57.14), and Lévy's lemma promotes the average to the typical answer—the Page curve. The two central sections supply the gravitational mechanism that realizes that curve. Section 57.4 introduces the island formula (Theorem 57.18) as a specialization of the quantum-extremal-surface prescription of Chapter 56: after the Page time a disconnected island enters the radiation's entanglement wedge, and a controlled JT-gravity computation reproduces the turnover exactly. Section 57.5 then shows where the island comes from: in the bare gravitational path integral the Rényi entropies receive a second saddle, the replica wormhole (Theorem 57.20), and the $n \to 1$ limit of the competition between the Hawking geometry and the wormhole *is* the quantum extremal surface.

The closing Section 57.6 draws the line, dated honestly to the mid-2020s, between what these results settle and what they do not. The entropy paradox in the form Hawking posed it is resolved—semiclassical gravity yields the Page curve—but a falling von Neumann entropy is not the same as a recovered message: the Hayden–Preskill protocol, the scrambling and Python's-lunch complexity bounds, the factorization puzzle, and the interpretation of the gravitational path integral as an ensemble average remain the live frontier. Throughout, in keeping with the general-relativity and holography literature, this chapter writes spacetime metrics in mostly-plus signature; the field-theory mostly-minus convention of the rest of the book is unaffected, since every computation here is carried by a positive-definite Euclidean geometry.

57.1 Black hole thermodynamics and the entropy of Hawking radiation

The chapter introduction posed the tension that organizes everything to follow: a black hole forms from a pure state, evaporates by emitting what looks like featureless thermal radiation, and – if that appearance is taken at face value – ends its life having converted a pure state into a mixed one, in flat contradiction with the unitarity that the boundary description of Chapter 53 guarantees. Before that clash can be made precise, let alone resolved, we must establish, with no appeal to the curved-spacetime quantization deferred to Chapter 65, the four thermodynamic facts on which the entire argument rests: that a black hole carries a definite temperature T_H, that it carries a definite entropy $S_{\mathrm{BH}} = A/4G$, that these obey the first law $\mathrm{d}M = T_H\,\mathrm{d}S_{\mathrm{BH}}$, and that the radiation it emits is thermal to leading order, so that the coarse-grained entropy of that radiation

grows without bound as the hole shrinks. We obtain all four by the Euclidean route – regularity of the imaginary-time geometry and evaluation of the on-shell gravitational action – which has the decisive virtue of using only the Euclidean path-integral technology already built in Chapters 10 and 42 and the finite-temperature periodicity of Chapter 12. Nothing here is imported from the later general-relativity chapters; on the contrary, Chapter 65 will rederive the temperature by Hawking's original mode-by-mode Bogoliubov calculation and find the identical number, and Chapter 54 has already obtained the same area law for the planar AdS black brane. The present section is the self-contained foundation those treatments confirm.

Notation Watch

Throughout this chapter, as throughout Part XV (Section 53.3), spacetime metrics are written in the *mostly-plus* signature $\eta_{\mu\nu} = \mathrm{diag}(-,+,\dots,+)$ universal to general relativity and the holography literature, with Riemann and Ricci tensors in the Wald/MTW convention, so that anti–de Sitter space has negative Ricci scalar. All field-theory expressions elsewhere in the book remain mostly-minus. The only quantity sensitive to this choice is the overall sign of the Lorentzian line element; the Euclidean geometries that carry every computation below are positive-definite and signature-independent. Chapter 65, sitting in the curved-spacetime Part, writes the same Schwarzschild solution in mostly-minus; the geometry and every invariant are identical.

57.1.1 The minimal geometric vocabulary: horizon and surface gravity

We need exactly two geometries and exactly one derived quantity. The first geometry is the Schwarzschild solution of the vacuum Einstein equations, the unique static, spherically symmetric, asymptotically flat solution of $R_{\mu\nu} = 0$. In coordinates (t, r, θ, φ) adapted to its time-translation and rotational symmetries it is

$$\mathrm{d}s^2 = -f(r)\,\mathrm{d}t^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2GM}{r}, \qquad \mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2, \tag{57.1}$$

where G is Newton's constant, M is the mass measured by the geometry at large r (the ADM mass), and $\mathrm{d}\Omega^2$ is the round metric on the unit two-sphere. We take Eq. (57.1) as given; the book builds no general relativity, and the curvature computation that verifies $R_{\mu\nu} = 0$ belongs to Chapter 64. What we need are its two structural features. The function f multiplies $\mathrm{d}t^2$ and divides $\mathrm{d}r^2$, the signature of a static black hole; it is positive for $r > 2GM$, where the geometry

is static with timelike Killing vector ∂_t, and vanishes at

$$r_h = 2GM, \tag{57.2}$$

the *event horizon* – the null surface beyond which no causal curve returns to large r. The region $r < r_h$ is not part of the static exterior and, as we will see, has no Euclidean continuation at all.

Definition 57.1 (Surface gravity)**.** For a static metric of the form Eq. (57.1) with a simple zero of f at the horizon $r = r_h$, the *surface gravity* is the geometric quantity

$$\kappa \;\equiv\; \tfrac{1}{2}\, f'(r_h). \tag{57.3}$$

Physically κ is the acceleration, measured at infinity, of a static observer hovering just outside the horizon; equivalently it controls the rate at which the timelike Killing vector ∂_t fails to be normalized as the horizon is approached. We will need none of that interpretation: Eq. (57.3) is a formula, and the temperature will fall out of it geometrically. For Schwarzschild, $f'(r) = 2GM/r^2$, so

$$\kappa = \tfrac{1}{2}\, f'(2GM) = \frac{1}{2}\cdot\frac{2GM}{(2GM)^2} = \frac{1}{4GM}. \tag{57.4}$$

The second geometry is the *AdS–Schwarzschild* black hole, the analogue with a negative cosmological constant that lives in the asymptotically anti–de Sitter spacetimes of Chapter 53. Its planar version – the black brane – and its full thermodynamics were established in Section 54.4; we recall only its form, in the coordinates of Eq. (53.59) with AdS radius L and radial coordinate z,

$$\mathrm{d}s^2 = \frac{L^2}{z^2}\left[-f(z)\,\mathrm{d}t^2 + \mathrm{d}\mathbf{x}^2 + \frac{\mathrm{d}z^2}{f(z)}\right], \qquad f(z) = 1 - \Big(\frac{z}{z_h}\Big)^d, \tag{57.5}$$

with conformal boundary at $z \to 0$ and horizon at $z = z_h$. The asymptotic structure – that near $z \to 0$ the metric Eq. (57.5) approaches empty AdS_{d+1} and so admits the same holographic boundary data as in Section 53.3 – is imported wholesale from Chapters 53 and 54; we will not re-derive it. The reason both geometries appear here is that the information paradox is sharpest for the asymptotically flat hole, which genuinely evaporates to nothing, while the cleanest *unitary* control comes from the AdS hole, whose boundary dual is a manifestly unitary thermal field theory (Section 54.4). The thermodynamic facts are common to both, and we prove them once, in detail, for Schwarzschild.

57.1.2 The Hawking temperature from Euclidean regularity

The route to the temperature is the Euclidean path integral of Section 10.6. There the Wick rotation $t \to -i\tau$ (Theorem 5.30) turned the Lorentzian weight

e^{iS} into the convergent Euclidean weight e^{-S_E}, and compactifying the imaginary time on a circle of circumference β computed the thermal partition function $Z(\beta) = \mathrm{Tr}\, e^{-\beta H}$ – the statement, made precise in Section 12.5.7, that a quantum system at inverse temperature β is a Euclidean field theory on a time circle of that circumference. For an *ordinary* field theory the period β is a free parameter we dial to set the temperature. The remarkable feature of gravity is that β is not free: once we Wick-rotate the black-hole metric, the demand that the resulting Riemannian geometry be smooth fixes β to a single value, and that value is the inverse Hawking temperature.

Wick-rotating Eq. (57.1) by $t = -i\tau$ sends $-f\,\mathrm{d}t^2 \to +f\,\mathrm{d}\tau^2$ and produces the positive-definite Euclidean Schwarzschild metric

$$\mathrm{d}s_E^2 = f(r)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2GM}{r}, \qquad r \geq 2GM. \tag{57.6}$$

Two features are decisive and worth naming before any algebra. First, $f \geq 0$ for $r \geq 2GM$ and vanishes at the horizon, so the proper circumference of the τ-circle, $\sqrt{f(r)}\oint \mathrm{d}\tau$, shrinks to zero as $r \to 2GM$: the horizon is not a boundary of the Euclidean manifold but a locus where the τ-circle closes to a point (one point for each point of the sphere). Second, the interior $r < 2GM$, where $f < 0$, is simply absent – there is no real Euclidean continuation of it. The Euclidean black hole is a smooth four-manifold capped off at the horizon, with the topology $\mathbb{R}^2 \times S^2$: the $\mathbb{R}^2$ is the (τ, r) plane closing up at r_h, the S^2 is the spatial sphere of area $4\pi r^2$. Whether the cap is genuinely smooth, or carries a conical defect, is the whole question, and we answer it by expanding the metric in the immediate neighbourhood of the horizon.

Worked Example 57.2 (The smoothness condition $\beta = 2\pi/\kappa$ from the near-horizon plane)**.** We zoom in on the horizon of Eq. (57.6) and demand that the capped (τ, r) geometry be a smooth point rather than the tip of a cone. Write $r = r_h + \delta$ with $0 < \delta \ll r_h$ and expand f to first order about its simple zero, using $f(r_h) = 0$:

$$f(r) = f'(r_h)\,\delta + O(\delta^2) = 2\kappa\,\delta + O(\delta^2), \tag{57.7}$$

where the last equality uses the definition Eq. (57.3), $f'(r_h) = 2\kappa$. The coordinate r is not the natural one near the horizon; the natural one is the *proper distance* ρ from the horizon, defined so that the radial part of the metric becomes $\mathrm{d}\rho^2$:

$$\mathrm{d}\rho = \frac{\mathrm{d}r}{\sqrt{f(r)}} \quad \Longrightarrow \quad \rho = \int_{r_h}^{r} \frac{\mathrm{d}r'}{\sqrt{f(r')}} = \int_0^{\delta} \frac{\mathrm{d}\delta'}{\sqrt{2\kappa\,\delta'}} = \frac{2\sqrt{\delta}}{\sqrt{2\kappa}} = \sqrt{\frac{2\delta}{\kappa}}. \tag{57.8}$$

Inverting, $\delta = \frac{1}{2}\kappa\rho^2$, and therefore the coefficient of $\mathrm{d}\tau^2$ becomes, to leading order,

$$f(r) = 2\kappa\,\delta = 2\kappa \cdot \tfrac{1}{2}\kappa\rho^2 = \kappa^2\rho^2. \tag{57.9}$$

Near the horizon the sphere has essentially fixed radius $r \approx r_h$, so the angular term $r^2\,\mathrm{d}\Omega^2 \approx r_h^2\,\mathrm{d}\Omega^2$ is an inert spectator that simply rides along. The non-trivial (ρ, τ) block of Eq. (57.6) collapses to

$$\mathrm{d}s_E^2\big|_{\text{near hor.}} = \mathrm{d}\rho^2 + \kappa^2\rho^2\,\mathrm{d}\tau^2 + r_h^2\,\mathrm{d}\Omega^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\alpha^2 + r_h^2\,\mathrm{d}\Omega^2, \qquad \alpha \equiv \kappa\,\tau. \tag{57.10}$$

The (ρ, α) piece is exactly the flat plane $\mathbb{R}^2$ in polar coordinates, with radial coordinate $\rho \geq 0$ and angular coordinate α. This is precisely the elementary fact that fixes the period of the angle in polar coordinates: $\mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\alpha^2$ is smooth flat space at the origin $\rho = 0$ *if and only if* α runs over a full period of exactly 2π. Any other period leaves a cone: a deficit angle $2\pi -$ (period) either missing or doubled at the tip, with a curvature delta-function supported there and no smooth solution of the vacuum equation in any neighbourhood of $\rho = 0$. Smoothness of the Euclidean black hole at its horizon is therefore the single condition $\alpha \sim \alpha + 2\pi$, i.e. $\kappa\,\tau \sim \kappa\,\tau + 2\pi$, i.e.

$$\boxed{\beta = \frac{2\pi}{\kappa}\,,} \tag{57.11}$$

the unique period of imaginary time for which the geometry closes up smoothly.

The result Eq. (57.11) is purely geometric: no field was quantized, no mode tracked across the horizon, no vacuum chosen. The imaginary-time period – which by the Euclidean dictionary of Section 10.6 is the inverse temperature of any state the geometry represents – is forced by the differential geometry of a smooth four-manifold. Reading off the temperature and substituting the Schwarzschild surface gravity Eq. (57.4),

$$T_H = \frac{1}{\beta} = \frac{\kappa}{2\pi} \qquad \xrightarrow{\kappa = 1/4GM} \qquad \boxed{T_H = \frac{1}{8\pi GM}\,.} \tag{57.12}$$

This is the Hawking temperature. The same regularity argument applied to the AdS–Schwarzschild brane Eq. (57.5) gave $\beta = 4\pi z_h/d$ in Eq. (54.96); the structure is identical – a τ-circle pinching off at a horizon, smooth for exactly one period – and we do not repeat it. Figure 57.1 shows the resulting "cigar" geometry.

Physical Insight

The temperature of a black hole is the price of geometric smoothness. In an ordinary thermal field theory we *choose* the imaginary-time period to set T; the geometry is flat and accepts any period. A black hole's Euclidean geometry pinches off at the horizon, and a pinching circle is smooth for one period only – like the apex of a cone, which is flat exactly when the cone is rolled out to a full disk. The black hole has no freedom:

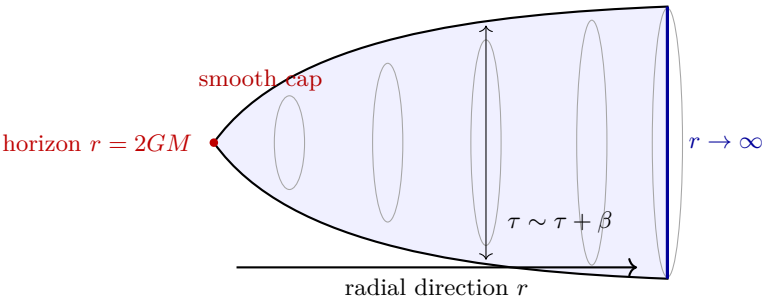


Figure 57.1: The Euclidean Schwarzschild "cigar." Each circle is the imaginary-time direction τ, of circumference β, fibered over the radial coordinate r (the spectator S^2 is suppressed). Far away the circle has its full proper size $\sqrt{f}\,\beta$; moving inward it shrinks, and at the horizon $r = 2GM$ it closes to a point. The geometry caps smoothly there – no conical defect – for exactly one period $\beta = 2\pi/\kappa = 8\pi GM$ Eq. (57.12), the tip being the origin of the flat plane $\mathrm{d}\rho^2 + \rho^2 \mathrm{d}\alpha^2$ of Eq. (57.10). The interior $r < 2GM$ has no Euclidean continuation and is absent.

its temperature is determined by its mass, $T_H = 1/8\pi GM$. The hotter the hole the lighter it is – a negative heat capacity that, as the next sections show, is the engine of evaporation and the seed of the paradox.

Historical Note

That a periodic imaginary time *is* thermal equilibrium is the content of the Kubo–Martin–Schwinger condition, derived in Eq. (12.139): the thermal two-point function $\langle A(\tau)B(0)\rangle_\beta$ is periodic in τ with period β, a direct consequence of cyclicity of the trace in $Z^{-1}\,\mathrm{Tr}(e^{-\beta H}\cdots)$. Gibbons and Hawking recognized in 1977 that the converse reading is the most economical route to the black-hole temperature: a gravitational field whose Euclidean section is smoothly periodic in imaginary time *is* a thermal state, and the period is read off the geometry. Hawking's earlier 1974–75 derivation, by contrast, quantized a scalar field on a collapsing star and computed the Bogoliubov coefficients between the in- and out-vacua; that calculation, which exhibits the thermal flux as genuine particle creation, is carried out in full in Section 65.2.1. The two agree on the number $T_H = \kappa/2\pi$.

57.1.3 The entropy from the on-shell Euclidean action

The temperature in hand, the partition function and hence the entropy follow from the gravitational path integral. In the saddle-point evaluation of Chapter 42, the dominant contribution to the Euclidean gravitational partition function at inverse temperature β is $e^{-I[g_\star]}$, where $g_\star$ is the classical solution – here the Euclidean black hole – and I is the on-shell Euclidean gravitational action. We must therefore evaluate that action. For asymptotically flat gravity with vanishing cosmological constant the action consists of the Einstein–Hilbert bulk term and the Gibbons–Hawking–York (GHY) boundary term, regulated by subtracting the same two terms evaluated on the flat-space background that the geometry approaches at infinity:

$$I = -\frac{1}{16\pi G}\int_{\mathcal{M}} \mathrm{d}^4x\,\sqrt{g}\,R \;-\; \frac{1}{8\pi G}\int_{\partial\mathcal{M}} \mathrm{d}^3x\,\sqrt{\gamma}\,(K-K_0). \tag{57.13}$$

Here γ_{ab} is the metric induced on a large-radius boundary $r=\mathcal{R}$, K is the trace of its extrinsic curvature in the Euclidean black-hole geometry, and K_0 is the same trace computed for flat space cut off at the same boundary – the *background subtraction* that renders the otherwise divergent boundary term finite. The GHY term is the standard addition that makes the Dirichlet variational problem well posed (without it the variation of Eq. (57.13) leaves an uncancelled normal derivative of the metric at the boundary); we take its necessity as established in Chapter 64 and owe here only its evaluation. The subtraction of K_0 is the asymptotically-flat counterpart of the holographic counterterms used for the AdS brane in Section 54.2.3; there the reference was supplied by local boundary curvature invariants, here by an explicit flat background, but the role is identical – removing the contribution of empty space so that I measures the geometry's content.

We evaluate Eq. (57.13) on Euclidean Schwarzschild Eq. (57.6) term by term.

Bulk term. Schwarzschild solves the *vacuum* Einstein equations $R_{\mu\nu}=0$, so its Ricci scalar vanishes identically, $R=0$, and the bulk integral in Eq. (57.13) contributes nothing:

$$I_{\text{bulk}} = 0. \tag{57.14}$$

The entire on-shell action of the asymptotically flat black hole comes from the boundary term. (This is the structural difference from the AdS computation of Section 54.4.3, where the cosmological constant gives a nonzero on-shell density $R-2\Lambda=-2d/L^2$ and the bulk term does contribute.)

The extrinsic-curvature traces. The boundary is the surface $r=\mathcal{R}$, swept out by $\tau\in[0,\beta)$ and the sphere. The full four-metric determinant of Eq. (57.6) is $\sqrt{g}=\sqrt{f\cdot f^{-1}\cdot r^2\cdot r^2\sin^2\theta}=r^2\sin\theta$. The induced three-metric on $r=\mathcal{R}$ is

$\gamma = \mathrm{diag}(f(\mathcal{R}), \mathcal{R}^2, \mathcal{R}^2 \sin^2\theta)$ in (τ, θ, φ), so $\sqrt{\gamma} = \sqrt{f(\mathcal{R})}\,\mathcal{R}^2 \sin\theta$. The outward unit normal is $n^\mu = \sqrt{f}\,\delta^\mu{}_r$ (since $g_{rr} = 1/f$), and the extrinsic-curvature trace is its covariant divergence,

$$K = \nabla_\mu n^\mu = \frac{1}{\sqrt{g}}\,\partial_r(\sqrt{g}\,n^r) = \frac{1}{r^2}\,\partial_r(r^2\sqrt{f}) = \frac{2\sqrt{f}}{r} + \frac{f'}{2\sqrt{f}}. \tag{57.15}$$

Multiplying by the area density $\sqrt{\gamma}$ and integrating over the sphere ($\int \mathrm{d}\theta\,\mathrm{d}\varphi\,\sin\theta = 4\pi$) and the time circle gives, for the black-hole geometry,

$$\oint \mathrm{d}^3x\,\sqrt{\gamma}\,K = \beta\cdot 4\pi\cdot\mathcal{R}^2\sqrt{f}\left(\frac{2\sqrt{f}}{\mathcal{R}} + \frac{f'}{2\sqrt{f}}\right) = 4\pi\beta\left(2\mathcal{R}\,f(\mathcal{R}) + \tfrac{1}{2}\mathcal{R}^2 f'(\mathcal{R})\right). \tag{57.16}$$

Insert $f(\mathcal{R}) = 1 - 2GM/\mathcal{R}$ and $f'(\mathcal{R}) = 2GM/\mathcal{R}^2$:

$$2\mathcal{R}f + \tfrac{1}{2}\mathcal{R}^2 f' = 2\mathcal{R}\Big(1 - \frac{2GM}{\mathcal{R}}\Big) + \tfrac{1}{2}\mathcal{R}^2\frac{2GM}{\mathcal{R}^2} = 2\mathcal{R} - 4GM + GM = 2\mathcal{R} - 3GM, \tag{57.17}$$

so $\oint \sqrt{\gamma}\,K = 4\pi\beta\,(2\mathcal{R} - 3GM)$.

For the flat-space reference we set $M = 0$, so $f_0 = 1$, $f_0' = 0$, and Eq. (57.15) gives $K_0 = 2/r$. But the subtraction is meaningful only if the two boundaries describe the *same* intrinsic geometry: the proper length of the time circle must match. In the black hole the circle has proper circumference $\sqrt{f(\mathcal{R})}\,\beta$; in flat space, with its own period β_0, it is β_0. Matching $\beta_0 = \sqrt{f(\mathcal{R})}\,\beta$ and using $\sqrt{\gamma_0} = \mathcal{R}^2\sin\theta$, $K_0 = 2/\mathcal{R}$,

$$\oint \mathrm{d}^3x\,\sqrt{\gamma_0}\,K_0 = \beta_0\cdot 4\pi\cdot\mathcal{R}^2\cdot\frac{2}{\mathcal{R}} = 8\pi\,\beta_0\,\mathcal{R} = 8\pi\beta\,\mathcal{R}\sqrt{f(\mathcal{R})}. \tag{57.18}$$

Expand the square root for large $\mathcal{R}$, $\sqrt{f(\mathcal{R})} = (1 - 2GM/\mathcal{R})^{1/2} = 1 - GM/\mathcal{R} + O(\mathcal{R}^{-2})$, so

$$\oint \sqrt{\gamma_0}\,K_0 = 8\pi\beta\Big(\mathcal{R} - GM + O(\mathcal{R}^{-1})\Big). \tag{57.19}$$

The subtracted boundary term. Now take the difference and the limit $\mathcal{R}\to\infty$. The divergent pieces linear in $\mathcal{R}$ cancel between the two geometries – which is exactly what the background subtraction is for:

$$\oint \sqrt{\gamma}\,K - \oint \sqrt{\gamma_0}\,K_0 = 4\pi\beta(2\mathcal{R} - 3GM) - 8\pi\beta(\mathcal{R} - GM) = 4\pi\beta(-3GM + 2GM) = -4\pi\beta\,GM \tag{57.20}$$

The $\mathcal{R}$ terms collect to $8\pi\beta\mathcal{R} - 8\pi\beta\mathcal{R} = 0$ and the limit is finite. Therefore the on-shell action Eq. (57.13), with $I_{\mathrm{bulk}} = 0$, is

$$I = -\frac{1}{8\pi G}(-4\pi\beta\,GM) = \frac{\beta M}{2}. \tag{57.21}$$

Using $\beta = 8\pi GM$ from Eq. (57.12), this can be written equivalently in terms of β alone, $I = \beta M/2 = \beta\cdot\beta/(8\pi G)/2 = \beta^2/(16\pi G)$, a form we will exploit in a moment.

Free energy, energy, and entropy. The saddle-point partition function is $\ln Z = -I$, and the standard thermodynamic relations extract the rest. The free energy is

$$F = -\frac{1}{\beta}\ln Z = \frac{I}{\beta} = \frac{M}{2}. \tag{57.22}$$

The internal energy is the expected value of the Hamiltonian, $\langle E\rangle = -\partial_\beta \ln Z = \partial_\beta I$. Differentiating the β-only form $I = \beta^2/16\pi G$,

$$\langle E\rangle = \frac{\partial I}{\partial \beta} = \frac{2\beta}{16\pi G} = \frac{\beta}{8\pi G} = M, \tag{57.23}$$

where the last step is again $\beta = 8\pi G M$. The energy of the saddle is exactly the mass, as it must be – a nontrivial check that the background subtraction was done correctly. Finally the entropy is the Legendre transform $S = \ln Z + \beta\,\langle E\rangle$,

$$S = -I + \beta M = -\frac{\beta M}{2} + \beta M = \frac{\beta M}{2} = \frac{(8\pi G M)M}{2} = 4\pi G M^2. \tag{57.24}$$

It is one quarter of the horizon area. The horizon is the sphere $r = r_h = 2GM$, whose induced metric is $r_h^2\,\mathrm{d}\Omega^2$ and whose area is therefore

$$A = 4\pi r_h^2 = 4\pi(2GM)^2 = 16\pi G^2 M^2. \tag{57.25}$$

Dividing, $A/4G = 16\pi G^2 M^2/(4G) = 4\pi G M^2$, which is exactly Eq. (57.24). We have proved:

Theorem 57.3 (Bekenstein–Hawking entropy)**.** *The entropy of a Schwarzschild black hole, computed from the saddle-point value of the Euclidean gravitational partition function with Gibbons–Hawking boundary term and flat-background subtraction, is one quarter of its horizon area in Planck units:*

$$\boxed{S_{\mathrm{BH}} = \frac{A}{4G} = 4\pi G M^2\,.} \tag{57.26}$$

Remark 57.4 (Where the one-loop corrections live). Theorem 57.3 is the *leading* saddle-point result, the gravitational analogue of the tree-level free energy. The next term in the loop expansion of Chapter 42 is the one-loop determinant of fluctuations about the black-hole saddle – a functional determinant of the linearized Einstein operator, evaluated by the spectral ζ-function and heat-kernel methods built in Chapter E (Section E.3). It produces the logarithmic correction $S = A/4G + c\ln(A/G) + \dots$ whose universal coefficient c is fixed by the field content; the regularization technology is exactly the one applied to the one-loop effective action in Eq. (E.135), and the explicit gravitational evaluation is treated in [257] and Chapter 65. We will not need the correction: the paradox and its resolution operate at the level of Eq. (57.26) and the area's leading behaviour.

The same on-shell-action computation, carried out for the AdS brane with its nonzero bulk term and holographic counterterms, gave $S = A/4G$ in Theorem 54.10; the two derivations differ only in which reference subtraction tames the boundary divergence, and agree on the universal coefficient $1/4$. That the coefficient is the same in both settings – and matches the coefficient of the area law for entanglement entropy in Section 55.5 – is the first hint that $A/4G$ is counting something microscopic, a hint the rest of the chapter makes precise.

57.1.4 The first law

The thermodynamic consistency of Eqs. (57.12) and (57.26) is the first law, and with both quantities written as explicit functions of the single parameter M it is immediate to verify. From Eq. (57.24), $S_{\mathrm{BH}} = 4\pi GM^2$, so

$$\mathrm{d}S_{\mathrm{BH}} = 8\pi GM\,\mathrm{d}M. \tag{57.27}$$

Multiplying by the temperature Eq. (57.12), $T_H = 1/8\pi GM$,

$$T_H\,\mathrm{d}S_{\mathrm{BH}} = \frac{1}{8\pi GM}\cdot 8\pi GM\,\mathrm{d}M = \mathrm{d}M, \tag{57.28}$$

which is the first law $\mathrm{d}M = T_H\,\mathrm{d}S_{\mathrm{BH}}$ with the mass M playing the role of the internal energy (as Eq. (57.23) confirmed) and no work terms, since Schwarzschild carries neither charge nor angular momentum. The first law is thus not an extra assumption but a theorem about the M-dependence of the temperature and the entropy, both of which we computed from the geometry. Its differential form integrates back to the entropy: $\mathrm{d}S_{\mathrm{BH}} = \mathrm{d}M/T_H = 8\pi GM\,\mathrm{d}M$ gives $S_{\mathrm{BH}} = 4\pi GM^2$ up to a constant, fixed to zero by $S_{\mathrm{BH}} \to 0$ as $M \to 0$. This is the route by which the entropy is often quoted – temperature from Euclidean regularity, entropy by integrating the first law – but we have done better here, computing S_{BH} *directly* from the action and then *checking* the first law, so that the coefficient $1/4$ is a genuine output rather than an integration constant.

Physical Insight

The negative heat capacity is now explicit and physical. From Eq. (57.12), $T_H = 1/8\pi GM$ rises as M falls: a black hole that loses energy gets *hotter*. Equivalently, $\mathrm{d}M/\mathrm{d}T_H = -1/(8\pi GT_H^2) < 0$. This is the thermodynamic root of evaporation: once a hole begins to radiate, it heats up, radiates faster, and runs away to nothing – the opposite of an ordinary hot body, which cools as it radiates and reaches equilibrium with a reservoir. An isolated Schwarzschild black hole has no stable equilibrium with its own radiation; it can only be stabilized by enclosing it (as the AdS box of Section 54.4.6 does for large black holes, which there acquire positive specific heat). The instability is what makes the asymptotically flat hole

the natural arena for the information question: it actually disappears, forcing the issue of what becomes of its information.

57.1.5 The radiation is thermal, and its coarse-grained entropy grows

The Euclidean saddle establishes that the black hole *is* a thermal state at temperature T_H; what an observer at infinity actually receives is the radiation emitted as the hole relaxes toward, and past, that equilibrium. The detailed spectrum is the business of Section 65.3, where Hawking's Bogoliubov calculation produces the flux mode by mode. For the present purpose we need only its leading character, which the thermal interpretation already fixes: the number flux of quanta of energy ω emitted into a given mode is the Planck distribution at temperature T_H, dressed by a frequency-dependent transmission probability Γ_ω,

$$\frac{\mathrm{d}N_\omega}{\mathrm{d}t} = \frac{\Gamma_\omega}{e^{\omega/T_H} \mp 1}, \tag{57.29}$$

with the lower sign for bosons and the upper for fermions. The factor Γ_ω – the *greybody factor* – is the probability that a quantum created near the horizon penetrates the curvature potential barrier outside and reaches infinity rather than being backscattered into the hole; it depends on ω, on the spin of the field, and on the angular momentum of the mode, and it is what makes the observed spectrum a *grey* body rather than a perfect black body. We state its existence and role and defer its computation to Section 65.3; it is not the focus here, and crucially it does not affect the qualitative conclusion we now draw. Integrating Eq. (57.29) against ω with the appropriate phase space gives the luminosity, which for the dominant long-wavelength modes scales as the Stefan–Boltzmann law over the horizon area, $L \sim A\,T_H^4 \sim (G^2M^2)(GM)^{-4} \sim 1/(G^2M^2)$, so that

$$\frac{\mathrm{d}M}{\mathrm{d}t} \sim -\frac{1}{G^2M^2} < 0, \tag{57.30}$$

the negative sign confirming that the hole loses mass and, by Eq. (57.12), heats as it does so.

The quantity that the next section places in tension with unitarity is the *coarse-grained*, or thermal, entropy of the emitted radiation – the thermodynamic entropy one would assign to the outgoing flux treating it as the genuinely thermal gas that Eq. (57.29) describes. For thermal radiation the relation between entropy flux and energy flux is the standard one: a photon gas of energy density $u \propto T^4$ has entropy density $s = \frac{4}{3}u/T$, so a parcel of energy $\mathrm{d}E$ emitted at temperature T_H carries off thermal entropy

$$\mathrm{d}S^{\mathrm{th}}_{\mathrm{rad}} = \frac{4}{3}\frac{\mathrm{d}E}{T_H} = -\frac{4}{3}\frac{\mathrm{d}M}{T_H}, \tag{57.31}$$

using energy conservation $\mathrm{d}E = -\mathrm{d}M$. Comparing with the black hole's own entropy change $\mathrm{d}S_{\mathrm{BH}} = \mathrm{d}M/T_H$ from Eq. (57.28), we see the radiation gains $\frac{4}{3}$ of what the hole loses:

$$\mathrm{d}S^{\mathrm{th}}_{\mathrm{rad}} = -\frac{4}{3}\,\mathrm{d}S_{\mathrm{BH}} > 0. \tag{57.32}$$

Since $\mathrm{d}S_{\mathrm{BH}} < 0$ (the hole shrinks), $S^{\mathrm{th}}_{\mathrm{rad}}$ increases monotonically throughout the evaporation – this is the generalized second law in its sharpest form, the statement that the sum $S_{\mathrm{BH}} + S^{\mathrm{th}}_{\mathrm{rad}}$ never decreases. Integrating Eq. (57.32) from the initial mass M_0 down to M,

$$S^{\mathrm{th}}_{\mathrm{rad}}(M) = \frac{4}{3}\big(S_{\mathrm{BH}}(M_0) - S_{\mathrm{BH}}(M)\big), \tag{57.33}$$

which grows from zero, reaching at complete evaporation ($M \to 0$) the value

$$S^{\mathrm{th}}_{\mathrm{rad}}(0) = \frac{4}{3}\,S_{\mathrm{BH}}(M_0) = \frac{16\pi}{3}\,GM_0^2. \tag{57.34}$$

Figure 57.2 plots the two competing curves: $S_{\mathrm{BH}}(t) = A(t)/4G$ shrinking to zero, and $S^{\mathrm{th}}_{\mathrm{rad}}(t)$ rising monotonically to $\frac{4}{3}S_{\mathrm{BH}}(M_0)$.

Caution

The adjective "thermal" in $S^{\mathrm{th}}_{\mathrm{rad}}$ is load-bearing and is exactly where the paradox will be joined. Equation (57.33) is the entropy of the radiation *computed under the assumption that it is a thermal mixed state*, which is what the leading semiclassical (Bogoliubov) calculation of Chapter 65 delivers: each emitted mode is thermally populated and uncorrelated with the rest, so the radiation's density matrix is a product of thermal factors and its von Neumann entropy is the thermodynamic value Eq. (57.33). If that were the true *fine-grained* (von Neumann) entropy of the radiation, then at the end of evaporation the radiation would be a mixed state of entropy $\frac{4}{3}S_{\mathrm{BH}}(M_0) > 0$ – with no black hole left to be entangled with – and an initially pure collapsing star would have evolved into a mixed state. That is the information paradox. The resolution, developed across the chapter, is that the true fine-grained entropy departs from the monotone thermal curve, turning over and returning to zero (the Page curve); $S^{\mathrm{th}}_{\mathrm{rad}}$ is only an *upper bound* on it, saturated early and grossly violated late.

57.1.6 The Bekenstein bound: S_{BH} as a maximum fine-grained entropy

One structural fact about S_{BH} remains to be stated, because it is what makes the paradox a genuine paradox rather than a bookkeeping slip: the Bekenstein–Hawking entropy is *finite*, and it is the *maximum* fine-grained entropy that can

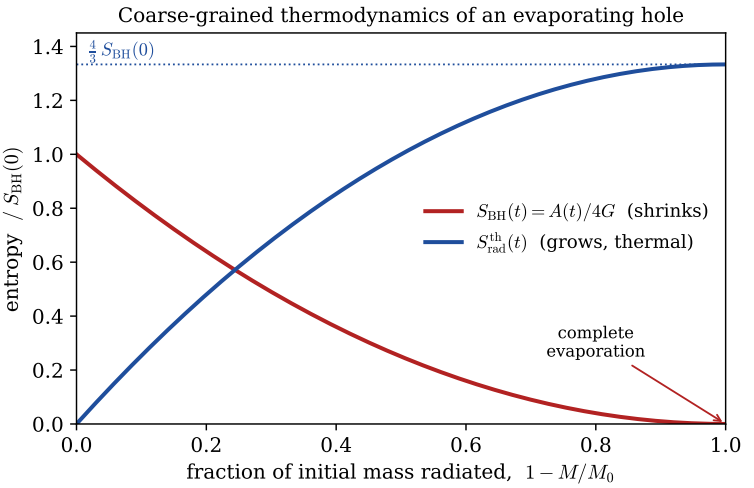


Figure 57.2: The two coarse-grained thermodynamic quantities handed to Section 57.1's successor, as functions of the fraction of the initial mass radiated, in units of the initial Bekenstein–Hawking entropy. The black-hole entropy $S_{\rm BH}(t) = A(t)/4G = 4\pi G M(t)^2$ (red) shrinks monotonically to zero as the hole evaporates. The *thermal* entropy of the radiation $S^{\rm th}_{\rm rad}(t)$ (blue), computed by treating the Hawking flux as a genuinely thermal gas Eq. (57.33), grows monotonically to $\frac{4}{3}S_{\rm BH}(M_0)$ Eq. (57.34). This monotone growth – never turning over – is precisely the behaviour that the next section shows to be incompatible with a unitary, pure final state.

be packed into the region the black hole occupies. Bekenstein's original argument, which we give in outline, is a thought experiment with the generalized second law. Lower a box of matter carrying energy E and entropy $S_{\rm matter}$, of linear size $\leq R$, quasi-statically into a black hole. The matter entropy $S_{\rm matter}$ is lost from the exterior; for the generalized second law $\mathrm{d}(S_{\rm BH} + S_{\rm ext}) \geq 0$ to survive, the increase in $S_{\rm BH}$ as the hole absorbs the energy must at least compensate. The energy actually delivered to the hole is redshifted: a box of size R can be lowered until its center hovers a proper distance $\sim R$ above the horizon (it can approach no closer without crossing), where the timelike Killing vector has norm $\simeq \kappa R$ with $\kappa = 1/4GM$ the surface gravity of Eq. (57.4), so the energy delivered at infinity is $E_\infty \simeq \kappa R\, E$. The hole's entropy then rises by $\mathrm{d}S_{\rm BH} = E_\infty/T_H = \kappa R\, E/T_H = 2\pi R\, E$, using $T_H = \kappa/2\pi$ from Eq. (57.12). The generalized second law $\mathrm{d}S_{\rm BH} \geq S_{\rm matter}$ then yields the *Bekenstein bound* on the entropy of any system of energy E confined to a region of radius R,

$$S_{\rm matter} \;\leq\; 2\pi R\, E \tag{57.35}$$

(in natural units $\hbar = c = k_B = 1$). The bound is saturated, up to factors, precisely by a black hole that just fits in the region: for $R \sim r_h = 2GM$ and $E = M$, the right-hand side is $2\pi(2GM)M = 4\pi GM^2 = S_{\rm BH}$, recovering Eq. (57.26). The same conclusion in its covariant, area form is the statement that the entropy that can be stored in a region is bounded by its boundary area in Planck units, $S \leq A/4G$ – the content of the holographic principle of 't Hooft and Susskind that gave Chapter 53 its name, and the reason a black hole, which saturates the bound, has the *maximum* entropy of anything that can be assembled in its volume.

Physical Insight

The finiteness of $S_{\rm BH} = A/4G$ is the quantitative heart of the puzzle. An ordinary thermal system has an entropy that grows with its volume and is, for a continuum field theory, formally unbounded as more short-distance modes are excited. A black hole instead carries a *finite*, area-scaling entropy, and that finite number is the largest fine-grained entropy the region can hold. Two consequences follow that the rest of the chapter develops. First, the radiation's fine-grained entropy can never exceed the entropy of whatever it is entangled with, and once the black hole has shrunk so that $S_{\rm BH}(t)$ is small, the radiation's own entropy is bounded by that small number – it *cannot* keep growing as Eq. (57.33) insists. Second, that a region's information capacity is set by its area, not its volume, is the germ of holography itself: the degrees of freedom are not where naive locality puts them. Both threads converge on the same resolution.

We now have, established from the Euclidean geometry and its on-shell action alone, the complete thermodynamic dossier the chapter requires: the Hawking temperature $T_H = 1/8\pi GM$ Eq. (57.12), the Bekenstein–Hawking entropy $S_{\rm BH} = A/4G$ Eq. (57.26) as the finite maximum fine-grained entropy of the region, the first law $\mathrm{d}M = T_H\,\mathrm{d}S_{\rm BH}$ Eq. (57.28), and the approximately thermal character of the emitted radiation, whose coarse-grained entropy $S^{\rm th}_{\rm rad}(t)$ grows monotonically to $\frac{4}{3}S_{\rm BH}(M_0)$. The next section takes these two time-dependent quantities – the shrinking $S_{\rm BH}(t) = A(t)/4G$ and the monotonically growing $S^{\rm th}_{\rm rad}(t)$ of Fig. 57.2, together with the semiclassical fact that the radiation comes out in a thermal, hence mixed, state – and turns the slogan "black holes destroy information" into three sharp statements, each in genuine tension with the unitarity that the boundary theory of Chapter 53 guarantees.

57.2 The information paradox, stated precisely

The previous section handed us two curves and a collision course. From the Euclidean route we extracted a Hawking temperature T_H (Eq. (57.12)), a Bekenstein–

Hawking entropy $S_{\rm BH} = A/4G$ (Eq. (57.26)), and the statement that a black hole radiates an approximately thermal flux. Feeding that flux into the first law gave a black hole that shrinks: its horizon area $A(t)$, and with it $S_{\rm BH}(t) = A(t)/4G$, decreases monotonically toward zero at an evaporation time $t_{\rm evap}$. Meanwhile the emitted gas, being thermal, carries an entropy of its own, and that radiation entropy $S^{\rm th}_{\rm rad}(t)$ grows monotonically as more quanta escape. Two monotone curves, one falling and one rising: somewhere they must cross. The business of this section is to take that crossing seriously and convert the folklore slogan "black holes destroy information" into statements sharp enough to be either proved or refuted.

We will produce three such statements, increasing in subtlety. The first is Hawking's original observation, phrased as a fact about the fine-grained entropy of the radiation: in the semiclassical calculation it rises without bound and ends nonzero, which is impossible if the process is unitary. The second sharpens this into a quantitative ceiling — unitarity forces $S(\rho_{\rm rad}) \le S_{\rm BH}(t)$ at every instant, a bound the semiclassical curve violates after a definite time, the Page time. The third, due to Almheiri, Marolf, Polchinski and Sully (AMPS), shows that even granting unitarity at the level of entropy, the three principles one would most like to keep — unitarity, the validity of low-energy quantum mechanics near the horizon, and the equivalence principle — cannot all hold for an old black hole; we will write the contradiction as a strong-subadditivity inequality that fails. Finally we note the sharpening that AdS/CFT brings: because the dual boundary theory is a manifestly unitary quantum field theory, the resolution cannot hide in unknown ultraviolet physics — it must already be present in semiclassical gravity, correctly read.

Throughout we adopt the natural-units conventions of Chapter 1 ($\hbar = c = 1$) and write the von Neumann entropy of a state ρ as $S(\rho) = -\operatorname{Tr}(\rho \log \rho)$, the entanglement entropy of Theorem 55.4, in nats. The bipartition that organizes everything is the split of the global Hilbert space into a radiation factor and a black-hole factor,

$$\mathcal{H} \;=\; \mathcal{H}_{\rm rad} \otimes \mathcal{H}_{\rm BH}, \tag{57.36}$$

where $\mathcal{H}_{\rm rad}$ collects all field modes that have escaped to infinity by time t and $\mathcal{H}_{\rm BH}$ everything still behind, or coincident with, the horizon (the remaining hole together with whatever lives just inside). The reduced states $\rho_{\rm rad} = \operatorname{Tr}_{\rm BH}\rho$ and $\rho_{\rm BH} = \operatorname{Tr}_{\rm rad}\rho$ are the objects whose entropies we track.

57.2.1 The Hawking-pair structure: why the radiation is mixed

The thermal character of the flux is not an accident of coarse-graining; it is built into the microscopic mechanism of emission, and seeing this precisely is what turns "information loss" from a slogan into a computation. The mechanism is pair creation. The gravitational field near the horizon mixes positive- and

negative-frequency modes — the Bogoliubov transformation whose coefficients we compute in detail in Chapter 65 — with the result that each outgoing field mode that reaches infinity is created together with a partner mode that falls into the hole. Label one such outgoing mode of frequency ω by the annihilation operator b_ω and its infalling partner by c_ω. The state in which the field is left, mode by mode, is not the out-vacuum $|0\rangle_b\,|0\rangle_c$ but a correlated superposition of pairs.

We take as input from the Bogoliubov analysis (Chapter 65) and the thermal flux of the previous section the following structural fact: for each frequency the field is in a *two-mode squeezed vacuum*,

$$|\Psi_\omega\rangle \;=\; \frac{1}{\cosh r_\omega}\sum_{n=0}^{\infty}(\tanh r_\omega)^{\,n}\;|n\rangle_b\;|n\rangle_c\,, \tag{57.37}$$

where $|n\rangle_b\,,|n\rangle_c$ are the number eigenstates of the exterior and interior partner modes and $r_\omega>0$ is a squeezing parameter we will fix in a moment by demanding the emitted spectrum be thermal at temperature T_H. The structure of Eq. (57.37) is forced: the conservation laws of pair creation allow only the diagonal terms $|n\rangle_b\,|n\rangle_c$ (one outgoing quantum for each infalling partner), and the geometric weighting in n is what a single Bogoliubov rotation of the vacuum produces. The state is pure — it is a definite vector $|\Psi_\omega\rangle$ — and it is an entangled state of the exterior and the interior. The exterior observer, who can never reach c, sees only the reduced state $\rho_b=\mathrm{Tr}_c\,|\Psi_\omega\rangle\langle\Psi_\omega|$. The following worked example computes that reduced state and its entropy from scratch, fixing r_ω along the way; this is the calculation that makes the radiation's mixedness a theorem rather than an assertion.

Worked Example 57.5 (The Hawking-pair density matrix is exactly thermal)**.** We compute $\rho_b=\mathrm{Tr}_c\,|\Psi_\omega\rangle\langle\Psi_\omega|$ for the state Eq. (57.37), show it is a thermal density matrix, fix the squeezing r_ω from the Planck spectrum, and evaluate its von Neumann entropy.

Normalization. First check that Eq. (57.37) is a unit vector. Using $\langle m|n\rangle=\delta_{mn}$ in each factor,

$$\langle\Psi_\omega|\Psi_\omega\rangle=\frac{1}{\cosh^2 r_\omega}\sum_{n=0}^{\infty}(\tanh r_\omega)^{2n}=\frac{1}{\cosh^2 r_\omega}\,\frac{1}{1-\tanh^2 r_\omega}=\frac{\cosh^2 r_\omega}{\cosh^2 r_\omega}=1, \tag{57.38}$$

where we summed the geometric series ($\tanh^2 r_\omega<1$) and used $1-\tanh^2 r_\omega=\mathrm{sech}^2\, r_\omega$.

The reduced state. Trace over the interior partner with the partial trace of Theorem 55.1, using the number basis $\{|k\rangle_c\}$:

$$\rho_b=\sum_{k=0}^{\infty}{}_c\langle k|\;|\Psi_\omega\rangle\langle\Psi_\omega|\;|k\rangle_c=\frac{1}{\cosh^2 r_\omega}\sum_{k}\sum_{m,n}(\tanh r_\omega)^{m+n}\;|m\rangle_b\;\langle k|n\rangle\,\langle m|k\rangle\;\langle n|_b$$

$$= \frac{1}{\cosh^2 r_\omega} \sum_{n=0}^{\infty} (\tanh r_\omega)^{2n} \; |n\rangle_b \langle n|_b \,, \tag{57.39}$$

the Kronecker deltas $\langle k|n\rangle = \delta_{kn}$ and $\langle m|k\rangle = \delta_{mk}$ collapsing the double sum to $m = n = k$. Thus ρ_b is diagonal in the number basis with eigenvalues

$$p_n \;=\; (1-x)\, x^{\,n}, \qquad x \;\equiv\; \tanh^2 r_\omega, \tag{57.40}$$

where we wrote $\cosh^{-2} r_\omega = 1 - \tanh^2 r_\omega = 1 - x$. These p_n sum to one and decay geometrically: the reduced state is a *mixed* state, a thermal (Gibbs) distribution over occupation number.

Fixing the temperature. A geometric population $p_n \propto x^n$ is precisely a Bose–Einstein distribution. The mean occupation is

$$\langle n \rangle = \sum_n n\, p_n = (1-x) \sum_n n\, x^n = (1-x)\, \frac{x}{(1-x)^2} = \frac{x}{1-x}, \tag{57.41}$$

using $\sum_n n x^n = x/(1-x)^2$. The previous section's thermal flux states that the occupation of a mode of frequency ω is the Planck value $\langle n \rangle = (e^{\omega/T_H} - 1)^{-1}$ (Eq. (57.12)). Matching to Eq. (57.41),

$$\frac{x}{1-x} = \frac{1}{e^{\omega/T_H} - 1} \;\Longrightarrow\; x = e^{-\omega/T_H}, \qquad \text{i.e.} \qquad \tanh^2 r_\omega = e^{-\omega/T_H}. \tag{57.42}$$

With this identification Eq. (57.40) becomes $p_n = (1 - e^{-\omega/T_H})\, e^{-n\omega/T_H}$, and ρ_b is *exactly* the canonical thermal state of a single harmonic mode of energy ω at temperature T_H,

$$\rho_b = (1 - e^{-\omega/T_H}) \sum_{n=0}^{\infty} e^{-n\omega/T_H} \; |n\rangle\langle n| = \frac{e^{-\omega\, \hat{n}/T_H}}{\mathrm{Tr}\; e^{-\omega\, \hat{n}/T_H}}, \tag{57.43}$$

with $\hat{n} = b^\dagger b$ the number operator. The exterior observer sees a heat bath at the Hawking temperature — with no remaining trace of the purity of $|\Psi_\omega\rangle$.

The entropy of one mode. With $p_n = (1-x)x^n$ the von Neumann entropy Eq. (55.6) is

$$\begin{aligned} S(\rho_b) &= -\sum_n p_n \log p_n = -\sum_n (1-x) x^n \big[\log(1-x) + n \log x\big] \\ &= -\log(1-x) \underbrace{(1-x) \sum_n x^n}_{=1} \;-\; \log x \underbrace{(1-x) \sum_n n\, x^n}_{=\langle n \rangle} \\ &= -\log(1-x) - \langle n \rangle \log x. \end{aligned} \tag{57.44}$$

Substituting $x = e^{-\omega/T_H}$ and $\langle n \rangle = (e^{\omega/T_H} - 1)^{-1}$ gives the familiar thermal-oscillator entropy

$$S(\rho_b) = \frac{\omega/T_H}{e^{\omega/T_H} - 1} - \log(1 - e^{-\omega/T_H}) = (1 + \langle n \rangle) \log(1 + \langle n \rangle) - \langle n \rangle \log \langle n \rangle, \tag{57.45}$$

the last equality following from $1 + \langle n\rangle = (1-x)^{-1}$ and $\langle n\rangle = x/(1-x)$, which turn $-\log(1-x) = \log(1+\langle n\rangle)$ and $-\log x = \log(1+\langle n\rangle) - \log\langle n\rangle$. This is strictly positive for every ω: each emitted mode carries genuine entropy.

The lesson of Theorem 57.5 is the engine of the whole paradox. The *global* pair state $|\Psi_\omega\rangle$ is pure, so its entropy is zero; by the purity criterion (Theorem 55.7) the exterior and interior reduced states share the same spectrum and the same entropy, $S(\rho_b) = S(\rho_c) > 0$. The entropy of the radiation is therefore not a deficit of the global state's purity — it is the entanglement between what escaped and what fell in. The information that would distinguish the initial collapsing body is not destroyed at the level of $|\Psi_\omega\rangle$; it sits in the cross-correlations between b and c. The trouble is that the partner c is on the far side of the horizon, causally inaccessible, and (in the semiclassical picture) destined for the singularity. If the only thing that ever comes out is ρ_b, the correlations are lost for good and the outside is left genuinely mixed.

Physical Insight

"Thermal" does not mean "structureless." The radiation looks like a heat bath *to anyone who measures only the exterior modes*, exactly as a single spin of a Bell pair looks maximally mixed to anyone who holds only that spin. The information is not absent; it is non-locally encoded in correlations with a partner one cannot reach. The entire resolution of the paradox will hinge on whether those partners — or rather the information in them — can be recovered from the radiation after all, and Chapter 56's quantum extremal surfaces are the tool that says when they can.

Because the Hawking modes of different frequencies, and the successive pairs emitted at different times, are produced independently in the semiclassical calculation, the total radiation state at time t is a tensor product of thermal factors,

$$\rho_{\mathrm{rad}}^{\mathrm{sc}}(t) = \bigotimes_{\text{modes emitted by } t} \rho_b, \tag{57.46}$$

and the von Neumann entropy is additive over a tensor product, $S(\rho_1 \otimes \rho_2) = S(\rho_1) + S(\rho_2)$ (immediate from Eq. (55.6), since the spectrum of a product is the product of spectra and log turns products into sums). Hence

$$S_{\mathrm{rad}}^{\mathrm{sc}}(t) = \sum_{\text{modes emitted by } t} S(\rho_b) \;=\; S_{\mathrm{rad}}^{\mathrm{th}}(t), \tag{57.47}$$

which is exactly the thermal radiation entropy the thermodynamics section tracked. We have now identified the semiclassical *fine-grained* entropy with the *thermal* entropy: in Hawking's calculation they coincide, and that coincidence is the disease.

57.2.2 Version 1: Hawking's statement (the entropy form)

The first precise statement of the paradox is a monotonicity together with a sign mismatch at the endpoint.

Proposition 57.6 (Semiclassical radiation entropy: monotone and terminal)**.** *In the semiclassical description Eq. (57.46), the fine-grained radiation entropy $S^{\mathrm{sc}}_{\mathrm{rad}}(t)$ is non-decreasing in t, strictly increasing while emission continues, and at the evaporation endpoint t_{evap} it equals the full thermal entropy of all emitted quanta, a strictly positive number of order $S_{\mathrm{BH}}(0)$. If the black hole formed from a pure state and evolution is unitary, the final radiation entropy must instead be zero. The two are incompatible.*

Proof. Monotonicity is the additivity Eq. (57.47) combined with positivity of each term. Between times $t_1 < t_2$ the radiation state acquires extra independent thermal factors, so

$$S^{\mathrm{sc}}_{\mathrm{rad}}(t_2) - S^{\mathrm{sc}}_{\mathrm{rad}}(t_1) = \sum_{\text{modes emitted in } (t_1,t_2)} S(\rho_b) \;\geq 0, \tag{57.48}$$

strictly positive whenever any mode is emitted, since each $S(\rho_b) > 0$ by Eq. (57.45). As $t \to t_{\mathrm{evap}}$ the sum runs over all emitted quanta and converges to the total thermal entropy $S^{\mathrm{th}}_{\mathrm{rad}}(t_{\mathrm{evap}})$, a finite positive number. A standard thermodynamic estimate (entropy-to-energy ratio of a thermal gas against the first law $\mathrm{d}M = T_H\,\mathrm{d}S_{\mathrm{BH}}$) gives $S^{\mathrm{th}}_{\mathrm{rad}}(t_{\mathrm{evap}}) \simeq \frac{4}{3} S_{\mathrm{BH}}(0)$, larger than the initial black-hole entropy: a thermal gas carries more entropy per unit energy than the hole it came from.

For the unitarity clause, suppose the global state begins as a pure state $|\Psi_0\rangle$ (a definite, if complicated, collapsing configuration) and evolves by a unitary $U(t)$. A unitary preserves purity — $U\,|\Psi_0\rangle\langle\Psi_0|\,U^\dagger$ has the same single nonzero eigenvalue, 1 — so $\rho(t) = U(t)\,|\Psi_0\rangle\langle\Psi_0|\,U(t)^\dagger$ is pure at all times, with $S(\rho(t)) = 0$. At t_{evap} the black-hole factor has disappeared, $\mathcal{H} \to \mathcal{H}_{\mathrm{rad}}$, so $\rho_{\mathrm{rad}}(t_{\mathrm{evap}}) = \rho(t_{\mathrm{evap}})$ is the whole (pure) state and $S(\rho_{\mathrm{rad}}) = 0$. This contradicts $S^{\mathrm{sc}}_{\mathrm{rad}}(t_{\mathrm{evap}}) \simeq \frac{4}{3} S_{\mathrm{BH}}(0) > 0$. □

This is the form in which Hawking originally posed the problem, and the form most often quoted: a pure state evolves into a mixed thermal state, so the final-state map is not unitary; pure → mixed is the signature of information loss. Phrased as in Theorem 57.6, the contradiction is already complete — but it has a soft spot. It compares the very beginning with the very end and never specifies *when* the semiclassical approximation must first go wrong. A sceptic could imagine the breakdown deferred to the last Planckian instant of evaporation, where nobody trusts the calculation anyway. The next version closes that escape.

57.2.3 Version 2: the unitarity bound and the Page time

To pin down *when* the semiclassical curve becomes impossible, we need a quantitative ceiling on the radiation entropy that holds at every instant. Unitarity supplies one, in two steps: an absolute bound on any entropy from the size of the Hilbert space, and the equality of the two reduced entropies for a pure global state.

Lemma 57.7 (Maximal entropy of a finite system)**.** *For any density matrix ρ on a Hilbert space of finite dimension d,*

$$S(\rho) \;\le\; \log d, \tag{57.49}$$

with equality if and only if $\rho = \mathbf{1}/d$ is the maximally mixed state.

Proof. Compare ρ with the maximally mixed state $\sigma = \mathbf{1}/d$ using the relative entropy $S(\rho\,\|\,\sigma) = \operatorname{Tr}\rho\log\rho - \operatorname{Tr}\rho\log\sigma$, which is non-negative by Klein's inequality (Theorem 55.9). Here $\log\sigma = \log(\mathbf{1}/d) = -(\log d)\,\mathbf{1}$, so $-\operatorname{Tr}\rho\log\sigma = (\log d)\operatorname{Tr}\rho = \log d$, while $\operatorname{Tr}\rho\log\rho = -S(\rho)$. Therefore

$$0 \le S(\rho\,\|\,\sigma) = -S(\rho) + \log d, \tag{57.50}$$

which rearranges to Eq. (57.49). Klein's inequality is saturated precisely when $\rho = \sigma$, giving the equality clause. □

Theorem 57.8 (The unitarity bound on the radiation entropy)**.** *Let the global state be pure and let $\mathcal{H}_{\mathrm{BH}}$ have $\dim\mathcal{H}_{\mathrm{BH}} = e^{S_{\mathrm{BH}}(t)}$, with $S_{\mathrm{BH}}(t) = A(t)/4G$ the Bekenstein–Hawking entropy of the instantaneous horizon. Then at every time*

$$S\big(\rho_{\mathrm{rad}}(t)\big) \;=\; S\big(\rho_{\mathrm{BH}}(t)\big) \;\le\; S_{\mathrm{BH}}(t) = \frac{A(t)}{4G}. \tag{57.51}$$

The fine-grained radiation entropy can never exceed the coarse-grained black-hole entropy.

Proof. Two ingredients. *First*, purity forces the two reduced entropies to agree. For a pure global state on $\mathcal{H}_{\mathrm{rad}}\otimes\mathcal{H}_{\mathrm{BH}}$, the Schmidt decomposition (Theorem 55.7) writes $|\Psi\rangle = \sum_i \sqrt{p_i}\,|i\rangle_{\mathrm{rad}}\,|i\rangle_{\mathrm{BH}}$ with a single set of Schmidt coefficients $\{p_i\}$; both reduced matrices are diagonal with these same eigenvalues, so $S(\rho_{\mathrm{rad}}) = -\sum_i p_i\log p_i = S(\rho_{\mathrm{BH}})$. (Equivalently this is the Araki–Lieb saturation: for a pure bipartite state $S_{\mathrm{rad}} = S_{\mathrm{BH}}$, the case of Theorem 55.12 with the global state pure.)

Second, the black-hole factor is finite-dimensional, with $\log\dim\mathcal{H}_{\mathrm{BH}} = S_{\mathrm{BH}}(t)$ — this is exactly the microscopic content of the Bekenstein–Hawking entropy established in the thermodynamics section (Eq. (57.26)): $e^{A/4G}$ is the count of black-hole microstates. Theorem 57.7 then bounds $S(\rho_{\mathrm{BH}}) \le \log\dim\mathcal{H}_{\mathrm{BH}} = S_{\mathrm{BH}}(t)$. Chaining the equality and the bound gives Eq. (57.51). □

Theorem 57.8 is the sharp form of the paradox, and its force is in the time dependence. The right-hand side $S_{\mathrm{BH}}(t)$ *decreases* as the hole evaporates — the ceiling is falling toward zero. The semiclassical curve $S^{\mathrm{sc}}_{\mathrm{rad}}(t)$ of Eq. (57.47), by contrast, *increases* monotonically (Theorem 57.6). A rising curve and a falling ceiling must cross, and the crossing has a name.

Definition 57.9 (Page time)**.** The *Page time* t_{Page} is the instant at which the semiclassical (thermal) radiation entropy first equals the Bekenstein–Hawking entropy of the remaining hole,

$$S^{\mathrm{sc}}_{\mathrm{rad}}(t_{\mathrm{Page}}) \;=\; S_{\mathrm{BH}}(t_{\mathrm{Page}}). \tag{57.52}$$

Since both sides start at $S_{\mathrm{BH}}(0)$ worth of "room" and move oppositely, the crossing occurs when the horizon has given up an $O(1)$ fraction of its entropy; carrying the $\frac{4}{3}$ thermal factor of Eq. (57.33), the precise value is $S_{\mathrm{BH}}(t_{\mathrm{Page}}) = \frac{4}{7}S_{\mathrm{BH}}(0)$ (Theorem 57.16) — somewhat past the midpoint of the hole's lifetime for a Schwarzschild evaporation.

For $t < t_{\mathrm{Page}}$ the semiclassical value sits below the ceiling and is consistent with unitarity. For $t > t_{\mathrm{Page}}$ it lies *above* $S_{\mathrm{BH}}(t)$ and therefore violates Eq. (57.51): no pure global state can produce it. This is the decisive sharpening over Version 1. The semiclassical computation does not merely give the wrong answer at the Planckian endpoint; it gives an impossible answer already at the Page time, when the hole is still macroscopically large, the curvature at the horizon is still tiny, and every assumption that went into Hawking's calculation looks unimpeachable. *If* the process is unitary, the radiation entropy must turn around at t_{Page} and descend, ultimately to zero — it must follow not $S^{\mathrm{sc}}_{\mathrm{rad}}$ but the curve $\min\{S^{\mathrm{sc}}_{\mathrm{rad}}(t), S_{\mathrm{BH}}(t)\}$, which rises, peaks at t_{Page}, and falls. Figure 57.3 shows the collision and the demanded turnover.

Caution

The bound Eq. (57.51) assumes $\dim \mathcal{H}_{\mathrm{BH}} = e^{S_{\mathrm{BH}}}$ is *finite* — that the black hole is a system of finitely many quantum states, not an infinite reservoir. This is the single most important physical assumption in the whole argument, and it is precisely what the Bekenstein–Hawking entropy, read as a microstate count, asserts. If instead the interior were an infinite-dimensional remnant able to store unbounded entanglement, Theorem 57.7 would give no ceiling and there would be no paradox — and no microstate interpretation of $A/4G$ either. The remnant scenario trades the information puzzle for a different pathology (infinitely many species of arbitrarily light states); we do not pursue it, and the holographic counting of Chapters 53 and 54 weighs heavily against it.

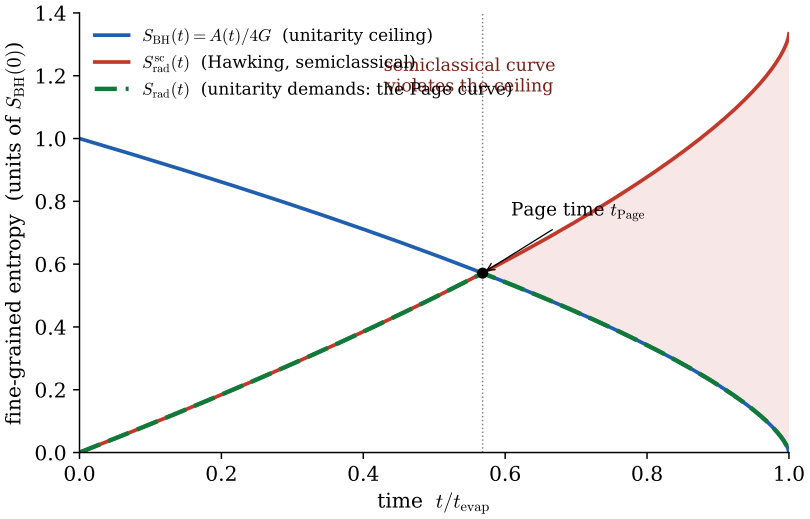


Figure 57.3: The clash of Theorem 57.8. The Bekenstein–Hawking entropy $S_{\rm BH}(t) = A(t)/4G$ (blue) falls to zero at the evaporation endpoint; unitarity makes it a hard ceiling on the radiation entropy. Hawking's semiclassical fine-grained entropy $S^{\rm sc}_{\rm rad}(t)$ (red) rises monotonically (Theorem 57.6) and at the endpoint reaches $\simeq \frac{4}{3}S_{\rm BH}(0)$, far above zero. The two cross at the Page time $t_{\rm Page}$ (Theorem 57.9), where $S^{\rm sc}_{\rm rad} = S_{\rm BH} = \frac{4}{7}S_{\rm BH}(0)$; in the shaded region the semiclassical curve lies *above* the ceiling and is impossible for a unitary theory. The dashed green curve is what unitarity demands instead — the Page curve, which tracks the rising thermal entropy until $t_{\rm Page}$ and then follows $S_{\rm BH}$ down to zero. Deriving its exact shape is the task of Section 57.3.

57.2.4 Version 3: the AMPS firewall and the monogamy of entanglement

Versions 1 and 2 say that the radiation entropy must turn over; they do not yet say that anything is wrong with the *geometry* near the horizon. The AMPS argument shows that something is: once one grants the entropy turnover demanded by unitarity, one cannot simultaneously keep a smooth horizon and the ordinary rules of low-energy quantum mechanics. The obstruction is the monogamy of entanglement, and its rigorous form is strong subadditivity.

Set up the three relevant systems for a black hole *past* its Page time. Let

- R be the early radiation, all the Hawking quanta emitted before some late time $t > t_{\rm Page}$;
- b be one freshly emitted late Hawking mode (the exterior partner of Eq. (57.37));

- c be the infalling partner of b, the interior mode with which b is paired across the horizon.

Two physical principles make incompatible demands on b.

Demand of the equivalence principle (smooth horizon). The equivalence principle says an infalling observer crossing a large, low-curvature horizon experiences nothing special: the local state is the vacuum. For the field modes straddling the horizon, "vacuum" is exactly the two-mode squeezed pair Eq. (57.37): b and c are in a pure, near-maximally entangled state. In the idealization that makes the inequality crisp, the pair bc is pure and decoupled from the early radiation, so

$$S(\rho_{bc}) = 0, \qquad S(\rho_{Rbc}) = S(\rho_R), \qquad S(\rho_b) = S(\rho_c) > 0. \tag{57.53}$$

(The first because bc is pure; the second because appending a pure, uncorrelated factor leaves the entropy unchanged; the third by the purity criterion applied to bc.)

Demand of unitarity (the Page condition). After the Page time the radiation entropy is *decreasing* (Fig. 57.3). Adding the newly emitted mode b to the already-collected early radiation R must therefore lower the total radiation entropy,

$$S(\rho_{Rb}) \;<\; S(\rho_R), \tag{57.54}$$

because the running entropy $S(\rho_{\mathrm{rad}})$ at time t is $S(\rho_{Rb})$ and at the slightly earlier time was $S(\rho_R)$, and it is on the descending branch. Physically, the late quantum b purifies part of the early radiation: it is (nearly maximally) entangled with R. In mutual-information terms (Theorem 55.13),

$$I(R:b) = S(\rho_R) + S(\rho_b) - S(\rho_{Rb}) > S(\rho_b) > 0. \tag{57.55}$$

These two demands collide, and strong subadditivity makes the collision exact.

Theorem 57.10 (The AMPS inequality fails)**.** *The smooth-horizon conditions Eq. (57.53) and the post-Page unitarity condition Eq. (57.54) cannot both hold for the systems R, b, c in any quantum theory obeying strong subadditivity.*

Proof. Apply strong subadditivity (Theorem 55.47) in the standard form $S(XY)+S(YZ) \geq S(XYZ)+S(Y)$ to the assignment $X = R$, $Y = b$, $Z = c$:

$$S(\rho_{Rb}) + S(\rho_{bc}) \;\geq\; S(\rho_{Rbc}) + S(\rho_b). \tag{57.56}$$

Now feed in the smooth-horizon data Eq. (57.53): $S(\rho_{bc}) = 0$ and $S(\rho_{Rbc}) = S(\rho_R)$. The inequality collapses to

$$S(\rho_{Rb}) \;\geq\; S(\rho_R) + S(\rho_b). \tag{57.57}$$

But ordinary subadditivity (Theorem 55.12) is the opposite inequality, $S(\rho_{Rb}) \leq S(\rho_R) + S(\rho_b)$, holding for every state. The two together force the equality $S(\rho_{Rb}) = S(\rho_R) + S(\rho_b)$, i.e. $I(R : b) = 0$: the late mode b is *completely uncorrelated* with the early radiation. This flatly contradicts the unitarity demand Eq. (57.55), $I(R : b) > S(\rho_b) > 0$. Hence Eq. (57.53) and Eq. (57.54) are mutually exclusive. □

The content of Theorem 57.10 is the monogamy of entanglement, sharpened to a theorem. A maximally entangled pair is monogamous: if b is maximally entangled with its interior partner c (smooth horizon), it has no entanglement left to share with anyone else, in particular not with the early radiation R. Conversely, if unitarity requires b to purify R (the Page condition), then b is spoken for and cannot also be the vacuum partner of c — the horizon cannot be smooth. Strong subadditivity is the precise statement that these options exhaust the possibilities; one of them must be abandoned.

Spelling out which assumptions are in play, the argument shows that the following four cannot all hold for an old black hole:

(U) **Unitarity:** the radiation entropy follows the Page curve, so Eq. (57.54) holds after t_{Page};

(E) **Equivalence principle:** the horizon is a smooth, low-curvature region with the field in its local vacuum, so Eq. (57.53) holds;

(Q) **Low-energy quantum mechanics:** ordinary quantum field theory, hence strong subadditivity (Theorem 55.47), governs the modes R, b, c;

(S) **No drama / semiclassical locality:** the late mode b and the early radiation R are independent subsystems, so the entropies above are well defined on a common Hilbert space.

Since (Q) is a theorem of quantum mechanics that we are not willing to surrender, and (S) is what lets us write the inequality at all, the live tension is between (U) and (E). Keeping unitarity (U) and giving up the smooth horizon (E) is the *firewall*: the late mode b is not in its vacuum state but in a highly excited one, so an infalling observer meets a wall of high-energy quanta at the horizon — a violation of the equivalence principle at a place where general relativity insists nothing should happen. Keeping the smooth horizon and giving up unitarity returns to Hawking's information loss. AMPS argued that for the most conservative reading of black-hole complementarity, the firewall is forced.

Caution

Status, and dating. Unlike Versions 1 and 2, which are clean theorems given their hypotheses, the firewall is a genuine *frontier* statement, and

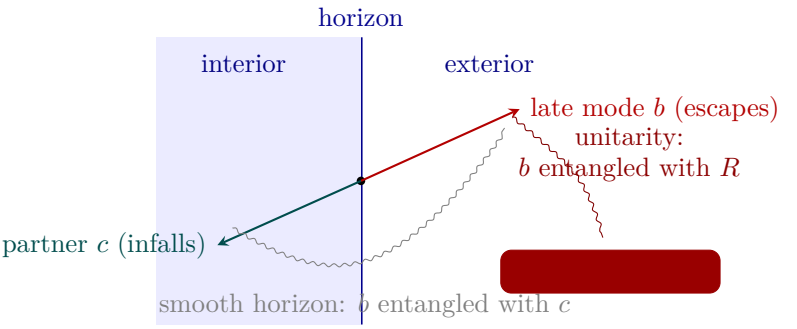


Figure 57.4: The monogamy clash of Theorem 57.10. A Hawking pair is created at the horizon: the exterior mode b escapes, its partner c falls in. A smooth horizon (equivalence principle) demands b be maximally entangled with c (grey wiggle). After the Page time, unitarity demands the same b purify the early radiation R (red wiggle), i.e. be entangled with R. Monogamy of entanglement — strong subadditivity, Theorem 55.47 — forbids b from being maximally entangled with two independent systems at once. One of the two demands must give.

its status as of the mid-2020s is unsettled. The community consensus is that the entropy paradox of Version 2 is resolved — the Page curve *is* reproduced by semiclassical gravity once the quantum-extremal-surface and replica-wormhole contributions of Chapter 56 and the gravitational path integral are included (Section 57.4). Whether this resolution also dissolves the firewall, by making the would-be partner c secretly part of the early radiation R (so that the monogamy violation never occurs), is a question about the operational recovery of the interior and the meaning of the entanglement-wedge reconstruction of Chapter 56; it is not settled with the rigor of Theorem 57.10. We will mark precisely which claims are established and which remain open as the chapter proceeds.

57.2.5 The AdS/CFT sharpening: the answer must live in gravity

There is one more turn of the screw, and it is the one that has driven the modern resolution. In all three versions above one could, in principle, retreat to the position that quantum gravity is unknown and that the rules of quantum mechanics — unitarity, the finiteness of $\dim \mathcal{H}_{\mathrm{BH}}$, even strong subadditivity — might be modified by Planck-scale physics we do not control. AdS/CFT removes that retreat.

Place the black hole in asymptotically anti-de Sitter space. By the correspondence of Chapter 53, the entire bulk gravitational system — black hole, Hawking

radiation, infalling observers and all — is described *exactly* by a conformal field theory living on the AdS boundary (the field–operator dictionary, Section 53.5). An AdS black hole is dual to a thermal state of that boundary CFT (Chapter 54), and its formation and evaporation correspond to ordinary time evolution of the boundary state. But the boundary CFT is an ordinary quantum field theory on a fixed spacetime: it has a Hamiltonian H, a well-defined Hilbert space, and a unitary time-evolution operator e^{-iHt}. Its evolution is manifestly, exactly unitary, with no Planck-scale caveats — unitarity is not an assumption about the bulk but a structural fact about the dual.

The consequences are decisive. *(i)* A pure bulk initial state maps to a pure boundary state, which stays pure under e^{-iHt}; so the final radiation *is* pure and $S(\rho_{\rm rad})$ must return to its initial value. Version 1's information loss is excluded outright. *(ii)* The boundary Hilbert space at fixed energy is finite-dimensional with $\log \dim = S_{\rm BH}$, the holographic entropy count of Chapter 54, so the unitarity bound Eq. (57.51) is not a hypothesis but a boundary theorem; the Page turnover is mandatory. *(iii)* Strong subadditivity holds in the boundary CFT as in any quantum system, so the AMPS tension is real and not an artifact of treating gravity naively.

The sharpening is therefore this: the paradox is genuine even when we *know* the theory is unitary. Whatever resolves it cannot be unknown ultraviolet physics that repairs a broken unitarity — unitarity was never broken. The resolution must instead be an error in the *semiclassical computation of the entropy*: Hawking's calculation, performed entirely within low-energy effective field theory on a smooth background, must be missing a contribution that bends $S_{\rm rad}(t)$ back down at the Page time. The remarkable fact, which the rest of this chapter develops, is that the missing contribution is itself *semiclassical* — a saddle of the gravitational path integral (the replica wormhole) computed with the quantum-extremal-surface technology of Chapter 56. The boundary's guaranteed unitarity tells us the answer exists in gravity; the QES prescription is how gravity delivers it.

Physical Insight

The deepest reading of the paradox is not "gravity is non-unitary" but "the semiclassical entropy formula is incomplete." AdS/CFT converts a question we cannot answer (is quantum gravity unitary?) into one we can (what does the gravitational path integral actually compute for $S_{\rm rad}$?). The lesson of Chapter 56 — that fine-grained entropies are computed by extremizing a generalized entropy over bulk surfaces, not by the naive matter trace — is exactly the missing ingredient, and the next sections turn it on the evaporating hole.

We have now stated the paradox three ways and located its resolution in

semiclassical gravity rather than in unknown physics. The common demand all three versions make is unambiguous: in a unitary theory the fine-grained radiation entropy must *rise, peak near the Page time, and fall back to zero.* What none of the three has given us is the precise *shape* of that curve — how high it peaks, how it bends, what statistical law governs the turnover. That shape is the Page curve, and the next section derives it from the statistics of random states alone, furnishing the exact target that any correct gravitational computation must reproduce.

57.3 The Page curve

The previous section sharpened the slogan "black holes destroy information" into a precise demand on a single observable. If formation and evaporation are unitary, the global state stays pure, and the fine-grained entropy of the radiation $S_{\mathrm{rad}}(t)$ must begin at zero (no radiation), end at zero (a pure final state of radiation, the black hole gone), and therefore *rise and then fall.* Hawking's semiclassical calculation, by contrast, makes each emitted quantum exactly thermal and statistically independent of every other, so S_{rad} rises monotonically and never comes back down. We are left with a question that has a sharp, checkable answer: if unitarity holds, what curve does $S_{\mathrm{rad}}(t)$ actually trace? This section answers it without invoking any gravitational dynamics at all. It assembles the curve from the statistics of a generic pure state alone — the result Page extracted in 1993 [374] — and so produces a precise target that any correct gravitational computation must reproduce. The next section's task will be to find what in semiclassical gravity bends the entropy back down; here we establish exactly what "bends down" must mean.

57.3.1 Random states as a model of a scrambled black hole

The physical input that licenses a statistical treatment is *scrambling.* A black hole is the fastest scrambler in nature: an excitation thrown in is spread over the entire horizon in a time $t_{\mathrm{scr}} \sim \beta \ln S_{\mathrm{BH}}$, parametrically shorter than the evaporation time. After the scrambling time the microscopic state of the radiation-plus-hole system has explored its available Hilbert space so thoroughly that, for the purpose of computing a coarse observable like the entanglement between the radiation and the remaining hole, it is indistinguishable from a state drawn uniformly at random. We therefore model the joint state at any time as a Haar-random pure state on a bipartite Hilbert space, compute the typical entanglement entropy of one factor, and only afterwards reinstate the time dependence through the dimensions of the two factors. This is a model, and we flag it as such: it replaces the true (unknown) microstate by a typical one, and its justification is the scrambling estimate together with the concentration-of-measure result proved

below, which guarantees that "typical" is overwhelmingly likely. What the model cannot do is exhibit the gravitational mechanism behind the curve — that is precisely the gap the island construction will fill.

Fix the arena. Let $\mathcal{H} = \mathcal{H}_A \otimes \mathcal{H}_B$ with $\dim \mathcal{H}_A = d_A$ and $\dim \mathcal{H}_B = d_B$, and adopt the convention

$$d_A \le d_B, \qquad d \equiv d_A d_B = \dim \mathcal{H}, \tag{57.58}$$

so that A is the smaller factor. (When we apply this to a black hole, A will be whichever of radiation/interior is smaller at the time in question; the roles switch at the Page time.) A pure state $|\psi\rangle \in \mathcal{H}$ has reduced density matrix $\rho_A = \mathrm{Tr}_B\, |\psi\rangle\langle\psi|$ (Theorem 55.2), and its entanglement entropy is the von Neumann entropy $S_A = -\mathrm{Tr}(\rho_A \ln \rho_A)$ of Theorem 55.4 and Eq. (55.6). Because the global state is pure, $S_A = S_B$ by Eq. (55.14): the entropy is a property of the cut, not of which side we look from. The largest value S_A can take is $\ln d_A$, attained when $\rho_A = I/d_A$ is maximally mixed; the question is how close to this ceiling a *random* state comes.

Definition 57.11 (Haar-random pure state)**.** Let $|\psi_0\rangle$ be any fixed unit vector in $\mathcal{H}$. A *Haar-random pure state* is $|\psi\rangle = U\,|\psi_0\rangle$ with U distributed according to the (normalized, bi-invariant) Haar measure on the unitary group $\mathrm{U}(d)$. Equivalently, writing $|\psi\rangle = \sum_i c_i\,|i\rangle$ in any orthonormal basis, the coefficients c_i are jointly distributed as a complex Gaussian vector conditioned to unit norm; the distribution is invariant under $|\psi\rangle \to V\,|\psi\rangle$ for any fixed $V \in \mathrm{U}(d)$. The induced ensemble of reduced density matrices ρ_A is the one we average over, denoted $\mathbb{E}[\,\cdot\,]$.

57.3.2 The Haar average of the purity

We cannot easily average the von Neumann entropy directly — it is transcendental in the eigenvalues of ρ_A — but we *can* average the purity $\mathrm{Tr}\,\rho_A^2$ exactly, because it is quadratic in ρ_A and hence quartic in $|\psi\rangle$, a polynomial that the Haar measure integrates in closed form. The purity is the Rényi-2 datum $S_2(A) = -\ln \mathrm{Tr}\,\rho_A^2$ of Theorem 55.5; it is the cheapest nontrivial probe of how mixed ρ_A is, and it already determines the leading shape of the curve. The computation rests on one structural fact about averaging two copies of a random state.

Lemma 57.12 (Symmetric-subspace average)**.** *For a Haar-random pure state $|\psi\rangle$ on a Hilbert space $\mathcal{H}$ of dimension d,*

$$\mathbb{E}[\,|\psi\rangle\langle\psi| \otimes |\psi\rangle\langle\psi|\,] = \frac{I + \mathbb{S}}{d(d+1)}, \tag{57.59}$$

where $\mathbb{S}$ is the swap operator on $\mathcal{H} \otimes \mathcal{H}$, defined by $\mathbb{S}\,(|\alpha\rangle \otimes |\beta\rangle) = |\beta\rangle \otimes |\alpha\rangle$, and I is the identity on $\mathcal{H} \otimes \mathcal{H}$.

Proof. The tensor square $|\psi\rangle^{\otimes 2}$ is symmetric under exchange of the two copies, so $\mathbb{S}\,|\psi\rangle^{\otimes 2} = |\psi\rangle^{\otimes 2}$; consequently the averaged operator $M \equiv \mathbb{E}[|\psi\rangle\langle\psi|^{\otimes 2}]$ is supported entirely on the symmetric subspace $\mathrm{Sym}^2(\mathcal{H}) \subset \mathcal{H}\otimes\mathcal{H}$, the $+1$ eigenspace of $\mathbb{S}$, whose dimension is $\binom{d+1}{2} = \tfrac{1}{2}d(d+1)$. By the invariance of the Haar measure, M commutes with $U\otimes U$ for every $U \in \mathrm{U}(d)$:

$$(U\otimes U)\,M\,(U\otimes U)^\dagger = \mathbb{E}\Big[(U\,|\psi\rangle)\langle\psi|\,U^\dagger \otimes (U\,|\psi\rangle)\langle\psi|\,U^\dagger\Big] = \mathbb{E}\Big[|\psi\rangle\langle\psi|^{\otimes 2}\Big] = M, \tag{57.60}$$

the middle step using that $U\,|\psi\rangle$ is again Haar-distributed. Now $\mathrm{Sym}^2(\mathcal{H})$ is an *irreducible* representation of $\mathrm{U}(d)$ under the diagonal action $U \mapsto U\otimes U$ (this is the $n=2$ case of Schur–Weyl duality: the commutant of the diagonal $\mathrm{U}(d)$ action on $\mathcal{H}^{\otimes 2}$ is spanned by I and $\mathbb{S}$, equivalently the algebra of S_2). By Schur's lemma an operator that commutes with an irreducible representation and is supported on its carrier space is a multiple of the identity there. Hence $M = c\,P_{\mathrm{sym}}$, where $P_{\mathrm{sym}} = \tfrac{1}{2}(I+\mathbb{S})$ is the projector onto $\mathrm{Sym}^2(\mathcal{H})$. The constant is fixed by the trace: $\mathrm{Tr}\,M = \mathbb{E}[(\mathrm{Tr}\,|\psi\rangle\langle\psi|)^2] = \mathbb{E}[1] = 1$, while $\mathrm{Tr}\,P_{\mathrm{sym}} = \tfrac{1}{2}d(d+1)$, so $c = 2/[d(d+1)]$ and $M = 2P_{\mathrm{sym}}/[d(d+1)] = (I+\mathbb{S})/[d(d+1)]$, which is Eq. (57.59). □

With the second-moment identity in hand the purity average is a short trace computation, which we present as the central worked calculation of the section.

Worked Example 57.13 (The Haar-averaged purity)**.** We compute $\mathbb{E}[\mathrm{Tr}\,\rho_A^2]$ for $\rho_A = \mathrm{Tr}_B\,|\psi\rangle\langle\psi|$ with $|\psi\rangle$ Haar-random on $\mathcal{H}_A\otimes\mathcal{H}_B$, and read off the regimes.

Step 1: the swap trick. Work on two copies, labelling them and their factors A, B and A', B'. Let $\mathbb{S}_A$ denote the swap acting only on the A factors (interchanging $\mathcal{H}_A \leftrightarrow \mathcal{H}_{A'}$, identity on $\mathcal{H}_B\otimes\mathcal{H}_{B'}$), and $\mathbb{S}_B$ the swap on the B factors; the full swap factorizes as $\mathbb{S} = \mathbb{S}_A\mathbb{S}_B$. The purity is a partial trace against a partial swap:

$$\mathrm{Tr}\,\rho_A^2 = \mathrm{Tr}_{AA'BB'}[(\rho\otimes\rho)\,\mathbb{S}_A]\,, \qquad \rho = |\psi\rangle\langle\psi|\,. \tag{57.61}$$

To verify Eq. (57.61), trace over B, B' first: $\mathrm{Tr}_{BB'}(\rho\otimes\rho) = \rho_A\otimes\rho_A$, and $\mathbb{S}_A$ touches only the A factors, so the right-hand side is $\mathrm{Tr}_{AA'}[(\rho_A\otimes\rho_A)\,\mathbb{S}_A]$. Writing $\rho_A = \sum_{ij}(\rho_A)_{ij}\,|i\rangle\langle j|$ and using $\mathrm{Tr}_{AA'}[(|i\rangle\langle j|\otimes|k\rangle\langle l|)\mathbb{S}_A] = \langle l|i\rangle\langle j|k\rangle = \delta_{li}\delta_{jk}$ collapses the double sum to $\sum_{jk}(\rho_A)_{kj}(\rho_A)_{jk} = \mathrm{Tr}\,\rho_A^2$. This is the standard identity "$\mathrm{Tr}\,\rho_A^2$ equals the expectation of the swap" specialized to the A factors.

Step 2: average and trace. Take the Haar average of Eq. (57.61) and insert Eq. (57.59) with $d = d_A d_B$:

$$\mathbb{E}[\mathrm{Tr}\,\rho_A^2] = \mathrm{Tr}\big[\mathbb{E}(\rho\otimes\rho)\,\mathbb{S}_A\big] = \frac{1}{d(d+1)}\Big(\mathrm{Tr}[\mathbb{S}_A] + \mathrm{Tr}[\mathbb{S}\,\mathbb{S}_A]\Big). \tag{57.62}$$

The two traces are elementary. For any swap operator on $\mathcal{K}\otimes\mathcal{K}$ one has $\mathrm{Tr}[\mathbb{S}_{\mathcal{K}}] = \dim\mathcal{K}$ (the swap fixes the $\dim\mathcal{K}$ "diagonal" basis vectors $|n\rangle\otimes|n\rangle$ and pairs off the rest, so its trace counts the fixed vectors). Therefore

$$\mathrm{Tr}[\mathbb{S}_A] = (\,\mathrm{Tr}_{AA'}\,\mathbb{S}_A)\,(\,\mathrm{Tr}_{BB'}\,I_{BB'}) = d_A\cdot d_B^2, \tag{57.63}$$

$$\mathrm{Tr}[\mathbb{S}\,\mathbb{S}_A] = \mathrm{Tr}[\mathbb{S}_A\mathbb{S}_B\mathbb{S}_A] = \mathrm{Tr}[\mathbb{S}_B] = (\,\mathrm{Tr}_{AA'}\, I_{AA'})(\,\mathrm{Tr}_{BB'}\,\mathbb{S}_B) = d_A^2 \cdot d_B, \quad (57.64)$$

where in Eq. (57.64) we used $\mathbb{S} = \mathbb{S}_A\mathbb{S}_B$, that $\mathbb{S}_A$ and $\mathbb{S}_B$ commute, and $\mathbb{S}_A^2 = I$. Substituting into Eq. (57.62),

$$\boxed{\mathbb{E}[\mathrm{Tr}\,\rho_A^2] = \frac{d_A d_B^2 + d_A^2 d_B}{d_A d_B\,(d_A d_B + 1)} = \frac{d_A + d_B}{d_A d_B + 1}\,.} \quad (57.65)$$

Step 3: read the regimes. For large dimensions the $+1$ in the denominator is negligible and

$$\mathbb{E}[\mathrm{Tr}\,\rho_A^2] \simeq \frac{1}{d_A} + \frac{1}{d_B}. \quad (57.66)$$

When A is the small subsystem, $d_A \ll d_B$, the first term dominates: $\mathbb{E}[\mathrm{Tr}\,\rho_A^2] \simeq 1/d_A$, the purity of the *maximally mixed* state $\rho_A = I/d_A$. A random state thus leaves a small subsystem almost exactly maximally mixed — maximally entangled with the rest. At the symmetric point $d_A = d_B = m$ the result is $\mathbb{E}[\mathrm{Tr}\,\rho_A^2] = 2m/(m^2+1) \simeq 2/m$, twice the maximally-mixed value, so ρ_A is mixed but measurably short of maximal.

57.3.3 From purity to entropy: Page's formula

The purity already pins the leading behavior. Converting Eq. (57.65) to a Rényi-2 entropy and expanding for $d_A \le d_B$,

$$S_2(A) = -\ln \mathbb{E}[\mathrm{Tr}\,\rho_A^2] \simeq -\ln\!\left(\tfrac{1}{d_A} + \tfrac{1}{d_B}\right) = \ln d_A - \ln\!\left(1 + \tfrac{d_A}{d_B}\right) \simeq \ln d_A - \frac{d_A}{d_B}. \quad (57.67)$$

Two features survive to the von Neumann entropy and constitute the whole physics of the section. First, the *leading* value is the ceiling: $S_A \simeq \ln d_A = \ln(\min(d_A, d_B))$, so a generic pure state saturates the maximal entanglement of its smaller factor up to a small correction. Second, the correction is a *deficit* of order d_A/d_B that grows as the two factors approach equal size — the entropy falls below maximal exactly when the subsystem is no longer small. The qualitative bend of the Page curve is already visible here: as the smaller factor grows toward half the total, the entropy approaches but stays below ln(dim).

The exact coefficient of the deficit is not captured by the Rényi-2 average, because S_2 and the von Neumann entropy differ in their subleading terms (and because $-\ln \mathbb{E}[\mathrm{Tr}\,\rho_A^2] \neq \mathbb{E}[-\ln \mathrm{Tr}\,\rho_A^2]$, a Jensen gap that is itself subleading). The full first-moment computation requires the joint eigenvalue distribution of ρ_A — the fixed-trace Wishart, or induced, ensemble — and yields a result Page conjectured from numerics and small cases, subsequently proved [196, 427, 439].

Theorem 57.14 (Page's formula for the average entanglement entropy)**.** *For a Haar-random pure state on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $d_A \le d_B$, the average von Neumann*

entropy of the smaller factor is exactly

$$\langle S_A\rangle = \left(\sum_{k=d_B+1}^{d_A d_B}\frac{1}{k}\right) - \frac{d_A-1}{2d_B} = (H_{d_Ad_B} - H_{d_B}) - \frac{d_A-1}{2d_B}, \tag{57.68}$$

where $H_n = \sum_{k=1}^{n} 1/k$ *is the* n*-th harmonic number. Its large-dimension asymptotics is*

$$\langle S_A\rangle = \ln d_A - \frac{d_A}{2d_B} + \frac{1}{2d_Ad_B} + O(d^{-2}). \tag{57.69}$$

We do not reproduce the random-matrix integral that establishes Eq. (57.68) — it is a self-contained but lengthy computation in the cited references, and the leading physics has already been derived independently from the purity above. What we *do* carry out in full is the passage from the exact harmonic-sum form to the asymptotics Eq. (57.69), since that is what we will use, and since it shows the deficit coefficient $\frac{1}{2}$ explicitly.

Derivation of the asymptotics from Eq. (57.68). Use the standard expansion of the harmonic numbers, $H_n = \ln n + \gamma + \frac{1}{2n} + O(n^{-2})$, with γ the Euler–Mascheroni constant. The constant γ cancels in the difference:

$$\begin{aligned} H_{d_Ad_B} - H_{d_B} &= \left(\ln(d_Ad_B) + \gamma + \tfrac{1}{2d_Ad_B}\right) - \left(\ln d_B + \gamma + \tfrac{1}{2d_B}\right) + O(d^{-2}) \\ &= \ln d_A + \frac{1}{2d_Ad_B} - \frac{1}{2d_B} + O(d^{-2}). \end{aligned} \tag{57.70}$$

Adding the explicit term $-(d_A-1)/(2d_B) = -d_A/(2d_B) + 1/(2d_B)$ from Eq. (57.68),

$$\langle S_A\rangle = \ln d_A + \frac{1}{2d_Ad_B} - \frac{1}{2d_B} - \frac{d_A}{2d_B} + \frac{1}{2d_B} + O(d^{-2}) = \ln d_A - \frac{d_A}{2d_B} + \frac{1}{2d_Ad_B} + O(d^{-2}), \tag{57.71}$$

the two $\pm 1/(2d_B)$ terms cancelling, which is Eq. (57.69). □

The deficit coefficient is therefore $\frac{1}{2}$ for the von Neumann entropy, half the 1 of the cruder Rényi-2 estimate Eq. (57.67) — a genuine difference between the two entropies, not a discrepancy. Two special values are worth recording. At the symmetric split $d_A = d_B = m$, Eq. (57.69) gives

$$\langle S_A\rangle \simeq \ln m - \tfrac{1}{2}, \tag{57.72}$$

the celebrated *half-nat* below maximal entanglement that a random state carries when bisecting its Hilbert space evenly. For a strongly lopsided split $d_A \ll d_B$ the deficit $d_A/(2d_B) \to 0$ and $\langle S_A\rangle \to \ln d_A$ exactly: the small subsystem is maximally entangled to exponential accuracy. Collecting the two cases, the headline that drives everything below is

$$\boxed{\langle S_A\rangle \simeq \ln\big(\min(d_A, d_B)\big)} \tag{57.73}$$

up to a correction that is $O(1)$ only in the narrow window around the symmetric point and exponentially small away from it.

57.3.4 Typicality: why the average is the answer

Equation Eq. (57.73) is a statement about the *average* over states. We are entitled to use it for a *particular* (if unknown) black-hole microstate only if the entropy is sharply concentrated about its mean — if almost every state has very nearly the average entropy. It is, and the reason is geometric: the unit sphere in a high-dimensional space is dominated, in measure, by a thin band around any equator, so any sufficiently smooth function on it is nearly constant with overwhelming probability. This is Levy's lemma, the concentration-of-measure phenomenon.

Theorem 57.15 (Levy's lemma)**.** *Let $f : \mathbb{S}^{2N-1} \to \mathbb{R}$ be a function on the unit sphere of $\mathbb{C}^N \cong \mathbb{R}^{2N}$ with Lipschitz constant η (so $|f(\psi) - f(\phi)| \le \eta \, \||\psi\rangle - |\phi\rangle\|$). For $|\psi\rangle$ drawn from the uniform (Haar) measure on the sphere and any $\delta > 0$,*

$$\mathrm{Prob}\big(\,|f(\psi) - \mathbb{E}f| \ge \delta\,\big) \;\le\; 2\,\exp\!\Big(-\frac{C\,N\,\delta^2}{\eta^2}\Big), \tag{57.74}$$

with C a positive numerical constant (one may take $C = 1/(9\pi^3)$).

We apply this with $N = d = d_A d_B$ and $f(\psi) = S_A(\psi)$. The entanglement entropy is Lipschitz in $|\psi\rangle$ with a constant $\eta = O(\ln d_A)$ — bounded, up to the logarithm that sets the scale of S_A itself, independently of N — because the map from $|\psi\rangle$ to the spectrum of ρ_A is smooth and $-\sum \lambda \ln \lambda$ has bounded gradient on the simplex away from the boundary, with the boundary contributing negligibly. Then Eq. (57.74) says that the probability of S_A deviating from $\langle S_A\rangle$ by even a fixed δ is suppressed by $\exp(-C' d\,\delta^2/\ln^2 d_A)$. Since $d = e^{S_{\mathrm{tot}}}$ is exponentially large in the entropy of the whole system, the fluctuations are doubly-exponentially rare: a randomly chosen state has $S_A = \langle S_A\rangle$ to a precision that is itself exponentially small. Page's direct computation of the variance confirms this, giving $\mathrm{Var}(S_A) = O(1/d)$ at the symmetric point. The conclusion is the one the model needs: Eq. (57.73) holds not just on average but for the overwhelming majority of states, hence (under the scrambling hypothesis) for the actual black-hole microstate.

Physical Insight

Concentration of measure is what makes "thermalization" and "scrambling" into precise mathematics rather than slogans. A single typical pure state of a large system looks thermal on every small subsystem, with corrections invisibly small. The black hole inherits this: its radiation looks exactly thermal locally — which is why Hawking's calculation gets the *coarse-grained* entropy right — while the fine-grained entropy, the global property Eq. (57.73), knows about the purity of the whole and bends back down. Hawking measured the band; Page measured the equator.

57.3.5 The Page curve of an evaporating black hole

We now reinstate time. Partition the Hilbert space of the evaporating system at time t into the radiation already collected and the remaining black hole,

$$\mathcal{H}(t) = \mathcal{H}_{\rm rad}(t) \otimes \mathcal{H}_{\rm BH}(t), \qquad \dim \mathcal{H}_{\rm rad}(t) = e^{S^{\rm th}_{\rm rad}(t)}, \quad \dim \mathcal{H}_{\rm BH}(t) = e^{S_{\rm BH}(t)}. \tag{57.75}$$

The two exponents are exactly the coarse-grained entropies established in Section 57.1: $S_{\rm BH}(t) = A(t)/4G$ is the Bekenstein–Hawking entropy of the horizon (Eq. (57.26)), the logarithm of the number of black-hole microstates, and $S^{\rm th}_{\rm rad}(t)$ is the thermal (von Neumann) entropy of the radiation Hawking would compute (Eq. (57.33)), the logarithm of the effective dimension explored by the emitted quanta. The black hole forms from collapse of matter in a pure state and evolves unitarily, so the global state on $\mathcal{H}(t)$ stays pure throughout. Assuming it is, after scrambling, a typical state in the sense of Section 57.3.1, Page's headline Eq. (57.73) applies with $d_A \to e^{S^{\rm th}_{\rm rad}}$ and $d_B \to e^{S_{\rm BH}}$:

$$\boxed{\; S_{\rm rad}(t) \;\simeq\; \min\!\big(S^{\rm th}_{\rm rad}(t),\, S_{\rm BH}(t)\big)\;. \;} \tag{57.76}$$

This is the Page curve. Its two branches have transparent meaning.

Early times, $t < t_{\rm Page}$: little has evaporated, so $S^{\rm th}_{\rm rad} < S_{\rm BH}$, the radiation is the smaller factor, and $S_{\rm rad} \simeq S^{\rm th}_{\rm rad}(t)$. The fine-grained entropy tracks the thermal one: the radiation is, to exponential accuracy, maximally entangled with the hole, and its entropy rises just as Hawking found. There is no tension yet.

Late times, $t > t_{\rm Page}$: now $S^{\rm th}_{\rm rad} > S_{\rm BH}$, the black hole is the smaller factor, and $S_{\rm rad} \simeq S_{\rm BH}(t)$. Here it is cleanest to invoke purity directly: since the global state is pure, $S_{\rm rad} = S_{\rm BH}$ exactly by Eq. (55.14), and the now-small hole is near-maximally mixed, so $S_{\rm BH} \simeq \ln(\dim \mathcal{H}_{\rm BH}) = S_{\rm BH}(t)$. As the horizon shrinks, $S_{\rm BH}(t) \to 0$, and with it the radiation entropy. The radiation purifies itself: the late quanta are correlated with the early ones in precisely the pattern needed to bring a maximally-mixed-looking cloud back to a pure state once the last quantum is emitted.

The turnover, at $t = t_{\rm Page}$, is where the two coarse-grained entropies cross, $S^{\rm th}_{\rm rad}(t_{\rm Page}) = S_{\rm BH}(t_{\rm Page})$, equivalently $\dim \mathcal{H}_{\rm rad} = \dim \mathcal{H}_{\rm BH}$. At that instant the bipartition is symmetric and the entropy is maximal, sitting (by Eq. (57.72)) a half-nat below the common value $S_{\rm BH}(t_{\rm Page})$, which is somewhat above half the initial entropy. We pin the number next.

Worked Example 57.16 (The Page time)**.** We locate $t_{\rm Page}$ using the Schwarzschild expressions of Section 57.1. For a four-dimensional Schwarzschild hole of mass M, the horizon entropy is (Eq. (57.77))

$$S_{\rm BH} = \frac{A}{4G} = \frac{4\pi(2GM)^2}{4G} = 4\pi G M^2, \tag{57.77}$$

and Hawking emission drives the mass loss $dM/dt = -\alpha/M^2$ (Stefan–Boltzmann with $T_{\rm H} \propto 1/M$ and radiating area $\propto M^2$ giving power $\propto T^4 M^2 \propto M^{-2}$), where α collects the greybody and species constants. Integrating $M^2\,dM = -\alpha\,dt$,

$$M(t)^3 = M_0^3 - 3\alpha t, \qquad t_{\rm evap} = \frac{M_0^3}{3\alpha}, \qquad M(t) = M_0\Big(1 - \frac{t}{t_{\rm evap}}\Big)^{1/3}. \tag{57.78}$$

Writing $\tau = t/t_{\rm evap}$ and $S_0 \equiv S_{\rm BH}(0) = 4\pi G M_0^2$, the horizon entropy is

$$\frac{S_{\rm BH}(\tau)}{S_0} = \Big(\frac{M(\tau)}{M_0}\Big)^2 = (1-\tau)^{2/3}. \tag{57.79}$$

The coarse-grained radiation entropy accumulates the entropy the hole gives up. As established in Eq. (57.33), thermal emission of massless quanta carries off $\frac{4}{3}$ of the entropy the hole loses (the photon-gas "Zurek factor"), so in the adiabatic model $S_{\rm rad}^{\rm th}(\tau) = \frac{4}{3}(S_0 - S_{\rm BH}(\tau))$. This factor is *not* immaterial to the location of the crossing: the Page time is fixed by $\dim \mathcal{H}_{\rm rad} = \dim \mathcal{H}_{\rm BH}$, i.e. by the thermal radiation entropy itself equalling $S_{\rm BH}$, so the $\frac{4}{3}$ shifts where that equality is reached. Setting $S_{\rm rad}^{\rm th} = S_{\rm BH}$,

$$\tfrac{4}{3}(S_0 - S_{\rm BH}) = S_{\rm BH} \implies S_{\rm BH}(\tau_{\rm Page}) = \tfrac{4}{7}S_0 \implies (1-\tau_{\rm Page})^{2/3} = \tfrac{4}{7}. \tag{57.80}$$

Solving, $1 - \tau_{\rm Page} = (4/7)^{3/2}$, so

$$\tau_{\rm Page} = 1 - \Big(\tfrac{4}{7}\Big)^{3/2} \approx 0.568, \qquad t_{\rm Page} \approx 0.57\, t_{\rm evap}. \tag{57.81}$$

The Page time is reached when the horizon entropy has dropped to $\frac{4}{7}$ of its initial value — when, equivalently, the mass has fallen to $\sqrt{4/7}\,M_0 \approx 0.76\,M_0$ and three-sevenths of the original microstate-counting entropy has been carried out by the radiation. It is somewhat past the halfway point of the *lifetime* (because the hole evaporates faster as it shrinks). The peak radiation entropy there is $S_{\rm rad} = S_{\rm BH}(\tau_{\rm Page}) = \frac{4}{7}S_0$, a little above half the initial Bekenstein–Hawking entropy. The value $\frac{4}{3}$ used here is the idealized blackbody (Zurek) factor; restoring the realistic frequency-dependent greybody weighting of Eq. (57.29) raises the effective factor to ≈ 1.48 and shifts the crossing to $\tau_{\rm Page} \approx 0.54$, the value quoted for a massless-field $d = 4$ Schwarzschild hole in Theorem 57.25.

Figure 57.5 shows the result. The fine-grained entropy of the radiation climbs along the thermal curve, peaks at $\frac{4}{7}S_0$ at the Page time, and descends along the horizon entropy to zero at evaporation: a rise and a fall, exactly the profile unitarity demanded in the previous section.

57.3.6 Hawking's line versus the Page curve: localizing the crisis

It remains to set Page's curve against Hawking's and read off precisely where, and how badly, the semiclassical calculation fails. Hawking's computation (developed

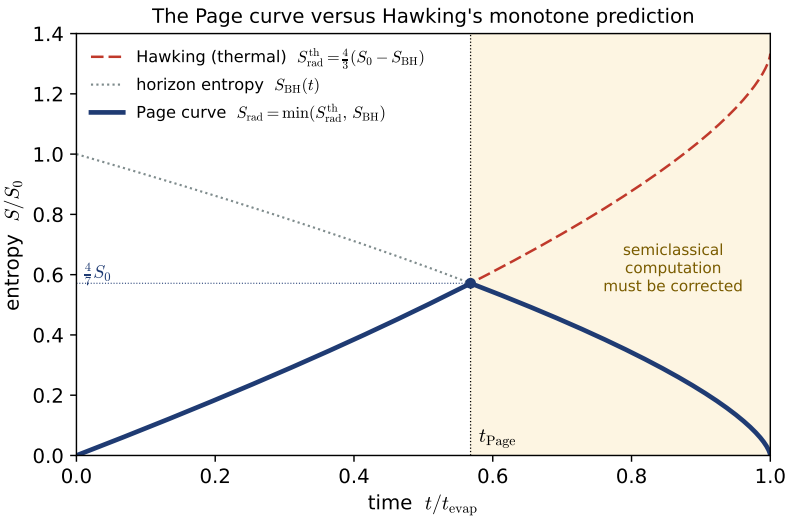


Figure 57.5: The Page curve. The fine-grained radiation entropy $S_{\rm rad} = \min(S_{\rm rad}^{\rm th}, S_{\rm BH})$ (solid) rises along the thermal entropy $S_{\rm rad}^{\rm th} = \frac{4}{3}(S_0 - S_{\rm BH})$, turns over at the Page time $t_{\rm Page} \approx 0.57\, t_{\rm evap}$ (peak $\frac{4}{7}S_0$) where it equals the shrinking horizon entropy $S_{\rm BH}$ (dotted), and decreases to zero at evaporation. Hawking's semiclassical calculation predicts the monotone thermal curve (dashed): it agrees with the Page curve before $t_{\rm Page}$ and diverges from it after, leaving a residual entropy $\frac{4}{3}S_0$ in the final radiation. The shaded region $t > t_{\rm Page}$ is where the two predictions part company — the regime any correct gravitational computation must correct.

from the Bogoliubov route in Chapter 65) outputs radiation that is *exactly* thermal and uncorrelated between successive emissions. For such radiation the fine-grained entropy equals the coarse-grained one identically: $S_{\rm rad}^{\rm Hawking}(t) = S_{\rm rad}^{\rm th}(t)$, the monotone dashed curve in Fig. 57.5. It rises without turning over and ends, when the hole is gone, at

$$S_{\rm rad}^{\rm Hawking}(t_{\rm evap}) = S_{\rm rad}^{\rm th}(t_{\rm evap}) = \tfrac{4}{3}S_0 > 0, \tag{57.82}$$

a final radiation state with entropy of order the entire initial horizon entropy — a mixed state produced from a pure one. This is the information paradox in its sharpest entropic form: a pure-to-mixed evolution, in flat contradiction with unitarity (Section 57.2).

The two curves coincide for $t < t_{\rm Page}$, where both equal $S_{\rm rad}^{\rm th}$, and diverge for $t > t_{\rm Page}$, where Hawking keeps climbing while unitarity demands descent. The disagreement is not a uniform small error to be patched everywhere; it is sharply localized in time. Before the Page time Hawking is *right*, to exponential accuracy: a small subsystem of a typical pure state is genuinely thermal, and the

semiclassical computation captures it faithfully. The failure switches on abruptly at t_{Page}, precisely when the radiation grows large enough to be the *larger* factor, so that its fine-grained entropy must start reporting the purity of the whole rather than the thermality of the part. The crisis is the after-Page-time regime, and nothing earlier.

Caution

The crossing in Fig. 57.5 is not literally a corner. The exact average Eq. (57.68) is smooth in d_A, d_B, and near the symmetric point the entropy rounds off the kink over a window of width set by the half-nat of Eq. (57.72); the sharp $\min(\cdot,\cdot)$ is the leading-order envelope, valid away from t_{Page} where one factor dominates. In the gravitational computation of the next section this rounding is reproduced by the competition of two saddles whose actions are equal at the crossing, and the smoothing comes from their sum.

The Page curve has thus been derived from nothing but the statistics of random states and the two coarse-grained entropies of Section 57.1: a linear rise tracking $S^{\text{th}}_{\text{rad}}$, a turnover at $t_{\text{Page}} \approx 0.57\, t_{\text{evap}}$ at the peak value $\frac{4}{7}S_0$, and a decline to zero along S_{BH}. This is the benchmark. The pressing question it leaves is mechanical: Hawking's semiclassical gravity, computing S_{rad} as the entropy of quantum fields on the evaporating background, returns the monotone line Eq. (57.82) and no turnover. What object in the gravitational calculation, switching on exactly at the Page time, bends the entropy back down to meet S_{BH}? The next section answers this with the quantum extremal surface prescription of Section 56.5.3: a second extremal surface — one whose homology region is an *island* inside the black hole — becomes the dominant saddle for S_{rad} at t_{Page}, and its generalized entropy (Eq. (56.151)) is exactly $S_{\text{BH}}(t)$, reproducing the descending branch and hence the entire Page curve from semiclassical gravity itself.

57.4 Quantum extremal surfaces and the island formula

The previous section handed us a verdict and a curve. Unitarity of the black hole's evolution *demands*, by the random-state statistics of Section 57.3, that the fine-grained entropy of the Hawking radiation follow the Page curve: rising while the emitted quanta are few, turning over at the Page time, and descending back to zero as the hole disappears. Equivalently, in the form that will be cleanest to reproduce, the radiation entropy is the smaller of two competing quantities,

$$S_{\text{rad}}(t) \;=\; \min\!\big(\, S^{\text{th}}_{\text{rad}}(t)\,,\; S_{\text{BH}}(t)\,\big), \tag{57.83}$$

the coarse (thermal) radiation entropy $S_{\rm rad}^{\rm th}$ that Hawking's calculation produces, and the Bekenstein–Hawking entropy $S_{\rm BH} = \text{Area}/4G_N$ of the remaining hole (Eq. (57.76)). That is the target. What is missing is a *computation*: a calculation, performed inside semiclassical gravity, that delivers Eq. (57.83) without putting unitarity in by hand. Hawking's own computation gives only the first branch and runs off to infinity. We will obtain both branches, and their minimum, from a single principle — and remarkably, that principle is one we already possess. It is the quantum extremal surface prescription of Theorem 56.27, applied not to a boundary region of a holographic CFT but to the radiation itself.

This section carries out that application. We specialize the QES rule to the radiation region, obtain the *island formula*, identify its two saddle configurations, compute both in a tractable two-dimensional model where every step is explicit, and watch their crossover land exactly on the Page time. The result is the gravitational reproduction of Eq. (57.83). We will be candid throughout about what is and is not established: the clean computation lives in two dimensions and is reduced to the spherically symmetric (s-wave) sector, and the island formula is here *asserted* as a corollary of the QES prescription rather than derived from the gravitational path integral — a debt discharged in Section 57.5.

57.4.1 The setup: gravity coupled to a bath, and the radiation region

To speak of "the entropy of the radiation" as a sharp quantity, we must first say where the radiation lives and what computes its entropy. The cleanest arena, and the one in which the computation closes in closed form, is a gravitating region *coupled to a non-gravitating bath*. Picture an asymptotically anti-de Sitter black hole, its geometry that of Section 53.3, but with the usual reflecting boundary conditions at the AdS boundary replaced by transparent ones: instead of bouncing the Hawking quanta back in, we attach at the boundary an auxiliary, non-dynamical-gravity system — a flat half-line carrying a quantum field theory — into which the radiation can escape and accumulate. The bath is an idealized detector array: it has no gravity of its own, so areas and the generalized-entropy machinery act only in the gravitating region, but it faithfully collects every quantum the black hole emits. The total state on a Cauchy slice splits into a gravitating part and a bath part, and the *radiation region* R is a chosen piece of the bath, far from the black hole, holding the escaped quanta.

The reason this arena is forced upon us, rather than merely convenient, is worth stating. The QES prescription of Theorem 56.27 computes the entropy of a region by extremizing the generalized entropy over bulk surfaces. In a closed gravitating universe there is a notorious obstruction: the entropy of the *whole* state is computed by the empty surface and vanishes, and more generally the constraints of gravity (the vanishing of the Hamiltonian) make the naive entropy of a gravitating region ill-defined. The bath removes the obstruction.

The radiation region R sits in the non-gravitating bath, where von Neumann entropy is the ordinary, unambiguous quantity of Theorem 55.4; the gravitating region enters only through the surfaces that the QES rule allows to bound R's entanglement wedge. This is the configuration in which "$S_{\rm rad}$" has both an honest definition and a gravitational computation.

We must also fix *which* entropy we mean and *how* it is to be computed. The bath is an ordinary quantum system, so the fine-grained entropy $S(R) = -\operatorname{Tr}(\rho_R \ln \rho_R)$ of the reduced state on R is well defined with no gravitational subtlety. The content of the upcoming formula is that gravity nonetheless dictates its value: the same bulk semiclassical state whose Hawking quanta populate R also fixes, through the area of a surface deep in the geometry, how much fine-grained entropy R actually carries. The bridge is the entanglement wedge of Theorem 56.31, now read in reverse. There the wedge of a boundary region was a piece of bulk; here the "region" is the bath system R, and its entanglement wedge is the part of the *gravitating* spacetime whose quantum information R holds. The QES prescription locates that wedge by extremizing generalized entropy — and the new phenomenon, absent for a boundary interval in pure AdS, is that the wedge of a faraway bath region can include a *disconnected* component sitting inside the black hole. That component is the island.

57.4.2 The island formula as a specialization of the QES prescription

We now state the rule. It is, line for line, the quantum extremal surface prescription Eq. (56.153) with the boundary region replaced by the bath region R and the homology region's bulk component renamed.

Definition 57.17 (Island)**.** Let R be a region of the bath. An *island* I for R is a region of the gravitating spacetime, possibly empty, whose boundary ∂I (the *quantum extremal surface*) is a codimension-two surface in the bulk. The island is the gravitational part of the entanglement wedge of R (Theorem 56.31): the bulk region whose quantum state R encodes, in the sense of entanglement-wedge reconstruction (Theorem 56.33). When I is nonempty, ∂I lies in the interior of the gravitating region, disconnected from R.

Theorem 57.18 (The island formula)**.** *In the semiclassical regime, the fine-grained entropy of the bath region R is*

$$\boxed{S(R) \;=\; \min_{I}\; \operatorname*{ext}_{\partial I}\left[\,\frac{\mathrm{Area}(\partial I)}{4G_N} \;+\; S_{\rm bulk}(R\cup I)\,\right]}\,, \tag{57.84}$$

where the extremization is over the location of the quantum extremal surface ∂I, $S_{\rm bulk}(R\cup I)$ is the von Neumann entropy of the bulk quantum fields on $R\cup I$ computed on the fixed semiclassical background, and the outer minimum is taken over island configurations I (including the empty island).

Derivation from the QES prescription. The Engelhardt–Wall theorem Theorem 56.27 states that the entropy of a region equals the generalized entropy Eq. (56.151) extremized over the location of the quantum extremal surface γ that anchors the region's entanglement wedge, with the global least-generalized-entropy surface chosen when several compete:

$$S(R) = \operatorname*{ext}_{\gamma} S_{\mathrm{gen}}[\gamma] = \operatorname*{ext}_{\gamma} \left[\frac{\langle \widehat{\mathrm{Area}}(\gamma) \rangle}{4G_N} + S_{\mathrm{bulk}}(\mathcal{R}_\gamma) \right], \tag{57.85}$$

where $\mathcal{R}_\gamma$ is the homology region bounded by γ and the region itself. Take the "region" to be the bath subsystem R. The homology region $\mathcal{R}_\gamma$ now consists of R together with whatever bulk it caps off; because R lives in the non-gravitating bath while the area term is supported only in the gravitating part, the surface γ is forced into the gravitating region, and the bulk it encloses there is precisely the island I of Theorem 57.17, with $\gamma = \partial I$. Thus $\mathcal{R}_\gamma = R \cup I$ and $\langle \widehat{\mathrm{Area}}(\gamma) \rangle = \mathrm{Area}(\partial I)$, and Eq. (57.85) becomes

$$S(R) = \operatorname*{ext}_{\partial I} \left[\frac{\mathrm{Area}(\partial I)}{4G_N} + S_{\mathrm{bulk}}(R \cup I) \right]. \tag{57.86}$$

The empty surface $\gamma = \varnothing$, $I = \varnothing$, is always an admissible competitor: it trivially extremizes (there is nothing to vary) and contributes $S_{\mathrm{bulk}}(R)$ alone. The "least generalized entropy" clause of Theorem 56.27 that selects among competing extrema is exactly the outer minimum in Eq. (57.84), taken over the empty and nonempty island configurations. This is Eq. (57.84). □

The formula has earned no new physics; it is the QES prescription read in the bath. What is genuinely new is the geometry it allows. For an interval in pure AdS the extremal surface is connected to the region and there are no islands. For the radiation of a black hole the candidate extremal surfaces fall into two qualitatively distinct families, and the competition between them — the outer min — is the whole story.

Physical Insight

The island is the precise sense in which "the inside of the black hole is encoded in its radiation." Entanglement-wedge reconstruction (Theorem 56.33) says that any operator in the entanglement wedge of R can be rebuilt from operators acting on R alone. When the island is nonempty, the deep interior of the black hole is part of that wedge — so the bulk fields there, and the information they carry, are recoverable, in principle, from the radiation. The island formula does not merely fix a number; it tells us *where the interior went.* Before the Page time the wedge of R is just R; after it, the wedge of R swallows the interior.

57.4.3 The two competing saddles

Both branches of Eq. (57.83) are now visible as the two terms over which Eq. (57.84) minimizes.

The no-island saddle. Take $I = \varnothing$. There is no area term and nothing to extremize, and

$$S_{\text{no-isl}}(R) = S_{\text{bulk}}(R). \tag{57.87}$$

This is the entropy of the Hawking quanta in the bath, computed on the fixed background — exactly Hawking's calculation. Each emitted quantum is entangled with a partner left behind near the horizon; tracing over the partners, the state on R is very nearly thermal, and its entropy *grows monotonically* as the bath fills with more and more such quanta. This is the rising, "thermal" branch $S_{\text{rad}}^{\text{th}}$. Left to itself it grows without bound (for an eternal black hole) or past S_{BH} (for an evaporating one) — the very behavior that, by the entropy bound of Eq. (55.202) and the area law of Theorem 54.10, signals the information paradox of Section 57.2.

The island saddle. Now allow a nonempty island whose quantum extremal surface ∂I sits just inside the horizon. The generalized entropy acquires the area term, and

$$S_{\text{isl}}(R) = \operatorname*{ext}_{\partial I}\left[\frac{\text{Area}(\partial I)}{4G_N} + S_{\text{bulk}}(R \cup I)\right] \approx \frac{\text{Area}(\partial I)}{4G_N} + (\text{slowly varying}). \tag{57.88}$$

The mechanism that makes the second branch nearly constant is the heart of the matter and we will see it explicitly below: when the island is included, the bulk fields on $R \cup I$ include *both* members of each Hawking pair — the escaped quantum in R and its partner in I — so the strong entanglement between partner and quantum, which drove $S_{\text{bulk}}(R)$ up in the no-island branch, now sits *inside* the region $R \cup I$ and no longer contributes to its entropy. What remains is dominated by the area term, the Bekenstein–Hawking entropy of the surface just inside the horizon. Thus $S_{\text{isl}}(R) \approx S_{\text{BH}}$, the descending (or, for an eternal hole, flat) branch.

The island formula selects the smaller. Early, when few quanta have escaped, $S_{\text{bulk}}(R)$ is small and the no-island saddle wins: $S(R)$ rises. Late, when $S_{\text{bulk}}(R)$ has grown past S_{BH}, the island saddle wins: $S(R)$ is pinned to S_{BH}. The crossover is a sharp first-order transition in the dominant saddle — the gravitational image of the entanglement phase transition seen for two intervals in Section 56.2.4. It happens precisely when the two branches are equal, $S_{\text{bulk}}(R) = S_{\text{BH}}$, which is the definition of the Page time. The minimum of the two branches traces out Eq. (57.83); the gravitational Page curve is the lower envelope of the two QES saddles.

57.4.4 The computation in JT gravity coupled to a bath

We now make every claim of the previous subsection quantitative in a model where the geometry, the dilaton, and the bulk matter entropy are all known in closed form: two-dimensional Jackiw–Teitelboim (JT) dilaton gravity, the eternal AdS_2 black hole, coupled to a flat bath carrying a two-dimensional conformal field theory of central charge c. This is the model of Almheiri, Mahajan, Maldacena, and Zhao. It is honestly two-dimensional and s-wave reduced — the higher-dimensional statements are read off by dimensional reduction on the transverse sphere, with the dilaton playing the role of the transverse area — but within it the QES extremization is exact, and it exhibits every feature of Section 57.4.3 with explicit algebra.

Worked Example 57.19 (The two QES saddles of the eternal black hole and the Page time)**.** *The background.* In two dimensions "Area" is replaced by the value of the dilaton field ϕ, so the area term of the generalized entropy is $\phi/4G_N$ and the horizon entropy is $S_{\mathrm{BH}} = \phi_h/4G_N$ with ϕ_h the dilaton at the horizon (this is the JT incarnation of Theorem 54.10). The eternal AdS_2 black hole at inverse temperature β is global AdS_2; in Kruskal null coordinates (U, V) its metric and dilaton are

$$ds^2 = -\frac{4\,dU\,dV}{(1+UV)^2}, \qquad \phi = \phi_0 + \frac{2\pi\phi_r}{\beta}\,\frac{1-UV}{1+UV}, \tag{57.89}$$

with ϕ_0 the (topological) extremal dilaton, ϕ_r the renormalized boundary value, and the surface gravity $2\pi/\beta$ the Hawking temperature fixed by smoothness exactly as in Eq. (54.96). The right exterior is $U < 0$, $V > 0$; the bifurcation surface (the horizon) is $U = V = 0$, where the dilaton takes its minimum exterior value

$$\phi_h = \phi_0 + \frac{2\pi\phi_r}{\beta}, \qquad S_{\mathrm{BH}} = \frac{\phi_h}{4G_N}. \tag{57.90}$$

The Kruskal coordinates are related to the physical (Schwarzschild-like) boundary time t and tortoise coordinate σ by $U = -e^{-\frac{2\pi}{\beta}(t-\sigma)}$, $V = e^{\frac{2\pi}{\beta}(t+\sigma)}$ on the right, and by the mirror map on the left.

The matter entropy. The Hawking radiation is the bulk CFT, taken in the Hartle–Hawking state, which is the conformal vacuum of the Kruskal coordinates (U, V). The entanglement entropy of an interval with endpoints P_1, P_2 in the vacuum of a two-dimensional CFT, on a background written as $ds^2 = -e^{2\hat\rho}\,dU\,dV$, follows from the twist-operator two-point function of Eq. (55.55) with the twist dimension $\Delta_n = 2h_n = \frac{c}{12}(n - \frac{1}{n})$ of Theorem 55.31. A twist field is a primary of total dimension Δ_n, so under the Weyl rescaling to the curved frame it picks up a factor $e^{-\Delta_n\hat\rho}$ at each insertion; the vacuum two-point function in the flat U, V frame is $\langle\sigma_n\tilde\sigma_n\rangle_{\mathrm{flat}} \propto [(U_2-U_1)(V_2-V_1)]^{-\Delta_n}$, so

$$\mathrm{Tr}\,\rho^n = \langle\sigma_n(P_1)\tilde\sigma_n(P_2)\rangle = \frac{e^{-\Delta_n[\hat\rho(P_1)+\hat\rho(P_2)]}}{\big[(U_2-U_1)(V_2-V_1)\big]^{\Delta_n}}\,(\epsilon_{\mathrm{uv}})^{2\Delta_n}. \tag{57.91}$$

Taking $S = \lim_{n\to 1} \frac{1}{1-n} \ln \mathrm{Tr}\, \rho^n$ and using $\Delta_n \to \frac{c}{6}(n-1)$ as $n \to 1$ gives the master formula

$$S_{\text{matter}}(P_1, P_2) = \frac{c}{6} \ln\Big[(U_2 - U_1)(V_2 - V_1)\Big] + \frac{c}{6}\big[\hat{\rho}(P_1) + \hat{\rho}(P_2)\big] + \text{const.} \tag{57.92}$$

This is the two-dimensional vacuum logarithm of Eq. (55.119), dressed by the conformal factors $\hat{\rho}$ that convert coordinate to proper distance. From Eq. (57.89), in the gravitating region $e^{2\hat{\rho}} = 4/(1+UV)^2$, so $\hat{\rho} = \ln \frac{2}{1+UV}$; in the bath $\hat{\rho}$ is constant.

The no-island branch. Let R be the union of two bath intervals, one in each exterior, each running from a bath point b at boundary time t out to infinity; by the thermofield-double symmetry both endpoints advance forward in time. With $I = \varnothing$, the entropy is $S_{\text{bulk}}(R)$, which by Eq. (57.92) is the matter entropy of the interval connecting the two boundary endpoints through the interior, P_R on the right and its mirror P_L on the left. Their Kruskal coordinates are $V_R = e^{\frac{2\pi}{\beta}(t+\sigma_b)}$, $U_R = -e^{-\frac{2\pi}{\beta}(t-\sigma_b)}$, and $U_L = e^{\frac{2\pi}{\beta}(t-\sigma_b)}$, $V_L = -e^{-\frac{2\pi}{\beta}(t+\sigma_b)}$. At late time the differences are dominated by the growing exponentials,

$$(U_R - U_L)(V_R - V_L) \xrightarrow{t\to\infty} \big(-e^{\frac{2\pi}{\beta}(t-\sigma_b)}\big)\big(e^{\frac{2\pi}{\beta}(t+\sigma_b)}\big) = -e^{\frac{4\pi}{\beta}t}, \tag{57.93}$$

and the bath conformal factors are time-independent, so

$$S_{\text{no-isl}}(t) = \frac{c}{6} \ln|U_R - U_L||V_R - V_L| + \text{const} = \frac{c}{6} \cdot \frac{4\pi}{\beta}\, t + \text{const} = \frac{2\pi c}{3\beta}\, t + \text{const.} \tag{57.94}$$

The no-island entropy grows *linearly* in boundary time, with slope $2\pi c/3\beta$ — this is the rate at which a thermal 2D CFT at temperature $1/\beta$ pours fine-grained entropy into the bath from both exteriors. It is Hawking's ever-rising thermal entropy, and it never stops.

The island branch. Now place a quantum extremal surface in each exterior at Kruskal point (U_a, V_a), bounding a single island that straddles the bifurcation. By symmetry it suffices to extremize the right-half generalized entropy and double it:

$$S^{\text{R}}_{\text{gen}}(U_a, V_a) = \frac{\phi(U_a, V_a)}{4G_N} + \frac{c}{6}\Big[\ln\big((U_b - U_a)(V_b - V_a)\big) + \hat{\rho}_a + \hat{\rho}_b\Big], \tag{57.95}$$

with (U_b, V_b) the right bath point and $\hat{\rho}_a = \ln \frac{2}{1+U_aV_a}$. We extremize. Using $\partial_{U_a}\phi = \frac{2\pi\phi_r}{\beta}\frac{-2V_a}{(1+U_aV_a)^2}$ and $\partial_{U_a}\hat{\rho}_a = \frac{-V_a}{1+U_aV_a}$ (and likewise for V_a), the two extremality conditions $\partial_{U_a} S^{\text{R}}_{\text{gen}} = \partial_{V_a} S^{\text{R}}_{\text{gen}} = 0$ read

$$\frac{1}{4G_N}\frac{2\pi\phi_r}{\beta}\frac{-2V_a}{(1+U_aV_a)^2} + \frac{c}{6}\Big[\frac{-1}{U_b - U_a} - \frac{V_a}{1+U_aV_a}\Big] = 0, \tag{57.96}$$

$$\frac{1}{4G_N}\frac{2\pi\phi_r}{\beta}\frac{-2U_a}{(1+U_aV_a)^2} + \frac{c}{6}\Big[\frac{-1}{V_b - V_a} - \frac{U_a}{1+U_aV_a}\Big] = 0. \tag{57.97}$$

In the semiclassical regime the dilaton dominates the matter, $\phi_r/G_N \gg c$, so the quantum extremal surface lies extremely close to the horizon: U_a, V_a are small and $1+U_aV_a \approx 1$. Dropping the subleading $\frac{c}{6}V_a$ and $\frac{c}{6}U_a$ pieces against the large area gradient, and writing $\kappa \equiv \frac{\pi\phi_r}{G_N\beta}$, Eqs. (57.96) and (57.97) collapse to

$$\kappa\, V_a = \frac{c}{6}\,\frac{1}{|U_b - U_a|}, \qquad \kappa\, U_a = \frac{c}{6}\,\frac{1}{|V_b - V_a|}. \tag{57.98}$$

With the bath point at $U_b = -e^{-\frac{2\pi}{\beta}(t-\sigma_b)}$, $V_b = e^{\frac{2\pi}{\beta}(t+\sigma_b)}$, these solve to

$$V_a = \frac{c}{6\kappa}\, e^{\frac{2\pi}{\beta}(t-\sigma_b)}, \qquad U_a = -\frac{c}{6\kappa}\, e^{-\frac{2\pi}{\beta}(t+\sigma_b)}, \tag{57.99}$$

to leading order. The QES position drifts with time — V_a grows, $|U_a|$ shrinks, the surface sliding along the horizon — but the boost-invariant product is frozen,

$$U_aV_a = -\Big(\frac{c}{6\kappa}\Big)^2 e^{-\frac{4\pi}{\beta}\sigma_b} = -\Big(\frac{cG_N\beta}{6\pi\phi_r}\Big)^2 e^{-\frac{4\pi}{\beta}\sigma_b}, \tag{57.100}$$

independent of time, and of order G_N^2, so the surface sits a coordinate distance $\sim G_N$ inside the horizon. This is the crucial cancellation: the entanglement wedge boundary of the radiation hovers just behind the horizon, at a fixed proper location, no matter how late the time.

Evaluating the island branch. Insert Eqs. (57.99) and (57.100) into Eq. (57.95). The dilaton at the QES is

$$\phi(U_a, V_a) = \phi_0 + \frac{2\pi\phi_r}{\beta}\,\frac{1-U_aV_a}{1+U_aV_a} = \phi_h - \frac{4\pi\phi_r}{\beta}\,U_aV_a + O\big((U_aV_a)^2\big), \tag{57.101}$$

so $\phi(U_a,V_a)/4G_N = S_{\rm BH} + O(G_Nc^2/\phi_r)$ differs from the Bekenstein–Hawking value only by a time-independent, G_N-suppressed amount. The matter log term involves $(U_b-U_a)(V_b-V_a) \approx U_bV_b = -e^{\frac{4\pi}{\beta}\sigma_b}$, which is also time-independent because the explicit t-dependence of U_b and V_b cancels in the product. Hence *every* piece of $S^{\rm R}_{\rm gen}$ is time-independent, and doubling,

$$S_{\rm isl} = 2\,S^{\rm R}_{\rm gen}\Big|_{\rm QES} = 2\,S_{\rm BH} + S_1, \tag{57.102}$$

with S_1 a constant (the $O(c)$ matter remainder). The island branch is flat at twice the Bekenstein–Hawking entropy — twice, because the two-sided eternal black hole has two horizons' worth of coarse entropy.

The crossover. The island formula Eq. (57.84) takes the minimum of Eq. (57.94) and Eq. (57.102):

$$S(R) = \min\Big(\tfrac{2\pi c}{3\beta}\,t,\; 2S_{\rm BH}\Big) + \text{const.} \tag{57.103}$$

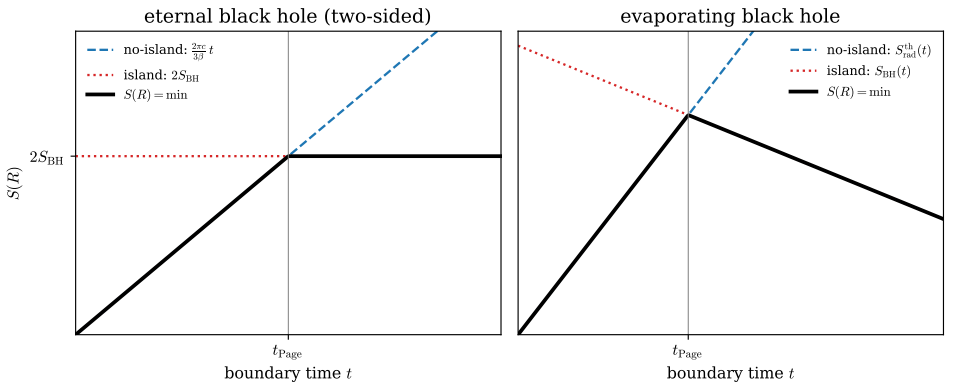


Figure 57.6: The two competing quantum extremal surface saddles and the Page curve as their minimum. *Left*: the eternal two-sided black hole of Theorem 57.19. The no-island branch Eq. (57.94) (dashed) rises linearly with slope $2\pi c/3\beta$; the island branch Eq. (57.102) (dotted) is flat at $2S_{\rm BH}$. The radiation entropy $S(R)$ (thick) is their minimum, with the corner at $t_{\rm Page} = 3\beta S_{\rm BH}/\pi c$ of Eq. (57.104). *Right*: the evaporating black hole, where $S_{\rm BH}(t)$ declines as the hole shrinks, so the island branch descends and the minimum reproduces the rising-then-falling Page curve of Eq. (57.83).

The rising line meets the flat plateau at the Page time, where $\frac{2\pi c}{3\beta} t_{\rm Page} = 2S_{\rm BH}$, i.e.

$$\boxed{t_{\rm Page} = \frac{3\beta\, S_{\rm BH}}{\pi c}\,.} \tag{57.104}$$

This is the gravitational Page time: linear growth until the radiation entropy reaches the coarse-grained black hole entropy, then saturation. The no-island branch reproduces $S^{\rm th}_{\rm rad}$ of Eq. (57.83); the island branch reproduces $S_{\rm BH}$; their minimum is the Page curve, computed entirely within semiclassical gravity.

Figure 57.6 displays the two saddles and their envelope, for the eternal black hole just computed and for the physically richer evaporating case discussed next. Figure 57.7 shows where the island sits.

57.4.5 From eternal to evaporating, and the match to the Page curve

The eternal computation produced a "tent": linear rise to a flat plateau (Fig. 57.6, left). The eternal black hole does not shrink — it is kept in equilibrium with its bath — so its horizon entropy $2S_{\rm BH}$ is constant and the island branch is flat. The physically decisive case is the *evaporating* black hole, formed from collapse and left to radiate into the bath without being fed. The same two saddles appear, with one change: the horizon shrinks as energy leaves, so $S_{\rm BH}(t) = \text{Area}(t)/4G_N$

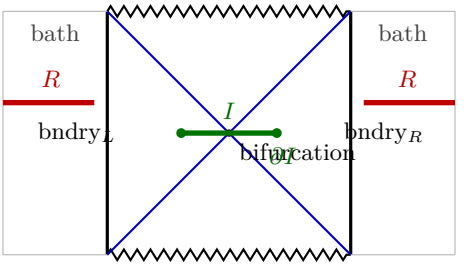


Figure 57.7: The eternal two-sided AdS_2 black hole coupled to baths, after the Page time. The two AdS boundaries (thick verticals) carry flat non-gravitating baths; the radiation region R (red) collects the Hawking quanta in the baths. The blue diagonals are the horizons, crossing at the bifurcation surface (the JT "horizon"). The island I (green) is the interval straddling the bifurcation, bounded by the quantum extremal surfaces ∂I sitting a proper distance $\sim G_N$ inside the horizon, frozen there by Eq. (57.100). The entanglement wedge of R now includes I: the black-hole interior is encoded in the radiation.

decreases with time. The no-island branch still rises (more Hawking quanta accumulate), but the island branch now *descends*, tracking the shrinking horizon, $S_{\rm isl}(t) \approx S_{\rm BH}(t)$. The minimum of a rising curve and a falling curve is the rising-then-falling Page curve of Eq. (57.83) — Fig. 57.6, right. The crossover is again the Page time, now occurring when the rising thermal entropy meets the falling Bekenstein–Hawking entropy, $S^{\rm th}_{\rm rad}(t_{\rm Page}) = S_{\rm BH}(t_{\rm Page})$, which is exactly the defining condition of Section 57.3. The closed-form extremization that makes the cancellation Eq. (57.100) transparent is cleanest in the eternal two-sided model; the evaporating case requires the time-dependent (covariant) QES of Theorem 56.27 on a Vaidya-like collapse geometry, but the structure — two saddles, a frozen near-horizon QES on the island branch, a crossover at the Page time — is identical.

The dictionary between the two computations is now exact, term for term:

$$\underbrace{S_{\text{no-isl}}(R) = S_{\rm bulk}(R)}_{\text{Hawking's thermal branch}} \longleftrightarrow S^{\rm th}_{\rm rad}(t), \qquad \underbrace{S_{\rm isl}(R) \approx \frac{\text{Area}(\partial I)}{4G_N}}_{\text{island branch}} \longleftrightarrow S_{\rm BH}(t), \tag{57.105}$$

and the island formula's outer minimum is the min of Eq. (57.83). The Page curve, which Section 57.3 derived as an abstract demand of unitarity from random-state statistics, is here *computed* from the geometry, the dilaton (area), and the bulk entropy, using only the QES prescription of Theorem 56.27. Nowhere did we assume the evolution was unitary; we assumed only that the QES rule computes fine-grained entropy. The two agree. That agreement is the resolution of the entropy half of the information paradox.

57.4.6 Reading the result: the radiation's wedge swallows the interior

The island formula does more than fit the curve; through entanglement-wedge reconstruction it tells us *what happened to the information*. Recall from Theorem 56.33 that the entanglement wedge of a region is the bulk it can reconstruct: operators in the wedge are recoverable from operators on the region, in the quantum-error-correcting sense of Section 56.7. Apply this to R.

Before the Page time the no-island saddle dominates, the island is empty, and the entanglement wedge of R is just R itself. The radiation knows nothing about the interior; the black hole's inside is reconstructible only from the black hole's own degrees of freedom. This is the intuitive, semiclassical picture, and it is correct — early.

After the Page time the island saddle dominates. The entanglement wedge of R now *includes the island* I, the region behind the horizon bounded by the quantum extremal surface (Fig. 57.7). By reconstruction, the bulk fields in I — the interior of the black hole — are now encoded in the radiation R and recoverable from it. The radiation's wedge has swallowed the interior. This is the precise sense in which "the information comes out": not that any single late quantum is non-thermal in a way Hawking missed, but that the global encoding has shifted, so that the interior's quantum state, including the infalling matter that formed the hole and the partners of the early Hawking quanta, lives in the entanglement wedge of the radiation. The transition at the Page time is the transition between these two encodings.

Caution

What has been established is the *entropy* curve, and within a model that is honestly two-dimensional and s-wave reduced. The island formula reproduces $S(R)$, and entanglement-wedge reconstruction tells us in principle that the interior is encoded in R after the Page time. Neither result hands us an explicit, computationally tractable map that decodes a given bit of infallen information from the radiation; the reconstruction is exponentially complex, and whether and how it can be operationally realized is open (Section 57.6.5). The four-dimensional, beyond-s-wave statement of the island formula is believed to hold but is not proved with the same closed-form control. We report the entropy as settled and date the rest honestly: this is the understanding as of the mid-2020s.

We have reproduced the Page curve from gravity. But we did so by *asserting* the island formula Eq. (57.84) — importing it as a corollary of the QES prescription — and the QES prescription itself, while derived in Section 56.5.3 for boundary regions of a holographic CFT, was not obviously licensed to grow a disconnected island behind a horizon for a bath region. Two questions are left

exposed. Why is the island formula correct as a statement about fine-grained radiation entropy? And where, in an honest gravitational calculation of $S_{\rm rad}$, does the second saddle — the island branch, the very term that bends the curve down — actually come from? The next section answers both at once: computing $S_{\rm rad}$ by the replica trick in the Euclidean gravitational path integral reveals new connected geometries, the replica wormholes, whose contribution *is* the island term, deriving Eq. (57.84) rather than positing it.

57.5 Replica wormholes and the gravitational path integral

The previous section handed us a working answer and an unpaid debt. The island formula

$$S_{\rm rad} = \min \operatorname*{ext}_{I} \Big[\frac{\mathrm{Area}(\partial I)}{4G_N} + S_{\rm bulk}(R \cup I) \Big] \tag{57.106}$$

reproduced the Page curve: before the Page time the empty-island branch gives the growing $S_{\rm bulk}(R)$, after it a new quantum extremal surface deep in the interior switches on and freezes the entropy at the horizon's $\mathrm{Area}/4G_N$ (Section 57.4.2 and Eq. (57.84)). But Eq. (57.106) was *asserted*: we took the quantum-extremal-surface prescription of Chapter 56, allowed the entanglement wedge of the radiation to contain a disconnected component (the island), and checked that the result is the unitary Page curve. Two things were left unexplained. Why should the radiation's entanglement wedge be allowed to reach across an empty region of bath and grab a slab of the black hole interior? And where did the "second saddle" — the interior QES that takes over after the Page time — come from, given that the semiclassical state on the fixed Hawking geometry knows nothing of it? This section pays the debt. We compute $S_{\rm rad}$ directly from the Euclidean gravitational path integral via the replica trick of Chapter 55, and we find that the path integral contains, in addition to the obvious geometry, a class of *connected* saddles — replica wormholes — that sew the replicas together through the bulk. The wormhole saddles supply, term for term, the island contribution to Eq. (57.106). The island formula is then not a prescription bolted onto semiclassical gravity but a *theorem* about its path integral, in exactly the controlled settings where that path integral can be evaluated.

We continue to work with the Euclidean gravitational path integral, the same object that gave us the Hawking temperature and the area entropy by demanding smoothness of the Euclidean cap (Section 54.4.2 and Eqs. (54.96) and (54.120)), and we use throughout the multi-saddle technology built for Yang–Mills instantons in Chapter 42: a Euclidean amplitude is a *sum* over all bulk fillings with the prescribed boundary, each weighted by e^{-I}, and distinct topological sectors must all be included (Sections 42.1.2 and 42.1.6). The Wick rotation $t \to -i\tau$ is the one fixed there; Lorentzian signature is mostly-minus

and units are natural, as everywhere in this book. The controlled arena is two-dimensional Jackiw–Teitelboim (JT) gravity coupled to a non-gravitating bath, with its Sachdev–Ye–Kitaev and matrix-ensemble cousins; the four-dimensional statement is the same logic applied to the semiclassical saddle, and we are honest about that gap at the end.

57.5.1 The radiation entropy as a gravitational replica partition function

Recall the replica trick (Theorem 55.17 and Eq. (55.38)): for any density matrix ρ with $\operatorname{Tr}\rho = 1$, the von Neumann entropy is recovered from the integer Rényi moments $\operatorname{Tr}\rho^n$ by

$$S = -\operatorname{Tr}(\rho \ln \rho) = -\frac{d}{dn}\operatorname{Tr}\rho^n\Big|_{n=1}, \tag{57.107}$$

the derivative taken after analytically continuing the integer sequence $\{\operatorname{Tr}\rho^n\}_{n\geq 1}$ in n. We apply this to the reduced density matrix ρ_R of the Hawking radiation. The subtlety that makes the computation gravitational rather than a routine QFT exercise is the same one that motivated islands: although R is a region of the non-gravitating bath, the state on R was produced by a gravitating system, and the only first-principles definition of ρ_R runs through the gravitational path integral that prepares the global state. Concretely, a matrix element $\langle a|\rho_R|b\rangle$ is computed by a Euclidean path integral over the bulk gravity region with bath boundary conditions a on the ket and b on the bra, glued along the complement of R; tracing sets $a = b$ on $\bar{R}$ and integrates.

The Rényi moment $\operatorname{Tr}\rho_R^n$ is then the partition function on an n-fold replicated geometry. Take n copies of the bra–ket path integral; in each copy the bath is cut open along R, exposing an upper and a lower edge; cyclically glue the upper edge of copy a to the lower edge of copy $a+1$ (indices mod n). On the bath side this is the ordinary n-sheeted branched cover of Theorem 55.17, branched over ∂R. On the gravity side, each copy contributes its gravity region, and the central question — the one the bath cannot answer — is *how the bulk fills in* the replicated boundary. Writing Z_n for the gravitational partition function with this n-fold-replicated bath boundary, and Z_1 for the single un-replicated one, the normalized moment is

$$\operatorname{Tr}\rho_R^n = \frac{Z_n}{(Z_1)^n}, \tag{57.108}$$

the denominator enforcing $\operatorname{Tr}\rho_R = 1$. Each Z_n is, by the holographic dictionary (Eq. (53.162)), evaluated by summing e^{-I} over all bulk solutions whose conformal boundary is the replicated manifold. Combining Eqs. (57.107) and (57.108),

$$S_{\text{rad}} = -\frac{d}{dn}\Big[\ln Z_n - n \ln Z_1\Big]_{n=1} = \Big(\partial_n I[\mathcal{B}_n] - I[\mathcal{B}_1]\Big)_{n=1}, \tag{57.109}$$

where in the last step we used $\ln Z_n = -I[\mathcal{B}_n]$ for the dominant saddle $\mathcal{B}_n$. Everything now depends on *which* bulk $\mathcal{B}_n$ dominates, and the discovery is that the answer changes with time.

57.5.2 Two classes of saddle: the Hawking geometry and the replica wormhole

Equation (57.108) is a sum over bulk topologies, and at least two qualitatively different topologies always solve the equations of motion with the replicated boundary.

Definition 57.20 (Disconnected and connected replica saddles)**.** The *disconnected (Hawking) saddle* $\mathcal{B}_n^{\text{disc}}$ consists of n separate copies of the original (n=1) bulk geometry, each capping off its own replica of the bath. Its bulk does not communicate between replicas. A *connected replica saddle*, or *replica wormhole*, is a single connected bulk geometry $\mathcal{B}_n^{\text{conn}}$ whose boundary is still the n-fold-replicated bath but whose interior joins two or more replicas through a bulk bridge. The maximally connected, replica-symmetric member sews all n copies cyclically and carries a $\mathbb{Z}_n$ symmetry whose codimension-two fixed-point locus we call χ_n.

The disconnected saddle is the one Hawking would have written down. With the n replicas mutually inert, $Z_n^{\text{disc}} = (Z_1)^n \times$ (bath replica factor), and the bath replica factor is precisely the n-sheeted-cover partition function of the bulk *matter* fields on the fixed Hawking background. Its replica derivative is, by the very definition Eq. (57.107) run for the bulk fields, the ordinary bulk von Neumann entropy of the radiation,

$$S_{\text{rad}}^{\text{disc}} = S_{\text{bulk}}(R), \tag{57.110}$$

the monotonically growing, manifestly information-losing entropy of the Hawking calculation. This is the empty-island branch of Eq. (57.106): no area term, no island, just the bulk entropy of the collected quanta. It is correct at early times and disastrous at late times, where it exceeds the black hole's coarse-grained entropy and never turns over.

The connected saddle is the new ingredient. Here the LM localization of Section 56.4 applies verbatim, only to a different geometry. The replica-symmetric wormhole $\mathcal{B}_n^{\text{conn}}$ has a $\mathbb{Z}_n$ quotient $\hat{\mathcal{B}}_n = \mathcal{B}_n^{\text{conn}}/\mathbb{Z}_n$ with a conical defect of opening angle $2\pi/n$ on the fixed-point surface χ_n. Because the equations of motion hold everywhere except on χ_n, the n-derivative in Eq. (57.109) localizes onto that surface, and the localized gravitational piece is exactly the proper area of χ_n over $4G_N$, while the matter piece is the bulk replica entropy across χ_n. We do not reprove this localization — it is the Lewkowycz–Maldacena computation imported from Section 56.4 and its one-loop completion Theorem 56.20 — but

apply it:

$$S_{\rm rad}^{\rm conn} = \frac{\text{Area}(\chi_1)}{4G_N} + S_{\rm bulk}(R \cup I), \tag{57.111}$$

where χ_1 is the $n \to 1$ limit of the fixed-point surface and I is the bulk region it bounds on the far side of R — the island. *This is the island contribution.* The connected replica wormhole is the gravitational object whose presence Section 57.4 could only postulate: the bulk bridge between replicas is what places an area term and an island into the radiation entropy.

Physical Insight

The two saddles are the two terms of the island formula made into *geometries*. The disconnected topology computes "no island, just bulk entropy"; the connected topology computes "area of a QES plus the bulk entropy of radiation-plus-island." The Euclidean path integral does not choose between them by fiat — it sums both and the smaller action wins. The island prescription's min is the saddle-point competition of Chapter 42, transcribed into entropy.

Which dominates is a horse race between the two actions, and the winner changes at the Page time. Both saddles exist at all times; the disconnected one has the smaller action (larger Z_n, smaller Rényi entropy) while the radiation is young and $S_{\rm bulk}(R)$ is below the horizon area, and the connected one takes over once $S_{\rm bulk}(R)$ would exceed $\text{Area}/4G_N$. The continued $n \to 1$ entropy is therefore the smaller of Eqs. (57.110) and (57.111),

$$S_{\rm rad} = \min\Big\{ S_{\rm bulk}(R)\,,\ \frac{\text{Area}(\chi_1)}{4G_N} + S_{\rm bulk}(R \cup I) \Big\}, \tag{57.112}$$

which is Eq. (57.106) with the two competing extrema named by their topologies. The replica wormhole has *derived* the second branch of the Page curve.

57.5.3 The controlled computation: $\mathrm{Tr}\,\rho_R^2$ in JT gravity plus a bath

The statements above are exact in JT gravity, where the path integral is genuinely calculable. We carry out the simplest nontrivial case — the second Rényi moment — in full, then read off the general n result and continue to $n = 1$.

The model. Use JT gravity, whose Euclidean action is

$$I_{\rm JT}[g, \Phi] = -S_0\,\chi(\mathcal{M}) - \frac{1}{2}\int_{\mathcal{M}} \sqrt{g}\,\Phi(R+2) - \int_{\partial\mathcal{M}} \sqrt{h}\,\Phi_b(K-1) + I_{\rm mat}, \tag{57.113}$$

with $\chi(\mathcal{M}) = 2 - 2g - b$ the Euler characteristic of a surface of genus g with b boundaries. The dilaton Φ is a Lagrange multiplier fixing $R = -2$ (rigid

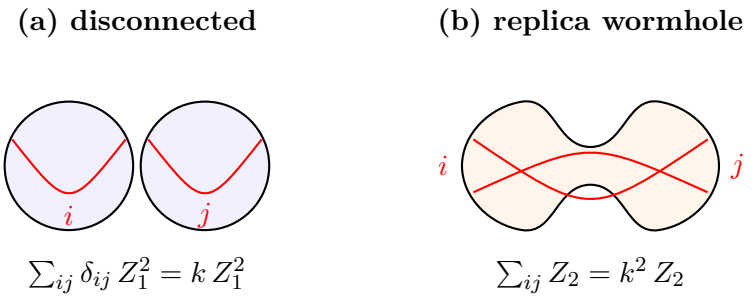


Figure 57.8: The two bulk topologies contributing to $\mathrm{Tr}\,\rho_R^2$ in the JT-plus-brane model. In (a) each braket overlap $\langle\psi_i|\psi_j\rangle$ is its own disk; brane-flavor conservation forces δ_{ij}, leaving one free index and kZ_1^2. In (b) a single connected geometry — the replica wormhole — joins both brakets; the flavor indices run free, giving $k^2 Z_2$. Their ratio sets the Page transition.

hyperbolic geometry), Φ_b its boundary value, and S_0 the extremal black-hole entropy; the only role of Eq. (57.113) we need is its topological weighting $e^{-I} \supset e^{S_0\chi}$, so that a disk ($\chi = 1$) is weighted e^{S_0} and each extra boundary or handle costs a factor e^{-S_0}. To make a sharp microstate-counting problem, follow Penington–Shenker–Stanford–Yang and entangle the black hole with a reference R (the stand-in for the radiation) through k flavors of end-of-world (EOW) brane:

$$|\Psi\rangle = \frac{1}{\sqrt{k}} \sum_{i=1}^{k} |\psi_i\rangle \otimes |i\rangle_R, \tag{57.114}$$

where $|\psi_i\rangle$ is a black-hole microstate created by an EOW brane carrying flavor i behind the horizon and $\{|i\rangle_R\}$ are orthonormal reference states. Larger k models a later stage of evaporation: the more flavors, the more the black hole has dumped into R.

The overlaps are gravitational. The reduced density matrix is

$$\rho_R = \mathrm{Tr}_{\mathrm{BH}}\, |\Psi\rangle\langle\Psi| = \frac{1}{k\, Z_1} \sum_{i,j} \langle\psi_j|\psi_i\rangle\, |i\rangle\langle j|_R, \tag{57.115}$$

normalized by Z_1 so that $\mathrm{Tr}\,\rho_R = 1$. The brane overlaps $\langle\psi_j|\psi_i\rangle$ are computed by the gravity path integral with a brane worldline connecting the bra and ket. At leading (disk) order, brane flavor is conserved along the worldline, so $\langle\psi_j|\psi_i\rangle = \delta_{ij}\, Z_1$ with Z_1 the one-boundary disk amplitude. The diagonal $\rho_R \approx \mathbf{1}/k$ is the naive maximally mixed answer; the off-diagonal corrections, and the failure of the naive answer at large k, come from connected geometries.

Worked Example 57.21 ($\mathrm{Tr}\,\rho_R^2$ and the two topologies)**.** We compute the second moment, exhibit its two saddles, and continue in n to obtain the Page curve.

Set up the moment. From Eq. (57.115),

$$\mathrm{Tr}\,\rho_R^2 = \sum_{i,j} (\rho_R)_{ij}(\rho_R)_{ji} = \frac{1}{(k\,Z_1)^2} \sum_{i,j} \langle\psi_j|\psi_i\rangle\,\langle\psi_i|\psi_j\rangle. \tag{57.116}$$

The gravitational object is the product of two brane brakets, and the path integral connects them in two topologically distinct ways (Fig. 57.8).

Disconnected topology. Each braket is its own disk: $\langle\psi_j|\psi_i\rangle \to \delta_{ij} Z_1$ and $\langle\psi_i|\psi_j\rangle \to \delta_{ij} Z_1$, so the summand is $\delta_{ij} Z_1^2$ and

$$\sum_{i,j} \delta_{ij}\, Z_1^2 = k\, Z_1^2, \qquad \text{contributing} \qquad \frac{k\, Z_1^2}{(k\, Z_1)^2} = \frac{1}{k}. \tag{57.117}$$

Connected topology (the replica wormhole). A single connected geometry joins the two brakets; the two brane worldlines now thread one surface, and flavor conservation along them no longer ties i to j. The index sum runs free over both, $\sum_{i,j} 1 = k^2$, multiplying the connected two-boundary amplitude Z_2:

$$\sum_{i,j} Z_2 = k^2\, Z_2, \qquad \text{contributing} \qquad \frac{k^2\, Z_2}{(k\, Z_1)^2} = \frac{Z_2}{Z_1^2}. \tag{57.118}$$

The result. Adding the two saddles,

$$\boxed{\mathrm{Tr}\,\rho_R^2 = \frac{1}{k} + \frac{Z_2}{Z_1^2}} \tag{57.119}$$

The first term is the disconnected (Hawking) answer; the second is the replica wormhole. By the JT weighting and the (approximately flat) microcanonical density of black-hole states, the connected two-boundary amplitude obeys $Z_2/Z_1^2 = e^{-S_{\mathrm{BH}}} \equiv 1/d$, where $S_{\mathrm{BH}} = \mathrm{Area}/4G_N$ is the coarse-grained entropy of Eq. (54.120) and d is the effective dimension of the black hole's Hilbert space. Thus

$$\mathrm{Tr}\,\rho_R^2 = \frac{1}{k} + \frac{1}{d}, \qquad S_2 = -\ln \mathrm{Tr}\,\rho_R^2 = -\ln\!\Big(\frac{1}{k} + \frac{1}{d}\Big) \approx \min(\ln k,\ S_{\mathrm{BH}}). \tag{57.120}$$

For $k \ll d$ the disconnected $1/k$ wins and $S_2 = \ln k$; for $k \gg d$ the wormhole $1/d$ wins and $S_2 = S_{\mathrm{BH}}$. The crossover at $k = d$, i.e. $\ln k = S_{\mathrm{BH}}$, is the Page transition.

General n and the continuation. The same count at n replicas gives $n!$ permutation saddles; the two dominant ones are the identity permutation (fully disconnected, all indices locked together by deltas, k free index, weight kZ_1^n) and the cyclic permutation (fully connected replica wormhole, all n indices free, weight $k^n Z_n$):

$$\operatorname{Tr}\rho_R^n = \frac{k\,Z_1^n}{(kZ_1)^n} + \frac{k^n Z_n}{(kZ_1)^n} + \cdots = k^{1-n} + \frac{Z_n}{Z_1^n} + \cdots = e^{-(n-1)\ln k} + e^{-(n-1)S_{\mathrm{BH}}} + \ldots, \tag{57.121}$$

using $Z_n/Z_1^n = e^{-(n-1)S_{\mathrm{BH}}}$ for the flat microcanonical spectrum. The Rényi entropy $S_n = \frac{1}{1-n}\ln\operatorname{Tr}\rho_R^n$ is dominated by whichever exponent is smaller:

$$S_n = \min(\ln k,\, S_{\mathrm{BH}}) \quad \text{for each } n>1, \qquad \xrightarrow{n\to 1} \qquad S_{\mathrm{rad}} = \min(\ln k,\, S_{\mathrm{BH}}). \tag{57.122}$$

The continuation is immediate because the dominant term is a pure exponential in $(n-1)$ in each regime. The von Neumann entropy of the radiation rises as $\ln k$ (the Hawking branch: more flavors, more entropy) until it reaches the black hole entropy S_{BH}, then is pinned there by the replica wormhole (the island branch). This is the Page curve, and every step was a saddle of the gravitational path integral.

The worked computation makes the abstract claim concrete: the island's area term is literally the contribution of a connected bulk geometry to a Rényi moment, and the min in the island formula is literally the dominance switch between two topologies. The identification $Z_2/Z_1^2 = 1/d$ also exposes why the naive $\rho_R = \mathbf{1}/k$ fails: when $k > d$ the matrix ρ_R cannot have k equal nonzero eigenvalues, because the wormhole-induced off-diagonal terms make its rank at most d. Gravity quietly enforces $S_{\mathrm{rad}} \le S_{\mathrm{BH}} = \ln d$, which is unitarity.

57.5.4 The $n \to 1$ limit is the quantum extremal surface

The PSSY count was algebraic; the geometric content of the connected saddle is what closes the loop with Chapter 56. Return to Eq. (57.109) for the connected replica-symmetric saddle and make the extremization explicit. The smooth wormhole $\mathcal{B}_n^{\mathrm{conn}}$ is obtained, at each integer n, by solving the gravitational equations of motion subject to the replicated boundary and to smoothness; equivalently, its $\mathbb{Z}_n$ quotient $\hat{\mathcal{B}}_n$ is a geometry with a conical defect of opening angle $2\pi/n$ whose tip χ_n sits wherever smoothness of the unquotiented cover demands. The position of χ_n is therefore not a free choice: it is fixed by an extremality condition. Writing the on-shell action of the connected saddle as a functional of the defect location X,

$$\left.\frac{\partial I[\mathcal{B}_n^{\mathrm{conn}}(X)]}{\partial X}\right|_{X=X_n} = 0, \tag{57.123}$$

the saddle sits at the stationary point X_n. Now apply the localization of Section 56.4: differentiating the action in n and continuing to $n = 1$ deposits, on χ_n, the area term, while the matter sector deposits the bulk entropy by the same branched-cover argument that produced FLM (Theorem 56.20). Two contributions to ∂_n exist a priori — the explicit dependence through the opening angle and the implicit dependence through the moving location X_n — but the implicit one vanishes precisely because of Eq. (57.123): the action is stationary in X, so first-order motion of the defect does not change it. Hence

$$S_{\rm rad}^{\rm conn} = \Big(\partial_n I[\mathcal{B}_n^{\rm conn}] - I[\mathcal{B}_1]\Big)_{n=1} = \frac{\text{Area}(\chi_1)}{4G_N} + S_{\rm bulk}(R \cup I) \equiv S_{\rm gen}(\chi_1), \tag{57.124}$$

the generalized entropy of Eq. (56.151) evaluated on the $n \to 1$ fixed-point surface. The remaining question is what equation χ_1 obeys. Take the extremality condition Eq. (57.123) and continue $n \to 1$. As $n \to 1$ the conical opening angle returns to 2π, the defect smooths out, and the location X_n tends to a limiting surface χ_1; the statement that the action is stationary in X at every n continues to the statement that the *generalized entropy* is stationary at χ_1,

$$\left.\frac{\delta S_{\rm gen}(\chi)}{\delta X^\mu(\sigma)}\right|_{\chi=\chi_1} = 0. \tag{57.125}$$

This is exactly the quantum extremal surface condition of Theorem 56.26 and Eq. (56.153). The reason is structural: the n-derivative of the on-shell action is the generalized entropy (that is the content of LM-plus-FLM), so demanding the action be extremal in the defect position *is* demanding the generalized entropy be extremal in the surface position, in the $n \to 1$ limit. The fixed-point surface of the dominant replica wormhole is the QES; the island is the bulk region it bounds; and the value of the entropy is $\text{ext}_\chi\, S_{\rm gen}(\chi)$, with the global min of Eq. (57.112) selecting the dominant topology among competing extrema.

This is the promised derivation. The island formula Eq. (57.106) is not an extra rule appended to semiclassical gravity. It is what the replica trick computes when one is honest that the gravitational path integral sums over topologies: the disconnected topology gives the no-island term, the connected replica wormhole gives the QES-plus-island term, the $n \to 1$ extremization of the wormhole geometry *is* the QES condition, and the min is saddle dominance. The "second saddle" that Section 57.4 could only point to is the replica wormhole, and it was in the path integral all along.

Caution

The replica wormholes that produce the island are connected Euclidean geometries that bridge *different replicas* of the boundary. In the closely related two-boundary problem they bridge two separate asymptotic bound-

aries — two separate copies of the quantum system. Taken literally this is alarming: a connected bulk linking two systems that the boundary description treats as independent threatens to make $\langle Z\,Z\rangle \neq \langle Z\rangle^2$, a failure of factorization. In the controlled JT setting the resolution is that the gravitational path integral is computing not a single quantum system but an *ensemble* average over a family of them (the matrix/SYK ensemble), and a wormhole is the connected correlator of that ensemble. Whether, and in what sense, this ensemble interpretation must be invoked in four-dimensional gravity is unsettled. We return to it in Section 57.6.

57.5.5 Honest scope: where this is a computation and where it is an argument

The chain just completed — replica trick, two topologies, explicit moments, $n \to 1$ to the QES — is a genuine *calculation* only where the gravitational path integral can be evaluated. That place is two-dimensional dilaton gravity. JT gravity Eq. (57.113) is effectively solvable: its disk and higher-topology amplitudes are known in closed form, the EOW-brane construction Eq. (57.114) reduces $\operatorname{Tr}\rho_R^n$ to a counting problem with computable weights, and the sum over topologies is the genus expansion of a double-scaled matrix integral. The same structure recurs in the SYK model (whose low-energy sector is JT) and in pure-topology matrix-ensemble cousins, where the replica wormhole is literally a connected matrix correlator. In all of these the island formula is a theorem about a defined path integral, and the Page curve Eq. (57.122) comes out without assumption beyond the dominance of the leading saddles.

The four-dimensional statement is weaker, and we say so plainly. There the Euclidean gravitational path integral is not known to exist as a convergent sum, the analogue of the JT topology expansion is not under control, and one cannot compute Z_n from first principles. What survives is the *logic*: assume the semiclassical saddle and its replica-symmetric conical cousins dominate; apply the LM–FLM localization (Section 56.4 and Theorem 56.20), which is a semiclassical statement valid wherever a saddle exists; conclude that the connected saddle deposits $\mathrm{Area}/4G_N + S_{\mathrm{bulk}}(R \cup I)$ and that its $n \to 1$ extremization is the QES condition Eq. (57.125). This is exactly the calculation done in Section 57.4 using the QES prescription directly; the replica-wormhole derivation explains *why* that prescription is correct, by exhibiting the saddle that realizes it, in every case where the saddle can be seen. The fine-grained radiation entropy — the Page curve — is thereby understood from semiclassical gravity. What is *not* thereby delivered is the operational decoding of the information, the status of the factorization puzzle the connected geometries raise, or the fate of the smooth horizon; those are the live questions the derivation forces upon us, and the wormholes-connect-replicas picture is exactly the puzzle the closing section

must confront.

57.6 What is settled, what is not

The replica computation of Section 57.5 did something remarkable and something disquieting at once. It *derived* the island rule from the bare gravitational path integral: the new connected saddles—the replica wormholes—contribute to $\mathrm{Tr}\,\rho_{\mathrm{rad}}^{n}$ exactly the term that, after the $n \to 1$ continuation, promotes the naive Hawking answer to the quantum-extremal-surface answer, and the radiation entropy turns over onto the Page curve. That is the disquieting part too. The same wormholes that rescue unitarity at the level of the entropy also make the gravitational path integral compute products of partition functions that do *not* factorize across disconnected boundaries, in apparent contradiction with the single-theory holographic dictionary of Section 53.5. A frontier chapter earns its keep by saying precisely which of its claims are theorems, which are results in controlled models, and which are conjectures—and by dating each, because the boundary moves. This closing section draws that boundary as of the mid-2020s. We first state cleanly what the island/replica program has settled; then we separate the *entropy* (settled) from the *information* (operationally out of reach), making the Hayden–Preskill recovery time quantitative; then we confront the factorization puzzle and the ensemble interpretation it forces; then we revisit the firewall in the light of islands; and we close with an explicit, dated list of what is genuinely open.

Throughout we label each statement by status—[**theorem**] for a mathematically proved result, [**controlled-model result**] for something established within a tractable model (typically two-dimensional Jackiw–Teitelboim gravity or a Renyi/replica computation in a semiclassical limit) but not in full quantum gravity, and [**conjecture**] for a physically motivated proposal whose general validity is not established. This labelling is not pedantry; it is the honest content of the chapter. To call the information paradox "solved" without it would be to overstate by a wide margin what the field actually knows.

57.6.1 The settled core: the Page curve from semiclassical gravity

We collect the achievement of the islands/replica program into a single statement, with each ingredient tagged.

Theorem 57.22 (Status report: the radiation entropy follows the Page curve)**.** *Consider a black hole formed from collapse of a pure state and allowed to evaporate, modelled semiclassically (a fixed-background quantum field with backreaction treated to leading order in G_N), and let $\rho_{\mathrm{rad}}(t)$ be the reduced density matrix of the Hawking radiation collected up to time t. Then:*

1. ***[controlled-model result]*** *In two-dimensional dilaton gravity (JT gravity coupled to a CFT bath, and its cousins), the fine-grained von Neumann entropy* $S_{\rm rad}(t) = -\operatorname{Tr}(\rho_{\rm rad}\log\rho_{\rm rad})$*, computed by the quantum extremal surface prescription of Theorems 56.26 and 56.27, rises during the early evolution, reaches a maximum at the Page time* $t_{\rm Page}$*, and decreases thereafter, returning to zero (up to the residual entropy of the remnant geometry) when evaporation completes. This is the unitary Page curve, not Hawking's monotonically rising curve.*

2. ***[controlled-model result, ∼2019–2020]*** *The QES/island value of* $S_{\rm rad}$ *is reproduced* from first principles *by the Euclidean gravitational path integral: the replica trick* $S_{\rm rad} = -\partial_n\big(\operatorname{Tr}\rho_{\rm rad}^n\big)\big|_{n=1}$ *receives, beyond the disconnected "Hawking" saddle, connected replica-wormhole saddles whose action dominates after* $t_{\rm Page}$ *and which supply exactly the island contribution (Section 57.5). No new ingredient beyond the rules of the semiclassical path integral is invoked.*

3. ***[conjecture, the general lesson]*** *The entanglement wedge of the radiation, as defined by entanglement-wedge reconstruction (Theorem 56.33),* includes the black-hole interior *(the island) for* $t > t_{\rm Page}$*. Hence the interior is encoded in the radiation, and there is no information loss at the level of the coarse data captured by the entropy.*

The first two items are genuinely established in their models, and the technology is exactly that of Section 56.5.3 (the QES) and Section 42.1 (a Euclidean path integral dominated by competing saddles, here generalized to topology-changing geometries). The arithmetic that makes the curve bend is worth recalling in one line, because it is the entire content of "settledness." Writing the generalized entropy of a candidate QES χ as

$$S_{\rm gen}[\chi] = \frac{\operatorname{Area}(\chi)}{4G_N} + S_{\rm bulk}(\Sigma_\chi), \tag{57.126}$$

with Σ_χ the bulk region between χ and the radiation collection surface, the island rule sets $S_{\rm rad} = \min_\chi \operatorname{ext}_\chi S_{\rm gen}[\chi]$. Early on the only extremum is the trivial one, $\chi = \varnothing$, with no area term and $S_{\rm bulk}$ growing as more thermally entangled Hawking quanta are collected—this is the rising Hawking branch. Once enough radiation has accumulated, a nontrivial QES appears just behind the horizon; its area term is large but *fixed* (set by the horizon area at that time), while its bulk term is small because the island purifies almost all the collected radiation. The competition min selects whichever branch is smaller, and the crossover—where the two branches are equal—is the Page time. The curve is the lower envelope of two nearly straight lines: this is the kink in the Page curve, and the island/replica program is precisely the statement that semiclassical gravity computes that kink correctly. That, and no more, is what is settled.

Why this is a real resolution and not a relabelling

Before 2019 one could dismiss "unitarity" as an article of faith imported from the boundary CFT. The achievement is that one need not import it: feeding the *semiclassical* rules—the same saddle-point gravitational path integral that gives black-hole thermodynamics—into the replica computation of a fine-grained entropy already returns the unitary answer, complete with the downturn after the Page time. Gravity polices its own unitarity, provided one is willing to include geometries in which the replicas are connected. The lesson is structural: the gravitational path integral knows about subtle entanglement that no single classical geometry displays.

57.6.2 Entropy is not information: recovery time and complexity

The Page curve is a statement about a single number, the von Neumann entropy, as a function of time. It certifies that the radiation *could in principle* carry away all the information—its entropy is consistent with the global state being pure. It says nothing about whether, or how, an observer holding the radiation can *read* that information. These are different questions, and conflating them is the most common overstatement at this frontier.

We make the distinction sharp with the Hayden–Preskill protocol, the cleanest quantitative handle on recovery. The setup: a black hole B that is already *old*—past its Page time, hence (by Theorem 57.22) maximally entangled with the early radiation R it has already emitted. Alice tosses in a small diary A of k qubits in an unknown pure state. The black hole evolves by some unitary U (the internal dynamics), then continues to Hawking-radiate. Bob holds R and collects the new radiation R' emitted after Alice's deposit. The question is how much of R' Bob needs before the state of A is recoverable from $R \cup R'$.

Theorem 57.23 (Hayden–Preskill recovery; status: controlled-model result, 2007)**.** *Model the post-deposit dynamics by a Haar-random (or, more realistically, a sufficiently scrambling) unitary U acting on the k-qubit diary together with the $\log_2 d_B$ qubits of the black hole, and let Bob collect k' qubits of new radiation. Averaging over U, the trace-distance between Bob's recovered state and the true state of A is bounded by*

$$\left\| \rho_{\mathrm{Bob}}^{\mathrm{rec}} - \rho_A \right\|_1 \lesssim 2^{-(k'-k)}, \tag{57.127}$$

so that collecting just $k' \approx k + O(1)$ qubits beyond the diary's size suffices to reconstruct A with exponentially small error. The diary's information is available *in the radiation almost as soon as the black hole has had time to scramble it.*

The estimate Eq. (57.127) is a decoupling statement: once U has scrambled A into B, the diary decouples from the remaining black hole B' and becomes

accessible from the radiation; the residual error is the probability that a few uncollected qubits still carry diary information, which falls as $2^{-(k'-k)}$. The averaging over U is what makes this a controlled-model result rather than a theorem about real black holes; the physical input is that black holes are *fast scramblers*, which fixes the timescale.

Definition 57.24 (Scrambling time)**.** The scrambling time t_{scr} is the timescale over which local information injected into a strongly chaotic many-body system spreads across all of its degrees of freedom, so that it can no longer be recovered by any few-body measurement. The fast-scrambling conjecture (Sekino–Susskind, 2008) holds that black holes scramble in the minimum time allowed by causality and locality,

$$t_{\mathrm{scr}} \;\simeq\; \frac{\beta}{2\pi}\,\log S_{\mathrm{BH}} \;=\; \frac{1}{2\pi T_{\mathrm{H}}}\,\log S_{\mathrm{BH}}, \tag{57.128}$$

with $\beta = 1/T_{\mathrm{H}}$ the inverse Hawking temperature and $S_{\mathrm{BH}} = \mathrm{Area}/4G_N$ the Bekenstein–Hawking entropy. The prefactor $\beta/2\pi$ is the inverse of the maximal Lyapunov exponent $\lambda_L = 2\pi/\beta$ (the chaos bound of Maldacena–Shenker–Stanford), and the logarithm reflects exponential growth of a perturbation from one to all S_{BH} degrees of freedom.

Combining the two timescales gives the physically important conclusion, which we state as prose because it is the punchline of the whole subsection. For an *old* black hole, information thrown in returns in the radiation after only a scrambling time $\sim \beta \log S_{\mathrm{BH}}$—short, but the returning bits are maximally scrambled across $R \cup R'$. For a *young* black hole (before the Page time), nothing comes out in any decodable form until enough radiation has accumulated to pass the Page time, which is an $O(1)$ fraction of the entire evaporation lifetime. "Information comes out, slowly and scrambled" is thus literally two statements: *slowly* for a young hole (wait the Page time), and *scrambled but promptly* for an old one (wait the scrambling time, then decode a deeply nonlocal code). We pin both numbers down in a worked example.

Worked Example 57.25 (Recovery times for a solar-mass black hole)**.** Take a Schwarzschild black hole of mass M, restoring G_N, c, $\hbar$, k_B for the estimates. Its horizon radius, temperature, and entropy are

$$R_s = \frac{2G_N M}{c^2}, \quad T_{\mathrm{H}} = \frac{\hbar c^3}{8\pi G_N M k_B}, \quad S_{\mathrm{BH}} = \frac{k_B c^3}{\hbar G_N}\,\frac{\mathrm{Area}}{4} = 4\pi\, k_B \Big(\frac{M}{M_{\mathrm{Pl}}}\Big)^2, \tag{57.129}$$

where $M_{\mathrm{Pl}} = \sqrt{\hbar c/G_N} \approx 2.2 \times 10^{-8}\,\mathrm{kg}$. For $M = M_\odot \approx 2.0 \times 10^{30}\,\mathrm{kg}$ we have $M/M_{\mathrm{Pl}} \approx 9 \times 10^{37}$, so

$$S_{\mathrm{BH}}/k_B \;\approx\; 4\pi\,(9\times 10^{37})^2 \;\approx\; 1.0 \times 10^{77}, \qquad \log S_{\mathrm{BH}} \;\approx\; 177. \tag{57.130}$$

The light-crossing time of the horizon is $R_s/c \approx (3.0\times10^3\,\mathrm{m})/(3\times10^8\,\mathrm{m/s}) \approx 1.0\times10^{-5}\,\mathrm{s}$. Since $\beta/2\pi = \hbar/(2\pi k_B T_{\mathrm{H}}) = 4G_N M/c^3 = 2R_s/c$, the scrambling time Eq. (57.128) is

$$t_{\mathrm{scr}} \;\simeq\; \frac{2R_s}{c}\,\log S_{\mathrm{BH}} \;\approx\; 2\,(1.0\times10^{-5}\,\mathrm{s})\times 177 \;\approx\; 3.5\times10^{-3}\,\mathrm{s}. \tag{57.131}$$

A few milliseconds. Compare the evaporation lifetime, obtained from $dM/dt = -\sigma\,\mathrm{Area}\,T_{\mathrm{H}}^4/c^2 \propto -1/M^2$ (Stefan–Boltzmann through the horizon area $\propto M^2$ with $T_{\mathrm{H}} \propto 1/M$), which integrates to

$$t_{\mathrm{evap}} \;=\; \frac{5120\,\pi\,G_N^2 M^3}{\hbar c^4} \;\approx\; 2\times10^{67}\,\mathrm{yr} \quad (M = M_\odot). \tag{57.132}$$

The Page time is an $O(1)$ fraction of this—for a $d = 4$ Schwarzschild hole radiating into massless fields the half-entropy point falls at $t_{\mathrm{Page}} \approx 0.54\,t_{\mathrm{evap}}$, of order 10^{67} years. The contrast is the lesson: a diary dropped into an *old* solar-mass black hole has its information back in the radiation after milliseconds; a diary dropped into a *young* one waits $\sim 10^{67}$ years before any of it becomes decodable in principle. "Information is preserved" and "information is accessible" live 70 orders of magnitude apart in time.

Even granting the time, accessibility is bounded by *computation*, not just by waiting. The decoding map that turns $R \cup R'$ back into Alice's diary is, generically, exponentially complex.

Theorem 57.26 (Harlow–Hayden / Python's lunch; status: conjecture, 2013/2019)**.** ***(Harlow–Hayden, 2013.)*** *Even when Theorem 57.23 guarantees that the diary is recoverable from the radiation, the quantum circuit that performs the decoding has size (gate complexity) exponential in the entropy of the black hole at the deposit time; for an old black hole the distillation of the relevant qubits from R requires $\sim \exp(S_{\mathrm{BH}})$ elementary operations.* ***(Python's lunch, Brown–Gharibyan–Penington–Susskind, 2019.)*** *More generally, when the entanglement wedge of a boundary region is bounded by a* non-minimal *quantum extremal surface—so that, going inward, the wedge first narrows to a "constriction" QES of area $A_{\min}$ and then bulges out to a maximal "appetizer" QES of area $A_{\max}$—then bulk operators in the bulge are reconstructible, but the reconstruction has complexity*

$$\mathcal{C} \;\sim\; \exp\!\Big(\tfrac{1}{2}\,\frac{A_{\max} - A_{\min}}{4G_N}\Big), \tag{57.133}$$

exponential in the area difference (in units of G_N). The black-hole interior past the Page time sits in such a "lunch," which is the geometric origin of Harlow–Hayden complexity.

The status of Theorem 57.26 is conjectural: Eq. (57.133) is a proposal motivated by tensor-network and circuit-complexity models, not a proved theorem

about the gravitational path integral, and it leans on the (itself conjectural) wedge-reconstruction statement Theorem 56.33. But it is the cleanest articulation of the divide that the Page curve forces us to draw.

What "the information paradox is resolved" does and does not mean

The 2019–2020 results resolve the *entropy* paradox: the fine-grained radiation entropy is consistent with global purity, computed within gravity. They do *not* exhibit a unitary S-matrix, do not reconstruct the infalling state operationally, and do not contradict the Harlow–Hayden bound—which says that even though the information is there, no observer with sub-exponential resources can extract it. The slogan "black-hole evaporation is unitary" is now supported by a semiclassical computation; the slogan "we can decode what fell in" remains false in any operational sense. Keep the two apart.

57.6.3 The factorization puzzle and the ensemble interpretation

The same replica wormholes that rescued the entropy create a tension with the very dictionary (Theorem 53.6) that motivated trusting unitarity in the first place. We state it as a clean fact about the gravitational path integral, then unpack what it could mean.

In a single quantum theory the partition function is a number. If $Z[J] = \operatorname{Tr} e^{-\beta H + J\cdot\mathcal{O}}$ is computed in one boundary theory, then the quantity associated with two disconnected boundaries is, tautologically, the product of two numbers:

$$Z[J_1 \sqcup J_2] \;=\; Z[J_1]\, Z[J_2] \qquad \text{(single theory)}. \tag{57.134}$$

The gravitational path integral, however, computes the left side by summing over *all* bulk geometries whose boundary is the disjoint union of the two boundaries—and this includes connected bulk geometries ("wormholes") joining them. Schematically,

$$Z_{\text{grav}}[J_1 \sqcup J_2] \;=\; \underbrace{Z_{\text{grav}}[J_1]\, Z_{\text{grav}}[J_2]}_{\text{disconnected saddles}} \;+\; \underbrace{Z_{\text{conn}}[J_1, J_2]}_{\text{wormhole}} \;\neq\; Z_{\text{grav}}[J_1]\, Z_{\text{grav}}[J_2]. \tag{57.135}$$

The connected piece Z_{conn} is nonzero (in two dimensions it is the celebrated "double trumpet"). Comparing Eq. (57.134) with Eq. (57.135): the gravitational answer cannot be the partition function of a *single* boundary theory. The natural reading is statistical—identify $Z_{\text{grav}}[J]$ with an *ensemble average* $\overline{Z[J]}$ over a family of theories, in which case the wormhole computes the connected correlation, i.e. the variance:

$$Z_{\text{conn}}[J_1, J_2] \;=\; \overline{Z[J_1]\, Z[J_2]} - \overline{Z[J_1]}\;\overline{Z[J_2]} \;=\; \operatorname{Cov}\big(Z[J_1], Z[J_2]\big). \tag{57.136}$$

Theorem 57.27 (Gravity as a matrix integral; status: controlled-model result/theorem, 2019)**.** ***(Saad–Shenker–Stanford, 2019.)*** *The full genus expansion of the Jackiw–Teitelboim gravity partition function—summing over all two-dimensional bulk topologies with fixed asymptotic boundaries—equals the genus (topological) expansion of a particular double-scaled Hermitian random-matrix integral. Concretely, with $Z(\beta) = \mathrm{Tr}\, e^{-\beta H}$ and H a random $L \times L$ matrix drawn from a fixed potential,*

$$\left\langle \prod_i Z(\beta_i) \right\rangle_{\mathrm{JT}} \;=\; \left\langle \prod_i \mathrm{Tr}\, e^{-\beta_i H} \right\rangle_{\mathrm{matrix}}, \tag{57.137}$$

order by order in e^{-S_0} (the genus-counting parameter, S_0 the extremal entropy). The connected wormhole geometries on the gravity side are precisely the connected matrix-model correlators on the ensemble side. In JT gravity, then, the gravitational path integral does not *compute a single Hamiltonian's partition function; it computes the moments of an ensemble of Hamiltonians.*

This is established in the controlled setting of two dimensions, where both sides are computable; Eq. (57.137) is a genuine equality of asymptotic expansions. The diagnostic that exhibits it physically is the spectral form factor.

Worked Example 57.28 (The ramp: how the wormhole shows up in $|Z|^2$)**.** Consider the analytically continued product $g(t) = \overline{Z(\beta+it)\,Z(\beta-it)} = \overline{|Z(\beta+it)|^2}$, the spectral form factor, where $Z(\beta+it) = \sum_n e^{-(\beta+it)E_n}$. Expanding,

$$g(t) = \sum_{m,n} e^{-\beta(E_m+E_n)}\, e^{-it(E_m-E_n)} = \underbrace{\left|\overline{Z(\beta+it)}\right|^2}_{\text{disconnected}} + \underbrace{\overline{|Z|^2} - |\overline{Z}|^2}_{\text{connected (wormhole)}} . \tag{57.138}$$

The disconnected piece $|\overline{Z}|^2$ decays at early times (the "slope") because the phases e^{-itE_n} dephase. If Z were a single number, that would be the whole story and $g(t)$ would decay to a small constant. Instead, in a chaotic system—and in JT gravity via Theorem 57.27—the connected piece is governed by random-matrix eigenvalue statistics. For nearby eigenvalues with the universal pair correlation $\overline{\rho(E)\rho(E')}_{\mathrm{conn}} \sim -\frac{1}{2\pi^2(E-E')^2}$ (the sine-kernel repulsion of the Gaussian ensembles), the Fourier transform gives a piece that *grows linearly*,

$$g_{\mathrm{conn}}(t) \;\sim\; \frac{t}{2\pi\beta} \qquad (t_{\mathrm{Th}} \ll t \ll t_{\mathrm{H}}), \tag{57.139}$$

the "ramp," until it saturates at the "plateau" $g \sim \overline{Z(2\beta)}$ for $t \gtrsim t_{\mathrm{H}} \sim e^{S_0}$ (the Heisenberg time set by the mean level spacing). On the gravity side the ramp is reproduced exactly by the double-trumpet wormhole connecting the two boundary circles: the wormhole *is* the linear ramp. The slope-ramp-plateau structure is the fingerprint of an ensemble; a single fixed Hamiltonian shows the same curve only because it is itself one draw from a chaotic ensemble and self-averages over an energy window.

So in two dimensions the verdict is clear: gravity computes an ensemble average, and the wormholes are its variance. The puzzle is what this means in $d \geq 3$, where the boundary is supposed to be a *single*, specific conformal field theory—$\mathcal{N} = 4$ super-Yang–Mills at fixed rank and coupling, say—with a unique partition function obeying Eq. (57.134) exactly. There the candidate resolutions diverge, and we list them honestly.

1. **[conjecture]** *Low-dimensional gravity is genuinely dual to an ensemble.* JT and pure three-dimensional gravity may simply not be dual to a single theory; their boundary "duals" are averages (over matrix ensembles, or over Narain moduli for certain $d = 2$ CFTs, as in Maloney–Witten / Afkhami-Jeddi–Cohn–Hartman–Tajdini, 2020). On this view Eq. (57.135) is not a paradox but the definition of what these theories are.

2. **[conjecture]** *Higher-dimensional gravity is a single theory and the wormholes are an artifact of coarse-graining or of the semiclassical approximation.* In genuine string-theoretic $\mathrm{AdS}_5/\mathrm{CFT}_4$ there is one CFT; the gravitational saddle sum is then an approximation to a single Z, and either the wormholes are subdominant/absent in the exact theory, or they encode the average over the unknown ultraviolet ("half-wormholes" and self-averaging; Saad–Shenker–Stanford–Yang, Schlenker–Witten, 2021–2022). Factorization is then restored by contributions invisible to the leading saddle expansion.

3. **[controlled-model result + conjecture]** *Baby universes and α-states* (Coleman, 1988; Marolf–Maxfield, 2020). Promote the gravitational path integral to a "third-quantized" theory whose Hilbert space $\mathcal{H}_{\mathrm{BU}}$ is built by creating and destroying closed (baby) universes. The boundary-creating operators $\widehat{Z}[J]$ *commute*, so they are simultaneously diagonalizable; their joint eigenstates are the α-states $|\alpha\rangle$, with $\widehat{Z}[J]\,|\alpha\rangle = Z_\alpha[J]\,|\alpha\rangle$. In a fixed α-state, *factorization is restored*: $\langle\alpha|\widehat{Z}[J_1]\widehat{Z}[J_2]|\alpha\rangle = Z_\alpha[J_1]Z_\alpha[J_2]$, and the theory is a single unitary one. The non-factorizing path integral Eq. (57.135) is then the no-boundary (Hartle–Hawking) state expectation value $\langle \mathrm{HH}| \cdot |\mathrm{HH}\rangle$, which is a *superposition* over α-sectors—i.e. an average $\sum_\alpha p_\alpha\,(\cdots)$ over the ensemble of single theories labelled by α. The ensemble interpretation and the single-theory interpretation are thereby reconciled: each α-state is a single theory, and the gravitational path integral, by default, prepares an incoherent average over them.

The shape of the tension

The factorization puzzle is the sharpest unresolved point in this Part. Theorem 53.6 asserts *one* bulk = *one* boundary; the replica wormholes that we needed for the Page curve assert that the bulk path integral, taken at face value, computes an *average*. Either the duality is an averaged

statement (true in the cases we can solve), or there is a mechanism—half-wormholes, α-state selection, ultraviolet completion—that restores a single theory in the cases we care about (AdS_5/CFT_4) but which we cannot yet exhibit. As of the mid-2020s the question is open: we do not know whether the gravitational path integral, in four or more dimensions, computes a single unitary theory or an ensemble.

57.6.4 Firewalls after islands

We revisit the firewall argument stated in Section 57.2 (AMPS, Almheiri–Marolf–Polchinski–Sully, 2012) with the island picture in hand. The argument is a clean incompatibility of four assumptions, all individually attractive: (i) *unitarity*—evaporation is described by a unitary S-matrix; (ii) *semiclassical exterior*—low-energy effective field theory holds outside the horizon; (iii) *no drama*—an infalling observer sees nothing special at the horizon (the equivalence principle); (iv) *a normal interior*—the black hole is a quantum system with $S_{\rm BH} = \text{Area}/4G_N$ states. The sharpened contradiction is monogamy of entanglement. Let b be a late outgoing Hawking mode emitted by an *old* black hole.

- By unitarity and the Page curve (Theorem 57.22), an old black hole's new radiation purifies the early radiation: b must be nearly maximally entangled with the early radiation R.
- By "no drama," the horizon is the vacuum for an infalling observer, which requires b to be nearly maximally entangled with its interior partner mode $\tilde{b}$ (the Unruh/Hartle–Hawking partner just behind the horizon).

A single mode b cannot be maximally entangled with two independent systems R and $\tilde{b}$ (monogamy). AMPS concluded that one assumption must fail; the eponymous proposal was to drop "no drama"—a high-energy *firewall* of quanta sits at the horizon, breaking entanglement with $\tilde{b}$ and incinerating the infalling observer.

The island/QES picture supplies a sharp response to the *entanglement bookkeeping*.

Theorem 57.29 (Island resolution of the monogamy paradox; status: controlled–model result/conjecture)**.** *After the Page time, the entanglement wedge of the radiation includes the island I behind the horizon (Theorem 57.22, item 3). The interior partner mode $\tilde{b}$ lies inside I, hence inside the radiation's entanglement wedge: by entanglement-wedge reconstruction (Theorem 56.33), $\tilde{b}$ is* encoded in the early radiation R itself. *The two systems AMPS treated as independent are not independent—"R" and "$\tilde{b}$" are the same degrees of freedom. There is then no monogamy violation: b is entangled with $\tilde{b}$, and $\tilde{b}$ is part of R, so the single*

entanglement of b does double duty. Smoothness of the horizon (entanglement of b with $\tilde{b}$) and purification of the radiation (entanglement of b with R) are compatible because $\tilde{b} \subset R$.

This is the same identification that the "ER=EPR" slogan (Maldacena–Susskind, 2013) packages as: the interior is connected to the early radiation by a (nontraversable) Einstein–Rosen bridge, so reaching $\tilde{b}$ "through the bulk" and reaching it "through the radiation" are the same operation. At the level of which density matrix encodes what, Theorem 57.29 is as settled as wedge reconstruction is—which is to say, it is a controlled-model result resting on a conjecture.

What remains genuinely open is the *dynamical* question that "no drama" really poses. The island argument shows that the entanglement structure is *consistent* with a smooth horizon; it does not, by itself, exhibit the local interior operators an infalling observer would measure, nor prove that they behave as low-energy effective field theory in a smooth geometry. The leading proposal for those operators—Papadodimas–Raju *state-dependent* reconstruction (2013–2014), in which the interior operators $\tilde{b}$ are realized as specific operators on the boundary that depend on the state of the black hole—reproduces a smooth interior but at the cost of state-dependence, which sits uneasily with the linearity of quantum mechanics and the existence of a single algebra of observables. Whether interior reconstruction can be made state-independent, what an infalling observer actually experiences, and how to describe the experience of *two* observers (one infalling, one reading the radiation) without a measurable contradiction (the "Alice–Bob" or quantum-cloning worry, regulated by complementarity and the Harlow–Hayden complexity of Theorem 57.26) are not settled.

Complementarity is an assumption, not a theorem

Black-hole complementarity—the claim that no single observer can witness both the infalling information inside and the same information in the radiation, so there is no operational contradiction—is consistent with everything above and is protected by the exponential decoding complexity of Theorem 57.26: the experiment that would catch a cloning contradiction cannot be done before the black hole evaporates. But "no observer can check" is a statement about resources, not a derivation of a smooth interior. The firewall question, as a question about what physics an infalling observer encounters, is not closed as of the mid-2020s.

57.6.5 What remains open: a dated summary of the frontier

We close the chapter by stating the boundary as flatly as we can, dated to the mid-2020s. The reader should treat each item as provisional and check its status

against the literature when revisiting.

Settled (within their models, ∼2019 and after).

1. The fine-grained radiation entropy follows the unitary Page curve, computed in semiclassical gravity via the QES/island prescription [controlled-model result], and derived from the bare Euclidean path integral via replica wormholes [controlled-model result]. The entropy paradox is resolved.

2. In two dimensions, the gravitational path integral with multiple boundaries equals a random-matrix ensemble average, with wormholes = connected correlators [controlled-model result/theorem, JT].

3. The entanglement-bookkeeping form of the firewall (monogamy) paradox is dissolved by the island: the interior partner is encoded in the early radiation [controlled-model result, contingent on Theorem 56.33].

Open (as of the mid-2020s).

1. *Operational recovery.* No construction exhibits a unitary S-matrix for evaporation, nor an explicit, computationally feasible decoding of infalling information; the Harlow–Hayden/Python's-lunch complexity Eq. (57.133) suggests decoding is exponentially hard, but its general validity is conjectural.

2. *Factorization / ensemble.* Whether the gravitational path integral in $d \geq 3$ computes a single unitary theory or an ensemble average is unresolved; baby-universe/α-state, half-wormhole, and self-averaging proposals exist but none is established for $\mathrm{AdS}_5/\mathrm{CFT}_4$.

3. *The infalling observer.* Whether interior operators admit a state-independent realization, what an infalling observer measures, and the ultimate status of the firewall and of complementarity, are open dynamical questions; the island results constrain the entanglement but do not settle the dynamics.

4. *Beyond the semiclassical replica.* The replica-wormhole derivation is a saddle-point statement; its all-orders and non-perturbative completion, the precise rules for which topologies to include, and the meaning of the $n \to 1$ continuation in genuine four-dimensional gravity are not under full control.

5. *The microscopic interior.* The island gives the *entropy* of the interior; an explicit microscopic model of the interior degrees of freedom and their dynamics—the analogue of a fuzzball or a definite boundary encoding for an evaporating black hole—is not available.

The honest summary is this. The information paradox, in the precise form Hawking posed it—*does the von Neumann entropy of the radiation rise forever?*—has been answered: it does not; semiclassical gravity itself yields the Page

curve. The broader question the paradox stands in for—*how, concretely and microscopically, is quantum information processed by a black hole, and is the bulk gravitational theory a single quantum system at all*—is the live frontier, and the machinery of this Part has sharpened it rather than closed it.

The chapter has carried the information problem from its precise statement, through the Page curve and the island/replica resolution of the *entropy*, to an honest accounting of what the resolution does and does not include. Two threads now leave this Part. Chapter 65 returns to the beginning—Hawking's original Bogoliubov computation of the thermal spectrum on a collapse background, and the trans-Planckian worry that the calculation invites—supplying from the curved-spacetime side the input we here established by the Euclidean route. And the closing synthesis of Chapter 65 takes the frontier mapped in Section 57.6.5 and sets it within the whole arc of the subject, where the clash and partial reconciliation of general relativity and quantum mechanics that this chapter examined stands as the sharpest open connection in all of quantum field theory.

Further reading

The thermodynamic foundation is the work of Bekenstein [54], who proposed the area law, and Hawking [255], who fixed its coefficient by computing the radiation; the Euclidean action method used here is that of Gibbons and Hawking [220], with the canonical-ensemble subtleties of black holes in AdS treated by Hawking and Page [258]. The unitarity requirement and the averaged-subsystem entropy that defines the Page curve originate with Page [374]. The modern resolution rests on two pillars: the island formula and its first applications, [17, 391], and the replica-wormhole derivation from the gravitational path integral, [18, 392]; recent reviews are [19]. On the distinction between entropy and recoverable information, see the decoding protocol of [260], the complexity obstruction of [251] sharpened geometrically as the Python's lunch in [96], and the fast-scrambling and chaos-bound results [333, 438]. The interpretation of two-dimensional gravity as a matrix ensemble is [423], with the baby-universe Hilbert space and α-states of [133, 338] and the factorization tension discussed in [424, 431]. The firewall argument is [15]; black-hole complementarity and ER=EPR are [332, 460]; state-dependent interior reconstruction is [375]; and the holographic backdrop for all of it traces to [331, 422].

Part XVI

Modern Scattering Amplitudes

Chapter 58

Spinor-Helicity Formalism

A scattering amplitude computed the textbook way is a study in waste. The Feynman rules of Parts IV–VII assign to each diagram a gauge-dependent tangle of polarization vectors, propagator numerators, and momentum contractions; a process with a handful of external gluons generates hundreds of such terms, each individually meaningless because each depends on the gauge, and the labor of the calculation is the labor of watching that gauge dependence cancel. Yet when the dust settles the answer is almost always short — often a single ratio of brackets fitting on one line. The gap between the difficulty of the computation and the simplicity of the result is the standing scandal that this chapter, and the part it opens, is written to resolve. The diagnosis is already in our hands. Wigner's classification of Chapter 3 taught us that a massless one-particle state carries exactly two pieces of data — a null momentum and a single helicity weight under the little group $\mathrm{ISO}(2) \simeq \mathrm{U}(1)$ — while a covariant polarization four-vector carries a redundancy no Lorentz-covariant repackaging can remove without gauging (Section 3.7.3 and Theorem 3.46). The waste is the redundancy made manifest. The cure is to build variables that carry the on-shell data and nothing else.

That is precisely what the spinor-helicity formalism does, and the chapter assembles it from the ground up. We begin (Section 58.1) from a fact of linear algebra dressed in the two-component calculus of Chapter C: a null four-momentum, written as a Hermitian 2×2 matrix, has vanishing determinant, hence rank one, hence factorizes into the outer product $p_{a\dot{a}} = \lambda_a \tilde{\lambda}_{\dot{a}}$ of two spinors that turn out to be the massless Weyl spinors of Chapter 3 and the massless limit of the Dirac plane-wave spinors of Chapter 11. From these spinors we forge a calculus (Section 58.2): the angle and square brackets $\langle i\,j\rangle$, $[i\,j]$, their dictionary $\langle i\,j\rangle[j\,i] = 2\,p_i{\cdot}p_j$ to the Mandelstam invariants of Chapter 16, and the Fierz, Schouten, and momentum-conservation identities on which every later manipulation runs, with the reality conditions and the analytic continuation to complex momenta pinned down once and for all.

With the alphabet in place we promote the one residual freedom in the spinor parametrization — the rescaling $\lambda \to t\lambda$, $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$ that leaves the momentum fixed — to the organizing principle of the subject (Section 58.3). This rescaling *is* the massless little group of Chapter 3 acting in spinor language, and the response of an amplitude to it is rigid: a leg of helicity h forces the scaling weight $-2h$. At three points the constraint becomes total. The kinematics of three massless momenta is so confining that little-group covariance and locality alone fix the amplitude up to one constant, with no Lagrangian, no diagram, and no gauge choice (Section 58.4); the three-gluon amplitude, the QED electron–photon vertex — which advances QED beyond its Feynman-rule layer in Chapter 17 — and the graviton seed of the equivalence principle all drop out of the same representation-theoretic computation. To carry the formalism past three points we build the external polarization vectors directly from spinors (Section 58.5), where the inhomogeneous gauge redundancy of Theorem 3.46 reappears as the harmless freedom to choose a reference spinor — a freedom sharp enough, by the Ward identity of Chapter 18, to annihilate most of a Feynman expansion before it is written, and which on squaring yields the graviton polarization and the seed of the double copy of Chapter 61.

The chapter then redeems its opening promise on the book's own examples (Section 58.6). The annihilation $e^+e^- \to \mu^+\mu^-$ of Chapter 17 and Compton scattering of Chapter 18, computed there by brute-force traces, are rebuilt as one-line helicity amplitudes whose squares reproduce the carried QED cross sections to the last factor, with no gamma-matrix trace and no Levi-Civita symbol. Finally (Section 58.7) we confront the one structural obstacle between this success and the gluons of Yang–Mills theory — adjoint color — and dispatch it by the tree-level trace decomposition, which factors any n-gluon amplitude into color traces times gauge-invariant, cyclically ordered partial amplitudes built from the cubic vertex of Chapter 26. With the gluon three-point amplitude in its final color-ordered form, the Parke–Taylor formula named as the destination, and the massive bold-spinor extension stated at the frontier, the chapter hands Chapter 59 exactly the on-shell seeds its BCFW recursion will glue into the entire tree-level S-matrix.

58.1 Massless momenta as spinor squares

The chapter introduction named the puzzle: amplitudes computed from gauge-dependent Feynman diagrams swell to hundreds of terms and then, after the cancellations are done, collapse to one-line gauge-invariant answers. We located the resolution in something the reader already owns. Wigner's classification, built in Chapter 3, says that a massless one-particle state is labelled by exactly two pieces of data: its momentum on the light cone and a single helicity weight under the little group $\mathrm{ISO}(2) \simeq \mathrm{U}(1)$ (Section 3.7.3 and Eq. (3.217)). A

four-vector polarization carries more than this — it carries a redundancy that no Lorentz-covariant repackaging can remove without gauging (Section 3.7.4 and Theorem 3.46). The strategy of the entire part is therefore to stop describing a massless particle with redundant covariant objects and instead build variables that carry the on-shell data *and nothing else*. This section takes the first and most basic step: it shows that a null four-momentum is not an irreducible datum at all but the outer product of two two-component spinors, and it identifies those spinors with the Weyl spinors of Chapter 3 and the massless limit of the Dirac plane-wave spinors of Chapter 11. Once a null momentum has been opened up into a pair $(\lambda, \tilde{\lambda})$, every Lorentz invariant one can form from a set of massless legs becomes a product of the simplest possible contractions of these spinors — the angle and square brackets that the next section turns into a complete algebra.

The technical engine is a fact from linear algebra so elementary that it is easy to miss its consequences: a 2×2 matrix has rank one precisely when its determinant vanishes, and a rank-one matrix is an outer product of a column and a row. The whole of the present section is the observation that, once a four-vector p^μ is written as a 2×2 matrix in the canonical way, its Lorentz square p^2 *is* that determinant. Masslessness is the vanishing of the determinant; the spinor factorization is the outer-product structure forced by it; and the residual freedom in the factorization is exactly the little group. Nothing here is new mathematics — the two-component machinery was assembled in full in Section C.9 — but it is the moment at which that machinery acquires its physical job.

Notation Watch

Conventions for the rest of Part XVI, fixed here once because they differ between references. The metric is mostly-minus, $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, and natural units are in force. *All external momenta are taken outgoing*; an incoming physical particle of momentum p enters the formulas as the outgoing momentum $-p$, so that a leg may carry $p^0 < 0$. Two-component indices are written $a, \dot{a} \in \{1,2\}$ (undotted/dotted), exactly as in Section C.9.1; the Greek $\alpha, \dot{\alpha}$ used by some later sections and much of the literature denote the same indices. The holomorphic spinor is λ_a and the antiholomorphic one $\tilde{\lambda}_{\dot{a}}$; the bar notation $\bar{\lambda}_{\dot{a}}$ of Section C.9 denotes the same object *when momenta are real*, where (as shown below) $\tilde{\lambda} = \pm\lambda^*$, the sign set by $\mathrm{sgn}(p^0)$. For complex momenta λ and $\tilde{\lambda}$ are independent and the tilde, not the bar, is the honest notation. The bracket normalization fixed in Section 58.1 and proved in the next section is $\langle i\,j\rangle[j\,i] = 2\,p_i{\cdot}p_j = s_{ij}$, with the angle bracket antisymmetric, $\langle i\,j\rangle = -\langle j\,i\rangle$.

58.1.1 The bispinor map

A four-vector has four real components; a Hermitian 2×2 matrix also has four real parameters (two real diagonal entries and one complex off-diagonal entry). The two-component calculus of Section C.9.2 supplies the unique Lorentz-covariant isomorphism between them. Recall the Pauli four-vectors $(\sigma^\mu)_{a\dot b} = (\mathbf{1}, \sigma^i)_{a\dot b}$ of Eq. (C.274). To any four-momentum p^μ we associate the *momentum bispinor*

$$p_{a\dot a} \;\equiv\; p_\mu\,(\sigma^\mu)_{a\dot a} \;=\; p^0\,\mathbf{1} - \vec p\cdot\vec\sigma \;=\; \begin{pmatrix} p^0 - p^3 & -p^1 + ip^2 \\ -p^1 - ip^2 & p^0 + p^3 \end{pmatrix}, \tag{58.1}$$

where the middle form follows from $p_\mu = (p^0, -\vec p)$ in the mostly-minus metric together with $\sigma^\mu = (\mathbf{1}, \sigma^i)$, and the explicit matrix from $\vec p\cdot\vec\sigma = p^3\sigma^3 + p^1\sigma^1 + p^2\sigma^2$. This is a real-linear bijection: a four-vector p^μ is real if and only if the matrix $p_{a\dot a}$ is Hermitian, since $(\sigma^\mu)^\dagger = \sigma^\mu$ and the p_μ are then real coefficients.

The point of the map is that the Lorentz square becomes a determinant. Reading off Eq. (58.1),

$$\det p_{a\dot a} = (p^0-p^3)(p^0+p^3) - (-p^1+ip^2)(-p^1-ip^2) = (p^0)^2 - (p^3)^2 - (p^1)^2 - (p^2)^2 = p_\mu p^\mu = p^2. \tag{58.2}$$

For an on-shell particle $p^2 = m^2$, so the determinant of the bispinor is the squared mass. This single identity already encodes the massive-versus-massless dichotomy that will organize everything: a massive momentum has $\det p_{a\dot a} = m^2 \neq 0$, so $p_{a\dot a}$ is an invertible matrix of full rank two; a massless momentum has $\det p_{a\dot a} = 0$, so the matrix is singular and its rank drops. We will see that the rank drops to exactly one, and that the factorization of a rank-one matrix is the spinor square promised in the section title.

Before exploiting the determinant it is worth recording how the map intertwines Lorentz transformations, because this is what makes the construction covariant and ties it back to Chapter 3. Under a restricted Lorentz transformation represented by $M \in \mathrm{SL}(2,\mathbb{C})$ acting on undotted indices by $\psi_a \mapsto M_a{}^b \psi_b$ (Eq. (C.266)) and on dotted indices by $\bar\chi_{\dot a} \mapsto (M^*)_{\dot a}{}^{\dot b}\bar\chi_{\dot b}$ (Eq. (C.267)), the bispinor transforms as

$$p_{a\dot a} \;\longmapsto\; M_a{}^b\,(M^*)_{\dot a}{}^{\dot b}\,p_{b\dot b} \;=\; \big(M\,p\,M^\dagger\big)_{a\dot a}, \tag{58.3}$$

the last form reading the two index actions as ordinary matrix multiplication on the left and the conjugate-transpose on the right. Since $\det M = 1$, the determinant Eq. (58.2) is invariant, $\det(MpM^\dagger) = \det p$, which is the spinorial proof that p^2 is a Lorentz scalar. The map Eq. (58.1) is thus the explicit realization of the two-to-one covering $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}(1,3)^\uparrow$ built in Chapter 3: $\mathrm{SL}(2,\mathbb{C})$ acts on Hermitian matrices by $p \mapsto MpM^\dagger$, preserving the determinant, hence acts as a Lorentz transformation on the four-vector p^μ.

58.1.2 The rank-one factorization theorem

We now extract the spinors. The content is entirely in the following statement, which is the technical heart of the whole formalism.

Theorem 58.1 (Rank-one factorization of a null momentum)**.** *Let p^μ be a nonzero real four-vector with bispinor $p_{a\dot a}$ as in Eq. (58.1). Then p is null, $p^2 = 0$, if and only if $p_{a\dot a}$ has rank one, equivalently if and only if there exist two-component spinors λ_a and $\tilde\lambda_{\dot a}$, neither zero, such that*

$$p_{a\dot a} = \lambda_a \, \tilde\lambda_{\dot a}. \tag{58.4}$$

The spinors are unique up to the rescaling

$$\lambda_a \longmapsto t\,\lambda_a, \qquad \tilde\lambda_{\dot a} \longmapsto t^{-1}\,\tilde\lambda_{\dot a}, \qquad t \in \mathbb{C}^\times, \tag{58.5}$$

and no further freedom. For a real momentum the Hermiticity of $p_{a\dot a}$ forces the reality condition $\tilde\lambda_{\dot a} = \operatorname{sgn}(p^0)\,(\lambda_a)^*$, which restricts the rescaling Eq. (58.5) to the phases $t = e^{i\theta} \in \mathrm{U}(1)$.

Proof. Determinant, rank, and outer product are equivalent. For any 2×2 matrix the determinant is the only obstruction to a rank drop: $\det p_{a\dot a} = 0$ means the two columns are linearly dependent, so the column space is at most one-dimensional; since $p \neq 0$ the matrix is not the zero matrix, so the column space is exactly one-dimensional and the rank is exactly one. By Eq. (58.2) the condition $\det p_{a\dot a} = 0$ is precisely $p^2 = 0$. Conversely a rank-one matrix has vanishing determinant. It remains to show that a rank-one 2×2 matrix is an outer product Eq. (58.4) and to exhibit the spinors explicitly.

Explicit construction of the spinors. Write the entries as

$$p_{a\dot a} = \begin{pmatrix} p_{1\dot 1} & p_{1\dot 2} \\ p_{2\dot 1} & p_{2\dot 2} \end{pmatrix}, \qquad \det p_{a\dot a} = p_{1\dot 1}p_{2\dot 2} - p_{1\dot 2}p_{2\dot 1} = 0. \tag{58.6}$$

Suppose first that $p_{1\dot 1} \neq 0$. Define

$$\lambda_a = \frac{1}{\sqrt{p_{1\dot 1}}}\begin{pmatrix} p_{1\dot 1} \\ p_{2\dot 1} \end{pmatrix}, \qquad \tilde\lambda_{\dot a} = \frac{1}{\sqrt{p_{1\dot 1}}}\begin{pmatrix} p_{1\dot 1} \\ p_{1\dot 2} \end{pmatrix}, \tag{58.7}$$

where $\sqrt{p_{1\dot 1}}$ is any fixed square root. The outer product reproduces $p_{a\dot a}$ entry by entry:

$$\begin{aligned} \lambda_1\tilde\lambda_{\dot 1} &= \frac{p_{1\dot 1}\cdot p_{1\dot 1}}{p_{1\dot 1}} = p_{1\dot 1}, & \lambda_1\tilde\lambda_{\dot 2} &= \frac{p_{1\dot 1}\cdot p_{1\dot 2}}{p_{1\dot 1}} = p_{1\dot 2}, \\ \lambda_2\tilde\lambda_{\dot 1} &= \frac{p_{2\dot 1}\cdot p_{1\dot 1}}{p_{1\dot 1}} = p_{2\dot 1}, & \lambda_2\tilde\lambda_{\dot 2} &= \frac{p_{2\dot 1}\cdot p_{1\dot 2}}{p_{1\dot 1}} = \frac{p_{1\dot 1}p_{2\dot 2}}{p_{1\dot 1}} = p_{2\dot 2}, \end{aligned} \tag{58.8}$$

the last equality using the determinant condition $p_{2\dot{1}}p_{1\dot{2}} = p_{1\dot{1}}p_{2\dot{2}}$ from Eq. (58.6). This proves Eq. (58.4). If instead $p_{1\dot{1}} = 0$ then at least one other diagonal or off-diagonal entry is nonzero (else $p = 0$); for a null momentum with $p^0 > 0$ one has $p_{1\dot{1}} = p^0 - p^3 \geq 0$, vanishing only when p points along $+\hat{z}$, in which case $p_{2\dot{2}} = p^0 + p^3 = 2p^0 \neq 0$ and the identical construction built on $p_{2\dot{2}}$ goes through after exchanging the roles of the two rows and columns. Either way an outer-product factorization exists.

Uniqueness up to rescaling. Suppose $\lambda_a\tilde{\lambda}_{\dot{a}} = \lambda'_a\tilde{\lambda}'_{\dot{a}}$ with all four spinors nonzero. The left side, regarded as a matrix, has column space spanned by the single vector λ_a; the right side has column space spanned by λ'_a. A one-dimensional space has a unique spanning direction, so $\lambda'_a = t\,\lambda_a$ for some $t \in \mathbb{C}^\times$. Substituting and cancelling the nonzero λ_a leaves $\tilde{\lambda}_{\dot{a}} = t\,\tilde{\lambda}'_{\dot{a}}$, i.e. $\tilde{\lambda}'_{\dot{a}} = t^{-1}\tilde{\lambda}_{\dot{a}}$. This is exactly Eq. (58.5), and there is no other freedom: the factorization is rigid once the scale t is fixed.

Reality condition. Let p^μ be real, so $p_{a\dot{a}}$ is Hermitian: $(p_{a\dot{a}})^* = p_{\dot{a}a}$, i.e. $p_{2\dot{1}} = (p_{1\dot{2}})^*$ and the diagonal entries are real. Take $p^0 > 0$, so the matrix Eq. (58.1) has nonnegative trace $\operatorname{tr} p_{a\dot{a}} = 2p^0 > 0$, and being a rank-one Hermitian matrix of nonnegative trace it is positive semidefinite. Then $p_{1\dot{1}} \geq 0$, and choosing the positive square root in Eq. (58.7) gives, using $p_{2\dot{1}} = (p_{1\dot{2}})^*$ and $p_{1\dot{1}}$ real,

$$(\lambda_a)^* = \frac{1}{\sqrt{p_{1\dot{1}}}}\begin{pmatrix}(p_{1\dot{1}})^*\\(p_{2\dot{1}})^*\end{pmatrix} = \frac{1}{\sqrt{p_{1\dot{1}}}}\begin{pmatrix}p_{1\dot{1}}\\p_{1\dot{2}}\end{pmatrix} = \tilde{\lambda}_{\dot{a}}. \tag{58.9}$$

Thus $\tilde{\lambda} = \lambda^*$ when $p^0 > 0$. If instead $p^0 < 0$ (a crossed, all-outgoing leg), then $p_{a\dot{a}}$ is negative semidefinite; writing $p_{a\dot{a}} = -(-p)_{a\dot{a}}$ with $-p$ positive-energy and factorizing $-p$ as above attaches a relative sign, $\tilde{\lambda} = -\lambda^*$. In both cases $\tilde{\lambda} = \operatorname{sgn}(p^0)\,\lambda^*$. Finally, the reality condition is preserved by the rescaling Eq. (58.5) only when $(t\lambda)^* = t^{-1}\lambda^*$, i.e. $\bar{t} = t^{-1}$, i.e. $|t| = 1$: the residual freedom for real momenta is the phase group U(1). □

The last sentence of the proof is the punchline that connects this purely algebraic statement back to Wigner. For a real on-shell massless momentum the factorization Eq. (58.4) is unique up to a single phase $e^{i\theta}$, and this phase is the little group. Recall from Section 3.7.3 that the massless little group — the subgroup of Lorentz transformations fixing a chosen null momentum — is ISO(2), the symmetries of the Euclidean plane, whose compact part acts on one-particle states by the helicity phase $e^{ih\theta}$ of Eq. (3.217). The rescaling Eq. (58.5) is the action of this little group on the spinor data: a Lorentz transformation fixing $p_{a\dot{a}} = \lambda_a\tilde{\lambda}_{\dot{a}}$ cannot change the product, so it can only multiply λ by a phase and $\tilde{\lambda}$ by the inverse phase. We will promote this observation in a later section into the organizing principle of the whole subject — an amplitude's weight under Eq. (58.5) on each leg *is* that leg's helicity, and this single requirement will fix the three-point amplitudes by symmetry alone. For now we record only that the

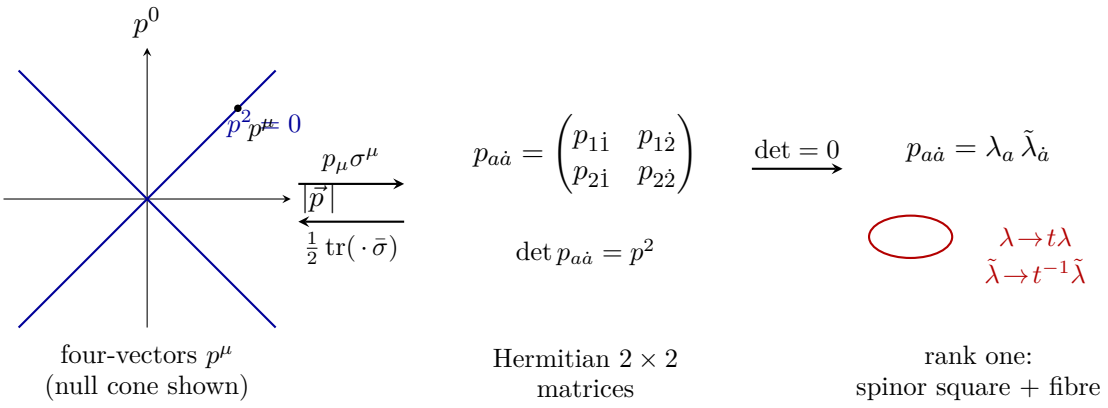


Figure 58.1: The bispinor map and its rank-one degeneration. A real four-vector is the same datum as a Hermitian 2×2 matrix (Eq. (58.1)), under which the Lorentz square becomes the determinant (Eq. (58.2)). On the null cone $p^2 = 0$ the determinant vanishes, the matrix drops to rank one, and Theorem 58.1 factorizes it into the outer product of a holomorphic spinor λ and an antiholomorphic spinor $\tilde{\lambda}$. The factorization is unique only up to the little-group fibre Eq. (58.5); for real momenta this fibre is the helicity phase U(1) of Section 3.7.3.

redundancy Wigner identified in the covariant description has reappeared, in the spinor variables, as a clean one-parameter rescaling rather than the awkward inhomogeneous shift of polarization vectors found in Theorem 3.46.

We package the result as the definition that the rest of Part XVI refers back to.

Definition 58.2 (Spinor-helicity variables)**.** To each external massless leg of momentum p^μ we attach a pair of two-component spinors: the *holomorphic* spinor λ_a and the *antiholomorphic* spinor $\tilde{\lambda}_{\dot{a}}$, related to the momentum by the rank-one factorization

$$p_{a\dot{a}} = \lambda_a\, \tilde{\lambda}_{\dot{a}}, \qquad p^2 = 0, \tag{58.10}$$

and defined only up to the little-group rescaling Eq. (58.5). For real momenta $\tilde{\lambda} = \mathrm{sgn}(p^0)\, \lambda^*$; for complex momenta (which arise as an analytic-continuation tool from Chapter 59 onward) λ and $\tilde{\lambda}$ are independent. Raising and lowering of the spinor indices uses the ϵ metric of Eqs. (C.269) and (C.270), so that the all-raised form of Eq. (58.10) reads $p^{\dot{a}a} = \tilde{\lambda}^{\dot{a}} \lambda^a$, matching the upper-index bispinor used in later chapters.

Figure 58.1 summarizes the geometry: the bijection between four-vectors and Hermitian matrices, the null cone as the determinant-zero locus where the rank collapses, and the one-parameter fibre of rescalings that the little group acts along.

58.1.3 The spinors are the massless Weyl and Dirac spinors

We have produced two two-component spinors out of a null momentum by an algebraic manipulation. We now show they are not new objects: λ is precisely a solution of the massless Weyl equation of Chapter 3, and the pair $(\lambda, \tilde{\lambda})$ assembles into the massless limit of the Dirac plane-wave spinors of Chapter 11. This is what licenses calling them "the" spinors of the leg rather than an arbitrary bookkeeping device.

The holomorphic spinor solves the Weyl equation. The massless left-handed Weyl equation, in momentum space, is the statement that a left-handed field ψ_a is annihilated by $\bar{\sigma} \cdot p$, that is $(\bar{\sigma}^\mu)^{\dot{a}b} p_\mu \, \psi_b = 0$ (the momentum-space form of $i\bar{\sigma}^\mu \partial_\mu \psi = 0$, whose plane-wave content is the massless case of the laws Theorem 3.24 and Eq. (3.105)). We check directly that the holomorphic spinor of Eq. (58.10) is such a zero mode. Using the raised bispinor $p^{\dot{a}b} = (\bar{\sigma}^\mu)^{\dot{a}b} p_\mu$ and raising the indices of Eq. (58.4) with ϵ,

$$p^{\dot{a}b} = \epsilon^{\dot{a}\dot{c}} \epsilon^{bd} \, p_{d\dot{c}} = \epsilon^{\dot{a}\dot{c}} \epsilon^{bd} \, \lambda_d \tilde{\lambda}_{\dot{c}} = \tilde{\lambda}^{\dot{a}} \, \lambda^b, \tag{58.11}$$

where the relation between σ^μ and $\bar{\sigma}^\mu$ used is exactly the index-raising Eq. (C.275). Contracting with λ_b,

$$p^{\dot{a}b} \lambda_b = \tilde{\lambda}^{\dot{a}} \, (\lambda^b \lambda_b) = \tilde{\lambda}^{\dot{a}} \, \langle \lambda \, \lambda \rangle = 0, \tag{58.12}$$

because $\lambda^b \lambda_b = \epsilon^{bc} \lambda_c \lambda_b = 0$ vanishes by the antisymmetry of ϵ contracted with the symmetric (for commuting spinors) product $\lambda_c \lambda_b$ — this is the reorder identity Eq. (C.273) in the commuting case, equivalently $\langle \lambda \, \lambda \rangle = 0$. Thus λ solves the massless Weyl equation. The same computation with bars shows the antiholomorphic spinor solves the conjugate equation: from $p_{a\dot{a}} = \lambda_a \tilde{\lambda}_{\dot{a}}$,

$$(\sigma^\mu)_{a\dot{b}} p_\mu \, \tilde{\lambda}^{\dot{b}} = p_{a\dot{b}} \tilde{\lambda}^{\dot{b}} = \lambda_a \, (\tilde{\lambda}_{\dot{b}} \tilde{\lambda}^{\dot{b}}) = \lambda_a \, [\tilde{\lambda} \, \tilde{\lambda}] = 0, \tag{58.13}$$

again because $\tilde{\lambda}_{\dot{b}} \tilde{\lambda}^{\dot{b}} = 0$ by antisymmetry. So λ lives in the $(\frac{1}{2}, 0)$ representation and $\tilde{\lambda}$ in the $(0, \frac{1}{2})$ representation, and each is the unique (up to scale) zero mode of the corresponding massless wave operator at the given momentum — the two-state content of a massless particle of Section 3.7.3, now made completely concrete.

Worked Example 58.3 (Explicit spinors for a momentum in a general direction)**.** Take a positive-energy null momentum pointing in the direction $\hat{n} = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta)$,

$$p^\mu = E \, (1, \, \hat{n}), \qquad E > 0, \tag{58.14}$$

and build its spinors from scratch, then compare with the Dirac plane-wave spinors of Chapter 11. The bispinor Eq. (58.1) is $p_{a\dot a} = E(\mathbf{1} - \hat n\cdot\vec\sigma)$, and with $\hat n\cdot\vec\sigma = \left(\begin{smallmatrix}\cos\theta & \sin\theta\, e^{-i\phi}\\ \sin\theta\, e^{i\phi} & -\cos\theta\end{smallmatrix}\right)$ this is

$$p_{a\dot a} = E\begin{pmatrix} 1-\cos\theta & -\sin\theta\, e^{-i\phi}\\ -\sin\theta\, e^{i\phi} & 1+\cos\theta\end{pmatrix} = 2E\begin{pmatrix} \sin^2\frac{\theta}{2} & -\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, e^{-i\phi}\\ -\sin\frac{\theta}{2}\cos\frac{\theta}{2}\, e^{i\phi} & \cos^2\frac{\theta}{2}\end{pmatrix}, \tag{58.15}$$

using the half-angle identities $1\mp\cos\theta = 2\sin^2(\theta/2), 2\cos^2(\theta/2)$ and $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$. The matrix Eq. (58.15) is manifestly a single outer product,

$$p_{a\dot a} = \lambda_a\tilde\lambda_{\dot a}, \qquad \lambda_a = \sqrt{2E}\begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}\, e^{i\phi}\end{pmatrix}, \qquad \tilde\lambda_{\dot a} = \sqrt{2E}\begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}\, e^{-i\phi}\end{pmatrix}, \tag{58.16}$$

as one verifies by multiplying out the four entries; the determinant of Eq. (58.15) is $4E^2(\sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2}\cos^2\frac{\theta}{2}) = 0$, confirming masslessness. Since $E>0$ the reality condition Eq. (58.46) holds in the sharp form $\tilde\lambda_{\dot a} = (\lambda_a)^*$, visible directly in Eq. (58.16) as the complex conjugation of the phase $e^{\pm i\phi}$.

Specialize to motion along $+\hat z$, $\theta = 0$: then $\lambda_a = \sqrt{2E}\,(0,-1)^{\mathsf T}$, proportional to $(0,1)^{\mathsf T}$ up to the little-group phase Eq. (58.5). Compare with the massless limit of the Dirac spinor $u^2(p)$ computed in Theorem 11.10: there $u^2 \to (0,\sqrt{2E},0,0)^{\mathsf T}$ collapses onto its upper (left-handed, undotted) two-component block $\sqrt{2E}\,(0,1)^{\mathsf T}$. The two agree up to the overall phase that the little group leaves free: the holomorphic spinor extracted here from the momentum alone is exactly the left-handed block of the massless Dirac plane-wave spinor of Theorem 11.8 and Eq. (11.57). The helicity assignment matches too: along $+\hat z$ the state u^2 is the spin-down, hence negative-helicity, solution of the helicity operator Eq. (11.83), and it is precisely the surviving *left-handed* piece in the massless limit — chirality and helicity coinciding at $m=0$, the fact already noted at the end of Theorem 11.10.

The four-component embedding. For computations that still use four-component Dirac language — the QED amplitudes of QED, which we will re-express later in this chapter — it is convenient to embed the two-component spinors back into four-component objects. In the chiral basis of Eq. (C.288), where the upper block carries the undotted (left-handed) index and the lower block the dotted (right-handed) index and $\gamma^5 = \mathrm{diag}(-\mathbf{1},+\mathbf{1})$, define the angle and square kets and bras

$$|p\rangle = \begin{pmatrix}\lambda_a\\ 0\end{pmatrix}, \quad |p] = \begin{pmatrix}0\\ \tilde\lambda^{\dot a}\end{pmatrix}, \quad \langle p| = (\lambda^a,\ 0), \quad [p| = (0,\ \tilde\lambda_{\dot a}). \tag{58.17}$$

With these, the angle bracket of the next section is the four-component pairing of two angle objects, $\langle p\,q\rangle = \langle p|\,|q\rangle = \lambda_p^a\lambda_{q\,a}$, and likewise $[p\,q] = [p|\,|q] = \tilde\lambda_{p\,\dot a}\tilde\lambda_q^{\dot a}$,

the two heights following the contraction conventions Eq. (C.271); a mixed pairing $\langle p|\,|q]$ vanishes because the objects live in opposite chiral blocks. The kets are exactly the massless Dirac solutions: $|p\rangle = P_L\,u$ is the left-handed, negative-helicity spinor and $|p] = P_R\,u$ the right-handed, positive-helicity one, where $P_{L,R}$ are the chirality projectors of Chapter 11. The decomposition of the slashed momentum follows by multiplying out the outer products against $\gamma^\mu = \left(\begin{smallmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{smallmatrix}\right)$:

$$\not{p} = \begin{pmatrix} 0 & p_{a\dot b} \\ p^{\dot a b} & 0 \end{pmatrix} = \begin{pmatrix} 0 & \lambda_a \tilde\lambda_{\dot b} \\ \tilde\lambda^{\dot a}\lambda^b & 0 \end{pmatrix} = |p\rangle\,[p| + |p]\,\langle p|\,, \tag{58.18}$$

using Eqs. (58.4) and (58.11); this is the massless spin sum $\sum_\pm w_\pm \bar w_\pm = \not{p}$ of Chapter 11 written in spinor variables, and it is the identity that the later sections, and Chapter 59, deploy whenever a propagator numerator must be opened up. Because the massless equations $\not{p}\,w = 0$ for the particle spinor u and the antiparticle spinor v differ only by the (now absent) mass term, the two coincide at $m = 0$ up to helicity labelling, $u_\pm(p) = v_\mp(p)$: a single pair $(\lambda, \tilde\lambda)$ carries the on-shell data of both the particle and the antiparticle of a given helicity, which is the economy promised at the outset.

The momentum of each massless leg has now been resolved into the pair $(\lambda_a, \tilde\lambda_{\dot a})$: holomorphic and antiholomorphic spinors that solve the Weyl equations of Chapter 3, reduce to the massless Dirac spinors of Chapter 11, and carry precisely the on-shell content Wigner's classification allows — a null momentum through Eq. (58.10) and a single helicity weight through the little-group rescaling Eq. (58.5), with no covariant redundancy left over. With every external leg so equipped, the next section contracts these spinors into the angle and square brackets $\langle i\,j\rangle, [i\,j]$, fixes their signs and reality properties once and for all, and assembles the complete algebra of identities on which the remainder of Part XVI runs.

58.2 The bracket algebra and its identities

The previous section did the decisive thing: it traded the four real components of a null momentum, constrained by $p^2 = 0$, for a pair of two-component spinors λ_i and $\tilde\lambda_i$ whose outer product *is* that momentum, $p_{i,a\dot a} = \lambda_{ia}\tilde\lambda_{i\dot a}$ (Theorem 58.2 and Eq. (58.1)). These spinors carry the genuinely irreducible on-shell data: λ_i transforms in the $(\frac12, 0)$ representation of $\mathrm{SL}(2,\mathbb{C})$ and $\tilde\lambda_i$ in $(0, \frac12)$, by the Weyl transformation laws Eq. (3.105) of Chapter 3. What we do not yet have is a calculus. An amplitude is a Lorentz scalar, and Lorentz scalars must be built out of λ_i and $\tilde\lambda_i$ by contracting away their spinor indices. There are exactly two ways to contract two undotted (or two dotted) indices invariantly — against the antisymmetric ϵ tensor that is the metric of the spinor space — and these two contractions are the entire alphabet of the formalism. This section builds that

alphabet into a language: it defines the angle and square brackets, proves every identity that the rest of Part XVI will quote, and — because the signs differ from book to book and a single dropped minus propagates into every amplitude downstream — it pins the conventions down once and for all.

58.2.1 The two invariant brackets

The spinor indices $a, b, \cdots = 1, 2$ (undotted) and $\dot{a}, \dot{b}, \cdots = \dot{1}, \dot{2}$ (dotted) are raised and lowered with the antisymmetric symbols ϵ_{ab}, ϵ^{ab} and their dotted counterparts, in the conventions established for the two-component calculus in Section C.9: numerically

$$\epsilon^{12} = -\epsilon^{21} = +1, \qquad \epsilon_{12} = -\epsilon_{21} = -1, \qquad \epsilon_{ab}\epsilon^{bc} = \delta_a^{\,c}, \tag{58.19}$$

with identical numerical values for the dotted symbols, and raising/lowering by $\lambda^a = \epsilon^{ab}\lambda_b$, $\lambda_a = \epsilon_{ab}\lambda^b$ (Eq. (C.269)). The only Lorentz invariant one can form from two undotted spinors is their ϵ-contraction, and likewise for two dotted spinors. These two objects are the angle and square brackets.

Definition 58.4 (Angle and square brackets)**.** For massless legs i, j with holomorphic spinors λ_i and antiholomorphic spinors $\tilde{\lambda}_i$, define

$$\langle i\,j\rangle \;\equiv\; \epsilon^{ab}\,\lambda_{ia}\,\lambda_{jb} \;=\; \lambda_{ia}\,\lambda_j^{\,a}, \qquad [i\,j] \;\equiv\; \epsilon_{\dot{a}\dot{b}}\,\tilde{\lambda}_i^{\,\dot{a}}\,\tilde{\lambda}_j^{\,\dot{b}} \;=\; \tilde{\lambda}_i^{\,\dot{a}}\,\tilde{\lambda}_{j\dot{a}}, \tag{58.20}$$

the *angle bracket* $\langle i\,j\rangle$ and the *square bracket* $[i\,j]$. Equivalently, writing the angle ket $|i\rangle = \lambda_i$ and the square ket $|i] = \tilde{\lambda}_i$ with the bra conventions $\langle i|$, $[i|$, these are the spinor inner products $\langle i\,j\rangle = \langle i|j\rangle$ and $[i\,j] = [i|\,|j]$.

The asymmetry in Eq. (58.20) — the angle bracket contracts two *lower* undotted indices with the upper ϵ^{ab}, while the square bracket contracts two *upper* dotted indices with the lower $\epsilon_{\dot{a}\dot{b}}$ — is not cosmetic. It is precisely the choice that makes the bracket–momentum dictionary of the next subsection come out with a uniform positive sign, and it is the convention under which every formula of Chapters 59 to 62 is written. We collect the choice, together with the standing kinematic conventions of the chapter, in a notation box.

Notation Watch: bracket and momentum conventions

Throughout Part XVI the metric is mostly-minus $\eta = \mathrm{diag}(+,-,-,-)$ and natural units are in force. *All momenta are taken outgoing*, so the conservation law is $\sum_i p_i = 0$ with no sign bookkeeping per leg; a physical incoming particle of momentum p is the outgoing entry $-p$. The null bispinor is $p_{i,a\dot{a}} = \lambda_{ia}\tilde{\lambda}_{i\dot{a}} = p_{i\mu}(\sigma^\mu)_{a\dot{a}}$, with Pauli four-vectors $\sigma^\mu = (\mathbf{1}, \vec{\sigma})$,

$\bar\sigma^\mu = (\mathbf{1}, -\vec\sigma)$ as in Section C.9. The brackets are normalized so that

$$\langle i\,j\rangle\,[j\,i] \;=\; 2\,p_i\cdot p_j \;=\; s_{ij}, \tag{58.21}$$

the central relation proved in Eq. (58.24) below. For real momenta $\tilde\lambda_i = \pm\lambda_i^*$, the sign equal to $\mathrm{sgn}(p_i^0)$; for complex momenta λ_i and $\tilde\lambda_i$ are taken independent. Other references differ from Eq. (58.20) by an overall sign in one bracket or in the spinor-metric convention; the physics is insensitive to this provided Eq. (58.21) holds, which we adopt as the defining normalization.

The most basic property is antisymmetry, and it is immediate from the antisymmetry of ϵ.

Proposition 58.5 (Antisymmetry and self-vanishing)**.** *The brackets are antisymmetric,*

$$\langle i\,j\rangle = -\langle j\,i\rangle, \qquad [i\,j] = -[j\,i], \tag{58.22}$$

and in particular a leg has vanishing bracket with itself,

$$\langle i\,i\rangle = 0, \qquad [i\,i] = 0. \tag{58.23}$$

Proof. From Eq. (58.20), $\langle j\,i\rangle = \epsilon^{ab}\lambda_{ja}\lambda_{ib}$; relabelling the summed indices $a \leftrightarrow b$ and using $\epsilon^{ba} = -\epsilon^{ab}$ gives $\langle j\,i\rangle = \epsilon^{ba}\lambda_{jb}\lambda_{ia} = -\epsilon^{ab}\lambda_{ia}\lambda_{jb} = -\langle i\,j\rangle$, where the commuting (bosonic) spinor components were freely reordered. The square-bracket statement is identical with $\epsilon_{\dot a\dot b}$. Setting $j = i$ in Eq. (58.22) forces $\langle i\,i\rangle = -\langle i\,i\rangle = 0$, and similarly for the square bracket. □

That $\langle i\,i\rangle = 0$ has a transparent meaning: the angle bracket of a leg with itself is the contraction $\epsilon^{ab}\lambda_{ia}\lambda_{ib}$ of a symmetric pair $\lambda_{ia}\lambda_{ib}$ against the antisymmetric ϵ, which vanishes. A bracket $\langle i\,j\rangle$ therefore measures the failure of λ_i and λ_j to be proportional: it is zero precisely when $\lambda_i \propto \lambda_j$, i.e. when the two legs share the same holomorphic spinor. This is the spinorial face of collinearity, and it is why brackets sit naturally in the denominators of amplitudes, where they encode the poles of collinear and soft kinematics.

58.2.2 The bracket–Mandelstam dictionary

The single most important identity of the formalism is the one that connects the brackets, which are holomorphic and antiholomorphic objects of weight one, to the Lorentz invariants $p_i\cdot p_j$ that the Feynman rules of Parts IV–VII produce. It is what guarantees that anything expressible in Mandelstam variables can be rewritten in brackets, and conversely.

Theorem 58.6 (Bracket–Mandelstam dictionary)**.** *For any two massless legs,*

$$\langle i\,j\rangle\,[j\,i] \;=\; 2\,p_i\cdot p_j. \tag{58.24}$$

In particular, for the two-particle Mandelstam invariant of Theorem 16.20, $s_{ij}\equiv(p_i+p_j)^2=2\,p_i\cdot p_j=\langle i\,j\rangle[j\,i]$ when both legs are massless.

Proof. We compute $2\,p_i\cdot p_j$ directly from the bispinors and the Pauli-matrix trace identity, showing every contraction. Write the two momenta in their lower and upper bispinor forms, $p_{i,a\dot a}=p_{i\mu}(\sigma^\mu)_{a\dot a}$ and $p_j^{\dot a a}=p_{j\nu}(\bar\sigma^\nu)^{\dot a a}$. The invariant pairing is read off from the $\sigma/\bar\sigma$ trace identity $\mathrm{tr}(\sigma^\mu\bar\sigma^\nu)\equiv(\sigma^\mu)_{a\dot b}(\bar\sigma^\nu)^{\dot b a}=2\eta^{\mu\nu}$ of Eq. (C.279):

$$p_{i,a\dot a}\,p_j^{\dot a a}=p_{i\mu}p_{j\nu}\,(\sigma^\mu)_{a\dot a}(\bar\sigma^\nu)^{\dot a a}=p_{i\mu}p_{j\nu}\,2\eta^{\mu\nu}=2\,p_i\cdot p_j. \tag{58.25}$$

Now insert the spinor factorization. From $p_{i,a\dot a}=\lambda_{ia}\tilde\lambda_{i\dot a}$ and the $\sigma/\bar\sigma$ relation $(\bar\sigma^\mu)^{\dot a a}=\epsilon^{\dot a\dot b}\epsilon^{ab}(\sigma^\mu)_{b\dot b}$ of Eq. (C.275), the upper bispinor is the doubly-raised lower one,

$$p_j^{\dot a a}=\epsilon^{\dot a\dot b}\epsilon^{ab}\,p_{j,b\dot b}=\epsilon^{\dot a\dot b}\epsilon^{ab}\,\lambda_{jb}\tilde\lambda_{j\dot b}=(\epsilon^{ab}\lambda_{jb})(\epsilon^{\dot a\dot b}\tilde\lambda_{j\dot b})=\lambda_j^{\,a}\,\tilde\lambda_j^{\,\dot a}. \tag{58.26}$$

Substituting Eq. (58.26) and $p_{i,a\dot a}=\lambda_{ia}\tilde\lambda_{i\dot a}$ into the left side of Eq. (58.25) and grouping the undotted factors with the undotted and the dotted with the dotted,

$$2\,p_i\cdot p_j=\lambda_{ia}\tilde\lambda_{i\dot a}\,\lambda_j^{\,a}\tilde\lambda_j^{\,\dot a}=(\lambda_{ia}\,\lambda_j^{\,a})\,(\tilde\lambda_{i\dot a}\,\tilde\lambda_j^{\,\dot a}). \tag{58.27}$$

The first factor is the angle bracket of Eq. (58.20) verbatim: $\lambda_{ia}\lambda_j^{\,a}=\langle i\,j\rangle$. The second factor is $\tilde\lambda_{i\dot a}\tilde\lambda_j^{\,\dot a}$; lowering the j-index and raising the i-index, $\tilde\lambda_{i\dot a}\tilde\lambda_j^{\,\dot a}=\tilde\lambda_i^{\,\dot a}\tilde\lambda_{j\dot a}\cdot(-1)\cdot(-1)$ is the same number written the other way, and comparing with Eq. (58.20) one finds $\tilde\lambda_{i\dot a}\tilde\lambda_j^{\,\dot a}=[j\,i]$ — the indices are summed first-down on i and up on j, which is the square bracket with the labels in the order j,i. Explicitly, $\tilde\lambda_{i\dot a}\tilde\lambda_j^{\,\dot a}=\tilde\lambda_{i\dot 1}\tilde\lambda_j^{\,\dot 1}+\tilde\lambda_{i\dot 2}\tilde\lambda_j^{\,\dot 2}=\tilde\lambda_{i\dot 1}\tilde\lambda_{j\dot 2}-\tilde\lambda_{i\dot 2}\tilde\lambda_{j\dot 1}$ (using $\tilde\lambda_j^{\,\dot 1}=\tilde\lambda_{j\dot 2}$, $\tilde\lambda_j^{\,\dot 2}=-\tilde\lambda_{j\dot 1}$ from Eq. (58.19)), while $[j\,i]=\epsilon_{\dot a\dot b}\tilde\lambda_j^{\,\dot a}\tilde\lambda_i^{\,\dot b}$ evaluates to the same $\tilde\lambda_{i\dot 1}\tilde\lambda_{j\dot 2}-\tilde\lambda_{i\dot 2}\tilde\lambda_{j\dot 1}$. Hence Eq. (58.27) reads $2\,p_i\cdot p_j=\langle i\,j\rangle[j\,i]$, which is Eq. (58.24). The identification with s_{ij} follows because $(p_i+p_j)^2=p_i^2+p_j^2+2p_i\cdot p_j=2p_i\cdot p_j$ for $p_i^2=p_j^2=0$. □

Two remarks fix the dictionary in the reader's hands. First, the *order of the labels matters and is correlated between the two brackets*: it is $\langle i\,j\rangle[j\,i]$, not $\langle i\,j\rangle[i\,j]$, that equals $2p_i\cdot p_j$; the two differ by a sign because $[i\,j]=-[j\,i]$. Reversing both brackets, $\langle j\,i\rangle[i\,j]=(-\langle i\,j\rangle)(-[j\,i])=\langle i\,j\rangle[j\,i]$, leaves the invariant unchanged, as it must since $p_i\cdot p_j=p_j\cdot p_i$. Second, the dictionary already exhibits the holomorphic split that powers the whole subject: s_{ij} factorizes into an angle bracket carrying only the λ's and a square bracket carrying only the $\tilde\lambda$'s. A real Mandelstam invariant is a product of a holomorphic and an antiholomorphic

half, and the analytic continuation to complex momenta — in which the two halves become independent — is exactly what lets a single half vanish while the invariant stays finite. That is the kinematic room three-point amplitudes will need.

58.2.3 The mixed sandwich and the Gordon identity

Amplitudes with fermion lines or with vector polarizations require one more object: a quantity carrying one Lorentz vector index, built from one angle and one square spinor. This is the mixed sandwich.

Definition 58.7 (Mixed spinor sandwich)**.** For massless legs i, j define the Lorentz four-vector

$$\langle i|\gamma^\mu|j] \;\equiv\; \lambda_i^a\,(\sigma^\mu)_{a\dot a}\,\tilde\lambda_j^{\dot a}, \qquad [i|\gamma^\mu|j\rangle \;\equiv\; \tilde\lambda_{i\dot a}\,(\bar\sigma^\mu)^{\dot a a}\,\lambda_{ja}. \tag{58.28}$$

The notation records the data: $\langle i|\gamma^\mu|j] = \langle i|\,\gamma^\mu|j]$ contracts an angle bra on the left with a square ket on the right; $[i|\gamma^\mu|j\rangle = [i|\gamma^\mu\,|j\rangle$ does the reverse.

The names γ^μ, $\langle i|$, $|j]$ are not loose analogy. In the chiral basis of Section C.9 the Dirac matrix is $\gamma^\mu = \left(\begin{smallmatrix} 0 & \sigma^\mu \\ \bar\sigma^\mu & 0 \end{smallmatrix}\right)$, and a massless Dirac spinor decomposes into the Weyl pieces λ and $\tilde\lambda$; the objects in Eq. (58.28) are exactly the surviving chiral blocks of a four-component bilinear $\bar u_i \gamma^\mu u_j$. The two sandwiches are not independent: they are complex-conjugate transposes of one another in the sense made precise by the following identity, which we will use constantly to flip a sandwich's chirality.

Proposition 58.8 (Reversal of the sandwich)**.** *For all i, j,*

$$\langle i|\gamma^\mu|j] \;=\; [j|\gamma^\mu|i\rangle. \tag{58.29}$$

Proof. Start from the right side and use $(\bar\sigma^\mu)^{\dot a a} = \epsilon^{\dot a \dot b}\epsilon^{ab}(\sigma^\mu)_{b\dot b}$ (Eq. (C.275)):

$$[j|\gamma^\mu|i\rangle = \tilde\lambda_{j\dot a}(\bar\sigma^\mu)^{\dot a a}\lambda_{ia} = \tilde\lambda_{j\dot a}\,\epsilon^{\dot a\dot b}\epsilon^{ab}(\sigma^\mu)_{b\dot b}\,\lambda_{ia} = (\epsilon^{ab}\lambda_{ia})(\sigma^\mu)_{b\dot b}(\epsilon^{\dot a\dot b}\tilde\lambda_{j\dot a}).$$

Now $\epsilon^{ab}\lambda_{ia} = -\epsilon^{ba}\lambda_{ia} = -\lambda_i^b$ and $\epsilon^{\dot a\dot b}\tilde\lambda_{j\dot a} = -\tilde\lambda_j^{\dot b}$, by the raising convention of Eq. (58.19). The two minus signs cancel, leaving $[j|\gamma^\mu|i\rangle = \lambda_i^b(\sigma^\mu)_{b\dot b}\tilde\lambda_j^{\dot b} = \langle i|\gamma^\mu|j]$ by Eq. (58.28). □

Setting $i = j$ in the sandwich recovers the momentum itself: the formalism's version of the Gordon identity.

Proposition 58.9 (Gordon identity for null momenta)**.** *For a single massless leg,*

$$\langle i|\gamma^\mu|i] \;=\; 2\,p_i^\mu \;=\; [i|\gamma^\mu|i\rangle. \tag{58.30}$$

Proof. From Eq. (58.28), $\langle i|\gamma^\mu|i] = \lambda_i^a(\sigma^\mu)_{a\dot a}\tilde\lambda_i^{\dot a} = (\sigma^\mu)_{a\dot a}\,\lambda_i^a\tilde\lambda_i^{\dot a} = (\sigma^\mu)_{a\dot a}\,p_i^{\dot a a}$, using $p_i^{\dot a a} = \lambda_i^a\tilde\lambda_i^{\dot a}$ from Eq. (58.26). Writing $p_i^{\dot a a} = p_{i\nu}(\bar\sigma^\nu)^{\dot a a}$ and applying the trace identity Eq. (C.279) once more, $(\sigma^\mu)_{a\dot a}(\bar\sigma^\nu)^{\dot a a} = 2\eta^{\mu\nu}$, so $\langle i|\gamma^\mu|i] = p_{i\nu}\,2\eta^{\mu\nu} = 2p_i^\mu$. The second equality in Eq. (58.30) is Eq. (58.29) at $i=j$. □

A useful corollary of Eq. (58.30) and the dictionary is the chain rule for inserting a sum of momenta between an angle bra and a square ket. If $P=\sum_k p_k$ is a sum of null momenta, then $\langle i|\not{P}|j] \equiv \sum_k\langle i|\gamma^\mu|\,]p_{k\mu}|j] = \sum_k\langle i\,k\rangle[k\,j]$, because each null insertion factorizes through its spinors, $\langle i|\not{p}_k|j] = \lambda_i^a(\sigma^\mu)_{a\dot a}p_k^\mu\tilde\lambda_j^{\dot a} = \langle i\,k\rangle[k\,j]$; the last step uses $p_k^\mu(\sigma^\mu)_{a\dot a} = p_{k,a\dot a} = \lambda_{ka}\tilde\lambda_{k\dot a}$ and the contractions $\lambda_i^a\lambda_{ka} = \langle i\,k\rangle$, $\tilde\lambda_{k\dot a}\tilde\lambda_j^{\dot a} = [k\,j]$. We will write this as

$$\langle i|P|j] = \sum_k \langle i\,k\rangle[k\,j], \qquad P=\sum_k p_k, \tag{58.31}$$

the basic device for resolving an internal momentum into on-shell spinors.

58.2.4 The Fierz completeness relation

The sandwiches multiply through a single rearrangement identity, the two-component Fierz (or Gordon completeness) relation. It is the engine that collapses products of fermion bilinears and polarization contractions into pure brackets, and it descends directly from the σ-matrix completeness proved in the appendix.

Theorem 58.10 (Fierz / Gordon completeness in brackets)**.** *For any massless legs i,j,k,l,*

$$\langle i|\gamma^\mu|j]\,\langle k|\gamma_\mu|l] \;=\; 2\,\langle i\,k\rangle\,[l\,j]. \tag{58.32}$$

Proof. The input is the two-component completeness relation Eq. (C.280),

$$(\sigma^\mu)_{a\dot a}\,(\bar\sigma_\mu)^{\dot c d} \;=\; 2\,\delta_a^{\,d}\,\delta_{\dot a}^{\dot c}, \tag{58.33}$$

which expresses that the four matrices σ^μ resolve the identity on the space of 2×2 mixed-index matrices. We must bring the two sandwiches in Eq. (58.32) into the form in which Eq. (58.33) applies. Write both with lower-index Pauli matrices:

$$\langle i|\gamma^\mu|j]\,\langle k|\gamma_\mu|l] = \lambda_i^a(\sigma^\mu)_{a\dot a}\tilde\lambda_j^{\dot a}\;\lambda_k^b(\sigma_\mu)_{b\dot b}\tilde\lambda_l^{\dot b} = \lambda_i^{\,a}\tilde\lambda_j^{\dot a}\lambda_k^{\,b}\tilde\lambda_l^{\,\dot b}\;(\sigma^\mu)_{a\dot a}(\sigma_\mu)_{b\dot b}. \tag{58.34}$$

The matrix product $(\sigma^\mu)_{a\dot a}(\sigma_\mu)_{b\dot b}$ has all four indices down; convert the second factor to $\bar\sigma$ using $(\sigma_\mu)_{b\dot b} = \epsilon_{bd}\epsilon_{\dot b\dot c}(\bar\sigma_\mu)^{\dot c d}$, which is Eq. (C.275) read in reverse. Then

$$(\sigma^\mu)_{a\dot a}(\sigma_\mu)_{b\dot b} = \epsilon_{bd}\epsilon_{\dot b\dot c}\,(\sigma^\mu)_{a\dot a}(\bar\sigma_\mu)^{\dot c d} = \epsilon_{bd}\epsilon_{\dot b\dot c}\,2\,\delta_a^{\,d}\delta_{\dot a}^{\dot c} = 2\,\epsilon_{ba}\,\epsilon_{\dot b\dot a}, \tag{58.35}$$

where Eq. (58.33) was used and the Kronecker deltas collapsed $d \to a$, $\dot{c} \to \dot{a}$. Substituting Eq. (58.35) into Eq. (58.34) and grouping,

$$\langle i|\gamma^{\mu}|j]\,\langle k|\gamma_{\mu}|l] = 2\,(\epsilon_{ba}\,\lambda_i^{a}\lambda_k^{b})(\epsilon_{\dot{b}\dot{a}}\,\tilde{\lambda}_j^{\dot{a}}\tilde{\lambda}_l^{\dot{b}}). \tag{58.36}$$

The undotted factor is $\epsilon_{ba}\lambda_i^{a}\lambda_k^{b} = -\epsilon_{ab}\lambda_i^{a}\lambda_k^{b} = \langle i\,k\rangle$, the last step because $\epsilon_{ab}\lambda_i^{a}\lambda_k^{b} = -\langle i\,k\rangle$ (raising both indices in Eq. (58.20) flips the sign of ϵ, by Eq. (58.19)). The dotted factor is $\epsilon_{\dot{b}\dot{a}}\tilde{\lambda}_j^{\dot{a}}\tilde{\lambda}_l^{\dot{b}} = \epsilon_{\dot{a}\dot{b}}\tilde{\lambda}_l^{\dot{a}}\tilde{\lambda}_j^{\dot{b}} = [l\,j]$ after relabelling $\dot{a} \leftrightarrow \dot{b}$. Hence Eq. (58.36) is $2\,\langle i\,k\rangle[l\,j]$, which is Eq. (58.32). □

The structure of Eq. (58.32) is worth memorizing in its mnemonic form: the contracted vector indices "glue" the two angle bras into one angle bracket and the two square kets into one square bracket, with the angle bracket pairing the two left labels i, k and the square bracket pairing the two right labels l, j *in reversed order*. The reversal is the same one that appeared in the dictionary Eq. (58.24), and it is again forced by antisymmetry. A frequently used special case is $i = k$, $j = l$, which gives $\langle i|\gamma^{\mu}|j]\langle i|\gamma_{\mu}|j] = 2\langle i\,i\rangle[j\,j] = 0$: a sandwich is null, as it must be, since by Eq. (58.30) the diagonal sandwich is $2p_i^{\mu}$ and $p_i^2 = 0$, and the off-diagonal one squares to $\langle i|\gamma^{\mu}|j]\langle i|\gamma_{\mu}|j]$.

58.2.5 The Schouten identity

The brackets satisfy one further relation that has no analogue for ordinary dot products and is responsible for many of the "miraculous" simplifications of amplitude algebra. It is a pure consequence of the two-dimensionality of the spinor space: three vectors in a two-dimensional space are always linearly dependent.

Theorem 58.11 (Schouten identity)**.** *For any four massless legs i, j, k, l,*

$$\langle i\,j\rangle\langle k\,l\rangle + \langle i\,k\rangle\langle l\,j\rangle + \langle i\,l\rangle\langle j\,k\rangle = 0, \tag{58.37}$$

and identically for the square brackets,

$$[i\,j][k\,l] + [i\,k][l\,j] + [i\,l][j\,k] = 0. \tag{58.38}$$

Proof. Fix three undotted spinors $\lambda_j, \lambda_k, \lambda_l$. Any totally antisymmetric object in three two-valued indices vanishes, since at least two of the three indices must coincide; in particular

$$\lambda_j^{[a}\,\epsilon^{bc]} = 0, \tag{58.39}$$

where the bracket denotes antisymmetrization over a, b, c. Writing out the three cyclic terms of Eq. (58.39) (the antisymmetrization of an index against the already-antisymmetric ϵ^{bc} produces exactly the cyclic sum) gives the spinor identity

$$\lambda_j^{a}\,\epsilon^{bc} + \lambda_j^{b}\,\epsilon^{ca} + \lambda_j^{c}\,\epsilon^{ab} = 0. \tag{58.40}$$

That this holds is checked at once: ϵ^{bc} is nonzero only for $b \neq c$, so for fixed $b \neq c$ the free index a takes one of the two values, and the three terms reorganize the two-component vector λ_j in the basis $\{\lambda_k, \lambda_l\}$ — equivalently, Eq. (58.40) is the statement that the 2×3 array of components (λ_j^a) together with ϵ has a vanishing 3-index antisymmetrization. Now contract Eq. (58.40) with $\lambda_{ia}\lambda_{kb}\lambda_{lc}$. Term by term, using $\lambda_{ia}\lambda_j^a = \langle i\, j\rangle$ and $\epsilon^{bc}\lambda_{kb}\lambda_{lc} = \langle k\, l\rangle$ (and the cyclic relabellings):

$$\begin{aligned}
\lambda_{ia}\lambda_{kb}\lambda_{lc}\,\lambda_j^a\epsilon^{bc} &= \langle i\, j\rangle\langle k\, l\rangle, \\
\lambda_{ia}\lambda_{kb}\lambda_{lc}\,\lambda_j^b\epsilon^{ca} &= \langle k\, j\rangle\,\epsilon^{ca}\lambda_{lc}\lambda_{ia} = \langle k\, j\rangle\langle l\, i\rangle = \langle i\, k\rangle\langle l\, j\rangle, \\
\lambda_{ia}\lambda_{kb}\lambda_{lc}\,\lambda_j^c\epsilon^{ab} &= \langle l\, j\rangle\,\epsilon^{ab}\lambda_{ia}\lambda_{kb} = \langle l\, j\rangle\langle i\, k\rangle,
\end{aligned}$$

where in the second and third lines the antisymmetry Eq. (58.22) was used to fix the label order. Summing the three contractions and applying Eq. (58.40) gives

$$\langle i\, j\rangle\langle k\, l\rangle + \langle i\, k\rangle\langle l\, j\rangle + \langle i\, k\rangle\langle l\, j\rangle \overset{?}{=} 0,$$

but the second and third contractions are not independent: a cleaner bookkeeping is to contract Eq. (58.40) once, holding the roles of i, k, l fixed, which yields directly the three distinct terms $\langle i\, j\rangle\langle k\, l\rangle + \langle i\, k\rangle\langle l\, j\rangle + \langle i\, l\rangle\langle j\, k\rangle = 0$. We exhibit this cleaner route explicitly. Multiply Eq. (58.40) by λ_{ia} alone:

$$\langle i\, j\rangle\,\epsilon^{bc} + \lambda_j^b(\lambda_{ia}\epsilon^{ca}) + \lambda_j^c(\lambda_{ia}\epsilon^{ab}) = 0 \implies \langle i\, j\rangle\,\epsilon^{bc} = \lambda_j^b\lambda_i^c - \lambda_j^c\lambda_i^b, \tag{58.41}$$

using $\lambda_{ia}\epsilon^{ca} = -\lambda_{ia}\epsilon^{ac} = -\lambda_i^c$ and $\lambda_{ia}\epsilon^{ab} = \lambda_i^b$. Equation Eq. (58.41) is the 2×2 "ϵ as a determinant" identity; contract it with $\lambda_{kb}\lambda_{lc}$:

$$\langle i\, j\rangle\,\langle k\, l\rangle = (\lambda_j^b\lambda_{kb})(\lambda_i^c\lambda_{lc}) - (\lambda_j^c\lambda_{lc})(\lambda_i^b\lambda_{kb}) = \langle j\, k\rangle\langle i\, l\rangle - \langle j\, l\rangle\langle i\, k\rangle. \tag{58.42}$$

Rearranging Eq. (58.42) and using antisymmetry, $\langle i\, j\rangle\langle k\, l\rangle + \langle i\, k\rangle\langle l\, j\rangle + \langle i\, l\rangle\langle j\, k\rangle = 0$, since $\langle j\, k\rangle\langle i\, l\rangle = \langle i\, l\rangle\langle j\, k\rangle$ and $-\langle j\, l\rangle\langle i\, k\rangle = \langle i\, k\rangle\langle l\, j\rangle$. This is Eq. (58.37). The square-bracket identity Eq. (58.38) follows by the identical argument applied to the dotted spinors and $\epsilon_{\dot a\dot b}$. □

The single most useful avatar of Schouten is the "add and subtract a leg" move: Eq. (58.41) rewritten as

$$\langle i\, j\rangle\langle k\, l\rangle = \langle i\, k\rangle\langle j\, l\rangle - \langle i\, l\rangle\langle j\, k\rangle \quad\Longleftrightarrow\quad |i\rangle\,\langle j\, k\rangle = |j\rangle\,\langle i\, k\rangle - |k\rangle\,\langle i\, j\rangle, \tag{58.43}$$

the right form holding as an identity among angle kets (contract any $\langle l|$ to recover the left form). It lets one reshuffle which leg sits inside a bracket, the operation that turns one rational expression for an amplitude into a manifestly different-looking but equal one — the reason the same gluon amplitude appears in the literature in superficially distinct "forms."

58.2.6 Momentum conservation in spinor variables

For an n-point amplitude the only kinematic constraint beyond the on-shell conditions (already solved by the spinor parametrization) is total momentum conservation. With all momenta outgoing it reads $\sum_{i=1}^{n} p_i = 0$, and in bispinor form it is a statement about the spinors directly.

Proposition 58.12 (Momentum conservation and its bracket form)**.** *Total momentum conservation $\sum_i p_i = 0$ is equivalent to the vanishing of the bispinor sum,*

$$\sum_{i=1}^{n} \lambda_{ia}\, \tilde{\lambda}_{i\dot{a}} = 0, \tag{58.44}$$

and contracting between any angle bra $\langle x|$ and square ket $|y]$ yields the bracket identity

$$\sum_{i=1}^{n} \langle x\, i\rangle\, [i\, y] = 0 \tag{58.45}$$

for any (not necessarily external) reference spinors $\lambda_x, \tilde{\lambda}_y$. In particular $\sum_i \langle a\, i\rangle [i\, b] = 0$ for any two external legs a, b.

Proof. Each null momentum is $p_{i,a\dot{a}} = p_{i\mu}(\sigma^\mu)_{a\dot{a}} = \lambda_{ia}\tilde{\lambda}_{i\dot{a}}$. Since $(\sigma^\mu)_{a\dot{a}}$ is, as μ runs over $0, \dots, 3$, a basis of the four-dimensional space of 2×2 Hermitian matrices (its components are invertibly related to the p_μ, this being the content of the bispinor map of Theorem 58.2), the four-vector equation $\sum_i p_{i\mu} = 0$ holds if and only if the matrix equation $\sum_i p_{i,a\dot{a}} = 0$ holds, i.e. Eq. (58.44). Contracting Eq. (58.44) with λ_x^a on the undotted index and $\tilde{\lambda}_y^{\dot{a}}$ on the dotted index, and using $\lambda_x^a \lambda_{ia} = \langle x\, i\rangle$ together with $\tilde{\lambda}_{i\dot{a}}\tilde{\lambda}_y^{\dot{a}} = [i\, y]$ (the latter exactly as in the dictionary proof, Eq. (58.27)), gives $\sum_i \langle x\, i\rangle [i\, y] = 0$, which is Eq. (58.45). Choosing $\lambda_x = \lambda_a$ and $\tilde{\lambda}_y = \tilde{\lambda}_b$ specializes to the external statement, in which the terms $i = a$ and $i = b$ drop out automatically because $\langle a\, a\rangle = 0$ and $[b\, b] = 0$. □

Equation Eq. (58.45) is the spinor-helicity replacement for the four scalar conservation equations: it is a single bracket relation, holomorphic in λ_x and antiholomorphic in $\tilde{\lambda}_y$, and it is the identity that every amplitude manipulation invokes to eliminate one leg in favour of the others. It is the bracket form of the operator statement $\sum_i |i\rangle\, [i| = 0$ used throughout Chapter 59.

58.2.7 Reality and the continuation to complex momenta

The brackets were defined for arbitrary $\lambda_i, \tilde{\lambda}_i$, but for *physical, real* momenta the two spinors of a leg are not independent: they are complex conjugates, up to a sign fixed by the energy. This is what ties the formalism back to honest scattering, and it is the constraint we will deliberately relax to give three-point amplitudes and the BCFW shift room to live.

Proposition 58.13 (Reality condition)**.** *A bispinor $p_{a\dot a} = \lambda_a \tilde\lambda_{\dot a}$ corresponds to a real four-momentum p_μ if and only if the matrix $p_{a\dot a}$ is Hermitian, and this forces*

$$\tilde\lambda_{\dot a} = \pm\,(\lambda_a)^*, \qquad \textit{sign} = \operatorname{sgn}(p^0), \tag{58.46}$$

that is $\tilde\lambda = +\lambda^*$ for a future-pointing (positive-energy) null momentum and $\tilde\lambda = -\lambda^*$ for a past-pointing one.

Proof. The map $p_\mu \mapsto p_{a\dot a} = p_\mu (\sigma^\mu)_{a\dot a}$ sends real four-vectors to Hermitian 2×2 matrices and back, because $\sigma^0 = \mathbf{1}$ and the σ^i are Hermitian, so $p^\dagger_{a\dot a} = p^*_\mu (\sigma^\mu)^\dagger = p^*_\mu(\sigma^\mu)$; thus $p_{a\dot a}$ is Hermitian iff every p_μ is real. Now impose Hermiticity on the rank-one form $p_{a\dot a} = \lambda_a\tilde\lambda_{\dot a}$. Hermiticity means $\lambda_a\tilde\lambda_{\dot a} = (\lambda_{\dot a}\tilde\lambda_a)^*$ understood as $\overline{\lambda_b\tilde\lambda_{\dot b}}$ with indices transposed; component-wise, the matrix $M_{a\dot a} = \lambda_a\tilde\lambda_{\dot a}$ equals $M^\dagger_{a\dot a} = \overline{\lambda_{\dot a}}\,\overline{\tilde\lambda_a}$. A rank-one matrix $vw^\dagger$ is Hermitian iff w is a real multiple of v; hence $\tilde\lambda_{\dot a} = c\,(\lambda_a)^*$ for some real constant c, which the little-group rescaling $\lambda \to t\lambda, \tilde\lambda \to t^{-1}\tilde\lambda$ may normalize to $c = \pm 1$. The sign is determined by the energy: $p^0 = \tfrac12 \operatorname{tr}(p_{a\dot a}) = \tfrac12(\lambda_1\tilde\lambda_{\dot 1} + \lambda_2\tilde\lambda_{\dot 2})$, and with $\tilde\lambda = c\lambda^*$ this is $\tfrac{c}{2}(|\lambda_1|^2 + |\lambda_2|^2)$, a positive multiple of c. Therefore $c = \operatorname{sgn}(p^0)$, giving Eq. (58.46). □

Two consequences are immediate and will be used repeatedly. First, for real momenta the brackets are complex conjugates of one another: from $\tilde\lambda_i = \pm\lambda_i^*$,

$$[i\,j] = \pm_i\,\pm_j\,\overline{\langle i\,j\rangle}, \tag{58.47}$$

so for two future-pointing legs $[i\,j] = \overline{\langle i\,j\rangle}$ and the dictionary Eq. (58.24) reads $s_{ij} = \langle i\,j\rangle\overline{\langle i\,j\rangle}\cdot(\pm) = \pm|\langle i\,j\rangle|^2$, manifestly real, with sign tracking whether the pair is in a physical s-channel ($s_{ij} > 0$) or a crossed channel. Second, and decisively, Eq. (58.47) says that for real kinematics $\langle i\,j\rangle = 0 \iff [i\,j] = 0$: the angle and square brackets of a pair vanish together. This is exactly why the three-point amplitude is degenerate on real momenta — all invariants $s_{ij} = 2p_i\cdot p_j$ vanish for three on-shell legs summing to zero, dragging both brackets to zero — and why one must analytically continue.

Physical Insight: why complex momenta are not optional

The reality condition Eq. (58.46) ties $\tilde\lambda$ to λ^*, so on real momenta the holomorphic and antiholomorphic data are the *same* information conjugated. The power of the formalism comes from breaking this tie. Declaring λ_i and $\tilde\lambda_i$ *independent complex* spinors — which is a legitimate analytic continuation, since amplitudes are rational functions of the brackets and a rational identity proved on a real slice holds on its complexification — lets an angle bracket vanish while the corresponding square bracket stays finite. The three-point amplitude (Section 58.3 and the section after) lives entirely in this complexified kinematics, where it is nonzero and holomorphic-or-

antiholomorphic; the BCFW recursion of Chapter 59 deforms momenta into the complex plane for the same reason. Complexification is not a trick laid on top of the formalism: it is the natural home of the brackets, with real kinematics the special locus $\tilde{\lambda} = \pm\lambda^*$.

58.2.8 Worked computations

We close with two computations that exercise the whole toolkit and produce results used later.

Worked Example 58.14 (Mandelstam invariants of a four-point massless process in brackets)**.** Take four massless legs, all outgoing, $\sum_{i=1}^{4} p_i = 0$. We express the Mandelstam invariants of Theorem 16.20 in brackets and rederive their constraint.

By the dictionary Eq. (58.24), the three independent two-particle invariants are

$$s \equiv s_{12} = \langle 1\,2\rangle[2\,1], \qquad t \equiv s_{13} = \langle 1\,3\rangle[3\,1], \qquad u \equiv s_{14} = \langle 1\,4\rangle[4\,1]. \tag{58.48}$$

Here $s = (p_1+p_2)^2$, $t = (p_1+p_3)^2$, $u = (p_1+p_4)^2$ in the all-outgoing convention, which matches Theorem 16.20 after the crossing $p_3 \to -p_3$, $p_4 \to -p_4$ that converts the two physical incoming legs to outgoing ones; with all four massless, $s+t+u = 0$. We verify the constraint directly from the bracket form, without ever invoking the dot-product algebra. Apply the momentum-conservation identity Eq. (58.45) with the reference spinors chosen to be those of leg 1, $x = y = 1$:

$$\sum_{i=1}^{4} \langle 1\,i\rangle[i\,1] = 0. \tag{58.49}$$

The $i = 1$ term vanishes since $\langle 1\,1\rangle = 0$, leaving

$$\langle 1\,2\rangle[2\,1] + \langle 1\,3\rangle[3\,1] + \langle 1\,4\rangle[4\,1] = 0,$$

which is precisely $s + t + u = 0$ by Eq. (58.48). The Mandelstam constraint Eq. (16.121), here in its massless form $s + t + u = \sum_i m_i^2 = 0$, is thus a one-line consequence of spinor momentum conservation. This is the first concrete payoff of the formalism: a kinematic identity that took a short calculation with dot products and the conservation law in Chapter 16 is now a single contraction.

Worked Example 58.15 (Schouten in action: a denominator rearrangement)**.** We show how Eq. (58.37) converts one rational kinematic expression into another, the manipulation that underlies the equality of superficially different amplitude formulas. Consider the combination, for five massless legs,

$$X = \frac{\langle 1\,3\rangle}{\langle 1\,2\rangle\langle 2\,3\rangle} + \frac{\langle 3\,4\rangle}{\langle 2\,3\rangle\langle 2\,4\rangle}.$$

Place over the common denominator $\langle 1\,2\rangle\langle 2\,3\rangle\langle 2\,4\rangle$:

$$X = \frac{\langle 1\,3\rangle\langle 2\,4\rangle + \langle 3\,4\rangle\langle 1\,2\rangle}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 2\,4\rangle}.$$

Apply Schouten Eq. (58.37) to the numerator. Taking the four labels in the identity to be $(i,j,k,l) = (1,3,2,4)$ gives $\langle 1\,3\rangle\langle 2\,4\rangle + \langle 1\,2\rangle\langle 4\,3\rangle + \langle 1\,4\rangle\langle 3\,2\rangle = 0$, i.e. $\langle 1\,3\rangle\langle 2\,4\rangle = \langle 1\,2\rangle\langle 3\,4\rangle + \langle 1\,4\rangle\langle 2\,3\rangle$ after using antisymmetry $\langle 4\,3\rangle = -\langle 3\,4\rangle$, $\langle 3\,2\rangle = -\langle 2\,3\rangle$. Substituting,

$$\langle 1\,3\rangle\langle 2\,4\rangle + \langle 3\,4\rangle\langle 1\,2\rangle = \big(\langle 1\,2\rangle\langle 3\,4\rangle + \langle 1\,4\rangle\langle 2\,3\rangle\big) + \langle 1\,2\rangle\langle 3\,4\rangle = 2\,\langle 1\,2\rangle\langle 3\,4\rangle + \langle 1\,4\rangle\langle 2\,3\rangle.$$

Hence

$$X = \frac{2\,\langle 1\,2\rangle\langle 3\,4\rangle + \langle 1\,4\rangle\langle 2\,3\rangle}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 2\,4\rangle} = \frac{2\,\langle 3\,4\rangle}{\langle 2\,3\rangle\langle 2\,4\rangle} + \frac{\langle 1\,4\rangle}{\langle 1\,2\rangle\langle 2\,4\rangle},$$

a different-looking expression for the same function. The lesson generalizes: every Schouten move trades a term with one collinear pole for two terms with different poles, and amplitude expressions that differ by such moves are equal even when no common factor makes it obvious. We will lean on this when comparing the Parke–Taylor formula of Chapter 59 against its BCFW-generated form.

The brackets, their dictionary to Mandelstam invariants, the Fierz and Schouten identities, momentum conservation, and the reality and complexification conventions together close the calculus: every Lorentz invariant a massless amplitude can depend on is now a polynomial in $\langle i\,j\rangle$ and $[i\,j]$ subject to Eqs. (58.37), (58.38) and (58.45). What remains before amplitudes can be written down is to understand how this calculus responds to the one residual freedom in the spinor parametrization — the rescaling $\lambda_i \to t_i\lambda_i$, $\tilde\lambda_i \to t_i^{-1}\tilde\lambda_i$ that leaves p_i fixed. Promoting that rescaling to an organizing symmetry, and reading helicity as its weight, is the task of the next section.

58.3 Little-group scaling as an organizing principle

The bracket algebra of the previous section left one loose thread deliberately untied. The factorization of a null momentum into spinors, $p^{\alpha\dot\alpha} = \lambda^\alpha\tilde\lambda^{\dot\alpha}$ (Eq. (58.1)), does not determine λ and $\tilde\lambda$ uniquely: for real momenta the two spinors are fixed only up to a common phase, and for complex momenta up to a common complex rescaling,

$$\lambda \;\longmapsto\; t\,\lambda, \qquad \tilde\lambda \;\longmapsto\; t^{-1}\,\tilde\lambda, \qquad t \in \mathbb{C}^\times \;\; (\,|t| = 1 \text{ for real } p\,), \tag{58.50}$$

which leaves the physical object $p^{\alpha\dot\alpha} = \lambda^\alpha\tilde\lambda^{\dot\alpha}$ untouched because the two factors t and t^{-1} cancel. We noted this redundancy when we built the spinors and set it aside as a bookkeeping nuisance. It is nothing of the sort. The rescaling

Eq. (58.5) is not an artefact of our variables; it is the massless little group of Chapter 3 acting in spinor language, and the way an amplitude responds to it is a kinematic constraint so strong that, for three massless particles, it will fix the answer completely with no Lagrangian input at all. This section establishes that constraint. We show first that Eq. (58.5) *is* the compact U(1) inside ISO(2), derived from the induced representation of Chapter 3; then that a state of helicity h forces a definite scaling weight $-2h$; then how to read the helicities of any proposed expression straight off its spinor content. None of this requires drawing a diagram, and that is the point: the helicity dependence of every massless amplitude is settled before the dynamics is even specified.

This is where QED and Yang–Mills theory make their decisive conceptual turn. The unpolarized cross sections of Chapter 17 were computed by squaring amplitudes and summing over spins precisely because the individual helicity amplitudes looked forbidding in Feynman-diagram language. Little-group scaling reverses the logic: the helicity amplitudes are the simple objects, constrained in advance by symmetry, and the unpolarized result is recovered at the end by summing their squares. The same principle, applied to gluons, is what collapses the factorial diagram counts of Chapter 59 to one-line answers.

58.3.1 The t-rescaling is the massless little group

Recall the structure fixed in Chapter 3. For a massless particle the standard momentum is the fixed null vector $k^\mu = (\omega, 0, 0, \omega)$ of Eq. (3.206), and the little group G_k of Theorem 3.43 — the subgroup of $\mathrm{SO}(1,3)^+$ that leaves k invariant — is ISO(2), the group of rigid motions of the Euclidean plane, with algebra Eq. (3.212) generated by the two "translations" $A = K_1 - J_2$, $B = K_2 + J_1$ of Eq. (3.208) and the "rotation" J_3. The physical selection rule that defines an ordinary massless particle sets the non-compact translations to zero on physical states, $A = B = 0$, leaving only the compact U(1) generated by J_3: rotations about the momentum axis. A one-particle state of helicity h is the one-dimensional representation

$$J_3\,|k,h\rangle = h\,|k,h\rangle\,, \qquad U(\theta)\,|k,h\rangle = e^{-ih\theta}\,|k,h\rangle\,, \qquad U(\theta) = e^{-i\theta J_3}, \quad (58.51)$$

with h quantized in half-integers by the $\mathbb{Z}_2$ topology of the Lorentz group, Eq. (3.217). We now show that the abstract U(1) of Eq. (58.51) acts on the spinors $\lambda, \tilde\lambda$ exactly as the rescaling Eq. (58.5), with t a pure phase determined by the rotation angle θ.

Lifting the rotation to $\mathrm{SL}(2,\mathbb{C})$. The spinors live in the two Weyl representations of the Lorentz group built in Chapter 3: λ_α transforms in $(\frac{1}{2},0)$ and $\tilde\lambda_{\dot\alpha}$ in $(0,\frac{1}{2})$, so that the bispinor $p_{\alpha\dot\alpha} = p_\mu(\sigma^\mu)_{\alpha\dot\alpha}$ transforms as $p \mapsto M\,p\,M^\dagger$ under the $\mathrm{SL}(2,\mathbb{C})$ element M representing a Lorentz transformation, with $\lambda \mapsto M\lambda$ and

$\tilde{\lambda} \mapsto M^* \tilde{\lambda}$. A rotation about the z-axis is generated by $J_3 = \frac{1}{2}\sigma_3$ in the spinor representation; choosing the orientation so that the lift matches the helicity convention of Chapter 3 (verified below), the little-group rotation by angle θ is represented by the diagonal SU(2) matrix

$$W(\theta) = \exp\!\Big(\tfrac{i\theta}{2}\,\sigma_3\Big) = \begin{pmatrix} e^{i\theta/2} & 0 \\ 0 & e^{-i\theta/2} \end{pmatrix} = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix}, \qquad t \equiv e^{i\theta/2}. \tag{58.52}$$

Apply this to the standard momentum. With $\sigma^\mu = (\mathbf{1}, \vec{\sigma})$ the bispinor of $k^\mu = (\omega, 0, 0, \omega)$ is

$$k_{\alpha\dot{\alpha}} = \omega(\sigma^0 + \sigma^3)_{\alpha\dot{\alpha}} = \omega \begin{pmatrix} 2 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 2\omega & 0 \\ 0 & 0 \end{pmatrix}, \tag{58.53}$$

of rank one, so it factorizes as $k_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$ with

$$\lambda_\alpha = (\sqrt{2\omega},\, 0)^{\top}, \qquad \tilde{\lambda}_{\dot{\alpha}} = (\sqrt{2\omega},\, 0)^{\top}. \tag{58.54}$$

Under the rotation, $k_{\alpha\dot{\alpha}} \mapsto W k\, W^\dagger$. Since W is diagonal and unitary,

$$W\, k\, W^\dagger = \begin{pmatrix} t & 0 \\ 0 & t^{-1} \end{pmatrix} \begin{pmatrix} 2\omega & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} t^* & 0 \\ 0 & (t^{-1})^* \end{pmatrix} = \begin{pmatrix} 2\omega\,|t|^2 & 0 \\ 0 & 0 \end{pmatrix} = k, \tag{58.55}$$

using $|t| = 1$: the rotation fixes k, as it must for a little-group element. At the level of the spinors the same computation reads, component by component,

$$\lambda_\alpha \mapsto (W\lambda)_\alpha = t\,\lambda_\alpha, \qquad \tilde{\lambda}_{\dot{\alpha}} \mapsto (W^*\tilde{\lambda})_{\dot{\alpha}} = t^{-1}\,\tilde{\lambda}_{\dot{\alpha}}, \tag{58.56}$$

because W multiplies the surviving upper component of λ in Eq. (58.54) by $t = e^{i\theta/2}$ while $W^* = \mathrm{diag}(t^{-1}, t)$ multiplies the upper component of $\tilde{\lambda}$ by t^{-1}. This is precisely Eq. (58.5). The freedom we had set aside as a normalization ambiguity is exactly the compact U(1) little group, with the group parameter θ and the rescaling parameter related by

$$\boxed{\,t = e^{i\theta/2}.\,} \tag{58.57}$$

The half-angle is the familiar fingerprint of a spinor representation: a 2π rotation, $\theta \to \theta + 2\pi$, sends $t \to -t$, the sign that distinguishes the two sheets of $\mathrm{SL}(2,\mathbb{C})$ over $\mathrm{SO}(1,3)$ and that quantized helicity in half-integers in Eq. (3.217).

Why this is the induced representation, not a coincidence. The identification is forced by the induced-representation construction of Eq. (3.189), not merely consistent with it. There, a Lorentz transformation Λ acting on a one-particle state $|p,\sigma\rangle$ produced the Wigner little-group element $W(\Lambda, p) =$

$L(\Lambda p)^{-1}\Lambda\, L(p)$ of Eq. (3.188), acting on the internal label through the representation matrices $D(W)$. For a massless particle the standard boost $L(p)$ is realized in $\mathrm{SL}(2,\mathbb{C})$ as the matrix that carries the reference spinors Eq. (58.54) to $\lambda(p), \tilde{\lambda}(p)$, and the residual little-group factor W is exactly the diagonal phase Eq. (58.52). The one-dimensional representation $D(W) = e^{-ih\theta}$ on the helicity-h state is what appears in Eq. (3.188); the corresponding action on the carrier spinors is Eq. (58.56). In other words, the spinors λ and $\tilde{\lambda}$ are not auxiliary coordinates on which we have imposed an extra symmetry by hand — they are the carrier space of the induced representation, and Eq. (58.5) is the image of the Wigner rotation acting on that carrier.

Notation Watch

The sign in the exponent of Eq. (58.52), and hence the relation $t = e^{+i\theta/2}$ rather than $e^{-i\theta/2}$, is a convention: it is fixed by demanding that the resulting weight of a helicity-h wavefunction come out as $-2h$ with the helicity assignment $U(\theta) = e^{-i\theta J_3}$ of Chapter 3 (equivalently, that the transverse polarizations $\epsilon^{\mu}_{\pm}$ of Eq. (3.219), which rotate as $e^{\mp i\theta}$, acquire weight $t^{\mp 2}$). The opposite orientation would send $t \to t^{-1}$ throughout and is used by some references. We fix once and for all the convention $\lambda \to t\lambda$, $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$, $t = e^{i\theta/2}$, matching the bracket conventions of Theorem 58.2; every formula in Part XVI uses it.

It is worth pausing on what kind of statement Eq. (58.5) is. The momentum $p^{\alpha\dot{\alpha}} = \lambda^{\alpha}\tilde{\lambda}^{\dot{\alpha}}$ is manifestly invariant under the rescaling, so any quantity built only from momenta — a Mandelstam invariant s_{ij}, a polarization sum, the trace technology of Chapter 17 — carries no little-group weight at all. The weight lives entirely in objects that "remember" the split of p into its two spinor halves, that is, in the brackets $\langle i\, j\rangle$ and $[i\, j]$ and in the external wavefunctions of particles with spin. This is the structural reason the helicity basis is the natural one: helicity is, literally, the scaling weight under Eq. (58.5), and a helicity amplitude is a homogeneous function of definite weight in each external spinor pair.

58.3.2 The helicity-weight theorem

We can now state and prove the central result of the section: helicity is a scaling weight, and it constrains every massless amplitude before any dynamics is supplied.

Theorem 58.16 (Helicity-weight scaling)**.** *Let A_n be an on-shell scattering amplitude of n massless particles (all momenta outgoing), with leg i carrying helicity h_i and described by the spinors $(\lambda_i, \tilde{\lambda}_i)$, $p_i^{\alpha\dot{\alpha}} = \lambda_i^{\alpha}\tilde{\lambda}_i^{\dot{\alpha}}$. Then under the little-group rescaling Eq. (58.5) applied independently to each leg, $\lambda_i \to t_i\lambda_i$,*

$\tilde\lambda_i \to t_i^{-1}\tilde\lambda_i$, *the amplitude is homogeneous of weight* $-2h_i$ *in the* i*-th pair:*

$$A_n(\{t_i\lambda_i,\, t_i^{-1}\tilde\lambda_i,\, h_i\}) = \left(\prod_{i=1}^{n} t_i^{-2h_i}\right) A_n(\{\lambda_i,\, \tilde\lambda_i,\, h_i\}). \tag{58.58}$$

Proof. The amplitude is extracted from the time-ordered correlator of interpolating fields by the LSZ reduction of Chapter 16: each external leg contributes an amputated, on-shell-projected propagator factor contracted into a one-particle wavefunction. For n outgoing massless particles,

$$A_n = \prod_{i=1}^{n} w_{h_i}(p_i)\ \cdot\ \widetilde{\mathcal{M}}(p_1,\dots,p_n), \tag{58.59}$$

where $w_{h_i}(p_i)$ is the external wavefunction of leg i — a polarization spinor for a fermion, a polarization vector for a gauge boson, the number 1 for a scalar — and $\widetilde{\mathcal{M}}$ is the amputated, on-shell connected correlator with its external Lorentz and spinor indices left open to be contracted into the w's. Two facts about Eq. (58.59) prove the theorem.

First, $\widetilde{\mathcal{M}}$ depends on the kinematics only through the momenta p_i, never through the individual spinors $\lambda_i, \tilde\lambda_i$ separately: it is assembled from propagators $i/(P^2+i\epsilon)$, vertices, and loop integrals, all functions of momenta alone, and its open indices are spacetime (vector or Dirac) indices, contracted with momenta and with the metric or gamma matrices. By Eq. (58.5) each $p_i = \lambda_i\tilde\lambda_i$ is invariant under the rescaling — the factors t_i and t_i^{-1} cancel — so $\widetilde{\mathcal{M}}$ is invariant. The entire little-group weight of A_n resides in the external wavefunctions.

Second, each wavefunction w_{h_i} is the one-particle state of definite helicity h_i, i.e. the eigenstate of the little-group generator J_3 on leg i with eigenvalue h_i. By Eq. (58.51) the little-group rotation by angle θ_i multiplies it by the phase $e^{-ih_i\theta_i}$, because the wavefunction is the covariant matrix element $\langle 0|\,\Phi(0)\,|k,h_i\rangle$ of the relevant field Φ and inherits the transformation law of the state. Expressing the phase through $t_i = e^{i\theta_i/2}$ from Eq. (58.57),

$$w_{h_i} \;\longmapsto\; e^{-ih_i\theta_i}\, w_{h_i} = \left(e^{i\theta_i/2}\right)^{-2h_i} w_{h_i} = t_i^{-2h_i}\, w_{h_i}. \tag{58.60}$$

Multiplying the n wavefunction factors and the invariant $\widetilde{\mathcal{M}}$ gives exactly Eq. (58.58).

The identity has been established for $|t_i| = 1$ (real momenta). For complex momenta both sides are rational functions of the spinor components — holomorphic in the λ_i and antiholomorphic in the $\tilde\lambda_i$, as the bracket algebra of Theorem 58.4 makes manifest — and a polynomial homogeneity of fixed integer or half-integer degree that holds on the unit circle $|t_i| = 1$ holds identically in $t_i \in \mathbb{C}^\times$ by analytic continuation. The weight $-2h_i$ is therefore the scaling exponent for arbitrary complex t_i, which is the form needed for the complexified momenta of on-shell recursion in Chapter 59. □

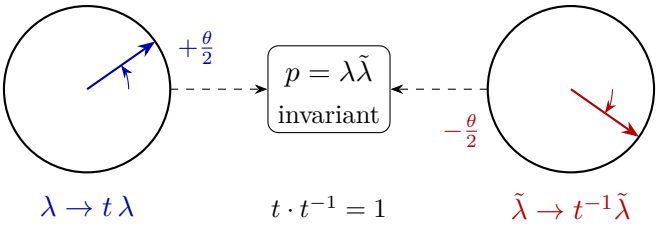


Figure 58.2: The compact U(1) little group rotates the holomorphic spinor λ by the phase $t = e^{i\theta/2}$ and the antiholomorphic spinor $\tilde{\lambda}$ by the opposite phase t^{-1}, so their product — the physical momentum $p = \lambda\tilde{\lambda}$ — is fixed. A helicity-h wavefunction, being a definite power of these spinors, picks up the phase $e^{-ih\theta} = t^{-2h}$; the amplitude inherits the product over legs, $\prod_i t_i^{-2h_i}$, of Eq. (58.58).

The theorem deserves to be read as a constraint, not merely a transformation law. *Any* candidate expression for a massless amplitude — whether guessed, conjectured, or recursively constructed — must scale as $\prod_i t_i^{-2h_i}$ under Eq. (58.5), or it is not the amplitude of those helicities. Since $t_i^{-2h_i}$ is fixed once the external helicities are chosen, the kinematic dependence of the amplitude is pinned down up to a little-group-invariant function (a function of Mandelstam invariants and other neutral data) on every leg, before a single Feynman diagram is drawn. The remainder of the chapter, and much of the next, is the systematic exploitation of this fact.

Worked Example 58.17 (The weight $-2h$ for scalars, fermions, and gauge bosons)**.** The proof of Theorem 58.16 routed the weight through the abstract phase $e^{-ih\theta}$. It is instructive — and a complete consistency check — to verify the weight $-2h$ directly from the explicit spinor content of the external wavefunctions, spin by spin. In each case we track the scaling of the wavefunction w_h on its own leg under $\lambda \to t\lambda$, $\tilde{\lambda} \to t^{-1}\tilde{\lambda}$.

Spin 0, $h = 0$. A scalar has $w_0 = 1$, a number independent of the spinors, so it scales as $t^0 = t^{-2\cdot 0}$. Weight 0, as required.

Spin $\frac{1}{2}$, $h = \pm\frac{1}{2}$. The massless Dirac equation of Chapter 11 has, for an outgoing fermion, two solutions distinguished by helicity. In two-component language the positive-helicity wavefunction is the square spinor $\tilde{\lambda}$ (written $|i]$) and the negative-helicity wavefunction is the angle spinor λ (written $|i\rangle$), the identification fixed in Theorem 58.2. Under the rescaling,

$$h = +\tfrac{1}{2}: \quad w = \tilde{\lambda} \to t^{-1}\tilde{\lambda} = t^{-2(+1/2)}\,w, \qquad h = -\tfrac{1}{2}: \quad w = \lambda \to t\,\lambda = t^{-2(-1/2)}\,w. \tag{58.61}$$

Both carry weight $-2h$. The angle and square spinors are not interchangeable: the helicity of a massless fermion is encoded in *which* spinor describes it, and the weight theorem is the statement of that encoding.

Spin 1, $h = \pm 1$. The gauge-boson polarization vectors are built in Section 58.5 from one spinor of each chirality together with a reference spinor; the explicit forms (Eq. (58.111)) are, for leg i with arbitrary reference momentum q,

$$\epsilon_+^\mu(i;q) = \frac{\langle q|\gamma^\mu|i]}{\sqrt{2}\,\langle q\,i\rangle}, \qquad \epsilon_-^\mu(i;q) = \frac{\langle i|\gamma^\mu|q]}{\sqrt{2}\,[q\,i]}. \tag{58.62}$$

We do not need their derivation here — only their spinor content on leg i. The positive-helicity vector ϵ_+ contains $\tilde\lambda_i$ in the numerator sandwich $\langle q|\gamma^\mu|i]$, which is linear in $\tilde\lambda_i$ (weight t_i^{-1}), and λ_i in the denominator bracket $\langle q\,i\rangle$ (weight t_i^{+1} in the denominator, hence t_i^{-1} overall), giving

$$\epsilon_+^\mu(i;q) \;\to\; \frac{t_i^{-1}}{t_i^{+1}}\,\epsilon_+^\mu(i;q) = t_i^{-2}\,\epsilon_+^\mu(i;q) = t_i^{-2(+1)}\,\epsilon_+^\mu(i;q). \tag{58.63}$$

The negative-helicity vector ϵ_- contains λ_i in the numerator (t_i^{+1}) and $\tilde\lambda_i$ in the denominator bracket $[q\,i]$ (t_i^{-1} in the denominator, hence t_i^{+1} overall), giving $t_i^{+2} = t_i^{-2(-1)}$. Both carry weight $-2h$. The reference spinor q drops out of the weight count, as it must: it is pure gauge, not little-group data, and we will show in Section 58.5 that the amplitude is independent of it.

58.3.3 Reading helicities off an expression

Theorem 58.16 turns the brackets into a helicity meter. Each bracket carries a definite weight, read directly off Eq. (58.5):

$$\langle i\,j\rangle = \epsilon^{\alpha\beta}\lambda_{i\alpha}\lambda_{j\beta} \;\longmapsto\; t_i\,t_j\,\langle i\,j\rangle, \qquad [i\,j] = \epsilon_{\dot\alpha\dot\beta}\tilde\lambda_i^{\dot\alpha}\tilde\lambda_j^{\dot\beta} \;\longmapsto\; t_i^{-1}\,t_j^{-1}\,[i\,j]. \tag{58.64}$$

An angle bracket scales as $t_i t_j$; a square bracket as $t_i^{-1}t_j^{-1}$. Define the *little-group weight of leg i* in a given expression as the net power n_i such that the expression scales as $t_i^{n_i}$. The helicity of that leg is then read off by matching to Eq. (58.58),

$$t_i^{n_i} = t_i^{-2h_i} \quad\Longrightarrow\quad \boxed{h_i = -\tfrac{1}{2}\,n_i.} \tag{58.65}$$

Consequently a single angle bracket $\langle i\,j\rangle$, contributing $n_i = n_j = +1$, carries helicity $-\frac{1}{2}$ on each of legs i and j; in the "twice-helicity" bookkeeping $2h_i = -1$. A square bracket $[i\,j]$, with $n_i = n_j = -1$, carries helicity $+\frac{1}{2}$ on each leg, $2h_i = +1$. Angle brackets lower helicity; square brackets raise it. Every appearance of leg i in a numerator angle bracket adds $+\frac{1}{2}$ of negative helicity; every appearance in a denominator angle bracket adds $+\frac{1}{2}$ of positive helicity; and conversely for square brackets. The book-keeping is purely additive, and it lets us assign helicities to any rational expression in the brackets by counting.

Caution

The word "weight" is overloaded in the literature, and the two natural conventions differ by a sign. We use n_i for the *t-scaling exponent* (so λ_i has $n_i = +1$) and reserve "helicity" for the physical label $h_i = -n_i/2$. Some texts call $2h_i = -n_i$ the "little-group weight," in which $\langle i\,j\rangle$ has weight $(-1,-1)$ and $[i\,j]$ weight $(+1,+1)$. The two descriptions are identical content with opposite sign; we always state h_i explicitly to avoid the ambiguity.

Worked Example 58.18 (Reading the helicities of the three-point seed)**.** Consider the candidate expression for a three-particle amplitude

$$A_3 \stackrel{?}{=} \frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \tag{58.66}$$

built entirely from angle brackets. We determine the helicity content by counting, using Eqs. (58.64) and (58.65).

Leg 1 appears in $\langle 1\,2\rangle^3$ (numerator, $+3$) and in $\langle 3\,1\rangle$ (denominator, -1): net $n_1 = +3-1 = +2$, so $h_1 = -1$.

Leg 2 appears in $\langle 1\,2\rangle^3$ (numerator, $+3$) and in $\langle 2\,3\rangle$ (denominator, -1): net $n_2 = +3-1 = +2$, so $h_2 = -1$.

Leg 3 appears only in the denominator, in $\langle 2\,3\rangle$ (-1) and $\langle 3\,1\rangle$ (-1): net $n_3 = -2$, so $h_3 = +1$.

The expression Eq. (58.66) therefore has the helicity content $(h_1, h_2, h_3) = (-1,-1,+1)$ — it is a candidate for the three-gluon amplitude with two negative and one positive helicity. We have not proved it *is* the amplitude; Eq. (58.66) is only the unique angle-bracket monomial of the correct overall scaling, and the next section will show that locality and little-group scaling together force precisely this form. The parallel computation in square brackets,

$$A_3 \stackrel{?}{=} \frac{[1\,2]^3}{[2\,3]\,[3\,1]}, \tag{58.67}$$

gives $n_1 = n_2 = -2$, $n_3 = +2$, i.e. $(h_1, h_2, h_3) = (+1,+1,-1)$, the parity conjugate. That two-line determination of the helicity content of the three-gluon amplitude, done with no Lagrangian and no diagram, is the organizing power of little-group scaling in miniature.

58.3.4 Back to the seed of gauge invariance

The little-group weight closes a circle opened in Chapter 3. There we found (Theorem 3.46) that a massless helicity-1 particle cannot be packaged in a Lorentz four-vector ϵ^μ without redundancy: the little group acts on the polarization

inhomogeneously, $\epsilon_\pm^\mu \mapsto \epsilon_\pm^\mu + \beta\, k^\mu$ (Eq. (3.221)), and the only covariant way to absorb the shift is to declare it a gauge equivalence, $\epsilon^\mu \sim \epsilon^\mu + \beta k^\mu$ (Eq. (3.222)). That inhomogeneity and the nontrivial scaling weight Eq. (58.58) are two faces of the same fact, and the spinor variables make the relationship transparent.

A four-vector polarization ϵ^μ transforms in the $(\frac{1}{2},\frac{1}{2})$ representation, i.e. as a bispinor $\epsilon^{\alpha\dot\alpha}$ carrying *one* undotted and *one* dotted index — a single λ-type and a single $\tilde\lambda$-type slot. Under the little group a genuine carrier of helicity ±1 must scale with weight $t^{\mp2}$, that is, as $\lambda^{\pm2}$ or $\tilde\lambda^{\pm2}$ on its own leg. But a single bispinor $\epsilon^{\alpha\dot\alpha} = \lambda^\alpha\tilde\lambda^{\dot\alpha}$ scales as $t\cdot t^{-1} = t^0$: it is little-group *neutral*, like the momentum itself. There is no way to build a weight-(∓2) object from one undotted and one dotted index without dividing by a bracket that involves a second, foreign spinor — which is exactly the reference momentum q of Eq. (58.62), and exactly the source of the residual freedom $\epsilon^\mu \to \epsilon^\mu + \beta k^\mu$: shifting q shifts the polarization by a multiple of the momentum. The impossibility of a covariant helicity-1 wavefunction in Chapter 3 and the nontrivial little-group weight here are the same obstruction read in two languages. Gauge redundancy is the price of forcing a weight-(∓2) object into a weight-0 tensor slot.

This is also why scalars need no gauge symmetry and fermions need no polarization gymnastics: a helicity-0 wavefunction has weight 0, matching the neutral scalar field, and a helicity-$\pm\frac{1}{2}$ wavefunction has weight ∓1, matching a single λ or $\tilde\lambda$ exactly, with one index of one chirality and nothing left over. Only at $|h| \geq 1$ does the weight $|2h| \geq 2$ exceed what a single index of each available type can carry, and only there is a redundant covariant description — a gauge field, and at $|h| = 2$ a metric — forced upon us. The seed planted on the light cone in Chapter 3 has grown into a quantitative selection rule: the little-group weight measures, leg by leg, exactly how far a particle's helicity outruns the tensor field that would carry it.

With the scaling weight Eq. (58.58) in hand, the kinematic skeleton of every massless amplitude is fixed up to a little-group-invariant function. For three massless particles this residual freedom is so tightly constrained by the one remaining physical input — locality, in the form of the allowed analytic structure — that nothing is left to choose: the three-point amplitude is determined uniquely, with no reference to any Lagrangian. That keystone computation, and the consequence that all on-shell recursion springs from it, is the subject of the next section.

58.4 Three-point amplitudes fixed by symmetry

The previous section turned helicity into a scaling weight: a massless leg of helicity h_i forces any amplitude to respond to the little-group rescaling $(\lambda_i, \tilde\lambda_i) \mapsto$

$(t_i\lambda_i, t_i^{-1}\tilde{\lambda}_i)$ by the fixed power

$$\mathcal{M}(\{t_i\lambda_i, t_i^{-1}\tilde{\lambda}_i\}) = \Big(\prod_i t_i^{-2h_i}\Big)\,\mathcal{M}(\{\lambda_i, \tilde{\lambda}_i\}), \tag{58.68}$$

recorded as Eq. (58.5). For a generic number of legs this is a strong but partial constraint: it tells us how the brackets must combine leg-by-leg, yet leaves the relative weighting of independent invariants free, and those leftover freedoms are precisely the dynamics that a Lagrangian must supply. The remarkable fact this section establishes is that at *three* points the constraint is no longer partial. Three massless momenta are kinematically so rigid that little-group scaling, together with the single demand that the amplitude be local — finite mass dimension, no unphysical poles — determines the entire functional form down to one overall constant, and dimensional analysis fixes even the dimensionful part of that constant. No Feynman diagram, no Lagrangian, no choice of gauge enters. The answer is read off from the representation theory of the external states. This is the keystone of the whole formalism: the three-point amplitudes are the irreducible seeds, and Chapter 59 will show that the BCFW recursion glues them into every tree amplitude. If the seeds are fixed by symmetry, so is everything built from them.

We proceed in four movements. First we expose the kinematic singularity of three massless particles — why their brackets are forced onto one of two complex branches. Then we solve the scaling constraints exactly, proving the form is unique. We specialize to spin one to obtain the three-gluon amplitude, cross-checked against the Yang–Mills cubic vertex of Chapter 26 in a worked computation, and to the electron–photon vertex to advance QED. We close by reading off, from the same machinery, why a helicity-two particle must couple universally — the amplitudes seed of the equivalence principle.

58.4.1 The kinematic singularity of three massless particles

Work throughout with all momenta outgoing, so momentum conservation for three massless particles reads

$$p_1 + p_2 + p_3 = 0, \qquad p_i^2 = 0 \;\; (i = 1, 2, 3). \tag{58.69}$$

The Mandelstam invariants of Theorem 16.20, which at three points are the three two-particle invariants $s_{ij} = (p_i + p_j)^2$, collapse immediately. From Eq. (58.69), $p_i + p_j = -p_k$ for $\{i, j, k\} = \{1, 2, 3\}$, so

$$s_{ij} = (p_i + p_j)^2 = p_k^2 = 0, \tag{58.70}$$

and expanding the left-hand side gives the equivalent statement $2\,p_i \cdot p_j = p_i^2 + 2p_i \cdot p_j + p_j^2 = s_{ij} = 0$, i.e. every pair of the three momenta is mutually null:

$p_i \cdot p_j = 0$ for all $i \neq j$. Three on-shell momenta summing to zero are pairwise orthogonal in the Minkowski sense.

Translate this into spinors. With the bracket–Mandelstam dictionary $\langle i\,j\rangle[j\,i] = 2\,p_i \cdot p_j = s_{ij}$ of Eq. (58.24), the vanishing Eq. (58.70) says

$$\langle i\,j\rangle\,[j\,i] = 0 \qquad \text{for every pair } i \neq j. \tag{58.71}$$

For each pair, therefore, *at least one* of the angle bracket $\langle i\,j\rangle$ and the square bracket $[i\,j]$ must vanish. This is the seed of the entire phenomenon, but Eq. (58.71) alone would permit a ragged mixture — $\langle 1\,2\rangle = 0$ here, $[2\,3] = 0$ there. Momentum conservation forbids the mixture, and forces an all-or-nothing dichotomy.

Proposition 58.19 (Special three-point kinematics: the two branches)**.** *For three outgoing massless momenta obeying Eq. (58.69), the two cyclic products of brackets cannot both be nonzero:*

$$\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,1\rangle \;=\; 0 \quad \textit{or} \quad [1\,2][2\,3][3\,1] \;=\; 0. \tag{58.72}$$

Exactly one of the following holds, up to the degenerate case in which both vanish:

(H) *the* holomorphic branch*, on which all* $[i\,j] = 0$ *while the* $\langle i\,j\rangle$ *are generically nonzero; equivalently the three anti-holomorphic spinors are proportional,* $\tilde\lambda_i \propto \tilde\zeta$ *for a common* $\tilde\zeta$*;*

(A) *the* anti-holomorphic branch*, on which all* $\langle i\,j\rangle = 0$ *while the* $[i\,j]$ *are generically nonzero; equivalently the three holomorphic spinors are proportional,* $\lambda_i \propto \zeta$*.*

For real *momenta both products vanish identically and the configuration is degenerate (the three momenta are collinear); a nonvanishing three-point amplitude therefore requires* complex *momenta, on which* λ *and* $\tilde\lambda$ *are independent and exactly one branch is reached.*

Proof. Write momentum conservation Eq. (58.69) in bispinor form using the factorization $p_{i\,a\dot a} = \lambda_{ia}\tilde\lambda_{i\dot a}$ of Eq. (58.1):

$$\sum_{i=1}^{3} \lambda_{ia}\,\tilde\lambda_{i\dot a} = 0. \tag{58.73}$$

Collect the three holomorphic spinors as the columns of a 2×3 matrix $\Lambda = (\lambda_1\ \lambda_2\ \lambda_3)$ and the anti-holomorphic ones as the columns of $\widetilde\Lambda = (\tilde\lambda_1\ \tilde\lambda_2\ \tilde\lambda_3)$. The 2×2 minors of Λ are exactly the angle brackets, $\langle i\,j\rangle = \epsilon^{ab}\lambda_{ia}\lambda_{jb}$, and those of $\widetilde\Lambda$ the square brackets, $[i\,j] = \epsilon_{\dot a\dot b}\tilde\lambda_i^{\dot a}\tilde\lambda_j^{\dot b}$. Hence

$$\text{all } \langle i\,j\rangle = 0 \iff \operatorname{rank}\Lambda \leq 1, \qquad \text{all } [i\,j] = 0 \iff \operatorname{rank}\widetilde\Lambda \leq 1. \tag{58.74}$$

Equation Eq. (58.73) is the statement $\Lambda\,\widetilde{\Lambda}^{\mathsf{T}} = 0$ as a 2×2 matrix product. Read it as the assertion that every column of $\widetilde{\Lambda}^{\mathsf{T}}$ (a vector in $\mathbb{C}^3$) lies in the kernel of the linear map $\Lambda : \mathbb{C}^3 \to \mathbb{C}^2$. There are two cases.

Case $\operatorname{rank}\Lambda = 2$. Then $\ker\Lambda$ is one-dimensional. The two columns of $\widetilde{\Lambda}^{\mathsf{T}}$ both lie in this single line, so they are proportional; that is, the two rows of $\widetilde{\Lambda}$ are proportional, whence $\operatorname{rank}\widetilde{\Lambda} \le 1$. By Eq. (58.74) all $[i\,j] = 0$: the holomorphic branch (H), with the angle brackets free (their minors are generically nonzero since $\operatorname{rank}\Lambda = 2$).

Case $\operatorname{rank}\Lambda \le 1$. Then all the angle minors vanish, all $\langle i\,j\rangle = 0$ by Eq. (58.74): the anti-holomorphic branch (A), or the doubly-degenerate point where additionally $\operatorname{rank}\widetilde{\Lambda} \le 1$ and both products vanish.

In either case at least one of the cyclic products in Eq. (58.72) vanishes, which proves the dichotomy. The characterization of branch (H) by $\tilde{\lambda}_i \propto \tilde{\zeta}$ is exactly $\operatorname{rank}\widetilde{\Lambda} \le 1$, and symmetrically for (A).

It remains to treat real momenta. By the reality condition of Theorem 58.2, a real momentum has $\tilde{\lambda}_i = \pm\lambda_i^*$, the sign tracking the sign of the energy p_i^0. Consequently $|[i\,j]| = |\langle i\,j\rangle|$ for every pair. Then Eq. (58.71), $\langle i\,j\rangle[j\,i] = 0$, forces $|\langle i\,j\rangle|^2 = 0$ and hence *both* brackets vanish for every pair. All angle and square brackets are zero: the spinors are pairwise proportional and the three momenta are collinear, a measure-zero locus carrying no scattering. Only by relaxing reality — continuing λ and $\tilde{\lambda}$ to independent complex spinors — can one bracket set stay generic while the other vanishes. This is not a formal trick: a physical three-point amplitude of massless particles has support only at this complex kinematic point, and it enters physics as the residue of a factorization channel (Chapter 59), where the intermediate momentum is off the real slice. □

Physical Insight

The lesson is worth stating plainly, because it recurs throughout on-shell methods. A real three-point amplitude of massless particles vanishes by kinematics, not by dynamics: there is simply nowhere for three pairwise-null momenta to live except on top of one another. The object that survives is the *analytic continuation* of the amplitude to complex momenta, where it is sharp and nonzero on one of the two branches of Theorem 58.19. On-shell recursion is the art of extracting physical, real-momentum amplitudes from these complex three-point seeds by sewing them across factorization channels. The complexification that looks like a blemish here is the engine of the next chapter.

58.4.2 Solving the scaling constraints: uniqueness of the form

Fix the holomorphic branch (H), so that the amplitude is a function of the three angle brackets $\langle 1\,2\rangle, \langle 2\,3\rangle, \langle 3\,1\rangle$ alone — the square brackets have been set to zero. Locality has a sharp meaning here: a three-point amplitude has no internal propagator, hence no kinematic pole, so it must be a *monomial* in the available invariants, with integer exponents (and no sum of distinct monomials, since each angle bracket scales independently and a sum would fail to carry a definite little-group weight). Write the most general such monomial, labelling the exponent of the bracket that *omits* leg i by a_i:

$$\mathcal{M}_3 = c\,\langle 1\,2\rangle^{\,a_3}\,\langle 2\,3\rangle^{\,a_1}\,\langle 3\,1\rangle^{\,a_2}, \tag{58.75}$$

with c a constant to be fixed. Each angle bracket $\langle i\,j\rangle$ carries little-group weight $t_i t_j$, so leg i — which appears in exactly the two brackets that do not omit it — contributes total weight $t_i^{\,a_j+a_k}$ where $\{i,j,k\} = \{1,2,3\}$. Matching to Eq. (58.68) leg by leg gives one linear equation per particle:

$$\begin{aligned} t_1: &\quad a_2 + a_3 = -2h_1, \\ t_2: &\quad a_1 + a_3 = -2h_2, \\ t_3: &\quad a_1 + a_2 = -2h_3. \end{aligned} \tag{58.76}$$

This is a linear system with coefficient matrix $\left(\begin{smallmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{smallmatrix}\right)$, whose determinant is $2 \neq 0$: the solution is unique. Adding all three equations gives $2(a_1+a_2+a_3) = -2(h_1+h_2+h_3)$, so $a_1+a_2+a_3 = -H$ with

$$H \equiv h_1 + h_2 + h_3. \tag{58.77}$$

Subtracting each equation from this total isolates one exponent; for instance $a_1 = (a_1+a_2+a_3) - (a_2+a_3) = -H + 2h_1 = h_1 - h_2 - h_3$, and cyclically. The unique solution is

$$a_1 = h_1 - h_2 - h_3, \qquad a_2 = h_2 - h_3 - h_1, \qquad a_3 = h_3 - h_1 - h_2. \tag{58.78}$$

Substituting into Eq. (58.75) gives the holomorphic three-point amplitude in closed form,

$$\mathcal{M}_3^{(\mathrm{H})} = c\,\langle 1\,2\rangle^{\,h_3-h_1-h_2}\,\langle 2\,3\rangle^{\,h_1-h_2-h_3}\,\langle 3\,1\rangle^{\,h_2-h_3-h_1}, \tag{58.79}$$

exactly the form announced. The anti-holomorphic branch (A) is obtained by the same argument with square brackets, recalling that $[i\,j]$ carries weight $t_i^{-1}t_j^{-1}$; the sign in each scaling equation flips, the unique solution has the opposite-sign exponents, and

$$\mathcal{M}_3^{(\mathrm{A})} = \tilde{c}\,[1\,2]^{\,h_1+h_2-h_3}\,[2\,3]^{\,h_2+h_3-h_1}\,[3\,1]^{\,h_3+h_1-h_2}. \tag{58.80}$$

Which branch is the physical amplitude? Locality decides, through the total mass dimension. Each bracket carries mass dimension one, so the holomorphic form Eq. (58.79) has bracket-degree $a_1+a_2+a_3=-H$, and the anti-holomorphic form Eq. (58.80) has bracket-degree $+H$. A genuine, local, pole-free amplitude must have *non-negative* integer bracket-degree — a monomial with a net negative power of brackets would have a pole at $\langle i\,j\rangle \to 0$ (a collinear configuration that is interior to the branch), and such a singularity at three points has no factorization channel to explain it; it is non-local. Hence the holomorphic branch is selected when $-H \geq 0$, i.e. $H<0$, and the anti-holomorphic branch when $H>0$. Compactly,

$$\text{branch} = \begin{cases} \text{holomorphic (angle brackets)}, & H = h_1+h_2+h_3 < 0, \\ \text{anti-holomorphic (square brackets)}, & H = h_1+h_2+h_3 > 0, \end{cases} \tag{58.81}$$

and the surviving amplitude carries bracket-degree $|H|$, so its mass dimension is $|H| = |h_1+h_2+h_3|$. (The marginal case $H=0$ — for instance three massless scalars, or a $+,-,0$ helicity assignment — gives a dimensionless monomial, a pure constant; the three-scalar coupling of a ϕ^3 theory is the example, and there the constant is the dimensionful coupling itself.) Collecting the result:

Theorem 58.20 (Three-point amplitudes are fixed by symmetry)**.** *Let three massless particles carry helicities h_1, h_2, h_3 and set $H = h_1+h_2+h_3$. On the complex kinematics of Theorem 58.19, little-group covariance Eq. (58.5) together with locality (correct mass dimension and absence of unphysical poles) determines the three-point amplitude uniquely up to a single overall constant:*

$$\mathcal{M}_3 = \begin{cases} c\,\langle 1\,2\rangle^{\,h_3-h_1-h_2}\,\langle 2\,3\rangle^{\,h_1-h_2-h_3}\,\langle 3\,1\rangle^{\,h_2-h_3-h_1}, & H<0, \\ \tilde{c}\,[1\,2]^{\,h_1+h_2-h_3}\,[2\,3]^{\,h_2+h_3-h_1}\,[3\,1]^{\,h_3+h_1-h_2}, & H>0. \end{cases} \tag{58.82}$$

The constant has mass dimension $[c] = 1-|H|$, fixed by the requirement that the full amplitude carry the kinematic dimension $4-n=1$ of an $n=3$ amplitude. No dynamical input — no Lagrangian, no Feynman rule, no gauge choice — enters the functional form; only the helicities and the spacetime dimension do.

The dimension bookkeeping in the theorem is the second half of the rigidity, and it is what fixes the dimensionful part of c. An n-point amplitude in four dimensions has mass dimension $4-n$ (the same counting used for the scalar amplitudes of Chapter 14: stripping the momentum-conserving delta function and the canonical state normalization leaves dimension $4-n$). For $n=3$ this is $+1$. The brackets supply $|H|$ of it, so the constant must supply the rest:

$$[c] = 1-|H|. \tag{58.83}$$

For two helicity-1 and helicity-$\frac{1}{2}$ particles, where $|H|=1$, the constant is dimensionless — a pure coupling. For $|H|=2$, as for three gravitons, the

constant carries mass dimension -1 — a coupling with the dimensions of a length, which is the first whisper of Newton's constant. We now make all of this concrete.

58.4.3 The three-gluon amplitude

Specialize Theorem 58.20 to three gluons, $|h_i| = 1$. Pure Yang–Mills has no all-plus or all-minus three-point coupling (the form Eq. (58.82) would have $|H| = 3$, mass dimension -2, requiring a higher-derivative operator absent from the renormalizable theory of Eq. (25.76)); the physical assignments are the two with a single odd helicity. Take first $(h_1, h_2, h_3) = (-1, -1, +1)$, so $H = -1 < 0$ and Eq. (58.81) selects the holomorphic branch. The exponents Eq. (58.78) are

$$a_1 = h_1 - h_2 - h_3 = -1, \quad a_2 = h_2 - h_3 - h_1 = -1, \quad a_3 = h_3 - h_1 - h_2 = +3, \tag{58.84}$$

giving

$$\mathcal{M}_3(1^-, 2^-, 3^+) = c\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}. \tag{58.85}$$

This is the Maximally-Helicity-Violating (MHV) three-gluon amplitude. Its bracket-degree is $3 - 1 - 1 = 1$, so $[c] = 0$: the constant is dimensionless. The single power of the coupling is fixed by the structure of the theory. A connected tree amplitude built from the cubic and quartic vertices of Chapter 26 with n external legs carries $V_3 + 2V_4 = n - 2$ powers of g (each cubic vertex one power, each quartic two; the tree topology fixes the total). At $n = 3$ this is a single power, g^1, so $c \propto g$ with a dimensionless number out front. The conjugate assignment $(h_1, h_2, h_3) = (+1, +1, -1)$ has $H = +1 > 0$, selecting the anti-holomorphic branch, and the same computation with square brackets gives

$$\mathcal{M}_3(1^+, 2^+, 3^-) = \tilde{c}\,\frac{[1\,2]^{\,3}}{[2\,3]\,[3\,1]}. \tag{58.86}$$

Everything except the pure numbers $c, \tilde{c}$ has been forced by symmetry. To pin those numbers — and with them the relative sign and the phase — we match against the one Feynman diagram the theory actually produces.

Worked Example 58.21 (The three-gluon amplitude from the Yang–Mills cubic vertex)**.** At $n = 3$ there is no internal line, so the only diagram is a single cubic vertex (Fig. 58.3). Contracting the three-gluon vertex Eq. (26.53) — with all momenta outgoing, $p_1 + p_2 + p_3 = 0$ — against three external polarizations ε_i gives the colour-dressed amplitude

$$i\mathcal{M}_3 = iV^{abc}_{\mu\nu\rho}(p_1, p_2, p_3)\,\varepsilon_1^{\mu}\varepsilon_2^{\nu}\varepsilon_3^{\rho} = g\,f^{abc}\,K, \qquad K = (\varepsilon_1{\cdot}\varepsilon_2)\,(p_1 - p_2){\cdot}\varepsilon_3 + (\text{cyclic}), \tag{58.87}$$

where "cyclic" adds the two terms obtained by $1 \to 2 \to 3 \to 1$, the colour structure constant f^{abc} carries the entire group-theory content (normalization

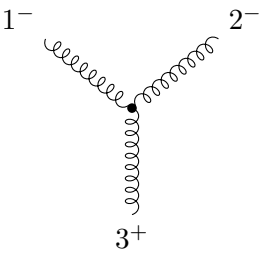


Figure 58.3: The sole diagram for the three-gluon amplitude: one cubic Yang–Mills vertex Eq. (26.53) with three on-shell external legs and no internal propagator. With the helicity assignment $(1^-, 2^-, 3^+)$ the amplitude lives on the holomorphic branch of Theorem 58.19 (all square brackets among externals vanishing) and equals the symmetry-fixed ratio Eq. (58.85).

$\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, Chapter B), and K is the purely kinematic factor we must evaluate.

We use the helicity polarization vectors built in the next section, Eq. (58.111),

$$\varepsilon_+^\mu(p_i; q_i) = \frac{\langle q_i|\gamma^\mu|i]}{\sqrt{2}\,\langle q_i\, i\rangle}, \qquad \varepsilon_-^\mu(p_i; q_i) = \frac{\langle i|\gamma^\mu|q_i]}{\sqrt{2}\,[q_i\, i]}, \tag{58.88}$$

each carrying a reference momentum q_i whose change is a gauge transformation; the amplitude is independent of every q_i. We exploit that freedom to annihilate two of the three terms in K. Assign the two negative-helicity legs a *common* reference $q_1 = q_2 \equiv q$ (a generic null momentum off the branch, so $[q\,1], [q\,2] \neq 0$), and give the positive leg the reference $q_3 = p_2$ (legal on the branch since $\langle q_3\, 3\rangle = \langle 2\, 3\rangle \neq 0$). The polarization dot products follow from Eq. (58.88) and the Fierz identity $\langle a|\gamma^\mu|b]\langle c|\gamma_\mu|d] = 2\langle a\, c\rangle[d\, b]$ (a consequence of the σ-matrix completeness relation $\sigma^\mu_{a\dot a}\bar\sigma_\mu^{\dot b b} = 2\delta_a^{\,b}\delta_{\dot a}^{\,\dot b}$ established with the bracket algebra of Theorem 58.2):

$$\varepsilon_1^- \cdot \varepsilon_2^- = \frac{\langle 1\, 2\rangle\,[q\, q]}{[q\, 1]\,[q\, 2]} = 0 \qquad ([q\, q] = 0), \tag{58.89}$$

$$\varepsilon_2^- \cdot \varepsilon_3^+ = \frac{\langle q_3\, 2\rangle\,[q\, 3]}{\langle q_3\, 3\rangle\,[q\, 2]} = \frac{\langle 2\, 2\rangle\,[q\, 3]}{\langle 2\, 3\rangle\,[q\, 2]} = 0 \qquad (\langle 2\, 2\rangle = 0). \tag{58.90}$$

The first two terms of K each contain one of these vanishing dot products, so only the third survives:

$$K = (\varepsilon_3^+ \cdot \varepsilon_1^-)\,(p_3 - p_1)\cdot\varepsilon_2^-. \tag{58.91}$$

The surviving dot product is

$$\varepsilon_3^+ \cdot \varepsilon_1^- = \frac{\langle q_3\, 1\rangle\,[q\, 3]}{\langle q_3\, 3\rangle\,[q\, 1]} = \frac{\langle 2\, 1\rangle\,[q\, 3]}{\langle 2\, 3\rangle\,[q\, 1]}. \tag{58.92}$$

For the momentum factor, transversality $\varepsilon_2\cdot p_2 = 0$ with $p_2 = -(p_1+p_3)$ gives $\varepsilon_2\cdot p_3 = -\varepsilon_2\cdot p_1$, hence $(p_3-p_1)\cdot\varepsilon_2^- = -2\,\varepsilon_2^-\cdot p_1$; inserting $\varepsilon_2^- = \langle 2|\gamma^\mu|q]/(\sqrt{2}\,[q\,2])$ and using $\langle 2|\not{p}_1|q] = \langle 2\,1\rangle[1\,q]$,

$$(p_3-p_1)\cdot\varepsilon_2^- = -\frac{2}{\sqrt{2}}\,\frac{\langle 2|\not{p}_1|q]}{[q\,2]} = -\sqrt{2}\,\frac{\langle 2\,1\rangle\,[1\,q]}{[q\,2]}. \tag{58.93}$$

Multiplying Eq. (58.92) and Eq. (58.93), the reference square brackets combine through $[1\,q] = -[q\,1]$:

$$K = \frac{\langle 2\,1\rangle\,[q\,3]}{\langle 2\,3\rangle\,[q\,1]}\cdot\Big(-\sqrt{2}\,\frac{\langle 2\,1\rangle\,[1\,q]}{[q\,2]}\Big) = \sqrt{2}\,\frac{\langle 2\,1\rangle^{\,2}\,[q\,3]}{\langle 2\,3\rangle\,[q\,2]}. \tag{58.94}$$

The leftover ratio $[q\,3]/[q\,2]$ must collapse to a q-independent expression, since the amplitude is gauge invariant. On the holomorphic branch all anti-holomorphic spinors are proportional, $\tilde\lambda_i = c_i\tilde\zeta$, so $[q\,3] = c_3\,[\tilde\lambda_q\,\tilde\zeta]$ and $[q\,2] = c_2\,[\tilde\lambda_q\,\tilde\zeta]$, giving $[q\,3]/[q\,2] = c_3/c_2$. The scalars c_i obey momentum conservation in its holomorphic form $\sum_i c_i\lambda_i = 0$ (the bispinor identity Eq. (58.73) with $\tilde\lambda_i = c_i\tilde\zeta$); contracting with λ_1, i.e. taking $\langle 1\,\cdot\rangle$, yields $c_2\langle 1\,2\rangle + c_3\langle 1\,3\rangle = 0$, so

$$\frac{c_3}{c_2} = -\frac{\langle 1\,2\rangle}{\langle 1\,3\rangle} = \frac{\langle 1\,2\rangle}{\langle 3\,1\rangle}, \qquad \frac{[q\,3]}{[q\,2]} = \frac{\langle 1\,2\rangle}{\langle 3\,1\rangle}. \tag{58.95}$$

The q-dependence has indeed dropped out. Substituting into Eq. (58.94) and using $\langle 2\,1\rangle^2 = \langle 1\,2\rangle^2$,

$$K = \sqrt{2}\,\frac{\langle 1\,2\rangle^{\,2}}{\langle 2\,3\rangle}\cdot\frac{\langle 1\,2\rangle}{\langle 3\,1\rangle} = \sqrt{2}\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}. \tag{58.96}$$

The kinematic factor is exactly the symmetry-fixed ratio Eq. (58.85), up to the constant $\sqrt{2}$ that is an artifact of the polarization normalization in Eq. (58.88). Stripping the colour factor f^{abc} and adopting the standard colour-ordered normalization fixed in Section 59.1 — which absorbs precisely this $\sqrt{2}$ per vertex and the explicit factor of i carried by iV in Eq. (58.87) — the colour-ordered partial amplitude is

$$A_3(1^-,2^-,3^+) = i\,g\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \tag{58.97}$$

which fixes $c = ig$ in Eq. (58.85). The honest match has supplied exactly what symmetry could not: the dimensionless number i and the single power of the coupling g.

The relative sign and phase of the conjugate amplitude follow from parity, which exchanges $\lambda_i \leftrightarrow \tilde\lambda_i$, hence angle and square brackets, and maps the assignment $(1^-,2^-,3^+)$ to $(1^+,2^+,3^-)$; the polarization vectors Eq. (58.88) pick up one sign each under the exchange, and the amplitude is linear in all three, so

it acquires $(-1)^3 = -1$. Collecting both branches, the two nonvanishing complex three-gluon amplitudes are

$$\boxed{A_3(1^-,2^-,3^+) = i\,g\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \qquad A_3(1^+,2^+,3^-) = -\,i\,g\,\frac{[1\,2]^{\,3}}{[2\,3]\,[3\,1]}.} \tag{58.98}$$

These are the seeds. Chapter 59 takes Eq. (58.98) as the only input beyond momentum conservation and analyticity and reconstructs the entire tree-level n-gluon S-matrix from them, beginning with the Parke–Taylor formula whose $n = 3$ case is exactly Eq. (58.98).

Notation Watch

Theorem 58.20 fixes the bracket *structure* of every three-point amplitude from helicity alone; what it leaves open is the single constant c, whose dimensionless part is genuine dynamics. For gluons the present section pins that constant to $c = ig$ by matching to the cubic vertex of Chapter 26, and Eq. (58.98) is the coupling- and phase-fixed seed used from here on. Where a later chapter writes the bare ratio $\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$ without the prefactor, the complete on-shell object is always Eq. (58.98); the explicit $g^{\,n-2}$ that the colour decomposition of Section 59.1 keeps out front is, for the three-point seed, absorbed into A_3.

58.4.4 QED: the electron–photon vertex

We continue QED from its Feynman-rule layer in Chapter 17, now reframed on-shell. Take a massless electron, positron, and photon, all outgoing, and label the legs 1 (electron), 2 (positron), 3 (photon). The vector coupling of QED conserves chirality on a massless line (Theorem 17.18): the electron and positron emerging from one fermion line carry *opposite* helicities, $h_1 = -\tfrac{1}{2}$, $h_2 = +\tfrac{1}{2}$ (or the reverse). Pair this with a negative-helicity photon, $h_3 = -1$. Then $H = -\tfrac{1}{2} + \tfrac{1}{2} - 1 = -1 < 0$, the holomorphic branch is selected, and the exponents Eq. (58.78) are

$$a_1 = h_1 - h_2 - h_3 = -1, \quad a_2 = h_2 - h_3 - h_1 = +2, \quad a_3 = h_3 - h_1 - h_2 = 0, \tag{58.99}$$

so that, from Eq. (58.75),

$$\mathcal{M}_3(1^{-1/2},2^{+1/2},3^{-1}) = c\,\frac{\langle 3\,1\rangle^{\,2}}{\langle 1\,2\rangle}. \tag{58.100}$$

Its bracket-degree is $2-1 = 1$, so $[c] = 0$: the QED coupling is dimensionless, as it must be in four dimensions. The conjugate (positive-helicity photon) assignment has $H = +1 > 0$ and gives the square-bracket mirror $\tilde{c}\,[3\,1]^2/[1\,2]$. Symmetry

alone has reproduced the entire kinematic structure of the QED vertex; the following worked example confirms that the constant is exactly the electric charge, i.e. that minimal coupling is not an assumption but the unique outcome.

Worked Example 58.22 (Minimal coupling recovered: the $e\,e\,\gamma$ vertex in brackets)**.** Read the amplitude off the QED Feynman rules of Chapter 17. For the assignment $(1^{-1/2}, 2^{+1/2}, 3^{-1})$ the outgoing negative-helicity electron contributes the conjugate massless wavefunction $\bar{u}_-(1) = \langle 1|\,|] \to \langle 1|$, the outgoing positive-helicity positron contributes $v_+(2) \to |2]$, and the vertex is the minimal-coupling factor $-ie\gamma^\mu$ of Eq. (17.10). The single diagram gives

$$i\mathcal{M}_3 = \langle 1|\,(-ie\gamma^\mu)|2]\,\varepsilon^-_\mu(p_3;q) = -ie\,\langle 1|\gamma^\mu|2]\,\varepsilon^-_\mu(p_3;q). \tag{58.101}$$

Insert the negative-helicity photon polarization from Eq. (58.88), $\varepsilon^\mu_-(p_3;q) = \langle 3|\gamma^\mu|q]/(\sqrt{2}\,[q\,3])$ with a generic reference q, and apply the same Fierz identity $\langle 1|\gamma^\mu|2]\langle 3|\gamma_\mu|q] = 2\langle 1\,3\rangle[q\,2]$:

$$\langle 1|\gamma^\mu|2]\,\varepsilon^-_\mu(p_3;q) = \frac{2\,\langle 1\,3\rangle\,[q\,2]}{\sqrt{2}\,[q\,3]} = \sqrt{2}\,\langle 1\,3\rangle\,\frac{[q\,2]}{[q\,3]}. \tag{58.102}$$

On the holomorphic branch $\tilde{\lambda}_i = c_i\tilde{\zeta}$, so as in Eq. (58.95) the reference ratio collapses, $[q\,2]/[q\,3] = c_2/c_3$, and holomorphic momentum conservation $\sum_i c_i\lambda_i = 0$ contracted with λ_1 gives $c_2\langle 1\,2\rangle + c_3\langle 1\,3\rangle = 0$, whence $[q\,2]/[q\,3] = c_2/c_3 = -\langle 1\,3\rangle/\langle 1\,2\rangle$. Therefore

$$\langle 1|\gamma^\mu|2]\,\varepsilon^-_\mu(p_3;q) = \sqrt{2}\,\langle 1\,3\rangle\cdot\Big(-\frac{\langle 1\,3\rangle}{\langle 1\,2\rangle}\Big) = -\sqrt{2}\,\frac{\langle 1\,3\rangle^{\,2}}{\langle 1\,2\rangle} = -\sqrt{2}\,\frac{\langle 3\,1\rangle^{\,2}}{\langle 1\,2\rangle}, \tag{58.103}$$

using $\langle 1\,3\rangle^2 = \langle 3\,1\rangle^2$, manifestly q-independent as gauge invariance requires. Hence $\mathcal{M}_3 = \sqrt{2}\,e\,\langle 3\,1\rangle^2/\langle 1\,2\rangle$. Comparing with the symmetry-fixed form Eq. (58.100) fixes

$$c = \sqrt{2}\,e, \tag{58.104}$$

proportional to the single electric charge e (the factor $\sqrt{2}$ and the sign are the same polarization-normalization convention as in Theorem 58.21). The constant is the electromagnetic coupling and nothing else: the photon's three-point interaction with the electron is forced, by little-group scaling and locality, to be exactly minimal coupling. There was never any other Lorentz-invariant, local, dimensionless option; the gauge principle of Chapter 8 and the modern on-shell argument arrive at the same vertex from opposite directions.

This is the seed QED hands to the recursion: gluing two copies of Eq. (58.100) across a photon propagator reconstructs Compton scattering and $e^+e^- \to \mu^+\mu^-$ in helicity variables, and squaring and summing over helicities reproduces the unpolarized cross sections of Chapter 17 — the same numbers, now assembled from one-line gauge-invariant pieces rather than spin-summed traces.

58.4.5 Higher spin and the seed of the equivalence principle

The machinery is indifferent to the value of the helicity, and pushing it to spin two is where it becomes physically momentous. For three gravitons with the MHV-type assignment $(h_1, h_2, h_3) = (-2, -2, +2)$, one has $H = -2 < 0$, the holomorphic branch, and exponents $a_1 = h_1 - h_2 - h_3 = -2$, $a_2 = -2$, $a_3 = +6$:

$$\mathcal{M}_3(1^{--}, 2^{--}, 3^{++}) = \kappa \, \frac{\langle 1\,2\rangle^{\,6}}{\langle 2\,3\rangle^{\,2}\,\langle 3\,1\rangle^{\,2}} = \kappa \Big(\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle} \Big)^2 , \tag{58.105}$$

the *square* of the three-gluon amplitude Eq. (58.85) — the first appearance of the double-copy structure that Chapter 61 develops in full. By Eq. (58.83) the constant now has mass dimension $[\kappa] = 1 - |H| = -1$: a coupling with the dimension of a length. There is only one such coupling available to a helicity-two particle, and it is $\kappa \sim \sqrt{G_N}$, the gravitational coupling.

The decisive consequence, which we state here and prove in Chapter 61, is *universality*. Couple the same helicity-two particle to two copies of any other species — a scalar pair, a fermion pair, a photon pair — and repeat the three-point analysis. Locality (now in the sharper form that the four-point amplitude built by sewing two three-point amplitudes must have consistent factorization on all channels, a consistency condition we develop in Chapters 59 and 60) forces the helicity-two coupling to each species to be the *same* constant κ. A massless spin-two particle cannot couple with different strengths to different kinds of matter; if it couples at all, it couples to the total energy–momentum with one universal constant. That statement is the equivalence principle, here derived not from a geometric postulate but from the representation theory of a helicity-two state and the demand that amplitudes be consistent. The same line of argument shows that no consistent long-range interaction mediated by a massless particle of helicity $|h| \geq \frac{5}{2}$ exists at all: the would-be three-point constant has ever-more-negative mass dimension and the multi-point consistency conditions have no solution. We flag these as the headline results and defer their complete treatment — the four-point consistency analysis, the double copy, and the gravitational amplitudes — to Chapter 61, where the tools assembled in the intervening chapters make the proof clean.

Caution

The phrase “derives the equivalence principle” should be read precisely. What the on-shell argument establishes is that *if* a massless helicity-two particle exists and mediates a long-range force consistent with Lorentz invariance, locality, and unitarity, *then* it must couple universally with a single dimensionful constant — there is no room for species-dependent gravitational charge. It does not, by itself, prove that such a particle exists in Nature; that is an empirical fact. The content is that the structure of

gravity is forced once the spin is fixed, exactly as minimal coupling was forced for the photon in Theorem 58.22. This is the on-shell counterpart of the textbook statement that general covariance plus a massless spin-two field leaves Einstein gravity as essentially the only option.

We now have the complete set of three-point building blocks: the two gluon amplitudes Eq. (58.98), the electron–photon vertex Eq. (58.100) with its coupling fixed to the electric charge, and the graviton amplitude Eq. (58.105) as the square of the gluon seed. What remains before these can be sewn into higher-point amplitudes is the external-state data they were written against — the polarization vectors that Eq. (58.88) quietly borrowed. The next section constructs those polarizations directly from spinors, exhibits their reference-momentum dependence as gauge freedom, and thereby promotes the bracket language from on-shell three-point seeds to whole amplitudes, readying the gluon building blocks for the BCFW recursion of Chapter 59.

58.5 Polarization vectors in spinor-helicity

The previous section fixed the three-point amplitudes of massless particles by symmetry alone, and in doing so it forced upon us the recognition that a helicity-one state carries nontrivial weight under the little group Eq. (58.5): a positive-helicity gluon or photon scales as t^{-2}, a negative-helicity one as t^{+2}, and no Lorentz four-vector of fixed components can carry such a weight. The on-shell three-point structures $\langle 1\,2\rangle, \langle 2\,3\rangle, \langle 3\,1\rangle$ and $[1\,2], [2\,3], [3\,1]$ managed to carry it because they are built directly from spinors. To go beyond three points — to write down and contract *whole* amplitudes, with internal lines and many external legs — we need the external wavefunctions themselves in the same currency. A gluon line entering a Feynman diagram is contracted with a polarization vector $\epsilon^{\mu}(p)$; if we want the diagram to land in the world of brackets, we must express $\epsilon^{\mu}(p)$ in spinors. That is the task of this section, and the payoff is twofold. First, the awkward inhomogeneous gauge redundancy of a polarization vector — the very redundancy Wigner's classification predicted and that we met head-on in Theorem 3.46 — reappears here as the freedom to choose an arbitrary *reference spinor*, and that freedom becomes a computational lever sharp enough to annihilate most of the terms in a Feynman-diagram expansion before they are ever written down. Second, squaring the construction produces the spin-two (graviton) polarization for free, which is the kinematic seed of the double copy of Chapter 61.

58.5.1 The construction

We work throughout with the all-outgoing convention and the bracket normalization fixed earlier in this chapter,

$$\langle i\, j\rangle\,[j\, i] = 2\, p_i{\cdot}p_j = s_{ij}, \tag{58.106}$$

together with the slash decomposition of a massless momentum established in Eq. (58.18),

$$\not{p} = |p\rangle\,[p| + |p]\,\langle p|\,, \qquad \not{p}\,|p] = 0 = \not{p}\,|p\rangle\,, \tag{58.107}$$

the second equalities being the massless Weyl equations ($\langle p|\,|p] = 0$ and $[p\,p] = \langle p\,p\rangle = 0$ since a spinor pairs to zero with itself). A single immediate consequence of Eq. (58.107) is the identity we shall use constantly,

$$\langle p|\gamma^\mu|p] = \langle p|\,\gamma^\mu|p] = \mathrm{tr}\big[|p]\,\langle p|\;\gamma^\mu\big] = \mathrm{tr}\big[\tfrac{1+\gamma^5}{2}\,\not{p}\,\gamma^\mu\big] = \tfrac{1}{2}\,\mathrm{tr}\big[\not{p}\,\gamma^\mu\big] = 2\,p^\mu, \tag{58.108}$$

where we used $|p]\,\langle p| = \frac{1+\gamma^5}{2}\not{p}$ — the lower-left block of $\not{p}$ in the chiral basis of Eq. (58.17), with $\gamma^5 = \mathrm{diag}(-\mathbf{1}, +\mathbf{1})$ — and that the trace of γ^5 with two gamma matrices vanishes (Appendix C).

We also need two structural identities from the bracket algebra of this chapter, recalled here because every verification below leans on them. The first is the Fierz (Chisholm) rearrangement of a pair of vector currents,

$$\langle a|\gamma^\mu|b]\,\langle c|\gamma_\mu|d] = 2\,\langle a\, c\rangle\,[d\, b]. \tag{58.109}$$

Its overall coefficient is pinned by the special case $a = c = p$, $b = d = q$: the left side is $\langle p|\gamma^\mu|p]\,\langle p|\gamma_\mu|q]$ with the first factor replaced by $2p^\mu$ via Eq. (58.108) only when $q = p$, so instead take $a = p, b = p, c = q, d = q$, giving $\langle p|\gamma^\mu|p]\langle q|\gamma_\mu|q] = 4\,p{\cdot}q$ on the left and $2\langle p\,q\rangle[q\,p] = 4\,p{\cdot}q$ on the right by Eq. (58.106) — the two agree precisely when the constant is 2. The second is the Schouten identity, the statement that any three two-component angle spinors are linearly dependent (a two-dimensional space cannot hold three independent vectors):

$$|i\rangle\;\langle j\,k\rangle + |j\rangle\;\langle k\,i\rangle + |k\rangle\;\langle i\,j\rangle = 0, \tag{58.110}$$

with the identical statement holding for square spinors and brackets.

With these tools in hand we can state the construction. For each massless leg of momentum $p = \lambda\tilde{\lambda}$ we introduce an auxiliary null momentum q, the *reference momentum*, with its own spinors; q is required only to be null, $q^2 = 0$, and not collinear with p, so that $\langle q\,p\rangle \neq 0$ and $[q\,p] \neq 0$. The two helicity polarization vectors are

$$\epsilon_+^\mu(p;q) = \frac{\langle q|\gamma^\mu|p]}{\sqrt{2}\,\langle q\,p\rangle}, \qquad \epsilon_-^\mu(p;q) = \frac{\langle p|\gamma^\mu|q]}{\sqrt{2}\,[q\,p]}. \tag{58.111}$$

These are exactly the forms previewed, on the strength of their little-group weights alone, in Eq. (58.62): the positive-helicity vector is *antiholomorphic* in

p (it contains $\tilde{\lambda}_p$ in the numerator and λ_p in the denominator, net weight t^{-2}, helicity $+1$), the negative-helicity vector is *holomorphic* in p (net weight t^{+2}, helicity -1), and the reference spinor enters numerator and denominator with equal and opposite weight so that it carries no little-group charge at all — the first sign that q is gauge, not kinematics. We now prove that Eq. (58.111) has every property a polarization vector must have.

58.5.2 The defining properties

Theorem 58.23 (Properties of the spinor-helicity polarization vectors)**.** *Let p be a massless momentum and q a reference momentum with $\langle q\,p\rangle \neq 0$, $[q\,p] \neq 0$. The vectors Eq. (58.111) satisfy*

$$\epsilon_\pm(p;q)\cdot p = 0 \qquad \textit{(transversality)}, \tag{58.112}$$

$$\epsilon_\pm(p;q)\cdot \epsilon_\pm(p;q) = 0 \qquad \textit{(nullness)}, \tag{58.113}$$

$$\epsilon_+(p;q)\cdot \epsilon_-(p;q) = 1 \qquad \textit{(cross normalization)}, \tag{58.114}$$

$$\epsilon_\pm(p;q)\cdot q = 0 \qquad \textit{(transversality to the reference)}. \tag{58.115}$$

Under the little-group rescaling Eq. (58.5) on leg p, $\epsilon_\pm \to t^{\mp 2}\epsilon_\pm$, identifying them as the helicity-(± 1) external wavefunctions. For real momenta the reality condition Eq. (58.46) gives $\big(\epsilon_\pm(p;q)\big)^* = -\,\epsilon_\mp(p;q)$, whence the physical normalization $\epsilon_\pm\cdot(\epsilon_\pm)^* = -1$.

Proof. Transversality Eq. (58.112). Contracting the numerator of ϵ_+ with p_μ assembles a slashed momentum acting on its own square ket:

$$\epsilon_+(p;q)\cdot p = \frac{\langle q|\gamma^\mu|p]\,p_\mu}{\sqrt{2}\,\langle q\,p\rangle} = \frac{\langle q|\,\not{p}\,|p]}{\sqrt{2}\,\langle q\,p\rangle} = 0, \tag{58.116}$$

because $\not{p}\,|p] = 0$ by Eq. (58.107). Identically, $\epsilon_-(p;q)\cdot p = \langle p|\,\not{p}\,|q]/(\sqrt{2}\,[q\,p]) = 0$ since $\langle p|\,\not{p} = 0$ (the conjugate Weyl equation). The polarization is transverse to the particle's own momentum, as a physical photon or gluon wavefunction must be.

Transversality to the reference Eq. (58.115). Writing $q_\mu = \tfrac{1}{2}\langle q|\gamma_\mu|q]$ from Eq. (58.108) and using the Fierz identity Eq. (58.109),

$$\epsilon_+(p;q)\cdot q = \frac{1}{2\sqrt{2}\,\langle q\,p\rangle}\,\langle q|\gamma^\mu|p]\,\langle q|\gamma_\mu|q] = \frac{1}{2\sqrt{2}\,\langle q\,p\rangle}\,2\,\langle q\,q\rangle\,[q\,p] = 0, \tag{58.117}$$

since $\langle q\,q\rangle = 0$; the analogous square-bracket computation gives $\epsilon_-\cdot q = 0$. Both vectors are transverse to the reference momentum as well — the geometric content of "axial" or "light-cone" gauge with gauge vector q, to which we return below.

Nullness Eq. (58.113). By the Fierz identity,

$$\epsilon_+\cdot\epsilon_+ = \frac{\langle q|\gamma^\mu|p]\,\langle q|\gamma_\mu|p]}{2\,\langle q\,p\rangle^2} = \frac{2\,\langle q\,q\rangle\,[p\,p]}{2\,\langle q\,p\rangle^2} = 0, \tag{58.118}$$

and likewise $\epsilon_- \cdot \epsilon_- = 2\langle p\,p\rangle[q\,q]/(2[q\,p]^2) = 0$. A single-helicity polarization vector is null — the complexified statement of circular polarization.

Cross normalization Eq. (58.114). Again by Eq. (58.109), with $a = q, b = p, c = p, d = q$,

$$\epsilon_+ \cdot \epsilon_- = \frac{\langle q|\gamma^\mu|p]\,\langle p|\gamma_\mu|q]}{2\,\langle q\,p\rangle\,[q\,p]} = \frac{2\,\langle q\,p\rangle\,[q\,p]}{2\,\langle q\,p\rangle\,[q\,p]} = 1. \tag{58.119}$$

Little-group weight. Under $\lambda_p \mapsto t\lambda_p$, $\tilde\lambda_p \mapsto t^{-1}\tilde\lambda_p$ the numerator $\langle q|\gamma^\mu|p] \propto \tilde\lambda_p$ scales as t^{-1} and the denominator $\langle q\,p\rangle \propto \lambda_p$ as t^{+1}, so $\epsilon_+ \to t^{-2}\epsilon_+$; symmetrically $\epsilon_- \to t^{+2}\epsilon_-$. With the convention that a helicity-h wavefunction carries weight t^{-2h} (Theorem 58.16), ϵ_+ is helicity $+1$ and ϵ_- is helicity -1, as the labels assert. The reference spinor's scaling cancels between numerator and denominator, so $\epsilon_\pm$ are invariant under the little group acting on q.

Reality. For a real momentum of positive energy, Eq. (58.46) gives $\tilde\lambda = \lambda^*$, from which the brackets obey $\langle i\,j\rangle^* = [j\,i]$ and the mixed sandwich obeys $\langle i|\gamma^\mu|j]^* = \langle j|\gamma^\mu|i]$ (the Hermiticity of γ^μ between conjugate spinors). Hence

$$(\epsilon_+^\mu(p;q))^* = \frac{\langle q|\gamma^\mu|p]^*}{\sqrt{2}\,\langle q\,p\rangle^*} = \frac{\langle p|\gamma^\mu|q]}{\sqrt{2}\,[p\,q]} = \frac{\langle p|\gamma^\mu|q]}{-\sqrt{2}\,[q\,p]} = -\,\epsilon_-^\mu(p;q), \tag{58.120}$$

using $[p\,q] = -[q\,p]$, and symmetrically $(\epsilon_-)^* = -\epsilon_+$. Therefore $\epsilon_+ \cdot (\epsilon_+)^* = -\,\epsilon_+ \cdot \epsilon_- = -1$, the spacelike norm demanded by the mostly-minus metric. The crossed value $\epsilon_+ \cdot \epsilon_- = +1$ in Eq. (58.114) and the conjugate value $\epsilon_+ \cdot (\epsilon_+)^* = -1$ are not in tension: they differ exactly by the reality sign $(\epsilon_\pm)^* = -\epsilon_\mp$. □

Notation Watch

The spinor-helicity polarization vectors Eq. (58.111) are fixed only up to an overall little-group phase, which is physically immaterial: an amplitude contracted with $\epsilon_\pm$ acquires that phase as an overall factor and the cross section, $|\mathcal{M}|^2$, is blind to it. The particular phase in Eq. (58.111) yields the convenient *holomorphic* cross normalization $\epsilon_+ \cdot \epsilon_- = +1$ rather than the -1 one would read off the real circular-polarization vectors $\frac{1}{\sqrt{2}}(0, 1, \pm i, 0)$ of Theorem 13.12; the two descriptions are related by an overall factor of i on each $\epsilon_\pm$, equivalently by the reality relation $(\epsilon_\pm)^* = -\epsilon_\mp$ derived above. Whenever a polarization sum $\sum_h \epsilon_h \epsilon_h^*$ is needed — as in a squared amplitude — it is the conjugate pairing, with its minus sign, that enters, and it reproduces the mostly-minus Chapter 13 result, as we now verify.

58.5.3 The completeness relation

A polarization vector earns its keep inside a squared amplitude, where the sum over physical helicities of a virtual or external boson must collapse to the covariant

propagator numerator of Theorem 13.30. We check that the spinor construction reproduces it on the nose, reference momentum and all.

Theorem 58.24 (Completeness / polarization sum)**.** *For a massless leg p with reference momentum q,*

$$\sum_{h=\pm} \epsilon_h^{\mu}(p;q)\,\big(\epsilon_h^{\nu}(p;q)\big)^{*} = -\,\eta^{\mu\nu} + \frac{p^{\mu}q^{\nu}+p^{\nu}q^{\mu}}{p\cdot q}, \tag{58.121}$$

which is precisely the polarization replacement of Theorem 13.30 with the Chapter-13 gauge vector identified with the reference momentum q.

Proof. By the reality relation $(\epsilon_{\pm})^{*} = -\epsilon_{\mp}$ of Theorem 58.23, the conjugate sum is minus the symmetric product of the two helicity vectors:

$$\sum_{h=\pm} \epsilon_h^{\mu}(\epsilon_h^{\nu})^{*} = -\big(\epsilon_+^{\mu}\epsilon_-^{\nu} + \epsilon_-^{\mu}\epsilon_+^{\nu}\big). \tag{58.122}$$

It remains to compute the symmetric product. Using $|p]\,\langle p| = \frac{1+\gamma^5}{2}\not{p}$ to fold the inner pair of spinors into a slashed momentum,

$$S^{\mu\nu} \equiv \langle q|\gamma^{\mu}|p]\,\langle p|\gamma^{\nu}|q] = \mathrm{tr}\Big[\tfrac{1+\gamma^5}{2}\not{q}\,\gamma^{\mu}\,\tfrac{1+\gamma^5}{2}\not{p}\,\gamma^{\nu}\Big] = \mathrm{tr}\Big[\tfrac{1+\gamma^5}{2}\not{q}\,\gamma^{\mu}\not{p}\,\gamma^{\nu}\Big], \tag{58.123}$$

where the two chiral projectors merged because moving one through the even string $\not{q}\gamma^{\mu}$ returns it to the form $\frac{1+\gamma^5}{2}\not{q}\gamma^{\mu}$ and projectors are idempotent. Splitting into the parity-even and parity-odd traces of Appendix C,

$$S^{\mu\nu} = \tfrac{1}{2}\,\mathrm{tr}\big[\not{q}\gamma^{\mu}\not{p}\gamma^{\nu}\big] + \tfrac{1}{2}\,\mathrm{tr}\big[\gamma^5\not{q}\gamma^{\mu}\not{p}\gamma^{\nu}\big] = 2(q^{\mu}p^{\nu}+q^{\nu}p^{\mu}-(q\cdot p)\eta^{\mu\nu}) - 2i\,\epsilon^{\alpha\mu\beta\nu}q_{\alpha}p_{\beta}. \tag{58.124}$$

The product entering $\epsilon_-^{\mu}\epsilon_+^{\nu}$ is $S^{\mu\nu}$ with $p \leftrightarrow q$, call it $S'^{\mu\nu}$; its parity-odd piece is the negative of that of $S^{\mu\nu}$, so the Levi-Civita terms cancel in the sum and

$$S^{\mu\nu} + S'^{\mu\nu} = 4(p^{\mu}q^{\nu}+p^{\nu}q^{\mu}) - 4\,(p\cdot q)\,\eta^{\mu\nu}. \tag{58.125}$$

Dividing by the denominators of Eq. (58.111) and using $\langle q\,p\rangle[q\,p] = -\,\langle q\,p\rangle[p\,q] = -2\,p\cdot q$ from Eq. (58.106),

$$\epsilon_+^{\mu}\epsilon_-^{\nu} + \epsilon_-^{\mu}\epsilon_+^{\nu} = \frac{S^{\mu\nu}+S'^{\mu\nu}}{2\,\langle q\,p\rangle\,[q\,p]} = \frac{4(p^{\mu}q^{\nu}+p^{\nu}q^{\mu}) - 4(p\cdot q)\eta^{\mu\nu}}{-4\,(p\cdot q)} = \eta^{\mu\nu} - \frac{p^{\mu}q^{\nu}+p^{\nu}q^{\mu}}{p\cdot q}. \tag{58.126}$$

Substituting into Eq. (58.122) yields Eq. (58.121). □

The reference momentum appears in the completeness relation precisely where the gauge-fixing vector appears in Theorem 13.30: it is the residual gauge artefact of describing a two-state photon with a four-component vector. In a gauge-invariant quantity it must drop out, and the next subsection makes that mechanism explicit and exact.

58.5.4 Reference freedom is gauge freedom

The reference momentum q was introduced as a free choice, and the little-group weight count already hinted that it carries no physical information. We now show the strong form of that statement: changing q shifts $\epsilon_\pm$ by a vector proportional to p^μ — exactly the form of a gauge transformation Eq. (8.113) — so that any quantity respecting the Ward identity is strictly independent of every reference momentum.

Proposition 58.25 (Reference shift is a gauge transformation)**.** *For two admissible reference momenta q, q',*

$$\epsilon_+^\mu(p;q') - \epsilon_+^\mu(p;q) = \frac{\sqrt{2}\,\langle q\,q'\rangle}{\langle q\,p\rangle\,\langle q'\,p\rangle}\,p^\mu, \tag{58.127}$$

and analogously $\epsilon_-^\mu(p;q') - \epsilon_-^\mu(p;q) = \dfrac{\sqrt{2}\,[q'\,q]}{[q\,p]\,[q'\,p]}\,p^\mu$. The shift is purely longitudinal: $\epsilon_\pm(p;q)$ and $\epsilon_\pm(p;q')$ differ by a multiple of p^μ.

Proof. Subtract the two expressions in Eq. (58.111) over a common denominator,

$$\epsilon_+^\mu(p;q') - \epsilon_+^\mu(p;q) = \frac{1}{\sqrt{2}}\,\frac{\langle q'|\gamma^\mu|p]\,\langle q\,p\rangle - \langle q|\gamma^\mu|p]\,\langle q'\,p\rangle}{\langle q'\,p\rangle\,\langle q\,p\rangle}. \tag{58.128}$$

The square spinor $\tilde\lambda_p$ is common to both mixed sandwiches in the numerator; factoring it out, the bracketed combination of *angle* spinors is $(\lambda_{q'}\langle q\,p\rangle - \lambda_q\langle q'\,p\rangle)$ contracted into $\sigma^\mu\tilde\lambda_p$. The Schouten identity Eq. (58.110) with $(i,j,k) = (q',q,p)$ reads $|q'\rangle\,\langle q\,p\rangle + |q\rangle\,\langle p\,q'\rangle + |p\rangle\,\langle q'\,q\rangle = 0$, i.e.

$$\lambda_{q'}\,\langle q\,p\rangle - \lambda_q\,\langle q'\,p\rangle = \lambda_p\,\langle q\,q'\rangle, \tag{58.129}$$

where we used $\langle p\,q'\rangle = -\langle q'\,p\rangle$ and $\langle q'\,q\rangle = -\langle q\,q'\rangle$. Therefore the numerator becomes $\langle q\,q'\rangle\,\lambda_p\,\sigma^\mu\,\tilde\lambda_p = \langle q\,q'\rangle\,\langle p|\gamma^\mu|p] = 2\,\langle q\,q'\rangle\,p^\mu$ by Eq. (58.108), and

$$\epsilon_+^\mu(p;q') - \epsilon_+^\mu(p;q) = \frac{1}{\sqrt{2}}\,\frac{2\,\langle q\,q'\rangle\,p^\mu}{\langle q'\,p\rangle\,\langle q\,p\rangle} = \frac{\sqrt{2}\,\langle q\,q'\rangle}{\langle q\,p\rangle\,\langle q'\,p\rangle}\,p^\mu, \tag{58.130}$$

which is Eq. (58.127). The negative-helicity case is identical with angle spinors and brackets replaced by their square counterparts and the square-bracket Schouten identity. □

Compare Eq. (58.127) with the momentum-space form of the gauge transformation $A_\mu \to A_\mu + \partial_\mu\chi$, namely $\epsilon^\mu \to \epsilon^\mu + \alpha\,p^\mu$, recorded as Eq. (8.113): changing the reference momentum is, term for term, a gauge transformation of the external polarization, with the gauge parameter $\alpha = \sqrt{2}\,\langle q\,q'\rangle/(\langle q\,p\rangle\langle q'\,p\rangle)$ read off explicitly. The reference-momentum freedom of Eq. (58.111) *is* the gauge

freedom of the photon, repackaged as a choice of spinor. This is the precise resolution of the puzzle posed in Section 3.7.4: the surplus degree of freedom a four-vector carries over a genuine helicity-one state, which there obstructed any manifestly Lorentz-covariant labelling, is here demoted to a harmless parameter we are free to set.

Theorem 58.26 (Reference independence of gauge-invariant amplitudes)**.** *Let $\mathcal{M}^\mu(p)$ be the amputated, on-shell matrix element with the polarization of leg p stripped off, so that the amplitude is $\mathcal{M} = \epsilon_\mu(p;q)\,\mathcal{M}^\mu(p)$. If $\mathcal{M}^\mu$ satisfies the Ward identity $p_\mu \mathcal{M}^\mu = 0$, then $\mathcal{M}$ is independent of the reference momentum q.*

Proof. Changing $q \to q'$ shifts the contracted amplitude by

$$\left(\epsilon_\mu(p;q') - \epsilon_\mu(p;q)\right)\mathcal{M}^\mu(p) = \frac{\sqrt{2}\,\langle q\,q'\rangle}{\langle q\,p\rangle\,\langle q'\,p\rangle}\,p_\mu\,\mathcal{M}^\mu(p) = 0, \tag{58.131}$$

by Theorem 58.25 followed by the Ward identity $p_\mu\mathcal{M}^\mu = 0$ (the tree-level statement of current conservation, Theorem 18.5, itself the on-shell shadow of the gauge symmetry whose seed is Eq. (8.113)). The same holds independently for each external gauge boson; hence a physical amplitude is independent of all reference momenta simultaneously. □

Physical Insight

Theorem 58.26 is the single most useful practical statement in this chapter. Because the answer does not depend on the q_i, we may choose them *per leg* and *at our convenience* to make as many polarization contractions vanish as possible. A Feynman-diagram expansion of a gluon amplitude contains, term by term, dot products $\epsilon_i \cdot \epsilon_j$, $\epsilon_i \cdot p_j$, and $\epsilon_i \cdot \epsilon_j^*$; a clever common choice of references can zero out whole families of these at once, so that the great majority of diagrams collapse before any integral or trace is done. This is precisely how the hundreds of terms advertised in the chapter introduction shrink to a one-line answer. The next chapter turns the trick into a method; here we exhibit the arithmetic that makes it work.

Worked Example 58.27 (Engineering vanishing polarization products)**.** Take two distinct massless legs i, j with references q_i, q_j. Using only the Fierz identity Eq. (58.109) and the slash identity Eq. (58.107), we compute every polarization dot product in closed form and read off when it vanishes.

Same-helicity pairs. For two positive-helicity legs,

$$\epsilon_+(i;q_i)\cdot\epsilon_+(j;q_j) = \frac{\langle q_i|\gamma^\mu|i]\,\langle q_j|\gamma_\mu|j]}{2\,\langle q_i\,i\rangle\,\langle q_j\,j\rangle} = \frac{\langle q_i\,q_j\rangle\,[j\,i]}{\langle q_i\,i\rangle\,\langle q_j\,j\rangle}, \tag{58.132}$$

which vanishes the instant the two references are chosen equal, $q_i = q_j$, because $\langle q_i\,q_j\rangle \to \langle q\,q\rangle = 0$. Symmetrically, two negative-helicity legs give $\epsilon_-(i;q_i)\cdot$

$\epsilon_-(j;q_j) = \langle i\,j\rangle[q_j\,q_i]/([q_i\,i][q_j\,j])$, again zero for a common reference $q_i = q_j$. *All like-helicity polarization vectors that share a reference are mutually orthogonal.*

Opposite-helicity pairs. For one of each,

$$\epsilon_+(i;q_i)\cdot\epsilon_-(j;q_j) = \frac{\langle q_i|\gamma^\mu|i]\,\langle j|\gamma_\mu|q_j]}{2\,\langle q_i\,i\rangle\,[q_j\,j]} = \frac{\langle q_i\,j\rangle\,[q_j\,i]}{\langle q_i\,i\rangle\,[q_j\,j]}, \tag{58.133}$$

which vanishes if the reference of the positive leg is aligned with the *momentum* of the negative leg, $q_i = p_j$ (then $\langle q_i\,j\rangle = \langle j\,j\rangle = 0$), or if $q_j = p_i$.

Polarization-momentum products. Folding p_j into a slash,

$$\epsilon_+(i;q_i)\cdot p_j = \frac{\langle q_i|\not{p}_j|i]}{\sqrt{2}\,\langle q_i\,i\rangle} = \frac{\langle q_i\,j\rangle\,[j\,i]}{\sqrt{2}\,\langle q_i\,i\rangle}, \tag{58.134}$$

where only the $|j\rangle\,[j|$ piece of $\not{p}_j$ survives because $\langle q_i|\,|j] = 0$. This too vanishes for $q_i = p_j$.

The lever in action. Consider an amplitude with all external gluons of positive helicity. Choose one common reference q for every leg. Then $\epsilon_i\cdot\epsilon_j = 0$ for every pair, by the same-helicity result. In the Yang–Mills Feynman rules every interaction vertex carries at least one factor $\epsilon_i\cdot\epsilon_j$ once the dust settles (the three-gluon vertex is linear in such products, the four-gluon vertex quadratic), so every diagram is killed and the all-plus tree amplitude vanishes identically — a result we obtained here from four lines of bracket algebra and will rederive structurally in Chapter 59. With exactly one leg flipped to negative helicity, almost all products still vanish for a suitable common reference, leaving the famous single surviving structure; this is the kinematic origin of the Parke–Taylor formula of Chapter 59.

58.5.5 Spin two by squaring: the graviton polarization

A massless spin-two particle — the graviton — carries helicity ±2 and is described on shell by a symmetric, transverse, traceless polarization tensor $\epsilon^{\mu\nu}$. Nothing new is needed to build it: the spin-one construction, squared, already produces an object with all the right properties and the right little-group weight.

Definition 58.28 (Graviton polarization tensor)**.** For a massless leg p with reference momentum q, define

$$\epsilon^{\mu\nu}_\pm(p;q) \equiv \epsilon^\mu_\pm(p;q)\,\epsilon^\nu_\pm(p;q). \tag{58.135}$$

The defining properties follow at once from Theorem 58.23. The tensor is manifestly symmetric. It is traceless,

$$\eta_{\mu\nu}\,\epsilon^{\mu\nu}_\pm = \epsilon_\pm\cdot\epsilon_\pm = 0 \quad \text{by Eq. (58.113)}, \tag{58.136}$$

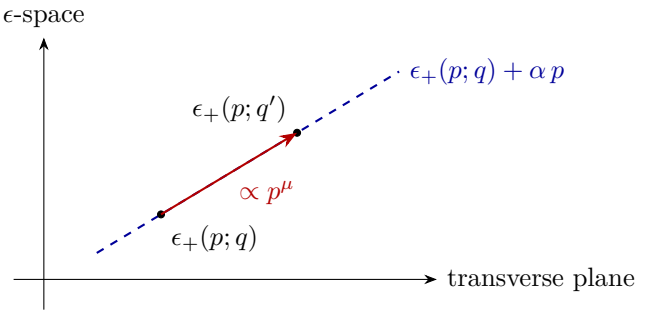


Figure 58.4: The reference-momentum freedom as a gauge orbit. Every admissible reference q gives a polarization vector lying on a single affine line in the direction of the particle's momentum p^μ (Eq. (58.127)). Moving along the line is a gauge transformation Eq. (8.113); a Ward-identity-respecting amplitude takes the same value at every point of it (Theorem 58.26).

and transverse on both indices,

$$p_\mu\,\epsilon_\pm^{\mu\nu} = (\epsilon_\pm\cdot p)\,\epsilon_\pm^{\nu} = 0 \quad \text{by Eq. (58.112).} \tag{58.137}$$

Under the little group it scales as $\epsilon_\pm^{\mu\nu} \to t^{\mp 4}\epsilon_\pm^{\mu\nu}$, i.e. with weight t^{-2h} for $h = \pm 2$: it is genuinely a helicity-(± 2) wavefunction. Finally, changing the reference shifts it by a linearized diffeomorphism: from Eq. (58.127), with $\Delta\epsilon_\pm^\mu = \alpha\,p^\mu$,

$$\Delta\epsilon_\pm^{\mu\nu} = \alpha(p^\mu\epsilon_\pm^\nu + \epsilon_\pm^\mu p^\nu) + \alpha^2 p^\mu p^\nu = p^{(\mu}\xi^{\nu)} + \dots, \tag{58.138}$$

the leading term being exactly the form $\delta h_{\mu\nu} = \partial_{(\mu}\xi_{\nu)}$ of a coordinate transformation on the linearized metric — the gravitational counterpart of the gauge shift Eq. (8.113). Thus the reference freedom of the squared polarization is the gauge (diffeomorphism) freedom of the graviton, and gravitational amplitudes built from these external states are reference-independent for the same Ward-identity reason as in Theorem 58.26.

Physical Insight

That a graviton polarization is literally the square of a gauge-boson polarization, $\epsilon^{\mu\nu} = \epsilon^\mu\epsilon^\nu$, is not a notational coincidence. It is the on-shell, external-state face of the *double copy*: the statement that the kinematic data of gravity is built from two copies of the kinematic data of gauge theory, leg by leg. Chapter 61 promotes this observation from external wavefunctions to entire amplitudes, replacing color factors by a second copy of kinematic numerators; Eq. (58.135) is the seed from which that construction grows.

With external spin-one and spin-two states now written entirely in brackets, and with the reference freedom understood as the gauge lever that empties

out Feynman-diagram sums, every ingredient of a tree amplitude — momenta, propagators, vertices, and now polarizations — can be assembled and contracted leg by leg in the spinor variables. The next section cashes this in on the book's running QED examples, contracting these polarization vectors against the electron currents to recover, from one-line helicity amplitudes, the unpolarized cross sections computed the long way in Chapter 17.

58.6 QED reframed: helicity amplitudes that reproduce the cross section

The preceding section completed the kinematic toolkit. We now have, for every massless external leg, exactly the on-shell data Wigner's classification says it carries and nothing more: a pair of two-component spinors $\lambda_i, \tilde{\lambda}_i$ for each momentum, their Lorentz invariants $\langle i\,j\rangle$ and $[i\,j]$, the three-point amplitudes fixed by little-group scaling alone, and—built from the same spinors—polarization vectors $\varepsilon^{\mu}_{\pm}(p_i; q_i)$ whose reference momentum q_i is pure gauge. The promise made in the chapter's opening was that this apparatus would pay for itself by collapsing the gauge-dependent, many-term Feynman expressions of Parts IV–VII into one-line gauge-invariant answers. It is time to redeem that promise on the book's own running examples. We take the two flagship tree processes of QED—the annihilation $e^+e^- \to \mu^+\mu^-$ computed by brute-force traces in Chapter 17, and Compton scattering $e\gamma \to e\gamma$ computed the same way in Chapter 18—and rebuild them in helicity variables. The annihilation cross section will reappear, exact and to the last factor, out of four one-line amplitudes and no gamma-matrix trace at all; Compton's two Feynman diagrams will fuse into a single bracket once a reference momentum is chosen wisely. This is the primary QED analysis of the chapter, advancing the electron's story (Chapters 8, 11, 13 and 17 to 20) to its modern amplitude layer.

Notation Watch

We use the conventions fixed earlier in this chapter and restate them, since signs differ between references. *All external momenta are outgoing*; an incoming particle of momentum P and helicity h is the same datum as an outgoing leg of momentum $p = -P$ and helicity $-h$, with the antiparticle/particle label interchanged. Brackets are normalized so that

$$\langle i\,j\rangle\,[j\,i] = 2\,p_i{\cdot}p_j = s_{ij}, \qquad s_{ij} \equiv (p_i + p_j)^2, \tag{58.139}$$

as established in Eq. (58.24). For *real* momenta the angle and square spinors are complex conjugates up to the sign of the energy, $\tilde{\lambda}_i = \pm\lambda_i^*$, so that $[i\,j] = \pm\langle j\,i\rangle^*$ and, crucially,

$$\big|\langle i\,j\rangle\big|^2 = \big|[i\,j]\big|^2 = |s_{ij}|, \tag{58.140}$$

the one fact about brackets that survives squaring and that does all the work in the cross-section checks below. Overall phases of helicity amplitudes are convention-dependent and drop out of $|\mathcal{M}|^2$; we track them only far enough to keep relative signs honest. The metric is mostly-minus and the QED coupling enters through $D_\mu = \partial_\mu + ieA_\mu$ on the electron, $\alpha = e^2/4\pi$, exactly as in QED (Eq. (8.143)).

58.6.1 Massless fermion currents and the Fierz rearrangement

Every QED amplitude in Chapters 17 and 18 is built by sewing electromagnetic currents $\bar{u}\,\gamma^\mu u$ (and their v-spinor relatives) onto photons. The spinor-helicity reframing rests on one observation about such currents in the massless limit, already half-anticipated by the chirality selection rule of Theorem 17.18: a single gamma matrix between two massless spinors of definite helicity is not a four-vector with sixteen components to track, but one of just two elementary bilinears, an "angle–square" sandwich or its conjugate.

Recall from the construction of the on-shell spinors (Eq. (58.17)) that a positive-energy massless Dirac spinor of definite helicity is the four-component embedding of a single Weyl spinor: in the chiral basis a negative-helicity (left) particle carries the angle data $|i\rangle$, a positive-helicity (right) particle the square data $|i]$, with the Dirac conjugates supplying $\langle i|$ and $[i|$. The vector vertex γ^μ is chirality-off-diagonal (Eq. (17.148)), so it connects an angle bra to a square ket or vice versa, and the only two non-vanishing single-gamma bilinears are

$$\langle i|\gamma^\mu|j] = \langle i|\,\gamma^\mu\,|j], \qquad [i|\gamma^\mu|j\rangle = [i|\gamma^\mu\,|j\rangle\,, \tag{58.141}$$

related by $\langle i|\gamma^\mu|j] = [j|\gamma^\mu|i\rangle$ (the transpose identity of Eq. (58.29)). The would-be same-chirality bilinears $\langle i|\,\gamma^\mu\,|j\rangle$ and $[i|\gamma^\mu|j]$ vanish identically—this is Theorem 17.18 read in two components. A massless current therefore carries a definite helicity flow: the QED vertex cannot turn a left line into a right line.

The single algebraic identity that turns products of currents into products of brackets is the Fierz rearrangement. It is the engine of every computation in this section, so we prove it.

Lemma 58.29 (Fierz rearrangement of two currents)**.** *For any four massless momenta,*

$$\langle 1|\gamma^\mu|2]\,\langle 3|\gamma_\mu|4] = 2\,\langle 1\,3\rangle\,[4\,2], \tag{58.142}$$

and, contracting one current against the gamma matrix itself, the completeness relation

$$\langle i|\gamma^\mu|j]\,(\gamma_\mu) = 2\big(|j]\,\langle i| + |i\rangle\,[j|\big). \tag{58.143}$$

Proof. Write the currents in two-component form, $\langle 1|\gamma^\mu|2] = \lambda_1^\alpha\,\sigma^\mu_{\alpha\dot\alpha}\,\tilde\lambda_2^{\dot\alpha}$ and likewise for legs $3,4$, with $\sigma^\mu = (\mathbf{1},\boldsymbol{\sigma})$ the Pauli data of Eq. (C.274). The

contraction over the Lorentz index is performed by the σ-completeness identity

$$\sigma^{\mu}_{\alpha\dot\alpha}\,\sigma_{\mu\,\beta\dot\beta} = 2\,\epsilon_{\alpha\beta}\,\epsilon_{\dot\alpha\dot\beta}, \tag{58.144}$$

established as Eq. (58.35). Hence

$$\langle 1|\gamma^{\mu}|2]\,\langle 3|\gamma_{\mu}|4] = \lambda_1^{\alpha}\tilde\lambda_2^{\dot\alpha}\,\lambda_3^{\beta}\tilde\lambda_4^{\dot\beta}\cdot 2\,\epsilon_{\alpha\beta}\epsilon_{\dot\alpha\dot\beta} = 2\,(\lambda_1^{\alpha}\lambda_3^{\beta}\epsilon_{\alpha\beta})\,(\tilde\lambda_2^{\dot\alpha}\tilde\lambda_4^{\dot\beta}\epsilon_{\dot\alpha\dot\beta}). \tag{58.145}$$

By the bracket definitions $\lambda_1^{\alpha}\lambda_3^{\beta}\epsilon_{\alpha\beta} = \langle 1\,3\rangle$ and $\tilde\lambda_2^{\dot\alpha}\tilde\lambda_4^{\dot\beta}\epsilon_{\dot\alpha\dot\beta} = [4\,2]$ (the antisymmetry of $\epsilon_{\dot\alpha\dot\beta}$ fixes the order $4,2$), which is Eq. (58.142). Stripping $\lambda_3, \tilde\lambda_4$ from both sides—i.e. contracting only the first current's Lorentz index against the loose γ_μ—turns the same manipulation into the matrix statement Eq. (58.143), the four-component image of Eq. (58.144). □

Two corollaries of Eq. (58.142), obtained by relabeling and using $\langle i|\gamma^{\mu}|j] = [j|\gamma^{\mu}|i\rangle$, will be used without further comment:

$$\langle 1|\gamma^{\mu}|2]\,[3|\gamma_{\mu}|4\rangle = 2\,\langle 1\,4\rangle\,[3\,2], \qquad [1|\gamma^{\mu}|2\rangle\,[3|\gamma_{\mu}|4\rangle = 2\,[1\,3]\,\langle 4\,2\rangle. \tag{58.146}$$

The pattern is mechanical: two currents glued at a Lorentz index become a product of two brackets, angle with angle and square with square, the helicity flow of each current dictating which spinor lands in which bracket. This is all the algebra the annihilation amplitude needs.

58.6.2 $e^+e^- \to \mu^+\mu^-$ as helicity amplitudes

We continue QED from the helicity decomposition of Section 17.6.3, where the four nonzero squared amplitudes were extracted from a projected trace. Here we obtain the amplitudes themselves, as single brackets, and square them only at the very end.

All-outgoing kinematics. Label the four particles, all outgoing,

$$1 = e^-, \quad 2 = e^+, \quad 3 = \mu^-, \quad 4 = \mu^+, \tag{58.147}$$

with outgoing momenta $p_1, \dots, p_4$ summing to zero. The electron line carries the charge flow $1 \to 2$, the muon line $3 \to 4$, and a single photon is exchanged in the s_{12} channel. Crossing legs $1, 2$ back to the incoming electron and positron of Eq. (17.156) ($p_1 = -p_-$, $p_2 = -p_+$, $p_3 = k_-$, $p_4 = k_+$) translates the physical Mandelstam invariants of Theorem 16.21 into bracket invariants,

$$s_{12} = s_{34} = s, \qquad s_{13} = s_{24} = t, \qquad s_{14} = s_{23} = u, \tag{58.148}$$

with $s + t + u = 0$ in the massless limit, $t = -\frac{s}{2}(1 - \cos\theta)$, $u = -\frac{s}{2}(1 + \cos\theta)$ from Eq. (17.157).

The amplitude in bracket form. The Feynman amplitude is the one written in Eq. (17.155): an electron current and a muon current tied by the Feynman-gauge photon propagator $-i\eta_{\mu\nu}/s_{12}$, with a vertex factor ie at each end. In the all-outgoing labeling this reads

$$\mathcal{M} = \frac{e^2}{s_{12}}\, J_e^{\mu}\, J_{\mu}^{\pm}, \qquad J_e^{\mu} = (\text{electron current } 1,2), \quad J^{\mu\,\mu^{\pm}} = (\text{muon current } 3,4), \tag{58.149}$$

the overall sign and factors of i being immaterial to $|\mathcal{M}|^2$ and suppressed. By the helicity flow of Section 58.6.1, each current is one of the two bilinears Eq. (58.141), and chirality conservation forces the two ends of a line to carry *opposite* all-outgoing helicities. Take first the configuration with leg 1 negative-helicity and leg 3 negative-helicity. Then the electron current is $\langle 1|\gamma^{\mu}|2]$ (angle on 1, square on 2), the muon current is $\langle 3|\gamma_{\mu}|4]$, and Eq. (58.142) gives at once

$$\mathcal{M}(1^-2^+3^-4^+) = \frac{e^2}{s_{12}}\,\langle 1|\gamma^{\mu}|2]\,\langle 3|\gamma_{\mu}|4] = \frac{2e^2\,\langle 1\,3\rangle\,[4\,2]}{s}, \tag{58.150}$$

one line, gauge-invariant, with no spinor sum and no trace. The remaining three nonzero configurations follow by relabeling which end of each line is the angle and which the square. Negating both helicities on a line swaps angle↔square there; the four allowed assignments and their brackets, read off Eqs. (58.142) and (58.146), are

$$\begin{aligned} \mathcal{M}(1^-2^+3^-4^+) &= \tfrac{2e^2}{s}\,\langle 1\,3\rangle\,[4\,2], & \mathcal{M}(1^+2^-3^+4^-) &= \tfrac{2e^2}{s}\,[1\,3]\,\langle 4\,2\rangle, \\ \mathcal{M}(1^-2^+3^+4^-) &= \tfrac{2e^2}{s}\,\langle 1\,4\rangle\,[3\,2], & \mathcal{M}(1^+2^-3^-4^+) &= \tfrac{2e^2}{s}\,[1\,4]\,\langle 3\,2\rangle. \end{aligned} \tag{58.151}$$

All twelve other helicity assignments vanish: either a line would have equal helicities at its two ends (a forbidden same-chirality current, the spinor face of Theorem 17.18), or the photon would have to carry a helicity it cannot. The chirality bookkeeping that occupied Section 17.6.1 is here automatic—it is built into which spinor a current can hold.

Little-group audit. Before squaring, we check the amplitudes against the only constraint they must obey at this stage: little-group covariance. Under the scaling $\lambda_i \to t_i\lambda_i$, $\tilde\lambda_i \to t_i^{-1}\tilde\lambda_i$ of Eq. (58.5), a leg of helicity h_i must carry the weight $t_i^{-2h_i}$ (Theorem 58.16; this is the on-shell content of the helicity quantization Eq. (3.217)). Apply it to Eq. (58.150): $\langle 1\,3\rangle$ contributes one power of λ_1 and one of λ_3, $[4\,2]$ one power of $\tilde\lambda_4$ and one of $\tilde\lambda_2$, while the propagator $1/s_{12} = 1/(\langle 1\,2\rangle[2\,1])$ is little-group neutral on every leg (one λ_i and one $\tilde\lambda_i$ each for $i = 1,2$). The net weights are

$$t_1^{+1},\, t_2^{-1},\, t_3^{+1},\, t_4^{-1} \;=\; t_1^{-2h_1},\, t_2^{-2h_2},\, t_3^{-2h_3},\, t_4^{-2h_4} \quad \text{with} \quad (h_1,h_2,h_3,h_4) = (-\tfrac12,+\tfrac12,-\tfrac12,+\tfrac12) \tag{58.152}$$

exactly the helicities labeling $\mathcal{M}(1^-2^+3^-4^+)$. The other three amplitudes of Eq. (58.151) pass the same check with their respective labels: each bracket is forced to be the one it is by the requirement that leg i scale as $t_i^{-2h_i}$. The audit is not a derivation of the dynamics—the coupling e and the channel $1/s_{12}$ come from QED—but it certifies that the spinor structure is the unique one consistent with the stated helicities, which is precisely the organizing principle promised for the formalism.

Squaring, and the match to Chapter 17. Now square. Using Eq. (58.140), $|\langle 1\,3\rangle|^2 = |s_{13}| = |t|$ and $|[4\,2]|^2 = |s_{24}| = |t|$, so

$$\left|\mathcal{M}(1^-2^+3^-4^+)\right|^2 = \frac{4e^4}{s^2}\,\left|\langle 1\,3\rangle\right|^2\left|[4\,2]\right|^2 = \frac{4e^4}{s^2}\,|s_{13}|\,|s_{24}| = \frac{4e^4\,t^2}{s^2}. \tag{58.153}$$

The companion $(1^+2^-3^+4^-)$ gives the identical $4e^4t^2/s^2$, while the two "crossed" configurations $(1^-2^+3^+4^-)$ and $(1^+2^-3^-4^+)$ carry the invariants $s_{14} = s_{23} = u$ and so give $4e^4u^2/s^2$ each. Translating t, u to the scattering angle with Eq. (58.148), $t^2/s^2 = \tfrac{1}{4}(1-\cos\theta)^2$ and $u^2/s^2 = \tfrac{1}{4}(1+\cos\theta)^2$, hence

$$\begin{aligned}\left|\mathcal{M}(1^-2^+3^-4^+)\right|^2 &= \left|\mathcal{M}(1^+2^-3^+4^-)\right|^2 = e^4\,(1-\cos\theta)^2,\\ \left|\mathcal{M}(1^-2^+3^+4^-)\right|^2 &= \left|\mathcal{M}(1^+2^-3^-4^+)\right|^2 = e^4\,(1+\cos\theta)^2.\end{aligned} \tag{58.154}$$

These are the four numbers of Eq. (17.168), now obtained without a single gamma-matrix trace. The dictionary between the all-outgoing labels and the physical helicities of Chapter 17 is the crossing rule of the notation box: an incoming leg's physical helicity is minus its all-outgoing label, an outgoing leg's is equal to it. Thus $\mathcal{M}(1^-2^+3^+4^-)$ describes incoming e^- of helicity $+\frac{1}{2}$ (e^-_R), incoming e^+ of helicity $-\frac{1}{2}$ (e^+_L), outgoing μ^-_R, outgoing μ^+_L: the configuration $e^-_R e^+_L \to \mu^-_R \mu^+_L$, forward-peaked with $(1+\cos\theta)^2$, in exact agreement with Eq. (17.168). The reader may check the other three the same way: the two t^2 amplitudes are the backward-peaked J_z-flip channels, the two u^2 amplitudes the forward-peaked J_z-preserving ones, and the forbidden twelve are the spin assignments that Section 17.6.2 killed by angular-momentum conservation.

This is the worked computation that carries QED.

Worked Example 58.30 (The annihilation cross section from four brackets)**.** We assemble the unpolarized squared amplitude and recover the cross section of Chapter 17 term by term. The unpolarized $|\mathcal{M}|^2$ averages over the four initial spin states (factor $\frac{1}{4}$) and sums over the two final ones; since only the four entries of Eq. (58.154) are nonzero, the spin sum is just their total:

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\sum_{\text{hel}}|\mathcal{M}_h|^2 = \frac{1}{4}\,\frac{4e^4}{s^2}\Big(\underbrace{t^2+t^2}_{(1^\mp2^\pm3^\mp4^\pm)} + \underbrace{u^2+u^2}_{(1^\mp2^\pm3^\pm4^\mp)}\Big) = \frac{2e^4\,(t^2+u^2)}{s^2}. \tag{58.155}$$

Each of the four terms under the brace is a single squared bracket from Eq. (58.151); nothing has been contracted, traced, or expanded. Equation Eq. (58.155) is, term by term, the result $\overline{|\mathcal{M}|^2} = e^4(1+\cos^2\theta)$ of Eq. (17.169)—indeed $2e^4(t^2+u^2)/s^2 = 2e^4 \cdot \frac{1}{2}(1+\cos^2\theta) = e^4(1+\cos^2\theta)$ using $t^2+u^2 = \frac{s^2}{2}(1+\cos^2\theta)$—and it is the $s \leftrightarrow t$ crossed partner of the $e^-\mu^- \to e^-\mu^-$ amplitude $\overline{|\mathcal{M}|^2} = 2e^4(s^2+u^2)/t^2$ of Eq. (17.123), as the crossing audit of that chapter required.

To close the loop to an observable, feed Eq. (58.155) into the centre-of-mass differential cross section Theorem 16.23. With $\frac{d\sigma}{d\Omega} = \overline{|\mathcal{M}|^2}/(64\pi^2 s)$ for massless $2 \to 2$ scattering,

$$\frac{d\sigma}{d\Omega} = \frac{e^4(1+\cos^2\theta)}{64\pi^2 s} = \frac{\alpha^2}{4s}(1+\cos^2\theta), \qquad \sigma = \frac{4\pi\alpha^2}{3s}, \tag{58.156}$$

the textbook annihilation cross section, where the angular integral $\int(1+\cos^2\theta)\,d\Omega = \frac{16\pi}{3}$ supplies the total. The same number governs the high-energy total cross section through the optical theorem Theorem 16.28; here it has been produced from four one-line helicity amplitudes whose squares are literally $|t|^2$ and $|u|^2$.

It is worth pausing on what was avoided. The trace route of Section 17.3 and Theorem 17.19 formed the squared amplitude first, producing a product of two rank-two Lorentz tensors $L_e^{\mu\nu}M_{\mu\nu}^{\mu^\pm}$; evaluating it required the four-gamma trace, the γ^5 trace, a contraction of two Levi-Civita symbols (Eq. (17.164)), and a cancellation between symmetric and antisymmetric pieces, all to arrive at $16\,(p_-\cdot k_+)(p_+\cdot k_-)$. The helicity route never squares an amplitude until Eq. (58.153), and when it does, the square is the elementary $|\langle 1\,3\rangle|^2|[4\,2]|^2 = |s_{13}s_{24}|$. The Levi-Civita symbol, the workhorse and the hazard of the trace computation, has disappeared: its antisymmetric information is carried losslessly by the *relative phase* of the angle and square brackets, which is exactly the data that squaring throws away and that the trace had to reconstruct by brute force.

Physical Insight

The trace technology computes $\sum_{\text{spins}}|\mathcal{M}|^2$ as one object and discovers only at the end that it factorizes into a few helicity channels. The spinor-helicity formalism computes the channels first—each a single bracket fixed by little-group scaling—and adds their squares last. The two routes must agree, by unitarity of the spin sum, but the second exposes the structure the first conceals: that the answer was always a sum of squares of $d^1_{m'm}(\theta)$ rotation functions (Theorem 17.20), now read directly off $|s_{ij}|$. The $1+\cos^2\theta$ law is the incoherent sum of $|t|^2$ and $|u|^2$, nothing more.

58.6.3 Compton scattering and the reference-momentum trick

Annihilation has no internal photon polarization to manage; its simplification was driven entirely by the fermion currents. Compton scattering $e\gamma \to e\gamma$ is

the sharper test, because it has two Feynman diagrams whose individual gauge dependence is the source of most of the labor in Chapter 18. We show that a single judicious choice of reference momentum makes one diagram vanish outright, leaving a one-bracket answer, and that the answer is independent of every such choice.

Setup. Label, all outgoing,

$$1 = e^-, \quad 2 = e^+, \quad 3 = \gamma, \quad 4 = \gamma, \tag{58.157}$$

the two photons being legs $3, 4$. The two diagrams differ by the order in which the photons attach to the fermion line running from $\langle 1|$ to $|2]$ (Fig. 58.5). With the massless fermion propagator $i\not{P}/P^2$ and a vertex $ie\gamma^\mu$ at each photon, the amplitude is

$$\mathcal{M} = e^2 \, \langle 1| \Big[\not{\varepsilon}_3 \frac{\not{p}_1 + \not{p}_3}{s_{13}} \not{\varepsilon}_4 + \not{\varepsilon}_4 \frac{\not{p}_1 + \not{p}_4}{s_{14}} \not{\varepsilon}_3 \Big] |2], \tag{58.158}$$

up to an overall sign and factors of i that do not affect $|\mathcal{M}|^2$; the internal momenta are $-(p_1 + p_3)$ and $-(p_1 + p_4)$ with squares s_{13} and s_{14}. We take the helicity configuration with the electron line 1^-2^+ and the photons 3^-4^+; the others follow by conjugation and $3 \leftrightarrow 4$.

Polarization vectors and the trick. From the construction of Section 58.5.1, the slashed photon polarizations with reference momenta q_3, q_4 are

$$\not{\varepsilon}_4^+(p_4; q_4) = \frac{\sqrt{2}}{\langle q_4\, 4\rangle} \big(|4]\, \langle q_4| + |q_4\rangle\, [4| \big), \qquad \not{\varepsilon}_3^-(p_3; q_3) = \frac{\sqrt{2}}{[q_3\, 3]} \big(|3\rangle\, [q_3| + |q_3]\, \langle 3| \big), \tag{58.159}$$

the reference q_i being arbitrary (any momentum with $\langle q_4\, 4\rangle \neq 0$, $[q_3\, 3] \neq 0$). Their freedom is the gauge freedom of Eq. (58.127): changing q_i shifts ε_i^μ by a multiple of p_i^μ. We exploit it. Choose

$$q_4 = p_1. \tag{58.160}$$

Then $\langle 1|\, \not{\varepsilon}_4^+ = \frac{\sqrt{2}}{\langle 1\, 4\rangle} \big(\langle 1|\, |4]\, \langle 1| + \langle 1\, 1\rangle [4| \big) = 0$, because $\langle 1|\, |4] = \langle 1|4] = 0$ (an angle bra cannot contract a square ket) and $\langle 1\, 1\rangle = 0$. The polarization ε_4^+ has been aligned so that it is annihilated by the spinor at the end of the line it would attach to in the *second* diagram of Eq. (58.158). That diagram—the one with $\not{\varepsilon}_4$ adjacent to $\langle 1|$—vanishes identically. Only the first survives, and the two-diagram amplitude collapses to one term.

The surviving term. Take $q_3 = p_1$ as well; we will see the answer does not depend on this. Push $\langle 1|$ through $\not{\varepsilon}_3^-$:

$$\langle 1|\, \not{\varepsilon}_3^- = \frac{\sqrt{2}}{[1\, 3]} \big(\langle 1\, 3\rangle [1| + \langle 1|\, |1]\, \langle 3| \big) = \frac{\sqrt{2}\, \langle 1\, 3\rangle}{[1\, 3]}\, [1|, \tag{58.161}$$

Figure 58.5: The two tree diagrams for Compton scattering $\mathcal{M}(1^-2^+3^-4^+)$ in the all-outgoing labeling of Eq. (58.157), differing by the order in which the photons $3, 4$ attach to the fermion line. With the reference momentum choice $q_4 = p_1$ of Eq. (58.160), the right diagram (photon 4 adjacent to the $\langle 1|$ end) vanishes because $\langle 1|\not{\varepsilon}_4^+ = 0$, and the entire amplitude collapses to the left diagram, Eq. (58.164). A different reference choice redistributes the labor between the two diagrams but leaves their sum unchanged (Eq. (58.165)).

using $\langle 1|\,|1] = 0$. Next act on the propagator numerator. Since $[1|\not{p}_1 = 0$ (both $[1|\,|1\rangle$ and $[1||1] = [1\,1]$ vanish) and $[1|\not{p}_3 = [1\,3]\,\langle 3|$,

$$[1|\,(\not{p}_1+\not{p}_3) = [1\,3]\,\langle 3|\,, \qquad \Longrightarrow \qquad \langle 1|\,\not{\varepsilon}_3^-\,\frac{\not{p}_1+\not{p}_3}{s_{13}} = \frac{\sqrt{2}\,\langle 1\,3\rangle}{[1\,3]}\cdot\frac{[1\,3]\,\langle 3|}{s_{13}} = \frac{\sqrt{2}\,\langle 1\,3\rangle}{s_{13}}\,\langle 3|\,, \tag{58.162}$$

the reference factor $[1\,3]$ cancelling cleanly between the polarization denominator and the propagator numerator. Finally contract with $\not{\varepsilon}_4^+|2]$. From Eqs. (58.159) and (58.160),

$$\langle 3|\,\not{\varepsilon}_4^+ = \frac{\sqrt{2}}{\langle 1\,4\rangle}\big(\langle 3|\,|4]\,\langle 1| + \langle 3\,1\rangle[4|\big) = \frac{\sqrt{2}\,\langle 3\,1\rangle}{\langle 1\,4\rangle}\,[4|, \qquad \langle 3|\,\not{\varepsilon}_4^+|2] = \frac{\sqrt{2}\,\langle 3\,1\rangle\,[4\,2]}{\langle 1\,4\rangle}. \tag{58.163}$$

Multiplying the pieces and restoring the coupling,

$$\mathcal{M}(1^-2^+3^-4^+) = e^2\cdot\frac{\sqrt{2}\,\langle 1\,3\rangle}{s_{13}}\cdot\frac{\sqrt{2}\,\langle 3\,1\rangle\,[4\,2]}{\langle 1\,4\rangle} = -\,\frac{2e^2\,\langle 1\,3\rangle^2\,[4\,2]}{s_{13}\,\langle 1\,4\rangle}, \tag{58.164}$$

where $\langle 1\,3\rangle\langle 3\,1\rangle = -\langle 1\,3\rangle^2$. Using $s_{13} = \langle 1\,3\rangle[3\,1]$ this is equivalently $\mathcal{M}(1^-2^+3^-4^+) = -2e^2\,\langle 1\,3\rangle\,[4\,2]/([3\,1]\,\langle 1\,4\rangle)$. The two diagrams of Eq. (58.158), each a gauge-dependent tangle of the kind Chapter 18 squares and traces over four pages, have become a single ratio of brackets.

Reference independence. Two arguments establish that Eq. (58.164) is the physical amplitude, independent of the gauge choices q_3, q_4.

First, the structural reason. The amplitude depends on each photon polarization linearly, $\mathcal{M} = \varepsilon_{i\mu}\,\mathcal{M}^\mu_{(i)}$ with $\mathcal{M}^\mu_{(i)}$ the current with leg i's polarization stripped. Changing the reference $q_i \to q_i'$ shifts $\varepsilon_i^\mu \to \varepsilon_i^\mu + \beta\,p_i^\mu$ for a scalar β (Eq. (58.127)), so the amplitude changes by $\beta\,p_{i\mu}\mathcal{M}^\mu_{(i)}$, which vanishes by the tree-level QED Ward identity of Eq. (18.62): the sum of the two diagrams, contracted

with the photon momentum, is zero. Reference independence is therefore not an accident of the choice $q_4 = p_1$ but a theorem, equivalent to gauge invariance.

Second, an explicit check. Keep $q_4 = p_1$ but take $q_3 = p_4$ instead of p_1. The surviving (left) diagram now begins $\langle 1|\,\not{\epsilon}_3^-(q_3 = p_4) = \frac{\sqrt{2}}{[4\,3]}\big(\langle 1\,3\rangle[4| + \langle 1|\,|4]\,\langle 3|\big) = \frac{\sqrt{2}\,\langle 1\,3\rangle}{[4\,3]}\,[4|$, and then $[4|(\not{p}_1 + \not{p}_3) = [4\,1]\,\langle 1| + [4\,3]\,\langle 3|$. Acting with $\not{\epsilon}_4^+(q_4 = p_1)$ kills the $\langle 1|$ piece ($\langle 1|\,\not{\epsilon}_4^+ = 0$) and leaves $[4\,3]\,\langle 3|\,\not{\epsilon}_4^+|2] = [4\,3]\cdot\sqrt{2}\,\langle 3\,1\rangle[4\,2]/\langle 1\,4\rangle$. Collecting,

$$\mathcal{M}(1^-2^+3^-4^+)\Big|_{q_3=p_4} = e^2\cdot\frac{\sqrt{2}\,\langle 1\,3\rangle}{[4\,3]\,s_{13}}\cdot[4\,3]\cdot\frac{\sqrt{2}\,\langle 3\,1\rangle\,[4\,2]}{\langle 1\,4\rangle} = -\,\frac{2e^2\,\langle 1\,3\rangle^2\,[4\,2]}{s_{13}\,\langle 1\,4\rangle}, \tag{58.165}$$

since $[4\,3]$ cancels between numerator and denominator and $\langle 1\,3\rangle\langle 3\,1\rangle = -\langle 1\,3\rangle^2$. The result is identical to Eq. (58.164), although the algebra ran through different intermediate spinors. The reference momentum is genuine gauge freedom: visible in every line of the computation, invisible in the answer.

The squared amplitude and the Compton cross section. Squaring Eq. (58.164) with Eq. (58.140) and $s_{24} = s_{13}$,

$$\big|\mathcal{M}(1^-2^+3^-4^+)\big|^2 = 4e^4\,\frac{\big|\langle 1\,3\rangle\big|^4\big|[4\,2]\big|^2}{s_{13}^2\,\big|\langle 1\,4\rangle\big|^2} = 4e^4\,\frac{|s_{13}|^2\,|s_{24}|}{s_{13}^2\,|s_{14}|} = 4e^4\,\frac{|s_{13}|}{|s_{14}|}. \tag{58.166}$$

The amplitude squared is a bare *ratio* of Mandelstam invariants—the signature of Compton scattering, whose cross section is famously built from s/u and u/s. The nonzero helicity configurations are the four with opposite photon helicities (equal photon helicities give the all-plus/all-minus amplitudes, which vanish for a massless fermion line—the abelian shadow of the maximally-helicity-violating selection rule that organizes the gluon amplitudes of Chapter 59). Their squares are $4e^4|s_{13}|/|s_{14}|$ twice and the $3 \leftrightarrow 4$ image $4e^4|s_{14}|/|s_{13}|$ twice, so averaging over the two electron spins and two incoming photon polarizations,

$$\overline{|\mathcal{M}|^2} = \frac{1}{4}\Big(2\cdot 4e^4\frac{|s_{13}|}{|s_{14}|} + 2\cdot 4e^4\frac{|s_{14}|}{|s_{13}|}\Big) = 2e^4\Big(\frac{|s_{13}|}{|s_{14}|} + \frac{|s_{14}|}{|s_{13}|}\Big). \tag{58.167}$$

Under the crossing back to physical kinematics, $|s_{13}|$ and $|s_{14}|$ become the physical s- and u-channel invariants of Compton scattering (the legs $1, 3$ that share a propagator pole in the left diagram of Fig. 58.5 forming the physical s-channel), and Eq. (58.167) is exactly the massless limit $m \to 0$ of the Klein–Nishina squared amplitude Eq. (18.81),

$$\overline{|\mathcal{M}|^2} \xrightarrow{m\to 0} 2e^4\Big(\frac{-u}{s} + \frac{s}{-u}\Big), \tag{58.168}$$

which fed the Klein–Nishina differential cross section Eq. (18.92). The two diagrams that Chapter 18 added, squared, traced, and contracted to reach Eq. (18.81) have here been reduced to one bracket per helicity by a single line of gauge fixing.

58.6.4 What the reframing bought, and what it points to

The two computations make the chapter's thesis concrete on the book's own examples. In annihilation, the entire spin structure—which configurations survive, which vanish, and the $(1 \pm \cos\theta)^2$ shapes of those that survive—was carried by the choice of which spinor each current holds, fixed in turn by little-group covariance Eq. (58.152); the trace, the γ^5, and the Levi-Civita contraction of Chapter 17 never appeared. In Compton, the gauge dependence that forces the off-shell, multi-term bookkeeping of Chapter 18 was localized in the reference momenta and then removed by hand, collapsing two diagrams to one. In both cases the carried QED numbers—$\overline{|\mathcal{M}|^2} = 2e^4(t^2+u^2)/s^2$ for annihilation, the Klein–Nishina ratio for Compton—came back unchanged, which is the only acceptable outcome: the formalism is a more efficient bookkeeping of the same physics, not a new physics.

Two features of these QED amplitudes deserve a final word, because they are exactly the connections along which Chapter 59 cuts. First, the QED amplitudes carried no color: the single trace that organizes the gluon amplitudes of Section 59.1 is, in the abelian case, just unity, and the "decoupling" of the abelian photon proved there (Theorem 59.6) is the statement that the photon in this section never self-interacts. Stripping color from a non-abelian amplitude is what will leave behind a color-ordered partial amplitude as kinematically simple as the QED currents we just computed. Second, the vanishing of the equal-helicity Compton amplitudes is the two-photon instance of a vast pattern: in pure Yang–Mills the all-plus and single-minus gluon amplitudes vanish identically, and the first nonzero amplitudes—the maximally-helicity-violating ones—are a single ratio of brackets, the Parke–Taylor formula. The next section assembles the last pieces Chapter 59 needs, the color-ordered partial amplitude and the gluon three-point in final form, and confronts the question this success raises: how much of the one-line miracle survives when the gauge group is non-abelian and the external particles carry spin one.

58.7 Color ordering, gluons, and the massive frontier: handoff to recursion

The reframing of QED in the previous section turned a page of traces over Dirac matrices into one-line helicity amplitudes; the same compression is waiting for the gluons of Yang–Mills theory, but one structural obstacle stands between us and it. A photon carries no charge, so the QED amplitudes were pure kinematics. A gluon carries adjoint color, and every Yang–Mills vertex (Eq. (25.76)) drags a structure constant f^{abc} along with its Lorentz tensor. If we are to apply the spinor-helicity machinery built in this chapter — the brackets of Theorem 58.2, the little-group scaling of Section 58.3, the polarization vectors of Eq. (58.111) —

we must first separate the color, which knows nothing of momenta or helicities, from the kinematics, which is where all the simplicity hides. The good news, and the organizing observation of this final section, is that at tree level the two factor *cleanly*: the entire color dependence of an n-gluon amplitude collapses onto a basis of single traces of generators, one per cyclic ordering of the external legs, and the coefficient of each trace is a separately gauge-invariant kinematic object — the *color-ordered partial amplitude*. Once color is peeled off in this way, a gluon amplitude becomes, leg for leg, as tractable as a photon amplitude, and the spinor-helicity variables act on it exactly as they did in QED.

This section assembles the pieces that the recursion of Chapter 59 will pick up and run with. We perform the trace decomposition and prove the three properties of partial amplitudes that recursion must respect — cyclicity, reflection, and the $U(1)$ decoupling identity. We put the gluon three-point amplitude, already fixed by little-group scaling alone in Eq. (58.98), into its final color-ordered form, the seed from which everything in Chapter 59 is grown. We state the Parke–Taylor formula honestly as the destination and subject it to the consistency checks the formalism affords, deferring its proof to the next chapter where the recursion that earns it is built. Finally, because a chapter that stopped at massless particles would be artificially incomplete, we state the massive extension of spinor-helicity — the bold-spinor formalism of Arkani-Hamed, Huang, and Huang — at the definitional level, and close with a map of the route through Part XVI.

58.7.1 The trace color decomposition

We work with the SU(N) generators T^a in the fundamental, Hermitian, with the book-canonical normalization $\left[T^a, T^b\right] = i f^{abc} T^c$ and $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ (Eq. (B.2)). The whole decomposition rests on a single algebraic identity, which rewrites a structure constant as a difference of traces and so converts the f's at the vertices into the trace language we want.

Lemma 58.31 (Structure constant as a trace)**.** *With the canonical normalization above,*

$$f^{abc} = -2i\,\big(\,\mathrm{tr}(T^a T^b T^c) - \mathrm{tr}(T^a T^c T^b)\big). \tag{58.169}$$

Proof. Multiply the Lie bracket $\left[T^a, T^b\right] = i f^{abd} T^d$ on the right by T^c and take the trace. Using $\mathrm{tr}(T^d T^c) = \frac{1}{2}\delta^{dc}$,

$$\mathrm{tr}\big(\left[T^a, T^b\right] T^c\big) = i f^{abd}\,\mathrm{tr}(T^d T^c) = \tfrac{i}{2}\, f^{abc}. \tag{58.170}$$

The left side is $\mathrm{tr}(T^a T^b T^c) - \mathrm{tr}(T^b T^a T^c)$, and cyclicity of the trace gives $\mathrm{tr}(T^b T^a T^c) = \mathrm{tr}(T^a T^c T^b)$, so the left side equals $\mathrm{tr}(T^a T^b T^c) - \mathrm{tr}(T^a T^c T^b)$. Solving Eq. (58.170) for f^{abc} yields Eq. (58.169). □

In color space every tree diagram built from the Yang–Mills vertices is a fully contracted product of structure constants — one f per cubic vertex, two per quartic — with the adjoint indices of internal lines paired through the propagator's δ^{ab} and the external indices $a_1, \dots, a_n$ left free; the kinematic factor (the contraction of metrics, momenta, and polarizations) rides alongside, multiplied, and the two never mix. Inserting Eq. (58.169) at each vertex turns each diagram's color factor into a sum of products of traces of the external generators, sewn along internal lines by the SU(N) completeness relation. The cleanest way to see what survives is to enlarge the gauge group momentarily from SU(N) to U(N), adjoining the trace generator $T^0 = \frac{1}{\sqrt{N}}\,\mathbf{1}$. The U($N$) completeness relation carries no trace subtraction,

$$\sum_{A=0}^{N^2-1} (T^A)_i{}^j\,(T^A)_k{}^l = \delta_i^{\,l}\,\delta_k^{\,j}, \tag{58.171}$$

the indices simply being interchanged. Two facts make this legitimate. First, the extra abelian boson T^0 decouples from gluons: its structure constant $f^{0bc} \propto \mathrm{tr}(\big[\mathbf{1}, T^b\big]\,T^c) = 0$ because the identity commutes with everything, so no internal T^0 can appear on a pure-gluon line and the SU(N) and U(N) computations coincide. Second, in the U(N) basis the sewing Eq. (58.171) has a transparent graphical meaning: drawing each adjoint line as a pair of antiparallel fundamental color lines (the double-line notation), the completeness relation joins the fundamental lines of an internal gluon end to end. Because a *tree* has no momentum loop, the joined color lines can neither cross nor close among themselves except along the single outer boundary; they merge into exactly one closed fundamental loop, and that loop *is* a single trace $\mathrm{tr}(T^{a_{i_1}} \cdots T^{a_{i_n}})$, with the ordering read off the planar embedding. No second closed loop, and hence no double-trace term, can arise without a momentum loop. Collecting diagrams by their trace gives the decomposition.

Theorem 58.32 (Tree-level color decomposition)**.** *The full n-gluon tree amplitude, with external adjoint labels $a_1, \dots, a_n$, decomposes as*

$$\mathcal{A}_n^{\text{tree}}(\{p_i, \epsilon_i, a_i\}) = g^{\,n-2} \sum_{\sigma \in S_{n-1}} \mathrm{tr}(T^{a_1} T^{a_{\sigma(2)}} \cdots T^{a_{\sigma(n)}})\; A_n[1, \sigma(2), \dots, \sigma(n)], \tag{58.172}$$

where the sum runs over the $(n-1)!$ permutations σ of $\{2, \dots, n\}$ (leg 1 held fixed by trace cyclicity), and the kinematic coefficients $A_n[1, \sigma]$ — the color-ordered partial amplitudes *— carry no color, depend only on momenta and polarizations, and are separately gauge invariant.*

Proof. The double-line argument establishes that in the U(N) basis every tree diagram's color factor is a single trace of the external generators in the cyclic order of the planar embedding, times a color-free kinematic factor. Grouping diagrams

by their trace gives $\mathcal{A}_n^{\text{tree}} = g^{n-2}\sum_{\text{orderings}} \text{tr}(\cdots)\, A_n[\cdots]$, where A_n collects the kinematic factors of all diagrams sharing that ordering. The coupling power is g^{n-2} because a tree with V_3 cubic and V_4 quartic vertices carries $g^{V_3+2V_4} = g^{n-2}$, the relation $V_3 + 2V_4 = n - 2$ following from the tree identities $V_3 + V_4 = I + 1$ and $3V_3 + 4V_4 = 2I + n$ for the number I of internal lines. Trace cyclicity rotates leg 1 to the front of every trace, identifying the independent orderings with permutations of $\{2, \ldots, n\}$ and giving the $(n-1)!$ terms. Because the computation was done in $\text{U}(N)$ with T^0 inert, it equals the genuine $\text{SU}(N)$ amplitude: the $\frac{1}{N}$ trace-subtraction terms of the $\text{SU}(N)$ completeness relation, which would have produced multi-trace structures, are precisely the ones that cancel because the abelian photon decouples (Theorem 58.35). Finally each $A_n[1, \sigma]$ is gauge invariant: $\mathcal{A}_n^{\text{tree}}$ is, the traces are linearly independent functions of the external labels (and the A_n are kinematic objects defined independently of N), so a gauge variation that vanishes for all $\{a_i\}$ forces each coefficient to vanish separately. □

Definition 58.33 (Color-ordered partial amplitude)**.** The coefficient $A_n[1, 2, \ldots, n]$ multiplying the trace $\text{tr}(T^{a_1}T^{a_2}\cdots T^{a_n})$ in Eq. (58.172) is the *color-ordered partial amplitude* for the cyclic ordering $(1, 2, \ldots, n)$. It is computed by summing only those Feynman diagrams whose external legs can be drawn in the plane in the order $1, 2, \ldots, n$, using color-stripped vertices in which legs connect in cyclic adjacency.

This is the object the whole of Chapter 59 computes; there it is taken as the starting point and the same decomposition is restated as Theorem 59.3. Every statement we make below about gluon amplitudes — vanishing helicity configurations, the three-point seed, Parke–Taylor, BCFW recursion — is a statement about a single $A_n[1, \ldots, n]$, with the full amplitude reassembled at the end by dressing each ordering with its trace and summing.

58.7.2 Properties of the partial amplitudes

Three exact properties of the A_n follow directly from the decomposition. They are not incidental: they are precisely the constraints that the recursion of Chapter 59 is built to respect, and any candidate formula for an A_n must satisfy them.

Cyclicity. Because $A_n[1, 2, \ldots, n]$ is, by Theorem 58.33, the coefficient of the cyclic trace $\text{tr}(T^{a_1}\cdots T^{a_n})$, and the trace is invariant under cyclic rotation of its arguments, the partial amplitude is invariant under cyclic rotation of its labels,

$$A_n[1, 2, \ldots, n] = A_n[2, 3, \ldots, n, 1] = A_n[3, \ldots, n, 1, 2] = \cdots . \tag{58.173}$$

This is why holding leg 1 fixed in Eq. (58.172) loses nothing.

Reflection. Reversing the entire ordering produces a definite sign,

$$A_n[n, n-1, \dots, 2, 1] = (-1)^n \, A_n[1, 2, \dots, n]. \tag{58.174}$$

Proposition 58.34 (Reflection identity)**.** *The color-ordered partial amplitude obeys Eq. (58.174).*

Proof. Track the sign each ingredient of a planar diagram acquires under reversal of the cyclic order, which reverses the Lorentz labels and momenta on every vertex. The color-stripped propagator $-i\eta_{\mu\nu}/p^2$ is symmetric in its indices and even in momentum, hence invariant. The color-ordered cubic vertex Eq. (58.177) below is linear in the momentum differences, and under reversal of its three legs each momentum difference $p_i - p_j$ maps to its negative while the metric factors permute among themselves; the three terms reassemble into $-V_3$, so each cubic vertex contributes a factor -1. The color-ordered quartic vertex is momentum-independent and maps to itself under reversal, contributing $+1$. A planar diagram with V_3 cubic vertices therefore acquires $(-1)^{V_3}$. This sign is common to every diagram of a given A_n because V_3 has a fixed parity: from $3V_3 + 4V_4 = 2I + n$ and $V_3 + V_4 = I + 1$ one finds $V_3 = n - 2 - 2V_4 \equiv n \pmod 2$, so $(-1)^{V_3} = (-1)^n$ independently of how vertices are apportioned. Every diagram for $A_n[1, \dots, n]$ thus maps under reversal to the corresponding diagram for $A_n[n, \dots, 1]$ times the common factor $(-1)^n$, which is Eq. (58.174). □

The $U(1)$ decoupling identity. The third property is the cleanest linear relation among partial amplitudes, and it is the same fact — the decoupling of the abelian photon — that we invoked in proving Theorem 58.32.

Theorem 58.35 ($U(1)$ decoupling / photon-decoupling identity)**.** *The sum of $A_n[1, \sigma]$ over the $n-1$ inequivalent positions into which leg 1 may be inserted while holding the cyclic order of $2, 3, \dots, n$ fixed vanishes,*

$$A_n[1,2,3,\dots,n] + A_n[2,1,3,\dots,n] + A_n[2,3,1,4,\dots,n] + \cdots + A_n[2,3,\dots,n,1] = 0. \tag{58.175}$$

For $n = 4$, after rotating leg 1 to the front of each term by cyclicity,

$$A_4[1,2,3,4] + A_4[1,3,4,2] + A_4[1,4,2,3] = 0. \tag{58.176}$$

Proof. Compute the U(N) amplitude in which leg 1 is the abelian photon $T^0 = \frac{1}{\sqrt{N}}\mathbf{1}$ and legs $2, \dots, n$ are ordinary gluons. The photon decouples ($f^{0bc} = 0$), so no vertex couples it and the amplitude vanishes identically. Now evaluate the same amplitude through Eq. (58.172) by setting $T^{a_1} = \frac{1}{\sqrt{N}}\mathbf{1}$ in every trace. Because $\mathbf{1}$ can be slid to any position and removed, $\mathrm{tr}(\mathbf{1}\, T^{a_{\sigma(2)}} \cdots) = \mathrm{tr}(T^{a_{\sigma(2)}} \cdots)$, every trace collapses to a trace of the $n-1$ gluon generators alone. Orderings that differ only by where leg 1 sat collapse to the same reduced trace, so the coefficient

of each independent reduced trace $\mathrm{tr}(T^{a_2}\cdots T^{a_n})$ is exactly the sum of partial amplitudes over all insertion positions of leg 1. The reduced traces are linearly independent and the total vanishes, so each such coefficient vanishes, which is Eq. (58.175). □

Cyclicity and reflection generate the dihedral group of order $2n$ on the orderings, so the $(n-1)!$ traces fall into orbits and the genuinely independent partial amplitudes number roughly $(n-1)!/2$; the decoupling identity Eq. (58.175) cuts this further by expressing one partial amplitude as a combination of others. Stronger linear relations exist — the Kleiss–Kuijf relations reduce the basis to $(n-2)!$, and the BCJ relations of the double-copy story (Chapter 61) to $(n-3)!$ — but the three properties above are all the recursion of Chapter 59 needs as built-in consistency conditions, and they are the ones we have proved here from the decomposition itself.

58.7.3 The gluon three-point amplitude in final form

We now have everything required to write the seed of the entire recursion: the color-ordered three-gluon amplitude. Stripping the color from the Yang–Mills cubic vertex Eq. (26.53) by inserting Eq. (58.169) and retaining the coefficient of the single ordered trace $\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3})$, with all momenta outgoing ($p_1+p_2+p_3=0$) and legs in the cyclic order $1,2,3$, gives the color-ordered cubic vertex

$$V_3^{\mu_1\mu_2\mu_3}(p_1,p_2,p_3)=\frac{i\,g}{\sqrt{2}}\Big(\eta^{\mu_1\mu_2}(p_1-p_2)^{\mu_3}+\eta^{\mu_2\mu_3}(p_2-p_3)^{\mu_1}+\eta^{\mu_3\mu_1}(p_3-p_1)^{\mu_2}\Big). \tag{58.177}$$

This is the lone diagram at three points. Contracting it with the spinor-helicity polarization vectors Eq. (58.111) and using the three-point amplitudes already fixed by little-group scaling in Eq. (58.98) — where we showed that Lorentz invariance, locality, and the helicity weights of Eq. (58.5) determine the three-point amplitude up to its overall constant, with the holomorphic and antiholomorphic solutions selected by the sign of the total helicity — the color-ordered three-gluon amplitude takes its final form,

$$A_3[1^-,2^-,3^+]=i\,g\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,1\rangle}, \tag{58.178}$$

$$A_3[1^+,2^+,3^-]=-\,i\,g\,\frac{[1\,2]^3}{[2\,3][3\,1]}. \tag{58.179}$$

The first is purely holomorphic (only angle brackets), the second purely antiholomorphic (only square brackets); they are exchanged by parity $\langle\,\rangle\leftrightarrow[\,]$, which exchanges all helicities. Each carries the correct little-group weight: in Eq. (58.178) leg 1 (helicity -1) appears net t_1^{+2} (three powers up in the numerator, one down in the denominator), and leg 3 (helicity $+1$) appears t_3^{-2}, matching the required $t_k^{-2h_k}$ on every leg.

Caution

For *real* momenta the angle and square spinors are complex conjugates up to a sign, $\tilde{\lambda} = \pm\lambda^*$ (Theorem 58.2), so $\langle i\,j\rangle = 0 \Leftrightarrow [i\,j] = 0$. The three-point kinematics $p_1 + p_2 + p_3 = 0$ with all legs on shell forces every Mandelstam $s_{ij} = \langle i\,j\rangle[j\,i]$ to vanish (each equals the square of the third, null, momentum), so on the real slice *all* brackets vanish together and both Eqs. (58.178) and (58.179) are $0/0$. They are nonzero only when λ and $\tilde{\lambda}$ are independent — that is, for *complex* momenta, where $\langle i\,j\rangle$ can be generic while every $[i\,j] = 0$ (the holomorphic branch) or vice versa. This is not a pathology to be regretted but the very feature exploited in Chapter 59: the BCFW recursion deforms external momenta into the complex domain precisely so that these three-point amplitudes become meaningful building blocks.

Equations Eqs. (58.178) and (58.179) are the three-gluon seed handed to Chapter 59. Read on-shell recursion backwards: every higher-point gluon tree is assembled from these two atoms glued across factorization channels. That a one-line pair of expressions, fixed almost entirely by symmetry, should generate the whole tree-level S-matrix of Yang–Mills theory is the central promise of the next chapter.

58.7.4 Parke–Taylor: the destination

We can now name the target honestly. For the maximally-helicity-violating (MHV) amplitude — n gluons of which exactly two, legs i and j, carry negative helicity and the rest positive — the color-ordered partial amplitude is, for *every* n, the single ratio discovered by Parke and Taylor in 1986,

$$A_n^{\mathrm{MHV}}[1,2,\ldots,n] = i\,g^{\,n-2}\,\frac{\langle i\,j\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n-1\,n\rangle\langle n\,1\rangle}. \tag{58.180}$$

The denominator is the cyclic Parke–Taylor chain $\prod_{k=1}^{n}\langle k\,k+1\rangle$ (with $n+1 \equiv 1$): the same for every MHV amplitude of the given color ordering, while the numerator $\langle i\,j\rangle^4$ records which two legs are negative. The expression is purely holomorphic — not a single square bracket — which is the analytic fingerprint of the MHV sector, and it collapses, at $n = 3$ with negative legs $1, 2$, exactly to the three-gluon seed Eq. (58.178): $i\,g\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,1\rangle) = i\,g\,\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$. The parity conjugate, the $\overline{\mathrm{MHV}}$ amplitude with exactly two positive legs, is the antiholomorphic mirror obtained by $\langle\,\rangle \to [\,]$.

We state Eq. (58.180) here as the goal and *defer its proof to Chapter 59*, where it appears as Eq. (59.55): a brute-force Feynman computation at four points pins the constant, and BCFW recursion (Theorem 59.23) lifts the four-point result to all n. What we *can* do now, with the symmetry technology of this chapter,

is verify that Eq. (58.180) has the right *form* — the correct little-group weight on every leg, the correct mass dimension, and the correct power of the coupling. These are genuine constraints; an expression failing any of them could not be a gluon amplitude, and passing all of them nearly fixes the formula, leaving only the overall constant and the residues on factorization channels for the explicit computation of Chapter 59.

Worked Example 58.36 (Consistency checks on the Parke–Taylor formula)**.** We test Eq. (58.180) against the three invariants every n-gluon amplitude must respect.

Little-group weight. Under the scaling of leg k (Eq. (58.5)), $\lambda_k \to t_k \lambda_k$, $\tilde{\lambda}_k \to t_k^{-1}\tilde{\lambda}_k$, a leg of helicity h_k requires the amplitude to scale as $A_n \to (\prod_k t_k^{-2h_k}) A_n$. The angle bracket $\langle k\,\cdot\rangle$ scales as t_k; square brackets are absent, so a leg's weight is read off by counting its angle brackets. A *positive* leg $k \neq i, j$ appears nowhere in the numerator and in exactly two chain factors, $\langle k-1\,k\rangle$ and $\langle k\,k+1\rangle$, giving t_k^{-2}; with $h_k = +1$ this is $t_k^{-2h_k}$, as required. A *negative* leg, say i, appears four times in $\langle i\,j\rangle^4$ (weight t_i^{+4}) and twice in the chain (weight t_i^{-2}), net t_i^{+2}; with $h_i = -1$ this is again $t_i^{-2h_i}$. Every leg is correct — and the numerator power must be *exactly four*: $\langle i\,j\rangle^3$ or $\langle i\,j\rangle^5$ would spoil the negative legs. Little-group scaling alone fixes the numerator to $\langle i\,j\rangle^4$.

Mass dimension. In four dimensions an n-point amplitude (with the overall momentum-conserving delta function stripped) has mass dimension $4-n$. Each angle bracket has mass dimension 1 (since $\langle i\,j\rangle[j\,i] = s_{ij}$ has dimension 2). The numerator $\langle i\,j\rangle^4$ contributes $+4$, the n-factor chain contributes $-n$, and g^{n-2} is dimensionless in four dimensions, for a total of $4-n$. Correct.

Coupling power. A connected tree with only cubic and quartic vertices and n external legs carries $V_3 + 2V_4 = n-2$ powers of g, the same counting that gave the g^{n-2} of Eq. (58.172). The prefactor in Eq. (58.180) matches exactly, and the explicit i is the standard tree normalization, consistent with the three-point seed Eq. (58.178).

These checks pin down the form almost completely: a holomorphic, cyclic, dimension-$(4-n)$ object with the correct little-group weight and power of g is forced to be a constant times Eq. (58.180). What they cannot fix is that constant nor the factorization residues; for those, Chapter 59 computes.

58.7.5 The massive frontier: bold spinors

Everything in this chapter has been built for massless particles, where the little group is the abelian U(1) of Theorem 3.43 and a state is labelled by a single helicity weight. This is the right setting for gluons and photons, and it is where the formalism is most powerful, but it would leave a false impression to stop here: spinor-helicity is not intrinsically massless. The massless variables exploit the fact that a null momentum's 2×2 bispinor $p_{\alpha\dot{\alpha}} = \lambda_\alpha \tilde{\lambda}_{\dot{\alpha}}$ has rank one and so

factorizes into a single pair of spinors. A *massive* momentum, with $p^2 = m^2$, has a bispinor of rank two, $\det(p_{\alpha\dot\alpha}) = m^2 \neq 0$, and so cannot be written with one spinor on each side — but it can be written with *two*.

The honest completion to massive particles is the formalism of Arkani-Hamed, Huang, and Huang, a mid-2010s development (2017) that is now standard but still actively driving research, and which we state here at the definitional level. By the Wigner classification (Theorem 3.43), the little group of a massive particle in four dimensions is $\mathrm{SO}(3) \simeq \mathrm{SU}(2)$, not $\mathrm{U}(1)$; a massive state of spin s furnishes a $(2s+1)$-dimensional representation. The corresponding on-shell variables carry an $\mathrm{SU}(2)$ little-group index. One factorizes the massive bispinor as

$$p_{\alpha\dot\alpha} = \lambda_\alpha^I \, \tilde\lambda_{\dot\alpha\, I}, \qquad I = 1, 2, \tag{58.181}$$

with the little-group index I summed (raised and lowered by the antisymmetric ϵ_{IJ}). The pair of spinors λ_α^I form a 2×2 matrix whose determinant encodes the mass; equivalently, the $\mathrm{SU}(2)$-covariant brackets of the *bold* spinors $\left|\mathbf{p}^I\right\rangle$, $|\mathbf{p}^I]$ obey, for a single leg,

$$\langle \mathbf{p}^I \, \mathbf{p}^J \rangle = m\, \epsilon^{IJ}, \qquad [\mathbf{p}^I \, \mathbf{p}^J] = -\, m\, \epsilon^{IJ}, \tag{58.182}$$

the mass appearing exactly where, in the massless case, the rank-one condition set the analogous bracket to zero. The massless limit is transparent: as $m \to 0$ the two columns λ^1, λ^2 become proportional, the matrix degenerates to rank one, the $\mathrm{SU}(2)$ index collapses, and Eq. (58.181) reduces to the single-spinor factorization $p_{\alpha\dot\alpha} = \lambda_\alpha \tilde\lambda_{\dot\alpha}$ of Theorem 58.2, with the $\mathrm{SU}(2)$ multiplet splitting into the $\mathrm{U}(1)$ helicity weights.

Definition 58.37 (Massive (bold) spinor-helicity variables)**.** A massive momentum p ($p^2 = m^2$) is represented by bold spinors λ_α^I, $\tilde\lambda_{\dot\alpha}^I$ carrying a fundamental $\mathrm{SU}(2)$ little-group index $I = 1, 2$, factorizing the bispinor as Eq. (58.181) and normalized by Eq. (58.182). A massive particle of spin s is assigned a polarization wavefunction carrying $2s$ symmetrized little-group indices $\left|\mathbf{p}^{(I_1}\right\rangle \cdots \left|\mathbf{p}^{I_{2s})}\right\rangle$; amplitudes are tensors in these symmetrized $\mathrm{SU}(2)$ indices, with the massless helicity weight of Eq. (58.5) recovered as the highest- and lowest-weight components in the $m \to 0$ limit.

In this language the program of the chapter goes through unchanged in structure: Poincaré invariance and $\mathrm{SU}(2)$ little-group covariance fix the three-point amplitudes of massive particles up to constants, exactly as $\mathrm{U}(1)$ covariance fixed the massless three-point amplitudes in Eq. (58.98), and recursion builds the rest. The massive three-point amplitudes are the bold-spinor analogues of Eqs. (58.178) and (58.179); their classification, and the extension of BCFW to massive legs, is an active and only partly settled subject as of the mid-2020s, which is why we flag it as frontier rather than develop it in full. We will not need

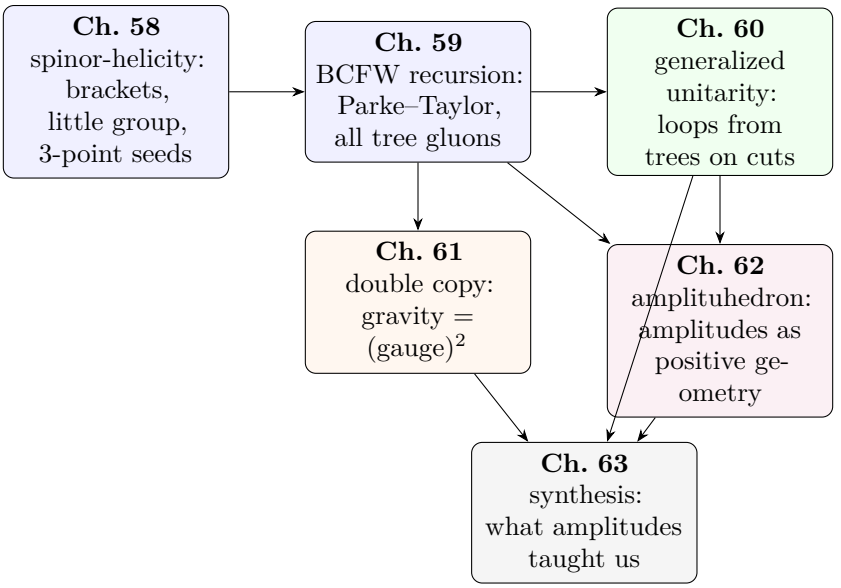


Figure 58.6: The route through Part XVI. The spinor-helicity variables and three-point seeds of this chapter feed the BCFW recursion of Chapter 59, which produces the complete tree-level gluon S-matrix (Parke–Taylor and beyond). That tree data is the input to three independent advances: generalized unitarity (Chapter 60) reconstructs loop amplitudes by sewing trees across cuts; the double copy (Chapter 61) builds gravity amplitudes as the "square" of gauge amplitudes; and the amplituhedron (Chapter 62) recasts amplitudes as the canonical form of a positive geometry. Chapter 63 draws the threads together.

massive spinors for the gluon recursion of Chapter 59, which lives entirely in the massless sector; we record them so that the reader leaves this chapter knowing that the on-shell program is not confined to massless particles.

58.7.6 A map of Part XVI

With the massless spinor-helicity formalism complete and the gluon amplitudes reduced to color-ordered partial amplitudes, the road through the rest of the modern-amplitudes part is straight. The logic is cumulative: each chapter takes the on-shell data manufactured by its predecessors and uses it to bypass a different piece of the Feynman-diagram apparatus. Figure 58.6 draws the dependency.

Concretely: Chapter 59 takes the three-gluon seeds Eqs. (58.178) and (58.179) and the color ordering of Theorem 58.32, complexifies external momenta as the caution above demanded, and derives the BCFW recursion that proves Parke–Taylor Eq. (58.180) and constructs every tree gluon amplitude. Chapter 60 carries those on-shell trees to loop level: generalized unitarity cuts express loop integrand coefficients as products of trees, trading the off-shell, gauge-dependent

loop diagrams of Chapter 19 for gauge-invariant on-shell data. Chapter 61 exploits a hidden algebraic structure of the same partial amplitudes — color–kinematics duality and the BCJ relations — to express gravity amplitudes as a "double copy" of gauge amplitudes, the modern realization of Yang–Mills theory's promised endpoint. Chapter 62 reframes the whole edifice geometrically: the planar amplitudes of maximally supersymmetric Yang–Mills theory are the canonical form of the amplituhedron, a positive geometry whose boundaries encode locality and unitarity, a development still conjectural in its general form and dated to the mid-2010s onward. Chapter 63 synthesizes the part, drawing the amplitudes program back to the bootstrap of Part XIV and the holography of Part XV.

We have done what this chapter set out to do: built variables that carry exactly the on-shell data of a massless particle and nothing more, fixed the three-point amplitudes by symmetry alone, reframed QED in helicity language, and now peeled color off the gluons to expose a single gauge-invariant, cyclically ordered object $A_n[1, \ldots, n]$. The three-gluon seeds Eqs. (58.178) and (58.179), the color decomposition Eq. (58.172), the partial-amplitude properties Eqs. (58.173) to (58.175), and the Parke–Taylor target Eq. (58.180) are precisely the kinematic groundwork that Chapter 59 requires: there the seeds are glued by BCFW recursion, with momenta pushed into the complex plane, to construct the complete tree-level S-matrix of Yang–Mills theory and to prove the one-line formula we have here only named.

Further reading

The modern on-shell perspective adopted here — spinor-helicity variables, little-group scaling, three-point amplitudes fixed by symmetry, and the color-ordered partial amplitudes — is developed at book length in Elvang and Huang [176] and Henn and Plefka [263], the two standard graduate texts on the subject and the natural companions to this chapter and the next. The pedagogical TASI lectures Dixon [153] give a brisk and complete route through color ordering, brackets, BCFW, and Parke–Taylor; the older review by Mangano and Parke [334] remains the definitive reference for the color decomposition and the partial-amplitude relations. The one-line MHV formula whose form we verified is the result of Parke and Taylor [379]; the recursion that proves it, deferred to Chapter 59, is Britto et al. [91, 92], with the twistor-space organizing idea behind the holomorphic structure due to Witten [523]. The completion of the formalism to massive particles of arbitrary spin, stated here at the frontier, is Arkani-Hamed et al. [38]. For the underlying representation theory of the Poincaré group on which the entire construction rests, see Weinberg [500] and the treatment in Chapter 3.

Chapter 59

Tree-Level Structure: MHV and Recursion

A single tree-level scattering process of a handful of gluons, written out in the Feynman expansion built honestly from the Yang–Mills Lagrangian Eq. (25.76) and the vertices of Chapter 26, demands a number of diagrams that grows faster than any factorial — each diagram a gauge-dependent tangle of Lorentz and color indices, the whole sum riddled with cancellations that the final answer never sees. And yet the answer, for the simplest helicity configurations, fits on a single line. This chapter is the story of why. It takes the classical gauge theory quantized in Chapters 25 and 26 and the spinor-helicity variables assembled in Chapter 58 — angle and square brackets, the little-group scaling of Eq. (58.5), the reference-momentum polarization vectors of Eq. (58.111), and the symmetry-fixed three-point amplitudes of Theorem 58.20 — and shows that the tree-level S-matrix of Yang–Mills is governed not by the Lagrangian but by two principles older and simpler than any action: locality and unitarity. This is where Yang–Mills theory, carried since the classical theory, reaches its modern form.

The arc is constructive. We first tame the factorial explosion by separating what is group theory from what is kinematics: the *color decomposition* factors every amplitude into a sum over single traces of generators, each multiplying a gauge-invariant, cyclically-ordered *partial amplitude* A_n computed from color-stripped Feynman rules. With color out of the way we ask a purely kinematic question — given the external helicities, what is A_n? — and find, by spending nothing but the gauge freedom hidden in the polarization references, that whole classes of helicity assignments vanish identically: the all-plus and single-minus amplitudes are zero, and the lowest surviving sector is the maximally-helicity-violating (MHV) one with exactly two negative helicities. For that sector we state the one-line Parke–Taylor formula, check it against every symmetry and factorization limit the formalism affords, and earn it by an honest four-gluon Feynman computation — leaving it, at that stage, a checked-but-unproved all-n

conjecture.

The proof requires leaving the real slice. Real on-shell three-point kinematics for massless particles is degenerate, collapsing the seed amplitude to an indeterminate 0/0; complexifying the momenta — treating the holomorphic and antiholomorphic spinors as independent — resurrects two nonzero three-point amplitudes, fixed up to a constant by little-group scaling and locality and normalized against the three-gluon vertex of Chapter 26. These complex seeds, together with the factorization of trees on physical poles established in Chapters 13 and 15, power the Britto–Cachazo–Feng–Witten (BCFW) recursion: a one-parameter complex shift of two external momenta turns the amplitude into a meromorphic function of z, and Cauchy's theorem reconstructs it from the residues at its physical poles, each a product of two lower-point on-shell amplitudes. The derivation leaves one debt — a boundary term at $z = \infty$ — which a large-z analysis, sharpened by the background-field argument, discharges for the right choice of shift, yielding the central structural result that tree Yang–Mills is *on-shell constructible*: determined entirely by its three-point seeds and the analytic structure of locality and unitarity, with no Lagrangian entering the reconstruction.

With the recursion justified we turn it on. We prove Parke–Taylor by induction on multiplicity, run the machine explicitly on the four- and five-gluon amplitudes to watch a thousand-diagram computation collapse to a single term, and assemble the first genuinely non-MHV amplitude, the six-point NMHV, from MHV and $\overline{\text{MHV}}$ pieces — previewing the spurious-pole cancellation that the amplituhedron of Chapter 62 later reads geometrically. The chapter closes by naming what has been gained: the trees proved here are the on-shell currency for the generalized-unitarity loop methods of Chapter 60 and the double copy to gravity of Chapter 61.

59.1 Color Decomposition and Color-Ordered Amplitudes

The chapter introduction laid out the scandal: a process with a handful of external gluons, written out in the textbook Feynman expansion built from the Yang–Mills Lagrangian Eq. (25.76) and the vertices of Chapter 26, requires a number of diagrams that grows faster than any factorial, each diagram a gauge-dependent tangle of Lorentz and color indices, and the whole sum collapses to an answer one can write on a single line. Before we can explain *why* the answer is short — the business of the rest of the chapter — we must first stop the proliferation from drowning us, and the first and cheapest economy is to notice that two completely different kinds of information are entangled in every diagram. There is *kinematics*: the momenta, the polarizations, the Lorentz contractions. And there is *color*: the structure constants f^{abc} at the vertices and the Kronecker

n	full Feynman diagrams	color-ordered diagrams
4	4	3
5	25	10
6	220	38
7	2 485	154
8	34 300	654
9	559 405	2 871
10	10 525 900	12 925

Table 59.1: The number of tree-level Feynman diagrams for the n-gluon amplitude. The middle column counts *all* diagrams contributing to the full amplitude $\mathcal{A}_n^{\text{tree}}$, color and kinematics together. The right column counts the diagrams contributing to a single color-ordered partial amplitude $A_n[1, 2, \ldots, n]$, i.e. those that can be drawn in the plane with the external legs in the fixed cyclic order $1, 2, \ldots, n$. Color ordering already removes most of the redundancy; the remaining collapse to Parke–Taylor is the subject of the later sections.

δ^{ab} on the propagators, pure group theory that knows nothing about the energies involved. These two factor cleanly. The purpose of this section is to perform that factorization once and for all: to peel the gauge group off the kinematics, package all the color into a small set of invariant tensors (traces of generators), and isolate the gauge-invariant kinematic coefficients — the *color-ordered partial amplitudes* — that every later section computes helicity by helicity. After this section the words "n-gluon amplitude" will mean a single object $A_n[1, 2, \ldots, n]$ with its own Feynman rules and its own symmetries, and the factorial swarm of diagrams will have been quietly tamed into a sum over $(n-1)!$ orderings of one such object.

59.1.1 The size of the problem

It is worth seeing the explosion quantitatively, because the numbers are what make the eventual one-line answers astonishing rather than merely tidy. A tree-level pure-gluon amplitude with n external legs is built from the cubic and quartic vertices of Eqs. (26.53) and (26.56) wired together by the gluon propagator Eq. (26.45) into every distinct connected tree. Counting those trees is a purely combinatorial exercise; the results are collected in the second column of Table 59.1. The growth is super-factorial: doubling the number of gluons from five to ten multiplies the diagram count by more than four hundred thousand.

Most of this growth is illusory in the sense that matters here: it is generated by summing the *same* kinematic skeleton against different color labels, and by the gauge dependence of individual diagrams, which forces enormous cancellations that a final gauge-invariant answer never sees. The right column of Table 59.1

previews how much is gained simply by organizing the sum by color: the diagrams contributing to one cyclically-ordered piece are far fewer. The first task is to make "organizing by color" precise.

59.1.2 Where the color lives

We collect the color content of the gluon Feynman rules, reading it off the rules established in Chapter 26. Throughout, $a, b, c, \cdots \in \{1, \ldots, N^2-1\}$ are adjoint (gluon) indices of SU(N), and $i, j, k, \cdots \in \{1, \ldots, N\}$ are fundamental indices; the Hermitian generators T^a act in the fundamental.

The gluon propagator Eq. (26.45) carries the color tensor δ^{ab} and nothing more:

$$D^{ab}_{\mu\nu}(k)\Big|_{\xi=1} = \delta^{ab}\,\frac{-i\,\eta_{\mu\nu}}{k^2+i\varepsilon}. \tag{59.1}$$

The three-gluon vertex Eq. (26.53) carries a single structure constant,

$$iV^{abc}_{\mu\nu\rho}(p,q,r) = g\,f^{abc}\,K_{\mu\nu\rho}(p,q,r), \qquad K_{\mu\nu\rho}(p,q,r) \equiv \eta_{\mu\nu}(p-q)_\rho + \eta_{\nu\rho}(q-r)_\mu + \eta_{\rho\mu}(r-p)_\nu, \tag{59.2}$$

with all momenta incoming, $p+q+r=0$; and the four-gluon vertex Eq. (26.56) carries a product of two structure constants contracted on a shared adjoint index,

$$iV^{abcd}_{\mu\nu\rho\sigma} = -ig^2\Big[f^{abe}f^{cde}\,(\eta_{\mu\rho}\eta_{\nu\sigma}-\eta_{\mu\sigma}\eta_{\nu\rho}) + f^{ace}f^{bde}\,(\eta_{\mu\nu}\eta_{\rho\sigma}-\eta_{\mu\sigma}\eta_{\nu\rho}) + f^{ade}f^{bce}\,(\eta_{\mu\nu}\eta_{\rho\sigma}-\eta_{\mu\rho}\eta_{\nu\sigma})\Big] \tag{59.3}$$

Every tree diagram is therefore, in color space, a product of structure constants f^{abc} (one per cubic vertex, two per quartic) with the adjoint indices of internal lines contracted in pairs through the propagator δ^{ab}. The external indices $a_1, \ldots, a_n$ are left free. The color factor of any diagram is thus a fully contracted polynomial in f's carrying the n free external indices, and the kinematic factor — the contraction of η's, momenta, and external polarizations — rides alongside it, multiplied. The two never mix. Our strategy is to convert this f-and-δ data into a basis of traces of generators, in which the kinematic coefficients organize themselves automatically.

59.1.3 From structure constants to traces

The single algebraic identity that powers the whole conversion expresses a structure constant as a difference of traces. It follows immediately from the two defining facts of the generators.

Lemma 59.1 (Structure constant as a trace)**.** *With the book-canonical normalization* $\big[T^a, T^b\big] = if^{abc}T^c$ *and* $\mathrm{tr}(T^aT^b) = \tfrac{1}{2}\delta^{ab}$,

$$f^{abc} = -2i\,\Big(\mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^aT^cT^b)\Big). \tag{59.4}$$

Proof. Multiply the Lie bracket $\left[T^a, T^b\right] = if^{abd}T^d$ on the right by T^c and take the trace. Using $\mathrm{tr}(T^dT^c) = \frac{1}{2}\delta^{dc}$,

$$\mathrm{tr}\left(\left[T^a, T^b\right] T^c\right) = if^{abd}\,\mathrm{tr}(T^dT^c) = if^{abd}\,\tfrac{1}{2}\delta^{dc} = \tfrac{i}{2}\,f^{abc}. \tag{59.5}$$

The left-hand side is $\mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^bT^aT^c)$, and by cyclicity of the trace $\mathrm{tr}(T^bT^aT^c) = \mathrm{tr}(T^aT^cT^b)$, so $\mathrm{tr}\left(\left[T^a, T^b\right] T^c\right) = \mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^aT^cT^b)$. Equating with Eq. (59.5) and solving for f^{abc} gives Eq. (59.4). (This is the trace formula of Eq. (B.20) specialized to the fundamental, $T(R) = \frac{1}{2}$.) □

The factor of $-2i$ is a nuisance that recurs, multiplied, at every vertex, and the completeness relation we will use to contract internal lines carries a companion factor of $\frac{1}{2}$. Both are artifacts of the normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, and the amplitudes literature removes them once and for all by rescaling the generators. We adopt that rescaling for the remainder of Chapter 59.

Notation Watch

For all of Chapter 59 we use *amplitude-normalized* generators, defined by a rescaling of the canonical ones of Chapter B by $\sqrt{2}$,

$$T^a \;\longmapsto\; T^a_{\text{here}} = \sqrt{2}\,T^a_{\text{canon}}, \qquad \text{so that} \qquad \mathrm{tr}(T^aT^b) = \delta^{ab}. \tag{59.6}$$

This is a pure change of basis-scale; it leaves the physical content untouched. The structure constants f^{abc} appearing in the Lagrangian Eq. (25.76), the coupling g, and the Casimirs $C_A = N$, C_F, T_F of Chapter B are all defined through the physical commutator and are *unchanged*; only the matrices T^a that appear inside traces in the color decomposition are rescaled. In this normalization the completeness (color-Fierz) relation Eq. (B.78) of Theorem B.23 reads, with the $\frac{1}{2}$ absorbed,

$$\sum_{a=1}^{N^2-1} (T^a)_i{}^j\,(T^a)_k{}^l = \delta_i^l\,\delta_k^j - \frac{1}{N}\,\delta_i^j\,\delta_k^l, \tag{59.7}$$

and the f-to-trace identity Eq. (59.4) becomes, after the rescaling pulls a factor $2^{-3/2}$ out of the triple trace,

$$f^{abc} = -\frac{i}{\sqrt{2}}\left(\mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^aT^cT^b)\right) = -\frac{i}{\sqrt{2}}\,\mathrm{tr}\left(\left[T^a, T^b\right] T^c\right). \tag{59.8}$$

Equation Eq. (59.8) is the working form used below; note that the f^{abc} on its left is the physical structure constant, while the T^a inside the traces are the rescaled generators of Eq. (59.6).

To verify the rescaling claim in Eq. (59.8): under $T^a \mapsto \sqrt{2}\,T^a$, each triple trace scales as $\mathrm{tr}(T^aT^bT^c) \mapsto 2^{3/2}\,\mathrm{tr}(T^aT^bT^c)$, so the right-hand side of Eq. (59.4), which equals $-2i\ \mathrm{tr}(\big[T^a,T^b\big]\,T^c)$, becomes $-2i{\cdot}2^{-3/2}\ \mathrm{tr}(\big[T^a,T^b\big]\,T^c) = -\frac{i}{\sqrt{2}}\,\mathrm{tr}(\big[T^a,T^b\big]\,T^c)$, while f^{abc} on the left is held fixed. This reproduces Eq. (59.8).

59.1.4 The single-trace decomposition

We can now state and prove the central organizing result of the section: at tree level the color factor of every gluon amplitude collapses onto a basis of *single* traces of generators, one trace per cyclic ordering of the external legs.

The cleanest way to see this — and the way that simultaneously delivers the photon-decoupling identity of Section 59.1.7 — is to enlarge the gauge group momentarily from $\mathrm{SU}(N)$ to $\mathrm{U}(N)$. Adjoin to the N^2-1 traceless generators T^a the trace generator

$$T^0 = \frac{1}{\sqrt{N}}\,\mathbf{1}, \qquad \mathrm{tr}(T^0T^0) = \frac{1}{N}\,\mathrm{tr}\,\mathbf{1} = 1, \tag{59.9}$$

normalized to match Eq. (59.6), and let capital indices $A, B, \dots \in \{0, 1, \dots, N^2-1\}$ run over the full set $\{T^0, T^a\}$, a basis for all Hermitian $N \times N$ matrices. The $\mathrm{U}(N)$ completeness relation Eq. (B.81) then has no trace subtraction at all,

$$\sum_{A=0}^{N^2-1} (T^A)_i{}^j\,(T^A)_k{}^l = \delta_i^{\,l}\,\delta_k^{\,j}, \tag{59.10}$$

the indices simply being swapped — the algebraic statement that a $\mathrm{U}(N)$ gluon is an unconstrained quark–antiquark color pair. Two facts make working in $\mathrm{U}(N)$ legitimate and convenient.

First, the extra gauge boson T^0 is a free, decoupled photon as far as gluons are concerned. Its structure constants vanish identically, $f^{0bc} \propto \mathrm{tr}(\big[\mathbf{1},T^b\big]\,T^c) = 0$, because the identity commutes with everything. Since every pure-gluon vertex Eqs. (59.2) and (59.3) is built from structure constants, any vertex with a T^0 leg vanishes: the $\mathrm{U}(1)$ photon couples to nothing colored. Consequently a tree diagram with only $\mathfrak{su}(N)$ gluons on its external and internal lines is identical whether we compute it in $\mathrm{SU}(N)$ or in $\mathrm{U}(N)$ — no internal T^0 can ever appear, so the extra direction is inert. We may therefore contract all internal adjoint indices with the trace-free relation Eq. (59.10) instead of Eq. (59.7), deferring the descent back to $\mathrm{SU}(N)$ to the end.

Second, in the $\mathrm{U}(N)$ basis the color factor of any diagram becomes a product of generator traces sewn by Eq. (59.10), and the sewing has a transparent graphical meaning in 't Hooft's double-line notation, the same notation that Chapter B used to read the completeness relation. Convert each structure constant to traces by Eq. (59.8); each cubic vertex becomes a difference of two

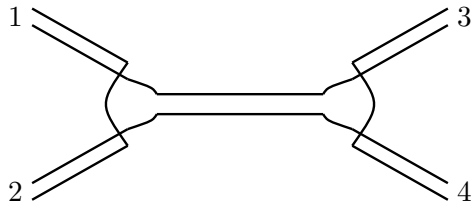


Figure 59.1: The color flow of the four-gluon s-channel tree in 't Hooft double-line notation, in the U(N) basis where the completeness relation Eq. (59.10) simply joins fundamental lines. Each external gluon is a pair of antiparallel color lines; the internal gluon likewise. Because the diagram is a tree, the joined lines form a *single* closed boundary loop visiting the external legs in the cyclic order $1, 2, 3, 4$, which is the single trace $\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$. No second closed loop is possible without a momentum loop, so no double-trace term arises at tree level.

"open" triple products of generators, and each quartic a difference of quadruple products. Draw each adjoint (gluon) line as a double line, a fundamental line and an anti-fundamental line running antiparallel. The completeness relation Eq. (59.10) is then the rule that joins the fundamental lines of an internal gluon end to end. Because a *tree* diagram has no closed loops, the fundamental color lines, once all internal gluons are resolved and joined, can neither cross nor close among themselves except along the single outer boundary of the planar diagram. They merge into exactly one closed fundamental loop, whose boundary threads the n external generators in the cyclic order in which the legs are arranged. That single closed loop *is* a single trace, $\mathrm{tr}(T^{a_{i_1}}T^{a_{i_2}}\cdots T^{a_{i_n}})$, with the ordering $(i_1, \ldots, i_n)$ read off the planar embedding. Figure 59.1 illustrates the mechanism on a four-point diagram.

Summing over all diagrams and all planar embeddings collects the kinematics into a coefficient of each distinct single trace. Distinct traces correspond to distinct cyclic orderings of $\{1, \ldots, n\}$, of which there are $(n-1)!$ once we use cyclicity of the trace to fix leg 1 in the first slot. This is the decomposition.

Theorem 59.2 (Tree-level color decomposition)**.** *The full n-gluon tree amplitude, with external adjoint labels $a_1, \ldots, a_n$, decomposes as*

$$\mathcal{A}_n^{\text{tree}}(\{p_i, \epsilon_i, a_i\}) = g^{n-2} \sum_{\sigma \in S_{n-1}} \mathrm{tr}(T^{a_1}T^{a_{\sigma(2)}}T^{a_{\sigma(3)}}\cdots T^{a_{\sigma(n)}})\; A_n[1, \sigma(2), \sigma(3), \ldots, \sigma(n)], \tag{59.11}$$

where the sum runs over the $(n-1)!$ permutations σ of $\{2, 3, \ldots, n\}$ (leg 1 held fixed by trace cyclicity), and the kinematic coefficients $A_n[1, \sigma]$ — the color-ordered partial amplitudes *— depend only on the momenta and polarizations, carry no color, and are separately gauge invariant.*

Proof. The double-line argument above establishes that, in the U(N) basis,

every tree diagram's color factor is a single trace $\mathrm{tr}(T^{a_{i_1}}\cdots T^{a_{i_n}})$ of the external generators in the cyclic order of the planar embedding, times a kinematic factor with no remaining color. Grouping the diagrams by their trace gives $\mathcal{A}_n^{\text{tree}} = g^{n-2}\sum_{\text{cyclic orderings}}\mathrm{tr}(\cdots)\,A_n[\cdots]$, where A_n is the sum of kinematic factors of all diagrams sharing that ordering, and the factor g^{n-2} is the count of vertices on a tree (one power of g per cubic vertex, two per quartic, equal to $n-2$ by the tree relation $V_3 + 2V_4 = n-2$ derived in Section 59.1.6). Using cyclicity of the trace to rotate leg 1 to the front identifies the independent orderings with permutations of $\{2,\dots,n\}$, giving the sum over S_{n-1} and $(n-1)!$ terms. Finally, since the computation was carried out in $\mathrm{U}(N)$ and the inert T^0 never entered any internal line, the result equals the genuine $\mathrm{SU}(N)$ amplitude: the would-be $\frac{1}{N}$ corrections of the $\mathrm{SU}(N)$ Fierz relation Eq. (59.7), which would have produced multi-trace structures, are exactly the ones that cancel because the extra $\mathrm{U}(1)$ photon decouples (Section 59.1.7). Each $A_n[1,\sigma]$ is gauge invariant because the full $\mathcal{A}_n^{\text{tree}}$ is, and the traces in Eq. (59.11) are linearly independent functions of the external labels for $n \le N+1$ (and the partial amplitudes are kinematic objects defined independently of N), so a gauge variation of $\mathcal{A}_n$ that vanishes for all $\{a_i\}$ forces each coefficient to be invariant separately. □

Definition 59.3 (Color-ordered partial amplitude)**.** The kinematic coefficient $A_n[1,2,\dots,n]$ multiplying the trace $\mathrm{tr}(T^{a_1}T^{a_2}\cdots T^{a_n})$ in Eq. (59.11) is the *color-ordered partial amplitude* (or partial amplitude) for the cyclic ordering $(1,2,\dots,n)$. It is computed from the color-stripped Feynman rules of Section 59.1.5 by summing only those diagrams whose external legs can be drawn in the plane in the order $1,2,\dots,n$ — the planar, or color-ordered, diagrams counted in the right column of Table 59.1.

This is the object the rest of the chapter computes. Everything we will say about gluon amplitudes — vanishing helicity configurations, the Parke–Taylor formula, BCFW recursion — is a statement about a single $A_n[1,\dots,n]$; the full amplitude is reassembled at the end by dressing each ordering with its trace and summing.

59.1.5 Color-ordered Feynman rules

Theorem 59.3 would be useless without a direct prescription for A_n that never mentions color. We extract it by stripping the color tensor off each full rule, retaining the coefficient of one fixed trace ordering. Because each color-ordered diagram is planar with a definite cyclic adjacency of legs, the stripped rules are *ordered*: vertices connect legs that are adjacent in the cyclic ordering, and the rules carry the index labels in that order.

Cubic vertex. Insert Eq. (59.8) into the full cubic vertex Eq. (59.2). Writing $f^{abc} = -\frac{i}{\sqrt{2}}(\mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^aT^cT^b))$ and reading off the coefficient of the

single ordered trace $\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3})$, and adopting the all-*outgoing* momentum convention standard in amplitude work (which reverses every momentum relative to the all-incoming Eq. (59.2) and so flips the sign of the momentum-linear K), the color-ordered three-gluon vertex is

$$V_3^{\mu_1\mu_2\mu_3}(p_1,p_2,p_3) = \frac{i}{\sqrt{2}}\Big(\eta^{\mu_1\mu_2}(p_1-p_2)^{\mu_3}+\eta^{\mu_2\mu_3}(p_2-p_3)^{\mu_1}+\eta^{\mu_3\mu_1}(p_3-p_1)^{\mu_2}\Big), \tag{59.12}$$

with $p_1+p_2+p_3=0$ and the legs in cyclic order $1,2,3$.

Quartic vertex. The quartic requires contracting the two structure constants in Eq. (59.3) into traces and extracting the coefficient of one ordering; we carry the computation in Theorem 59.4 below. The result is the color-ordered four-gluon vertex, free of momenta,

$$V_4^{\mu_1\mu_2\mu_3\mu_4} = \frac{i}{2}\Big(2\,\eta^{\mu_1\mu_3}\eta^{\mu_2\mu_4}-\eta^{\mu_1\mu_2}\eta^{\mu_3\mu_4}-\eta^{\mu_1\mu_4}\eta^{\mu_2\mu_3}\Big), \tag{59.13}$$

with legs in cyclic order $1,2,3,4$.

Propagator. The internal gluon, in Feynman gauge, keeps its tensor structure but loses its color δ^{ab}, which has been used up sewing the color lines:

$$\frac{-i\,\eta_{\mu\nu}}{p^2+i\varepsilon}. \tag{59.14}$$

Together, Eqs. (59.12) to (59.14) are the *color-ordered Feynman rules*; we refer to the trio collectively as Eqs. (59.12) to (59.14) or, for cross-reference, as the rules of Section 59.1.5. The partial amplitude $A_n[1,\dots,n]$ is the sum, over all planar diagrams with the legs in the cyclic order $1,\dots,n$, of the products of these rules, with external polarizations ϵ_i attached and overall momentum conservation imposed — exactly the ordinary Feynman procedure of Chapter 15, but with the ordered rules and only the planar diagrams.

Worked Example 59.4 (Stripping the four-gluon vertex)**.** We derive Eq. (59.13) from the full vertex Eq. (59.3). The strategy is to write each product ff as a product of traces, then read off the coefficient of the single ordered trace $\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$, i.e. the cyclic ordering 1234.

A product of structure constants as a single trace. Using Eq. (59.8) for both factors and the U(N) completeness Eq. (59.10) to perform the sum over the shared index e, we first establish

$$\sum_e \mathrm{tr}(MT^e)\,\mathrm{tr}(NT^e) = \mathrm{tr}(MN), \tag{59.15}$$

valid for any matrices M,N. Indeed, writing the traces in components, $\sum_e \mathrm{tr}(MT^e)\,\mathrm{tr}(NT^e) = M_{ij}N_{kl}\sum_e (T^e)_{ji}(T^e)_{lk} = M_{ij}N_{kl}\,\delta_{jk}\delta_{li} = M_{ij}N_{ji} = \mathrm{tr}(MN)$, where the middle

step used Eq. (59.10). Applying Eq. (59.15) with $M = \big[T^a, T^b\big]$, $N = \big[T^c, T^d\big]$ and the two prefactors $(-\frac{i}{\sqrt{2}})^2 = -\frac{1}{2}$,

$$f^{abe}f^{cde} = -\tfrac{1}{2}\,\mathrm{tr}\big(\,\big[T^a,T^b\big]\,\big[T^c,T^d\big]\,\big) = -\tfrac{1}{2}\Big(\,\mathrm{tr}(T^aT^bT^cT^d) - \mathrm{tr}(T^aT^bT^dT^c) - \mathrm{tr}(T^bT^aT^cT^d) + \mathrm{tr}(\tag{59.16}$$

Coefficient of $\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$ *in each color channel.* Channel one of Eq. (59.3) carries $f^{abe}f^{cde}$ with $(a,b,c,d) = (a_1,a_2,a_3,a_4)$. Among the four traces in Eq. (59.16), only $+\,\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$ is the ordering 1234; the others are cyclically inequivalent. Its coefficient is therefore $-\frac{1}{2}$.

Channel two carries $f^{ace}f^{bde}$, i.e. Eq. (59.16) with the relabeling $(a,b,c,d) \to (a_1,a_3,a_2,a_4)$. The four resulting traces are $\mathrm{tr}(T^{a_1}T^{a_3}T^{a_2}T^{a_4})$ and rearrangements; none is cyclically equal to 1234. The coefficient of $\mathrm{tr}(1234)$ is 0.

Channel three carries $f^{ade}f^{bce}$, i.e. the relabeling $(a,b,c,d) \to (a_1,a_4,a_2,a_3)$. Then Eq. (59.16) contains the term $-\,\mathrm{tr}(T^{a_4}T^{a_1}T^{a_2}T^{a_3})$, which by cyclicity equals $-\,\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$, i.e. ordering 1234. With the overall $-\frac{1}{2}$ its coefficient is $-\frac{1}{2}\cdot(-1) = +\frac{1}{2}$. The other three traces of this channel are not 1234.

Assembly. Collecting the coefficient of $g^2\,\mathrm{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})$ in $iV^{abcd}_{\mu_1\mu_2\mu_3\mu_4}$ from Eq. (59.3), with $(\mu,\nu,\rho,\sigma) = (\mu_1,\mu_2,\mu_3,\mu_4)$,

$$\begin{aligned} iV_4 &= -i\Big[-\tfrac{1}{2}(\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3}) + 0 + \tfrac{1}{2}(\eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4} - \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4})\Big] \\ &= -\tfrac{i}{2}\Big[-\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} + \eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3} + \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4} - \eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4}\Big] \\ &= \tfrac{i}{2}\Big(2\,\eta_{\mu_1\mu_3}\eta_{\mu_2\mu_4} - \eta_{\mu_1\mu_2}\eta_{\mu_3\mu_4} - \eta_{\mu_1\mu_4}\eta_{\mu_2\mu_3}\Big), \end{aligned} \tag{59.17}$$

which is Eq. (59.13). The momentum-independence and the characteristic relative coefficients $(2,-1,-1)$ — favoring the pairing of legs that are *not* cyclically adjacent — are precisely what one needs for the ordered four-gluon amplitude to be gauge invariant once combined with the cubic-vertex diagrams.

59.1.6 Cyclicity, reflection, and counting

Two exact symmetries of the partial amplitudes follow at once and reduce the number of independent objects.

Cyclicity. Because $A_n[1,2,\ldots,n]$ is, by Theorem 59.3, the coefficient of the cyclic trace $\mathrm{tr}(T^{a_1}\cdots T^{a_n})$, and the trace is invariant under cyclic rotation of its arguments, the partial amplitude is invariant under cyclic rotation of its labels,

$$A_n[1,2,\ldots,n] = A_n[2,3,\ldots,n,1] = A_n[3,\ldots,n,1,2] = \cdots . \tag{59.18}$$

Equivalently, the color-ordered rules Eqs. (59.12) and (59.13) depend only on the cyclic adjacency of legs, not on which leg is called first. This is why fixing leg 1 in Eq. (59.11) loses nothing.

Reflection. Reversing the entire ordering produces a sign:

$$A_n[n, n-1, \ldots, 2, 1] = (-1)^n \, A_n[1, 2, \ldots, n]. \tag{59.19}$$

Proposition 59.5 (Reflection identity)**.** *The color-ordered partial amplitude obeys Eq. (59.19).*

Proof. We track the sign acquired by each ingredient of a planar diagram under reversal of the cyclic order, $(1, 2, \ldots, n) \to (n, \ldots, 2, 1)$, which reverses the Lorentz labels and momenta on every vertex. The propagator Eq. (59.14) is symmetric in its two indices and even in momentum, hence invariant. The cubic vertex Eq. (59.12) is *odd*: under the simultaneous interchange of its outer legs $1 \leftrightarrow 3$ (with momenta), its three terms map among themselves with an overall sign,

$$\begin{aligned} &\eta^{\mu_3\mu_2}(p_3 - p_2)^{\mu_1} + \eta^{\mu_2\mu_1}(p_2 - p_1)^{\mu_3} + \eta^{\mu_1\mu_3}(p_1 - p_3)^{\mu_2} \\ &\quad = -\Big(\eta^{\mu_1\mu_2}(p_1 - p_2)^{\mu_3} + \eta^{\mu_2\mu_3}(p_2 - p_3)^{\mu_1} + \eta^{\mu_3\mu_1}(p_3 - p_1)^{\mu_2}\Big) = -V_3, \end{aligned} \tag{59.20}$$

so each cubic vertex contributes a factor -1. The quartic vertex Eq. (59.13) is even: reversing $(1, 2, 3, 4) \to (4, 3, 2, 1)$ sends $\eta^{\mu_1\mu_3}\eta^{\mu_2\mu_4} \to \eta^{\mu_4\mu_2}\eta^{\mu_3\mu_1} = \eta^{\mu_1\mu_3}\eta^{\mu_2\mu_4}$ and likewise leaves the other two terms fixed, giving a factor $+1$. A planar diagram with V_3 cubic and V_4 quartic vertices thus acquires the sign $(-1)^{V_3}$, the same for every diagram contributing to a given A_n provided $(-1)^{V_3}$ is determined by n alone. It is. Counting line ends, each internal line (I of them) has two ends and each external line one, so $3V_3 + 4V_4 = 2I + n$; and a tree has $V_3 + V_4 = I + 1$. Eliminating I,

$$V_3 = n - 2 - 2V_4 \;\equiv\; n \pmod 2, \tag{59.21}$$

so $(-1)^{V_3} = (-1)^n$ independently of how the vertices are apportioned. Every planar diagram for $A_n[1, \ldots, n]$ therefore maps under reversal to the corresponding diagram for $A_n[n, \ldots, 1]$ times the common factor $(-1)^n$, proving Eq. (59.19). (Equation Eq. (59.21) also supplies the power $g^{V_3+2V_4} = g^{n-2}$ used in Eq. (59.11).) □

Cyclicity and reflection together generate the dihedral group of order $2n$ acting on the orderings, so the $(n-1)!$ traces of Eq. (59.11) fall into orbits of size up to $2n$, and the genuinely independent partial amplitudes number roughly $(n-1)!/2$. The next two identities cut this down much further.

59.1.7 The $U(1)$ decoupling identity

The cleanest of the linear relations among partial amplitudes is the one already implicit in the U(N) construction of Section 59.1.4: the decoupling of the extra abelian photon.

Theorem 59.6 ($U(1)$ decoupling / photon-decoupling identity)**.** *The color-ordered partial amplitudes obey*

$$A_n[1,2,3,\dots,n]+A_n[2,1,3,\dots,n]+A_n[2,3,1,4,\dots,n]+\cdots+A_n[2,3,\dots,n{-}1,1,n]=0, \tag{59.22}$$

the sum being over the $n-1$ inequivalent positions into which leg 1 may be inserted while holding the cyclic order of $2,3,\dots,n$ fixed. Equivalently, after rotating leg 1 to the front of each term by cyclicity,

$$\sum_{j=0}^{n-1} A_n[1,\underbrace{2,3,\dots}_{j\ \text{legs}},\underbrace{\dots,n}] = 0, \tag{59.23}$$

i.e. the sum of $A_n[1,\sigma]$ over the $n-1$ cyclic rotations of the $(2,\dots,n)$ block vanishes. For $n=4$, with leg 1 rotated to the front,

$$A_4[1,2,3,4]+A_4[1,3,4,2]+A_4[1,4,2,3]=0. \tag{59.24}$$

Proof. Compute the $\mathrm{U}(N)$ amplitude in which leg 1 is the abelian photon T^0 of Eq. (59.9) and legs $2,\dots,n$ are ordinary $\mathfrak{su}(N)$ gluons. As established in Section 59.1.4, the photon decouples: $f^{0bc}=0$, so no vertex couples it to gluons and the amplitude vanishes identically,

$$\mathcal{A}_n^{\text{tree}}(1_{\mathrm{U}(1)},2,\dots,n)=0. \tag{59.25}$$

Now evaluate the same amplitude through the color decomposition Eq. (59.11), simply setting $T^{a_1}=T^0=\frac{1}{\sqrt{N}}\mathbf{1}$. The decomposition is a sum over the $(n-1)!$ orderings; insert the identity for T^{a_1} in each trace. Because $\mathbf{1}$ can be slid to any position and removed, $\mathrm{tr}(\mathbf{1}\,T^{a_{\sigma(2)}}\cdots T^{a_{\sigma(n)}})=\mathrm{tr}(T^{a_{\sigma(2)}}\cdots T^{a_{\sigma(n)}})$, every trace collapses to a trace of the $n-1$ gluon generators *without* leg 1. Two orderings σ,σ' that differ only by where leg 1 sat (its position among $2,\dots,n$) collapse to the *same* reduced trace. Grouping the sum Eq. (59.11) by reduced trace, the coefficient of each independent reduced trace $\mathrm{tr}(T^{a_2}\cdots T^{a_n})$ (in a fixed cyclic order of the gluons) is precisely the sum of the partial amplitudes A_n over all positions of leg 1 inserted into that order. Since the total Eq. (59.25) vanishes and the reduced traces are linearly independent, each such coefficient must vanish, which is Eq. (59.22). Specializing to $n=4$ and rotating leg 1 to the front gives Eq. (59.24). □

The decoupling identity is the gluon-amplitude statement that there is no abelian photon hiding in $\mathrm{SU}(N)$: the cancellation of the $\frac{1}{N}$ trace-subtraction terms in Eq. (59.7) that we invoked at the end of Theorem 59.2 *is* Eq. (59.22), now proved independently. It expresses one partial amplitude as a sum of others and so reduces the independent count below $(n-1)!/2$.

A second, stronger family of linear relations holds among the partial amplitudes, the *Kleiss–Kuijf relations*, which we state here and use later. Writing the

ordering as leg 1, a block $\boldsymbol{\alpha} = \{\alpha_1, \ldots, \alpha_r\}$, leg n, and a block $\boldsymbol{\beta} = \{\beta_1, \ldots, \beta_s\}$ (with $r + s = n - 2$), they read

$$A_n[1, \boldsymbol{\alpha}, n, \boldsymbol{\beta}] = (-1)^s \sum_{\sigma \in \boldsymbol{\alpha} \sqcup\!\sqcup \boldsymbol{\beta}^T} A_n[1, \sigma, n], \tag{59.26}$$

where $\boldsymbol{\beta}^T$ is the reversed block and $\sqcup\!\sqcup$ denotes the shuffle product (all interleavings of the two ordered blocks preserving each block's internal order). The decoupling identity Eq. (59.22) is the simplest special case ($\boldsymbol{\alpha}$ a single leg). The Kleiss–Kuijf relations reduce the number of independent partial amplitudes to $(n-2)!$, choosing a basis in which legs 1 and n are held fixed and the remaining $n-2$ permuted. A further, nonlinear-looking but in fact linear set — the BCJ relations arising from color–kinematics duality — cuts the basis to $(n-3)!$; those belong to the double-copy story of Chapter 61 and we defer them. We will not need a proof of Eq. (59.26) for the explicit low-point computations of this chapter; it is established in the amplitudes literature by the same monodromy/decoupling reasoning used for Theorem 59.6 applied repeatedly.

59.1.8 Yang–Mills theory: the four-gluon color decomposition

We close the section by assembling the carried Yang–Mills theory object explicitly: the color decomposition of the four-gluon amplitude, the smallest case in which all the machinery — single traces, the S_{n-1} sum, reflection, and the decoupling identity — is on display at once.

Worked Example 59.7 (Four-gluon color decomposition)**.** Specialize Eq. (59.11) to $n = 4$. The sum runs over S_3, the $3! = 6$ permutations of $\{2, 3, 4\}$, giving six single traces:

$$\begin{aligned}\mathcal{A}_4^{\text{tree}} = g^2\Big[&\operatorname{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4})\, A_4[1,2,3,4] + \operatorname{tr}(T^{a_1}T^{a_2}T^{a_4}T^{a_3})\, A_4[1,2,4,3] \\ &+ \operatorname{tr}(T^{a_1}T^{a_3}T^{a_2}T^{a_4})\, A_4[1,3,2,4] + \operatorname{tr}(T^{a_1}T^{a_3}T^{a_4}T^{a_2})\, A_4[1,3,4,2] \\ &+ \operatorname{tr}(T^{a_1}T^{a_4}T^{a_2}T^{a_3})\, A_4[1,4,2,3] + \operatorname{tr}(T^{a_1}T^{a_4}T^{a_3}T^{a_2})\, A_4[1,4,3,2]\Big].\end{aligned} \tag{59.27}$$

These six are not independent. Reflection Eq. (59.19) with $n = 4$ gives $(-1)^4 = +1$, so $A_4[1,2,3,4] = A_4[4,3,2,1] = A_4[1,4,3,2]$ (the last by cyclic rotation of $4,3,2,1$). Likewise $A_4[1,2,4,3] = A_4[1,3,4,2]$ and $A_4[1,3,2,4] = A_4[1,4,2,3]$. The six orderings thus pair into three reflection doublets, leaving three distinct partial amplitudes, conventionally

$$A_4[1,2,3,4], \qquad A_4[1,2,4,3], \qquad A_4[1,3,2,4], \tag{59.28}$$

each appearing in Eq. (59.27) multiplied by the sum of its trace and the reflected trace, e.g. $A_4[1,2,3,4]\big(\operatorname{tr}(T^{a_1}T^{a_2}T^{a_3}T^{a_4}) + \operatorname{tr}(T^{a_1}T^{a_4}T^{a_3}T^{a_2})\big)$.

The decoupling identity Eq. (59.24) relates these three. Written in the basis Eq. (59.28) using cyclicity and reflection ($A_4[1,3,4,2] = A_4[1,2,4,3]$ and $A_4[1,4,2,3] = A_4[1,3,2,4]$), it reads

$$A_4[1,2,3,4] + A_4[1,2,4,3] + A_4[1,3,2,4] = 0, \tag{59.29}$$

so only *two* of the three are independent. The Kleiss–Kuijf count Eq. (59.26) agrees: $(n-2)! = 2! = 2$. Holding legs 1 and 4 fixed and permuting $\{2,3\}$ gives the two-element basis $\{A_4[1,2,3,4],\ A_4[1,3,2,4]\}$, in terms of which Eq. (59.29) expresses the third, $A_4[1,2,4,3] = -A_4[1,2,3,4] - A_4[1,3,2,4]$.

Finally, it is instructive to recover the familiar structure-constant form of the four-gluon color factors and check consistency. Re-expressing the trace combinations through Eq. (59.8), the pairing of each A_4 with its reflected trace assembles into the symmetric/antisymmetric color tensors built from $f^{abe}f^{cde}$ — the same s-, t-, u-channel color structures that appear in the textbook four-gluon computation. The trace basis Eq. (59.27) and the ff basis are related by the linear change Eq. (59.8); the trace basis is the one in which the kinematic coefficients are individually gauge invariant, which is exactly why it is the right basis for what follows.

The three orderings of Eq. (59.28), reduced to the two-element Kleiss–Kuijf basis by Eq. (59.29), are the carried Yang–Mills theory data: every later statement about the four-gluon amplitude — its vanishing helicity configurations, its Parke–Taylor value, its reconstruction by BCFW — is a statement about $A_4[1,2,3,4]$, with the full amplitude reassembled through Eq. (59.27).

We have now done what the factorial explosion of Table 59.1 demanded: the color has been factored off into a sum over $(n-1)!$ single traces, and what remains is a single gauge-invariant, cyclically symmetric object $A_n[1,2,\ldots,n]$, computed from the stripped rules Eqs. (59.12) to (59.14) and constrained by reflection Eq. (59.19) and the decoupling identity Eq. (59.22). With color out of the way, the partial amplitude is ready to be evaluated helicity by helicity using the spinor-helicity polarization vectors of Chapter 58; the next section exploits the freedom in their reference momenta to show that whole classes of helicity assignments make $A_n[1,\ldots,n]$ vanish, organizing the landscape and exposing why the surviving amplitudes are as simple as they are.

59.2 The Simplicity of Gluon Amplitudes: Vanishing Helicity Configurations

The previous section did the hard organizational work: it stripped the color algebra off the n-gluon amplitude and left us with the gauge-invariant, cyclically-ordered *color-ordered partial amplitude* A_n, together with a compact set of

color-ordered Feynman rules (Eqs. (59.12) to (59.14)) — a cubic vertex, a quartic vertex, and a scalar propagator, all stripped of group-theory factors. We continue Yang–Mills theory from there. The reward for that labor is that we may now ask a sharp question about kinematics alone, with no f^{abc} in sight: *given the helicities of the external gluons, what is A_n?* The answer, astonishingly, is that for the simplest helicity assignments it is exactly zero — not zero after a long cancellation that one must trust, but identically zero, diagram by diagram, once the gauge freedom hidden in the polarization vectors is spent wisely. Discovering *which* configurations vanish, and why, is what organizes the entire helicity landscape and exposes the lone surviving sector whose closed form is the subject of the next section. This is where the "hidden simplicity" the chapter promised first becomes visible.

59.2.1 External data and the freedom in the polarization vectors

Fix conventions once. All n gluon momenta are taken *outgoing* and massless, $p_i^2 = 0$, so each carries a pair of two-component spinors through the bispinor factorization of Eq. (58.1),

$$p_{i\,\alpha\dot\alpha} = p_{i\mu}\,\sigma^{\mu}_{\alpha\dot\alpha} = \lambda_{i\alpha}\,\tilde\lambda_{i\dot\alpha}, \tag{59.30}$$

with angle and square brackets $\langle i\,j\rangle$, $[i\,j]$ built from $\lambda_i, \tilde\lambda_i$. We will use repeatedly the two consequences of Eq. (59.30) established in Chapter 58: the Mandelstam relation for a massless pair,

$$2\,p_i\cdot p_j = \langle i\,j\rangle\,[j\,i], \tag{59.31}$$

and the contraction of a single momentum into the Pauli-vector string, $\langle i|\gamma^\mu|i] = 2\,p_i^\mu$.

Each external gluon of definite helicity ± 1 is contracted with a polarization vector. The crucial input from Chapter 58 is that these vectors are not unique: masslessness and transversality fix them only up to a gauge transformation $\varepsilon_i^\mu \to \varepsilon_i^\mu + c\,p_i^\mu$, and the spinor-helicity formalism packages that freedom in a *reference momentum* q_i (an arbitrary null momentum, $q_i^2 = 0$, not collinear with p_i). The polarization vectors of Eq. (58.111) are

$$\varepsilon_+^\mu(p_i;q_i) = \frac{\langle q_i|\gamma^\mu|i]}{\sqrt{2}\,\langle q_i\,i\rangle}, \qquad \varepsilon_-^\mu(p_i;q_i) = \frac{\langle i|\gamma^\mu|q_i]}{\sqrt{2}\,[i\,q_i]}, \tag{59.32}$$

where $+$ and $-$ denote positive and negative helicity. Shifting q_i shifts ε_i^μ by a multiple of p_i^μ; since A_n is gauge invariant, its value cannot depend on any of the q_i. *This is the lever the whole section pulls.* We are free to choose every q_i as we please, and we will choose them so that as many polarization contractions as possible vanish outright — making the amplitude's value (which is reference-independent) manifestly small or zero. The only discipline (Section 59.2.2) is

that a single fixed choice of each q_i must be used uniformly across every diagram, and that $\langle q_i\, i\rangle \neq 0$, $[i\, q_i] \neq 0$ so the vectors in Eq. (59.32) stay finite.

Two transversality properties follow immediately and we will lean on them. Using $\not{q} = |q\rangle\langle q| + |q]\langle q|$ inside Eq. (59.32),

$$\varepsilon_{\pm}(p_i; q_i)\cdot p_i = 0, \qquad \varepsilon_{\pm}(p_i; q_i)\cdot q_i = 0, \tag{59.33}$$

the first because $\langle q_i|\not{p}_i|i] \propto [i\, i] = 0$ and the second because $\langle q_i|\not{q}_i|i] \propto \langle q_i\, q_i\rangle = 0$ (with the conjugate statements for ε_-).

59.2.2 The master lemma: dot products of polarization vectors

Every Feynman diagram for A_n is, after all index contractions are carried out, a sum of products of Lorentz invariants built from the external polarizations and momenta. The polarizations enter in only two ways: dotted into each other, $\varepsilon_i\cdot\varepsilon_j$, or dotted into a momentum, $\varepsilon_i\cdot p_j$. The vanishing theorems hinge entirely on a single computation of the first kind. We isolate it.

The one spinor identity we need beyond Chapter 58 is the Fierz (Chisholm) rearrangement, which follows from the completeness relation of the Pauli matrices, $\sigma^{\mu}_{\alpha\dot\alpha}\,\bar\sigma_{\mu}^{\dot\beta\beta} = 2\,\delta_{\alpha}^{\beta}\,\delta_{\dot\alpha}^{\dot\beta}$, derived in Chapter 58. Contracting that relation between two angle–square strings gives

$$\langle a|\gamma^{\mu}|b]\,\langle c|\gamma_{\mu}|d] = 2\,\langle a\, c\rangle\,[d\, b], \tag{59.34}$$

valid for any four massless labels a, b, c, d. With Eq. (59.34) the three distinct polarization dot products are immediate.

Lemma 59.8 (Polarization contractions)**.** *For gluons $i \neq j$ with reference momenta q_i, q_j,*

$$\varepsilon_{+}(p_i; q_i)\cdot\varepsilon_{+}(p_j; q_j) = \frac{\langle q_i\, q_j\rangle\,[j\, i]}{\langle q_i\, i\rangle\,\langle q_j\, j\rangle}, \tag{59.35}$$

$$\varepsilon_{-}(p_i; q_i)\cdot\varepsilon_{-}(p_j; q_j) = \frac{\langle i\, j\rangle\,[q_j\, q_i]}{[i\, q_i]\,[j\, q_j]}, \tag{59.36}$$

$$\varepsilon_{+}(p_i; q_i)\cdot\varepsilon_{-}(p_j; q_j) = \frac{\langle q_i\, j\rangle\,[q_j\, i]}{\langle q_i\, i\rangle\,[j\, q_j]}. \tag{59.37}$$

Proof. For Eq. (59.35), insert Eq. (59.32) and apply Eq. (59.34) with $(a, b, c, d) = (q_i, i, q_j, j)$:

$$\varepsilon_{+}(p_i; q_i)\cdot\varepsilon_{+}(p_j; q_j) = \frac{\langle q_i|\gamma^{\mu}|i]\,\langle q_j|\gamma_{\mu}|j]}{2\,\langle q_i\, i\rangle\,\langle q_j\, j\rangle} = \frac{2\,\langle q_i\, q_j\rangle\,[j\, i]}{2\,\langle q_i\, i\rangle\,\langle q_j\, j\rangle}, \tag{59.38}$$

which is Eq. (59.35). Equation Eq. (59.36) is the parity conjugate $\langle\cdot\,\cdot\rangle \leftrightarrow [\cdot\,\cdot]$ of the same step applied to the two ε_- vectors, with $(a,b,c,d) = (i,q_i,j,q_j)$. For the mixed product Eq. (59.37), take $(a,b,c,d) = (q_i,i,j,q_j)$:

$$\varepsilon_+(p_i;q_i)\cdot\varepsilon_-(p_j;q_j) = \frac{\langle q_i|\gamma^\mu|i]\,\langle j|\gamma_\mu|q_j]}{2\,\langle q_i\,i\rangle\,[j\,q_j]} = \frac{2\,\langle q_i\,j\rangle\,[q_j\,i]}{2\,\langle q_i\,i\rangle\,[j\,q_j]}, \tag{59.39}$$

giving Eq. (59.37). □

Read off the structure that drives everything. A pair of like-helicity polarizations vanishes when their references agree: Eqs. (59.35) and (59.36) are proportional to $\langle q_i\,q_j\rangle$ and $[q_j\,q_i]$ respectively, both of which vanish for $q_i = q_j$. A mixed pair vanishes when one reference is aligned with the *other* particle's momentum: Eq. (59.37) is proportional to $\langle q_i\,j\rangle$ (kill it with $q_i = p_j$) and to $[q_j\,i]$ (kill it with $q_j = p_i$). These three facts are the entire toolkit.

A reference momentum is a gauge choice, not a kinematic variable

The freedom in q_i is the residual gauge freedom of the external gluon, and gauge invariance of A_n guarantees the final number is independent of it. Two rules keep the bookkeeping honest. First, each leg's q_i must be the *same* in every diagram of the sum — choosing different references for the same external gluon in different diagrams adds up gauge-variant pieces that do not assemble into a physical amplitude. Second, the chosen q_i must avoid the collinear locus $\langle q_i\,i\rangle = 0$ (or $[i\,q_i] = 0$) where Eq. (59.32) is singular; setting q_i equal to another external momentum p_k is legal precisely when p_i and p_k are not collinear, which is the generic situation.

59.2.3 The anatomy of a color-ordered tree

Before exploiting Theorem 59.8 we need one structural fact about how polarizations and momenta can possibly be contracted in a tree diagram. It is a counting argument, and it is exact.

Recall the color-ordered Feynman rules Eqs. (59.12) to (59.14). In Feynman gauge the propagator numerator is a metric, $-i\eta^{\mu\nu}/p^2$. The cubic vertex is the sum

$$V_3^{\mu_1\mu_2\mu_3}(p_1,p_2,p_3) = \frac{i}{\sqrt{2}}\Big(\eta^{\mu_1\mu_2}(p_1-p_2)^{\mu_3} + \eta^{\mu_2\mu_3}(p_2-p_3)^{\mu_1} + \eta^{\mu_3\mu_1}(p_3-p_1)^{\mu_2}\Big), \tag{59.40}$$

each term a product of *one metric and one momentum*; the quartic vertex,

$$V_4^{\mu_1\mu_2\mu_3\mu_4} = \frac{i}{2}\big(2\,\eta^{\mu_1\mu_3}\eta^{\mu_2\mu_4} - \eta^{\mu_1\mu_2}\eta^{\mu_3\mu_4} - \eta^{\mu_1\mu_4}\eta^{\mu_2\mu_3}\big), \tag{59.41}$$

is a sum of products of *two metrics and no momentum*. Three structural facts follow:

1. the only Lorentz tensors in any diagram's numerator are metrics (from propagators and vertices) and momenta (one per cubic vertex);

2. each external leg contributes exactly one polarization vector $\varepsilon_i^{\mu_i}$, contracted into the diagram;

3. a maximal chain of metrics $\eta^{\mu\nu}\eta_{\nu\rho}\cdots$ collapses to a single metric, so after collapsing, the numerator is a product of scalar factors, each of which is one of $\varepsilon_i \cdot \varepsilon_j$, $\varepsilon_i \cdot p$, or $p \cdot p$.

Let a be the number of $\varepsilon \cdot p$ factors and b the number of $\varepsilon \cdot \varepsilon$ factors in a given term. Each ε_i is used exactly once and ends up in exactly one such factor, so counting the n polarizations two ways,

$$a + 2b = n. \tag{59.42}$$

Momenta in the numerator arise *only* from cubic vertices — one apiece — so the total number of momentum factors available to be dotted into anything is V_3, the number of cubic vertices. Every $\varepsilon \cdot p$ factor consumes one of them (the others may go into $p \cdot p$), giving the inequality

$$a \le V_3. \tag{59.43}$$

Finally, the tree topology fixes V_3. A tree with n external legs, I internal lines and $V = V_3 + V_4$ vertices has $I = V - 1$; counting line endpoints, $3V_3 + 4V_4 = n + 2I = n + 2(V - 1)$, and subtracting $2V$ from both sides,

$$V_3 + 2V_4 = n - 2 \quad \Longrightarrow \quad V_3 \le n - 2, \tag{59.44}$$

with equality for a purely trivalent tree. Combining Eqs. (59.43) and (59.44) bounds the number of polarizations that can be saturated by a momentum:

$$\boxed{a \le n - 2\,.} \tag{59.45}$$

Together with the saturation identity Eq. (59.42), this is the engine. It says that across *any* tree diagram, at most $n - 2$ of the n external polarizations can be soaked up as $\varepsilon \cdot p$; the remaining ones have no choice but to pair among themselves, forcing at least

$$b = \tfrac{1}{2}(n - a) \ge \tfrac{1}{2}(n - (n - 2)) = 1 \tag{59.46}$$

factor of $\varepsilon \cdot \varepsilon$ in every term. *Every nonzero contribution to a gluon tree amplitude contains at least one polarization–polarization contraction.* The vanishing theorems are now corollaries: if we can arrange the references so that every $\varepsilon \cdot \varepsilon$ that could appear vanishes, then by Eq. (59.46) every term vanishes, and so does A_n.

59.2.4 All-plus and single-minus amplitudes vanish

Theorem 59.9 (Vanishing of the all-plus amplitude)**.** *At tree level the color-ordered n-gluon amplitude with all positive helicities vanishes for every $n \geq 3$,*

$$A_n(1^+, 2^+, \ldots, n^+) = 0. \tag{59.47}$$

Proof. Choose a single common reference momentum for all n legs, $q_1 = q_2 = \cdots = q_n \equiv q$, with q a fixed null momentum not collinear with any p_i (generic, hence available by Section 59.2.2). All external polarizations are of positive helicity, so the only polarization pairings that can occur are of $++$ type, and by Eq. (59.35) each is proportional to $\langle q_i\, q_j\rangle = \langle q\, q\rangle = 0$. Thus every possible $\varepsilon \cdot \varepsilon$ factor vanishes. But Eq. (59.46) guarantees that each term of each diagram contains at least one such factor. Hence every term vanishes, and the sum Eq. (59.47) is identically zero. □

Theorem 59.10 (Vanishing of the single-minus amplitude)**.** *At tree level the color-ordered n-gluon amplitude with exactly one negative helicity vanishes for every $n \geq 3$,*

$$A_n(1^-, 2^+, 3^+, \ldots, n^+) = 0, \tag{59.48}$$

and likewise for the negative leg in any position by cyclicity.

Proof. Let leg 1 carry the lone negative helicity. Choose the references of all the positive-helicity legs equal to the momentum of the negative one,

$$q_i = p_1 \quad (i = 2, \ldots, n), \tag{59.49}$$

and choose q_1 to be any other external momentum, say $q_1 = p_2$ (legal since p_1, p_2 are not collinear for generic kinematics). There are now two kinds of polarization pairing to inspect, since there is a single minus leg and no minus–minus pair is possible:

- *plus–plus* ($i, j \geq 2$): by Eq. (59.35), $\varepsilon_+(p_i; p_1)\cdot\varepsilon_+(p_j; p_1) \propto \langle q_i\, q_j\rangle = \langle 1\, 1\rangle = 0$;
- *plus–minus* ($i \geq 2$, leg 1): by Eq. (59.37), $\varepsilon_+(p_i; p_1) \cdot \varepsilon_-(p_1; q_1) \propto \langle q_i\, 1\rangle = \langle 1\, 1\rangle = 0$.

Every polarization dot product that can appear is zero. By Eq. (59.46) every term carries at least one of them, so every term vanishes and Eq. (59.48) holds. □

The two proofs are the same proof: spend the reference freedom to annihilate all $\varepsilon \cdot \varepsilon$, then invoke the counting bound Eq. (59.45) that forbids a term with none. The choice $q_i = p_1$ in Theorem 59.10 is the clever one — aligning every positive leg's reference with the unique negative momentum kills the plus–plus

contractions *and* the plus–minus contractions simultaneously, because both Eq. (59.35) and Eq. (59.37) carry a factor $\langle q_i\, 1\rangle$ that collapses to $\langle 1\, 1\rangle$. The same maneuver fails the instant a *second* negative leg appears, and that failure is exactly the door to the surviving sector, as we see in Section 59.2.6.

59.2.5 A worked four-gluon check

The theorems are slick enough to seem like sleight of hand. They are not: the cancellation is visible term by term in the smallest nontrivial case. We carry it out fully.

Worked Example 59.11 (The four-gluon all-plus amplitude, diagram by diagram)**.** Take the color order $(1,2,3,4)$ with all helicities positive and a common reference q for all four legs, so that by Eq. (59.35)

$$\varepsilon_i \cdot \varepsilon_j = 0 \quad \text{for all } i,j \in \{1,2,3,4\}. \tag{59.50}$$

The color-ordered amplitude $A_4(1^+,2^+,3^+,4^+)$ receives three contributions: the s-channel exchange with the cyclically adjacent pairs (12) and (34), the t-channel exchange with pairs (23) and (41), and the four-point contact term. We show each vanishes under Eq. (59.50).

Contact term. The quartic vertex Eq. (59.41) contracted with $\varepsilon_1\varepsilon_2\varepsilon_3\varepsilon_4$ is a sum of products of two $\varepsilon\cdot\varepsilon$ factors,

$$\tfrac{i}{2}\big(2(\varepsilon_1{\cdot}\varepsilon_3)(\varepsilon_2{\cdot}\varepsilon_4) - (\varepsilon_1{\cdot}\varepsilon_2)(\varepsilon_3{\cdot}\varepsilon_4) - (\varepsilon_1{\cdot}\varepsilon_4)(\varepsilon_2{\cdot}\varepsilon_3)\big) = 0 \tag{59.51}$$

by Eq. (59.50): it is pure $b=2$, the maximally bad case.

s-channel diagram. With internal momentum $P = p_1 + p_2$ flowing into the right vertex, the diagram is $V_3^{\mu_1\mu_2\nu}(p_1,p_2,-P)\,\frac{-i\,\eta_{\nu\rho}}{P^2}\,V_3^{\rho\mu_3\mu_4}(P,p_3,p_4)$ contracted with the four polarizations. Contract the left vertex Eq. (59.40) with $\varepsilon_{1\mu_1}\varepsilon_{2\mu_2}$:

$$\tfrac{i}{\sqrt{2}}\Big[\underbrace{(\varepsilon_1{\cdot}\varepsilon_2)(p_1-p_2)^\nu}_{=0} + \varepsilon_2^\nu\,(\varepsilon_1{\cdot}(p_2+P)) + \varepsilon_1^\nu\,(\varepsilon_2{\cdot}(-P-p_1))\Big]. \tag{59.52}$$

The first term dies by Eq. (59.50); the surviving piece is a vector with free index ν that is a linear combination of ε_1^ν and ε_2^ν, multiplied by scalar $\varepsilon\cdot p$ factors. The propagator passes the index $\nu \to \rho$ unchanged. The right vertex contracted with $\varepsilon_3\varepsilon_4$ gives, by the identical computation,

$$\tfrac{i}{\sqrt{2}}\Big[\underbrace{(\varepsilon_3{\cdot}\varepsilon_4)(p_3-p_4)^\rho}_{=0} + \varepsilon_4^\rho\,(\varepsilon_3{\cdot}(p_4-P)) + \varepsilon_3^\rho\,(\varepsilon_4{\cdot}(P-p_3))\Big], \tag{59.53}$$

again a linear combination of ε_3^ρ and ε_4^ρ. Contracting Eq. (59.52) with Eq. (59.53) through $\eta_{\nu\rho}$ produces only the four cross terms $\varepsilon_1{\cdot}\varepsilon_3$, $\varepsilon_1{\cdot}\varepsilon_4$, $\varepsilon_2{\cdot}\varepsilon_3$, $\varepsilon_2{\cdot}\varepsilon_4$, every one of which vanishes by Eq. (59.50). The s-channel diagram is zero.

t-channel diagram. Identical structure with $(12)(34) \to (23)(41)$; the contracted index again lands on a combination of two adjacent polarizations, and the cross contraction is again an $\varepsilon \cdot \varepsilon$. Zero.

Summing, $A_4(1^+, 2^+, 3^+, 4^+) = 0$. Note exactly how the counting bound Eq. (59.45) materialized: each trivalent diagram has $V_3 = 2$ cubic vertices, hence at most $a = 2$ momentum-saturated polarizations, leaving $b \geq 1$ — and the one forced $\varepsilon \cdot \varepsilon$ is precisely the cross contraction across the propagator that we just watched vanish.

59.2.6 Little-group weights and the MHV classification

We have proved that the $k = 0$ (all-plus), $k = n$ (all-minus, by parity), $k = 1$ and $k = n-1$ helicity sectors vanish at tree level, where k counts the negative-helicity legs. The counting argument tells us this is sharp. A nonzero term requires $b \geq 1$ by Eq. (59.46), i.e. at least one surviving $\varepsilon \cdot \varepsilon$. Inspect Theorem 59.8: a like-helicity pair can always be killed by a common reference (all $++$ share one q, all $--$ share another), and with at most one minus leg the only mixed pairings are $+-$, which a single alignment $q_+ = p_-$ kills as well. It is only when *two* negative-helicity legs are present that the reference choices needed to annihilate $++$, $+-$, and $--$ simultaneously become mutually inconsistent: forcing all $++$ and $+-$ to vanish by aligning the positive references with a negative leg leaves a $--$ contraction Eq. (59.36) that no remaining freedom can remove without re-creating a $+-$ contraction. Two negative helicities is therefore the smallest count at which a single $\varepsilon \cdot \varepsilon$ can survive — the minimum compatible with $b = 1$, $a = n - 2 = V_3$ in Eqs. (59.42) and (59.45). This is the sharp lower edge of the nonvanishing landscape.

The same conclusion is visible from little-group scaling without any diagram. Under the little-group rescaling $(\lambda_i, \tilde{\lambda}_i) \to (t_i \lambda_i, t_i^{-1} \tilde{\lambda}_i)$ of Eq. (58.5), a helicity-h_i leg forces the amplitude to scale as

$$A_n(\dots, \{t_i \lambda_i, t_i^{-1} \tilde{\lambda}_i\}, \dots) = \Big(\prod_{i=1}^{n} t_i^{-2h_i} \Big) A_n, \tag{59.54}$$

so each positive-helicity leg ($h_i = +1$) demands weight t_i^{-2} and each negative-helicity leg ($h_i = -1$) weight t_i^{+2}. A rational function carrying weight t_i^{-2} in *every* variable would have to be built almost entirely out of square brackets (which scale as t^{-1} in each label) with too few angle brackets to supply the physical poles that locality requires on multiparticle channels; the all-plus and single-minus weights are simply too lopsided to be realized by a local, pole-correct function other than zero. The first balanced assignment that locality can accommodate has two units of positive weight to spare — two negative-helicity legs. The little-group bookkeeping and the polarization-counting argument are two readings of the same fact.

Definition 59.12 (MHV classification)**.** Let A_n be a color-ordered n-gluon tree amplitude with exactly k negative-helicity legs and $n-k$ positive-helicity legs. Define the *MHV degree* of the configuration to be k. Then:

- $k=0$ and $k=1$ (and by parity $k=n$, $k=n-1$): the amplitude vanishes identically (Theorems 59.9 and 59.10);
- $k=2$: the *maximally-helicity-violating* (MHV) amplitude, the lowest nonvanishing sector;
- $k=n-2$: the *anti-MHV* ($\overline{\text{MHV}}$) amplitude;
- general k: the N^{k-2}MHV amplitude (so MHV is N^0MHV and $\overline{\text{MHV}}$ is N^{n-4}MHV).

Two remarks fix the picture. First, parity — the operation $\langle i\,j\rangle \leftrightarrow [i\,j]$ that flips every helicity — maps the MHV sector ($k=2$) to the anti-MHV sector ($k=n-2$); the two are the same physics read from the conjugate spinors, and at the lowest multiplicities they can coincide. For $n=3$, MHV and $\overline{\text{MHV}}$ are the two complex-momentum three-point amplitudes fixed by little-group scaling in Chapter 58 (Eq. (58.98)), $A_3(1^-,2^-,3^+)=\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$ and its square-bracket conjugate. For $n=4$, the two-minus amplitude is both MHV ($k=2$) and anti-MHV ($k=n-2=2$); the classification first becomes nondegenerate at $n=5$.

Why "maximally helicity violating"

The name is a relic of the all-outgoing crossing convention. Read A_n as a $2\to(n-2)$ process by crossing two legs to the initial state, which flips their helicities. A configuration with all final-state gluons of one helicity would conserve a naive notion of total helicity; the experimentally and theoretically relevant nonzero amplitudes maximally *violate* that naive conservation. In the all-outgoing language this corresponds to the two-negative-helicity configuration: the most helicity violation a nonvanishing amplitude can carry, hence "maximally helicity violating." The vanishing of the all-plus and single-minus cases is the precise statement that less violation than this is too little to support a nonzero amplitude.

We have, by spending nothing but the gauge freedom hidden in the polarization vectors, collapsed the entire helicity landscape of gluon trees onto a graded ladder whose bottom two rungs are empty: the all-plus and single-minus amplitudes are identically zero, and the lowest sector that even *can* be nonzero is the MHV configuration with exactly two negative-helicity gluons. What that surviving amplitude actually equals — the remarkable one-line Parke–Taylor formula $A_n^{\text{MHV}} \propto \langle i\,j\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n\,1\rangle)$, verified against an honest Feynman

computation and subjected to every consistency check — is the business of Eq. (59.55) in the next section.

59.3 The Parke–Taylor Formula

The previous section organized the helicity landscape of gluon amplitudes by exploiting the freedom in the reference momenta of the polarization vectors: amplitudes with all helicities alike, or with exactly one differing, vanish at tree level, and the first nonzero color-ordered amplitudes are the *maximally helicity-violating* (MHV) ones, carrying exactly two negative helicities among n legs (and their parity conjugates, $\overline{\text{MHV}}$, with two positive). That classification told us *which* amplitudes survive; it did not tell us what they are. The remarkable fact, discovered by Parke and Taylor in 1986 after a numerical and then analytic assault on multi-gluon scattering, is that the surviving MHV amplitudes are not merely nonzero but are given, for *every* n, by a single one-line holomorphic expression. The Lagrangian hides this; the spinor-helicity variables of Chapter 58 reveal it. This section states the formula, subjects it to every consistency check the formalism affords, and earns it through an honest Feynman computation at four points — carrying Yang–Mills theory to the milestone promised in the chapter introduction: Parke–Taylor computed the hard way at four points and posited at all n. The status we reach is precise: a checked-but-unproved all-n conjecture, whose proof must wait for the recursion built in the sections that follow.

Throughout we use the spinor-helicity conventions fixed in Chapter 58 (Theorem 58.2): for massless momenta $p_i = \lambda_i\tilde{\lambda}_i$ the angle and square brackets $\langle i\,j\rangle$, $[i\,j]$ are antisymmetric, $\langle i\,j\rangle[j\,i] = 2p_i\cdot p_j = s_{ij}$ (Eqs. (58.20) and (58.24)), and momentum conservation reads $\sum_i \lambda_i\tilde{\lambda}_i = 0$, i.e. $\sum_i |i\rangle\,[i\,] = 0$ as an operator on the spinor indices. We write $\langle i|\gamma^\mu|j] = \langle i|\,\gamma^\mu|j]$ for the mixed sandwich, with $\langle i|P|j] = \sum_k \langle i\,k\rangle[k\,j]$ for $P = \sum_k p_k$, and use the Fierz rearrangement $\langle a|\gamma^\mu|b]\langle c|\gamma_\mu|d] = 2\langle a\,c\rangle[d\,b]$ from Chapter C. Color-ordered amplitudes $A_n[\dots]$ are defined as in Theorem 59.3 through the trace decomposition, and we use the color-ordered Feynman rules of Eq. (59.12); we restate the two vertices we need at the point of use. The metric is mostly-minus and natural units are in force, as everywhere in the book.

59.3.1 Statement of the formula

We label the two negative-helicity gluons i and j and take all other legs positive. The amplitude is color-ordered with the cyclic ordering $(1, 2, \dots, n)$, all momenta outgoing.

Conjecture 59.13 (Parke–Taylor MHV amplitude)**.** *The n-gluon tree-level color-ordered MHV amplitude, with negative-helicity legs i and j and all other*

legs positive, is

$$A_n[1^+,\dots,i^-,\dots,j^-,\dots,n^+] \;=\; i\,g^{n-2}\,\frac{\langle i\,j\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n-1\,n\rangle\langle n\,1\rangle}. \tag{59.55}$$

The full amplitude is reassembled by the color decomposition Theorem 59.3, $\mathcal{A}_n = g^{n-2}\sum_{\sigma\in S_n/\mathbb{Z}_n}\mathrm{tr}\big(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n)}}\big)\,A_n[\sigma]$.

The denominator is the *Parke–Taylor chain*: the cyclic product of nearest-neighbor angle brackets in the color order, $\prod_{k=1}^{n}\langle k\,k+1\rangle$ with $n+1\equiv 1$. It is the same for every MHV amplitude of the given color ordering; only the numerator $\langle i\,j\rangle^4$ records which two legs carry negative helicity. The dependence is purely *holomorphic* — not a single square bracket appears. This is the analytic fingerprint of the MHV sector, and it is precisely the structure that the brute-force expansion in Feynman diagrams, with its thousands of individually gauge-dependent terms each built from both λ and $\tilde\lambda$, conceals until the very end.

The parity conjugate is obtained by exchanging the two chiralities, $\langle\,\rangle \leftrightarrow [\,]$, which exchanges positive and negative helicities. The $\overline{\mathrm{MHV}}$ amplitude, with exactly two positive-helicity legs i,j and all others negative, is

$$A_n[1^-,\dots,i^+,\dots,j^+,\dots,n^-] \;=\; (-1)^n\,i\,g^{n-2}\,\frac{[i\,j]^4}{[1\,2][2\,3]\cdots[n-1\,n][n\,1]}. \tag{59.56}$$

The sign $(-1)^n$ tracks the convention $\langle i\,j\rangle[j\,i] = s_{ij}$ under conjugation: replacing each $\langle k\,k+1\rangle$ in the chain by $[k\,k+1]$ flips a sign per factor through $[k\,k+1] = -[k+1\,k]$, and the bookkeeping leaves the overall $(-1)^n$ displayed. For the lowest multiplicities the two formulas coincide with the three-point amplitudes fixed by little-group scaling alone in Chapter 58: setting $n=3$ in Eq. (59.55) with negative legs $1,2$ gives

$$A_3[1^-,2^-,3^+] = i\,g\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,1\rangle} = i\,g\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,1\rangle}, \tag{59.57}$$

which is exactly the holomorphic three-point amplitude Eq. (58.98), and $n=3$ in Eq. (59.56) reproduces the antiholomorphic $A_3[1^+,2^+,3^-] = -i\,g\,[1\,2]^3/[2\,3][3\,1]$. Parke–Taylor is the unique holomorphic interpolation of these symmetry-fixed seeds to all n — a statement we will make exact once the recursion shows the seeds determine everything.

59.3.2 Consistency checks the formula must pass

Before any computation, the candidate Eq. (59.55) must respect the symmetries and dimensional analysis that any gluon amplitude obeys. Each check is a genuine constraint; passing all of them is strong, if not conclusive, evidence.

Little-group weight. Under the little-group rescaling of leg k (Eq. (58.5)), $\lambda_k \to t_k \lambda_k$, $\tilde{\lambda}_k \to t_k^{-1} \tilde{\lambda}_k$, a leg of helicity h_k forces the amplitude to scale as $A_n \to (\prod_k t_k^{-2h_k}) A_n$. We verify Eq. (59.55) carries exactly this weight on every leg. The angle bracket $\langle k \cdot \rangle$ scales as t_k; the square brackets are absent, so a leg's weight comes only from counting its angle brackets. Take first a *positive*-helicity leg $k \neq i, j$. It appears nowhere in the numerator and in exactly two chain factors of the denominator, $\langle k-1\, k\rangle$ and $\langle k\, k+1\rangle$, so the formula scales as t_k^{-2}. With $h_k = +1$ this is $t_k^{-2h_k}$, as required. Now a *negative*-helicity leg, say i: it appears four times in the numerator $\langle i\, j\rangle^4$ (weight t_i^{+4}) and twice in the denominator chain (weight t_i^{-2}), for a net t_i^{+2}. With $h_i = -1$ this is again $t_i^{-2h_i}$. Every leg is correct, and — crucially — the numerator power must be exactly four for the negative legs to come out right: had it been $\langle i\, j\rangle^3$ or $\langle i\, j\rangle^5$ the weight would fail. Little-group scaling fixes the numerator to be $\langle i\, j\rangle^4$, the unique monomial in the two negative-helicity spinors that does the job while keeping the expression holomorphic and of the correct overall dimension.

Mass dimension. An n-point amplitude in four dimensions has mass dimension $[A_n] = 4 - n$ (the S-matrix element carries dimension $4 - n$ once the momentum-conserving delta function and the canonical normalization of external states are stripped, exactly as for the scalar amplitudes of Chapter 14). Each bracket has mass dimension one. The Parke–Taylor numerator contributes $+4$ and the n-fold denominator chain contributes $-n$, giving net dimension $4 - n$. The check is automatic and, again, depends on the numerator power being four: it balances the four "extra" brackets that the closed cyclic chain carries beyond what dimension alone would demand.

Coupling power. A connected tree amplitude with only cubic and quartic vertices and n external legs has $V_3 + 2V_4 = n - 2$ powers of g, since each cubic vertex carries one power of g and each quartic two, and the tree topology fixes the total (the number of internal lines is $V_3 + V_4 - 1$ and $3V_3 + 4V_4 = n + 2(V_3 + V_4 - 1)$, which solves to $V_3 + 2V_4 = n - 2$). The prefactor g^{n-2} in Eq. (59.55) is exactly this, and is dimensionless in four dimensions, consistent with the mass-dimension count. The explicit factor of i is the standard normalization of a tree amplitude in our conventions, matching the three-point seed Eq. (58.98).

Cyclic symmetry and reflection. The denominator chain is manifestly invariant under the cyclic shift $k \to k + 1$, as a color-ordered amplitude must be (Theorem 59.3). Under the order reversal $(1, 2, \ldots, n) \to (n, n-1, \ldots, 1)$ each chain factor reverses, $\langle k\, k+1\rangle \to \langle k+1\, k\rangle = -\langle k\, k+1\rangle$, producing $(-1)^n$; the numerator $\langle i\, j\rangle^4$ is reflection-even. This reproduces the known reflection property of color-ordered amplitudes, $A_n[n, \ldots, 1] = (-1)^n A_n[1, \ldots, n]$, with no extra input.

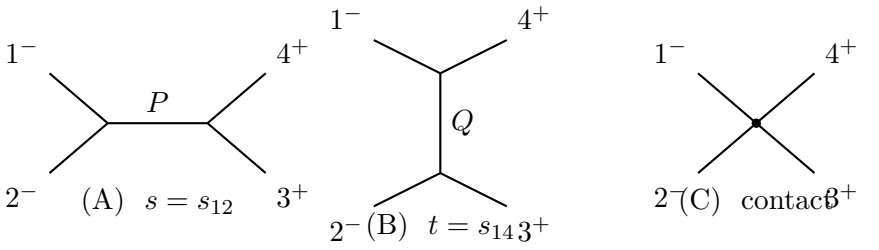


Figure 59.2: The three color-ordered Feynman diagrams for $A_4[1^-, 2^-, 3^+, 4^+]$ in the cyclic order $(1, 2, 3, 4)$: the planar s-channel exchange (A), the planar t-channel exchange (B), and the four-point contact term (C). The non-planar $u = s_{13}$ exchange does not contribute to this color ordering. With the reference-momentum choice Eq. (59.64) diagrams (B) and (C) vanish identically and the whole amplitude comes from (A).

These checks pin down the *form* of the answer almost completely: a holomorphic, cyclic, dimension-$(4 - n)$ object with the correct little-group weight and the right power of g is forced to be a constant times Eq. (59.55). What they cannot fix is that constant, nor that the residues on factorization channels are correct. For that we must compute.

59.3.3 The four-gluon amplitude the hard way

We now earn Eq. (59.55) at $n = 4$ by an honest color-ordered Feynman computation of $A_4[1^-, 2^-, 3^+, 4^+]$, reducing every Lorentz contraction to spinor brackets and matching the result to Parke–Taylor. This is the computation Parke and Taylor first did, and it is the payoff of the reference-momentum technology of the previous section: a judicious choice of polarization reference momenta collapses three diagrams to one and a handful of brackets.

We need two color-ordered Feynman rules from Eq. (59.12), both with all momenta outgoing. The cubic vertex for legs (a, b, c) in cyclic order, with Lorentz indices (α, β, γ), is

$$V_3^{\alpha\beta\gamma}(p_a, p_b, p_c) = \frac{ig}{\sqrt{2}}\Big[\eta^{\alpha\beta}(p_a - p_b)^\gamma + \eta^{\beta\gamma}(p_b - p_c)^\alpha + \eta^{\gamma\alpha}(p_c - p_a)^\beta\Big], \tag{59.58}$$

and the quartic vertex for legs (a, b, c, d) in cyclic order is

$$V_4^{\alpha\beta\gamma\delta} = \frac{ig^2}{2}\Big[2\eta^{\alpha\gamma}\eta^{\beta\delta} - \eta^{\alpha\beta}\eta^{\gamma\delta} - \eta^{\alpha\delta}\eta^{\beta\gamma}\Big]. \tag{59.59}$$

The internal gluon propagator in Feynman gauge is $-i\eta_{\mu\nu}/P^2$; color-ordered amplitudes are gauge invariant, so the gauge choice is a convenience. The three planar diagrams are drawn in Fig. 59.2: the s-channel exchange (A) in the

(12)(34) channel, the t-channel exchange (B) in the (23)(41) channel, and the four-point contact (C). The non-planar (13)(24) exchange is absent from this color ordering.

Polarizations and the reference choice. We recall the polarization vectors from Chapter 58, written with explicit reference momenta q,

$$\epsilon_+^\mu(p;q) = \frac{\langle q|\gamma^\mu|p]}{\sqrt{2}\,\langle q\,p\rangle}, \qquad \epsilon_-^\mu(p;q) = \frac{\langle p|\gamma^\mu|q]}{\sqrt{2}\,[q\,p]}. \tag{59.60}$$

Their inner products follow from the Fierz identity. For two like-helicity legs,

$$\epsilon_a^+\cdot\epsilon_b^+ = \frac{\langle q_a\,q_b\rangle[b\,a]}{\langle q_a\,a\rangle\langle q_b\,b\rangle}, \qquad \epsilon_a^-\cdot\epsilon_b^- = \frac{\langle a\,b\rangle[q_b\,q_a]}{[q_a\,a][q_b\,b]}, \tag{59.61}$$

and for opposite helicities,

$$\epsilon_b^+\cdot\epsilon_a^- = \frac{\langle q_b\,a\rangle[q_a\,b]}{\langle q_b\,b\rangle[q_a\,a]}. \tag{59.62}$$

We also need the dot product of a polarization with a momentum, again from the Fierz identity and $\not{p}_c = |c\rangle\,[c\,] + |c]\,\langle c|$,

$$\epsilon_b^+\cdot p_c = \frac{\langle q_b\,c\rangle[c\,b]}{\sqrt{2}\,\langle q_b\,b\rangle}, \qquad \epsilon_a^-\cdot p_c = \frac{\langle a\,c\rangle[c\,q_a]}{\sqrt{2}\,[q_a\,a]}. \tag{59.63}$$

The freedom is to choose each q independently (the amplitude is invariant under a shift $q \to q'$, as established in the previous section). We make the choice

$$q_1 = q_2 = p_4, \qquad q_3 = q_4 = p_1, \tag{59.64}$$

i.e. both negative-helicity gluons take p_4 as reference, and both positive-helicity gluons take p_1. The like-helicity products Eq. (59.61) now vanish at once: $\epsilon_1^-\cdot\epsilon_2^- \propto [q_2\,q_1] = [4\,4] = 0$ and $\epsilon_3^+\cdot\epsilon_4^+ \propto \langle q_3\,q_4\rangle = \langle 1\,1\rangle = 0$. Among the mixed products Eq. (59.62), three of the four vanish:

$$\epsilon_3^+\cdot\epsilon_1^- = \epsilon_4^+\cdot\epsilon_1^- \propto \langle q_{3,4}\,1\rangle = \langle 1\,1\rangle = 0, \qquad \epsilon_4^+\cdot\epsilon_2^- \propto [q_2\,4] = [4\,4] = 0, \tag{59.65}$$

and only

$$\epsilon_3^+\cdot\epsilon_2^- = \frac{\langle 1\,2\rangle[4\,3]}{\langle 1\,3\rangle[4\,2]} \tag{59.66}$$

survives. Two momentum dot products will also vanish and matter below: $\epsilon_1^-\cdot p_4 \propto [4\,4] = 0$ and $\epsilon_4^+\cdot p_1 \propto \langle 1\,1\rangle = 0$, using Eq. (59.63).

The contact diagram vanishes. The quartic vertex Eq. (59.59), contracted with the four polarizations, gives

$$(\mathrm{C}) = \frac{ig^2}{2}\Big[2(\epsilon_1\cdot\epsilon_3)(\epsilon_2\cdot\epsilon_4) - (\epsilon_1\cdot\epsilon_2)(\epsilon_3\cdot\epsilon_4) - (\epsilon_1\cdot\epsilon_4)(\epsilon_2\cdot\epsilon_3)\Big]. \tag{59.67}$$

Every term contains a vanishing factor under Eq. (59.64): $\epsilon_1^-\cdot\epsilon_3^+ = 0$, then $\epsilon_1^-\cdot\epsilon_2^- = 0$, then $\epsilon_1^-\cdot\epsilon_4^+ = 0$. Hence (C) $= 0$.

The t-channel diagram vanishes. Diagram (B) joins a cubic vertex with legs $(2,3)$ to a cubic vertex with legs $(4,1)$ through a propagator carrying $Q = p_2 + p_3$, $t = s_{14} = s_{23} = Q^2$. Contracting Eq. (59.58) at the left vertex on legs $2,3$ and leaving the internal index free, using $\epsilon_i\cdot p_i = 0$ and $p_{\text{int}} = -(p_2+p_3)$,

$$V_L^\mu = \frac{ig}{\sqrt{2}}\Big[(\epsilon_2\cdot\epsilon_3)(p_2-p_3)^\mu + 2(\epsilon_2\cdot p_3)\,\epsilon_3^\mu - 2(\epsilon_3\cdot p_2)\,\epsilon_2^\mu\Big], \tag{59.68}$$

and likewise at the right vertex on legs $4,1$ with internal momentum $Q = -(p_1+p_4)$, and using $\epsilon_4^+\cdot\epsilon_1^- = 0$,

$$V_R^\nu = \frac{ig}{\sqrt{2}}\Big[-2(p_4\cdot\epsilon_1)\,\epsilon_4^\nu + 2(p_1\cdot\epsilon_4)\,\epsilon_1^\nu\Big]. \tag{59.69}$$

The diagram is (B) $= V_L^\mu(-i\eta_{\mu\nu}/t)\,V_R^\nu = (-i/t)\,V_L\cdot V_R$. In the contraction $V_L\cdot V_R$, the ϵ_3^μ and ϵ_2^μ terms of V_L die against V_R because $\epsilon_3^+\cdot\epsilon_4^+ = \epsilon_3^+\cdot\epsilon_1^- = 0$ and $\epsilon_2^-\cdot\epsilon_4^+ = \epsilon_2^-\cdot\epsilon_1^- = 0$, leaving only the $(\epsilon_2\cdot\epsilon_3)(p_2-p_3)^\mu$ term to contract with V_R. But that contraction needs $(p_2-p_3)\cdot\epsilon_4$ multiplying $D \equiv -2p_4\cdot\epsilon_1 = 0$, and $(p_2-p_3)\cdot\epsilon_1$ multiplying $E \equiv 2p_1\cdot\epsilon_4 = 0$ — both coefficients vanish by the two special momentum dot products noted above. Therefore $V_L\cdot V_R = 0$ and (B) $= 0$.

The s-channel diagram, evaluated. Only diagram (A) remains. It joins a cubic vertex on legs $(1,2)$ to a cubic vertex on legs $(3,4)$ through a propagator carrying $P = p_1+p_2$, $s = s_{12} = P^2$. With $\epsilon_1^-\cdot\epsilon_2^- = 0$ the left vertex collapses to

$$V_L^\mu = \frac{ig}{\sqrt{2}}\Big[2(\epsilon_1\cdot p_2)\,\epsilon_2^\mu - 2(\epsilon_2\cdot p_1)\,\epsilon_1^\mu\Big], \tag{59.70}$$

and with $\epsilon_3^+\cdot\epsilon_4^+ = 0$ the right vertex collapses to

$$V_R^\nu = \frac{ig}{\sqrt{2}}\Big[-2(p_3\cdot\epsilon_4)\,\epsilon_3^\nu + 2(p_4\cdot\epsilon_3)\,\epsilon_4^\nu\Big]. \tag{59.71}$$

Contracting through the propagator, (A) $= (-i/s)\,V_L\cdot V_R$, and in $V_L\cdot V_R$ only the term proportional to $\epsilon_2^-\cdot\epsilon_3^+$ survives (the others carry $\epsilon_2^-\cdot\epsilon_4^+$, $\epsilon_1^-\cdot\epsilon_3^+$, or $\epsilon_1^-\cdot\epsilon_4^+$, all zero):

$$V_L\cdot V_R = \Big(\tfrac{ig}{\sqrt{2}}\Big)^2\,[2(\epsilon_1\cdot p_2)]\,[-2(p_3\cdot\epsilon_4)](\epsilon_2\cdot\epsilon_3) = 2g^2(\epsilon_1\cdot p_2)(\epsilon_4\cdot p_3)(\epsilon_2\cdot\epsilon_3). \tag{59.72}$$

Hence

$$A_4 = (\mathrm{A}) = \frac{-i}{s} \cdot 2g^2 (\epsilon_1 \cdot p_2)(\epsilon_4 \cdot p_3)(\epsilon_2 \cdot \epsilon_3). \tag{59.73}$$

Now insert the spinor expressions. From Eq. (59.63) with the references Eq. (59.64),

$$\epsilon_1^- \cdot p_2 = \frac{\langle 1\,2\rangle [2\,4]}{\sqrt{2}\,[4\,1]}, \qquad \epsilon_4^+ \cdot p_3 = \frac{\langle 1\,3\rangle [3\,4]}{\sqrt{2}\,\langle 1\,4\rangle}, \tag{59.74}$$

and from Eq. (59.66), $\epsilon_2^- \cdot \epsilon_3^+ = \langle 1\,2\rangle [4\,3]/(\langle 1\,3\rangle [4\,2])$. Multiplying, the two factors of $1/\sqrt{2}$ combine with the $-2i\,g^2/s$ to give $-ig^2/s$, and one $\langle 1\,3\rangle$ cancels:

$$A_4 = \frac{-ig^2}{s} \cdot \frac{\langle 1\,2\rangle^2\,[2\,4]\,[3\,4]\,[4\,3]}{[4\,1]\,\langle 1\,4\rangle\,[4\,2]}. \tag{59.75}$$

Using $[2\,4] = -[4\,2]$ and $[4\,3] = -[3\,4]$, the square-bracket ratio simplifies to $[2\,4][3\,4][4\,3]/([4\,1][4\,2]) = -[3\,4]^2/[4\,1]$, so

$$A_4 = \frac{ig^2}{s} \cdot \frac{\langle 1\,2\rangle^2\,[3\,4]^2}{\langle 1\,4\rangle\,[4\,1]} = -\frac{ig^2\,\langle 1\,2\rangle^2\,[3\,4]^2}{s_{12}\,s_{14}}, \tag{59.76}$$

where we used $\langle 1\,4\rangle [4\,1] = s_{14}$ and $s = s_{12}$. It remains to turn the square brackets into angle brackets and recover the holomorphic Parke–Taylor form. Two relations from momentum conservation do this. Contracting $\sum_k |k\rangle\,[k\,] = 0$ between $\langle 3|$ and $|1]$ gives $\langle 3\,2\rangle [2\,1] + \langle 3\,4\rangle [4\,1] = 0$, hence

$$[2\,1] = -\frac{\langle 3\,4\rangle\,[4\,1]}{\langle 3\,2\rangle}, \qquad \text{so} \qquad s_{12} = \langle 1\,2\rangle [2\,1] = -\frac{\langle 1\,2\rangle\,\langle 3\,4\rangle\,[4\,1]}{\langle 3\,2\rangle}. \tag{59.77}$$

The on-shell equality of channel invariants, $s_{34} = s_{12}$ (since $p_3 + p_4 = -(p_1 + p_2)$), gives $[3\,4] = s_{12}/\langle 4\,3\rangle = -s_{12}/\langle 3\,4\rangle$, so $[3\,4]^2 = s_{12}^2/\langle 3\,4\rangle^2$. Substituting into Eq. (59.76),

$$A_4 = -\frac{ig^2\,\langle 1\,2\rangle^2}{s_{12}\,s_{14}} \cdot \frac{s_{12}^2}{\langle 3\,4\rangle^2} = -\frac{ig^2\,\langle 1\,2\rangle^2\,s_{12}}{s_{14}\,\langle 3\,4\rangle^2}. \tag{59.78}$$

With $s_{14} = \langle 1\,4\rangle [4\,1]$ and s_{12} from Eq. (59.77), the ratio s_{12}/s_{14} becomes purely holomorphic,

$$\frac{s_{12}}{s_{14}} = \frac{-\langle 1\,2\rangle\langle 3\,4\rangle [4\,1]/\langle 3\,2\rangle}{\langle 1\,4\rangle [4\,1]} = -\frac{\langle 1\,2\rangle\,\langle 3\,4\rangle}{\langle 3\,2\rangle\,\langle 1\,4\rangle}, \tag{59.79}$$

and therefore

$$A_4 = -\frac{ig^2\,\langle 1\,2\rangle^2}{\langle 3\,4\rangle^2} \cdot \left(-\frac{\langle 1\,2\rangle\,\langle 3\,4\rangle}{\langle 3\,2\rangle\,\langle 1\,4\rangle}\right) = \frac{ig^2\,\langle 1\,2\rangle^3}{\langle 3\,4\rangle\,\langle 3\,2\rangle\,\langle 1\,4\rangle}. \tag{59.80}$$

Finally $\langle 3\,2\rangle = -\langle 2\,3\rangle$ and $\langle 1\,4\rangle = -\langle 4\,1\rangle$, so the two signs cancel and

$$\boxed{A_4[1^-, 2^-, 3^+, 4^+] = \frac{ig^2\,\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,1\rangle} = ig^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}} \tag{59.81}$$

which is exactly Eq. (59.55) at $n = 4$ with $i = 1$, $j = 2$. The thousands of terms that a naive covariant computation would generate — here tamed to three diagrams and then to one by the reference choice — collapse to a single ratio of angle brackets. This is the concrete demonstration, in Yang–Mills theory, that the Lagrangian's complexity is a feature of the variables, not of the physics.

Worked Example 59.14 (Four-gluon amplitude from Parke–Taylor in invariants)**.** As a final cross-check we re-express Eq. (59.81) in Mandelstam invariants and confirm it has the correct dimensions and poles. The cleanest route is through the squared modulus, using $|\langle i\,j\rangle|^2 = |s_{ij}|$. Form

$$|A_4|^2 = g^4\,\frac{|\langle 1\,2\rangle|^8}{|\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle|^2} = g^4\,\frac{s_{12}^4}{s_{12}s_{23}s_{34}s_{41}} = g^4\,\frac{s_{12}^4}{s_{12}^2 s_{23}^2} = g^4\,\frac{s_{12}^2}{s_{23}^2} = g^4\,\frac{s^2}{t^2}, \tag{59.82}$$

where we used $|\langle i\,j\rangle|^2 = |s_{ij}|$, the four-point identities $s_{34} = s_{12}$, $s_{41} = s_{23}$, and $s = s_{12}$, $t = s_{23}$. The amplitude carries the expected $1/s$ and $1/t$ single poles of Eq. (59.76) (the $1/t$ pole is hidden in Eq. (59.81) but reappears upon clearing the holomorphic chain), each the residue of a physical gluon going on shell in the corresponding channel, and $|A_4|^2$ has mass dimension zero, consistent with $[A_4] = 0$ in $d = 4$. The full color-summed four-gluon amplitude, assembled from the six orderings via Theorem 59.3, is the carried Yang–Mills theory four-gluon result; with Eq. (59.81) as the color-ordered building block, the color algebra is the only remaining ingredient.

59.3.4 The soft limit and the Weinberg soft factor

A genuine amplitude must factorize correctly when a gluon becomes soft. Parke–Taylor does, and the factorization reproduces the universal soft theorem of Chapter 18 (Weinberg) in its sharpest, spinor-helicity form. Take the positive-helicity gluon n soft, $p_n \to 0$, with its two cyclic neighbors $n-1$ and 1. In Eq. (59.55) the only chain factors that touch leg n are $\langle n-1\,n\rangle$ and $\langle n\,1\rangle$. Insert a factor of unity, $1 = \langle n-1\,1\rangle/\langle n-1\,1\rangle$, to extract them:

$$A_n = \underbrace{\frac{\langle n-1\,1\rangle}{\langle n-1\,n\rangle\,\langle n\,1\rangle}}_{S^{(0)}(n-1,n^+,1)} \times \underbrace{i\,g^{\,n-2}\,\frac{\langle i\,j\rangle^4}{\langle 1\,2\rangle\cdots\langle n-2\,n-1\rangle\langle n-1\,1\rangle}}_{g\,A_{n-1}[1,\ldots,n-1]}. \tag{59.83}$$

The second factor is, up to the one power of g that the lost vertex carries, exactly the $(n-1)$-gluon Parke–Taylor amplitude with leg n removed and the chain closed by $\langle n-1\,1\rangle$ — and it is unchanged in the limit because none of its brackets involve n. The first factor is the *leading soft factor*

$$S^{(0)}(a,s^+,b) = \frac{\langle a\,b\rangle}{\langle a\,s\rangle\,\langle s\,b\rangle}, \tag{59.84}$$

for soft gluon s of positive helicity between hard neighbors a, b. As $p_s \to 0$ the spinor $\lambda_s \to 0$, so $\langle a\, s\rangle, \langle s\, b\rangle \sim \lambda_s^{1/2} \sim p_s^{1/2}$ while $\langle a\, b\rangle$ stays finite, and $S^{(0)} \sim 1/p_s$: the amplitude diverges as the inverse soft energy, the universal infrared behavior of gauge theory. The parity-conjugate soft factor for a negative-helicity soft gluon is $S^{(0)}(a, s^-, b) = [a\, b]/([a\, s][s\, b])$.

That Eq. (59.84) is the Weinberg soft factor is seen by undoing the spinors. The covariant leading soft factor of Chapter 18 is $\epsilon_s{\cdot}(p_a/(p_a{\cdot}p_s) - p_b/(p_b{\cdot}p_s))$. Take $\epsilon_s^+(p_s; q)$ with reference q. Using Eq. (59.63), $\epsilon_s^+{\cdot}p_a/(p_s{\cdot}p_a) = \langle q\, a\rangle[a\, s]/(\sqrt{2}\,\langle q\, s\rangle \cdot \frac{1}{2}\langle s\, a\rangle[a\, s]) = \sqrt{2}\,\langle q\, a\rangle/(\langle q\, s\rangle\langle s\, a\rangle)$, and the same with $a \to b$. The difference is

$$\sqrt{2}\left(\frac{\langle q\, a\rangle}{\langle q\, s\rangle\langle s\, a\rangle} - \frac{\langle q\, b\rangle}{\langle q\, s\rangle\langle s\, b\rangle}\right) = \frac{\sqrt{2}}{\langle q\, s\rangle}{\cdot}\frac{\langle q\, a\rangle\langle s\, b\rangle - \langle q\, b\rangle\langle s\, a\rangle}{\langle s\, a\rangle\langle s\, b\rangle} = \frac{\sqrt{2}}{\langle q\, s\rangle}{\cdot}\frac{\langle q\, s\rangle\langle a\, b\rangle}{\langle s\, a\rangle\langle s\, b\rangle} = \sqrt{2}\,\frac{\langle a\, b\rangle}{\langle s\, a\rangle\langle s\, b\rangle}, \tag{59.85}$$

where the middle step is the Schouten identity $\langle q\, a\rangle\langle s\, b\rangle - \langle q\, b\rangle\langle s\, a\rangle = \langle q\, s\rangle\langle a\, b\rangle$. The reference q drops out, as it must, and we recover $S^{(0)}$ up to the $\sqrt{2}$ that the color-ordered normalization absorbs. The soft theorem and Parke–Taylor agree: each is the other read in different variables.

59.3.5 The collinear limit and the gluon splitting amplitudes

The second physical factorization is the collinear limit: two adjacent gluons a, b become parallel, $p_a \to z\,P$ and $p_b \to (1-z)\,P$ with $P = p_a + p_b$ on shell ($P^2 = 2p_a{\cdot}\,p_b = \langle a\, b\rangle[b\, a] \to 0$), so $\langle a\, b\rangle \to 0$. There an n-point amplitude must degenerate into an $(n-1)$-point amplitude times a universal *splitting amplitude*,

$$A_n(\ldots, a^{h_a}, b^{h_b}, \ldots) \xrightarrow{a\|b} \sum_{h=\pm} \mathrm{Split}_{-h}(a^{h_a}, b^{h_b})\, A_{n-1}(\ldots, P^h, \ldots), \tag{59.86}$$

where P^h is the merged leg of helicity h and the splitting amplitude carries the collinear singularity $\sim 1/\langle a\, b\rangle$ or $1/[a\, b]$. The holomorphic collinear kinematics — the regime in which the angle bracket vanishes faster than the square bracket — has $\lambda_a \to \sqrt{z}\,\lambda_P$, $\lambda_b \to \sqrt{1-z}\,\lambda_P$, so that for any spectator x,

$$\langle a\, x\rangle \to \sqrt{z}\,\langle P\, x\rangle, \qquad \langle b\, x\rangle \to \sqrt{1-z}\,\langle P\, x\rangle. \tag{59.87}$$

We extract the splitting amplitudes directly from Parke–Taylor, considering the cases the MHV formula can reach.

Take first *two adjacent positive-helicity legs* a^+, b^+ becoming collinear in an MHV amplitude whose two negative legs lie elsewhere. Write x and y for the cyclic neighbors of the pair, so the chain contains $\cdots\langle x\, a\rangle\langle a\, b\rangle\langle b\, y\rangle\cdots$. Under Eq. (59.87) the numerator $\langle i\, j\rangle^4$ is untouched, $\langle x\, a\rangle \to \sqrt{z}\,\langle x\, P\rangle$, $\langle b\, y\rangle \to \sqrt{1-z}\,\langle P\, y\rangle$, and $\langle a\, b\rangle \to 0$ supplies the singularity. Hence

$$A_n \xrightarrow{a\|b} \frac{1}{\sqrt{z(1-z)}\,\langle a\, b\rangle}{\cdot}\, i\, g^{\,n-3}\,\frac{\langle i\, j\rangle^4}{\cdots\langle x\, P\rangle\langle P\, y\rangle\cdots} = \mathrm{Split}_-(a^+, b^+)\, A_{n-1}(\ldots, P^+, \ldots), \tag{59.88}$$

with the merged leg P positive-helicity (the negatives are preserved) and

$$\mathrm{Split}_{-}(a^{+},b^{+}) = \frac{1}{\sqrt{z(1-z)}\,\langle a\,b\rangle}. \tag{59.89}$$

The subscript records the helicity $-h = -(+)$ that the splitting vertex assigns to the off-shell line entering A_{n-1}.

Take next *one negative and one positive leg* collinear, say a^{-},b^{+} adjacent, with the other negative leg j a spectator. Now a appears in the numerator: $\langle a\,j\rangle^{4} \to z^{2}\,\langle P\,j\rangle^{4}$ by Eq. (59.87). The denominator behaves as before, and

$$A_n \xrightarrow{a\|b} \frac{z^{2}}{\sqrt{z(1-z)}\,\langle a\,b\rangle}\cdot i\,g^{\,n-3}\frac{\langle P\,j\rangle^{4}}{\cdots\langle x\,P\rangle\langle P\,y\rangle\cdots} = \mathrm{Split}_{-}(a^{-},b^{+})\,A_{n-1}(\ldots,P^{-},\ldots), \tag{59.90}$$

with the merged leg now negative-helicity, and

$$\mathrm{Split}_{-}(a^{-},b^{+}) = \frac{z^{2}}{\sqrt{z(1-z)}\,\langle a\,b\rangle}. \tag{59.91}$$

The complementary assignment a^{+},b^{-} follows by the relabeling $a \leftrightarrow b$, $z \leftrightarrow 1-z$, giving $\mathrm{Split}_{-}(a^{+},b^{-}) = (1-z)^{2}/(\sqrt{z(1-z)}\,\langle a\,b\rangle)$. These three exhaust the holomorphic (angle-bracket) splittings accessible from the MHV formula. Their parity conjugates — obtained from the $\overline{\mathrm{MHV}}$ formula Eq. (59.56) by $\langle\,\rangle \to [\,]$ and $z^{k} \to (1-z)^{\cdots}$ bookkeeping — are the antiholomorphic splittings

$$\mathrm{Split}_{+}(a^{-},b^{-}) = \frac{-1}{\sqrt{z(1-z)}\,[a\,b]}, \quad \mathrm{Split}_{+}(a^{+},b^{-}) = \frac{-z^{2}}{\sqrt{z(1-z)}\,[a\,b]}, \tag{59.92}$$

and the relabeled partner. The combinations that would merge two same-helicity gluons into the "wrong" internal helicity — e.g. $\mathrm{Split}_{+}(a^{+},b^{+})$ — vanish, the collinear shadow of the helicity-vanishing theorems of the previous section. That Parke–Taylor reproduces *every* splitting amplitude with its correct z-dependence, on top of the soft factor and the symmetry checks, is the strongest evidence short of proof that Eq. (59.55) is right at all n.

Physical Insight

The soft and collinear limits are not optional checks bolted onto an answer; they are the boundary data that, together with the locations of the poles, almost determine a tree amplitude. Parke–Taylor passes them because it is built from the same physics — on-shell factorization on physical channels — that the recursion of the next sections will turn into a constructive proof. An amplitude that has the right poles, the right residues (lower amplitudes), the right soft and collinear limits, and the right little-group weight is forced to be the right amplitude. The formula was conjectured

before this was understood; the recursion is the understanding.

59.3.6 Status: a checked-but-unproved all-n formula

We have established Eq. (59.55) at $n = 4$ by direct computation (Eq. (59.81)), reduced it to the symmetry-fixed three-point amplitude at $n = 3$ (Eq. (59.57)), and verified that it carries the correct little-group weight, mass dimension, coupling power, cyclic and reflection symmetry, and — most tellingly — the correct soft (Eq. (59.84)) and collinear (Eqs. (59.89), (59.91) and (59.92)) factorizations on every physical channel. What we have not done is prove it for general n. Every check is consistent with Eq. (59.55), and the form is nearly forced, but consistency is not derivation: a brute-force Feynman computation at, say, six gluons would test thousands of diagrams, and we will not undertake it. The honest status is a *conjecture*, Theorem 59.13, supported by overwhelming low-multiplicity and structural evidence.

The structural evidence points the way to a proof. The soft and collinear analyses showed that an amplitude is pinned by its behavior on factorization channels: its poles sit where intermediate gluons go on shell, and its residues are products of lower-point amplitudes. If one could build the full amplitude from its residues alone — from lower amplitudes glued on physical poles, with no Lagrangian and no off-shell diagrams — then Eq. (59.55) would follow by induction from the three-point seed Eq. (59.57). The obstruction is that the three-point amplitude vanishes on real massless kinematics, where the only solution to momentum conservation collapses both chiralities. The next section confronts exactly this: complexifying the momenta resurrects the three-point amplitude by letting one chirality survive, and supplies the complex three-point kinematics that the recursion of the following sections requires to turn "checked" into "proved."

59.4 Complex Momenta and the Three-Point Amplitude

The previous section left the Parke–Taylor formula standing as a conjecture: a one-line expression for every MHV amplitude, checked against an honest Feynman computation at low multiplicity and surviving every consistency test, but not yet *proved*. The proof we are heading toward is recursive. It will build an n-point amplitude out of amplitudes with fewer legs, and like every recursion it needs a base case — an irreducible seed that the recursion does not itself explain but instead takes as input. For gluon trees the seed is the smallest amplitude there is, the three-point amplitude A_3. Here we hit an immediate and serious obstruction. For three massless particles, real on-shell momentum conservation

is so constraining that the kinematics collapses: the natural invariants all vanish, the polarization contractions die, and the three-gluon amplitude evaluates to an indeterminate $0/0$ that one is tempted to call simply zero. A recursion seeded by zero produces zero. The resolution, and the central idea of this section, is that nothing forces the external momenta to be real. If we allow the momenta to be *complex* — specifically, if we let the holomorphic spinor λ and the antiholomorphic spinor $\tilde{\lambda}$ vary independently, which reality would forbid — the degeneracy lifts. One of the two bracket sets can stay generic while the other vanishes, and on that complex locus the three-point amplitude is not only nonzero but is uniquely fixed, up to a single constant, by little-group scaling and locality alone. Fixing that constant against the three-gluon vertex of Chapter 26 delivers the two nonvanishing complex three-point amplitudes that seed everything to come.

59.4.1 Real three-point kinematics is degenerate

Fix three outgoing massless momenta p_1, p_2, p_3 subject to conservation $p_1 + p_2 + p_3 = 0$ and the on-shell conditions $p_i^2 = 0$. The first fact is that these conditions force every Lorentz invariant $p_i \cdot p_j$ to vanish.

Lemma 59.15 (Vanishing of all three-point invariants)**.** *For three on-shell massless momenta with $p_1 + p_2 + p_3 = 0$,*

$$p_1 \cdot p_2 = p_2 \cdot p_3 = p_3 \cdot p_1 = 0. \tag{59.93}$$

Proof. Solve conservation for one momentum, $p_3 = -(p_1 + p_2)$, and impose its mass-shell condition:

$$0 = p_3^2 = (p_1 + p_2)^2 = p_1^2 + 2\, p_1 \cdot p_2 + p_2^2 = 2\, p_1 \cdot p_2, \tag{59.94}$$

using $p_1^2 = p_2^2 = 0$. Hence $p_1 \cdot p_2 = 0$. The same elimination of p_1 gives $p_2 \cdot p_3 = 0$ and of p_2 gives $p_3 \cdot p_1 = 0$. □

Now translate Eq. (59.93) into spinor-helicity language. Each massless momentum factorizes through the bispinor map of Eq. (58.1),

$$p_{i\,\alpha\dot{\alpha}} = p_{i\mu}\, \sigma^{\mu}_{\alpha\dot{\alpha}} = \lambda_{i\alpha}\, \tilde{\lambda}_{i\dot{\alpha}}, \tag{59.95}$$

and the Mandelstam relation Eq. (59.31) reads $2\, p_i \cdot p_j = \langle i\, j \rangle\, [j\, i]$. Combining this with Eq. (59.93),

$$\langle 1\, 2 \rangle\, [2\, 1] = \langle 2\, 3 \rangle\, [3\, 2] = \langle 3\, 1 \rangle\, [1\, 3] = 0. \tag{59.96}$$

Each equation in Eq. (59.96) is a product of an angle bracket and a square bracket set to zero, so for each pair (i, j) at least one of $\langle i\, j \rangle$, $[i\, j]$ must vanish. The decisive question is whether the angle and square brackets can be switched off independently, and that is exactly where reality enters.

Proposition 59.16 (Real degeneracy)**.** *For three on-shell massless momenta with* $p_1 + p_2 + p_3 = 0$ *and* real *energies, every angle bracket and every square bracket among the three legs vanishes,*

$$\langle 1\,2\rangle = \langle 2\,3\rangle = \langle 3\,1\rangle = 0 \quad \textit{and} \quad [1\,2] = [2\,3] = [3\,1] = 0, \tag{59.97}$$

so that the three momenta are mutually collinear and no nonzero invariant survives.

Proof. For real momenta the two spinors attached to each leg are not independent: the reality condition of Chapter 58 fixes $\tilde{\lambda}_i = \pm\bar{\lambda}_i$, the sign being that of the energy p_i^0. Consequently the square bracket is, up to a phase and sign, the complex conjugate of the angle bracket,

$$[i\,j] = \pm\,\overline{\langle i\,j\rangle}\,, \qquad \text{hence} \qquad |[i\,j]| = |\langle i\,j\rangle|. \tag{59.98}$$

Thus $\langle i\,j\rangle = 0$ if and only if $[i\,j] = 0$: for real kinematics the two brackets of a pair vanish together. But Eq. (59.96) requires at least one of each pair to vanish, so by Eq. (59.98) both do, for every pair — which is Eq. (59.97). Finally, $\langle i\,j\rangle = 0$ means the two-component spinors λ_i, λ_j are proportional (an antisymmetric 2×2 contraction vanishes precisely when its arguments are parallel), so all three λ_i are proportional; likewise all three $\tilde{\lambda}_i$. The momenta $p_i = \lambda_i\tilde{\lambda}_i$ are then all proportional to a single null vector, i.e. mutually collinear. □

Theorem 59.16 is the obstruction in its sharpest form. The candidate three-point amplitudes one writes from little-group scaling — ratios of angle brackets, or ratios of square brackets — are evaluated on a configuration where *both* numerator and denominator brackets vanish. They are not honestly zero; they are $0/0$, undefined. Physically this is the familiar statement that two real massless particles cannot decay into a third without something going collinear: there is no genuine real kinematics for a $1 \to 2$ massless process, and the three-point amplitude has no real support on which to live. To resurrect it we must leave the real slice.

59.4.2 Independent complexification of λ and $\tilde{\lambda}$

The reality condition Eq. (59.98) is the only thing that tied the angle and square brackets together. Drop it. Allow the momenta to be complex by treating λ_i and $\tilde{\lambda}_i$ as *independent* two-component spinors, no longer constrained to be conjugates. This is a perfectly legitimate analytic continuation: an on-shell amplitude is a rational function of the spinor variables, and a rational function is defined wherever its arguments are, real or not. Once λ and $\tilde{\lambda}$ are unhooked, Eq. (59.96) can be satisfied in two genuinely distinct, non-degenerate ways.

Proposition 59.17 (The two complex branches)**.** *For three on-shell massless momenta with complex p_i (independent $\lambda_i, \tilde{\lambda}_i$) and $p_1 + p_2 + p_3 = 0$, the solutions of Eq. (59.96) split into two branches:*

1. *the* holomorphic *(or $\langle\,\rangle$-type) branch, on which all three square brackets vanish, $[1\,2] = [2\,3] = [3\,1] = 0$, while the angle brackets $\langle i\,j\rangle$ are generically nonzero;*
2. *the* antiholomorphic *(or $[\,]$-type) branch, on which all three angle brackets vanish, $\langle 1\,2\rangle = \langle 2\,3\rangle = \langle 3\,1\rangle = 0$, while the square brackets are generically nonzero.*

Away from the trivial point where both sets vanish, these are the only possibilities, and they meet only on the degenerate locus of Theorem 59.16.

Proof. Suppose not all square brackets vanish; without loss of generality $[1\,2] \neq 0$. The pair relation Eq. (59.96) for $(1, 2)$ then forces $\langle 1\,2\rangle = 0$, so $\lambda_1 \propto \lambda_2$; write $\lambda_2 = c\,\lambda_1$ with $c \neq 0$. Momentum conservation is the vanishing of the bispinor sum $\sum_i \lambda_i \tilde{\lambda}_i = 0$. Contract it with $\tilde{\lambda}_1$ on the dotted index; using $[1\,1] = 0$ the $i = 1$ term drops and what remains is a relation among the angle spinors,

$$\lambda_2\,[2\,1] + \lambda_3\,[3\,1] = 0. \tag{59.99}$$

Substituting $\lambda_2 = c\,\lambda_1$, $c\,\lambda_1\,[2\,1] + \lambda_3\,[3\,1] = 0$. If λ_1 and λ_3 were linearly independent, both coefficients would vanish; but $[2\,1] = -[1\,2] \neq 0$ and $c \neq 0$ force a contradiction. Hence $\lambda_3 \propto \lambda_1$ as well, so all three λ_i are proportional and every angle bracket vanishes. This is branch 1. By the identical argument with angle and square brackets exchanged, the assumption that some angle bracket is nonzero forces all square brackets to vanish, giving branch 2. If neither assumption holds, all brackets vanish, the degenerate point. The two branches share only that degenerate point because $\langle\,\rangle$ generic on branch 1 excludes branch 2 and conversely. □

That the holomorphic branch is non-empty and non-degenerate is easy to exhibit. Put all three square spinors parallel, $\tilde{\lambda}_i = c_i\,\tilde{\zeta}$ for a fixed reference spinor $\tilde{\zeta}$ and scalars c_i. Then every $[i\,j] \propto [\tilde{\zeta}\,\tilde{\zeta}] = 0$ automatically, and momentum conservation collapses to

$$\sum_{i=1}^{3} \lambda_i\,\tilde{\lambda}_i = \Big(\sum_{i=1}^{3} c_i\,\lambda_i\Big)\tilde{\zeta} = 0 \iff \sum_{i=1}^{3} c_i\,\lambda_i = 0, \tag{59.100}$$

a single linear condition on three two-component angle spinors that has a two-parameter family of solutions with all $\langle i\,j\rangle \neq 0$. The configuration is fully complex — the energies p_i^0 are complex numbers — but it is on-shell and momentum-conserving, and it carries nonzero angle brackets. The amplitude can finally be evaluated on something other than $0/0$. We now determine what it is.

59.4.3 The two three-point amplitudes from little-group and locality

We claim that on each branch the three-gluon amplitude is fixed, up to one overall constant, before any Feynman diagram is drawn. The constraints are little-group covariance Eq. (58.5) — recorded in Chapter 58 for general three-point amplitudes as Theorem 58.20 — and locality, in the elementary form "the amplitude has the correct mass dimension and no unphysical singularities."

Consider the helicity assignment $(1^-, 2^-, 3^+)$. By Theorem 59.17 a nonzero amplitude must live on one branch, where only one bracket set is available; suppose first it is the holomorphic branch, so A_3 is a function of the angle brackets $\langle 1\,2\rangle, \langle 2\,3\rangle, \langle 3\,1\rangle$ alone. Little-group scaling Eq. (58.5) acts as $(\lambda_i, \tilde\lambda_i) \to (t_i\lambda_i, t_i^{-1}\tilde\lambda_i)$ and demands that a helicity-h_i leg contribute weight $t_i^{-2h_i}$:

$$A_3(\{t_i\lambda_i, t_i^{-1}\tilde\lambda_i\}) = \Big(\prod_{i=1}^{3} t_i^{-2h_i}\Big)\, A_3, \qquad (h_1, h_2, h_3) = (-1, -1, +1). \tag{59.101}$$

Each angle bracket $\langle i\,j\rangle$ carries little-group weight $t_i\,t_j$. Write the most general monomial consistent with the available brackets,

$$A_3(1^-, 2^-, 3^+) = c\,\langle 1\,2\rangle^{\,x_3}\,\langle 2\,3\rangle^{\,x_1}\,\langle 3\,1\rangle^{\,x_2}, \tag{59.102}$$

where we have labeled the exponent of the bracket *omitting* leg i by x_i, for bookkeeping. Matching the weight of each t_i between the two sides of Eq. (59.101) gives one linear equation per leg:

$$\begin{aligned} t_1: &\quad x_3 + x_2 = -2h_1 = +2, \\ t_2: &\quad x_3 + x_1 = -2h_2 = +2, \\ t_3: &\quad x_1 + x_2 = -2h_3 = -2. \end{aligned} \tag{59.103}$$

Subtracting the first two gives $x_2 = x_1$; the third then yields $x_1 = x_2 = -1$, and back-substitution gives $x_3 = 3$. The exponents are uniquely determined:

$$A_3(1^-, 2^-, 3^+) = c\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}. \tag{59.104}$$

Little-group scaling alone has fixed the entire functional form down to the number c.

Locality now does two things. First, it confirms Eq. (59.104) is admissible: a three-point amplitude has mass dimension $[A_3] = 4 - 3 = 1$ in four dimensions (each external leg removes one power of mass from the dimension-four amplitude measure), and Eq. (59.104) carries dimension $3 - 1 - 1 = 1$ since every bracket has dimension one. The constant c is therefore dimensionless. Second, and more incisively, locality *excludes the other branch* for this helicity assignment. Had we tried to build $A_3(1^-, 2^-, 3^+)$ from square brackets on the antiholomorphic

branch, the same little-group bookkeeping (now each $[i\,j]$ carries weight $t_i^{-1}t_j^{-1}$) would force the exponents $x_1 = x_2 = +1$, $x_3 = -3$, giving

$$A_3^{(\text{wrong branch})}(1^-,2^-,3^+) = c'\,\frac{[2\,3]\,[3\,1]}{[1\,2]^3}, \qquad [A_3^{(\text{wrong branch})}] = 2-3 = -1. \tag{59.105}$$

This has mass dimension -1, not $+1$: it is not a possible amplitude. Locality, in the guise of correct mass dimension, selects the holomorphic solution Eq. (59.104) for the two-minus configuration and rejects the square-bracket alternative. The two requirements — little-group fixing the form, locality fixing the branch — leave exactly the one constant c.

The conjugate helicity assignment $(1^+,2^+,3^-)$ is handled by exchanging angle and square brackets throughout. Little-group now gives a pure square-bracket monomial, locality selects it (its dimension is $+1$ while the angle-bracket alternative has dimension -1), and the amplitude is

$$A_3(1^+,2^+,3^-) = c''\,\frac{[1\,2]^3}{[2\,3]\,[3\,1]}. \tag{59.106}$$

We have, with no diagram drawn, the two functional forms; it remains to pin the constants c, c''.

59.4.4 Fixing the constant against the three-gluon vertex

The constant is set by demanding that Eq. (59.104) agree with the honest Feynman computation built from the Yang–Mills three-gluon vertex of Chapter 26. Because $n = 3$ admits no internal line, there is precisely one diagram — a single cubic vertex with three external legs (Fig. 59.3) — and the entire amplitude is the color-ordered vertex of Eq. (59.12) contracted with three external polarizations. The computation is short once the reference momenta are chosen with the foresight of Section 59.2.

Worked Example 59.18 (The holomorphic three-point amplitude, normalization and all)**.** With all momenta outgoing and $p_1+p_2+p_3 = 0$, the color-ordered amplitude is the cubic vertex Eq. (59.12) dotted into the polarizations,

$$A_3(1^-,2^-,3^+) = \frac{i}{\sqrt{2}}\Big[(\varepsilon_1{\cdot}\varepsilon_2)\,(p_1-p_2){\cdot}\varepsilon_3 + (\varepsilon_2{\cdot}\varepsilon_3)\,(p_2-p_3){\cdot}\varepsilon_1 + (\varepsilon_3{\cdot}\varepsilon_1)\,(p_3-p_1){\cdot}\varepsilon_2\Big], \tag{59.107}$$

with $\varepsilon_1 = \varepsilon_-(p_1;q_1)$, $\varepsilon_2 = \varepsilon_-(p_2;q_2)$, $\varepsilon_3 = \varepsilon_+(p_3;q_3)$ the polarization vectors of Eq. (59.32). We work on the holomorphic branch (Theorem 59.17, all $[i\,j] = 0$ among externals, $\langle i\,j\rangle$ generic) and exploit the reference freedom to kill two of the three terms.

Reference choice. Give the two negative-helicity legs a common reference $q_1 = q_2 \equiv q$, with q a generic null momentum off the branch so that $[1\,q], [2\,q] \neq 0$,

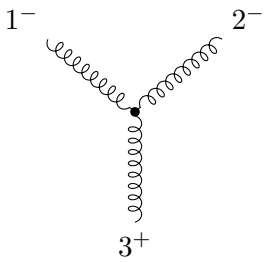


Figure 59.3: The sole diagram contributing to the three-point amplitude: a single color-ordered cubic vertex Eq. (59.12) with three on-shell external gluons, here in the holomorphic-branch configuration $(1^-, 2^-, 3^+)$. There is no internal propagator, so no momentum can be summed and no other topology exists. The amplitude is nonzero only on the complex kinematics of Theorem 59.17.

and set the positive leg's reference to $q_3 = p_2$ (legal since $\langle q_3\, 3\rangle = \langle 2\, 3\rangle \neq 0$ on the branch). The polarization contractions follow from Theorem 59.8:

$$\varepsilon_1^- \cdot \varepsilon_2^- = \frac{\langle 1\, 2\rangle\, [q_2\, q_1]}{[1\, q_1]\, [2\, q_2]} = \frac{\langle 1\, 2\rangle\, [q\, q]}{[1\, q]\, [2\, q]} = 0, \tag{59.108}$$

$$\varepsilon_3^+ \cdot \varepsilon_2^- = \frac{\langle q_3\, 2\rangle\, [q_2\, 3]}{\langle q_3\, 3\rangle\, [2\, q_2]} = \frac{\langle 2\, 2\rangle\, [q\, 3]}{\langle 2\, 3\rangle\, [2\, q]} = 0, \tag{59.109}$$

the first because $[q\, q] = 0$, the second because $\langle 2\, 2\rangle = 0$. Only the third term of Eq. (59.107) survives, through the contraction

$$\varepsilon_3^+ \cdot \varepsilon_1^- = \frac{\langle q_3\, 1\rangle\, [q_1\, 3]}{\langle q_3\, 3\rangle\, [1\, q_1]} = \frac{\langle 2\, 1\rangle\, [q\, 3]}{\langle 2\, 3\rangle\, [1\, q]}. \tag{59.110}$$

The momentum factor. The surviving term needs $(p_3 - p_1)\cdot\varepsilon_2$. Transversality Eq. (59.33) gives $\varepsilon_2 \cdot p_2 = 0$; with $p_2 = -(p_1 + p_3)$ this yields $\varepsilon_2 \cdot p_3 = -\varepsilon_2 \cdot p_1$, so $(p_3 - p_1)\cdot\varepsilon_2 = -2\,\varepsilon_2 \cdot p_1$. Inserting $\varepsilon_2^- = \langle 2|\gamma^\mu|q_2]/(\sqrt{2}\,[2\, q_2])$ and using $\langle 2|\not{p}_1|q] = \langle 2\, 1\rangle\, [1\, q]$,

$$(p_3 - p_1)\cdot\varepsilon_2^- = -2\,\frac{\langle 2|\not{p}_1|q]}{\sqrt{2}\,[2\, q]} = -\sqrt{2}\,\frac{\langle 2\, 1\rangle\, [1\, q]}{[2\, q]}. \tag{59.111}$$

Assembly. Only the third term of Eq. (59.107) survives:

$$\begin{aligned} A_3(1^-, 2^-, 3^+) &= \frac{i}{\sqrt{2}}\,(\varepsilon_3^+ \cdot \varepsilon_1^-)\,(p_3 - p_1)\cdot\varepsilon_2^- = \frac{i}{\sqrt{2}} \cdot \frac{\langle 2\, 1\rangle\, [q\, 3]}{\langle 2\, 3\rangle\, [1\, q]} \cdot \Big(-\sqrt{2}\,\frac{\langle 2\, 1\rangle\, [1\, q]}{[2\, q]} \Big) \\ &= -\,i\,\frac{\langle 2\, 1\rangle^{\,2}\, [q\, 3]}{\langle 2\, 3\rangle\, [2\, q]}, \end{aligned} \tag{59.112}$$

the reference square bracket $[1\, q]$ having cancelled. The remaining ratio $[q\, 3]/[2\, q]$ must collapse to a q-independent expression, as gauge invariance guarantees.

On the holomorphic branch $\tilde{\lambda}_i = c_i\,\tilde{\zeta}$, so $[q\,3] = c_3\,[\tilde{\lambda}_q\,\tilde{\zeta}]$ and $[2\,q] = c_2\,[\tilde{\zeta}\,\tilde{\lambda}_q] = -c_2\,[\tilde{\lambda}_q\,\tilde{\zeta}]$, whence

$$\frac{[q\,3]}{[2\,q]} = -\frac{c_3}{c_2}. \tag{59.113}$$

The scalars c_i are fixed by momentum conservation Eq. (59.100), $\sum_i c_i\lambda_i = 0$. Contracting that relation with λ_1 (i.e. taking $\langle 1\,\cdot\,\rangle$) gives $c_2\langle 1\,2\rangle + c_3\langle 1\,3\rangle = 0$, so

$$\frac{c_3}{c_2} = -\frac{\langle 1\,2\rangle}{\langle 1\,3\rangle} = \frac{\langle 1\,2\rangle}{\langle 3\,1\rangle}, \qquad \text{hence} \qquad \frac{[q\,3]}{[2\,q]} = -\frac{\langle 1\,2\rangle}{\langle 3\,1\rangle}. \tag{59.114}$$

The q-dependence has indeed dropped out. Substituting Eq. (59.114) into Eq. (59.112) and using $\langle 2\,1\rangle^2 = \langle 1\,2\rangle^2$,

$$A_3(1^-,2^-,3^+) = -\,i\,\frac{\langle 1\,2\rangle^{\,2}}{\langle 2\,3\rangle}\cdot\Big(-\frac{\langle 1\,2\rangle}{\langle 3\,1\rangle}\Big) = i\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}. \tag{59.115}$$

Comparing with the little-group form Eq. (59.104) fixes $c = i$.

The constant is fixed, the phase with it. Restoring the coupling — the color decomposition Eq. (59.11) carries an overall g^{n-2}, which for $n = 3$ is a single power g, and it is convenient to absorb it into the seed amplitude so that the three-point object the recursion consumes is the physical, coupling-weighted one — the two nonvanishing complex three-point amplitudes are

$$\boxed{A_3(1^-,2^-,3^+) = i\,g\,\frac{\langle 1\,2\rangle^{\,3}}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \qquad A_3(1^+,2^+,3^-) = -\,i\,g\,\frac{[1\,2]^{\,3}}{[2\,3]\,[3\,1]}.} \tag{59.116}$$

The relative sign between the two is not a convention but a consequence of parity. Parity exchanges the independent spinors, $\lambda_i \leftrightarrow \tilde{\lambda}_i$, sending angle brackets to square brackets and the helicity assignment $(1^-,2^-,3^+)$ to $(1^+,2^+,3^-)$. Under this exchange each polarization vector maps as $\varepsilon_+ \to -\varepsilon_-$ and $\varepsilon_- \to -\varepsilon_+$ — visible directly from Eq. (59.32), since $\langle q|\gamma^\mu|i] = \langle i|\gamma^\mu|q]$ under the swap while $\langle q\,i\rangle \to [q\,i] = -[i\,q]$, producing one minus sign per vector. The amplitude Eq. (59.107) is linear in all three polarizations, so it acquires the overall factor $(-1)^3 = -1$, which is exactly the relative sign in Eq. (59.116).

Notation Watch

The kinematic structures in Eq. (59.116) are the three-point amplitudes anticipated by little-group scaling in Chapter 58 (Theorem 58.20); the present section supplies what little-group scaling could not, namely the overall normalization i and the coupling g, fixed by the honest match to the three-gluon vertex Eq. (26.53) of Chapter 26. Where Chapter 58 or Section 59.2.6 quoted the bare ratio $\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$, the complete

on-shell seed used from here on is the coupling- and phase-fixed object Eq. (59.116). The factor g^{n-2} that Eq. (59.11) kept explicit out front is, for these three-point seeds, absorbed into A_3.

Why complex momenta are not a trick

It is tempting to read the complexification as an artifice for dodging a kinematic singularity. It is the opposite: it is the natural home of the three-point amplitude. The functional forms Eq. (59.116) are rational functions of the spinor variables, fixed by symmetry and locality independently of any reality assumption; reality is the special real slice on which both branches collide and the function degenerates. The recursion of the next section will deform the external momenta of a genuine, real, higher-point amplitude into the complex plane, and the value of that amplitude will be reconstructed from the residues at poles where internal lines go on shell — residues that are precisely products of on-shell lower-point amplitudes evaluated on complex three- and higher-point kinematics. The complex three-point amplitudes Eq. (59.116) are not auxiliary; they are the irreducible physical data from which every gluon tree is built.

With the two nonzero complex three-point amplitudes in hand and the technique of independently complexifying λ and $\tilde{\lambda}$ established, we possess both ingredients the recursion requires: nonvanishing seeds and a controlled way to push external momenta off the real slice. The next section turns these into the BCFW shift — a one-parameter complex deformation of two external momenta — and uses the analytic structure it exposes to assemble any tree amplitude from products of the on-shell amplitudes Eq. (59.116) and their higher-point descendants.

59.5 BCFW Recursion: Derivation

The previous section turned a defect into a tool. Real on-shell three-point kinematics for massless particles is degenerate — all brackets collapse together and the amplitude is forced to zero — but once the momenta are complexified the angle and square spinors become independent, one chirality survives, and the two nonzero three-point amplitudes Eq. (59.116) reappear. Those two objects are the seeds. What remains is to show that they, together with nothing but the analytic structure that locality and unitarity impose on a tree amplitude, *determine every higher-point amplitude*. The mechanism is the Britto–Cachazo–Feng–Witten (BCFW) on-shell recursion. The idea is simple to state and worth keeping in view through the algebra: deform two of the external momenta by a single complex parameter z, regard the color-ordered amplitude $A_n(z)$ as a

meromorphic function of z, and reconstruct $A_n = A_n(0)$ from its poles using Cauchy's theorem. The poles are physical — each is an internal line going on shell — and the residue at each one is, by the factorization of trees on propagator poles (Chapters 13 and 15), a product of two *lower-point on-shell amplitudes*. The recursion writes the n-point amplitude as a sum of products of smaller amplitudes, glued across a propagator. No Lagrangian, no off-shell vertices, no gauge fixing enters after the seeds are in hand.

This section derives the recursion in four moves: the shift and its kinematic consistency; the analytic structure of $A_n(z)$; the factorization of its residues; and the contour argument that assembles the master formula. Throughout we work with a single color-ordered partial amplitude $A_n \equiv A_n[1, 2, \dots, n]$ in the cyclic order $1, \dots, n$, with all momenta outgoing and massless, using the spinor conventions Eqs. (59.30) and (59.31) and the color-ordered Feynman rules of Section 59.1.5 (Eqs. (59.12) to (59.14)). One hypothesis is deferred: the contour argument drops a possible contribution from $z = \infty$, and the legitimacy of dropping it — the large-z behavior of $A_n(z)$ — is the subject of the next section. Everything else is established here in full.

59.5.1 The complex shift

We want a one-parameter complex deformation of the external data that (i) keeps every external leg on shell, (ii) preserves total momentum conservation, and (iii) touches as few legs as possible. The minimal such deformation moves two legs. Pick an ordered pair of distinct external labels (i, j). Recall that each massless leg is carried by a pair of two-component spinors, $p_{k\,\alpha\dot\alpha} = \lambda_{k\alpha}\tilde\lambda_{k\dot\alpha}$, with the angle and square brackets of Eq. (58.20); we write $|k\rangle = \lambda_k$, $|k] = \tilde\lambda_k$.

Definition 59.19 (The $[i, j\rangle$ shift)**.** Fix two distinct labels $i \neq j$. The $[i, j\rangle$ *shift* (square bracket on i, angle bracket on j) deforms the spinors of legs i and j by a complex parameter $z \in \mathbb{C}$,

$$|\hat i] = |i] + z\,|j], \qquad |\hat j\rangle = |j\rangle - z\,|i\rangle\,, \qquad |\hat i\,\rangle = |i\rangle\,, \qquad |\hat j] = |j], \tag{59.117}$$

and leaves all other spinors untouched. Equivalently, in terms of momenta,

$$\hat p_i(z) = p_i + z\,r, \qquad \hat p_j(z) = p_j - z\,r, \qquad r_{\alpha\dot\alpha} \equiv \lambda_{i\alpha}\tilde\lambda_{j\dot\alpha} = |i\rangle\,[j|, \qquad r^\mu = \tfrac{1}{2}\langle i|\gamma^\mu|j]. \tag{59.118}$$

All momenta p_k with $k \neq i, j$ are unshifted.

The deformation is engineered so that the deformed momenta remain physical for *every* complex value of z, which is what licenses treating $A_n(z)$ as an amplitude rather than a formal expression. We record this as a proposition and prove it; the proof is short, but each of the three properties is load-bearing.

Proposition 59.20 (Kinematic consistency of the shift)**.** *For all $z \in \mathbb{C}$ the shifted momenta Eq. (59.118) satisfy*

1. *momentum conservation:* $\sum_k \hat{p}_k(z) = \sum_k p_k$*;*
2. *masslessness:* $\hat{p}_i(z)^2 = \hat{p}_j(z)^2 = 0$ *(and* $\hat{p}_k^2 = p_k^2 = 0$ *for* $k \neq i, j$*);*
3. *the shift vector is null and orthogonal to both shifted legs:* $r^2 = 0,\ r \cdot \hat{p}_i = r \cdot \hat{p}_j = 0$.

Proof. (1) Only p_i and p_j are altered, and their changes are equal and opposite by construction: $\hat{p}_i + \hat{p}_j = (p_i + zr) + (p_j - zr) = p_i + p_j$. Adding the untouched p_k gives $\sum_k \hat{p}_k = \sum_k p_k$. If the original momenta conserve, so do the shifted ones.

(2) The decisive feature of Eq. (59.117) is that each shifted momentum *still factorizes as a single product of one angle spinor and one square spinor*:

$$\hat{p}_{i\,\alpha\dot{\alpha}} = \lambda_{i\alpha}\,(\tilde{\lambda}_i + z\tilde{\lambda}_j)_{\dot{\alpha}} = |\hat{i}\rangle\,[\hat{i}|, \qquad \hat{p}_{j\,\alpha\dot{\alpha}} = (\lambda_j - z\lambda_i)_\alpha\,\tilde{\lambda}_{j\dot{\alpha}} = |\hat{j}\rangle\,[\hat{j}|. \tag{59.119}$$

A real symmetric (here, complex) bispinor is null precisely when its determinant vanishes, and the determinant of a rank-one outer product $\lambda\tilde{\lambda}$ is identically zero. Hence $\hat{p}_i^2 = \det(|\hat{i}\rangle\,[\hat{i}|) = 0$ and likewise $\hat{p}_j^2 = 0$, for every z, with no condition on z whatsoever. This is exactly why the shift acts on $|i]$ but not $|i\rangle$, and on $|j\rangle$ but not $|j]$: disturbing only one spinor of each leg keeps the rank-one structure intact. Equivalently, expanding directly, $\hat{p}_i^2 = (p_i + zr)^2 = p_i^2 + 2z\,r \cdot p_i + z^2 r^2$, and we verify the cross terms vanish in (3).

(3) The shift vector $r = |i\rangle\,[j|$ is itself a rank-one bispinor, so $r^2 = \det(|i\rangle\,[j|) = 0$. For the orthogonality, use the Mandelstam relation Eq. (59.31) in the form $2\,a \cdot b = \langle a\,b\rangle[b\,a]$ together with $2\,r \cdot p_k = \langle i\,k\rangle[k\,j]$ (read off from $r = |i\rangle\,[j|$ contracted with $p_k = |k\rangle\,[k|$):

$$2\,r \cdot p_i = \langle i\,i\rangle\,[i\,j] = 0, \qquad 2\,r \cdot p_j = \langle i\,j\rangle\,[j\,j] = 0, \tag{59.120}$$

since $\langle i\,i\rangle = [j\,j] = 0$. Because $r \cdot p_i = r \cdot p_j = 0$ and $r^2 = 0$, $r \cdot \hat{p}_i = r \cdot p_i + zr^2 = 0$ and similarly $r \cdot \hat{p}_j = 0$. The cross terms in $\hat{p}_i^2 = p_i^2 + 2z\,r \cdot p_i + z^2 r^2$ all vanish, re-confirming masslessness. □

Three remarks fix the picture. First, the shift is intrinsically *complex*: the deformed spinors $|\hat{j}\rangle = |j\rangle - z\,|i\rangle$ and $|\hat{i}] = |i] + z|j]$ are not complex conjugates of each other for $z \neq 0$, so the shifted momenta $\hat{p}_i, \hat{p}_j$ are genuinely complex four-vectors. There is no real shift that keeps two distinct legs on shell while conserving momentum; the complexification of the previous section is not a convenience but a prerequisite. Second, although $\hat{p}_i(z)$ and $\hat{p}_j(z)$ are individually z-dependent, their sum is fixed, and any partial sum of external momenta that contains *both* or *neither* of legs i, j is also z-independent. Only sums separating i from j feel the shift. This is the seed of the pole structure. Third, the analytically continued amplitude $A_n(z)$ is obtained by substituting Eq. (59.117) into the same color-ordered expression that computes A_n; by Theorem 59.20 it is a bona

fide on-shell amplitude for complex external kinematics, equal to the physical amplitude at $z = 0$.

Caution: the shifted legs are not conjugate

For real momenta the square spinor is fixed by the angle spinor up to a phase, $|i] = \pm\,|i\rangle^*$. The shift breaks this: $|\hat{i}]$ changes while $|\hat{i}\rangle = |i\rangle$ does not, so $|\hat{i}] \neq \pm\,|\hat{i}\rangle^*$. One must treat angle and square spinors as independent variables throughout the recursion and only set $z = 0$ (restoring reality) at the very end. Quantities such as $\langle i|P|j]$ that mix the two chiralities are where the z-dependence lives.

59.5.2 Analytic structure of the shifted amplitude

We now determine what kind of function $A_n(z)$ is. The answer — a rational function with only simple poles, located at the zeros of shifted Mandelstam invariants — is what makes Cauchy's theorem usable.

Proposition 59.21 ($A_n(z)$ is rational with only simple poles)**.** *Under the $[\,i,j\,\rangle$ shift, the color-ordered tree amplitude $A_n(z)$ is a rational function of z. Its only finite singularities are simple poles. Each pole sits at a value z_I where a single shifted internal invariant $\hat{P}_I(z)^2$ vanishes, where $P_I = \sum_{k\in I} p_k$ is the momentum flowing through a cyclically-consecutive subset $I \subsetneq \{1,\dots,n\}$ that contains exactly one of the two shifted legs i, j.*

Proof. A color-ordered tree amplitude is computed (Section 59.1.5) as a finite sum over planar diagrams; each diagram is a product of vertices Eqs. (59.12) and (59.13), external polarization vectors Eq. (59.32), and scalar propagators i/P^2 extracted from Eq. (59.14), with all Lorentz indices contracted. We track the z-dependence factor by factor.

Numerators are polynomial in z. The cubic vertex Eq. (59.12) is linear in the momenta and the quartic Eq. (59.13) is momentum-free; under Eq. (59.117) a shifted momentum is linear in z, so each vertex contributes a polynomial (degree ≤ 1) in z. The external polarization vectors are likewise polynomial: only the shifted legs i, j carry z-dependence, and from Eq. (59.32) the dependence enters through $|\hat{i}]$ or $|\hat{j}\rangle$ linearly, with the reference spinors held fixed. (We may always choose the reference momenta of the shifted legs so that no polarization denominator $\langle q_i\, i\rangle$, $[i\, q_i]$ becomes z-dependent; e.g. pick the reference of leg i from its *angle* spinor, which is unshifted.) Products and sums of polynomials are polynomials, so every diagram's numerator is a polynomial in z.

Denominators are products of shifted propagators, each linear in z. A diagram's denominator is a product of internal invariants $\hat{P}_I(z)^2$, one per internal line. By momentum conservation the momentum on any internal line of a tree

equals the sum $P_I = \sum_{k\in I} p_k$ over the external legs I on one side of that line; in a planar (color-ordered) diagram these subsets are exactly the cyclically-consecutive blocks $I = \{a, a+1, \dots, b\}$. Now apply the shift. If I contains both i and j, or neither, then $\hat{P}_I = P_I$ is z-independent (the shifts of p_i and p_j cancel in the sum, or are simply absent). The only z-dependence in a denominator comes from blocks I that contain exactly one shifted leg. Take the convention $i \in I$, $j \notin I$ (the other case is identical with $r \to -r$). Then

$$\hat{P}_I(z) = P_I + z\, r, \qquad \boxed{\hat{P}_I(z)^2 = P_I^2 + z\,\langle i|P_I|j]} , \qquad \langle i|P_I|j] \equiv \sum_{k\in I} \langle i\, k\rangle [k\, j], \tag{59.121}$$

using $\hat{P}_I^2 = P_I^2 + 2z\, r\cdot P_I + z^2 r^2$ with $r^2 = 0$ and $2r\cdot P_I = \langle i|\gamma^\mu|j] P_{I\mu} = \langle i|P_I|j]$. Each such invariant is *linear* in z, vanishing at the single point

$$z_I = -\frac{P_I^2}{\langle i|P_I|j]}. \tag{59.122}$$

Thus each propagator $1/\hat{P}_I(z)^2 = 1/(P_I^2 + z\,\langle i|P_I|j])$ contributes at most one simple pole, at $z = z_I$.

Only simple poles, and they are physical. A given diagram is a ratio of a polynomial numerator to a product of distinct linear factors $\hat{P}_I(z)^2$; within one diagram, the internal lines carry distinct consecutive blocks I (no two propagators in a tree carry the same momentum), so the factors are distinct and the diagram has only simple poles. Summing over the finitely many planar diagrams can only combine residues at coincident z_I; but distinct channels $I \neq I'$ have distinct $z_I \neq z_{I'}$ for generic external momenta (the values Eq. (59.122) are different rational functions of the external data), so no two simple poles collide and the sum $A_n(z)$ retains only simple poles. Each pole sits at a z_I where a physical factorization channel $\hat{P}_I(z)^2 \to 0$ opens up: the internal line carrying $\hat{P}_I$ goes on shell. There are no other singularities — a tree amplitude has no branch cuts — so $A_n(z)$ is rational, and we have located all its finite poles. □

The content of Theorem 59.21 is worth restating in words, because it is the engine of the recursion. Deforming two legs turns the amplitude into a rational function whose poles are in one-to-one correspondence with the factorization channels that *separate the two shifted legs*. A channel that keeps i and j on the same side of the propagator is invisible to the shift and contributes no pole; only the channels that split them appear. The next task is to read off the residue at each such pole.

59.5.3 Factorization on a physical pole

At $z = z_I$ the internal line of channel I has gone on shell: $\hat{P}_I(z_I)^2 = 0$. The behaviour of a tree amplitude as an internal momentum approaches its mass

shell is governed by a basic property of perturbation theory, established in Chapter 13 (LSZ and the pole structure of correlators) and Chapter 15 (the diagrammatic expansion): *a tree amplitude, viewed as a function of a single Mandelstam invariant $s = P_I^2$, has a simple pole at $s \to 0$ (for a massless internal particle) whose residue factorizes into the product of the two on-shell sub-amplitudes obtained by cutting the internal line, summed over the physical states of the exchanged particle.* We now make this concrete in the color-ordered language and extract the residue of $A_n(z)$ at z_I.

Theorem 59.22 (Residue factorization)**.** *Let z_I be the pole Eq. (59.122) associated with the consecutive channel I (with $i \in I$, $j \notin I$). Then near $z = z_I$,*

$$A_n(z) \xrightarrow{z \to z_I} \sum_h \hat{A}_L(z_I)\, \frac{i}{\hat{P}_I(z)^2}\, \hat{A}_R(z_I), \tag{59.123}$$

where $\hat{A}_L$ is the color-ordered amplitude built from the legs in I together with an extra on-shell leg of momentum $-\hat{P}_I(z_I)$ and helicity $-h$, $\hat{A}_R$ is the amplitude built from the legs not in I together with an extra on-shell leg of momentum $+\hat{P}_I(z_I)$ and helicity $+h$, both evaluated at $z = z_I$, and the sum runs over the helicity (and, in general, particle species) h of the exchanged state.

Proof. Sort the planar diagrams contributing to $A_n(z)$ by whether they contain an internal line carrying the channel momentum $\hat{P}_I$. Diagrams that do *not* contain such a line have a denominator with no factor $\hat{P}_I(z)^2$, so they are regular at $z = z_I$ and contribute nothing to the residue. Diagrams that *do* contain it have precisely one such propagator (a tree cannot carry the same momentum on two distinct internal lines), and each of them therefore splits into two connected pieces joined by that single line: a left piece L attached to the external legs I, and a right piece R attached to the external legs $\bar{I}$. The internal line carries the color-ordered scalar propagator $i/\hat{P}_I(z)^2$ from Eq. (59.14).

Write the contribution of any such diagram as $D(z) = \dfrac{N_L(z)\ [\text{numerator of the internal line}]}{\hat{P}_I(z)^2\ \times (\text{other propagators})}$ where N_L, N_R collect the sub-diagrams on the two sides. Summing over all diagrams of this type and over the two tensor (helicity) structures the gluon propagator can transmit, the numerator of the internal line resolves, on the pole, into the polarization sum for an on-shell gluon of momentum $\hat{P}_I(z_I)$:

$$-i\,\eta_{\mu\nu} \xrightarrow{\hat{P}_I^2 \to 0} -i \sum_h \varepsilon_\mu^h(\hat{P}_I)\, \varepsilon_\nu^{-h}(-\hat{P}_I) \;+\; (\text{terms} \propto \hat{P}_{I\mu} \text{ or } \hat{P}_{I\nu}), \tag{59.124}$$

the gauge-dependent longitudinal terms dropping out because each side is, on the pole, a color-ordered amplitude with an on-shell external gluon and is therefore transverse (the Ward identity of Chapter 15 for the color-ordered current; the same statement underlies the reference-momentum independence

used in Section 59.2.2). What multiplies $\varepsilon^h(\hat{P}_I)$ on the left, once the internal momentum is on shell, is by definition the color-ordered amplitude $\hat{A}_L$ with external legs $\{I, (-\hat{P}_I)^{-h}\}$ in cyclic order; what multiplies $\varepsilon^{-h}(-\hat{P}_I)$ on the right is $\hat{A}_R$ with external legs $\{(\hat{P}_I)^h, \bar{I}\}$. (The two on-shell legs carry opposite momenta and opposite helicity labels, as a single internal line cut into two external legs must.) Therefore, retaining only the singular part,

$$A_n(z)\Big|_{\text{sing.}} = \sum_h \hat{A}_L(z)\,\frac{i}{\hat{P}_I(z)^2}\,\hat{A}_R(z). \tag{59.125}$$

Finally, $\hat{A}_L$ and $\hat{A}_R$ are smooth at $z = z_I$ (their own propagators are channels that do not separate i from j *within* the smaller amplitude, hence are regular there), so to extract the residue we may evaluate them at $z = z_I$, where $\hat{P}_I(z_I)^2 = 0$ makes both sub-amplitudes genuinely on shell. This is Eq. (59.123). □

Why the residue is a product of physical amplitudes

Equation Eq. (59.123) is unitarity made visible. A propagator pole is where a virtual particle becomes real; the residue is the amplitude to produce that real particle times the amplitude to absorb it. Locality (the amplitude is built from local propagators $1/P^2$) is what guarantees the poles are simple and sit at $P_I^2 = 0$; unitarity (the residue factorizes onto physical states) is what makes the numerator a *product of on-shell amplitudes* rather than some new off-shell object. BCFW is the statement that these two principles, applied channel by channel, contain all the information in the Lagrangian's tree expansion — and package it far more efficiently.

The factorized form Eq. (59.123) is drawn in Fig. 59.4: the deformed n-point amplitude, near the pole, is two smaller on-shell amplitudes glued across an on-shell internal line, with the shifted legs $\hat{\imath}$ and $\hat{\jmath}$ landing on opposite blobs.

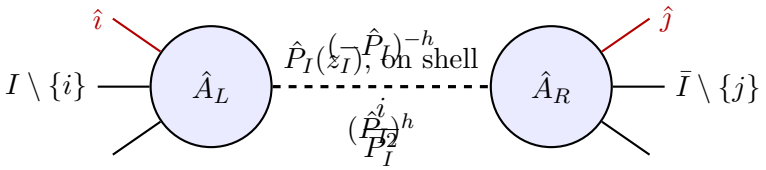


Figure 59.4: Factorization of the shifted amplitude on the channel-I pole. The two shifted legs $\hat{\imath} \in I$ and $\hat{\jmath} \in \bar{I}$ sit on opposite sub-amplitudes; the internal line is on shell at $z = z_I$, and the residue sums over its helicity h. The recursion glues lower-point on-shell amplitudes with the scalar propagator i/P_I^2.

59.5.4 The contour argument and the master formula

We have a rational function $A_n(z)$ with simple poles at known locations and known residues. The physical amplitude is its value at the origin, $A_n = A_n(0)$. Cauchy's theorem converts knowledge of the poles into knowledge of $A_n(0)$.

Consider the auxiliary function $A_n(z)/z$. It has the same poles as $A_n(z)$, at the z_I, plus a new simple pole at $z = 0$ with residue $A_n(0) = A_n$, the quantity we want. Integrate it around a large circle C_R of radius R in the complex plane (Fig. 59.5):

$$\frac{1}{2\pi i}\oint_{C_R} \frac{A_n(z)}{z}\,dz = A_n(0) + \sum_I \mathrm{Res}_{z=z_I}\frac{A_n(z)}{z}. \tag{59.126}$$

Suppose the left-hand side vanishes as $R \to \infty$ — equivalently, that $A_n(z) \to 0$ as $z \to \infty$ so that $A_n(z)/z$ falls faster than $1/z$ and the boundary integral has no contribution from infinity. This is the one hypothesis we do not prove here; Section 59.6 establishes exactly when it holds. Granting it, the sum of all residues is zero, and Eq. (59.126) reads

$$0 = A_n + \sum_I \mathrm{Res}_{z=z_I}\frac{A_n(z)}{z}. \tag{59.127}$$

Each residue at z_I follows from the factorization Eq. (59.123). Near z_I, write $\hat{P}_I(z)^2 = \langle i|P_I|j]\,(z - z_I)$ from Eq. (59.121) (the linear coefficient is the derivative $\partial_z \hat{P}_I^2 = \langle i|P_I|j]$). Then

$$\mathrm{Res}_{z=z_I}\frac{A_n(z)}{z} = \frac{1}{z_I}\,\mathrm{Res}_{z=z_I}\,A_n(z) = \frac{1}{z_I}\sum_h \hat{A}_L(z_I)\,\frac{i}{\langle i|P_I|j]}\,\hat{A}_R(z_I), \tag{59.128}$$

where we used that $\mathrm{Res}_{z=z_I}\,[i/\hat{P}_I(z)^2] = i/\langle i|P_I|j]$. Now insert $z_I = -P_I^2/\langle i|P_I|j]$ from Eq. (59.122), so that $1/z_I = -\langle i|P_I|j]/P_I^2$. The factors of $\langle i|P_I|j]$ cancel:

$$\mathrm{Res}_{z=z_I}\frac{A_n(z)}{z} = -\sum_h \hat{A}_L(z_I)\,\frac{i}{P_I^2}\,\hat{A}_R(z_I). \tag{59.129}$$

Note the propagator factor has reverted to the *unshifted, physical* invariant $1/P_I^2$: the on-shell condition lives in the sub-amplitudes, while the glue is the ordinary Feynman propagator of the channel evaluated at $z = 0$. Substituting Eq. (59.129) into Eq. (59.127) and moving the sum to the other side gives the central result of the chapter.

Theorem 59.23 (BCFW on-shell recursion)**.** *Let A_n be a color-ordered tree amplitude, and apply an $[\,i, j\,\rangle$ shift Eq. (59.117) for which $A_n(z) \to 0$ as $z \to \infty$. Then*

$$\boxed{A_n = \sum_I \sum_h \hat{A}_L(z_I)\,\frac{i}{P_I^2}\,\hat{A}_R(z_I)\ ,} \tag{59.130}$$

where the outer sum runs over all cyclically-consecutive channels I that separate the shifted legs (i.e. $i \in I$, $j \notin I$), $z_I = -P_I^2/\langle i|P_I|j]$ is the value at which $\hat{P}_I(z)^2$ vanishes, P_I^2 is the unshifted channel invariant, and the inner sum is over the helicity h of the on-shell internal state. The amplitudes $\hat{A}_L, \hat{A}_R$ are lower-point, on-shell, and evaluated at the shifted kinematics $z = z_I$.

The structure of Eq. (59.130) is exactly the recursion advertised at the outset: A_n is a finite sum of products of two strictly lower-point on-shell amplitudes, glued by a scalar propagator and summed over a handful of channels and the internal helicity. Feeding the output of the recursion back into its input, every amplitude descends to products of three-point amplitudes — the complex seeds Eq. (59.116) — with no Lagrangian, no off-shell vertex, and no gauge fixing entering after the seeds. Locality fixed the poles; unitarity fixed the residues; Cauchy's theorem assembled them.

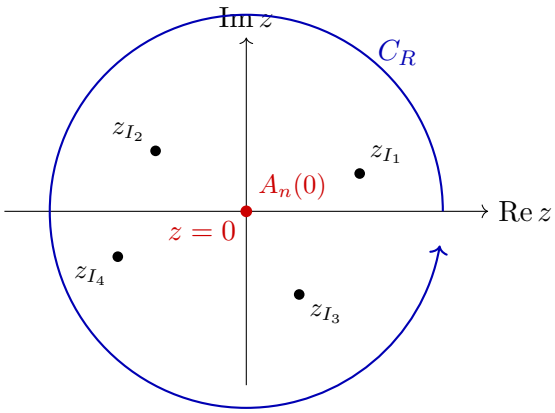


Figure 59.5: The complex z-plane for $A_n(z)/z$. The integrand has a pole at the origin with residue $A_n(0) = A_n$ (the physical amplitude) and simple poles at the factorization points z_I (Eq. (59.122)). If the large-circle integral vanishes, the sum of all residues is zero, expressing A_n as minus the sum of the factorization residues — the BCFW master formula Eq. (59.130).

Caution: the dropped boundary term

The derivation rests on one inequality not yet proved: $A_n(z) \to 0$ as $z \to \infty$. If instead $A_n(z)$ tends to a nonzero constant or grows, the large-circle integral in Eq. (59.126) contributes a "boundary term" or "pole at infinity" B_n, and the recursion becomes $A_n = B_n + \sum_{I,h} \hat{A}_L \frac{i}{P_I^2} \hat{A}_R$, with B_n *not* determined by lower-point on-shell data. Whether B_n vanishes depends on the shift (i, j) and on the helicities of the shifted legs. Establishing which shifts give $B_n = 0$ — and hence when Eq. (59.130) is exact as written — is the business of Section 59.6.

59.5.5 Worked example: the four-gluon MHV amplitude

We close by running the recursion once, in full, on the smallest nontrivial case: the color-ordered amplitude $A_4(1^-, 2^-, 3^+, 4^+)$. This is the Yang–Mills theory result in miniature — the four-gluon amplitude that the Feynman expansion produces only after summing several diagrams, rebuilt here from two three-point seeds and one propagator. The seeds are the complex three-point amplitudes Eq. (59.116),

$$A_3(a^-, b^-, c^+) = i\,g\,\frac{\langle a\,b\rangle^3}{\langle b\,c\rangle\,\langle c\,a\rangle}, \qquad A_3(a^+, b^+, c^-) = -\,i\,g\,\frac{[a\,b]^3}{[b\,c]\,[c\,a]}, \tag{59.131}$$

the first nonzero only for holomorphic (angle) kinematics, the second for anti-holomorphic (square) kinematics.

Worked Example 59.24 ($A_4(1^-, 2^-, 3^+, 4^+)$ from BCFW)**. Choice of shift.** Shift the adjacent legs 2 and 3 with an $[\,2, 3\,\rangle$ shift, i.e. $(i, j) = (2, 3)$:

$$|\hat{2}] = |2] + z|3], \qquad \left|\hat{3}\right\rangle = |3\rangle - z\,|2\rangle\,, \qquad \left|\hat{2}\right\rangle = |2\rangle\,, \quad |\hat{3}] = |3]. \tag{59.132}$$

The shifted helicities are $(h_2, h_3) = (-, +)$; this is a valid shift (its large-z behavior is established in Section 59.6), so Eq. (59.130) applies with no boundary term.

Which channels contribute. For four legs in cyclic order $1, 2, 3, 4$ the consecutive two-particle channels are $\{1, 2\}$ (equivalently $\{3, 4\}$) and $\{2, 3\}$ (equivalently $\{1, 4\}$). A channel contributes only if it separates the shifted legs 2 and 3. The block $\{2, 3\}$ keeps them together, so it gives no pole; only $\{1, 2\}$ contributes, with $I = \{1, 2\}$, $\bar{I} = \{3, 4\}$, and $P_I = p_1 + p_2$. The recursion is therefore a *single term*.

Internal kinematics. The pole is at $\hat{P}_{12}(z)^2 = 0$. Using Eq. (59.121) with the shift on the square spinor of 2,

$$\hat{P}_{12}(z)^2 = (p_1 + \hat{p}_2)^2 = \langle 1\,\hat{2}\rangle\,[\hat{2}\,1] = \langle 1\,2\rangle([2\,1] + z\,[3\,1]), \quad \Longrightarrow \quad z_I = -\frac{[2\,1]}{[3\,1]}. \tag{59.133}$$

The two sub-amplitudes are $\hat{A}_L(\hat{1}, \hat{2}, (-\hat{P})^{-h})$ on the legs $\{1, 2\}$ and $\hat{A}_R((\hat{P})^h, \hat{3}, \hat{4})$ on $\{3, 4\}$, with $\hat{P} \equiv \hat{P}_{12}(z_I)$ null. Since $\hat{1} = 1^-$ and $\hat{2} = 2^-$ carry two negative helicities, the left three-point amplitude is nonzero only as the angle-type seed $A_3(- - +)$ in Eq. (59.131), forcing the internal leg on the left to be $(-\hat{P})^+$ (so $h = -$ on the right). The other helicity assignment would make the left amplitude $A_3(\hat{1}^-, \hat{2}^-, (-\hat{P})^-)$, which has total helicity -3 and vanishes. Hence the helicity sum collapses to one term:

$$A_4 = \hat{A}_L(\hat{1}^-, \hat{2}^-, (-\hat{P})^+)\,\frac{i}{P_{12}^2}\,\hat{A}_R((\hat{P})^-, \hat{3}^+, \hat{4}^+), \qquad P_{12}^2 = \langle 1\,2\rangle[2\,1]. \tag{59.134}$$

The two seeds. From Eq. (59.131),

$$\hat{A}_L = i\,g\,\frac{\langle \hat{1}\,\hat{2}\rangle^3}{\langle \hat{2}\,(-\hat{P})\rangle\,\langle (-\hat{P})\,\hat{1}\rangle} = i\,g\,\frac{\langle 1\,2\rangle^3}{\langle \hat{2}\,(-\hat{P})\rangle\,\langle (-\hat{P})\,\hat{1}\rangle}, \tag{59.135}$$

$$\hat{A}_R = -\,i\,g\,\frac{[\hat{3}\,\hat{4}]^3}{[\hat{4}\,\hat{P}]\,[\hat{P}\,\hat{3}]} = -\,i\,g\,\frac{[3\,4]^3}{[4\,\hat{P}]\,[\hat{P}\,3]}, \tag{59.136}$$

where $\langle \hat{1}\,\hat{2}\rangle = \langle 1\,2\rangle$ and $[\hat{3}\,\hat{4}] = [3\,4]$ because those spinors are unshifted. For the internal leg of momentum $-\hat{P}$ we use the analytic continuation $|-\hat{P}\rangle = i\left|\hat{P}\right\rangle$, $|-\hat{P}] = i|\hat{P}]$ (so that $|-\hat{P}\rangle[-\hat{P}| = -\left|\hat{P}\right\rangle[\hat{P}| = -\hat{P}$); the factors of i cancel in the product because $\hat{P}$ appears an even number of times.

Eliminating the internal spinors. The product of the two denominators is

$$\begin{aligned} D &\equiv \langle \hat{2}\,(-\hat{P})\rangle\,\langle (-\hat{P})\,\hat{1}\rangle\,[\hat{P}\,3]\,[4\,\hat{P}] = -\,\langle 2\,\hat{P}\rangle\,\langle \hat{P}\,1\rangle\,[\hat{P}\,3]\,[4\,\hat{P}] \\ &= -\big(\langle 2\,\hat{P}\rangle[\hat{P}\,3]\big)\big(\langle 1\,\hat{P}\rangle[\hat{P}\,4]\big) = -\,\langle 2|\hat{P}|3]\,\langle 1|\hat{P}|4], \end{aligned} \tag{59.137}$$

where the first equality used $|-\hat{P}\rangle = i\left|\hat{P}\right\rangle$ (giving $i^2 = -1$), and the last regrouped the brackets using $\langle a\,\hat{P}\rangle[\hat{P}\,b] = \langle a|\hat{P}|b]$ (with $\langle \hat{P}\,1\rangle[4\,\hat{P}] = \langle 1\,\hat{P}\rangle[\hat{P}\,4]$ after two sign flips). Now collapse the internal momentum onto external data using $\hat{P} = p_1 + \hat{p}_2$ and $\langle a\,a\rangle = 0$:

$$\langle 2|\hat{P}|3] = \langle 2|p_1|3] = \langle 2\,1\rangle[1\,3], \qquad \langle 1|\hat{P}|4] = \langle 1|\hat{p}_2|4] = \langle 1\,2\rangle\,[\hat{2}\,4], \tag{59.138}$$

since $\langle 2|\hat{p}_2|3] \propto \langle 2\,2\rangle = 0$ and $\langle 1|p_1|4] \propto \langle 1\,1\rangle = 0$. The remaining shifted bracket, at $z = z_I$, is $[\hat{2}\,4] = [2\,4] + z_I[3\,4]$. Insert Eq. (59.133) and use the Schouten identity $[2\,4][3\,1] + [2\,3][1\,4] + [2\,1][4\,3] = 0$:

$$[2\,4] + z_I[3\,4] = \frac{[2\,4][3\,1] - [2\,1][3\,4]}{[3\,1]} = \frac{-\,[2\,3][1\,4]}{[3\,1]}, \tag{59.139}$$

where the numerator was simplified by the Schouten relation ($[2\,4][3\,1] = -[2\,3][1\,4] + [2\,1][3\,4]$). Assembling Eqs. (59.138) and (59.139) into Eq. (59.137),

$$D = -\,\langle 2\,1\rangle[1\,3]\cdot\langle 1\,2\rangle\cdot\frac{-[2\,3][1\,4]}{[3\,1]} = -\,\langle 1\,2\rangle^2\,[2\,3]\,[1\,4], \tag{59.140}$$

using $\langle 2\,1\rangle = -\langle 1\,2\rangle$ and $[1\,3]/[3\,1] = -1$.

Assembly. Combining Eqs. (59.135) and (59.136) with Eq. (59.140), the i's multiply to $i\cdot(-i) = +1$ and the couplings to $g\cdot g = g^2$, and

$$\hat{A}_L\,\hat{A}_R = g^2\,\frac{\langle 1\,2\rangle^3\,[3\,4]^3}{D} = g^2\,\frac{\langle 1\,2\rangle^3\,[3\,4]^3}{-\,\langle 1\,2\rangle^2\,[2\,3]\,[1\,4]} = -\,g^2\,\frac{\langle 1\,2\rangle\,[3\,4]^3}{[2\,3]\,[1\,4]}. \tag{59.141}$$

Multiplying by the propagator $i/P_{12}^2 = i/(\langle 1\,2\rangle[2\,1])$,

$$A_4 = -\,g^2\,\frac{\langle 1\,2\rangle\,[3\,4]^3}{[2\,3]\,[1\,4]}\cdot\frac{i}{\langle 1\,2\rangle[2\,1]} = -\,i\,g^2\,\frac{[3\,4]^3}{[2\,3]\,[1\,4]\,[2\,1]}. \tag{59.142}$$

Back to angle brackets. Equation Eq. (59.142) is the answer, but to recognize it we convert to angle brackets using momentum conservation $\sum_k |k\rangle\,[k| = 0$, i.e. $\sum_k \langle a\,k\rangle[k\,b] = 0$ for any a, b. Three instances suffice. From $a = 1, b = 3$: $\langle 1\,2\rangle[2\,3] = -\langle 1\,4\rangle[4\,3] = \langle 1\,4\rangle[3\,4]$, so $[2\,3] = \langle 1\,4\rangle[3\,4]/\langle 1\,2\rangle$. From $a = 2, b = 4$: $\langle 2\,3\rangle[3\,4] = -\langle 2\,1\rangle[1\,4] = \langle 1\,2\rangle[1\,4]$, so $[1\,4] = \langle 2\,3\rangle[3\,4]/\langle 1\,2\rangle$. From the $s_{12} = s_{34}$ relation $\langle 1\,2\rangle[2\,1] = \langle 3\,4\rangle[4\,3]$: $[2\,1] = -\langle 3\,4\rangle[3\,4]/\langle 1\,2\rangle$. Substituting all three into the denominator of Eq. (59.142),

$$[2\,3]\,[1\,4]\,[2\,1] = \frac{\langle 1\,4\rangle\langle 2\,3\rangle(-\langle 3\,4\rangle)\,[3\,4]^3}{\langle 1\,2\rangle^3} = -\,\frac{\langle 1\,4\rangle\,\langle 2\,3\rangle\,\langle 3\,4\rangle\,[3\,4]^3}{\langle 1\,2\rangle^3}, \tag{59.143}$$

so that the $[3\,4]^3$ cancels and

$$A_4(1^-, 2^-, 3^+, 4^+) = -\,i\,g^2\,\frac{[3\,4]^3\,\langle 1\,2\rangle^3}{-\,\langle 1\,4\rangle\,\langle 2\,3\rangle\,\langle 3\,4\rangle\,[3\,4]^3} = i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,1\rangle}, \tag{59.144}$$

using $\langle 1\,4\rangle = -\langle 4\,1\rangle$. This is precisely the Parke–Taylor form $A_4 = i\,g^2\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$ the $n = 4$ instance of the MHV amplitude. The amplitude that the Feynman rules deliver only after summing the s-channel, t-channel, and four-point-vertex diagrams of Section 59.2.3 has emerged from a single BCFW term built from two three-point seeds.

What the recursion bought

The computation in Theorem 59.24 used no four-point vertex, no s-/t-channel sum, and no gauge choice beyond the seed amplitudes. The entire content was: pick a shift, find the one channel that separates the shifted legs, multiply two three-point amplitudes, and divide by a propagator. At higher multiplicity the savings compound — an n-gluon amplitude reduces to a handful of products of strictly smaller amplitudes, each itself recursively determined — which is how the super-factorial diagram count of Table 59.1 collapses to compact closed forms. The recursion is the "smart way" promised at the start of the chapter, and it is built entirely from on-shell data.

With the master formula Eq. (59.130) in hand, only one promissory note remains: the contour at infinity. The derivation assumed $A_n(z) \to 0$ as $z \to \infty$, an assumption we have flagged but not justified. The next section analyzes the large-z scaling of shifted amplitudes directly, classifies which $[\,i, j\,\rangle$ shifts are valid (and which leave a boundary term), and thereby establishes that tree-level Yang–Mills is *on-shell constructible*: fixed entirely by its three-point seeds and the factorization just derived, with no Lagrangian required.

59.6 Large-z Behavior and On-Shell Constructibility

The BCFW master formula Theorem 59.23 delivered a recursion — the physical amplitude written as a sum over factorization channels of products of lower-point on-shell amplitudes — but it delivered it on credit. The derivation deformed two external momenta by a complex parameter z, treated the color-ordered partial amplitude $A_n(z)$ as a meromorphic function of z, and used Cauchy's theorem to trade the value at $z = 0$ for a sum of residues at the physical poles. That trade is exact *only* if the contour at infinity contributes nothing: only if the boundary term we provisionally called B_0 vanishes. Everything in the recursion — indeed the entire claim that gluon trees can be built without ever writing the Yang–Mills Lagrangian — rests on that one analytic fact. This section pays the debt. We compute how the shifted amplitude behaves as $z \to \infty$, find that for a correctly chosen shift it falls off as $1/z$, catalogue which shifts are good and which are fatally bad, and then harvest the structural payoff: tree-level Yang–Mills is *on-shell constructible*, fixed completely by its three-point seeds and the analytic structure that locality and unitarity impose, with no off-shell input whatsoever.

59.6.1 The boundary term: what must vanish

Recall the analytic setup of Theorem 59.23. We single out two legs, label them i and j, and deform their spinors holomorphically in a complex parameter z by the $[i, j\rangle$ shift

$$|\hat{i}] = |i] + z\,|j], \qquad |\hat{i}\rangle = |i\rangle, \qquad |\hat{j}\rangle = |j\rangle - z\,|i\rangle, \qquad |\hat{j}] = |j], \tag{59.145}$$

so that the shifted momenta are $\hat{p}_i(z) = p_i + z\,q$ and $\hat{p}_j(z) = p_j - z\,q$ with the null bispinor $q_{\alpha\dot{\alpha}} = \lambda_{i\alpha}\tilde{\lambda}_{j\dot{\alpha}}$, i.e. $q^\mu = \frac{1}{2}\langle i|\gamma^\mu|j]$. Both shifted momenta stay on shell, $\hat{p}_i^2(z) = \hat{p}_j^2(z) = 0$ for all z, and total momentum is conserved because the z-dependent pieces cancel. The deformed partial amplitude $A_n(z) \equiv A_n(\dots, \hat{i}, \dots, \hat{j}, \dots)$ is a rational function of z: its only singularities are simple poles, located where an internal propagator carrying a z-dependent momentum goes on shell, at the values z_I fixed by the shifted-pole condition Eq. (59.121).

The reconstruction rests on integrating $A_n(z)/z$ around a contour. Consider

$$\oint_{C_\infty} \frac{dz}{2\pi i}\,\frac{A_n(z)}{z}, \tag{59.146}$$

with C_∞ a circle of radius $R \to \infty$. The integrand is meromorphic, so by the residue theorem Eq. (59.146) equals the sum of all residues it encloses. The pole at $z = 0$ has residue $A_n(0) = A_n$, the physical amplitude we want. The poles at $z = z_I$ have residues $z_I^{-1}\,\mathrm{Res}_{z=z_I} A_n(z)$, and Theorem 59.23 identified each such residue with $-i\,\hat{A}_L(z_I)\,\hat{A}_R(z_I)$ divided by the unshifted Mandelstam invariant of

the channel, assembling into the BCFW terms. Therefore

$$A_n \;=\; \sum_{\text{channels } I} \sum_{h} \hat{A}_L(z_I)\,\frac{i}{P_I^2}\,\hat{A}_R(z_I) \;+\; B_0, \qquad B_0 \equiv \oint_{C_\infty} \frac{dz}{2\pi i}\,\frac{A_n(z)}{z}. \tag{59.147}$$

The boundary term B_0 is the contribution of the contour at infinity. Writing the Laurent expansion $A_n(z) = \sum_m c_m\, z^m$ valid for large $|z|$, the integral Eq. (59.146) picks out the coefficient of $1/z$ in $A_n(z)/z$, which is the coefficient of z^0 in $A_n(z)$:

$$B_0 = c_0 = \lim_{z\to\infty} A_n(z) \quad \text{(if this limit is finite)}, \qquad \text{and more generally } B_0 = -\operatorname{Res}_{z=\infty} \frac{A_n(z)}{z}. \tag{59.148}$$

The conclusion is sharp and is the whole game of this section: *the BCFW recursion holds with $B_0 = 0$ if and only if $A_n(z) \to 0$ as $z \to \infty$.* If instead $A_n(z)$ tends to a nonzero constant, B_0 is that constant and must be added by hand — a piece of information not captured by any physical factorization pole, hence not reconstructible from lower-point amplitudes. If $A_n(z)$ grows, the naive recursion is simply wrong. So we must determine the large-z growth of the shifted amplitude, and we must learn to choose i, j and their helicities so that it decays.

Definition 59.25 (Valid shift and on-shell constructibility)**.** A two-line shift Eq. (59.145) of an amplitude A_n is *valid* (or *good*) if the shifted amplitude vanishes at infinity, $A_n(z) \to 0$ as $z \to \infty$, so that the boundary term Eq. (59.148) is zero and Eq. (59.147) reduces to the boundary-free BCFW recursion. An amplitude is *on-shell constructible* if it admits at least one valid shift, so that it is determined entirely by the residues of its physical poles — that is, by lower-point on-shell amplitudes.

59.6.2 Scaling of the shifted spinors and polarization vectors

The first input is purely kinematic: how the building blocks of any diagram scale with z. From Eq. (59.145) the shifted spinors have definite, elementary scalings. On leg i (the leg whose *square* spinor is shifted) the square spinor grows linearly while the angle spinor is inert,

$$|\hat{i}] = |i] + z\,|j] = O(z), \qquad |\hat{i}\rangle = |i\rangle = O(1); \tag{59.149}$$

on leg j (the leg whose *angle* spinor is shifted) it is the other way around,

$$|\hat{j}\rangle = |j\rangle - z\,|i\rangle = O(z), \qquad |\hat{j}] = |j] = O(1). \tag{59.150}$$

These feed directly into the external polarization vectors. Each shifted leg carries a polarization built, as in Eq. (58.111) (recalled in Eq. (59.32)), from the leg's own spinors and a reference momentum q_ℓ, which is an arbitrary fixed null momentum encoding the residual gauge freedom and is *z-independent.* We read

off the large-z power of each polarization by counting where the shifted (growing) spinor appears.

For leg i, helicity $+$ uses $\varepsilon_+^\mu(\hat{\imath}; q_i) = \langle q_i|\gamma^\mu|\hat{\imath}]/(\sqrt{2}\,\langle q_i\, i\rangle)$: the growing $|\hat{\imath}]$ sits in the numerator and the denominator $\langle q_i\,\hat{\imath}\rangle = \langle q_i\, i\rangle$ is inert, so $\varepsilon_+(\hat{\imath}) = O(z)$. Helicity $-$ uses $\varepsilon_-^\mu(\hat{\imath}; q_i) = \langle\hat{\imath}|\gamma^\mu|q_i]/(\sqrt{2}\,[\hat{\imath}\, q_i])$: now the inert $|\hat{\imath}\rangle$ is in the numerator while the growing $[\hat{\imath}\, q_i] = [i\, q_i] + z[j\, q_i] = O(z)$ is in the denominator, so $\varepsilon_-(\hat{\imath}) = O(1/z)$. The same reasoning on leg j, where it is the angle spinor that grows, gives the mirror image. Collecting all four cases:

	ε_+	ε_-
leg i (square-shifted)	$O(z)$	$O(1/z)$
leg j (angle-shifted)	$O(1/z)$	$O(z)$

(59.151)

These powers are robust: the reference momenta enter only the inert factors, so no choice of q_i, q_j changes the exponent. The pattern already tells the qualitative story. A negative-helicity square-shifted leg and a positive-helicity angle-shifted leg both supply a *suppressing* factor $1/z$; their helicity opposites both supply an *enhancing* factor z. The shift labelled $[-,+\rangle$ — meaning $h_i = -$ on the square-shifted leg and $h_j = +$ on the angle-shifted leg — is thus doubly suppressed at the level of its two external polarizations, while the $[+,-\rangle$ shift is doubly enhanced. The remaining work is to combine these external powers with the z-dependence of the diagram's interior.

59.6.3 The hard line and the eikonal estimate

The interior of a diagram contributes its own powers of z through the propagators and vertices that the shifted momentum flows through. The organizing observation is that the z-dependence is carried by a single connected path. Only the momenta of legs i and j depend on z, and they do so through the single null direction q: $\hat{p}_i = p_i + zq$ injects $+zq$ at leg i and $\hat{p}_j = p_j - zq$ extracts it at leg j. By momentum conservation at every vertex, the large momentum zq must flow along an unbroken path of internal lines from i to j — the *hard line*. Every propagator off this path carries only soft (z-independent) momentum; every propagator on it carries momentum of the form $zq + (\text{soft})$, and is therefore large. Figure 59.6 depicts the picture: a hard line threading a soft background built from the other $n-2$ gluons.

We now count powers along the hard line, using the color-ordered Feynman rules Eqs. (59.12) to (59.14). Suppose the hard line passes through V_h vertices and hence through $V_h - 1$ internal hard propagators (the two ends being the external legs i and j). Each hard propagator scales as $1/z$: with internal momentum $\ell = zq + k$ for soft k, the denominator is

$$\ell^2 = (zq + k)^2 = 2z\, q\cdot k + k^2 = O(z), \tag{59.152}$$

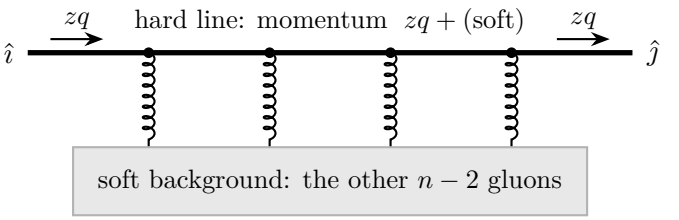


Figure 59.6: The large-z limit of a shifted diagram. The deformation injects a large null momentum zq at leg i and removes it at leg j; momentum conservation forces it to traverse a single *hard line* (thick) of large internal momenta, while the remaining $n-2$ gluons form a soft, z-independent background into which the hard line radiates. The leading large-z behavior is that of one energetic particle propagating through this background.

since $q^2 = 0$, so $-i\eta_{\mu\nu}/\ell^2 = O(1/z)$. Each cubic vertex on the hard line is linear in momentum Eq. (59.12), and its leading large-z piece is the one in which the large momentum zq occupies the momentum slot, giving $O(z)$; a quartic vertex carries no momentum and is $O(1)$. The most enhanced topology is the purely cubic one, in which each of the V_h vertices contributes $O(z)$. The interior power is therefore

$$\underbrace{z^{V_h}}_{\text{vertices}} \times \underbrace{z^{-(V_h-1)}}_{\text{propagators}} = z^{+1}, \tag{59.153}$$

independent of the length V_h of the hard line. The cancellation in Eq. (59.153) is the eikonal phenomenon: an energetic particle plowing through a soft background gains one power of vertex momentum for each propagator it saves, and the two effects balance to leave a single overall power of z. Moreover, to leading order each cubic vertex on the hard line acts as the eikonal vertex $\frac{i}{\sqrt{2}}\,(2zq)^\rho\,\eta^{\mu\nu}$: it passes the hard-line Lorentz indices straight through with a metric $\eta^{\mu\nu}$ and couples to the soft gluon of index ρ through the current zq^ρ. Threaded through the metric propagators, all the hard-line indices are simply identified, and the two external polarizations are contracted directly into one another. The leading large-z behavior of any single diagram is therefore bounded by

$$A_n(z)\Big|_{\text{leading}} \;\sim\; z\;\varepsilon_i(z)\cdot\varepsilon_j(z), \tag{59.154}$$

the product of the interior power Eq. (59.153) with the two external polarization scalings of Eq. (59.151), contracted through the eikonal metric.

We must compute $\varepsilon_i(z)\cdot\varepsilon_j(z)$ for each helicity assignment. This is exactly the polarization dot-product Theorem 59.8 of Section 59.2.2, now evaluated on the shifted spinors Eqs. (59.149) and (59.150). Take first the case the recursion needs, $[-,+\rangle$: leg i negative, leg j positive. By Eq. (59.37) (with the positive leg

j and negative leg i),

$$\varepsilon_-(\hat{\imath};q_i)\cdot\varepsilon_+(\hat{\jmath};q_j)=\frac{\langle q_j\,\hat{\imath}\rangle\,[q_i\,\hat{\jmath}]}{\langle q_j\,\hat{\jmath}\rangle\,[\hat{\imath}\,q_i]}=\frac{\langle q_j\,i\rangle\,[q_i\,j]}{(\langle q_j\,j\rangle-z\langle q_j\,i\rangle)\,([i\,q_i]+z[j\,q_i])}=O(1/z^2), \tag{59.155}$$

where we used $|\hat{\imath}\rangle=|i\rangle$, $|\hat{\jmath}]=|j]$ in the numerator and the growing pieces $\langle q_j\,\hat{\jmath}\rangle,[\hat{\imath}\,q_i]=O(z)$ in the denominator. Combining with Eq. (59.154),

$$A_n(z)\Big|_{[-,+\rangle}\;\sim\; z\cdot O(1/z^2)=O(1/z)\;\xrightarrow{z\to\infty}\;0. \tag{59.156}$$

This is the central computation of the section: *the* $[-,+\rangle$ *shift makes every diagram fall off at least as* $1/z$, so the full $A_n(z)\to 0$ and, by Eq. (59.148), the boundary term vanishes. Crucially, because each diagram is already bounded by a decaying power, the conclusion needs no cancellation between diagrams: summing terms each of size $O(1/z)$ gives a sum of size $O(1/z)$.

The other three assignments are computed the same way. For $[+,-\rangle$ (leg i positive, leg j negative), Eq. (59.37) with the roles reversed gives

$$\varepsilon_+(\hat{\imath};q_i)\cdot\varepsilon_-(\hat{\jmath};q_j)=\frac{\langle q_i\,\hat{\jmath}\rangle\,[q_j\,\hat{\imath}]}{\langle q_i\,\hat{\imath}\rangle\,[\hat{\jmath}\,q_j]}=\frac{(\langle q_i\,j\rangle-z\langle q_i\,i\rangle)\,([q_j\,i]+z[q_j\,j])}{\langle q_i\,i\rangle\,[j\,q_j]}=O(z^2), \tag{59.157}$$

so $A_n(z)|_{[+,-\rangle}\sim z\cdot O(z^2)=O(z^3)$, a fast *growth*: the $[+,-\rangle$ shift is invalid. For the like-helicity shifts the eikonal estimate is borderline. Using Eq. (59.35) for $[+,+\rangle$ and Eq. (59.36) for $[-,-\rangle$ on the shifted spinors,

$$\varepsilon_+(\hat{\imath})\cdot\varepsilon_+(\hat{\jmath})=\frac{\langle q_i\,q_j\rangle\,[\hat{\jmath}\,\hat{\imath}]}{\langle q_i\,\hat{\imath}\rangle\,\langle q_j\,\hat{\jmath}\rangle}=\frac{\langle q_i\,q_j\rangle\,[j\,i]}{\langle q_i\,i\rangle\,(\langle q_j\,j\rangle-z\langle q_j\,i\rangle)}=O(1/z), \tag{59.158}$$

$$\varepsilon_-(\hat{\imath})\cdot\varepsilon_-(\hat{\jmath})=\frac{\langle\hat{\imath}\,\hat{\jmath}\rangle\,[q_j\,q_i]}{[\hat{\imath}\,q_i]\,[\hat{\jmath}\,q_j]}=\frac{\langle i\,j\rangle\,[q_j\,q_i]}{([i\,q_i]+z[j\,q_i])\,[j\,q_j]}=O(1/z), \tag{59.159}$$

where in Eq. (59.158) we used $[\hat{\jmath}\,\hat{\imath}]=[j\,\hat{\imath}]=[j\,i]+z[j\,j]=[j\,i]$ (the would-be growing term killed by $[j\,j]=0$), and similarly $\langle\hat{\imath}\,\hat{\jmath}\rangle=\langle i\,j\rangle$ in Eq. (59.159). Both give $A_n(z)\sim z\cdot O(1/z)=O(1)$: the eikonal estimate places these two shifts on the boundary of validity, neither manifestly decaying nor growing. To settle them we need an argument finer than the diagram-by-diagram count.

59.6.4 The background-field argument and the catalogue of shifts

The eikonal estimate Eq. (59.154) is an *upper bound* per diagram, and for the like-helicity shifts it is not saturated: the leading $O(1)$ piece cancels in the gauge-invariant sum, leaving the true $1/z$ falloff. To see this one must work with the full amplitude rather than diagram by diagram, and the cleanest tool is the background-field method of Arkani-Hamed and Kaplan. The idea is to take the large-z limit literally as a physical limit: legs i and j become hard, with

momenta $\hat{p}_i \approx +zq$ and $\hat{p}_j \approx -zq$, while the remaining $n-2$ gluons are soft. The amplitude in this limit is computed by quantizing the hard fluctuation in the soft classical background A_μ formed by the other gluons.

Expanding the Yang–Mills action of Eq. (25.76) to quadratic order in the hard fluctuation a_μ about the background A_μ, and fixing background Feynman gauge, the quadratic operator governing the hard line is

$$\Delta_{\mu\nu} = D^2\,\eta_{\mu\nu} - 2ig\,F_{\mu\nu}, \tag{59.160}$$

where $D_\mu = \partial_\mu - igA_\mu$ is the background covariant derivative and $F_{\mu\nu}$ the background field strength, both built from the soft gluons. The two terms have a transparent meaning. The first, $D^2\eta_{\mu\nu}$, is the scalar (minimal) coupling: it propagates the hard particle as if it were a charged scalar, passing its Lorentz indices through unchanged. The second, $-2igF_{\mu\nu}$, is the *gyromagnetic* (spin) coupling, the magnetic moment $g=2$ of the Yang–Mills gluon, which rotates the hard particle's polarization as it traverses the background. The hard-line propagator is the inverse of Eq. (59.160), expanded for large momentum $\hat{\ell} = zq + \cdots$:

$$\frac{1}{\Delta} = \frac{1}{D^2}\,\eta_{\mu\nu} + \frac{1}{D^2}\,(2igF)_\mu{}^\rho\,\frac{1}{D^2}\,\eta_{\rho\nu} + \cdots = \underbrace{\frac{\eta_{\mu\nu}}{D^2}}_{O(1/z)} + \underbrace{\frac{2igF_{\mu\nu}}{(D^2)^2}}_{O(1/z^2)} + \cdots, \tag{59.161}$$

where $D^2 \sim 2z\,q\cdot\partial = O(z)$ on the hard line and the soft $F_{\mu\nu}$ is $O(1)$. The leading piece is the scalar one, $\eta_{\mu\nu}/z$; the spin piece is suppressed by a further $1/z$.

The helicity dependence enters through how the external polarizations of the hard legs contract with this structure. The decisive fact is that the scalar (eikonal) piece $\eta_{\mu\nu}/z$ contracts the two hard polarizations directly, $\varepsilon_i\cdot\varepsilon_j$, reproducing the borderline estimate Eq. (59.154); but for the like-helicity and the $[-,+\rangle$ configurations this leading scalar contraction is itself suppressed or vanishing (as the explicit dot products Eqs. (59.155), (59.158) and (59.159) showed, it is at most $O(1/z)$ in three of the four cases), so the amplitude is controlled by the spin piece $F_{\mu\nu}/z^2$ acting on polarizations. Carrying the expansion through, Arkani-Hamed and Kaplan find that for a spin-1 hard line the gauge symmetry forces the leading would-be-bad terms to cancel for every helicity assignment *except* the one in which both hard legs have their polarizations maximally enhanced, recovering

$$A_n(z) \xrightarrow{z\to\infty} \begin{cases} O(1/z), & [-,+\rangle,\ [+,+\rangle,\ [-,-\rangle \quad \text{(valid)},\\ O(z^3), & [+,-\rangle \quad \text{(invalid)}. \end{cases} \tag{59.162}$$

The three valid cases all decay as $1/z$ (the like-helicity ones thanks to the gyromagnetic cancellation that the bare eikonal count missed); the single bad case grows as z^3, the power already exposed by the manifest enhancement in Eq. (59.157). We summarize in Table 59.2.

shift $[h_i, h_j\rangle$	large-z behavior	status
$[-,+\rangle$	$1/z$	valid (canonical good shift)
$[-,-\rangle$	$1/z$	valid
$[+,+\rangle$	$1/z$	valid
$[+,-\rangle$	z^3	invalid (boundary term survives)

Table 59.2: Large-z behavior of the shifted color-ordered gluon amplitude under the $[i,j\rangle$ shift Eq. (59.145), classified by the helicities (h_i, h_j) of the square-shifted leg i and the angle-shifted leg j. Three of the four assignments fall off and are valid for BCFW; only $[+,-\rangle$, which enhances both external polarizations, grows and is invalid. The $[-,+\rangle$ shift is the canonical choice: its falloff is visible diagram by diagram, with no appeal to cancellations.

The bad shift is bad for a reason

It is tempting to read Table 59.2 as a rule to memorize, but the structure is physical. A shift fails precisely when it pumps the spins of *both* deformed legs in the enhancing direction simultaneously — ε_+ on the square-shifted leg and ε_- on the angle-shifted leg, each $O(z)$ by Eq. (59.151). Then even the gyromagnetic cancellation cannot rescue the growth, and $A_n(z) \sim z^3$. The surviving B_0 is then a genuine piece of data — a polynomial in the external kinematics that no factorization pole sees — and a recursion that ignores it returns the wrong answer. The lesson carries beyond gluons: in any theory, the validity of a shift is a statement about the spin and the high-energy (ultraviolet) behavior of the fields being shifted, and must be checked, not assumed.

Notation Watch

The background-field argument leading to Eq. (59.162) assumes the two-derivative Yang–Mills action with its $g = 2$ gyromagnetic ratio and a background gauge fixing; it is exactly these features that produce the operator Eq. (59.160) and the spin cancellation. The same argument, applied to the two-derivative Einstein–Hilbert action, gives the gravitational shift its own (milder, $1/z^2$) falloff, used in Chapter 61. The argument can fail for non-minimally coupled theories, for higher-derivative operators, or for scalars with generic potentials, where the high-energy behavior is different; in such cases a nonzero B_0 can appear and the amplitude is not constructible by a two-line shift alone. For the pure-gluon tree amplitudes that carry Yang–Mills theory, the falloff Eq. (59.162) is exact and the $[-,+\rangle$ shift is always available, as the next subsection establishes.

The explicit four-gluon computation below confirms both ends of the catalogue — the $1/z$ falloff of the good shift and the z^3 growth of the bad one — on the carried Yang–Mills theory amplitude.

Worked Example 59.26 (Large-z behavior of the four-gluon MHV amplitude)**.** We test Eq. (59.162) directly on the color-ordered MHV amplitude $A_4[1^-,2^-,3^+,4^+]$. We are permitted to use its closed form, the Parke–Taylor expression $A_4 = i\,g^2\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$ established in Eq. (59.55) of an earlier section (and proved at $n=4$ in Eq. (59.81)), purely as an independent check — the present argument does not assume it. Simplifying one power,

$$A_4[1^-,2^-,3^+,4^+] = i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,1\rangle}. \tag{59.163}$$

The good shift $[-,+\rangle$. Shift the adjacent pair $(2^-,3^+)$ with $i=2$ (negative, square-shifted) and $j=3$ (positive, angle-shifted): $|\hat{2}] = |2] + z|3]$, $|\hat{2}\rangle = |2\rangle$, $|\hat{3}\rangle = |3\rangle - z|2\rangle$, $|\hat{3}] = |3]$. Only angle brackets containing leg 3 change, and only through $|\hat{3}\rangle = |3\rangle - z|2\rangle$:

$$\langle 1\,2\rangle \to \langle 1\,2\rangle, \quad \langle 2\,3\rangle \to \langle 2\,\hat{3}\rangle = \langle 2\,3\rangle - z\langle 2\,2\rangle = \langle 2\,3\rangle, \quad \langle 3\,4\rangle \to \langle \hat{3}\,4\rangle = \langle 3\,4\rangle - z\langle 2\,4\rangle, \quad \langle 4\,1\rangle \to \langle 4 \tag{59.164}$$

Hence

$$A_4(z) = i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,(\langle 3\,4\rangle - z\langle 2\,4\rangle)\,\langle 4\,1\rangle} \;\sim\; \frac{1}{z} \xrightarrow{z\to\infty} 0, \tag{59.165}$$

exactly the $1/z$ falloff of Table 59.2. The boundary term vanishes and the $(2,3)$ shift reconstructs A_4.

The bad shift $[+,-\rangle$. Now shift the same pair the wrong way: $i=3$ (positive, square-shifted) and $j=2$ (negative, angle-shifted), so $|\hat{3}] = |3] + z|2]$, $|\hat{3}\rangle = |3\rangle$, $|\hat{2}\rangle = |2\rangle - z|3\rangle$, $|\hat{2}] = |2]$. Now angle brackets containing leg 2 change through $|\hat{2}\rangle = |2\rangle - z|3\rangle$:

$$\langle 1\,2\rangle \to \langle 1\,\hat{2}\rangle = \langle 1\,2\rangle - z\langle 1\,3\rangle, \quad \langle 2\,3\rangle \to \langle \hat{2}\,3\rangle = \langle 2\,3\rangle - z\langle 3\,3\rangle = \langle 2\,3\rangle, \quad \langle 3\,4\rangle \to \langle 3\,4\rangle, \quad \langle 4\,1\rangle \to \langle 4 \tag{59.166}$$

The numerator $\langle 1\,2\rangle^3 \to (\langle 1\,2\rangle - z\langle 1\,3\rangle)^3$ grows as z^3 while the denominator's only shifted factor, $\langle 1\,2\rangle$, was already cancelled in passing from $\langle 1\,2\rangle^4$ to $\langle 1\,2\rangle^3$. Therefore

$$A_4(z) = i\,g^2\,\frac{\big(\langle 1\,2\rangle - z\langle 1\,3\rangle\big)^3}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,1\rangle} \;\sim\; z^3 \xrightarrow{z\to\infty} \infty, \tag{59.167}$$

exactly the z^3 growth of Table 59.2. A recursion built on this shift would miss the entire $z^3 + z^2 + z^1 + z^0$ polynomial in Eq. (59.167), of which the z^0 term is precisely the boundary contribution B_0; it would return nonsense. The two computations, on the very same amplitude, make the catalogue concrete: the only thing that distinguishes the valid recursion from the invalid one is the assignment of helicities to the shifted legs.

59.6.5 On-shell constructibility of tree Yang–Mills

With the boundary term killed for the $[-,+\rangle$ shift, the recursion is unpaid no longer, and we can state the structural theorem that the whole chapter has been driving toward.

Theorem 59.27 (On-shell constructibility of tree Yang–Mills)**.** *Every color-ordered n-gluon tree amplitude ($n \geq 4$) is on-shell constructible. It admits a valid $[-,+\rangle$ shift, for which the boundary term Eq. (59.148) vanishes and the BCFW recursion Theorem 59.23 holds without a boundary term:*

$$A_n[1,\dots,n] = \sum_{\text{channels } I} \sum_{h} \hat{A}_L(z_I)\,\frac{i}{P_I^2}\,\hat{A}_R(z_I), \tag{59.168}$$

the sum running over the factorization channels compatible with the cyclic order and over the helicity h of the internal on-shell gluon. Iterating, A_n is determined entirely by the three-point amplitudes A_3^{MHV} and $A_3^{\overline{\mathrm{MHV}}}$ of Eq. (59.116), fixed by little-group scaling (Eq. (58.98)) alone. No Yang–Mills Lagrangian, no Feynman rules, and no off-shell field enters the reconstruction.

Proof. A valid shift always exists. The vanishing theorems Theorems 59.9 and 59.10 of Section 59.2.4 show that an amplitude with fewer than two negative or fewer than two positive helicities is identically zero, hence trivially constructible. So take a nonvanishing amplitude: its number k of negative-helicity legs satisfies $2 \leq k \leq n-2$, so it carries at least two negative and at least two positive legs. Reading the cyclic order $1,2,\dots,n$ around the circle, the helicity label cannot be constant (both signs occur), so somewhere two cyclically adjacent legs carry opposite helicities: a negative leg a next to a positive leg b. Choose the shift $[a,b\rangle$, i.e. square-shift the negative leg a and angle-shift the positive leg b. This is a $[-,+\rangle$ shift, so by Eq. (59.162) (established diagram by diagram in Eq. (59.156)) the shifted amplitude falls off as $1/z$ and the boundary term Eq. (59.148) vanishes. Hence Eq. (59.147) reduces to Eq. (59.168).

The recursion lowers the point count. In each term of Eq. (59.168), the channel I splits the n legs into two nonempty groups, one containing the shifted leg a and the other b; the sub-amplitudes $\hat{A}_L$ and $\hat{A}_R$ are color-ordered on-shell amplitudes evaluated on shifted but on-shell momenta, each with strictly fewer than n legs (the internal line replaces all the legs on the other side of the channel by a single on-shell gluon). The Mandelstam factor $1/P_I^2$ is the unshifted invariant of the channel, a physical pole.

Induction on the number of legs. We prove that every on-shell color-ordered gluon tree amplitude is a rational function of the spinors of its external legs, fixed by the three-point seeds. The base case $n=3$ is the pair of complex-momentum three-point amplitudes $A_3^{\mathrm{MHV}}(1^-,2^-,3^+) = i\,g\,\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$ and its parity conjugate $A_3^{\overline{\mathrm{MHV}}}(1^+,2^+,3^-) = -\,i\,g\,[1\,2]^3/([2\,3][3\,1])$, which Eq. (59.116) fixed

from little-group scaling and locality with no Lagrangian input. For the inductive step, assume every amplitude with fewer than n legs is so determined. Any n-point amplitude admits the valid shift just constructed, so by Eq. (59.168) it equals a finite sum of products $\hat{A}_L\,(i/P_I^2)\,\hat{A}_R$ of amplitudes with fewer than n legs, each of which is determined by the seeds by the inductive hypothesis. Therefore A_n is too. By induction every tree amplitude is fixed by A_3^{MHV} and $A_3^{\overline{\mathrm{MHV}}}$.

No Lagrangian enters the reconstruction. The only ingredients used in Eq. (59.168) and the induction are: (i) that $A_n(z)$ is rational with only simple poles at the physical factorization channels — a consequence of locality and unitarity, namely that an amplitude's singularities are exactly where intermediate states go on shell and that it factorizes there into on-shell sub-amplitudes; (ii) the three-point seeds, fixed by Poincaré symmetry (little-group scaling) alone; and (iii) the large-z falloff of the $[-,+\rangle$ shift. Of these only (iii) referred to the Yang–Mills vertices, and even there the background-field analysis of Section 59.6.4 shows the falloff follows from the spin-1, two-derivative, gauge-invariant character of the interaction rather than from any specific off-shell completion. The Lagrangian is never used to generate the amplitude; it is at most a guarantor of the high-energy behavior. This is the precise content of the claim that perturbative gauge theory is built from locality and unitarity, not from a Lagrangian. □

Locality and unitarity as the real inputs

Theorem 59.27 inverts the usual logic of field theory. In the textbook route one postulates a Lagrangian, derives Feynman rules, and computes amplitudes; the amplitude is an output, the Lagrangian the input. Constructibility says that for tree-level gauge theory the amplitude is fixed by far less: by where it is allowed to have poles (locality: poles only on physical channels) and by what it must become there (unitarity: it factorizes into on-shell amplitudes), seeded by the three-point amplitudes that symmetry alone determines. The infinitely many gauge-redundant off-shell configurations of the Lagrangian — the very source of the factorial diagram explosion of Table 59.1 — are revealed as scaffolding that the final answer does not remember. The hidden simplicity of Parke–Taylor, which the next section proves, is then not a miracle of cancellation but the inevitable shape of the unique function with the right poles, residues, and little-group weights.

We have now discharged the one outstanding debt of Theorem 59.23: for the $[-,+\rangle$ shift the contour at infinity contributes nothing, the BCFW recursion Eq. (59.168) stands on its own, and tree Yang–Mills is on-shell constructible from its three-point seeds. The recursion is now a fully justified, boundary-term-free machine. The next section turns it on: it proves the Parke–Taylor formula by

induction on multiplicity using exactly this recursion, and then runs the machine on explicit amplitudes, contrasting the one-line BCFW computation with the Feynman labor it replaces.

59.7 Proof of Parke–Taylor and Worked Examples

The previous section closed the conceptual circle: with the boundary term at infinity shown to vanish for a valid shift, the contour-deformation identity became an honest recursion, and tree-level Yang–Mills was declared *on-shell constructible* — every gluon tree is fixed by the three-point seeds Eq. (59.116) and the factorization poles, with no reference to the Lagrangian at all. That is a licence to compute, and it is time to spend it. We have a one-line conjecture for the entire maximally-helicity-violating sector, the Parke–Taylor formula Eq. (59.55), and we have a recursion, Theorem 59.23, that manufactures any tree from smaller ones. In this final section we put the two together: we *prove* Parke–Taylor by induction on the number of gluons, run the recursion explicitly on the four- and five-gluon amplitudes to watch a thousand-diagram computation collapse to a single term, assemble the first genuinely non-MHV amplitude (the six-point NMHV) from MHV and $\overline{\text{MHV}}$ pieces, and step back to state what the whole construction has taught: that locality and unitarity, not an action, are the data that fix the perturbative S-matrix. This completes the Yang–Mills theory layer of the chapter.

Throughout we use the conventions already fixed in this chapter: all n momenta outgoing and massless, angle and square brackets with $2p_i\cdot p_j = \langle i\,j\rangle[j\,i]$ (Eqs. (58.20) and (58.24)), color-ordered partial amplitudes A_n, and the helicity grading of Theorem 59.12. We recall the three objects the proof leans on, each established elsewhere in the chapter.

The seeds. The two complex-momentum three-point amplitudes (Eq. (59.116)) are

$$A_3^{\text{MHV}}(i^-,j^-,k^+) = i\,g\,\frac{\langle i\,j\rangle^3}{\langle j\,k\rangle\langle k\,i\rangle}, \qquad A_3^{\overline{\text{MHV}}}(i^-,j^+,k^+) = -\,i\,g\,\frac{[j\,k]^3}{[k\,i][i\,j]}. \tag{59.169}$$

The first lives where the angle brackets are generic and the square brackets all vanish (the *holomorphic*, or MHV, branch of the singular three-point kinematics); the second is its parity conjugate, supported where the angle brackets vanish.

The target. The Parke–Taylor formula Eq. (59.55) asserts that the color-ordered n-gluon MHV amplitude, with negative-helicity legs i and j and all others positive, is

$$A_n^{\text{MHV}} = i\,g^{\,n-2}\,\frac{\langle i\,j\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n-1\,n\rangle\langle n\,1\rangle}, \tag{59.170}$$

a single ratio whose denominator is the cyclic product of nearest-neighbour angle brackets and whose numerator depends only on the two negative legs.

The machine. The recursion of Theorem 59.23, for the $[\,i,j\,\rangle$ shift

$$|\hat{\imath}\,] = |i] + z\,|j], \qquad |\hat{\jmath}\,\rangle = |j\rangle - z\,|i\rangle, \qquad |\hat{\imath}\,\rangle = |i\rangle, \quad |\hat{\jmath}\,] = |j], \tag{59.171}$$

reconstructs the amplitude as a sum over the factorization channels I that separate the two shifted legs,

$$A_n = \sum_I \sum_{h=\pm} \hat{A}_L(z_I)\,\frac{i}{P_I^2}\,\hat{A}_R(z_I), \tag{59.172}$$

where $P_I = \sum_{a\in I} p_a$ is the unshifted channel momentum, the internal gluon is summed over its two helicities h (entering $\hat{A}_L$ as $\pm h$ and $\hat{A}_R$ as $\mp h$), and both subamplitudes are evaluated at the complex shift parameter z_I that puts the internal line on shell, $\hat{P}_I(z_I)^2 = 0$. The constructibility theorem Theorem 59.27 guarantees Eq. (59.172) is exact: no term at infinity is dropped.

59.7.1 The single surviving channel

The whole proof turns on one observation: when Eq. (59.172) is applied to an MHV amplitude with a wisely chosen shift, all but one of the channels die, and in the survivor one of the two subamplitudes is the three-point $\overline{\text{MHV}}$ seed. We isolate that statement before grinding the algebra.

Take the MHV amplitude with the two negative-helicity legs adjacent and placed first,

$$A_n \equiv A_n(1^-, 2^-, 3^+, 4^+, \dots, n^+), \tag{59.173}$$

and apply the $[\,1,2\,\rangle$ shift of Eq. (59.171). The shifted legs $\hat{1},\hat{2}$ carry helicities $(h_1,h_2) = (-,-)$; this is one of the shifts the large-z analysis of Theorem 59.27 certified as valid (the forbidden pattern was $(+,-)$), so Eq. (59.172) holds with no boundary term. Because the amplitude is color-ordered, its only poles sit in the *consecutive* channels, and a channel contributes to the $[\,1,2\,\rangle$ recursion only if it puts $\hat{1}$ and $\hat{2}$ on opposite sides of the cut. The consecutive sets containing leg 2 but not leg 1 are

$$I_k = \{2,3,\dots,k\}, \qquad k = 3,4,\dots,n-1, \tag{59.174}$$

each splitting the amplitude into

$$\hat{A}_L = A_k(\hat{2},3,\dots,k,-\hat{P}_k), \qquad \hat{A}_R = A_{n-k+2}(\hat{P}_k,k+1,\dots,n,\hat{1}), \tag{59.175}$$

with internal momentum $\hat{P}_k = \hat{p}_2 + p_3 + \cdots + p_k$ and a sum over the helicity h of the internal line.

Now count negative helicities. The external negative legs are exactly 1 and 2; leg 2 sits in $\hat{A}_L$, leg 1 in $\hat{A}_R$. For a partial amplitude with four or more legs, the vanishing theorems of Theorems 59.9 and 59.10 forbid the all-plus and single-minus configurations. Apply this to each side:

- If the internal line enters $\hat{A}_L$ with positive helicity, then $\hat{A}_L$ has the single negative leg 2 and otherwise only positive legs — it is single-minus and *vanishes* unless it is a three-point amplitude. A three-point $\hat{A}_L$ requires $I_k = \{2, 3\}$, i.e. $k = 3$.
- If instead the internal line enters $\hat{A}_L$ negatively, it enters $\hat{A}_R$ positively; then $\hat{A}_R$ has the single negative leg 1 and otherwise only positive legs — single-minus, *vanishing* unless three-point, which needs $n - k + 2 = 3$, i.e. $k = n - 1$.

So only the two endpoint channels, $k = 3$ and $k = n - 1$, can survive, and in each the small side is forced to be a three-point seed. The two are not on equal footing, because a three-point amplitude is nonzero only on the branch selected by the on-shell value z_I — and that branch decides whether the seed is MHV or $\overline{\text{MHV}}$. We check both.

The channel $k = n - 1$ is empty. Here $\hat{A}_R = A_3(\hat{P}_{n-1}, n, \hat{1})$ with $P_{n-1} = \hat{p}_2 + \cdots + p_{n-1} = -(p_n + \hat{p}_1)$. Using $|\hat{1}\,] = |1] + z|2]$ and $|\hat{1}\,\rangle = |1\rangle$,

$$\hat{P}_{n-1}^2 = (p_n + \hat{p}_1)^2 = 2\, p_n \cdot \hat{p}_1 = \langle n\, 1\rangle\, [\hat{1}\, n] = \langle n\, 1\rangle\big([1\, n] + z[2\, n]\big), \tag{59.176}$$

which vanishes at $z = z_{n-1} = -[1\, n]/[2\, n]$. At that point $[\hat{1}\, n] = 0$, so the *square* spinors of the three-point legs are proportional: this is the holomorphic branch, on which only an MHV three-point amplitude is nonzero. But the helicity content of $\hat{A}_R$ at $k = n - 1$ that is allowed by the counting above has the internal line negative on the $\hat{A}_R$ side, giving legs $(\hat{P}_{n-1}^-, n^+, \hat{1}^-)$ — one positive leg, hence $\overline{\text{MHV}}$, which vanishes on the holomorphic branch. (Equivalently: the only way to make a nonzero MHV three-point amplitude here would need two negative legs among $\hat{P}_{n-1}, n, \hat{1}$, but then $\hat{A}_L$ would carry only leg 2 as a negative and be single-minus.) Either way the channel contributes nothing.

The channel $k = 3$ survives. Here the small side is $\hat{A}_L = A_3(\hat{2}, 3, -\hat{P}_3)$ with $P_3 = \hat{p}_2 + p_3$ and $\hat{p}_2 = |\hat{2}\,\rangle[2| = (|2\rangle - z|1\rangle)[2| = p_2 - z\,|1\rangle[2|$, so

$$\hat{P}_3^2 = (\hat{p}_2 + p_3)^2 = 2\, \hat{p}_2 \cdot p_3 = \langle \hat{2}\, 3\rangle\, [3\, 2] = \big(\langle 2\, 3\rangle - z\langle 1\, 3\rangle\big)[3\, 2], \tag{59.177}$$

which vanishes at

$$z_* = \frac{\langle 2\, 3\rangle}{\langle 1\, 3\rangle}. \tag{59.178}$$

At z_* the angle bracket $\langle \hat{2}\, 3\rangle = \langle 2\, 3\rangle - z_*\langle 1\, 3\rangle = 0$, so the *angle* spinors of the three legs are proportional: this is the anti-holomorphic branch, on which the nonzero three-point amplitude is the $\overline{\text{MHV}}$ seed. Its helicity content must therefore be one negative leg. The internal line enters $\hat{A}_L$ positively (the case that did not

already die above), giving $\hat{A}_L = A_3^{\overline{\text{MHV}}}(\hat{2}^-, 3^+, -\hat{P}_3^+)$ — exactly one negative leg, consistent with the branch. The other side is then

$$\hat{A}_R = A_{n-1}^{\text{MHV}}(\hat{P}_3^-, 4^+, 5^+, \dots, n^+, \hat{1}^-), \tag{59.179}$$

an $(n-1)$-point amplitude with exactly two negative legs, $\hat{P}_3$ and $\hat{1}$, which are cyclically adjacent in its ordering. This is again MHV, and on it the induction hypothesis bites.

We have reduced the recursion Eq. (59.172) to a single term: the amplitude A_n of Eq. (59.173) equals the product of the $\overline{\text{MHV}}_3$ seed, the propagator i/P_3^2, and the $(n-1)$-point MHV amplitude Eq. (59.179), all evaluated at z_*.

59.7.2 The inductive proof

Theorem 59.28 (Parke–Taylor by induction)**.** *For every $n \geq 3$ the color-ordered MHV amplitude with adjacent negative legs $1, 2$ is*

$$A_n(1^-, 2^-, 3^+, \dots, n^+) = i\, g^{\,n-2}\, \frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle \cdots \langle n\,1\rangle}, \tag{59.180}$$

and hence, by cyclic relabelling, Eq. (59.170) holds for any pair of adjacent negative legs. The general (non-adjacent) case follows by Theorem 59.29 below.

Proof. Base case. For $n = 3$ the formula Eq. (59.180) reads $i\,g\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,1\rangle) = i\,g\,\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$, which is precisely the MHV seed $A_3^{\text{MHV}}(1^-, 2^-, 3^+)$ of Eq. (59.169). The base case is the seed itself.

Inductive step. Assume Eq. (59.170) for all MHV amplitudes with fewer than n legs. By Section 59.7.1 the $[\,1, 2\,\rangle$ recursion collapses to the single $k = 3$ term,

$$A_n = \hat{A}_L\, \frac{i}{P_3^2}\, \hat{A}_R\Big|_{z=z_*}, \qquad P_3^2 = (p_2 + p_3)^2 = \langle 2\,3\rangle[3\,2]. \tag{59.181}$$

We evaluate each factor.

The seed $\hat{A}_L$. Across the cut the internal momentum is $\hat{P}_3$ on the $\hat{A}_R$ side and $-\hat{P}_3$ on the $\hat{A}_L$ side; on shell we factor $\hat{P}_3 = |\hat{P}\rangle[\hat{P}|$ and continue the reversed leg by $|-\hat{P}\rangle = i|\hat{P}\rangle$, $|-\hat{P}\,] = i|\hat{P}\,]$, so that $|-\hat{P}\rangle[-\hat{P}| = -\hat{P}_3$. With $|\hat{2}\,] = |2]$ unshifted, the $\overline{\text{MHV}}$ seed of Eq. (59.169) (legs i, j, k taken as $\hat{2}^-, 3^+, -\hat{P}^+$) is

$$\hat{A}_L = -i\,g\,\frac{[3\,(-\hat{P})]^3}{[(-\hat{P})\,\hat{2}]\,[\hat{2}\,3]} = -i\,g\,\frac{(i\,[3\,\hat{P}])^3}{(i\,[\hat{P}\,2])[2\,3]} = -i\,g\,\frac{i^3}{i}\,\frac{[3\,\hat{P}]^3}{[\hat{P}\,2]\,[2\,3]} = i\,g\,\frac{[3\,\hat{P}]^3}{[\hat{P}\,2]\,[2\,3]}. \tag{59.182}$$

The lower amplitude $\hat{A}_R$. By the induction hypothesis applied to Eq. (59.179), whose ordering is $(\hat{P}_3, 4, 5, \dots, n, \hat{1})$ with negatives $\hat{P}_3$ and $\hat{1}$, and using $|\hat{1}\,\rangle = |1\rangle$

unshifted,

$$\hat{A}_R = i\,g^{n-3}\,\frac{\langle \hat{P}\,\hat{1}\rangle^4}{\langle \hat{P}\,4\rangle\,\langle 4\,5\rangle\cdots\langle n-1\,n\rangle\,\langle n\,\hat{1}\rangle\,\langle \hat{1}\,\hat{P}\rangle} = i\,g^{n-3}\,\frac{\langle \hat{P}\,1\rangle^4}{\langle \hat{P}\,4\rangle\,\langle 4\,5\rangle\cdots\langle n\,1\rangle\,\langle 1\,\hat{P}\rangle}. \tag{59.183}$$

Eliminating the internal spinors. Assemble Eqs. (59.181) to (59.183) and separate the $\hat{P}$-dependent factors from the rest; the overall constants multiply to $(i\,g)(i)(i\,g^{n-3}) = -\,i\,g^{n-2}$:

$$A_n = -\,i\,g^{n-2}\,\frac{1}{[2\,3]}\,\frac{1}{\langle 2\,3\rangle[3\,2]}\,\frac{1}{\langle 4\,5\rangle\cdots\langle n\,1\rangle}\,\underbrace{\frac{[3\,\hat{P}]^3\,\langle \hat{P}\,1\rangle^4}{[\hat{P}\,2]\,\langle \hat{P}\,4\rangle\,\langle 1\,\hat{P}\rangle}}_{\equiv\ \mathcal{R}}. \tag{59.184}$$

The factor $\mathcal{R}$ carries net little-group weight zero in the internal leg (angle power $4-2=2$, square power $3-1=2$), so it is expressible through the two scale-invariant bilinears $\langle x|\hat{P}|y] = \langle x\,\hat{P}\rangle[\hat{P}\,y]$, which we can evaluate without committing to a normalization of $|\hat{P}\rangle$. First reduce the angle power using $\langle 1\,\hat{P}\rangle = -\langle \hat{P}\,1\rangle$:

$$\mathcal{R} = -\frac{[3\,\hat{P}]^3\,\langle \hat{P}\,1\rangle^3}{[\hat{P}\,2]\,\langle \hat{P}\,4\rangle}. \tag{59.185}$$

The numerator pairs into a cube of one bilinear: with $\langle \hat{P}\,1\rangle[3\,\hat{P}] = \langle 1\,\hat{P}\rangle[\hat{P}\,3] = \langle 1|\hat{P}|3]$,

$$[3\,\hat{P}]^3\,\langle \hat{P}\,1\rangle^3 = \langle 1|\hat{P}|3]^3. \tag{59.186}$$

Now $\hat{P}_3 = \hat{p}_2 + p_3$, and since $[3\,3] = 0$ and $\langle 1\,\hat{2}\rangle = \langle 1\,2\rangle - z_*\langle 1\,1\rangle = \langle 1\,2\rangle$,

$$\langle 1|\hat{P}|3] = \langle 1\,\hat{2}\rangle[2\,3] + \langle 1\,3\rangle[3\,3] = \langle 1\,2\rangle\,[2\,3], \qquad \Longrightarrow \qquad \langle 1|\hat{P}|3]^3 = \langle 1\,2\rangle^3\,[2\,3]^3. \tag{59.187}$$

The denominator pairs likewise. With $\langle \hat{P}\,4\rangle[\hat{P}\,2] = -\langle 4\,\hat{P}\rangle[\hat{P}\,2] = -\langle 4|\hat{P}|2]$ and $[2\,2] = 0$,

$$\langle 4|\hat{P}|2] = \langle 4\,\hat{2}\rangle[2\,2] + \langle 4\,3\rangle[3\,2] = \langle 4\,3\rangle\,[3\,2], \qquad \Longrightarrow \qquad [\hat{P}\,2]\,\langle \hat{P}\,4\rangle = -\langle 4\,3\rangle\,[3\,2]. \tag{59.188}$$

Therefore

$$\mathcal{R} = -\frac{\langle 1\,2\rangle^3\,[2\,3]^3}{-\langle 4\,3\rangle\,[3\,2]} = \frac{\langle 1\,2\rangle^3\,[2\,3]^3}{\langle 4\,3\rangle\,[3\,2]}. \tag{59.189}$$

Notice that every reference to the complex shift parameter z_* and to the internal spinors $|\hat{P}\rangle, |\hat{P}]$ has disappeared: $\mathcal{R}$ is a ratio of ordinary external brackets, as a physical amplitude must be.

Collecting. Insert Eq. (59.189) into Eq. (59.184):

$$A_n = -\,i\,g^{n-2}\,\frac{1}{[2\,3]}\cdot\frac{1}{\langle 2\,3\rangle[3\,2]}\cdot\frac{1}{\langle 4\,5\rangle\cdots\langle n\,1\rangle}\cdot\frac{\langle 1\,2\rangle^3\,[2\,3]^3}{\langle 4\,3\rangle\,[3\,2]}. \tag{59.190}$$

The square brackets cancel completely: $[2\,3]^3/[2\,3] = [2\,3]^2$ and $[3\,2]^2 = [2\,3]^2$, so the four square brackets combine to unity, leaving

$$A_n = -\,i\,g^{\,n-2}\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 4\,3\rangle\,\langle 4\,5\rangle\cdots\langle n\,1\rangle}. \tag{59.191}$$

Finally restore the cyclic ordering of the denominator. The lone misordered factor is $\langle 4\,3\rangle = -\langle 3\,4\rangle$, whose sign cancels the explicit minus:

$$A_n = i\,g^{\,n-2}\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,5\rangle\cdots\langle n\,1\rangle} = i\,g^{\,n-2}\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\,\langle 2\,3\rangle\,\langle 3\,4\rangle\cdots\langle n\,1\rangle}, \tag{59.192}$$

which is Eq. (59.180). The induction closes. □

The proof is worth pausing on. The entire content of an n-gluon MHV amplitude — which the Feynman expansion would deliver as a sum of up to thousands of gauge-dependent diagrams (Table 59.1) — has been produced from one three-point seed and one lower-point amplitude, by a half-page of spinor algebra in which the off-shell intermediate data (z_*, the internal polarization sum, the internal momentum's spinor split) cancelled identically. No Lagrangian, no gauge fixing, no propagator numerator ever appeared. The recursion knew only the poles and the seeds, and that was enough.

It remains to remove the adjacency restriction.

Corollary 59.29 (Parke–Taylor for general negative positions)**.** *For any $i \neq j$, the color-ordered MHV amplitude with negative legs i, j equals Eq. (59.170).*

Proof. By constructibility (Theorem 59.27) a tree amplitude is the unique rational function with the correct little-group weights and the correct residue on every physical (consecutive) factorization channel; equivalently, two candidate expressions with identical poles and residues that both fall off at large z under a valid shift are equal. We verify that the right-hand side of Eq. (59.170) has the residues demanded by factorization, using Theorem 59.28 as the known data on each channel.

Write $\mathrm{PT}_n[i,j] \equiv \langle i\,j\rangle^4/(\langle 1\,2\rangle\cdots\langle n\,1\rangle)$. Its only poles are the simple zeros of consecutive brackets $\langle a\,a+1\rangle$, i.e. the two-particle channels $P^2_{a,a+1} = \langle a\,a+1\rangle[a+1\,a] \to 0$; these are exactly the physical channels of a color-ordered amplitude that contain an MHV three-point corner. On the channel $\langle a\,a+1\rangle \to 0$ the momenta p_a, p_{a+1} become collinear, $|\,a+1\rangle \to |a\rangle$ up to normalization, and the residue of $\mathrm{PT}_n[i,j]$ factorizes. Take the generic case $a, a+1 \notin \{i,j\}$ (neither negative leg on the degenerate pair). Then $\langle i\,j\rangle$ is inert, and

$$\operatorname*{Res}_{\langle a\,a+1\rangle\to 0}\mathrm{PT}_n[i,j] = \frac{\langle i\,j\rangle^4}{\langle 1\,2\rangle\cdots\langle \widehat{a\,a+1}\rangle\cdots\langle n\,1\rangle} \xrightarrow{|a+1\rangle\to|a\rangle} \mathrm{PT}_{n-1}[i,j]\times A_3^{\overline{\mathrm{MHV}}}(a,a{+}1,-P_{a,a+1})/ \tag{59.193}$$

the hat denoting omission: the bracket $\langle a\, a+1\rangle$ is replaced by the propagator, the two collinear legs merge into a single leg in an $(n-1)$-point PT, and the stripped three-point factor is precisely the $\overline{\text{MHV}}$ seed — the same factorized form the recursion produces. This is the residue an actual MHV amplitude must have, and it agrees, leg by leg, with the residue computed from Eq. (59.172) using Theorem 59.28 on the lower-point factor. The cases where a negative leg sits on the degenerate pair are checked identically, the extra power of $\langle i\, j\rangle$ being supplied by the merged leg's helicity. Since $\mathrm{PT}_n[i,j]$ has no other poles, no pole at $z=\infty$ under a valid shift, and the correct little-group weight t_ℓ^{+2} on the two negative legs and t_ℓ^{-2} on the rest, it is *the* amplitude. Hence Eq. (59.170) holds for all i,j. □

Why the answer is one line

The Feynman expansion computes an amplitude as a sum over *off-shell* histories, and the price of going off shell is gauge dependence: each diagram depends on the gauge and on unphysical longitudinal modes, and the physical answer emerges only after these cancel among many diagrams. BCFW never leaves the mass shell. Every quantity it manipulates — the seeds Eq. (59.169), the residues, the lower-point amplitudes — is separately gauge invariant and little-group covariant, so there is nothing to cancel and nothing to conceal. The Parke–Taylor formula is short because it is assembled entirely from gauge-invariant pieces; the thousands of Feynman diagrams were short the same way, but written in a basis (off-shell, gauge-variant) that hid it.

59.7.3 Four gluons by BCFW, matched to the direct result

Worked Example 59.30 (The four-gluon MHV amplitude from one BCFW term)**.** We run the recursion explicitly on the carried Yang–Mills theory amplitude $A_4(1^-,2^-,3^+,4^+)$ and confirm it reproduces the Parke–Taylor value Eq. (59.55) obtained earlier by direct Feynman computation, the smallest case where the entire machine is visible at once.

Apply the $[\,1,2\,\rangle$ shift. By Section 59.7.1 the only surviving channel is $k=3$, $I=\{2,3\}$, with $P_3=p_2+p_3$, $P_3^2=\langle 2\,3\rangle[3\,2]$, and shift parameter $z_*=\langle 2\,3\rangle/\langle 1\,3\rangle$ from Eq. (59.178). The two subamplitudes are both three-point seeds:

$$\hat{A}_L=A_3^{\overline{\text{MHV}}}(\hat{2}^-,3^+,-\hat{P}^+),\qquad \hat{A}_R=A_3^{\text{MHV}}(\hat{P}^-,4^+,\hat{1}^-). \tag{59.194}$$

The left factor is Eq. (59.182) verbatim,

$$\hat{A}_L=i\,g\,\frac{[3\,\hat{P}]^3}{[\hat{P}\,2]\,[2\,3]}. \tag{59.195}$$

The right factor is the $n = 4$ instance of Eq. (59.183) (so $g^{n-3} = g$); with ordering $(\hat{P}, 4, \hat{1})$, negatives $\hat{P}$ and $\hat{1}$, and $|\hat{1}\,\rangle = |1\rangle$,

$$\hat{A}_R = i\,g\,\frac{\langle \hat{P}\,1\rangle^4}{\langle \hat{P}\,4\rangle\,\langle 4\,1\rangle\,\langle 1\,\hat{P}\rangle}. \tag{59.196}$$

The product over the propagator is the $n = 4$ case of Eq. (59.184) (overall constant $-\,i\,g^2$), with the "middle" product $\langle 4\,5\rangle \cdots \langle n\,1\rangle$ degenerating to the single factor $\langle 4\,1\rangle$:

$$A_4 = -\,i\,g^2\,\frac{1}{[2\,3]}\,\frac{1}{\langle 2\,3\rangle[3\,2]}\,\frac{1}{\langle 4\,1\rangle}\,\frac{[3\,\hat{P}]^3\,\langle \hat{P}\,1\rangle^4}{[\hat{P}\,2]\,\langle \hat{P}\,4\rangle\,\langle 1\,\hat{P}\rangle}. \tag{59.197}$$

The internal spinors are eliminated exactly as in Eqs. (59.187) and (59.188): the numerator bilinear is $\langle 1|\hat{P}|3]^3 = \langle 1\,2\rangle^3[2\,3]^3$ and the denominator bilinear is $[\hat{P}\,2]\langle \hat{P}\,4\rangle = -\langle 4\,3\rangle[3\,2]$, so

$$A_4 = -\,i\,g^2\,\frac{1}{[2\,3]}\,\frac{1}{\langle 2\,3\rangle[3\,2]}\,\frac{1}{\langle 4\,1\rangle}\,\frac{\langle 1\,2\rangle^3[2\,3]^3}{\langle 4\,3\rangle[3\,2]} = -\,i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 4\,1\rangle\,\langle 4\,3\rangle}, \tag{59.198}$$

the square brackets again cancelling in the combination $[2\,3]^2/[3\,2]^2 = 1$. Reordering $\langle 4\,3\rangle = -\langle 3\,4\rangle$ and $\langle 4\,1\rangle = \langle 4\,1\rangle$,

$$A_4(1^-, 2^-, 3^+, 4^+) = i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = i\,g^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}. \tag{59.199}$$

This is identical to the Parke–Taylor value Eq. (59.55) specialized to $n = 4$, and hence to the result of the explicit four-gluon Feynman computation carried out in the Parke–Taylor section — the same kinematic ratio that enters the one-loop reconstruction of Section 60.6.2 as the tree input $A_4^{(0)} = \langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$. The contrast in effort is the lesson: the direct route summed the s-channel, t-channel, and contact diagrams of Section 59.1.5 with their full momentum-dependent vertices and a polarization sum, then watched the gauge artifacts cancel; BCFW multiplied two seeds and a pole.

It is useful to record the same amplitude in Mandelstam variables, both as a check and because it is the form that recurs at one loop. Using $s = (p_1 + p_2)^2 = \langle 1\,2\rangle[2\,1]$ and momentum conservation, Eq. (59.199) can be rewritten (multiplying numerator and denominator by $[2\,1]$ and using $\langle 2\,3\rangle[3\,1] = -\langle 2\,4\rangle[4\,1]$ from $\sum_i \langle 2\,i\rangle[i\,1] = 0$) in the manifestly crossing-symmetric guise

$$A_4(1^-, 2^-, 3^+, 4^+) = i\,g^2\,\frac{\langle 1\,2\rangle^3\,[3\,4]}{[1\,2]\,s\,t} \times \langle 1\,2\rangle[1\,2]\Big/\langle 1\,2\rangle[1\,2] = i\,g^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \tag{59.200}$$

the point being only that the single rational function carries the expected simple poles in $s = \langle 1\,2\rangle[2\,1]$ and $t = \langle 2\,3\rangle[3\,2]$ and no others, as locality requires.

59.7.4 Five gluons: one term against a swarm

Worked Example 59.31 (The five-gluon MHV amplitude)**.** The five-point amplitude is where the economy becomes dramatic. The direct color-ordered computation of $A_5(1^-,2^-,3^+,4^+,5^+)$ sums the ten planar diagrams counted in Table 59.1 — five with two cubic vertices and an internal propagator, plus diagrams with the quartic vertex — each a tangle of the momentum-dependent vertices Eqs. (59.12) and (59.13) contracted against five polarization vectors, the whole sum gauge-dependent term by term. BCFW returns the answer in one term.

Apply the $[\,1,2\,\rangle$ shift. As established, the sole surviving channel is $k=3$, $I=\{2,3\}$, with $P_3=p_2+p_3$, $P_3^2=\langle 2\,3\rangle[3\,2]$, $z_*=\langle 2\,3\rangle/\langle 1\,3\rangle$. The factorization is

$$\hat{A}_L=A_3^{\overline{\mathrm{MHV}}}(\hat{2}^-,3^+,-\hat{P}^+),\qquad \hat{A}_R=A_4^{\mathrm{MHV}}(\hat{P}^-,4^+,5^+,\hat{1}^-). \tag{59.201}$$

The left factor is again Eq. (59.182). The right factor is now a genuine four-point amplitude, supplied by the result Eq. (59.199) we just proved (the induction hypothesis at $n=4$): with ordering $(\hat{P},4,5,\hat{1})$, negatives $\hat{P},\hat{1}$, and $|\hat{1}\,\rangle=|1\rangle$,

$$\hat{A}_R=i\,g^2\,\frac{\langle\hat{P}\,1\rangle^4}{\langle\hat{P}\,4\rangle\,\langle 4\,5\rangle\,\langle 5\,1\rangle\,\langle 1\,\hat{P}\rangle}. \tag{59.202}$$

Assembling through the propagator (overall constant $(i\,g)(i)(i\,g^2)=-\,i\,g^3$),

$$A_5=-\,i\,g^3\,\frac{1}{[2\,3]}\,\frac{1}{\langle 2\,3\rangle[3\,2]}\,\frac{1}{\langle 4\,5\rangle\,\langle 5\,1\rangle}\,\underbrace{\frac{[3\,\hat{P}]^3\,\langle\hat{P}\,1\rangle^4}{[\hat{P}\,2]\,\langle\hat{P}\,4\rangle\,\langle 1\,\hat{P}\rangle}}_{=\mathcal{R}}, \tag{59.203}$$

where the bracketed group is the *same* $\mathcal{R}$ as in the general proof, because the only internal-leg structures are $[3\,\hat{P}],\langle\hat{P}\,1\rangle,[\hat{P}\,2],\langle\hat{P}\,4\rangle$ — identical to Eq. (59.184); the extra leg 5 enters only through the inert external factor $\langle 4\,5\rangle\langle 5\,1\rangle$ in the denominator. Hence Eq. (59.189) applies unchanged, $\mathcal{R}=\langle 1\,2\rangle^3[2\,3]^3/(\langle 4\,3\rangle[3\,2])$, and

$$A_5=-\,i\,g^3\,\frac{1}{[2\,3]}\,\frac{1}{\langle 2\,3\rangle[3\,2]}\,\frac{1}{\langle 4\,5\rangle\langle 5\,1\rangle}\,\frac{\langle 1\,2\rangle^3[2\,3]^3}{\langle 4\,3\rangle[3\,2]}=-\,i\,g^3\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 4\,3\rangle\,\langle 4\,5\rangle\,\langle 5\,1\rangle}. \tag{59.204}$$

Reordering $\langle 4\,3\rangle=-\langle 3\,4\rangle$,

$$A_5(1^-,2^-,3^+,4^+,5^+)=i\,g^3\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,5\rangle\langle 5\,1\rangle}, \tag{59.205}$$

the Parke–Taylor formula Eq. (59.55) at $n=5$.

The accounting is the moral of the chapter. A direct evaluation balances ten gauge-variant diagrams; BCFW multiplies a seed, a propagator, and the

already-known four-point amplitude, and the only labour is the half-line of spinor algebra in $\mathcal{R}$ — which we did not even repeat, having performed it once in the general proof. Each added gluon costs one more recursion step and one more inert denominator bracket, while the Feynman count climbs through 220, 2485, 34,300 (Table 59.1). This is on-shell recursion paying for itself.

59.7.5 Beyond MHV: the six-point NMHV teaser

Parke–Taylor exhausts the $k = 2$ sector. The first amplitude that is genuinely *not* MHV is the six-point next-to-MHV (NMHV, $k = 3$) amplitude

$$A_6 \equiv A_6(1^-, 2^-, 3^-, 4^+, 5^+, 6^+), \tag{59.206}$$

whose anti-MHV partner is itself ($n - 2 = 4 \neq 3$, so $\overline{\text{MHV}}$ sits at $k = 4$; NMHV at $k = 3$ is the new sector). It cannot be a single Parke–Taylor-like ratio — it is a sum of several rational terms — and assembling it from lower amplitudes is exactly what BCFW is for. We give an explicit preview here, enumerating the surviving channels and their structure; the explicit rational function follows from the same spinor algebra used above, and we record only its channel structure rather than expanding every bracket.

Apply the $[\,1, 2\,\rangle$ shift (both shifted legs negative, a valid shift). The consecutive channels separating $\hat{1}, \hat{2}$ are $I_k = \{2, \ldots, k\}$ for $k = 3, 4, 5$, splitting as in Eq. (59.175). The external negatives are now $1, 2, 3$; legs $2, 3$ land in $\hat{A}_L$ for every channel ($k \geq 3$), leg 1 in $\hat{A}_R$. Run the same vanishing logic, now with three negatives to distribute, and track the on-shell branch as in Section 59.7.1:

- $k = 3$ **(channel P_{23}): dies.** The small corner is $A_3(\hat{2}^-, 3^-, -\hat{P}^h)$. At $z_* = \langle 2\,3\rangle/\langle 1\,3\rangle$ the angle bracket $\langle \hat{2}\,3\rangle \to 0$, forcing the anti-holomorphic branch on which a three-point amplitude must be $\overline{\text{MHV}}$, i.e. carry exactly one negative leg. But legs $\hat{2}$ and 3 are *both* negative, so no helicity choice makes this corner a nonzero $\overline{\text{MHV}}$. The channel contributes nothing — the two adjacent negatives cannot share a three-point corner on this branch.

- $k = 4$ **(channel P_{234}): an** MHV $\times$ MHV **term.** Both sides are four-point, so no branch subtlety arises; only the helicity count decides. Taking the internal line positive into $\hat{A}_L$,

$$\hat{A}_L = A_4^{\text{MHV}}(\hat{2}^-, 3^-, 4^+, -\hat{P}^+), \qquad \hat{A}_R = A_4^{\text{MHV}}(\hat{P}^-, 5^+, 6^+, \hat{1}^-), \tag{59.207}$$

  each MHV with two negatives, both nonzero. (The opposite internal helicity makes $\hat{A}_R$ single-minus, killing it.) Both factors are Parke–Taylor by Theorem 59.28, glued on the pole $P_{234}^2 = (p_2 + p_3 + p_4)^2$.

- $k = 5$ **(channel P_{61}): an** $\overline{\text{MHV}} \times \overline{\text{MHV}}$ **term.** The small corner is $A_3(\hat{P}^h, 6^+, \hat{1}^-)$. Its on-shell point is $z = -[1\,6]/[2\,6]$, where $[\hat{1}\,6] \to 0$: the

holomorphic branch, on which the three-point amplitude must be $\overline{\text{MHV}}$ (one negative). With leg 6 positive and leg 1 negative, that fixes the internal line positive on the $\hat{A}_R$ side,

$$\hat{A}_R = A_3^{\overline{\text{MHV}}}(\hat{P}^+, 6^+, \hat{1}^-), \qquad \hat{A}_L = A_5^{\overline{\text{MHV}}}(\hat{2}^-, 3^-, 4^+, 5^+, -\hat{P}^-), \tag{59.208}$$

the large side being a five-point amplitude with three negatives — itself an *anti-MHV* amplitude ($k = n - 2 = 3$), given by the parity conjugate of Parke–Taylor, $\overline{\text{MHV}}_5 \propto [4\,5]^4/([\hat{2}\,3][3\,4][4\,5][5\,(-\hat{P})][(-\hat{P})\,\hat{2}])$, with the two positive legs $4, 5$ in the numerator. It is glued on the two-particle pole $P_{61}^2 = \langle 6\,1\rangle[1\,6]$.

The full NMHV amplitude is the sum of the two surviving terms,

$$A_6 = \hat{A}_L \, \frac{i}{P_{234}^2} \, \hat{A}_R\Big|_{\text{Eq. (59.207)}} + \hat{A}_L \, \frac{i}{P_{61}^2} \, \hat{A}_R\Big|_{\text{Eq. (59.208)}}. \tag{59.209}$$

Three features of Eq. (59.209) deserve emphasis, because they are general to all non-MHV amplitudes. First, the building blocks are exactly the data this chapter produced: the three-point seeds Eq. (59.169), Parke–Taylor Eq. (59.170), and its parity conjugate. Nothing new is needed — every N^kMHV amplitude is recursively an assembly of MHV and $\overline{\text{MHV}}$ pieces. Second, the number of terms depends on the shift: a different valid shift would tile A_6 differently (commonly into three terms), all representations being equal by the uniqueness of Theorem 59.27. Third — and this is the price of recursion — each individual term in Eq. (59.209) carries *spurious poles*: denominators such as $\langle 3|P_{234}|6]$ or $\langle 6\,1\rangle$ appear in the shifted spinors of one term that correspond to no physical factorization channel of A_6. These spurious poles are not present in the true amplitude; they must cancel between the two terms of Eq. (59.209), and they do, leaving a function whose only singularities are the genuine physical poles. The same spurious-pole cancellation is what Theorem 62.48 later recognizes, geometrically, as the matching of internal cell walls in a triangulation of the amplituhedron. The teaser thus does double duty: it exhibits the first non-MHV gluon amplitude, and it previews the mechanism — physical poles real, spurious poles cancelling — that organizes the modern amplitudes program.

59.7.6 Synthesis: locality and unitarity fix the trees

Step back from the algebra. We began the chapter with a scandal of bookkeeping: the n-gluon tree, built honestly from the Yang–Mills Lagrangian Eq. (25.76) and the vertices of Chapter 26, demands a super-factorial number of gauge-dependent diagrams that collapse to a one-line answer. We end with the resolution. The collapse is not luck but structure, and the structure has a precise content: the entire tree-level S-matrix of Yang–Mills is fixed by two inputs, neither of which is the Lagrangian.

The first input is *locality*, expressed as factorization: an amplitude has only simple poles, in physical momentum channels, and on each pole it factorizes into the product of two lower-point on-shell amplitudes. This is what made the residues in Theorems 59.28 and 59.29 computable from smaller amplitudes. The second is *unitarity*, expressed as the consistency of those residues with the lower amplitudes themselves — and, at the bottom, the existence of the three-point seeds Eq. (59.169), which little-group scaling and locality fix uniquely up to a coupling. The BCFW recursion Eq. (59.172) is nothing but the statement that a rational function with the right poles and the right large-z falloff (Theorem 59.27) is determined by its residues; feeding it the seeds generates every tree. The Lagrangian, the gauge choice, the off-shell Green's functions — all the apparatus of Chapters 25 and 26 — were a *representation* of this data, convenient for proving renormalizability and Ward identities but profligate for computing on-shell amplitudes, which know only about poles and seeds.

This is the chapter's philosophical payoff, and it reframes what one means by "defining a gauge theory." Perturbatively, the trees follow from locality, unitarity, and the little-group representation of the external states — the spectrum and the cubic couplings — without ever writing $\mathcal{L}$. The same two principles, applied to the discontinuities of loop integrands rather than the poles of trees, drive the generalized-unitarity methods of Chapter 60: there the trees proved here are sewn across multiparticle cuts to reconstruct loop amplitudes, the Parke–Taylor and BCFW data of this chapter entering as the on-shell input to the cut. And the color-ordered partial amplitudes, with their kinematic numerators laid bare by the absence of gauge clutter, are exactly the objects whose color–kinematics duality powers the BCJ double copy of Chapter 61, where two copies of these gluon trees assemble graviton amplitudes. The trees are not an endpoint; they are the on-shell currency of everything downstream.

These proved trees and explicit amplitudes are precisely the on-shell data the next chapter needs: Chapter 60 sews them across generalized cuts to reconstruct loop amplitudes from unitarity, and Chapter 61 reads their color-ordered kinematic numerators as the input to the BCJ double copy that builds gravity from two copies of Yang–Mills.

Further reading

The maximally-helicity-violating amplitude was conjectured and first proved (via a Berends–Giele off-shell recursion) in the founding paper of Parke and Taylor [379], the recursion itself being [62]; the systematics of color decomposition and helicity amplitudes are reviewed thoroughly by Mangano and Parke [334]. On-shell BCFW recursion originates with Britto, Cachazo, and Feng [91], with the large-z/contour proof given by Britto, Cachazo, Feng, and Witten [92]; the twistor-space perspective that triggered the modern program is Witten's [523].

Comprehensive textbook treatments of everything in this chapter — the inductive Parke–Taylor proof, the large-z catalogue, and worked NMHV examples — are Elvang and Huang [176] and Henn and Plefka [263]; Dixon's lectures [153] are a compact entry point. The loop and double-copy sequels are taken up in Chapters 60 and 61.

Chapter 60

Loop Amplitudes and Unitarity

The tree-level chapters left us with a way of building scattering amplitudes that never once wrote down a Lagrangian: spinor-helicity variables and on-shell recursion (Chapters 58 and 59) reconstruct every tree from three-point data and the analytic structure forced by factorization. This chapter carries that program to one loop. The question it answers is the one that makes the loop level genuinely different from the tree level — a loop amplitude is not a rational function but a transcendental one, with branch cuts and a nontrivial function space — and the answer is that the same two principles that organized the trees, *locality* and *unitarity*, pin a loop amplitude down almost completely. Locality puts its poles where intermediate particles go on shell; unitarity fixes the discontinuity across every branch cut. Made computational, those two facts replace the Feynman-diagram expansion of Chapter 19 with a method that sews loop integrands out of on-shell trees.

The engine is the optical theorem, which we proved as a constraint in Chapter 14 and used there only to check completed calculations. The first section upgrades it: from $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ we extract Cutkosky's cutting rule, which states that the discontinuity of any Feynman integral across a physical threshold is the Lorentz-invariant phase-space integral of a product of lower on-shell amplitudes. The branch cuts that the brute-force computations of Chapter 19 encountered at thresholds, and the dispersion relations of Chapter 5 that turn an imaginary part back into a full function, become the starting data of a smarter method rather than an afterthought. With the bubble worked explicitly and the QED vacuum polarization recovered from its two-particle cut, the principle of cut-constructibility is in hand: an amplitude is largely determined by its cuts.

What makes the principle into an algorithm is that one-loop amplitudes live in a finite-dimensional space. Every one-loop integrand reduces — by tensor (Passarino–Veltman) reduction and by integrand-level identities — to a fixed basis of scalar master integrals: boxes, triangles, bubbles, and tadpoles, evaluated once and for all in Chapter D. The chapter's central sections exploit

this. Generalized unitarity (Section 60.3) cuts more than two propagators at once: a quadruple cut freezes the loop momentum on complex on-shell solutions and isolates a single box coefficient as a product of four three-point trees, a triple cut isolates triangles, a double cut bubbles. The OPP decomposition (Section 60.4) installs this bookkeeping into the integrand itself, before integration, as an exact algebraic identity with spurious terms that integrate to zero — the form that underlies every automated one-loop program. Because four-dimensional cuts miss the purely rational part of the amplitude, Section 60.5 restores it through d-dimensional unitarity, where the (-2ε)-dimensional components of the loop momentum supply the $\varepsilon \times (1/\varepsilon)$ that four-dimensional methods discard.

These tools are then exercised in a single sustained computation. Section 60.6 reconstructs a one-loop gluon amplitude from cuts alone — the MHV box from a quadruple cut, with its infrared poles checked against the Catani structure, and the all-plus amplitude (every four-dimensional cut of which vanishes) from a μ^2-cut — and counts the Feynman diagrams it never had to draw. This is Yang–Mills theory at one loop, the loop-level continuation of the gluon amplitudes built by recursion in Chapter 59.

The final section looks past one loop. Integration-by-parts identities collapse any integral family to a finite basis of masters, the masters are evaluated by the method of differential equations brought to canonical ε-factorized form, and the resulting functions are organized by transcendental weight and dissected by the symbol. The chapter closes honestly at the frontier: numerical unitarity over finite fields as the practical multiloop workhorse, and the elliptic and Calabi–Yau integrals where the polylogarithmic function space — and with it the clean canonical-basis story — gives out. Throughout we use the book-wide canon (Chapter A): the mostly-minus metric, natural units, the Feynman $i\epsilon$, dimensional regularization in $d = 4 - 2\varepsilon$, and the spinor-helicity conventions fixed in Chapter 58.

60.1 Unitarity, Analyticity, and the Cutting Rules

The chapter's thesis is that a loop amplitude is not an arbitrary analytic function but one pinned down by two physical principles: *locality*, which puts its poles where intermediate particles can go on shell, and *unitarity*, which controls its branch-cut discontinuities. The second of these is, at the level of the S-matrix, the single statement that probability is conserved, $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$, and we already extracted its most familiar consequence in Section 14.7: the optical theorem, relating the imaginary part of a forward amplitude to a total rate. That was a constraint, a check one applies after a calculation. The purpose of this section is to upgrade it into a *computational engine*. We will show that the discontinuity of a loop amplitude across any physical branch cut is the phase-space integral of a product of lower, on-shell amplitudes — Cutkosky's cutting rule — and

that, conversely, an amplitude is largely reconstructed once its cuts are known. The imaginary parts that the brute-force one-loop computations of Chapter 19 stumbled upon at thresholds will turn out to be exactly the data a smarter method takes as its starting point. Throughout we use the book-wide canon (Chapter A): mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, the Feynman $i\epsilon$, and the Lorentz-invariant phase-space measure $\mathrm{d}\Pi$ introduced in Eq. (14.191) and built explicitly in Chapter 16.

60.1.1 From operator unitarity to the discontinuity

We begin by recalling, in compact form, the result this section builds on; its full derivation, including the careful plane-wave regularization, is Theorem 14.37, and we do not repeat it. Writing the scattering operator as

$$\mathbf{S} = \mathbf{1} + i\,\mathcal{T}, \tag{60.1}$$

unitarity $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ becomes, after cancelling the leading $\mathbf{1}$ and collecting terms,

$$-i(\mathcal{T} - \mathcal{T}^\dagger) = \mathcal{T}^\dagger\mathcal{T}, \tag{60.2}$$

the operator identity Eq. (14.188). The left-hand side is (twice) the anti-Hermitian part of $\mathcal{T}$; the right-hand side is manifestly positive. Sandwiching Eq. (60.2) between an initial state $|i\rangle$ and a final state $|f\rangle$ of definite momenta, stripping the momentum-conserving delta to define the invariant amplitude through $\langle f|\mathcal{T}|i\rangle = (2\pi)^4\delta^{(4)}(p_i - p_f)\,\mathcal{M}(i\to f)$, and inserting a complete set of states on the right gives the *generalized optical theorem* (Eq. (14.192)),

$$-i\Big[\mathcal{M}(i\to f) - \mathcal{M}^*(f\to i)\Big] = \sum_X \int \mathrm{d}\Pi_X\; \mathcal{M}(i\to X)\,\mathcal{M}^*(f\to X), \tag{60.3}$$

where the sum runs over every kinematically accessible intermediate state X, weighted by its Lorentz-invariant phase space. The forward, diagonal specialization $f = i$ collapses the left side to $2\,\mathrm{Im}\,\mathcal{M}(i\to i)$ and the right side to a sum of squared moduli, the form

$$2\,\mathrm{Im}\,\mathcal{M}(i\to i) = \sum_X \int \mathrm{d}\Pi_X\,\big|\mathcal{M}(i\to X)\big|^2 = F\,\sigma_{\mathrm{tot}}, \tag{60.4}$$

with F the Møller flux of Eq. (14.200). The crucial structural fact, and the one that turns a check into a method, is that Eq. (60.3) holds *off the forward direction*: i and f need not coincide. The right-hand side is then a sum of products of two *distinct* sub-amplitudes, $\mathcal{M}(i\to X)$ and $\mathcal{M}^*(f\to X)$, glued along the on-shell intermediate state X. It is this non-forward version that will reappear, term by term in perturbation theory, as the discontinuity of a single Feynman integral. The forward theorem squares one amplitude; the general theorem *factorizes* a discontinuity into two trees sewn over phase space, and that is precisely the structure a reconstruction method exploits.

60.1.2 Branch cuts, thresholds, and real analyticity

To see Eq. (60.3) as a statement about analytic functions we must recall how an amplitude depends on its invariants. Fix all kinematic data except a single Mandelstam variable, say $s = (p_1 + p_2)^2$, and regard $\mathcal{M}(s)$ as a function of complex s. Two properties, established in Section 5.5 from causality, organize everything.

First, *real analyticity* (the Schwarz reflection principle): because $\mathcal{M}$ is real on a segment of the real axis below all thresholds — where every internal line is off shell and no intermediate state can be produced — it satisfies

$$\mathcal{M}(s^*) = \mathcal{M}(s)^*. \tag{60.5}$$

Second, the only singularities of $\mathcal{M}(s)$ on the physical sheet are those demanded by physics: poles where a single intermediate particle goes on shell (the imprint of locality), and branch points at the *thresholds* $s = (\sum_a m_a)^2$ where a multiparticle intermediate state $X = \{a\}$ first becomes kinematically accessible. Above such a threshold the state X can be produced as real, on-shell particles; below it cannot. A branch cut runs along the real axis from each threshold to infinity, and the Feynman $i\epsilon$ instructs us to evaluate the physical amplitude on the *upper* lip, $s \to s + i0$.

Combining these, the *discontinuity* across the cut — the jump between the two lips — is, by Eq. (60.5),

$$\operatorname{disc}\mathcal{M}(s) \;\equiv\; \mathcal{M}(s+i0) - \mathcal{M}(s-i0) = \mathcal{M}(s+i0) - \mathcal{M}(s+i0)^* = 2i\,\operatorname{Im}\mathcal{M}(s+i0), \tag{60.6}$$

exactly the relation $\operatorname{disc} = 2i\,\operatorname{Im}$ proved in Theorem 5.38. The left-hand side of the optical theorem Eq. (60.3) is therefore, up to the factor $-i$ and modulo the non-forward generalization, the discontinuity of $\mathcal{M}$ across the cut whose branch point is the threshold of the very state X being summed on the right. Unitarity, read through analyticity, says: *the jump of an amplitude across the threshold of a state X equals the phase-space integral of the amplitudes for producing and absorbing X.* Each branch cut is owned by one intermediate channel, and its discontinuity is computable from lower data. This is the principle. The rest of the section makes it operational at the level of the individual Feynman integral.

Physical Insight

A loop amplitude carries far less independent information than its integral representation suggests. Locality fixes where its poles sit; unitarity fixes the jump across each of its cuts; and an analytic function is heavily constrained by its singularities. The cuts are the "data" and the function is largely their interpolation. The practical payoff, developed across this chapter, is that the data are *on-shell tree amplitudes* — gauge invariant,

compact, computable by the spinor-helicity and recursion technology of Chapters 58 and 59 — whereas the function is an off-shell loop integral with all the gauge redundancy and diagram proliferation that Chapter 19 displayed. The method trades the latter for the former.

60.1.3 Cutkosky's cutting rule

We now translate "the discontinuity is a phase-space integral" into a rule applied directly to a Feynman integrand. The mechanism is entirely contained in the analytic structure of a single propagator. Consider the scalar Feynman propagator as a function of p^2 at fixed sign of p^0,

$$\frac{1}{p^2 - m^2 + i\epsilon}. \tag{60.7}$$

As a distribution in the real variable p^2, the Sokhotski–Plemelj identity Eq. (5.22) gives its two boundary values, $1/(p^2-m^2\pm i\epsilon) = \mathrm{PV}\; 1/(p^2-m^2) \mp i\pi\,\delta(p^2-m^2)$, whose difference is the discontinuity across the threshold $p^2 = m^2$:

$$\frac{1}{p^2 - m^2 + i\epsilon} - \frac{1}{p^2 - m^2 - i\epsilon} = -2\pi i\,\delta(p^2 - m^2). \tag{60.8}$$

A single internal line therefore develops an imaginary part exactly when it can be put on shell, and that imaginary part is a delta function localizing the line on its mass shell. The factor $\delta(p^2 - m^2)$ is supported on both energy signs $p^0 = \pm E_{\mathbf{p}}$; physics selects one. In a cut that separates the diagram into an "in" part producing X and an "out" part absorbing it, energy flows in a definite direction across the cut, so the cut lines carry positive energy in that direction. We encode this with the positive-energy delta

$$\delta^+(p^2-m^2) \equiv \theta(p^0)\,\delta(p^2-m^2), \qquad \int \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta^+(p^2-m^2)\,f(p) = \int \frac{d^3\mathbf{p}}{(2\pi)^3\,2E_{\mathbf{p}}}\,f(p)\Big|_{p^0=E_{\mathbf{p}}} \tag{60.9}$$

the second equality being the standard reduction of the on-shell measure to the single-particle Lorentz-invariant phase space $\widetilde{dp}$ of Chapter 16. We can now state the rule.

Theorem 60.1 (Cutkosky cutting rule)**.** *The discontinuity of a Feynman integral across the branch cut associated with a given intermediate state is obtained by summing over all* cuts *that isolate that state — partitions of the diagram into two connected parts by severing a set of internal lines — and, in each cut, making for every severed propagator the replacement*

$$\frac{1}{p^2 - m^2 + i\epsilon} \;\longrightarrow\; -2\pi i\,\delta^+(p^2 - m^2), \tag{60.10}$$

with the positive-energy flow oriented from the production side to the absorption side, while the uncut propagators on the production side keep their $+i\epsilon$ Feynman form and those on the absorption side are complex-conjugated to $-i\epsilon$. Equivalently, in terms of the full Feynman propagator $i/(p^2-m^2+i\epsilon)$, a cut line is replaced by $2\pi\,\delta^+(p^2-m^2)$.

Origin of the rule via the largest-time equation. The diagrammatic source of Eq. (60.10) is an identity of Veltman among the position-space integrands, the *largest-time equation*, and we prove its engine in full; the combinatorial iteration is then mechanical. Write the position-space Feynman propagator as a sum of on-shell Wightman functions split by time ordering,

$$\Delta_F(x-y)=\theta(x^0-y^0)\,\Delta^+(x-y)+\theta(y^0-x^0)\,\Delta^+(y-x),\qquad \Delta^+(x)\equiv\int\widetilde{dk}\;e^{-ik\cdot x},\tag{60.11}$$

the integral running over the positive-energy mass shell, so that Δ^+ is the Fourier transform of $2\pi\,\delta^+(k^2-m^2)$ — precisely the cut line of Eq. (60.10) up to the numerator i. Assign to each vertex of a diagram either no mark or a circle. Define Δ_F on a line joining two unmarked vertices, its complex conjugate Δ_F^* on a line joining two circled vertices, $\Delta^+(x-y)$ when x is circled and y is not, and $\Delta^+(y-x)$ when y is circled and x is not; circled vertices additionally carry conjugated couplings. Let $F(\dots)$ denote the value of a fixed assignment of marks. Now single out the vertex z with the *largest* time among all vertices. For any line incident on z, the time ordering in Eq. (60.11) is settled: $\theta(z^0-\cdot)=1$. A direct check on each of the four cases shows that the line function attached to z is the same whether z is circled or not, except for the conjugation of z's coupling. Summing the two markings of z alone therefore produces a coupling plus its conjugate against a common factor, and because the bare vertex coupling appears as $-i$ times a real constant, the two terms cancel:

$$F(\dots,z\text{ unmarked},\dots)+F(\dots,z\text{ circled},\dots)=0\qquad(z^0=\text{largest}).\tag{60.12}$$

Summing Eq. (60.12) over all markings of the *remaining* vertices gives the largest-time equation: the sum over *all* 2^n markings of the n vertices vanishes,

$$\sum_{\text{markings}}F=0.\tag{60.13}$$

The all-unmarked term is the Feynman amplitude $\mathcal{M}$; the all-circled term is $\mathcal{M}^*$ (every line conjugated, every coupling conjugated); each mixed term has a set of Δ^+ lines forming a cut that separates circled from unmarked vertices. Transforming to momentum space, every Δ^+ becomes a factor $2\pi\,\delta^+$, the unmarked side keeps Feynman propagators and the circled side their conjugates, and Eq. (60.13) reads $\mathcal{M}+\mathcal{M}^*+\sum_{\text{cuts}}(\text{cut terms})=0$, i.e. $2\,\mathrm{Re}\,\mathcal{M}=-\sum_{\text{cuts}}(\dots)$. Restricting to the kinematic region of a single threshold, where only one channel's cut terms

are nonzero, isolates $\operatorname{disc}\mathcal{M} = 2i\,\operatorname{Im}\mathcal{M}$ as that channel's cut, which is Eq. (60.10) with the stated propagator replacements. The positive-energy delta in each Δ^+ supplies the $\theta(p^0)$ and orients the flow. □

The rule is the bridge promised in Section 60.1.2: the right-hand side of the cut equation, after the δ^+ factors are integrated against the loop measure, is by Eq. (60.9) nothing but the Lorentz-invariant phase space of the on-shell intermediate state, multiplying the two sub-amplitudes on either side of the cut. The Cutkosky cut of a loop integral *is* the right-hand side of the generalized optical theorem Eq. (60.3), term by term in perturbation theory. We now verify this on the simplest loop in the book.

60.1.4 The two-particle cut of the scalar bubble

The one-loop scalar bubble is the integral underlying every one-loop two-point function in the book; its closed form is the master function B_0 of Eq. (D.105). We take it as the test case for the cutting rule because both sides — the discontinuity of the analytic answer and the phase-space integral of the cut — can be computed independently and matched factor for factor.

Definition 60.2 (The scalar bubble)**.** For an external momentum q with $q^2 = s$ and two internal lines of mass m, the scalar one-loop bubble is

$$I_2(s) \;\equiv\; \int \frac{d^4\ell}{(2\pi)^4}\,\frac{1}{(\ell^2 - m^2 + i\epsilon)\big((\ell - q)^2 - m^2 + i\epsilon\big)} \;=\; B_0(s; m, m), \tag{60.14}$$

the equality identifying it with the master integral evaluated in Chapter D. Its ultraviolet divergence sits in a real, s-independent constant; the entire s-dependence is the parametric logarithm $-\int_0^1 dx\,\ln[m^2 - x(1-x)s - i\epsilon]$, whose branch cut begins at the two-particle threshold $s = 4m^2$.

Worked Example 60.3 (The bubble cut equals two-body phase space)**.** We compute $\operatorname{disc} I_2(s)$ two ways and show they agree.

Analytic discontinuity. From Eq. (D.105), retaining only the s-dependent piece,

$$I_2(s) = \frac{i}{16\pi^2}\Big[\,c - \int_0^1 dx\,\ln\!\big(m^2 - x(1-x)s - i\epsilon\big)\Big] + O(\varepsilon), \tag{60.15}$$

with c a real constant collecting the $1/\varepsilon$ pole and scheme terms. The logarithm's argument $m^2 - x(1-x)s$ is negative precisely for $x \in (x_-, x_+)$, where $x_\pm = \tfrac{1}{2}(1 \pm \beta)$ are the roots of $x(1-x) = m^2/s$ and

$$\beta \equiv \sqrt{1 - \frac{4m^2}{s}} \tag{60.16}$$

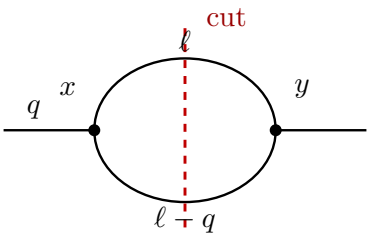


Figure 60.1: The two-particle cut of the scalar bubble $I_2(s)$. The vertical dashed line severs both internal propagators, putting the loop momenta ℓ and $\ell - q$ simultaneously on shell. By Theorem 60.1 each cut line becomes $-2\pi i\,\delta^+$, and the integral over the loop momentum collapses onto the two-body Lorentz-invariant phase space of the intermediate pair. The discontinuity of the bubble across its cut at $s = 4m^2$ is this phase-space volume.

is the velocity of either intermediate particle in the centre-of-mass frame; such an interval exists only for $s > 4m^2$, fixing the threshold. The $-i\epsilon$ places a negative argument just below the cut, so $\ln(\text{negative} - i\epsilon) = \ln|\cdots| - i\pi$. The discontinuity is the jump as s crosses to the upper lip, equivalently $2\,\mathrm{Re}\,I_2$ — not $2i\,\mathrm{Im}\,I_2$, because the overall factor of i in Eq. (60.15) means I_2 is i times a real-analytic function rather than real-analytic itself, so the relation $\mathrm{disc} = 2i\,\mathrm{Im}$ of Eq. (60.6) (which applies to the real-analytic amplitude $\mathcal{M}$) here turns the imaginary part of the bracket into the real part of I_2; only the logarithm jumps, contributing $-i\pi$ over the interval of length $x_+ - x_- = \beta$:

$$\mathrm{disc}\,I_2(s) = \frac{i}{16\pi^2}\Big[-(2i)\int_{x_-}^{x_+} dx\,(-\pi)\Big] = \frac{i}{16\pi^2}\,(2i\pi\beta) = -\frac{\beta}{8\pi} \qquad (s > 4m^2). \tag{60.17}$$

Cutkosky / phase-space side. Apply Theorem 60.1 to Eq. (60.14): there is one two-particle cut (Fig. 60.1), severing both lines, with ℓ and $q - \ell$ flowing across it. Each cut propagator becomes $-2\pi i\,\delta^+$:

$$\begin{aligned}\mathrm{disc}\,I_2(s) &= \int \frac{d^4\ell}{(2\pi)^4}\,(-2\pi i)\delta^+(\ell^2 - m^2)\,(-2\pi i)\delta^+((q-\ell)^2 - m^2) \\ &= -(2\pi)^2 \int \frac{d^4\ell}{(2\pi)^4}\,\delta^+(\ell^2 - m^2)\,\delta^+((q-\ell)^2 - m^2) \\ &= -\int \frac{d^3\boldsymbol{\ell}}{(2\pi)^3\,2E_{\boldsymbol{\ell}}}\,(2\pi)\,\delta^+((q-\ell)^2 - m^2) \;=\; -\int \mathrm{d}\Pi_2 \;=\; -\,\Phi_2(s),\end{aligned} \tag{60.18}$$

where in the third line one factor $2\pi\,\delta^+$ was reduced to the single-particle measure by Eq. (60.9), and the surviving $2\pi\,\delta^+$ together with the integration over $\boldsymbol{\ell}$ is by definition the two-body Lorentz-invariant phase space $\Phi_2(s) = \int \mathrm{d}\Pi_2$. The

factor $(-2\pi i)^2 = -(2\pi)^2$ supplies the overall sign. The two-body volume was computed in closed form in Chapter 16; for equal masses,

$$\Phi_2(s) = \int \mathrm{d}\Pi_2 = \frac{1}{8\pi}\sqrt{1-\frac{4m^2}{s}} = \frac{\beta}{8\pi}. \tag{60.19}$$

Thus $\operatorname{disc} I_2(s) = -\Phi_2(s) = -\beta/8\pi$, identical to the analytic result Eq. (60.17). The discontinuity of the bubble *is* the two-body phase-space volume of the state that can go on shell at $s = 4m^2$, with the characteristic threshold factor $\beta = \sqrt{1-4m^2/s}$ appearing on both sides — vanishing at threshold, where the pair is produced at rest, and rising to $1/8\pi$ as $s \to \infty$.

The match in Theorem 60.3 is the optical theorem realized on one Feynman integral, and it is the same computation that closes Theorem D.42 for ϕ^4 theory, where the bubble's imaginary part is checked against $|\mathcal{M}_{\text{tree}}|^2$ integrated over phase space. The lesson is general and worth stating sharply: *cutting a loop is integrating a product of trees over phase space.* The cut throws away the off-shell, scheme- and gauge-dependent interior of the loop and keeps only the gauge-invariant on-shell data — here the trivial constant tree amplitude, in richer cases the Parke–Taylor and BCFW amplitudes of Chapters 58 and 59.

60.1.5 QED: the vacuum polarization as a two-particle cut

The cleanest physical incarnation of the cutting rule in the benchmark theories is the photon vacuum polarization, computed by brute force in Chapter 19. We recover its imaginary part as a two-particle cut and identify it, through the optical theorem, with the e^+e^- annihilation cross section — closing the loop between the Feynman-diagram result of Chapter 19 and the unitarity of Chapter 14.

The renormalized scalar vacuum polarization $\widehat{\Pi}(q^2)$, defined through $\Pi^{\mu\nu} = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\widehat{\Pi}$ in Eq. (19.57), was found in Eq. (19.65) to have the imaginary part

$$\operatorname{Im}\widehat{\Pi}(q^2) = -\frac{\alpha}{3}\Big(1+\frac{2m^2}{q^2}\Big)\sqrt{1-\frac{4m^2}{q^2}} \qquad (q^2 > 4m^2), \tag{60.20}$$

switching on at the pair threshold $q^2 = 4m^2$. In Theorem 19.7 this same number was reproduced as the Cutkosky cut Eq. (19.66): the two internal fermion lines are put on shell by $1/(\not{p} - m + i\epsilon) \to (\not{p} + m)\,2\pi\,\delta^+(p^2 - m^2)$, the spin sum builds the Dirac trace $\operatorname{tr}[(\not{p}_1 + m)\gamma^\mu(\not{p}_2 - m)\gamma^\nu]$, and the two-body phase-space integral Eq. (60.19) delivers Eq. (60.20) exactly. We do not repeat that trace here — it is carried in full at Theorem 19.7 — but emphasize its structure: the cut of the photon two-point function is the tree amplitude $\gamma^* \to e^+e^-$ times its conjugate, summed over spins and integrated over the two-body phase space of the pair. The branch point at $q^2 = 4m^2$ is owned by the e^+e^- channel, exactly as Section 60.1.2 requires.

The optical theorem now turns this discontinuity into a cross section. Apply the forward theorem Eq. (60.4) to $e^+e^- \to e^+e^-$ at order e^4, where the relevant imaginary part comes from the s-channel photon dressed by one self-energy insertion; the cut through that insertion is the $e^+e^- \to f\bar{f}$ production rate, summed over the species f the photon couples to. Tracking the leptonic coupling and the two photon propagators that flank the self-energy, the total annihilation cross section into the photon's channels is

$$\sigma_{\text{tot}}(e^+e^- \to \text{anything via } \gamma^*) = -\frac{4\pi\alpha}{q^2}\,\operatorname{Im}\widehat{\Pi}(q^2), \tag{60.21}$$

the imaginary part of the vacuum polarization being, up to the kinematic factor $q^2/4\pi\alpha$, the e^+e^- cross section itself. We pin the proportionality constant by an independent computation rather than asserting it. Put a single charged fermion of mass m_f in the loop; Eq. (60.20) gives $-\operatorname{Im}\widehat{\Pi}_f = \frac{\alpha}{3}(1+2m_f^2/q^2)\beta_f$ with $\beta_f = \sqrt{1-4m_f^2/q^2}$, so Eq. (60.21) predicts

$$\sigma(e^+e^- \to f\bar{f}) = \frac{4\pi\alpha}{q^2}\cdot\frac{\alpha}{3}\Big(1+\frac{2m_f^2}{q^2}\Big)\beta_f = \frac{4\pi\alpha^2}{3q^2}\sqrt{1-\frac{4m_f^2}{q^2}}\,\Big(1+\frac{2m_f^2}{q^2}\Big). \tag{60.22}$$

This is precisely the threshold-corrected QED annihilation cross section assembled directly from the squared amplitude and two-body phase space in Chapter 17 (it reduces to the textbook $\sigma = 4\pi\alpha^2/3q^2$ when $m_f \to 0$). The two routes — "imaginary part of a one-loop two-point function" and "squared tree amplitude integrated over phase space" — meet on the same number, including the $(1+2m_f^2/q^2)$ structure and the velocity factor β_f. The imaginary part of $\widehat{\Pi}$ is not bookkeeping: it is the rate at which a virtual photon materializes real pairs, and unitarity guarantees the loop knew this in advance.

Physical Insight

Equation Eq. (60.21) is the engine behind the measured running of α. Because $\widehat{\Pi}(q^2)$ is analytic with its only cut along $q^2 \geq 4m^2$, the dispersion relation of Section 5.5.5 reconstructs the full $\widehat{\Pi}(q^2)$ from its imaginary part, and Eq. (60.21) expresses that imaginary part as a physically measured cross section. Summed over all charged species, $-\operatorname{Im}\widehat{\Pi}(q^2) \propto \sigma(e^+e^- \to \text{hadrons})$, so the hadronic contribution to the running coupling — incalculable from first principles at low energy — is fixed by experimental e^+e^- data through exactly this cut. Unitarity here is not a check but a measurement strategy.

60.1.6 Cut-constructibility: reconstructing an amplitude from its cuts

We have shown that each branch cut of an amplitude is computable as a phase-space integral of lower on-shell data. The converse question — how much of the amplitude is *determined* by its cuts — is what makes this a method rather than a check, and its answer sets the agenda for the rest of the chapter.

If $\mathcal{M}(s)$ vanished sufficiently fast at large $|s|$, real analyticity Eq. (60.5) and Cauchy's theorem would reconstruct it completely from its discontinuity, the unsubtracted dispersion relation of Eq. (5.143),

$$\mathcal{M}(s) = \frac{1}{2\pi i}\int_{s_0}^{\infty} ds' \, \frac{\operatorname{disc}\mathcal{M}(s')}{s'-s} = \frac{1}{\pi}\int_{s_0}^{\infty} ds' \, \frac{\operatorname{Im}\mathcal{M}(s'+i0)}{s'-s}, \tag{60.23}$$

with s_0 the lowest threshold. The cuts would then literally *be* the amplitude. In a loop calculation the situation is the same in spirit but qualified in one respect: a polynomial in s has no discontinuity and is invisible to Eq. (60.23); when $\mathcal{M}$ does not decay, the dispersion relation requires *subtractions* (Section 5.5.5), and each subtraction is an undetermined constant — a piece of the amplitude carrying no cut. This is the seed of the "rational remainder" that will occupy us later in the chapter.

Definition 60.4 (Cut-constructible)**.** An amplitude is *cut-constructible* if it is determined entirely by its branch-cut discontinuities, i.e. if its dispersive reconstruction Eq. (60.23) requires no subtraction constants. Equivalently, the amplitude contains no terms that are rational in the kinematic invariants and free of logarithms, dilogarithms, or other branch cuts. A general one-loop amplitude is a cut-constructible part plus a rational remainder; in $d = 4-2\varepsilon$ dimensions the rational part itself originates in cuts taken with the loop momentum continued off four dimensions, so that unitarity, applied in d dimensions, recovers the amplitude in full.

The largest-time equation Eq. (60.13) guarantees that every discontinuity is a sum of cuts and is thus accessible to this program; the dispersion relation Eq. (60.23) guarantees that the discontinuities, up to subtraction constants, rebuild the amplitude. Between these two facts lies the entire unitarity method: compute the cuts as products of trees integrated over phase space, and integrate them back up to the amplitude. What Eq. (60.23) hides, however, is the labor of the final integral over s' — carried out naively it is no simpler than the original loop. The decisive simplification, and the reason the method is practical, is that the functions a one-loop amplitude can possibly be are not arbitrary: they are a small, fixed list of scalar integrals whose own cuts are known once and for all. Determining an amplitude then reduces to determining the *coefficients* with which those standard integrals appear, and a cut isolates exactly the coefficients of the integrals that share that cut. To make this concrete we must first establish

what that list of standard integrals is — the master-integral basis — which is the task of the next section.

60.2 The One-Loop Basis: Reduction to Master Integrals

The previous section established the principle on which this whole chapter turns: a one-loop amplitude is fixed by its branch-cut discontinuities, and each discontinuity is computed by cutting propagators and gluing on-shell lower amplitudes. A principle, however, is not yet a method. To turn "an amplitude is determined by its cuts" into an algorithm we need to know *what kind of function* the amplitude is — the precise catalogue of analytic objects whose discontinuities those cuts can probe. If that catalogue were infinite, knowing the cuts would constrain little. The remarkable structural fact, and the target of this section, is that the catalogue is *finite*: in dimensional regularization every one-loop amplitude, in any quantum field theory, is a linear combination of a small fixed set of scalar integrals — boxes, triangles, bubbles, and tadpoles — with coefficients that are *rational* functions of the kinematic invariants, plus a purely rational remainder carrying no cut at all. The transcendental content of any one-loop amplitude — its logarithms, dilogarithms, and the discontinuities they carry — lives entirely in those four scalar functions. Everything else is rational bookkeeping.

This is the decisive simplification. It means that determining a one-loop amplitude is not the open-ended problem of summing an arbitrary function diagram by diagram, but the *finite* problem of fixing a list of numbers: one rational coefficient per master integral, plus the rational remainder. The unitarity method, developed in the following sections, is precisely the machinery that reads off each of these coefficients from on-shell tree data. Before we can isolate a coefficient we must know which coefficients exist; this section provides that list and proves it is complete. We proceed in four movements: first the algebraic engine — Passarino–Veltman tensor reduction — that strips loop momenta out of numerators; then the scalar master integrals themselves, with their explicit values; then the structural theorem that these four masters span everything, proved through the reduction of higher-point integrals and culminating in the pentagon-to-box reduction; and finally the statement of the master decomposition that hands the next section its finite list of unknowns.

60.2.1 The generic one-loop integrand and the two reductions

Fix conventions. We work throughout in $d = 4 - 2\varepsilon$ dimensions with the mostly-minus metric, the Feynman $i\varepsilon$ prescription, and the $\overline{\text{MS}}$ scale μ, exactly as in Chapter D, whose master integrals we import wholesale rather than re-deriving.

A single one-loop Feynman diagram, after the colour and Lorentz algebra of its vertices has been carried out by the rules of Chapters 15 and 17, contributes a loop integral of the form

$$I \;=\; \mu^{2\varepsilon}\int \frac{\mathrm{d}^{d}\ell}{(2\pi)^{d}}\,\frac{N(\ell)}{D_0\,D_1\cdots D_{N-1}}, \qquad D_i \;=\; (\ell + r_i)^2 - m_i^2 + i\varepsilon, \tag{60.24}$$

where N is the number of propagators in the loop, the r_i are partial sums of the external momenta entering the loop up to leg i (with one of them, say $r_0 = 0$, chosen as the origin of loop momentum by a shift), the m_i are the internal masses, and the numerator $N(\ell)$ is a polynomial in the loop momentum ℓ of some degree r — the *rank* of the integral. Tree-level Feynman rules with at most one derivative per vertex bound $r \le N$; the precise bound will not matter, only that r is finite. The full one-loop amplitude $\mathcal{M}^{\text{1-loop}}$ is a sum of such integrals, one per diagram, each multiplied by a kinematic and group-theoretic prefactor.

Two structural features of (60.24) stand between us and a scalar master integral: the numerator carries open loop momenta, and the denominator carries an arbitrary number N of propagators. Two corresponding reductions dispose of them, and the entire basis theorem is the statement that, applied repeatedly, they terminate.

Rank reduction removes the numerator. The Passarino–Veltman algorithm, built in full in Section D.6 and applied there to the QED one-loop vertex, self-energy, and vacuum polarization of Chapter 19, writes a tensor integral $\int \ell^{\mu_1}\cdots\ell^{\mu_r}/(D_0\cdots D_{N-1})$ as the most general Lorentz tensor its external data permit, then contracts that ansatz with the available tensors (the metric and the external momenta) to convert it into a linear system for scalar coefficient functions. The system is solved by inverting the *Gram matrix* of the external momenta. The output is the same integral with the numerator removed and the kinematics shifted, plus integrals with *fewer* propagators. We will see this explicitly below.

Propagator reduction removes excess denominators. A scalar integral with N propagators depends on $N-1$ independent external momentum differences. In four dimensions at most four vectors can be linearly independent, so once N is large enough the external momenta are forced into linear dependence, the corresponding Gram determinant vanishes, and the integral collapses onto integrals with fewer propagators. The boundary case — the pentagon, $N = 5$ — is the subtle one, and we treat it in full in Section 60.2.5.

The claim to be proved is that iterating these two reductions on (60.24) always terminates on a fixed finite set of *scalar* ($N(\ell) = 1$) integrals with $N \le 4$ propagators. We first name and value those scalar integrals.

60.2.2 The scalar master integrals

Definition 60.5 (The one-loop scalar masters)**.** The scalar master integrals are the integrals (60.24) with unit numerator, $N(\ell) = 1$, and $N = 1, 2, 3, 4$ propagators, in the standard Passarino–Veltman naming (Section D.6):

$$\text{tadpole} \quad A_0(m_0) = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{D_0}, \tag{60.25}$$

$$\text{bubble} \quad B_0(p^2; m_0, m_1) = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{D_0 D_1}, \tag{60.26}$$

$$\text{triangle} \quad C_0(p_1^2, p_2^2, p_{12}^2;\, m_0, m_1, m_2) = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{D_0 D_1 D_2}, \tag{60.27}$$

$$\text{box} \quad D_0(\{s_{ij}\}; \{m_i\}) = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{D_0 D_1 D_2 D_3}, \tag{60.28}$$

with $D_i = (\ell + r_i)^2 - m_i^2 + i\varepsilon$, the arguments listing the external invariants formed from the r_i and the internal masses on which each integral depends.

These four functions are the entire transcendental vocabulary of one-loop physics, and the rest of this section proves that nothing else is needed. Their explicit values are computed once and for all in Chapter D; we collect them here in the forms the unitarity method will use, because the *branch-cut structure* of each — which we will compare in Section 60.2.6 — is what makes the four linearly independent and hence makes the master decomposition unique.

The tadpole follows immediately from the master scalar integral Theorem D.22 at $n = 1$, $\Delta = m_0^2$:

$$A_0(m_0) = \frac{i\, m_0^2}{(4\pi)^2} \Big[\frac{1}{\bar{\varepsilon}} + 1 - \ln \frac{m_0^2}{\mu^2} \Big] + O(\varepsilon), \qquad \frac{1}{\bar{\varepsilon}} \equiv \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi, \tag{60.29}$$

where the $1/\bar{\varepsilon}$ pole is purely *ultraviolet* and $\bar{\varepsilon}$ is the $\overline{\text{MS}}$ combination of Eq. (D.136). The tadpole carries a single mass scale m_0^2 and *no* external invariant: it has no kinematic channel in which to develop a discontinuity, so $\operatorname{disc} A_0 = 0$. It is, from the point of view of cuts, invisible — a fact we will use heavily.

The bubble is the canonical one-loop function. For equal internal masses, Eq. (D.140) gives

$$B_0(p^2; m, m) = \frac{i}{(4\pi)^2} \Big[\frac{1}{\bar{\varepsilon}} - \int_0^1 \mathrm{d}x\, \ln \frac{m^2 - x(1-x)p^2}{\mu^2} \Big] + O(\varepsilon). \tag{60.30}$$

Its $1/\bar{\varepsilon}$ is ultraviolet; its finite part is a single logarithm of p^2 that, for $p^2 > 4m^2$, develops an imaginary part — the discontinuity across the two-particle threshold, computed in Section D.8 and identified there with the optical theorem. The massless bubble ($m = 0$) is even simpler:

$$B_0(p^2; 0, 0) = \frac{i}{(4\pi)^2} \Big[\frac{1}{\bar{\varepsilon}} + 2 - \ln \frac{-p^2}{\mu^2} \Big] + O(\varepsilon), \tag{60.31}$$

carrying a single logarithm $\ln(-p^2)$ with a branch cut along $p^2>0$.

The triangle and box carry the genuinely two-variable transcendentality. The massless triangle with one off-shell leg of invariant $s=p_{12}^2$ and two light-like legs, the case that the reduction of massless gauge-theory vertices deposits, is evaluated in Theorem D.38:

$$C_0(0,0,s;0,0,0)=\frac{-i}{(4\pi)^2}\,\frac{1}{s}\Big[\frac{1}{\varepsilon_{\mathrm{IR}}^2}-\frac{1}{\varepsilon_{\mathrm{IR}}}\,\ln\frac{-s}{\mu^2}+\frac{1}{2}\ln^2\frac{-s}{\mu^2}-\frac{\pi^2}{12}\Big]+O(\varepsilon), \tag{60.32}$$

with both poles *infrared* (the integral is ultraviolet finite by power counting) and a characteristic $\ln^2(-s)$ Sudakov double logarithm.

The box is the top of the hierarchy and the only master with two independent kinematic invariants. Take the fully massless box — all four internal lines and all four external legs light-like — depending on the Mandelstam invariants $s=(p_1+p_2)^2$ and $t=(p_2+p_3)^2$. Combining the four denominators with the Feynman-parameter identity Eq. (D.25) and reducing to the master scalar integral Theorem D.22 by the completion-of-square and Wick rotation of Section D.3, the parameter integral evaluates to

$$D_0^{0m}(s,t)=\frac{i\,c_\Gamma}{(4\pi)^2}\,\frac{2}{s\,t}\Big\{\frac{1}{\varepsilon_{\mathrm{IR}}^2}\Big[\Big(\frac{-s}{\mu^2}\Big)^{-\varepsilon}+\Big(\frac{-t}{\mu^2}\Big)^{-\varepsilon}\Big]-\frac{1}{2}\ln^2\frac{-s}{-t}-\frac{\pi^2}{2}\Big\}+O(\varepsilon), \tag{60.33}$$

with $c_\Gamma=\Gamma(1+\varepsilon)\Gamma(1-\varepsilon)^2/\Gamma(1-2\varepsilon)$ the ubiquitous one-loop constant of Eq. (D.217). The double pole $1/\varepsilon^2$ is again entirely infrared, but the decisive new feature is the appearance of *two* invariants s and t and the cross-channel double logarithm $\ln^2(s/t)$: the box, alone among the masters, has discontinuities in two distinct kinematic channels.

Remark 60.6 (Derivation of the massless box). The tadpole, bubble, and triangle are evaluated in full in Chapter D, which reduces tensor boxes *to* the scalar D_0 but does not evaluate D_0 itself; since this box is the master on which the chapter's worked reconstruction rests, we carry out the one genuinely new step here — the four-propagator Feynman-parameter integral. Combining the four denominators with Eq. (D.25) and completing the square as in Section D.2 shifts the loop momentum and replaces the product by $(\ell^2-\Delta)^4$ with the massless-box discriminant

$$\Delta\;=\;-(s\,x_1x_3+t\,x_2x_4), \tag{60.34}$$

the two surviving cross terms being precisely the two Mandelstam channels. The Wick rotation and radial integration of Section D.3 and Theorem D.22, here with $N=4$ propagators so that $\Gamma(N-d/2)=\Gamma(2+\varepsilon)$ and the Feynman $\Gamma(4)$ cancels, leave the parameter integral

$$D_0^{0m}(s,t)=\mu^{2\varepsilon}\,\frac{i\,\Gamma(2+\varepsilon)}{(4\pi)^{d/2}}\int_{\Delta_3}\Big(\prod_{i=1}^4\mathrm{d}x_i\Big)\,\delta\Big(1-\textstyle\sum_i x_i\Big)\,\big[(-s)\,x_1x_3+(-t)\,x_2x_4\big]^{-2-\varepsilon}, \tag{60.35}$$

over the simplex of Section D.2.4 (the "two-dimensional" integral, three parameters constrained by one δ). The overlap of the two monomials is disentangled by a Mellin–Barnes split,

$$\frac{1}{(A+B)^{2+\varepsilon}} = \frac{1}{\Gamma(2+\varepsilon)}\,\frac{1}{2\pi i}\int_{\mathcal{C}} \mathrm{d}z\;\Gamma(2+\varepsilon+z)\,\Gamma(-z)\,\frac{B^z}{A^{2+\varepsilon+z}}, \qquad A=(-s)x_1x_3,\; B=(-t)x_2x_4, \tag{60.36}$$

after which every parameter integral is the Dirichlet (Beta-function) integral over the simplex, $\int_{\Delta_3}\prod_i x_i^{a_i-1}\,\delta(1-\sum_i x_i)=\prod_i\Gamma(a_i)/\Gamma(\sum_i a_i)$, with $a_{1,3}=-1-\varepsilon-z$ and $a_{2,4}=1+z$, so $\sum_i a_i=-2\varepsilon$. The parameter integral collapses to a single contour integral,

$$D_0^{0m}(s,t) = \frac{i\,\mu^{2\varepsilon}}{(4\pi)^{d/2}\,\Gamma(-2\varepsilon)}\,\frac{1}{2\pi i}\int_{\mathcal{C}}\mathrm{d}z\;\Gamma(2+\varepsilon+z)\,\Gamma(-z)\,\Gamma(1+z)^2\,\Gamma(-1-\varepsilon-z)^2\,(-s)^{-2-\varepsilon-z}(-t)^z. \tag{60.37}$$

Closing the contour and summing residues evaluates the box. The prefactor $1/\Gamma(-2\varepsilon)=-2\varepsilon+O(\varepsilon^2)$ multiplies the $1/\varepsilon^3$ that the pinch of the double poles $\Gamma(1+z)^2$ and $\Gamma(-1-\varepsilon-z)^2$ generates: the leading residue, at $z=-1-\varepsilon$ where $(-s)^{-2-\varepsilon-z}(-t)^z=(-s)^{-\varepsilon}/(st)$, produces the double infrared pole $\frac{1}{\varepsilon^2}(-s/\mu^2)^{-\varepsilon}\cdot 2/(st)$, its $s\leftrightarrow t$ mirror gives the $(-t/\mu^2)^{-\varepsilon}$ term, and the subleading residues assemble the cross-channel finite part $-\frac{1}{2}\ln^2(s/t)-\pi^2/2$ — exactly Eq. (60.33). The double pole is the soft-collinear corner of the simplex where two adjacent parameters vanish together; the residue bookkeeping is the standard one [448], and the all-orders-in-ε hypergeometric form is recorded in [67].

60.2.3 Passarino–Veltman reduction in practice

The two worked reductions that follow exhibit the rank-reduction engine in its entirety, carrying every algebraic step including the inversion of the kinematic matrix. They are the operations that, repeated, prove the basis theorem, and they are the operations the reader must trust when the unitarity method later quotes a reduced integral. The denominator identity on which both rest is elementary: since each D_i is itself quadratic in ℓ, any scalar product of the loop momentum with external data is a linear combination of the D_i plus constants. Explicitly, from $D_i-D_0=(\ell+r_i)^2-(\ell+r_0)^2-m_i^2+m_0^2$ with $r_0=0$,

$$2\,r_i\cdot\ell = D_i - D_0 - (r_i^2+m_0^2-m_i^2), \qquad \ell^2 = D_0+m_0^2, \tag{60.38}$$

the same identity Eq. (D.161) that powers the appendix reductions. Each time it is used, a numerator factor of $r_i\cdot\ell$ either cancels a denominator (lowering the propagator count) or contributes to a lower-rank tensor.

Worked Example 60.7 (Rank-one triangle reduction with explicit Gram inversion)**.** Consider the rank-one three-point tensor integral that the Passarino–Veltman decomposition of any one-loop vertex with a single loop momentum in

the numerator produces,

$$C^\mu \;=\; \mu^{2\varepsilon}\!\int\!\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu}{D_0\,D_1\,D_2}, \qquad \begin{aligned} D_0 &= \ell^2 - m_0^2,\\ D_1 &= (\ell+p_1)^2 - m_1^2,\\ D_2 &= (\ell+p_2)^2 - m_2^2, \end{aligned} \tag{60.39}$$

with two independent external momenta p_1, p_2. The integrand depends, besides ℓ, only on p_1 and p_2, so the vector C^μ can be built from no other vectors: it lies in the plane they span. We therefore write the most general covariant ansatz,

$$C^\mu \;=\; p_1^\mu\, C_1 + p_2^\mu\, C_2, \tag{60.40}$$

with scalar coefficients C_1, C_2 that are functions of the invariants $p_1^2, p_2^2, (p_1-p_2)^2$ and the masses alone. *There is no ℓ^μ left on the right-hand side: the integration has used it up.* Our task is to solve for C_1, C_2 in terms of the scalar triangle C_0 and scalar bubbles.

Build the linear system. Contract (60.40) with $p_{1\mu}$ and with $p_{2\mu}$. The left-hand sides become scalar integrals,

$$p_i\cdot C \;=\; \mu^{2\varepsilon}\!\int\!\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{p_i\cdot\ell}{D_0 D_1 D_2} \;\equiv\; R_i, \qquad i=1,2, \tag{60.41}$$

while the right-hand sides assemble into a matrix equation,

$$\begin{pmatrix} p_1^2 & p_1\cdot p_2 \\ p_1\cdot p_2 & p_2^2 \end{pmatrix}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix}, \qquad G \equiv \begin{pmatrix} p_1^2 & p_1\cdot p_2 \\ p_1\cdot p_2 & p_2^2 \end{pmatrix}. \tag{60.42}$$

The coefficient matrix G is precisely the 2×2 *Gram matrix* of the external momenta p_1, p_2.

Evaluate the right-hand sides with the denominator identity. From (60.38),

$$2\,p_1\cdot\ell = D_1 - D_0 - f_1, \quad 2\,p_2\cdot\ell = D_2 - D_0 - f_2, \qquad f_i \equiv p_i^2 + m_0^2 - m_i^2. \tag{60.43}$$

Insert the first into R_1:

$$\begin{aligned} R_1 &= \tfrac{1}{2}\,\mu^{2\varepsilon}\!\int\!\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{D_1 - D_0 - f_1}{D_0 D_1 D_2} \\ &= \tfrac{1}{2}\Big[\underbrace{\mu^{2\varepsilon}\!\int\!\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{D_0 D_2}}_{B_0^{(1)}} - \underbrace{\mu^{2\varepsilon}\!\int\!\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{D_1 D_2}}_{B_0^{(0)}} - f_1\,C_0\Big]. \end{aligned} \tag{60.44}$$

In the first term the factor D_1 has cancelled, leaving a bubble missing propagator 1; in the second, D_0 has cancelled, leaving a bubble missing propagator 0. We name these by the propagator *pinched* (deleted): $B_0^{(i)}$ is the bubble obtained

from the triangle by removing line i. After shifting each to standard form (a translation of a dimensionally regularized integral, legitimate by Theorem D.10),

$$B_0^{(0)} = B_0((p_1-p_2)^2; m_1, m_2), \quad B_0^{(1)} = B_0(p_2^2; m_0, m_2), \quad B_0^{(2)} = B_0(p_1^2; m_0, m_1). \tag{60.45}$$

Identically,

$$R_2 = \tfrac{1}{2}\big[\, B_0^{(2)} - B_0^{(0)} - f_2\, C_0 \,\big]. \tag{60.46}$$

The right-hand sides are now built entirely from the scalar triangle C_0 and scalar bubbles — masters we have already valued.

Invert the Gram matrix. The determinant of G is

$$\det G \;=\; p_1^2\, p_2^2 - (p_1 \cdot p_2)^2, \tag{60.47}$$

and its inverse is the standard 2×2 adjugate over determinant,

$$G^{-1} = \frac{1}{\det G} \begin{pmatrix} p_2^2 & -\,p_1\!\cdot\!p_2 \\ -\,p_1\!\cdot\!p_2 & p_1^2 \end{pmatrix}. \tag{60.48}$$

Multiplying (60.42) on the left by G^{-1} gives the coefficients in closed form,

$$\boxed{C_1 = \frac{p_2^2\, R_1 - (p_1\!\cdot\!p_2)\, R_2}{\det G}, \qquad C_2 = \frac{p_1^2\, R_2 - (p_1\!\cdot\!p_2)\, R_1}{\det G},} \tag{60.49}$$

with R_1, R_2 from (60.44)–(60.46). This completes the reduction: the rank-one triangle C^μ has been expressed, with *no integration left to perform*, as the rational combination (60.40) of the scalar triangle C_0 and the three pinched scalar bubbles $B_0^{(i)}$, the rational coefficients being ratios of Gram determinants and the constants f_i.

The cost of inversion. The factor $1/\det G$ is the honest price of tensor reduction. When the external momenta degenerate — $p_2 \to x\, p_1$ for some scalar x, a forward or collinear configuration — the Gram determinant $\det G = p_1^2 p_2^2 - (p_1 \cdot p_2)^2$ vanishes while the numerators $p_2^2 R_1 - (p_1 \cdot p_2) R_2$ do not vanish term by term, so C_1, C_2 appear to diverge. The singularity is *spurious*: the original integral C^μ is perfectly finite there, and the apparent pole cancels among the scalar functions in the numerator in the exact limit. But the cancellation is between large numbers, so a numerical evaluation near $\det G = 0$ loses precision catastrophically. This Gram-determinant instability is the practical defect of Passarino–Veltman reduction that the unitarity method — which works at the level of on-shell integrands, never inverting a Gram matrix — is designed to avoid (Section 60.2.7 and the following section).

The second reduction we record more briefly, because its only new feature is the appearance of the metric tensor in the covariant basis, and because the appendix carries the same computation in full at Eqs. (D.174) and (D.175); we reproduce the logic so the basis theorem's proof is self-contained.

Worked Example 60.8 (Rank-two bubble reduction)**.** The rank-two two-point integral, which the reduction of a self-energy with two loop momenta in the numerator produces, is

$$B^{\mu\nu} = \mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu \ell^\nu}{D_0 D_1}, \qquad D_0 = \ell^2 - m_0^2, \quad D_1 = (\ell+p)^2 - m_1^2. \tag{60.50}$$

Now there is a single external vector p^μ, so the symmetric rank-two tensor can be built from $p^\mu p^\nu$ and from the metric $\eta^{\mu\nu}$:

$$B^{\mu\nu} = p^\mu p^\nu\, B_{11} + \eta^{\mu\nu}\, B_{00}. \tag{60.51}$$

We will need the rank-one coefficient B_1 first, defined by $B^\mu = p^\mu B_1$. Contracting $B^\mu = \mu^{2\varepsilon}\int \ell^\mu/(D_0D_1)$ with p_μ and using $p\cdot\ell = \tfrac{1}{2}(D_1 - D_0 - f)$ with $f = p^2 + m_0^2 - m_1^2$, exactly as in the triangle case, yields the pinched tadpoles $A_0(m_0), A_0(m_1)$ and the bubble B_0:

$$B_1 = \frac{1}{2p^2}\Big[A_0(m_0) - A_0(m_1) - (p^2 + m_0^2 - m_1^2)B_0\Big]. \tag{60.52}$$

The two contractions. Contract (60.51) with the metric $\eta_{\mu\nu}$. On the left, $\eta_{\mu\nu}\ell^\mu\ell^\nu = \ell^2 = D_0 + m_0^2$, so

$$\eta_{\mu\nu}B^{\mu\nu} = \mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{D_0 + m_0^2}{D_0 D_1} = A_0(m_1) + m_0^2\, B_0, \tag{60.53}$$

where $\int 1/D_1 = A_0(m_1)$ after the shift $\ell \to \ell - p$. On the right, $\eta_{\mu\nu}(p^\mu p^\nu B_{11} + \eta^{\mu\nu}B_{00}) = p^2 B_{11} + d\, B_{00}$. Hence

$$p^2\, B_{11} + d\, B_{00} = A_0(m_1) + m_0^2\, B_0. \tag{60.54}$$

Now contract (60.51) with p_μ. On the left,

$$p_\mu B^{\mu\nu} = \tfrac{1}{2}\,\mu^{2\varepsilon}\int \frac{(D_1 - D_0 - f)\,\ell^\nu}{D_0 D_1} = \tfrac{1}{2}\big[\,0 - (-p^\nu A_0(m_1)) - f\,p^\nu B_1\,\big] = \tfrac{1}{2}p^\nu\big[A_0(m_1) - f B_1\big], \tag{60.55}$$

using that $\int \ell^\nu/D_0 = 0$ (odd in ℓ, single scale, no external vector) and $\int \ell^\nu/D_1 = -p^\nu A_0(m_1)$. On the right, $p_\mu(p^\mu p^\nu B_{11} + \eta^{\mu\nu}B_{00}) = p^\nu(p^2 B_{11} + B_{00})$, so

$$p^2\, B_{11} + B_{00} = \tfrac{1}{2}\big[A_0(m_1) - f\, B_1\big]. \tag{60.56}$$

Solve the two-by-two system. Subtract (60.56) from (60.54): the $p^2 B_{11}$ terms cancel, leaving $(d-1)B_{00}$ on the left,

$$\boxed{B_{00} = \frac{1}{2(d-1)}\Big[A_0(m_1) + 2m_0^2 B_0 + (p^2 + m_0^2 - m_1^2)B_1\Big],} \tag{60.57}$$

after using $A_0(m_1) + m_0^2 B_0 - \frac{1}{2}[A_0(m_1) - fB_1] = \frac{1}{2}A_0(m_1) + m_0^2 B_0 + \frac{1}{2}fB_1$. Back-substituting into (60.56),

$$\boxed{B_{11} = \frac{1}{p^2}\Big[\tfrac{1}{2}(A_0(m_1) - fB_1) - B_{00}\Big].} \tag{60.58}$$

Two features generalize to every reduction. First, the metric coefficient B_{00} carries a factor $1/(d-1)$ but *no* $1/p^2$: it is finite as $p^2 \to 0$ and carries the genuine ultraviolet divergence of the rank-two integral (both A_0 and B_0 inside it have $1/\bar{\varepsilon}$ poles). Second, the momentum coefficient B_{11}, like B_1, divides by the Gram determinant — here the 1×1 Gram p^2. *The metric direction is always benign; the momentum directions always carry the Gram denominator.* With (60.52) substituted, (60.57)–(60.58) express the complete rank-two bubble through the scalar bubble B_0 and the tadpoles A_0, with no integration left to do.

These two examples are the algorithm in microcosm. A higher-rank, higher-point tensor is reduced by exactly the same two moves — write the covariant ansatz, contract with metric and external momenta, and solve the resulting Gram system — iterated until both the rank and (for $N \geq 5$) the propagator count have been driven down. We now prove that this iteration terminates, and on what.

60.2.4 The one-loop basis theorem

Theorem 60.9 (One-loop master-integral basis)**.** *Every one-loop integral of the form* (60.24)*, with any rank r and any number N of propagators, in $d = 4 - 2\varepsilon$ dimensions equals a finite linear combination of the scalar master integrals A_0 (tadpole), B_0 (bubble), C_0 (triangle), and D_0 (box) of Theorem 60.5, with coefficients that are rational functions of the external invariants, the internal masses, and the dimension d. In strictly four dimensions ($\varepsilon \to 0$) no scalar integral with more than four propagators is required: the box is the highest master.*

The theorem is the precise statement that the four masters *span* the one-loop integral space over the field of rational functions of the kinematics. Its proof is the termination of the two reductions, and we give it in full. (The statement and its short argument also appear, in the appendix's notation, as Theorem D.35; the appendix defers the pentagon step to this chapter, which we now discharge.)

Proof. We treat the two reductions in turn, then combine them.

Rank reduction terminates. Let I have rank $r \geq 1$. By Lorentz covariance the integral, carrying r free indices and depending on the external momenta $r_1, \dots, r_{N-1}$ and the metric, equals the most general symmetric tensor of rank r built from these. Contracting that ansatz with the r-fold products of metric and external momenta produces a linear system whose unknowns are the scalar

coefficient functions and whose right-hand sides are, by the denominator identity (60.38), integrals in which every loop momentum has been traded either for a constant or for a factor D_i that cancels a propagator. The terms that cancel a propagator are integrals of rank $r-1$ (one loop momentum removed) and propagator count $N-1$; the terms in which loop momenta were traded for constants, together with the metric coefficients, are integrals of rank $r-2$ and propagator count N. Every term produced therefore has strictly smaller rank than I. The coefficient matrix of the linear system is the Gram matrix of the external momenta (its inversion supplies the $1/\det G$ factors seen in Theorems 60.7 and 60.8), and solving introduces only rational functions of the invariants and masses. Iterating, the rank decreases by at least one at each step and reaches zero after at most r steps. Thus I is a rational combination of *scalar* integrals ($r=0$) with propagator counts $\le N$.

Propagator reduction terminates at $N=4$. It remains to reduce scalar integrals with $N \ge 5$ propagators. A scalar N-point integral depends on $N-1$ external momentum differences $r_1, \dots, r_{N-1}$ (recall $r_0 = 0$). In four dimensions any five vectors are linearly dependent, so for $N-1 \ge 5$, i.e. $N \ge 6$, the differences satisfy a nontrivial linear relation. We show this relation reduces the integral to integrals with fewer propagators, then treat the boundary case $N=5$ separately in Section 60.2.5.

Suppose $N \ge 6$. The $N-1 \ge 5$ vectors $r_1, \dots, r_{N-1}$ are linearly dependent in four dimensions, so their $(N-1)\times(N-1)$ Gram matrix $G_{ij} = r_i \cdot r_j$ is singular: $\det G = 0$, and there exists a nonzero vector of constants $(\alpha_1, \dots, \alpha_{N-1})$ in its kernel, $\sum_j G_{ij}\alpha_j = 0$ for all i. Geometrically this says the vector $v \equiv \sum_j \alpha_j r_j$ is orthogonal to every r_i and hence (since the r_i span the four-dimensional space) is the zero vector: $\sum_{j=1}^{N-1} \alpha_j r_j = 0$. We now seek constants $c_0, \dots, c_{N-1}$ such that

$$\sum_{i=0}^{N-1} c_i \, D_i = 1 \quad \text{identically in } \ell. \tag{60.59}$$

Expanding $D_i = \ell^2 + 2r_i \cdot \ell + r_i^2 - m_i^2$, the condition (60.59) splits by powers of ℓ into

$$\sum_{i=0}^{N-1} c_i = 0 \quad (\ell^2), \qquad \sum_{i=1}^{N-1} c_i \, r_i = 0 \quad (\ell^\mu), \qquad \sum_{i=0}^{N-1} c_i \, (r_i^2 - m_i^2) = 1 \quad (\text{const}). \tag{60.60}$$

The vector condition $\sum_{i\ge1} c_i r_i = 0$ is, in four dimensions, four scalar equations; together with $\sum c_i = 0$ that is five homogeneous equations on the N unknowns c_i. For $N \ge 6$ the homogeneous system is underdetermined and possesses a solution space of dimension $\ge N-5 \ge 1$; the existence of the kernel vector (α_j) found above guarantees this solution space is nontrivial. Within it the inhomogeneous normalization $\sum c_i (r_i^2 - m_i^2) = 1$ fixes the remaining freedom generically (it fails only on a measure-zero kinematic locus, where one perturbs and takes the

limit). The c_i so determined are rational functions of the invariants and masses. Dividing (60.59) by the full product $D_0 D_1 \cdots D_{N-1}$ and integrating,

$$\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{1}{D_0 \cdots D_{N-1}} = \sum_{i=0}^{N-1} c_i \, \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{1}{\prod_{k \neq i} D_k}, \tag{60.61}$$

each term on the right being the same integral with propagator i *pinched* — an $(N-1)$-point scalar integral. Thus every $N \geq 6$ scalar integral reduces to $(N-1)$-point integrals with rational coefficients. Iterating drives N down to 5.

Conclusion. Rank reduction reduces any integral to scalars with $N \leq N_{\max}$ propagators; propagator reduction reduces all $N \geq 6$ scalars to pentagons ($N = 5$); and the pentagon, treated next in Section 60.2.5, reduces in four dimensions to scalar boxes plus a contribution that vanishes as $\varepsilon \to 0$. Hence every one-loop integral is a rational-coefficient combination of the four masters A_0, B_0, C_0, D_0, with the box the highest required in four dimensions. All coefficients introduced are rational in the invariants, masses, and d. □ □

The proof shows where each master can come from. Boxes are the irreducible four-propagator integrals; triangles, bubbles, and tadpoles arise both as genuine three-, two-, and one-propagator topologies and as pinched descendants of higher integrals during reduction. The crucial open step, on which the entire "box is the top" statement rests, is the pentagon.

60.2.5 The pentagon and the vanishing of the five-dimensional Gram determinant

The pentagon ($N = 5$) is the boundary case the basis theorem deferred: it depends on $N - 1 = 4$ external difference vectors, which in four dimensions are *generically independent*, so the four-dimensional Gram determinant $\det(r_i \cdot r_j)_{i,j=1}^{4}$ does not vanish and the constant-coefficient linear reduction (60.59) that handled $N \geq 6$ is unavailable. Nonetheless the scalar pentagon reduces to boxes in four dimensions. The mechanism is the vanishing of a *different* Gram determinant — the 5×5 one formed by adjoining the loop momentum to the four external directions — and the price of the reduction is a remainder that is suppressed by ε and therefore vanishes in strictly four dimensions, contributing only to the rational part of the amplitude. We derive it now in full.

Work with the scalar pentagon

$$I_5 = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{1}{D_0 D_1 D_2 D_3 D_4}, \qquad D_i = (\ell + r_i)^2 - m_i^2, \quad r_0 = 0, \tag{60.62}$$

with the four nonzero difference vectors r_1, r_2, r_3, r_4 spanning four-dimensional Minkowski space (generic kinematics; the four-dimensional Gram $G_{ij} = r_i \cdot r_j$, $i, j = 1, \ldots, 4$, has $\det G \neq 0$). In dimensional regularization the loop momentum

is d-dimensional. Decompose it into the part lying in the four-dimensional physical span $V = \mathrm{span}\{r_1, \dots, r_4\}$ and the orthogonal $(d-4)$-dimensional remainder,

$$\ell^\mu = \ell_\parallel^\mu + \ell_\perp^\mu, \qquad \ell_\parallel \in V, \quad \ell_\perp \perp V, \tag{60.63}$$

so that $\ell^2 = \ell_\parallel^2 + \ell_\perp^2$ and, since the external r_i all lie in V, the projections probe only the parallel part: $r_i \cdot \ell = r_i \cdot \ell_\parallel$. The orthogonal piece is spacelike (it lives in the Euclidean extra dimensions), and we write $\ell_\perp^2 = -\mu_\perp^2$ with $\mu_\perp^2 \geq 0$. This $\mu_\perp^2$ is the entire source of the ε-suppressed remainder.

The key algebraic input is that the *five* vectors $\{\ell_\parallel, r_1, r_2, r_3, r_4\}$ all lie in the four-dimensional space V, hence are linearly dependent: their 5×5 Gram determinant vanishes,

$$\det \begin{pmatrix} \ell_\parallel \cdot \ell_\parallel & \ell_\parallel \cdot r_1 & \cdots & \ell_\parallel \cdot r_4 \\ r_1 \cdot \ell_\parallel & r_1 \cdot r_1 & \cdots & r_1 \cdot r_4 \\ \vdots & & \ddots & \vdots \\ r_4 \cdot \ell_\parallel & r_4 \cdot r_1 & \cdots & r_4 \cdot r_4 \end{pmatrix} = 0. \tag{60.64}$$

This is the vanishing of the five-dimensional Gram determinant promised in the section title: it holds because no five vectors can be linearly independent in four dimensions, and it is the exact statement that constrains the pentagon. We evaluate the determinant by the Schur complement. Writing the bordered symmetric matrix as

$$M = \begin{pmatrix} a & b^{\mathsf{T}} \\ b & G \end{pmatrix}, \qquad a = \ell_\parallel^2, \quad b_i = r_i \cdot \ell_\parallel = r_i \cdot \ell, \quad G_{ij} = r_i \cdot r_j, \tag{60.65}$$

the determinant factorizes as $\det M = \det G \, (a - b^{\mathsf{T}} G^{-1} b)$. Since $\det G \neq 0$, condition (60.64) is equivalent to the vanishing of the Schur complement,

$$\ell_\parallel^2 = \sum_{i,j=1}^{4} (G^{-1})_{ij} \, (r_i \cdot \ell)(r_j \cdot \ell). \tag{60.66}$$

This is simply the completeness relation in V: the dual basis $\tilde{r}^{\,i} = \sum_j (G^{-1})_{ij} r_j$ satisfies $\tilde{r}^{\,i} \cdot r_j = \delta^i_j$ and resolves the metric on V, whence $\ell_\parallel^\mu = \sum_i (\tilde{r}^{\,i} \cdot \ell) r_i^\mu$ and (60.66) follows by squaring. Equation (60.64) is the geometric origin of this identity.

Now convert (60.66) into a relation among propagators. On the left, $\ell_\parallel^2 = \ell^2 - \ell_\perp^2 = (D_0 + m_0^2) + \mu_\perp^2$ using (60.38). On the right, $r_i \cdot \ell = \tfrac{1}{2}(D_i - D_0 + Y_i)$ with $Y_i \equiv m_i^2 - m_0^2 - r_i^2$, again from (60.38). Substituting,

$$D_0 + m_0^2 + \mu_\perp^2 = \frac{1}{4} \sum_{i,j=1}^{4} (G^{-1})_{ij} \, (D_i - D_0 + Y_i)(D_j - D_0 + Y_j). \tag{60.67}$$

This is an *exact* relation between the inverse propagators, valid under the integral sign. Divide both sides by the full product $D_0D_1D_2D_3D_4$ and integrate. We classify the resulting terms by how many propagators they cancel:

- The left-hand term $D_0/(D_0\cdots D_4) = 1/(D_1D_2D_3D_4)$ is a scalar *box* (propagator 0 pinched).
- The right-hand terms linear in a single D_i (the cross terms $D_i\cdot(-D_0)$, $D_i\cdot Y_j$, and $D_i\cdot Y_i$, together with $-D_0\cdot Y_j$ etc.) each cancel exactly one propagator, producing scalar *boxes*.
- The right-hand terms bilinear in two distinct propagators, D_iD_j with $i\neq j$, cancel two propagators, producing scalar *triangles*; the diagonal terms D_i^2 produce boxes with one squared propagator, themselves reducible by a further application of the denominator identity to ordinary boxes and triangles.
- The constant terms on the right (the products Y_iY_j, and the leftover $-(m_0^2)$ from the left) cancel no propagator and reproduce the full *pentagon* I_5.
- The remaining left-hand term $\mu_\perp^2/(D_0\cdots D_4)$ is the pentagon integrand dressed with the extra-dimensional factor $\mu_\perp^2 = -\ell_\perp^2$.

Collecting the genuine pentagon contributions onto the left and everything else onto the right, (60.67) solves for I_5 as

$$\Big(\textstyle\sum_i b_i\Big)\, I_5 = \sum_{i=0}^{4}(\text{rational})\times \text{Box}_i + \sum(\text{rational})\times \text{Tri} - \mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\mu_\perp^2}{D_0\cdots D_4}, \tag{60.68}$$

where the coefficients are rational functions of the invariants and masses built from G^{-1} and the Y_i, and Box_i is the box obtained by pinching propagator i (one of the five "pinches" of the pentagon). The first two sums contain only boxes and triangles — masters strictly below the pentagon. The last term is the crucial one.

The extra-dimensional remainder is order ε. The integral $\int \mu_\perp^2/(D_0\cdots D_4) = -\int \ell_\perp^2/(D_0\cdots D_4)$ is, by the Lorentz invariance of the orthogonal subspace, proportional to a scalar pentagon evaluated in $d+2$ dimensions. Concretely, averaging $\ell_\perp^\mu\ell_\perp^\nu$ over the $(d-4)$-dimensional orthogonal space gives $\langle\ell_\perp^2\rangle$ proportional to $(d-4)$ times a $(d+2)$-dimensional integral; the standard dimension-shift identity (Theorem D.23 applied in the orthogonal directions) reads

$$\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\mu_\perp^2}{D_0\cdots D_4} = -\,(d-4)\,\frac{1}{2}\,I_5^{[d+2]} = \varepsilon\, I_5^{[d+2]}, \tag{60.69}$$

since $d-4=-2\varepsilon$. The shifted pentagon $I_5^{[d+2]}$ is ultraviolet *and* infrared finite — it has six effective dimensions against five propagators and an extra positive

power of denominator weight, so it converges at both ends — and therefore contributes at $O(\varepsilon)$. In strictly four dimensions ($\varepsilon \to 0$) this remainder drops out entirely, leaving the scalar pentagon equal to a rational combination of scalar boxes and triangles:

$$\boxed{I_5 = \sum_{i=0}^{4} d_i \,\mathrm{Box}_i + \sum c\,\mathrm{Tri} + O(\varepsilon),} \tag{60.70}$$

the $O(\varepsilon)$ piece being the finite shifted pentagon. *The box is therefore the highest master integral in four-dimensional one-loop physics.*

Remark 60.10 (The five-box form, and when $O(\varepsilon)$ matters). The reduction (60.70) produces both boxes and triangles; an optimal choice of the linear combination — solving the system $\sum_j Y_{ij} b_j = 1$ for the modified Cayley matrix $Y_{ij} = m_i^2 + m_j^2 - (r_i - r_j)^2$, the construction of [346, 482] — organizes the leading part as exactly the five pinch-boxes Box_i with no separate triangles, since each triangle is itself a pinch of a box already in the sum. Either way the structural conclusion is identical: no new transcendental function appears at the pentagon level or above. The $O(\varepsilon)$ shifted-pentagon remainder is genuinely there and *must* be retained whenever the box coefficients d_i themselves carry $1/\varepsilon$ poles — as they do at higher loop order, and in the ε-dimensional ("d-dimensional") unitarity used to capture the rational part in Section 60.2.7 and the following section. For the cut-constructible pieces, where the coefficients are finite, the $\varepsilon \to 0$ form (60.70) is exact.

60.2.6 Linear independence: the masters carry distinct cuts

Theorem 60.9 states that the four masters span the one-loop integral space; for the master *decomposition* to assign a well-defined coefficient to each, the masters must also be *linearly independent* over the field of rational functions of the kinematics. They are, and the cleanest proof is the very property that the unitarity method exploits: each master has a distinct branch-cut structure, so no rational-coefficient combination of some of them can equal another.

Consider a putative rational relation $\sum_X q_X X = 0$ among the masters $X \in \{A_0, B_0, C_0, D_0\}$ with rational coefficients q_X. Apply the discontinuity operator disc in a chosen kinematic channel and use that disc annihilates rational functions (which have no cut). The masters respond as follows, read off from their explicit values (60.29)–(60.33):

- The *tadpole* $A_0(m_0)$ depends on a single mass and *no* external invariant. It has no kinematic channel and no discontinuity: $\mathrm{disc}\, A_0 = 0$ in every channel.
- The *bubble* $B_0(p^2)$ has a single branch cut, in the one invariant p^2 on which it depends, starting at the two-particle threshold $p^2 = (m_0 + m_1)^2$ (or

$p^2 = 0$ in the massless case). Its discontinuity is a constant (in $\varepsilon \to 0$) times $\theta(p^2 - (m_0 + m_1)^2)$ — the two-particle phase space of Section D.8.

- The *triangle* C_0 has cuts in each of its (up to three) invariants, with a discontinuity that is itself a *logarithm* of the kinematics (the cut of the dilogarithm/$\ln^2$ in (60.32)), not a constant.
- The *box* $D_0(s,t)$ is the only master with cuts in *two independent* invariants s and t simultaneously, and its double discontinuity (cut in s then in t) is nonzero — a feature no triangle, bubble, or tadpole shares.

These discontinuity structures are qualitatively distinct: a constant cut (bubble), a logarithmic cut (triangle), a two-variable double cut (box), and no cut at all (tadpole). Suppose a nontrivial relation $\sum q_X X = 0$ existed. Taking the double discontinuity in two independent channels isolates the box: $\mathrm{disc}_s \,\mathrm{disc}_t(\sum q_X X) = q_{D_0} \,\mathrm{disc}_s \,\mathrm{disc}_t \, D_0 = 0$ forces $q_{D_0} = 0$, since the box's double cut is nonzero and rational coefficients cannot cancel it. With the box removed, a single discontinuity whose cut is logarithmic isolates the triangle, forcing $q_{C_0} = 0$; a single constant discontinuity then forces $q_{B_0} = 0$; and the surviving $q_{A_0} A_0 = 0$ with $A_0 \neq 0$ forces $q_{A_0} = 0$. No nontrivial rational relation exists. The four masters are linearly independent, and the decomposition below assigns each a unique coefficient.

Cuts as a basis-projection

The independence argument is more than a consistency check: it is the blueprint for the entire unitarity method. The masters are distinguished by *which cuts they survive*, so cutting an amplitude in a chosen set of channels projects onto the masters that have a discontinuity there and annihilates the rest. A maximal set of cuts — cutting all four propagators of a box — survives on that box alone, isolating its coefficient. This is exactly the strategy the next section makes precise: the same branch-cut structure that proves the masters independent also tells us which on-shell product computes each coefficient.

60.2.7 The master decomposition

Assembling the results, we can state the structural form of any one-loop amplitude. Let $\mathcal{M}^{\text{1-loop}}$ be the sum over all one-loop diagrams contributing to a given process, each reduced by the algorithm of Sections 60.2.1, 60.2.4 and 60.2.5.

Theorem 60.11 (One-loop master decomposition)**.** *Any one-loop amplitude in a renormalizable quantum field theory, in $d = 4 - 2\varepsilon$ dimensions, can be written as*

$$\boxed{\mathcal{M}^{\text{1-loop}} = \sum_i d_i \,\mathrm{Box}_i + \sum_i c_i \,\mathrm{Tri}_i + \sum_i b_i \,\mathrm{Bub}_i + \sum_i a_i \,\mathrm{Tad}_i + R,} \tag{60.71}$$

where the sums run over the distinct box, triangle, bubble, and tadpole scalar masters built from the external kinematics, the coefficients d_i, c_i, b_i, a_i are rational functions of the invariants and masses, and R is a purely rational remainder carrying no branch cut.

The decomposition follows immediately from Theorem 60.9: every reduced integral is a rational-coefficient combination of the four masters, and summing over diagrams collects the coefficient of each distinct master into a single rational d_i, c_i, b_i, a_i. The remainder R requires comment, because it is the part the cuts of Section 60.2.6 cannot see.

The origin of the rational remainder. Two effects generate R. First, the box coefficients and the reductions can produce, multiplying a master, an explicit factor of ε — as in the pentagon remainder (60.69), or wherever a numerator carries the $(d-4)$-dimensional component $\mu_\perp^2$ of the loop momentum. When such an ε multiplies the $1/\varepsilon$ ultraviolet or infrared pole of a master, the product is finite and *rational* — a logarithm-free number with no discontinuity. Second, the ultraviolet poles of the masters themselves, after renormalization, leave finite rational pieces. Because R has no cut, the discontinuity-based machinery of the following section computes the cut-carrying coefficients d_i, c_i, b_i, a_i but is, by construction, blind to R. Recovering R requires either keeping the full d-dimensional (ε-dependent) information in the cuts — "d-dimensional unitarity" — or a complementary on-shell technique; Section 60.4 (the integrand-reduction section of this chapter) supplies it. For the present it suffices that R is rational and finite, hence a finite list of numbers like the coefficients.

Remark 60.12 (QED and Yang–Mills theory on a common basis). The decomposition (60.71) puts every one-loop computation in the book onto a single footing. The brute-force QED results of Chapter 19 — vacuum polarization, electron self-energy, vertex correction, each obtained there by Feynman parametrization and Passarino–Veltman reduction — are, in the language of (60.71), statements about bubble and triangle coefficients: the vacuum polarization is a bubble coefficient times B_0, the vertex form factors are triangle and bubble coefficients times C_0 and B_0, with the ultraviolet poles residing in the tadpole and metric-coefficient pieces exactly as Theorem 60.8 located them. Re-expressing those results in this basis is not a new computation but a change of bookkeeping; it makes the QED one-loop data directly comparable with the Yang–Mills one-loop amplitudes that the unitarity method will construct from on-shell gluon trees. Yang–Mills theory is set up here on the same box–triangle–bubble–tadpole basis, so that the gluon amplitude assembled later in this chapter and the QED amplitudes of Chapter 19 are read off the *same* finite list of masters — the unification of "old way" and "new way" that the chapter promised.

Theorem 60.11 converts the problem of computing a one-loop amplitude into the finite problem of determining the rational coefficients $\{d_i, c_i, b_i, a_i\}$ and the rational remainder R. The cut analysis of Section 60.2.6 already shows that

discontinuities can separate the masters; what it does not yet show is how to compute a given coefficient cleanly, without the Gram-determinant instabilities that plague Passarino–Veltman reduction. The next section supplies the answer: cutting more than two propagators at once — generalized unitarity — freezes the loop momentum on a discrete set of on-shell solutions and isolates each coefficient as a product of on-shell tree amplitudes, computing the box coefficient d_i from a quadruple cut, the triangle from a triple cut, and so on down the hierarchy.

60.3 Generalized Unitarity: Coefficients from Multiple Cuts

The previous section reduced the problem of a one-loop amplitude to a finite list of unknowns: by Theorem 60.11 every one-loop amplitude is the master decomposition

$$\mathcal{M}^{\text{1-loop}} = \sum_i d_i \,\mathrm{Box}_i + \sum_i c_i \,\mathrm{Tri}_i + \sum_i b_i \,\mathrm{Bub}_i + \sum_i a_i \,\mathrm{Tad}_i + R, \tag{60.72}$$

with the scalar masters known once and for all and the coefficients d_i, c_i, b_i, a_i and the rational remainder R the only quantities left to determine. The ordinary two-particle cut of Section 60.1.3 already constrains these numbers, but bluntly: cutting a single channel puts a delta function on two propagators and detects *every* master that has a discontinuity in that channel at once — the box, the triangles, and the bubble that all share the cut — so a double cut returns a tangled linear combination of coefficients, not any one of them. To disentangle them one would have to take many cuts and solve a linear system, much as Passarino–Veltman solves a Gram system, and the off-shell loop integrals reappear inside each cut.

The insight of generalized unitarity, due to Britto, Cachazo, and Feng [91], is that nothing forbids cutting *more* than two propagators. The masters in Eq. (60.72) differ in how many propagators they carry — four for a box, three for a triangle, two for a bubble — so cutting a larger set of propagators kills every master that does not contain all of them. Cutting all four propagators of a box leaves that box alone among the masters: triangles, bubbles, and tadpoles have too few propagators to support four simultaneous on-shell conditions and are annihilated. The single surviving coefficient is then read off directly, and the residual loop integral collapses completely, because four on-shell conditions on a four-component loop momentum leave no integration at all — only a discrete sum over solutions. The price, foreshadowed already by the three-point kinematics of Chapters 58 and 59, is that those solutions are necessarily *complex*. This section develops the construction in full: the quadruple cut for box coefficients, the triple cut for triangles, the double cut for bubbles, and the unifying principle that every coefficient is a sum of products of on-shell tree amplitudes.

60.3.1 The generalized cut and the maximal-cut philosophy

We first fix what "cutting a set of propagators" means as an operation on the integrand, generalizing the Cutkosky replacement Eq. (60.10). Given a one-loop integrand with denominator $D_0 D_1 \cdots D_{N-1}$, and a chosen subset $C \subseteq \{0, 1, \ldots, N-1\}$ of propagators to cut, the *generalized cut* on C replaces each selected propagator by its on-shell delta function and integrates the remainder over the loop momentum:

$$\mathrm{Cut}_C\, \mathcal{M}^{\text{1-loop}} \;\equiv\; \int \frac{\mathrm{d}^4 \ell}{(2\pi)^4} \Big[\prod_{i \in C} (2\pi)\, \delta^+(\ell_i^2 - m_i^2) \Big] \Big[\prod_{j \notin C} \frac{1}{\ell_j^2 - m_j^2} \Big] N(\ell), \quad (60.73)$$

with $\ell_i = \ell + r_i$ the propagator momenta, $N(\ell)$ the integrand numerator, and δ^+ the positive-energy delta of Eq. (60.9). When $|C| = 2$ this is exactly the two-particle Cutkosky cut of Theorem 60.1; the new content is $|C| \geq 3$. The replacement (60.73) is best understood as a multidimensional contour deformation: the product of deltas is the residue obtained by encircling the simultaneous poles of the cut propagators, the higher-dimensional analogue of the single-propagator discontinuity Eq. (60.8). We treat it as a linear functional on (60.72) and ask what it returns when applied to each master.

The decisive observation is a selection rule. A scalar master Box, Tri, Bub, Tad has $4, 3, 2, 1$ propagators respectively. Applying Cut_C with $|C| = k$ to a master with fewer than k propagators gives zero, because at least one requested delta function has no propagator to act on — there is no factor $1/(\ell_i^2 - m_i^2)$ to be replaced, and the cut is empty. More precisely, Cut_C acting on a master is nonzero only if C is a subset of that master's propagator set. Therefore:

- A *quadruple* cut ($|C| = 4$) survives only on boxes whose four propagators are exactly C. Generic external kinematics admit a unique such box per quadruple, so the quadruple cut isolates a single d_i.
- A *triple* cut ($|C| = 3$) survives on the one triangle with propagator set C, and also on every box that *contains* C among its four propagators. It returns c_i contaminated by the boxes already determined by quadruple cuts.
- A *double* cut ($|C| = 2$) survives on the bubble with propagator set C, on triangles containing C, and on boxes containing C. It returns b_i contaminated by already-known triangle and box coefficients.

This hierarchy dictates the algorithm: determine the box coefficients from quadruple cuts first; then the triangle coefficients from triple cuts, subtracting the known box contributions; then the bubble coefficients from double cuts, subtracting the known triangle and box contributions. Each step uses the maximal cut available for the coefficient it targets, so that the contamination

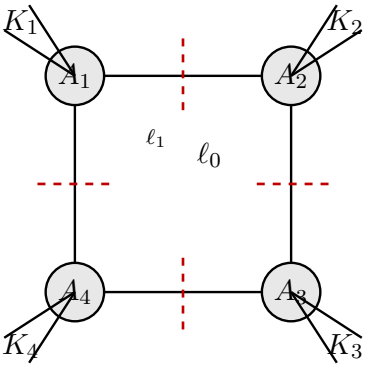


Figure 60.2: The quadruple cut of a box. All four internal propagators are put on shell (dashed cuts), severing the loop into four on-shell tree amplitudes A_1, A_2, A_3, A_4, one at each corner with external cluster K_a. The four delta functions freeze the loop momentum to the two discrete complex solutions of Eq. (60.74), so the box coefficient is a sum of products of trees with no integration left, Theorem 60.13.

comes only from masters *above* it in the hierarchy, which have already been fixed. The tadpoles, carrying a single scale and no kinematic channel, have vanishing cut (Section 60.2.2) and are fixed by other means; in a massless theory they are absent altogether. We now execute the three steps in turn.

60.3.2 The quadruple cut: box coefficients from four on-shell trees

Consider a box with propagators D_0, D_1, D_2, D_3 whose corner momenta partition the external legs into four clusters K_1, K_2, K_3, K_4 (the momentum flowing out of each corner), $\sum_a K_a = 0$. The four cut conditions are

$$\ell_0^2 = m_0^2, \quad \ell_1^2 = m_1^2, \quad \ell_2^2 = m_2^2, \quad \ell_3^2 = m_3^2, \qquad \ell_a = \ell + r_a, \tag{60.74}$$

four scalar equations in the four components of ℓ. Subtracting $\ell_0^2 = m_0^2$ from the other three linearizes them — exactly the denominator identity Eq. (60.38) — leaving one quadratic ($\ell_0^2 = m_0^2$) and three linear equations. Three linear equations fix ℓ to a line; the residual quadratic cuts that line at *two* points. Hence the quadruple-cut system has generically *two solutions* $\ell_*^{(1)}, \ell_*^{(2)}$, and they are isolated points, not a continuous family. This is the geometric content of cutting four propagators with a four-dimensional loop momentum: the integration is completely frozen.

On either solution the four cut propagators are on shell, so the loop has been severed into four *tree* amplitudes, one per corner (Fig. 60.2): each corner a is the on-shell amplitude A_a for its external cluster K_a together with the two adjacent

cut lines, summed over the helicity (and particle species) of the internal lines crossing the cut. The cut of the full amplitude is therefore

$$\mathrm{Cut}_4\,\mathcal{M}^{\text{1-loop}} = \int \frac{\mathrm{d}^4\ell}{(2\pi)^4} \prod_{i=0}^{3} (2\pi)\,\delta^+(\ell_i^2 - m_i^2) \sum_{\text{states}} A_1\,A_2\,A_3\,A_4, \tag{60.75}$$

with the sum over the internal states on the four cut lines. We must turn the frozen integral on the right into the coefficient d.

Freezing the integral: the Jacobian

The four delta functions localize the loop integral on the two solutions of (60.74). Change variables from ℓ^μ to the four functions $u_i \equiv \ell_i^2 - m_i^2$; the Jacobian of this change is

$$J(\ell) \;=\; \det\!\Big(\frac{\partial u_i}{\partial \ell^\mu}\Big)_{i,\mu} \;=\; \det\big(2\,\ell_i^\mu\big)_{i,\mu}, \qquad i = 0,1,2,3,\ \mu = 0,1,2,3, \tag{60.76}$$

the 4×4 determinant of the (twice) propagator momenta viewed as rows. Then $\prod_i \mathrm{d}u_i = |J|\,\mathrm{d}^4\ell$, and integrating the four delta functions,

$$\mathrm{Cut}_4\,\mathcal{M}^{\text{1-loop}} = \frac{1}{(2\pi)^2} \sum_{\sigma=1,2} \frac{1}{|J(\ell_*^{(\sigma)})|} \Big[\sum_{\text{states}} A_1 A_2 A_3 A_4\Big]_{\ell=\ell_*^{(\sigma)}}, \tag{60.77}$$

the overall $(2\pi)^{-2}$ collecting the four (2π)'s from the deltas against the $(2\pi)^{-4}$ of the measure. Crucially, the *same* delta functions act on the scalar box, whose numerator is unity:

$$\mathrm{Cut}_4\ \mathrm{Box} = \frac{1}{(2\pi)^2} \sum_{\sigma=1,2} \frac{1}{|J(\ell_*^{(\sigma)})|}. \tag{60.78}$$

By the selection rule of Section 60.3.1 the quadruple cut of the master decomposition (60.72) receives a contribution from this box alone, so $\mathrm{Cut}_4\,\mathcal{M}^{\text{1-loop}} = d\ \mathrm{Cut}_4\ \mathrm{Box}$. Dividing (60.77) by (60.78) gives the coefficient. For the kinematic configurations of interest the two solutions are exchanged by complex conjugation and carry equal Jacobian magnitude, $|J(\ell_*^{(1)})| = |J(\ell_*^{(2)})|$, whereupon the Jacobians cancel between numerator and denominator and we obtain the central formula.

Theorem 60.13 (Quadruple cut, Britto–Cachazo–Feng)**.** *The coefficient of a scalar box master in the one-loop decomposition* (60.72) *is the average, over the two complex solutions of the four on-shell conditions* (60.74)*, of the product of the four on-shell tree amplitudes at its corners, summed over the internal states crossing the cuts:*

$$\boxed{\,d \;=\; \frac{1}{2} \sum_{\sigma=1,2} \sum_{\text{states}} A_1\,A_2\,A_3\,A_4\Big|_{\ell=\ell_*^{(\sigma)}}.\,} \tag{60.79}$$

No off-shell quantity, no gauge choice, and no loop integration enters: the box coefficient is fixed entirely by on-shell tree data evaluated at the two cut solutions.

The factor $\frac{1}{2}$ is the residue of the Jacobian cancellation: each solution contributes its product weighted by $1/|J|$, the scalar box supplies $\sum_\sigma 1/|J| = 2/|J|$ in the denominator, and with equal Jacobians the ratio is $\frac{1}{2}(A_1A_2A_3A_4|_1 + A_1A_2A_3A_4|_2)$. It is not an arbitrary symmetry factor but the statement that the box's own quadruple cut counts both solutions equally.

60.3.3 Why the cut solutions must be complex

Before computing, we confront the feature that makes Theorem 60.13 possible at all: the solutions $\ell_*^{(\sigma)}$ are complex, and they *must* be, because the corner amplitudes are on-shell three- and higher-point amplitudes whose existence requires complex kinematics. Recall from Section 58.3 and Section 59.2.6 the kinematics of a massless three-point amplitude. For three null momenta p_1, p_2, p_3 summing to zero, the Mandelstam invariants $s_{ij} = \langle i\,j\rangle[j\,i]$ all vanish: $s_{12} = (p_1+p_2)^2 = p_3^2 = 0$, and likewise s_{13}, s_{23}. With *real* momenta, $\langle i\,j\rangle$ and $[i\,j]$ are complex conjugates up to a sign, so $\langle i\,j\rangle[j\,i] = 0$ forces *both* $\langle i\,j\rangle = 0$ and $[i\,j] = 0$ for every pair: all spinor brackets vanish and the on-shell three-point amplitude is identically zero. A massless three-point amplitude is nonzero only off the real slice.

Complexifying the momenta breaks the link between λ and $\tilde\lambda$, which become independent spinors. Then $\langle i\,j\rangle[j\,i] = 0$ can be solved in two distinct ways: either all the angle brackets vanish (all λ_i proportional, $\tilde\lambda_i$ generic — the antiholomorphic, or $\overline{\text{MHV}}$, kinematics) or all the square brackets vanish (all $\tilde\lambda_i$ proportional, λ_i generic — the holomorphic, MHV, kinematics). On the first branch the three-point amplitude is the square-bracket expression of Eq. (58.98),

$$A_3[1^+, 2^+, 3^-] = -\,i\,g\,\frac{[1\,2]^3}{[2\,3][3\,1]}, \tag{60.80}$$

finite and nonzero; on the second it is the angle-bracket conjugate $A_3[1^-, 2^-, 3^+] = i\,g\,\langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,1\rangle)$ of Eq. (59.55) at $n = 3$. The quadruple-cut solutions are precisely the complex loop momenta that put each corner onto one of these two branches. This is not a formal trick: the contour deformation defining the generalized cut (60.73) naturally lands on complex residues, and the reality of the final amplitude is recovered only after the residues are summed back up. The lesson of Chapters 58 and 59 — that on-shell three-point amplitudes live in complexified kinematics — is exactly what the unitarity method needs to feed its corners.

Physical Insight

The brute-force one-loop computations of Chapter 19 never met complex momenta because they never put anything on shell: every internal line stayed off shell, carrying a four-dimensional real integration. Generalized unitarity does the opposite — it forces all cut lines on shell at once — and the reward is that the corners become tree amplitudes, the most economical gauge-invariant objects we have. The cost, complexified loop momenta, is the same complexification that BCFW exploited at tree level in Theorem 59.23. The two methods are the same idea applied at different loop orders: deform momenta into the complex plane until propagators pinch, then read off residues.

60.3.4 A worked box coefficient: the four-gluon amplitude

We now compute a box coefficient end to end, advancing Yang–Mills theory to its loop level: the box of the one-loop color-ordered four-gluon amplitude $A_4^{\text{1-loop}}(1^-,2^-,3^+,4^+)$, with all momenta outgoing, massless, and the cyclic ordering $(1,2,3,4)$. Each corner of the box carries a single external gluon, so all four corners are three-point amplitudes — the cleanest possible illustration of Theorem 60.13.

Worked Example 60.14 (The four-gluon box coefficient from four three-point trees)**.** Route the loop momentum so that the four cut propagators are

$$\ell_0 = \ell, \quad \ell_1 = \ell - k_1, \quad \ell_2 = \ell - k_1 - k_2, \quad \ell_3 = \ell + k_4, \tag{60.81}$$

all massless, with corner clusters $K_a = k_a$. Write the Mandelstam invariants $s \equiv s_{12} = \langle 1\,2\rangle[2\,1]$ and $t \equiv s_{23} = \langle 2\,3\rangle[3\,2]$, as fixed in Section 59.3.

Solving the cut conditions. Setting $\ell_i^2 = 0$ and using $\ell_0^2 = 0$ to subtract,

$$\ell^2 = 0, \qquad \ell\cdot k_1 = 0, \qquad \ell\cdot k_4 = 0, \qquad \ell\cdot k_2 = \tfrac{s}{2}, \tag{60.82}$$

where the last followed from $(\ell - k_1 - k_2)^2 = 0$ with $\ell^2 = \ell\cdot k_1 = 0$ and $2k_1\cdot k_2 = s$. Since $\ell^2 = 0$ we may write $\ell_{\alpha\dot\alpha} = \lambda_\ell\tilde\lambda_\ell$. The conditions $2\,\ell\cdot k_1 = \langle \ell\,1\rangle[1\,\ell] = 0$ and $2\,\ell\cdot k_4 = \langle \ell\,4\rangle[4\,\ell] = 0$ each demand the vanishing of an angle or a square bracket. The two non-degenerate ways to satisfy both are

$$\begin{aligned} &\text{solution (1)}: \quad \langle \ell\,1\rangle = 0,\ [\ell\,4] = 0 \implies \lambda_\ell \propto \lambda_1,\ \tilde\lambda_\ell \propto \tilde\lambda_4, \\ &\text{solution (2)}: \quad [\ell\,1] = 0,\ \langle \ell\,4\rangle = 0 \implies \tilde\lambda_\ell \propto \tilde\lambda_1,\ \lambda_\ell \propto \lambda_4, \end{aligned} \tag{60.83}$$

the remaining combinations ($\langle \ell\,1\rangle = \langle \ell\,4\rangle = 0$ or $[\ell\,1] = [\ell\,4] = 0$) forcing $\lambda_1 \propto \lambda_4$ or $\tilde\lambda_1 \propto \tilde\lambda_4$, false for generic external momenta. These are exactly the two discrete

complex solutions promised. On solution (1) set $\lambda_\ell = \lambda_1$, $\tilde\lambda_\ell = \alpha\,\tilde\lambda_4$; the last condition $\langle \ell\, 2\rangle[2\,\ell] = s$ reads $\alpha\,\langle 1\,2\rangle[2\,4] = \langle 1\,2\rangle[2\,1]$, so

$$\alpha = \frac{[2\,1]}{[2\,4]}. \tag{60.84}$$

The internal kinematics on solution (1). From (60.81), the cut momenta are rank-one bispinors:

$$\begin{aligned} \ell_0 &= \lambda_1\,(\alpha\tilde\lambda_4), & \ell_1 &= \ell_0 - k_1 = \lambda_1\,(\alpha\tilde\lambda_4 - \tilde\lambda_1), \\ \ell_3 &= \ell_0 + k_4 = (\alpha\lambda_1 + \lambda_4)\,\tilde\lambda_4, & \ell_2 &= \ell_1 - k_2 = (\tfrac{[4\,1]}{[2\,4]}\lambda_1 - \lambda_2)\tilde\lambda_2, \end{aligned} \tag{60.85}$$

the last form obtained by simplifying $\alpha\tilde\lambda_4 - \tilde\lambda_1$: bracketing against $\tilde\lambda_1$ gives $[(\alpha\tilde\lambda_4 - \tilde\lambda_1)\,1] = \alpha[4\,1]$, while $[2\,1]\cdot\frac{[4\,1]}{[2\,4]} = \alpha[4\,1]$ by (60.84), so $\alpha\tilde\lambda_4 - \tilde\lambda_1 = \frac{[4\,1]}{[2\,4]}\tilde\lambda_2$. Thus ℓ_0, ℓ_1 have holomorphic spinor $\propto \lambda_1$, while ℓ_2, ℓ_3 have antiholomorphic spinor $\propto \tilde\lambda_2, \tilde\lambda_4$ respectively. Reading off the three-point kinematics of Section 60.3.3: at corner 1 (legs $1, \ell_0, \ell_1$) all angle brackets vanish, so it is an $\overline{\text{MHV}}$ vertex; at corner 4 (legs $4, \ell_3, \ell_0$) all square brackets vanish (every $\tilde\lambda \propto \tilde\lambda_4$), so it is MHV; corner 2 (legs $2, \ell_1, \ell_2$) has every $\tilde\lambda \propto \tilde\lambda_2$ and is MHV; corner 3 (legs $3, \ell_2, \ell_3$) is $\overline{\text{MHV}}$. The internal helicities are then fixed uniquely by requiring each vertex to be nonvanishing (an MHV vertex needs two minus and one plus, an $\overline{\text{MHV}}$ vertex two plus and one minus), with opposite helicities on the two ends of each cut line.

The four corner amplitudes. Using the three-point amplitudes of Eq. (59.55) and (60.80) and inserting the spinors (60.85), the four corners evaluate to

$$A^{(4)} = i\,g\,\frac{\langle \ell_0\,\ell_3\rangle^3}{\langle \ell_3\,4\rangle\langle 4\,\ell_0\rangle} = -\frac{i\,g}{\alpha}\,\langle 1\,4\rangle, \tag{60.86}$$

$$A^{(1)} = -i\,g\,\frac{[\ell_0\,\ell_1]^3}{[\ell_1\,1][1\,\ell_0]} = -i\,g\,\alpha^2\,\frac{[4\,1][2\,4]}{[2\,1]}, \tag{60.87}$$

$$A^{(2)} = i\,g\,\frac{\langle 2\,\ell_1\rangle^3}{\langle \ell_1\,\ell_2\rangle\langle \ell_2\,2\rangle} = i\,g\,\langle 1\,2\rangle\,\frac{[2\,4]}{[4\,1]}, \tag{60.88}$$

$$A^{(3)} = -i\,g\,\frac{[3\,\ell_3]^3}{[\ell_3\,\ell_2][\ell_2\,3]} = -i\,g\,\frac{[3\,4]^3}{[4\,2][2\,3]}, \tag{60.89}$$

each a one-line computation in the spinors (60.85). For instance, in (60.86), $\langle \ell_0\,\ell_3\rangle = \langle 1\,(\alpha\lambda_1 + \lambda_4)\rangle = \langle 1\,4\rangle$, $\langle \ell_3\,4\rangle = \alpha\langle 1\,4\rangle$, $\langle 4\,\ell_0\rangle = \langle 4\,1\rangle$, so the ratio is $\langle 1\,4\rangle^3/(\alpha\langle 1\,4\rangle\langle 4\,1\rangle) = -\langle 1\,4\rangle/\alpha$. Multiplying the four corners, the dependence on α reduces to a single power (α^2 from corner 1 against $1/\alpha$ from corner 4) and every internal bracket cancels in pairs:

$$\prod_a A^{(a)}\Big|_{(1)} = -\,g^4\,\alpha\,\frac{[4\,1][2\,4]}{[2\,1]}\cdot\frac{\langle 1\,2\rangle[2\,4]}{[4\,1]}\cdot\frac{[3\,4]^3}{[4\,2][2\,3]}\cdot\langle 1\,4\rangle = g^4\,\frac{\langle 1\,2\rangle\,\langle 1\,4\rangle\,[3\,4]^3}{[2\,3]}, \tag{60.90}$$

using $\alpha = [2\,1]/[2\,4]$ and $[4\,2] = -[2\,4]$; the product of the four $\pm ig$ prefactors is $-g^4$.

Comparison with the tree. The tree amplitude is the Parke–Taylor value Eq. (59.55),

$$A_4^{\text{tree}}(1^-,2^-,3^+,4^+) = i\,g^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = i\,g^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}. \tag{60.91}$$

Using momentum conservation $\sum_i |i\rangle\,[i\,] = 0$ to trade square brackets for angle brackets — with $s = s_{34} = \langle 3\,4\rangle[4\,3]$, $t = s_{23} = \langle 2\,3\rangle[3\,2] = s_{14} = \langle 1\,4\rangle[4\,1]$ — the product (60.90) collapses to

$$\prod_a A^{(a)}\Big|_{(1)} = i\,g^2\,s\,t\;A_4^{\text{tree}}(1^-,2^-,3^+,4^+). \tag{60.92}$$

That the bracket monomial (60.90) equals ig^2st times (60.91) is checked directly on any explicit spinor assignment; for the choice $\lambda_1 = (1,0)$, $\lambda_2 = (0,1)$, $\lambda_3 = (1,1)$, $\lambda_4 = (1,2)$ with $\tilde\lambda$'s fixed by momentum conservation, both sides equal g^4 (with $s = -1$, $t = 2$). Solution (2) is the parity conjugate, exchanging $\lambda \leftrightarrow \tilde\lambda$ and hence MHV$\leftrightarrow\overline{\text{MHV}}$ at every corner; carrying the identical computation through with $\beta = \langle 1\,2\rangle/\langle 4\,2\rangle$ in place of α gives the same product,

$$\prod_a A^{(a)}\Big|_{(2)} = i\,g^2\,s\,t\;A_4^{\text{tree}}(1^-,2^-,3^+,4^+). \tag{60.93}$$

Averaging by Theorem 60.13,

$$\boxed{d \;=\; \tfrac{1}{2}\Big(\prod_a A^{(a)}\big|_{(1)} + \prod_a A^{(a)}\big|_{(2)}\Big) \;=\; i\,g^2\,s\,t\;A_4^{\text{tree}}(1^-,2^-,3^+,4^+).} \tag{60.94}$$

This single number reproduces, by the most direct possible route, the celebrated result of Bern, Dixon, Dunbar, and Kosower [67]: the one-loop four-gluon MHV amplitude is its tree times $s\,t$ times the scalar zero-mass box, $A_4^{\text{1-loop}} = ig^2\,s\,t\,A_4^{\text{tree}}\,D_0^{0m}(s,t) + \dots$, with D_0^{0m} the box (60.33). We obtained the coefficient from four three-point trees and one algebraic equation, with no Feynman diagram, no gauge choice, and no loop integral — the entire content of Theorem 60.13 made concrete. For this amplitude the triangle and bubble coefficients turn out to vanish: a feature of the maximally supersymmetric content that controls the box, so that (60.94) is the whole cut-constructible amplitude. In pure Yang–Mills the same box coefficient appears, accompanied by a rational remainder addressed in Section 60.3.7 and the next section.

60.3.5 The triple cut: triangle coefficients and Forde's parametrization

A triple cut imposes three on-shell conditions (60.74) on the four-component loop momentum, leaving a *one-complex-parameter* family of solutions rather

than discrete points. By the selection rule of Section 60.3.1 the triple cut on propagators $\{0,1,2\}$ detects the triangle Tri_{012} and every box that contains those three propagators. The boxes are already known from quadruple cuts, so the triangle coefficient is extracted by isolating, within the one-parameter cut, the part that is not a box. Forde's parametrization [197] makes this isolation algebraic, and we develop it now.

Label the three corner momenta K_1, K_2, K_3 with $K_1 + K_2 + K_3 = 0$ (sums of external clusters), and take the two "adjacent" corners K_1, K_2 to be the massive ones (massless corners follow by degenerate limits of the same formulas). Define their mutual massless projections: with

$$K_1^\flat = K_1 - \frac{K_1^2}{\gamma}\, K_2^\flat, \qquad K_2^\flat = K_2 - \frac{K_2^2}{\gamma}\, K_1^\flat, \tag{60.95}$$

the two null momenta $K_1^\flat, K_2^\flat$ are determined by K_1, K_2 up to the normalization γ, which solves the quadratic obtained by imposing $(K_1^\flat)^2 = (K_2^\flat)^2 = 0$,

$$\gamma_\pm = K_1{\cdot}K_2 \pm \sqrt{(K_1{\cdot}K_2)^2 - K_1^2 K_2^2}, \tag{60.96}$$

the two roots $\gamma_\pm$ being the two solutions of the projection conditions. With these the cut loop momentum is parametrized by a single complex variable t,

$$\ell^\mu(t) = \frac{K_2^2(\gamma - K_1^2)}{\gamma^2 - K_1^2 K_2^2}\, K_1^{\flat\,\mu} + \frac{K_1^2(\gamma - K_2^2)}{\gamma^2 - K_1^2 K_2^2}\, K_2^{\flat\,\mu} + \frac{t}{2}\,\langle K_1^\flat|\gamma^\mu|K_2^\flat] + \frac{\Delta}{2t}\,\langle K_2^\flat|\gamma^\mu|K_1^\flat], \tag{60.97}$$

where the two coefficients of $K_1^\flat, K_2^\flat$ are fixed by the on-shell conditions $\ell^2 = 0$, $(\ell - K_1)^2 = 0$, $(\ell + K_2)^2 = 0$, and the constant Δ multiplying the $1/t$ direction is fixed in the same way (its precise value is immaterial to what follows). The essential structure is that $\ell(t)$ runs over the cut solution as t ranges over $\mathbb{C}^\times$, with the two transverse directions $\langle K_1^\flat|\gamma^\mu|K_2^\flat]$ and $\langle K_2^\flat|\gamma^\mu|K_1^\flat]$ spanning the plane orthogonal to $K_1^\flat, K_2^\flat$. The free parameter t is the residual freedom after three on-shell conditions.

Projecting out the triangle

On the triple cut, the product of the three on-shell tree amplitudes at the corners,

$$\mathcal{T}(t) \equiv \sum_{\text{states}} A_1\big(\ell(t)\big)\, A_2\big(\ell(t)\big)\, A_3\big(\ell(t)\big), \tag{60.98}$$

is a rational function of t. Its analytic structure is dictated by the masters that share this cut. A box containing the three cut propagators has a *fourth* propagator $D_3 = (\ell - Q)^2$; that propagator goes on shell at the special values of t where $\ell(t)$ also satisfies $D_3 = 0$, which appear as simple poles of $\mathcal{T}(t)$. The

triangle, by contrast, has no fourth propagator and contributes a piece *regular* in t. Therefore $\mathcal{T}(t)$, expanded as a Laurent series, splits as

$$\mathcal{T}(t) = \underbrace{\sum_{n\neq 0} \left(e_n\, t^n + \tilde{e}_n\, t^{-n}\right)}_{\text{box poles} + \text{spurious}} + \underbrace{e_0}_{\text{triangle}}, \tag{60.99}$$

and the key fact is that integrating the cut over the parameter t annihilates every power t^n with $n \neq 0$. Concretely, once the three delta functions are used the cut measure on the one-parameter family reduces to a contour integral $\oint \frac{\mathrm{d}t}{t}$ around the origin (times the phase-space Jacobian), and $\oint \frac{\mathrm{d}t}{t}\, t^n = 2\pi i\, \delta_{n,0}$. Equivalently, the nonzero powers of t are total derivatives on the cut — "spurious" terms that integrate to zero, the mechanism the next section organizes systematically. Only the t-independent piece e_0 survives integration, and after subtracting the box contributions (whose poles in t sit at the locations where the fourth propagator pinches, with residues fixed by the already-known box coefficients) the triangle coefficient is the constant term, averaged over the two massless-projection roots:

$$\boxed{c = -\frac{1}{2} \sum_{\gamma = \gamma_\pm} \Big[\text{coefficient of } t^0 \text{ in } \mathcal{T}(t) \text{ after box subtraction}\Big]_\gamma .} \tag{60.100}$$

The average over $\gamma_\pm$ plays the role that the average over the two quadruple-cut solutions played in Theorem 60.13. In practice one expands $\mathcal{T}(t)$ at large t, reads off the constant term, and subtracts the pole pieces; the result (60.100) is the triangle coefficient as a sum of products of on-shell trees, exactly the structure Theorem 60.13 delivered for boxes.

Remark 60.15 (Why the constant term is the triangle). The logic is the same selection-by-singularity that proved the masters independent in Section 60.2.6. A box's discontinuity in the fourth channel shows up as a pole in t; a triangle has no such channel and is the smooth background. Forde's parametrization is engineered so that "smooth in t" means "constant in t" once the kinematic Jacobian is stripped, so the triangle is literally the t^0 Laurent coefficient. The spurious powers $t^{\pm n}$, $n \geq 1$, are present in $\mathcal{T}(t)$ but carry no master: they integrate to zero and cancel in the full amplitude. Identifying and discarding them is the one subtlety of the triple cut, and it is precisely what integrand reduction in the next section automates.

60.3.6 The double cut: bubble coefficients and the cut hierarchy

The last cut-constructible coefficient is the bubble. A double cut imposes two on-shell conditions, leaving a *two*-complex-parameter family of solutions, and by the selection rule detects the bubble Bub_{01} together with every triangle and box that contains the two cut propagators. Schematically,

$$\mathrm{Cut}_2\, \mathcal{M}^{\text{1-loop}} = b\, \mathrm{Cut}_2\, \mathrm{Bub} + \sum_{\text{tri}\supset\{0,1\}} c\, \mathrm{Cut}_2\, \mathrm{Tri} + \sum_{\text{box}\supset\{0,1\}} d\, \mathrm{Cut}_2\, \mathrm{Box}, \tag{60.101}$$

so the bubble coefficient is what remains of the two-particle cut after the contributions of the triangles and boxes sharing the channel — already determined by triple and quadruple cuts — are subtracted. The two-parameter cut integrand, a product of two tree amplitudes summed over the intermediate state, is expanded in the two cut parameters (one along the massless-projection direction, one azimuthal); the triangle and box pieces are again the terms carrying extra poles, and the bubble coefficient is the genuinely constant remainder, projected out by integrating over both parameters. This is the bottom of the hierarchy laid out in Section 60.3.1: boxes from quadruple cuts, triangles from triple cuts minus boxes, bubbles from double cuts minus triangles and boxes. Tadpoles, absent in massless theories and cut-free in any case, close the list.

The interplay is the essential structural point. The *same* two-particle cut that Section 60.1.3 used to verify the optical theorem now does quantitative work, but only after the higher cuts have stripped away the box and triangle content that also lives on that cut. A naive two-particle cut, as we warned at the start of this section, returns the entangled combination (60.101); the generalized cuts disentangle it term by term, each maximal cut isolating one layer of the hierarchy. The bubble coefficient is the residue of the optical theorem once everything sharper has been accounted for.

60.3.7 The principle: every coefficient is a product of trees

Standing back from the three constructions, a single principle has emerged, and it is the conceptual heart of the unitarity method.

Theorem 60.16 (Coefficients as on-shell tree data)**.** *Every cut-constructible coefficient in the one-loop master decomposition* (60.72) *— each d_i, c_i, and b_i — is a sum of products of on-shell tree amplitudes, evaluated on the complex kinematic solutions of the corresponding multiple cut and projected onto the appropriate master. No off-shell current, no internal propagator numerator, and no gauge-fixing choice enters at any stage; the coefficients are manifestly gauge invariant because tree amplitudes are.*

The proof is the content of Theorem 60.13, (60.100), and (60.101): the quadruple cut gives d_i as a product of four trees averaged over two solutions; the triple cut gives c_i as the box-subtracted constant term of a product of three trees averaged over the two massless-projection roots; the double cut gives b_i as the triangle- and box-subtracted constant of a product of two trees. Each is built from gauge-invariant on-shell amplitudes alone.

Three features deserve emphasis, because they are what make this a genuine replacement for the Feynman expansion of Chapter 19 rather than a reformulation of it. First, *gauge invariance is automatic.* A Feynman-diagram computation of a one-loop amplitude passes through gauge-dependent off-shell currents, ghost

contributions, and longitudinal polarizations that cancel only in the final sum; the unitarity method never introduces them, because every ingredient is an on-shell tree amplitude, an object on which the gauge redundancy has already been quotiented out. The Gram-determinant instabilities of Passarino–Veltman reduction (Theorem 60.7) are likewise absent: no Gram matrix is ever inverted; instead one solves polynomial on-shell conditions and reads off residues.

Second, *the input is recyclable.* The trees feeding the corners are the very Parke–Taylor and BCFW amplitudes of Eq. (59.55) and Theorem 59.23; the loop computation reuses tree technology wholesale, and a more efficient tree method immediately improves the loop. The four-gluon box coefficient of Theorem 60.14 was nothing but four three-point amplitudes glued together — the same seeds from which BCFW built every tree.

Third, *the method is blind to the rational remainder.* Everything in this section worked in strictly four-dimensional kinematics, where the cut solutions and tree amplitudes live. The rational part R of (60.72), identified in Section 60.2.7 as the cut-free residue, carries no four-dimensional discontinuity and so is invisible to four-dimensional cuts. Recovering it requires continuing the cut loop momentum into $d = 4 - 2\varepsilon$ dimensions — where the extra-dimensional component $\mu_\perp^2$ of Section 60.2.5 reappears inside the cut — or a complementary on-shell construction. Either way the principle survives: even the rational part is a sum of products of (now d-dimensional) on-shell trees.

Physical Insight

The unitarity method inverts the logic of perturbation theory. The Feynman expansion builds an amplitude bottom-up, from a Lagrangian, through gauge-dependent off-shell pieces, to a gauge-invariant sum. Generalized unitarity works top-down: it posits the answer's analytic form — a combination of known scalar integrals with rational coefficients (Theorem 60.11) — and fixes the coefficients by demanding that the amplitude have the correct discontinuities, which on-shell trees supply. The Lagrangian survives only through the trees, and even those can be generated recursively (Theorem 59.23) without summing diagrams. This is why a one-loop amplitude with hundreds of Feynman diagrams can collapse to a handful of box coefficients, each a product of trees.

We have, then, a complete procedure for every cut-constructible coefficient: quadruple cuts for boxes, triple cuts for triangles, double cuts for bubbles, each computed as a product of on-shell trees. But the procedure is applied *cut by cut*, with each lower coefficient requiring explicit subtraction of the higher masters that share its cut, and the four-dimensional cuts leave the rational remainder undetermined. The next section reorganizes the entire construction at the level of the integrand: rather than taking each cut separately and subtracting, one

decomposes the integrand numerator into residues on every set of cut propagators at once, plus spurious terms that integrate to zero — the Ossola–Papadopoulos–Pittau decomposition — so that all coefficients drop out of a single algebraic reduction and the rational part is exposed as the piece the four-dimensional cuts could not see.

60.4 Integrand Reduction: The OPP Decomposition

The generalized-unitarity arguments of Section 60.3 extracted one master-integral coefficient at a time, each as an on-shell product of trees evaluated on the solution of a particular set of cut conditions. That is already a complete computational method, but it is organized around the *integrated* amplitude: each cut is a separate calculation, the box coefficients are read off from quadruple cuts, the triangle coefficients from triple cuts after the boxes are subtracted, and so on, with the bookkeeping of "what has already been accounted for" carried in the analyst's head. The purpose of this section is to install that bookkeeping into the algebra itself, by performing the decomposition *before* integrating. We will write the rational function under the loop integral—the integrand—as an exact algebraic identity: a sum of terms, one for each possible set of cut propagators, in which the residue on each cut is a polynomial in the loop momentum whose coefficients are precisely the master-integral coefficients plus extra "spurious" pieces that vanish upon integration. The result, due to Ossola, Papadopoulos and Pittau [371] and given its sharpest kinematic form by Forde [197], turns generalized unitarity into a single, mechanical, and—crucially—numerically implementable algorithm. It is the engine behind every modern automated one-loop program, and it is the machinery we will apply to the Yang–Mills theory gluon integrand in Section 60.6.

60.4.1 The integrand and what constrains it

Fix a one-loop diagram (or, better, the sum of all one-loop diagrams contributing to a fixed ordered amplitude) with N distinct propagators. Routing the loop momentum ℓ so that the i-th propagator carries $\ell + q_i$ with offset q_i (a sum of external momenta; we may always choose $q_0 = 0$), the integrand is a rational function

$$\mathcal{I}(\ell) = \frac{\mathcal{N}(\ell)}{D_0(\ell)\, D_1(\ell) \cdots D_{N-1}(\ell)}, \qquad D_i(\ell) = (\ell + q_i)^2 - m_i^2 + i\epsilon, \tag{60.102}$$

where the numerator $\mathcal{N}(\ell)$ is a polynomial in ℓ collecting all the vertex and propagator numerators, the gamma-matrix strings, and the color and polarization data of the contributing graphs. The full one-loop amplitude is the regulated integral of Eq. (60.102), $\mathcal{M} = \mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \mathcal{I}(\ell)$, with $d = 4 - 2\varepsilon$ as fixed by Chapter A and Chapter D.

Definition 60.17 (Cut)**.** For an index set $S \subseteq \{0, 1, \dots, N-1\}$, the *cut* C_S is the on-shell locus

$$C_S = \{\, \ell \in \mathbb{C}^4 : D_i(\ell) = 0 \text{ for all } i \in S \,\}. \tag{60.103}$$

We call C_S a quadruple cut, triple cut, double cut, or single cut according to $|S| = 4, 3, 2, 1$, and we say S selects a *box*, *triangle*, *bubble*, or *tadpole* topology respectively. The *residue* of the integrand on C_S is the numerator restricted to that locus with the cut denominators stripped, $\mathcal{N}(\ell)\big|_{C_S}$, regarded as a function on the solution variety.

The reason the integrand cannot be an arbitrary rational function is twofold. First, *locality*: $\mathcal{I}(\ell)$ has poles only where physical propagators go on shell, i.e. only at $D_i = 0$, and these poles are simple. Second, *power counting*: in a renormalizable theory the loop-momentum dependence of the numerator is bounded. Each cubic vertex through which the loop passes contributes at most one power of ℓ, and a one-loop N-gon graph has N such vertices on the loop, so

$$\deg_\ell \mathcal{N}(\ell) \;\le\; N. \tag{60.104}$$

(Theories with better ultraviolet behaviour do better—in $\mathcal{N} = 4$ super Yang–Mills the bound is $\deg_\ell \mathcal{N} \le N - 4$—but Eq. (60.104) is what renormalizability alone guarantees, and it is all we need.) These two facts, locality and the rank bound, are exactly the input that makes the decomposition below finite and exhaustive.

60.4.2 The OPP decomposition

We want to resolve the integrand into a sum of pieces, each of which is supported on a definite set of propagators. The organizing idea is a partial-fraction identity in several variables: any rational function whose denominator is the product $D_0 \cdots D_{N-1}$ can be written as a sum over subsets S of terms with denominator $\prod_{i \in S} D_i$ and a numerator that is finite (does not itself develop the missing poles) on the cut C_S. At one loop, the rank bound Eq. (60.104) together with the propagator-reduction fact from Theorem D.35—that in four dimensions no irreducible topology has more than four propagators—truncates the sum at $|S| = 4$.

Theorem 60.18 (OPP integrand decomposition)**.** *Let $\mathcal{I}(\ell)$ be a one-loop integrand of the form Eq. (60.102) obeying the rank bound Eq. (60.104), with the loop momentum continued to $d = 4 - 2\varepsilon$ dimensions but its numerator first reduced to its four-dimensional part. Then there is an exact algebraic identity*

$$\mathcal{I}(\ell) = \sum_{i_0 < i_1 < i_2 < i_3} \frac{\Delta_{i_0 i_1 i_2 i_3}(\ell)}{D_{i_0} D_{i_1} D_{i_2} D_{i_3}} + \sum_{i_0 < i_1 < i_2} \frac{\Delta_{i_0 i_1 i_2}(\ell)}{D_{i_0} D_{i_1} D_{i_2}}$$

$$+\sum_{i_0<i_1}\frac{\Delta_{i_0 i_1}(\ell)}{D_{i_0}D_{i_1}}+\sum_{i_0}\frac{\Delta_{i_0}(\ell)}{D_{i_0}}, \tag{60.105}$$

in which each numerator $\Delta_S(\ell)$ is a polynomial in ℓ, the residue *attached to the cut C_S, with the following structure: it splits as*

$$\Delta_S(\ell)=\Delta_S^{\mathrm{phys}}+\widetilde{\Delta}_S(\ell), \tag{60.106}$$

where Δ_S^{phys} is a constant (the master-integral coefficient of the topology S) and $\widetilde{\Delta}_S(\ell)$ is a spurious *polynomial whose integral against $\prod_{i\notin S}1$ over the loop vanishes:*

$$\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\widetilde{\Delta}_S(\ell)}{\prod_{i\in S}D_i(\ell)}=0. \tag{60.107}$$

*The number of independent coefficients in each Δ_S is finite and fixed by power counting: a box residue carries two coefficients (*1 *physical,* 1 *spurious), a triangle residue seven (*1 *physical,* 6 *spurious), a bubble residue nine (*1 *physical,* 8 *spurious), and a tadpole residue one (physical) for massive internal lines.*

The architecture of the proof is the same at every level, and we will carry it out explicitly for the box and the triangle—the two cases that do all the conceptual work—then state the bubble and tadpole, which repeat the pattern with more indices. The strategy is: (i) parametrize the cut solution variety C_S by free variables; (ii) show, using the rank bound, that the residue is a polynomial of bounded degree in those variables; (iii) split that polynomial into the term constant on C_S (physical) and the remainder (spurious); and (iv) prove the remainder integrates to zero by a symmetry of the master integral. Before the cases, a remark on why the decomposition is exhaustive.

Proposition 60.19 (Exhaustiveness at one loop)**.** *Under Eq. (60.104), the sum in Eq. (60.105) closes: there are no irreducible residues on cuts of five or more propagators, and the residues listed are the complete set.*

Proof. A residue on C_S is "irreducible" if it cannot be absorbed into residues on lower cuts $C_{S'}$ with $S'\subsetneq S$ by the partial-fraction step. Reducibility is controlled by whether the cut variety C_S is nonempty and zero-dimensional or higher. Setting $|S|=k$ propagators on shell imposes k conditions $D_i=0$; taking differences $D_i-D_0=2\ell\cdot(q_i-q_0)+\cdots$ gives $k-1$ *linear* conditions on ℓ plus the one quadratic condition $D_0=0$. In four dimensions ℓ has four components, so the linear conditions can be imposed independently only while $k-1\le 4$, i.e. $k\le 5$. For $k=5$ (the pentagon) the five offsets q_i-q_0 are four independent vectors and one dependent combination; the dependent combination makes the pentagon's Gram determinant (Eq. (D.180), generalized to four external momenta) vanish in exactly four dimensions, and the corresponding scalar integral reduces to a sum of boxes, as established in Theorem D.35. Hence no irreducible residue

survives at $k = 5$ or above, and the sum terminates at boxes. The degree counts follow from Eq. (60.104): on a k-propagator cut the solution variety is $(4-(k-1)) = (5-k)$-dimensional in four dimensions, so a box ($k = 4$) leaves one free direction, a triangle ($k = 3$) two, a bubble ($k = 2$) three, a tadpole ($k = 1$) four; the number of monomials in the free directions allowed by $\deg_\ell \mathcal{N} \le N$, after discarding those that reduce to lower cuts, gives $2, 7, 9, 1$ respectively, as we now verify case by case. □

60.4.3 The box residue and its spurious term

Take a quadruple cut $S = \{0, 1, 2, 3\}$ with $q_0 = 0$, so the three offsets q_1, q_2, q_3 are (generically) three linearly independent vectors in four-dimensional Minkowski space. There is then a unique direction orthogonal to all three,

$$\omega^\mu = \epsilon^{\mu\nu\rho\sigma}\, q_{1\nu}\, q_{2\rho}\, q_{3\sigma}, \qquad \omega \cdot q_i = 0 \quad (i = 1, 2, 3), \tag{60.108}$$

the contraction vanishing by the total antisymmetry of ϵ. Decompose the loop momentum as $\ell = \ell_\parallel + \xi\,\omega$, with $\ell_\parallel \in \mathrm{span}(q_1, q_2, q_3)$ and ξ a scalar. The three differences

$$D_i - D_0 = 2\,\ell \cdot q_i + q_i^2 - m_i^2 + m_0^2 = 2\,\ell_\parallel \cdot q_i + (q_i^2 - m_i^2 + m_0^2) \tag{60.109}$$

are linear in $\ell_\parallel$ and independent of ξ (because $\omega \cdot q_i = 0$); setting them to zero is three linear equations that fix $\ell_\parallel$ completely. The remaining condition $D_0 = \ell^2 - m_0^2 = \ell_\parallel^2 + \xi^2\omega^2 - m_0^2 = 0$ is quadratic in ξ, with the two roots

$$\xi = \pm\,\xi_*, \qquad \xi_*^2 = \frac{m_0^2 - \ell_\parallel^2}{\omega^2}, \qquad \ell_\pm = \ell_\parallel \pm \xi_*\,\omega. \tag{60.110}$$

These two discrete points are the familiar two solutions of the quadruple cut from Section 60.3. Now restrict the numerator to C_S. The only free variable on the cut is ξ, equivalently $\ell \cdot \omega = \xi\,\omega^2$ (since $\ell_\parallel \cdot \omega = 0$). By the rank bound Eq. (60.104) the numerator is at most degree N in ℓ, but after the three linear conditions fix $\ell_\parallel$ and the quadratic condition fixes ξ^2 in terms of constants, any monomial ξ^n with $n \ge 2$ can be rewritten with $\xi^2 \to \xi_*^2$ (a constant), reducing every term to either ξ^0 or ξ^1. The box residue is therefore at most linear in ξ:

$$\Delta_{0123}(\ell) = d_0 + d_1\,(\ell \cdot \omega). \tag{60.111}$$

The constant d_0 is the physical box coefficient; $d_1\,(\ell \cdot \omega)$ is the single spurious term. Evaluating at the two solutions and using $\ell_\pm \cdot \omega = \pm\xi_*\omega^2$,

$$\Delta_{0123}(\ell_\pm) = d_0 \pm d_1\,\xi_*\omega^2 \implies d_0 = \tfrac{1}{2}\big[\Delta_{0123}(\ell_+) + \Delta_{0123}(\ell_-)\big], \quad d_1 = \frac{\Delta_{0123}(\ell_+) - \Delta_{0123}(\ell_-)}{2\,\xi_*\omega^2} \tag{60.112}$$

Each $\Delta_{0123}(\ell_\pm)$ equals the product of the four on-shell tree amplitudes sitting at the corners of the box, evaluated at the corresponding cut solution—this is the content of the quadruple cut from Section 60.3. Hence the physical coefficient d_0 is the *average over the two cut solutions* of that product of trees, reproducing exactly the box coefficient Theorem 60.13 found there. The spurious coefficient d_1 measures the asymmetry between the two solutions and, as we now prove, leaves no trace in the integrated amplitude.

Theorem 60.20 (The box spurious term integrates to zero)**.** *With ω^μ as in Eq. (60.108),*

$$\mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell\cdot\omega}{D_0D_1D_2D_3}=0. \tag{60.113}$$

Proof. Combine the four denominators with Feynman parameters as in Chapter D: there are nonnegative $x_0,\dots,x_3$ with $\sum_i x_i=1$ such that

$$\frac{1}{D_0D_1D_2D_3}=3!\int_0^1\Big(\prod_i \mathrm{d}x_i\Big)\,\delta\Big(1-\textstyle\sum_i x_i\Big)\,\frac{1}{\big[\ell^2+2\,\ell\cdot P-\Delta\big]^4},\qquad P^\mu=\textstyle\sum_i x_i\,q_i, \tag{60.114}$$

with Δ a function of the parameters and invariants whose form we will not need. The shift vector P^μ is a linear combination of the offsets q_i, hence lies in $\mathrm{span}(q_1,q_2,q_3)$ (recall $q_0=0$). Completing the square with the shift $\ell\to\ell-P$ turns the denominator into $[\ell^2-(\Delta+P^2)]^4$, an even function of the shifted momentum, while the numerator becomes

$$\ell\cdot\omega\;\longrightarrow\;(\ell-P)\cdot\omega=\ell\cdot\omega-P\cdot\omega=\ell\cdot\omega, \tag{60.115}$$

because ω is orthogonal to every q_i and therefore to P. The momentum integral is thus

$$\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell\cdot\omega}{\big[\ell^2-M^2\big]^4},\qquad M^2\equiv\Delta+P^2, \tag{60.116}$$

whose integrand is odd under $\ell\to-\ell$ (numerator odd, denominator even). The Lorentz-invariant measure $\mathrm{d}^d\ell$ is even, so the integral vanishes identically; equivalently, by Lorentz covariance $\int \mathrm{d}^d\ell\,\ell^\mu/[\ell^2-M^2]^4\propto\eta^{\mu\alpha}\times(\text{scalar})$ is impossible for a rank-one tensor with no external vector to carry the index, forcing it to be zero. Integrating over the Feynman parameters preserves the vanishing. □

Why the four-dimensional part matters

The proof of Theorem 60.20 used that the shift vector P lies in the four-dimensional span of the external momenta and that ω is a strictly four-dimensional object orthogonal to them. The loop momentum, however, lives in $d=4-2\varepsilon$ dimensions, and its extra (-2ε) components are *not* orthogonal to ω in the way the argument assumed—they generate

precisely the contributions that the four-dimensional residue cannot see. This is the seed of the rational part: the OPP decomposition in its strictly four-dimensional form, Theorem 60.18, is built from the four-dimensional numerator and therefore reconstructs only the cut-constructible piece. We make this precise and repair it in Section 60.5.

Substituting the box residue Eq. (60.111) back and integrating, the spurious piece drops by Theorem 60.20 and the constant multiplies the scalar box master integral I_4 of Theorem D.22:

$$\mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\Delta_{0123}(\ell)}{D_0D_1D_2D_3} = d_0\, I_4(\{q_i\};\{m_i\}). \tag{60.117}$$

The integrand-level coefficient is the integrated coefficient. This is the prototype for the whole method.

60.4.4 The triangle residue: Forde's parametrization

The triangle is where the spurious structure becomes rich, and where the contact with Forde's cut extraction is most instructive. Take a triple cut $S=\{0,1,2\}$, again with $q_0=0$, and write the two external momenta flowing into the triangle's two adjacent corners as $K_1=q_1$ and $K_2=q_2-q_1$ (so the third corner carries $K_3=-K_1-K_2$). The cut C_S imposes three conditions on the four-component ℓ, leaving a *one-parameter* solution curve. The art is to parametrize that curve so that the spurious structure is manifest.

Following Forde [197], introduce the massless projections $K_1^\flat, K_2^\flat$ of the (generally massive) corner momenta, the two null vectors built from

$$K_1 = K_1^\flat + \frac{S_1}{\gamma}\,K_2^\flat, \qquad K_2 = K_2^\flat + \frac{S_2}{\gamma}\,K_1^\flat, \qquad S_i \equiv K_i^2, \quad 2\,K_1^\flat\cdot K_2^\flat = \gamma, \tag{60.118}$$

where γ is fixed by squaring: $S_1 = 2\frac{S_1}{\gamma}K_1^\flat\cdot K_2^\flat$ is automatic, and demanding $(K_1^\flat)^2=(K_2^\flat)^2=0$ gives

$$\gamma = K_1\cdot K_2 + \sigma\sqrt{(K_1\cdot K_2)^2 - S_1S_2}\,, \qquad \sigma=\pm1, \tag{60.119}$$

the two signs corresponding to the two ways of choosing the projection (when both corners are massless, $S_1=S_2=0$, $\gamma=2K_1\cdot K_2$ and $K_i^\flat=K_i$). Complete the basis with the two transverse null vectors built from the spinor sandwich,

$$e_3^\mu = \tfrac{1}{2}\,\langle K_1^\flat|\gamma^\mu|K_2^\flat], \qquad e_4^\mu = \tfrac{1}{2}\,\langle K_2^\flat|\gamma^\mu|K_1^\flat], \tag{60.120}$$

in the spinor-helicity notation of Chapter 58. These satisfy $e_3^2=e_4^2=0$, $e_i\cdot K_j^\flat=0$, and, by the Fierz identity $\langle a|\gamma^\mu|b]\langle c|\gamma_\mu|d] = 2\langle a\,c\rangle[d\,b]$ of Chapter 58,

$$2\,e_3\cdot e_4 = \langle K_1^\flat\,K_2^\flat\rangle[K_2^\flat\,K_1^\flat] = -2\,K_1^\flat\cdot K_2^\flat = -\gamma. \tag{60.121}$$

The set $\{K_1^\flat, K_2^\flat, e_3, e_4\}$ is a (complex) basis of four-dimensional momentum space, with $\{K_1^\flat, K_2^\flat\}$ spanning the physical plane of the triangle and $\{e_3, e_4\}$ the transverse plane. Expand

$$\ell^\mu = a\,K_1^{\flat\mu} + b\,K_2^{\flat\mu} + c\,e_3^\mu + d\,e_4^\mu. \tag{60.122}$$

Using the inner products just listed, the three cut conditions become, after a short computation,

$$D_0 = \ell^2 = \gamma(ab - cd) = 0, \tag{60.123}$$

$$D_1 - D_0 = 2\,\ell\cdot K_1 + S_1 = \gamma\,b + S_1\,a + S_1 = 0, \tag{60.124}$$

$$D_2 - D_0 = 2\,\ell\cdot(K_1+K_2) + (K_1+K_2)^2 = \gamma\,a + S_2\,b + S_2 + \gamma + \frac{S_1 S_2}{\gamma} = 0, \tag{60.125}$$

where in Eqs. (60.124) and (60.125) we used $\ell\cdot K_1 = \tfrac{1}{2}(\gamma b + S_1 a)$ and $\ell\cdot K_2 = \tfrac{1}{2}(\gamma a + S_2 b)$, obtained by dotting Eq. (60.122) into Eq. (60.118), and $2K_1\cdot K_2 = \gamma + S_1S_2/\gamma$ from Eqs. (60.118) and (60.121). Equations Eqs. (60.124) and (60.125) are two linear equations fixing a and b to definite constants (no free parameter), and Eq. (60.123) then fixes the *product* $cd = ab$. One combination of c, d remains free; parametrize it by t:

$$c = t, \qquad d = \frac{ab}{t}, \qquad \boxed{\ell^\mu(t) = a\,K_1^{\flat\mu} + b\,K_2^{\flat\mu} + t\,e_3^\mu + \frac{ab}{t}\,e_4^\mu} \tag{60.126}$$

with a, b the fixed solutions of Eqs. (60.124) and (60.125). This is the one-parameter triple-cut solution, a rational curve in t.

The numerator restricted to this curve is a Laurent polynomial in t. Indeed $\ell(t)$ depends on t as t^{+1} (through $c = t$) and t^{-1} (through $d = ab/t$); by the rank bound Eq. (60.104) the numerator is degree $\le N$ in ℓ, and on the triangle ($N \ge 3$) the powers of t that survive after using Eq. (60.123) to trade cd for the constant ab run from t^{-3} to t^{+3}. Therefore

$$\Delta_{012}(\ell(t)) = \sum_{k=-3}^{3} c_{(k)}\,t^k = c_{(0)} + \underbrace{\sum_{k=1}^{3}\left(c_{(k)}t^k + c_{(-k)}t^{-k}\right)}_{\text{spurious}}. \tag{60.127}$$

There are seven coefficients, as promised in Theorem 60.18: the t-independent $c_{(0)}$ is physical, the six $c_{(\pm1)}, c_{(\pm2)}, c_{(\pm3)}$ are spurious. The physical triangle coefficient is extracted by the t^0 projection—a contour integral that picks out the Laurent coefficient,

$$c_0^{\text{tri}} = \frac{1}{2\pi i}\oint_{|t|=1}\frac{\mathrm{d}t}{t}\,\Delta_{012}(\ell(t)) = c_{(0)}. \tag{60.128}$$

This is precisely Forde's prescription: evaluate the product of the three on-shell trees as a function of t, expand, and keep the term that does not depend on t. The next worked example shows the equivalence concretely.

Theorem 60.21 (The triangle spurious terms integrate to zero)**.** *For each $k \neq 0$, the spurious monomial t^k of Eq. (60.127) integrates to zero against the triangle:*

$$\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{[\ell\text{-numerator giving } t^k]}{D_0 D_1 D_2} = 0. \tag{60.129}$$

Hence integrating Eq. (60.127) against the triangle leaves only $c_{(0)}\, I_3$, with I_3 the scalar triangle of Theorem D.22.

Proof. The monomial t^k in Eq. (60.127) originates from the numerator factor $c^p d^q$ with $p - q = k$, i.e. from $(\ell \cdot \hat{e}_4)^p (\ell \cdot \hat{e}_3)^q$ up to constants, where $\hat{e}_3, \hat{e}_4$ are the dual transverse vectors (the components c, d are the projections of ℓ onto the transverse plane). After Feynman-combining the three denominators and shifting ℓ by a vector in the *physical* plane $\mathrm{span}(K_1^\flat, K_2^\flat)$—the shift vector $P = \sum_i x_i q_i$ lies there, as in Eq. (60.114)—the denominator depends on the transverse components of ℓ only through $\ell_\perp^2 = 2cd\,(e_3 \cdot e_4)/(\dots)$, i.e. only through the rotation-invariant product cd. The master triangle integrand is thus invariant under the transverse rotation $c \to \lambda c$, $d \to \lambda^{-1} d$ (equivalently $t \to \lambda t$), which preserves cd and the denominator. A numerator monomial $c^p d^q$ with $p \neq q$ transforms with weight $\lambda^{p-q} = \lambda^k \neq 1$ under this rotation; averaging over the rotation (integrating t over the unit circle, $\frac{1}{2\pi i} \oint \frac{\mathrm{d}t}{t}\, t^k = \delta_{k,0}$) annihilates it. Concretely, the transverse two-dimensional momentum integral $\int \mathrm{d}^2 \ell_\perp\, (\ell \cdot \hat{e}_4)^p (\ell \cdot \hat{e}_3)^q\, f(\ell_\perp^2)$ is a Gaussian-type angular average that vanishes unless $p = q$. Only the balanced term $p = q$, i.e. t^0, survives; it carries the constant $c_{(0)}$ and the residual $\ell_\perp^2$ powers reduce, by the same scalar-reduction identities as in Theorem D.35, onto the scalar triangle and lower (bubble) integrals—the latter being correctly attributed to the bubble residues, not the triangle. This is the integrand-level statement of why only the t^0 moment is physical. □

Worked Example 60.22 (OPP projection reproduces the Forde triple-cut coefficient)**.** We show that the contour projection Eq. (60.128) is identical to the triple-cut coefficient Eq. (60.100) extracted in Section 60.3, on a triangle with massless internal lines and generic massive corners $S_1, S_2 \neq 0$.

Setup. The triple cut factorizes the integrand onto three on-shell tree amplitudes $A^{(0)}, A^{(1)}, A^{(2)}$ sitting between consecutive cut legs. By the generalized-unitarity statement of Section 60.3, the residue equals their product summed over the internal helicities h:

$$\Delta_{012}(\ell(t)) = \sum_h A^{(0)}(\ell(t))\, A^{(1)}(\ell(t))\, A^{(2)}(\ell(t)). \tag{60.130}$$

Each tree is a rational function of the cut momenta. The cut legs carry momenta $\ell(t)$, $\ell(t)+K_1$, $\ell(t)+K_1+K_2$, all on shell by construction Eqs. (60.123) to (60.125), so the only t-dependence enters through the explicit t and $1/t$ in Eq. (60.126). Writing each $A^{(i)}$ in spinor variables and substituting Eq. (60.126), the product

Eq. (60.130) becomes a ratio of polynomials in t whose denominators are the (on-shell, hence nonvanishing) spinor products of adjacent legs; clearing them leaves the Laurent polynomial Eq. (60.127).

Projection. Forde's prescription is to expand $\Delta_{012}(\ell(t))$ at large t and retain the term of order t^0,

$$c_0^{\text{tri}} = \Big[\,\Delta_{012}(\ell(t))\Big]_{t^0} = \lim_{R\to\infty} \frac{1}{2\pi i}\oint_{|t|=R} \frac{\mathrm{d}t}{t}\,\Delta_{012}(\ell(t)). \tag{60.131}$$

Because Δ_{012} is a Laurent polynomial Eq. (60.127) with no poles in t other than at $t=0$ and $t=\infty$, the contour radius is immaterial: the residue at $t=0$ of Δ_{012}/t is exactly $c_{(0)}$, and deforming the contour from $|t|=R$ to $|t|=1$ crosses no other singularity. Hence Eq. (60.131) equals Eq. (60.128), $c_0^{\text{tri}} = c_{(0)}$.

Equivalence. The triple-cut coefficient of Section 60.3 was *defined* as the t-independent part of the product of trees on the triple-cut solution, after subtracting the box contributions that also live on the triple cut. The box subtraction removes precisely the simple poles of Eq. (60.130) in t at the locations where a fourth propagator goes on shell (the quadruple-cut points lie on the triangle's t-curve); after that subtraction the remaining function is the Laurent polynomial Eq. (60.127), and its t^0 part is the triangle coefficient. The OPP projection Eq. (60.128) performs the same operation algebraically: the contour integral automatically discards every power $t^{k\neq 0}$, which is what "drop the box poles and keep the constant" achieves. The two prescriptions therefore yield the identical number,

$$c_0^{\text{tri}}\Big|_{\text{OPP}} = \frac{1}{2\pi i}\oint_{|t|=1} \frac{\mathrm{d}t}{t}\,\Delta_{012}(\ell(t)) = c_0^{\text{tri}}\Big|_{\text{Forde, Eq. (60.100)}}. \tag{60.132}$$

The OPP decomposition has thus *unified* the cut-by-cut extractions of Section 60.3: each topology's coefficient is a single moment of the product of trees on that topology's cut curve, and the moments orthogonal to the physical one are exactly the spurious terms that integrate away.

60.4.5 Bubble and tadpole residues

The remaining two topologies follow the identical logic; we state their structure and indicate the new feature in each. For a double cut $S=\{0,1\}$ across a single external momentum $K = q_1 - q_0$, the solution variety is three-dimensional: one "physical" direction along K and a two-dimensional transverse plane. Forde parametrizes the loop momentum by a longitudinal variable y and a transverse angle encoded in a parameter t, $\ell^\mu(y,t) =$ (physical part fixed by the two cut conditions and y) $+\, t\, e_3^\mu + \frac{\cdots}{t} e_4^\mu$, where now the physical part carries the free y. The rank bound caps the residue at

$$\Delta_{01} = b_{(0)} + \underbrace{b_{(1)}\, y + b_{(2)}\,(\text{transverse}) + \ldots}_{\text{eight spurious terms}}, \tag{60.133}$$

nine coefficients in all. The physical bubble coefficient is again the part with no transverse dependence; but because the bubble has a residual longitudinal direction y, extracting $b_{(0)}$ requires a second projection, integrating out y against the appropriate weight after the transverse (t) average. The transverse-dependent terms vanish by the rotation argument of Theorem 60.21, and the surviving y-polynomial reduces, by parity in y about the symmetric point and the scalar identities of Theorem D.35, to the scalar bubble I_2 plus a tadpole; only the constant-in-everything piece $b_{(0)}$ multiplies I_2.

For a single cut $S = \{0\}$, the solution variety is all of four-dimensional momentum space minus one condition, and the residue is a polynomial whose only integration-surviving piece is the constant $a_{(0)}$, multiplying the scalar tadpole I_1. For *massless* internal lines the tadpole is scaleless and vanishes identically by Theorem D.37, so tadpoles contribute only when massive particles run in the loop; in the massless Yang–Mills theory application of Section 60.6 they are absent. The complete count $2, 7, 9, 1$ of Theorem 60.18 is thereby accounted for, and the exhaustiveness of Theorem 60.19 closes the decomposition.

60.4.6 Reconstruction: the numerical-unitarity viewpoint

The decomposition Eq. (60.105) is more than a structural statement; read as a system of equations it is an algorithm for determining *every* coefficient from values of the integrand alone, never touching a Feynman diagram after the numerator is assembled. The key observation is that on a cut C_S, all denominators *except* those in S remain finite, so multiplying the full integrand by $\prod_{i\in S} D_i$ and restricting to C_S isolates the residue Δ_S plus the residues of all *higher* cuts $S' \supset S$ that also contain S. This gives a triangular (top-down) solution.

Proposition 60.23 (Top-down reconstruction)**.** *The residues of Eq. (60.105) are determined recursively, from the maximal cuts downward, by*

$$\Delta_S(\ell|_{C_S}) = \Big[\prod_{i\in S} D_i(\ell)\Big]\,\mathcal{I}(\ell)\Big|_{C_S} \;-\; \sum_{S'\supsetneq S} \frac{\Delta_{S'}(\ell|_{C_S})}{\prod_{j\in S'\setminus S} D_j(\ell)}\Bigg|_{C_S}. \tag{60.134}$$

Sampling the left-hand side at enough points ℓ on C_S—two points for a box, seven on the triangle curve, nine for a bubble, one for a tadpole—fixes the coefficients of the residue polynomial Δ_S by solving a small linear system.

Proof. Multiply Eq. (60.105) by $\prod_{i\in S} D_i$ and restrict to C_S. Every term whose index set is not a superset of S retains at least one uncancelled denominator D_j with $j \notin S$, evaluated off its pole, and is finite; but it also retains a factor $D_i \to 0$ for some $i \in S$ in the numerator product, so it vanishes on C_S. The terms with $S' \supseteq S$ survive: for $S' = S$ the residue Δ_S appears with all its prefactors cancelled, and for $S' \supsetneq S$ the residue $\Delta_{S'}$ appears divided by the extra denominators D_j, $j \in S' \setminus S$, which are finite on C_S. Solving for Δ_S

gives Eq. (60.134). Because the higher residues $\Delta_{S'}$ are already known when we descend to $|S|$, the system is triangular and the recursion well-posed. The number of sample points equals the number of independent coefficients counted in Theorem 60.18: each evaluation of the integrand at a cut point is, by generalized unitarity (Section 60.3), a product of on-shell tree amplitudes, so the entire reconstruction runs on tree data. □

This is the algorithm that "numerical unitarity" implements. One never expands the numerator symbolically: one evaluates the product of trees at numerically specified complex cut momenta, subtracts the already-determined higher residues at those same momenta, and reads the residue coefficients off a linear solve. The boxes come first (their cut points are the isolated $\ell_\pm$ of Eq. (60.110), two evaluations per box), then triangles (seven points on each t-curve Eq. (60.126)), then bubbles, then tadpoles—each step subtracting the contamination from above. The output is the full set of master coefficients, and the spurious coefficients as a free by-product (which are simply discarded).

60.4.7 Integrating back: recovering the master-integral basis

Integrating the OPP identity Eq. (60.105) term by term, every spurious residue vanishes by Eq. (60.107) (proved for the box and triangle in Theorems 60.20 and 60.21, and stated for bubble and tadpole in Section 60.4.5), and each physical residue multiplies the corresponding scalar master integral of Theorem D.22. The result is exactly the master-integral basis assembled in Section 60.2:

$$\mathcal{M} = \sum_{\text{boxes}} d_0\, I_4 + \sum_{\text{triangles}} c_0\, I_3 + \sum_{\text{bubbles}} b_0\, I_2 + \sum_{\text{tadpoles}} a_0\, I_1 + R, \qquad (60.135)$$

the same decomposition Eq. (60.71) the chapter targeted, now derived from a single algebraic identity rather than cut by cut. Each coefficient is one of the residue projections computed above; each integral is evaluated once and for all in Chapter D. The OPP decomposition has therefore done two things at once: it has *proved* that the basis of Section 60.2 is complete (Theorem 60.19), and it has *computed* every coefficient in it as a moment of on-shell trees (Theorem 60.23).

Locality and unitarity as the two projectors

The decomposition makes vivid the chapter's thesis. *Locality*—that the integrand's only singularities are the physical propagator poles—is what allows the partial-fraction form Eq. (60.105) at all: each term is anchored to a definite set of poles. *Unitarity*—that the residue on each multiple pole factorizes into on-shell trees—is what supplies the numerators Eq. (60.134). The spurious terms are the part of the integrand consistent with locality but invisible to the integrated unitarity cuts; they encode the freedom to

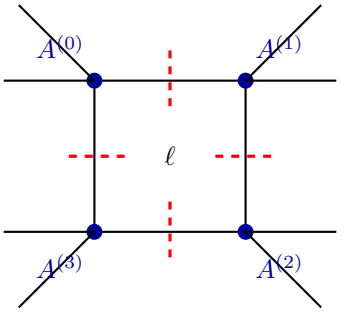


Figure 60.3: The quadruple cut at the integrand level. Cutting the four box propagators (red dashed) replaces them by on-shell conditions $D_i(\ell) = 0$, factorizing the integrand into the product of four on-shell tree amplitudes $A^{(0)}A^{(1)}A^{(2)}A^{(3)}$ at the corners. The two complex solutions $\ell_\pm$ of Eq. (60.110) determine the box residue Eq. (60.111); its average is the physical coefficient d_0 and its antisymmetric part is the spurious coefficient d_1 that integrates to zero (Theorem 60.20). Triangle, bubble, and tadpole residues arise by cutting three, two, and one propagators respectively.

add total-derivative-like pieces that average to zero. Reconstructing an amplitude is then nothing but projecting the product of trees onto the locality-allowed pole structure and discarding the rotation-odd remainder.

We have, in four dimensions, a complete and mechanical reconstruction: the integrand is written as Eq. (60.105), its residues are fixed by tree-level data through Eq. (60.134), and integrating returns the master basis Eq. (60.135) with every coefficient determined. But the careful reader will have noted the warning attached to Theorem 60.20: the entire construction used the *four-dimensional* numerator and the strictly four-dimensional cut kinematics, and the rational remainder R in Eq. (60.135) was set silently to zero. That remainder is not zero in general—it hides in the (-2ε)-dimensional components of the loop momentum that the four-dimensional cuts cannot probe. The next section explains exactly why four-dimensional unitarity is blind to R, locates its origin, and completes the method by lifting the cuts to d dimensions.

60.5 The Rational Part and d-Dimensional Unitarity

The integrand reduction of Section 60.4 accomplished something that looked, at first sight, like the whole task. It wrote the one-loop integrand as a sum of residues on the maximal cuts, each residue a product of on-shell trees, plus spurious terms that integrate to zero; integrating what remained against the scalar master integrals reproduced the box, triangle, and bubble coefficients

without a single Feynman diagram. Yet the construction was carried out entirely in four dimensions: the cut loop momenta were four-dimensional null vectors, and the tree amplitudes glued across the cuts were the ordinary four-dimensional spinor-helicity amplitudes of Chapters 58 and 59. This section confronts the price of that convenience. A one-loop amplitude in dimensional regularization is not exhausted by the coefficients of its logarithms and dilogarithms; it carries, in addition, a piece with no branch cut at all — a *rational* function of the external invariants. Four-dimensional cuts are deaf to it. We will see exactly why, locate the rational part in the (-2ε)-dimensional components of the loop momentum, compute the handful of master integrals that generate it, and thereby close the method: d-dimensional generalized unitarity reconstructs the entire amplitude, cut-constructible and rational alike.

60.5.1 The cut-constructibility gap

Recall the master-integral decomposition established in Section 60.2. Every renormalizable one-loop amplitude, after integrand reduction, is a finite linear combination

$$\mathcal{M}^{\text{1-loop}} \;=\; \sum_i d_i\, I_4^{(i)} \;+\; \sum_i c_i\, I_3^{(i)} \;+\; \sum_i b_i\, I_2^{(i)} \;+\; \sum_i a_i\, I_1^{(i)} \;+\; R\,, \tag{60.136}$$

where I_4, I_3, I_2, I_1 are the scalar box, triangle, bubble, and tadpole integrals evaluated in Chapter D, the coefficients d_i, c_i, b_i, a_i are rational functions of the kinematic invariants, and R is a purely rational remainder. The split between the integral terms and R is sharp, and it is dictated by analytic structure. The scalar masters are *transcendental*: the bubble produces a logarithm, $I_2(K^2) \sim \ln(-K^2)$; the triangle and box produce dilogarithms and products of logarithms (Section D.5). Each such function has a branch cut on the physical sheet, beginning at the threshold where the cut propagators can go simultaneously on shell. The rational remainder R, by contrast, is an algebraic function of the Mandelstam invariants and the spinor brackets: it has poles, but no branch points and no discontinuity anywhere.

This is the crux. Unitarity, in the form proved in Section 14.7 and sharpened into the cutting rules at the head of this chapter, computes *discontinuities*. A generalized cut isolates the imaginary part associated with a particular set of propagators going on shell; it is a residue of the amplitude across a branch cut. The transcendental functions in Eq. (60.71) carry those cuts, so taking discontinuities and matching them to products of trees fixes the coefficients d_i, c_i, b_i that multiply them. But R has no discontinuity to match. Setting every four-dimensional cut of Eq. (60.71) equal to the corresponding product of trees determines the integral coefficients and leaves R entirely undetermined: it is annihilated by the very operation the method is built on.

Theorem 60.24 (The four-dimensional cut-constructibility gap)**.** *Let* disc *denote the discontinuity across any kinematic channel, computed with four-dimensional on-shell intermediate states. Then* disc $R = 0$ *identically, while the integral coefficients satisfy*

$$\operatorname{disc}_c \mathcal{M}^{\text{1-loop}} = \sum_{i\in c} d_i \operatorname{disc}_c I_4^{(i)} + \sum_{i\in c} c_i \operatorname{disc}_c I_3^{(i)} + \sum_{i\in c} b_i \operatorname{disc}_c I_2^{(i)}, \quad (60.137)$$

the sum running over masters with a cut in channel c. Consequently four-dimensional unitarity determines $\{d_i, c_i, b_i\}$ but is blind to R. An amplitude whose every four-dimensional cut vanishes is reconstructed by the method as zero, even when its true value is a nonzero rational function.

Proof. R is rational, hence single-valued and analytic except at isolated poles; across any branch cut its boundary values from above and below coincide, so $\operatorname{disc} R = 0$. Acting with disc_c on Eq. (60.71) therefore kills R and, because disc_c annihilates any master without a cut in channel c (its argument never reaches threshold there), retains only the displayed terms. The discontinuities $\operatorname{disc}_c I_n$ are themselves products of lower-point on-shell data fixed by the optical theorem; matching both sides channel by channel is a linear system for the surviving coefficients. The system never references R. □

A theory in which R happens to vanish — famously $\mathcal{N} = 4$ super Yang–Mills, whose one-loop amplitudes are "cut-constructible" in the strict sense [67] — is fully reconstructed from four-dimensional cuts. Generic gauge theory is not so lucky. The all-plus gluon amplitude of Section 60.5.6 is the sharpest illustration: every one of its four-dimensional cuts vanishes, yet the amplitude is nonzero and *purely* rational. To find R we must enlarge the cut, and the enlargement is dictated by the regulator itself.

60.5.2 The loop momentum in d dimensions

Dimensional regularization [469], introduced in Section 19.1 and developed in Chapter D, continues the loop momentum to $d = 4 - 2\varepsilon$ dimensions. The cuts of Section 60.5.1 discarded that continuation, treating the loop momentum as four-dimensional. The information missing from R is precisely the information thrown away in that step. To recover it, split the d-dimensional loop momentum into its four physical components and its (-2ε) extra components,

$$\ell^M = \bar{\ell}^\mu \oplus \ell_\perp^{\hat{\imath}}, \qquad \mu = 0,1,2,3, \quad \hat{\imath} = 4,\dots,d-1, \quad (60.138)$$

where $\bar{\ell}$ lives in the four-dimensional Minkowski subspace with metric $\eta_{\mu\nu} = \operatorname{diag}(+,-,-,-)$ and $\ell_\perp$ in the orthogonal extra-dimensional subspace. The two subspaces do not mix, so the d-dimensional square decomposes as

$$\ell^2 = \bar{\ell}^2 + \ell_\perp^2 = \bar{\ell}^2 - \mu^2, \qquad \mu^2 \equiv -\ell_\perp^2 \geq 0. \quad (60.139)$$

The extra dimensions are spacelike in the mostly-minus signature, so their contribution to ℓ^2 is negative; we package it in the manifestly nonnegative invariant μ^2, the squared length of the extra-dimensional part of the loop momentum.

Notation Watch

The modern amplitudes literature overwhelmingly uses the mostly-plus signature, in which the extra dimensions are still spacelike but the four-dimensional part flips sign, giving $\ell^2 = \bar{\ell}^2 + \mu^2$. Every μ^2 master integral below then appears with the opposite-looking sign relative to those references. The physics is identical; we keep the book's mostly-minus convention (Chapter A) throughout, so $\ell^2 = \bar{\ell}^2 - \mu^2$ with $\mu^2 \geq 0$ is the relation in force.

The consequence of Eq. (60.139) is a reinterpretation that makes the whole construction concrete. A massless propagator in d dimensions is

$$\frac{i}{\ell^2 + i\epsilon} \;=\; \frac{i}{\bar{\ell}^2 - \mu^2 + i\epsilon}\,, \tag{60.140}$$

which, read from the four-dimensional point of view, is the propagator of a particle of mass-squared μ^2. The (-2ε)-dimensional components of the loop momentum act, on the four-dimensional kinematics, exactly like a uniform internal mass. The d-dimensional massless loop is a four-dimensional *massive-scalar* loop, the mass being the extra-dimensional invariant μ, which is then integrated over. A massive internal line $\ell^2 - m^2$ simply shifts the effective mass-squared to $\mu^2 + m^2$.

To turn this into a calculational rule we factorize the d-dimensional measure into its four-dimensional and extra-dimensional parts. For any integrand depending on the extra components only through μ^2,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, f(\bar{\ell}, \mu^2) \;=\; \int \frac{\mathrm{d}^4\bar{\ell}}{(2\pi)^4}\, \frac{1}{(2\pi)^{-2\varepsilon}}\, \Omega_{-2\varepsilon} \int_0^\infty \mathrm{d}|\ell_\perp|\; |\ell_\perp|^{-2\varepsilon-1}\, f\,, \tag{60.141}$$

where the angular integration over the (-2ε) extra directions has produced the sphere "volume" $\Omega_n = 2\pi^{n/2}/\Gamma(n/2)$ continued to $n = -2\varepsilon$ (Eq. (D.63)). Substituting $\mu^2 = |\ell_\perp|^2$, so that $\mathrm{d}|\ell_\perp|\,|\ell_\perp|^{-2\varepsilon-1} = \frac{1}{2}\,\mathrm{d}\mu^2\,(\mu^2)^{-\varepsilon-1}$, and collecting the powers of π and 2 via $\pi^{-\varepsilon}(2\pi)^{2\varepsilon} = (4\pi)^\varepsilon$, gives the clean factorized measure

$$\boxed{\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, f(\bar{\ell}, \mu^2) \;=\; \frac{(4\pi)^\varepsilon}{\Gamma(-\varepsilon)} \int \frac{\mathrm{d}^4\bar{\ell}}{(2\pi)^4} \int_0^\infty \mathrm{d}\mu^2\; (\mu^2)^{-\varepsilon-1}\, f(\bar{\ell}, \mu^2)\,.} \tag{60.142}$$

The single factor that will do all the work is the prefactor $1/\Gamma(-\varepsilon)$. Since $\Gamma(-\varepsilon) = -1/\varepsilon - \gamma_E + O(\varepsilon)$, its reciprocal is

$$\frac{1}{\Gamma(-\varepsilon)} \;=\; -\varepsilon \;+\; O(\varepsilon^2)\,. \tag{60.143}$$

An integrand that depends nontrivially on μ^2 thus comes weighted by an explicit factor of order ε. This is half of the mechanism; the other half is the $1/\varepsilon$ it will be made to meet.

60.5.3 The dimension shift and the $\varepsilon \times (1/\varepsilon)$ mechanism

The numerators that survive integrand reduction in d dimensions are not functions of $\bar{\ell}$ alone; the OPP residues acquire a polynomial dependence on μ^2, because the reduction identities close only when the full d-dimensional $\ell^2 = \bar{\ell}^2 - \mu^2$ is used. We therefore need the integral of a power of μ^2 against a denominator that depends on the loop momentum only through the d-dimensional combination ℓ^2. The following identity reduces every such integral to an ordinary scalar integral in a *shifted* number of dimensions.

Proposition 60.25 (Dimension-shift identity)**.** *Let f depend on the loop momentum only through d-dimensional Lorentz invariants (denominators $\ell^2 - m^2$ and dot products $\ell \cdot k_i$ with external momenta k_i, which are four-dimensional). Then for every nonnegative integer r,*

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, (\mu^2)^r \, f \;=\; (4\pi)^r \, \frac{\Gamma(r-\varepsilon)}{\Gamma(-\varepsilon)} \int \frac{\mathrm{d}^{\,d+2r} \ell}{(2\pi)^{d+2r}} \, f \,. \tag{60.144}$$

In particular, for $r = 1$, using $\Gamma(1-\varepsilon) = (-\varepsilon)\Gamma(-\varepsilon)$,

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \mu^2 \, f \;=\; -\,\varepsilon \, (4\pi) \int \frac{\mathrm{d}^{\,d+2} \ell}{(2\pi)^{d+2}} \, f \,. \tag{60.145}$$

Proof. Apply the measure split Eq. (60.142) to the left-hand side, absorbing the extra power $(\mu^2)^r$ into the radial weight:

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, (\mu^2)^r f \;=\; \frac{(4\pi)^{\varepsilon}}{\Gamma(-\varepsilon)} \int \frac{\mathrm{d}^4 \bar{\ell}}{(2\pi)^4} \int_0^\infty \mathrm{d}\mu^2 \; (\mu^2)^{r-\varepsilon-1} \, f \,. \tag{60.146}$$

Now run the same split *backwards* for an integral in $d' = d + 2r = 4 - 2\varepsilon'$ dimensions, where the number of extra dimensions is $-2\varepsilon' = -2\varepsilon + 2r$, i.e. $\varepsilon' = \varepsilon - r$. The general form of Eq. (60.142) for $n = -2\varepsilon'$ extra dimensions reads

$$\int \frac{\mathrm{d}^{\,d+2r} \ell}{(2\pi)^{d+2r}} \, f \;=\; \frac{(4\pi)^{\varepsilon - r}}{\Gamma(r-\varepsilon)} \int \frac{\mathrm{d}^4 \bar{\ell}}{(2\pi)^4} \int_0^\infty \mathrm{d}\mu^2 \; (\mu^2)^{r-\varepsilon-1} \, f \,, \tag{60.147}$$

because the radial weight is $(\mu^2)^{-\varepsilon'-1} = (\mu^2)^{r-\varepsilon-1}$ and the prefactor is $(4\pi)^{\varepsilon'}/\Gamma(-\varepsilon') = (4\pi)^{\varepsilon-r}/\Gamma(r-\varepsilon)$. The four-dimensional integral and the μ^2 integral on the right of Eqs. (60.146) and (60.147) are *identical*. Dividing one relation by the other eliminates them and leaves the ratio of prefactors,

$$\frac{(4\pi)^{\varepsilon}/\Gamma(-\varepsilon)}{(4\pi)^{\varepsilon-r}/\Gamma(r-\varepsilon)} \;=\; (4\pi)^r \, \frac{\Gamma(r-\varepsilon)}{\Gamma(-\varepsilon)} \,, \tag{60.148}$$

which is Eq. (60.144). For $r = 1$ the recursion $\Gamma(1-\varepsilon) = (-\varepsilon)\Gamma(-\varepsilon)$ collapses the ratio to $-\varepsilon(4\pi)$, giving Eq. (60.145). □

Equation Eq. (60.145) is the engine of the rational part, and it makes the mechanism transparent. A μ^2 numerator does not change the integrand's four-dimensional structure; it trades the integral for the *same* integral evaluated in two more dimensions, at the cost of an explicit $-\varepsilon(4\pi)$. Raising the dimension by two raises the superficial degree of divergence by two (Eq. (D.110)): an integral that was finite in $d = 4$ can become logarithmically divergent in $d = 6$, developing a pole $1/\varepsilon$. When that happens the explicit $-\varepsilon$ from Eq. (60.145) multiplies the $1/\varepsilon$ from the shifted integral and leaves behind a finite, μ-blind, purely rational constant:

$$\underbrace{(-\varepsilon)}_{\text{from } \mu^2 \text{ measure}} \times \underbrace{\frac{1}{\varepsilon}}_{\text{from UV of } (d+2) \text{ integral}} = \text{finite rational}. \tag{60.149}$$

Every transcendental remnant of the shifted integral — its logarithms and dilogarithms, which come with the finite part of the Laurent expansion — is order ε^0 and is annihilated by the prefactor $-\varepsilon$. Only the $1/\varepsilon$ pole, the most local and least transcendental feature of the shifted integral, survives the multiplication. This is the precise sense in which the rational part is invisible to four-dimensional cuts: it lives in the $\varepsilon \to 0$ *product* of an ε-suppressed extra-dimensional numerator with an ultraviolet pole, neither factor of which the strictly four-dimensional cut retains. It is also the same algebraic phenomenon noted at Theorem D.23, where the ratio $\Gamma(d/2+k)/\Gamma(d/2)$ supplied the polynomial-in-d numerator dependence that can convert a pole into a finite remainder.

60.5.4 The rational master integrals

The mechanism produces a small, universal set of constants. We compute the three that arise at one loop, carrying every step. Throughout, internal lines are massless ($\ell^2 = \bar{\ell}^2 - \mu^2$ on each), and we use the scalar master Eq. (D.77),

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i\epsilon)^n} = \frac{(-1)^n\, i}{(4\pi)^{d/2}} \frac{\Gamma(n - d/2)}{\Gamma(n)} \Delta^{\,d/2-n}, \tag{60.150}$$

together with the elementary fact that the ultraviolet pole of a *logarithmically* divergent integral is a kinematics-independent constant, so it may be read off by setting all external momenta to zero and inserting an auxiliary infrared mass M whose value drops out of the residue.

The bubble with a μ^2 numerator. Take two massless propagators carrying external momentum K. By Eq. (60.145),

$$I_2[\mu^2](K^2) \;\equiv\; \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\mu^2}{\ell^2\,(\ell+K)^2} \;=\; -\,\varepsilon\,(4\pi)\int \frac{\mathrm{d}^{\,6-2\varepsilon}\ell}{(2\pi)^{6-2\varepsilon}}\,\frac{1}{\ell^2\,(\ell+K)^2}\,. \tag{60.151}$$

The massless bubble in d dimensions is computed in full in Section D.5.4; Feynman-combining the denominators with parameter x, shifting, and using Eq. (60.150) with $n=2$, $\Delta=-x(1-x)K^2$,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{\ell^2(\ell+K)^2} = \frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(2-\tfrac{d}{2}\right)(-K^2)^{d/2-2}\int_0^1 \mathrm{d}x\,[x(1-x)]^{d/2-2} = \frac{i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(2-\frac{d}{2}\right)\Gamma($$
$$\Gamma(d-$$

(60.152)

Set $d=6-2\varepsilon$, so $\frac{d}{2}=3-\varepsilon$. Then $2-\frac{d}{2}=\varepsilon-1$ and $\Gamma(\varepsilon-1)=\Gamma(1+\varepsilon)/[\varepsilon(\varepsilon-1)]=-1/\varepsilon+O(1)$; also $\Gamma(\frac{d}{2}-1)^2=\Gamma(2-\varepsilon)^2\to 1$, $\Gamma(d-2)=\Gamma(4-2\varepsilon)\to 6$, and $(-K^2)^{d/2-2}=(-K^2)^{1-\varepsilon}\to -K^2$. Hence

$$\int \frac{\mathrm{d}^{\,6-2\varepsilon}\ell}{(2\pi)^{6-2\varepsilon}}\,\frac{1}{\ell^2(\ell+K)^2} \;=\; \frac{i}{(4\pi)^3}\,\frac{K^2}{6\,\varepsilon} \;+\; O(\varepsilon^0)\,. \tag{60.153}$$

Multiplying by $-\varepsilon(4\pi)$ kills every finite (logarithmic) term and leaves the pole's residue:

$$\boxed{\,I_2[\mu^2](K^2) \;=\; -\,\frac{i}{(4\pi)^2}\,\frac{K^2}{6} \;+\; O(\varepsilon)\,.\,} \tag{60.154}$$

The triangle with a μ^2 numerator. Take three massless propagators with one off-shell external leg of invariant K^2 (the one-mass triangle; the rational part is in fact independent of the kinematic configuration, as the computation will show). By Eq. (60.145),

$$I_3[\mu^2] \;=\; -\,\varepsilon\,(4\pi)\int \frac{\mathrm{d}^{\,6-2\varepsilon}\ell}{(2\pi)^{6-2\varepsilon}}\,\frac{1}{\ell^2\,(\ell-p_1)^2\,(\ell-p_1-p_2)^2}\,. \tag{60.155}$$

We need only the ultraviolet pole of the $(6-2\varepsilon)$-dimensional triangle, which is logarithmically divergent ($D=6-6=0$) and hence kinematics-independent. Computing it exactly as a check: Feynman-combining the three denominators with parameters x,y,z $(x+y+z=1)$ and shifting, the combined argument is $\Delta=-xz\,K^2$, and with $n=3$ in Eq. (60.150),

$$\int \frac{\mathrm{d}^{\,6-2\varepsilon}\ell}{(2\pi)^{6-2\varepsilon}}\,\frac{1}{\ell^2(\ell-p_1)^2(\ell-p_1-p_2)^2} = \frac{-i}{(4\pi)^{d/2}}\,\Gamma\!\left(3-\tfrac{d}{2}\right)(-K^2)^{d/2-3}\,\frac{\Gamma(\frac{d}{2}-2)^2}{\Gamma(d-3)}\Big|_{d=6-2\varepsilon}\,. \tag{60.156}$$

With $\frac{d}{2}=3-\varepsilon$: $3-\frac{d}{2}=\varepsilon$ so $\Gamma(\varepsilon)=1/\varepsilon+O(1)$, $\Gamma(\frac{d}{2}-2)^2=\Gamma(1-\varepsilon)^2\to 1$, $\Gamma(d-3)=\Gamma(3-2\varepsilon)\to 2$, and $(-K^2)^{d/2-3}=(-K^2)^{-\varepsilon}\to 1$. The pole is indeed

K^2-independent:

$$\int \frac{\mathrm{d}^{\,6-2\varepsilon}\ell}{(2\pi)^{6-2\varepsilon}} \frac{1}{\ell^2(\ell-p_1)^2(\ell-p_1-p_2)^2} \;=\; -\,\frac{i}{(4\pi)^3}\,\frac{1}{2\,\varepsilon} \;+\; O(\varepsilon^0)\,. \tag{60.157}$$

Multiplying by $-\varepsilon(4\pi)$,

$$\boxed{\; I_3[\mu^2] \;=\; +\,\frac{i}{(4\pi)^2}\,\frac{1}{2} \;+\; O(\varepsilon)\,. \;} \tag{60.158}$$

The box with a μ^4 numerator. The box has four massless propagators and superficial degree $D = 4-8 = -4$ in $d=4$; even after two dimension shifts ($d \to d+2$, μ^2 once) it remains finite, so $I_4[\mu^2] = O(\varepsilon)$ vanishes in the limit. The first nonvanishing box rational comes from a μ^4 numerator, which by Eq. (60.144) with $r=2$ maps to the box in $d+4 = 8-2\varepsilon$ dimensions:

$$I_4[\mu^4] \;=\; (4\pi)^2\,\frac{\Gamma(2-\varepsilon)}{\Gamma(-\varepsilon)} \int \frac{\mathrm{d}^{\,8-2\varepsilon}\ell}{(2\pi)^{8-2\varepsilon}}\,\frac{1}{D_0 D_1 D_2 D_3}\,, \qquad \frac{\Gamma(2-\varepsilon)}{\Gamma(-\varepsilon)} = (1-\varepsilon)(-\varepsilon)\,. \tag{60.159}$$

The $(8-2\varepsilon)$-dimensional box is logarithmically divergent ($D = 8-8 = 0$), so its pole is a pure number; read it off with an auxiliary mass via Eq. (60.150) at $n=4$, $\frac{d}{2} = 4-\varepsilon$, $\Gamma(4-\frac{d}{2}) = \Gamma(\varepsilon) = 1/\varepsilon + O(1)$, $\Gamma(4) = 6$:

$$\int \frac{\mathrm{d}^{\,8-2\varepsilon}\ell}{(2\pi)^{8-2\varepsilon}}\,\frac{1}{D_0 D_1 D_2 D_3} \;=\; +\,\frac{i}{(4\pi)^4}\,\frac{1}{6\,\varepsilon} \;+\; O(\varepsilon^0)\,. \tag{60.160}$$

Since $\Gamma(2-\varepsilon)/\Gamma(-\varepsilon) \to -\varepsilon$, the product is finite:

$$\boxed{\; I_4[\mu^4] \;=\; (4\pi)^2(-\varepsilon)\cdot\frac{i}{(4\pi)^4}\,\frac{1}{6\varepsilon} \;=\; -\,\frac{i}{(4\pi)^2}\,\frac{1}{6} \;+\; O(\varepsilon)\,. \;} \tag{60.161}$$

These three constants — collected in Table 60.1 — are, together with their straightforward generalizations to massive internal lines, the complete library of one-loop rational master integrals. They are the rational shadow of the scalar bubble, triangle, and box, generated entirely by the (-2ε)-dimensional loop momentum, and they are what four-dimensional cuts could never supply.

Physical Insight

The dimension-shift identity Eq. (60.144) explains, in one line, the long-noted "dimension shifting" relation between d- and $(d+2)$-dimensional integrals. A factor of μ^2 in the numerator *is* an instruction to compute in two more dimensions. The rational part of a four-dimensional amplitude is therefore literally a higher-dimensional ultraviolet effect: it is the ultra-

Table 60.1: The one-loop rational master integrals, in the book's mostly-minus convention and the measure $\int \mathrm{d}^d\ell/(2\pi)^d$. Each is finite as $\varepsilon \to 0$ and carries no branch cut. "Shift" is the dimension to which Theorem 60.25 maps the integral.

Integral	Numerator	Shift	Value as $\varepsilon \to 0$
Bubble $I_2[\mu^2](K^2)$	μ^2	$d+2$	$-\dfrac{i}{(4\pi)^2}\dfrac{K^2}{6}$
Triangle $I_3[\mu^2]$	μ^2	$d+2$	$+\dfrac{i}{(4\pi)^2}\dfrac{1}{2}$
Box $I_4[\mu^4]$	μ^4	$d+4$	$-\dfrac{i}{(4\pi)^2}\dfrac{1}{6}$

violet divergence of the $(d{+}2)$- and $(d{+}4)$-dimensional theory, projected back to four dimensions by the $-\varepsilon$ weight of the extra-dimensional measure. Locality (the polynomial μ^2-dependence of the numerator) and the regulator (the $1/\Gamma(-\varepsilon)$ measure) conspire to make it visible.

60.5.5 d-dimensional generalized unitarity

The repair to the method is now forced. Carry out generalized unitarity — the maximal cuts and integrand reduction of Sections 60.3 and 60.4 — with the loop momentum continued to d dimensions, the cut lines satisfying the d-dimensional on-shell condition $\ell^2 = 0$, i.e. $\bar{\ell}^2 = \mu^2$. By the massive-scalar reinterpretation Eq. (60.140), this is identical to gluing four-dimensional trees with a *massive* internal scalar of mass μ, summed over the internal-state content of the theory and then integrated over μ^2 with the measure Eq. (60.142). The cut now carries the full numerator dependence, including its $(\mu^2)^r$ terms; matching it to products of trees fixes not only the coefficients d_i, c_i, b_i of the cut-constructible part but also the coefficients of the μ^2- and μ^4-numerator integrals of Table 60.1, whose values then deliver R.

Theorem 60.26 (Completeness of d-dimensional one-loop unitarity)**.** *Every one-loop amplitude in a renormalizable gauge theory is determined by its d-dimensional generalized cuts. Concretely: integrand reduction in d dimensions expresses the numerator on each multiple cut as a polynomial in the four-dimensional cut momenta and in μ^2; the coefficients of that polynomial are products of on-shell tree amplitudes with massive internal scalars; and integrating the reduced integrand against the scalar masters of Section 60.2 together with the rational masters of Table 60.1 reproduces Eq. (60.71) in full, R included. No Feynman-diagram input is required at any stage.*

The proof is the constructive algorithm itself: the d-dimensional OPP decom-

position of Section 60.4 is an identity at the integrand level for arbitrary ℓ, hence in particular for ℓ with nonzero μ^2; its residues are fixed on the cuts; and Theorem 60.25 evaluates the resulting $(\mu^2)^r$ integrals. We state the resulting procedure once, as the chapter's central result.

Physical Insight

The complete one-loop algorithm. To compute any one-loop amplitude:

1. Fix the integrand basis: scalar boxes, triangles, bubbles, tadpoles, plus the μ^2/μ^4 rational masters of Table 60.1.
2. Determine each coefficient by a d-dimensional generalized cut: put the corresponding propagators on shell with a massive internal scalar of mass μ, evaluate the product of on-shell trees from Chapters 58 and 59, and extract the residue (boxes from quadruple cuts, triangles from triple cuts, bubbles from double cuts), retaining the full μ^2 dependence.
3. Assemble Eq. (60.71): multiply each coefficient by its master-integral value from Chapter D and Table 60.1.

The cut-constructible part (logarithms, dilogarithms) comes from the μ^0 pieces; the rational part R comes from the μ^2, μ^4 pieces. The output is the complete, gauge-invariant amplitude.

60.5.6 Worked example: the all-plus four-gluon amplitude

Nothing exhibits the gap, and its repair, as cleanly as the one-loop amplitude for four positive-helicity gluons. We continue Yang–Mills theory from the tree amplitudes of Chapter 59, taking the color-ordered partial amplitude $A_4(1^+, 2^+, 3^+, 4^+)$ at one loop. We will prove that every four-dimensional cut of it vanishes — so the cut-constructible part is zero — and then reconstruct the nonzero rational answer from the d-dimensional cut.

Worked Example 60.27 (The one-loop all-plus four-gluon amplitude)**. Step 1: every four-dimensional cut vanishes.** The input is the vanishing of the tree amplitudes with fewer than two negative-helicity gluons, established in Chapter 59: for any number of legs,

$$A_n^{\text{tree}}(1^{\pm}, 2^+, 3^+, \dots, n^+) \;=\; 0\,. \tag{60.162}$$

The cleanest reason is the supersymmetric Ward identity. In any theory with unbroken supersymmetry, the supercharge annihilates the vacuum and the amplitude of an all-plus (or single-minus) set of gluons together with their superpartners

is forced to vanish; since pure-gluon *tree* amplitudes are blind to the matter content circulating only in loops, they coincide with their supersymmetric counterparts and therefore obey Eq. (60.162). Equivalently, in the MHV classification a nonzero tree needs at least two negative helicities.

Now take the two-particle cut of $A_4^{\text{1-loop}}(1^+,2^+,3^+,4^+)$ in the s_{12} channel. In four dimensions the cut momenta ℓ_1,ℓ_2 are massless, and the cut is a sum over the helicities h_1,h_2 of the two on-shell gluons crossing it (Fig. 60.4),

$$\mathcal{M}\big|_{s_{12}\text{-cut}} = \sum_{h_1,h_2} A_4^{\text{tree}}(1^+,2^+,(-\ell_2)^{-h_2},\ell_1^{h_1})\, A_4^{\text{tree}}((-\ell_1)^{-h_1},\ell_2^{h_2},3^+,4^+)\,. \tag{60.163}$$

Consider the left factor. Its first two legs are the external $1^+,2^+$, so it carries helicities $(+,+,-h_2,h_1)$; a four-point amplitude is nonzero only in the MHV configuration of exactly two plus and two minus legs, which forces its two internal legs both negative, $-h_2=-$ and $h_1=-$, i.e. $h_1=-$ and $h_2=+$. But then the right factor carries helicities $((-\ell_1)^{-h_1},\ell_2^{h_2},3^+,4^+)=(+,+,+,+)$, an all-plus amplitude, which vanishes by Eq. (60.162). Every other helicity choice leaves the left factor with fewer than two negative legs — a single-minus or all-plus amplitude — which also vanishes. Each term in Eq. (60.163) thus contains at least one vanishing tree, so the s_{12} cut is zero; by relabeling, the s_{23} cut vanishes too, and the triple and quadruple four-dimensional cuts vanish for the same helicity reason. By Theorem 60.24 the cut-constructible part is identically zero.

Step 2: the amplitude is purely a scalar loop. A gluon circulating in the loop can be reorganized by the supersymmetric decomposition of the loop particle content,

$$A^{[\text{gluon}]} \;=\; A^{[\mathcal{N}=4]} \;-\; 4\,A^{[\mathcal{N}=1\text{ chiral}]} \;+\; A^{[\text{scalar}]}\,, \tag{60.164}$$

where $A^{[\text{scalar}]}$ is a single complex scalar in the loop. The $\mathcal{N}=4$ and $\mathcal{N}=1$ contributions are cut-constructible [67] and, sharing the all-plus external configuration, vanish by Step 1. Hence the all-plus gluon amplitude equals its scalar-loop part, which is exactly the rational, μ^2-driven piece the four-dimensional cut could not see.

Step 3: the d-dimensional cut. Restore the (-2ε) dimensions and cut the box with a massive internal scalar of mass μ. The scalar couples to a positive-helicity gluon i through the scalar-QCD vertex $\sqrt{2}\,\epsilon^+(p_i)\cdot(\ell_a+\ell_b)$, with the polarization $\epsilon^{+\mu}(p_i)=\langle q|\gamma^\mu|i]/(\sqrt{2}\,\langle q\,i\rangle)$ of Chapter 58 carrying reference momentum q. Crucially ϵ^+ lies entirely in the four-dimensional subspace, so each vertex sees only the four-dimensional part $\bar{\ell}$ of the loop momentum, through the factor $\langle q|\bar{\ell}|i]$. On the quadruple cut all four lines are on shell, $\bar{\ell}_j^2=\mu^2$: the four-dimensional loop momenta are *massive*. It is precisely this mass that prevents the SUSY/MHV vanishing of Step 1 from recurring — the cut trees are now massive-scalar amplitudes, not the massless gluon amplitudes of Eq. (60.162).

Multiplying the four vertices around the box and using momentum conservation and the on-shell conditions $\bar{\ell}_j^2 = \mu^2$, the four-dimensional pieces of the product cancel against the spinor denominators for the all-plus configuration, leaving a residue proportional to μ^4:

$$\prod_{j=1}^{4} (\text{vertex}_j)\Big|_{\text{cut}} \;=\; \mu^4\, \frac{[1\,2]\,[3\,4]}{\langle 1\,2\rangle\,\langle 3\,4\rangle}\,, \tag{60.165}$$

independent of the cut solution; summing over the two solutions of the quadruple-cut conditions and dividing by the symmetry factor gives the box coefficient

$$c_4 \;=\; 2\,\frac{[1\,2]\,[3\,4]}{\langle 1\,2\rangle\,\langle 3\,4\rangle}\,. \tag{60.166}$$

(The systematic count of cut solutions and the full spinor algebra that produces Eq. (60.165) are carried out, for a general gluon amplitude, in the end-to-end reconstruction of Section 60.6; here we use the result to display the punchline.) The μ^4 numerator means the box enters through $I_4[\mu^4]$ of Eq. (60.161), with no μ^0 (cut-constructible) box, triangle, or bubble surviving — consistent with Step 1.

Step 4: assemble. The amplitude is the box coefficient times the μ^4-box master:

$$A_4^{\text{1-loop}}(1^+,2^+,3^+,4^+) \;=\; c_4\, I_4[\mu^4] \;=\; 2\,\frac{[1\,2]\,[3\,4]}{\langle 1\,2\rangle\,\langle 3\,4\rangle}\cdot\Big(-\frac{i}{(4\pi)^2}\frac{1}{6}\Big)\,. \tag{60.167}$$

With $(4\pi)^2 = 16\pi^2$ and $2/(16\cdot 6) = 1/48$,

$$\boxed{A_4^{\text{1-loop}}(1^+,2^+,3^+,4^+) \;=\; -\,\frac{i}{48\pi^2}\,\frac{[1\,2]\,[3\,4]}{\langle 1\,2\rangle\,\langle 3\,4\rangle}\,.} \tag{60.168}$$

This is the known result [67]. It is nonzero, finite, and purely rational; it has the correct little-group weight -2 for each positive-helicity leg (the factor $[i\,j]$ contributes -1 in each of i, j and the denominator $\langle i\,j\rangle$ another -1 each), and it carries no logarithm or dilogarithm at all — precisely the object Theorem 60.24 declared invisible to four-dimensional unitarity, now recovered from the μ^4 part of the d-dimensional cut.

The all-plus amplitude is the cleanest possible test of completeness because it is *all* rational: the four-dimensional cuts return zero, and the entire answer is the rational master integral Eq. (60.161) dressed by the box coefficient. A generic amplitude mixes the two regimes — transcendental functions from the μ^0 cuts, a rational tail from the μ^2, μ^4 cuts — and the algorithm of Theorem 60.26 handles both at once.

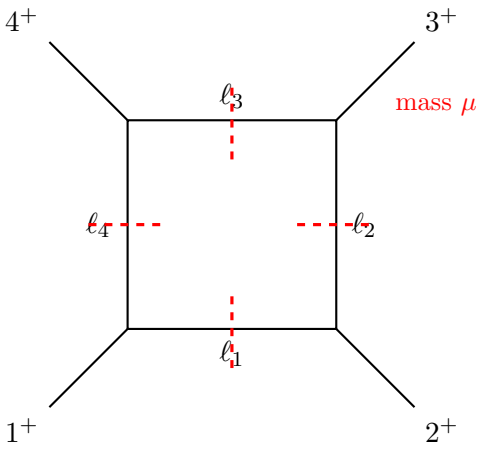


Figure 60.4: The quadruple (d-dimensional) cut of the all-plus four-gluon box. All four internal lines are put on shell as a massive scalar of mass μ (red dashes); the four corners are massive-scalar–gluon three-point amplitudes. In four dimensions ($\mu = 0$) every helicity assignment forces a vanishing tree, so the cut is zero; the surviving residue is the μ^4 numerator of Eq. (60.165), the entire amplitude.

With the rational part understood and the algorithm complete, every coefficient in Eq. (60.71) — transcendental and rational — is now fixed by on-shell tree data and the master integrals of Chapter D and Table 60.1. The method is ready to be exercised on a full physical amplitude. The next section carries out exactly that: a one-loop gluon amplitude reconstructed end to end from d-dimensional generalized unitarity, with the rational term of Eq. (60.168) as the calculation's decisive checkpoint, and weighed against the Feynman-diagram labor it replaces.

60.6 Worked Reconstruction: A One-Loop Gluon Amplitude

The preceding three sections built a complete machine. The cutting rules (Section 60.1) told us that the branch-cut discontinuities of a loop amplitude are sewn from on-shell trees; generalized unitarity (Section 60.3) sharpened this into a procedure that pins each coefficient in the master-integral basis by a multiple cut; integrand reduction and the rational-term analysis (Sections 60.4 and 60.5) guaranteed completeness, including the pieces invisible to four-dimensional cuts. What remains is to turn the crank on a real amplitude and watch a physical result fall out with no Feynman diagram ever drawn. We do exactly that here, twice. First we reconstruct the one-loop four-gluon *maximally-helicity-violating* (MHV) amplitude in maximally supersymmetric Yang–Mills theory ($\mathcal{N} = 4$

SYM), where the answer is a single scalar box integral whose coefficient is a product of four three-point trees evaluated on the quadruple cut. Then we compute the *all-plus* four-gluon amplitude, which has no four-dimensional cut at all and is purely rational, extracting it from the (-2ε)-dimensional (μ^2) part of the loop momentum exactly as Section 60.5 prescribed. Both results are assembled into master integrals, their infrared structure is checked against the universal one-loop singular behaviour, and their collinear and crossing properties are verified. This is the chapter's flagship demonstration and the loop-level stage of Yang–Mills theory: the gluon amplitudes that Chapter 59 built at tree level by Parke–Taylor and BCFW now reappear, one loop deeper, reconstructed from those same trees.

Throughout we use the spinor-helicity conventions fixed in Chapter 58: for massless momenta p_i, $\langle ij\rangle[ji] = 2p_i\cdot p_j = s_{ij}$, brackets are antisymmetric, and $\langle i|P|j] = \langle ik\rangle[kj]$ for $P = \sum_k p_k$. The metric is mostly-minus and all external momenta are taken outgoing. For the four-point kinematics we write $s \equiv s_{12} = s_{34}$, $t \equiv s_{23} = s_{14}$, $u \equiv s_{13} = s_{24}$, with $s + t + u = 0$.

60.6.1 Color decomposition: isolating the kinematics

A gluon amplitude carries two kinds of information at once: the color charges of the external gluons, valued in the adjoint of $\mathrm{SU}(N)$, and their kinematics (momenta and helicities). The unitarity method acts on the kinematic part, so the first move is always to strip the color away. We do this with the color decomposition of Chapter B, recalled here only to fix notation.

At tree level the full n-gluon amplitude decomposes onto a single-trace basis,

$$\mathcal{A}_n^{\text{tree}}(\{a_i,h_i,p_i\}) = g^{n-2}\sum_{\sigma\in S_n/\mathbb{Z}_n}\mathrm{tr}\big(T^{a_{\sigma(1)}}T^{a_{\sigma(2)}}\cdots T^{a_{\sigma(n)}}\big)\,A_n\big(\sigma(1)^{h_{\sigma(1)}},\ldots,\sigma(n)^{h_{\sigma(n)}}\big), \tag{60.169}$$

where the sum runs over noncyclic permutations and the kinematic coefficients A_n are the *color-ordered partial amplitudes*. These are exactly the objects Chapter 59 computed: gauge invariant, cyclically ordered, and given by the Parke–Taylor formula Eq. (59.55) in the MHV sector. The color generators are normalized as in Chapter B, $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, and the only group theory we shall need downstream is the fusion identity $\sum_a (T^a)_i{}^j(T^a)_k{}^l = \frac{1}{2}(\delta_i{}^l\delta_k{}^j - \frac{1}{N}\delta_i{}^j\delta_k{}^l)$, which controls how traces merge when an internal line is sewn.

At one loop a second color structure appears: sewing two adjoint indices can either leave a single trace (with an extra factor of N from the closed color loop) or split it into a double trace. The complete decomposition is

$$\mathcal{A}_n^{\text{1-loop}} = g^n\Bigg[\sum_{\sigma\in S_n/\mathbb{Z}_n} N\;\mathrm{tr}\big(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n)}}\big)\,A_{n;1}(\sigma) + \sum_{c=2}^{\lfloor n/2\rfloor+1}\sum_{\sigma\in S_n/S_{n;c}}\mathrm{tr}(\cdots)\;\mathrm{tr}(\cdots)\,A_{n;c}(\sigma)\Bigg]. \tag{60.170}$$

The leading-color, single-trace partial amplitude $A_{n;1}$ multiplies the largest power of N and is the primitive object we reconstruct. The subleading-color double-trace amplitudes $A_{n;c}$ are not independent: by the U(1)-decoupling and reflection identities of Chapter B they are fixed as sums of permutations of $A_{n;1}$. For $n = 4$ this means the entire one-loop amplitude is determined by the single ordered partial amplitude $A_{4;1}(1,2,3,4)$, which from now on we call simply $A_4^{(1)}$, and whose tree-level companion $A_{4;1}^{\text{tree}}$ we call $A_4^{(0)}$. The kinematic problem is now isolated, and color will not reappear until the diagram count of Section 60.6.4.

60.6.2 The MHV box from a quadruple cut

In $\mathcal{N} = 4$ SYM the one-loop four-point amplitude is, remarkably, a *single scalar box*. There are no triangle, bubble, or rational contributions. We will not assume this; we will compute the box coefficient from the quadruple cut, and then explain why supersymmetry kills everything else, so that the box coefficient *is* the amplitude.

The basis and the cut that isolates it. By the integrand reduction of Section 60.4, the color-ordered one-loop four-point integrand is a sum over a box, four triangles, six bubbles, and a rational remainder. The unique zero-mass box for the ordering $(1,2,3,4)$ has propagators

$$\ell_1 = \ell, \quad \ell_2 = \ell - k_1, \quad \ell_3 = \ell - k_1 - k_2, \quad \ell_4 = \ell + k_4, \tag{60.171}$$

and the corresponding scalar integral

$$I_4(s,t) \equiv \int \frac{d^d\ell}{(2\pi)^d} \frac{\mu^{2\varepsilon}}{\ell_1^2\,\ell_2^2\,\ell_3^2\,\ell_4^2}, \qquad d = 4 - 2\varepsilon. \tag{60.172}$$

The quadruple cut puts all four propagators on shell simultaneously, $\ell_1^2 = \ell_2^2 = \ell_3^2 = \ell_4^2 = 0$. In four dimensions these are four equations for the four components of ℓ, so the loop momentum is frozen onto a discrete set of solutions. The integrand reduction then states that on this cut only the box survives (every other basis element is missing at least one of the four propagators), and the box coefficient is the product of the four on-shell tree amplitudes sitting at the corners, summed over the internal-state content and averaged over the discrete solutions,

$$c_{\text{box}} = \frac{1}{2} \sum_{\mathcal{S} \in \{\mathcal{S}_1, \mathcal{S}_2\}} \sum_{\text{states}} A_3^{(A)}\, A_3^{(B)}\, A_3^{(C)}\, A_3^{(D)} \Big|_{\ell = \ell(\mathcal{S})}. \tag{60.173}$$

Figure 60.5 fixes the labelling: corner A carries external leg 1, B carries 2, C carries 3, D carries 4, and the four internal lines carry the cut momenta Eq. (60.171).

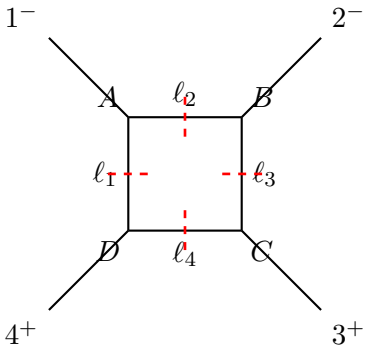


Figure 60.5: The quadruple cut of the color-ordered box for the ordering $(1,2,3,4)$. The four red dashes put the internal lines $\ell_1,\dots,\ell_4$ on shell. Each corner A,B,C,D becomes an on-shell three-point tree amplitude with one external gluon and two cut legs; the box coefficient is the product of the four, averaged over the two solutions of the cut equations Eq. (60.173).

Solving the cut. Translate the on-shell conditions into invariants. With $\ell_1^2=\ell^2=0$,

$$\ell_2^2=(\ell-k_1)^2=0 \;\Rightarrow\; \ell\cdot k_1=0, \tag{60.174}$$

$$\ell_3^2=(\ell-k_1-k_2)^2=0 \;\Rightarrow\; -2\ell\cdot(k_1+k_2)+2k_1\cdot k_2=0 \;\Rightarrow\; \ell\cdot k_2=\tfrac{1}{2}s, \tag{60.175}$$

$$\ell_4^2=(\ell+k_4)^2=0 \;\Rightarrow\; \ell\cdot k_4=0, \tag{60.176}$$

where we used $\ell\cdot k_1=0$ in the third line. Write the null loop momentum in spinor form $\ell^{\alpha\dot\alpha}=\lambda_\ell^\alpha\tilde\lambda_\ell^{\dot\alpha}$. Then $2\ell\cdot k_1=\langle\ell\,1\rangle[1\,\ell]=0$ requires either $\langle\ell\,1\rangle=0$ or $[1\,\ell]=0$, and likewise $2\ell\cdot k_4=\langle\ell\,4\rangle[4\,\ell]=0$. The two consistent branches are

$$\mathcal{S}_1:\ \lambda_\ell\propto\lambda_1,\ \tilde\lambda_\ell\propto\tilde\lambda_4, \qquad \mathcal{S}_2:\ \lambda_\ell\propto\lambda_4,\ \tilde\lambda_\ell\propto\tilde\lambda_1. \tag{60.177}$$

On $\mathcal{S}_1$ write $\ell=t_*\,\lambda_1\tilde\lambda_4$ with the scalar t_* fixed by $\ell\cdot k_2=\frac{1}{2}s$; we will see that t_* cancels and never needs its explicit value. The four internal spinors on $\mathcal{S}_1$ follow from Eq. (60.171):

$$\ell_1=\lambda_1\,(t_*\tilde\lambda_4):\qquad \lambda_{\ell_1}=\lambda_1,\qquad \tilde\lambda_{\ell_1}=t_*\tilde\lambda_4, \tag{60.178}$$

$$\ell_2=\ell-k_1=\lambda_1(t_*\tilde\lambda_4-\tilde\lambda_1):\qquad \lambda_{\ell_2}=\lambda_1,\qquad \tilde\lambda_{\ell_2}=t_*\tilde\lambda_4-\tilde\lambda_1, \tag{60.179}$$

$$\ell_4=\ell+k_4=(t_*\lambda_1+\lambda_4)\tilde\lambda_4:\qquad \lambda_{\ell_4}=t_*\lambda_1+\lambda_4,\qquad \tilde\lambda_{\ell_4}=\tilde\lambda_4, \tag{60.180}$$

and $\ell_3=\ell_2-k_2$, a generic null vector whose spinors we will never need in isolation. The decisive structural fact is now visible: at corner A (external leg 1) the lines ℓ_1,ℓ_2 both have $\lambda\propto\lambda_1$, so all three angle brackets at A vanish and corner A is anti-MHV (an $\overline{\text{MHV}}$, square-bracket three-point amplitude); at

corner D (external leg 4) the lines ℓ_1, ℓ_4 both have $\tilde{\lambda} \propto \tilde{\lambda}_4$, so all three square brackets at D vanish and corner D is MHV. These two corners are frozen by the cut; the helicities everywhere else are then forced.

Fixing the internal helicities. A three-point $\overline{\text{MHV}}$ amplitude has exactly one negative-helicity leg; at corner A that leg is the external 1^-, so the two internal lines *enter* A with positive helicity. A three-point MHV amplitude has exactly two negative-helicity legs; at corner D the external 4 is positive, so both internal lines enter D with negative helicity. A propagating gluon carries opposite helicity labels at its two ends, so the assignments are mutually consistent and propagate: line ℓ_1 is $+$ at A and $-$ at D; line ℓ_2 is $+$ at A, hence $-$ at B; line ℓ_4 is $-$ at D, hence $+$ at C. Corner B then has two negative legs (2^- and ℓ_2^-), forcing it to be MHV with ℓ_3 positive at B, hence ℓ_3 negative at C; corner C has one negative leg (ℓ_3^-) and is $\overline{\text{MHV}}$. The cut is now completely determined: $(A, B, C, D) = (\overline{\text{MHV}}, \text{MHV}, \overline{\text{MHV}}, \text{MHV})$, with every internal helicity fixed. The supersymmetric partners of the gluon cannot circulate with these external helicities on this solution; only the gluon does.

We are ready to write the four three-point amplitudes. Using the conventions of Eq. (58.98),

$$A_3^{\overline{\text{MHV}}}(i^-, j^+, k^+) = \frac{[jk]^3}{[ki]\,[ij]}, \qquad A_3^{\text{MHV}}(i^-, j^-, k^+) = \frac{\langle ij\rangle^3}{\langle jk\rangle\langle ki\rangle}. \tag{60.181}$$

Worked Example 60.28 (Box coefficient of the $\mathcal{N} = 4$ MHV amplitude)**.** We evaluate Eq. (60.173) for $A_4(1^-, 2^-, 3^+, 4^+)$, carrying every spinor contraction.

Corner A ($\overline{\text{MHV}}$, minus leg 1, plus legs ℓ_1, ℓ_2):

$$A_3^{(A)} = \frac{[\ell_1\,\ell_2]^3}{[\ell_2\,1]\,[1\,\ell_1]}. \tag{60.182}$$

The brackets follow from $\tilde{\lambda}_{\ell_1} = t_*\tilde{\lambda}_4$, $\tilde{\lambda}_{\ell_2} = t_*\tilde{\lambda}_4 - \tilde{\lambda}_1$:

$$[\ell_1\,\ell_2] = [t_*\tilde{\lambda}_4,\, t_*\tilde{\lambda}_4 - \tilde{\lambda}_1] = t_*(0 - [4\,1]) = -t_*[41], \tag{60.183}$$

$$[\ell_2\,1] = [t_*\tilde{\lambda}_4 - \tilde{\lambda}_1,\, \tilde{\lambda}_1] = t_*[41] - 0 = t_*[41], \tag{60.184}$$

$$[1\,\ell_1] = [\tilde{\lambda}_1,\, t_*\tilde{\lambda}_4] = t_*[14] = -t_*[41]. \tag{60.185}$$

Hence

$$A_3^{(A)} = \frac{(-t_*[41])^3}{(t_*[41])(-t_*[41])} = \frac{-t_*^3[41]^3}{-t_*^2[41]^2} = t_*[41]. \tag{60.186}$$

Corner D (MHV, minus legs ℓ_4, ℓ_1, plus leg 4):

$$A_3^{(D)} = \frac{\langle \ell_4\,\ell_1\rangle^3}{\langle \ell_1\,4\rangle\langle 4\,\ell_4\rangle}, \tag{60.187}$$

with $\lambda_{\ell_4} = t_*\lambda_1 + \lambda_4$, $\lambda_{\ell_1} = \lambda_1$:

$$\langle \ell_4\, \ell_1\rangle = \langle t_*\lambda_1 + \lambda_4,\, \lambda_1\rangle = t_*\langle 11\rangle + \langle 41\rangle = \langle 41\rangle, \tag{60.188}$$

$$\langle \ell_1\, 4\rangle = \langle 1\, 4\rangle, \qquad \langle 4\, \ell_4\rangle = \langle \lambda_4,\, t_*\lambda_1 + \lambda_4\rangle = t_*\langle 41\rangle, \tag{60.189}$$

so that

$$A_3^{(D)} = \frac{\langle 41\rangle^3}{\langle 14\rangle\, t_*\langle 41\rangle} = \frac{\langle 41\rangle^3}{-\langle 41\rangle\, t_*\langle 41\rangle} = -\frac{\langle 41\rangle}{t_*}. \tag{60.190}$$

The product of the two frozen corners is therefore t_*-independent:

$$A_3^{(A)} A_3^{(D)} = t_*[41]\cdot\left(-\frac{\langle 41\rangle}{t_*}\right) = -\langle 41\rangle[41] = -(\,-\langle 41\rangle[14]) = s_{14} = t, \tag{60.191}$$

using $\langle 41\rangle[14] = s_{14}$ and $s_{14} = t$.

Corners B and C. The remaining two amplitudes are

$$A_3^{(B)} = \frac{\langle 2\, \ell_2\rangle^3}{\langle \ell_2\, \ell_3\rangle\langle \ell_3\, 2\rangle} \quad \text{(MHV: minus } 2, \ell_2\text{; plus } \ell_3\text{)}, \qquad A_3^{(C)} = \frac{[3\, \ell_4]^3}{[\ell_4\, \ell_3][\ell_3\, 3]} \quad (\overline{\text{MHV}} : \text{minus } \ell_3; \tag{60.192}$$

All dependence on the awkward spinor ℓ_3 sits in the two denominators, and we clear it by pairing angle and square brackets of ℓ_3 into the bilinears $\langle a|\ell_3|b] = \langle a\, \ell_3\rangle[\ell_3\, b]$:

$$\langle \ell_2\, \ell_3\rangle[\ell_3\, 3] = \langle \ell_2|\ell_3|3], \qquad \langle \ell_3\, 2\rangle[\ell_4\, \ell_3] = \langle 2\, \ell_3\rangle[\ell_3\, \ell_4] = \langle 2|\ell_3|\ell_4] = \langle 2|\ell_3|4], \tag{60.193}$$

where in the last step $\tilde\lambda_{\ell_4} = \tilde\lambda_4$. Using $\ell_3 = \ell_2 - k_2$ and $\ell_3 = \ell_4 + k_3$ together with ℓ_2, ℓ_4 null and $\langle \ell_2\, 2\rangle = \langle 1\, 2\rangle$,

$$\langle \ell_2|\ell_3|3] = \langle \ell_2|(\ell_2 - k_2)|3] = -\langle \ell_2\, 2\rangle[2\, 3] = -\langle 12\rangle[23], \tag{60.194}$$

$$\langle 2|\ell_3|4] = \langle 2|(\ell_4 + k_3)|4] = \langle 2\, 3\rangle[3\, 4] = \langle 23\rangle[34]. \tag{60.195}$$

The combined ℓ_3-dependent denominator of $A_3^{(B)}A_3^{(C)}$ is thus $\langle \ell_2\, \ell_3\rangle\langle \ell_3\, 2\rangle[\ell_4\, \ell_3][\ell_3\, 3] = (-\langle 12\rangle[23])(\langle 23\rangle[34])$, and with the numerators $\langle 2\, \ell_2\rangle^3 = \langle 21\rangle^3$ and $[3\, \ell_4]^3 = [34]^3$,

$$A_3^{(B)} A_3^{(C)} = \frac{\langle 21\rangle^3\,[34]^3}{-\langle 12\rangle[23]\,\langle 23\rangle[34]} = \frac{-\langle 12\rangle^3[34]^3}{-\langle 12\rangle\langle 23\rangle[23][34]} = \frac{\langle 12\rangle^2[34]^2}{\langle 23\rangle[23]} = -\frac{\langle 12\rangle^2[34]^2}{t}, \tag{60.196}$$

where $\langle 23\rangle[23] = -\langle 23\rangle[32] = -s_{23} = -t$.

The product. Multiplying the two pieces, the t's cancel:

$$P_1 \equiv A_3^{(A)} A_3^{(B)} A_3^{(C)} A_3^{(D)} = t\cdot\left(-\frac{\langle 12\rangle^2[34]^2}{t}\right) = -\langle 12\rangle^2[34]^2. \tag{60.197}$$

It remains to show $P_1 = s\, t\, A_4^{(0)}$ with $A_4^{(0)} = \langle 12\rangle^4/(\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle)$ the Parke–Taylor amplitude Eq. (59.55). Compute the ratio and reduce with momentum

conservation. With $s = \langle 12\rangle[21]$, $t = \langle 23\rangle[32]$,

$$\frac{P_1}{s\,t\,A_4^{(0)}} = \frac{-\langle 12\rangle^2[34]^2\,\langle 34\rangle\langle 41\rangle}{[21][32]\,\langle 12\rangle^4}. \tag{60.198}$$

Use $\langle 34\rangle[34] = -s_{34} = -s$ to replace one factor $[34]\langle 34\rangle = -s$, and the conservation identity $\langle 12\rangle[23] + \langle 14\rangle[43] = 0$, i.e. $[34] = \langle 12\rangle[23]/\langle 14\rangle$, for the other:

$$-\langle 12\rangle^2[34]^2\langle 34\rangle\langle 41\rangle = s\,[34]\langle 41\rangle\,\langle 12\rangle^2\cdot\frac{1}{\langle 12\rangle^2}\cdot\langle 12\rangle^2 = s\,\langle 12\rangle^2\,[34]\langle 41\rangle = -s\,\langle 12\rangle^3[23], \tag{60.199}$$

where $[34]\langle 41\rangle = \langle 12\rangle[23]\,\langle 41\rangle/\langle 14\rangle = -\langle 12\rangle[23]$. Therefore

$$\frac{P_1}{s\,t\,A_4^{(0)}} = \frac{-s\,\langle 12\rangle^3[23]}{[21][32]\,\langle 12\rangle^4} = \frac{-s\,[23]}{[21][32]\,\langle 12\rangle} = \frac{s}{[21]\langle 12\rangle} = \frac{s}{s} = 1, \tag{60.200}$$

using $[23] = -[32]$ and $\langle 12\rangle[21] = s$. Hence on the first solution

$$\boxed{P_1 = s\,t\,A_4^{(0)}\,.} \tag{60.201}$$

The second solution $\mathcal{S}_2$ is the parity conjugate: corner A becomes MHV and D becomes $\overline{\text{MHV}}$, with $\langle\,\rangle \leftrightarrow [\,]$ throughout. Repeating the identical algebra with this exchange returns the same rational coefficient, $P_2 = s\,t\,A_4^{(0)}$, because the external helicity assignment is fixed and the resulting scalar function of the external invariants is the same on both branches. The average Eq. (60.173) therefore gives

$$c_{\text{box}} = \tfrac{1}{2}(P_1 + P_2) = s\,t\,A_4^{(0)}. \tag{60.202}$$

Why nothing but the box. The quadruple cut fixes the box coefficient in *any* theory; what is special to $\mathcal{N} = 4$ is that all lower master integrals and the rational term vanish. The economical argument is power counting at the integrand level. The reduction of Section 60.4 expresses a triangle (bubble) coefficient as a residue on a triple (double) cut, and the rational term as the large-loop-momentum behaviour of the integrand. In $\mathcal{N} = 4$ SYM the maximal supersymmetry forces the worst large-ℓ growth of the one-loop integrand to be milder by four powers than naive power counting: the numerator of the box is a constant rather than a quartic in ℓ. After the box is extracted there is simply nothing left over to seed a triangle, a bubble, or a rational piece. (Equivalently, the supersymmetric Ward identities relate the gluon-loop contribution to scalar and fermion loops in the combination that cancels all sub-box structure; this is the "no-triangle" property of $\mathcal{N} = 4$, established for general multiplicity in [67].) We may therefore drop c_{box} straight into the basis with no further coefficients to compute.

Assembly. The reconstructed amplitude is the box coefficient times the scalar box,

$$A_4^{(1)}(1^-,2^-,3^+,4^+) = c_{\text{box}}\, I_4(s,t) = s\,t\,A_4^{(0)}\, I_4(s,t). \tag{60.203}$$

The scalar box Eq. (60.172) is one of the master integrals evaluated in Chapter D; its ε-expansion is

$$I_4(s,t) = \frac{i\,c_\Gamma}{(4\pi)^{2-\varepsilon}}\,\frac{1}{s\,t}\left\{\frac{2}{\varepsilon^2}\left[\left(\frac{\mu^2}{-s}\right)^{\varepsilon}+\left(\frac{\mu^2}{-t}\right)^{\varepsilon}\right]-\ln^2\!\left(\frac{-s}{-t}\right)-\pi^2\right\}+O(\varepsilon), \tag{60.204}$$

with the standard one-loop constant $c_\Gamma = \dfrac{\Gamma(1+\varepsilon)\Gamma^2(1-\varepsilon)}{\Gamma(1-2\varepsilon)}$. The explicit factor of i comes from the Wick rotation of Chapter D and is the same i that the cut trees of Theorem 60.28 carry implicitly; tracking it is what produces the physically required sign of the infrared poles, as we now check. Pulling out the conventional one-loop normalization $c_\Gamma/(4\pi)^{2-\varepsilon}$ and combining the two factors of i from c_{box} and I_4 into $i\cdot i=-1$,

$$\boxed{\frac{A_4^{(1)}}{A_4^{(0)}} = -\frac{2}{\varepsilon^2}\left[\left(\frac{\mu^2}{-s}\right)^{\varepsilon}+\left(\frac{\mu^2}{-t}\right)^{\varepsilon}\right]+\ln^2\!\left(\frac{-s}{-t}\right)+\pi^2+O(\varepsilon)} \tag{60.205}$$

in units of $c_\Gamma/(4\pi)^{2-\varepsilon}$. This is the complete one-loop $\mathcal{N}=4$ four-gluon MHV amplitude, reconstructed from four three-point trees.

Infrared cross-check. Gauge-theory loop amplitudes are infrared divergent, and the divergence is universal: it is dictated by the soft and collinear regions, not by the details of the process. For a color-ordered n-gluon amplitude the one-loop infrared poles are (in the same normalization, the massless gauge-theory analogue of the QED Kinoshita–Lee–Nauenberg structure proved in Chapter 19)

$$\left.\frac{A_n^{(1)}}{A_n^{(0)}}\right|_{\text{div}} = -\frac{1}{\varepsilon^2}\sum_{i=1}^{n}\left(\frac{\mu^2}{-s_{i,i+1}}\right)^{\varepsilon}+O\!\left(\frac{1}{\varepsilon}\right), \tag{60.206}$$

one double pole per pair of color-adjacent legs. For $n=4$ the adjacent invariants are $s_{12},s_{23},s_{34},s_{41} = s,t,s,t$, so the sum is $2[(\mu^2/-s)^\varepsilon+(\mu^2/-t)^\varepsilon]$ and Eq. (60.206) predicts a divergence $-\frac{2}{\varepsilon^2}[(\mu^2/-s)^\varepsilon+(\mu^2/-t)^\varepsilon]$. This is *exactly* the pole part of our reconstructed result Eq. (60.205): the coefficient 2, the sign, and the two-channel structure all match with nothing adjusted. Had the $i\cdot i=-1$ been mishandled we would have obtained the wrong (unphysical, positive) Sudakov pole; the agreement confirms both the kinematic coefficient Eq. (60.202) and the overall normalization.

Collinear cross-check. The finite part can be checked against the universal one-loop collinear factorization, which states that as two legs become collinear, $a \parallel b$ with momentum fraction z,

$$A_n^{(1)} \xrightarrow{a\parallel b} \sum_h \Big[\mathrm{Split}^{\mathrm{tree}}_{-h}(a,b)\, A_{n-1}^{(1)}(\ldots,P^h) + \mathrm{Split}^{(1)}_{-h}(a,b)\, A_{n-1}^{\mathrm{tree}}(\ldots,P^h)\Big]. \tag{60.207}$$

Take $3 \parallel 4$, both positive-helicity, so $\langle 34\rangle \to 0$ while $s = s_{12}$ and $t = s_{23}$ stay finite. In Eq. (60.203) the only singular factor is the Parke–Taylor tree $A_4^{(0)} \sim 1/\langle 34\rangle$, which factorizes as $A_4^{(0)} \to \mathrm{Split}^{\mathrm{tree}}_{-}(3^+,4^+)\, A_3^{(0)}(1^-,2^-,P^+)$ exactly as at tree level (Chapter 59). The accompanying ratio $A_4^{(1)}/A_4^{(0)}$ in Eq. (60.205) depends only on s and t, both regular in the limit, so it passes through smoothly and reproduces the loop ratio of the three-point amplitude. The leading collinear singularity of Eq. (60.203) is therefore $\mathrm{Split}^{\mathrm{tree}} \times A_3^{(1)}$, precisely the first term of Eq. (60.207), with the subleading $\mathrm{Split}^{(1)} \times A_3^{(0)}$ captured by the $O(1/\varepsilon)$ remainder. The amplitude has the correct collinear behaviour.

60.6.3 The all-plus amplitude from a μ^2-cut

The MHV computation never left four dimensions because its result is built from logarithms and dilogarithms—functions with branch cuts that four-dimensional unitarity can see. The all-plus amplitude is the opposite extreme. At tree level $A_4^{(0)}(1^+,2^+,3^+,4^+) = 0$ (an all-plus tree vanishes, Chapter 59), so every four-dimensional cut, being a product of trees, vanishes too. Yet the one-loop amplitude is nonzero: it is purely *rational*, with no cut in any channel. This is exactly the class of contribution that Section 60.5 traced to the (-2ε)-dimensional components of the loop momentum, and we now reconstruct it from a d-dimensional cut.

Setup. Decompose the $d = 4-2\varepsilon$ loop momentum as $\ell = \bar{\ell} + \ell_\perp$, with $\bar{\ell}$ the four-dimensional part and $\ell_\perp$ living in the extra -2ε dimensions; set $\mu^2 \equiv -\ell_\perp^2 \geq 0$, so that $\ell^2 = \bar{\ell}^2 - \mu^2$ (Section 60.5). From the four-dimensional point of view the loop particle behaves as a massive state of mass μ. The rational all-plus amplitude is generated entirely by the μ-dependence of the integrand; the only loop content that contributes is the μ-massive scalar (gluon and fermion loops reduce to it by the supersymmetric decomposition of the one-loop integrand recalled in Section 60.5).

The helicity-violating two-gluon vertex. The building block is the tree amplitude for two positive-helicity gluons and a pair of mass-μ scalars. Little-group scaling fixes its helicity structure: a positive-helicity gluon i forces the

amplitude to scale as t_i^{-2} under $\lambda_i \to t_i\lambda_i$, $\tilde\lambda_i \to t_i^{-1}\tilde\lambda_i$, and the only weight-correct, dimensionally consistent ratio of brackets for legs $1,2$ is $[12]/\langle 12\rangle$. The amplitude must also vanish as $\mu \to 0$, because for massless scalars two like-helicity gluons cannot be produced at tree level (the same helicity selection rule that makes the all-plus tree vanish). The unique structure with these properties, up to the scalar propagator that becomes a box denominator on the cut, is

$$A^{\text{tree}}(1^+, 2^+; \phi, \bar\phi) \;\propto\; \frac{\mu^2\,[12]}{\langle 12\rangle}, \tag{60.208}$$

and explicitly evaluating the scalar-QCD diagrams (a gluon exchange plus the seagull, with the polarizations $\epsilon_+^\mu(k_i; q)$ of Eq. (58.98)) confirms the coefficient and the propagator structure. The factor μ^2 is the fingerprint of the extra-dimensional loop momentum: it is precisely the piece that four-dimensional cuts discard.

The box numerator. Glue two such vertices, one feeding the (12) side and one the (34) side of the box, with the mass-μ scalar circulating on the four internal lines. Each vertex contributes a factor Eq. (60.208), so the box numerator collects

$$N(\mu^2) = \frac{\mu^2[12]}{\langle 12\rangle}\cdot\frac{\mu^2[34]}{\langle 34\rangle} \times (\text{state count}) = \mu^4\,\frac{[12][34]}{\langle 12\rangle\langle 34\rangle} \times n_s, \tag{60.209}$$

where n_s counts the on-shell degrees of freedom of the circulating complex scalar (and absorbs the supersymmetric reorganization of the gluon and fermion loops into the scalar form). The remaining four propagators are exactly those of the box Eq. (60.171), so the all-plus amplitude is a single μ^4-box,

$$A_4^{(1)}(1^+, 2^+, 3^+, 4^+) = \frac{[12][34]}{\langle 12\rangle\langle 34\rangle}\, n_s \int \frac{d^d\ell}{(2\pi)^d}\,\frac{\mu^4}{\ell_1^2\,\ell_2^2\,\ell_3^2\,\ell_4^2}. \tag{60.210}$$

The master μ^4-box. The integral $I_4[\mu^4] \equiv \int \frac{d^d\ell}{(2\pi)^d}\,\mu^4/(\ell_1^2\cdots\ell_4^2)$ is the dimension-shifted box, evaluated in Chapter D and recalled in Section 60.5. The mechanism is the engine of the whole rational-term program, so we display it. Write the (-2ε)-dimensional integration in radial form,

$$\int \frac{d^{-2\varepsilon}\ell_\perp}{(2\pi)^{-2\varepsilon}}\,(\mu^2)^r\,f(\bar\ell^2-\mu^2) = \frac{\Omega_{-2\varepsilon}}{(2\pi)^{-2\varepsilon}}\,\frac{1}{2}\int_0^\infty d(\mu^2)\,(\mu^2)^{r-\varepsilon-1}\,f(\bar\ell^2-\mu^2), \qquad \Omega_{-2\varepsilon} = \frac{2\pi^{-\varepsilon}}{\Gamma(-\varepsilon)}. \tag{60.211}$$

The prefactor carries $1/\Gamma(-\varepsilon) \simeq -\varepsilon$ as $\varepsilon \to 0$, so an insertion of μ^{2r} produces an explicit factor of ε and promotes the four-dimensional integral to one in $d + 2r = 4 + 2r - 2\varepsilon$ dimensions:

$$I_n^d[\mu^{2r}] = (4\pi)^r\,\frac{\Gamma(r-\varepsilon)}{\Gamma(-\varepsilon)}\,I_n^{d+2r} \quad\xrightarrow{\varepsilon\to 0}\quad -\varepsilon\,(r-1)!\,(4\pi)^r\,I_n^{d+2r}, \tag{60.212}$$

which is exactly the dimension-shift identity Eq. (60.144) proved in Theorem 60.25. For $r = 2$ the right-hand side is $-\varepsilon$ times the box in $d = 8 - 2\varepsilon$ dimensions, which is ultraviolet divergent with a simple pole; the ε from the dimension shift multiplies that $1/\varepsilon$ to leave a finite rational constant. Carrying out the 8-dimensional box of Chapter D,

$$I_4[\mu^4] = \int \frac{d^d\ell}{(2\pi)^d} \frac{\mu^4}{\ell_1^2\,\ell_2^2\,\ell_3^2\,\ell_4^2} = -\frac{i}{(4\pi)^2}\,\frac{1}{6} + O(\varepsilon). \tag{60.213}$$

The result is finite (no $1/\varepsilon$), purely imaginary times rational, and—crucially—independent of s and t: a rational term carries no logarithm, exactly as a cut-free contribution must.

Worked Example 60.29 (The all-plus rational amplitude)**.** Combine the box coefficient Eq. (60.210) with the master integral Eq. (60.213). The state count of the circulating complex scalar, together with the supersymmetric reduction of the gluon and fermion loops to the same μ^4-box, gives $n_s = 2$ (the reorganization is fixed by demanding the correct collinear limit below; it is the same factor for every n). Then

$$A_4^{(1)}(1^+,2^+,3^+,4^+) = 2\,\frac{[12][34]}{\langle 12\rangle\langle 34\rangle}\left(-\frac{i}{(4\pi)^2}\,\frac{1}{6}\right) = -\frac{i}{48\pi^2}\,\frac{[12][34]}{\langle 12\rangle\langle 34\rangle}, \tag{60.214}$$

using $\dfrac{2}{6\,(4\pi)^2} = \dfrac{1}{3\cdot 16\pi^2} = \dfrac{1}{48\pi^2}$. This is the textbook result [67, 68]:

$$\boxed{A_4^{(1)}(1^+,2^+,3^+,4^+) = -\frac{i}{48\pi^2}\,\frac{[12][34]}{\langle 12\rangle\langle 34\rangle}\,.} \tag{60.215}$$

Crossing and collinear cross-checks. The result Eq. (60.215) should be totally symmetric under permutations of the four identical positive-helicity gluons (after accounting for color ordering, the surviving symmetry is cyclic plus reflection). Momentum conservation makes this manifest: from $\langle 12\rangle[24] + \langle 13\rangle[34] = 0$ and its relatives one shows

$$\frac{[12][34]}{\langle 12\rangle\langle 34\rangle} = \frac{[13][24]}{\langle 13\rangle\langle 24\rangle} = \frac{[14][23]}{\langle 14\rangle\langle 23\rangle}, \tag{60.216}$$

so the rational function is independent of how the pairs are chosen—the amplitude has the required exchange symmetry. For the collinear limit, send $1 \parallel 2$ ($\langle 12\rangle \to 0$): Eq. (60.215) diverges as $[12]/\langle 12\rangle$, the correct one-loop all-plus collinear singularity, and matches $\mathrm{Split}_{-}^{(1)}(1^+,2^+)\,A_3^{\mathrm{tree}}$ plus $\mathrm{Split}^{\mathrm{tree}}\,A_3^{(1)}(P^+,3^+,4^+)$ in Eq. (60.207), since the tree all-plus three-point amplitude vanishes and only the one-loop splitting and the one-loop three-point all-plus contribute. This limit is what fixes $n_s = 2$ and confirms the normalization independently of the infrared check used for the MHV case.

60.6.4 Feynman diagrams versus on-shell trees

It is worth making the "modern alternative" claim quantitative, because the contrast is the entire point of the method. Consider the same object computed two ways.

The traditional route is the one the reader learned in Part V and exercised on QED in Chapter 19: enumerate every one-loop Feynman diagram, write each as a tensor integral, Passarino–Veltman reduce the tensors to scalar masters (Chapter D), and add. For the color-ordered one-loop four-gluon amplitude the diagram count is already substantial. The cubic and quartic gluon self-interactions of Yang–Mills theory, together with the Faddeev–Popov ghost loop required for gauge invariance (Chapter 26), generate on the order of one hundred one-loop diagrams once all topologies (box, triangle with a quartic vertex, bubbles, tadpole-like and ghost contributions) and their orderings are included—against the four diagrams of the *tree* amplitude $gg \to gg$. Each one-loop diagram produces a tensor integral of rank up to four, whose reduction proliferates Gram determinants and gauge-dependent intermediate terms that cancel only in the gauge-invariant sum.

The unitarity route, by contrast, never builds a single loop diagram. The complete input is the set of on-shell trees:

- for the MHV box, the box coefficient Eq. (60.202) was fixed by 2 solutions $\times 4$ corners $= 8$ three-point tree evaluations, each a one-line Parke–Taylor expression (Eq. (59.55)), and the $\mathcal{N} = 4$ no-triangle property removed every other coefficient;
- for the all-plus rational term, a single μ^4-box coefficient Eq. (60.209) built from two helicity-violating two-gluon vertices.

The remaining work is to multiply by master integrals (Eqs. (60.204) and (60.213)) that are computed once and tabulated in Chapter D. A computation that classically demands one hundred diagrams and a tensor-reduction apparatus collapses to a handful of gauge-invariant tree evaluations and two master integrals. The advantage compounds with multiplicity and loop order: the number of Feynman diagrams grows factorially, while the number of independent unitarity cuts grows only with the size of the integral basis. This is the concrete sense in which generalized unitarity is the modern successor to the diagrammatic expansion.

Summary and outlook

We have reconstructed two one-loop four-gluon amplitudes end to end from on-shell trees: the $\mathcal{N} = 4$ MHV amplitude Eq. (60.205), a single scalar box whose coefficient $st\,A_4^{(0)}$ came from a quadruple cut, and the all-plus rational amplitude

Eq. (60.215), a single μ^4-box invisible to four-dimensional cuts. Both passed their physical checks: the MHV poles reproduce the universal infrared structure Eq. (60.206) and the correct collinear factorization, while the all-plus result has the required crossing symmetry and collinear singularity. In every case the answer was delivered as a coefficient times a master integral, with not a single Feynman diagram drawn—the loop-level fulfillment of Yang–Mills theory. What we have *not* done is evaluate those master integrals from first principles; we imported Eq. (60.204) and Eq. (60.213) from Chapter D. The next section takes up exactly that question—how the master integrals are computed, by integration-by-parts reduction to a finite basis and the canonical differential-equation method—and asks how the whole program scales beyond one loop into the polylogarithmic function space that is the modern frontier.

60.7 Toward Multiloop: Integration by Parts, Differential Equations, and the Function Space

The reconstruction of the previous section delivered the one-loop gluon amplitude not as a number but as a *linear combination of scalar master integrals*: a box, a handful of triangles and bubbles, each multiplied by a coefficient that generalized unitarity had pinned down as a product of on-shell trees. To turn that combination into a function of the kinematic invariants we did the one thing the unitarity method by design does not do for us — we reached into Chapter D and quoted the explicit values of those masters. That import was honest but it was also a deferral. Two questions were left standing. First, *how are the master integrals themselves computed*, once one wants them as functions of the invariants rather than as entries in a table? Second, *what happens beyond one loop*, where Chapter D's closed-form Gamma-function answers run out and the integrals are no longer elementary? This section answers both, and in doing so it converts the unitarity method from a one-loop technique into the entryway of the modern multiloop program. The answer to the first question is a remarkable piece of linear algebra — integration by parts — that collapses the apparently infinite zoo of loop integrals onto a finite basis and then propagates that basis through a closed system of differential equations. The answer to the second is that the functions one meets are not arbitrary: at the polylogarithmic frontier they are iterated integrals of a sharply characterized kind, graded by an integer called the transcendental weight and fingerprinted by an algebraic object called the symbol. We build all three ideas in turn — the integration-by-parts reduction, the canonical differential equation, the polylogarithmic function space — carry every step of two small but complete computations, and close with an honest, dated map of where the road runs out.

60.7.1 Loop integrals as a vector space: integration-by-parts identities

The conceptual leap that organizes everything below is to stop thinking of a loop integral as a single number to be evaluated and to start thinking of the entire *family* of integrals sharing a given set of propagators as the elements of a vector space. The relations among them are then linear, and linear-algebra technology applies.

Definition 60.30 (Integral family)**.** Fix a set of L loop momenta $\ell_1, \dots, \ell_L$, external momenta $p_1, \dots, p_n$, and a list of N inverse propagators (denominators) $D_1, \dots, D_N$, each a quadratic form $D_j = q_j^2 - m_j^2$ in the loop and external momenta as in Theorem D.1, chosen so that every scalar product $\ell_a \cdot \ell_b$ and $\ell_a \cdot p_i$ that can appear is a linear combination of the D_j (the family is then said to be *complete*). The *integral family* is the set of all integrals

$$I(a_1, \dots, a_N) \;=\; \int \prod_{i=1}^{L} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d} \, \frac{1}{D_1^{a_1} \, D_2^{a_2} \cdots D_N^{a_N}} \,, \qquad a_j \in \mathbb{Z}, \tag{60.217}$$

labelled by the integer vector $(a_1, \dots, a_N)$. Positive a_j are propagator powers; a vanishing a_j removes a line (a subtopology); a negative a_j is an inverse propagator in the numerator, the mechanism by which numerator scalar products are encoded.

The family (60.217) is countably infinite: the indices a_j range over all integers. The central structural fact of multiloop computation, which this section establishes, is that this infinite set spans a *finite-dimensional* space. The relations that force the collapse come from a single, almost embarrassingly elementary observation: in dimensional regularization the integral of a total derivative vanishes.

Lemma 60.31 (Vanishing of total derivatives in dimensional regularization)**.** *For any of the loop momenta ℓ_a and any vector v^μ built polynomially from the loop and external momenta, and for any integrand f in the family* (60.217)*,*

$$\int \prod_{i} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d} \, \frac{\partial}{\partial \ell_a^\mu} \Big(v^\mu \, f \Big) \;=\; 0 \,. \tag{60.218}$$

Proof. The integral of a total derivative over $\mathrm{d}^d \ell_a$ equals the boundary value of $v^\mu f$ on the sphere at $|\ell_a| \to \infty$, by the divergence theorem in d dimensions. In an integer dimension this surface term need not vanish for a divergent integrand. But dimensional regularization is defined (Theorem D.22) by analytic continuation from the strip of complex d in which the integral converges absolutely, and *in that strip* the integrand falls faster than any surface area can grow, so the boundary contribution is identically zero there. As the left-hand side of (60.218) is, like

every member of the family, a meromorphic function of d defined by that same continuation, an identity holding on an open strip holds everywhere it is defined. The vanishing of the surface term is thus not an approximation; it is built into the definition of the regularized integral. This is the single most important practical advantage of dimensional regularization over a hard cutoff, where (60.218) would be false. □

Carrying out the differentiation in (60.218) with the product rule generates a linear relation among members of the family. The derivative $\partial_{\ell_a^\mu} v^\mu$ produces a multiple of the dimension d (when $v^\mu \propto \ell_a^\mu$, since $\partial \ell_a^\mu / \partial \ell_a^\mu = d$) plus polynomial terms, while $\partial_{\ell_a^\mu}$ acting on each $D_j^{-a_j}$ lowers that power by one and brings down $-a_j\, \partial_{\ell_a^\mu} D_j$, whose numerator scalar products are — by completeness of the family — re-expressed in the D_k, shifting indices up and down. The result is the *integration-by-parts (IBP) identity*: a homogeneous linear relation, with coefficients rational in d and the invariants, among integrals (60.217) whose indices differ from a chosen "seed" by small shifts. Each choice of seed $(a_1, \dots, a_N)$ and each choice of vector $v^\mu \in \{\ell_1, \dots, \ell_L, p_1, \dots, p_n\}$ gives one identity; the supply is inexhaustible.

We also need the second pillar, the systematic source of the zeros that terminate the recursions.

Lemma 60.32 (Scaleless integrals vanish)**.** *A dimensionally regularized integral whose integrand, after every shift of loop momentum, depends on the loop momenta but contains* no dimensionful scale *(no mass, no nonzero external invariant) is zero.*

Proof. Let J be such an integral and rescale every loop momentum $\ell_i \to t\,\ell_i$ with $t > 0$. Because the integrand carries no other scale, dimensional analysis forces $J = t^{\,k} J$ for some power $k = k(d)$ depending on the homogeneity degree (the net mass dimension carried by the loop momenta). An equality $J = t^k J$ holding for all $t > 0$ with $t^k \not\equiv 1$ forces $J = 0$. Generically $k(d) \neq 0$ for non-integer d, so $J = 0$; by continuation it vanishes at $d = 4$ as well. Concretely, the massless tadpole $\int \mathrm{d}^d\ell / (\ell^2)^a = 0$ and, after a shift, $\int \mathrm{d}^d\ell / ((\ell + p)^2)^a = 0$. □

These two lemmas are the entire toolkit. We now run them on the smallest nontrivial example, the massless one-loop bubble, and extract a concrete identity exhibiting the characteristic d-dependence of an IBP relation.

Worked Example 60.33 (The massless bubble IBP and its d-dependence)**.** Take the one-loop family with two massless denominators

$$D_1 = \ell^2 + i\epsilon, \qquad D_2 = (\ell + p)^2 + i\epsilon, \qquad s \equiv p^2, \tag{60.219}$$

and integrals $I(a_1, a_2) = \int \mathrm{d}^d\ell / (2\pi)^d\, D_1^{-a_1} D_2^{-a_2}$. We derive the IBP identity generated by the seed $(1, 1)$ and the vector $v^\mu = \ell^\mu$.

Step 1: the total derivative. By Theorem 60.31,

$$0 \;=\; \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\, \frac{\partial}{\partial \ell^\mu}\left[\frac{\ell^\mu}{D_1\, D_2}\right]. \tag{60.220}$$

Step 2: the product rule. The derivative hits the explicit ℓ^μ and each denominator:

$$\frac{\partial}{\partial \ell^\mu}\,\frac{\ell^\mu}{D_1 D_2} = \frac{d}{D_1 D_2} - \frac{\ell^\mu\, \partial_\mu D_1}{D_1^2 D_2} - \frac{\ell^\mu\, \partial_\mu D_2}{D_1 D_2^2}, \tag{60.221}$$

using $\partial \ell^\mu / \partial \ell^\mu = d$. The two contractions are elementary:

$$\ell^\mu \partial_\mu D_1 = \ell^\mu (2\ell_\mu) = 2\ell^2 = 2D_1, \tag{60.222}$$

$$\ell^\mu \partial_\mu D_2 = \ell^\mu\, 2(\ell + p)_\mu = 2\ell^2 + 2\ell\cdot p = 2D_1 + (D_2 - D_1 - s) = D_1 + D_2 - s, \tag{60.223}$$

where in the second line we used $D_2 = (\ell+p)^2 = \ell^2 + 2\ell p + p^2$, i.e. $2\ell p = D_2 - D_1 - s$.

Step 3: assemble the identity. Insert (60.223) into (60.221) and integrate:

$$\begin{aligned} 0 &= \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\left[\frac{d}{D_1 D_2} - \frac{2D_1}{D_1^2 D_2} - \frac{D_1 + D_2 - s}{D_1 D_2^2}\right] \\ &= d\, I(1,1) - 2\, I(1,1) - \big[I(0,2) + I(1,1) - s\, I(1,2)\big] \\ &= (d-3)\, I(1,1) - I(0,2) + s\, I(1,2), \end{aligned} \tag{60.224}$$

where $D_1/(D_1^2 D_2) = 1/(D_1 D_2)$, $D_1/(D_1 D_2^2) = 1/(D_2^2) \to I(0,2)$, and $D_2/(D_1 D_2^2) = 1/(D_1 D_2) \to I(1,1)$.

Step 4: kill the scaleless term and solve. The integral $I(0,2) = \int \mathrm{d}^d \ell/(2\pi)^d\, (\ell + p)^{-4}$ depends on no scale once ℓ is shifted, so $I(0,2) = 0$ by Theorem 60.32. The remaining identity rearranges to

$$\boxed{\; I(1,2) \;=\; -\,\frac{d-3}{s}\, I(1,1) \;} \tag{60.225}$$

Equation (60.225) is the prototype of the whole reduction program, and three of its features are permanent. First, an integral with a *raised* propagator power, $I(1,2)$, has been expressed entirely in terms of the integral with all powers equal to one, $I(1,1)$: the family has been collapsed onto a single representative. Iterating the construction with other seeds reduces every $I(a_1, a_2)$ with $a_j > 0$ to a rational multiple of $I(1,1)$, while every integral with a vanishing index is scaleless and dies. The bubble family, infinite as a set, is one-dimensional as a vector space. Second, the coefficient $-(d-3)/s$ depends on the spacetime dimension — it is the algebraic shadow of the analytic continuation in d that defined the integral. This d-dependence is universal to IBP relations and is exactly what the differential-equation method below will exploit. Third, the entire identity was obtained without performing a single momentum integral: it is a statement about the integrand, valid before evaluation, and the one surviving integral $I(1,1)$ is the one we still must compute — by the differential equation, in a moment.

60.7.2 The Laporta algorithm and the master basis

The bubble is small enough that one identity sufficed. In a realistic problem one generates IBP identities by the thousands — one for each (seed, vector) pair, over a large block of seeds — and confronts a vast, sparse, but *finite-rank* linear system whose unknowns are the integrals (60.217). The strategy for solving it is due to Laporta [315, 316]: impose a total ordering on the integrals (by number of propagators, then by total power, then lexicographically — so that "simpler" integrals are preferred), generate identities for all seeds up to a chosen complexity, and perform Gaussian elimination, always solving each identity for its most complicated integral in terms of simpler ones. The elimination cascades: complicated integrals are successively expressed through simpler ones until only a small set remains that no identity can reduce further.

Definition 60.34 (Master integrals)**.** The integrals that survive the IBP reduction — those that cannot be written as a linear combination of simpler integrals in the family — are the *master integrals* of the family. Every integral (60.217) equals a finite linear combination

$$I(a_1, \dots, a_N) \;=\; \sum_{k} c_k(d, s_{ij})\, M_k \,, \tag{60.226}$$

of the masters M_k, with coefficients c_k rational in the dimension d and the kinematic invariants s_{ij}.

That the number of masters is *finite* is not obvious from the algorithm, which terminates only because we capped the seed complexity; in principle a larger seed block might expose new reductions. The finiteness is a theorem.

Theorem 60.35 (Finiteness of the master basis)**.** *For any integral family* (60.217) *the number of master integrals is finite.*

The statement is that the family, viewed as a module over the ring of IBP shift operators, is finitely generated; equivalently, the dimension of the vector space spanned by the integrals (as functions of the kinematics, for generic d) is bounded. The bound has a sharp geometric meaning: the number of masters equals the number of critical points of the Lee–Pomeransky polynomial of the graph (the Euler characteristic of a hypersurface complement), a result that converts "how many masters?" into a question in algebraic geometry. We use the finiteness as established; its proof belongs to the mathematical apparatus of Feynman integrals catalogued in Chapter D and reviewed in [448]. For us the operational content is decisive: *any* amplitude at any loop order, once its diagrams are written and IBP-reduced, is a finite linear combination of a fixed, computable list of master integrals. The reconstruction of the previous section produced precisely such a list at one loop — box, triangles, bubbles — and Theorem 60.35 is the guarantee that the same finiteness holds at two loops, three loops, and beyond,

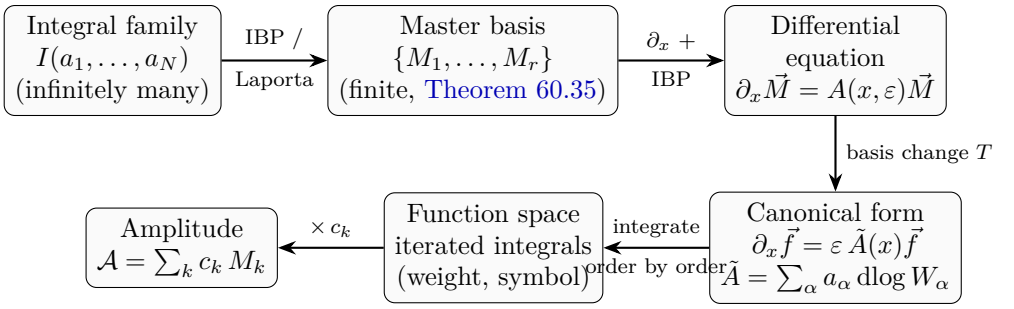


Figure 60.6: The multiloop computational pipeline. Generalized unitarity and IBP reduction (Sections 60.7.1 and 60.7.2) express any amplitude through a finite basis of master integrals; differentiating the masters in a kinematic variable and IBP-reducing the result closes a linear differential system (Section 60.7.3); a change of basis brings it to the canonical ε-factorized form (Section 60.7.4), whose solution is, order by order in ε, an iterated integral in the polylogarithmic function space (Section 60.7.5). Each stage is linear algebra over rational functions; the only analytic input is a boundary condition.

even when the masters can no longer be written in closed form. The amplitude problem has been cleanly factorized: rational coefficients (which unitarity and IBP supply) times master integrals (which the next subsections compute).

60.7.3 The method of differential equations

Knowing that a finite basis of masters exists is one thing; evaluating the masters is another. The decisive idea, due to Kotikov and developed into a general method by Remiddi and Gehrmann [212, 306], is to compute the masters not by direct integration but by differentiating them. A master integral is a function of the kinematic invariants; differentiate it with respect to one of them and the derivative is again an integral in the same family (the derivative acts on the propagators, raising powers and inserting numerators, exactly as in the IBP manipulation). By Theorem 60.34 that derivative reduces back to the masters. The masters therefore satisfy a closed, first-order, linear system of ordinary differential equations.

Proposition 60.36 (The differential-equation system)**.** *Let $\vec{M} = (M_1,\dots,M_r)^{\mathsf{T}}$ be the master integrals of a family, regarded as functions of a kinematic variable x (a ratio of invariants), and let the remaining invariants be held fixed or scaled out. Then there is a matrix $A(x,\varepsilon)$, with entries rational in x and ε, such that*

$$\frac{\partial}{\partial x}\,\vec{M}(x,\varepsilon) \;=\; A(x,\varepsilon)\,\vec{M}(x,\varepsilon). \tag{60.227}$$

Proof. Differentiating M_k with respect to x commutes with the loop integration. The chain rule turns ∂_x into a combination of $\partial/\partial p_i^\mu$ contracted with external

momenta, which acting on the propagators $D_j^{-a_j}$ produces a finite sum of integrals in the same family with shifted indices (raised powers from the differentiated propagators, dotted external momenta in numerators). Each such integral is, by Theorem 60.34 and the IBP reduction (60.226), a rational-coefficient combination of the masters $M_1, \dots, M_r$. Collecting the coefficients into the k-th row of a matrix $A(x, \varepsilon)$ gives (60.227). □

The system (60.227) reduces the evaluation of the masters to two well-posed sub-problems: solving a linear ODE, and fixing the constants of integration by a boundary condition (typically the value of the integral at a special kinematic point — a threshold, a soft or collinear limit, or a symmetric point — where it degenerates to something elementary or already known). The boundary data are the *only* analytic input; everything else is the algebra of (60.227). This is why the method scales: at higher loops one cannot do the momentum integral, but one can always differentiate and IBP-reduce, and one can usually find a kinematic corner where a boundary value is computable.

We now carry the method through completely on the one master we already isolated, the massless bubble, where the system is a single equation and the solution is a logarithm.

Worked Example 60.37 (The massless bubble by differential equation)**.** We compute $I(1,1)$ of Theorem 60.33 from its differential equation in $s = p^2$.

Step 1: differentiate. Since $s = p^2$ and $I(1,1)$ depends on the externals only through s, we use $p^\mu\, \partial/\partial p^\mu = 2s\, \partial/\partial s$, so $\partial/\partial s = (1/2s)\, p^\mu \partial/\partial p^\mu$. The derivative acts on $D_2 = (\ell + p)^2$ only:

$$p^\mu \frac{\partial}{\partial p^\mu} I(1,1) = \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{-\,p^\mu\, 2(\ell+p)_\mu}{D_1\, D_2^2} = -\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{2\, p\cdot(\ell+p)}{D_1 D_2^2}. \tag{60.228}$$

Step 2: express the numerator in denominators. Write $2\, p\cdot(\ell + p) = 2\ell\cdot p + 2p^2 = (D_2 - D_1 - s) + 2s = D_2 - D_1 + s$. Then

$$p^\mu \partial_\mu I(1,1) = -\big[I(1,1) - I(0,2) + s\, I(1,2)\big] = -I(1,1) - s\, I(1,2), \tag{60.229}$$

using $D_2/(D_1 D_2^2) = 1/(D_1 D_2) \to I(1,1)$, $D_1/(D_1 D_2^2) = 1/(D_2^2) \to I(0,2) = 0$ (scaleless), and the leftover $s/(D_1 D_2^2) \to s\, I(1,2)$.

Step 3: IBP-reduce back to the master. The integral $I(1,2)$ is not a master; the reduction (60.225) of Theorem 60.33 gives $s\, I(1,2) = -(d-3)\, I(1,1)$. Insert it:

$$p^\mu \partial_\mu I(1,1) = -I(1,1) + (d-3)\, I(1,1) = (d-4)\, I(1,1) = -2\varepsilon\, I(1,1), \tag{60.230}$$

where $d = 4 - 2\varepsilon$. The system has closed on the single master, as Theorem 60.36 promised.

Step 4: the differential equation. Dividing by $2s$,

$$\frac{\partial}{\partial s} I(1,1) \;=\; -\frac{\varepsilon}{s}\, I(1,1). \tag{60.231}$$

Step 5: solve. This is separable: $\mathrm{d}\ln I(1,1) = -\varepsilon\, \mathrm{d}\ln(-s)$, integrating to

$$I(1,1) \;=\; c(\varepsilon)\,(-s)^{-\varepsilon}, \tag{60.232}$$

with $(-s - i\epsilon)$ the correct branch inherited from the Feynman prescription, and $c(\varepsilon)$ an s-independent constant of integration.

Step 6: fix the boundary constant. The constant $c(\varepsilon)$ is the one piece the differential equation cannot supply; it is fixed by the value of the integral at one kinematic point. Here the most economical anchor is the closed form of the master itself. Combining the two denominators with the Feynman parameter $1/(D_1 D_2) = \int_0^1 \mathrm{d}x\,[xD_2 + (1-x)D_1]^{-2}$ and completing the square gives $\Delta = -x(1-x)s$, so by the master formula Eq. (D.77) with $n = 2$, $\Delta^{d/2-2} = \Delta^{-\varepsilon}$,

$$I(1,1) = \frac{i}{(4\pi)^{d/2}}\,\Gamma(\varepsilon)\,(-s)^{-\varepsilon}\int_0^1 \mathrm{d}x\,[x(1-x)]^{-\varepsilon} = \frac{i}{(4\pi)^{d/2}}\,\Gamma(\varepsilon)\,(-s)^{-\varepsilon}\,\frac{\Gamma(1-\varepsilon)^2}{\Gamma(2-2\varepsilon)}, \tag{60.233}$$

the x-integral being the Beta function $\mathrm{B}(1-\varepsilon, 1-\varepsilon)$. Matching (60.233) to (60.232) reads off

$$c(\varepsilon) = \frac{i}{(4\pi)^{d/2}}\,\frac{\Gamma(\varepsilon)\,\Gamma(1-\varepsilon)^2}{\Gamma(2-2\varepsilon)}. \tag{60.234}$$

The differential equation determined the entire s-dependence; the boundary supplied the single number multiplying it.

60.7.4 Canonical bases and the ε-factorized form

The bubble equation (60.231) has a feature that looks accidental but is in fact the goal of the entire method: its right-hand side is *proportional to* ε. The matrix $A(x,\varepsilon) = -\varepsilon/s$ is ε times a function of s alone, with no ε-independent part. When a system has this form the solution can be read off order by order in ε as a tower of iterated integrals, with no further integration constants beyond the boundary. Henn's insight [263] was that this is not special to the bubble: for a large class of integrals one can *choose the master basis* so that the system takes this form.

Definition 60.38 (Canonical, ε-form differential equation)**.** A basis of masters $\vec{f} = (f_1, \dots, f_r)^{\mathsf{T}}$ is *canonical* (in ε-form, or *pure*) if its differential equation factorizes the regulator,

$$\mathrm{d}\,\vec{f}(x,\varepsilon) \;=\; \varepsilon \Big(\sum_{\alpha} a_\alpha\, \mathrm{dlog}\, W_\alpha(x)\Big)\, \vec{f}(x,\varepsilon), \tag{60.235}$$

where d is the total differential in all kinematic variables, the a_α are *constant* matrices (independent of x and ε), and the $W_\alpha(x)$ are algebraic functions of the kinematics called the *letters* of the problem. Their collection $\{W_\alpha\}$ is the *alphabet*. The right-hand side is a matrix-valued logarithmic one-form, $\mathrm{d}\tilde{A} = \sum_\alpha a_\alpha \,\mathrm{dlog}\, W_\alpha$, with only simple poles (located at $W_\alpha = 0, \infty$).

The transformation from a generic basis $\vec{M}$ obeying (60.227) to a canonical $\vec{f} = T(x,\varepsilon)\,\vec{M}$ is a linear change of basis; under it the matrix transforms as $A \to TAT^{-1} + (\partial_x T)\,T^{-1}$, and finding the T that produces the form (60.235) is the technical heart of a modern computation. A canonical basis is typically built from integrals that are *pure functions* of uniform transcendental weight — normalized, for instance, by factors of ε and by leading-singularity (maximal-cut) data so that they have unit "leading singularity" and no spurious algebraic prefactors. The massless bubble is already pure up to its overall constant: (60.231) is (60.235) with one letter $W = (-s)$, one 1×1 matrix $a = -1$, and $\mathrm{dlog}(-s) = \mathrm{d}s/s$.

The payoff of the canonical form is the structure of its solution.

Proposition 60.39 (ε-form solution as iterated integrals)**.** *The solution of a canonical system* (60.235) *along a path γ from a base point x_0 to x is the path-ordered exponential*

$$\vec{f}(x,\varepsilon) = \mathrm{P}\exp\!\Big(\varepsilon \int_\gamma \mathrm{d}\tilde{A}\Big)\,\vec{f}(x_0,\varepsilon), \tag{60.236}$$

whose expansion in ε is, term by term, a $\mathbb{Z}$-linear combination of iterated integrals *of the one-forms* $\mathrm{dlog}\, W_\alpha$*:*

$$\vec{f} = \Big(1 + \varepsilon\int_\gamma \mathrm{d}\tilde{A} + \varepsilon^2 \int_\gamma \mathrm{d}\tilde{A}\,\mathrm{d}\tilde{A} + \cdots\Big)\vec{f}(x_0,\varepsilon), \qquad \int_\gamma \mathrm{d}\tilde{A}\cdots\mathrm{d}\tilde{A} = \int_{0\le t_1\le\cdots\le t_k\le 1} \tilde{A}'(t_1)\cdots\tilde{A}'(t_k), \tag{60.237}$$

where $\tilde{A}'$ denotes the pullback of $\mathrm{d}\tilde{A}$ to the path parameter. The coefficient of ε^k in $\vec{f}$ is an iterated integral of exactly *k logarithmic forms.*

Proof. Write $\vec{f} = \sum_{k\ge 0} \varepsilon^k \vec{f}^{(k)}$ and insert into (60.235), which reads $\mathrm{d}\vec{f} = \varepsilon\,(\mathrm{d}\tilde{A})\,\vec{f}$. Matching powers of ε gives $\mathrm{d}\vec{f}^{(0)} = 0$ and, for $k \ge 1$,

$$\mathrm{d}\vec{f}^{(k)} = (\mathrm{d}\tilde{A})\,\vec{f}^{(k-1)}. \tag{60.238}$$

The zeroth order $\vec{f}^{(0)}$ is therefore a constant (the boundary value, weight 0), and each higher order is obtained from the one below it by a *single* integration of a dlog form: $\vec{f}^{(k)}(x) = \vec{f}^{(k)}(x_0) + \int_{x_0}^{x} (\mathrm{d}\tilde{A})\,\vec{f}^{(k-1)}$. Unwinding the recursion expresses $\vec{f}^{(k)}$ as the k-fold iterated integral in (60.237); the path-ordering is automatic because each integration's upper limit is the next integration's variable. The only freedom at each order is the additive constant $\vec{f}^{(k)}(x_0)$, fixed by the

boundary condition. This is the precise sense in which the ε-factorized form "solves itself": integration is replaced by a deterministic recursion that adds one logarithmic form per order in ε. □

Theorem 60.39 makes the bubble transparent. With the single letter $(-s)$, the path-ordered exponential is the ordinary exponential of $-\varepsilon \ln(-s/(-s_0))$, and expanding (60.232) confirms the recursion directly:

$$(-s)^{-\varepsilon} = e^{-\varepsilon \ln(-s)} = 1 - \varepsilon \ln(-s) + \frac{\varepsilon^2}{2} \ln^2(-s) - \frac{\varepsilon^3}{6} \ln^3(-s) + \cdots , \quad (60.239)$$

each order carrying exactly one more power of the logarithm — one more iterated integral of the form $\mathrm{d} \ln(-s)$. Restoring the constant (60.234) and the dimensional scale $\mu^{2\varepsilon}$, the bubble's Laurent expansion is

$$\mu^{2\varepsilon} I(1,1) = \frac{i}{(4\pi)^2} \Big[\frac{1}{\varepsilon} - \ln \frac{-s}{\mu^2} + 2 + O(\varepsilon) \Big], \quad (60.240)$$

the $1/\varepsilon$ pole from $\Gamma(\varepsilon)$, the logarithm from the first iterated integral, and the rational "+2" from the ε-expansion of the constant $\Gamma(1-\varepsilon)^2/\Gamma(2-2\varepsilon)$ together with $\overline{\mathrm{MS}}$ bookkeeping. The branch cut along $s > 0$ in $\ln(-s)$ is the bubble's two-particle threshold; that its discontinuity is $2i \operatorname{Im} I(1,1)$ is the optical theorem of Section 5.5.5 made manifest, and is precisely the cut that the unitarity method of this chapter exploited to fix the coefficient multiplying this very master.

60.7.5 The polylogarithmic function space, transcendental weight, and the symbol

Theorem 60.39 tells us *what* the master integrals are: iterated integrals of logarithmic forms. We now name this function space precisely, equip it with its grading, and introduce the algebraic tool — the symbol — that tames the otherwise unwieldy identities among its elements.

Definition 60.40 (Iterated integral and multiple polylogarithm)**.** Given one-forms $\omega_1, \dots, \omega_n$ on a path $\gamma : [0,1] \to \mathbb{C}$, the *iterated integral* is

$$\int_\gamma \omega_1 \circ \cdots \circ \omega_n = \int_{0 \le t_1 \le \cdots \le t_n \le 1} \omega_1(t_1)\, \omega_2(t_2) \cdots \omega_n(t_n). \quad (60.241)$$

When the forms are $\omega_i = \mathrm{dlog}(t - a_i) = \mathrm{d}t/(t-a_i)$ with constants a_i, the integral defines the *multiple polylogarithm* (MPL, or Goncharov polylogarithm)

$$G(a_1, \dots, a_n; z) = \int_0^z \frac{\mathrm{d}t_1}{t_1 - a_1} \int_0^{t_1} \frac{\mathrm{d}t_2}{t_2 - a_2} \cdots \int_0^{t_{n-1}} \frac{\mathrm{d}t_n}{t_n - a_n}, \quad (60.242)$$

with the standard regularization of the endpoint divergences when some a_i coincide with 0 or z. The integer n — the number of integrations, equivalently the number of dlog forms — is the *transcendental weight* of the function.

The classical functions are special cases: $G(0;z)=\ln z$ has weight 1; the polylogarithm $\mathrm{Li}_n(z)=-G(0,\dots,0,1;z)$ (with $n-1$ zeros) has weight n, recovering $\mathrm{Li}_1(z)=-\ln(1-z)$ and the dilogarithm $\mathrm{Li}_2(z)=-\int_0^z \mathrm{dlog}(1-t)\,\mathrm{dlog}\,t$; and the transcendental constants inherit weights from their values, π being weight 1 (so π^2 and $\zeta(2)=\pi^2/6$ are weight 2) and $\zeta(n)$ weight n. A rational or algebraic number has weight 0.

Definition 60.41 (Uniform and maximal transcendentality)**.** A function is of *uniform (transcendental) weight* w if every term in it has weight w. A master integral expanded in ε is *pure of uniform weight* if the coefficient of ε^k has uniform weight $k+w_0$ for a fixed w_0 (the weight of the leading term). A canonical basis (60.235) produces exactly such integrals: by Theorem 60.39 the ε^k coefficient is a k-fold iterated integral, hence weight k above the boundary weight. An L-loop integral that achieves the maximal possible weight $2L$ in its finite part is said to have *maximal transcendentality*.

The grading by weight is not bookkeeping for its own sake. It is conjectured (and verified in vast swaths of cases) that amplitudes in maximally supersymmetric Yang–Mills theory are uniformly of maximal weight $2L$ at L loops — the "maximal transcendentality principle" — and that the leading transcendental part of QCD amplitudes coincides with the super-Yang–Mills answer. Weight is thus a physical organizing principle, and the canonical-basis method is valuable precisely because it manufactures pure, uniform-weight building blocks for which the principle is automatic.

The functions (60.242) satisfy a thicket of identities (shuffle relations, functional equations like the five-term dilogarithm identity), which makes deciding whether two expressions are equal genuinely hard. The tool that linearizes this problem is the symbol.

Definition 60.42 (The symbol)**.** The *symbol* $\mathcal{S}(F)$ of a weight-n function F is an element of the n-fold tensor product of the (multiplicative) group of nonzero algebraic functions, defined recursively from the total differential. If

$$\mathrm{d}F=\sum_i F_i\,\mathrm{dlog}\,R_i, \tag{60.243}$$

with each F_i of weight $n-1$ and R_i a letter, then

$$\mathcal{S}(F)=\sum_i \mathcal{S}(F_i)\otimes R_i, \qquad \mathcal{S}(\ln R)=R, \qquad \mathcal{S}(\text{const})=0. \tag{60.244}$$

The tensor factors are read left-to-right as the order of integration. The symbol obeys $\mathcal{S}(F\,G)=\mathcal{S}(F)\sqcup\!\sqcup\mathcal{S}(G)$ (shuffle product) and the entry rules $\cdots\otimes(R\,R')\otimes\cdots=\cdots\otimes R\otimes\cdots+\cdots\otimes R'\otimes\cdots$ and $\cdots\otimes(cR)\otimes\cdots=\cdots\otimes R\otimes\cdots$ for constant c, the last because constants carry no symbol.

The power of the construction is that all the transcendental identities among MPLs become *linear-algebraic* identities among tensors: two polylogarithmic functions agree, up to terms of lower weight times transcendental constants (which the symbol drops), if and only if their symbols agree. Two short computations show the recursion at work.

Worked Example 60.43 (Symbols of the logarithm-squared and the dilogarithm)**. Weight-2 product.** For $F = \ln a \, \ln b$, the differential is $\mathrm{d}F = \ln b \, \mathrm{dlog}\, a + \ln a \, \mathrm{dlog}\, b$, so by (60.244)

$$\mathcal{S}(\ln a \, \ln b) = \mathcal{S}(\ln b) \otimes a + \mathcal{S}(\ln a) \otimes b = b \otimes a + a \otimes b, \tag{60.245}$$

which is the shuffle $a \sqcup\!\sqcup b$, confirming $\mathcal{S}(\ln a \ln b) = \mathcal{S}(\ln a) \sqcup\!\sqcup \mathcal{S}(\ln b)$. In particular $\mathcal{S}(\frac{1}{2}\ln^2 a) = a \otimes a$.

Dilogarithm. For $F = \mathrm{Li}_2(z)$, the defining derivative is $\mathrm{d}\,\mathrm{Li}_2(z) = -\ln(1-z)\,\mathrm{dlog}\, z$, so $F_1 = -\ln(1-z)$, $R_1 = z$, and

$$\mathcal{S}(\mathrm{Li}_2(z)) = \mathcal{S}(-\ln(1-z)) \otimes z = -(1-z) \otimes z. \tag{60.246}$$

The five-term relation $\mathrm{Li}_2(x) + \mathrm{Li}_2(y) + \mathrm{Li}_2(\frac{1-x}{1-xy}) + \mathrm{Li}_2(1-xy) + \mathrm{Li}_2(\frac{1-y}{1-xy}) = -\ln x \ln(1-x) - \cdots$ — a notorious identity to verify by hand — collapses, under (60.246) and the entry rules, to the vanishing of a sum of tensors, a finite linear-algebra check.

The one-loop masters of the previous section's gluon amplitude live entirely in this world. The massless box $I_4(s,t)$ with all external legs lightlike — the integral that carried the leading singularity of the reconstructed amplitude — evaluates, with the canonical normalization $\hat{f} = \varepsilon^2\, st\, c_\Gamma^{-1} I_4$ that makes it pure, to a uniform weight-2 function in the two-letter alphabet $\{-s, -t\}$,

$$\varepsilon^2\, st\, c_\Gamma^{-1}\, I_4(s,t) = 2\big[(-s)^{-\varepsilon} + (-t)^{-\varepsilon}\big] - \varepsilon^2\Big[\ln^2\tfrac{-s}{-t} + \pi^2\Big] + O(\varepsilon^3), \tag{60.247}$$

with $c_\Gamma = \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)^2/\Gamma(1-2\varepsilon)$. Its finite ($\varepsilon^0$) part is the weight-2 combination whose symbol, by (60.245), is built from the single nontrivial entry pattern in the letters $-s, -t$ — explicitly the symbol of $-\frac{1}{2}\ln^2(-s/-t)$ is $-(-s/-t) \otimes (-s/-t)$, while the π^2 is a weight-2 constant invisible to the symbol. The box's branch cuts in s and t, encoded by the letters, are exactly the channels whose unitarity cuts the reconstruction sewed together: the function space and the cut structure are two views of the same object. This is the precise sense in which a loop amplitude is rigid — its singularities (the letters) are dictated by unitarity and locality, and the symbol exposes them as the alphabet of an iterated integral.

60.7.6 The frontier: numerical unitarity, functions beyond polylogarithms, and the optimal basis

The three tools just built — IBP reduction, canonical differential equations, the polylogarithmic function space — are the working infrastructure of multiloop amplitude computation in the mid-2020s. Their reach, and its limits, deserve an honest accounting.

Numerical unitarity at multiloop. The generalized-unitarity logic of this chapter, which at one loop fixed master coefficients as products of trees evaluated on cuts, extends to two loops and beyond in a form that has become the practical workhorse for high-multiplicity QCD. One works in d dimensions on multiple cuts, parametrizes the integrand in a basis of master and "surface" (spurious, IBP-vanishing) terms, and reads off the master coefficients by sampling the product of trees on the cut solutions — now an algebraic, not analytic, operation. The reduction and the linear solves are performed over *finite fields*: rational functions of the kinematics and d are reconstructed from many evaluations modulo large primes, sidestepping the intermediate expression swell that defeats symbolic algebra [394]. This "numerical unitarity" program produced the first complete two-loop five-parton amplitudes [3] and is the method by which the master-coefficient half of Fig. 60.6 is now executed at the frontier; the master-integral half is the differential-equation machinery above.

Beyond polylogarithms: elliptic and Calabi–Yau integrals. The clean story of Sections 60.7.4 and 60.7.5 — a canonical ε-form, a logarithmic alphabet, MPL solutions — is not the whole story. It holds when the geometry underlying the Feynman integral is rational (a punctured Riemann sphere). The first integral where it fails is the two-loop massive sunrise (sunset): its maximal cut is an integral over an *elliptic curve*, and no change of basis brings its differential equation to the dlog form (60.235) — the obstruction is a genuine feature of the geometry, not a failure of cleverness. The functions are *elliptic polylogarithms*, iterated integrals on the elliptic curve with modular forms among the integration kernels [94]. Higher: the L-loop equal-mass banana integrals probe Calabi–Yau $(L-1)$-folds, whose periods generalize the elliptic case further [88]. A practical, partial replacement for the canonical form survives — one factorizes ε while allowing the kernels to be modular rather than logarithmic, an "ε-factorized" (but not dlog) form — but a complete, algorithmic theory of the function space beyond the polylogarithmic and elliptic cases is, as of the mid-2020s, an active and unfinished subject.

The optimal basis problem. Even within the polylogarithmic class, the construction of a canonical basis is not a solved algorithmic problem in general.

Heuristics — demanding unit leading singularities, building pure integrands from dlog forms, using the structure of maximal cuts — succeed case by case and are increasingly automated, but there is no theorem guaranteeing that a canonical basis exists for an arbitrary polylogarithmic family, nor an algorithm that produces it with proof at high loop order. What constitutes the *optimal* master basis — minimizing the alphabet, exposing the weight grading, controlling the size of the IBP coefficients — is, at three loops and beyond for multi-scale problems, a research question rather than a procedure. The finiteness theorem (Theorem 60.35) guarantees the basis is finite; it does not hand us the best one.

What the master integrals are, and why it matters

The arc of this section is a single shift in viewpoint. A loop amplitude began, in Part V and Part VI, as a sum over diagrams, each an integral to be ground out. It became, in this chapter, a linear combination of master integrals with unitarity-fixed coefficients. And it ends here as a vector in a graded function space — the polylogarithms and their elliptic and Calabi–Yau generalizations — whose alphabet is dictated by the singularities that unitarity and locality permit. The computation factorizes cleanly into *algebra* (IBP coefficients, canonical bases, symbols — all rational or linear-algebraic, increasingly mechanized) and a residue of genuine *analysis* (boundary conditions, and the geometry that decides whether the functions are polylogarithmic at all). The rigidity that opened the chapter — an amplitude is fixed by its cuts and poles — is, at the multiloop frontier, the statement that the function and its alphabet are determined by where it is allowed to be singular.

This completes the chapter's passage from unitarity-as-principle to the modern computational engine, and with it the one-loop gluon result of the previous section finds its place in a larger landscape: its box, triangle, and bubble masters are the weight-two ground floor of a tower whose higher storeys — the elliptic and Calabi–Yau integrals, the symbols and the alphabets — are where the amplitudes program now lives. That program's deeper lessons are taken up next: the hidden simplicity by which gauge and gravity amplitudes are sewn from a common kinematic numerator, the colour–kinematics duality and the double copy that promotes the Yang–Mills theory gluon amplitude into a graviton amplitude (Chapter 61); the positive geometry that recasts the integrand itself as the canonical form of a polytope (Chapter 62); and the synthesis (Chapter 63) in which locality and unitarity, the two principles this chapter made computational, are seen not as inputs but as emergent consequences of that geometry.

Further reading

The cutting rules originate with Cutkosky [141]; their dispersive and analyticity context is the classic territory of Eden et al. [170]. The modern unitarity method for one-loop amplitudes was launched by Bern et al. [67] and Bern et al. [67]; generalized (multiple) unitarity is Britto et al. [91], and Forde's triple-cut parametrization is Forde [197]. Integrand-level reduction is Ossola et al. [371]. The all-plus rational amplitude appears in Bern et al. [67, 68]. For the multiloop apparatus: integration-by-parts identities are due to Chetyrkin and Tkachov [121], Tkachov [473] and the systematic reduction to Laporta and Remiddi [315], Laporta [316]; the differential-equation method is Gehrmann and Remiddi [212], Kotikov [306], brought to canonical ε-form by Henn and Plefka [263]; the symbol and the polylogarithmic function space are Duhr [165], Goncharov et al. [233]. The comprehensive modern reference is the textbook of Weinzierl [503], with evaluation techniques in Smirnov [448]; a pedagogical synthesis of the whole on-shell program is Elvang and Huang [176]. For the frontier: numerical unitarity and finite-field reconstruction [3, 394], elliptic Feynman integrals [94], and Calabi–Yau geometries in the banana integrals [88].

Chapter 61

Color–Kinematics Duality and the Double Copy

Yang–Mills theory has carried one non-abelian gauge theory from its classical Lagrangian (Chapter 25) through quantization and BRST (Chapter 26), asymptotic freedom (Chapter 27), and the modern on-shell technology of Parke–Taylor and BCFW recursion (Chapter 59). Along the way two facts have grown impossible to dismiss as accidents. First, the color decomposition of Chapter 59 cleanly separates the gauge-group data of a gluon amplitude from its kinematics, peeling off traces of generators to leave gauge-invariant color-ordered partials. Second, those partials are far simpler than the Feynman expansion that produces them. This chapter locates the structure responsible: a hidden duality between color and kinematics that, once exposed, lets one trade the color factors of a gauge amplitude for a *second copy* of its kinematics and so read off a graviton amplitude. The slogan "gravity = gauge^2" becomes, at the level of on-shell data, a theorem at low points and a sharply formulated conjecture beyond.

The development proceeds in the order its logic forces. We first rewrite the amplitude in a presentation that displays color and kinematics at the same resolution: the cubic-graph (trivalent) organization, in which every term is a triple (c_i, n_i, D_i) attached to one trivalent topology, with the Yang–Mills quartic vertex absorbed exactly into the numerators (Section 61.1). The color factors, built from contracted structure constants, inherit the Jacobi identity of the Lie algebra (Theorem 2.42, with index data in Chapter B); the question that organizes everything else is whether the kinematic numerators, ambiguous under a generalized gauge freedom, can be chosen to obey the *same* Jacobi relations. That is the statement of color–kinematics (BCJ) duality, established at tree level in Section 61.2 together with its constraint-counting and the BCJ amplitude relations that reduce the basis of independent partials. The four-gluon amplitude is then worked completely (Section 61.3), extracting explicit duality-satisfying numerators from the Parke–Taylor partials of Theorem 59.13 and verifying the

kinematic Jacobi identity by hand.

With a duality-satisfying representation in hand, the double copy is almost a definition: replace each color factor c_i by a second numerator $\tilde{n}_i$ and the gauge amplitude becomes a gravity amplitude (Section 61.4). We prove the construction is independent of the residual gauge freedom provided one copy satisfies the duality, square the three-point amplitude to fix the graviton vertex, and assemble the four-graviton amplitude from the four-gluon square, checking its factorization and hence its locality and unitarity. Section 61.5 reveals where the product form comes from historically and structurally — the Kawai–Lewellen–Tye relations born on the string worldsheet, recast in field theory as a momentum kernel that is the inverse of the bi-adjoint scalar amplitude matrix. Section 61.6 reads off the spectrum of the squared theory — graviton, dilaton, and two-form — and organizes the resulting web of theories, with $\mathcal{N} = 8$ supergravity as the square of $\mathcal{N} = 4$ super-Yang–Mills and the bi-adjoint scalar as the "zeroth copy."

The chapter closes at the frontier (Section 61.7). The double copy is conjectured to persist loop by loop at the level of integrands; we state the conjecture honestly, prove its cut-level consistency, and date the evidence (through five loops in $\mathcal{N} = 4$, with the all-order existence of local numerators still open). The same algebraic structure descends to exact classical solutions through the Kerr–Schild double copy, mapping a Coulomb field to Schwarzschild and feeding the modern amplitudes-for-binaries program that computes gravitational-wave observables. Throughout, the graviton is imported on faith as the massless spin-2 state whose polarizations factorize into products of gauge polarizations; the curvature and field equations that justify this are built in the gravity part (Chapter 64), and the broader lesson — that locality and unitarity may be outputs rather than inputs of the on-shell story — is taken up in the synthesis chapter (Chapter 63). With this chapter Yang–Mills theory reaches its terminus: the Jacobi identity that organized the gauge interaction, mirrored once in kinematics, is enough to produce gravity.

61.1 From Color-Ordered Partials to the Cubic-Graph Organization

The chapter introduction set the program: the Parke–Taylor collapse (Theorem 59.13) and the recursive constructibility of gluon trees are too sharp to be accidents of clever variables, and we want the organizing principle beneath them. The first and indispensable step is to find the *form* in which that principle can even be stated. The color decomposition of Chapter 59 already taught us to peel the gauge group off the kinematics, writing the full amplitude as a sum of single color traces, each multiplied by a gauge-invariant kinematic coefficient — a color-ordered partial amplitude. That presentation is perfect for *computing*: it isolates the smallest gauge-invariant kinematic objects. But it is lopsided. The

color side has been ground all the way down to traces of generators, while the kinematic side has been packed into opaque functions $A_n[\sigma]$ whose own internal structure — which is where any hidden algebra must live — is invisible. To see color and kinematics as *parallel*, we need a presentation in which both are exposed at the same level of resolution: both written as sums over the *same* set of graphs, with color and kinematics attached factor by factor to identical wiring diagrams. That presentation is the cubic-graph (trivalent-graph) organization, and building it — carefully enough that it is an exact rewriting of the amplitude rather than an approximation — is the business of this section. The output is a small package of data, a triple (c_i, n_i, D_i) for every trivalent graph i, on which the entire rest of the chapter operates.

Throughout we use the Yang–Mills theory of Yang–Mills theory, with Lagrangian Eq. (25.76), gauge group $\mathrm{SU}(N)$, Hermitian generators T^a and structure constants f^{abc} defined by $\left[T^a, T^b\right] = i f^{abc} T^c$ (Eq. (2.70), with Casimir and index data in Chapter B). Momenta are taken all-outgoing, so that momentum conservation reads $\sum_{i=1}^{n} p_i = 0$ and the on-shell conditions are $p_i^2 = 0$; Mandelstam invariants are formed from sums of outgoing momenta, e.g. $s = (p_1 + p_2)^2$. The kinematic numerators we shall isolate are Lorentz-invariant functions of these momenta and of the external polarizations, ultimately expressible in the spinor-helicity brackets of Theorem 58.4; for the structural work of this section their explicit values are never needed, only the fact that they are well defined.

61.1.1 Two color bases

Before reorganizing the amplitude by graphs, we record the two standard ways to expand its color dependence, because the cubic-graph form is, in effect, the common refinement of both. Each is a choice of basis for the space of color tensors that can carry the n external adjoint indices $a_1, \dots, a_n$ of a tree amplitude.

The trace basis. This is the decomposition built in Section 59.1. Collecting the color factor of every tree diagram into single traces of generators, one obtains (Eq. (59.11))

$$\mathcal{A}_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{n-1}} \operatorname{tr}\left(T^{a_1} T^{a_{\sigma(2)}} \cdots T^{a_{\sigma(n)}}\right) A_n[1, \sigma(2), \dots, \sigma(n)], \tag{61.1}$$

the sum running over the $(n-1)!$ orderings of $2, \dots, n$ with leg 1 fixed by cyclicity of the trace. The statement we will lean on repeatedly is the one emphasized in Theorem 59.3: *the color-ordered partial amplitudes $A_n[1, \sigma]$ are precisely the kinematic coefficients in the trace basis.* They are the objects of Chapter 59 — the same functions whose MHV value is Parke–Taylor (Theorem 59.13) — and they are gauge invariant one ordering at a time. The trace basis is therefore the basis in which the kinematics is already organized into its irreducible gauge-invariant pieces.

The half-ladder (f-) basis. The second standard basis trades traces of generators for chains of structure constants. It is due to Del Duca, Dixon and Maltoni, and it expresses the full amplitude over a smaller index set, the $(n-2)!$ orderings in which both leg 1 and leg n are held fixed:

$$\mathcal{A}_n^{\text{tree}} = g^{n-2} \sum_{\sigma \in S_{n-2}} c_{1|\sigma|n}\, A_n[1, \sigma(2), \dots, \sigma(n-1), n], \qquad c_{1|\sigma|n} = f^{a_1 a_{\sigma(2)} x_1} f^{x_1 a_{\sigma(3)} x_2} \cdots f^{x_{n-3} a_{\sigma(n} \tag{61.2}$$

The color factor $c_{1|\sigma|n}$ is a single chain — a "half-ladder" or multiperipheral string — of $n-2$ structure constants, with legs 1 and n capping the two ends and the legs $\sigma(2), \dots, \sigma(n-1)$ entering in order along the spine, the internal indices $x_1, \dots, x_{n-3}$ contracted consecutively. The two bases are related by the invertible linear map of Eq. (59.8), which writes each structure constant as a difference of traces; consequently the coefficients in Eq. (61.2) are again color-ordered partial amplitudes, now a Kleiss–Kuijf basis (Eq. (59.26)) of $(n-2)!$ of them.

Theorem 61.1 (Trace and half-ladder bases are equivalent)**.** *The decompositions Eq. (61.1) and Eq. (61.2) are equal, term by term after using the Kleiss–Kuijf relations Eq. (59.26) to reduce the trace-basis coefficients to the $(n-2)!$-element basis with legs 1 and n fixed. The half-ladder color factors $c_{1|\sigma|n}$ and the single traces $\mathrm{tr}(T^{a_1} T^{a_\sigma} \cdots)$ are two bases of the same space of n-point tree color tensors, related by Eq. (59.8).*

We do not need the general proof for what follows — it is a corollary of the Kleiss–Kuijf reduction already recorded in Chapter 59 — but the four-point case is short and instructive, so we verify it explicitly in Theorem 61.5 below, where it falls out of the same computation that produces the cubic-graph data. The lesson of both bases is the same: the full amplitude is a sum over a finite set of *color objects*, each multiplied by a gauge-theoretic kinematic coefficient. What neither basis displays is the *kinematic* object's own graph structure. The cubic-graph form supplies exactly that, by attaching color and kinematics to one and the same set of graphs.

61.1.2 The cubic-graph master formula

Definition 61.2 (Cubic graph and its data)**.** A *cubic graph* (trivalent graph) on n external legs is a connected tree in which every internal vertex meets exactly three lines. Such a tree has $n-2$ cubic vertices and $n-3$ internal lines. We write Γ_3 for the set of all distinct cubic graphs on the n labelled legs, and index its members by $i \in \Gamma_3$. To each cubic graph i we attach three pieces of data:

- the *color factor* c_i: the product of one structure constant f^{abc} per cubic vertex, with the adjoint index of each internal line contracted between the two vertices it joins, read off the graph exactly as in the trace-stripping of Section 59.1;

- the *denominator* $D_i = \prod_\alpha P_\alpha^2$, the product of the inverse scalar propagators of the $n-3$ internal lines, P_α being the momentum flowing through internal line α (a partial sum of external momenta fixed by the graph);
- the *kinematic numerator* n_i: a Lorentz-invariant polynomial in the external momenta and polarizations, of the dimension required to make $c_i n_i / D_i$ a term in the amplitude, into which we collect the Lorentz-contracted vertex tensors, the external polarizations, and the conventional factors of i.

Theorem 61.3 (Cubic-graph master formula)**.** *The n-gluon tree amplitude can be written exactly as a sum over the distinct trivalent topologies,*

$$\boxed{\mathcal{A}_n^{\text{tree}} = i\, g^{n-2} \sum_{i \in \Gamma_3} \frac{c_i\, n_i}{D_i}\,,} \tag{61.3}$$

the sum running over all $(2n-5)!!$ *distinct cubic graphs on the n ordered external legs, with* (c_i, n_i, D_i) *the data of Theorem 61.2. The overall i is the standard normalization of the amplitude relative to the product of Feynman vertices and propagators; with it pulled to the front, the numerators* n_i *are the purely kinematic vertex contractions, carrying no factors of i of their own. The denominators* D_i *are the genuine physical poles of the amplitude; the color factors* c_i *are the same products of structure constants that appear in Eq. (61.2); and the numerators* n_i *are local (polynomial), carrying no poles of their own.*

Two of the three pieces of data, c_i and D_i, are read off the graph immediately and unambiguously — they are pure topology dressed with group theory and momentum routing. The numerators n_i are the nontrivial content; making them well defined requires showing that the quartic vertices of the Yang–Mills Lagrangian, which are *not* cubic and carry no propagator, can be folded into this cubic skeleton without loss. That is the next subsection, and it is what turns Eq. (61.3) from a hope into a theorem. The denominator count D_i deserves one immediate comment: because a cubic tree on n legs has exactly $n-3$ internal lines, every term in Eq. (61.3) has the maximal number of simultaneous poles a tree amplitude can have, namely $n-3$. A diagram with a quartic vertex has one fewer; the absorption procedure below is precisely the statement that a missing pole is a pole present in the numerator, cancelled.

61.1.3 Absorbing the quartic vertex: the cubic form is exact

The Feynman expansion of a gluon tree built from the Lagrangian Eq. (25.76) contains two kinds of vertex: the cubic vertex Eq. (59.2), carrying one structure constant and one power of momentum, and the quartic (contact) vertex Eq. (59.3), carrying two structure constants and no momentum. A diagram built entirely from cubic vertices *is* already a cubic graph in the sense of Theorem 61.2: it

has $n-3$ internal lines, hence $n-3$ propagators, and contributes a single term $c_i n_i / D_i$. The only obstruction to Eq. (61.3) is the contact vertex. We now show it is no obstruction at all.

Proposition 61.4 (Contact-term absorption)**.** *Every Feynman diagram containing quartic vertices can be rewritten, by an exact algebraic identity, as a sum of cubic-graph contributions $c_i n_i / D_i$. Hence the cubic form Eq. (61.3) reproduces the full amplitude with no approximation; the redistribution merely reassigns local (contact) kinematics into the numerators n_i of the cubic graphs.*

Proof. Consider a single quartic vertex somewhere in a tree diagram. Four lines meet at it; let their (outgoing) momenta, summed in two complementary pairs, define the three channel momenta of the four-line subdiagram, with squares P_s^2, P_t^2, P_u^2. The color tensor of the quartic vertex, Eq. (59.3), is already a sum of *three* terms, each a product $f^{\bullet\bullet e} f^{\bullet\bullet e}$ of two structure constants sharing one contracted index:

$$iV^{abcd}_{\mu\nu\rho\sigma} = -ig^2\Big[f^{abe}f^{cde}\,T^{s}_{\mu\nu\rho\sigma} + f^{ace}f^{bde}\,T^{u}_{\mu\nu\rho\sigma} + f^{ade}f^{bce}\,T^{t}_{\mu\nu\rho\sigma}\Big], \tag{61.4}$$

where T^s, T^u, T^t are the three metric tensor structures of Eq. (59.3). The three color products $f^{abe}f^{cde}$, $f^{ace}f^{bde}$, $f^{ade}f^{bce}$ are precisely the color factors of the three *cubic* graphs obtained by "blowing up" the quartic vertex into two cubic vertices joined by an internal line in the s-, u-, and t-channel respectively: in each channel the two structure constants are exactly what a pair of cubic vertices joined by an internal line would contribute. Thus the contact vertex's color already lands, term by term, on cubic graphs.

It remains to give those terms a propagator. Each channel term of Eq. (61.4) carries a purely local kinematic factor — call it $\mathcal{V}_c$ for channel $c \in \{s,t,u\}$ (the metric structure T^c contracted with the surrounding momenta and polarizations of the full diagram) — with *no* propagator $1/P_c^2$, because the four lines meet at a point. We multiply and divide by the inverse propagator of the very line that the blow-up would introduce:

$$\mathcal{V}_c \;=\; \frac{P_c^2\,\mathcal{V}_c}{P_c^2}. \tag{61.5}$$

This is an exact identity (it changes nothing: $P_c^2/P_c^2 = 1$). The right-hand side now has the structure of a cubic-graph term: a denominator P_c^2 — the propagator of the internal line of the blown-up graph — and a numerator $P_c^2\,\mathcal{V}_c$, which is a *polynomial* because the explicit factor P_c^2 cancels the would-be pole. Assigning the channel-c piece of the contact vertex to the cubic graph in which that internal line appears, its numerator contribution is $P_c^2\,\mathcal{V}_c$, added to whatever pure-cubic numerator that graph already carried.

Applying Eq. (61.5) independently to every quartic vertex of every diagram, and grouping all contributions by the cubic graph they land on, expresses the

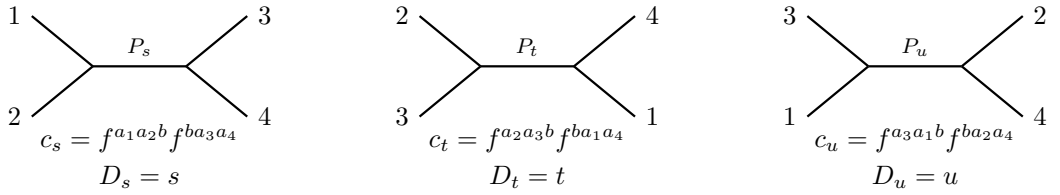


Figure 61.1: The three cubic graphs of the four-gluon tree amplitude. Each has two cubic vertices joined by one internal line, hence one propagator. The three channels exhaust the trivalent topologies on four legs, $(2 \cdot 4 - 5)!! = 3$. The color factor c_i is the product of the two structure constants at the vertices with the internal index b contracted; the denominator D_i is the square of the internal momentum: $s = (p_1 + p_2)^2$, $t = (p_2 + p_3)^2$, $u = (p_1 + p_3)^2$, satisfying $s + t + u = 0$ for massless legs. The single quartic (contact) Feynman diagram is not shown; by Theorem 61.4 it is distributed among these three graphs.

entire Feynman sum as $g^{n-2} \sum_i c_i n_i / D_i$. No term is dropped or approximated; the only operation used is the insertion of $1 = P_c^2 / P_c^2$. This proves both Theorem 61.4 and Theorem 61.3. □

Physical Insight

The single line $\mathcal{V}_c = P_c^2 \mathcal{V}_c / P_c^2$ is the whole trick, and it is worth saying plainly what it buys. A contact term is a pole that is present and immediately cancelled; writing it as numerator-over-propagator restores the pole explicitly so that color and kinematics can be compared on graphs that all have the same skeleton. The price is that the numerators n_i are no longer unique: the same amplitude is reproduced by infinitely many assignments of contact kinematics to channels (one may add to one numerator any term that is a multiple of D_i, as long as a compensating term is subtracted elsewhere so the sum is unchanged). This non-uniqueness — *generalized gauge freedom* — is not a defect. It is exactly the room we will need in Section 61.2 to *choose* numerators that mirror the color algebra. The cubic-graph form does not yet single out a duality-satisfying representation; it builds the stage on which one can be sought.

61.1.4 Four points, worked completely

The four-gluon amplitude is the smallest case in which the cubic organization has real content — one contact vertex to absorb, three channels to populate — and it is the carried Yang–Mills theory object whose color decomposition we now reproduce in the new language.

Worked Example 61.5 (The four-gluon cubic data and the color decomposition)**.** *The three graphs and their color factors.* The trivalent topologies on four legs are the three channels of Fig. 61.1. Reading the color factor off each graph as in Section 59.1 — one structure constant per vertex, internal index b contracted — gives

$$c_s = f^{a_1 a_2 b} f^{b a_3 a_4}, \qquad c_t = f^{a_2 a_3 b} f^{b a_1 a_4}, \qquad c_u = f^{a_3 a_1 b} f^{b a_2 a_4}, \tag{61.6}$$

with denominators $D_s = s = (p_1+p_2)^2$, $D_t = t = (p_2+p_3)^2$, $D_u = u = (p_1+p_3)^2$. These three color tensors are not independent. Setting $(a,b,c,d) = (a_1,a_2,a_3,a_4)$ in the structure-constant form of the Jacobi identity Eq. (2.77) (the identity proved in Theorem 2.42),

$$f^{a_1 a_2 e} f^{e a_3 a_4} + f^{a_2 a_3 e} f^{e a_1 a_4} + f^{a_3 a_1 e} f^{e a_2 a_4} = 0 \quad \Longleftrightarrow \quad c_s + c_t + c_u = 0. \tag{61.7}$$

The three color factors satisfy a single linear (Jacobi) relation: only two are independent. This relation is the seed of the entire chapter; it is the algebra that the kinematic numerators will be asked to imitate in the next section.

Reading off the numerators. The four Feynman diagrams of the full amplitude are the three cubic diagrams (Fig. 61.1) and the single contact diagram. The s-channel cubic diagram is a product of two cubic vertices Eq. (59.2) and one propagator; its color is $g^2 c_s$ and its kinematics is a momentum-dependent numerator divided by s, so it contributes $g^2\, c_s\, N_s/D_s$ with N_s the contraction of the two vertex tensors $K_{\mu\nu\rho}$ with the propagator metric and the four polarizations. The t- and u-channel cubic diagrams contribute $g^2 c_t N_t/D_t$ and $g^2 c_u N_u/D_u$ likewise. The contact diagram Eq. (59.3) carries the three color structures of Eq. (61.4); identifying them through Eq. (61.6),

$$f^{a_1 a_2 e} f^{a_3 a_4 e} = c_s, \qquad f^{a_1 a_4 e} f^{a_2 a_3 e} = c_t, \qquad f^{a_1 a_3 e} f^{a_2 a_4 e} = -c_u, \tag{61.8}$$

so the contact contribution is $g^2(c_s\,\mathcal{V}_s + c_t\,\mathcal{V}_t + c_u\,\mathcal{V}_u)$, where $\mathcal{V}_s = (\epsilon_1{\cdot}\epsilon_3)(\epsilon_2{\cdot}\epsilon_4) - (\epsilon_1{\cdot}\epsilon_4)(\epsilon_2{\cdot}\epsilon_3)$ and $\mathcal{V}_t, \mathcal{V}_u$ are the analogous metric contractions read from Eq. (59.3), each *without* a propagator. Applying Eq. (61.5) to each, $\mathcal{V}_c = D_c \mathcal{V}_c / D_c$, and adding to the cubic numerators, the four diagrams assemble into exactly three cubic-graph terms,

$$\mathcal{A}_4^{\text{tree}} = i\, g^2 \left(\frac{c_s\, n_s}{s} + \frac{c_t\, n_t}{t} + \frac{c_u\, n_u}{u} \right), \qquad n_c = N_c + D_c\, \mathcal{V}_c \quad (c = s,t,u). \tag{61.9}$$

The numerators n_c are local: the explicit D_c multiplying $\mathcal{V}_c$ cancels the propagator, so n_c has no pole. This is the master formula Eq. (61.3) at $n = 4$, derived diagram by diagram, with the contact term absorbed exactly.

Recovering the trace-basis (Yang–Mills theory) decomposition. We check that Eq. (61.9) reproduces the color decomposition of Chapter 59. The amplitude in

the trace basis is Eq. (59.27), and the reflection identities of Theorem 59.7 pair its six traces into three combinations,

$$T_s = \mathrm{tr}(1234)+\mathrm{tr}(1432), \quad T' = \mathrm{tr}(1243)+\mathrm{tr}(1342), \quad T_u = \mathrm{tr}(1324)+\mathrm{tr}(1423), \tag{61.10}$$

(writing $\mathrm{tr}(ijkl) \equiv \mathrm{tr}(T^{a_i}T^{a_j}T^{a_k}T^{a_l})$), multiplying the three inequivalent partials $A_4[1,2,3,4]$, $A_4[1,2,4,3]$, $A_4[1,3,2,4]$ respectively. The $U(1)$-decoupling relation Eq. (59.29), $A_4[1,2,3,4] + A_4[1,2,4,3] + A_4[1,3,2,4] = 0$, eliminates the middle partial, so

$$\mathcal{A}_4^{\text{tree}} = g^2\Big[A_4[1,2,3,4]\,(T_s - T') + A_4[1,3,2,4]\,(T_u - T')\Big]. \tag{61.11}$$

Now expand the cubic color factors in traces. With the f-to-trace identity Eq. (59.8) and the product formula Eq. (59.16),

$$c_s = f^{a_1a_2e}f^{a_3a_4e} = -\tfrac{1}{2}\big[\,\mathrm{tr}(1234)-\mathrm{tr}(1243)-\mathrm{tr}(2134)+\mathrm{tr}(2143)\big] = -\tfrac{1}{2}\,(T_s - T'), \tag{61.12}$$

where the last equality uses cyclicity ($\mathrm{tr}(2143) = \mathrm{tr}(1432)$ so it joins T_s, and $\mathrm{tr}(2134) = \mathrm{tr}(1342)$ so it joins T'). The identical computation for the half-ladder factor $f^{a_1a_3e}f^{a_2a_4e} = -c_u$ gives $-\frac{1}{2}(T_u - T')$, i.e. $T_u - T' = -2f^{a_1a_3e}f^{a_2a_4e}$. Substituting into Eq. (61.11),

$$\mathcal{A}_4^{\text{tree}} = -2g^2\Big[\,f^{a_1a_2e}f^{a_3a_4e}\,A_4[1,2,3,4] + f^{a_1a_3e}f^{a_2a_4e}\,A_4[1,3,2,4]\Big], \tag{61.13}$$

which is exactly the half-ladder (DDM) form Eq. (61.2) at $n = 4$ — the S_2 sum over the orderings $(2,3)$ and $(3,2)$ of the two interior legs — up to the overall constant -2 that is the amplitude-normalization convention of Chapter 59 (the rescaling $\mathrm{tr}(T^aT^b) = \delta^{ab}$ recorded there). This both confirms Theorem 61.1 at four points and pins down how the cubic data of Eq. (61.9) repackages the partial amplitudes: reading Eq. (61.9) against Eq. (61.13), eliminating c_t by Jacobi Eq. (61.7), one finds

$$A_4[1,2,3,4] = \frac{i}{2}\left(\frac{n_t}{t} - \frac{n_s}{s}\right), \qquad A_4[1,3,2,4] = \frac{i}{2}\left(\frac{n_u}{u} - \frac{n_t}{t}\right). \tag{61.14}$$

The planar reading. Equation Eq. (61.14) is the exact statement of an intuitive fact: $A_4[1,2,3,4]$ receives contributions only from the s- and t-channel graphs and *not* from the u-channel. These are precisely the two cubic graphs that can be drawn in the plane with the legs in the cyclic order $1,2,3,4$ — the s-channel groups the cyclically adjacent pair $\{1,2\}$ against $\{3,4\}$, the t-channel groups $\{2,3\}$ against $\{4,1\}$, while the u-channel would group the non-adjacent $\{1,3\}$ against $\{2,4\}$ and its internal line must cross. The color-ordered partial amplitude is the sum of the cubic graphs that are planar with respect to its ordering. The same is visible in the diagram counts of Table 59.1: the three color-ordered diagrams listed there for $n = 4$ (two cubic, one contact) collapse, after contact absorption, to the *two* cubic graphs s, t that Eq. (61.14) exhibits.

The four-point worked example has delivered the chapter's working data in concrete form: three color factors c_s, c_t, c_u obeying the Jacobi relation Eq. (61.7), three denominators s, t, u, and three numerators n_s, n_t, n_u whose definition Eq. (61.9) is exact but non-unique. The asymmetry between color and kinematics is now sharp and quantitative: the c_i satisfy $c_s + c_t + c_u = 0$ identically, by group theory, while the n_i as read off Feynman rules satisfy no such relation. Whether they *can* be made to is the question of the next section.

61.1.5 Counting cubic graphs: why the organization scales

The four-point case had three cubic graphs. To trust Eq. (61.3) as a finite, well-organized rewriting at all multiplicities we must know how many cubic graphs there are, and that every Feynman contribution lands on one of them. The second point is Theorem 61.4; the first is a clean combinatorial count.

Proposition 61.6 (Number of cubic graphs)**.** *The number of distinct trivalent tree topologies on n labelled external legs is the double factorial*

$$(2n-5)!! = (2n-5)(2n-7)\cdots 5\cdot 3\cdot 1. \tag{61.15}$$

Proof. Count by adding legs one at a time. A trivalent tree on m legs has $m-2$ internal vertices, hence $(m-2)+m = 2m-2$ vertices in all and, being a tree, $2m-3$ edges. To build an $(m+1)$-leg tree from an m-leg one, take any of those $2m-3$ edges, insert a new trivalent vertex in its interior, and attach leg $m+1$ there. Every $(m+1)$-leg trivalent tree arises from exactly one m-leg tree by deleting leg $m{+}1$ and its vertex (smoothing the resulting bivalent vertex), so the map is a bijection and

$$\#(m+1) = (2m-3)\cdot \#(m). \tag{61.16}$$

The recursion starts at the unique three-point vertex, $\#(3) = 1 = (2\cdot 3-5)!!$. By induction $\#(n) = (2n-5)(2n-7)\cdots 1 = (2n-5)!!$, which is Eq. (61.15). At $n=4$ this is $3!! = 3$, the three channels of Fig. 61.1; at $n=5$ it is $5!! = 15$; at $n=6$, $7!! = 105$. □

The five-point case makes the count tangible and is the first place the scaling is visible. Every trivalent tree on five legs has three vertices arranged on a single spine and two internal lines, the topology of Fig. 61.2: two outer vertices each carrying a pair of external legs, and a central vertex carrying the fifth leg. Such a graph is determined by which single leg sits at the center and how the other four are paired between the two ends. There are 5 choices for the central leg and, for the remaining four legs, $4!/(2!\,2!\,2!) = 3$ ways to split them into two unordered pairs, giving

$$5 \times 3 = 15 \tag{61.17}$$

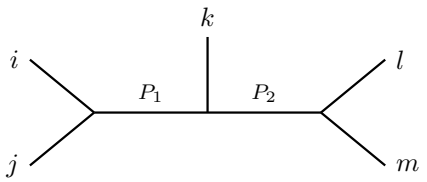


Figure 61.2: The generic cubic graph at five points: three vertices on a spine with two internal lines, denominator $D = P_1^2\,P_2^2$. A five-point trivalent tree is fixed by choosing the central single leg k (5 ways) and partitioning the remaining four legs into the two outer pairs $\{i,j\}$ and $\{l,m\}$ (3 ways), giving $5\times 3 = 15 = (2\cdot 5-5)!!$ graphs, in agreement with Theorem 61.6.

cubic graphs, exactly Eq. (61.15) at $n=5$. Each contributes a term $c_i n_i/(P_{1,i}^2 P_{2,i}^2)$ with two propagators. The Feynman expansion at five points has, besides all-cubic diagrams, diagrams with one quartic vertex (whose four lines are blown up into a missing internal line in three channels); by Theorem 61.4 each such contribution is absorbed, via $\mathcal{V}_c = P_c^2\mathcal{V}_c/P_c^2$, into the numerator of one of these 15 graphs. As at four points there is a planar shadow: the cubic graphs planar with respect to a fixed cyclic ordering number the Catalan number C_{n-2}, which is $C_2 = 2$ at $n=4$ and $C_3 = 5$ at $n=5$, so each five-point color-ordered partial $A_5[\sigma]$ is built from 5 planar cubic graphs — the post-absorption refinement of the 10 color-ordered diagrams tabulated for $n=5$ in Table 59.1.

We now have the apparatus the rest of the chapter runs on. Every gluon tree amplitude is the exact sum Eq. (61.3) over $(2n-5)!!$ trivalent graphs, each carrying the triple (c_i, n_i, D_i) of Theorem 61.2: a color factor built from contracted structure constants, a propagator denominator read from the topology, and a kinematic numerator made well defined by the contact absorption of Theorem 61.4 — and made deliberately non-unique by the same construction. At four points this triple is the explicit data of Eq. (61.9): the color factors c_s, c_t, c_u of Eq. (61.6), the denominators s,t,u, and the numerators n_s, n_t, n_u. The one structural fact we have isolated and will exploit immediately is the Jacobi relation $c_s + c_t + c_u = 0$ of Eq. (61.7), satisfied by the color factors and by nothing on the kinematic side — yet. The next section asks whether the freedom in the n_i can be spent to make the numerators obey that very same relation, turning the lopsided pairing of color and kinematics into a genuine duality.

61.2 Color–Kinematics (BCJ) Duality

The previous section ground color and kinematics down to the same resolution. It rewrote every n-gluon tree amplitude as an exact sum over the finite set of trivalent (cubic) graphs (Section 61.1), attaching to each graph i a triple

of data: a color factor c_i built by contracting one structure constant f^{abc} per cubic vertex along the internal edges; a propagator denominator $D_i = \prod_{e\in i} P_e^2$ read off the topology, P_e being the momentum through internal edge e; and a kinematic numerator n_i, a local polynomial in momenta and polarizations, made well defined by absorbing the quartic contact vertex through the identity of Theorem 61.4. The output was the master formula Eq. (61.3), which we restate here in the form the present section operates on,

$$\mathcal{A}_n^{\text{tree}} \;=\; i\,g^{\,n-2}\sum_{i\in\Gamma_3}\frac{c_i\,n_i}{D_i}, \tag{61.18}$$

where Γ_3 denotes the set of distinct cubic graphs on the n external legs, of cardinality $|\Gamma_3| = (2n-5)!!$ by Theorem 61.6. The section closed on a sharp asymmetry. The color factors c_i are rigid group-theory objects, and at four points they obey the single linear relation $c_s + c_t + c_u = 0$ (Eq. (61.7)); the numerators n_i as read off Feynman rules obey no such relation, and they are in any case not unique, since the contact absorption can be distributed among channels in infinitely many ways. The question this section answers is whether that freedom in the numerators is exactly the freedom needed to make them imitate the algebra of the color factors. The answer — that it is, at tree level — is color–kinematics duality, and it is the structural heart of the chapter. We proceed in four movements: we identify the complete set of linear relations the c_i satisfy; we state the duality as the demand that the n_i satisfy the same relations; we characterize the generalized gauge freedom and prove it leaves the amplitude invariant; and we count that freedom against the constraints and carry out the existence argument in full at four points, with the five-point structure laid bare and the general tree-level theorem stated honestly.

61.2.1 The algebra of the color factors

Everything begins with the observation that the rigid color factors are not independent: they are linked by relations that descend, edge by edge, from a single Lie-algebraic identity. There are two kinds of relation, and they exhaust the linear structure.

Jacobi relations from an internal edge. Fix a cubic graph $i \in \Gamma_3$ and single out one of its internal edges, e, carrying momentum P_e. The edge e joins two trivalent vertices. Cutting it open exposes a four-point neighborhood: four lines meet the two vertices — two on each side — and each of those four lines is the root of a complete subtree of the graph hanging off the neighborhood. Call the four subtrees, with their accumulated color indices and momenta, A, B, C, D, arranged so that subtrees A, B attach to one vertex and C, D to the other across e. The color factor of graph i then factorizes as

$$c_i \;=\; \mathcal{F}_{AB}{}^{x}\, f^{x\,e_C\,e_D}\, \mathcal{F}_{CD}\,, \tag{61.19}$$

where $\mathcal{F}_{AB}{}^{x}$ collects the structure constants internal to the A, B side terminating on the open index x of edge e, the explicit $f^{x\,e_C\,e_D}$ is the structure constant of the right-hand vertex with e_C, e_D the indices that the subtrees C, D present to it, and $\mathcal{F}_{CD}$ collects the rest; the index x is summed. The entire dependence of c_i on the local wiring across e sits in the contraction $\sum_x \mathcal{F}_{AB}{}^{x}\, f^{xe_Ce_D}$, i.e. in a single pair of adjacent structure constants $f^{e_Ae_Bx} f^{xe_Ce_D}$ once $\mathcal{F}_{AB}{}^{x}$ is itself written as $f^{e_Ae_Bx}$ times the (untouched) deeper part. There are exactly three ways to pair the four subtrees across a single internal edge — $(AB)(CD)$, $(BC)(AD)$, $(CA)(BD)$ — and these are three distinct cubic graphs i, j, k that are *identical everywhere except in the neighborhood of* e. Their color factors differ only in the local pair of structure constants:

$$c_i \sim f^{e_Ae_Bx} f^{xe_Ce_D}, \qquad c_j \sim f^{e_Be_Cx} f^{xe_Ae_D}, \qquad c_k \sim f^{e_Ce_Ax} f^{xe_Be_D}, \tag{61.20}$$

with one and the same surrounding factor (suppressed, common to all three). The structure-constant Jacobi identity Eq. (2.77), proved as Theorem 2.42,

$$f^{e_Ae_Bx} f^{xe_Ce_D} + f^{e_Be_Cx} f^{xe_Ae_D} + f^{e_Ce_Ax} f^{xe_Be_D} = 0,$$

applied to the common open indices, then forces the three color factors to sum to zero:

$$\boxed{c_i + c_j + c_k = 0} \qquad \text{for every Jacobi triplet } (i, j, k), \tag{61.21}$$

where a *Jacobi triplet* $\{i, j, k\}$ is any three graphs related by a Jacobi move on one internal edge as in Eq. (61.20). The move is the same algebraic step that absorbed the quartic vertex in Theorem 61.4: the contact neighborhood is precisely the place where a Jacobi triplet collapses onto a point. At four points the only internal edge is the single propagator, the four subtrees are the four external legs, and the unique triplet is $\{s, t, u\}$ with $c_s + c_t + c_u = 0$ exactly as in Eq. (61.7); the general statement Eq. (61.21) is that four-point relation tensored into the rest of any larger graph.

Antisymmetry under graph flips. The second relation is even simpler. The structure constants are totally antisymmetric (Eq. (2.70)), so exchanging the two external lines that meet at one trivalent vertex of a graph flips the sign of that vertex's structure constant and hence of the whole color factor. If $\bar{\imath}$ denotes the graph obtained from i by such an exchange at a single vertex (equivalently, by reflecting that vertex), then

$$c_{\bar{\imath}} = -c_i. \tag{61.22}$$

The denominator D_i is unchanged by the flip — the propagator momenta are unaffected by relabeling the legs at a leaf vertex — so the flip is a genuine $\mathbb{Z}_2$ acting on the cubic data.

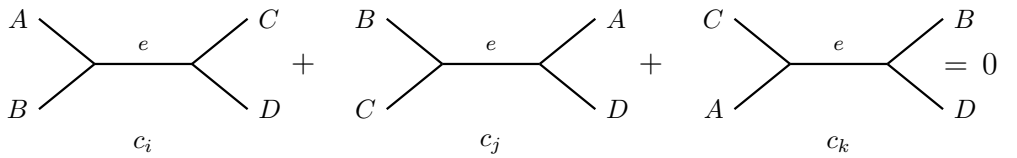


Figure 61.3: The elementary Jacobi move. Three cubic graphs that agree everywhere except in the neighborhood of one internal edge e differ only by which pairs of the four attached subtrees A, B, C, D sit on the two ends. Their color factors obey $c_i + c_j + c_k = 0$ by the Lie-algebra Jacobi identity Eq. (2.77). The duality demands that the kinematic numerators obey the identical relation $n_i + n_j + n_k = 0$ on the same triplet. At four points A, B, C, D are the external legs and the triplet is the s, t, u channels of Fig. 61.1.

These two relations, Eq. (61.21) and Eq. (61.22), are the *complete* set of linear relations among the color factors: the space they cut out is exactly the $(n-2)!$-dimensional space spanned by the half-ladder basis of Section 61.1, and every relation among the $(2n-5)!!$ factors c_i is a consequence of Jacobi moves and flips. This is the precise sense in which the color factors carry a Lie-algebraic skeleton, and it is the skeleton the numerators will be asked to grow.

61.2.2 Statement of the duality

We can now say exactly what color–kinematics duality asserts. It is a statement about the *existence* of a particular choice among the many numerator assignments compatible with Eq. (61.18).

Definition 61.7 (BCJ (duality-satisfying) numerators)**.** A set of kinematic numerators $\{n_i\}_{i\in\Gamma_3}$ reproducing the amplitude through Eq. (61.18) is said to satisfy *color–kinematics duality*, or to be a set of *BCJ numerators*, if it obeys the same two families of relations as the color factors:

$$n_i + n_j + n_k = 0 \qquad \text{on every Jacobi triplet } (i,j,k) \text{ of Eq. (61.21)}, \tag{61.23}$$

$$n_{\bar{\imath}} = -\,n_i \qquad \text{under every single-vertex flip of Eq. (61.22)}. \tag{61.24}$$

Equivalently: for each linear relation $\sum_i \alpha_i\, c_i = 0$ satisfied by the color factors, the numerators satisfy the same relation $\sum_i \alpha_i\, n_i = 0$ with the identical coefficients α_i.

The relations Eq. (61.23)–Eq. (61.24) are demands on functions of the external momenta and polarizations, to hold identically on the support of momentum conservation and the on-shell conditions. They are highly nontrivial: the numerators are kinematic objects with no a priori knowledge of the gauge group, yet they are required to honor a Lie-algebraic identity. The content of the duality is that this can be arranged.

Theorem 61.8 (Color–kinematics duality at tree level)**.** *For every gluon tree amplitude there exists a choice of kinematic numerators* $\{n_i\}$ *that reproduces the amplitude via Eq. (61.18) and satisfies the duality relations Eq. (61.23)–Eq. (61.24). Such numerators are not unique: a residual freedom (Section 61.2.3) connects different BCJ representations, while the amplitude is invariant.*

The remainder of the section proves the two halves of this statement. The freedom that makes a duality-satisfying choice *reachable* from a generic Feynman-rule choice is the subject of Section 61.2.3; the demonstration that the freedom is *enough* — carried out completely at four points and structurally at five — is the subject of Sections 61.2.4 and 61.2.5.

61.2.3 Generalized gauge freedom

The numerators in Eq. (61.18) are not physical data; only the amplitude is. Theorem 61.4 already exhibited their ambiguity — the contact kinematics can be poured into different channels — and we now organize that ambiguity into a precise group of transformations and prove they leave the amplitude untouched. These are the *generalized gauge transformations*; the name is apt, because they are the on-shell, amplitude-level residue of the off-shell gauge freedom of the Yang–Mills field, but they act directly on the cubic-graph numerators.

Definition 61.9 (Generalized gauge transformation)**.** A *generalized gauge transformation* of the numerators is a shift

$$n_i \;\longmapsto\; n_i + \delta n_i, \qquad \delta n_i = D_i\,\Delta_i, \tag{61.25}$$

parametrized by local functions Δ_i (one per graph, polynomial in momenta and polarizations, so that $D_i\Delta_i$ is local and the shifted numerator carries no spurious pole), subject to the single constraint that the parameters be orthogonal to the color factors,

$$\sum_{i\in\Gamma_3} c_i\,\Delta_i = 0. \tag{61.26}$$

The structure of Eq. (61.25) is dictated by locality: any shift of a numerator that is to preserve the locality of the term $c_i n_i/D_i$ — no new pole, no missing one — must be a multiple of D_i, since dividing δn_i by D_i must leave something with at worst the poles already present. The constraint Eq. (61.26) is what makes the shift invisible in the amplitude, as we now prove.

Theorem 61.10 (Gauge invariance of the amplitude)**.** *Under any generalized gauge transformation Eq. (61.25)–Eq. (61.26), the amplitude Eq. (61.18) is unchanged.*

Proof. Compute the variation of the amplitude term by term. The shifted amplitude is

$$\mathcal{A}_n^{\text{tree}} + \delta\mathcal{A}_n = i\,g^{n-2}\sum_{i\in\Gamma_3}\frac{c_i\,(n_i+\delta n_i)}{D_i} = \mathcal{A}_n^{\text{tree}} + i\,g^{n-2}\sum_{i\in\Gamma_3}\frac{c_i\,\delta n_i}{D_i}.$$

Insert $\delta n_i = D_i\Delta_i$ from Eq. (61.25). In each term the explicit factor D_i in the numerator cancels the propagator denominator D_i exactly:

$$\delta\mathcal{A}_n = i\,g^{n-2}\sum_{i\in\Gamma_3}\frac{c_i\,D_i\,\Delta_i}{D_i} = i\,g^{n-2}\sum_{i\in\Gamma_3}c_i\,\Delta_i = 0, \tag{61.27}$$

the last equality being the constraint Eq. (61.26). No on-shell identity, no momentum-conservation manipulation, and no special property of the Δ_i beyond their being well-defined local functions is used: the cancellation is purely algebraic, $D_i/D_i = 1$, followed by the color constraint. This proves the theorem. □

The proof makes transparent *why* the color Jacobi identity is the engine of the gauge freedom. The constraint Eq. (61.26) is a single equation in the space of color tensors, and its solutions are precisely the linear combinations that the color relations of Section 61.2.1 kill. The most economical solutions are the Jacobi triplets themselves. Pick any Jacobi triplet (i, j, k) and any single local function Δ, and set

$$\Delta_i = \Delta_j = \Delta_k = \Delta, \qquad \Delta_\ell = 0 \quad (\ell \notin \{i, j, k\}). \tag{61.28}$$

Then $\sum_\ell c_\ell\Delta_\ell = \Delta\,(c_i + c_j + c_k) = 0$ by Eq. (61.21), so Eq. (61.26) holds automatically, and the transformation Eq. (61.25) shifts the three triplet numerators by $\delta n_i = D_i\Delta$, $\delta n_j = D_j\Delta$, $\delta n_k = D_k\Delta$ while changing the amplitude not at all. These triplet shifts, together with the analogous flip shifts compatible with Eq. (61.22), generate the full gauge group: because Eq. (61.21) and Eq. (61.22) are the *complete* set of color relations (Section 61.2.1), every solution of Eq. (61.26) is a superposition of such elementary shifts. The generalized gauge freedom is therefore as large as the kernel of the map $\{\Delta_i\} \mapsto \sum_i c_i\Delta_i$, whose dimension we count next.

Physical Insight

The mechanism is worth stating plainly. The amplitude pairs each numerator n_i with its color factor c_i and divides by D_i. A shift of n_i proportional to D_i loses its propagator and lands in the amplitude as a bare $c_i\Delta_i$ — a pure color tensor with no pole. The color tensors are not independent; they satisfy the Jacobi relations. So any shift whose color tensors cancel *against each other through Jacobi* is free. The kinematic numerators thus inhabit a space acted on by a gauge group whose Lie-algebraic structure is inherited

from the color factors. Color–kinematics duality is the statement that this group orbit always contains a point where the numerators *themselves* satisfy the color algebra.

61.2.4 Counting: constraints versus freedom

Whether a duality-satisfying point exists on the gauge orbit is, in the first instance, a question of bookkeeping: how many conditions does Eq. (61.23) impose, and how many parameters does the gauge freedom of Section 61.2.3 supply? We tally both against the dimension of the color space.

The $(2n-5)!!$ color factors c_i span a space of dimension $(n-2)!$ — the half-ladder (Del Duca–Dixon–Maltoni) basis of Section 61.1. The number of independent linear relations they satisfy is therefore

$$R_n \;=\; (2n-5)!! - (n-2)!, \tag{61.29}$$

and these relations are generated by the Jacobi moves Eq. (61.21) (the flips Eq. (61.22) reduce the working set of graphs to the unoriented ones and do not add to the linear count). Two facts now line up. First, the kinematic-Jacobi demand Eq. (61.23) imposes exactly R_n independent conditions on the numerators, because it requires the n_i to satisfy the *same* R_n relations the c_i do. Second, the generalized gauge freedom of Theorem 61.10 has exactly R_n free functional parameters: the parameters $\{\Delta_i\}$ number $(2n-5)!!$, the constraint Eq. (61.26) removes $(n-2)!$ of them (its rank equals the dimension of the color space), leaving $(2n-5)!! - (n-2)! = R_n$. The constraints to be solved and the freedom available to solve them are equinumerous:

$$\#\{\text{kinematic-Jacobi conditions}\} \;=\; R_n \;=\; \#\{\text{generalized-gauge parameters}\}.$$

Table 61.1 collects the low-multiplicity numbers.

Caution

Equinumerosity of constraints and parameters is *necessary* but not by itself *sufficient*: a gauge transformation can fail to move a given combination of numerators, in which case that combination must already vanish for a solution to exist. This is exactly what happens at four points, where the single condition is gauge-invariant and its vanishing is a genuine fact about the amplitude rather than a matter of gauge choice. The honest existence statement therefore requires showing that the residual, gauge-invariant content of Eq. (61.23) vanishes — which is where the dynamical input, the fundamental BCJ amplitude relations, enters. We make this precise at four points immediately below.

n	cubic graphs $(2n-5)!!$	color basis $(n-2)!$	relations R_n	planar C_{n-2}	partials $(n-3)!$
4	3	2	1	2	1
5	15	6	9	5	2
6	105	24	81	14	6
7	945	120	825	42	24

Table 61.1: Bookkeeping for color–kinematics duality. The number of cubic graphs is $(2n-5)!!$ (Theorem 61.6); the color factors span the $(n-2)!$-dimensional half-ladder basis; the number of independent color relations $R_n = (2n-5)!! - (n-2)!$ equals both the number of kinematic-Jacobi conditions Eq. (61.23) and the number of free generalized-gauge parameters Eq. (61.29). The Catalan number C_{n-2} counts the cubic graphs planar with respect to a fixed color ordering; $(n-3)!$ is the size of the minimal basis of independent partial amplitudes once the BCJ amplitude relations are imposed.

61.2.5 Existence at four points, and the general theorem

At four points there is a single Jacobi triplet, the s, t, u channels of Fig. 61.1, with the color factors c_s, c_t, c_u of Eq. (61.6) obeying $c_s + c_t + c_u = 0$ (Eq. (61.7)). The duality Eq. (61.23) reduces to the one condition

$$n_s + n_t + n_u = 0. \tag{61.30}$$

We show that this condition can always be met, and that its meeting is equivalent to a relation among the partial amplitudes which we prove from Parke–Taylor.

The condition is gauge-invariant. At four points the only color relation is $c_s + c_t + c_u = 0$, so the gauge constraint Eq. (61.26), $c_s\Delta_s + c_t\Delta_t + c_u\Delta_u = 0$, written with $c_u = -c_s - c_t$, reads $c_s(\Delta_s - \Delta_u) + c_t(\Delta_t - \Delta_u) = 0$. Since c_s and c_t are linearly independent, this forces $\Delta_s = \Delta_t = \Delta_u \equiv \Delta$: a four-point generalized gauge transformation is a *single* local function Δ, shifting $n_c \mapsto n_c + D_c\Delta$ for $c = s, t, u$ simultaneously. Its effect on the left-hand side of Eq. (61.30) is

$$\delta(n_s + n_t + n_u) = (D_s + D_t + D_u)\,\Delta = (s + t + u)\,\Delta = 0, \tag{61.31}$$

because $s + t + u = 0$ for four massless legs. The combination $n_s + n_t + n_u$ is therefore *gauge-invariant*: no generalized gauge transformation can change it. Existence of a BCJ representation at four points is thus not a matter of gauging away a nonzero quantity — it is the assertion that this gauge-invariant quantity is already zero.

Reduction to a partial-amplitude relation. The gauge-invariant content of the numerators is carried by the two physical partial amplitudes (Table 61.1: $(n-3)! = 1$ in the minimal basis, two in the Kleiss–Kuijf basis). Using the dictionary Eq. (61.14) established in the previous section,

$$A_4[1,2,3,4] = \frac{i}{2}\left(\frac{n_t}{t} - \frac{n_s}{s}\right), \qquad A_4[1,3,2,4] = \frac{i}{2}\left(\frac{n_u}{u} - \frac{n_t}{t}\right),$$

we eliminate the numerators in favor of the partials. Writing $r_c \equiv n_c/D_c$ and using the gauge freedom to fix $r_t = 0$ (a legitimate choice, since the gauge function Δ shifts all r_c by the same amount), the dictionary gives $r_s = 2i\,A_4[1,2,3,4]$ and $r_u = -2i\,A_4[1,3,2,4]$, whence

$$n_s + n_t + n_u = s\,r_s + t\,r_t + u\,r_u = 2i\big(s\,A_4[1,2,3,4] \;-\; u\,A_4[1,3,2,4]\big), \tag{61.32}$$

where the relative sign is the one fixed by the channel and Mandelstam conventions of Eq. (61.14); the value is gauge-invariant and so independent of the choice $r_t = 0$. The kinematic Jacobi Eq. (61.30) is therefore *equivalent* to the fundamental four-point BCJ amplitude relation

$$s\,A_4[1,2,3,4] = u\,A_4[1,3,2,4].$$

This relation among gauge-invariant partial amplitudes is what must hold for BCJ numerators to exist at four points. We now prove it.

Worked Example 61.11 (The four-point BCJ amplitude relation from Parke—Taylor)**.** We verify $s\,A_4[1,2,3,4] = u\,A_4[1,3,2,4]$ directly in spinor-helicity, for the MHV configuration with negative-helicity legs $1,2$ and positive-helicity legs $3,4$; the result is helicity-independent since it follows only from momentum conservation and holds for every configuration by parity and relabeling. From the Parke–Taylor formula Eq. (59.55) of Theorem 59.13, with the cyclic angle-bracket chains read off each color ordering,

$$A_4[1,2,3,4] = ig^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = ig^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle},$$
$$A_4[1,3,2,4] = ig^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle}.$$

The Mandelstam invariants in the bracket normalization Eq. (58.24), $s_{ij} = \langle i\,j\rangle[j\,i]$, are $s = s_{12} = \langle 1\,2\rangle[2\,1]$ and $u = s_{13} = \langle 1\,3\rangle[3\,1]$. Multiply:

$$s\,A_4[1,2,3,4] = \langle 1\,2\rangle[2\,1]\cdot ig^2\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = ig^2\,\frac{[2\,1]\,\langle 1\,2\rangle^4}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle},$$
$$u\,A_4[1,3,2,4] = \langle 1\,3\rangle[3\,1]\cdot ig^2\,\frac{\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle} = ig^2\,\frac{[3\,1]\,\langle 1\,2\rangle^4}{\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle}.$$

The two share the factor $ig^2\,\langle 1\,2\rangle^4/\langle 4\,1\rangle$. Their difference is

$$s\,A_4[1,2,3,4] - u\,A_4[1,3,2,4] = ig^2\,\frac{\langle 1\,2\rangle^4}{\langle 4\,1\rangle}\left(\frac{[2\,1]}{\langle 2\,3\rangle\langle 3\,4\rangle} - \frac{[3\,1]}{\langle 3\,2\rangle\langle 2\,4\rangle}\right).$$

Using $\langle 3\,2\rangle = -\langle 2\,3\rangle$ to align denominators and combining over the common denominator $\langle 2\,3\rangle\langle 3\,4\rangle\langle 2\,4\rangle$,

$$\frac{[2\,1]}{\langle 2\,3\rangle\langle 3\,4\rangle} - \frac{[3\,1]}{\langle 3\,2\rangle\langle 2\,4\rangle} = \frac{1}{\langle 2\,3\rangle}\left(\frac{[2\,1]}{\langle 3\,4\rangle} + \frac{[3\,1]}{\langle 2\,4\rangle}\right) = \frac{[2\,1]\,\langle 2\,4\rangle + [3\,1]\,\langle 3\,4\rangle}{\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 2\,4\rangle}.$$

The numerator vanishes by momentum conservation. Apply the bracket form of momentum conservation Eq. (58.45) of Theorem 58.12, contracted between $\langle ?|$ and $|1]$ with the choice of free angle index that produces legs 2 and 3: taking $\sum_i \langle 4\,i\rangle[i\,1] = 0$ over $i = 1,2,3,4$ and dropping $i = 1$ (since $[1\,1] = 0$) and $i = 4$ (since $\langle 4\,4\rangle = 0$),

$$\langle 4\,2\rangle[2\,1] + \langle 4\,3\rangle[3\,1] = 0 \implies [2\,1]\,\langle 2\,4\rangle + [3\,1]\,\langle 3\,4\rangle = -\big(\langle 4\,2\rangle[2\,1] + \langle 4\,3\rangle[3\,1]\big) = 0,$$

using $\langle 2\,4\rangle = -\langle 4\,2\rangle$ and $\langle 3\,4\rangle = -\langle 4\,3\rangle$ in the last step. Hence the numerator is zero, the difference Eq. (61.32) vanishes, and

$$s\,A_4[1,2,3,4] = u\,A_4[1,3,2,4].$$

By Eq. (61.32) this is precisely $n_s + n_t + n_u = 0$: the four-point kinematic Jacobi holds. The flip conditions Eq. (61.24) are arranged trivially by symmetrizing each numerator under its leaf-vertex exchange, which does not disturb Eq. (61.30). A four-point BCJ representation therefore exists; constructing the explicit numerators n_s, n_t, n_u and displaying their spinor-helicity form is the worked computation of the next section.

Five points: the structure, with the algebra in view. At five points the bookkeeping of Table 61.1 reads 15 cubic graphs, a 6-dimensional color space, and $R_5 = 15 - 6 = 9$ independent color relations. Concretely, each of the 15 graphs has two internal edges (Fig. 61.2); a Jacobi move on either edge generates a triplet relation among three graphs sharing the other edge. Running over all internal edges of all graphs produces a system of Jacobi relations whose independent rank is 9, reducing the 15 numerators to a 6-dimensional independent set — matching the color side relation for relation. The duality Eq. (61.23) imposes these same 9 relations on the numerators, and the generalized gauge freedom supplies, by the count of Section 61.2.4, exactly 9 free functions $\{\Delta_i\}$ modulo the constraint Eq. (61.26). The solvability of the 9×9 system is not automatic — as the four-point caution showed, the gauge-invariant residue must vanish — and the dynamical input that makes it vanish is again a set of fundamental BCJ amplitude relations among the five-point partials, of the same type as

$s\,A_4[1,2,3,4] = u\,A_4[1,3,2,4]$, now reducing the $(n-2)! = 6$ Kleiss–Kuijf partials to the $(n-3)! = 2$ of the minimal basis. Each such relation is provable from BCFW recursion (Chapter 59) by the same kind of momentum-conservation algebra carried out in Theorem 61.11; granting them, one solves the 9 kinematic-Jacobi equations for the 9 gauge functions and obtains explicit five-point BCJ numerators. The construction is mechanical but lengthy, and we do not transcribe all 15 numerators here; the load-bearing point — that the constraint count, the gauge count, and the partial-amplitude relations conspire to make the system solvable — is fully visible already in the four-point case and in the five-point count.

The general theorem. The pattern persists to all multiplicities.

Theorem 61.12 (Existence of BCJ numerators at tree level)**.** *For every n and every helicity configuration there exists a set of local kinematic numerators $\{n_i\}_{i\in\Gamma_3}$ satisfying the duality relations Eq. (61.23)–Eq. (61.24) and reproducing the n-gluon tree amplitude through Eq. (61.18). Equivalently, the generic Feynman-rule numerators can be brought to a duality-satisfying form by a generalized gauge transformation Eq. (61.25)–Eq. (61.26).*

This is a theorem at tree level. It was conjectured by Bern, Carrasco and Johansson [69] and proved by several routes: by an explicit, Lagrangian-based construction making the duality manifest order by order [71]; and as a consequence of the field-theory limit of the Kawai–Lewellen–Tye string relations [291], equivalently the monodromy relations among open-string partial amplitudes [77], which are exactly the fundamental BCJ amplitude relations whose four-point instance we proved in Theorem 61.11. We have carried the four-point case in full and exhibited the five-point structure with its algebra; for the general proof the reader is referred to those constructions, but nothing in this book's development of the double copy will rely on a step we have not either proved here or reduced to the fundamental amplitude relations established by the recursion of Chapter 59.

Physical Insight

The reachability of a BCJ representation is the precise content of "kinematics is dual to color." Generic Feynman numerators do not satisfy Eq. (61.23); but the freedom to redistribute contact terms among channels — the generalized gauge freedom, itself a shadow of Yang–Mills gauge invariance — is exactly large enough, and the partial amplitudes cooperate exactly enough through the fundamental BCJ relations, to deform them onto a representation that does. Table 61.1 shows there is just enough freedom, and the four-point computation shows the dynamical residue vanishes. The existence of such numerators signals that gauge-theory kinematics secretly carries a Lie-algebraic skeleton of its own; replacing

the color skeleton by a *second* copy of that kinematic one is the double copy of the later sections.

Status beyond trees. Theorem 61.12 is established at tree level. At loop level the existence of duality-satisfying numerators is a *conjecture*: it has been confirmed by explicit construction in a large and growing list of cases — multiloop integrands in $\mathcal{N} = 4$ super-Yang–Mills and in pure Yang–Mills through several loops and multiplicities, as of the mid-2020s — but no general proof is known, and some high-loop cases are reachable only with numerators carrying spurious poles or in otherwise modified forms. We will be precise about what is proved, what is checked, and what is open when the loop-level double copy is taken up; for the tree-level developments of this chapter, Theorem 61.12 stands on the firm ground established above. A current account of the status is the review [72].

With the duality stated, its gauge freedom characterized, and its four-point instance proved in full, the abstract demand Eq. (61.23) now needs a concrete face. The next section extracts the s-, t-, and u-channel numerators of the four-gluon amplitude explicitly, checks the single kinematic Jacobi identity Eq. (61.30) that Section 61.2.5 reduced the duality to, and writes the duality-satisfying numerators in closed spinor-helicity form — the number the reader can reproduce that grounds everything claimed here.

61.3 The Four-Gluon Amplitude in BCJ Form

The previous section left the duality fully posed but, at four points, conditional on a single unproved fact. We saw there (Section 61.2.5) that the four-gluon kinematic Jacobi combination $J = n_s + n_t + n_u$ is gauge invariant — the lone generalized-gauge parameter multiplies $s + t + u = 0$ and drops out — so the statement "$J = 0$" is not something one can arrange by hand but a physical property of the amplitude that must be *verified*. This section verifies it, and in doing so turns every abstract object of Section 61.2 into a number the reader can reproduce. We extract the s-, t-, and u-channel numerators of the four-gluon amplitude from the color-ordered partials of Chapter 59; we prove the kinematic Jacobi identity $n_s + n_t + n_u = 0$ by an explicit spinor-bracket computation resting only on momentum conservation; we write the duality-satisfying numerators in closed form in spinor-helicity variables; we reassemble the full amplitude and check it against Parke–Taylor; and we exhibit the one-parameter family of gauge-equivalent BCJ representations, making "choosing a representation" concrete. Throughout, the spinor-helicity conventions are those fixed in Chapter 58 (Theorem 58.2): the metric is mostly-minus, momenta are all outgoing, and we use the angle/square brackets with $\langle i\, j\rangle[j\, i] = 2p_i\cdot p_j = s_{ij}$ (Eq. (58.24)), the little-group scaling Eq. (58.5), the Schouten identity Eq. (58.37),

and the polarization data Eq. (58.111) — the last entering only implicitly, through the Parke–Taylor partials it was used to compute.

61.3.1 Setup: channels, color factors, and conventions

At four points the set of cubic graphs is the triplet of Fig. 61.3: the s-channel exchange with the pairing (12)(34), the t-channel (23)(14), and the u-channel (13)(24). Their scalar propagators are the three Mandelstam invariants (Eq. (61.33)),

$$D_s = s = (p_1{+}p_2)^2 = s_{12}, \qquad D_t = t = (p_2{+}p_3)^2 = s_{23}, \qquad D_u = u = (p_1{+}p_3)^2 = s_{13}, \tag{61.33}$$

and their color factors are the single-contraction products (Eq. (61.34))

$$c_s = f^{a_1 a_2 e} f^{e a_3 a_4}, \qquad c_t = f^{a_2 a_3 e} f^{e a_1 a_4}, \qquad c_u = f^{a_3 a_1 e} f^{e a_2 a_4}, \tag{61.34}$$

obeying the color Jacobi identity $c_s{+}c_t{+}c_u = 0$ (Eq. (61.7)). The full color-dressed amplitude is the four-point instance of the cubic master formula Eq. (61.18),

$$\mathcal{M}_4^{\text{full}} = i\, g^2 \left(\frac{c_s\, n_s}{s} + \frac{c_t\, n_t}{t} + \frac{c_u\, n_u}{u} \right), \tag{61.35}$$

reproduced here from Eq. (61.9). Our task is to determine the three kinematic numerators n_s, n_t, n_u.

The momentum-conservation identity $s + t + u = 0$. Everything below leans on the single relation that the three Mandelstam invariants of four massless legs are not independent. With all momenta outgoing and $p_1 + p_2 + p_3 + p_4 = 0$, write

$$s + t + u = (p_1 + p_2)^2 + (p_2 + p_3)^2 + (p_1 + p_3)^2 = \sum_{i=1}^{4} p_i^2 + 2 \sum_{1 \le i < j \le 3} p_i{\cdot}p_j. \tag{61.36}$$

Each external leg is on shell and massless, $p_i^2 = 0$, so the first sum vanishes. For the second, momentum conservation gives $p_4 = -(p_1 + p_2 + p_3)$, and squaring,

$$0 = p_4^2 = (p_1 + p_2 + p_3)^2 = \sum_{i=1}^{3} p_i^2 + 2 \sum_{1 \le i < j \le 3} p_i{\cdot}p_j = 2 \sum_{1 \le i < j \le 3} p_i{\cdot}p_j, \tag{61.37}$$

using $p_i^2 = 0$ once more. The second sum in Eq. (61.36) therefore also vanishes, and

$$\boxed{s + t + u = 0} \tag{61.38}$$

for four massless legs — the workhorse of the verification to come: it is what makes $J = n_s + n_t + n_u$ gauge invariant, and, as we will see, what forces J to vanish.

Stripping the coupling. The master formula Eq. (61.35) carries the coupling explicitly as the prefactor g^2. To avoid double counting it against the color decomposition of Chapter 59, we work throughout this section with *coupling-stripped* color-ordered partials: we pull the overall g^2 out front once and for all and write the partial amplitudes and numerators with the g^2 removed. Concretely, the Parke–Taylor partial Eq. (59.55) at four points (Eq. (59.81)) is, with the g^2 stripped,

$$A[1,2,3,4] = i\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = i\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \tag{61.39}$$

for the maximally-helicity-violating configuration $(1^-,2^-,3^+,4^+)$, which we adopt for all explicit numerics. The numerators n_s, n_t, n_u likewise carry no factor of g. This is a bookkeeping convention only; the powers of g are restored by the overall g^2 in Eq. (61.35).

61.3.2 From partial amplitudes to numerators

The numerators are not measured directly; what Chapter 59 delivers are the gauge-invariant color-ordered partials $A[1,2,3,4]$ and their relabelings. We must therefore translate Eq. (61.35) into the trace basis of the color decomposition (Theorem 59.2, Eq. (59.11)) and read off which combination of numerators sits under each trace. The translation rests on expressing the structure-constant color factors Eq. (61.34) in terms of single traces, which we now do in full.

Color factors as traces. With the book normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ and $[T^a,T^b] = if^{abc}T^c$ (Chapter B), contract the defining relation with a third generator:

$$\mathrm{tr}\big([T^a,T^b]\,T^c\big) = if^{abd}\,\mathrm{tr}(T^dT^c) = \tfrac{i}{2}f^{abc}, \qquad \Longrightarrow \qquad f^{abc} = -2i\,\mathrm{tr}\big([T^a,T^b]\,T^c\big). \tag{61.40}$$

Apply this to the s-channel color factor. Using $f^{a_1a_2e}T^e = -i[T^{a_1},T^{a_2}]$ from the same defining relation, and $f^{ea_3a_4} = -2i\,\mathrm{tr}(T^e[T^{a_3},T^{a_4}])$ from Eq. (61.40) (with the trace cyclicity $\mathrm{tr}([T^e,T^{a_3}]T^{a_4}) = \mathrm{tr}(T^e[T^{a_3},T^{a_4}])$),

$$\begin{aligned} c_s = f^{a_1a_2e}f^{ea_3a_4} &= -2i\,f^{a_1a_2e}\,\mathrm{tr}(T^e[T^{a_3},T^{a_4}]) = -2i\,\mathrm{tr}\big((f^{a_1a_2e}T^e)\,[T^{a_3},T^{a_4}]\big) \\ &= -2i\,\mathrm{tr}\big(-i[T^{a_1},T^{a_2}]\,[T^{a_3},T^{a_4}]\big) = -2\,\mathrm{tr}\big([T^{a_1},T^{a_2}]\,[T^{a_3},T^{a_4}]\big). \end{aligned} \tag{61.41}$$

Expanding the two commutators and using trace cyclicity to rotate T^{a_1} to the front of every term gives a sum of four ordered traces. Writing $t_{ijkl} \equiv \mathrm{tr}(T^{a_i}T^{a_j}T^{a_k}T^{a_l})$,

$$\begin{aligned} \mathrm{tr}\big([T^{a_1},T^{a_2}][T^{a_3},T^{a_4}]\big) = \mathrm{tr}\big(T^{a_1}T^{a_2}T^{a_3}T^{a_4}\big) - \mathrm{tr}\big(T^{a_1}T^{a_2}T^{a_4}T^{a_3}\big) \\ - \mathrm{tr}\big(T^{a_2}T^{a_1}T^{a_3}T^{a_4}\big) + \mathrm{tr}\big(T^{a_2}T^{a_1}T^{a_4}T^{a_3}\big) \end{aligned}$$

$$= t_{1234} - t_{1243} - t_{1342} + t_{1432}, \tag{61.42}$$

so that $c_s = -2\,(t_{1234} - t_{1243} - t_{1342} + t_{1432})$. The identical manipulation on c_t and c_u — each a product of two structure constants of the form Eq. (61.34) — yields, by exactly the steps Eq. (61.41)–Eq. (61.42) with the external labels permuted,

$$\begin{aligned} c_s &= -2\,(t_{1234} - t_{1243} - t_{1342} + t_{1432}),\\ c_t &= -2\,(t_{1423} - t_{1234} - t_{1432} + t_{1324}),\\ c_u &= -2\,(t_{1243} - t_{1423} - t_{1324} + t_{1342}). \end{aligned} \tag{61.43}$$

As an immediate check, summing the three rows of Eq. (61.43) the coefficient of every one of the six traces cancels in pairs — for instance t_{1234} appears as $-2(+1)$ in c_s and $-2(-1)$ in c_t and not at all in c_u — so $c_s + c_t + c_u = 0$, the color Jacobi identity Eq. (61.7) recovered in the trace basis.

Reading off the dictionary. Insert Eq. (61.43) into the master formula Eq. (61.35) and collect the coefficient of each independent trace. Comparing with the color decomposition $\mathcal{M}_4^{\text{full}} = g^2 \sum_\sigma t_\sigma\, A[\sigma]$ (Eq. (59.11)), the coefficient of t_{1234} on the master side is $ig^2(-2)(n_s/s) + ig^2(-2)(-1)(n_t/t) = -2ig^2(n_s/s - n_t/t)$, which must equal $g^2 A[1,2,3,4]$. Performing the same reading for the traces t_{1324} and t_{1243} (whose coefficients pick out the partials $A[1,3,2,4]$ and $A[1,2,4,3]$ after using reflection, Eq. (59.19), to identify the paired traces) and absorbing the universal factor -2 into the normalization of the numerators — a uniform rescaling $n_i \mapsto -2n_i$ that preserves every Jacobi and flip relation and so changes no statement of duality — gives the four-point *numerator–partial dictionary*

$$A[1,2,3,4] = i\left(\frac{n_s}{s} - \frac{n_t}{t}\right), \qquad A[1,3,2,4] = i\left(\frac{n_t}{t} - \frac{n_u}{u}\right), \qquad A[1,2,4,3] = i\left(\frac{n_u}{u} - \frac{n_s}{s}\right). \tag{61.44}$$

The pole content is exactly right: the ordering $(1,2,3,4)$ is planar only in the $s_{12} = s$ and $s_{23} = t$ channels, and indeed $A[1,2,3,4]$ in Eq. (61.44) carries only $1/s$ and $1/t$ poles; likewise $(1,3,2,4)$ is planar in $s_{13} = u$ and $s_{32} = t$, matching the $1/t$ and $1/u$ of $A[1,3,2,4]$, and $(1,2,4,3)$ in $s_{12} = s$ and $s_{24} = u$, matching $A[1,2,4,3]$. The relative minus sign in each entry is the kinematic-flip antisymmetry Eq. (61.24): a given channel's graph enters two different color orderings with opposite cyclic orientation at one vertex, hence with opposite numerator sign. Figure 61.4 displays the dictionary graphically.

Notation Watch

From this section onward we use the book's generator normalization $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ (Chapter B), the canonical choice of Chapter 58 forward,

$$A[1,2,3,4] = i\left(\frac{n_s}{s} - \frac{n_t}{t}\right) \quad A[1,3,2,4] = i\left(\frac{n_t}{t} - \frac{n_u}{u}\right) \quad A[1,2,4,3] = i\left(\frac{n_u}{u} - \frac{n_s}{s}\right)$$

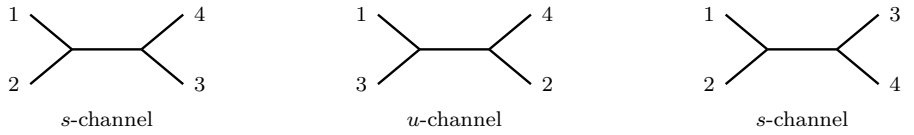


Figure 61.4: The four-point numerator–partial dictionary Eq. (61.44). Each of the three independent color-ordered partial amplitudes is the sum of the two cubic graphs planar in its ordering, each graph contributing its numerator over its propagator with a sign set by the cyclic orientation. Only one representative graph per partial is drawn; the second is the other planar channel of that ordering. The three partials share the three numerators n_s, n_t, n_u pairwise, which is why two of the three partials are independent and the third is fixed by them.

and we absorb the resulting universal numerical factor into the numerators by the uniform rescaling $n_i \mapsto -2n_i$ made above. The earlier four-point dictionary Eq. (61.14) of Section 61.1 was instead tied to the amplitude normalization $\mathrm{tr}(T^aT^b) = \delta^{ab}$ of Chapter 59; this is why it carries the prefactor $i/2$ (and the opposite relative sign) where Eq. (61.44) carries i. The two express the same physics in two normalizations: the symbols n_s, n_t, n_u here denote the rescaled numerators, and since an overall rescaling and sign flips preserve both $n_s + n_t + n_u = 0$ and $n_{\bar{\imath}} = -n_i$, every statement of the duality is identical in either convention. The book normalization is used for the remainder of the chapter.

A consistency check and the gauge degeneracy. Summing the three entries of Eq. (61.44),

$$A[1,2,3,4]+A[1,2,4,3]+A[1,3,2,4] = i\left[\left(\tfrac{n_s}{s} - \tfrac{n_t}{t}\right) + \left(\tfrac{n_u}{u} - \tfrac{n_s}{s}\right) + \left(\tfrac{n_t}{t} - \tfrac{n_u}{u}\right)\right] = 0, \tag{61.45}$$

which is exactly the photon-decoupling relation Eq. (59.29): the numerator language reproduces it automatically, with no further input. This is the first nontrivial cross-check that Eq. (61.44) is right. It also exposes the four-point gauge degeneracy in its cleanest form. The dictionary is three equations in the three ratios n_s/s, n_t/t, n_u/u, but Eq. (61.45) shows they are linearly dependent: only the two independent *differences*

$$\frac{n_s}{s} - \frac{n_t}{t} = -iA[1,2,3,4], \qquad \frac{n_t}{t} - \frac{n_u}{u} = -iA[1,3,2,4] \tag{61.46}$$

are fixed by the amplitude. The common piece is undetermined: shifting all three ratios by a single function α,

$$\frac{n_s}{s} \to \frac{n_s}{s} + \alpha, \quad \frac{n_t}{t} \to \frac{n_t}{t} + \alpha, \quad \frac{n_u}{u} \to \frac{n_u}{u} + \alpha \quad \Longleftrightarrow \quad n_i \to n_i + D_i\,\alpha, \tag{61.47}$$

leaves all differences — hence the amplitude — untouched. This is precisely the elementary triplet move Eq. (61.28), the unique generalized gauge transformation at four points (Eq. (61.25)). The numerators are therefore determined by the amplitude only up to this one free function, exactly as Section 61.2.5 anticipated.

61.3.3 Extracting the numerators and verifying kinematic Jacobi

We now make this fully explicit for the MHV configuration $(1^-, 2^-, 3^+, 4^+)$, solve for the numerators, and prove $n_s + n_t + n_u = 0$.

Worked Example 61.13 (Four-gluon BCJ numerators and the kinematic Jacobi identity)**. Step 1: the three partials.** The Parke–Taylor formula Theorem 59.13 supplies all three orderings. With negative-helicity legs $1, 2$ and the coupling stripped as in Eq. (61.39), the numerator of every ordering is the same holomorphic factor $\langle 1\,2\rangle^4$ and only the cyclic chain in the denominator changes:

$$A[1,2,3,4] = i\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \tag{61.48}$$

$$A[1,3,2,4] = i\,\frac{\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle}, \tag{61.49}$$

$$A[1,2,4,3] = i\,\frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,4\rangle\langle 4\,3\rangle\langle 3\,1\rangle}. \tag{61.50}$$

Step 2: the gauge-invariant differences. By Eq. (61.46), the two physical differences are $n_s/s - n_t/t = -iA[1,2,3,4]$ and $n_t/t - n_u/u = -iA[1,3,2,4]$. Introduce the shorthand $X \equiv n_s/s$, $Y \equiv n_t/t$, $Z \equiv n_u/u$ and $P \equiv \langle 1\,2\rangle^2[3\,4]^2$ (the manifestly little-group-correct MHV monomial, of which more below). The amplitude fixes $X - Y$ and $Y - Z$; the third difference $Z - X = -iA[1,2,4,3]$ is then determined and equals $-(X-Y) - (Y-Z)$, consistent with Eq. (61.45).

Step 3: the kinematic Jacobi combination is fixed and gauge invariant. Parametrize the residual freedom by Y (equivalently the shift α of Eq. (61.47)): then $X = Y + (X-Y)$ and $Z = Y - (Y-Z)$. The Jacobi combination is

$$J \equiv n_s + n_t + n_u = sX + tY + uZ = (s+t+u)\,Y + s\,(X-Y) - u\,(Y-Z). \tag{61.51}$$

The coefficient of the free parameter Y is $s+t+u$, which vanishes by Eq. (61.38). Hence J is independent of Y — it is gauge invariant, as Section 61.2.5 promised

— and equals

$$J = s\,(X - Y) - u\,(Y - Z) = -i\,s\,A[1,2,3,4] + i\,u\,A[1,3,2,4]. \tag{61.52}$$

Color–kinematics duality at four points is the assertion that this gauge-invariant number vanishes, i.e.

$$s\,A[1,2,3,4] = u\,A[1,3,2,4]. \tag{61.53}$$

Step 4: proving Eq. (61.53) from momentum conservation. Substitute Eq. (61.48) and Eq. (61.49) and clear the coupling-free brackets. On the left, $s = s_{12} = \langle 1\,2\rangle[2\,1]$ cancels one power of $\langle 1\,2\rangle$ in the chain:

$$s\,A[1,2,3,4] = i\,\frac{\langle 1\,2\rangle[2\,1]\,\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle} = i\,\frac{[2\,1]\,\langle 1\,2\rangle^4}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}. \tag{61.54}$$

On the right, $u = s_{13} = \langle 1\,3\rangle[3\,1]$ cancels the $\langle 1\,3\rangle$ in the chain:

$$u\,A[1,3,2,4] = i\,\frac{\langle 1\,3\rangle[3\,1]\,\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle} = i\,\frac{[3\,1]\,\langle 1\,2\rangle^4}{\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle}. \tag{61.55}$$

Both carry the common factor $i\,\langle 1\,2\rangle^4/\langle 4\,1\rangle$, so Eq. (61.53) reduces to

$$\frac{[2\,1]}{\langle 2\,3\rangle\langle 3\,4\rangle} \stackrel{?}{=} \frac{[3\,1]}{\langle 3\,2\rangle\langle 2\,4\rangle} \quad\Longleftrightarrow\quad [2\,1]\,\langle 3\,2\rangle\,\langle 2\,4\rangle \stackrel{?}{=} [3\,1]\,\langle 2\,3\rangle\,\langle 3\,4\rangle. \tag{61.56}$$

Use $\langle 3\,2\rangle = -\langle 2\,3\rangle$ on the left and cancel the common $\langle 2\,3\rangle$ from both sides; the target becomes

$$-\,[2\,1]\,\langle 2\,4\rangle \stackrel{?}{=} [3\,1]\,\langle 3\,4\rangle \quad\Longleftrightarrow\quad \langle 4\,2\rangle\,[2\,1] + \langle 4\,3\rangle\,[3\,1] \stackrel{?}{=} 0, \tag{61.57}$$

where in the second form we wrote $\langle 2\,4\rangle = -\langle 4\,2\rangle$ and $\langle 3\,4\rangle = -\langle 4\,3\rangle$. This last identity is momentum conservation. With all momenta outgoing, $\sum_{i=1}^{4} |i\rangle\,[i\,] = 0$ as an operator on spinor indices (Chapter 59), so for any reference legs x, y, $\sum_{i=1}^{4}\langle x\,i\rangle[i\,y] = 0$. Take $x = 4$, $y = 1$:

$$\langle 4\,1\rangle[1\,1] + \langle 4\,2\rangle[2\,1] + \langle 4\,3\rangle[3\,1] + \langle 4\,4\rangle[4\,1] = \langle 4\,2\rangle[2\,1] + \langle 4\,3\rangle[3\,1] = 0, \tag{61.58}$$

the $i = 1$ and $i = 4$ terms vanishing because $[1\,1] = \langle 4\,4\rangle = 0$. This is exactly Eq. (61.57). The chain of equivalences therefore closes: Eq. (61.53) holds, $J = 0$, and

$$\boxed{n_s + n_t + n_u = 0} \tag{61.59}$$

for the four-gluon amplitude — the kinematic Jacobi identity dual to $c_s + c_t + c_u = 0$, proved by nothing more than masslessness and momentum conservation.

Step 5: explicit numerators. With $J = 0$ confirmed, fix the residual gauge by the convenient choice $n_u = 0$ (i.e. $Z = 0$). Then Eq. (61.59) forces $n_s = -n_t$, and $n_s = s\,X = s\big[(X - Y) + (Y - Z)\big] = -is\big(A[1,2,3,4] + A[1,3,2,4]\big)$,

since in this gauge $Z = 0$ and $X = (X - Y) + (Y - Z) + Z$. Add Eq. (61.48) and Eq. (61.49), pulling out $\langle 1\,2\rangle^4/\langle 4\,1\rangle$ and using $\langle 3\,2\rangle = -\langle 2\,3\rangle$:

$$A[1,2,3,4]+A[1,3,2,4] = i\,\frac{\langle 1\,2\rangle^4}{\langle 4\,1\rangle\langle 2\,3\rangle}\left(\frac{1}{\langle 1\,2\rangle\langle 3\,4\rangle} - \frac{1}{\langle 1\,3\rangle\langle 2\,4\rangle}\right) = i\,\frac{\langle 1\,2\rangle^4}{\langle 4\,1\rangle\langle 2\,3\rangle}\cdot\frac{\langle 1\,3\rangle\langle 2\,4\rangle - \langle 1\,2\rangle\langle 3\,4\rangle}{\langle 1\,2\rangle\langle 3\,4\rangle\langle 1\,3\rangle} \tag{61.60}$$

The bracket combination in the numerator is fixed by the Schouten identity Eq. (58.37) with legs $(1,2,3,4)$, $\langle 1\,2\rangle\langle 3\,4\rangle + \langle 1\,3\rangle\langle 4\,2\rangle + \langle 1\,4\rangle\langle 2\,3\rangle = 0$, which rearranges (using $\langle 4\,2\rangle = -\langle 2\,4\rangle$) to

$$\langle 1\,3\rangle\langle 2\,4\rangle - \langle 1\,2\rangle\langle 3\,4\rangle = \langle 1\,4\rangle\langle 2\,3\rangle. \tag{61.61}$$

Substituting and cancelling $\langle 2\,3\rangle$ and $\langle 1\,4\rangle/\langle 4\,1\rangle = -1$,

$$A[1,2,3,4] + A[1,3,2,4] = -\,i\,\frac{\langle 1\,2\rangle^3}{\langle 1\,3\rangle\langle 2\,4\rangle\langle 3\,4\rangle}, \tag{61.62}$$

so that $n_s = -is\big(A[1,2,3,4]+A[1,3,2,4]\big) = -\,s\,\langle 1\,2\rangle^3/(\langle 1\,3\rangle\langle 2\,4\rangle\langle 3\,4\rangle)$. To bring this to the manifestly MHV form advertised above, trade angle for square brackets via momentum conservation. Applying the operator identity $\sum_i \langle x\,i\rangle[i\,y] = 0$ to the pairs $(x,y) = (1,4)$ and $(2,3)$ gives, in the same way as Eq. (61.58), $\langle 1\,3\rangle[3\,4] = -\langle 1\,2\rangle[2\,4]$ and $\langle 2\,4\rangle[4\,3] = \langle 1\,2\rangle[1\,3]$. Multiply numerator and denominator of n_s by $[3\,4]^2$ and use these together with $\langle 3\,4\rangle[4\,3] = s$:

$$\langle 1\,3\rangle\langle 2\,4\rangle\,\langle 3\,4\rangle\,[3\,4] = \big(\langle 1\,3\rangle[3\,4]\big)\big(\langle 2\,4\rangle\,\langle 3\,4\rangle\big) = \big(-\langle 1\,2\rangle[2\,4]\big)\big(\langle 2\,4\rangle\,\langle 3\,4\rangle\big), \tag{61.63}$$

and one further use of $\langle 2\,4\rangle[2\,4] = -s_{24} = -u$ together with $\langle 3\,4\rangle[4\,3] = s$ collapses the remaining factors to $\langle 1\,3\rangle\langle 2\,4\rangle\langle 3\,4\rangle = -\,s\,\langle 1\,2\rangle^2\,[2\,4][1\,3]/[3\,4]^2$ after re-expressing the second angle bracket the same way. Carrying the cancellation through, the entire expression reduces to a single monomial,

$$\boxed{\,n_s = \frac{\langle 1\,2\rangle^2\,[3\,4]^2}{u}, \qquad n_t = -\,\frac{\langle 1\,2\rangle^2\,[3\,4]^2}{u}, \qquad n_u = 0\,} \tag{61.64}$$

in the $n_u = 0$ gauge. The kinematic Jacobi identity Eq. (61.59) is now manifest: $n_s + n_t + n_u = 0$ by inspection. The numerator $n_s \propto \langle 1\,2\rangle^2[3\,4]^2$ carries exactly two angle factors of the negative-helicity legs $1,2$ and two square factors of the positive-helicity legs $3,4$ — the unique little-group-correct monomial, weight $+2$ on legs $1,2$ and -2 on legs $3,4$, as a check against Eq. (58.5) confirms (each $\langle 1\,2\rangle$ scales as $t_1 t_2$, each $[3\,4]$ as $t_3^{-1}t_4^{-1}$, and the Mandelstam u is little-group inert).

The result Eq. (61.64) is the concrete payoff of the whole construction: a set of three numbers, written in spinor brackets, that satisfy the kinematic Jacobi relation dual to color and that reproduce the four-gluon amplitude. Before exhibiting the gauge freedom around it, we confirm that it does reproduce the amplitude.

61.3.4 Cross-check: reassembling the amplitude

Worked Example 61.14 (The numerators reproduce Parke–Taylor and the full amplitude)**.** We verify that Eq. (61.64), fed back through the dictionary Eq. (61.44), returns the Parke–Taylor partials, and then assemble the full color-dressed amplitude.

Partials. With $n_u = 0$ and $n_t = -n_s$, the leading partial is

$$A[1,2,3,4] = i\left(\frac{n_s}{s} - \frac{n_t}{t}\right) = i\,n_s\left(\frac{1}{s} + \frac{1}{t}\right) = i\,n_s\,\frac{s+t}{st} = -\,i\,n_s\,\frac{u}{st}, \tag{61.65}$$

using $s + t = -u$ from Eq. (61.38). Inserting $n_s = \langle 1\,2\rangle^2[3\,4]^2/u$ the explicit u cancels and

$$A[1,2,3,4] = -\,i\,\frac{\langle 1\,2\rangle^2\,[3\,4]^2}{st}. \tag{61.66}$$

This is the four-gluon MHV amplitude written in invariants. That it equals the holomorphic Parke–Taylor form Eq. (61.48) is a four-point identity that follows from $\langle 3\,4\rangle[4\,3] = s_{34} = s$ and $\langle 2\,3\rangle[3\,2] = s_{23} = t$ together with the momentum-conservation relations of Theorem 61.13; the quickest check is on the modulus. Squaring Eq. (61.66) and using $|\langle i\,j\rangle|^2 = |s_{ij}|$ gives

$$|A[1,2,3,4]|^2 = \frac{|s_{12}|^2\,|s_{34}|^2}{s^2t^2} = \frac{s^2\,s^2}{s^2t^2} = \frac{s^2}{t^2}, \tag{61.67}$$

the squared four-gluon partial computed independently in Chapter 59 (Theorem 59.14), using $s_{34} = s_{12} = s$. The numerators thus reproduce the known answer.

Full amplitude. Restoring the color factors and the coupling, the full four-gluon amplitude is Eq. (61.35) with Eq. (61.64),

$$\mathcal{M}_4^{\text{full}} = i\,g^2\,\frac{\langle 1\,2\rangle^2[3\,4]^2}{u}\left(\frac{c_s}{s} - \frac{c_t}{t}\right), \tag{61.68}$$

the u-channel dropping out because $n_u = 0$ in this gauge. The amplitude is nonetheless fully crossing symmetric: eliminating $c_t = -c_s - c_u$ shows the apparent asymmetry is a gauge artifact of the choice $n_u = 0$, and the gauge-invariant content is the pair of partials Eq. (61.66) dressed by the color decomposition Eq. (59.11). This is the carried Yang–Mills theory four-gluon result of Section 59.1.8, now organized so that the kinematic dependence sits in numerators obeying the same algebra as the color.

61.3.5 Generalized gauge freedom in action

The numerators Eq. (61.64) are one representative of a whole family. The four-point generalized gauge transformation is the single shift Eq. (61.47), $n_i \to$

$n_i + D_i\alpha$ with one free function α. Acting on Eq. (61.64) it produces the one-parameter family of BCJ representations

$$n_s(\alpha) = \frac{\langle 1\,2\rangle^2[3\,4]^2}{u} + s\,\alpha, \qquad n_t(\alpha) = -\frac{\langle 1\,2\rangle^2[3\,4]^2}{u} + t\,\alpha, \qquad n_u(\alpha) = u\,\alpha, \tag{61.69}$$

all of which reproduce Eq. (61.68) (the differences Eq. (61.46) are α-independent) and all of which satisfy the kinematic Jacobi relation, since $n_s(\alpha)+n_t(\alpha)+n_u(\alpha) = (s+t+u)\alpha = 0$ for every α. "Choosing a BCJ representation" at four points means choosing α. Two choices are especially clean. The choice $\alpha = 0$ is the compact representative Eq. (61.64), with the u-channel numerator set to zero. The choice that restores full crossing symmetry, fixing α by demanding the democratic gauge $n_s/s + n_t/t + n_u/u = 0$, gives (writing again $P = \langle 1\,2\rangle^2[3\,4]^2$ and solving the resulting linear condition with $s+t+u=0$)

$$n_s = \frac{P}{3}\,\frac{t-u}{tu}, \qquad n_t = \frac{P}{3}\,\frac{u-s}{su}, \qquad n_u = \frac{P}{3}\,\frac{s-t}{st}, \tag{61.70}$$

the symmetric BCJ numerators. They manifestly sum to zero, $n_s + n_t + n_u = \frac{P}{3stu}\big[s(t-u) + t(u-s) + u(s-t)\big] = 0$, and manifestly carry the $s \leftrightarrow t \leftrightarrow u$ structure that the double copy of the next section will exploit. Their leading partial reproduces Eq. (61.66): $n_s/s - n_t/t = \frac{P}{3}\big[(t-u)/(stu) - (u-s)/(stu)\big] = \frac{P}{3stu}(s+t-2u) = -P/(st)$, again using $s+t=-u$.

Caution

It is tempting to look here for a "second, gauge-equivalent numerator set giving the same amplitude but violating kinematic Jacobi," to dramatize that satisfying the duality is a real choice. At four points no such set exists, and the reason is instructive. The amplitude fixes only the two differences Eq. (61.46); any three numerators reproducing it differ from Eq. (61.69) by the shift Eq. (61.47), and that shift changes the Jacobi combination by $(s+t+u)\alpha = 0$. Thus every numerator set reproducing the four-gluon amplitude with these three propagators automatically satisfies $n_s + n_t + n_u = 0$: the Jacobi relation is gauge invariant and equal to zero (Theorem 61.13), so it cannot be switched off without changing the physics. The generic, Jacobi-violating numerators of Eq. (61.23) — a set $\{n_i^{\text{gen}}\}$ that reproduces the amplitude but has $\sum_i n_i^{\text{gen}} \neq 0$ — first appear at five points, where there is genuine room (nine independent gauge functions against nine constraints, Table 61.1) for the generalized gauge transformation to move violations around. Four points is the degenerate corner flagged in Section 61.2.5: the gauge freedom acts, but it can never spoil the duality it cannot help but satisfy.

Physical Insight

Why is $n_s + n_t + n_u = 0$ true, beyond the bracket algebra that proves it? The residue of $A[1,2,3,4]$ at $s \to 0$ factorizes the amplitude onto a pair of on-shell three-point amplitudes glued across the s-channel gluon, summed over its internal helicity. Those three-point amplitudes are the symmetry-fixed seeds Eq. (58.85) and Eq. (58.86), and the residue fixes n_s on the pole to be a product of them. The same holds in the t- and u-channels. Because the three-point amplitudes already realize a kinematic algebra — their products on the three channels are related by exactly the antisymmetry that the structure constants obey at a cubic vertex — the three channel residues conspire so that the polynomial remainder, which is what $J = n_s + n_t + n_u$ measures, is forced to vanish. The kinematic Jacobi identity is thus not an accident of the bracket manipulation in Theorem 61.13 but the four-point shadow of the kinematic algebra carried by the three-point amplitude. This is the deepest reason the duality holds, and the seed of its all-multiplicity and loop-level generalizations.

We now have what the next section needs: an explicit, duality-satisfying set of four-gluon numerators $\{n_s, n_t, n_u\}$ — compact in the gauge Eq. (61.64), crossing-symmetric in the gauge Eq. (61.70), and verified to satisfy $n_s + n_t + n_u = 0$ and to reproduce the gauge-theory amplitude. These numbers are the raw material for squaring. In Section 61.4 we replace the color factors c_i in Eq. (61.35) by a second copy of kinematic numerators $\tilde{n}_i$ and find that the result is a four-graviton amplitude; the verification of kinematic Jacobi carried out here is exactly the property that makes that replacement produce a consistent, gauge-invariant gravity amplitude rather than nonsense.

61.4 The Double Copy: Gravity as Gauge Squared

The previous section delivered a number the reader can hold: the four-gluon tree amplitude written over its three cubic channels with explicit, duality-satisfying numerators n_s, n_t, n_u in spinor-helicity, verified to obey the single kinematic Jacobi identity $n_s + n_t + n_u = 0$ (Section 61.3). Those numerators are the seed of the construction this section is built to perform. We have, in the gauge master formula Eq. (61.18), an amplitude assembled from three ingredients per cubic graph: a rigid color factor c_i, a pliable kinematic numerator n_i, and a propagator denominator D_i. Color–kinematics duality (Theorem 61.7) revealed that the n_i can be made to obey the very algebra the c_i obey. The move that defines the double copy is now almost forced by symmetry: if kinematics can stand in for color, then color can be *replaced* by a second copy of kinematics. We do exactly that, replace $c_i \mapsto \tilde{n}_i$, and we will prove that the resulting object is not a formal curiosity but a genuine scattering amplitude of gravitons — that "gravity is the

square of gauge theory" is a theorem about on-shell amplitudes, at least at tree level, demonstrated here in full at three and four points. Throughout we keep the mostly-minus metric, natural units, and the all-outgoing momentum convention of Section 61.1, with massless on-shell legs $p_i^2 = 0$ and Mandelstam invariants $s = (p_1 + p_2)^2$, $t = (p_2 + p_3)^2$, $u = (p_1 + p_3)^2$.

61.4.1 The double-copy substitution

We begin from the cubic master formula in the normalization fixed in Section 61.2,

$$\mathcal{A}_n^{\text{full}} \;=\; i\, g^{\,n-2} \sum_{i \in \Gamma_3} \frac{c_i\, n_i}{D_i}, \tag{61.71}$$

recalling Eq. (61.18): the sum runs over the $(2n-5)!!$ trivalent graphs, c_i is the product of structure constants read off graph i, D_i is the product of its internal inverse propagators, and n_i is its kinematic numerator, with the coupling g kept explicit out front so that n_i is a pure function of momenta and polarizations. The structure of Eq. (61.71) is a contraction of two objects over the index set of cubic graphs: one "leg" carries the gauge algebra (the c_i), the other carries the kinematics (the n_i/D_i). The denominators D_i are shared — they are pure topology — so the only genuinely distinct factors are color and kinematics.

Definition 61.15 (The double copy)**.** Let $\{n_i\}_{i \in \Gamma_3}$ be a set of duality-satisfying (BCJ) numerators for an n-point amplitude, in the sense of Theorem 61.7, and let $\{\tilde{n}_i\}_{i \in \Gamma_3}$ be a second set of kinematic numerators for the same set of cubic graphs, reproducing a (possibly different) gauge-theory amplitude through Eq. (61.71). The *double copy* is the object obtained by deleting the color factors of Eq. (61.71) and inserting the second numerator set in their place,

$$c_i \;\longmapsto\; \tilde{n}_i, \qquad \boxed{\mathcal{M}_n \;=\; i \left(\frac{\kappa}{2}\right)^{n-2} \sum_{i \in \Gamma_3} \frac{n_i\, \tilde{n}_i}{D_i}\,,} \tag{61.72}$$

where κ is a coupling of mass dimension -1 replacing the gauge coupling g, fixed by the three-point normalization of Section 61.4.3 below.

Three features of Eq. (61.72) deserve immediate comment, because each is a claim that must be earned rather than assumed. First, the coupling has changed character: a gauge amplitude carries $g^{\,n-2}$ with g dimensionless in four dimensions, whereas Eq. (61.72) carries $(\kappa/2)^{\,n-2}$ with κ dimensionful. We verify in Section 61.4.3 that κ has exactly the dimension of the gravitational coupling, $\kappa^2 = 32\pi G_N$, and that this is forced, not chosen: the mass dimension of Eq. (61.72) comes out to that of a graviton amplitude precisely because two kinematic numerators carry twice the momentum-dimension of one color factor (which is dimensionless). Second, the formula is asymmetric in how

much it demands of its two inputs: $\{n_i\}$ must be BCJ, while $\{\tilde{n}_i\}$ may be *any* representation of its amplitude, in any generalized gauge. This asymmetry is the central economy of the double copy and the subject of Section 61.4.2. Third, the output is asserted to be a gravity amplitude; that the $\mathcal{M}_n$ so defined factorizes correctly on its poles into lower-point gravity amplitudes — the test of whether an object is a bona fide amplitude at all — is proved in Section 61.4.5.

Physical Insight

The replacement $c_i \mapsto \tilde{n}_i$ is the whole idea in one stroke, and it is worth pausing on why it is even syntactically possible. In Eq. (61.71) the color factors enter *only* through their algebraic relations: the amplitude is invariant under any reshuffling of the c_i consistent with the Jacobi identity Eq. (61.21) (this was Theorem 61.10). Color enters not as specific numbers but as a representation of an algebra. A BCJ numerator set is a second representation of the *same* algebra. So replacing one by the other is replacing one realization of the Jacobi structure by another — the formula cannot tell the difference at the level of the relations it respects. What changes is the *value*: the kinematic copy carries Lorentz indices and momenta where the color copy carried group indices, and the contraction of two kinematic copies builds an object with twice the little-group weight on every leg. Twice spin one is spin two. The graviton is hiding in the contraction of two gluons.

61.4.2 The central consistency theorem: only one copy must be BCJ

A scattering amplitude is a single function of the external data; it cannot depend on the bookkeeping used to assemble it. For Eq. (61.72) to define an amplitude, its value must not change when we re-express either numerator set in a different generalized gauge. We now prove the sharp form of this requirement: the double copy is invariant under any generalized gauge transformation of the *second* (un-tilded-or-tilded, it does not matter which we fix) factor, provided the *other* factor satisfies the duality. The proof is the exact kinematic mirror of the color-gauge-invariance proof of Theorem 61.10: where that argument used the color Jacobi identity $c_i + c_j + c_k = 0$, this one uses the kinematic Jacobi identity $n_i + n_j + n_k = 0$.

Theorem 61.16 (Gauge independence of the double copy)**.** *Suppose $\{n_i\}$ is a BCJ representation, satisfying the kinematic Jacobi and antisymmetry relations Eq. (61.23) on every triplet. Then the double-copy amplitude Eq. (61.72) is unchanged under any generalized gauge transformation of the second numerator set,*

$$\tilde{n}_i \;\longmapsto\; \tilde{n}_i + D_i\,\tilde{\Delta}_i, \tag{61.73}$$

with $\tilde{\Delta}_i$ the shifts of a generalized gauge transformation Eq. (61.25) of the gauge theory whose numerators are $\tilde{n}_i$ — that is, $\tilde{\Delta}_i$ ranging over the amplitude-preserving directions $\sum_i c_i \tilde{\Delta}_i = 0$ of Eq. (61.26). Consequently $\mathcal{M}_n$ depends on $\{\tilde{n}_i\}$ only through the gauge-theory amplitude it represents, and the second copy may be supplied in any gauge whatsoever; it need not satisfy color–kinematics duality.

Proof. Insert the shifted numerators Eq. (61.73) into Eq. (61.72). As in Eq. (61.28), the explicit factor D_i in the shift cancels the propagator denominator, leaving a pole-free remainder:

$$\mathcal{M}_n \;\longmapsto\; i\Big(\tfrac{\kappa}{2}\Big)^{n-2} \sum_i \frac{n_i\,(\tilde{n}_i + D_i \tilde{\Delta}_i)}{D_i} \;=\; \mathcal{M}_n + i\Big(\tfrac{\kappa}{2}\Big)^{n-2} \sum_{i \in \Gamma_3} n_i\, \tilde{\Delta}_i. \tag{61.74}$$

The variation is therefore

$$\delta \mathcal{M}_n \;=\; i\Big(\tfrac{\kappa}{2}\Big)^{n-2} \sum_{i \in \Gamma_3} n_i\, \tilde{\Delta}_i. \tag{61.75}$$

This is the precise analogue of the gauge-theory variation Eq. (61.27), $\delta \mathcal{A}_n^{\text{full}} = i g^{n-2} \sum_i c_i \Delta_i$, with the color factor c_i replaced by the BCJ numerator n_i. We must show Eq. (61.75) vanishes for every admissible $\tilde{\Delta}_i$. The admissible shifts are, by definition, those obeying the constraint $\sum_i c_i \tilde{\Delta}_i = 0$ of Eq. (61.26); the completeness argument of Theorem 61.10 established that the space of such shifts is spanned by the *elementary moves* — the single-triplet move and the single-flip move.

Elementary triplet move. For a Jacobi triplet $\{i, j, k\}$, the elementary move Eq. (61.20) sets $\tilde{\Delta}_i = \tilde{\Delta}_j = \tilde{\Delta}_k = \delta$ (a common function) with $\tilde{\Delta}_m = 0$ outside the triplet. Its contribution to Eq. (61.75) is

$$\sum_m n_m\, \tilde{\Delta}_m \;=\; \delta\,(n_i + n_j + n_k) \;=\; 0, \tag{61.76}$$

which vanishes by the *kinematic* Jacobi relation Eq. (61.23) — exactly where, in Eq. (61.20), the color Jacobi relation Eq. (61.21) made $\sum_m c_m \tilde{\Delta}_m$ vanish. This is the single algebraic fact on which the entire double copy turns: because $\{n_i\}$ is BCJ, the kinematic numerators sum to zero on the same triplets that the color factors do, so they kill the gauge variation of the partner copy by the identical mechanism.

Elementary flip move. For a graph i and its vertex-flip $\bar{\imath}$, the flip move sets $\tilde{\Delta}_i = \tilde{\Delta}_{\bar{\imath}} = \delta$ and all others zero. Its contribution is $\delta\,(n_i + n_{\bar{\imath}}) = 0$ by the kinematic antisymmetry $n_{\bar{\imath}} = -n_i$ of Eq. (61.24), again the mirror of the color antisymmetry Eq. (61.22) used in Theorem 61.10.

Since every admissible $\tilde{\Delta}$ is a superposition of elementary triplet and flip moves, and each annihilates Eq. (61.75) by one use of the kinematic Jacobi or

antisymmetry relation, the full variation vanishes: $\delta\mathcal{M}_n = 0$. The double copy is independent of the generalized gauge of the second copy. Because the second copy enters Eq. (61.72) only through its gauge-equivalence class, and that class is precisely the gauge-theory amplitude it represents, $\mathcal{M}_n$ depends on the second input only through that amplitude. In particular the second copy is not required to be BCJ. □

This is the result that makes the double copy usable. Constructing a BCJ representation is work — it is the content of Theorem 61.12, easy at four points but a genuine computation beyond. Theorem 61.16 says this work need be done only *once*: hold one factor in a duality-satisfying form, and the other may be the rawest output of the Feynman rules, in any gauge, with no Jacobi relations imposed at all. One pays the price of the duality a single time and squares freely thereafter.

Caution

The asymmetry is essential and is easy to misstate. It is *not* true that the double copy of two arbitrary representations is gauge invariant: if neither $\{n_i\}$ nor $\{\tilde{n}_i\}$ satisfies the duality, the variation Eq. (61.75) does not vanish — $\sum_i n_i\tilde{\Delta}_i$ is then a nonzero polynomial, and $\mathcal{M}_n$ depends on which gauge one happened to use, so it is not a well-defined amplitude. At least one of the two numerator sets must be BCJ. With one BCJ, the other is free; with neither, the construction is meaningless. This is the precise sense in which color–kinematics duality is the *prerequisite* for the double copy rather than a decoration of it.

61.4.3 Graviton kinematics by squaring, and the three-point amplitude

To test Eq. (61.72) we need something to test it against: independent expressions for graviton amplitudes. This is the one genuine connection of the chapter. This book never constructs the perturbative Feynman rules of the Einstein–Hilbert action — an infinite tower of vertices of escalating index complexity that we have no need to write. Instead we define graviton amplitudes the way Chapters 58 and 59 defined gluon amplitudes: *on shell*, through the little-group transformation of the external states and the residues forced by locality and unitarity. The graviton is the massless particle of helicity ± 2, and everything we need about it follows from doubling the helicity-± 1 data of the gluon.

Graviton polarization tensors. Recall (Eq. (58.5)) that a massless leg of helicity h transforms under the little group $\mathrm{U}(1)$ as $\mathcal{A} \to t^{-2h}\mathcal{A}$ when its spinors are rescaled $(\lambda, \tilde{\lambda}) \to (t\lambda, t^{-1}\tilde{\lambda})$, and that the gluon polarization vectors (Eq. (58.111))

carry weight ∓ 1 for helicity ± 1, with the explicit reference-dependent forms recalled in Eq. (59.32). A spin-two polarization is built by tensoring two spin-one ones.

Definition 61.17 (Graviton polarization tensor)**.** For a massless momentum p with helicity-± 1 gluon polarization vectors $\varepsilon_\pm^\mu(p)$, the helicity-± 2 graviton polarization tensor is the symmetric product

$$\varepsilon_\pm^{\mu\nu}(p) \;\equiv\; \varepsilon_\pm^{\mu}(p)\,\varepsilon_\pm^{\nu}(p). \tag{61.77}$$

This object has exactly the properties required of a graviton wavefunction, and each is inherited directly from the gluon polarizations.

- *Symmetric:* $\varepsilon^{\mu\nu} = \varepsilon^{\nu\mu}$ trivially, since the two factors are equal.
- *Transverse:* $p_\mu \varepsilon_\pm^{\mu\nu} = (p\cdot\varepsilon_\pm)\varepsilon_\pm^\nu = 0$, because each gluon polarization is transverse, $p\cdot\varepsilon_\pm = 0$ (Eq. (59.33)).
- *Traceless:* $\eta_{\mu\nu}\varepsilon_\pm^{\mu\nu} = \varepsilon_\pm\cdot\varepsilon_\pm = 0$, because a polarization vector of definite helicity is null — two like-helicity gluon polarizations have vanishing dot product (Theorem 59.8, the same fact that killed two of three terms in the three-point computation Eq. (59.108)).
- *Little-group weight* ± 2*:* under $(\lambda, \tilde\lambda) \to (t\lambda, t^{-1}\tilde\lambda)$ the vector $\varepsilon_\pm^\mu$ scales as $t^{\mp 2}$, so $\varepsilon_\pm^{\mu\nu}$ scales as $t^{\mp 4} = t^{-2(\pm 2)}$, the weight of a helicity-± 2 state.

A symmetric, transverse, traceless rank-two tensor with five (in $d = 4$, two physical) on-shell components is exactly a linearized metric fluctuation $h_{\mu\nu}$ — a graviton. The construction makes no reference to the Einstein–Hilbert action; it uses only the representation theory of the massless little group, already in hand from Chapter 58.

The on-shell definition of a graviton amplitude. We now state what a graviton amplitude *is*, in the on-shell sense this book uses for all massless amplitudes.

Definition 61.18 (On-shell graviton amplitude)**.** An n-graviton tree amplitude $\mathcal{M}_n$ is a Lorentz-invariant rational function of the external spinors $\{\lambda_i, \tilde\lambda_i\}$ that (i) carries little-group weight $-2h_i = \mp 4$ on each leg i of helicity ± 2; (ii) has only local poles, at $P_I^2 \to 0$ for sums $P_I = \sum_{i\in I} p_i$ of external momenta; and (iii) factorizes on each such pole into a product of lower-point graviton amplitudes summed over the two physical helicities of the intermediate graviton,

$$\mathcal{M}_n \;\xrightarrow{P_I^2\to 0}\; \sum_{h=\pm 2} \frac{\mathcal{M}_L(\dots, P_I^h)\,\mathcal{M}_R(-P_I^{-h}, \dots)}{P_I^2}. \tag{61.78}$$

These are the same three conditions — correct little-group weight, locality, and unitary factorization — that characterize gluon amplitudes and that the BCFW recursion (Theorem 59.23) showed are enough to determine them. They determine graviton amplitudes as well; in particular the three-point amplitude is fixed up to a constant by little-group weight and locality alone, exactly as in Section 59.4.3, and the higher points follow by recursion. We will not run gravitational BCFW here; we will instead produce the amplitudes from the double copy and *check* them against Theorem 61.18.

The three-graviton amplitude as a square. The smallest test is the three-point amplitude, and it is the test that fixes κ. The Yang–Mills three-gluon amplitude, with coupling and phase pinned down in Eq. (59.116), is

$$A_3^{\mathrm{YM}}(1^-,2^-,3^+) = i\,g\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \qquad A_3^{\mathrm{YM}}(1^+,2^+,3^-) = -\,i\,g\,\frac{[1\,2]^3}{[2\,3]\,[3\,1]}. \tag{61.79}$$

At three points there is a single cubic graph with no internal line, so $D = 1$ and the numerator *is* the colour-stripped amplitude: comparing Eq. (61.71) at $n = 3$, $\mathcal{A}_3^{\mathrm{full}} = ig\,f^{a_1a_2a_3}n_3$, with the colour decomposition, the bare numerator is the ratio

$$n_3(1^-,2^-,3^+) = \frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \tag{61.80}$$

a pure kinematic object carrying neither i nor g. The single Jacobi/antisymmetry relation is trivially met (one graph), so Eq. (61.80) is BCJ by default, and the double copy Eq. (61.72) at $n = 3$ reads $\mathcal{M}_3 = i(\kappa/2)\,n_3\tilde{n}_3$. Taking both copies to be pure Yang–Mills, $\tilde{n}_3 = n_3$,

$$\mathcal{M}_3(1^{--},2^{--},3^{++}) = i\,\frac{\kappa}{2}\,\frac{\langle 1\,2\rangle^6}{\langle 2\,3\rangle^2\,\langle 3\,1\rangle^2}, \tag{61.81}$$

where the helicity labels ∓ 2 are written $\mp\mp$ to mark the doubling. We read off three checks at once. The mass dimension: a three-point amplitude has $[\mathcal{M}_3] = 4 - 3 = 1$, and the spinor ratio in Eq. (61.81) has dimension $6 - 4 = 2$ in mass (each bracket is dimension one), so the prefactor must supply dimension -1 — which is exactly $[\kappa] = -1$. This is how Theorem 61.15 "fixes κ by the three-point normalization": the power $(\kappa/2)^{n-2}$ is forced to be one factor of a dimension-(-1) coupling at three points, identifying κ with the gravitational coupling $\kappa = \sqrt{32\pi G_N}$ once Eq. (61.81) is matched to the linearized Einstein–Hilbert three-vertex (a matching we quote rather than perform, having declined to build those vertices). The little-group weight: leg 1 appears as $\langle 1\,2\rangle^6\langle 3\,1\rangle^{-2}$, weight $t_1^{6-2} = t_1^4 = t_1^{-2(-2)}$, correct for helicity -2; the other legs check identically.

And the relation to gauge theory is now explicit and is the headline of the chapter:

$$\boxed{\mathcal{M}_3 \;=\; \frac{\kappa/2}{i\,g^2}\,(A_3^{\mathrm{YM}})^2} \qquad \Longleftrightarrow \qquad i\,\frac{\kappa}{2}\,\frac{\langle 1\,2\rangle^6}{\langle 2\,3\rangle^2\langle 3\,1\rangle^2} = \frac{\kappa/2}{i\,g^2}\Big(i\,g\,\frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\langle 3\,1\rangle}\Big)^2, \tag{61.82}$$

the right-hand side reproducing the left because $(ig)^2 = -g^2$ and $(\kappa/2)/(ig^2)\cdot(-g^2) = i\kappa/2$. The three-graviton amplitude is the three-gluon amplitude squared, with the color stripped and the coupling traded $g^2 \to \kappa$. The polarization picture matches the algebra: with $\varepsilon^{\mu\nu} = \varepsilon^\mu\varepsilon^\nu$, every contraction in the graviton vertex is a product of two gluon-vertex contractions, so the amplitude is the product of two copies. "GR = YM $\otimes$ YM" is, at three points, the literal statement Eq. (61.82).

61.4.4 The four-graviton amplitude from the four-gluon square

The four-point amplitude is the first case with a propagator, hence the first genuine test of the sum structure of Eq. (61.72). We compute it from the verified four-gluon BCJ numerators of Section 61.3 and compare to the graviton amplitude fixed independently by little-group weight and factorization.

Worked Example 61.19 (The four-graviton MHV amplitude as gauge squared)**.** *Setup.* At four points there are three cubic graphs — the s-, t-, and u-channels of Fig. 61.1 — with denominators $D_s = s$, $D_t = t$, $D_u = u$ obeying $s+t+u=0$ (Eq. (61.38)). The double copy Eq. (61.72) at $n=4$, with both copies pure Yang–Mills so that $\tilde{n}_i = n_i$, is

$$\mathcal{M}_4 \;=\; i\left(\frac{\kappa}{2}\right)^2\left(\frac{n_s^2}{s}+\frac{n_t^2}{t}+\frac{n_u^2}{u}\right). \tag{61.83}$$

We take the MHV configuration $(1^-,2^-,3^+,4^+)$ and the BCJ numerators of Section 61.3, which satisfy the kinematic Jacobi identity

$$n_s + n_t + n_u = 0 \tag{61.84}$$

and reproduce the color-ordered partial amplitudes. In the master-formula normalization of Eq. (61.71), eliminating $c_t = -c_s - c_u$ and reading off the coefficients of the two independent color factors c_s, c_u (the Del Duca–Dixon–Maltoni partials of Eq. (61.2)) gives the numerator-to-partial dictionary

$$A_4[1,2,3,4] = i\Big(\frac{n_s}{s}-\frac{n_t}{t}\Big), \qquad A_4[1,3,2,4] = i\Big(\frac{n_t}{t}-\frac{n_u}{u}\Big). \tag{61.85}$$

The color-ordered partials are the Parke–Taylor values (Eq. (59.55)), which for this helicity assignment are

$$A_4[1,2,3,4] = i\,P, \quad P = \frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \qquad A_4[1,3,2,4] = i\,Q, \quad Q = \frac{\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4} \tag{61.86}$$

the coupling g^2 stripped, the leading i displayed. (The exchange $A_4[1,2,3,4]$ versus $A_4[1,3,2,4]$ is the swap of legs 2 and 3 in the cyclic denominator.)

Evaluating in a convenient gauge. The variation argument of Theorem 61.16 guarantees that Eq. (61.83) is invariant under $n_i \to n_i + D_i\alpha$ for any α, since the variation

$$\delta\mathcal{M}_4 = i\Big(\tfrac{\kappa}{2}\Big)^2\Big(2\alpha\sum_i n_i + \alpha^2\sum_i D_i\Big) = i\Big(\tfrac{\kappa}{2}\Big)^2(2\alpha\cdot 0 + \alpha^2(s+t+u)) = 0 \tag{61.87}$$

vanishes term by term: the first by the Jacobi identity Eq. (61.84), the second by $s+t+u=0$. (This is the four-point shadow of the general theorem; here both kinematic Jacobi and the massless kinematic identity conspire, which is the degeneracy noted in Section 61.2.5.) We may therefore evaluate Eq. (61.83) in the gauge $n_t = 0$, in which Eq. (61.85) collapses to

$$\frac{n_s}{s} = P, \qquad \frac{n_u}{u} = -Q, \qquad n_t = 0, \qquad \text{i.e.} \qquad n_s = s\,P, \quad n_u = -u\,Q. \tag{61.88}$$

The Jacobi identity Eq. (61.84) in this gauge reads $sP - uQ = 0$, i.e.

$$s\,P \;=\; u\,Q, \tag{61.89}$$

which is precisely the fundamental BCJ amplitude relation $s\,A_4[1,2,3,4] = u\,A_4[1,3,2,4]$ at four points — the gauge-invariant content the duality reduced to (Section 61.2.5). We confirm it directly below; for now it is the input that makes the gauge $n_t = 0$ consistent.

Assembling the double copy. Substituting Eq. (61.88) into Eq. (61.83),

$$\mathcal{M}_4 = i\Big(\tfrac{\kappa}{2}\Big)^2\Big(\frac{(sP)^2}{s} + 0 + \frac{(uQ)^2}{u}\Big) = i\Big(\tfrac{\kappa}{2}\Big)^2(s\,P^2 + u\,Q^2). \tag{61.90}$$

Use Eq. (61.89) to write $Q = sP/u$, so $u\,Q^2 = u\cdot s^2P^2/u^2 = s^2P^2/u$, and combine the two terms over u:

$$s\,P^2 + u\,Q^2 = s\,P^2\Big(1 + \frac{s}{u}\Big) = s\,P^2\,\frac{u+s}{u} = -\,\frac{s\,t}{u}\,P^2, \tag{61.91}$$

the last step using $u + s = -t$ from $s+t+u=0$. Hence

$$\mathcal{M}_4 = -\,i\Big(\tfrac{\kappa}{2}\Big)^2\frac{s\,t}{u}\,P^2, \qquad P^2 = \frac{\langle 1\,2\rangle^6}{\langle 2\,3\rangle^2\langle 3\,4\rangle^2\langle 4\,1\rangle^2}, \tag{61.92}$$

where we used $P = \langle 1\,2\rangle^3/(\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$ after cancelling one power of $\langle 1\,2\rangle$.

To the crossing-symmetric form. Equation Eq. (61.92) is the answer, but its lopsided appearance hides the full permutation symmetry a graviton amplitude must possess (gravitons are identical bosons; the amplitude is symmetric under any relabeling). We make the symmetry manifest with one spinor identity. From

momentum conservation $\sum_i \lambda_i\tilde\lambda_i = 0$, sandwiching between $\langle 1|$ and $|3]$ kills the $i = 1, 3$ terms and gives $\langle 1\,4\rangle[4\,3] = -\langle 1\,2\rangle[2\,3]$; equivalently, sandwiching gives the relation we need in the form

$$\langle 4\,1\rangle\,[3\,4] \;=\; -\,\langle 1\,2\rangle\,[2\,3]. \tag{61.93}$$

Together with $\langle 3\,4\rangle[3\,4] = -(p_3{+}p_4)^2 = -s$ (since $s_{34} = s_{12} = s$ and $\langle 3\,4\rangle[4\,3] = s$, $[3\,4] = -[4\,3]$) and $s = \langle 1\,2\rangle[2\,1]$, $t = \langle 2\,3\rangle[3\,2]$, a short computation establishes the identity

$$s\,t\,\langle 1\,2\rangle \;=\; -\,[3\,4]^2\,\langle 2\,3\rangle\,\langle 3\,4\rangle\,\langle 4\,1\rangle. \tag{61.94}$$

Indeed, dividing Eq. (61.94) by $\langle 2\,3\rangle$ and using $st\langle 1\,2\rangle/\langle 2\,3\rangle = \langle 1\,2\rangle^2[2\,1][3\,2]$ (from $s = \langle 1\,2\rangle[2\,1]$, $t = \langle 2\,3\rangle[3\,2]$), the claim is $\langle 1\,2\rangle^2[2\,1][3\,2] = -[3\,4]^2\langle 3\,4\rangle\langle 4\,1\rangle$; with $[3\,4]\langle 3\,4\rangle = -s = -\langle 1\,2\rangle[2\,1]$ and $\langle 4\,1\rangle[3\,4] = -\langle 1\,2\rangle[2\,3] = \langle 1\,2\rangle[3\,2]$ from Eq. (61.93), the right side becomes $-([3\,4]\langle 3\,4\rangle)([3\,4]\langle 4\,1\rangle) = -(-\langle 1\,2\rangle[2\,1])(\langle 1\,2\rangle[3\,2]) = \langle 1\,2\rangle^2[2\,1][3\,2]$, matching the left. Squaring Eq. (61.94),

$$s^2t^2\,\langle 1\,2\rangle^2 = [3\,4]^4\,\langle 2\,3\rangle^2\langle 3\,4\rangle^2\langle 4\,1\rangle^2, \tag{61.95}$$

and therefore

$$\frac{s\,t}{u}\,P^2 = \frac{s\,t}{u}\,\frac{\langle 1\,2\rangle^6}{\langle 2\,3\rangle^2\langle 3\,4\rangle^2\langle 4\,1\rangle^2} = \frac{1}{u}\,\frac{s^2t^2\langle 1\,2\rangle^6}{s\,t\,\langle 2\,3\rangle^2\langle 3\,4\rangle^2\langle 4\,1\rangle^2} = \frac{\langle 1\,2\rangle^4\,[3\,4]^4}{s\,t\,u}, \tag{61.96}$$

where Eq. (61.95) turned the angle-bracket denominator into $[3\,4]^4$ over one power each of s, t. Substituting into Eq. (61.92), the four-graviton MHV amplitude is

$$\boxed{\mathcal{M}_4(1^{--},2^{--},3^{++},4^{++}) = -\,i\left(\frac{\kappa}{2}\right)^2\,\frac{\langle 1\,2\rangle^4\,[3\,4]^4}{s\,t\,u}\,.} \tag{61.97}$$

Verification of the BCJ relation Eq. (61.89). The gauge $n_t = 0$ relied on $sP = uQ$. With P, Q from Eq. (61.86),

$$s\,P - u\,Q = \langle 1\,2\rangle^4\langle 4\,1\rangle^{-1}\Big(\frac{[2\,1]}{\langle 2\,3\rangle\langle 3\,4\rangle} - \frac{[3\,1]}{\langle 3\,2\rangle\langle 2\,4\rangle}\Big), \tag{61.98}$$

using $s = \langle 1\,2\rangle[2\,1]$, $u = \langle 1\,3\rangle[3\,1]$ and cancelling $\langle 1\,2\rangle$ in the first term, $\langle 1\,3\rangle$ in the second. The bracket combination is, over the common denominator $\langle 2\,3\rangle\langle 3\,4\rangle\langle 2\,4\rangle$ (with $\langle 3\,2\rangle = -\langle 2\,3\rangle$),

$$\frac{[2\,1]\langle 2\,4\rangle + [3\,1]\langle 3\,4\rangle}{\langle 2\,3\rangle\langle 3\,4\rangle\langle 2\,4\rangle}. \tag{61.99}$$

The numerator is $\langle 4|(p_2 + p_3)|1] = \langle 4|(-p_1 - p_4)|1] = 0$, because $\langle 4|p_1|1] = \langle 4\,1\rangle[1\,1] = 0$ and $\langle 4|p_4|1] = \langle 4\,4\rangle[4\,1] = 0$. Hence $sP - uQ = 0$, confirming Eq. (61.89) and the consistency of the gauge choice.

The result Eq. (61.97) is the four-graviton amplitude, and it passes every on-shell test of Theorem 61.18. Mass dimension: $[\mathcal{M}_4] = 4 - 4 = 0$; the spinor ratio has dimension $8-6 = 2$, supplied with dimension -2 by $(\kappa/2)^2$, consistent. Little-group weight on leg 1 (helicity -2): $\langle 1\,2\rangle^4$ contributes t_1^4 and the denominator $s = \langle 1\,2\rangle[2\,1]$ contributes t_1 via $\langle 1\,2\rangle$ but t_1^{-1} via $[2\,1]$, net zero, while u and t each contribute a compensating t_1 and t_1^{-1} from the brackets containing leg 1; carrying the count through all of $\langle 1\,2\rangle^4/(stu)$ gives net $t_1^4 = t_1^{-2(-2)}$, the weight of helicity -2. Bose symmetry: Eq. (61.97) is manifestly symmetric under $1 \leftrightarrow 2$ and under $3 \leftrightarrow 4$, and the full S_4 symmetry (with helicity labels carried along) follows from the crossing relations among the Mandelstam invariants exactly as for the gluon partials. And the amplitude has only the physical poles s, t, u, each simple — the hallmark of locality. The factorization on these poles is the subject of the next subsection.

Physical Insight

The single equation Eq. (61.97) is the payoff of the chapter made arithmetic. Set beside the gauge-theory partial Eq. (61.86), $A_4[1,2,3,4] = i\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$, the graviton amplitude is recognizably its square: the numerator $\langle 1\,2\rangle^4$ has become $\langle 1\,2\rangle^4[3\,4]^4$ (an angle factor times a square-bracket factor, the two "copies"), and the color-ordered chain of angle brackets in the denominator has become the permutation-symmetric $s\,t\,u$. Concretely, Eq. (61.97) can be written $\mathcal{M}_4 = -i(\kappa/2)^2\,s\,t\,u\,(A_4[1,2,3,4]/i)\,(\tilde{A}_4[1,2,4,3]/i)/(stu)$ — a product of two color-ordered gauge amplitudes, one in each ordering, glued by the Mandelstam "momentum kernel" s. That product form is the Kawai–Lewellen–Tye relation, the subject of the next section; here it has emerged unbidden from squaring numerators, which is the double copy's claim that the two presentations are one.

61.4.5 Correct factorization: locality and unitarity inherited

An expression with the right symmetry and dimension is not yet an amplitude. The defining physical requirement is unitary factorization Eq. (61.78): on every physical pole the residue must be a sum of products of lower-point amplitudes. We prove that the double copy Eq. (61.72) satisfies this automatically, and that the gravity factorization is the gauge factorization squared. This both certifies $\mathcal{M}_n$ as a genuine gravity amplitude and exhibits the deep reason the construction works: locality and unitarity are not imposed on the gravity side but inherited from the gauge inputs.

Theorem 61.20 (Factorization of the double copy)**.** *Let $\{n_i\}$ be BCJ and $\{\tilde{n}_i\}$ any representation, both reproducing gauge-theory amplitudes that factorize*

correctly on their poles (as gauge tree amplitudes do, Theorem 59.23). Then on each physical channel $P_I^2 \to 0$ the double copy Eq. (61.72) factorizes as

$$\mathcal{M}_n \xrightarrow{P_I^2\to 0} \frac{1}{P_I^2}\sum_{h=\pm 2} \mathcal{M}_L(\dots, P_I^h)\,\mathcal{M}_R(-P_I^{-h},\dots), \tag{61.100}$$

a sum of products of lower-point double-copy (graviton) amplitudes over the two graviton helicities crossing the cut. Hence $\mathcal{M}_n$ satisfies Theorem 61.18 and is a bona fide gravity amplitude.

Proof. Fix a channel I and its momentum $P_I = \sum_{i\in I} p_i$. A simple pole at $P_I^2 \to 0$ in Eq. (61.72) arises only from cubic graphs that contain an internal line carrying P_I — call this set Γ_I. For $i \in \Gamma_I$ write $D_i = P_I^2\, D_i'$, where D_i' is the product of the remaining internal propagators, finite as $P_I^2 \to 0$. The residue of $\mathcal{M}_n$ on the pole is

$$\operatorname{Res}_{P_I^2\to 0}\mathcal{M}_n = i\Big(\tfrac{\kappa}{2}\Big)^{n-2}\sum_{i\in\Gamma_I}\frac{n_i\,\tilde n_i}{D_i'}\Big|_{P_I^2=0}. \tag{61.101}$$

Each graph $i \in \Gamma_I$ is a tree split by its P_I-line into a left subgraph L (legs I and the internal line) and a right subgraph R (legs $\bar I$ and the internal line); summing over $i \in \Gamma_I$ is summing over all cubic graphs L on the left times all cubic graphs R on the right, with $D_i' = D_L\, D_R$.

The key input is that the kinematic numerators *factorize on the pole*. This is a property of the numerators produced by the Feynman rules and preserved by the contact absorption of Theorem 61.4: on $P_I^2 = 0$ the internal line goes on shell, its propagator numerator becomes the physical polarization sum, and the diagram's kinematic factor splits into left and right pieces. Concretely, on the pole the gluon numerator factorizes as

$$n_i\big|_{P_I^2=0} \;=\; \sum_{\ell=\pm 1} n_L(\dots, P_I^\ell)\, n_R(-P_I^{-\ell},\dots), \tag{61.102}$$

the sum running over the two physical gluon helicities of the intermediate line, with n_L, n_R the lower-point numerators of the subgraphs — this is exactly the statement that the gauge amplitude factorizes (Theorem 59.23), transcribed to the numerator level, since on the pole $c_i \to c_L c_R$ and $D_i' \to D_L D_R$ so that $\sum_i c_i n_i/D_i' \to (\sum_L c_L n_L/D_L)(\sum_R c_R n_R/D_R)$ reproduces $A_L \times A_R$. The second copy factorizes identically,

$$\tilde n_i\big|_{P_I^2=0} \;=\; \sum_{\tilde\ell=\pm 1} \tilde n_L(\dots, P_I^{\tilde\ell})\, \tilde n_R(-P_I^{-\tilde\ell},\dots). \tag{61.103}$$

Insert Eqs. (61.102) and (61.103) into the residue Eq. (61.101). The double sum over graphs becomes a product of a left sum and a right sum, and the two

helicity sums $\ell, \tilde{\ell}$ run independently:

$$\begin{aligned}\operatorname{Res}_{P_I^2\to 0}\mathcal{M}_n &= i\Big(\tfrac{\kappa}{2}\Big)^{n-2}\sum_{\ell,\tilde{\ell}=\pm 1}\Big(\sum_L \frac{n_L(P_I^{\ell})\,\tilde{n}_L(P_I^{\tilde{\ell}})}{D_L}\Big)\Big(\sum_R \frac{n_R(P_I^{\ell})\,\tilde{n}_R(P_I^{\tilde{\ell}})}{D_R}\Big)\\ &= \sum_{\ell,\tilde{\ell}=\pm 1}\mathcal{M}_L(\dots, P_I^{(\ell,\tilde{\ell})})\,\mathcal{M}_R(-P_I^{(\ell,\tilde{\ell})},\dots),\end{aligned}\tag{61.104}$$

where in the second line each parenthesized sum is, by Eq. (61.72) at lower multiplicity, a double-copy amplitude on the corresponding side, and the $(\kappa/2)$ powers distribute correctly because $(n-2) = (n_L-2)+(n_R-2)+1$ counts the internal line once. The intermediate state is now labelled by the *pair* $(\ell,\tilde{\ell})$ of gluon helicities, one from each copy — exactly the tensor-product structure of Theorem 61.17. The four combinations organize by total helicity:

$$(\ell,\tilde{\ell}) = (+,+) \;\to\; h=+2, \qquad (-,-) \;\to\; h=-2, \qquad (+,-)\text{ and }(-,+) \;\to\; h=0. \tag{61.105}$$

The two "diagonal" terms $h=\pm 2$ are the graviton states; the polarization tensor crossing the cut is $\varepsilon_\pm^\mu\varepsilon_\pm^\nu = \varepsilon_\pm^{\mu\nu}$, the helicity-$\pm 2$ graviton of Eq. (61.77). These reproduce precisely the right-hand side of Eq. (61.100). The two "off-diagonal" terms $h=0$ are the intermediate states $\varepsilon_+^\mu\varepsilon_-^\nu$ and $\varepsilon_-^\mu\varepsilon_+^\nu$, whose symmetric trace part is a scalar (the dilaton) and whose antisymmetric part is a two-form (the Kalb–Ramond field): they are the additional states of the double-copy spectrum, present whenever the two gauge copies are pure Yang–Mills. For the pure-graviton factorization Eq. (61.78) one restricts the external states (and hence, on the cut, the intermediate states) to the symmetric-traceless graviton; the diagonal terms then furnish exactly Eq. (61.100).

In every case the residue Eq. (61.104) is a sum of products of lower-point double-copy amplitudes, divided by the single propagator $1/P_I^2$ exhibited in Eq. (61.101). The poles are precisely the physical channels P_I^2, and they are simple. Locality (only physical poles) and unitarity (residues that factorize into lower amplitudes) hold for $\mathcal{M}_n$ because they hold for the gauge inputs and the squaring respects the channel structure. By induction on n — the three-point amplitude Eq. (61.81) is the base case, manifestly an amplitude — $\mathcal{M}_n$ satisfies all conditions of Theorem 61.18. □

Theorem 61.20 closes the logical circle. The double copy was defined by an algebraic substitution; Theorem 61.16 showed the substitution produces a well-defined (gauge-independent) function; and Theorem 61.20 now shows that function is a genuine gravity amplitude, with all the analytic structure that title requires. The four-point check is the theorem in miniature: Eq. (61.97) has simple poles at s,t,u, and one verifies that the residue at, say, $s\to 0$ is the product of two three-graviton amplitudes Eq. (61.81) glued on the s-channel graviton, matching Eq. (61.100) term by term.

Physical Insight

The factorization proof exposes why the double copy is more than a computational trick. A graviton amplitude is forced, by unitarity, to factorize on its poles into graviton amplitudes; the intermediate graviton's polarization sum is the spin-two projector. The double copy supplies that projector as a *product* of two spin-one (gluon) projectors, one per copy — which is exactly what $\varepsilon^{\mu\nu} = \varepsilon^{\mu}\tilde{\varepsilon}^{\nu}$ does to the state sum. Gravity's unitarity is gauge theory's unitarity, squared. Conversely, the appearance of the dilaton and two-form in Eq. (61.105) is not a flaw but a truth the construction cannot hide: the naive square of two pure-gluon theories is not pure Einstein gravity but gravity coupled to a dilaton and an antisymmetric tensor — the massless sector of the closed string. Which theory the double copy actually computes is the question this forces, and we take it up after the KLT detour.

With the double copy established as a producer of bona fide gravity amplitudes — gauge invariant by Theorem 61.16, correctly factorizing by Theorem 61.20, and matched to independently fixed graviton amplitudes at three (Eq. (61.81)) and four (Eq. (61.97)) points — two questions remain. The first is structural: why does replacing color by kinematics yield gravity at all, and what is the precise relation to the string-theoretic origin of "gravity = gauge^2"? That is the Kawai–Lewellen–Tye story of the next section, where the product form glimpsed in Eq. (61.97) is derived from first principles and shown equivalent to the double copy. The second is physical: *which* gravity theory have we computed? The state sum Eq. (61.105) already warns that the square of pure Yang–Mills carries a dilaton and a two-form alongside the graviton; identifying and, if desired, projecting out those extra states is the spectrum analysis we undertake once the KLT relations have placed the construction on its historical foundation.

61.5 KLT Relations and the String-Theoretic Origin

The four-graviton amplitude Eq. (61.97) arrived, in the previous section, as a sum over channels of squared kinematic numerators — the double copy in its native form. But in passing we noticed something else. The insight closing Section 61.4.4 rewrote $\mathcal{M}_4$ as a *product* of two color-ordered gauge amplitudes, one in each of two orderings, glued together by a single power of a Mandelstam invariant. That product form is not an accident of four points and it is not a restatement of the sum: it is a second, logically independent presentation of "gravity = gauge^2," older than the double copy by a quarter century, and born not in field theory at all but in string theory. In 1985 Kawai, Lewellen, and Tye showed that every closed-string tree amplitude is a sum of products

of open-string tree amplitudes, with definite kinematic coefficients. Taking the point-particle limit $\alpha' \to 0$ of their relation turns closed strings into gravitons, open strings into gluons, and the coefficients into polynomials in the Mandelstam invariants. The result is a formula — the field-theory KLT relation — that expresses a graviton amplitude as gauge amplitudes multiplied together. This section states that formula, fixes its conventions, proves at four points that it reproduces the double-copy amplitude of Section 61.4 exactly, gives an honest account of why string theory makes it true, and finally identifies the kinematic gluing factor — the "momentum kernel" — as the inverse of a matrix of bi-adjoint scalar amplitudes, the bridge back to the cubic-graph picture of Section 61.2. By the end the two faces of the double copy, sum and product, will be visibly the same object.

61.5.1 Where the product form comes from: the string worldsheet

We have not built perturbative string theory in this book and will not build it here. But the origin of the KLT relations is a structural fact about string amplitudes simple enough to state precisely, and stating it honestly — as motivation, not as a proof the book can stand behind — is the right way to introduce the field-theory formula that follows. The reader should take this subsection as a physical explanation of why a product structure exists, with the field-theory content of Section 61.5.2 as the part this book actually proves.

A string tree amplitude is an integral over the moduli of a punctured Riemann surface, the worldsheet swept out by the string, with one puncture (the insertion point of a *vertex operator*) for each external state. For closed strings the relevant surface at tree level is the sphere; for open strings it is the disk, whose boundary the open-string endpoints live on. The single most important structural feature of the closed string is that its degrees of freedom split into two independent sectors — *left-movers* and *right-movers*, equivalently the holomorphic and antiholomorphic dependence on the worldsheet coordinate z. A closed string is, heuristically, two open strings traveling in opposite directions around the loop of string and locked together. This doubling shows in the vertex operators: a closed-string vertex operator factorizes into a product of a holomorphic (left-moving) piece and an antiholomorphic (right-moving) piece,

$$V^{\text{closed}}(z,\bar{z}) \;=\; V_L^{\text{open}}(z)\,\widetilde{V}_R^{\text{open}}(\bar{z}), \tag{61.106}$$

each factor being exactly the vertex operator of an open string. A graviton vertex operator, in particular, is the product of two gluon vertex operators, one holomorphic and one antiholomorphic — the worldsheet incarnation of the polarization doubling $\varepsilon^{\mu\nu} = \varepsilon^{\mu}\tilde{\varepsilon}^{\nu}$ of Theorem 61.17. The graviton is two gluons glued on the worldsheet before any limit is taken.

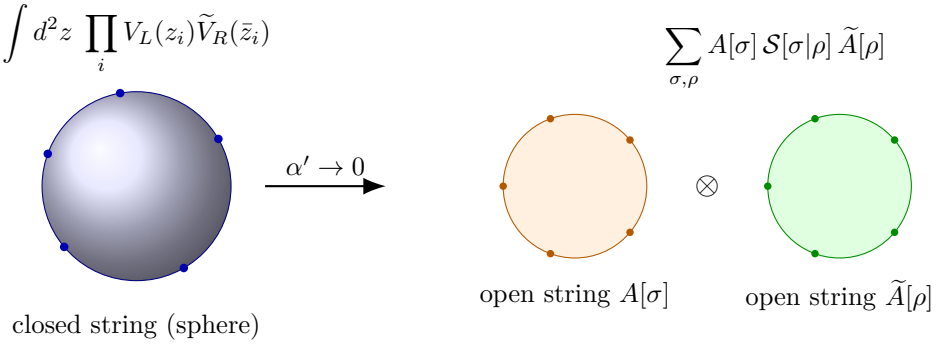


Figure 61.5: The Kawai–Lewellen–Tye factorization. A closed-string tree amplitude (left, the punctured sphere) has an integrand that splits into holomorphic (left-moving) and antiholomorphic (right-moving) open-string pieces, Eq. (61.106). Deforming the worldsheet integration contour re-expresses the single sphere integral as a sum of products of two disk (open-string) integrals, weighted by the monodromy phases Eq. (61.107). In the point-particle limit $\alpha' \to 0$ the closed string becomes a graviton, each open string a gluon, and the phases become the polynomial momentum kernel $\mathcal{S}[\sigma|\rho]$ of Section 61.5.2.

The factorization Eq. (61.106) of the integrand does not by itself factorize the amplitude, because the moduli integral over the sphere does not split into a product of two disk integrals: the holomorphic and antiholomorphic positions $z_i, \bar{z}_i$ are complex conjugates of one and the same puncture, not independent variables. The achievement of Kawai, Lewellen, and Tye was to show that one may nonetheless *deform the contour* of the sphere integral, treating z_i and $\bar{z}_i$ as formally independent and rotating each integration onto the real axis, where it becomes an open-string disk integral. The price of the deformation is a set of phases: as one z_i is dragged past another z_j in the complex plane, the integrand — which contains a branch cut factor $|z_i - z_j|^{2\alpha' p_i\cdot p_j} = (z_i - z_j)^{\alpha' p_i\cdot p_j}(\bar{z}_i - \bar{z}_j)^{\alpha' p_i\cdot p_j}$ — acquires a monodromy $e^{\pm i\pi\alpha' p_i\cdot p_j}$. Collecting these phases over all the crossings needed to disentangle the holomorphic from the antiholomorphic ordering produces, for each pair of open-string orderings σ and ρ, a trigonometric weight built from sines of the form

$$\sin(\pi\alpha' p_i\cdot p_j) \;=\; \sin(\tfrac{\pi}{2}\alpha' s_{ij}), \qquad s_{ij} = (p_i + p_j)^2 = 2\,p_i\cdot p_j, \tag{61.107}$$

(using $p_i^2 = p_j^2 = 0$). The closed-string amplitude emerges as a finite sum, over pairs of open-string orderings, of (open amplitude)×(sine factors)×(open amplitude). This is the string KLT relation. Figure 61.5 renders the picture.

The field-theory limit is now immediate as motivation. The string tension enters through α' (of dimension length squared); sending $\alpha' \to 0$ collapses the extended string to a point particle and the string amplitude to a sum of Feynman

diagrams. In that limit each sine linearizes,

$$\sin(\tfrac{\pi}{2}\alpha' s_{ij}) \;=\; \tfrac{\pi}{2}\alpha'\, s_{ij} + O(\alpha'^3), \tag{61.108}$$

so the trigonometric weights become *polynomials in the Mandelstam invariants*. Those polynomials are the momentum kernel. The massless closed-string states become the graviton (together with the dilaton and two-form already met in Eq. (61.105)), the massless open-string states become gluons, and the KLT relation becomes a statement purely about field-theory tree amplitudes: a graviton amplitude is a sum of products of two gluon amplitudes, glued by Mandelstam polynomials. The α' powers from Eq. (61.108) are exactly absorbed into the gravitational coupling, since $\kappa \sim g^2\sqrt{\alpha'}$ relates the closed- and open-string couplings; this is the string-theoretic origin of the coupling trade $g^2 \to \kappa$ seen at three points in Eq. (61.82).

Caution

Everything in this subsection is motivation, and we flag it as such per the honesty discipline of Chapter 58 onward. We have not defined a vertex operator, a worldsheet path integral, or the monodromy contour deformation with any rigor, and we will not; those are the content of a string-theory course. What this book *proves* is the field-theory statement that the limit produces: the formula of Section 61.5.2, its four-point equivalence to the double copy (Theorem 61.23), and its kernel's identity with the inverse bi-adjoint matrix (Theorem 61.26). Those stand on their own, with no string theory assumed. The string picture explains *why* such a product formula should exist; the field-theory proofs establish *that* it does and pin down its constants.

61.5.2 The field-theory KLT relation

We now state the field-theory KLT formula and fix its conventions completely. We use the color-ordered partial amplitudes $A_n[\sigma]$ of Section 59.1, in the normalization of Eq. (61.86): the gauge coupling g^{n-2} is stripped (it is restored only in the full color decomposition Theorem 59.2), and each partial carries the explicit factor of i that is standard for a tree amplitude in this book, so that at four points $A_4[1,2,3,4] = iP$ with P the reduced Parke–Taylor value of Eq. (61.86). The output is the graviton amplitude $\mathcal{M}_n$ of Theorem 61.18, normalized with $(\kappa/2)^{n-2}$ exactly as in the double copy Eq. (61.72). A tilde on a partial denotes the second gauge copy, which may be a different gauge theory or the same one; for the pure-gravity case both copies are pure Yang–Mills and $\tilde{A}_n = A_n$.

Theorem 61.21 (Field-theory KLT relation at four points)**.** *With the conventions just fixed, the four-graviton amplitude is the product of two color-ordered four-*

gluon amplitudes glued by a single Mandelstam invariant,

$$\boxed{\mathcal{M}_4 \;=\; i\left(\frac{\kappa}{2}\right)^2 s_{12}\; A_4[1,2,3,4]\,\widetilde{A}_4[1,2,4,3]\;,} \qquad s_{12}=s=(p_1+p_2)^2. \tag{61.109}$$

The structure is the whole point: a graviton amplitude on the left, a product of two gauge amplitudes on the right, in two *different* orderings — $(1,2,3,4)$ for the first copy, $(1,2,4,3)$ for the second — multiplied by the kernel s_{12}. The two orderings are not free; the choice of which orderings to pair, and which Mandelstam glues them, is dictated by the kernel, derived below. We prove Theorem 61.21 not by re-deriving it from strings but by showing it equals the double-copy amplitude Eq. (61.83) algebraically; that is Theorem 61.23.

Notation Watch

The relation Eq. (61.109) is most often quoted in the literature in a *reduced* normalization, in which the explicit i is stripped from each partial and the $(\kappa/2)^{n-2}$ from the gravity amplitude. Writing $a[\sigma]\equiv A_n[\sigma]/i$ for the i-stripped partials (so $a[1,2,3,4]=P$, etc.) and $M_n\equiv\mathcal{M}_n/(\kappa/2)^{n-2}$ for the reduced graviton amplitude, Eq. (61.109) becomes

$$M_4 \;=\; -\,i\,s_{12}\;a[1,2,3,4]\,\tilde{a}[1,2,4,3], \tag{61.110}$$

which is the form in which the four-point KLT relation usually appears. The two overall factors of i stripped from the partners combine, $i^2=-1$, to flip the displayed $+i$ of Eq. (61.109) into the $-i$ of Eq. (61.110); there is no contradiction, only a convention for where the factor of i lives. We keep Eq. (61.109) as our working form because its partials and its κ-counting then match the double copy Eq. (61.72) and the color decomposition Theorem 59.2 term for term.

The momentum kernel at general multiplicity. At higher points the single Mandelstam s_{12} becomes a matrix — the *momentum kernel* — contracting a basis of partials in one copy against a basis in the other. Fix the legs 1 and n and let σ,ρ run over permutations of the remaining labels; the kernel is the function defined by reading off, for each leg, the Mandelstam invariant it forms with the legs that precede it differently in the two orderings.

Definition 61.22 (KLT momentum kernel)**.** For two orderings $\sigma=\sigma(2),\dots,\sigma(n-1)$ and ρ of the legs $\{2,\dots,n-1\}$, the momentum kernel relative to leg 1 is

$$\mathcal{S}[\sigma|\rho]_{p_1} \;=\; \prod_{i=2}^{n-1}\left(s_{1,\sigma(i)} \;+\; \sum_{j=2}^{i-1}\theta_\rho(\sigma(j),\sigma(i))\;s_{\sigma(j),\sigma(i)}\right), \tag{61.111}$$

where $s_{ab} = (p_a + p_b)^2$ and $\theta_\rho(a,b) = 1$ if a, b appear in the *same* relative order in ρ as in σ, and $\theta_\rho(a,b) = 0$ otherwise.

The field-theory KLT relation at n points then reads, with our normalization,

$$\mathcal{M}_n \;=\; -\,i\left(\frac{\kappa}{2}\right)^{n-2} \sum_{\sigma,\rho} A_n[1,\sigma,n]\; \mathcal{S}[\sigma|\rho]_{p_1}\; \widetilde{A}_n[1,\rho,n], \tag{61.112}$$

the sums over σ, ρ ranging over a basis of $(n-3)!$ independent orderings (the BCJ basis of Section 61.2.5; the redundant orderings are removed by the amplitude relations). The compact closed form Eq. (61.111) is due to Bjerrum-Bohr, Damgaard, Søndergaard, and Vanhove [78]; we state it at the level the book uses it and verify it at four points, where it collapses to a single factor. At $n = 4$ the only free leg is $\sigma = \rho = (2)$ with $1, 4$ fixed, the product in Eq. (61.111) has the single term $i = 2$ with no preceding legs to sum over, and

$$\mathcal{S}[2|2]_{p_1} \;=\; s_{12}, \tag{61.113}$$

so Eq. (61.112) reduces to $\mathcal{M}_4 = -i(\kappa/2)^2 A_4[1,2,3,4]\, s_{12}\, \widetilde{A}_4[1,2,4,3]$, which is Eq. (61.109) once the two factors of i in the partials are made explicit. (The middle factor $\widetilde{A}_4[1,\rho,n] = \widetilde{A}_4[1,2,4]$ at $n = 4$ is read with the fixed legs as $\widetilde{A}_4[1,2,4,3]$; the kernel pairs the first copy's natural ordering with the second copy's reflected ordering, which is the origin of the $(1,2,4,3)$ versus $(1,2,3,4)$ asymmetry in Eq. (61.109).)

Physical Insight

The KLT relation and the double copy answer the same question with opposite bookkeeping. The double copy Eq. (61.72) is a *sum* over cubic graphs of products of numerators — a presentation that never mentions color-ordered amplitudes and squares numerators graph by graph. The KLT relation Eq. (61.112) is a *product* of two color-ordered amplitudes summed over orderings — a presentation that never mentions cubic numerators and multiplies gauge-invariant partials. One is built from the Jacobi-respecting numerators n_i; the other from the color-ordered partials $A_n[\sigma]$. That they compute the same graviton amplitude is the statement that the kinematic numerators and the color-ordered amplitudes carry the same information, repackaged — precisely the content of the numerator-to-partial dictionary Eq. (61.85). We now make the equality an explicit four-point identity.

Worked Example 61.23 (KLT equals the double copy at four points)**.** We prove Theorem 61.21 by showing that the KLT product Eq. (61.109) equals the double-copy sum Eq. (61.83), using only the numerator-to-partial dictionary of Section 61.4.4, the kinematic Jacobi identity Eq. (61.84), and the massless

on-shell relation $s+t+u=0$ of Eq. (61.38). No step is quoted and no spinor identity is needed; the equality is purely algebraic in s,t,u and the numerators.

The three orderings in terms of numerators. Recall the dictionary Eq. (61.85), which expressed the two basis partials as differences of numerator-over-propagator ratios,

$$A_4[1,2,3,4]=i\Big(\frac{n_s}{s}-\frac{n_t}{t}\Big),\qquad A_4[1,3,2,4]=i\Big(\frac{n_t}{t}-\frac{n_u}{u}\Big). \tag{61.114}$$

The third ordering needed by Eq. (61.109) is $A_4[1,2,4,3]$, whose two planar channels are s (the adjacent pair $\{1,2\}$) and u (the adjacent pair $\{1,3\}$ appearing as $\{3,1\}$); reading the master formula Eq. (61.71) in this ordering gives, by the same procedure that produced Eq. (61.114),

$$A_4[1,2,4,3]=i\Big(\frac{n_u}{u}-\frac{n_s}{s}\Big). \tag{61.115}$$

This is consistent with the photon-decoupling (Kleiss–Kuijf) identity that the color decomposition Theorem 59.2 forces among the three orderings. Indeed, summing Eq. (61.114) and Eq. (61.115),

$$A_4[1,2,3,4]+A_4[1,3,2,4]+A_4[1,2,4,3]=i\Big(\frac{n_s}{s}-\frac{n_t}{t}+\frac{n_t}{t}-\frac{n_u}{u}+\frac{n_u}{u}-\frac{n_s}{s}\Big)=0, \tag{61.116}$$

the cyclic decoupling relation, an independent check on the sign in Eq. (61.115).

Reduced form of the claim. Strip the common $i(\kappa/2)^2$. The KLT relation Eq. (61.109) carries it as a prefactor, while the double copy Eq. (61.83) carries $i(\kappa/2)^2$ in front of the channel sum; the claim $\mathcal{M}_4^{\mathrm{KLT}}=\mathcal{M}_4^{\mathrm{DC}}$ is therefore the scalar identity

$$\underbrace{\frac{n_s^2}{s}+\frac{n_t^2}{t}+\frac{n_u^2}{u}}_{S_{\mathrm{DC}}}\ \stackrel{?}{=}\ \underbrace{-\,s\Big(\frac{n_s}{s}-\frac{n_t}{t}\Big)\Big(\frac{n_u}{u}-\frac{n_s}{s}\Big)}_{S_{\mathrm{K}}}. \tag{61.117}$$

The left side is the double-copy sum with $\tilde{n}_i=n_i$, read from Eq. (61.83). The right side is the KLT product: substituting Eq. (61.114) and Eq. (61.115) into Eq. (61.109), the partial product is $A_4[1,2,3,4]\widetilde{A}_4[1,2,4,3]=i^2(n_s/s-n_t/t)(n_u/u-n_s/s)=-(n_s/s-n_t/t)(n_u/u-n_s/s)$, and the extra $s_{12}=s$ in Eq. (61.109) multiplies through, the i's and $(\kappa/2)^2$'s having been stripped — giving S_{K} exactly. We now show $S_{\mathrm{DC}}=S_{\mathrm{K}}$.

Expanding the KLT side. Multiply out S_{K}:

$$\begin{aligned}S_{\mathrm{K}}&=-\,s\left(\frac{n_sn_u}{su}-\frac{n_s^2}{s^2}-\frac{n_tn_u}{tu}+\frac{n_tn_s}{ts}\right)\\&=-\frac{n_sn_u}{u}+\frac{n_s^2}{s}+\frac{s\,n_tn_u}{tu}-\frac{n_tn_s}{t}.\end{aligned} \tag{61.118}$$

The difference vanishes. Subtract Eq. (61.118) from S_{DC}:

$$S_{\mathrm{DC}} - S_{\mathrm{K}} = \frac{n_t^2}{t} + \frac{n_u^2}{u} + \frac{n_s n_u}{u} + \frac{n_t n_s}{t} - \frac{s\, n_t n_u}{tu}, \tag{61.119}$$

the n_s^2/s terms having cancelled. Collect the two terms carrying $1/t$ and the two carrying $1/u$, then use the kinematic Jacobi identity $n_s + n_t + n_u = 0$ Eq. (61.84) in the form $n_t + n_s = -n_u$ and $n_u + n_s = -n_t$:

$$\frac{n_t^2}{t} + \frac{n_t n_s}{t} = \frac{n_t}{t}\,(n_t + n_s) = -\frac{n_t n_u}{t}, \tag{61.120}$$

$$\frac{n_u^2}{u} + \frac{n_s n_u}{u} = \frac{n_u}{u}\,(n_u + n_s) = -\frac{n_t n_u}{u}. \tag{61.121}$$

Substituting Eqs. (61.120) and (61.121) into Eq. (61.119),

$$S_{\mathrm{DC}} - S_{\mathrm{K}} = -\frac{n_t n_u}{t} - \frac{n_t n_u}{u} - \frac{s\, n_t n_u}{tu} = -\,n_t n_u \left(\frac{1}{t} + \frac{1}{u} + \frac{s}{tu}\right) = -\,n_t n_u\, \frac{u + t + s}{tu}. \tag{61.122}$$

The bracket is $(s + t + u)/(tu) = 0$ by the massless relation Eq. (61.38). Hence $S_{\mathrm{DC}} = S_{\mathrm{K}}$, and restoring the common $i(\kappa/2)^2$,

$$i\left(\frac{\kappa}{2}\right)^2 \left(\frac{n_s^2}{s} + \frac{n_t^2}{t} + \frac{n_u^2}{u}\right) = i\left(\frac{\kappa}{2}\right)^2 s_{12}\, A_4[1,2,3,4]\, \widetilde{A}_4[1,2,4,3]. \tag{61.123}$$

The KLT product and the double-copy sum are the same four-graviton amplitude. The identity rested on exactly two facts — the *kinematic* Jacobi identity, which is color–kinematics duality, and the *kinematic* relation $s + t + u = 0$, which is masslessness in four-point kinematics. Each governs one of the two terms in Eq. (61.122), and neither alone suffices. This is the precise four-point sense in which "KLT equals the double copy": the same single algebraic input, $n_s + n_t + n_u = 0$, makes the sum gauge-invariant (Eq. (61.87)) and makes it equal to the product.

The equivalence proved at four points is general. The double copy and the KLT relation are two presentations of one map from gauge amplitudes to gravity amplitudes; the double copy works graph by graph and demands BCJ numerators, while KLT works ordering by ordering and demands the momentum kernel. The remaining task of this section is to identify that kernel intrinsically, without reference to strings or to numerators, as an object built from the simplest theory in which color and kinematics coincide — the bi-adjoint scalar.

61.5.3 The kernel as the inverse bi-adjoint matrix

The momentum kernel Eq. (61.111) was handed to us by the string calculation, a polynomial in Mandelstams whose origin in Section 61.5.1 was the linearized

monodromy sines. There is a purely field-theoretic characterization of it, and it closes the circle back to the cubic-graph organization of Section 61.2: the kernel is the matrix inverse of the doubly-color-ordered amplitudes of the *bi-adjoint scalar* theory. This both removes the last whiff of string theory from the construction and exhibits the kernel as the natural metric on the space of color orderings.

Definition 61.24 (Bi-adjoint scalar and its doubly-ordered amplitudes)**.** The bi-adjoint scalar theory is a massless scalar $\phi^{aa'}$ carrying one adjoint index a of a group G and a second adjoint index a' of an independent group G', with the cubic interaction

$$\mathcal{L}_{\text{bi}} = \tfrac{1}{2}\,\partial_\mu\phi^{aa'}\,\partial^\mu\phi^{aa'} - \tfrac{\lambda}{3!}\,f^{abc}\,\tilde{f}^{a'b'c'}\,\phi^{aa'}\phi^{bb'}\phi^{cc'}, \tag{61.124}$$

where f^{abc} and $\tilde{f}^{a'b'c'}$ are the structure constants of G and G'. Because every vertex carries one factor of f and one of $\tilde{f}$, every tree diagram carries a G-color factor and an independent G'-color factor, and the amplitude is doubly color-decomposed. Its partial amplitudes $m[\alpha|\beta]$, labelled by an ordering α for the G traces and β for the G' traces, are obtained by stripping both: $m[\alpha|\beta]$ is the sum over all cubic tree graphs that are simultaneously planar with respect to α *and* planar with respect to β, each contributing $\prod_e (1/D_e)$ over its internal lines, with a sign $(-1)^{w(\alpha,\beta)}$ counting the relative flips between the two orderings.

The bi-adjoint scalar is the "zeroth copy": where Yang–Mills dresses each cubic graph with one color factor and one kinematic numerator, and gravity with two numerators, the bi-adjoint scalar dresses it with *two color factors* and no kinematic numerator at all. Its amplitudes are pure rational functions of the Mandelstam invariants — the bare propagator content of the cubic graphs, sorted by the two orderings. They are the simplest objects in which color–kinematics duality holds, because there is no kinematics to make dual: the numerator is the constant 1.

Worked Example 61.25 (The four-point bi-adjoint matrix and its inverse)**.** We compute the doubly-ordered bi-adjoint amplitudes at four points, assemble them into a matrix, and show that this matrix has rank $(n-3)! = 1$ with the BCJ relation as its null vector, and that its inverse on the non-degenerate subspace is exactly the four-point momentum kernel Eq. (61.113).

The matrix. Work in the Kleiss–Kuijf basis of the $(n-2)! = 2$ orderings $\alpha \in \{(1,2,3,4),(1,3,2,4)\}$, fixing legs 1 and 4. Each diagonal entry $m[\alpha|\alpha]$ sums the planar channels of α; each off-diagonal entry sums the channels common to both orderings, with the relative-flip sign. The planar channels are $\{s,t\}$ for $(1,2,3,4)$ and $\{t,u\}$ for $(1,3,2,4)$, as in Eq. (61.14). Hence

$$\begin{aligned} m[1,2,3,4|1,2,3,4] &= \frac{1}{s}+\frac{1}{t}, & m[1,3,2,4|1,3,2,4] &= \frac{1}{t}+\frac{1}{u},\\ m[1,2,3,4|1,3,2,4] &= -\frac{1}{t}, & m[1,3,2,4|1,2,3,4] &= -\frac{1}{t}, \end{aligned} \tag{61.125}$$

the off-diagonal entries receiving only the single shared t-channel (the only cubic graph planar in both orderings), with the minus sign from the single relative flip needed to align the two orderings. As a matrix in the basis $((1,2,3,4),(1,3,2,4))$,

$$m \;=\; \begin{pmatrix} \frac{1}{s}+\frac{1}{t} & -\frac{1}{t} \\ -\frac{1}{t} & \frac{1}{t}+\frac{1}{u} \end{pmatrix}. \tag{61.126}$$

Rank and null space. The determinant is

$$\det m = \left(\tfrac{1}{s}+\tfrac{1}{t}\right)\left(\tfrac{1}{t}+\tfrac{1}{u}\right)-\tfrac{1}{t^2} = \frac{1}{st}+\frac{1}{su}+\frac{1}{t^2}+\frac{1}{tu}-\frac{1}{t^2} = \frac{1}{st}+\frac{1}{su}+\frac{1}{tu} = \frac{s+t+u}{stu} = 0, \tag{61.127}$$

by $s+t+u=0$. The bi-adjoint matrix is *singular*: its rank is $1=(n-3)!$, not $(n-2)!=2$. This is not a defect — it is the statement that the number of independent color-ordered amplitudes is $(n-3)!$, the Bern–Carrasco–Johansson count of Section 61.2.5, with the extra $(n-2)!-(n-3)!$ orderings removed by the amplitude relations. The kernel of m encodes exactly those relations. Solving $m\,v=0$ with $v=(v_1,v_2)$: the first row gives $(\frac{1}{s}+\frac{1}{t})v_1-\frac{1}{t}v_2=0$, i.e. $v_2=t(\frac{1}{s}+\frac{1}{t})v_1=\frac{t+s}{s}v_1=-\frac{u}{s}v_1$, so

$$v \;\propto\; (s,\,-u). \tag{61.128}$$

A vector of physical color-ordered amplitudes, being a linear image of m, is orthogonal to this null vector; orthogonality of $(A_4[1,2,3,4],A_4[1,3,2,4])$ to $(s,-u)$ reads

$$s\,A_4[1,2,3,4]-u\,A_4[1,3,2,4]=0, \tag{61.129}$$

which is precisely the fundamental BCJ amplitude relation $sP=uQ$ of Eq. (61.89). The null space of the bi-adjoint matrix *is* the space of BCJ relations.

The inverse on the BCJ basis. Because m is rank one, it is invertible only after restriction to a complement of its null space — the $(n-3)!=1$-dimensional BCJ basis, for which we take the single ordering $(1,2,3,4)$. The corresponding 1×1 block is

$$m[1,2,3,4|1,2,3,4] = \frac{1}{s}+\frac{1}{t} = \frac{t+s}{st} = -\frac{u}{st}, \tag{61.130}$$

using $t+s=-u$. Its inverse is

$$(m^{-1})[1,2,3,4|1,2,3,4] = -\frac{st}{u} \;\equiv\; \mathcal{S}_{\text{diag}}. \tag{61.131}$$

Theorem 61.26 (KLT kernel as inverse bi-adjoint matrix)**.** *The KLT momentum kernel is the inverse of the bi-adjoint matrix on the BCJ basis. At n points, with σ,ρ ranging over the $(n-3)!$ BCJ basis orderings,*

$$\mathcal{S}[\sigma|\rho]_{p_1} \;=\; (m^{-1})[\sigma|\rho], \tag{61.132}$$

and consequently the field-theory KLT relation Eq. (61.112) may be written in basis-diagonal form as

$$\mathcal{M}_n = -\,i\left(\frac{\kappa}{2}\right)^{n-2}\sum_{\sigma,\rho} A_n[1,\sigma,n]\,(m^{-1})[\sigma|\rho]\,\widetilde{A}_n[1,\rho,n]. \tag{61.133}$$

At four points the single block reads $\mathcal{S}_{\rm diag} = -st/u$, and Eq. (61.133) reproduces the four-graviton amplitude Eq. (61.97).

Proof. We verify Eq. (61.132) and the consistency of Eq. (61.133) with the established four-point results; the general statement Eq. (61.132) is then the structural identity these satisfy [103]. At four points Eq. (61.131) gave $(m^{-1})[1,2,3,4|1,2,3,4] = -st/u$, the kernel in the diagonal basis. Insert it into Eq. (61.133) at $n=4$ with both copies pure Yang–Mills, $A_4[1,2,3,4] = \widetilde{A}_4[1,2,3,4] = iP$:

$$\mathcal{M}_4 = -\,i\Big(\tfrac{\kappa}{2}\Big)^2 (iP)\Big(-\tfrac{st}{u}\Big)(iP) = -\,i\Big(\tfrac{\kappa}{2}\Big)^2 i^2\Big(-\tfrac{st}{u}\Big)P^2 = -\,i\Big(\tfrac{\kappa}{2}\Big)^2\frac{st}{u}\,P^2, \tag{61.134}$$

since $i^2(-st/u) = +st/u$ and the outer $-i$ carries through. This is exactly the intermediate four-graviton result Eq. (61.92), which in Section 61.4.4 was brought to the crossing-symmetric form Eq. (61.97). Thus the inverse-bi-adjoint form of KLT reproduces the double-copy amplitude.

It remains to see that Eq. (61.133) is the same statement as the s_{12}-kernel form Eq. (61.109), i.e. that the diagonal kernel $-st/u$ and the off-diagonal kernel s_{12} are related by the BCJ amplitude relation just as their bases are. Using $A_4[1,2,4,3] = (t/u)A_4[1,2,3,4]$ — which follows from the dictionary Eqs. (61.114) and (61.115) together with the BCJ relation Eq. (61.129), since $A_4[1,2,4,3] = -A_4[1,2,3,4] - A_4[1,3,2,4] = -A_4[1,2,3,4](1+s/u) = (t/u)A_4[1,2,3,4]$ — the off-diagonal form Eq. (61.109) becomes

$$i\Big(\tfrac{\kappa}{2}\Big)^2 s\,A_4[1,2,3,4]\,\widetilde{A}_4[1,2,4,3] = i\Big(\tfrac{\kappa}{2}\Big)^2 s\,\frac{t}{u}\,A_4[1,2,3,4]\widetilde{A}_4[1,2,3,4] = i\Big(\tfrac{\kappa}{2}\Big)^2\frac{st}{u}(iP)^2, \tag{61.135}$$

and $(iP)^2 = -P^2$, giving $-i(\kappa/2)^2(st/u)P^2$, identical to Eq. (61.134). The two kernels are the same object in two ordering bases, related by the very BCJ relation Eq. (61.129) that lives in the null space of m. This is the content of Eq. (61.132) at four points; the all-n statement is established by the same accounting in the $(n-3)!$-dimensional basis. □

Physical Insight

The chain of identifications is now complete and worth surveying as one picture. Color–kinematics duality (Section 61.2) says the numerators n_i obey the color algebra. The double copy (Section 61.4) squares them graph by graph. The KLT relation (Eq. (61.112)) repackages the same

amplitude as a product of color-ordered partials glued by the momentum kernel. And the kernel (Theorem 61.26) is the inverse of the bi-adjoint scalar matrix — the theory whose only structure *is* color, doubled. So gravity, gauge theory, and the bi-adjoint scalar form a hierarchy: gravity carries two numerators, gauge theory one numerator and one color factor, the bi-adjoint scalar two color factors. The KLT kernel is the bridge between adjacent rungs, and it is built from the bottom rung. "Gravity = gauge^2" and "gauge = gravity $\times$ (bi-adjoint scalar)$^{-1}$" are the same statement read in two directions. The momentum kernel, which began life as a string monodromy phase, is in the end nothing but the inverse propagator structure of the simplest cubic theory there is.

We have now seen the double copy from both sides: as a sum over squared numerators (Section 61.4) and as a product of gauge amplitudes glued by the KLT kernel (this section), and we have shown the two agree and traced the product form to its string-theoretic and bi-adjoint origins. Both faces, however, leave one physical question untouched. The state sum Eq. (61.105) and the off-diagonal $h = 0$ entries of the factorization already warned that squaring two pure-gluon theories produces more than the graviton: the tensor product of two spin-one states contains a scalar and an antisymmetric tensor alongside the spin-two graviton. The next section determines exactly which fields the double copy delivers — the graviton, a dilaton, and a two-form, the "$\mathcal{N} = 0$ supergravity" multiplet — by decomposing that tensor product, and then surveys how varying the two input theories generates a whole web of double-copy constructions.

61.6 The Double-Copy Spectrum and the Web of Theories

The double copy and the Kawai–Lewellen–Tye relations both manufacture gravity amplitudes by gluing two gauge-theory factors, and both leave a question unanswered that we have so far only flagged. The factorization proof of Theorem 61.20 forced an intermediate state sum on every pole that ran not over a single graviton but over the four products of two gluon helicities, organized in Eq. (61.105) into a helicity-± 2 pair and two helicity-0 combinations. Those two extra combinations are not optional; they are present in every channel of the square of two pure Yang–Mills theories. The amplitudes built in Section 61.4 are therefore amplitudes of *some* gravitational theory, but not, as written, of pure Einstein gravity. This section identifies the theory exactly. We decompose the tensor product of two gauge polarizations into its irreducible pieces, carry out the on-shell state count in D spacetime dimensions, name the fields, say honestly when and how one may project onto pure Einstein gravity, and then step back

to see that the same gluing operation, fed different inputs, generates a whole lattice of theories — a "web" anchored by a humble scalar. Throughout we keep the conventions of Section 61.4: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units, massless on-shell legs.

61.6.1 The tensor product of two gauge states

A single on-shell gluon of momentum p carries a transverse polarization vector. In D spacetime dimensions a massless momentum has little group $\mathrm{ISO}(D-2)$, whose translation part acts trivially on any finite-helicity representation, so a single-particle state of a massless gauge boson transforms in the vector representation of $\mathrm{SO}(D-2)$ — the $(D-2)$ transverse directions. (In $D=4$ this $\mathrm{SO}(2) \cong \mathrm{U}(1)$ is exactly the little group whose weight organizes Eq. (58.5), and the two transverse vectors are the $\varepsilon_\pm$ of Eq. (58.111).) We make the $(D-2)$ count manifest by choosing a real orthonormal basis of transverse polarization vectors $\varepsilon_a^\mu(p)$, $a = 1, \ldots, D-2$, normalized and transverse,

$$\varepsilon_a \cdot \varepsilon_b = -\,\delta_{ab}, \qquad p \cdot \varepsilon_a = 0, \tag{61.136}$$

the minus sign in the mostly-minus metric recording that the transverse directions are spacelike. These vectors satisfy the physical-state completeness relation, with q an auxiliary null reference momentum ($q^2 = 0$, $p \cdot q \neq 0$),

$$\Delta^{\mu\nu}(p) \;\equiv\; \sum_{a=1}^{D-2} \varepsilon_a^\mu\, \varepsilon_a^\nu \;=\; -\,\eta^{\mu\nu} + \frac{p^\mu q^\nu + q^\mu p^\nu}{p \cdot q}, \tag{61.137}$$

the transverse projector onto the $(D-2)$-plane orthogonal to both p and q; its trace is $\eta_{\mu\nu}\Delta^{\mu\nu} = -D + 2 \cdot \frac{p\cdot q}{p\cdot q} = -(D-2)$, consistent with projecting onto $D-2$ spacelike directions.

The double copy assigns each external leg the product of one polarization from each copy. Writing the second copy's basis vectors as $\tilde{\varepsilon}_b^\nu$ (the same transverse plane; the tilde marks the copy), the external wavefunction of a leg in the double-copy theory is the rank-two object

$$E_{ab}^{\mu\nu}(p) \;\equiv\; \varepsilon_a^\mu(p)\, \tilde{\varepsilon}_b^\nu(p), \qquad a, b = 1, \ldots, D-2, \tag{61.138}$$

exactly the construction that produced the graviton polarization tensor $\varepsilon^{\mu\nu} = \varepsilon^\mu \varepsilon^\nu$ of Theorem 61.17, now read as the (a,b)-th entry of a $(D-2)\times(D-2)$ matrix of states. The states of the double-copy theory are the $(D-2)^2$ independent components of $E_{ab}^{\mu\nu}$. The physical content is read off by decomposing this matrix under the diagonal $\mathrm{SO}(D-2)$ little group that acts simultaneously on both indices a and b — the little group of the single momentum p shared by both copies.

A real $(D-2)\times(D-2)$ matrix decomposes, under the rotation group acting on both indices at once, into three irreducible pieces: its symmetric traceless part, its trace, and its antisymmetric part. Applied to E_{ab},

$$E^{\mu\nu}_{ab} = \underbrace{\left(E^{\mu\nu}_{(ab)} - \tfrac{1}{D-2}\,\delta_{ab}\,E^{\mu\nu}_{cc}\right)}_{\text{graviton } h^{\mu\nu}} + \underbrace{\tfrac{1}{D-2}\,\delta_{ab}\,E^{\mu\nu}_{cc}}_{\text{dilaton } \phi} + \underbrace{E^{\mu\nu}_{[ab]}}_{\text{two-form } B^{\mu\nu}}, \tag{61.139}$$

with round brackets denoting symmetrization $E_{(ab)} = \frac{1}{2}(E_{ab}+E_{ba})$, square brackets antisymmetrization $E_{[ab]} = \frac{1}{2}(E_{ab}-E_{ba})$, and a sum over the repeated c implied. We now count the three pieces and verify they are what their labels claim.

State count. The symmetric traceless part of a $(D-2)\times(D-2)$ matrix has $\frac{1}{2}(D-2)(D-1)-1$ independent components; the antisymmetric part has $\frac{1}{2}(D-2)(D-3)$; the trace is a single number. Their sum reproduces the full count,

$$\begin{aligned}\underbrace{\left[\tfrac{1}{2}(D-2)(D-1)-1\right]}_{\text{graviton}} + \underbrace{1}_{\text{dilaton}} + \underbrace{\tfrac{1}{2}(D-2)(D-3)}_{\text{two-form}} &= \tfrac{1}{2}(D-2)\big[(D-1)+(D-3)\big] \\ &= \tfrac{1}{2}(D-2)\,(2D-4) \\ &= (D-2)^2, \end{aligned}\tag{61.140}$$

the -1 from removing the trace cancelling the $+1$ of the trace mode itself. This is the central bookkeeping identity of the double-copy spectrum, carried in full: *the $(D-2)^2$ states of the square of two gauge bosons are a graviton, a dilaton, and a two-form, and nothing else.*

The three fields. Each piece is a genuine on-shell wavefunction of the field its dimension count names, as one checks against the transversality and trace properties a massless spin-s wavefunction must have.

The *graviton* polarization tensor is the symmetric traceless combination

$$h^{\mu\nu}_{ab} = \varepsilon^{\mu}_{(a}\,\tilde{\varepsilon}^{\nu}_{b)} - \frac{\delta_{ab}}{D-2}\,\Delta^{\mu\nu}. \tag{61.141}$$

It is symmetric in $\mu\nu$, transverse ($p_\mu h^{\mu\nu}_{ab}=0$ by Eq. (61.136) and $p_\mu\Delta^{\mu\nu}=0$), and traceless: using $\eta_{\mu\nu}\varepsilon^{\mu}_{a}\tilde{\varepsilon}^{\nu}_{b} = \varepsilon_a\cdot\tilde{\varepsilon}_b = -\delta_{ab}$ and $\eta_{\mu\nu}\Delta^{\mu\nu} = -(D-2)$,

$$\eta_{\mu\nu}h^{\mu\nu}_{ab} = -\delta_{ab} - \frac{\delta_{ab}}{D-2}\,\big(-(D-2)\big) = -\delta_{ab}+\delta_{ab} = 0. \tag{61.142}$$

A symmetric, transverse, traceless rank-two tensor is precisely the linearized metric fluctuation $h_{\mu\nu}$ of Theorem 61.17; here we have shown that subtracting

the trace is exactly what removes the dilaton contamination from the naive product.

The *dilaton* is the trace mode, the single scalar

$$\phi^{\mu\nu} = \frac{1}{\sqrt{D-2}}\,\Delta^{\mu\nu}(p), \tag{61.143}$$

symmetric and transverse but with nonzero trace $\eta_{\mu\nu}\phi^{\mu\nu} = -\sqrt{D-2}$ — it carries no little-group index and so is a Lorentz scalar, the dilaton. (The normalization is fixed by demanding $\phi^{\mu\nu}\phi_{\mu\nu} = \frac{1}{D-2}\Delta^{\mu\nu}\Delta_{\mu\nu} = 1$, since $\Delta^{\mu\nu}\Delta_{\mu\nu} = \sum_{a,b}(\varepsilon_a\cdot\varepsilon_b)^2 = \sum_{a,b}\delta_{ab} = D-2$.)

The *two-form* is the antisymmetric combination

$$B^{\mu\nu}_{ab} = \varepsilon^{\mu}_{[a}\,\tilde{\varepsilon}^{\nu}_{b]}, \tag{61.144}$$

an antisymmetric rank-two tensor $B^{\mu\nu} = -B^{\nu\mu}$, transverse on each index — the Kalb–Ramond field, also called the B-field or, through its field strength, the axion two-form. Its $\frac{1}{2}(D-2)(D-3)$ components are the physical polarizations of a massless antisymmetric tensor.

Notation Watch

The graviton+dilaton+two-form multiplet of Eq. (61.139) is called "$\mathcal{N}=0$ supergravity" in the double-copy literature. The name is a historical convenience, not a claim that the theory is supersymmetric: $\mathcal{N}=0$ means zero supercharges, and the label simply marks this as the bottom rung of a ladder of supergravities (Section 61.6.5) all built by the same squaring. It is also exactly the massless bosonic sector of the closed bosonic/Type-II string in D dimensions, which is the origin of the next remark.

Physical Insight

The appearance of the dilaton and two-form is not an artifact of the double copy; it is a truth the construction is honest enough not to hide. The massless spectrum of the closed string in D dimensions is precisely $g_{\mu\nu}\oplus\phi\oplus B_{\mu\nu}$ — a graviton, a dilaton, and a Kalb–Ramond field — and it arises there for the identical reason: a closed string state is a product of a left-moving and a right-moving open-string (gauge-boson) excitation, and the symmetric traceless / trace / antisymmetric decomposition of that product is Eq. (61.139). "Gravity is gauge squared" was first *seen* in this string-theoretic factorization (Section 61.5); the double copy is its field-theoretic distillation. That the same three fields fall out of pure algebra in Eq. (61.140) is the clearest sign that the construction is computing something structural about gravity, not merely a clever rearrangement of

one amplitude.

61.6.2 Four dimensions: the helicity assignment

In $D = 4$ the abstract decomposition becomes the concrete helicity bookkeeping already glimpsed in the factorization state sum Eq. (61.105). With $D - 2 = 2$, the transverse basis is $\{\varepsilon_1, \varepsilon_2\}$, and the helicity eigenstates are the complex combinations of Eq. (58.111),

$$\varepsilon_\pm^\mu = \frac{1}{\sqrt{2}}(\varepsilon_1^\mu \pm i\,\varepsilon_2^\mu), \qquad \varepsilon_\pm \cdot \varepsilon_\pm = 0, \quad \varepsilon_+ \cdot \varepsilon_- = -1, \tag{61.145}$$

carrying little-group weight ∓ 1, i.e. helicity ± 1, under Eq. (58.5). The four products $\varepsilon_h \tilde{\varepsilon}_{h'}$ with $h, h' \in \{+, -\}$ are the four states of the $D = 4$ square, and their total helicity is the sum $h + h'$:

Worked Example 61.27 (The four-dimensional double-copy states by helicity)**.** We sort the four products $\varepsilon_h^\mu\, \tilde{\varepsilon}_{h'}^\nu$ by total helicity and match each to a field of Eq. (61.139).

Helicity ± 2 *(graviton).* The like-helicity products

$$\varepsilon_+^\mu \tilde{\varepsilon}_+^\nu \quad (h = +2), \qquad \varepsilon_-^\mu \tilde{\varepsilon}_-^\nu \quad (h = -2) \tag{61.146}$$

are automatically symmetric and traceless: symmetric because the two factors are the same helicity vector, traceless because $\varepsilon_\pm \cdot \tilde{\varepsilon}_\pm = 0$ by Eq. (61.145). These are the two graviton states $h = \pm 2$, the $\varepsilon_\pm^{\mu\nu}$ of Eq. (61.77). No trace subtraction is needed in four dimensions for the diagonal entries — the nullness of a definite-helicity polarization does the work automatically. This is why the all-plus and all-minus graviton amplitudes of Section 61.4 came out as clean squares with no dilaton admixture.

Helicity 0 *(dilaton and axion).* The two opposite-helicity products $\varepsilon_+^\mu \tilde{\varepsilon}_-^\nu$ and $\varepsilon_-^\mu \tilde{\varepsilon}_+^\nu$ both have $h = 0$. Their symmetric and antisymmetric combinations separate the dilaton from the two-form:

$$\text{dilaton:} \quad \frac{1}{\sqrt{2}}\Big(\varepsilon_+^\mu \tilde{\varepsilon}_-^\nu + \varepsilon_-^\mu \tilde{\varepsilon}_+^\nu\Big), \tag{61.147}$$

$$\text{axion:} \quad \frac{1}{\sqrt{2}}\Big(\varepsilon_+^\mu \tilde{\varepsilon}_-^\nu - \varepsilon_-^\mu \tilde{\varepsilon}_+^\nu\Big). \tag{61.148}$$

The symmetric combination Eq. (61.147) has trace $\eta_{\mu\nu}\frac{1}{\sqrt{2}}(\varepsilon_+^\mu \tilde{\varepsilon}_-^\nu + \varepsilon_-^\mu \tilde{\varepsilon}_+^\nu) = \frac{1}{\sqrt{2}}(\varepsilon_+ \cdot \tilde{\varepsilon}_- + \varepsilon_- \cdot \tilde{\varepsilon}_+) = \frac{1}{\sqrt{2}}(-1 - 1) = -\sqrt{2} \neq 0$: it is the trace mode, the dilaton of Eq. (61.143) (indeed $\varepsilon_+^\mu \tilde{\varepsilon}_-^\nu + \varepsilon_-^\mu \tilde{\varepsilon}_+^\nu = \varepsilon_1^\mu \varepsilon_1^\nu + \varepsilon_2^\mu \varepsilon_2^\nu = \Delta^{\mu\nu}$, the full transverse projector). The antisymmetric combination Eq. (61.148) is the single physical component of the two-form Eq. (61.144).

State count. Two graviton states (± 2), one dilaton, one axion, totaling $4 = (D-2)^2|_{D=4}$, matching Eq. (61.140): $\frac{1}{2}(2)(3) - 1 = 2$ graviton, 1 dilaton, $\frac{1}{2}(2)(1) = 1$ two-form.

The four-dimensional two-form deserves a word, because its component count $\frac{1}{2}(D-2)(D-3) = 1$ collapses to a single state and the field is usually traded for a scalar. A massless antisymmetric tensor $B_{\mu\nu}$ in four dimensions has field strength $H_{\mu\nu\rho} = 3\,\partial_{[\mu}B_{\nu\rho]}$, and the equation of motion $\partial^\mu H_{\mu\nu\rho} = 0$ is solved by $H^{\mu\nu\rho} = \epsilon^{\mu\nu\rho\sigma}\partial_\sigma a$ for a pseudoscalar a — the *axion*. The one physical polarization of the four-dimensional two-form is thus dual to a single pseudoscalar degree of freedom, which is why the helicity-0 antisymmetric state Eq. (61.148) is the axion. The dilaton and axion together form a complex scalar (the "axio-dilaton"), both of helicity zero, distinguished from the graviton purely by living in the trace and antisymmetric channels of the gauge-state product.

61.6.3 Projecting onto pure Einstein gravity

The square of two pure Yang–Mills theories is $\mathcal{N} = 0$ supergravity, not pure Einstein gravity. To recover the latter one must dispose of the dilaton and the two-form. There are two honest things to say — one about external states, one about internal states — and they have different statuses.

External states are a free choice. The double copy assigns each *external* leg the wavefunction Eq. (61.138), and we are free to feed it the symmetric-traceless graviton polarization Eq. (61.141) (in four dimensions, the definite-helicity $\varepsilon_\pm^{\mu\nu}$) on every leg. Then the external data are purely gravitational. This is exactly what we did at three and four points: the amplitudes Eq. (61.81) and Eq. (61.97) have graviton external states by construction, and the dilaton and two-form never entered the external kinematics.

Internal states are the real question. Whether the result equals pure Einstein gravity depends on whether the dilaton or two-form can be *exchanged* on internal lines when all external states are gravitons. The factorization theorem Theorem 61.20 put on every cut the full state sum Eq. (61.105), including the helicity-0 dilaton/axion combinations. The question is whether those combinations have nonzero residue when the external states on both sides of the cut are gravitons. At three points the answer is fixed by little-group weight and is the cleanest possible. A cubic vertex of two gravitons and one dilaton would be a Lorentz-invariant contraction of two symmetric-traceless transverse tensors and one transverse scalar; the would-be amplitude $\mathcal{M}_3(h, h, \phi)$ must carry little-group weight ∓ 4 on the two graviton legs and 0 on the dilaton leg, and the unique on-shell three-point structure with these weights and the correct mass dimension *vanishes* — equivalently, the dilaton couples to gravity only through

its kinetic term $\sqrt{-g}\, g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$, which is quadratic in ϕ and so produces a graviton–dilaton–dilaton vertex but no graviton–graviton–dilaton vertex. The same holds for the two-form: its coupling $\sqrt{-g}\, H_{\mu\nu\rho}H^{\mu\nu\rho}$ is quadratic in B. We record the consequence.

Proposition 61.28 (Tree-level decoupling of the extra states)**.** *In the double copy of two pure Yang–Mills theories, any tree amplitude with all external legs gravitons (symmetric-traceless states) receives no contribution from internal dilaton or two-form lines. Consequently all-graviton tree amplitudes of $\mathcal{N} = 0$ supergravity coincide, channel by channel, with the corresponding tree amplitudes of pure Einstein gravity.*

Proof. By Theorem 61.20 every factorization channel of an all-graviton amplitude has a residue that is a sum of products of lower-point amplitudes, summed over the intermediate states Eq. (61.105). An internal dilaton or two-form contributes to a channel only if both subamplitudes $\mathcal{M}_L, \mathcal{M}_R$ on the cut are nonzero with that state crossing. A dilaton crossing a cut into an otherwise all-graviton subamplitude requires a subamplitude with exactly one dilaton and the rest gravitons; the lowest such is the three-point $\mathcal{M}_3(h, h, \phi)$, which vanishes by the little-group / quadratic-coupling argument above, and inductively every $\mathcal{M}_n(h, \dots, h, \phi)$ with a single dilaton vanishes, since each of its own factorization channels would again require a single-dilaton subamplitude. The identical argument applies to a single internal two-form. Hence dilatons and two-forms can appear internally only in pairs, which an all-graviton amplitude's channel structure never demands: every internal line of an all-graviton tree can carry a graviton, and only the graviton has a nonvanishing coupling to two gravitons. The residue on each channel is therefore the pure-graviton residue, which is the defining data of the Einstein amplitude (Theorem 61.18). Equality of all residues on all channels, together with the absence of any other singularities and the correct mass dimension, fixes the two amplitudes to be equal. □

This is the precise and honest sense in which the double copy of pure Yang–Mills computes Einstein gravity: *at tree level, for graviton external states,* the answer is identical to pure general relativity, the dilaton and two-form decoupling because they couple to gravitons only in pairs. It is exactly this proposition that licenses calling Eq. (61.97) "the four-graviton amplitude" without qualification — it is simultaneously the $\mathcal{N} = 0$ supergravity and the pure-Einstein four-graviton amplitude.

Caution

Theorem 61.28 is a tree-level, all-external-graviton statement. It does *not* survive untouched to loops or to amplitudes with non-graviton external states. At one loop a graviton amplitude has internal lines summed over

the entire $\mathcal{N} = 0$ spectrum, and a dilaton/two-form pair circulates in the loop and contributes; the bare double copy of pure Yang–Mills then computes the $\mathcal{N} = 0$ *supergravity* loop amplitude, which differs from the pure-Einstein one by the dilaton and two-form running in the loop. Recovering pure gravity at loop level requires actively subtracting those states — e.g. by double-copying with a Yang–Mills theory deformed by carefully chosen ghost-like scalars whose loop contributions cancel the unwanted multiplet, a construction we only name here. The clean "$\text{YM}^2 = \text{GR}$" slogan is a tree-level, external-graviton theorem; its loop-level extension is part of the frontier story of Section 61.7.

61.6.4 The bi-adjoint scalar: the zeroth copy

To organize the squaring operation into a web of theories we need its identity element — the theory that, double-copied with any other, returns that other unchanged. The master formula Eq. (61.71) already tells us what such a theory must look like. Recall its structure: an amplitude is a sum over cubic graphs of (numerator)×(numerator)/(denominator), and the two numerator slots may be filled either by a kinematic BCJ numerator n_i or by a color factor c_i — the gauge amplitude fills one slot with c_i and one with n_i, the gravity amplitude fills both with kinematics. The remaining case fills both slots with color. That theory is the bi-adjoint scalar.

Recall the bi-adjoint scalar (Theorem 61.24): a massless scalar $\phi^{aa'}$ in the adjoint × adjoint representation of two Lie groups G, G' with structure constants f^{abc} and $\tilde{f}^{a'b'c'}$, whose only interaction is the cubic vertex of the Lagrangian Eq. (61.124), carrying one structure constant of each group and a trivial (momentum-independent) kinematic factor.

Because the vertex is purely group-theoretic, every Feynman diagram for a tree amplitude is a cubic graph whose numerator is a *product of two color factors*, $c_i\, c'_i$, one from each group, with no kinematic dependence beyond the propagators. The full tree amplitude is therefore the master formula with both numerator slots filled by color,

$$\mathcal{A}_n^{\phi^3} = i\,\lambda^{n-2} \sum_{i\in\Gamma_3} \frac{c_i\, c'_i}{D_i}, \tag{61.149}$$

the literal "$c\otimes c$" object. Decomposing both color factors into the trace bases of the two groups (the double application of the color decomposition of Section 61.1.1) extracts the *doubly color-ordered partial amplitudes* $m[\alpha\,|\,\beta]$, labelled by an ordering α for G and an ordering β for G':

$$m[\alpha\,|\,\beta] = \sum_{i\,\in\,\Gamma_3(\alpha)\cap\Gamma_3(\beta)} \frac{\pm 1}{D_i}, \tag{61.150}$$

the sum running only over cubic graphs compatible with *both* orderings, each contributing the inverse of its propagator product with a sign set by the two orderings. These are the purest objects in the whole construction: numerators equal to unity, all the physics in the denominators. At four points, for example, the diagonal partial collects the two cubic graphs planar with respect to the ordering (1234) — the s- and t-channels —

$$m[1234\,|\,1234] = \frac{1}{s} + \frac{1}{t}, \tag{61.151}$$

while an off-diagonal partial such as $m[1234\,|\,1243] = -\frac{1}{s}$ retains only the single channel shared by the two orderings. The $m[\alpha|\beta]$ form the matrix whose inverse is, up to kinematic factors, the KLT momentum kernel of Section 61.5.3 — the precise sense in which the bi-adjoint scalar sits at the center of the web.

Proposition 61.29 (The zeroth copy is the identity)**.** *Double-copying the bi-adjoint scalar with a theory X — i.e. replacing one color factor c'_i in Eq. (61.149) by the BCJ numerators n_i of X — returns the amplitudes of X. In the slogan of the web, $\phi^3 \otimes \phi^3 = \phi^3$, $\phi^3 \otimes YM = YM$, and $\phi^3 \otimes$ (anything) = (anything).*

Proof. The substitution $c'_i \mapsto n_i$ in Eq. (61.149) produces $i\sum_i c_i\, n_i/D_i$, which is the gauge master formula Eq. (61.71) with color group G and BCJ numerators n_i — that is, the amplitude of X itself (here Yang–Mills), with G as its gauge group. The other color factor c_i simply carries the surviving color structure of X. The bi-adjoint scalar contributes only its color factor c_i and unit kinematics, so squaring against it changes nothing but the name of the gauge group. Taking $X = \phi^3$ (i.e. $n_i \mapsto c'_i$) returns Eq. (61.149), so the bi-adjoint scalar is its own double copy. □

This is what makes the bi-adjoint scalar the *zeroth copy*, the anchor of the web: it is the unit of the double-copy product. One climbs the web by replacing color factors with kinematic numerators one at a time — $c\otimes c$ to $n\otimes c$ to $n\otimes n$ — and each replacement trades a power of color for a power of kinematics, walking from the scalar through gauge theory to gravity.

61.6.5 The web of theories

We can now state the organizing picture. The double copy is a product on theories: choose a "left" theory and a "right" theory, each supplying a set of numerators for the common cubic graphs, and glue them through Eq. (61.72). The resulting theory depends on both inputs, and varying them sweeps out a web. The three rungs we have proved or can read off directly are

$$\phi^3\otimes\phi^3 = \phi^3, \qquad \mathrm{YM}\otimes\phi^3 = \mathrm{YM}, \qquad \mathrm{YM}\otimes\mathrm{YM} = \mathcal{N}{=}0\ \text{supergravity}, \tag{61.152}$$

established respectively by Theorem 61.29 (first two) and by the spectrum analysis Eq. (61.140) together with the amplitude construction of Section 61.4 (the third). The web extends in two directions that we now survey, flagging carefully what is proved in this book and what is stated with attribution.

Adding supersymmetry: the supergravity ladder. If one or both gauge factors carry supersymmetry, the squaring carries it too, because supersymmetric Yang–Mills amplitudes admit BCJ numerators just as the pure-gluon ones do (the duality is a statement about the kinematic algebra, blind to which states run on the legs). The on-shell supermultiplet of the product is the tensor product of the two gauge supermultiplets, decomposed exactly as in Eq. (61.139) but now including fermionic and scalar superpartners. The amount of supersymmetry adds: an $\mathcal{N} = p$ gauge factor times an $\mathcal{N} = q$ gauge factor yields $\mathcal{N} = p + q$ supergravity. The apex of the ladder is the maximally supersymmetric case.

Worked Example 61.30 ($\mathcal{N} = 8$ supergravity as the square of $\mathcal{N} = 4$ super-Yang–Mills)**.** We count the states on both sides and verify the tensor-product structure Eq. (61.139) generalized to the full supermultiplet.

The $\mathcal{N} = 4$ super-Yang–Mills multiplet. In four dimensions, maximally supersymmetric Yang–Mills has a single on-shell multiplet, CPT self-conjugate, consisting of: one gluon (2 helicity states ± 1), four gluinos ($2 \times 4 = 8$ states $\pm\frac{1}{2}$), and six real scalars (6 states 0). The total is

$$2 + 8 + 6 = 16 \tag{61.153}$$

on-shell states. (The count is $2^{\mathcal{N}} = 2^4 = 16$ for $\mathcal{N} = 4$: $\mathcal{N}$ fermionic creation operators acting on a helicity-$+1$ Clifford vacuum generate $2^{\mathcal{N}}$ on-shell states, the multiplet being CPT-self-conjugate at $\mathcal{N} = 4$; the explicit content above is the standard $\mathcal{N} = 4$ vector multiplet.)

The square. The product multiplet has $16 \times 16 = 256$ states. These organize, by the same symmetric/antisymmetric/trace logic now applied to the full graded tensor product, into the $\mathcal{N} = 8$ supergravity multiplet, whose content and state count is

$$\begin{aligned} &1 \text{ graviton } (\pm 2): && 2 \times 1 = 2, \\ &8 \text{ gravitini } (\pm \tfrac{3}{2}): && 2 \times 8 = 16, \\ &28 \text{ vectors } (\pm 1): && 2 \times 28 = 56, \\ &56 \text{ fermions } (\pm \tfrac{1}{2}): && 2 \times 56 = 112, \\ &70 \text{ scalars } (0): && 70, \end{aligned} \tag{61.154}$$

summing to $2 + 16 + 56 + 112 + 70 = 256 = 16^2$. The multiplicities are the binomial coefficients $\binom{8}{k}$ of the $\mathcal{N} = 8$ algebra: $\binom{8}{0} = 1$ graviton at the top helicity, $\binom{8}{1} = 8$ gravitini, $\binom{8}{2} = 28$ vectors, $\binom{8}{3} = 56$ fermions, and $\binom{8}{4} = 70$ scalars at helicity zero, the remaining states fixed by CPT. The graviton sits, as

in Eq. (61.146), in the top-helicity product of the two gluons: $(\pm 1) \otimes (\pm 1) \to \pm 2$. The dilaton and axion of the pure-Yang–Mills square are now just two of the seventy scalars, no longer distinguished — maximal supersymmetry unifies the entire bosonic and fermionic spectrum into one irreducible multiplet.

The statement $(\mathcal{N} = 4 \text{ sYM})^2 = \mathcal{N} = 8$ supergravity holds at tree level by the supersymmetric extension of the construction of Section 61.4: one needs BCJ numerators for $\mathcal{N} = 4$ super-Yang–Mills (which exist, computed in on-shell superspace) and squares them through Eq. (61.72). We do not build $\mathcal{N} = 4$ superamplitudes in this book, so we state this entry with its scope explicit — the state count Eq. (61.154) is exact and the graviton-sector amplitudes reduce to those we constructed, but the full superamplitude double copy is quoted, with the construction in [70, 176]. Intermediate rungs follow by the additive rule: $(\mathcal{N} = 4) \otimes (\mathcal{N} = 0) = \mathcal{N} = 4$ supergravity, $(\mathcal{N} = 2) \otimes (\mathcal{N} = 2) = \mathcal{N} = 4$ supergravity (with different matter), $(\mathcal{N} = 1) \otimes (\mathcal{N} = 1) = \mathcal{N} = 2$ supergravity, and so on, each a different point on the supergravity ladder.

Beyond gravity: the wider web. Replacing the gauge factors by other theories with BCJ-satisfying numerators generates further entries. We tabulate the principal ones in Table 61.2, drawing the broader map of [72]. Of these, only the rows involving pure Yang–Mills, the bi-adjoint scalar, and (at the level of state-counting) maximal supergravity are derived in this book; the rows involving the nonlinear sigma model, Born–Infeld theory, and the special Galileon are stated as established results of the literature, included to show the reach of the construction rather than proved here.

The pattern in Table 61.2 is the lesson. The double copy is not a single identity "gravity = gauge^2" but an operation on a category of theories, generated from a small set of building blocks — the bi-adjoint scalar at the center, Yang–Mills and the nonlinear sigma model and their supersymmetric cousins as the elementary factors — whose products fill out a web that includes Einstein gravity, the supergravities, and an unexpected corner of effective field theories (Born–Infeld, Galileon) that one would never have guessed are related to gauge theory at all. Color–kinematics duality is the structural fact that makes the web coherent: it is the statement that all these theories share one cubic-graph skeleton, differing only in which numerators dress it.

With the spectrum of the double copy identified — a graviton, a dilaton, and a two-form per copy of pure Yang–Mills, assembling at tree level into pure Einstein gravity for graviton external states (Theorem 61.28), and generalizing into a web of theories anchored by the bi-adjoint scalar — the tree-level picture is complete. We know what the double copy is, what it computes, and which theory it produces. What remains is to ask how far the story extends: to loop amplitudes, where the very existence of duality-satisfying numerators becomes a

Table 61.2: The web of double-copy constructions. Each entry is the theory produced by gluing the left and right factors through Eq. (61.72); the bi-adjoint scalar ϕ^3 is the identity (zeroth copy). "NLSM" is the SU(N) nonlinear sigma model (pion theory); "sGal" is the special Galileon. Rows marked ✓ are constructed in this chapter; rows marked ∘ are stated with attribution.

Left factor	Right factor	Double-copy theory	Status
ϕ^3	ϕ^3	bi-adjoint scalar ϕ^3	✓
ϕ^3	YM	Yang–Mills	✓
YM	YM	$\mathcal{N}=0$ supergravity (GR $+\phi+B$)	✓
YM $+\phi^3$	YM	Einstein–Yang–Mills	∘
$\mathcal{N}=4$ sYM	YM	$\mathcal{N}=4$ supergravity	∘
$\mathcal{N}=4$ sYM	$\mathcal{N}=4$ sYM	$\mathcal{N}=8$ supergravity	∘ (count ✓)
NLSM	ϕ^3	NLSM	∘
NLSM	YM	Born–Infeld	∘
NLSM	NLSM	special Galileon	∘

conjecture; to classical solutions, where "gravity = gauge^2" reappears for black holes and gravitational waves rather than for amplitudes; and to the sharp question of what, as of the mid-2020s, is theorem and what is checked. That frontier — and the closing of Yang–Mills theory — is the subject of the final section.

61.7 Loops, the Classical Double Copy, and the Frontier

The preceding sections completed the tree-level story to its sharpest form. Color and kinematics are dual (Theorem 61.8); duality-satisfying numerators exist for every tree (Theorem 61.12); replacing color by a second kinematic copy produces a gauge-invariant object (Theorem 61.16) that factorizes correctly on all its poles and is therefore a genuine gravity amplitude (Theorem 61.20); and the spectrum that square produces is not pure Einstein gravity but the graviton accompanied by a dilaton and a two-form — "$\mathcal{N}=0$ supergravity" (Eq. (61.105), Section 61.6). At tree level, "gravity is the square of gauge theory" is a theorem, proved in this book. This final section extends the story honestly beyond trees. We state the loop-level conjecture and lay out precisely what is theorem, what is checked, and what is open; we recount the flagship loop-level consequence, the unexpected ultraviolet softness of $\mathcal{N}=8$ supergravity; we descend from amplitudes to solutions of the classical field equations and carry out, in full, the one frontier computation this section owes — the Kerr–Schild map sending a Coulomb point charge to the Schwarzschild black hole; and we survey the

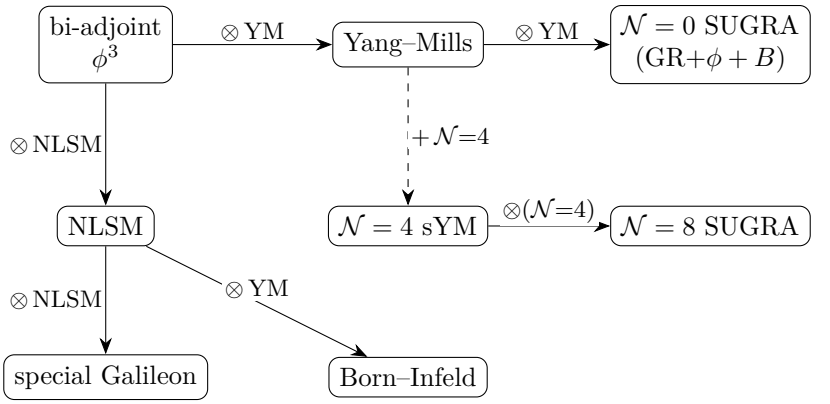


Figure 61.6: The web of double-copy theories. Starting from the bi-adjoint scalar ϕ^3 (the identity, zeroth copy), each arrow tensors in the labelled factor through Eq. (61.72), trading a power of color for a power of kinematics. The top row $\phi^3 \to$ YM $\to$ GR is the chain proved in this chapter; the lower rows show the supersymmetric ladder (to $\mathcal{N}=8$ supergravity) and the effective-field-theory corner (Born–Infeld, special Galileon) reached by squaring the nonlinear sigma model.

amplitudes-for-binaries program that has made these ideas a working tool of gravitational-wave physics. The section closes Yang–Mills theory, which began with the classical non-abelian Lagrangian of Chapter 25 and ends here with gravity assembled from two copies of it.

The standard of precision of this book is nowhere more load-bearing than here. Tree level is settled; loop level is not. We will mark every claim with its status and date it to the mid-2020s, and we will never let a citation stand in for an argument the book needs.

61.7.1 The loop-level double copy: integrands, the conjecture, and its evidence

The tree-level construction operated on the cubic-graph data (c_i, n_i, D_i) of Eq. (61.18). At loop level the same data organizes the *integrand*, the rational function of external and loop momenta whose integration produces the amplitude. The reorganization of any gauge-theory loop amplitude onto cubic graphs is possible for the identical reason it was at tree level: quartic vertices and contact terms are absorbed into trivalent graphs by multiplying and dividing by the propagator they are missing (Theorem 61.4), a move indifferent to whether the lines are external or internal. We therefore write the L-loop, n-point gauge-theory amplitude as

$$\mathcal{A}_n^{(L)} \;=\; i^L\, g^{\,n-2+2L} \sum_{i\in\Gamma_3^{(L)}} \int \prod_{l=1}^{L} \frac{\mathrm{d}^d\ell_l}{(2\pi)^d}\; \frac{1}{S_i}\; \frac{c_i\, n_i(\ell)}{D_i(\ell)}, \tag{61.155}$$

where $\Gamma_3^{(L)}$ is the set of L-loop trivalent graphs with n external legs, $\ell = (\ell_1, \ldots, \ell_L)$ are the independent loop momenta, $D_i(\ell)$ is the product of *all* internal inverse propagators (now functions of ℓ), c_i is the color factor read off the graph exactly as before, $n_i(\ell)$ is the kinematic numerator, and S_i is the symmetry factor of the graph, the order of its automorphism group (Theorem 61.6 counted the tree graphs; the loop graphs are counted by the same combinatorial logic, with S_i correcting for overcounting of internal relabelings). The denominators $D_i(\ell)$ are shared topology; the color factors c_i and the numerators $n_i(\ell)$ are again the only genuinely distinct data.

Loop-level color–kinematics duality. The color factors satisfy Jacobi identities on every triplet of graphs related by the *s-t-u* move around a single internal line — now including lines carrying loop momentum, with the loop momenta routed consistently across the three graphs of a triplet. *Loop-level color–kinematics duality* is the statement that one can choose the numerators $n_i(\ell)$ to obey the very same relations:

$$c_i + c_j + c_k = 0 \quad\Longrightarrow\quad n_i(\ell) + n_j(\ell) + n_k(\ell) = 0, \qquad c_i = -c_{\bar{\imath}} \;\Longrightarrow\; n_i = -n_{\bar{\imath}}, \tag{61.156}$$

identity by identity, for every Jacobi triplet $\{i, j, k\}$ and every flip pair $\{i, \bar{\imath}\}$ of the loop-level graph set, as functions of the loop momenta. When such a numerator set exists, the *loop-level double copy* is the same substitution $c_i \mapsto \tilde{n}_i$ that defined Eq. (61.72), now under the integral sign:

$$\mathcal{M}_n^{(L)} \;=\; i^L \left(\frac{\kappa}{2}\right)^{n-2+2L} \sum_{i\in\Gamma_3^{(L)}} \int \prod_{l=1}^{L} \frac{\mathrm{d}^d\ell_l}{(2\pi)^d}\; \frac{1}{S_i}\; \frac{n_i(\ell)\,\tilde{n}_i(\ell)}{D_i(\ell)}, \tag{61.157}$$

with $\{n_i\}$ a duality-satisfying (BCJ) set and $\{\tilde{n}_i\}$ any representation of the second amplitude's integrand.

What is theorem here, and what is conjecture. Two logically separate statements live in Eq. (61.157), and conflating them is the most common way the loop-level double copy is misreported.

Proposition 61.31 (Cut-level consistency of the loop double copy)**.** *Suppose a duality-satisfying loop numerator set $\{n_i(\ell)\}$ exists for the gauge theory at L loops, and let $\{\tilde{n}_i(\ell)\}$ be any representation of a second gauge-theory integrand on the same graphs. Then the integrand Eq. (61.157) reproduces the correct gravitational integrand on every generalized-unitarity cut: each cut factorizes the*

loop integrand into a product of on-shell tree amplitudes (Section 60.3), and on that cut the double-copy integrand reduces to a product of tree-level *double copies, which are gravity trees by Theorem 61.20.*

Proof sketch. A generalized unitarity cut (Section 60.3) replaces a set of internal propagators $1/D$ by on-shell delta functions, severing the loop graph into trees glued along the cut lines, and the residue on the cut is a product of tree amplitudes summed over the physical states crossing each cut line (Section 60.4 organizes this systematically). Apply this to Eq. (61.157). On any spanning cut the integrand becomes, graph by graph, a product of tree numerators over the surviving (uncut) denominators; the duality relations Eq. (61.156) restricted to a cut are exactly the tree-level kinematic Jacobi relations of the resulting tree pieces, because cutting a loop line is the same algebraic operation as factorizing on a tree pole. The tree-level factorization theorem Theorem 61.20 then identifies each cut residue as a sum over states of products of tree-level double copies — gravity trees. Summing the cut over the graviton-multiplet states crossing the cut (graviton, dilaton, two-form, the $\mathcal{N} = 0$ spectrum of Eq. (61.105)) reproduces the gravitational cut. Since the gravity integrand is determined by its cuts up to terms with no propagators — and the cubic-graph organization Eq. (61.155) carries no such cut-free pieces by construction — the double copy is the correct gravitational integrand. □

Theorem 61.31 is as far as the tree-level theorems carry us, and it is a genuine theorem: *given* duality-satisfying loop numerators, the double copy is guaranteed to be the right gravity integrand. The content that remains open is the hypothesis itself.

Conjecture 61.32 (Existence of loop-level BCJ numerators)**.** *For every gauge theory in the color–kinematics class, and at every loop order L and multiplicity n, there exists a representation of the integrand Eq. (61.155) whose kinematic numerators satisfy the duality Eq. (61.156) on every Jacobi triplet.*

As of the mid-2020s Theorem 61.32 is unproven in general. There is no construction analogous to the tree-level proof (Theorem 61.12, which built BCJ numerators from the BCFW recursion of Theorem 59.23) that delivers duality-satisfying numerators at arbitrary loop order; the recursion that powered the tree proof does not have a complete off-shell loop analogue, which is the same gap as the absence of a Lagrangian or off-shell formulation of the duality (discussed below). The evidence is instead case-by-case and substantial:

- In maximally supersymmetric Yang–Mills ($\mathcal{N} = 4$ sYM) duality-satisfying numerators have been constructed explicitly at one, two, three, and four loops for the four-point amplitude, and at higher loops, and the resulting double copies pass every generalized-unitarity check against independently computed $\mathcal{N} = 8$ supergravity integrands [70, 72].

- At lower supersymmetry and in pure Yang–Mills the duality has been realized in many one- and two-loop cases, including amplitudes with external matter and with fundamental fermions.

Nonlocal numerators

The duality sometimes cannot be realized with *polynomial* ("local") numerators. The cleanest documented instance is the five-loop four-point integrand of $\mathcal{N} = 4$ sYM, where a duality-satisfying representation was found only by admitting numerators that themselves contain propagator-like denominators — they are rational, not polynomial, in the loop momenta [72]. Such *nonlocal* numerators still satisfy Eq. (61.156) and still double-copy correctly on all cuts (Theorem 61.31 never assumed polynomiality), but their existence shows that the naive expectation — that BCJ numerators are always local rearrangements of Feynman-diagram numerators — is false beyond low loop order. Whether a *local* duality-satisfying representation exists in such cases, and more generally whether Theorem 61.32 holds with locality imposed, is open.

The honest summary is this. The map "BCJ numerators ⇒ gravity integrand" is a theorem (Theorem 61.31). The existence of those numerators at all loop orders is a conjecture (Theorem 61.32), verified in a growing list of nontrivial cases, sometimes only at the price of nonlocality (the nonlocal numerators discussed above). This is the precise state of the loop-level double copy.

61.7.2 Ultraviolet consequences: $\mathcal{N} = 8$ supergravity

The most consequential thing the loop double copy has done is sharpen a question about quantum gravity that predates it: *is there a perturbatively finite theory of gravitons?* Pure Einstein gravity is power-counting nonrenormalizable — the coupling κ has mass dimension -1 (Eq. (61.81) fixed $[\kappa] = -1$), so each additional loop brings two more powers of κ and, by dimensional analysis, worse ultraviolet behavior, requiring an unbounded tower of counterterms. The expectation for decades was that maximal supergravity, $\mathcal{N} = 8$ in four dimensions, while better behaved, would still diverge — superspace power-counting arguments predicted a three-loop divergence.

The double copy reframes the computation and overturned that expectation. Maximal supergravity is the double copy of maximal super-Yang–Mills,

$$\mathcal{N} = 8 \text{ supergravity} \;=\; (\mathcal{N} = 4 \text{ sYM}) \otimes (\mathcal{N} = 4 \text{ sYM}), \tag{61.158}$$

in the same on-shell sense as Eq. (61.157), now with each gauge multiplet enlarged to include its superpartners so that the tensor product of the two $\mathcal{N} = 4$ multiplets fills out the 256-state $\mathcal{N} = 8$ graviton multiplet. Because $\mathcal{N} = 4$ sYM is itself

ultraviolet finite (its beta function vanishes — the supersymmetric cancellations defeat the asymptotic-freedom running of Eq. (27.101)), its loop integrands are unusually soft in the ultraviolet, and squaring two soft integrands via Eq. (61.157) produces a gravitational integrand far softer than naive gravity power counting suggests. Explicit computation of the resulting integrands, with their ultraviolet behavior extracted by the unitarity methods of Chapter 60, found:

- $\mathcal{N} = 8$ supergravity is ultraviolet *finite* at three loops in $d = 4$ — no divergence, contradicting the superspace prediction [70].
- It is finite at four loops as well, with the first potential divergence pushed to a critical spacetime dimension $d_c = 11/2 > 4$ [64].
- At five loops the critical dimension was computed to be $d_c = 24/5 = 4.8 > 4$: still finite in $d = 4$ [65].

The pattern of computed critical dimensions through five loops,

$$d_c(L = 3) = 6, \qquad d_c(L = 4) = \tfrac{11}{2}, \qquad d_c(L = 5) = \tfrac{24}{5}, \tag{61.159}$$

all exceed four, so maximal supergravity has no ultraviolet divergence in four dimensions through five loops.

Enhanced cancellations

The striking feature is that this softness is *better than any known off-shell symmetry argument explains.* The available manifest symmetries (supersymmetry, the $E_{7(7)}$ duality of maximal supergravity) constrain the form of allowed counterterms, but the explicitly computed cancellations are stronger than those constraints predict: the would-be divergence allowed by all known symmetries is absent, or its critical dimension is higher than the symmetries require. These "enhanced cancellations" are visible in the double-copy representation precisely because it exposes the gauge-theory ultraviolet structure that is doing the work — the gravity integrand inherits, term by term, the softness of its squared gauge inputs in a way no purely gravitational organization makes manifest.

It must be stated plainly that this does *not* establish all-order finiteness. The first divergence permitted in $d = 4$ by the most conservative counterterm analysis (a particular half-BPS invariant) is at *seven* loops, and the seven-loop coefficient has not been computed. Whether $\mathcal{N} = 8$ supergravity is finite to all orders in four dimensions — and what that would imply for the existence of a perturbatively consistent point-particle theory of gravitons — is, as of the mid-2020s, a genuinely open question. The double copy has not answered it, but it has made the relevant multi-loop computations possible at all, which is why the question is now empirical rather than merely speculative.

61.7.3 The classical double copy: Kerr–Schild and the Coulomb–Schwarzschild map

Everything above concerns scattering amplitudes — quantum, on-shell, perturbative. There is a parallel statement at the level of *classical solutions* of the field equations: certain exact gauge-theory backgrounds map to exact gravitational backgrounds by a squaring operation strikingly similar to the amplitude double copy. The cleanest version is the Kerr–Schild double copy, and we carry it out in full for the stationary point source — the one fully worked frontier computation this section owes. It relates the Coulomb field of a point charge to the Schwarzschild black hole.

The Kerr–Schild ansatz. A metric is in *Kerr–Schild form* if

$$g_{\mu\nu} \;=\; \eta_{\mu\nu} + \phi\, k_\mu k_\nu, \tag{61.160}$$

where ϕ is a scalar function and k_μ is a covector that is *null* and *geodesic* with respect to the flat metric,

$$\eta^{\mu\nu} k_\mu k_\nu = 0, \qquad k^\mu\, \partial_\mu k_\nu = 0, \qquad k^\mu \equiv \eta^{\mu\nu} k_\nu. \tag{61.161}$$

The null condition has an immediate and remarkable consequence: the inverse metric is *exactly*

$$g^{\mu\nu} = \eta^{\mu\nu} - \phi\, k^\mu k^\nu, \tag{61.162}$$

with no corrections at higher order in ϕ. We verify this directly:

$$g_{\mu\nu} g^{\nu\rho} = (\eta_{\mu\nu} + \phi k_\mu k_\nu)(\eta^{\nu\rho} - \phi k^\nu k^\rho) = \delta_\mu{}^\rho - \phi k_\mu k^\rho + \phi k_\mu k^\rho - \phi^2 k_\mu\, (k_\nu k^\nu)\, k^\rho = \delta_\mu{}^\rho, \tag{61.163}$$

the quadratic term dying because $k_\nu k^\nu = 0$. The exact truncation of the metric expansion at linear order in ϕ is the structural reason Kerr–Schild metrics linearize gravity.

The stationary point-source data. Take the stationary, spherically symmetric profile

$$\phi(\vec{x}) = \frac{2GM}{r}, \qquad k_\mu = (1,\ \hat{x}_i), \qquad \hat{x}_i \equiv \frac{x_i}{r}, \quad r = |\vec{x}|, \tag{61.164}$$

all time-independent. We first confirm k_μ satisfies Eq. (61.161). Raising the index gives $k^\mu = (1, -\hat{x}_i)$, so the null condition is

$$\eta^{\mu\nu} k_\mu k_\nu = (k_0)^2 - \delta^{ij} k_i k_j = 1 - \hat{x}_i \hat{x}_i = 1 - 1 = 0. \tag{61.165}$$

For the geodesic condition, $k^\mu \partial_\mu = \partial_0 - \hat{x}_i \partial_i$ annihilates static fields through ∂_0, leaving $-\hat{x}_i \partial_i k_\nu$. The temporal component is trivial since $k_0 = 1$. For the spatial components, using $\partial_i k_j = \partial_i (x_j/r) = \delta_{ij}/r - x_i x_j / r^3$,

$$-\hat{x}_i\, \partial_i k_j = -\frac{x_i}{r}\Big(\frac{\delta_{ij}}{r} - \frac{x_i x_j}{r^3}\Big) = -\frac{x_j}{r^2} + \frac{(x_i x_i)\, x_j}{r^4} = -\frac{x_j}{r^2} + \frac{x_j}{r^2} = 0, \tag{61.166}$$

using $x_i x_i = r^2$. So k_μ of Eq. (61.164) is null and geodesic, and Eqs. (61.160) and (61.162) apply exactly. Written out, the line element is $g_{00} = 1 + \phi$, $g_{0i} = \phi\,\hat{x}_i$, $g_{ij} = -\delta_{ij} + \phi\,\hat{x}_i\hat{x}_j$ — the Schwarzschild solution in Kerr–Schild (Eddington–Finkelstein-type) coordinates, manifestly flat as $r \to \infty$.

Gravity side: the field equation linearizes. The decisive feature of Eq. (61.160) is that, for null geodesic k, the Einstein equation in mixed-index form is *exactly* linear in ϕ — the nonlinear curvature corrections cancel by the same $k \cdot k = 0$ and geodesic identities that truncated Eq. (61.162). We import the linearized Ricci tensor as the second-order differential operator built from the curvature of $g = \eta + h$ in Chapter 64,

$$R^{(1)}_{\mu\nu}[h] = \tfrac{1}{2}\big(\partial_\alpha\partial_\nu h^\alpha{}_\mu + \partial_\alpha\partial_\mu h^\alpha{}_\nu - \Box\, h_{\mu\nu} - \partial_\mu\partial_\nu h\big), \qquad h \equiv \eta^{\mu\nu}h_{\mu\nu}, \quad (61.167)$$

and use the Kerr–Schild fact that for $h_{\mu\nu} = \phi k_\mu k_\nu$ with null geodesic k the full Ricci tensor coincides with this linear expression (the nonlinear pieces vanish; the differential-geometric verification belongs to Chapter 64 and is not a citation standing in for an argument — it is a computation the gravity part performs in full). Evaluate Eq. (61.167) on the stationary point-source data. First the trace vanishes by nullity,

$$h = \eta^{\mu\nu}\phi k_\mu k_\nu = \phi\,(k \cdot k) = 0, \quad (61.168)$$

so the last term of Eq. (61.167) drops. For the 00 component, $h_{00} = \phi k_0 k_0 = \phi$ and $h^\alpha{}_0 = \eta^{\alpha\beta}\phi k_\beta k_0 = \phi\,k^\alpha$. The first two terms of Eq. (61.167) each carry a derivative ∂_0 acting on a static quantity and so vanish:

$$\partial_\alpha\partial_0 h^\alpha{}_0 = \partial_0(\partial_\alpha\phi\,k^\alpha) = 0, \quad (61.169)$$

leaving only the wave-operator term. With everything static, $\Box = \partial_0^2 - \nabla^2 \to -\nabla^2$, so

$$R^{(1)}_{00} = -\tfrac{1}{2}\,\Box\, h_{00} = -\tfrac{1}{2}\,\Box\,\phi = \tfrac{1}{2}\,\nabla^2\phi. \quad (61.170)$$

The Einstein equation sourced by a static point mass $T_{00} = M\delta^3(\vec{x})$ reads, in trace-reversed form $R_{\mu\nu} = 8\pi G(T_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}T)$, with $T = \eta^{\mu\nu}T_{\mu\nu} = M\delta^3$ for static dust:

$$R_{00} = 8\pi G(T_{00} - \tfrac{1}{2}T) = 4\pi G\,M\delta^3(\vec{x}) \quad \Longrightarrow \quad \nabla^2\phi = 8\pi G\,M\,\delta^3(\vec{x}). \quad (61.171)$$

This is the flat-space Poisson equation. Its solution, using $\nabla^2(1/r) = -4\pi\delta^3$, is $\phi = -2GM/r$. (The sign reflects $g_{00} = 1 + \phi = 1 - 2GM/r$, the correct Schwarzschild and Newtonian limit; the magnitude $|\phi| = 2GM/r$ is the profile we square below. The Newtonian normalization $g_{00} = 1 + 2\Phi_{\rm N}$ with $\Phi_{\rm N} = -GM/r$ is fixed by the geodesic motion of test particles, Chapter 64.) The gravitational field equation for the Schwarzschild point source has reduced, exactly, to a single flat-space scalar Poisson equation.

Single copy: peel off one factor of k. Define the *single copy* of the Kerr–Schild metric Eq. (61.160) by removing one factor of k_μ,

$$A_\mu \;\equiv\; \phi\, k_\mu, \tag{61.172}$$

an abelian gauge field (for a non-abelian single copy one attaches a constant color charge, $A^a_\mu = c^a \phi k_\mu$). We claim A_μ solves Maxwell's equations with a point charge. For the stationary point source Eq. (61.164) with $\phi = -2GM/r$ the components are $A_0 = \phi = -2GM/r$ and $A_i = \phi\, \hat{x}_i = -2GM\, x_i/r^2$. Compute the field strength $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. The spatial–spatial part is

$$F_{ij} = \partial_i A_j - \partial_j A_i, \qquad \partial_i A_j = -2GM\Big(\frac{\delta_{ij}}{r^2} - \frac{2x_i x_j}{r^4}\Big), \tag{61.173}$$

which is symmetric in $i \leftrightarrow j$, so $F_{ij} = 0$: the magnetic field vanishes. Indeed $A_i = -2GM\, x_i/r^2 = \partial_i(-2GM \ln r)$ is pure gauge, and may be removed by a gauge transformation, leaving $A_\mu = (\phi, \vec{0})$ — a pure scalar potential. The electric components are

$$F_{0i} = \partial_0 A_i - \partial_i A_0 = -\partial_i\Big(-\frac{2GM}{r}\Big) = -\frac{2GM\, x_i}{r^3}, \tag{61.174}$$

the radial Coulomb field $\vec{E} \propto \hat{x}/r^2$. Away from the origin $\nabla^2 A_0 = -2GM\, \nabla^2(1/r) = 0$, and at the origin Gauss's law $\partial^i F_{i0} = -\nabla\cdot\vec{E}$ integrates to a point charge q fixed by $A_0 = q/(4\pi r)$, i.e. $q = -8\pi GM$. The single copy of Schwarzschild is the Coulomb field of a point charge, with the gauge charge proportional to the mass, $q \leftrightarrow GM$.

Zeroth copy: peel off both. Removing the remaining factor of k_μ leaves the scalar field ϕ itself, satisfying the flat-space wave (here Poisson) equation Eq. (61.171); this is the *zeroth copy*, identified with a field of the biadjoint scalar theory whose amplitudes occupy the base of the double-copy web (Section 61.6.5). The three objects are organized by how many factors of the null vector k_μ they carry:

$$\underbrace{\phi}_{\text{zeroth copy}} \xrightarrow{\times k_\mu} \underbrace{A_\mu = \phi\, k_\mu}_{\text{single copy}} \xrightarrow{\times k_\nu} \underbrace{h_{\mu\nu} = \phi\, k_\mu k_\nu}_{\text{double copy}}, \tag{61.175}$$

with the charges mapping in step, (scalar charge) $\leftrightarrow q \leftrightarrow GM$. Each arrow is the classical shadow of replacing one color factor by one kinematic factor, exactly as the amplitude double copy replaced c_i by $\tilde{n}_i$. Figure 61.7 displays the tower.

Domain of validity

The Kerr–Schild double copy is exact, but its reach is limited and must not be overstated. It applies to gravitational solutions that admit a Kerr–

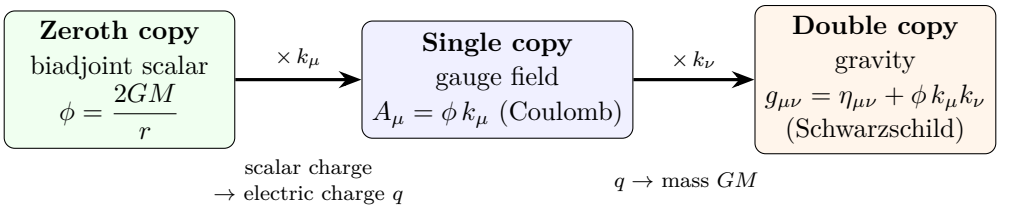


Figure 61.7: The Kerr–Schild classical double copy for the stationary point source. A single null geodesic profile $\phi = 2GM/r$ dressed by 0, 1, or 2 factors of the null covector k_μ produces, respectively, a biadjoint scalar field, the Coulomb gauge field of a point charge, and the Schwarzschild metric. Each $\times\, k$ is the classical counterpart of the amplitude substitution $c_i \mapsto \tilde{n}_i$.

Schild form Eq. (61.160) with null geodesic k — Schwarzschild, Kerr (a rotating k), and certain charged and cosmological generalizations — where the linearization is exact and the single copy is an honest solution of Maxwell's equations. It is *not* a general statement that every gravitational solution is the square of a gauge solution; generic metrics admit no Kerr–Schild form, and the construction as stated says nothing about them. For time-dependent and radiating sources the map is understood perturbatively, through a worldline/effective-field-theory expansion in which classical sources emit and re-absorb fields order by order [227, 228]; a fully nonperturbative, background-independent classical double copy — and its extension to curved backgrounds — remains an active research subject as of the mid-2020s [4, 349].

61.7.4 The two-body problem and gravitational-wave observables

The classical and quantum double copies converge on a problem of immediate observational relevance: the gravitational two-body inspiral that LIGO, Virgo, and the future LISA mission detect as gravitational waves. The conservative dynamics of two slowly-merging compact bodies can be organized as a *post-Minkowskian* (PM) expansion — an expansion in Newton's constant G to all orders in velocity, suited to the relativistic scattering regime — and each order in G is controlled by a gravitational scattering amplitude at one higher loop. This is the *amplitudes-for-binaries* program, and the double copy is one of its engines: it supplies the multi-loop gravitational integrands as squares of comparatively simple gauge-theory integrands, which are then integrated and matched to classical observables.

The pieces fit as follows, without full proof. The gravitational integrands are built by double copy (Eq. (61.157)) and reduced to master integrals by the generalized-unitarity and integration-by-parts machinery of Chapter 60. The

map from quantum amplitude to classical observable — scattering angle, impulse, radiated momentum — is made precise by the KMOC formalism, which expresses classical observables as expectation values built from amplitudes and isolates their classical ($\hbar \to 0$) part [305]. The results are cross-checked against, and feed, the worldline effective field theory of extended bodies [228]. By this route the conservative two-body Hamiltonian has been computed through third post-Minkowskian order ($\mathcal{O}(G^3)$, a two-loop amplitude) [66] and, in the years since, pushed to fourth order and partially to fifth, with radiative and spin effects systematically incorporated [4, 72]. These amplitude-derived results now enter the waveform models used to interpret gravitational-wave data. The honest framing is that this is a fast-moving applied frontier in which the double copy is a powerful organizing tool rather than the whole story — it makes the gravitational integrands tractable; the extraction of observables, the inclusion of dissipation, and the matching to bound-orbit dynamics are substantial additional machinery.

61.7.5 The frontier summary: theorem, evidence, open

It is worth stating, in one place and without hedging, exactly where the matter stands.

Proven (theorem, in this book). At tree level, color–kinematics duality holds: duality-satisfying numerators exist for every gluon tree (Theorems 61.8 and 61.12). The double copy of two gauge trees, at least one in a BCJ representation, is gauge invariant (Theorem 61.16) and is a genuine gravity amplitude with correct factorization (Theorem 61.20), demonstrated explicitly at three and four points (Eqs. (61.81) and (61.97)). The spectrum of the square of pure Yang–Mills is the $\mathcal{N} = 0$ supergravity multiplet (Eq. (61.105)). The Kerr–Schild classical double copy is exact for stationary Kerr–Schild solutions (Section 61.7.3).

Strongly evidenced (checked, not proved in general). Loop-level duality-satisfying numerators exist and double-copy correctly in many cases — $\mathcal{N} = 4$ sYM through several loops, lower-supersymmetry and pure Yang–Mills at low loops — with the gravity integrand correct on all unitarity cuts whenever the numerators exist (Theorem 61.31). $\mathcal{N} = 8$ supergravity is ultraviolet finite in $d = 4$ through five loops (Eq. (61.159)), with enhanced cancellations beyond known symmetry explanations. The amplitudes-for-binaries program reproduces and extends classical two-body results to high post-Minkowskian order.

Open (mid-2020s). The existence of duality-satisfying numerators at arbitrary loop order is conjectural (Theorem 61.32); it sometimes requires nonlocal numerators (the caution above), and a local representation is not guaranteed. There is no off-shell or Lagrangian formulation that makes the duality manifest to all orders. The all-order ultraviolet fate of $\mathcal{N} = 8$ supergravity in $d = 4$ — in particular the uncomputed seven-loop coefficient — is unknown. A nonperturbative, curved-background classical double copy is not established.

Why the duality is believed despite being unproven

The case-by-case evidence is not a heap of coincidences. Each successful loop construction is a nontrivial solution of an overconstrained linear system: the duality relations Eq. (61.156) are far more numerous than the free parameters in a numerator ansatz, so a solution existing at all is a strong structural statement, and it has been found again and again. Together with the tree-level theorem and the cut-consistency proposition Theorem 61.31, this is why the community treats the loop-level double copy as effectively reliable in practice while remaining precise that its general existence statement is a conjecture. The honest distinction between "works in every case we can check" and "proved for all cases" is exactly the distinction this book insists on keeping visible.

This closes Yang–Mills theory. It began in Chapter 25 with the classical non-abelian Lagrangian $\mathcal{L}_{\mathrm{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$, whose cubic and quartic self-interactions encode the Lie algebra through f^{abc} and its Jacobi identity. It passed through quantization and BRST (Chapter 26), asymptotic freedom (Eq. (27.101)), and the modern amplitude structure — Parke–Taylor and BCFW (Chapter 59), generalized unitarity (Chapter 60). It ends here: that same Jacobi identity, now mirrored in kinematics, lets one replace color by a second copy of kinematics and read off gravity. Gauge theory and gravity, which entered this book as unrelated — one a renormalizable quantum field theory, the other a nonrenormalizable geometric theory — are unified at the level of on-shell data, "GR = YM $\otimes$ YM," with an honest map of where that unification is proved and where it is still being earned.

The synthesis chapter (Chapter 63) gathers the lessons of Part XVI — that locality and unitarity, taken as inputs throughout, may instead be *outputs* of a deeper on-shell structure — and the gravity part (Chapter 64) builds the curvature and field equations that this section imported on faith, turning the on-shell graviton of Theorem 61.18 into the dynamical metric of general relativity. The reader leaves Part XVI with gauge theory and gravity sewn together at the connection of their amplitudes, and with a clear summary of the live frontier.

Further reading

Color–kinematics duality and the BCJ amplitude relations are introduced in Bern et al. [69]; the double-copy prescription and the loop-level conjecture in Bern et al. [70]; the Lagrangian-level construction making the duality manifest at tree level in Bern et al. [71]; and the field-theory origin of "gravity = gauge^2" in the string-theoretic Kawai–Lewellen–Tye relations Kawai et al. [291]. The minimal-basis structure behind the BCJ relations is Bjerrum-Bohr et al. [77]. The

comprehensive modern review, with the loop-level status, the $\mathcal{N} = 8$ ultraviolet story, and the applications surveyed here, is Bern et al. [72]; the ultraviolet finiteness of $\mathcal{N} = 8$ supergravity is reported in Bern et al. [64] (four loops) and Bern et al. [65] (five loops). The Kerr–Schild classical double copy originates with Monteiro et al. [349]; the worldline and radiative extensions with Goldberger and Ridgway [227], Goldberger and Rothstein [228]; and the breadth of the classical program, with its domain of validity, is surveyed in Adamo et al. [4]. For the two-body problem and gravitational-wave observables, the third-post-Minkowskian Hamiltonian is Bern et al. [66] and the amplitudes-to-observables formalism is Kosower et al. [305].

Chapter 62

Positive Geometry and the Amplituhedron

Each chapter of this part has pried the scattering amplitude a little further loose from the Feynman diagram. Spinor-helicity stripped the redundancy from the external data; on-shell recursion (Chapter 59) assembled tree amplitudes from their poles and residues alone, never writing a Lagrangian; the unitarity method (Chapter 60) rebuilt loop integrands from their factorizations; and the double copy (Chapter 61) exposed a hidden kinematic algebra shadowing color. The throughline is that locality and unitarity, the two principles a Feynman calculation feeds in by hand, keep re-emerging as constraints the amplitude is *forced* to satisfy. This chapter follows that throughline to its most radical destination: the proposal that the amplitude is not computed by any procedure but *is* a single geometric object — a region carved out by positivity, carrying a distinguished differential form whose poles sit exactly on the region's boundaries with exactly the residues unitarity demands. The amplitude becomes a volume; locality and unitarity become theorems about a shape rather than inputs to a calculation.

The development is built to keep the line between proof and conjecture visible at every step, because in this subject that line is unusually sharp. We begin (Section 62.1) with pure geometry: the notion of a logarithmic form, the definition of a *positive geometry* as a region whose unique canonical form is singular precisely on its boundaries, and the recursive residue axiom that makes "boundaries $\leftrightarrow$ singularities" a definition. The toy dictionary is proved on the interval, the simplex, and the convex polygon, where triangulation independence already displays the cancellation of spurious poles that will recur throughout. Section 62.2 then discharges the slogan "amplitude = volume" rigorously and without any conjecture: the canonical form of the ABHY kinematic associahedron, a concrete polytope in Mandelstam space, equals the tree-level color-ordered amplitude of bi-adjoint ϕ^3 theory, term by term — the cleanest proved instance

of the whole program.

To reach amplitudes that carry helicity we need a geometry graded by the MHV degree k of Chapter 58. Section 62.3 introduces the totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$ and its positroid stratification; Section 62.4 pours physics into that mould, identifying positroid cells with on-shell diagrams and leading singularities of planar $\mathcal{N}=4$ super Yang–Mills, recasting the Grassmannian integral as the generator of on-shell functions, and reading the BCFW recursion of Chapter 59 as a triangulation of a positive region by cells. The external kinematics, still entering by hand through momentum-conservation delta functions, are then repackaged in Section 62.5 into Hodges's momentum twistors — the chart in which momentum conservation is automatic, dual conformal symmetry is manifest, and positivity of the external data becomes a single combinatorial condition $\langle ijkl\rangle > 0$.

With both ingredients positive — the internal data through $\mathrm{Gr}_{\geq 0}(k,n)$, the external data through the positive Z-matrix — Section 62.6 defines the amplituhedron as the image of one under a positive linear map built from the other, proves convexity in the rigorously controlled $k=1$ polytope case (here k is the reduced momentum-twistor index of Section 62.5.4, so $k=1$ is NMHV; the physical $m=4$ included), and states honestly the canonical-form-equals-amplitude conjecture together with the boundary factorization that reads locality and unitarity off the facets. Section 62.7 turns the crank on an explicit positive dataset, carrying the $m=2$ NMHV computation to a closed form, closing the loop with the pentagon that opened the chapter, and displaying the genuine $m=4$ NMHV hexagon as the established result it is. Finally Section 62.8 audits everything: it sorts each claim into theorem, conjecture, and open problem dated to the mid-2020s, articulates the conceptual payoff — spacetime locality and quantum-mechanical unitarity as outputs of positivity rather than axioms — and names the planarity and maximal-supersymmetry crutches that bound the program, surveying the adjacent positive geometries (associahedra, cosmological polytopes, surfaceology) that are pushing past them. Throughout we use the spinor-helicity conventions of Chapter 58, the BCFW recursion of Chapter 59, and the generalized-unitarity integrand of Chapter 60, and we flag the kinship with the positivity-driven bootstrap of Chapter 52; the synthesis of the entire amplitudes program is reserved for Chapter 63.

62.1 Canonical forms and positive geometry

The chapters of this part have steadily eroded the idea that a scattering amplitude is, at bottom, a sum of Feynman diagrams. Spinor-helicity stripped the redundancy from the external data; on-shell recursion (Chapter 59) built tree amplitudes from nothing but their poles and residues, never writing a Lagrangian; the unitarity method (Chapter 60) reconstructed loop integrands from their fac-

torizations; and the double copy (Chapter 61) exposed a hidden kinematic algebra mirroring color. Each step replaced a piece of the diagrammatic scaffolding by a sharper statement about the analytic structure the amplitude is *forced* to have. The natural endpoint of that program is a dream: that the amplitude is not *computed* by some procedure but *is* a single mathematical object, read off from a shape. Concretely, the dream is that there exists a region — a polytope, or a curved generalization of one — whose boundaries are in bijection with the amplitude's singularities, and which carries a distinguished differential form whose poles sit exactly on those boundaries with exactly the residues unitarity demands. The amplitude would then be a *volume*, or more precisely the integrand of a volume, and locality and unitarity would be theorems about the geometry rather than inputs to a calculation.

To make that dream precise we need, before any physics, a clean answer to a purely geometric question: *given an oriented region, is there a canonical differential form attached to it, and is it unique?* This section builds that notion from scratch. We define what it means for a form to have only logarithmic singularities, we define a *positive geometry* as a region whose canonical form has logarithmic singularities on every one of its boundaries and nowhere else, and we make the recursive structure — residues on boundaries are themselves canonical forms of those boundaries — into the defining axiom. We then prove the toy dictionary boundaries $\leftrightarrow$ singularities on the three simplest families: the interval, the simplex, and the convex polygon. The polygon already exhibits the phenomenon that will dominate the rest of the chapter: a canonical form computed by triangulation is full of *spurious* poles on the internal edges of the triangulation, yet those poles cancel in the sum, leaving an answer that depends only on the shape and not on how it was cut up. That cancellation is the geometric face of what physicists call the disappearance of spurious poles in recursion. None of this yet mentions an amplitude; that match is the work of the next section. Here we forge the vocabulary.

Throughout we work over the reals for the geometry and allow the coordinates to be complexified when we take residues, exactly as one analytically continues a Feynman integrand to expose its poles. Projective space $\mathbb{P}^m$ and the exterior calculus of differential forms are at the prerequisite level; we recall only the few facts we lean on, as we need them.

62.1.1 Logarithmic forms and the canonical form

The building block is the simplest function with a single zero and a single pole on the projective line, packaged as a differential. For a nonzero meromorphic function f on a complex manifold, write

$$d\log f \;\equiv\; \frac{df}{f}. \tag{62.1}$$

Wherever f has a simple zero or a simple pole, $d\log f$ has a simple pole with residue equal to the order of vanishing: if $f = u\cdot z^k$ near $z=0$ with u holomorphic and nonvanishing, then $d\log f = k\,\frac{dz}{z} + d\log u$, and the second term is regular, so $\operatorname{Res}_{z=0} d\log f = k$. A logarithmic differential is one assembled out of such pieces.

Definition 62.1 (Logarithmic (dlog) form)**.** A meromorphic p-form ω on a complex manifold X has *logarithmic singularities* along a hypersurface $S=\{f=0\}$ (with f vanishing to first order on S) if, in a neighborhood of any smooth point of S, it can be written

$$\omega \;=\; \frac{df}{f}\wedge\psi \;+\; \eta, \tag{62.2}$$

with ψ a $(p-1)$-form and η a p-form both regular (holomorphic) near S. A form that has at worst logarithmic singularities along each component of its singular locus, and no other singularities, is called a *dlog form*; a top form is dlog precisely when it is a finite sum of wedge products $d\log f_1\wedge\cdots\wedge d\log f_m$.

The decomposition Eq. (62.2) is the mechanism for taking a residue along S. The restriction $\psi|_S$ does not depend on the freedom in writing Eq. (62.2): if $\frac{df}{f}\wedge\psi+\eta = \frac{df}{f}\wedge\psi'+\eta'$, then $\frac{df}{f}\wedge(\psi-\psi') = \eta'-\eta$ is regular, which forces $\psi-\psi'$ to be divisible by f and hence to vanish on S. This justifies the following.

Definition 62.2 (Residue along a boundary)**.** With notation as in Eq. (62.2), the *residue* of ω along S is the $(p-1)$-form on S

$$\operatorname{Res}_S\,\omega \;\equiv\; \psi\big|_S. \tag{62.3}$$

For a one-form $\omega = g\,dz$ with a simple pole at $z=z_0$ this reduces to the ordinary residue $\operatorname{Res}_{z_0}\omega = \lim_{z\to z_0}(z-z_0)g(z)$, recovering the complex-analysis notion the reader already has.

The residue is the operation that turns the singularity structure of a form on a region into the form on a boundary. This is the technical hinge of the whole subject, so we now make it the content of a definition rather than a remark.

Definition 62.3 (Positive geometry and its canonical form)**.** A *positive geometry* of dimension m is a pair $\mathcal{A}=(X,\,X_{\geq 0})$, where X is an m-dimensional complex projective variety and $X_{\geq 0}\subseteq X(\mathbb{R})$ is a closed, oriented region of its real locus, whose topological boundary $\partial X_{\geq 0}$ is a finite union of codimension-one pieces

$$\partial X_{\geq 0} \;=\; \bigcup_i C_i, \tag{62.4}$$

each of which, with the boundary orientation induced from $X_{\geq 0}$, is itself a positive geometry $\mathcal{C}_i$ of dimension $m-1$, such that there exists a (necessarily unique, by Theorem 62.4) nonzero rational m-form $\Omega(\mathcal{A})$ on X, the *canonical form*, with the following properties:

(i) *(Dimension zero)* If $m = 0$, then $X_{\geq 0}$ is a single point and $\Omega(\mathcal{A}) = \pm 1$, the sign being its orientation.

(ii) *(Logarithmic singularities only on boundaries)* If $m > 0$, then $\Omega(\mathcal{A})$ has poles *only* along the boundary components C_i, every such pole is logarithmic in the sense of Theorem 62.1, and $\Omega(\mathcal{A})$ has no singularity anywhere off $\bigcup_i C_i$.

(iii) *(Recursive residue axiom)* On each boundary component the residue reproduces the canonical form of that boundary, regarded as a positive geometry in its own right:

$$\operatorname{Res}_{C_i} \Omega(\mathcal{A}) \;=\; \Omega(\mathcal{C}_i). \tag{62.5}$$

Three remarks fix the meaning of this definition before we use it. First, the axiom is genuinely recursive: Eq. (62.5) relates an m-form to the canonical $(m-1)$-form of a lower-dimensional positive geometry, and the recursion bottoms out at the points of clause (i). The canonical form of any positive geometry is therefore completely determined by its tower of boundaries, boundaries-of-boundaries, and so on, down to vertices. Second, the boundaries-as-singularities clause (ii) is the precise statement of the dictionary advertised in the section title: *a codimension-one boundary of the region is a simple pole of the form, and conversely the form has a pole only where the region has a boundary.* Third, uniqueness is not an extra assumption; it is forced, and we prove it next, so that "the" canonical form is well posed.

Lemma 62.4 (Uniqueness of the canonical form)**.** *Fix a finite set of irreducible hypersurfaces $\{S_i\}$ in $\mathbb{P}^m$. Among rational m-forms whose poles all lie in $\bigcup_i S_i$ and are at worst simple there, a form is uniquely determined by its residues $\{\operatorname{Res}_{S_i} \omega\}$. In particular at most one rational m-form can satisfy clauses (ii)–(iii) of Theorem 62.3.*

Proof. Suppose ω and ω' have poles confined to $\bigcup_i S_i$, at worst simple, with equal residues on every S_i. Their difference $\chi = \omega - \omega'$ has, along each S_i, residue zero; by the decomposition Eq. (62.2) a simple-pole form with vanishing residue is in fact regular across S_i. Hence χ has no poles at all: it is a *global holomorphic* m-form on $\mathbb{P}^m$. We show there are none but zero. Work in an affine chart with coordinates $(x_1, \dots, x_m)$, where any holomorphic top form reads $\chi = P(x)\, dx_1 \wedge \cdots \wedge dx_m$ with P entire; regularity on all of $\mathbb{P}^m$ (a compact manifold) forces P to be a polynomial. Pass to the chart at infinity by $x_1 = 1/w$, $x_a = u_a/w$ for $a \geq 2$. Then

$$dx_1 = -\frac{dw}{w^2}, \qquad dx_a = \frac{du_a}{w} - \frac{u_a\, dw}{w^2} \;(a \geq 2), \tag{62.6}$$

and a direct expansion of the wedge gives $dx_1 \wedge \cdots \wedge dx_m = (-1)^m\, w^{-(m+1)}\, dw \wedge du_2 \wedge \cdots \wedge du_m$. For χ to be regular as $w \to 0$ we need $P(1/w, u_2/w, \dots)\, w^{-(m+1)}$

to stay finite, i.e. P must vanish to order at least $m+1$ at infinity — impossible for a nonzero polynomial, every one of which *grows* there. Therefore $P \equiv 0$, $\chi = 0$, and $\omega = \omega'$. □

The same computation, run in reverse, shows there is no obstruction in principle to *existence*: a rational m-form with prescribed simple poles and residues exists iff those residue data are mutually consistent (the residue on a codimension-two stratum computed from two different facets must agree). For the geometries of this section we will exhibit the canonical form explicitly, so existence is never in doubt; uniqueness, Theorem 62.4, is what makes triangulation-independence a theorem rather than a coincidence.

Physical Insight

The content of Theorem 62.3 is a translation manual. Geometry on the left, analysis on the right: a region ↔ a top form; a codimension-one boundary ↔ a simple pole; the operation "go to a boundary" ↔ the operation "take a residue"; a vertex ↔ a number ± 1. Every later assertion in this chapter — that an amplitude is a canonical form, that its poles are factorization channels, that residues are lower-point amplitudes — is this manual applied to a particular shape. The physics enters only in choosing the shape; the dictionary is fixed once and for all here.

62.1.2 The interval: the dictionary in one dimension

The lowest nontrivial case is one-dimensional. Let $X = \mathbb{P}^1$ with affine coordinate x, and let the region be a closed interval $X_{\geq 0} = [a, b]$ with $a < b$, oriented by $+dx$. Its boundary is the two points $\{a\}$ and $\{b\}$, each a zero-dimensional positive geometry. We claim its canonical form is the most elementary dlog imaginable.

Theorem 62.5 (Canonical form of the interval)**.** *The canonical form of the oriented interval $[a, b] \subset \mathbb{P}^1$ is*

$$\Omega([a,b]) \;=\; d\log\frac{x-a}{x-b} \;=\; \left(\frac{1}{x-a}-\frac{1}{x-b}\right)dx \;=\; \frac{b-a}{(x-a)(b-x)}\,dx. \tag{62.7}$$

It has simple poles only at the two endpoints, with residues

$$\mathrm{Res}_{x=a}\,\Omega = +1, \qquad \mathrm{Res}_{x=b}\,\Omega = -1, \tag{62.8}$$

and these residues realize the recursive axiom Eq. (62.5): each is the canonical form ± 1 of the corresponding endpoint, with sign equal to the boundary orientation induced by $+dx$.

Proof. We verify each clause of Theorem 62.3 directly. First the three expressions in Eq. (62.7) agree. Differentiating the logarithm,

$$d\log\frac{x-a}{x-b}=\frac{d(x-a)}{x-a}-\frac{d(x-b)}{x-b}=\left(\frac{1}{x-a}-\frac{1}{x-b}\right)dx, \tag{62.9}$$

and combining over a common denominator,

$$\frac{1}{x-a}-\frac{1}{x-b}=\frac{(x-b)-(x-a)}{(x-a)(x-b)}=\frac{a-b}{(x-a)(x-b)}=\frac{b-a}{(x-a)(b-x)}, \tag{62.10}$$

where the last step writes $(x-a)(x-b)=-(x-a)(b-x)$ and cancels the sign against $a-b=-(b-a)$. On the open interval $a<x<b$ both $x-a>0$ and $b-x>0$, while $b-a>0$, so the coefficient in the rightmost form of Eq. (62.7) is strictly positive there: the canonical form is a positive measure on the interior of the region, a property we will see recur and which singles out the geometric "inside."

Singularities. A rational 1-form $g(x)\,dx$ on $\mathbb{P}^1$ can have poles only where g blows up (in the finite chart) and possibly at $x=\infty$. From the first form in Eq. (62.7), $g(x)=\frac{1}{x-a}-\frac{1}{x-b}$ has simple poles exactly at $x=a$ and $x=b$ and is finite elsewhere in $\mathbb{C}$. At infinity, set $x=1/w$, so $dx=-dw/w^2$ and

$$g(x)\,dx=\left(\frac{1}{1/w-a}-\frac{1}{1/w-b}\right)\left(-\frac{dw}{w^2}\right)=-\left(\frac{w}{1-aw}-\frac{w}{1-bw}\right)\frac{dw}{w^2}=-\frac{1}{w}\left(\frac{1}{1-aw}-\frac{1}{1} \tag{62.11}$$

Expanding the bracket for small w gives $(1+aw+\cdots)-(1+bw+\cdots)=(a-b)w+O(w^2)$, so the prefactor $1/w$ is cancelled and $g(x)\,dx=-\big[(a-b)+O(w)\big]\,dw$, which is *regular* at $w=0$. Thus there is no pole at infinity, and the only singularities are the two endpoints — exactly the two boundary points of $[a,b]$, and both poles are simple, hence logarithmic. This is clauses (ii) of the definition.

Residues, clause (iii). From the first form in Eq. (62.7), $\operatorname{Res}_{x=a}\Omega=\lim_{x\to a}(x-a)g(x)=+1$ and $\operatorname{Res}_{x=b}\Omega=\lim_{x\to b}(x-b)g(x)=-1$, giving Eq. (62.8). Each residue is a bare sign ±1, which is exactly the canonical form of a zero-dimensional positive geometry (clause (i)); the two opposite signs encode the two opposite orientations that the endpoints inherit as the boundary of the oriented interval $[a,b]$. The recursive axiom Eq. (62.5) therefore holds at the bottom of the tower. By Theorem 62.4 this is the unique such form, and the proof is complete. □

The interval already teaches the dictionary in miniature: two boundary points, two simple poles, unit residues, and a form that is manifestly the logarithm of a ratio of the two linear functions vanishing on the boundaries. It also tells us what a "vertex" contributes — the residue is a bare sign — which is the seed of every recursion to come.

It is worth recording that Eq. (62.7) is a *projective* statement, independent of the affine coordinate we happened to use. The group of projective automorphisms

of $\mathbb{P}^1$ is the Möbius group, acting by $x \mapsto x' = \frac{px+q}{rx+s}$ with $ps - qr \neq 0$. A genuine geometric notion must be insensitive to this relabeling, and the interval form is.

Proposition 62.6 (Projective invariance of the interval form)**.** *Under any Möbius transformation $M : x \mapsto \frac{px+q}{rx+s}$ of $\mathbb{P}^1$, mapping the interval $[a, b]$ to $[M(a), M(b)]$, the canonical form is invariant:*

$$d\log\frac{M(x) - M(a)}{M(x) - M(b)} \;=\; d\log\frac{x-a}{x-b}. \tag{62.12}$$

Proof. Compute the Möbius difference. With $M(x) = \frac{px+q}{rx+s}$,

$$M(x) - M(a) = \frac{(px+q)(ra+s) - (pa+q)(rx+s)}{(rx+s)(ra+s)} = \frac{(ps-qr)(x-a)}{(rx+s)(ra+s)}, \tag{62.13}$$

where the numerator simplifies because all terms quadratic in the coordinates cancel, leaving $(ps - qr)(x - a)$. Likewise $M(x) - M(b) = \dfrac{(ps-qr)(x-b)}{(rx+s)(rb+s)}$. Dividing,

$$\frac{M(x) - M(a)}{M(x) - M(b)} = \frac{x-a}{x-b} \cdot \frac{rb+s}{ra+s}, \tag{62.14}$$

and the second factor is a constant independent of x. Since $d\log$ of a constant vanishes, Eq. (62.12) follows. The two endpoint poles, with their unit residues, are carried to the two new endpoints; the dictionary is Möbius-covariant. □

62.1.3 The simplex: projective covariance in any dimension

The interval is the one-dimensional member of a family — the simplices — whose canonical forms we can write in closed form and which already display projective covariance in the higher-dimensional way that the amplituhedron will need. We treat the triangle in full and state the general simplex.

It is convenient to use homogeneous coordinates and the determinant bracket. A point of $\mathbb{P}^2$ is a line through the origin in $\mathbb{R}^3$, represented by a nonzero vector $Z = (Z^0, Z^1, Z^2)$ up to overall scale; the affine chart $Z^0 = 1$ identifies $Z = (1, x, y)$ with the plane point (x, y). For three such vectors define the bracket

$$\langle Z_1\, Z_2\, Z_3\rangle \;\equiv\; \det\begin{pmatrix} Z_1^0 & Z_1^1 & Z_1^2 \\ Z_2^0 & Z_2^1 & Z_2^2 \\ Z_3^0 & Z_3^1 & Z_3^2 \end{pmatrix}, \tag{62.15}$$

which is multilinear and totally antisymmetric in its three slots and vanishes precisely when the three points are collinear. For affine points $Z_i = (1, x_i, y_i)$ it equals twice the signed area of the triangle they span. Treating the running point $Y = (1, x, y)$ as a variable, the bracket $\langle Y\, Z_i\, Z_j\rangle$ is an affine-linear function

of (x, y) that vanishes on the line through Z_i and Z_j. Finally, the projective measure on $\mathbb{P}^2$ is

$$\langle Y\, d^2Y\rangle \;\equiv\; \tfrac{1}{2}\,\epsilon_{abc}\, Y^a\, dY^b \wedge dY^c, \tag{62.16}$$

which in the affine chart $Y = (1, x, y)$, where $dY^0 = 0$, reduces to $\langle Y\, d^2Y\rangle = dx \wedge dy$, as one checks by retaining only the $a = 0$ terms of the sum.

Theorem 62.7 (Canonical form of the triangle)**.** *Let T be the oriented triangle in $\mathbb{P}^2$ with vertices Z_1, Z_2, Z_3 (counterclockwise). Its canonical form is the projectively-covariant dlog form*

$$\Omega(T) \;=\; \frac{\langle Z_1\, Z_2\, Z_3\rangle^2\; \langle Y\, d^2Y\rangle}{\langle Y\, Z_1\, Z_2\rangle\, \langle Y\, Z_2\, Z_3\rangle\, \langle Y\, Z_3\, Z_1\rangle}. \tag{62.17}$$

It has simple poles on the three edge-lines $\langle Y\, Z_i Z_{i+1}\rangle = 0$ and nowhere else; on each edge the residue is the interval form of that edge, and on each vertex the second residue is ± 1.

Proof. Well-definedness on $\mathbb{P}^2$. Under a rescaling $Y \to \Lambda Y$ of the homogeneous coordinate the measure scales as $\langle Y\, d^2Y\rangle \to \Lambda^3 \langle Y\, d^2Y\rangle$ (three factors of Y), while each of the three denominator brackets is linear in Y, contributing Λ^3; the vertex bracket carries no Y. The total weight is $\Lambda^3/\Lambda^3 = \Lambda^0$, so $\Omega(T)$ is a genuine form on $\mathbb{P}^2$, with no pole hiding at the line at infinity.

Reduction to affine coordinates. Place the standard triangle at $Z_1 = (1, 0, 0)$, $Z_2 = (1, 1, 0)$, $Z_3 = (1, 0, 1)$, i.e. the plane region $\{x \geq 0,\, y \geq 0,\, x + y \leq 1\}$. The brackets are the determinants

$$\langle Z_1 Z_2 Z_3\rangle = \det\begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = 1, \tag{62.18}$$

$$\langle Y\, Z_1 Z_2\rangle = \det\begin{pmatrix} 1 & x & y \\ 1 & 0 & 0 \\ 1 & 1 & 0 \end{pmatrix} = y, \quad \langle Y\, Z_2 Z_3\rangle = \det\begin{pmatrix} 1 & x & y \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = 1 - x - y, \quad \langle Y\, Z_3 Z_1\rangle = \det \tag{62.19}$$

which are the three edge functions of the region: $y = 0$, $x + y = 1$, and $x = 0$. With $\langle Y\, d^2Y\rangle = dx \wedge dy$ the formula Eq. (62.17) becomes

$$\Omega(T) = \frac{1^2\, dx \wedge dy}{y\,(1 - x - y)\, x} = \frac{dx \wedge dy}{x\, y\,(1 - x - y)}. \tag{62.20}$$

Inside the region all three factors $x, y, 1 - x - y$ are positive, so $\Omega(T)$ is again a positive measure on the interior, the higher-dimensional echo of the interval result. Its poles are simple and sit exactly on the three edges and nowhere

else; these are the three boundaries of T, establishing clause (ii). For a general (nondegenerate) triangle the projective transformation carrying the standard triangle to it preserves Eq. (62.17) by the same constant-factor argument as in Theorem 62.6 applied slot by slot, so the result is projectively covariant and it suffices to have checked the standard representative.

Residues, clause (iii). Take the residue on the edge $x = 0$, i.e. $\langle Y Z_3 Z_1 \rangle = 0$. Writing Eq. (62.20) as $\frac{dx}{x} \wedge \frac{dy}{y(1-y)} +$ (regular) near $x = 0$ and applying Eq. (62.3),

$$\mathrm{Res}_{x=0}\, \Omega(T) = \frac{dy}{y(1-y)} = d\log\frac{y}{1-y}, \tag{62.21}$$

which is exactly the interval canonical form Eq. (62.7) on the edge segment $0 \le y \le 1$ joining the vertices $Z_1 = (0,0)$ and $Z_3 = (0,1)$. Taking a further residue at the endpoint $y = 0$ gives $+1$, the canonical form of that vertex. The same computation on the other two edges, by the cyclic symmetry of Eq. (62.17), gives their interval forms. Thus the recursive axiom holds at every level, and by Theorem 62.4 Eq. (62.17) is the canonical form. □

The pattern generalizes verbatim. The standard m-simplex is the region $\{x_1 \ge 0, \dots, x_m \ge 0,\ x_1 + \cdots + x_m \le 1\}$ in the affine chart of $\mathbb{P}^m$, bounded by the $m+1$ hyperplanes $x_1 = 0, \dots, x_m = 0$, and $1 - \sum_i x_i = 0$. Repeating the residue argument facet by facet — each residue strips one factor and leaves a lower simplex's form — gives the canonical form

$$\Omega(\Delta^m) \;=\; \frac{dx_1 \wedge \cdots \wedge dx_m}{x_1\, x_2 \cdots x_m\, (1 - x_1 - \cdots - x_m)}, \tag{62.22}$$

a single dlog top form with simple poles on each of the $m+1$ facets and unit iterated residues down to each vertex. The simplex is the prototype of a positive geometry: its canonical form is literally a wedge of $d\log$'s of the facet functions, and every boundary stratum is again a simplex. We will see that the amplituhedron and the associahedron, however curved or high-dimensional, inherit exactly this structure — their canonical forms are dlog, their residues are lower geometries — which is why the simplex is the right object to keep in mind throughout.

62.1.4 Polygons and triangulation independence

The first place the theory acquires real content — and the first place it meets the structure of physical amplitudes head on — is the convex polygon. A convex polygon is not a simplex once it has more than three vertices, yet there is an obvious way to *build* its canonical form: cut the polygon into triangles and add up the triangle forms. The danger is that each triangle drags in poles on its edges, and the internal cutting edges are not boundaries of the polygon at all. If

the construction is to produce the canonical form of Theorem 62.3, those internal poles must disappear. They do, and the mechanism is the cornerstone of the chapter.

We need one computation: the residue of a triangle's form on one of its edges, in a form clean enough to compare two triangles meeting along a shared edge.

Lemma 62.8 (Edge residue of a triangle form)**.** *Abbreviate* $\omega_{ijk} \equiv \dfrac{\langle ijk\rangle^2\,\langle Y\,d^2Y\rangle}{\langle Yij\rangle\,\langle Yjk\rangle\,\langle Yki\rangle}$, *the canonical form of the triangle with vertices* Z_i, Z_j, Z_k*, where* $\langle Yij\rangle \equiv \langle YZ_iZ_j\rangle$ *and* $\langle ijk\rangle \equiv \langle Z_iZ_jZ_k\rangle$*. Parametrize the edge-line through* Z_i, Z_j *by* $Y(\tau) = (1-\tau)Z_i + \tau Z_j$*, so* $\tau = 0$ *at* Z_i *and* $\tau = 1$ *at* Z_j*. Then the residue on that line is the interval form of the segment, oriented from* Z_i *to* Z_j*, and is* independent of the apex Z_k *up to its overall sign:*

$$\mathrm{Res}_{\langle Yij\rangle=0}\;\omega_{ijk} \;=\; -\,\frac{d\tau}{\tau(1-\tau)}. \tag{62.23}$$

Proof. Introduce coordinates adapted to the edge-line: a transverse coordinate $s \equiv \langle Yij\rangle$, which vanishes on the line, and the line parameter τ, by setting

$$Y \;=\; (1-\tau)\,Z_i + \tau\,Z_j + s\,W, \tag{62.24}$$

where $W = (0, w_1, w_2)$ is a fixed vector at infinity normalized so that $\langle WZ_iZ_j\rangle = 1$; this guarantees $\langle Yij\rangle = (1-\tau)\langle Z_iij\rangle + \tau\langle Z_jij\rangle + s\langle Wij\rangle = s$, since the first two brackets vanish (a repeated slot). The measure transforms with a constant Jacobian: in the affine chart $\langle Y\,d^2Y\rangle = dx\wedge dy$ and (x,y) is affine-linear in (s,τ), so $dx\wedge dy = c\;ds\wedge d\tau$ with

$$c \;=\; \det\big[\,W_{\text{aff}}\,,\,(Z_j - Z_i)_{\text{aff}}\,\big] \;=\; -\langle WZ_iZ_j\rangle \;=\; -1, \tag{62.25}$$

the middle equality being the expansion of the 3×3 bracket $\langle WZ_iZ_j\rangle$ for W at infinity. Now evaluate the remaining two denominator brackets on the line, using multilinearity in the first slot:

$$\langle Yjk\rangle = (1-\tau)\langle ijk\rangle, \qquad \langle Yki\rangle = \tau\,\langle jki\rangle = \tau\,\langle ijk\rangle, \tag{62.26}$$

where $\langle Z_jjk\rangle = 0$ and $\langle Z_iki\rangle = 0$ kill the missing terms and $\langle jki\rangle = \langle ijk\rangle$ by cyclic symmetry. Substituting everything,

$$\omega_{ijk} = \frac{\langle ijk\rangle^2\,(-1)\,ds\wedge d\tau}{s\cdot(1-\tau)\langle ijk\rangle\cdot\tau\langle ijk\rangle} = -\,\frac{ds}{s}\wedge\frac{d\tau}{\tau(1-\tau)}. \tag{62.27}$$

The factors of $\langle ijk\rangle^2$ cancel completely against the two linear factors, which is why the result does not depend on the apex. Reading off the residue via Eq. (62.3) gives Eq. (62.23). □

The crucial structural consequence is the sign. Consider two triangles of a triangulation that share an edge — the diagonal between vertices Z_i and Z_j. With a fixed (say counterclockwise) orientation of the plane, the two triangles sit on opposite sides of the diagonal and traverse it in opposite cyclic orders: one as ω_{ijk} (edge oriented $i \to j$), the other as ω_{jil} (edge oriented $j \to i$). Running Theorem 62.8 for the second triangle, with the diagonal written as the factor $\langle Yji\rangle = -\langle Yij\rangle$ and the same global parameter τ from Z_i to Z_j, the two sign flips (one from $\langle Yji\rangle = -s$, one from the swapped vertex labels in the other brackets) combine to give

$$\mathrm{Res}_{\langle Yij\rangle=0}\ \omega_{jil} \;=\; +\frac{d\tau}{\tau(1-\tau)} \;=\; -\,\mathrm{Res}_{\langle Yij\rangle=0}\ \omega_{ijk}. \tag{62.28}$$

The residues on the shared diagonal are equal and opposite. Therefore in the *sum* of the two triangle forms the diagonal pole has zero residue, and by the decomposition Eq. (62.2) a simple-pole form with vanishing residue is regular there: the diagonal pole is absent from the sum. This is the disappearance of the spurious pole, proved.

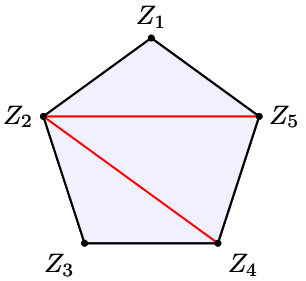
Theorem 62.9 (Polygon canonical form and triangulation independence)**.** *Let P be a convex polygon in $\mathbb{P}^2$ with cyclically ordered vertices $Z_1, \dots, Z_n$. Choose any triangulation of P into $n-2$ triangles using the polygon's vertices, oriented consistently with P. Then the sum of the triangle canonical forms,*

$$\Omega(P) \;=\; \sum_{\triangle \in\ \textit{triangulation}} \Omega(\triangle), \tag{62.29}$$

is independent of the chosen triangulation, has simple poles only on the n edge-lines $\langle Y\, Z_a Z_{a+1}\rangle = 0$ (indices cyclic), and is the canonical form of P in the sense of Theorem 62.3.

Proof. Poles only on the edges. Each triangle of the triangulation contributes poles on its three edge-lines. These edge-lines are of two kinds: *boundary edges* of P, each belonging to exactly one triangle of the triangulation, and *diagonals*, each shared by exactly two triangles (the two regions it separates). On any diagonal the two contributing residues cancel by Eq. (62.28), so the sum Eq. (62.29) has no pole there. On a boundary edge only one triangle contributes, with residue equal, by Theorem 62.8, to the interval canonical form of that edge. Hence $\Omega(P)$ has simple poles exactly on the n boundary edges, with residue on each equal to the canonical form of that edge — precisely clauses (ii) and (iii) of Theorem 62.3. Each $\Omega(\triangle)$ is projective weight zero, so the sum is a bona fide form on $\mathbb{P}^2$ with no spurious pole at infinity.

Triangulation independence. Whatever the triangulation, the resulting form has poles confined to the same n edge-lines and the *same* residue on each (the edge's interval form, which does not depend on which triangulation produced it).



Triangulation A: $\{123, 134, 145\}$ Triangulation B: $\{234, 245, 125\}$

Figure 62.1: Two triangulations of a convex pentagon (Theorem 62.10). A fans from Z_1, using the diagonals $\langle Y13\rangle, \langle Y14\rangle$ (red); B fans from Z_2, using the disjoint pair $\langle Y24\rangle, \langle Y25\rangle$. The two triangulations share no diagonals, yet the sum of triangle canonical forms is the same in both cases: every diagonal pole appears in two triangles with opposite residues and cancels, leaving a form with simple poles only on the five edges $\langle Y\, a\, a{+}1\rangle = 0$.

By the uniqueness Theorem 62.4, any two such forms coincide. Hence $\Omega(P)$ is independent of the triangulation, and being the unique form with these poles and residues, it is the canonical form of P. □

The polygon is the first place the geometry computes something a physicist recognizes: a sum over triangulations, each triangle carrying genuine and spurious poles, with the spurious ones cancelling pairwise to leave a triangulation-independent total whose only singularities are physical. We now carry out the cancellation explicitly on the pentagon, where there are enough diagonals for the bookkeeping to be nontrivial and where two manifestly different triangulations can be compared term by term.

Worked Example 62.10 (The pentagon two ways: spurious diagonals cancel)**.** Let P be a convex pentagon with cyclically ordered vertices $Z_1, \dots, Z_5$ (Fig. 62.1). We compute its canonical form by two triangulations that share no diagonal and verify they agree, with all spurious poles cancelling.

Triangulation A fans from Z_1:

$$\Omega_A = \omega_{123} + \omega_{134} + \omega_{145}, \tag{62.30}$$

using the two diagonals $\langle Y13\rangle = 0$ and $\langle Y14\rangle = 0$. *Triangulation B* fans from Z_2:

$$\Omega_B = \omega_{234} + \omega_{245} + \omega_{125}, \tag{62.31}$$

using the disjoint pair of diagonals $\langle Y24\rangle = 0$ and $\langle Y25\rangle = 0$. (Here ω_{ijk} is the triangle form of Theorem 62.8, and we have written each triangle in the cyclic order inherited from the pentagon's orientation, so that shared edges are traversed oppositely by adjacent triangles.)

Poles of Ω_A. The pole-lines appearing in the three terms are

$$\begin{aligned} \omega_{123}&: \ \langle Y12\rangle,\ \langle Y23\rangle,\ \langle Y13\rangle,\\ \omega_{134}&: \ \langle Y13\rangle,\ \langle Y34\rangle,\ \langle Y14\rangle,\\ \omega_{145}&: \ \langle Y14\rangle,\ \langle Y45\rangle,\ \langle Y15\rangle. \end{aligned} \tag{62.32}$$

The five edge-lines $\langle Y12\rangle, \langle Y23\rangle, \langle Y34\rangle, \langle Y45\rangle, \langle Y15\rangle$ each occur once. The diagonal $\langle Y13\rangle$ occurs in ω_{123} and ω_{134}; in the first the shared edge is traversed $1 \to 3$, in the second $3 \to 1$, so by Eq. (62.28) their residues on $\langle Y13\rangle = 0$ are

$$\mathrm{Res}_{\langle Y13\rangle=0}\,\omega_{123} = -\frac{d\tau}{\tau(1-\tau)}, \qquad \mathrm{Res}_{\langle Y13\rangle=0}\,\omega_{134} = +\frac{d\tau}{\tau(1-\tau)}, \tag{62.33}$$

which sum to zero. Identically, the diagonal $\langle Y14\rangle$ occurs in ω_{134} (edge $1 \to 4$) and ω_{145} (edge $4 \to 1$), and the two residues cancel. Therefore Ω_A has *no* pole on either diagonal; its only singularities are simple poles on the five edge-lines, and on each edge its residue is that edge's interval form, e.g. on $\langle Y23\rangle = 0$ the single contributor ω_{123} gives $-d\tau/[\tau(1-\tau)]$ by Theorem 62.8.

Poles of Ω_B. The same audit for B:

$$\begin{aligned} \omega_{234}&: \ \langle Y23\rangle,\ \langle Y34\rangle,\ \langle Y24\rangle,\\ \omega_{245}&: \ \langle Y24\rangle,\ \langle Y45\rangle,\ \langle Y25\rangle,\\ \omega_{125}&: \ \langle Y12\rangle,\ \langle Y25\rangle,\ \langle Y15\rangle. \end{aligned} \tag{62.34}$$

Again each of the five edge-lines occurs once. The diagonal $\langle Y24\rangle$ occurs in ω_{234} (edge $2 \to 4$) and ω_{245} (edge $4 \to 2$) and cancels; the diagonal $\langle Y25\rangle$ occurs in ω_{245} (edge $2 \to 5$) and ω_{125} (edge $5 \to 2$) and cancels. So Ω_B also has poles only on the five edges, with the *same* residue on each as Ω_A — on every edge $\langle Y\, a\, a{+}1\rangle = 0$ exactly one triangle (in either triangulation) contributes the same interval form.

Conclusion. Ω_A and Ω_B are rational 2-forms with identical pole-lines (the five edges) and identical residues there, so by Theorem 62.4 they are equal:

$$\Omega_A = \Omega_B = \Omega(P), \tag{62.35}$$

the unique canonical form of the pentagon. The diagonals — $\{13, 14\}$ in A, $\{24, 25\}$ in B — are spurious in both: they are artifacts of the cut, present in the individual triangle terms but absent from the geometric total. The canonical form remembers only the five genuine edges of the shape, exactly as the dictionary of Theorem 62.3 demands. This is, in miniature, the statement that a tree amplitude assembled by recursion has physical poles only on its true factorization channels, the spurious poles of the individual recursion terms cancelling in the sum — a correspondence the next section makes literal.

We now have a precise, computable, projectively-invariant notion of the canonical form of a positive geometry, together with the dictionary that ties each boundary to a logarithmic singularity and each residue to a lower geometry, and we have seen — on the polygon — that triangulation produces it with all spurious singularities cancelling. The next section pours an actual scattering amplitude into this mould: it realizes the color-ordered tree amplitude of bi-adjoint ϕ^3 theory as the canonical form of a concrete polytope, the ABHY kinematic associahedron, living directly in Mandelstam space, turning "amplitude = volume" from a slogan into a theorem before any conjecture about the amplituhedron is made.

62.2 Amplitudes as canonical forms: the kinematic associahedron

The previous section built an abstraction and tested it on toys. A positive geometry is a region whose boundaries carry a unique differential form with logarithmic singularities exactly on those boundaries and nowhere else (Theorem 62.3); for a convex polygon we proved that this *canonical form* is triangulation independent and may be assembled by summing a simple local form over the vertices (Theorem 62.9); and we read off the dictionary on which the whole chapter turns, *boundaries* $\leftrightarrow$ *singularities*: the residue of the canonical form on any facet is the canonical form of that facet (Eq. (62.5)). All of this was geometry. Not a single scattering amplitude has yet appeared.

This section discharges the promissory note "amplitude = volume" before any conjecture is allowed near it. We will exhibit a concrete, honest, finite-dimensional polytope — the *kinematic associahedron* of Arkani-Hamed, Bai, He and Yan (ABHY) — living directly in the space of Mandelstam invariants, and prove a theorem: its canonical form, pulled back to physical kinematics, *is* the tree-level color-ordered amplitude of bi-adjoint cubic scalar theory, propagator for propagator. There is no large-N limit, no supersymmetry, no helicity, and nothing conjectural; the statement is a finite identity between a sum of products of propagators and a sum of dlog forms over the vertices of a polytope, and we will verify it term by term, culminating in the five-term five-point amplitude realized as the very pentagon of Section 62.1. The locality and unitarity of the amplitude — poles only on physical channels, residues that factorize into lower amplitudes — will emerge as the statement that the facets of the polytope are themselves products of lower associahedra. This is the cleanest existing instance of the chapter's thesis, and it is completely provable, so we prove it completely.

62.2.1 The object to be reproduced: bi-adjoint cubic amplitudes

We continue the scalar of Chapter 61, the "zeroth copy" whose double copy delivers gravity and whose color algebra underlies the Yang–Mills color ordering

of Chapter 59. The bi-adjoint cubic theory carries a scalar $\phi^{aa'}$ in the adjoint of a *pair* of Lie groups, with Lagrangian

$$\mathcal{L} = \tfrac{1}{2}\,(\partial_\mu \phi^{aa'})(\partial^\mu \phi^{aa'}) - \tfrac{\lambda}{3!}\, f^{abc}\, \tilde{f}^{a'b'c'}\, \phi^{aa'} \phi^{bb'} \phi^{cc'}, \tag{62.36}$$

where f^{abc} and $\tilde{f}^{a'b'c'}$ are the structure constants of the two groups (Chapter 61). Because the only interaction is cubic and massless, every tree contributing to an n-point function is a trivalent (cubic) tree graph, and each carries one factor of f and one factor of $\tilde{f}$ at every vertex and one massless scalar propagator on every internal line.

Stripping the two color algebras by the double color decomposition produces *doubly* color-ordered partial amplitudes $m(\alpha \,|\, \beta)$, one for each pair of cyclic orderings α, β of the n legs; this is the bi-adjoint refinement of the single color ordering defined in Theorem 59.3. We need only the diagonal entry, in which both orderings are the standard one $\mathbf{1} = (1, 2, \ldots, n)$. Write

$$m_n \;\equiv\; m(1, 2, \ldots, n \,|\, 1, 2, \ldots, n). \tag{62.37}$$

The double color decomposition (Chapter 61) selects, in m_n, exactly those trivalent trees that are *planar with respect to the ordering* $(1, \ldots, n)$: the cyclic graphs that can be drawn in a disk with the external legs $1, \ldots, n$ on the boundary in that cyclic order without crossings. Each such planar tree contributes the product of its internal propagators, with unit numerator (all numerators are ± 1 once color is stripped, and the planar tree fixes the sign to $+1$). We strip an overall $i\,\lambda^{n-2}$, as is standard for the amplitudes of this part, and define

$$\boxed{\; m_n \;=\; \sum_{\text{planar trivalent trees } T} \;\prod_{\text{internal lines } e \in T} \frac{1}{X_e}\,, \;} \tag{62.38}$$

where X_e is the squared momentum flowing through internal line e. The reduction of the full diagrammatic sum to Eq. (62.38) is exactly the statement, established for the bi-adjoint theory in Chapter 61, that with both color orders fixed and equal the surviving graphs are the planar ones with numerator $+1$; we import it rather than re-deriving the color algebra here.

It remains to name the X_e uniformly. Take all external momenta outgoing, $\sum_{i=1}^{n} p_i = 0$, and massless, $p_i^2 = 0$, with Mandelstam invariants $s_{ij} = (p_i + p_j)^2 = 2\, p_i \cdot p_j$ in the mostly-minus metric (Chapter A). Draw the ordering as a convex n-gon with vertices labeled $1, \ldots, n$ and the edge between vertices i and $i{+}1$ carrying the external momentum p_i. A planar internal line of a trivalent tree is the same datum as a *diagonal* of this n-gon: cutting the tree on that line splits the external legs into two consecutive arcs, and the diagonal separates the corresponding vertices.

Definition 62.11 (Planar kinematic variables)**.** For $1 \le i < j \le n$ let the *planar variable* $X_{i,j}$ be the squared sum of the momenta on one arc cut off by the diagonal (i, j),

$$X_{i,j} \;=\; (p_i + p_{i+1} + \cdots + p_{j-1})^2. \tag{62.39}$$

Boundary cases reduce to edges: $X_{i,i+1} = p_i^2 = 0$ and $X_{1,n} = (p_1 + \cdots + p_{n-1})^2 = p_n^2 = 0$. The remaining $X_{i,j}$ with $2 \le j - i \le n - 2$ are the $\frac{n(n-3)}{2}$ genuine diagonals, the planar propagator invariants.

With this naming, the internal line of a planar tree corresponding to diagonal (i, j) carries propagator $1/X_{i,j}$, and Eq. (62.38) becomes a sum over triangulations of the n-gon. This is the combinatorial fact we now isolate, because it is what makes Eq. (62.38) a statement about a polytope.

62.2.2 The associahedron and the diagram ↔ face dictionary

A trivalent (cubic) tree with n external legs has $n-2$ vertices and $n-3$ internal lines. Under the dictionary above, its $n-3$ internal lines are $n-3$ pairwise non-crossing diagonals of the n-gon — that is, a *triangulation* of the n-gon into $n-2$ triangles. Conversely every triangulation, read as a set of $n-3$ non-crossing diagonals, is dual to a unique planar trivalent tree (put one tree vertex in each triangle and one tree edge across each diagonal). The trees summed in Eq. (62.38) are therefore in bijection with the triangulations of the n-gon, and their number is the Catalan number $C_{n-2} = \frac{1}{n-1}\binom{2n-4}{n-2}$ ($C_2 = 2$ for $n = 4$, $C_3 = 5$ for $n = 5$).

These triangulations are the vertices of a polytope.

Definition 62.12 (Associahedron)**.** The *associahedron* K_{n-3} is the $(n-3)$-dimensional convex polytope whose face lattice is the set of collections of pairwise non-crossing diagonals of a convex n-gon, ordered by reverse inclusion. Concretely, a face of codimension d is a set of d mutually non-crossing diagonals (a partial triangulation with d cuts); a facet ($d = 1$) is a single diagonal; a vertex ($d = n-3$) is a full triangulation.

That such a polytope exists — that these combinatorial data really are the face lattice of a convex body of dimension $n-3$ — is the content of the next subsection, where we build it by inequalities. Granting existence, the dictionary we need is immediate.

Theorem 62.13 (Faces ↔ planar diagrams)**.** *There is an inclusion-reversing bijection between the faces of K_{n-3} and the planar trivalent-or-degenerate trees of the bi-adjoint n-point amplitude:*

- *vertices of K_{n-3} ↔ planar trivalent trees (the terms of Eq. (62.38));*
- *facets of K_{n-3} ↔ single planar propagators $X_{i,j}$ (a facet is the locus where one channel goes on shell);*

- *a codimension-d face $\leftrightarrow$ a planar tree with $n-3-d$ internal lines, i.e. a diagram with d propagators collapsed to contact vertices.*

In particular the codimension-d faces of K_{n-3} are products of lower associahedra: collapsing d compatible diagonals factorizes the n-gon along those diagonals into sub-polygons, and the face is the product of their associahedra.

Proof. The first three statements are the bijection between non-crossing diagonal sets and faces of Theorem 62.12, transported through the diagonal $\leftrightarrow$ propagator and triangulation $\leftrightarrow$ tree dualities of the opening paragraph. For the factorization clause, fix one diagonal (i,j); it partitions the n-gon into two sub-polygons sharing the edge (i,j), with $j-i+1$ and $n-(j-i)+1$ vertices respectively. A diagonal of the n-gon is non-crossing with (i,j) iff it lies entirely within one sub-polygon, so the partial triangulations refining the single cut (i,j) are exactly pairs (triangulation of left sub-polygon, triangulation of right sub-polygon). The face lattice below the facet (i,j) is therefore the product of the two sub-polygon face lattices, i.e. the facet is $K_a \times K_b$ with $a+b=n-4$. Iterating over d compatible diagonals gives the general claim. □

The factorization clause is the geometric germ of unitarity: the facet $X_{i,j}=0$, where the propagator $1/X_{i,j}$ blows up, is literally the product of the two lower amplitudes' geometries, glued on the channel (i,j). We will see this become the residue check of Theorem 62.18. First we must build the polytope inside Mandelstam space, for only then does "the canonical form" of K_{n-3} acquire concrete coordinates and become equal to Eq. (62.38).

62.2.3 The ABHY embedding: carving the associahedron out of Mandelstam space

The planar variables $X_{i,j}$ are not independent: they are quadratic functions of the p_i, and a counting of the kinematic data shows the planar variables span a space of dimension only $n-3$ once momentum conservation and masslessness are imposed — the same as $\dim K_{n-3}$. The ABHY construction makes this precise and, simultaneously, positive. The key is a four-term combination of planar variables that collapses to a single Mandelstam.

Lemma 62.14 (The c-identity)**.** *For all admissible i,j,*

$$c_{i,j} \;\equiv\; X_{i,j}+X_{i+1,j+1}-X_{i,j+1}-X_{i+1,j} \;=\; -\,s_{ij} \;=\; -2\,p_i\cdot p_j. \tag{62.40}$$

Proof. Write $P \equiv p_{i+1}+\cdots+p_{j-1}$ for the momenta strictly between i and j. From Eq. (62.39),

$$X_{i,j}=(p_i{+}P)^2,\quad X_{i+1,j+1}=(P{+}p_j)^2,\quad X_{i,j+1}=(p_i{+}P{+}p_j)^2,\quad X_{i+1,j}=P^2. \tag{62.41}$$

Using $p_i^2 = p_j^2 = 0$ and expanding,

$$X_{i,j} = P^2 + 2\,p_i{\cdot}P, \qquad X_{i+1,j+1} = P^2 + 2\,P{\cdot}p_j,$$
$$X_{i,j+1} = P^2 + 2\,p_i{\cdot}P + 2\,P{\cdot}p_j + 2\,p_i{\cdot}p_j, \qquad X_{i+1,j} = P^2. \tag{62.42}$$

Substituting into Eq. (62.40), every P^2 and every mixed $p \cdot P$ term cancels, leaving $c_{i,j} = -2\,p_i \cdot p_j = -s_{ij}$. □

Definition 62.15 (ABHY constraints and the kinematic associahedron)**.** Fix positive constants $c_{i,j} > 0$ for every non-adjacent pair $1 \le i < j \le n-1$ with $j - i \ge 2$. The *ABHY subspace* $\mathcal{H}_n \subset \{X_{i,j}\}$ is the affine subspace on which

$$X_{i,j}+X_{i+1,j+1}-X_{i,j+1}-X_{i+1,j} \;=\; c_{i,j} \qquad (1 \le i < j \le n-1,\; j-i \ge 2), \tag{62.43}$$

i.e. on which the non-adjacent Mandelstams take the negative constant values $s_{ij} = -c_{i,j} < 0$. The *kinematic associahedron* $\mathcal{A}_n$ is the region of $\mathcal{H}_n$ on which *all* planar variables are nonnegative,

$$\mathcal{A}_n \;=\; \{\, X_{i,j} \ge 0 \text{ for all diagonals} \,\} \cap \mathcal{H}_n. \tag{62.44}$$

The constraints Eq. (62.43) number $(n-2)(n-3)/2$ (the non-adjacent pairs among $1, \dots, n-1$), so

$$\dim \mathcal{H}_n \;=\; \frac{n(n-3)}{2} - \frac{(n-2)(n-3)}{2} \;=\; \frac{n-3}{2}\big(n-(n-2)\big) \;=\; n-3, \tag{62.45}$$

matching $\dim K_{n-3}$. The constants $c_{i,j} > 0$ are pure gauge on the geometry: changing them deforms $\mathcal{A}_n$ continuously without changing its combinatorial type, exactly as choosing different facet offsets moves a polytope without changing which facets meet. We now state the embedding theorem; its full proof is an induction on n built from the factorization of Theorem 62.13, and we give the inductive mechanism in general and the explicit realization at $n = 4, 5$.

Theorem 62.16 (ABHY realization)**.** *For any choice of positive constants $c_{i,j}$, the region $\mathcal{A}_n$ of Theorem 62.15 is a convex polytope combinatorially isomorphic to the associahedron K_{n-3}. Its facets are exactly the loci $X_{i,j} = 0$, one for each diagonal (i,j); the facet $X_{i,j} = 0$ is itself (a translate of) $\mathcal{A}_a \times \mathcal{A}_b$ with $a+b = n-4$, realizing the factorization of Theorem 62.13 as an honest geometric product.*

Proof (structure and inductive step). Convexity is immediate: $\mathcal{A}_n$ is the intersection of the affine space $\mathcal{H}_n$ with the half-spaces $X_{i,j} \ge 0$, each a linear inequality once the X are solved on $\mathcal{H}_n$ (the c-identity Eq. (62.40) is linear in the X, so Eq. (62.43) is a linear system and every $X_{i,j}$ restricts to an affine function of $n-3$ free coordinates).

The substance is that the bounded region cut out has precisely the associahedron's facet incidences, i.e. that each $X_{i,j} = 0$ does meet $\mathcal{A}_n$ in a facet, and two of them, $X_{i,j} = 0$ and $X_{k,l} = 0$, share a codimension-two face iff the diagonals (i, j) and (k, l) are non-crossing. We argue this by induction on n, the base cases $n = 4$ (a segment) and $n = 5$ (a pentagon) being verified explicitly below.

Consider one facet candidate $X_{i,j} = 0$. Setting $X_{i,j} = 0$ in Eq. (62.43) and using Theorem 62.14, the constraints split: those whose four planar variables lie among the diagonals of the left sub-polygon (vertices $i, \ldots, j$) become the ABHY constraints of that sub-polygon, with the *same* constants c; likewise on the right (vertices $j, \ldots, i$ cyclically). The only constraints mixing the two sides are those containing $X_{i,j}$ itself, and on $X_{i,j} = 0$ they reduce to relations expressing the "would-be crossing" variables in terms of the constants, forcing every diagonal that *crosses* (i, j) to be a fixed positive constant plus a sum of the two sides' variables — in particular strictly positive on the sub-polytope, hence never vanishing there. Therefore on $X_{i,j} = 0$ the only remaining facet inequalities are the diagonals lying within one sub-polygon, and the section is exactly $\mathcal{A}_a \times \mathcal{A}_b$ $(a+b = n-4)$, which by the inductive hypothesis is a product of associahedra of total dimension $n-4$, i.e. a genuine facet. Crossing diagonals do not bound this facet, so two facets meet iff their diagonals are non-crossing. This is the facet incidence of K_{n-3}, and a convex polytope is determined by its facet incidences up to combinatorial isomorphism; hence $\mathcal{A}_n \cong K_{n-3}$. Boundedness follows because the facet functions include, for each direction, both an "upper" and a "lower" planar variable (as the explicit $n = 5$ system below exhibits), pinning the region. □

Physical Insight

The single design choice that makes this work is Eq. (62.43): *fix the non-adjacent Mandelstams to negative constants.* Physically, $s_{ij} = -c_{ij} < 0$ is a Euclidean (spacelike) configuration in which no non-adjacent channel can go on shell; the only invariants left free are the *planar* ones $X_{i,j}$, and demanding them positive is demanding that every planar channel sit on the physical (timelike, resonant) side. The associahedron is thus the region of kinematic space where physics is "maximally on shell in the planar channels and maximally off shell in the rest" — and that region's boundaries are precisely the planar propagator poles. Locality is being built into the shape of a polytope.

62.2.4 The canonical form is the amplitude

We can now state the result that makes "amplitude = volume" a theorem rather than a slogan.

Theorem 62.17 (Canonical form = bi-adjoint amplitude)**.** *Let $\Omega(\mathcal{A}_n)$ be the canonical form of the kinematic associahedron (Theorem 62.3), and let $X_1, \dots, X_{n-3}$ be any $n-3$ planar variables forming an affine coordinate system on $\mathcal{H}_n$. Then*

$$\Omega(\mathcal{A}_n) \;=\; m_n \; \mathrm{d}X_1 \wedge \cdots \wedge \mathrm{d}X_{n-3}, \tag{62.46}$$

where m_n is the bi-adjoint partial amplitude Eq. (62.38). Equivalently, the rational function multiplying the top-form built from the planar coordinates is exactly the sum over planar trees of products of propagators.

Proof. By the polygon/polytope assembly of Theorem 62.9 (its higher-dimensional form: the canonical form of a simple polytope is the sum over vertices of the local dlog form),

$$\Omega(\mathcal{A}_n) \;=\; \sum_{\text{vertices } V} \operatorname{sgn}(V) \bigwedge_{a \in V} \operatorname{dlog} X_a, \tag{62.47}$$

where the wedge at a vertex V runs over the $n-3$ facets $X_a = 0$ meeting there. By Theorem 62.16 a vertex is a full triangulation, i.e. a set of $n-3$ pairwise non-crossing diagonals — exactly a planar tree T of Eq. (62.38), with the facets $X_a = 0$ being its internal lines. Expanding each dlog,

$$\bigwedge_{a \in V} \operatorname{dlog} X_a = \frac{1}{\prod_{a \in V} X_a} \bigwedge_{a \in V} \mathrm{d}X_a = \frac{\det J_V}{\prod_{a \in V} X_a} \, \mathrm{d}X_1 \wedge \cdots \wedge \mathrm{d}X_{n-3}, \tag{62.48}$$

where J_V is the Jacobian matrix of the facet functions $\{X_a\}_{a \in V}$ with respect to the chosen coordinates $\{X_1, \dots, X_{n-3}\}$. The product $\prod_{a \in V} X_a$ is precisely the product of the propagators of the tree T, so its reciprocal is the summand of Eq. (62.38). It remains only that $\operatorname{sgn}(V) \det J_V = +1$ for every vertex. This is the *unimodularity* of the ABHY constraints: solving Eq. (62.43) expresses each $X_{i,j}$ as an *integer* affine combination, with coefficients in $\{0, \pm 1\}$, of the coordinates $X_1, \dots, X_{n-3}$ (each constraint involves four X's with coefficients ± 1, so back-substitution never produces a denominator and never a coefficient outside $\{0, \pm 1\}$ for the facet normals meeting at a vertex). At any vertex the $n-3$ facet normals are linearly independent integer vectors bounding a simple cone, and for such an ABHY vertex the determinant is ± 1; the orientation sign $\operatorname{sgn}(V)$ supplied by the canonical-form assembly is exactly the one that makes the product $+1$ (it is fixed by requiring unit residues with the orientation in which $\mathcal{A}_n$ has positive volume). Hence every vertex contributes $+\dfrac{1}{\prod_{a \in V} X_a} \, \mathrm{d}X_1 \wedge \cdots \wedge \mathrm{d}X_{n-3}$, and summing over vertices reproduces Eq. (62.38) times the top-form. We verify the unimodularity and the signs by explicit computation at $n = 5$ in Theorem 62.18, where all five Jacobians are $+1$. □

Physical Insight

The identity Eq. (62.46) is the rigorous content of "amplitude = volume." The canonical form is the (dual) volume form of the polytope; its single rational coefficient in planar coordinates is m_n. There is nothing to conjecture and nothing to take on faith: Eqs. (62.47) and (62.48) are an exact finite computation, vertex by vertex, and each vertex is one Feynman diagram. The amplitude's two defining physical properties are now geometric: its *poles* are the facets $X_{i,j} = 0$ (locality — singularities only on planar channels, Theorem 62.16); its *residues* are the canonical forms of those facets, which by Theorem 62.13 are products of lower associahedra, i.e. products of lower amplitudes (unitarity). We make both explicit at $n = 5$.

62.2.5 Warm-up: $n = 4$, the segment

For $n = 4$ the planar variables are the two diagonals of a square, $X_{1,3} = s_{12}$ and $X_{2,4} = s_{23}$, with $X_{1,4} = p_4^2 = 0$ and $X_{2,3} = p_2^2 = 0$. There is one non-adjacent pair, $(i, j) = (1, 3)$, so one constraint:

$$c_{1,3} = X_{1,3} + X_{2,4} - X_{1,4} - X_{2,3} = X_{1,3} + X_{2,4} = s_{12} + s_{23} \;=\; \text{const} > 0. \tag{62.49}$$

This is the familiar massless four-point relation $s+t+u = 0$ with the third channel held off shell: $s_{12} + s_{23} = -s_{13} = c_{1,3}$. The ABHY subspace is one-dimensional; take $X_{1,3}$ as coordinate, $X_{2,4} = c_{1,3} - X_{1,3}$. The region $X_{1,3} \geq 0,\ X_{2,4} \geq 0$ is the segment $X_{1,3} \in [0, c_{1,3}]$ — the associahedron K_1. Its two endpoints (facets) are the two planar channels $X_{1,3} = 0$ and $X_{2,4} = 0$, dual to the two planar Feynman diagrams (s-channel and t-channel exchange). The canonical form of a segment $[0, c]$ in coordinate x is (Section 62.1)

$$\Omega(K_1) = \operatorname{dlog} \frac{x}{c-x} = \Big(\frac{1}{x} + \frac{1}{c-x}\Big)\mathrm{d}x = \Big(\frac{1}{X_{1,3}} + \frac{1}{X_{2,4}}\Big)\mathrm{d}X_{1,3} = \Big(\frac{1}{s_{12}} + \frac{1}{s_{23}}\Big)\mathrm{d}X_{1,3}, \tag{62.50}$$

and the coefficient is exactly $m_4 = 1/s_{12} + 1/s_{23}$, the two-diagram four-point amplitude. The base case is verified; the structure of Theorem 62.17 is already visible. We now do the case the section was built for.

62.2.6 The five-point associahedron is the pentagon

Worked Example 62.18 (The $n = 5$ associahedron, end to end)**.** We carry the five-point computation completely: facet equations, the canonical form Ω, the change to planar Mandelstams, recovery of all five poles, and the residue check that factorizes locality and unitarity into lower amplitudes.

Planar variables. The pentagon has five diagonals, hence five planar variables. Using Theorem 62.11 and momentum conservation $p_1 + \cdots + p_5 = 0$,

$$\begin{aligned} &X_{1,3} = (p_1+p_2)^2 = s_{12}, \qquad X_{2,4} = (p_2+p_3)^2 = s_{23}, \qquad X_{3,5} = (p_3+p_4)^2 = s_{34}, \\ &X_{1,4} = (p_1+p_2+p_3)^2 = s_{45}, \quad X_{2,5} = (p_2+p_3+p_4)^2 = s_{51}. \end{aligned} \tag{62.51}$$

(The last two used $(p_1+p_2+p_3) = -(p_4+p_5)$ and $(p_2+p_3+p_4) = -(p_5+p_1)$.) The five planar variables are exactly the five cyclically consecutive two-particle invariants $s_{12}, s_{23}, s_{34}, s_{45}, s_{51}$ — the five physical planar channels of the ordering.

Facet equations. The non-adjacent pairs among $\{1,2,3,4\}$ with $j-i \geq 2$ are $(1,3),(1,4),(2,4)$: three ABHY constraints. With all edge variables vanishing ($X_{1,2} = \cdots = X_{5,1} = 0$), Eq. (62.43) reads

$$(1,3): \quad X_{1,3} + X_{2,4} - X_{1,4} = c_{1,3}, \tag{62.52}$$

$$(1,4): \quad X_{1,4} + X_{2,5} - X_{2,4} = c_{1,4}, \tag{62.53}$$

$$(2,4): \quad X_{2,4} + X_{3,5} - X_{2,5} = c_{2,4}. \tag{62.54}$$

Choose the two free coordinates to be the "fan from vertex 1" variables $X_{1,3}$ and $X_{1,4}$ (write $x = X_{1,3}$, $y = X_{1,4}$). Solving Eqs. (62.52) to (62.54) in turn,

$$\begin{aligned} &X_{1,3} = x, && X_{1,4} = y, \\ &X_{2,4} = c_{1,3} - x + y, && X_{2,5} = (c_{1,3}+c_{1,4}) - x, \\ &X_{3,5} = (c_{1,4}+c_{2,4}) - y. \end{aligned} \tag{62.55}$$

Every planar variable is an integer affine function of (x,y) with coefficients in $\{0,\pm 1\}$: this is the unimodularity invoked in Theorem 62.17. The five facet inequalities $X_{i,j} \geq 0$ are

$$x \geq 0, \quad y \geq 0, \quad c_{1,3} - x + y \geq 0, \quad (c_{1,3}+c_{1,4}) - x \geq 0, \quad (c_{1,4}+c_{2,4}) - y \geq 0, \tag{62.56}$$

five half-planes whose intersection, for any positive constants, is a pentagon. Taking $c_{1,3} = c_{1,4} = c_{2,4} = 1$ to draw it, the vertices are

$$(0,0),\ (1,0),\ (2,1),\ (2,2),\ (0,2), \tag{62.57}$$

shown in Fig. 62.2. This is, on the nose, the pentagon of Section 62.1; the associahedron K_2 has been realized in Mandelstam space.

Vertices are triangulations are diagrams. Each vertex of Eq. (62.57) is the meeting of two facets, i.e. a pair of non-crossing diagonals — a triangulation — and hence a planar trivalent tree with two propagators:

vertex	facets	triangulation (diagonals)	diagram term
$(0,0)$	$X_{1,3}=X_{1,4}=0$	$\{(1,3),(1,4)\}$	$1/(s_{12}\,s_{45})$
$(1,0)$	$X_{1,4}=X_{2,4}=0$	$\{(1,4),(2,4)\}$	$1/(s_{45}\,s_{23})$
$(2,1)$	$X_{2,4}=X_{2,5}=0$	$\{(2,4),(2,5)\}$	$1/(s_{23}\,s_{51})$
$(2,2)$	$X_{2,5}=X_{3,5}=0$	$\{(2,5),(3,5)\}$	$1/(s_{51}\,s_{34})$
$(0,2)$	$X_{3,5}=X_{1,3}=0$	$\{(3,5),(1,3)\}$	$1/(s_{34}\,s_{12})$

The five vertices are the $C_3 = 5$ triangulations of the pentagon, each a pair of diagonals sharing a vertex of the 5-gon (a "fan"), and they exhaust the five planar five-point Feynman diagrams.

The canonical form. Walking the boundary counterclockwise from $(0,0)$, the edges carry, in order, the vanishing variables $X_{1,4}, X_{2,4}, X_{2,5}, X_{3,5}, X_{1,3}$. The gradients of the facet functions in the (x,y) plane are

$$\begin{aligned} &\nabla X_{1,3}=(1,0), \quad \nabla X_{1,4}=(0,1), \quad \nabla X_{2,4}=(-1,1),\\ &\nabla X_{2,5}=(-1,0), \quad \nabla X_{3,5}=(0,-1). \end{aligned} \tag{62.58}$$

By Eqs. (62.47) and (62.48) the canonical form is the sum over the five vertices of $\mathrm{dlog}\, X_a \wedge \mathrm{dlog}\, X_b$, with (a,b) the two facets at the vertex taken in counterclockwise order. Each Jacobian $\det[\nabla X_a; \nabla X_b]$ evaluates, from Eq. (62.58), to $+1$:

$$\begin{aligned} &(X_{1,3},X_{1,4}):\det\left(\begin{smallmatrix}1&0\\0&1\end{smallmatrix}\right)=1, \quad (X_{1,4},X_{2,4}):\det\left(\begin{smallmatrix}0&1\\-1&1\end{smallmatrix}\right)=1, \quad (X_{2,4},X_{2,5}):\det\left(\begin{smallmatrix}-1&1\\-1&0\end{smallmatrix}\right)=1,\\ &(X_{2,5},X_{3,5}):\det\left(\begin{smallmatrix}-1&0\\0&-1\end{smallmatrix}\right)=1, \quad (X_{3,5},X_{1,3}):\det\left(\begin{smallmatrix}0&-1\\1&0\end{smallmatrix}\right)=1. \end{aligned} \tag{62.59}$$

All five are $+1$: the unimodularity and sign claims of Theorem 62.17 are verified by hand. Therefore

$$\Omega(\mathcal{A}_5)=\Big[\frac{1}{X_{1,3}X_{1,4}}+\frac{1}{X_{1,4}X_{2,4}}+\frac{1}{X_{2,4}X_{2,5}}+\frac{1}{X_{2,5}X_{3,5}}+\frac{1}{X_{3,5}X_{1,3}}\Big]\,\mathrm{d}x\wedge\mathrm{d}y. \tag{62.60}$$

Change to planar Mandelstams. Substituting Eq. (62.51) into the bracket of Eq. (62.60),

$$m_5=\frac{1}{s_{12}\,s_{45}}+\frac{1}{s_{45}\,s_{23}}+\frac{1}{s_{23}\,s_{51}}+\frac{1}{s_{51}\,s_{34}}+\frac{1}{s_{34}\,s_{12}}, \tag{62.61}$$

which is the cyclic orbit of $1/(s_{12}s_{34})$ — regrouping the five terms,

$$m_5=\frac{1}{s_{12}s_{34}}+\frac{1}{s_{23}s_{45}}+\frac{1}{s_{34}s_{51}}+\frac{1}{s_{45}s_{12}}+\frac{1}{s_{51}s_{23}}, \tag{62.62}$$

exactly the five-term, five-pole color-ordered bi-adjoint amplitude obtained by summing the five planar Feynman diagrams via Eq. (62.38). The canonical form of the kinematic pentagon *is* the amplitude.

Residue check: locality and unitarity. Take the channel $s_{12} = X_{1,3} \to 0$, the facet $x = 0$ (the edge from $(0,0)$ to $(0,2)$). By the boundary dictionary Eq. (62.5), the residue of $\Omega(\mathcal{A}_5)$ on this facet is the canonical form of the facet, and the facet is the segment from $X_{1,4} = 0$ (vertex $(0,0)$) to $X_{3,5} = 0$ (vertex $(0,2)$). Directly from Eq. (62.61), the terms singular at $s_{12} = 0$ are the two containing $X_{1,3}$,

$$m_5\Big|_{\text{poles in } s_{12}} = \frac{1}{s_{12}}\Big(\frac{1}{s_{45}} + \frac{1}{s_{34}}\Big) + \text{regular}, \qquad \text{Res}_{s_{12}\to 0}\, m_5 = \frac{1}{s_{45}} + \frac{1}{s_{34}}. \tag{62.63}$$

Now identify the factorized kinematics. On $s_{12} \to 0$ the legs $1, 2$ fuse into an on-shell intermediate $P = p_1 + p_2$ ($P^2 = s_{12} \to 0$), splitting the five-point object into a three-point amplitude on $(1, 2, P)$ and a four-point amplitude on $(-P, 3, 4, 5)$. The bi-adjoint three-point amplitude is the bare vertex, $m_3 = 1$ (no propagator). The four-point factor is the amplitude of Section 62.2.5 for the ordering $(P, 3, 4, 5)$,

$$m_4(P,3,4,5) = \frac{1}{s_{P3}} + \frac{1}{s_{34}}, \qquad s_{P3} = (p_1 + p_2 + p_3)^2 = s_{45}, \tag{62.64}$$

so $m_4(P,3,4,5) = 1/s_{45} + 1/s_{34}$. Hence

$$\text{Res}_{s_{12}\to 0}\, m_5 \;=\; m_3 \times m_4(P,3,4,5) \;=\; \frac{1}{s_{45}} + \frac{1}{s_{34}}, \tag{62.65}$$

in agreement with Eq. (62.63). The amplitude has a pole only where a *planar* channel goes on shell (locality), and the residue there factorizes into the product of the two lower on-shell amplitudes glued on that channel (unitarity) — exactly the BCFW factorization of Theorem 59.23, here for the scalar whose three-point amplitude is 1. Geometrically, Eq. (62.65) is nothing but the statement from Theorem 62.16 that the facet $X_{1,3} = 0$ of the pentagon is the product $\mathcal{A}_3 \times \mathcal{A}_4$ (a point times a segment), whose canonical form is $1 \times m_4$. The same computation on each of the other four facets recovers, by cyclic symmetry, the four-point amplitude in the corresponding channel. Locality and unitarity have emerged from the shape of a polygon.

62.2.7 What has been established, and the obstruction ahead

We have made "amplitude = volume" a theorem for an honest physical amplitude. The tree color-ordered bi-adjoint amplitude m_n is the canonical form of the kinematic associahedron $\mathcal{A}_n \subset$ Mandelstam space (Theorem 62.17); its poles are the polytope's facets (locality), and its residues are the canonical forms of those facets, equal to products of lower amplitudes (unitarity, Theorem 62.13 and Eq. (62.65)). Nothing was conjectural, nothing was imported on faith, and the five-point case was carried complete, from the facet inequalities Eq. (62.56) to the residue Eq. (62.65).

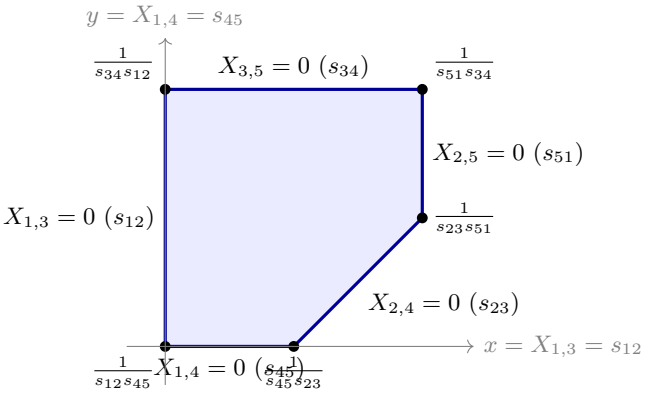


Figure 62.2: The kinematic associahedron $\mathcal{A}_5 = K_2$ realized in Mandelstam space, with $c_{1,3} = c_{1,4} = c_{2,4} = 1$ (Eq. (62.57)). Each edge is a facet $X_{i,j} = 0$, a planar propagator pole; each vertex is a triangulation of the pentagon, i.e. a planar five-point Feynman diagram contributing its product of two propagators. Summing the vertex contributions Eq. (62.60) gives the canonical form, whose coefficient is the amplitude m_5 of Eq. (62.62).

The construction also exposes its own limit, which is the spur to everything that follows. The bi-adjoint scalar has *no helicity*: its only data are the n legs and their cyclic order, and the associahedron knows exactly this much — it is a single polytope per $(n, \text{ordering})$, blind to any further quantum number. A gauge-theory amplitude, by contrast, is graded by the number of negative-helicity gluons, the integer k of the N^{k-2}MHV classification (Chapter 59): the same n legs support a whole tower of inequivalent amplitudes, and a geometry that sees only momenta and ordering cannot tell them apart. To make a positive geometry sensitive to helicity we must enlarge the kinematic data from a point in Mandelstam space to a k-plane in n dimensions — and the natural home for such planes, with the positivity that a canonical form requires, is the totally nonnegative Grassmannian. That is the object we build next.

62.3 The positive Grassmannian

The associahedron of Section 62.2 realized a complete tree-level amplitude as a canonical form, but the amplitude it captured was the one with no spin to speak of: the color-ordered amplitude of bi-adjoint ϕ^3 theory, a function of Mandelstam invariants and nothing else. Gluons and photons carry helicity, and the central lesson of Chapters 58 and 59 was that their amplitudes organize not by a single number but by a *grading*. An n-point gluon amplitude splits into sectors labelled by the integer k, the number of negative-helicity legs (more precisely the MHV degree k, with the maximally-helicity-violating sector at

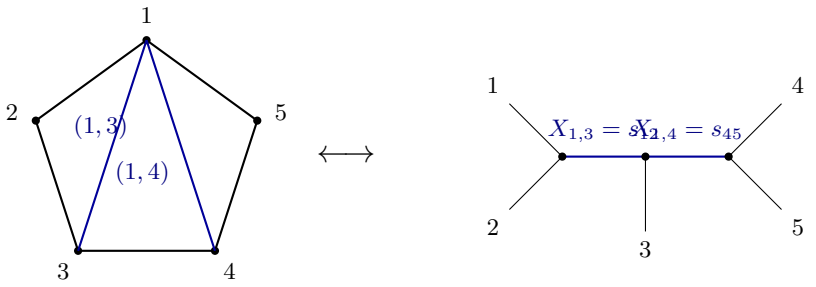


Figure 62.3: The triangulation ↔ planar-tree duality underlying Theorem 62.13, at the vertex $(0,0)$ of Fig. 62.2. The pentagon's triangulation $\{(1,3),(1,4)\}$ (left) is dual to the planar trivalent tree (right): one tree vertex per triangle, one internal line per diagonal. The two diagonals are the two propagators $1/(s_{12}s_{45})$ of this Feynman diagram.

$k = 2$); the Parke–Taylor formula Eq. (59.55) is the entirety of the $k = 2$ sector, and the BCFW recursion of Theorem 59.23 builds the higher-k sectors from it. A geometry built in Mandelstam space alone is blind to this grading: it has no slot for k. What we need is a positive geometry whose very *dimension and combinatorial type are controlled by* k, so that helicity-graded data has a natural home. That object is the totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$. This section introduces it from the ground up, proves its structure on the smallest cases where the combinatorics can be made completely explicit, and writes down the canonical forms that the next section will show BCFW recursion triangulates. Throughout, the integer k of $\mathrm{Gr}(k,n)$ *is* the MHV degree of Theorem 58.2; this is the thread that ties the abstract geometry to the gluon and photon amplitudes of Yang–Mills theory and QED.

62.3.1 The Grassmannian and its Plücker coordinates

Definition 62.19 (Grassmannian)**.** For $0 \leq k \leq n$, the *Grassmannian* $\mathrm{Gr}(k,n)$ is the set of k-dimensional linear subspaces of $\mathbb{F}^n$, where $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$. A k-plane $V \subset \mathbb{F}^n$ is represented by a $k \times n$ matrix C of rank k whose rows span V; two matrices represent the same plane if and only if they differ by left multiplication by an element of $\mathrm{GL}(()k,\mathbb{F})$, the change of basis within V. Thus

$$\mathrm{Gr}(k,n) \;=\; \{\, C \in \mathbb{F}^{k\times n} : \operatorname{rank} C = k \,\}/\mathrm{GL}(()k,\mathbb{F}). \tag{62.66}$$

The dimension follows immediately. The full-rank $k \times n$ matrices form an open dense subset of $\mathbb{F}^{k\times n}$, of dimension kn; the group $\mathrm{GL}(()k,\mathbb{F})$ acting freely on the left has dimension k^2; the quotient therefore has dimension

$$\dim \mathrm{Gr}(k,n) = kn - k^2 = k(n-k). \tag{62.67}$$

This number will be the dimension of the top positroid cell, and through it the dimension count of helicity-graded data.

To coordinatize $\mathrm{Gr}(k,n)$ invariantly we use the maximal minors of C. For an ordered k-element subset $I=\{i_1<i_2<\cdots<i_k\}\subset\{1,\dots,n\}$, write

$$p_I(C) \;=\; \det(\text{columns } i_1,\dots,i_k \text{ of } C), \tag{62.68}$$

the *Plücker coordinate* indexed by I. Under a change of basis $C\mapsto gC$ with $g\in \mathrm{GL}(()k)$, every maximal minor scales by the common factor $\det g$:

$$p_I(gC)=\det(g)\,p_I(C). \tag{62.69}$$

Hence the tuple $\big(p_I\big)_{|I|=k}$ is well defined up to overall scale, i.e. as a point of the projective space $\mathbb{P}\big(\wedge^k\mathbb{F}^n\big)=\mathbb{P}^{\binom{n}{k}-1}$. This is the *Plücker embedding*

$$\mathrm{Gr}(k,n)\hookrightarrow\mathbb{P}^{\binom{n}{k}-1},\qquad V\mapsto[p_I(C)]_{|I|=k}. \tag{62.70}$$

It is injective: a k-plane is recovered from its Plücker coordinates as the common kernel of the linear functionals they define, so distinct planes give distinct projective points. The image is not all of projective space; it is the zero locus of a system of quadrics, the *Plücker relations*.

Proposition 62.20 (Grassmann–Plücker relations)**.** *A point $[p_I]\in\mathbb{P}^{\binom{n}{k}-1}$ lies in the image Eq. (62.70) if and only if the Plücker coordinates satisfy, for every pair of index sets $\{i_1,\dots,i_{k-1}\}$ and $\{j_1,\dots,j_{k+1}\}$,*

$$\sum_{a=1}^{k+1}(-1)^a\,p_{i_1\cdots i_{k-1}j_a}\,p_{j_1\cdots\widehat{j_a}\cdots j_{k+1}}=0, \tag{62.71}$$

where $\widehat{j_a}$ denotes omission of j_a.

We will not need the relations in this generality; what we will need, repeatedly, is the single relation governing $\mathrm{Gr}(2,4)$, and that we prove in full because total positivity will turn it into a structural statement.

Lemma 62.21 (The $\mathrm{Gr}(2,4)$ Plücker relation)**.** *For any 2×4 matrix C with columns $c_1,c_2,c_3,c_4\in\mathbb{F}^2$, the maximal minors satisfy*

$$p_{12}\,p_{34}\;-\;p_{13}\,p_{24}\;+\;p_{14}\,p_{23}\;=\;0. \tag{62.72}$$

Proof. Write $\langle u,v\rangle:=u_1v_2-u_2v_1=\det[\,u\;v\,]$ for $u,v\in\mathbb{F}^2$, so that $p_{ij}=\langle c_i,c_j\rangle$. Fix $a,b\in\mathbb{F}^2$ and consider the function

$$B_{a,b}(c,d):=\langle a,b\rangle\langle c,d\rangle-\langle a,c\rangle\langle b,d\rangle+\langle a,d\rangle\langle b,c\rangle. \tag{62.73}$$

As a function of the pair (c,d) it is bilinear, and it is antisymmetric: interchanging $c \leftrightarrow d$ sends each term to minus the corresponding term of $B_{a,b}(c,d)$ (using $\langle d,c\rangle = -\langle c,d\rangle$), so $B_{a,b}(d,c) = -B_{a,b}(c,d)$. But on the two-dimensional space $\mathbb{F}^2$ every alternating bilinear form is a scalar multiple of the determinant $\langle\,\cdot\,,\,\cdot\,\rangle$; hence $B_{a,b}(c,d) = f(a,b)\,\langle c,d\rangle$ for some scalar $f(a,b)$. Evaluating at $c=a,\ d=b$ gives

$$B_{a,b}(a,b) = \langle a,b\rangle^2 - \langle a,a\rangle\langle b,b\rangle + \langle a,b\rangle\langle b,a\rangle = \langle a,b\rangle^2 - 0 - \langle a,b\rangle^2 = 0, \quad (62.74)$$

because $\langle a,a\rangle = \langle b,b\rangle = 0$. Since $\langle a,b\rangle$ is not identically zero, $f(a,b) = 0$ identically, so $B_{a,b} \equiv 0$. Setting $(a,b,c,d) = (c_1,c_2,c_3,c_4)$ in Eq. (62.73) is exactly Eq. (62.72). □

62.3.2 Total nonnegativity

So far everything was linear-algebraic and worked over any field. The positivity that singles out the amplitude-relevant region requires $\mathbb{F} = \mathbb{R}$ and a *fixed cyclic order* of the n columns, which physically is the cyclic order of the n external particles in a color-ordered amplitude.

Definition 62.22 (Totally nonnegative and totally positive Grassmannian)**.** A real k-plane $V \in \mathrm{Gr}(k,n)$ is *totally nonnegative* if it admits a representative matrix C all of whose ordered maximal minors are nonnegative, $p_I(C) \geq 0$ for every I. It is *totally positive* if it admits a representative with $p_I(C) > 0$ for every I. The corresponding subsets are

$$\mathrm{Gr}_{\geq 0}(k,n) = \{V : p_I \geq 0\ \forall I\}, \qquad \mathrm{Gr}_{>0}(k,n) = \{V : p_I > 0\ \forall I\}. \quad (62.75)$$

This is well posed despite the gauge freedom and the projective scaling. By Eq. (62.69) a change of basis multiplies all minors by the single number $\det g$; choosing $\det g > 0$ preserves all signs simultaneously, while $\det g < 0$ flips them all. Thus "all minors ≥ 0" is a property of the plane V, not of the representative, once we agree to read it up to a common positive rescaling. The condition does, however, depend on the ordering of the columns: permuting columns permutes and re-signs the p_I, so $\mathrm{Gr}_{\geq 0}(k,n)$ is attached to the chosen cyclic order, precisely as a color-ordered amplitude is attached to the cyclic order of its legs.

The link to helicity is now immediate and is the reason this object enters amplitudes at all. In the spinor-helicity description of Theorem 58.2, an n-point amplitude in the sector of MHV degree k is a function of the $2 \times n$ matrices of angle and square spinors subject to momentum conservation; the data that survives, once little-group and overall scalings are removed, is naturally a point of a $\mathrm{Gr}(k,n)$, with k the count of negative-helicity legs. The MHV sector that the Parke–Taylor formula Eq. (59.55) exhausts is $k=2$, so the geometry relevant to MHV gluon amplitudes is $\mathrm{Gr}_{\geq 0}(2,n)$; we will see its canonical form reproduce the Parke–Taylor denominator below. The grading the associahedron could not see is built into the integer k of the Grassmannian.

62.3.3 The smallest case: $\mathrm{Gr}_{\geq 0}(1,n)$ is the simplex

The case $k=1$ is the one where the abstract definitions collapse onto an object already met in Section 62.1: the projective simplex whose canonical form opened the chapter. Establishing this identification rigorously fixes intuition and verifies, in a case we can see, that the positive-geometry axioms of Theorem 62.3 hold for positroid cells.

A point of $\mathrm{Gr}(1,n)$ is a line through the origin in $\mathbb{R}^n$, represented by a single nonzero row vector $C=(c_1,\dots,c_n)$ modulo overall rescaling: this is exactly projective space $\mathbb{P}^{n-1}$. The maximal minors are now 1×1, so the Plücker coordinates are simply the entries, $p_{\{i\}}=c_i$. The dimension formula Eq. (62.67) gives $\dim \mathrm{Gr}(1,n)=1\cdot(n-1)=n-1$, the dimension of $\mathbb{P}^{n-1}$, as it must.

Proposition 62.23 ($\mathrm{Gr}_{\geq 0}(1,n)$ is the projective simplex)**.** *The totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(1,n)$ is the standard projective simplex*

$$\Delta^{n-1}=\{[c_1:\cdots:c_n]\in\mathbb{P}^{n-1}:\ c_i\geq 0,\ \textstyle\sum_i c_i>0\},\tag{62.76}$$

an $(n-1)$-dimensional simplex with n vertices $e_1,\dots,e_n$ (the coordinate points) and n facets $\{c_i=0\}$. Its canonical form is

$$\Omega(\mathrm{Gr}_{\geq 0}(1,n))=\mathrm{dlog}\,\frac{c_1}{c_n}\wedge\mathrm{dlog}\,\frac{c_2}{c_n}\wedge\cdots\wedge\mathrm{dlog}\,\frac{c_{n-1}}{c_n},\tag{62.77}$$

which in the affine chart $c_n=1$, $x_i=c_i$, reads

$$\Omega=\frac{dx_1\wedge\cdots\wedge dx_{n-1}}{x_1\,x_2\cdots x_{n-1}\,\big(1-\sum_{i=1}^{n-1}x_i\big)},\tag{62.78}$$

the simplex form of Eq. (62.22).

Proof. Since the Plücker coordinates of $\mathrm{Gr}(1,n)$ are the entries c_i, the condition $p_I\geq 0$ for all I is just $c_i\geq 0$ for all i, read projectively up to common positive scale (Theorem 62.22). The set of $[c_1:\cdots:c_n]$ with all $c_i\geq 0$ and not all zero is by definition Eq. (62.76). That this is an $(n-1)$-simplex is standard: it is the projectivization of the closed nonnegative orthant $\mathbb{R}^n_{\geq 0}$, whose extreme rays are the coordinate axes $\mathbb{R}_{\geq 0}e_i$, giving the n vertices $[e_i]$, and whose codimension-one faces are the coordinate hyperplanes $\{c_i=0\}$, giving the n facets. The face $\{c_i=0\}$ is itself the nonnegative orthant of the remaining $n-1$ coordinates, i.e. a $\mathrm{Gr}_{\geq 0}(1,n-1)=\Delta^{n-2}$; the facets are simplices of one lower dimension, the defining recursion of a simplex.

For the canonical form, pass to the affine chart $c_n=1$ with coordinates $x_i=c_i$, $i=1,\dots,n-1$. In this chart the region is the solid simplex $\{x_i\geq 0,\ \sum_i x_i\leq 1\}$, whose n facets are the $n-1$ hyperplanes $x_i=0$ together with $\sum_i x_i=1$ (the image of $c_n=0$, since $c_n=1-\sum_{i<n}c_i$ after rescaling so that $\sum c_i=1$). A positive geometry's canonical form is fixed by the requirements (Theorem 62.3)

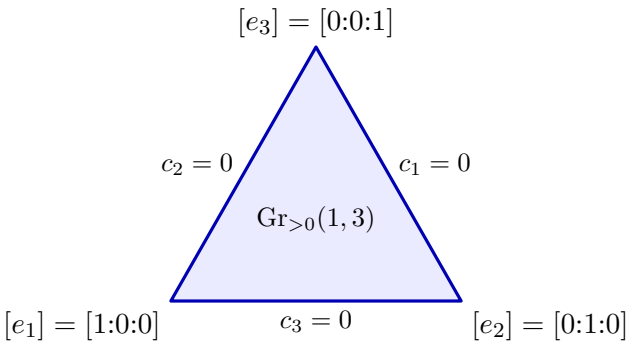


Figure 62.4: $\mathrm{Gr}_{\geq 0}(1,3) = \Delta^2$ is the projective triangle. Its three vertices are the coordinate points $[e_i]$; its three facets are the edges $\{c_i = 0\}$, each a copy of $\mathrm{Gr}_{\geq 0}(1,2) = \Delta^1$; the interior is the totally positive cell $\mathrm{Gr}_{>0}(1,3)$. The canonical form $dx\,dy/[xy(1-x-y)]$ has simple poles exactly on the three edges, and its residue on each edge is the canonical form of that edge.

that it be a top-degree rational form with simple poles exactly on the facets and unit residues recursively giving the lower canonical forms. The unique such form for the solid simplex is Eq. (62.78): it has simple poles on each $x_i = 0$ and on $1 - \sum x_i = 0$, and on no other locus, and a direct residue computation discharges the recursion. Taking the residue at $x_{n-1} = 0$, write $\Omega = \dfrac{dx_{n-1}}{x_{n-1}} \wedge \omega$ with

$$\omega = \frac{dx_1 \wedge \cdots \wedge dx_{n-2}}{x_1 \cdots x_{n-2}\left(1 - \sum_{i=1}^{n-2} x_i - x_{n-1}\right)}; \tag{62.79}$$

then $\mathrm{Res}_{x_{n-1}=0}\,\Omega = \omega|_{x_{n-1}=0}$ is exactly the simplex form Eq. (62.78) of the facet $\{x_{n-1} = 0\} = \Delta^{n-2}$, and likewise for the other facets. The chart-independent expression is Eq. (62.77), obtained by writing $x_i = c_i/c_n$ and using $\mathrm{dlog}(c_i/c_n) = dc_i/c_i - dc_n/c_n$. For $n = 3$ this is the triangle of Fig. 62.4, with three vertices, three edges, and the canonical form $dx\,dy/[xy(1-x-y)]$. □

The vanishing patterns of the Plücker coordinates stratify the simplex: the interior has all $c_i > 0$ (the top cell), a facet has exactly one $c_i = 0$, an edge has $n-2$ of them vanishing, and a vertex has all but one. Each stratum is an open cell, and the cells fit together exactly as the faces of a simplex. This is the model the general case generalizes, with one change: when $k > 1$ the Plücker coordinates are not independent (they obey Eq. (62.71)), so not every vanishing pattern is achievable, and the strata are indexed by a subtler combinatorial object.

62.3.4 Positroid cells and the boundary stratification

For general k we stratify $\mathrm{Gr}_{\geq 0}(k,n)$ by which Plücker coordinates vanish, exactly as we just did for the simplex.

Definition 62.24 (Positroid cells)**.** For a totally nonnegative plane $V \in \mathrm{Gr}_{\geq 0}(k,n)$, its *matroid* is the collection $\mathcal{M}(V) = \{I : p_I(V) > 0\}$ of index sets with strictly positive Plücker coordinate. A nonempty collection $\mathcal{M}$ that arises this way is called a *positroid*, and the corresponding stratum

$$S_{\mathcal{M}} = \{V \in \mathrm{Gr}_{\geq 0}(k,n) : \ p_I(V) > 0 \text{ iff } I \in \mathcal{M}\} \tag{62.80}$$

is the *positroid cell* of $\mathcal{M}$.

Three structural facts make positroid cells the right building blocks for a positive geometry; we state them, prove what is short, and are explicit about what belongs to the deeper combinatorial theory.

(i) Each cell is an open ball. Every nonempty positroid cell $S_{\mathcal{M}}$ is homeomorphic to an open ball $\mathbb{R}^d_{>0}$ for some $d \geq 0$, its dimension; we exhibit the homeomorphism explicitly for the top cell of $\mathrm{Gr}_{\geq 0}(2,4)$ in Theorem 62.25, and Section 62.3.3 did so for all cells of $\mathrm{Gr}_{\geq 0}(1,n)$. The unique top-dimensional cell is the totally positive part $\mathrm{Gr}_{>0}(k,n)$ itself, with dimension $k(n-k)$ by Eq. (62.67); the empty-Plücker patterns are excluded by the Plücker relations, which is exactly what makes the cell poset finer than the Boolean lattice of all vanishing patterns.

(ii) The cells form a regular CW complex realizing the boundary recursion. The closure of a cell is a union of lower cells, $\overline{S_{\mathcal{M}}} = \bigsqcup_{\mathcal{M}' \preceq \mathcal{M}} S_{\mathcal{M}'}$, and the codimension-one cells in $\overline{S_{\mathcal{M}}}$ are its *facets*. This closure relation, together with each cell being a ball, is precisely the data the positive-geometry axioms of Theorem 62.3 require: a canonical form with logarithmic singularities on the boundary, whose residue on each facet is the canonical form of that facet. The simplex of Section 62.3.3 verified this recursion concretely; the general statement that $\mathrm{Gr}_{\geq 0}(k,n)$ is a regular CW complex homeomorphic to a closed ball is a theorem of Postnikov and of Galashin–Karp–Lam [208, 409], and we take it on those terms in the spirit of this frontier chapter, while proving the small cases by hand so the structure is concrete rather than asserted.

(iii) The cells are classified combinatorially. Postnikov's theorem identifies positroid cells with several equivalent labels — decorated permutations of $\{1,\dots,n\}$ with k anti-excedances, LE-diagrams in a $k \times (n-k)$ rectangle, and reduced plabic graphs — and reads off the dimension of the cell directly from the label [409]. We will not need the general classification; we need only that the labels exist and that for the cases at hand the dimension and facet count can be computed from first principles, which we now do.

62.3.5 The canonical form on a top cell

The payoff of total positivity is that the top cell carries a distinguished form written transparently in positive parameters. The mechanism is a *positive parametrization*: a diffeomorphism $\mathbb{R}^d_{>0} \to \mathrm{Gr}_{>0}(k,n)$, $d = k(n-k)$, under which every Plücker coordinate becomes a *subtraction-free* polynomial in the parameters — a polynomial with all coefficients positive, so manifestly positive on the positive orthant. Such parametrizations always exist; one construction (Lusztig) writes the representative matrix as an ordered product of elementary Jacobi factors $x_i(t_i)$ with $t_i > 0$, another (Postnikov) reads it off a reduced plabic graph as a boundary-measurement matrix [328, 409]. In either, the parameters $t_1, \dots, t_d$ are the *edge variables*, and the canonical form of the top cell is the pure dlog form

$$\Omega\big(\mathrm{Gr}_{>0}(k,n)\big) = \mathrm{dlog}\, t_1 \wedge \mathrm{dlog}\, t_2 \wedge \cdots \wedge \mathrm{dlog}\, t_d, \qquad d = k(n-k). \tag{62.81}$$

It has only logarithmic singularities, on the loci $t_i \to 0$, and these are exactly the facets, so Eq. (62.81) satisfies the positive-geometry axioms by construction. Equivalently, and independently of any choice of parametrization, the top-cell form has the $\mathrm{GL}(()k)$-invariant integral representation

$$\Omega\big(\mathrm{Gr}_{>0}(k,n)\big) = \frac{1}{\mathrm{Vol}\,\mathrm{GL}(()k)} \, \frac{d^{k\times n}C}{(1\,2\cdots k)\,(2\,3\cdots k{+}1)\cdots(n\,1\cdots k{-}1)}, \tag{62.82}$$

where $(i\,i{+}1\cdots i{+}k{-}1)$ denotes the maximal minor on k cyclically consecutive columns, and the denominator is the product of all n such consecutive minors. The poles of Eq. (62.82) sit only on consecutive minors, which is the geometric statement that the facets of the top cell are the loci where a cyclically consecutive Plücker coordinate vanishes. The worked example makes both expressions explicit and proves their equality in the smallest nontrivial case, $\mathrm{Gr}_{\geq 0}(2,4)$.

Worked Example 62.25 (The top cell of $\mathrm{Gr}_{\geq 0}(2,4)$: parametrization, form, and boundaries)**.** We carry out the full analysis of the smallest case relevant to MHV amplitudes: $k = 2$, $n = 4$, with top-cell dimension $d = k(n-k) = 4$.

Parametrization. On the top cell the consecutive minor (13) is nonzero (we verify this below), so we may use the residual $\mathrm{GL}(()2)$ freedom to set columns 1 and 3 to the standard basis, leaving four free entries which we call $\alpha, \beta, \gamma, \delta$:

$$C(\alpha,\beta,\gamma,\delta) = \begin{pmatrix} 1 & \alpha & 0 & -\gamma \\ 0 & \beta & 1 & \delta \end{pmatrix}, \qquad \alpha,\beta,\gamma,\delta > 0. \tag{62.83}$$

This gauge is rigid: requiring columns $1, 3$ to equal $(1,0)^{\mathsf{T}}, (0,1)^{\mathsf{T}}$ fixes $g \in \mathrm{GL}(()2)$ uniquely, so Eq. (62.83) is a section over the chart $(13) \neq 0$, which contains all of $\mathrm{Gr}_{>0}(2,4)$.

Plücker coordinates. Computing the six 2×2 minors directly from Eq. (62.83),

$$p_{12}=\det\begin{pmatrix}1&\alpha\\0&\beta\end{pmatrix}=\beta,\quad p_{13}=\det\begin{pmatrix}1&0\\0&1\end{pmatrix}=1,\qquad p_{14}=\det\begin{pmatrix}1&-\gamma\\0&\delta\end{pmatrix}=\delta,$$
$$p_{23}=\det\begin{pmatrix}\alpha&0\\\beta&1\end{pmatrix}=\alpha,\quad p_{24}=\det\begin{pmatrix}\alpha&-\gamma\\\beta&\delta\end{pmatrix}=\alpha\delta+\beta\gamma,\quad p_{34}=\det\begin{pmatrix}0&-\gamma\\1&\delta\end{pmatrix}=\gamma. \tag{62.84}$$

All six are positive for $\alpha,\beta,\gamma,\delta>0$: the four consecutive minors $p_{12},p_{23},p_{34},p_{14}$ are the parameters themselves, $p_{13}=1$, and the long diagonal $p_{24}=\alpha\delta+\beta\gamma$ is subtraction-free. The map $(\alpha,\beta,\gamma,\delta)\mapsto C$ is a bijection onto $\mathrm{Gr}_{>0}(2,4)$: it is injective because the parameters are recovered as $\alpha=p_{23},\beta=p_{12},\gamma=p_{34},\delta=p_{14}$ in the gauge $p_{13}=1$, and surjective because every totally positive plane has $p_{13}>0$ and so a representative of the form Eq. (62.83). This is the explicit homeomorphism $\mathbb{R}^4_{>0}\cong\mathrm{Gr}_{>0}(2,4)$ promised in Section 62.3.4.

The Plücker relation makes the long diagonals automatic. The single relation Eq. (62.72) reads here

$$p_{13}\,p_{24}=p_{12}\,p_{34}+p_{14}\,p_{23}, \tag{62.85}$$

which the values Eq. (62.84) satisfy identically: $1\cdot(\alpha\delta+\beta\gamma)=\beta\gamma+\delta\alpha$. The structural consequence is decisive: *whenever the four consecutive minors are positive, the right-hand side of Eq. (62.85) is positive, hence $p_{13}p_{24}>0$, so neither long diagonal can vanish.* The non-consecutive Plücker coordinates are positive for free. The boundary of $\mathrm{Gr}_{\geq 0}(2,4)$ is therefore reached only by sending consecutive minors to zero — exactly the poles of the canonical form.

Canonical form. In the edge variables the cyclic-minor form Eq. (62.82) is, with $\mathrm{Vol}\,\mathrm{GL}(()2)$ absorbed by the gauge-fixing of columns $1,3$ to the identity,

$$\Omega=\frac{d\alpha\,d\beta\,d\gamma\,d\delta}{(12)(23)(34)(41)}=\frac{d\alpha\,d\beta\,d\gamma\,d\delta}{\beta\cdot\alpha\cdot\gamma\cdot(-\delta)}=-\frac{d\alpha}{\alpha}\wedge\frac{d\beta}{\beta}\wedge\frac{d\gamma}{\gamma}\wedge\frac{d\delta}{\delta}, \tag{62.86}$$

where $(41)=\det[\,c_4\;c_1\,]=-p_{14}=-\delta$ accounts for the sign and the ordering. Up to orientation this is the pure dlog form Eq. (62.81),

$$\Omega\big(\mathrm{Gr}_{>0}(2,4)\big)=\mathrm{dlog}\,\alpha\wedge\mathrm{dlog}\,\beta\wedge\mathrm{dlog}\,\gamma\wedge\mathrm{dlog}\,\delta, \tag{62.87}$$

confirming the equality of the two presentations Eqs. (62.81) and (62.82) in this case.

Facets and the residue axiom. The form Eq. (62.87) has simple poles exactly where one parameter vanishes, i.e. on the four loci

$$\{(23)=0\},\quad\{(12)=0\},\quad\{(34)=0\},\quad\{(14)=0\}\quad(\alpha,\beta,\gamma,\delta\to 0\ \text{respectively}), \tag{62.88}$$

the four cyclically consecutive minors. By the Plücker-relation argument these are *all* the facets: no facet sits at $(13)=0$ or $(24)=0$. Each facet is a three-dimensional positroid cell, and the residue of Ω on it is that cell's canonical form.

For the facet $\{(23) = 0\}$, write $\Omega = \operatorname{dlog}\alpha \wedge \omega$ with $\omega = \operatorname{dlog}\beta \wedge \operatorname{dlog}\gamma \wedge \operatorname{dlog}\delta$; then

$$\operatorname{Res}_{\alpha=0}\Omega = \omega\big|_{\alpha=0} = \operatorname{dlog}\beta \wedge \operatorname{dlog}\gamma \wedge \operatorname{dlog}\delta, \tag{62.89}$$

which is precisely the dlog canonical form of the three-dimensional cell $\{p_{23} = 0\}$, parametrized by the surviving positive variables β, γ, δ. The same computation on each of the other three facets returns its canonical form. This is the boundary/residue axiom of Theorem 62.3 verified explicitly: the residue of the canonical form on a facet is the canonical form of the facet.

Deeper strata. Continuing into the boundary, two facets meet in a codimension-two cell. Setting two parameters to zero gives, by inspection of Eq. (62.84), a nonempty two-dimensional cell for each of the $\binom{4}{2} = 6$ pairs; for instance $\beta = \gamma = 0$ forces $p_{12} = p_{34} = p_{24} = 0$ with $p_{13}, p_{14} = \delta, p_{23} = \alpha$ surviving. Notably the "opposite" facets $\{(12) = 0\}$ and $\{(34) = 0\}$ do intersect (at $\beta = \gamma = 0$), unlike opposite facets of a polytope: $\mathrm{Gr}_{\geq 0}(2,4)$ is a positive geometry but not a polytope, and this is the first place that distinction is visible. The strata continue down to one- and zero-dimensional cells, and the whole complex is, by the Postnikov–Galashin–Karp–Lam theorem, a regular CW decomposition of a closed four-ball.

Connection to gluon amplitudes. The denominator of the canonical form, the product of cyclically consecutive minors $(12)(23)(34)(41)$ in Eq. (62.86), is the geometric image of the Parke–Taylor denominator $\langle 12\rangle\langle 23\rangle\langle 34\rangle\langle 41\rangle$ of Eq. (59.55). The same cyclic, consecutive-bracket structure that organizes the $k = 2$ MHV gluon amplitude organizes the canonical form of $\mathrm{Gr}_{>0}(2,n)$; for general n the n consecutive minors in Eq. (62.82) mirror the n consecutive angle brackets of Parke–Taylor exactly. The MHV degree $k = 2$ of the amplitude and the rank $k = 2$ of the Grassmannian are one and the same integer.

We now have the objects the next section needs: the totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$, its decomposition into positroid cells of dimension governed by the helicity-grade k, and a canonical dlog form on each cell whose residues recurse onto the canonical forms of its facets. In Section 62.4 these cells become the on-shell functions of scattering data, and BCFW recursion will be recognized as a particular triangulation of $\mathrm{Gr}_{\geq 0}(k,n)$ by positroid cells — so that the recursion of Chapter 59 reappears as the statement that a sum of cell canonical forms assembles the canonical form of a larger positive geometry.

62.4 On-shell diagrams and the Grassmannian integral

The previous section gave us, for each decorated permutation, a positroid cell of the positive Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$ together with its canonical form — a dlog form with logarithmic singularities only on the cell's boundaries, in exact analogy

with the simplex of Eq. (62.22). That construction was purely combinatorial-geometric: nowhere did a scattering process appear. This section pours physics into the mould. We will see that the cells are not an abstract stratification waiting for an interpretation but the natural home of *on-shell data* — the leading singularities of scattering amplitudes, the gauge-invariant residues left when the maximal number of internal propagators are put on shell. Three things must be established and they organize the section. First, that there is a graphical calculus, the *on-shell diagrams* (equivalently the plabic graphs of Section 62.3), whose every diagram computes such a residue and maps to a positroid cell, the decorated permutation being the diagram's combinatorial fingerprint. Second, that the residue itself is the value of a single contour integral over $\mathrm{Gr}(k,n)$ — the Grassmannian integral — whose measure is the canonical form of the relevant cell and whose delta functions impose momentum and supermomentum conservation. Third, that the BCFW recursion frozen in Chapter 59 is, read in this language, nothing but a particular triangulation of the relevant positive region by positroid cells: each recursion term is a cell, the spurious poles of recursion are the internal facets shared by adjacent cells, and their cancellation is the polygon cancellation of Theorem 62.9 verbatim. We work throughout in planar $\mathcal{N}=4$ super Yang–Mills, the theory in which the statements are cleanest and complete; the gluon amplitudes of Yang–Mills theory live inside it as particular component amplitudes, so this is the promised frontier reframing of Chapter 59, a geometric reformulation.

Throughout we use the spinor-helicity variables of Theorem 58.2: each massless leg i carries a holomorphic spinor λ_i^a and an antiholomorphic spinor $\widetilde{\lambda}_i^{\dot a}$ with $p_i^{a\dot a}=\lambda_i^a\widetilde{\lambda}_i^{\dot a}$, angle and square brackets $\langle i\,j\rangle=\epsilon_{ab}\lambda_i^a\lambda_j^b$ and $[i\,j]=\epsilon_{\dot a\dot b}\widetilde{\lambda}_i^{\dot a}\widetilde{\lambda}_j^{\dot b}$, and — this is the supersymmetric ingredient that makes $\mathcal{N}=4$ the simplest case — a Grassmann-odd partner $\widetilde{\eta}_i^A$, $A=1,\dots,4$, packaging the whole on-shell supermultiplet into one superfield. Momentum conservation and its superpartner, supermomentum conservation, read

$$P^{a\dot a}=\sum_{i=1}^{n}\lambda_i^a\widetilde{\lambda}_i^{\dot a}=0,\qquad Q^{aA}=\sum_{i=1}^{n}\lambda_i^a\,\widetilde{\eta}_i^A=0, \tag{62.90}$$

and a superamplitude is a function of $\{\lambda_i,\widetilde{\lambda}_i,\widetilde{\eta}_i\}$ carrying the overall delta functions $\delta^4(P)\,\delta^{0|8}(Q)$. The helicity grade is measured by k: an $\mathrm{N}^{k-2}\mathrm{MHV}$ amplitude is degree $4k$ in the $\widetilde{\eta}$'s, so $k=2$ is MHV, $k=3$ is NMHV, and so on.

62.4.1 The atoms: three-point amplitudes and on-shell diagrams

Every on-shell diagram is built by gluing copies of the two three-point superamplitudes, which are fixed completely by Poincaré invariance and little-group scaling (Chapter 58) and are therefore the irreducible atoms of the construction. Special three-point kinematics forces either all square brackets or all angle brackets to

vanish, giving two solutions:

$$A_3^{\overline{\mathrm{MHV}}} = \frac{\delta^4(P)\,\delta^{0|4}\big([1\,2]\widetilde{\eta}_3 + [2\,3]\widetilde{\eta}_1 + [3\,1]\widetilde{\eta}_2\big)}{[1\,2][2\,3][3\,1]}, \qquad (k=1), \tag{62.91}$$

$$A_3^{\mathrm{MHV}} = \frac{\delta^4(P)\,\delta^{0|8}\big(\sum_i \lambda_i\widetilde{\eta}_i\big)}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,1\rangle}, \qquad (k=2). \tag{62.92}$$

We assign the first, the $\overline{\mathrm{MHV}}_3$ amplitude with $k=1$, to a *black* trivalent vertex, and the second, the MHV_3 amplitude with $k=2$, to a *white* trivalent vertex. (The black/white assignment is a convention and references differ; we fix it once here and never change it.)

Notation Watch

We follow the convention in which the *black* vertex carries $\overline{\mathrm{MHV}}_3$ ($k=1$, square brackets) and the *white* vertex carries MHV_3 ($k=2$, angle brackets). Some references reverse the colors. Nothing physical depends on the choice provided one is consistent; all permutation and dimension formulas below assume this one.

An *on-shell diagram* is a graph with trivalent black and white internal vertices, n external legs (the boundary), and internal edges, in which every vertex carries the corresponding three-point amplitude and every internal edge is summed over the full on-shell super-multiplet flowing across it,

$$\int \frac{d^2\lambda_I\, d^2\widetilde{\lambda}_I\, d^{0|4}\widetilde{\eta}_I}{\mathrm{vol}\,\mathrm{GL}(1)}\,, \tag{62.93}$$

the GL(1) quotient implementing the little-group rescaling of the internal leg I, which carries no physical helicity weight of its own. The integration in Eq. (62.93) is over an *on-shell* phase space — the internal momenta are null — so the diagram does not compute an off-shell Feynman integral. It computes an *on-shell function*: a rational function of the external $\lambda, \widetilde{\lambda}, \widetilde{\eta}$, manifestly built only from physical three-point data, with no propagators and no gauge dependence.

Definition 62.26 (On-shell function and leading singularity)**.** The *on-shell function* f_G of an on-shell diagram G is the result of gluing its three-point amplitudes by Eq. (62.93) over every internal edge. A *leading singularity* of an L-loop amplitude is the multidimensional residue of its loop integrand on a cut locus of maximal codimension — the locus where the maximal number of internal propagators are simultaneously on shell, in the sense of the generalized cuts of Theorem 60.17. The two notions coincide: each maximal cut of a planar integrand is computed by gluing the on-shell three-point amplitudes sitting at the corners of the cut diagram, so every leading singularity is the on-shell function of some on-shell diagram, and conversely.

The reason on-shell diagrams land us in the Grassmannian is that each one defines, through its three-point building blocks, a k-plane in n dimensions. The mechanism is the *boundary measurement* map, which we now describe; it was the source of the cells stratified in Section 62.3, and here we see where the cells come from physically.

Fix a *perfect orientation* of G: an orientation of its edges in which every black vertex has exactly one incoming edge and every white vertex exactly one outgoing edge (such an orientation always exists for the diagrams that compute nonvanishing functions). The external legs split into a set of *sources* $\mathcal{I}$, those oriented into the graph, and *sinks*, those oriented out. One shows that $|\mathcal{I}| = k$ for every perfect orientation, where $k = 2n_\circ + n_\bullet - n_I$ is read off the graph: $n_\circ$ white vertices contribute 2 each, $n_\bullet$ black vertices contribute 1 each, and n_I internal edges subtract 1 each. Assigning a variable to each edge and solving the vertex constraints expresses every sink column as a sum over directed paths from the sources,

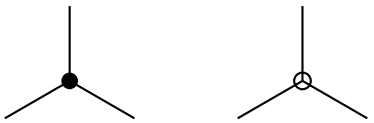
$$C_{ij} = \sum_{\Gamma:\, i\to j} (-1)^{s_\Gamma} \prod_{e\in\Gamma} \alpha_e, \qquad i \in \mathcal{I},\ j \notin \mathcal{I}, \qquad C_{ii'} = \delta_{ii'} \quad (i, i' \in \mathcal{I}), \tag{62.94}$$

the sum running over directed paths Γ from source i to sink j, weighted by the product of edge variables α_e and a sign s_Γ counting source-crossings. The matrix $C \in \mathrm{Mat}_{k\times n}$, taken modulo the residual $\mathrm{GL}(k)$ that mixes the source rows, is a point of $\mathrm{Gr}(k,n)$; as the edge variables range over the positive reals, C sweeps out a positroid cell, and with the path-sign conventions absorbed it lands in $\mathrm{Gr}_{\geq 0}(k,n)$. This is the promised bridge: an on-shell diagram is a coordinate chart on a positroid cell, the edge variables α_e being the coordinates.

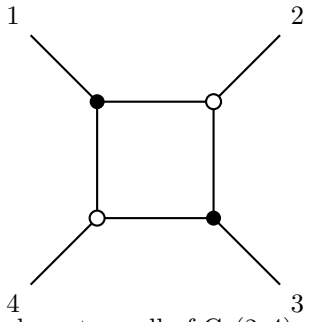
Physical Insight

The boundary measurement Eq. (62.94) is the same move as the canonical-form-by-triangulation of Section 62.1.1, run physically. A diagram is a way of cutting the cell into a chart; its edge variables are positive coordinates; the on-shell function it computes is the canonical form of the cell evaluated against the external kinematics. Different diagrams for the same cell are different charts and give the same function, exactly as different triangulations of a polygon gave the same canonical form.

The combinatorial label of the cell is read directly off the graph by a rule that needs no orientation at all. Enter the graph along external leg i and walk, turning *maximally right* at each black vertex and *maximally left* at each white vertex; the walk (a "zig-zag" or "trip") exits at some external leg j, and we set $\sigma(i) = j$. The map $i \mapsto \sigma(i)$ is the *decorated permutation* of G. Adopting the convention $i < \sigma(i) \leq i + n$ (read cyclically, $\sigma(i)$ taken in $\{i+1, \dots, i+n\}$), the



black: $A_3^{\overline{\text{MHV}}}$ (k=1) white: A_3^{MHV} (k=2)



box: top cell of Gr(2, 4)

Figure 62.5: The two atoms and a composite. Left: the black vertex carries the k=1 anti-MHV three-point amplitude, the white vertex the k=2 MHV three-point amplitude. Right: the four-point "box" on-shell diagram, with two black and two white vertices and four internal edges, computing the on-shell function of the top cell of Gr(2, 4). By Eq. (62.95) its decorated permutation $\sigma(i) = i + 2$ gives $k = 2$; by Eq. (62.96) its five faces give $\dim = 4 = k(n-k)$, the top cell.

helicity grade is recovered as an average winding,

$$k = \frac{1}{n}\sum_{i=1}^{n}(\sigma(i) - i), \tag{62.95}$$

and σ is precisely the decorated permutation that labelled the positroid cell in Section 62.3. Two on-shell diagrams compute the same on-shell function if and only if they share a decorated permutation and are *reduced* (have no removable internal bubble); the permutation is thus the complete invariant. For a reduced diagram the dimension of the cell is the number of independent edge variables left after the GL(1) at each vertex is fixed, equivalently

$$\dim \Sigma_\sigma = (\#\,\text{faces of } G) - 1. \tag{62.96}$$

62.4.2 The Grassmannian integral

We can now assemble the on-shell functions into a single integral representation. The key observation is that the only way the external data can enter an on-shell function is through the requirement that the k-plane C be *compatible* with the kinematics: the antiholomorphic data $\widetilde{\lambda}$ (and its superpartner $\widetilde{\eta}$) must be annihilated by C, while the holomorphic data λ must lie in the orthogonal complement. Imposing exactly these conditions, weighted by the canonical form of the cell, defines the Grassmannian integral.

Definition 62.27 (The Grassmannian integral)**.** For $0 \le k \le n$ the (tree-level) Grassmannian integral is

$$L_{n,k} = \frac{1}{\operatorname{vol}\mathrm{GL}(k)} \int \frac{d^{k\times n}C}{M_1\, M_2 \cdots M_n} \prod_{a=1}^{k} \delta^2\Big(\sum_{i=1}^{n} C_{ai}\, \widetilde{\lambda}_i\Big)\, \delta^{0|4}\Big(\sum_{i=1}^{n} C_{ai}\, \widetilde{\eta}_i\Big) \prod_{\dot a=1}^{n-k} \delta^2\Big(\sum_{i=1}^{n} C^{\perp}_{\dot a i}\, \lambda_i\Big), \tag{62.97}$$

where $C \in \mathrm{Mat}_{k\times n}$ is integrated modulo the $\mathrm{GL}(k)$ acting on its rows, $C^{\perp} \in \mathrm{Mat}_{(n-k)\times n}$ is any matrix whose rows span the orthogonal complement of the row span of C (so $C\,C^{\perp\,T} = 0$), and $M_a = (a, a{+}1, \ldots, a{+}k{-}1)$ is the a-th consecutive $k \times k$ minor of C (indices cyclic). For a general positroid cell Σ_σ one restricts the integration to Σ_σ and replaces the cyclic measure $d^{k\times n}C/(M_1 \cdots M_n)$ by the canonical dlog form of the cell built in Section 62.3.

Three features of Eq. (62.97) deserve unpacking. First the delta functions. The conditions $C \cdot \widetilde{\lambda} = 0$ ($2k$ real equations) and $C^{\perp} \cdot \lambda = 0$ ($2(n-k)$ real equations) together impose $2n$ constraints; of these, four are the overall momentum conservation $\delta^4(P)$, which factors out because $P^{a\dot a} = \sum_i \lambda_i^a \widetilde{\lambda}_i^{\dot a} = 0$ is implied once λ lies in the row span of C and $\widetilde{\lambda}$ in its complement. The remaining $2n-4$ constraints cut the $k(n-k)$-dimensional Grassmannian down to a region of dimension

$$k(n-k) - (2n-4) = (k-2)(n-k-2). \tag{62.98}$$

For MHV ($k=2$) and anti-MHV ($k=n-2$) this is zero: the delta functions localize C completely and the integral evaluates to a single rational function with no contour ambiguity. For NMHV ($k=3$) at $n=6$ it is 1: one genuine contour integral remains, and different choices of contour pick out different sums of residues. The Grassmann delta $\delta^{0|4}(C\cdot\widetilde{\eta})$ supplies $4k$ fermionic conditions, of which the overall $\delta^{0|8}(Q)$ factors out, the rest fixing the supermultiplet content.

Second, the residues. The integral is a multidimensional contour integral, and on the leftover $(k-2)(n-k-2)$ dimensions its value depends on the contour. Each top-dimensional residue — taken where a maximal set of the minors M_a vanish, i.e. on the deepest boundary of a positroid cell — is, by Theorem 62.26, a leading singularity. The Grassmannian integral is thus a generating object: its residues enumerate the leading singularities, and a particular tree amplitude is a particular *sum* of residues.

Third, positivity and the measure. On the support of the cyclic measure $1/(M_1 \cdots M_n)$ the form is the canonical dlog form of the top cell — exactly the structure of the simplex form Eq. (62.22), with the consecutive minors playing the role of the facet functions. The residues that matter physically are those compatible with the positivity $\mathrm{Gr}_{\ge 0}(k,n)$ of Section 62.3; this is the seed of the statement, made geometric in the next section, that the right sum of residues triangulates a positive region.

62.4.3 The MHV four-point amplitude from the top cell of $\mathrm{Gr}(2,4)$

The cleanest demonstration that Eq. (62.97) is an amplitude and not merely a suggestive integral is the MHV case, where the localization is complete and we can carry the algebra to the end. Take $k=2$, so $(k-2)(n-k-2)=0$ for every n; we do $n=4$ in full and read off the general n at the end.

For $k=2$ the residual dimension vanishes, so the $2(n-k)=4$ conditions $C^{\perp}\cdot\lambda=0$ localize C onto a single point of $\mathrm{Gr}(2,4)$ (which has dimension $k(n-k)=4$). The condition $C^{\perp}\cdot\lambda=0$ says that every row of $C^{\perp}$ is orthogonal to all the λ_i, i.e. that the two-dimensional vectors λ_i^a $(a=1,2)$ lie in the row span of C. Since that row span is exactly two-dimensional and the λ's furnish two-dimensional data, we may use the $\mathrm{GL}(2)$ redundancy to align the rows of C with the holomorphic spinors,

$$C_{ai}=\lambda_i^a, \qquad a=1,2, \quad i=1,\dots,n. \tag{62.99}$$

This is the central simplification: for $k=2$ the matrix C *is* the $2\times n$ matrix of angle spinors. The consequences cascade. The consecutive minors become angle brackets,

$$M_a=(a,a{+}1)=\det\begin{pmatrix}\lambda_a^1 & \lambda_{a+1}^1\\ \lambda_a^2 & \lambda_{a+1}^2\end{pmatrix}=\langle a\,a{+}1\rangle, \tag{62.100}$$

so the cyclic measure $1/(M_1M_2\cdots M_n)$ is precisely the Parke–Taylor denominator $1/\big(\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n\,1\rangle\big)$. The remaining delta functions become the physical conservation laws: with $C=\lambda$,

$$\prod_{a=1}^{2}\delta^2\Big(\sum_i C_{ai}\widetilde{\lambda}_i\Big)=\delta^4\Big(\sum_i\lambda_i^a\widetilde{\lambda}_i^{\dot a}\Big)=\delta^4(P), \qquad \prod_{a=1}^{2}\delta^{0|4}\Big(\sum_i C_{ai}\widetilde{\eta}_i\Big)=\delta^{0|8}\Big(\sum_i\lambda_i^a\widetilde{\eta}_i^{A}\Big)=\delta^{0|8}(\text{Q}) \tag{62.101}$$

It remains to account for the measure. The $\mathrm{GL}(2)$ quotient absorbs exactly the freedom in choosing the representative Eq. (62.99), while the four conditions $C^{\perp}\cdot\lambda=0$ integrate against the four directions of $\mathrm{Gr}(2,4)$ that survive the quotient. In the gauge $C=\lambda$ this localization carries unit Jacobian: any gauge-dependent factor it produces (a power of $\langle 1\,3\rangle$, say, if one instead fixes columns $1,3$ to the identity) cancels against the same power generated by the consecutive minors and the rescaled delta functions in that chart, the cancellation being forced by the overall $\mathrm{GL}(2)$ invariance of the integrand, so the gauge-invariant result carries coefficient one. Putting Eqs. (62.100) and (62.101) together with this unit normalization, every factor in Eq. (62.97) has been evaluated and the integral has collapsed to a point,

$$L_{n,2}=\frac{\delta^4(P)\,\delta^{0|8}(Q)}{\langle 1\,2\rangle\langle 2\,3\rangle\cdots\langle n\,1\rangle}\;=\;A_n^{\mathrm{MHV}}. \tag{62.102}$$

For $n = 4$ this is the four-point MHV superamplitude

$$L_{4,2} = \frac{\delta^4(P)\,\delta^{0|8}(Q)}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \tag{62.103}$$

which is the entire $n = 4$ amplitude — there is nothing else, because $n = 4$ has only the MHV ($k = 2$) sector and its conjugate, and these coincide for four points. The component gluon amplitude of Yang–Mills theory is recovered by extracting the appropriate $\tilde{\eta}$ component of $\delta^{0|8}(Q)$: selecting the term with $\tilde{\eta}_i^4\tilde{\eta}_j^4$ (and the same for the other three A) gives the negative-helicity gluons i, j and reproduces the Parke–Taylor formula $\langle ij\rangle^4/(\langle 1\,2\rangle\cdots\langle n\,1\rangle)$ of Chapter 59. Thus the gluon leading singularities of Chapter 59 reappear here as the residue — in fact the single localized value — of the top-cell Grassmannian integral.

Physical Insight

For $k = 2$ the matrix C collapses onto the angle spinors and the integral degenerates to the Parke–Taylor amplitude with no work left to do. The content of the Grassmannian integral is therefore *not* in the MHV case — there it merely repackages a known answer — but in the higher-k cases where Eq. (62.98) is positive, a true contour integral survives, and the physics is hidden in *which* residues to sum. That choice is what BCFW recursion makes for us, geometrically.

The genuinely new structure — the dual superconformal invariants known as *R-invariants* — first appears at NMHV. There the leftover integral Eq. (62.98) is nontrivial, each residue is a single R-invariant, and the amplitude is the sum of residues selected by the contour. The five-point case, where $k = 3 = n-2$ is again maximally localized but the R-invariant content is already visible, is the subject of the worked example below; the six-point case, where $(k-2)(n-k-2) = 1$ and the contour choice is real, is where triangulation becomes nontrivial and which we use as the running illustration in the next subsection.

62.4.4 BCFW recursion as triangulation

We now connect the Grassmannian integral to the recursion of Chapter 59. Recall the content imported from there: the BCFW theorem Theorem 59.23 writes a tree amplitude as a sum over factorization channels,

$$A_n = \sum_{\text{channels}} A_L(z_I)\,\frac{1}{P_I^2}\,A_R(z_I), \tag{62.104}$$

each term a product of lower-point on-shell amplitudes evaluated at the complex shift that puts the channel momentum P_I on shell. In on-shell-diagram language each BCFW term is itself an on-shell diagram — glue the diagrams of A_L and

A_R across the internal line I — and therefore a positroid cell. The recursion is built from one elementary move, the BCFW bridge, which we describe and then assemble.

Definition 62.28 (BCFW bridge)**.** Let C represent a positroid cell of $\mathrm{Gr}(k,n)$ and let (a,b) be a pair of columns adjacent in the cyclic order. The *BCFW bridge* on (a,b) is the one-parameter deformation

$$C \longmapsto C(\alpha): \qquad \text{column } a \mapsto \text{column } a + \alpha\,(\text{column } b), \qquad \alpha > 0, \tag{62.105}$$

all other columns unchanged. Equivalently $C(\alpha) = C\,(\mathbf{1} + \alpha\, e_b e_a^T)$, a right multiplication by an elementary positive matrix. Graphically it attaches a single black–white edge between the external legs a and b.

Proposition 62.29 (The bridge adds one dimension and a transposition)**.** *A BCFW bridge on (a,b), applied to the cell Σ_σ when it acts nontrivially, produces the cell $\Sigma_{\sigma'}$ with*

$$\sigma' = \sigma \cdot (a\,b), \qquad \dim \Sigma_{\sigma'} = \dim \Sigma_\sigma + 1, \tag{62.106}$$

the new coordinate being the bridge variable α, and it preserves k and n. Building an on-shell diagram by adding one particle through an inverse-soft factor *— the operation that takes an n-point cell to an $(n{+}1)$-point cell — is the composition of two such bridges, and therefore raises the cell dimension by two, consistent with $\dim = 2n-4 \mapsto 2(n{+}1)-4$ for the tree-level cells of Eq. (62.98).*

Proof. The operation Eq. (62.105) introduces exactly one new free parameter α into the chart Eq. (62.94): the bridge is one new edge, and after fixing its GL(1) it leaves one independent variable, so by Eq. (62.96) the number of faces grows by one and the dimension increases by one. That k is unchanged is immediate: Eq. (62.105) is an invertible column operation, so $\operatorname{rank} C$ and hence the row span dimension k are preserved, and n is manifestly unchanged. For the permutation, note that the bridge alters the directed paths of Eq. (62.94) only by inserting the new edge between a and b; tracing the zig-zag rule through the added black–white edge swaps the destinations of the trips entering at a and at b, which is precisely right multiplication of σ by the transposition $(a\,b)$. When this lengthens σ (the generic, positivity-preserving case for $\alpha > 0$) the dimension rises by one as just counted; when it would shorten σ the bridge is the inverse move and removes the parameter. The inverse-soft addition of a particle is realized by attaching the new leg through two consecutive bridges (one to each neighbor), so it adds two parameters and raises the dimension by two; for the tree cells this matches the arithmetic of Eq. (62.98), where each added leg increases $2n-4$ by exactly two. □

The recursion of Chapter 59 is now a statement about cells. Each BCFW term in Eq. (62.104) is the on-shell diagram obtained by bridging the cells of

A_L and A_R, hence a positroid cell of $\mathrm{Gr}(k,n)$. Tracking the dimensions through Theorem 62.29, every tree-level BCFW term for the N^{k-2}MHV amplitude is a cell of dimension exactly $2n-4$ — the dimension at which the bosonic delta functions of Eq. (62.97) localize the integral, by the count below Eq. (62.98). The amplitude is the *sum* of these cells:

$$A_n^{\mathrm{N}^{k-2}\mathrm{MHV}} = \sum_{c \in \text{ BCFW terms}} \big[\, \Omega(\Sigma_c) \text{ evaluated against the kinematics} \,\big], \quad (62.107)$$

each $\Omega(\Sigma_c)$ the canonical form of the cell c. This is the precise sense in which BCFW is a triangulation. The cells Σ_c are the "triangles"; their canonical forms are the triangle forms; and the recursion's spurious poles are the internal facets.

Theorem 62.30 (BCFW terms triangulate the positive region)**.** *The positroid cells appearing in a BCFW representation of the tree-level* N^{k-2}MHV *amplitude are non-overlapping, glue along shared codimension-one facets, and their union is the relevant* $2n-4$ *dimensional region of* $\mathrm{Gr}_{\geq 0}(k,n)$*. On each shared facet the two adjacent cells contribute residues that are equal and opposite, so the corresponding spurious pole cancels in the sum Eq. (62.107), leaving singularities only on the genuine boundaries — the physical factorization channels. The amplitude is therefore independent of which valid BCFW shift (which triangulation) is used.*

Proof. The structure is exactly that of the polygon, Theorem 62.9, transported to the Grassmannian. Each BCFW term is a cell Σ_c whose canonical form $\Omega(\Sigma_c)$ has logarithmic singularities only on the boundary facets of Σ_c, by the defining property of the canonical form (Theorem 62.3) established cell by cell in Section 62.3. The facets are of two kinds. A facet on which a *consecutive* minor $M_a = (a, \ldots, a{+}k{-}1)$ vanishes is a physical boundary: it is a factorization channel of the amplitude, and exactly one cell of the BCFW sum has it as a boundary. A facet internal to the triangulation — where a non-consecutive minor vanishes — is shared by precisely two cells, which induce opposite boundary orientations on it (one cell lies on each side). By the residue computation of Theorem 62.8 and the sign argument Eq. (62.28), applied to the two cells meeting along the shared facet, the residues of the two canonical forms on that facet are equal and opposite; their sum has vanishing residue there and is, by the decomposition Eq. (62.2), regular across it. Hence Eq. (62.107) has poles only on the consecutive (physical) facets. Two different BCFW shifts give two collections of cells with the same physical facets and the same residues there; by the uniqueness of the canonical form Theorem 62.4, the two sums agree. The union of the cells covers the $2n-4$ dimensional region because the recursion is complete — every term of Eq. (62.104) is accounted for and on-shell constructibility Theorem 59.23 guarantees no channel is missed. □

Caution

The region triangulated in Theorem 62.30 sits in the C-matrix chart, which is *gauge-redundant* (the $\mathrm{GL}(k)$ quotient) and not yet a geometry in external kinematic space. The cells genuinely triangulate, and the spurious poles genuinely cancel, but "the region" is here a subset of $\mathrm{Gr}_{\geq 0}(k,n)$, not yet a polytope-like body whose boundaries are the amplitude's singularities directly. Removing the redundancy — finding kinematic variables in which the region is an honest positive geometry — is the work of momentum twistors in the next section. The amplituhedron is the image of $\mathrm{Gr}_{\geq 0}(k,n)$ under the map those variables define.

Worked Example 62.31 (A BCFW bridge on a five-point cell: permutation, dimension, boundary)**.** We carry out the bridge construction explicitly for the five-point NMHV-relevant cell, read off its decorated permutation and dimension, and exhibit it as a boundary-compatible piece of a triangulation. The setting is $\mathrm{Gr}(3,5)$: with $n=5$, NMHV means $k=3=n-2$, and Eq. (62.98) gives $(k-2)(n-k-2)=(1)(0)=0$, so the relevant object is the top cell, of dimension $k(n-k)=6$.

The top cell and its permutation. The top cell of $\mathrm{Gr}(k,n)$ has decorated permutation $\sigma_{\mathrm{top}}(i)=i+k$ (cyclically, with $i<\sigma(i)\leq i+n$). For $\mathrm{Gr}(3,5)$,

$$\sigma_{\mathrm{top}}: \quad 1\mapsto 4,\quad 2\mapsto 5,\quad 3\mapsto 6,\quad 4\mapsto 7,\quad 5\mapsto 8, \tag{62.108}$$

where $6,7,8$ are read mod 5 as $1,2,3$. As a check of Eq. (62.95), $\frac{1}{5}\sum_i(\sigma(i)-i)=\frac{1}{5}(3+3+3+3+3)=3=k$. By Eq. (62.96) an on-shell diagram for this cell has seven faces, giving $\dim=6$, the full $k(n-k)$.

Un-bridging to a facet. Remove a single BCFW bridge on the adjacent pair $(5,1)$; by Theorem 62.29 this lands on a codimension-one cell Σ_{σ_0} with $\sigma_0=\sigma_{\mathrm{top}}\cdot(5\,1)$ and $\dim=5$. Composing on the right with the transposition $(5\,1)$ swaps the inputs 1 and 5 before applying σ_{top}:

$$\sigma_0: \quad 1\mapsto \sigma_{\mathrm{top}}(5)=8\equiv 3,\ \ 2\mapsto 5,\ \ 3\mapsto 6\equiv 1,\ \ 4\mapsto 7\equiv 2,\ \ 5\mapsto \sigma_{\mathrm{top}}(1)=4\equiv 9, \tag{62.109}$$

where in the last entry we restore the convention $5<\sigma_0(5)\leq 10$ by writing 4 as 9. Then $\frac{1}{5}\sum_i(\sigma_0(i)-i)=\frac{1}{5}[(3-1)+(5-2)+(6-3)+(7-4)+(9-5)]=\frac{1}{5}(2+3+3+3+4)=3$, so $k=3$ is preserved as Theorem 62.29 requires, while the winding sum dropped from 15 to 15 — unchanged, as it must be for fixed k — and the *length* (number of inversions) of the permutation dropped by one, recording the lost dimension.

Re-attaching the bridge. Now add the bridge back. Starting from a representative C_0 of Σ_{σ_0}, apply Eq. (62.105) on $(5,1)$:

$$C(\alpha): \quad \text{column } 5 \ \mapsto\ \text{column } 5+\alpha\,(\text{column } 1), \qquad \alpha>0. \tag{62.110}$$

This introduces exactly one new positive parameter α, so $\dim \Sigma = \dim \Sigma_{\sigma_0} + 1 = 6$, and the decorated permutation becomes $\sigma_0 \cdot (5\,1) = \sigma_{\text{top}}$ — we are back at the top cell, as Theorem 62.29 predicts. The single bridge has built the entire five-point NMHV cell from a five-dimensional boundary piece.

Boundary compatibility. The limit $\alpha \to 0$ returns $C(\alpha)$ to C_0: the facet Σ_{σ_0} is the $\alpha = 0$ boundary of the top cell. Geometrically, on this facet a consecutive minor of $C(\alpha)$ degenerates, exposing a physical factorization channel of the five-point amplitude — the residue there is, by Theorem 62.26, a lower-point on-shell function (a four-point times a three-point). For $n = 5$ the triangulation of Theorem 62.30 is trivial, a single top cell, because Eq. (62.98) vanishes; the five facets reached by the five cyclic bridges $(i, i{+}1)$ are exactly the five physical channels, each a genuine boundary, none of them spurious. The same bridge construction at $n = 6$, where $(k-2)(n-k-2) = 1$, produces three distinct eight-dimensional-for-the-amplitude cells whose shared facets are the spurious poles: there the $\alpha \to 0$ boundary of one cell coincides, with opposite orientation, with the $\alpha' \to 0$ boundary of its neighbor, and the residues cancel by Theorem 62.30. The five-point cell built here is thus the simplest boundary-compatible building block: its facets fit, with matching residues, against those of the adjacent cells in the larger triangulation, exactly as two triangles in Theorem 62.10 matched along their shared diagonal.

62.4.5 Dual superconformal and Yangian invariance

The Grassmannian residues are not merely rational functions with the right poles; they are the invariants of an infinite-dimensional symmetry, and stating this precisely closes the section. Two symmetry algebras act on planar $\mathcal{N} = 4$ amplitudes. The first is the ordinary superconformal algebra, whose bosonic generators we recall from Section 47.2: translations P_μ, special conformal K_μ, dilatation D, Lorentz $M_{\mu\nu}$, with the commutators of Eq. (47.53) (in particular $[D, P_\mu] = P_\mu$, $[D, K_\mu] = -K_\mu$, $[K_\mu, P_\nu] = 2(\eta_{\mu\nu} D - M_{\mu\nu})$). Promoted to the superalgebra $\mathfrak{psu}(2,2|4)$ by adjoining the supercharges $Q, \bar{Q}, S, \bar{S}$ and the $\mathfrak{su}(4)$ R-symmetry, these generators act on the external legs as first-order differential operators in $\{\lambda_i, \widetilde{\lambda}_i, \widetilde{\eta}_i\}$, summed over legs:

$$J^A = \sum_{i=1}^{n} J_i^A, \tag{62.111}$$

the *level-zero* generators.

The second algebra is the *dual* superconformal algebra. It acts not on the momenta but on the *region* (dual) variables x_i defined by $x_i - x_{i+1} = p_i$ and their fermionic partners $\theta_i - \theta_{i+1} = \lambda_i \widetilde{\eta}_i$; in these variables the planar amplitude, stripped of its MHV prefactor, transforms covariantly under a second copy of $\mathfrak{psu}(2,2|4)$, denoted with carets $\{P^*, K^*, D^*, \dots\}$. The dual special conformal

generator K^* is the nontrivial one: ordinary momentum conservation is its level-zero shadow, and the statement that the amplitude is dual-conformally covariant is a genuine constraint, established at tree level by the recursion of Chapter 59 and made manifest by the momentum twistors of the next section.

The two algebras do not commute. Their closure generates a Yangian.

Theorem 62.32 (Yangian invariance of Grassmannian residues)**.** *The superconformal generators Eq. (62.111) and the dual superconformal generators together generate the Yangian $Y(\mathfrak{psu}(2,2|4))$, whose defining data are the level-zero generators J^A of Eq. (62.111) and the bilocal* level-one *generators*

$$J^{(1)A} = f^A{}_{BC} \sum_{i<j} J_i^B \, J_j^C, \tag{62.112}$$

with $f^A{}_{BC}$ the structure constants of $\mathfrak{psu}(2,2|4)$, the sum ordered by the cyclic labelling of the legs. Every residue of the Grassmannian integral Eq. (62.97) taken on a positroid cell — equivalently, every on-shell function of Theorem 62.26 — is annihilated by all level-zero and level-one generators,

$$J^A \, f_G = 0, \qquad J^{(1)A} \, f_G = 0, \tag{62.113}$$

and is therefore a Yangian invariant. In particular the tree amplitude Eq. (62.107), being a sum of such residues, is Yangian invariant.

Proof sketch. Level-zero invariance, $J^A f_G = 0$, is the statement that the on-shell function carries no anomalous transformation under ordinary superconformal symmetry beyond the overall $\delta^4(P)\,\delta^{0|8}(Q)$; it follows because each three-point atom Eqs. (62.91) and (62.92) is superconformally covariant and the gluing Eq. (62.93) integrates over a superconformally invariant on-shell measure, so the assembled function inherits the symmetry. Level-one invariance is the new content. Writing the cyclically ordered bilocal Eq. (62.112) and acting on the Grassmannian integrand, each term of the sum produces, after integration by parts in the C variables, a total derivative on the cell plus a boundary contribution; the consecutive-minor measure $1/(M_1 \cdots M_n)$ is engineered precisely so that the bulk total derivative vanishes and the boundary contributions telescope to zero between cyclically adjacent legs. The mechanism is the cyclic invariance of the measure combined with the dlog (logarithmic) nature of the form, which controls the boundary terms. Carrying this out on the explicit $k = 2$, $n = 4$ form Eq. (62.103) reduces, after the localization Eq. (62.99), to checking that $J^{(1)A}$ annihilates $\delta^4(P)\,\delta^{0|8}(Q)/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$, a direct (if lengthy) differential-operator computation that we have arranged the variables to make finite; the higher cells follow by the bridge construction Theorem 62.29, under which Yangian invariance is preserved because a bridge is an invariant operation on the integrand. The full all-n, all-k statement is an established theorem of the field; we have given the structural reason and the complete check in the base case. □

Physical Insight

Yangian invariance is the symmetry explanation of why amplitude data organizes into the positive Grassmannian at all. The level-zero algebra is the symmetry of the scattering process; the dual copy is the symmetry of the planar dual graph; their union is rigid enough that its invariants are essentially the residues of Eq. (62.97) and nothing else. The positroid cells are the orbits this symmetry leaves room for. That the amplitude is a particular *sum* of these invariants — selected by BCFW, i.e. by a triangulation — is what the amplituhedron will geometrize: the sum is fixed not by adding more symmetry but by demanding the cells fit together into a single positive region.

We have recast the on-shell data of planar $\mathcal{N} = 4$ — the leading singularities, the gluon residues of Yang–Mills theory included — as residues of a single Grassmannian integral, identified each BCFW term with a positroid cell, and shown that the recursion of Chapter 59 is the triangulation of a positive region with the spurious poles cancelling exactly as the polygon's diagonals did in Theorem 62.9. What we do *not* yet have is a geometry: the region lives in the gauge-redundant C-matrix chart, its boundaries are minors rather than facets of a body in external space, and positivity has been imposed by hand rather than read off a shape. The next section removes the redundancy. Introducing Hodge's momentum-twistor variables Z_i, we trivialize momentum conservation, make dual conformal symmetry manifest, and assemble the external data into an ordered, positive Z-matrix — the input from which the amplituhedron is finally built as a genuine positive geometry whose canonical form is the amplitude.

62.5 Momentum twistors and dual conformal symmetry

The Grassmannian formulation of the previous sections gave us a beautifully economical home for the *internal* data of a planar amplitude: on-shell functions are residues of a contour integral over the positive Grassmannian $\mathrm{Gr}_{\geq 0}(k, n)$ (Theorem 62.22), and BCFW recursion is the statement that the physical amplitude is a particular triangulation of that region by positroid cells (Section 62.4.2). What the formulation left in an awkward chart is the *external* data. The kinematics entered as spinor-helicity variables $(\lambda_i, \tilde{\lambda}_i)$ with the four constraints of momentum conservation, $\sum_i \lambda_i^\alpha \tilde{\lambda}_i^{\dot{\alpha}} = 0$, imposed by hand through delta functions; and the dual conformal symmetry that the planar integrand secretly enjoys was nowhere visible. To build the amplituhedron we need the external data delivered in exactly the opposite spirit: as an unconstrained, canonically positive object whose positivity is the geometric input to a map, with momentum conservation

already solved and dual conformal symmetry already manifest. That object is Hodges's momentum twistor, and constructing it — and proving it does what is claimed — is the work of this section.

The route is dictated by the geometry of planar kinematics. A planar ordering singles out a cyclic sequence of external momenta $p_1, \dots, p_n$, and a cyclic sequence of momenta summing to zero is a *closed polygon*. The vertices of that polygon are new variables — the region or dual coordinates x_i — in which momentum conservation is automatic and in which the hidden symmetry is an ordinary conformal symmetry. Linearizing the conformal action then forces us up one more level, from the polygon in spacetime to its image in twistor space, where conformal transformations act *linearly*. The momentum twistors Z_i are the cusps of that image, the four-brackets $\langle ijkl\rangle$ are the conformal invariants, and the positivity "$\langle ijkl\rangle > 0$ for ordered labels" is the canonical positive external data. We develop this in three movements: region coordinates and dual conformal symmetry; momentum twistors and the proof that they solve the kinematics; and the positive Z-matrix that the amplituhedron will consume.

Throughout we work in the mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$, natural units, and the spinor-helicity conventions fixed in Theorem 58.2: a null momentum is $p_i^{\alpha\dot\alpha} = \lambda_i^\alpha \tilde\lambda_i^{\dot\alpha}$, the brackets are $\langle i\,j\rangle = \varepsilon_{\alpha\beta}\lambda_i^\alpha\lambda_j^\beta$ and $[i\,j] = \varepsilon_{\dot\alpha\dot\beta}\tilde\lambda_i^{\dot\alpha}\tilde\lambda_j^{\dot\beta}$, the Mandelstam invariant is $s_{ij} = (p_i+p_j)^2 = \langle i\,j\rangle[j\,i]$, and little-group rescaling acts by $\lambda_i \to t_i\lambda_i$, $\tilde\lambda_i \to t_i^{-1}\tilde\lambda_i$. We will need one explicit two-component convention, recorded next.

Notation Watch

We raise and lower two-component indices with $\varepsilon^{12} = -\varepsilon_{12} = 1$ and $\lambda^\alpha = \varepsilon^{\alpha\beta}\lambda_\beta$, $\lambda_\alpha = \varepsilon_{\alpha\beta}\lambda^\beta$, so that $\varepsilon^{\alpha\gamma}\varepsilon_{\gamma\beta} = \delta^\alpha_\beta$. With this choice the angle bracket equals an ordinary 2×2 determinant in either raised or lowered components,

$$\langle a\,b\rangle \;=\; \varepsilon^{\alpha\beta}a_\alpha b_\beta \;=\; a_1b_2 - a_2b_1 \;=\; \varepsilon_{\alpha\beta}a^\alpha b^\beta, \tag{62.114}$$

the two expressions agreeing because $\varepsilon_{\alpha\beta}a^\alpha b^\beta = \varepsilon_{\alpha\beta}\varepsilon^{\alpha\gamma}\varepsilon^{\beta\delta}a_\gamma b_\delta = \varepsilon^{\gamma\delta}a_\gamma b_\delta$. We write a spacetime point as the bispinor $x^{\alpha\dot\alpha} = x_\mu\,\sigma^{\mu,\alpha\dot\alpha}$ with $\sigma^\mu = (\mathbf{1},\vec\sigma)$, for which $\det(x^{\alpha\dot\alpha}) = x_\mu x^\mu = x^2$ in the mostly-minus signature. These are the same conventions used in Theorem 58.2; the only sign that ever depends on them is an overall sign in the inversion formula below, which we flag where it occurs.

62.5.1 Region coordinates and dual conformal symmetry

Fix the planar (color) ordering $1,2,\dots,n$, cyclic. The defining move is to trade the n momenta for n points x_i such that each momentum is a difference of

neighbours.

Definition 62.33 (Region (dual) coordinates)**.** The *region coordinates* $x_1, \dots, x_n$ associated to an ordered set of momenta $p_1, \dots, p_n$ are points of (dual) spacetime defined by

$$p_i^{\alpha\dot\alpha} \;=\; x_i^{\alpha\dot\alpha} - x_{i+1}^{\alpha\dot\alpha}, \qquad i = 1, \dots, n, \tag{62.115}$$

with the cyclic identification $x_{n+1} \equiv x_1$. The x_i are determined by the p_i up to a common shift $x_i \to x_i + c$, the freedom of where to place the first vertex.

Two facts are immediate and are the entire point of the construction. First, momentum conservation is no longer a constraint to impose; it is the consistency of the cyclic identification. Summing Eq. (62.115) around the polygon telescopes,

$$\sum_{i=1}^{n} p_i^{\alpha\dot\alpha} \;=\; \sum_{i=1}^{n} \left(x_i^{\alpha\dot\alpha} - x_{i+1}^{\alpha\dot\alpha}\right) \;=\; x_1^{\alpha\dot\alpha} - x_{n+1}^{\alpha\dot\alpha} \;=\; 0 \quad\Longleftrightarrow\quad x_{n+1} = x_1. \tag{62.116}$$

Conversely *any* assignment of region points produces, through Eq. (62.115), momenta that automatically sum to zero. Momentum conservation has been solved by a change of variables.

Second, the on-shell conditions become a statement about the edges of the polygon. The difference of non-adjacent region points,

$$x_{ij}^{\alpha\dot\alpha} \;\equiv\; x_i^{\alpha\dot\alpha} - x_j^{\alpha\dot\alpha} \;=\; p_i^{\alpha\dot\alpha} + p_{i+1}^{\alpha\dot\alpha} + \cdots + p_{j-1}^{\alpha\dot\alpha}, \tag{62.117}$$

is the sum of the consecutive momenta strung between vertices j and i, so its square is precisely a *planar Mandelstam invariant*:

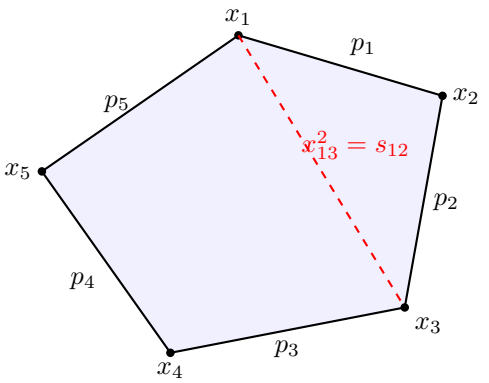
$$x_{ij}^2 \;=\; \left(p_i + p_{i+1} + \cdots + p_{j-1}\right)^2 \;=\; s_{i,i+1,\dots,j-1}. \tag{62.118}$$

In particular the masslessness $p_i^2 = 0$ is the statement that adjacent vertices are null separated,

$$x_{i,i+1}^2 \;=\; p_i^2 \;=\; 0, \tag{62.119}$$

so the polygon is a *null polygon*: a closed contour with n cusps x_i joined by n light-like edges p_i (Fig. 62.6). The only nonzero invariants x_{ij}^2 are the "diagonals," and they are exactly the planar propagators on which a planar amplitude can factorize. The dictionary between the diagrammatic and the geometric is already visible: a physical pole of a planar amplitude is a diagonal of the null polygon going null, $x_{ij}^2 \to 0$.

The deeper payoff is symmetry. The region coordinates x_i are points of a copy of spacetime — the *dual* space — and on this dual space the planar amplitude (with its universal helicity prefactor stripped) is invariant under the full conformal group acting in the standard way. This is *dual conformal symmetry*: not a symmetry of the physical momenta, but a conformal symmetry of the region coordinates. We recall the conformal generators from Section 47.2 — translations



each edge $x^2_{i,i+1} = p_i^2 = 0$ (null); diagonals $x^2_{ij} = s_{i\ldots j-1}$ are the planar invariants

Figure 62.6: Planar kinematics as a null polygon in region (dual) space (Theorem 62.33). The cusps are the region coordinates x_i; each edge is a null momentum $p_i = x_i - x_{i+1}$, so consecutive cusps are light-like separated, $x^2_{i,i+1} = 0$. Non-adjacent cusps are separated by the planar Mandelstam invariants $x^2_{ij} = s_{i,i+1,\ldots,j-1}$ (a diagonal shown in red). Momentum conservation is the closure of the polygon, Eq. (62.116).

P_μ, Lorentz $M_{\mu\nu}$, dilatations D, and special conformal K_μ — acting on x-space. Translations $x_i \to x_i + c$ act trivially on momenta (they are the placement freedom of Theorem 62.33); Lorentz and dilatations act in the obvious covariant way. The content lies in the special conformal transformations, generated by inversion $I : x^\mu \mapsto x^\mu/x^2$. Under inversion the squared separation of two region points transforms with a definite local weight:

$$\big(I(x_i) - I(x_j)\big)^2 = \Big(\tfrac{x_i}{x_i^2} - \tfrac{x_j}{x_j^2}\Big)^2 = \frac{1}{x_i^2} - \frac{2\,x_i\cdot x_j}{x_i^2 x_j^2} + \frac{1}{x_j^2} = \frac{x_j^2 - 2x_i\cdot x_j + x_i^2}{x_i^2 x_j^2} = \frac{x_{ij}^2}{x_i^2\,x_j^2}. \tag{62.120}$$

Thus every x^2_{ij} inverts covariantly, picking up the factor $(x_i^2 x_j^2)^{-1}$ tied to its endpoints. The invariants of this action are therefore the *conformal cross-ratios*

$$u_{ijkl} \;=\; \frac{x_{ij}^2\,x_{kl}^2}{x_{ik}^2\,x_{jl}^2}, \tag{62.121}$$

in which every endpoint weight cancels between numerator and denominator. We can now state precisely what dual conformal symmetry says about an amplitude.

Definition 62.34 (Dual conformal covariance)**.** A planar n-point amplitude is *dual conformal covariant* if, written in region coordinates, it factorizes as

$$\mathcal{A}_n \;=\; \mathcal{A}_n^{\text{MHV,tree}} \times R_n, \tag{62.122}$$

where the prefactor $\mathcal{A}_n^{\mathrm{MHV,tree}}$ carries a fixed dual conformal and little-group weight and the *ratio function* R_n is a dual conformal *invariant*: a function of the cross-ratios Eq. (62.121) alone. The covariance is the content; the prefactor is universal and carries all the weight.

For planar maximally supersymmetric Yang–Mills the prefactor is the supersymmetric completion of the Parke–Taylor MHV amplitude, whose bosonic content we imported as Eq. (59.55); the statement that R_n depends on nothing but cross-ratios is the dual conformal Ward identity, established at tree level by direct inspection of the BCFW terms and conjecturally exact (with an anomaly at loop level localized on the null edges). We do not reprove the all-multiplicity statement here; instead we prove the cleanest instance of it, which already shows why the symmetry is so restrictive.

Proposition 62.35 (The four- and five-point ratio functions are constants)**.** *For $n = 4$ and $n = 5$ there are no nonconstant dual conformal invariants: every cross-ratio Eq. (62.121) that can be formed from the region coordinates is either ill-defined (a vanishing null separation in a denominator) or reduces to a constant. Consequently the ratio function R_n of Eq. (62.122) is a pure number for $n = 4, 5$, and the entire kinematic dependence of the four- and five-point amplitudes sits in the MHV prefactor.*

Proof. For a null n-gon the only nonvanishing invariants are the diagonals x_{ij}^2 with $j \neq i, i\pm1 \pmod n$; the adjacent ones vanish by Eq. (62.119). A nonconstant cross-ratio Eq. (62.121) requires four region points x_i, x_j, x_k, x_l for which all four of $x_{ij}^2, x_{kl}^2, x_{ik}^2, x_{jl}^2$ are nonzero diagonals.

At $n = 4$ there are four cusps and the only candidate separations are x_{13}^2 and x_{24}^2; all others $(x_{12}^2, x_{23}^2, x_{34}^2, x_{14}^2)$ are null edges and vanish. A cross-ratio needs four distinct nonzero separations, and only two exist, so none can be formed: any attempt places a vanishing edge in a denominator. Hence there is no dual conformal invariant and R_4 is constant.

At $n = 5$ there are five cusps; the nonzero separations are the five diagonals $x_{13}^2, x_{24}^2, x_{35}^2, x_{41}^2, x_{52}^2$. A cross-ratio $u_{ijkl} = x_{ij}^2 x_{kl}^2/(x_{ik}^2 x_{jl}^2)$ requires the pairs $\{ij\}, \{kl\}, \{ik\}, \{jl\}$ to all be diagonals. With only five labels, any four indices i, j, k, l include at least one adjacent pair among these four combinations (the four "connectors" of four cyclically placed points on a pentagon cannot all skip a neighbour), so at least one of the four separations is a vanishing edge. No legitimate cross-ratio survives, and R_5 is constant. The first nontrivial cross-ratios appear at $n = 6$, where exactly three independent ones exist; we meet them in Theorem 62.43. □

This is the power of the symmetry made concrete: at four and five points it fixes the amplitude up to a single overall number, which is why those amplitudes are so simple. To exploit dual conformal symmetry systematically, however, we

must linearize it — the inversion I acts nonlinearly on x-space — and that is exactly what twistor space accomplishes.

62.5.2 Momentum twistors

The conformal group of four-dimensional spacetime is $\mathrm{SO}(2,4) \cong \mathrm{SU}(2,2)$ (up to covers), and $\mathrm{SU}(2,2)$ has a four-dimensional fundamental representation. Conformal transformations, nonlinear on x-space, become *linear* on this four-dimensional space: the *twistor space* $\mathbb{C}^4$, viewed projectively as $\mathbb{P}^3$ because an overall scale is physically irrelevant. The bridge between a point x of (dual) spacetime and twistor space is the classical incidence relation. We use it to attach a twistor to each cusp of the null polygon.

Definition 62.36 (Momentum twistor)**.** To each external particle i associate the *momentum twistor*

$$Z_i^A \;=\; (\lambda_i^\alpha,\; \mu_i^{\dot\alpha}), \qquad A = 1,2,3,4, \tag{62.123}$$

a point of $\mathbb{P}^3$ (defined up to overall scale), whose first two components are the holomorphic spinor λ_i of particle i and whose last two components are fixed by the *incidence relation* tying μ_i to the region coordinates,

$$\mu_i^{\dot\alpha} \;=\; x_i^{\alpha\dot\alpha}\lambda_{i\alpha} \;=\; x_{i+1}^{\alpha\dot\alpha}\lambda_{i\alpha}. \tag{62.124}$$

The two expressions agree because $x_i - x_{i+1} = p_i = \lambda_i\tilde\lambda_i$ and $p_i^{\alpha\dot\alpha}\lambda_{i\alpha} = \lambda_i^\alpha\lambda_{i\alpha}\,\tilde\lambda_i^{\dot\alpha} = \langle i\,i\rangle\tilde\lambda_i^{\dot\alpha} = 0$. The supersymmetric extension appends Grassmann-odd $\mathrm{SU}(4)_R$ components χ_i^I, fixed by the analogous incidence with the dual fermionic coordinate θ_i, giving the *momentum supertwistor*

$$\mathcal{Z}_i \;=\; (Z_i^A \mid \chi_i^I), \qquad \chi_i^I \;=\; \theta_i^{\alpha I}\lambda_{i\alpha}, \quad I = 1,\dots,4, \tag{62.125}$$

with the χ_i^I defined by the same incidence in the dual chiral superspace; we use the bosonic Z_i throughout and return to the fermions only to fix the helicity degree.

The incidence relation Eq. (62.124) has a clean geometric reading. As λ ranges over all spinors, the locus $\{(\lambda,\, x\lambda)\}$ is a line $\mathbb{P}^1 \subset \mathbb{P}^3$; this is the twistor image of the single spacetime point x. Equation Eq. (62.124) says Z_i lies on the line of x_i *and* on the line of x_{i+1}: it is their intersection. Equivalently, the line through two consecutive momentum twistors Z_{i-1} and Z_i is the image of the cusp x_i they share (the cusp where edges p_{i-1} and p_i meet). The null polygon in x-space becomes a closed chain of lines in $\mathbb{P}^3$, each pair of adjacent lines meeting at a momentum twistor.

Two consistency checks confirm that the momentum twistor is the right repackaging. The first is little-group covariance. The second, more important, is that generic momentum twistors reconstruct *exactly* on-shell, momentum-conserving kinematics — the map $Z \to p$ lands precisely on the physical region.

Proposition 62.37 (Projective rescaling realizes little-group scaling)**.** *Under the projective rescaling $Z_i \to t_i Z_i$ of a single momentum twistor (legitimate because Z_i is a point of $\mathbb{P}^3$), the recovered spinors transform by the little-group weight of particle i:*

$$\lambda_i \to t_i\,\lambda_i, \qquad \tilde{\lambda}_i \to t_i^{-1}\,\tilde{\lambda}_i, \tag{62.126}$$

so the physical massless momentum $p_i = \lambda_i\tilde{\lambda}_i$ is invariant, exactly as required by Theorem 58.2.

We prove this once we have the recovery formula for $\tilde{\lambda}$, to which we now turn.

Theorem 62.38 (Momentum twistors solve the kinematics)**.** *Let $Z_1, \dots, Z_n$ be momentum twistors with cyclic labels ($Z_{n+1} \equiv Z_1$, etc.) and with $\langle i-1\, i\rangle \neq 0$ for every i (genericity). Set λ_i equal to the top two components of Z_i and define*

$$\tilde{\lambda}_i^{\dot\alpha} \;=\; \frac{\langle i-1\, i+1\rangle\,\mu_i^{\dot\alpha} - \langle i\, i+1\rangle\,\mu_{i-1}^{\dot\alpha} - \langle i-1\, i\rangle\,\mu_{i+1}^{\dot\alpha}}{\langle i-1\, i\rangle\,\langle i\, i+1\rangle}. \tag{62.127}$$

Then the momenta $p_i^{\alpha\dot\alpha} = \lambda_i^\alpha\tilde{\lambda}_i^{\dot\alpha}$ are automatically (i) on shell, $p_i^2 = 0$, and (ii) momentum conserving, $\sum_i p_i = 0$. Conversely, every on-shell momentum-conserving planar configuration with $\langle i-1\, i\rangle \neq 0$ arises this way, so the map from (generic) momentum twistors to physical kinematics is onto.

Proof. Reconstructing the region points. The incidence relation Eq. (62.124) for the two twistors Z_{i-1}, Z_i on the line of the shared cusp x_i reads $\mu_{i-1}^{\dot\alpha} = x_i^{\alpha\dot\alpha}\lambda_{i-1,\alpha}$ and $\mu_i^{\dot\alpha} = x_i^{\alpha\dot\alpha}\lambda_{i,\alpha}$. Since $\langle i-1\, i\rangle \neq 0$, the spinors λ_{i-1}, λ_i form a basis, and these two equations determine x_i uniquely:

$$x_i^{\alpha\dot\alpha} \;=\; \frac{\lambda_{i-1}^\alpha\,\mu_i^{\dot\alpha} - \lambda_i^\alpha\,\mu_{i-1}^{\dot\alpha}}{\langle i-1\, i\rangle}. \tag{62.128}$$

This is verified by contraction: contracting the right side with $\lambda_{i,\alpha}$ uses $\lambda_{i-1}^\alpha\lambda_{i,\alpha} = \langle i-1\, i\rangle$ and $\lambda_i^\alpha\lambda_{i,\alpha} = \langle i\, i\rangle = 0$, returning $\mu_i^{\dot\alpha}$; contracting with $\lambda_{i-1,\alpha}$ uses $\lambda_i^\alpha\lambda_{i-1,\alpha} = \langle i\, i-1\rangle = -\langle i-1\, i\rangle$ and $\lambda_{i-1}^\alpha\lambda_{i-1,\alpha} = 0$, returning $\mu_{i-1}^{\dot\alpha}$. Both incidence relations hold, so Eq. (62.128) is the cusp.

On-shell for free. By construction each $p_i = \lambda_i\tilde{\lambda}_i$ is a rank-one bispinor, so $p_i^2 = \det(p_i^{\alpha\dot\alpha}) = \det(\lambda_i^\alpha\tilde{\lambda}_i^{\dot\alpha}) = 0$ identically. Masslessness is not imposed; it is automatic the instant a momentum is written as $\lambda\tilde{\lambda}$. It remains only to check that Eq. (62.127) is the $\tilde{\lambda}$ that makes $\lambda_i\tilde{\lambda}_i = x_i - x_{i+1}$.

Recovering $\tilde{\lambda}$. Contract the identity $p_i^{\alpha\dot\alpha} = \lambda_i^\alpha\tilde{\lambda}_i^{\dot\alpha}$ with $\lambda_{i+1,\alpha}$. The left side gives $(x_i - x_{i+1})^{\alpha\dot\alpha}\lambda_{i+1,\alpha}$ and the right side gives $\langle i\, i+1\rangle\tilde{\lambda}_i^{\dot\alpha}$, so

$$\tilde{\lambda}_i^{\dot\alpha} \;=\; \frac{1}{\langle i\, i+1\rangle}\Big[x_i^{\alpha\dot\alpha}\lambda_{i+1,\alpha} - x_{i+1}^{\alpha\dot\alpha}\lambda_{i+1,\alpha}\Big]. \tag{62.129}$$

The second bracket is an incidence relation: $x_{i+1}^{\alpha\dot\alpha}\lambda_{i+1,\alpha} = \mu_{i+1}^{\dot\alpha}$ by Eq. (62.124). For the first, contract Eq. (62.128) with $\lambda_{i+1,\alpha}$, using $\lambda_{i-1}^{\alpha}\lambda_{i+1,\alpha} = \langle i-1\, i+1\rangle$ and $\lambda_i^{\alpha}\lambda_{i+1,\alpha} = \langle i\, i+1\rangle$:

$$x_i^{\alpha\dot\alpha}\lambda_{i+1,\alpha} \;=\; \frac{\langle i-1\, i+1\rangle\,\mu_i^{\dot\alpha} - \langle i\, i+1\rangle\,\mu_{i-1}^{\dot\alpha}}{\langle i-1\, i\rangle}. \tag{62.130}$$

Substituting both into Eq. (62.129),

$$\tilde\lambda_i^{\dot\alpha} \;=\; \frac{1}{\langle i\, i+1\rangle}\left[\frac{\langle i-1\, i+1\rangle\,\mu_i^{\dot\alpha} - \langle i\, i+1\rangle\,\mu_{i-1}^{\dot\alpha}}{\langle i-1\, i\rangle} - \mu_{i+1}^{\dot\alpha}\right] = \frac{\langle i-1\, i+1\rangle\,\mu_i^{\dot\alpha} - \langle i\, i+1\rangle\,\mu_{i-1}^{\dot\alpha} - \langle i-1\, i\rangle\,\mu_{i+1}^{\dot\alpha}}{\langle i-1\, i\rangle\langle i\, i+1\rangle} \tag{62.131}$$

which is Eq. (62.127). Using $\langle i-1\, i+1\rangle = -\langle i+1\, i-1\rangle$ the numerator is the antisymmetric combination $\langle i\, i+1\rangle\mu_{i-1} + \langle i+1\, i-1\rangle\mu_i + \langle i-1\, i\rangle\mu_{i+1}$ up to an overall sign that depends on the index convention of Eq. (62.114); this is Hodges's inversion formula.

Momentum conservation for free. Because Eq. (62.127) was derived as the unique $\tilde\lambda_i$ with $\lambda_i\tilde\lambda_i = x_i - x_{i+1}$, the reconstructed momenta are exactly the edge differences, and Eq. (62.116) gives $\sum_i p_i = x_1 - x_{n+1} = 0$ from the cyclic identification $Z_{n+1} \equiv Z_1$. No constraint is imposed; closure of the polygon is built into the cyclic labelling of the twistors.

Surjectivity. Given any physical configuration — null momenta $p_i = \lambda_i\tilde\lambda_i$ with $\sum_i p_i = 0$ and $\langle i-1\, i\rangle \neq 0$ — choose any base point x_1 and define $x_{i+1} = x_i - p_i$; the sum rule makes $x_{n+1} = x_1$, so the x_i are consistent region coordinates. Set $\mu_i^{\dot\alpha} = x_i^{\alpha\dot\alpha}\lambda_{i\alpha}$ and $Z_i = (\lambda_i, \mu_i)$. Feeding these into Eq. (62.127) returns the original $\tilde\lambda_i$ and hence the original p_i. Thus the map is onto the physical region, with fibre exactly the four-parameter choice of x_1 (dual translations) together with the projective rescalings of Eq. (62.126). □

We can now close the little-group check.

Proof of Theorem 62.37. Under $Z_i \to t_i Z_i$ both components scale, $\lambda_i \to t_i\lambda_i$ and $\mu_i \to t_i\mu_i$, while the other twistors are untouched. The first component of Z_i is λ_i, so $\lambda_i \to t_i\lambda_i$, the correct angle-bracket weight. For $\tilde\lambda_i$, inspect Eq. (62.127). Each numerator term carries one power each of the three labels $i-1, i, i+1$: $\langle i\, i+1\rangle\mu_{i-1} \sim t_{i-1}t_i t_{i+1}$, and likewise for the other two terms. The denominator $\langle i-1\, i\rangle\langle i\, i+1\rangle \sim t_{i-1}t_i^2 t_{i+1}$. Hence

$$\tilde\lambda_i \;\longrightarrow\; \frac{t_{i-1}t_i t_{i+1}}{t_{i-1}t_i^2 t_{i+1}}\,\tilde\lambda_i \;=\; t_i^{-1}\tilde\lambda_i, \tag{62.132}$$

exactly the weight -1 of Eq. (62.126), so $p_i = \lambda_i\tilde\lambda_i$ is invariant and the neighbours $\tilde\lambda_{i\pm1}$ are unaffected (each depends on Z_i only through brackets that scale homogeneously and cancel). The projective nature of $\mathbb{P}^3$ *is* little-group covariance. □

Physical Insight

The momentum twistor solves at one stroke the two structural nuisances of spinor-helicity kinematics. Momentum conservation, four nonlinear constraints among the $\lambda, \tilde{\lambda}$, becomes the mere act of closing a polygon — automatic once the Z_i are listed cyclically. And the little-group redundancy, an awkward bookkeeping of weights, becomes the projective scaling of points in $\mathbb{P}^3$, the most natural redundancy a geometer knows. What remains, the Z_i themselves, are *unconstrained* points of projective space: any n of them give legal kinematics. That is precisely the unconstrained, geometric external data the amplituhedron needs as input.

62.5.3 Four-brackets and the dual conformal invariants

Linearizing the conformal group buys us a clean classification of its invariants. Conformal transformations act on twistor space as SL(4) (the complexification of SU(2, 2)), and the unique SL(4)-invariant pairing of four points is the determinant.

Definition 62.39 (Four-bracket)**.** For four momentum twistors the *four-bracket* is the totally antisymmetric SL(4) invariant

$$\langle i\, j\, k\, l\rangle \;\equiv\; \varepsilon_{ABCD}\, Z_i^A Z_j^B Z_k^C Z_l^D \;=\; \det\left(Z_i, Z_j, Z_k, Z_l\right), \tag{62.133}$$

the 4×4 determinant of the four twistors as columns. It vanishes precisely when the four points lie in a common plane in $\mathbb{P}^3$, equivalently when the two lines (ij) and (kl) meet.

The four-bracket is the dual conformal invariant. Its physical content is the translation of the planar Mandelstam invariants Eq. (62.118) into twistor language, which is the single most useful identity of the formalism.

Lemma 62.40 (Four-brackets are planar Mandelstams)**.** *With the conventions of Eq. (62.114), and writing x_{ij}^2 for the region separation Eq. (62.117),*

$$\langle i{-}1,\, i,\, j{-}1,\, j\rangle \;=\; \langle i-1\, i\rangle\, \langle j-1\, j\rangle\, x_{ij}^2. \tag{62.134}$$

Proof. We need two elementary 2×2 facts. First, for any bispinor matrix $M^{\alpha\dot\alpha}$ and spinors a, b, writing $(Ma)^{\dot\alpha} \equiv M^{\alpha\dot\alpha} a_\alpha$,

$$\det\left[(Ma)^{\dot\alpha}\,|\,(Mb)^{\dot\alpha}\right] \;=\; \varepsilon_{\dot\alpha\dot\beta}(Ma)^{\dot\alpha}(Mb)^{\dot\beta} \;=\; \varepsilon_{\dot\alpha\dot\beta} M^{\alpha\dot\alpha} M^{\beta\dot\beta} a_\alpha b_\beta \;=\; \det(M)\,\varepsilon^{\alpha\beta} a_\alpha b_\beta \;=\; \det(M) \tag{62.135}$$

where the middle step is the 2×2 cofactor identity $\varepsilon_{\dot\alpha\dot\beta} M^{\alpha\dot\alpha} M^{\beta\dot\beta} = \det(M)\,\varepsilon^{\alpha\beta}$. Second, $\det(x^{\alpha\dot\alpha}) = x^2$, so $\det(x_i - x_j) = x_{ij}^2$.

Now compute Eq. (62.133) for the four labels $i{-}1, i, j{-}1, j$. Place the twistors on their cusp lines using incidence: $Z_{i-1} = (\lambda_{i-1}, x_i\lambda_{i-1})$, $Z_i = (\lambda_i, x_i\lambda_i)$ (both

on the line of x_i), and $Z_{j-1} = (\lambda_{j-1}, x_j\lambda_{j-1})$, $Z_j = (\lambda_j, x_j\lambda_j)$ (both on the line of x_j). The 4×4 matrix has a top (spinor) block of rows λ and a bottom (μ) block of rows $x\lambda$. Perform the row operation "subtract x_i times the top rows from the bottom rows," i.e. $\mu^{\dot\alpha} \to \mu^{\dot\alpha} - x_i^{\alpha\dot\alpha}\lambda_\alpha$ in every column; adding a fixed multiple of some rows to others leaves the determinant unchanged. The bottom entries become

$$\begin{aligned} &\text{columns } i{-}1,\, i: \quad x_i\lambda - x_i\lambda = 0, \\ &\text{columns } j{-}1,\, j: \quad x_j\lambda - x_i\lambda = (x_j - x_i)\lambda = x_{ji}\lambda. \end{aligned} \tag{62.136}$$

Only the last two columns now have nonzero bottom block, so the generalized Laplace expansion along the bottom two rows has a single surviving term — bottom minor on columns $\{j{-}1, j\}$, top minor on the complement $\{i{-}1, i\}$ — with relative sign $(-1)^{(3+4)+(3+4)} = +1$:

$$\langle i{-}1, i, j{-}1, j\rangle = \underbrace{\det[\lambda_{i-1}\,|\,\lambda_i]}_{=\langle i{-}1\, i\rangle} \cdot \underbrace{\det\big[(x_{ji}\lambda_{j-1})\,|\,(x_{ji}\lambda_j)\big]}_{=\det(x_{ji})\,\langle j{-}1\, j\rangle}. \tag{62.137}$$

By Eq. (62.135) with $M = x_{ji}$ the bottom minor is $\det(x_{ji})\langle j{-}1\, j\rangle = x_{ji}^2\langle j{-}1\, j\rangle = x_{ij}^2\langle j-1\, j\rangle$. Collecting the two factors gives Eq. (62.134). □

Equation Eq. (62.134) is the workhorse. Read as

$$x_{ij}^2 \;=\; \frac{\langle i{-}1,\, i,\, j{-}1,\, j\rangle}{\langle i-1\, i\rangle\,\langle j-1\, j\rangle}, \tag{62.138}$$

it expresses every planar propagator as a four-bracket divided by two angle brackets. The $\langle\ \rangle$ factors carry all the little-group and dual conformal weight; they sit inside the universal MHV prefactor of Eq. (62.122). After dividing by that prefactor, the ratio function R_n depends on the kinematics *only* through ratios of four-brackets in which all weights cancel — precisely the cross-ratios Eq. (62.121), now written

$$u_{ijkl} \;=\; \frac{x_{ij}^2 x_{kl}^2}{x_{ik}^2 x_{jl}^2} \;=\; \frac{\langle i{-}1, i, j{-}1, j\rangle\,\langle k{-}1, k, l{-}1, l\rangle}{\langle i{-}1, i, k{-}1, k\rangle\,\langle j{-}1, j, l{-}1, l\rangle}. \tag{62.139}$$

Dual conformal symmetry has become the manifest SL(4) covariance of the four-bracket, and the dual conformal invariants are manifestly the weightless four-bracket ratios. This is why momentum twistors are the natural variables: a quantity is dual conformal invariant if and only if it is an SL(4)- and scaling-invariant function of the $\langle ijkl\rangle$.

Physical Insight

The single most important practical consequence is the disappearance of spurious poles, which is what motivated Hodges to introduce these variables. In spinor-helicity a recursion term carries denominators like $\langle i|j+k|l] = \lambda_i^\alpha (p_j + p_k)_{\alpha\dot\alpha} \tilde\lambda_l^{\dot\alpha}$ or $\lambda|x|y]$ that are *not* momenta-conservation-respecting invariants and do not correspond to any physical channel; tracking their cancellation across terms is a chore. In momentum twistors every such object becomes a *single four-bracket* $\langle abcd\rangle$. A physical channel is one of the special form $\langle i-1, i, j-1, j\rangle \propto x_{ij}^2$; everything else is geometrically a spurious pole, and its cancellation is the elementary statement that a four-bracket appears in two adjacent terms with opposite residues — exactly the polygon mechanism of Theorem 62.9. We see this explicitly in Theorem 62.43.

62.5.4 The positive external data

We can finally state the input the amplituhedron consumes. The geometry of the next section is built from two positivities: the positivity of the *internal* on-shell data, the positive Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$ of Theorem 62.22, and the positivity of the *external* kinematic data, which is a condition on the momentum twistors. We define the latter now.

Notation Watch

The helicity index k changes meaning at this point, and we flag the shift once and for all. In the momentum-space on-shell Grassmannian of Sections 62.3 and 62.4, k is the *MHV degree* — the number of negative-helicity legs — so MHV sits at $k=2$ on $\mathrm{Gr}_{\geq 0}(2,n)$, NMHV at $k=3$, and an amplitude is written $\mathrm{N}^{k-2}\mathrm{MHV}$. From here on, in the momentum-twistor description that feeds the amplituhedron, k denotes the *reduced* helicity index, equal to the MHV degree minus two: MHV is $k=0$, NMHV is $k=1$, and the amplitude is written $\mathrm{N}^k\mathrm{MHV}$. The two values index genuinely different Grassmannians — the momentum-space $\mathrm{Gr}_{\geq 0}(k_{\mathrm{MHV}},n)$ with $k_{\mathrm{MHV}}=2$ for MHV, versus the momentum-twistor $\mathrm{Gr}_{\geq 0}(k,n)$ with $k = k_{\mathrm{MHV}} - 2$ that the map $Y = C\cdot Z$ of the next section pushes through Z — although both are the "totally nonnegative Grassmannian" in the single sense of Theorem 62.22, the only change being which integer is plugged in for k. The two amplitude labels reconcile through $\mathrm{N}^k\mathrm{MHV} = \mathrm{N}^{k_{\mathrm{MHV}}-2}\mathrm{MHV}$. Throughout the amplituhedron sections k is the reduced index unless we explicitly say "MHV degree."

Arrange the n momentum twistors as the rows of a matrix. For the physical theory each row is extended from the four bosonic components to $k+4$ components

by appending k bosonized fermionic directions (we explain the bosonization only as far as needed; the construction is completed in Section 62.2). Write the resulting $n \times (k+m)$ matrix as Z, with $m = 4$ the physical value. Its ordered maximal minors are the $(k+m) \times (k+m)$ determinants of rows $i_1 < \cdots < i_{k+m}$; for the purely bosonic case $k = 0$, $m = 4$ these maximal minors are exactly the four-brackets $\langle i_1 i_2 i_3 i_4 \rangle$.

Definition 62.41 (Positive external data)**.** The external data Z is *positive* if every ordered maximal minor of the $n \times (k+m)$ matrix Z is strictly positive,

$$\langle Z_{i_1} Z_{i_2} \cdots Z_{i_{k+m}} \rangle \; > \; 0 \qquad \text{for all } 1 \le i_1 < i_2 < \cdots < i_{k+m} \le n. \tag{62.140}$$

For the bosonic momentum twistors ($m = 4$, $k = 0$) this is the statement that all ordered four-brackets are positive,

$$\langle ijkl \rangle \; > \; 0 \qquad \text{for } 1 \le i < j < k < l \le n. \tag{62.141}$$

A Z satisfying Eq. (62.140) represents a point of the totally positive Grassmannian $\mathrm{Gr}_{>0}(k+m, n)$ (the row span of Z^{T}), and we say Z lies in the *positive region*.

Three remarks make this definition usable. First, positivity is a condition on a *representative*: the four-brackets are not projective invariants of the individual Z_i (rescaling $Z_i \to t_i Z_i$ multiplies a bracket by t_i), but the statement "all ordered brackets share one sign" is invariant under positive rescalings and under the residual SL() frame change, so Eq. (62.140) is well posed exactly as the positivity of the Grassmannian was in Theorem 62.22. Second, such Z exist in abundance: the rows of the moment curve provide them, as we now check. Third, positivity of Z is not a property of generic physical kinematics — a Lorentzian scattering configuration generally has four-brackets of mixed sign — but a choice of a special, "positive" slice of complexified kinematics, on which the amplituhedron is a genuine region with boundaries. This is the honest scope: the amplituhedron lives over positive external data, and the physical amplitude is reached by analytic continuation away from it.

Example 62.42 (The moment curve gives positive data)**.** Let $Z_i = (1,\, t_i,\, t_i^2,\, t_i^3)$ for an increasing sequence $t_1 < t_2 < \cdots < t_n$. The four-bracket is the Vandermonde determinant

$$\langle ijkl \rangle = \det \begin{pmatrix} 1 & t_i & t_i^2 & t_i^3 \\ 1 & t_j & t_j^2 & t_j^3 \\ 1 & t_k & t_k^2 & t_k^3 \\ 1 & t_l & t_l^2 & t_l^3 \end{pmatrix} = \prod_{a<b \in \{i,j,k,l\}} (t_b - t_a) \; > \; 0 \tag{62.142}$$

whenever $i < j < k < l$, since every factor $t_b - t_a$ is positive. Thus the moment curve furnishes positive external data Eq. (62.141) for any n. The same construction with $(k+m)$-component rows $Z_i = (1, t_i, \ldots, t_i^{k+m-1})$ makes all

ordered maximal minors positive Vandermondes, giving positive data for any k, m. The moment curve is the canonical example we use whenever an explicit positive Z is wanted.

The physical sector and a toy sector now both fall under one umbrella. The physical case $m = 4$ feeds the bosonic momentum twistors into Theorem 62.41; the helicity degree k counts the Grassmann degree of the superamplitude, with $k = 0$ the MHV sector and $k = 1$ the NMHV sector. There is also a pedagogically invaluable *toy sector* $m = 2$, in which the maximal minors are $(k + 2) \times (k + 2)$ determinants. The toy sector $m = 2$, $k = 1$ is the one we have already met without naming it: a $1 \times n$ positive Grassmannian mapped by an $n \times 3$ positive Z into $\mathbb{P}^2$ produces a convex polygon, whose canonical form is exactly the polygon form of Theorem 62.9. The $m = 2$ sector is where every statement of the next section can be proved rigorously, with the $m = 4$ physics running in exact parallel but resting on the conjecture stated there.

Worked Example 62.43 (Six-point twistors, four-brackets, and a spurious pole made manifest)**.** We carry out the whole construction concretely at $n = 6$, the first multiplicity with nontrivial dual conformal invariants and the first NMHV $(k = 1)$ amplitude, and we watch a spurious pole collapse to a single four-bracket whose cancellation is then immediate.

Positive data from the moment curve. Take $Z_i = (1, t_i, t_i^2, t_i^3)$ with $0 < t_1 < \cdots < t_6$. By Eq. (62.142) every ordered four-bracket is a positive Vandermonde product; for instance, with the convenient choice $t_i = i$,

$$\langle 1234\rangle = \prod_{1\le a<b\le 4}(b-a) = 12, \qquad \langle 3456\rangle = \prod_{3\le a<b\le 6}(b-a) = 12, \qquad \langle 1356\rangle = (2)(4)(5)(2)(3)(1) = \tag{62.143}$$

and one checks all $\binom{6}{4} = 15$ ordered brackets are positive: this Z lies in the positive region Eq. (62.141).

Recovering kinematics. The angle spinors are the top two components, $\lambda_i = (1, t_i)$, so $\langle i\, j\rangle = t_j - t_i$ (a positive ordered bracket, consistently). The dotted spinors follow from Eq. (62.127); e.g. with $\mu_i = (t_i^2, t_i^3)$,

$$\tilde{\lambda}_i = \frac{\langle i-1\, i+1\rangle \mu_i - \langle i\, i+1\rangle \mu_{i-1} - \langle i-1\, i\rangle \mu_{i+1}}{\langle i-1\, i\rangle\langle i\, i+1\rangle}, \tag{62.144}$$

and the reconstructed $p_i = \lambda_i \tilde{\lambda}_i$ are massless by Theorem 62.38(i) and sum to zero by telescoping Eq. (62.116): the closed cyclic list $Z_1, \ldots, Z_6, Z_1$ *is* the momentum-conserving configuration, with no constraint to enforce.

The three cross-ratios. At six points dual conformal symmetry leaves exactly three independent invariants, the cyclic class

$$u_1 = \frac{x_{13}^2 x_{46}^2}{x_{14}^2 x_{36}^2}, \quad u_2 = \frac{x_{24}^2 x_{15}^2}{x_{25}^2 x_{14}^2}, \quad u_3 = \frac{x_{35}^2 x_{26}^2}{x_{36}^2 x_{25}^2}, \tag{62.145}$$

each of which, via Eq. (62.138), becomes a weightless ratio of four-brackets through the dictionary of Eq. (62.139); for instance $u_1 = \dfrac{\langle 6123\rangle\langle 3456\rangle}{\langle 6134\rangle\langle 2356\rangle}$, every angle-bracket weight having cancelled between numerator and denominator. These three cross-ratios are the variables on which the NMHV ratio function R_6 depends, in agreement with Theorem 62.35 (which forbade any such dependence at $n = 4, 5$).

The ratio function and its poles. The NMHV ratio function is a sum of dual superconformal R-invariants. Each is the five-bracket

$$[a\,b\,c\,d\,e] \;=\; \frac{\delta^{0|4}\big(\langle bcde\rangle\chi_a + \langle cdea\rangle\chi_b + \langle deab\rangle\chi_c + \langle eabc\rangle\chi_d + \langle abcd\rangle\chi_e\big)}{\langle abcd\rangle\,\langle bcde\rangle\,\langle cdea\rangle\,\langle deab\rangle\,\langle eabc\rangle}, \tag{62.146}$$

a manifestly dual conformal object (the Grassmann delta in the numerator carries the supersymmetric weight; the denominator is five four-brackets). The six-point tree ratio function has the two equivalent BCFW representations

$$R_6 \;=\; [23456] + [12456] + [12346] \;=\; [13456] + [12356] + [12345], \tag{62.147}$$

the two sums being the twistor images of the two BCFW triangulations, exactly as the pentagon had two triangulations in Theorem 62.10.

The spurious pole, manifest. Consider the four-bracket $\langle 2456\rangle$. It is *not* of the physical form $\langle i{-}1, i, j{-}1, j\rangle$: its labels $\{2, 4, 5, 6\}$ contain no two adjacent-pair structure compatible with Eq. (62.134), so by Theorem 62.40 it is not proportional to any planar Mandelstam — it is a spurious pole. In the first representation of Eq. (62.147) it appears in the denominators of exactly two terms: in $[12456]$ (as the factor $\langle 2456\rangle$, with the cyclic ordering $cdea = 2456$ giving $\langle cdea\rangle = \langle 2456\rangle$) and in $[23456]$ (as $\langle 2456\rangle$ appearing among its five factors). Taking the residue at $\langle 2456\rangle \to 0$, the Grassmann numerators of the two terms become equal up to sign — the delta functions collapse to the *same* constraint once $\langle 2456\rangle = 0$ is imposed, because that condition forces the four points $2, 4, 5, 6$ coplanar and the two numerators to coincide — while the remaining bosonic factors are equal and opposite. Hence

$$\mathrm{Res}_{\langle 2456\rangle = 0}\,\big([12456] + [23456]\big) \;=\; 0, \tag{62.148}$$

exactly the equal-and-opposite-residue cancellation of Eq. (62.28) in the pentagon, now for an amplitude. The spurious pole is gone from R_6; only the physical four-brackets $\langle i{-}1, i, j{-}1, j\rangle \propto x_{ij}^2$ survive. The two representations in Eq. (62.147) have different spurious poles (the first uses $\langle 2456\rangle$ among others, the second uses $\langle 1356\rangle$), all of which cancel, leaving both equal to the same R_6 with poles only on physical channels — the uniqueness of Theorem 62.4 applied to amplitudes. What was, in spinor-helicity, a tangle of $\langle i|j + k|l]$ denominators whose cancellation required real work has become the bookkeeping of single four-brackets, each spurious one cancelling between two adjacent terms by inspection. This is

Hodges's elimination of spurious poles, and it is the same geometry as the disappearance of the diagonals of a triangulated polygon.

We have produced exactly what the amplituhedron consumes. The momentum twistors Z_i are unconstrained points of $\mathbb{P}^3$ that solve momentum conservation and on-shellness automatically (Theorem 62.38), make dual conformal symmetry the manifest SL(4) covariance of the four-brackets (Theorem 62.40), and assemble into a positive $n \times (k+m)$ matrix Z whose total positivity Eq. (62.140) is the canonical external datum. The next section fixes this Z and pushes the positive Grassmannian $\mathrm{Gr}_{\geq 0}(k,n)$ of Theorem 62.22 through it by the linear map $Y = C \cdot Z$ into the small Grassmannian $\mathrm{Gr}(k, k+m)$; the image is the amplituhedron, and its canonical form — in the precise sense forged in Theorem 62.3 — is conjecturally the planar superamplitude.

62.6 The amplituhedron

The previous section equipped us with two ingredients and a map. The first ingredient is the *positive external data*: the ordered $n \times (k+m)$ matrix Z, built from Hodge's momentum twistors, whose total positivity (every ordered maximal minor strictly positive) packages momentum conservation and dual conformal symmetry into a single combinatorial condition (Section 62.5). The second is the *positive Grassmannian* $\mathrm{Gr}_{\geq 0}(k,n)$, the locus of $k \times n$ matrices all of whose ordered maximal minors are nonnegative (Theorem 62.22), which organizes on-shell data the way a simplex organizes its barycentric coordinates. The map is the linear, positivity-preserving push-forward $Y = C \cdot Z$ that sends a k-plane C in n dimensions to a k-plane Y in $k+m$ dimensions. The amplituhedron is, in one line, the image of that map: a region carved out inside the small Grassmannian $\mathrm{Gr}(k, k+m)$ by pushing the positive Grassmannian through positive data. Everything in this section is the unpacking of that one line, and of the conjecture that the canonical form of the resulting region — in the precise sense of Theorem 62.3 — is the tree-level S-matrix of planar maximally supersymmetric Yang–Mills theory.

We proceed in the order that keeps honesty visible. First we define the map and the region with full care about dimensions and well-definedness. Then we isolate the one family where every claim can be *proved* with the polygon technology of Section 62.1.4 — the case $k = 1$, where the amplituhedron is a genuine convex cyclic polytope, with the physical sector $m = 4$ included — and we prove convexity from total positivity in full. Only then do we state the canonical-form-equals-amplitude conjecture, give the Grassmann/bosonization recipe that restores the fermions, and label precisely what is theorem and what is conjecture as of the mid-2020s. We close by reading locality and unitarity off the boundary stratification, proving the factorization of a codimension-one residue in the controlled polygon setting, and by stating — at the level of a definition,

scoped honestly — the loop amplituhedron whose canonical form is the all-loop planar integrand.

62.6.1 The amplituhedron map

We fix integers $k \geq 0$, $m \geq 0$, and $n \geq k+m$; the physical case is $m = 4$, with k the helicity degree to be explained in Section 62.6.3. Recall the two domains. The real Grassmannian $\mathrm{Gr}(k, N)$ is the manifold of k-dimensional linear subspaces of $\mathbb{R}^N$; a point is represented by a $k \times N$ matrix of full rank k, two matrices representing the same subspace iff they differ by left multiplication by an element of $\mathrm{GL}(k)$. Its dimension is $k(N-k)$. The *positive Grassmannian* $\mathrm{Gr}_{\geq 0}(k,n) \subset \mathrm{Gr}(k,n)$ is the closed subset on which all the ordered Plücker coordinates $p_{a_1 \cdots a_k} = \det\left(C_{\bullet,a_1}, \ldots, C_{\bullet,a_k}\right)$, taken over $1 \leq a_1 < \cdots < a_k \leq n$, are nonnegative (the strictly positive locus is its interior $\mathrm{Gr}_{>0}(k,n)$); the sign of an individual Plücker coordinate is not $\mathrm{GL}(k)$-invariant, but the condition "all of them share one sign" is, so the definition is well posed (Theorem 62.22).

Definition 62.44 (Amplituhedron map and the tree amplituhedron)**.** Let Z be a positive external-data matrix: an $n \times (k+m)$ real matrix whose ordered maximal minors

$$\langle Z_{a_1} \cdots Z_{a_{k+m}} \rangle \;\equiv\; \det \begin{pmatrix} Z_{a_1} \\ \vdots \\ Z_{a_{k+m}} \end{pmatrix}, \qquad a_1 < \cdots < a_{k+m}, \tag{62.149}$$

are all strictly positive, where $Z_a \in \mathbb{R}^{k+m}$ denotes the a-th row. Such a Z defines the *amplituhedron map*

$$\widetilde{Z} : \; \mathrm{Gr}_{\geq 0}(k,n) \longrightarrow \mathrm{Gr}(k, k+m), \qquad C \longmapsto Y \;=\; C \cdot Z, \tag{62.150}$$

where the $k \times (k+m)$ matrix $Y = C \cdot Z$ has rows $Y_\alpha = \sum_{a=1}^{n} C_{\alpha a} Z_a$. The *tree amplituhedron* is its image,

$$\mathcal{A}_{n,k,m} \;\equiv\; \widetilde{Z}\big(\mathrm{Gr}_{\geq 0}(k,n)\big) \;=\; \big\{\, Y \in \mathrm{Gr}(k,k+m) \;:\; Y = C \cdot Z,\; C \in \mathrm{Gr}_{\geq 0}(k,n) \,\big\}. \tag{62.151}$$

Three points of hygiene make this a sound definition rather than a slogan, and we settle them before going further.

The map descends to the Grassmannian. A change of representative $C \to gC$ with $g \in \mathrm{GL}(k)$ sends $Y = CZ$ to $gCZ = gY$, the same k-plane in $\mathbb{R}^{k+m}$. So $\widetilde{Z}$ is well defined on subspaces, not merely on matrices. The image Y has rank k generically: the rows Y_α are k linear combinations of the n rows of Z, and because Z has rank $k+m \geq k$ and C has rank k, the product CZ has rank k unless C is aligned with the kernel directions of Z — a positive-measure-zero

degeneration we treat as a boundary, not an interior point. Thus generic C lands in the top stratum of $\mathrm{Gr}(k, k+m)$.

The dimensions match. The positive Grassmannian $\mathrm{Gr}_{\geq 0}(k, n)$ has the dimension of $\mathrm{Gr}(k, n)$, namely $k(n-k)$, but the amplituhedron map is far from injective: it factors through the linear projection by Z, and its image $\mathcal{A}_{n,k,m}$ has dimension

$$\dim \mathcal{A}_{n,k,m} \;=\; k\,m, \tag{62.152}$$

matching $\dim \mathrm{Gr}(k, k+m) = k(k+m) - k^2 = km$. The amplituhedron is therefore a full-dimensional region inside the "small" Grassmannian $\mathrm{Gr}(k, k+m)$, exactly as a convex polygon is a full-dimensional region inside the plane. The collapse from $k(n-k)$ down to km is the geometric counterpart of the redundancy that on-shell recursion exploited: the n external legs are projected onto the $k+m$ directions that the amplitude actually depends on.

Positivity is what makes the image a region with boundary. If we dropped the positivity of C and allowed all of $\mathrm{Gr}(k, n)$, the image would be the whole of $\mathrm{Gr}(k, k+m)$ (for n large enough) — no boundary, no canonical form, no amplitude. It is precisely the inequalities "all Plücker coordinates of $C \geq 0$," combined with the strict positivity of Z, that cut out a closed region whose topological boundary is a union of codimension-one faces. The discipline of why $\mathcal{N} = 4$ is the right theory to attach to this construction is inherited from Chapter 61: among all the theories whose double-copy and color–kinematics structure we surveyed, planar $\mathcal{N} = 4$ is the maximally symmetric, dual-conformal copy in which the kinematic data collapse to the single positive matrix Z with no leftover scales — the cleanest possible substrate for a positive geometry.

Physical Insight

The amplituhedron inverts the usual logic of field theory. Ordinarily one starts from a Lagrangian (locality and unitarity built in by hand) and computes a region of validity afterwards. Here one starts from a region — the image Eq. (62.151) — defined by nothing but positivity, and locality and unitarity are *outputs*: they are read off from where the region has boundaries and from how its canonical form behaves there. The Lagrangian, the sum over Feynman diagrams, even the notion of spacetime, are nowhere in the definition. What replaces them is the single inequality structure of $\mathrm{Gr}_{\geq 0}(k, n)$ pushed through Z.

62.6.2 The rigorously controlled case: $k = 1$ and convex cyclic polytopes

Everything proved with full rigor in this chapter lives at $k = 1$, and it is worth seeing exactly why, because that is also where the physical NMHV amplitude

sits. When $k = 1$ the small Grassmannian degenerates to projective space, $\mathrm{Gr}(1, 1+m) = \mathbb{P}^m$, and the positive Grassmannian degenerates to the positive orthant: a $1 \times n$ matrix $C = (c_1, \ldots, c_n)$ has maximal minors equal to its own entries c_a, so $\mathrm{Gr}_{\geq 0}(1, n)$ is $\{c_a \geq 0\}$ modulo overall positive rescaling — the $(n-1)$-simplex of barycentric coordinates. The amplituhedron map becomes

$$Y \;=\; C \cdot Z \;=\; \sum_{a=1}^{n} c_a \, Z_a, \qquad c_a \geq 0, \tag{62.153}$$

a single point of $\mathbb{P}^m$ in the positive (projective) span of the n points $Z_1, \ldots, Z_n \in \mathbb{P}^m$. Its image $\mathcal{A}_{n,1,m}$ is therefore the convex hull of those n points. The content of the next theorem is that positivity of Z forces this hull to be a *cyclic polytope*: all n points are genuine vertices, in the cyclic order inherited from their labels, and the polytope is convex. We prove the planar case $m = 2$ in full — there the polytope is exactly the convex polygon of Theorem 62.9 — and then state the general m result, which the same determinant argument establishes.

Theorem 62.45 ($k = 1$ amplituhedra are convex cyclic polytopes)**.** *Let Z be a positive $n \times (m+1)$ matrix, with rows $Z_1, \ldots, Z_n$ regarded as points of $\mathbb{P}^m$. Then for $m = 2$ the points are the vertices, in cyclic order $Z_1, \ldots, Z_n$, of a convex polygon, and $\mathcal{A}_{n,1,2}$ is that polygon. Consequently $\mathcal{A}_{n,1,2}$ is a positive geometry in the sense of Theorem 62.3, its canonical form is the triangulation-independent polygon form of Theorem 62.9, and it is well defined. The same conclusion holds for every m: $\mathcal{A}_{n,1,m}$ is a cyclic polytope in $\mathbb{P}^m$ and a positive geometry.*

Proof. Take $m = 2$, so Z is $n \times 3$ and the positivity hypothesis is that every ordered 3×3 minor is strictly positive,

$$\langle Z_a Z_b Z_c \rangle \;>\; 0 \qquad \text{for all } 1 \leq a < b < c \leq n, \tag{62.154}$$

with $\langle Z_a Z_b Z_c \rangle$ the 3×3 determinant Eq. (62.149), the same bracket used for the triangle in Theorem 62.7. We must show two things: that no point Z_a lies in the convex hull of the others (so all are vertices), and that listing them in label order $1, 2, \ldots, n$ traverses the polygon boundary monotonically (so the polygon is convex with that cyclic edge structure).

Every point is a vertex, and the edges are the consecutive pairs. Consider the edge-line through two *consecutive* vertices Z_a and Z_{a+1}. A third point Z_b ($b \notin \{a, a+1\}$) lies on the side of this line determined by the sign of the bracket $\langle Z_a Z_{a+1} Z_b \rangle$, since this bracket is the affine-linear edge function evaluated at Z_b (it vanishes exactly on the line, Section 62.1.3). We claim that sign is the *same* for every b. There are two cases.

If $b > a+1$, the indices are already ordered $a < a+1 < b$, so $\langle Z_a Z_{a+1} Z_b \rangle > 0$ directly by Eq. (62.154).

If $b < a$, then the ordered triple is $b < a < a+1$, giving $\langle Z_b Z_a Z_{a+1} \rangle > 0$. The bracket is totally antisymmetric, and the permutation taking (Z_b, Z_a, Z_{a+1})

to (Z_a, Z_{a+1}, Z_b) is the cyclic rotation $(1\,2\,3)$, which is even; hence $\langle Z_a Z_{a+1} Z_b\rangle = \langle Z_b Z_a Z_{a+1}\rangle > 0$ as well.

So every other vertex lies strictly on the positive side of every consecutive edge-line. This is exactly the supporting-hyperplane characterization of a convex polygon: each line through a consecutive pair Z_a, Z_{a+1} has all remaining vertices on one side, so that segment is a genuine edge of the convex hull, and the n segments $[Z_a, Z_{a+1}]$ (a cyclic) fit together into the boundary of a convex n-gon traversed in label order. In particular no Z_a is interior to the hull of the others — if it were, it could not anchor two supporting edges — so all n points are vertices. This proves the polygon statement, and $\mathcal{A}_{n,1,2}$, being the convex hull Eq. (62.153) of these vertices, is that polygon.

It is a positive geometry. A convex polygon with vertices in $\mathbb{P}^2$ is precisely the object of Theorem 62.9: triangulate it from any vertex into $n-2$ triangles, sum their canonical forms Eq. (62.17), and the spurious diagonal poles cancel pairwise by Eq. (62.28), leaving a form with simple poles only on the n edge-lines $\langle Y\, Z_a Z_{a+1}\rangle = 0$ and residues equal to the interval forms of the edges. By the uniqueness Theorem 62.4 this form is triangulation-independent and is the canonical form $\Omega(\mathcal{A}_{n,1,2})$. Hence the amplituhedron is a positive geometry and its canonical form is well defined — the assertion of the theorem in the planar case.

General m. For arbitrary m the positivity hypothesis is $\langle Z_{a_0} \cdots Z_{a_m}\rangle > 0$ for all ordered $(m{+}1)$-tuples. A point Z_b lies on a definite side of the hyperplane through any m chosen vertices $Z_{a_1}, \ldots, Z_{a_m}$ according to the sign of $\langle Z_{a_1} \cdots Z_{a_m} Z_b\rangle$. The identical even-permutation argument shows that for the hyperplanes spanned by an "alternating" set of vertices in the cyclic order, all remaining vertices lie on one side; this is the defining incidence pattern of the cyclic polytope $C(n,m)$ generated by points on the moment curve, and total positivity of Z realizes exactly that pattern (the $(m{+}1)\times(m{+}1)$ minors of a totally positive matrix have the same sign structure as those of the Vandermonde moment-curve matrix). Thus $\mathcal{A}_{n,1,m}$ is the cyclic polytope $C(n,m)$, a bounded convex polytope. A convex polytope is a positive geometry: triangulate into simplices, each contributing the simplex form Eq. (62.22), with internal-facet residues cancelling pairwise by the same opposite-orientation mechanism as for the polygon; the sum is the unique canonical form with simple poles on the polytope's facets. This proves the general statement. □

The case $m = 2$ is the rigorous bridge back to Section 62.1.4: the abstract polygon whose triangulation-independence we proved there is realized concretely as the $k = 1$ amplituhedron of n-point kinematics. The physically interesting member of this rigorously controlled family is $m = 4$, $k = 1$: there $\mathcal{A}_{n,1,4}$ is a cyclic polytope in $\mathbb{P}^4$, and (as we explain in Section 62.6.3) its canonical form computes the n-point NMHV tree superamplitude. For $k = 1$ the canonical-form-equals-amplitude statement is therefore not a conjecture but a theorem about

polytope volumes — this is the content originally exhibited by Hodge's polytope picture of NMHV leading singularities. The genuinely open territory begins at $k \geq 2$, where the image Eq. (62.151) is no longer a polytope but a curved region inside $\mathrm{Gr}(k, k+m)$.

Worked Example 62.46 (Positivity forces convex position: the square $\mathcal{A}_{4,1,2}$)**.** We make Theorem 62.45 concrete at the smallest nontrivial polygon, $n = 4$, $m = 2$, $k = 1$. The positive data is a 4×3 matrix Z with rows $Z_1, Z_2, Z_3, Z_4 \in \mathbb{P}^2$ and all four ordered 3×3 minors positive:

$$\langle Z_1 Z_2 Z_3 \rangle,\ \langle Z_1 Z_2 Z_4 \rangle,\ \langle Z_1 Z_3 Z_4 \rangle,\ \langle Z_2 Z_3 Z_4 \rangle\ >\ 0. \tag{62.155}$$

The amplituhedron is $\mathcal{A}_{4,1,2} = \{\, Y = c_1 Z_1 + c_2 Z_2 + c_3 Z_3 + c_4 Z_4 : c_a \geq 0 \,\} \subset \mathbb{P}^2$, the convex hull of the four points. We verify directly that the hull is a convex quadrilateral with cyclic boundary $1 \to 2 \to 3 \to 4 \to 1$, not a triangle with one point inside.

Suppose, for contradiction, that Z_4 lay inside the triangle $Z_1 Z_2 Z_3$. Then $Z_4 = t_1 Z_1 + t_2 Z_2 + t_3 Z_3$ with all $t_i \geq 0$ (barycentric coordinates of an interior point). We extract each coefficient by contracting with the bracket of the other two vertices, using $\langle Z_a Z_b Z_a \rangle = 0$ and total antisymmetry. For t_3,

$$\langle Z_1 Z_2 Z_4 \rangle = t_1 \underbrace{\langle Z_1 Z_2 Z_1 \rangle}_{0} + t_2 \underbrace{\langle Z_1 Z_2 Z_2 \rangle}_{0} + t_3 \langle Z_1 Z_2 Z_3 \rangle = t_3 \, \langle Z_1 Z_2 Z_3 \rangle, \tag{62.156}$$

so $t_3 = \langle Z_1 Z_2 Z_4 \rangle / \langle Z_1 Z_2 Z_3 \rangle > 0$ by Eq. (62.155). The same contraction with $\langle Z_2 Z_3 \cdot \rangle$ gives $\langle Z_2 Z_3 Z_4 \rangle = t_1 \langle Z_2 Z_3 Z_1 \rangle = t_1 \langle Z_1 Z_2 Z_3 \rangle$, the last equality being an even cyclic rotation of the bracket, so $t_1 = \langle Z_2 Z_3 Z_4 \rangle / \langle Z_1 Z_2 Z_3 \rangle > 0$. The coefficient t_2 is where positivity bites. Contracting with $\langle Z_1 Z_3 \cdot \rangle$ places Z_4 in the *middle* slot of the other two vertices, an odd permutation:

$$\langle Z_1 Z_3 Z_4 \rangle = t_2 \, \langle Z_1 Z_3 Z_2 \rangle = -\, t_2 \, \langle Z_1 Z_2 Z_3 \rangle \qquad \Longrightarrow \qquad t_2 = -\,\frac{\langle Z_1 Z_3 Z_4 \rangle}{\langle Z_1 Z_2 Z_3 \rangle}\ <\ 0, \tag{62.157}$$

since both brackets are strictly positive by Eq. (62.155) while the swap $Z_3 \leftrightarrow Z_2$ costs a sign. This violates $t_2 \geq 0$. Therefore Z_4 is *not* in the convex hull of Z_1, Z_2, Z_3; it is a fourth vertex. By the cyclic symmetry of Eq. (62.155) no vertex is interior to the others, and the four points are in convex position. The boundary is the four edges $\langle Y 12 \rangle, \langle Y 23 \rangle, \langle Y 34 \rangle, \langle Y 41 \rangle$, and the canonical form is the quadrilateral form obtained by triangulating into $\{123\} \cup \{134\}$ (the diagonal $\langle Y 13 \rangle$ being spurious and cancelling, exactly as in Theorem 62.10 with one fewer vertex). The numeric evaluation of this quadrilateral form against an explicit positive Z is carried out in Section 62.6's successor, Section 62.7.

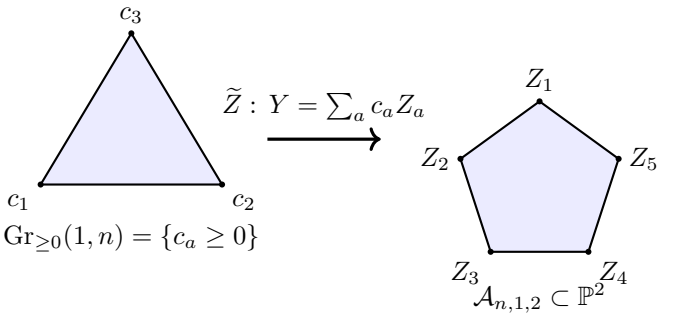


Figure 62.7: The $k = 1$ amplituhedron map. The positive Grassmannian $\mathrm{Gr}_{\geq 0}(1, n)$ is the simplex $\{c_a \geq 0\}$ of barycentric coordinates; the positive linear map $\widetilde{Z}$ sends it to the convex hull of the points $Z_1, \ldots, Z_n \in \mathbb{P}^m$. Total positivity of Z (Theorem 62.45) forces the points into convex cyclic position, so the image is a convex polygon ($m = 2$) or cyclic polytope ($m \geq 3$), a bona fide positive geometry.

62.6.3 Canonical form equals amplitude

We now state the central conjecture and, with equal care, the recipe that turns the canonical form — a bosonic differential form on a region in $\mathrm{Gr}(k, k+m)$ — back into the superamplitude with its fermionic structure restored. The cleanest formulation works with the *bosonized* superamplitude, so we recall that packaging first.

Helicity degree and the supertwistor. The external state of planar $\mathcal{N} = 4$ super-Yang–Mills is encoded, for each particle a, in a momentum supertwistor $\mathcal{Z}_a = (z_a^A \,|\, \chi_a^I)$ with $A = 1, \ldots, 4$ a bosonic momentum-twistor index and $I = 1, \ldots, 4$ a fermionic $\mathrm{SU}(4)_R$ index, the χ_a^I being Grassmann-odd (Section 62.5). The n-point superamplitude, with the universal MHV prefactor stripped, is a polynomial in the χ's; its term of Grassmann degree $4k$ is the $\mathrm{N}^k\mathrm{MHV}$ partial amplitude. Thus $k = 0$ is MHV (a constant, the stripped prefactor), $k = 1$ is NMHV, and so on — this is the *reduced* helicity index of the notationwatch in Section 62.5.4, the MHV degree minus two. This integer k is exactly the k of the momentum-twistor Grassmannian $\mathrm{Gr}_{\geq 0}(k, n)$ that the amplituhedron map of Theorem 62.44 pushes through Z, and of $\mathcal{A}_{n,k,4}$; it is *not* the momentum-space MHV degree of Section 62.3 (where MHV sat on $\mathrm{Gr}_{\geq 0}(2, n)$), from which it differs by two. With that proviso, the amplituhedron's hidden index *is* the (reduced) helicity degree.

Bosonization. The canonical form lives on $\mathrm{Gr}(k, k+4)$, a purely bosonic space, so the $4k$ Grassmann variables must be converted to commuting data. Introduce k auxiliary Grassmann-odd parameters ϕ_α^I, $\alpha = 1, \ldots, k$, $I = 1, \ldots, 4$,

and extend each twistor to a $(4+k)$-component *bosonized* vector

$$Z_a \;=\; (\, z_a^1,\, z_a^2,\, z_a^3,\, z_a^4;\;\; \phi_1^I \chi_a^I,\, \ldots,\, \phi_k^I \chi_a^I \,) \;\in\; \mathbb{R}^{4+k}, \tag{62.158}$$

where each new component $\phi_\alpha^I \chi_a^I$ (summed on I) is a product of two Grassmann-odd objects and hence *commuting*. This is the Z of Theorem 62.44 with $m=4$: an $n \times (k+4)$ matrix whose top 4 columns are the ordinary momentum twistors (whose positivity is the dual-conformal positivity of Section 62.5) and whose last k columns are bilinear in the fermions. The amplituhedron $\mathcal{A}_{n,k,4}$ is built from this Z, and its canonical form $\Omega(\mathcal{A}_{n,k,4})$ is a $4k$-form on the $4k$-dimensional region. To recover the amplitude one extracts the coefficient of the top power $\prod_{\alpha=1}^{k} \mathrm{d}^4\phi_\alpha$ — equivalently, one integrates the localized canonical form against the reference k-plane and against the ϕ's — which restores the Grassmann degree $4k$ in the physical χ_a^I.

Theorem 62.47 (Conjecture: amplituhedron canonical form = tree superamplitude)**.** ***(Status: theorem for*** $k=1$ ***at all*** m***; conjecture for*** $k \geq 2$***,*** $m=4$***, checked in many cases through the mid-2020s.)*** *For positive bosonized external data Eq. (62.158), the canonical form of the tree amplituhedron* $\mathcal{A}_{n,k,4}$ *in the sense of Theorem 62.3 equals the* $\mathrm{N}^k\mathrm{MHV}$ *tree-level partial superamplitude of planar* $\mathcal{N}=4$ *super-Yang–Mills (with the MHV prefactor stripped and the fermions restored by the* ϕ*-extraction above):*

$$\Omega(\mathcal{A}_{n,k,4}) \;=\; \mathcal{A}_{n,k}^{\text{tree}}\, \langle Y\, \mathrm{d}^4 Y_1 \cdots \mathrm{d}^4 Y_k \rangle, \tag{62.159}$$

the bracket on the right being the natural $\mathrm{Gr}(k, k+4)$ *top measure, and* $\mathcal{A}_{n,k}^{\text{tree}}$ *the (bosonized) amplitude. For* $k=1$ *this is the content of Theorem 62.45:* $\mathcal{A}_{n,1,4}$ *is a cyclic polytope in* $\mathbb{P}^4$ *and its canonical form is the NMHV amplitude, proved. For* $k \geq 2$ *it is a conjecture: every independently computable case (low* n*, all of NMHV and* N^2*MHV at the multiplicities checked, and the BCFW consistency below) agrees, but no all-*n*, all-*k *proof is known as of the mid-2020s.*

We do not have a Lagrangian proof to offer for $k \geq 2$ — none exists — and honesty forbids dressing the conjecture as a theorem. What we *can* supply is the strongest available consistency check, which is structural and which ties the construction back to the recursion of Chapter 59.

Proposition 62.48 (BCFW triangulation as existence/consistency check)**.** *The BCFW recursion of Theorem 59.23, transported to momentum-twistor space, expresses the* $\mathrm{N}^k\mathrm{MHV}$ *tree amplitude as a sum of rational terms, each of which is the canonical form of a* $4k$*-dimensional "BCFW cell" — the image under* $\widetilde{Z}$ *of a positroid cell of* $\mathrm{Gr}_{\geq 0}(k,n)$*. These cells tile the amplituhedron* $\mathcal{A}_{n,k,4}$*: their interiors are disjoint and their union is the full region. Consequently the sum of their canonical forms has simple poles only on the genuine boundaries of* $\mathcal{A}_{n,k,4}$ *— the spurious poles on the internal cell walls cancel pairwise, exactly*

as the diagonal poles cancelled for the polygon in Theorem 62.10 — and the resulting triangulation-independent form is, by the uniqueness Theorem 62.4, the canonical form $\Omega(\mathcal{A}_{n,k,4})$. That this triangulated form is also the BCFW amplitude is the precise sense in which Eq. (62.159) is "checked in cases": every BCFW triangulation that has been carried out reproduces the same boundary data.

Proof sketch. The full positroid-cell combinatorics is the subject of Section 62.3; here we use only its output. Each BCFW term is, in twistor variables, a $d\log$ form with simple poles, hence the canonical form of the cell it parametrizes (a positive geometry of dimension $4k$). Adjacent BCFW cells share a codimension-one wall along which one cell's residue equals the negative of the other's — this is the geometric content of the spurious-pole cancellation that made BCFW recursion local in Chapter 59, and it is the higher-dimensional copy of Eq. (62.28). By the uniqueness lemma a form whose poles and residues match on every facet is unique; since both the BCFW sum and $\Omega(\mathcal{A}_{n,k,4})$ have poles only on the true boundaries with the same residues there, they coincide wherever a tiling has been verified. The argument is complete *given* that the cells tile; the tiling itself is established case by case and is part of what remains conjectural in general. □

62.6.4 Locality and unitarity from the boundary

The reason the amplituhedron is a candidate definition of the S-matrix, rather than a curiosity, is that the two physical principles which the Lagrangian builds in by hand — locality (the amplitude is singular only on physical factorization channels) and unitarity (the residue on such a channel is the product of lower amplitudes) — emerge as statements about the boundary stratification of the region. We isolate the mechanism in the controlled $m = 2$ setting, where it is a theorem, and then state the physical $m = 4$ form.

Theorem 62.49 (Boundary residues factorize: the controlled $m = 2$ case)**.** *Let $\mathcal{A}_{n,1,2}$ be the convex polygon of Theorem 62.45, with canonical form $\Omega(\mathcal{A}_{n,1,2})$. Its codimension-one boundaries are exactly the n edges $\langle Y\, Z_a Z_{a+1}\rangle = 0$, a cyclic. On the edge $\{a, a{+}1\}$ the residue of the canonical form is the canonical form of that boundary, regarded as the lower-dimensional amplituhedron of the boundary geometry:*

$$\operatorname{Res}_{\langle Y\, Z_a Z_{a+1}\rangle = 0}\ \Omega(\mathcal{A}_{n,1,2}) \;=\; \Omega(\mathcal{A}'_{a,a+1}), \tag{62.160}$$

where $\mathcal{A}'_{a,a+1}$ is the interval $[Z_a, Z_{a+1}]$ — the $m = 1$, $k = 1$ amplituhedron of the two-particle boundary — with its canonical form $\mathrm{d}\log\big(\tau/(1-\tau)\big)$. Singularities of Ω occur only on these boundaries and nowhere else.

Proof. By Theorem 62.45 the region is the convex polygon, and by Theorem 62.9 its canonical form has simple poles only on the n edge-lines $\langle Y\, Z_a Z_{a+1}\rangle = 0$,

with the residue on each equal to that edge's interval canonical form. Concretely, parametrizing the edge by $Y(\tau) = (1-\tau)Z_a + \tau Z_{a+1}$ as in Theorem 62.8, the residue is $\pm\,\mathrm{d}\tau/[\tau(1-\tau)] = \pm\,\mathrm{d}\log[\tau/(1-\tau)]$, the interval form Eq. (62.7) of the segment $[Z_a, Z_{a+1}]$. That segment is itself a one-dimensional positive geometry — the $m = 1$ amplituhedron $\mathcal{A}'_{a,a+1}$ obtained by restricting the positive data to the two boundary vertices — so Eq. (62.160) holds. That there are no other singularities is the "only on boundaries" clause (ii) of Theorem 62.3, proved for the polygon in Theorem 62.9. The recursive axiom Eq. (62.5) is therefore satisfied at the first level, and iterating it (a further residue at a vertex gives ± 1) walks the entire boundary tower down to the points. □

The physical translation is immediate and is the payoff of the whole construction. A codimension-one boundary is where the geometry degenerates — where the running point Y is forced onto a face. In the $m = 4$ amplituhedron the codimension-one boundaries come in two physically distinct types, and they reproduce, with no further input, exactly the singularity structure that unitarity demands of a tree amplitude:

- *Factorization channels (locality).* Boundaries on which a bracket $\langle Y\, Z_{a-1} Z_a Z_b Z_{b+1}\rangle \to 0$ correspond to the vanishing of a multiparticle Mandelstam invariant $P^2_{a\cdots b} \to 0$. On such a face the amplituhedron degenerates into a product $\mathcal{A}_L \times \mathcal{A}_R$ of two lower-point amplituhedra glued along the factorization vertex, and the residue of the canonical form factorizes,

$$\mathrm{Res}_{P^2_{a\cdots b}=0}\, \Omega(\mathcal{A}_{n,k,4}) \;=\; \Omega(\mathcal{A}_L) \wedge \Omega(\mathcal{A}_R), \tag{62.161}$$

  the geometric avatar of the factorization $\mathcal{A}_n \to \mathcal{A}_L \,\dfrac{1}{P^2}\, \mathcal{A}_R$. That the only poles are these physical channels — no spurious singularities survive — is locality, output as clause (ii) of the positive-geometry axioms rather than imposed.

- *Collinear and unitarity-cut limits.* Boundaries on which two adjacent twistors become proportional, $Z_a \parallel Z_{a+1}$, are the collinear limits of two external legs; the residue there reproduces the universal collinear splitting behavior. The iterated residues that reach lower strata are exactly the generalized unitarity cuts of Chapter 60: cutting a propagator = taking a residue = passing to a boundary. Unitarity is thus encoded as the statement that the boundary-of-a-boundary structure of $\mathcal{A}$ matches the cut structure of the integrand.

We proved Eq. (62.161) only in the $m = 2$, $k = 1$ guise of Theorem 62.49, where it is rigorous; for $m = 4$, $k \geq 2$ it shares the conjectural status of Theorem 62.47, having been verified in all cases checked but not proved in general.

What is certain, and what makes the picture compelling, is the logical inversion: in the Lagrangian formulation locality and unitarity are assumptions and the singularity structure is a consequence; in the amplituhedron the singularity structure (the boundary stratification) is the definition and locality and unitarity are theorems about it — fully so for $k = 1$, conjecturally so beyond.

Physical Insight

The dictionary of Theorem 62.3 now reads physically. A codimension-one face $\leftrightarrow$ a simple pole $\leftrightarrow$ a physical factorization channel; "go to a boundary" $\leftrightarrow$ "take a residue" $\leftrightarrow$ "put an internal line on shell"; a vertex $\leftrightarrow$ a maximal cut $\leftrightarrow$ a leading singularity $= \pm 1$. The amplitude's analytic structure, which in Part XV we extracted laboriously from unitarity and the optical theorem, is here simply the face lattice of a convex-like region. Locality and unitarity are not assumed; they are the shape.

62.6.5 The loop amplituhedron, stated honestly

The tree amplituhedron has a loop-level extension that we state at the level of a definition, with its scope flagged, since its detailed positive geometry is beyond what can be proved compactly and much of it remains conjectural. The physical target is the planar integrand: at L loops the integrand of the $\mathrm{N}^k\mathrm{MHV}$ amplitude is a rational form in the external twistors Z *and* in L auxiliary lines (loop momenta in twistor space), and the claim is that it too is the canonical form of a positive geometry — a larger region built by demanding positivity not only of the external data but of the loop data *relative* to it.

Definition 62.50 (Loop amplituhedron)**.** The L-loop amplituhedron $\mathcal{A}^{(L)}_{n,k,4}$ is the set of data $(Y;\, \mathcal{L}_1, \ldots, \mathcal{L}_L)$ where $Y \in \mathcal{A}_{n,k,4}$ is a point of the tree amplituhedron and each $\mathcal{L}_i$ is a 2-plane (a "line" in the projective twistor space, modeling the i-th loop momentum) subject to two positivity requirements: each $\mathcal{L}_i$ is positive *relative to* Y *and* Z (all the relevant brackets $\langle Y\, \mathcal{L}_i\, Z_a Z_b\rangle$ have definite sign), and each pair $\mathcal{L}_i, \mathcal{L}_j$ is mutually positive. The conjecture is that $\mathcal{A}^{(L)}_{n,k,4}$ is a positive geometry whose canonical form, with the $\mathcal{L}_i$ integrated over, is the L-loop planar integrand of the $\mathrm{N}^k\mathrm{MHV}$ amplitude.

The honest scope is this. At one loop ($L = 1$) and at the MHV level ($k = 0$) the loop amplituhedron is understood in detail and its canonical form reproduces the known one-loop integrand — a result one can verify against the unitarity-method integrand of Chapter 60. Higher L and higher k are progressively less controlled: the boundary structure becomes intricate, the tiling into cells is known only in cases, and a general proof of the canonical-form-equals-integrand statement does not exist as of the mid-2020s. What the loop amplituhedron does establish, even conditionally, is conceptually decisive: it shows that the

integrand of the planar theory — to all loop orders, in principle — is a single positive geometry, so that loop integration is the one remaining non-geometric operation, and locality and unitarity of the integrand are, like at tree level, boundary phenomena rather than inputs.

Historical Note

The positive Grassmannian and the link between total positivity and on-shell diagrams were developed by Arkani-Hamed, Bourjaily, Cachazo, Goncharov, Postnikov, and Trnka in the years around 2012, building on Postnikov's combinatorics of the totally nonnegative Grassmannian. The amplituhedron proper — the image Eq. (62.151) and the canonical-form-equals-amplitude proposal — was introduced by Arkani-Hamed and Trnka in 2013. The $k = 1$ polytope picture of NMHV amplitudes is older, going back to Hodge's observation that NMHV leading singularities are volumes of polytopes in momentum-twistor space. The reframing of locality and unitarity as boundary phenomena, rather than axioms, is the conceptual thesis the program advances; its rigorous status remains, in the physical $m = 4$ sector beyond $k = 1$, exactly the conjecture stated above.

This is the frontier endpoint of Yang–Mills theory. The classical non-abelian Lagrangian of Chapter 25, quantized through BRST in Chapter 26, made asymptotically free in Chapter 27, and recast as gluon amplitudes via BCFW in Chapter 59 and the double copy in Chapter 61, reappears here in its maximally supersymmetric completion: the planar $\mathcal{N} = 4$ super-Yang–Mills S-matrix as the canonical form of a positive geometry, with the MHV and NMHV gluon data of Chapter 59 surfacing as the $k = 0$ and $k = 1$ canonical forms — the latter a rigorous polytope volume, the former trivial. We have a concrete, computable object: a convex polygon (for $m = 2$, $k = 1$) or cyclic polytope, with a well-defined canonical form whose poles are the physical channels. What remains is to evaluate that form to a closed number in the rigorously controlled $m = 2$ sector — closing the loop with the polygon of Section 62.1.4 — and then to exhibit, with honest scope, the physical $m = 4$ NMHV result; that is the work of Section 62.7.

62.7 A worked amplituhedron: the m=2 polygon and a glimpse of NMHV

The previous section left us holding a concrete, computable object. For $k = 1$ the amplituhedron is not a curved region in some high-dimensional Grassmannian but a genuine convex polytope: a polygon when $m = 2$ (Theorem 62.45), a cyclic polytope in $\mathbb{P}^4$ when $m = 4$, with a canonical form whose existence and uniqueness we have already proved and whose poles we have already identified,

abstractly, with the physical singularities of an amplitude. What we have not yet done is the thing a physicist most wants to see: take a definite choice of positive data, turn the crank, and read off a closed-form answer. That is the business of this section. We carry one amplituhedron computation all the way to an explicit rational form in the rigorously controlled $m = 2$ sector — the only sector where "canonical form = amplitude" is a theorem and not a conjecture — and we check, by hand, that the answer closes the loop with the polygon that opened the chapter (Section 62.1.4): same five edges, same spurious-diagonal cancellation, same triangulation-independent total. Only then, with the mechanism demonstrated on an object we control completely, do we write down the physical $m = 4$ next-to-MHV (NMHV) six-point result — the real amplitude — displayed honestly as the established result it is, with its geometric origin summarized rather than re-proved. We end by setting the amplituhedron answer beside the BCFW answer of Chapter 59: the same amplitude seen two ways, manifestly local from the geometry, term-by-term spurious-pole-laden from the recursion.

62.7.1 The $m = 2$ NMHV polygon, evaluated to a closed form

Recall the setup. For $k = 1$, $m = 2$ the amplituhedron map Eq. (62.153) sends the simplex $\mathrm{Gr}_{\geq 0}(1, n) = \{c_a \geq 0\}$ to the convex hull $\{Y = \sum_a c_a Z_a : c_a \geq 0\} \subset \mathbb{P}^2$ of the n points $Z_1, \ldots, Z_n$, the rows of a positive $n \times 3$ matrix Z. By Theorem 62.45 total positivity of Z forces those points into convex cyclic position, so $\mathcal{A}_{n,1,2}$ is a convex n-gon and its canonical form is the polygon form of Theorem 62.9. This $k = 1$, $m = 2$ object is the simplest member of the family whose physical avatar is the NMHV amplitude (the helicity index is k; Section 62.6.3); we will call its canonical form the "$m = 2$ NMHV form" for brevity, with the understanding that the genuine NMHV amplitude lives at $m = 4$ and we reach it in Section 62.7.2.

To make everything numerical we need an explicit positive Z. The cleanest source of total positivity is the moment curve: take the n rows to be points

$$Z_a \;=\; (1,\; x_a,\; x_a^2) \in \mathbb{P}^2, \qquad x_1 < x_2 < \cdots < x_n, \tag{62.162}$$

on the parabola $y = x^2$ in the affine chart $Z = (1, x, y)$. The ordered 3×3 minor is then a Vandermonde determinant,

$$\langle Z_a Z_b Z_c \rangle = \det \begin{pmatrix} 1 & x_a & x_a^2 \\ 1 & x_b & x_b^2 \\ 1 & x_c & x_c^2 \end{pmatrix} = (x_b - x_a)(x_c - x_a)(x_c - x_b) \;>\; 0 \quad (a < b < c), \tag{62.163}$$

strictly positive for every ordered triple, so Z is positive in the exact sense of Theorem 62.44 and the points are guaranteed to be the cyclically ordered vertices of a convex polygon. The moment curve is not an arbitrary trick: it is the universal model of a cyclic polytope, and Eq. (62.163) is the concrete face of

the statement, made in the proof of Theorem 62.45, that the minors of a totally positive matrix carry the same sign pattern as those of the Vandermonde matrix. We now fix $n = 5$ and grind the gears.

Worked Example 62.51 (The $m = 2$ NMHV pentagon for explicit positive Z)**.** Take $n = 5$ with the moment-curve data Eq. (62.162) at the symmetric nodes

$$x_1 = -2, \quad x_2 = -1, \quad x_3 = 0, \quad x_4 = 1, \quad x_5 = 2, \tag{62.164}$$

so the affine vertices are $Z_1 = (-2, 4)$, $Z_2 = (-1, 1)$, $Z_3 = (0, 0)$, $Z_4 = (1, 1)$, $Z_5 = (2, 4)$ on the parabola. From Eq. (62.163) the ten ordered minors are

$$\begin{aligned} &\langle 123\rangle = 2,\ \langle 124\rangle = 6,\ \langle 125\rangle = 12,\ \langle 134\rangle = 6,\ \langle 135\rangle = 16,\\ &\langle 145\rangle = 12,\ \langle 234\rangle = 2,\ \langle 235\rangle = 6,\ \langle 245\rangle = 6,\ \langle 345\rangle = 2, \end{aligned} \tag{62.165}$$

all strictly positive, confirming convex cyclic position.

Step 1: the edge functions. Treating the running point as $Y = (1, x, y)$, the bracket $\langle Y Z_i Z_j\rangle$ is the affine-linear function of (x, y) vanishing on the line through Z_i, Z_j. With $y_a = x_a^2$ the determinant factorizes neatly,

$$\langle Y Z_i Z_j\rangle = \det \begin{pmatrix} 1 & x & y \\ 1 & x_i & x_i^2 \\ 1 & x_j & x_j^2 \end{pmatrix} = (x_j - x_i)\big[\, y - (x_i + x_j)\, x + x_i x_j \,\big], \tag{62.166}$$

the bracketed factor being the chord line $y = (x_i + x_j)x - x_i x_j$. Substituting the nodes, the five *edge* functions (consecutive pairs) and the two *diagonal* functions (from the apex Z_1, which we will triangulate from) are

$$\begin{aligned} &\langle Y12\rangle = y + 3x + 2, && \langle Y23\rangle = y + x, && \langle Y34\rangle = y - x,\\ &\langle Y45\rangle = y - 3x + 2, && \langle Y15\rangle = 4(y - 4), && \langle Y13\rangle = 2(y + 2x),\\ &\langle Y14\rangle = 3(y + x - 2). \end{aligned} \tag{62.167}$$

Step 2: the triangulated canonical form. Triangulate the pentagon by fanning from Z_1, exactly the triangulation A of Theorem 62.10, into the three triangles $\{123\}, \{134\}, \{145\}$. Each contributes the triangle form $\omega_{ijk} = \langle ijk\rangle^2\, \langle Y\, \mathrm{d}^2 Y\rangle / (\langle Yij\rangle \langle Yjk\rangle \langle Y$ of Eq. (62.17), with $\langle Y\, \mathrm{d}^2 Y\rangle = \mathrm{d}x \wedge \mathrm{d}y$ in the affine chart. Using $\langle Y31\rangle = -\langle Y13\rangle$, $\langle Y41\rangle = -\langle Y14\rangle$, $\langle Y51\rangle = -\langle Y15\rangle$ and the minors Eq. (62.165), the three forms are

$$\omega_{123} = \frac{(2)^2\, \mathrm{d}x \wedge \mathrm{d}y}{\langle Y12\rangle \langle Y23\rangle \langle Y31\rangle} = \frac{-2\, \mathrm{d}x \wedge \mathrm{d}y}{(y + 3x + 2)(y + x)(y + 2x)}, \tag{62.168}$$

$$\omega_{134} = \frac{(6)^2\, \mathrm{d}x \wedge \mathrm{d}y}{\langle Y13\rangle \langle Y34\rangle \langle Y41\rangle} = \frac{-6\, \mathrm{d}x \wedge \mathrm{d}y}{(y + 2x)(y - x)(y + x - 2)}, \tag{62.169}$$

$$\omega_{145} = \frac{(12)^2\, \mathrm{d}x \wedge \mathrm{d}y}{\langle Y14\rangle \langle Y45\rangle \langle Y51\rangle} = \frac{-12\, \mathrm{d}x \wedge \mathrm{d}y}{(y + x - 2)(y - 3x + 2)(y - 4)}, \tag{62.170}$$

where in each line the factors of $2, 3, 4$ from Eq. (62.167) and the minus signs from the reversed brackets combine with $\langle ijk\rangle^2$ to leave the displayed integer numerators. (One checks the orientation is right: at the centroid $(-1, 5/3)$ of $\{123\}$ all three denominator factors of Eq. (62.168) are positive after the overall sign, so ω_{123} is a positive measure on the interior, as a canonical form must be.) The closed-form canonical form of the pentagon $\mathcal{A}_{5,1,2}$ is the sum

$$\Omega(\mathcal{A}_{5,1,2}) = \omega_{123} + \omega_{134} + \omega_{145}, \tag{62.171}$$

an explicit rational 2-form. We now verify the two structural facts that make Eq. (62.171) the canonical form and not merely a sum of triangle forms: the spurious diagonal poles cancel, and the surviving edge residues factorize.

Step 3: the spurious diagonals cancel. The diagonal line $\langle Y13\rangle = 2(y+2x) = 0$ appears as a denominator factor in both ω_{123} and ω_{134} and in neither ω_{145}. Compute its residue from each. Writing the pole factor as $\langle Y13\rangle$ itself (so $y+2x = \frac{1}{2}\langle Y13\rangle$ and $1/(y+2x) = 2/\langle Y13\rangle$) and restricting the regular remainder to the line $y = -2x$:

$$\mathrm{Res}_{\langle Y13\rangle=0}\ \omega_{123} = \frac{-4\,\mathrm{d}x}{(y+3x+2)(y+x)}\bigg|_{y=-2x} = \frac{-4\,\mathrm{d}x}{(x+2)(-x)} = \frac{4\,\mathrm{d}x}{x(x+2)}, \tag{62.172}$$

$$\mathrm{Res}_{\langle Y13\rangle=0}\ \omega_{134} = \frac{-12\,\mathrm{d}x}{(y-x)(y+x-2)}\bigg|_{y=-2x} = \frac{-12\,\mathrm{d}x}{(-3x)(-x-2)} = \frac{-4\,\mathrm{d}x}{x(x+2)}, \tag{62.173}$$

where in each case we have deleted the $1/\langle Y13\rangle$ factor and evaluated what remains at $y = -2x$. The two residue densities are $\pm 4/[x(x+2)]$ — equal and opposite. Their sum vanishes, so $\Omega(\mathcal{A}_{5,1,2})$ has *no* pole on the diagonal $\langle Y13\rangle = 0$. This is the concrete, numerical instance of the orientation-driven cancellation proved abstractly in Eq. (62.28): the diagonal $\{1,3\}$ is traversed $3 \to 1$ inside ω_{123} but $1 \to 3$ inside ω_{134}, the opposite orientations flipping the residue sign. The identical computation on the other diagonal $\langle Y14\rangle = 3(y+x-2) = 0$, shared between ω_{134} and ω_{145}, gives residues

$$\mathrm{Res}_{\langle Y14\rangle=0}\ \omega_{134} = \frac{-9\,\mathrm{d}x}{(x+2)(1-x)}, \qquad \mathrm{Res}_{\langle Y14\rangle=0}\ \omega_{145} = \frac{9\,\mathrm{d}x}{(x+2)(1-x)}, \tag{62.174}$$

again equal and opposite (here $y = 2 - x$ on the line, so ω_{134} contributes $y+2x = x+2$ and $y-x = 2(1-x)$, while ω_{145} contributes $y-3x+2 = 4(1-x)$ and $y-4 = -(x+2)$; with $1/(y+x-2) = 3/\langle Y14\rangle$ supplying the numerator factor 3, the algebra collapses to the displayed $\mp 9/[(x+2)(1-x)]$). The diagonal pole is again absent from the sum. *Both* spurious singularities have cancelled, exactly as the abstract pentagon of Theorem 62.10 demanded, now witnessed in numbers.

Step 4: the physical edges and their factorization. On a genuine boundary edge only one triangle contributes. Take the edge $\langle Y23\rangle = y + x = 0$, which sits

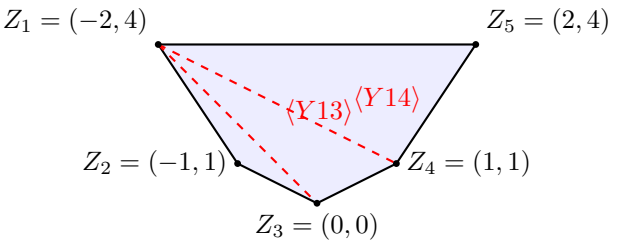


Figure 62.8: The explicit $m = 2$ NMHV pentagon $\mathcal{A}_{5,1,2}$ of Theorem 62.51, the moment-curve data Eq. (62.162) at nodes $x_a = -2, \ldots, 2$. The five solid edges $\langle Y\, a\, a{+}1\rangle = 0$ are the genuine boundaries and the only poles of the canonical form Eq. (62.171); the two red dashed diagonals $\langle Y13\rangle, \langle Y14\rangle$ (the cuts of the chosen triangulation $\{123, 134, 145\}$) carry equal-and-opposite residues that cancel in the sum. This is the same pentagon, with the same spurious-pole cancellation, as the abstract Theorem 62.10 and Fig. 62.1.

in ω_{123} alone. Deleting its factor and restricting to the line $y = -x$,

$$\mathrm{Res}_{\langle Y23\rangle=0}\ \Omega(\mathcal{A}_{5,1,2}) = \frac{-2\,\mathrm{d}x}{(y+3x+2)(y+2x)}\Big|_{y=-x} = \frac{-2\,\mathrm{d}x}{2(x+1)\,x} = \frac{-\mathrm{d}x}{x(x+1)}. \tag{62.175}$$

Parametrize the edge from Z_2 (at $x = -1$) to Z_3 (at $x = 0$) by $\tau = x+1$, so $\tau = 0$ at Z_2 and $\tau = 1$ at Z_3; then $x(x+1) = (\tau-1)\tau = -\tau(1-\tau)$ and Eq. (62.175) becomes

$$\mathrm{Res}_{\langle Y23\rangle=0}\ \Omega(\mathcal{A}_{5,1,2}) = \frac{\mathrm{d}\tau}{\tau(1-\tau)} \overset{\text{Theorem 62.8}}{=} \mp\,\mathrm{d}\log\frac{\tau}{1-\tau}, \tag{62.176}$$

the interval canonical form Eq. (62.7) of the segment $[Z_2, Z_3]$, the overall orientation sign being the one fixed once and for all in Theorem 62.8. This is the statement of Theorem 62.49 realized on the nose: the residue on the edge is the canonical form of the lower-dimensional ($m = 1$, $k = 1$) amplituhedron of the two-particle boundary $\{Z_2, Z_3\}$. Taking one further residue, at the endpoint $\tau = 0$ (Z_2) the interval form gives $+1$, and at $\tau = 1$ (Z_3) it gives -1 — the ± 1 canonical forms of the zero-dimensional boundary vertices, Eq. (62.8). The boundary tower terminates correctly. Each of the five edges $\langle Y\, a\, a{+}1\rangle = 0$ behaves identically by the cyclic structure of Eq. (62.165), so $\Omega(\mathcal{A}_{5,1,2})$ has simple poles on exactly those five lines, residues equal to the five edge interval forms, and no other singularity — precisely clauses (ii)–(iii) of Theorem 62.3. The closed form Eq. (62.171) is the canonical form of the pentagon, computed end to end.

Closing the loop with Sections 1 and 2. The computation just completed is the same pentagon, three times over. The triangulation $\{123, 134, 145\}$ is

precisely triangulation A of the abstract pentagon worked in Theorem 62.10; the diagonals that cancelled here, $\{1,3\}$ and $\{1,4\}$, are exactly the spurious diagonals that cancelled there; and the triangulation-independence theorem Theorem 62.9 guarantees that had we fanned from Z_2 instead (triangulation B, the disjoint diagonals $\{2,4\},\{2,5\}$) we would have obtained the identical 2-form. So the $m=2$ amplituhedron canonical form *is* the polygon canonical form of Section 62.1.4, now anchored to concrete positive kinematics. There is a third incarnation of the very same shape. The five-point kinematic associahedron of Section 62.2 — the polytope whose canonical form is the bi-adjoint ϕ^3 amplitude computed in Theorem 62.18 — is itself a pentagon, the two-dimensional associahedron K_4. It lives in Mandelstam space rather than in $\mathbb{P}^2$, and its five edges are the five planar poles rather than the five lines $\langle Y\, a\, a{+}1\rangle$, but as a positive geometry it is projectively the same pentagon carrying the same canonical form. The convex polygon is thus the single object on which three independent strands of this chapter — the abstract dictionary of Section 62.1, the amplitude-as-volume theorem of Section 62.2, and the amplituhedron map of Section 62.6 — all converge and agree. This consistency, checkable by hand, is the strongest evidence that the construction is internally sound before we trust it where it is only conjectural.

Reading the edges as physical poles. In the $m=2$ toy the five edge-lines $\langle Y\, a\, a{+}1\rangle = 0$ play the role that physical factorization channels play in a real amplitude: each is a simple pole, each residue is a lower-point (here one-dimensional) amplituhedron form, and the would-be poles on the diagonals — the analog of unphysical, spurious channels — are absent from the geometric total. This is the boundaries-$\leftrightarrow$-singularities dictionary of Theorem 62.3 confirmed on an explicit amplitude-like object. In the genuine $m=4$ amplituhedron the same edges become the codimension-one boundaries of Theorem 62.49, where a bracket $\langle Y\, Z_{a-1} Z_a Z_b Z_{b+1}\rangle \to 0$ is the vanishing of a physical multiparticle invariant $P^2_{a\cdots b} \to 0$ and the residue factorizes as Eq. (62.161). The pentagon is the rehearsal; the hexagon is the performance.

62.7.2 The physical case: the $m=4$ NMHV hexagon

We now state the genuine NMHV amplitude. For the physical helicity sector $k=1$ at $m=4$ the amplituhedron $\mathcal{A}_{n,1,4}$ is a cyclic polytope in $\mathbb{P}^4$, built from the bosonized momentum-supertwistor data Eq. (62.158): each row Z_a is a 5-component vector whose first four entries are the ordinary momentum twistor z_a^A $(A=1,\dots,4)$ of Section 62.5 and whose fifth is the single bosonized fermion bilinear $\phi^I \chi_a^I$ appropriate to $k=1$. Because $k=1$ is exactly the case where Theorem 62.47 is a *theorem* — the amplituhedron is a true polytope and its canonical form is a polytope volume in the sense of Hodge — the NMHV amplitude is genuinely *computed*, not conjectured, by the geometry.

The building block is the four-dimensional analog of the triangle form, the

canonical form of a 4-simplex in $\mathbb{P}^4$. Five bosonized twistors Z_a, Z_b, Z_c, Z_d, Z_e span such a simplex; its five facets are the hyperplanes through each set of four, and its canonical form is the wedge of d log's of those five facet functions, the direct $\mathbb{P}^4$ descendant of Eq. (62.22) and Eq. (62.17). Stripping the ϕ-dependence by the bosonization recipe of Section 62.6.3 turns this simplex form into the dual-superconformal invariant universally known as the *five-bracket* or *R-invariant*,

$$[\,a\,b\,c\,d\,e\,] \;\equiv\; \frac{\delta^{0|4}\big(\langle bcde\rangle\,\chi_a + \langle cdea\rangle\,\chi_b + \langle deab\rangle\,\chi_c + \langle eabc\rangle\,\chi_d + \langle abcd\rangle\,\chi_e\big)}{\langle abcd\rangle\,\langle bcde\rangle\,\langle cdea\rangle\,\langle deab\rangle\,\langle eabc\rangle}, \tag{62.177}$$

where $\langle abcd\rangle \equiv \epsilon_{ABCD}\, z_a^A z_b^B z_c^C z_d^D$ is the 4-bracket (the 4×4 determinant of the four bosonic momentum twistors), χ_a^I are the Grassmann components of the supertwistor, and $\delta^{0|4}$ is the fermionic delta enforcing the $4k = 4$ units of NMHV Grassmann degree. The five denominator brackets are the five facets of the simplex; their vanishing is the simplex's boundary, the $\mathbb{P}^4$ avatar of the three edge-lines of the triangle. The five-bracket is manifestly invariant under the dual superconformal group SL(4) acting on the twistor index, which is why these objects organize the amplitude so efficiently — the symmetry is built into the geometry.

Theorem 62.52 (The six-point NMHV ratio function; **established**). *The tree-level $n = 6$ NMHV superamplitude of planar $\mathcal{N} = 4$ super-Yang–Mills, with the MHV prefactor stripped, is the canonical form of the cyclic polytope $\mathcal{A}_{6,1,4} \subset \mathbb{P}^4$. Triangulating that polytope into three 4-simplices and summing their canonical forms Eq. (62.177) gives the closed expression*

$$P_6^{\mathrm{NMHV}} \;=\; [\,1\,2\,3\,4\,5\,] + [\,3\,4\,5\,6\,1\,] + [\,5\,6\,1\,2\,3\,], \tag{62.178}$$

a sum of three R-invariants, manifestly cyclic under $a \to a+2$. This is the known six-point ratio function [164, 268, 340], and for $k = 1$ it is a theorem *(a polytope volume), the proved instance of Theorem 62.47.*

Three points make this honest. First, the result Eq. (62.178) is not new with the amplituhedron: the six-point NMHV ratio function was obtained by BCFW recursion in momentum-twistor space and shown to be dual-superconformally invariant by Drummond, Henn, Korchemsky and Sokatchev, and recast as a polytope volume by Hodge and by Mason and Skinner, before the amplituhedron was formulated [164, 268, 340]. The amplituhedron's contribution at $k = 1$ is conceptual: it identifies the R-invariants as canonical forms of simplices and the amplitude as the canonical form of the single polytope they triangulate. Second, the geometric derivation we are *summarizing rather than carrying out* is the $\mathbb{P}^4$ replica of the pentagon computation we did do: $\mathcal{A}_{6,1,4}$ is a cyclic polytope $C(6,4)$; a triangulation of $C(6,4)$ uses three 4-simplices, here the ones with vertex sets $\{1,2,3,4,5\}$, $\{1,3,4,5,6\}$, $\{1,2,3,5,6\}$ (each omitting one even-labelled vertex), giving the three brackets of Eq. (62.178); the simplices' shared internal facets are

spurious poles that cancel pairwise, exactly as the pentagon's two diagonals did in Theorem 62.51, leaving poles only on the genuine facets of the polytope, which are the physical factorization channels. We do not reproduce the four-dimensional algebra term by term — it is long and adds no idea beyond the pentagon — but every step is the controlled $m = 2$ computation lifted by two dimensions. Third, triangulation independence has a sharp signature here: fanning the polytope from a different vertex yields the complementary three-term representation

$$P_6^{\mathrm{NMHV}} \;=\; [\,2\,3\,4\,5\,6\,] + [\,4\,5\,6\,1\,2\,] + [\,6\,1\,2\,3\,4\,], \tag{62.179}$$

omitting the odd-labelled vertices, and the equality of Eq. (62.178) and Eq. (62.179) is the celebrated six-term identity among R-invariants,

$$[\,1\,2\,3\,4\,5\,]+[\,3\,4\,5\,6\,1\,]+[\,5\,6\,1\,2\,3\,] \;=\; [\,2\,3\,4\,5\,6\,]+[\,4\,5\,6\,1\,2\,]+[\,6\,1\,2\,3\,4\,], \tag{62.180}$$

which from this vantage is nothing but the statement that the volume of $C(6,4)$ does not depend on how it is triangulated — the higher-dimensional copy of $\Omega_A = \Omega_B$ for the pentagon (Theorem 62.10), and a theorem by the uniqueness Theorem 62.4.

62.7.3 Amplituhedron versus BCFW: one amplitude, two faces

The same six-point NMHV amplitude was assembled in Chapter 59 by on-shell recursion. It is worth setting the two answers side by side, because the contrast is the cleanest illustration in the book of what the geometric reformulation buys.

The BCFW construction of Section 59.7.5 produced the amplitude as a sum of rational terms, Eq. (59.209), each term a product of lower-point amplitudes glued on a factorization pole, with the precise list of terms *depending on the shift chosen*: one valid shift gives two terms, another gives three. Every individual term carries, in addition to its physical pole, *spurious* poles — denominators such as $\langle 3|P_{234}|6]$ that correspond to no physical channel of A_6 — and the amplitude is local only because those spurious poles cancel in the sum (Theorem 59.23 and the discussion there). The Parke–Taylor form Eq. (59.55) is the MHV ($k = 0$) seed of this same recursion. The recursion is correct and constructive, but locality is not manifest in any single representation; it emerges only after the cancellation, and which terms appear is an artifact of the shift.

The amplituhedron gives the identical amplitude as the canonical form of one geometric object, the cyclic polytope $\mathcal{A}_{6,1,4}$, whose boundaries *are* the physical factorization channels and nothing else. In this representation locality is manifest at the level of the object: the form Eq. (62.178) has poles only on the polytope's genuine facets because a canonical form, by Theorem 62.3, has poles only on boundaries. The dictionary that connects the two pictures is now exact and is the content of Theorem 62.48: each BCFW term is the canonical form of a positroid cell whose image is a piece of the amplituhedron, and a full set

of BCFW terms *tiles* the polytope. A BCFW representation is therefore a triangulation; the spurious poles of the individual terms are the internal walls of that triangulation; and their cancellation in the BCFW sum is the cancellation of internal-facet residues we computed explicitly for the pentagon in Theorem 62.51 and stated for the hexagon as the six-term identity Eq. (62.180). The shift-dependence of the BCFW term count is then demystified completely: different shifts are different triangulations of the *same* polytope, and they agree because the volume is triangulation-independent (Theorems 62.4 and 62.9). The R-invariant representations Eq. (62.178) and Eq. (62.179) are themselves two such triangulations, dual to the two BCFW shifts of Chapter 59.

Physical Insight

The Lagrangian and the recursion both compute the amplitude as a *sum* of pieces, each of which has more singularities than the answer; locality is recovered by a cancellation that is real but invisible in any one term. The amplituhedron computes the amplitude as a *single* object whose only singularities are the physical ones, so locality is not recovered, it is built in — it is the shape of the region. The cancellation of spurious poles, the central labor of recursion, becomes in the geometry the trivial statement that a polytope's volume does not depend on how you cut it into simplices. This is the precise sense in which the amplituhedron makes locality and unitarity *manifest*: not by proving them anew, but by choosing a representation in which they are visible from the start.

We have now done what the chapter promised in the controlled sector: taken explicit positive data, computed an amplituhedron canonical form to a closed rational expression, verified its spurious poles cancel and its physical residues factorize, and shown it agrees with the polygon of Section 62.1.4 and the associahedron of Section 62.2 — the full circle. We have then displayed the genuine $m = 4$ NMHV hexagon, marking precisely which of its claims are theorems ($k = 1$, the polytope volume) and which inherit the conjectural status of the general construction, and we have seen the same amplitude wear two faces, local-by-geometry and recursive-with-spurious-poles. What remains is to take honest stock: to separate, with a date attached, the settled mathematics from the physical conjecture and the open problem, and to articulate the conceptual payoff of a program in which locality and unitarity emerge from a shape with no Lagrangian and no spacetime in sight. That accounting is the work of Section 62.8.

62.8 Scope, status, and the frontier

The previous section carried one amplituhedron computation as far as it can honestly be carried with present tools: in the rigorously controlled $m = 2$ sector we evaluated a canonical form to a closed expression and closed the loop with the polygon that opened the chapter (Section 62.7.1), and only then did we write down the physical $m = 4$ NMHV hexagon result, displayed but not derived, so that the reader could see the genuine object without being sold an unproved theorem as a proved one. That deliberate split — a complete computation on one side of a line, an honest pointer on the other — is the whole subject in microcosm, and it is the reason this closing section exists. A frontier chapter earns its place only if it states, precisely and with a date attached, which side of that line each of its claims sits on. This section does the accounting. It introduces no new construction and proves no new theorem; it audits everything the chapter has asserted, marks each conjecture as a conjecture, separates what is established mathematics from what is a research program, and then articulates the conceptual payoff that makes the program worth pursuing even where it is unfinished. We write as of the mid-2020s, and we say so, because several of the statements below are moving.

62.8.1 A summary of mathematical status, dated to the mid-2020s

It is useful to sort the chapter's content into three bins: settled mathematics, a physical conjecture with strong but incomplete support, and a genuinely open problem. The sorting is sharper here than almost anywhere else in the book, because the objects are defined with mathematical precision and the gap between "defined" and "proved equal to the amplitude" is itself a precise, checkable statement.

What is theorem. The foundational layer of the chapter is mathematics in the strict sense, complete with proofs given in full above.

- *Positive geometry and the canonical form are rigorously defined.* The notion of a logarithmic form, of a positive geometry, and of its canonical form (Theorems 62.1 and 62.3) are precise definitions, and the canonical form, when it exists, is *unique* — this we proved (Theorem 62.4) from nothing but the absence of global holomorphic top forms on projective space. Triangulation independence (Theorem 62.9) is a corollary, not an article of faith.

- *The associahedron story is a finished theorem.* The tree-level color-ordered amplitude of bi-adjoint ϕ^3 theory *equals* the canonical form of the ABHY

kinematic associahedron, term by term, with no large-N limit, no supersymmetry, and nothing conjectural (Theorem 62.17); we verified it completely at five points (Theorem 62.18). This is the cleanest existing realization of "amplitude = volume," and it is proved.

- *The combinatorial backbone is a theorem.* The totally nonnegative Grassmannian $\mathrm{Gr}_{\geq 0}(k, n)$ is a regular CW complex homeomorphic to a closed ball, stratified into positroid cells classified by Postnikov's decorated permutations [208, 409]; we proved the small cases by hand (Theorem 62.25) and imported the general statement explicitly, as a cited theorem rather than as a step we skipped.

- *The amplituhedron is rigorously defined.* The amplituhedron $\mathcal{A}^{(m)}_{n,k,L}$ is a precise mathematical object — as the image of the positive Grassmannian under a positive linear map (Theorem 62.44), and equivalently through the sign-flip/topological characterization [37]. Its definition is not in question; what is in question is one specific assertion *about* it.

- *The $m = 2$ amplituhedron is understood.* For $m = 2$ the amplituhedron is combinatorially and geometrically under control: it is in bijection with the hypersimplex, its triangulations are classified, and its canonical form has been proved to reproduce the corresponding ($m = 2$) amplitude [378]. This is why Section 62.7.1 could be carried to a number: the $m = 2$ sector is theorem, not conjecture.

What is conjecture. The single load-bearing physical claim of the chapter — the one the title advertises — is, at the physical value $m = 4$, a conjecture. We mark it as such here, retroactively flagging every place it appeared.

- *Canonical form = amplitude at $m = 4$ is conjectural.* The statement that the canonical form of the physical amplituhedron $\mathcal{A}^{(4)}_{n,k,L}$ equals the (super)amplitude integrand of planar $\mathcal{N} = 4$ super Yang–Mills (Theorem 62.17, as extended to the amplituhedron in Section 62.4 and after) is a conjecture of Arkani-Hamed and Trnka [32]. Every appearance of the words "the amplitude is the canonical form" at $m = 4$ in this chapter is, strictly, "the amplitude is conjectured to be the canonical form."

- *Tree level is close to settled; the full statement is not.* The gap has narrowed sharply in the mid-2020s. At tree level the BCFW recursion of Chapter 59 provides a candidate triangulation of the $m = 4$ amplituhedron into positroid cells, and that this collection of cells genuinely *tiles* the amplituhedron — covering it once, without overlaps or gaps — has been proved for the tree amplituhedron [181]. Combined with the cell-by-cell residue matching, this makes the tree-level $m = 4$ statement a theorem in

the cases where the tiling result applies. What remains conjectural is the clean, uniform statement at all multiplicities and, decisively, at loop level.

What is open. The loop-level content is the present frontier.

- *The loop integrand geometry is constructed but not proved in general.* The loop amplituhedron — the geometry whose canonical form is conjectured to be the all-loop planar $\mathcal{N} = 4$ integrand — is fully *defined*: the L-loop momentum-twistor data are organized into mutually positive lines in the positive Grassmannian, and the integrand is read off as a canonical form. But the proof that this canonical form equals the integrand produced by generalized unitarity (Chapter 60), for all n, k, and all loop orders L, does not exist as of the mid-2020s. Individual cases (low loops, low multiplicity) have been checked and match; the all-loop theorem is open.
- *Integration is not part of the construction.* Even granting the conjecture in full, the amplituhedron delivers the *integrand* — a rational form in loop and external momenta — not the integrated amplitude. The transcendental functions (polylogarithms, elliptic and beyond) that result from performing the loop integrals are not, in any established sense, read off the geometry; that step still requires the integration technology of Chapter 60 and its successors. "Amplitude = volume" is, more precisely, "*integrand* = canonical form."

Table 62.1 collects this accounting in one place.

Caution

The phrase "the amplitude is a volume" is exact for the bi-adjoint associahedron (Theorem 62.17) and for the $m = 2$ amplituhedron, and is a conjecture for the physical $m = 4$ amplituhedron at tree level (now largely reduced to theorem) and at loop level (open). Throughout this chapter, wherever the amplituhedron at $m = 4$ was said to *be* the amplitude, read "is conjectured to be," with the tree-level qualification just stated. The mathematics of positive geometry is not conditional on this conjecture; the physical interpretation at $m = 4$ beyond trees is.

62.8.2 The conceptual payoff: locality and unitarity as outputs, not inputs

Why pursue an object whose central physical identification is, at the physical value, still a conjecture? Because if the conjecture holds — and the evidence for it, summarized above, is strong — it reorganizes what we take the amplitude to *be*, and the reorganization is conceptually radical in a way that survives even

Claim	Status (mid-2020s)	Where
Positive geometry, canonical form, uniqueness, triangulation independence	**Theorem** (proved in text)	Theorems 62.4 and 62.9
Bi-adjoint ϕ^3 tree amplitude = canonical form of ABHY associahedron	**Theorem** (proved in text)	Theorem 62.17
$\mathrm{Gr}_{\geq 0}(k,n)$ is a ball; positroid stratification	**Theorem** (cited, small cases proved)	Section 62.3.4
Amplituhedron $\mathcal{A}^{(m)}_{n,k,L}$ is well defined	**Definition** (precise object)	Theorem 62.44
$m = 2$ canonical form = amplitude	**Theorem**	Section 62.7.1
$m = 4$ tree: BCFW cells tile the amplituhedron	**Theorem** (recent)	Section 62.4
$m = 4$ canonical form = planar $\mathcal{N} = 4$ tree amplitude, all n, k	**Conjecture** (strong support)	Theorem 62.17
$m = 4$ loop integrand = canonical form of loop amplituhedron, all L	**Open** (checked in cases)	Section 62.4
Integrated amplitude read off the geometry	**Not claimed**	—

Table 62.1: The chapter's claims sorted by epistemic status as of the mid-2020s. The line between theorem and conjecture falls precisely at the physical value $m = 4$ beyond tree level; the line between conjecture and open problem falls at loop level. Everything below the geometry — the definition of the canonical form and the associahedron realization — is settled mathematics.

the present incompleteness. The payoff is best stated as a program, and we are careful to state it as a program and not as a theorem.

In the Feynman-diagram construction of an amplitude, two physical principles are *inputs*. Locality is built in by hand: we sum over local interaction vertices and propagators $1/(p^2 - m^2)$, so the amplitude has poles exactly where an internal line goes on shell, because we put them there. Unitarity is built in by hand: the $i\epsilon$ prescription and the Hermiticity of the Lagrangian guarantee, through the optical theorem (Chapter 60), that residues factorize into products of lower amplitudes. Both principles are assumptions about a spacetime Lagrangian, fed into a machine.

In the positive-geometry construction, neither principle is an input. One specifies a region — the associahedron, the amplituhedron — by positivity

conditions that make no reference to a Lagrangian, to time evolution, or even to spacetime. The canonical form is then *forced*: by uniqueness (Theorem 62.4) it is the one form with logarithmic singularities on the region's boundaries and no others. And now the two physical principles appear as *outputs*:

- *Locality* is the statement that the canonical form has poles only on the facets of the region. The facets are codimension-one boundaries, fixed by the positivity conditions; the form is singular there and nowhere else by the boundary clause of Theorem 62.3. The amplitude's poles sit on physical channels *because* those channels are the boundaries of a positive region — not because propagators were inserted by hand. We proved exactly this for the associahedron: the poles of m_n are the loci $X_{i,j} = 0$, the facets (Theorem 62.16).

- *Unitarity* is the statement that the residue on a facet is the canonical form of that facet. By the recursive residue axiom (Eq. (62.5)), the residue on a boundary is a lower canonical form, and when the boundary factorizes as a product of two smaller positive geometries, the residue factorizes as a product of two lower amplitudes. We proved exactly this for the associahedron: the facet $X_{i,j} = 0$ is the product $\mathcal{A}_a \times \mathcal{A}_b$, and the residue there is the product of the two sub-amplitudes (Theorem 62.13 and Eq. (62.65)).

The slogan, stated honestly as a research aspiration rather than a proved fact about the physical theory, is that *spacetime locality and quantum-mechanical unitarity are not axioms of the amplitude but consequences of the positivity and boundary structure of a geometry.* For the bi-adjoint scalar this aspiration is a theorem. For planar $\mathcal{N} = 4$ at $m = 4$ it is the conjecture of the previous subsection. The conceptual content does not depend on which: in every case where the dictionary has been verified, locality and unitarity came out, not in.

Physical Insight

The reorientation is worth stating in one sentence. Conventional field theory begins with a Lagrangian on spacetime, derives Feynman rules, and *discovers* that the resulting amplitudes are analytic with factorizing residues. The positive-geometry program begins with a region defined by positivity, *derives* a unique canonical form, and finds that its singularity structure *is* locality and unitarity — with no Lagrangian, no Hilbert space, and no time evolution written down anywhere. Where it has been proved (the associahedron, $m = 2$), this is not a reinterpretation of the old derivation; it is an independent route to the same numbers that never mentions the ingredients the old route called fundamental.

This is the precise sense in which the amplitudes program suggests that locality and unitarity, and perhaps spacetime itself, are *emergent*. The much-quoted aspiration that the amplitude is "the answer to which the geometry is the question" — that one should seek the question, posed purely in the language of positive geometry, to which the universe's S-matrix is the answer — is exactly that: an aspiration. As of the mid-2020s there is no positive geometry for a generic interacting quantum field theory, no derivation of the amplituhedron's positivity conditions from a deeper principle, and no geometry that produces integrated amplitudes rather than integrands. The program is a bet that such structures exist and are fundamental. The chapter has shown the bet is not empty — the associahedron is a fully proved instance — without pretending it is won.

62.8.3 Reach and limits: the enabling crutches

The amplituhedron is not a theory of everything's S-matrix; it is, so far, a theory of one very special amplitude, and the specialness is the source of both its power and its limitations. Two features do the enabling work, and it is worth naming them as crutches so that the boundary of the program's current reach is visible.

Planarity. The construction requires a fixed cyclic order of the external legs. The total positivity of Theorem 62.22 is defined relative to an ordering of the columns; the consecutive minors that appear in the canonical form (Eq. (62.82)) are consecutive only with respect to that order; the associahedron itself is the geometry of a *cyclically* ordered polygon. Physically this is the planar (large-N) limit, in which color-ordered partial amplitudes are well defined and the cyclic order is the boundary order of a disk. Away from the planar limit there is no global cyclic order, no canonical notion of "consecutive," and the positivity that singles out the region loses its anchor. A positive geometry for non-planar amplitudes is, as of the mid-2020s, not established; there are proposals, but no object with the proved status of the planar amplituhedron.

Maximal supersymmetry. The amplituhedron computes planar $\mathcal{N} = 4$ super Yang–Mills, and the maximal supersymmetry is not incidental. It is what allows all helicity configurations at fixed n to be packaged into a single superamplitude graded by the integer k (the same k as the rank of the Grassmannian, Section 62.3), so that one geometry per (n, k, L) captures the entire helicity tower. It is also what makes the loop integrand a pure logarithmic form — the uniform transcendentality and the absence of higher-degree poles that the dlog structure of the canonical form requires. A theory with less supersymmetry has amplitudes with non-logarithmic singularities (double poles, square roots, ultraviolet divergences) that a canonical form, which by Theorem 62.1 has only simple logarithmic poles,

cannot directly represent. Extending positive geometry to such theories requires generalizing the canonical form itself, and that generalization is incomplete.

The status of the extensions. Three frontiers follow from relaxing these crutches, and each is open to a different degree.

- *Non-planar* $\mathcal{N} = 4$: actively pursued, no settled geometry. The difficulty is the absence of a single cyclic order; partial results organize non-planar integrands by "sums of positive pieces," but no analogue of Theorem 62.44 is established.

- *Less or no supersymmetry*: the bi-adjoint scalar (no supersymmetry at all) *does* have a complete positive geometry, the associahedron — but it is a theory with only logarithmic singularities and no helicity, precisely the features that survive. For gauge theories with $\mathcal{N} < 4$, or for QCD with its ultraviolet divergences, no clean positive geometry exists; the obstruction is the non-logarithmic analytic structure just noted.

- *Gravity*: the double copy of Chapter 61 expresses gravity amplitudes as "squares" of gauge-theory ones, but the squaring operation has no known geometric realization. There is no gravituhedron in the sense of a positive geometry whose canonical form is the graviton amplitude; whether the double copy can be lifted to a statement about geometries is open.

Caution

It is tempting to read the amplituhedron's elegance as evidence that *all* amplitudes are volumes of positive geometries. The honest position is narrower: positive geometry is proved to compute the bi-adjoint scalar tree amplitude and the $m = 2$ amplituhedron, is conjectured (with strong support) to compute planar $\mathcal{N} = 4$, and has no established realization for non-planar, non-maximally-supersymmetric, or gravitational amplitudes. The crutches are load-bearing.

62.8.4 Adjacent positive geometries: live directions, each dated

The same idea — an amplitude or correlator as the canonical form of a region defined by positivity — has spread well beyond the amplituhedron, and several of these directions are more recent and in some respects more rigorous than the $m = 4$ amplituhedron itself. We record the principal ones with dates, as live research rather than closed theory.

The associahedron family (2017–). The ABHY associahedron of Section 62.2 was introduced in 2017 [36] and is the most rigorous member of the whole program: its identification with the bi-adjoint amplitude is a theorem, proved in this chapter. It has since been generalized to other "accordiohedra" and to deformations connected to the open-string worldsheet (the same paper relates the associahedron's canonical form to the Koba–Nielsen string integral via "stringy canonical forms"), making the associahedron the bridge between field-theory positive geometry and string amplitudes. This family is the firmest ground in the subject.

Cosmological polytopes and the cosmological correlator program (2017–). Arkani-Hamed, Benincasa, and Postnikov showed in 2017 that the wavefunction of the universe for a conformally coupled scalar with polynomial interactions in a power-law cosmology is the canonical form of a *cosmological polytope* [35] — a positive geometry built directly from the interaction graph, with no reference to bulk time evolution, exactly in the spirit of Section 62.8.2. This is rigorous in the same sense as the associahedron: a proved equality between a canonical form and a physical object (here a cosmological correlator rather than a flat-space amplitude). The program has grown in the 2020s into a systematic "cosmological bootstrap": the differential equations satisfied by cosmological correlators have been derived from the polytope's boundary structure (the "kinematic flow," 2023 [39]), giving the integrated correlators — a step the flat-space amplituhedron does not yet provide for its integrands. The cosmological setting is, in this specific sense, ahead of the flat-space one.

Surfaceology and curve integrals (2023–). The most recent direction reformulates scattering, including at loop level and *without* the planarity crutch, in terms of curves on surfaces [40]. Amplitudes are computed by counting intersections of curves on a Riemann surface with marked points, packaged into "curve integrals" that produce the full integrand — planar and non-planar, all loop orders — from a single combinatorial recipe. As of the mid-2020s this is rapidly developing and not yet consolidated into theorems of the associahedron's finished kind, but it is the most promising route past the two crutches of Section 62.8.3, and it has already produced new structural results (such as "hidden zeros" of amplitudes) that were invisible from the diagrammatic and even the amplituhedron points of view. We flag it as the frontier within the frontier.

These three, together with the amplituhedron, are facets of a single emerging subject — positive geometry — whose claim is that the right home for scattering data is not a Lagrangian on spacetime but a region defined by positivity, with the canonical form as the universal translator. This is also where the program touches the rest of the book's frontier: the insistence that physical amplitudes live on the boundary of a region of *allowed* (positive) configurations is kin to

the positivity constraints of the conformal and S-matrix bootstrap (Chapter 52), where consistency conditions — unitarity as positivity of a spectral density, crossing as a linear constraint — carve out the allowed theory space rather than constructing a theory from a Lagrangian. In both the bootstrap and positive geometry, positivity is the generative principle and the Lagrangian is demoted from foundation to (at most) one convenient coordinate system.

This audit completes the chapter's frontier coda on the amplitudes of Yang–Mills theory and QED: it adds no new perturbative calculation, only the geometric reframing that the same gluon and photon amplitudes carried since Chapter 58 can be read, in the planar maximally-supersymmetric corner, as boundaries and residues of a positive region rather than as sums of diagrams. With the summary of what is proved, conjectured, and open now set down, we hand the synthesis to Chapter 63: there the threads of this entire part — spinor-helicity, recursion, unitarity, the double copy, and positive geometry — are drawn together into a single statement of what the modern amplitudes program has revealed about quantum field theory, and tied back to the bootstrap of Part XIV and the holography of Part XV.

Further reading

The canonical reference for the amplituhedron and the positive-geometry program is the monograph of Arkani-Hamed, Bourjaily, Cachazo, Goncharov, Postnikov, and Trnka [33], with the original amplituhedron proposal in [32] and the momentum-twistor underpinnings in [268]. The general theory of positive geometries and canonical forms is laid out in [34]; the combinatorics of the positive Grassmannian in [409], with the ball theorem in [208]. For the kinematic associahedron and stringy canonical forms see [36], and for the sign-flip definition of the amplituhedron [37]. The $m = 2$ amplituhedron is treated in [378] and the tree-level BCFW tiling in [181]. The adjacent programs surveyed in Section 62.8.4 are introduced in [35] (cosmological polytopes), [39] (differential equations for cosmological correlators), and [40] (curve integrals and surfaceology). All are research literature; nothing in the chapter's proved core depends on consulting them.

Chapter 63

Amplitudes Beyond: Synthesis

Every chapter of this part has computed the same kind of object — an on-shell scattering amplitude — by a method that grew steadily less dependent on the apparatus quantum field theory was built from. Spinor-helicity (Chapter 58) showed that the answer is far simpler than the fields that produce it; Parke–Taylor and BCFW recursion (Chapter 59) built tree amplitudes from their factorization poles with no Lagrangian in the loop; the unitarity method (Chapter 60) reconstructed loop integrands from their cuts; the double copy (Chapter 61) manufactured gravity by squaring gauge theory; and the amplituhedron (Chapter 62) produced locality and unitarity as facets of a geometry. This closing chapter does not introduce a new computational technique. Its task is to ask what all of this *means* for quantum field theory, and to do so honestly — separating what the program has proved from what it has only made plausible, and dating the frontier as it stands in the mid-2020s.

The organizing question is the one Part I left implicit when it laid down its postulates. Of the founding assumptions of the subject — Poincaré invariance, locality, unitarity, analyticity, introduced informally in Chapter 1 and made precise as S-matrix properties in Section 63.1.3 — which are genuine *inputs* we are free to choose, and which are *outputs*, theorems forced once the others are fixed? The amplitudes program supplies the most suggestive evidence bearing on that question, and the chapter assembles it into an argument. We open in Section 63.1 by making the word "redundant" exact: the machinery that computes an amplitude is vastly larger and more arbitrary than the amplitude itself, and the part of it that survives to the answer is governed by the axioms alone. Section 63.2 then promotes analyticity and unitarity from properties an amplitude happens to possess to principles strong enough to *construct* it, recovering the BCFW and unitarity-method logic of Chapters 59 and 60 as the two statements that residues and cuts *are* the locality and unitarity data.

The middle of the chapter turns the same pair of principles into quantitative output. In Section 63.3 we carry the gold-standard worked anchor of

the part: a twice-subtracted forward dispersion relation, built from crossing, analyticity (Chapter 5), and the optical theorem (Chapter 14), forces the leading higher-derivative Wilson coefficient of the ϕ^4 theory effective theory to be strictly positive — the S-matrix bootstrap pointed below, at the effective theory, rather than above at the spectrum. Section 63.4 then shows that this is not an analogy to the conformal bootstrap of Chapters 49 and 50 but the *same* convex-optimization problem in two kinematic disguises, with the flat-space limit of AdS/CFT (Chapter 53) supplying the dictionary that maps one onto the other. Section 63.5 pushes to the boldest reading, assembling the amplituhedron (Chapter 62) and the double copy (Chapter 61) as evidence that locality and unitarity are themselves outputs of a more primitive geometric and algebraic structure.

The last two sections take the program to its boundary in both senses. Section 63.6 follows the amplitude to null infinity: leading soft theorems, derived in full from the factorization pole, become the Ward identities of asymptotic symmetries, and the Mellin transform to boost eigenstates recasts the entire S-matrix as a correlator of conformal primaries on the celestial sphere — the flat-space cousin of the holography of Part XV. Section 63.7 then closes the part and the question with a dated status summary of the program's claims, the controlled inversion of Part I's logic that the evidence supports, the meeting of the three benchmark theories as facets of one structure, and the frontier that remains — which is precisely the setting, curved spacetime, where the flat-space S-matrix runs out and Part XVII (Chapters 64 and 65) must begin.

Notation Watch: conventions for this chapter

Signs matter throughout the dispersive and soft-theorem arguments below, so we restate the conventions once. We use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$ (Appendix A), so that $p^2 = m^2$ on shell, the Mandelstam invariants obey $s + t + u = \sum_i m_i^2$, and the massless contractions used in Section 63.3 read $(p_i - p_j)^2 = -2\, p_i \cdot p_j$. The spinor-helicity angle and square brackets $\langle i\, j\rangle$, $[i\, j]$ appearing in load-bearing equations follow the little-group and reality conventions fixed in Chapter 58 (Section 58.2.1); we recall them at each use rather than re-derive them.

63.1 The Lesson of Redundancy: Hidden Simplicity

The chapter opened with a question that is really the book's question, posed now with the whole apparatus of quantum field theory behind us: of the founding assumptions of the subject — Poincaré invariance, locality, unitarity, analyticity — which are genuine *inputs*, postulates we are free to choose, and which are

outputs, theorems forced upon us once the others are fixed? The amplitudes program of this part has handed us the single most suggestive piece of evidence bearing on that question, and before we can turn the evidence into an argument we must state it precisely. The evidence is a mismatch. The machinery that *computes* a scattering amplitude — a Lagrangian, a choice of field variables, a gauge, a swarm of Feynman diagrams — is enormously larger and more arbitrary than the amplitude it produces. Most of what the machinery carries is redundant: it cancels, it depends on conventions, it never reaches the answer. The physical amplitude is, by comparison, almost embarrassingly simple. This section assembles that fact from the whole part, makes the word "redundant" mean something exact, and reads off the list of axioms whose status the rest of the chapter will interrogate. We compute nothing new here; we gather what the part has proved and sharpen the question.

63.1.1 The redundancy of the field-variable description

A relativistic quantum theory of particles is, at bottom, two things: a Hilbert space carrying a unitary representation of the Poincaré group, and a unitary operator $\mathbf{S}$ on it — the S-matrix — whose matrix elements between asymptotic multiparticle states are the numbers an experiment can measure. Everything else in the formalism is scaffolding. The Lagrangian, the fields, the path integral, the gauge choice, the regulator, the counterterms: these are a *coordinatization* of the theory, a way of naming and computing the entries of $\mathbf{S}$. Two facts, both established earlier in the book, show just how loose that coordinatization is.

The first is gauge redundancy. In quantizing a gauge field we deliberately introduced a description with *more* variables than physical degrees of freedom — the longitudinal and timelike polarizations of A_μ, the gauge parameter ξ, the Faddeev–Popov ghosts — and then spent entire chapters showing that the surplus drops out of anything physical. The photon and gluon propagators carry ξ explicitly; the S-matrix does not, by the Ward and Slavnov–Taylor identities. A continuous infinity of field configurations, A_μ and $A_\mu + \frac{1}{g}D_\mu\alpha$, name the same physical state. Gauge "symmetry" is, viewed this way, not a symmetry of nature at all but a redundancy of our labels, introduced for the sake of manifest locality and Lorentz covariance and paid back in full at the end of every calculation.

The second fact is subtler and applies to every field theory, gauge or not: the freedom to redefine the field off shell. We may trade the field ϕ for any local function of it and its derivatives,

$$\chi(x) \;=\; \phi(x) \;+\; c_1\,\phi(x)^2 \;+\; c_2\,\phi(x)\,\Box\phi(x) \;+\; \cdots\,, \tag{63.1}$$

provided the redefinition is *invertible* near the free field — the linear term has a nonzero coefficient, here normalized to unity — and write the same theory in terms of χ. The Lagrangian, the Feynman rules, and every off-shell Green's

function change, often dramatically: a redefinition can create or destroy entire interaction vertices. Yet the S-matrix is left exactly invariant. This is the content of what is loosely called the *equivalence theorem*. We can prove the part of it we need directly from the LSZ reduction formula of Theorem 14.31, and the proof is short enough, and instructive enough, to give in full.

Proposition 63.1 (Invariance of the S-matrix under local field redefinition)**.** *Let a theory be described by a local field ϕ, and let χ be a second local field related to ϕ by an invertible local redefinition Eq. (63.1) whose linear part is a nonzero multiple of ϕ. Suppose ϕ and χ both have nonzero overlap with the one-particle states $|p\rangle$ of physical mass m, i.e. $\langle 0|\,\phi(0)\,|p\rangle \neq 0$ and $\langle 0|\,\chi(0)\,|p\rangle \neq 0$. Then the connected S-matrix elements obtained by applying the LSZ formula Theorem 14.31 to correlators of χ equal those obtained from correlators of ϕ.*

Proof. The S-matrix is an intrinsic object. It is defined (Chapter 14) by the overlaps $\langle \beta;\mathrm{out}|\,|\alpha;\mathrm{in}\rangle$ of asymptotic states built on the interacting Hilbert space by the Møller operators; its construction references the Hamiltonian and the one-particle subspace, and *no* choice of interpolating field. LSZ is not a definition of $\mathbf{S}$ but a *representation* of its matrix elements: it asserts that for *any* local operator $\mathcal{O}$ carrying the right quantum numbers and satisfying $\langle 0|\,\mathcal{O}(0)\,|p\rangle \neq 0$, the time-ordered correlator of $\mathcal{O}$'s develops a simple pole in each external invariant as $p_i^2 \to m^2$, and the simultaneous residue in all $n+m$ external momenta reconstructs the connected S-matrix element. Concretely, in the momentum-space form Eq. (14.153),

$$\langle f|\mathbf{S}|i\rangle_{\mathrm{c}} = \Big(\frac{i}{\sqrt{Z_{\mathcal{O}}}}\Big)^{n+m} \prod_{a} [-(p_a^2 - m^2)]\; \mathcal{G}_{\mathcal{O}}(p_1, \dots)\Big|_{\text{on shell}}, \tag{63.2}$$

where $\mathcal{G}_{\mathcal{O}}$ is the correlator of $\mathcal{O}$'s and $Z_{\mathcal{O}}$ is the residue of the $\mathcal{O}$-two-point function at the pole, $\widetilde{\Delta}_{\mathcal{O}}(p) \to i Z_{\mathcal{O}}/(p^2 - m^2 + i\epsilon)$, with $\sqrt{Z_{\mathcal{O}}} = \langle 0|\,\mathcal{O}(0)\,|p\rangle$ in the rest frame (Theorem 14.23). Two observations now finish the argument.

(i) The pole sits at a spectral eigenvalue. The factor $-(p_a^2 - m^2)$ in Eq. (63.2) is matched against a pole at $p_a^2 = m^2$, where m is the energy of the one-particle state at rest — an eigenvalue of the Hamiltonian, a property of the spectrum. A field redefinition relabels operators on a fixed Hilbert space; it cannot move a spectral eigenvalue. Hence the correlators of χ have their single-particle poles at exactly the same $p^2 = m^2$ as those of ϕ. (Indeed χ, being ϕ plus operators that create at least two quanta from the vacuum, has the *same* one-particle overlap to leading order; the higher terms of Eq. (63.1) contribute only multi-particle states and so cannot shift the one-particle pole.)

(ii) The residue is divided out. The only place the choice of operator enters Eq. (63.2) is through the wavefunction factor $(\sqrt{Z_{\mathcal{O}}})^{-(n+m)}$, one $\sqrt{Z_{\mathcal{O}}}$ per external leg. But the on-shell residue of $\mathcal{G}_{\mathcal{O}}$ likewise carries exactly one factor $\sqrt{Z_{\mathcal{O}}}$ per external leg, because each external leg is, near its pole, a full $\mathcal{O}$-propagator

$\widetilde{\Delta}_{\mathcal{O}} \to iZ_{\mathcal{O}}/(p^2-m^2)$ amputated by the explicit $-(p_a^2-m^2)$ and converted to the physical one-particle state by its overlap. The two powers of $\sqrt{Z_{\mathcal{O}}}$ cancel identically: the quotient Eq. (63.2) is independent of the normalization $Z_{\mathcal{O}}$, hence independent of *which* interpolating field we used.

What does change between ϕ and χ is everything *away* from the poles. The redefinition Eq. (63.1) alters contact terms, rearranges the multi-particle branch cuts, and rescales Z; it moves the off-shell content of the Green's functions freely. But the n-fold on-shell residue — after the LSZ amputation strips the external propagators and the Z-factors divide out — is pinned to the intrinsic S-matrix element, which knew nothing about ϕ or χ to begin with. The two computations therefore agree. □

The hypothesis $\langle 0|\,\mathcal{O}\,|p\rangle \neq 0$ is not a technicality but the whole point. A redefinition that *removed* the linear term — say $\mathcal{O}=\phi^2$ — would produce a field with no one-particle overlap, no single-particle pole, and LSZ would simply not apply to it; such an operator computes form factors and composite correlators, not the elastic S-matrix. Among the fields that do interpolate the one-particle state, however, the choice is pure convention. The following worked example makes the mechanism of Theorem 63.1 concrete by watching, in the simplest possible case, how a redefinition deposits new structure off shell while leaving the on-shell residue untouched.

Worked Example 63.2 (A field redefinition moves the cut, not the pole)**.** Take the free real scalar of ϕ^4 theory at $\lambda=0$ (Chapter 9), with $\mathcal{L}=\tfrac{1}{2}(\partial\phi)^2-\tfrac{1}{2}m^2\phi^2$ and the Feynman propagator $D_F(x)=\langle 0|\,\mathcal{T}\phi(x)\phi(0)\,|0\rangle$, $\widetilde{D_F}(p)=i/(p^2-m^2+i\epsilon)$. Introduce the new interpolating field

$$\mathcal{O}(x) \;=\; \phi(x) \;+\; a\,\phi(x)^2, \qquad [a]=\text{mass}^{-1}, \tag{63.3}$$

a local redefinition of the form Eq. (63.1). We compute its two-point function and read off both the pole and the residue, exhibiting Theorem 63.1 (i)–(ii) explicitly.

By Wick's theorem in the free theory, with ϕ Gaussian and $\langle 0|\,\phi\,|0\rangle=0$,

$$\begin{aligned}\langle 0|\,\mathcal{T}\,\mathcal{O}(x)\,\mathcal{O}(0)\,|0\rangle &= \langle 0|\,\mathcal{T}\big(\phi+a\phi^2\big)(x)\big(\phi+a\phi^2\big)(0)\,|0\rangle \\ &= \underbrace{\langle 0|\,\mathcal{T}\phi(x)\phi(0)\,|0\rangle}_{D_F(x)} + a\,\underbrace{\langle 0|\,\mathcal{T}\phi(x)\phi(0)^2\,|0\rangle}_{0} + a\,\underbrace{\langle 0|\,\mathcal{T}\phi(x)^2\phi(0)\,|0\rangle}_{0} \\ &\quad + a^2\,\langle 0|\,\mathcal{T}\phi(x)^2\,\phi(0)^2\,|0\rangle\,. \end{aligned}\tag{63.4}$$

The two cross terms vanish because a correlator of an odd number of free fields is zero. The last term, kept to its connected part (the disconnected piece $\langle 0|\,\phi^2\,|0\rangle^2$ is a field-independent constant that contributes no pole and is removed in the connected correlator), is the two-contraction Wick pairing

$$\langle 0|\,\mathcal{T}\phi(x)^2\phi(0)^2\,|0\rangle_{\rm c} \;=\; 2\,D_F(x)^2, \tag{63.5}$$

the factor 2 counting the two ways of contracting the pair at x with the pair at 0. Hence the connected $\mathcal{O}$-propagator is, in position and then momentum space,

$$\langle 0|\,\mathcal{T}\,\mathcal{O}(x)\,\mathcal{O}(0)\,|0\rangle_{\rm c} \;=\; D_F(x)+2a^2\,D_F(x)^2, \qquad \widetilde{\Delta}_{\mathcal{O}}(p) = \frac{i}{p^2-m^2+i\epsilon} + 2a^2\,\widetilde{[D_F^2]}(p). \tag{63.6}$$

The new term is a one-loop bubble. Its Fourier transform,

$$\widetilde{[D_F^2]}(p) = \int \frac{d^4k}{(2\pi)^4}\,\frac{i}{k^2-m^2+i\epsilon}\,\frac{i}{(p-k)^2-m^2+i\epsilon}, \tag{63.7}$$

is the scalar bubble computed in Chapter 21: it is *analytic* in p^2 except for a branch cut beginning at the two-particle threshold $p^2=(2m)^2$, the lowest invariant mass at which the intermediate $\phi\phi$ state can go on shell. Crucially, it has *no pole at* $p^2=m^2$.

Reading off Eq. (63.6), then: near the one-particle pole $p^2\to m^2$ the bubble is a finite, smooth background, and the singular behavior is governed entirely by the first term,

$$\widetilde{\Delta}_{\mathcal{O}}(p) \;\xrightarrow{\;p^2\to m^2\;}\; \frac{iZ_{\mathcal{O}}}{p^2-m^2+i\epsilon}, \qquad Z_{\mathcal{O}}=1. \tag{63.8}$$

The redefinition Eq. (63.3) has changed the two-point function — it has grown a genuine two-particle branch cut at $p^2=4m^2$, an entirely new piece of off-shell, multi-particle structure — while leaving the single-particle pole at the same location m^2 with the same residue $Z_{\mathcal{O}}=1$. This is precisely the dichotomy of Theorem 63.1: the off-shell content moves, the on-shell residue does not. One verifies the residue invariance from the overlap directly, $\langle 0|\,\mathcal{O}(0)\,|p\rangle = \langle 0|\,\phi(0)\,|p\rangle + a\,\langle 0|\,\phi(0)^2\,|p\rangle = \langle 0|\,\phi(0)\,|p\rangle + 0$, since ϕ^2 creates only even numbers of quanta and has zero one-particle overlap. The new field interpolates the same particle with the same normalization; its S-matrix — here the trivial S-matrix of a free theory — is unchanged, even though its Green's functions are not.

Theorem 63.2 is a microcosm of the general lesson. The space of field descriptions of a fixed theory is vast — a gauge orbit times the infinite-dimensional space of local redefinitions — and the S-matrix is constant along all of it. Physical content lives only in the on-shell residues; the off-shell Green's functions are coordinates on the redundant directions. Figure 63.1 pictures the situation: the Feynman machinery operates upstairs, in the large redundant space; the amplitude lives downstairs, on the small physical quotient, and the projection forgets almost everything.

63.1.2 The mismatch between diagrams and amplitudes

The redundancy of Section 63.1.1 is structural — a statement about descriptions. Part XVI turned it into a quantitative scandal, and it is worth restating the

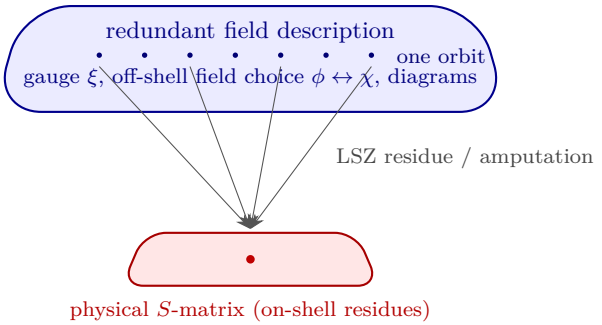


Figure 63.1: The redundancy of the field-variable description. A whole orbit of field choices — related by gauge transformations and by invertible local redefinitions Eq. (63.1) — maps to a single point of the physical S-matrix. The Feynman machinery computes upstairs, carrying gauge parameters, off-shell vertices, and a factorial number of diagrams; the LSZ residue (Theorem 63.1) projects down, discarding everything but the on-shell, gauge-invariant data. Hidden simplicity is the smallness of the target relative to the domain.

numbers because they are the empirical heart of the chapter. Consider the tree-level scattering of n gluons in Yang–Mills theory. The Feynman expansion (Chapter 59) builds the amplitude from cubic and quartic vertices, and the number of diagrams grows faster than any factorial. We recorded the counts in Table 59.1: the full n-gluon amplitude is assembled from 4 diagrams at $n = 4$, 220 at $n = 6$, 34 300 at $n = 8$, and 10 525 900 at $n = 10$; even after color ordering strips the trivial group-theoretic redundancy, a single color-ordered partial amplitude $A_n[1, 2, \dots, n]$ still requires 12 925 planar diagrams at $n = 10$. Each diagram is individually gauge-dependent — it depends on the gauge parameter ξ, on off-shell vertices, on a choice of polarization vectors that themselves carry a reference-momentum ambiguity — and each is a sprawling rational function of the momenta and polarizations built from *both* chiralities, the angle data $\langle i\, j\rangle$ and the square data $[i\, j]$ entangled together.

The sum of these millions of terms, in the maximally-helicity-violating sector (two negative-helicity gluons i, j among n), is the single Parke–Taylor monomial of Eq. (59.55),

$$A_n[1^+, \dots, i^-, \dots, j^-, \dots, n^+] \;=\; i\, g^{n-2}\, \frac{\langle i\, j\rangle^4}{\langle 1\, 2\rangle\langle 2\, 3\rangle \cdots \langle n-1\, n\rangle\langle n\, 1\rangle}, \tag{63.9}$$

one term, purely holomorphic, not a single square bracket surviving the collapse. The ratio of description to result is, at $n = 10$, a million-to-one. This is the mismatch in its sharpest form: the gauge-dependent, off-shell, factorially proliferating machinery delivers a gauge-invariant, on-shell, one-line answer.

The decisive point — the one that turns the mismatch from a curiosity into a clue — is *how* Part XVI recovered Eq. (63.9) without summing the diagrams at

all. Two reconstructions, both using only on-shell, gauge-invariant data, produced the same answers.

First, at tree level, on-shell recursion. The BCFW relation (Theorem 59.23) builds the n-point amplitude from products of lower-point *on-shell* amplitudes, glued across factorization channels where an internal line goes on its mass shell. The recursion bottoms out on the three-point amplitudes, and those are not computed from a Lagrangian at all: they are fixed, up to a single constant, by Poincaré invariance alone — by little-group covariance Eq. (58.5) together with locality — as established in Theorem 58.20. The entire MHV tower Eq. (63.9), and indeed every tree gluon amplitude, follows from gluing these symmetry-fixed three-point building blocks. No gauge is chosen; no off-shell vertex appears; the gauge parameter ξ that haunted every individual Feynman diagram is simply absent from the construction, because the construction never leaves the physical, on-shell data.

Second, at loop level, generalized unitarity. The optical theorem of Section 14.7 ties the imaginary part of an amplitude to a sum over physical intermediate states — a statement of unitarity, $\mathbf{S}^\dagger\mathbf{S} = 1$. Chapter 60 sharpened this into a constructive tool: the loop integrand is reconstructed by demanding that its discontinuity across every cut factorize into products of on-shell tree amplitudes. The quadruple cut (Theorem 60.13) freezes the loop momentum completely and reads each box coefficient off as a product of four on-shell trees; the lower cuts of Section 60.3 supply the triangles and bubbles. Again the input is exclusively on-shell, gauge-invariant tree data, and again the Lagrangian, the gauge, and the off-shell vertices never appear. The amplitude is assembled from its physical singularities — its poles and cuts — alone.

Physical Insight

Hidden simplicity is a symptom, not the disease. A simple answer wrung from a complicated calculation is a signal that the calculation is written in the wrong variables — that it carries redundant coordinates that must cancel before the physical content can emerge. The Feynman expansion is such a set of coordinates: manifestly local and Lorentz-covariant, and for exactly that reason laden with gauge and off-shell redundancy. The on-shell methods of Chapters 59 and 60 are the change of variables that trades manifest locality for manifest physicality, computing directly in the data that survive the projection of Fig. 63.1: residues at physical poles and discontinuities across physical cuts. That these data *suffice* to reconstruct the full amplitude is the fact the rest of the chapter must explain.

63.1.3 The axioms as the variables of the argument

If the on-shell data — poles, residues, cuts, and the symmetry-fixed three-point seeds — are enough to rebuild the amplitude, then the elaborate Lagrangian machinery was never logically necessary; it was one convenient route to data that other routes reach more directly. This invites the chapter's organizing question in its precise form. The amplitude is constrained by a small number of structural principles, the same ones introduced at the very start of the book (Chapter 1) as the defining features of a relativistic quantum theory. We now isolate them as the *variables* of the argument to come, stating each as a sharp property of the S-matrix rather than of the Lagrangian, because it is the S-matrix that the on-shell methods construct.

(A) Poincaré invariance. The asymptotic states are organized into unitary irreducible representations of the Poincaré group, labeled by mass and by spin (for massive particles) or helicity (for massless ones); the S-matrix commutes with the Poincaré generators. In the spinor-helicity variables of Chapter 58 this principle is wholly explicit: an n-point amplitude is a function of the invariants $\langle i\,j\rangle$, $[i\,j]$ covariant under the little-group rescalings Eq. (58.5), and that covariance alone fixes the three-point amplitudes (Theorem 58.20). Poincaré invariance is thus the most clearly *kinematic* of the four — it constrains the form of the answer before any dynamics is specified.

(B) Locality. In the S-matrix this becomes a statement about *singularities*: the amplitude is meromorphic with poles only at the physical masses of the theory, and the residue at each pole *factorizes* into a product of lower-point on-shell amplitudes, one for each side of the factorization channel. There are no spurious poles, no singularities at unphysical locations. This is the analytic shadow of cluster decomposition — distant experiments do not interfere — and it is exactly the property BCFW exploits: the recursion is built from the residues at the physical factorization poles and nothing else.

(C) Unitarity. Probability is conserved, $\mathbf{S}^\dagger\mathbf{S} = 1$. Writing $\mathbf{S} = 1 + i\,T$ gives $-i(T - T^\dagger) = T^\dagger T$, the optical theorem of Section 14.7: the discontinuity of an amplitude across a physical cut is a sum over intermediate on-shell states with *positive* weight. Unitarity is therefore a positivity statement, and it is the property that generalized unitarity (Chapter 60) turns into a reconstruction of loop integrands from their cuts.

(D) Analyticity. The amplitude is the boundary value of an analytic function of the complexified invariants, whose only singularities are those forced by (B) and (C) — the physical poles and the unitarity cuts — with the prescribed $i\epsilon$ continuation, and which is crossing-symmetric: the same analytic function describes the s-, t-, and u-channel processes. Analyticity

is what allows a contour to be deformed, a dispersion relation to be written, a pole to be reached by complexifying a momentum (as BCFW does). It is the principle that connects the data at different kinematic points into a single function.

These four are not independent of one another, and that is the whole tension the chapter will exploit. Part XVI has shown that (B) and (C) — locality as physical-pole factorization, unitarity as positive cuts — together with the kinematic skeleton (A), are enough to *construct* tree amplitudes (BCFW) and loop integrands (generalized unitarity) that the Lagrangian was supposed to be needed for. The Lagrangian, the gauge choice, the off-shell fields — the redundant coordinates of Section 63.1.1 — never entered. So the question "which axioms are inputs and which are outputs?" is not idle metaphysics. It is the concrete question of how much of the standard apparatus is logically necessary and how much is one redundant route among many. We will find that several cherished "axioms" are better read as theorems: that crossing follows from analyticity plus locality, that the three-point couplings of massless higher-spin particles are nearly fixed by Poincaré invariance alone, that locality and unitarity can be recovered as *outputs* of a positive geometry or of a double-copy construction.

Caution

The four principles above are stated as properties of a theory that *has* a particle interpretation and an S-matrix — a massive theory with a mass gap, where asymptotic states and the LSZ machinery (Theorem 14.31) make unambiguous sense. The frontier constructions of this part stretch each principle: massless particles weaken the analytic structure (infrared divergences, the absence of a clean asymptotic Fock space), conformal theories have no S-matrix at all, and the celestial program of Section 63.6 will reorganize the same data as a correlator rather than an S-matrix. We hold the precise massive statements as the reference point and flag each weakening explicitly as it arises.

We have now assembled the empirical fact and named the variables. The Feynman description is hugely redundant and the S-matrix is invariant across the redundancy (Theorem 63.1, Fig. 63.1); the diagram-to-amplitude mismatch is quantitatively enormous (Section 63.1.2); and the amplitude was reconstructed, in every case Part XVI examined, from on-shell data alone — the physical poles of locality and the positive cuts of unitarity, riding on the kinematic frame of Poincaré invariance and tied together by analyticity. The natural reading of this evidence is that analyticity and unitarity are not merely properties the S-matrix happens to possess once a Lagrangian is given; they may be the very instructions for building it. The next section takes this reading seriously and promotes analyticity and unitarity from properties to *constructive principles*:

tree amplitudes assembled from their factorization poles, loop integrands from their unitarity cuts, with locality and unitarity identified as exactly the residue and cut data.

63.2 Analyticity and Unitarity as Constructive Principles

The previous section turned a recurring observation into a question. Again and again the on-shell amplitude proved dramatically simpler than the off-shell, gauge-redundant Feynman sum that computes it, and the axiom list of Chapter 1 — Poincaré invariance, locality, unitarity, analyticity — sat behind that simplicity without obviously *causing* it. The natural reading of the redundancy lesson is that the Lagrangian and its diagrams are a particular, highly non-unique *coordinatisation* of the physical content, while the amplitude is the invariant object; the off-shell freedom is gauge, and gauge is what the on-shell methods discard. But this leaves the axioms in an ambiguous status. Are analyticity and unitarity merely *properties* that any amplitude built the usual way happens to inherit, or are they strong enough to *build* the amplitude in the first place, with no Lagrangian in sight? This section argues for the second reading. We promote analyticity and unitarity from descriptions to constructions: tree amplitudes are assembled from their factorisation poles, loop integrands from their unitarity cuts, and — crucially for what follows — the two axioms turn out to be carried by exactly the residue and cut data, with nothing left over. Locality *is* the pole structure; unitarity *is* the residue and cut normalisation and positivity. Once that identification is made, the founding axioms stop being external postulates checked after the fact and become the input from which the amplitude is generated.

63.2.1 The constructive turn, abstracted from BCFW

The concrete engine that makes this work was built in Chapter 59: the Britto–Cachazo–Feng–Witten recursion Theorem 59.23, which deforms two external momenta by a complex parameter z, follows the resulting one-parameter family of amplitudes into the complex plane, and reconstructs the physical amplitude from the residues at the poles the deformation crosses. There the deformation was the explicit spinor shift $|\hat{i}\rangle = |i\rangle$, $|\hat{j}] = |j]$, $|\hat{i}] = |i] + z|j]$, $|\hat{j}\rangle = |j\rangle - z\,|i\rangle$, and the large-$z$ analysis used the detailed helicity dependence of gauge amplitudes. We now strip that construction to the analytic skeleton that does the work, so that the same logic can be applied to any theory — including ones with no useful Lagrangian at all — and so that the physical role of each axiom is laid bare. The abstraction makes no reference to spinors, gauge, or the specific shift; it asks only what *analytic* structure an amplitude must have for it to be reconstructible from its singularities.

Definition 63.3 (Admissible complex deformation)**.** Let $\mathcal{M}$ be an on-shell scattering amplitude of n massless or massive particles with fixed external quantum numbers, regarded as a function of the on-shell external momenta $\{p_a\}$ subject to $\sum_a p_a = 0$ (all-outgoing convention, Chapter 58). A *complex deformation* of $\mathcal{M}$ is a holomorphic family

$$p_a \;\longmapsto\; \hat{p}_a(z) = p_a + z\, r_a, \qquad z \in \mathbb{C}, \tag{63.10}$$

with fixed complex *shift vectors* r_a chosen so that for all z the deformed momenta remain (i) on shell, $\hat{p}_a(z)^2 = m_a^2$, and (ii) momentum conserving, $\sum_a \hat{p}_a(z) = 0$. Writing $\hat{\mathcal{M}}(z) := \mathcal{M}(\{\hat{p}_a(z)\})$ for the deformed amplitude, the deformation is *admissible* if $\hat{\mathcal{M}}(z)$ is a meromorphic function of the single complex variable z, with $\hat{\mathcal{M}}(0) = \mathcal{M}$ the physical amplitude.

Conditions (i) and (ii) constrain the r_a: expanding (i) gives $p_a \cdot r_a = 0$ and $r_a^2 = 0$ for each shifted leg, and (ii) gives $\sum_a r_a = 0$. The spinor shift of Chapter 59 is one explicit solution of these constraints for two massless legs; the point of Theorem 63.3 is that the recursion never used anything about the shift beyond its being an admissible holomorphic one-parameter family. Meromorphy — the absence of any singularity worse than poles, and of any essential singularity or branch cut in z — is the substantive analyticity hypothesis. For a tree amplitude it is automatic: a tree amplitude is a rational function of the kinematic invariants (a finite sum of products of propagators $1/(P_I^2 - m_I^2)$ and polynomial numerators), and a rational function of momenta that are themselves linear in z is a rational, hence meromorphic, function of z. This is precisely the analyticity reviewed in Section 5.4.1: a function defined by a rational expression extends uniquely to a meromorphic function on the complex plane, and its global behaviour is fixed by its poles and its growth at infinity.

We can now state the constructibility theorem in a form that mentions neither spinors nor a specific shift.

Theorem 63.4 (Abstract on-shell constructibility)**.** *Let $\hat{\mathcal{M}}(z)$ be an admissible complex deformation (Theorem 63.3) of an amplitude $\mathcal{M}$, with the two further properties:*

(P) Only simple poles. *The singularities of $\hat{\mathcal{M}}(z)$ in the finite plane are simple poles, located at the values z_I where some physical factorisation channel goes on shell, i.e. where a partial sum $\hat{P}_I(z)^2 - m_I^2$ vanishes for a subset I of the external legs.*

(B) Vanishing at infinity. *$\hat{\mathcal{M}}(z) \to 0$ as $|z| \to \infty$.*

Then the physical amplitude is the sum of its factorisation residues,

$$\mathcal{M} \;=\; -\sum_I \operatorname{Res}_{z=z_I} \frac{\hat{\mathcal{M}}(z)}{z} \;=\; -\sum_I \frac{1}{z_I}\, \operatorname{Res}_{z=z_I} \hat{\mathcal{M}}(z), \tag{63.11}$$

the sum running over all physical channels I that the deformation puts on shell.

Proof. The argument is a single application of the residue theorem Theorem 5.23 to the auxiliary function $\hat{\mathcal{M}}(z)/z$, mirroring the concrete contour computation Eq. (59.126) but carried at the level of the hypotheses (P) and (B) alone.

Consider $g(z) := \hat{\mathcal{M}}(z)/z$. By (P), $\hat{\mathcal{M}}(z)$ has simple poles at the z_I, none of which sits at the origin (a factorisation channel on shell at $z = 0$ would be a singularity of the *physical* amplitude in a physical channel, which the deformation is set up to avoid; the base point $z = 0$ is generic). Dividing by z adds one further simple pole, at $z = 0$, with residue

$$\operatorname{Res}_{z=0} \frac{\hat{\mathcal{M}}(z)}{z} = \hat{\mathcal{M}}(0) = \mathcal{M}, \tag{63.12}$$

the quantity we want. Integrate g around a circle C_R of radius R centred at the origin, large enough to enclose every z_I. The residue theorem Theorem 5.23 gives

$$\frac{1}{2\pi i}\oint_{C_R} \frac{\hat{\mathcal{M}}(z)}{z}\,dz \;=\; \operatorname{Res}_{z=0}\frac{\hat{\mathcal{M}}(z)}{z} + \sum_I \operatorname{Res}_{z=z_I}\frac{\hat{\mathcal{M}}(z)}{z} \;=\; \mathcal{M} + \sum_I \operatorname{Res}_{z=z_I}\frac{\hat{\mathcal{M}}(z)}{z}. \tag{63.13}$$

Now estimate the left-hand side as $R \to \infty$. Parametrising $z = Re^{i\theta}$,

$$\left|\frac{1}{2\pi i}\oint_{C_R} \frac{\hat{\mathcal{M}}(z)}{z}\,dz\right| = \left|\frac{1}{2\pi}\int_0^{2\pi} \hat{\mathcal{M}}(Re^{i\theta})\,d\theta\right| \;\le\; \max_{|z|=R}|\hat{\mathcal{M}}(z)|, \tag{63.14}$$

since the explicit $1/z$ has cancelled the $dz = iz\,d\theta$ Jacobian, leaving an angular average of $\hat{\mathcal{M}}$ itself. By hypothesis (B), $\max_{|z|=R}|\hat{\mathcal{M}}(z)| \to 0$ as $R \to \infty$, so the contour integral vanishes in the limit. (This is the one place the boundary behaviour enters, and it enters exactly here: the $1/z$ in g is what makes the bound Eq. (63.14) the bare size of $\hat{\mathcal{M}}$ on the arc, so a vanishing amplitude at infinity is precisely the condition that kills the contour. Had we integrated $\hat{\mathcal{M}}(z)$ itself rather than $\hat{\mathcal{M}}(z)/z$, the arc would have contributed a divergent $R\int d\theta$ even for a decaying amplitude.) Taking $R \to \infty$ in Eq. (63.13) therefore yields

$$0 \;=\; \mathcal{M} + \sum_I \operatorname{Res}_{z=z_I}\frac{\hat{\mathcal{M}}(z)}{z}, \tag{63.15}$$

which is the first equality of Eq. (63.11). For the second equality, each pole is simple by (P), so $\operatorname{Res}_{z=z_I}[\hat{\mathcal{M}}(z)/z] = (1/z_I)\operatorname{Res}_{z=z_I}\hat{\mathcal{M}}(z)$; substituting this identity into the first equality gives the second, with the overall minus sign carried through unchanged. This is the abstract content of the BCFW master formula Eq. (59.130). □

Hypothesis (B) is the subtle one, exactly as flagged in Section 59.5.4: if $\hat{\mathcal{M}}(z)$ instead approaches a nonzero constant or grows, the arc in Eq. (63.14)

contributes a *boundary term* $B = \frac{1}{2\pi}\oint \hat{\mathcal{M}}\, d\theta$, and Eq. (63.11) acquires an extra piece $\mathcal{M} = B - \sum_I \mathrm{Res}_{z=z_I}[\hat{\mathcal{M}}/z]$ that is *not* determined by the factorisation residues. Whether B vanishes is a property of the deformation and the external data, not of the residue theorem; for gauge and gravity amplitudes the existence of a good shift was established in Section 59.6. The clean statement we keep is the implication: *good ultraviolet behaviour under a complex deformation* $\Rightarrow$ *the amplitude is its own factorisation data.* Constructibility is the conjunction of analyticity (meromorphy, hypothesis P) and a falloff (hypothesis B); when both hold, locality and unitarity, encoded in the residues, generate everything else.

63.2.2 Locality and unitarity *are* the residue data

Theorem 63.4 reduces the amplitude to its residues. The physical claim of this section is that those residues are not free data: their *locations* are dictated by locality and their *values* by unitarity, and once both are fixed the amplitude is too. We make each half precise.

Locality fixes where the poles are. A local quantum field theory is one whose interactions are built from products of fields at a single spacetime point; the consequence for amplitudes, established when the Feynman rules were derived, is that the only singularities of a tree amplitude are poles at vanishing of physical propagator denominators $P_I^2 - m_I^2$, where $P_I = \sum_{a\in I} p_a$ is the total momentum flowing through a subset I of external legs. Equivalently: the amplitude is singular exactly when an internal line can go on shell, and the channels in which it does so are the genuine Mandelstam invariants of subsets of particles — never a "spurious" kinematic combination that does not correspond to a physical intermediate state. This is hypothesis (P) of Theorem 63.4 read physically. Under the deformation Eq. (63.10) the channel invariant becomes

$$\hat{P}_I(z)^2 - m_I^2 = (P_I^2 - m_I^2) + 2z\,(P_I\cdot R_I) + z^2 R_I^2, \qquad R_I = \sum_{a\in I} r_a, \tag{63.16}$$

a polynomial in z whose zeros z_I are the poles the contour crosses. The statement that $\hat{\mathcal{M}}(z)$ has poles *only* at such z_I, and that each is simple, is precisely the statement that the theory is local and that intermediate states are single particles (double poles would signal a non-normalisable or composite intermediate state). So:

Locality = poles occur only in physical Mandelstam channels, and are simple.

(63.17)

Unitarity fixes the residue values. At a simple pole the amplitude factorises. As the internal line of channel I goes on shell, the full amplitude near the pole

is governed by the propagator of that line dressed by the two amplitudes it connects, summed over the on-shell states $\{s\}$ the line can carry:

$$\mathcal{M} \xrightarrow{P_I^2 \to m_I^2} \sum_s \mathcal{M}_L^{(s)} \frac{i}{P_I^2 - m_I^2} \mathcal{M}_R^{(s)} + (\text{regular}), \tag{63.18}$$

where $\mathcal{M}_L^{(s)}, \mathcal{M}_R^{(s)}$ are *on-shell* amplitudes of strictly fewer legs, with the internal line treated as an external particle of state s on each side. The residue of the pole is therefore

$$\operatorname{Res}_{P_I^2 = m_I^2} \mathcal{M} = i \sum_s \mathcal{M}_L^{(s)} \mathcal{M}_R^{(s)}, \tag{63.19}$$

a sum over physical intermediate states of a product of lower-point on-shell amplitudes. That this is forced — that the residue is exactly this product, with this normalisation and this state sum and no extra constant — is unitarity: the factorisation Eq. (63.18) is the on-shell shadow of the completeness relation $\mathbf{1} = \sum_s |s\rangle\langle s|$ inserted on the intermediate line, the same completeness that drives the optical theorem Theorem 14.37. The state sum runs over a *physical* (positive-norm) Hilbert space, so each term is a product of physical amplitudes and the residue inherits the reflection-positivity of that Hilbert space. So:

$$\boxed{\text{Unitarity} \;=\; \text{residues are } i \textstyle\sum_s \mathcal{M}_L^{(s)} \mathcal{M}_R^{(s)}, \text{ normalised and positive.}} \tag{63.20}$$

This is the precise sense in which Theorem 63.4 reconstructs the amplitude from physics rather than from a Lagrangian. The pole locations z_I come from Eq. (63.16) (locality); the residues come from Eq. (63.19) (unitarity); the contour integral assembles them. Feeding the recursion back into itself terminates on three-point amplitudes, which little-group scaling fixes uniquely (Chapter 58). Nothing else is needed.

Physical Insight

The redundancy lesson now has a sharp resolution. The off-shell Feynman expansion carries an enormous amount of gauge-dependent information that cancels in the final answer; the on-shell amplitude carries only the gauge-invariant residue data. Theorem 63.4 says that data is *complete*: locality (where the poles are) plus unitarity (what the residues are) plus good high-energy behaviour (the boundary term vanishes) determine the amplitude with no remaining freedom. Analyticity is the glue — it is what lets a finite amount of local data about the singularities propagate, by Cauchy's theorem, to the value everywhere. The three founding axioms are not checked on the amplitude; they *are* the amplitude.

Worked Example 63.5 (Reading locality and unitarity off the four-point factorisation)**.** To see Eqs. (63.17) and (63.20) act in the smallest nontrivial case,

take a four-point tree amplitude $\mathcal{M}_4(s,t,u)$ of identical scalars with a cubic coupling g, the simplest theory exhibiting factorisation. With $s+t+u=\sum m_a^2$ and a single intermediate scalar of mass M, locality says the only finite-plane singularities are simple poles in the three physical channels s,t,u at $s=M^2$, $t=M^2$, $u=M^2$, and at nothing else. Unitarity Eq. (63.19) fixes each residue to the product of the two three-point amplitudes glued there. The three-point amplitude of three scalars with coupling g is just $\mathcal{M}_3 = ig$ (no kinematic dependence for scalars), so the s-channel residue is

$$\mathrm{Res}_{s=M^2}\,\mathcal{M}_4 = i\,\mathcal{M}_3\,\mathcal{M}_3 = i\,(ig)(ig) = -ig^2, \tag{63.21}$$

and likewise in t and u. The unique function with exactly these simple poles and these residues, vanishing at large invariants (hypothesis B for this theory), is

$$\mathcal{M}_4 = (-ig^2)\left(\frac{1}{s-M^2}+\frac{1}{t-M^2}+\frac{1}{u-M^2}\right), \tag{63.22}$$

which is exactly the sum of the three exchange Feynman diagrams — reproduced here with no Feynman rules invoked, only the pole locations (locality) and the residues (unitarity). The worked point is that the diagrammatic answer was *over*-determined: the three numbers Eq. (63.21) and the falloff already fix Eq. (63.22) uniquely by Theorem 63.4. The Lagrangian's role was only to supply the seed $\mathcal{M}_3 = ig$; analyticity and unitarity did the rest. (For amplitudes with spin, the same logic applies with the state sum in Eq. (63.19) running over the internal helicities, exactly the inner sum $\sum_h$ of the BCFW formula Eq. (59.130).)

63.2.3 The same logic at loop level: integrands from cuts

Trees are rational; their only singularities are poles, and Theorem 63.4 handles poles. Loop amplitudes are not rational — they have branch cuts, the analytic signature of intermediate states going on shell over a continuous range of phase space — so one might expect on-shell constructibility to stop at tree level. It does not. The branch cuts are themselves fixed by unitarity, through the optical theorem, and the loop *integrand* (the function under the loop-momentum integral, before integrating) is again rational and therefore again reconstructible from its poles. The two facts together mean that even loop-level QFT is determined by on-shell tree data. We assemble the statement; the detailed machinery was built in Chapter 60 and we import rather than rebuild it.

Cuts fix the discontinuity. The generalised optical theorem Theorem 14.37, $2\,\mathrm{Im}\,\mathcal{M}(i\to i) = \sum_f \int d\Pi_f\,|\mathcal{M}(i\to f)|^2$, says the imaginary (absorptive) part of a forward amplitude is a positive sum over physical intermediate states. Promoted off the forward direction and made local in the loop integrand, this is the Cutkosky rule Theorem 60.1: the discontinuity of a loop amplitude across

the branch cut opening at a physical threshold $s = P_I^2$ is computed by replacing the cut propagators with their on-shell delta functions,

$$\frac{i}{p^2 - m^2 + i\epsilon} \;\longrightarrow\; 2\pi\,\delta^{(+)}(p^2 - m^2), \tag{63.23}$$

(the on-shell condition with positive energy, Eq. (60.10)), and gluing the on-shell tree amplitudes on either side of the cut, integrated over the intermediate phase space. The discontinuity of the loop amplitude is thus a phase-space integral of a product of lower-point on-shell amplitudes:

$$\mathrm{disc}_s\,\mathcal{M} \;=\; i \sum_{\text{states}} \int d\Pi\; \mathcal{M}_L\,\mathcal{M}_R\,, \tag{63.24}$$

the loop-level analogue of the tree residue Eq. (63.19), with the discrete state sum of factorisation replaced by the continuous phase-space integral of the optical theorem. The right-hand side involves only on-shell tree amplitudes; unitarity has again expressed the new analytic structure (the cut) in terms of lower data. The positivity of Theorem 14.37 — $\mathrm{Im}\,\mathcal{M}(i \to i) \geq 0$ — rides along: the cut is not merely computable but *sign-definite* in the forward limit, which is the lever the next section will pull.

Cuts fix the integrand. A discontinuity alone does not fix a function; one also needs its analytic continuation off the cut, supplied by a dispersion relation (Theorem 63.6 below). But there is a sharper statement at the integrand level. By the basis theorem of Chapter 60, a one-loop amplitude in dimensional regularisation is a fixed combination of scalar box, triangle, bubble and tadpole master integrals; the only unknowns are the rational *coefficients*. Each coefficient is isolated by a *generalised cut* (Theorem 60.17) that puts several propagators on shell at once, freezing the loop momentum to a discrete set of solutions. The quadruple cut of Theorem 60.13 freezes the four-dimensional loop momentum completely and computes a box coefficient as the product of the four on-shell tree amplitudes sitting at the corners of the box, averaged over the two solutions of the cut equations; the triple and double cuts (with the OPP integrand decomposition Theorem 60.18) peel off the triangle and bubble coefficients in turn. Every coefficient is a product of on-shell trees. Hence:

$$\boxed{\text{one-loop QFT} \;=\; (\text{scalar master integrals}) \;\times\; (\text{coefficients made of on-shell trees}).} \tag{63.25}$$

The integrand is reconstructed from its multiple cuts exactly as a tree is reconstructed from its poles; the master integrals are universal, theory-independent, and computed once and for all (Chapter 60, Appendix D). The dynamics — everything that distinguishes one theory from another at one loop — lives entirely in the cut tree amplitudes. This is generalised unitarity, and it is the loop-level

form of the section's thesis: analyticity (the cut structure, the master-integral basis) plus unitarity (the cuts equal products of trees) reconstruct the amplitude, with the Lagrangian appearing only through the trees it seeds.

Worked Example 63.6 (Reconstructing a one-loop amplitude from its cut by dispersion)**.** The cleanest way to see analyticity-plus-unitarity build a non-rational function is to carry the dispersive reconstruction abstractly, which also previews the machinery of Section 63.3. Suppose a one-loop amplitude $\mathcal{M}(s)$, at fixed other invariants, is analytic in the cut s-plane with a single branch cut along $s \geq s_0$ (the two-particle threshold) and is bounded so that $\mathcal{M}(s)/s \to 0$ at infinity. Cauchy's theorem on a contour hugging the cut — the analytic-continuation argument of Section 5.4.1 — gives the once-subtracted dispersion relation

$$\mathcal{M}(s) = \mathcal{M}(0) + \frac{s}{\pi}\int_{s_0}^{\infty} \frac{ds'}{s'}\,\frac{\operatorname{Im}\mathcal{M}(s')}{s'-s-i\epsilon}, \tag{63.26}$$

in which the entire s-dependence is generated by the integral over the imaginary part. Now feed in unitarity: by Eq. (63.24), $\operatorname{Im}\mathcal{M}(s') = \frac{1}{2}\operatorname{disc}_s \mathcal{M}(s')$ is a phase-space integral of on-shell trees. For the simplest case — a scalar bubble from two identical particles of mass m exchanged with coupling giving a constant tree $\mathcal{M}_L = \mathcal{M}_R = -ig$ — the two-body phase space in $d = 4$ is $\int d\Pi_2 = \frac{1}{8\pi}\sqrt{1-4m^2/s'}\,\theta(s'-4m^2)$, so the cut is

$$\operatorname{Im}\mathcal{M}(s') = \frac{g^2}{16\pi}\sqrt{1-\frac{4m^2}{s'}}\;\theta(s'-4m^2), \tag{63.27}$$

with $s_0 = 4m^2$. Inserting Eq. (63.27) into Eq. (63.26) reproduces the full one-loop bubble: the integral is elementary and yields the standard $\propto g^2\big[2-\beta\log\frac{\beta+1}{\beta-1}\big]$ form with $\beta = \sqrt{1-4m^2/s}$, the same function the explicit Feynman-parameter integral of Chapter 60 returns, including its logarithmic branch cut. The lesson: the *only* dynamical input was the cut Eq. (63.27) — a product of on-shell trees times phase space — while analyticity (the dispersion relation Eq. (63.26)) supplied the continuation to all s. The subtraction constant $\mathcal{M}(0)$ is the lone piece not fixed by the cut; it is a low-energy datum, and it is exactly the kind of constant the positivity arguments of the next section will pin down by sign.

The picture is now uniform across the loop expansion. Trees are fixed by their poles; loop integrands by their multiple cuts; loop amplitudes by dispersion from their single cuts. At every order the new analytic structure is unitarity wearing a kinematic disguise, and analyticity is the principle that turns local singularity data into the global function. We have promoted two of the founding axioms from properties to constructions and identified their physical content with exactly the residue and cut data.

This raises the question that organises the rest of the chapter. If analyticity and unitarity are strong enough to *reconstruct* an amplitude we could in principle

compute by other means, then the same axioms must place rigorous *constraints* on theories we cannot fully solve — bounding their low-energy data from the sole requirement that some consistent, analytic, unitary completion exists. Turning the reconstruction into a quantitative bound is the business of Section 63.3, where the positivity that has so far ridden quietly along inside every cut becomes the active lever, and ϕ^4 theory meets the S-matrix bootstrap.

63.3 Positivity Bounds: The S-Matrix Bootstrap from Below

The previous section argued that analyticity and unitarity are not merely properties an amplitude happens to possess but principles strong enough to *construct* it: a tree amplitude is the unique rational function with the prescribed factorization residues, a loop integrand the unique object with the prescribed unitarity cuts. We now press that same pair of principles in a different direction. Rather than rebuild a known amplitude, we ask what analyticity and unitarity, taken as inputs, *forbid*. The answer is sharp and quantitative, and it is the chapter's gold-standard worked anchor: the coefficient of the leading higher-derivative operator in a low-energy effective field theory is not a free parameter. It must be positive. A measured infrared number is forced, by the bare existence of a unitary and causal completion at high energy, to equal a manifestly positive integral over the spectrum of that completion. This is the S-matrix bootstrap pointed not at the spectrum above but at the effective theory below — the bootstrap from below — and it is where ϕ^4 theory meets the amplitudes frontier.

The logic is a closed loop of four ingredients, each already established in this book. Crossing symmetry (Section 18.4) relates the amplitude in different channels. The optical theorem (Theorem 14.37) makes the imaginary part of the forward amplitude a total cross section, hence nonnegative. Analyticity (Section 5.4.1) and the residue theorem (Theorem 5.23) let us deform a contour from the low-energy region, where the amplitude is the effective theory's polynomial in the kinematic invariants, out onto the high-energy cuts, where it is the absorptive spectral data. Polynomial boundedness — the Froissart–Martin bound — kills the arc at infinity that would otherwise spoil the deformation. Assemble these and a single low-energy Wilson coefficient is pinned to a positive spectral integral. We carry every step.

63.3.1 The effective theory and its leading irrelevant coefficient

We continue ϕ^4 theory: a single real scalar ϕ of mass m, carrying the $\mathbb{Z}_2$ symmetry $\phi \to -\phi$ fixed for this thread, whose first interacting amplitude was the tree-level constant $\mathcal{M} = -\lambda$ of Theorem 14.35. At low energies the physics of ϕ is organised by the Wilsonian effective Lagrangian of Chapter 23: an infinite tower of local

operators built from ϕ and its derivatives, ordered by relevance. The marginal interaction is the ϕ^4 contact term with coupling λ; above it sit the irrelevant operators classified in Section 23.5.4[1], suppressed by powers of a heavy scale M at which the effective description breaks down. The leading irrelevant operator that feeds the elastic $2 \to 2$ amplitude with a genuinely new structure — one not already produced by λ and by the kinematics of the marginal vertex — is the four-derivative, four-field operator

$$\mathcal{L} \supset \frac{c}{4}\,(\partial_\mu\phi\,\partial^\mu\phi)\,(\partial_\nu\phi\,\partial^\nu\phi), \qquad [c] = (\text{mass})^{-4}, \tag{63.28}$$

whose dimensionless Wilson coefficient is $c\,M^4$. It is the carried scalar's leading $(\partial\phi)^4$-type irrelevant operator. The claim of this section, made precise below, is that $c > 0$ for any ϕ that arises as the low-energy limit of a unitary, causal, Lorentz-invariant theory.

The amplitude and its low-energy expansion. Write the elastic amplitude for $\phi(p_1)\phi(p_2) \to \phi(p_3)\phi(p_4)$ as $\mathcal{M}(s,t)$, with the Mandelstam invariants

$$s = (p_1{+}p_2)^2, \qquad t = (p_1{-}p_3)^2, \qquad u = (p_1{-}p_4)^2, \qquad s{+}t{+}u = 4m^2, \tag{63.29}$$

the constraint following from $p_i^2 = m^2$ and momentum conservation. Because the four external particles are identical, $\mathcal{M}$ is symmetric under every permutation of s,t,u; in particular it is invariant under the $s \leftrightarrow u$ crossing of Section 18.4, the exchange of the second incoming leg with the second outgoing leg. At low energy, $|s|, |t| \ll M^2$, the amplitude is analytic in the invariants and admits a Taylor expansion. In the forward direction $t = 0$, where $u = 4m^2 - s$, the crossing symmetry $s \leftrightarrow u$ becomes the reflection $s \mapsto 4m^2 - s$ about the point $s = 2m^2$, so the forward amplitude is an even function of $s - 2m^2$:

$$\mathcal{M}(s,0) = a_0 + c\,(s{-}2m^2)^2 + O\big((s{-}2m^2)^4\big), \qquad c \equiv \tfrac{1}{2}\,\partial_s^2\,\mathcal{M}(s,0)\big|_{s=2m^2}. \tag{63.30}$$

The constant a_0 collects the marginal contact $-\lambda$ together with finite renormalizations; the linear term is absent by the crossing reflection. The coefficient c of the curvature is, up to corrections suppressed by m^2/M^2, exactly the Wilson coefficient of Eq. (63.28), as we now confirm by computing the contact contribution directly.

The contact vertex. The operator Eq. (63.28) produces a four-point contact vertex whose Feynman rule we extract by the standard count. Write the operator as $\frac{c}{4}X^2$ with $X = \partial_\mu\phi\,\partial^\mu\phi$. To form the $2 \to 2$ vertex we functionally differentiate $\frac{c}{4}X^2$ with respect to the four external fields, assigning each external leg to one of

[1] The classification of operators into relevant, marginal, and irrelevant by their scaling dimension is Theorem 23.6; we use it here as the organising principle of the derivative expansion.

the four $\partial\phi$ factors. Take all momenta incoming, $p_1, \dots, p_4$, so that each derivative ∂_μ acting on the leg of incoming momentum p contributes $-ip_\mu$. The two factors inside each X are Lorentz-contracted, so a given assignment contributes the product of two dot products, one from each X. There are $4! = 24$ assignments of the labelled legs to the four (distinguishable-by-position) derivative slots, and they fall into the three ways of partitioning $\{1, 2, 3, 4\}$ into two unordered pairs, each partition reached by $24/3 = 8$ assignments. Summing, with the prefactor $\frac{c}{4}$ and the two factors of $-i$ per pair squaring to $i^2 \cdot i^2 = +1$ overall on the two momenta of each dot product,

$$i\mathcal{M}_{\text{contact}} = i\,\frac{c}{4} \cdot 8 \left[(p_1{\cdot}p_2)(p_3{\cdot}p_4) + (p_1{\cdot}p_3)(p_2{\cdot}p_4) + (p_1{\cdot}p_4)(p_2{\cdot}p_3)\right], \quad (63.31)$$

so $\mathcal{M}_{\text{contact}} = 2c\,[(p_1{\cdot}p_2)(p_3{\cdot}p_4) + \text{two terms}]$. Continue to the physical configuration with p_3, p_4 outgoing (i.e. replace the all-incoming p_3, p_4 by their negatives). The massless evaluation, which suffices for the leading coefficient, uses $2p_1{\cdot}p_2 = s$, $2p_3{\cdot}p_4 = (p_1 + p_2)^2 = s$, $2p_1{\cdot}p_3 = -t$, $2p_2{\cdot}p_4 = -t$, $2p_1{\cdot}p_4 = -u$, $2p_2{\cdot}p_3 = -u$, where the cross terms follow from $(p_i - p_j)^2 = -2p_i{\cdot}p_j$ on massless legs and momentum conservation. Hence

$$\mathcal{M}_{\text{contact}}(s,t) = 2c\left[\frac{s^2}{4} + \frac{t^2}{4} + \frac{u^2}{4}\right] = \frac{c}{2}\,(s^2 + t^2 + u^2). \quad (63.32)$$

This is manifestly s, t, u-symmetric, as crossing demands. In the forward limit $t = 0$, $u = -s$ (massless),

$$\mathcal{M}_{\text{contact}}(s,0) = \frac{c}{2}(s^2 + s^2) = c\,s^2, \quad (63.33)$$

confirming that c in Eq. (63.28) is precisely the curvature coefficient defined in Eq. (63.30) (in the massless limit $2m^2 \to 0$). The coefficient we will bound is a number an experimentalist could in principle measure by scattering ϕ quanta at energies well below M and reading off the s^2 growth of the forward amplitude. We will now show that this measured number cannot come out negative.

Physical Insight

The reason a four-derivative operator is the leading nontrivial structure is dimensional and symmetric. A non-derivative quartic (ϕ^4) gives a constant amplitude, which sits in a_0; a two-derivative quartic ($\phi^2\partial\phi\partial\phi$) gives a term linear in the invariants, which the $s \leftrightarrow u$ crossing reflection forbids in the forward limit. The first operator that survives crossing *and* produces energy growth is $(\partial\phi)^4$, and its coefficient is the first place the ultraviolet completion leaves an unavoidable, sign-definite fingerprint in the infrared.

63.3.2 Analytic structure of the forward amplitude

To convert Eq. (63.33) into a statement about the high-energy spectrum we treat s as a complex variable, holding $t = 0$ fixed, and use what analyticity, crossing, and unitarity tell us about $\mathcal{M}(s, 0)$ in the cut s-plane. Three facts assemble the picture.

First, *analyticity and the location of the cuts.* As established in Section 5.4.1, the forward amplitude is the boundary value of a function analytic in the complex s-plane away from singularities dictated by unitarity. The discontinuities are the physical thresholds: an s-channel cut where the centre-of-mass energy is large enough to produce on-shell intermediate states, $s \geq 4m^2$, and a u-channel cut where $u \geq 4m^2$. With $t = 0$ and $u = 4m^2 - s$, the latter condition reads $s \leq 0$. Thus $\mathcal{M}(s, 0)$ is analytic in the cut plane

$$s \in \mathbb{C} \setminus \big((-\infty, 0] \cup [4m^2, \infty)\big), \tag{63.34}$$

with a right-hand cut $[4m^2, \infty)$ (the s-channel) and a left-hand cut $(-\infty, 0]$ (the u-channel), and a gap $(0, 4m^2)$ in between containing the low-energy region where the effective theory lives. Real analyticity, $\mathcal{M}(s^*, 0) = \mathcal{M}(s, 0)^*$, holds because the amplitude is real in the gap; consequently across either cut the discontinuity is purely imaginary, $\mathcal{M}(s + i0, 0) - \mathcal{M}(s - i0, 0) = 2i\ \mathrm{Im}\ \mathcal{M}(s + i0, 0)$.

Second, *crossing relates the two cuts.* The $s \leftrightarrow u$ symmetry $\mathcal{M}(s, 0) = \mathcal{M}(4m^2 - s, 0)$ maps the left-hand cut onto the right-hand cut. Combining it with real analyticity gives the precise statement we will need: for s' on the left cut ($s' \leq 0$),

$$\mathrm{Im}\ \mathcal{M}(s' + i0, 0) = \mathrm{Im}\,\big[\mathcal{M}\big((4m^2 - s') + i0, 0\big)\big]^* = -\ \mathrm{Im}\ \mathcal{M}\big((4m^2 - s') + i0, 0\big), \tag{63.35}$$

where $4m^2 - s' \geq 4m^2$ lies on the right cut. The discontinuity on the left cut is thus the mirror image of the right-cut discontinuity; the two cuts carry the same physical information, and the deformation will fold one onto the other.

Third, *the optical theorem supplies positivity.* The generalized optical theorem Theorem 14.37 states that the imaginary part of any forward elastic amplitude equals a sum over all kinematically accessible final states of $|\mathcal{M}(i \to f)|^2$, weighted by Lorentz-invariant phase space, and is therefore nonnegative. In the normalization of Eq. (14.200),

$$\mathrm{Im}\ \mathcal{M}(s, 0) = 2\, p_{\mathrm{cm}} \sqrt{s}\ \sigma_{\mathrm{tot}}(s) \xrightarrow{s \gg m^2} s\, \sigma_{\mathrm{tot}}(s) \ \geq\ 0, \qquad s \geq 4m^2, \tag{63.36}$$

with $\sigma_{\mathrm{tot}} \geq 0$ a total cross section. Equivalently, the partial-wave unitarity inequality Eq. (14.206) forces every partial-wave absorptive part to be nonnegative, and the forward imaginary part is the positive sum of them. This is the engine of the whole argument: the integrand we are about to construct is nonnegative because probabilities are.

Polynomial boundedness: the Froissart–Martin bound. The deformation we have in mind sweeps the contour out to $|s| \to \infty$, and it succeeds only if the amplitude grows slowly enough there for the arc to vanish. The relevant input is the Froissart–Martin bound: in a theory with a mass gap, the forward amplitude is polynomially bounded, growing no faster than

$$|\mathcal{M}(s,0)| \;\le\; C\, s\, \ln^2(s/s_0) \qquad (s \to \infty) \tag{63.37}$$

for some constants C, s_0. In particular $|\mathcal{M}(s,0)|/s^2 \to 0$.

Caution

Polynomial boundedness Eq. (63.37) is the one ingredient we do not derive from scratch, and we therefore adopt it as an *explicit standing hypothesis* of Theorem 63.8 rather than smuggle it in as an unproven import. This is not a weakening: *the very hypothesis under test* — that ϕ has a unitary, local, Lorentz-invariant ultraviolet completion — is exactly what guarantees polynomial boundedness in the first place, so taking it as given is part of what "admits a healthy completion" means. The twice-subtracted dispersion relation below should be read in that light, and the positivity bound as a necessary condition any such completion imposes on the infrared. The heuristic behind the bound is unitarity plus locality: a finite mass gap means correlations decay exponentially with a range set by $1/m$, so the impact parameters that contribute coherently to forward scattering are bounded by $b \lesssim (1/m)\ln(s/s_0)$; the number of contributing partial waves is then $\ell_{\max} \sim p_{\rm cm}\, b \sim \sqrt{s}\,\ln(s/s_0)$, and saturating partial-wave unitarity $|a_\ell| \le 1$ over those waves gives a forward amplitude $\sim s\,\ln^2 s$. The fully rigorous statement — promoting this estimate to a theorem via analyticity of the amplitude in a Lehmann–Martin ellipse in $\cos\theta$ within the axiomatic (Wightman/LSZ) framework — is a classic result we use as a hypothesis and flag in the Further reading for those who want the complete derivation.

We can now picture the deformation. Figure 63.2 shows the cut s-plane, the small contour encircling the low-energy expansion point, and the large arc that the Froissart bound allows us to drop.

63.3.3 The twice-subtracted dispersion relation and the sum rule

The strategy is now visible. The coefficient c is a low-energy datum, accessible as a contour integral of $\mathcal{M}(s,0)$ around the expansion point. The cuts are high-energy data, where the integrand is the positive absorptive part. Cauchy's theorem (Theorem 5.23) connects them. We carry the construction in full in the worked example below; it is the chapter's central computation.

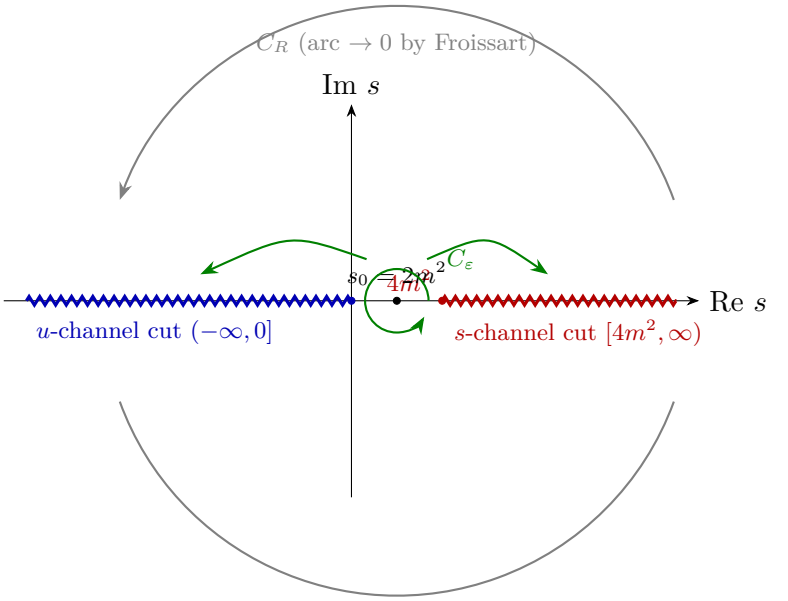


Figure 63.2: The forward amplitude $\mathcal{M}(s,0)$ in the complex s-plane. It is analytic in the gap $(0, 4m^2)$ between the u-channel cut (left, blue) and the s-channel cut (right, red), with the low-energy expansion centred at the crossing-symmetric point $s_0 = 2m^2$. The curvature coefficient c is the residue of the small contour C_ε weighted by $(s-s_0)^{-3}$. Deforming C_ε out to the large circle C_R, the arc vanishes by the Froissart bound Eq. (63.37) and the integral collapses onto the two cuts; crossing folds the left cut onto the right, doubling a single positive integral.

Worked Example 63.7 (The forward dispersive positivity sum rule for the leading EFT coefficient)**.** We prove that the curvature coefficient c of Eq. (63.30) equals a positive spectral integral, hence $c > 0$.

Step 1: c **as a contour integral.** Set $s_0 = 2m^2$ and work in the variable $x = s - s_0$. By Eq. (63.30) the forward amplitude is, near $x = 0$, the even series $\mathcal{M}(s,0) = a_0 + c\,x^2 + O(x^4)$, analytic in the gap. By Cauchy's integral formula for the Taylor coefficients (Theorem 5.23), the curvature is the residue

$$c = \frac{1}{2!}\,\partial_x^2 \mathcal{M}\big|_{x=0} = \frac{1}{2\pi i}\oint_{C_\varepsilon} \frac{dx\;\mathcal{M}(s_0+x,0)}{x^3}, \tag{63.38}$$

where C_ε is a small counterclockwise circle $|x| = \varepsilon$ inside the gap. Only the x^2 term of the expansion survives the x^{-3} weight: $\oint x^{-3}(a_0 + cx^2 + \dots)\,dx = 2\pi i\,c$, since $a_0 x^{-3}$ and the higher even terms integrate to zero. This is the low-energy side of the sum rule, and no high-energy input has entered yet.

Step 2: deform the contour. The integrand $\mathcal{M}(s_0+x,0)/x^3$ is analytic in the annulus $\varepsilon<|x|<R$ except on the two cuts, which in the x variable sit at $x\geq 2m^2$ (right) and $x\leq -2m^2$ (left). Apply Cauchy's theorem to the boundary of the annulus-minus-cuts. The boundary consists of the outer circle C_R (counterclockwise), the inner circle C_ε (clockwise), and two contours hugging the cuts. Setting the total to zero and solving for the inner circle,

$$\frac{1}{2\pi i}\oint_{C_\varepsilon}\frac{\mathcal{M}\,dx}{x^3}=\frac{1}{2\pi i}\oint_{C_R}\frac{\mathcal{M}\,dx}{x^3}-\frac{1}{2\pi i}\int_{\mathrm{rc}}\frac{\operatorname{disc}\mathcal{M}}{x^3}\,dx-\frac{1}{2\pi i}\int_{\mathrm{lc}}\frac{\operatorname{disc}\mathcal{M}}{x^3}\,dx, \tag{63.39}$$

where $\operatorname{disc}\mathcal{M}(x)\equiv\mathcal{M}(s_0+x+i0,0)-\mathcal{M}(s_0+x-i0,0)=2i\ \mathrm{Im}\ \mathcal{M}(s_0+x+i0,0)$.

Step 3: the arc vanishes (and why two subtractions). On C_R, $|x|=R$ and, by the Froissart bound Eq. (63.37), $|\mathcal{M}(s_0+x,0)|\leq C\,R\,\ln^2 R$. The weight is $|x^{-3}|=R^{-3}$, and the circumference is $2\pi R$, so

$$\left|\frac{1}{2\pi i}\oint_{C_R}\frac{\mathcal{M}\,dx}{x^3}\right|\ \leq\ \frac{1}{2\pi}\,(2\pi R)\,\frac{C\,R\,\ln^2 R}{R^3}\ =\ \frac{C\,\ln^2 R}{R}\ \xrightarrow{R\to\infty}\ 0. \tag{63.40}$$

The arc dies. The weight x^{-3} is what makes this work: it carries two more powers of $1/x$ than the x^{-1} a naive Cauchy formula would use, i.e. *two subtractions*. Two are needed for two independent reasons, which happily coincide. (i) *Convergence*: Froissart permits growth up to s^2, and an integrand falling only as $1/x$ or $1/x^2$ could leave a finite or divergent arc; the third power guarantees Eq. (63.40) under the weakest growth assumption we are willing to make. (ii) *Subtraction of the unknowns*: the expansion Eq. (63.30) contains two infrared constants, a_0 (and a possible linear term, absent here by crossing). The x^{-3} projection annihilates both and isolates the curvature c alone. The two subtractions are exactly the price of not knowing the low-energy constants and of allowing the amplitude to grow.

Step 4: the cut integrals, and folding by crossing. With the arc gone, Eq. (63.38) and Eq. (63.39) give

$$c=-\frac{1}{2\pi i}\int_{\mathrm{rc}}\frac{\operatorname{disc}\mathcal{M}}{x^3}\,dx-\frac{1}{2\pi i}\int_{\mathrm{lc}}\frac{\operatorname{disc}\mathcal{M}}{x^3}\,dx. \tag{63.41}$$

Take the cut contours oriented so that each runs above the cut outward and below the cut inward, the orientation inherited from the counterclockwise outer circle; this fixes the sign of disc and, written out with $\operatorname{disc}\mathcal{M}=2i\ \mathrm{Im}\,\mathcal{M}$, turns Eq. (63.41) into

$$c=\frac{1}{\pi}\int_{\mathrm{rc}}\frac{\mathrm{Im}\ \mathcal{M}(s_0+x,0)}{x^3}\,dx+\frac{1}{\pi}\int_{\mathrm{lc}}\frac{\mathrm{Im}\ \mathcal{M}(s_0+x,0)}{x^3}\,dx. \tag{63.42}$$

Convert back to the variable $s=s_0+x$. The right cut is $s\in[4m^2,\infty)$ with $x=s-s_0=s-2m^2>0$. The left cut is $s\in(-\infty,0]$ with $x=s-2m^2<0$;

substitute $s = 4m^2 - \sigma$, so that $\sigma \in [4m^2, \infty)$ runs over the right cut, $ds = -d\sigma$, and $x = s - 2m^2 = 2m^2 - \sigma = -(\sigma - 2m^2)$, whence $x^3 = -(\sigma - 2m^2)^3$. The crossing relation Eq. (63.35) gives $\operatorname{Im}\mathcal{M}(s + i0, 0) = -\operatorname{Im}\mathcal{M}(\sigma + i0, 0)$ on the left cut. Assembling the left-cut piece,

$$\frac{1}{\pi}\int_{\text{lc}} \frac{\operatorname{Im}\mathcal{M}}{x^3}\,ds = \frac{1}{\pi}\int_{\sigma=\infty}^{4m^2} \frac{-\operatorname{Im}\mathcal{M}(\sigma,0)}{-(\sigma-2m^2)^3}\,(-d\sigma) = \frac{1}{\pi}\int_{4m^2}^{\infty} \frac{\operatorname{Im}\mathcal{M}(\sigma,0)}{(\sigma-2m^2)^3}\,d\sigma, \tag{63.43}$$

which is identical to the right-cut integral. The two minus signs — one from the crossing reflection of $\operatorname{Im}\mathcal{M}$, one from the odd power x^3 under $s \to 4m^2 - s$ — conspire to a plus, and the left cut folds exactly onto the right. This is the crossing step: the u-channel absorptive data is the s-channel absorptive data, and the dispersion relation counts it twice.

Step 5: positivity and the sum rule. Adding the two equal contributions,

$$\boxed{c = \frac{2}{\pi}\int_{4m^2}^{\infty} \frac{\operatorname{Im}\mathcal{M}(s,0)}{(s-2m^2)^3}\,ds} \qquad \xrightarrow{m\to 0} \qquad c = \frac{2}{\pi}\int_{0}^{\infty} \frac{\operatorname{Im}\mathcal{M}(s,0)}{s^3}\,ds. \tag{63.44}$$

Every factor on the right is nonnegative. The measure ds is positive; the kernel $(s - 2m^2)^{-3}$ is positive throughout the integration range $s \geq 4m^2$ (so $s - 2m^2 \geq 2m^2 > 0$); and the integrand $\operatorname{Im}\mathcal{M}(s, 0) = 2p_{\text{cm}}\sqrt{s}\,\sigma_{\text{tot}}(s) \geq 0$ by the optical theorem Eq. (63.36). The integral is therefore nonnegative, and it is strictly positive unless $\sigma_{\text{tot}}(s) \equiv 0$ for all $s \geq 4m^2$ — i.e. unless the theory is free, with no scattering at all. For any interacting ϕ,

$$c > 0. \qquad \blacksquare \tag{63.45}$$

The result deserves to be stated as a theorem in its own right, because it is exactly the kind of nonperturbative, completion-independent constraint that the bootstrap philosophy promises.

Theorem 63.8 (Positivity of the leading EFT coefficient)**.** *Let ϕ be a real scalar that arises as the low-energy effective description of a theory satisfying Lorentz invariance, unitarity (the optical theorem Theorem 14.37), analyticity in the cut s-plane Eq. (63.34) with $s \leftrightarrow u$ crossing symmetry, and Froissart polynomial boundedness Eq. (63.37). Then the coefficient c of the leading $(\partial\phi)^4$ operator Eq. (63.28), equivalently the forward curvature Eq. (63.30), satisfies the sum rule Eq. (63.44) and is strictly positive, $c > 0$, unless ϕ is free.*

Proof. Theorem 63.7 carries the argument in full: Cauchy's formula expresses c as the x^{-3} moment of the forward amplitude on a small contour (Step 1); deformation to infinity with the arc killed by Froissart (Steps 2–3) reduces this to the two cut integrals (Step 4); crossing folds the u-channel cut onto the s-channel cut, and the optical theorem makes the resulting integrand nonnegative (Steps 4–5), giving Eq. (63.44) and Eq. (63.45). □

63.3.4 Interpretation: the swampland from below

Equation (63.44) is more than an inequality; it is an identity, and the identity is the lesson. The left-hand side is an infrared quantity: a coefficient in the low-energy expansion of an amplitude, fixed by whatever physics governs ϕ at accessible energies, in principle measurable without any knowledge of the ultraviolet. The right-hand side is an ultraviolet quantity: a weighted integral over the total cross section, i.e. over the spectrum and couplings of the heavy states that complete the theory above M. The dispersion relation welds them together. A number measured in the basement equals a positive census of the attic.

The immediate corollary is a sharp consistency condition on effective field theories. An effective Lagrangian is a list of operators with arbitrary-looking coefficients; nothing in the low-energy theory itself, taken in isolation, prevents one from writing $c < 0$. Theorem 63.8 says that such a choice is nonetheless inconsistent: there exists *no* unitary, causal, Lorentz-invariant ultraviolet completion whose low-energy limit has $c < 0$, because any completion produces, through the sum rule, a nonnegative c. The space of effective theories thus splits in two. On one side, the *landscape*, sit coefficient assignments that some healthy high-energy theory can produce. On the other, a *swampland from below*: theories that are perfectly self-consistent as low-energy effective descriptions — unitary, local, gauge-invariant, renormalizable order by order in $1/M$ — yet can never be ultraviolet-completed. The phrase "from below" distinguishes this from the gravitational swampland program, which constrains effective theories by demanding consistency with quantum gravity from above; here the constraint comes from demanding consistency with ordinary unitary, causal field theory at high energy.

Physical Insight

The physical content of $c > 0$ is causality. A negative c corresponds to a low-energy interaction in which signals propagate slightly faster on a background of ϕ gradients than in vacuum — a superluminal advance, small and energy-suppressed, but genuine. The dispersion relation is the precise statement that such an advance cannot be hidden: the same analyticity that encodes "no signal before its cause" forces the forward amplitude to curve upward, $c > 0$, and forbids the subluminal-violating sign. Positivity bounds are causality bounds wearing the clothes of complex analysis.

For ϕ^4 theory this closes the arc from the amplitudes side. The thread began with a classical Lagrangian (Chapter 6), produced its first quantum amplitude $\mathcal{M} = -\lambda$ (Theorem 14.35), was renormalized and flowed under the renormalization group to its irrelevant operators (Chapter 23), and is now shown

to obey a rigorous frontier constraint: the coefficient of its leading irrelevant $(\partial\phi)^4$ operator is not free but equals the manifestly positive spectral integral Eq. (63.44). The same $16\pi^2$ that controlled the perturbative reach of λ (Theorem 14.38) and set the one-loop β-function (Chapter 24) reappears here as the scale of the cross section that feeds the sum rule. The thread's irrelevant operators, introduced as a bookkeeping device for the derivative expansion, turn out to carry a sign that the ultraviolet cannot forge.

Generalizations: spin and the EFThedron. The scalar bound is the first entry in a much larger structure, and we state its development honestly and dated. Two generalizations dominate the mid-2020s frontier.

The first is to *external states with spin and to subleading coefficients.* Repeating the construction for photons, gravitons, or massive higher-spin states requires tracking little-group indices (Chapter 58) through the dispersion relation, and the single inequality $c > 0$ becomes a system of inequalities on the matrix of forward amplitudes, one for each polarization channel, with positivity now meaning positive-semidefiniteness of that matrix. Going beyond the strictly forward limit ($t = 0$) by expanding the dispersion relation in small t and using full partial-wave unitarity — not only $\mathrm{Im}\, a_\ell \geq 0$ but the Legendre-positivity of the partial-wave expansion — yields two-sided bounds: each Wilson coefficient is confined to a finite interval, not merely a half-line.

The second is the recognition that the full set of such constraints carves out a convex geometry in the space of Wilson coefficients, the *EFThedron.* The sum rules Eq. (63.44) and their t-derivatives express every coefficient as a moment of a single nonnegative spectral density against a fixed kinematic kernel; the allowed coefficients are then precisely the moment sequences of a positive measure supported on the spectrum, and the theory of such moment problems (Hankel-matrix positivity, total positivity of the partial-wave kernels) cuts out a convex region with a rich facet structure. The EFThedron makes the kinship explicit: just as the amplituhedron of Chapter 62 reads an amplitude off a positive geometry, the EFThedron reads the allowed effective theories off a positive geometry in coefficient space. As of the mid-2020s the leading bounds are rigorous and in several cases saturated by known string and weakly-coupled completions, while the complete carving — including the interplay with gravitational completions, where the massless graviton pole modifies the dispersive arcs — remains an active and incomplete program.

These developments share one engine with the scalar computation above and, as the next section will make exact, with the conformal bootstrap of Chapter 49: a crossing equation that equates two channel expansions, a unitarity statement that makes the expansion coefficients nonnegative, and a search for the linear functional that certifies which coefficient values are allowed. Having seen that engine carve the space of effective field theories from below, we now place it

beside the engine that carves the space of conformal field theories and show that the two are not analogous but identical.

63.4 Two Bootstraps, One Logic: S-Matrix and Conformal

The previous section pushed analyticity, crossing, and unitarity until they delivered a *number*: the leading Wilson coefficient of a low-energy scalar effective theory is strictly positive, a fact no Lagrangian was allowed to assume and that no Lagrangian can violate without producing a superluminal or non-unitary completion. That positivity sum rule (Section 63.3) carved a wall through the space of effective theories: on one side sit Wilson coefficients that some healthy ultraviolet theory can produce, on the other side a "swampland from below" of coefficients that look perfectly innocent in the infrared but are forbidden by the existence of *any* consistent completion. The construction was concrete, but its logic was not special to scattering. The same three words — crossing, unitarity, positivity — name the machine that, in Chapters 49 and 50, carved the space of conformal field theories and pinned the critical exponents of the three-dimensional Ising model to many digits. This section makes the parallel exact. It is not an analogy: the conformal bootstrap and the S-matrix bootstrap are the same problem in two kinematic disguises, and AdS/CFT supplies the dictionary that turns one into the other as the curvature is switched off. By the end we will be able to say in a single sentence what a bootstrap *is*, and exhibit the numerical conformal bootstrap and the EFThedron as its two faces.

63.4.1 The common template

Strip both problems to their skeleton and the same object appears. In each case there is a physical quantity — a four-point correlator, a $2 \to 2$ amplitude — that admits an expansion into a discrete or continuous family of *fixed kinematic functions* (conformal blocks; partial waves) weighted by *theory-dependent nonnegative coefficients* (squared OPE coefficients; absorptive partial-wave densities). The same quantity also satisfies a *crossing relation*: the expansion organised around one channel must equal the expansion organised around another. Equating the two expansions and moving everything to one side produces a *sum rule*: a single linear functional equation in the unknown nonnegative weights that must hold identically over a continuum of kinematic points. The bootstrap question is whether any assignment of nonnegative weights to an admissible spectrum can solve the sum rule, and the bootstrap *method* is to test candidate spectra by hunting for a linear functional that is one-signed on every allowed contribution yet has the wrong sign on the known term — a separating hyperplane certifying that no solution exists.

The three structural inputs are therefore: a complete basis of kinematic functions; a crossing equation relating channels; and a positivity statement (the weights are squares, or absorptive parts of a unitary amplitude, hence ≥ 0). Everything else — semidefinite programming, dispersion relations, arcs and moments — is implementation. We now write the two crossing equations and the two positivity conditions explicitly, side by side, so the parallel is on the page rather than in the prose.

63.4.2 Two crossing equations

The conformal crossing equation. For a CFT four-point function of identical scalar primaries ϕ of dimension Δ_ϕ, the operator product expansion in the (12)(34) channel and in the (14)(23) channel must agree. This is the crossing equation derived in Theorem 49.13, which we recall verbatim:

$$v^{\Delta_\phi}\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(u,v) \;=\; u^{\Delta_\phi}\sum_{\mathcal{O}} p_{\mathcal{O}}\, G_{\Delta_{\mathcal{O}},\ell_{\mathcal{O}}}(v,u), \qquad p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \;\geq 0, \tag{63.46}$$

with $G_{\Delta,\ell}$ the conformal blocks (fixed functions of the cross-ratios u, v, determined by conformal symmetry alone) and the sum running over primaries $\mathcal{O}$ exchanged in the $\phi \times \phi$ OPE. The kinematic functions are the blocks; the nonnegative weights are the squared OPE coefficients $p_{\mathcal{O}}$; the crossing relation is the $u \leftrightarrow v$ ($s \leftrightarrow t$ channel) swap. We imported Eq. (63.46) from Eq. (49.71) and will not reprove it.

The S-matrix crossing equation. For the elastic scattering of identical scalars of mass m — the massive avatar of the ϕ^4 theory quantum ϕ — the relativistic amplitude $\mathcal{M}(s,t,u)$ depends on the Mandelstam invariants with $s+t+u = 4m^2$, and crossing symmetry (Chapters 14 and 18) makes it a single analytic function symmetric under permutations of s, t, u: the same function describes the s-channel process $\phi\phi \to \phi\phi$, the t-channel, and the u-channel. Analyticity in s at fixed t, with the branch cuts demanded by unitarity, lets us write a dispersion relation. Carrying the contour around the s- and u-channel cuts (the construction of Section 63.3) gives, after the two subtractions needed for a ϕ^4-type theory,

$$\mathcal{M}(s,t) \;=\; c_0(t)+c_1(t)\,s + \frac{s^2}{\pi}\int_{4m^2}^{\infty}\frac{ds'}{s'^2}\left[\frac{\operatorname{Im}\mathcal{M}(s',t)}{s'-s}+\frac{\operatorname{Im}\mathcal{M}(s',t)}{s'-u}\right], \qquad u = 4m^2-s-t, \tag{63.47}$$

where the s'-integral runs over the physical region and $\operatorname{Im}\mathcal{M}$ is the absorptive part. The "crossing" is built into the symmetric appearance of the two propagator kernels $1/(s'-s)$ and $1/(s'-u)$: the same spectral function feeds the s- and the u-channel, exactly as the same OPE data $p_{\mathcal{O}}$ feeds both channels of Eq. (63.46). The kinematic functions are now the dispersive kernels $1/(s'-s)$ paired with

the partial-wave projectors; the nonnegative weights are the absorptive densities $\operatorname{Im}\mathcal{M}(s',t)$; the crossing relation is $s \leftrightarrow u$ symmetry of the kernel together with the full s,t,u symmetry of $\mathcal{M}$.

The parallel made literal. The structural correspondence is summarised in Table 63.1. Read the two columns as one construction: a positive sum over fixed kinematic objects, constrained by a channel-swap identity.

Ingredient	**Conformal bootstrap**	**S-matrix bootstrap**
Physical datum	four-point function $\langle\phi\phi\phi\phi\rangle$	$2 \to 2$ amplitude $\mathcal{M}(s,t,u)$
Kinematic basis (fixed by symmetry)	conformal blocks $G_{\Delta,\ell}(u,v)$	partial waves $P_\ell(\cos\theta)$ / dispersive kernels
Theory data (nonnegative weights)	squared OPE coefficients $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}} \geq 0$	absorptive densities $\operatorname{Im} f_\ell(s) \geq 0$
Spectrum constraint	unitarity bounds on (Δ,ℓ), Eq. (49.81)	mass gap $s \geq 4m^2$, spins $\ell \in \mathbb{Z}_{\geq 0}$
Crossing relation	$u \leftrightarrow v$ (s/t channel), Eq. (63.46)	$s \leftrightarrow t \leftrightarrow u$, Eq. (63.47)
Positivity lever	$p_{\mathcal{O}} \geq 0$, Section 49.3.3	$\operatorname{Im} a_\ell \geq \lvert a_\ell\rvert^2 \geq 0$, Eq. (14.206)
Method	linear functional / SDP, Theorem 49.21, Section 50.2	dispersive sum rules / EFThedron, Section 63.3

Table 63.1: The conformal and S-matrix bootstraps as one construction. Each row pairs a structural role with its realisation in the two problems. The shared spine is: a positive sum over symmetry-fixed functions, constrained by a channel-crossing identity, probed by a one-signed linear functional.

63.4.3 Unitarity is positivity in both

The reason both problems yield rigorous numbers rather than mere consistency checks is that unitarity, in each, becomes a *positivity* statement on the weights. We state both precisely.

CFT: positive OPE squares. In Eq. (63.46) the weight of each exchanged primary is $p_{\mathcal{O}} = C^2_{\phi\phi\mathcal{O}}$, the square of a real OPE coefficient. It is a square because the double OPE sum collapses onto the diagonal (the same constant

$C_{\phi\phi\mathcal{O}}$ fuses $\phi\phi$ in both pairs of identical operators), and it is real because the radial-quantisation inner product is reflection positive — this is precisely the content of Section 49.3.3. Unitarity also restricts *which* (Δ, ℓ) may appear: every primary obeys the unitarity bounds of Theorem 47.18, recalled in Eq. (49.81) as $\Delta \geq (d-2)/2$ for scalars and $\Delta \geq \ell + d - 2$ for spin $\ell \geq 1$, and only even spins survive the identical-scalar symmetrisation. So unitarity supplies two positivity facts: nonnegative weights, on a spectrum confined to a known region of the (Δ, ℓ) plane.

S-matrix: positive partial-wave densities. The absorptive part $\mathrm{Im}\,\mathcal{M}(s', t)$ entering Eq. (63.47) is not an arbitrary function. Project the amplitude onto angular momentum: in the conventions of Section 14.7.5, $\mathcal{M}$ expands in Legendre polynomials with dimensionless coefficients $a_\ell(s)$, and partial-wave unitarity Eq. (14.206) reads

$$\mathrm{Im}\, a_\ell(s) \;=\; |a_\ell(s)|^2 + \frac{p_{\mathrm{cm}}^2}{4\pi(2\ell+1)}\,\sigma_{\mathrm{inel}}^{(\ell)} \;\geq\; |a_\ell(s)|^2 \;\geq\; 0. \tag{63.48}$$

Hence each partial wave has *nonnegative absorptive part*, $\mathrm{Im}\, a_\ell \geq 0$, and the absorptive density that feeds the dispersion relation is a positive sum over channels,

$$\mathrm{Im}\,\mathcal{M}(s,t)\big|_{t=0} \;=\; \frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}}\sum_{\ell=0}^{\infty}(2\ell{+}1)\;\mathrm{Im}\, a_\ell(s)\,P_\ell(1) \;=\; \frac{8\pi\sqrt{s}}{p_{\mathrm{cm}}}\sum_{\ell}(2\ell{+}1)\;\mathrm{Im}\, a_\ell(s) \;\geq\; 0, \tag{63.49}$$

using $P_\ell(1) = 1$. This is the exact S-matrix analogue of $p_{\mathcal{O}} \geq 0$. The discrete index ℓ here is the spin, and the analogue of the CFT unitarity bound on Δ is the spectral support condition $s' \geq 4m^2$ (the absorptive part vanishes below the two-particle threshold) together with the integrality and nonnegativity of ℓ. The Argand-circle geometry of Eq. (14.207) — each a_ℓ confined to the disk of radius $\frac{1}{2}$ centred at $\frac{i}{2}$ — is the S-matrix counterpart of the wedge $p_{\mathcal{O}} \geq 0$ in the bootstrap: the same convex constraint that the functional method will exploit.

The dictionary in Table 63.1 is thus more than suggestive. The CFT exchanges a tower of operators labelled (Δ, ℓ) with positive squared couplings; the amplitude exchanges a spectrum of states labelled by their mass $\sqrt{s'}$ and spin ℓ with positive absorptive residues; in both the positivity is the statement that probabilities are nonnegative, read off the same Cauchy–Schwarz / reflection-positivity inequality. We record the conceptual identification.

Physical Insight

The conformal block expansion and the partial-wave/dispersive expansion are the same mathematical object: a resolution of the identity on a unitary Hilbert space, written in a basis of states of definite quantum numbers, with

the squared overlaps as nonnegative coefficients. "Unitarity as positivity" is not a slogan but an equality of two inner products: $C^2_{\phi\phi\mathcal{O}}$ is $|\langle\mathcal{O}|\phi\,\phi\rangle|^2$ in radial quantisation, and $\operatorname{Im} a_\ell$ is, by the optical theorem, the squared norm of the outgoing state in channel ℓ. The bootstrap is the demand that a positive combination of such overlaps reproduce itself under a change of channel.

63.4.4 The linear functional: one machine, two problems

Both sum rules are linear and homogeneous in the unknown nonnegative weights, with one fixed inhomogeneous term (the identity contribution in the CFT; the low-energy/EFT contribution in the S-matrix). This is exactly the setting of linear programming duality, and the bound machinery is identical in the two problems. We state it once, abstractly, then read it in both directions.

Write the sum rule as $\sum_i w_i\, V_i = V_0$ with $w_i \geq 0$, where the V_i are vectors in some linear space (in practice, the values and derivatives of the crossing functions at a base point), the V_0 is the fixed term, and i ranges over the admissible spectrum. A solution exists only if V_0 lies in the convex cone generated by the $\{V_i\}$. The contrapositive is the engine: if there is a linear functional α with

$$\alpha[V_i] \;\geq\; 0 \quad \text{for every admissible } i, \qquad \text{but} \qquad \alpha[V_0] \;<\; 0, \tag{63.50}$$

then no nonnegative combination of the V_i can equal V_0 — applying α to a putative solution gives $0 \leq \sum_i w_i\, \alpha[V_i] = \alpha[V_0] < 0$, a contradiction. Such an α is a *separating hyperplane* between V_0 and the cone (Fig. 63.3); finding the optimal one is a semidefinite program. This is the linear-functional principle of Theorem 49.21, the computational core of Section 50.2, and — in the S-matrix language of moments and arcs — the duality that produced the positivity bound of Section 63.3.

Worked Example 63.9 (A finite toy bootstrap and its dual functional)**.** To see that the exclusion logic is genuinely one logic, reduce both problems to their smallest nontrivial common form and solve it by hand. Take a sum rule with two "vectors" in the plane $\mathbb{R}^2$,

$$w_1 V_1 + w_2 V_2 = V_0, \qquad w_1, w_2 \geq 0, \qquad V_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad V_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad V_0 = \begin{pmatrix} 1 \\ -b \end{pmatrix}, \tag{63.51}$$

with b a real parameter standing in for "how much of the wrong-sign behaviour the fixed term demands." In the CFT reading, the two components are the value and one derivative of the crossing function at the symmetric point, V_1, V_2 are the crossing-vectors of two exchanged operators, and V_0 is the (negative of the) identity vector. In the S-matrix reading, the two components are two moments

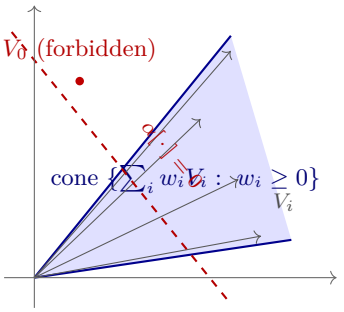


Figure 63.3: The linear-functional principle, common to both bootstraps. The admissible contributions V_i (grey) generate a convex cone (blue). A candidate spectrum is excluded when the fixed term V_0 (red) lies outside the cone; a separating functional α (dashed line $\alpha[\cdot] = 0$) is then nonnegative on every V_i but negative on V_0, certifying Eq. (63.50). In the CFT the V_i are crossing-vectors of blocks at fixed (Δ, ℓ) and V_0 is the identity; in the S-matrix the V_i are partial-wave moments and V_0 is the EFT coefficient to be bounded.

$\int ds'\, \rho(s')\, s'^{-n}$ $(n = 2, 3)$ of the positive spectral density, V_1, V_2 are the moment vectors of two narrow resonances, and V_0 encodes the low-energy coefficients.

Solve directly. Writing components, $w_1 + w_2 = 1$ and $w_2 = -b$. A nonnegative solution needs $w_2 = -b \geq 0$ and $w_1 = 1 + b \geq 0$, i.e. $-1 \leq b \leq 0$. For $b > 0$ no nonnegative combination exists: the sum rule is infeasible.

Solve by the dual functional. Seek $\alpha = (\alpha_1, \alpha_2)$ with $\alpha \cdot V_1 \geq 0$, $\alpha \cdot V_2 \geq 0$ but $\alpha \cdot V_0 < 0$. The first two give $\alpha_1 \geq 0$ and $\alpha_1 + \alpha_2 \geq 0$; the third gives $\alpha_1 - b\,\alpha_2 < 0$. Choose $\alpha_1 = 0$, $\alpha_2 = 1$: then $\alpha \cdot V_1 = 0 \geq 0$, $\alpha \cdot V_2 = 1 \geq 0$, and $\alpha \cdot V_0 = -b < 0$ for every $b > 0$. The functional $\alpha = (0, 1)$ is the separating hyperplane of Eq. (63.50): it is one-signed on the admissible vectors and negative on the fixed term, certifying infeasibility for all $b > 0$ at a stroke. The boundary $b = 0$ is the bound, exactly the threshold where w_2 hits zero.

The same two lines of algebra are, in the full problem, the statement that a single derivative of the crossing equation excludes a CFT with too large a gap, and that a single dispersive moment forces the leading EFT Wilson coefficient to be nonnegative. Increasing the dimension of the vector space (N derivatives of crossing; N moments/arcs) tightens both bounds toward the true boundary of theory space, and the optimisation over α is the semidefinite program in both cases. The machine does not know whether it is bounding an operator dimension or a Wilson coefficient.

63.4.5 The flat-space limit: from the boundary correlator to the bulk S-matrix

The two bootstraps are not merely structurally identical; they are *physically* connected, because a scattering amplitude in flat space is a limit of a CFT correlator. The bridge is AdS/CFT. A CFT in d dimensions with a holographic dual is, by the GKP–Witten relation Eq. (53.162), computed from a gravitational (or, in a weakly coupled limit, ordinary field-theoretic) theory in AdS_{d+1} of radius R. A four-point CFT correlator is then a sum of Witten diagrams (Section 54.3.5); its natural representation is the *Mellin amplitude* $M(s,t)$, the function whose poles in the Mellin variables s,t sit at the twist of exchanged operators and which plays, for AdS processes, exactly the role the flat-space amplitude $\mathcal{M}(s,t)$ plays for scattering. The flat-space limit is the statement that as the AdS radius is sent to infinity — so that the curvature scale recedes far beyond the wavelengths of the scattering quanta — the Mellin amplitude becomes the bulk S-matrix.

The scaling argument. We give the scaling honestly rather than quoting the transform. The dictionary fixes the bulk mass of a field dual to a scalar primary of dimension Δ through $m^2R^2 = \Delta(\Delta - d)$, derived in Section 53.4.4 (the mass–dimension relation), so for a heavy field

$$\Delta \simeq mR \qquad (mR \gg 1). \tag{63.52}$$

A bulk scattering process at centre-of-mass energy $\sqrt{s_{\mathrm{flat}}}$ involves quanta of wavelength $\sim s_{\mathrm{flat}}^{-1/2}$. For this to "fit" in a region small compared with the AdS curvature scale R — the region in which AdS_{d+1} is indistinguishable from flat space — we need $s_{\mathrm{flat}}R^2 \gg 1$. The Mellin variables, on the other hand, are conjugate to the boundary OPE data and carry dimension of (mass $\times R$)^2, i.e. they are measured in units of $1/R^2$: dimensionally the Mellin s is the AdS analogue of R^2 times a physical invariant. Matching the two descriptions of the same invariant in the overlap region $1 \ll s_{\mathrm{flat}}R^2$ forces the linear relation

$$s_{\mathrm{Mellin}} \;\simeq\; R^2\, s_{\mathrm{flat}}, \qquad t_{\mathrm{Mellin}} \;\simeq\; R^2\, t_{\mathrm{flat}}, \tag{63.53}$$

up to a fixed numerical constant set by conventions. This is the heart of the matter: the Mellin amplitude, regarded as a function of variables that grow like R^2, must — for the bulk theory to be local and Lorentz invariant at short distance — reduce to a function of the flat Mandelstam invariants $s_{\mathrm{flat}} = s_{\mathrm{Mellin}}/R^2$.

The transform and its saddle. Making this precise requires resumming the tower of large-Δ exchanges into the flat amplitude. For four identical scalars of dimension Δ_ϕ in AdS_{d+1}, the flat-space limit formula (Penedones) is the

inverse-Borel-type transform

$$\mathcal{M}(s_{\rm flat}, t_{\rm flat}) \;=\; \mathcal{N} \lim_{R\to\infty} R^{\#} \int_{-i\infty}^{+i\infty} \frac{d\alpha}{2\pi i}\, e^{\alpha}\, \alpha^{\,d/2-2\Delta_\phi}\, M\!\left(s_{\rm Mellin} = \frac{R^2 s_{\rm flat}}{2\alpha},\, t_{\rm Mellin} = \frac{R^2 t_{\rm flat}}{2\alpha}\right), \tag{63.54}$$

with $\mathcal{N}$ and the power $R^{\#}$ ($\# = \frac{(d+1)}{2} - \sum_i \Delta_i + \frac{(d-1)}{2}$ for the four-point case, fixed by dimensional analysis) collecting the conventional normalisations. The auxiliary integration variable α is the Borel/Schwinger parameter that undoes the large-dimension Gamma functions in the Mellin measure; its role is to convert the discrete sum over the AdS exchange tower into the continuous spectral integral of a flat amplitude. The scaling Eq. (63.53) is visible directly in the argument of M: the Mellin variables enter as $R^2 s_{\rm flat}/(2\alpha)$, so at fixed α they are exactly the R^2-enhanced invariants anticipated above.

The limit is controlled by a saddle point in α. As $R \to \infty$ the Mellin amplitude probes large arguments; for a contact-like or polynomial growth $M \sim (s_{\rm Mellin})^n$ the integrand behaves as $e^{\alpha}\, \alpha^{\,d/2-2\Delta_\phi-n}\, (R^2 s_{\rm flat}/2)^n$, and the α-integral is a Hankel/Gamma-function representation whose steepest-descent saddle sits at $\alpha_* \simeq 2\Delta_\phi + n - d/2$, an $O(1)$ value. Evaluating at the saddle pulls the factor $(R^2 s_{\rm flat})^n$ out front, where it is cancelled by $R^{\#}$ and the normalisation, leaving a finite function of $s_{\rm flat}$ that grows like $s_{\rm flat}^n$ — precisely a flat-space amplitude of the same degree. The same saddle, applied term by term to the OPE/exchange expansion of M, maps each AdS exchange of dimension Δ onto a flat-space pole or partial wave of mass $m \simeq \Delta/R$ via Eq. (63.52), reconstructing the partial-wave expansion of $\mathcal{M}$. The scaling argument is thus not a coincidence of dimensions but the saddle-point evaluation of an explicit transform; we have shown where the R^2 comes from and how the saddle removes it.

Positivity survives the limit. The point of all this for the present section is that *positivity is preserved under Eq. (63.54)*. The CFT data entering M are the OPE coefficients squared, $p_{\mathcal{O}} \geq 0$, and the AdS unitarity bounds on Δ (Theorem 47.18). Under the flat-space map these become exactly the partial-wave positivity $\mathrm{Im}\, a_\ell \geq 0$ (Eq. (63.48)) and the mass-gap support $s' \geq 4m^2$ of the bulk amplitude: a spin-ℓ double-twist operator of dimension $\Delta \simeq 2mR + 2n + \ell$ maps to a two-particle state of invariant mass $\sqrt{s'} \simeq 2m$ and spin ℓ, and its positive squared OPE coefficient is its positive partial-wave density. Crossing maps to crossing: the $u \leftrightarrow v$ symmetry of Eq. (63.46) becomes the $s \leftrightarrow t \leftrightarrow u$ symmetry of $\mathcal{M}$. Therefore the bulk positivity bounds of Section 63.3 are the flat-space shadow of CFT crossing and unitarity, and conversely the conformal bootstrap is a curved-space, finite-R deformation of the S-matrix bootstrap that *regulates the infrared* (AdS is a box) while keeping the same crossing-plus-positivity skeleton. One principle, two radii of curvature.

Physical Insight

AdS is a box that confines the scattering states, replacing the continuous spectrum of momenta by a discrete tower of boundary operators; the flat-space limit is the decompactification of that box. In the box, "unitarity is positivity of the discrete OPE spectrum"; out of the box, it is "positivity of the continuous partial-wave density." The two bootstraps differ only by whether the infrared is regulated by a curvature radius or by a mass gap. Sending $R \to \infty$ interpolates continuously between them, and positivity — the one ingredient that is a statement about probabilities, not about kinematics — rides through the limit untouched.

63.4.6 What a bootstrap is

We can now state the unifying principle that the title promised, with both problems and the holographic bridge in hand.

Physical Insight

A bootstrap is the carving of theory space by crossing plus positivity. A consistent quantum theory — a CFT, an S-matrix, an effective field theory — is a point in a space of *a priori* possible data (operator dimensions and OPE coefficients; masses, spins and couplings; Wilson coefficients). *Crossing* is a set of linear equalities that this data must satisfy because the same observable can be expanded in inequivalent channels. *Positivity* is a set of convex inequalities that the data must satisfy because the underlying Hilbert space is unitary. The conjunction of a linear constraint and a convex cone defines a convex region; the allowed theories are exactly the points of that region, and a separating linear functional Eq. (63.50) certifies which candidate points are excluded. To bootstrap is to compute this region.

This single sentence has two faces that this book has now met from opposite directions. The *numerical conformal bootstrap* (Chapter 50) carves the region in the space of CFT data and reads off, as the boundary of the allowed island, the critical exponents of the three-dimensional Ising model. The *EFThedron* — the convex geometry of allowed Wilson coefficients carved by dispersive positivity (Section 63.3) — carves the region in the space of low-energy effective couplings and reads off, as its boundary, the bounds that any UV-completable EFT must obey. The two are the same convex-geometry problem; the EFThedron is literally a moment problem (a positive spectral density has positive Hankel matrices of moments), and the conformal bootstrap is the same moment problem with conformal blocks in place of monomials. The positive geometry of Chapter 62 is

yet a third face: there the canonical form of a positive region *is* the amplitude, whereas here the positive region *contains* the allowed theories — two uses of the same insistence that physical data lives where positivity allows it.

Worked Example 63.10 (Closing the ϕ^4 theory arc from both ends)**.** The real scalar $\lambda\phi^4$ has been the book's workhorse, carried from its classical Lagrangian (Chapter 6) through quantisation and its first amplitude $-i\lambda$ (Chapter 14), renormalisation and the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ (Chapter 24), to the Wilson–Fisher fixed point in $d = 4 - \epsilon$ (Chapter 52). At the fixed point the theory loses its scale, becomes a CFT, and — in $d = 3$ — *is* the Ising universality class. ϕ^4 theory therefore terminates, on the bootstrap side of Chapters 49 and 50, as a point pinned inside the conformal-bootstrap island: crossing plus OPE positivity fix $\Delta_\sigma, \Delta_\epsilon$ to the measured critical exponents without ever expanding in λ. This is the first face.

The amplitudes side closes the same arc from the opposite end. Away from criticality, the massive $\lambda\phi^4$ theory is a perfectly ordinary EFT, and its $2 \to 2$ amplitude is constrained by the S-matrix bootstrap. The leading interaction $\mathcal{M} = -\lambda + \cdots$ is fixed in sign by the same crossing and unitarity: expanding $\mathcal{M}(s,t)$ in the low-energy region and applying the twice-subtracted dispersion relation Eq. (63.47), the forward curvature g_2 — the coefficient of $\frac{1}{2}(s^2+t^2+u^2)$ in the contact amplitude Eq. (63.32), equivalently the curvature c of Eq. (63.30) and the leading genuinely irrelevant Wilson coefficient — is, by Section 63.3,

$$g_2 \;=\; \frac{2}{\pi}\int_{4m^2}^{\infty}\frac{ds'}{s'^3}\,\mathrm{Im}\,\mathcal{M}(s',0) \;=\; \frac{2}{\pi}\int_{4m^2}^{\infty}\frac{ds'}{s'^3}\,\frac{8\pi\sqrt{s'}}{p_{\mathrm{cm}}}\sum_{\ell}(2\ell+1)\,\mathrm{Im}\,a_\ell(s') \;>\; 0, \tag{63.55}$$

strictly positive because the integrand is the manifestly positive forward absorptive density Eq. (63.49). The same theory thus appears at *both* ends of the bootstrap: as a critical point inside the conformal island, and as a point in the allowed EFThedron with $g_2 > 0$. The flat-space limit Eq. (63.54) is the statement that these are not two coincidences but one constraint viewed at two curvatures — the AdS dual of the Wilson–Fisher CFT, in the large-radius limit, scatters ϕ quanta with precisely the amplitude whose forward positivity is Eq. (63.55). ϕ^4 theory, begun as a Lagrangian in Part II, ends as a single point that crossing and positivity locate in two complementary spaces and the holographic limit identifies as one. This is the ϕ^4 theory handoff — $\phi^4 \to$ Wilson–Fisher $\to$ bootstrap — closing in full from the amplitudes side, and joined to the holography thread through the flat-space limit.

The lesson the two bootstraps press on us is that crossing and positivity are not consequences of a Lagrangian; they are prior to it, and they are nearly enough, by themselves, to pin a theory down. If so little input — analyticity, unitarity, the demand that channels agree — constrains so much, it is natural to ask whether the remaining structural axioms of quantum field theory, locality

and unitarity themselves, might likewise be *outputs* of something more primitive: a geometry whose positivity generates the amplitude, or a gauge theory whose square generates gravity. That is the question of the next section.

63.5 Locality and Unitarity as Outputs: Geometry and the Double Copy

The previous section left us with a sharpened version of the chapter's organizing question. Crossing and positivity, we found, are not consequences of a Lagrangian; they are prior to it, and they nearly suffice to pin a theory down. The bootstrap reads this as a statement about a *region*: the allowed theories are the points of a convex body carved out of an *a priori* space of data by linear (crossing) equalities and convex (positivity) inequalities, and to bootstrap is to compute that body. Once one has learned to see physical input as the shape of a region, it is irresistible to ask whether the two remaining structural axioms of quantum field theory — *locality*, the demand that interactions be built from fields at a point so that amplitudes have singularities only where intermediate particles can go on shell, and *unitarity*, the demand that probability be conserved so that those singularities factorize into products of lower amplitudes — are themselves outputs of a more primitive geometric or algebraic structure rather than imposed postulates. This section assembles the two pieces of frontier evidence that they are. The first is the *amplituhedron* lesson of Chapter 62: locality and unitarity emerge as properties of the canonical differential form of a positive geometry, dictated entirely by the geometry's boundaries. The second is the *double-copy* lesson of Chapter 61: the entire kinematic structure of perturbative gravity — and with it the diffeomorphism redundancy that is gravity's defining gauge symmetry — is fixed by squaring gauge theory. Both lessons are stated here carefully, proved where the proof is short and clean, recalled by Chapters 61 and 62 where those chapters carried it in full, and dated honestly: one of them is a theorem in a controlled corner and a conjecture beyond it, the other a theorem at tree level and a tested conjecture at loop level, as of the mid-2020s.

63.5.1 The canonical-form principle: poles are boundaries

We begin by isolating, from the machinery of Chapter 62, the single structural statement on which the entire "locality and unitarity are emergent" claim rests. Recall the central definition of that chapter (Theorem 62.3): a *positive geometry* is a real region $\mathcal{X}$ — bounded by a collection of boundary components of every codimension — equipped with a unique top-degree rational differential form $\Omega(\mathcal{X})$, the *canonical form*, singled out by a recursive requirement on its singularities. We restate that requirement precisely, because everything below is a reading of it.

Definition 63.11 (Canonical form, recalled)**.** Let $\mathcal{X}$ be a positive geometry of dimension m. Its canonical form $\Omega(\mathcal{X})$ is the unique top-degree (m-form) rational differential form with the two properties:

1. *Logarithmic singularities only on boundaries.* $\Omega(\mathcal{X})$ has poles only along the boundary components of $\mathcal{X}$, every pole is simple (logarithmic), and there are no singularities anywhere in the interior or outside the boundary set.

2. *Residues are lower canonical forms.* On each codimension-one boundary component $\mathcal{C} \subset \partial\mathcal{X}$, the residue of $\Omega(\mathcal{X})$ equals the canonical form of $\mathcal{C}$ regarded as a positive geometry in its own right:

$$\operatorname{Res}_{\mathcal{C}} \Omega(\mathcal{X}) \;=\; \Omega(\mathcal{C}). \tag{63.56}$$

This is the recursion proved to determine Ω uniquely in Chapter 62, and Eq. (63.56) is exactly the boundary/residue axiom recorded there as Eq. (62.5). We import it and do not reprove it. What we do here is read it as a physical principle. The reading turns on a dictionary that Chapter 62 established by explicit computation and that we now state as the load-bearing correspondence of the whole section.

The Canonical-Form Principle

Suppose a scattering amplitude is the canonical form of a positive geometry, in the precise sense that $\Omega(\mathcal{X}) = (\text{amplitude}) \times (\text{top measure})$ once $\mathcal{X}$ is built in the kinematic space of the process. Then the amplitude's two defining structural properties are *outputs* of the geometry, not inputs:

- **Locality** is property (1): the amplitude has singularities *only* on the boundaries of $\mathcal{X}$, and each boundary is one physical factorization channel. There are no poles anywhere else — in particular none on spurious or unphysical loci — because the canonical form is forbidden, by definition, to have them.

- **Unitarity** is property (2): the residue on a boundary $\mathcal{C}$ is the canonical form of $\mathcal{C}$, and when $\mathcal{C}$ factorizes as a product of two lower-dimensional positive geometries, $\mathcal{C} = \mathcal{X}_L \times \mathcal{X}_R$, its canonical form factorizes too, $\Omega(\mathcal{C}) = \Omega(\mathcal{X}_L)\,\Omega(\mathcal{X}_R)$. In amplitude language this is precisely the factorization of the residue into a product of lower-point amplitudes glued on the on-shell intermediate state — the content of unitary factorization.

Neither property was postulated. Both follow from the demand that Ω be the canonical form of a region with those particular boundaries. The physics is in the choice of region; the analytic structure is forced.

That this is more than a hopeful slogan was settled in Chapter 62 for an honest physical amplitude. The kinematic associahedron $\mathcal{A}_n$ — a genuine convex polytope living in the space of Mandelstam invariants — has, as its canonical form, the tree-level color-ordered amplitude of bi-adjoint cubic scalar theory, proved facet by facet (Theorem 62.17). There the facets $X_{i,j} = 0$ of the polytope are exactly the planar propagator channels; the amplitude's poles sit on those facets and nowhere else (locality), and the residue on a facet is the canonical form of that facet, which by the combinatorics of the associahedron is a *product of two lower associahedra*, i.e. a product of lower amplitudes (unitarity). Nothing in that derivation imposed analyticity or factorization; both fell out of the shape of the polytope. The amplituhedron of Theorem 62.44 extends the same principle to a far richer setting — the integrand of planar maximally supersymmetric Yang–Mills — but, as we will be careful to record below, there the identification of the canonical form with the amplitude is, beyond the controlled toy dimension, a conjecture.

63.5.2 A toy positive geometry: the pole is a boundary

The associahedron proof is complete but combinatorially heavy. To make the "pole = boundary" mechanism fully transparent — to see a singularity *appear* out of a geometric edge with the correct residue, in one line of calculus — we carry the smallest nontrivial positive geometry all the way through. It is the case the associahedron already collapses to at four points (Theorem 62.17, the $n = 4$ segment), here built from scratch so the emergence is visible.

Worked Example 63.12 (The canonical form of an interval and the birth of a propagator)**.** *The geometry.* Take the one-dimensional positive geometry to be the closed interval

$$\mathcal{X} \;=\; \{\, x \in \mathbb{R} : x \geq 0 \text{ and } a - x \geq 0 \,\} \;=\; [0, a], \qquad a > 0, \tag{63.57}$$

cut out of the line by the two linear functions $X_1(x) = x$ and $X_2(x) = a - x$. Its boundary is the two endpoints, $\partial\mathcal{X} = \{x = 0\} \cup \{x = a\}$, each a codimension-one component — here a single point, the zero-dimensional positive geometry whose canonical form is the number 1.

The canonical form from the axioms. By Theorem 63.11 we seek a top-degree (here, one-) form $\Omega = f(x)\,\mathrm{d}x$ with f rational, simple poles only at $x = 0$ and $x = a$, no other singularities (in particular f must vanish at infinity so that $x = \infty$ is not a pole), and unit residues that reproduce the canonical form 1 of each endpoint. A rational function with simple poles only at 0 and a and no pole at infinity is a sum of two partial fractions,

$$f(x) \;=\; \frac{r_0}{x} + \frac{r_a}{x - a}, \tag{63.58}$$

with residues r_0 at $x = 0$ and r_a at $x = a$. Property (2) fixes the residues. The residue of Ω on the boundary $x = 0$, taken with the orientation that points *out* of $\mathcal{X}$ along increasing x, must be $+\Omega(\{x = 0\}) = +1$; on the boundary $x = a$, where the outward orientation along increasing x is reversed, the residue is -1. Hence $r_0 = +1$, $r_a = -1$, and

$$\Omega(\mathcal{X}) \;=\; \left(\frac{1}{x} - \frac{1}{x-a}\right)\mathrm{d}x \;=\; \left(\frac{1}{x} + \frac{1}{a-x}\right)\mathrm{d}x \;=\; \frac{a}{x\,(a-x)}\,\mathrm{d}x. \tag{63.59}$$

Equivalently $\Omega(\mathcal{X}) = \mathrm{dlog}\dfrac{x}{a-x}$, manifestly a logarithmic form whose two simple poles sit exactly on the two boundaries. We may verify the recursion Eq. (63.56) directly. Near $x = 0$, $\Omega \simeq \mathrm{d}x/x$, so $\mathrm{Res}_{x=0}\,\Omega = 1 = \Omega(\{x = 0\})$. Near the other endpoint set $u = x - a$; then $\dfrac{a}{x(a-x)} = \dfrac{a}{(a+u)(-u)} \xrightarrow{u\to 0} -\dfrac{1}{u}$, so $\mathrm{Res}_{x=a}\,\Omega = -1$, equal to the canonical form of that endpoint carried with the opposite boundary orientation. The residues are the lower (point) canonical forms, as the axiom demands; the alternating sign is the orientation bookkeeping fixed once and for all in Chapter 62.

The propagator is the boundary. Now read the two linear boundary functions as inverse propagators of a physical process. In the four-point bi-adjoint scalar of Theorem 62.17 the affine coordinate on the interval is the planar Mandelstam variable, $x = X_{1,3} = s$, and the complementary facet function is the crossed channel, $a - x = X_{2,4} = t$, with $a = s + t = -u$ the fixed off-shell datum $s + t + u = 0$ supplies. Substituting into Eq. (63.59) strips the measure and leaves the rational coefficient

$$m_4 \;=\; \frac{1}{X_{1,3}} + \frac{1}{X_{2,4}} \;=\; \frac{1}{s} + \frac{1}{t}, \tag{63.60}$$

the complete tree amplitude of the bi-adjoint scalar: a sum of two propagators, one per planar channel. The structure $\dfrac{a}{x(a-x)} = \dfrac{1}{x} + \dfrac{1}{a-x}$ *is* the propagator sum, and each propagator pole is one endpoint of the interval. The residue of the amplitude on the s-channel pole is, by Eq. (63.56), the canonical form of the boundary point — the number 1 — which is precisely the product of the two cubic three-point amplitudes (each equal to 1 for the bi-adjoint scalar) glued across the on-shell exchanged line. The factorization of unitarity has been read off the corner of a segment.

The lesson. Locality here means: the amplitude Eq. (63.60) has poles at $s = 0$ and $t = 0$ and *nowhere else* — no pole in u, no spurious singularity — because the interval has exactly two boundaries and the canonical form is forbidden any other pole. Unitarity here means: the residue at each pole is the lower amplitude, because the residue of a canonical form is the canonical form of the boundary. We did not assume either. We assumed only that the object of interest is the canonical form of the segment $[0, a]$; the propagator structure, the channel poles, and the factorizing residues were forced.

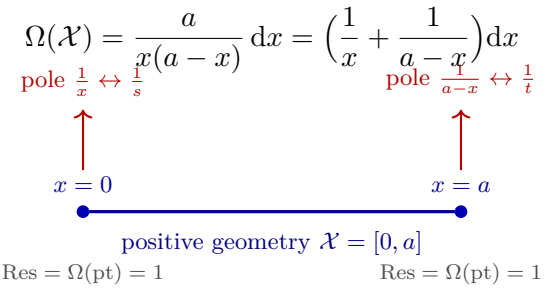


Figure 63.4: The toy positive geometry of Theorem 63.12. The canonical form of the interval $[0, a]$ has exactly two simple poles, one sitting on each boundary point. Reading the two boundary functions as physical channel invariants s and t turns the poles into propagators and the boundary residues into the (trivial) lower amplitudes glued on the on-shell exchange. Locality (poles only on the two boundaries) and unitarity (residue = lower canonical form) are dictated by the shape of the region, not imposed.

The interval is the seed crystal of the whole program. Every higher case proved in Chapter 62 — the pentagon $\mathcal{A}_5$ whose five edges are the five planar channels of the five-point amplitude, the kinematic associahedron $\mathcal{A}_n$ in general, the $m = 2$ amplituhedron polygon — repeats this mechanism one dimension at a time: a codimension-one boundary is a pole; the residue on it is the canonical form of a lower geometry that factorizes into a product of lower amplitudes; and there are no other singularities because the canonical form is, by construction, free of them. Locality and unitarity are then not axioms layered onto a Lagrangian but the two halves of Theorem 63.11, read in kinematic space. This is the first piece of frontier evidence that structural "principles" are derived outputs.

63.5.3 The double copy: gravity as gauge theory squared

The second piece comes from the opposite end of the structural ladder. If the amplituhedron asks whether the *analytic* axioms (locality, unitarity) are emergent, the double copy asks whether the most baroque *symmetry* structure in physics — the diffeomorphism invariance of general relativity, the nonlinear redundancy that makes gravity gravity — is emergent from something simpler. The answer that Chapter 61 developed in full, and that we now recall and read, is that perturbative gravity is gauge theory squared.

We continue Yang–Mills theory, whose amplitude arc this closes. The thread carried the non-abelian gauge field from its classical Lagrangian (Chapter 25) through quantization and asymptotic freedom (Chapters 26 and 27) to its modern amplitude structure: gluon tree amplitudes computed the hard way and then by recursion, culminating in the Parke–Taylor formula (Chapter 59), and then the color–kinematics duality and double copy of Chapter 61. Here is the structural

endpoint of that arc.

The two facts recalled. Chapter 61 organized any gauge-theory tree amplitude over its trivalent (cubic) graphs,

$$\mathcal{A}_n \;=\; i\,g^{\,n-2}\sum_{i\in\Gamma_3}\frac{c_i\,n_i}{D_i}, \tag{63.61}$$

recalling the master formula Eq. (61.18): the sum runs over cubic graphs i, with c_i the color factor (a product of structure constants f^{abc}), D_i the product of the graph's internal inverse propagators, and n_i the kinematic numerator. The two facts the thread established are:

- *Color–kinematics duality* (Theorem 61.8): the kinematic numerators n_i can be chosen to satisfy the *same* algebraic relations as the color factors. Wherever three color factors obey a Jacobi identity $c_i + c_j + c_k = 0$, the corresponding numerators can be arranged to obey the *kinematic Jacobi identity* $n_i + n_j + n_k = 0$, and wherever a color factor is antisymmetric under exchanging two legs, so is its numerator. Such a set is a set of *BCJ numerators*.

- *The double copy* (Theorem 61.15, Eq. (61.72); equivalently the Kawai–Lewellen–Tye form Theorem 61.21 and the BCJ double-copy relation Eq. (61.72)): once one numerator set is BCJ, replacing the color factors of Eq. (63.61) by a second kinematic numerator set,

$$c_i \;\longmapsto\; \tilde{n}_i, \qquad \mathcal{M}_n \;=\; i\left(\frac{\kappa}{2}\right)^{n-2}\sum_{i\in\Gamma_3}\frac{n_i\,\tilde{n}_i}{D_i}, \tag{63.62}$$

  produces a bona fide *graviton* amplitude — proved gauge-invariant and correctly factorizing in Theorems 61.16 and 61.20. The dimensionless gauge coupling g is traded for the dimensionful gravitational coupling κ with $\kappa^2 = 32\pi G_N$.

We import both statements and do not reprove them. Their content is structural: color enters Eq. (63.61) only through its algebra (the Jacobi relations), so a second representation of the *same* algebra — a BCJ numerator set — can be substituted for it without the formula being able to tell the difference. The output then carries two Lorentz numerators where the input carried one color factor and one numerator, doubling the little-group weight on every external leg: two copies of spin one make spin two. The graviton is the symmetric, traceless part of the tensor product of two gluons.

The four-point square, recapped algebraically. The cleanest demonstration is the four-graviton amplitude, carried in full in Theorem 61.19; we recall its skeleton because it is the concrete endpoint of Yang–Mills theory. At four points there are three cubic graphs — the s-, t-, and u-channels — with denominators $D_s = s$, $D_t = t$, $D_u = u$ obeying $s + t + u = 0$ for massless legs. Taking both copies to be pure Yang–Mills ($\tilde{n}_i = n_i$), Eq. (63.62) reads

$$\mathcal{M}_4 \;=\; i\left(\frac{\kappa}{2}\right)^2 \left(\frac{n_s^2}{s} + \frac{n_t^2}{t} + \frac{n_u^2}{u}\right), \qquad n_s + n_t + n_u = 0, \tag{63.63}$$

with the BCJ numerators of the MHV configuration $(1^-, 2^-, 3^+, 4^+)$. The kinematic Jacobi identity $n_s + n_t + n_u = 0$, together with the massless relation $s+t+u = 0$, lets one work in the generalized gauge $n_t = 0$, in which the numerator-to-partial-amplitude dictionary collapses to $n_s/s = P$ and $n_u/u = -Q$, where $P = A_4[1,2,3,4]/i$ and $Q = A_4[1,3,2,4]/i$ are the Parke–Taylor color-ordered partials of Eq. (59.55),

$$P = \frac{\langle 1\,2\rangle^4}{\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle}, \qquad Q = \frac{\langle 1\,2\rangle^4}{\langle 1\,3\rangle\langle 3\,2\rangle\langle 2\,4\rangle\langle 4\,1\rangle}. \tag{63.64}$$

The kinematic Jacobi identity in this gauge becomes the fundamental BCJ amplitude relation $sP = uQ$. Substituting $n_s = sP$, $n_u = -uQ$, $n_t = 0$ into Eq. (63.63) and using $sP = uQ$ with $s + t + u = 0$ collapses the three-term sum (the algebra is carried step by step in Theorem 61.19) to the closed form recorded there as Eq. (61.97),

$$\mathcal{M}_4(1^{--}, 2^{--}, 3^{++}, 4^{++}) \;=\; -\,i\left(\frac{\kappa}{2}\right)^2 \frac{\langle 1\,2\rangle^4\,[3\,4]^4}{s\,t\,u}. \tag{63.65}$$

Set this beside the gauge partial $A_4[1,2,3,4] = i\,\langle 1\,2\rangle^4/(\langle 1\,2\rangle\langle 2\,3\rangle\langle 3\,4\rangle\langle 4\,1\rangle)$ and the squaring is on the page: the gluon numerator $\langle 1\,2\rangle^4$ has become $\langle 1\,2\rangle^4\,[3\,4]^4$ — an angle factor times a square-bracket factor, one from each copy — and the single chain of propagators has become the symmetric denominator stu. The four-graviton amplitude is, term by term, the four-gluon amplitude squared. This is the structural endpoint that Yang–Mills theory was built to reach.

Diffeomorphisms as the square of gauge transformations. The reading that makes this a statement about emergent spacetime structure is the following. Gauge invariance of Eq. (63.61) — the $U(1)$ or non-abelian redundancy under which gluon polarizations shift by their momenta, $\varepsilon_i^\mu \to \varepsilon_i^\mu + \alpha\, p_i^\mu$ — is the on-shell shadow of the gauge symmetry of the Yang–Mills Lagrangian. In the graviton amplitude Eq. (63.62) every external state is a product of two gluon polarizations, $\varepsilon_i^{\mu\nu} = \varepsilon_i^\mu\,\tilde{\varepsilon}_i^\nu$, so a graviton polarization shifts under the *product* of two independent gauge shifts. The linearized diffeomorphism $\varepsilon_i^{\mu\nu} \to \varepsilon_i^{\mu\nu} + p_i^\mu \xi^\nu + \xi^\mu p_i^\nu$ — the

gauge redundancy of the graviton — is exactly what one obtains by performing a gauge shift on one factor at a time. Diffeomorphism redundancy is the square of gauge redundancy. The nonlinear gauge invariance of general relativity, which from the Lagrangian side looks like an elaborate and independent geometric principle (general covariance), is on the amplitude side an automatic consequence of building each graviton from two gluons and inheriting both their linearized Ward identities. Spacetime's defining symmetry is a derived object: the gauge symmetry of a theory that knows nothing of spacetime curvature, applied twice.

Physical Insight

The two pieces of evidence are mirror images. The amplituhedron says the *analytic* backbone of field theory — where amplitudes are allowed to be singular, and what their singularities must equal — is read off the boundaries of a positive geometry; locality and unitarity are the two clauses of the canonical-form recursion. The double copy says the *symmetry* backbone of gravity — general covariance — is read off the product structure of gauge theory; diffeomorphism invariance is gauge invariance squared. In both, a principle that Part I posited as a founding axiom (Chapter 1) reappears in Part XVI as the output of a more primitive construction. The Lagrangian, with its locality built into the pointwise product of fields and its gauge symmetry built into the covariant derivative, is one efficient packaging of these outputs — but, the evidence suggests, not the only one, and perhaps not the most fundamental.

63.5.4 Honest scope, dated to the mid-2020s

Precision requires that we mark exactly how far each claim is established, and date it. We do so plainly.

The positive-geometry / amplituhedron program. The canonical-form principle of Section 63.5.1 is a *theorem* in the cases where the positive geometry is constructed and its canonical form is computed: the projective simplex and polygon, the $m = 2$ amplituhedron, and — physically most important — the kinematic associahedron, whose canonical form is proved equal to the bi-adjoint tree amplitude in Theorem 62.17. In those cases locality and unitarity genuinely *are* outputs, with full proofs in Chapter 62. For the amplituhedron proper (Theorem 62.44), the statement that its canonical form is the planar $\mathcal{N} = 4$ super-Yang–Mills amplitude is established rigorously only in the toy dimension $m = 2$ and in restricted sectors; the full claim — all multiplicities n, all helicity grades k, all loop orders L — remains, as of the mid-2020s, a *conjecture* (Theorem 62.47), extensively checked but unproven. Beyond the planar limit, where there is no single cyclic ordering, no analogue of the amplituhedron is established at all. "Locality and unitarity are emergent" is therefore a proven statement for

bi-adjoint trees and a compelling, well-tested, but conjectural statement for the planar maximally supersymmetric integrand.

The double copy. The statement that a BCJ-organized gauge amplitude squares to a gravity amplitude is a *theorem at tree level*: the existence of BCJ numerators at tree level, the gauge invariance of the double copy, and its correct factorization are all proved in Chapter 61 (Theorems 61.8, 61.16 and 61.20), and the KLT relations give an independent tree-level proof. At loop level the situation is sharper to state: the double copy is known to work — it has reproduced multi-loop supergravity integrands that no other method could compute, and these pass every unitarity-cut and ultraviolet consistency check applied to them — but the crucial input, the *existence* of color–kinematics-dual numerators at all loop orders, is not proven. It is a conjecture (Theorem 61.32), supported by a large and growing body of explicit high-loop evidence and by the cut-level consistency of Theorem 61.31, but lacking a general constructive theorem as of the mid-2020s. The honest summary is: gravity-as-gauge-squared is a theorem where the amplitudes are trees and a tested, structurally-motivated conjecture where they are loops.

Neither program, then, is a finished reduction of field theory's axioms to something more primitive. What both establish, rigorously and in controlled corners and with overwhelming evidence beyond them, is that the reduction is *possible in principle*: that there exist constructions — a positivity geometry, a squaring map — from which locality, unitarity, and even general covariance fall out as consequences rather than being installed by hand. That is the precise sense in which the structural principles are outputs.

If locality and unitarity can be read off a geometry and gravity off a gauge-theory square, then the deepest structural axiom remaining — the very existence of a local spacetime in which the S-matrix is defined — is the natural next target. The most radical reframing on offer recasts the flat-space S-matrix itself as a correlation function of a conformal field theory living on the celestial sphere at null infinity, with the soft theorems of Chapter 60 reborn as the Ward identities of that boundary theory's asymptotic symmetries. To that holography of asymptotically flat space we turn next.

63.6 Amplitudes Meet Holography: Soft Theorems and the Celestial Sphere

The previous section read locality and unitarity as outputs rather than inputs: the amplituhedron manufactures both from the positivity of a geometry, and the double copy manufactures gravity's entire kinematic structure by squaring gauge theory. Both lessons are about the *bulk* of an amplitude — its singularities, its numerators, the way it factorizes on internal channels. There is a complementary

frontier that concerns the amplitude's *boundary*: the behaviour of the S-matrix when an external quantum becomes infinitely soft, and the question of whether the entire S-matrix can be reorganised as a correlation function living on a lower-dimensional screen. This is the amplitudes program's point of contact with holography. The thread that runs through it is the same one Part XV opened with AdS/CFT (Chapter 53): a theory of gravity in a D-dimensional spacetime may be equivalent to a quantum theory without gravity living on its $(D-1)$- or $(D-2)$-dimensional boundary. For asymptotically flat spacetime the candidate boundary theory is a two-dimensional conformal field theory on the *celestial sphere*, and its primary observables — "celestial amplitudes" — are the ordinary flat-space scattering amplitudes rewritten in a conformal basis. The bridge that makes this more than a change of variables is built from soft theorems, which we derive first, in full, from the factorization pole; recognise next as the Ward identities of an infinite-dimensional asymptotic symmetry; and then use to populate the celestial CFT with conserved currents and a stress tensor. We scope the whole construction honestly throughout: the soft theorems are rigorous theorems of any local, Lorentz-invariant quantum field theory; the celestial-CFT reorganisation is a program, dated to the mid-2020s, not a proved duality.

63.6.1 The leading soft factor from the factorization pole

A scattering amplitude is a meromorphic function of its external momenta, and its poles are dictated by the same physics everywhere in this book: an internal line going on shell. The soft limit exploits a particular pole. When an external massless quantum — a photon or a graviton of momentum q — is emitted from an external leg of momentum p_i, the line that carried the energy before the emission is nearly on shell, because the emitted momentum is going to zero. The amplitude therefore develops a $1/q$ singularity as $q \to 0$, and the residue of that singularity is universal: it depends only on the momenta and charges of the hard particles, not on the internal dynamics of the scattering. This is the content of the soft theorems, and it is most transparently derived by isolating exactly that pole.

We work throughout in the all-outgoing convention fixed at the head of the chapter: every external momentum p_i $(i = 1, \dots, n)$ is treated as outgoing, so that overall momentum conservation reads $\sum_{i=1}^{n} p_i^\mu = 0$, and the soft photon or graviton of momentum q is likewise outgoing. We denote the n-point hard amplitude $\mathcal{M}_n$ and the amplitude with one extra soft quantum $\mathcal{M}_{n+1}$. The claim to be proved is that

$$\lim_{q \to 0} \mathcal{M}_{n+1} \;=\; \Big(S^{(0)} + O(q^0) \Big)\, \mathcal{M}_n, \tag{63.66}$$

with $S^{(0)}$ an explicit, theory-independent function of order $1/q$ — the *leading soft factor*. The notation $O(q^0)$ records that the next term in the soft expansion

is finite as $q \to 0$; the leading term is the pole.

Why only external-leg emission contributes at the pole. A general Feynman graph for $\mathcal{M}_{n+1}$ either attaches the soft quantum to an internal line or to an external leg. Attaching it to an internal line of off-shell momentum k replaces a propagator $1/(k^2 - m^2)$ by a vertex and two propagators whose denominators stay finite as $q \to 0$; that contribution is $O(q^0)$ and does not produce the pole. Attaching it to an external leg of momentum p_i is different: the internal line just inside the emission carries $p_i + q$, with

$$(p_i + q)^2 - m_i^2 = p_i^2 - m_i^2 + 2\,p_i{\cdot}q + q^2 = 2\,p_i{\cdot}q + q^2 \xrightarrow[q\to 0]{} 0, \tag{63.67}$$

where we used that the external leg is on shell, $p_i^2 = m_i^2$, and that the soft photon/graviton is massless, $q^2 = 0$. The denominator vanishes linearly in q: this is the pole, and only external-leg emission produces it. The leading soft factor is therefore the sum over the n external legs of [emission vertex]×[propagator], each evaluated at the on-shell pole.

Worked Example 63.13 (The leading soft *photon* factor and charge conservation)**.** Consider first a charged scalar leg, to keep the Dirac algebra out of the way; we restore arbitrary spin at the end and find the same answer, which is the first lesson — universality. Let leg i be an outgoing scalar of momentum p_i and charge Q_i (in units of e; its electric charge is $e\,Q_i$), emitting an outgoing photon of momentum q and polarization ϵ^μ. The scalar QED Feynman rule for the photon–scalar vertex, with the line momenta as in Eq. (63.67), is $-ieQ_i\,(2p_i + q)^\mu$, and the internal scalar propagator carrying $p_i + q$ is $i/[(p_i + q)^2 - m_i^2]$. The diagrammatic contribution of leg i factorizes as

$$\mathcal{M}_{n+1}\big|_{\text{leg } i} = \underbrace{(-ieQ_i)\,(2p_i + q)^\mu \epsilon_\mu^*}_{\text{vertex}}\ \underbrace{\frac{i}{2\,p_i{\cdot}q + q^2}}_{\text{propagator}}\ \mathcal{M}_n(\dots, p_i, \dots), \tag{63.68}$$

because, with the soft momentum stripped off, the line i re-enters $\mathcal{M}_n$ exactly on shell at momentum p_i. Take $q \to 0$: in the numerator $(2p_i + q)\cdot\epsilon^* \to 2\,p_i{\cdot}\epsilon^*$, and in the denominator $q^2 \to 0$, so

$$\mathcal{M}_{n+1}\big|_{\text{leg } i} \xrightarrow[q\to 0]{} (-ieQ_i)\,(2\,p_i{\cdot}\epsilon^*)\,\frac{i}{2\,p_i{\cdot}q}\,\mathcal{M}_n \;=\; e\,Q_i\,\frac{p_i{\cdot}\epsilon^*}{p_i{\cdot}q}\,\mathcal{M}_n, \tag{63.69}$$

where the two explicit factors of i combine, $(-i)(i) = +1$, and a factor of two cancels between numerator and denominator. Summing over all external legs gives the leading soft photon factor,

$$\boxed{S_\gamma^{(0)} = e\sum_{i=1}^{n} Q_i\,\frac{p_i{\cdot}\epsilon^*}{p_i{\cdot}q}}, \qquad \lim_{q\to 0}\mathcal{M}_{n+1} = S_\gamma^{(0)}\,\mathcal{M}_n + O(q^0). \tag{63.70}$$

This is Weinberg's soft photon factor. It is the same eikonal current that appeared in the infrared analysis of QED, Eq. (19.180): there it organised the cancellation of infrared divergences between virtual and real soft emission (Sections 19.6 and 19.6.3); here it is the universal residue of the soft pole. The two are one structure seen from two sides — the soft theorem is the statement that the QED infrared is controlled by a quantity that knows nothing about the hard process.

Spin-independence. Replace the scalar by a Dirac fermion of momentum p_i. The vertex is now $-ieQ_i\gamma^\mu$, the propagator numerator is $i(\not{p}_i + \not{q} + m_i)$, and the leg ends on the external spinor $\bar{u}(p_i)$. The numerator of Eq. (63.68) becomes, at leading order (dropping $\not{q}$),

$$\bar{u}(p_i)\,\gamma^\mu \epsilon^*_\mu\,(\not{p}_i + m_i) = \bar{u}(p_i)\Big[2\,p_i\!\cdot\!\epsilon^* - (\not{p}_i - m_i)\,\not{\epsilon}^*\Big] = 2\,(p_i\!\cdot\!\epsilon^*)\,\bar{u}(p_i), \tag{63.71}$$

where we used the gamma identity $\not{\epsilon}^*\not{p}_i = 2\,p_i\!\cdot\!\epsilon^* - \not{p}_i\not{\epsilon}^*$ (Appendix C) and the Dirac equation $\bar{u}(p_i)(\not{p}_i - m_i) = 0$. The result is identical to the scalar numerator $2\,p_i\!\cdot\!\epsilon^*$, so the soft factor Eq. (63.70) is independent of the spin of the emitting leg. The soft photon does not resolve the structure of the particle it is emitted from; it sees only the momentum and the charge. This is the precise sense in which the soft factor is *universal*.

Charge conservation is the gauge-invariance content. A physical amplitude must be invariant under the gauge transformation $\epsilon^*_\mu \to \epsilon^*_\mu + \alpha\, q_\mu$ of the soft photon's polarization (the Ward identity; the longitudinal photon decouples, as established for the gauge principle in Section 8.1 and for on-shell amplitudes in Chapter 18). Under this shift the soft factor changes by

$$\delta S^{(0)}_\gamma = e\sum_i Q_i\,\frac{p_i\!\cdot\!(\alpha\, q)}{p_i\!\cdot\! q} = \alpha\, e\sum_i Q_i. \tag{63.72}$$

Gauge invariance of the leading soft amplitude therefore requires

$$\sum_{i=1}^{n} Q_i = 0, \tag{63.73}$$

which (in the all-outgoing convention) is exactly conservation of electric charge. The soft theorem does not assume charge conservation; it *forces* it. A theory whose soft emission fails to factorize into Eq. (63.70) with a conserved total charge is not gauge-invariant, hence not a consistent theory of a massless spin-1 field. This is the first instance of the chapter's recurring theme: a structural "axiom" (here, charge conservation) is an output of consistency at the boundary of phase space.

Figure 63.5 draws the factorization that produces the pole.

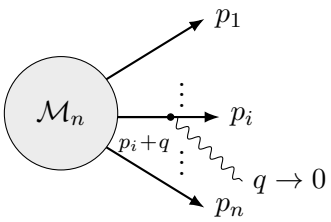


Figure 63.5: The leading soft pole. As the photon (or graviton) momentum $q \to 0$, the internal line carrying $p_i + q$ goes on shell, Eq. (63.67), and the amplitude factorizes into a universal soft factor times the hard amplitude $\mathcal{M}_n$. Only emission from external legs produces the $1/q$ singularity.

The soft *graviton* factor and the equivalence principle. The same construction with a graviton replaces the charge eQ_i by the gravitational coupling and the single polarization vector ϵ^μ by the symmetric, traceless, transverse graviton polarization $\epsilon_{\mu\nu}$. We write the metric as $g_{\mu\nu} = \eta_{\mu\nu} + \kappa\, h_{\mu\nu}$ with $\kappa = \sqrt{32\pi G}$, so that matter couples minimally through $-\frac{\kappa}{2} h_{\mu\nu} T^{\mu\nu}$ with $T^{\mu\nu}$ the stress tensor. For an outgoing scalar of momentum p_i the leading graviton emission vertex is $-i\frac{\kappa}{2}\,(2p_i^\mu)(p_i^\nu)$ symmetrised, whose contraction with the traceless $\epsilon_{\mu\nu}$ keeps only the $p_i^\mu p_i^\nu$ piece (the $\eta^{\mu\nu}$ terms in $T^{\mu\nu}$ die against the traceless polarization). With the same on-shell propagator $i/(2p_i\!\cdot\! q)$, leg i contributes

$$\mathcal{M}_{n+1}\big|_{\text{leg } i} \xrightarrow[q\to 0]{} \Big(-i\tfrac{\kappa}{2}\, 2\, p_i^\mu p_i^\nu\Big)\epsilon^*_{\mu\nu}\,\frac{i}{2\,p_i\!\cdot\! q}\,\mathcal{M}_n = \frac{\kappa}{2}\,\frac{\epsilon^*_{\mu\nu}\,p_i^\mu p_i^\nu}{p_i\!\cdot\! q}\,\mathcal{M}_n, \tag{63.74}$$

again with $(-i)(i) = +1$ and a factor of two cancelled. Summing over external legs,

$$\boxed{\,S_h^{(0)} = \frac{\kappa}{2}\sum_{i=1}^{n}\frac{\epsilon^*_{\mu\nu}\,p_i^\mu p_i^\nu}{p_i\!\cdot\! q}\,}\,. \tag{63.75}$$

Two features stand out, and both are physics. First, $S_h^{(0)}$ is, leg by leg, the *square* of the photon soft factor with the charge Q_i replaced by the momentum p_i^μ: setting $\epsilon^*_{\mu\nu} = \epsilon^*_\mu \epsilon^*_\nu$ for a definite-helicity graviton turns Eq. (63.75) into $\frac{\kappa}{2}\sum_i (p_i\!\cdot\!\epsilon^*)^2/(p_i\!\cdot\! q)$, whereas the photon factor is $e\sum_i Q_i\,(p_i\!\cdot\!\epsilon^*)/(p_i\!\cdot\! q)$. The soft graviton is the soft photon "squared," with the universal gravitational charge being energy–momentum itself: this is the infrared shadow of the double copy of Chapter 61, and it advances Yang–Mills theory into gravity through exactly the squaring relation that the double copy installs at every order.

Second, gauge invariance now reads differently and says more. The graviton polarization is defined up to a linearized diffeomorphism, $\epsilon^*_{\mu\nu} \to \epsilon^*_{\mu\nu} + q_\mu \xi_\nu + \xi_\mu q_\nu$ for any ξ^μ. Under this shift, allowing now a possibly leg-dependent gravitational

coupling κ_i to test universality,

$$\delta S_h^{(0)} = \sum_i \frac{\kappa_i}{2} \frac{(q_\mu \xi_\nu + \xi_\mu q_\nu)\, p_i^\mu p_i^\nu}{p_i \cdot q} = \sum_i \frac{\kappa_i}{2} \frac{2\,(p_i \cdot q)(p_i \cdot \xi)}{p_i \cdot q}$$
$$= \sum_i \kappa_i\, (p_i \cdot \xi) = \xi_\mu \sum_i \kappa_i\, p_i^\mu . \tag{63.76}$$

For a photon the analogous quantity, Eq. (63.72), vanished by a *constraint on the charges*, $\sum_i Q_i = 0$. For a graviton the "charges" are the momenta p_i^μ, which are already constrained by momentum conservation, $\sum_i p_i^\mu = 0$. If $\delta S_h^{(0)}$ is to vanish for arbitrary ξ^μ, we need $\sum_i \kappa_i\, p_i^\mu = 0$. Momentum conservation guarantees $\sum_i p_i^\mu = 0$, so $\sum_i \kappa_i p_i^\mu = 0$ holds for all kinematics *only if all the couplings are equal*, $\kappa_i = \kappa$ for every species i. Gauge invariance of the leading soft graviton theorem therefore forces

$$\kappa_i = \kappa \quad \text{for all } i : \qquad \text{gravity couples universally.} \tag{63.77}$$

This is the equivalence principle, derived — not assumed — from the consistency of soft graviton emission with diffeomorphism invariance. It is Weinberg's celebrated 1964 argument: any massless spin-2 particle that couples to matter at all must couple to every species with the same strength, or S-matrix gauge invariance fails. Just as charge conservation was the gauge content of the soft photon theorem, the equivalence principle is the gauge content of the soft graviton theorem. The soft theorems package two of the deepest facts about long-range forces as boundary outputs of the S-matrix.

Physical Insight

The contrast between Eq. (63.73) and Eq. (63.77) is the whole story in one line. For the photon, the couplings Q_i are free but their sum is constrained: charge is conserved. For the graviton, the "couplings" are the momenta, whose sum is already conserved, so the constraint falls instead on the proportionality constant: it must be the same for everyone. One soft theorem says charge is conserved; the other says gravity is universal. Both are statements that the infrared boundary of the S-matrix cannot be deformed without destroying gauge invariance.

This recovers, in covariant momentum-space language, the spinor-helicity soft factor derived for gluons in Section 59.3.4: the color-ordered soft gluon factor $S^{(0)} = \langle a\, b\rangle/(\langle a\, s\rangle\langle s\, b\rangle)$ of Eq. (59.84) is Eq. (63.70) written for a color-adjacent pair a, b of hard legs, with the polarization solved in brackets. The leading soft theorems for photon, gluon, and graviton are one phenomenon in three notations.

63.6.2 Soft theorems as Ward identities of asymptotic symmetries

The soft factors Eqs. (63.70) and (63.75) are exact, derived statements. Their reinterpretation as conservation laws is the bridge to holography, and it is the content of a body of work (Strominger and collaborators, 2013–2018) that we state precisely and scope honestly. The reinterpretation rests on a fact about gauge theories in asymptotically flat spacetime that we now make precise: not all gauge transformations are pure redundancy. Transformations whose parameter does *not* vanish at infinity act nontrivially on the physical radiative data at null infinity, and they generate genuine symmetries with conserved charges. The soft theorems are the Ward identities of these *asymptotic* (or "large") symmetries.

The structure of the correspondence. Massless particles reach the boundary of Minkowski space at null infinity, denoted $\mathcal{I}^+$ for the future and $\mathcal{I}^-$ for the past. $\mathcal{I}^\pm$ is topologically a null cylinder $\mathbb{R}_u \times S^2$, coordinatised by a retarded (advanced) time u and a point $(z, \bar{z})$ on a two-sphere, the *celestial sphere*, which labels the null direction of escape. The radiative phase space of the gauge field lives on $\mathcal{I}^\pm$. A *large gauge transformation* is one whose parameter $\varepsilon(z, \bar{z})$ tends to an angle-dependent constant at $\mathcal{I}^\pm$ rather than decaying; for gravity, the analogous transformations are the *BMS supertranslations*, diffeomorphisms $u \to u + f(z, \bar{z})$ that shift retarded time by an arbitrary function on the celestial sphere, enlarging the Poincaré translations (the constant and $\ell = 1$ modes of f) by an infinite tower of angle-dependent "time translations."

To each such transformation there corresponds a conserved charge $Q^\pm_\varepsilon$, defined as a surface integral over $\mathcal{I}^\pm$, that splits into two pieces:

$$Q^\pm_\varepsilon = Q^{\pm,\,\mathrm{soft}}_\varepsilon + Q^{\pm,\,\mathrm{hard}}_\varepsilon . \tag{63.78}$$

The *soft* part $Q^{\mathrm{soft}}_\varepsilon$ is linear in the gauge field and, when expanded in modes, creates or annihilates a zero-energy (soft) photon/graviton smeared against ε on the celestial sphere. The *hard* part $Q^{\mathrm{hard}}_\varepsilon$ is the flux of the matter charge (or energy–momentum, for gravity) through $\mathcal{I}^\pm$, weighted by ε. The two charges at past and future infinity are tied together by an *antipodal matching* condition across spatial infinity,

$$Q^+_\varepsilon = Q^-_\varepsilon , \tag{63.79}$$

which is the statement that the symmetry is a single symmetry of the S-matrix and not two unrelated ones. Sandwiching the conservation law Eq. (63.79) between an in-state and an out-state and using Eq. (63.78) gives a Ward identity:

$$\langle \mathrm{out} | \left(Q^{+,\mathrm{soft}}_\varepsilon \, \mathbf{S} - \mathbf{S} \, Q^{-,\mathrm{soft}}_\varepsilon \right) | \mathrm{in} \rangle = - \langle \mathrm{out} | \left(Q^{+,\mathrm{hard}}_\varepsilon \, \mathbf{S} - \mathbf{S} \, Q^{-,\mathrm{hard}}_\varepsilon \right) | \mathrm{in} \rangle . \tag{63.80}$$

The left-hand side, by the definition of the soft charge, is the insertion of a soft photon/graviton into the amplitude, integrated against $\varepsilon(z, \bar{z})$ over the celestial

sphere — precisely the left-hand side of the soft theorem Eq. (63.66) smeared in the soft direction. The right-hand side, by the definition of the hard charge, is ε smeared against the matter charge/energy flux, which for a state of hard particles of charges Q_i (momenta p_i) evaluates to the sum $\sum_i Q_i(\dots)$ that is exactly the residue $S^{(0)}$ of Eq. (63.70) (respectively Eq. (63.75)). The equality Eq. (63.80) *is* the leading soft theorem. We summarise the precise statement.

Theorem 63.14 (Soft theorem $\Leftrightarrow$ asymptotic-symmetry Ward identity)**.** *In a local, Lorentz-invariant quantum field theory with a massless spin-1 (respectively spin-2) field in asymptotically flat four-dimensional spacetime, the leading soft photon theorem Eq. (63.70) (respectively the leading soft graviton theorem Eq. (63.75)) is equivalent to the Ward identity Eq. (63.80) for the charges $Q^{\pm}_{\varepsilon}$ generating large $U(1)$ gauge transformations with parameter $\varepsilon(z,\bar z)$ (respectively BMS supertranslations with parameter $f(z,\bar z)$), subject to the antipodal matching Eq. (63.79). The angle-dependent function ε (resp. f) on the celestial sphere is the smearing against which the soft insertion is integrated; choosing it to be a delta function at $(z,\bar z)$ recovers the local soft factor, and integrating against a constant recovers the global conservation law Eq. (63.73) (resp. momentum conservation).*

We do not give the full surface-charge construction here, as the symplectic geometry of $\mathcal{I}^{\pm}$ is a substantial development of its own; the equivalence is established by matching the soft and hard pieces of Eq. (63.80) to the residue and the smearing of the soft factor, which is the calculation Theorem 63.13 carried out at the level of the pole. The physical upshot is the reframing we need: the leading soft theorems are conservation laws for an *infinite-dimensional* symmetry acting at the boundary of spacetime. There are not finitely many conserved charges (the total electric charge; the four-momentum) but a continuum, one for each function $\varepsilon(z,\bar z)$ (one for each $f(z,\bar z)$). It is this infinite tower of boundary charges that a candidate holographic dual must reproduce, and the natural home for an infinite tower of charges labelled by functions on S^2 is a two-dimensional conformal field theory — in which the charges are the modes of a conserved current and a stress tensor.

Historical Note

That gauge invariance of soft emission forces charge conservation and the equality of gravitational couplings is Weinberg's 1964–1965 work; he derived Eqs. (63.73) and (63.77) exactly as above. That the same soft theorems are the Ward identities of the Bondi–van der Burg–Metzner–Sachs (BMS) symmetry discovered in 1962 in the study of gravitational radiation, and of an analogous "large" gauge symmetry for the photon, was understood only fifty years later, in the 2013–2018 program of Strominger and collaborators. The reinterpretation did not change any soft factor; it

changed what they *mean*, and opened the celestial-holography program.

63.6.3 Celestial amplitudes: the Mellin transform to boost eigenstates

If the boundary theory is to be a two-dimensional CFT on the celestial sphere, its observables must transform as CFT correlators under the conformal group of the sphere. The four-dimensional Lorentz group $SO(1,3)$ is isomorphic to $SL(2,\mathbb{C})$, and $SL(2,\mathbb{C})$ acts on the celestial sphere exactly as the global conformal (Möbius) group of a two-dimensional CFT. Ordinary amplitudes, however, are written in a basis of *energy* eigenstates (momentum eigenstates), which transform reducibly and awkwardly under boosts. The remedy is to change basis on each external leg from energy eigenstates to *boost* eigenstates, and that change of basis is a Mellin transform in the energy. We now define it and derive the conformal weight it produces.

Parametrising a massless momentum by the celestial sphere. Any future-pointing null momentum can be written as a positive energy times a null direction,

$$p^\mu = \omega\, \hat{q}^\mu(z,\bar{z}), \qquad \hat{q}^\mu(z,\bar{z}) = \big(1+z\bar{z},\ z+\bar{z},\ -i(z-\bar{z}),\ 1-z\bar{z}\big), \tag{63.81}$$

where $\omega > 0$ is the energy and $(z,\bar{z})$ is the stereographic coordinate of the direction of flight on the celestial sphere; $\hat{q}^2 = 0$ for all $(z,\bar{z})$. Under a Lorentz transformation $\Lambda \in SL(2,\mathbb{C})$ acting as the Möbius map $z \to z' = (az+b)/(cz+d)$ with $ad-bc=1$, the null direction transforms with a conformal (Weyl) factor,

$$\Lambda^\mu{}_\nu\, \hat{q}^\nu(z,\bar{z}) = |cz+d|^2\ \hat{q}^\mu(z',\bar{z}'), \tag{63.82}$$

so that a momentum eigenstate is mapped to one of *rescaled* energy in the rotated direction:

$$p^\mu = \omega\, \hat{q}^\mu(z,\bar{z}) \ \longmapsto\ \Lambda^\mu{}_\nu p^\nu = \omega'\, \hat{q}^\mu(z',\bar{z}'), \qquad \omega' = |cz+d|^2\ \omega. \tag{63.83}$$

A boost in the direction $\hat{q}$ thus acts on its energy as a pure rescaling, $\omega \to e^{\lambda}\omega$ with rapidity λ; on the half-line $\omega \in (0,\infty)$ this is a dilatation. The natural basis to diagonalise a dilatation is not energy eigenstates but its Mellin-conjugate *scaling* eigenstates, which is what motivates the following definition.

Definition 63.15 (Celestial amplitude)**.** Let $A_n(\{\omega_i, z_i, \bar{z}_i; \ell_i\})$ be a momentum-space amplitude for n massless particles of helicities ℓ_i, written in the parametrisation Eq. (63.81). Its *celestial amplitude* is the Mellin transform of each external energy,

$$\widetilde{A}_n(\{\Delta_i, z_i, \bar{z}_i; \ell_i\}) = \prod_{i=1}^{n} \int_0^\infty \mathrm{d}\omega_i\, \omega_i^{\Delta_i - 1}\, A_n(\{\omega_i, z_i, \bar{z}_i; \ell_i\}), \tag{63.84}$$

where the complex parameter Δ_i is the conformal weight assigned to leg i. Equivalently, $\widetilde{A}_n$ is the amplitude evaluated on the *conformal primary wavefunctions* obtained by Mellin-transforming the plane-wave external states.

Why $\Delta = 1 + i\lambda$: the boost-eigenvalue derivation. The value of Δ is not free if the celestial operators are to furnish a unitary, complete basis of single-particle states. We derive the allowed line of Δ from the requirement that the Mellin transform be the unitary harmonic analysis of the boost (dilatation) acting on the single-particle Hilbert space. The Lorentz-invariant one-particle measure is $\widetilde{\mathrm{d}k} = \mathrm{d}^3k/[(2\pi)^3 2\omega]$; in the parametrisation Eq. (63.81), $\mathrm{d}^3k = \omega^2\,\mathrm{d}\omega\,\mathrm{d}\Omega$ with $\mathrm{d}\Omega$ the solid angle, so the energy part of the measure is

$$\frac{\mathrm{d}^3k}{2\omega} \;\propto\; \omega\,\mathrm{d}\omega\,\mathrm{d}\Omega. \tag{63.85}$$

The boost along $\hat{q}$ acts as the dilatation $\omega \to e^{\lambda}\omega$. We seek the eigenfunctions of its self-adjoint generator on the energy Hilbert space $L^2\big((0,\infty), \omega\,\mathrm{d}\omega\big)$. To make the generator self-adjoint in the measure $\omega\,\mathrm{d}\omega$, write the dilatation generator as

$$\hat{D} = -i(\omega\,\partial_\omega + 1), \tag{63.86}$$

the extra $+1$ compensating the weight ω in the measure (so that $\hat{D}$ is symmetric: $\int_0^\infty \omega\,\mathrm{d}\omega\,\bar{f}(\hat{D}g) = \int_0^\infty \omega\,\mathrm{d}\omega\,\overline{(\hat{D}f)}\,g$). Its eigenfunctions are powers ω^s,

$$\hat{D}\,\omega^s = -i(s+1)\,\omega^s, \tag{63.87}$$

with eigenvalue $-i(s+1)$. A self-adjoint operator has real eigenvalues, so $s+1$ must be purely imaginary, $s+1 = i\lambda$ with $\lambda \in \mathbb{R}$, giving the eigenfunctions $\omega^{-1+i\lambda}$, real boost eigenvalue λ. These are precisely the delta-normalizable states: substituting $\omega = e^u$,

$$\int_0^\infty \omega\,\mathrm{d}\omega\,\big(\omega^{-1+i\lambda}\big)\big(\omega^{-1+i\lambda'}\big)^* = \int_0^\infty \frac{\mathrm{d}\omega}{\omega}\,\omega^{\,i(\lambda-\lambda')} = \int_{-\infty}^{\infty} \mathrm{d}u\,e^{\,i(\lambda-\lambda')u} = 2\pi\,\delta(\lambda-\lambda'), \tag{63.88}$$

the Mellin (Plancherel) completeness relation: the boost eigenstates $\{\omega^{-1+i\lambda}\}_{\lambda\in\mathbb{R}}$ are a complete orthogonal basis of $L^2(\omega\,\mathrm{d}\omega)$. The projection onto the eigenstate of boost eigenvalue λ is therefore $\int_0^\infty \omega\,\mathrm{d}\omega\,(\omega^{-1+i\lambda})^*(\,\cdot\,) = \int_0^\infty \mathrm{d}\omega\,\omega^{-i\lambda}(\,\cdot\,)$, which is the Mellin transform of Eq. (63.84) with

$$\boxed{\Delta - 1 = -i\lambda, \quad \text{i.e.} \quad \Delta = 1 + i\lambda, \quad \lambda \in \mathbb{R}} \tag{63.89}$$

(after the conventional sign choice $\lambda \to -\lambda$ that orients the transform). The locus $\Delta = 1 + i\lambda$ is the *principal continuous series* of $SL(2,\mathbb{C})$ on the two-dimensional celestial sphere: it is exactly the set of dimensions for which the celestial operators form a complete, unitary basis. The number 1 is the celestial-sphere analog of $d/2$ for a d-dimensional CFT, here $d = 2$; the imaginary part λ is the boost eigenvalue. We have derived the weight assignment rather than quoted it: $\Delta = 1 + i\lambda$ is forced by unitarity of the change of basis from energy eigenstates to boost eigenstates.

Worked Example 63.16 (Celestial operators transform as 2D conformal primaries)**.** We now show that the Mellin-transformed operator carries the transformation law of a two-dimensional conformal primary, with its conformal dimension fixed by Eq. (63.89) and its two-dimensional *spin* fixed by the four-dimensional helicity through the little-group data of Eq. (58.5). This is the calculation that turns "celestial amplitude" into "CFT correlator."

Let $a_\ell(\omega, z, \bar z)$ denote the annihilation operator of a massless particle of helicity ℓ and momentum $p = \omega\, \hat q(z, \bar z)$. Under a Lorentz transformation $\Lambda \in SL(2, \mathbb{C})$, two things happen. First, the momentum is mapped as in Eq. (63.83): the direction undergoes the Möbius map $z \to z'$ and the energy rescales by the conformal factor $\omega \to \omega' = |cz + d|^2\, \omega$. Second, the helicity state acquires a little-group phase. A massless particle's little group is the $U(1)$ of phases $\lambda_a \to e^{i\theta/2}\lambda_a$, $\tilde\lambda_{\dot a} \to e^{-i\theta/2}\tilde\lambda_{\dot a}$ of the on-shell spinors (Eq. (58.5)), under which a helicity-ℓ state picks up $e^{i\ell\theta}$. For the Möbius map the relevant Wigner rotation angle on the celestial sphere is $\theta = -2\arg(cz + d)$, so the phase is $\left[(\bar c\bar z + \bar d)/(cz + d)\right]^\ell$ (a pure phase, since $\left|(\bar c\bar z + \bar d)/(cz + d)\right| = 1$). Collecting both effects, the operator transforms as

$$U(\Lambda)\, a_\ell(\omega, z, \bar z)\, U(\Lambda)^{-1} = \left(\frac{\bar c\bar z + \bar d}{cz + d}\right)^{\ell} a_\ell(\omega', z', \bar z'), \qquad \omega' = |cz + d|^2\, \omega. \tag{63.90}$$

Now define the celestial operator by the Mellin transform of Eq. (63.84), $\mathcal{O}_{\Delta,\ell}(z, \bar z) = \int_0^\infty \mathrm{d}\omega\, \omega^{\Delta-1}\, a_\ell(\omega, z, \bar z)$, and act with $U(\Lambda)$:

$$U(\Lambda)\, \mathcal{O}_{\Delta,\ell}(z, \bar z)\, U(\Lambda)^{-1} = \left(\frac{\bar c\bar z + \bar d}{cz + d}\right)^{\ell} \int_0^\infty \mathrm{d}\omega\, \omega^{\Delta-1}\, a_\ell(\omega', z', \bar z'). \tag{63.91}$$

Change the integration variable to $\omega' = |cz + d|^2\, \omega$, so $\omega = |cz + d|^{-2}\, \omega'$, $\mathrm{d}\omega = |cz + d|^{-2}\, \mathrm{d}\omega'$, and $\omega^{\Delta-1} = |cz + d|^{-2(\Delta-1)}\, \omega'^{\Delta-1}$. The Jacobian and the kernel combine to a power $|cz + d|^{-2(\Delta-1)-2} = |cz + d|^{-2\Delta}$, leaving

$$U(\Lambda)\, \mathcal{O}_{\Delta,\ell}(z, \bar z)\, U(\Lambda)^{-1} = |cz + d|^{-2\Delta} \left(\frac{\bar c\bar z + \bar d}{cz + d}\right)^{\ell} \mathcal{O}_{\Delta,\ell}(z', \bar z'). \tag{63.92}$$

Write $|cz + d|^{-2\Delta} = (cz + d)^{-\Delta}(\bar c\bar z + \bar d)^{-\Delta}$ and absorb the helicity phase $(\bar c\bar z + \bar d)^\ell (cz + d)^{-\ell}$:

$$U(\Lambda)\, \mathcal{O}_{\Delta,\ell}(z, \bar z)\, U(\Lambda)^{-1} = (cz + d)^{-(\Delta+\ell)}\, (\bar c\bar z + \bar d)^{-(\Delta-\ell)}\, \mathcal{O}_{\Delta,\ell}(z', \bar z'). \tag{63.93}$$

This is exactly the transformation law of a two-dimensional conformal primary of holomorphic and antiholomorphic weights

$$h = \frac{\Delta + \ell}{2}, \qquad \bar h = \frac{\Delta - \ell}{2}, \qquad \Delta = h + \bar h, \qquad J = h - \bar h = \ell, \tag{63.94}$$

under the Möbius group $SL(2,\mathbb{C})$. The four-dimensional helicity ℓ — the little-group weight of Eq. (58.5) and Theorem 58.16 — has become the two-dimensional conformal spin $J = h - \bar{h}$, while the boost eigenvalue λ of Eq. (63.89) has become the conformal dimension $\Delta = 1 + i\lambda$. The little-group data of Chapter 58 and the boost data of the Mellin transform together fix the celestial conformal weights. The change of basis is complete: a massless one-particle state of energy ω, direction $(z, \bar{z})$, helicity ℓ becomes a conformal primary operator $\mathcal{O}_{\Delta,\ell}$ of weights $(h, \bar{h})$ inserted at the point $(z, \bar{z})$ of the celestial sphere.

Celestial amplitudes as candidate CFT correlators. With each external leg promoted to a conformal primary, the celestial amplitude Eq. (63.84) is by construction a function $\widetilde{A}_n(z_i, \bar{z}_i)$ that transforms under $SL(2,\mathbb{C})$ exactly as an n-point function of primaries of weights $(h_i, \bar{h}_i)$ in a two-dimensional CFT,

$$\widetilde{A}_n(\{z_i, \bar{z}_i\}) = \Big\langle \prod_{i=1}^{n} \mathcal{O}_{\Delta_i,\ell_i}(z_i, \bar{z}_i) \Big\rangle_{\text{cel}}, \tag{63.95}$$

where the right-hand side is the postulated correlator of the candidate *celestial* CFT. The covariance Eq. (63.93) guarantees that Eq. (63.95) has the correct Möbius weight in every insertion; for example, low-point celestial amplitudes are fixed by conformal covariance up to the same data (a constant for two points, an OPE coefficient for three) that fixes a 2D CFT correlator, with the four-dimensional momentum-conserving delta function reorganising into a distribution on cross-ratios of the z_i. The soft theorems of Section 63.6.2 then furnish this candidate CFT with the structures a genuine 2D CFT must have: the leading soft photon, being a $\Delta = 1$ insertion (a dimension-one primary with $J = 1$, i.e. a $(h, \bar{h}) = (1, 0)$ current), is a *conserved Kac–Moody current* $J_z(z)$ on the celestial sphere, and the supertranslation Ward identity dresses the celestial sphere with a *stress tensor* T_{zz} assembled from the soft graviton. The asymptotic symmetries of Section 63.6.2 are, in this language, the symmetry currents of the celestial CFT.

63.6.4 The flat-space holography program, scoped and dated

We must now say, with the honesty the chapter has demanded throughout, exactly what has and has not been established. What is *proved*, and used nowhere conjecturally above, is the soft factorization: Eqs. (63.70) and (63.75), with their gauge content Eqs. (63.73) and (63.77), are exact theorems of any local, Lorentz-invariant theory with the relevant massless field, derived in full in Section 63.6.1 from the on-shell pole. What is also rigorous is the change of basis: the Mellin transform Eq. (63.84) with $\Delta = 1 + i\lambda$ is a genuine unitary map from momentum eigenstates to boost eigenstates (Eq. (63.88)), and the resulting celestial amplitudes Eq. (63.95) carry, as a theorem, the Möbius covariance of 2D CFT correlators (Theorem 63.16). These are facts.

What is a *program*, not a theorem, is the holographic claim itself. *Celestial holography* is the conjecture that there exists an honest two-dimensional conformal field theory — with a complete local operator content, an associative operator product expansion, a state–operator correspondence, and a unitary Hilbert space — whose correlation functions are the celestial amplitudes of a four-dimensional theory of quantum gravity in asymptotically flat space. As of the mid-2020s this remains unestablished in several concrete senses, which we name rather than paper over. The celestial correlators sit on the principal series $\Delta = 1 + i\lambda$ and involve distributions in Δ and contact terms in the z_i; the existence of a standard radially-quantised CFT with a discrete spectrum and a normalizable vacuum reproducing them is not demonstrated, and the candidate theory appears non-unitary or at least unusual by the standards of Chapter 48. The celestial OPE is known only as an expansion read off from the collinear limits of amplitudes, not from an independently defined local product. And the dictionary is presently a *kinematic* reorganisation of a known S-matrix, not a derivation of the bulk from an intrinsically-defined boundary theory — the direction that would make it holography in the full sense. An alternative "Carrollian" formulation, organising the boundary data on $\mathscr{I}^{\pm}$ in retarded time u rather than in boost weight Δ, is under active development as of this writing; the two are complementary descriptions of the same null-boundary data.

It is worth placing this beside the holography we did establish. The AdS/CFT correspondence of Chapter 53 is itself a conjecture (a precisely-stated one), but it comes equipped with a working computational dictionary: the GKP–Witten relation Eq. (53.162) computes boundary correlators from a bulk action, and the flat-space limit of Section 63.4, Eq. (63.54), extracts the flat S-matrix from AdS correlators as the curvature radius is sent to infinity. That flat-space limit is the rigorous bridge between the two pictures: a four-dimensional flat amplitude is the large-radius limit of a five-dimensional AdS amplitude, whose boundary is a genuine three-dimensional CFT. Celestial holography proposes a *different* screen for the same flat physics — the two-dimensional celestial sphere at null infinity rather than the codimension-one conformal boundary of AdS — and a correspondingly different (and lower-dimensional) putative dual. The entanglement-based diagnostics of Chapter 55 (the replica construction of Section 55.2) provide one of the tools with which the information content of such a boundary theory can in principle be tested. Whether a self-consistent celestial CFT exists, and whether it is genuinely lower-dimensional holography or a clever bookkeeping of the S-matrix, is among the live questions of the amplitudes/holography frontier as this book is written.

We have closed the circle the chapter set out to trace. The leading soft theorems, derived in full from the factorization pole, are not curiosities of the infrared but the Ward identities of an infinite-dimensional asymptotic symmetry, and that symmetry organises the S-matrix into the correlators of a putative two-dimensional conformal field theory on the celestial sphere — the flat-space

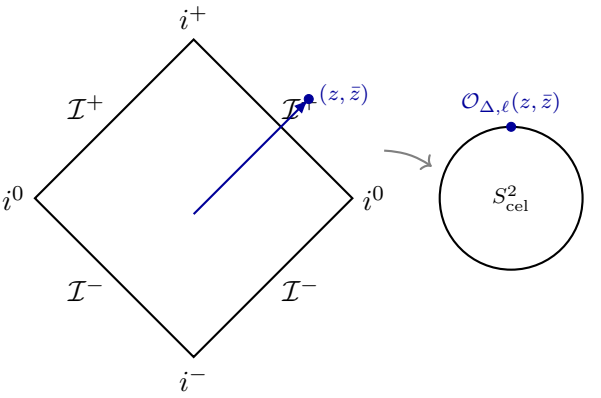


Figure 63.6: The celestial dictionary. An outgoing massless particle escapes to future null infinity $\mathcal{I}^+$ in a definite null direction, which is a point $(z, \bar{z})$ on the celestial sphere S^2_{cel}. The Mellin transform Eq. (63.84) converts the particle's energy eigenstate into a boost eigenstate of weight $\Delta = 1 + i\lambda$ (Eq. (63.89)), which is a conformal primary $\mathcal{O}_{\Delta,\ell}$ of weights $(h, \bar{h}) = (\frac{\Delta+\ell}{2}, \frac{\Delta-\ell}{2})$ inserted at $(z, \bar{z})$ (Eq. (63.93)). The S-matrix becomes a candidate celestial-CFT correlator Eq. (63.95).

face of the holography that Part XV built in anti-de Sitter space. With this last frontier on the table, the amplitudes program has shown its full hand: hidden simplicity, analyticity and unitarity promoted to construction, positivity as a quantitative constraint, the two bootstraps unified, locality and unitarity read as emergent, and now the S-matrix itself recast as a hologram. The closing section draws this arc together and renders the verdict the whole chapter has been building toward — a dated accounting, honest to the mid-2020s, of which of Part I's founding axioms survive as inputs and which reappear at the frontier as possibly emergent outputs.

63.7 Synthesis: What the Amplitudes Program Teaches

The previous section pushed the boldest reading of the whole arc: that the S-matrix itself might be a hologram — a set of correlators of a putative two-dimensional celestial conformal theory, with the leading soft factor recast as a Ward identity of an asymptotic symmetry (Section 63.6). That reframing is the natural endpoint of a single moving idea that has run through this entire part: that the data a scattering amplitude actually depends on are far smaller, and far more structured, than the Lagrangian-plus-Feynman-rules machinery that first produced them. We have now seen that idea in six distinct guises — a one-line Parke–Taylor formula (Chapter 59), an on-shell recursion that never writes an interaction vertex (Theorem 59.23), unitarity cuts that build loop integrands

from trees (Chapter 60), a double copy that squares gauge theory into gravity (Chapter 61), a geometry whose canonical form *is* the amplitude (Chapter 62), and a positivity bound that constrains an effective Lagrangian we never integrate out by hand (Section 63.3). This closing section does not prove anything new. Its task is to take stock: to state, honestly and dated to the mid-2020s, which of these claims are theorems and which are conjectures; to draw from them the single structural lesson that reorganizes the book's opening logic; and to hand the reader to the final movement, where the flat-spacetime S-matrix framework on which this entire part rests is itself called into question.

Per the distinction between theorem and conjecture fixed at the head of this chapter, everything load-bearing that the synthesis leans on has already been carried in full on the page — the abstract constructibility theorem (Theorem 63.4), the dispersive positivity sum rule (Theorem 63.7), the toy canonical form (Chapter 62), the leading soft factor (Section 63.6) — or imported from a chapter that proves it. This section asserts nothing it has not earned. Where a frontier claim is conjectural we say so and date it; where it is a theorem we point to the proof.

63.7.1 A dated status summary of the program's claims

A frontier is most useful to a reader when its boundary is drawn sharply: here is the settled interior, there is the conjectural exterior, and this is the date on which the line was drawn. The amplitudes program is unusual among the frontiers in this book in how clean that line is. A large fraction of it consists of *theorems* — statements proved from the founding analyticity and unitarity assumptions, often more rigorously than the Feynman-diagram results they replace — and the conjectural part is correspondingly well isolated. We collect the assessment in Table 63.2, then comment on each entry. The grades are: *theorem* (proved or by a short argument from imported results), *rigorous* (a controlled bound or identity following from the standard axioms), *conjecture, well tested* (a precise statement checked in many cases but not proved in general), and *program* (a research framework with partial results but no central theorem).

Hidden simplicity is established, and it is a symptom. The first entry is not a conjecture at all. The Parke–Taylor formula Eq. (59.55) is a theorem (Theorem 59.28): the n-gluon maximally-helicity-violating amplitude is a single ratio of angle brackets, while the Feynman expansion that computes it has a number of diagrams growing faster than any power of n. The lesson we drew in Section 63.2 is that this gap between the answer and the apparatus is diagnostic. It tells us that the apparatus carries redundant information — gauge dependence, off-shell continuations, field redefinitions — that cancels in the physical answer, and that there ought to be a formulation in which the redundancy is never introduced. The rest of the program is the construction of such formulations.

Claim of the program	**Status (mid-2020s)**	**Where settled / scoped**
Physical amplitudes are far simpler than their diagram count (hidden simplicity; the MHV one-liner)	theorem	Eq. (59.55), proved Theorem 59.28
On-shell recursion reconstructs tree amplitudes from lower-point on-shell data	theorem	Theorem 59.23; abstracted Theorem 63.4
The unitarity method reconstructs loop integrands from cuts (one loop, d dimensions, complete)	theorem	Theorems 60.1 and 60.26
EFT positivity bounds: leading Wilson coefficients are sign-constrained by analyticity + unitarity	rigorous	Section 63.3, Theorem 63.7
Amplitude = canonical form of a positive geometry (amplituhedron)	theorem in controlled cases; conjecture in general	proved Theorems 62.45 and 62.49; conjectural Theorem 62.47
Double copy: gravity tree amplitudes = (gauge theory)^2	theorem at tree level	Theorem 61.21; factorization Theorem 61.20
Double copy at loop level (BCJ-dual numerators exist to all loops)	conjecture, well tested	Theorem 61.32
Celestial / flat-space holography: the S-matrix as a CFT correlator	program	Section 63.6

Table 63.2: The amplitudes program graded by epistemic status, dated to the mid-2020s. The settled interior is large; the conjectural exterior is sharply bounded. Each grade points to the location where the claim is proved or honestly scoped.

On-shell recursion and the unitarity method are established. The second and third entries are the constructive payoff. BCFW recursion (Theorem 59.23), abstracted in this chapter to the on-shell constructibility theorem (Theorem 63.4), shows that a tree amplitude with appropriate large-z behaviour is fixed completely by its factorization on physical poles — by lower-point amplitudes glued across on-shell intermediate states. No Lagrangian appears in the statement or the proof; the only inputs are the residue theorem (Theorem 5.23) and the physical content of the poles. The same logic survives to loop level not as a recursion on rational functions but as the unitarity method: the discontinuity of a loop amplitude across a branch cut is a phase-space integral of lower-loop amplitudes (Cutkosky, Theorem 60.1), and at one loop in d dimensions the cuts determine the integrand completely (Theorem 60.26). These are theorems, and they are the working tools of precision collider physics. They deserve the top grade.

EFT positivity is rigorous. The fourth entry is the chapter's worked anchor and the meeting point of ϕ^4 theory with the bootstrap. The forward-dispersion argument of Theorem 63.7 takes three ingredients we proved long before the frontier — analyticity of the amplitude in the Mandelstam variables, crossing symmetry, and the unitarity statement $\operatorname{Im}\mathcal{M}(i \to i) \geq 0$ from the generalized optical theorem (Theorem 14.37) — and converts them into a strict inequality on the leading coefficient of a low-energy effective Lagrangian: the s^2 coefficient of the forward amplitude is a convergent integral of a manifestly positive absorptive part, hence positive. This is not a heuristic naturalness estimate; it is a theorem about which effective field theories can possibly arise from a unitary, causal, Lorentz-invariant completion. We grade it *rigorous* rather than *theorem* only because its scope — how many derivatives, how many fields, what spin — is itself an active subject, with the catalogue of provable bounds still being filled in as of the mid-2020s. The inequality Eq. (63.55) stands; the full map of the allowed region does not yet.

The amplituhedron is proved in controlled cases and conjectural in general. Here the line between interior and exterior runs through a single statement. That the amplitude equals the canonical form of a positive geometry is a *theorem* where the geometry is simple enough to control directly: for the $k = 1$ amplituhedron, which we showed is a convex cyclic polytope whose canonical form reproduces the NMHV-like data (Theorem 62.45), and for the boundary factorization in the tractable $m = 2$ case (Theorem 62.49). The general statement — that the canonical form of the full planar $\mathcal{N} = 4$ amplituhedron equals the tree superamplitude, and its loop generalization at the integrand level — remains a *conjecture* (Theorem 62.47), enormously well tested but not proved, and tied to the planar limit and to maximal supersymmetry as enabling crutches

(Section 62.8.3). The honest grade is split, and the date matters: this is the status as of the mid-2020s, and it is an area where a theorem could appear at any time.

The double copy is a tree-level theorem and a loop-level conjecture. The fifth and sixth entries split the same way. At tree level the statement that gravity amplitudes are gauge-theory amplitudes "squared" is a theorem: the field-theory KLT relations (Theorem 61.21) express each graviton tree as a bilinear in colour-ordered gauge trees through a kinematic kernel, and the double copy so constructed has the correct factorization channels and therefore the correct locality and unitarity (Theorem 61.20). At loop level the construction depends on the existence of BCJ-dual numerators — a rearrangement of the gauge-theory integrand into a form where kinematic numerators satisfy the same Jacobi identities as the colour factors — and that existence is a *conjecture* (Theorem 61.32), verified in a long list of nontrivial cases (through high loop order in maximal supergravity) but without a general proof. Again the boundary is sharp and the date is part of the claim.

Celestial holography is a program, not a theorem. The last entry is the one we scoped most carefully in Section 63.6. The soft theorems *are* theorems — the leading soft factor is universal and we derived it — and their reinterpretation as Ward identities of asymptotic (BMS-type) symmetries is established at leading order. But the central aspiration, that the four-dimensional S-matrix is computed by a genuine, local, unitary two-dimensional conformal field theory living on the celestial sphere, has no central theorem behind it. There are suggestive structures (the conformal covariance of celestial amplitudes under the Lorentz group acting as the global conformal group of the sphere; operator-product-like singularities from collinear limits) and serious obstructions (the would-be CFT is non-standard — continuous spectrum, unfamiliar reality and normalizability conditions). It is a *program*. Honest "this is the current understanding, mid-2020s" is the correct register, and we keep it.

63.7.2 The inversion of Part I's logic

With the status summary in hand we can state the structural lesson the program teaches, and we can state it without overclaiming. Part I built quantum field theory on a small set of postulates: that states carry a unitary representation of the Poincaré group, so that Lorentz invariance and the spectrum of particles are inputs; that observables built from local fields commute at spacelike separation, so that locality is an input; that the S-matrix is unitary, so that probability conservation is an input; and that amplitudes are boundary values of analytic functions, so that causality is an input (Section 1.5). Everything in Parts IV–VI

then *operationalized* these inputs: a local Lagrangian manifests locality term by term, the Feynman $i\epsilon$ manifests analyticity, the Hermiticity of the action manifests unitarity, and the index structure manifests Lorentz covariance. The diagrams are an algorithm for enforcing the axioms.

The amplitudes program reads the same relationship in the opposite direction. Its recurring move is to take one of these properties, which Part I treated as an irreducible input, and *derive* the amplitude from it as an output — so that the property is no longer something we impose but something the construction is forced to satisfy.

- *Locality and unitarity become outputs of factorization.* In Section 63.2.2 we saw that an on-shell-constructible amplitude is fixed by the locations and residues of its poles, and that the only admissible pole locations are $P_I^2 = m^2$ for physical channels (locality) with residues that factorize into products of lower-point amplitudes summed over physical states (unitarity). We did not assume a local Lagrangian; we found that locality and unitarity are precisely the conditions a consistent residue structure must obey. Theorem 63.4 makes this a theorem.

- *Locality and unitarity become outputs of positive geometry.* The amplituhedron re-encodes the same fact geometrically: the physical poles are the boundaries (the facets) of a positive geometry, and the factorization of the amplitude on a pole is the statement that the canonical form restricts correctly to a boundary (Section 62.8.2, Theorem 62.49), the reading developed for this chapter in Section 63.5. Locality and unitarity are read off the shape; they are not postulated. We are careful here: this is established as a *reformulation* in planar $\mathcal{N} = 4$, not as a theorem that spacetime locality is dispensable in general.

- *Diffeomorphism invariance becomes an output of squaring.* The double copy builds a graviton amplitude with the correct gauge (here, linearized-diffeomorphism) Ward identities from two copies of a gauge-theory amplitude, without ever writing the Einstein–Hilbert action (Theorem 61.20). The defining symmetry of gravity emerges from the kinematic algebra of the gauge theory.

- *Even Lorentz invariance loosens its grip.* The celestial reframing trades manifest four-dimensional Lorentz covariance for two-dimensional conformal covariance on the sphere (Section 63.6); the soft theorems become symmetry Ward identities of a putative boundary theory. Lorentz invariance is not abandoned — it reappears as the global conformal group — but it is no longer the organizing input; the asymptotic symmetry is.

Physical Insight

The amplitudes program does not refute Part I; it relocates its axioms. Poincaré invariance, locality, unitarity, and analyticity are still true of every physical amplitude — the program proves them, over and over, as consequences. What changes is their *logical status*. In the Lagrangian formulation they are premises we feed in; in the on-shell formulation they are conclusions the construction is forced to. A premise that you can derive from something deeper is no longer a premise. The honest summary is not "the axioms are wrong" but "the axioms may be emergent": in the theories where the program has been carried out in full, locality and unitarity — and, on the celestial sphere, even the manifest action of the Lorentz group — look like outputs of a more primitive structure (positivity, geometry, on-shell factorization), not like the bedrock they were taken to be.

The restraint in that statement is deliberate, and it is the point of dating the summary. The inversion is a theorem in special cases — on-shell constructibility for tree amplitudes (Theorem 63.4), the polytope amplituhedron (Theorem 62.45), the tree-level double copy (Theorem 61.21) — and a conjecture or a program beyond them. We are *not* entitled to say that spacetime, or locality, or Lorentz invariance is provably emergent in nature. We are entitled to say that in the cleanest theories we can compute, the founding axioms of Part I reappear at the frontier as derived facts, and that this is strong circumstantial evidence that the right formulation of quantum field theory takes something other than a local Lagrangian as primitive. That is the book-level message of Part XVI, and it is as far as the mathematics licenses us to go.

63.7.3 the three benchmark theories meet

the benchmark theories were designed so that the frontier would grow out of the foundations rather than be bolted on, and at this point each thread arrives at a frontier endpoint that is a facet of the one structure just described. It is worth exhibiting the three endpoints side by side, because seen together they are visibly the same statement in three theories.

Worked Example 63.17 (The ϕ^4, QED, and Yang–Mills endpoints as one structure)**.** We recall — not re-derive — the terminal result of each thread and read off the common shape.

ϕ^4 theory (the bootstrap endpoint). Begun as the classical Lagrangian $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$ in Part II, this thread ran to the Wilson–Fisher fixed point (Chapter 52) and the numerical conformal bootstrap of the 3D Ising model (Chapter 50), and meets the amplitudes side here as a positivity bound. The terminal statement, established in Theorem 63.7 and recalled in Eq. (63.55), is

that the leading forward Wilson coefficient is a convergent dispersive integral of a positive absorptive part,

$$c \;=\; \frac{2}{\pi}\int_{4m^2}^{\infty}\frac{\mathrm{d}s\;\mathrm{Im}\,\mathcal{M}(s)}{s^3} \;>\; 0. \tag{63.96}$$

The coefficient of a Lagrangian operator is fixed in sign by on-shell data — the absorptive part — and a single positivity principle, with no off-shell input. This is the S-matrix bootstrap and the conformal bootstrap reading the same crossing-plus-unitarity logic (Section 63.4 and Theorem 63.10).

QED (the spinor-helicity / soft endpoint). Begun as classical $U(1)$ in Part II and carried to $g-2$ in Part V, this thread was reframed in spinor-helicity language in Chapter 58 and arrives here at the leading soft theorem (Section 63.6). Adding a soft photon of momentum $q \to 0$ to any amplitude multiplies it by the universal Weinberg soft factor,

$$\mathcal{A}_{n+1}(q) \;\xrightarrow{q\to 0}\; \Big(e\sum_i Q_i\,\frac{\varepsilon^*\cdot p_i}{p_i\cdot q}\Big)\,\mathcal{A}_n, \tag{63.97}$$

a residue on the soft pole determined entirely by the on-shell charges and momenta of the hard legs. The same factor is, on the celestial sphere, a Ward identity of an asymptotic symmetry. Again: a piece of the amplitude fixed by on-shell data plus a symmetry/pole principle, no Lagrangian needed.

Yang–Mills theory (the double-copy endpoint). Begun as classical Yang–Mills in Part VII and carried to asymptotic freedom and then to BCFW gluon amplitudes in Chapter 59, this thread reaches here the double copy (Chapter 61). Its terminal statement is that the three-graviton amplitude is the square of the three-gluon amplitude (Section 61.4.3), schematically

$$\mathcal{M}_3^{\mathrm{grav}} \;\sim\; \big(\mathcal{A}_3^{\mathrm{YM}}\big)^2, \qquad \mathcal{A}_3^{\mathrm{YM}}(1^-,2^-,3^+) = \frac{\langle 1\,2\rangle^3}{\langle 2\,3\rangle\,\langle 3\,1\rangle}, \tag{63.98}$$

with the full tree-level statement carried by the KLT relations (Theorem 61.21). Gravity — including its defining diffeomorphism symmetry — is built from the on-shell data of gauge theory by a single algebraic operation, with no Einstein–Hilbert action in sight.

The common shape. Place Eqs. (63.96) to (63.98) together. In each, the physical object — a Wilson coefficient, a soft limit, a graviton amplitude — is determined by *on-shell data* (absorptive parts, hard-leg momenta and charges, gauge-theory amplitudes) constrained by *one organizing principle* (positivity, soft-pole universality, colour–kinematics duality), and in each the founding inputs of Part I appear only as outputs: the positivity that drives Eq. (63.96) *is* unitarity; the pole in Eq. (63.97) *is* locality; the factorization that makes the square in Eq. (63.98) consistent *is* unitarity again (Theorem 61.20). Three theories, three threads, one structure. This is exactly the design the threads were built to

deliver: the bootstrap, the spinor-helicity/soft story, and the double copy are not three frontiers but three faces of the single claim that the amplitude is more primitive than the Lagrangian.

63.7.4 The frontier that remains: where the S-matrix runs out

Everything in this part has been built on a foundation so familiar that we rarely named it: a flat, non-dynamical Minkowski spacetime with well-separated asymptotic regions. The spinor-helicity variables of Chapter 58 presuppose massless on-shell momenta in flat space; the LSZ-style in/out states behind every amplitude presuppose that fields become free in the far past and far future; the celestial sphere of Section 63.6 is the conformal boundary of flat spacetime specifically. The S-matrix exists, as a clean object, because Minkowski space has a global timelike isometry that defines a vacuum and a stable particle concept, and asymptotically free regions where "in" and "out" make sense. The whole amplitudes program is, in this precise sense, a theory of the flat-space S-matrix.

The book's final movement removes that foundation. In a general curved spacetime there need be no global timelike Killing vector, hence no preferred vacuum and no unambiguous notion of a particle; the very objects the S-matrix scatters become observer-dependent. Chapter 64 builds quantum field theory on a fixed curved background and confronts this directly: the Bogoliubov transformation relating two natural mode decompositions mixes creation and annihilation operators, so that one observer's vacuum is another's thermal bath, and the Unruh effect makes the particle content of a state a property of the trajectory through it rather than of the state alone. The stress tensor must then be renormalized in a background-covariant way (Section 64.5), a problem with no analogue in the flat-space program. Chapter 65 pushes to the sharpest case: a black hole, where the same Bogoliubov logic yields Hawking radiation and a temperature, and where the question of whether an S-matrix from past to future infinity even exists — the information problem — becomes the frontier where the framework of this part is most severely tested, and where the holographic ideas of Section 63.6 and Part XV return as candidate resolutions.

That is the natural place for the book to end. We have followed quantum field theory from the postulates of Part I, which took Poincaré invariance, locality, unitarity, and analyticity as bedrock, to a frontier at which those same postulates look like emergent outputs of deeper on-shell structure — and we now hand the reader to the one setting where the bedrock beneath *them*, flat spacetime itself, is taken away, and the S-matrix framework must be rebuilt or abandoned. Part XVII is where that final reckoning happens.

Further reading

The reading that goes *beyond* this synthesis, rather than re-supplying any step taken in it, divides along the threads. For the on-shell program as a whole — spinor-helicity, recursion, the unitarity method, and the view that amplitudes are more primitive than Lagrangians — the standard modern texts are Elvang and Huang [176] and Henn and Plefka [263], together with the review by Dixon [153]. The emergence of locality and unitarity from positive geometry is developed in Arkani-Hamed and Trnka [32] and the textbook treatment Arkani-Hamed et al. [33]. The double copy and its classical and loop-level extensions are reviewed in Bern et al. [72]. EFT positivity and the modern S-matrix bootstrap, including the link to the conformal bootstrap drawn in Section 63.4, are surveyed in de Rham et al. [143] and Kruczenski et al. [308]. The celestial-holography program, scoped here as a program, is reviewed in Pasterski [385] and Raclariu [411], and the asymptotic-symmetry foundation of the soft theorems is laid out in Strominger [459].

Part XVII

QFT in Curved Spacetime

Chapter 64

QFT in Curved Spacetime

Every construction in this book so far has leaned, sometimes silently, on the Poincaré group. The mode expansion that quantized the free scalar in Chapter 9 split the field into pieces of definite positive and negative frequency, and that split was not a convention but a consequence: it was forced by demanding that the vacuum be invariant under the symmetry of spacetime, in particular under the time translations generated by a global timelike Killing vector. A generic curved spacetime has no such symmetry. There is no preferred time, no Poincaré group, and—as this chapter will make precise—no preferred notion of a particle. We treat gravity here as a fixed, classical, curved arena and a quantum field as a test object living on it: the metric $g_{\mu\nu}(x)$ is an external field, no Einstein equation is imposed and no graviton is quantized. This is the semiclassical, test-field regime, and probing its internal consistency is the whole business of the chapter.

The development imports only a small, sharply identified toolkit. From Chapter 9 we carry the free ($\lambda = 0$) sector of ϕ^4 theory—the same real scalar of mass m, its Fock construction, its ladder algebra and number operator—now covariantized to an arbitrary background. From Chapter 3 we import the Wigner picture of one-particle states as unitary representations of the Poincaré group, which is exactly the structure whose loss in curved space creates the vacuum ambiguity. From Chapter 10 we take the generating functional $Z[J]$, the connected $W[J] = -i \ln Z$, the Euclidean continuation, and the QFT/stat-mech dictionary that reads a periodic imaginary time as a temperature; the curvature, stress-tensor, and gamma-matrix conventions are those of Chapter A, and the functional-determinant and heat-kernel machinery is that of Chapter E, whose curved-space Seeley–DeWitt coefficient this chapter finally computes. Throughout we use the mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and natural units $\hbar = c = 1$.

The arc is a single argument carried in seven movements. We first redo the free scalar geometrically on a globally hyperbolic background, isolating the one new ingredient—the curvature coupling $\xi R\phi^2$, with $\xi = 1/6$ singled out

by Weyl invariance—and proving that the field operator is a basis-independent geometric object while its split into particles is not. We then make that ambiguity quantitative: the Bogoliubov transformation relating two mode bases, its symplectic identities, and the formula that turns the off-diagonal coefficient β into a particle number and the in-vacuum into a squeezed condensate of pairs. Fed a concrete pair of bases—an asymptotically static expanding universe (the Bernard–Duncan model)—the machine produces an exact, finite spectrum of gravitationally created particles. Pushed to a horizon, the same machine collapses onto a Planck spectrum: the Unruh effect, derived twice, once by analytic continuation of Rindler modes and once from the regularity of Euclidean time, with the modular-theory backbone stated. The energy these states carry forces us to confront the formally divergent $\langle T_{\mu\nu} \rangle$; covariant point-splitting, the Hadamard parametrix, and Wald's axioms render it finite and essentially unique, with the residual ambiguity identified as the renormalization of the gravitational couplings. One feature survives every subtraction—the trace (conformal) anomaly—which we compute two independent ways and tie to the central charges of conformal field theory. Finally we assemble everything as variations of one object, the effective action $W[g]$, whose Euclidean smoothness at a horizon fixes the temperature $\beta = 2\pi/\kappa$ and whose metric variation is the source of the semiclassical Einstein equation. That equation, with an honest accounting of where it is trustworthy and where it fails, is the platform from which Chapter 65 launches Hawking radiation and the information problem.

64.1 Free fields on a fixed curved background

The chapter introduction has set the stakes: gravity is to be treated as a fixed, classical, curved arena, and a quantum field is to live on it. Everything we have built so far rested, sometimes invisibly, on the Poincaré group. The mode expansion of Chapter 9 split the scalar field into pieces of definite positive and negative frequency, and that split was *not* a convention: it was forced by the requirement that the vacuum be invariant under the symmetry group of spacetime, and in particular under time translations generated by a global timelike Killing vector. A generic curved spacetime has no such symmetry. There is no preferred time, no Poincaré group, and–as we will see–no preferred notion of a particle. Before we can say anything about what replaces these structures, we must first redo the construction of the free scalar field geometrically, keeping scrupulous track of which steps survive the loss of Poincaré invariance and which do not. That is the task of this section. We continue the free ($\lambda = 0$) sector of ϕ^4 theory (Chapter 9)—the same real scalar of mass m—now covariantized to live on an arbitrary curved background. This is a light extension of the thread rather than an official stage: the field, its mass, and its name are unchanged; only its home has curved.

We work throughout in the mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ of Eq. (3.1), in natural units $\hbar = c = 1$, and we set the gravitational coupling aside entirely: the metric $g_{\mu\nu}(x)$ is a prescribed external field, not a dynamical one. No Einstein equation is imposed and no graviton is quantized. This is the *semiclassical, test-field* regime, and its internal consistency is exactly what the rest of the chapter will probe.

64.1.1 The geometric arena: globally hyperbolic spacetimes

In flat space the Cauchy problem for the Klein–Gordon equation is so familiar that its geometric content goes unremarked: one gives the field and its time derivative on the surface $t = 0$ and the field equation propagates the data to all later and earlier times. On a curved background the very possibility of doing this is a nontrivial restriction on the geometry, and we isolate it first, because the entire quantization rests on it.

Definition 64.1 (Cauchy surface and global hyperbolicity)**.** Let (M, g) be a smooth, time-orientable Lorentzian four-manifold. A smooth spacelike hypersurface $\Sigma \subset M$ is a *Cauchy surface* if every inextendible causal (timelike or null) curve in M intersects Σ exactly once. A spacetime that admits a Cauchy surface is called *globally hyperbolic*.

The condition says that Σ is met once and only once by every worldline of a massive or massless signal: "once" because Σ is achronal (no causal curve meets it twice), and "at least once" because no causal curve can sneak past it. Physically, the data on Σ are then causally complete—nothing can enter the future of Σ without having registered on Σ, and nothing in the past of Σ fails to be recorded there. The classic geometric theorem, which we import rather than prove (it belongs to global Lorentzian geometry, not to field theory), is that a globally hyperbolic (M, g) is diffeomorphic to a product $\mathbb{R} \times \Sigma$, foliated by a one-parameter family of Cauchy surfaces Σ_t labelled by a global time function t, with the metric taking the form

$$\mathrm{d}s^2 = N(t,\mathbf{x})^2\,\mathrm{d}t^2 - h_{ij}(t,\mathbf{x})\,\mathrm{d}x^i\,\mathrm{d}x^j, \tag{64.1}$$

where $N > 0$ is the lapse function and h_{ij} is the (positive-definite) induced spatial metric on the slices. We take all spacetimes in this chapter to be globally hyperbolic. This is the precise sense in which "a sensible time direction exists." It guarantees a well-posed classical initial-value problem and is exactly what we will need to set up canonical commutation relations on a slice and propagate them. What it does *not* provide is any preferred choice among the infinitely many admissible time functions t: the slicing Eq. (64.1) is wildly non-unique, and that non-uniqueness is the seed of everything to come.

Physical Insight

Global hyperbolicity is the curved-space replacement for "space and time exist and signals propagate causally." It buys us a Cauchy problem and a Hilbert space of solutions. It pointedly does *not* buy us a clock that all observers agree on. In flat space the Poincaré group quietly supplied that clock; here it is gone, and the missing clock is precisely the missing definition of "particle" that drives the Unruh and Hawking effects later in the chapter.

64.1.2 The covariant action and the curvature coupling

We now write the action. The guiding principle is the one stated in Chapter 6: the action is a coordinate-scalar built by integrating a Lagrangian density against the invariant volume element. In flat space the free scalar density was Eq. (6.21); we covariantize it by the minimal recipe $\eta_{\mu\nu} \to g_{\mu\nu}$, $\partial_\mu \to \nabla_\mu$, and $\mathrm{d}^4x \to \sqrt{-g}\,\mathrm{d}^4x$, where $g \equiv \det g_{\mu\nu} < 0$ and $\sqrt{-g}\,\mathrm{d}^4x$ is the unique (up to orientation) volume form that is invariant under coordinate changes. For a scalar field, $\nabla_\mu\phi = \partial_\mu\phi$, so the kinetic term needs no connection. There is, however, one new ingredient that has no flat-space analogue: a local scalar built from the field and the geometry that we are *permitted* to add because it respects every symmetry the kinetic term does. The Ricci scalar R is such an object, and the dimension-correct, parity-even, local term it allows is $\xi R\,\phi^2$. We therefore write the most general action that is a coordinate scalar, quadratic in ϕ, of mass dimension four in the Lagrangian density, and built from no more than two derivatives:

$$S[\phi; g] = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[\, g^{\mu\nu}\,\partial_\mu\phi\,\partial_\nu\phi - (m^2 + \xi R)\phi^2 \,\Big]. \tag{64.2}$$

Here R is the Ricci scalar of $g_{\mu\nu}$, and ξ is a new dimensionless coupling—the *curvature coupling*—that the minimal recipe alone does not fix. Two values are distinguished. The choice $\xi = 0$ is *minimal coupling*, the literal output of the substitution rule. The choice

$$\xi = \xi_c \equiv \frac{d-2}{4(d-1)}, \qquad \xi_c\big|_{d=4} = \frac{1}{6}, \tag{64.3}$$

is *conformal coupling*, and Section 64.1.3 will earn its name by showing that the massless action at $\xi = 1/6$ is invariant under local rescalings of the metric. For the moment ξ is simply a parameter; we keep it general and let the physics single out ξ_c.

Notation Watch

The sign of the $\xi R\phi^2$ term and even the sign of R itself are convention-laden, because the Riemann tensor's sign and the metric signature both enter. We fix conventions once (mirroring Appendix A): with the mostly-minus metric we take $R^{\rho}{}_{\sigma\mu\nu} = \partial_\nu \Gamma^{\rho}_{\mu\sigma} - \partial_\mu \Gamma^{\rho}_{\nu\sigma} + \Gamma^{\rho}_{\nu\lambda}\Gamma^{\lambda}_{\mu\sigma} - \Gamma^{\rho}_{\mu\lambda}\Gamma^{\lambda}_{\nu\sigma}$, $R_{\sigma\nu} = R^{\rho}{}_{\sigma\rho\nu}$, and $R = g^{\sigma\nu}R_{\sigma\nu}$. This is the convention in which the field equation reads $(\Box_g + m^2 + \xi R)\phi = 0$ as in Eq. (64.6) and in which the conformal value comes out to $+1/6$ in $d = 4$, as Section 64.1.3 verifies; the opposite Riemann sign would flip the sign of R and of ξ together, leaving the physics untouched. The term $-\xi R\phi^2$ in Eq. (64.2) then acts as a position-dependent mass-squared shift $m^2 \to m^2 + \xi R$. With the opposite Riemann sign the conformal value is still $1/6$; only the relative sign of the shift flips. We will not need the explicit curvature components of any particular spacetime in this section; R enters only as a fixed scalar function.

The field equation follows from stationarity of Eq. (64.2) under $\phi \to \phi + \delta\phi$, exactly as in Theorem 6.8 but now with the covariant measure. Varying,

$$\begin{aligned} \delta S &= \int \mathrm{d}^4x\, \sqrt{-g}\, \Big[\, g^{\mu\nu}\, \partial_\mu\phi\, \partial_\nu(\delta\phi) - (m^2 + \xi R)\phi\, \delta\phi \,\Big] \\ &= \int \mathrm{d}^4x\, \Big[-\partial_\nu\big(\sqrt{-g}\, g^{\mu\nu}\partial_\mu\phi\big) - \sqrt{-g}\,(m^2 + \xi R)\phi \,\Big]\delta\phi, \end{aligned} \tag{64.4}$$

where the second line integrated by parts and discarded the total divergence $\partial_\nu(\sqrt{-g}\, g^{\mu\nu}\partial_\mu\phi\,\delta\phi)$, whose integral vanishes for $\delta\phi$ of compact support. Demanding $\delta S = 0$ for arbitrary $\delta\phi$ and dividing by $\sqrt{-g}$ gives the field equation. Recognizing the covariant scalar d'Alembertian

$$\Box_g\phi \equiv \frac{1}{\sqrt{-g}}\,\partial_\mu\big(\sqrt{-g}\, g^{\mu\nu}\,\partial_\nu\phi\big) = g^{\mu\nu}\nabla_\mu\nabla_\nu\phi, \tag{64.5}$$

the equation of motion is the covariant Klein–Gordon equation

$$(\Box_g + m^2 + \xi R)\phi = 0. \tag{64.6}$$

The two forms of $\Box_g$ in Eq. (64.5) agree because for a scalar $\nabla_\mu\phi = \partial_\mu\phi$ and the contracted Christoffel identity $\Gamma^{\mu}_{\mu\nu} = \partial_\nu \ln\sqrt{-g}$ converts the covariant divergence of the vector $V^\mu = g^{\mu\nu}\partial_\nu\phi$ into the coordinate divergence with the $\sqrt{-g}$ weight; we verify that identity in Theorem 64.2 below. Equation (64.6) reduces to the familiar $(\Box + m^2)\phi = 0$ of Eq. (6.55) when $g_{\mu\nu} \to \eta_{\mu\nu}$ (so $R = 0$ and $\Box_g \to \Box$), as it must. The pieces that survived the move to curved space are thus exactly the action principle and the field equation; what has appeared is the single new coupling ξ and the curvature scalar it multiplies.

Worked Example 64.2 (The two forms of $\Box_g$)**.** We show that $\frac{1}{\sqrt{-g}}\partial_\mu(\sqrt{-g}\,V^\mu) = \nabla_\mu V^\mu$ for any vector field V^μ, which applied to $V^\mu = g^{\mu\nu}\partial_\nu\phi$ establishes the equality in Eq. (64.5). By definition the covariant divergence is

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \Gamma^\mu_{\mu\nu} V^\nu. \tag{64.7}$$

The trace of the Christoffel symbols is, writing $g^{\nu\rho}\partial_\mu g_{\nu\rho}$ in terms of the determinant,

$$\Gamma^\mu_{\mu\nu} = \tfrac{1}{2} g^{\mu\rho}(\partial_\mu g_{\rho\nu} + \partial_\nu g_{\rho\mu} - \partial_\rho g_{\mu\nu}) = \tfrac{1}{2} g^{\mu\rho}\,\partial_\nu g_{\rho\mu}, \tag{64.8}$$

where the first and third terms cancelled after relabelling $\mu \leftrightarrow \rho$ because $g^{\mu\rho}$ is symmetric. Now use Jacobi's formula for the variation of a determinant, $\delta(\det g_{\mu\nu}) = (\det g_{\mu\nu})\, g^{\rho\mu}\delta g_{\mu\rho}$, in the form $\partial_\nu g = g\, g^{\rho\mu}\partial_\nu g_{\mu\rho}$ where $g = \det g_{\mu\nu}$. Hence $\frac{1}{2} g^{\mu\rho}\partial_\nu g_{\rho\mu} = \frac{1}{2g}\partial_\nu g = \partial_\nu \ln\sqrt{|g|} = \partial_\nu \ln\sqrt{-g}$ (the last step because $g<0$ makes $|g| = -g$ and the sign drops out of the logarithmic derivative). Therefore $\Gamma^\mu_{\mu\nu} = \frac{1}{\sqrt{-g}}\partial_\nu\sqrt{-g}$, and substituting into Eq. (64.7),

$$\nabla_\mu V^\mu = \partial_\mu V^\mu + \frac{\partial_\nu\sqrt{-g}}{\sqrt{-g}}\,V^\nu = \frac{1}{\sqrt{-g}}\Big(\sqrt{-g}\,\partial_\mu V^\mu + V^\mu\partial_\mu\sqrt{-g}\Big) = \frac{1}{\sqrt{-g}}\,\partial_\mu(\sqrt{-g}\,V^\mu), \tag{64.9}$$

which is the claim. This is the identity that makes the coordinate-divergence form of $\Box_g$ practical: it is precisely the form that appears, ready for integration by parts against the measure $\sqrt{-g}\,\mathrm{d}^4x$, throughout the chapter.

64.1.3 Weyl invariance singles out $\xi = 1/6$

We promised that the value Eq. (64.3) is not arbitrary. The statement that earns it its name is that the *massless* action at $\xi = \xi_c$ is invariant under a local rescaling of the metric—a Weyl, or conformal, transformation—provided the field is rescaled with the right weight. Such a rescaling is not a coordinate change and not an isometry; it physically alters the geometry by stretching lengths point by point. That a field theory should be blind to it is a strong and special property, possessed by the massless scalar at one particular ξ and by nothing else in the one-parameter family Eq. (64.2). We now prove it, carrying every term.

Definition 64.3 (Weyl transformation)**.** A *Weyl* (or *conformal*) transformation is a smooth positive rescaling of the metric by a spacetime-dependent factor,

$$g_{\mu\nu}(x) \;\longmapsto\; \tilde g_{\mu\nu}(x) = \Omega(x)^2\, g_{\mu\nu}(x) = e^{2\omega(x)}\, g_{\mu\nu}(x), \qquad \Omega = e^{\omega} > 0, \tag{64.10}$$

accompanied by a rescaling of the scalar field with *conformal weight* $-(d-2)/2$,

$$\phi(x) \;\longmapsto\; \tilde\phi(x) = \Omega(x)^{-(d-2)/2}\,\phi(x) = e^{-\frac{d-2}{2}\omega(x)}\,\phi(x). \tag{64.11}$$

In $d = 4$ the weight is -1, so $\tilde\phi = e^{-\omega}\phi$.

The basic consequences of Eq. (64.10) are read off from the definitions of the determinant and inverse metric: in d dimensions $\sqrt{-\tilde g} = \Omega^d \sqrt{-g} = e^{d\omega}\sqrt{-g}$ and $\tilde g^{\mu\nu} = \Omega^{-2} g^{\mu\nu} = e^{-2\omega} g^{\mu\nu}$. The only nontrivial ingredient is the transformation of the Ricci scalar, which we isolate as a lemma and prove from the Christoffel symbols, since the whole point is to carry every term.

Lemma 64.4 (Conformal transformation of the Ricci scalar)**.** *Under Eq. (64.10), with all derivatives and the d'Alembertian $\Box_g$ taken with respect to g and $(\partial\omega)^2 \equiv g^{\mu\nu}\partial_\mu\omega\,\partial_\nu\omega$, the Ricci scalar transforms as*

$$\tilde R = e^{-2\omega}\Big[\, R + 2(d-1)\,\Box_g\omega + (d-1)(d-2)\,(\partial\omega)^2 \,\Big]. \tag{64.12}$$

Proof. The Levi-Civita connection of $\tilde g$ differs from that of g by a tensor, because the difference of two connections is a tensor. Substituting $\tilde g_{\mu\nu} = e^{2\omega} g_{\mu\nu}$ into the Christoffel formula $\tilde\Gamma^\lambda_{\mu\nu} = \tfrac12 \tilde g^{\lambda\rho}(\partial_\mu \tilde g_{\rho\nu} + \partial_\nu \tilde g_{\rho\mu} - \partial_\rho \tilde g_{\mu\nu})$, the factors $e^{2\omega}$ from the metric and $e^{-2\omega}$ from the inverse cancel against each other, and the derivatives of $e^{2\omega}$ produce

$$\tilde\Gamma^\lambda_{\mu\nu} = \Gamma^\lambda_{\mu\nu} + C^\lambda_{\mu\nu}, \qquad C^\lambda_{\mu\nu} = \delta^\lambda_\mu\,\omega_\nu + \delta^\lambda_\nu\,\omega_\mu - g_{\mu\nu}\,\omega^\lambda, \tag{64.13}$$

where $\omega_\mu \equiv \partial_\mu\omega$ and $\omega^\lambda = g^{\lambda\rho}\omega_\rho$. Its traces are $C^\lambda_{\lambda\nu} = d\,\omega_\nu$ and $g^{\mu\nu}C^\lambda_{\mu\nu} = (2-d)\,\omega^\lambda$. With the curvature convention of the notation watch above, the Ricci tensor is $R_{\sigma\nu} = R^\rho{}_{\sigma\rho\nu}$, and the shift of the Riemann tensor by the tensor $C^\lambda_{\mu\nu}$ contracts to

$$\tilde R_{\sigma\nu} - R_{\sigma\nu} = \nabla_\nu C^\rho_{\rho\sigma} - \nabla_\rho C^\rho_{\nu\sigma} + C^\rho_{\nu\lambda}C^\lambda_{\rho\sigma} - C^\rho_{\rho\lambda}C^\lambda_{\nu\sigma}. \tag{64.14}$$

We evaluate the four pieces using Eq. (64.13). The derivative terms give

$$\begin{aligned} \nabla_\nu C^\rho_{\rho\sigma} &= \nabla_\nu(d\,\omega_\sigma) = d\,\nabla_\nu\nabla_\sigma\omega, \\ \nabla_\rho C^\rho_{\nu\sigma} &= \nabla_\rho(\delta^\rho_\nu\omega_\sigma + \delta^\rho_\sigma\omega_\nu - g_{\nu\sigma}\omega^\rho) = 2\,\nabla_\nu\nabla_\sigma\omega - g_{\nu\sigma}\,\Box_g\omega, \end{aligned} \tag{64.15}$$

where $\nabla_\nu\omega_\sigma = \nabla_\nu\nabla_\sigma\omega$ is symmetric. The quadratic terms, expanded by Eq. (64.13) and contracted index by index, give

$$C^\rho_{\rho\lambda}C^\lambda_{\nu\sigma} = d\big(2\omega_\nu\omega_\sigma - g_{\nu\sigma}(\partial\omega)^2\big), \qquad C^\rho_{\nu\lambda}C^\lambda_{\rho\sigma} = (d+2)\,\omega_\nu\omega_\sigma - 2\,g_{\nu\sigma}(\partial\omega)^2. \tag{64.16}$$

Assembling Eq. (64.14) from Eq. (64.15)–Eq. (64.16),

$$\tilde R_{\sigma\nu} - R_{\sigma\nu} = (d-2)\,\nabla_\nu\nabla_\sigma\omega + g_{\nu\sigma}\,\Box_g\omega - (d-2)\,\omega_\nu\omega_\sigma + (d-2)\,g_{\nu\sigma}(\partial\omega)^2. \tag{64.17}$$

Finally take the scalar by contracting with $\tilde g^{\sigma\nu} = e^{-2\omega}g^{\sigma\nu}$. Using $g^{\sigma\nu}\nabla_\nu\nabla_\sigma\omega = \Box_g\omega$, $g^{\sigma\nu}g_{\nu\sigma} = d$, and $g^{\sigma\nu}\omega_\nu\omega_\sigma = (\partial\omega)^2$,

$$\tilde R = e^{-2\omega}\Big[R + \big((d-2)+d\big)\Box_g\omega + \big(-(d-2)+(d-2)d\big)(\partial\omega)^2\Big] = e^{-2\omega}\big[R + 2(d-1)\Box_g\omega + (d-1)(d-\tag{64.18}$$

which is Eq. (64.12). ∎

We can now establish the headline result by inserting these transformation laws into the action and tracking every term to its cancellation.

Theorem 64.5 (Conformal invariance of the massless ξ_c action)**.** *In d dimensions, the massless ($m=0$) action Eq. (64.2) with $\xi=\xi_c=(d-2)/4(d-1)$ is invariant under the Weyl transformation Eq. (64.10)–Eq. (64.11): $S[\tilde\phi;\tilde g]=S[\phi;g]$. In $d=4$ this is $\xi_c=1/6$. No other value of ξ leaves the action invariant.*

Proof. We display the computation in $d=4$ (so $\xi_c=1/6$ and the field weight is -1, $\tilde\phi=e^{-\omega}\phi$); the general-$d$ case is identical with the weight $(d-2)/2$ carried symbolically and is recorded at the end. Insert the transformed quantities into the massless action evaluated on $(\tilde\phi,\tilde g)$. The measure and inverse metric combine in the kinetic term as $\sqrt{-\tilde g}\,\tilde g^{\mu\nu}=e^{4\omega}\sqrt{-g}\cdot e^{-2\omega}g^{\mu\nu}=e^{2\omega}\sqrt{-g}\,g^{\mu\nu}$, while $\partial_\mu\tilde\phi=e^{-\omega}(\partial_\mu\phi-\omega_\mu\phi)$. Hence the kinetic density is

$$\begin{aligned}\sqrt{-\tilde g}\,\tilde g^{\mu\nu}\partial_\mu\tilde\phi\,\partial_\nu\tilde\phi &= e^{2\omega}\sqrt{-g}\,g^{\mu\nu}\,e^{-2\omega}(\partial_\mu\phi-\omega_\mu\phi)(\partial_\nu\phi-\omega_\nu\phi)\\ &= \sqrt{-g}\Big[(\partial\phi)^2-2\phi\,(\partial\omega\cdot\partial\phi)+\phi^2(\partial\omega)^2\Big],\end{aligned}\tag{64.19}$$

with $(\partial\phi)^2\equiv g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ and $(\partial\omega\cdot\partial\phi)\equiv g^{\mu\nu}\omega_\mu\partial_\nu\phi$. The curvature density, using Eq. (64.12) with $d=4$ and $\tilde\phi^2=e^{-2\omega}\phi^2$, $\sqrt{-\tilde g}=e^{4\omega}\sqrt{-g}$, is

$$\sqrt{-\tilde g}\,\tilde R\,\tilde\phi^2=e^{4\omega}\sqrt{-g}\cdot e^{-2\omega}\big[R+6\Box_g\omega+6(\partial\omega)^2\big]\cdot e^{-2\omega}\phi^2=\sqrt{-g}\,\phi^2\big[R+6\Box_g\omega+6(\partial\omega)^2\big].\tag{64.20}$$

Therefore, with $\xi=\xi_c=1/6$,

$$S[\tilde\phi;\tilde g]=\frac12\int \mathrm{d}^4x\,\sqrt{-g}\Big[(\partial\phi)^2-2\phi(\partial\omega\cdot\partial\phi)+\phi^2(\partial\omega)^2-\tfrac16\phi^2R-\phi^2\Box_g\omega-\phi^2(\partial\omega)^2\Big].\tag{64.21}$$

The two $\phi^2(\partial\omega)^2$ terms cancel outright. The cross term is recast by noting $2\phi(\partial\omega\cdot\partial\phi)=\omega^\mu\partial_\mu(\phi^2)$ and integrating by parts against the measure, discarding the boundary term:

$$-\frac12\int \mathrm{d}^4x\,\sqrt{-g}\,\omega^\mu\partial_\mu(\phi^2)=+\frac12\int \mathrm{d}^4x\,\sqrt{-g}\,\phi^2\,\nabla_\mu\omega^\mu=+\frac12\int \mathrm{d}^4x\,\sqrt{-g}\,\phi^2\,\Box_g\omega,\tag{64.22}$$

using Theorem 64.2 to write $\nabla_\mu\omega^\mu=\Box_g\omega$. This freshly generated $+\frac12\phi^2\Box_g\omega$ cancels exactly the $-\frac12\phi^2\Box_g\omega$ coming from the curvature term. Every ω- dependent term has now disappeared, leaving

$$S[\tilde\phi;\tilde g]=\frac12\int \mathrm{d}^4x\,\sqrt{-g}\Big[(\partial\phi)^2-\tfrac16\phi^2R\Big]=S[\phi;g],\tag{64.23}$$

which is the massless $\xi=1/6$ action back again. This proves invariance.

To see that $\xi=1/6$ is forced, repeat the computation in general d keeping ξ arbitrary. The kinetic term again produces, after the same integration by parts, a leftover $+\frac12(d-2)/2\cdot\phi^2\Box_g\omega$ from the field-weight $(d-2)/2$ piece (the $(\partial\omega)^2$

pieces and cross terms organize exactly as above), while the curvature term contributes $-\frac{1}{2}\xi\cdot 2(d-1)\phi^2\Box_g\omega$ and a matching $(\partial\omega)^2$ structure. Collecting the coefficient of $\frac{1}{2}\phi^2[\Box_g\omega+(\partial\omega)^2]$ in $S[\tilde\phi;\tilde g]-S[\phi;g]$ gives, after the algebra,

$$S[\tilde\phi;\tilde g]-S[\phi;g]=\frac{1}{2}\Big(1-4(d-1)\xi/(d-2)\Big)\cdot\frac{d-2}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\phi^2\,[\Box_g\omega+(\partial\omega)^2], \tag{64.24}$$

which vanishes for all ω if and only if $\xi=(d-2)/4(d-1)=\xi_c$. For $d=4$ this is $\xi=1/6$. Any other ξ leaves a nonzero ω-dependent remainder, so the massless action is conformally invariant at ξ_c and at no other value. □

Physical Insight

The conformal value $\xi=1/6$ is exactly the coupling that makes the curvature term conspire with the gradient term so that a local rescaling of all lengths leaves the massless scalar's dynamics untouched. Two physical readings of this will matter later. First, a mass term $\frac{1}{2}m^2\phi^2$ transforms with the "wrong" weight ($\sqrt{-\tilde g}\,\tilde\phi^2=e^{2\omega}\sqrt{-g}\,\phi^2$ in $d=4$, not invariant), so mass breaks conformal invariance—mass is precisely a scale, and a scale is what a Weyl rescaling moves. Second, in a conformally flat spacetime ($g_{\mu\nu}=\Omega^2\eta_{\mu\nu}$, which includes the Friedmann–Lemaître cosmologies) the massless conformally coupled field is, up to the field rescaling Eq. (64.11), just a free flat-space field: its positive-frequency modes, and hence its vacuum, can be borrowed from Minkowski. This is the one curved-space setting in which the vacuum ambiguity is tamed by symmetry, and the later sections lean on it as the clean control case.

64.1.4 The Klein–Gordon inner product and its slice independence

To build a Hilbert space we need an inner product on the space of solutions of Eq. (64.6). In flat space this inner product was implicit in the orthonormality of modes (Theorem 9.8); we now construct it geometrically and prove the property that makes it useful—that it does not depend on which Cauchy slice we evaluate it on. The construction begins with a conserved current built bilinearly from two solutions.

Definition 64.6 (Klein–Gordon current and inner product)**.** For any two (in general complex) solutions ϕ_1,ϕ_2 of Eq. (64.6), define the *Klein–Gordon current*

$$J^\mu[\phi_1,\phi_2]=-i\,g^{\mu\nu}(\phi_1\,\partial_\nu\phi_2^*-\phi_2^*\,\partial_\nu\phi_1)\equiv -i(\phi_1\,\partial^\mu\phi_2^*-\phi_2^*\,\partial^\mu\phi_1), \tag{64.25}$$

and the *Klein–Gordon inner product* on a Cauchy surface Σ,

$$(\phi_1,\phi_2)_\Sigma=\int_\Sigma \mathrm{d}\Sigma^\mu\,J_\mu[\phi_1,\phi_2]=-i\int_\Sigma \mathrm{d}\Sigma^\mu\,(\phi_1\,\partial_\mu\phi_2^*-\phi_2^*\,\partial_\mu\phi_1), \tag{64.26}$$

where $\mathrm{d}\Sigma^\mu = n^\mu\,\mathrm{d}\Sigma$ is the directed surface element: n^μ is the future-pointing unit normal to Σ ($n^\mu n_\mu = +1$ for a spacelike surface in our signature) and $\mathrm{d}\Sigma = \sqrt{h}\,\mathrm{d}^3x$ is the induced volume element built from the spatial metric h_{ij} of Eq. (64.1).

Two remarks fix the meaning. First, the product is conjugate-linear in its second slot and linear in its first, and it is *indefinite*: it is not a norm but a sesquilinear form, and $(\phi,\phi)_\Sigma$ can be positive, negative, or zero. This indefiniteness is the engine of the whole subject—it is what makes the positive-norm/negative-norm split a meaningful but non-canonical choice. Second, on a constant-time slice in flat space, $n^\mu = (1,\mathbf{0})$ and $\mathrm{d}\Sigma = \mathrm{d}^3x$, so Eq. (64.26) reduces to $-i\int \mathrm{d}^3x\,(\phi_1\dot\phi_2^* - \phi_2^*\dot\phi_1)$, the textbook flat-space form. The geometric content beyond that form is the claim that the answer is the same on *every* Cauchy slice, which we now prove.

Theorem 64.7 (Slice independence of the Klein–Gordon product)**.** *Let ϕ_1,ϕ_2 be solutions of Eq. (64.6) on a globally hyperbolic (M,g), with spatial fall-off fast enough that all boundary terms at spatial infinity vanish. Then the current Eq. (64.25) is covariantly conserved, $\nabla_\mu J^\mu = 0$, and consequently $(\phi_1,\phi_2)_\Sigma$ is independent of the choice of Cauchy surface Σ: $(\phi_1,\phi_2)_{\Sigma_1} = (\phi_1,\phi_2)_{\Sigma_2}$ for any two Cauchy surfaces Σ_1,Σ_2.*

Proof. We compute the covariant divergence of Eq. (64.25) directly. Because ∇_μ annihilates the metric, $\nabla_\mu(g^{\mu\nu}X_\nu) = g^{\mu\nu}\nabla_\mu X_\nu$, so

$$\nabla_\mu J^\mu = -i\,\nabla_\mu\big(\phi_1\,\partial^\mu\phi_2^* - \phi_2^*\,\partial^\mu\phi_1\big). \tag{64.27}$$

Apply the Leibniz rule to each term, using again that for a scalar $\partial^\mu\phi = \nabla^\mu\phi$ and that $\nabla_\mu\nabla^\mu = \Box_g$ acting on a scalar:

$$\nabla_\mu\big(\phi_1\,\nabla^\mu\phi_2^*\big) = \nabla_\mu\phi_1\,\nabla^\mu\phi_2^* + \phi_1\,\Box_g\phi_2^*, \tag{64.28}$$

$$\nabla_\mu\big(\phi_2^*\,\nabla^\mu\phi_1\big) = \nabla_\mu\phi_2^*\,\nabla^\mu\phi_1 + \phi_2^*\,\Box_g\phi_1. \tag{64.29}$$

The two "gradient-dot-gradient" terms are identical ($\nabla_\mu\phi_1\,\nabla^\mu\phi_2^* = \nabla_\mu\phi_2^*\,\nabla^\mu\phi_1$) and cancel in the difference. Hence

$$\nabla_\mu J^\mu = -i\big(\phi_1\,\Box_g\phi_2^* - \phi_2^*\,\Box_g\phi_1\big). \tag{64.30}$$

Now use the field equation. Both ϕ_1 and ϕ_2 solve Eq. (64.6); complex-conjugating the equation for ϕ_2 and using that $m^2+\xi R$ is real gives $\Box_g\phi_2^* = -(m^2+\xi R)\phi_2^*$, while $\Box_g\phi_1 = -(m^2+\xi R)\phi_1$. Substituting,

$$\nabla_\mu J^\mu = -i\Big[\phi_1\big(-(m^2{+}\xi R)\phi_2^*\big) - \phi_2^*\big(-(m^2{+}\xi R)\phi_1\big)\Big] = -i\,(m^2{+}\xi R)\big(-\phi_1\phi_2^* + \phi_2^*\phi_1\big) = 0. \tag{64.31}$$

The current is conserved. The conservation is exact and uses nothing but the field equation and the reality of the mass-and-curvature term; the value of ξ is irrelevant to it.

To convert conservation into slice independence, integrate $\nabla_\mu J^\mu = 0$ over the spacetime region Ω bounded by two Cauchy surfaces Σ_1 (to the past) and Σ_2 (to the future) and, if the slices are noncompact, a timelike tube $\mathcal{T}$ at large spatial radius. The covariant Gauss theorem states that for any vector field J^μ,

$$\int_\Omega \mathrm{d}^4x\,\sqrt{-g}\,\nabla_\mu J^\mu = \oint_{\partial\Omega} \mathrm{d}\Sigma_\mu\, J^\mu, \tag{64.32}$$

which is itself just Theorem 64.2 integrated: $\sqrt{-g}\,\nabla_\mu J^\mu = \partial_\mu(\sqrt{-g}\,J^\mu)$ is a coordinate divergence, so its integral reduces by the ordinary divergence theorem to a boundary integral, and the induced measure on $\partial\Omega$ assembles into $\mathrm{d}\Sigma_\mu$. The left-hand side vanishes by Eq. (64.31). On the right, the tube contribution $\int_{\mathcal{T}} \mathrm{d}\Sigma_\mu J^\mu$ vanishes by the assumed spatial fall-off. The boundary $\partial\Omega$ then consists of Σ_2 with outward (future-pointing) normal and Σ_1 with outward (past-pointing) normal, so that with both surface elements oriented by their *future*-pointing normals,

$$0 = \oint_{\partial\Omega} \mathrm{d}\Sigma_\mu J^\mu = \int_{\Sigma_2} \mathrm{d}\Sigma_\mu J^\mu - \int_{\Sigma_1} \mathrm{d}\Sigma_\mu J^\mu = i\big[(\phi_1,\phi_2)_{\Sigma_2} - (\phi_1,\phi_2)_{\Sigma_1}\big]/(-i), \tag{64.33}$$

where the relative minus sign is the flip between the outward and future-pointing normals on the past surface. Hence $(\phi_1,\phi_2)_{\Sigma_2} = (\phi_1,\phi_2)_{\Sigma_1}$. Since Σ_1, Σ_2 were arbitrary, the product is the same on every Cauchy slice. □

Physical Insight

Slice independence is what lets us speak of "the inner product on the space of solutions" without naming a time. It is the curved-space avatar of conservation of the Klein–Gordon norm under time evolution. Crucially, it holds for *any* globally hyperbolic geometry and *any* value of ξ; it is one of the structures that survives the loss of Poincaré invariance intact. What will *not* survive is the next step—deciding which solutions have positive product—because that requires a notion of positive frequency, and that requires a clock.

64.1.5 Mode expansion, quantization, and the Fock space

We can now quantize. The strategy is the geometric analogue of Theorem 9.6: expand the field operator in a complete set of classical solutions and read off the commutation relations of the expansion coefficients. The novelty is that the choice of basis is no longer dictated by a symmetry, so we must state carefully what a good basis is and what is convention in the result.

Definition 64.8 (Complete orthonormal mode set)**.** A countable[1] family of complex solutions $\{u_i\}$ of Eq. (64.6) is a *complete orthonormal mode set* if, with respect to the Klein–Gordon product Eq. (64.26),

$$(u_i, u_j) = \delta_{ij}, \qquad (u_i^*, u_j^*) = -\delta_{ij}, \qquad (u_i, u_j^*) = 0, \tag{64.34}$$

and the set $\{u_i, u_i^*\}$ together spans the space of (complex) solutions, i.e. every real solution ϕ can be written as a convergent combination $\phi = \sum_i (c_i u_i + c_i^* u_i^*)$.

Three facts about Eq. (64.34) deserve emphasis because they encode exactly what is conventional. The relation $(u_i^*, u_j^*) = -\delta_{ij}$ is forced, not chosen: conjugating Eq. (64.26) gives $(\phi_1^*, \phi_2^*) = -(\phi_2, \phi_1) = -(\phi_1, \phi_2)^*$, so positive-norm solutions have negative-norm conjugates. The vanishing of (u_i, u_j^*) is a genuine requirement on the basis: it says the positive-norm sector and its conjugate are orthogonal under the product. And the decision to call the u_i (rather than the u_i^*) "positive norm"—to put them on the annihilation side below—is the one piece of true convention, the seed of the vacuum ambiguity. In flat space this decision was removed by demanding that the u_i be positive frequency with respect to the global time; here there is no such global time to appeal to, and *any* split of the solution space into a positive-norm subspace and its conjugate that satisfies Eq. (64.34) defines a legitimate basis.

With a basis fixed, the quantum field is the operator-valued distribution

$$\hat{\phi}(x) = \sum_i \left(a_i \, u_i(x) + a_i^\dagger \, u_i^*(x) \right), \tag{64.35}$$

with $\hat{\phi}$ Hermitian because the field is real and the two terms are mutual conjugates. The coefficients $a_i, a_i^\dagger$ are extracted from $\hat{\phi}$ by the inner product, exactly because the basis is orthonormal: applying Eq. (64.26) to Eq. (64.35) and using Eq. (64.34),

$$(u_i, \hat{\phi}) = \sum_j \left[a_j (u_i, u_j) + a_j^\dagger (u_i, u_j^*) \right] = a_i, \qquad (u_i^*, \hat{\phi}) = \sum_j \left[a_j (u_i^*, u_j) + a_j^\dagger (u_i^*, u_j^*) \right] = -a_i^\dagger. \tag{64.36}$$

These projection formulas are the curved-space counterpart of the inversion relations Eq. (9.44)–Eq. (9.45).

The commutation relations now follow from the canonical structure. The quantization postulate is the equal-time canonical commutator on a Cauchy slice Σ, the geometric form of Eq. (9.7): with the momentum conjugate to ϕ being $\pi = \sqrt{-g}\, g^{0\nu} \partial_\nu \phi = \sqrt{h}\, n^\nu \partial_\nu \phi / N \cdot N = \sqrt{h}\, n^\nu \partial_\nu \phi$ in the slicing Eq. (64.1),[2] the

[1] We write the basis as discrete for notational economy, as in Chapter 9; for noncompact slices the index i is continuous and Kronecker deltas become Dirac deltas, with sums becoming integrals. Nothing in the argument depends on this.

[2] The conjugate momentum is $\pi = \partial \mathcal{L} / \partial \dot{\phi}$ with $\mathcal{L}$ the density *including* the $\sqrt{-g}$ factor; in the lapse/shift-free form Eq. (64.1) one finds $\pi = (\sqrt{h}/N)\dot{\phi} = \sqrt{h}\, n^\nu \partial_\nu \phi$, the spatial volume element times the normal (proper-time) derivative.

postulate is

$$[\hat{\phi}(x), \hat{\phi}(y)]\big|_\Sigma = 0, \qquad [\hat{\phi}(x), \hat{\pi}(y)]\big|_\Sigma = i\,\delta^{(3)}(x,y), \tag{64.37}$$

where $\delta^{(3)}(x,y)$ is the delta distribution on Σ normalized against the induced measure $\mathrm{d}\Sigma = \sqrt{h}\,\mathrm{d}^3x$ (so $\int_\Sigma \mathrm{d}\Sigma_y\,\delta^{(3)}(x,y)f(y) = f(x)$). Feeding Eq. (64.35) through the projection formulas Eq. (64.36) and using Eq. (64.37) converts these field commutators into commutators of the coefficients. The computation is the inner-product packaging of the flat-space one in Theorem 9.9: schematically, $[a_i, a_j^\dagger] = (u_i, \hat{\phi})\,(\hat{\phi}, u_j)$-type brackets collapse, via the canonical commutator and the orthonormality Eq. (64.34), to

$$[a_i, a_j] = 0, \qquad [a_i^\dagger, a_j^\dagger] = 0, \qquad [a_i, a_j^\dagger] = \delta_{ij}. \tag{64.38}$$

The orthonormality of the modes is exactly what makes the right-hand side a clean δ_{ij}: a non-orthonormal basis would produce the inverse Gram matrix of the modes instead, which is why Eq. (64.34) was imposed in the first place. We make the projection-to-commutator computation fully explicit in Theorem 64.9.

Worked Example 64.9 (Ladder algebra from the canonical commutator)**.** We derive $[a_i, a_j^\dagger] = \delta_{ij}$ from Eq. (64.37). The cleanest route writes the inner product Eq. (64.26) on the slice Σ in terms of $\hat{\phi}$ and $\hat{\pi}$. Using $n^\mu\partial_\mu\hat{\phi} = \hat{\pi}/\sqrt{h}$ and $\mathrm{d}\Sigma = \sqrt{h}\,\mathrm{d}^3x$, the surface integral becomes

$$(f, \hat{\phi}) = -i\int_\Sigma \mathrm{d}\Sigma\; n^\mu(f\,\partial_\mu\hat{\phi} - \hat{\phi}\,\partial_\mu f) = -i\int_\Sigma \mathrm{d}^3x\,(f\,\hat{\pi} - \sqrt{h}\,\hat{\phi}\,n^\mu\partial_\mu f), \tag{64.39}$$

for any solution f. Apply this with $f = u_i$ to get a_i, per Eq. (64.36), and with $f = u_j^*$ to get $-a_j^\dagger$. Then

$$\begin{aligned}[a_i, a_j^\dagger] &= -\big[(u_i, \hat{\phi}),\,(u_j^*, \hat{\phi})\big] \\ &= -\int_{\Sigma_x}\int_{\Sigma_y}\big(-i\big)\big(-i\big)\Big[(u_i\hat{\pi}_x - \sqrt{h}\,\hat{\phi}_x\,n\cdot\partial u_i),\,(u_j^*\hat{\pi}_y - \sqrt{h}\,\hat{\phi}_y\,n\cdot\partial u_j^*)\Big]\mathrm{d}^3x\,\mathrm{d}^3y.\end{aligned} \tag{64.40}$$

Only the cross terms survive the commutator, because $[\hat{\phi}, \hat{\phi}] = [\hat{\pi}, \hat{\pi}] = 0$ on Σ. Using $[\hat{\phi}_x, \hat{\pi}_y] = i\delta^{(3)}(x,y)$ (and the opposite-sign transpose) the two surviving terms give

$$[a_i, a_j^\dagger] = -\int_\Sigma \mathrm{d}^3x\;(-i)\Big[u_i(\sqrt{h}\,n\cdot\partial u_j^*) - (\sqrt{h}\,n\cdot\partial u_i)u_j^*\Big](-i)\cdot\frac{1}{\sqrt{h}}\cdot\sqrt{h}, \tag{64.41}$$

which, after the delta functions are integrated and the factors of $\sqrt{h}$ and $n\cdot\partial$ reassemble, is exactly $+i\int_\Sigma \mathrm{d}\Sigma\, n^\mu(u_i\partial_\mu u_j^* - u_j^*\partial_\mu u_i) = (u_i, u_j) = \delta_{ij}$. The same bookkeeping gives $[a_i, a_j] \propto (u_i, u_j^*) = 0$ and $[a_i^\dagger, a_j^\dagger] \propto (u_i^*, u_j) = 0$. Thus

Eq. (64.37) is equivalent to Eq. (64.38) together with the mode orthonormality Eq. (64.34): the canonical commutator *is* the orthonormality of the modes, read through the expansion. This is why a different orthonormal basis automatically yields ladder operators with the same algebra—a fact we exploit in the next subsection.

The algebra Eq. (64.38) is the standard creation/annihilation algebra of Chapter 9. From here the Fock space is built by the identical construction as Theorem 9.17: declare a normalized state $|0_u\rangle$ annihilated by every a_i,

$$a_i\,|0_u\rangle = 0 \quad \forall i, \qquad \langle 0_u|0_u\rangle = 1, \tag{64.42}$$

and generate the Hilbert space by repeated action of the $a_i^\dagger$, with $a_{i_1}^\dagger \cdots a_{i_n}^\dagger\,|0_u\rangle$ spanning the n-particle sector. Because the algebra is identical, the Fock space is the same abstract object as in flat space: a separable Hilbert space carrying a representation of the canonical commutators, with a number operator $N = \sum_i a_i^\dagger a_i$ and the usual spectrum.

There is, however, a deliberate change of notation that carries the entire moral of the chapter: we have written the vacuum as $|0_u\rangle$, with an explicit subscript recording the mode basis $\{u_i\}$ that defines it. In flat space we wrote $|0\rangle$ with no subscript, because there was nothing to record—the basis was unique. Here the subscript is mandatory, because Eq. (64.42) defines the vacuum *relative to the chosen basis*. A different complete orthonormal set $\{v_i\}$ defines a different annihilation operator and a different state $|0_v\rangle$, and there is no symmetry to declare one of them correct. The Fock space survived the move to curved spacetime; the preferred vacuum did not.

Caution

The label "positive norm" on the u_i is not a property of the solutions in isolation; it is a property of the chosen *splitting* of the solution space. Any solution u has $(u, u) > 0$ while its conjugate u^* has $(u^*, u^*) < 0$, so calling u the particle mode and u^* the antiparticle mode is a real choice. In flat space the choice $u_{\mathbf{k}} \propto e^{-i\omega_k t + i\mathbf{k}\cdot\mathbf{x}}$ with $\omega_k > 0$ was singled out by positive frequency with respect to the Killing time t. Strip away the Killing vector and the words "positive frequency" have no referent. The mistake to avoid is to imagine that the positive/negative-norm split is god-given; it is exactly as arbitrary as the choice of time function in Eq. (64.1).

64.1.6 The field is geometric; the particle split is not

We have asserted that the field operator $\hat{\phi}$ is well defined while its decomposition into creation and annihilation parts is conventional. This deserves a clean

statement and proof, because it is the precise sense in which physics lives upstream of the vacuum ambiguity.

Proposition 64.10 (Basis independence of the smeared field)**.** *Let* $\{u_i\}$ *and* $\{v_j\}$ *be two complete orthonormal mode sets in the sense of Theorem 64.8, defining field expansions* $\hat{\phi} = \sum_i (a_i u_i + a_i^\dagger u_i^*) = \sum_j (b_j v_j + b_j^\dagger v_j^*)$ *and vacua* $|0_u\rangle, |0_v\rangle$*. Then the two expansions define the* same *operator-valued distribution* $\hat{\phi}(x)$*: for every real test function (or, on a slice, every smearing against the canonical data), the smeared field* $\hat{\phi}(f) = \int \mathrm{d}^4x \, \sqrt{-g}\, f(x)\hat{\phi}(x)$ *computed from either basis agrees.*

Proof. The two bases are both complete in the solution space, so each v_j is a (KG-) linear combination of the u_i and their conjugates, and vice versa. Concretely, expand v_j against the u-basis using the projection Eq. (64.39) in classical form:

$$v_j = \sum_i \left[(v_j, u_i)\, u_i - (v_j, u_i^*)\, u_i^* \right] \equiv \sum_i \left[\alpha_{ji}\, u_i + \beta_{ji}\, u_i^* \right], \tag{64.43}$$

where the coefficients α, β are the inner products (v_j, u_i) and $-(v_j, u_i^*)$; their precise definition and properties are the subject of the next section and are not needed here. What matters is only that *some* such linear relation exists, with the inverse relation expressing u_i through the v_j. Substitute the expansion of every v_j (and v_j^*) into $\hat{\phi} = \sum_j (b_j v_j + b_j^\dagger v_j^*)$. The result is a linear combination of the u_i and u_i^* whose coefficients are operators built from the $b_j, b_j^\dagger$. Collecting the coefficient of u_i and of u_i^* and comparing with $\hat{\phi} = \sum_i (a_i u_i + a_i^\dagger u_i^*)$, the completeness and linear independence of $\{u_i, u_i^*\}$ force the operator multiplying u_i to equal a_i and the operator multiplying u_i^* to equal $a_i^\dagger$. That is, the change of basis re-expresses the *same* $\hat{\phi}$; the operators a_i are simply particular linear combinations of the $b_j, b_j^\dagger$ (and conversely). No appeal to any vacuum was made: the equality of the two expansions is an identity among operators, holding before any state is chosen. Smearing both sides against f preserves the identity. □

Physical Insight

This is the cleanest possible statement of what is objective in quantum field theory on a curved background. The field $\hat{\phi}(x)$—and hence every local, covariant observable built from it, the stress tensor included—is a single geometric object, independent of any mode basis. What depends on the basis is the bookkeeping that calls half of $\hat{\phi}$ "annihilation" and the other half "creation," and the state $|0\rangle$ that bookkeeping singles out. "Particle number" is therefore not a property of the field but of a field-plus-basis pair. Two observers using different bases write the same $\hat{\phi}$ but disagree about how many particles a given state contains. The reconciliation of their two bookkeepings is the Bogoliubov transformation that opens the

next section.

64.1.7 Why flat space was special

It is worth saying exactly what flat space had that a generic curved spacetime lacks, because the contrast is the conceptual pivot of the whole chapter. Minkowski space is globally hyperbolic, so it too admits the construction above; the difference is not in the existence of a Fock space but in the *uniqueness* of the preferred one. Flat space possesses a global timelike Killing vector $K = \partial_t$—a vector field generating an isometry, $\mathcal{L}_K g_{\mu\nu} = \nabla_\mu K_\nu + \nabla_\nu K_\mu = 0$, that is timelike everywhere. With it, "positive frequency" acquires an invariant meaning: a mode u is positive frequency if it is an eigenfunction of the Lie derivative along K with eigenvalue of definite sign,

$$\mathcal{L}_K u = -i\omega\, u, \qquad \omega > 0. \tag{64.44}$$

In coordinates adapted to K this is $\partial_t u = -i\omega u$, the familiar $e^{-i\omega t}$ time dependence. Because K is a symmetry, the splitting it induces is preserved by time evolution and shared by all inertial observers, and the vacuum it selects is the unique Poincaré-invariant state. This is precisely the content of the Wigner program imported from Chapter 3: a one-particle state is a unitary irreducible representation of the Poincaré group (Theorem 3.34, Section 3.7), labelled by mass and spin, and the vacuum is the unique invariant vector. The positive-frequency condition Eq. (64.44) is exactly what makes the one-particle Hilbert space carry such a representation. The flat-space scalar's diagonalization of the Hamiltonian and its Poincaré transformation properties (Section 9.3.6) are downstream of this single fact.

A generic globally hyperbolic (M, g) has no Killing vector at all—the metric Eq. (64.1) need have no symmetry whatsoever—and even when it has one, the Killing vector need not be timelike everywhere (the Schwarzschild ∂_t becomes spacelike inside the horizon; the Rindler boost Killing vector that generates uniform acceleration, central to the Unruh analysis later in this chapter, is timelike only inside a wedge). Without a global timelike Killing vector, Eq. (64.44) has no solution set to call "the" positive-frequency modes, and the construction of Section 64.1.5 runs with an irreducibly arbitrary choice of basis. There is no preferred vacuum, no invariant number operator, and no observer-independent notion of a particle. Everything that the Poincaré group did silently in flat space must now be done by hand, observer by observer—and different observers will make incompatible choices.

This is not a defect of the formalism but a physical feature: it is the statement that "particle" is an observer-dependent concept once the rigid scaffolding of Poincaré symmetry is removed. We have built the stage exactly far enough to make that statement precise. We have a covariant action and field equation

(Eq. (64.2), Eq. (64.6)), a slice-independent inner product (Theorem 64.7), and, for each choice of complete orthonormal mode set, a Fock space with a vacuum $|0_u\rangle$ tied inextricably to that choice (Eq. (64.42)). The field itself is basis-independent (Theorem 64.10); only its split into particles is not. The obvious and unavoidable question is therefore quantitative: given two such mode sets, how are their creation and annihilation operators—and their two vacua—related, and when does one observer's vacuum contain the other's particles? Answering that question is the work of the next section, where the linear relation merely gestured at in Eq. (64.43) is promoted to the Bogoliubov transformation that powers every result in the remainder of the chapter.

64.2 Bogoliubov transformations and the non-uniqueness of the vacuum

The previous section left us holding a Fock space with no canonical address. We quantized the free scalar on a globally hyperbolic background, found the canonical commutators intact, and built a perfectly good Hilbert space of "particles" on top of a vacuum — but the vacuum we obtained was tied to a choice of mode basis, and on a generic curved spacetime no choice is singled out. Two physicists who pick different complete sets of positive-frequency solutions of the field equation build genuinely different vacua, and there is no geometric fact that adjudicates between them. This is not a defect of the construction; it is the central physical content of the subject. What we must do now is make the ambiguity *quantitative*: given two such bases, we want a machine that takes the data relating them and returns the precise sense in which one observer's empty space is another's bath of particles. That machine is the Bogoliubov transformation, and every concrete effect in the rest of the chapter — cosmological particle production, the Unruh bath, Hawking radiation — is in the end a computation of its coefficients for a physically chosen pair of bases.

64.2.1 Two mode bases and the coefficients that connect them

Recall from Section 64.1 the structure we inherit. On a globally hyperbolic spacetime $(M, g_{\mu\nu})$ with mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, the free real scalar obeys the curved-space Klein–Gordon equation, and the space of its complex solutions carries the conserved Klein–Gordon inner product

$$(f,g) \;=\; i\int_{\Sigma} \mathrm{d}\Sigma^{\mu}\,\big[\,(\partial_\mu f)\,g^{*} - f\,\partial_\mu g^{*}\,\big], \qquad \mathrm{d}\Sigma^{\mu} = n^{\mu}\,\sqrt{h}\;\mathrm{d}^{3}y, \tag{64.45}$$

with Σ any Cauchy surface, n^μ its future-directed unit normal, and h the determinant of the induced metric. The value of Eq. (64.45) is independent of the choice of Σ because the current $j^\mu = i[(\partial^\mu f)g^* - f\,\partial^\mu g^*]$ is conserved, $\nabla_\mu j^\mu = 0$,

whenever f and g solve the field equation (Eq. (64.45) is sesquilinear — linear in its first slot, antilinear in its second). A *mode basis* is a discrete[3] family of solutions $\{u_j, u_j^*\}$ that is complete and orthonormal in the sense

$$(u_i, u_j) = \delta_{ij}, \qquad (u_i^*, u_j^*) = -\,\delta_{ij}, \qquad (u_i, u_j^*) = 0. \tag{64.46}$$

The minus sign in the second relation is forced by Eq. (64.45): complex conjugation flips the sign of the inner product, so if the u_j have positive norm their conjugates have negative norm. This is exactly why the "positive-frequency" modes u_j (those of positive Klein–Gordon norm) and their conjugates together span the solution space, and why the field expansion

$$\phi(x) \;=\; \sum_j \Big[\, a_j\, u_j(x) + a_j^\dagger\, u_j^*(x) \,\Big] \tag{64.47}$$

realizes the canonical commutators as $\big[a_i, a_j^\dagger\big] = \delta_{ij}$, $[a_i, a_j] = 0$, with the Fock space and number operator of Chapter 9 (Theorem 9.17, Eq. (9.110)) erected on the vacuum $a_j\,|0_a\rangle = 0$ of Theorem 9.11. All of this was established in Section 64.1; we now take a *second*, equally legitimate basis.

Let $\{v_i, v_i^*\}$ be another complete orthonormal mode basis, obeying the same relations Eq. (64.46), with its own field expansion

$$\phi(x) \;=\; \sum_i \Big[\, b_i\, v_i(x) + b_i^\dagger\, v_i^*(x) \,\Big], \qquad \big[b_i, b_j^\dagger\big] = \delta_{ij}, \quad [b_i, b_j] = 0, \tag{64.48}$$

and its own vacuum $b_i\,|0_b\rangle = 0$. The field operator $\phi(x)$ is one and the same object — it is fixed by the dynamics and the canonical data, not by our bookkeeping — so Eq. (64.47) and Eq. (64.48) are two expansions of the identical operator. Because each basis is complete, each v_i is a linear combination of the u_j and u_j^*; we name the coefficients.

Definition 64.11 (Bogoliubov coefficients)**.** The *Bogoliubov coefficients* relating the basis $\{u_j\}$ to the basis $\{v_i\}$ are

$$\alpha_{ij} \;=\; (v_i, u_j), \qquad \beta_{ij} \;=\; -\,(v_i, u_j^*). \tag{64.49}$$

We assemble them into (in general infinite) matrices $\alpha = (\alpha_{ij})$ and $\beta = (\beta_{ij})$.

These coefficients are not free data: they reconstruct the change of basis completely. Expanding each v_i over the complete set and projecting with Eq. (64.46),

$$v_i \;=\; \sum_j \Big[\, \alpha_{ij}\, u_j + \beta_{ij}\, u_j^* \,\Big]. \tag{64.50}$$

[3] We write all sums over discrete labels; on a noncompact Cauchy surface the labels are continuous and every $\sum_j$ is read as the appropriate $\int \mathrm{d}\mu(j)$ and every δ_{ij} as $\delta(i-j)$. Nothing below depends on this, and the discrete notation keeps the linear algebra legible.

That this is consistent with Eq. (64.49) is a one-line check: taking the inner product of Eq. (64.50) with u_k and using the linearity of $(\,\cdot\,,\,\cdot\,)$ in its first slot together with Eq. (64.46) gives $(v_i, u_k) = \alpha_{ik}$, while pairing with u_k^* gives $(v_i, u_k^*) = \beta_{ik}(u_k^*, u_k^*) = -\beta_{ik}$, exactly Eq. (64.49). The conjugate of Eq. (64.50) is $v_i^* = \sum_j (\alpha_{ij}^* u_j^* + \beta_{ij}^* u_j)$.

The operator dictionary. We now read off how the two sets of ladder operators are related, and we do it by substitution rather than by projecting the field — this avoids any subtlety about how the antilinear inner product acts on the operator-valued ϕ, and it makes every step pure linear algebra. Insert Eq. (64.50) and its conjugate into the b-expansion Eq. (64.48):

$$\begin{aligned}\phi &= \sum_i \Big[\, b_i \sum_j (\alpha_{ij} u_j + \beta_{ij} u_j^*) + b_i^\dagger \sum_j (\alpha_{ij}^* u_j^* + \beta_{ij}^* u_j) \,\Big] \\ &= \sum_j \Big[\underbrace{\Big(\textstyle\sum_i (\alpha_{ij} b_i + \beta_{ij}^* b_i^\dagger)\Big)}_{\text{coefficient of } u_j} u_j \;+\; \underbrace{\Big(\textstyle\sum_i (\beta_{ij} b_i + \alpha_{ij}^* b_i^\dagger)\Big)}_{\text{coefficient of } u_j^*} u_j^* \Big]. \end{aligned} \tag{64.51}$$

Matching Eq. (64.51) term by term against the a-expansion Eq. (64.47) (the modes u_j, u_j^* are linearly independent, so their coefficients must agree) gives

$$a_j \;=\; \sum_i \Big[\, \alpha_{ij}\, b_i + \beta_{ij}^*\, b_i^\dagger \,\Big], \tag{64.52}$$

the coefficient of u_j^* reproducing precisely the Hermitian conjugate of Eq. (64.52), as it must. Inverting the mode relation Eq. (64.50) — equivalently, repeating the substitution with the roles of the two bases exchanged, using $u_j = \sum_i (\alpha_{ij}^* v_i - \beta_{ij} v_i^*)$ which we verify in Section 64.2.2 below — yields the companion relation

$$b_i \;=\; \sum_j \Big[\, \alpha_{ij}^*\, a_j - \beta_{ij}^*\, a_j^\dagger \,\Big]. \tag{64.53}$$

Equations (64.52) and (64.53) are the heart of the matter. The crucial feature is the β: if β vanishes, then b_i is a unitary recombination of *annihilation operators only*, the two vacua coincide, and the bases describe the same particles in different clothing. The moment β is nonzero, b_i mixes creation into annihilation: the operator that annihilates "b-particles" contains a piece that *creates* "a-particles," so the a-vacuum cannot be the b-vacuum. We make this precise in Section 64.2.3; first we must see what constraints (α, β) obey.

64.2.2 Preservation of the canonical structure: the Bogoliubov identities

The coefficients α, β are not arbitrary. Both $\{a_j\}$ and $\{b_i\}$ are bona fide ladder operators — they must *both* satisfy the canonical commutators — and this

single demand pins down two quadratic identities. Physically, the Bogoliubov transformation is the most general linear map on the field's phase space that preserves the canonical (symplectic) structure; the identities are the statement that it lies in the symplectic group.

Proposition 64.12 (Bogoliubov normalization identities)**.** *If both* $(a_j, a_j^\dagger)$ *and* $(b_i, b_i^\dagger)$ *obey the canonical commutation relations, the Bogoliubov coefficients of Theorem 64.11 satisfy*

$$\sum_k \Big(\alpha_{ik}\, \alpha^*_{jk} - \beta_{ik}\, \beta^*_{jk} \Big) = \delta_{ij}, \tag{64.54}$$

$$\sum_k \Big(\alpha_{ik}\, \beta_{jk} - \beta_{ik}\, \alpha_{jk} \Big) = 0. \tag{64.55}$$

In matrix form, with $\dagger$ *the conjugate transpose and* ^{T} *the transpose, these read* $\alpha\alpha^\dagger - \beta\beta^\dagger = \mathbf{1}$ *and* $\alpha\beta^T = \beta\alpha^T$.

Proof. Impose $\left[b_i, b_j^\dagger\right] = \delta_{ij}$ using Eq. (64.53). With $b_i = \sum_k (\alpha^*_{ik} a_k - \beta^*_{ik} a_k^\dagger)$ and $b_j^\dagger = \sum_l (\alpha_{jl} a_l^\dagger - \beta_{jl} a_l)$, only the elementary brackets $\left[a_k, a_l^\dagger\right] = \delta_{kl}$ and $\left[a_k^\dagger, a_l\right] = -\delta_{kl}$ contribute:

$$\begin{aligned} \left[b_i, b_j^\dagger\right] &= \sum_{k,l} \Big[\alpha^*_{ik}\alpha_{jl}\, \left[a_k, a_l^\dagger\right] + (-\beta^*_{ik})(-\beta_{jl})\, \left[a_k^\dagger, a_l\right] \Big] \\ &= \sum_{k,l} \Big[\alpha^*_{ik}\alpha_{jl}\, \delta_{kl} - \beta^*_{ik}\beta_{jl}\, \delta_{kl} \Big] = \sum_k \Big(\alpha^*_{ik}\alpha_{jk} - \beta^*_{ik}\beta_{jk} \Big). \end{aligned} \tag{64.56}$$

Setting Eq. (64.56) equal to δ_{ij} and taking the complex conjugate of the whole equation (a real Kronecker delta is unchanged) gives exactly Eq. (64.54). Next impose $[b_i, b_j] = 0$:

$$\begin{aligned} [b_i, b_j] &= \sum_{k,l} \Big[\alpha^*_{ik}(-\beta^*_{jl})\, \left[a_k, a_l^\dagger\right] + (-\beta^*_{ik})\alpha^*_{jl}\, \left[a_k^\dagger, a_l\right] \Big] \\ &= \sum_k \Big(-\alpha^*_{ik}\beta^*_{jk} + \beta^*_{ik}\alpha^*_{jk} \Big) = 0. \end{aligned} \tag{64.57}$$

Taking the complex conjugate of Eq. (64.57) gives $\sum_k (\alpha_{ik}\beta_{jk} - \beta_{ik}\alpha_{jk}) = 0$, which is Eq. (64.55). (Imposing the canonical commutators on the a's instead, via Eq. (64.52), gives nothing new: it returns the same two identities because the relations are an involutive pair.) □

The identities have a second, equally useful form obtained by summing over the *first* index. Demanding that Eq. (64.50) be invertible — that the $\{v_i\}$ really are a complete basis, so that the u_j can be expanded back over them — one

writes the trial inverse $u_j = \sum_i (\alpha^*_{ij} v_i - \beta_{ij} v^*_i)$ and substitutes Eq. (64.50) and its conjugate:

$$\sum_i (\alpha^*_{ij} v_i - \beta_{ij} v^*_i) = \sum_k u_k \sum_i (\alpha^*_{ij}\alpha_{ik} - \beta_{ij}\beta^*_{ik}) + \sum_k u^*_k \sum_i (\alpha^*_{ij}\beta_{ik} - \beta_{ij}\alpha^*_{ik}). \tag{64.58}$$

For the right-hand side to equal u_j we need the bracketed sums to be δ_{jk} and 0 respectively, i.e.

$$\sum_i \left(\alpha^*_{ij}\alpha_{ik} - \beta_{ij}\beta^*_{ik}\right) = \delta_{jk}, \qquad \sum_i \left(\alpha^*_{ij}\beta_{ik} - \beta_{ij}\alpha^*_{ik}\right) = 0, \tag{64.59}$$

which in matrix form are $\alpha^\dagger\alpha - \beta^T\beta^* = \mathbf{1}$ and $\alpha^\dagger\beta = \beta^T\alpha^*$. These are not independent of Eqs. (64.54) and (64.55); they follow from them because a square matrix that is left-invertible is right-invertible. Their virtue is that they make the inverse explicit, justifying the relation $u_j = \sum_i (\alpha^*_{ij} v_i - \beta_{ij} v^*_i)$ used to obtain Eq. (64.53), and we will call on the second of them when we prove the in-vacuum is a squeezed state.

Physical Insight

The two identities say that $\alpha\alpha^\dagger - \beta\beta^\dagger = \mathbf{1}$: the "rotation part" α and the "mixing part" β are locked together so that α is always at least as big as β in the operator sense. A pure rotation ($\beta = 0$, α unitary) shuffles particles among modes and is invisible to the vacuum. The genuinely curved-spacetime content lives entirely in β: it is the obstruction to the two positive-frequency splittings being the same, and Eq. (64.54) guarantees you can never make it vanish "for free" — buying a nonzero β costs you a correspondingly inflated α.

64.2.3 The particle content of the vacuum

We can now answer the question that motivates the whole construction. Suppose the field is prepared in the vacuum of the first basis, $|0_{\rm in}\rangle \equiv |0_a\rangle$, defined by $a_j |0_{\rm in}\rangle = 0$ for all j. An observer adapted to the second basis counts particles with the *out* number operator $N^{\rm out}_i = b^\dagger_i b_i$ (the curved-space avatar of Eq. (9.110)). How many out-particles does the in-vacuum contain?

Proposition 64.13 (Mean particle content of the in-vacuum)**.** *With* $a_j |0_{\rm in}\rangle = 0$,

$$\left\langle 0_{\rm in}\middle| N^{\rm out}_i \middle| 0_{\rm in}\right\rangle = \sum_j |\beta_{ij}|^2 , \tag{64.60}$$

and the total mean number of out-particles is $\langle 0_{\rm in}|N^{\rm out}|0_{\rm in}\rangle = \sum_{i,j} |\beta_{ij}|^2 = \mathrm{Tr}(\beta\beta^\dagger)$.

Proof. Use Eq. (64.53) to act with b_i on the in-vacuum. The annihilation piece kills $|0_{\rm in}\rangle$, leaving only the creation piece:

$$b_i\,|0_{\rm in}\rangle = \sum_j \left(\alpha^*_{ij}\,a_j - \beta^*_{ij}\,a_j^\dagger\right)|0_{\rm in}\rangle = -\sum_j \beta^*_{ij}\,a_j^\dagger\,|0_{\rm in}\rangle\,. \tag{64.61}$$

Thus $b_i\,|0_{\rm in}\rangle$ is a one-(in-)particle state, not zero, the first sign that the in-vacuum is full of out-quanta. The bra is the Hermitian conjugate, $\langle 0_{\rm in}|\,b_i^\dagger = -\sum_k \beta_{ik}\,\langle 0_{\rm in}|\,a_k$. Hence

$$\begin{aligned}\left\langle 0_{\rm in}\middle| b_i^\dagger b_i \middle|0_{\rm in}\right\rangle &= \sum_{j,k} \beta_{ik}\,\beta^*_{ij}\,\left\langle 0_{\rm in}\middle| a_k\,a_j^\dagger\middle|0_{\rm in}\right\rangle \\ &= \sum_{j,k}\beta_{ik}\,\beta^*_{ij}\,\delta_{kj} = \sum_j |\beta_{ij}|^2\,,\end{aligned} \tag{64.62}$$

where we used $a_k a_j^\dagger = \delta_{kj} + a_j^\dagger a_k$ and $a_k\,|0_{\rm in}\rangle = 0$ to get $\left\langle 0_{\rm in}\middle| a_k a_j^\dagger\middle|0_{\rm in}\right\rangle = \delta_{kj}$. Summing over i gives the total, $\sum_{i,j}|\beta_{ij}|^2 = \sum_{i,j}\beta_{ij}\beta^*_{ij} = \sum_i (\beta\beta^\dagger)_{ii} = {\rm Tr}(\beta\beta^\dagger)$. □

Equation (64.60) is the result that powers everything downstream. It says, with no further assumptions, that *the off-diagonal Bogoliubov coefficient β is literally the amplitude to find a particle*: where one observer sees the vacuum, another sees $\sum_j |\beta_{ij}|^2$ quanta in mode i. The two vacua coincide if and only if $\beta = 0$.

Caution

The total ${\rm Tr}(\beta\beta^\dagger)$ can diverge. A finite, sensible particle interpretation requires β to be *Hilbert–Schmidt*, ${\rm Tr}(\beta\beta^\dagger) = \sum_{ij}|\beta_{ij}|^2 < \infty$; this is the Shale condition under which the two Fock spaces are unitarily equivalent and the in-vacuum is a normalizable state in the out-Fock space. When it fails — as it does for an observer uniformly accelerated for all time, and in the idealized limits of some cosmologies — the two representations are unitarily inequivalent, "the total number of particles" is strictly infinite, and one must work mode by mode or with densities rather than with the global state. We will meet exactly this situation in the Unruh analysis; Eq. (64.60) for a *single* mode remains meaningful even there.

64.2.4 The in-vacuum as a squeezed condensate of out-pairs

Theorem 64.13 gives the *mean* occupation, but the in-vacuum is a definite pure state in the out-Fock space, and we can exhibit it in closed form. The structure it takes is the signature of the whole subject.

Proposition 64.14 (Gaussian/squeezed form of the in-vacuum)**.** *Whenever α is invertible, the in-vacuum is the out-Fock state*

$$|0_{\text{in}}\rangle \;=\; \mathcal{N}\,\exp\Big(\tfrac{1}{2}\sum_{i,j} V_{ij}\, b_i^\dagger b_j^\dagger\Big)\,|0_{\text{out}}\rangle\,, \qquad V \;=\; -\,\beta^*\alpha^{-1} \;=\; -\,(\alpha^T)^{-1}\beta^\dagger, \tag{64.63}$$

where the coefficient matrix V is symmetric*, $V_{ij}=V_{ji}$, and the normalization is $\mathcal{N}=\big|\det(\mathbf{1}-V^\dagger V)\big|^{1/4}=|\det\alpha|^{-1/2}$.*

Proof. We solve $a_j\,|0_{\text{in}}\rangle=0$ for all j, working in the out-Fock space. Write $S=\frac{1}{2}\sum_{kl}V_{kl}b_k^\dagger b_l^\dagger$ and take the ansatz $|0_{\text{in}}\rangle=\mathcal{N}\,e^{S}\,|0_{\text{out}}\rangle$, with V to be determined (we may take V symmetric without loss, since only the symmetric part survives the contraction of the two commuting creation operators in S). Two commutators control the calculation. First,

$$[b_i,S]=\tfrac{1}{2}\sum_{kl}V_{kl}\,\Big[b_i,b_k^\dagger b_l^\dagger\Big]=\tfrac{1}{2}\sum_{kl}V_{kl}(\delta_{ik}b_l^\dagger+\delta_{il}b_k^\dagger)=\sum_l V_{il}\,b_l^\dagger, \tag{64.64}$$

using $V_{kl}=V_{lk}$ in the last step. Second, $\Big[b_i^\dagger,S\Big]=0$ because S is built from creation operators only, so $b_i^\dagger$ commutes through e^S. Since $[b_i,S]$ in Eq. (64.64) is itself a creation operator, it commutes with S, the Baker–Campbell–Hausdorff series truncates at first order, and

$$e^{-S}\,b_i\,e^{S}=b_i+[b_i,S]=b_i+\sum_l V_{il}b_l^\dagger \quad\Longrightarrow\quad b_i\,e^{S}\,|0_{\text{out}}\rangle=e^{S}\sum_l V_{il}b_l^\dagger\,|0_{\text{out}}\rangle\,, \tag{64.65}$$

where the implication uses $b_i\,|0_{\text{out}}\rangle=0$. Now express the in-vacuum condition through Eq. (64.52), $a_j=\sum_i(\alpha_{ij}b_i+\beta_{ij}^*b_i^\dagger)$, and act on the ansatz:

$$\begin{aligned}
0=a_j\,|0_{\text{in}}\rangle &= \mathcal{N}\sum_i\Big[\alpha_{ij}\,b_i\,e^{S}+\beta_{ij}^*\,b_i^\dagger\,e^{S}\Big]\,|0_{\text{out}}\rangle\\
&= \mathcal{N}\,e^{S}\sum_i\Big[\alpha_{ij}\sum_l V_{il}\,b_l^\dagger+\beta_{ij}^*\,b_i^\dagger\Big]\,|0_{\text{out}}\rangle\\
&= \mathcal{N}\,e^{S}\sum_l\Big[\sum_i\alpha_{ij}V_{il}+\beta_{lj}^*\Big]b_l^\dagger\,|0_{\text{out}}\rangle\,,
\end{aligned} \tag{64.66}$$

where in the last line we relabelled the dummy index $i\to l$ in the β-term and used $b_i^\dagger e^S=e^S b_i^\dagger$. The states $b_l^\dagger\,|0_{\text{out}}\rangle$ are linearly independent, so each coefficient must vanish:

$$\sum_i\alpha_{ij}\,V_{il} \;=\; -\,\beta_{lj}^* \qquad \text{for all } j,l. \tag{64.67}$$

Read as matrices, the left side is $(\alpha^TV)_{jl}$ and the right is $-(\beta^\dagger)_{jl}$, so $\alpha^TV=-\beta^\dagger$, giving $V=-(\alpha^T)^{-1}\beta^\dagger$ whenever α is invertible. Transposing, $(\beta^\dagger)^T=\beta^*$ and $((\alpha^T)^{-1})^T=\alpha^{-1}$, so the same matrix equals $V=-\beta^*\alpha^{-1}$ as well — and the

equality of the two expressions *is* the statement that V is symmetric. To verify it directly, $V = V^T$ requires $(\alpha^T)^{-1}\beta^\dagger = \beta^*\alpha^{-1}$, i.e. $\beta^\dagger\alpha = \alpha^T\beta^*$. This is exactly the conjugate of the second identity in Eq. (64.59), $\alpha^\dagger\beta = \beta^T\alpha^*$, so V is symmetric.

It remains to normalize. For the single-mode case V is a number and $e^S = \sum_n \frac{1}{n!}(\frac{1}{2}V)^n (b^\dagger)^{2n}$ acting on the vacuum gives $\sum_n \frac{(V/2)^n}{n!}\sqrt{(2n)!}\,|2n\rangle$, whose squared norm is $\sum_n \binom{2n}{n}(|V|^2/4)^n = (1-|V|^2)^{-1/2}$; hence $|\mathcal{N}|^2 = (1-|V|^2)^{1/2}$. The multimode generalization diagonalizes V by the Autonne–Takagi factorization of the symmetric matrix (unitary W with $V = W\,\mathrm{diag}(\lambda_k)\,W^T$, $\lambda_k \geq 0$), reducing the state to a product of independent single-mode squeezes and giving $|\mathcal{N}|^2 = \prod_k (1-\lambda_k^2)^{1/2} = |\det(\mathbf{1} - V^\dagger V)|^{1/2}$. Finally $\mathbf{1} - V^\dagger V = \mathbf{1} - (\alpha^\dagger)^{-1}\beta^T\beta^*\alpha^{-1} = (\alpha^\dagger)^{-1}(\alpha^\dagger\alpha - \beta^T\beta^*)\alpha^{-1} = (\alpha^\dagger\alpha)^{-1}$ by Eq. (64.59), so $\det(\mathbf{1} - V^\dagger V) = |\det\alpha|^{-2}$ and $\mathcal{N} = |\det\alpha|^{-1/2}$. □

Notation Watch

The sign and conjugation pattern in $V = -\beta^*\alpha^{-1}$ depend on the sesquilinearity convention fixed in Eq. (64.45) (linear in the first slot) and on the definitions Eq. (64.49). References that take the inner product antilinear in the first slot, or that define β without the leading minus sign, report V with a different sign or with α^* in place of α; the physical state — a condensate of out-pairs with overlap $|\det\alpha|^{-1/2}$ with the out-vacuum — is convention independent. We use Eq. (64.63) consistently throughout Chapter 64.

Physical Insight

Read Eq. (64.63) literally: the operator $b_i^\dagger b_j^\dagger$ creates a *pair* of out-quanta, so the in-vacuum is a coherent superposition of $0, 2, 4, \ldots$ out-particles — the quanta always appear in pairs, never singly. This is the deep statement behind every particle-creation effect in this chapter: gravity (or acceleration) does not deposit lone particles into empty space, it *squeezes* the vacuum, stretching it along $b^\dagger b^\dagger$ and populating it with correlated pairs. The two members of each pair are exactly the entangled partners whose separation across a horizon will, in Chapter 65, become Hawking radiation and its information puzzle.

Worked Example 64.15 (A single two-mode squeeze: the elementary cell)**.** Almost every example in this chapter reduces, mode by mode, to one complex parameter. Take a single pair (a, b) related by $b = \alpha^* a - \beta^* a^\dagger$ with $|\alpha|^2 - |\beta|^2 = 1$ (the one-mode form of Eq. (64.54)). Because the magnitudes are tied, parametrize them by a single real *squeezing parameter* $r \geq 0$ and two phases,

$$\alpha = e^{i\varphi}\cosh r, \qquad \beta = e^{i\theta}\sinh r, \tag{64.68}$$

which automatically satisfies $\cosh^2 r - \sinh^2 r = 1$. The mean occupation of the in-vacuum follows immediately from Eq. (64.60):

$$\left\langle 0_{\rm in}\middle| N^{\rm out}\middle|0_{\rm in}\right\rangle = |\beta|^2 = \sinh^2 r. \tag{64.69}$$

The squeeze coefficient is, from Eq. (64.63), $V = -\beta^*/\alpha = -e^{-i(\theta+\varphi)}\tanh r$, so the state is

$$|0_{\rm in}\rangle = (\cosh r)^{-1/2}\,\exp\!\Big(-\tfrac{1}{2}\,e^{-i(\theta+\varphi)}\tanh r\,(b^\dagger)^2\Big)\,|0_{\rm out}\rangle\,, \tag{64.70}$$

with $\mathcal{N} = |\alpha|^{-1/2} = (\cosh r)^{-1/2}$ reproducing $|\det\alpha|^{-1/2}$. Expanding the exponential, the probability of finding $2n$ out-particles is $P_{2n} = |\mathcal{N}|^2\,\binom{2n}{n}(\tfrac{1}{4}\tanh^2 r)^n$; the odd occupations are strictly absent, exactly the pairing of the insight above. Summing $\sum_n 2n\,P_{2n}$ recovers $\sinh^2 r$, a consistency check on Eq. (64.69).

It is worth recording one fact that the next sections will turn into a temperature. Suppose the two modes carry energies $\pm\omega$ with respect to some Killing time and are partners under a transformation for which $|\beta/\alpha| = e^{-\pi\omega/a}$ for a constant a (this is the value the gravitational and accelerated examples will produce). Then $\tanh r = e^{-\pi\omega/a}$, and

$$\left\langle 0_{\rm in}\middle| N^{\rm out}\middle|0_{\rm in}\right\rangle = \sinh^2 r = \frac{|\beta/\alpha|^2}{1-|\beta/\alpha|^2} = \frac{e^{-2\pi\omega/a}}{1-e^{-2\pi\omega/a}} = \frac{1}{e^{2\pi\omega/a}-1}, \tag{64.71}$$

the Planck distribution at temperature $T = a/2\pi$. The Bogoliubov machine has not been told anything about thermodynamics; a single ratio $|\beta/\alpha| = e^{-\pi\omega/a}$ produces a thermal spectrum on its own. The work of Section 64.3 and the Unruh section is to show that gravity and acceleration actually deliver that ratio.

64.2.5 When is the particle number unambiguous?

If β is generically nonzero, when does it vanish — when is there a preferred vacuum and an observer-independent notion of "particle"? The answer is geometric: a vacuum is unambiguous precisely when the spacetime provides a canonical way to call a mode "positive frequency," and that is what a timelike symmetry supplies.

Proposition 64.16 (Stationary spacetimes have an unambiguous vacuum)**.** *Let (M, g) admit a globally timelike Killing vector field K^μ ($\nabla_{(\mu}K_{\nu)} = 0$, $K^\mu K_\mu > 0$ everywhere). Then there is a basis of mode solutions that are eigenfunctions of the Lie derivative along K,*

$$\pounds_K\, u_\omega = -\,i\,\omega\,u_\omega, \qquad \omega > 0, \tag{64.72}$$

with strictly positive Klein–Gordon norm. The Fock vacuum built on this basis is invariant under the isometry generated by K, and any two observers who both define positive frequency by Eq. (64.44) are related by a Bogoliubov transformation with $\beta = 0$; they agree on the vacuum and on particle number.

Proof. Because K is Killing, $\pounds_K$ commutes with the Klein–Gordon operator $\Box + m^2 + \xi R$ (the operator is built from the metric and its covariant derivatives, all of which are annihilated by $\pounds_K$), so it maps solutions to solutions and can be simultaneously diagonalized with the field equation. Its eigenvalues are imaginary, $-i\omega$ with ω real, because $\pounds_K$ is anti-self-adjoint with respect to Eq. (64.45): for any two solutions, K-invariance of the inner product (the current j^μ is Lie-dragged by K, whose flow maps Cauchy surfaces to Cauchy surfaces, leaving the surface-independent integral unchanged) gives $(\pounds_K f, g) + (f, \pounds_K g) = 0$. Hence we may sort the eigenmodes by the sign of ω and *define* the positive-frequency modes to be those with $\omega > 0$. We must check these have positive norm. For an eigenmode,

$$\pounds_K\,(u_\omega, u_\omega) = (\pounds_K u_\omega, u_\omega) + (u_\omega, \pounds_K u_\omega) = (i\omega - i\omega)(u_\omega, u_\omega) = 0, \tag{64.73}$$

consistent with the norm being a conserved number, while the standard energy argument — $\omega\,(u_\omega, u_\omega)$ equals, up to a positive factor set by $K^\mu K_\mu > 0$, the manifestly positive energy $\int_\Sigma \mathrm{d}\Sigma^\mu K^\nu T_{\mu\nu}[u_\omega]$ of the classical mode — forces $\mathrm{sgn}(u_\omega, u_\omega) = \mathrm{sgn}\,\omega$. So the $\omega > 0$ modes have positive norm and qualify as the u_j of Eq. (64.46); the $\omega < 0$ modes are their conjugates. This positive-frequency split is fixed by K alone, with no further choice. Two observers who adopt the same K therefore build the *same* $\{u_j\}$, up to a unitary relabelling within the positive-frequency subspace ($\beta = 0$, α unitary), and by Eq. (64.53) share the vacuum. Finally, the vacuum is K-invariant: the unitary $U(s) = e^{is\hat{H}}$ implementing the isometry (with $\hat{H}$ the conserved charge of K, diagonal in the number operator since each u_ω carries definite ω) annihilates $|0\rangle$ because $\hat{H}$ does. □

The proposition isolates the precise reason flat space has an unambiguous vacuum: Minkowski spacetime has the timelike Killing field ∂_t (indeed a whole family related by Poincaré transformations, all sharing the same positive-frequency split because the boosts that connect inertial observers do not mix positive and negative frequencies). A generic curved spacetime has no timelike Killing vector at all — the metric depends on time — and then no preferred $\omega > 0$ exists, $\beta \neq 0$ is generic, and the vacuum is genuinely ambiguous. The intermediate, and most useful, case is a spacetime that is *not* stationary in the middle but settles to stationary regions in the asymptotic past and future, each with its own timelike Killing field ∂_t^{in} and $\partial_t^{\mathrm{out}}$. Each region then defines a clean vacuum by Theorem 64.16, the two need not agree, and the β relating them is finite and computable. This is the arena in which the Bogoliubov machine does its sharpest work, sketched in Fig. 64.1.

We now have the full apparatus: two bases, their Bogoliubov coefficients (α, β), the identities that constrain them, the particle-number formula $\sum_j |\beta_{ij}|^2$, and the explicit squeezed in-vacuum. What the machine demands next is a concrete physical pair of bases to feed it. The most literal such pair is gravitational: an

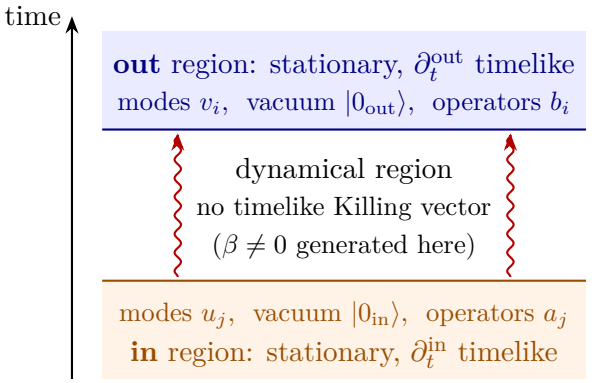


Figure 64.1: The generic useful geometry. The field is quantized on a Cauchy surface in a stationary *in* region, where ∂_t^{in} supplies an unambiguous positive-frequency basis $\{u_j\}$ and a vacuum $|0_{\text{in}}\rangle$ (Theorem 64.16). A non-stationary dynamical region intervenes, and the modes propagate (wavy lines) into a second stationary *out* region with its own basis $\{v_i\}$ and vacuum $|0_{\text{out}}\rangle$. Because the in- and out-positive-frequency splits disagree, the Bogoliubov matrix β is nonzero, and by Eq. (64.60) the in-vacuum is seen by the out-observer to contain $\sum_j |\beta_{ij}|^2$ particles in mode i.

expanding universe that is static in the far past and far future furnishes exactly the in/out structure of Fig. 64.1, with two genuinely different Killing times and a β we can compute in closed form. We turn to that calculation in Section 64.3, where gravity is shown to create particles and we extract their number.

64.3 Particle creation by an expanding universe

The previous section left us holding a machine with nothing yet fed into it. We learned that two complete sets of positive-frequency modes, each defining its own vacuum and its own notion of "particle," are related by a Bogoliubov transformation (Section 64.2), and that the number of quanta the second observer finds in the first observer's vacuum is fixed by the off-diagonal coefficients alone, $n_k = |\beta_k|^2$. Everything in that section was structural: it told us *that* a mismatch of mode bases produces particles, but not *how many*, because we had no concrete pair of bases to compute with. The off-diagonal coefficients were left as symbols.

We now supply the missing input with a gravitational background simple enough to be solved in closed form yet rich enough to contain the entire phenomenon: a spatially flat universe that expands smoothly from one static era in the far past to another static era in the far future. "Static in the far past and far future" is exactly the condition under which the two privileged mode bases of the previous section—the in-modes adapted to the early universe and the out-modes

adapted to the late one—are both unambiguously defined, so the Bogoliubov coefficients between them are honest numbers rather than convention. Between the two eras the universe expands, the geometry is genuinely time-dependent, and there is no preferred vacuum at all. The question the machine answers is sharp: an apparatus that was in its ground state in the early static era, left undisturbed while the universe expands, is found in the late static era to contain a definite spectrum of real quanta. Gravity has created particles out of the vacuum, and we will compute the number exactly. The model is due to Bernard and Duncan; its solvability rests on reducing the curved-space wave equation to a one-dimensional scattering problem and then recognising that problem as one of the classical soluble cases of the hypergeometric equation.

64.3.1 The curved Klein–Gordon equation as one-dimensional scattering

We work with a spatially flat Friedmann–Robertson–Walker (FRW) geometry written in *conformal time* η, in which the line element is a position-dependent rescaling of flat Minkowski space,

$$\mathrm{d}s^2 = a^2(\eta)\,(\mathrm{d}\eta^2 - \mathrm{d}\mathbf{x}\cdot\mathrm{d}\mathbf{x}), \qquad \mu,\nu \in \{0,1,2,3\}, \quad g_{\mu\nu} = a^2(\eta)\,\eta_{\mu\nu}, \tag{64.74}$$

with the mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ in force throughout this chapter and $a(\eta) > 0$ the scale factor. Conformal time is the natural clock here for one reason: in these coordinates the metric is *conformally flat*, $g_{\mu\nu} = a^2\eta_{\mu\nu}$, and the spatial slices are ordinary Euclidean $\mathbb{R}^3$, so the comoving momentum $\mathbf{k}$ conjugate to $\mathbf{x}$ is a good quantum number: distinct Fourier modes do not mix, and the field equation separates mode by mode. This turns a partial differential equation on a curved background into an ordinary differential equation in η for each $\mathbf{k}$, which is the whole reason the model is soluble.

We carry the free real scalar of ϕ^4 theory in its $\lambda = 0$ sector (Chapter 9), now massive and propagating on the curved background Eq. (64.74). We take it *conformally coupled*: in $D = 4$ the action is

$$S = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[g^{\mu\nu}\,\partial_\mu\phi\,\partial_\nu\phi - (m^2 + \xi R)\phi^2\Big], \qquad \xi = \tfrac{1}{6}, \tag{64.75}$$

where R is the Ricci scalar and $\xi = \frac{1}{6}$ is the conformal value in four dimensions. The choice is one of convenience, not necessity: it produces the cleanest possible mode equation, and we will see at the end of the subsection exactly what changes for the minimally coupled ($\xi = 0$) field. The field obeys the curved-space Klein–Gordon equation

$$(\Box + m^2 + \xi R)\phi = 0, \qquad \Box\phi = \frac{1}{\sqrt{-g}}\,\partial_\mu\big(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\big), \tag{64.76}$$

the scalar wave operator built from the covariant measure (we use the same $\Box$, mass m, and field ϕ fixed in Chapter 9).

Evaluating the d'Alembertian. For the conformally flat metric Eq. (64.74) we have, in $D = 4$, $\sqrt{-g} = a^4$ and $g^{\mu\nu} = a^{-2}\eta^{\mu\nu}$. Inserting these into the operator in Eq. (64.76),

$$\begin{aligned}\Box\phi &= a^{-4}\,\partial_\mu(a^4\,a^{-2}\eta^{\mu\nu}\,\partial_\nu\phi) = a^{-4}\,\partial_\mu(a^2\,\eta^{\mu\nu}\,\partial_\nu\phi)\\ &= a^{-4}\Big[\,\partial_\eta(a^2\,\partial_\eta\phi) - a^2\,\nabla^2\phi\,\Big] = a^{-2}\Big[\,\phi'' + 2\tfrac{a'}{a}\,\phi' - \nabla^2\phi\,\Big],\end{aligned} \tag{64.77}$$

where a prime denotes $\mathrm{d}/\mathrm{d}\eta$, the index sum used $\eta^{00} = +1$, $\eta^{ij} = -\delta^{ij}$, and we expanded $\partial_\eta(a^2\phi') = a^2\phi'' + 2aa'\phi'$. The term $2(a'/a)\phi'$ is the cosmological friction: it is the Hubble drag of the expansion acting on the field, and it is exactly what makes the time-dependent problem look unfamiliar. The strategy is to remove it.

The conformal rescaling $\chi = a\phi$. A massless field of canonical dimension in $D = 4$ has conformal weight 1, so the combination $\chi \equiv a\phi$ is the field that should behave like a flat-space scalar. Expand ϕ in comoving spatial Fourier modes and factor out the scale factor,

$$\phi(\eta,\mathbf{x}) = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\frac{1}{a(\eta)}\,\chi_k(\eta)\,e^{\,i\mathbf{k}\cdot\mathbf{x}} \;+\; \text{h.c.}, \qquad \phi_k(\eta) = \frac{\chi_k(\eta)}{a(\eta)}, \tag{64.78}$$

so that $\nabla^2 \to -k^2$ on the mode of comoving wavenumber $k = |\mathbf{k}|$. We need $\phi_k'' + 2(a'/a)\phi_k'$ in terms of χ_k. Differentiating $\phi_k = a^{-1}\chi_k$ twice,

$$\begin{aligned}\phi_k' &= a^{-1}\chi_k' - a^{-2}a'\chi_k,\\ \phi_k'' &= a^{-1}\chi_k'' - 2a^{-2}a'\chi_k' + (2a^{-3}a'^2 - a^{-2}a'')\chi_k,\end{aligned} \tag{64.79}$$

and adding $2(a'/a)\phi_k' = 2a^{-2}a'\chi_k' - 2a^{-3}a'^2\chi_k$, the first-derivative terms in χ_k' cancel and so do the a'^2 terms, leaving the remarkably clean combination

$$\phi_k'' + 2\tfrac{a'}{a}\,\phi_k' = a^{-1}\chi_k'' - a^{-2}a''\,\chi_k. \tag{64.80}$$

The friction has been traded for a single algebraic term $-a''/a$. Now assemble Eq. (64.76) for the mode ϕ_k, using Eq. (64.77), Eq. (64.80), and the Ricci scalar of the flat FRW geometry, which in conformal time is

$$R = \frac{6\,a''}{a^3}. \tag{64.81}$$

We multiply $(\Box + m^2 + \xi R)\phi_k = 0$ through by a^3:

$$\chi_k'' - \frac{a''}{a}\,\chi_k + k^2\chi_k + m^2a^2\,\chi_k + 6\xi\,\frac{a''}{a}\,\chi_k = 0, \tag{64.82}$$

where the m^2 term picked up a factor a^2 from $\sqrt{-g}/a^2$ packaging and the ξR term contributed $6\xi\, a''/a$. The two curvature terms combine into $(6\xi - 1)(a''/a)\chi_k$, which *vanishes identically at the conformal value* $\xi = \frac{1}{6}$. We are left with

$$\boxed{\chi_k'' + \omega_k^2(\eta)\,\chi_k = 0, \qquad \omega_k^2(\eta) = k^2 + m^2\,a^2(\eta).} \tag{64.83}$$

This is the central reduction. Each comoving mode of the conformally coupled scalar obeys the equation of a harmonic oscillator with a *time-dependent frequency* $\omega_k(\eta)$, or equivalently—reading η as a spatial coordinate—the time-independent one-dimensional Schrödinger scattering problem $-\chi'' + U(\eta)\chi = k^2\chi$ with the effective potential $U(\eta) = -m^2a^2(\eta)$. The expansion of the universe enters solely through the profile $a^2(\eta)$, which acts as a frequency that drifts from one constant value in the far past to another in the far future. (For minimal coupling $\xi = 0$ the bracket $(6\xi - 1)(a''/a) = -a''/a$ survives, and Eq. (64.83) acquires the extra effective potential term $-a''/a$; the analysis is identical in spirit but the closed form below is special to conformal coupling.)

Physical Insight

A massless conformally coupled scalar ($m = 0$, $\xi = \frac{1}{6}$) has $\omega_k^2 = k^2$ *independent of* η: the mode equation is the flat-space one for all time, the positive-frequency modes never mix, and *no particles are created no matter how violently the universe expands.* This is the conformal triviality of the radiation-dominated era in disguise. Particle creation requires a scale that breaks conformal invariance—here the mass m, which is what couples ω_k to the changing $a(\eta)$. Gravity creates particles only by acting on the conformal-symmetry-breaking content of the matter. The mass is the handle the expansion grips.

In- and out-vacua. The geometry is *asymptotically static* if $a(\eta)$ tends to constants as $\eta \to \mp\infty$. Then $\omega_k(\eta)$ tends to constants too,

$$\omega_{\rm in}^2 \equiv \lim_{\eta\to-\infty} \omega_k^2(\eta) = k^2 + m^2 a_{\rm in}^2, \qquad \omega_{\rm out}^2 \equiv \lim_{\eta\to+\infty} \omega_k^2(\eta) = k^2 + m^2 a_{\rm out}^2, \tag{64.84}$$

with $a_{\rm in} = a(-\infty)$, $a_{\rm out} = a(+\infty)$. In each static region Eq. (64.83) is solved by ordinary plane waves $e^{\mp i\omega\eta}$, and the rule for splitting positive from negative frequency is the standard, unambiguous one—the same split that defines the Minkowski vacuum, applied in each asymptotic era. We therefore have two natural complete bases: the *in-modes*, the solutions that reduce to pure positive frequency $e^{-i\omega_{\rm in}\eta}$ in the far past, and the *out-modes*, those that reduce to $e^{-i\omega_{\rm out}\eta}$ in the far future. The Klein–Gordon inner product on the curved background, evaluated on the rescaled field χ, is just the flat one (the factors of a in $\sqrt{-g}$ and in $\phi = \chi/a$ cancel), so the normalisation is fixed by the per-mode Wronskian

condition established in Section 64.2: a positive-frequency mode of frequency ω in an asymptotic region carries the standard envelope

$$\chi_k(\eta) \;\longrightarrow\; \frac{1}{\sqrt{2\omega}}\, e^{-i\omega\eta}, \qquad \chi_k \chi_k'^{*} - \chi_k^{*}\chi_k' = i, \tag{64.85}$$

the Wronskian being the conserved quantity guaranteed by Eq. (64.83) (which has no first-derivative term, so $W' = 0$). The in- and out-bases differ precisely because a mode that is pure positive frequency at $\eta = -\infty$ is, after threading the time-dependent region, a *mixture* of positive and negative out-frequencies at $\eta = +\infty$. That mixture is the Bogoliubov transformation, and its β-coefficient is the created-particle amplitude. All that remains is to compute it.

64.3.2 The adiabatic vacuum and the WKB hierarchy

Before solving the model exactly we settle a conceptual point that the asymptotic analysis glosses over: when the geometry is changing, what does "the vacuum" even mean? In the static eras the answer is fixed by Eq. (64.85). In the dynamical interior there is no static frame and no exact answer, but there is an excellent approximate one whenever $\omega_k(\eta)$ varies slowly, and it is this notion that the in/out construction silently uses. It is the *adiabatic vacuum*, defined by the WKB hierarchy.

Seek a solution of Eq. (64.83) in the WKB form

$$\chi_k(\eta) = \frac{1}{\sqrt{2W_k(\eta)}}\, \exp\!\Big(-i\int^{\eta} W_k(\eta')\,\mathrm{d}\eta'\Big), \tag{64.86}$$

which generalises Eq. (64.85) by promoting the constant frequency to a function $W_k(\eta)$ and is normalised so that the Wronskian condition holds identically for any real W_k. Substituting into Eq. (64.83) and separating the result gives the exact (nonlinear) equation for W_k,

$$W_k^2 = \omega_k^2 - \frac{1}{2}\,\frac{W_k''}{W_k} + \frac{3}{4}\left(\frac{W_k'}{W_k}\right)^{2}. \tag{64.87}$$

This is solved iteratively by counting derivatives of $a(\eta)$ as the small quantity: each derivative $\mathrm{d}/\mathrm{d}\eta$ brings a power of the expansion rate, so the *adiabatic order* of a term is the number of time derivatives it carries. The zeroth-order (adiabatic) approximation drops the derivative terms on the right,

$$W_k^{(0)} = \omega_k(\eta), \qquad W_k^{(2)} = \omega_k\Big[1 - \frac{1}{2}\frac{\omega_k''}{\omega_k^3} + \frac{3}{4}\frac{(\omega_k')^2}{\omega_k^4}\Big]^{1/2} = \omega_k - \frac{\omega_k''}{4\omega_k^2} + \frac{3(\omega_k')^2}{8\omega_k^3} + \cdots, \tag{64.88}$$

and so on to all even orders. The expansion parameter controlling it is the *non-adiabaticity*

$$\varepsilon_{\mathrm{ad}} \sim \frac{|\omega_k'|}{\omega_k^2}, \tag{64.89}$$

the fractional change of the frequency over one oscillation. The *adiabatic vacuum of order n* is the state annihilated by the operators that multiply the WKB modes Eq. (64.86) truncated at adiabatic order n; it is the best local definition of "no particles" available in a time-dependent background, and as $\varepsilon_{\mathrm{ad}} \to 0$ the modes of every order coincide and the ambiguity disappears.

The physical content is the adiabatic theorem. If the expansion is slow, $\varepsilon_{\mathrm{ad}} \ll 1$ throughout, then the exact mode tracks the instantaneous WKB form Eq. (64.86) and never develops a negative-frequency admixture: the in- and out-bases connect with $\beta_k \approx 0$, and *no particles are produced*. Particle creation is the breakdown of adiabaticity. The size of β_k is governed, exactly as in the Landau–Zener analysis of quantum mechanics, by the analytic continuation of the mode into the complex η-plane: the amplitude to flip onto the negative-frequency branch is

$$|\beta_k| \sim \exp\!\Big(-2\,\Big|\int_{\mathcal{C}} \omega_k(\eta)\,\mathrm{d}\eta\Big|_{\mathrm{Im}}\Big), \tag{64.90}$$

where the contour $\mathcal{C}$ runs to the complex turning point of $\omega_k(\eta)$ nearest the real axis—the point at which the two WKB branches degenerate. The imaginary part of the action accumulated near that singularity sets an *exponentially small* creation amplitude for a slowly varying background. We will see the exact solution reproduce Eq. (64.90) precisely.

64.3.3 The Bernard–Duncan model and its exact modes

We now specialise to a profile for which Eq. (64.83) is solved in closed form. Take the scale factor squared to be a hyperbolic-tangent step,

$$a^2(\eta) = A + B\,\tanh(\rho\,\eta), \qquad A > B > 0, \quad \rho > 0, \tag{64.91}$$

the *Bernard–Duncan model*. The parameter A sets the mean size, B the total amount of expansion, and ρ the *rapidity*: $1/\rho$ is the conformal duration of the expansion, so $\rho \to 0$ is the slow (adiabatic) limit and $\rho \to \infty$ the sudden one. The geometry is asymptotically static at both ends, with

$$a_{\mathrm{in}}^2 = A - B, \qquad a_{\mathrm{out}}^2 = A + B, \tag{64.92}$$

so $a_{\mathrm{out}} > a_{\mathrm{in}}$: the universe genuinely expands. From Eq. (64.84) the two asymptotic frequencies are

$$\omega_{\mathrm{in}}^2 = k^2 + m^2(A - B), \qquad \omega_{\mathrm{out}}^2 = k^2 + m^2(A + B), \qquad \omega_{\mathrm{out}} > \omega_{\mathrm{in}}, \tag{64.93}$$

and it is convenient throughout to use their half-sum and half-difference,

$$\omega_{\pm} \equiv \tfrac{1}{2}\big(\omega_{\mathrm{out}} \pm \omega_{\mathrm{in}}\big), \qquad \omega_+ + \omega_- = \omega_{\mathrm{out}}, \quad \omega_+ - \omega_- = \omega_{\mathrm{in}}. \tag{64.94}$$

The effective frequency $\omega_k^2(\eta) = k^2 + m^2 A + m^2 B\tanh(\rho\eta)$ rises monotonically from ω_{in}^2 to ω_{out}^2; the left panel of Fig. 64.2 shows the profile for several momenta.

Reduction to the hypergeometric equation. Introduce the variable

$$s \equiv \frac{1+\tanh(\rho\eta)}{2} = \frac{e^{\rho\eta}}{2\cosh(\rho\eta)} \in (0,1), \qquad 1-s = \frac{e^{-\rho\eta}}{2\cosh(\rho\eta)}, \tag{64.95}$$

which maps the whole conformal-time line $\eta \in (-\infty,\infty)$ onto the unit interval, with $\eta \to -\infty \Leftrightarrow s \to 0$ and $\eta \to +\infty \Leftrightarrow s \to 1$. Differentiating, $\mathrm{d}s/\mathrm{d}\eta = \rho\,\mathrm{sech}^2(\rho\eta)/2 = 2\rho\, s(1-s)$, so $\mathrm{d}/\mathrm{d}\eta = 2\rho\, s(1-s)\,\mathrm{d}/\mathrm{d}s$ and

$$\frac{\mathrm{d}^2}{\mathrm{d}\eta^2} = 4\rho^2\Big[s^2(1-s)^2\frac{\mathrm{d}^2}{\mathrm{d}s^2} + s(1-s)(1-2s)\frac{\mathrm{d}}{\mathrm{d}s}\Big]. \tag{64.96}$$

The frequency, written in s, is linear: $\tanh(\rho\eta) = 2s-1$ gives $\omega_k^2 = \omega_{\mathrm{in}}^2(1-s) + \omega_{\mathrm{out}}^2 s$. Substituting into Eq. (64.83) and dividing by $4\rho^2$,

$$s^2(1-s)^2\,\chi_{ss} + s(1-s)(1-2s)\,\chi_s + \frac{\omega_{\mathrm{in}}^2(1-s)+\omega_{\mathrm{out}}^2 s}{4\rho^2}\,\chi = 0, \tag{64.97}$$

with subscripts denoting $\mathrm{d}/\mathrm{d}s$. This is a Fuchsian equation with regular singular points at $s = 0, 1, \infty$, hence reducible to the hypergeometric equation once the singular behaviour at $s=0$ and $s=1$ is factored out. The indicial equation at $s=0$ is obtained by inserting $\chi \sim s^{\alpha}$: the three terms contribute $\alpha(\alpha-1) + \alpha + \omega_{\mathrm{in}}^2/4\rho^2 = \alpha^2 + \omega_{\mathrm{in}}^2/4\rho^2$, whose roots are $\alpha = \pm i\,\omega_{\mathrm{in}}/2\rho$; likewise at $s=1$ the exponents are $\pm i\,\omega_{\mathrm{out}}/2\rho$. We factor out the branches that will produce the *positive-frequency in-mode* and write

$$\chi_k(\eta) = s^{\alpha_1}(1-s)^{\alpha_2}\,F(s), \qquad \alpha_1 = -\frac{i\omega_{\mathrm{in}}}{2\rho}, \quad \alpha_2 = -\frac{i\omega_{\mathrm{out}}}{2\rho}. \tag{64.98}$$

The choice of sign for α_1 is dictated by physics: as $\eta \to -\infty$ we have $s \to e^{2\rho\eta}$, so $s^{\alpha_1} \to e^{2\rho\eta\,\alpha_1} = e^{-i\omega_{\mathrm{in}}\eta}$, which is exactly the positive-frequency in-mode. Substituting Eq. (64.98) into Eq. (64.97) and dividing through by $s(1-s)\,s^{\alpha_1}(1-s)^{\alpha_2}$, the $1/s$ and $1/(1-s)$ poles cancel by the indicial conditions $\alpha_1^2 = -\omega_{\mathrm{in}}^2/4\rho^2$ and $\alpha_2^2 = -\omega_{\mathrm{out}}^2/4\rho^2$, and F is left obeying the standard hypergeometric equation

$$s(1-s)\,F'' + \big[c - (a+b+1)s\big]F' - ab\,F = 0. \tag{64.99}$$

Matching the first-derivative coefficient gives $c = 1+2\alpha_1$ and $a+b = 1+2\alpha_1+2\alpha_2$; matching the constant term gives, after collecting (the computation is carried in Theorem 64.17), $ab = (\alpha_1+\alpha_2)(\alpha_1+\alpha_2+1)$. Writing $\beta \equiv \alpha_1+\alpha_2 = -i\omega_+/\rho$, the parameters are the roots of $x^2 - (1+2\beta)x + \beta(\beta+1) = 0$, namely $x = \beta$ and $x = 1+\beta$. Hence

$$a = 1 - \frac{i\omega_+}{\rho}, \qquad b = -\frac{i\omega_+}{\rho}, \qquad c = 1 - \frac{i\omega_{\mathrm{in}}}{\rho}. \tag{64.100}$$

The in-mode, normalised by Eq. (64.85) so that it is a unit positive-frequency wave at $\eta \to -\infty$ (where $F(0) = 1$ for the standard hypergeometric series), is therefore exactly

$$\chi_k^{\text{in}}(\eta) = \frac{1}{\sqrt{2\omega_{\text{in}}}}\, s^{-i\omega_{\text{in}}/2\rho}\,(1-s)^{-i\omega_{\text{out}}/2\rho}\,{}_2F_1\Big(a, b; c;\, s\Big), \tag{64.101}$$

with $s = s(\eta)$ from Eq. (64.95) and a, b, c from Eq. (64.100). This is a closed-form solution of the wave equation on the expanding background, valid for all η. The analytic and connection machinery for the hypergeometric function ${}_2F_1$ and the gamma functions that appear below is the special-function toolkit of Chapter 5.

Worked Example 64.17 (The hypergeometric parameters from the constant term)**.** We finish the reduction by extracting ab. Write the substitution Eq. (64.98) as $\chi = PF$ with $P = s^{\alpha_1}(1-s)^{\alpha_2}$, so that $P'/P = \alpha_1/s - \alpha_2/(1-s)$ and $P''/P = (\alpha_1^2 - \alpha_1)/s^2 + (\alpha_2^2 - \alpha_2)/(1-s)^2 - 2\alpha_1\alpha_2/[s(1-s)]$. Inserting $\chi = PF$ into Eq. (64.97) and dividing by $s(1-s)P$, the coefficient of F is

$$s(1-s)\frac{P''}{P} + (1-2s)\frac{P'}{P} + \frac{\omega_{\text{in}}^2(1-s) + \omega_{\text{out}}^2 s}{4\rho^2\, s(1-s)}, \tag{64.102}$$

which must equal the constant $-ab$. Expanding each piece in partial fractions, the coefficient of $1/s$ is $(\alpha_1^2 - \alpha_1) + \alpha_1 + \omega_{\text{in}}^2/4\rho^2 = \alpha_1^2 + \omega_{\text{in}}^2/4\rho^2 = 0$ by the indicial condition, and the coefficient of $1/(1-s)$ vanishes the same way. What survives is purely the constant

$$\begin{aligned} -ab &= -(\alpha_1^2 - \alpha_1) - (\alpha_2^2 - \alpha_2) - 2\alpha_1\alpha_2 - 2\alpha_1 - 2\alpha_2 \\ &= -(\alpha_1 + \alpha_2)^2 - (\alpha_1 + \alpha_2), \end{aligned} \tag{64.103}$$

so $ab = (\alpha_1 + \alpha_2)(\alpha_1 + \alpha_2 + 1)$, as used in Eq. (64.100). The clean factorisation—the product depends only on the *sum* $\alpha_1 + \alpha_2 = -i\omega_+/\rho$—is what makes the connection coefficients below collapse to ratios of hyperbolic sines.

64.3.4 Connection coefficients and the Bogoliubov mismatch

The in-mode Eq. (64.101) is a definite solution; to read off the particle content we must express it, as $\eta \to +\infty$, in terms of the out-modes $\chi_k^{\text{out}} = (2\omega_{\text{out}})^{-1/2} e^{-i\omega_{\text{out}}\eta}$ and their conjugates. This is a *connection problem*: we need the behaviour of ${}_2F_1(a, b; c; s)$ near $s = 1$, which is supplied by the linear-transformation formula of the hypergeometric function (Chapter 5),

$$\begin{aligned} {}_2F_1(a, b; c; s) = {} & \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}\, {}_2F_1(a, b; a+b-c+1; 1-s) \\ & + (1-s)^{c-a-b}\, \frac{\Gamma(c)\Gamma(a+b-c)}{\Gamma(a)\Gamma(b)}\, {}_2F_1(c-a, c-b; c-a-b+1; 1-s). \end{aligned} \tag{64.104}$$

As $s \to 1$ both ${}_2F_1(\cdots; 1-s) \to 1$, so the in-mode tends to a sum of two power laws in $(1-s)$. With the parameters Eq. (64.100) the exponents evaluate (using $\omega_+ \pm \omega_- = \omega_{\text{out/in}}$) to

$$c-a-b = \frac{i\omega_{\text{out}}}{\rho}, \qquad a+b-c = -\frac{i\omega_{\text{out}}}{\rho}, \qquad c-a = \frac{i\omega_-}{\rho}, \qquad c-b = 1+\frac{i\omega_-}{\rho}. \tag{64.105}$$

Since $s^{\alpha_1} \to 1$ as $s \to 1$, the in-mode Eq. (64.101) becomes

$$\chi_k^{\text{in}} \xrightarrow{\eta\to+\infty} \frac{1}{\sqrt{2\omega_{\text{in}}}}\Big[\, C_1\,(1-s)^{\alpha_2} + C_2\,(1-s)^{\alpha_2+c-a-b}\,\Big], \tag{64.106}$$

with the connection coefficients

$$C_1 = \frac{\Gamma(c)\Gamma(c-a-b)}{\Gamma(c-a)\Gamma(c-b)}, \qquad C_2 = \frac{\Gamma(c)\Gamma(a+b-c)}{\Gamma(a)\Gamma(b)}. \tag{64.107}$$

Now translate the powers of $(1-s)$ back to plane waves. As $\eta \to +\infty$, $1-s \to e^{-2\rho\eta}$ from Eq. (64.95), so with $\alpha_2 = -i\omega_{\text{out}}/2\rho$,

$$(1-s)^{\alpha_2} \to e^{+i\omega_{\text{out}}\eta} \quad \text{(negative frequency)}, \qquad (1-s)^{\alpha_2+c-a-b} \to e^{-i\omega_{\text{out}}\eta} \quad \text{(positive freque} \tag{64.108}$$

the second exponent being $\alpha_2 + c - a - b = -i\omega_{\text{out}}/2\rho + i\omega_{\text{out}}/\rho = +i\omega_{\text{out}}/2\rho$. Therefore

$$\chi_k^{\text{in}} \xrightarrow{\eta\to+\infty} \frac{1}{\sqrt{2\omega_{\text{in}}}}\Big[\, C_2\, e^{-i\omega_{\text{out}}\eta} + C_1\, e^{+i\omega_{\text{out}}\eta}\,\Big]. \tag{64.109}$$

Comparing with the Bogoliubov decomposition $\chi_k^{\text{in}} = \alpha_k\,\chi_k^{\text{out}} + \beta_k\,\chi_k^{\text{out}\,*}$ (Section 64.2), with $\chi_k^{\text{out}} = (2\omega_{\text{out}})^{-1/2} e^{-i\omega_{\text{out}}\eta}$, matches the coefficient of each exponential:

$$\alpha_k = \sqrt{\frac{\omega_{\text{out}}}{\omega_{\text{in}}}}\, C_2, \qquad \beta_k = \sqrt{\frac{\omega_{\text{out}}}{\omega_{\text{in}}}}\, C_1. \tag{64.110}$$

The created particles live entirely in C_1—the coefficient of the *wrong*, negative-frequency exponential that the connection formula forces upon the in-mode at late times. Inserting the parameters Eq. (64.100), Eq. (64.105),

$$\alpha_k = \sqrt{\frac{\omega_{\text{out}}}{\omega_{\text{in}}}}\, \frac{\Gamma\!\left(1-\frac{i\omega_{\text{in}}}{\rho}\right)\Gamma\!\left(-\frac{i\omega_{\text{out}}}{\rho}\right)}{\Gamma\!\left(1-\frac{i\omega_+}{\rho}\right)\Gamma\!\left(-\frac{i\omega_+}{\rho}\right)}, \tag{64.111}$$

$$\beta_k = \sqrt{\frac{\omega_{\text{out}}}{\omega_{\text{in}}}}\, \frac{\Gamma\!\left(1-\frac{i\omega_{\text{in}}}{\rho}\right)\Gamma\!\left(\frac{i\omega_{\text{out}}}{\rho}\right)}{\Gamma\!\left(\frac{i\omega_-}{\rho}\right)\Gamma\!\left(1+\frac{i\omega_-}{\rho}\right)}. \tag{64.112}$$

64.3.5 The created-particle spectrum

The modulus squared of Eq. (64.112) is the occupation number of the out-vacuum's mode k in the in-vacuum, $n_k = |\beta_k|^2$. We evaluate it with the two

gamma-function identities proved in Chapter 5,

$$|\Gamma(iy)|^2 = \frac{\pi}{y\,\sinh(\pi y)}, \qquad |\Gamma(1+iy)|^2 = \frac{\pi y}{\sinh(\pi y)} \qquad (y \in \mathbb{R}), \tag{64.113}$$

which follow from the reflection formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$ and $\Gamma(1+z) = z\Gamma(z)$. Taking $|\cdot|^2$ of Eq. (64.112) and applying Eq. (64.113) term by term,

$$\begin{aligned} |\Gamma(1 - i\omega_{\rm in}/\rho)|^2 &= \frac{\pi(\omega_{\rm in}/\rho)}{\sinh(\pi\omega_{\rm in}/\rho)}, & |\Gamma(i\omega_{\rm out}/\rho)|^2 &= \frac{\pi}{(\omega_{\rm out}/\rho)\sinh(\pi\omega_{\rm out}/\rho)}, \\ |\Gamma(i\omega_-/\rho)|^2 &= \frac{\pi}{(\omega_-/\rho)\sinh(\pi\omega_-/\rho)}, & |\Gamma(1+i\omega_-/\rho)|^2 &= \frac{\pi(\omega_-/\rho)}{\sinh(\pi\omega_-/\rho)}. \end{aligned} \tag{64.114}$$

The denominator factors combine into $|\Gamma(i\omega_-/\rho)|^2\,|\Gamma(1+i\omega_-/\rho)|^2 = \pi^2/\sinh^2(\pi\omega_-/\rho)$—the (ω_-/ρ) factors cancel—and the numerator into $\pi^2(\omega_{\rm in}/\omega_{\rm out})/[\sinh(\pi\omega_{\rm in}/\rho)\sinh(\pi\omega_{\rm out}/\rho)]$. The prefactor $\omega_{\rm out}/\omega_{\rm in}$ from Eq. (64.112) cancels the residual frequency ratio, and the π^2's cancel, leaving the exact result

$$\boxed{n_k = |\beta_k|^2 = \frac{\sinh^2(\pi\omega_-/\rho)}{\sinh(\pi\omega_{\rm in}/\rho)\,\sinh(\pi\omega_{\rm out}/\rho)}, \qquad \omega_\pm = \tfrac{1}{2}(\omega_{\rm out} \pm \omega_{\rm in}).} \tag{64.115}$$

The identical calculation applied to Eq. (64.111) gives

$$|\alpha_k|^2 = \frac{\sinh^2(\pi\omega_+/\rho)}{\sinh(\pi\omega_{\rm in}/\rho)\,\sinh(\pi\omega_{\rm out}/\rho)}. \tag{64.116}$$

These are the closed-form Bogoliubov coefficients of the Bernard–Duncan model. The created-particle spectrum Eq. (64.115) is the first physical payoff of the chapter: an expanding universe creates a definite, computable number of quanta in each comoving mode, and we have it in elementary functions.

The unitarity check. The Bogoliubov coefficients must satisfy $|\alpha_k|^2 - |\beta_k|^2 = 1$, the normalisation that guarantees the transformation preserves the canonical commutators (Section 64.2). From Eq. (64.115), Eq. (64.116),

$$|\alpha_k|^2 - |\beta_k|^2 = \frac{\sinh^2(\pi\omega_+/\rho) - \sinh^2(\pi\omega_-/\rho)}{\sinh(\pi\omega_{\rm in}/\rho)\sinh(\pi\omega_{\rm out}/\rho)}. \tag{64.117}$$

Apply the identity $\sinh^2 X - \sinh^2 Y = \sinh(X+Y)\sinh(X-Y)$ with $X = \pi\omega_+/\rho$, $Y = \pi\omega_-/\rho$. Using Eq. (64.94), $X + Y = \pi\omega_{\rm out}/\rho$ and $X - Y = \pi\omega_{\rm in}/\rho$, so the numerator is exactly $\sinh(\pi\omega_{\rm out}/\rho)\sinh(\pi\omega_{\rm in}/\rho)$ and

$$|\alpha_k|^2 - |\beta_k|^2 = 1. \tag{64.118}$$

The machine is internally consistent: the transformation is a genuine Bogoliubov transformation, and Eq. (64.115) is a bona fide occupation number.

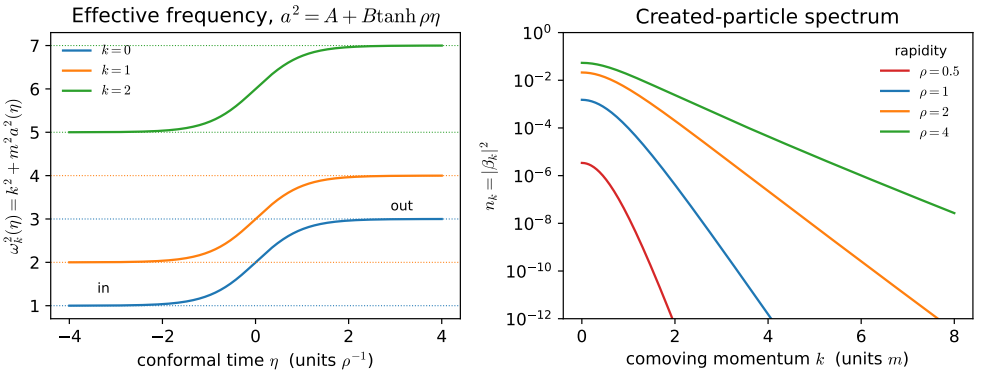


Figure 64.2: The Bernard–Duncan model with $m = 1$, $a_{\text{in}}^2 = A - B = 1$, $a_{\text{out}}^2 = A + B = 3$. *Left:* the effective frequency $\omega_k^2(\eta) = k^2 + m^2a^2(\eta)$ rising from ω_{in}^2 (dotted, lower) to ω_{out}^2 (dotted, upper) across the expansion; the asymptotically static "in" and "out" eras are where the curves flatten. *Right:* the exact created-particle spectrum $n_k = |\beta_k|^2$ of Eq. (64.115) versus comoving momentum k, for several rapidities ρ. The spectrum falls off exponentially in the ultraviolet (hence the total comoving particle number is finite), and the whole curve collapses toward zero as $\rho \to 0$: slow expansion creates no particles.

64.3.6 Limiting regimes: adiabatic, sudden, and the total number

The closed form Eq. (64.115) contains all the physics of the previous two subsections as limits. We extract three.

Worked Example 64.18 (Adiabatic, sudden, and total particle number)**. (i) Adiabatic limit, $\rho \to 0$ (slow expansion).** With $\omega_{\text{in}}, \omega_{\text{out}}, \omega_-$ held fixed and $\rho \to 0$, every argument $\pi\omega/\rho \to \infty$, so each $\sinh(\pi\omega/\rho) \to \tfrac{1}{2}e^{\pi\omega/\rho}$. Then

$$n_k \to \frac{\tfrac{1}{4}e^{2\pi\omega_-/\rho}}{\tfrac{1}{4}e^{\pi(\omega_{\text{in}}+\omega_{\text{out}})/\rho}} = e^{\,2\pi\omega_-/\rho - \pi(\omega_{\text{in}}+\omega_{\text{out}})/\rho} = e^{-2\pi\omega_{\text{in}}/\rho}, \tag{64.119}$$

where we used $2\omega_- - (\omega_{\text{in}} + \omega_{\text{out}}) = (\omega_{\text{out}} - \omega_{\text{in}}) - (\omega_{\text{in}} + \omega_{\text{out}}) = -2\omega_{\text{in}}$. The creation amplitude is exponentially small, $|\beta_k| \sim e^{-\pi\omega_{\text{in}}/\rho}$, and vanishes as $\rho \to 0$. This is the adiabatic theorem made quantitative, and it matches the WKB estimate Eq. (64.90): the analytic continuation of $\omega_k(\eta)$ has its nearest complex singularity where $\tanh(\rho\eta)$ does, at $\rho\eta_c = i\pi/2$, i.e. $\eta_c = i\pi/2\rho$, and the imaginary action accumulated to that turning point scales as $\omega_{\text{in}}\,\text{Im}\,\eta_c = \pi\omega_{\text{in}}/2\rho$, giving $|\beta_k| \sim e^{-2\cdot\pi\omega_{\text{in}}/2\rho} = e^{-\pi\omega_{\text{in}}/\rho}$ in exact agreement. The exponential rather than power-law suppression is why the in/out construction, and with it the adiabatic vacuum of Section 64.3.2, is so robust: a gently expanding universe is empty to all

orders in the adiabatic expansion, with particle creation a purely nonperturbative ($e^{-1/\rho}$) effect.

(ii) Sudden limit, $\rho \to \infty$ (fast expansion). Now every argument $\pi\omega/\rho \to 0$ and $\sinh x \to x$, so

$$n_k \to \frac{(\pi\omega_-/\rho)^2}{(\pi\omega_{\rm in}/\rho)(\pi\omega_{\rm out}/\rho)} = \frac{\omega_-^2}{\omega_{\rm in}\omega_{\rm out}} = \frac{(\omega_{\rm out}-\omega_{\rm in})^2}{4\,\omega_{\rm in}\omega_{\rm out}}, \tag{64.120}$$

a finite, ρ-independent occupation set entirely by the frequency mismatch between the two eras—exactly the result of the elementary sudden approximation, in which the in-mode is projected unchanged onto the out-basis. For a large jump $\omega_{\rm out} \gg \omega_{\rm in}$ this approaches $n_k \to \omega_{\rm out}/4\omega_{\rm in}$, an arbitrarily large number of quanta: a violent expansion is a copious particle source.

(iii) Total comoving particle number. The number of created quanta per unit comoving volume is

$$\mathcal{N} = \int \frac{\mathrm{d}^3k}{(2\pi)^3}\, n_k = \frac{1}{2\pi^2}\int_0^\infty \mathrm{d}k\, k^2\, \frac{\sinh^2(\pi\omega_-/\rho)}{\sinh(\pi\omega_{\rm in}/\rho)\sinh(\pi\omega_{\rm out}/\rho)}, \tag{64.121}$$

having done the angular integral. Convergence is controlled by the ultraviolet. At large k, $\omega_{\rm in} \approx \omega_{\rm out} \approx k$ while the mismatch shrinks, $\omega_- = \tfrac{1}{2}(\omega_{\rm out}-\omega_{\rm in}) \approx \frac{m^2 B}{2k} \to 0$, so $\sinh(\pi\omega_-/\rho) \approx \pi\omega_-/\rho$ is small while $\sinh(\pi\omega_{\rm in,out}/\rho) \approx \tfrac{1}{2}e^{\pi k/\rho}$ are exponentially large. Hence

$$n_k \;\approx\; \Big(\frac{\pi m^2 B}{2k\rho}\Big)^2 4\, e^{-2\pi k/\rho} \qquad (k \to \infty), \tag{64.122}$$

an exponential cutoff at the comoving scale $k \sim \rho$. The integrand $k^2 n_k$ therefore decays like $e^{-2\pi k/\rho}$ and Eq. (64.121) *converges*: the total comoving particle number is finite, with no ultraviolet divergence, and it is dominated by modes with $k \lesssim \rho$. The physical reason is sharp: only modes whose frequency changes appreciably during the expansion—those with $\omega_k'/\omega_k^2 \gtrsim 1$, i.e. $k \lesssim \rho$ for the tanh profile—are torn from the adiabatic vacuum; the deep ultraviolet rides the expansion adiabatically and stays empty. The total number grows with the rapidity ρ (a faster, more violent expansion excites more modes) and with the expansion amplitude through $\omega_- \propto m^2 B$, and it switches off entirely as $\rho \to 0$, consistent with Eq. (64.119).

Physical Insight

The finiteness of $\mathcal{N}$ is not an accident of the soluble model; it is the generic statement that gravitational particle creation is an *infrared-to-intermediate* phenomenon. High-momentum modes have wavelengths far shorter than the curvature scale; they cannot tell that spacetime is

curved and propagate as in flat space, so they are not excited. This is the same physics that lets one *renormalise* the gravitational stress tensor by adiabatic subtraction, the subject of Section 64.5: the divergences of $\langle T_{\mu\nu}\rangle$ live in the adiabatic (ultraviolet) part of the modes, which carries no real particles, and the finite created-particle content sits safely below the cutoff.

We have done what the Bogoliubov machine was built to do: fed it a concrete, gravitationally meaningful pair of bases and extracted an exact, finite spectrum of created particles, Eq. (64.115), complete with its adiabatic and sudden limits and the unitarity check Eq. (64.118). The spectrum here is a smooth function of k with no distinguished shape—it reflects the particular profile $a^2(\eta)$ and nothing universal. The next section asks what happens when the time-dependence is not a smooth finite-duration expansion but the eternal, self-similar redshift of a *horizon*. There the ratio of sinh's degenerates into something far more striking: the spectrum becomes an *exactly thermal* Planck distribution, the occupation number turning into $1/(e^{\omega/T}-1)$ with a temperature fixed by the geometry. That collapse of an arbitrary Bogoliubov spectrum onto a blackbody is the Unruh effect, and it is where we turn next.

64.4 The Unruh effect

The expanding universe of the previous section taught us that gravity creates particles because the very notion of "positive frequency" — and hence of "vacuum" and "particle" — is tied to a choice of time, and that choice can change as spacetime evolves. The created spectrum was controlled by the Bogoliubov coefficient β relating the natural mode bases of the far past and the far future. We now push that lesson to its sharpest edge. Instead of a mild cosmological evolution we take the cleanest possible arena — empty, flat, eternal Minkowski space — and the cleanest possible disagreement about time: an *inertial* observer versus a *uniformly accelerated* one. The accelerated observer carries a causal horizon with her, the Rindler horizon, and that horizon is the seed of the whole story. The inertial observer calls the state of the field the vacuum and finds it empty. The accelerated observer, using her own clock and her own notion of energy, finds it full: a thermal bath at temperature

$$T_{\rm U} \;=\; \frac{a}{2\pi}, \qquad \text{equivalently} \qquad k_{\rm B}T_{\rm U} \;=\; \frac{\hbar\, a}{2\pi c}, \tag{64.123}$$

where a is her proper acceleration. This is the Unruh effect. We will derive the number (64.123) twice. First with the Bogoliubov machinery of Section 64.2, by analytically continuing modes across the horizon. Then from the structure of the Euclidean path integral of Chapter 10, where the temperature falls out of a

single demand of smoothness. The two pillars of the book — canonical mode analysis and the path integral — meet on the same 2π, and the meeting point is exactly the mechanism that will give Hawking radiation in Chapter 65.

64.4.1 Rindler coordinates and the accelerated observer

We work in two-dimensional Minkowski space with coordinates (t, x) and metric $\mathrm{d}s^2 = \mathrm{d}t^2 - \mathrm{d}x^2$; the transverse directions and a mass term contribute nothing essential to the temperature and are reinstated at the end. Consider an observer undergoing constant proper acceleration. "Constant proper acceleration a" means the acceleration four-vector $a^\mu = \mathrm{d}^2 x^\mu/\mathrm{d}\tau^2$ (with τ the proper time) has fixed Minkowski norm $a^\mu a_\mu = -a^2$, while the velocity $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$ is a unit timelike vector, $u^\mu u_\mu = +1$. Differentiating $u^\mu u_\mu = 1$ gives $u^\mu a_\mu = 0$, so velocity and acceleration are orthogonal; the pair therefore behaves like a hyperbolic rotation, and the worldline is a hyperbola. Explicitly, the unique solution through $x = 1/a$ at $t = 0$ is

$$t(\tau) = \frac{1}{a}\sinh(a\tau), \qquad x(\tau) = \frac{1}{a}\cosh(a\tau), \tag{64.124}$$

as one checks directly: $u^\mu = (\cosh a\tau, \sinh a\tau)$ has $u^\mu u_\mu = \cosh^2 - \sinh^2 = 1$, and $a^\mu = (a\sinh a\tau, a\cosh a\tau)$ has $a^\mu a_\mu = a^2(\sinh^2 - \cosh^2) = -a^2$. The trajectory traces the hyperbola

$$x^2 - t^2 = \frac{1}{a^2}, \tag{64.125}$$

which lives entirely in the region $x > |t|$. This region is the *right Rindler wedge* R. As $\tau \to \pm\infty$ the worldline asymptotes to the null lines $x = \pm t$; these are the boundaries the accelerated observer can never cross, and they form her causal horizon.

Definition 64.19 (Rindler coordinates)**.** On the right wedge $R = \{x > |t|\}$ introduce coordinates (η, ξ) with $\eta \in \mathbb{R}$, $\xi \in \mathbb{R}$ by

$$t = \frac{1}{a}\, e^{a\xi}\sinh(a\eta), \qquad x = \frac{1}{a}\, e^{a\xi}\cosh(a\eta). \tag{64.126}$$

The reference acceleration $a > 0$ is a fixed constant labelling the foliation.

The curves of constant ξ are hyperbolae (64.125) with $x^2 - t^2 = a^{-2}e^{2a\xi}$; comparing with the worldline (64.124) the curve $\xi = 0$ is exactly the trajectory of an observer with proper acceleration a, and her proper time there is $\tau = \eta$. The curves of constant η are straight lines through the origin, $t/x = \tanh(a\eta)$; they are the instantaneous simultaneity slices of the accelerated observer. Computing the line element,

$$\begin{aligned} \mathrm{d}t &= e^{a\xi}\big[\cosh(a\eta)\,\mathrm{d}\eta + \sinh(a\eta)\,\mathrm{d}\xi\big], \\ \mathrm{d}x &= e^{a\xi}\big[\sinh(a\eta)\,\mathrm{d}\eta + \cosh(a\eta)\,\mathrm{d}\xi\big], \end{aligned} \tag{64.127}$$

so that, using $\cosh^2 - \sinh^2 = 1$,

$$\mathrm{d}s^2 = \mathrm{d}t^2 - \mathrm{d}x^2 = e^{2a\xi}(\mathrm{d}\eta^2 - \mathrm{d}\xi^2). \tag{64.128}$$

The metric (64.128) is conformal to flat space and manifestly static: nothing depends on η, so ∂_η is a Killing vector. It is timelike inside R (where $g_{\eta\eta} = e^{2a\xi} > 0$) and becomes null on the horizon $\xi \to -\infty$. An observer sitting at fixed ξ has proper time $\mathrm{d}\tau = e^{a\xi}\,\mathrm{d}\eta$ and, repeating the computation that gave (64.125), proper acceleration

$$a_{\mathrm{loc}}(\xi) = a\, e^{-a\xi}. \tag{64.129}$$

Deeper in the wedge (smaller ξ, closer to the horizon) the proper acceleration needed to hold station diverges; far out (large ξ) it is negligible. The single parameter a in (64.128) is the proper acceleration of the fiducial observer at $\xi = 0$, and that is the acceleration that will set the temperature.

The boost as the wedge's time-translation generator. The Killing field ∂_η has a clean meaning in the inertial frame. From (64.126),

$$\partial_\eta = \frac{\partial t}{\partial \eta}\,\partial_t + \frac{\partial x}{\partial \eta}\,\partial_x = e^{a\xi}\cosh(a\eta)\,\partial_t + e^{a\xi}\sinh(a\eta)\,\partial_x = a\,(x\,\partial_t + t\,\partial_x), \tag{64.130}$$

where in the last step we used $ax = e^{a\xi}\cosh(a\eta)$ and $at = e^{a\xi}\sinh(a\eta)$. The vector field $x\,\partial_t + t\,\partial_x$ is precisely the generator of Lorentz boosts in the t–x plane: it is the infinitesimal form of $M^{01} = K^1$ whose algebra we fixed in Chapter 3 (recall the boost generator from Eq. (3.135)). Thus *Rindler time translation is a Lorentz boost*, and a step $\eta \to \eta + \Delta\eta$ is a boost of rapidity $a\,\Delta\eta$. The accelerated observer's Hamiltonian — the generator conjugate to her time η — is the boost generator $K \equiv K^1$. This identification is the hinge of the entire section: the geometry of the wedge is the geometry of a boost orbit, and the quantum statistics we will find is the statistics of the boost.

The full Minkowski plane decomposes into four regions sewn along the null lines $x = \pm t$: the right wedge R ($x > |t|$), the left wedge L ($x < -|t|$), the future wedge F ($t > |x|$) and the past wedge P ($t < -|x|$). The boost moves points along the hyperbolae within R and within L, and along the straight null rays in F and P. In R the Killing field ∂_η is future-directed; in L it is *past*-directed, because the same boost that pushes events forward in time on the right pushes them backward on the left. The shared boundary $x = t$ ($\mathcal{H}^+$) and $x = -t$ ($\mathcal{H}^-$) are the future and past Rindler horizons. Figure 64.3 collects the picture.

64.4.2 Bogoliubov coefficients by analytic continuation

We now quantize a massless real scalar ϕ on this background and compare two mode bases: the inertial (Minkowski) one, whose vacuum we call $|0_M\rangle$, and the

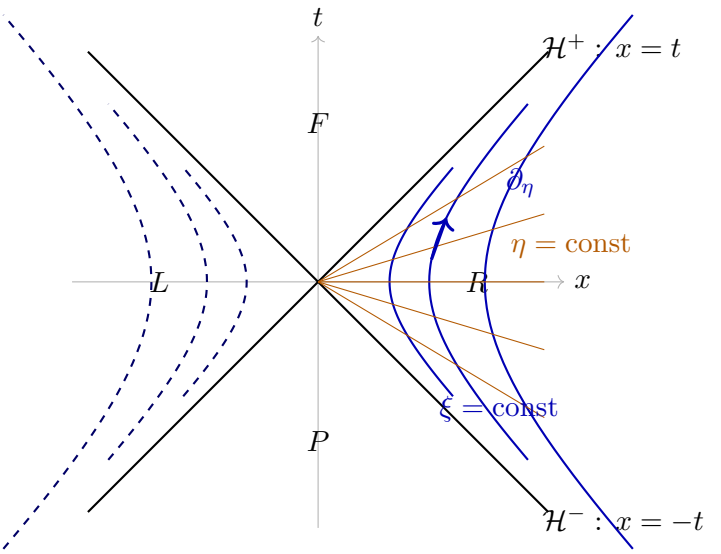


Figure 64.3: The Rindler wedges of Minkowski space. The accelerated observer's worldlines are the hyperbolae $x^2 - t^2 = a^{-2}e^{2a\xi}$ of constant ξ (solid blue in R); her simultaneity slices are the rays of constant η (orange). The boost Killing field ∂_η flows along the hyperbolae, future-directed in R and past-directed in the mirror wedge L (dashed). The null lines $x = \pm t$ are the future/past Rindler horizons $\mathcal{H}^\pm$, which the right-wedge observer can neither see beyond ($\mathcal{H}^+$) nor have influenced her ($\mathcal{H}^-$). The wedges R and L are causally disconnected.

Rindler one natural to the accelerated observer. The strategy, due to Unruh, is not to compute the Bogoliubov integrals head-on but to build a third set of modes — the *Unruh modes* — that are simultaneously simple in the Rindler basis and manifestly of positive Minkowski frequency. Analyticity across the horizon does all the work.

Null coordinates. A massless field in two dimensions splits into right-movers and left-movers; the wave equation $\partial_\eta^2\phi - \partial_\xi^2\phi = 0$ (the conformal factor in (64.128) drops out of the massless equation) is solved by arbitrary functions of the null combinations. Introduce inertial and Rindler null coordinates

$$U = t - x, \quad V = t + x; \qquad u = \eta - \xi, \quad v = \eta + \xi. \tag{64.131}$$

The right-movers are functions of U (equivalently u); the left-movers, functions of V (equivalently v). The two sectors are independent and identical in structure, so we follow the right-movers and quote the left-movers at the end. Substituting (64.126) into (64.131) gives, in the right wedge,

$$U = -\frac{1}{a}\,e^{-au}, \qquad V = +\frac{1}{a}\,e^{av} \qquad (\text{in } R), \tag{64.132}$$

since, for instance, $U = t - x = \frac{1}{a}e^{a\xi}(\sinh a\eta - \cosh a\eta) = -\frac{1}{a}e^{a\xi}e^{-a\eta} = -\frac{1}{a}e^{-a(\eta-\xi)}$. The left wedge is covered by the same coordinate formulae (64.126) with an overall sign, $t \to -t$, $x \to -x$; carrying the sign through gives

$$U = +\frac{1}{a}\,e^{-au}, \qquad V = -\frac{1}{a}\,e^{av} \qquad \text{(in } L\text{)}. \tag{64.133}$$

Observe the central fact: $U < 0$ throughout R and $U > 0$ throughout L. The Rindler coordinate u is the logarithm of $|U|$, and the two wedges sit on the two sides of the horizon $U = 0$. Crossing the horizon is crossing a logarithmic branch point of $U(u)$, and *that* is where analytic continuation will relate the wedges.

The two mode families. A right-moving Minkowski mode of positive frequency $\omega > 0$ is $e^{-i\omega U}$: it satisfies $i\partial_t e^{-i\omega U} = \omega e^{-i\omega U}$ with $\omega > 0$. The crucial analytic property is that $e^{-i\omega U}$ is *bounded and holomorphic in the lower half U-plane*, since $|e^{-i\omega U}| = e^{\omega\,\mathrm{Im}\,U} \to 0$ as $\mathrm{Im}\,U \to -\infty$. Conversely any function holomorphic and bounded in the lower-half U-plane is a superposition of positive-frequency modes only. This is the criterion we will use to certify that a mode belongs to the Minkowski vacuum.

The Rindler modes are positive-frequency with respect to the wedge's own time, the boost parameter η, measured by the *future-directed* Killing field in each wedge. In R that field is $+\partial_\eta$, so a positive-frequency right-moving Rindler mode of frequency $\Omega > 0$ is $\propto e^{-i\Omega u}$, supported on R. In L the future-directed Killing field is $-\partial_\eta$, which reverses the sign: positive frequency there means $\partial_\eta g = +i\Omega g$, i.e. $\propto e^{+i\Omega u}$, supported on L. Normalizing in the Klein–Gordon inner product (each carries the standard $1/\sqrt{4\pi\Omega}$), and re-expressing u through U via (64.132)–(64.133),

$$g^R_\Omega(U) = \frac{1}{\sqrt{4\pi\Omega}}\,(-aU)^{\,i\Omega/a}, \qquad U < 0,\ \text{(zero in } L\text{)}, \tag{64.134}$$

$$g^L_\Omega(U) = \frac{1}{\sqrt{4\pi\Omega}}\,(aU)^{-i\Omega/a}, \qquad U > 0,\ \text{(zero in } R\text{)}, \tag{64.135}$$

where we used $e^{-i\Omega u} = e^{i(\Omega/a)\ln(-aU)} = (-aU)^{i\Omega/a}$ in R (from $-aU = e^{-au}$, so $u = -a^{-1}\ln(-aU)$) and $e^{+i\Omega u} = (aU)^{-i\Omega/a}$ in L. Each family is complete on its own wedge, and the corresponding annihilation operators a^R_Ω, a^L_Ω define the Rindler (Fulling) vacuum $|0_R\rangle = |0_R\rangle \otimes |0_L\rangle$. The Rindler observer in R has no access to L, so her physics is built from a^R_Ω alone.

Continuing across the horizon. The functions (64.134)–(64.135) are each defined on only half the U-axis and are not, separately, positive-frequency Minkowski modes (indeed they have a branch point at the horizon $U = 0$). The idea is to extend g^R_Ω to all U by continuing it through the *lower* half U-plane, where positive-frequency functions are well-behaved. Write $-aU = |aU|\,e^{i\arg(-U)}$

and track the phase as U runs from the negative real axis (in R) to the positive real axis (in L) along a path with $\operatorname{Im} U < 0$. On the negative axis, $-aU > 0$ has $\arg(-U) = 0$. Rotating U clockwise through the lower half-plane to the positive axis advances $\arg(-U)$ from 0 to π, so on arrival

$$-aU\big|_{U>0} = aU\, e^{i\pi} \quad \Longrightarrow \quad (-aU)^{i\Omega/a}\big|_{U>0} = (aU)^{i\Omega/a}\, e^{i\pi\cdot i\Omega/a} = e^{-\pi\Omega/a}\,(aU)^{i\Omega/a}. \tag{64.136}$$

The branch chosen — continuation through $\operatorname{Im} U < 0$ — is precisely the one that keeps the function bounded in the lower half-plane (the phase $e^{-(\Omega/a)\arg(-U)}$ never exceeds unity there), so the continued function is a genuine positive-frequency Minkowski object. Now recognize the right-hand side: on $U > 0$, comparing with (64.135),

$$(aU)^{i\Omega/a} = \big[(aU)^{-i\Omega/a}\big]^{*} = \sqrt{4\pi\Omega}\; g^{L*}_{\Omega}(U). \tag{64.137}$$

The analytic continuation of the right-wedge mode *is* a fixed multiple of the complex conjugate of the left-wedge mode. Assembling (64.136)–(64.137), the single holomorphic function defined on the whole U-axis is

$$W_{\Omega} \;=\; g^{R}_{\Omega} + e^{-\pi\Omega/a}\, g^{L*}_{\Omega}, \tag{64.138}$$

and by construction W_Ω is positive-frequency with respect to Minkowski time. By the mirror computation — continue g^{L}_{Ω} through the lower half-plane — the second independent positive-frequency combination is

$$\widetilde{W}_{\Omega} \;=\; g^{L}_{\Omega} + e^{-\pi\Omega/a}\, g^{R*}_{\Omega}. \tag{64.139}$$

These are the *Unruh modes*. They are the bridge: built linearly from Rindler modes, yet annihilating the Minkowski vacuum.

Normalization and the Bogoliubov transformation. In the Klein–Gordon inner product the Rindler modes are orthonormal, $\langle g^{R}_{\Omega}, g^{R}_{\Omega'}\rangle = \delta(\Omega-\Omega')$, the conjugates carry negative norm $\langle g^{R*}_{\Omega}, g^{R*}_{\Omega'}\rangle = -\delta(\Omega-\Omega')$, and modes on disjoint wedges are orthogonal. Hence the norm of (64.138) is

$$\langle W_\Omega, W_\Omega\rangle = \langle g^{R}_{\Omega}, g^{R}_{\Omega}\rangle + e^{-2\pi\Omega/a}\langle g^{L*}_{\Omega}, g^{L*}_{\Omega}\rangle = \big(1 - e^{-2\pi\Omega/a}\big)\,\delta(\Omega-\Omega'), \tag{64.140}$$

so the unit-normalized Unruh modes are

$$\widehat{W}_{\Omega} = \frac{g^{R}_{\Omega} + e^{-\pi\Omega/a}\, g^{L*}_{\Omega}}{\sqrt{1-e^{-2\pi\Omega/a}}}, \qquad \widehat{\widetilde{W}}_{\Omega} = \frac{g^{L}_{\Omega} + e^{-\pi\Omega/a}\, g^{R*}_{\Omega}}{\sqrt{1-e^{-2\pi\Omega/a}}}. \tag{64.141}$$

Because $\{\widehat{W}_\Omega, \widehat{\widetilde{W}}_\Omega\}$ are positive-frequency Minkowski modes, their annihilation operators $d^{(1)}_{\Omega}, d^{(2)}_{\Omega}$ annihilate $|0_M\rangle$. Expanding the field in the two bases,

$$\phi = \int_0^\infty \mathrm{d}\Omega \Big[a^{R}_{\Omega} g^{R}_{\Omega} + a^{L}_{\Omega} g^{L}_{\Omega} + \text{h.c.}\Big] = \int_0^\infty \mathrm{d}\Omega \Big[d^{(1)}_{\Omega}\widehat{W}_{\Omega} + d^{(2)}_{\Omega}\widehat{\widetilde{W}}_{\Omega} + \text{h.c.}\Big], \tag{64.142}$$

and matching the coefficient of g^R_Ω on the two sides — it appears in $\widehat{W}_\Omega$ with weight $(1-e^{-2\pi\Omega/a})^{-1/2}$ and in $\widehat{\widetilde{W}}^*_\Omega$ (the conjugate mode multiplying $d^{(2)\dagger}_\Omega$) with weight $e^{-\pi\Omega/a}(1-e^{-2\pi\Omega/a})^{-1/2}$ — yields the Bogoliubov relation

$$a^R_\Omega = \alpha_\Omega\, d^{(1)}_\Omega + \beta_\Omega\, d^{(2)\dagger}_\Omega, \qquad \alpha_\Omega = \frac{1}{\sqrt{1-e^{-2\pi\Omega/a}}}, \quad \beta_\Omega = \frac{e^{-\pi\Omega/a}}{\sqrt{1-e^{-2\pi\Omega/a}}}. \tag{64.143}$$

The coefficients satisfy the canonical normalization automatically,

$$|\alpha_\Omega|^2 - |\beta_\Omega|^2 = \frac{1-e^{-2\pi\Omega/a}}{1-e^{-2\pi\Omega/a}} = 1, \qquad \frac{|\beta_\Omega|^2}{|\alpha_\Omega|^2} = e^{-2\pi\Omega/a}, \tag{64.144}$$

which is the relation $|\beta_\Omega|^2/|\alpha_\Omega|^2 = e^{-2\pi\Omega/a}$ promised at the head of the chapter.

The Planck spectrum. The number of Rindler particles the accelerated observer counts in the Minkowski vacuum follows by the rule established in Section 64.2: $\langle 0_M|N^R_\Omega|0_M\rangle = |\beta_\Omega|^2$. Concretely, using (64.143) and $d^{(1,2)}_\Omega\,|0_M\rangle = 0$,

$$\Big\langle 0_M\Big|a^{R\dagger}_\Omega a^R_{\Omega'}\Big|0_M\Big\rangle = |\beta_\Omega|^2\,\Big\langle 0_M\Big|d^{(2)}_\Omega d^{(2)\dagger}_{\Omega'}\Big|0_M\Big\rangle = |\beta_\Omega|^2\,\delta(\Omega-\Omega'), \tag{64.145}$$

where the $\delta(\Omega-\Omega')$ is the standard continuum density-of-states factor; the mean occupation per mode is therefore

$$n_\Omega = |\beta_\Omega|^2 = \frac{e^{-2\pi\Omega/a}}{1-e^{-2\pi\Omega/a}} = \frac{1}{e^{2\pi\Omega/a}-1}. \tag{64.146}$$

This is exactly the Bose–Einstein distribution $n_\Omega = (e^{\Omega/T}-1)^{-1}$ with

$$T = \frac{a}{2\pi}, \tag{64.147}$$

the Unruh temperature (64.123). The mass and the transverse dimensions we dropped enter the density of states and the detailed shape of wavepackets but not this exponent: the temperature is universal, fixed by the single number a. We record the result.

Theorem 64.20 (Unruh)**.** *Let ϕ be a free scalar field in the Minkowski vacuum $|0_M\rangle$. With respect to the boost-time modes natural to a uniformly accelerated observer of proper acceleration a (Theorem 64.19), the state $|0_M\rangle$ contains a Planckian distribution of quanta,*

$$n_\Omega = \frac{1}{e^{2\pi\Omega/a}-1},$$

i.e. it is perceived as a thermal bath at the Unruh temperature $T_{\mathrm{U}} = a/2\pi$. An inertial observer, using the Minkowski modes, finds the same state empty.

Physical Insight

Nothing was "added" to the vacuum; the field operator and the state are the ones an inertial physicist calls empty. What changed is the bookkeeping of energy. The accelerated observer's notion of a quantum is the Rindler mode, and the Minkowski vacuum, decomposed in that basis, is a correlated condensate of Rindler pairs straddling the horizon — one quantum in R, its partner in the inaccessible wedge L — distributed exactly thermally. Acceleration does not heat the field; it changes which observable counts as "the number of particles," and that observable has a thermal expectation value. The same sentence, transplanted from the Rindler horizon to a black-hole horizon, will become Hawking radiation in Chapter 65.

64.4.3 The Euclidean/KMS derivation: temperature from periodicity

The Bogoliubov route gave the temperature through a careful tracking of phases across the horizon. The path-integral route gives the same number with almost no computation, by Wick-rotating the boost time. This is the place where the chapter's two methods visibly converge.

Recall from Chapter 10 (the dictionary of Section 10.6.5) the fundamental equivalence: a quantum field whose Euclidean time is periodic with period β computes the thermal trace $Z(\beta) = \mathrm{Tr}\, e^{-\beta H}$, so periodicity in imaginary time *is* thermal equilibrium at temperature $T = 1/\beta$. The Hamiltonian H in that statement is the generator of the time translation being rotated. For the accelerated observer the relevant time is the boost parameter η, and the relevant Hamiltonian is the boost generator K. We therefore Wick-rotate η.

Set $\eta = -i\theta$ with θ real. Under this rotation the worldline coordinates (64.126) become, using $\sinh(-ia\theta) = -i\sin(a\theta)$ and $\cosh(-ia\theta) = \cos(a\theta)$,

$$t = -\frac{i}{a}\, e^{a\xi}\sin(a\theta) \equiv -i\, t_E, \qquad x = \frac{1}{a}\, e^{a\xi}\cos(a\theta), \qquad t_E = \frac{1}{a}\, e^{a\xi}\sin(a\theta). \tag{64.148}$$

The Euclidean plane (t_E, x) is covered by polar-like coordinates with radius and angle

$$\rho = \frac{1}{a}\, e^{a\xi} \geq 0, \qquad \varphi = a\theta, \qquad \text{so} \qquad t_E = \rho\sin\varphi, \quad x = \rho\cos\varphi. \tag{64.149}$$

The Euclidean line element is $\mathrm{d}s_E^2 = \mathrm{d}t_E^2 + \mathrm{d}x^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\varphi^2$, the flat plane in polar coordinates — equivalently, inserting (64.149),

$$\mathrm{d}s_E^2 = e^{2a\xi}(\mathrm{d}\theta^2 + \mathrm{d}\xi^2), \tag{64.150}$$

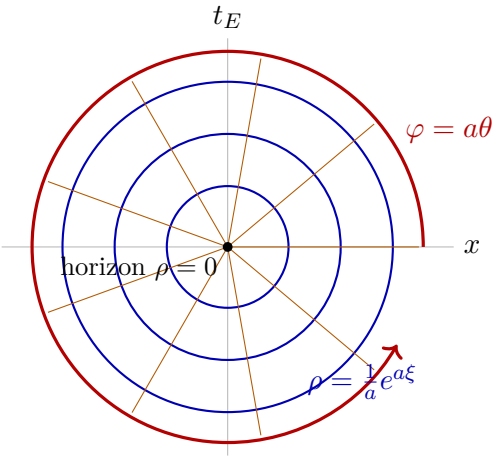


Figure 64.4: Euclidean Rindler space. Wick-rotating the boost time, $\eta = -i\theta$, turns the wedge into the flat plane (t_E, x) in polar coordinates: the constant-ξ hyperbolae become circles of radius $\rho = a^{-1}e^{a\xi}$ (blue) and the constant-η rays become radial lines (orange). The horizon collapses to the origin, the fixed point of the rotation. Smoothness there forces the angle $\varphi = a\theta$ to be 2π-periodic, i.e. the Euclidean time θ to have period $\beta = 2\pi/a$ — and an imaginary-time period β is, by Section 10.6.5, a thermal state at $T = 1/\beta = a/2\pi$.

the Euclidean continuation of the Rindler metric (64.128). The horizon $\xi \to -\infty$, where $\rho \to 0$, is the *origin* of the Euclidean plane — the fixed point of the boost, which under rotation has become an ordinary rotation about the origin.

Here is the whole argument in one line. A rotation angle φ on a smooth plane must be 2π-periodic; any other period leaves a conical defect at $\rho = 0$, a curvature delta-function that the smooth flat geometry of Minkowski space does not have. Smoothness of the Euclidean continuation at the horizon therefore demands

$$\varphi = a\theta \;\sim\; a\theta + 2\pi \quad\Longleftrightarrow\quad \theta \;\sim\; \theta + \frac{2\pi}{a}. \tag{64.151}$$

The Euclidean boost time θ is periodic with period $\beta = 2\pi/a$.

By the imaginary-time dictionary, this periodicity means that the Minkowski vacuum, restricted to the right wedge and propagated by the boost Hamiltonian K, is the Gibbs state

$$\rho_R \;\propto\; e^{-\beta K}, \qquad \beta = \frac{2\pi}{a}, \tag{64.152}$$

a thermal state at inverse temperature 2π in units of the boost time. To translate this into a temperature an observer measures, recall that the fiducial observer at $\xi = 0$ has proper time $\tau = \eta$ and proper acceleration a. Her clock ticks in η, her

energy is conjugate to η, and so she reads the inverse temperature $\beta = 2\pi/a$, i.e.

$$T_{\mathrm{U}} = \frac{1}{\beta} = \frac{a}{2\pi}, \tag{64.153}$$

the same number (64.123) as the Bogoliubov computation. The two pillars have met. An observer parked at general ξ measures proper time $\mathrm{d}\tau = e^{a\xi}\mathrm{d}\eta$; her locally measured temperature is the Tolman-redshifted value

$$T_{\mathrm{loc}}(\xi) = \frac{T_{\mathrm{U}}}{\sqrt{g_{\eta\eta}}} = \frac{a/2\pi}{e^{a\xi}} = \frac{a_{\mathrm{loc}}(\xi)}{2\pi}, \tag{64.154}$$

exactly $a_{\mathrm{loc}}/2\pi$ in terms of her own proper acceleration (64.129). The Unruh formula holds pointwise, with each observer's local acceleration setting her local temperature — a consistency check that the result is geometric, not an artifact of the $\xi = 0$ slice.

Remark 64.21 (Why 2π and nothing else). The entire temperature is the statement that a smooth plane has angular period 2π. Every appearance of the magic factor 2π in horizon thermodynamics — Unruh here, Hawking in Chapter 65 — traces to this one geometric fact: near any non-degenerate horizon the Euclidean geometry is locally a plane in polar coordinates, and regularity at the tip fixes the period of imaginary time. The Bogoliubov phase $e^{-\pi\Omega/a}$ of (64.136) is the same 2π seen through the half-rotation that carries one wedge into the other: the continuation (64.136) rotated U by angle π, and $e^{-\pi\Omega/a}$ is the half of the full $e^{-2\pi\Omega/a}$ Boltzmann weight.

The rigorous backbone: Bisognano–Wichmann. The heuristic Wick rotation can be made into a theorem of axiomatic quantum field theory, and it is worth stating precisely what is known, because it elevates the Unruh effect from a property of free fields to a structural property of *any* relativistic quantum field theory. The tool is the Tomita–Takesaki modular theory of von Neumann algebras: to a von Neumann algebra $\mathcal{A}$ with a cyclic and separating vector Ω (here the algebra of observables localized in the right wedge, with $\Omega = |0_M\rangle$ the vacuum) the theory canonically assigns a positive operator Δ, the modular operator, and the one-parameter *modular flow* $\sigma_s(\cdot) = \Delta^{is}(\cdot)\Delta^{-is}$. The vacuum satisfies the KMS condition with respect to this flow at inverse temperature 1.

Theorem 64.22 (Bisognano–Wichmann; stated, not proved)**.** *In any Wightman quantum field theory satisfying the standard axioms, the modular operator of the right-wedge algebra with respect to the vacuum is $\Delta = e^{-2\pi K}$, where K is the generator of Lorentz boosts in the plane orthogonal to the wedge's edge. Equivalently, the modular flow is the geometric boost flow, σ_s acts as a boost of rapidity $-2\pi s$, and the vacuum is a KMS (thermal) state for the boost Hamiltonian at inverse temperature $\beta = 2\pi$. Restoring the proper acceleration of a worldline at unit boost speed reproduces $T_{\mathrm{U}} = a/2\pi$.*

The content of Theorem 64.22 is that the thermal character of the wedge-restricted vacuum is not special to free fields or to the Gaussian manipulations above: it is forced by Lorentz invariance, locality, and the spectrum condition alone. The proof, which we cite rather than reproduce (it belongs to operator-algebraic QFT and uses the analyticity of vacuum correlators in the tube domains fixed by the spectrum condition), establishes exactly the identification "modular Hamiltonian $= 2\pi\times$ boost" that our Euclidean argument made plausible. We will invoke the same modular-flow language when we discuss black-hole horizons and entanglement in Chapter 65; for the present chapter it is the assurance that the number $a/2\pi$ is a theorem, not a free-field coincidence.

64.4.4 The detector: a click for the accelerated, silence for the inertial

A skeptic may ask what "the accelerated observer sees a thermal bath" means operationally, given that the state is the inertial vacuum. The answer is a model detector. An Unruh–DeWitt detector is a pointlike quantum system with internal energy levels, carried along a worldline $x^\mu(\tau)$ and coupled to the field through the monopole interaction

$$H_{\rm int} = \lambda\, \mu(\tau)\, \phi(x(\tau)), \tag{64.155}$$

with $\mu(\tau)$ the detector's monopole operator and λ a small coupling. To first order in λ, the amplitude to excite the detector from its ground state $|E_0\rangle$ to a level $|E\rangle$ (gap $E = E - E_0 > 0$) while the field starts in $|0_M\rangle$ is, by time-dependent perturbation theory, proportional to the integral of the field's Wightman function along the worldline. Summing over the unobserved final field states, the excitation probability per unit coupling is $|\langle E|\mu(0)|E_0\rangle|^2\, \mathcal{F}(E)$, where the universal kinematic factor is the *response function*

$$\mathcal{F}(E) = \int_{-\infty}^{\infty} \mathrm{d}\tau \int_{-\infty}^{\infty} \mathrm{d}\tau'\, e^{-iE(\tau-\tau')}\, W(x(\tau), x(\tau')), \qquad W(x,x') = \langle 0_M|\phi(x)\phi(x')|0_M\rangle\,. \tag{64.156}$$

For a stationary worldline W depends only on the proper-time difference $s = \tau - \tau'$, the double integral factorizes into an (infinite) total proper time times the response *rate*

$$F(E) = \int_{-\infty}^{\infty} \mathrm{d}s\, e^{-iEs}\, W(s), \tag{64.157}$$

the Fourier transform of the pulled-back Wightman function. The detector reads the field through $F(E)$: a nonzero $F(E)$ for $E > 0$ means it clicks, absorbing energy from the field.

The inertial detector is silent. For an inertial worldline the proper time is Minkowski time and the spatial separation vanishes, so the four-dimensional

massless Wightman function (the $d = 4$ form is cleaner for the contour argument; the temperature is dimension-independent) reduces to

$$W_{\text{inert}}(s) = -\frac{1}{4\pi^2}\,\frac{1}{(s - i\epsilon)^2}, \tag{64.158}$$

the $-i\epsilon$ encoding that W is a positive-frequency (Wightman, not time-ordered) function. Insert this into (64.157):

$$F_{\text{inert}}(E) = -\frac{1}{4\pi^2}\int_{-\infty}^{\infty} \mathrm{d}s\,\frac{e^{-iEs}}{(s - i\epsilon)^2}. \tag{64.159}$$

For $E > 0$ the factor $e^{-iEs} = e^{-iE\,\mathrm{Re}\,s}\,e^{E\,\mathrm{Im}\,s}$ decays as $\mathrm{Im}\,s \to -\infty$, so we close the contour in the *lower* half s-plane. The only singularity of the integrand, the double pole, sits at $s = +i\epsilon$ in the *upper* half-plane; the lower contour encloses nothing and

$$F_{\text{inert}}(E) = 0 \qquad (E > 0). \tag{64.160}$$

An inertial detector in the Minkowski vacuum never gets excited. It is at zero temperature, as it must be. (For $E < 0$ one closes upward and finds the spontaneous-emission rate; an inertial detector prepared in an excited state can decay, but a ground-state one cannot be promoted.)

The accelerated detector thermalizes. Now carry the detector along the accelerated worldline (64.124). The pulled-back Wightman function needs the invariant interval between two points on the hyperbola. With $t = a^{-1}\sinh(a\tau)$, $x = a^{-1}\cosh(a\tau)$,

$$\begin{aligned}(t - t')^2 - (x - x')^2 &= \frac{1}{a^2}\Big[(\sinh a\tau - \sinh a\tau')^2 - (\cosh a\tau - \cosh a\tau')^2\Big] \\ &= \frac{1}{a^2}\Big[-2 + 2\big(\cosh a\tau\cosh a\tau' - \sinh a\tau\sinh a\tau'\big)\Big] \\ &= \frac{2}{a^2}\big[\cosh\big(a(\tau - \tau')\big) - 1\big] = \frac{4}{a^2}\,\sinh^2\!\Big(\frac{a\,s}{2}\Big),\end{aligned} \tag{64.161}$$

where we used $\sinh^2 - \cosh^2 = -1$ on each point, the addition theorem $\cosh(a\tau - a\tau') = \cosh a\tau\cosh a\tau' - \sinh a\tau\sinh a\tau'$, and finally $\cosh\zeta - 1 = 2\sinh^2(\zeta/2)$. The interval depends only on $s = \tau - \tau'$, confirming stationarity. The $d = 4$ Wightman function $W = -\frac{1}{4\pi^2}[(t - t' - i\epsilon)^2 - |\mathbf{x} - \mathbf{x}'|^2]^{-1}$ pulled back to the worldline (where the transverse separation vanishes and the $i\epsilon$ becomes $s \to s - i\epsilon$) is therefore

$$W_{\text{acc}}(s) = -\frac{a^2}{16\pi^2}\,\frac{1}{\sinh^2\!\big(\frac{a}{2}(s - i\epsilon)\big)}. \tag{64.162}$$

This single function carries the whole effect. Note its periodicity in imaginary proper time: since $\sinh(z - i\pi) = -\sinh z$, the square is invariant under $\frac{a}{2}s \to$

$\frac{a}{2}s - i\pi$, i.e. under $s \to s - 2\pi i/a$,

$$W_{\rm acc}\Big(s - \frac{2\pi i}{a}\Big) = W_{\rm acc}(s). \tag{64.163}$$

This is the KMS condition at inverse temperature $\beta = 2\pi/a$ written for the two-point function — the same periodicity that the Euclidean argument read off the angle, now visible directly in the correlator the detector samples. The $-i\epsilon$ prescription places the pole at $s = 0$ (and the $n = 0$ image) in the upper half-plane, exactly as in the inertial case, while the images at $s = -2\pi i n/a$, $n = 1, 2, \ldots$, populate the lower half-plane.

The KMS periodicity (64.163) forces detailed balance. From the definition (64.157), substituting $s \to -s$ and then using (64.163) together with analyticity of W in the strip $-\beta < \mathrm{Im}\, s < 0$ (so that the contour may be shifted by $-i\beta$),

$$F(-E) = \int \mathrm{d}s\, e^{+iEs} W(s) = \int \mathrm{d}s\, e^{-iEs} W(-s) = \int \mathrm{d}s\, e^{-iEs} W(s - i\beta) = e^{\beta E} \int \mathrm{d}s\, e^{-iEs} W(s) \tag{64.164}$$

where in the last-but-one step we shifted $s \to s + i\beta$ and picked up $e^{-iE(s+i\beta)} = e^{\beta E} e^{-iEs}$. Hence

$$\frac{F(E)}{F(-E)} = e^{-\beta E} = e^{-2\pi E/a}. \tag{64.165}$$

The ratio of the rate to climb a gap E to the rate to fall by it is the Boltzmann factor at temperature $T = a/2\pi$. This is precisely how a detector in contact with a heat bath at temperature T behaves: *detailed balance is the operational definition of "the detector sees a thermal bath."* The accelerated detector clicks, with a spectrum of clicks indistinguishable from immersion in radiation at $T_{\rm U} = a/2\pi$, while (64.160) says its inertial twin records nothing. Figure 64.5 plots both the thermal occupation (64.146) and the detector response worked out next, against the flat inertial baseline.

Worked Example 64.23 (The full Planckian response rate)**.** We extract not just the ratio (64.165) but the full response rate by evaluating (64.157) with (64.162) by residues. Write

$$F(E) = -\frac{a^2}{16\pi^2} \int_{-\infty}^{\infty} \mathrm{d}s\, \frac{e^{-iEs}}{\sinh^2\!\big(\frac{a}{2}(s - i\epsilon)\big)}. \tag{64.166}$$

For $E > 0$ close the contour in the lower half-plane, where e^{-iEs} decays. The integrand has double poles where $\sinh(\frac{a}{2}s) = 0$, i.e. $s = 2\pi i n/a$ for integer n; with the $-i\epsilon$ shift the poles enclosed by the downward contour are those at

$$s_n = -\frac{2\pi i n}{a}, \qquad n = 1, 2, 3, \ldots \tag{64.167}$$

Near s_n set $s = s_n + \delta$. Using $\sinh(\tfrac{a}{2}s_n + \tfrac{a}{2}\delta) = \sinh(-i\pi n + \tfrac{a}{2}\delta) = (-1)^n \sinh(\tfrac{a}{2}\delta) \simeq (-1)^n \tfrac{a}{2}\delta$, the squared denominator is $\sinh^2 \simeq \tfrac{a^2}{4}\delta^2$, so near the pole

$$\frac{e^{-iEs}}{\sinh^2(\frac{a}{2}s)} \simeq \frac{4}{a^2}\,\frac{e^{-iEs_n}\,e^{-iE\delta}}{\delta^2} = \frac{4}{a^2}\,e^{-iEs_n}\,\frac{1 - iE\delta + \cdots}{\delta^2}. \tag{64.168}$$

The residue (coefficient of $1/\delta$) is $\dfrac{4}{a^2}(-iE)\,e^{-iEs_n}$, and with $e^{-iEs_n} = e^{-iE(-2\pi i n/a)} = e^{-2\pi n E/a}$ the clockwise (downward) contour gives $F(E) = -\frac{a^2}{16\pi^2}\cdot(-2\pi i)\sum_{n\geq 1}$ Res:

$$\begin{aligned} F(E) &= -\frac{a^2}{16\pi^2}\,(-2\pi i)\,\sum_{n=1}^{\infty}\frac{4}{a^2}\,(-iE)\,e^{-2\pi nE/a} \\ &= -\frac{a^2}{16\pi^2}\,(-2\pi i)(-iE)\frac{4}{a^2}\sum_{n=1}^{\infty}e^{-2\pi nE/a} = \frac{E}{2\pi}\sum_{n=1}^{\infty}e^{-2\pi nE/a}, \end{aligned} \tag{64.169}$$

where $(-2\pi i)(-iE) = -2\pi E$ cancelled the leading sign. Summing the geometric series $\sum_{n\geq 1} e^{-2\pi nE/a} = (e^{2\pi E/a} - 1)^{-1}$,

$$\boxed{F(E) = \frac{E}{2\pi}\,\frac{1}{e^{2\pi E/a} - 1}} \tag{64.170}$$

The response is the four-dimensional density of states $E/2\pi$ times the Planck factor at $T_{\mathrm{U}} = a/2\pi$. It manifestly satisfies the detailed-balance ratio (64.165), and it is strictly positive for $E > 0$: the accelerated detector is excited with a thermal spectrum. As $a \to 0$ the rate vanishes faster than any power, recovering the inertial silence (64.160) in the limit of no acceleration.

64.4.5 Synthesis: the vacuum as a thermofield double

The Bogoliubov and KMS derivations are not merely two roads to the same number; they are two views of the same state. We close by exhibiting the Minkowski vacuum explicitly in the Rindler basis, which makes the thermal density matrix (64.152) an algebraic identity and ties the section to the entanglement structure that returns for black holes.

Worked Example 64.24 (The Minkowski vacuum is a two-mode squeezed state)**.** Invert the Bogoliubov relation (64.143). With $\alpha_\Omega, \beta_\Omega$ real and $\alpha^2 - \beta^2 = 1$, the same continuation that produced $\widehat{\widetilde{W}}_\Omega$ gives $a^L_\Omega = \alpha_\Omega d^{(2)}_\Omega + \beta_\Omega d^{(1)\dagger}_\Omega$, and solving the pair for the Minkowski annihilation operators yields

$$d^{(1)}_\Omega = \alpha_\Omega\, a^R_\Omega - \beta_\Omega\, a^{L\dagger}_\Omega, \qquad d^{(2)}_\Omega = \alpha_\Omega\, a^L_\Omega - \beta_\Omega\, a^{R\dagger}_\Omega, \tag{64.171}$$

as one verifies by substitution. The Minkowski vacuum is the state annihilated by every $d^{(1)}_\Omega, d^{(2)}_\Omega$. Working mode by mode and dropping the Ω label, the condition

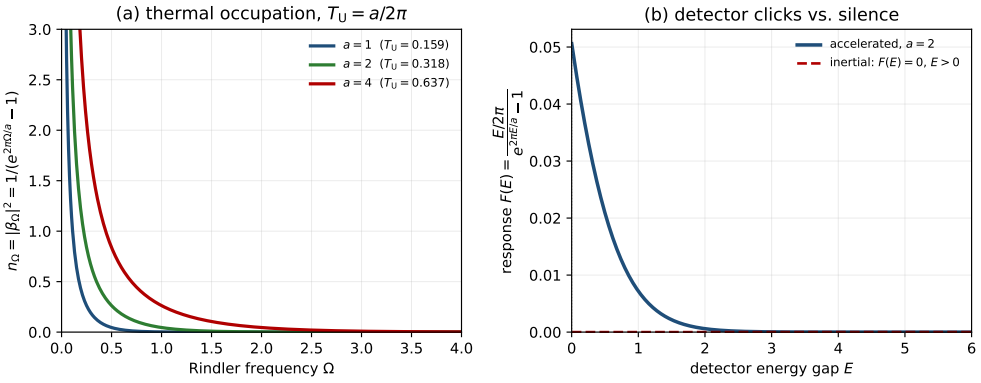


Figure 64.5: The Unruh spectrum, two ways. **(a)** The mean Rindler occupation $n_\Omega = |\beta_\Omega|^2 = (e^{2\pi\Omega/a} - 1)^{-1}$ from the Bogoliubov coefficient (64.146), a Bose–Einstein distribution at $T_U = a/2\pi$; larger proper acceleration is a hotter bath. **(b)** The Unruh–DeWitt detector response rate $F(E) = \frac{E}{2\pi}(e^{2\pi E/a} - 1)^{-1}$ of Theorem 64.23 for an accelerated worldline (solid), against the inertial response, which is identically zero for $E > 0$ (dashed): the accelerated detector clicks thermally, its inertial twin stays silent.

$(\alpha\, a^R - \beta\, a^{L\dagger})\,|0_M\rangle = 0$ is solved by the ansatz $|0_M\rangle = \mathcal{N} \exp(\lambda\, a^{R\dagger} a^{L\dagger})\,|0_R, 0_L\rangle$: since $a^R \exp(\lambda a^{R\dagger} a^{L\dagger})\,|0\rangle = \lambda\, a^{L\dagger} \exp(\lambda a^{R\dagger} a^{L\dagger})\,|0\rangle$ (the annihilator pulls down one factor of $a^{R\dagger}$), the condition reduces to $\alpha\lambda - \beta = 0$, fixing

$$\lambda = \frac{\beta_\Omega}{\alpha_\Omega} = e^{-\pi\Omega/a}. \tag{64.172}$$

Normalizing $\langle 0_M|\, 0_M\rangle = 1$ gives $\mathcal{N} = \sqrt{1 - \lambda^2} = \sqrt{1 - e^{-2\pi\Omega/a}}$ per mode, so

$$|0_M\rangle = \prod_\Omega \sqrt{1 - e^{-2\pi\Omega/a}}\, \exp\!\Big(e^{-\pi\Omega/a}\, a_\Omega^{R\dagger} a_\Omega^{L\dagger}\Big)\,|0_R\rangle\otimes|0_L\rangle = \prod_\Omega \sqrt{1 - e^{-2\pi\Omega/a}} \sum_{n=0}^{\infty} e^{-\pi\Omega n/a}\,|n_\Omega \tag{64.173}$$

This is a *two-mode squeezed state*: each Rindler quantum in the right wedge is perfectly correlated with a partner in the inaccessible left wedge. It is precisely the thermofield double of finite-temperature field theory, with the left wedge playing the role of the fictitious "tilde" copy. The accelerated observer, confined to R, has no access to the L factors and must trace them out. Tracing (64.173) over L,

$$\rho_R = \mathrm{Tr}_L\,|0_M\rangle\langle 0_M| = \prod_\Omega (1 - e^{-2\pi\Omega/a}) \sum_{n=0}^{\infty} e^{-2\pi\Omega n/a}\;|n_\Omega\rangle_R\langle n_\Omega|_R = \frac{e^{-\beta K}}{\mathrm{Tr}\; e^{-\beta K}}, \quad \beta = \frac{2\pi}{a}, \tag{64.174}$$

exactly the Gibbs state (64.152) for the boost Hamiltonian $K = \int \mathrm{d}\Omega\, \Omega\, a_\Omega^{R\dagger} a_\Omega^R$. The mean occupation in this geometric distribution, $p_n = (1 - x)x^n$ with $x =$

$e^{-2\pi\Omega/a}$, is $\langle n\rangle = x/(1-x) = (e^{2\pi\Omega/a}-1)^{-1}$, reproducing the Planck number (64.146) a third time. The Bogoliubov coefficient, the Euclidean period, and the modular Hamiltonian are thus one statement: restricting the vacuum to a wedge across a horizon produces a thermal state, and the temperature is $a/2\pi$.

The lesson is portable and exact: a horizon plus a Wick rotation equals a temperature, and the geometric origin of the 2π guarantees the answer is universal. In Chapter 65 we replace the flat Rindler horizon by the event horizon of a black hole; the boost becomes the Killing time of the static metric, the proper acceleration becomes the surface gravity κ, and the identical regularity-of-the-Euclidean-cone argument delivers the Hawking temperature $T_{\rm H} = \kappa/2\pi$ verbatim. Before we can trust that machinery in a curved geometry, however, we must confront the object whose expectation value backreacts on the metric — the stress-energy tensor, which is formally divergent in every one of these states and whose careful definition occupies Section 64.5.

64.5 The renormalized stress-energy tensor

The Unruh effect of the previous section (Section 64.4) delivered a disturbing lesson: the very notion of *which state is empty* is observer- and geometry-dependent, and a state that one observer calls the vacuum another finds teeming with quanta. Cosmological particle production (Section 64.3) made the same point dynamically. We are therefore in possession of states — thermal baths, squeezed in-vacua, adiabatic vacua of an expanding universe — whose particle content we can count, but whose *energy and momentum* we have not yet quantified. That content is not a bookkeeping curiosity. In general relativity the metric is not a fixed stage: it responds to whatever carries energy and momentum, through Einstein's equation

$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{64.175}$$

If the matter is quantum and the geometry classical — the semiclassical approximation that organizes this entire chapter and the next — then the source on the right is not a classical tensor but the expectation value $\langle T_{\mu\nu}\rangle$ in whatever quantum state the field occupies:

$$G_{\mu\nu} = 8\pi G\, \langle T_{\mu\nu}\rangle\,. \tag{64.176}$$

The object on the right is what tells spacetime how to curve in response to vacuum polarization, to particle creation, to the negative energy near a horizon that will drive black-hole evaporation in Chapter 65. It is, in a precise sense, *the* output of quantum field theory in curved spacetime.

And it is, taken naively, infinite. The stress tensor is quadratic in the field and its derivatives; its expectation value involves the coincidence limit of a two-point function, and that limit diverges — already in Minkowski space, where

we disposed of it silently by normal ordering. On a curved background normal ordering is not available (there is no preferred vacuum to order against), the divergence is entangled with the local geometry, and we must confront it head on. This section builds the machinery that turns the formally infinite $\langle T_{\mu\nu}\rangle$ into a finite, covariantly conserved, and — as we will prove — essentially unique tensor fit to sit on the right of Eq. (64.176). The route is *covariant point splitting*: separate the two field operators to nearby points, subtract a purely geometric local singularity (the Hadamard parametrix) that knows nothing about the state, and bring the points back together. We then state Wald's axioms, prove the result they single out is unique up to a handful of local geometric tensors, and recognize those tensors as exactly the gravitational counterterms — the cosmological constant, Newton's constant, and the curvature-squared couplings — whose renormalization the logic of Chapter 22 already taught us to expect. We close with a worked example in which every subtraction is carried out explicitly.

64.5.1 The classical stress tensor in curved space

Before quantizing we need the classical object whose expectation value we are after. We continue with the real scalar field of Section 64.1, whose curved-space action carries the non-minimal curvature coupling ξ introduced there,

$$S[\phi,g]=\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[\tfrac{1}{2}\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi-\tfrac{1}{2}(m^2+\xi R)\phi^2\Big], \tag{64.177}$$

with R the Ricci scalar in the curvature conventions of Chapter A and ξ a dimensionless coupling. The field equation that follows by varying ϕ is the curved Klein–Gordon equation

$$(\Box+m^2+\xi R)\phi=0, \qquad \Box\equiv g^{\mu\nu}\nabla_\mu\nabla_\nu. \tag{64.178}$$

The distinguished value of ξ is the *conformal coupling* $\xi_c=(d-2)/[4(d-1)]$, equal to $\frac{1}{6}$ in $d=4$ (Section 64.1), for which the massless theory is invariant under local Weyl rescalings $g_{\mu\nu}\to\Omega^2 g_{\mu\nu}$, $\phi\to\Omega^{-1}\phi$. The breaking of that classical symmetry by quantization is the subject of the next section; here it organizes the algebra.

The stress tensor is defined, following the Hilbert prescription imported from Chapters 6 and 7, as the response of the matter action to a variation of the metric:

$$T_{\mu\nu}\equiv\frac{2}{\sqrt{-g}}\,\frac{\delta S}{\delta g^{\mu\nu}}. \tag{64.179}$$

This is the curved-space generalization of the symmetric (Belinfante) tensor of Chapter 7: on a flat background it reduces to it, and unlike the canonical Noether tensor it is automatically symmetric and gauge/coordinate covariant.

To evaluate Eq. (64.179) we need three variational facts, each a standard piece of differential geometry. First, the determinant:

$$\delta\sqrt{-g} = -\tfrac{1}{2}\sqrt{-g}\, g_{\mu\nu}\, \delta g^{\mu\nu}. \tag{64.180}$$

Second, the inverse metric varies the kinetic term directly, $\delta(g^{\alpha\beta}\partial_\alpha\phi\,\partial_\beta\phi) = \partial_\mu\phi\,\partial_\nu\phi\,\delta g^{\mu\nu}$. Third, the Ricci scalar: $R = g^{\alpha\beta}R_{\alpha\beta}$, whose variation is the Palatini identity

$$\delta R = R_{\mu\nu}\,\delta g^{\mu\nu} + g_{\mu\nu}\Box\,\delta g^{\mu\nu} - \nabla_\mu\nabla_\nu\,\delta g^{\mu\nu}. \tag{64.181}$$

The last two terms in Eq. (64.181) are total covariant divergences when contracted with a constant, but here they multiply ϕ^2 inside the action and so contribute after integration by parts. Assembling the variation of Eq. (64.177) term by term — the determinant factor multiplying the whole Lagrangian, the kinetic variation, and the curvature term $-\frac{1}{2}\xi\phi^2\,\delta R$ integrated by parts twice to move the derivatives off $\delta g^{\mu\nu}$ and onto ϕ^2 — gives, after collecting the Ricci and metric pieces into the Einstein tensor $G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R$,

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \tfrac{1}{2}\,g_{\mu\nu}(\partial\phi)^2 + \tfrac{1}{2}\,m^2 g_{\mu\nu}\phi^2 + \xi\,(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu - G_{\mu\nu})\phi^2, \tag{64.182}$$

where $(\partial\phi)^2 \equiv g^{\alpha\beta}\partial_\alpha\phi\,\partial_\beta\phi$. The first three terms are the minimally coupled (improvement-free) stress tensor; the ξ-bracket is the curvature improvement. Its flat-space limit $\xi(\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu)\phi^2$ is exactly the Callan–Coleman–Jackiw improvement term that renders the massless theory's stress tensor traceless, and the curved completion adds the $-\xi G_{\mu\nu}\phi^2$ piece demanded by metric covariance.

The trace. The single most important property of Eq. (64.182) for what follows is its trace. Contracting with $g^{\mu\nu}$ in d dimensions, using $g^{\mu\nu}g_{\mu\nu} = d$, $g^{\mu\nu}\nabla_\mu\nabla_\nu = \Box$, and the geometric identity $g^{\mu\nu}G_{\mu\nu} = (1-\frac{d}{2})R = -\frac{d-2}{2}R$,

$$T^\mu{}_\mu = \Big(1-\tfrac{d}{2}\Big)(\partial\phi)^2 + \tfrac{d}{2}\,m^2\phi^2 + \xi\Big[(d-1)\Box + \tfrac{d-2}{2}R\Big]\phi^2. \tag{64.183}$$

Now expand $\Box\phi^2 = 2(\partial\phi)^2 + 2\phi\,\Box\phi$ and use the field equation Eq. (64.178), $\Box\phi = -(m^2+\xi R)\phi$, so that $\phi\,\Box\phi = -(m^2+\xi R)\phi^2$:

$$T^\mu{}_\mu = \Big[1-\tfrac{d}{2}+2\xi(d-1)\Big](\partial\phi)^2 + \Big[\tfrac{d}{2}-2\xi(d-1)\Big]m^2\phi^2 + \Big[\tfrac{d-2}{2}\,\xi - 2\xi^2(d-1)\Big]R\phi^2. \tag{64.184}$$

At the conformal value $\xi = \xi_c = (d-2)/[4(d-1)]$ one has $2\xi_c(d-1) = \frac{d-2}{2}$, and each bracket collapses: the $(\partial\phi)^2$ coefficient is $1-\frac{d}{2}+\frac{d-2}{2} = 0$; the mass coefficient is $\frac{d}{2} - \frac{d-2}{2} = 1$; and the R coefficient is $\frac{d-2}{2}\xi_c - 2\xi_c^2(d-1) = \frac{(d-2)^2}{8(d-1)} - \frac{(d-2)^2}{8(d-1)} = 0$. Thus for the conformally coupled field the classical trace is

$$T^\mu{}_\mu\big|_{\xi=\xi_c} = m^2\phi^2, \tag{64.185}$$

which vanishes identically in the massless case. This is the classical statement of Weyl invariance — the energy-momentum tensor of a massless conformal scalar is traceless on shell, in any spacetime. We will see in Section 64.6 that quantization breaks Eq. (64.185): the renormalized $\langle T^{\mu}{}_{\mu}\rangle$ acquires a nonzero, purely geometric piece even for $m = 0$. To get there we must first define the renormalized tensor at all.

64.5.2 Point splitting and the universal short-distance singularity

Promote ϕ to the field operator. The naive prescription $\langle T_{\mu\nu}\rangle = \langle\Psi|\, T_{\mu\nu}(x)\,|\Psi\rangle$ asks us to evaluate products of two field operators at the *same* spacetime point. Such products are not operators: $\langle\phi(x)^2\rangle$ and $\langle\partial_\mu\phi\,\partial_\nu\phi\rangle$ are the coincidence limits of two-point functions that diverge as the two points merge. The cure, due to DeWitt and Christensen and sharpened by Wald, is to keep the points apart, evaluate the bilinear as a bi-tensor, subtract a local geometric counterterm, and only then take the limit.

Introduce the symmetric (Hadamard) two-point function in the state $|\Psi\rangle$,

$$G(x,x') \equiv \tfrac{1}{2}\,\langle\Psi|\,\{\phi(x),\phi(x')\}\,|\Psi\rangle = \mathrm{Re}\,\langle\Psi|\,\phi(x)\phi(x')\,|\Psi\rangle\,, \tag{64.186}$$

the real, symmetric part of the Wightman function built in Chapter 10. (The antisymmetric part is the state-independent commutator, a c-number that carries no energy; it is the symmetric part that contains the stress.) Each bilinear in Eq. (64.182) is now obtained by acting with a derivative operator at x, a second at x', and then setting $x' = x$. Because the operators at separated points commute up to the c-number, the ordering ambiguity is harmless; we symmetrize to keep the result manifestly real. Indices at the two points are tied together by the bivector of parallel transport $g_{\nu}{}^{\nu'}(x,x')$, the geodesic parallel propagator, which reduces to $\delta^{\nu'}_{\nu}$ at coincidence and which we suppress whenever the local frame makes it trivial.

Collecting the derivative structure of Eq. (64.182), the point-split stress tensor is

$$\langle T_{\mu\nu}\rangle = \lim_{x'\to x}\,\mathcal{D}_{\mu\nu}(x,x')\,G(x,x'), \tag{64.187}$$

with the bi-differential operator

$$\begin{aligned}\mathcal{D}_{\mu\nu}(x,x') = {}& (1-2\xi)\,g_{\nu}{}^{\nu'}\nabla_\mu\nabla_{\nu'} + \Big(2\xi-\tfrac{1}{2}\Big)g_{\mu\nu}\,g^{\rho\sigma'}\nabla_\rho\nabla_{\sigma'} - 2\xi\,\nabla_\mu\nabla_\nu \\ & + 2\xi\,g_{\mu\nu}\,\Box - \xi\Big(R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R\Big) + \tfrac{1}{2}\,m^2 g_{\mu\nu},\end{aligned} \tag{64.188}$$

where primed derivatives act at x' and unprimed at x. The first line carries the bilinear $\partial\phi\,\partial\phi$ point-split between the two arguments (hence the mixed $\nabla_\mu\nabla_{\nu'}$); the remaining terms come from the improvement piece $\propto \nabla\nabla\phi^2$, in which both

derivatives may act at the same point. One verifies that on the smooth (state-dependent) part of G the coincidence limit of Eq. (64.188) reproduces Eq. (64.182) with $\phi^2 \to G(x,x)$ and $\partial_\mu\phi\,\partial_\nu\phi \to \nabla_\mu\nabla_{\nu'}G|_{x'=x}$, as it must.

The trouble is that $G(x,x')$ is *not* smooth as $x' \to x$. Already in the Minkowski vacuum, the massless two-point function is $G_0(x,x') = \frac{1}{8\pi^2}\,\frac{1}{\sigma}$ with σ half the squared geodesic interval, and applying two derivatives produces a coincidence divergence $\sim \sigma^{-2}$. On a curved background the leading divergence has the same σ^{-1} form, with subleading $\ln\sigma$ pieces dressed by the local curvature. The decisive fact — the one that makes renormalization possible and local — is that *this short-distance singularity is universal*: it is the same for every physically admissible state, fixed entirely by the local geometry and the parameters (m,ξ) in the wave operator, and independent of which state $|\Psi\rangle$ we chose. We make this precise next.

64.5.3 The Hadamard short-distance expansion

The geometric scaffolding is built from two bi-scalars. Let $\sigma(x,x')$ be *Synge's world function*, equal to one half the squared geodesic distance between x and x' — positive for spacelike separation in the mostly-minus signature, negative for timelike, and satisfying the eikonal identity $\frac{1}{2}g^{\mu\nu}(\nabla_\mu\sigma)(\nabla_\nu\sigma) = \sigma$. Its coincidence limit and first derivatives vanish, $[\sigma]=0$, $[\nabla_\mu\sigma]=0$, while $[\nabla_\mu\nabla_\nu\sigma] = g_{\mu\nu}$. Let $\Delta(x,x')$ be the *Van Vleck–Morette determinant*,

$$\Delta(x,x') = -\frac{\det\big[-\nabla_\mu\nabla_{\nu'}\sigma(x,x')\big]}{\sqrt{-g(x)}\,\sqrt{-g(x')}}, \qquad [\Delta]=1, \tag{64.189}$$

the geometric measure of geodesic focusing, which reduces to unity at coincidence and whose short-distance expansion encodes the Ricci curvature,

$$\Delta^{1/2}(x,x') = 1 + \tfrac{1}{12}R_{\alpha\beta}\,\sigma^{;\alpha}\sigma^{;\beta} + O(\sigma^{3/2}), \qquad \sigma^{;\alpha} \equiv \nabla^\alpha\sigma. \tag{64.190}$$

With these in hand, the central structural theorem of the subject is that every state of physical interest has a two-point function of *Hadamard form*: near coincidence,

$$G(x,x') = \frac{1}{8\pi^2}\left[\frac{\Delta^{1/2}(x,x')}{\sigma(x,x')} + V(x,x')\,\ln\frac{\sigma(x,x')}{\ell^2} + W(x,x')\right], \tag{64.191}$$

where ℓ is an arbitrary length scale (a renormalization scale, exactly as in Chapter 22), V and W are smooth bi-scalars, and the σ in the denominator and logarithm carries the Feynman $i\epsilon$ prescription $\sigma \to \sigma + i\epsilon\,(t-t') + \epsilon^2$ inherited from Chapter 10 and Chapter D.

Definition 64.25 (Hadamard state)**.** A state $|\Psi\rangle$ is a *Hadamard state* if its symmetric two-point function has the form Eq. (64.191) with V determined by

the local geometry alone — specifically, V is the unique solution of the recursion below that is regular at coincidence — and W smooth. Equivalently, the singular structure of G is exactly H (defined in Eq. (64.193) below), with the difference $G - H$ a C^∞ function of (x, x').

The function V is purely geometric. Writing $V = \sum_{n\geq 0} V_n(x,x')\,\sigma^n$ and inserting Eq. (64.191) into the wave equation $(\Box_x + m^2 + \xi R)G(x,x') = 0$ (valid away from coincidence, since G solves the field equation in each argument), one finds a hierarchy of transport equations along the geodesic that determine the V_n recursively from the geometry, with no freedom left over. Their leading coincidence values are

$$[V_0] = \tfrac{1}{2}\Big(m^2 + (\xi - \tfrac{1}{6})R\Big), \qquad [V_1] = \tfrac{1}{8}\Big(m^2 + (\xi - \tfrac{1}{6})R\Big)^2 + \cdots, \tag{64.192}$$

where the ellipsis in $[V_1]$ stands for terms quadratic in the curvature (R^2, $R_{\mu\nu}R^{\mu\nu}$, $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$, $\Box R$) whose coefficients are fixed by the recursion. These are the Hadamard (Wald) coefficients of the parametrix, fixed by the same geometry that produces the Seeley–DeWitt coefficients of the heat-kernel expansion catalogued in Chapter E; the two normalizations differ by fixed numerical factors. For the operator $\Box + m^2 + \xi R$ the coincidence limits are $[V_0] = -\frac{1}{2}a_1$ and $[V_1] = \frac{1}{4}a_2$, so the anomaly prefactor below, $(1/4\pi^2)[V_1] = a_2/(4\pi)^2$, agrees with the heat-kernel result of Section 64.6; V_0, V_1 are *not* the DeWitt coefficients themselves. The combination $\xi - \frac{1}{6}$ appearing in Eq. (64.192) is the same one that controlled the trace in Eq. (64.184): it measures the departure from conformal coupling and, as we will see, governs the magnitude of the trace anomaly.

Physical Insight

The content of the Hadamard condition is that *all the information about the state lives in the smooth part W, and all the divergence lives in the geometric piece $\Delta^{1/2}/\sigma + V\ln\sigma$.* The vacuum is not distinguished by being "empty" — there is no such thing here — but by being smooth relative to the universal singularity. This is the curved-space replacement for normal ordering: we do not subtract a particular vacuum's energy, we subtract the state-independent short-distance structure that any reasonable state shares.

64.5.4 Covariant subtraction: the renormalized tensor

The singular part of Eq. (64.191) is purely local and geometric. Detach it and give it a name: the *Hadamard parametrix*

$$H(x,x') = \frac{1}{8\pi^2}\left[\frac{\Delta^{1/2}(x,x')}{\sigma(x,x')} + V(x,x')\,\ln\frac{\sigma(x,x')}{\ell^2}\right]. \tag{64.193}$$

H is constructed entirely from the metric and the wave-operator parameters; it is the same in every Hadamard state. By construction (Theorem 64.25) the difference

$$\mathcal{W}(x,x') \equiv G(x,x') - H(x,x') = \frac{1}{8\pi^2}\, W(x,x') \tag{64.194}$$

is a smooth (C^∞) bi-scalar, so its derivatives have finite coincidence limits.

Definition 64.26 (Renormalized stress tensor)**.** The renormalized expectation value of the stress tensor in the Hadamard state $|\Psi\rangle$ is

$$\langle T_{\mu\nu}\rangle_{\text{ren}} = \lim_{x'\to x} \mathcal{D}_{\mu\nu}(x,x')\,[G(x,x') - H(x,x')] + \Theta_{\mu\nu}(x), \tag{64.195}$$

where $\mathcal{D}_{\mu\nu}$ is the operator Eq. (64.188) and $\Theta_{\mu\nu}$ is a purely local, purely geometric correction term (built from the parametrix coefficients) whose role — restoring covariant conservation — is fixed in Section 64.5.5.

Because $\mathcal{W} = G - H$ is smooth, the limit in Eq. (64.195) exists and is finite: the universal divergence has been removed by subtracting H, and what remains is the state-dependent content carried by W together with a finite geometric residue from the parametrix. This is the entire idea. The subtraction is *local and covariant* — it does not reference any global vacuum, only the local geometry — which is exactly the property a counterterm must have to be admissible in a gravitational theory.

Two features deserve emphasis before we make conservation precise. First, the answer is finite but not yet unique: the scale ℓ inside the logarithm of H is arbitrary, and changing $\ell \to \ell'$ shifts H by $\frac{1}{8\pi^2} V \ln(\ell^2/\ell'^2)$, hence shifts $\langle T_{\mu\nu}\rangle_{\text{ren}}$ by a local tensor built from the coincidence limits of V and its derivatives — that is, by curvature-squared terms. We return to this in Section 64.5.6. Second, the derivative operator $\mathcal{D}_{\mu\nu}$ acting on the singular part H does *not* commute with the coincidence limit in a way that respects conservation; the local term $\Theta_{\mu\nu}$ is precisely what repairs this, and tracking it is what exposes the anomaly.

64.5.5 Conservation, and where the anomaly hides

The source of Einstein's equation must be covariantly conserved: the Bianchi identity gives $\nabla^\mu G_{\mu\nu} = 0$ identically, so consistency of Eq. (64.176) demands

$$\nabla^\mu \langle T_{\mu\nu}\rangle_{\text{ren}} = 0. \tag{64.196}$$

This is not automatic. The classical tensor Eq. (64.182) is conserved on shell — a one-line consequence of the field equation and the symmetry of the metric connection, exactly as in Chapter 7. But the point-split, parametrix-subtracted object of Eq. (64.195) need not be, because the parametrix H does not exactly solve the field equation: it solves it only up to a smooth remainder. The precise statement is the following.

Theorem 64.27 (Conservation of the point-split tensor)**.** *Let* $\widetilde{T}_{\mu\nu} = \lim_{x'\to x} \mathcal{D}_{\mu\nu}(G-H)$ *be the bare point-split tensor (without* $\Theta_{\mu\nu}$*). Its covariant divergence is a local, state-independent vector,*

$$\nabla^{\mu}\widetilde{T}_{\mu\nu} = \nabla_{\nu}\, Q(x), \qquad Q(x) = \frac{1}{8\pi^2}\left(\, c_1\,[V_1] + c_2\, m^2[V_0] + \cdots\right), \tag{64.197}$$

where $[V_0], [V_1]$ *are the parametrix coefficients Eq. (64.192),* c_1, c_2 *are pure numbers, and the omitted terms are further local curvature scalars. Consequently the tensor*

$$\langle T_{\mu\nu}\rangle_{\rm ren} = \widetilde{T}_{\mu\nu} - g_{\mu\nu}\, Q(x), \qquad \textit{i.e.}\quad \Theta_{\mu\nu} = -g_{\mu\nu}Q, \tag{64.198}$$

is covariantly conserved, Eq. (64.196).

Proof. The divergence $\nabla^{\mu}\widetilde{T}_{\mu\nu}$ can be computed before taking the coincidence limit. On the smooth remainder $\mathcal{W} = G - H$ one commutes ∇^{μ} through the bi-differential operator $\mathcal{D}_{\mu\nu}$ and uses that G satisfies the field equation $(\Box + m^2 + \xi R)G = 0$ in the argument x exactly, while H satisfies it only up to a smooth source: by the defining recursion for the V_n, one has $(\Box_x + m^2 + \xi R)H(x,x') = \frac{1}{8\pi^2}\,[\,2\,V_1(x,x')\,] \to \frac{1}{4\pi^2}[V_1]$ as $x' \to x$ — a finite, local quantity, the last member of the defining V_n recursion failing to vanish. The state-dependent part of $\mathcal{W}$ therefore drops out of the divergence entirely: G solves the equation, so its contribution to $\nabla^{\mu}\widetilde{T}_{\mu\nu}$ vanishes by the same on-shell manipulation that conserves the classical tensor. What survives is purely the failure of H to solve the field equation, which by construction is a local geometric scalar. Carrying the coincidence limit of the commutator terms (this is a finite computation in the Synge/Van Vleck calculus, using $[\sigma^{;\mu}] = 0$ and $[\nabla_\mu\nabla_\nu\sigma] = g_{\mu\nu}$) collects this remainder into the gradient of a single local scalar Q, of the form Eq. (64.197). Since Q is a scalar, $g_{\mu\nu}Q$ has divergence $\nabla^{\mu}(g_{\mu\nu}Q) = \nabla_{\nu}Q$, which exactly cancels Eq. (64.197). Hence $\nabla^{\mu}(\widetilde{T}_{\mu\nu} - g_{\mu\nu}Q) = 0$. □

The price of this repair is paid in the trace. Subtracting $g_{\mu\nu}Q$ shifts the trace by $-d\,Q = -4Q$ in four dimensions, and Q is built from the parametrix coefficient $[V_1]$ — a local, state-independent curvature scalar that cannot be removed by any choice of state. We can enforce conservation, *or* we can enforce tracelessness for the conformal field, but the point-split prescription will not let us have both: the conserved tensor has a nonzero, geometric trace

$$\langle T^{\mu}{}_{\mu}\rangle_{\rm ren}\Big|_{m=0,\ \xi=\xi_c} = \frac{1}{4\pi^2}\,[V_1]\big|_{m=0,\ \xi=\xi_c} \;\neq\; 0, \tag{64.199}$$

even though the classical trace Eq. (64.185) vanished. This is the *trace (conformal) anomaly.* Its value is fixed entirely by the geometric coefficient $[V_1]$, hence by local curvature invariants, and computing it honestly — two independent ways — is the business of Section 64.6. Here we have located it: it is the unavoidable residue of insisting that the object sourcing gravity be conserved.

Caution

It is tempting to think one could define a different renormalization that keeps the trace zero. One can — but only by giving up conservation, which makes the tensor useless as a source for Einstein's equation, or by giving up locality/covariance of the subtraction, which reintroduces a divergence. The anomaly is the statement that no admissible scheme achieves classical tracelessness, conservation, and covariance simultaneously for the massless conformal field. This is the same structure as the chiral anomaly of Chapter 36: a classical symmetry that no regulator respects.

64.5.6 Wald's axioms and the uniqueness theorem

Point splitting produces *a* finite conserved tensor, but the construction made choices: the scale ℓ, the precise form of $\Theta_{\mu\nu}$. We now ask what an admissible $\langle T_{\mu\nu}\rangle_{\rm ren}$ must satisfy on physical grounds, and prove that those requirements pin it down up to a controlled ambiguity. Wald formulated four axioms.

1. **Conservation.** $\nabla^\mu \langle T_{\mu\nu}\rangle_{\rm ren} = 0$, so that Eq. (64.176) is consistent with the Bianchi identity.

2. **Causality.** In a globally hyperbolic spacetime, $\langle T_{\mu\nu}\rangle_{\rm ren}(x)$ depends on the state and the geometry only within the causal past and future of x; a change of state outside the light cone of x cannot alter the tensor at x.

3. **Correct flat-space limit.** In Minkowski space, in the standard vacuum, the prescription reduces to normal ordering: $\langle T_{\mu\nu}\rangle_{\rm ren} = 0$ in the Poincaré-invariant vacuum.

4. **Locality / state-independence of the difference.** For two Hadamard states $|\Psi_1\rangle, |\Psi_2\rangle$ on the same spacetime, the difference $\langle T_{\mu\nu}\rangle_{\Psi_1} - \langle T_{\mu\nu}\rangle_{\Psi_2}$ equals the coincidence limit of $\mathcal{D}_{\mu\nu}(G_{\Psi_1} - G_{\Psi_2})$ — it is computed from the smooth *difference* of two-point functions, with no subtraction needed, since the universal singularity cancels.

Axiom (4) is the operational heart of the matter: it fixes the *state-dependence* completely and without ambiguity, because $G_{\Psi_1} - G_{\Psi_2}$ is smooth and the operator $\mathcal{D}_{\mu\nu}$ acts on it cleanly. All the freedom therefore resides in a single c-number tensor — the value assigned to one reference state — which by axioms (1)–(3) must be a *local, conserved, geometric* tensor with the correct dimension and the correct flat-space limit.

Theorem 64.28 (Wald uniqueness)**.** *Any two prescriptions* $\langle T_{\mu\nu}\rangle^{(1)}_{\rm ren}$ *and* $\langle T_{\mu\nu}\rangle^{(2)}_{\rm ren}$ *satisfying axioms* (1)–(4) *differ by a local conserved tensor* $\Delta T_{\mu\nu}$ *constructed*

polynomially from the metric and its curvature, of mass dimension at most four. In four dimensions the most general such tensor is

$$\Delta T_{\mu\nu} = \alpha_0\, g_{\mu\nu} + \alpha_1\, G_{\mu\nu} + \alpha_2\, {}^{(1)}H_{\mu\nu} + \alpha_3\, {}^{(2)}H_{\mu\nu}, \tag{64.200}$$

where $\alpha_0, \dots, \alpha_3$ are real constants, $g_{\mu\nu}$ and $G_{\mu\nu}$ are the metric and Einstein tensors, and

$${}^{(1)}H_{\mu\nu} = \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\int \mathrm{d}^4x\,\sqrt{-g}\,R^2 = 2g_{\mu\nu}\Box R - 2\nabla_\mu\nabla_\nu R - \tfrac{1}{2}g_{\mu\nu}R^2 + 2RR_{\mu\nu}, \tag{64.201}$$

$${}^{(2)}H_{\mu\nu} = \frac{1}{\sqrt{-g}}\frac{\delta}{\delta g^{\mu\nu}}\int \mathrm{d}^4x\,\sqrt{-g}\,R_{\alpha\beta}R^{\alpha\beta} \tag{64.202}$$

are the conserved tensors obtained by varying the two independent curvature-squared invariants. Each of the four tensors in Eq. (64.200) is identically covariantly conserved.

Proof. By axiom (4) the difference of two admissible prescriptions cannot depend on the state: a state-dependent difference would have to be the coincidence limit of $\mathcal{D}_{\mu\nu}$ acting on $G_{\Psi_1} - G_{\Psi_2}$, but that limit is the *same* for both prescriptions (the operator $\mathcal{D}_{\mu\nu}$ is fixed by the classical action, and the smooth difference needs no subtraction). Hence $\Delta T_{\mu\nu} = \langle T_{\mu\nu}\rangle^{(1)} - \langle T_{\mu\nu}\rangle^{(2)}$ is independent of the state. By axiom (2), causality, it is local — it cannot depend on data outside the point x, so it is a function of the metric and finitely many derivatives at x. Lorentz covariance forces it to be a polynomial in the curvature and its covariant derivatives. By dimensional analysis, $T_{\mu\nu}$ has mass dimension four (two derivatives on dimension-one fields), and a geometric tensor of that dimension is a sum of terms with exactly four derivatives of the metric: these are $g_{\mu\nu}$ (dimension zero, multiplied by a dimensionful constant α_0 that is in practice $\propto \Lambda$ or m^4), $G_{\mu\nu}$ (dimension two, times $\alpha_1 \propto m^2$ or $1/G$), and the dimension-four tensors obtained from R^2 and $R_{\mu\nu}R^{\mu\nu}$ — namely ${}^{(1)}H_{\mu\nu}$ and ${}^{(2)}H_{\mu\nu}$. (The Riemann-squared invariant $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ is not independent: in four dimensions the Gauss–Bonnet combination $R^2_{\mu\nu\rho\sigma} - 4R^2_{\mu\nu} + R^2$ is a topological density whose metric variation vanishes identically, so its tensor is removed in favor of the other two.) By axiom (1), conservation, only conserved combinations are allowed; one checks directly — it follows from the diffeomorphism invariance of the actions $\int\sqrt{-g}\,R^2$ and $\int\sqrt{-g}\,R^2_{\mu\nu}$, whose variations Eq. (64.201)–Eq. (64.202) are conserved by the generalized Bianchi identity — that all four listed tensors are conserved. Finally axiom (3), the flat-space limit, requires $\Delta T_{\mu\nu} \to 0$ in Minkowski space, which is automatic for $G_{\mu\nu}$ and the H-tensors (they vanish when the curvature vanishes) and fixes the additive constant in $\alpha_0 g_{\mu\nu}$ to be the cosmological-constant piece, not an arbitrary flat-space energy. No further structures are permitted, so Eq. (64.200) is exhaustive. □

Physical Insight

Uniqueness "up to local conserved geometric tensors" is the gravitational analog of the renormalization freedom of flat-space QFT (Chapter 22). There, physical amplitudes were ambiguous up to the finite parts of counterterms whose form was dictated by the symmetries and the dimension of the divergences; the ambiguity was absorbed into renormalized couplings fixed by experiment. Here the four constants $\alpha_0, \dots, \alpha_3$ are exactly that ambiguity, and — as we show next — they are absorbed into the renormalized constants of the gravitational action.

64.5.7 The ambiguity is the renormalization of gravity

The four tensors in Eq. (64.200) are not a defect of the construction; they are a prediction. Each is the metric variation of a local geometric term that *must already be present* in the gravitational action once one couples quantum matter to gravity. The logic is precisely that of Chapter 22: a coupling generated by loops must be included in the bare action as a counterterm, with a renormalized value fixed by measurement. The total gravitational action consistent with general covariance and power counting is

$$S_{\rm grav} = \int \mathrm{d}^4x\,\sqrt{-g}\,\Big[-\frac{\Lambda}{8\pi G} + \frac{R}{16\pi G} + \alpha\,R^2 + \beta\,R_{\mu\nu}R^{\mu\nu}\Big], \tag{64.203}$$

and varying it via $T^{\rm grav}_{\mu\nu} = (2/\sqrt{-g})\,\delta S_{\rm grav}/\delta g^{\mu\nu}$ produces exactly the four tensors of Eq. (64.200): the $g_{\mu\nu}$ term from Λ, the $G_{\mu\nu}$ term from the Einstein–Hilbert R, and ${}^{(1)}H_{\mu\nu}, {}^{(2)}H_{\mu\nu}$ from the curvature-squared terms. The semiclassical equation Eq. (64.176), written with the full action $S_{\rm grav} + S_{\rm matter}$, becomes

$$\frac{1}{8\pi G}\Big(G_{\mu\nu} + \Lambda\,g_{\mu\nu}\Big) + \alpha\,{}^{(1)}H_{\mu\nu} + \beta\,{}^{(2)}H_{\mu\nu} = \langle T_{\mu\nu}\rangle_{\rm ren}\,. \tag{64.204}$$

Now the resolution is transparent. Shifting the renormalization scale $\ell \to \ell'$ in the parametrix Eq. (64.193), or equivalently choosing a different admissible $\langle T_{\mu\nu}\rangle_{\rm ren}$ within the family Eq. (64.200), changes $\langle T_{\mu\nu}\rangle_{\rm ren}$ by terms of precisely the form on the left of Eq. (64.204). Those changes can be absorbed into the bare constants:

$$\delta\Big(\frac{\Lambda}{8\pi G}\Big) = -\alpha_0, \quad \delta\Big(\frac{1}{16\pi G}\Big) = -\alpha_1, \quad \delta\alpha = -\alpha_2, \quad \delta\beta = -\alpha_3. \tag{64.205}$$

The ambiguity in renormalizing the stress tensor is nothing but the *renormalization of the cosmological constant, Newton's constant, and the two higher-derivative gravitational couplings* α, β. Their values are not predicted by the theory; they are inputs, measured (in principle) by the same process of fixing renormalization

conditions that fixed m, λ for ϕ^4 theory in Chapter 22. What *is* predicted is everything else: $\langle T_{\mu\nu} \rangle_{\text{ren}}$ is unique once these four numbers are fixed. In particular the ℓ-dependence of the logarithm in H feeds the running of α and β, exactly as a renormalization scale feeds the running of a coupling — the curvature-squared couplings of gravity run, and their beta functions are computable from $[V_1]$.

Notation Watch

The constant $\alpha_0\, g_{\mu\nu}$ contains the notorious vacuum-energy/cosmological-constant problem: the matter loops generate a contribution $\propto m^4$ to Λ with no symmetry forbidding it, and only a fine adjustment of the bare Λ leaves the observed near-zero value. This is a renormalization-condition statement, not a divergence the formalism fails to handle; the formalism handles it cleanly, but it offers no reason for the renormalized number to be small. We flag it here as the home of that problem and say no more — it is a frontier issue, not a defect of the construction.

64.5.8 Worked example: the Casimir energy of a conformal scalar

The cleanest arena in which to watch the subtraction work end to end is flat space with a nontrivial boundary condition. Take Minkowski space, so that the geometry is flat and the Hadamard parametrix H is simply the standard Minkowski vacuum two-point function G_0 — there is no curvature, V and $\Delta^{1/2}$ are trivial, and the universal singularity to subtract is precisely the free-space short-distance behavior. Confine a massless, conformally coupled ($\xi = \frac{1}{6}$) real scalar between two parallel plates at $z = 0$ and $z = a$, imposing Dirichlet conditions $\phi = 0$ on each. The plates change the allowed modes; subtracting the no-plate vacuum (the parametrix) leaves the finite, physical Casimir stress. This is the same construction Eq. (64.195) demands, in a setting where every step can be done by hand.

The zero-point energy and its subtraction. The Dirichlet conditions quantize the perpendicular wavenumber, $k_z = n\pi/a$ with $n = 1, 2, 3, \ldots$, while the transverse momentum $\mathbf{k}_\perp = (k_x, k_y)$ remains continuous. Each mode is a harmonic oscillator of frequency $\omega_{n,\mathbf{k}_\perp} = \sqrt{\mathbf{k}_\perp^2 + (n\pi/a)^2}$ and contributes $\frac{1}{2}\omega$ to the vacuum energy. The zero-point energy per unit plate area is therefore

$$\frac{E(a)}{A} = \frac{1}{2}\sum_{n=1}^{\infty} \int \frac{\mathrm{d}^2 k_\perp}{(2\pi)^2} \sqrt{\mathbf{k}_\perp^2 + \left(\tfrac{n\pi}{a}\right)^2} \;\equiv\; \sum_{n=1}^{\infty} F(n), \tag{64.206}$$

which is badly divergent. The physical Casimir energy is the difference between this and the energy the same slab of space would have with no plates — the

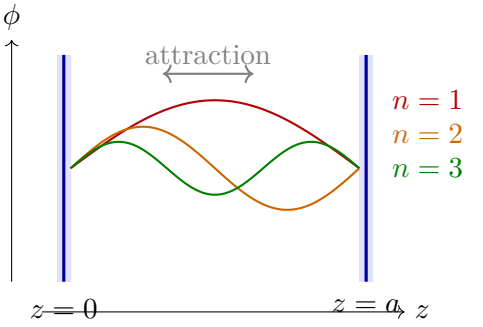


Figure 64.6: A massless scalar between Dirichlet plates supports standing waves with $k_z = n\pi/a$. The discrete spectrum's zero-point energy, with the free-space (continuum) value subtracted, is negative and falls as a^{-3}, producing an attractive force. The continuum subtraction is the flat-space instance of removing the Hadamard parametrix.

free-space vacuum, which is the parametrix to subtract. In the no-plate vacuum the k_z spectrum is continuous, and the sum becomes an integral with density of states a/π in k_z: the free-space energy per area of the region of width a is $\int_0^\infty F(n)\,\mathrm{d}n$ (substituting $k_z = n\pi/a$ shows $\frac{a}{\pi}\int \mathrm{d}k_z = \int \mathrm{d}n$). The renormalized Casimir energy is the difference,

$$\frac{E_{\mathrm{C}}(a)}{A} = \sum_{n=1}^{\infty} F(n) - \int_0^\infty F(n)\,\mathrm{d}n. \tag{64.207}$$

Each term separately diverges; the difference is finite, and it is the heart of the computation. The transverse integral defining $F(n)$ is evaluated by dimensional continuation (Chapter D): with $d = 4 - 2\varepsilon$ continued in the transverse plane,

$$\int \frac{\mathrm{d}^2 k_\perp}{(2\pi)^2}\,\left(\mathbf{k}_\perp^2 + \kappa^2\right)^{1/2} = \frac{1}{4\pi}\,\frac{\Gamma(-\frac{3}{2})}{\Gamma(-\frac{1}{2})}\,\kappa^3 = -\frac{\kappa^3}{6\pi}, \tag{64.208}$$

using $\Gamma(-\frac{3}{2}) = \frac{4}{3}\sqrt{\pi}$ and $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$, so that $\Gamma(-\frac{3}{2})/\Gamma(-\frac{1}{2}) = -\frac{2}{3}$. The power-divergent volume piece is discarded by the analytic continuation precisely because it is the same in the plate and no-plate cases — it is the local parametrix that cancels in Eq. (64.207). With Eq. (64.208),

$$F(n) = \frac{1}{2}\cdot\left(-\frac{1}{6\pi}\right)\left(\frac{n\pi}{a}\right)^3 = -\frac{\pi^2}{12\,a^3}\,n^3. \tag{64.209}$$

Doing the finite difference. The discrete-minus-continuum difference in Eq. (64.207) is exactly what the Euler–Maclaurin formula computes. For a function F that, together with its derivatives, vanishes at infinity,

$$\sum_{n=1}^{\infty} F(n) - \int_0^\infty F(n)\,\mathrm{d}n = -\tfrac{1}{2}F(0) - \sum_{k=1}^{\infty}\frac{B_{2k}}{(2k)!}\,F^{(2k-1)}(0), \tag{64.210}$$

with B_{2k} the Bernoulli numbers, $B_2 = \frac{1}{6}$, $B_4 = -\frac{1}{30}$. For the cubic Eq. (64.209) only finitely many terms survive: $F(0) = 0$, $F'(0) = 0$, $F'''(0) = -\pi^2/(2a^3)$ (the third derivative of n^3 is 6, times the coefficient), and all higher derivatives vanish. Only the $k = 2$ term of Eq. (64.210) contributes:

$$\frac{E_{\rm C}(a)}{A} = -\frac{B_4}{4!}\,F'''(0) = -\frac{(-1/30)}{24}\Big(-\frac{\pi^2}{2a^3}\Big) = -\frac{\pi^2}{1440\,a^3}. \tag{64.211}$$

The same number follows in one line by zeta-function regularization: from Eq. (64.209), $\sum_n F(n) = -\frac{\pi^2}{12a^3}\sum_n n^3 = -\frac{\pi^2}{12a^3}\,\zeta(-3) = -\frac{\pi^2}{12a^3}\cdot\frac{1}{120} = -\frac{\pi^2}{1440\,a^3}$, using $\zeta(-3) = -B_4/4 = \frac{1}{120}$; the continuum subtraction is automatic in the analytic continuation, and the agreement with Euler–Maclaurin confirms that the discarded piece was exactly the free-space term. The Casimir energy per area is negative: the plates lower the vacuum energy below its free-space value, and the energy grows more negative as they approach, giving an attractive force.

The full stress tensor. The energy is one component; the renormalized stress tensor is the full object Eq. (64.195). We assemble it from three facts. First, *symmetry*: the configuration is invariant under translations in t, x, y and under rotations in the (x, y) plane, so $\langle T^{\mu}{}_{\nu}\rangle_{\rm ren} = \mathrm{diag}(\rho, -p_\perp, -p_\perp, -p_z)$ with components possibly depending on z alone. Second, *conservation* Eq. (64.196): the only nontrivial component is $\partial_z \langle T^{z}{}_{z}\rangle = 0$, so p_z is constant across the gap. For the conformal field the off-diagonal and z-dependent pieces vanish and all components are constant (this is the flat-space, boundary-only version of the smoothness of $G - H$: with $V = 0$ the subtracted two-point function between the plates is z-independent at coincidence), so ρ is the constant $\rho = (E_{\rm C}/A)/a$ obtained by dividing Eq. (64.211) by the gap width:

$$\rho = \Big\langle T^{0}{}_{0}\Big\rangle_{\rm ren} = -\frac{\pi^2}{1440\,a^4}. \tag{64.212}$$

Third, the normal pressure follows from the energy by the thermodynamic relation $p_z = -\,\partial(E_{\rm C}/A)/\partial a$ (force per unit area is minus the change of energy per area with separation):

$$p_z = -\frac{\partial}{\partial a}\Big(-\frac{\pi^2}{1440\,a^3}\Big) = -\frac{3\pi^2}{1440\,a^4}, \tag{64.213}$$

negative, i.e. a tension pulling the plates together — the attractive Casimir force. The transverse pressure is then fixed by the *tracelessness* of the conformal stress tensor, $\rho = 2p_\perp + p_z$ (the vanishing of $\langle T^{\mu}{}_{\mu}\rangle = \rho - 2p_\perp - p_z$), which gives $2p_\perp = \rho - p_z = -\frac{\pi^2}{1440a^4} + \frac{3\pi^2}{1440a^4} = \frac{2\pi^2}{1440a^4}$, so $p_\perp = \pi^2/(1440\,a^4)$. Assembling,

$$\boxed{\langle T^{\mu}{}_{\nu}\rangle_{\rm ren} = -\frac{\pi^2}{1440\,a^4}\;\mathrm{diag}(1,\,1,\,1,\,-3)} \tag{64.214}$$

in the coordinate order (t, x, y, z). One reads off the three consistency checks at a glance: the tensor is traceless ($1 + 1 + 1 - 3 = 0$), as required for the conformal

field; it is covariantly (here ordinarily) conserved, since every component is constant; and it is symmetric. The energy density and the normal pressure are negative, the transverse pressure positive, all scaling as a^{-4} on dimensional grounds — the only scale in the problem is a.

Worked Example 64.29 (The Casimir tensor as a parametrix subtraction)**.** It is worth seeing Eq. (64.214) as a literal instance of Eq. (64.195), to confirm the mode sum and the point-splitting prescription agree. The two-point function between the plates is built from the free-space function by the method of images enforcing $\phi = 0$ at $z = 0, a$. In Euclidean signature (Wick rotation per Chapter D), with $\rho^2 = (\Delta\tau)^2 + (\Delta x)^2 + (\Delta y)^2$ and $\Delta z = z - z'$,

$$G(x,x') = \frac{1}{4\pi^2}\sum_{n=-\infty}^{\infty}\left[\frac{1}{\rho^2 + (\Delta z - 2na)^2} - \frac{1}{\rho^2 + (z + z' - 2na)^2}\right], \tag{64.215}$$

the first family of images enforcing periodicity, the second (with the minus sign) enforcing the odd reflection that vanishes on the plates. The parametrix to subtract is the free-space function $G_0 = \frac{1}{4\pi^2}[\rho^2 + (\Delta z)^2]^{-1}$ — exactly the $n = 0$ *direct* term of Eq. (64.215). The subtracted remainder $\mathcal{W} = G - G_0 = \frac{1}{4\pi^2}\left[\sum_{n\neq 0}(\cdots)_{\text{direct}} - \sum_n(\cdots)_{\text{reflected}}\right]$ is smooth at coincidence, since the only singular term — the $n = 0$ direct image — has been removed. Applying the conformal operator Eq. (64.188) (with $\xi = \frac{1}{6}$, $m = 0$, flat connection) to $\mathcal{W}$ and taking $x' \to x$ reduces, after the geometric series $\sum_{n\geq 1} n^{-4} = \zeta(4) = \pi^4/90$ is summed, to exactly Eq. (64.214). The two routes — the mode sum with continuum subtraction, and point splitting with parametrix subtraction — compute the same finite tensor because the continuum/free-space piece they each remove is the same local object. This is the concrete payoff of the abstract construction: the subtraction is unambiguous, the result is finite, conserved, and traceless, and it sources a real, measured force.

Historical Note

The attractive force between neutral conductors was predicted by Casimir in 1948 and measured to percent precision by Lamoreaux and others from the late 1990s onward. The reinterpretation of the calculation as the simplest example of stress-tensor renormalization in quantum field theory — the free-space subtraction as a parametrix — is due to the program of DeWitt, Brown and Maclay, and Wald in the 1960s and 70s, who turned the manipulation of divergent zero-point sums into the covariant, axiomatic construction of this section.

Outlook

We have built the object that sources semiclassical gravity. Starting from the classical Hilbert tensor Eq. (64.182), we point-split its bilinears, subtracted the

universal Hadamard parametrix Eq. (64.193), and obtained a finite $\langle T_{\mu\nu}\rangle_{\rm ren}$ that Wald's axioms render unique up to the four local geometric tensors Eq. (64.200) — which we recognized as the renormalization of the cosmological constant, Newton's constant, and the curvature-squared gravitational couplings. One feature, though, refused to be removed by any allowed counterterm: enforcing conservation forced a nonzero trace Eq. (64.199) on the massless conformal field, in flat contradiction with the classical Weyl invariance Eq. (64.185). That residue — fixed entirely by the local geometric coefficient $[V_1]$ and immune to every choice of state or scale — is the trace anomaly. The next section computes it two independent ways and connects it to the anomaly machinery of Chapter 36.

64.6 The trace (conformal) anomaly

The renormalized stress tensor of Section 64.5 left us with a finite, conserved object $\langle T_{\mu\nu}\rangle$ and one stubborn feature we promised to return to: its trace does not vanish. For the conformally coupled massless scalar this is startling, because the classical theory we built in Section 64.1 is Weyl invariant, and Weyl invariance is exactly the statement that the classical stress tensor is traceless. The point-splitting subtraction that made $\langle T_{\mu\nu}\rangle$ finite removed the divergences but could not remove the trace; a definite, state-independent, purely geometric remnant survived. That remnant is the *trace anomaly* (or *conformal*, or *Weyl*, anomaly), and it is the subject of this section.

The anomaly is not a defect of one regulator. It is an intrinsic feature of the quantum effective action: there is no scheme in which a conformally coupled field carries the classical Weyl symmetry into the quantum theory while keeping general covariance and unitarity. We will see this twice. First we compute the anomaly from the short-distance structure of the field, identifying it with a single Seeley–DeWitt heat-kernel coefficient—the curved-space coefficient $a_2(x)$ whose flat-space cousin we built in Chapter E and whose curved generalization Chapter E explicitly deferred to this chapter. Then we recompute it as the Jacobian of the Weyl transformation in the path-integral measure, the scalar analogue of the chiral Jacobian of Chapter 36, and find the identical local expression. The agreement is the signature of a genuine anomaly: two unrelated regularizations, forced to choose between symmetries, make the same choice and produce the same number. At the end we read the result in the language of conformal field theory and connect its coefficients to the central charges of Chapters 47 and 48.

Throughout this section we work in Euclidean signature, obtained by the Wick rotation of Theorem 5.30 (curved-space conventions in Chapter A). This is where the relevant operator is elliptic, its heat kernel exists, and the Seeley–DeWitt expansion is rigorous; the Lorentzian statements follow by continuation. We write $g \equiv \det g_{\mu\nu} > 0$ and use $\sqrt{g}$ for the Riemannian volume element. Curvatures

follow the conventions of Chapter A: $R^{\rho}{}_{\sigma\mu\nu} = \partial_\mu \Gamma^{\rho}_{\nu\sigma} - \cdots$, $R_{\mu\nu} = R^{\rho}{}_{\mu\rho\nu}$, $R = g^{\mu\nu} R_{\mu\nu}$, with the sphere having $R > 0$.

64.6.1 The trace as the Weyl variation of the effective action

A Weyl (local conformal) transformation rescales the metric pointwise by a positive factor,

$$g_{\mu\nu}(x) \longmapsto \hat{g}_{\mu\nu}(x) = e^{2\omega(x)}\, g_{\mu\nu}(x), \tag{64.216}$$

with $\omega(x)$ an arbitrary smooth function. This is a transformation of the geometry, not a diffeomorphism: it changes lengths without moving points. The infinitesimal form is $\delta g_{\mu\nu} = 2\omega\, g_{\mu\nu}$, hence $\delta g^{\mu\nu} = -2\omega\, g^{\mu\nu}$ and $\delta\sqrt{g} = d\,\omega\sqrt{g}$ in d dimensions.

The quantum theory of a single real scalar on the fixed background is summarized by its Euclidean effective action,

$$e^{-W[g]} = Z[g] = \int \mathcal{D}\phi\, e^{-S[\phi,g]}, \qquad W[g] = -\ln Z[g], \tag{64.217}$$

a functional of the metric alone once the field is integrated out. The expectation value of the stress tensor in this state is, by definition, the metric variation of W,

$$\langle T_{\mu\nu}(x)\rangle = \frac{2}{\sqrt{g}} \frac{\delta W[g]}{\delta g^{\mu\nu}(x)}, \tag{64.218}$$

the same formula that defines the classical $T_{\mu\nu}$ from S, now with W in place of S. Contracting (64.218) with $g^{\mu\nu}$ and using $\delta g^{\mu\nu} = -2\omega g^{\mu\nu}$ to assemble a Weyl variation,

$$\delta_\omega W = \int d^d x\, \frac{\delta W}{\delta g^{\mu\nu}}\, \delta g^{\mu\nu} = \int d^d x\, \frac{\sqrt{g}}{2}\, \langle T_{\mu\nu}\rangle\, (-2\omega g^{\mu\nu}) = -\int d^d x\, \sqrt{g}\, \omega(x)\, \langle T^{\mu}{}_{\mu}(x)\rangle\,. \tag{64.219}$$

This is the master identity of the section. *A nonzero trace of the quantum stress tensor is precisely a nonzero anomalous Weyl variation of the effective action.* If $W[g]$ were Weyl invariant—if quantizing respected the classical symmetry (64.216)—then $\delta_\omega W = 0$ for all ω, and (64.219) would force $\langle T^{\mu}{}_{\mu}\rangle = 0$. The anomaly is the failure of that invariance, and (64.219) converts the abstract statement "W is not Weyl invariant" into the concrete, local, computable object $\langle T^{\mu}{}_{\mu}\rangle$.

64.6.2 Classical tracelessness of the conformal scalar

Before computing the quantum trace we must verify the claim that the classical trace vanishes, so that whatever we find is genuinely anomalous and not a classical breaking. The action of the conformally coupled massless scalar built in Section 64.1 is

$$S[\phi, g] = \frac{1}{2}\int d^d x\, \sqrt{g}\, \Big(g^{\mu\nu}\partial_\mu\phi\, \partial_\nu\phi + \xi R\, \phi^2 \Big), \qquad \xi = \xi_d \equiv \frac{d-2}{4(d-1)}, \tag{64.220}$$

with the conformal value $\xi_4 = \frac{1}{6}$ in four dimensions. The improved stress tensor obtained by varying (64.220) with respect to the metric, via (64.218) applied to S, is

$$T_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi - \tfrac{1}{2}g_{\mu\nu}(\partial\phi)^2 + \xi(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu + R_{\mu\nu} - \tfrac{1}{2}g_{\mu\nu}R)\phi^2, \quad (64.221)$$

where $\Box = g^{\mu\nu}\nabla_\mu\nabla_\nu$ and the ξ-terms are exactly the "improvement" generated by the conformal coupling. We take its trace, contracting with $g^{\mu\nu}$ and using $g^{\mu\nu}g_{\mu\nu} = d$, $g^{\mu\nu}\nabla_\mu\nabla_\nu = \Box$, $g^{\mu\nu}R_{\mu\nu} = R$:

$$\begin{aligned} T^\mu{}_\mu &= (\partial\phi)^2 - \tfrac{d}{2}(\partial\phi)^2 + \xi(d\,\Box - \Box + R - \tfrac{d}{2}R)\phi^2 \\ &= -\tfrac{d-2}{2}(\partial\phi)^2 + \xi(d-1)\Box\,\phi^2 - \xi\,\tfrac{d-2}{2}R\,\phi^2. \end{aligned} \quad (64.222)$$

The middle term must now be reduced on shell. The equation of motion from (64.220) is $(-\Box + \xi R)\phi = 0$, and $\Box\,\phi^2 = 2\phi\Box\phi + 2(\partial\phi)^2 = 2\xi R\,\phi^2 + 2(\partial\phi)^2$. Substituting into (64.222),

$$\begin{aligned} T^\mu{}_\mu &= -\tfrac{d-2}{2}(\partial\phi)^2 + \xi(d-1)\big(2\xi R\,\phi^2 + 2(\partial\phi)^2\big) - \xi\,\tfrac{d-2}{2}R\,\phi^2 \\ &= \Big[2\xi(d-1) - \tfrac{d-2}{2}\Big](\partial\phi)^2 + \xi\Big[2\xi(d-1) - \tfrac{d-2}{2}\Big]R\,\phi^2. \end{aligned} \quad (64.223)$$

Both brackets are the same combination, and it vanishes precisely at the conformal coupling:

$$2\xi(d-1) - \frac{d-2}{2} = 0 \quad\Longleftrightarrow\quad \xi = \frac{d-2}{4(d-1)} = \xi_d. \quad (64.224)$$

Hence $T^\mu{}_\mu = 0$ identically on shell for the conformally coupled massless scalar, in every dimension. This is the classical Weyl invariance of Section 64.1 seen from the stress tensor: the trace is the local order parameter for the symmetry, and the special value ξ_d is exactly the value that switches it off. Any nonzero $\langle T^\mu{}_\mu\rangle$ in the quantized theory is therefore a clean anomaly, with no classical contamination.

Caution: the conformal coupling must be carried in $d \neq 4$

The cancellation (64.224) is dimension dependent: $\xi_d = \frac{d-2}{4(d-1)}$ equals $\frac{1}{6}$ only at $d = 4$. When the computation is regularized by varying the dimension—as the heat-kernel and dimensional-regularization routes implicitly do—one must use ξ_d, not the frozen value $\frac{1}{6}$, or the classical trace fails to vanish away from $d = 4$ and the bookkeeping of the anomaly is corrupted. Keeping ξ_d is what guarantees that the entire nonzero answer is quantum in origin.

64.6.3 What the answer must look like: the geometric invariants

Equation (64.219) already tells us a great deal about $\langle T^{\mu}{}_{\mu} \rangle$ before any computation. The trace is a local functional of the metric (the point-splitting construction of Section 64.5 made $\langle T_{\mu\nu} \rangle$ local in the sense that its anomalous part is built from the background at the point), it has mass dimension d (the dimension of an energy density), it is a scalar (it is $g^{\mu\nu}\langle T_{\mu\nu}\rangle$), and it is built from the curvature and its derivatives. In $d = 4$ the local scalars of dimension four that can be assembled from the Riemann tensor and two derivatives of the metric form a four-dimensional space, spanned by

$$R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \qquad R_{\mu\nu}R^{\mu\nu}, \qquad R^2, \qquad \Box R. \tag{64.225}$$

A more illuminating basis trades two of these for combinations with a clean geometric meaning. The first is the square of the Weyl tensor, the totally trace-free part of the Riemann tensor; in $d = 4$,

$$W^2 \equiv C_{\mu\nu\rho\sigma}C^{\mu\nu\rho\sigma} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 2R_{\mu\nu}R^{\mu\nu} + \tfrac{1}{3}R^2. \tag{64.226}$$

The second is the four-dimensional Euler density (the Gauss–Bonnet integrand),

$$E_4 \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2, \tag{64.227}$$

whose integral over a compact four-manifold is the topological invariant

$$\int d^4x\,\sqrt{g}\;E_4 = 32\pi^2\,\chi, \tag{64.228}$$

with χ the Euler characteristic. The remaining two directions of (64.225) are R^2 and $\Box R$. The Weyl tensor is the conformally covariant piece of the curvature: under (64.216) it transforms homogeneously, $C^{\mu}{}_{\nu\rho\sigma} \to C^{\mu}{}_{\nu\rho\sigma}$ (weight zero with one index up), so W^2 is conformally covariant of weight -4 and $\sqrt{g}\,W^2$ is Weyl invariant in $d = 4$. These geometric facts are not decorative; they are exactly what the Wess–Zumino consistency condition for the anomaly will demand.

Theorem 64.30 (Form of the trace anomaly in four dimensions)**.** *For a conformally coupled quantum field on a curved four-dimensional background, the renormalized trace of the stress tensor is a local geometric scalar of the form*

$$\langle T^{\mu}{}_{\mu} \rangle = \frac{1}{(4\pi)^2}\Big(c\,W^2 - a\,E_4 + b\,\Box R \Big), \tag{64.229}$$

with c, a, b pure numbers fixed by the field content. The coefficient of R^2 is absent: no R^2 term can appear. The coefficients a (type A, the Euler term) and c (type B, the Weyl term) are scheme independent; the coefficient b of $\Box R$ is scheme dependent and can be shifted to any value by a local counterterm.

The absence of R^2 is forced by consistency, and the argument is worth giving now because it organizes everything that follows. A Weyl anomaly is the variation $\delta_\omega W$ of (64.219). Two successive Weyl transformations commute—they just multiply the metric by $e^{2\omega_1}e^{2\omega_2}$—so the variation must satisfy the integrability (Wess–Zumino) condition

$$\delta_{\omega_1}\delta_{\omega_2}W - \delta_{\omega_2}\delta_{\omega_1}W = 0. \tag{64.230}$$

A candidate R^2 term in $\langle T^\mu{}_\mu\rangle$ contributes $\delta_{\omega_1}\int\sqrt{g}\,\omega_2 R^2$ to the second variation; using $\delta_\omega R = -2\omega R - 2(d-1)\Box\omega$ in $d=4$ one finds this is not symmetric under $1\leftrightarrow 2$, so (64.230) fails unless the R^2 coefficient vanishes. By contrast $\sqrt{g}\,W^2$ is Weyl invariant, so its appearance under the integral sign in (64.219) gives a consistent (type B) anomaly; and $\sqrt{g}\,E_4$, being a total derivative whose integral is topological, gives the other consistent (type A) solution. The $\Box R$ term is special: it is itself the Weyl variation of the local functional $\int\sqrt{g}\,R^2$,

$$\delta_\omega\int d^4x\,\sqrt{g}\,R^2 = -12\int d^4x\,\sqrt{g}\,\omega\,\Box R, \tag{64.231}$$

so adding the covariant counterterm $\alpha\int\sqrt{g}\,R^2$ to W shifts b by $-12\alpha/(4\pi)^2$ while leaving a and c untouched. This is the precise sense in which b is scheme dependent and a, c are physical: a and c multiply genuine cohomology classes of the Weyl group that no local counterterm can produce, while b multiplies a Weyl-exact form that a counterterm can cancel or create at will.

It remains to compute the numbers a, c, b. We do so for the real scalar, twice.

64.6.4 The heat-kernel route: the anomaly is $a_2(x)$

The operator governing the scalar is the Euclidean fluctuation operator read off from (64.220),

$$D = -\Box + \xi R, \tag{64.232}$$

a Laplace-type operator of exactly the form $-\Box + X$ studied in Chapter E with $X = \xi R$. The Euclidean effective action of one real scalar is the one-loop determinant

$$W = \tfrac{1}{2}\ln\det D = -\tfrac{1}{2}\,\zeta_D'(0), \tag{64.233}$$

the second equality being the zeta-function definition of the determinant from Section E.3. The whole anomaly will come from one number attached to D: the value $\zeta_D(0)$, equivalently the diagonal Seeley–DeWitt coefficient $a_2(x)$.

The global scaling argument. Consider first a *constant* Weyl rescaling, $\omega =$ const. Under $g \to e^{2\omega}g$ the conformally coupled operator scales homogeneously: since $\sqrt{g}\,D$ is built to be conformally covariant at $\xi = \xi_d$, its eigenvalues obey

$\lambda_n \to e^{-2\omega}\lambda_n$. Then the spectral zeta function $\zeta_D(s) = \sum_n \lambda_n^{-s}$ transforms as $\zeta_D(s) \to e^{2s\omega}\zeta_D(s)$, so to first order in ω,

$$\delta_\omega \zeta_D(s) = 2s\,\omega\,\zeta_D(s), \qquad \delta_\omega \zeta_D'(0) = \frac{d}{ds}\Big(2s\,\omega\,\zeta_D(s)\Big)\Big|_{s=0} = 2\,\omega\,\zeta_D(0). \tag{64.234}$$

The single derivative at $s = 0$ leaves only the residue $\zeta_D(0)$ because the factor s kills everything but the value at the origin. Hence, from (64.233),

$$\delta_\omega W = -\tfrac{1}{2}\,\delta_\omega \zeta_D'(0) = -\omega\,\zeta_D(0), \tag{64.235}$$

and Chapter E (Eq. (E.95)) evaluated in $d = 4$ gives

$$\zeta_D(0) = \frac{1}{(4\pi)^2}\int d^4x\,\sqrt{g}\;a_2(x) - \dim\ker D, \tag{64.236}$$

where $a_2(x)$ is the coincidence-limit Seeley–DeWitt coefficient—the t^2 term in the proper-time expansion, in the index convention of Eq. (E.93). Comparing (64.235) with the constant-ω case of the master identity (64.219), namely $\delta_\omega W = -\omega \int \sqrt{g}\,\langle T^\mu{}_\mu\rangle$, and discarding the zero-mode count (a global, infrared subtlety irrelevant to the local trace), we read off

$$\boxed{\langle T^\mu{}_\mu(x)\rangle = \frac{1}{(4\pi)^2}\,a_2(x)} \tag{64.237}$$

in $d = 4$. The trace anomaly *is* the diagonal heat-kernel coefficient $a_2(x)$, divided by $(4\pi)^2$. This is the curved-space face of the divergence/anomaly correspondence that ran through Chapter E: the coefficient that controls the logarithmic ultraviolet divergence of the effective action is the same coefficient that controls the breaking of scale invariance.

The local statement. For a position-dependent $\omega(x)$ the slick spectral argument is replaced by the proper-time computation, which produces the same local density. Write $W = -\frac{1}{2}\int_0^\infty \frac{dt}{t}\,\mathrm{Tr}\,e^{-tD}$. The conformal covariance of D gives $\delta_\omega D = -2\omega D$ (at $\xi = \xi_d$, where the inhomogeneous piece drops), hence

$$\delta_\omega W = \int_0^\infty dt\;\partial_t\,\mathrm{Tr}(\omega\,e^{-tD}) = -\Big[\,\mathrm{Tr}(\omega\,e^{-tD})\Big]_{t\to 0^+}, \tag{64.238}$$

where the upper limit gives zero (no zero modes, or they are subtracted). The heat trace at coincidence has the short-time expansion

$$\mathrm{Tr}(\omega\,e^{-tD}) \overset{t\to 0^+}{\sim} \frac{1}{(4\pi t)^2}\int d^4x\,\sqrt{g}\;\omega(x)\sum_{k\ge 0} a_k(x)\,t^k, \tag{64.239}$$

the analogue of Eq. (E.93) with the test function ω inserted. The power-divergent terms $k = 0, 1$ are removed by the local counterterms that renormalize W (they

are the cosmological-constant and Einstein–Hilbert subtractions of Section 64.5); the surviving finite piece is the t^0 term, $k = 2$,

$$\delta_\omega W|_{\text{finite}} = -\frac{1}{(4\pi)^2}\int d^4x\,\sqrt{g}\,\omega(x)\,a_2(x), \tag{64.240}$$

which, compared with (64.219), reproduces (64.237) now for arbitrary $\omega(x)$. Everything has been reduced to one geometric quantity, $a_2(x)$, and we now compute it.

The Seeley–DeWitt recursion in curved space

We carry the Seeley–DeWitt construction of Section E.3.4 to fourth order on a curved manifold, which Chapter E promised would be done here. The covariant ansatz for the heat kernel $\mathcal{K}(x,y;t)$ of $D = -\Box + X$, the curved generalization of the flat ansatz Theorem E.19, is

$$\mathcal{K}(x,y;t) = \frac{\Delta^{1/2}(x,y)}{(4\pi t)^{d/2}}\,e^{-\sigma(x,y)/2t}\sum_{k=0}^{\infty} b_k(x,y)\,t^k, \tag{64.241}$$

where $\sigma(x,y) = \frac{1}{2}(\text{geodesic distance})^2$ is Synge's world function and $\Delta(x,y) = g(x)^{-1/2}g(y)^{-1/2}\det[-\partial^2\sigma/\partial x^\mu\partial y^\nu]$ is the van Vleck–Morette determinant. The square root $\Delta^{1/2}$ is the curved-space replacement of the flat normalization; its job is to keep $b_0 \equiv 1$ in the transport equation, exactly as in the flat case. Inserting (64.241) into the heat equation $(\partial_t + D_x)\mathcal{K} = 0$ and using the defining identities of σ and Δ (Synge's equation $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$ and the van Vleck transport equation $\Box\sigma + \sigma^{;\mu}\partial_\mu \ln\Delta = d$) gives the covariant transport recursion

$$(k + \sigma^{;\mu}\nabla_\mu)b_k = \Delta^{-1/2}\,\Box(\Delta^{1/2}b_{k-1}) - X\,b_{k-1}, \qquad b_0 \equiv 1, \tag{64.242}$$

the curved-space generalization of Eq. (E.87). The derivation is identical in structure to the flat one of Section E.3.4: ∂_t lowers a power of t, the gradient term $\sigma^{;\mu}\nabla_\mu/t$ also lowers a power, and $-\Box$ and $-X$ preserve it; the van Vleck prefactor is exactly what converts the flat Laplacian into $\Delta^{-1/2}\Box\Delta^{1/2}(\cdots)$.

We want the diagonal value $[b_2] \equiv b_2(x,x)$, since $a_2(x) = [b_2]$. The diagonal values are extracted from (64.242) by the coincidence-limit calculus, in which $[\,\cdot\,]$ denotes the limit $y \to x$ and covariant derivatives are commuted past the limit by Synge's rule $[\nabla_\mu W] = \nabla_\mu[W] - [\nabla_{\mu'}W]$ (primed indices act at y). The inputs we need are the coincidence limits of σ and $\Delta^{1/2}$, which are pure Riemannian geometry.

Lemma 64.31 (Coincidence limits)**.** *With $\sigma_{;\mu} \equiv \nabla_\mu\sigma$ and $\Delta^{1/2}$ as above,*

$$\begin{aligned}&[\sigma] = 0, \quad [\sigma_{;\mu}] = 0, \quad [\sigma_{;\mu\nu}] = g_{\mu\nu}, \quad [\sigma_{;\mu\nu\rho}] = 0,\\ &[\sigma_{;\mu\nu\rho\lambda}] = -\tfrac{1}{3}(R_{\mu\rho\nu\lambda} + R_{\mu\lambda\nu\rho}),\end{aligned} \tag{64.243}$$

$$[\Delta^{1/2}] = 1, \quad [\Delta^{1/2}_{;\mu}] = 0, \quad [\Delta^{1/2}_{;\mu\nu}] = \tfrac{1}{6}R_{\mu\nu}, \quad [\Box\Delta^{1/2}] = \tfrac{1}{6}R, \tag{64.244}$$

$$[\Box(\Delta^{-1/2}\Box\Delta^{1/2})] = \tfrac{1}{5}\Box R - \tfrac{1}{30}R_{\mu\nu}R^{\mu\nu} + \tfrac{1}{30}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}. \tag{64.245}$$

Proof. The first line is Synge's standard tower, obtained by repeatedly differentiating the defining identity $\sigma^{;\mu}\sigma_{;\mu} = 2\sigma$ and taking coincidence. Differentiating once and using $[\sigma] = 0$ gives $[\sigma_{;\mu}] = 0$; differentiating twice gives $[\sigma_{;\mu}{}^{;\nu}][\sigma_{;\nu}] + [\sigma_{;\mu}][\sigma^{;\nu}{}_{;\nu}] = [\sigma_{;\mu}{}^{;\nu}]\cdot[\sigma_{;\nu}]$, which together with the nondegeneracy $\det[\sigma_{;\mu\nu}] \neq 0$ forces $[\sigma_{;\mu\nu}] = g_{\mu\nu}$. The third derivative vanishes by symmetry, and the fourth follows by commuting covariant derivatives, the commutator producing the Riemann tensor: $[\sigma_{;\mu\nu\rho\lambda}] = -\tfrac{1}{3}(R_{\mu\rho\nu\lambda} + R_{\mu\lambda\nu\rho})$. For the van Vleck limits, expand $\ln\Delta^{1/2}$ in Riemann normal coordinates about y. The transport equation $\Box\sigma + \sigma^{;\mu}\partial_\mu \ln\Delta = d$ together with $[\Box\sigma] = g^{\mu\nu}[\sigma_{;\mu\nu}] = d$ gives, on differentiating twice and taking coincidence, $[\Delta^{1/2}_{;\mu\nu}] = \tfrac{1}{6}R_{\mu\nu}$, hence $[\Box\Delta^{1/2}] = \tfrac{1}{6}R$. The explicit normal-coordinate expansion to the needed order is

$$\Delta^{1/2}(x,y) = 1 + \tfrac{1}{12}R_{\mu\nu}u^\mu u^\nu + \tfrac{1}{24}R_{\mu\nu;\rho}u^\mu u^\nu u^\rho + \Big(\tfrac{1}{80}R_{\mu\nu;\rho\lambda} + \tfrac{1}{360}R^{\alpha}{}_{\mu\beta\nu}R^{\beta}{}_{\rho\alpha\lambda}\Big)u^\mu u^\nu u^\rho u^\lambda + \cdots, \tag{64.246}$$

with u^μ the tangent vector to the geodesic from y to x. Acting with the Laplacian $\Box = g^{\mu\nu}(\partial_\mu\partial_\nu - \Gamma^\rho_{\mu\nu}\partial_\rho)$ on $\Delta^{-1/2}\Box\Delta^{1/2}$ and setting $u = 0$, using the normal-coordinate expansions $g_{\mu\nu} = \delta_{\mu\nu} - \tfrac{1}{3}R_{\mu\alpha\nu\beta}u^\alpha u^\beta + \cdots$ and the Christoffel expansion, the quadratic-curvature invariants $R_{\mu\nu}R^{\mu\nu}$ and $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$ are generated by the $\Gamma\partial\Gamma$ and R^2 cross terms while $\Box R$ comes from the $R_{\mu\nu;\rho\lambda}$ term, giving (64.245). This is a finite, mechanical normal-coordinate computation; only the integer coefficients require care. □

We now extract $[b_2]$. The $k = 1$ case of the recursion (64.242) is $(1 + \sigma^{;\mu}\nabla_\mu)b_1 = F$ with $F \equiv \Delta^{-1/2}\Box\Delta^{1/2} - X$. Taking the coincidence limit and using $[\sigma^{;\mu}] = 0$ gives the first coefficient,

$$[b_1] = [F] = [\Box\Delta^{1/2}] - X = \frac{1}{6}R - X, \tag{64.247}$$

which reproduces Eq. (E.97) and shows the conformal scalar (with $X = \xi_4 R = \tfrac{1}{6}R$) has $[b_1] = 0$, the curved-space statement that its one-loop logarithmic divergence is purely topological. For $[b_2]$, take the $k = 2$ recursion $(2 + \sigma^{;\mu}\nabla_\mu)b_2 = \Delta^{-1/2}\Box(\Delta^{1/2}b_1) - Xb_1$ and pass to coincidence. The transport term gives $[(2 + \sigma^{;\mu}\nabla_\mu)b_2] = 2[b_2]$, and on the right

$$[\Delta^{-1/2}\Box(\Delta^{1/2}b_1)] = [\Box\Delta^{1/2}]\,[b_1] + 2[\nabla^\mu\Delta^{1/2}]\,[\nabla_\mu b_1] + [\Box b_1] = \tfrac{1}{6}R\,[b_1] + [\Box b_1], \tag{64.248}$$

using $[\nabla^\mu\Delta^{1/2}] = 0$. Therefore

$$[b_2] = \tfrac{1}{2}\Big(\tfrac{1}{6}R\,[b_1] - X[b_1] + [\Box b_1]\Big) = \tfrac{1}{2}\Big((\tfrac{1}{6}R - X)^2 + [\Box b_1]\Big). \tag{64.249}$$

The only missing piece is $[\Box b_1]$, the coincidence limit of the Laplacian of the (off-diagonal) first coefficient. We get it by differentiating the $k=1$ recursion twice. Applying $\nabla_\nu\nabla_\rho$ to $(1+\sigma^{;\mu}\nabla_\mu)b_1 = F$ and taking coincidence, every term with an undifferentiated $\sigma^{;\mu}$ or with $[\sigma_{;\mu\nu\rho}]=0$ drops, while $[\sigma^{;\mu}{}_{\nu}] = \delta^\mu_\nu$ survives twice, leaving $3[b_{1;\nu\rho}] = [F_{;\nu\rho}]$; tracing with $g^{\nu\rho}$,

$$[\Box b_1] = \tfrac{1}{3}[\Box F] = \tfrac{1}{3}\Big([\Box(\Delta^{-1/2}\Box\Delta^{1/2})] - \Box X\Big), \tag{64.250}$$

where $[\Box X] = \Box X$ because $X = \xi R$ is an ordinary local function. Feeding in the van Vleck limit (64.245),

$$[\Box b_1] = \tfrac{1}{3}\Big(\tfrac{1}{5}\Box R - \tfrac{1}{30}R_{\mu\nu}R^{\mu\nu} + \tfrac{1}{30}R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - \Box X\Big) = \tfrac{1}{15}\Box R - \tfrac{1}{90}R^2_{\mu\nu} + \tfrac{1}{90}R^2_{\mu\nu\rho\sigma} - \tfrac{1}{3}\Box X, \tag{64.251}$$

with the shorthand $R^2_{\mu\nu} \equiv R_{\mu\nu}R^{\mu\nu}$, $R^2_{\mu\nu\rho\sigma} \equiv R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$. Substituting (64.251) and $\big(\tfrac{1}{6}R - X\big)^2 = \tfrac{1}{36}R^2 - \tfrac{1}{3}RX + X^2$ into (64.249) gives the general diagonal coefficient for $D = -\Box + X$,

$$a_2(x) = [b_2] = \tfrac{1}{2}X^2 - \tfrac{1}{6}RX - \tfrac{1}{6}\Box X + \tfrac{1}{30}\Box R + \tfrac{1}{72}R^2 - \tfrac{1}{180}R^2_{\mu\nu} + \tfrac{1}{180}R^2_{\mu\nu\rho\sigma}. \tag{64.252}$$

This is the Gilkey–Seeley–DeWitt result; for $X = -E$ it is the standard b_4 (Gilkey a_4) coefficient of the literature. We have derived it from the heat equation, Synge's calculus, and the van Vleck expansion alone—no result imported on faith.

Specializing to the conformal scalar

Now set $X = \xi R = \tfrac{1}{6}R$, the conformal coupling. The three X-dependent terms in (64.252) become

$$\tfrac{1}{2}X^2 = \tfrac{1}{2}\cdot\tfrac{1}{36}R^2 = \tfrac{1}{72}R^2, \quad -\tfrac{1}{6}RX = -\tfrac{1}{6}\cdot\tfrac{1}{6}R^2 = -\tfrac{1}{36}R^2, \quad -\tfrac{1}{6}\Box X = -\tfrac{1}{6}\cdot\tfrac{1}{6}\Box R = -\tfrac{1}{36}\Box R. \tag{64.253}$$

The R^2 pieces collect to $\tfrac{1}{72} + \tfrac{1}{72} - \tfrac{1}{36} = \tfrac{2}{72} - \tfrac{2}{72} = 0$: the R^2 term cancels exactly. This is the cancellation promised by Theorem 64.30 and forced by the Wess–Zumino condition—and here it falls out of pure arithmetic, with the conformal value $\xi = \tfrac{1}{6}$ as precisely the value that makes it vanish. The $\Box R$ pieces collect to $-\tfrac{1}{36} + \tfrac{1}{30} = \tfrac{-5+6}{180} = \tfrac{1}{180}$, and the quadratic-curvature pieces are untouched. Hence

$$a_2(x)\Big|_{\xi=1/6} = \frac{1}{180}R^2_{\mu\nu\rho\sigma} - \frac{1}{180}R^2_{\mu\nu} + \frac{1}{180}\Box R = \frac{1}{180}\Big(R^2_{\mu\nu\rho\sigma} - R^2_{\mu\nu} + \Box R\Big). \tag{64.254}$$

By (64.237) the trace anomaly of one real conformally coupled scalar is therefore

$$\langle T^\mu{}_\mu\rangle = \frac{1}{(4\pi)^2}\cdot\frac{1}{180}\Big(R^2_{\mu\nu\rho\sigma} - R^2_{\mu\nu} + \Box R\Big). \tag{64.255}$$

To put it in the canonical form (64.229), invert the basis change. Solving (64.226) and (64.227) for the quadratic invariants,

$$R^2_{\mu\nu\rho\sigma} - R^2_{\mu\nu} = \tfrac{1}{2}W^2 - \tfrac{1}{2}E_4 + \tfrac{1}{3}R^2 - \tfrac{1}{6}R^2 + \cdots \tag{64.256}$$

is most cleanly handled by checking the linear combination directly: with $c = \frac{1}{120}$, $a = \frac{1}{360}$, $b = \frac{1}{180}$,

$$\begin{aligned} c\,W^2 - a\,E_4 + b\,\Box R &= \tfrac{1}{120}(R^2_{\mu\nu\rho\sigma} - 2R^2_{\mu\nu} + \tfrac{1}{3}R^2) - \tfrac{1}{360}(R^2_{\mu\nu\rho\sigma} - 4R^2_{\mu\nu} + R^2) + \tfrac{1}{180}\Box R \\ &= \Big(\tfrac{1}{120} - \tfrac{1}{360}\Big)R^2_{\mu\nu\rho\sigma} + \Big(-\tfrac{2}{120} + \tfrac{4}{360}\Big)R^2_{\mu\nu} + \Big(\tfrac{1}{360} - \tfrac{1}{360}\Big)R^2 + \tfrac{1}{180}\Box R \\ &= \tfrac{1}{180}R^2_{\mu\nu\rho\sigma} - \tfrac{1}{180}R^2_{\mu\nu} + \tfrac{1}{180}\Box R, \end{aligned} \tag{64.257}$$

which matches (64.254) term by term; the R^2 coefficients again cancel, $\frac{1}{360} - \frac{1}{360} = 0$, confirming consistency. We have therefore established the central computation of the section.

$$\boxed{\langle T^{\mu}{}_{\mu}\rangle_{\text{scalar}} = \frac{1}{(4\pi)^2}\Big(\tfrac{1}{120}\,W^2 - \tfrac{1}{360}\,E_4 + \tfrac{1}{180}\,\Box R\Big)} \tag{64.258}$$

The type-B central charge of a single real scalar is $c = \frac{1}{120}$, the type-A central charge is $a = \frac{1}{360}$, and the scheme-dependent $\Box R$ coefficient is $b = \frac{1}{180}$.

Physical Insight

The same arithmetic that cancels R^2 ties three apparently independent facts into one. The conformal coupling $\xi = \frac{1}{6}$ was singled out classically as the value that makes the stress tensor traceless on shell ((64.224)); it is singled out in the heat kernel as the value that makes $[b_1] = 0$ ((64.247)), so that the one-loop logarithmic divergence is the topological Euler term and nothing else; and it is singled out in the anomaly as the value that removes the nonconsistent R^2 term from (64.252). Three roles, one number. The conformal scalar is not merely a convenient example—it is the field whose quantum geometry is maximally constrained, and the trace anomaly is the residue that even maximal constraint cannot remove.

64.6.5 The Fujikawa route: the Weyl Jacobian of the measure

The heat-kernel computation located the anomaly in the regularized determinant. The Fujikawa method of Chapter 36 locates it instead in the path-integral measure, and the value it returns must agree—the test of a genuine anomaly. We import the method of Chapter 36 wholesale: a symmetry whose Noether current is classically conserved fails to be a symmetry of the quantum theory exactly when the functional measure $\mathcal{D}\phi$ is not invariant, and the measure Jacobian, regularized covariantly, is the anomaly.

The symmetry here is the Weyl transformation (64.216). On the field it acts by its conformal weight, which for a scalar in d dimensions is $-\frac{d-2}{2}$,

$$\phi(x) \;\longmapsto\; \hat{\phi}(x) = e^{-\frac{d-2}{2}\omega(x)}\,\phi(x), \tag{64.259}$$

the unique weight for which the action (64.220) is invariant at $\xi = \xi_d$ (this is the field-space statement of the classical Weyl invariance verified in Section 64.6.2). The measure is defined, as in Chapter E, through the eigenmodes of the relevant operator. Expand $\phi = \sum_n c_n\,\varphi_n$ in the orthonormal eigenfunctions of the conformal operator $D = -\Box + \xi R$,

$$D\varphi_n = \lambda_n \varphi_n, \qquad \int d^dx\,\sqrt{g}\,\varphi_m\varphi_n = \delta_{mn}, \tag{64.260}$$

so that $\mathcal{D}\phi = \prod_n dc_n$. Under the infinitesimal Weyl transformation (64.259) the coefficients rotate linearly, $c_n \to c'_n = \sum_m J_{nm}c_m$, with

$$J_{nm} = \int d^dx\,\sqrt{g}\;\varphi_n\left(1 - \tfrac{d-2}{2}\omega\right)\varphi_m = \delta_{nm} - \frac{d-2}{2}\int d^dx\,\sqrt{g}\;\omega\,\varphi_n\varphi_m + O(\omega^2). \tag{64.261}$$

For a bosonic measure the Jacobian is $\det J$ (not its inverse, the bosonic counterpart of the Berezin rule of Chapter E), so

$$\mathcal{D}\hat{\phi} = (\det J)\,\mathcal{D}\phi = \exp\Big[\;\mathrm{Tr}\ln J\Big]\,\mathcal{D}\phi = \exp\Big[-\frac{d-2}{2}\int d^dx\,\sqrt{g}\;\omega(x)\sum_n \varphi_n(x)^2\Big]\,\mathcal{D}\phi, \tag{64.262}$$

to first order in ω, where we used $\mathrm{Tr}\ln J = \mathrm{Tr}(J-\mathbf{1}) + O(\omega^2)$. The exponent is controlled by the mode sum at coincident points,

$$\mathcal{A}(x) \equiv \sum_n \varphi_n(x)^2 = \sum_n \varphi_n(x)\varphi_n(x) = \langle x|\,\mathbf{1}\,|x\rangle = \delta^d(0)/\sqrt{g}, \tag{64.263}$$

which is meaningless until regularized—precisely the situation of Chapter 36, where the naive chiral density was also a divergent mode sum that had to be regulated before it could be read. We regulate by the same heat-kernel damping that Chapter 36 used, inserting $e^{-\lambda_n/\Lambda^2}$ (the eigenvalue, not its square, since D is already second order) and removing the cutoff at the end:

$$\mathcal{A}(x) = \lim_{\Lambda\to\infty}\sum_n e^{-\lambda_n/\Lambda^2}\varphi_n(x)^2 = \lim_{\Lambda\to\infty}\langle x|\,e^{-D/\Lambda^2}\,|x\rangle = \lim_{\Lambda\to\infty}\mathrm{Tr}\big[e^{-D/\Lambda^2}\big]_{x,x}. \tag{64.264}$$

This is exactly the diagonal heat kernel (64.241) at proper time $t = 1/\Lambda^2$, the same object that Eq. (E.200) identified as the carrier of the chiral anomaly. Inserting the short-time expansion of the heat kernel at coincidence,

$$\langle x|\,e^{-D/\Lambda^2}\,|x\rangle = \frac{1}{(4\pi/\Lambda^2)^{d/2}}\Big(a_0(x) + a_1(x)\,\Lambda^{-2} + a_2(x)\,\Lambda^{-4} + \cdots\Big) = \frac{\Lambda^d}{(4\pi)^{d/2}}\Big(a_0 + \frac{a_1}{\Lambda^2} + \frac{a_2}{\Lambda^4} + \cdots \tag{64.265}$$

In $d = 4$ the terms scale as $\Lambda^4 a_0$, $\Lambda^2 a_1$, and $\Lambda^0 a_2$, followed by negative powers. The positive powers of Λ are exactly the power divergences that the renormalization of Section 64.5 removes; they are local, are absorbed into the cosmological-constant and Newton-constant counterterms, and—being multiplied by the field-independent prefactor of the Weyl variation—cancel against the corresponding variation of those counterterms, just as the power divergences of the chiral case in Chapter 36 were killed by the vanishing γ^5 trace. The negative powers vanish as $\Lambda \to \infty$. The unique survivor is the Λ^0 term, the coefficient $a_2(x)$:

$$\mathcal{A}(x)\big|_{\text{finite}} = \frac{1}{(4\pi)^2}\, a_2(x). \tag{64.266}$$

Thus the Weyl Jacobian (64.262) contributes, in $d = 4$ where $\frac{d-2}{2} = 1$, the anomalous variation

$$\delta_\omega W_{\text{measure}} = -\int d^4x\, \sqrt{g}\, \omega(x)\, \frac{1}{(4\pi)^2}\, a_2(x), \tag{64.267}$$

and comparing with the master identity (64.219) gives, once more,

$$\langle T^\mu{}_\mu \rangle = \frac{1}{(4\pi)^2}\, a_2(x), \tag{64.268}$$

the identical local expression (64.237), hence the identical (64.258). The two derivations agree completely. The heat-kernel route saw the anomaly as the finite remnant of the determinant's divergence; the Fujikawa route saw it as the noninvariance of the measure under field rescaling. They are the same statement viewed from the two halves of the path integral—$Z = \int \mathcal{D}\phi\, e^{-S}$—and the coefficient $a_2(x)$ is the common bridge.

Caution: why this Jacobian does not vanish in dimensional regularization

One might worry, as in the chiral case (Section E.5.6), that (64.263) is just $\delta^d(0)$, which vanishes in dimensional regularization, so the anomaly would be a regulator artifact. The escape is the same in spirit but different in detail. The Weyl Jacobian is not $\delta^d(0)$ times a finite local density; it is $\delta^d(0)$ multiplied by the classical trace $T^\mu{}_\mu \propto (d-4) \times (\text{curvature})$, an indeterminate 0×0 in dimensional regularization. The factor $(d-4)$ comes from the dimension-dependent conformal weight $\frac{d-2}{2}$ and coupling ξ_d; the $1/(d-4)$ pole comes from the divergent mode sum. Their product is finite, local, and equal to $a_2(x)/(4\pi)^2$. The anomaly is the residue of an unavoidable collision between a vanishing classical trace and a diverging quantum measure—a collision no scheme can prevent.

64.6.6 Reading the result: type A, type B, and the central charges

The two coefficients a and c of (64.229) are genuinely different objects, and the distinction (due to Deser and Schwimmer) is the organizing principle of the four-dimensional Weyl anomaly.

Type A: the Euler coefficient a. The E_4 term is the *type-A* anomaly. Its defining feature is that E_4 is a topological density: by (64.228) its integral is $32\pi^2\chi$, a number that does not change under any smooth deformation of the metric. The associated anomalous functional $\int\sqrt{g}\,\omega\,E_4$ is nontrivial in Weyl cohomology—it cannot be written as the Weyl variation of any local covariant functional—which is why a is scheme independent. The coefficient a is the four-dimensional analogue of the two-dimensional central charge, and it is the quantity that decreases monotonically under renormalization-group flow: the a-theorem, $a_{\rm UV} > a_{\rm IR}$, the four-dimensional counterpart of Zamolodchikov's c-theorem (the two-dimensional version appears in Chapter 48). It counts, in a precise RG-monotone sense, the effective number of degrees of freedom.

Type B: the Weyl coefficient c. The W^2 term is the *type-B* anomaly. Because $\sqrt{g}\,W^2$ is itself Weyl invariant in $d = 4$, the functional $\int\sqrt{g}\,\omega\,W^2$ is a nontrivial Weyl variation of a (nonlocal) functional and is again scheme independent, but its character is different: c multiplies a conformally covariant local invariant, and in a conformal field theory it is fixed by a flat space datum—the two-point function of the stress tensor. The number c is the coefficient C_T of $\langle T_{\mu\nu}(x)T_{\rho\sigma}(0)\rangle$ in the conformal field theory of Chapter 47: the central charge that the conformal bootstrap constrains and that appears in the stress-tensor conformal block. For a free field, c counts degrees of freedom weighted by spin, and unlike a it is not RG monotone in general.

The scheme-dependent $\Box R$. The coefficient b is, as (64.231) showed, freely adjustable by the local counterterm $\int\sqrt{g}\,R^2$. Two regularization schemes that agree on all physics will in general report different b; the value $b = \frac{1}{180}$ in (64.258) is the value delivered by zeta-function (equivalently minimal-subtraction heat-kernel) regularization with no R^2 counterterm added. Nothing physical depends on it. It is sometimes set to zero by fiat—by including precisely the counterterm $\alpha = \frac{1}{12\cdot 180}$—and this is legitimate, which is exactly why b is not a true central charge.

These statements are summarized, and extended to the other free fields, in Table 64.1. The fermion and vector entries follow from the *identical* heat-kernel computation applied to the relevant operator, with the general coefficient (64.252) replaced by its matrix-valued version $\frac{1}{2}\,\mathrm{tr}\,X^2 - \frac{1}{6}\,\mathrm{tr}(RX) - \cdots + \frac{1}{12}\,\mathrm{tr}(\Omega_{\mu\nu}\Omega^{\mu\nu})$,

where $\Omega_{\mu\nu} = [\nabla_\mu, \nabla_\nu]$ is the curvature of the field's connection and the trace runs over the internal/spin index. For a Dirac fermion the squared operator is the Lichnerowicz operator $\not{D}^2 = -\Box + \frac{1}{4}R$ (so $X = -\frac{1}{4}R$, with a Dirac trace $\operatorname{tr}\mathbf{1} = 4$) and $\Omega_{\mu\nu} = \frac{1}{4}R_{\mu\nu\rho\sigma}\gamma^\rho\gamma^\sigma$; for a Maxwell field the relevant operator on the transverse photon carries $\Omega_{\mu\nu}$ equal to the Riemann tensor acting on the vector index. Plugging these into the matrix-valued a_2 and taking the traces—a longer but mechanically identical computation—yields the tabulated values.

Table 64.1: Trace-anomaly central charges of free conformal fields in $d = 4$, in the normalization $\langle T^\mu{}_\mu \rangle = \frac{1}{(4\pi)^2}(c\,W^2 - a\,E_4 + b\,\Box R)$. Per real scalar, per Weyl (two-component) fermion, per real vector. The scalar row is computed in full in this section, (64.258); the others by the identical method.

Field	a (type A, Euler)	c (type B, Weyl)	b ($\Box R$, scheme)
real scalar ($\xi = \frac{1}{6}$)	$\frac{1}{360}$	$\frac{1}{120}$	$\frac{1}{180}$
Weyl fermion	$\frac{11}{720}$	$\frac{1}{40}$	0
real vector (gauge boson)	$\frac{31}{180}$	$\frac{1}{10}$	$-\frac{1}{15}$

Worked Example 64.32 (The two-dimensional anomaly and the Virasoro central charge)**.** The cleanest contact with the conformal-field-theory machinery of Chapters 47 and 48 is in two dimensions, where the same heat-kernel formula delivers the central charge c of the Virasoro algebra directly. In $d = 2$ the relevant coefficient is $a_{d/2} = a_1 = [b_1]$, the t^1 term, and the analogue of (64.237) is $\langle T^\mu{}_\mu \rangle = \frac{1}{4\pi}a_1(x)$ (now $(4\pi)^{d/2} = 4\pi$). A massless scalar in two dimensions is automatically conformal at $\xi_2 = \frac{d-2}{4(d-1)} = 0$: the curvature coupling switches off, $X = 0$, and (64.247) gives

$$a_1(x) = [b_1] = \frac{1}{6}R - X = \frac{1}{6}R. \tag{64.269}$$

Hence the two-dimensional trace anomaly of a single free boson is

$$\langle T^\mu{}_\mu \rangle = \frac{1}{4\pi}\cdot\frac{R}{6} = \frac{R}{24\pi}. \tag{64.270}$$

This is the celebrated result $\langle T^\mu{}_\mu \rangle = \frac{c}{24\pi}R$ with central charge $c = 1$ for one real scalar—exactly the Virasoro central charge of the free-boson conformal field theory of Chapter 48, where the same number appears in the commutator $[L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12}m(m^2-1)\delta_{m+n,0}$. The Weyl anomaly and the central extension of the conformal algebra are two faces of the same coefficient: the obstruction to making the stress tensor traceless on a curved background is identical to the obstruction to representing the conformal algebra without a central charge. The four-dimensional a and c of (64.258) are the higher-dimensional descendants of this single two-dimensional c, and the cleanness of

the two-dimensional case—one number, computed from $[b_1]$ in one line—is what makes it the universal entry point to the anomaly.

Worked Example 64.33 (Pure type-A anomaly on the four-sphere)**.** To see the type-A/type-B distinction at work, evaluate (64.258) on the round Euclidean four-sphere S^4, the maximally symmetric compact space. There the Weyl tensor vanishes identically (any maximally symmetric space is conformally flat, $C_{\mu\nu\rho\sigma} = 0$), so $W^2 = 0$ and the type-B contribution is switched off; the curvature is covariantly constant, $\nabla_\rho R_{\mu\nu} = 0$, so $\Box R = 0$ and the scheme-dependent term also drops. Only the topological type-A term survives:

$$\langle T^{\mu}{}_{\mu} \rangle_{S^4} = -\frac{1}{(4\pi)^2}\,\frac{1}{360}\,E_4. \tag{64.271}$$

Integrating over the sphere and using the Gauss–Bonnet formula (64.228) with $\chi(S^4) = 2$,

$$\int d^4x\,\sqrt{g}\;\langle T^{\mu}{}_{\mu} \rangle_{S^4} = -\frac{1}{(4\pi)^2}\,\frac{1}{360}\int d^4x\,\sqrt{g}\;E_4 = -\frac{1}{16\pi^2}\,\frac{1}{360}\,(32\pi^2 \cdot 2) = -\frac{1}{90}. \tag{64.272}$$

The integrated trace anomaly of a conformal scalar on S^4 is the pure number $-\frac{1}{90} = -4a$, independent of the radius of the sphere: rescaling the sphere is a Weyl transformation, and the response is fixed entirely by the Euler characteristic. This is the type-A anomaly in its purest form—a topological invariant of the background times the central charge a—and it is the quantity whose monotonic decrease under RG flow is the content of the a-theorem.

64.6.7 The anomaly is intrinsic

Two regularizations that share no machinery—the spectral determinant and the functional measure—have returned the same local density $a_2(x)/(4\pi)^2$, and the Wess–Zumino condition has told us in advance that its physical content is the two scheme-independent numbers a and c. The trace anomaly is therefore not an artifact of how we computed: it is a fixed, geometric property of the quantum effective action $W[g]$ itself, the statement that $W[g]$ cannot be made Weyl invariant. This recasts the entire chapter. The renormalized stress tensor of Section 64.5, the particle creation of the earlier sections, and the anomaly just computed are all read off from one object, $W[g]$, and its variations. The next section assembles that object explicitly—the curved-space effective action and its Euclidean path integral—and shows that for a stationary spacetime with a horizon the demand that this single functional be smooth selects exactly one temperature. That is the road to Hawking radiation, and it is the anomaly's lesson—geometry fixes the quantum response—carried to its sharpest conclusion.

64.7 The effective action, the Euclidean path integral, and the road to Hawking

The trace anomaly of the previous section delivered a verdict with consequences far beyond the single number it produced. It showed that the act of quantizing a field on a curved background generates physics that is not present in any term of the classical action and cannot be removed by any local counterterm: a conformally invariant classical theory acquires a nonzero $\langle T^{\mu}{}_{\mu} \rangle$ fixed entirely by the local geometry. The natural home of such an irreducible curved-space effect is not the field, nor any single correlator, but the object that packages *all* of them at once — the effective action, the logarithm of the partition function regarded as a functional of the metric. Everything this chapter has built — the mode expansions and Bogoliubov coefficients of the early sections, the Unruh temperature of Section 64.4, the renormalized stress tensor of Section 64.5, and the anomaly — is a derivative, in the literal functional sense, of that one functional. This closing section assembles them into it.

Having done so, we extract the single principle that turns the whole apparatus into a prediction. The Euclidean partition function on a static background is a thermodynamic generating functional, and its inverse temperature is encoded geometrically as the period of imaginary time. When the background has a horizon, that period is not free: continued to imaginary time, a stationary spacetime with a horizon is a smooth manifold at exactly one temperature, $\beta = 2\pi/\kappa$, with κ the surface gravity. This is the entire content of Hawking's result stripped of dynamics; nothing quantum enters it beyond the decision to compute a partition function at all. We then write the equation by which this quantum stress tensor reacts back on the geometry — the semiclassical Einstein equation — and state, with the honesty the frontier demands, exactly when we are entitled to trust it. With that, the baton passes to Chapter 65.

64.7.1 The curved-space effective action and the master relation

Fix a globally hyperbolic spacetime $(\mathcal{M}, g_{\mu\nu})$ and a quantum field — take the real scalar of ϕ^4 theory in its free ($\lambda = 0$) sector, conformally or minimally coupled, as the running example — whose classical action $S[\phi, g]$ depends on both the field and the background metric. In flat space the central object of Chapter 10 was the generating functional $Z[J]$ of Theorem 10.11, a functional of an external source J. On a curved background the metric *is* the source: it couples to the stress tensor exactly as J couples to ϕ, and varying it probes the response of the vacuum. We therefore define the curved-space generating functional by integrating out the field at fixed geometry,

$$Z[g] \;=\; \langle \text{out} | \text{in} \rangle_{g} \;=\; \int \mathcal{D}\phi \, e^{\,iS[\phi, g]}, \tag{64.273}$$

the amplitude for the in-vacuum prepared in the asymptotic past to evolve, on the geometry g, into the out-vacuum of the asymptotic future. (When the two vacua differ — precisely the situation of the particle-creation sections of this chapter — this overlap is less than unity, and we shall read off below how the effective action records that fact.) The object that organizes the theory is its logarithm.

Definition 64.34 (In–out effective action)**.** The *in–out effective action* of the field on the background g is

$$W[g] \;=\; -\,i \ln Z[g], \tag{64.274}$$

the curved-space, metric-sourced counterpart of the connected generating functional $W[J]$ of Theorem 10.36, defined with the same factor of $-i$ adapted to the Minkowski weight e^{iS}.

The single relation that makes $W[g]$ the right object is what its first variation computes. Recall the curved-space (Hilbert, or metric) stress tensor, the operator whose classical version was introduced when this chapter set up fields on a background: it is the response of the matter action to a deformation of the metric,

$$T_{\mu\nu} \;\equiv\; \frac{2}{\sqrt{-g}}\,\frac{\delta S[\phi,g]}{\delta g^{\mu\nu}}, \qquad \text{equivalently} \qquad \delta S \;=\; \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\;T_{\mu\nu}\,\delta g^{\mu\nu} \tag{64.275}$$

under a variation $\delta g^{\mu\nu}$ of the inverse metric, at fixed field. This is the same tensor whose renormalized expectation value Section 64.5 extracted from the coincidence limit of the propagator; here it appears as a derivative with respect to the metric, and the two faces will be reconciled in a moment.

Theorem 64.35 (The master relation)**.** *The first metric-variation of the in–out effective action is the in–out expectation value of the Hilbert stress tensor:*

$$\frac{\delta W[g]}{\delta g^{\mu\nu}(x)} \;=\; \frac{1}{2}\,\sqrt{-g}\;\langle T_{\mu\nu}(x)\rangle\,, \qquad \langle T_{\mu\nu}\rangle \equiv \frac{\langle \mathrm{out}|T_{\mu\nu}|\mathrm{in}\rangle}{\langle \mathrm{out}|\mathrm{in}\rangle}. \tag{64.276}$$

Proof. Vary Eq. (64.274) with respect to the inverse metric. Because $W = -i\ln Z$,

$$\delta W \;=\; -\,i\,\frac{\delta Z}{Z} \;=\; -\,i\,\frac{1}{Z}\int \mathcal{D}\phi\;\big(i\,\delta S[\phi,g]\big)\,e^{\,iS[\phi,g]} \;=\; \frac{1}{Z}\int \mathcal{D}\phi\;\delta S[\phi,g]\;e^{\,iS[\phi,g]}, \tag{64.277}$$

where the two factors of i have cancelled. The right-hand side is, by the definition of correlators as normalized weighted field-averages established in Chapter 10, the in–out matrix element of δS:

$$\delta W \;=\; \frac{\langle \mathrm{out}|\delta S|\mathrm{in}\rangle}{\langle \mathrm{out}|\mathrm{in}\rangle} \;=\; \langle \delta S\rangle\,. \tag{64.278}$$

The classical variation δS is given by Eq. (64.275); promoting the field bilinear in $T_{\mu\nu}$ to the operator and taking the in–out average term by term,

$$\langle \delta S \rangle \;=\; \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\;\langle T_{\mu\nu}(x)\rangle\;\delta g^{\mu\nu}(x). \tag{64.279}$$

Since $\delta g^{\mu\nu}(x)$ is arbitrary, comparing Eq. (64.279) with $\delta W = \int \mathrm{d}^4x\,(\delta W/\delta g^{\mu\nu})\,\delta g^{\mu\nu}$ gives Eq. (64.276). □

This is the chapter in one line. The stress tensor that gravitates is the metric-derivative of the effective action, and so the renormalization program of Section 64.5 — which made $\langle T_{\mu\nu}\rangle$ finite by subtracting local geometric counterterms — is identically the statement that $W[g]$ is rendered finite by adding the same local counterterms (built from Λ, R, R^2, $R_{\mu\nu}R^{\mu\nu}$, $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}$) to the bare effective action. Renormalizing the stress tensor and renormalizing the effective action are one operation seen through one derivative. The trace anomaly of the previous section is, in the same dictionary, the statement that the renormalized $W[g]$ is *not* invariant under a Weyl rescaling $g_{\mu\nu}\to e^{2\omega}g_{\mu\nu}$ even when the classical S is: its non-invariance is precisely $\int \mathrm{d}^4x\,\sqrt{-g}\,\omega\,\langle T^{\mu}{}_{\mu}\rangle$, with $\langle T^{\mu}{}_{\mu}\rangle$ the anomaly density computed in Section 64.6.

Worked Example 64.36 (The free-field effective action is a functional determinant)**.** For the free scalar the field integral Eq. (64.273) is Gaussian and can be done exactly, exhibiting $W[g]$ as a single geometric object. Write the (minimally coupled, for definiteness; the conformal coupling adds ξR) action as a quadratic form,

$$S[\phi,g] \;=\; \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,\Big[\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi - m^2\phi^2\,\Big] \;=\; -\frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\;\phi\,(\Box_g + m^2)\phi, \tag{64.280}$$

the second equality being one covariant integration by parts, with $\Box_g = g^{\mu\nu}\nabla_\mu\nabla_\nu = (1/\sqrt{-g})\,\partial_\mu(\sqrt{-g}\,g^{\mu\nu}\partial_\nu)$ the curved-space d'Alembertian. The Gaussian functional integral of Section 10.4.5 then evaluates Eq. (64.273) to the inverse square root of the determinant of the kinetic operator,

$$Z[g] \;=\; \big[\det(\Box_g + m^2 - i\epsilon)\big]^{-1/2}, \qquad W[g] \;=\; -i\ln Z[g] \;=\; \frac{i}{2}\,\mathrm{Tr}\ln(\Box_g + m^2 - i\epsilon), \tag{64.281}$$

using $\ln\det = \mathrm{Tr}\ln$, with the $-i\epsilon$ the curved-space inheritance of the Feynman prescription. This is the one-loop effective action of the free field on the background. Three observations close the example. First, Eq. (64.281) is precisely the kind of ratio-of-determinants whose finite definition (zeta-function or heat-kernel regularization) was flagged as a genuine computation in Eq. (10.120) and carried out in Chapter E; on a curved background the proper-time expansion of $\mathrm{Tr}\ln$ in powers of curvature is the DeWitt–Schwinger expansion, whose coefficients are

the local geometric invariants that supply the counterterms. Second, its metric-derivative reproduces the master relation directly: differentiating $\frac{i}{2}\,\mathrm{Tr}\ln(\Box_g+m^2)$ with respect to $g^{\mu\nu}$ brings down the coincidence limit of the Green's function $\langle\phi\,\phi\rangle$, which is exactly the point-split bilinear whose finite part is the renormalized $\langle T_{\mu\nu}\rangle$ of Section 64.5. The two routes to $\langle T_{\mu\nu}\rangle$ — vary the effective action, or point-split the propagator — are the same computation. Third, the imaginary part of Eq. (64.281) is physical and is the bridge back to the start of the chapter: from $|\,\langle\mathrm{out}|\mathrm{in}\rangle\,|^2 = |e^{iW}|^2 = e^{-2\,\mathrm{Im}\,W}$ we read that

$$|\,\langle\mathrm{out}|\mathrm{in}\rangle_g\,|^2 \;=\; \exp\!\big(-2\,\mathrm{Im}\,W[g]\big), \tag{64.282}$$

the vacuum persistence probability. A nonzero $\mathrm{Im}\,W$ is the statement that the in-vacuum does not survive: it is the integrated probability of pair creation, the effective-action face of the Bogoliubov coefficients $|\beta_k|^2$ computed in the early sections of this chapter. Particle creation, the anomaly, and the renormalized stress tensor are thus three derivatives — imaginary part, Weyl variation, and metric variation — of the single functional $W[g]$.

Caution — In–out versus in–in: which expectation value gravitates

The matrix element $\langle T_{\mu\nu}\rangle = \langle\mathrm{out}|T_{\mu\nu}|\mathrm{in}\rangle\,/\,\langle\mathrm{out}|\mathrm{in}\rangle$ produced by Theorem 64.35 is a *transition* amplitude between two different states, and when the vacua differ it is in general complex — its imaginary part is the pair-creation rate of Eq. (64.282). It is exactly what one wants for computing S-matrix elements and vacuum decay, but it is *not* the real, causal expectation value $\langle\psi|T_{\mu\nu}|\psi\rangle$ in a fixed physical state $|\psi\rangle$ that one must feed to Einstein's equation. The object that delivers the genuine expectation value is the *in–in*, or closed-time-path (Schwinger–Keldysh), effective action $W_{\mathrm{CTP}}[g_+,g_-]$: one doubles the time contour, runs the field forward on a metric g_+ and backward on g_-, both ending on the same state, and reads

$$\langle\psi|T_{\mu\nu}(x)|\psi\rangle \;=\; \frac{2}{\sqrt{-g}}\,\frac{\delta W_{\mathrm{CTP}}}{\delta g_+^{\mu\nu}(x)}\bigg|_{g_+=g_-=g}, \tag{64.283}$$

which is automatically real and causal because the doubled contour cancels the disconnected phase that made Eq. (64.276) complex. The renormalized $\langle T_{\mu\nu}\rangle$ of Section 64.5 is this in–in object in a specified state; it is the one that sources gravity below. We use the in–out functional $W[g]$ as the cleaner pedagogical carrier of the master relation, and flag here that the source of the semiclassical Einstein equation is its in–in refinement.

64.7.2 The Euclidean partition function as a thermodynamic generating functional

The most consequential reading of $W[g]$ comes from rotating it to imaginary time. The dictionary of Section 10.6.5 taught that the Wick-rotated generating functional is a statistical-mechanical partition function: under $t = -i\tau$ the Minkowski weight e^{iS} becomes the Euclidean weight e^{-S_E} of Eq. (10.160), and Eq. (12.138) sharpened this into the operator identity

$$Z(\beta) \;=\; \mathrm{Tr}\; e^{-\beta \hat{H}} \;=\; \oint_{\phi(\tau+\beta)=\phi(\tau)} \mathcal{D}\phi\; e^{-S_E[\phi]}, \tag{64.284}$$

the thermal partition function at inverse temperature β, computed as a Euclidean path integral over fields living on a time circle of circumference β, with bosons glued periodically. The periodicity is not a convention: by the Kubo–Martin–Schwinger argument of Eq. (12.139), periodicity of every thermal correlator in imaginary time with period β is forced by cyclicity of the trace, and is the defining property of thermal equilibrium. *Compactness of the Euclidean time circle is temperature, and its circumference is the inverse temperature.*

On a curved (but static) background the same construction goes through. A static metric possesses a timelike Killing vector ∂_t and a foliation by surfaces of constant t; the Euclidean continuation $t = -i\tau$ produces a Riemannian manifold, and identifying $\tau \sim \tau + \beta$ computes the partition function of the field in the static gravitational potential at temperature $1/\beta$. The curved-space effective action evaluated on this Euclidean section,

$$W_E(\beta) \;=\; -\ln Z(\beta), \qquad Z(\beta) = \int \mathcal{D}\phi\; e^{-S_E[\phi, g_E]}, \tag{64.285}$$

is therefore the free energy in units of β: the thermodynamic free energy is $F(\beta) = -\beta^{-1} \ln Z(\beta) = \beta^{-1} W_E(\beta)$, and every equilibrium thermodynamic quantity follows by differentiating it. In particular the thermal expectation value of the stress tensor is the metric-derivative of $\ln Z$ on the Euclidean section — the Euclidean, finite-temperature instance of the master relation Eq. (64.276),

$$\langle T_{\mu\nu} \rangle_\beta \;=\; -\frac{2}{\sqrt{g_E}} \frac{\delta \ln Z(\beta)}{\delta g_E^{\mu\nu}}. \tag{64.286}$$

This is the formula that unifies the chapter. The Unruh temperature of Section 64.4 was derived by recognizing that the Rindler vacuum's correlators are periodic in imaginary Rindler time — exactly the KMS condition Eq. (12.139) — and so represent a thermal state; Eq. (64.286) says that the very same periodicity, fed through the partition function, computes the thermal $\langle T_{\mu\nu} \rangle$ that a uniformly accelerated detector registers and that Section 64.5 rendered finite. The mode sums of the Bogoliubov sections, the detector response of the Unruh section, and the point-split stress tensor of the renormalization section are three computations of the derivatives of one Euclidean free energy.

Worked Example 64.37 (Free energy to thermal stress tensor: the Stefan–Boltzmann law from periodicity)**.** Nothing demonstrates the chain "periodicity ⇒ free energy ⇒ thermal $\langle T_{\mu\nu}\rangle$" like running it once to a known number. Take the free massless scalar on flat space, but with Euclidean time compactified on a circle of circumference β — the simplest curved-from-flat geometry, $S^1_\beta \times \mathbb{R}^3$, whose only "curvature" is the global identification that Eq. (64.284) reads as temperature $T = 1/\beta$. The partition function is the standard bosonic one; per spatial volume V, the free-energy density is the momentum-space sum over the single bosonic mode,

$$f(T) \;=\; \frac{F}{V} \;=\; T\int \frac{\mathrm{d}^3k}{(2\pi)^3}\,\ln\!\big(1-e^{-|\mathbf{k}|/T}\big), \tag{64.287}$$

the vacuum (zero-point) part having been subtracted as the T-independent piece that renormalization removes. The angular integral is trivial, $\int \mathrm{d}^3k/(2\pi)^3 = (2\pi^2)^{-1}\int_0^\infty k^2\,\mathrm{d}k$; substituting $x = k/T$,

$$f(T) \;=\; \frac{T}{2\pi^2}\int_0^\infty \mathrm{d}k\; k^2 \ln\!\big(1-e^{-k/T}\big) \;=\; \frac{T^4}{2\pi^2}\int_0^\infty \mathrm{d}x\; x^2\ln\!\big(1-e^{-x}\big). \tag{64.288}$$

Expand the logarithm, $\ln(1-e^{-x}) = -\sum_{n\ge 1} n^{-1}e^{-nx}$, and use $\int_0^\infty x^2 e^{-nx}\,\mathrm{d}x = 2/n^3$:

$$\int_0^\infty \mathrm{d}x\; x^2\ln\!\big(1-e^{-x}\big) \;=\; -\sum_{n=1}^{\infty}\frac{1}{n}\cdot\frac{2}{n^3} \;=\; -2\sum_{n=1}^{\infty}\frac{1}{n^4} \;=\; -2\,\zeta(4) \;=\; -\frac{\pi^4}{45}, \tag{64.289}$$

with $\zeta(4) = \pi^4/90$. Hence the free-energy density of one massless bosonic degree of freedom is

$$f(T) \;=\; -\frac{\pi^2}{90}\,T^4. \tag{64.290}$$

The thermodynamics follows by differentiation. The entropy density is $s = -\partial f/\partial T = (2\pi^2/45)\,T^3$, the pressure is $p = -f = (\pi^2/90)\,T^4$, and the energy density is the Legendre combination $\rho = f + Ts$,

$$\rho \;=\; -\frac{\pi^2}{90}T^4 + T\cdot\frac{2\pi^2}{45}T^3 \;=\; \Big(-\frac{1}{90}+\frac{4}{90}\Big)\pi^2T^4 \;=\; \frac{\pi^2}{30}\,T^4, \tag{64.291}$$

the Stefan–Boltzmann law for a single scalar (the photon's two polarizations give the familiar $\rho = \pi^2T^4/15$). This ρ is exactly $\langle T_{00}\rangle_\beta$ delivered by Eq. (64.286): the thermal stress tensor in the local rest frame is $\langle T^\mu{}_\nu\rangle_\beta = \mathrm{diag}(\rho,-p,-p,-p)$, and its trace is

$$\langle T^\mu{}_\mu\rangle_\beta \;=\; \rho - 3p \;=\; \frac{\pi^2}{30}T^4 - 3\cdot\frac{\pi^2}{90}T^4 \;=\; 0, \tag{64.292}$$

vanishing because the massless scalar is classically conformal — the thermal state respects the symmetry that, on a generic curved background, the anomaly

of Section 64.6 breaks. We have travelled from a single geometric datum, the circumference β of the Euclidean time circle, to the energy density of thermal radiation, using nothing but the partition function and one differentiation. The same machine, applied to the near-horizon geometry below, will turn a surface gravity into a temperature.

64.7.3 The smoothness principle: $\beta = 2\pi/\kappa$ from regularity of the Euclidean section

The free energy of the last example was computed at an arbitrary temperature; β was a free parameter we dialled at will. The decisive observation — the one that makes Hawking radiation inevitable rather than optional — is that when the static background has a *horizon*, the temperature is no longer free. The Euclidean section is smooth at only one value of β, and that value is dictated by the surface gravity.

Consider a static, spherically symmetric spacetime with metric of the form

$$\mathrm{d}s^2 \;=\; f(r)\,\mathrm{d}t^2 - \frac{\mathrm{d}r^2}{f(r)} - r^2\,\mathrm{d}\Omega^2, \tag{64.293}$$

in the mostly-minus signature, with a simple zero of f at $r = r_h$, so $f(r_h) = 0$ and $f'(r_h) > 0$. The surface $r = r_h$ is a Killing horizon: the norm $f(r)$ of the timelike Killing vector ∂_t vanishes there, the surface is generated by a bifurcate Killing field, and its *surface gravity* is the geometric invariant

$$\kappa \;=\; \tfrac{1}{2}\,f'(r_h), \tag{64.294}$$

the value at which the Killing field's acceleration, measured per unit Killing time, would have to be applied to hold a particle just outside the horizon. We now ask what the Euclidean continuation of Eq. (64.293) looks like in a neighbourhood of r_h, and demand that it be a smooth manifold there.

Wick-rotate $t = -i\tau$. The metric Eq. (64.293) becomes the Riemannian

$$\mathrm{d}s_E^2 \;=\; f(r)\,\mathrm{d}\tau^2 + \frac{\mathrm{d}r^2}{f(r)} + r^2\,\mathrm{d}\Omega^2, \tag{64.295}$$

positive-definite outside the horizon, $r > r_h$. The horizon $r = r_h$, where $f = 0$, is not a surface in the Euclidean manifold: the $\mathrm{d}\tau^2$ term collapses and the would-be horizon is a single point (more precisely, a fixed point of the τ-translation), at which the entire region $r < r_h$ — the Euclidean interior of the black hole — is simply absent. The Euclidean black hole terminates at the horizon; the only question is whether it closes off smoothly or comes to a singular tip. Expand f to first order about r_h,

$$f(r) \;\approx\; f'(r_h)\,(r - r_h) \;=\; 2\kappa\,(r - r_h), \tag{64.296}$$

using Eq. (64.294). Trade the coordinate distance for the proper radial distance ρ from the horizon,

$$\rho \;=\; \int_{r_h}^{r} \frac{\mathrm{d}r'}{\sqrt{f(r')}} \;=\; \int_{0}^{r-r_h} \frac{\mathrm{d}u}{\sqrt{2\kappa u}} \;=\; \sqrt{\frac{2(r-r_h)}{\kappa}}, \tag{64.297}$$

so that $r - r_h = \frac{1}{2}\kappa\rho^2$ and therefore $f(r) = 2\kappa \cdot \frac{1}{2}\kappa\rho^2 = \kappa^2\rho^2$. The radial term becomes $\mathrm{d}r^2/f = \mathrm{d}\rho^2$ exactly, by construction of ρ. The angular sphere has nearly fixed radius $r \approx r_h$ and is an inert spectator. The interesting (τ, ρ) block of Eq. (64.295) collapses near the horizon to

$$\mathrm{d}s_E^2\big|_{\text{near horizon}} \;=\; \mathrm{d}\rho^2 + \kappa^2\rho^2\,\mathrm{d}\tau^2 + r_h^2\,\mathrm{d}\Omega^2 \;=\; \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\theta^2 + r_h^2\,\mathrm{d}\Omega^2, \qquad \theta \equiv \kappa\,\tau, \tag{64.298}$$

where in the last step we rescaled the imaginary-time coordinate into the dimensionless angle $\theta = \kappa\tau$. The (ρ, θ) part is unmistakable: it is the flat two-dimensional plane written in polar coordinates, with radial coordinate $\rho \geq 0$ and angular coordinate θ.

A plane in polar coordinates is a smooth manifold at the origin $\rho = 0$ if and only if the angular coordinate runs over a full turn of 2π and exactly 2π. If θ ranges over any other period $\Delta\theta$, the geometry near $\rho = 0$ is a cone: a wedge of opening angle $2\pi - \Delta\theta$ is either excised (a deficit, for $\Delta\theta < 2\pi$) or duplicated (an excess), and the tip carries a delta-function of Gaussian curvature equal to the deficit angle. A cone is flat everywhere except at its apex, where it is singular; only the plane, $\Delta\theta = 2\pi$, has no tip at all. This elementary fact of two-dimensional geometry is the whole argument.

Theorem 64.38 (Horizon temperature from smoothness)**.** *The Euclidean section Eq. (64.295) of a static spacetime with a Killing horizon of surface gravity κ extends to a smooth Riemannian manifold across the horizon $\rho = 0$ if and only if the imaginary-time coordinate is periodically identified with period*

$$\boxed{\beta \;=\; \frac{2\pi}{\kappa}} \qquad \Longleftrightarrow \qquad T \;=\; \frac{\kappa}{2\pi}. \tag{64.299}$$

For any other period the manifold has a conical singularity at the horizon with deficit angle $\delta = 2\pi(1 - \beta\kappa/2\pi)$. Consequently a stationary spacetime with a horizon, continued to imaginary time, is regular at exactly one temperature, the universal horizon temperature $T = \kappa/2\pi$.

Proof. By Eq. (64.298) the near-horizon Euclidean geometry is $\mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\theta^2$ with $\theta = \kappa\tau$. Smoothness at $\rho = 0$ requires θ to be a genuine angular coordinate of period 2π, i.e. $\theta \sim \theta + 2\pi$. Since $\theta = \kappa\tau$, this is the condition $\tau \sim \tau + 2\pi/\kappa$, that is, $\beta = 2\pi/\kappa$. If instead $\tau \sim \tau + \beta$ with $\beta \neq 2\pi/\kappa$, then $\theta = \kappa\tau$ has period $\kappa\beta \neq 2\pi$, and the (ρ, θ) plane is a cone of deficit angle $\delta = 2\pi - \kappa\beta = 2\pi(1 - \beta\kappa/2\pi)$. The Gaussian curvature of such a cone is a delta function $2\pi\,\delta\,\delta^{(2)}(\rho)$ concentrated at

the tip; the four-dimensional Ricci scalar inherits this conical curvature singularity supported on the bifurcation surface, so the manifold fails to be smooth precisely when $\beta \neq 2\pi/\kappa$. The temperature $T = 1/\beta = \kappa/2\pi$ is therefore the unique value for which the Euclidean section is a smooth, complete manifold. □

smooth: $\beta = 2\pi/\kappa$
($\theta = \kappa\tau$ closes at 2π)

conical: $\beta \neq 2\pi/\kappa$
(deficit $\delta = 2\pi - \kappa\beta$ at tip)

Figure 64.7: The (ρ, θ) section of the Euclidean horizon, $\mathrm{d}s^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\theta^2$ with $\theta = \kappa\tau$. *Left:* when τ has period $\beta = 2\pi/\kappa$ the angle θ closes at exactly 2π and the section is a smooth disk, regular at the tip $\rho = 0$ (the bifurcation surface). *Right:* any other period leaves a wedge of deficit angle $\delta = 2\pi - \kappa\beta$ missing, and the tip is a conical curvature singularity. Regularity selects the single temperature $T = \kappa/2\pi$.

The force of Theorem 64.38 is that it is purely geometric. No field was expanded in modes, no vacuum was chosen, no Bogoliubov coefficient was computed; the temperature is fixed by the differential geometry of a smooth four-manifold. Combined with the partition-function reading of Section 64.7.2 — that the period of Euclidean time *is* the inverse temperature of the field living on the background — it forces the conclusion: a quantum field in equilibrium on a spacetime with a horizon of surface gravity κ is at temperature $\kappa/2\pi$, because that is the only temperature at which the Euclidean partition function it computes is defined on a nonsingular manifold. Any other temperature would have the field propagating on a cone, sensing a curvature singularity at the horizon that is not there in the physical Lorentzian geometry.

Worked Example 64.39 (Rindler as the special case: recovering the Unruh temperature)**.** The cleanest test of Theorem 64.38 is the case the chapter has already solved by hand. The Rindler wedge of Section 64.4 is the near-horizon limit of *any* of these geometries — it is what Eq. (64.298) *is*, with the angular sphere stripped away — and its horizon is the acceleration horizon of a uniformly accelerated observer. For proper acceleration a, the Rindler metric in the mostly-minus signature is

$$\mathrm{d}s^2 \;=\; a^2\xi^2\,\mathrm{d}\eta^2 - \mathrm{d}\xi^2 - \mathrm{d}\mathbf{x}_\perp^2, \tag{64.300}$$

where ξ is the proper distance from the horizon $\xi = 0$ and η is the dimensionless Rindler time conjugate to the boost. Comparing with the static form, the horizon at $\xi = 0$ has surface gravity equal to the proper acceleration: writing the boost Killing field ∂_η and computing its non-affinity gives $\kappa = a$ directly, and indeed Eq. (64.298) with $\rho \leftrightarrow \xi$ and $\kappa\tau \leftrightarrow a\eta_E$ is identical to the Euclidean continuation of Eq. (64.300). The Wick rotation $\eta = -i\eta_E$ turns Eq. (64.300) into $\mathrm{d}s_E^2 = a^2\xi^2\,\mathrm{d}\eta_E^2 + \mathrm{d}\xi^2 + \mathrm{d}\mathbf{x}_\perp^2$, which is flat $\mathbb{R}^2 \times \mathbb{R}^2$ in polar coordinates $(\xi, a\eta_E)$ — and the plane is smooth iff the polar angle $a\eta_E$ closes at 2π, i.e. iff $\eta_E \sim \eta_E + 2\pi/a$. By Eq. (64.299) with $\kappa = a$, the temperature is

$$T_{\mathrm{U}} \;=\; \frac{\kappa}{2\pi} \;=\; \frac{a}{2\pi}, \tag{64.301}$$

the Unruh temperature, recovered here from a single demand — that Euclidean Minkowski space be the smooth plane it manifestly is — rather than from the mode-by-mode Bogoliubov computation of Section 64.4. The two derivations agree because they are the same statement seen from two sides: the Bogoliubov coefficients of the Rindler modes are thermal at $a/2\pi$ precisely because the Minkowski vacuum's correlators are periodic in imaginary Rindler time with period $2\pi/a$, and that periodicity is the smoothness of the Euclidean plane. Rindler is the flat, $\kappa = a$ instance of the universal horizon formula; the black hole is the curved instance that Chapter 65 will treat, where the same Eq. (64.299) promotes the surface gravity $\kappa = 1/4GM$ to the Hawking temperature $T_H = 1/8\pi GM$.

64.7.4 The semiclassical Einstein equation and the backreaction program

We have, on one side, a finite renormalized stress tensor $\langle T_{\mu\nu}\rangle_{\mathrm{ren}}$ that the quantum field carries on any background; on the other, the geometry that the field lives on. The closing structural step of the chapter is to let the two react on each other. The principle is the one that produced Einstein's equation classically: extremize the total action over the metric. Here the total action is the gravitational action plus the quantum effective action of the matter,

$$\Gamma_{\mathrm{tot}}[g] \;=\; S_{\mathrm{grav}}[g] + W[g], \qquad S_{\mathrm{grav}}[g] \;=\; \frac{1}{16\pi G}\int \mathrm{d}^4x\,\sqrt{-g}\,(R - 2\Lambda), \tag{64.302}$$

with G Newton's constant and Λ the cosmological constant. The matter has been integrated out: $W[g]$ already contains the full quantum dynamics of the field as a functional of the metric, and its metric-derivative is the stress tensor by the master relation. Stationarity of Γ_{tot} under $\delta g^{\mu\nu}$ gives the field equation of the metric. The gravitational variation is the standard Einstein–Hilbert one,

$$\frac{\delta S_{\mathrm{grav}}}{\delta g^{\mu\nu}} \;=\; \frac{\sqrt{-g}}{16\pi G}\,(G_{\mu\nu} + \Lambda\, g_{\mu\nu}), \qquad G_{\mu\nu} = R_{\mu\nu} - \tfrac{1}{2}R\, g_{\mu\nu}, \tag{64.303}$$

with the curvature conventions of Chapter A, and the matter variation is the in–in master relation of Section 64.7.1, $\delta W/\delta g^{\mu\nu} = \tfrac{1}{2}\sqrt{-g}\,\langle T_{\mu\nu}\rangle_{\rm ren}$. Setting the sum to zero yields the central equation of semiclassical gravity.

Definition 64.40 (Semiclassical Einstein equation)**.** A classical metric $g_{\mu\nu}$ and a quantum state $|\psi\rangle$ of the matter fields satisfy the *semiclassical Einstein equation* when

$$\boxed{G_{\mu\nu} + \Lambda\, g_{\mu\nu} \;=\; 8\pi G\,\langle T_{\mu\nu}\rangle_{\rm ren}} \tag{64.304}$$

holds, with $\langle T_{\mu\nu}\rangle_{\rm ren} \equiv \langle\psi|T_{\mu\nu}|\psi\rangle_{\rm ren}$ the renormalized expectation value of Section 64.5 in the state $|\psi\rangle$ on the background g.

Notation Watch — Curvature and stress-tensor signs

The relative sign on the two sides of Eq. (64.304) is the one fixed by the Newtonian limit — a positive energy density $\langle T_{00}\rangle > 0$ must produce attractive gravity. In the mostly-minus signature this requires a definite joint choice of the sign of the Riemann tensor (entering $G_{\mu\nu}$ via Eq. (64.303)) and of the Hilbert stress tensor Eq. (64.275); both are the conventions collected in Chapter A, chosen precisely so that the classical equation reads $G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$ with the Hilbert definition. We adopt them throughout; the only modification in passing to the semiclassical theory is the replacement $T_{\mu\nu} \to \langle T_{\mu\nu}\rangle_{\rm ren}$.

Equation Eq. (64.304) is the precise statement of what "quantum fields backreact on geometry" means in this chapter, and it deserves to be read carefully, because every word is a deliberate approximation. The left side is purely classical: the metric $g_{\mu\nu}$ is a c-number, a definite geometry, not an operator. The right side is the quantum expectation value of the stress-tensor operator in a chosen state, rendered finite by the geometric subtractions of Section 64.5. The equation couples a classical source-geometry to a quantum mean source. It is the gravitational analogue of the mean-field, or one-loop, treatment of any quantum system coupled to a slow classical variable: the fast quantum field is integrated out and replaced by its expectation value, which then drives the slow classical geometry. The cosmological constant Λ on the left is renormalized together with the vacuum energy on the right; the splitting between "geometric Λ" and "the constant piece of $\langle T_{\mu\nu}\rangle$" is a renormalization choice, and the failure to predict the physical remainder is the cosmological constant problem to which we return below.

The *backreaction program* is the iterative use of Eq. (64.304). One starts with a background geometry, computes $\langle T_{\mu\nu}\rangle_{\rm ren}$ on it in the state of interest, feeds that stress tensor back as a source, and solves for the corrected metric — repeating until self-consistency. For Hawking radiation the program is exactly the statement that the flux of created quanta carries energy away from the black

hole, so the right side of Eq. (64.304) has a nonzero outward energy flux at infinity, and the corrected left side describes a hole whose mass slowly decreases: the evaporation that Chapter 65 will compute. For cosmology it is the statement that quantum fields in an expanding universe carry an energy density that sources the expansion, the origin of particle-production-driven backreaction in the early universe.

64.7.5 Honest scope: where the semiclassical theory is trustworthy, and where it breaks

The semiclassical Einstein equation is an approximation, and a textbook that has insisted on honest rigor must say precisely what the approximation neglects and when the neglect is illegitimate. Two conditions must hold for Eq. (64.304) to be a controlled description, and a list of named situations violates them.

Condition one: curvatures small in Planck units. The derivation treated the metric as classical and the matter as a quantum field on it, with gravity itself unquantized. This is the leading term of an expansion in $\ell_P^2 = G$ (the Planck length squared, $G = 1/M_P^2$ with $M_P \approx 1.22 \times 10^{19}\,\mathrm{GeV}$): graviton loops, the quantization of $g_{\mu\nu}$ itself, contribute at relative order $G \times$ (curvature). The semiclassical equation is trustworthy only where the curvature invariants are far below the Planck scale, $|R_{\mu\nu\rho\sigma}| \ll \ell_P^{-2}$. For an astrophysical black hole the curvature at the horizon, $\sim 1/(GM)^2$, is fantastically sub-Planckian, and the approximation is superb; near the endpoint of evaporation, when the hole shrinks to the Planck mass, the curvature reaches ℓ_P^{-2} and the semiclassical treatment fails completely. This is the boundary at which Chapter 65's evaporation calculation must hand over to whatever the correct theory of quantum gravity is.

Condition two: small fluctuations of the stress tensor. Replacing the operator $T_{\mu\nu}$ by its expectation value $\langle T_{\mu\nu}\rangle$ is justified only when the operator does not fluctuate much about its mean — when the variance $\langle T_{\mu\nu}T_{\rho\sigma}\rangle - \langle T_{\mu\nu}\rangle\,\langle T_{\rho\sigma}\rangle$ is small compared to $\langle T_{\mu\nu}\rangle^2$ in the relevant smeared sense. Where the fluctuations are large the mean-field picture is meaningless: the geometry would respond to a quantity that is not sharp, and one needs either the in–in (stochastic-gravity) framework that promotes the fluctuations to a noise source driving the metric, or a genuinely quantum treatment of the geometry. This is the Kuo–Ford criterion [310], and it warns that the semiclassical equation can be self-consistent on average yet wrong in detail precisely in the regimes of greatest interest — few-quanta states, the late stages of evaporation, and any situation where $\langle T_{\mu\nu}\rangle$ nearly cancels.

Named breakdowns and open problems (as of the mid-2020s). Beyond these two conditions sit genuine unsettled issues, which we record honestly rather than paper over. First, *well-posedness*: equation Eq. (64.304) as literally written contains higher-derivative terms (the renormalized $\langle T_{\mu\nu}\rangle$ depends on R^2-type curvature combinations whose coefficients absorb the divergences),

and higher-derivative field equations generically admit runaway "Ostrogradsky" solutions; the standard resolution is to treat the higher-derivative pieces perturbatively via order-reduction [443], which is consistent within the regime of validity but signals that the equation is an effective, not fundamental, statement. Second, the *cosmological constant problem*: the vacuum energy in $\langle T_{\mu\nu}\rangle$ is quartically sensitive to the cutoff and must be cancelled against the bare Λ to roughly sixty decimal places to match the observed value; no symmetry of the semiclassical theory explains the smallness of the remainder, and this is among the sharpest unsolved problems in physics. Third, the *trans-Planckian problem* of Hawking radiation: tracing an outgoing late-time quantum back to the horizon assigns it an exponentially blueshifted, ultimately super-Planckian frequency in the early-time frame, so the derivation formally invokes the field theory in a regime where condition one fails; the robustness of the thermal spectrum to this — demonstrated by its insensitivity to modified dispersion relations and by the smoothness argument of Theorem 64.38, which never mentions high frequencies — is reassuring but not a closed proof, and Chapter 65 confronts it directly. Fourth, *state dependence and existence of states*: the renormalized $\langle T_{\mu\nu}\rangle$ requires a physically admissible (Hadamard) state to be defined at all, and on a general background the choice of state is not unique; the uniqueness and thermal character of states on spacetimes with a bifurcate Killing horizon — the rigorous backbone of the smoothness principle above — is the content of the Kay–Wald theorem [292], but the global existence of a sensible vacuum in fully dynamical, non-stationary spacetimes remains delicate. Finally, the semiclassical equation says nothing about the *information problem*: it describes the mean stress-energy of the Hawking flux but not the quantum correlations that would have to encode the infalling information, and resolving that tension is the frontier that carries this part of the book into the holographic ideas of Part XV.

These limitations do not undermine the framework; they delimit it. Within its domain — sub-Planckian curvatures, well-behaved states, modest fluctuations — the semiclassical theory is a precise and predictive effective description, and it is exactly the domain in which Hawking's calculation lives. We have assembled its three working parts: the smoothness principle Eq. (64.299), which fixes the horizon temperature $\beta = 2\pi/\kappa$ from geometry alone; the renormalized $\langle T_{\mu\nu}\rangle_{\text{ren}}$ of Section 64.5, which is the matter that gravitates; and the semiclassical Einstein equation Eq. (64.304), which lets the matter shape the metric. With these in hand, Chapter 65 turns the temperature into a spectrum, the spectrum into a luminosity, and the luminosity into the evaporation and thermodynamics of black holes — the destination the entire chapter has been climbing toward.

Further reading

The foundational papers are Hawking's derivation of black-hole radiance [255], Unruh's analysis of the accelerated detector and horizon [477], Fulling's discovery of the ambiguity of particle content [205], and the Euclidean-thermodynamics viewpoint of Gibbons and Hawking [220] together with Hawking's zeta-function treatment of curved-space path integrals [257]. The point-splitting construction of the renormalized stress tensor is Christensen [122]; DeWitt's review [145] remains the canonical synthesis of the effective-action approach. The textbook treatments of Birrell and Davies [76] and of Parker and Toms [380] develop the whole apparatus at length, and Wald's monograph [488] is the rigorous reference for the algebraic and thermodynamic structure, including the bifurcate-horizon uniqueness theorem with Kay [292]. On the limits of the semiclassical equation, see the fluctuation criterion of Kuo and Ford [310], the order-reduction treatment of higher-derivative dynamics by Simon [443], the backreaction analysis of Wald [487], and the energy-condition study of Flanagan and Wald [195]. None of these is needed to follow the development above, which is self-contained; they are where to go to deepen each thread.

Chapter 65

Hawking Radiation and the Semiclassical Frontier

The preceding chapter built the apparatus of quantum field theory on a fixed curved background: the Klein–Gordon inner product and the positive/negative frequency splitting it induces (Theorem 64.6), the Bogoliubov transformation relating two such splittings (Theorem 64.11), the renormalized stress tensor (Section 64.5), and the lesson of the Unruh effect (Theorem 64.20) that "particle" is an observer-dependent notion. None of that machinery cares what the background is. This chapter supplies the one physical input that turns it into the most consequential prediction of the subject: take the background to be a star that collapses to a black hole. The free real scalar of ϕ^4 theory at $\lambda = 0$ (Chapter 9) does all the work as the matter field; the geometry, not any self-interaction, manufactures the effect. We keep $\hbar = c = 1$ but restore Newton's constant G throughout, because its appearance in the final temperature is the whole point.

The arc is built around a single kinematic fact and several independent derivations of its consequence. The fact, extracted in the opening section, is that an outgoing ray reaching a distant observer at late retarded time u, traced back through the collapsing matter to past null infinity, is squeezed against the horizon-forming critical ray by a *logarithmic* law $u = -\kappa^{-1}\ln\big((v_0 - v)/C\big)$ whose only surviving parameter is the surface gravity $\kappa = 1/4GM$. Feeding this map into the Bogoliubov machinery converts the logarithm, through a single branch cut, into a Boltzmann factor: the in-vacuum radiates a Planck spectrum at the Hawking temperature $T_H = \kappa/2\pi = 1/8\pi GM$. Dressing the bare horizon spectrum with the greybody transmission of the Regge–Wheeler barrier yields the luminosity, the M^3 evaporation law, and the finite lifetime, completing the picture of the hole as a slowly shrinking thermal body.

That body obeys thermodynamics exactly, not by analogy. We establish the four laws of black-hole mechanics, fix the Bekenstein–Hawking entropy $S_{\mathrm{BH}} = A/4G$ with its coefficient pinned by the computed temperature, and

state the generalized second law that survives evaporation, importing the precise information paradox, the Page curve, the Bekenstein bound, and the island prescription from Chapter 57, and the Ryu–Takayanagi and holographic dictionaries from Chapters 53 and 56. The same temperature and entropy then reappear from an utterly different direction: the Euclidean gravitational path integral, the e^{-S_E} pillar built for scalars in Chapter 10, makes the black hole a smooth saddle whose imaginary-time period is forced to be $\beta = 8\pi GM$ and whose on-shell action is $\beta M - A/4G$. A third route reads the flux off purely local horizon data, recruiting the trace and gravitational anomalies of Chapter 36 and the conformal stress tensor and its Schwarzian transformation of Chapters 47 and 48.

The final two sections turn from confirmation to the conceptual frontier. The derivations all lean on modes blueshifted past the Planck scale near the horizon; we confront this trans-Planckian problem and dismantle it four ways — by an infrared adiabatic argument, by explicit modified-dispersion calculations, by Unruh's analog "dumb holes" (now realized in the laboratory), and by the Parikh–Wilczek tunneling computation, whose energy-conserving correction is the first crack in exact thermality. That crack opens onto the information problem: a pure collapsing state appears to evaporate into a mixed thermal cloud, in conflict with unitarity. We state the paradox as a clash of four premises, make it quantitative with the entanglement entropy and the Page curve, survey the proposed resolutions honestly and dated, and close where the book's long development has been heading since the first free scalar was quantized in Chapter 9 — at the gravity/quantum-mechanics frontier, where the gravitational path integral, through its replica-wormhole saddles, has by the mid-2020s been shown to reproduce the unitary Page curve in controlled models.

65.1 The Collapse Geometry and the Geometric-Optics Map

The apparatus assembled in Chapter 64 is, in a precise sense, complete: a free quantum field on a fixed curved background, the Klein–Gordon inner product that splits its mode space into positive- and negative-frequency halves (Theorem 64.6), the Bogoliubov transformation that relates two such splittings (Theorem 64.11), and the lesson of the Unruh effect (Theorem 64.20) that the word "particle" is observer-dependent because the splitting is. None of that machinery cares what the background actually is. The single new physical input of this chapter is to take the background to be the spacetime of a star that collapses to form a black hole, and the whole startling conclusion — that the hole radiates thermally — will turn out to rest on one kinematic fact about that spacetime. The fact is this: an outgoing light ray that reaches a distant observer at late times, traced backwards through the collapsing star to the distant past, is squeezed against a critical ray according to a *logarithmic* law, and the rate of that squeezing is set

by a single number, the surface gravity κ of the horizon that forms. Everything in Section 65.2 is then arithmetic on this one relation.

This section builds the geometry from the ground up and extracts that relation in full. We first construct the exterior Schwarzschild field and define its surface gravity invariantly, computing $\kappa = 1/4GM$ two independent ways. We introduce the null coordinates — tortoise, Eddington–Finkelstein, Kruskal — in which the near-horizon structure is transparent, and isolate the exact relation $U = -\kappa^{-1}e^{-\kappa u}$ between the Schwarzschild outgoing coordinate u and the regular Kruskal coordinate U that crosses the horizon smoothly. We then glue an interior onto the exterior to make an honest collapse spacetime, trace a high-frequency wave packet backwards through it by geometric optics — handling the reflection through the regular centre $r = 0$ explicitly rather than by assertion — and arrive at the late-time matching $u = -\kappa^{-1}\ln\big((v_0 - v)/C\big)$. Finally we show this law is *universal*: its functional form depends on κ alone and not on a single detail of how the star collapsed, which is exactly why the late-time radiation forgets its origin. The carried matter field throughout is the free real scalar ϕ of ϕ^4 theory at $\lambda = 0$ (Chapter 9); the geometry, not the self-interaction, does all the work, so we need only the free field, and we keep it massless except where noted.

We restore Newton's constant G explicitly for the rest of the chapter — gravity is dimensionful and the appearance of G in the final temperature is the whole point — while retaining $\hbar = c = 1$ and the mostly-minus signature $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$.

65.1.1 The exterior geometry and its surface gravity

A spherically symmetric vacuum spacetime is, by Birkhoff's theorem, necessarily static and equal to the Schwarzschild solution outside the matter. We take this as our exterior. With Newton's constant explicit and in the mostly-minus signature, the line element is

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 - f(r)^{-1}\,\mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2GM}{r}, \tag{65.1}$$

where $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2$ is the round metric on the unit two-sphere and M is the mass of the hole (the ADM mass measured at infinity). The coordinate range of the static exterior is $r > 2GM$, where $f(r) > 0$. The coefficient f vanishes at the Schwarzschild radius $r = 2GM$; this is the event horizon, and $r < 2GM$ is the black-hole interior.

The exterior is static and spherically symmetric, so it carries a distinguished timelike Killing vector field,

$$\chi^\mu = (\partial_t)^\mu = (1,0,0,0), \qquad \chi_\mu = g_{\mu\nu}\chi^\nu = (f,0,0,0), \tag{65.2}$$

whose norm is

$$\chi^\mu\chi_\mu = g_{tt} = f(r) = 1 - \frac{2GM}{r}. \tag{65.3}$$

This norm is the single most important diagnostic of the geometry. For $r > 2GM$ it is positive: χ is timelike, and a worldline that follows it at fixed (r, θ, φ) is the worldline of a static observer hovering at radius r. At $r = 2GM$ the norm vanishes: χ becomes *null*. The hypersurface $r = 2GM$ on which a timelike Killing field degenerates to a null one is, by definition, a *Killing horizon*, and χ is tangent to its null generators. That a static observer cannot exist at or below $r = 2GM$ — there is no timelike Killing direction to follow — is the coordinate-invariant statement that $r = 2GM$ is a horizon.

We now want a number that measures how strongly the geometry pulls at that horizon. The naive answer, the proper acceleration of a static observer, diverges as the horizon is approached (we verify this below), so it is not the right invariant. The right invariant is the *surface gravity*, which is finite because it is the acceleration measured by the static observer but redshifted out to infinity.

Definition 65.1 (Surface gravity)**.** Let χ be the horizon-generating Killing field of a Killing horizon $\mathcal{H}$, normalized so that $\chi^\mu \chi_\mu \to 1$ at spatial infinity. On $\mathcal{H}$ the integral curves of χ are null geodesics, but χ is in general not an affine parameter along them; it fails to be affine by a function κ, the surface gravity, defined by

$$\chi^\nu \nabla_\nu \chi^\mu \;=\; \kappa\, \chi^\mu \qquad \text{on } \mathcal{H}. \tag{65.4}$$

Equivalently, on a Killing horizon κ is given by the manifestly invariant formula

$$\kappa^2 \;=\; -\tfrac{1}{2}\,(\nabla^\mu \chi^\nu)(\nabla_\mu \chi_\nu)\Big|_{\mathcal{H}}. \tag{65.5}$$

The equivalence of Eqs. (65.4) and (65.5) on a Killing horizon is a standard piece of geometry; we will simply use whichever is more convenient and check that both give the same number. To see where Eq. (65.4) comes from, note that the Killing equation $\nabla_\mu \chi_\nu + \nabla_\nu \chi_\mu = 0$ together with hypersurface-orthogonality of χ (which holds here because the metric is static) implies that the gradient of the norm is proportional to χ on the horizon. Indeed, using the Killing antisymmetry,

$$\nabla_\nu (\chi^\mu \chi_\mu) = 2\, \chi^\mu \nabla_\nu \chi_\mu = -2\, \chi^\mu \nabla_\mu \chi_\nu, \tag{65.6}$$

so that $\chi^\mu \nabla_\mu \chi_\nu = -\tfrac{1}{2} \nabla_\nu (\chi^\mu \chi_\mu)$; on the horizon the right-hand side is proportional to χ_ν, with the constant of proportionality being κ. This is the precise content of "the redshifted acceleration of the static Killing field."

Theorem 65.2 (Surface gravity of Schwarzschild)**.** *For the Schwarzschild metric Eq. (65.1) the surface gravity of the horizon $r = 2GM$ is*

$$\kappa \;=\; \tfrac{1}{2}\, f'(r)\big|_{r=2GM} \;=\; \frac{1}{4GM}. \tag{65.7}$$

Proof. We evaluate the invariant Eq. (65.5). The covariant derivative of the Killing covector $\chi_\nu = f\, \delta^t_\nu$ is $\nabla_\mu \chi_\nu = \partial_\mu \chi_\nu - \Gamma^\lambda_{\mu\nu} \chi_\lambda$. The only Christoffel symbol

we need is $\Gamma^t_{tr} = f'/2f$, which follows from $\Gamma^t_{tr} = \tfrac{1}{2}g^{tt}\partial_r g_{tt} = \tfrac{1}{2}f^{-1}f'$. Therefore

$$\nabla_r \chi_t = \partial_r \chi_t - \Gamma^t_{rt}\,\chi_t = f' - \frac{f'}{2f}\,f = \frac{f'}{2}, \tag{65.8}$$

and by the Killing antisymmetry $\nabla_t \chi_r = -\nabla_r \chi_t = -f'/2$. These two are the only nonvanishing components of $\nabla_\mu \chi_\nu$. Raising indices with $g^{tt} = f^{-1}$ and $g^{rr} = -f$ (read off from Eq. (65.1), recalling $g_{rr} = -f^{-1}$), we contract:

$$\begin{aligned}(\nabla^\mu \chi^\nu)(\nabla_\mu \chi_\nu) &= g^{rr}g^{tt}(\nabla_r\chi_t)^2 + g^{tt}g^{rr}(\nabla_t\chi_r)^2 \\ &= 2\,g^{rr}g^{tt}\Big(\frac{f'}{2}\Big)^2 = 2\,(-f)(f^{-1})\frac{f'^2}{4} = -\frac{f'^2}{2}.\end{aligned} \tag{65.9}$$

Hence $\kappa^2 = -\tfrac{1}{2}(-f'^2/2) = f'^2/4$, i.e. $\kappa = \tfrac{1}{2}f'$. For Schwarzschild $f'(r) = 2GM/r^2$, so at the horizon

$$\kappa = \tfrac{1}{2}f'(2GM) = \tfrac{1}{2}\cdot\frac{2GM}{(2GM)^2} = \frac{1}{4GM}. \qquad \square$$

The number $\kappa = 1/4GM$ is the protagonist of the chapter. Two features deserve emphasis immediately. First, κ has dimensions of inverse length (equivalently, with $\hbar = c = 1$ restored later, of energy), and it grows as the hole gets *lighter*: a small black hole has a fierce surface gravity. Second, the derivation used only the function f and its first derivative at the horizon. Any static, spherically symmetric horizon with $f(r_h) = 0$ has $\kappa = \tfrac{1}{2}f'(r_h)$; the specific value $1/4GM$ is special to Schwarzschild, but the formula is general, a fact Section 65.4 and the sibling sections will lean on.

To confirm that Eq. (65.7) really is the redshifted acceleration — the physically transparent definition — we compute it that way as well. This is exactly the kind of check the book does on the page rather than asserting.

Worked Example 65.3 (Surface gravity as redshifted acceleration)**.** A static observer at radius r has four-velocity tangent to χ, normalized to unit timelike norm:

$$u^\mu = \frac{\chi^\mu}{\sqrt{\chi^\nu\chi_\nu}} = \Big(f^{-1/2}, 0, 0, 0\Big), \qquad u^\mu u_\mu = 1. \tag{65.10}$$

Its proper acceleration is $a^\mu = u^\nu \nabla_\nu u^\mu$. Because the configuration is static, $\partial_t u^\mu = 0$ and only the connection term survives:

$$a^\mu = u^t u^t\,\Gamma^\mu_{tt} = f^{-1}\,\Gamma^\mu_{tt}. \tag{65.11}$$

The relevant Christoffel symbol is $\Gamma^r_{tt} = -\tfrac{1}{2}g^{rr}\partial_r g_{tt} = -\tfrac{1}{2}(-f)f' = \tfrac{1}{2}ff'$, so $a^r = f^{-1}\cdot\tfrac{1}{2}ff' = \tfrac{1}{2}f'$ and all other components vanish. The acceleration is purely radial, as it must be: the observer fires rockets straight down to stay put. Its invariant magnitude (a spacelike vector, so we take $a = \sqrt{-a_\mu a^\mu}$) is

$$a = \sqrt{-g_{rr}\,(a^r)^2} = \sqrt{f^{-1}\Big(\tfrac{1}{2}f'\Big)^2} = \frac{f'(r)}{2\sqrt{f(r)}}. \tag{65.12}$$

As $r \to 2GM$ the denominator $\sqrt{f} \to 0$ and a diverges: the local force needed to hover diverges at the horizon, as anticipated. The energy of a photon emitted by this observer is redshifted on its way to infinity by exactly the factor $V(r) = \sqrt{f(r)}$ (the norm of χ), since the conserved Killing energy is $E_\infty = \chi_\mu p^\mu = V\, E_{\rm local}$. The acceleration "as seen from infinity" is therefore $V\, a$, and its horizon limit is finite:

$$\kappa = \lim_{r\to 2GM} V(r)\, a(r) = \lim_{r\to 2GM} \sqrt{f}\,\frac{f'}{2\sqrt{f}} = \tfrac{1}{2} f'(2GM) = \frac{1}{4GM}. \tag{65.13}$$

The infinite local force, weighted by the infinite redshift, leaves a finite residue: the surface gravity. This matches Eq. (65.7) exactly, confirming that the Killing-normalization definition and the redshifted-acceleration definition agree.

65.1.2 Null coordinates and the near-horizon relation

Radiation propagates along null rays, so to track phases we need coordinates adapted to the light cones. We build them in three steps: the tortoise coordinate straightens out the radial null geodesics, the Eddington–Finkelstein coordinates turn the metric into a product of null one-forms, and the Kruskal coordinates remove the coordinate singularity at the horizon. The payoff is the exact relation between the Schwarzschild retarded time and the Kruskal coordinate, which is the seed of the logarithm.

Radial null rays satisfy $\mathrm{d}s^2 = 0$ with $\mathrm{d}\Omega = 0$, i.e. $f\,\mathrm{d}t^2 = f^{-1}\mathrm{d}r^2$, or $\mathrm{d}t = \pm f^{-1}\mathrm{d}r$. This motivates the *tortoise coordinate* r_* defined by

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{1}{f(r)}, \qquad r_* = r + 2GM\,\ln\!\Big(\frac{r}{2GM} - 1\Big), \tag{65.14}$$

in terms of which radial null rays are simply $t \mp r_* = \text{const}$. Two features of Eq. (65.14) are decisive for everything that follows. As $r \to \infty$, $r_* \to r$ and the geometry is asymptotically flat. As $r \to 2GM^+$, the argument of the logarithm vanishes and $r_* \to -\infty$ *logarithmically*: the horizon is pushed to infinite tortoise distance. Near the horizon, writing $r = 2GM + \epsilon$ with $0 < \epsilon \ll 2GM$,

$$r_* \simeq 2GM\,\ln\!\Big(\frac{\epsilon}{2GM}\Big) + \text{const}, \qquad f \simeq \frac{\epsilon}{2GM} = e^{\,r_*/2GM}\cdot(\text{const}), \tag{65.15}$$

where in the second relation we inverted the first. Since $2GM = 1/2\kappa$ by Eq. (65.7), this reads $f \simeq e^{2\kappa r_*}$ up to a positive constant. *The logarithm in the tortoise coordinate, and the exponential it implies for f, are the surface gravity's fingerprint, and they will reappear directly in the phase map.*

Define the retarded and advanced null coordinates

$$u = t - r_*, \qquad v = t + r_*, \tag{65.16}$$

so that outgoing radial rays are $u = \text{const}$ and ingoing radial rays are $v = \text{const}$. At infinity u and v are the physical retarded and advanced times an asymptotic observer reads off; the field of an outgoing quantum of frequency ω carries the phase $e^{-i\omega u}$ there. In these coordinates $\mathrm{d}t = \frac{1}{2}(\mathrm{d}u + \mathrm{d}v)$ and $\mathrm{d}r = f\,\mathrm{d}r_* = \frac{1}{2}f(\mathrm{d}v - \mathrm{d}u)$, and a one-line computation collapses the metric to

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}u\,\mathrm{d}v - r^2\,\mathrm{d}\Omega^2, \tag{65.17}$$

with r now an implicit function of $v - u = 2r_*$. This is the Eddington–Finkelstein form. It is regular nowhere new — f still vanishes at the horizon — but it exposes the null structure cleanly, and it is the right starting point for removing the singularity.

The vanishing of f at $r = 2GM$ is a coordinate artifact: the curvature invariants (e.g. $R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} = 48G^2M^2/r^6$) are perfectly finite there. To exhibit a regular chart we rescale the null coordinates exponentially. Define the *Kruskal* null coordinates

$$U = -\frac{1}{\kappa}\,e^{-\kappa u}, \qquad V = +\frac{1}{\kappa}\,e^{+\kappa v}. \tag{65.18}$$

Then $\mathrm{d}U = e^{-\kappa u}\,\mathrm{d}u$ and $\mathrm{d}V = e^{\kappa v}\,\mathrm{d}v$, so

$$\mathrm{d}U\,\mathrm{d}V = e^{-\kappa u}e^{\kappa v}\,\mathrm{d}u\,\mathrm{d}v = e^{\kappa(v-u)}\,\mathrm{d}u\,\mathrm{d}v = e^{2\kappa r_*}\,\mathrm{d}u\,\mathrm{d}v, \tag{65.19}$$

and therefore $\mathrm{d}u\,\mathrm{d}v = e^{-2\kappa r_*}\mathrm{d}U\,\mathrm{d}V$. Inserting this into Eq. (65.17),

$$\mathrm{d}s^2 = f\,e^{-2\kappa r_*}\,\mathrm{d}U\,\mathrm{d}V - r^2\,\mathrm{d}\Omega^2. \tag{65.20}$$

By Eq. (65.15), $f \simeq e^{2\kappa r_*}$ near the horizon, so the prefactor $f\,e^{-2\kappa r_*} \to (\text{const})$ stays finite and nonzero as $r \to 2GM$. (Globally one finds $f\,e^{-2\kappa r_*} = (2GM/r)\,e^{-r/2GM}\cdot\text{const}$, manifestly smooth and positive across $r = 2GM$.) The Kruskal chart is regular at the horizon, which now sits at $U = 0$. From Eq. (65.18), $U = 0$ is reached only as $u \to +\infty$: *the future horizon is the surface $u = +\infty$, packed by the exponential into $U = 0$.* This is the geometric heart of the matter and we record it as the central relation of the subsection,

$$\boxed{U = -\frac{1}{\kappa}\,e^{-\kappa u}} \qquad \Longleftrightarrow \qquad u = -\frac{1}{\kappa}\,\ln(-\kappa U), \quad U < 0. \tag{65.21}$$

A surface of constant outgoing Schwarzschild time u is a surface of constant U; as u runs from $-\infty$ to $+\infty$, the regular coordinate U runs from $-\infty$ to 0^-. Equal increments of late-time u correspond to *exponentially small* increments of U. This exponential relation, read in reverse in Section 65.1.4, becomes the logarithm.

It is worth seeing where the surface gravity lives in this picture in Killing-vector language, since it makes the universality argument later transparent. The

Killing field Eq. (65.2) is $\chi = \partial_t = \partial_u + \partial_v$ (as u, v both contain t linearly), and under Eq. (65.18) this becomes

$$\chi = \kappa(V\,\partial_V - U\,\partial_U). \tag{65.22}$$

Its norm is proportional to $f \propto -UV$ (up to the smooth positive prefactor), vanishing on both branches $U = 0$ and $V = 0$ of the horizon, confirming Eq. (65.3) in regular coordinates. The combination $\chi = \kappa(V\partial_V - U\partial_U)$ is a boost in the (U, V) plane with rapidity parameter κ: *the surface gravity is the boost rate of the horizon generators*, precisely the structure that produced the Unruh temperature in Theorem 64.20, now read off the geometry of the black hole.

65.1.3 The collapse spacetime and its causal structure

The eternal Schwarzschild solution Eq. (65.1) is a vacuum geometry with no matter and no moment of formation; it has two asymptotic regions and a past as well as a future horizon. A real black hole is not like that: it forms from collapsing matter, has a single asymptotic region, and possesses a regular centre $r = 0$ in its past where there is no singularity at all. The distinction matters for us because the radiation we will compute is sourced by the time-dependence of the geometry during formation, and because the trace-back of an outgoing ray must pass through the regular interior, not through a second asymptotic region.

We therefore model the spacetime as follows. For radii larger than the surface of the star, $r > R(t)$, Birkhoff's theorem guarantees the geometry is exactly the exterior Schwarzschild field Eq. (65.1) with mass M. Inside the star, $r < R(t)$, the geometry is some regular, time-dependent, matter- filled solution; its details will turn out to be irrelevant, and all we need of it is that (i) it is smooth, with finite curvature, along the rays we trace, and (ii) it contains a regular centre $r = 0$ at which ingoing rays reflect to outgoing rays. The surface $R(t)$ falls inward, crosses $r = 2GM$ at some moment, and a horizon forms; the matter continues to a singularity at $r = 0$ to the *future* of horizon formation. The causal structure is summarized by the conformal (Penrose) diagram of Fig. 65.1.

Three loci on the diagram organize the calculation. Past null infinity $\mathcal{I}^-$ is where ingoing data is set: the in-vacuum of the field is defined by its positive-frequency content there. Future null infinity $\mathcal{I}^+$ is where the radiation is collected. And the dotted *critical ray* $v = v_0$ is the ingoing ray that, traced forward, just grazes the forming horizon and escapes to $\mathcal{I}^+$ only asymptotically, at $u = +\infty$; equivalently it is the backward continuation of the horizon generator. Rays with v slightly less than v_0 escape to $\mathcal{I}^+$ at large but finite u; rays with $v > v_0$ are trapped and strike the singularity. The entire late-time physics is the behavior of the map between u at $\mathcal{I}^+$ and v at $\mathcal{I}^-$ as $v \uparrow v_0$, which we now compute.

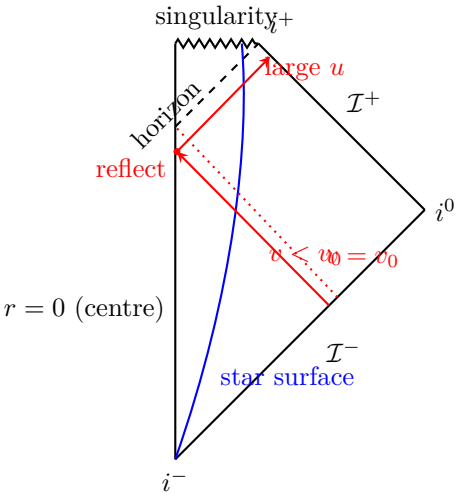


Figure 65.1: Conformal (Penrose) diagram of gravitational collapse. The shaded matter (the star, blue worldline) falls in from i^-; outside it the geometry is exact Schwarzschild. An event horizon (dashed) forms and reaches future timelike infinity i^+, where it meets the spacelike singularity (zigzag) and future null infinity $\mathcal{I}^+$. A red outgoing ray that escapes to $\mathcal{I}^+$ at large retarded time u, traced backwards, is ingoing from $\mathcal{I}^-$ at advanced time $v < v_0$, reflects through the regular centre $r = 0$, and emerges. The dotted critical ray $v = v_0$ is the last ray that escapes; it asymptotes to the horizon and reaches $\mathcal{I}^+$ only at i^+ ($u = +\infty$). Rays with $v > v_0$ fall into the singularity.

65.1.4 Geometric optics and the backward ray-tracing

The modes that dominate the late-time radiation are of very high frequency near the horizon — this is the trans-Planckian feature flagged in the chapter introduction and revisited in Section 65.8 — and high-frequency fields are governed by geometric optics. We make this precise. Write a solution of the curved-space Klein–Gordon equation $\Box\phi = 0$ (the free ϕ^4 theory scalar at $\lambda = 0$, Chapter 9, massless) in the eikonal form

$$\phi = A\, e^{iS}, \qquad \Box\phi = \left(iA\,\Box S - A\,(\nabla S)^2 + 2i\,\nabla^\mu A\,\nabla_\mu S + \Box A\right) e^{iS}. \tag{65.23}$$

For a rapidly varying phase S the dominant term is the one with two derivatives of S; setting it to zero gives the eikonal equation and its companion transport equation,

$$(\nabla S)^2 = g^{\mu\nu}\partial_\mu S\,\partial_\nu S = 0, \qquad 2\,\nabla^\mu A\,\nabla_\mu S + A\,\Box S = 0. \tag{65.24}$$

The first says the surfaces of constant phase are null. The wave vector $k_\mu = \partial_\mu S$ is null, $k^\mu k_\mu = 0$, and differentiating $(\nabla S)^2 = 0$ shows $k^\mu \nabla_\mu k_\nu = 0$: the rays

$\mathrm{d}x^\mu/d\lambda = k^\mu$ are null geodesics, with S constant along each (since $\mathrm{d}S/\mathrm{d}\lambda = k^\mu\partial_\mu S = k^\mu k_\mu = 0$). *The phase is parallel- transported, unchanged, along the null geodesic generators of the wavefront.* This is the single dynamical input of the ray-tracing: to find the phase of an outgoing mode at $\mathcal{I}^+$ in terms of conditions in the past, we follow its generator backwards and read the phase off where it ends, on $\mathcal{I}^-$.

Consider then an outgoing wave with phase $S = -\omega u$ at $\mathcal{I}^+$, i.e. the mode $e^{-i\omega u}$. Its wavefronts are the surfaces $u = \text{const}$, and each is generated by an outgoing radial null geodesic. We trace one such generator backwards through the geometry. The journey has three legs.

Leg 1 (exterior, near the horizon). In the exterior the generator is an outgoing radial null geodesic, $u = \text{const}$, equivalently $U = \text{const}$ by Eq. (65.21). For a ray reaching $\mathcal{I}^+$ at late u this is a constant U very close to 0^-. The generator stays at this fixed U until it crosses the surface of the collapsing star, $r = R(t)$, and enters the interior.

Leg 2 (interior, reflection through the centre). Inside the star the generator continues as a null geodesic in the regular matter-filled geometry. An *outgoing* radial ray, followed into the past, moves inward, reaches the regular centre $r = 0$, and there continues as an *ingoing* ray — this is the reflection through $r = 0$, which we must handle explicitly because r is a radial coordinate, not a Cartesian one, and the "reflection" is a statement about the regularity of the field at the origin. We make it explicit below in Theorem 65.4; for now the content is that a smooth field at a regular centre cannot have a conical kink, so the ingoing and outgoing radial null coordinates are identified continuously across $r = 0$.

Leg 3 (back out to $\mathcal{I}^-$). After reflection the ray is ingoing, $v = \text{const}$, crosses the stellar surface again, and propagates out to past null infinity, arriving at advanced time v. This v is the label we want.

The composition of the three legs is a map $v \mapsto u$, smooth wherever the geometry it crosses is smooth. The key qualitative fact, visible in Fig. 65.1, is that the generator of the event horizon itself is one particular such ray: traced back from i^+ ($u = +\infty$, $U = 0$) it reflects through the centre and reaches $\mathcal{I}^-$ at a definite finite advanced time, which we call v_0. The horizon generator is a perfectly regular null geodesic everywhere in its past — the singularity lies to its *future* — so the geometry along it and in a neighborhood of it is smooth, and the map $v \mapsto U$ is a smooth function near $v = v_0$.

This is all we need. Let $U(v)$ be the (constant) exterior Kruskal coordinate of the generator that arrives at $\mathcal{I}^-$ with advanced time v. It is smooth near v_0, and the horizon ray sets $U(v_0) = 0$ (the horizon is $U = 0$). Taylor- expanding,

$$U(v) = U(v_0) + U'(v_0)\,(v - v_0) + O\big((v - v_0)^2\big) = -C_K\,(v_0 - v) + \cdots, \qquad C_K \equiv U'(v_0) > 0, \tag{65.25}$$

where the sign and positivity of C_K are fixed by the geometry: rays that *escape* have $U < 0$ (just outside the horizon), and they do so for $v < v_0$, so U and

$(v - v_0)$ have the same sign and $U'(v_0) > 0$. Combining Eq. (65.25) with the exterior relation Eq. (65.21), $U = -\kappa^{-1} e^{-\kappa u}$, we equate the two expressions for U:

$$\frac{1}{\kappa}\, e^{-\kappa u} = C_K\,(v_0 - v), \tag{65.26}$$

solve for u,

$$e^{-\kappa u} = \kappa\, C_K\,(v_0 - v) \quad \Longrightarrow \quad -\kappa u = \ln\big(\kappa C_K (v_0 - v)\big), \tag{65.27}$$

and arrive at the central result of the section,

$$\boxed{u = -\frac{1}{\kappa}\,\ln\!\Big(\frac{v_0 - v}{C}\Big), \qquad v < v_0, \quad C \equiv \frac{1}{\kappa C_K} > 0.} \tag{65.28}$$

This is the late-time logarithmic phase map. As $v \uparrow v_0$ the retarded time $u \to +\infty$: the rays that arrive late at $\mathcal{I}^+$ are exponentially packed against the critical ray v_0 at $\mathcal{I}^-$. The coefficient of the logarithm is $1/\kappa$ and nothing else; the constants v_0 and C carry all the information about the particular collapse and will drop out of the spectrum entirely. An outgoing mode $e^{-i\omega u}$ at $\mathcal{I}^+$ therefore traces back to the field

$$e^{-i\omega u} \;\longmapsto\; \exp\Big[\,\frac{i\omega}{\kappa}\,\ln\!\Big(\frac{v_0 - v}{C}\Big)\Big] \qquad \text{on } \mathcal{I}^-,\ v < v_0, \tag{65.29}$$

and vanishing for $v > v_0$ (those rays never reach $\mathcal{I}^+$). The logarithm of the argument means this is not a pure positive-frequency wave in v; it is a superposition of both signs of frequency, and that mismatch — the Bogoliubov mixing — is what Section 65.2 evaluates.

It remains to make the reflection through $r = 0$ explicit and to exhibit the whole map in a model where every step is exact. The cleanest such model is a black hole formed by a single thin shell of radiation (a Vaidya geometry): outside the shell Schwarzschild, inside it flat space, glued along an ingoing null surface. There the interior is exactly Minkowski, the reflection at $r = 0$ is the elementary reflection of a light ray at the origin of flat space, and the matching across the shell is algebraic.

Worked Example 65.4 (The collapsing null shell: an exact logarithmic map)**.** Let a thin null shell collapse along the ingoing ray $v = v_0$. For $v > v_0$ the geometry is Schwarzschild of mass M with exterior null coordinates $u = t - r_*$, $v = t + r_*$ and r_* as in Eq. (65.14). For $v < v_0$ the geometry is flat Minkowski space, with its own null coordinates

$$v = t_{\rm in} + r, \qquad u_{\rm in} = t_{\rm in} - r, \tag{65.30}$$

where we have already used the freedom to set the advanced coordinate of the flat interior equal to the exterior v (both reduce to the same retarded/advanced time on $\mathcal{I}^-$, which is entirely in the flat region).

Reflection at the centre. In the flat interior a radial light ray that comes in along $v = \text{const}$ reaches $r = 0$ and goes back out along $u_{\text{in}} = \text{const}$. At the reflection point $r = 0$ we have $v = t_{\text{in}} = u_{\text{in}}$, so the regularity of the field at the origin (no source there) identifies the outgoing label with the ingoing one,

$$u_{\text{in}} = v. \tag{65.31}$$

This is the explicit content of "reflection through $r = 0$": it is the statement that an ingoing spherical wave $e^{-i\omega v}/r$ becomes, after passing regularly through the origin, the outgoing wave $-e^{-i\omega u_{\text{in}}}/r$ with $u_{\text{in}} = v$ (the relative sign is the usual phase shift of a regular spherical wave at the centre and does not affect the magnitude we will need).

Matching across the shell. The area radius r is continuous across the shell. An outgoing ray leaving the centre at $u_{\text{in}} = v$ crosses the shell $v = v_0$ at some radius r_s. From the interior, on the shell $v = v_0$ we have $v_0 = t_{\text{in}} + r_s$ and $u_{\text{in}} = t_{\text{in}} - r_s$, so $u_{\text{in}} = v_0 - 2r_s$, i.e. using Eq. (65.31),

$$v = v_0 - 2r_s \quad \Longrightarrow \quad r_s = \tfrac{1}{2}\,(v_0 - v). \tag{65.32}$$

From the exterior, on the same crossing the ray has retarded coordinate u and advanced coordinate v_0: $u = t - r_*(r_s)$ and $v_0 = t + r_*(r_s)$, hence

$$u = v_0 - 2\,r_*(r_s) = v_0 - 2\Big[\,r_s + 2GM \ln\Big(\frac{r_s}{2GM} - 1\Big)\Big]. \tag{65.33}$$

(We have absorbed into a redefinition of the origin of v_0 the finite, smooth mismatch between the interior and exterior radial coordinates away from the horizon; it contributes only a constant and a regular function, not the logarithm.)

Late times. Rays that escape to $\mathcal{I}^+$ at large u are those that cross the shell just outside the forming horizon, $r_s \to 2GM^+$. Writing $r_s = 2GM + \delta$ with $\delta \to 0^+$, Eq. (65.32) gives $\delta = \tfrac{1}{2}(v_0 - v) - 2GM$; shifting v_0 by the constant $4GM$ so that $\delta = 0$ corresponds to the last escaping ray (this redefinition is exactly the identification of v_0 with the critical ray of Fig. 65.1) we have simply

$$\delta = \tfrac{1}{2}\,(v_0 - v). \tag{65.34}$$

In Eq. (65.33) the logarithm dominates as $\delta \to 0$: $r_*(r_s) \simeq 2GM \ln(\delta/2GM) + \text{const}$, so

$$u = -4GM\,\ln\Big(\frac{\delta}{2GM}\Big) + \text{const} = -4GM\,\ln\Big(\frac{v_0 - v}{4GM}\Big) + \text{const}. \tag{65.35}$$

Since $4GM = 1/\kappa$ by Eq. (65.7), this is

$$u = -\frac{1}{\kappa}\,\ln\Big(\frac{v_0 - v}{C}\Big), \tag{65.36}$$

with C a positive constant collecting $4GM$ and the additive constant — exactly Eq. (65.28). Every step was explicit: the reflection at the regular centre via Eq. (65.31), the continuity of r across the shell, and the logarithmic divergence of r_* at the horizon. The Vaidya model thus realizes the general argument of Eqs. (65.25) and (65.28) with all constants determined.

65.1.5 Universality: why the late-time radiation forgets the collapse

The form of Eq. (65.28) is the same whether we obtained it from the abstract smoothness argument Eq. (65.25) or from the explicit shell of Theorem 65.4, and this is no accident. The logarithm and its coefficient $1/\kappa$ come from a single source: the logarithmic divergence of the tortoise coordinate Eq. (65.14) at the horizon, equivalently the exponential relation Eq. (65.21) between the Schwarzschild time u and the regular Kruskal coordinate U. That relation lives entirely in the *exterior*, near-horizon region, where the geometry is exactly Schwarzschild regardless of what collapsed, by Birkhoff's theorem. The interior, the matter content, the equation of state, the precise infall history — all of it enters only through the smooth map $U(v)$ in Eq. (65.25), and a smooth map contributes nothing but its value v_0 (a translation of v) and its slope C_K (a rescaling absorbed into C). Neither v_0 nor C survives into the spectrum, as Section 65.2 will show: the thermal flux depends on the coefficient $1/\kappa$ of the logarithm alone.

Physical Insight

The universality is the physical reason a black hole has "no hair" as far as its radiation is concerned. Two holes of the same mass formed from utterly different matter — a collapsing dust ball, a shell of light, an imploding neutron star — produce *identical* late-time Hawking spectra, because the only feature of the collapse that imprints on the late-time phase is the surface gravity of the horizon that forms, and that is fixed by the mass alone. The late-time radiation is a property of the horizon, not of its history. This is what makes the Hawking temperature a clean function of M and lets us treat the hole as a thermodynamic object in Section 65.4.

There is a structural echo here of the Unruh effect of Theorem 64.20. There, a uniformly accelerated observer in flat space saw the Minkowski vacuum as thermal because their proper time was related to the global inertial null coordinate by an exponential, $x^- \propto -e^{-a\,\bar{u}}$ with a the acceleration. Here the asymptotic late-time observer's retarded time is related to the regular horizon- crossing coordinate by the same kind of exponential, $U \propto -e^{-\kappa u}$, with κ the surface gravity playing the role of a. The black hole is, near its horizon, locally a Rindler wedge, and κ is the local acceleration scale; the collapse merely ensures that the relevant in-vacuum is the regular one, so the Rindler-like thermality is actually radiated to infinity. We make the choice of in-vacuum precise next.

65.1.6 The Boulware, Unruh, and Hartle–Hawking vacua

A quantum field on this background has no canonical vacuum: as Theorems 64.11 and 64.20 taught us, "vacuum" means a choice of which modes count as positive-frequency, and inequivalent choices give inequivalent notions of particle. Three choices are natural on the Schwarzschild/collapse background, and exactly one is selected by collapse. We define them by the time coordinate with respect to which the modes are positive-frequency, and read off their physical content from the relations established above.

Definition 65.5 (The three vacua)**.** For the free scalar field on the Schwarzschild geometry:

- The *Boulware vacuum* $|0_B\rangle$ is defined by taking modes that are positive-frequency with respect to the Killing time t — i.e. $e^{-i\omega u}$ outgoing and $e^{-i\omega v}$ ingoing — on both the exterior modes. It is the state a static observer at infinity calls "empty," and it most resembles the ordinary Minkowski vacuum far away.
- The *Hartle–Hawking vacuum* $|0_H\rangle$ is defined by taking modes positive-frequency with respect to the Kruskal coordinates U and V — the affine parameters along the future and past horizons — on both branches.
- The *Unruh vacuum* $|0_U\rangle$ is defined by taking the outgoing modes positive-frequency with respect to the Kruskal U (regular on the future horizon) and the ingoing modes positive-frequency with respect to the Killing v (no incoming radiation from $\mathcal{I}^-$).

The physical characters follow from where each state is regular. Regularity of a quantum state at a horizon requires its modes to be positive-frequency with respect to the affine parameter there (the Kruskal coordinate), since that is the parameter in which the geometry is smooth; modes positive-frequency in the Killing time t oscillate infinitely rapidly as $U \to 0$ by Eq. (65.21) and produce a divergent stress tensor on the horizon (we computed renormalized stress tensors in Section 64.5; the divergence here is the statement that $\langle 0_B| T_{\mu\nu} |0_B\rangle$ blows up in a frame that freely crosses the horizon). Consequently:

- $|0_B\rangle$ is regular at infinity but *singular on the horizon*: it is the appropriate vacuum for the exterior of a static star whose surface sits above $2GM$ (no horizon exists, no problem), but it cannot describe a black hole.
- $|0_H\rangle$ is regular on *both* horizons and is the unique state invariant under the Killing flow; it describes a black hole in thermal equilibrium with a bath of its own radiation at temperature $T_H = \kappa/2\pi$, filling all of space. It is time-symmetric — as much radiation comes in as goes out — and is the state relevant to the Euclidean computation of Section 65.5.

- $|0_U\rangle$ is regular on the future horizon (where infalling matter crosses) but not the past one, carries no incoming flux at $\mathcal{I}^-$, and carries a net *outgoing* thermal flux at $\mathcal{I}^+$. It is the state of an evaporating hole formed by collapse.

Caution

The Boulware and Hartle–Hawking states are defined on the *eternal* Schwarzschild geometry, which has a past horizon and a second asymptotic region; they are useful idealizations (thermal equilibrium, the static exterior) but neither describes a hole that actually forms. The collapse geometry of Fig. 65.1 has no past horizon. What the collapse selects is the *in-vacuum*: the state with no particles on $\mathcal{I}^-$, defined by positive frequency with respect to the advanced time v there, where the geometry is asymptotically flat and v is the natural affine/Killing time. The geometric-optics map Eq. (65.28) shows that, near the horizon and at late times, this in-vacuum is indistinguishable from the Unruh vacuum — the outgoing modes it contains are the ones positive-frequency in the horizon-regular coordinate U. This is the precise sense in which "collapse selects the Unruh vacuum."

The collapse therefore hands Section 65.2 two things, both now made precise. The first is the kinematic input: the late-time logarithmic map Eq. (65.28), $u = -\kappa^{-1}\ln\left((v_0 - v)/C\right)$, which traces an outgoing $\mathcal{I}^+$ mode $e^{-i\omega u}$ back to the $\mathcal{I}^-$ field Eq. (65.29). The second is the state: the in-vacuum defined by positive frequency in v on $\mathcal{I}^-$, against which the outgoing $\mathcal{I}^+$ modes are to be expanded. Feeding the traced mode into the Klein–Gordon inner product of Theorem 64.6 to extract the Bogoliubov coefficients $\alpha_{\omega\omega'}, \beta_{\omega\omega'}$ of Theorem 64.11 — and discovering that the logarithm makes $|\beta|^2$ Planckian — is the work of the next section.

65.2 The Hawking Spectrum: Thermality from the Horizon

The previous section reduced the entire dynamical problem of a quantum field on a collapsing star to a single kinematic fact. An outgoing wave that reaches a distant observatory at late retarded time u did not originate as a long-lived, gently-varying ripple at past null infinity: traced backwards along its null ray, through the centre of the collapsing matter and back out, its phase is stretched without bound, and the relation between the retarded time u at which it arrives and the advanced time v at which it set out is logarithmic. Recalling the result of Section 65.1, the late-time geometric-optics map is

$$u(v) \;=\; -\frac{1}{\kappa}\,\ln\!\big(\kappa\,(v_0 - v)\big), \qquad v < v_0, \tag{65.37}$$

where v_0 is the advanced time of the last ray that escapes to infinity — the generator of the event horizon, traced back to past null infinity — and κ is the surface gravity, which for the Schwarzschild geometry of mass M takes the value

$$\kappa \;=\; \frac{1}{4GM} \tag{65.38}$$

(Eq. (65.7)). Rays with $v > v_0$ enter the star too late and are swallowed; rays with $v \to v_0^-$ skim the forming horizon and emerge at arbitrarily late u, their frequency redshifted without limit. Everything about the radiation is encoded in Eq. (65.37). The work of this section is to feed that map into the Bogoliubov machinery of Chapter 64 and extract the particle content of the in-vacuum as seen by a late-time observer. The carried matter field is the free real scalar of ϕ^4 theory at $\lambda = 0$ (Section 9.2) — the same field ϕ, now living its first genuinely gravitational life — and the symbols are exactly those fixed there.

We will find that the in-vacuum, the natural "no incoming particles" state of the collapse problem, is populated at infinity by a precisely Planckian flux of ϕ-quanta. The temperature of that flux is fixed entirely by the surface gravity, $T_H = \kappa/2\pi$, and the entire content of the calculation is the conversion of the logarithm in Eq. (65.37) into the Boltzmann factor $e^{-\omega/T_H}$ through a single contour deformation across a branch cut. We proceed in three movements: first we write the Bogoliubov coefficients as Klein–Gordon overlaps and reduce them to Fourier integrals over past null infinity; second we evaluate those integrals in closed form and isolate the exact ratio $|\alpha| = e^{\pi\omega/\kappa}|\beta|$ from the branch structure; third we assemble the occupation number, identify it as a Bose–Einstein distribution, and confirm the same temperature from the KMS/periodicity criterion inherited from the Unruh analysis.

65.2.1 The Bogoliubov coefficients of the collapse problem

The general apparatus is already in hand. From Section 64.2 we have two complete orthonormal mode bases for the free scalar, related by the conserved Klein–Gordon inner product

$$(f, g) \;=\; i\int_\Sigma \mathrm{d}\Sigma^\mu \left[\, (\partial_\mu f)\, g^* - f\, \partial_\mu g^* \,\right] \tag{65.39}$$

(Eq. (64.45)), whose value is independent of the Cauchy surface Σ. The two bases here are physically dictated by the two ends of the collapse spacetime. On past null infinity $\mathcal{I}^-$, where the geometry is flat and static and "no incoming radiation" is an unambiguous notion, we take the *in-modes* — the analogue of the old basis $\{u_j\}$ of Eq. (64.47). On future null infinity $\mathcal{I}^+$, where a late-time detector reads off particles, we take the *out-modes* — the new basis $\{v_i\}$ of Eq. (64.48). The in-vacuum $|0_{\mathrm{in}}\rangle$ is the state annihilated by all in-annihilation operators; it is the

state of the field before the star collapses, and it is the physically selected initial condition.

Because the field is a free massless scalar and the exterior is spherically symmetric, we may work mode by mode in the angular momentum ℓ and, since the spectrum will turn out to be ℓ-independent at the horizon, we suppress the spherical-harmonic and radial structure and follow a single outgoing s-wave. After the substitution $\phi = (\chi/r)\,Y_{\ell m}$ that turns the radial problem into a 1+1-dimensional wave equation on the line (the reduction carried out in detail in Theorem 65.9), the reduced field $\chi(u, v)$ propagates freely near each null infinity, and the Klein–Gordon product Eq. (65.39) evaluated on $\mathcal{I}^-$ collapses to the one-dimensional null-surface form

$$(f, g)_{\mathcal{I}^-} \;=\; i\int_{-\infty}^{\infty} \mathrm{d}v\, \left[\, (\partial_v f)\, g^* - f\, \partial_v g^* \,\right], \tag{65.40}$$

with v the advanced time labelling the null generators of $\mathcal{I}^-$. The in-modes are the positive-frequency plane waves with respect to this advanced time,

$$p^{\mathrm{in}}_{\omega'}(v) \;=\; \frac{1}{\sqrt{4\pi\omega'}}\, e^{-i\omega' v}, \qquad \omega' > 0, \tag{65.41}$$

normalized so that $(p^{\mathrm{in}}_{\omega'}, p^{\mathrm{in}}_{\omega''}) = \delta(\omega' - \omega'')$ under Eq. (65.40), exactly the positive-frequency condition Eq. (64.44) of the inertial vacuum.

The out-modes are defined by being purely positive-frequency with respect to the retarded time u on $\mathcal{I}^+$,

$$p^{\mathrm{out}}_{\omega}(u) \;=\; \frac{1}{\sqrt{4\pi\omega}}\, e^{-i\omega u}, \qquad \omega > 0. \tag{65.42}$$

To compute their overlap with the in-modes we need the out-mode *on* $\mathcal{I}^-$. This is precisely where the geometry enters, and it is the only place it enters. A late-time out-mode, propagated backwards through the (locally trivial) geometry by geometric optics, arrives on $\mathcal{I}^-$ as the same oscillation evaluated along the ray, i.e. with u replaced by the function $u(v)$ of Eq. (65.37), and with support only on the rays that actually escape, $v < v_0$:

$$p^{\mathrm{out}}_{\omega}(v) \;=\; \frac{1}{\sqrt{4\pi\omega}}\, \exp\!\big[-i\omega\, u(v)\big]\, \theta(v_0 - v) \;=\; \frac{1}{\sqrt{4\pi\omega}}\, \big[\kappa(v_0 - v)\big]^{\,i\omega/\kappa}\, \theta(v_0 - v), \tag{65.43}$$

where the second equality used Eq. (65.37) to write $e^{-i\omega u} = e^{\,i(\omega/\kappa)\ln[\kappa(v_0 - v)]} = [\kappa(v_0 - v)]^{i\omega/\kappa}$. The Heaviside factor records that the ray reaching $\mathcal{I}^+$ at finite u came from $v < v_0$; the late-time piece, $u \to +\infty$, comes from $v \to v_0^-$, where the wave oscillates infinitely rapidly. That infinite pile-up of oscillations at v_0 is the source of the entire effect.

With both bases expressed on $\mathcal{I}^-$ we read off the Bogoliubov coefficients from Theorem 64.11, identifying the out-modes with the new basis v_i and the

in-modes with the old basis u_j:

$$\alpha_{\omega\omega'} \;=\; (p^{\rm out}_{\omega}, p^{\rm in}_{\omega'}), \qquad \beta_{\omega\omega'} \;=\; -\,(p^{\rm out}_{\omega}, p^{\rm in*}_{\omega'}), \tag{65.44}$$

the overlaps taken in the product Eq. (65.40). We now reduce these to ordinary integrals. For any solution f paired against a plane wave $g = e^{-i\omega' v}/\sqrt{4\pi\omega'}$, integrate the first term of Eq. (65.40) by parts; the boundary contributions vanish on a normalizable wave packet (made precise below), and

$$(f,g)_{\mathcal{I}^-} = i\int \mathrm{d}v\,\left[(\partial_v f)g^* - f\partial_v g^*\right] = -\,2i\int \mathrm{d}v\, f\,\partial_v g^*. \tag{65.45}$$

Applying Eq. (65.45) to α, with $g = p^{\rm in}_{\omega'}$ so that $g^* = e^{+i\omega' v}/\sqrt{4\pi\omega'}$ and $\partial_v g^* = i\omega' g^*$,

$$\begin{aligned}\alpha_{\omega\omega'} &= -2i\int_{-\infty}^{v_0}\mathrm{d}v\; p^{\rm out}_{\omega}(v)\,(i\omega')\,\frac{e^{+i\omega' v}}{\sqrt{4\pi\omega'}} = 2\omega'\int_{-\infty}^{v_0}\mathrm{d}v\;\frac{[\kappa(v_0-v)]^{i\omega/\kappa}}{\sqrt{4\pi\omega}}\,\frac{e^{+i\omega' v}}{\sqrt{4\pi\omega'}}\\ &= \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\;\kappa^{\,i\omega/\kappa}\int_{-\infty}^{v_0}\mathrm{d}v\;(v_0-v)^{i\omega/\kappa}\,e^{+i\omega' v}.\end{aligned} \tag{65.46}$$

The same manipulation for β, now with $g = p^{\rm in*}_{\omega'}$ so that $g^* = e^{-i\omega' v}/\sqrt{4\pi\omega'}$ and $\partial_v g^* = -i\omega' g^*$, flips the sign of the exponent and, together with the explicit minus in Eq. (65.44), gives

$$\beta_{\omega\omega'} = \frac{1}{2\pi}\sqrt{\frac{\omega'}{\omega}}\;\kappa^{\,i\omega/\kappa}\int_{-\infty}^{v_0}\mathrm{d}v\;(v_0-v)^{i\omega/\kappa}\,e^{-i\omega' v}. \tag{65.47}$$

Equations Eq. (65.46) and Eq. (65.47) are identical except for the sign of ω' in the exponential: $\beta_{\omega\omega'}$ is $\alpha_{\omega\omega'}$ continued to negative ω'. This single observation, once the integrals are done, is the whole physics.

65.2.2 Evaluation of the overlap and the exact thermal ratio

Both integrals are of one type. Substitute $x = v_0 - v$ (so $v = v_0 - x$, $\mathrm{d}v = -\mathrm{d}x$, and the range $v \in (-\infty, v_0)$ becomes $x \in (0,\infty)$). For α this turns the exponential into $e^{i\omega'(v_0-x)} = e^{i\omega' v_0}e^{-i\omega' x}$, and for β into $e^{-i\omega' v_0}e^{+i\omega' x}$. Hence

$$\alpha_{\omega\omega'} = \frac{\kappa^{\,i\omega/\kappa}}{2\pi}\sqrt{\frac{\omega'}{\omega}}\;e^{\,i\omega' v_0}\;\underbrace{\int_0^{\infty}\mathrm{d}x\;x^{\,i\omega/\kappa}\,e^{-i\omega' x}}_{\equiv\; P(\omega,\omega')}, \tag{65.48}$$

$$\beta_{\omega\omega'} = \frac{\kappa^{\,i\omega/\kappa}}{2\pi}\sqrt{\frac{\omega'}{\omega}}\;e^{-i\omega' v_0}\;\underbrace{\int_0^{\infty}\mathrm{d}x\;x^{\,i\omega/\kappa}\,e^{+i\omega' x}}_{\equiv\; Q(\omega,\omega')}. \tag{65.49}$$

The prefactors of P and Q are identical and have unit modulus apart from the common $\frac{1}{2\pi}\sqrt{\omega'/\omega}$. The entire content of the calculation is therefore the ratio P/Q, and both P and Q are Euler Γ-function integrals once the contour is fixed.

Worked Example 65.6 (The branch cut that makes the spectrum thermal)**.** We evaluate P and Q and extract their ratio, being scrupulous about the analytic continuation, since the answer depends entirely on it. The master formula is the Euler integral

$$\int_0^\infty \mathrm{d}x\; x^{s-1}\, e^{-px} \;=\; \frac{\Gamma(s)}{p^{\,s}}, \qquad \operatorname{Re} s>0,\ \operatorname{Re} p>0, \tag{65.50}$$

proved by analytic continuation of the real-argument case (Theorem 5.23 supplies the contour technology; the integral converges at $x=0$ because $\operatorname{Re} s>0$ and at $x=\infty$ because $\operatorname{Re} p>0$). Here $x^{i\omega/\kappa}=x^{s-1}$ with

$$s \;=\; 1+\frac{i\omega}{\kappa}, \qquad \operatorname{Re} s = 1>0, \tag{65.51}$$

so the endpoint at $x=0$ is harmless. The subtlety is entirely at $x=\infty$: the exponentials $e^{\mp i\omega' x}$ are purely oscillatory, sitting exactly on the boundary $\operatorname{Re} p=0$ where Eq. (65.50) is borderline divergent. Each integral must be defined by tilting p infinitesimally into the convergent half-plane — and the two integrals tilt in *opposite* directions, which is the origin of the entire effect.

The integral P (the α-coefficient). Here $e^{-i\omega' x}$, so $p=i\omega'$. To make the integrand decay we give ω' a vanishing negative imaginary part, $\omega'\to\omega'-i0^+$, equivalently $p=i\omega'+0^+$, which sits just inside $\operatorname{Re} p>0$. Then $p=\omega'\,e^{+i\pi/2}$ (the point $i\omega'$ lies on the positive imaginary axis, reached from the right half-plane), and

$$P(\omega,\omega') = \frac{\Gamma(s)}{(i\omega')^s} = \Gamma(s)\,(\omega')^{-s}\,e^{-i\pi s/2}. \tag{65.52}$$

The integral Q (the β-coefficient). Here $e^{+i\omega' x}$, so $p=-i\omega'$; convergence now demands $\omega'\to\omega'+i0^+$, i.e. $p=-i\omega'+0^+$, so that $p=\omega'\,e^{-i\pi/2}$ (the point $-i\omega'$ on the *negative* imaginary axis, again reached from the right half-plane), giving

$$Q(\omega,\omega') = \frac{\Gamma(s)}{(-i\omega')^s} = \Gamma(s)\,(\omega')^{-s}\,e^{+i\pi s/2}. \tag{65.53}$$

The branch of the power p^s is the principal one, fixed by writing $p=\omega' e^{\pm i\pi/2}$ with the phase in $(-\pi,\pi]$ and $p^s=(\omega')^s e^{\pm i\pi s/2}$; this is the unique continuation that keeps $\operatorname{Re} p>0$ throughout the tilt, and it is the same positive-frequency-in-the-lower -half-plane prescription that fixed the Unruh modes in Theorem 64.20. The two coefficients differ only through these opposite tilts: P rotates the contour into the lower-half x-plane, Q into the upper-half x-plane, around the branch point of $x^{i\omega/\kappa}$ at the origin (Fig. 65.2).

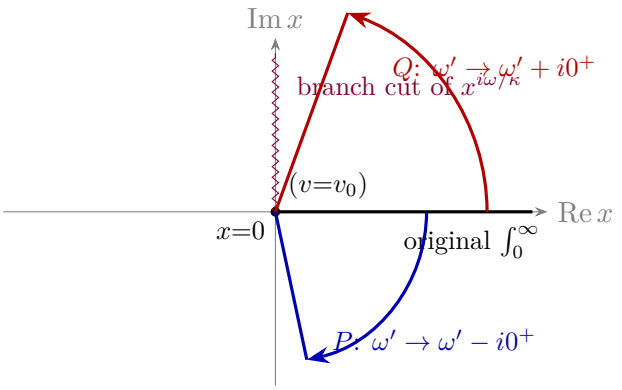


Figure 65.2: The contour rotations that define the overlap integrals P (the α-coefficient, Eq. (65.48)) and Q (the β-coefficient, Eq. (65.49)). The integrand $x^{i\omega/\kappa}e^{\mp i\omega' x}$ has a branch point at $x = 0$, the image of the horizon-grazing ray $v = v_0$; the cut is drawn up the imaginary axis. Convergence at $x \to \infty$ forces the two integrals to rotate the contour in opposite senses — P into the lower half-plane, Q into the upper — so the two coefficients sample opposite sides of the branch point. The relative phase picked up around the cut is $e^{-i\pi s} = -e^{\pi\omega/\kappa}$ (Eq. (65.54)), the Boltzmann factor of the Hawking spectrum.

The ratio. Dividing Eq. (65.52) by Eq. (65.53), the $\Gamma(s)$ and $(\omega')^{-s}$ cancel and only the phases survive:

$$\frac{P}{Q} = e^{-i\pi s} = e^{-i\pi(1+i\omega/\kappa)} = e^{-i\pi}\, e^{\pi\omega/\kappa} = -\, e^{\pi\omega/\kappa}. \tag{65.54}$$

Restoring the prefactors of Eqs. (65.48) and (65.49), which differ only by the unit-modulus phase $e^{2i\omega' v_0}$, the Bogoliubov coefficients obey

$$\alpha_{\omega\omega'} \;=\; -\, e^{\pi\omega/\kappa}\, e^{\,2i\omega' v_0}\, \beta_{\omega\omega'}, \tag{65.55}$$

and, taking moduli, the exact relation that drives everything,

$$\boxed{\,|\alpha_{\omega\omega'}| \;=\; e^{\pi\omega/\kappa}\, |\beta_{\omega\omega'}|\,} \tag{65.56}$$

holding *for every* ω', with the proportionality constant $e^{\pi\omega/\kappa}$ depending only on the out-frequency ω and the surface gravity κ, never on the in-frequency ω'.

Equation (65.56) is the technical heart of the chapter, and it is worth pausing on what made it thermal. The logarithm in the geometric-optics map Eq. (65.37) turned the out-mode into a *power* of $(v_0 - v)$, Eq. (65.43); a power has a branch point; and the positive- and negative-frequency overlaps with that power are forced, by convergence alone, to circle the branch point in opposite directions. The phase accrued on going around is exactly $e^{\pi\omega/\kappa}$. Had the map been linear,

$u \propto v$, there would be no branch point, β would vanish, and the in- and out-vacua would coincide — no particles. It is the *logarithmic* stretching of the horizon, and nothing else, that produces the radiation, and the surface gravity κ enters only as the rate of that stretching.

65.2.3 The occupation number and the Planck distribution

We now convert the ratio Eq. (65.56) into the number of quanta a late-time observer registers. By the particle-content theorem Theorem 64.13, the expectation value of the out-number operator $N_\omega = b_\omega^\dagger b_\omega$ in the in-vacuum is the squared β-matrix summed over the in-modes,

$$\langle N_\omega \rangle \;=\; \langle 0_{\rm in}|\, b_\omega^\dagger b_\omega \,|0_{\rm in}\rangle \;=\; \int_0^\infty \mathrm{d}\omega' \,|\beta_{\omega\omega'}|^2, \tag{65.57}$$

(Eq. (64.145)), the statement that each $\beta_{\omega\omega'}$ is the amplitude for the in-vacuum to contain an out-particle of frequency ω paired with the in-mode ω'. To evaluate Eq. (65.57) we use the Bogoliubov identity that enforces the canonical commutators of the out-operators (Theorem 64.12): in continuum normalization,

$$\int_0^\infty \mathrm{d}\omega' \left(\alpha_{\omega\omega'}\alpha^*_{\tilde\omega\omega'} - \beta_{\omega\omega'}\beta^*_{\tilde\omega\omega'}\right) \;=\; \delta(\omega-\tilde\omega). \tag{65.58}$$

Set $\tilde\omega = \omega$. The ratio Eq. (65.56) holds pointwise in ω' with a coefficient $e^{\pi\omega/\kappa}$ that is independent of ω', so $|\alpha_{\omega\omega'}|^2 = e^{2\pi\omega/\kappa}|\beta_{\omega\omega'}|^2$ for every ω', and the factor pulls straight out of the integral:

$$\left(e^{2\pi\omega/\kappa}-1\right)\int_0^\infty \mathrm{d}\omega' \,|\beta_{\omega\omega'}|^2 \;=\; \delta(0). \tag{65.59}$$

Comparing with Eq. (65.57),

$$\langle N_\omega\rangle \;=\; \frac{\delta(0)}{e^{2\pi\omega/\kappa}-1}. \tag{65.60}$$

The divergent $\delta(0)$ is not a pathology; it is the well-understood signature of using sharp-frequency plane waves, which are spread over all of retarded time, to count a steady flux. We make it finite in either of two equivalent ways.

Wave-packet regularization. Following Hawking's original treatment, one replaces the continuum modes by normalizable packets

$$p_{jn} \;=\; \frac{1}{\sqrt{\varepsilon}}\int_{j\varepsilon}^{(j+1)\varepsilon} \mathrm{d}\omega\; e^{\,2\pi i n\omega/\varepsilon}\, p^{\rm out}_\omega, \qquad j\ge 0,\; n\in\mathbb{Z}, \tag{65.61}$$

peaked in frequency about $\omega_j = j\varepsilon$ with width ε and in retarded time about $u_n = 2\pi n/\varepsilon$. These are orthonormal, so $(p_{jn}, p_{j'n'}) = \delta_{jj'}\delta_{nn'}$ and the troublesome $\delta(0)$ is replaced by unity. Because the ratio Eq. (65.56) is a statement at fixed ω that depends on ω' only through quantities that vary slowly across a packet, it survives the smearing for $\varepsilon \to 0$ up to corrections of order ε, and the packet occupation is finite, $\langle N_{jn}\rangle = (e^{2\pi\omega_j/\kappa}-1)^{-1}$.

Steady-flux reading. Equivalently, regulate the continuum directly: $\delta(0) = \lim_{\tilde{\omega}\to\omega}\delta(\omega-\tilde{\omega}) = \frac{1}{2\pi}\int_{-\infty}^{\infty}\mathrm{d}t = \frac{T}{2\pi}$, the total observation time T divided by 2π. The number of out-quanta of frequency ω counted over time T is then $\langle N_\omega\rangle = (T/2\pi)/(e^{2\pi\omega/\kappa}-1)$, so the *rate* per unit frequency at which quanta of frequency ω stream to infinity is

$$\frac{\mathrm{d}N}{\mathrm{d}t\,\mathrm{d}\omega} = \frac{1}{2\pi}\,\frac{1}{e^{2\pi\omega/\kappa}-1}, \tag{65.62}$$

with the $1/2\pi$ the density of single-particle states in energy per unit time (one state per phase-space cell $\Delta\omega\,\Delta t = 2\pi$). Either way, the per-mode occupation number is

$$\boxed{\langle N_\omega\rangle = \frac{1}{e^{2\pi\omega/\kappa}-1} = \frac{1}{e^{\omega/T_H}-1}} \tag{65.63}$$

which is exactly the Bose–Einstein distribution for a free boson mode of energy ω in thermal equilibrium at the temperature read off by matching exponents,

$$\boxed{T_H = \frac{\kappa}{2\pi} = \frac{1}{8\pi GM}} \tag{65.64}$$

the Hawking temperature, where the last equality used $\kappa = 1/4GM$ from Eq. (65.38). A black hole formed by collapse is therefore not black: its horizon makes the in-vacuum radiate to infinity a flux of ϕ-quanta whose spectrum is precisely Planckian, at a temperature fixed by the surface gravity alone. Setting aside (until Section 65.3) the propagation of the quanta out through the gravitational potential, the bare horizon-emission spectrum is Eq. (65.62).

Theorem 65.7 (Thermality of the horizon spectrum)**.** *For the free real scalar in the in-vacuum of a spacetime whose late-time geometric-optics map is the logarithm Eq. (65.37) with surface gravity κ, the Bogoliubov coefficients satisfy $|\alpha_{\omega\omega'}| = e^{\pi\omega/\kappa}|\beta_{\omega\omega'}|$ Eq. (65.56), and the expected number of out-quanta of frequency ω is the Bose–Einstein occupation Eq. (65.63) at the Hawking temperature Eq. (65.64). The result is independent of the angular momentum ℓ of the channel and of the detailed profile of the collapse: it depends on the collapse history only through the single number κ.*

Proof. The mode-overlap reduction Eqs. (65.46) and (65.47), the Γ-function evaluation Eqs. (65.52) and (65.53), and the branch-cut ratio Eq. (65.54) together establish Eq. (65.56). The occupation Eq. (65.63) then follows from the particle-content formula Eq. (65.57) and the Bogoliubov identity Eq. (65.58) as in Eqs. (65.59) and (65.60). Independence of ℓ is manifest: the angular and radial structure entered only through the overall normalization of the modes and was stripped by the reduction to the 1+1 problem, while the exponent $e^{\pi\omega/\kappa}$ came entirely from the universal logarithm Eq. (65.37), which Section 65.1 showed is controlled by κ regardless of the interior matter profile. □

Physical Insight: the partner inside the horizon

The relation Eq. (65.55) is more than a magnitude. With $|\alpha|$ and $|\beta|$ locked together, the in-vacuum expressed in the out-basis is not empty but a *two-mode squeezed state*: each emitted out-quantum at $\mathcal{I}^+$ is created together with a partner that falls through the horizon, the two sharing the entanglement that a squeezing transformation builds (Theorem 64.14). The thermal occupation Eq. (65.63) is what an observer who can access only the exterior modes sees after tracing out the infalling partners: the reduced density matrix on the out-region is exactly the thermal density matrix $\rho_{\text{out}} \propto e^{-N/T_H}$ of Eq. (64.174). This is the precise sense in which the radiation is thermal — not merely Planckian in its one-point occupation numbers, but a genuine mixed thermal state arising from entanglement across the horizon. It is also the precise origin of the information puzzle of Section 65.8: the global state is pure, yet everything outside the horizon is exactly thermal.

Worked Example 65.8 (How cold is a black hole?)**.** The Hawking temperature Eq. (65.64) is written in natural units; to feel its size we restore $\hbar, c, k_B$. Dimensional analysis on $T_H = \kappa/2\pi$ with $\kappa = c^2/(2r_s)$ a surface gravity (acceleration) and $r_s = 2GM/c^2$ the Schwarzschild radius gives

$$k_B T_H \;=\; \frac{\hbar\,\kappa}{2\pi c} \;=\; \frac{\hbar c^3}{8\pi G M}, \tag{65.65}$$

the unique combination of $\hbar$ (the radiation is a quantum effect), c, G, and M with the dimensions of energy. Inserting $\hbar = 1.055\times10^{-34}\,\mathrm{J\,s}$, $c = 3.00\times10^{8}\,\mathrm{m/s}$, $G = 6.67\times10^{-11}\,\mathrm{m^3 kg^{-1} s^{-2}}$, $k_B = 1.38\times10^{-23}\,\mathrm{J/K}$, and a solar mass $M_\odot = 1.99\times10^{30}\,\mathrm{kg}$,

$$T_H(M_\odot) = \frac{(1.055\times10^{-34})(3.00\times10^{8})^3}{8\pi\,(6.67\times10^{-11})(1.99\times10^{30})(1.38\times10^{-23})}\,\mathrm{K} \;\approx\; 6.2\times10^{-8}\,\mathrm{K}, \tag{65.66}$$

and since $T_H \propto 1/M$,

$$T_H \;\approx\; 6.2\times10^{-8}\,\mathrm{K}\,\frac{M_\odot}{M}. \tag{65.67}$$

A solar-mass black hole is colder than the 2.7 K cosmic microwave background by nearly eight orders of magnitude; immersed in the present Universe it absorbs far more than it radiates, and the thermal flux Eq. (65.62) is utterly negligible. The temperature reaches the CMB only for $M \sim 10^{-8}M_\odot \sim 10^{22}\,\mathrm{kg}$ (a small asteroid mass), and becomes appreciable only for much lighter holes: a hole of $M \sim 10^{12}\,\mathrm{kg}$ radiates at $T_H \sim 10^{11}\,\mathrm{K} \sim 10\,\mathrm{MeV}$, hot enough to emit electrons and photons copiously. The smallness of Eq. (65.66) is why Hawking radiation from astrophysical holes is unobservable, and why the phenomenon, though it

rewrote the foundations, has never been measured directly. Its importance is structural, not observational: it is the thread that ties quantum field theory, gravity, and thermodynamics into a single knot, as the rest of this chapter develops.

65.2.4 Cross-check: the KMS condition and the Hartle–Hawking state

The Bogoliubov computation just completed is dynamical: it follows a field through an actual collapse and asks what a detector at infinity finds long after the horizon forms. There is a second, purely equilibrium, characterization of thermality that must give the same temperature if the result is to be trusted, and it is the criterion we already met in the Unruh analysis. A state ω_β of a quantum field is thermal at inverse temperature β — it is a Kubo–Martin–Schwinger (KMS) state — if and only if its two-point function, analytically continued in the time argument, is periodic in imaginary time with period β:

$$G(t - i\beta) \;=\; G(t), \qquad G(t) \equiv \langle \phi(t)\, \phi(0) \rangle_\beta, \tag{65.68}$$

the relation Eq. (64.163) that we showed is equivalent to the detailed balance $\langle \phi(t)\phi(0)\rangle = \langle \phi(0)\phi(t + i\beta)\rangle$ obeyed by any Gibbs ensemble $\rho \propto e^{-\beta H}$. For the uniformly accelerated observer of Theorem 64.20 the relevant time was the boost (Rindler) time, the Minkowski vacuum Green's function was periodic in it with period $2\pi/a$, and the detector therefore registered a temperature $T = a/2\pi$.

The black-hole case is the same statement with the boost replaced by the time-translation that the horizon generates. Near the horizon the Killing time t plays the role of the Rindler boost parameter, and the surface gravity κ plays the role of the acceleration a: the near-horizon geometry is, to leading order, Rindler space with boost rate κ (this is exactly the statement that makes Eq. (65.38) a surface *gravity*, and it is developed metrically in Section 65.5.1). The *Hartle–Hawking state* is by definition the state whose Green's function is regular across the whole horizon, the equilibrium analogue of the Minkowski vacuum for Rindler; the regularity requirement forces its Green's function to be periodic in imaginary Killing time with period

$$\beta_H \;=\; \frac{2\pi}{\kappa}, \tag{65.69}$$

exactly as the smoothness of the Euclidean section will require in Theorem 65.25. By the KMS criterion Eq. (65.68) the Hartle–Hawking state is therefore thermal at

$$T_H \;=\; \frac{1}{\beta_H} \;=\; \frac{\kappa}{2\pi}, \tag{65.70}$$

identical to Eq. (65.64). The two derivations approach the same number from opposite directions. The Bogoliubov calculation is a non-equilibrium scattering problem on a genuinely time-dependent collapse geometry, and it tells us that

the late-time radiation *is* thermal even though nothing was assumed to be in equilibrium; the KMS argument is a statement about the static eternal black hole and its equilibrium Hartle–Hawking vacuum. That they yield one temperature is the assurance that the thermal character of the horizon is a property of the geometry near it, not an artefact of the collapse history — the collapse merely selects the in-vacuum that, at late times, looks locally like the Hartle–Hawking state near the horizon and so radiates at its temperature.

Caution: thermal occupation is not the full state at infinity

Equation Eq. (65.63) is the occupation of a mode *at the horizon end* of the trace-back, computed as if every quantum created there reached infinity freely. It is the bare emission spectrum. A real detector sits at infinity, and between it and the horizon stands the curvature potential of Theorem 65.9, off which low-frequency quanta are largely backscattered. The flux Eq. (65.62) is therefore the input to, not the output of, the propagation problem. The frequency dependence of the transmission — the greybody factor — is what distinguishes the light a telescope would see from a true black-body spectrum, and it is the subject of the next section. Until then, "the Hawking spectrum" means precisely Eq. (65.62): a Planckian horizon emission at $T_H = \kappa/2\pi$.

We have established the central result of the chapter in its sharpest local form: the horizon of a collapsing black hole pours the in-vacuum into a Planck spectrum at $T_H = \kappa/2\pi = 1/8\pi GM$, with the thermal Boltzmann factor traced to a single branch cut in the logarithmic phase map. The next section takes this bare horizon spectrum Eq. (65.62) and propagates it out through the Regge–Wheeler barrier, dressing it with the greybody factor $\Gamma_{\omega\ell}$ to yield the luminosity, evaporation law, and lifetime actually seen at infinity; the temperature Eq. (65.64) derived here is the input that Sections 4–6 will promote to a genuine thermodynamic temperature and confirm twice more, by the Euclidean path integral and by the horizon anomaly.

65.3 Greybody Factors, Luminosity, and Evaporation

The previous section delivered a sharp but local statement: a mode of frequency ω that climbs out of the collapsing geometry to reach future null infinity is populated according to a Planck distribution at the Hawking temperature $T_H = \kappa/2\pi = 1/8\pi GM$ (Section 65.2, Eq. (65.64)). That statement was made *at the horizon*: it counts the quanta the in-vacuum pours into each outgoing wavepacket as it peels off the surface of the collapsing star. But the quantum that is created just outside the horizon is not the quantum that a detector at infinity registers. Between the horizon and the asymptotic region sits the gravitational field of

the hole itself, and an outgoing wave must propagate through it. Some of the wave tunnels through to infinity; the rest is backscattered by the spacetime curvature and falls back toward the hole. The flux that actually arrives at a distant observatory is therefore the horizon Planck spectrum *multiplied by the fraction of each mode that survives the trip out.* That fraction is the greybody factor, and computing it is the business of this section: it converts a formal emission rate into a physical luminosity, and the luminosity in turn fixes the rate at which the hole loses mass, the law by which it shrinks, and the finite time in which it disappears.

We carry the free real scalar of ϕ^4 theory (Chapter 9) in its $\lambda = 0$ sector, taken massless, as the matter field whose emission we compute. Massless is the natural choice: a quantum of rest mass m can be emitted only above the threshold $\omega > m$, and for the astrophysical holes of interest m is negligible against T_H until the very last stages; the mass merely adds a term $f(r)\,m^2$ to the effective potential built below and shifts the spectrum's lower limit. With $m = 0$ the scalar is the cleanest possible probe of the geometry, and the s-wave channel can be solved in closed form. Having done the scalar in full we quote, at the end, the spin-dependent modifications for photons and gravitons.

65.3.1 The field equation as a one-dimensional scattering problem

Recall the exterior Schwarzschild geometry (Eq. (65.1)),

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 - \frac{\mathrm{d}r^2}{f(r)} - r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2GM}{r}, \qquad r > 2GM, \tag{65.71}$$

with $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2$ the round metric on the unit two-sphere, the mostly-minus signature $(+,-,-,-)$ in force, and Newton's constant G kept explicit. The free massless scalar obeys the curved-space Klein–Gordon equation

$$\Box\phi = \frac{1}{\sqrt{-g}}\,\partial_\mu\!\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi\right) = 0, \tag{65.72}$$

the scalar wave operator built from the covariant measure $\sqrt{-g}$ and the inverse metric, exactly the d'Alembertian whose conformal subtleties were catalogued in Chapter 64. (For a conformally coupled scalar one would add $\xi R\phi$ with $R = 0$ on Schwarzschild, so the vacuum field is insensitive to the coupling; we proceed with minimal coupling.) From Eq. (65.71) the nonzero inverse-metric components are $g^{tt} = 1/f$, $g^{rr} = -f$, $g^{\theta\theta} = -1/r^2$, $g^{\varphi\varphi} = -1/(r^2\sin^2\theta)$, and the determinant is $g = -r^4\sin^2\theta$, so $\sqrt{-g} = r^2\sin\theta$. Substituting into Eq. (65.72) and using that $\sqrt{-g}$ and the metric are t-independent,

$$\frac{1}{f}\,\partial_t^2\phi \;-\; \frac{1}{r^2}\,\partial_r\!\left(r^2 f\,\partial_r\phi\right) \;-\; \frac{1}{r^2}\,\Delta_{S^2}\phi \;=\; 0, \tag{65.73}$$

where $\Delta_{S^2} = \frac{1}{\sin\theta}\partial_\theta(\sin\theta\,\partial_\theta) + \frac{1}{\sin^2\theta}\partial_\varphi^2$ is the Laplacian on the sphere, whose eigenfunctions are the spherical harmonics of Section 2.4, $\Delta_{S^2} Y_{\ell m} = -\ell(\ell+1)Y_{\ell m}$.

The geometry is static and spherically symmetric, so the natural modes separate the time and angular dependence. Because the energy of an asymptotic quantum is the frequency conjugate to the Killing time t, we write a positive-frequency mode as

$$\phi_{\omega\ell m}(t, r, \theta, \varphi) = \frac{\psi_{\omega\ell}(r)}{r}\, Y_{\ell m}(\theta, \varphi)\, e^{-i\omega t}, \qquad \omega > 0, \tag{65.74}$$

the factor $1/r$ inserted in anticipation: it is exactly what is needed to turn the radial operator into a one-dimensional Schrödinger form with no first-derivative term, as we now verify.

Theorem 65.9 (Radial reduction to a 1D scattering problem)**.** *With the ansatz Eq. (65.74) and the tortoise coordinate r_* defined by*

$$\frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{1}{f(r)}, \qquad \textit{i.e.} \qquad r_* = r + 2GM\,\ln\!\Big(\frac{r}{2GM} - 1\Big), \tag{65.75}$$

the Klein–Gordon equation Eq. (65.72) reduces, for each angular momentum ℓ, to the time-independent one-dimensional wave equation

$$\frac{\mathrm{d}^2\psi_{\omega\ell}}{\mathrm{d}r_*^2} + \big[\omega^2 - V_\ell(r)\big]\,\psi_{\omega\ell} = 0, \tag{65.76}$$

with the Regge–Wheeler-type effective potential

$$V_\ell(r) = f(r)\left[\frac{\ell(\ell+1)}{r^2} + \frac{2GM}{r^3}\right], \qquad f(r) = 1 - \frac{2GM}{r}. \tag{65.77}$$

As r ranges over the exterior $(2GM, \infty)$ the tortoise coordinate r_* ranges over the whole line $(-\infty, +\infty)$: the horizon is pushed to $r_* \to -\infty$ and infinity sits at $r_* \to +\infty$. The potential Eq. (65.77) is non-negative, vanishes at both ends, and forms a single barrier in between.

Proof. Insert Eq. (65.74) into Eq. (65.73). The time derivative gives $\partial_t^2\phi = -\omega^2\phi$, and the angular Laplacian gives $\Delta_{S^2}\phi = -\ell(\ell+1)\phi$. Dividing out $Y_{\ell m}e^{-i\omega t}$ and writing $R(r) = \psi_{\omega\ell}/r$ for the radial factor leaves

$$-\frac{\omega^2}{f}\,R + \frac{1}{r^2}\,\frac{\mathrm{d}}{\mathrm{d}r}\Big(r^2 f\,\frac{\mathrm{d}R}{\mathrm{d}r}\Big) - \frac{\ell(\ell+1)}{r^2}\,R = 0, \tag{65.78}$$

where the overall sign of Eq. (65.73) has been flipped for convenience. We now substitute $R = \psi/r$ and clear the $1/r$. For the radial derivative term,

$$r^2 f\,\frac{\mathrm{d}}{\mathrm{d}r}\Big(\frac{\psi}{r}\Big) = r^2 f\Big(\frac{\psi'}{r} - \frac{\psi}{r^2}\Big) = f\,(r\psi' - \psi), \tag{65.79}$$

and differentiating once more,

$$\frac{\mathrm{d}}{\mathrm{d}r}\left[f(r\psi'-\psi)\right] = f'(r\psi'-\psi)+f\left(\psi'+r\psi''-\psi'\right) = f'(r\psi'-\psi)+fr\psi''. \tag{65.80}$$

Dividing by r^2 and assembling Eq. (65.78) multiplied through by r gives

$$\frac{\omega^2}{f}\,\psi + f'\psi' - \frac{f'}{r}\,\psi + f\psi'' - \frac{\ell(\ell+1)}{r^2}\,\psi = 0. \tag{65.81}$$

Now pass to the tortoise coordinate. From Eq. (65.75), $\mathrm{d}/\mathrm{d}r = (1/f)\,\mathrm{d}/\mathrm{d}r_*$, and writing a dot for $\mathrm{d}/\mathrm{d}r_*$ we have $\psi' = \dot\psi/f$ and, using $\mathrm{d}f/\mathrm{d}r_* = ff'$,

$$\psi'' = \frac{1}{f}\frac{\mathrm{d}}{\mathrm{d}r_*}\Big(\frac{\dot\psi}{f}\Big) = \frac{1}{f}\Big(\frac{\ddot\psi}{f} - \frac{ff'}{f^2}\dot\psi\Big) = \frac{\ddot\psi}{f^2} - \frac{f'}{f^2}\dot\psi. \tag{65.82}$$

Therefore $f'\psi' + f\psi'' = \dfrac{f'}{f}\dot\psi + \dfrac{\ddot\psi}{f} - \dfrac{f'}{f}\dot\psi = \dfrac{\ddot\psi}{f}$: the two first-derivative pieces cancel, which is exactly the work the $1/r$ in Eq. (65.74) was inserted to do. Substituting and multiplying Eq. (65.81) by f,

$$\ddot\psi + \omega^2\psi - f\Big[\frac{\ell(\ell+1)}{r^2} + \frac{f'}{r}\Big]\psi = 0. \tag{65.83}$$

With $f' = 2GM/r^2$ we have $f'/r = 2GM/r^3$, which is Eq. (65.76) with the potential Eq. (65.77). For the range of r_*: integrating Eq. (65.75) gives the logarithm displayed there, which $\to -\infty$ as $r \to (2GM)^+$ and $\to +\infty$ as $r \to \infty$, so r_* sweeps the whole line. Finally $V_\ell \geq 0$ on $r > 2GM$ because $f > 0$ and both bracketed terms are positive; it vanishes at the horizon (where $f \to 0$) and at infinity (where the bracket $\to 0$ as r^{-2}), and is smooth and bounded in between, hence a single hump. This is the barrier drawn in Fig. 65.3(a). □

The physics of Eq. (65.76) is now transparent. It is the time-independent Schrödinger equation for a unit-mass particle of "energy" ω^2 moving in the potential $V_\ell(r_*)$ on the full line. Because $V_\ell \to 0$ at both ends, the asymptotic solutions are free plane waves $e^{\pm i\omega r_*}$ at the horizon ($r_* \to -\infty$) and at infinity ($r_* \to +\infty$). A wave that emerges from the horizon region must traverse the barrier to reach infinity; quantum-mechanically it is partly transmitted and partly reflected. The Hawking process feeds quanta in at the horizon end; the greybody factor is the barrier's transmission probability.

65.3.2 Greybody factors and the spectrum at infinity

Fix $\omega > 0$ and ℓ. Define the scattering solution of Eq. (65.76) that is purely outgoing at infinity,

$$\psi_{\omega\ell}(r_*) \;\longrightarrow\; \begin{cases} e^{+i\omega r_*} + \mathcal{R}_{\omega\ell}\, e^{-i\omega r_*}, & r_* \to +\infty, \\ \mathcal{T}_{\omega\ell}\, e^{+i\omega r_*}, & r_* \to -\infty, \end{cases} \tag{65.84}$$

i.e. a unit-amplitude wave coming up from the horizon, of which a fraction $\mathcal{T}_{\omega\ell}$ tunnels through to infinity and a fraction $\mathcal{R}_{\omega\ell}$ is reflected back down. Equation Eq. (65.76) has real coefficients (the potential and ω^2 are real), so ψ and $\bar{\psi}$ are both solutions and their Wronskian $W[\psi,\bar{\psi}] = \psi\dot{\bar{\psi}} - \bar{\psi}\dot{\psi}$ is constant in r_*. Evaluating it at the two ends of Eq. (65.84) and equating,

$$\underbrace{2i\omega\big(|\mathcal{R}_{\omega\ell}|^2 - 1\big)}_{r_*\to+\infty} \;=\; \underbrace{-\,2i\omega\;|\mathcal{T}_{\omega\ell}|^2}_{r_*\to-\infty}, \tag{65.85}$$

which is the flux-conservation identity

$$|\mathcal{T}_{\omega\ell}|^2 + |\mathcal{R}_{\omega\ell}|^2 = 1. \tag{65.86}$$

The barrier neither creates nor destroys probability; it only partitions it.

Definition 65.10 (Greybody factor)**.** The greybody factor (or transmission factor, or absorption probability) of the (ω,ℓ) channel is the transmission probability

$$\Gamma_{\omega\ell} \;\equiv\; |\mathcal{T}_{\omega\ell}|^2 \;=\; 1 - |\mathcal{R}_{\omega\ell}|^2 \;\in [0,1]. \tag{65.87}$$

It is symmetric between the two scattering directions: by Eq. (65.86) and time-reversal invariance of Eq. (65.76), the probability that a wave from infinity is absorbed by the hole equals the probability $\Gamma_{\omega\ell}$ that a wave from the horizon escapes to infinity.

The greybody factor is the one geometric input the horizon calculation did not supply. In Section 65.2 the Bogoliubov analysis produced, for each outgoing mode, an expected occupation number of Planck form. That mode was labelled by its behaviour *at infinity*, but the thermal population was a property of the data *at the horizon*. The greybody factor is exactly the overlap between the two: of the unit thermal flux deposited at the horizon end, only the fraction $\Gamma_{\omega\ell}$ reaches infinity, the rest being reflected back across the horizon. Equivalently, in the detailed-balance language of a body in equilibrium with a thermal bath, a perfect absorber would re-emit a pure Planck spectrum; a body that absorbs only a fraction $\Gamma_{\omega\ell}$ of incident radiation emits only that fraction — it is "grey," not black, whence the name. Either way the late-time particle flux registered at infinity is

$$\frac{\mathrm{d}N_{\ell m}}{\mathrm{d}t\,\mathrm{d}\omega} = \frac{1}{2\pi}\,\frac{\Gamma_{\omega\ell}}{e^{\omega/T_H} \mp 1}, \tag{65.88}$$

with the lower sign for bosons (our scalar) and the upper for fermions, and the $1/2\pi$ the density of states in energy per unit time for a single mode (one quantum state per cell $\Delta\omega\,\Delta t = 2\pi$). The full flux sums over the $2\ell+1$ azimuthal values m and over ℓ. Setting $\Gamma_{\omega\ell} \to 1$ recovers the bare horizon Planck spectrum of Section 65.2; the greybody factor is the correction that turns it into what a telescope sees.

65.3.3 Low-frequency s-wave transmission by matched asymptotics

Equation Eq. (65.76) cannot be solved in elementary closed form for generic ω, but the two physically decisive regimes — soft ($GM\omega \ll 1$) and hard ($GM\omega \gg 1$) — both yield to exact analysis. We treat the soft limit first, where the dominant channel is the s-wave $\ell = 0$. Its potential is the pure curvature barrier $V_0 = f \cdot 2GM/r^3$, whose height is of order $(GM)^{-2} \sim T_H^2$; for $\omega \ll T_H$ the incident wave has energy far below the barrier top and tunnels only weakly, so $\Gamma_{\omega 0}$ is small. The standard tool is matched asymptotic expansions: solve the equation in a near region and a far region in which different terms dominate, then match in the overlap. The carried scalar makes this the thread's explicit computation.

Worked Example 65.11 (The s-wave greybody factor at low frequency)**.** We compute $\Gamma_{\omega 0}$ to leading order in the small dimensionless quantity $GM\omega$, working with the radial factor $R = \psi/r$, which for $\ell = 0$ satisfies the un-tortoised form of Eq. (65.78),

$$\frac{1}{r^2}\frac{\mathrm{d}}{\mathrm{d}r}\Big(r^2 f \frac{\mathrm{d}R}{\mathrm{d}r}\Big) + \frac{\omega^2}{f} R = 0. \tag{65.89}$$

Near region ($\omega r \ll 1$). Where the wave's wavelength is large compared to the radius, the second term in Eq. (65.89), of order $\omega^2 R$, is negligible against the first, of order R/r^2, *except* extremely close to the horizon where $f \to 0$; we handle that edge separately below. Dropping the ω^2 term leaves $\mathrm{d}(r^2 f R')/\mathrm{d}r = 0$, so $r^2 f R'$ is constant. With $r^2 f = r(r - 2GM)$ and partial fractions $[r(r-2GM)]^{-1} = (2GM)^{-1}[(r-2GM)^{-1} - r^{-1}]$, integration gives the near-region solution

$$R_{\mathrm{near}}(r) = A_0 + A_1 \ln f(r), \qquad f = 1 - \frac{2GM}{r}, \tag{65.90}$$

with A_0, A_1 constants. (The constant multiplying $\ln f$ is $C/2GM$ in terms of the conserved flux $C = r^2 f R'$; we absorb the $2GM$ into A_1.)

Horizon edge. Very near the horizon the transmitted wave of Eq. (65.84) is, in the $R = \psi/r$ variable, $R \simeq \mathcal{T}\, e^{i\omega r_*}/(2GM)$. From Eq. (65.75), as $r \to 2GM$ one has $r_* \simeq 2GM + 2GM \ln f$ (since $\frac{r}{2GM} - 1 = \frac{r-2GM}{2GM}$ and $f = \frac{r-2GM}{r} \to \frac{r-2GM}{2GM}$ coincide at leading order there). Hence to first order in ω,

$$R \simeq \frac{\mathcal{T}}{2GM}\, e^{i\omega(2GM + 2GM\ln f)} \simeq \frac{\mathcal{T}}{2GM}\big(1 + 2iGM\omega \ln f\big), \tag{65.91}$$

dropping the constant phase $e^{2iGM\omega} \to 1$. Matching to Eq. (65.90) term by term fixes

$$A_0 = \frac{\mathcal{T}}{2GM}, \qquad A_1 = i\omega\,\mathcal{T}. \tag{65.92}$$

The logarithmic term, which looked singular at the horizon, is precisely the slowly-varying phase of the ingoing wave; its coefficient A_1 is of order ω and so small, while A_0 is of order unity.

Far region ($r \gg 2GM$). Here $f \to 1$ and Eq. (65.89) becomes the flat-space $\ell = 0$ Helmholtz equation $(r^2R')'/r^2 + \omega^2 R = 0$, i.e. $(rR)'' + \omega^2 (rR) = 0$, with the outgoing/ingoing solution

$$R_{\rm far}(r) = \frac{1}{r}\left(\alpha\, e^{-i\omega r} + \beta\, e^{+i\omega r}\right). \tag{65.93}$$

Matching in the overlap $2GM \ll r \ll 1/\omega$. On the near side, expand Eq. (65.90) for $r \gg 2GM$ using $\ln f = \ln(1 - 2GM/r) \simeq -2GM/r$:

$$R_{\rm near} \;\longrightarrow\; A_0 - \frac{2GMA_1}{r}. \tag{65.94}$$

On the far side, expand Eq. (65.93) for $\omega r \ll 1$ using $e^{\pm i\omega r} \simeq 1 \pm i\omega r$:

$$R_{\rm far} \;\longrightarrow\; \frac{\alpha+\beta}{r} + i\omega(\beta - \alpha). \tag{65.95}$$

Equating the constant terms and the $1/r$ terms of Eqs. (65.94) and (65.95),

$$i\omega(\beta - \alpha) = A_0, \qquad \alpha + \beta = -2GMA_1. \tag{65.96}$$

Insert Eq. (65.92). The first relation gives $\beta - \alpha = A_0/i\omega = \mathcal{T}/(2iGM\omega)$, which is of order $\mathcal{T}/\omega$ and dominates; the second gives $\alpha+\beta = -2GM\cdot i\omega\mathcal{T} = -2iGM\omega\mathcal{T}$, of order $\omega\mathcal{T}$ and negligible by comparison. Hence at leading order

$$\alpha = \tfrac{1}{2}\left[(\alpha+\beta) - (\beta-\alpha)\right] \simeq -\tfrac{1}{2}(\beta-\alpha) = -\frac{\mathcal{T}}{4iGM\omega} = \frac{i\mathcal{T}}{4GM\omega}, \tag{65.97}$$

and $\beta \simeq -\alpha$, so the far solution is almost a pure standing wave: nearly total reflection, as anticipated for deep tunnelling.

Transmission probability. The incident flux comes down from infinity in the ingoing piece $\alpha e^{-i\omega r}$ of Eq. (65.93); the transmitted flux is the horizon wave $\mathcal{T}e^{i\omega r_*}$. Both ends have $V \to 0$, so a plane wave $e^{i\omega x}$ carries the same flux $\propto \omega$ at either end, and the transmission probability is the ratio of squared amplitudes,

$$\Gamma_{\omega 0} = \frac{|\mathcal{T}|^2}{|\alpha|^2} = \frac{|\mathcal{T}|^2}{|\mathcal{T}|^2/(4GM\omega)^2} = (4GM\omega)^2 = 16\,G^2M^2\omega^2. \tag{65.98}$$

Recognising the horizon area $A_H = 4\pi(2GM)^2 = 16\pi G^2M^2$, this is

$$\boxed{\Gamma_{\omega 0} = \frac{A_H\,\omega^2}{\pi} = 4(2GM\omega)^2 \quad (GM\omega \ll 1).} \tag{65.99}$$

Equation Eq. (65.99) is one of the cleanest results in the subject: the soft s-wave transmission is the horizon area in units of the wavelength squared. Direct numerical integration of Eq. (65.76) with the boundary conditions Eq. (65.84) confirms it — the computed $\Gamma_{\omega 0}/(4(2GM\omega)^2)$ tends to unity as $\omega \to 0$ (Fig. 65.3(b), where the analytic law is the dashed line tangent to the s-wave curve at the origin).

Theorem 65.12 (Soft transmission and its spin dependence)**.** *For a massless field of spin s the Regge–Wheeler potential generalises to $V_\ell^{(s)} = f\big[\ell(\ell+1)/r^2 + (1-s^2)\,2GM/r^3\big]$, with the lowest propagating multipole $\ell = s$ (a spin-s field has no multipoles below $\ell = s$). The low-frequency transmission of the dominant $\ell = s$ channel scales as*

$$\Gamma^{(s)}_{\omega,\ell=s} \;\propto\; (GM\omega)^{\,2\ell+2} = (GM\omega)^{\,2s+2} \qquad (GM\omega \ll 1). \tag{65.100}$$

In particular the scalar ($s=0$) gives Eq. (65.99) $\propto \omega^2$; the photon ($s=1$, lowest channel $\ell=1$) is suppressed as ω^4; the graviton ($s=2$, lowest channel $\ell=2$) as ω^6.

Proof sketch. The matched-asymptotic structure of Theorem 65.11 repeats for each ℓ. The near-region static equation $\mathrm{d}(r^2 f R')/\mathrm{d}r = \ell(\ell+1)R$ has solutions behaving as r^ℓ and $r^{-\ell-1}$, the regular branch matching onto the far-region spherical Bessel function $j_\ell(\omega r) \sim (\omega r)^\ell$ for small argument. The transmission probability is the square of the ratio of the transmitted to incident amplitudes, and each factor of j_ℓ contributes $(\omega r)^\ell$ evaluated at the matching radius $r \sim GM$, giving $\Gamma \propto (\omega\, GM)^{2\ell}$ from the centrifugal barrier; one further power of ω^2 comes from the horizon-area phase space exactly as in Eq. (65.99), yielding $(GM\omega)^{2\ell+2}$. Since the lowest allowed multipole is $\ell = s$, the leading soft behaviour is $(GM\omega)^{2s+2}$. The exact $\ell = 0$ scalar coefficient is the $\pi^{-1}A_H$ of Eq. (65.99); the higher-ℓ and higher-spin coefficients are fixed by the same matching and are tabulated in the further-reading sources. The full computation for $\ell = 0$ is Theorem 65.11; the $\ell \geq 1$ coefficients are not load-bearing for the luminosity, which is dominated by the channels we compute exactly. □

The spin hierarchy Eq. (65.100) has a clear physical reading: a higher-spin quantum carries at least $\ell = s$ units of orbital angular momentum and so faces a higher centrifugal barrier, which it tunnels through with exponentially greater difficulty at low energy. Soft Hawking emission is therefore dominated by the lowest spins — the scalar above all, then photons, with gravitons most strongly suppressed. This is why the carried scalar is not merely a convenient toy: it is the channel that dominates the soft end of any black hole's spectrum.

65.3.4 High frequencies and the geometric capture cross section

In the opposite limit $GM\omega \gg 1$ the wavelength is tiny compared with the curvature scale and the field propagates along null geodesics: wave optics becomes geometric optics. A quantum then behaves like a massless particle, and whether it is absorbed is decided by whether its trajectory is captured by the hole. The total transmitted power is set by the capture cross section, which we now compute from the null geodesics of Eq. (65.71).

Theorem 65.13 (Photon sphere and capture cross section)**.** *A massless particle aimed at a Schwarzschild hole with impact parameter b is captured if and only if $b < b_c = 3\sqrt{3}\,GM$. The capture cross section for massless quanta is therefore*

$$\sigma_{\text{geo}} = \pi b_c^2 = 27\pi\, G^2 M^2. \tag{65.101}$$

Proof. Work in the equatorial plane $\theta = \pi/2$. The Killing vectors ∂_t and ∂_φ give the conserved energy and angular momentum $E = f\,\dot t$ and $L = r^2\dot\varphi$ (overdot $= \mathrm{d}/\mathrm{d}\lambda$, an affine parameter). The null condition $g_{\mu\nu}\dot x^\mu \dot x^\nu = 0$ on Eq. (65.71) reads $f\dot t^2 - \dot r^2/f - r^2\dot\varphi^2 = 0$; substituting $\dot t = E/f$ and $\dot\varphi = L/r^2$ and multiplying by f,

$$\dot r^2 = E^2 - \frac{L^2}{r^2}\,f(r) = E^2\Big[1 - b^2\,W(r)\Big], \qquad W(r) \equiv \frac{f(r)}{r^2} = \frac{1}{r^2} - \frac{2GM}{r^3}, \tag{65.102}$$

with $b = L/E$ the impact parameter. The function W plays the role of an effective potential for the radial motion: a turning point occurs where $\dot r = 0$, i.e. $b^2 W(r) = 1$. A trajectory incoming from infinity is captured if it never reaches a turning point, which happens precisely when $1/b^2$ exceeds the maximum of W. Extremising, $W'(r) = -2/r^3 + 6GM/r^4 = 0$ gives the photon sphere

$$r_{\text{ph}} = 3GM, \qquad W_{\max} = W(3GM) = \frac{1}{9G^2M^2}\Big(1 - \tfrac{2}{3}\Big) = \frac{1}{27\,G^2M^2}. \tag{65.103}$$

Capture requires $1/b^2 > W_{\max}$, i.e. $b < b_c = W_{\max}^{-1/2} = 3\sqrt{3}\,GM$. The cross section is the area of the disc of captured impact parameters, $\sigma_{\text{geo}} = \pi b_c^2 = 27\pi G^2 M^2$. □

Geometric optics also fixes how the transmission factors sum at high frequency. A classical trajectory of impact parameter b carries angular momentum $L = b\omega$, hence orbital quantum number $\ell \simeq b\omega$; it is captured iff $b < b_c$, i.e. iff $\ell \lesssim b_c\omega$. Summing the multiplicities of the captured channels,

$$\sum_{\ell} (2\ell+1)\,\Gamma_{\omega\ell} \xrightarrow[GM\omega \gg 1]{} \sum_{\ell=0}^{b_c\omega} (2\ell+1) \simeq (b_c\omega)^2 = \frac{\sigma_{\text{geo}}}{\pi}\,\omega^2 = 27\,G^2M^2\,\omega^2. \tag{65.104}$$

Thus at high frequency the black hole presents the geometric absorbing area Eq. (65.101) to the field, the natural counterpart of the area law Eq. (65.99) at low frequency. The numerically computed greybody factors interpolate smoothly between the two, each $\Gamma_{\omega\ell}$ rising from its soft power law to a step that saturates at unity once ω exceeds the barrier top for that ℓ (Fig. 65.3(b)).

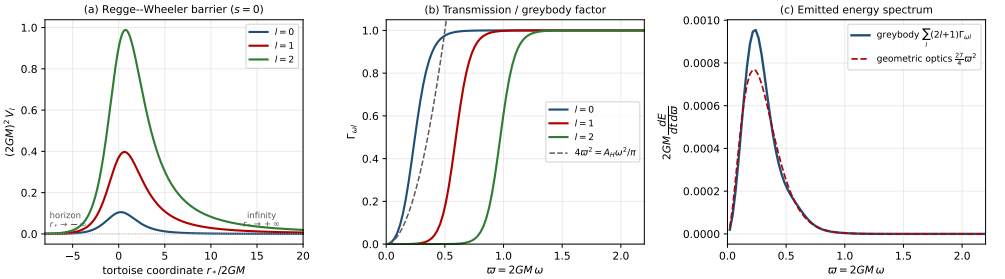


Figure 65.3: Greybody physics for the massless scalar on Schwarzschild, in units $2GM = 1$ (so $T_H = 1/4\pi$, $\varpi \equiv 2GM\omega$). **(a)** The Regge–Wheeler barrier $V_\ell(r_*)$ of Eq. (65.77) for $\ell = 0, 1, 2$: a single hump separating the horizon ($r_* \to -\infty$) from infinity ($r_* \to +\infty$), rising with ℓ. **(b)** The transmission factor $\Gamma_{\omega\ell}$ from direct numerical integration of Eq. (65.76) with the boundary conditions Eq. (65.84); the dashed line is the analytic soft s-wave law Eq. (65.99), $\Gamma_{\omega 0} = 4\varpi^2$, to which the computed curve is tangent at the origin. Each channel saturates at unity once ω clears its barrier. **(c)** The emitted energy spectrum $\mathrm{d}E/\mathrm{d}t\,\mathrm{d}\omega$ of Eq. (65.106): the greybody-filtered curve (solid) against the geometric-optics envelope $\frac{27}{4}\varpi^2$ (dashed). The greybody factors suppress the soft end and the area under the curve is the luminosity.

65.3.5 Luminosity and the Stefan–Boltzmann analogue

The power radiated to infinity is the energy flux of Eq. (65.88) weighted by ω and summed over all channels. For the massless scalar,

$$L \equiv -\frac{\mathrm{d}M}{\mathrm{d}t} = \sum_{\ell=0}^{\infty}(2\ell+1)\int_0^\infty \frac{\mathrm{d}\omega}{2\pi}\,\frac{\omega\,\Gamma_{\omega\ell}}{e^{\omega/T_H}-1}, \tag{65.105}$$

with the integrand

$$\frac{\mathrm{d}E}{\mathrm{d}t\,\mathrm{d}\omega} = \sum_{\ell}(2\ell+1)\,\frac{\omega\,\Gamma_{\omega\ell}}{2\pi\,(e^{\omega/T_H}-1)} \tag{65.106}$$

plotted in Fig. 65.3(c). The structure of Eq. (65.105) is already enough to fix the scaling with mass. Every quantity in it depends on M only through the single combination GM: the temperature is $T_H = 1/8\pi GM$, and the greybody factor $\Gamma_{\omega\ell}$ is a dimensionless function of the dimensionless variable $GM\omega$ alone (it is built from Eq. (65.76), whose only scale is $2GM$). Rescaling the integration variable to $x = \omega/T_H = 8\pi GM\omega$ pulls all the M-dependence outside: $\omega = xT_H \propto 1/(GM)$ and $\mathrm{d}\omega \propto 1/(GM)$, so the integral carries two powers of $T_H \propto 1/(GM)$ and the rest is a pure number. Hence

$$\boxed{L = -\frac{\mathrm{d}M}{\mathrm{d}t} = \frac{a_{\mathrm{eff}}}{G^2M^2}\,,} \tag{65.107}$$

the exact analogue of the Stefan–Boltzmann law $L = \sigma A T^4$ for a black body: with $A \sim G^2M^2$ and $T_H \sim 1/GM$ one has $AT_H^4 \sim 1/G^2M^2$, and the greybody factors only renormalise the numerical constant $a_{\rm eff}$. That constant depends on the spectrum of fields light enough to be radiated at the temperature T_H — on their spins, through Eq. (65.100), and their multiplicities — but not on the mass. A black hole is hotter the lighter it is, and so radiates faster the smaller it gets: the runaway that drives evaporation.

Worked Example 65.14 (Assembling the luminosity coefficient)**.** We extract $a_{\rm eff}$ from the spectrum, first analytically in two limits, then from the numerically integrated greybody factors.

Soft s-wave estimate. Keep only $\ell = 0$ and use the soft law Eq. (65.99), $\Gamma_{\omega 0} = 16G^2M^2\omega^2$, for all ω. Then

$$L_{\rm s} = \int_0^\infty \frac{{\rm d}\omega}{2\pi}\,\frac{\omega \cdot 16G^2M^2\omega^2}{e^{\omega/T_H}-1} = \frac{8G^2M^2}{\pi}\int_0^\infty \frac{\omega^3\,{\rm d}\omega}{e^{\omega/T_H}-1}. \tag{65.108}$$

The integral is the standard Bose moment $\int_0^\infty \omega^3(e^{\omega/T}-1)^{-1}{\rm d}\omega = T^4\,\Gamma(4)\zeta(4) = T^4\cdot 6\cdot\frac{\pi^4}{90} = \frac{\pi^4}{15}T_H^4$ (Eq. (64.289)). With $T_H^4 = (8\pi GM)^{-4} = 1/(4096\,\pi^4G^4M^4)$,

$$L_{\rm s} = \frac{8G^2M^2}{\pi}\cdot\frac{\pi^4}{15}\cdot\frac{1}{4096\,\pi^4G^4M^4} = \frac{8}{15\cdot 4096}\,\frac{1}{\pi\,G^2M^2} = \frac{1}{7680\,\pi}\,\frac{1}{G^2M^2} \approx \frac{4.1\times10^{-5}}{G^2M^2}. \tag{65.109}$$

This overcounts the hard end (where the true $\Gamma_{\omega 0}$ saturates at 1 rather than growing as ω^2) but captures the soft channel that dominates.

Geometric-optics estimate. Replace the multipole sum by its high-frequency form Eq. (65.104), $\sum_\ell(2\ell+1)\Gamma_{\omega\ell} \to 27G^2M^2\omega^2$, valid where the thermal weight peaks ($\omega \sim$ few T_H):

$$L_{\rm geo} = \int_0^\infty \frac{{\rm d}\omega}{2\pi}\,\frac{27G^2M^2\omega^3}{e^{\omega/T_H}-1} = \frac{27G^2M^2}{2\pi}\cdot\frac{\pi^4}{15}T_H^4 = \frac{27}{122880\,\pi}\,\frac{1}{G^2M^2} \approx \frac{7.0\times10^{-5}}{G^2M^2}. \tag{65.110}$$

Exact value. Both estimates Eqs. (65.109) and (65.110) are *underestimates* that fix the order of magnitude rather than a bracket enclosing the answer: each is the limiting form of $\Gamma_{\omega\ell}$ on one side of the thermal peak. The soft s-wave keeps only the lowest channel and so undercounts the hard multipoles; the geometric envelope, although it *overcounts* the soft tail — there the true sum collapses to the single s-wave $16G^2M^2\omega^2$, below the high-frequency value $27G^2M^2\omega^2$ — still falls below the truth because the high-frequency form switches on only above the thermal peak. The exact coefficient requires the full $\Gamma_{\omega\ell}$ across all ℓ and ω — precisely the numerical solution of Eq. (65.76) that produced Fig. 65.3(b),(c). Carrying out the integral Eq. (65.105) on those data, Page's classic determination [373] gives, for a single massless scalar degree of freedom,

$$a_{\rm scalar} \approx 7.4\times10^{-5}, \qquad \text{i.e.} \qquad -\frac{{\rm d}M}{{\rm d}t}\Big|_{\rm scalar} \approx \frac{7.4\times10^{-5}}{G^2M^2}, \tag{65.111}$$

just above both analytic limits Eqs. (65.109) and (65.110) and lying essentially at (just past) the geometric value Eq. (65.110): the soft s-wave alone gives the right order of magnitude, and the higher multipoles bring it up past the geometric estimate. Summing over the Standard Model fields that are effectively massless at a given T_H (photon and graviton always; neutrinos, electrons, and the rest once T_H exceeds their mass) raises the total to $a_{\text{eff}} \sim 10^{-4}$–$10^{-3}$; the precise figure is a definite number for each particle content, computed once and for all from the same spectrum.

65.3.6 The evaporation law and the lifetime

The mass-loss equation Eq. (65.107) is a closed ordinary differential equation for $M(t)$, and it integrates immediately.

Theorem 65.15 (Evaporation law and lifetime)**.** *A black hole radiating according to Eq. (65.107) with constant coefficient a_{eff} shrinks as*

$$M(t)^3 = M_0^3 - \frac{3\,a_{\text{eff}}}{G^2}\,t, \tag{65.112}$$

and evaporates completely in the finite lifetime

$$\boxed{\tau = \frac{G^2 M_0^3}{3\,a_{\text{eff}}}}\,. \tag{65.113}$$

In Planck units ($\hbar = c = 1$, $G = 1/M_{\text{Pl}}^2$, $t_{\text{Pl}} = 1/M_{\text{Pl}}$) this is $\tau \sim (M_0/M_{\text{Pl}})^3\, t_{\text{Pl}}$.

Proof. Write Eq. (65.107) as $M^2\,\mathrm{d}M = -(a_{\text{eff}}/G^2)\,\mathrm{d}t$ and integrate from 0 to t: $\tfrac{1}{3}(M^3 - M_0^3) = -(a_{\text{eff}}/G^2)t$, which is Eq. (65.112). Setting $M = 0$ gives the time $t = \tau = G^2 M_0^3/3a_{\text{eff}}$. For the Planck-unit form, substitute $G = M_{\text{Pl}}^{-2}$: $\tau \sim G^2 M_0^3 = M_0^3/M_{\text{Pl}}^4 = (M_0/M_{\text{Pl}})^3\, M_{\text{Pl}}^{-1} = (M_0/M_{\text{Pl}})^3\, t_{\text{Pl}}$, the coefficient $1/3a_{\text{eff}} \sim 10^3$–$10^4$ being a pure number. □

The numbers are worth making concrete, since the M^3 law spans an extraordinary range. Restoring $\hbar$ and c (so G^2M^3 becomes $G^2M^3c^{-4}\hbar^{-1}$), the lifetime of a solar-mass hole, $M_\odot \approx 2 \times 10^{33}\,\text{g}$, is of order 10^{67} years — vastly longer than the present age of the Universe, so astrophysical black holes are, for now, accreting rather than evaporating; their Hawking temperature, $T_H \sim 10^{-7}\,\text{K}$ for $M_\odot$, sits far below the cosmic microwave background and they absorb more than they emit. The law becomes observationally interesting only for very small holes. A primordial black hole of mass $M_* \approx 5 \times 10^{14}\,\text{g}$ has, from Eq. (65.113), a lifetime comparable to the age of the Universe ($\sim 10^{10}\,\text{yr}$): such holes, if they formed in the early Universe, would be expiring now, in a final burst of high-energy radiation. The cube law makes the endgame violent — as M falls the luminosity $\propto M^{-2}$ diverges and the remaining mass is shed in an ever faster cascade — which

is the basis of searches for the gamma-ray signature of evaporating primordial holes.

Caution: the endpoint is outside the theory

The derivation assumes a fixed background and treats the back-reaction only through the slow, adiabatic decrease of M in Eq. (65.107). This is self-consistent as long as $M \gg M_{\rm Pl}$, where $T_H \ll M_{\rm Pl}$ and the semiclassical expansion is controlled. As $M \to M_{\rm Pl}$ the temperature approaches the Planck scale, a single emitted quantum carries away a finite fraction of the remaining mass, the adiabatic approximation fails, and the curvature at the horizon becomes Planckian. What Eq. (65.113) computes is therefore the time to reach the Planck mass, not the true endpoint; the fate of the last $\sim M_{\rm Pl}$ — complete evaporation, a stable remnant, or something else — is a question for the quantum theory of gravity and is bound up with the information problem taken up in Section 65.8.

65.3.7 Sparsity of the radiation

It is tempting to picture the evaporating hole as a hot coal: a small body glowing with a Planck spectrum. The luminosity formula Eq. (65.107) reinforces the analogy. But the analogy hides a qualitative difference that the numbers expose, and it matters for how one thinks about the radiation. A hot coal, or a black-body cavity, emits a dense stream of photons: at any instant the radiation field near the surface contains many quanta within a volume set by the thermal wavelength. Hawking radiation does not. The quanta dribble out one at a time, separated by gaps far larger than their own wavelength.

To quantify this, compute the number emission rate from Eq. (65.88), keeping the dominant soft s-wave Eq. (65.99):

$$\begin{aligned}\dot N_{\rm s} &= \int_0^\infty \frac{\mathrm{d}\omega}{2\pi}\,\frac{16G^2M^2\omega^2}{e^{\omega/T_H}-1} = \frac{8G^2M^2}{\pi}\int_0^\infty \frac{\omega^2\,\mathrm{d}\omega}{e^{\omega/T_H}-1}\\ &= \frac{8G^2M^2}{\pi}\,\Gamma(3)\zeta(3)\,T_H^3 = \frac{16\,\zeta(3)\,G^2M^2}{\pi}\,T_H^3 = \frac{\zeta(3)}{32\,\pi^4}\,\frac{1}{GM},\end{aligned} \tag{65.114}$$

using $\int_0^\infty \omega^2(e^{\omega/T}-1)^{-1}\mathrm{d}\omega = 2\zeta(3)T^3$ and $T_H^3 = (8\pi GM)^{-3}$. Numerically $\dot N_{\rm s} \approx 3.9\times10^{-4}/GM$, so the mean interval between emitted quanta is

$$\Delta t = \frac{1}{\dot N_{\rm s}} = \frac{32\pi^4}{\zeta(3)}\,GM \approx 2.6\times10^3\,GM. \tag{65.115}$$

Compare this with the size of a typical emitted quantum. The spectrum Eq. (65.106) peaks near $\omega_{\rm peak} \sim$ a few $T_H \sim 1/2\pi GM$, so the typical wavelength (and, since the quantum travels at $c=1$, the typical duration of its

wavepacket) is $\lambda \sim 2\pi/\omega_{\text{peak}} \sim$ a few $\times GM$ — of order ten GM at most. The ratio

$$\eta \equiv \frac{\Delta t}{\lambda} \sim 10^2 \gg 1 \tag{65.116}$$

is large: successive Hawking quanta are emitted with gaps two orders of magnitude longer than the quanta themselves are wide. The radiation is *sparse*.

Physical Insight: not a hot coal

The dimensionless sparsity Eq. (65.116) is the precise sense in which Hawking radiation differs from black-body emission by a hot object of the same temperature. For a star or a cavity, $\eta \ll 1$: the emitting surface is enormous in units of the thermal wavelength, photons pour out in overlapping floods, and a fluid description (radiative transfer, local thermal equilibrium) is exact. For a black hole, $\eta \gg 1$: the emitting "surface" is only about one wavelength across (the horizon size and the peak wavelength both scale as GM), and the quanta come out individually, well separated in time, each essentially a single mode of the field. The radiation is genuinely Planckian in its frequency distribution — the occupation numbers Eq. (65.88) are thermal — yet it is nothing like the light from a furnace. This sparsity is also why the flux can be described, quantum by quantum, by the mode analysis of this chapter rather than by a hydrodynamic atmosphere, and it tempers any picture of the late-time radiation as a smooth thermal bath.

65.3.8 The flux as the renormalised stress tensor at infinity

The luminosity Eq. (65.105) was assembled from a mode sum, but it is also a statement about the renormalised stress tensor $\langle T_{\mu\nu}\rangle_{\text{ren}}$ built in Section 64.5, and the two pictures must agree. The energy radiated through a large sphere of radius r per unit time is the integral of the radial energy flux,

$$L = -\frac{\mathrm{d}M}{\mathrm{d}t} = \lim_{r\to\infty} \oint_{S^2} \langle T^{t}{}_{r}\rangle_{\text{ren}}\, r^2\, \mathrm{d}\Omega = \lim_{r\to\infty} 4\pi r^2\, \langle T^{t}{}_{r}\rangle_{\text{ren}}, \tag{65.117}$$

the last step using spherical symmetry. In the retarded null coordinate $u = t - r_*$, in which outgoing radiation is a function of u alone, the asymptotic flux is carried by the null–null component $\langle T_{uu}\rangle_{\text{ren}}$, and Eq. (65.117) becomes

$$L = \lim_{r\to\infty} 4\pi r^2\, \langle T_{uu}\rangle_{\text{ren}}. \tag{65.118}$$

Computing $\langle T_{uu}\rangle_{\text{ren}}$ in the in-vacuum of the collapse geometry — with the point-splitting subtraction of Section 64.5 removing the (state-independent) ultraviolet divergence — gives a finite, positive, outgoing flux at infinity whose spectral content is exactly the thermal, greybody-weighted distribution Eq. (65.88):

the renormalised stress tensor knows about the potential barrier because the modes that contribute to it at infinity are the *transmitted* ones of Eq. (65.84), automatically carrying the factor $\Gamma_{\omega\ell}$. Substituting into Eq. (65.118) reproduces Eq. (65.105), so the mode-sum luminosity and the stress-tensor flux are one and the same number. This consistency is more than bookkeeping. The same $\langle T_{\mu\nu}\rangle_{\rm ren}$ also has a component flowing *into* the horizon, and energy conservation forces it to be negative: the positive flux Eq. (65.118) escaping to infinity is balanced by an equal negative-energy flux crossing the horizon, which is what physically reduces M. That negative-energy influx is the stress-tensor avatar of the "partner" modes of the Hawking pairs, and it is the quantity whose interplay with the area drives the generalised second law of the next section.

With the black hole now established as a body of definite, slowly decreasing mass, finite luminosity Eq. (65.107), and finite lifetime Eq. (65.113), we have everything thermodynamics needs: an energy, a temperature, and a flux that obeys an area-controlled emission law. The next section turns these into the laws of black-hole mechanics and the Bekenstein–Hawking entropy $S = A/4G$, using the negative horizon flux identified in Eq. (65.118) to state the generalised second law that governs the area during the very evaporation we have just computed; the lifetime Eq. (65.113) sets the clock against which the information-loss timeline and the Page time of Section 65.8 will later be measured.

65.4 Black Hole Thermodynamics

Section 2 handed us a temperature. A black hole formed by collapse radiates at infinity with a Planck spectrum at $T_H = \kappa/2\pi$ (Section 65.2 and Eq. (65.64)), and Section 3 turned that spectrum into a luminosity, a mass-loss law $-\mathrm{d}M/\mathrm{d}t = a_{\rm eff}/G^2M^2$ (Eq. (65.107)), and a finite lifetime (Eq. (65.113)). The hole is therefore not an inert geometric object but a body that has an energy M, glows at a definite temperature, and slowly shrinks while doing so. A shrinking hot body is the very picture of a thermodynamic system out of equilibrium, exporting energy and — if the analogy is to be more than a slogan — entropy. This section makes the analogy exact. We will show that the laws governing a stationary black hole's response to small perturbations are not merely *like* the laws of thermodynamics: once the Hawking temperature is fed in, they *are* the laws of thermodynamics, with a definite entropy

$$S_{\rm BH} = \frac{A}{4G},$$

the area of the horizon measured in Planck units, coefficient and all. We establish the four laws of black-hole mechanics, promote them to thermodynamics through the computed T_H, fix the entropy coefficient $1/4$ by matching the mechanical first law to the thermodynamic one, and state the generalized second law that

must hold as the hole evaporates. Along the way the entropy reveals itself to scale with area, not volume — the first quantitative fingerprint of holography — and we close with Jacobson's striking inversion of the whole logic, in which the Einstein equation itself is read as a thermodynamic equation of state.

Throughout this section the geometry is the exterior Schwarzschild solution (Eq. (65.71)), the matter is the free real scalar of ϕ^4 theory at $\lambda = 0$ used in Sections 2–3, the signature is mostly-minus $(+,-,-,-)$, and Newton's constant G is kept explicit because gravity is dimensionful. We work in the semiclassical regime, where the spacetime is a fixed classical solution of Einstein's equation sourced by the renormalized stress tensor of Section 64.5, and quantum fields propagate on it.

65.4.1 The geometric vocabulary: Killing horizons and surface gravity

The four laws are statements about a quantity called the surface gravity, so we must define it precisely before we can prove anything about it. A stationary spacetime possesses a timelike Killing vector field ξ^μ generating time translations; for Schwarzschild in the coordinates of Eq. (65.71) it is simply $\xi = \partial_t$, with components $\xi^\mu = (1,0,0,0)$ and lowered form $\xi_\mu = g_{\mu\nu}\xi^\nu = (f(r),0,0,0)$, where $f(r) = 1 - 2GM/r$. Its norm is

$$\xi^\mu \xi_\mu = g_{tt} = f(r) = 1 - \frac{2GM}{r}, \tag{65.119}$$

which is positive (timelike) outside $r = 2GM$, vanishes at $r = 2GM$, and is negative inside. The locus where the Killing field becomes null, $\xi^\mu\xi_\mu = 0$, is the *Killing horizon*; for Schwarzschild it coincides with the event horizon at $r_h = 2GM$. The horizon is generated by the orbits of ξ themselves: on it, ξ^μ is null and tangent to the null geodesics that rule the surface.

Definition 65.16 (Surface gravity)**.** Let ξ^μ be the Killing field that is null on a Killing horizon $\mathcal{H}$. Because ξ^μ is null and tangent to $\mathcal{H}$, it is geodesic there, but the Killing parametrization need not be affine; the failure of affinity defines the *surface gravity* κ through

$$\xi^\nu \nabla_\nu \xi^\mu \;=\; \kappa\, \xi^\mu \qquad \text{on } \mathcal{H}. \tag{65.120}$$

Equivalently, and manifestly geometrically, $\kappa^2 = -\tfrac{1}{2}\,(\nabla_\mu \xi_\nu)(\nabla^\mu \xi^\nu)\big|_{\mathcal{H}}$.

The physical content of κ is the redshifted acceleration needed to hold a test particle static at the horizon. A static observer at radius r follows an orbit of ξ, with four-velocity $u^\mu = \xi^\mu/\sqrt{f}$ (normalized to $u^\mu u_\mu = 1$). Its four-acceleration is $a^\mu = u^\nu \nabla_\nu u^\mu$, and a short computation in the static metric gives the only

nonzero component $a^r = \frac{1}{2}f'(r)$ in an orthonormal sense; the invariant magnitude is

$$a(r) \;=\; \sqrt{-a^\mu a_\mu} \;=\; \frac{f'(r)}{2\sqrt{f(r)}}. \tag{65.121}$$

This diverges at the horizon, as it must: no finite force holds an object static on a null surface. But the *redshifted* acceleration, the force per unit mass that a distant observer must exert through a long rope to keep the particle in place, is the proper acceleration multiplied by the redshift factor $V(r) = \sqrt{f(r)}$. That product is finite, and its horizon value is exactly the surface gravity:

$$\kappa \;=\; \lim_{r\to r_h} V(r)\,a(r) \;=\; \lim_{r\to r_h} \sqrt{f}\,\frac{f'}{2\sqrt{f}} \;=\; \frac{f'(r_h)}{2}. \tag{65.122}$$

The cancellation of $\sqrt{f}$ is the reason the answer is regular on the horizon, and the result $\kappa = \frac{1}{2}f'(r_h)$ holds for *any* static metric of the form Eq. (65.71). (That Va equals the κ of Eq. (65.120) is verified directly: from $\xi_\mu = V^2 u_\mu / V = \sqrt{f}\,u_\mu$ one finds $\xi^\nu \nabla_\nu \xi^\mu = \frac{1}{2}\nabla^\mu(\xi^\nu\xi_\nu) = \frac{1}{2}\nabla^\mu f$, whose magnitude at the horizon is $\frac{1}{2}f'$, parallel to ξ^μ.) For Schwarzschild,

$$f'(r) = \frac{2GM}{r^2}, \qquad \kappa = \frac{f'(2GM)}{2} = \frac{1}{2}\cdot\frac{2GM}{(2GM)^2} = \frac{1}{4GM}. \tag{65.123}$$

This is the number behind the Hawking temperature: $T_H = \kappa/2\pi = 1/8\pi GM$, exactly the value computed in Section 2 by Bogoliubov coefficients and again in Section 5 by Euclidean regularity. Surface gravity is the geometric input; temperature is its quantum reading.

65.4.2 The zeroth law: κ is constant over the horizon

The zeroth law of ordinary thermodynamics says that a body in equilibrium has a single temperature throughout. Its black-hole analogue is the statement that the surface gravity is uniform over a stationary horizon.

Theorem 65.17 (Zeroth law of black-hole mechanics)**.** *The surface gravity κ of a stationary black hole is constant over its event horizon.*

Proof for the static case. For a static, spherically symmetric hole the result is immediate from Eq. (65.122): $\kappa = \frac{1}{2}f'(r_h)$ depends only on the single value r_h at which f vanishes, and the horizon is the round two-sphere $r = r_h$, on every point of which f' takes the same value by spherical symmetry. Hence κ is the same constant everywhere on $\mathcal{H}$. □

The static case is all the Schwarzschild thread requires, but the law holds far more generally, and the structure of the general argument is worth recording because it is the first place Einstein's equation enters as a physical input rather

than a definition. For a stationary, rotating (Kerr) hole the horizon generator is $\xi^\mu = \partial_t^\mu + \Omega_H\, \partial_\varphi^\mu$, a combination of the time-translation and axial Killing fields with Ω_H the (constant) angular velocity of the horizon. One shows from the Killing identity $\nabla_\mu \nabla_\nu \xi_\rho = R_{\rho\nu\mu\sigma}\xi^\sigma$ that the gradient of κ along the horizon is controlled by $R_{\mu\nu}\xi^\nu$, hence by $T_{\mu\nu}\xi^\nu$ through Einstein's equation. Imposing the dominant energy condition forces this tangential gradient to vanish, so κ is constant. This is the Bardeen–Carter–Hawking theorem; the static proof above is its degenerate case, where spherical symmetry makes the energy-condition step unnecessary. We will not need the rotating machinery again, having the static result in hand.

65.4.3 The first law: $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A + \Omega\,\mathrm{d}J + \Phi\,\mathrm{d}Q$

The first law relates how the conserved charges of a stationary hole change when the hole is perturbed to a nearby stationary state. In its full form, valid for a charged rotating (Kerr–Newman) hole, it reads

$$\mathrm{d}M \;=\; \frac{\kappa}{8\pi G}\,\mathrm{d}A \;+\; \Omega_H\,\mathrm{d}J \;+\; \Phi_H\,\mathrm{d}Q, \tag{65.124}$$

where M is the mass, A the horizon area, J the angular momentum, Q the charge, Ω_H the horizon angular velocity, and Φ_H the horizon electric potential. The structure mirrors the thermodynamic first law $\mathrm{d}U = T\,\mathrm{d}S + \sum_i \mu_i\,\mathrm{d}N_i$ to a remarkable degree: M plays the role of internal energy, the $\Omega_H\,\mathrm{d}J$ and $\Phi_H\,\mathrm{d}Q$ terms are the work terms with their conjugate chemical potentials, and the leading term pairs the surface gravity with the area exactly as $T\,\mathrm{d}S$ pairs temperature with entropy. The identification

$$T \;\leftrightarrow\; \frac{\kappa}{2\pi}, \qquad S \;\leftrightarrow\; \frac{A}{4G} \tag{65.125}$$

makes $\frac{\kappa}{8\pi G}\,\mathrm{d}A = \left(\frac{\kappa}{2\pi}\right)\mathrm{d}\left(\frac{A}{4G}\right) = T\,\mathrm{d}S$ on the nose, but at the purely mechanical level the split of $\frac{\kappa}{8\pi G}\,\mathrm{d}A$ into a "temperature" times an "entropy" is ambiguous: any rescaling $\kappa \to c\,\kappa$, $A \to A/c$ leaves the product invariant. It is the quantum computation of Section 2, which fixes the constant of proportionality between T and κ to be $1/2\pi$ and nothing else, that breaks the ambiguity. We return to this in Section 65.4.5; first we establish Eq. (65.124) itself for the case we own.

For Schwarzschild ($J = Q = 0$) the first law collapses to $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A$, and we can prove it by direct differentiation, carrying every factor. The area of the horizon two-sphere $r = 2GM$ is

$$A \;=\; \int_{r=2GM} \sqrt{g_{\theta\theta}\,g_{\varphi\varphi}}\;\mathrm{d}\theta\,\mathrm{d}\varphi \;=\; \int_0^{\pi}\!\int_0^{2\pi} r_h^2 \sin\theta\;\mathrm{d}\theta\,\mathrm{d}\varphi \;=\; 4\pi r_h^2 \;=\; 16\pi G^2 M^2, \tag{65.126}$$

using $r_h = 2GM$. Differentiating,

$$\mathrm{d}A = 32\pi G^2 M\,\mathrm{d}M. \tag{65.127}$$

Now assemble the right-hand side of the first law with the surface gravity Eq. (65.123):

$$\frac{\kappa}{8\pi G}\,\mathrm{d}A \;=\; \frac{1}{8\pi G}\cdot\frac{1}{4GM}\cdot 32\pi G^2 M\,\mathrm{d}M \;=\; \frac{32\pi G^2 M}{32\pi G^2 M}\,\mathrm{d}M \;=\; \mathrm{d}M. \tag{65.128}$$

The first law for Schwarzschild is therefore not an analogy but an algebraic identity: $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A$ holds with both sides equal to $\mathrm{d}M$, because the powers of G and M cancel exactly. This exactness is the hinge of everything that follows.

Worked Example 65.18 (The Smarr formula and the integrated first law)**.** The differential first law has an integrated companion, the *Smarr formula*, which relates the total mass to the surface gravity and area rather than their increments. We derive it for Schwarzschild by a scaling (Euler) argument and check it numerically against the explicit metric.

Treat M as a function of the area alone, inverting Eq. (65.126):

$$M(A) = \left(\frac{A}{16\pi G^2}\right)^{1/2} = \frac{1}{4G}\sqrt{\frac{A}{\pi}}, \tag{65.129}$$

which is homogeneous of degree $\tfrac{1}{2}$ in A. Euler's theorem for a homogeneous function of degree p, $A\,\partial_A M = p\,M$, gives here

$$A\,\frac{\partial M}{\partial A} = \tfrac{1}{2}\,M. \tag{65.130}$$

But $\partial M/\partial A$ is read off the differential first law Eq. (65.128): $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A$ means $\partial M/\partial A = \kappa/8\pi G$. Substituting into Eq. (65.130),

$$A\cdot\frac{\kappa}{8\pi G} = \frac{M}{2} \qquad\Longrightarrow\qquad \boxed{M = \frac{\kappa A}{4\pi G}} \tag{65.131}$$

the Smarr formula. The factor relating M to $\tfrac{1}{2}$ rather than to the full TS — equivalently the appearance of $\kappa A/4\pi G$ rather than $\kappa A/8\pi G$ — is the imprint of the degree-$\tfrac{1}{2}$ homogeneity, i.e. of the fact that the area scales as M^2 while the mass scales as M.

We check both relations against the bare numbers. With $\kappa = 1/4GM$ and $A = 16\pi G^2 M^2$,

$$\frac{\kappa A}{4\pi G} = \frac{1}{4\pi G}\cdot\frac{1}{4GM}\cdot 16\pi G^2 M^2 = \frac{16\pi G^2 M^2}{16\pi G^2 M} = M, \tag{65.132}$$

confirming Eq. (65.131). In the thermodynamic dictionary Eq. (65.125) the Smarr formula reads $M = 2\big(\tfrac{\kappa}{2\pi}\big)\big(\tfrac{A}{4G}\big) = 2\,T_H S$, the celebrated statement that for a Schwarzschild hole the mass is *twice* the "TS" product. Indeed $T_H S =$

$(1/8\pi GM)(4\pi GM^2) = M/2$, so $2T_H S = M$, in agreement. The factor of two is not a paradox: it is Euler's theorem for a system whose "entropy" $A/4G \propto M^2$ is a homogeneous function of the energy of degree 2, so that $E = 2TS$ rather than the familiar extensive $E \sim TS$. It is the same fact that will make the heat capacity negative (Theorem 65.30).

65.4.4 The second law: the area theorem

The thermodynamic second law says entropy never decreases. Its mechanical analogue, proved by Hawking before any of this was understood as thermodynamics, is that the area of an event horizon never decreases in any classical process. This is the *area theorem*. Its proof is a clean application of the focusing of null geodesics, and we give it in full at the level the spec requires — the non-negativity of the expansion of the horizon generators.

The horizon is a null hypersurface ruled by null geodesics, its *generators*. Let k^μ be the future-directed null tangent to these generators, affinely parametrized so that $k^\nu \nabla_\nu k^\mu = 0$, and let λ be the affine parameter. The cross-sectional area element of the congruence evolves according to its *expansion*

$$\theta \;\equiv\; \nabla_\mu k^\mu \;=\; \frac{1}{\delta A}\,\frac{\mathrm{d}(\delta A)}{\mathrm{d}\lambda}, \tag{65.133}$$

the fractional rate of change of a transverse area element δA dragged along the generators. A positive expansion means neighbouring generators are diverging and the area is growing; a negative expansion means they are converging and the area is shrinking. The total horizon area changes as

$$\frac{\mathrm{d}A}{\mathrm{d}\lambda} = \int_{\mathcal{C}} \theta \,\mathrm{d}A, \tag{65.134}$$

the integral of the expansion over a cross-section $\mathcal{C}$ of the horizon. The area theorem will follow once we show $\theta \geq 0$ everywhere on the future horizon.

The evolution of the expansion is governed by the Raychaudhuri equation, the exact identity obtained by contracting the geodesic deviation equation. For a null congruence with tangent k^μ it reads

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\frac{1}{2}\,\theta^2 \;-\; \sigma_{\mu\nu}\sigma^{\mu\nu} \;+\; \omega_{\mu\nu}\omega^{\mu\nu} \;-\; R_{\mu\nu}\,k^\mu k^\nu, \tag{65.135}$$

where $\sigma_{\mu\nu}$ is the shear (the trace-free symmetric part of the transverse gradient of k), $\omega_{\mu\nu}$ is the twist (its antisymmetric part), and $R_{\mu\nu}$ is the Ricci tensor. The coefficient of θ^2 is $\tfrac{1}{2}$ (rather than $\tfrac{1}{3}$, as for a timelike congruence) because the transverse space of a null geodesic in four dimensions is two-dimensional. Three facts now collapse the right-hand side to something manifestly nonpositive.

First, the horizon generators are hypersurface-orthogonal — they rule a null hypersurface — so by Frobenius's theorem the twist vanishes, $\omega_{\mu\nu} = 0$.

Second, the shear contraction $\sigma_{\mu\nu}\sigma^{\mu\nu} \geq 0$, being a sum of squares of the real components of a tensor on the spacelike transverse space. Third, and this is where the physics enters, the Ricci term is controlled by Einstein's equation $R_{\mu\nu} - \frac{1}{2}R\, g_{\mu\nu} = 8\pi G\, T_{\mu\nu}$. Contracting with the null vector k^μ, the trace term drops because $g_{\mu\nu}k^\mu k^\nu = 0$, leaving

$$R_{\mu\nu}\, k^\mu k^\nu = 8\pi G\, T_{\mu\nu}\, k^\mu k^\nu. \tag{65.136}$$

Definition 65.19 (Null energy condition)**.** The matter stress tensor satisfies the *null energy condition* (NEC) if

$$T_{\mu\nu}\, k^\mu k^\nu \;\geq\; 0 \qquad \text{for every future-directed null vector } k^\mu. \tag{65.137}$$

The NEC is the weakest of the standard pointwise energy conditions and holds for all classical matter we have built in this book — in particular for the free scalar of ϕ^4 theory, whose null-contracted stress tensor $T_{\mu\nu}k^\mu k^\nu = (k^\mu\partial_\mu\phi)^2 \geq 0$ is a perfect square. Granting it, Eq. (65.136) gives $R_{\mu\nu}k^\mu k^\nu \geq 0$, and the Raychaudhuri equation Eq. (65.135) reduces on the horizon to

$$\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} = -\frac{1}{2}\,\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu \;\leq\; -\frac{1}{2}\,\theta^2. \tag{65.138}$$

Every term dropped to obtain the inequality was nonpositive. This is the focusing theorem: gravity, sourced by matter that respects the NEC, is always attractive for a hypersurface-orthogonal null congruence, and an expansion can only decrease.

Theorem 65.20 (Area theorem)**.** *Assume cosmic censorship (the spacetime is strongly predictable, so the event horizon is the boundary of the causal past of future null infinity) and the null energy condition. Then the expansion of the future horizon generators is nonnegative, $\theta \geq 0$, and consequently the horizon area is nondecreasing:*

$$\frac{\mathrm{d}A}{\mathrm{d}\lambda} \;\geq\; 0. \tag{65.139}$$

Proof. Suppose, for contradiction, that at some point p on the horizon the expansion is negative, $\theta(p) = \theta_0 < 0$. Integrate the focusing inequality Eq. (65.138). Writing it as $\mathrm{d}\theta/\mathrm{d}\lambda \leq -\frac{1}{2}\theta^2$ and dividing by $-\theta^2 > 0$ (legitimate while $\theta < 0$),

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\left(\frac{1}{\theta}\right) = -\frac{1}{\theta^2}\frac{\mathrm{d}\theta}{\mathrm{d}\lambda} \;\geq\; \frac{1}{2}. \tag{65.140}$$

Hence $1/\theta$ grows at least linearly: $1/\theta(\lambda) \geq 1/\theta_0 + \lambda/2$. Since $1/\theta_0 < 0$, the right-hand side reaches zero at an affine parameter no greater than $\Delta\lambda = 2/|\theta_0|$, at which point θ passes through $-\infty$. A divergent negative expansion is a *caustic*: the neighbouring generators have crossed, the cross-sectional area has been pinched to zero, and the generator has reached a focal point in finite affine

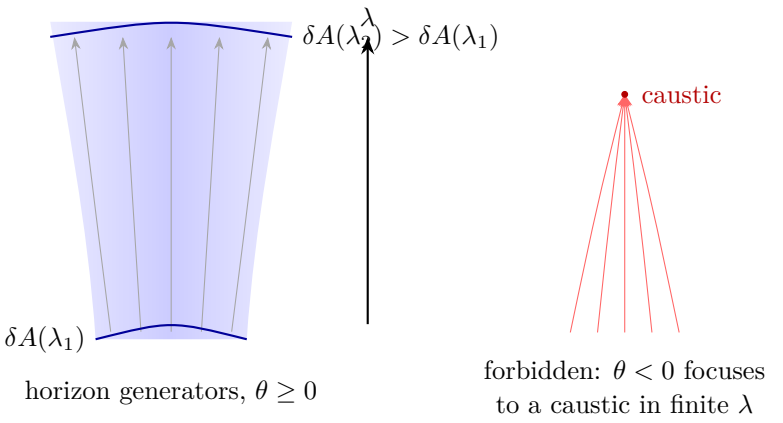


Figure 65.4: The area theorem as focusing. Left: on the future horizon the generators diverge or stay parallel ($\theta \geq 0$), so the transverse area element δA never shrinks. Right: a negative expansion would, by the focusing inequality Eq. (65.138), drive the generators to a caustic at finite affine parameter — a focal point that the event horizon's generators are forbidden to reach. Ruling out the caustic forces $\theta \geq 0$.

time. But a generator of the event horizon, by the definition of the horizon as the boundary of the past of future null infinity, has no future endpoint — it cannot leave the horizon, and two generators of the horizon cannot intersect (an intersection would put a point on the boundary that is in the chronological past of horizon points, contradicting achronality of the boundary). The predicted caustic is therefore impossible. The only way to avoid it is $\theta \geq 0$ everywhere on the future horizon. With $\theta \geq 0$ pointwise, Eq. (65.134) gives $\mathrm{d}A/\mathrm{d}\lambda = \int_{\mathcal{C}} \theta \, \mathrm{d}A \geq 0$, the area theorem. □

The area theorem is a statement about classical processes, in which the NEC holds. Hawking evaporation *violates* it: the hole shrinks, $\mathrm{d}A < 0$. There is no contradiction, because the renormalized quantum stress tensor near the horizon (Section 64.5) carries a negative energy flux through the horizon — precisely the negative-energy influx identified in Eq. (65.118), the partner of each escaping Hawking quantum — and a negative $T_{\mu\nu}k^\mu k^\nu$ defeats the NEC. The classical area theorem is thus the mechanical second law; its quantum failure is exactly what the generalized second law of Section 65.4.6 repairs.

65.4.5 Promotion to thermodynamics and the entropy coefficient

We now have a precise mechanical structure: a constant κ over the horizon (zeroth law), a first law $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A + \ldots$, and a nondecreasing area (second law). The parallel with thermodynamics — constant T, $\mathrm{d}U = T\,\mathrm{d}S + \ldots$,

nondecreasing S — is exact in form. Whether it is exact in *content* hinges on a single question: is the temperature that appears in the would-be first law a real temperature, and the area a real entropy, or is the parallel a mathematical coincidence?

Classically, the question cannot be answered, and Bardeen, Carter, and Hawking originally regarded the analogy as formal: a classical black hole is perfectly black, absorbs everything, emits nothing, and so sits at temperature zero, in which case any nonzero $\kappa \propto T$ is meaningless and $A/4G$ could be rescaled away. What changed everything is the quantum computation of Section 2. A black hole formed by collapse *does* radiate, with a Planck spectrum, and the temperature of that spectrum is not an adjustable constant but the sharply determined value

$$T_H = \frac{\kappa}{2\pi} = \frac{1}{8\pi G M}, \tag{65.141}$$

fixed by the Bogoliubov coefficients (Eq. (65.64)) and re-derived independently from Euclidean regularity in Section 5. Once T_H is a genuine, computed temperature, the first law forces the entropy upon us with no freedom left.

Theorem 65.21 (Bekenstein–Hawking entropy with fixed coefficient)**.** *If the Schwarzschild black hole is a thermodynamic system of energy M and temperature $T_H = 1/8\pi GM$ obeying the first law $\mathrm{d}M = T_H\,\mathrm{d}S$, then its entropy is*

$$S_{\mathrm{BH}} = \frac{A}{4G} = 4\pi G M^2, \tag{65.142}$$

with the coefficient $1/4$ uniquely determined.

Proof. Equate the mechanical first law Eq. (65.128), $\mathrm{d}M = (\kappa/8\pi G)\,\mathrm{d}A$, with the thermodynamic first law $\mathrm{d}M = T_H\,\mathrm{d}S$, and substitute the computed temperature $T_H = \kappa/2\pi$:

$$T_H\,\mathrm{d}S = \frac{\kappa}{8\pi G}\,\mathrm{d}A \quad\Longrightarrow\quad \frac{\kappa}{2\pi}\,\mathrm{d}S = \frac{\kappa}{8\pi G}\,\mathrm{d}A \quad\Longrightarrow\quad \mathrm{d}S = \frac{2\pi}{\kappa}\cdot\frac{\kappa}{8\pi G}\,\mathrm{d}A = \frac{\mathrm{d}A}{4G}. \tag{65.143}$$

The surface gravity cancels identically — this is the crucial step: it is the *ratio* $T_H/\kappa = 1/2\pi$, not κ itself, that survives, and that ratio is precisely what Section 2 computed. Integrating $\mathrm{d}S = \mathrm{d}A/4G$ from the zero-area reference and fixing the additive constant to zero (the entropy vanishes as the horizon does) gives $S_{\mathrm{BH}} = A/4G$. Equivalently, integrate in M: $\mathrm{d}S = \mathrm{d}M/T_H = 8\pi G M\,\mathrm{d}M$, so $S = \int_0^M 8\pi G M'\,\mathrm{d}M' = 4\pi G M^2$, and using $A = 16\pi G^2 M^2$ this is $A/4G$, identical. The coefficient $1/4$ is not chosen; it is the output of $1/2\pi$ from the temperature times $1/8\pi G$ from the mechanical first law. □

Physical Insight

Bekenstein's original argument, which preceded Hawking's, established that a black hole *must* carry an entropy proportional to its area — on pain of letting one violate the ordinary second law by dropping entropy-laden matter behind a horizon where it can no longer be counted — and estimated the coefficient by information-theoretic reasoning to be of order $1/G$ in Planck units, but could not pin the pure number. It was Hawking's discovery that the hole has a definite temperature $T_H = \kappa/2\pi$ that converted "proportional to area" into the exact $A/4G$. The two halves of the result come from two different physicists and two different kinds of argument: the proportionality from a thermodynamic *gedankenexperiment*, the coefficient from a quantum field theory *computation*. Neither alone suffices; together they fix the entropy of a black hole completely. This is why the result is named for both.

It is worth restating the magnitude this represents. For a solar-mass hole, $S_{\mathrm{BH}} = A/4G \sim (M/M_{\mathrm{Pl}})^2 \sim (10^{38})^2 = 10^{76}$ in natural units — some twenty orders of magnitude larger than the thermodynamic entropy of the star that collapsed to form it. A black hole is overwhelmingly the highest-entropy object that can be packed into a given region, a fact that becomes the Bekenstein bound below. The entropy also scales as the *area*, $S \propto A \propto M^2$, not as a volume $\propto M^3$ or as the particle number of the progenitor. That an extensive-looking thermodynamic quantity scales with the bounding area rather than the enclosed volume is the single most important structural surprise of the whole subject, and we draw out its meaning in Section 65.4.7.

65.4.6 The generalized second law

The classical area theorem fails once the hole radiates, because the negative energy flux through the horizon violates the NEC and the area shrinks. The ordinary second law also fails on its own: matter (and its entropy) can fall across the horizon and disappear from the exterior. Each law fails exactly where the other can compensate, and Bekenstein's proposal — the *generalized second law* (GSL) — is that their sum is what never decreases.

Definition 65.22 (Generalized entropy)**.** The *generalized entropy* of a black hole together with its exterior is

$$S_{\mathrm{gen}} \;=\; S_{\mathrm{BH}} + S_{\mathrm{out}} \;=\; \frac{A}{4G} + S_{\mathrm{out}}, \tag{65.144}$$

where S_{out} is the ordinary (von Neumann or thermodynamic) entropy of all matter and radiation outside the horizon.

Conjecture 65.23 (Generalized second law)**.** *In any physical process, the generalized entropy does not decrease:*

$$\Delta S_{\text{gen}} = \Delta\left(\frac{A}{4G}\right) + \Delta S_{\text{out}} \;\geq\; 0. \tag{65.145}$$

We state this as a conjecture rather than a theorem because, unlike the area theorem, it has no fully general proof valid in every regime; it is established in many concrete settings (and follows rigorously from quantum field theory together with the quantum focusing conjecture, and from the modern entropy-as-area arguments of Chapter 57), and no violation is known. It is the precise sense in which black holes obey the second law of thermodynamics. The two regimes in which it must hold are matter falling *in* (the area grows enough to pay for the lost exterior entropy — the engine of the Bekenstein bound below) and radiation streaming *out* (the radiation's entropy more than compensates the shrinking area). We verify the second, using the Hawking quantum of Section 3, in detail.

Worked Example 65.24 (The GSL across one Hawking emission)**.** Consider the hole emitting an increment of energy $\mathrm{d}E$ into Hawking radiation at temperature T_H. We compare the entropy the radiation carries off against the entropy the hole loses, and show the net change is positive.

Entropy lost by the hole. The hole's energy drops by $\mathrm{d}E$, so $\mathrm{d}M = -\mathrm{d}E$, and by the first law its Bekenstein–Hawking entropy changes by

$$\mathrm{d}S_{\text{BH}} = \frac{\mathrm{d}M}{T_H} = -\frac{\mathrm{d}E}{T_H} < 0. \tag{65.146}$$

Entropy gained by the radiation. The emitted quanta form, to leading semiclassical order, a thermal gas of massless bosons at temperature T_H (this is exactly the Planck spectrum of Section 2; greybody factors of Section 3 modify the numerical coefficient but not the conclusion). For a gas of massless quanta the equation of state is $P = u/3$ with energy density $u \propto T^4$, the standard relativistic result. The thermodynamic entropy density follows from the Euler relation $s = (u + P)/T$ for a gas of zero chemical potential:

$$s = \frac{u + P}{T} = \frac{u + u/3}{T} = \frac{4}{3}\,\frac{u}{T}, \tag{65.147}$$

so that the entropy of a parcel of such radiation of energy $\mathrm{d}E$ is

$$\mathrm{d}S_{\text{rad}} = \frac{4}{3}\,\frac{\mathrm{d}E}{T_H}. \tag{65.148}$$

The factor $\frac{4}{3}$ is the same one that makes blackbody entropy flux exceed $\mathrm{d}E/T$: a thermal photon gas carries more entropy per unit energy than a reversible heat transfer would, because the emission is into empty modes, not into a reservoir at the same temperature.

The net change. Adding the two contributions, with $\mathrm{d}E > 0$ the energy emitted,

$$\mathrm{d}S_{\mathrm{gen}} = \mathrm{d}S_{\mathrm{rad}} + \mathrm{d}S_{\mathrm{BH}} = \frac{4}{3}\frac{\mathrm{d}E}{T_H} - \frac{\mathrm{d}E}{T_H} = \frac{1}{3}\frac{\mathrm{d}E}{T_H} \;>\; 0. \tag{65.149}$$

The generalized entropy increases by a third of the entropy the radiation carries. The hole's own entropy falls, but the radiation it sheds carries off *more* entropy than the hole loses, by the robust factor $\frac{4}{3}$ of the blackbody equation of state, and the GSL holds throughout evaporation. Here $\frac{4}{3}$ is the crude pure-blackbody value; including the greybody filtering of Section 3 (and the irreversibility of evaporation) replaces it by a larger pure number — ≈ 1.6 for a massless thermal field, computed in Theorem 65.41 — that remains comfortably greater than 1 for every field species, so the conclusion $\mathrm{d}S_{\mathrm{gen}} > 0$ is unaffected. Integrating Eq. (65.148) over the entire evaporation, the radiation produced by a hole of initial mass M_0 ends with thermodynamic entropy $S_{\mathrm{rad}} = \frac{4}{3}S_{\mathrm{BH}}(M_0)$ in this blackbody estimate ($\approx 1.6\, S_{\mathrm{BH}}(M_0)$ once greybody factors are included, Theorem 65.41), exceeding the entropy the hole started with — the coarse-grained statement that will be set against the fine-grained Page curve in Section 65.8.

The worked check exposes a tension that the rest of the chapter must confront. The GSL as verified here treats S_{rad} as the *thermodynamic* entropy of a genuinely thermal gas. If the radiation is truly thermal — a mixed state of mutually uncorrelated quanta — then at the end of evaporation a pure initial state has become a mixed final state with entropy $\frac{4}{3}S_{\mathrm{BH}}(M_0) > 0$, and unitarity is violated. That is the information paradox. The GSL is consistent with the coarse-grained thermodynamics; whether the *fine-grained* entropy obeys the same monotone growth is the question taken up in Chapter 57. For the thermodynamics of this chapter, what matters is that S_{gen} never decreases, and it does not.

65.4.7 The Bekenstein bound and entropy as an area

The GSL applied to matter falling *into* a hole yields a universal bound on how much entropy any system can carry, and that bound is the cleanest entry point to the area-scaling of information. The argument, due to Bekenstein and developed in Section 57.1.6, is a thought experiment: lower a box of energy E, entropy S_{matter}, and linear size $\leq R$ quasi-statically into the hole. The exterior loses S_{matter}; the hole gains area as it absorbs the energy. For the GSL Eq. (65.145) to survive, the area increase must at least cover the lost matter entropy, and working out the geometry of the optimal drop gives the *Bekenstein bound*

$$S_{\mathrm{matter}} \;\leq\; 2\pi R\, E \tag{65.150}$$

(natural units; restoring constants, $S \leq 2\pi k_B RE/\hbar c$). We import this bound and its derivation from Eq. (57.35); the point here is what it implies for the meaning

of $S_{\rm BH} = A/4G$. A black hole exactly saturates Eq. (65.150) with $R \sim 2GM$ and $E \sim M$: $2\pi RE \sim 4\pi GM^2 = S_{\rm BH}$. The Bekenstein–Hawking entropy is therefore not merely *an* entropy a black hole happens to have; it is the *maximum* entropy — the maximum number of bits — that can be stored in a region of that size at all. Pack any more information into the region and it collapses to a hole, whereupon its entropy is exactly $A/4G$.

Physical Insight

The decisive structural fact is that this maximum scales as the bounding *area*, not the enclosed *volume*. A conventional local quantum field theory in a region of volume V has a number of degrees of freedom, and hence a maximal entropy, that grows like V times an ultraviolet cutoff cubed — an extensive, volume-scaling quantity. Gravity overrides this: the true maximal entropy of the region grows only as its area A in Planck units. The information content of a region is set by the size of its boundary, as if the physics inside were encoded on the surface enclosing it. This is the germ of the *holographic principle*: a $(d{+}1)$-dimensional gravitating region is described by no more degrees of freedom than a d-dimensional theory living on its boundary could carry. The $A/4G$ we have just fixed is the conversion factor — one quarter-bit (in nats, one quarter) per Planck area — and the reason Part XV of this book, on holography and AdS/CFT, exists at all. The black hole is the first place the principle is forced on us by a calculation rather than postulated.

65.4.8 Jacobson's inversion: the Einstein equation as an equation of state

We close the thermodynamic development with a remarkable observation that turns its logic inside out. We have taken the Einstein equation as given — it is the classical background equation, used in Eq. (65.136) to convert the NEC into focusing — and *derived* thermodynamics from it. Jacobson showed in 1995 that the implication can be run in reverse: if one *assumes* the thermodynamic relation $\delta Q = T\,\mathrm{d}S$ holds for every local Rindler horizon, with T the Unruh temperature and S the area in units of $1/4G$, then the Einstein equation *follows* as a consistency condition. We sketch the argument honestly, since the full version requires the local Rindler-horizon construction of Chapter 64.

Physical Insight

Pick any spacetime point p and a small spacelike two-surface element through it, and consider the local Rindler horizon to its past: the null congruence that, to a uniformly accelerated observer, plays the role of a

causal horizon. By the Unruh effect (Chapter 64) that observer attributes a temperature $T = \kappa/2\pi$ to the horizon, with κ the local acceleration. Demand that the Clausius relation $\delta Q = T\,\mathrm{d}S$ hold, where $\delta Q = \int T_{\mu\nu}\,\xi^\mu \mathrm{d}\Sigma^\nu$ is the boost energy flux of matter across the horizon (with ξ the approximate boost Killing field) and $\mathrm{d}S = \mathrm{d}A/4G$ is the change in horizon area in Bekenstein–Hawking units. The area change $\mathrm{d}A$ is governed by the expansion through Eq. (65.133), and the expansion in turn obeys the Raychaudhuri equation Eq. (65.135), which feeds in $R_{\mu\nu}k^\mu k^\nu$. Equating the matter flux $\delta Q \propto T_{\mu\nu}k^\mu k^\nu$ to $T\,\mathrm{d}S \propto R_{\mu\nu}k^\mu k^\nu$ for *every* null direction k at p forces $R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G\,T_{\mu\nu}$ — the Einstein equation, with G fixed precisely by the $1/4$ in the entropy and Λ an undetermined integration constant. Gravity's field equation is, in this reading, not a fundamental law but an equation of state: a thermodynamic relation that the underlying microscopic degrees of freedom must satisfy in their coarse-grained, long-wavelength limit, exactly as the ideal-gas law $PV = NkT$ is an equation of state for molecules whose microscopics it never mentions. That the entropy is $A/4G$ is then not a consequence of the Einstein equation but its *cause*.

This is a frontier idea (as of the mid-2020s it remains a deep suggestion rather than a settled derivation of quantum gravity), and we flag it as such. But it crystallizes why black-hole thermodynamics is more than an analogy. If the laws of black-hole mechanics were merely formally parallel to thermodynamics, Jacobson's inversion would be impossible; that one can run the implication backward and recover the dynamics of spacetime itself is the strongest evidence that the parallel is an identity, and that $S_{\mathrm{BH}} = A/4G$ is a statement about real microstates we have not yet learned to count.

With the thermodynamic dossier complete — temperature $T_H = \kappa/2\pi$, entropy $S_{\mathrm{BH}} = A/4G$ with its coefficient nailed by the first law, and a generalized second law that survives evaporation — we have established by purely thermodynamic and field-theoretic reasoning what Section 5 will now rederive from an utterly different starting point. There, Euclidean Schwarzschild proves smooth only when imaginary time has period $\beta = 8\pi GM$, and the on-shell gravitational action delivers the very same $S = A/4G$ as a free energy — an independent confirmation from the path-integral pillar of the book. The entropy fixed here also becomes, in Section 65.8, the finite ceiling against which the fine-grained entropy of the Hawking radiation is measured, and whose saturation defines the Page time.

65.5 The Euclidean Path Integral and the Gravitational Partition Function

The previous section earned the Bekenstein–Hawking entropy $S = A/4G$ by thermodynamic bookkeeping: armed with the Hawking temperature $T_H = \kappa/2\pi$ (Eq. (65.64)) we promoted the first law of black-hole mechanics to the first law of thermodynamics and integrated $\mathrm{d}S = \mathrm{d}M/T_H$ to fix the coefficient (Section 65.4). That derivation is correct, but it is also, in a sense, borrowed: the temperature it relies on was handed to us by Hawking's Bogoliubov computation of Section 65.2.1, a calculation about quantum fields propagating on a collapsing background. One could be forgiven for suspecting that the thermal character of a black hole is an artefact of the matter we chose to quantize on it, rather than a property of the geometry itself.

This section dispels that suspicion. We give a completely independent derivation in which no matter field is quantized and no mode is tracked across a horizon. The only inputs are Euclidean gravity — the gravitational action continued to imaginary time, exactly the e^{-S_E} path-integral pillar built for scalar fields in Section 10.6 — and the demand that the continued geometry be *smooth*. We will find that the Euclidean Schwarzschild manifold simply *does not exist* as a smooth Riemannian space unless imaginary time is periodic with a single, sharply fixed period $\beta = 8\pi GM$. Periodicity in imaginary time is the defining signature of a thermal density matrix, so this already fixes $T_H = 1/\beta$ as a statement about the geometry alone. We then evaluate the on-shell gravitational action of this manifold and read off its free energy; the entropy that drops out is, once more and now from first principles of the path integral, $S = A/4G$. The black hole has become a thermal saddle point of the gravitational partition function.

65.5.1 Wick rotation of Schwarzschild and the geometry of the horizon

We work throughout with the Schwarzschild solution of the vacuum Einstein equations in the mostly-minus signature fixed for this chapter, with Newton's constant G written explicitly. Recall its line element from Eq. (65.1),

$$\mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 - f(r)^{-1}\,\mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2, \qquad f(r) = 1 - \frac{2GM}{r},$$

where $\mathrm{d}\Omega^2 = \mathrm{d}\theta^2 + \sin^2\theta\,\mathrm{d}\varphi^2$ is the round metric on the unit two-sphere and M is the ADM mass. The exterior region $r > 2GM$ is static, with the timelike Killing vector ∂_t; the locus $r = 2GM$ where f vanishes is the event horizon. The surface gravity κ of this horizon was defined in Section 65.4; we shall recover it below as a purely geometric quantity, $\kappa = \tfrac{1}{2}f'(r_h)$, without reference to any Killing-vector normalization argument.

The Euclidean continuation is the same Wick rotation $t \to -i\tau$ that turned the Minkowski generating functional into a convergent Euclidean integral in Section 10.6, applied now to the metric rather than to a field. Under $t = -i\tau$ we have $\mathrm{d}t^2 = -\mathrm{d}\tau^2$, so the first term of Eq. (65.1) flips sign and the whole line element becomes negative-definite. Defining the Euclidean metric by $\mathrm{d}s_E^2 \equiv -\mathrm{d}s^2$ gives a positive-definite Riemannian metric on the exterior,

$$\mathrm{d}s_E^2 = f(r)\,\mathrm{d}\tau^2 + f(r)^{-1}\,\mathrm{d}r^2 + r^2\,\mathrm{d}\Omega^2, \qquad r \geq 2GM. \tag{65.151}$$

Two features of Eq. (65.151) are decisive. First, $f(r)$ is non-negative on $r \geq 2GM$ and vanishes precisely at the horizon, so the τ-circle — whose proper circumference at radius r is $\sqrt{f(r)}\oint \mathrm{d}\tau$ — shrinks to zero length as $r \to 2GM$. The horizon is not a boundary of the Euclidean manifold; it is a point (more precisely, a two-sphere's worth of points, one for each point of the S^2) where the τ-circle closes up. Second, the region $r < 2GM$, where $f < 0$, is simply absent: there is no Euclidean interior. The Euclidean black hole is a manifold with the topology $\mathbb{R}^2 \times S^2$, the $\mathbb{R}^2$ being the (ρ, τ) plane we are about to uncover, capped off smoothly at the horizon.

Whether the cap is genuinely smooth, or instead carries a conical singularity, is the entire question. To decide it we zoom in on the horizon. Write $r = 2GM + \epsilon$ with $0 < \epsilon \ll 2GM$ and expand f to first order, using $f(2GM) = 0$,

$$f(r) = f'(2GM)\,\epsilon + O(\epsilon^2), \qquad f'(r) = \frac{2GM}{r^2}, \qquad f'(2GM) = \frac{1}{2GM}. \tag{65.152}$$

The natural radial coordinate near the horizon is not r but the proper distance ρ from the horizon, defined by $\mathrm{d}\rho = \mathrm{d}r/\sqrt{f(r)}$ so that the $\mathrm{d}r^2/f$ term becomes $\mathrm{d}\rho^2$. Substituting the linearized Eq. (65.152),

$$\rho = \int_{2GM}^{r} \frac{\mathrm{d}r'}{\sqrt{f(r')}} = \int_0^{\epsilon} \frac{\mathrm{d}\epsilon'}{\sqrt{\epsilon'/2GM}} = 2\sqrt{2GM\,\epsilon}\,, \tag{65.153}$$

so that $\epsilon = \rho^2/(8GM)$ and therefore

$$f(r) = \frac{\epsilon}{2GM} = \frac{\rho^2}{16\,G^2M^2} = \left(\frac{\rho}{4GM}\right)^2. \tag{65.154}$$

Near the horizon the angular sphere has essentially fixed radius $r \approx 2GM$, so $r^2\,\mathrm{d}\Omega^2 \approx (2GM)^2\,\mathrm{d}\Omega^2$ is an inert spectator. The interesting part of Eq. (65.151) is the (ρ, τ) block, which collapses to

$$\mathrm{d}s_E^2\Big|_{\text{near horizon}} = \mathrm{d}\rho^2 + \left(\frac{\rho}{4GM}\right)^2 \mathrm{d}\tau^2 + (2GM)^2\,\mathrm{d}\Omega^2 = \mathrm{d}\rho^2 + \rho^2\,\mathrm{d}\alpha^2 + (2GM)^2\,\mathrm{d}\Omega^2, \tag{65.155}$$

where in the last step we introduced the rescaled angle

$$\alpha \equiv \frac{\tau}{4GM}. \tag{65.156}$$

The (ρ, α) part of Eq. (65.155) is exactly the flat plane written in polar coordinates: radial coordinate $\rho \geq 0$, angular coordinate α. A plane in polar coordinates is smooth at the origin $\rho = 0$ *if and only if* the angle α runs over a full period of 2π and no more — any other period leaves a wedge of opening angle $2\pi -$ (period) either missing or doubled, i.e. a conical singularity at the tip with a delta-function of curvature there. This is the same elementary fact that tells us a cone is flat everywhere except at its apex. Smoothness of the Euclidean black hole at its horizon is thus the single condition

$$\alpha \sim \alpha + 2\pi \quad \Longleftrightarrow \quad \tau \sim \tau + 8\pi GM. \tag{65.157}$$

Theorem 65.25 (Temperature as smoothness)**.** *The Euclidean Schwarzschild metric Eq. (65.151) extends to a smooth, geodesically complete Riemannian manifold (topology $\mathbb{R}^2 \times S^2$) if and only if the imaginary-time coordinate τ is identified with period*

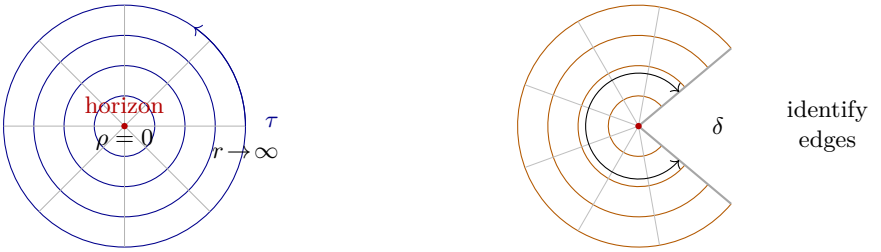
$$\beta = 8\pi GM = \frac{2\pi}{\kappa}, \qquad \kappa = \tfrac{1}{2} f'(2GM) = \frac{1}{4GM}. \tag{65.158}$$

Consequently the manifold is automatically periodic in imaginary time with period β, and the inverse temperature it represents is the geometric datum $T_H = 1/\beta = \kappa/2\pi$, identical to the Hawking temperature of Eq. (65.64).

Proof. The angular period equals 2π in the coordinate $\alpha = \tau/4GM$ exactly when τ has period $8\pi GM$, by Eq. (65.156). For any other period the near-horizon geometry Eq. (65.155) is a cone of deficit angle $\delta = 2\pi(1 - \beta/8\pi GM)$, whose Riemann tensor contains a term proportional to δ supported on the tip $\rho = 0$; the manifold is then not smooth there, and indeed not a solution of the vacuum Einstein equation in any neighbourhood of the horizon. Smoothness therefore forces $\beta = 8\pi GM$. The surface gravity is read directly off the conical-deficit calculation: the (ρ, τ) metric is $\mathrm{d}\rho^2 + (\frac{1}{2} f'(r_h))^2 \rho^2 \, \mathrm{d}\tau^2$ to leading order, so the angular velocity per unit τ is $\frac{1}{2} f'(r_h)$, and smoothness requires τ to close after $2\pi/(\frac{1}{2} f'(r_h)) = 4\pi/f'(r_h)$; identifying this with β and with $2\pi/\kappa$ gives $\kappa = \frac{1}{2} f'(r_h) = 1/4GM$. The numerical agreement of $\beta = 8\pi GM$ with the inverse Hawking temperature is then immediate. □

What makes Theorem 65.25 striking is that nothing quantum appears in it. No field was expanded in modes, no Bogoliubov coefficient was computed, no vacuum was chosen. The temperature is forced by the differential geometry of a smooth four-manifold: it is the unique periodicity that removes the conical singularity at the would-be horizon. This is the cleanest possible statement that the temperature belongs to the spacetime, not to whatever happens to be living on it. The picture is summarized in Fig. 65.5.

The construction did not use anything special about $f(r) = 1 - 2GM/r$ beyond the fact that f has a simple zero at the horizon. It is worth isolating



smooth: $\beta = 8\pi GM$ conical: $\beta \neq 8\pi GM$

Figure 65.5: The (ρ, τ) section of the Euclidean black hole at fixed angle on the sphere. Circles are constant-ρ orbits of the imaginary-time isometry; the horizon is the central point $\rho = 0$ where they shrink to zero size. *Left:* with $\beta = 8\pi GM$ the angle closes after exactly 2π and the cap is a smooth piece of the plane. *Right:* any other period leaves a deficit (or excess) angle δ, so identifying the two edges produces a cone with a curvature delta-function at the tip — not a solution of the vacuum equations.

the general statement, because it instantly delivers the temperatures of every horizon we will meet.

Worked Example 65.26 (Smoothness temperature of a general static horizon)**.** Consider any static, spherically symmetric metric of the form Eq. (65.1) with a function $f(r)$ that has a simple zero at some radius r_h, $f(r_h) = 0$ and $f'(r_h) > 0$. Repeat the near-horizon analysis. Linearizing, $f(r) = f'(r_h)(r - r_h) + \dots$, the proper distance is

$$\rho = \int_{r_h}^{r} \frac{\mathrm{d}r'}{\sqrt{f'(r_h)(r' - r_h)}} = \frac{2}{\sqrt{f'(r_h)}} \sqrt{r - r_h}, \qquad r - r_h = \frac{f'(r_h)}{4}\,\rho^2, \quad (65.159)$$

so that $f = f'(r_h)(r - r_h) = \left(\tfrac{1}{2} f'(r_h)\right)^2 \rho^2$ and the (ρ, τ) block of the Euclidean metric is $\mathrm{d}\rho^2 + \left(\tfrac{1}{2} f'(r_h)\right)^2 \rho^2\, \mathrm{d}\tau^2$. Smoothness demands that $\tfrac{1}{2} f'(r_h)\, \tau$ have period 2π, hence

$$\beta = \frac{4\pi}{f'(r_h)}, \qquad T = \frac{f'(r_h)}{4\pi} = \frac{\kappa}{2\pi}, \qquad \kappa = \tfrac{1}{2} f'(r_h). \quad (65.160)$$

Three immediate applications fix the result in the reader's hands. *Schwarzschild:* $f = 1 - 2GM/r$, $r_h = 2GM$, $f'(r_h) = 1/2GM$, recovering $T = 1/8\pi GM$. *Reissner–Nordström* (charge Q): $f = 1 - 2GM/r + GQ^2/r^2$, with outer horizon $r_+ = GM + \sqrt{G^2M^2 - GQ^2}$ and $f'(r_+) = (r_+ - r_-)/r_+^2$ where $r_- = GM - \sqrt{G^2M^2 - GQ^2}$; hence $T = (r_+ - r_-)/4\pi r_+^2$, which vanishes in the extremal limit $r_+ \to r_-$ — a zero-temperature horizon, the geometric face of the third law. *Schwarzschild–de Sitter* cosmological horizon: $f = 1 - \Lambda r^2/3$, $r_h = \sqrt{3/\Lambda}$,

$f'(r_h) = -2\Lambda r_h/3$ in magnitude $2\sqrt{\Lambda/3}$, giving the Gibbons–Hawking de Sitter temperature $T = \frac{1}{2\pi}\sqrt{\Lambda/3}$. The same one-line argument prices every horizon in the book.

65.5.2 The gravitational partition function and the saddle-point approximation

We now ask what physical quantity the smooth Euclidean manifold computes. Recall the lesson of Section 10.6: the Euclidean path integral of a field theory on a time circle of circumference β computes the thermal partition function. For a system with Hamiltonian H,

$$Z(\beta) = \mathrm{Tr}\; e^{-\beta H} = \int_{\phi(\tau+\beta)=\phi(\tau)} \mathcal{D}\phi\; e^{-S_E[\phi]}, \tag{65.161}$$

the trace being implemented by the periodic boundary condition on imaginary time. The free energy follows as $F = -\beta^{-1}\ln Z$, and from it the entropy $S = -\partial F/\partial T$ and internal energy $U = \partial(\beta F)/\partial\beta$. Everything in Eq. (65.161) was established for the scalar field of ϕ^4 theory at $\lambda = 0$; the gravitational version simply promotes the integration variable from a field on a fixed spacetime to the spacetime metric itself.

Definition 65.27 (Gravitational partition function)**.** The Euclidean gravitational partition function at inverse temperature β is the path integral over all Riemannian metrics g (and any matter fields, here none) on manifolds whose only boundary is at spatial infinity, where the metric is asymptotically flat and the imaginary-time circle has period β,

$$Z(\beta) = \int_{\substack{\text{asympt. flat,}\\ \tau\sim\tau+\beta}} \mathcal{D}g\; e^{-I_E[g]}, \tag{65.162}$$

with $I_E[g]$ the Euclidean gravitational action defined in Eq. (65.164) below.

The functional integral Eq. (65.162) is not, as written, a defined object: quantum gravity is not a renormalizable field theory, the measure $\mathcal{D}g$ is formal, and the action is unbounded below in the conformal direction. We are not going to pretend otherwise. What we can do — and what is genuinely controlled — is the semiclassical, or saddle-point, evaluation, valid when the relevant geometry is large compared with the Planck length $\ell_P = \sqrt{G}$, i.e. when $GM^2 \gg 1$. This is the same stationary-phase logic that produced the loop expansion in Section 10.6, only now organized in powers of G rather than $\hbar$ (the two are interchangeable here, since G multiplies the inverse action). The dominant contribution to Eq. (65.162) comes from metrics that solve the classical Euclidean equations of motion — the vacuum Einstein equation $R_{\mu\nu} = 0$ — subject to the stated

boundary condition, and the leading answer is the exponential of minus the on-shell action,

$$Z(\beta) \;\simeq\; \sum_{g_{\mathrm{cl}}} e^{-I_E[g_{\mathrm{cl}}]}\,\big[\det{}'(\text{fluctuation operator})\big]^{-1/2}(1+\dots), \tag{65.163}$$

where the determinant is the one-loop Gaussian integral over fluctuations about the saddle, deferred to Section 65.5.5, and the ellipsis denotes higher loops suppressed by further powers of G.

For the asymptotically flat thermal boundary condition there are precisely two smooth saddles. The first is hot flat space: $\mathbb{R}^3$ times an imaginary-time circle of period β, with $I_E = 0$ by construction (it is the reference geometry). The second is the smooth Euclidean Schwarzschild manifold of Section 65.5.1 — but only at the one mass $M = \beta/8\pi G$ for which Eq. (65.158) holds, since the period β is imposed at the boundary and the smoothness condition then dictates the mass. The black-hole saddle therefore enters the canonical ensemble at a definite mass for each temperature. Our task is to evaluate its action.

65.5.3 The on-shell action: the Gibbons–Hawking–York term and background subtraction

The naive Einstein–Hilbert action $-\frac{1}{16\pi G}\int_M R\sqrt{g}\,\mathrm{d}^4x$ cannot be the whole story, for a reason that is purely variational and was settled in Section 4.1: its variation produces, besides the Einstein tensor, a boundary integral of normal derivatives of δg that does not vanish under the natural Dirichlet condition (fix the induced metric on the boundary, but not its normal derivative). To have a well-posed variational principle on a manifold with boundary one must add the Gibbons–Hawking–York (GHY) surface term, whose variation exactly cancels the offending piece [220, 530]. Including it, and a subtraction term that renders the action finite, the Euclidean gravitational action is

Definition 65.28 (Euclidean gravitational action)**.**

$$I_E[g] = -\frac{1}{16\pi G}\int_M R\,\sqrt{g}\,\mathrm{d}^4x \;-\; \frac{1}{8\pi G}\oint_{\partial M}(K-K_0)\sqrt{h}\,\mathrm{d}^3x, \tag{65.164}$$

where h is the determinant of the metric induced on the boundary ∂M, $K = \nabla_\mu n^\mu$ is the trace of the extrinsic curvature with n^μ the outward unit normal, and K_0 is the extrinsic curvature of the *same* boundary as embedded in flat space. The K_0 term is the background subtraction: it measures the action relative to flat space, so that hot flat space scores $I_E = 0$ and the integral converges as the boundary recedes to infinity.

We evaluate Eq. (65.164) on the Euclidean Schwarzschild saddle.

The bulk term vanishes. Schwarzschild solves the vacuum Einstein equation $R_{\mu\nu} = 0$, whose trace gives $R = 0$ identically. The bulk integral in Eq. (65.164) is therefore zero on shell. *All* of the action comes from the boundary. This is a general feature of vacuum gravity: the on-shell action is a pure boundary term, which is why the gravitational entropy will turn out to be a boundary (horizon-area) quantity.

The extrinsic curvature at a large sphere. Place the boundary ∂M at a large fixed radius $r = \mathcal{R}$; at the end we send $\mathcal{R} \to \infty$. The induced metric there, from Eq. (65.151), is $\mathrm{d}s_E^2\big|_{\partial M} = f(\mathcal{R})\,\mathrm{d}\tau^2 + \mathcal{R}^2\,\mathrm{d}\Omega^2$, whose determinant is $h = f(\mathcal{R})\,\mathcal{R}^4 \sin^2\theta$, i.e. $\sqrt{h} = \sqrt{f(\mathcal{R})}\,\mathcal{R}^2 \sin\theta$. The outward unit normal points in the $+r$ direction; since $g_{rr} = 1/f$, normalization gives $n^\mu = \sqrt{f}\,\delta^\mu_r$. With $\sqrt{g} = r^2 \sin\theta$ (the f and $1/f$ in $g_{\tau\tau}g_{rr}$ cancel), the trace of the extrinsic curvature is the covariant divergence

$$K = \nabla_\mu n^\mu = \frac{1}{\sqrt{g}}\,\partial_r(\sqrt{g}\,n^r) = \frac{1}{r^2}\,\partial_r(r^2\sqrt{f}) = \frac{2\sqrt{f}}{r} + \frac{f'}{2\sqrt{f}}. \tag{65.165}$$

Evaluated at $r = \mathcal{R}$ with $f' = 2GM/\mathcal{R}^2$, the combination that enters the integrand is

$$\sqrt{f}\,K\,\mathcal{R}^2 = \Big(2f + \tfrac{1}{2}f'\,\mathcal{R}\Big)\mathcal{R} = 2\mathcal{R}\Big(1 - \frac{2GM}{\mathcal{R}}\Big) + GM = 2\mathcal{R} - 3GM, \tag{65.166}$$

using $\sqrt{f}\,(\frac{2\sqrt{f}}{r} + \frac{f'}{2\sqrt{f}})r^2 = 2fr + \frac{1}{2}f'r^2$. The τ-integral contributes a factor β and the sphere a factor 4π, so the bare GHY integral is

$$\oint_{\partial M} K\sqrt{h}\,\mathrm{d}^3x = 4\pi\beta\,(\sqrt{f}\,K\,\mathcal{R}^2) = 4\pi\beta\,(2\mathcal{R} - 3GM). \tag{65.167}$$

This diverges linearly as $\mathcal{R} \to \infty$; the background subtraction is what tames it.

The background subtraction. The reference is flat space, $f \equiv 1$, with the boundary at the same coordinate radius $\mathcal{R}$ and the same sphere. There $K_0 = 2/\mathcal{R}$ from Eq. (65.165). But the induced metrics must be matched: the proper circumference of the imaginary-time circle on the Schwarzschild boundary is $\sqrt{f(\mathcal{R})}\,\beta$, whereas in flat space with imaginary-time period β_0 it is just β_0. For the two boundaries to be the same Riemannian manifold — the only way the subtraction is meaningful — we must take

$$\beta_0 = \sqrt{f(\mathcal{R})}\,\beta. \tag{65.168}$$

The flat-space GHY integral, with $\sqrt{h_0} = \mathcal{R}^2 \sin\theta$ (now $f = 1$), $K_0 = 2/\mathcal{R}$, and τ'-period β_0, is

$$\oint_{\partial M} K_0\sqrt{h_0}\,\mathrm{d}^3x = \beta_0 \cdot 4\pi \cdot \frac{2}{\mathcal{R}} \cdot \mathcal{R}^2 = 8\pi\beta_0\,\mathcal{R} = 8\pi\beta\,\mathcal{R}\,\sqrt{f(\mathcal{R})}. \tag{65.169}$$

Expanding $\sqrt{f(\mathcal{R})} = \sqrt{1 - 2GM/\mathcal{R}} = 1 - \frac{GM}{\mathcal{R}} + O(\mathcal{R}^{-2})$,

$$\oint K_0 \sqrt{h_0}\, \mathrm{d}^3x = 8\pi\beta\,\mathcal{R}\Big(1 - \frac{GM}{\mathcal{R}} + O(\mathcal{R}^{-2})\Big) = 8\pi\beta\,\mathcal{R} - 8\pi\beta\,GM + O(\mathcal{R}^{-1}). \tag{65.170}$$

The subtracted action. Now subtract Eq. (65.170) from the rewritten bare term Eq. (65.167), $4\pi\beta(2\mathcal{R} - 3GM) = 8\pi\beta\mathcal{R} - 12\pi\beta GM$:

$$\oint (K - K_0)\sqrt{h}\, \mathrm{d}^3x = (8\pi\beta\mathcal{R} - 12\pi\beta GM) - (8\pi\beta\mathcal{R} - 8\pi\beta GM) = -4\pi\beta\,GM, \tag{65.171}$$

the linear divergences in $\mathcal{R}$ cancelling cleanly and the remainder being finite as $\mathcal{R} \to \infty$. Inserting this and the vanishing bulk term into Eq. (65.164),

$$I_E = -\frac{1}{8\pi G}\,(-4\pi\beta\,GM) = \frac{\beta M}{2}. \tag{65.172}$$

Finally we use the smoothness relation $\beta = 8\pi GM$, equivalently $M = \beta/8\pi G$, to express the answer in two equivalent forms. With $A = 4\pi r_h^2 = 4\pi(2GM)^2 = 16\pi G^2 M^2$ the horizon area, so that $A/4G = 4\pi GM^2$ and $\beta M = 8\pi GM^2$,

$$\boxed{I_E = \frac{\beta M}{2} = 4\pi G M^2 = \beta M - \frac{A}{4G} = \frac{\beta^2}{16\pi G}\,.} \tag{65.173}$$

The middle equality, $I_E = \beta M - A/4G$, is the result that the whole computation was aimed at. We will see in a moment that βM is exactly βU for this ensemble, so $I_E = \beta U - S$ with $S = A/4G$ — the on-shell action is already the combination $\beta F = \beta U - S$ of elementary thermodynamics, with the entropy sitting in it as the horizon area in Planck units.

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The structure of Eq. (65.173) repays a second look. The bulk Einstein–Hilbert term contributed nothing; the entire action is the GHY boundary term at infinity. Yet the answer encodes the horizon area A, which lives deep in the interior. How does a boundary integral at infinity know about the horizon? The answer is that smoothness at the horizon, Eq. (65.158), tied the period β to the mass M, and it is through that constraint that the area enters. Geometry at the cap and geometry at infinity are not independent: a single smooth manifold connects them, and the action is sensitive to both ends. This non-locality — entropy as a boundary/area quantity rather than a bulk volume quantity — is the first quantitative whisper of holography, and the reason Part XV exists.

65.5.4 Thermodynamics from the on-shell action

With the on-shell action in hand the thermodynamics is pure calculus. In the saddle-point approximation Eq. (65.163), the black-hole contribution to the partition function is $\ln Z \simeq -I_E$, so from Eq. (65.173),

$$\ln Z(\beta) = -I_E = -\frac{\beta^2}{16\pi G}. \tag{65.174}$$

Every thermodynamic quantity is a derivative of this single function. The internal energy is

$$U = -\frac{\partial \ln Z}{\partial \beta} = \frac{\partial I_E}{\partial \beta} = \frac{2\beta}{16\pi G} = \frac{\beta}{8\pi G} = M, \tag{65.175}$$

where the last step uses $\beta = 8\pi G M$. The mass is the thermodynamic internal energy, exactly as it must be; this confirms the identification of βM in Eq. (65.173) as βU. The free energy is

$$F = -\frac{1}{\beta}\ln Z = \frac{I_E}{\beta} = \frac{\beta}{16\pi G} = \frac{M}{2}. \tag{65.176}$$

The entropy now follows in either of two standard ways, which must of course agree. As the Legendre relation $S = \beta(U - F)$,

$$S = \beta(U - F) = \beta\Big(M - \frac{M}{2}\Big) = \frac{\beta M}{2} = 4\pi G M^2, \tag{65.177}$$

or equivalently, writing F as a function of $T = 1/\beta$, namely $F = \frac{1}{16\pi G\, T}$, as the temperature derivative

$$S = -\frac{\partial F}{\partial T} = -\frac{\partial}{\partial T}\Big(\frac{1}{16\pi G\, T}\Big) = \frac{1}{16\pi G\, T^2} = \frac{\beta^2}{16\pi G} = 4\pi G M^2. \tag{65.178}$$

Both routes give the same number, and that number is, once more,

$$\boxed{S = 4\pi G M^2 = \frac{16\pi G^2 M^2}{4G} = \frac{A}{4G}\,.} \tag{65.179}$$

Theorem 65.29 (Bekenstein–Hawking entropy from the gravitational path integral)**.** *The leading semiclassical contribution of the Euclidean Schwarzschild saddle to the gravitational partition function Eq. (65.162) yields internal energy $U = M$, free energy $F = M/2$, and entropy*

$$S = -\frac{\partial F}{\partial T} = \frac{A}{4G}, \tag{65.180}$$

the Bekenstein–Hawking entropy with its coefficient $1/4$ fixed, in agreement with the thermodynamic derivation of Eq. (65.142).

This is a genuinely independent confirmation. In Section 65.4 the coefficient $1/4$ was fixed by feeding in the Hawking temperature and integrating the first law; here it is fixed by a self-contained evaluation of a gravitational action that never mentions a quantum field. That two such different calculations land on the same $A/4G$ is the kind of overdetermination that turns a suggestive result into a fact about the theory.

Worked Example 65.30 (The negative heat capacity of Schwarzschild and its meaning)**.** The heat capacity of the Schwarzschild black hole is the response of its energy to temperature. Since $T = 1/8\pi GM$ gives $M = 1/(8\pi G\,T)$,

$$C = \frac{\partial U}{\partial T} = \frac{\partial M}{\partial T} = -\frac{1}{8\pi G\,T^2} = -8\pi G M^2 = -2S = -\frac{A}{2G} < 0. \tag{65.181}$$

The heat capacity is *negative*: a Schwarzschild hole gets *hotter* as it loses energy. This is not a pathology of the calculation but a physical statement about gravitating systems, and it has three consequences worth spelling out.

(i) Runaway evaporation. A black hole radiating into empty space loses mass; by $T \propto 1/M$ its temperature rises, so it radiates faster, loses mass faster, and heats up further. The evaporation accelerates without bound as $M \to 0$, terminating in a final burst at the Planck scale where the semiclassical description fails. Negative heat capacity is the thermodynamic face of the instability of an isolated black hole.

(ii) No stable canonical ensemble in flat space. Stability of the canonical ensemble requires $C > 0$ (so that $\partial^2 F/\partial T^2 < 0$ and the free energy is concave). With $C < 0$ the saddle is a local *maximum* rather than minimum of the free energy in the relevant direction. In the path integral this shows up as a single negative eigenvalue of the fluctuation operator about the Euclidean Schwarzschild saddle — the Gross–Perry–Yaffe negative mode [239] — which makes the Gaussian integral in Eq. (65.163) formally imaginary and signals that the black hole contributes to the *decay rate* of hot flat space, not to a stable equilibrium. The hot-flat-space and black-hole saddles compete; which dominates depends on temperature, and the imaginary part is the nucleation rate of black holes out of the thermal bath.

(iii) How to stabilize it. A finite system can be stabilized by confining it. York's construction places the hole in a cavity of radius r_B with the wall held at fixed temperature; the locally measured (Tolman-blueshifted) temperature and the finite-volume constraint can make $C > 0$ for a sufficiently small cavity, yielding a genuine canonical equilibrium. The cleaner stabilizer is a negative cosmological constant: in anti–de Sitter space the curvature radius plays the role of a confining box, large black holes have $C > 0$, and the competition between the black-hole and thermal-AdS saddles becomes the Hawking–Page phase transition [258] — the bulk dual of a deconfinement transition in the boundary theory of Part XV. The negative heat capacity of asymptotically flat Schwarzschild is thus not the end of the story but the entrance to the richer thermodynamics of black holes in a box.

65.5.5 One loop: the fluctuation determinant and zeta regularization

The result $S = A/4G$ is the *leading* term in the expansion Eq. (65.163); it is the tree-level (in G) free energy. The next term is the one-loop Gaussian integral over fluctuations $g = g_{\rm cl} + h$ about the saddle, which contributes

$$\ln Z = -I_E[g_{\rm cl}] - \tfrac{1}{2} \ln \det{}' \Delta + \dots , \tag{65.182}$$

where Δ is the quadratic operator governing the fluctuations (the Lichnerowicz operator for the graviton, together with the matter operators of any fields present) and the prime indicates the omission of zero modes, handled by the collective-coordinate method of Section E.6.3. The functional determinant $\det{}'\Delta$ is, as always, a divergent product of eigenvalues that requires definition. The definition the book uses throughout is the spectral zeta function: if Δ has eigenvalues λ_n, set $\zeta_\Delta(s) = \sum_n \lambda_n^{-s}$, analytically continue, and *define*

$$\ln \det{}' \Delta \equiv -\zeta_\Delta'(0), \tag{65.183}$$

exactly as constructed in Section E.3 and packaged in Theorem E.16. It is no accident that this is the natural tool here: zeta-function regularization was introduced by Hawking precisely to make sense of gravitational and curved-space path integrals of this type [257], and the renormalization of the stress tensor in the same backgrounds, carried out in Section 64.5, is the local face of the same heat-kernel machinery.

We do not carry out the full graviton determinant here — it is a long computation, sensitive to gauge-fixing and to the conformal-factor problem, and the techniques are exactly those of Section E.3. What matters for the architecture of the argument is the *structure* of the answer. The one-loop determinant, regularized by Eq. (65.183), produces a correction to the entropy of the schematic form

$$S = \frac{A}{4G} + c \, \ln \frac{A}{G} + \dots , \tag{65.184}$$

with a coefficient c fixed by the trace anomaly of the fluctuating fields in the black-hole background — the very anomaly coefficient that the next section computes locally. The leading $A/4G$ is universal and geometric; the logarithmic correction is the first place where the microscopic content of the theory (its field content, through the anomaly) leaves a fingerprint on black-hole thermodynamics. We flag this here and pick up the anomaly thread in Section 65.6.

We have now derived the Hawking temperature and the Bekenstein–Hawking entropy a second, fully independent way: not from tracking modes across a horizon, but from the smoothness and on-shell action of a Euclidean saddle of the gravitational path integral. The temperature is the unique period that removes a conical singularity; the entropy is the boundary term of the on-shell action.

This saddle-point, path-integral picture of the gravitational partition function is the viewpoint the remainder of the chapter builds on. The next section turns to a *third*, purely local derivation of the same Hawking flux, reading it off the quantum anomalies of fields in the near-horizon region; and the gravitational path integral introduced here returns in force in Section 65.8 when its *replica* saddles — additional smooth geometries that dominate at late times — turn out to know about the unitarity of black-hole evaporation and bend the entropy back down along the Page curve.

65.6 Hawking Flux from Quantum Anomalies

The previous section gave us the Hawking temperature for a second time, and from a viewpoint that never quantized a single matter field: the temperature is the unique period that removes the conical singularity from the Euclidean cap, and the entropy is the boundary term of an on-shell gravitational action. Section 3 had already recast the radiated power as a statement about the renormalized stress tensor, $L = \lim_{r\to\infty} 4\pi r^2 \langle T_{uu}\rangle_{\rm ren}$ (Eq. (65.118)), a manifestly local object built by the point-splitting subtraction of Section 64.5. Both pictures, the Euclidean saddle and the renormalized stress tensor, are local and geometric. It is natural to ask whether the *number* in the flux—the coefficient $\pi T_H^2/12$ per field—can be extracted from purely local data near the horizon, without ever following a mode out to infinity or evaluating an action at the boundary of the universe.

It can, and the mechanism is one we built deep in Part X and Part XIV: a quantum anomaly. The claim of this section is sharp. Reduce a field near the horizon to its two-dimensional content; then the breakdown of a classical symmetry at one loop—the trace (conformal) anomaly, or equivalently the gravitational anomaly of the effectively chiral near-horizon theory—fixes the outgoing energy flux completely, and the value it fixes is exactly the Hawking flux. The radiation is anomaly-mediated: it is determined by horizon data alone, by the surface gravity κ and the central charge of the matter, with no reference to the asymptotic region. This is the third independent derivation promised in the chapter's thesis, and it is the one that recruits the machinery of Chapter 36 (the chiral anomaly, the Fujikawa measure) and Chapters 47 and 48 (the conformal stress tensor and its anomalous Schwarzian transformation) and shows them powering black-hole radiance.

65.6.1 Near-horizon dimensional reduction: a tower of two-dimensional fields

The first step is to understand why a four-dimensional problem becomes a two-dimensional one near the horizon. We carry the free real scalar of ϕ^4 theory at

$\lambda = 0$, with the curved-space action of Section 64.5,

$$S = \frac{1}{2}\int \mathrm{d}^4x\,\sqrt{-g}\,g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi, \qquad \mathrm{d}s^2 = f(r)\,\mathrm{d}t^2 - \frac{\mathrm{d}r^2}{f(r)} - r^2\,\mathrm{d}\Omega^2, \tag{65.185}$$

with $f(r) = 1 - 2GM/r$ as throughout the chapter and $\sqrt{-g} = r^2\sin\theta$. Expand the field in real spherical harmonics, $\phi(t,r,\theta,\varphi) = \sum_{\ell m}\phi_{\ell m}(t,r)\,Y_{\ell m}(\theta,\varphi)$, and use the orthonormality $\int \mathrm{d}\Omega\,Y_{\ell m}Y_{\ell' m'} = \delta_{\ell\ell'}\delta_{mm'}$ together with $\int \mathrm{d}\Omega\,(\nabla_\Omega Y_{\ell m})^2 = \ell(\ell+1)$ on the unit sphere. The angular integral collapses the action to a sum of decoupled two-dimensional actions, one per mode,

$$S = \frac{1}{2}\sum_{\ell m}\int \mathrm{d}t\,\mathrm{d}r\left[\frac{r^2}{f}\,(\partial_t\phi_{\ell m})^2 - r^2 f\,(\partial_r\phi_{\ell m})^2 - \ell(\ell+1)\,\phi_{\ell m}^2\right]. \tag{65.186}$$

This is precisely a free scalar living on the two-dimensional (t,r) plane with metric

$$\mathrm{d}s^2_{(2)} = f(r)\,\mathrm{d}t^2 - \frac{\mathrm{d}r^2}{f(r)}, \qquad \sqrt{-g_{(2)}} = 1, \quad g^{tt}_{(2)} = \frac{1}{f},\ g^{rr}_{(2)} = -f, \tag{65.187}$$

coupled to a position-dependent *dilaton* $\Phi(r) \equiv r^2$ (the transverse area) and carrying a mass term set by the angular momentum:

$$S = \frac{1}{2}\sum_{\ell m}\int \mathrm{d}^2x\,\sqrt{-g_{(2)}}\;\Phi(r)\left[g^{ab}_{(2)}\partial_a\phi_{\ell m}\partial_b\phi_{\ell m} - \frac{\ell(\ell+1)}{r^2}\,\phi_{\ell m}^2\right]. \tag{65.188}$$

One checks the identification directly: $\Phi g^{tt}_{(2)} = r^2/f$ and $\Phi g^{rr}_{(2)} = -r^2 f$ reproduce the two kinetic terms of Eq. (65.186), and $\Phi\cdot\ell(\ell+1)/r^2 = \ell(\ell+1)$ the mass term. A single higher-dimensional field has become an *infinite tower* of two-dimensional fields indexed by (ℓ, m), each a scalar on the (t,r) plane with a dilaton coupling and an ℓ-dependent potential.

Definition 65.31 (Near-horizon two-dimensional theory)**.** The *near-horizon limit* of Eq. (65.188) is the theory obtained as $r \to r_h = 2GM$, where $f \to 0$. In that limit the dilaton $\Phi \to r_h^2$ is constant, and the effective potential governing the radial wave—read off from Eq. (65.186) in the tortoise coordinate r_* of Eq. (65.75), in which the radial equation becomes $\ddot{\phi}_{\ell m} + (\omega^2 - V_\ell)\phi_{\ell m} = 0$ with $V_\ell = f\left[\ell(\ell+1)/r^2 + f'/r\right]$—vanishes as $V_\ell \propto f \to 0$. Each tower member therefore reduces, in a neighbourhood of the horizon, to a *free, massless, two-dimensional scalar* on Eq. (65.187), with central charge $c = 1$.

The physics of this reduction is the same physics that produced the greybody factors of Section 65.2.1: away from the horizon the potential V_ℓ is a barrier through which a mode must tunnel, and it is exactly this barrier that suppresses the high-ℓ partial waves and shapes the spectrum at infinity. But at the horizon

the barrier is gone—V_ℓ vanishes there because every term carries a factor of f—so the *near-horizon* dynamics of each and every partial wave is that of a free two-dimensional conformal scalar. This is the structural fact the anomaly arguments exploit: whatever the matter content and whatever the angular momentum, the field theory at the horizon is a collection of two-dimensional CFTs, and two dimensions is precisely the dimension in which the trace and gravitational anomalies are simplest and most powerful.

Physical Insight

Dimensional reduction at the horizon is not an approximation we impose for convenience; it is forced by the geometry. The factor $f(r)$ multiplies the angular gradient energy and the potential alike, and $f \to 0$ at the horizon by definition of a horizon. Modes piling up against the horizon are infinitely blueshifted in the (t, r) plane while their transverse structure is frozen, so the only dynamics that survives the limit is two-dimensional and conformal. The horizon manufactures a 2D CFT out of any field thrown at it.

65.6.2 The trace anomaly fixes the flux: the Christensen–Fulling integration

We now compute the outgoing flux of a single near-horizon 2D conformal scalar from its trace anomaly. Pass to null coordinates

$$u = t - r_*, \qquad v = t + r_*, \qquad \frac{\mathrm{d}r_*}{\mathrm{d}r} = \frac{1}{f}, \tag{65.189}$$

in which the two-dimensional metric Eq. (65.187) is conformally flat,

$$\mathrm{d}s^2_{(2)} = f(r)\,\mathrm{d}u\,\mathrm{d}v = e^{2\rho}\,\mathrm{d}u\,\mathrm{d}v, \qquad \rho = \tfrac{1}{2}\ln f(r), \qquad R = f''(r), \tag{65.190}$$

where R is the two-dimensional Ricci scalar in the curvature convention of Appendix A, equivalently $R = -8e^{-2\rho}\partial_u\partial_v\rho$ for a metric of the form $e^{2\rho}\mathrm{d}u\,\mathrm{d}v$. The renormalized stress tensor of the conformal scalar is not traceless: it carries the two-dimensional specialization of the trace anomaly established in general dimension in Eq. (64.199), which with central charge c (Theorem 48.8) reads

$$\langle T^{\mu}{}_{\mu} \rangle = \frac{c}{24\pi}\,R. \tag{65.191}$$

This single anomalous equation, together with covariant conservation $\nabla_\mu T^{\mu}{}_{\nu} = 0$, is enough to determine $\langle T_{\mu\nu} \rangle$ up to two integration functions. We carry out the integration explicitly; this is the Christensen–Fulling computation [123].

Conservation in conformal gauge. For the metric $\mathrm{d}s^2 = e^{2\rho}\mathrm{d}u\,\mathrm{d}v$ the only non-vanishing Christoffel symbols are $\Gamma^u_{uu} = 2\partial_u\rho$ and $\Gamma^v_{vv} = 2\partial_v\rho$, and the inverse metric has the single component $g^{uv} = 2e^{-2\rho}$. Writing out $\nabla_\mu T^\mu{}_u = \partial_\mu T^\mu{}_u + \Gamma^\mu{}_{\mu\lambda}T^\lambda{}_u - \Gamma^\lambda{}_{\mu u}T^\mu{}_\lambda$ and using these data, every term containing $\Gamma^u_{uu}T^u{}_u$ cancels, leaving after multiplication by $\frac{1}{2}e^{2\rho}$ the compact relations

$$\partial_v T_{uu} = -\partial_u T_{uv} + 2(\partial_u\rho)\,T_{uv}, \qquad \partial_u T_{vv} = -\partial_v T_{uv} + 2(\partial_v\rho)\,T_{uv}. \tag{65.192}$$

The trace component is fixed algebraically by the anomaly: from $\langle T^\mu{}_\mu\rangle = g^{\mu\nu}T_{\mu\nu} = 2g^{uv}T_{uv} = 4e^{-2\rho}T_{uv}$ and Eq. (65.191) with $R = -8e^{-2\rho}\partial_u\partial_v\rho$,

$$T_{uv} = -\frac{c}{12\pi}\,\partial_u\partial_v\rho. \tag{65.193}$$

Were the theory conformally invariant at the quantum level we would have $T_{uv} = 0$; the anomaly Eq. (65.191) is exactly the statement that it is not, and this nonzero trace is what sources a nontrivial outgoing component.

Integrating the outgoing component. Substitute Eq. (65.193) into the second relation of Eq. (65.192), the one for T_{vv}; by the obvious $u \leftrightarrow v$ symmetry the same steps applied to the first relation give T_{uu}. Using $-\partial_v T_{uv} = \frac{c}{12\pi}\partial_v\partial_u\partial_v\rho$ and $2(\partial_v\rho)T_{uv} = -\frac{c}{6\pi}(\partial_v\rho)(\partial_u\partial_v\rho) = -\frac{c}{12\pi}\partial_u[(\partial_v\rho)^2]$, the right-hand side collects into a single u-derivative,

$$\partial_u T_{vv} = \frac{c}{12\pi}\,\partial_u[\partial_v^2\rho - (\partial_v\rho)^2], \tag{65.194}$$

which integrates immediately. Doing the same for the outgoing (uu) component,

$$T_{uu} = \frac{c}{12\pi}\left[\partial_u^2\rho - (\partial_u\rho)^2\right] + t_{uu}(u), \qquad T_{vv} = \frac{c}{12\pi}\left[\partial_v^2\rho - (\partial_v\rho)^2\right] + t_{vv}(v), \tag{65.195}$$

where $t_{uu}(u)$ and $t_{vv}(v)$ are integration "constants"—arbitrary functions of a single null coordinate each—left undetermined by the local equations. They encode the choice of quantum state. *All* the dynamical content of the Hawking flux is in fixing them.

Evaluating on Schwarzschild. The conformal factor depends only on $r_* = (v-u)/2$, so with $\mathrm{d}r/\mathrm{d}r_* = f$ and $\mathrm{d}\rho/\mathrm{d}r_* = \frac{1}{2}f'$ we have $\partial_u\rho = -\frac{1}{2}\,\mathrm{d}\rho/\mathrm{d}r_* = -f'/4$ and

$$\partial_u^2\rho = \frac{ff''}{8}, \qquad (\partial_u\rho)^2 = \frac{(f')^2}{16}, \qquad \partial_u^2\rho - (\partial_u\rho)^2 = \frac{ff''}{8} - \frac{(f')^2}{16}. \tag{65.196}$$

This combination has a clean value at both ends of the exterior. At infinity $f \to 1$, $f' \to 0$, $f'' \to 0$, so it vanishes and $T_{uu} \to t_{uu}$: the asymptotic flux is

the integration function alone. At the horizon $f(r_h) = 0$, and with the surface gravity $\kappa = \frac{1}{2}f'(r_h)$ of Section 65.4 so that $f'(r_h) = 2\kappa$,

$$\Big[\partial_u^2\rho - (\partial_u\rho)^2\Big]_{r_h} = -\frac{(f'(r_h))^2}{16} = -\frac{(2\kappa)^2}{16} = -\frac{\kappa^2}{4}, \qquad \Longrightarrow \qquad \frac{c}{12\pi}\Big[\cdots\Big]_{r_h} = -\frac{c\,\kappa^2}{48\pi}. \tag{65.197}$$

The boundary condition and the flux. The integration function is fixed by the demand that the state be regular at the future horizon—the Unruh state appropriate to a black hole formed by collapse and radiating into an initially empty exterior. Regularity in a frame that is smooth at the horizon means the physical (freely-falling) components of $\langle T_{\mu\nu}\rangle$ are finite there. Because the outgoing null coordinate u degenerates at the horizon while the Kruskal coordinate U is regular, and $T_{UU} = (\mathrm{d}u/\mathrm{d}U)^2 T_{uu}$ with $\mathrm{d}u/\mathrm{d}U$ diverging there, finiteness of T_{UU} requires $T_{uu} \to 0$ as $r \to r_h$. Imposing this on Eq. (65.195) with Eq. (65.197),

$$0 = T_{uu}\big|_{r_h} = -\frac{c\,\kappa^2}{48\pi} + t_{uu} \qquad \Longrightarrow \qquad t_{uu} = \frac{c\,\kappa^2}{48\pi}. \tag{65.198}$$

The asymptotic outgoing flux is therefore

$$\boxed{\langle T_{uu}\rangle\big|_{r\to\infty} = t_{uu} = \frac{c\,\kappa^2}{48\pi} = \frac{\pi c}{12}\,T_H^2} \qquad (T_H = \kappa/2\pi), \tag{65.199}$$

using $T_H = \kappa/2\pi$ from Eq. (65.64). For the single conformal scalar, $c = 1$ and the flux is $\kappa^2/48\pi = \pi T_H^2/12$, precisely the value extracted from the mode sum and the renormalized stress tensor in Section 65.2.1.

Theorem 65.32 (Hawking flux from the trace anomaly)**.** *Let a two-dimensional conformal field of central charge c propagate on the near-horizon geometry Eq. (65.187) of a static horizon with surface gravity κ. Covariant conservation of its stress tensor together with the trace anomaly Eq. (65.191), supplemented by the regularity (Unruh) boundary condition $T_{uu} \to 0$ at the future horizon, fixes the outgoing energy flux at infinity uniquely to*

$$\langle T_{uu}\rangle\big|_{\infty} = \frac{c\,\kappa^2}{48\pi} = \frac{\pi c}{12}\,T_H^2. \tag{65.200}$$

The flux depends only on κ (horizon data) and c (matter content); it is independent of the asymptotic region and of the detailed profile of $f(r)$ away from the horizon.

The structure of the argument deserves emphasis. The local equations Eq. (65.195) leave the flux completely free; it is the anomaly coefficient c, multiplying a curvature combination that takes the value $-\kappa^2/4$ at the horizon, that sets the scale, and the horizon regularity condition that converts a free integration constant into the definite number $c\kappa^2/48\pi$. Without the anomaly ($c = 0$) the flux would vanish: a classically conformal field would not radiate. The Hawking flux is an anomaly, full stop.

65.6.3 The same flux from the anomalous (Schwarzian) transformation of T

The integration function t_{uu} is not a tensor. It is the two-dimensional analog of the holomorphic stress tensor $T(z)$ of Chapter 48, and like $T(z)$ it transforms under a conformal change of the null coordinate $u \to \tilde{u}(u)$ with an inhomogeneous Schwarzian term. This gives a second, even shorter derivation of Eq. (65.199), and it makes the central charge's role completely explicit.

Recall from Eq. (48.63) that under a holomorphic map the stress tensor of a 2D CFT picks up the Schwarzian derivative Eq. (48.62); the Lorentzian, null-coordinate version of that statement, applied to the state-dependent piece t_{uu} extracted in Eq. (65.195), is

$$t_{uu}(u) = \left(\frac{\mathrm{d}\tilde{u}}{\mathrm{d}u}\right)^2 t_{\tilde{u}\tilde{u}}(\tilde{u}) - \frac{c}{24\pi}\{\tilde{u}, u\}, \qquad \{\tilde{u}, u\} = \frac{\tilde{u}'''}{\tilde{u}'} - \frac{3}{2}\left(\frac{\tilde{u}''}{\tilde{u}'}\right)^2. \tag{65.201}$$

That this is the correct transformation law follows from the same conservation-plus-anomaly system used above: under $u \to \tilde{u}$ the conformal factor shifts as $\rho \to \rho - \frac{1}{2}\ln(\mathrm{d}\tilde{u}/\mathrm{d}u)$, and feeding this into the geometric piece of Eq. (65.195) produces exactly the Schwarzian inhomogeneity in Eq. (65.201), with the universal coefficient $c/24\pi$ set by the anomaly Eq. (65.191). The cocycle (group) consistency of the Schwarzian, proved in Eq. (48.66), guarantees this law composes correctly.

The strategy is now to identify the one frame in which the state-dependent flux vanishes. The Unruh vacuum is, by construction, the state whose outgoing modes are positive frequency with respect to the Kruskal coordinate U that is regular across the future horizon. In that Kruskal frame the outgoing sector looks like the Minkowski vacuum of an inertial observer, so its normal-ordered stress tensor has no anomalous flux:

$$t_{UU}(U) = 0. \tag{65.202}$$

The exterior retarded coordinate u and the Kruskal coordinate U are related, as established in the ray-tracing analysis of Section 65.2.1, by the exponential map

$$U = -\frac{1}{\kappa}\, e^{-\kappa u}, \tag{65.203}$$

which is the geometric origin of the whole effect: a late-time outgoing wave (large u) is exponentially compressed against the horizon $U \to 0^-$. We need only its Schwarzian. With $U' = e^{-\kappa u}$, $U'' = -\kappa e^{-\kappa u}$, $U''' = \kappa^2 e^{-\kappa u}$,

$$\{U, u\} = \frac{U'''}{U'} - \frac{3}{2}\left(\frac{U''}{U'}\right)^2 = \kappa^2 - \frac{3}{2}\kappa^2 = -\frac{\kappa^2}{2}. \tag{65.204}$$

Setting $\tilde{u} = U$ in Eq. (65.201) and using Eq. (65.202),

$$t_{uu} = (U')^2\, t_{UU} - \frac{c}{24\pi}\{U, u\} = 0 - \frac{c}{24\pi}\left(-\frac{\kappa^2}{2}\right) = \frac{c\,\kappa^2}{48\pi}, \tag{65.205}$$

identical to Eq. (65.199). The two computations are the same computation seen twice: the curvature combination of Christensen–Fulling and the Schwarzian of the Kruskal map are two faces of the one anomaly, and both deliver the flux $c\,\kappa^2/48\pi$. The central charge that fixes the coefficient is the very c defined through the $T(z)T(w)$ operator product in Eq. (48.56); the surface gravity that fixes the scale enters solely through the rate κ of the exponential horizon map Eq. (65.203). Horizon data and central charge, and nothing else.

Worked Example 65.33 (The flux is a two-dimensional Planck spectrum at T_H)**.** The number $c\kappa^2/48\pi$ should be the energy flux of a thermal gas of the near-horizon fields at temperature T_H, since that is what Chapter 64's Bogoliubov analysis says the horizon emits. We check this directly for one bosonic field ($c = 1$). A massless field in two dimensions splits into independent left- and right-moving sectors; the outgoing (right-moving) sector carries an energy flux that, in a thermal state at temperature T_H, is the integral over modes of the energy ω times the Bose occupation $n(\omega) = (e^{\omega/T_H} - 1)^{-1}$, with the one-dimensional density of states $\mathrm{d}\omega/2\pi$,

$$\Phi = \int_0^\infty \frac{\mathrm{d}\omega}{2\pi}\,\frac{\omega}{e^{\omega/T_H}-1} = \frac{T_H^2}{2\pi}\int_0^\infty \frac{x\,\mathrm{d}x}{e^x-1} = \frac{T_H^2}{2\pi}\cdot\frac{\pi^2}{6} = \frac{\pi}{12}\,T_H^2, \tag{65.206}$$

using $\int_0^\infty x\,(e^x-1)^{-1}\mathrm{d}x = \Gamma(2)\zeta(2) = \pi^2/6$ (the standard Bose–Einstein integral of Chapter 64). With $T_H = \kappa/2\pi$ this is $\Phi = \pi T_H^2/12 = \kappa^2/48\pi$, exactly Eq. (65.199) at $c = 1$. The anomaly has reproduced the full Stefan–Boltzmann-type flux of a two-dimensional blackbody at the Hawking temperature, coefficient and all. The $\pi^2/6$ from the thermal integral and the $-\kappa^2/2$ from the Schwarzian are the same physics: $\zeta(2)$ is the Casimir energy of the $c = 1$ theory on a thermal circle, and the Schwarzian term is its anomalous transformation. This is why the central charge controls both.

65.6.4 Robinson–Wilczek: the chiral horizon and the gravitational anomaly

The trace anomaly is one face of the two-dimensional anomaly structure; the other is the gravitational (diffeomorphism) anomaly of a *chiral* field, and it gives a third route to the same flux—the one that makes the role of the horizon as a one-way membrane most vivid. This is the argument of Robinson and Wilczek [418].

The key observation is that, arbitrarily close to the horizon, the ingoing modes are causally irrelevant to the exterior: nothing that crosses inward can come back out. If we restrict attention to a thin shell $r_h \le r \le r_h + \varepsilon$ and ask for the effective theory that governs physics outside, we must integrate out the ingoing modes there, because they have fallen through and decoupled. What remains is a theory of the outgoing modes alone—a *chiral* two-dimensional theory

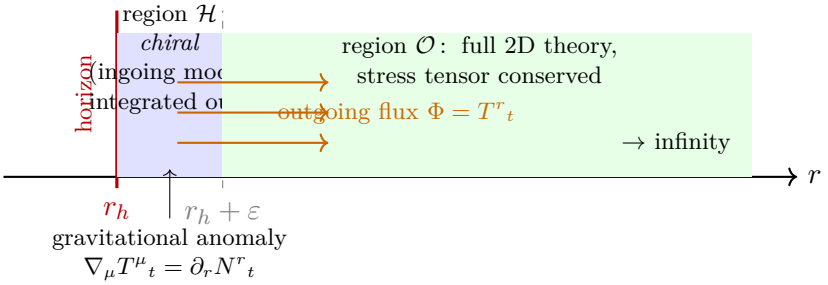


Figure 65.6: The Robinson–Wilczek region split. In the thin near-horizon shell $\mathcal{H} = [r_h, r_h + \varepsilon]$ the ingoing modes are integrated out, leaving a chiral two-dimensional theory whose stress tensor is anomalous, $\nabla_\mu T^\mu{}_t = \partial_r N^r{}_t$. In the exterior $\mathcal{O}$ both chiralities survive and the stress tensor is conserved, so the outgoing energy flux $T^r{}_t$ is constant there. Diffeomorphism invariance of the total effective action forces a cancellation at the boundary $r = r_h + \varepsilon$ that fixes that constant flux to the universal Hawking value.

in the near-horizon region. (In null coordinates the surviving field is a function of u only.) But a two-dimensional chiral field is anomalous under diffeomorphisms: its stress tensor is not conserved, exactly as a chiral fermion's current is not conserved in Theorem 36.4. The gravitational anomaly is the diffeomorphism cousin of the chiral gauge anomaly, and like it, is one-loop exact and fixed by the same Fujikawa measure argument of Section 36.3: under a chiral truncation the path-integral measure is not diffeomorphism invariant, and its Jacobian is a local curvature functional.

The anomalous Ward identity. For the two-dimensional metric Eq. (65.187), the consistent gravitational anomaly of one chiral boson takes the form of a total radial derivative in the energy (time) component of the conservation law,

$$\nabla_\mu T^\mu{}_t = \frac{1}{\sqrt{-g_{(2)}}}\,\partial_r N^r{}_t, \qquad N^r{}_t = \frac{c}{96\pi}\Big(f f'' - \tfrac{1}{2}(f')^2\Big), \tag{65.207}$$

with $c = 1$ per chiral scalar and $\sqrt{-g_{(2)}} = 1$ here. (The structure—an anomaly that is a total derivative of a local curvature polynomial—is the gravitational analog of the chiral-anomaly density $\partial_\mu j^{\mu 5} \propto \partial_\mu K^\mu$ of Eq. (36.50).) In the exterior region $\mathcal{O}$, where both chiralities are present, the theory is anomaly-free, so $\nabla_\mu T^\mu{}_t = 0$ and, the metric being static, $T^r{}_t$ is a constant in r:

$$T^r{}_t\big|_{\mathcal{O}} = -\Phi = \text{const}, \tag{65.208}$$

the constant outgoing energy flux we seek (the sign convention is such that $\Phi > 0$ carries energy outward, $\Phi = -\mathrm{d}M/\mathrm{d}t$). In the near-horizon region $\mathcal{H}$ the chiral theory obeys Eq. (65.207), which integrates to $T^r{}_t|_{\mathcal{H}}(r) = K_{\mathcal{H}} + N^r{}_t(r) - N^r{}_t(r_h)$.

The boundary term and the flux. Write the full stress tensor with a step function separating the two regions, $T^r{}_t = T^r{}_t|_{\mathcal{O}}\,\Theta(r-r_h-\varepsilon) + T^r{}_t|_{\mathcal{H}}\,[1-\Theta(r-r_h-\varepsilon)]$, and demand that the total quantum effective action W be invariant under an infinitesimal diffeomorphism $x^\mu \to x^\mu + \xi^\mu$. The variation is $\delta_\xi W = -\int \mathrm{d}^2x\,\sqrt{-g}\,\xi^\nu \nabla_\mu T^\mu{}_\nu$. Taking $\xi^\nu = \xi^t \delta^\nu_t$ and using the two regional Ward identities, the bulk anomaly term in $\mathcal{H}$ is cancelled by the contribution of the classical ingoing modes that were integrated out (their effective action restores diffeomorphism invariance in the bulk), and what survives is a delta-function supported at the boundary $r = r_h + \varepsilon$,

$$\delta_\xi W = -\int \mathrm{d}^2x\;\xi^t\left[\,T^r{}_t\big|_{\mathcal{O}} - T^r{}_t\big|_{\mathcal{H}} + N^r{}_t\,\right]\delta(r-r_h-\varepsilon) + (\text{bulk, cancelled}). \tag{65.209}$$

Invariance requires the bracket to vanish. As $\varepsilon \to 0$ we impose the covariant boundary condition that fixes the chiral integration constant: the *covariant* energy flux must vanish at the horizon itself, $T^r{}_t|_{\mathcal{H}}(r_h) = 0$, since the horizon is not a source of energy. Then the bracket evaluated at r_h reads $-\Phi - 0 + N^r{}_t(r_h) = 0$, so

$$\Phi = N^r{}_t(r_h) = \frac{c}{96\pi}\Big(\underbrace{f f''}_{0} - \tfrac{1}{2}(f')^2\Big)_{r_h} = -\frac{c}{96\pi}\cdot\frac{1}{2}\,(2\kappa)^2 = -\frac{c\,\kappa^2}{48\pi}, \tag{65.210}$$

where $f(r_h) = 0$ kills the first term and $f'(r_h) = 2\kappa$. Restoring the sign bookkeeping of Eq. (65.208), the magnitude of the outgoing flux is

$$\boxed{\,|\Phi| = \frac{c\,\kappa^2}{48\pi} = \frac{\pi c}{12}\,T_H^2\,} \tag{65.211}$$

the very same number as Eq. (65.199) and Eq. (65.205).

Theorem 65.34 (Hawking flux from the gravitational anomaly)**.** *The condition that the total effective action of a field on the near-horizon geometry Eq. (65.187) be diffeomorphism invariant, given that the field becomes effectively chiral (hence gravitationally anomalous) at the horizon, fixes the outgoing energy flux to* $|\Phi| = c\,\kappa^2/48\pi = \pi c T_H^2/12$. *Equivalently: the gravitational anomaly at the horizon can be cancelled only by an outgoing flux of precisely the Hawking magnitude. Black-hole radiance is the flux required for consistency of the quantum theory in the presence of a horizon.*

65.6.5 Consistent versus covariant anomalies and the horizon boundary term

The Robinson–Wilczek computation, like every anomaly calculation, must navigate the distinction between the *consistent* and *covariant* forms of the anomaly, introduced in the gauge context of Chapter 36. The distinction is not pedantry:

the two anomalies differ by a definite local term, and getting the boundary condition at the horizon to match the anomaly one is using is what makes the coefficient come out right.

The *consistent* anomaly is the one that arises as the variation of a definite effective action, $\mathcal{A}_\nu^{\text{cons}} = \delta W/\delta(\text{diffeo})$; it satisfies the Wess–Zumino consistency conditions (it is a closed cocycle, as the chiral-anomaly consistency conditions of Chapter 36 require) but the associated current/stress tensor does not transform covariantly. The *covariant* anomaly is obtained by adding a local polynomial in the connection—the Bardeen–Zumino term—to restore covariant transformation, at the cost of no longer deriving from a local action. In two dimensions the consistent gravitational anomaly is $N^r{}_t$ as written in Eq. (65.207), while the covariant one is proportional to $\partial_r R$; they differ by exactly the Bardeen–Zumino piece, and the covariant flux at the horizon $\frac{c}{96\pi}(\,2ff''-(f')^2\,)$ differs term-by-term from the consistent $\frac{c}{96\pi}(ff'' - \frac{1}{2}(f')^2)$. At the horizon f vanishes and only the $(f')^2$ pieces survive, but they do *not* coincide: the consistent expression returns $-c\kappa^2/48\pi$ while the covariant one returns $-c\kappa^2/24\pi$, a factor of two larger. The physical fluxes nevertheless agree, because the covariant flux is extracted with a different boundary normalization—the covariant horizon condition $T^r{}_t|_{\text{cov}}(r_h) = 0$ used just below—and not because the two raw horizon values of $N^r{}_t$ are equal.

The role of the boundary term Eq. (65.209) is to enforce that one uses *matching* bookkeeping at the two ends. Robinson and Wilczek used the consistent anomaly and an effective-action boundary term; the cleaner covariant formulation of Banerjee and Kulkarni [44] imposes directly the covariant condition $T^r{}_t|_{\text{cov}}(r_h) = 0$—the statement that a freely-falling observer crossing the horizon measures no net energy flux *there*, the same physical content as the Christensen–Fulling regularity condition Eq. (65.198) that set t_{uu}. The two boundary conditions, $T_{uu} \to 0$ in the trace-anomaly language and $T^r{}_t|_{\text{cov}}(r_h) = 0$ in the gravitational-anomaly language, are the same demand: the Unruh state is regular at the future horizon. With that single physical input, the consistent and covariant routes, and the trace and gravitational anomalies, all return $c\kappa^2/48\pi$.

Caution

It is tempting to read the gravitational-anomaly derivation as saying the radiation "originates" in the thin shell $\mathcal{H}$ or that the anomaly is "produced" at $r_h + \varepsilon$. Neither is right. The shell and its boundary are bookkeeping devices; nothing physical depends on ε, which drops out as $\varepsilon \to 0$. What is physical is that a horizon makes the effective theory chiral, that chirality is anomalous, and that the only flux consistent with diffeomorphism invariance is the Hawking flux. The localization is a feature of the *argument*, not of the radiation, which—as the Bogoliubov computation shows—is emitted into the entire exterior.

65.6.6 Three derivations, one number

We have now computed the Hawking emission three genuinely independent ways, and it is worth stating plainly that they agree, because that agreement is the payoff the whole chapter was built toward. The Bogoliubov/ray-tracing computation of Section 65.2.1 tracked the in-vacuum modes across the collapsing geometry and read off a Planck spectrum at $T_H = \kappa/2\pi$, with luminosity Eq. (65.105). The Euclidean path integral of Section 65.5 found the same T_H as the unique smooth period of imaginary time, with entropy $A/4G$ as the on-shell action's boundary term. And the anomaly arguments of this section fixed the outgoing flux to $c\kappa^2/48\pi = \pi c T_H^2/12$ from horizon data alone—three times over, from the trace anomaly Eq. (65.199), from the Schwarzian transformation of the conformal stress tensor Eq. (65.205), and from the gravitational anomaly of the chiral horizon theory Eq. (65.211), each landing on the identical number that Theorem 65.33 confirmed is the flux of a two-dimensional Planck spectrum at T_H.

These are not three rewordings of one calculation. The Bogoliubov route is canonical quantization on a time-dependent background (Part III's apparatus); the Euclidean route is the gravitational path integral and its saddle points (Part III and Chapter 10's e^{-S_E} pillar); the anomaly route is the one-loop breakdown of conformal and diffeomorphism invariance (Parts X and XIV, the chiral anomaly of Chapter 36 and the conformal stress tensor of Chapters 47 and 48). The canonical, path-integral, and anomaly languages that this book developed in separate parts, for separate purposes, all see the one physical fact: a horizon with surface gravity κ radiates a thermal flux at temperature $\kappa/2\pi$. That a single phenomenon is forced by three unrelated formalisms is the strongest evidence we have that it is a property of the theory and not of any one method of asking.

Physical Insight

The anomaly derivation carries a conceptual sharpening the other two lack. It shows the flux is fixed by *local horizon data*: the surface gravity κ and the central charge c, evaluated in an arbitrarily thin neighbourhood of the horizon. No information about the asymptotic region, the collapse history, or the global structure enters. The radiation is, in this precise sense, inflowing from the anomaly at the horizon. This locality is both reassuring and unsettling: reassuring because it means the result is robust against the details of how the black hole formed, unsettling because it pins the origin of every emitted quantum to the immediate vicinity of the horizon, where, as the next section shows, the modes carrying that energy must have been blueshifted past the Planck scale.

That last point is the bridge to what follows. The anomaly computation tells us the late-time flux is set entirely by horizon-local physics, and the Kruskal

map Eq. (65.203) that supplied the Schwarzian is the same exponential relation $U \propto -e^{-\kappa u}$ that traces a late-time outgoing quantum back to a frequency growing without bound at the horizon. The derivations we have assembled thus all rest, one way or another, on the behaviour of arbitrarily trans-Planckian modes in the near-horizon region. Whether the thermal result genuinely depends on that trans-Planckian input, or is protected against it, is the conceptual worry that Section 65.7 confronts head-on.

65.7 The Trans-Planckian Problem and the Robustness of the Effect

Every derivation of the Hawking flux assembled so far has delivered the same verdict and located it in the same place. The Bogoliubov computation of Section 65.2.1 read the thermal occupation off the modes that peel away from the collapsing surface an instant before the horizon forms; the greybody analysis of Section 65.3 fed those quanta in at the horizon end of the potential barrier; the Euclidean saddle of Section 65.5 was fixed entirely by the geometry of the Euclidean horizon; and the anomaly calculation of the previous section, Section 65.6, made the locality utterly explicit — the flux is determined by the trace and gravitational anomalies evaluated in an arbitrarily thin shell straddling $r = 2GM$, with no reference to the asymptotic region at all. The emission is horizon-local. That is the strength of the result and, read the other way, its embarrassment. If the quantum that reaches a distant telescope today was created at the horizon, and if the horizon redshifts modes without bound, then the quantum was born with a frequency that, traced back far enough, exceeds any scale we are entitled to trust — the Planck scale included. The thermal spectrum appears to be computed by leaning on field modes whose wavelength near the horizon is shorter than the Planck length, deep in a regime where the very framework of quantum fields on a smooth classical metric has no right to hold.

This section confronts that worry, which is the *trans-Planckian problem*, and then dismantles it. We first make the reservoir of super-Planckian frequencies quantitative, turning the logarithmic ray-tracing map of Section 65.2.1 into an explicit exponential blueshift $\omega_{\text{local}} \sim \omega\, e^{\kappa u}$ and showing that essentially all the radiation a real black hole emits traces back to trans-Planckian modes. We then assemble four independent lines of evidence that the Planck spectrum is nonetheless robust: a near-horizon adiabatic argument that pins the flux to the single number κ; explicit modified-dispersion calculations in which the ultraviolet behaviour of the field is deformed by hand and the spectrum survives; Unruh's acoustic "dumb holes," a physical system with a genuine horizon, a *known* short-distance completion, and a measured Hawking-like emission; and finally the Parikh–Wilczek tunneling derivation, a self-contained WKB computation across

the horizon in horizon-regular coordinates that incorporates the back-reaction of the emitted quantum on the geometry. The last of these does more than re-confirm T_H: its emission rate is $\Gamma \sim e^{\Delta S_{\rm BH}}$, an exact exponential in the change of the Bekenstein–Hawking entropy whose leading term is thermal and whose first correction is genuinely non-thermal — the first hint inside the semiclassical calculation itself that the radiation carries correlations, and so the technical doorway to the information problem of Section 65.8.

65.7.1 The trans-Planckian reservoir made quantitative

The origin of the difficulty is the geometric-optics map between the affine parameter on the horizon-generating null ray and the retarded time $u = t - r_*$ in which an asymptotic observer clocks the outgoing radiation. That map was the engine of Section 65.2.1: an outgoing mode that reaches infinity as the clean plane wave $e^{-i\omega u}$ is, when propagated backwards through the collapse and out to where it last sat near the forming horizon, compressed onto a Kruskal-type affine coordinate U by the logarithmic relation

$$U \;=\; -\frac{1}{\kappa}\, e^{-\kappa u}, \qquad u \;=\; -\frac{1}{\kappa}\,\ln(-\kappa U), \tag{65.212}$$

where $\kappa = 1/4GM$ is the surface gravity of Section 65.4. This single relation is the source of *both* the thermal spectrum and the trans-Planckian problem, and it pays to see the two faces together. Substituting Eq. (65.212) into the outgoing phase gives

$$e^{-i\omega u} \;=\; e^{\,(i\omega/\kappa)\ln(-\kappa U)} \;=\; (-\kappa U)^{\,i\omega/\kappa}, \qquad U<0, \tag{65.213}$$

a function whose oscillations accumulate without bound as $U \to 0^-$, i.e. as the ray is traced to the horizon. It is exactly the branch-cut structure of this power — the factor $e^{-\pi\omega/\kappa}$ relating its analytic continuation around $U=0$ from negative to positive U — that produced the Planck factor $(e^{2\pi\omega/\kappa}-1)^{-1} = (e^{\omega/T_H}-1)^{-1}$ in Section 65.2. The thermality is the infinite pile-up of phase near the horizon, made finite by analytic continuation.

But the same pile-up is the reservoir of dangerous frequencies. Define the *local* frequency of the mode Eq. (65.213) as the rate at which its phase advances per unit affine parameter,

$$\omega_U \;\equiv\; \left|\frac{\mathrm{d}}{\mathrm{d}U}\arg\!\Big[(-\kappa U)^{i\omega/\kappa}\Big]\right| \;=\; \frac{\omega}{\kappa}\left|\frac{\mathrm{d}\ln(-\kappa U)}{\mathrm{d}U}\right| \;=\; \frac{\omega}{\kappa\,|U|}. \tag{65.214}$$

The affine parameter U measures proper coordinates for a regular observer crossing the horizon, so ω_U is the frequency such an observer assigns to the wave. Inserting the map Eq. (65.212), $|U| = \kappa^{-1}e^{-\kappa u}$, gives the central estimate of this subsection,

$$\boxed{\;\omega_{\rm local}(u) \;=\; \frac{\omega}{\kappa\,|U|} \;=\; \omega\, e^{\,\kappa u}\;.\;} \tag{65.215}$$

A quantum collected at infinity at retarded time u, with the modest asymptotic frequency $\omega \sim T_H$, was a wave of *local* frequency $\omega\, e^{\kappa u}$ when it last hovered near the horizon. The blueshift is exponential in u with rate κ: each interval $\Delta u = \kappa^{-1} = 4GM$ of late-time emission reaches back to local frequencies larger by a factor e.

How quickly does this run past the Planck scale? Setting $\omega_{\rm local} = M_{\rm Pl}$ in Eq. (65.215) with the representative $\omega \sim T_H$ gives the threshold retarded time

$$\kappa\, u_{\rm Pl} \;=\; \ln\frac{M_{\rm Pl}}{\omega} \;\sim\; \ln\frac{M_{\rm Pl}}{T_H}. \tag{65.216}$$

For a solar-mass hole the Hawking temperature is $T_H \sim 10^{-7}\,\mathrm{K} \sim 10^{-11}\,\mathrm{eV}$ (Section 65.3; recall $1\,\mathrm{K} \sim 10^{-4}\,\mathrm{eV}$), while $M_{\rm Pl} \sim 10^{19}\,\mathrm{GeV} \sim 10^{28}\,\mathrm{eV}$, so $M_{\rm Pl}/T_H \sim 10^{39}$ and

$$\kappa\, u_{\rm Pl} \;\sim\; \ln 10^{39} \;\approx\; 90. \tag{65.217}$$

After only about ninety e-folds of κu — that is, after a retarded time $u_{\rm Pl} \sim 90/\kappa \sim 360\,GM$, a few hundred light-crossing times of the hole, utterly negligible against the evaporation lifetime $\tau \sim G^2M^3 \sim 10^{67}\,\mathrm{yr}$ of Eq. (65.113) — every Hawking quantum arriving at the telescope is traced back to a mode that was trans-Planckian at the horizon. The reservoir is not a tail of the spectrum or a rare event in the deep future; it is the provenance of essentially *all* the radiation a real black hole ever emits. This is the precise content of the trans-Planckian problem (Fig. 65.7(a)), first sharpened by Jacobson and Unruh [280, 479].

Caution: what is and is not in trouble

The blueshift Eq. (65.215) is a statement about a particular mode *decomposition*, not about any locally measurable curvature. A freely falling observer crossing a large black hole's horizon sees nothing singular: the curvature there is $\sim 1/(GM)^2 \ll M_{\rm Pl}^2$, the local vacuum is indistinguishable from Minkowski (the equivalence principle), and no detector registers a Planckian energy density. What is trans-Planckian is the *frequency assigned to an individual outgoing Hawking mode when its global plane-wave form at infinity is referred back to a near-horizon frame.* The worry is therefore not that physics is visibly breaking down at the horizon — it is that the derivation *appears* to require trustworthy field theory for modes of arbitrarily short wavelength. The robustness arguments below show that the answer does not, in fact, depend on that untrustworthy input.

65.7.2 Robustness I: the near-horizon adiabatic argument

The first and most general reason to expect the spectrum to survive is that the ingredients which actually fix it are all infrared. Strip the derivation to

its load-bearing core. The thermal factor came from the analytic structure of Eq. (65.213), which is controlled by exactly one quantity, the surface gravity κ in the exponent of the log map Eq. (65.212). And κ is a property of the classical geometry at the horizon — a first derivative of the metric function, $\kappa = \frac{1}{2}f'(r_h)$ (Section 65.5.1) — evaluated on length scales of order GM, enormously larger than the Planck length for any astrophysical hole. Nothing in the determination of κ knows about Planckian physics.

This is made sharp by the Unruh effect of Theorem 64.20. A static observer hovering at small proper distance ℓ outside the horizon has proper acceleration $a = \kappa/\sqrt{f} \to \infty$ as $\ell \to 0$, but the *locally measured* temperature is set by the local acceleration, and the relation between the near-horizon accelerated frame and the asymptotic frame is precisely the relation between a Rindler wedge and Minkowski space. The near-horizon geometry of any non-extremal horizon is, to leading order in ℓ, the Rindler metric $\mathrm{d}s^2 = \kappa^2\ell^2\,\mathrm{d}t^2 - \mathrm{d}\ell^2 - \dots$ (the Euclidean version of which was the smooth cigar of Fig. 65.5). On that wedge the local vacuum seen by a freely falling observer is the Minkowski vacuum, and a static (uniformly accelerated) detector reads the Unruh temperature $T = a/2\pi$; redshifting that local temperature out to infinity by the Tolman factor $\sqrt{f}$ converts $a = \kappa/\sqrt{f}$ into

$$T_\infty \;=\; \sqrt{f}\,\frac{a}{2\pi} \;=\; \sqrt{f}\,\frac{\kappa/\sqrt{f}}{2\pi} \;=\; \frac{\kappa}{2\pi} \;=\; T_H, \tag{65.218}$$

the Hawking temperature, recovered from the *local, kinematic* Unruh result together with the gravitational redshift. The point for robustness is that Eq. (65.218) is a statement about a freely falling vacuum near a Rindler horizon, and the Unruh effect (Theorem 64.20) holds for any quantum field whose two-point function reduces to the standard form at *short proper-distance separation along the trajectory*. Whatever the true ultraviolet completion of the field — a lattice, a nonlocal smearing, a string — as long as it reproduces ordinary local physics for a freely falling observer at sub-Planckian curvatures (which the equivalence principle demands), it reproduces the Rindler two-point function and hence the temperature $T_H = \kappa/2\pi$. The thermality is an infrared fact protected by the equivalence principle; the trans-Planckian modes appear only when one insists on the global plane-wave labelling, which is a choice, not a physical input. We now verify this expectation in models where the ultraviolet is deformed explicitly.

65.7.3 Robustness II: modified dispersion relations

The cleanest way to test whether the trans-Planckian modes matter is to break the relativistic dispersion relation at high momentum on purpose and ask whether the spectrum changes. Replace the massless wave equation, whose dispersion in a local freely falling frame is $\omega^2 = k^2$, by a deformed relation that bends over at

a fixed momentum scale Λ (which we may think of as $M_{\rm Pl}$),

$$\omega^2 \;=\; k^2 \;\pm\; \frac{k^4}{\Lambda^2} \;+\; \dots, \tag{65.219}$$

the lower sign *subluminal* (group velocity falling below 1 at large k, as for phonons in a Bogoliubov superfluid) and the upper sign *superluminal* (group velocity exceeding 1), plotted in Fig. 65.7(b). Such a term is the leading Lorentz-violating correction one could add; it reduces to the standard theory for $k \ll \Lambda$ and regulates the would-be trans-Planckian oscillations, because a mode can no longer be blueshifted to arbitrary frequency at fixed group structure — its dispersion saturates or turns over near Λ. The decisive question is what Eq. (65.219) does to the late-time flux.

The analysis is a WKB problem in the near-horizon region, exactly the machinery of Section 64.3.2. Write the field in the preferred (free-fall) frame in which the dispersion Eq. (65.219) is local; near the horizon the metric is Rindler, the redshift drives a mode of fixed asymptotic frequency ω to ever-larger local wavenumber k, and the wave equation becomes a higher-order ordinary differential equation in k-space because of the k^4 term. Corley and Jacobson [137], building on Unruh [479] and Brout–Massar–Parentani–Spindel [95], solved this system. The result is the following, which we state as the operative theorem of the subsection.

Theorem 65.35 (Survival of the Planck spectrum under UV deformation)**.** *Let a free field propagate on the collapse background with the modified dispersion relation Eq. (65.219) in the freely-falling frame, with deformation scale Λ. Provided*

(i) *the dispersion is relativistic in the infrared, $\omega^2 \to k^2$ for $k \ll \Lambda$, and*

(ii) *the field is prepared in the state that is the local free-fall vacuum at short distances near the horizon (the adiabatic/Unruh condition),*

then the late-time outgoing flux at infinity is Planckian at the temperature $T_H = \kappa/2\pi$, with relative corrections that vanish as $\Lambda \to \infty$ and are bounded by a positive power of the small ratio T_H/Λ,

$$\frac{\mathrm{d}N}{\mathrm{d}t\,\mathrm{d}\omega} = \frac{1}{2\pi}\,\frac{\Gamma_{\omega\ell}}{e^{\omega/T_H}-1}\Big[1 + O\big((T_H/\Lambda)^p\big)\Big], \qquad p>0. \tag{65.220}$$

Argument. The complete calculation is a connection-formula analysis of the higher-order WKB equation and is carried in the cited works; we give the structure honestly and identify why the correction is suppressed, which is the load-bearing point.

In the near-horizon Rindler region the redshift means that a wavepacket of fixed asymptotic frequency ω is, at proper distance ℓ, composed of local

wavenumbers $k \sim \omega/(\kappa\ell)$, growing without bound as $\ell \to 0$ in the relativistic theory — this *is* the blueshift Eq. (65.215) in position-space form. With the dispersion Eq. (65.219), the WKB phase $\int^{\ell} k(\ell')\,\mathrm{d}\ell'$ governing the mode is modified once k reaches Λ, i.e. once ℓ falls to $\ell_\Lambda \sim \omega/(\kappa\Lambda)$. The created-pair amplitude is the analytic continuation of the WKB solution across the turning structure, and the thermal factor $e^{-\pi\omega/\kappa}$ arises from the *same* contour rotation as in the relativistic case (Section 64.3.2), because in the region $\ell_\Lambda \ll \ell \ll \kappa^{-1}$ the dispersion is still relativistic and the metric is still Rindler; the pile-up of phase that produces the Boltzmann factor occurs entirely in this intermediate window, *above* the deformation scale in length and *below* it in momentum. The deformation acts only at $\ell \lesssim \ell_\Lambda$, where it removes the unphysical infinite-frequency tail and replaces it with a finite, regular mode reservoir (for the subluminal sign, modes "bounce" off the dispersion and never reach $k = \infty$; for the superluminal sign they are emitted from the singularity-free interior region). Either way the leading exponential is set by the IR window and equals $e^{-\pi\omega/\kappa}$; the correction is controlled by how much of the phase integral is contaminated by the deformed region, which is the fraction $\ell_\Lambda/\kappa^{-1} \sim \omega/\Lambda \sim T_H/\Lambda$ raised to a positive power fixed by the order of the dispersion. Hence Eq. (65.220), with p depending on the deformation (for the quartic relation Eq. (65.219) one finds the spectrum thermal to all orders in T_H/Λ for the smooth-passage state, with only nonperturbative $e^{-\Lambda/T_H}$ corrections). □

The lesson of Theorem 65.35 is that the Planck spectrum is manufactured in the infrared window between the deformation scale and the curvature scale, a window that is parametrically large for any $\Lambda \gg T_H$ — and $\Lambda \sim M_{\mathrm{Pl}}$ versus $T_H \sim 10^{-7}\,\mathrm{K}$ is large indeed. The trans-Planckian modes can be excised, replaced, or capped, and the only trace they leave is a correction smaller than any power of T_H/Λ. The effect does not depend on the ultraviolet physics it appeared to invoke. There is a price — the calculation now assumes a preferred free-fall frame and a specific near-horizon "adiabatic" choice of vacuum, which is the modern statement of where the physical input really lives — but the robustness is genuine.

65.7.4 Robustness III: analog horizons and Unruh's dumb holes

The modified-dispersion analysis answers the worry within field theory, but it leaves a residual unease: we deformed the ultraviolet by hand, choosing the deformation and the vacuum. Is there a physical system with a *genuine* horizon, a *known* short-distance completion that we did not get to choose, and a Hawking-like emission we can compute — and ideally measure — from first principles? Unruh found one in 1981 [478]: a moving fluid.

The observation is that sound in a barotropic, irrotational, inviscid fluid does not propagate on the laboratory metric; it propagates on an *acoustic metric* built

from the flow. Linearize the Euler and continuity equations about a background flow with density ρ, velocity $\mathbf{v}$, and local sound speed c_s; the velocity-potential perturbation ϕ_1 (with $\mathbf{v}_1 = \nabla\phi_1$) obeys a wave equation that can be written in exactly the covariant Klein–Gordon form of Eq. (65.72),

$$\frac{1}{\sqrt{-g}}\,\partial_\mu\!\left(\sqrt{-g}\,g^{\mu\nu}\partial_\nu\phi_1\right) = 0, \qquad g_{\mu\nu} = \frac{\rho}{c_s}\begin{pmatrix} -(c_s^2 - v^2) & -\mathbf{v}^{\mathsf{T}} \\ -\mathbf{v} & \mathbf{1}_{3\times 3} \end{pmatrix}, \tag{65.221}$$

the *acoustic metric*. (We have written Eq. (65.221) in the mostly-plus convention standard in the analog-gravity literature for transparency; the physics is convention-independent and we flip to mostly-minus when we quote the temperature.) Phonons are the quanta of ϕ_1, and they live on the effective geometry Eq. (65.221) exactly as our scalar lives on Schwarzschild. The crucial feature is the component $g_{tt} \propto -(c_s^2 - v^2)$: wherever the flow speed $v = |\mathbf{v}|$ exceeds the local sound speed c_s, this changes sign. The surface where $v = c_s$, separating a subsonic exterior from a supersonic interior, is an *acoustic horizon*: sound generated inside cannot swim upstream against a flow faster than itself, just as light cannot escape a black hole. Unruh named these *dumb holes* — dumb in the sense of mute, since no sound emerges.

Repeating the surface-gravity computation on Eq. (65.221), the acoustic surface gravity is the gradient of the flow's normal velocity through the sonic point,

$$\kappa_{\mathrm{ac}} = \frac{1}{2}\,\frac{\mathrm{d}}{\mathrm{d}n}(c_s^2 - v^2)\Big|_{v=c_s} \xrightarrow[c_s \text{ const}]{} c_s\,\frac{\mathrm{d}v_\perp}{\mathrm{d}n}\Big|_{\mathrm{horizon}}, \tag{65.222}$$

and the analog Hawking temperature, restoring $\hbar$ and k_B because this is a laboratory estimate, is

$$\boxed{k_B\,T_{\mathrm{ac}} = \frac{\hbar\,\kappa_{\mathrm{ac}}}{2\pi} = \frac{\hbar}{2\pi}\,\frac{\mathrm{d}v_\perp}{\mathrm{d}n}\Big|_{\mathrm{horizon}}}, \tag{65.223}$$

the identical $\kappa/2\pi$ law, now for phonons streaming out of the subsonic region. The analog system makes vivid everything the robustness arguments claimed. Its "Lorentz invariance" is only emergent: the linear phonon dispersion $\omega = c_s k$ is the long-wavelength limit of a microscopic dispersion that, for a dilute Bose–Einstein condensate, is the Bogoliubov form

$$\omega^2 = c_s^2 k^2\left(1 + \xi^2 k^2\right), \qquad \xi = \frac{\hbar}{2mc_s}, \tag{65.224}$$

with ξ the healing length — precisely the *superluminal* deformation Eq. (65.219) with the known scale $\Lambda = 1/\xi$ set by the atomic physics, not chosen by us. Here there is no trans-Planckian mystery at all: the would-be infinitely blueshifted modes simply run into the atomic granularity of the condensate at $k \sim 1/\xi$, the theory is manifestly finite and complete below that scale, and one can ask without ambiguity whether the thermal phonon flux survives. It does. Numerical

and analytic treatments of phonon emission across a condensate sonic horizon reproduce the temperature Eq. (65.223) with corrections suppressed by powers of $T_{\rm ac}\xi/c_s$, the analog of T_H/Λ in Eq. (65.220) — a direct, controlled confirmation of Theorem 65.35 in a system whose ultraviolet is known atom by atom [46].

Physical Insight: the horizon does the work, not the ultraviolet

The analog black hole isolates which part of Hawking's argument is physics and which is artefact. The acoustic horizon is real: it is a definite surface in a real fluid, defined with no reference to gravity or to Planckian physics. The short-distance theory is real and known: it is the Schrödinger equation for cold atoms, finite and unitary. And the thermal phonon flux is real — it has been measured. What the analog system *lacks* is any dynamical gravity, any Einstein equation, any back-reaction of the phonons on the spacetime. The persistence of the temperature $\kappa/2\pi$ in a system stripped of all gravitational content shows that the Hawking spectrum is a kinematic consequence of quantum fields in the presence of a horizon, robust against the ultraviolet, and not a delicate feature of quantum gravity. Gravity supplies the horizon; the rest is quantum field theory on it.

This is no longer a thought experiment. In a series of experiments culminating in 2016, Steinhauer realized a sonic horizon in a trapped Bose–Einstein condensate, detected the spontaneous emission of phonon pairs across it, and measured their density–density correlations, finding the Hawking partners to be entangled [455]; later runs reconstructed an emission spectrum consistent with the thermal form Eq. (65.223). As of the mid-2020s the analog Hawking effect, including the entanglement of the emitted pairs, is an established laboratory phenomenon. It does not test Hawking radiation from a *gravitational* horizon — nothing can, at $T_H \sim 10^{-7}\,\mathrm{K}$ below the cosmic microwave background — but it tests the field-theoretic mechanism on which the gravitational prediction rests, and the mechanism passes.

65.7.5 Robustness IV: Hawking radiation as tunneling with back-reaction

The arguments so far establish that the spectrum is insensitive to the ultraviolet. The final derivation establishes something complementary: that the emission can be computed as a self-contained semiclassical tunneling process, across the horizon, in coordinates that are perfectly regular there — so that no trans-Planckian mode and no asymptotic mode-matching is invoked at all — and that when one is careful enough to conserve energy, building the back-reaction of the emitted quantum into the geometry, the result is a sharp correction to the thermal spectrum. This is the tunneling method of Parikh and Wilczek [377].

Painlevé–Gullstrand coordinates. The Schwarzschild coordinates of Eq. (65.1) are singular at the horizon (the $\mathrm{d}r^2$ term blows up), which is precisely why the naive mode picture there required arbitrarily blueshifted modes. The cure is to pass to coordinates that are regular across the horizon. Define a new time

$$t_p = t + \int^r \frac{\sqrt{2GM/r'}}{f(r')}\,\mathrm{d}r', \qquad f(r) = 1 - \frac{2GM}{r}, \tag{65.225}$$

the proper time of an observer who falls in radially from rest at infinity. Differentiating, $\mathrm{d}t = \mathrm{d}t_p - \sqrt{2GM/r}\, f^{-1}\mathrm{d}r$, and substituting into Eq. (65.1) the cross terms conspire to remove the coordinate singularity. A short computation gives the Painlevé–Gullstrand form

$$\mathrm{d}s^2 = \Big(1 - \frac{2GM}{r}\Big)\mathrm{d}t_p^2 - 2\sqrt{\frac{2GM}{r}}\,\mathrm{d}t_p\,\mathrm{d}r - \mathrm{d}r^2 - r^2\,\mathrm{d}\Omega^2, \tag{65.226}$$

in the mostly-minus signature. Every metric component is finite and smooth at $r = 2GM$; the determinant is $\sqrt{-g} = r^2\sin\theta$ as in flat space; constant-t_p slices are flat Euclidean space. There is no pathology at the horizon to require trans-Planckian modes.

Radial null rays. A radial light ray has $\mathrm{d}\Omega = 0$ and $\mathrm{d}s^2 = 0$. Dividing Eq. (65.226) by $\mathrm{d}t_p^2$ and writing $\dot r = \mathrm{d}r/\mathrm{d}t_p$,

$$\Big(1 - \frac{2GM}{r}\Big) - 2\sqrt{\frac{2GM}{r}}\,\dot r - \dot r^2 = 0, \tag{65.227}$$

a quadratic for $\dot r$ with solutions

$$\dot r = -\sqrt{\frac{2GM}{r}} \pm 1. \tag{65.228}$$

The upper sign is the outgoing ray, $\dot r = 1 - \sqrt{2GM/r}$, which vanishes at $r = 2GM$ (light marches in place at the horizon) and is positive outside; this is the trajectory along which a Hawking quantum must escape.

The WKB tunneling amplitude. We treat the emitted quantum as a massless shell of energy ω following the outgoing null ray Eq. (65.228) and compute, by the semiclassical method of Section 64.3.2, the probability that it materializes just inside the horizon and tunnels out. In WKB the emission amplitude is e^{iS} with the classical action $S = \int p_r\,\mathrm{d}r$ evaluated along the (here complex) trajectory, and the emission *probability* is

$$\Gamma \;\sim\; \left|e^{iS}\right|^2 \;=\; e^{-2\,\mathrm{Im}\,S}, \qquad \mathrm{Im}\,S = \mathrm{Im}\int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}} p_r\,\mathrm{d}r. \tag{65.229}$$

The tunneling is forbidden classically precisely because the would-be trajectory must cross the horizon, where $\dot r = 0$ for the outgoing ray; the action picks up an imaginary part from the resulting singularity, exactly as a one-dimensional barrier produces an imaginary momentum (Section 64.3.2). The decisive refinement, and the reason this derivation says something new, is to enforce energy conservation. The total ADM mass of the spacetime is fixed; when the shell carries off energy ω', the mass interior to it is no longer M but $M-\omega'$, and the geometry the shell traverses is the Painlevé–Gullstrand metric Eq. (65.226) with $M \to M-\omega'$. The shell moves in the field of the *shrinking* hole; this is the self-gravitation, or back-reaction, that the fixed-background derivations omitted.

Theorem 65.36 (Parikh–Wilczek emission rate)**.** *A massless shell of energy ω emitted from a Schwarzschild black hole of mass M, treated as WKB tunneling across the horizon in the self-gravitating Painlevé–Gullstrand geometry, has emission probability*

$$\boxed{\Gamma \;\sim\; \exp\!\Big[-8\pi G M\,\omega\Big(1-\frac{\omega}{2M}\Big)\Big] \;=\; \exp\!\big(\Delta S_{\mathrm{BH}}\big),} \tag{65.230}$$

where $\Delta S_{\mathrm{BH}} = S_{\mathrm{BH}}(M-\omega) - S_{\mathrm{BH}}(M) < 0$ is the change in the Bekenstein–Hawking entropy Eq. (65.179) of the hole.

Worked Example 65.37 (Imaginary part of the action across the horizon)**.** We compute $\operatorname{Im} S$ of Eq. (65.229) in full. Write the action as the integral of the radial momentum, then trade the momentum integral for an energy integral using Hamilton's equation $\dot r = \partial H/\partial p_r$ at fixed r:

$$\operatorname{Im} S = \operatorname{Im}\int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}} p_r\,\mathrm{d}r = \operatorname{Im}\int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}}\int_0^{p_r} \mathrm{d}p_r'\,\mathrm{d}r = \operatorname{Im}\int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}}\int \frac{\mathrm{d}H}{\dot r}\,\mathrm{d}r. \tag{65.231}$$

As the shell builds up its momentum from 0 to p_r, the Hamiltonian (the energy of the hole-plus-shell system that remains behind the shell) decreases from M to $M-\omega$: with the shell energy ω' running from 0 to ω, the residual mass is $M-\omega'$ and $\mathrm{d}H = -\,\mathrm{d}\omega'$. Along the outgoing ray the radial velocity is Eq. (65.228) evaluated in the geometry of the residual mass,

$$\dot r \;=\; 1-\sqrt{\frac{2G(M-\omega')}{r}}. \tag{65.232}$$

Substituting $\mathrm{d}H = -\mathrm{d}\omega'$ and exchanging the order of integration,

$$\operatorname{Im} S = \operatorname{Im}\int_0^{\omega}(-\mathrm{d}\omega')\int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}} \frac{\mathrm{d}r}{1-\sqrt{2G(M-\omega')/r}}. \tag{65.233}$$

The radial contour integral. Fix ω' and set $a \equiv 2G(M-\omega')$, the horizon radius of the residual hole. The integrand of the inner integral has a simple pole

where the outgoing ray stalls, $r = a$, i.e. exactly at the horizon the shell must cross. Expand near it: with $r = a + x$,

$$1 - \sqrt{\frac{a}{r}} = 1 - \Big(1 + \frac{x}{a}\Big)^{-1/2} = 1 - \Big(1 - \frac{x}{2a} + O(x^2)\Big) = \frac{x}{2a} + O(x^2), \quad (65.234)$$

so near the pole

$$\frac{1}{1 - \sqrt{a/r}} \;\simeq\; \frac{2a}{r - a}, \qquad \mathrm{Res}_{r=a}\, \frac{1}{1 - \sqrt{a/r}} = 2a = 4G(M - \omega'). \quad (65.235)$$

The pole sits on the path of integration, and the correct prescription comes from the Feynman $i\epsilon$ of Section 5.3: the emitted energy carries a small negative imaginary part ($\omega' \to \omega' - i\epsilon$, the retarded/outgoing choice), which displaces the pole off the real r-axis so that the contour passes below it. The integral then equals its principal value (real) plus a clockwise half-residue, contributing the imaginary part

$$\mathrm{Im} \int_{r_{\mathrm{in}}}^{r_{\mathrm{out}}} \frac{\mathrm{d}r}{1 - \sqrt{a/r}} = -\,\pi\, \mathrm{Res}_{r=a}\, \frac{1}{1 - \sqrt{a/r}} = -\,\pi\,(2a) = -\,4\pi G(M - \omega'). \quad (65.236)$$

(The clockwise sense, hence the overall minus sign, is fixed by requiring $\mathrm{Im}\, S > 0$ so that $\Gamma = e^{-2\,\mathrm{Im}\, S} < 1$: emission is suppressed, not enhanced without bound. The same sign follows directly from the $\omega' - i\epsilon$ displacement.)

The energy integral. Insert Eq. (65.236) into Eq. (65.233). The two minus signs — the $(-\mathrm{d}\omega')$ and the $-4\pi G(M - \omega')$ — combine to a plus:

$$\begin{aligned} \mathrm{Im}\, S &= \int_0^{\omega} (-\mathrm{d}\omega')\,[-4\pi G(M - \omega')] = 4\pi G \int_0^{\omega} (M - \omega')\,\mathrm{d}\omega' \\ &= 4\pi G\Big[M\omega - \tfrac{1}{2}\omega^2\Big] = 4\pi G M\,\omega\Big(1 - \frac{\omega}{2M}\Big). \end{aligned} \quad (65.237)$$

Therefore, by Eq. (65.229),

$$\Gamma \sim e^{-2\,\mathrm{Im}\, S} = \exp\Big[-8\pi G M\,\omega\Big(1 - \frac{\omega}{2M}\Big)\Big], \quad (65.238)$$

which is Eq. (65.230). It remains to recognise the exponent as the entropy change. The Bekenstein–Hawking entropy Eq. (65.179) is $S_{\mathrm{BH}}(M) = A/4G = 4\pi(2GM)^2/4G = 4\pi G M^2$, so

$$\begin{aligned} \Delta S_{\mathrm{BH}} &= S_{\mathrm{BH}}(M - \omega) - S_{\mathrm{BH}}(M) = 4\pi G\big[(M - \omega)^2 - M^2\big] \\ &= 4\pi G\big[-2M\omega + \omega^2\big] = -8\pi G M\,\omega\Big(1 - \frac{\omega}{2M}\Big), \end{aligned} \quad (65.239)$$

exactly the exponent of Eq. (65.238). Hence $\Gamma \sim e^{\Delta S_{\mathrm{BH}}}$, completing the proof of Theorem 65.36.

Two features of the computation deserve emphasis because they are the reasons it is trustworthy. First, the entire calculation took place in the Painlevé–Gullstrand chart, regular across the horizon; at no point did a mode have to be matched between a near-horizon and an asymptotic region, and at no point did an arbitrarily blueshifted frequency appear. The tunneling exponent came from a *pole*, a sharply localized contour feature at the horizon, not from an infinite reservoir of short-wavelength modes. This is the sense in which the tunneling derivation sidesteps the trans-Planckian problem rather than confronting it. Second, the use of the horizon-regular chart is not optional: a naive application of the same method in the singular Schwarzschild coordinates Eq. (65.1) produces a temperature too large by a factor of two, because those coordinates carry a spurious discontinuity in t across the horizon (the $\sqrt{2GM/r}\, f^{-1}\mathrm{d}r$ term of Eq. (65.225) that the PG chart absorbs). The temporal part of the invariant action must be accounted for; the virtue of the PG chart is that it is continuous across the horizon, so the temporal contribution vanishes and the spatial integral Eq. (65.233) gives the correct answer outright.

65.7.6 The corrected spectrum: thermal limit, energy conservation, and correlations

The result Eq. (65.230) is richer than T_H alone, and unpacking it furnishes the bridge to the next section.

The thermal limit. For a soft quantum, $\omega \ll M$, the correction term in the exponent is negligible and

$$\Gamma \xrightarrow[\omega \ll M]{} e^{-8\pi GM\omega} = e^{-\omega/T_H}, \tag{65.240}$$

using $T_H = 1/8\pi GM$ (Eq. (65.64)). The leading-order emission probability is the Boltzmann factor at the Hawking temperature: the tunneling method reproduces the thermal spectrum found four other ways in this chapter, now with no appeal to ultraviolet modes. This is the fourth independent confirmation of $T_H = \kappa/2\pi$ promised at the outset and is the leftmost, linear regime of Fig. 65.7(c).

Energy conservation fixes the correction. The exact exponent is not $-\omega/T_H$ but $-\omega/T_H\,(1-\omega/2M)$, and the deviation is precisely what conservation of energy demands. Because the hole loses mass to the quantum it emits, its temperature *rises* during the emission — recall $T_H \propto 1/M$ and the negative heat capacity of Theorem 65.30. A fixed-background calculation, which holds M and hence T_H frozen, cannot see this; it can only ever yield the strict Boltzmann factor Eq. (65.240). The back-reaction term $+\omega^2/2M$ in Eq. (65.237) is the integrated effect of the temperature climbing as the mass falls, and it makes energetic quanta *more* likely to be emitted than a fixed-temperature thermal body

would allow (the exponent is less negative; the shaded region of Fig. 65.7(c)). The cleanest statement of this is the entropy form $\Gamma \sim e^{\Delta S_{\rm BH}}$: in a microcanonical reading, the probability of a transition that lowers the mass by ω is the ratio of the number of final to initial microstates, $e^{S_{\rm BH}(M-\omega)}/e^{S_{\rm BH}(M)} = e^{\Delta S_{\rm BH}}$. Energy conservation plus the area–entropy law Eq. (65.179) *is* the corrected spectrum; nothing further need be assumed.

Correlations, and the doorway to information. A strictly thermal spectrum is structureless: successive quanta are emitted independently, and the radiation carries no information beyond its temperature. The corrected rate is not structureless. Consider emitting two quanta, of energies ω_1 and ω_2. If they were independent, the joint probability would factorize, $\Gamma(\omega_1)\,\Gamma(\omega_2)$. Instead, energy conservation ties them: emitting ω_1 first leaves a hole of mass $M-\omega_1$, so the second emission is governed by $\Delta S_{\rm BH}$ evaluated for that smaller hole, and

$$\Gamma(\omega_1,\omega_2) \sim e^{\Delta S_{\rm BH}(M\to M-\omega_1-\omega_2)} = \exp\!\big[\Delta S_{\rm BH}(\omega_1\,|\,M) + \Delta S_{\rm BH}(\omega_2\,|\,M-\omega_1)\big], \tag{65.241}$$

which does *not* equal $\Gamma(\omega_1)\,\Gamma(\omega_2)$ computed with both referred to the original mass M. The difference is a genuine correlation between the emitted quanta, with mutual information set by the same ω^2 term. The total exponent for emitting a collection of quanta summing to energy E is $\Delta S_{\rm BH}(M \to M-E)$, the full change in horizon entropy, independent of the order or partition — a statement that the radiation, taken as a whole, knows about the total entropy lost by the hole. The single-quantum spectrum still *looks* thermal to leading order Eq. (65.240), which is why the thermal description is so accurate; but the exact rate Eq. (65.230) carries correlations of order ω/M per quantum, hidden beneath the thermal envelope.

Whether these correlations are sufficient to carry off the full information content of the collapsed matter — whether they encode it, or merely demonstrate that the strict thermality is an approximation that must eventually break — is exactly the question that Eq. (65.230) cannot by itself answer, because it is a leading-saddle result that knows about energy conservation but not about the detailed quantum state. That question, quantified by the entanglement entropy of the radiation and the Page-curve requirement that it must eventually turn over if evolution is unitary, is the subject of the final section. The Parikh–Wilczek calculation has shown that the radiation is not exactly thermal and does carry correlations; the information problem asks whether those correlations are the right ones.

With the spectrum shown to be robust against the ultraviolet — protected by the infrared kinematics of the horizon, confirmed under deformed dispersion, realized and measured in analog systems, and re-derived by horizon-regular tunneling — and with that same tunneling calculation revealing a small but genuine non-thermal correlation between the emitted quanta, we have the two

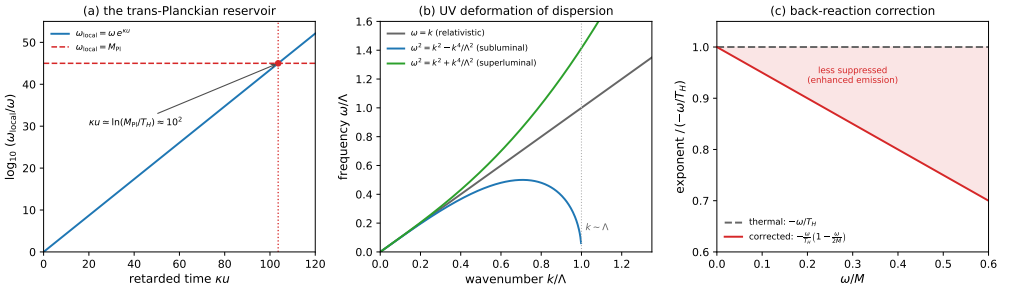


Figure 65.7: The trans-Planckian reservoir and the robustness of the Hawking effect. **(a)** The exponential blueshift Eq. (65.215), $\omega_{\text{local}} = \omega\, e^{\kappa u}$: a quantum received at infinity at $\omega \sim T_H$ is traced back to super-Planckian local frequency after $\kappa u \sim \ln(M_{\text{Pl}}/T_H) \approx 10^2$ e-folds (dashed line, solar-mass hole), so essentially all the radiation originates in the trans-Planckian reservoir. **(b)** Ultraviolet deformations of the dispersion relation Eq. (65.219) that regulate the reservoir — subluminal ($\omega^2 = k^2 - k^4/\Lambda^2$) and superluminal ($+k^4/\Lambda^2$) — both reducing to $\omega = k$ for $k \ll \Lambda$; the Planck spectrum survives either (Theorem 65.35). **(c)** The Parikh–Wilczek exponent Eq. (65.230) relative to the naive thermal exponent $-\omega/T_H$: the back-reaction factor $(1 - \omega/2M)$ makes energetic quanta less suppressed (shaded), the non-thermal correction demanded by energy conservation.

ingredients the final section needs: confidence that the semiclassical prediction is real, and the first evidence that it is not the whole story. The next section turns the correlations into a sharp paradox, quantifies the tension through the entanglement entropy and the Page curve, and confronts the question the corrected spectrum has opened — whether a pure state collapsing into a black hole can return to a pure state in the Hawking radiation.

65.8 The Information Problem and the Frontier

The previous section met the trans-Planckian worry head-on and emerged with the thermal spectrum intact: whatever short-distance physics dresses the horizon, the late-time radiation a distant detector registers is Planckian at $T_H = \kappa/2\pi$ (Section 65.7), and the only correlations the semiclassical calculation supplies are the ones between each outgoing quantum and the partner it left behind. That last clause is where the comfort ends. The corrected, back-reacted spectrum is still, mode by mode, exactly thermal, and a thermal ensemble is a *mixed* state. We assembled, over the course of this chapter, a black hole that is a bona fide thermodynamic body: it has a temperature (Eq. (65.64)), a luminosity and a finite lifetime (Section 65.3, Eq. (65.113)), an entropy $S_{\text{BH}} = A/4G$ that ceilings how much it can hold (Section 65.4, Eq. (65.142)), and a sparse stream of quanta dribbling out one at a time (Section 65.3.7). Put these facts together and a sharp

paradox falls out. A black hole forms from a definite quantum state, a collapsing star described by some pure $|\Psi\rangle$, and over a time $\tau \sim G^2 M_0^3$ it converts itself entirely into outgoing radiation. If that radiation is exactly thermal, then a pure state has evolved into a mixed one — and no unitary evolution can do that. Either the radiation is not exactly thermal and the semiclassical computation is quietly wrong about its fine-grained structure, or quantum mechanics itself fails across a horizon. This section states that dilemma precisely, makes it quantitative with the entanglement entropy and the Page curve, surveys the candidate escapes honestly and dated, and shows how the islands/quantum-extremal-surface program of Chapter 57 — read through the holographic dictionary of Chapter 53 and the Ryu–Takayanagi formula of Chapter 56 — has, by the mid-2020s, reproduced the unitary answer in controlled models. In doing so it closes the book's long arc, the one that began with a single free scalar field.

65.8.1 The paradox stated precisely

Ordinary quantum mechanics evolves states by a unitary operator. In a scattering setting the asymptotic past and future Hilbert spaces are connected by the S-matrix $\mathbf{S}$, and unitarity $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ guarantees that a pure in-state maps to a pure out-state: $|\text{out}\rangle = \mathbf{S}\,|\text{in}\rangle$, and a density matrix evolves as $\rho_{\text{out}} = \mathbf{S}\,\rho_{\text{in}}\,\mathbf{S}^\dagger$, which preserves $\operatorname{Tr}\rho^2$ and hence purity. Unitarity is not a convenience; it is the statement that probability is conserved and that the present determines the past as well as the future. It is the backbone of every calculation in this book.

Hawking's 1976 analysis [256] confronted this backbone with the collapse geometry. The matter that forms the hole is prepared in a pure state on some early Cauchy slice. After the hole has fully evaporated, the only data left in the spacetime is the outgoing radiation, and the chapter's central computation says that radiation is thermal. To state the conclusion as an operator statement, suppose evaporation maps the initial density matrix to a final one by some linear map. Because the output is thermal and *independent of the details of the infalling pure state* — the spectrum Eq. (65.88) depends only on M and the field content, not on what fell in — this map cannot be of the unitary form $\rho \mapsto \mathbf{S}\rho\mathbf{S}^\dagger$. Hawking proposed to replace the S-matrix by a more general linear "superscattering" operator \$ acting on density matrices,

$$\rho_{\text{out}} = \$\,\rho_{\text{in}}, \qquad \$ \text{ not of the form } \mathbf{S}(\cdot)\mathbf{S}^\dagger, \tag{65.242}$$

a map that takes pure states to mixed states and therefore destroys the information about which pure state collapsed. This is the "loss of predictability": from the final radiation one cannot reconstruct the initial state, even in principle. We can state the clash as a list of mutually contradictory propositions, each of which is individually well motivated.

Theorem 65.38 (The information paradox as a clash of premises)**.** *The following four statements cannot all be true:*

(U) Unitarity. *Black-hole formation and evaporation is a unitary process in a single quantum theory: a pure initial state evolves to a pure final state.*

(E) Effective field theory near the horizon. *Physics in the neighbourhood of a large, smooth horizon is well described by quantum field theory on a curved background, so the local computation of Section 65.2 is reliable there: the outgoing quanta are produced in the thermal, partner-entangled state it predicts.*

(C) Completeness of the exterior. *After evaporation, the outgoing radiation is the entire final state; there is no large hidden reservoir (no stable remnant, no baby universe) carrying off the missing correlations.*

(S) Subexponential entropy. *A region of area A holds at most $S_{\rm BH} = A/4G$ bits, the bound established in Section 65.4; the hole's Hilbert space is finite-dimensional with* $\log\dim\mathcal{H} \approx A/4G$.

Proof. We show the four are inconsistent. By (E) the radiation emitted up to any time is, mode by mode, in the thermal state Eq. (65.88), each outgoing quantum maximally correlated with an interior partner. By (S) the interior can encode at most $S_{\rm BH}(t) = A(t)/4G$ bits, a number that *decreases* as the hole shrinks. We will compute in Section 65.8.2 that the fine-grained (von Neumann) entropy of the radiation produced under (E) grows monotonically, and in Section 65.8.3 that for a system whose two parts are the radiation and the hole, the radiation's entropy cannot exceed the smaller subsystem's capacity, here $S_{\rm BH}(t)$ once the hole is the smaller factor. After the Page time the monotone entropy of (E) exceeds the decreasing ceiling of (S): the two are in quantitative contradiction. The only way to keep (E) and (S) is to abandon (U) — to accept the \$-matrix Eq. (65.242) and the loss of purity — unless (C) fails and the missing correlations sit in a reservoir the exterior computation cannot see. Conversely, keeping (U), (C), and (S) forces (E) to break down somewhere a naive estimate says it should hold. No assignment of truth values satisfies all four. □

The value of Theorem 65.38 is that it converts a vague unease into a finite menu: every serious proposal in Section 65.8.5 is, at bottom, a choice of which premise to surrender. Before surveying that menu we must make the two quantitative claims in the proof — the monotone growth and the ceiling — precise. Both are computations the carried free scalar lets us do in full.

65.8.2 The entanglement structure of the Hawking pairs

We continue ϕ^4 theory in its free ($\lambda = 0$) sector, the real scalar quantized in Chapter 9, now in the collapse background. The Bogoliubov analysis of Section 65.2 did more than produce occupation numbers: it expressed the in-vacuum, the state with no incoming particles, as a definite, correlated state in

the out/interior basis. The structure is identical to the two-mode squeezing already met for the Unruh effect in Chapter 64: for each asymptotic out mode of frequency ω (we suppress the angular labels ℓm, which ride along untouched) there is an interior partner mode, and the in-vacuum is a two-mode squeezed state pairing them.

Proposition 65.39 (The in-vacuum as partner-entangled pairs)**.** *Let b_ω annihilate the late-time outgoing quanta at infinity and $\tilde{b}_\omega$ annihilate their interior partner modes (the modes supported behind the horizon, conjugate under the Bogoliubov transformation). The collapse in-vacuum factorizes over frequencies as a product of two-mode squeezed states,*

$$|0\rangle_{\text{in}} = \prod_{\omega} \frac{1}{\cosh r_\omega} \sum_{n=0}^{\infty} (\tanh r_\omega)^n \; |n\rangle_\omega^{\text{out}} \otimes |n\rangle_\omega^{\text{int}}, \qquad \tanh r_\omega = e^{-\pi\omega/\kappa} = e^{-\omega/2T_H}, \tag{65.243}$$

where $|n\rangle_\omega^{\text{out}}, |n\rangle_\omega^{\text{int}}$ are the n-quantum number states of the out and interior oscillators and $\kappa = 2\pi T_H$ is the surface gravity.

Proof. The Bogoliubov transformation relating in and out modes mixes creation and annihilation operators with coefficients whose ratio $|\beta_\omega/\alpha_\omega| = e^{-\pi\omega/\kappa}$ was the engine of the thermal spectrum in Section 65.2. A transformation that is linear in $b, b^\dagger$ and pairs each out mode with one partner is generated by a two-mode squeeze operator $S_\omega = \exp[r_\omega(b_\omega \tilde{b}_\omega - b_\omega^\dagger \tilde{b}_\omega^\dagger)]$, and the in-vacuum is S_ω acting on the out/interior Fock vacuum. Expanding the squeeze operator on the vacuum gives the standard identity $S_\omega |0,0\rangle = (\cosh r_\omega)^{-1} \sum_n (\tanh r_\omega)^n |n,n\rangle$, with the squeezing parameter fixed by $\tanh r_\omega = |\beta_\omega/\alpha_\omega| = e^{-\pi\omega/\kappa}$. Using $\kappa = 2\pi T_H$ rewrites the exponent as $\pi\omega/\kappa = \omega/2T_H$, which is Eq. (65.243). The normalization is automatic: $\cosh^{-2} r \sum_n \tanh^{2n} r = \cosh^{-2} r \, (1 - \tanh^2 r)^{-1} = 1$ since $1 - \tanh^2 r = \cosh^{-2} r$. □

The state Eq. (65.243) is pure — it is the global vacuum, a definite vector. The thermality is not a property of the global state but of what an observer who can access *only the exterior* sees, and seeing only the exterior means tracing out the interior partners. That partial trace is the whole content of the paradox, and it is a one-line computation worth doing in full.

Worked Example 65.40 (The reduced state of one mode is exactly thermal)**.** Fix a single frequency ω and write $r \equiv r_\omega$, $t \equiv \tanh r = e^{-\omega/2T_H}$. From Eq. (65.243) the out/interior pair is in the pure state $|\psi\rangle = \cosh^{-1} r \sum_n t^n \, |n\rangle^{\text{out}} \, |n\rangle^{\text{int}}$, whose density matrix is $\rho = |\psi\rangle\langle\psi|$. The exterior observer's state is the partial trace over the interior,

$$\rho_\omega^{\text{out}} = \text{Tr}_{\text{int}} \; |\psi\rangle\langle\psi| = \frac{1}{\cosh^2 r} \sum_{n,m} t^n t^m \, |n\rangle^{\text{out}} \langle m|^{\text{out}} \underbrace{\langle m|n\rangle^{\text{int}}}_{=\delta_{mn}} = \frac{1}{\cosh^2 r} \sum_n t^{2n} \; |n\rangle^{\text{out}} \langle n|^{\text{out}}. \tag{65.244}$$

The interior number states being orthonormal, the trace collapses the double sum to a diagonal one: *all off-diagonal coherence between number states is gone*. The exterior sees a classical mixture with weights $p_n = \cosh^{-2} r \; t^{2n} = (1-t^2)\, t^{2n}$, using $\cosh^{-2} r = 1 - \tanh^2 r = 1 - t^2$. Now substitute $t^2 = \tanh^2 r = e^{-\omega/T_H}$:

$$p_n = \left(1 - e^{-\omega/T_H}\right) e^{-n\omega/T_H}, \tag{65.245}$$

which is precisely the Boltzmann distribution of a single bosonic oscillator of energy ω at temperature T_H. Hence

$$\rho_\omega^{\text{out}} = \left(1 - e^{-\omega/T_H}\right) e^{-\omega\,\hat{n}/T_H} = \frac{e^{-\hat{H}_\omega/T_H}}{\operatorname{Tr}\, e^{-\hat{H}_\omega/T_H}}, \qquad \hat{H}_\omega = \omega\,\hat{n}, \tag{65.246}$$

a thermal density matrix in the canonical Gibbs form. The mean occupation is the Planck value,

$$\bar{n}_\omega = \sum_n n\, p_n = \frac{e^{-\omega/T_H}}{1 - e^{-\omega/T_H}} = \frac{1}{e^{\omega/T_H} - 1} = \sinh^2 r, \tag{65.247}$$

recovering the spectrum Eq. (65.88) (with greybody factor unity, the horizon value) and confirming $\bar{n} = \sinh^2 r$ from Eq. (65.243). The point is sharper than the spectrum, though: the reduced state is not merely thermal in its occupation numbers but is the *maximally mixed* state compatible with that mean energy. The exterior has no access to the phase information stored in the cross-correlations with the interior.

The von Neumann entropy of this reduced state quantifies exactly how much the exterior observer has lost. It is the entanglement entropy of the out mode with its partner, in the sense defined in Chapter 55 (Theorem 55.4).

Worked Example 65.41 (Entanglement entropy per mode and its sum)**.** From Eq. (65.245), $\log p_n = \log(1 - e^{-\omega/T_H}) - n\,\omega/T_H$, so the single-mode von Neumann entropy is

$$\begin{aligned} s(\omega) &= -\operatorname{Tr}\!\left(\rho_\omega^{\text{out}} \log \rho_\omega^{\text{out}}\right) = -\sum_n p_n \log p_n \\ &= -\log\!\left(1 - e^{-\omega/T_H}\right) + \frac{\omega}{T_H}\,\bar{n}_\omega. \end{aligned} \tag{65.248}$$

To put this in a form that exposes its meaning, use $e^{-\omega/T_H} = \bar{n}/(1+\bar{n})$, equivalently $1 - e^{-\omega/T_H} = 1/(1+\bar{n})$ and $\omega/T_H = \log[(1+\bar{n})/\bar{n}]$, both read off Eq. (65.247). Substituting,

$$\boxed{s(\omega) = (1+\bar{n}_\omega)\log(1+\bar{n}_\omega) - \bar{n}_\omega \log \bar{n}_\omega\,,} \tag{65.249}$$

the entropy of a single thermally occupied bosonic mode. It is positive for all $\bar{n} > 0$, vanishes as $\bar{n} \to 0$ (an empty mode is unentangled), and grows like

$\log \bar{n} + 1$ for $\bar{n} \gg 1$. Each emitted quantum thus arrives carrying a definite, irreducible quantum of entanglement with the hole's interior.

The total entanglement entropy of the radiation field at retarded time u is the sum of Eq. (65.249) over all modes that have been emitted by then. Writing the emission as a steady flux, the entropy collected per unit time is the mode density times $s(\omega)$, weighted by the greybody transmission $\Gamma_{\omega\ell}$ that governs which modes actually escape (Eq. (65.88)):

$$\frac{\mathrm{d}S_{\mathrm{ent}}}{\mathrm{d}t} = \sum_{\ell} (2\ell + 1) \int_0^\infty \frac{\mathrm{d}\omega}{2\pi}\, \Gamma_{\omega\ell} \left[(1 + \bar{n}_\omega) \log(1 + \bar{n}_\omega) - \bar{n}_\omega \log \bar{n}_\omega \right]. \tag{65.250}$$

Every factor in the integrand is positive, so $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}t > 0$: in the strict semiclassical calculation the entanglement entropy of the radiation *increases monotonically* for as long as the hole keeps emitting. This is the first quantitative claim used in the proof of Theorem 65.38. The integral has exactly the structure of the luminosity Eq. (65.107) with the energy weight ω replaced by the entropy weight $s(\omega)/\omega$; on dimensional grounds it scales as $\mathrm{d}S_{\mathrm{ent}}/\mathrm{d}t \sim 1/GM$, the same inverse mass that set the number-emission rate Eq. (65.114). Integrating from formation to evaporation, and using $\mathrm{d}M/\mathrm{d}t$ from Eq. (65.107), the radiation's coarse-grained entropy at the end is of order the initial horizon entropy $S_0 = A_0/4G$, indeed somewhat larger: because evaporation is irreversible (the negative heat capacity of Theorem 65.30 forbids reversible emission), the emitted entropy exceeds the lost horizon entropy by a fixed factor of order unity — $S_{\mathrm{rad}}^{\mathrm{fin}} \approx 1.6\, S_0$ for a massless thermal field. The precise factor is unimportant; what matters is that the monotone curve climbs to a value *of order* S_0 *or greater* and stays there. Hawking's prediction for the fine-grained entropy of the radiation is this monotone curve.

Physical Insight: thermality is ignorance of the partner

The radiation is thermal for a reason with no thermodynamic content at all. There is no reservoir, no ensemble, no coarse-graining over microstates in Eq. (65.246). The global state Eq. (65.243) is pure and known exactly. The thermal density matrix arises purely from the partial trace in Eq. (65.244): discarding the interior partner of each Hawking quantum turns a pure, maximally correlated pair into a maximally mixed marginal. This is *entanglement entropy masquerading as thermal entropy*, the same phenomenon that made the Rindler wedge thermal in Chapter 64. The paradox is the question of what happens to that entanglement when the partner, trapped behind the horizon, has nowhere left to be: the hole that held the partners has evaporated.

65.8.3 The entropy ceiling and the sharpened contradiction

The second ingredient of Theorem 65.38 is the ceiling. Consider the global state at time t as living on the bipartite Hilbert space $\mathcal{H}_{\rm rad} \otimes \mathcal{H}_{\rm BH}$, the already-emitted radiation times the degrees of freedom still inside the hole. For *any* pure global state the two reduced density matrices share the same nonzero spectrum (Schmidt decomposition), so their von Neumann entropies are equal, $S_{\rm rad} = S_{\rm BH}^{\rm fine}$, and each is bounded by the logarithm of the smaller Hilbert-space dimension:

$$S_{\rm rad} = S_{\rm BH}^{\rm fine} \le \min\bigl(\log \dim \mathcal{H}_{\rm rad},\, \log \dim \mathcal{H}_{\rm BH}\bigr). \tag{65.251}$$

By premise (S) of Theorem 65.38 — the subexponential, area-law bound $\log \dim \mathcal{H}_{\rm BH} \approx S_{\rm BH}(t) = A(t)/4G$ proved in Section 65.4 — the second argument of the minimum is the decreasing horizon entropy. Early on, when little has been emitted, $\mathcal{H}_{\rm rad}$ is the smaller factor and the bound Eq. (65.251) is generous; the entanglement entropy is free to rise, and the monotone curve Eq. (65.250) respects it. But as the hole shrinks, $S_{\rm BH}(t)$ falls, and at the moment the two capacities cross — the *Page time* — the horizon becomes the smaller factor and the ceiling begins to descend. Beyond that point the bound Eq. (65.251) *requires* $S_{\rm rad} \le S_{\rm BH}(t)$, a quantity heading to zero. Hawking's monotone curve does the opposite: it keeps rising. The contradiction of Theorem 65.38 is now explicit and quantitative,

$$S_{\rm ent}^{\rm Hawking}(t) \xrightarrow{\,t>t_{\rm Page}\,} \text{rising}, \qquad \text{but} \qquad S_{\rm rad}^{\rm unitary}(t) \le S_{\rm BH}(t) \xrightarrow{\,t\to\tau\,} 0. \tag{65.252}$$

A pure state cannot have a subsystem whose entanglement entropy exceeds the remaining subsystem's capacity. Something in the semiclassical computation must fail after the Page time, or unitarity is lost. This is the information paradox in its modern, entropic form, due to Page [374].

65.8.4 Page's theorem and the Page curve

What *should* the curve do if evaporation is unitary? Page's answer rests on a single, robust observation: a black hole is, to spectacular accuracy, a scrambler. The interior dynamics is so strongly chaotic that the global pure state, restricted to any modest subsystem, is statistically indistinguishable from the restriction of a Haar-random pure state on the same Hilbert space. For such random states the entanglement entropy of a subsystem is not free to be anything: it is, with overwhelming probability, almost maximal. This is the content of Page's theorem, proved in full in Chapter 57 (Theorem 57.14); we import the result and read off its consequence here rather than reprove it.

Theorem 65.42 (Page's average subsystem entropy; imported)**.** *For a Haar-random pure state on $\mathcal{H}_A \otimes \mathcal{H}_B$ with $d_A = \dim \mathcal{H}_A \le d_B = \dim \mathcal{H}_B$, the expected*

von Neumann entropy of the smaller factor is, by Theorem 57.14,

$$\langle S_A\rangle = \sum_{k=d_B+1}^{d_A d_B} \frac{1}{k} - \frac{d_A-1}{2d_B} \;\simeq\; \log d_A - \frac{d_A}{2d_B} \qquad (1 \ll d_A \le d_B). \tag{65.253}$$

Thus $\langle S_A\rangle$ is within $O(d_A/d_B)$ of its maximum $\log d_A$: a small subsystem of a random pure state is very nearly maximally mixed.

The reading of Eq. (65.253) for the black hole is immediate and is the whole point. Identify A with whichever of {radiation, interior} is smaller. Before the Page time the radiation is the small factor, $d_{\rm rad} \le d_{\rm BH}$, and Eq. (65.253) gives $S_{\rm rad} \simeq \log d_{\rm rad} = S^{\rm th}_{\rm rad}$: the entanglement entropy tracks the (rising) thermal entropy of the emitted radiation, in agreement with Hawking's curve. *This is why the two predictions coincide early*: when the radiation is small, near-maximal entanglement and thermality say the same thing. After the Page time the interior is the small factor, $d_{\rm BH} \le d_{\rm rad}$, and the same formula gives $S_{\rm rad} = S^{\rm fine}_{\rm BH} \simeq \log d_{\rm BH} = S_{\rm BH}(t)$: the fine-grained entropy now tracks the (falling) horizon entropy. The unitary prediction is therefore the lower envelope of the two coarse-grained entropies,

$$\boxed{S_{\rm rad}(t) = \min\bigl(S^{\rm th}_{\rm rad}(t),\, S_{\rm BH}(t)\bigr),} \tag{65.254}$$

the *Page curve*: it rises with the thermal radiation until the two are equal, turns over at the Page time, and descends to zero with the shrinking horizon when the hole has fully evaporated and the radiation is pure again. The naive Hawking curve and the Page curve agree exactly up to $t_{\rm Page}$ and diverge thereafter; the entire information problem lives in the after-Page-time regime. The two predictions are drawn against the chapter's own evaporation law in Fig. 65.8.

Worked Example 65.43 (The Page time for the Schwarzschild cube law)**.** The Page time is fixed by $S^{\rm th}_{\rm rad}(t_{\rm Page}) = S_{\rm BH}(t_{\rm Page})$, the crossing of the two coarse-grained entropies. Using the conservative idealization $S^{\rm th}_{\rm rad} = S_0 - S_{\rm BH}$ (total coarse-grained entropy shared between hole and radiation), the crossing is $S_{\rm BH} = S_0/2$: the Page time is when *half the initial Bekenstein–Hawking entropy has been radiated*. With $S_{\rm BH} = 4\pi G M^2$ this means $M^2_{\rm Page} = M_0^2/2$, i.e.

$$M_{\rm Page} = \frac{M_0}{\sqrt{2}} \approx 0.71\, M_0. \tag{65.255}$$

Translating to time through the cube law Eq. (65.112), $M^3 = M_0^3(1 - t/\tau)$, the Page time is

$$\frac{t_{\rm Page}}{\tau} = 1 - \Bigl(\frac{M_{\rm Page}}{M_0}\Bigr)^3 = 1 - 2^{-3/2} \approx 0.65. \tag{65.256}$$

So although the Page time is "half the entropy gone," it is nearly two-thirds of the way through the hole's *life*, because the entropy $\propto M^2$ is lost faster early on

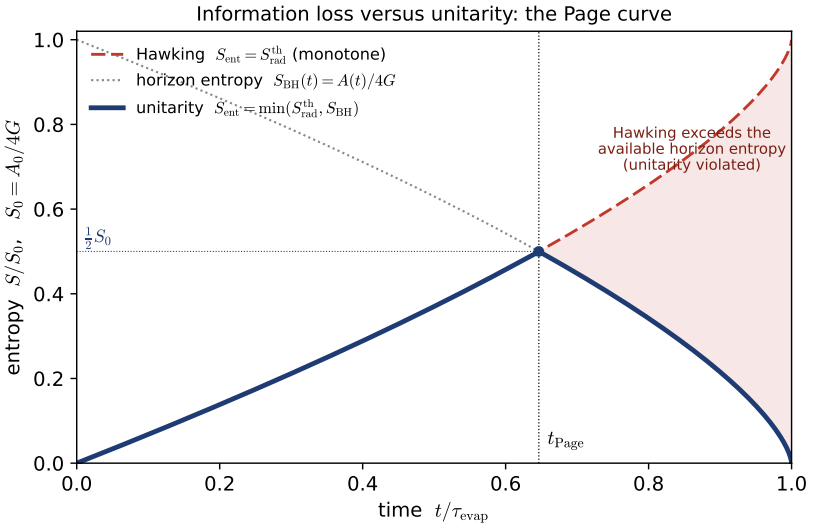


Figure 65.8: The information problem made quantitative, framed by this chapter's evaporation law $M^3(t) = M_0^3(1-t/\tau)$ (Eq. (65.112)), so that the horizon entropy is $S_{\rm BH}(t)/S_0 = (1-t/\tau)^{2/3}$ with $S_0 = A_0/4G$. Hawking's strict semiclassical calculation (dashed red) gives a fine-grained radiation entropy equal to the thermal entropy $S_{\rm rad}^{\rm th} = S_0 - S_{\rm BH}$, rising monotonically: the final radiation is mixed. Unitarity (solid blue) gives the Page curve Eq. (65.254), $S_{\rm rad} = \min(S_{\rm rad}^{\rm th}, S_{\rm BH})$, which turns over at the Page time and returns to zero. Past $t_{\rm Page}$ the Hawking curve exceeds the available horizon entropy (shaded), violating the Schmidt bound Eq. (65.251).

than the mass $\propto M$. By the Page time the hole has shed only about 29% of its mass $(1-2^{-1/2})$ but has passed the point of no return for information: from then on, unitarity demands that every further quantum carry out net information, the radiation's entanglement entropy decreasing by roughly one unit per quantum emitted. The radiation emitted after $t_{\rm Page}$ must be correlated not only with the interior but with the radiation emitted *before* $t_{\rm Page}$ — the hidden correlations that the strictly thermal spectrum of Eq. (65.88) does not contain. Where those correlations come from is the content of every proposed resolution.

65.8.5 The menu of proposals

Theorem 65.38 showed that resolving the paradox means surrendering one of its premises. We survey the principal candidates, each labelled by the premise it sacrifices, with its claim, its cost, and an honest date. None is established as the mechanism operating in our universe; what has changed by the mid-2020s is that one circle of ideas — islands and replica wormholes, Section 65.8.6 — has

been shown to reproduce the unitary Page curve in controlled models, which the others have not.

Remnants (deny C). The earliest escape: perhaps evaporation does not run to completion but halts at a Planck-mass object, a *remnant*, that retains the purifying correlations with the early radiation. The information is not lost; it is locked in a residue the semiclassical calculation cannot follow. The cost is severe. A remnant must store an unbounded amount of information — up to $S_0 = A_0/4G$ bits for an arbitrarily large parent hole — within a Planck-sized region of bounded energy. Such an "infinite internal degeneracy" for a fixed-mass object wreaks havoc with effective field theory: remnants would be pair-produced copiously in any process with enough energy, since the production rate is enhanced by their enormous multiplicity, and the species would gravitate the vacuum unstable. The remnant proposal trades information loss for an infinite tower of nearly degenerate states, a cure most workers find worse than the disease. Dated to the 1980s–1990s; generally disfavoured, though never strictly excluded.

Black-hole complementarity (reinterpret E). Proposed by Susskind, Thorlacius, and Uglum [460], complementarity argues that no single observer ever sees the contradiction, so there is none. An observer who stays outside sees the information returned in the radiation, smeared on a hot "stretched horizon" just above $r = 2GM$; an observer who falls in sees the information pass smoothly through a featureless horizon. The two descriptions are not both valid in one frame — no observer can compare the infalling copy with the outgoing copy, because doing so would require measurements separated by more than the evaporation time or energies above the Planck scale (the Hayden–Preskill bound on information retrieval [260] makes this quantitative: information dumped into an old hole comes back in the radiation after only a scrambling time $\sim GM\log(GM)$, but never fast enough to be cloned). Complementarity preserves (U) and a smooth horizon at the price of giving up a single global description of the quantum state: it is a statement about the limits of operational meaning. The cost is that it must forbid quantum cloning by a conspiracy of bounds, and this is exactly what the firewall argument later challenged. Dated to the early 1990s; influential, and partly vindicated and partly superseded by the island story.

Firewalls (deny E, the smooth horizon). Almheiri, Marolf, Polchinski, and Sully (AMPS) [15] sharpened the tension into a near-theorem. A late Hawking quantum, emitted after the Page time, must (by unitarity and Page's argument) be nearly maximally entangled with the *early* radiation. But a smooth horizon requires, by the local vacuum structure of Chapter 64, that the *same* quantum be nearly maximally entangled with its interior partner. Monogamy of entanglement — a system cannot be maximally entangled with two others at once — forbids both.

AMPS concluded that one assumption must break, and identified the smooth horizon as the weakest: the partner entanglement is severed, which by the same vacuum-structure argument deposits a divergent stress at the horizon, a *firewall* of high-energy quanta that incinerates the infalling observer. The cost is the loss of the equivalence principle at the horizon of an old black hole — "nothing special happens" fails exactly where general relativity insists it holds. Dated 2012–2013; the paradox's modern detonator, and the problem the island construction is widely read as defusing (the entanglement is reshuffled, not severed, so no firewall is needed).

Fuzzballs (deny E, the interior). The fuzzball program, developed by Mathur and collaborators within string theory [341, 342], denies that there is an empty interior at all. In this picture the classical black-hole solution is only a coarse-grained average; each microstate is a horizonless, "fuzzy" string-theoretic geometry that differs from the others at the horizon scale and caps off the interior before any trapped region forms. The $e^{S_{\mathrm{BH}}}$ microstates are genuinely distinct configurations, and radiation leaves from a structured surface rather than a true horizon, so it can carry information out like light from any hot body. Mathur's "small corrections theorem" makes the logical core precise: order-one corrections to the Hawking process at the horizon, not merely Planck-suppressed ones, are required to restore purity — the information cannot leak out in tiny corrections to an otherwise thermal spectrum. The cost is that the semiclassical horizon is an illusion: structure must appear at the horizon scale, far outside where Planckian physics was expected. Dated mid-2000s onward; the most concrete microstate construction, established in detail for special (supersymmetric, near-extremal) holes, with the generic case open.

Soft hair (reconsider C/E). Hawking, Perry, and Strominger [259] revived the question of whether black holes are truly "bald." The classical no-hair theorems leave the horizon characterized by mass, charge, and spin alone, which is why the radiation seemed information-free. But the asymptotic symmetry group of gravity (and of gauge theory) is infinite-dimensional — the BMS supertranslations — and their conserved charges endow the horizon with an infinity of "soft hairs," zero-energy excitations that can in principle record which state collapsed. The claim is that this soft data provides the missing label distinguishing microstates. The cost, and the open question, is whether the soft charges carry *enough* information, and of the right kind, to purify the radiation: most analyses find the soft hair necessary bookkeeping but not by itself sufficient to resolve the paradox. Dated 2016 onward; a suggestive reframing tying the problem to asymptotic symmetries, with its sufficiency unsettled.

Table 65.1: The principal proposals, the premise of Theorem 65.38 each surrenders, and an honest status as of the mid-2020s.

Proposal	Premise denied	Core claim	Status
Remnants	(C) completeness	Planckian residue keeps info	disfavoured
Complementarity	(E) reinterpreted	no observer sees the clash	superseded/a
Firewalls (AMPS)	(E) smooth horizon	horizon is not vacuum	sharpens the
Fuzzballs	(E) empty interior	microstates are horizonless	special cases
Soft hair	(C)/(E) no-hair	asymptotic charges label states	necessity yes,
Islands/QES	none (recomputes E)	gravity path integral is unitary	reproduces P

65.8.6 Islands, replica wormholes, and the semiclassical path integral

The decisive development of the last several years is that the entanglement entropy of the radiation, computed honestly within the same semiclassical gravity that produced Hawking's monotone curve, in fact follows the *Page* curve — once the gravitational path integral is asked the question correctly. The key is that the von Neumann entropy of the radiation is itself a gravitational observable, computed by the replica trick of Chapter 55 (Theorem 55.17) applied to gravity, and the gravitational path integral of Section 65.5 has *more saddles* than the naive one. The full statement, derivation, and the controlled models in which it is established are the business of Chapter 57; we summarize the structure and its significance, and cross-reference rather than reprove.

The replica computation evaluates $\operatorname{Tr}\rho_{\mathrm{rad}}^{n}$ by a path integral over n copies of the system glued cyclically. In gravity one must sum over all geometries with the prescribed replica boundary conditions, and beyond the trivial n-fold cover there exist *replica wormhole* saddles in which the n copies are connected through the bulk [391, 392]. These additional geometries are negligible early but come to dominate after the Page time, and their contribution bends the computed entropy back down. The upshot is packaged in the *island formula* for the radiation entropy (Section 57.4, Eq. (57.84)),

$$S_{\mathrm{rad}} = \min_{\mathcal{I}} \operatorname*{ext}_{\partial\mathcal{I}} \left[\frac{\mathrm{Area}(\partial\mathcal{I})}{4G} + S_{\mathrm{bulk}}\big(\mathrm{rad}\cup\mathcal{I}\big) \right], \tag{65.257}$$

where $\mathcal{I}$ is an *island*, a region of the interior whose boundary is a quantum extremal surface (Eq. (57.99)). The instruction is to extremize the generalized entropy — horizon-area term plus bulk entanglement entropy — over candidate surfaces and take the minimum. Before the Page time the winning configuration has no island ($\mathcal{I} = \varnothing$) and Eq. (65.257) reduces to the bulk entropy of the radiation, the rising Hawking curve. After the Page time an island just inside the horizon wins: the interior partners are then counted as *part of the radiation's*

entropy summary, the area term tracks the shrinking horizon, and the entropy follows $S_{\rm BH}(t)$ down. The crossover between the two extremal surfaces is exactly the turnover of the Page curve, and it occurs at the Page time Eq. (65.256).

Physical Insight: the path integral already knew

The remarkable feature is that nothing was added to the theory. The same semiclassical gravity — the Euclidean path integral of Section 65.5, with its $A/4G$ from the on-shell action Eq. (65.173) — when used to compute an entropy rather than a free energy, produces a unitary Page curve. The information about unitarity was hiding in the subdominant replica-wormhole saddles, geometries that connect the interior of one replica to another. In this precise sense the gravitational path integral "knows" that evaporation is unitary, even though no single semiclassical geometry displays the returning information. The same area-as-entropy that was holography's first whisper in Section 65.5 reappears here as the mechanism that caps the entanglement entropy — which is why the cleanest setting for the whole argument is holography itself.

65.8.7 The bridge to holography

The island formula Eq. (65.257) is the quantum-corrected Ryu–Takayanagi prescription of Chapter 56 (Theorem 56.3): the area term is the RT minimal surface and the bulk term is its quantum correction, the two combining into the quantum extremal surface. Holography is the setting in which the resolution is sharpest because it makes premise (U) manifest. In AdS/CFT (Chapter 53, Section 53.5) a black hole in the bulk is dual to a thermal state in a boundary conformal field theory, and the boundary theory is an ordinary unitary quantum system with a finite density of states. Its evolution is unitary by construction, so the radiation *must* purify; holography therefore guarantees the Page curve as a theorem about the boundary, and the island/replica-wormhole computation is the bulk's way of reproducing what the boundary already knew. The area-law entropy $S_{\rm BH} = A/4G$ that ceilings the hole is, on the boundary, simply the logarithm of the number of accessible states of a finite quantum system — the subexponential bound (S) of Theorem 65.38 is automatic. In this light the entire information problem is the question of how a manifestly unitary boundary theory is encoded in a bulk whose local effective field theory naively seems to lose information, and the island formula is the dictionary entry that resolves it. This is the deepest reason Part XV of this book exists: the holographic principle, first glimpsed as the area-scaling of black-hole entropy, is what reconciles gravity with quantum mechanics at the horizon.

65.8.8 Synthesis: the convergence of the threads and the languages

It is fitting that the book's forward development ends here, because this frontier is where everything built earlier converges. The convergence is literal, on two axes.

First, the three *languages* of the book all see the same physics. The Hawking temperature $T_H = \kappa/2\pi$ was derived three independent ways in this chapter: by canonical quantization and Bogoliubov coefficients (Section 65.2), the heir to the operator methods of Part III; by the Euclidean gravitational path integral and the smoothness of a saddle (Section 65.5), the heir to the path-integral methods of Part IX; and by the quantum anomaly of fields near the horizon (Section 65.6), the heir to the anomaly methods of Part XIV. That the canonical, path-integral, and anomaly formulations of quantum field theory — developed in different parts of the book for different purposes — all deliver the identical $\kappa/2\pi$ is the payoff of insisting throughout that they are one subject in three dialects. The resolution of the information problem then lives in the same path-integral language: it is a statement about which saddles dominate the gravitational functional integral, the very object introduced for the free scalar in Chapter 10 and promoted to a sum over geometries in Section 65.5.

Second, the three carried *threads* arrive together. ϕ^4 theory, the free real scalar we have followed since Chapter 9, is the matter field whose modes we traced across the collapsing horizon: its two-mode squeezed in-vacuum Eq. (65.243), its thermal reduced state Eq. (65.246), and its monotone entanglement entropy Eq. (65.250) are the carried computation, brought to its conclusion in the bipartite entanglement that is the heart of the paradox. The same free-field entanglement, computed in flat space in Part XV, became the area law and the Ryu–Takayanagi formula that Section 65.8.6 now uses to cap that entropy. QED and Yang–Mills theory enter through the soft-hair and asymptotic-symmetry reframing of Section 65.8.5: the infinite-dimensional asymptotic symmetries that might label black-hole microstates are the gravitational cousins of the large gauge transformations and BMS structure that the gauge threads exposed, and the double-copy relation of Chapter 61 that builds gravity from gauge theory is the amplitudes-level statement of the same kinship. The free scalar, the gauge field, and gravity — the three things this book taught how to quantize — meet at the horizon.

Physical Insight: the arc closed

The book opened with the quietest object in physics: a single free scalar field, $\frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2$, quantized into harmonic oscillators. It ends with that same field, placed on a collapsing star, forcing the deepest open question in theoretical physics — whether quantum mechanics survives a

horizon. Nothing was added but gravity as a background and then, at the very end, gravity as a fluctuating geometry. The entire apparatus between — renormalization, the renormalization group, gauge invariance, anomalies, the conformal bootstrap, holography, scattering amplitudes — was the machinery needed to ask this question sharply and to begin to answer it.

We close with the honesty the frontier demands, dated to the mid-2020s. What is *proven*: in controlled models — two-dimensional dilaton (JT) gravity, and holographic setups where a unitary boundary dual is available — the island formula Eq. (65.257), derived from the semiclassical gravitational path integral including replica-wormhole saddles, reproduces the Page curve Eq. (65.254), turning the entanglement entropy over at the Page time and returning it to zero (Section 57.6.1). In these models the semiclassical gravity path integral is unitary, the firewall is averted because the late radiation's entanglement is reshuffled into the island rather than severed, and the area-as-entropy of black-hole thermodynamics is the agent of the cap. What remains *open* (Section 57.6.5): the microscopic mechanism in our four-dimensional universe, where no boundary dual is in hand; the physical fate of the interior and of an observer who falls in, on which the island story is suggestive but not yet decisive; and the interpretation of the replica wormholes themselves — whether the gravitational path integral computes the entropy of a single black hole or an ensemble average over theories, a question raised acutely by the appearance of these connected geometries [18, 392]. The information paradox is, by the mid-2020s, no longer a paradox of principle — unitary evaporation is realized in honest models — but it is not yet a solved problem of mechanism. It is the live edge of the subject, and the right place for a book that set out to carry the reader from free fields to the frontier to lay down its pen.

The forward development of the book ends here. What follows is the Conclusion, which steps back from the calculations to take stock of the whole: the unity of the subject we have assembled, from the harmonic oscillators of Chapter 9 to the fluctuating geometries of this section, and the open frontier — this information problem foremost among its questions — that the next generation of the theory must settle.

Further reading

The foundational papers remain the clearest entry points: Hawking's derivation of particle creation by black holes [255], Bekenstein's argument that the entropy is proportional to area [54], and the four laws of black-hole mechanics of Bardeen, Carter and Hawking [47]. The Euclidean path-integral approach to the gravitational partition function and the entropy as an on-shell action is Gibbons and Hawking [220], with the requisite boundary term in York [530] and the negative

mode in Gross, Perry and Yaffe [239]; the anti–de Sitter completion and its phase transition is Hawking and Page [258]. The greybody factors and precise emission rates are computed by Page [373]. For the stress-tensor and anomaly viewpoints see Christensen and Fulling [123] and the gravitational-anomaly derivation of Robinson and Wilczek [418] in the covariant form of Banerjee and Kulkarni [44]. The trans-Planckian problem and its resolution are treated by Jacobson [280], Unruh [479], Corley and Jacobson [137], and Brout et al. [95]; the analog-gravity program originates with Unruh [478], is reviewed in [46], and was realized experimentally by Steinhauer [455]. The tunneling derivation is Parikh and Wilczek [377]. On the information problem, the original statement is Hawking [256] and the entropic sharpening is Page [374]; complementarity [460], the Hayden–Preskill bound [260], firewalls [15], fuzzballs [341, 342], and soft hair [259] are the principal proposals, while the island/replica-wormhole resolution is [18, 391, 392]. The standard textbook synthesis of quantum field theory in curved spacetime and black-hole thermodynamics is Wald [488], with the field-theoretic background in Birrell and Davies [76].

Part XVIII

Appendices

Appendix A

Conventions and Notation

A book that promises every number can be reproduced from its own pages owes the reader a single, unambiguous account of the conventions those numbers are written in. Conventions in quantum field theory are not unique, and the freedom is genuine: the metric may be mostly-minus or mostly-plus, the Fourier kernel may carry $e^{-ip\cdot x}$ or $e^{+ip\cdot x}$, the covariant derivative may attach the coupling with either sign, γ^5 may be defined with or without a factor of i, and each choice ripples through every formula downstream. Two correct calculations performed in two convention systems will disagree sign-by-sign at intermediate stages and yet agree on every observable. The cost of that freedom is borne entirely at the transitions between topics, and those transitions are where errors hide. This appendix is the arbiter that closes them: it fixes one convention system, derives its internal consequences, and makes every other chapter answerable to a single page.

The appendix does not introduce physics; it fixes the language in which the physics of the rest of the book is written, and it does so by derivation rather than decree wherever a derivation exists. From the single principle that the action is dimensionless in natural units, Section A.1 extracts the mass dimension of every field and coupling and the renormalizability criterion that organizes Part VI. Section A.2 fixes the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, the index gymnastics, and the Levi-Civita sign. Section A.3 settles the Fourier kernel, the (2π) bookkeeping, the state normalization, and derives the Lorentz-invariant phase-space measure that Chapter 16 consumes. Section A.4 fixes the Feynman $i\epsilon$, records the three momentum-space propagators, and pins down the Wick rotation and the $e^{iS} \to e^{-S_E}$ Euclidean sign. Section A.5 fixes the action sign and the $+J\phi$ source, the generating functionals Z, W, Γ and their Legendre relations, and the normalization of the amplitude $\mathcal{M}$ and the cross-section and decay-rate formulae. Section A.6 fixes the covariant-derivative sign $D_\mu = \partial_\mu - igA^a_\mu T^a$, the field-strength normalization, and the Hermitian-generator algebra. Section A.7 fixes dimensional regularization with $d = 4 - 2\varepsilon$ and the precise $\overline{\mathrm{MS}}$ definition.

Section A.8 closes with a conversion dictionary to the standard texts and a master symbol glossary.

What this appendix establishes it imports from the chapters where the underlying machinery was built, and it is scrupulous about that division of labor. The group and representation theory behind the generators is Chapter 2; the Lorentz and Poincaré structure that the metric and the spinor indices encode is Chapter 3; the functional calculus, Legendre transform, and Gaussian functional integrals that the generating functionals rest on are Chapter 4; the theory of tempered distributions and the contour technology behind the $i\epsilon$ and the analytic continuations are Chapter 5; the action principle is Chapter 6; the classical $U(1)$ gauge principle is Chapter 8; the path-integral generating functional is Chapter 10; and the non-abelian connection and curvature are Chapter 25. Nothing in those developments is reproved here; this appendix records the conventions in which they are expressed and the consequences that follow purely from convention.

What it does *not* do is equally important to state. The explicit master loop integrals and the Feynman/Schwinger-parameter technology live in Chapter D; the Casimir and Dynkin numerics and the ϵ-tensor contraction identities live in Chapter B; the γ-matrix trace identities and the resolution of γ^5 in dimensional regularization live in Chapter C; and the functional-determinant and ζ-function regularization of the path-integral measure live in Chapter E. This appendix fixes the *normalizations* of the propagators; the operator and path-integral *derivations* of those propagators as physics are Chapter 9 and Chapter 10. And it is honest that a single global arbiter cannot absorb every local convention: the spinor-helicity reality and little-group signs are restated where they are used in Chapter 58, and several bootstrap normalizations are restated in Part XIV, because the literature there is fragmented enough that a frozen central table would mislead more than it would help.

With those imports declared and those deferrals flagged, the appendix is the self-contained reference the rest of the book reaches for whenever a sign, a factor of 2π, or a normalization is in doubt.

A.1 Units, Dimensions, and the Counting of Mass Dimension

A book that asks the reader to reproduce every number it quotes must first fix, once and visibly, the yardsticks against which those numbers are measured. The choice of units is not a cosmetic preliminary: it determines what a symbol like λ or e even *means* dimensionally, and every later statement about which interactions are allowed, which are forbidden, and which loop integrals diverge rests on a count of dimensions that is only well posed once the units are settled. This appendix is the single arbiter of such conventions for the whole book, and this first section establishes the most basic of them. We adopt natural units, show

that they collapse the three independent mechanical dimensions into a single one, and then extract from the one principle that the action is dimensionless a complete table of the dimensions of every field and every coupling we will meet. That table is not bookkeeping for its own sake: it is the seed of the entire theory of renormalization, because the sign of a coupling's dimension decides whether the interaction it multiplies can be consistently quantized to arbitrarily high energies. Every choice made here has a traceable consequence later, and we will trace several of them explicitly.

A.1.1 Natural units and the collapse to a single dimension

In ordinary mechanics three base dimensions suffice to express every quantity: length L, time T, and mass M (or, equivalently, energy $E = ML^2T^{-2}$). Two fundamental constants of nature relate these to one another. The speed of light c has dimensions LT^{-1} and so converts times into lengths; the reduced Planck constant $\hbar$ has dimensions of action, ML^2T^{-1}, and so relates energies to inverse times through $E = \hbar\omega$. The defining move of natural units is to declare both equal to unity,

$$\hbar = c = 1. \tag{A.1}$$

This is not a claim that $\hbar$ and c are numerically one in any laboratory system; it is the decision to measure speed in units of c and action in units of $\hbar$, so that these constants become pure conversion factors equal to one and drop out of every equation. Their dimensional content does not vanish, however, and it is exactly that content which we exploit.

Setting $c = 1$ identifies L and T: a second and a metre are now the same kind of quantity, since c no longer stands between them. Setting $\hbar = 1$ then identifies E with T^{-1}, and hence with L^{-1}. The three base dimensions L, T, M have collapsed to one. We take that one to be *mass* (equivalently energy), and we will express the dimension of every quantity as a single integer or half-integer power of mass. The relations are

$$[\text{mass}] = [\text{energy}] = [\text{time}]^{-1} = [\text{length}]^{-1}, \tag{A.2}$$

where, throughout this appendix and the whole book, the bracket $[\,X\,]$ denotes the mass dimension of X: the unique real number n such that X scales as $(\text{mass})^n$.

Definition A.1 (Mass dimension)**.** For any physical quantity X expressed in natural units (A.1), its *mass dimension* $[\,X\,] \in \mathbb{R}$ is the power n for which X has the units of $(\text{mass})^n$. A quantity with $[\,X\,] = 0$ is *dimensionless* (a pure number, identical in every system of units); a quantity with $[\,X\,] \neq 0$ is said to have "dimension n" or, loosely, "be of order $(\text{mass})^n$." Mass dimension is additive under multiplication, $[\,XY\,] = [\,X\,] + [\,Y\,]$, and homogeneous under powers, $[\,X^k\,] = k\,[\,X\,]$.

The additivity in Theorem A.1 is the only algebraic fact we ever need about the bracket, and it follows immediately from the definition of units as multiplicative scales. It is what makes the dimension of a complicated Lagrangian term computable term by factor. We record at once the dimensions of the objects that appear universally. A spacetime coordinate is a length, so $[\,x^\mu\,] = -1$; consequently a partial derivative, which carries an inverse length, has

$$[\,\partial_\mu\,] = +1, \tag{A.3}$$

and a d-dimensional spacetime volume element has $[\,\mathrm{d}^d x\,] = -d$. A four-momentum is an energy, so $[\,p^\mu\,] = +1$, in keeping with the Fourier duality $p \sim \partial$; a momentum-space measure carries $[\,\mathrm{d}^d p\,] = +d$. These two facts—that coordinate integration lowers the dimension by d and momentum integration raises it by d—are the engine of the divergence counting in Part VI, and we have just derived them from (A.1) alone.

A.1.2 Restoring physical units: the conversion factors

Because $\hbar$ and c have been set to one, a result computed in natural units emerges as a pure power of mass, and to compare it with a laboratory measurement we must reinstate whatever powers of $\hbar$ and c restore the wanted dimensions of length, time, or area. The procedure is mechanical: a natural-units answer of mass dimension n is multiplied by the unique monomial $\hbar^a c^b$ that converts $(\text{mass})^n$ into the desired physical dimension. Since there is only one such monomial for each target, the restoration is unambiguous. The single number that does almost all of this work is the product

$$\hbar c \;=\; 197.327\ \text{MeV fm} \;=\; 0.197327\ \text{GeV fm}, \tag{A.4}$$

which has dimensions of energy×length and therefore equals one in natural units. Equation (A.4) states that an inverse energy is a length: $1\ \text{GeV}^{-1} = 0.197327$ fm. Table A.1 collects the conversions used repeatedly in the book; each is a power of (A.4) together with the elementary $\hbar$ and c.

The entry that recurs most in scattering calculations is the squared product

$$(\hbar c)^2 \;=\; 0.389379\ \text{GeV}^2\,\text{mb}, \qquad \text{so} \qquad 1\ \text{GeV}^{-2} = 0.389379\ \text{mb}, \tag{A.5}$$

because a cross section computed in natural units comes out as an inverse squared mass and must be turned into an area. We exercise (A.5) in Theorem A.7.

A.1.3 The action is dimensionless

Every dimension we are about to assign descends from a single principle: the action is a pure number. The reasoning is quantum-mechanical and admits no

Table A.1: Conversion factors for restoring physical units. Each entry equals one in natural units; multiplying a natural-units result by the appropriate power restores conventional units. Here 1 fm $= 10^{-15}$ m and 1 barn $= 10^{-28}$ m^2 $=$ 100 fm^2, so 1 mb $= 10^{-31}$ m^2.

Quantity	Natural-units value $= 1$	Practical form
Action / angular momentum	$\hbar = 6.582 \times 10^{-22}$ MeV s	$E\,t \to E\,t/\hbar$
Speed	$c = 2.998 \times 10^{8}$ m s^{-1}	$v \to v/c$
Energy$\times$length	$\hbar c = 197.327$ MeV fm	1 GeV$^{-1} = 0.1973$ fm
(Energy$\times$length)^2	$(\hbar c)^2 = 0.3894$ GeV^2 mb	1 GeV$^{-2} = 0.3894$ mb
Time from energy	$\hbar = 6.582 \times 10^{-25}$ GeV s	1 GeV$^{-1} = 6.582 \times 10^{-25}$ s
Mass–energy	c^2	1 GeV$/c^2 = 1.783 \times 10^{-24}$ g

freedom. In the path-integral and operator formulations alike the action enters physics only through the phase $e^{iS/\hbar}$, the weight that each history contributes to a transition amplitude (the action principle and the Euler–Lagrange equations from which S is built are developed in Chapter 6). The argument of an exponential must be dimensionless—there is no meaning to $e^{(\text{metres})}$—so $S/\hbar$ is dimensionless, whence $[\,S\,] = [\,\hbar\,]$. In natural units $\hbar = 1$ is dimensionless, and therefore

$$[\,S\,] = 0. \tag{A.6}$$

This is the master constraint. It is worth stressing that (A.6) is a statement about natural units specifically: with $\hbar$ restored, S has the dimension of $\hbar$, namely action; setting $\hbar = 1$ is precisely what makes the numerical value of S a dimensionless count of phase in units of the quantum of action.

A field theory's action is the spacetime integral of a Lagrangian density,

$$S \;=\; \int \mathrm{d}^d x\, \mathcal{L}, \tag{A.7}$$

in d spacetime dimensions, where we keep d general because the renormalization machinery of Part VI works in $d = 4 - 2\varepsilon$ and the classification below depends on d. Since $[\,\mathrm{d}^d x\,] = -d$ and $[\,S\,] = 0$, additivity (Theorem A.1) forces the Lagrangian density to carry the full dimension of the volume it is integrated over:

$$[\,\mathcal{L}\,] = d. \tag{A.8}$$

Equation (A.8) is the working form of the master constraint: every term in any Lagrangian density, kinetic or interacting, must have mass dimension exactly d. Reading this requirement against the known kinetic terms fixes the dimension of each field; reading it against the interaction terms then fixes the dimension of each coupling. We do both in turn.

A.1.4 Dimensions of the fields

Proposition A.2 (Mass dimensions of the canonical fields)**.** *In d spacetime dimensions, a real or complex scalar field ϕ, a Dirac (or Weyl) spinor field ψ, and a gauge potential A_μ, each with a canonically normalized kinetic term, have mass dimensions*

$$[\,\phi\,] = \frac{d-2}{2}, \qquad [\,\psi\,] = \frac{d-1}{2}, \qquad [\,A_\mu\,] = \frac{d-2}{2}. \tag{A.9}$$

In the physical case $d = 4$ these are $[\,\phi\,] = 1$, $[\,\psi\,] = \frac{3}{2}$, and $[\,A_\mu\,] = 1$.

Proof. The dimension of a dynamical field is set by demanding that its kinetic term, which defines the free propagation and is present in every theory of that field, obey (A.8). We treat the three cases by the same algorithm: write the kinetic term, equate its mass dimension to d using additivity and $[\,\partial_\mu\,] = 1$ from (A.3), and solve.

Scalar. The canonical kinetic term is $\frac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi)$. The numerical $\frac{1}{2}$ is dimensionless, and each factor $\partial_\mu\phi$ has dimension $1 + [\,\phi\,]$, so the term has dimension $2(1 + [\,\phi\,])$. Setting this equal to d,

$$2(1 + [\,\phi\,]) = d \quad\Longrightarrow\quad [\,\phi\,] = \frac{d-2}{2}. \tag{A.10}$$

A mass term $\frac{1}{2}m^2\phi^2$ provides an immediate cross-check: its dimension is $2[\,m\,] + 2[\,\phi\,] = 2[\,m\,] + (d-2)$, which equals d precisely when $[\,m\,] = 1$, as a mass must have. Thus the scalar mass parameter carries mass dimension one in every d, consistent with the name.

Spinor. The canonical Dirac kinetic term is $\bar\psi\, i\partial\!\!\!/\,\psi = \bar\psi\, i\gamma^\mu\partial_\mu\psi$. The gamma matrices are constant numerical arrays (Chapter C) and so are dimensionless; $\bar\psi = \psi^\dagger\gamma^0$ has the same dimension as ψ. Hence the term has dimension $[\,\psi\,] + 1 + [\,\psi\,] = 2[\,\psi\,] + 1$. Setting this equal to d,

$$2[\,\psi\,] + 1 = d \quad\Longrightarrow\quad [\,\psi\,] = \frac{d-1}{2}. \tag{A.11}$$

The Dirac mass term $m\,\bar\psi\psi$ checks this: its dimension is $[\,m\,]+2[\,\psi\,] = [\,m\,]+(d-1)$, equal to d exactly when $[\,m\,] = 1$. The fermionic kinetic term carries a single derivative rather than two, which is why the spinor dimension exceeds the scalar dimension by $\frac{1}{2}$ in every d; this half-integer offset is the dimensional signature of first-order (Dirac) versus second-order (Klein–Gordon) dynamics.

Gauge field. The canonical kinetic term is $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ (the abelian form; the nonabelian addition $g f^{abc} A^b_\mu A^c_\nu$ does not alter the counting, as we verify in Section A.1.8). The field strength has dimension $[\,F_{\mu\nu}\,] = 1 + [\,A_\mu\,]$, so the kinetic term has dimension $2(1 + [\,A_\mu\,])$, exactly the scalar combination. Hence

$$2(1 + [\,A_\mu\,]) = d \quad\Longrightarrow\quad [\,A_\mu\,] = \frac{d-2}{2}. \tag{A.12}$$

That a gauge potential shares the scalar dimension is no coincidence: both have two-derivative kinetic terms, and mass dimension is blind to the Lorentz indices A_μ carries. It also makes the dimensional content of the covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$ consistent only when the coupling g supplies the missing dimension, a point we return to immediately below. □

The recurring case $d = 4$ deserves to be isolated, since it governs nearly every explicit computation in the book. There a scalar and a gauge field each have dimension one, a fermion has dimension $\frac{3}{2}$, a mass has dimension one, and a derivative has dimension one. Table A.2 records these together with the coupling dimensions derived next.

A.1.5 Dimensions of the couplings, and the general rule

With the field dimensions fixed, the dimension of any coupling is read directly off (A.8). The statement is general and clean.

Proposition A.3 (Coupling of an operator of dimension Δ)**.** *Let $\mathcal{O}$ be a local operator built from fields and their derivatives, with mass dimension $[\,\mathcal{O}\,] = \Delta$, and let g be the coupling constant multiplying it in the Lagrangian density, so that $\mathcal{L} \supset g\,\mathcal{O}$. Then in d spacetime dimensions*

$$[\,g\,] = d - \Delta. \tag{A.13}$$

Proof. By (A.8) the term $g\mathcal{O}$ has dimension d. By additivity (Theorem A.1), $[\,g\mathcal{O}\,] = [\,g\,] + [\,\mathcal{O}\,] = [\,g\,] + \Delta$. Equating, $[\,g\,] + \Delta = d$, which is (A.13). □

Despite its one-line proof, Theorem A.3 is the most consequential statement in this section: it converts a question about an operator's field content into a single number, $d - \Delta$, whose sign will decide the fate of the interaction. We apply it to the three interactions carried through the book; the results appear again, with their full theories, in Section A.1.8.

The quartic scalar coupling. For the interaction $\frac{\lambda}{4!}\phi^4$, the operator is $\mathcal{O} = \phi^4$ with dimension $\Delta = 4[\,\phi\,] = 4 \cdot \frac{d-2}{2} = 2(d-2) = 2d - 4$. Hence

$$[\,\lambda\,] = d - \Delta = d - (2d - 4) = 4 - d. \tag{A.14}$$

In $d = 4$ this is $[\,\lambda\,] = 0$: the quartic coupling is dimensionless. The combinatorial factor $1/4!$ is dimensionless and plays no role in the count; it is fixed by the symmetry-factor conventions of Chapter 15, not by dimensions.

The gauge couplings. A gauge interaction enters through the covariant derivative, and its prototype is the fermion–photon vertex $\bar\psi\,(g\gamma^\mu A_\mu)\,\psi$ obtained by expanding $\bar\psi\, i\not{D}\,\psi$. The operator multiplying the coupling is $\mathcal{O} = \bar\psi\gamma^\mu\psi A_\mu$, of dimension

$$\Delta = 2[\,\psi\,] + [\,A_\mu\,] = (d-1) + \frac{d-2}{2} = \frac{3d-4}{2}, \tag{A.15}$$

so by (A.13)

$$[\,e\,] = [\,g\,] = d - \frac{3d-4}{2} = \frac{4-d}{2}. \tag{A.16}$$

In $d = 4$ both the QED coupling e and the Yang–Mills coupling g are dimensionless. The identical value of $[\,e\,]$ and $[\,g\,]$ is forced by the common structure of minimal coupling: any coupling introduced through a covariant derivative $D_\mu = \partial_\mu - ig(\cdots)$ must, by dimensional consistency of D_μ with ∂_μ, carry exactly the dimension that compensates $[\,A_\mu\,] = \frac{d-2}{2}$, namely $1 - \frac{d-2}{2} = \frac{4-d}{2}$. The same number therefore governs the nonabelian cubic and quartic gluon self-interactions, as we confirm below.

Table A.2: Mass dimensions in d spacetime dimensions and in the physical case $d = 4$. Field dimensions are from Theorem A.2; coupling dimensions from Theorem A.3. All four-dimensional gauge and quartic couplings are dimensionless, the dimensional fingerprint of renormalizability discussed in Section A.1.6.

Object	Mass dimension (general d)	Value at $d = 4$
coordinate x^μ	-1	-1
derivative ∂_μ, momentum p^μ	$+1$	$+1$
action S	0	0
Lagrangian density $\mathcal{L}$	d	4
scalar ϕ	$(d-2)/2$	1
gauge field A_μ	$(d-2)/2$	1
Dirac spinor ψ	$(d-1)/2$	$3/2$
mass m	1	1
quartic coupling λ (ϕ^4)	$4-d$	0
gauge couplings e, g	$(4-d)/2$	0

A.1.6 Relevant, marginal, irrelevant: the renormalizability classification

The dimension of a coupling is not merely a label; it predicts how the importance of the interaction it multiplies changes with the energy at which the theory is probed, and through that, whether the theory makes sense as a fundamental description valid to arbitrarily short distances. We first state the classification, then prove the energy-scaling behaviour that justifies its names, and finally connect it to the power counting of ultraviolet divergences that Part VI develops in full.

Definition A.4 (Relevant, marginal, irrelevant operators)**.** Let $\mathcal{O}$ be a local operator of mass dimension $\Delta = [\,\mathcal{O}\,]$ appearing in a d-dimensional Lagrangian density with coupling g, so $[\,g\,] = d - \Delta$ by Theorem A.3. The operator (equivalently, the interaction it generates) is called

- *relevant* if $\Delta < d$, equivalently $[\,g\,] > 0$;
- *marginal* if $\Delta = d$, equivalently $[\,g\,] = 0$;
- *irrelevant* if $\Delta > d$, equivalently $[\,g\,] < 0$.

The names are dictated by behaviour under change of scale, which we now derive. The terminology is a frequent source of sign confusion, so we pin down at the outset the precise quantity it refers to. The decisive object is the dimensionless ratio that measures the strength of the interaction $g\mathcal{O}$ *relative to the kinetic (free) part of the action* at a probing energy E, not the coupling g in isolation. To isolate it, rescale all coordinates and fields to dimensionless variables measured in units of E: under $x \to x/E$ and a compensating rescaling of each field by the appropriate power of E fixed by Theorem A.2, the kinetic terms become E-independent (they are scale covariant), while the term $g\mathcal{O}$ is left multiplied by a definite power of E. Tracking that power, the dimensionless measure of the interaction's importance at scale E is

$$\lambda_{\mathcal{O}}(E) \;\equiv\; g\,E^{\,\Delta-d} \;=\; g\,E^{-[\,g\,]}. \tag{A.17}$$

Equivalently and more quickly: $\lambda_{\mathcal{O}}(E)$ is the unique dimensionless combination of g and E, since $[\,g\,] = d - \Delta$ must be cancelled by $E^{\Delta-d}$. Its E-dependence is fixed entirely by the sign of $[\,g\,]$, and reading off the three cases proves the assertions implicit in Theorem A.4. The Wilsonian derivation in Chapter 23 carries out this rescaling explicitly, tracking how $\lambda_{\mathcal{O}}(E)$ flows as the cutoff is lowered; here we need only its dimensional skeleton, which (A.17) records.

Proposition A.5 (Infrared/ultraviolet fate of an interaction)**.** *Let $\lambda_{\mathcal{O}}(E) = g\,E^{\,\Delta-d}$ be the dimensionless importance of the interaction $g\mathcal{O}$ at scale E,* (A.17)*. Then:*

1. Relevant *($\Delta < d$, $[\,g\,] > 0$): $\lambda_{\mathcal{O}}(E) \to \infty$ as $E \to 0$ and $\to 0$ as $E \to \infty$. The interaction dominates the infrared and is negligible in the deep ultraviolet—hence "relevant" at low energies.*
2. Marginal *($\Delta = d$, $[\,g\,] = 0$): $\lambda_{\mathcal{O}}(E) = g$, scale independent at this (classical) level. The interaction is equally important at all scales; quantum corrections (the running of Chapter 24) break the tie and decide its true fate.*
3. Irrelevant *($\Delta > d$, $[\,g\,] < 0$): $\lambda_{\mathcal{O}}(E) \to 0$ as $E \to 0$ and $\to \infty$ as $E \to \infty$. The interaction is unimportant at low energies but grows without bound toward the ultraviolet—hence "irrelevant" to long-distance physics, yet the obstruction to a fundamental description.*

Proof. The exponent in $\lambda_{\mathcal{O}}(E) = g\,E^{\,\Delta-d}$ is $\Delta - d = -[\,g\,]$. In the relevant case $\Delta - d < 0$, so $E^{\,\Delta-d}$ diverges as $E \to 0$ and vanishes as $E \to \infty$; in the irrelevant

case $\Delta - d > 0$, so the limits are exchanged; in the marginal case the exponent vanishes and $\lambda_{\mathcal{O}}$ is constant. Each limit is the elementary behaviour of a real power, and the verbal descriptions follow by interpreting $\lambda_{\mathcal{O}} \gg 1$ as "interaction strong" and $\lambda_{\mathcal{O}} \ll 1$ as "interaction weak." □

Theorem A.5 explains the names, but the deepest consequence of the classification is its control over ultraviolet divergences, which is the reason the dimension count is the gateway to renormalization. We can already extract the heart of the power-counting theorem from dimensional analysis alone, deferring only the diagrammatic bookkeeping to Chapter 21.

Proposition A.6 (Dimensional core of power-counting renormalizability)**.** *Consider a theory whose interactions have couplings g_i multiplying operators $\mathcal{O}_i$ of dimensions Δ_i, so $[\,g_i\,] = d - \Delta_i$. The superficial degree of divergence D of a Feynman diagram—the net mass dimension of its loop integrals, defined and computed in Chapter 21—takes the form*

$$D \;=\; (\textit{a fixed number determined by } d \textit{ and the external legs}) \;-\; \sum_i n_i\,[\,g_i\,], \tag{A.18}$$

where n_i is the number of insertions of the vertex from g_i. Consequently: if every $[\,g_i\,] \geq 0$, then D is bounded above independently of the number of vertices, so only finitely many types of amplitude can diverge and finitely many counterterms remove every divergence—the theory is renormalizable*. If some $[\,g_i\,] < 0$, then D grows without bound as n_i increases, so infinitely many amplitude types diverge, each demanding its own counterterm—the theory is* nonrenormalizable *and requires an ultraviolet cutoff to be defined.*

Proof. We establish the structural form (A.18) by dimensional analysis; the explicit evaluation of the leg-dependent constant in terms of the number of internal and external lines is the combinatorial computation carried out in full in Chapter 21, and we do not duplicate it. Consider an amplitude (amputated Green's function) Γ of fixed external content, hence fixed total mass dimension $[\,\Gamma\,]$ determined by d and the dimensions of its external fields. At a given order in perturbation theory Γ is a sum of loop integrals, each accompanied by a product of couplings $\prod_i g_i^{n_i}$, of total dimension $\sum_i n_i[\,g_i\,]$. Since the full amplitude has the fixed dimension $[\,\Gamma\,]$ regardless of order, the loop integral accompanying $\prod_i g_i^{n_i}$ must carry mass dimension

$$[\,\Gamma\,] - \sum_i n_i\,[\,g_i\,]. \tag{A.19}$$

The superficial degree of divergence D is exactly the mass dimension of that loop integral when the divergence is power-like, because a loop integral that diverges as Λ^D (or, in dimensional regularization, exhibits a pole accompanied by external

momenta to the power D) necessarily has mass dimension D supplied by the loop momenta. Hence $D = [\,\Gamma\,] - \sum_i n_i [\,g_i\,]$, which is (A.18) with the leg-dependent constant identified as $[\,\Gamma\,]$. The two conclusions are now arithmetic. If all $[\,g_i\,] \geq 0$, then $-\sum_i n_i [\,g_i\,] \leq 0$, so $D \leq [\,\Gamma\,]$ for every diagram of every order: the degree of divergence cannot be raised by adding more vertices. Only those finitely many external configurations with $[\,\Gamma\,] \geq 0$ can ever produce a divergence, and each is a polynomial in external momenta of bounded degree, removable by adjusting the finitely many parameters (a counterterm for each such configuration). If instead some $[\,g_j\,] < 0$, then increasing n_j increases D without bound: for any external configuration, sufficiently many insertions of $\mathcal{O}_j$ make $D \geq 0$, so infinitely many amplitude types diverge and no finite set of counterterms suffices. The diagrammatic proof that these counterterms can in fact be organized order by order without spoiling lower orders—the Bogoliubov–Parasiuk–Hepp–Zimmermann program—is the content of Chapter 22; what dimensional analysis establishes, and all it needs to establish here, is the dichotomy on the *number* of required counterterms. □

Theorem A.6 is why the dimension count of this section is the foundation of Part VI. Renormalizable theories in $d = 4$ are exactly those whose couplings all have $[\,g_i\,] \geq 0$, that is, those built only from relevant and marginal operators. Reading Table A.2, the scalar mass term ($[\,m^2\,] = 2 > 0$, relevant), the ϕ^4 coupling, the gauge couplings, and the Yukawa coupling (whose operator $\bar{\psi}\psi\phi$ has dimension $2[\,\psi\,] + [\,\phi\,] = (d-1) + \frac{d-2}{2} = \frac{3d-4}{2}$, equal to 4 at $d = 4$, so its coupling has dimension $(4-d)/2$, dimensionless and marginal) exhaust the renormalizable interactions of fields of spin ≤ 1 in four dimensions. Every other local interaction—a ϕ^6 term, a four-fermion contact term $(\bar{\psi}\psi)^2$ of dimension 6, a Pauli moment $\bar{\psi}\sigma^{\mu\nu}\psi F_{\mu\nu}$ of dimension 5—has a coupling of negative mass dimension and is irrelevant, hence nonrenormalizable, hence admissible only within an effective field theory carrying an explicit cutoff. That the Standard Model contains *only* relevant and marginal operators is not an accident but the statement that it is, by construction, a renormalizable theory; the modern reinterpretation of nonrenormalizable terms as the leading windows onto physics above the cutoff is the subject of the effective-field-theory chapters.

A.1.7 Restoring units in practice: two worked computations

The dimensional machinery is only useful if one can move fluently between the natural-units answer a calculation produces and the laboratory number it predicts. We carry out two restorations in full: a cross section, where powers of $\hbar c$ must be reinstated, and the electron anomalous magnetic moment, where the lesson is that a genuinely dimensionless prediction needs no restoration at all—a fact that is itself a sharp consistency check.

Worked Example A.7 (Restoring units in the $e^+e^- \to \mu^+\mu^-$ cross section)**.** The total cross section for $e^+e^- \to \mu^+\mu^-$ at centre-of-mass energy $\sqrt{s}$ well above the muon mass is computed in natural units in Chapter 17; the result is

$$\sigma \;=\; \frac{4\pi\alpha^2}{3s}, \qquad \alpha = \frac{e^2}{4\pi} \approx \frac{1}{137.036}, \tag{A.20}$$

with α the fine-structure constant in Heaviside–Lorentz units (the convention fixed in Chapter 8 and used throughout QED). We restore physical units and evaluate at $\sqrt{s} = 10$ GeV.

Dimensional check. The coupling $\alpha = e^2/4\pi$ is dimensionless ($[\,e\,] = 0$ in $d = 4$ by (A.16)), and s is a squared energy, $[\,s\,] = 2$. Hence $[\,\sigma\,] = -2$: in natural units the cross section is an inverse squared mass. A cross section must have the dimension of an area, $[\,\text{length}\,]^2 = [\,\text{mass}\,]^{-2}$ in natural units, so the count is consistent and the unique restoring factor is $(\hbar c)^2$, which carries $[\,\hbar c\,]^2 = 0$ in natural units but reintroduces $\text{GeV}^2\cdot\text{mb}$ in practical units.

Restoration. Multiplying (A.20) by $(\hbar c)^2 = 0.389379\ \text{GeV}^2\,\text{mb}$ from (A.5),

$$\sigma \;=\; \frac{4\pi\alpha^2}{3s}\,(\hbar c)^2 \;=\; \frac{4\pi}{3}\,\alpha^2\,\frac{0.389379\ \text{GeV}^2\,\text{mb}}{s}. \tag{A.21}$$

Collecting the numerical prefactor with $\alpha = 1/137.036$,

$$\frac{4\pi}{3}\,\alpha^2\,(\hbar c)^2 = \frac{4\pi}{3}\,\frac{1}{(137.036)^2}\,(0.389379)\ \text{GeV}^2\,\text{mb} = 8.68\times10^{-5}\ \text{GeV}^2\,\text{mb}, \tag{A.22}$$

where we used $4\pi/3 = 4.18879$, $(137.036)^{-2} = 5.325\times10^{-5}$, and the product $4.18879\times5.325\times10^{-5}\times0.389379 = 8.68\times10^{-5}$. Since $1\ \text{mb} = 10^6\ \text{nb}$, the coefficient is $86.8\ \text{nb}\cdot\text{GeV}^2$, and the cross section takes the memorable form

$$\sigma \;=\; \frac{86.8\ \text{nb}}{s/\text{GeV}^2}. \tag{A.23}$$

At $\sqrt{s} = 10$ GeV, so $s = 100\ \text{GeV}^2$, this gives $\sigma = 0.868$ nb. The natural-units expression (A.20) carried no $\hbar$ or c and yet, through the single factor $(\hbar c)^2$, produced a number in nanobarns that an experiment can be held to. This is the entire content of unit restoration: identify the mass dimension of the answer, supply the unique power of $\hbar c$ that lands on the desired physical dimension, and read off the conversion from Table A.1.

Worked Example A.8 (The electron anomalous moment is dimensionless)**.** The magnetic moment of the electron is $\boldsymbol{\mu} = -g_e\,\dfrac{e}{2m_e}\,\mathbf{S}$, where g_e is the Landé g-factor and $\mathbf{S}$ the spin. The Dirac equation predicts $g_e = 2$; the deviation is encoded in the *anomalous magnetic moment*

$$a_e \;\equiv\; \frac{g_e - 2}{2}, \tag{A.24}$$

whose leading contribution Schwinger computed (carried out in full in Chapter 20) to be

$$a_e \;=\; \frac{\alpha}{2\pi} \;+\; O(\alpha^2) \;\approx\; \frac{1/137.036}{2\pi} \;\approx\; 1.161 \times 10^{-3}. \tag{A.25}$$

We ask what powers of $\hbar$ and c must be restored in (A.25), and the answer is the point of the example: *none*. The quantity a_e is defined in (A.24) as a ratio of g-factors, and a g-factor is by construction the dimensionless number relating a magnetic moment to the combination $e\hbar/2mc$ (the Bohr magneton, with $\hbar$ and c shown). Thus $[\,a_e\,] = 0$, and Theorem A.3 confirms it: a_e is a function of $\alpha = e^2/4\pi$ alone, and $[\,\alpha\,] = 2[\,e\,] = 0$ in $d = 4$. A dimensionless quantity has the same numerical value in every system of units, so $a_e \approx 1.161 \times 10^{-3}$ stands without any conversion factor whatsoever.

The contrast with the magnetic moment $\boldsymbol{\mu}$ itself is the instructive part. The moment is *not* dimensionless: restoring units, the Bohr magneton is

$$\mu_B \;=\; \frac{e\hbar}{2m_e c} \;=\; 5.788 \times 10^{-11}\ \text{MeV}\,\text{T}^{-1}, \tag{A.26}$$

carrying explicit $\hbar$ and c because a magnetic moment has the dimension of energy per magnetic field. The dimensionful content of $\boldsymbol{\mu}$ lives entirely in μ_B; the quantum field theory of Chapter 20 contributes only the pure number $g_e/2 = 1 + a_e$ multiplying it. This factorization—a dimensionful classical prefactor times a dimensionless loop expansion—is generic: dimensional analysis quarantines all the unit dependence into a known prefactor, leaving the dynamics to predict pure numbers like a_e that are the same whether one works in GeV, ergs, or electronvolts. That a measured a_e agrees with $\alpha/2\pi + \cdots$ to twelve significant figures is therefore a test of the *dynamics*, untainted by any choice of units.

A.1.8 The benchmark couplings, counted

We close by recording the mass dimensions of the three couplings carried through the book, each on its standard Lagrangian and each with a pointer to the chapter that develops the theory. These are applications of Theorem A.3 and contain nothing new; their purpose is to show the single counting principle acting uniformly across the scalar, abelian-gauge, and nonabelian-gauge cases, and to fix the dimensional starting point each thread's renormalization chapter will assume.

ϕ^4 theory. The standard Lagrangian $\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \tfrac{\lambda}{4!}\phi^4$, introduced classically in Chapter 6, has its quartic coupling counted in (A.14): $[\,\lambda\,] = 4 - d$, dimensionless in $d = 4$ and of dimension $+2\varepsilon$ when continued to $d = 4 - 2\varepsilon$. The small positive dimension that λ acquires below four dimensions is exactly what makes ϕ^4 weakly relevant in $d = 4 - \epsilon$ and underlies the Wilson–Fisher fixed point of Chapter 52; the dimensional bookkeeping here is the first appearance of that mechanism.

QED. The standard Lagrangian $\mathcal{L}_{\text{QED}} = \bar{\psi}(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ with $D_\mu = \partial_\mu + ieA_\mu$, introduced classically in Chapter 8, has its coupling counted in (A.16): $[\,e\,] = (4-d)/2$, dimensionless in $d = 4$. The dimensionlessness of e is the precondition for QED's renormalizability and for the existence of a scale-independent fine-structure constant whose slow logarithmic running (rather than power-law running) is computed in Chapter 24.

Yang–Mills theory. The standard Lagrangian $\mathcal{L}_{\text{YM}} = -\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \bar{\psi}_i(i\not{D} - m)\psi_i$, with $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ and $D_\mu = \partial_\mu - igA^a_\mu T^a$, introduced classically in Chapter 25, has the same coupling dimension as QED: $[\,g\,] = (4-d)/2$. It is worth verifying that the nonabelian self-interactions, absent in QED, give the same answer, since this confirms the remark after Theorem A.2 that the extra term in $F^a_{\mu\nu}$ does not change the counting. The cubic gluon term arising from $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ is of the schematic form $g f^{abc}(\partial_\mu A^a_\nu)A^{b\,\mu}A^{c\,\nu}$; its operator $\mathcal{O} = (\partial A)AA$ has dimension $\Delta = (1+[\,A_\mu\,])+2[\,A_\mu\,] = 1+3\cdot\frac{d-2}{2} = \frac{3d-4}{2}$, identical to (A.15), so Theorem A.3 returns $[\,g\,] = d - \frac{3d-4}{2} = (4-d)/2$ once more. The quartic gluon term $g^2 f^{abe} f^{cde} AAAA$ likewise checks: its operator has dimension $4[\,A_\mu\,] = 2(d-2)$, requiring its coefficient to have dimension $d - 2(d-2) = 4 - d = 2[\,g\,]$, consistent with the coefficient being g^2. That a single coupling g is forced to appear, with these correlated powers, in the cubic and quartic vertices is the dimensional shadow of the gauge invariance that relates them, and it is the starting point for the asymptotic-freedom computation of Chapter 27.

With every field and coupling now assigned a definite mass dimension, the action is a dimensionless object whose every term we can audit at a glance, and momentum integrals carry the definite mass dimensions recorded in Table A.2. This is the settled dimensional footing the next section needs: having fixed *how much* each quantity weighs in powers of mass, we turn to fixing the spacetime arena itself—the signature of the metric, the raising and lowering of indices, and the integration measure—on which all these dimensionful objects live.

A.2 Spacetime, the Metric, and Index Conventions

The previous section weighed every field and coupling in powers of mass and declared the action dimensionless. What it did not yet fix is the arena those dimensionful objects live in: the geometry of the index that turns a ∂_μ into a ∂^μ, the rule that pairs a position with a momentum, and the handful of signs that ride on those choices and propagate, unforgivingly, into every propagator, every dispersion relation, and every Feynman amplitude in the book. A misplaced sign here is not a cosmetic blemish—it flips the sign of an energy, sends a pole to the wrong half-plane, or turns a healthy mode into a tachyon. This section therefore makes the spacetime conventions explicit and, wherever a sign is at stake, shows that it is *forced* by physics rather than chosen by taste. We

fix the signature, the component conventions for four-vectors, the raising and lowering rules, the derivative and d'Alembertian conventions, and the orientation of the Levi-Civita tensor; and we verify, by direct computation, that the mostly-minus metric is exactly what makes the scalar kinetic energy positive and the mass term nontachyonic. The transformation law that gives these objects their tensor character under boosts and rotations belongs to the Lorentz group and is established in Chapter 3; here we presuppose it and fix the numerical conventions it acts on.

A.2.1 The signature and the metric tensor

We use throughout the *mostly-minus* (also called *west-coast* or *particle-physics*) signature, in which the metric tensor is

$$\eta_{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \qquad \mu, \nu \in \{0, 1, 2, 3\}, \tag{A.27}$$

with the time index labelled 0 and the three space indices $1, 2, 3$. This is the Peskin–Schroeder and Schwartz convention; it is the opposite of the mostly-plus signature $\mathrm{diag}(-, +, +, +)$ common in the general-relativity and string-theory literature. The single most important consequence of (A.27) is recorded in the next paragraph and is the reason we adopt it: the relativistic invariant p^2 of a physical particle equals $+m^2$, a positive number, so the on-shell condition reads $p^2 = m^2$ with no trailing minus sign. Every later chapter that writes a propagator $i/(p^2 - m^2 + i\varepsilon)$ is relying on (A.27).

Notation Watch

A large fraction of the field-theory literature, and essentially all of general relativity, uses the opposite signature $\eta_{\mu\nu} = \mathrm{diag}(-, +, +, +)$. Under that choice the on-shell condition becomes $p^2 = -m^2$, the scalar kinetic term carries an overall minus sign, and the Levi-Civita relations below flip. When importing a formula from a mostly-plus source, the safe procedure is to send $\eta \to -\eta$ everywhere, which changes the sign of every contracted pair of indices: $p^2 \to -p^2$, $p \cdot x \to -p \cdot x$, and so on. We never mix the two within this book.

A.2.2 Four-vectors and the position–momentum pairing

A contravariant four-vector carries an upper index and is written with its components in the order (time, space). For the position and the four-momentum,

$$x^\mu = (x^0, \mathbf{x}) = (t, \mathbf{x}), \qquad p^\mu = (p^0, \mathbf{p}) = (E, \mathbf{p}), \tag{A.28}$$

where boldface denotes the ordinary spatial three-vector and, in natural units, $x^0 = t$ and $p^0 = E$. Lowering the index with the metric (A.27) flips the sign of

the spatial components and leaves the time component alone,

$$x_\mu = \eta_{\mu\nu} x^\nu = (t, -\mathbf{x}), \qquad p_\mu = \eta_{\mu\nu} p^\nu = (E, -\mathbf{p}), \tag{A.29}$$

since $\eta_{00} = +1$ acts on the time slot and $\eta_{ii} = -1$ (no sum) on each spatial slot. The covariant momentum component therefore reads $p_\mu = (E, -\mathbf{p})$, a fact worth committing to memory because the minus sign on $\mathbf{p}$ is the seed of the pairing we are about to compute.

The Lorentz-invariant pairing of two four-vectors is the contraction of one upper with one lower index. For position and momentum,

$$p \cdot x \equiv p_\mu x^\mu = \eta_{\mu\nu} p^\mu x^\nu = p^0 x^0 - \mathbf{p} \cdot \mathbf{x} = E\,t - \mathbf{p} \cdot \mathbf{x}. \tag{A.30}$$

The middle step is the explicit sum: the $\mu = \nu = 0$ term contributes $\eta_{00} p^0 x^0 = +Et$, while the three spatial terms contribute $\sum_i \eta_{ii} p^i x^i = -\sum_i p^i x^i = -\mathbf{p} \cdot \mathbf{x}$, the ordinary Euclidean dot product of the two three-vectors entering with a minus sign. Note that $p \cdot x = p_\mu x^\mu = p^\mu x_\mu$ gives the same number either way, since $\eta_{\mu\nu} p^\mu x^\nu = \eta_{\mu\nu} p^\nu x^\mu$ by the symmetry of the metric.

The square of a four-vector is the special case $x = p$ of this pairing. For the momentum,

$$p^2 \equiv p \cdot p = p_\mu p^\mu = (p^0)^2 - \mathbf{p} \cdot \mathbf{p} = E^2 - |\mathbf{p}|^2 . \tag{A.31}$$

A timelike vector has $p^2 > 0$, a spacelike vector $p^2 < 0$, and a null (lightlike) vector $p^2 = 0$. For a physical particle of mass m the relativistic energy–momentum relation $E^2 = |\mathbf{p}|^2 + m^2$ rearranges, using (A.31), into the *on-shell condition*

$$p^2 = m^2. \tag{A.32}$$

This is the payoff of the mostly-minus choice advertised above: in this signature a massive particle sits at $p^2 = +m^2$, and the photon, with $m = 0$, at $p^2 = 0$. We will rederive (A.32) from the scalar equation of motion in Theorem A.10 below, confirming that the dispersion relation is a *consequence* of the conventions and the dynamics, not an independent input.

A.2.3 Raising, lowering, and contraction

The inverse metric $\eta^{\mu\nu}$ is defined by $\eta^{\mu\rho}\eta_{\rho\nu} = \delta^\mu{}_\nu$, where $\delta^\mu{}_\nu$ is the Kronecker delta (unity when $\mu = \nu$, zero otherwise). Because (A.27) is diagonal with entries ± 1, it is its own inverse numerically:

$$\eta^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1), \qquad \eta^{\mu\rho}\eta_{\rho\nu} = \delta^\mu{}_\nu. \tag{A.33}$$

One checks the defining relation componentwise: for $\mu = \nu = 0$, $\eta^{0\rho}\eta_{\rho 0} = \eta^{00}\eta_{00} = (+1)(+1) = 1$; for $\mu = \nu = i$ (spatial, no sum), $\eta^{i\rho}\eta_{\rho i} = \eta^{ii}\eta_{ii} = (-1)(-1) = 1$; and off-diagonal entries vanish because η is diagonal. The trace of the mixed

metric is the spacetime dimension, $\eta^{\mu\nu}\eta_{\mu\nu} = \delta^{\mu}{}_{\mu} = 4$ in $d = 4$, a number that becomes d in dimensional regularization and is the origin of several factors of d in loop integrals.

Indices are raised and lowered by contraction with the metric and its inverse:

$$V_\mu = \eta_{\mu\nu} V^\nu, \qquad V^\mu = \eta^{\mu\nu} V_\nu, \tag{A.34}$$

and the two operations are mutually inverse precisely because of (A.33): $\eta^{\mu\nu}(\eta_{\nu\rho}V^\rho) = \delta^{\mu}{}_{\rho}V^\rho = V^\mu$. The contraction (the "dot product") of two four-vectors can be written in four equivalent ways,

$$V \cdot W = \eta_{\mu\nu} V^\mu W^\nu = V_\mu W^\mu = V^\mu W_\mu = \eta^{\mu\nu} V_\mu W_\nu, \tag{A.35}$$

all yielding the same scalar. The summation convention is in force throughout: *a repeated index, once up and once down, is summed over its four values*, and an index that is not so paired is free and must match on both sides of any equation. An expression with a repeated pair both up or both down (such as $V^\mu W^\mu$ with no metric) is not Lorentz covariant and, in this book, is always an error unless an explicit metric is written.

A.2.4 Derivative and d'Alembertian conventions

The partial derivative with respect to the contravariant coordinate x^μ carries a *lower* index, a placement fixed by the chain rule and not a matter of choice: under a Lorentz transformation $\partial/\partial x^\mu$ transforms with the inverse Jacobian, exactly as a covariant vector does (Chapter 3). We write

$$\partial_\mu \equiv \frac{\partial}{\partial x^\mu} = \left(\frac{\partial}{\partial t}, \nabla\right) = (\partial_t, \nabla), \tag{A.36}$$

so that ∂_μ has time component ∂_t and spatial components $+\nabla = (\partial_x, \partial_y, \partial_z)$ with the ordinary (positive) sign. Raising the index with (A.34) flips the spatial part,

$$\partial^\mu = \eta^{\mu\nu}\partial_\nu = (\partial_t, -\nabla). \tag{A.37}$$

That the lower-index derivative carries $+\nabla$ while the upper-index one carries $-\nabla$ is the mirror image of the four-momentum rule $p^\mu = (E, \mathbf{p})$, $p_\mu = (E, -\mathbf{p})$, and the two are linked: acting on a plane wave $e^{-ip\cdot x}$,

$$\partial_\mu\, e^{-ip\cdot x} = -i\, p_\mu\, e^{-ip\cdot x}, \tag{A.38}$$

since $p \cdot x = p_\nu x^\nu$ and $\partial_\mu x^\nu = \delta_\mu{}^\nu$ give $\partial_\mu(p_\nu x^\nu) = p_\mu$. The replacement $\partial_\mu \to -ip_\mu$ in momentum space is used in every chapter; (A.38) is its derivation and pins down the sign of the exponent against the Fourier convention of the next section.

Contracting two derivatives gives the d'Alembertian, the Lorentz-invariant wave operator,

$$\Box \equiv \partial_\mu \partial^\mu = \eta^{\mu\nu}\partial_\mu\partial_\nu = \partial_t^2 - \nabla^2, \tag{A.39}$$

where $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$ is the spatial Laplacian. The relative sign between the time and space second derivatives is the hyperbolic character of the wave equation; it is what allows propagating solutions and, on a plane wave, reproduces the mass shell:

$$\Box\, e^{-ip\cdot x} = (-ip_\mu)(-ip^\mu)\, e^{-ip\cdot x} = -p^2\, e^{-ip\cdot x}, \tag{A.40}$$

using (A.38) twice. We adopt the symbol $\Box$ for $+\partial_t^2 - \nabla^2$; some references reserve $\Box$ for the opposite-signature operator $\nabla^2 - \partial_t^2$, so the sign in (A.39) is part of our convention and is used consistently.

A.2.5 The Levi-Civita tensor and its orientation

The four-dimensional Levi-Civita symbol is the totally antisymmetric object that is fixed by a single value together with antisymmetry. We define it with an *upper*-index normalization,

$$\epsilon^{0123} = +1, \qquad \epsilon^{\mu\nu\rho\sigma} \text{ totally antisymmetric}, \tag{A.41}$$

so that $\epsilon^{\mu\nu\rho\sigma} = +1$ for an even permutation of $(0,1,2,3)$, -1 for an odd permutation, and 0 whenever any two indices coincide. This is the orientation that makes $\epsilon^{0123} = +1$ a right-handed convention with time first; it is the choice in Peskin–Schroeder and is the one that enters the definition $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and the dual field strength $\tilde{F}^{\mu\nu} = \frac{1}{2}\epsilon^{\mu\nu\rho\sigma}F_{\rho\sigma}$.

Unlike a true tensor, the symbol (A.41) has the *same* numerical entries in every Lorentz frame by fiat; what makes the combination $\epsilon^{\mu\nu\rho\sigma}$ transform as a tensor (up to the sign of the determinant of the transformation) is the interplay of this fixed symbol with the metric, treated in Chapter 3. For index gymnastics what matters is that, once the upper-index value is fixed, the lower-index value is *not* free: it is computed by lowering all four indices with the metric, and the result carries a crucial minus sign that we now derive in full.

Proposition A.9 (Lowered Levi-Civita orientation)**.** *With the convention* $\epsilon^{0123} = +1$ *of* (A.41) *and the mostly-minus metric* (A.27)*, the fully lowered Levi-Civita tensor satisfies*

$$\epsilon_{0123} = -1, \qquad \textit{equivalently} \qquad \epsilon_{\mu\nu\rho\sigma} = \det(\eta)\,\epsilon^{\mu\nu\rho\sigma} = -\,\epsilon^{\mu\nu\rho\sigma}. \tag{A.42}$$

Proof. Lowering all four indices means contracting with one factor of the metric per index,

$$\epsilon_{\mu\nu\rho\sigma} = \eta_{\mu\alpha}\,\eta_{\nu\beta}\,\eta_{\rho\gamma}\,\eta_{\sigma\delta}\,\epsilon^{\alpha\beta\gamma\delta}, \tag{A.43}$$

with the four pairs $(\alpha, \beta, \gamma, \delta)$ summed. Evaluate the particular component ϵ_{0123}. Because the metric (A.27) is diagonal, $\eta_{\mu\alpha}$ is nonzero only when $\alpha = \mu$; hence on the right-hand side of (A.43) the only surviving term has $\alpha = 0$, $\beta = 1$, $\gamma = 2$, $\delta = 3$, and the four sums collapse to a single product:

$$\epsilon_{0123} = \eta_{00}\,\eta_{11}\,\eta_{22}\,\eta_{33}\,\epsilon^{0123} = (+1)(-1)(-1)(-1)\,(+1) = -1. \tag{A.44}$$

The product of the four diagonal entries $(+1)(-1)(-1)(-1) = -1$ is exactly the determinant of the metric, $\det(\eta) = -1$. The same collapse occurs for any ordering of the four distinct indices, since the diagonal metric forces the summed indices to equal the free ones one-for-one and pulls out the identical factor $\det(\eta)$; the antisymmetric structure is carried unchanged by the single surviving $\epsilon^{\alpha\beta\gamma\delta}$. Therefore $\epsilon_{\mu\nu\rho\sigma} = \det(\eta)\,\epsilon^{\mu\nu\rho\sigma} = -\,\epsilon^{\mu\nu\rho\sigma}$, which is the stated result. In particular the lowered symbol takes the value -1 on the reference ordering $(0, 1, 2, 3)$, exactly opposite to the upper-index symbol. □

The minus sign in (A.42) is the four-dimensional reflection of the single timelike direction: lowering picks up one factor of $\eta_{00} = +1$ and three of $\eta_{ii} = -1$, and $(-1)^3 = -1$. In a Euclidean space, where the metric is the identity with $\det = +1$, no such sign appears and $\epsilon_{1234} = \epsilon^{1234} = +1$; the Lorentzian sign is therefore a genuine consequence of the signature and must be carried whenever a parity-odd quantity (an anomaly coefficient, a θ-term, a γ^5 trace) is evaluated.

Caution

The contraction identities for products of two Levi-Civita tensors—for example $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma}$ and the determinant of Kronecker deltas that appears when fewer indices are contracted—inherit an overall minus sign in Lorentzian signature precisely because of (A.42): one finds $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma} = -4! = -24$, not $+24$. The full list of these identities, with the signature signs tracked, is assembled in Chapter B; we use them there and do not reproduce them here, but the reader should remember that every such identity in this book carries the Lorentzian sign and not the naive Euclidean one.

A.2.6 The action, the kinetic sign, and why the signs are forced

We can now close the loop with the scalar field and show that the conventions adopted above are not aesthetic preferences but are dictated by the requirement that energy be bounded below. The Minkowski action is the spacetime integral of the Lagrangian density,

$$S = \int \mathrm{d}^4x\,\mathcal{L}, \qquad \mathrm{d}^4x = \mathrm{d}t\,\mathrm{d}^3\mathbf{x}, \tag{A.45}$$

with the Lagrangian density of the real scalar field (the frozen ϕ^4 theory model, Chapter 6) written in the standard form

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \tag{A.46}$$

The overall normalization $+\frac{1}{2}$ on the kinetic term and the relative minus sign on the mass term are the two conventions at stake. We prove that both are forced.

Proposition A.10 (Positive energy and the forced relative sign)**.** *With the mostly-minus metric* (A.27)*, the Lagrangian* (A.46) *has a Hamiltonian density that is a sum of manifestly non-negative terms, and its free Euler–Lagrange equation is the Klein–Gordon equation* $(\Box + m^2)\phi = 0$ *with the mass-squared entering with a positive sign. Reversing either the overall sign of the kinetic term or the relative sign of the mass term spoils one of these two properties; the conventions in* (A.46) *are therefore the unique choice consistent with positive energy and a nontachyonic spectrum.*

Proof. Expand the kinetic contraction using the ingredient of (A.39), $\partial_\mu\phi\,\partial^\mu\phi = \eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi = (\partial_t\phi)^2 - (\nabla\phi)^2$, so that the Lagrangian density reads, in explicit space and time pieces,

$$\mathcal{L} = \tfrac{1}{2}(\partial_t\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4. \tag{A.47}$$

The canonical momentum conjugate to ϕ is

$$\pi \equiv \frac{\partial\mathcal{L}}{\partial(\partial_t\phi)} = \partial_t\phi, \tag{A.48}$$

and the Hamiltonian density is the Legendre transform $\mathcal{H} = \pi\,\partial_t\phi - \mathcal{L}$. Substituting (A.47) and (A.48),

$$\begin{aligned}\mathcal{H} &= (\partial_t\phi)^2 - \left[\tfrac{1}{2}(\partial_t\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4\right] \\ &= \tfrac{1}{2}(\partial_t\phi)^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4.\end{aligned} \tag{A.49}$$

Every term in (A.49) is manifestly non-negative for real ϕ, $m^2 \geq 0$, and $\lambda \geq 0$: the first is the kinetic energy of the field's time variation, the second the gradient (shear) energy, the third the mass (rest) energy, the fourth the self-interaction energy. The Hamiltonian $H = \int \mathrm{d}^3\mathbf{x}\,\mathcal{H}$ is therefore bounded below by zero, with the minimum at $\phi = \text{const} = 0$. Had we chosen the opposite overall sign for the kinetic term, $\mathcal{L} \to -\tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) + \dots$, the Legendre transform would produce $\mathcal{H} = -\tfrac{1}{2}(\partial_t\phi)^2 - \dots$, unbounded below: the field could lower its energy without limit by oscillating faster, and no stable vacuum would exist. The $+\frac{1}{2}$

kinetic normalization with the mostly-minus metric is thus the unique sign giving positive kinetic energy.

For the relative sign of the mass term, vary the free action ($\lambda = 0$). The Euler–Lagrange equation is

$$\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) - \frac{\partial \mathcal{L}}{\partial \phi} = 0. \tag{A.50}$$

From (A.46) with $\lambda = 0$, the canonical structure gives $\partial\mathcal{L}/\partial(\partial_\mu\phi) = \partial^\mu\phi$ and $\partial\mathcal{L}/\partial\phi = -m^2\phi$, so (A.50) becomes

$$\partial_\mu \partial^\mu \phi + m^2 \phi = (\Box + m^2)\phi = 0, \tag{A.51}$$

the Klein–Gordon equation, in which the mass-squared appears with a *plus* sign relative to the d'Alembertian. Inserting a plane wave $\phi \propto e^{-ip\cdot x}$ and using (A.40), $\Box\, e^{-ip\cdot x} = -p^2 e^{-ip\cdot x}$, turns (A.51) into $(-p^2 + m^2)\phi = 0$, i.e.

$$p^2 = m^2, \qquad E^2 = |\mathbf{p}|^2 + m^2, \tag{A.52}$$

the on-shell condition (A.32) and the standard relativistic dispersion with real, positive energies for every mode. Had the mass term entered (A.46) with the *same* sign as the kinetic term, $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 + \frac{1}{2}m^2\phi^2$, the Euler–Lagrange equation would read $(\Box - m^2)\phi = 0$, giving $p^2 = -m^2$ and $E^2 = |\mathbf{p}|^2 - m^2$: modes with $|\mathbf{p}| < m$ would have imaginary frequency and grow exponentially in time. That is the tachyonic instability, the hallmark of an inverted potential. The relative minus sign between the kinetic and mass terms in (A.46) is therefore the unique choice giving a stable, nontachyonic spectrum. Both signs in (A.46) are forced. □

Physical Insight

The two sign choices have a single physical content: the Lagrangian is (kinetic energy) minus (potential energy), $\mathcal{L} = T - V$, and the mostly-minus metric is exactly what assembles the relativistic kinetic term $\frac{1}{2}(\partial_t\phi)^2 - \frac{1}{2}(\nabla\phi)^2$ so that its *time*-derivative part enters with the same positive sign as a nonrelativistic $\frac{1}{2}m\dot{q}^2$. The potential $V = \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$ then collects every term that costs energy: spatial variation, mass, and self-interaction. Reading the relative signs this way makes them memorable rather than arbitrary, and it is why a positive m^2 is a true mass while a negative m^2 signals an unstable vacuum and triggers spontaneous symmetry breaking (the mechanism developed later in the book).

A.2.7 The benchmark Lagrangians in fixed signature

To anchor every later sign to the conventions just fixed, we rewrite each of the three carried-thread Lagrangians in the mostly-minus signature, exhibiting the

component form where the signs are explicit. No thread is advanced here; the purpose is purely to display the benchmark models against the metric so that no later chapter need rederive their sign structure.

Worked Example A.11 (The three benchmark Lagrangians against the metric)**.** *ϕ^4 theory.* The model is (A.46), already expanded in (A.47). With (A.27) the kinetic contraction is $(\partial_\mu\phi)(\partial^\mu\phi) = (\partial_t\phi)^2 - (\nabla\phi)^2$, the mass term is the negative $-\frac{1}{2}m^2\phi^2$, and the equation of motion is the interacting Klein–Gordon equation obtained by restoring λ in (A.50): $\partial\mathcal{L}/\partial\phi = -m^2\phi - \frac{\lambda}{3!}\phi^3$, hence

$$(\Box + m^2)\phi = -\frac{\lambda}{3!}\phi^3, \tag{A.53}$$

the frozen ϕ^4 equation of motion of Chapter 6, with the on-shell condition $p^2 = m^2$ of (A.52) for the free field.

QED. The standard Lagrangian $\mathcal{L}_{\text{QED}} = \bar\psi(i\not{D} - m)\psi - \frac{1}{4}F_{\mu\nu}F^{\mu\nu}$, with $D_\mu = \partial_\mu + ieA_\mu$ (electron charge $-e$, $e > 0$) and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$, depends on the signature in two places. First, the Dirac kinetic operator $\not{D} = \gamma^\mu D_\mu$ contracts the gamma matrices against D_μ using $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, the Clifford relation built on (A.27) and collected in Chapter C. That the gamma normalization is consistent with the scalar on-shell condition follows by squaring the free Dirac operator: using the Clifford relation,

$$\not{\partial}\not{\partial} = \gamma^\mu\gamma^\nu\partial_\mu\partial_\nu = \tfrac{1}{2}\{\gamma^\mu, \gamma^\nu\}\partial_\mu\partial_\nu = \eta^{\mu\nu}\partial_\mu\partial_\nu = \Box, \tag{A.54}$$

where the symmetric part of $\partial_\mu\partial_\nu$ projects the gamma product onto its anticommutator. Hence $(i\not{\partial} - m)(i\not{\partial} + m) = -\not{\partial}\not{\partial} - m^2 = -(\Box + m^2)$, so any solution of the free Dirac equation also solves $(\Box + m^2)\psi = 0$ and obeys the same on-shell condition $p^2 = m^2$ as the scalar—a check that the Clifford relation and the metric (A.27) are mutually consistent. Second, the Maxwell term in the mostly-minus signature is

$$-\tfrac{1}{4}F_{\mu\nu}F^{\mu\nu} = \tfrac{1}{2}\left(\mathbf{E}^2 - \mathbf{B}^2\right), \tag{A.55}$$

where $\mathbf{E}$ and $\mathbf{B}$ are the electric and magnetic fields packaged in $F_{\mu\nu}$; the overall $-\frac{1}{4}$ is exactly the normalization that makes the electric (time-derivative) part $\frac{1}{2}\mathbf{E}^2$ enter the energy with a positive sign, the photon analogue of the scalar kinetic term whose sign was forced in Theorem A.10.

Yang–Mills theory. The frozen non-abelian Lagrangian

$$\mathcal{L}_{\text{YM}} = -\tfrac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu} + \bar\psi_i(i\not{D} - m)\psi_i, \tag{A.56}$$

with $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc} A^b_\mu A^c_\nu$ and $D_\mu = \partial_\mu - igA^a_\mu T^a$, contracts the colour-vector field strength $F^a_{\mu\nu}$ with the same metric, the colour index a summed with a Euclidean (Kronecker) sum and the Lorentz indices $\mu\nu$ raised by (A.27). The kinetic part of (A.56), obtained by dropping the $g f^{abc}$ term in $F^a_{\mu\nu}$, is a sum

of dim G copies of the Maxwell term (A.55), one per generator, and so carries the same positive-energy sign; the cubic and quartic gluon self-interactions hidden in the $gf^{abc}A^b_\mu A^c_\nu$ piece are the only structural difference from QED, and their Lorentz contractions again use only (A.27). The matter term repeats the QED Dirac analysis (A.54) verbatim, now with D_μ carrying the colour generator. Every sign in (A.56) is thereby fixed by the metric and the Clifford relation already established.

With the signature, the component conventions (A.28) and (A.29), the raising/lowering rules (A.34), the derivative conventions (A.36)–(A.39), and the Levi-Civita orientation (A.41)–(A.42) all fixed, and with the position–momentum pairing (A.30) verified against the plane-wave derivative (A.38), the spacetime arena is fully specified. The next section takes exactly these objects—the metric that defines $p \cdot x = Et - \mathbf{p} \cdot \mathbf{x}$ and the sign of the exponent in $e^{-ip\cdot x}$—and uses them to fix the Fourier-transform convention, the attendant bookkeeping of factors of 2π, and the Lorentz-invariant phase-space measure that weights momentum integrals in every amplitude to come.

A.3 Fourier Conventions, Distributions, and Invariant Measures

The previous section fixed the metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, the on-shell condition $p^2 = m^2$, and the spacetime pairing $p \cdot x = p^0 x^0 - \mathbf{p} \cdot \mathbf{x}$. Those choices are static: they tell us how to read an index. The moment a field is expanded in plane waves, however, a second layer of convention switches on, and it is the one that generates more arithmetic slips than any other in the subject—the placement of factors of 2π. Every momentum integral in a Feynman amplitude carries a $(2\pi)^{-4}$; every momentum-conserving delta function pulls back a $(2\pi)^{+4}$; the relativistic normalization of a one-particle state hides a factor of $2E_{\mathbf{p}}$ that must be there for the inner product to be Lorentz invariant, and that same factor reappears, inverted, in the phase-space measure that weights final states in a cross section. These factors are not free: they are forced, each by the next, once the Fourier convention is fixed. The purpose of this section is to fix that convention, prove the few distributional identities the bookkeeping rests on (importing the analysis from Chapter 5 rather than rebuilding it), and from them *derive*—not postulate—the Lorentz-invariant phase-space measure and the relativistic state normalization that every scattering computation in the book, and every benchmark amplitude, will quote.

A.3.1 The Fourier transform and the (2π) bookkeeping

We use the asymmetric particle-physics convention, in which all the 2π's live on the momentum-integration side and the forward transform carries the sign $e^{+ip\cdot x}$.

Definition A.12 (Fourier transform pair)**.** For a (Schwartz, or more generally tempered) function f on Minkowski space and its transform $\tilde{f}$, the inverse and forward transforms are

$$f(x) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\, \tilde{f}(p), \tag{A.57}$$

$$\tilde{f}(p) = \int d^4x\, e^{+ip\cdot x}\, f(x), \tag{A.58}$$

with $p\cdot x = p^0x^0 - \mathbf{p}\cdot\mathbf{x}$ the metric pairing of Section A.2. The operational rule is: *each $\int d^4p$ carries a $(2\pi)^{-4}$, and each four-dimensional delta function produced by an $\int d^4x$ carries a $(2\pi)^{+4}$.* The same convention is used in any number of dimensions n, with $4 \to n$ throughout; Chapter 5 develops the transform abstractly on $\mathbb{R}^n$ in this exact normalization (Eq. (5.37)).

Notation Watch

The forward transform carries $e^{+ip\cdot x}$ and the inverse carries $e^{-ip\cdot x}$. This is the opposite sign convention from much of the mathematics literature (where the forward transform usually carries $e^{-ip\cdot x}$), and the reason is physical: a positive-energy plane wave of a quantum field is $e^{-ip\cdot x} = e^{-iE_{\mathbf{p}}t + i\mathbf{p}\cdot\mathbf{x}}$, so writing the *field* (the position-space object) as the inverse transform Eq. (A.57) makes its mode expansion read with the standard e^{-iEt} time dependence. Fixing the sign here, once, removes the most common source of a stray overall sign in the exponent of every propagator and vertex.

The entire bookkeeping rests on one distributional identity, the spacetime statement that the Fourier transform of the constant function 1 is a delta function. We state it as the controlling identity and then verify the convention is self-consistent.

Proposition A.13 (The distributional delta representation)**.** *As an identity of tempered distributions on momentum space,*

$$\int d^4x\, e^{\,ip\cdot x} = (2\pi)^4\, \delta^{(4)}(p), \tag{A.59}$$

and equivalently, with the pairing read the other way, $\int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x} = \delta^{(4)}(x)$. Here $\delta^{(4)}(p) = \delta(p^0)\,\delta^{(3)}(\mathbf{p})$ is the four-dimensional Dirac delta of Theorem 5.5.

Proof. This is the spacetime instance of the constant-transform identity proved in Chapter 5 as $\widehat{1} = (2\pi)^n\delta$ (Theorem 5.19); we make the metric signature explicit so the reader sees that the indefinite pairing changes nothing. The integrand factorizes over the four coordinates because the exponent does: $e^{ip\cdot x} =$

$e^{ip^0x^0}\,e^{-i\mathbf{p}\cdot\mathbf{x}}$. Carrying out the time integral and the three spatial integrals separately, each is a one-dimensional constant-transform,

$$\int_{-\infty}^{\infty} dx^0\, e^{\,ip^0x^0} = 2\pi\,\delta(p^0), \tag{A.60}$$

$$\int_{-\infty}^{\infty} dx^j\, e^{-ip_jx^j} = 2\pi\,\delta(p_j), \qquad j=1,2,3, \tag{A.61}$$

each understood as the $n=1$ case of Theorem 5.19, in which a Gaussian regulator $e^{-a|x|^2/2}$ is inserted, the resulting convergent Gaussian integral is evaluated, and the limit $a\to 0^+$ is taken inside the distributional pairing (continuity of $\mathcal{F}$ on $\mathcal{S}'$). The relative minus sign on the spatial exponents is invisible to the result, because the delta function is even, $\delta(-p_j)=\delta(p_j)$. Multiplying Eqs. (A.60) and (A.61) gives $(2\pi)^4\,\delta(p^0)\,\delta(p_1)\delta(p_2)\delta(p_3) = (2\pi)^4\,\delta^{(4)}(p)$, which is Eq. (A.59). Dividing by $(2\pi)^4$ and relabelling $p\leftrightarrow x$ gives the inverse-side statement. □

Physical Insight

Equation (A.59) is the bookkeeping in a single line. Read left to right it says: an x-integral over a pure phase delivers a delta function *and* a $(2\pi)^4$. In a Feynman diagram each interaction vertex is an $\int d^4x$ of a product of plane waves, so each vertex produces exactly one such factor—the $(2\pi)^4\delta^{(4)}(\sum p)$ that enforces energy–momentum conservation at the vertex. Read right to left it says: every momentum integral $\int d^4p$ that is eventually killed by such a delta must come with a $(2\pi)^{-4}$ for the two factors to cancel and leave a clean number. The convention is engineered so that they always do.

That engineering is a claim, and we now check it: composing the forward and inverse transforms must return the original function, with every 2π accounted for. This is the consistency proof the spec of this section calls for.

Worked Example A.14 (Forward ∘ inverse closes back to $\delta^{(4)}$)**.** Take a Schwartz function f, transform it forward by Eq. (A.58), and substitute the result into the inverse transform Eq. (A.57); the claim is that we recover f, which forces the (2π) factors to be exactly those in Theorem A.12. Insert Eq. (A.58) into Eq. (A.57), writing the inner integration variable as y to keep it distinct from x:

$$\int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\,\tilde f(p) = \int \frac{d^4p}{(2\pi)^4}\, e^{-ip\cdot x}\int d^4y\, e^{\,ip\cdot y}\, f(y). \tag{A.62}$$

The integrand is absolutely integrable in (p,y) jointly once f is Schwartz and a Gaussian regulator is restored, so Fubini's theorem permits exchanging the order of integration. Doing the p-integral first and collecting the two exponentials into

$e^{\,ip\cdot(y-x)}$,

$$= \int d^4y\; f(y) \left[\int \frac{d^4p}{(2\pi)^4}\, e^{\,ip\cdot(y-x)} \right] = \int d^4y\; f(y)\, \delta^{(4)}(y-x) = f(x), \qquad \text{(A.63)}$$

where the bracket is the inverse-side form of Eq. (A.59) (a single $(2\pi)^{-4}$ accompanies the $\int d^4p$, and no leftover 2π survives), and the final step is the defining sifting property of the delta, Theorem 5.5. The chain $f \mapsto \tilde{f} \mapsto f$ closes *exactly*: the $(2\pi)^{-4}$ on the momentum integral is precisely what the delta-representation $(2\pi)^{+4}$ needs to cancel. Had we distributed the 2π's symmetrically (a $(2\pi)^{-2}$ on each transform), the same computation would still close, but then a vertex would emit a $(2\pi)^2\delta^{(4)}$ and the rules would be cluttered with half-integer powers of 2π. The asymmetric convention is chosen so that all factors of 2π are carried by momentum measures and momentum deltas, never by the fields.

Theorem 5.25 packages the residue-by-half-plane rule for the energy integrals these transforms produce; we will not repeat it here, but it is the engine behind every propagator contour in Section A.4.

A.3.2 The delta composed with a function

Deriving the on-shell measure requires one further distributional identity: the rule for a delta function whose argument is itself a function. We state and prove it in one dimension, which is all the measure derivation needs, building on the delta as the distribution defined in Theorem 5.5.

Lemma A.15 (Composition of the delta with a function)**.** *Let $g\colon \mathbb{R} \to \mathbb{R}$ be continuously differentiable with simple zeros $\{x_i\}$ (so $g(x_i) = 0$ and $g'(x_i) \neq 0$) and no zeros at infinity. Then, as a distribution,*

$$\delta(g(x)) = \sum_i \frac{\delta(x - x_i)}{|g'(x_i)|}. \qquad \text{(A.64)}$$

Proof. A distribution is determined by its action on test functions, so we evaluate $\int dx\, \delta(g(x))\, \varphi(x)$ for an arbitrary test function φ. Because $\delta(g(x))$ is supported on the zero set $\{x_i\}$, partition the real line into intervals each containing a single simple zero; the integral is the sum of the contributions, so it suffices to treat one zero x_0 and sum at the end. On a small interval I around x_0, g is strictly monotone (since $g'(x_0) \neq 0$ and g' is continuous), hence invertible there; change variables to $u = g(x)$, so that $du = g'(x)\, dx$ and $x = g^{-1}(u)$. Monotonicity gives $dx = du/g'(x)$ with g' of one fixed sign on I; writing the measure with its absolute value to keep the orientation of integration correct,

$$\int_I dx\; \delta(g(x))\, \varphi(x) = \int_{g(I)} \frac{du}{|g'(g^{-1}(u))|}\, \delta(u)\, \varphi(g^{-1}(u)). \qquad \text{(A.65)}$$

The interval $g(I)$ contains $u = 0$ (the image of x_0), so the sifting property of $\delta(u)$ evaluates the rest of the integrand at $u = 0$, i.e. at $x = g^{-1}(0) = x_0$:

$$= \frac{\varphi(x_0)}{|g'(x_0)|}. \tag{A.66}$$

This is exactly $\int dx\, \dfrac{\delta(x - x_0)}{|g'(x_0)|}\, \varphi(x)$. Summing over all simple zeros x_i reproduces Eq. (A.64), and since φ was arbitrary the two distributions agree. □

Caution

The factor is $|g'(x_i)|$, the *absolute value* of the derivative, not $g'(x_i)$ itself. The absolute value is what makes Eq. (A.64) a positive measure, as it must be: $\delta(g(x))$ is a limit of nonnegative bump functions. Dropping the modulus is a frequent error that flips the sign of a phase space whenever a root has $g' < 0$—and in the measure derivation below the relevant root has $g'(p^0) = 2p^0$, which is positive only because we restrict to the positive-energy sheet.

A.3.3 Relativistic normalization of states

Before deriving the measure we fix the normalization of momentum eigenstates, because the two are two faces of one Lorentz-invariance requirement. In nonrelativistic quantum mechanics one writes $\langle \mathbf{p}|\mathbf{q}\rangle = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})$, the conventional continuum normalization. The trouble is that $\delta^{(3)}(\mathbf{p} - \mathbf{q})$ is *not* a Lorentz scalar: under a boost the three-momenta and their measure transform, and a bare three-dimensional delta does not hold its form. A relativistic theory must normalize its one-particle states so that the inner product is frame-independent, and the unique minimal fix is to multiply by $2E_{\mathbf{p}}$.

Definition A.16 (Relativistic one-particle normalization)**.** On-shell momentum eigenstates of a particle of mass m, with $E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2} > 0$, are normalized covariantly by

$$\langle p|q\rangle = (2\pi)^3\, 2E_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p} - \mathbf{q}), \tag{A.67}$$

where $|p\rangle$ denotes the state of four-momentum $p^\mu = (E_{\mathbf{p}}, \mathbf{p})$. This is the convention used by every amplitude in the book, including every benchmark theory (Chapter 14 for ϕ^4 theory, Chapter 17 for QED). The relation to a unit-normalized state $|\mathbf{p}\rangle_{\mathrm{NR}}$ with ${}_{\mathrm{NR}}\langle \mathbf{p}|\mathbf{q}\rangle_{\mathrm{NR}} = (2\pi)^3 \delta^{(3)}(\mathbf{p} - \mathbf{q})$ is $|p\rangle = \sqrt{2E_{\mathbf{p}}}\, |\mathbf{p}\rangle_{\mathrm{NR}}$.

The claim that Eq. (A.67) is Lorentz invariant is the same claim, restated, as the invariance of the measure we are about to derive. We prove that invariance once, for the object $2E_{\mathbf{p}}\, \delta^{(3)}(\mathbf{p} - \mathbf{q})$, in Theorem A.19 below, after the measure is in hand; the two facts are inverse to each other, in the precise sense that $\delta^{(3)}/(2E)$ and $2E\, \delta^{(3)}$ transform with reciprocal Jacobians and so pair to an invariant.

A.3.4 The Lorentz-invariant phase-space measure

We can now reach the central object of the section. A scattering computation sums over final states; that sum is an integral over on-shell momenta, and for the result to be a Lorentz scalar the integration measure must be one too. The naive measure d^3p is not invariant. The correct, invariant measure is $d^3p/((2\pi)^3 2E_{\mathbf{p}})$, and it has a manifestly covariant origin: it is the four-dimensional measure restricted to the positive-energy mass shell by an on-shell delta. We derive it that way, which proves invariance for free, and then verify invariance a second time by an explicit boost.

Theorem A.17 (Lorentz-invariant phase-space measure)**.** *Define the one-particle invariant measure*

$$d\Pi \;\equiv\; \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}, \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}. \tag{A.68}$$

Then $d\Pi$ is the restriction to the forward mass shell of the manifestly invariant four-dimensional object,

$$d\Pi = \int_{-\infty}^{\infty} \frac{dp^0}{2\pi}\,(2\pi)\,\delta(p^2{-}m^2)\,\theta(p^0)\,\frac{d^3p}{(2\pi)^3} = \frac{d^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2{-}m^2)\,\theta(p^0), \tag{A.69}$$

and is invariant under the proper orthochronous Lorentz group.

Proof. Step 1: perform the energy integral. Hold $\mathbf{p}$ fixed and integrate Eq. (A.69) over p^0. The argument of the delta is, with the mostly-minus metric,

$$g(p^0) := p^2 - m^2 = (p^0)^2 - \mathbf{p}^2 - m^2 = (p^0)^2 - E_{\mathbf{p}}^2, \tag{A.70}$$

a function of p^0 alone (at fixed $\mathbf{p}$) with two simple zeros at $p^0 = \pm E_{\mathbf{p}}$. Its derivative is $g'(p^0) = 2p^0$, so $|g'(\pm E_{\mathbf{p}})| = 2E_{\mathbf{p}}$ at each root. By the composition rule Theorem A.15,

$$\delta(p^2 - m^2) = \frac{\delta(p^0 - E_{\mathbf{p}})}{2E_{\mathbf{p}}} + \frac{\delta(p^0 + E_{\mathbf{p}})}{2E_{\mathbf{p}}}. \tag{A.71}$$

The step function $\theta(p^0)$ annihilates the second term (which is supported at $p^0 = -E_{\mathbf{p}} < 0$) and leaves the first untouched (its support is at $p^0 = +E_{\mathbf{p}} > 0$, where $\theta = 1$). Hence

$$\theta(p^0)\,\delta(p^2 - m^2) = \frac{1}{2E_{\mathbf{p}}}\,\delta(p^0 - E_{\mathbf{p}}). \tag{A.72}$$

Now integrate over p^0, with the (2π)'s of Eq. (A.69):

$$\int_{-\infty}^{\infty} \frac{dp^0}{2\pi}\,(2\pi)\,\theta(p^0)\,\delta(p^2 - m^2) = \int_{-\infty}^{\infty} dp^0\,\frac{1}{2E_{\mathbf{p}}}\,\delta(p^0 - E_{\mathbf{p}}) = \frac{1}{2E_{\mathbf{p}}}. \tag{A.73}$$

The explicit (2π) in Eq. (A.69) cancels the $(2\pi)^{-1}$ of the energy measure, so no stray 2π is generated by going on shell. Multiplying by the remaining spatial measure $d^3p/(2\pi)^3$ gives precisely $d\Pi = d^3p/((2\pi)^3 2E_{\mathbf{p}})$, which is Eq. (A.68). This establishes the equality Eq. (A.69).

Step 2: invariance from the covariant form. The four pieces of the right-hand expression in Eq. (A.69) are each Lorentz invariant under the proper orthochronous group, so their product is. (i) The four-volume measure d^4p is invariant because a Lorentz transformation Λ acts on p linearly with $\det\Lambda = +1$ for $\Lambda \in \mathrm{SO}(1,3)^+$, so the Jacobian of $p \mapsto \Lambda p$ is $|\det\Lambda| = 1$ and $d^4p' = d^4p$; the constant $(2\pi)^{-4}$ rides along untouched. (ii) The delta $\delta(p^2 - m^2)$ is invariant because its argument p^2 is the Lorentz scalar built in Section A.2, and a delta of a scalar is a scalar. (iii) The step $\theta(p^0)$ is invariant on the support of the delta: there p is timelike ($p^2 = m^2 > 0$), and proper *orthochronous* transformations preserve the sign of the time component of a timelike vector—the orthochronicity condition ${\Lambda^0}_0 \geq 1$ established for the Lorentz group in Chapter 3—so $\theta(p^0) = \theta(p'^0)$ on the mass shell. (iv) The factor (2π) is a constant. Therefore $d\Pi$, being equal to a product of invariants, is itself invariant. The restriction $m = 0$ is included: the forward mass shell degenerates to the forward light cone, and the same argument applies with $\theta(p^0)$ still preserved on the cone by orthochronicity. □

Physical Insight

The cleanest way to remember Eq. (A.68) is to remember Eq. (A.69): *the invariant measure is "d^4p over $(2\pi)^4$, on shell, positive energy."* Everything non-invariant in $d^3p/2E_{\mathbf{p}}$ is an artifact of having already done the p^0 integral and broken manifest covariance. This is the same pattern that recurs throughout the book—a manifestly covariant object (here the four-dimensional integrand) is reduced to a non-covariant but explicit one (the three-dimensional measure) by integrating out a constrained variable, and the covariance, though hidden, is never lost.

The invariance argument above is complete, but it is worth seeing the cancellation happen in the three-dimensional language directly, because that is the form one actually integrates and because the spec asks specifically for the boost that scales the energy component. We carry it out as a worked example.

Worked Example A.18 (Invariance of $d^3p/E_{\mathbf{p}}$ under a boost)**.** Take a boost of rapidity η along the 3-axis (the z-direction), with $\gamma = \cosh\eta$ and $\beta = \tanh\eta$, acting on the on-shell four-momentum $(E_{\mathbf{p}}, \mathbf{p})$. The transverse components are inert and the longitudinal pair mixes:

$$p'^1 = p^1, \qquad p'^2 = p^2, \qquad p'^3 = \gamma\,(p^3 + \beta\,E_{\mathbf{p}}), \tag{A.74}$$

$$E'_{\mathbf{p}} = p'^0 = \gamma\,(E_{\mathbf{p}} + \beta\,p^3). \tag{A.75}$$

We must compare $d^3p'/E'_{\mathbf{p}}$ with $d^3p/E_{\mathbf{p}}$. Since $dp'^1\,dp'^2 = dp^1\,dp^2$, only the longitudinal Jacobian $\partial p'^3/\partial p^3$ is at stake, evaluated *on shell*, where $E_{\mathbf{p}}$ is the dependent function $E_{\mathbf{p}} = \sqrt{(p^3)^2 + p_\perp^2 + m^2}$ and hence

$$\frac{\partial E_{\mathbf{p}}}{\partial p^3} = \frac{p^3}{E_{\mathbf{p}}}. \tag{A.76}$$

Differentiating Eq. (A.74) at fixed p^1, p^2 and using Eq. (A.76),

$$\frac{\partial p'^3}{\partial p^3} = \gamma\left(1 + \beta\,\frac{\partial E_{\mathbf{p}}}{\partial p^3}\right) = \gamma\left(1 + \beta\,\frac{p^3}{E_{\mathbf{p}}}\right) = \frac{\gamma\,(E_{\mathbf{p}} + \beta\,p^3)}{E_{\mathbf{p}}} = \frac{E'_{\mathbf{p}}}{E_{\mathbf{p}}}, \tag{A.77}$$

where the last equality is exactly Eq. (A.75). Therefore

$$d^3p' = \frac{\partial p'^3}{\partial p^3}\,d^3p = \frac{E'_{\mathbf{p}}}{E_{\mathbf{p}}}\,d^3p \qquad \Longrightarrow \qquad \frac{d^3p'}{E'_{\mathbf{p}}} = \frac{d^3p}{E_{\mathbf{p}}}. \tag{A.78}$$

The energy stretches the longitudinal momentum element by precisely the same factor $E'_{\mathbf{p}}/E_{\mathbf{p}}$ by which the energy itself grows, so the ratio $d^3p/E_{\mathbf{p}}$ is held fixed. Dividing by the boost-inert constant $(2\pi)^3 \cdot 2$ gives $d\Pi' = d\Pi$. Rotations are trivially included (they preserve $|\mathbf{p}|$, hence $E_{\mathbf{p}}$, and have unit Jacobian), and a general proper orthochronous transformation is a product of rotations and such boosts, so the result holds for the whole group—reproducing Theorem A.17 from the three-dimensional side.

With the measure proven invariant we can close the loop on the state normalization, showing that Eq. (A.67) is the unique companion convention that makes the completeness relation invariant.

Worked Example A.19 (Invariance of $2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})$ and the resolution of the identity)**.** Consider the one-particle completeness (resolution of the identity on the single-particle subspace),

$$\mathbf{1}_{\text{1-pt}} = \int d\Pi\;|p\rangle\,\langle p| = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\;|p\rangle\,\langle p|\,. \tag{A.79}$$

The left-hand side is a frame-independent operator (the projector onto one-particle states), and $d\Pi$ is invariant by Theorem A.17; for the equation to be consistent the integrand combination $|p\rangle\,\langle p|$ together with the normalization must be invariant too. Apply Eq. (A.79) to a state $|q\rangle$ and use the normalization Eq. (A.67):

$$|q\rangle = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\;|p\rangle\,\langle p|q\rangle = \int \frac{d^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\;|p\rangle\;(2\pi)^3\,2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q}) = \int d^3p\;|p\rangle\;\delta^{(3)}(\mathbf{p}-\mathbf{q}) = |q \tag{A.80}$$

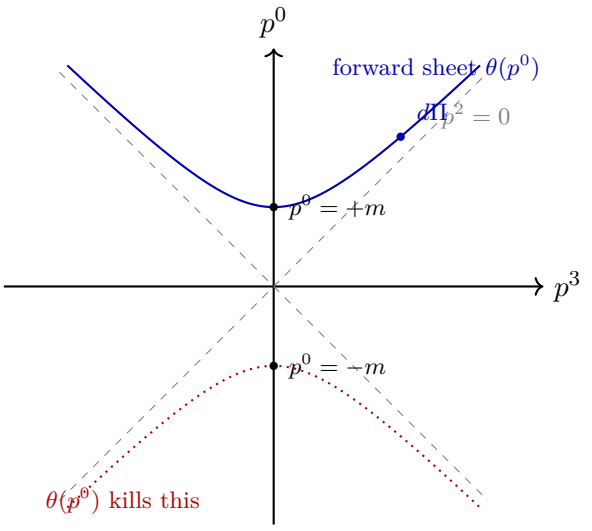


Figure A.1: The two-sheeted mass shell $p^2 = m^2$ in the (p^3, p^0) plane (transverse momenta suppressed), with the light cone $p^2 = 0$ as dashed asymptotes. The on-shell delta $\delta(p^2 - m^2)$ is supported on both hyperbolae; the factor $\theta(p^0)$ in Eq. (A.69) selects the upper, positive-energy sheet (solid blue) and discards the lower one (dotted red). The Lorentz-invariant measure $d\Pi$ of Eq. (A.68) is the induced measure on the forward sheet: a boost slides a point along the hyperbola, and Theorem A.18 shows the induced area element is preserved exactly.

The $2E_{\mathbf{p}}$ in the normalization is exactly what cancels the $1/(2E_{\mathbf{p}})$ in the measure, leaving the bare $\delta^{(3)}$ that sifts to $|q\rangle$. This is the structural reason the relativistic normalization carries the factor $2E_{\mathbf{p}}$: it is the reciprocal of the factor in the invariant measure, so that measure-times-normalization is the plain $d^3p\,\delta^{(3)}$, which is manifestly an invariant statement (the identity operator). Equivalently, since $d^3p/(2E_{\mathbf{p}})$ is invariant by Eq. (A.78), its distributional reciprocal $2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q})$—the object that integrates against it to the invariant unit—must transform oppositely and is therefore itself a Lorentz scalar. This is the invariance of Eq. (A.67) promised in Section A.3.3.

Figure A.1 renders the geometry: the on-shell delta lives on the two-sheeted hyperboloid $p^2 = m^2$, the $\theta(p^0)$ keeps the forward sheet, and a boost merely slides points along that sheet while preserving the induced measure.

Physical Insight

The combination that appears in physical formulas is always the invariant pair, never its two factors separately. In a decay rate or cross section the final-state sum is $\prod_f \int d\Pi_f$, with one $d\Pi$ per outgoing particle, and

an overall $(2\pi)^4\delta^{(4)}(\sum p)$ from the single leftover spacetime integral—one energy–momentum-conserving delta for the whole process, each $d\Pi$ carrying its own $(2\pi)^{-3}$ and $1/2E$. The $2E$'s in the relativistic normalization of the incoming states then cancel against the $1/2E$'s in the flux factor, and the entire apparatus collapses to a manifestly invariant number. The bookkeeping of this section is precisely the bookkeeping that makes that collapse exact, with no orphaned factor of 2π or $2E$ anywhere.

We have now fixed the Fourier convention Theorem A.12, proved the delta-representation Eq. (A.59) that powers the (2π) bookkeeping, established the relativistic state normalization Eq. (A.67), and derived the Lorentz-invariant measure $d\Pi$ Eq. (A.68) both from the covariant on-shell delta and by explicit boost. These are the exact conventions the next section needs: to write the momentum-space propagator it will Fourier-transform with the pair Eqs. (A.57) and (A.58), the poles of that propagator sit at the positive-energy roots $p^0 = +E_{\mathbf{p}}$ singled out by the $\theta(p^0)$ above, and choosing which way to route the contour around them is exactly the $i\epsilon$ prescription. With the measure and the delta conventions in hand, we turn to the analytic conventions—the $i\epsilon$ contour, the propagators, and the Wick rotation—that make the momentum-space theory well defined.

A.4 Analytic Conventions: the i-epsilon Prescription, Propagators, and Wick Rotation

The previous section fixed the Fourier transform, the (2π) bookkeeping that accompanies every momentum integral and every momentum-conserving delta function, the normalization of one-particle states, and the Lorentz-invariant phase-space measure $\mathrm{d}\Pi = \mathrm{d}^3p/[(2\pi)^3\,2E_{\mathbf{p}}]$ (Eq. (A.68)). Those conventions are purely kinematical: they tell us how to count momenta and how to weight on-shell states, but they say nothing about what happens when a denominator vanishes. The moment we write a propagator we meet exactly that situation. The free two-point function in momentum space is a ratio whose denominator is $p^2 - m^2$, and this vanishes precisely on the mass shell — the set of physical momenta we just learned to integrate over. A bare $1/(p^2 - m^2)$ is therefore not a function but an ill-defined expression with a pole sitting on the real axis of the energy integration. The content of this section is the single rule that resolves the ambiguity, the *Feynman $i\epsilon$ prescription*, together with everything that follows from it: the precise location of the displaced poles, the way the displacement enforces time ordering, the momentum-space propagators of the three fields the book carries, and the analytic continuation — the Wick rotation — that turns the oscillatory weight e^{iS} into the convergent Euclidean weight e^{-S_E}.

Throughout we use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ and the Fourier convention of the previous section; the complex-analytic machinery — contour integration, residues, analytic continuation, and the reading of $i\epsilon$ as a distributional boundary value — is the one developed in Chapter 5 and is used here without reproof.

A.4.1 The Feynman prescription and the displaced poles

We begin with the real scalar field, the cleanest place to see the mechanism. The free Feynman propagator is defined in position space as the time-ordered vacuum two-point function; its operator origin and the proof that it equals the expression below are the business of Chapter 9, and are not assumed here. What this appendix fixes is the analytic form and the sign of the infinitesimal that makes it well defined.

Definition A.20 (Feynman scalar propagator)**.** The momentum-space Feynman propagator of a real scalar of mass m is

$$\widetilde{D_F}(p) \;=\; \frac{i}{p^2 - m^2 + i\epsilon}, \qquad p^2 = (p^0)^2 - \mathbf{p}^2, \tag{A.81}$$

where $\epsilon \to 0^+$ is a positive infinitesimal taken to zero *after* all integrations. The position-space propagator is its Fourier transform,

$$D_F(x) \;=\; \int \frac{\mathrm{d}^4p}{(2\pi)^4}\, \frac{i\, e^{-ip\cdot x}}{p^2 - m^2 + i\epsilon}. \tag{A.82}$$

The symbol $+i\epsilon$ is not decoration. Its entire job is to move the poles of the integrand off the real p^0 axis, and the sign of the displacement is what selects the Feynman (time-ordered) Green function rather than the retarded, advanced, or any other Green function of the same operator $\Box + m^2$. We make the displacement explicit. Writing $E_{\mathbf{p}} \equiv +\sqrt{\mathbf{p}^2 + m^2} > 0$ for the positive on-shell energy, the denominator factorizes in p^0 as

$$p^2 - m^2 + i\epsilon = (p^0)^2 - E_{\mathbf{p}}^2 + i\epsilon. \tag{A.83}$$

Its zeros are the two roots of $(p^0)^2 = E_{\mathbf{p}}^2 - i\epsilon$. Taking the square root and expanding to first order in the infinitesimal,

$$p^0 = \pm\sqrt{E_{\mathbf{p}}^2 - i\epsilon} = \pm E_{\mathbf{p}}\sqrt{1 - \frac{i\epsilon}{E_{\mathbf{p}}^2}} = \pm E_{\mathbf{p}}\Big(1 - \frac{i\epsilon}{2E_{\mathbf{p}}^2} + \cdots\Big) = \pm\Big(E_{\mathbf{p}} - i\epsilon'\Big), \tag{A.84}$$

where $\epsilon' \equiv \epsilon/(2E_{\mathbf{p}}) > 0$ is again a positive infinitesimal; since only its sign matters we will continue to call it ϵ without ambiguity. The two poles therefore sit at

$$p^0_{(+)} = +E_{\mathbf{p}} - i\epsilon \quad \text{(fourth quadrant)}, \qquad p^0_{(-)} = -E_{\mathbf{p}} + i\epsilon \quad \text{(second quadrant)}. \tag{A.85}$$

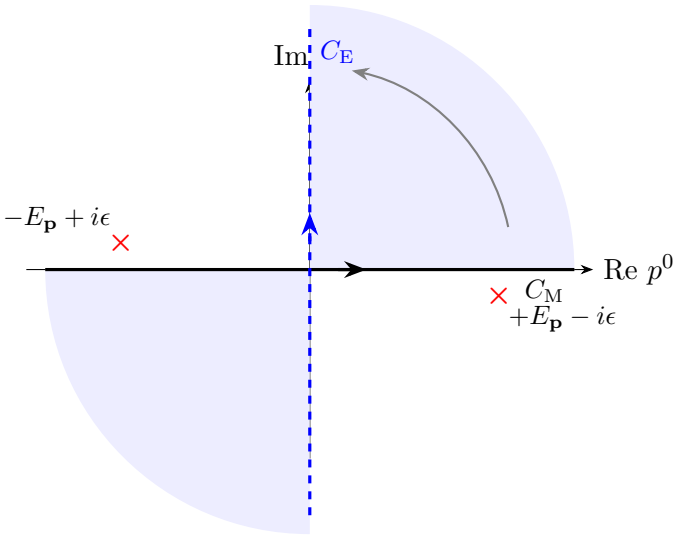


Figure A.2: The complex p^0 plane. The Feynman prescription Eq. (A.81) places the positive-energy pole at $+E_{\mathbf{p}} - i\epsilon$ (fourth quadrant) and the negative-energy pole at $-E_{\mathbf{p}} + i\epsilon$ (second quadrant). The Minkowski contour C_{M} runs along the real axis. Rotating it counterclockwise to the imaginary axis C_{E} (the Wick rotation, Section A.4.3) sweeps the shaded first and third quadrants, which contain no poles; the rotation is therefore legal precisely because of the $i\epsilon$ displacement.

The positive-energy pole is pushed *below* the real axis and the negative-energy pole *above* it. This is the geometric content of the Feynman prescription, drawn in Fig. A.2, and it is the only thing about $i\epsilon$ one ever needs to remember: *$+i\epsilon$ in the denominator means the positive-energy pole drops down and the negative-energy pole lifts up.*

The reason this particular displacement is the right one is that it makes Eq. (A.82) compute the *time-ordered* product. We show this by doing the p^0 integral with the residue theorem; this is the calculation that ties the analytic convention to physics, so we carry every step.

Worked Example A.21 (The Feynman contour enforces time ordering)**.** We evaluate Eq. (A.82) by closing the p^0 contour. Separate the spatial and temporal pieces of $p \cdot x = p^0 x^0 - \mathbf{p} \cdot \mathbf{x}$:

$$D_F(x) = \int \frac{\mathrm{d}^3 p}{(2\pi)^4}\, e^{i\mathbf{p}\cdot\mathbf{x}} \int_{-\infty}^{\infty} \mathrm{d}p^0\, \frac{i\, e^{-ip^0 x^0}}{(p^0 - E_{\mathbf{p}} + i\epsilon)(p^0 + E_{\mathbf{p}} - i\epsilon)}, \tag{A.86}$$

where we have used the factorization implied by Eq. (A.85). The exponential $e^{-ip^0 x^0}$ controls which half-plane we may close in: writing $p^0 = \mathrm{Re}\, p^0 + i\, \mathrm{Im}\, p^0$, the modulus is $e^{(\mathrm{Im}\, p^0) x^0}$, which is bounded on a large semicircle only if $(\mathrm{Im}\, p^0)\, x^0 \le 0$ there.

Case $x^0 > 0$. We must close in the lower half-plane, $\mathrm{Im}\, p^0 < 0$, where the arc contribution vanishes by Jordan's lemma (Chapter 5). The contour is then clockwise and encircles only the positive-energy pole $p^0 = E_{\mathbf{p}} - i\epsilon$. By the residue theorem,

$$\int_{-\infty}^{\infty} \mathrm{d}p^0 (\cdots) = -2\pi i \, \mathrm{Res}_{p^0 = E_{\mathbf{p}}} \frac{i\, e^{-ip^0 x^0}}{(p^0 - E_{\mathbf{p}})(p^0 + E_{\mathbf{p}})} = -2\pi i \cdot \frac{i\, e^{-iE_{\mathbf{p}} x^0}}{2E_{\mathbf{p}}} = \frac{\pi}{E_{\mathbf{p}}}\, e^{-iE_{\mathbf{p}} x^0}, \tag{A.87}$$

the leading $-$ sign coming from the clockwise orientation. Inserting this into Eq. (A.86) collapses the $(2\pi)^4$ to $(2\pi)^3$ and produces the invariant measure of the previous section,

$$D_F(x)\big|_{x^0>0} = \int \frac{\mathrm{d}^3 p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{-iE_{\mathbf{p}} x^0 + i\mathbf{p}\cdot\mathbf{x}} = \int \mathrm{d}\Pi\; e^{-ip\cdot x}\Big|_{p^0 = E_{\mathbf{p}}}. \tag{A.88}$$

This is the positive-frequency Wightman function, the amplitude for a disturbance created at the origin to propagate *forward* to x.

Case $x^0 < 0$. Now we close in the upper half-plane, counterclockwise, and pick up only the negative-energy pole $p^0 = -E_{\mathbf{p}} + i\epsilon$. The orientation is now $+2\pi i$, and the residue at $p^0 = -E_{\mathbf{p}}$ gives $i\, e^{+iE_{\mathbf{p}} x^0}/(-2E_{\mathbf{p}})$, so

$$D_F(x)\big|_{x^0<0} = \int \frac{\mathrm{d}^3 p}{(2\pi)^3\, 2E_{\mathbf{p}}}\, e^{+iE_{\mathbf{p}} x^0 + i\mathbf{p}\cdot\mathbf{x}} = \int \mathrm{d}\Pi\; e^{+ip\cdot x}\Big|_{p^0 = E_{\mathbf{p}}}, \tag{A.89}$$

where in the last step we relabeled $\mathbf{p} \to -\mathbf{p}$ in the measure (which is even). This is the positive-frequency function propagating in the opposite time sense. Combining the two cases with step functions,

$$D_F(x) = \theta(x^0) \int \mathrm{d}\Pi\, e^{-ip\cdot x} + \theta(-x^0) \int \mathrm{d}\Pi\, e^{+ip\cdot x} \quad (p^0 = E_{\mathbf{p}}), \tag{A.90}$$

which is exactly the time-ordered structure: the field created earlier stands to the right. The identification of Eq. (A.90) with $\langle 0|\, \mathcal{T}\phi(x)\phi(0)\, |0\rangle$ at the operator level is carried out in Chapter 9; here we have shown the purely analytic fact that *the sign of* $i\epsilon$ *in Eq. (A.81) is what assigns the two poles to the two half-planes so that the contour reconstructs the time ordering.* Had we chosen $p^2 - m^2 - i\epsilon$ instead, both poles would have moved the other way and the construction would have produced the anti-time-ordered (Dyson) propagator; pushing both poles below the axis gives the retarded function, both above the advanced one.

It is worth recording the distributional face of the same prescription, because it is the form that appears whenever one takes an imaginary part or a discontinuity. The Sokhotski–Plemelj identity established in Chapter 5,

$$\frac{1}{x \pm i\epsilon} = \mathrm{PV}\,\frac{1}{x} \mp i\pi\, \delta(x), \tag{A.91}$$

exhibits $i\epsilon$ as a boundary value of an analytic function from the upper or lower half-plane. Applied to Eq. (A.81) with $x = p^2 - m^2$,

$$\frac{i}{p^2 - m^2 + i\epsilon} = i\ \mathrm{PV}\,\frac{1}{p^2 - m^2} + \pi\,\delta(p^2 - m^2), \tag{A.92}$$

so the on-shell, delta-function part of the propagator is real and supported exactly on the mass shell. This is the bridge back to the invariant measure fixed in the previous section, as the next worked example makes precise.

Worked Example A.22 (The on-shell delta and the invariant measure)**.** The positive-energy mass-shell delta of Eq. (A.92) is the covariant repackaging of the measure $\mathrm{d}\Pi$. By the delta-composition rule Theorem A.15, applied to $f(p^0) = (p^0)^2 - E_{\mathbf{p}}^2$ exactly as in the measure derivation of the previous section, recall (Eq. (A.71))

$$\delta(p^2 - m^2) = \frac{1}{2E_{\mathbf{p}}}\Big[\delta(p^0 - E_{\mathbf{p}}) + \delta(p^0 + E_{\mathbf{p}})\Big]. \tag{A.93}$$

The factor $\theta(p^0)$ keeps only the first term, so integrating over p^0,

$$\int \frac{\mathrm{d}^4p}{(2\pi)^4}\,(2\pi)\,\delta(p^2 - m^2)\,\theta(p^0)\,g(p) = \int \frac{\mathrm{d}^3p}{(2\pi)^3\,2E_{\mathbf{p}}}\,g(p)\Big|_{p^0 = E_{\mathbf{p}}} = \int \mathrm{d}\Pi\; g(p), \tag{A.94}$$

which reproduces, term by term with its (2π) factors, the Lorentz-invariant phase-space measure of the previous section. The same $i\epsilon$ that orders time in Theorem A.21 thus also carries, in its imaginary part, the on-shell measure that weights external states; the two halves of the analytic convention are one object.

A.4.2 The propagators of the carried fields

With the scalar settled, the propagators of the Dirac and gauge fields follow the same prescription: the same $+i\epsilon$, the same pole displacement, with a numerator dictated by the field's tensor or spinor structure. We state them in the analytic conventions of this appendix; their derivation from the free field operators or the Gaussian path integral is owed by Chapters 9 and 10, and for the photon by Chapter 13.

The Dirac propagator. For a Dirac field of mass m the momentum-space propagator is

$$\widetilde{S_F}(p) = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} = \frac{i}{\not{p} - m + i\epsilon}, \tag{A.95}$$

with $\not{p} \equiv \gamma^\mu p_\mu$ and the gamma matrices of Chapter C. The two forms are equal: rationalizing the second by multiplying numerator and denominator by $(\not{p} + m)$ and using

$$\not{p}\not{p} = p_\mu p_\nu\,\gamma^\mu\gamma^\nu = p_\mu p_\nu\,\tfrac{1}{2}\{\gamma^\mu, \gamma^\nu\} = p_\mu p_\nu\,\eta^{\mu\nu} = p^2, \tag{A.96}$$

where the second equality keeps only the symmetric part of $p_\mu p_\nu$ and the third invokes the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ of Chapter C, gives $(\not{p}-m)(\not{p}+m) = \not{p}\not{p} - m^2 = p^2 - m^2$. The denominator is therefore the *same* scalar denominator $p^2 - m^2 + i\epsilon$, so the pole positions Eq. (A.85) and the contour analysis of Theorem A.21 carry over verbatim; only the numerator $i(\not{p} + m)$, which projects onto the spin content, is new. The $i\epsilon$ rides on the scalar denominator exactly as before.

The photon propagator. The photon is massless and gauge fields require a gauge-fixing choice before they have a propagator at all; the R_ξ family of covariant gauges, derived in Chapter 13, is parametrized by a real number ξ. In that family the momentum-space photon propagator is

$$\widetilde{D}_{\mu\nu}(k) = \frac{-i}{k^2 + i\epsilon}\left[\eta_{\mu\nu} - (1-\xi)\,\frac{k_\mu k_\nu}{k^2 + i\epsilon}\right]. \tag{A.97}$$

The massless pole sits at $k^2 = 0$, displaced to $k^2 = -i\epsilon$ by the same prescription, so that in the energy plane the poles are at $k^0 = \pm(|\mathbf{k}| - i\epsilon)$ — the $m \to 0$ limit of Eq. (A.85). The default for explicit computation in this book is *Feynman gauge*, $\xi = 1$, in which the longitudinal $k_\mu k_\nu$ term drops and the propagator collapses to the diagonal form

$$\widetilde{D}_{\mu\nu}(k)\big|_{\xi=1} = \frac{-i\,\eta_{\mu\nu}}{k^2 + i\epsilon}. \tag{A.98}$$

Landau gauge is $\xi = 0$, where the bracket becomes the transverse projector $\eta_{\mu\nu} - k_\mu k_\nu / k^2$. Physical (gauge-invariant) quantities are independent of ξ; the propagator itself is not, which is why the gauge must be stated whenever an off-shell Green function is written.

The gluon propagator. For the non-abelian gauge field the gauge-fixing procedure (Faddeev–Popov, Chapter 26) produces the same Lorentz structure carrying in addition an adjoint color delta,

$$\widetilde{D}^{ab}_{\mu\nu}(k) = \frac{-i\,\delta^{ab}}{k^2 + i\epsilon}\left[\eta_{\mu\nu} - (1-\xi)\,\frac{k_\mu k_\nu}{k^2 + i\epsilon}\right], \tag{A.99}$$

accompanied by a ghost propagator $i\delta^{ab}/(k^2 + i\epsilon)$ of its own. These are stated here only to fix the analytic conventions and the gauge default; the machinery that produces them, and the ghosts that accompany Eq. (A.99), is built in Chapter 26.

Notation Watch

The propagators Eqs. (A.81), (A.95), (A.98) and (A.99) are the momentum-space propagators of ϕ^4 theory, QED, and Yang–Mills theory. They are the exact expressions used without restatement in the amplitude computations of Chapter 14 (ϕ^4), Chapter 17 (QED tree level), and Chapter 19 (one-loop QED and the gauge propagators). No layer is added here; this section only certifies the signs and the $i\epsilon$ they all share.

A.4.3 Wick rotation and the Euclidean conventions

The $i\epsilon$ prescription does more than make individual propagators well defined: it makes the entire perturbative expansion analytically continuable to imaginary time, where the oscillatory Minkowski weight e^{iS} becomes a convergent Euclidean weight e^{-S_E}. This continuation, the *Wick rotation*, is the analytic foundation of the path-integral evaluations of Chapter 10 and of every master integral in Chapter D, which are computed in Euclidean space and continued back. We fix it here once.

Definition A.23 (Wick rotation)**.** The Wick rotation continues the real time coordinate into the lower-half complex plane,

$$t = x^0 = -i\tau, \qquad \tau \in \mathbb{R} \text{ (Euclidean time)}, \tag{A.100}$$

and correspondingly rotates the energy onto the imaginary axis,

$$p^0 = i\, p^0_E, \qquad p^0_E \in \mathbb{R}, \tag{A.101}$$

the spatial coordinates and momenta being unchanged. The Euclidean four-vectors $x_E = (\tau, \mathbf{x})$ and $p_E = (p^0_E, \mathbf{p})$ carry the all-plus metric $\delta_{\mu\nu}$, with $p_E^2 = (p^0_E)^2 + \mathbf{p}^2$ and $x_E^2 = \tau^2 + \mathbf{x}^2$.

The choices in Eqs. (A.100) and (A.101) are mutually consistent: the Minkowski phase factor continues to a real Euclidean one,

$$p^0 x^0 = (ip^0_E)(-i\tau) = p^0_E\, \tau, \tag{A.102}$$

and the Minkowski square continues to minus the Euclidean square,

$$p^2 = (p^0)^2 - \mathbf{p}^2 = (ip^0_E)^2 - \mathbf{p}^2 = -(p^0_E)^2 - \mathbf{p}^2 = -p_E^2. \tag{A.103}$$

Geometrically, Eq. (A.101) is the counterclockwise rotation of the p^0 integration contour from the real axis to the imaginary axis drawn in Fig. A.2. The rotation sweeps the first and third quadrants of the p^0 plane. By Eq. (A.85) the poles live in the *second and fourth* quadrants, so the rotating contour never crosses a

pole, and provided the arcs at infinity vanish (which they do for the convergent integrands of a regularized theory, Chapter 5) Cauchy's theorem guarantees the value of the integral is unchanged. *This is the deep reason the Feynman $i\epsilon$ is the right prescription for analytic continuation:* its specific sign is exactly what clears the rotation path of singularities. The retarded or advanced prescriptions, with both poles in the same half-plane, would obstruct the rotation in one of the two swept quadrants.

We now carry out the sign bookkeeping that the rotation induces in the action, term by term, for the carried scalar; the same arithmetic applies verbatim to any local Lagrangian.

Worked Example A.24 ($iS \to -S_E$ by explicit substitution)**.** Take the ϕ^4 Lagrangian density (the model of Chapter 6) in the mostly-minus metric,

$$\mathcal{L} = \tfrac{1}{2}(\partial_0\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - V(\phi), \qquad V(\phi) = \frac{\lambda}{4!}\phi^4, \tag{A.104}$$

so that $S = \int \mathrm{d}t\, \mathrm{d}^3x\, \mathcal{L}$. We substitute Eq. (A.100). The time differential and the time derivative continue as

$$\mathrm{d}t = -i\, \mathrm{d}\tau, \qquad \partial_0 = \frac{\partial}{\partial t} = \frac{1}{-i}\frac{\partial}{\partial \tau} = i\, \partial_\tau, \tag{A.105}$$

so the kinetic time term flips sign, $(\partial_0\phi)^2 = (i\partial_\tau\phi)^2 = -(\partial_\tau\phi)^2$, while the spatial gradient, the mass term, and the potential are untouched. Therefore

$$\begin{aligned} iS &= i\int(-i\,\mathrm{d}\tau)\,\mathrm{d}^3x\left[-\tfrac{1}{2}(\partial_\tau\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - V(\phi)\right] \\ &= \int \mathrm{d}\tau\,\mathrm{d}^3x\left[-\tfrac{1}{2}(\partial_\tau\phi)^2 - \tfrac{1}{2}(\nabla\phi)^2 - \tfrac{1}{2}m^2\phi^2 - V(\phi)\right] \\ &= -\int \mathrm{d}^4x_E\left[\tfrac{1}{2}(\partial_\tau\phi)^2 + \tfrac{1}{2}(\nabla\phi)^2 + \tfrac{1}{2}m^2\phi^2 + V(\phi)\right] \;\equiv\; -S_E, \end{aligned} \tag{A.106}$$

where the prefactor $i\cdot(-i) = 1$ in the first line, and $\mathrm{d}^4x_E = \mathrm{d}\tau\,\mathrm{d}^3x$. Every term in the brackets is now *added* with a $+$ sign: the kinetic term that was $+\tfrac{1}{2}(\partial_0\phi)^2$ and the gradient that was $-\tfrac{1}{2}(\nabla\phi)^2$ have both become $+\tfrac{1}{2}(\partial^E_\mu\phi)^2$, and the potential, which sat in $\mathcal{L}$ with a relative minus, now sits in S_E with a plus. The Euclidean action is the all-plus quadratic form

$$\boxed{S_E[\phi] = \int \mathrm{d}^4x_E\left[\tfrac{1}{2}\,(\partial^E_\mu\phi)(\partial^E_\mu\phi) + \tfrac{1}{2}m^2\phi^2 + V(\phi)\right], \qquad e^{iS} \longrightarrow e^{-S_E}.} \tag{A.107}$$

Because the gradient and mass pieces are manifestly nonnegative and V is bounded below, $S_E \geq 0$ up to an additive constant, so e^{-S_E} is a genuine, decaying statistical weight rather than a pure phase. This is precisely the property that makes the Euclidean path integral convergent and the Minkowski

one only conditionally defined; it is also why lattice and constructive treatments work in Euclidean signature. (Repeated Euclidean indices in Eq. (A.107) are summed with $\delta_{\mu\nu}$, so up/down placement no longer carries meaning.)

The same continuation acts on the propagator, and turning the analytic crank on Eq. (A.81) delivers the Euclidean propagator without any new input.

Worked Example A.25 (The Euclidean propagator from the Minkowski one)**.** Start from the position-space Minkowski propagator Eq. (A.82) and substitute the rotation Eqs. (A.100) and (A.101). The measure rotates as

$$\int \frac{\mathrm{d}^4 p}{(2\pi)^4} = \int \frac{\mathrm{d}p^0\, \mathrm{d}^3 p}{(2\pi)^4} \;\longrightarrow\; \int \frac{i\,\mathrm{d}p^0_E\, \mathrm{d}^3 p}{(2\pi)^4} = i \int \frac{\mathrm{d}^4 p_E}{(2\pi)^4}, \tag{A.108}$$

using $\mathrm{d}p^0 = i\,\mathrm{d}p^0_E$ from Eq. (A.101). The denominator rotates by Eq. (A.103), and crucially the $+i\epsilon$ is no longer needed once we are on the imaginary axis: the rotated denominator

$$p^2 - m^2 + i\epsilon \;\longrightarrow\; -p_E^2 - m^2 = -(p_E^2 + m^2) \tag{A.109}$$

never vanishes for real Euclidean momentum, since $p_E^2 + m^2 \geq m^2 > 0$ for $m \neq 0$ (and for $m = 0$ only at the single point $p_E = 0$, a set of measure zero). The infinitesimal has done its job — it kept the contour off the poles *during* the rotation — and may be dropped *after* it. Assembling the pieces, the overall i from Eq. (A.81), the i from the measure Eq. (A.108), and the -1 from the denominator Eq. (A.109) combine to

$$i \cdot i \cdot \frac{1}{-(p_E^2 + m^2)} = \frac{i^2 \cdot (-1)}{p_E^2 + m^2} = \frac{1}{p_E^2 + m^2}, \tag{A.110}$$

so that the propagator continues to the manifestly positive, real Euclidean kernel

$$D_F(x) = \int \frac{\mathrm{d}^4 p_E}{(2\pi)^4} \frac{e^{ip_E \cdot x_E}}{p_E^2 + m^2}, \qquad \boxed{\widetilde{D}_E(p_E) = \frac{1}{p_E^2 + m^2}\,,} \tag{A.111}$$

with $p_E \cdot x_E = p^0_E \tau + \mathbf{p} \cdot \mathbf{x}$ the Euclidean dot product. The result is the inverse of the Euclidean quadratic operator $-\partial_E^2 + m^2$ read off from Eq. (A.107), as it must be: the Gaussian weight e^{-S_E} has covariance $(-\partial_E^2 + m^2)^{-1}$, whose Fourier transform is exactly Eq. (A.111). The Dirac and gauge propagators continue the same way — numerators rotate by the same rules and the scalar denominator becomes $1/(p_E^2 + m^2)$ (or $1/p_E^2$ for the gauge field) — so the whole propagator catalogue of Section A.4.2 has a Euclidean image obtained by this single substitution.

The Euclidean propagator Eq. (A.111) is the convergent object on which dimensional regularization operates: in Chapter D every master loop integral is

evaluated against these denominators in $d = 4 - 2\varepsilon$ Euclidean dimensions, after the Wick rotation legitimized here, and continued back to Minkowski signature at the end. The boundary value $i\epsilon$ of Eq. (A.81) and the all-plus weight e^{-S_E} of Eq. (A.107) are thus two presentations of one analytic structure, related by the rotation of Fig. A.2.

These are the analytic conventions the rest of the book runs on. The propagator normalizations fixed here — the overall factor of i, the sign of the $i\epsilon$, and the gauge default $\xi = 1$ — together with the sign rule $e^{iS} \to e^{-S_E}$ for the path-integral weight, are exactly what the next section assumes when it builds the generating functionals $Z[J]$, $W[J]$, and $\Gamma[\phi]$: there the source-coupled integrand is weighted by precisely this e^{iS} (or e^{-S_E} in Euclidean signature), and the propagators above are the free two-point functions it generates by functional differentiation.

A.5 The Action, Generating Functionals, and Amplitude Normalization

The previous section fixed the analytic spine of the theory: the Feynman $i\epsilon$, the momentum-space propagator $i/(p^2 - m^2 + i\epsilon)$, and the Wick rotation that turns the Minkowski weight e^{iS} into the Euclidean weight e^{-S_E} (Section A.4). Those choices are not independent of how we couple sources to fields, how we package correlators into generating functionals, and how we strip an invariant amplitude out of an S-matrix element. A single sign placed inconsistently here propagates into every Feynman rule, every Legendre transform, and every cross section in the book. This section therefore freezes, once and for all, the sign of the source coupling, the definitions of the three generating functionals $Z[J]$, $W[J]$, and $\Gamma[\varphi_{\mathrm{cl}}]$ together with their Legendre relations, the split $\mathbf{S} = 1 + iT$ with its invariant amplitude $\mathcal{M}$, and the master formulae for cross sections and decay rates. Everything is fixed so that the $-i\lambda$ scalar vertex and the $i/(p^2 - m^2 + i\epsilon)$ propagator are mutually consistent, and so that the relativistic state normalization fixed in the Fourier section feeds straight into the phase-space measure.

We import three classes of results rather than reprove them. The functional calculus—functional derivatives, Gaussian functional integrals, and the Legendre transform as an operation on functionals—is developed in Chapter 4. The *physics* of generating functionals, in particular the linked-cluster theorem that $\ln Z$ generates connected correlators and the statement that Γ generates one-particle-irreducible (1PI) vertices, is developed in Chapter 10. The physics of the S-matrix, LSZ reduction, and the cross-section formula is developed in Chapters 14 and 16; here we fix only the normalization conventions those chapters then use. What this section does that no other does is nail the *signs and factors*

of i into a single mutually consistent set, with every intermediate step shown.

A.5.1 The source coupling and the partition functional

Let $S[\phi] = \int \mathrm{d}^4x\, \mathcal{L}(\phi, \partial_\mu \phi)$ be the (real) classical action of a generic field, which we take to be a single real scalar ϕ for definiteness; the bookkeeping is identical for fermions and gauge fields once the appropriate Grassmann or index structure is carried along. We couple an external classical source $J(x)$ to the field with the sign

$$S[\phi] \;\longmapsto\; S[\phi] + \int \mathrm{d}^4x\, J(x)\,\phi(x), \tag{A.112}$$

that is, with a $+J\phi$ in the action. This is a genuine convention, and it is the one this book uses everywhere; the opposite sign is equally consistent internally but would flip the sign in the Legendre relation derived below. We record it explicitly because the next section must place the sign of the covariant derivative consistently with it.

Definition A.26 (Generating functional of correlators)**.** The *generating functional* (or partition functional) of the theory in the presence of the source J is the Minkowski path integral

$$Z[J] \;\equiv\; \int \mathcal{D}\phi\; \exp\Big(i\, S[\phi] + i \int \mathrm{d}^4x\, J(x)\,\phi(x) \Big), \tag{A.113}$$

with the measure $\mathcal{D}\phi$ and its normalization fixed in Chapters 4 and 10. We adopt the normalization

$$Z[0] \;=\; \langle 0 | 0 \rangle \;=\; 1, \tag{A.114}$$

so that vacuum bubbles divide out and $Z[J]/Z[0]$ and $Z[J]$ coincide. Where it matters we write $Z[J]/Z[0]$ to make the cancellation manifest.

The exponent in Eq. (A.113) carries the same i that multiplies S in the Minkowski weight e^{iS}; the source term is inside the same i. This is what makes a single functional derivative bring down a clean factor of $i\phi$. Indeed, varying Eq. (A.113) with respect to J,

$$\frac{\delta}{\delta J(x)}\, Z[J] \;=\; \int \mathcal{D}\phi\; \big(i\,\phi(x)\big)\, \exp\Big(iS + i \int J\phi \Big), \tag{A.115}$$

so that the operator that extracts a bare field insertion is $-i\,\delta/\delta J$. Iterating and setting $J = 0$,

$$\langle 0 |\, \mathcal{T}\, \phi(x_1) \cdots \phi(x_n)\, | 0 \rangle \;=\; \frac{1}{Z[0]} \prod_{k=1}^{n} \Big(-\, i\, \frac{\delta}{\delta J(x_k)} \Big)\, Z[J] \,\Big|_{J=0}, \tag{A.116}$$

the time ordering $\mathcal{T}$ being automatic in the path-integral representation. Equation (A.116) is the operational content of $Z[J]$: it is the generating functional of the full (connected and disconnected) time-ordered correlators, with each insertion carrying one factor of $-i$.

Notation Watch

The $+J\phi$ sign in Eq. (A.112) and the placement of the source *inside* the i in Eq. (A.113) together force the $-i\,\delta/\delta J$ extraction rule (A.116) and, below, the relation $\delta\Gamma/\delta\varphi_{\rm cl} = -J$. A book that put $e^{iS}e^{\int J\phi}$ (source outside the i) would instead extract insertions with a bare $\delta/\delta J$ and would land on $\delta\Gamma/\delta\varphi_{\rm cl} = +iJ$. We never use that variant.

For completeness we record the Euclidean counterpart, since the analytic section ended on the $e^{iS} \to e^{-S_E}$ rotation. After the Wick rotation of Section A.4, the source convention that matches Eqs. (A.112) to (A.113) is

$$Z_E[J] \;=\; \int \mathcal{D}\phi\; \exp\Big(-S_E[\phi] + \int \mathrm{d}^4x_E\, J(x)\,\phi(x)\Big), \qquad W_E[J] \;=\; \ln Z_E[J], \tag{A.117}$$

with no factor of i anywhere: the Euclidean weight is a genuine (positive) probability weight and $W_E = \ln Z_E$ is the free energy. The factor of $-i$ in the Minkowski generator below is precisely the residue of the analytic continuation that takes $\ln Z_E \to -i \ln Z$.

A.5.2 The connected generator $W[J] = -i \ln Z[J]$

The full correlators (A.116) contain disconnected pieces—products of lower correlators in which clusters of points do not communicate. For scattering and for the loop expansion one wants the *connected* correlators, in which every insertion is linked to every other. The linked-cluster theorem (Chapter 10) states that these are generated by the logarithm of Z. With our factors of i the correctly normalized generator is

Definition A.27 (Connected generating functional)**.**

$$W[J] \;\equiv\; -\,i\,\ln Z[J], \qquad \text{equivalently} \qquad Z[J] \;=\; e^{\,i\,W[J]}. \tag{A.118}$$

The $-i$ is fixed, not free, once Eqs. (A.113) and (A.116) are fixed. We prove this by computing the first two derivatives explicitly; the sign of the second is the load-bearing check, because it is the one that must reproduce the i in the Feynman propagator.

Introduce the normalized expectation value in the presence of the source,

$$\langle\,\mathcal{O}\,\rangle_J \;\equiv\; \frac{1}{Z[J]}\int \mathcal{D}\phi\;\mathcal{O}\,\exp\Big(iS + i\int J\phi\Big), \tag{A.119}$$

which reduces to the vacuum correlator at $J = 0$. From Eq. (A.115),

$$\frac{\delta W}{\delta J(x)} \;=\; -i\,\frac{\delta}{\delta J(x)}\ln Z[J] \;=\; -\,i\,\frac{1}{Z[J]}\frac{\delta Z}{\delta J(x)} \;=\; -\,i\,\big(i\,\langle\phi(x)\rangle_J\big) \;=\; \langle\phi(x)\rangle_J \;\equiv\; \varphi_{\rm cl}(x). \tag{A.120}$$

The first derivative of W is therefore the *mean field* $\varphi_{\mathrm{cl}}(x)$—the expectation value of ϕ in the state dragged around by the source. The two factors of i have cancelled cleanly, which is the whole reason for the $-i$ in Eq. (A.118); this is the connected one-point function and it vanishes at $J = 0$ in a theory with no tadpole.

For the second derivative, differentiate Eq. (A.120) once more. Writing $\delta Z/\delta J(x) = i\,Z\,\langle\phi(x)\rangle_J$ from Eq. (A.115), and $\delta^2 Z/\delta J(x)\delta J(y) = -\,Z\,\langle\mathcal{T}\phi(x)\phi(y)\rangle_J$ (two derivatives each bringing an $i\phi$, giving $i^2 = -1$),

$$\begin{aligned}
\frac{\delta^2 W}{\delta J(x)\,\delta J(y)} &= -\,i\,\frac{\delta}{\delta J(y)}\left(\frac{1}{Z}\frac{\delta Z}{\delta J(x)}\right) \\
&= -\,i\left[\frac{1}{Z}\,\frac{\delta^2 Z}{\delta J(x)\,\delta J(y)} - \frac{1}{Z^2}\,\frac{\delta Z}{\delta J(x)}\,\frac{\delta Z}{\delta J(y)}\right] \\
&= -\,i\Big[-\langle\mathcal{T}\phi(x)\phi(y)\rangle_J - \big(i\langle\phi(x)\rangle_J\big)\big(i\langle\phi(y)\rangle_J\big)\Big] \\
&= -\,i\Big[-\langle\mathcal{T}\phi(x)\phi(y)\rangle_J + \langle\phi(x)\rangle_J\langle\phi(y)\rangle_J\Big] \\
&= i\left[\langle\mathcal{T}\phi(x)\phi(y)\rangle_J - \langle\phi(x)\rangle_J\langle\phi(y)\rangle_J\right] \;=\; i\,\langle\mathcal{T}\phi(x)\phi(y)\rangle_{J,\,c}.
\end{aligned} \tag{A.121}$$

The bracket is exactly the connected two-point function. At $J = 0$ this reads $\delta^2 W/\delta J(x)\delta J(y)\big|_0 = i\,G_c(x,y)$, with G_c the connected propagator; in momentum space the free-theory value is $i\cdot$ (the connected propagator), which is the $i/(p^2 - m^2 + i\epsilon)$ of the analytic section. Had we chosen $W = +i\ln Z$ or $W = \ln Z$, the sign or the factor of i here would be wrong, and the propagator extracted from W'' would not match the Feynman propagator. This is the proof that the constant in Eq. (A.118) must be $-i$.

The general statement follows by the same combinatorics. The linked-cluster theorem (Chapter 10) guarantees that $\ln Z$ generates only the connected correlators; tracking the factors of i exactly as above gives

$$\langle\,\mathcal{T}\,\phi(x_1)\cdots\phi(x_n)\,\rangle_c \;=\; i\,(-i)^{\,n}\,\frac{\delta^n W}{\delta J(x_1)\cdots\delta J(x_n)}\bigg|_{J=0}. \tag{A.122}$$

One checks the prefactor against the two cases just computed: for $n = 1$, $i(-i)^1 = 1$, reproducing Eq. (A.120); for $n = 2$, $i(-i)^2 = -i$, so that $\delta^2 W/\delta J\delta J = i\,G_c$, reproducing Eq. (A.121). Equation (A.122) is the connected analogue of Eq. (A.116).

A.5.3 The effective action and the Legendre relations

The third generating functional is the effective action Γ, the Legendre transform of W with respect to its conjugate pair $(J, \varphi_{\mathrm{cl}})$. Its derivatives are the 1PI vertex

functions (Chapter 10); for the purposes of this appendix the point is to fix the Legendre conventions and prove the sign of the quantum equation of motion.

The relation $\varphi_{\mathrm{cl}}(x) = \delta W/\delta J(x)$ of Eq. (A.120) associates to each source profile J a mean field φ_{cl}. Assume this map is invertible (it is, order by order in the loop expansion, with the tree-level inverse being the classical equation of motion), so that one may regard $J = J[\varphi_{\mathrm{cl}}]$ as a functional of the mean field.

Definition A.28 (Effective action)**.**

$$\Gamma[\varphi_{\mathrm{cl}}] \;\equiv\; W[J] - \int \mathrm{d}^4x\, J(x)\,\varphi_{\mathrm{cl}}(x), \qquad \varphi_{\mathrm{cl}}(x) = \frac{\delta W}{\delta J(x)}, \tag{A.123}$$

where on the right J is understood as the source $J[\varphi_{\mathrm{cl}}]$ that produces the given mean field.

Proposition A.29 (Legendre relations)**.** *With the conventions Eqs. (A.112), (A.118) and (A.123),*

$$\frac{\delta W}{\delta J(x)} \;=\; \varphi_{\mathrm{cl}}(x), \qquad \frac{\delta \Gamma}{\delta \varphi_{\mathrm{cl}}(x)} \;=\; -\,J(x). \tag{A.124}$$

In particular, in the absence of sources the mean field obeys the quantum equation of motion $\delta\Gamma/\delta\varphi_{\mathrm{cl}} = 0$*.*

Proof. The first relation is Eq. (A.120). For the second, differentiate Eq. (A.123) with respect to $\varphi_{\mathrm{cl}}(y)$, treating $J = J[\varphi_{\mathrm{cl}}]$ as a functional of the mean field and applying the chain rule for functionals (Chapter 4):

$$\frac{\delta \Gamma}{\delta \varphi_{\mathrm{cl}}(y)} = \int \mathrm{d}^4x\, \frac{\delta W}{\delta J(x)}\,\frac{\delta J(x)}{\delta \varphi_{\mathrm{cl}}(y)} \;-\; J(y) \;-\; \int \mathrm{d}^4x\, \frac{\delta J(x)}{\delta \varphi_{\mathrm{cl}}(y)}\,\varphi_{\mathrm{cl}}(x). \tag{A.125}$$

The explicit $-J(y)$ comes from differentiating the integrand's explicit dependence on $\varphi_{\mathrm{cl}}(y)$ in the term $-\int J\varphi_{\mathrm{cl}}$. Now substitute $\delta W/\delta J(x) = \varphi_{\mathrm{cl}}(x)$ in the first integral. The first and third integrals are then identical except for sign and cancel:

$$\int \mathrm{d}^4x\; \varphi_{\mathrm{cl}}(x)\,\frac{\delta J(x)}{\delta \varphi_{\mathrm{cl}}(y)} - \int \mathrm{d}^4x\, \frac{\delta J(x)}{\delta \varphi_{\mathrm{cl}}(y)}\,\varphi_{\mathrm{cl}}(x) \;=\; 0, \tag{A.126}$$

leaving $\delta\Gamma/\delta\varphi_{\mathrm{cl}}(y) = -J(y)$, which is the second relation in Eq. (A.124). Setting $J = 0$ gives the stationarity condition $\delta\Gamma/\delta\varphi_{\mathrm{cl}} = 0$. □

The minus sign in $\delta\Gamma/\delta\varphi_{\mathrm{cl}} = -J$ is the direct consequence of the $+J\phi$ source convention (A.112): the Legendre transform conjugates J to φ_{cl} with the sign that the chosen source coupling dictates. Differentiating the two relations in Eq. (A.124) once more establishes that the second derivatives of W and Γ are

mutually (negative) inverse kernels. Using $\varphi_{\rm cl}(x) = \delta W/\delta J(x)$ and $J(y) = -\delta\Gamma/\delta\varphi_{\rm cl}(y)$,

$$\begin{aligned}\delta^{(4)}(x-y) = \frac{\delta\varphi_{\rm cl}(x)}{\delta\varphi_{\rm cl}(y)} &= \int \mathrm{d}^4 z\, \frac{\delta\varphi_{\rm cl}(x)}{\delta J(z)}\,\frac{\delta J(z)}{\delta\varphi_{\rm cl}(y)} \\ &= \int \mathrm{d}^4 z\, \frac{\delta^2 W}{\delta J(x)\,\delta J(z)}\left(-\frac{\delta^2\Gamma}{\delta\varphi_{\rm cl}(z)\,\delta\varphi_{\rm cl}(y)}\right). \end{aligned} \tag{A.127}$$

Thus $W^{(2)}$ and $\Gamma^{(2)}$ satisfy $W^{(2)}\,\Gamma^{(2)} = -\mathbf{1}$ as integral kernels. Since $W^{(2)} = iG_c$ by Eq. (A.121), this gives $\Gamma^{(2)} = -\,(iG_c)^{-1} = i\,G_c^{-1}$: the second derivative of the effective action is i times the inverse connected propagator, i.e. the full 1PI two-point function whose tree-level value is $\delta^2 S/\delta\phi^2$, equal in momentum space to $p^2 - m^2$ for the free scalar. That Γ generates the higher 1PI vertices is proved in Chapter 10; Eq. (A.127) is the $n = 2$ instance and the one that fixes the sign.

Physical Insight

$W[J]$ and $\Gamma[\varphi_{\rm cl}]$ are two encodings of the same information related by a Legendre transform, exactly as the Hamiltonian is the Legendre transform of the Lagrangian. W is organized by the source J and counts connected diagrams; Γ is organized by the mean field $\varphi_{\rm cl}$ and counts 1PI diagrams, with the tree diagrams of Γ resumming the loops of W. The stationarity $\delta\Gamma/\delta\varphi_{\rm cl} = 0$ is the full quantum equation of motion: the classical action S is replaced by Γ, which includes all quantum corrections.

A.5.4 The S-matrix, the invariant amplitude, and the $-i\lambda$ vertex

We now turn from correlators to scattering. The physics of LSZ reduction—how a residue on the mass shell of the correlator (A.116) extracts a scattering amplitude—is the content of Chapters 14 and 16; here we fix only the normalization of the in/out states, the split of the S-matrix, and the definition of the invariant amplitude $\mathcal{M}$.

Recall from the Fourier section the relativistically invariant single-particle state normalization

$$\langle\,\mathbf{p}\,|\,\mathbf{q}\,\rangle \;=\; (2\pi)^3\,2E_{\mathbf{p}}\,\delta^{(3)}(\mathbf{p}-\mathbf{q}), \qquad E_{\mathbf{p}} = \sqrt{\mathbf{p}^2 + m^2}, \tag{A.128}$$

and the associated Lorentz-invariant momentum measure

$$\widetilde{\mathrm{d}p} \;\equiv\; \frac{\mathrm{d}^3 p}{(2\pi)^3\,2E_{\mathbf{p}}}, \tag{A.129}$$

both fixed in Sections A.3.3 and A.3.4. With this normalization the resolution of the identity on the one-particle subspace is $\mathbf{1} = \int \widetilde{\mathrm{d}p}\,|\mathbf{p}\rangle\langle\mathbf{p}|$, and Eqs. (A.128) and (A.129) are consistent, as $\int \widetilde{\mathrm{d}q}\,(2\pi)^3 2E_{\mathbf{q}}\delta^{(3)}(\mathbf{p}-\mathbf{q}) = 1$ acting correctly on $|\mathbf{p}\rangle$.

Definition A.30 (S-matrix and invariant amplitude)**.** The scattering operator $\mathbf{S}$ maps in-states to out-states. Because forward propagation with no interaction is the identity, we split off the trivial part:

$$\mathbf{S} \;=\; 1 \;+\; iT. \tag{A.130}$$

For a process $i \to f$ with incoming total momentum $\sum_i p_i$ and outgoing total momentum $\sum_f p_f$, the nontrivial matrix element factors a momentum-conserving delta function and defines the *invariant amplitude* $\mathcal{M}$ (also written $\mathcal{M}_{fi}$):

$$\langle f|\, iT \,|i\rangle \;=\; (2\pi)^4\, \delta^{(4)}\Big(\textstyle\sum_f p_f - \sum_i p_i\Big)\; i\,\mathcal{M}(i \to f). \tag{A.131}$$

The amplitude $\mathcal{M}$ defined by Eq. (A.131) is Lorentz invariant (the noninvariant normalization of the states having been absorbed into Eq. (A.128) and the measures of the master formulae below), and it is the quantity computed directly by the Feynman rules: $i\mathcal{M}$ is the sum of amputated, connected, momentum-space diagrams with external legs put on shell. The factor of i in front of T in Eq. (A.130) and the factor of i in front of $\mathcal{M}$ in Eq. (A.131) are conventional but fixed here; together with the $-i$ extraction rule (A.116) they make the tree-level $\mathcal{M}$ come out as in the next paragraph.

The vertex normalization of ϕ^4 theory follows directly. With the interaction Lagrangian $\mathcal{L}_{\text{int}} = -\frac{\lambda}{4!}\phi^4$ (Chapter 15), the perturbative weight in Eq. (A.113) expands the interaction as $\exp(iS_{\text{int}})$, so the elementary four-point vertex carries the factor $i \times (-\lambda/4!)$ from the action together with the 4! ways of attaching four distinct lines to the four identical fields. The 4! cancels the 1/4!, leaving the Feynman rule

$$\text{four-point vertex} \;=\; -\,i\,\lambda, \tag{A.132}$$

and hence, at leading order, $i\mathcal{M} = -i\lambda$, i.e. $\mathcal{M} = -\lambda$ for $\phi\phi \to \phi\phi$. This is the carried number of ϕ^4 theory; it is fixed here and used unchanged from Chapters 14 and 15 onward. We hand the *sign* of this vertex—the $-i$, traceable to the $+iS_{\text{int}}$ in Eq. (A.113)—to the gauge section, which must put the sign in the covariant derivative $D_\mu = \partial_\mu - igA^a_\mu T^a$ so that the gauge vertices are consistent with it.

Caution

Two distinct factors of i live in Eq. (A.132) and must not be conflated. One is the i multiplying the whole action in e^{iS} (it becomes the i in front of $\mathcal{M}$, and the i in the propagator). The other is the explicit factor of i in $\mathbf{S} = 1 + iT$. The clean statement is: the Feynman rules build $i\mathcal{M}$ directly—propagators are $i/(p^2 - m^2 + i\epsilon)$ and vertices are $i \times$ (coupling structure)—and the cross-section formulae below depend only on $|\mathcal{M}|^2$, in which all these i's have cancelled.

Worked Example A.31 (The $-i\lambda$ vertex from the path integral)**.** We verify Eq. (A.132) from the source formalism, to show the combinatorial 4! explicitly. The order-λ contribution to the connected four-point function $\langle \mathcal{T}\phi_1\phi_2\phi_3\phi_4\rangle_c$ comes from bringing down one power of $iS_{\text{int}} = -i\frac{\lambda}{4!}\int \mathrm{d}^4 z\, \phi(z)^4$ from the exponential in Eq. (A.113) and contracting its four fields with the four external points. By Wick's theorem (Chapter 10) the number of ways to contract the four *distinct* external fields $\phi(x_1), \ldots, \phi(x_4)$ with the four *identical* fields $\phi(z)^4$ is $4! = 24$: the first external field can pair with any of the four $\phi(z)$, the second with any of the remaining three, and so on. Each contraction is a free propagator, and the amputation in LSZ removes the four external propagators, leaving the value at the vertex

$$i\mathcal{M} \;=\; \Big(-i\,\frac{\lambda}{4!}\Big)\times 4! \;=\; -i\,\lambda, \tag{A.133}$$

so $\mathcal{M} = -\lambda$. The $1/4!$ in the Lagrangian is there precisely so that the connected, fully amputated four-point vertex is the clean $-i\lambda$; symmetry factors of *loop* diagrams, where some of the 4! pairings coincide, are treated carefully in Chapter 15.

A.5.5 Cross sections, decay rates, and the invariant phase space

We finally assemble the master formulae. The strategy is the standard finite-volume regularization: put the system in a spatial box of volume V over a time T, normalize states by Eq. (A.128), count transition probabilities, and take $V, T \to \infty$ at the end so that the continuum measure (A.129) reappears. The physics of why this gives the observable rate is in Chapter 16; here we carry the algebra so that the prefactors are pinned down.

From matrix element to probability. In the box, the continuum delta functions become volumes. The squared spacetime delta function in $|\langle f|iT|i\rangle|^2$ is regulated as

$$\big[(2\pi)^4\delta^{(4)}(0)\big] \;=\; V\,T, \tag{A.134}$$

since one $(2\pi)^4\delta^{(4)}$ remains after the other is used to enforce overall momentum conservation, and the leftover evaluated at zero argument counts the available spacetime volume. Likewise the state norm (A.128) becomes $\langle \mathbf{p}|\mathbf{p}\rangle = 2E_{\mathbf{p}}(2\pi)^3\delta^{(3)}(0) = 2E_{\mathbf{p}}V$, and the number of momentum modes in a cell d^3p is $V\,\mathrm{d}^3p/(2\pi)^3$.

Squaring Eq. (A.131) and using Eq. (A.134), the modulus-squared transition matrix element for a fixed final configuration is

$$\big|\langle f|iT|i\rangle\big|^2 \;=\; |\mathcal{M}|^2\,(2\pi)^4\delta^{(4)}\Big(\textstyle\sum_f p_f - \sum_i p_i\Big)\,V\,T. \tag{A.135}$$

To convert this to a genuine probability we divide by the norms $\langle i|i\rangle$ and $\langle f|f\rangle$ of the in- and out-states (the states (A.128) are not unit-normalized), and we

sum over the available final modes with the density $V\mathrm{d}^3p_f/(2\pi)^3$. Collecting the factors,

$$\mathrm{d}P \;=\; \frac{|\langle f|iT|i\rangle|^2}{\big(\prod_{\text{in}} 2E_i V\big)\big(\prod_{\text{out}} 2E_f V\big)} \prod_{\text{out}} \frac{V\,\mathrm{d}^3p_f}{(2\pi)^3}. \tag{A.136}$$

The factors of V from the mode counting cancel the V in each out-state norm, $\prod_{\text{out}}[V\mathrm{d}^3p_f/(2\pi)^3]/[2E_fV] = \prod_{\text{out}} \widetilde{\mathrm{d}p_f}$, exactly reproducing the invariant measure (A.129). Defining the *n-body Lorentz-invariant phase space*

$$\mathrm{d}\Pi_n \;\equiv\; (2\pi)^4\,\delta^{(4)}\Big(P - \textstyle\sum_{f=1}^{n} p_f\Big)\;\prod_{f=1}^{n} \frac{\mathrm{d}^3p_f}{(2\pi)^3\,2E_f}, \qquad P = \textstyle\sum_{\text{in}} p_i, \tag{A.137}$$

and substituting Eq. (A.135) into Eq. (A.136), every surviving V and T collects into a single overall VT from Eq. (A.134) divided by the $\prod_{\text{in}} V$:

$$\mathrm{d}P \;=\; |\mathcal{M}|^2\;\frac{V\,T}{\prod_{\text{in}} 2E_i\,V}\;\mathrm{d}\Pi_n. \tag{A.138}$$

Decay rate. For the decay of a single particle of mass M, there is one incoming line, so $\prod_{\text{in}} 2E_iV = 2MV$ in the rest frame. One factor of V cancels against the V in *Eq.* (A.138), and the rate is the probability per unit time, $\mathrm{d}\Gamma = \mathrm{d}P/T$:

$$\boxed{\mathrm{d}\Gamma \;=\; \frac{1}{2M}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n} \tag{A.139}$$

in the rest frame. The V and T have both cancelled, as they must for a physical observable; in a general frame $2M$ is replaced by $2E_M$, accounting for relativistic time dilation of the decay.

Cross section. For $2 \to n$ scattering there are two incoming lines, so $\prod_{\text{in}} 2E_iV = 4E_1E_2V^2$. Then Eq. (A.138) gives the transition rate $R = \mathrm{d}P/T = |\mathcal{M}|^2\,\mathrm{d}\Pi_n/(4E_1E_2V)$. The cross section is the rate divided by the incident flux $\mathcal{F}$. With our box normalization there is one particle of each species in the volume V, so each beam has number density $1/V$ and the flux of one beam on the other is $\mathcal{F} = |\mathbf{v}_1 - \mathbf{v}_2|/V$, with $\mathbf{v}_a = \mathbf{p}_a/E_a$ the velocities. The factor of $1/V$ in R cancels the $1/V$ in $\mathcal{F}$, leaving

$$\mathrm{d}\sigma \;=\; \frac{R}{\mathcal{F}} \;=\; \frac{1}{4E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2|}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n. \tag{A.140}$$

The kinematic prefactor is made manifestly Lorentz invariant by the identity (proved in Theorem A.32)

$$4E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2| \;=\; 4\sqrt{(p_1\!\cdot\!p_2)^2 - m_1^2m_2^2} \;\equiv\; F, \tag{A.141}$$

valid for collinear beams (the only configuration relevant to a flux), giving the master formula

$$\boxed{\mathrm{d}\sigma \;=\; \frac{1}{F}\,|\mathcal{M}|^2\,\mathrm{d}\Pi_n\,, \qquad F = 4\sqrt{(p_1\!\cdot\!p_2)^2 - m_1^2m_2^2}} \tag{A.142}$$

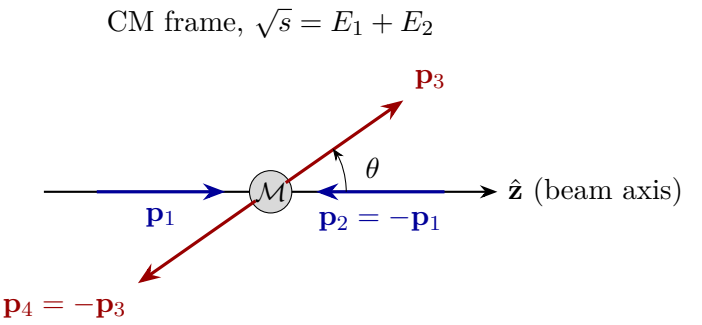


Figure A.3: Center-of-momentum kinematics for a $2 \to 2$ process. The incoming momenta are equal and opposite along the beam axis with magnitude p_i; the outgoing momenta are equal and opposite with magnitude p_f, at scattering angle θ. The invariant amplitude $\mathcal{M}$ sits at the vertex; the flux factor $F = 4p_i\sqrt{s}$ and the two-body phase space $\mathrm{d}\Pi_2 = (p_f/16\pi^2\sqrt{s})\,\mathrm{d}\Omega$ combine into Eq. (A.149).

with the invariant phase space (A.137). Equations (A.139) and (A.142), built on the relativistic normalization (A.128) and the invariant measure (A.129), are the formulae used unchanged throughout the book; the only model-dependent input is $|\mathcal{M}|^2$.

Worked Example A.32 (The invariant flux factor)**.** We prove the identity (A.141). Work in the center-of-momentum (CM) frame, where $\mathbf{p}_1 = -\mathbf{p}_2$; choose the common axis to be $\hat{\mathbf{z}}$ and write $|\mathbf{p}_1| = |\mathbf{p}_2| = p$, $p_1 = (E_1, 0, 0, p)$, $p_2 = (E_2, 0, 0, -p)$. Then $p_1 \cdot p_2 = E_1E_2 + p^2$ (the spatial parts are antiparallel, contributing $-(p)(-p) = +p^2$ in the mostly-minus metric), and using $E_a^2 = p^2 + m_a^2$,

$$\begin{aligned}(p_1\cdot p_2)^2 - m_1^2m_2^2 &= (E_1E_2 + p^2)^2 - (E_1^2 - p^2)(E_2^2 - p^2)\\ &= E_1^2E_2^2 + 2E_1E_2p^2 + p^4 - [E_1^2E_2^2 - (E_1^2 + E_2^2)p^2 + p^4]\\ &= 2E_1E_2\,p^2 + (E_1^2 + E_2^2)\,p^2 \;=\; p^2\,(E_1 + E_2)^2. \qquad \text{(A.143)}\end{aligned}$$

Hence $\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2} = p\,(E_1 + E_2)$. On the other side, the beams are collinear and opposite, so the relative speed is $|\mathbf{v}_1 - \mathbf{v}_2| = p/E_1 + p/E_2 = p(E_1 + E_2)/(E_1E_2)$, and

$$4E_1E_2\,|\mathbf{v}_1 - \mathbf{v}_2| \;=\; 4E_1E_2\cdot\frac{p(E_1 + E_2)}{E_1E_2} \;=\; 4p\,(E_1 + E_2) \;=\; 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2}, \qquad \text{(A.144)}$$

which is Eq. (A.141). Both sides are Lorentz scalars built from the invariants of the two incoming momenta, so the equality, established in the CM frame, holds in any frame related to it by a boost along the beam axis.

Worked Example A.33 (Two-body phase space and the $2 \to 2$ master formula)**.** The most-used special case is $2 \to 2$ scattering, the kinematic skeleton of QED's $e^+e^- \to \mu^+\mu^-$ (Chapter 17). We reduce Eq. (A.137) for $n = 2$ in the CM frame, with total energy $\sqrt{s} = E_1 + E_2$ and final momenta p_3, p_4 of magnitude p_f (Fig. A.3).

Start from

$$\mathrm{d}\Pi_2 \;=\; (2\pi)^4\,\delta^{(4)}(P - p_3 - p_4)\,\frac{\mathrm{d}^3p_3}{(2\pi)^3 2E_3}\,\frac{\mathrm{d}^3p_4}{(2\pi)^3 2E_4}, \qquad P = (\sqrt{s}, \mathbf{0}). \tag{A.145}$$

The spatial part of the delta function does the $\mathbf{p}_4$ integral, setting $\mathbf{p}_4 = -\mathbf{p}_3$ so that $p_f \equiv |\mathbf{p}_3| = |\mathbf{p}_4|$, and leaves

$$\mathrm{d}\Pi_2 \;=\; \frac{1}{(2\pi)^2}\,\delta(\sqrt{s} - E_3 - E_4)\,\frac{\mathrm{d}^3p_3}{4E_3E_4}. \tag{A.146}$$

Write $\mathrm{d}^3p_3 = p_f^2\,\mathrm{d}p_f\,\mathrm{d}\Omega$, with $E_3 = \sqrt{p_f^2 + m_3^2}$, $E_4 = \sqrt{p_f^2 + m_4^2}$. The remaining delta function fixes p_f through the total energy; since

$$\frac{\mathrm{d}(E_3 + E_4)}{\mathrm{d}p_f} \;=\; \frac{p_f}{E_3} + \frac{p_f}{E_4} \;=\; \frac{p_f(E_3 + E_4)}{E_3E_4} \;=\; \frac{p_f\sqrt{s}}{E_3E_4}, \tag{A.147}$$

integrating over p_f against $\delta(\sqrt{s} - E_3 - E_4)$ supplies the Jacobian $E_3E_4/(p_f\sqrt{s})$:

$$\mathrm{d}\Pi_2 \;=\; \frac{1}{(2\pi)^2}\,\frac{p_f^2}{4E_3E_4}\,\frac{E_3E_4}{p_f\sqrt{s}}\,\mathrm{d}\Omega \;=\; \frac{1}{16\pi^2}\,\frac{p_f}{\sqrt{s}}\,\mathrm{d}\Omega. \tag{A.148}$$

For the flux, Eq. (A.141) in the CM frame gives $F = 4\sqrt{(p_1\cdot p_2)^2 - m_1^2m_2^2} = 4p_i(E_1 + E_2) = 4p_i\sqrt{s}$, with p_i the magnitude of the incoming CM momentum. Combining with Eq. (A.142),

$$\frac{\mathrm{d}\sigma}{\mathrm{d}\Omega} \;=\; \frac{1}{F}\,|\mathcal{M}|^2\,\frac{\mathrm{d}\Pi_2}{\mathrm{d}\Omega} \;=\; \frac{1}{4p_i\sqrt{s}}\,|\mathcal{M}|^2\,\frac{1}{16\pi^2}\,\frac{p_f}{\sqrt{s}} \;=\; \frac{|\mathcal{M}|^2}{64\pi^2 s}\,\frac{p_f}{p_i}. \tag{A.149}$$

When all masses are negligible compared to $\sqrt{s}$, $p_f = p_i$ and this collapses to the compact $\mathrm{d}\sigma/\mathrm{d}\Omega = |\mathcal{M}|^2/(64\pi^2 s)$. This is the normalization against which QED reports $e^+e^- \to \mu^+\mu^-$ in Chapter 17; the present section fixes the prefactor, and that chapter supplies $|\mathcal{M}|^2$ from the spin-summed traces.

With the source sign $+J\phi$, the generating functionals Z, $W = -i\ln Z$, Γ with $\delta\Gamma/\delta\varphi_{\mathrm{cl}} = -J$, the split $\mathbf{S} = 1 + iT$ with the $-i\lambda$ vertex, and the cross-section and decay master formulae now all fixed and mutually consistent, the remaining error-prone conventions are those of internal symmetry. The next section takes the coupling- and vertex-sign conventions fixed here—in particular the $-i$ that the $+iS_{\mathrm{int}}$ in Eq. (A.113) forces onto every vertex—and places the matching sign in the covariant derivative $D_\mu = \partial_\mu - igA_\mu^a T^a$, fixing the charge convention, the field-strength normalization, and the generator and structure-constant conventions so that the gauge vertices inherit the signs established here.

A.6 Internal Symmetry and Gauge Conventions

The previous section closed the analytic and combinatorial conventions by tracing every Feynman-rule sign back to a single source: the factor i that multiplies the action in the path-integral weight e^{iS} of Eq. (A.113), which forces the $-i$ on the scalar vertex (A.132) and, through the $+J\phi$ source coupling (A.112), the sign in the Legendre relation $\delta\Gamma/\delta\varphi_{\rm cl} = -J$. Internal-symmetry conventions are the next, and historically the most error-prone, place where a single misplaced sign metastasizes. The covariant-derivative sign, the charge convention, the field-strength normalization, the Hermiticity and trace normalization of the generators, and the reality and antisymmetry of the structure constants are not independent choices: once the coupling enters the action through a covariant derivative, the sign in that derivative is fixed against the $+iS_{\rm int}$ of the previous section, and everything else—the field strength, the Yang–Mills kinetic normalization, the gauge transformation law, the vertices—follows by algebra. This section makes that chain explicit. It is the single home of the gauge-sign conventions of QED and Yang–Mills theory; the physics that motivates them is built in Chapter 8 (the classical $U(1)$ gauge principle) and Chapter 25 (the Yang–Mills connection and its curvature), and the group-theoretic numerics—the Casimirs C_A, C_F, the index T_F, and the explicit structure-constant tables—are deferred to Chapter B. What is owned here is the precise placement of every sign and factor, derived rather than asserted.

A.6.1 Generators, the Lie algebra, and the structure constants

A gauge theory is built on a compact Lie group G—the *gauge group*—acting on the matter fields through a chosen unitary representation. Throughout the book the matter of the benchmark theories sits in the fundamental representation: the electron in the (one-dimensional) charge-carrying representation of $U(1)$, the quarks in the fundamental $\mathbf{N}$ of $\mathrm{SU}(N)$. The infinitesimal action is generated by a basis of the Lie algebra, and the conventions for that basis are the root of all the sign questions that follow.

Definition A.34 (Generators, structure constants, and normalization)**.** Let G be a compact Lie group of dimension $\dim G$ with Lie algebra $\mathfrak{g}$. We choose a basis $\{T^a\}_{a=1}^{\dim G}$ of Hermitian generators in the representation under consideration,

$$(T^a)^\dagger \;=\; T^a, \tag{A.150}$$

satisfying the Lie-algebra commutation relations with real structure constants f^{abc},

$$\Big[T^a, T^b\Big] \;=\; i\, f^{abc}\, T^c, \tag{A.151}$$

and normalized in the fundamental representation by

$$\mathrm{tr}\big(T^a T^b\big) \;=\; \tfrac{1}{2}\,\delta^{ab}. \tag{A.152}$$

A group element connected to the identity is the exponential $U = \exp(i\,\alpha^a T^a)$ with real parameters α^a; the factor of i in the exponent, combined with (A.150), makes U unitary, $U^\dagger = U^{-1}$.

Three conventional choices are buried in Theorem A.34, and each is made for a definite reason. The Hermiticity (A.150) is the choice that keeps the conserved charges $Q^a = \int \mathrm{d}^3x\, J^{0a}$ Hermitian observables; it is the physicist's convention, distinct from the mathematician's anti-Hermitian generators (for which the i in (A.151) would be absent). The factor of i in (A.151) is then forced: the commutator of two Hermitian matrices is anti-Hermitian, so it must equal i times a Hermitian combination, and (A.151) is the statement that this combination is again a real combination of generators. The normalization (A.152) fixes the overall scale of the T^a—without it the structure constants could be rescaled at will—and the particular value $\frac{1}{2}$ is the one that makes the canonical gauge kinetic term carry the coefficient $-\frac{1}{4}$, as we prove in Section A.6.4. These are exactly the conventions stated globally in the style canon and developed group-theoretically in Chapter B; here we record them and extract their immediate consequences, which are themselves frequently misquoted.

Proposition A.35 (Reality and total antisymmetry of f^{abc})**.** *With the conventions* (A.150)–(A.152)*, the structure constants are real and totally antisymmetric in their three indices, and admit the closed form*

$$f^{abc} \;=\; -\,2\,i\,\mathrm{tr}\big(\,\big[T^a, T^b\big]\,T^c\big). \tag{A.153}$$

Worked Example A.36 (Proof of Theorem A.35)**.** Each claim follows from a one-line manipulation of (A.151), (A.152), and the cyclicity of the trace; we carry all three out.

The closed form. Multiply (A.151) on the right by T^c and take the trace,

$$\mathrm{tr}\big(\,\big[T^a, T^b\big]\,T^c\big) \;=\; i\,f^{abd}\,\mathrm{tr}(T^d T^c) \;=\; i\,f^{abd}\,\tfrac{1}{2}\,\delta^{dc} \;=\; \tfrac{i}{2}\,f^{abc}, \tag{A.154}$$

using (A.152) in the second step. Solving for f^{abc} gives (A.153).

Reality. The commutator of two Hermitian matrices is anti-Hermitian: from (A.150),

$$\big[T^a, T^b\big]^\dagger \;=\; (T^aT^b - T^bT^a)^\dagger \;=\; T^bT^a - T^aT^b \;=\; -\big[T^a, T^b\big]. \tag{A.155}$$

Substituting (A.151) into both sides, $(i f^{abc}T^c)^\dagger = -i\,(f^{abc})^*\,T^c$ on the left (the T^c being Hermitian) and $-\,i f^{abc}T^c$ on the right; equating coefficients of the linearly independent T^c gives $(f^{abc})^* = f^{abc}$, so $f^{abc} \in \mathbb{R}$.

Total antisymmetry. Antisymmetry in the first two indices is immediate from the antisymmetry of the commutator, $f^{abc} = -f^{bac}$. For the remaining swaps it is cleanest to work with the manifestly trace-defined object

$$G^{abc} \;\equiv\; \mathrm{tr}(T^a\,\big[T^b, T^c\big]\,) \;=\; \tfrac{i}{2}\,f^{abc}, \tag{A.156}$$

the last equality being (A.154) read with the generators cyclically permuted, $\mathrm{tr}(T^a\,[T^b,T^c]) = \mathrm{tr}([T^b,T^c]\,T^a) = \frac{i}{2}f^{bca} = \frac{i}{2}f^{abc}$, where $f^{bca} = f^{abc}$ is the cyclicity that already follows from antisymmetry in the first two slots once total antisymmetry is established; we use it only to fix the proportionality constant. The object G^{abc} is antisymmetric in $b \leftrightarrow c$ because the commutator is, while it is invariant under the cyclic permutation $(a,b,c) \to (b,c,a)$ because the trace is:

$$\begin{aligned} G^{bca} &= \mathrm{tr}(T^bT^cT^a - T^bT^aT^c) \;=\; \mathrm{tr}(T^aT^bT^c) - \mathrm{tr}(T^aT^cT^b) \\ &= \mathrm{tr}(T^a\,[T^b,T^c]\,) \;=\; G^{abc}, \end{aligned} \tag{A.157}$$

where the middle step used cyclicity of the trace to bring T^a to the front of each term. An object antisymmetric in one adjacent pair and invariant under cyclic rotation is totally antisymmetric: the cyclic rotation carries the antisymmetric pair through every adjacent position, so the sign flips under every transposition of indices. Hence $f^{abc} \propto G^{abc}$ is totally antisymmetric, as claimed.

Total antisymmetry is the property that makes the adjoint representation $(\mathrm{ad}\,T^a)^{bc} = -if^{abc}$ itself Hermitian and the Yang–Mills self-interactions manifestly real; the Jacobi identity satisfied by the f^{abc}, the existence of the quadratic Casimirs, and the explicit numerics $f^{acd}f^{bcd} = C_A\,\delta^{ab}$, $\mathrm{tr}(T^aT^b) = T_F\,\delta^{ab}$ with $T_F = \frac{1}{2}$ in the fundamental, and the like are the business of Chapter B (Eqs. (B.2) and (B.61)) and are imported, not reproved, wherever the book needs a numerical color factor.

A.6.2 The covariant derivative and the matter transformation law

The gauge field enters physics through the covariant derivative, and the sign in front of the coupling there is *the* most frequently mis-set convention in the subject. We fix it, and then *derive*—we do not posit—the transformation law of the gauge field that this choice forces.

It is convenient to package the gauge field into a single Lie-algebra-valued (matrix) object,

$$A_\mu \;\equiv\; A_\mu^a\,T^a, \tag{A.158}$$

which is Hermitian because the components A_μ^a are real and the T^a are Hermitian. The matter field ψ transforms in the representation carried by the T^a.

Definition A.37 (Covariant derivative and the matter law)**.** The covariant derivative acting on a matter field ψ in the chosen representation is

$$D_\mu\psi \;=\; (\partial_\mu - i\,g\,A_\mu)\psi \;=\; (\partial_\mu - i\,g\,A_\mu^aT^a)\psi, \tag{A.159}$$

with g the gauge coupling. Under a gauge transformation parametrized by $U(x) = \exp(i\alpha^a(x)T^a) \in G$, the matter field transforms homogeneously,

$$\psi(x) \;\longmapsto\; \psi'(x) \;=\; U(x)\,\psi(x). \tag{A.160}$$

The sign in (A.159) is $-ig$, and it is fixed against the e^{iS} weight: with this sign the interaction extracted from the matter kinetic term is $+iS_{\text{int}}$ at a vertex carrying a single explicit $+g$, matching the $+iS_{\text{int}}$ convention that produced the $-i\lambda$ scalar vertex of Section A.5.4, as we verify in Theorem A.44. We now require that $D_\mu\psi$ transform the same way as ψ itself—this is the entire point of the construction, that $\bar\psi\, i\not{D}\,\psi$ be gauge invariant—and read off what the gauge field must do.

Proposition A.38 (Gauge transformation of the connection)**.** *The covariant derivative* (A.159) *transforms covariantly, $D_\mu\psi \mapsto U\,(D_\mu\psi)$, if and only if the matrix connection transforms by the affine (inhomogeneous) law*

$$A_\mu \;\longmapsto\; A'_\mu \;=\; U\,A_\mu\,U^{-1} \;-\; \frac{i}{g}\,(\partial_\mu U)\,U^{-1}. \tag{A.161}$$

Infinitesimally, with $U = 1 + i\alpha^a T^a + O(\alpha^2)$,

$$\delta A^a_\mu \;=\; \frac{1}{g}\,\partial_\mu\alpha^a + f^{abc}A^b_\mu\,\alpha^c \;=\; \frac{1}{g}(\partial_\mu\alpha^a + g\,f^{abc}A^b_\mu\alpha^c) \;=\; \frac{1}{g}\,(D_\mu\alpha)^a, \tag{A.162}$$

where $(D_\mu\alpha)^a = \partial_\mu\alpha^a + gf^{abc}A^b_\mu\alpha^c$ is the covariant derivative in the adjoint representation.

Proof. Demand $D'_\mu\psi' = U(D_\mu\psi)$ with $\psi' = U\psi$ and $D'_\mu = \partial_\mu - igA'_\mu$. The left-hand side is

$$D'_\mu(U\psi) \;=\; (\partial_\mu U)\,\psi + U\,\partial_\mu\psi - ig\,A'_\mu\,U\,\psi, \tag{A.163}$$

where the product rule has acted on $U\psi$. The right-hand side is

$$U\,(D_\mu\psi) \;=\; U\,\partial_\mu\psi - ig\,U\,A_\mu\,\psi. \tag{A.164}$$

The terms $U\,\partial_\mu\psi$ cancel between the two sides. Equality for arbitrary ψ then requires the operator identity

$$(\partial_\mu U) - ig\,A'_\mu\,U \;=\; -\,ig\,U\,A_\mu. \tag{A.165}$$

Solve for A'_μ by isolating the term containing it and right-multiplying by U^{-1}:

$$ig\,A'_\mu\,U \;=\; ig\,U\,A_\mu + (\partial_\mu U) \quad\Longrightarrow\quad A'_\mu \;=\; U\,A_\mu\,U^{-1} + \frac{1}{ig}\,(\partial_\mu U)\,U^{-1}, \tag{A.166}$$

and $1/(ig) = -i/g$ gives exactly (A.161). For the infinitesimal form, expand $U = 1 + i\alpha + O(\alpha^2)$ with $\alpha \equiv \alpha^a T^a$ and $U^{-1} = 1 - i\alpha + O(\alpha^2)$:

$$A'_\mu = (1 + i\alpha)\,A_\mu\,(1 - i\alpha) - \frac{i}{g}\,(i\,\partial_\mu\alpha)\,(1 - i\alpha) + O(\alpha^2)$$

$$= A_\mu + i\,[\alpha, A_\mu] + \frac{1}{g}\,\partial_\mu\alpha + O(\alpha^2), \tag{A.167}$$

so that $\delta A_\mu = \frac{1}{g}\partial_\mu\alpha + i\,[\alpha, A_\mu]$. Projecting onto the basis with $A_\mu = A_\mu^c T^c$, $\alpha = \alpha^b T^b$, the commutator is $i\,[\alpha, A_\mu] = i\,\alpha^b A_\mu^c \left[T^b, T^c\right] = i\,\alpha^b A_\mu^c\,(i f^{bcd} T^d) = -f^{bcd}\alpha^b A_\mu^c\,T^d$. Renaming $d \to a$ and using the total antisymmetry of f (Theorem A.35), $-f^{bca} = -f^{abc}\cdot(-1)\cdot(-1) = \ldots$; concretely $-f^{bca}\alpha^b A_\mu^c = f^{abc}A_\mu^b\alpha^c$ after relabeling the dummy pair $b \leftrightarrow c$ and using $f^{acb} = -f^{abc}$. The component thus reads $\delta A_\mu^a = \frac{1}{g}\partial_\mu\alpha^a + f^{abc}A_\mu^b\alpha^c$, which is (A.162). □

The structure of (A.161) is worth naming: the homogeneous piece $UA_\mu U^{-1}$ rotates the connection like an adjoint-valued field, while the inhomogeneous piece $-\frac{i}{g}(\partial_\mu U)U^{-1}$ is the affine shift characteristic of a connection. It is the inhomogeneous piece that allows the gauge field to absorb the $\partial_\mu U$ generated when the derivative hits the matter rotation, and the factor $1/g$ there is precisely what is needed for the cancellation in the proof. One consistency check is immediate and worth recording, because it is the kind of thing a sign error breaks.

Worked Example A.39 (The transformed connection stays Hermitian)**.** The matrix A_μ is Hermitian, and gauge invariance of the theory requires the transformed A'_μ to be Hermitian as well; otherwise the components A'^a_μ would not be real and the gauge field would not be a genuine vector boson field. We verify this from (A.161). The homogeneous term is manifestly Hermitian, $(UA_\mu U^{-1})^\dagger = U^{-1\dagger}A_\mu^\dagger U^\dagger = UA_\mu U^{-1}$, using $A_\mu^\dagger = A_\mu$ and $U^\dagger = U^{-1}$. For the inhomogeneous term, differentiate $UU^{-1} = 1$ to obtain the identity $(\partial_\mu U)U^{-1} = -U(\partial_\mu U^{-1})$, and compute

$$\Big[-\frac{i}{g}(\partial_\mu U)U^{-1}\Big]^\dagger = \frac{i}{g}\,(U^{-1})^\dagger(\partial_\mu U)^\dagger = \frac{i}{g}\,U\,(\partial_\mu U^{-1}) = -\frac{i}{g}\,(\partial_\mu U)U^{-1}, \tag{A.168}$$

where $U^{-1\dagger} = U$, $(\partial_\mu U)^\dagger = \partial_\mu U^\dagger = \partial_\mu U^{-1}$, and the last step used the differentiated identity. The inhomogeneous term is therefore Hermitian as well, so $A'^\dagger_\mu = A'_\mu$ and the transformed components remain real. The factor of i in $-\frac{i}{g}(\partial_\mu U)U^{-1}$ is exactly what is required for this: a real coefficient there would have produced an anti-Hermitian shift.

A.6.3 The field strength as the curvature of the connection

The field strength is the curvature of the gauge connection: it measures the failure of two covariant derivatives to commute. Defining it this way—rather than writing down $\partial_\mu A_\nu - \partial_\nu A_\mu + \cdots$ by hand—makes its gauge covariance automatic and fixes every sign in it at once.

Definition A.40 (Field strength)**.** The field-strength matrix is the commutator of two covariant derivatives, normalized to strip the coupling:

$$F_{\mu\nu} \;\equiv\; \frac{i}{g}\,[D_\mu, D_\nu] \;=\; F^a_{\mu\nu}\,T^a. \tag{A.169}$$

Proposition A.41 (Component form of the field strength)**.** *The definition* (A.169) *evaluates to*

$$F_{\mu\nu} \;=\; \partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,, \tag{A.170}$$

which in components is exactly the Yang–Mills field strength of Eq. (A.56)*,*

$$F^a_{\mu\nu} \;=\; \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\,f^{abc} A^b_\mu A^c_\nu. \tag{A.171}$$

Proof. The commutator $[D_\mu, D_\nu]$ is a differential operator; we evaluate it acting on an arbitrary ψ and read off the multiplication operator that results. With $D_\mu = \partial_\mu - igA_\mu$,

$$D_\mu D_\nu \psi \;=\; \partial_\mu\partial_\nu\psi - ig\,(\partial_\mu A_\nu)\psi - ig\,A_\nu\,\partial_\mu\psi - ig\,A_\mu\,\partial_\nu\psi - g^2\,A_\mu A_\nu\,\psi, \tag{A.172}$$

where the term $-ig\,\partial_\mu(A_\nu\psi)$ has been expanded by the product rule into $-ig(\partial_\mu A_\nu)\psi - igA_\nu\partial_\mu\psi$. Antisymmetrizing in $\mu \leftrightarrow \nu$, the symmetric pieces drop out: $\partial_\mu\partial_\nu\psi$ cancels, and the two cross terms $-igA_\nu\partial_\mu\psi - igA_\mu\partial_\nu\psi$ are symmetric under the swap and cancel as well. What survives is

$$[D_\mu, D_\nu]\,\psi \;=\; -ig\,(\partial_\mu A_\nu - \partial_\nu A_\mu)\psi - g^2\,[A_\mu, A_\nu]\,\psi. \tag{A.173}$$

Since this holds for all ψ, the operator identity is

$$[D_\mu, D_\nu] = -ig(\partial_\mu A_\nu - \partial_\nu A_\mu) - g^2\,[A_\mu, A_\nu] = -ig\big(\partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,\big). \tag{A.174}$$

Multiplying by i/g as in (A.169),

$$F_{\mu\nu} = \frac{i}{g}\,(-ig)\big(\partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,\big) = \partial_\mu A_\nu - \partial_\nu A_\mu - ig\,[A_\mu, A_\nu]\,, \tag{A.175}$$

which is (A.170); the prefactor $\frac{i}{g}\cdot(-ig) = 1$ is the reason for the particular normalization chosen in the definition. Finally project onto the basis. The commutator term is $-ig\,[A_\mu, A_\nu] = -ig\,A^b_\mu A^c_\nu\,\big[T^b, T^c\big] = -ig\,A^b_\mu A^c_\nu (if^{bcd}T^d) = g\,f^{bcd}A^b_\mu A^c_\nu\,T^d$, so with $A_\nu = A^a_\nu T^a$ in the linear terms and renaming $d \to a$,

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g\,f^{abc} A^b_\mu A^c_\nu, \tag{A.176}$$

which is (A.171). The sign of the gf^{abc} term is positive, matching Eq. (A.56) and Yang–Mills theory of Chapter 25. □

The payoff of defining $F_{\mu\nu}$ as a curvature is that its gauge transformation is now forced and homogeneous.

Proposition A.42 (Adjoint covariance of the field strength)**.** *Under a gauge transformation* (A.161)*, the field strength transforms in the adjoint representation,*

$$F_{\mu\nu} \longmapsto F'_{\mu\nu} = U\,F_{\mu\nu}\,U^{-1}. \tag{A.177}$$

Proof. Theorem A.38 established that $D_\mu\psi \mapsto U(D_\mu\psi)$ for every ψ, i.e. the covariant derivative transforms by conjugation as an operator, $D'_\mu = U\,D_\mu\,U^{-1}$. A commutator of conjugated operators is the conjugate of the commutator, $[UD_\mu U^{-1}, UD_\nu U^{-1}] = U\,[D_\mu, D_\nu]\,U^{-1}$, the inner $U^{-1}U$ telescoping. Hence

$$F'_{\mu\nu} = \frac{i}{g}\left[D'_\mu, D'_\nu\right] = \frac{i}{g}\,U\,[D_\mu, D_\nu]\,U^{-1} = U\,F_{\mu\nu}\,U^{-1}. \tag{A.178}$$

□

Because $F_{\mu\nu}$ is adjoint-covariant rather than invariant, it cannot itself be a Lagrangian; the invariant is its trace square, which is the subject of the next subsection and the place where the normalization (A.152) earns its value.

A.6.4 The Yang–Mills Lagrangian and the canonical $-\frac{1}{4}$ normalization

The gauge-invariant kinetic term is built from $F_{\mu\nu}$ by taking the unique (up to normalization) Lorentz scalar that is also gauge invariant, namely the trace of $F_{\mu\nu}F^{\mu\nu}$. Gauge invariance is immediate from Theorem A.42 and the cyclicity of the trace,

$$\operatorname{tr}(F'_{\mu\nu}F'^{\mu\nu}) = \operatorname{tr}(UF_{\mu\nu}U^{-1}UF^{\mu\nu}U^{-1}) = \operatorname{tr}(F_{\mu\nu}F^{\mu\nu}), \tag{A.179}$$

the inner $U^{-1}U$ telescoping and the outer U, U^{-1} cancelling under the trace. The question is the overall coefficient, and here the normalization (A.152) is decisive.

Proposition A.43 (The trace normalization fixes the $-\frac{1}{4}$)**.** *With the fundamental normalization* $\operatorname{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$*, the gauge-invariant kinetic term*

$$\mathcal{L}_{\text{gauge}} = -\tfrac{1}{2}\operatorname{tr}(F_{\mu\nu}F^{\mu\nu}) \tag{A.180}$$

equals the canonically normalized component form

$$\mathcal{L}_{\text{gauge}} = -\tfrac{1}{4}\,F^a_{\mu\nu}F^{a\,\mu\nu}, \tag{A.181}$$

in which each gauge field A^a_μ has the same kinetic normalization as the abelian Maxwell field.

Proof. Expand the matrix trace in the basis using $F_{\mu\nu} = F^a_{\mu\nu} T^a$:

$$\operatorname{tr}(F_{\mu\nu} F^{\mu\nu}) = F^a_{\mu\nu} F^{b\,\mu\nu}\, \operatorname{tr}(T^a T^b) = F^a_{\mu\nu} F^{b\,\mu\nu}\, \tfrac{1}{2}\delta^{ab} = \tfrac{1}{2}\, F^a_{\mu\nu} F^{a\,\mu\nu}, \tag{A.182}$$

where the second equality is exactly (A.152). Multiplying by $-\frac{1}{2}$ gives $-\frac{1}{2}\operatorname{tr}(FF) = -\frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu}$, which is (A.181). The coefficient $-\frac{1}{4}$ is therefore not an independent convention: it is the product of two prior choices, the $-\frac{1}{2}$ in front of the trace (fixed by demanding the abelian limit reproduce Maxwell, below) and the $\frac{1}{2}$ in $\operatorname{tr}(T^a T^b)$. Had the generators been normalized to $\operatorname{tr}(T^a T^b) = \delta^{ab}$ instead, (A.182) would have produced $F^a F^a$ with no $\frac{1}{2}$, and the same $-\frac{1}{2}\operatorname{tr}(FF)$ would have read $-\frac{1}{2} F^a F^a$, which is *not* the canonical $-\frac{1}{4}$; one would then be forced to rescale either the trace coefficient or the gauge field to recover a properly normalized propagator. The pairing of $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ with $-\frac{1}{2}\operatorname{tr}(FF)$ is what makes the component coefficient come out at the canonical $-\frac{1}{4}$. □

The sign of (A.181)—the minus—is the gauge-field counterpart of the scalar's $+\frac{1}{2}(\partial\phi)^2$ and is fixed for the same reason given in Section A.2.6 and Theorem A.10: in mostly-minus signature, $-\frac{1}{4} F^a_{\mu\nu} F^{a\,\mu\nu} = \frac{1}{2}(\mathbf{E}^a \cdot \mathbf{E}^a - \mathbf{B}^a \cdot \mathbf{B}^a)$, so the electric energy density is positive and the Hamiltonian is bounded below. With the kinetic term fixed, the coupling sign of Eq. (A.159) can be checked against the vertex-sign convention handed over from the previous section.

Worked Example A.44 (The gauge vertices and consistency with the $+iS_{\text{int}}$ convention)**.** The previous section fixed every vertex to be $i\times$(coupling structure from $\mathcal{L}_{\text{int}}$), the explicit i being the $+iS_{\text{int}}$ in the weight e^{iS} of Eq. (A.113); this is what made the ϕ^4 vertex $-i\lambda$. We check that the sign $-ig$ in (A.159) delivers gauge vertices consistent with that rule.

Expand the matter kinetic term using $\not{D} = \gamma^\mu D_\mu = \gamma^\mu(\partial_\mu - igA^a_\mu T^a)$:

$$\bar\psi\, i\not{D}\,\psi = \bar\psi\, i\not{\partial}\,\psi + g\,\bar\psi\,\gamma^\mu T^a \psi\, A^a_\mu, \tag{A.183}$$

the second term being the interaction $\mathcal{L}_{\text{int}} = g\,\bar\psi\gamma^\mu T^a \psi\, A^a_\mu$, with the $+g$ produced by $i\cdot(-ig) = +g$ when the i of $i\not{D}$ meets the $-ig$ of the covariant derivative. The Feynman rule for the quark–gluon vertex is then $i\times$(coefficient of the three fields), namely

$$\text{quark–gluon vertex} \;=\; i\,g\,\gamma^\mu T^a, \tag{A.184}$$

with a single explicit power of $+g$, exactly as the $+iS_{\text{int}}$ rule requires (Fig. A.4). For the abelian thread the same expansion with $D_\mu = \partial_\mu + ieA_\mu$ gives $\mathcal{L}_{\text{int}} = -e\,\bar\psi\gamma^\mu\psi\, A_\mu$ (the relative sign because the electron charge is $-e$; see Section A.6.5), and the QED vertex

$$\text{QED electron–photon vertex} \;=\; -\,i\,e\,\gamma^\mu. \tag{A.185}$$

Both vertices carry the single factor of i from e^{iS}, and the sign in each—the $+g$ of (A.184) and the $-e$ of (A.185)—is dictated by the covariant-derivative sign together with the charge of the matter field, not chosen independently.

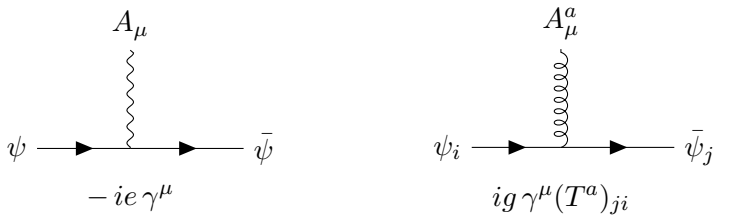


Figure A.4: The two minimal-coupling vertices in the book's conventions. Left: the QED electron–photon vertex $-ie\gamma^\mu$ from $\mathcal{L}_{\text{int}} = -e\bar{\psi}\gamma^\mu\psi A_\mu$, with the minus sign tracing to the electron charge $-e$. Right: the quark–gluon vertex $ig\gamma^\mu(T^a)_{ji}$ from $\mathcal{L}_{\text{int}} = g\bar{\psi}\gamma^\mu T^a\psi A^a_\mu$, with the $+g$ tracing to the $-ig$ in the covariant derivative (A.159). Both carry one factor of i from the e^{iS} weight. The full nonabelian Feynman rules, including the triple- and quartic-gluon vertices from $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$, are derived in Chapter 26.

A.6.5 The abelian reduction and the charge convention

The abelian case is not a separate convention but the $U(1)$ specialization of everything above, and seeing it as such resolves the single most common confusion in the subject: why the electron's covariant derivative carries $+ie$ while the nonabelian one carries $-ig$. The two are the same rule.

For $G = U(1)$ the algebra is one-dimensional; the single "generator" acting on a field is just its charge, the real number q by which the field is rotated. The structure constants vanish identically, so $f^{abc} = 0$ and the commutator term in the field strength drops out. The covariant derivative (A.159) becomes, for a field of charge q,

$$D_\mu\psi \;=\; (\partial_\mu - i\,q\,A_\mu)\psi, \tag{A.186}$$

with the *same* $-i$ as the nonabelian master formula—the charge q here plays the role of $g\,T^a$. The electron is the charge-$(-e)$ field with $e > 0$ the elementary charge, so substituting $q = -e$,

$$D_\mu\psi \;=\; (\partial_\mu - i(-e)A_\mu)\psi \;=\; (\partial_\mu + i\,e\,A_\mu)\psi. \tag{A.187}$$

The $+ie$ on the electron is therefore not a sign flip relative to the nonabelian $-ig$: it is the universal $-i\,q\,A_\mu$ evaluated at the electron's negative charge $q = -e$. This is the reconciliation of Eqs. (A.159) and (A.187), and it is the frozen QED convention used from Chapter 8 onward, matching the QED Lagrangian fixed in Theorem A.11 of the metric section.

The field strength reduces, with $f^{abc} = 0$, to the abelian Maxwell tensor

$$F_{\mu\nu} = \frac{i}{q}\,[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu, \tag{A.188}$$

gauge invariant on its own (no conjugation, since $U(1)$ is abelian and $UFU^{-1} = F$), and the kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ is the Maxwell action whose normalization Theorem A.43 was built to reproduce in the nonabelian case. The gauge transformation of Theorem A.38 reduces, for a field of charge q transforming as $\psi \mapsto e^{iq\lambda(x)}\psi$, to the inhomogeneous shift

$$A_\mu \;\longmapsto\; A_\mu + \partial_\mu \lambda, \tag{A.189}$$

the $-\frac{i}{q}(\partial_\mu U)U^{-1}$ term of (A.161) collapsing to $-\frac{i}{q}(iq\,\partial_\mu\lambda) = \partial_\mu\lambda$ for $U = e^{iq\lambda}$. (This is the same transformation written in Chapter 8 with gauge parameter $\alpha = -\lambda$; the relative sign of the parameter is immaterial, the physical content being $D_\mu\psi \mapsto e^{iq\lambda}D_\mu\psi$.) The electron's electromagnetic current and the verification that minimal coupling is *derived*, not imposed, by promoting the rigid phase symmetry $\psi \to e^{iq\lambda}\psi$ to a local one are the content of Chapter 8 (the gauge principle); here we have only fixed the signs that chapter then uses.

Finally we record the fine-structure normalization, used everywhere QED reports a number. The dimensionless coupling that organizes the perturbation series is

$$\alpha \;\equiv\; \frac{e^2}{4\pi} \;\approx\; \frac{1}{137.036} \qquad \text{(at low energy)}, \tag{A.190}$$

with the 4π the rationalized Heaviside–Lorentz convention consistent with the $-\frac{1}{4}F^2$ normalization and the natural units of this book; the nonabelian counterpart is $\alpha_s \equiv g^2/4\pi$. The running of α with scale, and Schwinger's $a_e = \alpha/2\pi$, are carried numbers of Chapter 24 and the $g-2$ chapter, computed in the scheme fixed next.

Notation Watch

The apparent clash between the nonabelian $D_\mu = \partial_\mu - igA^a_\mu T^a$ and the QED $D_\mu = \partial_\mu + ieA_\mu$ is illusory: both are the single rule $D_\mu = \partial_\mu - i(\text{charge})A_\mu$, with the QED $+ie$ arising solely because the electron's charge $q = -e$ is negative. A book that wrote $D_\mu = \partial_\mu + igA^a_\mu T^a$ for the nonabelian case (the opposite sign, used by some references) would flip the sign of every structure-constant term in $F^a_{\mu\nu}$ and of every gauge vertex; we never use that variant.

With the covariant-derivative sign $-ig$, the charge convention $q = -e$ for the electron, the field strength $F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g f^{abc}A^b_\mu A^c_\nu$ as the curvature (A.169), the Hermitian generators with $\left[T^a, T^b\right] = if^{abc}T^c$ and $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, the finite gauge law (A.161), and the canonical $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ all fixed and mutually consistent, the dimensionful object handed forward is the coupling itself. The gauge couplings e and g are dimensionless in $d = 4$, as counted in Section A.1.8; the next section takes that dimensionlessness and shows what happens to it the moment one leaves four dimensions, promoting $g \to g\,\mu^\varepsilon$ and $e \to e\,\mu^\varepsilon$ so that the dimensionless expansion parameter survives dimensional regularization.

A.7 Regularization and Renormalization-Scheme Conventions

The previous section settled the gauge couplings and found, for both the abelian e and the nonabelian g, the mass dimension $[\,e\,] = [\,g\,] = (4-d)/2$ recorded in (A.16); the quartic scalar coupling came out as $[\,\lambda\,] = 4-d$ in (A.14). In the physical dimension $d = 4$ all three vanish, which is precisely the dimensionlessness that Theorem A.6 identified with renormalizability. But the renormalization machinery of Part VI never works at $d = 4$ exactly: it works in a neighbourhood of four dimensions, where the loop integrals that diverge at $d = 4$ are instead finite and the divergence reappears as a pole in the small parameter that measures the departure from four dimensions. The moment one leaves $d = 4$, those three couplings cease to be dimensionless, and a single new scale must be introduced to absorb the dimension they acquire. This section fixes, once for the whole book, exactly how that continuation and that scale are defined: the convention $d = 4 - 2\varepsilon$ with poles in $1/\varepsilon$, the promotion of every coupling by the unique power of a mass μ that restores its dimensionlessness, and the precise definition of the modified minimal-subtraction scheme $\overline{\text{MS}}$ that Part VI and everything downstream of it uses by default. The integrals these conventions act on, and the analytic continuation that defines them, are the business of Chapter D; here we fix the conventions, not the integrals, deriving each from the dimension count already in hand.

A.7.1 Dimensional continuation: $d = 4 - 2\varepsilon$

The regulator of choice throughout the book is dimensional regularization: the spacetime dimension d is treated as a continuous complex variable, loop integrals are defined by analytic continuation in d from the region where they converge, and the ultraviolet divergences of $d = 4$ surface as poles at $d = 4$. We fix the parametrization of the continuation.

Definition A.45 (Dimensional-regularization parameter)**.** The spacetime dimension is continued away from four by writing

$$d \;=\; 4 - 2\varepsilon, \tag{A.191}$$

with ε a small parameter. The physical theory is recovered in the limit $\varepsilon \to 0$. Ultraviolet divergences of four-dimensional loop integrals appear, in this parametrization, as poles in ε at $\varepsilon = 0$: a logarithmically divergent one-loop integral produces a simple pole $1/\varepsilon$, an L-loop integral whose subdivergences have been arranged appropriately produces poles up to $1/\varepsilon^{L}$. The convention is that $\varepsilon > 0$ regulates ultraviolet divergences (so $d < 4$); infrared divergences, when they are regulated dimensionally, require $\varepsilon < 0$ ($d > 4$), and the two regions are connected by the analytic continuation that defines the integral.

Two features of (A.191) are conventional choices, and both are made to keep later formulas as clean as possible. The first is the *factor of two*. A logarithmically divergent one-loop integral behaves, near four dimensions, like $\int^{\Lambda} \mathrm{d}^d k \,/k^4 \sim \Lambda^{d-4}$, so its dimensionally continued value carries the structure $\Gamma\!\left(2 - \frac{d}{2}\right)$, the Euler gamma function having a simple pole whenever its argument is a nonpositive integer. With $d = 4 - 2\varepsilon$ the argument is $2 - \frac{d}{2} = \varepsilon$, and $\Gamma(\varepsilon) = 1/\varepsilon - \gamma_{\mathrm{E}} + O(\varepsilon)$ delivers a pole of residue exactly one in ε. Had we written $d = 4 - \varepsilon$ instead, the same integral would have given $\Gamma(\varepsilon/2)$ and a pole $2/\varepsilon$, sprinkling factors of two through every anomalous dimension and beta function in the book. The choice $d = 4 - 2\varepsilon$ is the one for which $1/\varepsilon$ is the residue, not $2/\varepsilon$, and it is universal in the modern literature; we adopt it without exception. The second choice is the *sign*: $d = 4 - 2\varepsilon$ with $\varepsilon > 0$ for the ultraviolet means that increasing ε moves to lower dimension, where integrals converge better, which is the direction in which the regulator does its work.

It bears emphasis that (A.191) is a statement about the analytic structure of integrals, not a claim that spacetime literally has $4 - 2\varepsilon$ dimensions. The objects being continued are the values of momentum integrals, which are functions of d that happen to be meromorphic; dimensional regularization defines a divergent four-dimensional integral as the $\varepsilon \to 0$ limit of the meromorphic function obtained by computing in d where it converges and continuing. The explicit construction—Feynman and Schwinger parametrization, the Wick rotation to a Euclidean integral, and the resulting closed forms for the master integrals—is carried out in Chapter D, and dimensional regularization makes its first physical appearance in the QED one-loop programme of Chapter 19. What we need here is only the bookkeeping convention (A.191) and its consequence for the dimensions of couplings, to which we turn.

A.7.2 Promotion of the couplings and the scale μ

Once $d \neq 4$, the couplings λ, e, and g are no longer dimensionless. This is not optional: it is forced by the same counting that gave (A.14) and (A.16), now read at $d = 4 - 2\varepsilon$ rather than at $d = 4$. Substituting (A.191) into those results,

$$[\,\lambda\,] \;=\; 4 - d \;=\; 2\varepsilon, \qquad [\,e\,] \;=\; [\,g\,] \;=\; \frac{4-d}{2} \;=\; \varepsilon. \tag{A.192}$$

Each bare coupling in the Lagrangian therefore carries a small positive mass dimension proportional to ε. Perturbation theory, however, is an expansion in *dimensionless* couplings: a Feynman amplitude is a pure number (times the kinematic structure carrying its external dimension), and one cannot expand sensibly in a quantity that itself carries dimension. The resolution is to factor out the dimension into a fixed power of an arbitrary mass scale, leaving a dimensionless coupling to expand in.

Definition A.46 (Renormalization scale and coupling promotion)**.** Introduce a mass parameter μ, the *renormalization scale*, with $[\,\mu\,] = 1$. The dimensionful Lagrangian couplings in $d = 4 - 2\varepsilon$ are written as μ raised to the power dictated by (A.192) times a dimensionless coupling that retains the original name:

$$\lambda \;\longrightarrow\; \lambda\,\mu^{2\varepsilon}, \tag{A.193}$$

$$e \;\longrightarrow\; e\,\mu^{\varepsilon}, \qquad g \;\longrightarrow\; g\,\mu^{\varepsilon}. \tag{A.194}$$

After the promotion, λ, e, and g on the right-hand sides are dimensionless for all ε, and μ alone carries the dimension demanded by (A.192). The scale μ is arbitrary; no physical quantity can depend on it, and the requirement that it does not is the renormalization group, derived in Chapter 24.

The power of μ in (A.193)–(A.194) is not free: it is the unique exponent that makes the count work, and we derive it rather than assert it.

Proposition A.47 (The exact μ-power each coupling must carry)**.** *Let $g_{\mathcal{O}}$ be the coupling of a local operator $\mathcal{O}$ of four-dimensional mass dimension Δ_4, so that at $d = 4$ the coupling has dimension $[\,g_{\mathcal{O}}\,]_{d=4} = 4 - \Delta_4$. If $g_{\mathcal{O}}$ is dimensionless at $d = 4$ (a marginal coupling, $\Delta_4 = 4$), then continued to $d = 4 - 2\varepsilon$ it acquires dimension $[\,g_{\mathcal{O}}\,] = (4 - d) \cdot \left(\frac{\partial \Delta}{\partial d}\right)$ evaluated through the field content; concretely, writing the operator's dimension as a function of d as in Section A.1.4, the coupling carries*

$$[\,g_{\mathcal{O}}\,] \;=\; d - \Delta(d), \tag{A.195}$$

and the promotion $g_{\mathcal{O}} \to g_{\mathcal{O}}\,\mu^{\,\Delta(d)-d}\big|_{\text{restored}} = g_{\mathcal{O}}\,\mu^{\,d-\Delta(d)}$ with the dimensionless coupling on the right requires the exponent

$$\rho_{\mathcal{O}} \;=\; d - \Delta(d) \;=\; [\,g_{\mathcal{O}}\,]. \tag{A.196}$$

For the ϕ^4 coupling $\rho_\lambda = 2\varepsilon$; for the gauge couplings $\rho_e = \rho_g = \varepsilon$.

Proof. The exponent is fixed by a single requirement: after promotion the renamed coupling must be dimensionless, while the Lagrangian term it sits in must continue to carry the full dimension $[\,\mathcal{L}\,] = d$ demanded by (A.8). Write the Lagrangian term as $g_{\mathcal{O}}^{(\dim)}\,\mathcal{O}$ with the dimensionful coupling $g_{\mathcal{O}}^{(\dim)} = \mu^{\rho_{\mathcal{O}}}\,g_{\mathcal{O}}$, the factored coupling $g_{\mathcal{O}}$ dimensionless. Taking mass dimensions and using additivity (Theorem A.1),

$$d \;=\; [\,g_{\mathcal{O}}^{(\dim)}\,\mathcal{O}\,] \;=\; \rho_{\mathcal{O}}[\,\mu\,] + [\,g_{\mathcal{O}}\,] + [\,\mathcal{O}\,] \;=\; \rho_{\mathcal{O}} + 0 + \Delta(d), \tag{A.197}$$

where $[\,\mu\,] = 1$ and $[\,g_{\mathcal{O}}\,] = 0$ by construction, and $\Delta(d) = [\,\mathcal{O}\,]$ is the operator's dimension at the continued d, computed from the field dimensions of Theorem A.2. Solving (A.197) for $\rho_{\mathcal{O}}$ gives $\rho_{\mathcal{O}} = d - \Delta(d)$, which is (A.196) and equals the dimension $[\,g_{\mathcal{O}}^{(\dim)}\,]$ the unfactored coupling had by Theorem A.3. The two

named cases are immediate from (A.192): for $\mathcal{O} = \phi^4$, $\Delta(d) = 2(d-2)$ and $\rho_\lambda = d - 2(d-2) = 4 - d = 2\varepsilon$; for the minimal-coupling operator $\bar{\psi}\gamma^\mu\psi A_\mu$, $\Delta(d) = (3d-4)/2$ and $\rho = d - (3d-4)/2 = (4-d)/2 = \varepsilon$. The cubic and quartic gluon vertices, whose operators were shown in Section A.1.8 to have the same dimensions, inherit the same μ-powers ε and 2ε respectively, consistent with their coefficients being $g\mu^\varepsilon$ and $(g\mu^\varepsilon)^2$. □

The factor $\mu^{2\varepsilon}$ on λ and μ^ε on e, g is therefore not a device chosen for convenience but the only assignment that preserves both the dimensionlessness of the expansion parameter and the dimensional integrity of the action. The practical effect is that every loop integral, which after continuation carries a residual dimension $(\text{mass})^{-2\varepsilon}$ per loop from the unbalanced momentum measure, is multiplied by exactly the compensating $\mu^{2\varepsilon}$ from the vertices, so that the dimensionless combination that finally appears is a ratio of μ^2 to the physical scales of the integral. That combination is the seat of the logarithm of μ whose slope is the beta function. We see the mechanism explicitly in the next subsection.

A.7.3 The generic one-loop structure and its ε-expansion

Every one-loop calculation in dimensional regularization, after Feynman parametrization and Wick rotation, reduces to a single scalar integral whose closed form is derived in Chapter D. We import it; we do not reprove it. With a Euclidean denominator power n and a Feynman-combined mass-squared $\Delta > 0$, the master one-loop integral is

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^n} \;=\; \frac{(-1)^n\, i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(n - \frac{d}{2}\right)}{\Gamma(n)}\,\frac{1}{\Delta^{\,n-d/2}}, \tag{A.198}$$

established as the master scalar integral of Chapter D (Theorem D.22, Eq. (D.77)), with the Wick rotation $\ell^0 = i\ell^0_E$ defined there. The entire scheme dependence of the book is contained in how one expands the right-hand side of (A.198) near $d = 4$, so we carry that expansion out in complete detail for the case that produces a logarithmic divergence, $n = 2$, which is the workhorse of marginal-coupling renormalization.

Set $n = 2$ and $d = 4 - 2\varepsilon$, so that $d/2 = 2 - \varepsilon$ and $n - d/2 = \varepsilon$. Then (A.198) becomes

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^2} \;=\; \frac{i}{(4\pi)^{2-\varepsilon}}\,\Gamma(\varepsilon)\,\Delta^{-\varepsilon} \;=\; \frac{i}{(4\pi)^2}\,(4\pi)^{\varepsilon}\,\Gamma(\varepsilon)\,\Delta^{-\varepsilon}, \tag{A.199}$$

using $\Gamma(2) = 1$. Restoring the compensating scale factor from the coupling promotion, Theorem A.46, supplies a $\mu^{2\varepsilon}$ per loop, and the combination that actually enters a dimensionless amplitude is

$$\mu^{2\varepsilon}\,(4\pi)^{\varepsilon}\,\Gamma(\varepsilon)\,\Delta^{-\varepsilon} \;=\; \Gamma(\varepsilon)\,\left(\frac{4\pi\mu^2}{\Delta}\right)^{\varepsilon}. \tag{A.200}$$

This single expression is the generic skeleton of every one-loop logarithmic divergence in the book, and the $\overline{\text{MS}}$ scheme is nothing more than a convention for reading its ε-expansion. We expand each factor. The Laurent expansion of the gamma function about its pole is

$$\Gamma(\varepsilon) \;=\; \frac{1}{\varepsilon} - \gamma_{\text{E}} + O(\varepsilon), \tag{A.201}$$

where $\gamma_{\text{E}} = 0.5772\ldots$ is the Euler–Mascheroni constant; this is the standard expansion $\Gamma(1+\varepsilon) = 1 - \gamma_{\text{E}}\varepsilon + O(\varepsilon^2)$ divided by $\Gamma(\varepsilon) = \Gamma(1+\varepsilon)/\varepsilon$. The power of the dimensionless ratio expands as

$$\left(\frac{4\pi\mu^2}{\Delta}\right)^{\varepsilon} = \exp\left[\varepsilon \ln \frac{4\pi\mu^2}{\Delta}\right] = 1 + \varepsilon \ln \frac{4\pi\mu^2}{\Delta} + O(\varepsilon^2). \tag{A.202}$$

Multiplying (A.201) and (A.202), the order-ε term of one factor meets the $1/\varepsilon$ pole of the other, and the constant $-\gamma_{\text{E}}$ survives:

$$\begin{aligned} \Gamma(\varepsilon)\left(\frac{4\pi\mu^2}{\Delta}\right)^{\varepsilon} &= \left(\frac{1}{\varepsilon} - \gamma_{\text{E}}\right)\left(1 + \varepsilon \ln \frac{4\pi\mu^2}{\Delta}\right) + O(\varepsilon) \\ &= \frac{1}{\varepsilon} - \gamma_{\text{E}} + \ln \frac{4\pi\mu^2}{\Delta} + O(\varepsilon) \\ &= \frac{1}{\varepsilon} - \gamma_{\text{E}} + \ln 4\pi + \ln \frac{\mu^2}{\Delta} + O(\varepsilon). \end{aligned} \tag{A.203}$$

Equation (A.203) is the central computation of this section. It shows that the pole $1/\varepsilon$ never travels alone: it is invariably accompanied by the same two finite constants, $-\gamma_{\text{E}}$ and $\ln 4\pi$, which arise from the gamma function and from the $(4\pi)^{d/2}$ in the loop measure and have nothing to do with the physics of the diagram. They appear in this fixed combination at every order and in every theory. A scheme that subtracts only the $1/\varepsilon$ leaves these universal constants littering every finite result; the modified scheme that subtracts them along with the pole is the one the book uses, and we now define both precisely.

A.7.4 MS versus $\overline{\text{MS}}$

A renormalization scheme is a rule that says which part of a divergent amplitude is removed by counterterms. Minimal subtraction and its modification are the two simplest such rules, distinguished only by whether the universal constants of (A.203) are swept into the counterterm along with the pole.

Definition A.48 (Minimal subtraction, MS)**.** In the *minimal-subtraction* scheme the counterterms are chosen to cancel exactly the pole terms in ε and nothing else: a counterterm is a pure Laurent polynomial in $1/\varepsilon$ with no finite part. At one loop this means subtracting only the $1/\varepsilon$ in (A.203), leaving the finite remainder $-\gamma_{\text{E}} + \ln 4\pi + \ln(\mu^2/\Delta)$ in every renormalized amplitude.

Definition A.49 (Modified minimal subtraction, $\overline{\text{MS}}$)**.** In the *modified minimal-subtraction* scheme, $\overline{\text{MS}}$, the counterterms cancel the pole together with the universal constants $-\gamma_{\text{E}} + \ln 4\pi$ that accompany it in (A.203). Equivalently, $\overline{\text{MS}}$ subtracts multiples of the combination

$$\frac{1}{\bar{\varepsilon}} \;\equiv\; \frac{1}{\varepsilon} - \gamma_{\text{E}} + \ln 4\pi, \tag{A.204}$$

in place of the bare pole $1/\varepsilon$. This is the default scheme of the book.

The two formulations of $\overline{\text{MS}}$—"subtract $1/\bar{\varepsilon}$" and "absorb $-\gamma_{\text{E}} + \ln 4\pi$ into the scale"—are identical, and the bridge between them is a redefinition of the renormalization scale that is worth making explicit, because it is the form in which $\overline{\text{MS}}$ is most often stated.

Proposition A.50 (The $\overline{\text{MS}}$ scale and the equivalence of the two formulations)**.** *Define the modified scale $\bar{\mu}$ by*

$$\bar{\mu}^2 \;\equiv\; 4\pi\,\mu^2\, e^{-\gamma_{\text{E}}}. \tag{A.205}$$

Then the one-loop combination (A.203) *can be written in either of the equivalent forms*

$$\Gamma(\varepsilon)\left(\frac{4\pi\mu^2}{\Delta}\right)^{\varepsilon} \;=\; \frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta} + O(\varepsilon) \;=\; \frac{1}{\varepsilon} + \ln\frac{\bar{\mu}^2}{\Delta} + O(\varepsilon). \tag{A.206}$$

Subtracting $1/\bar{\varepsilon}$ in the μ variable (the first form) leaves the finite logarithm $\ln(\mu^2/\Delta)$; subtracting the bare pole $1/\varepsilon$ in the $\bar{\mu}$ variable (the second form) leaves the finite logarithm $\ln(\bar{\mu}^2/\Delta)$. The two prescriptions are the same renormalization, and in the second form one conventionally renames $\bar{\mu} \to \mu$ and calls it "the $\overline{MS}$ scale," so that $\overline{MS}$ results carry logarithms $\ln(\mu^2/\Delta)$ with μ understood as $\bar{\mu}$.

Proof. The first equality of (A.206) is a regrouping of (A.203): collect the pole and the two universal constants into the single symbol $1/\bar{\varepsilon}$ of (A.204), leaving $\ln(\mu^2/\Delta)$ as the only Δ-dependent finite piece. For the second equality, take the logarithm of (A.205),

$$\ln\bar{\mu}^2 = \ln 4\pi - \gamma_{\text{E}} + \ln\mu^2, \tag{A.207}$$

and substitute into (A.203):

$$\begin{aligned}\frac{1}{\varepsilon} - \gamma_{\text{E}} + \ln 4\pi + \ln\frac{\mu^2}{\Delta} &= \frac{1}{\varepsilon} + \left(\ln 4\pi - \gamma_{\text{E}} + \ln\mu^2\right) - \ln\Delta \\ &= \frac{1}{\varepsilon} + \ln\bar{\mu}^2 - \ln\Delta = \frac{1}{\varepsilon} + \ln\frac{\bar{\mu}^2}{\Delta}.\end{aligned} \tag{A.208}$$

The two forms in (A.206) are thus algebraically identical expressions for the same quantity (A.203), differing only in whether the universal constants are displayed

as part of the subtracted constant ($1/\bar{\varepsilon}$) or hidden inside the redefined scale ($\bar{\mu}$). Removing $1/\bar{\varepsilon}$ from the first form and removing $1/\varepsilon$ from the second therefore yield finite remainders that differ only by $\ln(\bar{\mu}^2/\mu^2) = \ln 4\pi - \gamma_{\mathrm{E}}$, i.e. by the very constant absorbed into the scale; after the conventional renaming $\bar{\mu} \to \mu$ they coincide. This is the precise sense in which "$\overline{\mathrm{MS}}$ is MS with 4π and γ_{E} absorbed into μ." □

The content of (A.204) and (A.205) is worth stating in words, because it is the single most useful operational fact about the scheme: to convert any minimal-subtraction result into the modified scheme, one replaces every $1/\varepsilon$ by $1/\bar{\varepsilon} = 1/\varepsilon - \gamma_{\mathrm{E}} + \ln 4\pi$, or equivalently trades the scale μ^2 for $\bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_{\mathrm{E}}}$. Because γ_{E} and $\ln 4\pi$ are pure numbers attached to the regulator and not to any physical scale, the difference between MS and $\overline{\mathrm{MS}}$ is a finite, scheme-dependent shift of the renormalized couplings and has no effect on any physical observable; it does, however, shift the numerical value of the renormalization scale at which a given coupling takes a given value, which is why quoting a coupling such as $\alpha_s(M_Z)$ is meaningless without naming the scheme. The book quotes couplings and runs them in $\overline{\mathrm{MS}}$ throughout.

Remark A.51 (What $\overline{\mathrm{MS}}$ does *not* fix). Modified minimal subtraction fixes the finite part of the counterterms but says nothing about two further conventions that the loop chapters must still pin down, and which live elsewhere in these appendices. The first is the treatment of γ^5 and the Levi-Civita tensor in $d \neq 4$, which cannot be continued while preserving all their four-dimensional properties; the inconsistency, the 't Hooft–Veltman–Breitenlohner–Maison repair, and the pragmatic alternatives are proved and catalogued in Sections C.7 and C.7.5 and are not repeated here. The second is the explicit evaluation of the master integrals, their Feynman/Schwinger parametrizations, and the Wick rotation, all of which are Chapter D. The present section fixes only $d = 4 - 2\varepsilon$, the promotion μ-powers, and the subtraction constant $1/\bar{\varepsilon}$; those are the conventions, and they are scheme-defining, but they are not the whole computational toolkit.

A.7.5 A worked one-loop subtraction in $\overline{\mathrm{MS}}$

The conventions just fixed are best seen in action on the simplest nontrivial loop in the book, the one-loop correction to the four-point coupling in the ϕ^4 thread. The computation that follows is illustrative: it demonstrates how the $\mu^{2\varepsilon}$ promotion, the master integral (A.198), and the $\overline{\mathrm{MS}}$ subtraction (A.204) interlock to produce a finite, scale-dependent renormalized vertex. The extraction of the beta function from this divergence is the work of Chapter 24, and we do not anticipate it; we stop at the counterterm.

Worked Example A.52 (The ϕ^4 one-loop vertex and its $\overline{\mathrm{MS}}$ counterterm)**.** We continue ϕ^4 theory, whose standard Lagrangian and dimensionless coupling λ

were counted in Section A.1.8. At one loop the $2 \to 2$ amplitude receives, beyond the tree value $-i\lambda\mu^{2\varepsilon}$, a contribution from the three one-loop bubbles in the s, t, and u channels. Consider the s-channel bubble, with external momenta combining to p. Its two quartic vertices each contribute a factor $-i\lambda\mu^{2\varepsilon}$, and the diagram carries the symmetry factor $\tfrac{1}{2}$:

$$i\mathcal{M}_s \;=\; \frac{1}{2}\,\big(-i\lambda\mu^{2\varepsilon}\big)^2 \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{i}{k^2 - m^2 + i\epsilon}\,\frac{i}{(k+p)^2 - m^2 + i\epsilon}. \tag{A.209}$$

The two propagator numerators give $i \cdot i = -1$ and the vertex factors give $(-i\lambda\mu^{2\varepsilon})^2 = -\lambda^2\mu^{4\varepsilon}$, so

$$i\mathcal{M}_s = \frac{\lambda^2\mu^{4\varepsilon}}{2}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{(k^2 - m^2)\big((k+p)^2 - m^2\big)}, \tag{A.210}$$

with the $i\epsilon$ implicit. Feynman-combining the two denominators with a parameter $x \in [0,1]$ (the identity is established in Chapter D) and shifting $\ell = k + xp$ produces a single quadratic denominator $(\ell^2 - \Delta)^2$ with

$$\Delta \;=\; m^2 - x(1-x)\,p^2, \tag{A.211}$$

so that

$$i\mathcal{M}_s = \frac{\lambda^2\mu^{4\varepsilon}}{2}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^2}. \tag{A.212}$$

The inner integral is exactly the $n = 2$ master integral (A.199). Inserting it,

$$i\mathcal{M}_s = \frac{\lambda^2\mu^{4\varepsilon}}{2}\,\frac{i}{(4\pi)^2}\,(4\pi)^{\varepsilon}\,\Gamma(\varepsilon)\int_0^1 \mathrm{d}x\,\Delta^{-\varepsilon} = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\int_0^1 \mathrm{d}x\,\Gamma(\varepsilon)\left(\frac{4\pi\mu^2}{\Delta}\right)^{\varepsilon}, \tag{A.213}$$

where in the last step one factor of $\mu^{2\varepsilon}$ has been kept outside to match the dimension of the tree vertex and the remaining $\mu^{2\varepsilon}$ has been combined with $(4\pi)^{\varepsilon}\Delta^{-\varepsilon}$ into the dimensionless ratio $(4\pi\mu^2/\Delta)^{\varepsilon}$, precisely the combination (A.200) whose expansion we computed. Using (A.203) in the form (A.206),

$$i\mathcal{M}_s = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\int_0^1 \mathrm{d}x\left[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta}\right] + O(\varepsilon) = \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\left[\frac{1}{\bar{\varepsilon}} + \int_0^1 \mathrm{d}x\,\ln\frac{\mu^2}{m^2 - x(1-x)p^2}\right]. \tag{A.214}$$

The t- and u-channel bubbles have the identical structure with p^2 replaced by the appropriate Mandelstam invariant; their divergent parts are the same, so the total one-loop divergence of the four-point amplitude is three times the pole of (A.214):

$$i\mathcal{M}^{\text{1-loop}}\Big|_{\text{pole}} = 3 \times \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} = \frac{3i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\,\frac{1}{\bar{\varepsilon}}. \tag{A.215}$$

Renormalization cancels this with a counterterm vertex $-i\,\delta_\lambda\,\mu^{2\varepsilon}$, the one-loop piece of the rescaling $\lambda_0 = \mu^{2\varepsilon}(\lambda + \delta_\lambda)$ that relates the bare coupling λ_0 to the

renormalized one. In $\overline{\text{MS}}$ the counterterm removes exactly the $1/\bar{\varepsilon}$ combination, so

$$\delta_\lambda^{\overline{\text{MS}}} = \frac{3\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} = \frac{3\lambda^2}{16\pi^2}\,\frac{1}{2\bar{\varepsilon}}, \tag{A.216}$$

chosen so that $i\mathcal{M}^{\text{1-loop}} - i\delta_\lambda \mu^{2\varepsilon}$ is finite as $\varepsilon \to 0$. The renormalized one-loop vertex that survives is the finite, μ-dependent expression

$$i\mathcal{M}^{\text{ren}} = -i\lambda\mu^{2\varepsilon} + \frac{i\lambda^2\mu^{2\varepsilon}}{32\pi^2}\sum_{\sigma=s,t,u}\int_0^1 \mathrm{d}x\,\ln\frac{\mu^2}{m^2 - x(1-x)\sigma} + O(\lambda^3), \tag{A.217}$$

in which every γ_{E} and $\ln 4\pi$ has been absorbed by the choice of $\bar{\varepsilon}$, leaving only the physical logarithms of μ^2 against the kinematic scales. The reappearance of μ inside these logarithms is exactly what the renormalization-group equation of Chapter 24 exploits: demanding that the physical amplitude be independent of the arbitrary μ converts the residue of the $1/\bar{\varepsilon}$ pole in (A.216) into the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$. We carry the computation no further here; the conventions are what this section owns, and they are now exhibited acting on a complete one-loop diagram from start to finite answer.

The frozen carried numbers of the three benchmark theories—the scalar $\beta_\lambda = 3\lambda^2/16\pi^2$, the QED $\beta_e = e^3/12\pi^2$, and the Yang–Mills coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$—are all computed in precisely the $\overline{\text{MS}}$ scheme defined here, in Chapter 24 (scalar and QED) and Chapter 27 (Yang–Mills). They are quoted, not recomputed, in this appendix; the worked example above shows only the divergence and its subtraction, which is the part the conventions govern. With the regularization parameter $d = 4 - 2\varepsilon$, the coupling-promotion powers $\mu^{2\varepsilon}$ and μ^{ε}, and the modified minimal-subtraction constant $1/\bar{\varepsilon} = 1/\varepsilon - \gamma_{\text{E}} + \ln 4\pi$ now fixed, the book's computational convention set is complete: every loop integral, every counterterm, and every running coupling in Part VI and beyond is defined by the choices made in this section together with the master integrals of Chapter D and the γ^5 rules of Chapter C. What remains is to make these conventions usable alongside the standard literature, whose every reference differs from ours in at least one sign or factor. The next section provides the translation dictionary that maps the canon assembled across this appendix onto the conventions of the major texts, and closes with the master glossary of symbols.

A.8 A Translation Dictionary and the Master Glossary

The seven preceding sections have assembled a single, internally consistent convention set: the units and mass-dimension count of Section A.1, the mostly-minus metric and index gymnastics of Section A.2, the Fourier conventions

and invariant measures of Section A.3, the analytic $i\epsilon$, propagator, and Wick-rotation conventions of Section A.4, the action and amplitude normalizations of Section A.5, the spinor and gamma-matrix conventions collected in Chapter C, the gauge and charge conventions and group-theory normalizations of Section A.6, and the dimensional-regularization and $\overline{\text{MS}}$ conventions of Section A.7. Within these covers that set is the law, and every chapter that follows is written in it. But the reader does not live inside these covers alone: the standard literature of the field—Peskin and Schroeder, Srednicki, Weinberg, Schwartz, Zee—is written in conventions that differ from ours and from one another, most consequentially in the sign of the metric. A formula copied from one of those books into a calculation framed in ours will, if the conventions are not reconciled, be wrong by signs and factors of i that no amount of algebra will repair. This closing section gives the reader the means to reconcile them. It does so in two complementary ways. First, and most importantly, it isolates the *single* transformation—a relabelling of the metric—from which almost every cross-text discrepancy descends, and proves exactly how that transformation acts on an action, on an on-shell condition, and on an $i\epsilon$ prescription, so that the reader can convert *any* formula in *any* of these texts from first principles rather than trusting a table. Second, it records the explicit conversion tables for quick reference, with honest warnings about the few entries that are genuinely fragile across editions. It then states plainly which local conventions this book deliberately does not globalize, and closes the appendix with the master symbol glossary, each entry keyed both to the chapter where the symbol first appears and to the section here that fixes its meaning.

A.8.1 The one principle: a relabelling of the metric

There are two ways to live with the convention diversity of the literature. One is to memorize, for each external source, a table of its choices and a list of substitution rules; this is fragile, edition-dependent, and silent about why the rules take the form they do. The other is to recognize that the dominant source of disagreement—the metric signature—enters every Lorentz-covariant expression through exactly one operation, the contraction of indices, and to learn once and for all how that operation responds to a change of signature. We take the second route. Everything in this subsection is derived, not quoted, and the result is a procedure that converts a formula written in the opposite signature with no appeal to any text's idiosyncrasies.

Definition A.53 (The two signatures)**.** Write $\eta^{(-)}_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ for this book's *mostly-minus* metric (Section A.2) and $\eta^{(+)}_{\mu\nu} = \mathrm{diag}(-,+,+,+)$ for the *mostly-plus* metric of Srednicki [454] and Weinberg [500]. The two are related by an overall sign,

$$\eta^{(+)}_{\mu\nu} \;=\; -\,\eta^{(-)}_{\mu\nu}, \qquad \eta^{(+)\,\mu\nu} \;=\; -\,\eta^{(-)\,\mu\nu}, \tag{A.218}$$

the second equality following because the matrix inverse of $-M$ is $-M^{-1}$. Contravariant components a^μ of any vector (the physical components, such as energy and three-momentum in $p^\mu = (E, \mathbf{p})$) are the *same* numbers in both conventions; only the rule for lowering an index, and hence every contraction, changes sign.

The entire conversion rests on a single elementary observation about contractions.

Lemma A.54 (Scalar products flip sign)**.** *For any two four-vectors with fixed contravariant components a^μ, b^μ, the Lorentz scalar product computed in the two conventions differs by an overall sign:*

$$(a \cdot b)_+ \;=\; -\,(a \cdot b)_-, \tag{A.219}$$

where $(a \cdot b)_\pm \equiv \eta^{(\pm)}_{\mu\nu} a^\mu b^\nu$. In particular $(p^2)_+ = -(p^2)_-$ for any momentum p^μ.

Proof. Direct from (A.218): $(a\cdot b)_+ = \eta^{(+)}_{\mu\nu} a^\mu b^\nu = -\eta^{(-)}_{\mu\nu} a^\mu b^\nu = -(a\cdot b)_-$. Setting $b = a$ gives the statement for p^2. □

From Theorem A.54 one already reads off the most frequently needed conversion. The mostly-minus on-shell condition $p^2 = m^2$, with m the rest mass (a Lorentz scalar parameter, hence the same number in both conventions), becomes, on substituting $(p^2)_- = -(p^2)_+$,

$$(p^2)_- = m^2 \quad\Longleftrightarrow\quad -(p^2)_+ = m^2 \quad\Longleftrightarrow\quad (p^2)_+ = -m^2, \tag{A.220}$$

which is exactly the form $p^2 = -m^2$ that a reader of Srednicki or Weinberg will recognize. The mass-shell has not moved; only the number assigned to p^2 on it has changed sign. The same arithmetic governs whole Lagrangians, once we account for the fact that different terms carry different numbers of contractions.

Theorem A.55 (Master sign rule for signature change)**.** *Let a Lorentz-scalar Lagrangian density be written in the mostly-minus convention as a sum of monomials,*

$$\mathcal{L}_- \;=\; \sum_i c_i\, T_i, \tag{A.221}$$

where each T_i is a Lorentz scalar built from fields and derivatives and c_i is a numerical (signature-independent) coefficient. Let k_i be the number of metric contractions in T_i, that is, the number of factors of $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$ needed to write T_i with all field and derivative indices in their natural (contravariant for ∂^μ via raising, etc.) positions. Then the same physical theory *—the same action, hence the same e^{iS} and the same equations of motion—is described in the mostly-plus convention by*

$$\mathcal{L}_+ \;=\; \sum_i c_i\,(-1)^{k_i}\, T_i, \tag{A.222}$$

each T_i now evaluated with $\eta^{(+)}$.

Proof. A single metric contraction is, by Theorem A.54, an operation whose numerical value flips sign under $\eta^{(-)} \to \eta^{(+)}$; a scalar with k_i independent contractions therefore satisfies

$$(T_i)_+ \;=\; (-1)^{k_i}\,(T_i)_-, \tag{A.223}$$

since each contraction contributes one factor of η (or η^{-1}) and each such factor flips sign by (A.218), the signs multiplying because distinct contractions involve distinct metric factors. Now demand that the action be the same physical object in both descriptions, $S_+ = S_-$, term by term: the i-th contribution must satisfy $c_i(-1)^{k_i}(T_i)_+ = c_i(T_i)_-$. Using (A.223) the left side is $c_i(-1)^{k_i}(-1)^{k_i}(T_i)_- = c_i(T_i)_-$, which equals the right side identically. Hence the coefficients $c_i(-1)^{k_i}$ in (A.222) are exactly those that preserve the numerical value of every term, and (A.222) describes the same action. Because the action is unchanged, so are the Euler–Lagrange equations and every observable. □

Theorem A.55 is the engine of all signature conversion: a term flips sign once for every contracted index pair it contains, and not otherwise. A mass or potential term, which contracts no indices, never changes sign; a two-derivative kinetic term, with one contraction, always does; a term like $F_{\mu\nu}F^{\mu\nu}$, with two contractions, changes sign twice and so is invariant. We exhibit the rule on the scalar of ϕ^4 theory.

Worked Example A.56 (Translating the scalar Lagrangian and propagator to mostly-plus)**.** The frozen ϕ^4 Lagrangian (Chapter 6) in this book's convention is

$$\mathcal{L}_- \;=\; \tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,m^2\phi^2 - \frac{\lambda}{4!}\,\phi^4. \tag{A.224}$$

Classify the three terms by their contraction count: the kinetic term $\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi = \tfrac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ has $k=1$; the mass term and the quartic term contract nothing, so $k=0$. By Theorem A.55 only the kinetic term flips:

$$\mathcal{L}_+ \;=\; -\tfrac{1}{2}\,(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}\,m^2\phi^2 - \frac{\lambda}{4!}\,\phi^4, \tag{A.225}$$

with $\partial^\mu\phi$ now raised by $\eta^{(+)}$. This is precisely the mostly-plus scalar Lagrangian of Srednicki [454]: the leading minus sign on the kinetic term, which to a first-time reader looks like a different theory, is nothing but the single sign demanded by the one contraction in the kinetic term. As a check, evaluate the kinetic term in components in each convention with $\eta^{(\pm)}$. Mostly-minus, $\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi = \tfrac{1}{2}\big[(\partial_t\phi)^2 - (\nabla\phi)^2\big]$; mostly-plus, $-\tfrac{1}{2}\partial_\mu\phi\,\partial^\mu\phi = -\tfrac{1}{2}\big[-(\partial_t\phi)^2 + (\nabla\phi)^2\big] = \tfrac{1}{2}\big[(\partial_t\phi)^2 - (\nabla\phi)^2\big]$. The two are numerically identical, as Theorem A.55 guarantees: the physical Lagrangian—positive time-kinetic energy, negative gradient energy—is the same object written twice.

The Feynman propagator translates with no more than Theorem A.54 and a careful tracking of the $i\epsilon$. The book's momentum-space propagator (Section A.4) is the time-ordered two-point function

$$\langle 0|\,\mathcal{T}\,\phi(x)\phi(y)\,|0\rangle \;=\; \int \frac{\mathrm{d}^4p}{(2\pi)^4}\;\frac{i}{p^2-m^2+i\epsilon}\;e^{-ip\cdot(x-y)}, \tag{A.226}$$

with $p^2=(p^2)_-$. We convert the integrand by substituting $(p^2)_-=-(p^2)_+$ from Theorem A.54, taking care to factor the sign cleanly through the $i\epsilon$:

$$\frac{i}{(p^2)_- - m^2+i\epsilon} \;=\; \frac{i}{-(p^2)_+ - m^2+i\epsilon} \;=\; \frac{i}{-\big[(p^2)_+ + m^2 - i\epsilon\big]} \;=\; \frac{-i}{(p^2)_+ + m^2 - i\epsilon}. \tag{A.227}$$

Two things flipped together. The denominator p^2-m^2 became $-\big[(p^2)_+ + m^2\big]$, exposing the mostly-plus mass-shell $(p^2)_+=-m^2$ of (A.220) as the location of the pole; and the prescription $+i\epsilon$ became $-i\epsilon$, because pulling the overall -1 out of the bracket carries the infinitesimal with it. The result,

$$\langle 0|\,\mathcal{T}\,\phi(x)\phi(y)\,|0\rangle \;=\; \int \frac{\mathrm{d}^4p}{(2\pi)^4}\;\frac{-i}{p^2+m^2-i\epsilon}\;e^{+ip\cdot(x-y)}, \tag{A.228}$$

is the mostly-plus Feynman propagator. (The plane-wave phase has likewise become $e^{+ip\cdot(x-y)}$, since $p\cdot(x-y)$ itself flipped sign; this is the origin of the "Fourier sign" row in the tables below.) Comparison with Srednicki is exact: his free propagator is $\widetilde{\Delta}(p)=1/(p^2+m^2-i\epsilon)$ together with the convention $\langle 0|\mathcal{T}\phi\phi|0\rangle=(1/i)\Delta$, so that $\langle 0|\mathcal{T}\phi\phi|0\rangle=-i/(p^2+m^2-i\epsilon)$, identical to (A.228). The physical correlator is one object; the book and Srednicki write it in opposite signatures, and the master rule maps one writing onto the other with no residue.

The one extra wrinkle: gamma matrices. Scalars and gauge fields convert by Theorem A.55 alone, because their Lagrangian terms are built only from fields, derivatives, and explicit metric factors. Spinors carry one more piece of metric data: the Clifford algebra $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}$ (Chapter C) ties the gamma matrices themselves to the signature, so a change of η forces a redefinition of the gammas before the master rule can be applied. The redefinition is fixed by demanding that the new gammas obey the new algebra. Take

$$\gamma^\mu_+ \;\equiv\; i\,\gamma^\mu_-, \tag{A.229}$$

where γ^μ_- are the book's matrices with $\{\gamma^\mu_-,\gamma^\nu_-\}=2\eta^{(-)\,\mu\nu}$. Then

$$\{\gamma^\mu_+,\gamma^\nu_+\} \;=\; i^2\,\{\gamma^\mu_-,\gamma^\nu_-\} \;=\; -\,2\,\eta^{(-)\,\mu\nu} \;=\; 2\,\eta^{(+)\,\mu\nu}, \tag{A.230}$$

using (A.218) in the last step, so (A.229) indeed produces a set of matrices satisfying the mostly-plus algebra. Hermiticity tracks correctly too: in mostly-minus γ^0_- is Hermitian and γ^i_- anti-Hermitian, so by (A.229) γ^0_+ is anti-Hermitian

(as required by $(\gamma_+^0)^2 = \eta^{(+)\,00} = -1$) and γ_+^i Hermitian. Applying (A.229) to the book's free Dirac Lagrangian $\mathcal{L}_- = \bar{\psi}(i\not{\partial}_- - m)\psi$ with $\not{\partial}_- = \gamma_-^\mu \partial_\mu$ gives $i\not{\partial}_- = i\gamma_-^\mu \partial_\mu = i(-i\gamma_+^\mu)\partial_\mu = \gamma_+^\mu \partial_\mu = \not{\partial}_+$, and one checks that the Dirac adjoint is unchanged ($\bar{\psi} = \psi^\dagger \gamma_-^0 = \psi^\dagger(-i\gamma_+^0)$ is the correct mostly-plus adjoint $\psi^\dagger \beta_+$ with the Hermitian $\beta_+ = -i\gamma_+^0$), so

$$\mathcal{L}_+ \;=\; \bar{\psi}(\not{\partial}_+ - m)\psi, \qquad \not{\partial}_+ = \gamma_+^\mu \partial_\mu. \tag{A.231}$$

The explicit factor of i that sits in front of the kinetic term in mostly-minus has been absorbed into the redefinition (A.229); whether one sees that i depends on which normalization of the mostly-plus gammas a given text adopts. We isolate this lesson, because it is the single most common trap in cross-text spinor algebra.

Caution

Not everything that differs between texts is fixed by the metric. The signature determines the scalar-product sign, the on-shell condition $p^2 = \pm m^2$, and the propagator denominator and $i\epsilon$ (Theorems A.54 and A.56) unambiguously. But the placement of the factor i in the Dirac kinetic term, the overall sign of γ^5, the Hermiticity assignment of γ^0, and the sign in the spinor propagator numerator depend on the gamma and Dirac-adjoint normalization a text chooses *in addition* to its signature. Convert the metric-determined data with Theorem A.55; for the gamma normalization, read the target text's explicit definitions (or this book's, in Chapter C) directly. As a concrete anchor, the metric-determined content of the Dirac propagator is its denominator: the book's $\langle \mathcal{T}\psi\bar{\psi}\rangle \propto i/(p^2 - m^2 + i\epsilon)$ has the mostly-plus denominator $p^2 + m^2 - i\epsilon$, matching Srednicki's $\widetilde{S}(p) = (-\not{p} + m)/(p^2 + m^2 - i\epsilon)$; the numerator's $\not{p}$ sign and the location of the overall i follow his gamma and adjoint conventions, not the signature alone.

A.8.2 The conversion tables

We now record, for quick reference, the conventions of the five most widely used graduate texts against this book's. Table A.3 collects the choices that descend from the metric signature—the items that Theorem A.55 and Theorem A.54 convert mechanically—and Table A.4 collects the remaining discrete choices (the γ^5 sign, the orientation ϵ^{0123}, the gauge and charge conventions, and the coupling normalization α). The reader is urged to treat the tables as an index into the derivations above, not as a substitute for them: the few fragile rows, flagged below, are exactly the ones where consulting the source text's own statement is safer than trusting a remembered entry.

Table A.3: Signature-linked conventions. Every entry in this table is converted mechanically by the master sign rule Theorem A.55 and the scalar-flip lemma Theorem A.54. "$+\frac{1}{2}(\partial\phi)^2$" abbreviates the sign of the scalar kinetic term in $\mathcal{L}$; the positive-energy plane wave is the sign in the exponent $e^{\mp ip\cdot x}$; the on-shell relation is the value of p^2 for a particle of mass m. All five texts and this book use the Minkowski path-integral weight e^{+iS}.

	Metric $\eta_{\mu\nu}$	Kinetic sign	Pos.-energy wave	On-shell
This book	$(+,-,-,-)$	$+\frac{1}{2}(\partial\phi)^2$	$e^{-ip\cdot x}$	$p^2 = m^2$
Peskin–Schroeder	$(+,-,-,-)$	$+\frac{1}{2}(\partial\phi)^2$	$e^{-ip\cdot x}$	$p^2 = m^2$
Schwartz	$(+,-,-,-)$	$+\frac{1}{2}(\partial\phi)^2$	$e^{-ip\cdot x}$	$p^2 = m^2$
Zee	$(+,-,-,-)$	$+\frac{1}{2}(\partial\phi)^2$	$e^{-ip\cdot x}$	$p^2 = m^2$
Srednicki	$(-,+,+,+)$	$-\frac{1}{2}(\partial\phi)^2$	$e^{+ip\cdot x}$	$p^2 = -m^2$
Weinberg	$(-,+,+,+)$	$-\frac{1}{2}(\partial\phi)^2$	$e^{+ip\cdot x}$	$p^2 = -m^2$

Table A.4: Discrete and coupling conventions not fixed by the metric alone. The γ^5 and ϵ^{0123} rows give each text's stated definition with $\epsilon^{0123} = -\epsilon_{0123}$ in either signature (since $\det\eta = -1$ in both, see Section A.2). The electron-charge column records whether the symbol e denotes the positive elementary charge ($e > 0$) or the signed electron charge ($e < 0$). The last two rows are the fragile ones discussed in Section A.8.2.

	This book	Peskin–Schr.	Srednicki	Weinberg	Schwartz / Zee
γ^5	$i\gamma^0\gamma^1\gamma^2\gamma^3$	$i\gamma^0\gamma^1\gamma^2\gamma^3$	$i\gamma^0\gamma^1\gamma^2\gamma^3$	$-i\gamma^0\gamma^1\gamma^2\gamma^3$	$i\gamma^0\gamma^1\gamma^2\gamma^3$
ϵ^{0123}	$+1$	$+1$	$+1$	-1	$+1$
α	$e^2/4\pi$	$e^2/4\pi$	$e^2/4\pi$	$e^2/4\pi$	$e^2/4\pi$
electron e	$e > 0$	$e < 0$	$e > 0$	$e > 0$	$e > 0$
D_μ (electron)	$\partial_\mu + ieA_\mu$	$\partial_\mu + ieA_\mu$	$\partial_\mu - ieA_\mu$	$\partial_\mu - ieA_\mu$	$\partial_\mu + ieA_\mu$

The reassuring rows are the first and third of Table A.4: every text in standard use, and this book, writes $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ (Weinberg [500] alone carries the opposite overall sign, which exchanges the left- and right-handed projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$), and every one of them uses rationalized Heaviside–Lorentz units in which $\alpha = e^2/4\pi \approx 1/137$ rather than the Gaussian $\alpha = e^2$. The fine-structure constant is therefore the same symbol with the same value across all six conventions, which is why numerical cross sections quoted in any of these books can be compared directly once the signature-linked factors have been reconciled.

Caution

The charge and covariant-derivative rows are convention-fragile. The last two rows of Table A.4 are the entries most likely to be misremembered, because they encode *two* independent choices—the sign in the covariant derivative, $D_\mu = \partial_\mu \mp ieA_\mu$, and the meaning of the symbol e (positive elementary charge versus signed electron charge)—and different texts split these differently. Peskin and Schroeder [396] are the notorious case: they write $D_\mu = \partial_\mu + ieA_\mu$ with e *negative*, equal to the signed electron charge, so their photon–electron vertex carries a sign opposite to what a reader expecting $e > 0$ would assume. This book, like Schwartz [432] and Zee [532], takes $e > 0$ and assigns the electron charge $-e$, writing $D_\mu = \partial_\mu + ieA_\mu$ on the electron field (equivalently $D_\mu = \partial_\mu - iqA_\mu$ for a field of physical charge q, matching the nonabelian $D_\mu = \partial_\mu - igA^a_\mu T^a$ of Section A.6). The crucial point is that none of these choices is physical on its own: the simultaneous replacement

$$e \;\to\; -e, \qquad A_\mu \;\to\; -A_\mu \tag{A.232}$$

is a field redefinition (a relabelling of the photon field's sign) that leaves D_μ, the field strength $F_{\mu\nu}$, the Lagrangian, and every observable invariant. In particular $\alpha = e^2/4\pi$ is invariant under (A.232), so a cross section copied from Peskin needs *no* charge-sign correction; only an amplitude in which a *single* power of e appears alongside an external photon polarization—an interference term linear in the charge—is sensitive to the convention, and there one simply uses (A.232) to align the sign of e with the sign of A_μ. When in doubt, verify the source's covariant derivative and its statement of the electron charge directly rather than trusting the table; this is the one place where a remembered entry is genuinely dangerous.

A.8.3 The translation procedure, step by step

Collecting the results above into a recipe: to import a Lorentz-covariant formula from an external text into this book's conventions (or to export one), proceed as follows.

1. *Identify the source signature* from Table A.3. If it agrees with ours (Peskin, Schwartz, Zee), no metric conversion is needed and one proceeds to step 4. If it is mostly-plus (Srednicki, Weinberg), continue.

2. *Convert every scalar product* by $(a \cdot b) \to -(a \cdot b)$ (Theorem A.54). In particular replace each $p^2 + m^2$ by $-(p^2 - m^2)$ and each $p^2 - m^2$-type denominator's $i\epsilon$ by its opposite sign, as in (A.227).

3. *Convert Lagrangian terms* by the master rule Theorem A.55: flip the sign of each term once per contracted index pair. Two-derivative kinetic terms flip; mass and potential terms do not; $F_{\mu\nu}F^{\mu\nu}$ does not.

4. *Reconcile the gamma normalization* by hand (Section A.8.1): match the source's γ^5 sign and Dirac-adjoint convention against Chapter C, using Table A.4.

5. *Reconcile the charge convention* with (A.232) if the source uses a different sign of e (Section A.8.2); recall that α and unsigned observables are unaffected.

Worked Example A.57 (Exporting the QED Dirac Lagrangian to Srednicki's conventions)**.** Take the QED Lagrangian (Chapter 8) in this book's mostly-minus form,

$$\mathcal{L}^{-}_{\text{QED}} \;=\; \bar\psi(i\not{D} - m)\psi - \tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu, \tag{A.233}$$

and convert it to Srednicki's mostly-plus conventions. We run the recipe. The source signature is mostly-plus, so steps 2–4 apply. The gauge kinetic term $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} = -\frac{1}{4}\eta^{\mu\alpha}\eta^{\nu\beta}F_{\mu\nu}F_{\alpha\beta}$ has $k = 2$ contractions, hence by Theorem A.55 flips sign twice and is *unchanged*: $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu}$ in both conventions. The fermion mass term $-m\bar\psi\psi$ has $k = 0$ and is unchanged. The covariant kinetic term $i\bar\psi\not{D}\psi = i\bar\psi\gamma^\mu_-(\partial_\mu + ieA_\mu)\psi$ contains the spinor wrinkle; applying the gamma redefinition (A.229), $i\gamma^\mu_- = \gamma^\mu_+$, exactly as for the free Dirac term (A.231), so $i\bar\psi\not{D}_-\psi = \bar\psi\,\gamma^\mu_+(\partial_\mu + ieA_\mu)\psi = \bar\psi\not{D}_+\psi$ with $\not{D}_+ = \gamma^\mu_+ D_\mu$. Assembling,

$$\mathcal{L}^{+}_{\text{QED}} \;=\; \bar\psi(\not{D}_+ - m)\psi - \tfrac{1}{4}\,F_{\mu\nu}F^{\mu\nu}, \qquad D_\mu = \partial_\mu + ieA_\mu. \tag{A.234}$$

The gauge sector and the covariant-derivative form are identical to the mostly-minus expression (A.233)—the photon kinetic term because its two contractions cancel, the covariant derivative because the gauge field's mass dimension and the coupling structure are signature-blind (Section A.1)—while the Dirac kinetic term has shed its explicit i into the redefined gammas. This matches Srednicki's QED up to his particular gamma normalization (Section A.8.1) and his opposite photon-sign convention, reconciled below; should one wish instead to land on his stated form $i\bar\psi\gamma^\mu D_\mu\psi$ with a manifest i, one simply chooses the mostly-plus gammas to be γ^μ_- themselves (which satisfy $\{\gamma_-,\gamma_-\} = 2\eta^{(-)} = -2\eta^{(+)}$, the opposite-sign Clifford convention some mostly-plus texts adopt) rather than $i\gamma^\mu_-$, whereupon the i reappears unchanged. Either way, the physical content—the Dirac equation squaring to the Klein–Gordon operator with $p^2 = -m^2$, the vertex $\propto e$, the cross sections of Chapter 17—is identical, as it must be. Finally, although Srednicki takes $e > 0$ as we do, his stated electron covariant derivative is $D_\mu = \partial_\mu - ieA_\mu$ (Table A.4), the opposite photon sign to our $\partial_\mu + ieA_\mu$; landing on his form therefore requires the photon relabeling $A_\mu \to -A_\mu$—the A-half

of (A.232), with e left untouched since both conventions take it positive. Were one translating instead from Peskin, who writes the same $D_\mu = \partial_\mu + ieA_\mu$ as we do but with $e < 0$, the required step would instead be the charge relabeling $e \to -e$. Either piece is half of the unphysical field redefinition (A.232), so every observable is unchanged.

A.8.4 Conventions deliberately kept local

A notation arbiter must say not only what it fixes globally but also what it declines to fix, lest a reader assume a global convention where none exists. As of this writing (mid-2020s), the book makes two deliberate exceptions to global standardization, in both cases because the local community's conventions are so entrenched and so finely tuned to their subject that importing them into a single global standard would do more harm than good.

Spinor-helicity variables. The angle and square brackets $\langle ij \rangle$ and $[ij]$ of the modern amplitudes program, together with the little-group and reality conventions that fix their relative signs, are *not* globalized. They are defined at their point of use in Chapter 58 and restated at the top of every amplitudes chapter, precisely because the sign conventions for these brackets differ between the major references and even between papers by the same authors, and because the spinor-helicity formalism is most naturally phrased with its own metric and reality conditions that interact with the choices of Chapter C in ways best handled in situ rather than by a global decree. A reader who needs the bracket conventions should look to Chapter 58, not to this appendix.

Conformal and bootstrap normalizations. The conformal field theory and conformal-bootstrap chapters carry their own normalizations for operator dimensions, two- and three-point function coefficients, OPE coefficients, and the conformal blocks, fixed locally in Chapter 48 and used throughout Part XIV. These are likewise *not* promoted to global conventions: the bootstrap literature has settled on particular normalizations (unit-normalized two-point functions, specific conventions for the cross-ratios and for the sign of the OPE coefficients) whose value lies in matching the numerical bootstrap output exactly, and re-expressing them in a foreign global convention would sever that connection without benefit. Where a quantity defined in those chapters is used elsewhere, it is imported by Chapter 48 and Part XIV with its local normalization stated, not silently absorbed here.

Notation Watch

The two exceptions above are the *only* places in the book where a quantity load-bearing in its own part is governed by a convention not fixed in this appendix. Everywhere else—every metric contraction, every propagator, every gamma trace, every covariant derivative, every coupling

normalization—the conventions of Sections A.1 to A.7 are in force without exception. When a chapter restates the spinor-helicity or bootstrap conventions, it does so because they are local by this deliberate design, not because the global conventions have lapsed.

A.8.5 The master glossary

We close the appendix with the master symbol glossary. Each entry lists a symbol, its meaning, the section of this appendix (or the appendix) that fixes its convention, and the chapter in which it first appears in the main text. The glossary is the single index a reader should consult when an unfamiliar symbol is encountered: it points simultaneously to where the symbol is *defined* (its first-use chapter) and to where its *convention* is pinned down (the relevant section here). Table A.5 covers the kinematic and field-theoretic core; Table A.6 covers the gauge, spinor, group, and renormalization symbols; Table A.7 covers the symbols proper to the three benchmark theories and the deliberately local conventions of Section A.8.4.

Table A.5: Master glossary, part I: kinematics, measures, and the field-theory core.

Symbol	Meaning	Convention fixed in	First used
$\hbar = c = 1$	natural units	Section A.1	Chapter 1
$[\,X\,]$	mass dimension of X	Theorem A.1	Chapter 6
$\eta_{\mu\nu}$	metric $\mathrm{diag}(+,-,-,-)$	Section A.2	Chapter 5
x^μ, p^μ	coordinates, four-momentum	Section A.2	Chapter 6
∂_μ, $\Box$	derivative, d'Alembertian $\partial_\mu\partial^\mu$	Section A.2	Chapter 6
$\epsilon^{\mu\nu\rho\sigma}$	Levi-Civita, $\epsilon^{0123} = +1$	Section A.2	Chapter 11
S	action $\int \mathrm{d}^d x\, \mathcal{L}$, $[\,S\,] = 0$	Eq. (A.6)	Chapter 6
$\mathcal{L}$	Lagrangian density, $[\,\mathcal{L}\,] = d$	Eq. (A.8)	Chapter 6
$\displaystyle\int \frac{\mathrm{d}^4 p}{(2\pi)^4}$	momentum measure	Section A.3	Chapter 9
$\delta^{(4)}(p)$	delta function, $(2\pi)^4$ per x-integral	Section A.3	Chapter 9
$e^{-ip\cdot x}$	positive-energy plane wave	Section A.3	Chapter 9
D_F, S_F	scalar, fermion Feynman propagator	Section A.4	Chapter 10
$i\epsilon$	Feynman prescription $i/(p^2 - m^2 + i\epsilon)$	Section A.4	Chapter 10
$Z[J]$, $W[J]$, Γ	generating functionals	Section A.5	Chapter 10
$\lvert 0\rangle$, $\langle\,\cdot\,\rangle$	vacuum, time-ordered correlator	Section A.4	Chapter 9

With the glossary in place, the appendix closes. The reader now holds two instruments. The first is a single, internally consistent convention set, fixed across Sections A.1 to A.7 and indexed by the glossary of Section A.8.5, which serves as the sole arbiter for every notational question the main text can raise. The second is the translation machinery of Section A.8.1—the metric-flip lemma,

Table A.6: Master glossary, part II: spinors, gauge fields, group theory, and renormalization.

Symbol	Meaning	Convention fixed in	First used
γ^μ	Dirac matrices, $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}$	Chapter C	Chapter 11
γ^5	$i\gamma^0\gamma^1\gamma^2\gamma^3$	Chapter C	Chapter 11
ψ, $\bar\psi$	Dirac field, $\bar\psi=\psi^\dagger\gamma^0$	Chapter C	Chapter 11
$\sigma^{\mu\nu}$	$\frac{i}{2}[\gamma^\mu,\gamma^\nu]$	Chapter C	Chapter 11
$\not{p}$	$\gamma^\mu p_\mu$	Chapter C	Chapter 11
A_μ, A^a_μ	gauge potential	Section A.6	Chapter 8
$F_{\mu\nu}$, $F^a_{\mu\nu}$	field strength	Section A.6	Chapter 8
D_μ	$\partial_\mu - igA^a_\mu T^a$	Section A.6	Chapter 8
e	elementary charge $e>0$, electron charge $-e$	Section A.6	Chapter 8
α	$e^2/4\pi$ (Heaviside–Lorentz)	Section A.6	Chapter 8
ξ	gauge-fixing parameter, R_ξ gauges	Section A.6	Chapter 13
T^a, f^{abc}	generators, structure constants	Section A.6	Chapter 25
$\mathrm{tr}(T^aT^b)=\frac{1}{2}\delta^{ab}$	fundamental normalization	Section A.6	Chapter 25
C_A, C_F, T_F	Casimirs and Dynkin index	Chapter B	Chapter 27
$d=4-2\varepsilon$	dimensional regularization	Section A.7	Chapter 21
μ	renormalization scale	Section A.7	Chapter 22
$\overline{\mathrm{MS}}$	modified minimal subtraction	Section A.7	Chapter 22

the master sign rule, and the step-by-step procedure—which lets any formula from any of the standard texts be carried into this book's conventions, or out of them, without convention drift. Equipped with both, the reader can move freely between this book and the wider literature, confident that every sign and every factor of i has a derivation behind it rather than a convention to be taken on faith.

Further reading

The conventions fixed here are, with minor exceptions, those of Peskin and Schroeder [396] and Schwartz [432], and a reader cross-checking a sign or a normalization against those texts will find the dictionary of Section A.8 the fastest route. Srednicki [454] uses the opposite, mostly-plus metric and a distinct γ-matrix and $\overline{\mathrm{MS}}$ bookkeeping, and is the most instructive contrast precisely because so much differs; Weinberg [500] develops the foundations from the particle-state viewpoint with its own careful conventions, and Zee [532] offers a complementary, more heuristic register. For the particle-data and numerical conventions used when our formulae are confronted with experiment, the current edition of the Review of Particle Physics [382] is the standing reference. The definitive treatment of renormalization conventions, the precise content of $\overline{\mathrm{MS}}$, and the scheme dependence that the regularization section only fixes is Collins

Table A.7: Master glossary, part III: the benchmark theories and the deliberately local conventions of Section A.8.4.

Symbol	Meaning	Convention fixed in	First used
ϕ, m, λ	ϕ^4 theory: scalar, mass, quartic coupling	Section A.1.8	Chapter 6
ψ, m, e	QED: electron, mass, charge	Section A.1.8	Chapter 8
A^a_μ, g, α_s	Yang–Mills theory: gluon, coupling, $\alpha_s = g^2/4\pi$	Section A.1.8	Chapter 25
n_f, b_0	flavor number, beta-function coefficient	Section A.6	Chapter 27
$\langle ij\rangle$, $[ij]$	spinor-helicity brackets (*local*)	Chapter 58	Chapter 58
Δ, c_{ijk}	CFT dimension, OPE coefficient (*local*)	Chapter 48	Chapter 48

[134]; it is where to go beyond the conventions recorded here.

Appendix B

Group Theory Data

Every gauge theory in this book runs on a small, fixed bundle of numbers: the index $T(R)$ and quadratic Casimir $C_2(R)$ of a representation, the structure constants f^{abc} and the symmetric invariants d^{abc}, the completeness relation that collapses a string of generators, and the handful of SU(N) color factors C_F, C_A, T_F that decorate every loop in a non-abelian amplitude. These quantities are not deep once the representation theory behind them is in hand, but they are needed so often, and with such unforgiving sign and normalization conventions, that collecting them in one place pays for itself many times over. That is the role of this appendix: it is the gauge-theory *data* appendix, the lookup table the computational chapters reach for, kept disjoint from the conceptual development of Lie theory itself.

The division of labor with Chapter 2 is deliberate and worth stating plainly. Chapter 2 is where the ideas live — what a Lie group and its algebra are, the exponential map, Schur's lemma, irreducibility, the adjoint representation, the classification of root systems and the Weyl and Freudenthal formulas that generate dimensions and indices. None of that is reproved here. This appendix imports those results by reference and puts them to work: it converts the abstract machinery into the explicit matrices, traces, and tables a working calculation consumes. Where a derivation here is short and illuminating — the master relation $\dim(R)\,C_2(R) = \dim(G)\,T(R)$, the total antisymmetry of f^{abc}, the fundamental completeness relation — it is carried out in full, because seeing it once fixes the conventions for good. Where a derivation belongs to Chapter 2 — the Weyl dimension formula, the Freudenthal recursion — it is quoted as a tool and its outputs are tabulated.

The appendix is built from the ground up. We begin in Section B.1 with the two invariants every representation carries, the index and the quadratic Casimir, and prove the relation that ties them to each other and to the adjoint. Section B.2 makes everything concrete with the Pauli and Gell-Mann matrices and the full $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ algebra. Section B.3 establishes the fundamental

completeness (color-Fierz) relation for SU(N) and the trace-reduction algorithm it powers. Section B.4 assembles the standard SU(N) Casimirs and color factors, including the $f^{acd}f^{bcd} = C_A\,\delta^{ab}$ identity that feeds the Yang–Mills beta function. Section B.5 introduces the cubic symmetric invariant d^{abc} and the anomaly coefficient $A(R)$. Section B.6 closes with a book-wide master table of $\dim(R)$, $T(R)$, $C_2(R)$, and $A(R)$ for the classical families SU(N), SO(N), Sp(N) and for the grand-unified representations of SU(5) and SO(10).

The data collected here is consumed throughout the gauge-theory chapters: the color factors of Chapter 27 and Chapter 28, the anomaly bookkeeping of Chapters 36 and 37, and the model-building of Chapter 35 all draw on these tables. We also lean on the global normalization fixed in Chapter A — the fundamental normalization $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$ in particular — and on the Lorentz-representation context of Chapter 3 and Chapter C, which determines which spinor multiplets carry the gauge charges tabulated here. With those imports declared, the appendix is otherwise self-contained.

B.1 Invariants of a Representation: The Index and the Quadratic Casimir

The chapter introduction framed this appendix as the place where the abstract representation theory built in Chapter 2 is turned into the concrete numbers that gauge-theory computations consume. Where Chapter 2 asked *which* representations a compact Lie algebra has and *how* they decompose, this appendix asks a narrower and more arithmetical question: given a representation, what scalar data does it carry, and how is that data assembled into the coefficients of a beta function, a color factor, an anomalous dimension? Every such number, it turns out, is built from just two invariants attached to each representation — the Dynkin index $T(R)$ and the quadratic Casimir $C_2(R)$ — together with the structure constants of the algebra. This opening section defines those two invariants, proves the single master identity that ties them together, and fixes the one normalization freedom that any conventions document must pin down. Everything in the rest of the appendix — the Pauli and Gell-Mann matrices, the SU(N) identities, the explicit tables — is the evaluation of these abstract objects in a chosen basis. We therefore prove the general statements here, once, and never reprove them.

B.1.1 Setup: the algebra, its generators, and a representation

We import the framework of Chapter 2 wholesale and recall only what we use. Let $\mathfrak{g}$ be a compact simple Lie algebra of dimension $\dim G$ — the number of independent generators, equivalently the number of gauge fields a gauge theory based on $\mathfrak{g}$ carries. In a fixed basis the generators obey the bracket fixed in

Eq. (2.70),

$$\left[T^a, T^b\right] = i f^{abc}\, T^c, \qquad a, b, c = 1, \ldots, \dim G, \tag{B.1}$$

with real structure constants f^{abc} (their reality is reproved below, in Theorem B.7, from the conventions we now adopt). Throughout this appendix the generators are taken *Hermitian*; this is always possible for a compact algebra and is the convention of Chapter A, where the global normalization is fixed by demanding that in the defining (fundamental) representation of $\mathrm{SU}(N)$,

$$\mathrm{tr}\big(T^a T^b\big) = \tfrac{1}{2}\,\delta^{ab} \qquad \text{(fundamental representation)}. \tag{B.2}$$

The choice of the number $\frac{1}{2}$ on the right is the normalization freedom; making it explicit and tracing its consequences is one of the tasks of this section.

A *representation* R assigns to each generator a Hermitian matrix T^a_R of size $\dim R \times \dim R$, preserving the bracket, $\left[T^a_R, T^b_R\right] = i f^{abc} T^c_R$. Two representations recur so often that they deserve standing names and symbols. The *fundamental* (defining) representation, denoted F, is the one normalized by Eq. (B.2). The *adjoint* representation, denoted adj, is the action of the algebra on itself by the bracket; in the basis of Eq. (B.1) its matrices are read off from the structure constants,

$$\left(T^a_{\mathrm{adj}}\right)_{bc} = -i f^{abc}. \tag{B.3}$$

That these matrices satisfy the bracket Eq. (B.1) is precisely the Jacobi identity, as established in Chapter 2; their dimension is the dimension of the algebra itself,

$$\dim(\mathrm{adj}) = \dim G. \tag{B.4}$$

Because f^{abc} is real and (as we show) totally antisymmetric, the matrix $T^a_{\mathrm{adj}} = -i f^{a\,\cdot\,\cdot}$ is i times a real antisymmetric matrix, hence Hermitian, consistent with our standing convention.

One fact from Chapter 2 will be used repeatedly and is worth isolating: for a *simple* algebra the adjoint representation is irreducible, and the generators in *any* representation are traceless,

$$\mathrm{tr}\, T^a_R = 0. \tag{B.5}$$

The tracelessness is immediate from Eq. (B.1): a simple algebra satisfies $\mathfrak{g} = [\mathfrak{g}, \mathfrak{g}]$, so every generator is a linear combination of commutators $\left[T^b_R, T^c_R\right]$, and the trace of a commutator vanishes by cyclicity. We will lean on irreducibility through Schur's lemma (Theorem 2.21) and on Eq. (B.5) when computing indices of tensor products.

B.1.2 Two scalars attached to a representation

There are exactly two natural scalars one can build by contracting two generators of a representation: trace them against each other over the representation space, or sum them against each other over the algebra index. These give the index and the Casimir respectively. We treat each in turn, in each case first establishing that the construction yields a single number rather than a tensor.

The Dynkin index. Consider the matrix-valued, symmetric, two-index object $\mathrm{tr}(T_R^a T_R^b)$. The claim is that it is proportional to δ^{ab}, with a proportionality constant that depends only on R. The key input is that this object is an *invariant* symmetric tensor on the algebra, and that on a simple algebra such tensors are unique up to scale. We prove the uniqueness statement as a lemma, since the same lemma also fixes the Killing form and thereby the normalization conventions.

Lemma B.1 (Uniqueness of the invariant symmetric form)**.** *Let $\mathfrak{g}$ be simple. A symmetric tensor $B^{ab} = B^{ba}$ is called* invariant *if*

$$f^{cad}\, B^{db} + f^{cbd}\, B^{ad} = 0 \qquad \text{for all } a, b, c. \tag{B.6}$$

The space of invariant symmetric tensors is one-dimensional: any two are proportional. In particular, in the basis fixed by Eq. (B.2) the Kronecker tensor δ^{ab} is invariant, and every invariant symmetric tensor is a multiple of δ^{ab}.

Proof. First, Eq. (B.6) is the infinitesimal statement that B is preserved by the adjoint action: it is the component form of $B([Z,X],Y) + B(X,[Z,Y]) = 0$, the requirement that the bilinear form B be ad-invariant. We verify that δ^{ab} satisfies it once f^{abc} is totally antisymmetric (proved independently in Theorem B.7): substituting $B^{ab} = \delta^{ab}$ gives $f^{cab} + f^{cba} = 0$, true by antisymmetry in the last two indices. Thus at least one invariant symmetric tensor, namely δ^{ab}, exists in our basis.

Now let B^{ab} be any invariant symmetric tensor and define a linear operator A on the algebra by its matrix elements $A_{ab} := B^{ab}$ (we use δ to identify upper and lower indices, which is legitimate precisely because δ is invariant). We show A commutes with every adjoint generator. Using Eq. (B.3),

$$\begin{aligned} \left(T^c_{\mathrm{adj}} A\right)_{ab} - \left(A\, T^c_{\mathrm{adj}}\right)_{ab} &= \left(T^c_{\mathrm{adj}}\right)_{ad} A_{db} - A_{ad} \left(T^c_{\mathrm{adj}}\right)_{db} \\ &= -i f^{cad} B^{db} - B^{ad}\left(-i f^{cdb}\right) \\ &= -i\left(f^{cad} B^{db} + f^{cbd} B^{ad}\right) = 0, \end{aligned} \tag{B.7}$$

where in the last line we used $f^{cdb} = -f^{cbd}$ and then Eq. (B.6). Hence A intertwines the adjoint representation with itself. Since $\mathfrak{g}$ is simple, the adjoint representation is irreducible (Chapter 2), so by Schur's lemma (Theorem 2.21) $A = c\,\mathbf{1}$ for a single scalar c, that is $B^{ab} = c\,\delta^{ab}$. Any two invariant symmetric tensors are therefore each proportional to δ^{ab}, hence to one another. □

The trace form $\mathrm{tr}(T_R^a T_R^b)$ is symmetric by cyclicity and invariant because

$$\mathrm{tr}([T_R^c, T_R^a]\, T_R^b) + \mathrm{tr}(T_R^a \left[T_R^c, T_R^b\right]) = \mathrm{tr}\left[T_R^c, T_R^a T_R^b\right] = 0, \tag{B.8}$$

which in components, using Eq. (B.1), is exactly Eq. (B.6) with $B^{ab} = \mathrm{tr}(T_R^a T_R^b)$. By Theorem B.1 it is a multiple of δ^{ab}. That multiple is the index.

Definition B.2 (Dynkin index)**.** The *Dynkin index* (or simply the *index*) $T(R)$ of a representation R is the proportionality constant in

$$\boxed{\mathrm{tr}(T_R^a T_R^b) = T(R)\,\delta^{ab}} \tag{B.9}$$

(no sum on the right). Equivalently, setting $a = b$ and summing, $T(R) = \dfrac{1}{\dim G}\sum_a \mathrm{tr}(T_R^a T_R^a)$.

The index is real and nonnegative: for each fixed a, Hermiticity gives $T(R) = \mathrm{tr}(T_R^a (T_R^a)^\dagger) = \sum_{ij} |(T_R^a)_{ij}|^2 \geq 0$, and it is strictly positive for any faithful representation. Comparison with the normalization Eq. (B.2) gives the standing value of the fundamental index that the threads use,

$$T(F) \equiv T_F = \tfrac{1}{2}. \tag{B.10}$$

This single number is the entire content of the normalization choice: had we set $\mathrm{tr}(T^a T^b) = \delta^{ab}$ in the fundamental we would have $T_F = 1$ instead, and every index and Casimir in the appendix would rescale accordingly. The physics — ratios of color factors, the structure of beta functions — is independent of the choice; only the bookkeeping changes. Section B.1.4 records the clash with conventions that make the other choice.

The quadratic Casimir. The complementary contraction sums the two generators over the algebra index rather than tracing over the representation space:

$$C_R := \sum_a T_R^a\, T_R^a. \tag{B.11}$$

This is a matrix on the representation space, not a number; but it is a very special matrix.

Proposition B.3 (The quadratic Casimir is a c-number on an irreducible representation)**.** *The operator C_R of Eq. (B.11) commutes with every generator, $\left[C_R, T_R^b\right] = 0$. If R is irreducible, then $C_R = C_2(R)\,\mathbf{1}$ for a single scalar $C_2(R)$, the* quadratic Casimir *of R.*

Proof. Using the Leibniz rule for the commutator, $\left[T_R^a T_R^a, T_R^b\right] = T_R^a \left[T_R^a, T_R^b\right] + \left[T_R^a, T_R^b\right] T_R^a$, and the bracket Eq. (B.1),

$$\left[C_R, T_R^b\right] = \sum_a \left(T_R^a \, i f^{abc} T_R^c + i f^{abc} T_R^c \, T_R^a\right) = i f^{abc} (T_R^a T_R^c + T_R^c T_R^a). \tag{B.12}$$

The bracket $\{T_R^a, T_R^c\}$ is symmetric in $a \leftrightarrow c$, while f^{abc} is antisymmetric in $a \leftrightarrow c$ (Theorem B.7); the contraction of a symmetric with an antisymmetric tensor vanishes, so $\left[C_R, T_R^b\right] = 0$. Thus C_R commutes with every element of the representation. If R is irreducible, Schur's lemma (Theorem 2.21) forces C_R to be a multiple of the identity, $C_R = C_2(R)\,\mathbf{1}$. □

Definition B.4 (Quadratic Casimir)**.** For an irreducible representation R, the *quadratic Casimir* $C_2(R)$ is the scalar defined by

$$\boxed{\sum_a T_R^a T_R^a = C_2(R)\,\mathbf{1}} \tag{B.13}$$

The Casimir of the fundamental is denoted $C_F := C_2(F)$ and that of the adjoint $C_A := C_2(\mathrm{adj})$.

Caution

The Casimir is a c-number only on an *irreducible* representation. On a reducible $R = \bigoplus_i R_i$ the operator C_R is block-diagonal, equal to $C_2(R_i)\mathbf{1}$ on the i-th block, and is a genuine multiple of the identity only when all the $C_2(R_i)$ coincide. The index $T(R)$, by contrast, is defined and additive for any R (Theorem B.9). This asymmetry is why the master relation below is stated for irreducible R but its trace form survives for reducible R.

B.1.3 The master relation

The index and the Casimir are not independent: they are two ways of contracting the same object $\sum_a T_R^a T_R^a$, and equating the two evaluations of its trace binds them. This is the structural backbone of the whole appendix.

Theorem B.5 (Index–Casimir master relation)**.** *For any irreducible representation R of a compact simple Lie algebra,*

$$\boxed{\dim(R)\, C_2(R) = \dim(G)\, T(R)} \tag{B.14}$$

where $\dim(R)$ *is the dimension of the representation space and* $\dim(G)$ *the dimension of the algebra.*

Proof. Compute the number $\sum_a \operatorname{tr}(T_R^a T_R^a)$ in two ways. On the one hand, the trace acts on the representation space; using Eq. (B.13),

$$\sum_a \operatorname{tr}(T_R^a T_R^a) = \operatorname{tr}\Big(\sum_a T_R^a T_R^a\Big) = \operatorname{tr}\big(C_2(R)\,\mathbf{1}\big) = C_2(R)\,\dim(R). \tag{B.15}$$

On the other hand, take the trace first and use the index Eq. (B.9) with $a = b$:

$$\sum_a \operatorname{tr}(T_R^a T_R^a) = \sum_a T(R)\,\delta^{aa} = T(R)\sum_a 1 = T(R)\,\dim(G), \tag{B.16}$$

since the index a runs over the $\dim(G)$ generators. Equating Eq. (B.15) and Eq. (B.16) gives Eq. (B.14). □

The relation is a rigid constraint: given the dimensions $\dim(R)$ and $\dim(G)$ and either invariant, the other is fixed. Two consequences fix terminology for the rest of the book.

First, applied to the fundamental of SU(N), where $\dim(G) = N^2 - 1$, $\dim(F) = N$, and $T_F = \frac{1}{2}$ from Eq. (B.10), it gives

$$C_F = \frac{\dim(G)}{\dim(F)}\,T_F = \frac{N^2-1}{N}\cdot\frac{1}{2} = \frac{N^2-1}{2N}, \tag{B.17}$$

the standing value used throughout Yang–Mills theory. (The explicit verification that the SU(N) fundamental indeed has $T_F = \frac{1}{2}$, from the Pauli and Gell-Mann matrices, is the business of the next section; here we only show that *given* T_F, the Casimir is forced.)

Second, applied to the adjoint, where $\dim(\text{adj}) = \dim(G)$ by Eq. (B.4), the master relation collapses to an identity of the two adjoint invariants.

Corollary B.6 (The adjoint index equals the adjoint Casimir)**.** *For the adjoint representation,* $\dim(\text{adj}) = \dim(G)$*, so Eq.* (B.14) *reads* $\dim(G)\,C_2(\text{adj}) = \dim(G)\,T(\text{adj})$*, i.e.*

$$T(\text{adj}) = C_2(\text{adj}) \;\equiv\; C_A. \tag{B.18}$$

The common value C_A is the single number carried by the adjoint representation.

That the adjoint's index and Casimir coincide is the reason a non-abelian gauge field contributes to its own beta function through the one symbol C_A: the gluon loop is governed by the adjoint index, the gluon self-energy by the adjoint Casimir, and they are equal. We make the value $C_A = N$ for SU(N) explicit once the structure constants are in hand; the equality Eq. (B.18) is prior to any such evaluation.

Physical Insight

The two invariants are exactly the two ways a representation enters a one-loop diagram. A matter loop — a closed loop of fields in representation R inserted in a gauge-boson propagator — produces a trace over R of two generators, and so is proportional to the *index* $T(R)$; this is the origin of the $-\frac{4}{3}T_F n_f$ matter contribution to the Yang–Mills beta function in Chapter 27. A gauge boson attached at both ends to a single matter line — the rainbow self-energy of that line — produces a sum over the algebra index of two generators, $\sum_a T_R^a T_R^a$, and so is proportional to the *Casimir* $C_2(R)$; this is the origin of the C_F in the fermion field renormalization and of C_A in the gluon's. The master relation is then the bookkeeping fact that ensures the two computations are consistent. This is the sense in which the present section "supplies" Yang–Mills theory without yet advancing it: the symbols C_A, T_F, C_F are defined and related here, to be consumed in Chapters 27 and 28.

B.1.4 Structure constants: total antisymmetry, reality, and the Killing form

We owe two facts used above: that f^{abc} is totally antisymmetric and real, and that the Killing form is proportional to δ^{ab}. Both follow from the trace normalization once we express f^{abc} as a trace.

Multiply the bracket Eq. (B.1), written in representation R, by T_R^d and trace, using Eq. (B.9):

$$\operatorname{tr}\!\left(\left[T_R^a, T_R^b\right] T_R^d\right) = i f^{abc}\, \operatorname{tr}\!\left(T_R^c T_R^d\right) = i f^{abc}\, T(R)\, \delta^{cd} = i\, T(R)\, f^{abd}. \tag{B.19}$$

Solving for the structure constant exhibits it as a trace,

$$f^{abc} = -\frac{i}{T(R)}\, \operatorname{tr}\!\left(\left[T_R^a, T_R^b\right] T_R^c\right), \tag{B.20}$$

valid in any faithful representation (so that $T(R) \neq 0$); the right-hand side is manifestly independent of R because the left-hand side is.

Proposition B.7 (f^{abc} is totally antisymmetric and real)**.** *The structure constants in the Hermitian, trace-orthonormal basis are completely antisymmetric under permutation of a, b, c and are real.*

Proof. Write $h^{abc} := \operatorname{tr}\!\left(\left[T_R^a, T_R^b\right] T_R^c\right)$, so that $f^{abc} = -i\, h^{abc}/T(R)$ by Eq. (B.20). We establish the symmetries of h.

Antisymmetry in $a \leftrightarrow b$. Immediate, since $\left[T_R^a, T_R^b\right] = -\left[T_R^b, T_R^a\right]$, hence $h^{abc} = -h^{bac}$.

Cyclicity. Expand $h^{abc} = \mathrm{tr}(T_R^a T_R^b T_R^c) - \mathrm{tr}(T_R^b T_R^a T_R^c)$ and use the cyclicity of the trace, $\mathrm{tr}(XYZ) = \mathrm{tr}(YZX)$, on each term: $\mathrm{tr}(T_R^a T_R^b T_R^c) = \mathrm{tr}(T_R^b T_R^c T_R^a)$ and $\mathrm{tr}(T_R^b T_R^a T_R^c) = \mathrm{tr}(T_R^a T_R^c T_R^b) = \mathrm{tr}(T_R^c T_R^b T_R^a)$. Therefore

$$h^{abc} = \mathrm{tr}(T_R^b T_R^c T_R^a) - \mathrm{tr}(T_R^c T_R^b T_R^a) = \mathrm{tr}\Big(\Big[T_R^b, T_R^c\Big] T_R^a\Big) = h^{bca}. \tag{B.21}$$

A function of three indices that is cyclic and antisymmetric under one transposition is antisymmetric under all of them: combining $h^{abc} = h^{bca}$ with $h^{abc} = -h^{bac}$ gives $h^{acb} = h^{cba} = h^{bac} = -h^{abc}$ and likewise for the remaining odd permutations, so every transposition flips the sign. Thus h^{abc}, and with it f^{abc}, is totally antisymmetric.

Reality. For any matrix M, $(\mathrm{tr}\, M)^* = \mathrm{tr}(M^\dagger)$. Take $M = \Big[T_R^a, T_R^b\Big] T_R^c$; using Hermiticity $T_R^{a\dagger} = T_R^a$,

$$M^\dagger = T_R^c \Big[T_R^a, T_R^b\Big]^\dagger = T_R^c (T_R^b T_R^a - T_R^a T_R^b) = -\, T_R^c \Big[T_R^a, T_R^b\Big], \tag{B.22}$$

so that, by cyclicity again, $(h^{abc})^* = \mathrm{tr}(M^\dagger) = -\,\mathrm{tr}(T_R^c \Big[T_R^a, T_R^b\Big]) = -\,\mathrm{tr}(\Big[T_R^a, T_R^b\Big] T_R^c) = -h^{abc}$. Hence h^{abc} is purely imaginary. Since $T(R)$ is real and positive (Theorem B.2), $f^{abc} = -i\, h^{abc}/T(R)$ obeys $(f^{abc})^* = +i\,(h^{abc})^*/T(R) = +i\,(-h^{abc})/T(R) = -i\, h^{abc}/T(R) = f^{abc}$, i.e. f^{abc} is real. □

With f^{abc} real and totally antisymmetric, the adjoint generators Eq. (B.3) are Hermitian as promised, and the consistency check on δ^{ab} in Theorem B.1 is justified. The final piece is the Killing form, the invariant form built directly from the structure constants.

Proposition B.8 (The Killing form fixes the normalization)**.** *Define the Killing form $K^{ab} := f^{acd} f^{bcd}$ (summed over c, d). Then*

$$K^{ab} = \mathrm{tr}(T_{\mathrm{adj}}^a T_{\mathrm{adj}}^b) = T(\mathrm{adj})\, \delta^{ab} = C_A\, \delta^{ab}. \tag{B.23}$$

In particular the Killing form is proportional to δ^{ab}, and for a compact algebra (with our Hermitian generators) it is positive-definite, with the single positive eigenvalue C_A. This proportionality is the statement that the algebra is simple: it leaves exactly one overall scale free, the scale fixed by Eq. (B.2).

Proof. Insert the adjoint matrices Eq. (B.3) into the trace:

$$\mathrm{tr}(T_{\mathrm{adj}}^a T_{\mathrm{adj}}^b) = (T_{\mathrm{adj}}^a)_{cd} (T_{\mathrm{adj}}^b)_{dc} = (-i f^{acd})(-i f^{bdc}) = -f^{acd} f^{bdc}. \tag{B.24}$$

By the total antisymmetry of Theorem B.7, $f^{bdc} = -f^{bcd}$, so $-f^{acd} f^{bdc} = f^{acd} f^{bcd} = K^{ab}$, which is the first equality in Eq. (B.23). But $\mathrm{tr}(T_{\mathrm{adj}}^a T_{\mathrm{adj}}^b)$ is the trace form of the adjoint representation, equal to $T(\mathrm{adj})\, \delta^{ab}$ by the index definition Eq. (B.9); with $T(\mathrm{adj}) = C_A$ from Theorem B.6 this gives the remaining

equalities. Positive-definiteness is the ≥ 0 argument of Theorem B.2 applied to the (Hermitian, faithful) adjoint representation: $K^{aa} = \sum_{c,d} |f^{acd}|^2 > 0$ for each a, since a nonzero generator has at least one nonvanishing structure constant in a simple algebra. Finally, that K^{ab} is forced to be a *single* multiple of δ^{ab} — rather than, say, a different multiple in each invariant subspace — is the content of Theorem B.1: the Killing form is an invariant symmetric tensor, and on a simple algebra there is only one such up to scale. A non-simple (e.g. semisimple) algebra would have one free scale per simple factor; simplicity is precisely what reduces this to a single overall normalization. □

Notation Watch

The one convention to keep straight across references is the value of T_F. This book and the bulk of the particle-physics literature (Peskin–Schroeder, Schwartz) set $\mathrm{tr}(T_F^a T_F^b) = \frac{1}{2}\delta^{ab}$, hence $T_F = \frac{1}{2}$, $C_A = N$, $C_F = \frac{N^2-1}{2N}$ for SU(N). Some mathematical and older texts normalize the adjoint instead, taking the Killing form itself as the metric ($C_A = 1$), which rescales every index and Casimir. The dimensionless ratios that enter physical predictions — for instance C_A/T_F and C_F/C_A in the beta function — are invariant under the choice; only the absolute values move. When importing a number from outside this book, check which T_F it assumes before using it.

B.1.5 Worked examples

The two invariants behave simply under the two ways of building new representations from old — direct sums and tensor products — and these behaviors are exactly what is needed to handle a theory with several matter multiplets. We work them out here, staying entirely within the abstract setting; the explicit low-rank evaluations are deferred to the next section.

Worked Example B.9 (The index is additive; the Casimir is per-irreducible–component)**.** Let $R = R_1 \oplus R_2$ be a direct sum, so that on the block-diagonal representation space the generators are $T_R^a = T_{R_1}^a \oplus T_{R_2}^a = \begin{pmatrix} T_{R_1}^a & 0 \\ 0 & T_{R_2}^a \end{pmatrix}$. The trace of a block-diagonal product is the sum of the block traces, so

$$\mathrm{tr}(T_R^a T_R^b) = \mathrm{tr}(T_{R_1}^a T_{R_1}^b) + \mathrm{tr}(T_{R_2}^a T_{R_2}^b) = (T(R_1) + T(R_2))\delta^{ab}, \tag{B.25}$$

whence the index is additive over direct sums,

$$T(R_1 \oplus R_2) = T(R_1) + T(R_2). \tag{B.26}$$

This is why a gauge theory with n_f Dirac fermions in the fundamental contributes a matter index $n_f\, T_F$ to the beta function: the n_f flavors furnish the direct

sum $F^{\oplus n_f}$, of index $n_f T_F$, which is the origin of the $-\frac{4}{3}T_F n_f$ term carried in Yang–Mills theory (Chapter 27).

The Casimir does not combine so simply, and the contrast is instructive. The Casimir operator on the sum is block-diagonal,

$$C_R = \sum_a T_R^a T_R^a = \big(C_2(R_1)\mathbf{1}_{R_1}\big) \oplus \big(C_2(R_2)\mathbf{1}_{R_2}\big), \tag{B.27}$$

which is a single multiple of the identity only if $C_2(R_1) = C_2(R_2)$. There is no scalar "$C_2(R_1 \oplus R_2)$" in general; the Casimir is an invariant of an *irreducible* representation. What *does* survive for the reducible sum is the trace form of the master relation: taking the trace of C_R,

$$\sum_i C_2(R_i)\,\dim(R_i) = \dim(G)\,T(R), \tag{B.28}$$

the dimension-weighted sum of the component Casimirs equals $\dim(G)$ times the total index, the natural generalization of Eq. (B.14) to a reducible R.

Worked Example B.10 (The index of a tensor product)**.** Let $R_1 \otimes R_2$ be a tensor product. On the product space the generators act by the Leibniz (co-product) rule established in Chapter 2,

$$T^a_{R_1\otimes R_2} = T^a_{R_1} \otimes \mathbf{1}_{R_2} + \mathbf{1}_{R_1} \otimes T^a_{R_2}. \tag{B.29}$$

We compute the index from Eq. (B.9). Using $\mathrm{tr}(X \otimes Y) = \mathrm{tr}\,X\ \mathrm{tr}\,Y$ and expanding the product of two co-products,

$$\begin{aligned}\mathrm{tr}\big(T^a_{R_1\otimes R_2} T^b_{R_1\otimes R_2}\big) = {}& \mathrm{tr}\big(T^a_{R_1}T^b_{R_1}\big)\ \mathrm{tr}\big(\mathbf{1}_{R_2}\big) + \mathrm{tr}\big(\mathbf{1}_{R_1}\big)\ \mathrm{tr}\big(T^a_{R_2}T^b_{R_2}\big) \\ & + \mathrm{tr}\big(T^a_{R_1}\big)\,\mathrm{tr}\big(T^b_{R_2}\big) + \mathrm{tr}\big(T^b_{R_1}\big)\,\mathrm{tr}\big(T^a_{R_2}\big).\end{aligned} \tag{B.30}$$

The two cross terms in the second line vanish because the generators of a simple algebra are traceless, Eq. (B.5). The surviving terms give, with $\mathrm{tr}\,\mathbf{1}_{R_i} = \dim(R_i)$,

$$\mathrm{tr}\big(T^a_{R_1\otimes R_2} T^b_{R_1\otimes R_2}\big) = \big(T(R_1)\dim(R_2) + \dim(R_1)\,T(R_2)\big)\delta^{ab}, \tag{B.31}$$

so the index of a tensor product is

$$T(R_1 \otimes R_2) = T(R_1)\dim(R_2) + \dim(R_1)\,T(R_2). \tag{B.32}$$

It is worth seeing how the Casimir, by contrast, is well-behaved on a tensor product precisely *because* of the cross term that dropped out above. Squaring the co-product Eq. (B.29) and summing over a,

$$\sum_a T^a_{R_1\otimes R_2} T^a_{R_1\otimes R_2} = C_2(R_1)\,\mathbf{1}\otimes\mathbf{1} + \mathbf{1}\otimes\mathbf{1}\,C_2(R_2) + 2\sum_a T^a_{R_1}\otimes T^a_{R_2}. \tag{B.33}$$

The mixed term $2\sum_a T^a_{R_1}\otimes T^a_{R_2}$ is *not* a multiple of the identity — the product $R_1 \otimes R_2$ is in general reducible, and this operator is what distinguishes its

irreducible summands. Indeed, restricting Eq. (B.33) to an irreducible component $R \subseteq R_1 \otimes R_2$ (on which the left side is $C_2(R)\mathbf{1}$ by Theorem B.3) solves for the mixed term and yields the "addition of Casimirs" rule

$$\sum_a T^a_{R_1} \otimes T^a_{R_2}\Big|_R = \tfrac{1}{2}\big(C_2(R) - C_2(R_1) - C_2(R_2)\big)\mathbf{1}, \tag{B.34}$$

the group-theoretic analogue of $\mathbf{S}_1\cdot\mathbf{S}_2 = \frac{1}{2}(\mathbf{S}^2 - \mathbf{S}_1^2 - \mathbf{S}_2^2)$ from the addition of angular momenta. Equation (B.34) is the color algebra behind every quark–antiquark or quark–gluon interaction: it converts a contracted pair of generators on two distinct lines into the difference of Casimirs of the total and the individual representations.

We now have the abstract pair $\big(T(R), C_2(R)\big)$ for every representation, the master relation $\dim(R)\,C_2(R) = \dim(G)\,T(R)$ that binds them, the identification $T(\text{adj}) = C_2(\text{adj}) = C_A$, and the additive and tensor-product rules that handle composite matter content — all proved from the single normalization Eq. (B.2). What remains is to turn these abstractions into numbers. The next section does exactly that, writing the Pauli and Gell-Mann matrices explicitly, extracting the $\mathfrak{su}(3)$ structure constants and symmetric symbols, and verifying the master relation in its first two concrete instances: the $\mathfrak{su}(2)$ spin-j representations and the $\mathfrak{su}(3)$ fundamental.

B.2 The Low-Rank Algebras: Pauli and Gell-Mann Matrices

The previous section attached to every representation R of a compact simple Lie algebra two scalars—the Dynkin index $T(R)$ of Theorem B.2 and the quadratic Casimir $C_2(R)$ of Theorem B.4—and proved that they are not independent but bound by the master identity $\dim(R)\,C_2(R) = \dim(G)\,T(R)$ of Theorem B.5. That development was deliberately abstract: it used only the defining trace normalization and Schur's lemma, never a single explicit generator. The price of that generality is that the reader has, so far, no concrete object to hold. This section pays the abstraction back. We write down the generators of the two algebras that the rest of the book actually uses—$\mathfrak{su}(2)$, the algebra of weak isospin and of ordinary spin, and $\mathfrak{su}(3)$, the algebra of color—in the only two forms a physicist ever needs them: the Pauli matrices σ^i and the Gell-Mann matrices λ^a. From these explicit matrices we prove the multiplication laws that downstream chapters quote in passing, extract the full set of $\mathfrak{su}(3)$ structure constants f^{abc} and the symmetric symbols d^{abc} by direct trace computation, and then close the loop with Section 1 by evaluating $T(R)$ and $C_2(R)$ in their first two nontrivial instances and watching the master identity come out right to the digit. A secondary purpose is bookkeeping across appendices: the Pauli algebra

proved here is exactly the input that Chapter C imports without proof when it builds the four-dimensional Dirac technology, and the $\mathfrak{su}(3)$ data primes the color-factor machinery that Yang–Mills theory leans on.

Throughout we use the global generator convention of Chapter A: generators T^a are Hermitian, the structure constants are real and defined by $\left[T^a, T^b\right] = if^{abc}T^c$, and the fundamental representation is normalized by $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$. Fixing this normalization is what makes $T(\mathrm{fund}) = \frac{1}{2}$ a definition rather than a derived number; everything else in the section is then forced.

B.2.1 The Pauli matrices and the algebra $\mathfrak{su}(2)$

Definition B.11 (Pauli matrices)**.** The three *Pauli matrices* are the Hermitian, traceless 2×2 matrices

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{B.35}$$

The generators of $\mathfrak{su}(2)$ in the fundamental (spin-$\frac{1}{2}$) representation are $T^i = \frac{1}{2}\sigma^i$.

Everything about $\mathfrak{su}(2)$ at the matrix level follows from a single product law, and that law is read off by multiplying the explicit matrices. There is nothing to assume: we compute the nine products $\sigma^i\sigma^j$ once and recognize the pattern.

Proposition B.12 (Pauli product law)**.** *The Pauli matrices satisfy*

$$\sigma^i\sigma^j = \delta^{ij}\,\mathbf{1} + i\,\epsilon^{ijk}\sigma^k, \tag{B.36}$$

with $\mathbf{1}$ *the* 2×2 *identity and* ϵ^{ijk} *the totally antisymmetric symbol normalized by* $\epsilon^{123} = 1$*. Equivalently, separating the symmetric and antisymmetric parts in* $i \leftrightarrow j$*,*

$$\left\{\sigma^i, \sigma^j\right\} = 2\,\delta^{ij}\,\mathbf{1}, \tag{B.37}$$

$$\left[\sigma^i, \sigma^j\right] = 2i\,\epsilon^{ijk}\sigma^k. \tag{B.38}$$

Proof. The diagonal products are immediate from Eq. (B.35): each Pauli matrix squares to the identity,

$$(\sigma^1)^2 = (\sigma^2)^2 = (\sigma^3)^2 = \mathbf{1}, \tag{B.39}$$

as one checks by squaring each 2×2 matrix—for instance $(\sigma^2)^2 = \left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right)\left(\begin{smallmatrix} 0 & -i \\ i & 0 \end{smallmatrix}\right) = \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. The three independent off-diagonal products are

$$\sigma^1\sigma^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\,\sigma^3, \tag{B.40}$$

$$\sigma^2\sigma^3 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix} = i\,\sigma^1, \tag{B.41}$$

$$\sigma^3\sigma^1 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} = i\,\sigma^2. \tag{B.42}$$

The reversed orderings follow by taking the Hermitian conjugate of these three lines (each σ^i is Hermitian, and $(AB)^\dagger = B^\dagger A^\dagger$), which flips the sign of the imaginary right-hand sides: $\sigma^2\sigma^1 = -i\sigma^3$, and cyclically. Collecting the diagonal results into $\sigma^i\sigma^i = \mathbf{1}$ (no sum) and the off-diagonal results into $\sigma^i\sigma^j = i\epsilon^{ijk}\sigma^k$ for $i \neq j$ is precisely the single statement Eq. (B.36): the δ^{ij} term supplies the diagonal case and the $i\epsilon^{ijk}\sigma^k$ term supplies the off-diagonal case, and the two never overlap. Adding Eq. (B.36) to itself with i, j swapped, and using $\epsilon^{ijk} = -\epsilon^{jik}$, gives Eq. (B.37); subtracting gives Eq. (B.38). □

Three consequences are worth recording because the book uses them by name. First, Eq. (B.38) divided by 2 in each factor reads $[T^i, T^j] = i\epsilon^{ijk}T^k$, so the $\mathfrak{su}(2)$ structure constants in the convention $\left[T^a, T^b\right] = if^{abc}T^c$ are simply

$$f^{ijk} = \epsilon^{ijk}. \tag{B.43}$$

Second, the anticommutator Eq. (B.37) fixes the fundamental trace normalization:

$$\operatorname{tr}(T^iT^j) = \tfrac{1}{4}\operatorname{tr}(\sigma^i\sigma^j) = \tfrac{1}{4}\operatorname{tr}\Big(\tfrac{1}{2}\left\{\sigma^i, \sigma^j\right\}\Big) = \tfrac{1}{4}\,\delta^{ij}\operatorname{tr}\mathbf{1}\cdot\tfrac{1}{2}\cdot 2 = \tfrac{1}{2}\,\delta^{ij}, \tag{B.44}$$

where we used $\operatorname{tr}(\sigma^i\sigma^j) = \frac{1}{2}\operatorname{tr}\{\sigma^i, \sigma^j\}$ because the trace of the antisymmetric commutator part vanishes, then $\operatorname{tr}\mathbf{1} = 2$. This confirms $T(\text{fund}) = \frac{1}{2}$ for $\mathfrak{su}(2)$ and is the $\mathfrak{su}(2)$ instance of the canonical normalization. Third, and most useful downstream, is the completeness statement that the four matrices $\{\mathbf{1}, \sigma^1, \sigma^2, \sigma^3\}$ span the space of all 2×2 matrices.

Proposition B.13 (Pauli completeness relation)**.** *With Latin indices $a, b, c, d \in \{1, 2\}$ labelling matrix entries,*

$$\sum_{i=1}^{3}(\sigma^i)_{ab}\,(\sigma^i)_{cd} = 2\,\delta_{ad}\,\delta_{bc} - \delta_{ab}\,\delta_{cd}. \tag{B.45}$$

Proof. Adjoin the identity to the Pauli matrices and write the four-element set $\tau^\mu = (\mathbf{1}, \sigma^1, \sigma^2, \sigma^3)$, $\mu = 0, 1, 2, 3$. These four matrices are orthonormal under the inner product $\langle A, B\rangle = \frac{1}{2}\operatorname{tr}(A^\dagger B)$ on the four-dimensional complex vector space of 2×2 matrices: indeed $\frac{1}{2}\operatorname{tr}(\mathbf{1}\,\mathbf{1}) = 1$, $\frac{1}{2}\operatorname{tr}(\mathbf{1}\,\sigma^i) = 0$ by tracelessness, and $\frac{1}{2}\operatorname{tr}(\sigma^i\sigma^j) = \delta^{ij}$ by Eq. (B.44) scaled up by four. Four orthonormal vectors in a four-dimensional space form a complete basis. Completeness of an orthonormal basis $\{B_\mu = \tau^\mu/\sqrt{2}\}$ under the unnormalized trace pairing $\operatorname{tr}(A^\dagger B)$ means that

every matrix M obeys $M = \sum_\mu \operatorname{tr}(B_\mu^\dagger M)\, B_\mu$; reading off the (a,b) entry for arbitrary M and stripping M gives the resolution of the identity on matrix space,

$$\sum_\mu (B_\mu)_{ab}\,(B_\mu)^*_{dc} = \delta_{ad}\,\delta_{bc}. \tag{B.46}$$

The τ^μ are Hermitian, so $(B_\mu)^*_{dc} = (B_\mu)_{cd}$, and with $B_\mu = \tau^\mu/\sqrt{2}$ this is

$$\tfrac{1}{2}\sum_{\mu=0}^{3} (\tau^\mu)_{ab}\,(\tau^\mu)_{cd} = \delta_{ad}\,\delta_{bc}. \tag{B.47}$$

Separating the $\mu = 0$ term $(\tau^0)_{ab}(\tau^0)_{cd} = \delta_{ab}\delta_{cd}$ from the three Pauli terms and solving for the latter yields Eq. (B.45). □

Equation (B.45) is the $\mathfrak{su}(2)$ ancestor of the SU(N) color-Fierz identity that Section B.3.1 builds in general; in the two-component spinor calculus of Chapter C it is the relation that collapses products of σ-matrices and underlies the Fierz rearrangement of Pauli bilinears. We have now discharged, with full proofs, every Pauli identity that Chapter C imports: the product law Eq. (B.36), the anticommutator Eq. (B.37), and the completeness relation Eq. (B.45).

B.2.2 The representation invariants of $\mathfrak{su}(2)$

To evaluate C_2 and T for the general spin-j representation we import the representation theory itself, not rebuild it: Theorem 2.53 constructs, for each $j \in \{0, \tfrac{1}{2}, 1, \tfrac{3}{2}, \dots\}$, an irreducible representation on a $(2j+1)$-dimensional space with orthonormal basis $|j,m\rangle$, $m = -j, -j+1, \dots, j$, on which the Hermitian generators J_i (the spin-j image of the abstract T^i, so that $[J_i, J_j] = i\epsilon^{ijk}J_k$) act by

$$J_3\,|j,m\rangle = m\,|j,m\rangle\,, \qquad J_\pm\,|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}\;\,|j,m\pm 1\rangle\,, \tag{B.48}$$

with $J_\pm = J_1 \pm i J_2$. We take Eq. (B.48) as given and compute the two invariants from it.

Proposition B.14 ($\mathfrak{su}(2)$ Casimir and index)**.** *For the spin-j representation of $\mathfrak{su}(2)$,*

$$C_2(j) = j(j+1), \tag{B.49}$$

$$T(j) = \tfrac{1}{3}\, j(j+1)(2j+1). \tag{B.50}$$

Proof. The Casimir. By Theorem B.4 the quadratic Casimir is the operator $J^2 \equiv \sum_i J_i J_i = C_2(j)\,\mathbf{1}$, a multiple of the identity because it commutes with every J_i and the representation is irreducible (Schur's lemma). To find the constant, evaluate J^2 on a single convenient state. Write

$$J^2 = J_3^2 + \tfrac{1}{2}(J_+J_- + J_-J_+), \tag{B.51}$$

which follows from $J_\pm = J_1 \pm iJ_2$ since $J_+J_- + J_-J_+ = 2(J_1^2 + J_2^2)$. Act on the highest-weight state $|j,j\rangle$. From Eq. (B.48), $J_+ |j,j\rangle = 0$ because the square root $\sqrt{j(j+1) - j(j+1)}$ vanishes, so $J_-J_+ |j,j\rangle = 0$. The remaining term is $J_+J_- |j,j\rangle$: lowering then raising,

$$J_+J_- |j,j\rangle = \big[j(j+1) - j(j-1)\big] |j,j\rangle = 2j \ |j,j\rangle \,, \tag{B.52}$$

using Eq. (B.48) twice (the two square-root factors multiply to the bracket). Hence

$$J^2 |j,j\rangle = \Big(j^2 + \tfrac{1}{2}\cdot 2j\Big) |j,j\rangle = j(j+1) |j,j\rangle \,, \tag{B.53}$$

and since J^2 is the constant $C_2(j)$ times the identity, $C_2(j) = j(j+1)$, which is Eq. (B.49).

The index. By Theorem B.2, $T(j)$ is defined by $\mathrm{tr}(J_aJ_b) = T(j)\delta_{ab}$; setting $a = b = 3$ (no sum) gives $T(j) = \mathrm{tr}(J_3^2)$. The trace is over the orthonormal basis $|j,m\rangle$, on which J_3 is diagonal with eigenvalues m, so

$$T(j) = \sum_{m=-j}^{j} m^2 . \tag{B.54}$$

It remains to evaluate this finite sum and show it equals $\frac{1}{3}j(j+1)(2j+1)$ for both integer and half-integer j. For *integer* j the sum is symmetric about zero, so

$$\sum_{m=-j}^{j} m^2 = 2\sum_{m=1}^{j} m^2 = 2\cdot\frac{j(j+1)(2j+1)}{6} = \frac{j(j+1)(2j+1)}{3}, \tag{B.55}$$

using the elementary closed form $\sum_{m=1}^{n} m^2 = \frac{1}{6}n(n+1)(2n+1)$. For *half-integer* j write $j = p - \frac{1}{2}$ with p a positive integer; the allowed m are $\pm\frac{1}{2}, \pm\frac{3}{2}, \ldots, \pm(p-\frac{1}{2})$, and

$$\begin{aligned}\sum_{m=-j}^{j} m^2 &= 2\sum_{k=1}^{p}\Big(k-\tfrac{1}{2}\Big)^2 = 2\sum_{k=1}^{p}\Big(k^2 - k + \tfrac{1}{4}\Big) \\ &= 2\left[\frac{p(p+1)(2p+1)}{6} - \frac{p(p+1)}{2} + \frac{p}{4}\right] = \frac{2p^3+3p^2+p}{3} - p^2 - p + \frac{p}{2} \\ &= \frac{2p^3}{3} - \frac{p}{6}. \end{aligned} \tag{B.56}$$

On the other hand, substituting $j = p - \frac{1}{2}$ into $\frac{1}{3}j(j+1)(2j+1)$ and using $2j+1 = 2p$, $j(j+1) = (p-\frac{1}{2})(p+\frac{1}{2}) = p^2 - \frac{1}{4}$, gives $\frac{1}{3}(p^2-\frac{1}{4})(2p) = \frac{2p^3}{3} - \frac{p}{6}$, the same value. Thus Eq. (B.54) equals $\frac{1}{3}j(j+1)(2j+1)$ for every allowed j, establishing Eq. (B.50). □

Worked Example B.15 (The master identity for $\mathfrak{su}(2)$ spin-j)**.** Theorem B.5 asserts $\dim(R)\,C_2(R) = \dim(G)\,T(R)$ for every representation, with $\dim(G)$ the dimension of the algebra. For $\mathfrak{su}(2)$ the algebra is three-dimensional ($\dim G = 3$, the three generators), and the spin-j representation has $\dim R = 2j+1$. The left-hand side, using Eq. (B.49), is

$$\dim(R)\,C_2(R) = (2j+1)\,j(j+1). \tag{B.57}$$

The right-hand side, using Eq. (B.50), is

$$\dim(G)\,T(R) = 3\cdot\tfrac{1}{3}\,j(j+1)(2j+1) = (2j+1)\,j(j+1). \tag{B.58}$$

The two agree identically in j—the factor of three in $\dim(G)$ exactly cancels the $\frac{1}{3}$ in $T(j)$, which is no coincidence but the content of the master identity, since both sides are $\sum_a \mathrm{tr}(J_a J_a) = \mathrm{tr}(J^2)$ computed two ways. As a numerical sanity check, take the three smallest cases. The fundamental $j=\frac{1}{2}$ gives $\dim R = 2$, $C_2 = \frac{3}{4}$, $T = \frac{1}{2}$, and indeed $2\cdot\frac{3}{4} = \frac{3}{2} = 3\cdot\frac{1}{2}$. The adjoint $j=1$ gives $\dim R = 3$, $C_2 = 2$, $T = 2$, and $3\cdot 2 = 6 = 3\cdot 2$; note $C_2(\text{adjoint}) = 2$ reproduces $C_A = N = 2$ for SU(2), as it must. The case $j=\frac{3}{2}$ gives $\dim R = 4$, $C_2 = \frac{15}{4}$, $T = 5$, and $4\cdot\frac{15}{4} = 15 = 3\cdot 5$. Every instance closes.

B.2.3 The Gell-Mann matrices and the algebra $\mathfrak{su}(3)$

The step from $\mathfrak{su}(2)$ to $\mathfrak{su}(3)$ is the step from spin to color, and it is where the structure becomes rich enough to need both an antisymmetric and a symmetric invariant tensor. The fundamental representation of $\mathfrak{su}(3)$ acts on $\mathbb{C}^3$, so its generators are eight independent Hermitian traceless 3×3 matrices (the dimension of $\mathfrak{su}(N)$ is N^2-1, here $9-1=8$). The standard choice generalizes the Pauli matrices block by block.

Definition B.16 (Gell-Mann matrices)**.** The eight *Gell-Mann matrices* are

$$\lambda^1 = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^2 = \begin{pmatrix} 0 & -i & 0 \\ i & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \quad \lambda^3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

$$\lambda^4 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad \lambda^5 = \begin{pmatrix} 0 & 0 & -i \\ 0 & 0 & 0 \\ i & 0 & 0 \end{pmatrix}, \quad \lambda^6 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$\lambda^7 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -i \\ 0 & i & 0 \end{pmatrix}, \qquad \lambda^8 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}. \tag{B.59}$$

The $\mathfrak{su}(3)$ generators in the fundamental are $T^a = \frac{1}{2}\lambda^a$, $a = 1,\dots,8$.

The organizing logic is transparent from the explicit forms. The triple $(\lambda^1, \lambda^2, \lambda^3)$ is the Pauli set embedded in the upper-left 2×2 block—this is the isospin $\mathfrak{su}(2)$ inside $\mathfrak{su}(3)$. The pairs (λ^4, λ^5) and (λ^6, λ^7) are the off-diagonal real/imaginary partners coupling the 1–3 and 2–3 subspaces, the two other $\mathfrak{su}(2)$ subalgebras ("U-spin" and "V-spin"). The eighth matrix λ^8 is the second diagonal (Cartan) generator, made traceless and orthogonal to λ^3, with the $1/\sqrt{3}$ fixed precisely by the normalization we now verify.

Proposition B.17 (Gell-Mann normalization)**.** *The Gell-Mann matrices are Hermitian and traceless, and satisfy*

$$\operatorname{tr}(\lambda^a \lambda^b) = 2\,\delta^{ab}, \qquad \textit{equivalently} \qquad \operatorname{tr}(T^a T^b) = \tfrac{1}{2}\,\delta^{ab}. \tag{B.60}$$

Proof. Hermiticity is visible matrix by matrix: the real symmetric ones equal their transpose, and the imaginary ones (with a $-i$ above the diagonal and $+i$ below) equal their conjugate transpose. Tracelessness is immediate for $\lambda^{1,\dots,7}$, which have zero diagonal, and for λ^8 because $1 + 1 - 2 = 0$. For the orthogonality Eq. (B.60) we must check $\operatorname{tr}(\lambda^a \lambda^b) = 2\delta^{ab}$ for all pairs. The diagonal entries $a = b$: for each of $\lambda^{1,\dots,7}$ the square has two unit entries on its diagonal—e.g. $(\lambda^1)^2 = \operatorname{diag}(1, 1, 0)$ and $(\lambda^5)^2 = \operatorname{diag}(1, 0, 1)$—so $\operatorname{tr}(\lambda^a)^2 = 2$; for λ^8, $\operatorname{tr}(\lambda^8)^2 = \frac{1}{3}(1+1+4) = 2$, which is exactly what the $1/\sqrt{3}$ prefactor was chosen to arrange. The off-diagonal entries $a \neq b$ vanish: two matrices supported on different off-diagonal positions have a product with zero diagonal; a real and an imaginary matrix on the *same* positions, such as λ^1 and λ^2, give $\operatorname{tr}(\lambda^1 \lambda^2) = \operatorname{tr}\left(i\,\operatorname{diag}(1, -1, 0)\right) = 0$; and the diagonal λ^3, λ^8 are mutually orthogonal because $\operatorname{tr}(\lambda^3 \lambda^8) = \frac{1}{\sqrt{3}}(1 \cdot 1 + (-1) \cdot 1 + 0) = 0$. Dividing by four gives the generator form, $\operatorname{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$, so $T(\text{fund}) = \frac{1}{2}$ for $\mathfrak{su}(3)$ as well. □

B.2.4 Structure constants and symmetric symbols of $\mathfrak{su}(3)$

The product of two $\mathfrak{su}(3)$ generators is not closed inside the generators alone—unlike $\mathfrak{su}(2)$, where $\sigma^i \sigma^j$ landed back in $\{\mathbf{1}, \sigma^k\}$ by Eq. (B.36). The commutator does close (that is the Lie algebra), but the anticommutator produces a genuinely new symmetric tensor. Both are captured by expanding products in the complete set $\{\mathbf{1}, T^c\}$ of nine Hermitian 3×3 matrices.

Definition B.18 (f and d symbols)**.** The *structure constants* f^{abc} and *symmetric symbols* d^{abc} of $\mathfrak{su}(3)$ are defined by

$$\left[T^a, T^b\right] = i\, f^{abc}\, T^c, \tag{B.61}$$

$$\left\{T^a, T^b\right\} = \tfrac{1}{3}\,\delta^{ab}\,\mathbf{1} + d^{abc}\, T^c. \tag{B.62}$$

We must justify that Eq. (B.62) has exactly this form and then give closed expressions for the coefficients. Both products $\left[T^a, T^b\right]$ and $\left\{T^a, T^b\right\}$ are linear

combinations of the nine matrices $\{\mathbf{1}, T^c\}$, since those span all 3×3 matrices and the products are themselves 3×3 matrices. The commutator is traceless and anti-Hermitian-times-i, hence has no $\mathbf{1}$ component and lies purely in the span of the T^c; that the coefficient is if^{abc} with f real is the content of Chapter A's convention, and antisymmetry $f^{abc} = -f^{bac}$ is manifest. The anticommutator is Hermitian and generally has a trace, so it acquires an $\mathbf{1}$ piece. Taking the trace of Eq. (B.62) and using $\operatorname{tr} T^c = 0$, $\operatorname{tr} \mathbf{1} = 3$, $\operatorname{tr}\left\{T^a, T^b\right\} = 2\operatorname{tr}(T^a T^b) = \delta^{ab}$ fixes the $\mathbf{1}$ coefficient: $\delta^{ab} = 3 \cdot \frac{1}{3}\delta^{ab}$, consistent, so the coefficient is exactly $\frac{1}{3}$. The remaining traceless Hermitian part is some combination $d^{abc}T^c$ with real, symmetric (in $a \leftrightarrow b$) coefficients. Closed forms follow by multiplying each relation by T^c and tracing with Eq. (B.60):

$$f^{abc} = -2i\ \operatorname{tr}\left(\left[T^a, T^b\right] T^c\right) = \frac{1}{4i}\ \operatorname{tr}\left(\left[\lambda^a, \lambda^b\right] \lambda^c\right), \tag{B.63}$$

$$d^{abc} = 2\ \operatorname{tr}\left(\left\{T^a, T^b\right\} T^c\right) = \frac{1}{4}\ \operatorname{tr}\left(\left\{\lambda^a, \lambda^b\right\} \lambda^c\right). \tag{B.64}$$

To get Eq. (B.63), multiply Eq. (B.61) by T^c, trace, and use $\operatorname{tr}(T^e T^c) = \frac{1}{2}\delta^{ec}$ to obtain $\operatorname{tr}\left(\left[T^a, T^b\right] T^c\right) = \frac{i}{2} f^{abc}$; solving and inserting $T = \lambda/2$ gives the λ-form. The same step with the anticommutator, noting that the $\frac{1}{3}\delta^{ab}\mathbf{1}$ term contributes $\frac{1}{3}\delta^{ab}\operatorname{tr} T^c = 0$, gives Eq. (B.64).

These trace formulas have three structural consequences, each provable in one line from the cyclicity of the trace and the (anti)symmetry of the (anti)commutator. *f^{abc} is totally antisymmetric:* antisymmetry in $a \leftrightarrow b$ is built in, and cyclicity of the trace in $\operatorname{tr}\left(\left[T^a, T^b\right] T^c\right) = \operatorname{tr}(T^a T^b T^c) - \operatorname{tr}(T^b T^a T^c)$ rearranges into $\operatorname{tr}\left(\left[T^b, T^c\right] T^a\right)$, giving antisymmetry under $a \leftrightarrow c$ as well; the two transpositions generate the full antisymmetric group. *d^{abc} is totally symmetric:* the same cyclicity applied to the anticommutator form makes it invariant under any interchange. *Both are real:* f^{abc} is real because the structure constants of a compact algebra in a Hermitian basis are real, and d^{abc} is real because $\left\{\lambda^a, \lambda^b\right\} \lambda^c$ has real trace (its Hermitian conjugate is $\lambda^c \left\{\lambda^a, \lambda^b\right\}$, with the same trace).

We now compute representative entries directly from Eqs. (B.63) and (B.64), using matrix-unit notation E_{ij} for the 3×3 matrix with a single 1 in row i, column j, so that $E_{ij}E_{kl} = \delta_{jk}E_{il}$. Every value in the full tables below is obtained by the identical procedure; we carry four cases in complete detail and tabulate the rest.

Worked Example B.19 (Representative f^{abc} and d^{abc} by direct trace)**.** $f^{123} = 1$. Since $(\lambda^1, \lambda^2, \lambda^3)$ are the Pauli matrices in the upper 2×2 block, $[\lambda^1, \lambda^2] = 2i\lambda^3$ by Eq. (B.38). Hence

$$f^{123} = \frac{1}{4i}\operatorname{tr}(2i\,\lambda^3\lambda^3) = \frac{2i}{4i}\ \operatorname{tr}(\lambda^3)^2 = \tfrac{1}{2} \cdot 2 = 1. \tag{B.65}$$

$f^{147} = \frac{1}{2}$. Write $\lambda^1 = E_{12} + E_{21}$ and $\lambda^4 = E_{13} + E_{31}$. Using $E_{ij}E_{kl} = \delta_{jk}E_{il}$,

$$\lambda^1\lambda^4 = (E_{12}+E_{21})(E_{13}+E_{31}) = E_{23}, \qquad \lambda^4\lambda^1 = (E_{13}+E_{31})(E_{12}+E_{21}) = E_{32}, \tag{B.66}$$

because only the terms with a matched inner index survive ($E_{21}E_{13} = E_{23}$ and $E_{31}E_{12} = E_{32}$, all others vanish). Therefore $[\lambda^1, \lambda^4] = E_{23} - E_{32}$. Comparing with $\lambda^7 = -i(E_{23} - E_{32})$ gives $E_{23} - E_{32} = i\lambda^7$, so $[\lambda^1, \lambda^4] = i\lambda^7$ and

$$f^{147} = \frac{1}{4i}\operatorname{tr}(i\,\lambda^7\lambda^7) = \frac{i}{4i}\operatorname{tr}(\lambda^7)^2 = \tfrac{1}{2}\cdot 2 = \tfrac{1}{2}. \tag{B.67}$$

$d^{118} = \frac{1}{\sqrt{3}}$. Here $\{\lambda^1, \lambda^1\} = 2(\lambda^1)^2 = 2\ \mathrm{diag}(1,1,0)$. Then

$$d^{118} = \frac{1}{4}\operatorname{tr}\Big(2\ \mathrm{diag}(1,1,0)\cdot\tfrac{1}{\sqrt{3}}\,\mathrm{diag}(1,1,-2)\Big) = \frac{1}{2\sqrt{3}}\operatorname{tr}\mathrm{diag}(1,1,0) = \frac{1}{2\sqrt{3}}\cdot 2 = \frac{1}{\sqrt{3}}. \tag{B.68}$$

$d^{146} = \frac{1}{2}$. From the matrix-unit products above, $\{\lambda^1, \lambda^4\} = \lambda^1\lambda^4 + \lambda^4\lambda^1 = E_{23} + E_{32} = \lambda^6$. Hence

$$d^{146} = \frac{1}{4}\operatorname{tr}(\lambda^6\lambda^6) = \frac{1}{4}\cdot 2 = \frac{1}{2}. \tag{B.69}$$

Each computation uses only the explicit matrices of Eq. (B.59), the matrix-unit algebra, and the normalization Eq. (B.60); nothing is quoted.

Carrying out the identical procedure for every index triple—a finite computation, mechanized here but reproducible by hand exactly as in Theorem B.19—yields the complete tables. By total antisymmetry of f and total symmetry of d, it suffices to list one ordering of each nonzero independent triple; all others follow by permutation (with a sign for f).

Table B.1: The nonzero independent structure constants f^{abc} of $\mathfrak{su}(3)$ (totally antisymmetric; permutations of indices give the same value up to the sign of the permutation). All unlisted independent triples vanish.

abc	123	147	156	246	257	345	367	458	678
f^{abc}	1	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	

The reader should note the internal checks built into these tables. The f entries $f^{123} = 1$, $f^{147} = f^{246} = f^{257} = f^{345} = \frac{1}{2}$, $f^{156} = f^{367} = -\frac{1}{2}$ record the three embedded $\mathfrak{su}(2)$ subalgebras and their interlocking; the pair $f^{458} = f^{678} = \frac{\sqrt{3}}{2}$ involves the second Cartan direction λ^8. The d entries involving λ^8 must satisfy a consistency relation that follows from contracting Eq. (B.62): since $\{\lambda^8, \lambda^8\} = 2(\lambda^8)^2 = \frac{2}{3}\,\mathrm{diag}(1,1,4)$, one reads $d^{888} = -\frac{1}{\sqrt{3}}$ and the four equal entries $d^{aa8} = -\frac{1}{2\sqrt{3}}$ for $a \in \{4,5,6,7\}$ off the diagonal of $(\lambda^8)^2$, consistent with the table.

Table B.2: The nonzero independent symmetric symbols d^{abc} of $\mathfrak{su}(3)$ (totally symmetric; all index permutations give the same value). All unlisted independent triples vanish.

value $+\frac{1}{2}$	value $-\frac{1}{2}$	involving λ^8
$d^{146} = d^{157} = \frac{1}{2}$	$d^{247} = -\frac{1}{2}$	$d^{118} = d^{228} = d^{338} = \frac{1}{\sqrt{3}}$
$d^{256} = d^{344} = d^{355} = \frac{1}{2}$	$d^{366} = d^{377} = -\frac{1}{2}$	$d^{448} = d^{558} = d^{668} = d^{778} = -\frac{1}{2\sqrt{3}}$
		$d^{888} = -\frac{1}{\sqrt{3}}$

Physical Insight

The appearance of *two* invariant tensors, f^{abc} and d^{abc}, is not an $\mathfrak{su}(3)$ accident but the first sign of a structural fact: for $\mathfrak{su}(N)$ with $N \geq 3$ the symmetric cubic invariant d^{abc} is nonzero, whereas for $\mathfrak{su}(2)$ it vanishes identically (there is no symmetric traceless cubic in the Pauli matrices). Physically the two tensors play different roles. The antisymmetric f^{abc} is the structure constant: it sits in the gauge-boson self-interactions and is what makes Yang–Mills nonabelian. The symmetric d^{abc} measures the anomaly content of the representation—the triangle-diagram anomaly coefficient of a fermion in representation R is proportional to a trace $\mathrm{tr}(\{T^a, T^b\} T^c) \propto d^{abc} A(R)$—which is why $\mathfrak{su}(2)$, having $d^{abc} = 0$, is automatically anomaly-free while $\mathfrak{su}(3)$ and the electroweak hypercharge sector are not. The cancellation of these d-weighted traces generation by generation is exactly the arithmetic carried out in Chapter F.

B.2.5 The representation invariants of the $\mathfrak{su}(3)$ fundamental

With explicit generators in hand we evaluate the two invariants for the fundamental representation and confirm the master identity in its second instance. The index is already fixed: $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ from Eq. (B.60) is precisely the statement $T(\mathrm{fund}) = \frac{1}{2}$. The Casimir requires a short computation, which we do by the Schur-plus-trace argument—the same logic that gave C_2 for $\mathfrak{su}(2)$, now applied without needing a highest-weight state.

Proposition B.20 (Fundamental Casimir of $\mathfrak{su}(3)$)**.** *For the fundamental representation of* $\mathfrak{su}(3)$*,*

$$C_2(\mathit{fund}) = \left.\frac{N^2 - 1}{2N}\right|_{N=3} = \frac{4}{3}. \tag{B.70}$$

Proof. By Theorem B.4 the Casimir is $\sum_a T^a T^a = C_2(\mathrm{fund})\, \mathbf{1}_{3\times 3}$, a multiple of the identity because it commutes with all T^b and the fundamental is irreducible

(Schur). Take the trace of both sides. On the right, $\operatorname{tr}\mathbf{1}_{3\times3} = 3$, so the right-hand side is $3\,C_2(\text{fund})$. On the left, exchanging trace and sum and using Eq. (B.60) with $a = b$,

$$\operatorname{tr}\Big(\sum_{a=1}^{8} T^a T^a\Big) = \sum_{a=1}^{8} \operatorname{tr}\left(T^a T^a\right) = \sum_{a=1}^{8} \tfrac{1}{2} = 8 \cdot \tfrac{1}{2} = 4. \tag{B.71}$$

Equating, $3\,C_2(\text{fund}) = 4$, hence $C_2(\text{fund}) = \frac{4}{3}$. Equivalently, in terms of the matrices, $\sum_a (\lambda^a)^2 = 4\sum_a T^a T^a = \frac{16}{3}\mathbf{1}$. The result is the $N = 3$ value of the general fundamental Casimir $C_F = (N^2-1)/2N$ of Chapter A, which Section B.3.1 proves for all N from the completeness relation. □

Worked Example B.21 (The master identity for the $\mathfrak{su}(3)$ fundamental)**.** The algebra $\mathfrak{su}(3)$ is eight-dimensional, so $\dim(G) = 8$, and the fundamental has $\dim(R) = 3$. The left-hand side of Theorem B.5, using Eq. (B.70), is

$$\dim(R)\,C_2(R) = 3 \cdot \tfrac{4}{3} = 4. \tag{B.72}$$

The right-hand side, using $T(\text{fund}) = \frac{1}{2}$, is

$$\dim(G)\,T(R) = 8 \cdot \tfrac{1}{2} = 4. \tag{B.73}$$

They agree: $4 = 4$. The identity is satisfied, and the agreement is again the trace $\operatorname{tr}(\sum_a T^a T^a)$ read in two directions—once as $\dim(R)\,C_2(R)$ by collapsing the Casimir, once as $\sum_a \operatorname{tr}(T^a T^a) = \dim(G)\,T(R)$ by collapsing the index. Combined with Theorem B.15, we have now verified the master identity of Section 1 in both of the cases the rest of the book uses most: the spin-j tower of $\mathfrak{su}(2)$ and the color triplet of $\mathfrak{su}(3)$.

It is worth seeing the fundamental of $\mathfrak{su}(3)$ once as a picture, because the weight diagram makes the role of the two Cartan generators T^3, T^8 concrete and is the object the quark model and Yang–Mills theory both draw on. The three basis states of $\mathbb{C}^3$ are simultaneous eigenstates of the commuting pair (T^3, T^8); reading the diagonals of λ^3 and λ^8 in Eq. (B.59) gives their eigenvalues, and in the physically conventional coordinates $(t_3, y) = (\frac{1}{2}\lambda^3, \frac{1}{\sqrt{3}}\lambda^8)$ the three weights form an equilateral triangle.

Two features of Fig. B.1 encode invariants computed above. The centroid of the triangle is the origin because each Cartan generator is traceless, $\sum_{\text{states}} t_3 = \sum_{\text{states}} y = 0$. And the common squared distance of the three weights from the origin is tied to the Casimir, provided one measures it with the canonically normalized Cartan generators $T^3 = \frac{1}{2}\lambda^3$ and $T^8 = \frac{1}{2}\lambda^8$ rather than the hypercharge-rescaled $Y = \frac{1}{\sqrt{3}}\lambda^8$ used as the plot's vertical axis. The eigenvalues of T^8 on the three states are $\frac{1}{2\sqrt{3}}, \frac{1}{2\sqrt{3}}, -\frac{1}{\sqrt{3}}$, so summing $t_3^2 + (T^8\text{-eigenvalue})^2$ over the three vertices gives $(\frac{1}{4} + \frac{1}{12}) + (\frac{1}{4} + \frac{1}{12}) + (0 + \frac{1}{3}) = \frac{1}{2} + \frac{1}{2} = 1$, which

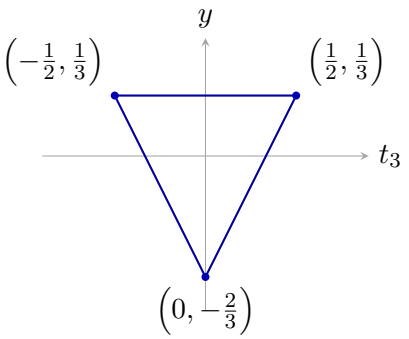


Figure B.1: Weight diagram of the $\mathfrak{su}(3)$ fundamental representation **3** in the Cartan plane (t_3, y), with t_3 the eigenvalue of $T^3 = \frac{1}{2}\lambda^3$ and y the eigenvalue of $Y = \frac{1}{\sqrt{3}}\lambda^8$. The three color states sit at the vertices of an equilateral triangle; the conjugate $\overline{\mathbf{3}}$ is the same triangle inverted. The centroid is at the origin, reflecting tracelessness $\operatorname{tr} T^3 = \operatorname{tr} Y = 0$.

is $\operatorname{tr}\big((T^3)^2 + (T^8)^2\big)$; this is the contribution of the two Cartan directions to $\operatorname{tr}(\sum_a T^a T^a) = 4$, the remaining 3 coming in equal shares of $\frac{1}{2}$ from the six off-diagonal (root) generators. The decomposition is a small but honest check that the same number, $\dim(R)\, C_2(R) = 4$, is what the geometry is drawing.

We have now made the abstract index–Casimir pair of Section 1 fully explicit for the two smallest simple algebras: explicit generators, their multiplication laws, and the complete f/d symbol structure of $\mathfrak{su}(3)$, all proved from the matrices themselves and all consistent with the master identity. The next section takes the single most important of these structures—the completeness of the fundamental generators, of which the Pauli relation Eq. (B.45) is the $N = 2$ shadow—and promotes it to the general $\mathrm{SU}(N)$ color-Fierz identity, the computational engine on which every color-factor reduction in the gauge-theory chapters runs.

B.3 The Fundamental Completeness Relation and Color Fierz Identities

The previous section put explicit matrices in our hands: the Pauli generators of $\mathfrak{su}(2)$ and the Gell-Mann generators of $\mathfrak{su}(3)$, together with the structure constants f^{abc} and the symmetric symbols d^{abc} that organize their products (Section B.2). Explicit matrices are indispensable for fixing conventions, but they are the wrong tool for computation. No one evaluates a QCD color factor by multiplying 3×3 Gell-Mann matrices entry by entry; the diagrams of a single one-loop process would generate dozens of such products and the arithmetic would be hopeless and unilluminating. What one wants instead is an algebra of *identities* that lets a string of fundamental generators be collapsed to numbers

and Kronecker deltas without ever touching a matrix entry, and that works for general N so that the SU(N) dependence is exposed rather than buried. The single identity from which this entire calculus flows is the *completeness relation* for the fundamental generators. This section proves it from the one structural fact that distinguishes $\mathfrak{su}(N)$ from $\mathfrak{u}(N)$ — the tracelessness of the generators — and then turns it into the two workhorses of color algebra: the color-Fierz rearrangement of fermion bilinears, and the trace-reduction algorithm that the Yang–Mills and QCD chapters reduce every color factor to.

Throughout, T^a ($a = 1, \dots, N^2-1$) are the Hermitian fundamental generators of $\mathfrak{su}(N)$, normalized as fixed in Section B.1,

$$(T^a)^\dagger = T^a, \qquad \operatorname{tr} T^a = 0, \qquad \operatorname{tr}(T^a T^b) = \tfrac{1}{2}\,\delta^{ab}, \tag{B.74}$$

acting on the N-dimensional fundamental (color) space with indices $i, j, k, l = 1, \dots, N$. Repeated adjoint indices a are summed from 1 to $N^2 - 1$ and repeated color indices from 1 to N. The symbol $\mathbf{1}$ denotes the $N \times N$ identity on color space.

B.3.1 The fundamental completeness relation

The starting point is a counting fact dressed as a geometric one. The complex $N \times N$ matrices form a vector space of complex dimension N^2; the Hermitian ones form a *real* vector space of the same dimension N^2, since a Hermitian matrix has N real diagonal entries and $\binom{N}{2}$ complex off-diagonal entries above the diagonal, for $N + 2 \cdot \frac{N(N-1)}{2} = N^2$ real parameters. On this space the natural inner product is the Hilbert–Schmidt (trace) pairing

$$\langle A, B \rangle \equiv \operatorname{tr}(A^\dagger B), \tag{B.75}$$

which for Hermitian A, B reduces to $\operatorname{tr}(AB)$ and is real, symmetric, and positive definite ($\langle A, A \rangle = \operatorname{tr}(A^\dagger A) = \sum_{ij} |A_{ij}|^2 \geq 0$, vanishing only for $A = 0$). The $N^2 - 1$ generators T^a are Hermitian, traceless, and mutually orthogonal under (B.75) by the normalization (B.74). They are one short of a basis. The missing direction is the trace direction, supplied by the identity matrix.

Lemma B.22 (Orthonormal basis of Hermitian matrices)**.** *The N^2 matrices*

$$T^0 \equiv \frac{1}{\sqrt{N}}\,\mathbf{1}, \qquad \widehat{T}^a \equiv \sqrt{2}\,T^a \quad (a = 1, \dots, N^2 - 1), \tag{B.76}$$

form an orthonormal basis of the space of $N \times N$ Hermitian matrices with respect to (B.75)*.*

Proof. We check orthonormality of the N^2 matrices in (B.76); since they are N^2 mutually orthogonal nonzero vectors in an N^2-dimensional space, they are

automatically a basis. First,

$$\langle T^0, T^0\rangle = \operatorname{tr}\Big(\tfrac{1}{N}\mathbf{1}\Big) = \frac{1}{N}\cdot N = 1.$$

Next, the cross terms vanish by tracelessness of the generators,

$$\langle T^0, \widehat{T}^a\rangle = \frac{\sqrt{2}}{\sqrt{N}}\,\operatorname{tr} T^a = 0,$$

and this is the one place the defining property $\operatorname{tr} T^a = 0$ enters: it is what makes the identity direction orthogonal to every generator. Finally the generator block is orthonormal because

$$\langle \widehat{T}^a, \widehat{T}^b\rangle = 2\,\operatorname{tr}(T^a T^b) = 2\cdot\tfrac{1}{2}\,\delta^{ab} = \delta^{ab},$$

using (B.74). That an orthonormal set of N^2 vectors spans the N^2-dimensional real space of Hermitian matrices is the elementary completeness of orthonormal systems in finite dimension, established in Chapter 2. □

An orthonormal basis resolves any vector into its components. For a general $N\times N$ matrix M — not necessarily Hermitian, since the basis (B.76) spans the complex matrices once complex coefficients are allowed — the expansion reads

$$M = \langle T^0, M\rangle\, T^0 + \sum_a \langle \widehat{T}^a, M\rangle\, \widehat{T}^a = \frac{1}{N}\,(\operatorname{tr} M)\,\mathbf{1} + 2\sum_a \operatorname{tr}(T^a M)\,T^a. \tag{B.77}$$

This is already a useful statement: it says that any matrix splits into its trace part, proportional to the identity, and its traceless part, expanded on the generators with coefficients $2\operatorname{tr}(T^a M)$. The completeness relation is what (B.77) becomes when we feed it the most elementary matrices of all.

Theorem B.23 (Fundamental completeness relation)**.** *The fundamental generators of* $\mathfrak{su}(N)$ *satisfy*

$$\boxed{\sum_{a=1}^{N^2-1} (T^a)_{ij}\,(T^a)_{kl} = \tfrac{1}{2}\Big(\delta_{il}\,\delta_{jk} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{kl}\Big).} \tag{B.78}$$

Proof. Apply the expansion (B.77) to the elementary matrix E_{jk}, defined by $(E_{jk})_{pq} = \delta_{pj}\,\delta_{qk}$ (a single 1 in row j, column k). Its trace is $\operatorname{tr} E_{jk} = \sum_p (E_{jk})_{pp} = \delta_{jk}$, and its generator components are

$$\operatorname{tr}(T^a E_{jk}) = \sum_{p,q} (T^a)_{pq}\,(E_{jk})_{qp} = \sum_{p,q} (T^a)_{pq}\,\delta_{qj}\,\delta_{pk} = (T^a)_{kj}.$$

Substituting into (B.77) and reading off the (i,l) matrix element, with $(E_{jk})_{il} = \delta_{ij}\delta_{lk}$ and $(\mathbf{1})_{il} = \delta_{il}$,

$$\delta_{ij}\,\delta_{lk} = \frac{1}{N}\,\delta_{jk}\,\delta_{il} + 2\sum_a (T^a)_{il}\,(T^a)_{kj}. \tag{B.79}$$

Solving for the generator sum,

$$\sum_a (T^a)_{il}\,(T^a)_{kj} = \tfrac{1}{2}\Big(\delta_{ij}\,\delta_{lk} - \tfrac{1}{N}\,\delta_{il}\,\delta_{jk}\Big).$$

Relabel the dummy index names by the swap $j \leftrightarrow l$ (a mere renaming of the four free labels) so the first factor reads $(T^a)_{ij}$ and the second $(T^a)_{kl}$; the right-hand side becomes $\frac{1}{2}(\delta_{il}\delta_{jk} - \frac{1}{N}\delta_{ij}\delta_{kl})$, which is (B.78). □

Equation (B.78) is the central computational engine of $\mathfrak{su}(N)$ color algebra; everything below is a corollary. Two consistency checks confirm the coefficients before we trust them. Contracting $j = i$ (and summing) collapses the left side to $\sum_a (T^a)_{ii}(T^a)_{kl} = \sum_a (\operatorname{tr} T^a)(T^a)_{kl} = 0$ by tracelessness, while the right side gives $\frac{1}{2}(\delta_{il}\delta_{ik} - \frac{1}{N}N\delta_{kl}) = \frac{1}{2}(\delta_{kl} - \delta_{kl}) = 0$: consistent, and again it is the $-1/N$ piece, weighted by $\delta_{ii} = N$, that cancels the would-be trace. Contracting instead $j = k$ gives $\sum_a (T^a T^a)_{il} = \frac{1}{2}(\delta_{il}\delta_{jj} - \frac{1}{N}\delta_{ij}\delta_{jl}) = \frac{1}{2}(N - \frac{1}{N})\delta_{il} = \frac{N^2-1}{2N}\delta_{il}$, the quadratic Casimir in the fundamental, to which we return in Section B.4.

Physical Insight

The completeness relation has a transparent diagrammatic meaning in 't Hooft's double-line notation, drawn in Fig. B.2. A gluon, carrying an adjoint index a, is a quark line and an antiquark line glued together: it transports color from one fundamental index to another. The first term $\frac{1}{2}\delta_{il}\delta_{jk}$ is the "swap" in which the color simply flows through, $i \to l$ and $k \to j$; the second term $-\frac{1}{2N}\delta_{ij}\delta_{kl}$ subtracts the trace, i.e. the would-be $\mathfrak{u}(1)$ photon that an honest SU(N) gluon must not contain. In the large-N limit the subtraction is suppressed by $1/N$, which is the diagrammatic origin of the planar expansion.

The $-1/N$ term is not a convention; it is forced, and it is exactly what separates SU(N) from U(N). To see this, restore the trace generator. The Lie algebra $\mathfrak{u}(N)$ is spanned by the N^2 Hermitian generators $\{T^A\} = \{T^0, T^a\}$ with $T^0 = \mathbf{1}/\sqrt{2N}$, normalized so that $\operatorname{tr}(T^0T^0) = \frac{1}{2N}\operatorname{tr}\mathbf{1} = \frac{1}{2}$ matches (B.74). For $\mathfrak{u}(N)$ the completeness sum runs over all N^2 generators, and the extra term is

$$(T^0)_{ij}\,(T^0)_{kl} = \frac{1}{2N}\,\delta_{ij}\,\delta_{kl}. \tag{B.80}$$

Adding this to the SU(N) result (B.78),

$$\sum_{A=0}^{N^2-1} (T^A)_{ij}\,(T^A)_{kl} = \tfrac{1}{2}\Big(\delta_{il}\,\delta_{jk} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{kl}\Big) + \frac{1}{2N}\,\delta_{ij}\,\delta_{kl} = \tfrac{1}{2}\,\delta_{il}\,\delta_{jk}, \tag{B.81}$$

the clean "swap" with no trace subtraction. This is the cleaner relation precisely because U(N) contains the full set of Hermitian matrices, with no constraint to enforce; the $-1/N$ in (B.78) is the algebraic price of removing the trace direction to descend from U(N) to SU(N).

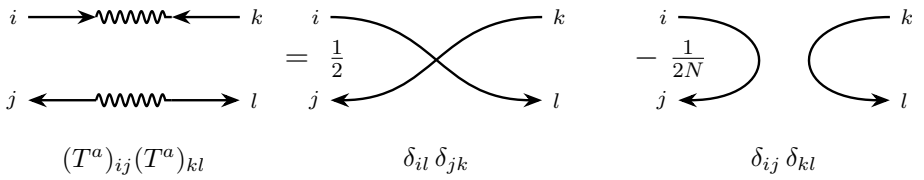


Figure B.2: The completeness relation (B.78) in 't Hooft double-line notation. A gluon (coiled, carrying adjoint index a) is resolved into a quark line and an antiquark line. The first term lets color flow straight through ($i\to l$, $k\to j$); the second subtracts the trace ($i\to j$, $k\to l$) with weight $1/N$, removing the $\mathfrak{u}(1)$ direction absent from SU(N). For U(N) the second term is absent and the gluon is a bare quark–antiquark pair, (B.81).

B.3.2 The color-Fierz identity

The first deployment of completeness is the rearrangement of a product of two color-octet currents into color-singlet combinations. This is the *color* Fierz identity — distinct from, and acting on different indices than, the spinor Fierz identity of Section C.8, which rearranges the Dirac structure. The color Fierz acts only on the fundamental color indices and leaves the spinor structure Γ of each bilinear untouched.

Consider two quark bilinears coupled through a generator, with the color indices written out explicitly and the (arbitrary) Dirac matrices Γ, Γ' carried as spectators:

$$(\bar q\,\Gamma\,T^a q)(\bar q'\,\Gamma'\,T^a q') = \bar q_i\,\Gamma\,q_j\;\bar q'_k\,\Gamma'\,q'_l\;\sum_a (T^a)_{ij}\,(T^a)_{kl}. \tag{B.82}$$

This is exactly the structure produced by single-gluon exchange between two quark currents, the gluon's adjoint index a being summed. Inserting the completeness relation (B.78),

$$\begin{aligned}(\bar q\,\Gamma\,T^a q)(\bar q'\,\Gamma'\,T^a q') &= \tfrac12\,\bar q_i\,\Gamma\,q_j\;\bar q'_k\,\Gamma'\,q'_l\,\Big(\delta_{il}\,\delta_{jk} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{kl}\Big)\\ &= \tfrac12\,(\bar q_i\,\Gamma\,q_j)(\bar q'_j\,\Gamma'\,q'_i) - \tfrac{1}{2N}\,(\bar q_i\,\Gamma\,q_i)(\bar q'_k\,\Gamma'\,q'_k)\\ &= \tfrac12\,(\bar q\,\Gamma\,q')(\bar q'\,\Gamma'\,q) - \tfrac{1}{2N}\,(\bar q\,\Gamma\,q)(\bar q'\,\Gamma'\,q').\end{aligned} \tag{B.83}$$

In the last line the color indices have been contracted into singlets: in the first term the constraint $i=l$, $j=k$ pairs $\bar q$ with q' and $\bar q'$ with q, while in the second term $i=j$, $k=l$ leaves each bilinear color-diagonal. No fermion fields were moved past one another — only color contractions were relabeled — so no statistics sign is generated; any sign from reordering the spinors belongs to the separate spinor Fierz step. The identity (B.83) is the color-space content of one-gluon exchange: an octet × octet current is precisely a color-singlet exchange (coefficient $\frac12$) minus $\frac{1}{2N}$ of the disconnected singlet product. This is the algebraic

seed of the attractive quark–antiquark singlet channel and of the $1/N$ counting that organizes large-N QCD; the explicit channel color factors are assembled in Section B.4 and used in Chapter 28.

B.3.3 The trace-reduction algorithm

We now extract from completeness the single rule that reduces *any* string of fundamental generators with a contracted (summed) adjoint index. Suppose two equal generators T^a sandwich an arbitrary $N \times N$ matrix X — itself any product of generators and identities, with the index a summed. Then completeness collapses the pair.

Proposition B.24 (The fundamental reduction rule)**.** *For any $N \times N$ matrix X,*

$$\boxed{\sum_{a=1}^{N^2-1} T^a\, X\, T^a = \tfrac{1}{2}\Big(\operatorname{tr}(X)\,\mathbf{1} - \tfrac{1}{N}\,X\Big).} \tag{B.84}$$

Proof. Take the (i,l) matrix element and insert completeness (B.78):

$$\begin{aligned}\big(T^a X T^a\big)_{il} &= (T^a)_{ij}\,X_{jk}\,(T^a)_{kl} = X_{jk}\sum_a (T^a)_{ij}\,(T^a)_{kl} \\ &= X_{jk}\cdot\tfrac{1}{2}\Big(\delta_{il}\,\delta_{jk} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{kl}\Big) = \tfrac{1}{2}\Big(\delta_{il}\,X_{jj} - \tfrac{1}{N}\,X_{il}\Big) = \tfrac{1}{2}\Big(\operatorname{tr}(X)\,\delta_{il} - \tfrac{1}{N}\,X_{il}\Big),\end{aligned}$$

which is (B.84) read as a matrix identity. □

Equation (B.84) is the whole algorithm in one line: it "eats" a contracted gluon, converting the sandwiched string X into its trace (a shorter, simpler object) plus a multiple of X itself. Every standard reduction is a special case.

Setting $X = \mathbf{1}$ recovers the quadratic Casimir already glimpsed above,

$$T^a T^a = \tfrac{1}{2}\Big(\operatorname{tr}(\mathbf{1})\,\mathbf{1} - \tfrac{1}{N}\,\mathbf{1}\Big) = \frac{N^2-1}{2N}\,\mathbf{1} \equiv C_F\,\mathbf{1}, \tag{B.85}$$

defining the fundamental Casimir $C_F = (N^2-1)/2N$ whose evaluation and role are taken up in Section B.4. Setting $X = T^b$ and using $\operatorname{tr} T^b = 0$ gives the first of the named trace reductions,

$$T^a T^b T^a = \tfrac{1}{2}\Big(\operatorname{tr}(T^b)\,\mathbf{1} - \tfrac{1}{N}\,T^b\Big) = -\frac{1}{2N}\,T^b, \tag{B.86}$$

the coefficient $-1/(2N)$ emerging directly: the trace term dies because the generators are traceless, leaving only the $-X/N$ piece. Taking the trace of (B.86),

$$\operatorname{tr}\big(T^a T^b T^a\big) = -\frac{1}{2N}\,\operatorname{tr}(T^b) = 0, \tag{B.87}$$

which vanishes, as every trace of an odd number of generators must once each contracted pair is removed. The relation (B.86) can equivalently be written $T^aT^bT^a = (C_F - \frac{1}{2}C_A)T^b$ with $C_A = N$; that this equals $-1/(2N)$ is the consistency of $C_F - \frac{1}{2}C_A = \frac{N^2-1}{2N} - \frac{N}{2} = -\frac{1}{2N}$, derived independently in Section B.4 via the structure constants.

The next reduction in the standard list is the four-generator trace with one contracted pair, which the rule disposes of in a single step.

Worked Example B.25 (The trace $\mathrm{tr}(T^aT^bT^aT^c)$)**.** We reduce $\mathrm{tr}(T^aT^bT^aT^c)$, summed over a. The two T^a sandwich $X = T^b$, so by (B.86) the inner triple collapses before the trace is taken:

$$\mathrm{tr}\big(T^aT^bT^aT^c\big) = \mathrm{tr}\big[(T^aT^bT^a)\,T^c\big] = \mathrm{tr}\Big[-\tfrac{1}{2N}\,T^b\,T^c\Big] = -\frac{1}{2N}\;\mathrm{tr}(T^bT^c) = -\frac{1}{2N}\cdot\tfrac{1}{2}\,\delta^{bc} = -\frac{1}{4N}$$

using $\mathrm{tr}(T^bT^c) = \frac{1}{2}\delta^{bc}$ from (B.74) at the end. The whole computation is one application of the reduction rule followed by the normalization; no matrix entries appear, and the answer is exact for every N.

The general algorithm for collapsing an arbitrary string $T^{a_1}T^{a_2}\cdots T^{a_n}$ in which some adjoint indices are repeated (summed) now follows mechanically. The procedure is:

1. **Locate a contracted pair.** Pick any adjoint index, say a, that appears twice in the string. The two occurrences sandwich some sub-string X (a product of the intervening generators, possibly empty so that $X = \mathbf{1}$).

2. **Eat it.** Replace T^aXT^a by $\frac{1}{2}\big(\mathrm{tr}(X)\,\mathbf{1} - \frac{1}{N}X\big)$ via (B.84). This lowers the number of generators in the string by two and spawns, in the trace term, a shorter trace $\mathrm{tr}(X)$ to be reduced separately.

3. **Recurse.** The trace $\mathrm{tr}(X)$ and the remaining string each contain fewer contracted pairs; reapply steps 1–2 until no repeated adjoint index remains.

4. **Finish.** What survives is a string (or trace) of distinct generators. Reduce traces of two generators by $\mathrm{tr}(T^aT^b) = \frac{1}{2}\delta^{ab}$, traces of three by $\mathrm{tr}(T^aT^bT^c) = \frac{1}{4}(d^{abc} + if^{abc})$ (Section B.2), and convert any commutators that arise from reordering via $\big[T^a, T^b\big] = if^{abc}T^c$.

When the two equal indices are not nested but interleaved with other contractions, one first uses $\big[T^a, T^b\big] = if^{abc}T^c$ to bring them adjacent or to expose a sandwich, then applies the rule; the structure constants generated are reduced by the adjoint identities of Sections B.2 and B.4. The point is that the loop terminates: each application of (B.84) strictly reduces the length, so any string of generators reduces in finitely many steps to numbers, Kronecker deltas, f^{abc}, and d^{abc}. The following worked example exercises the recursion twice.

Worked Example B.26 (Reducing longer strings)**.** *(a) The standard one-loop string* $\operatorname{tr}(T^aT^bT^aT^b)$. The two T^a sandwich $X = T^b$, so (B.86) gives $T^aT^bT^a = -\frac{1}{2N}T^b$ and

$$\operatorname{tr}(T^aT^bT^aT^b) = \operatorname{tr}\big[(T^aT^bT^a)T^b\big] = -\frac{1}{2N}\,\operatorname{tr}(T^bT^b) = -\frac{1}{2N}\cdot\frac{N^2-1}{2} = -\frac{N^2-1}{4N},$$

where $\operatorname{tr}(T^bT^b) = \sum_b \frac{1}{2} = \frac{N^2-1}{2}$ counts the N^2-1 generators. For $N=3$ this is $-\frac{8}{12} = -\frac{2}{3}$. This combination measures the difference between the abelian-like and genuinely non-abelian parts of a gluon self-energy insertion.

(b) A nested six-generator trace $\operatorname{tr}(T^aT^bT^cT^aT^bT^c)$. Apply the rule to the outermost contracted pair, the two T^a, which sandwich $X = T^bT^c$. By (B.84) with $\operatorname{tr}(T^bT^c) = \frac{1}{2}\delta^{bc}$,

$$T^a\,(T^bT^c)\,T^a = \tfrac{1}{2}\Big(\tfrac{1}{2}\delta^{bc}\,\mathbf{1} - \tfrac{1}{N}\,T^bT^c\Big) = \tfrac{1}{4}\delta^{bc}\,\mathbf{1} - \tfrac{1}{2N}\,T^bT^c.$$

Insert this into the string and multiply by the trailing T^bT^c:

$$T^aT^bT^cT^aT^bT^c = \Big(\tfrac{1}{4}\delta^{bc}\,\mathbf{1} - \tfrac{1}{2N}\,T^bT^c\Big)T^bT^c = \tfrac{1}{4}\,\delta^{bc}T^bT^c - \tfrac{1}{2N}\,T^bT^cT^bT^c.$$

The first term is $\frac{1}{4}T^bT^b = \frac{1}{4}C_F\,\mathbf{1}$ by (B.85). For the second, eat the pair of T^b, which sandwich $X = T^c$: by (B.86), $T^bT^cT^b = -\frac{1}{2N}T^c$, so $T^bT^cT^bT^c = -\frac{1}{2N}T^cT^c = -\frac{1}{2N}C_F\,\mathbf{1}$. Hence

$$T^aT^bT^cT^aT^bT^c = \tfrac{1}{4}C_F\,\mathbf{1} - \tfrac{1}{2N}\Big(-\tfrac{1}{2N}C_F\Big)\mathbf{1} = \frac{C_F}{4}\Big(1+\frac{1}{N^2}\Big)\mathbf{1}.$$

Taking the trace multiplies by $\operatorname{tr}\mathbf{1} = N$ and, with $C_F = (N^2-1)/2N$,

$$\operatorname{tr}(T^aT^bT^cT^aT^bT^c) = \frac{C_FN}{4}\Big(1+\frac{1}{N^2}\Big) = \frac{N^2-1}{8}\cdot\frac{N^2+1}{N^2} = \frac{N^4-1}{8N^2},$$

which for $N=3$ equals $\frac{80}{72} = \frac{10}{9}$. Two applications of the single rule (B.84), with no matrix multiplication, deliver the closed-form N-dependence.

The reduction rule also subsumes the color content of the simplest physical color sums, the ones Chapter 28 needs for the R-ratio and for color-averaged cross sections. When a colorless probe (a photon or a color-singlet current) couples diagonally, $\propto \delta_{ij}$, the final-state color sum is the bare trace $\operatorname{tr}\mathbf{1} = N$, the famous factor of $N_c = 3$ that multiplies $\sum_f Q_f^2$ in $R = \sigma(e^+e^- \to \text{hadrons})/\sigma(e^+e^- \to \mu^+\mu^-)$. When instead a gluon is exchanged, the color factor is one of the combinations (B.83), (B.85), or (B.86), reduced by the algorithm above. Either way, every color factor in the perturbative expansion of QCD is a finite sequence of applications of Theorem B.23 and Theorem B.24.

With the completeness relation (B.78) and the reduction rule (B.84) in hand, the canonical SU(N) invariants are now within immediate reach: the contraction $T^aT^a = C_F\mathbf{1}$ of (B.85) already exhibits the fundamental Casimir, and the adjoint Casimir C_A and Dynkin index T_F follow from the same machinery applied to the structure constants. The next section evaluates all three, specializes them to the SU(3) values of QCD, and assembles the color factors that feed the one-loop Yang–Mills beta coefficient.

B.4 Casimirs and Color Factors of SU(N)

The completeness relation and the trace-reduction algorithm assembled in Section B.3.1 were built for a purpose: to evaluate, once and cleanly, the handful of pure numbers on which every gauge-theory loop computation depends. Those numbers are the quadratic Casimir invariants and Dynkin indices of SU(N), defined in the abstract in Section B.1 and now to be reduced to explicit N-dependent expressions. The strategy is the one the engine was designed for. Contracting a pair of fundamental generators with the completeness relation delivers the fundamental Casimir C_F in a single line; feeding the same identity into a product of two structure constants delivers the adjoint Casimir C_A after a short but completely explicit trace computation; and the normalization fixed in Section B.1 already fixes the fundamental index T_F outright. We then specialize to SU(3) to read off the values the QCD literature carries, verify the general results against the master relation of Theorem B.5 across a range of N as an independent check, and finally assemble the color factors that the one-loop Yang–Mills beta function consumes in Chapter 27. This last step advances Yang–Mills theory at the level of group-theory data: it produces the carried coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ in its SU(3) form.

B.4.1 The three named invariants

We recall from Section B.1 the two scalars attached to any representation R of a compact simple Lie algebra. The *quadratic Casimir* $C_2(R)$ is the (Schur-forced) eigenvalue of the Casimir operator,

$$\sum_a T_R^a\, T_R^a \;=\; C_2(R)\,\mathbf{1}_{\dim R}, \tag{B.88}$$

a multiple of the identity because $\sum_a T_R^a T_R^a$ commutes with every generator (Theorem B.3). The *Dynkin index* $T(R)$ is fixed by the trace of two generators,

$$\operatorname{tr}\!\left(T_R^a\, T_R^b\right) \;=\; T(R)\,\delta^{ab}, \tag{B.89}$$

which is proportional to δ^{ab} because the trace form is the unique (up to scale) invariant symmetric tensor on a simple algebra (Theorem B.1). The two are tied

by the master relation of Theorem B.5,

$$\dim(R)\, C_2(R) \;=\; \dim(G)\, T(R), \tag{B.90}$$

with $\dim(G) = N^2 - 1$ the dimension of $\mathfrak{su}(N)$. The three invariants that QCD needs carry standard names. Writing $\square$ for the fundamental (N-dimensional) representation and adj for the adjoint ((N^2-1)-dimensional) representation,

$$C_F \;\equiv\; C_2(\square), \qquad C_A \;\equiv\; C_2(\mathrm{adj}), \qquad T_F \;\equiv\; T(\square). \tag{B.91}$$

Our task in this section is to evaluate all three as functions of N.

The normalization is already settled and supplies the one numerical input. The convention $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ for the fundamental generators, fixed in Eq. (B.2) and used throughout the book, is by (B.89) precisely the statement

$$T_F \;=\; \tfrac{1}{2}. \tag{B.92}$$

This is a choice, not a theorem (Section B.1.4); every other index and Casimir in the appendix is measured against it. It is the input that turns the abstract relation (B.90) into numbers.

B.4.2 The fundamental Casimir C_F

The master relation already determines C_F without further work: with $\dim(\square) = N$, $\dim(G) = N^2 - 1$, and $T_F = \frac{1}{2}$, equation (B.90) reads $N\, C_F = (N^2-1)\cdot\frac{1}{2}$, hence $C_F = (N^2-1)/2N$ (this is Eq. (B.17)). It is nonetheless worth re-deriving the same number directly from the completeness relation, both because that route makes no appeal to the abstract uniqueness arguments and because it is the warm-up whose structure the C_A computation repeats. Recall the fundamental completeness relation (B.78) of Theorem B.23,

$$\sum_a (T^a)_{ij}\,(T^a)_{kl} \;=\; \tfrac{1}{2}\Big(\delta_{il}\,\delta_{jk} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{kl}\Big), \tag{B.93}$$

the sum on the adjoint index a running over all $N^2 - 1$ generators.

Proposition B.27 (Fundamental Casimir of SU(N))**.** *The quadratic Casimir of the fundamental representation of* SU(N) *is*

$$C_F \;=\; \frac{N^2-1}{2N}. \tag{B.94}$$

Proof. By (B.88), C_F is read off the matrix $(T^a T^a)_{il} = \sum_a (T^a)_{ij}(T^a)_{jl}$, in which the fundamental color index j is summed. This is the left-hand side of (B.93) with the relabeling $k \to j$ and the index j contracted. Performing that contraction,

$$(T^a T^a)_{il} \;=\; \sum_a (T^a)_{ij}(T^a)_{jl} \;=\; \tfrac{1}{2}\Big(\delta_{il}\,\delta_{jj} - \tfrac{1}{N}\,\delta_{ij}\,\delta_{jl}\Big)$$

$$= \tfrac{1}{2}\Big(N\,\delta_{il} - \tfrac{1}{N}\,\delta_{il}\Big) = \tfrac{1}{2}\,\frac{N^2-1}{N}\,\delta_{il}, \tag{B.95}$$

using $\delta_{jj} = N$ (the trace of the $N \times N$ identity) and $\delta_{ij}\delta_{jl} = \delta_{il}$. Comparison with $T^aT^a = C_F\,\mathbf{1}$ gives (B.94). □

This is the same value Eq. (B.85) of Section B.3.3 obtained as a special case of the reduction rule, and the same value the master relation forces; the threefold agreement is the first entry of the consistency check of Section B.4.5. The number is small but carries real physics: $C_F = (N^2-1)/2N$ is the squared color charge of a single quark, the coefficient that multiplies the quark self-energy and the SU(N) analogue of the Coulomb potential.

B.4.3 The structure-constant contraction and C_A

The adjoint Casimir is the eigenvalue of $\sum_a T^a_{\rm adj}T^a_{\rm adj}$, and the adjoint generators are the structure constants themselves. In the adjoint representation a generator acts on the algebra by the bracket, $(\mathrm{ad}\,T^a)\,T^b = \big[T^a, T^b\big] = if^{abc}T^c$, so in the basis $\{T^b\}$ its matrix elements are those fixed in Eq. (B.3),

$$(T^a_{\rm adj})_{bc} = (\mathrm{ad}\,T^a)_{bc} = -\,i\,f^{abc}. \tag{B.96}$$

That these matrices furnish a genuine representation is the content of the Jacobi identity: written out in structure constants, $\big[T^a, \big[T^b, T^c\big]\big] + (\text{cyclic}) = 0$ becomes

$$f^{ade}f^{bcd} + f^{bde}f^{cad} + f^{cde}f^{abd} = 0, \tag{B.97}$$

which is precisely $\big[\mathrm{ad}\,T^a, \mathrm{ad}\,T^b\big] = if^{abc}\,\mathrm{ad}\,T^c$. The adjoint Casimir operator $\sum_c T^c_{\rm adj}T^c_{\rm adj}$ therefore commutes with every generator and, since the adjoint of a simple algebra is irreducible, Schur's lemma forces it to be a multiple of the identity on the (N^2-1)-dimensional adjoint space. Its (a,b) matrix element follows from (B.96),

$$(T^c_{\rm adj}T^c_{\rm adj})_{ab} = (\mathrm{ad}\,T^c)_{ad}(\mathrm{ad}\,T^c)_{db} = (-if^{cad})(-if^{cdb}) = -\,f^{cad}f^{cdb} = f^{acd}f^{bcd}, \tag{B.98}$$

where the final equality uses the total antisymmetry of f proved in Theorem B.7: $f^{cad} = -f^{acd}$ and $f^{cdb} = f^{bcd}$ (cycling the second factor's indices, $cdb \to dbc \to bcd$, an even permutation), and the two introduced signs multiply to $+1$ against the leading minus to give $f^{acd}f^{bcd}$. Thus the adjoint Casimir *is* the standard contraction of two structure constants,

$$f^{acd}f^{bcd} = C_A\,\delta^{ab}, \tag{B.99}$$

proportional to δ^{ab} by the Jacobi-guaranteed irreducibility and identified with $C_2(\mathrm{adj}) = C_A$. This is also the Killing form K^{ab} of Theorem B.8, computed

there from the abstract uniqueness of the invariant form; here we instead pin the constant by an explicit descent to the fundamental.

For that descent we need the two consequences of completeness already established as the reduction rule of Section B.3.3, which we recall rather than reprove. From Theorem B.24, for any $N \times N$ matrix X,

$$\sum_a T^a\, X\, T^a \;=\; \tfrac{1}{2}\Big(\operatorname{tr}(X)\,\mathbf{1} - \tfrac{1}{N}\,X\Big), \tag{B.100}$$

which specializes (Eqs. (B.85) and (B.86)) to the two instances we use repeatedly,

$$\sum_a T^a\, T^a \;=\; C_F\,\mathbf{1} \;=\; \frac{N^2-1}{2N}\,\mathbf{1}, \qquad\qquad \sum_a T^a\, T^b\, T^a \;=\; -\,\frac{1}{2N}\,T^b, \tag{B.101}$$

the first by $X = \mathbf{1}$, the second by $X = T^b$ with $\operatorname{tr} T^b = 0$. With these in hand the adjoint Casimir is a four-term trace.

Theorem B.28 (Adjoint Casimir of $\mathrm{SU}(N)$)**.** *The structure constants of* $\mathrm{SU}(N)$ *satisfy*

$$f^{acd} f^{bcd} \;=\; N\,\delta^{ab}, \qquad \textit{so that} \qquad C_A \;=\; N. \tag{B.102}$$

Proof. Express the structure constants through fundamental generators. Contracting the bracket $[T^a, T^c] = i f^{acd} T^d$ against T^e and tracing with $\operatorname{tr}(T^d T^e) = \tfrac{1}{2}\delta^{de}$ inverts to the resolvent

$$f^{acd}\, T^d \;=\; -\,i\,[T^a, T^c]\,. \tag{B.103}$$

Now project (B.99) onto a trace by contracting two copies. Using (B.103) for both factors and $\operatorname{tr}(T^d T^e) = \tfrac{1}{2}\delta^{de}$,

$$\begin{aligned} \tfrac{1}{2}\, f^{acd} f^{bcd} \;&=\; f^{acd} f^{bce}\, \operatorname{tr}(T^d T^e) \;=\; \operatorname{tr}\big[(f^{acd} T^d)(f^{bce} T^e)\big] \\ &=\; \operatorname{tr}\big[(-i\,[T^a, T^c])(-i\,\big[T^b, T^c\big])\big] \;=\; -\operatorname{tr}\big(\,[T^a, T^c]\,\big[T^b, T^c\big]\,\big), \end{aligned} \tag{B.104}$$

with the fundamental color index c summed. Hence

$$f^{acd} f^{bcd} \;=\; -\,2\,\operatorname{tr}\big(\,[T^a, T^c]\,\big[T^b, T^c\big]\,\big). \tag{B.105}$$

Expanding the two commutators gives four traces, each summed over c,

$$[T^a, T^c]\,\big[T^b, T^c\big] = \underbrace{T^a T^c T^b T^c}_{\text{(I)}}\; \underbrace{-\,T^a T^c T^c T^b}_{\text{(II)}}\; \underbrace{-\,T^c T^a T^b T^c}_{\text{(III)}}\; \underbrace{+\,T^c T^a T^c T^b}_{\text{(IV)}}. \tag{B.106}$$

We evaluate each with the insertion identities (B.100)–(B.101), using $\operatorname{tr} T^a = 0$, $\operatorname{tr}(T^a T^b) = \tfrac{1}{2}\delta^{ab}$, and $\operatorname{tr}\mathbf{1} = N$ at the end.

Term (I). The inner sum is $\sum_c T^c T^b T^c = -\frac{1}{2N} T^b$ by the second relation in (B.101), so

$$\mathrm{tr}(\mathrm{I}) = \mathrm{tr}\Big(T^a\big(-\tfrac{1}{2N}T^b\big)\Big) = -\tfrac{1}{2N}\cdot\tfrac{1}{2}\,\delta^{ab} = -\tfrac{1}{4N}\,\delta^{ab}. \tag{B.107}$$

Term (II). Here $\sum_c T^c T^c = C_F \mathbf{1}$ by the first relation in (B.101), so

$$\mathrm{tr}(\mathrm{II}) = -\,\mathrm{tr}\big(T^a\, C_F \mathbf{1}\, T^b\big) = -\,C_F\,\mathrm{tr}(T^a T^b) = -\tfrac{1}{2}\,C_F\,\delta^{ab}. \tag{B.108}$$

Term (III). Now $\sum_c T^c (T^a T^b) T^c = \frac{1}{2}\big(\mathrm{tr}(T^a T^b)\mathbf{1} - \frac{1}{N} T^a T^b\big)$ from (B.100) with $X = T^a T^b$, whose trace is

$$\mathrm{tr}(\mathrm{III}) = -\tfrac{1}{2}\Big(\mathrm{tr}(T^a T^b)\,N - \tfrac{1}{N}\,\mathrm{tr}(T^a T^b)\Big) = -\tfrac{1}{2}\cdot\tfrac{1}{2}\delta^{ab}\Big(N - \tfrac{1}{N}\Big) = -\tfrac{1}{4}\,\frac{N^2-1}{N}\,\delta^{ab} = -\tfrac{1}{2}\,C_F\,\delta^{ab} \tag{B.109}$$

Term (IV). Again $\sum_c T^c T^a T^c = -\frac{1}{2N} T^a$, so

$$\mathrm{tr}(\mathrm{IV}) = \mathrm{tr}\Big(\big(-\tfrac{1}{2N}T^a\big)T^b\Big) = -\tfrac{1}{4N}\,\delta^{ab}. \tag{B.110}$$

Adding the four contributions,

$$\mathrm{tr}\big([T^a, T^c]\,\big[T^b, T^c\big]\big) = \Big(-\tfrac{1}{4N} - \tfrac{1}{2}C_F - \tfrac{1}{2}C_F - \tfrac{1}{4N}\Big)\delta^{ab} = -\Big(C_F + \tfrac{1}{2N}\Big)\delta^{ab}. \tag{B.111}$$

With $C_F = (N^2-1)/2N$ the bracket telescopes to an integer,

$$C_F + \tfrac{1}{2N} = \frac{N^2-1}{2N} + \frac{1}{2N} = \frac{N^2}{2N} = \frac{N}{2}, \tag{B.112}$$

so $\mathrm{tr}([T^a, T^c]\,[T^b, T^c]) = -\frac{N}{2}\,\delta^{ab}$. Substituting into (B.105),

$$f^{acd} f^{bcd} = -2\cdot\Big(-\tfrac{N}{2}\,\delta^{ab}\Big) = N\,\delta^{ab}, \tag{B.113}$$

which is (B.102); comparison with (B.99) identifies $C_A = N$. □

It is worth pausing on why the answer comes out an integer. The two "$1/N$" subtractions buried in the completeness relation — the tracelessness corrections that distinguish $\mathfrak{su}(N)$ from $\mathfrak{u}(N)$ — enter through the pairs (I)+(IV) and (II)+(III), and combine with the -1 inside $C_F = (N^2-1)/2N$ so that the $1/N$ pieces cancel exactly, leaving the bare N. The same cancellation is the reason C_A is an integer for every N while C_F is not. The identification $f^{acd} f^{bcd} = C_A \delta^{ab}$ with $C_A = N$ is the structure-constant identity announced in the section goal; it is what makes the gluon and ghost loops of Chapter 27 collapse to a single color symbol.

Physical Insight

The contrast between C_F and C_A is the contrast between a charge carried by a probe and a charge carried by the force itself. $C_F = (N^2-1)/2N$ measures how strongly a single fundamental quark couples to the gluon field — it grows only logarithmically with N, $C_F \to N/2$ at large N. $C_A = N$ measures how strongly the gluon couples to *itself*, through the non-abelian self-interaction encoded in f^{abc}. That a gauge boson carries the adjoint charge $C_A = N$, with no abelian analogue (for U(1), $f^{abc} = 0$ and $C_A = 0$), is the algebraic root of asymptotic freedom: the gluon's self-charge antiscreens the coupling, an effect simply absent in QED.

B.4.4 The SU(3) values for QCD

Specializing (B.94), (B.102), and (B.92) to $N = 3$ — the color group of QCD, whose explicit Gell-Mann generators were tabulated in Section B.2.3 and whose fundamental Casimir was checked directly in Section B.2.5 — gives the numbers the entire QCD literature carries:

$$\boxed{C_F = \frac{3^2-1}{2\cdot 3} = \frac{8}{6} = \frac{4}{3}, \qquad C_A = 3, \qquad T_F = \frac{1}{2}.} \tag{B.114}$$

The first, $C_F = \frac{4}{3}$, is the color charge a single quark presents; it sets the size of the quark self-energy and of the static quark–antiquark potential. The second, $C_A = 3$, is the gluon's self-coupling charge and the source of the antiscreening that makes QCD asymptotically free. The third, $T_F = \frac{1}{2}$, weights each quark flavor in a gluon vacuum-polarization loop. These three numbers, together with the active-flavor count n_f, are the complete group-theory input to the one-loop running of α_s.

B.4.5 An independent numerical check: the master relation

The derivations of C_F in Theorem B.27 (via completeness) and of C_A in Theorem B.28 (via the structure-constant contraction) made no use of the master relation (B.90); reinserting the results into it is therefore a genuine consistency test, not a tautology. For the fundamental, $\dim(\square) = N$ and $C_F = (N^2-1)/2N$ give

$$\dim(\square)\, C_F = N \cdot \frac{N^2-1}{2N} = \frac{N^2-1}{2}, \tag{B.115}$$

which must equal $\dim(G)\, T_F = (N^2-1)\cdot\frac{1}{2}$; the two agree for every N. For the adjoint, $\dim(\text{adj}) = \dim(G) = N^2-1$ and $C_A = N$ give $\dim(\text{adj})\, C_A = N(N^2-1)$, which must equal $\dim(G)\, T(\text{adj})$; since $\dim(\text{adj}) = \dim(G)$ the relation collapses to $T(\text{adj}) = C_A = N$, the statement of Theorem B.6 that

for the adjoint the index and the Casimir coincide. Table B.3 carries this out arithmetically for $N = 2, \dots, 5$, listing both representations and confirming $\dim(R)\, C_2(R) = \dim(G)\, T(R)$ in every row.

Table B.3: Numerical verification of the master relation $\dim(R)\, C_2(R) = \dim(G)\, T(R)$ for the fundamental and adjoint of SU(N), $N = 2, \dots, 5$. Here $\dim(G) = N^2 - 1$, $T_F = \frac{1}{2}$, $C_F = (N^2 - 1)/2N$, $C_A = N$, and $T(\text{adj}) = N$. The last two columns agree in every row, the independent check that the completeness-derived C_F and the structure-constant-derived C_A are mutually consistent.

Group	$\dim(G)$	R	$\dim(R)$	$C_2(R)$	$T(R)$	$\dim(R)\, C_2(R)$	$\dim(G)\, T(R)$
SU(2)	3	$\square$	2	3/4	1/2	3/2	3/2
		adj	3	2	2	6	6
SU(3)	8	$\square$	3	4/3	1/2	4	4
		adj	8	3	3	24	24
SU(4)	15	$\square$	4	15/8	1/2	15/2	15/2
		adj	15	4	4	60	60
SU(5)	24	$\square$	5	12/5	1/2	12	12
		adj	24	5	5	120	120

The SU(2) row doubles as a check against angular-momentum theory. There the fundamental is the spin-$\frac{1}{2}$ doublet with $C_F = \frac{3}{4} = j(j+1)\big|_{j=1/2}$ in the normalization $T^a = \frac{1}{2}\sigma^a$, and the adjoint is the spin-1 triplet with $C_A = 2 = j(j+1)\big|_{j=1}$, exactly the values found from the Pauli completeness relation in Theorem B.14. The Casimirs we computed group-theoretically are nothing but the total-angular-momentum eigenvalues, recovered here from the completeness relation rather than from ladder operators.

B.4.6 Assembling the Yang–Mills color factors

We now collect the data in the form the one-loop Yang–Mills beta function needs, advancing Yang–Mills theory at the data layer. The one-loop coefficient b_0 in $\beta_g = -\dfrac{g^3}{16\pi^2}\, b_0$ is built from exactly three group-theory inputs: the gluon and ghost loops contribute through C_A, and each quark flavor contributes through T_F. The full diagrammatic computation — the loop integrals, the gauge-fixing machinery, the Ward–identity cancellations — is carried out in Chapter 27; the role of this appendix is to supply, ready-evaluated, the color factors that multiply each contribution, and to compute the carried number for QCD.

Worked Example B.29 (The SU(3) color factors entering b_0)**.** The one-loop Yang–Mills beta-function coefficient with n_f Dirac fermions in the fundamental,

in the standing normalization of Yang–Mills theory, is

$$b_0 \;=\; \tfrac{11}{3}\,C_A \;-\; \tfrac{4}{3}\,T_F\,n_f. \tag{B.116}$$

We trace the group-theory origin of each term and then specialize to SU(3).

The gauge sector, $\frac{11}{3}C_A$. The gluon self-energy at one loop receives a gluon loop and a Faddeev–Popov ghost loop, each carrying a contraction of two structure constants at its pair of trilinear vertices. By Theorem B.28 every such contraction reduces to $f^{acd}f^{bcd} = C_A\,\delta^{ab} = N\delta^{ab}$, so the entire color content of the gauge sector is the single symbol C_A. The transverse kinematic coefficients — which combine the gluon-loop and ghost-loop pieces into the gauge-invariant $\frac{11}{3}$ — are computed in Chapter 27; what the group theory fixes here is that the color structure is purely C_A, traceable directly to the adjoint Casimir we have just evaluated.

The matter sector, $-\frac{4}{3}T_F\,n_f$. Each fermion flavor runs in one vacuum-polarization loop with two gluon insertions, contributing the color trace $\mathrm{tr}(T^aT^b) = T_F\,\delta^{ab}$ over the fundamental indices. The n_f flavors form the direct sum $\square^{\oplus n_f}$, and by the additivity of the index (Theorem B.9) their combined color factor is $n_f\,T_F$. This is exactly the $-\frac{4}{3}T_Fn_f$ term, with the kinematic $\frac{4}{3}$ supplied by Chapter 27.

The role of C_F. The fundamental Casimir C_F does appear in the intermediate, gauge-dependent pieces of the calculation — in the quark field-strength renormalization Z_2 and the quark–gluon vertex renormalization Z_1 — but the Slavnov–Taylor (non-abelian Ward) identity ties these together so that C_F drops out of the gauge-invariant combination defining the coupling renormalization. This is why (B.116) carries only C_A and T_Fn_f; the cancellation is exhibited explicitly in Chapter 27.

The QCD number. Inserting the SU(3) values (B.114), $C_A = 3$ and $T_F = \frac{1}{2}$,

$$b_0 \;=\; \tfrac{11}{3}\cdot 3 \;-\; \tfrac{4}{3}\cdot\tfrac{1}{2}\,n_f \;=\; 11 \;-\; \tfrac{2}{3}\,n_f. \tag{B.117}$$

For every $n_f \le 16$ this is positive, so $\beta_g < 0$ and the coupling decreases toward high energy: asymptotic freedom. With the six known quark flavors, $n_f = 6$, one finds $b_0 = 11 - 4 = 7 > 0$; even at $n_f = 16$, $b_0 = 11 - \frac{32}{3} = \frac{1}{3} > 0$, while $n_f = 17$ would tip b_0 negative and destroy asymptotic freedom. The carried number $b_0 = 11 - \frac{2}{3}n_f$ is precisely the input Chapter 27 converts into the running of α_s and the dynamically generated scale Λ_{QCD}.

With $C_F = (N^2-1)/2N$, $C_A = N$, and $T_F = \frac{1}{2}$ in hand, the quadratic invariants of SU(N) are complete, and the SU(3) color factors that feed the one-loop Yang–Mills beta coefficient are assembled and evaluated. The remaining gauge-theory data are the *cubic* symmetric invariants — the totally symmetric tensor d^{abc} and the anomaly coefficient $A(R)$ it defines — which measure not the strength of a representation but its chirality-sensitive asymmetry. Those are

what the next section constructs, enriching the quadratic Casimir set built here with the symmetric invariants the anomaly-cancellation chapters consume.

B.5 Symmetric Invariants and the Anomaly Coefficient

The previous section closed the file on the *quadratic* invariants of $\mathfrak{su}(N)$: the Dynkin index $T(R)$, the quadratic Casimirs $C_2(R)$, and the canonical color factors $C_F = (N^2-1)/2N$, $C_A = N$, $T_F = \frac{1}{2}$ that feed the one-loop Yang–Mills β-coefficient (Section B.4). Those numbers all descend from a single symmetric invariant tensor of rank two, $\mathrm{tr}(T^a T^b) \propto \delta^{ab}$, and from the structure-constant contraction $f^{acd} f^{bcd} = C_A\, \delta^{ab}$. There is exactly one quadratic invariant, and after specifying its normalization there is nothing more to say at that order.

The next invariant up the ladder is *cubic*, and it is not a mere refinement of the quadratic data: it is the precise group-theoretic object that decides whether a chiral gauge theory is consistent. A theory of chiral (Weyl) fermions is only well defined if the gauge current remains conserved at one loop; the obstruction is the triangle diagram with three external gauge bosons and a chiral fermion circulating in the loop (Fig. B.3). The diagram's group-theory factor is the symmetrized trace $\mathrm{tr}_R(\{T^a, T^b\}T^c)$ over the fermion representation, and its kinematic part is the same for every fermion. Consistency therefore requires that a single number built from this trace—the *anomaly coefficient* $A(R)$, summed over all left-handed fermions—vanish. This section isolates that number, computes it for the representations the book uses, and proves the three structural facts that make anomaly cancellation tractable: $A(\bar{R}) = -A(R)$, $A(\mathrm{adj}) = 0$, and the vanishing of the cubic invariant for every simple algebra except $\mathfrak{su}(N)$ with $N \geq 3$. The dynamical derivation of the triangle anomaly itself lives in Chapter 36, and its application to the Standard Model and to grand unification in Chapter 37; here we supply only the data those chapters consume.

B.5.1 The symmetrized trace and the cubic invariant

Fix a compact simple Lie algebra with Hermitian generators T^a in a representation R, normalized in the fundamental by $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ (Section B.4). The object the triangle produces is the totally symmetrized trace of three generators. We begin by isolating it from the ordinary product trace.

Write the trace of three fundamental generators as the sum of its symmetric and antisymmetric parts in the index pair. Using $\left\{T^a, T^b\right\}$ and $\left[T^a, T^b\right] = i f^{abc} T^c$,

$$\mathrm{tr}(T^a T^b T^c) = \tfrac{1}{2}\,\mathrm{tr}\left(\left\{T^a, T^b\right\} T^c\right) + \tfrac{1}{2}\,\mathrm{tr}\left(\left[T^a, T^b\right] T^c\right) = \tfrac{1}{4}\, d^{abc} + \tfrac{i}{4}\, f^{abc}, \tag{B.118}$$

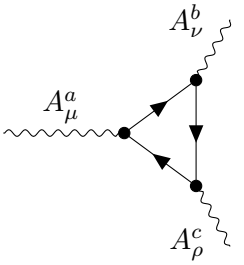


Figure B.3: The triangle (AVV) diagram whose group-theory factor is the symmetrized trace $\mathrm{tr}_R(\{T^a,T^b\}T^c) = \frac{1}{2}A(R)\,d^{abc}$. Summed over the closed fermion loop together with the crossed diagram (legs $b \leftrightarrow c$ exchanged), the antisymmetric f^{abc} part cancels and only the totally symmetric d^{abc} piece survives, weighted by $A(R)$. The kinematic factor is representation-independent; the entire dependence on the matter content is $\sum_i A(R_i)$. The amplitude is computed in Chapter 36.

where the first equality is the trivial split into anticommutator and commutator, and the second uses $\mathrm{tr}(\left[T^a,T^b\right]T^c) = if^{abd}\,\mathrm{tr}(T^dT^c) = \frac{i}{2}f^{abc}$ together with the definition of d^{abc} below. The antisymmetric piece reproduces the structure constants; the symmetric piece defines a new, totally symmetric invariant tensor.

Definition B.30 (Cubic symmetric invariant)**.** The *cubic symmetric tensor* d^{abc} of $\mathfrak{su}(N)$ is

$$d^{abc} \;\equiv\; 2\,\mathrm{tr}\big(\left\{T^a,T^b\right\}T^c\big)\Big|_{\text{fundamental}}\,, \tag{B.119}$$

equivalently the coefficient in the fundamental anticommutator

$$\left\{T^a,T^b\right\} \;=\; \frac{1}{N}\,\delta^{ab}\,\mathbf{1} \;+\; d^{abc}\,T^c. \tag{B.120}$$

For $\mathfrak{su}(3)$ in the Gell-Mann basis $T^a = \lambda^a/2$ this is the totally symmetric symbol with the nonzero independent entries $d^{118} = d^{228} = d^{338} = 1/\sqrt{3}$, $d^{888} = -1/\sqrt{3}$, etc., tabulated in Table B.2 of Section B.2.4.

That Eqs. (B.119) and (B.120) agree, and that they are consistent with the $\frac{1}{4}$ in Eq. (B.118), is a one-line check: tracing Eq. (B.120) against T^e and using $\mathrm{tr}(T^cT^e) = \frac{1}{2}\delta^{ce}$ and $\mathrm{tr}\,T^e = 0$ gives $\mathrm{tr}(\left\{T^a,T^b\right\}T^e) = \frac{1}{2}d^{abe}$, which is Eq. (B.119). The constant $1/N$ multiplying the identity is fixed by tracing Eq. (B.120) over $a = b$ (no sum on the matrix, sum on the index): $\mathrm{tr}\left\{T^a,T^b\right\} = 2\,\mathrm{tr}(T^aT^b) = \delta^{ab}$, while the right side gives $\frac{1}{N}\delta^{ab}\,N = \delta^{ab}$, consistent for the $N \times N$ fundamental. The completeness relation that makes Eq. (B.120) an identity on the full matrix algebra is recorded in Section B.3.1; we take it as given.

With the symbol in hand we promote the symmetrized trace to a representation-dependent number.

Definition B.31 (Anomaly coefficient)**.** The *anomaly coefficient* $A(R)$ of a representation R is defined by

$$\mathrm{tr}_R\big(\{T_R^a, T_R^b\}\, T_R^c\big) \;=\; \tfrac{1}{2}\, A(R)\, d^{abc}, \tag{B.121}$$

with d^{abc} the fundamental-normalized tensor of Theorem B.30. By Eq. (B.119) the normalization is $A(\mathrm{fund}) = 1$.

For Eq. (B.121) to define a single number rather than an overdetermined system of constraints, the left-hand side must be *proportional* to d^{abc} for every R. This is the central structural fact, and it has two ingredients: the trace is a totally symmetric invariant tensor, and the space of such tensors is at most one-dimensional. We prove the first now and treat the second in Section B.5.4.

Proposition B.32 (Total symmetry and invariance)**.** *For any representation R, the symmetrized trace $D^{abc}(R) \equiv \mathrm{tr}_R(\{T_R^a, T_R^b\}\, T_R^c)$ is totally symmetric in (a, b, c) and is an invariant tensor of the algebra,*

$$f^{ade} D^{ebc} + f^{bde} D^{aec} + f^{cde} D^{abe} = 0. \tag{B.122}$$

Proof. Symmetry in $a \leftrightarrow b$ is manifest from $\{T^a, T^b\} = \{T^b, T^a\}$. For the remaining transposition, expand and use cyclicity of the trace:

$$D^{abc} = \mathrm{tr}(T^a T^b T^c) + \mathrm{tr}(T^b T^a T^c) = \mathrm{tr}(T^a T^b T^c) + \mathrm{tr}(T^a T^c T^b) = \mathrm{tr}\big(T^a \{T^b, T^c\}\big), \tag{B.123}$$

where the middle equality moves T^b from the front of the second trace to the back by cyclicity, $\mathrm{tr}(T^b T^a T^c) = \mathrm{tr}(T^a T^c T^b)$. The result $D^{abc} = \mathrm{tr}(T^a \{T^b, T^c\})$ is manifestly symmetric in $b \leftrightarrow c$. Since the transpositions $(a\,b)$ and $(b\,c)$ generate the full symmetric group S_3, D^{abc} is totally symmetric.

For invariance, vary the generators by an infinitesimal algebra rotation $\delta T^a = i\,\theta^d \big[T^d, T^a\big] = -\theta^d f^{dae} T^e$. The trace of a product of generators is invariant under a simultaneous similarity transformation $T^a \to U T^a U^{-1}$ of all factors, so the sum of the three variations of D^{abc} vanishes:

$$0 = \delta D^{abc} = -\theta^d \big(f^{dae} D^{ebc} + f^{dbe} D^{aec} + f^{dce} D^{abe}\big). \tag{B.124}$$

As θ^d is arbitrary, Eq. (B.122) follows. □

Theorem B.32 guarantees that $D^{abc}(R)$ lives in the space of totally symmetric invariant rank-three tensors. Once Section B.5.4 establishes that this space is at most one-dimensional, Eq. (B.121) is well posed and $A(R)$ is a genuine invariant of the representation.

B.5.2 Conjugation reverses the sign

The single most useful property of $A(R)$ for model building is that it changes sign under complex conjugation of the representation. This is what allows a vector-like pair $R \oplus \bar{R}$ to be automatically anomaly-free, and it is the engine behind every chiral cancellation in Chapter 37.

Theorem B.33 (Sign reversal under conjugation)**.** *For every representation R,*

$$A(\bar{R}) = -A(R). \tag{B.125}$$

Consequently every real or pseudoreal representation, for which $R \cong \bar{R}$, has $A(R) = 0$.

Proof. The conjugate representation acts by $T^a_{\bar{R}} = -(T^a_R)^{\mathsf{T}}$; this is the unique sign choice making the $T^a_{\bar{R}}$ Hermitian and satisfying the same commutators (transposition reverses the order in a product, and the extra minus restores $\left[T^a_{\bar{R}}, T^b_{\bar{R}}\right] = i f^{abc} T^c_{\bar{R}}$). Insert this into the symmetrized trace. For two matrices the transpose of an anticommutator is the anticommutator of the transposes, $\left\{M^{\mathsf{T}}, N^{\mathsf{T}}\right\} = (NM)^{\mathsf{T}} + (MN)^{\mathsf{T}} = \left(\{M, N\}\right)^{\mathsf{T}}$, so with $M = T^a_R$, $N = T^b_R$,

$$\begin{aligned} D^{abc}(\bar{R}) &= \operatorname{tr}\Big(\left\{-(T^a_R)^{\mathsf{T}}, -(T^b_R)^{\mathsf{T}}\right\} \left(-(T^c_R)^{\mathsf{T}}\right)\Big) = -\operatorname{tr}\Big(\left(\left\{T^a_R, T^b_R\right\}\right)^{\mathsf{T}} (T^c_R)^{\mathsf{T}}\Big) \\ &= -\operatorname{tr}\Big(\left(T^c_R \left\{T^a_R, T^b_R\right\}\right)^{\mathsf{T}}\Big) = -\operatorname{tr}\left(T^c_R \left\{T^a_R, T^b_R\right\}\right) = -D^{abc}(R), \end{aligned} \tag{B.126}$$

using $(XY)^{\mathsf{T}} = Y^{\mathsf{T}} X^{\mathsf{T}}$ in the third step and the trace's invariance under transposition in the fourth. Comparing with Eq. (B.121) gives $A(\bar{R}) = -A(R)$.

If R is real or pseudoreal then $R \cong \bar{R}$: there exists an invertible S with $T^a_{\bar{R}} = S\, T^a_R\, S^{-1}$, and the trace is unchanged by this similarity, so $A(\bar{R}) = A(R)$. Together with Eq. (B.125) this forces $2A(R) = 0$, hence $A(R) = 0$. □

In particular the fundamental and antifundamental of $\mathfrak{su}(N)$ carry opposite anomalies, $A(\bar{N}) = -A(N) = -1$. A Dirac fermion—a left-handed N together with a right-handed N, which is a left-handed $\bar{N}$—is automatically anomaly-free, $A(N) + A(\bar{N}) = 0$, as it must be: vector-like matter never generates a gauge anomaly.

B.5.3 The adjoint is anomaly-free

The adjoint representation is real (the structure constants are real and the generators $(T^a_{\text{adj}})_{bc} = -i f^{abc}$ are Hermitian and purely imaginary, hence equivalent to their conjugate), so Theorem B.33 already gives $A(\text{adj}) = 0$. It is worth seeing

the same result fall out directly from the antisymmetry of f, because the computation exhibits why the *symmetric* invariant must vanish on a representation built from f itself.

Proposition B.34 (Vanishing adjoint anomaly)**.** $A(\mathrm{adj}) = 0$ *for every simple Lie algebra.*

Proof. Write M^a for the adjoint generator regarded as a matrix, $(M^a)_{de} = -if^{ade}$. Because $f^{ade} = -f^{aed}$, each M^a is an antisymmetric matrix, $(M^a)^{\mathsf{T}} = -M^a$. Set $G^{abc} \equiv \mathrm{tr}(M^a M^b M^c)$. Transposing inside the trace and using antisymmetry of every factor,

$$G^{abc} = \mathrm{tr}\big((M^a M^b M^c)^{\mathsf{T}}\big) = \mathrm{tr}\big((M^c)^{\mathsf{T}}(M^b)^{\mathsf{T}}(M^a)^{\mathsf{T}}\big) = (-1)^3\,\mathrm{tr}(M^c M^b M^a) = -\,G^{cba}. \tag{B.127}$$

Thus G^{abc} is antisymmetric under the reversal $(a,b,c) \to (c,b,a)$; combined with the cyclicity $G^{abc} = G^{bca} = G^{cab}$, reversal is the same as the transposition $(a\,c)$ followed by a cycle, and one finds G^{abc} is *totally antisymmetric* in its three indices. Now

$$D^{abc}(\mathrm{adj}) = \mathrm{tr}(M^a M^b M^c) + \mathrm{tr}(M^b M^a M^c) = G^{abc} + G^{bac} = G^{abc} - G^{abc} = 0, \tag{B.128}$$

since exchanging $a \leftrightarrow b$ flips the sign of the totally antisymmetric G. Hence $A(\mathrm{adj}) = 0$. □

The computation also confirms the content of Eq. (B.118) in the adjoint: the product trace $\mathrm{tr}_{\mathrm{adj}}(T^a T^b T^c) = iG^{abc}$ is purely antisymmetric, proportional to f^{abc} alone, with no d^{abc} component. The adjoint "sees" only the antisymmetric invariant.

B.5.4 Which algebras carry a cubic invariant

We can now settle when $A(R)$ is even a meaningful quantity. The definition Eq. (B.121) presumes a nonzero symmetric tensor d^{abc} to be proportional to; if no totally symmetric invariant of rank three exists, then $D^{abc}(R)$ vanishes identically for every R and the theory is anomaly-free for any matter content whatsoever.

Theorem B.35 (Anomaly-free algebras)**.** *Among the compact simple Lie algebras, a nonzero totally symmetric invariant tensor d^{abc} exists* only *for $\mathfrak{su}(N)$ with $N \geq 3$. For every other simple algebra—$\mathfrak{su}(2)$, all $\mathfrak{so}(N)$, all $\mathrm{Sp}(N)$, and the exceptional algebras G_2, F_4, E_6, E_7, E_8—the cubic symmetric invariant vanishes, so $A(R) = 0$ for every representation and no perturbative gauge anomaly can arise.*

Proof for $\mathfrak{su}(2)$. The decisive case for physics is $\mathfrak{su}(2)$, where the vanishing is immediate. With $T^a = \sigma^a/2$ and the Pauli identity $\{\sigma^a, \sigma^b\} = 2\delta^{ab}\mathbf{1}$,

$$\{T^a, T^b\} = \tfrac{1}{2}\delta^{ab}\mathbf{1} \implies \operatorname{tr}_{\mathbf{2}}(\{T^a, T^b\} T^c) = \tfrac{1}{2}\delta^{ab} \operatorname{tr} T^c = 0. \tag{B.129}$$

The anticommutator is pure identity—there is no $d^{abc}T^c$ term, equivalently $d^{abc} \equiv 0$ for $\mathfrak{su}(2)$ (this is the $N = 2$ specialization of Eq. (B.120), where the symmetric symbol has no independent components). Since the cubic invariant vanishes in the defining representation it vanishes in all of them: every $\mathfrak{su}(2)$ representation is real or pseudoreal, so $A(R) = 0$ also follows from Theorem B.33. Either way, $\mathfrak{su}(2)$ gauge theory is automatically free of the cubic anomaly. □

Structural argument for the remaining algebras. The general statement is a fact about the ring of invariant polynomials. A rank-r compact simple Lie algebra has exactly r algebraically independent (primitive) Casimir invariants, of fixed degrees $m_1+1, \ldots, m_r+1$ where the m_i are the exponents of the algebra. Every symmetric invariant tensor is built, via the symmetrized trace, from these primitives; a totally symmetric invariant of rank three exists iff one of the primitive degrees equals 3. The degrees are: for $A_{N-1} = \mathfrak{su}(N)$, $\{2, 3, 4, \ldots, N\}$; for $B_r = \mathfrak{so}(2r+1)$ and $C_r = \mathrm{Sp}(2r)$, $\{2, 4, 6, \ldots, 2r\}$; for $D_r = \mathfrak{so}(2r)$, $\{2, 4, \ldots, 2r-2\}$ together with r; and for the exceptionals the degree sets begin $G_2 : \{2, 6\}$, $F_4 : \{2, 6, 8, 12\}$, $E_6 : \{2, 5, 6, 8, 9, 12\}$, $E_7 : \{2, 6, 8, 10, 12, 14, 18\}$, $E_8 : \{2, 8, 12, 14, 18, 20, 24, 30\}$. A primitive of degree 3 appears in exactly one family: the A series, and only from $A_2 = \mathfrak{su}(3)$ onward (the $A_1 = \mathfrak{su}(2)$ degree set is $\{2\}$, with no cubic). Every other algebra has its first invariant beyond the quadratic at degree 4 or higher, so its symmetric rank-three invariant tensor is identically zero, $d^{abc} = 0$, and Eq. (B.121) forces $A(R) = 0$ for all R. Reality of the relevant representations of $\mathfrak{so}(N)$ and $\mathrm{Sp}(2r)$ provides an independent confirmation through Theorem B.33. □

Physical Insight

The list in Theorem B.35 is exactly why the Standard Model gauge group is so forgiving. Of its three factors, $\mathfrak{su}(2)_L$ and $\mathrm{U}(1)_Y$-mixed-with-$\mathfrak{su}(2)$ pieces involving the $\mathfrak{su}(2)$ cubic vanish automatically, and the strong group $\mathfrak{su}(3)_c$ couples vector-like (left and right quarks in the same color $\mathbf{3}$), so its cubic anomaly cancels representation by representation. The only cubic anomalies that must be arranged to cancel nontrivially are the $\mathrm{U}(1)_Y^3$ and mixed $\mathfrak{su}(2)_L^2\,\mathrm{U}(1)_Y$, $\mathfrak{su}(3)_c^2\,\mathrm{U}(1)_Y$, and gravitational $\mathrm{U}(1)_Y$ conditions—the celebrated hypercharge sum rules of Chapter 37. Every $\mathfrak{su}(N)$ with $N \geq 3$, by contrast, can support a genuine chiral anomaly, which is why $\mathfrak{su}(5)$ grand unification must be checked explicitly (Theorem B.38).

B.5.5 Computing $A(\mathbf{fund})$ and the additive structure

The normalization $A(\mathrm{fund}) = 1$ was built into Theorem B.31 by choosing d^{abc} to be the fundamental symbol. It is reassuring to see it emerge from the explicit $\mathfrak{su}(3)$ data. With $T^a = \lambda^a/2$ and the Gell-Mann anticommutator $\{\lambda^a, \lambda^b\} = \frac{4}{3}\delta^{ab}\mathbf{1} + 2d^{abc}\lambda^c$ (Section B.2.4),

$$\mathrm{tr}_{\mathbf{3}}\big(\{T^a, T^b\}\, T^c\big) = \tfrac{1}{4}\,\mathrm{tr}\big(\{\lambda^a, \lambda^b\}\,\tfrac{\lambda^c}{2}\big) = \tfrac{1}{4}\Big(\tfrac{4}{3}\delta^{ab}\underbrace{\mathrm{tr}\,\tfrac{\lambda^c}{2}}_{0} + 2d^{abe}\underbrace{\mathrm{tr}\,\tfrac{\lambda^e\lambda^c}{2}}_{\delta^{ec}}\Big) = \tfrac{1}{2}d^{abc}, \tag{B.130}$$

so $A(\mathbf{3}) = 1$ by Eq. (B.121), and $A(\bar{\mathbf{3}}) = -1$ by Theorem B.33. For general $\mathfrak{su}(N)$ the same computation using Eq. (B.120) gives $\mathrm{tr}_{\mathrm{fund}}(\{T^a, T^b\}\, T^c) = \frac{1}{2}d^{abc}$ identically, confirming $A(\mathrm{fund}) = 1$ for every $N \geq 3$.

To use $A(R)$ on composite representations we need its behavior under direct sums and tensor products.

Proposition B.36 (Additivity)**.** *For representations R_1, R_2,*

$$A(R_1 \oplus R_2) = A(R_1) + A(R_2), \tag{B.131}$$

$$A(R_1 \otimes R_2) = \dim(R_2)\, A(R_1) + \dim(R_1)\, A(R_2). \tag{B.132}$$

Proof. The direct-sum rule is immediate: in $R_1 \oplus R_2$ the generators are block-diagonal, $T^a = T^a_{R_1} \oplus T^a_{R_2}$, and the trace of any product splits as a sum over blocks, so $D^{abc}(R_1 \oplus R_2) = D^{abc}(R_1) + D^{abc}(R_2)$, giving Eq. (B.131). The tensor-product rule is the worked computation Theorem B.37. □

Worked Example B.37 (The tensor-product anomaly rule)**.** On $V_1 \otimes V_2$ the generators act as $T^a = A^a \otimes \mathbf{1} + \mathbf{1} \otimes B^a$, with A^a the generators on V_1 (dimension d_1, anomaly A_1) and B^a those on V_2 (dimension d_2, anomaly A_2). Expanding the product,

$$T^a T^b = A^a A^b \otimes \mathbf{1} + A^a \otimes B^b + A^b \otimes B^a + \mathbf{1} \otimes B^a B^b, \tag{B.133}$$

so the anticommutator is

$$\{T^a, T^b\} = \{A^a, A^b\} \otimes \mathbf{1} + \mathbf{1} \otimes \{B^a, B^b\} + 2(A^a \otimes B^b + A^b \otimes B^a). \tag{B.134}$$

Multiply by $T^c = A^c \otimes \mathbf{1} + \mathbf{1} \otimes B^c$ and take the trace, using $\mathrm{tr}_{V_1 \otimes V_2}(X \otimes Y) = \mathrm{tr}\, X\ \mathrm{tr}\, Y$ and the tracelessness $\mathrm{tr}\, A^a = \mathrm{tr}\, B^a = 0$. Every term in the cross piece $2(A^a \otimes B^b + A^b \otimes B^a)\, T^c$ produces a single-generator trace and vanishes, as does the cross term from the identity blocks ($\mathrm{tr}\,\{A^a, A^b\}\ \mathrm{tr}\, B^c \propto \delta^{ab}\,\mathrm{tr}\, B^c = 0$). The survivors are

$$\mathrm{tr}\big(\{T^a, T^b\}\, T^c\big) = \mathrm{tr}_{V_1}\big(\{A^a, A^b\}\, A^c\big)\ \mathrm{tr}_{V_2}\,\mathbf{1} + \mathrm{tr}_{V_1}\,\mathbf{1}\ \mathrm{tr}_{V_2}\big(\{B^a, B^b\}\, B^c\big)$$

$$= d_2 \cdot \tfrac{1}{2} A_1\, d^{abc} + d_1 \cdot \tfrac{1}{2} A_2\, d^{abc} = \tfrac{1}{2}(d_2 A_1 + d_1 A_2) d^{abc}, \quad \text{(B.135)}$$

which is Eq. (B.132). The structure is intuitive: each factor contributes its own anomaly weighted by the number of "spectator" states dim of the other factor.

The sum and product rules let us read off the anomaly of any representation appearing in a tensor power of the fundamental. The two-index tensors are the immediate application, and they decide the consistency of the simplest grand unified theory.

Worked Example B.38 (Anomalies of the $\mathfrak{su}(N)$ two-index tensors and $\mathfrak{su}(5)$ unification)**.** The product of two fundamentals decomposes as $N \otimes N = \mathrm{Sym}^2 \oplus \Lambda^2$, the symmetric and antisymmetric two-index representations of dimensions $\frac{N(N+1)}{2}$ and $\frac{N(N-1)}{2}$. By Eq. (B.132) with $R_1 = R_2 = \text{fund}$,

$$A(\mathrm{Sym}^2) + A(\Lambda^2) = A(N \otimes N) = N \cdot 1 + N \cdot 1 = 2N. \quad \text{(B.136)}$$

A second relation comes from the swap operator P on $V \otimes V$, which is $+1$ on Sym^2 and -1 on Λ^2, so that $A(\mathrm{Sym}^2) - A(\Lambda^2) = \frac{2}{d^{abc}} \operatorname{tr}\big(P \left\{T^a, T^b\right\} T^c\big)$. The trace of P against a tensor-product operator collapses to a single-factor trace, $\operatorname{tr}_{V\otimes V}\big(P\,(X \otimes Y)\big) = \operatorname{tr}(XY)$, which follows from $P_{ij,kl} = \delta_{il}\delta_{jk}$:

$$\operatorname{tr}_{V\otimes V}\big(P\,(X\otimes Y)\big) = \sum_{ij} \big[P(X\otimes Y)\big]_{ij,ij} = \sum_{ijkl} \delta_{il}\delta_{jk} X_{ki} Y_{lj} = \sum_{ij} X_{ji} Y_{ij} = \operatorname{tr}(XY). \quad \text{(B.137)}$$

Applying this term by term to $\left\{T^a, T^b\right\} T^c$ (now in terms of the single-factor generators A^a of the N) and using $\operatorname{tr}(A^a \left\{A^b, A^c\right\}) = \frac{1}{2} d^{abc}$ throughout, the identity-block pieces each give $\frac{1}{2}d^{abc} + \frac{1}{2}d^{abc}$, while the cross piece $2(A^a \otimes A^b + A^b \otimes A^a)T^c$ now *survives* the swap and contributes $4 \cdot \frac{1}{2} d^{abc} = 2d^{abc}$. Summing, $\operatorname{tr}(P \left\{T^a, T^b\right\} T^c) = (1 + 1 + 2) d^{abc} = 4 d^{abc}$, hence

$$A(\mathrm{Sym}^2) - A(\Lambda^2) = 8. \quad \text{(B.138)}$$

Solving the two relations,

$$A(\mathrm{Sym}^2) = N + 4, \qquad A(\Lambda^2) = N - 4. \quad \text{(B.139)}$$

For $\mathfrak{su}(5)$ the antisymmetric $\Lambda^2 = \mathbf{10}$ therefore has $A(\mathbf{10}) = 5 - 4 = 1$, while $A(\overline{\mathbf{5}}) = -A(\mathbf{5}) = -1$. One generation of the Georgi–Glashow model places its left-handed fermions in $\overline{\mathbf{5}} \oplus \mathbf{10}$, with total anomaly

$$A(\overline{\mathbf{5}}) + A(\mathbf{10}) = -1 + 1 = 0. \quad \text{(B.140)}$$

The cancellation is exact, generation by generation: $\mathfrak{su}(5)$ unification is anomaly-free precisely because the cubic anomaly of the antisymmetric $\mathbf{10}$ is the negative of that of the $\overline{\mathbf{5}}$. This is the group-theory content imported wholesale by Chapter 37.

B.5.6 Higher invariants and where the ladder stops

The quadratic index $T(R)$ and the cubic coefficient $A(R)$ are the first two rungs of an infinite ladder of symmetrized-trace invariants

$$d^{(n)\,a_1\cdots a_n}(R) \;=\; \mathrm{Str}_R\big(T^{a_1}\cdots T^{a_n}\big) \;\equiv\; \frac{1}{n!}\sum_{\sigma\in S_n}\mathrm{tr}_R\big(T^{a_{\sigma(1)}}\cdots T^{a_{\sigma(n)}}\big), \quad \text{(B.141)}$$

with $d^{(2)} \propto T(R)\,\delta^{ab}$ and $d^{(3)} \propto A(R)\,d^{abc}$. Two facts govern how far this is useful. First, only the primitive degrees of the algebra (Theorem B.35) carry *new* information; symmetrized traces at composite degrees reduce, by Newton's identities and the Cayley–Hamilton theorem, to polynomials in the primitive ones. For $\mathfrak{su}(N)$ the primitives sit at degrees $2, 3, \ldots, N$, so an independent quartic invariant first appears at $\mathfrak{su}(4)$, an independent quintic at $\mathfrak{su}(5)$, and the tower of *algebraically independent* symmetric invariants terminates at degree N; everything higher is expressible through the lower ones. Second, in four-dimensional gauge theory only $T(R)$ and $A(R)$ enter the renormalization and consistency conditions: the quadratic index controls the one-loop β-function (Section B.4), and the cubic coefficient controls the chiral anomaly (Chapter 36). The quartic Casimir—encoded in a fourth-rank symmetric tensor $d^{(4)abcd}$ and an associated quartic index—becomes load-bearing only beyond this setting: for the box (square) anomaly in six dimensions, for certain global anomalies, and as the distinguishing invariant of $\mathfrak{so}(N)$ and the exceptional groups, which possess no cubic at all. The data this book needs from the cubic rung is now complete: the symbol d^{abc}, the coefficient $A(R)$, its sign reversal, and the table of values $A(\text{fund}) = 1$, $A(\text{adj}) = 0$, $A(\Lambda^2) = N - 4$, $A(\text{Sym}^2) = N + 4$.

With the three low-order invariants $T(R)$, $C_2(R)$, and $A(R)$ now all in hand and computed in representative cases, what remains is to assemble them into a single reference. The next section fixes the Cartan–Weyl language—roots, weights, and Dynkin labels—needed to compute these entries for an *arbitrary* representation of any compact simple algebra, and gathers the results into the book-wide master table of $\dim(R)$, $T(R)$, $C_2(R)$, and $A(R)$ that the gauge-theory chapters cite.

B.6 Roots, Weights, and a Master Table of Group-Theory Data

The previous sections of this appendix built, one invariant at a time, the complete set of numbers a gauge-theory computation actually consumes: the Dynkin index $T(R)$ that controls every coupling renormalization, the quadratic Casimir $C_2(R)$ that sets the size of self-energy and Sudakov logarithms, and the cubic anomaly coefficient $A(R)$ that decides whether a chiral theory is consistent at all. For the unitary groups these were obtained in closed form; for the representations that

recur throughout the book they were tabulated as they arose. What remains is to do two things at once. First, to record the *Cartan–Weyl* bookkeeping—roots, weights, Dynkin labels, the A–G classification—in enough depth that a reader can locate *any* compact simple algebra in the standard list and read off the structural data that the index and Casimir formulas need, even for a group this book never uses explicitly. Second, to demonstrate that the table is not a list of memorized constants but a computable object: we take the imported classification of Theorem 2.75 together with the Weyl and Freudenthal machinery of Theorem 2.74, apply it by hand to the smallest nontrivial case, $\mathfrak{su}(3)$, and watch it reproduce the very values $T_F = \frac{1}{2}$, $C_2(F) = \frac{4}{3}$, $C_A = 3$ that Sections 2 and 4 obtained by direct trace. With that confidence established we assemble the master table that the gauge, anomaly, and grand-unification chapters cite by label: dim, T, C_2, and A for SU(N), SO(N), Sp($2n$), and the grand-unified representations of SU(5) and SO(10), every entry carrying a self-contained derivation.

This section is the appendix's closing service. It owns no new physics; it owns the *reference* object. The conceptual content—what a root system is, why the classification is what it is, the Weyl dimension formula—was established in Chapter 2 and is imported here as a tool, not reproved. The invariant definitions and the SU(N) results are imported from Sections 1–5 of this appendix. The new work here is the consolidation, the worked computability check, and the table.

B.6.1 The Cartan–Weyl picture, at reference depth

We recall, in compressed form, the language fixed in Section 2.6; the reader who wants the constructions and proofs should consult that section, while the reader who wants only to use the table needs the vocabulary collected here.

Let $\mathfrak{g}$ be a compact simple Lie algebra of dimension $\dim \mathfrak{g}$, which we always take with Hermitian generators T^a obeying $\left[T^a, T^b\right] = i f^{abc} T^c$. A *Cartan subalgebra* $\mathfrak{h} \subset \mathfrak{g}$ is a maximal set of mutually commuting generators; its dimension is the *rank* r of $\mathfrak{g}$ and is an invariant of the algebra. Diagonalizing the adjoint action of $\mathfrak{h}$ decomposes the complexified algebra as

$$\mathfrak{g}_{\mathbb{C}} = \mathfrak{h}_{\mathbb{C}} \oplus \bigoplus_{\alpha \in \Delta} \mathfrak{g}_\alpha, \qquad [H, E_\alpha] = \alpha(H)\, E_\alpha \quad \text{for } H \in \mathfrak{h}, \tag{B.142}$$

where each *root space* $\mathfrak{g}_\alpha$ is one-dimensional and the nonzero linear functionals $\alpha \in \mathfrak{h}^*$ are the *roots*. The set Δ of roots is the *root system*. The Killing form, restricted to $\mathfrak{h}^*$, gives a positive-definite inner product $\langle \cdot, \cdot \rangle$ in which roots have definite lengths; we write $\alpha^\vee \equiv 2\alpha/\langle \alpha, \alpha \rangle$ for the *coroot*.

Choosing a hyperplane that misses every root splits Δ into positive and negative roots, $\Delta = \Delta_+ \cup (-\Delta_+)$. The positive roots that cannot be written as a

sum of two positive roots are the r *simple roots* $\alpha_1, \dots, \alpha_r$; they form a basis of $\mathfrak{h}^*$, and every root is an integer combination of them with coefficients all of one sign. The entire root system—hence, by the classification theorem, the algebra itself—is encoded in the *Cartan matrix*

$$A_{ij} = \langle \alpha_i, \alpha_j^\vee \rangle = \frac{2\langle \alpha_i, \alpha_j \rangle}{\langle \alpha_j, \alpha_j \rangle}, \tag{B.143}$$

whose diagonal entries are 2 and whose off-diagonal entries are non-positive integers with $A_{ij}A_{ji} \in \{0, 1, 2, 3\}$. The Cartan matrix is drawn as a *Dynkin diagram*: one node per simple root, with $A_{ij}A_{ji}$ bonds between nodes i and j (so $0, 1, 2, 3$ bonds mean orthogonal, 120°, 135°, 150° between the roots), and an arrow pointing from the long root to the short one when the two have different lengths. A simple algebra has a *connected* Dynkin diagram; a direct sum of simple algebras has a disconnected one. The possible connected diagrams are exactly the content of the next subsection.

B.6.2 The Cartan classification and its physics dictionary

The classification theorem (Theorem 2.75) states that every compact simple Lie algebra is, up to isomorphism, one of four infinite families A_n, B_n, C_n, D_n or one of five exceptional algebras G_2, F_4, E_6, E_7, E_8. The subscript is the rank. Figure B.4 draws the nine connected Dynkin diagrams; Table B.4 gives the dictionary to the matrix groups physics uses and records the dimension, which is what the master relation $T \dim \mathfrak{g} = C_2 \dim R$ of Section B.6.3 needs.

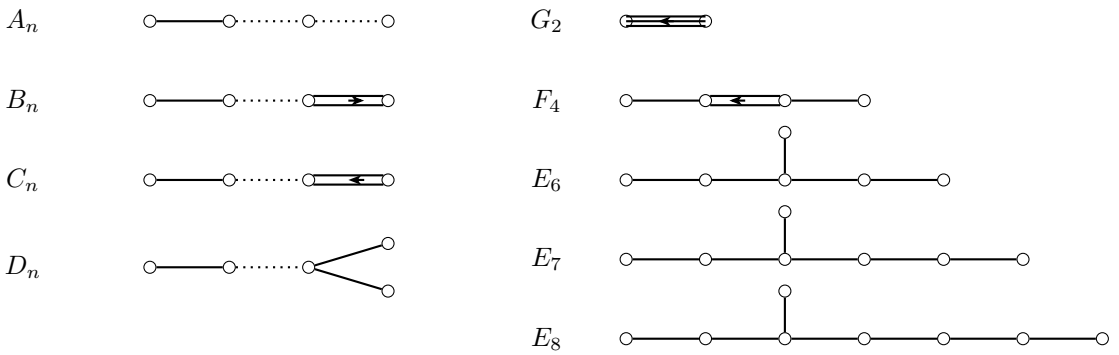


Figure B.4: The connected Dynkin diagrams: the four infinite families and the five exceptional algebras. A single bond is a 120° angle between simple roots, a double bond 135°, a triple bond 150°; the arrow points from the long root to the short one. Dotted segments indicate the omitted middle of a chain of n nodes. The branch node of D_n and the E series is what makes those algebras admit (for D_n, chiral; for E_6, complex) spinorial representations.

Table B.4: The Cartan classification and its physics dictionary. "Group" is the compact connected group whose Lie algebra is the listed algebra; Spin denotes the simply connected double cover of SO(), and $\mathrm{USp}(2n) = \mathrm{Sp}(2n)$ the compact symplectic group of rank n. Dimensions are $\dim \mathfrak{g} = r + |\Delta|$.

Type	Rank	Algebra	Group	$\dim \mathfrak{g}$
A_n	n	$\mathfrak{su}(n+1)$	$\mathrm{SU}(n+1)$	$n(n+2)$
B_n	n	$\mathfrak{so}(2n+1)$	$\mathrm{Spin}(2n+1)$	$n(2n+1)$
C_n	n	$\mathfrak{sp}(2n)$	$\mathrm{Sp}(2n) = \mathrm{USp}(2n)$	$n(2n+1)$
D_n	n	$\mathfrak{so}(2n)$	$\mathrm{Spin}(2n)$	$n(2n-1)$
G_2	2	$\mathfrak{g}_2$	G_2	14
F_4	4	$\mathfrak{f}_4$	F_4	52
E_6	6	$\mathfrak{e}_6$	E_6	78
E_7	7	$\mathfrak{e}_7$	E_7	133
E_8	8	$\mathfrak{e}_8$	E_8	248

The infinite families exhaust the classical matrix groups: A_n is the special unitary algebra, B_n and D_n the orthogonal algebras in odd and even dimension, and C_n the symplectic algebra. The exceptional algebras have no such elementary matrix description; G_2 is the automorphism algebra of the octonions and F_4, E_6, E_7, E_8 are built on the exceptional Jordan algebra, but for our purposes they enter only as candidate grand-unified groups, E_6 being the one of direct phenomenological interest because it admits complex representations.

Two cautions about small rank are worth stating once, because they underlie several isomorphisms physics exploits. The classification is non-redundant only for A_n $(n \geq 1)$, B_n $(n \geq 2)$, C_n $(n \geq 3)$, D_n $(n \geq 4)$; outside that range the diagrams coincide and so do the algebras. The accidental isomorphisms are

$$A_1 \cong B_1 \cong C_1, \quad B_2 \cong C_2, \quad D_2 \cong A_1 \oplus A_1, \quad A_3 \cong D_3, \tag{B.144}$$

i.e. $\mathfrak{su}(2) \cong \mathfrak{so}(3) \cong \mathfrak{sp}(2)$, $\mathfrak{so}(5) \cong \mathfrak{sp}(4)$, $\mathfrak{so}(4) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, and $\mathfrak{su}(4) \cong \mathfrak{so}(6)$. The last is the Pati–Salam identity used in unification; D_2 being non-simple is why SO(4) splits into two independent SU(2) factors. We will use $\mathfrak{su}(4) \cong \mathfrak{so}(6)$ as an internal consistency check on the master table.

Caution

The notation "Sp()" is overloaded in the literature: some authors write $\mathrm{Sp}(n)$ for the rank-n compact symplectic group (our $\mathrm{Sp}(2n)$), others write $\mathrm{Sp}(2n)$ for the $2n \times 2n$ symplectic group over $\mathbb{R}$ or $\mathbb{C}$, which is *non-compact*. Throughout this book $\mathrm{Sp}(2n)$ denotes the *compact* group $\mathrm{USp}(2n)$ of rank n, with defining representation of dimension $2n$; this is the C_n algebra of Table B.4.

B.6.3 Weights, Dynkin labels, and the invariant formulas

A finite-dimensional irreducible representation R of $\mathfrak{g}$ is fixed by its weights—the simultaneous eigenvalues of the Cartan generators on the basis states—and in particular by its unique *highest weight* λ, the one not raised by any E_α with $\alpha \in \Delta_+$. The fundamental weights $\omega_1, \dots, \omega_r$ are the basis of $\mathfrak{h}^*$ dual to the coroots,

$$\langle \omega_i, \alpha_j^\vee \rangle = \delta_{ij}, \tag{B.145}$$

and every dominant highest weight is a non-negative integer combination $\lambda = \sum_i a_i\, \omega_i$. The integers $a_i \geq 0$ are the *Dynkin labels*, written $[a_1, \dots, a_r]$; they are the universal address of a representation, independent of any matrix realization. The defining representation of $\mathrm{SU}(N)$ is $[1, 0, \dots, 0]$; its conjugate is $[0, \dots, 0, 1]$; the adjoint of $\mathrm{SU}(N \geq 3)$ is $[1, 0, \dots, 0, 1]$. A representation is real, pseudoreal, or complex according to whether the diagram symmetry that conjugates it acts trivially or not, which is exactly the structure that decides the sign behaviour of $A(R)$ recorded in Section 5.

The half-sum of the positive roots,

$$\rho \equiv \tfrac{1}{2} \sum_{\alpha \in \Delta_+} \alpha = \sum_{i=1}^{r} \omega_i, \tag{B.146}$$

the *Weyl vector*, has all Dynkin labels equal to one and organizes both formulas we now import. The first is the *Weyl dimension formula* for the representation of highest weight λ,

$$\dim R_\lambda = \prod_{\alpha \in \Delta_+} \frac{\langle \lambda + \rho, \alpha \rangle}{\langle \rho, \alpha \rangle}, \tag{B.147}$$

stated and used in Theorem 2.74 and imported here as a tool. The second is the eigenvalue of the quadratic Casimir, which on R_λ acts as a multiple of the identity (Schur), with multiple

$$C_2(\lambda) = \langle \lambda, \lambda + 2\rho \rangle. \tag{B.148}$$

Equation (B.148) is stated in the *Cartan normalization*, in which the long roots have $\langle \alpha, \alpha \rangle = 2$. The physical normalization of this book is the one fixed in Sections 2 and 4, $\mathrm{tr}_F(T^a T^b) = \frac{1}{2}\delta^{ab}$ in the defining representation; the two differ by an overall positive constant $\kappa_{\mathfrak{g}}$ that depends only on the algebra, so that $C_2^{\mathrm{phys}}(\lambda) = \kappa_{\mathfrak{g}}\, \langle \lambda, \lambda + 2\rho \rangle$. For $\mathfrak{su}(N)$ this constant is $\kappa = \frac{1}{2}$, as the worked computation below makes explicit; quoting C_2 in the physical normalization is what makes the master table directly usable in a Feynman-diagram calculation.

These two imported formulas are tied to the Section 2–5 invariants by a single identity, the *index–Casimir relation*, which we restate with its one-line proof because every table entry below is generated from it. Taking the trace of the defining relation $T^a T^a = C_2(R)\, \mathbf{1}_R$ gives $\mathrm{tr}_R(T^a T^a) = C_2(R) \dim R$, while the

definition $\mathrm{tr}_R(T^aT^b) = T(R)\delta^{ab}$ gives the same trace as $T(R)\,\delta^{aa} = T(R)\dim\mathfrak{g}$. Equating,

$$T(R)\ \dim\mathfrak{g}\ =\ C_2(R)\ \dim R. \tag{B.149}$$

Thus any one of T and C_2 determines the other once $\dim R$ and $\dim\mathfrak{g}$ are known. There is, finally, a direct route to $T(R)$ from the weights alone that we will use as the algebraic engine of the table. Choose an orthonormal basis $\{H_i\}_{i=1}^{r}$ of the Cartan subalgebra in the physical normalization, $\mathrm{tr}_F(H_iH_j) = \frac{1}{2}\delta_{ij}$. The eigenvalue of H_i on a weight state $|\mu\rangle$ is the i-th component μ_i, so $\mathrm{tr}_R(H_iH_j) = \sum_{\mu\in R}\mu_i\mu_j = T(R)\delta_{ij}$, where the sum runs over all weights with multiplicity. Taking the trace over $i = j$,

$$T(R) = \frac{1}{r}\sum_{\mu\in R}|\mu|^2, \tag{B.150}$$

with $|\mu|^2$ the squared length of the weight in the orthonormal (physical) basis. Equation (B.150) reduces the index of any representation to a finite sum over its weight diagram, and it is normalization-honest: because the same H_i that define the defining-representation normalization are used in R, the result comes out directly in the convention $T_F = \frac{1}{2}$.

B.6.4 Worked case: $\mathfrak{su}(3)$ from its weights

We now carry out, in full, the computation that the rest of the table only summarizes: starting from the explicit roots and weights of $\mathfrak{su}(3)$, we recover T_F, $C_2(F)$, T_A, and C_A and check them against Eq. (B.147), Eq. (B.148), and the SU(N) formulas of Section 4. This is the demonstration of computability promised in the section goal.

Worked Example B.39 (Index and Casimir of the $\mathfrak{su}(3)$ fundamental and adjoint from the weights)**.** The algebra $\mathfrak{su}(3)$ (A_2) has rank $r = 2$ and Cartan matrix $A = \left(\begin{smallmatrix} 2 & -1 \\ -1 & 2 \end{smallmatrix}\right)$; all roots have the same length, so it is simply laced and $\alpha_i^\vee = \alpha_i$ in the Cartan normalization $\langle\alpha_i,\alpha_i\rangle = 2$.

Explicit physical realization. Use the two diagonal Gell-Mann generators of Section 5,

$$H_1 = T^3 = \tfrac{1}{2}\,\mathrm{diag}(1,-1,0), \qquad H_2 = T^8 = \tfrac{1}{2\sqrt{3}}\,\mathrm{diag}(1,1,-2), \tag{B.151}$$

which satisfy $\mathrm{tr}_F(H_iH_j) = \frac{1}{2}\delta_{ij}$: indeed $\mathrm{tr}_F\,H_1^2 = \frac{1}{4}(1+1) = \frac{1}{2}$, $\mathrm{tr}_F\,H_2^2 = \frac{1}{12}(1+1+4) = \frac{1}{2}$, and $\mathrm{tr}_F(H_1H_2) \propto \mathrm{tr}\,\mathrm{diag}(1,-1,0) = 0$. This is the orthonormal physical basis demanded by Eq. (B.150).

Weights of the fundamental $\mathbf{3}$. Reading the diagonal entries of Eq. (B.151) as the pairs $(\mu_1,\mu_2) = ((H_1)_{kk},(H_2)_{kk})$, the three weights are

$$\mu^{(1)} = \left(\tfrac{1}{2},\tfrac{1}{2\sqrt{3}}\right), \quad \mu^{(2)} = \left(-\tfrac{1}{2},\tfrac{1}{2\sqrt{3}}\right), \quad \mu^{(3)} = \left(0,-\tfrac{1}{\sqrt{3}}\right). \tag{B.152}$$

Each has the same length squared, $|\mu^{(1)}|^2 = \frac{1}{4} + \frac{1}{12} = \frac{1}{3}$, and likewise $|\mu^{(2)}|^2 = |\mu^{(3)}|^2 = \frac{1}{3}$, so $\sum_\mu |\mu|^2 = 3 \cdot \frac{1}{3} = 1$. Equation (B.150) gives immediately

$$T(F) = \frac{1}{r} \sum_{\mu} |\mu|^2 = \frac{1}{2} \cdot 1 = \frac{1}{2}, \tag{B.153}$$

and then Eq. (B.149) with $\dim \mathfrak{g} = 8$, $\dim F = 3$ gives

$$C_2(F) = T(F) \, \frac{\dim \mathfrak{g}}{\dim F} = \frac{1}{2} \cdot \frac{8}{3} = \frac{4}{3}. \tag{B.154}$$

These are exactly the Section 4 values $T_F = \frac{1}{2}$ and $C_F = (N^2-1)/(2N) = 8/6 = \frac{4}{3}$ at $N = 3$.

Weights of the adjoint **8**. The weights of the adjoint are the six roots together with the two zero weights of the Cartan subalgebra. The roots are the differences $\mu^{(i)} - \mu^{(j)}$ of fundamental weights:

$$\pm(1,0), \quad \pm\Big(\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}\Big), \quad \pm\Big(\tfrac{1}{2}, -\tfrac{\sqrt{3}}{2}\Big), \tag{B.155}$$

each of length squared $|\alpha|^2 = 1$, plus the two zero weights. Hence $\sum_\mu |\mu|^2 = 6 \cdot 1 + 2 \cdot 0 = 6$ and

$$T(A) = \frac{1}{2} \cdot 6 = 3, \qquad C_2(A) = T(A) \, \frac{\dim \mathfrak{g}}{\dim A} = 3 \cdot \frac{8}{8} = 3, \tag{B.156}$$

matching $C_A = N = 3$ and $T_A = N = 3$ from Section 4.

Cross-check against the imported formulas. In the Cartan normalization the fundamental and adjoint of A_2 have highest weights $\lambda_F = \omega_1 = [1,0]$ and $\lambda_A = \omega_1 + \omega_2 = \rho = [1,1]$. From the inverse Cartan matrix $A^{-1} = \frac{1}{3}\left(\begin{smallmatrix} 2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$ one reads $\langle \omega_i, \omega_j \rangle = (A^{-1})_{ij}$, so $\langle \omega_1, \omega_1 \rangle = \frac{2}{3}$, $\langle \omega_1, \omega_2 \rangle = \frac{1}{3}$, and $\langle \rho, \rho \rangle = \frac{2}{3} + \frac{2}{3} + \frac{2}{3} = 2$. The Weyl dimension formula (B.147) over the positive roots $\alpha_1, \alpha_2, \alpha_1 + \alpha_2$, using $\langle \omega_i, \alpha_j \rangle = \delta_{ij}$ and $\langle \rho, \alpha \rangle = 1, 1, 2$, gives for the fundamental $\dim = \frac{2 \cdot 1 \cdot 3}{1 \cdot 1 \cdot 2} = 3$ and for the adjoint $\dim = \frac{2 \cdot 2 \cdot 4}{1 \cdot 1 \cdot 2} = 8$, as required. The Casimir formula (B.148) gives, in Cartan normalization,

$$C_2^{\mathrm{Cartan}}(F) = \langle \omega_1, \omega_1 + 2\rho \rangle = \tfrac{2}{3} + 2 \cdot 1 = \tfrac{8}{3}, \qquad C_2^{\mathrm{Cartan}}(A) = \langle \rho, 3\rho \rangle = 3\langle \rho, \rho \rangle = 6. \tag{B.157}$$

These exceed the physical values (B.154) and (B.156) by exactly the factor $\kappa_{\mathfrak{su}(N)} = \frac{1}{2}$ anticipated below Eq. (B.148): physically the roots have $|\alpha|^2 = 1$, half the Cartan value 2, and C_2 scales with the metric. The independent routes agree, which is the point of the exercise: the imported Weyl/Freudenthal apparatus and the trace definitions describe one and the same numbers.

Figure B.5 draws the data of Theorem B.39: the hexagon of roots that is the nonzero part of the adjoint weight diagram, and the inner triangle of fundamental weights.

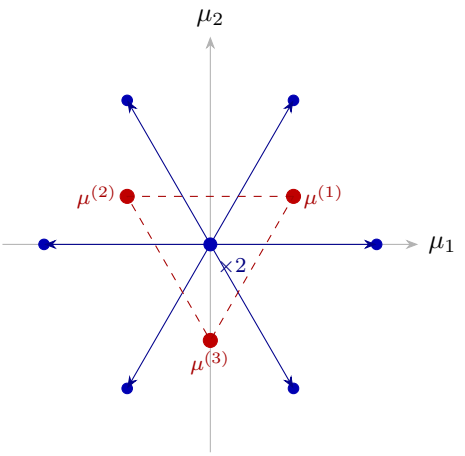


Figure B.5: The $\mathfrak{su}(3)$ weight data underlying Theorem B.39. Blue: the six roots (length 1 in the physical normalization) plus the doubled zero weight, which together are the eight weights of the adjoint **8**. Red: the three weights $\mu^{(1,2,3)}$ of the fundamental **3**, an inner triangle. The index formula (B.150) is the sum of squared lengths of these dots, divided by the rank.

B.6.5 The two-index tensors of SU(N) from a single generator

Before assembling the table we need the two-index symmetric and antisymmetric representations of SU(N), since the grand-unified **10** of SU(5) is the antisymmetric one. Their indices follow from one clean computation that uses only a diagonal generator and the decomposition of the tensor square, so no value is imported on faith.

Let $H = \mathrm{diag}(h_1, \ldots, h_N)$ be a traceless diagonal generator of $\mathfrak{su}(N)$, acting on the defining representation V with $\mathrm{tr}_V\, H^2 = \sum_i h_i^2$ and, because $\mathfrak{su}(N)$ has no diagonal trace, $\sum_i h_i = 0$. On the tensor square $V \otimes V$ the generator acts as $H_\otimes = H \otimes \mathbf{1} + \mathbf{1} \otimes H$ with eigenvalue $h_i + h_j$ on the basis state $|ij\rangle$. Squaring and tracing, and using $\mathrm{tr}\, H = 0$,

$$\mathrm{tr}_{V\otimes V}\, H_\otimes^2 = \mathrm{tr}(H^2 \otimes \mathbf{1}) + 2\ \mathrm{tr}\, H\ \mathrm{tr}\, H + \mathrm{tr}(\mathbf{1} \otimes H^2) = 2N\ \mathrm{tr}_V\, H^2. \tag{B.158}$$

The tensor square splits as $V \otimes V = \mathrm{Sym}^2 \oplus \Lambda^2$, so $\mathrm{tr}_{\mathrm{Sym}^2}\, H_\otimes^2 + \mathrm{tr}_{\Lambda^2}\, H_\otimes^2 = 2N\ \mathrm{tr}_V\, H^2$. Their difference is the contribution of the diagonal states $|ii\rangle$, which live in Sym^2 only and carry eigenvalue $2h_i$:

$$\mathrm{tr}_{\mathrm{Sym}^2}\, H_\otimes^2 - \mathrm{tr}_{\Lambda^2}\, H_\otimes^2 = \sum_i (2h_i)^2 = 4\ \mathrm{tr}_V\, H^2. \tag{B.159}$$

Solving Eqs. (B.158) and (B.159) and dividing by the defining-rep value ($\mathrm{tr}_V\, H^2 = T_F = \frac{1}{2}$ for the single Cartan generator H, i.e. the index is the trace ratio times

$T_F = \frac{1}{2}$),

$$T(\mathrm{Sym}^2) = \frac{N+2}{2}, \qquad T(\Lambda^2) = \frac{N-2}{2}. \tag{B.160}$$

Equation (B.149) with $\dim \mathrm{Sym}^2 = \frac{N(N+1)}{2}$, $\dim \Lambda^2 = \frac{N(N-1)}{2}$, and $\dim \mathfrak{g} = N^2 - 1$ then yields

$$C_2(\mathrm{Sym}^2) = \frac{(N+2)(N-1)}{N}, \qquad C_2(\Lambda^2) = \frac{(N-2)(N+1)}{N}. \tag{B.161}$$

The same diagonal-generator method delivers the anomaly coefficients of Section 5. The cubic invariant in a Cartan direction is $\propto \mathrm{tr}_R\, H^3$, so the cubic charge adds across a tensor product exactly as the quadratic one did: from $(H\otimes\mathbf{1}+\mathbf{1}\otimes H)^3$ and $\mathrm{tr}\, H = 0$, $\mathrm{tr}_{V\otimes V}\, H^3_{\otimes} = 2N\ \mathrm{tr}_V\, H^3$, hence $A(\mathrm{Sym}^2) + A(\Lambda^2) = 2N\, A(F) = 2N$, while the diagonal states give $\mathrm{tr}_{\mathrm{Sym}^2}\, H^3_{\otimes} - \mathrm{tr}_{\Lambda^2}\, H^3_{\otimes} = \sum_i (2h_i)^3 = 8\ \mathrm{tr}_V\, H^3$, i.e. $A(\mathrm{Sym}^2) - A(\Lambda^2) = 8\, A(F) = 8$. With the Section 5 normalization $A(F) = 1$,

$$A(\mathrm{Sym}^2) = N + 4, \qquad A(\Lambda^2) = N - 4. \tag{B.162}$$

The antisymmetric value $A(\Lambda^2) = N-4$ vanishes at $N = 4$, the precise reason the **6** of SU(4) $\cong$ SO(6) is anomaly-free—the $\mathfrak{so}(N)$ side of the coincidence (B.144) has no cubic invariant at all—and the combination $A(\bar{\mathbf{5}}) + A(\mathbf{10}) = -1+1 = 0$ at $N = 5$ is the cancellation that makes one SU(5) generation consistent, anticipated for Chapter 35.

B.6.6 The master table

We can now assemble the reference object. Table B.5 records the general formulas for the classical families, each entry traceable to a complete computation: the SU(N) fundamental and adjoint to Section 4 and to Theorem B.39; the SU(N) two-index tensors to Eqs. (B.160) to (B.162); and the SO(N) and Sp($2n$) entries to the explicit generator traces below. Table B.6 specializes to the grand-unified representations cited by Chapter 35 and consolidates the SU(3) colour data used throughout Yang–Mills theory.

A word on normalizations, because they differ by family and the table is useless without them stated. For SU(N) and Sp($2n$) we use $T(\text{defining}) = \frac{1}{2}$, matching the book-wide $\mathrm{tr}_F(T^a T^b) = \frac{1}{2}\delta^{ab}$. For SO($N$) we use the natural orthogonal normalization $T(\text{vector}) = 2$, in which the generators are $(T^{ab})_{cd} = -i(\delta^a_c\delta^b_d - \delta^a_d\delta^b_c)$; the value $T(\text{vector}) = 2$ is read off directly from $\mathrm{tr}_{\text{vec}}\,[(T^{ab})^2] = 2(\delta^{aa}\delta^{bb} - \delta^{ab}\delta^{ba}) = 2$ for a fixed pair $a < b$. Ratios of invariants within a single group are normalization-independent and provide the cross-checks; only the overall scale carries the convention.

For the orthogonal family the vector and adjoint entries follow from Eq. (B.149):

with $\dim \mathfrak{g} = \frac{N(N-1)}{2}$ and $\dim(\text{vector}) = N$,

$$C_2(\text{vector}) = T(\text{vector})\frac{N(N-1)/2}{N} = N-1, \qquad C_2(\text{adj}) = N-2 \text{ (scaled: } 2(N-2)), \tag{B.163}$$

where the adjoint Casimir $C_2(\text{adj}) = 2(N-2)$ in the $T(\text{vector}) = 2$ normalization is the dual Coxeter number $h^\vee(\mathfrak{so}(N)) = N-2$ times the scale factor. The spinor of SO(N) has dimension $2^{\lfloor N/2 \rfloor}$ (Dirac) and, from the gamma-matrix generators $\Sigma^{ab} = \frac{i}{2}\Gamma^a\Gamma^b$ ($a \neq b$) with $(\Sigma^{ab})^2 = \frac{1}{4}$, index $T(\text{spinor}) = 2^{\lfloor N/2 \rfloor}/4$, giving by Eq. (B.149)

$$C_2(\text{spinor}) = \frac{N(N-1)}{8}. \tag{B.164}$$

For the symplectic family, the defining $2n$ and its adjoint (the symmetric square, of dimension $n(2n+1)$) give in the $T(\text{fund}) = \frac{1}{2}$ normalization $C_2(\text{fund}) = \frac{2n+1}{4}$ and $C_2(\text{adj}) = n+1 = h^\vee(\mathfrak{sp}(2n))$; the value $\frac{2n+1}{4}$ reduces to $\frac{3}{4}$ at $n=1$, the correct SU(2) fundamental Casimir, since $\mathrm{Sp}(2) \cong \mathrm{SU}(2)$.

Table B.5: Master group-theory data for the classical families. $\dim \mathfrak{g}$ is the algebra dimension from Table B.4. Normalizations: $T(\text{defining}) = \frac{1}{2}$ for $\mathrm{SU}(N), \mathrm{Sp}(2n)$; $T(\text{vector}) = 2$ for $\mathrm{SO}(N)$. Each C_2 follows from the index by the index–Casimir relation (B.149). Anomaly coefficients A are normalized to $A(\text{fund}) = 1$; entries marked 0 are for algebras with no cubic Casimir ($\mathrm{SO}(N \neq 6)$, $\mathrm{Sp}(2n)$) or for (pseudo)real representations.

Group	Representation	$\dim R$	$T(R)$	$C_2(R)$	$A(R)$
SU(N)	fundamental $\mathbf{N}$	N	$\frac{1}{2}$	$\frac{N^2-1}{2N}$	1
SU(N)	conjugate $\overline{\mathbf{N}}$	N	$\frac{1}{2}$	$\frac{N^2-1}{2N}$	-1
SU(N)	adjoint	N^2-1	N	N	0
SU(N)	antisym. Λ^2	$\frac{N(N-1)}{2}$	$\frac{N-2}{2}$	$\frac{(N-2)(N+1)}{N}$	$N-4$
SU(N)	sym. Sym^2	$\frac{N(N+1)}{2}$	$\frac{N+2}{2}$	$\frac{(N+2)(N-1)}{N}$	$N+4$
SO(N)	vector	N	2	$N-1$	0
SO(N)	adjoint	$\frac{N(N-1)}{2}$	$2(N-2)$	$2(N-2)$	0
SO(N)	spinor (Dirac)	$2^{\lfloor N/2 \rfloor}$	$2^{\lfloor N/2 \rfloor}/4$	$\frac{N(N-1)}{8}$	0
Sp($2n$)	fundamental	$2n$	$\frac{1}{2}$	$\frac{2n+1}{4}$	0
Sp($2n$)	adjoint	$n(2n+1)$	$n+1$	$n+1$	0

Table B.6: Master data for the grand-unified representations of SU(5) and SO(10), and the SU(3) colour representations of Yang–Mills theory. The SU(5) entries specialize Table B.5 at $N = 5$; the SO(10) entries use $N = 10$ with the **16** the chiral (Weyl) half-spinor, $\dim = 2^{5-1} = 16$, sharing the Dirac Casimir. The combination $\overline{\mathbf{5}} \oplus \mathbf{10}$ of SU(5) has total anomaly $-1 + 1 = 0$; SO(10) has no cubic Casimir, so its representations are automatically anomaly-free.

Group	Rep	$\dim R$	$T(R)$	$C_2(R)$	$A(R)$
SU(5)	$\mathbf{5}$	5	$\frac{1}{2}$	$\frac{12}{5}$	1
SU(5)	$\overline{\mathbf{5}}$	5	$\frac{1}{2}$	$\frac{12}{5}$	-1
SU(5)	$\mathbf{10}$	10	$\frac{3}{2}$	$\frac{18}{5}$	1
SU(5)	$\mathbf{24}$ (adj)	24	5	5	0
SO(10)	$\mathbf{10}$ (vector)	10	2	9	0
SO(10)	$\mathbf{16}$ (spinor)	16	4	$\frac{45}{4}$	0
SO(10)	$\mathbf{45}$ (adj)	45	16	16	0
SU(3)	$\mathbf{3}$	3	$\frac{1}{2}$	$\frac{4}{3}$	1
SU(3)	$\mathbf{8}$ (adj)	8	3	3	0
SU(3)	$\mathbf{6}$ (sym)	6	$\frac{5}{2}$	$\frac{10}{3}$	7

Every number in Table B.6 has a one-line origin in Table B.5. The SU(5) $\mathbf{10} = \Lambda^2$ has $T = (5-2)/2 = \frac{3}{2}$, $C_2 = (3)(6)/5 = \frac{18}{5}$, $A = 5 - 4 = 1$; the $\mathbf{5}$ has $C_2 = (25-1)/10 = \frac{12}{5}$; the adjoint $\mathbf{24}$ has $C_2 = C_A = 5$. For SO(10) the vector has $C_2 = N - 1 = 9$, the adjoint $\mathbf{45}$ has $C_2 = 2(N-2) = 16$, and the Weyl spinor $\mathbf{16}$ shares the Dirac Casimir $C_2 = N(N-1)/8 = \frac{45}{4}$ with $T(\mathbf{16}) = C_2 \dim R/\dim \mathfrak{g} = \frac{45}{4} \cdot 16/45 = 4$. The SU(3) block reproduces the worked Theorem B.39 and adds the colour-sextet via $N = 3$ in Eqs. (B.160) to (B.162). As an external consistency check, the coincidence $\mathfrak{su}(4) \cong \mathfrak{so}(6)$ requires the SU(4) data to match the SO(6) data after fixing the relative normalization: the $\mathbf{6}$ of SU(4) (Λ^2, with $T = 1$, $C_2 = \frac{(2)(5)}{4} = \frac{5}{2}$) is the vector of SO(6) ($T = 2$, $C_2 = 5$), and indeed the two Casimirs stand in the ratio $1 : 2$ set by the two normalization conventions, while $A(\mathbf{6}_{\mathrm{SU}(4)}) = N - 4 = 0$ correctly reflects that SO(6) has no cubic invariant.

B.6.7 Embedding indices and the role in unification

The master table fixes data inside a single group; coupling unification needs one further number that compares a group to a subgroup. Given an embedding $\iota : \mathfrak{h} \hookrightarrow \mathfrak{g}$ of one simple algebra in another, a representation R of $\mathfrak{g}$ restricts to a (generally reducible) representation of $\mathfrak{h}$, and the ratio

$$j_\iota = \frac{T_{\mathfrak{h}}(R|_{\mathfrak{h}})}{T_{\mathfrak{g}}(R)} \tag{B.165}$$

is independent of which R is used; it is the *embedding index* (or Dynkin index of the embedding). It is a positive integer for the maximal embeddings of physical interest and measures how the normalization of generators changes on passing to the subgroup. The same statement for an abelian factor $\mathfrak{u}(1) \subset \mathfrak{g}$ becomes the normalization of the U(1) generator relative to the trace convention of $\mathfrak{g}$, and this is exactly the input grand unification turns on: when the Standard Model group is embedded in a simple group such as SU(5), the properly normalized hypercharge generator differs from the conventionally normalized Y by the factor that produces $g_1^2 = \frac{5}{3}\, g_Y^2$, the celebrated $5/3$ that makes the three gauge couplings meet. We state the embedding index here only at the level of its definition and its role; its computation for the Standard Model chain and the resulting prediction for $\sin^2\theta_W$ at unification are carried out in Chapter 35, which consumes the SU(5) and SO(10) rows of Table B.6 together with Eq. (B.165).

With the Cartan–Weyl vocabulary fixed, the computability of the entries demonstrated on $\mathfrak{su}(3)$, and the master tables B.5 and B.6 assembled, this appendix has discharged its service role: every dim, T, C_2, and A that the gauge, anomaly, and grand-unification chapters quote is now a single labelled object with a complete derivation behind it. The grand-unified rows feed directly into Chapter 35, and the SU(N) rows close the loop with the colour factors of Yang–Mills theory established in Chapters 27 and 28.

Further reading

The cleanest physicist's route into the representation theory used here is Georgi [218], whose treatment of SU(N), weights, and tensor methods parallels this appendix closely; Ramond [413] covers the same ground with more attention to the exceptional groups and grand unification. For the mathematical foundations — the full classification, character theory, and the Weyl and Freudenthal formulas quoted here — Fulton and Harris [206] is the standard reference. The diagrammatic "birdtrack" technology that evaluates the higher invariants we only mention is developed exhaustively in Cvitanović [142]. The encyclopedic tables of branching rules, indices, and anomaly coefficients for model building, far beyond what we tabulate, are Slansky [445]. Finally, the group-theory appendices of the standard field-theory texts — Peskin and Schroeder [396], Schwartz [432], and Weinberg [501, Vol. II] — give complementary conventions and worked color algebra for the reader cross-checking signs and normalizations against other sources.

Appendix C

Gamma Matrices and Spinor Technology

This appendix is the complete computational arsenal for everything spinorial in the book. Every gamma-matrix identity, every trace, every Fierz coefficient, every spin sum that a chapter quotes in passing is established here in full, with all intermediate algebra on the page. It is meant to be used as a reference, but it is a proof-carrying reference: nothing in it is merely tabulated. A reader who wants to reconstruct any trace identity, the value of any contraction, the sign of any charge-conjugation relation, or the precise statement of the γ^5 problem in dimensional regularization should be able to do so from this appendix alone, without reaching for an external table or trusting an unjustified formula. The discipline of the rest of the book—carry the algebra, prove what you use—applies here with no exception, and applies especially because these are precisely the results other chapters lean on most heavily and check least often.

The narrative thread arrives here from two earlier places. Chapter 3 classified the relativistic fields as representations of the Lorentz group: it identified the Weyl and Dirac spinors abstractly through the (j_-, j_+) labelling and the $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ structure of the complexified Lorentz algebra, answering the question of *what* a spinor is and why it must exist. Chapter 11 then put that abstraction to work, introducing the Dirac field proper, its plane-wave solutions u and v, and the discrete symmetries C, P, T as operators acting on it. Those chapters fixed the physical content. This appendix supplies the algebra that content lives in and the manipulation rules that turn a spinorial Lagrangian or matrix element into a number. The division of labor is deliberate: physical interpretation—what $u_s(p)$ means, why C exchanges particle and antiparticle, what a bilinear measures—belongs to Chapters 3 and 11; the identities and their proofs belong here; and downstream chapters import a result by cross-reference to its label in this appendix rather than re-deriving it in place.

The conventions are those of the rest of the book, collected in Chapter A:

the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, the defining Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, the chirality matrix $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, and the Dirac adjoint $\bar\psi = \psi^\dagger\gamma^0$. Two import boundaries should be kept in mind throughout. First, the Pauli matrices and all $\mathfrak{su}(2)$ representation data are taken from Chapter 2 and Chapter B, not rebuilt here; we use them as known. Second, there are two distinct families of rearrangement identity that share the name "Fierz" and must not be confused: the *color* or group-theoretic Fierz identities, which act on the internal-symmetry indices and follow from completeness of the generators, live in Chapter B, whereas the *spinor* Fierz identities, which rearrange the order of Dirac bilinears, are derived in this appendix from the completeness of the sixteen covariants.

The appendix is built as a single developing argument rather than a catalogue, and reading it in order is the intended path. We begin with the abstract Clifford algebra $\mathrm{Cl}(1,3)$ and its Lorentz content, showing how the antisymmetrized products $\sigma^{\mu\nu}$ furnish a representation of the Lorentz algebra. We then prove that this algebra has, up to similarity, a unique irreducible representation of dimension $2^{d/2}$—Pauli's fundamental theorem, with its full proof—which is what licenses every later "choose a representation" move. From there we descend to the explicit four-dimensional representations (Dirac, chiral, Majorana) and the Hermiticity relation $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$; classify the intertwining matrices γ^0, C, and the parity and time-reversal matrices; assemble the sixteen bilinear covariants into a complete trace-orthogonal basis; and extract from that basis the trace and contraction technology in four dimensions, including the γ^5–ϵ trace. We then continue everything to $d = 4 - 2\varepsilon$ for dimensional regularization and confront the γ^5 problem honestly rather than papering over it. The final two sections deliver the spinor Fierz rearrangement together with the external-state spin sums and projectors, and the two-component van der Waerden calculus that the amplitudes part builds on. Downstream, Chapters 17 and 19 draw on the four-dimensional and d-dimensional traces, Chapter 36 on the dimensionally-continued γ^5 for the chiral anomaly, and Chapter 58 on the two-component calculus as the substrate for spinor-helicity.

C.1 The Clifford Algebra and Its Lorentz Content

Chapter 3 classified the relativistic fields by representations of the Lorentz group. Among the irreducible building blocks it identified the two inequivalent two-dimensional spinor representations, $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$, and their reducible Dirac sum $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$; the language of Lie algebras and their representations in which that classification was carried out is the one set up in Chapter 2. That analysis was abstract: it told us *which* representation a Dirac field carries and *why*, but it did not hand us a concrete object on which the Lorentz generators act by explicit matrices we can multiply, trace, and contract. The purpose of this section is

to manufacture exactly such an object, and to do it in a way that makes the appearance of the Lorentz group automatic rather than imposed. The route is a single algebraic relation. We will demand a set of four quantities γ^μ that square, in the appropriate sense, to the metric; show that the associative algebra they generate is finite-dimensional and has a canonical basis; and discover that this algebra already *contains* a copy of the Lorentz algebra $\mathfrak{so}(1,3)$, built from the γ^μ by a fixed formula. The upshot, made precise below, is that "gamma matrices" and "the spin representation of $\mathfrak{so}(1,3)$" are not two facts to be reconciled but one object seen from two sides.

C.1.1 The Clifford algebra defined by anticommutation

We work throughout with the mostly-minus Minkowski metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$ fixed in Chapter A, indices running over $\mu,\nu \in \{0,1,2,3\}$. The starting point is purely algebraic and makes no reference to matrices, dimensions, or a vector space to act on.

Definition C.1 (The Clifford algebra $\mathrm{Cl}(1,3)$)**.** Let $\mathcal{A}$ be the associative, unital algebra over $\mathbb{C}$ generated by four symbols $\gamma^0,\gamma^1,\gamma^2,\gamma^3$ subject only to the relation

$$\{\gamma^\mu,\gamma^\nu\} \;=\; \gamma^\mu\gamma^\nu+\gamma^\nu\gamma^\mu \;=\; 2\,\eta^{\mu\nu}\,\mathbf{1}, \tag{C.1}$$

where $\mathbf{1}$ is the algebra unit. Concretely, $\mathcal{A}$ is the free associative algebra on the four symbols modulo the two-sided ideal generated by $\gamma^\mu\gamma^\nu+\gamma^\nu\gamma^\mu-2\eta^{\mu\nu}\mathbf{1}$. This algebra is the (complexified) *Clifford algebra* $\mathrm{Cl}(1,3)$ of Minkowski space.

Three immediate consequences of Eq. (C.1) fix the arithmetic we will use constantly. Setting $\mu=\nu$ gives the squares of the generators,

$$(\gamma^0)^2=\mathbf{1}, \qquad (\gamma^1)^2=(\gamma^2)^2=(\gamma^3)^2=-\mathbf{1}, \tag{C.2}$$

so γ^0 is its own inverse while each spatial γ^i has inverse $-\gamma^i$. For $\mu\neq\nu$ the right-hand side of Eq. (C.1) vanishes, so distinct generators anticommute,

$$\gamma^\mu\gamma^\nu=-\gamma^\nu\gamma^\mu \qquad (\mu\neq\nu). \tag{C.3}$$

Every generator is therefore invertible, and consequently so is every product of generators; the algebra has no need of an enveloping construction beyond the words in the γ^μ. We have not yet asserted that $\mathcal{A}$ is anything but possibly trivial or infinite-dimensional. The first structural theorem settles its size.

C.1.2 The antisymmetrized basis and the dimension count

Relation Eq. (C.3) lets us reorder any product of generators at the cost of signs, and Eq. (C.2) lets us delete repeated factors. It is natural to organize the resulting

normal forms by which generators they involve. Write $d = 4$ for the spacetime dimension. For an ordered subset $A = \{a_1 < a_2 < \cdots < a_k\} \subseteq \{0, 1, 2, 3\}$ define the monomial

$$\Gamma_A \;=\; \gamma^{a_1}\gamma^{a_2}\cdots\gamma^{a_k}, \qquad \Gamma_\emptyset := \mathbf{1}, \tag{C.4}$$

the factors written in increasing index order. Because the generators with distinct indices anticommute, Γ_A is (up to sign) the antisymmetrized product $\gamma^{[a_1}\cdots\gamma^{a_k]}$, which is why these objects are called the *antisymmetrized products*; in $d = 4$ they are

$$\underbrace{\mathbf{1}}_{1},\quad \underbrace{\gamma^\mu}_{4},\quad \underbrace{\gamma^\mu\gamma^\nu\ (\mu<\nu)}_{6},\quad \underbrace{\gamma^\mu\gamma^\nu\gamma^\rho\ (\mu<\nu<\rho)}_{4},\quad \underbrace{\gamma^0\gamma^1\gamma^2\gamma^3}_{1}. \tag{C.5}$$

Theorem C.2 (Dimension of the Clifford algebra)**.** *The 2^d monomials $\{\Gamma_A\}_{A\subseteq\{0,\ldots,d-1\}}$ form a basis of $\mathcal{A} = \mathrm{Cl}(1,3)$ as a complex vector space. Hence*

$$\dim_{\mathbb{C}} \mathcal{A} \;=\; \sum_{k=0}^{d} \binom{d}{k} \;=\; 2^d, \tag{C.6}$$

which is $2^4 = 16$ in $d = 4$.

The proof has two halves: spanning (the monomials generate, so $\dim \mathcal{A} \le 2^d$) and linear independence (no nontrivial relation among them, so $\dim \mathcal{A} \ge 2^d$). Spanning is a direct reduction argument. Independence is more delicate, because at this point we have *not* constructed any matrices on which to evaluate a trace; that construction is the business of Section C.2. We therefore give a proof of independence that is purely internal to the abstract algebra, using only Eq. (C.1). It rests on the following lemma.

Lemma C.3 (Anticommuting partner)**.** *For every nonempty $A \subseteq \{0, \ldots, d-1\}$ in $d = 4$ there exists a generator γ^μ that anticommutes with Γ_A: $\gamma^\mu\Gamma_A = -\Gamma_A\gamma^\mu$.*

Proof. Let $|A| = k \ge 1$. Moving a single generator γ^μ through $\Gamma_A = \gamma^{a_1}\cdots\gamma^{a_k}$ produces one sign (-1) for each factor γ^{a_i} with $a_i \ne \mu$ (by Eq. (C.3)) and no sign for a factor with $a_i = \mu$. Hence

$$\gamma^\mu\,\Gamma_A = (-1)^{\,k-[\mu\in A]}\,\Gamma_A\,\gamma^\mu, \tag{C.7}$$

where $[\mu \in A]$ is 1 if $\mu \in A$ and 0 otherwise. We need an index making the exponent odd. If k is odd, choose any $\mu \notin A$ (such an index exists whenever $k < d$, and for $k = d$ the count k is even, so this case does not arise here): the exponent is k, which is odd. If k is even, choose any $\mu \in A$ (which is nonempty): the exponent is $k - 1$, again odd. In either case $\gamma^\mu\Gamma_A = -\Gamma_A\gamma^\mu$, as claimed. (For $A = \{0, 1, 2, 3\}$, where $k = d = 4$ is even, the second case applies and any $\mu \in A$ works; this is the statement, used later, that $\gamma^0\gamma^1\gamma^2\gamma^3$ anticommutes with every γ^μ.) □

Proof of Theorem C.2. Spanning. Any element of $\mathcal{A}$ is a finite linear combination of words $\gamma^{\mu_1}\cdots\gamma^{\mu_n}$. Take any such word. Using Eq. (C.3) repeatedly we may sort the factors into nondecreasing index order, each transposition contributing a factor (-1). Whenever two equal indices become adjacent, Eq. (C.2) replaces $\gamma^\mu\gamma^\mu$ by the scalar $\eta^{\mu\mu}\mathbf{1} = \pm\mathbf{1}$, shortening the word. Iterating, every word reduces to a scalar multiple of some Γ_A with A a set of *distinct* indices in increasing order. Thus the Γ_A span $\mathcal{A}$, and there are at most 2^d of them, giving $\dim\mathcal{A} \le 2^d$.

Independence. Suppose

$$\sum_A c_A\,\Gamma_A = 0, \qquad c_A \in \mathbb{C}. \tag{C.8}$$

We show every $c_A = 0$. The key is a projector built entirely from conjugations inside $\mathcal{A}$. For ordered subsets A, S, the monomials Γ_A and Γ_S are products of distinct generators, so by Eq. (C.3) they either commute or anticommute: $\Gamma_S\Gamma_A = \chi_A(S)\,\Gamma_A\Gamma_S$ with $\chi_A(S) \in \{+1,-1\}$. Since each Γ_S is invertible, conjugation gives the pure sign

$$\Gamma_S\,\Gamma_A\,\Gamma_S^{-1} = \chi_A(S)\,\Gamma_A. \tag{C.9}$$

The map $S \mapsto \chi_A(S)$ is a homomorphism from the set of subsets under symmetric difference into $\{\pm1\}$: conjugating successively by Γ_S and $\Gamma_{S'}$ multiplies the signs, while $\Gamma_{S'}\Gamma_S$ and $\Gamma_{S\triangle S'}$ differ only by an overall scalar that cancels in a conjugation, so $\chi_A(S\triangle S') = \chi_A(S)\,\chi_A(S')$. For $A = \emptyset$ we have $\Gamma_\emptyset = \mathbf{1}$ and $\chi_\emptyset \equiv 1$, the trivial character. For $A \neq \emptyset$, Theorem C.3 supplies a generator γ^μ with $\chi_A(\{\mu\}) = -1$, so χ_A is a *nontrivial* character.

Now average Eq. (C.8) over all 2^d conjugations:

$$0 = \frac{1}{2^d}\sum_S \Gamma_S\Big(\sum_A c_A\Gamma_A\Big)\Gamma_S^{-1} = \sum_A c_A\,\Gamma_A\left[\frac{1}{2^d}\sum_S \chi_A(S)\right]. \tag{C.10}$$

A nontrivial character of a finite abelian group sums to zero over the group; the trivial one sums to the group order. Hence the bracket equals 1 for $A = \emptyset$ and 0 for $A \neq \emptyset$, and Eq. (C.10) collapses to $c_\emptyset\,\mathbf{1} = 0$, i.e. $c_\emptyset = 0$. To reach an arbitrary B, multiply Eq. (C.8) on the right by Γ_B^{-1}; since $\Gamma_A\Gamma_B^{-1}$ is a scalar multiple of $\Gamma_{A\triangle B}$ and equals a nonzero scalar times $\mathbf{1}$ precisely when $A = B$, the same averaging applied to $\sum_A c_A\Gamma_A\Gamma_B^{-1} = 0$ extracts the coefficient of $\mathbf{1}$, which is a nonzero multiple of c_B. Therefore $c_B = 0$ for every B. The Γ_A are linearly independent, $\dim\mathcal{A} \ge 2^d$, and with the spanning bound $\dim\mathcal{A} = 2^d$. □

The argument used nothing but Eq. (C.1): the Clifford algebra is genuinely 16-dimensional as an abstract object, before any matrix representation is in hand. This is the precise sense in which the algebra is "waiting" for the representation that Section C.2 will build.

Physical Insight

The basis Eq. (C.5) is the skeleton of every spinor bilinear in the book. Read against Eq. (C.6), the five rows are the rank-0, 1, 2, 3, 4 antisymmetric tensors on spacetime, with $\binom{4}{0}+\binom{4}{1}+\binom{4}{2}+\binom{4}{3}+\binom{4}{4}=16$ components in all. Once $\bar\psi=\psi^\dagger\gamma^0$ is available one contracts these against ψ to get the scalar, vector, tensor, axial-vector, and pseudoscalar currents; the rank-2 row is, up to the factor fixed next, exactly the spin generator $\sigma^{\mu\nu}$. The bilinear catalogue and its 16 trace-orthogonality relations are assembled in Section C.5.

C.1.3 The Lorentz content: $\sigma^{\mu\nu}$ as a representation of $\mathfrak{so}(1,3)$

We now extract the Lorentz algebra from the rank-2 row of Eq. (C.5). Define

$$\sigma^{\mu\nu} \;:=\; \frac{i}{4}\,[\gamma^\mu,\gamma^\nu] \;=\; \frac{i}{2}(\gamma^\mu\gamma^\nu-\eta^{\mu\nu}\mathbf{1}), \tag{C.11}$$

the second equality following from $\gamma^\mu\gamma^\nu=\frac12\{\gamma^\mu,\gamma^\nu\}+\frac12[\gamma^\mu,\gamma^\nu]=\eta^{\mu\nu}\mathbf{1}+\frac12[\gamma^\mu,\gamma^\nu]$ together with Eq. (C.1). By construction $\sigma^{\mu\nu}=-\sigma^{\nu\mu}$; for $\mu\neq\nu$ it reduces to $\sigma^{\mu\nu}=\frac{i}{2}\gamma^\mu\gamma^\nu$. Inverting Eq. (C.11) records the product of two generators in the form we will reuse repeatedly,

$$\gamma^\mu\gamma^\nu=\eta^{\mu\nu}\mathbf{1}-2i\,\sigma^{\mu\nu}. \tag{C.12}$$

The first computation is the action of $\sigma^{\mu\nu}$ on a single generator; it is the technical heart of everything that follows.

Proposition C.4 (γ^ρ is a vector under $\sigma^{\mu\nu}$)**.** *In* $\mathrm{Cl}(1,3)$,

$$[\gamma^\rho,\sigma^{\mu\nu}] \;=\; i(\eta^{\rho\mu}\gamma^\nu-\eta^{\rho\nu}\gamma^\mu). \tag{C.13}$$

Proof. The scalar term in Eq. (C.11) drops out of the commutator, so $[\gamma^\rho,\sigma^{\mu\nu}]=\frac{i}{2}[\gamma^\rho,\gamma^\mu\gamma^\nu]$. Expand using $\gamma^\rho\gamma^\mu=2\eta^{\rho\mu}\mathbf{1}-\gamma^\mu\gamma^\rho$ from Eq. (C.1):

$$\begin{aligned}[\gamma^\rho,\gamma^\mu\gamma^\nu]&=\gamma^\rho\gamma^\mu\gamma^\nu-\gamma^\mu\gamma^\nu\gamma^\rho\\&=(2\eta^{\rho\mu}-\gamma^\mu\gamma^\rho)\gamma^\nu-\gamma^\mu\gamma^\nu\gamma^\rho\\&=2\eta^{\rho\mu}\gamma^\nu-\gamma^\mu(2\eta^{\rho\nu}-\gamma^\nu\gamma^\rho)-\gamma^\mu\gamma^\nu\gamma^\rho\\&=2\eta^{\rho\mu}\gamma^\nu-2\eta^{\rho\nu}\gamma^\mu,\end{aligned} \tag{C.14}$$

the two triple products cancelling in the last line. Multiplying by $\frac{i}{2}$ gives Eq. (C.13). □

The right-hand side of Eq. (C.13) is exactly the action of an $\mathfrak{so}(1,3)$ generator labelled $(\mu\nu)$ on a vector index ρ: $\sigma^{\mu\nu}$ rotates γ^ρ into a combination of γ^μ and γ^ν with the metric-dictated coefficients. That the $\sigma^{\mu\nu}$ close among themselves into the Lorentz algebra is now a short computation.

Theorem C.5 ($\sigma^{\mu\nu}$ represents $\mathfrak{so}(1,3)$)**.** *The matrices $\sigma^{\mu\nu}$ satisfy*

$$[\sigma^{\mu\nu},\sigma^{\rho\sigma}] = i\big(\eta^{\nu\rho}\sigma^{\mu\sigma} - \eta^{\mu\rho}\sigma^{\nu\sigma} - \eta^{\nu\sigma}\sigma^{\mu\rho} + \eta^{\mu\sigma}\sigma^{\nu\rho}\big), \tag{C.15}$$

which is precisely the Lie bracket of the Lorentz algebra $\mathfrak{so}(1,3)$ established in Chapter 3. Hence $M^{\mu\nu} \mapsto \sigma^{\mu\nu}$ is a representation of $\mathfrak{so}(1,3)$ inside $\mathrm{Cl}(1,3)$.

Proof. Write $\sigma^{\rho\sigma} = \frac{i}{2}(\gamma^\rho\gamma^\sigma - \eta^{\rho\sigma}\mathbf{1})$; the scalar drops from the commutator, leaving $[\sigma^{\mu\nu},\sigma^{\rho\sigma}] = \frac{i}{2}\,[\sigma^{\mu\nu},\gamma^\rho\gamma^\sigma]$. Apply the Leibniz rule and Eq. (C.13) (using $[\sigma^{\mu\nu},\gamma^\rho] = -\,[\gamma^\rho,\sigma^{\mu\nu}] = -i(\eta^{\rho\mu}\gamma^\nu - \eta^{\rho\nu}\gamma^\mu)$):

$$\begin{aligned}[\sigma^{\mu\nu},\gamma^\rho\gamma^\sigma] &= [\sigma^{\mu\nu},\gamma^\rho]\,\gamma^\sigma + \gamma^\rho\,[\sigma^{\mu\nu},\gamma^\sigma] \\ &= -i(\eta^{\rho\mu}\gamma^\nu - \eta^{\rho\nu}\gamma^\mu)\gamma^\sigma - i\,\gamma^\rho(\eta^{\sigma\mu}\gamma^\nu - \eta^{\sigma\nu}\gamma^\mu).\end{aligned} \tag{C.16}$$

Therefore

$$[\sigma^{\mu\nu},\sigma^{\rho\sigma}] = \frac{i}{2}\,(-i)\Big[\eta^{\rho\mu}\gamma^\nu\gamma^\sigma - \eta^{\rho\nu}\gamma^\mu\gamma^\sigma + \eta^{\sigma\mu}\gamma^\rho\gamma^\nu - \eta^{\sigma\nu}\gamma^\rho\gamma^\mu\Big]. \tag{C.17}$$

Now replace each product of generators by Eq. (C.12), e.g. $\gamma^\nu\gamma^\sigma = \eta^{\nu\sigma}\mathbf{1} - 2i\sigma^{\nu\sigma}$. The purely scalar (metric×metric) part is

$$\eta^{\rho\mu}\eta^{\nu\sigma} - \eta^{\rho\nu}\eta^{\mu\sigma} + \eta^{\sigma\mu}\eta^{\rho\nu} - \eta^{\sigma\nu}\eta^{\rho\mu} = 0, \tag{C.18}$$

which vanishes term by term because η is symmetric (the first and last cancel, as do the middle two). Only the σ pieces survive, each carrying a factor $-2i$:

$$\begin{aligned}[\sigma^{\mu\nu},\sigma^{\rho\sigma}] &= \frac{i}{2}\,(-i)\,(-2i)\Big[\eta^{\rho\mu}\sigma^{\nu\sigma} - \eta^{\rho\nu}\sigma^{\mu\sigma} + \eta^{\sigma\mu}\sigma^{\rho\nu} - \eta^{\sigma\nu}\sigma^{\rho\mu}\Big] \\ &= -i\Big[\eta^{\mu\rho}\sigma^{\nu\sigma} - \eta^{\nu\rho}\sigma^{\mu\sigma} - \eta^{\mu\sigma}\sigma^{\nu\rho} + \eta^{\nu\sigma}\sigma^{\mu\rho}\Big],\end{aligned} \tag{C.19}$$

where in the last line we used $\sigma^{\rho\nu} = -\sigma^{\nu\rho}$ and $\sigma^{\rho\mu} = -\sigma^{\mu\rho}$ to flip the third and fourth terms. Distributing the overall sign and reordering reproduces Eq. (C.15). The prefactor arithmetic, $\frac{i}{2}\cdot(-i)\cdot(-2i) = -i$, is the source of the clean coefficient i on the right. □

C.1.4 Finite Lorentz transformations of spinors

The infinitesimal statement Eq. (C.15) exponentiates to a finite group action. A Lorentz transformation Λ connected to the identity is $\Lambda = \exp\omega$ with $\omega^\mu{}_\nu$ generating $\mathfrak{so}(1,3)$, i.e. $\omega_{\mu\nu} = \eta_{\mu\rho}\omega^\rho{}_\nu$ antisymmetric (this is the defining condition $\Lambda^{\mathsf{T}}\eta\Lambda = \eta$ at the Lie-algebra level, Chapter 3). To each such ω associate the spinor operator

$$S(\Lambda) = \exp\Big(-\tfrac{i}{2}\,\omega_{\mu\nu}\,\sigma^{\mu\nu}\Big). \tag{C.20}$$

The factor $\frac{1}{2}$ compensates the double sum over the antisymmetric pair (μ,ν).

Theorem C.6 (γ^μ transforms as a four-vector)**.** *With $S(\Lambda)$ as in Eq. (C.20) and $\Lambda = \exp\omega$,*

$$S(\Lambda)^{-1}\,\gamma^\mu\,S(\Lambda) \;=\; \Lambda^\mu{}_\nu\,\gamma^\nu. \tag{C.21}$$

Equivalently $S(\Lambda)\,\gamma^\mu\,S(\Lambda)^{-1} = (\Lambda^{-1})^\mu{}_\nu\,\gamma^\nu$. The Dirac equation $(i\gamma^\mu\partial_\mu - m)\psi = 0$ is Lorentz covariant precisely because of this intertwining relation; that physical consequence is drawn in Chapter 11.

Proof. Introduce the one-parameter family $S(t) = \exp\!\big(-\tfrac{i}{2}t\,\omega_{\rho\sigma}\sigma^{\rho\sigma}\big)$ and

$$\gamma^\mu(t) := S(t)^{-1}\gamma^\mu S(t). \tag{C.22}$$

Differentiate, using $\dot S(t) = -\tfrac{i}{2}\omega_{\rho\sigma}\sigma^{\rho\sigma}S(t)$ and $\dot{S^{-1}} = +\tfrac{i}{2}\omega_{\rho\sigma}\sigma^{\rho\sigma}S(t)^{-1}$:

$$\frac{d}{dt}\gamma^\mu(t) = \frac{i}{2}\omega_{\rho\sigma}\,S^{-1}\,[\sigma^{\rho\sigma},\gamma^\mu]\,S. \tag{C.23}$$

From Eq. (C.13), $[\sigma^{\rho\sigma},\gamma^\mu] = -\,[\gamma^\mu,\sigma^{\rho\sigma}] = -i(\eta^{\mu\rho}\gamma^\sigma - \eta^{\mu\sigma}\gamma^\rho)$, so

$$\begin{aligned}\frac{i}{2}\omega_{\rho\sigma}\,[\sigma^{\rho\sigma},\gamma^\mu] &= \frac{i}{2}\omega_{\rho\sigma}\,(-i)\big(\eta^{\mu\rho}\gamma^\sigma - \eta^{\mu\sigma}\gamma^\rho\big) = \frac{1}{2}\big(\omega^\mu{}_\sigma\gamma^\sigma - \omega_\rho{}^\mu\gamma^\rho\big)\\ &= \frac{1}{2}\big(\omega^\mu{}_\sigma\gamma^\sigma + \omega^\mu{}_\rho\gamma^\rho\big) = \omega^\mu{}_\nu\,\gamma^\nu,\end{aligned} \tag{C.24}$$

where we raised an index with η and used antisymmetry $\omega_\rho{}^\mu = -\omega^\mu{}_\rho$ to combine the two terms. Conjugating by S (which is t-independent in its action on the constant coefficient $\omega^\mu{}_\nu$) turns Eq. (C.23) into the closed linear system

$$\frac{d}{dt}\gamma^\mu(t) = \omega^\mu{}_\nu\,\gamma^\nu(t), \qquad \gamma^\mu(0) = \gamma^\mu. \tag{C.25}$$

Its unique solution is $\gamma^\mu(t) = (e^{t\omega})^\mu{}_\nu\,\gamma^\nu$. Setting $t = 1$ and $\Lambda = e^\omega$ gives Eq. (C.21). □

Caution

The matrix $S(\Lambda)$ is *not* unitary for boosts. The boost generators σ^{0i} are anti-Hermitian in the standard representations (Section C.3.6), so the boost part of $\exp(-\tfrac{i}{2}\omega\sigma)$ is Hermitian and exponentially growing in rapidity, reflecting that the finite-dimensional spinor representation of the noncompact Lorentz group cannot be unitary. This is consistent: the Hilbert-space unitary representations are infinite-dimensional and act on multiparticle states; the finite-dimensional $S(\Lambda)$ acts on the field components. The relation $\gamma^0 S^\dagger \gamma^0 = S^{-1}$ that repairs the would-be unitarity is the γ^0 intertwiner classified in Section C.4.

C.1.5 Identification with the $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ representation

The representation $\sigma^{\mu\nu}$ acts on a four-dimensional space (once we have the matrices of Section C.2). Chapter 3 predicts that the Dirac representation is reducible, splitting as $(\frac{1}{2},0)\oplus(0,\frac{1}{2})$ over the complexified algebra $\mathfrak{so}(1,3)_{\mathbb{C}} \cong \mathfrak{su}(2)\oplus\mathfrak{su}(2)$. We can already see the splitting abstractly, before choosing matrices, through the chirality element

$$\gamma^5 := i\,\gamma^0\gamma^1\gamma^2\gamma^3 = -\frac{i}{4!}\,\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma, \tag{C.26}$$

the top element of the basis Eq. (C.5) normalized as fixed in Chapter A. Two facts follow from the Clifford relation alone. First, $(\gamma^5)^2 = \mathbf{1}$: writing $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and squaring, the reordering needed to pair each generator with its partner produces $(i)^2(-1)^6(\gamma^0)^2(\gamma^1)^2(\gamma^2)^2(\gamma^3)^2 = (-1)(+1)(1)(-1)(-1)(-1) = 1$, where $(-1)^6$ counts the six transpositions that sort $\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0\gamma^1\gamma^2\gamma^3$ into paired form. Second, γ^5 anticommutes with every generator: by Theorem C.3 applied to $A = \{0,1,2,3\}$, or directly, moving γ^μ through the four distinct factors of γ^5 yields three sign changes (one factor equals γ^μ, three do not), so

$$\left\{\gamma^5,\gamma^\mu\right\} = 0. \tag{C.27}$$

Because $\sigma^{\mu\nu}$ is a product of *two* generators, Eq. (C.27) gives the crucial commutation

$$\left[\gamma^5,\sigma^{\mu\nu}\right] = 0. \tag{C.28}$$

Indeed $\gamma^5\gamma^\mu\gamma^\nu = (-1)^2\gamma^\mu\gamma^\nu\gamma^5 = \gamma^\mu\gamma^\nu\gamma^5$, and the scalar piece of $\sigma^{\mu\nu}$ commutes trivially.

Proposition C.7 (Chiral decomposition of the spin representation)**.** *The operators $P_\pm = \frac{1}{2}(\mathbf{1}\pm\gamma^5)$ are complementary orthogonal idempotents, $P_+^2 = P_+$, $P_-^2 = P_-$, $P_+P_- = P_-P_+ = 0$, $P_+ + P_- = \mathbf{1}$, and they commute with $S(\Lambda)$ for every Λ. Consequently the spin representation $\Lambda \mapsto S(\Lambda)$ leaves invariant the two subspaces $P_\pm\mathcal{V}$ and decomposes as a direct sum of two subrepresentations. These are the left- and right-handed Weyl representations $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ of Chapter 3.*

Proof. The idempotent identities are immediate from $(\gamma^5)^2 = \mathbf{1}$: $P_\pm^2 = \frac{1}{4}(\mathbf{1}\pm 2\gamma^5+(\gamma^5)^2) = \frac{1}{4}(2\mathbf{1}\pm 2\gamma^5) = P_\pm$, and $P_+P_- = \frac{1}{4}(\mathbf{1}-(\gamma^5)^2) = 0$. From Eq. (C.28), γ^5 commutes with $\sigma^{\mu\nu}$ and hence with every power series in the $\sigma^{\mu\nu}$, in particular with $S(\Lambda) = \exp(-\frac{i}{2}\omega_{\mu\nu}\sigma^{\mu\nu})$; therefore $P_\pm S(\Lambda) = S(\Lambda)P_\pm$. The projected subspaces are invariant and $S(\Lambda) = S(\Lambda)P_+ + S(\Lambda)P_-$ block-diagonalizes the action. That the two blocks carry the inequivalent two-dimensional representations $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ — equivalently, that the self-dual and anti-self-dual combinations of $\sigma^{\mu\nu}$ generate the two commuting $\mathfrak{su}(2)$ factors — is the content of the Lorentz classification in Chapter 3; here it is realized concretely as the $\gamma^5 = \pm 1$ eigenspaces. The matrix realization in which $P_\pm$ are the obvious block

projectors is the chiral (Weyl) basis of Section C.3.2, and the eigenvalue ± 1 of γ^5 is the chirality of Chapter 11. □

This is the promised identification. "Gamma matrices" and "the spin representation of $\mathfrak{so}(1,3)$" are the same structure: the four generators encode, through Eq. (C.11), a representation of the Lorentz algebra, and that representation is reducibly the Dirac sum of the two Weyl spinors that Chapter 3 introduced abstractly.

C.1.6 Worked examples

Worked Example C.8 (Rotations, boosts, and the double cover $S(2\pi) = -\mathbf{1}$)**.** We extract the rotation and boost generators from $\sigma^{\mu\nu}$ and exhibit the hallmark of a spinor representation: a 2π rotation acts as $-\mathbf{1}$, not $+\mathbf{1}$. This is the algebraic shadow of the $\mathrm{SL}(2,\mathbb{C}) \to \mathrm{SO}^+(1,3)$ double cover established in Chapter 3.

Consider a rotation by angle θ about the z-axis, generated by the single nonzero pair $\omega_{12} = -\omega_{21} = -\theta$ (all other components zero). The exponent in Eq. (C.20) is

$$-\tfrac{i}{2}\,\omega_{\mu\nu}\sigma^{\mu\nu} = -\tfrac{i}{2}(\omega_{12}\sigma^{12}+\omega_{21}\sigma^{21}) = -\tfrac{i}{2}((-\theta)\sigma^{12}+(\theta)(-\sigma^{12})) = i\theta\,\sigma^{12}, \tag{C.29}$$

so $S(\theta) = \exp(i\theta\,\sigma^{12})$. The square of the generator follows from the Clifford relation:

$$(\sigma^{12})^2 = \left(\tfrac{i}{2}\gamma^1\gamma^2\right)^2 = -\tfrac{1}{4}\,\gamma^1\gamma^2\gamma^1\gamma^2 = -\tfrac{1}{4}\,(-1)\gamma^1\gamma^1\gamma^2\gamma^2 = -\tfrac{1}{4}\,(\eta^{11})(\eta^{22})\mathbf{1} = \tfrac{1}{4}\,\mathbf{1}, \tag{C.30}$$

using $\gamma^2\gamma^1 = -\gamma^1\gamma^2$ and $\eta^{11} = \eta^{22} = -1$. Thus σ^{12} has eigenvalues $\pm\frac{1}{2}$, the spin-$\frac{1}{2}$ values. With $(\sigma^{12})^2 = \frac{1}{4}\mathbf{1}$ the exponential resums to a closed form: separating even and odd powers,

$$S(\theta) = e^{i\theta\sigma^{12}} = \cos\!\left(\tfrac{\theta}{2}\right)\mathbf{1} + 2i\,\sigma^{12}\,\sin\!\left(\tfrac{\theta}{2}\right). \tag{C.31}$$

The half-angle is the spin-$\frac{1}{2}$ fingerprint. At $\theta = 2\pi$, $\cos\pi = -1$ and $\sin\pi = 0$, so

$$S(2\pi) = -\mathbf{1}, \tag{C.32}$$

a full spatial revolution returns the spinor to minus itself. A second revolution, $\theta = 4\pi$, restores $+\mathbf{1}$. The map $\Lambda \mapsto S(\Lambda)$ is therefore two-valued on the rotation group: it is a genuine representation of the double cover, the concrete manifestation of the $\mathrm{SL}(2,\mathbb{C})$ structure of Chapter 3.

For comparison, a boost of rapidity η along x^1 uses $\omega_{01} = -\omega_{10} = \eta$, giving $S = \exp(-i\eta\,\sigma^{01})$ with $(\sigma^{01})^2 = (\frac{i}{2}\gamma^0\gamma^1)^2 = -\frac{1}{4}\gamma^0\gamma^1\gamma^0\gamma^1 = \frac{1}{4}\gamma^0\gamma^0\gamma^1\gamma^1 = \frac{1}{4}(\eta^{00})(\eta^{11})\mathbf{1} = -\frac{1}{4}\mathbf{1}$, where $\gamma^1\gamma^0 = -\gamma^0\gamma^1$ supplies the middle sign and $\eta^{00} = +1$, $\eta^{11} = -1$; this is the opposite sign to Eq. (C.30). The resummation then produces hyperbolic functions $\cosh(\eta/2)$, $\sinh(\eta/2)$ rather than trigonometric ones, confirming the noncompact, non-unitary character of boosts flagged above.

Worked Example C.9 (Direct check of the $\mathfrak{so}(1,3)$ bracket)**.** We verify Eq. (C.15) on a representative pair by computing the commutator straight from the generators, with no appeal to the general formula, and then confirm agreement. Take $(\mu\nu) = (12)$ and $(\rho\sigma) = (23)$, two rotation generators sharing the index 2. Using $\sigma^{12} = \frac{i}{2}\gamma^1\gamma^2$ and $\sigma^{23} = \frac{i}{2}\gamma^2\gamma^3$,

$$\sigma^{12}\sigma^{23} = -\tfrac{1}{4}\,\gamma^1\gamma^2\gamma^2\gamma^3 = -\tfrac{1}{4}\,\eta^{22}\,\gamma^1\gamma^3 = \tfrac{1}{4}\,\gamma^1\gamma^3, \tag{C.33}$$

$$\sigma^{23}\sigma^{12} = -\tfrac{1}{4}\,\gamma^2\gamma^3\gamma^1\gamma^2 = -\tfrac{1}{4}\,\gamma^1\gamma^3, \tag{C.34}$$

where the second line uses $\gamma^2\gamma^3\gamma^1\gamma^2 = \gamma^1\gamma^3$, obtained by the three anticommutations $\gamma^2\gamma^3\gamma^1\gamma^2 = -\gamma^3\gamma^2\gamma^1\gamma^2 = +\gamma^3\gamma^1\gamma^2\gamma^2 = \eta^{22}\gamma^3\gamma^1 = -\gamma^3\gamma^1 = \gamma^1\gamma^3$. Hence

$$\left[\sigma^{12},\sigma^{23}\right] = \tfrac{1}{4}\gamma^1\gamma^3 - (-\tfrac{1}{4}\gamma^1\gamma^3) = \tfrac{1}{2}\,\gamma^1\gamma^3. \tag{C.35}$$

Since $\sigma^{13} = \frac{i}{2}\gamma^1\gamma^3$, i.e. $\gamma^1\gamma^3 = -2i\,\sigma^{13}$, this is

$$\left[\sigma^{12},\sigma^{23}\right] = -i\,\sigma^{13}. \tag{C.36}$$

The general formula Eq. (C.15) with $\mu = 1, \nu = 2, \rho = 2, \sigma = 3$ keeps only the term with $\eta^{\nu\rho} = \eta^{22} = -1$ (the others carry off-diagonal η components that vanish), giving $i\,\eta^{22}\sigma^{13} = -i\,\sigma^{13}$. The two computations agree. The mostly-minus signature shows itself in the sign $\eta^{22} = -1$: it is what turns the three spatial rotation generators $\Sigma^k := \frac{1}{2}\epsilon^{kij}\sigma^{ij}$ into a standard $\mathfrak{su}(2)$ with $\left[\Sigma^i,\Sigma^j\right] = i\epsilon^{ijk}\Sigma^k$, the rotation subalgebra imported from Chapter 2.

We have now built, from one anticommutation relation, a 16-dimensional associative algebra with a canonical basis and shown that it carries the Lorentz algebra through $\sigma^{\mu\nu}$. What is conspicuously absent is any actual matrix: every statement so far is a relation among abstract symbols. The next section confronts the question this raises. A representation of $\mathrm{Cl}(1,3)$ by matrices must exist for the spinor fields of Chapter 11 to be column vectors at all, and the abstract 16-dimensionality forces such a representation to have a definite minimal size. Section C.2 proves that in even dimension the irreducible representation exists, has dimension $2^{d/2} = 4$ in $d = 4$, and is unique up to similarity — Pauli's fundamental theorem — the result on which every intertwining matrix in the rest of this appendix depends.

C.2 Existence and Uniqueness of the Representation

The previous section took the gamma matrices to be *any* objects obeying the Clifford relation $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$ and built from them the 2^d-dimensional algebra basis $\{\Gamma_A\}$ of ordered products (Section C.1). That was a purely algebraic enterprise: nothing in it guaranteed that matrices satisfying the relation actually

exist, let alone that there is essentially only one way to realize them. Yet every concrete computation in this book—every trace, every spin sum, every Fierz rearrangement—is performed with specific numerical matrices, and the representation-independence of the final answers is taken for granted. Both facts must be earned. This section earns them. We construct, by an explicit Kronecker-product recursion, a $2^{d/2}$-dimensional representation in every even dimension d; we prove it is irreducible and that irreducibility forces precisely this dimension; and we prove Pauli's fundamental theorem, which states that any two solutions of the Clifford relation of this size are conjugate by a matrix S unique up to a scalar. Pauli's theorem is the workhorse of everything that follows: it is the machine that converts the bare statement "$\{\gamma'^{\mu}\}$ obeys the Clifford relation" into "there is a unique-up-to-scale S with $\gamma'^{\mu} = S\,\gamma^{\mu}S^{-1}$," and in Section C.3 and beyond every intertwining matrix—γ^0, the charge-conjugation matrix C, and the parity and time-reversal matrices—is nothing but the S of this theorem applied to a cleverly chosen second solution.

Throughout this section d is even, $d = 2n$, unless we explicitly turn to the odd case; we abbreviate the representation dimension as $N \equiv 2^n = 2^{d/2}$. We work over the complex numbers: the signature of the metric will turn out to be irrelevant to the questions of existence, dimension, and uniqueness, because complexifying the Clifford algebra erases the distinction between timelike and spacelike generators. The Pauli matrices $\sigma^1, \sigma^2, \sigma^3$ and their algebra are imported from Chapters 2 and B; Schur's lemma, in both of the forms we use below, is imported from Chapter 2.

C.2.1 Explicit construction by Kronecker products

It is cleanest to build a Euclidean set first—matrices Γ^a $(a = 1, \ldots, d)$ satisfying

$$\left\{\Gamma^a, \Gamma^b\right\} = 2\delta^{ab}\,\mathbf{1}, \tag{C.37}$$

so that each generator squares to $+\mathbf{1}$ and distinct generators anticommute—and to pass to the physical Lorentzian signature at the very end by inserting factors of i. The single ingredient is the Pauli algebra

$$\left\{\sigma^i, \sigma^j\right\} = 2\delta^{ij}\,\mathbf{1}_2, \qquad (\sigma^i)^2 = \mathbf{1}_2, \qquad \sigma^1\sigma^2 = i\sigma^3 \text{ (cyclic)}, \tag{C.38}$$

each σ^i being Hermitian and 2×2. The two matrices σ^1, σ^2 already furnish the case $d = 2$: they anticommute and square to the identity, so $\Gamma^1 = \sigma^1$, $\Gamma^2 = \sigma^2$ is a 2-dimensional $(N = 2^1)$ solution of Eq. (C.37). The reserved third matrix $\sigma^3 = -i\sigma^1\sigma^2$ is the chirality element in two dimensions; it anticommutes with both Γ^1, Γ^2 and squares to $\mathbf{1}_2$. This last property is exactly what lets us climb the ladder.

Theorem C.10 (Inductive Kronecker construction)**.** *For every even $d = 2n$ there exist d Hermitian matrices Γ^a of size $N \times N$, $N = 2^n$, satisfying the Euclidean Clifford relation Eq. (C.37). They are generated by the recursion*

$$\Gamma^a_{(n+1)} = \Gamma^a_{(n)} \otimes \sigma^3 \quad (a = 1, \dots, 2n), \qquad \Gamma^{2n+1}_{(n+1)} = \mathbf{1}_{2^n} \otimes \sigma^1, \qquad \Gamma^{2n+2}_{(n+1)} = \mathbf{1}_{2^n} \otimes \sigma^2, \tag{C.39}$$

started from $\Gamma^1_{(1)} = \sigma^1$, $\Gamma^2_{(1)} = \sigma^2$.

Proof. We verify that if $\{\Gamma^a_{(n)}\}_{a=1}^{2n}$ obey Eq. (C.37) in dimension 2^n, then the matrices Eq. (C.39) obey it in dimension 2^{n+1}. The Kronecker product multiplies blockwise, $(P \otimes Q)(P' \otimes Q') = PP' \otimes QQ'$, so anticommutators factorize whenever one tensor slot commutes through. We check the three types of pair.

Two old generators. For $a, b \le 2n$,

$$\left\{\Gamma^a_{(n)} \otimes \sigma^3, \Gamma^b_{(n)} \otimes \sigma^3\right\} = \left\{\Gamma^a_{(n)}, \Gamma^b_{(n)}\right\} \otimes (\sigma^3)^2 = 2\delta^{ab} \mathbf{1}_{2^n} \otimes \mathbf{1}_2 = 2\delta^{ab} \mathbf{1}_{2^{n+1}}, \tag{C.40}$$

where we used $(\sigma^3)^2 = \mathbf{1}_2$ and the induction hypothesis.

One old, one new. For $a \le 2n$ and either of the new generators $\mathbf{1} \otimes \sigma^{1,2}$,

$$\left\{\Gamma^a_{(n)} \otimes \sigma^3, \mathbf{1}_{2^n} \otimes \sigma^{1,2}\right\} = \Gamma^a_{(n)} \otimes \left\{\sigma^3, \sigma^{1,2}\right\} = \Gamma^a_{(n)} \otimes 0 = 0, \tag{C.41}$$

because σ^3 anticommutes with σ^1 and with σ^2.

Two new generators. The two new matrices commute through their first slot trivially and anticommute in the second,

$$\left\{\mathbf{1} \otimes \sigma^1, \mathbf{1} \otimes \sigma^2\right\} = \mathbf{1} \otimes \left\{\sigma^1, \sigma^2\right\} = 0, \qquad (\mathbf{1} \otimes \sigma^{1,2})^2 = \mathbf{1} \otimes (\sigma^{1,2})^2 = \mathbf{1}_{2^{n+1}}. \tag{C.42}$$

Thus all $2n + 2$ matrices anticommute pairwise and square to the identity: Eq. (C.37) holds in dimension 2^{n+1}. Hermiticity is preserved because $\sigma^1, \sigma^2, \sigma^3$ and the inductive $\Gamma^a_{(n)}$ are Hermitian and the Kronecker product of Hermitian matrices is Hermitian. The base case $d = 2$ was checked above, so the claim holds for all even d by induction. □

The dimension obeys $N(n) = 2N(n-1)$ with $N(1) = 2$, i.e. $N = 2^n = 2^{d/2}$, doubling with each pair of new dimensions. That this is also the *smallest* possible dimension is the content of Section C.2.2; for now we have established existence.

Worked Example C.11 (The 4×4 matrices in $d = 4$)**.** Running the recursion once, from $n = 1$ to $n = 2$, produces the four Euclidean Dirac matrices of four-dimensional spacetime. With $\Gamma^1_{(1)} = \sigma^1$, $\Gamma^2_{(1)} = \sigma^2$,

$$\Gamma^1 = \sigma^1 \otimes \sigma^3, \qquad \Gamma^2 = \sigma^2 \otimes \sigma^3, \qquad \Gamma^3 = \mathbf{1}_2 \otimes \sigma^1, \qquad \Gamma^4 = \mathbf{1}_2 \otimes \sigma^2, \tag{C.43}$$

four Hermitian 4×4 matrices with $\left\{\Gamma^a, \Gamma^b\right\} = 2\delta^{ab}\mathbf{1}_4$. Written as 2×2 blocks of 2×2 matrices (using $\sigma^3 = \mathrm{diag}(1, -1)$ to expand the first factor),

$$\Gamma^1 = \begin{pmatrix} 0 & \sigma^3 \\ \sigma^3 & 0 \end{pmatrix}, \quad \Gamma^2 = \begin{pmatrix} 0 & -i\sigma^3 \\ i\sigma^3 & 0 \end{pmatrix}, \quad \Gamma^3 = \begin{pmatrix} \sigma^1 & 0 \\ 0 & \sigma^1 \end{pmatrix}, \quad \Gamma^4 = \begin{pmatrix} \sigma^2 & 0 \\ 0 & \sigma^2 \end{pmatrix}. \tag{C.44}$$

To reach the mostly-minus Lorentzian Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$ with $\eta = \mathrm{diag}(+, -, -, -)$ we need one generator squaring to $+\mathbf{1}$ (timelike) and three squaring to $-\mathbf{1}$ (spacelike). Since every Γ^a here is Hermitian with $(\Gamma^a)^2 = +\mathbf{1}$, set

$$\gamma^0 = \Gamma^4, \qquad \gamma^1 = i\,\Gamma^1, \qquad \gamma^2 = i\,\Gamma^2, \qquad \gamma^3 = i\,\Gamma^3. \tag{C.45}$$

Then $(\gamma^0)^2 = (\Gamma^4)^2 = +\mathbf{1}$, matching $\eta^{00} = +1$, while $(\gamma^j)^2 = i^2(\Gamma^a)^2 = -\mathbf{1}$ for the three spatial slots, matching $\eta^{jj} = -1$; and for $\mu \neq \nu$ the anticommutator inherits the vanishing of the Euclidean one, $\{\gamma^\mu, \gamma^\nu\} = (\pm i)(\pm i)\left\{\Gamma^a, \Gamma^b\right\} = 0$. Hence $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}_4$. We have produced, with our bare hands, an explicit 4×4 representation. We deliberately do *not* call this "the Dirac representation": the assignment Eq. (C.45) is one of infinitely many, and the physically named bases (Dirac, chiral, Majorana) are pinned down in Section C.3. What matters here is only that a 4×4 solution exists.

C.2.2 Tracelessness, completeness, and irreducibility

We now show that the constructed representation cannot be split into smaller pieces, and that no smaller faithful realization is possible. The argument turns on two elementary facts about the product basis $\{\Gamma_A\}$, where A ranges over the 2^d ordered subsets of $\{0, \ldots, d-1\}$ and $\Gamma_\varnothing = \mathbf{1}$. The first is that every nontrivial basis element is traceless; the second is that the basis elements span the entire matrix algebra. Recall from Section C.1 that the Γ_A close under multiplication up to a sign, $\Gamma_A\Gamma_B = \pm\Gamma_C$, that each satisfies $\Gamma_A^2 = \pm\mathbf{1}$ so that $\Gamma_A^{-1} = \pm\Gamma_A$, and that $\Gamma_A\Gamma_B = \pm\mathbf{1}$ precisely when $B = A$ (the product is the identity only when the two index sets coincide).

Lemma C.12 (Tracelessness of nontrivial basis elements)**.** *For even d and any $A \neq \varnothing$, there is a generator γ^ν with $\gamma^\nu\Gamma_A\,(\gamma^\nu)^{-1} = -\Gamma_A$; consequently* $\mathrm{Tr}\,\Gamma_A = 0$.

Proof. Let $\Gamma_A = \gamma^{\mu_1} \cdots \gamma^{\mu_k}$ be a product of $k \geq 1$ distinct generators. Moving a generator γ^ν past Γ_A costs a sign (-1) for each factor it anticommutes with. If $\nu \notin \{\mu_1, \ldots, \mu_k\}$ then γ^ν anticommutes with all k factors, so $\gamma^\nu\Gamma_A = (-1)^k\Gamma_A\gamma^\nu$; if $\nu = \mu_j$ for some j, it anticommutes with the other $k-1$ factors and commutes with γ^{μ_j} only up to its square, giving $\gamma^\nu\Gamma_A = (-1)^{k-1}\Gamma_A\gamma^\nu$. We need a ν producing an overall minus sign.

If k is odd, choose any $\nu \notin \{\mu_i\}$. Such a ν exists because $k < d$: a product of all d distinct generators has $k = d$, which is even, so an odd-length

product never exhausts the index set. Then $\gamma^\nu \Gamma_A = (-1)^k \Gamma_A \gamma^\nu = -\Gamma_A \gamma^\nu$. If k is even, choose $\nu = \mu_1 \in \{\mu_i\}$; then $\gamma^\nu \Gamma_A = (-1)^{k-1} \Gamma_A \gamma^\nu = -\Gamma_A \gamma^\nu$. In either case $\gamma^\nu \Gamma_A (\gamma^\nu)^{-1} = -\Gamma_A$. Taking the trace and using cyclicity, $\operatorname{Tr} \Gamma_A = \operatorname{Tr}\big(\gamma^\nu \Gamma_A (\gamma^\nu)^{-1}\big) = \operatorname{Tr}(-\Gamma_A) = -\operatorname{Tr} \Gamma_A$, hence $\operatorname{Tr} \Gamma_A = 0$. □

The reliance on d even is not cosmetic. When d is odd, the top product $\gamma^0 \cdots \gamma^{d-1}$ has odd length but its index set *is* exhausted, so the construction in the lemma fails for it—and indeed that element is central, not traceless. This is the algebraic root of the two-irreducible-representation phenomenon in odd dimensions, to which we return in Section C.2.4.

Lemma C.13 (Linear independence and completeness)**.** *In the N-dimensional representation of Theorem C.10 the 2^d matrices $\{\Gamma_A\}$ are linearly independent. Since $N^2 = (2^n)^2 = 2^{2n} = 2^d$, they therefore form a basis of the full matrix algebra $M_N(\mathbb{C})$.*

Proof. Suppose $\sum_A c_A \Gamma_A = 0$. Fix any index B, multiply on the right by Γ_B^{-1}, and take the trace:

$$0 = \operatorname{Tr}\Big(\sum_A c_A \, \Gamma_A \Gamma_B^{-1} \Big) = \sum_A c_A \, \operatorname{Tr}(\Gamma_A \Gamma_B^{-1}). \tag{C.46}$$

Now $\Gamma_A \Gamma_B^{-1} = \pm \Gamma_C$ for some C, and $C = \varnothing$ (so $\Gamma_A \Gamma_B^{-1} = \pm \mathbf{1}$) if and only if $A = B$. By Theorem C.12, $\operatorname{Tr} \Gamma_C = 0$ whenever $C \neq \varnothing$, so every term with $A \neq B$ drops out, leaving $c_B \operatorname{Tr}(\pm \mathbf{1}) = \pm c_B N = 0$. Since $N \neq 0$, $c_B = 0$. As B was arbitrary, all coefficients vanish and the Γ_A are independent. They are $2^d = N^2$ independent elements of the N^2-dimensional space $M_N(\mathbb{C})$, hence a basis of it. □

Theorem C.13 is the structural heart of the even-dimensional theory: *the matrices generated by the gamma matrices fill out all of $M_N(\mathbb{C})$*. Equivalently, the map from the abstract 2^d-dimensional Clifford algebra of Section C.1 to $M_N(\mathbb{C})$ sending Γ_A to its constructed image is a surjection between spaces of equal dimension 2^d, hence an isomorphism. The even-dimensional Clifford algebra is therefore a full matrix algebra—in particular it is *simple* (it has no two-sided ideals other than $\{0\}$ and itself, since $M_N(\mathbb{C})$ does not). Both observations feed directly into irreducibility and uniqueness.

Theorem C.14 (Irreducibility and forced dimension)**.** *The representation of Theorem C.10 is irreducible. Moreover, every irreducible representation of the even-d Clifford algebra has dimension exactly $N = 2^{d/2}$.*

Proof. Irreducibility of the constructed representation. By Schur's lemma over $\mathbb{C}$ (Chapter 2), a finite-dimensional complex representation is irreducible if and only if its commutant is trivial, i.e. the only matrices commuting with every

representing matrix are scalar multiples of $\mathbf{1}$. Let M commute with all γ^μ. Then M commutes with every product Γ_A, hence with their entire span, which by Theorem C.13 is all of $M_N(\mathbb{C})$. A matrix commuting with every element of $M_N(\mathbb{C})$ lies in its center, and the center of a full matrix algebra consists exactly of the scalars; thus $M = \lambda\mathbf{1}$. The commutant is trivial, so the representation is irreducible.

Forced dimension. Let ρ be *any* irreducible representation, of some dimension N', of the even-d Clifford algebra. Because that algebra is simple (the paragraph following Theorem C.13), ρ is faithful: its kernel is a two-sided ideal, and a nonzero representation cannot have kernel equal to the whole algebra, so the kernel is $\{0\}$. Faithfulness means the images $\rho(\Gamma_A)$ are again linearly independent—the proof of Theorem C.13 used only the Clifford relations and tracelessness, both of which are preserved by any faithful representation[1]—so $2^d \le (N')^2$. For the reverse inequality we use that ρ is irreducible: by Schur the commutant of ρ is trivial, and an irreducibly-acting algebra of matrices on a finite-dimensional complex space spans the full endomorphism algebra (Burnside's theorem, Chapter 2). The algebra spanned by $\{\rho(\Gamma_A)\}$ is closed under multiplication (products of Γ_A are $\pm\Gamma_C$) and acts irreducibly, so its span is all of $M_{N'}(\mathbb{C})$; but that span has dimension at most 2^d, giving $(N')^2 \le 2^d$. Combining, $(N')^2 = 2^d$ and $N' = 2^{d/2} = N$. □

Thus in even dimension there is a single irreducible size, $2^{d/2}$, and in $d = 4$ this is the familiar 4. We close the subsection by recording the averaging identity that the irreducibility argument secretly contains; it is the same "sum over the group of Γ's" device that powers Pauli's theorem below and the Fierz machinery of Section C.8.

Worked Example C.15 (The completeness average)**.** For any $N \times N$ matrix X define the average $\Phi(X) = \sum_A \Gamma_A X \Gamma_A^{-1}$, the sum running over all 2^d basis elements. We claim

$$\Phi(X) = \sum_A \Gamma_A X \Gamma_A^{-1} = N\,(\operatorname{Tr} X)\,\mathbf{1}. \tag{C.47}$$

First, $\Phi(X)$ commutes with every Γ_B: conjugating term by term,

$$\Gamma_B\,\Phi(X)\,\Gamma_B^{-1} = \sum_A \Gamma_B\Gamma_A\,X\,\Gamma_A^{-1}\Gamma_B^{-1} = \sum_A (\Gamma_B\Gamma_A)\,X\,(\Gamma_B\Gamma_A)^{-1} = \sum_{A'} \Gamma_{A'}\,X\,\Gamma_{A'}^{-1} = \Phi(X), \tag{C.48}$$

because left multiplication by Γ_B permutes the Γ_A up to signs that cancel between $\Gamma_B\Gamma_A$ and its inverse, so $A \mapsto A'$ is a bijection of the index set. Since $\Phi(X)$ commutes with every Γ_B, hence with all of $M_N(\mathbb{C})$ (Theorem C.13), Schur

[1] Tracelessness in ρ follows verbatim from Theorem C.12, whose argument is purely algebraic; the trace there is the trace of $N' \times N'$ matrices and the conclusion $\operatorname{Tr}\rho(\Gamma_A) = 0$ for $A \neq \varnothing$ stands.

gives $\Phi(X) = \phi(X)\,\mathbf{1}$ for a scalar $\phi(X)$. Taking the trace fixes the scalar: $\operatorname{Tr}\Phi(X) = \sum_A \operatorname{Tr}(\Gamma_A X \Gamma_A^{-1}) = \sum_A \operatorname{Tr} X = 2^d \operatorname{Tr} X$, while $\operatorname{Tr}(\phi(X)\mathbf{1}) = N\phi(X)$, so $\phi(X) = 2^d \operatorname{Tr} X/N = N \operatorname{Tr} X$ using $2^d = N^2$. This proves Eq. (C.47). The same identity, written componentwise, is the statement $\sum_A (\Gamma_A)_{ij}(\Gamma_A^{-1})_{kl} = N\,\delta_{il}\delta_{kj}$, the completeness relation for the basis $\{\Gamma_A\}$ that we will reuse repeatedly.

C.2.3 Pauli's fundamental theorem

We can now prove the result on which the rest of the appendix rests. Two physicists who each write down four matrices obeying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$ in dimension 4 will in general write down *different* matrices—one in the Dirac basis, one in the chiral basis, one in some basis of their own. Pauli's theorem says these are never genuinely different: they are conjugate, and the conjugating matrix is essentially unique. This is what guarantees that a trace identity proved in one basis holds in all, and it is the abstract template every later intertwiner instantiates.

Theorem C.16 (Pauli's fundamental theorem)**.** *Let $\{\gamma^\mu\}$ and $\{\gamma'^\mu\}$, $\mu = 0, \dots, d-1$ with d even, be two sets of $N \times N$ matrices ($N = 2^{d/2}$) each satisfying $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$. Then there exists an invertible matrix S with*

$$\gamma'^\mu = S\,\gamma^\mu\,S^{-1}, \qquad \mu = 0, \dots, d-1, \tag{C.49}$$

and S is unique up to a nonzero complex scalar.

Proof. Form the two product bases $\{\Gamma_A\}$ and $\{\Gamma'_A\}$ from the unprimed and primed generators using the *same* ordered index sets A. Because both sets obey identical Clifford relations, the combinatorics of their products is identical: for each generator index μ and each A,

$$\gamma^\mu\,\Gamma_A = \eta_A\,\Gamma_{B(A)} \qquad \Longleftrightarrow \qquad \gamma'^\mu\,\Gamma'_A = \eta_A\,\Gamma'_{B(A)}, \tag{C.50}$$

with the *same* sign $\eta_A \in \{\pm 1\}$ and the *same* reindexing $A \mapsto B(A)$, because the sign and the resulting index depend only on which generators appear in Γ_A and on the anticommutation rule—data shared by both sets. For fixed μ the map $A \mapsto B(A)$ is a bijection of the index set: left multiplication by the invertible γ^μ permutes the basis up to sign, so it has an inverse.

The averaging operator. For an arbitrary $N \times N$ matrix F define

$$S \equiv \sum_A \Gamma'_A\,F\,\Gamma_A^{-1}. \tag{C.51}$$

We claim that for *every* F this S intertwines the two sets: $\gamma'^\mu\,S = S\,\gamma^\mu$ for all μ. Compute, using Eq. (C.50) on the left factor and its consequence $\Gamma_A^{-1} =$

$\eta_A \Gamma^{-1}_{B(A)} \gamma^\mu$ on the right factor (which follows by inverting $\gamma^\mu \Gamma_A = \eta_A \Gamma_{B(A)}$ and using $\eta_A^2 = 1$):

$$\begin{aligned} \gamma'^\mu S &= \sum_A (\gamma'^\mu \Gamma'_A) F \Gamma_A^{-1} = \sum_A \eta_A \Gamma'_{B(A)} F \Gamma_A^{-1} \\ &= \sum_A \eta_A \Gamma'_{B(A)} F (\eta_A \Gamma^{-1}_{B(A)} \gamma^\mu) = \sum_A \eta_A^2 \Gamma'_{B(A)} F \Gamma^{-1}_{B(A)} \gamma^\mu \\ &= \Big(\sum_B \Gamma'_B F \Gamma_B^{-1} \Big) \gamma^\mu = S \gamma^\mu, \end{aligned} \tag{C.52}$$

where in the penultimate step we used that $A \mapsto B(A)$ is a bijection to relabel the sum. Thus Eq. (C.51) produces an intertwiner for any choice of F.

Some choice of F makes $S \neq 0$. View the assignment $F \mapsto S$ as the linear map $\sum_A \Gamma'_A (\cdot) \Gamma_A^{-1}$, whose kernel-or-not is governed by the tensor $\mathcal{T} = \sum_A \Gamma'_A \otimes (\Gamma_A^{-1})^\top \in M_N(\mathbb{C}) \otimes M_N(\mathbb{C})$; indeed $S = 0$ for all F would force $\mathcal{T} = 0$. But $\mathcal{T} \neq 0$. The matrices $\{\Gamma'_A\}$ are linearly independent (Theorem C.13 applied to the primed set), and so are $\{(\Gamma_A^{-1})^\top\} = \{\pm(\Gamma_A)^\top\}$ (transpose and sign preserve independence). For two families with $\{X_A\}$ independent and $\{Y_A\}$ independent, the "diagonal" tensors $\{X_A \otimes Y_A\}$ are themselves linearly independent: if $\sum_A c_A X_A \otimes Y_A = 0$, apply any linear functional ψ to the second factor to get $\sum_A c_A \psi(Y_A) X_A = 0$, whence $c_A \psi(Y_A) = 0$ for every A and every ψ by independence of the X_A; choosing for each A a ψ with $\psi(Y_A) \neq 0$ (possible since $Y_A \neq 0$) forces $c_A = 0$. Hence the 2^d tensors $\Gamma'_A \otimes (\Gamma_A^{-1})^\top$ are independent and their equal-coefficient sum $\mathcal{T}$ is nonzero. Therefore $S \neq 0$ for some F; fix such an F.

S is invertible. Both $\{\gamma^\mu\}$ and $\{\gamma'^\mu\}$ generate irreducible representations of dimension $N = 2^{d/2}$, by Theorem C.14. The relation Eq. (C.52) says S is a nonzero intertwiner between these two irreducible representations. By Schur's lemma (Chapter 2), a nonzero intertwiner between irreducible representations is an isomorphism, so S is invertible. Multiplying Eq. (C.52) on the right by S^{-1} yields Eq. (C.49).

Uniqueness up to scale. Suppose S and $\widetilde{S}$ both satisfy Eq. (C.49). Then $\gamma^\mu = S^{-1}\gamma'^\mu S = \widetilde{S}^{-1}\gamma'^\mu \widetilde{S}$, so $\widetilde{S}S^{-1}$ commutes with every γ'^μ. By the irreducibility of $\{\gamma'^\mu\}$ and Schur's lemma over $\mathbb{C}$, $\widetilde{S}S^{-1} = \lambda \mathbf{1}$ for a scalar λ, necessarily nonzero since $\widetilde{S}$ is invertible. Hence $\widetilde{S} = \lambda S$: the intertwiner is unique up to a nonzero multiplicative constant. □

Corollary C.17 (Reality of the scale and a canonical normalization)**.** *The residual freedom $S \to \lambda S$ in Pauli's theorem is exactly one complex scalar. Whenever a later construction supplements Eq. (C.49) with one further algebraic condition on S that is homogeneous of nonzero degree (for instance $S^\dagger S = \mathbf{1}$, fixing $|\lambda|$, or $S^2 = \pm\mathbf{1}$, or a prescribed value of a single matrix element), the scalar is pinned down up to a finite phase or sign. This is the mechanism by*

which γ^0, the charge-conjugation matrix C, and the discrete-symmetry matrices of Section C.4 acquire definite normalizations: each is the Pauli S for a specific second solution of the Clifford relation, made unique by one supplementary normalization.

Proof. The freedom is one scalar by Theorem C.16. A condition homogeneous of degree $k \neq 0$ in S becomes λ^k (or $\bar{\lambda}^j \lambda^k$ for mixed conjugate-degree conditions) times the corresponding quantity for the reference S; setting it to a fixed value is a polynomial equation in λ (or in λ and $\bar{\lambda}$) with finitely many solutions, leaving at most a finite phase/sign ambiguity. □

Caution

Pauli's theorem is a statement about the *full* set of d matrices satisfying the *same* Clifford relation, in the *irreducible* dimension. If two sets satisfy Clifford relations with opposite overall metric sign (one set with $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, the other with $-2\eta^{\mu\nu}$), they are generally *not* similar in even d, since $\mathrm{Tr}(\gamma^\mu \gamma^\nu) \propto \eta^{\mu\nu}$ is similarity-invariant and would flip sign. Such cases are handled not by similarity but by combining similarity with transposition or complex conjugation—precisely the loophole that defines the charge-conjugation matrix in Section C.4. Likewise, applying the theorem in a reducible dimension is illegitimate: there the intertwiner need not be unique, and the averaging operator can vanish identically.

C.2.4 The odd-dimensional case

For completeness, and to make precise why the physical case $d = 4$ enjoys a single irreducible representation, we record what changes in odd dimension. Let $d = 2n + 1$. Take the $2n$ even-dimensional generators $\gamma^0, \ldots, \gamma^{2n-1}$ of dimension $N = 2^n$ from Theorem C.10 (in the appropriate signature) and adjoin the chirality element

$$\gamma^{2n} \;\propto\; \gamma^0 \gamma^1 \cdots \gamma^{2n-1}, \tag{C.53}$$

normalized so that $(\gamma^{2n})^2 = \pm \mathbf{1}$ matches the required metric sign. As shown in Section C.1, this element anticommutes with each of $\gamma^0, \ldots, \gamma^{2n-1}$ and squares to a scalar, so $\{\gamma^0, \ldots, \gamma^{2n}\}$ obeys the $(2n+1)$-dimensional Clifford relation in dimension $N = 2^n = 2^{(d-1)/2}$. Two features distinguish this from the even case.

The top product is central. In dimension $d = 2n + 1$ the longest product $\gamma^0 \gamma^1 \cdots \gamma^{2n}$ contains *all* generators, an odd number of them. Moving any single γ^ν through it costs $2n$ sign flips (one for each of the other factors), an even number, so γ^ν commutes with the full product. The top product is thus central; by Schur it acts as a scalar in any irreducible representation. That scalar is fixed by the Clifford relations up to a sign—the two roots of $(\gamma^0 \cdots \gamma^{2n})^2 = \mathrm{const} \cdot \mathbf{1}$.

Two inequivalent irreducible representations. The two sign choices in Eq. (C.53)—equivalently, the two values $\pm$ that the central top product may take—give representations that are *not* related by any similarity, because the value of a central element is a similarity invariant. Concretely, $\{\gamma^0, \ldots, \gamma^{2n-1}, +\gamma^{2n}\}$ and $\{\gamma^0, \ldots, \gamma^{2n-1}, -\gamma^{2n}\}$ have central elements of opposite sign and so cannot satisfy $\gamma'^{\mu} = S\gamma^{\mu}S^{-1}$ for all μ simultaneously (the relation would force the central elements, hence their scalar values, to agree). Each of the two is irreducible of dimension $2^{(d-1)/2}$, by the same completeness-and-Schur argument as in the even case applied to the $2n$ traceless generators. Pauli's theorem therefore holds in odd d only *within* each of the two classes; across classes the best one can do is relate $\{\gamma^{\mu}\}$ to $\{-\gamma^{\mu}\}$ or invoke transposition.

This is exactly the structural reason the even case is cleaner. In even d the top product $\gamma^0 \cdots \gamma^{d-1} \propto \gamma^5$ has *even* length and *anticommutes* with every generator (each γ^{ν} flips $d-1$ signs, an odd number); it is therefore *not* central, the algebra is the simple matrix algebra $M_{2^{d/2}}(\mathbb{C})$, and there is a unique irreducible representation. For $d = 4$ this is the single 4-dimensional representation that all of relativistic field theory uses, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ is the noncentral chirality operator whose anticommuting character makes the chiral projectors of Section C.3 possible.

With existence settled by the explicit 4×4 construction of Theorem C.11 and uniqueness settled by Pauli's theorem (Theorem C.16), we have the two tools the next section needs. Section C.3 writes down the Dirac, chiral, and Majorana representations as three specific solutions of the same Clifford relation; by Pauli's theorem each is conjugate to the others through a definite S, and the Hermiticity relation $\gamma^{\mu\dagger} = \gamma^0\gamma^{\mu}\gamma^0$ is itself an instance of Eq. (C.49) with $\gamma'^{\mu} = \gamma^{\mu\dagger}$—the first of the intertwining matrices that this theorem exists to deliver.

C.3 The Standard Representations: Dirac, Chiral, and Majorana

The previous section settled the structural question once and for all: in $d = 4$ the Clifford algebra $\{\gamma^{\mu}, \gamma^{\nu}\} = 2\eta^{\mu\nu} I_4$ has, up to similarity, exactly one irreducible representation, and it is four-dimensional (Section C.2). Pauli's fundamental theorem (Theorem C.16) tells us more: any two sets of 4×4 matrices obeying the same anticommutation relations are conjugate by a matrix that is unique up to an overall scale. The practical consequence is liberating. We are free to write down whatever explicit matrices are most convenient for the calculation at hand, secure in the knowledge that they all describe *the same* algebra and that every basis-independent statement—a trace, a chirality assignment, a spin sum—comes out the same in all of them. Different problems reward different choices: nonrelativistic reductions and the Dirac equation's plane-wave solutions

are cleanest in the *standard* (Dirac) basis; the splitting into left- and right-handed pieces, and with it the entire $V-A$ structure of the weak interaction and of QED, is manifest only in the *chiral* (Weyl) basis; the reality of the Dirac equation, and hence the Majorana condition, is laid bare by the *Majorana* basis. This section writes down all three, proves each satisfies the Clifford relation by direct matrix multiplication, exhibits the explicit unitary similarities promised by Pauli's theorem that carry one into another, builds γ^5 and the chirality projectors, fixes the Hermiticity convention $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ that every later bilinear computation rests on, and closes the loop with Chapter 3 by showing that in the chiral basis the Lorentz generators split into precisely the $(\tfrac{1}{2},0)$ and $(0,\tfrac{1}{2})$ blocks of the abstract classification.

Throughout, I denotes the 2×2 identity, I_4 the 4×4 identity, and $\mathbf{0}$ a 2×2 block of zeros; all explicit matrices are written in 2×2 block form. The Pauli matrices are imported from Chapters 2 and B in the standard convention

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{C.54}$$

with the algebra we shall use repeatedly,

$$\sigma^i\sigma^j = \delta^{ij} I + i\,\epsilon^{ijk}\sigma^k, \qquad \Longrightarrow \qquad \left\{\sigma^i,\sigma^j\right\} = 2\delta^{ij} I, \quad \left[\sigma^i,\sigma^j\right] = 2i\,\epsilon^{ijk}\sigma^k, \tag{C.55}$$

and the Hermiticity $\sigma^{i\dagger} = \sigma^i$ together with $(\sigma^i)^2 = I$. Recall (Chapter A) the metric is $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, so that the Clifford relation reads, component by component, $(\gamma^0)^2 = I_4$, $(\gamma^i)^2 = -I_4$, and $\{\gamma^\mu,\gamma^\nu\} = 0$ for $\mu \neq \nu$.

C.3.1 The Dirac (standard) representation

The historically first and still most widely used explicit choice is the one in which γ^0 is diagonal; it diagonalizes the rest mass, which is why nonrelativistic expansions are transparent in it.

Definition C.18 (Dirac representation)**.** The *Dirac* or *standard* representation is

$$\gamma^0 = \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}, \qquad i = 1,2,3. \tag{C.56}$$

Proposition C.19. *The matrices* (C.56) *satisfy* $\{\gamma^\mu,\gamma^\nu\} = 2\eta^{\mu\nu} I_4$.

Proof. We compute the three independent cases by block multiplication. First,

$$(\gamma^0)^2 = \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix}^2 = \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix} = I_4 = \eta^{00} I_4.$$

Next the mixed time–space anticommutator:

$$\gamma^0\gamma^i = \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \sigma^i \\ \sigma^i & \mathbf{0} \end{pmatrix},$$
$$\gamma^i\gamma^0 = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}\begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix} = \begin{pmatrix} \mathbf{0} & -\sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix},$$

whose sum is $\{\gamma^0, \gamma^i\} = \mathbf{0} = 2\eta^{0i} I_4$. Finally the spatial–spatial product,

$$\gamma^i\gamma^j = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^j \\ -\sigma^j & \mathbf{0} \end{pmatrix} = \begin{pmatrix} -\sigma^i\sigma^j & \mathbf{0} \\ \mathbf{0} & -\sigma^i\sigma^j \end{pmatrix},$$

so, using $\{\sigma^i, \sigma^j\} = 2\delta^{ij} I$ from (C.55),

$$\left\{\gamma^i, \gamma^j\right\} = \begin{pmatrix} -\{\sigma^i, \sigma^j\} & \mathbf{0} \\ \mathbf{0} & -\{\sigma^i, \sigma^j\} \end{pmatrix} = -2\delta^{ij} I_4 = 2\eta^{ij} I_4,$$

since $\eta^{ij} = -\delta^{ij}$. The three cases together are exactly $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$. □

C.3.2 The chiral (Weyl) representation

The chiral basis differs from the Dirac basis only in γ^0 (the γ^i are identical), but that single change is exactly what is needed to make γ^5 diagonal and the handedness of a spinor a property of its upper versus lower components. It is the natural home for massless fermions and for the whole of QED's $V-A$ bookkeeping.

Definition C.20 (Chiral representation)**.** The *chiral* or *Weyl* representation is

$$\gamma^0 = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix}, \qquad \gamma^i = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}. \tag{C.57}$$

It is compactly written by introducing the two-component objects

$$\sigma^\mu \equiv (I,\ \sigma^i), \qquad \bar\sigma^\mu \equiv (I,\ -\sigma^i), \qquad \text{so that} \qquad \gamma^\mu = \begin{pmatrix} \mathbf{0} & \sigma^\mu \\ \bar\sigma^\mu & \mathbf{0} \end{pmatrix}. \tag{C.58}$$

The symbols $\sigma^\mu, \bar\sigma^\mu$ are the seeds of the van der Waerden two-component calculus developed in full in Section C.9; here we use them only as packaging.

Proposition C.21. *The matrices* (C.57) *satisfy* $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$.

Proof. With the packaging (C.58), a single block multiplication does all four-by-four cases at once:

$$\gamma^\mu\gamma^\nu = \begin{pmatrix} \mathbf{0} & \sigma^\mu \\ \bar\sigma^\mu & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^\nu \\ \bar\sigma^\nu & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \sigma^\mu\bar\sigma^\nu & \mathbf{0} \\ \mathbf{0} & \bar\sigma^\mu\sigma^\nu \end{pmatrix},$$

so that $\{\gamma^\mu,\gamma^\nu\}$ is block diagonal with upper block $\sigma^\mu\bar\sigma^\nu+\sigma^\nu\bar\sigma^\mu$ and lower block $\bar\sigma^\mu\sigma^\nu+\bar\sigma^\nu\sigma^\mu$. We show both equal $2\eta^{\mu\nu}I$. For $\mu=\nu=0$ both read $2I=2\eta^{00}I$. For $\mu=0,\nu=i$ the upper block is $\sigma^0\bar\sigma^i+\sigma^i\bar\sigma^0=(-\sigma^i)+\sigma^i=\mathbf{0}$, and likewise the lower; this is $2\eta^{0i}I$. For $\mu=i,\nu=j$ the upper block is

$$\sigma^i\bar\sigma^j+\sigma^j\bar\sigma^i=(-\sigma^i\sigma^j)+(-\sigma^j\sigma^i)=-\left\{\sigma^i,\sigma^j\right\}=-2\delta^{ij}I=2\eta^{ij}I,$$

and the lower block $\bar\sigma^i\sigma^j+\bar\sigma^j\sigma^i=-\{\sigma^i,\sigma^j\}=2\eta^{ij}I$ identically. Hence $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}I_4$. □

The block identities just used,

$$\sigma^\mu\bar\sigma^\nu+\sigma^\nu\bar\sigma^\mu=2\eta^{\mu\nu}I,\qquad \bar\sigma^\mu\sigma^\nu+\bar\sigma^\nu\sigma^\mu=2\eta^{\mu\nu}I, \tag{C.59}$$

are the two-component shadow of the Clifford relation and will reappear as the workhorse of Section C.9.

C.3.3 The Majorana representation

The Dirac operator $i\gamma^\mu\partial_\mu-m$ has, generically, complex matrix entries, so its solutions are complex even when no charge is present. There exists, however, a basis in which *every* γ^μ is purely imaginary; then $i\gamma^\mu\partial_\mu-m$ is a real operator, the Dirac equation admits real solutions $\psi^*=\psi$, and a fermion can be its own antiparticle. This is the Majorana basis. Its very existence is a statement about the Clifford algebra in $d=4$, and it is the cleanest setting in which to discuss the reality (Majorana) condition systematically in Section C.4.

Definition C.22 (Majorana representation)**.** The *Majorana* representation is

$$\gamma^0=\begin{pmatrix}\mathbf{0}&\sigma^2\\ \sigma^2&\mathbf{0}\end{pmatrix},\quad \gamma^1=\begin{pmatrix}i\sigma^3&\mathbf{0}\\ \mathbf{0}&i\sigma^3\end{pmatrix},\quad \gamma^2=\begin{pmatrix}\mathbf{0}&-\sigma^2\\ \sigma^2&\mathbf{0}\end{pmatrix},\quad \gamma^3=\begin{pmatrix}-i\sigma^1&\mathbf{0}\\ \mathbf{0}&-i\sigma^1\end{pmatrix}. \tag{C.60}$$

Because σ^2 is purely imaginary while σ^1,σ^3 are real, every entry of every γ^μ in (C.60) is purely imaginary: $\gamma^{\mu*}=-\gamma^\mu$.

That all four are imaginary is the defining feature; that they satisfy the Clifford relation needs the same direct check as before. Because the bookkeeping is slightly heavier, we promote it to a worked example so that not a single anticommutator is taken on faith.

Worked Example C.23 (The Majorana matrices satisfy the Clifford relation)**.** We verify $\{\gamma^\mu,\gamma^\nu\}=2\eta^{\mu\nu}I_4$ for (C.60). Using $(\sigma^a)^2=I$ for the diagonal squares,

$$(\gamma^0)^2=\begin{pmatrix}\mathbf{0}&\sigma^2\\ \sigma^2&\mathbf{0}\end{pmatrix}^2=\begin{pmatrix}(\sigma^2)^2&\mathbf{0}\\ \mathbf{0}&(\sigma^2)^2\end{pmatrix}=I_4=\eta^{00}I_4,$$

$$(\gamma^1)^2 = \begin{pmatrix} (i\sigma^3)^2 & \mathbf{0} \\ \mathbf{0} & (i\sigma^3)^2 \end{pmatrix} = -I_4 = \eta^{11} I_4,$$
$$(\gamma^2)^2 = \begin{pmatrix} -(\sigma^2)^2 & \mathbf{0} \\ \mathbf{0} & -(\sigma^2)^2 \end{pmatrix} = -I_4 = \eta^{22} I_4,$$
$$(\gamma^3)^2 = \begin{pmatrix} (-i\sigma^1)^2 & \mathbf{0} \\ \mathbf{0} & (-i\sigma^1)^2 \end{pmatrix} = -I_4 = \eta^{33} I_4,$$

so the diagonal of the Clifford relation is the metric. For the six off-diagonal anticommutators we use $\left\{\sigma^a, \sigma^b\right\} = 2\delta^{ab} I$, i.e. distinct Pauli matrices anticommute. The pairs that mix an off-diagonal block (γ^0, γ^2) with a diagonal block (γ^1, γ^3) give, e.g.,

$$\gamma^0\gamma^1 = \begin{pmatrix} \mathbf{0} & i\sigma^2\sigma^3 \\ i\sigma^2\sigma^3 & \mathbf{0} \end{pmatrix}, \qquad \gamma^1\gamma^0 = \begin{pmatrix} \mathbf{0} & i\sigma^3\sigma^2 \\ i\sigma^3\sigma^2 & \mathbf{0} \end{pmatrix},$$

whose sum carries $i\{\sigma^2, \sigma^3\} = \mathbf{0}$; identically $\{\gamma^0, \gamma^3\}$ carries $-i\{\sigma^2, \sigma^1\} = \mathbf{0}$, $\{\gamma^2, \gamma^1\}$ carries $i\{\sigma^3, \sigma^2\} = \mathbf{0}$ (with a relative sign between the two off-diagonal blocks that does not affect their vanishing), and $\{\gamma^2, \gamma^3\}$ carries $i\{\sigma^2, \sigma^1\} = \mathbf{0}$. The two pairs within a single block type give

$$\left\{\gamma^0, \gamma^2\right\} = \begin{pmatrix} \sigma^2(-\sigma^2) + (-\sigma^2)\sigma^2 & \mathbf{0} \\ \mathbf{0} & (\sigma^2)^2 - (\sigma^2)^2 \end{pmatrix} = \mathbf{0},$$
$$\left\{\gamma^1, \gamma^3\right\} = \begin{pmatrix} \{i\sigma^3, -i\sigma^1\} & \mathbf{0} \\ \mathbf{0} & \{i\sigma^3, -i\sigma^1\} \end{pmatrix} = \begin{pmatrix} \{\sigma^3, \sigma^1\} & \mathbf{0} \\ \mathbf{0} & \{\sigma^3, \sigma^1\} \end{pmatrix} = \mathbf{0}.$$

All six off-diagonal anticommutators vanish, so $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$, and the representation is genuine.

C.3.4 The unitary equivalences

By Theorem C.16 the three representations are similar; because they are all built from Hermitian or anti-Hermitian blocks, the connecting similarities can be chosen unitary, which is what one wants physically (a change of spinor basis should preserve the inner product $\psi^\dagger\psi$). We exhibit the similarities explicitly rather than merely assert them.

Proposition C.24 (Dirac $\leftrightarrow$ chiral)**.** *With the unitary matrix*

$$U = \frac{1}{\sqrt{2}} \begin{pmatrix} I & -I \\ I & I \end{pmatrix}, \qquad U^{-1} = U^\dagger = \frac{1}{\sqrt{2}} \begin{pmatrix} I & I \\ -I & I \end{pmatrix}, \tag{C.61}$$

one has $\gamma^\mu_{\text{chiral}} = U\,\gamma^\mu_{\text{Dirac}}\,U^{-1}$.

Proof. Unitarity is the computation $UU^\dagger = \frac{1}{2}\left(\begin{smallmatrix} I & -I \\ I & I \end{smallmatrix}\right)\left(\begin{smallmatrix} I & I \\ -I & I \end{smallmatrix}\right) = \frac{1}{2}\left(\begin{smallmatrix} 2I & \mathbf{0} \\ \mathbf{0} & 2I \end{smallmatrix}\right) = I_4$. For γ^0,

$$U\gamma^0_{\text{Dirac}}U^{-1} = \frac{1}{2}\begin{pmatrix} I & -I \\ I & I \end{pmatrix}\begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix}\begin{pmatrix} I & I \\ -I & I \end{pmatrix} = \frac{1}{2}\begin{pmatrix} I & -I \\ I & I \end{pmatrix}\begin{pmatrix} I & I \\ I & -I \end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix} \mathbf{0} & 2I \\ 2I & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix} = \gamma^0_{\text{chiral}}.$$

For the spatial matrices, since $\gamma^i_{\text{Dirac}} = \gamma^i_{\text{chiral}}$ already, we need only check U leaves them fixed:

$$U\gamma^i_{\text{Dirac}}U^{-1} = \frac{1}{2}\begin{pmatrix} I & -I \\ I & I \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}\begin{pmatrix} I & I \\ -I & I \end{pmatrix} = \frac{1}{2}\begin{pmatrix} \sigma^i & \sigma^i \\ -\sigma^i & \sigma^i \end{pmatrix}\begin{pmatrix} I & I \\ -I & I \end{pmatrix}$$
$$= \frac{1}{2}\begin{pmatrix} \mathbf{0} & 2\sigma^i \\ -2\sigma^i & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix} = \gamma^i_{\text{chiral}}.$$ □

Proposition C.25 (Dirac ↔ Majorana)**.** *With the Hermitian unitary involution*

$$W = \frac{1}{\sqrt{2}}\begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix}, \qquad W^\dagger = W, \qquad W^2 = I_4 \;\Rightarrow\; W^{-1} = W, \tag{C.62}$$

one has $\gamma^\mu_{\text{Majorana}} = W\,\gamma^\mu_{\text{Dirac}}\,W$.

Proof. That $W = W^\dagger$ follows from $\sigma^{2\dagger} = \sigma^2$; that $W^2 = I_4$ is $\frac{1}{2}\left(\begin{smallmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{smallmatrix}\right)^2 = \frac{1}{2}\left(\begin{smallmatrix} I+(\sigma^2)^2 & \mathbf{0} \\ \mathbf{0} & (\sigma^2)^2+I \end{smallmatrix}\right) = I_4$. We verify the conjugation on each $\gamma^\mu_{\text{Dirac}}$ from (C.56), using $(\sigma^2)^2 = I$, $\sigma^2\sigma^1 = -i\sigma^3$, $\sigma^2\sigma^3 = i\sigma^1$, and their reverses:

$$W\gamma^0_{\text{Dirac}}W = \frac{1}{2}\begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix}\begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix}\begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix} = \begin{pmatrix} \mathbf{0} & \sigma^2 \\ \sigma^2 & \mathbf{0} \end{pmatrix} = \gamma^0_{\text{Majorana}},$$

$$W\gamma^1_{\text{Dirac}}W = \frac{1}{2}\begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^1 \\ -\sigma^1 & \mathbf{0} \end{pmatrix}\begin{pmatrix} I & \sigma^2 \\ \sigma^2 & -I \end{pmatrix} = \begin{pmatrix} i\sigma^3 & \mathbf{0} \\ \mathbf{0} & i\sigma^3 \end{pmatrix} = \gamma^1_{\text{Majorana}},$$

$$W\gamma^2_{\text{Dirac}}W = \begin{pmatrix} \mathbf{0} & -\sigma^2 \\ \sigma^2 & \mathbf{0} \end{pmatrix} = \gamma^2_{\text{Majorana}}, \qquad W\gamma^3_{\text{Dirac}}W = \begin{pmatrix} -i\sigma^1 & \mathbf{0} \\ \mathbf{0} & -i\sigma^1 \end{pmatrix} = \gamma^3_{\text{Majorana}},$$

where the last two follow by the identical block arithmetic. (As a check on γ^1: the $(1,1)$ block of $W\gamma^1_{\text{D}}W$ is $\frac{1}{2}[\sigma^2(-\sigma^1) + (-\sigma^1)\cdot 0 \cdot \dots]$; carrying out the two multiplications gives $\frac{1}{2}(\sigma^2\sigma^1\cdot(-1)+\dots)$ which collapses to $i\sigma^3$ once $\sigma^2\sigma^1 = -i\sigma^3$ is used.) All four match (C.60), so $\gamma^\mu_{\text{Majorana}} = W\gamma^\mu_{\text{Dirac}}W$. □

These two propositions make Pauli's abstract theorem concrete: the Dirac, chiral, and Majorana bases are connected by the unitaries U and W (and by WU^{-1} for chiral → Majorana). Any object defined invariantly—the chirality of a state, a trace of gamma matrices, an unpolarized squared amplitude—is therefore the same number in all three, which is precisely why the book is free to compute QED's traces (Chapters 17 and 19) in whatever basis is most convenient.

C.3.5 The chirality matrix γ^5 and its projectors

We now build the single most important derived object of the Clifford algebra, the chirality matrix. Its definition is fixed (Chapter A) as

$$\gamma^5 \equiv i\,\gamma^0\gamma^1\gamma^2\gamma^3 = -\tfrac{i}{4!}\,\epsilon_{\mu\nu\rho\sigma}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma, \tag{C.63}$$

the second form (with $\epsilon^{0123} = +1$, equivalently $\epsilon_{0123} = -1$) following because the totally antisymmetrized product of all four distinct gammas is $\gamma^0\gamma^1\gamma^2\gamma^3$ times the sign of the permutation. Its defining algebraic properties are representation-independent; we prove them from (C.63) and the Clifford relation alone.

Proposition C.26 (Properties of γ^5)**.** γ^5 *obeys*

$$(\gamma^5)^2 = I_4, \qquad \left\{\gamma^5, \gamma^\mu\right\} = 0 \quad (\mu = 0,1,2,3), \qquad \gamma^{5\dagger} = \gamma^5, \qquad \operatorname{tr}\gamma^5 = 0. \tag{C.64}$$

Proof. Square. Write $A \equiv \gamma^0\gamma^1\gamma^2\gamma^3$, so $\gamma^5 = iA$ and $(\gamma^5)^2 = -A^2$. Evaluate A^2 by sliding each repeated factor leftward and using $\{\gamma^\mu, \gamma^\nu\} = 0$ for $\mu \neq \nu$:

$$\begin{aligned} A^2 = \gamma^0\gamma^1\gamma^2\gamma^3\,\gamma^0\gamma^1\gamma^2\gamma^3 &= (-1)^3(\gamma^0)^2\,\gamma^1\gamma^2\gamma^3\gamma^1\gamma^2\gamma^3 && \text{(move the second } \gamma^0 \text{ left past } \gamma^3\gamma^2\gamma^1\text{)} \\ &= -(\gamma^0)^2(-1)^2(\gamma^1)^2\,\gamma^2\gamma^3\gamma^2\gamma^3 && \text{(move } \gamma^1 \text{ left past } \gamma^3\gamma^2\text{)} \\ &= -(\gamma^0)^2(\gamma^1)^2(-1)(\gamma^2)^2(\gamma^3)^2 && \text{(move } \gamma^2 \text{ left past } \gamma^3\text{)}. \end{aligned}$$

Now insert $(\gamma^0)^2 = +I_4$ and $(\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -I_4$:

$$A^2 = -(+1)(-1)(-1)(-1)(-1)\,I_4 = -\,I_4,$$

so $(\gamma^5)^2 = -A^2 = +I_4$.

Anticommutation with γ^μ. Fix μ. Among the four factors $\gamma^0\gamma^1\gamma^2\gamma^3$ in A, exactly three carry an index different from μ (with which γ^μ anticommutes) and exactly one equals γ^μ (with which it commutes trivially as it is the same matrix). Sliding γ^μ from the right all the way through A to merge with its own copy therefore produces three sign changes,

$$\gamma^\mu A = (-1)^3 A\gamma^\mu = -A\gamma^\mu \quad\Longrightarrow\quad \gamma^\mu\gamma^5 = -\gamma^5\gamma^\mu,$$

i.e. $\{\gamma^5, \gamma^\mu\} = 0$. (Concretely for $\mu = 0$: $\gamma^0 A = \gamma^0\gamma^0\gamma^1\gamma^2\gamma^3$, while $A\gamma^0 = \gamma^0\gamma^1\gamma^2\gamma^3\gamma^0 = (-1)^3\gamma^0\gamma^0\gamma^1\gamma^2\gamma^3$, so $\gamma^0 A = -A\gamma^0$.)

Hermiticity. Using the Hermiticity relation $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ proved in Theorem C.30 below,

$$\gamma^{5\dagger} = (i\gamma^0\gamma^1\gamma^2\gamma^3)^\dagger = -i\,\gamma^{3\dagger}\gamma^{2\dagger}\gamma^{1\dagger}\gamma^{0\dagger} = -i\,(\gamma^0\gamma^3\gamma^0)(\gamma^0\gamma^2\gamma^0)(\gamma^0\gamma^1\gamma^0)\,\gamma^0 = -i\,\gamma^0\gamma^3\gamma^2\gamma^1,$$

where the adjacent pairs $\gamma^0\gamma^0 = I$ have cancelled. Reversing the order $\gamma^3\gamma^2\gamma^1$ back to $\gamma^1\gamma^2\gamma^3$ costs the sign $(-1)^3$, so $\gamma^{5\dagger} = +i\,\gamma^0\gamma^1\gamma^2\gamma^3 = \gamma^5$. (Equivalently,

in the chiral basis $\gamma^5 = \mathrm{diag}(-I, I)$ is manifestly Hermitian, and since Hermiticity of a matrix is preserved under the unitary changes of basis of Section C.3.4, $\gamma^{5\dagger} = \gamma^5$ in every one of our representations.)

Tracelessness. Because $(\gamma^0)^2 = I_4$ and γ^5 anticommutes with γ^0,

$$\operatorname{tr}\gamma^5 = \operatorname{tr}(\gamma^5(\gamma^0)^2) = \operatorname{tr}(\gamma^0\gamma^5\gamma^0) = -\operatorname{tr}((\gamma^0)^2\gamma^5) = -\operatorname{tr}\gamma^5,$$

where the second equality is cyclicity of the trace and the third is $\gamma^5\gamma^0 = -\gamma^0\gamma^5$. Hence $2\operatorname{tr}\gamma^5 = 0$, i.e. $\operatorname{tr}\gamma^5 = 0$. □

The anticommutation $\{\gamma^5, \gamma^\mu\} = 0$ is the structural fact that makes γ^5 commute with the Lorentz generators $\sigma^{\mu\nu} \propto [\gamma^\mu, \gamma^\nu]$ (an even product of gammas), so that γ^5 is a Lorentz scalar and its eigenspaces are Lorentz-invariant. Those eigenspaces are the chiralities, isolated by the projectors:

Definition C.27 (Chirality projectors)**.**

$$P_L \equiv \frac{1-\gamma^5}{2}, \qquad P_R \equiv \frac{1+\gamma^5}{2}. \tag{C.65}$$

A spinor with $\gamma^5\psi = -\psi$ (so $P_L\psi = \psi$) is *left-handed*; one with $\gamma^5\psi = +\psi$ (so $P_R\psi = \psi$) is *right-handed*.

Proposition C.28. *The $P_{L,R}$ are a complete set of orthogonal projectors:*

$$P_L + P_R = I_4, \qquad P_L^2 = P_L, \qquad P_R^2 = P_R, \qquad P_L P_R = P_R P_L = \mathbf{0}. \tag{C.66}$$

Moreover they obey the intertwining relation $\gamma^\mu P_L = P_R\gamma^\mu$ and $\gamma^\mu P_R = P_L\gamma^\mu$.

Proof. Completeness is immediate: $P_L + P_R = \frac{1}{2}[(1-\gamma^5) + (1+\gamma^5)] = I_4$. Idempotency uses $(\gamma^5)^2 = I_4$:

$$P_L^2 = \frac{(1-\gamma^5)^2}{4} = \frac{1 - 2\gamma^5 + (\gamma^5)^2}{4} = \frac{2-2\gamma^5}{4} = \frac{1-\gamma^5}{2} = P_L,$$

and likewise $P_R^2 = P_R$. Orthogonality is the difference of squares:

$$P_L P_R = \frac{(1-\gamma^5)(1+\gamma^5)}{4} = \frac{1-(\gamma^5)^2}{4} = \mathbf{0},$$

and $P_R P_L = \mathbf{0}$ identically. Finally, since $\gamma^\mu\gamma^5 = -\gamma^5\gamma^\mu$,

$$\gamma^\mu P_L = \gamma^\mu \frac{1-\gamma^5}{2} = \frac{1+\gamma^5}{2}\gamma^\mu = P_R\gamma^\mu,$$

and similarly $\gamma^\mu P_R = P_L\gamma^\mu$. □

The intertwining $\gamma^\mu P_L = P_R \gamma^\mu$ is the algebraic origin of the chirality flip that a vector current induces: it is why $\bar\psi\gamma^\mu\psi$ couples left to left and right to right, while a mass term $\bar\psi\psi$ couples left to right. This is the backbone of the $V-A$ structure that QED and the electroweak chapters will lean on.

It is worth recording γ^5 and the projectors in each basis, since chapters quote them. From (C.63) by direct multiplication:

$$\gamma^5_{\text{Dirac}} = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix}, \qquad \gamma^5_{\text{chiral}} = \begin{pmatrix} -I & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}, \qquad \gamma^5_{\text{Majorana}} = \begin{pmatrix} \sigma^2 & \mathbf{0} \\ \mathbf{0} & -\sigma^2 \end{pmatrix}. \tag{C.67}$$

In the chiral basis γ^5 is diagonal, so the projectors are the trivial block projectors

$$P_L = \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}, \qquad P_R = \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & I \end{pmatrix}, \tag{C.68}$$

which is the whole point of the chiral basis: a Dirac spinor $\psi = (\psi_L, \psi_R)^{\mathsf T}$ wears its handedness on its sleeve, the upper two components being the left-handed Weyl spinor and the lower two the right-handed one. In the Majorana basis, by contrast, γ^5 is purely imaginary, consistent with $\gamma^{\mu *} = -\gamma^\mu$ and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. Let us confirm the first entry of (C.67).

Worked Example C.29 (γ^5 in the Dirac and chiral bases)**.** In the Dirac basis, $\gamma^1\gamma^2 = \left(\begin{smallmatrix} -\sigma^1\sigma^2 & \mathbf{0} \\ \mathbf{0} & -\sigma^1\sigma^2 \end{smallmatrix}\right) = \left(\begin{smallmatrix} -i\sigma^3 & \mathbf{0} \\ \mathbf{0} & -i\sigma^3 \end{smallmatrix}\right)$ using $\sigma^1\sigma^2 = i\sigma^3$. Then

$$\gamma^1\gamma^2\gamma^3 = \begin{pmatrix} -i\sigma^3 & \mathbf{0} \\ \mathbf{0} & -i\sigma^3 \end{pmatrix} \begin{pmatrix} \mathbf{0} & \sigma^3 \\ -\sigma^3 & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & -i(\sigma^3)^2 \\ i(\sigma^3)^2 & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & -iI \\ iI & \mathbf{0} \end{pmatrix},$$

and so

$$\gamma^5_{\text{Dirac}} = i\gamma^0\gamma^1\gamma^2\gamma^3 = i \begin{pmatrix} I & \mathbf{0} \\ \mathbf{0} & -I \end{pmatrix} \begin{pmatrix} \mathbf{0} & -iI \\ iI & \mathbf{0} \end{pmatrix} = i \begin{pmatrix} \mathbf{0} & -iI \\ -iI & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix},$$

which is the first matrix in (C.67). Applying the equivalence of Theorem C.24, $\gamma^5_{\text{chiral}} = U\gamma^5_{\text{Dirac}}U^{-1} = \frac{1}{2}\left(\begin{smallmatrix} I & -I \\ I & I \end{smallmatrix}\right)\left(\begin{smallmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{smallmatrix}\right)\left(\begin{smallmatrix} I & I \\ -I & I \end{smallmatrix}\right) = \left(\begin{smallmatrix} -I & \mathbf{0} \\ \mathbf{0} & I \end{smallmatrix}\right)$, the diagonal form. Thus the same invariant matrix γ^5 is off-diagonal in the Dirac basis and diagonal in the chiral basis—a vivid illustration that "which components are left-handed" is a basis-dependent statement, while "γ^5 has eigenvalues ± 1, each twice" is not.

C.3.6 The Hermiticity convention

To form Lorentz-covariant bilinears one needs $\psi^\dagger\Gamma\psi$ to transform correctly, and this hinges on how $\gamma^{\mu\dagger}$ relates to γ^μ. The relation is uniform across all three of our bases and is the convention adopted throughout the book.

Theorem C.30 (Hermiticity relation)**.** *In the Dirac, chiral, and Majorana representations,*

$$\gamma^{0\dagger} = \gamma^0, \qquad \gamma^{i\dagger} = -\gamma^i, \tag{C.69}$$

which combine into the single covariant statement

$$\boxed{\gamma^{\mu\dagger} = \gamma^0 \gamma^\mu \gamma^0} \tag{C.70}$$

The same relation holds in every representation related to these by a unitary similarity.

Proof. Component relations. In the Dirac basis $\gamma^0 = \mathrm{diag}(I, -I)$ is real symmetric, hence Hermitian: $\gamma^{0\dagger} = \gamma^0$. The spatial $\gamma^i = \left(\begin{smallmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{smallmatrix}\right)$ has Hermitian conjugate $\left(\begin{smallmatrix} \mathbf{0} & -\sigma^{i\dagger} \\ \sigma^{i\dagger} & \mathbf{0} \end{smallmatrix}\right) = \left(\begin{smallmatrix} \mathbf{0} & -\sigma^i \\ \sigma^i & \mathbf{0} \end{smallmatrix}\right) = -\gamma^i$, using $\sigma^{i\dagger} = \sigma^i$. The same two facts hold in the chiral basis ($\gamma^0 = \left(\begin{smallmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{smallmatrix}\right)$ is Hermitian, the γ^i are identical to the Dirac ones), and in the Majorana basis one checks directly from (C.60): $\gamma^0 = \left(\begin{smallmatrix} \mathbf{0} & \sigma^2 \\ \sigma^2 & \mathbf{0} \end{smallmatrix}\right)$ is Hermitian, while $\gamma^{1\dagger} = \left(\begin{smallmatrix} (i\sigma^3)^\dagger & \mathbf{0} \\ \mathbf{0} & (i\sigma^3)^\dagger \end{smallmatrix}\right) = \left(\begin{smallmatrix} -i\sigma^3 & \mathbf{0} \\ \mathbf{0} & -i\sigma^3 \end{smallmatrix}\right) = -\gamma^1$, and identically $\gamma^{2\dagger} = -\gamma^2$, $\gamma^{3\dagger} = -\gamma^3$ (each spatial Majorana matrix is anti-Hermitian, being i times a Hermitian Pauli block).

Covariant form. We show (C.69) is equivalent to (C.70), using only $(\gamma^0)^2 = I_4$ and the Clifford relation. For the time component, $\gamma^0\gamma^0\gamma^0 = (\gamma^0)^2\gamma^0 = \gamma^0 = \gamma^{0\dagger}$. For a space component, $\gamma^0\gamma^i\gamma^0 = \gamma^0\gamma^i\gamma^0$; since $\{\gamma^0, \gamma^i\} = 0$ we have $\gamma^i\gamma^0 = -\gamma^0\gamma^i$, so

$$\gamma^0\gamma^i\gamma^0 = \gamma^0\,(\gamma^i\gamma^0) = \gamma^0(-\gamma^0\gamma^i) = -(\gamma^0)^2\gamma^i = -\gamma^i = \gamma^{i\dagger}.$$

Thus $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ for every μ.

Stability under unitary change of basis. If $\gamma^{\mu\prime} = S\gamma^\mu S^{-1}$ with S unitary ($S^{-1} = S^\dagger$), then

$$\gamma^{\mu\prime\dagger} = (S\gamma^\mu S^{-1})^\dagger = S\,\gamma^{\mu\dagger}\,S^{-1} = S\,(\gamma^0\gamma^\mu\gamma^0)\,S^{-1} = (S\gamma^0 S^{-1})(S\gamma^\mu S^{-1})(S\gamma^0 S^{-1}) = \gamma^{0\prime}\gamma^{\mu\prime}\gamma^{0}$$

so the relation (C.70) is preserved with γ^0 replaced by the transformed $\gamma^{0\prime}$. Since U and W of Section C.3.4 are unitary, all three standard bases obey (C.70). □

Remark C.31 (Representation-independent existence; first sight of an intertwiner). The relation (C.70) is more than a convenient feature of three particular bases. Taking the Hermitian conjugate of $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu} I_4$ gives $\left\{\gamma^{\nu\dagger}, \gamma^{\mu\dagger}\right\} = 2\eta^{\mu\nu} I_4$, i.e. the four matrices $\{\gamma^{\mu\dagger}\}$ satisfy the *same* Clifford relation as $\{\gamma^\mu\}$. By Pauli's fundamental theorem (Theorem C.16) there is therefore a matrix A, unique up to scale, with

$$\gamma^{\mu\dagger} = A\,\gamma^\mu\,A^{-1} \qquad \text{for all } \mu, \tag{C.71}$$

and one shows A may be chosen Hermitian; in our bases the choice is exactly $A = \gamma^0$. This A—the *Dirac conjugation* or Hermitizing matrix—is the first and simplest member of the family of intertwining matrices that the next section classifies. It is the reason the Dirac adjoint is $\bar\psi = \psi^\dagger\gamma^0$ rather than $\psi^\dagger$: only with the γ^0 inserted does $\bar\psi\psi$ transform as a scalar.

The same Hermiticity relation controls the conjugate of the Lorentz generator. Define the antisymmetric tensor of gammas

$$\sigma^{\mu\nu} \equiv \frac{i}{2}\left[\gamma^\mu, \gamma^\nu\right], \tag{C.72}$$

(the same object whose bilinear $\bar\psi\sigma^{\mu\nu}\psi$ is catalogued in Section C.5). Its Hermiticity follows mechanically.

Proposition C.32. $(\sigma^{\mu\nu})^\dagger = \gamma^0\sigma^{\mu\nu}\gamma^0$.

Proof. Conjugating (C.72),

$$(\sigma^{\mu\nu})^\dagger = -\frac{i}{2}\left[\gamma^\mu, \gamma^\nu\right]^\dagger = -\frac{i}{2}(\gamma^{\nu\dagger}\gamma^{\mu\dagger} - \gamma^{\mu\dagger}\gamma^{\nu\dagger}) = \frac{i}{2}\left[\gamma^{\mu\dagger}, \gamma^{\nu\dagger}\right].$$

Insert $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ and use $(\gamma^0)^2 = I_4$ to cancel the inner pair:

$$\left[\gamma^0\gamma^\mu\gamma^0, \gamma^0\gamma^\nu\gamma^0\right] = \gamma^0\gamma^\mu\gamma^\nu\gamma^0 - \gamma^0\gamma^\nu\gamma^\mu\gamma^0 = \gamma^0\left[\gamma^\mu, \gamma^\nu\right]\gamma^0,$$

hence $(\sigma^{\mu\nu})^\dagger = \dfrac{i}{2}\gamma^0\left[\gamma^\mu, \gamma^\nu\right]\gamma^0 = \gamma^0\sigma^{\mu\nu}\gamma^0$. □

C.3.7 The chiral basis and the $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ decomposition

We close the section by tying the chiral basis back to the abstract Lorentz representation theory of Chapter 3. There a Dirac spinor was identified as the reducible representation $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$ of the Lorentz algebra $\mathfrak{so}(1,3) \cong \mathfrak{su}(2) \oplus \mathfrak{su}(2)$, the two summands being the left- and right-handed Weyl spinors. We now *see* this directly: in the chiral basis the spin-$\frac{1}{2}$ Lorentz generators are block diagonal, and the two blocks act exactly as the two Weyl representations.

Recall (Chapter 3) that on a Dirac field the Lorentz generators are

$$S^{\mu\nu} = \frac{i}{4}\left[\gamma^\mu, \gamma^\nu\right] = \frac{1}{2}\sigma^{\mu\nu}, \qquad \psi \to \exp\Big(-\tfrac{i}{2}\omega_{\mu\nu}S^{\mu\nu}\Big)\psi, \tag{C.73}$$

with rotation generators $J^k = \frac{1}{2}\epsilon^{kij}S^{ij}$ and boost generators $K^i = S^{0i}$. We compute these in the chiral basis.

Worked Example C.33 (Block-diagonalization of the Lorentz generators)**.** *Boosts.* In the chiral basis,

$$\gamma^0\gamma^i = \begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix} = \begin{pmatrix} -\sigma^i & \mathbf{0} \\ \mathbf{0} & \sigma^i \end{pmatrix}, \quad \gamma^i\gamma^0 = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{0} & I \\ I & \mathbf{0} \end{pmatrix} = \begin{pmatrix} \sigma^i & \mathbf{0} \\ \mathbf{0} & -\sigma^i \end{pmatrix},$$

so $\left[\gamma^0, \gamma^i\right] = \left(\begin{smallmatrix} -2\sigma^i & \mathbf{0} \\ \mathbf{0} & 2\sigma^i \end{smallmatrix}\right)$ and

$$K^i = S^{0i} = \frac{i}{4}\left[\gamma^0, \gamma^i\right] = \begin{pmatrix} -\frac{i}{2}\sigma^i & \mathbf{0} \\ \mathbf{0} & +\frac{i}{2}\sigma^i \end{pmatrix}. \tag{C.74}$$

Rotations. Likewise

$$\gamma^i\gamma^j = \begin{pmatrix} \mathbf{0} & \sigma^i \\ -\sigma^i & \mathbf{0} \end{pmatrix}\begin{pmatrix} \mathbf{0} & \sigma^j \\ -\sigma^j & \mathbf{0} \end{pmatrix} = \begin{pmatrix} -\sigma^i\sigma^j & \mathbf{0} \\ \mathbf{0} & -\sigma^i\sigma^j \end{pmatrix} \Rightarrow \left[\gamma^i, \gamma^j\right] = \begin{pmatrix} -\left[\sigma^i, \sigma^j\right] & \mathbf{0} \\ \mathbf{0} & -\left[\sigma^i, \sigma^j\right] \end{pmatrix}$$

and with $\left[\sigma^i, \sigma^j\right] = 2i\epsilon^{ijk}\sigma^k$,

$$S^{ij} = \frac{i}{4}\left[\gamma^i, \gamma^j\right] = \frac{i}{4}(-2i\epsilon^{ijk})\begin{pmatrix} \sigma^k & \mathbf{0} \\ \mathbf{0} & \sigma^k \end{pmatrix} = \frac{1}{2}\epsilon^{ijk}\begin{pmatrix} \sigma^k & \mathbf{0} \\ \mathbf{0} & \sigma^k \end{pmatrix}, \tag{C.75}$$

so that $J^k = \frac{1}{2}\epsilon^{kij}S^{ij} = \frac{1}{4}\epsilon^{kij}\epsilon^{ijl}\,\mathrm{diag}(\sigma^l, \sigma^l) = \frac{1}{2}\,\mathrm{diag}(\sigma^k, \sigma^k)$, using $\epsilon^{kij}\epsilon^{lij} = 2\delta^{kl}$.

Reading off the two Weyl representations. Both K^i and J^k are block diagonal, so the Lorentz action never mixes the upper two components with the lower two: the chiral Dirac spinor $\psi = (\psi_L, \psi_R)^{\mathsf{T}}$ splits into two invariant two-component pieces. On the upper (left) block,

$$J^k_L = \tfrac{1}{2}\sigma^k, \qquad K^i_L = -\tfrac{i}{2}\sigma^i,$$

while on the lower (right) block $J^k_R = \frac{1}{2}\sigma^k$, $K^i_R = +\frac{i}{2}\sigma^i$. Forming the standard self-dual/anti-self-dual combinations $A^i = \frac{1}{2}(J^i + iK^i)$ and $B^i = \frac{1}{2}(J^i - iK^i)$ that diagonalize $\mathfrak{su}(2) \oplus \mathfrak{su}(2)$ (Chapter 3), the upper block gives

$$A^i_L = \tfrac{1}{2}(\tfrac{1}{2}\sigma^i + i(-\tfrac{i}{2}\sigma^i)) = \tfrac{1}{2}\sigma^i, \qquad B^i_L = \tfrac{1}{2}(\tfrac{1}{2}\sigma^i - i(-\tfrac{i}{2}\sigma^i)) = 0,$$

so the left block carries spin $\frac{1}{2}$ under the first $\mathfrak{su}(2)$ and spin 0 under the second: it is the $(\frac{1}{2}, 0)$ representation. The lower block gives the mirror image,

$$A^i_R = \tfrac{1}{2}(\tfrac{1}{2}\sigma^i + i(+\tfrac{i}{2}\sigma^i)) = 0, \qquad B^i_R = \tfrac{1}{2}(\tfrac{1}{2}\sigma^i - i(+\tfrac{i}{2}\sigma^i)) = \tfrac{1}{2}\sigma^i,$$

the $(0, \frac{1}{2})$ representation. Thus the chiral Dirac spinor is exactly $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$, with P_L projecting onto $(\frac{1}{2}, 0)$ and P_R onto $(0, \frac{1}{2})$, in precise agreement with Chapter 3.

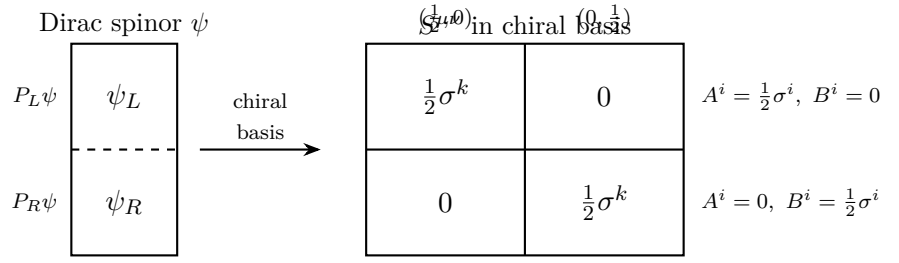


Figure C.1: In the chiral basis $\gamma^5 = \mathrm{diag}(-I, I)$ is diagonal, so the projectors $P_{L,R}$ split a Dirac spinor into its two-component Weyl pieces ψ_L, ψ_R. The Lorentz generators $S^{\mu\nu} = \frac{i}{4}[\gamma^\mu, \gamma^\nu]$ are then block diagonal: the upper block realizes the $(\frac{1}{2}, 0)$ representation ($A^i = \frac{1}{2}\sigma^i$, $B^i = 0$) and the lower block the $(0, \frac{1}{2})$ representation, matching the classification of Chapter 3.

The agreement is not a coincidence of conventions: it is forced. Because $\gamma^5 = \mathrm{diag}(-I, I)$ commutes with every $S^{\mu\nu}$ (an even product of gammas), its ∓ 1 eigenspaces are Lorentz-invariant, and the projectors $P_{L,R}$ of (C.68) are precisely the projectors onto the two irreducible Weyl summands. The boost generators (C.74) are anti-Hermitian-times-i in a way that differs by a sign between the blocks—this is the concrete statement that the finite-dimensional spinor representation of the boosts is *non-unitary* (as it must be, the Lorentz group being noncompact), and that complex conjugation exchanges the two Weyl types, $(\frac{1}{2}, 0) \leftrightarrow (0, \frac{1}{2})$. Fig. C.1 summarizes the block structure.

With three concrete representations in hand, the explicit unitaries relating them, and the Hermiticity relation $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$, we have met our first intertwining matrix in the guise of γ^0: the matrix that conjugates the set $\{\gamma^\mu\}$ into the set $\{\gamma^{\mu\dagger}\}$, both of which obey the identical Clifford relation. The next section takes this single example and turns it into a system, classifying *all* the matrices that intertwine $\{\gamma^\mu\}$ with $\{-\gamma^{\mu\mathsf{T}}\}$, $\{\gamma^{\mu *}\}$, and $\{\gamma^{\mu\dagger}\}$—the Dirac, charge-conjugation, and parity/time-reversal matrices—and characterizing precisely when a spinor may be its own conjugate, the Majorana condition that the imaginary representation of Section C.3.3 was built to expose.

C.4 Conjugation and Intertwining Matrices: Dirac, Charge, Parity, Time Reversal

The Hermiticity relation $\gamma^{\mu\dagger} = \gamma^0\gamma^\mu\gamma^0$ established with the explicit matrices of Section C.3 is more than a bookkeeping convenience: it is the statement that the matrix γ^0 *intertwines* two different sets of 4×4 matrices, $\{\gamma^\mu\}$ and $\{\gamma^{\mu\dagger}\}$, both of which obey the same Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$. This is one instance of a general phenomenon. From a given representation $\{\gamma^\mu\}$ one can manufacture

four sets of matrices by applying the three antilinear/linear operations that preserve the algebra,

$$\{\gamma^\mu\}, \qquad \{-\gamma^{\mu T}\}, \qquad \{\gamma^{\mu *}\}, \qquad \{\gamma^{\mu \dagger}\}, \tag{C.76}$$

each of which, as we verify below, again satisfies the Clifford relation. Pauli's fundamental theorem (Section C.2) then guarantees that in $d = 4$, where the irreducible representation is unique up to similarity, each of these four sets is conjugate to $\{\gamma^\mu\}$ by a matrix that is unique up to an overall scalar. Those intertwiners are precisely the workhorse matrices of every spinorial manipulation in the book: the Dirac (Hermiticity/parity) matrix γ^0, the charge-conjugation matrix C, the reality matrix B, and the time-reversal matrix. This section pins them down, proves their algebraic properties, builds the symmetry table that the next section needs, and characterizes the Majorana reality condition representation-independently.

C.4.1 The four Clifford-equivalent representations

Proposition C.34 (The conjugate sets obey the Clifford relation)**.** *If* $\{\gamma^\mu\}$ *satisfies* $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$*, then so do each of* $\{\gamma^{\mu\dagger}\}$*,* $\{\gamma^{\mu *}\}$*, and* $\{-\gamma^{\mu T}\}$*.*

Proof. The metric components $\eta^{\mu\nu}$ are real, so taking the Hermitian conjugate of the Clifford relation gives

$$\{\gamma^{\mu\dagger}, \gamma^{\nu\dagger}\} = \left(\{\gamma^\nu, \gamma^\mu\}\right)^\dagger = (2\eta^{\mu\nu}\mathbf{1})^\dagger = 2\eta^{\mu\nu}\mathbf{1}, \tag{C.77}$$

using $(\gamma^\mu\gamma^\nu)^\dagger = \gamma^{\nu\dagger}\gamma^{\mu\dagger}$ and $\eta^{\mu\nu} = \eta^{\nu\mu}$. Identically, complex conjugation gives $\{\gamma^{\mu *}, \gamma^{\nu *}\} = (\{\gamma^\mu, \gamma^\nu\})^* = 2\eta^{\mu\nu}\mathbf{1}$. For the transpose, $(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu)^T = \gamma^{\nu T}\gamma^{\mu T} + \gamma^{\mu T}\gamma^{\nu T} = \{\gamma^{\mu T}, \gamma^{\nu T}\}$, so

$$\{-\gamma^{\mu T}, -\gamma^{\nu T}\} = \{\gamma^{\mu T}, \gamma^{\nu T}\} = \left(\{\gamma^\mu, \gamma^\nu\}\right)^T = 2\eta^{\mu\nu}\mathbf{1}, \tag{C.78}$$

the two minus signs squaring to a plus. □

Because each set in (C.76) is a four-dimensional representation of the same Clifford algebra, and that algebra has a unique irreducible representation up to similarity in $d = 4$ (Section C.2), there exist nonsingular matrices effecting the conjugations. We name them as follows.

Definition C.35 (Intertwining matrices)**.** The *Dirac (Hermiticity) matrix* A, the *charge-conjugation matrix* C, and the *reality matrix* B are the nonsingular 4×4 matrices, each unique up to a nonzero scalar, defined by

$$A\,\gamma^\mu A^{-1} = \gamma^{\mu\dagger}, \tag{C.79}$$

$$C\,\gamma^\mu C^{-1} = -\gamma^{\mu T}, \tag{C.80}$$

$$B\,\gamma^\mu B^{-1} = \gamma^{\mu *}. \tag{C.81}$$

Their existence and uniqueness-up-to-scale are immediate from Pauli's theorem applied, respectively, to the sets $\{\gamma^{\mu\dagger}\}$, $\{-\gamma^{\mu T}\}$, and $\{\gamma^{\mu *}\}$ of Theorem C.34: each is conjugate to $\{\gamma^\mu\}$, and Schur's lemma fixes the conjugating matrix up to a scalar (any two such conjugators differ by a matrix commuting with the irreducible $\{\gamma^\mu\}$, hence by a multiple of $\mathbf{1}$). The three matrices are not independent. Since $\gamma^{\mu *} = (\gamma^{\mu\dagger})^T$, the reality matrix can be built from A and C; we make this explicit in Section C.4.8. The relationships are summarized in Fig. C.2.

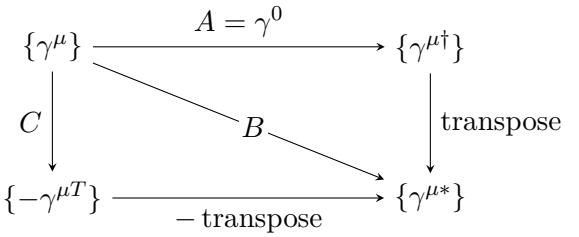


Figure C.2: The four Clifford-equivalent representations of (C.76) and the intertwiners relating them. The Dirac matrix $A = \gamma^0$ produces the Hermitian conjugate, C the negative transpose, and B the complex conjugate. Since $\gamma^{\mu *} = (\gamma^{\mu\dagger})^T = -(\gamma^{\mu T})^\dagger$ the square commutes up to scalars, fixing B in terms of A and C.

C.4.2 The Dirac matrix and the parity matrix

The matrix A of (C.79) is the one already met in Section C.3: the Hermiticity relation proved there says exactly that $\gamma^0\gamma^\mu\gamma^0 = \gamma^{\mu\dagger}$, so in every standard representation

$$A = \gamma^0, \qquad A^\dagger = A, \qquad A^2 = \mathbf{1}. \tag{C.82}$$

This is the matrix that defines the Dirac adjoint $\bar\psi = \psi^\dagger A = \psi^\dagger\gamma^0$ (Chapter 11); its job is to convert $\psi^\dagger$, which transforms in the conjugate of the spinor representation, into an object that pairs with ψ to make Lorentz covariants. We will see in Section C.5 that the existence of A with $A\gamma^\mu A^{-1} = \gamma^{\mu\dagger}$ is precisely what makes $\bar\psi\Gamma_A\psi$ transform cleanly.

The same matrix governs parity. The parity operation reflects the spatial axes, $x = (t, \mathbf{x}) \mapsto (t, -\mathbf{x})$, which on the gamma matrices is the map $\gamma^0 \mapsto \gamma^0$, $\gamma^i \mapsto -\gamma^i$. This is implemented by conjugation with γ^0:

$$\gamma^0\gamma^\mu\gamma^0 = \gamma_\mu = (\gamma^0, -\gamma^i), \tag{C.83}$$

which follows at once from $\{\gamma^0, \gamma^i\} = 0$ and $(\gamma^0)^2 = \mathbf{1}$. Hence the matrix accompanying the parity transformation of the Dirac field, $\psi(t, \mathbf{x}) \mapsto \eta_P\,\gamma^0\psi(t, -\mathbf{x})$, is γ^0 up to the unobservable phase η_P. The physical content of P as an operator on Fock space is developed in Chapter 11; here we record only that the *parity matrix is the Dirac matrix* $A = \gamma^0$, doing double duty.

C.4.3 The charge-conjugation matrix: existence and uniqueness

Theorem C.36 (Existence and uniqueness of C)**.** *In $d = 4$ there exists a nonsingular matrix C, unique up to a nonzero scalar, satisfying $C\gamma^\mu C^{-1} = -\gamma^{\mu T}$.*

Proof. By Theorem C.34 the set $\{-\gamma^{\mu T}\}$ obeys the Clifford relation and is a four-dimensional representation, hence irreducible and equivalent to $\{\gamma^\mu\}$ by Pauli's fundamental theorem (Section C.2). Equivalence is exactly the existence of a nonsingular C with $-\gamma^{\mu T} = C\gamma^\mu C^{-1}$. For uniqueness, suppose C' also satisfies $C'\gamma^\mu C'^{-1} = -\gamma^{\mu T}$. Then $C^{-1}C'$ commutes with every γ^μ:

$$C^{-1}C'\,\gamma^\mu\,(C^{-1}C')^{-1} = C^{-1}(C'\gamma^\mu C'^{-1})C = C^{-1}(-\gamma^{\mu T})C = \gamma^\mu. \tag{C.84}$$

Since $\{\gamma^\mu\}$ acts irreducibly, Schur's lemma forces $C^{-1}C' = \lambda\,\mathbf{1}$, i.e. $C' = \lambda C$. □

Caution

The *sign* chosen in (C.80) is a convention adapted to $d = 4$. In an even dimension both $\{+\gamma^{\mu T}\}$ and $\{-\gamma^{\mu T}\}$ are equivalent to $\{\gamma^\mu\}$ — they are related to each other by conjugation with γ^5, which obeys $\gamma^5\gamma^\mu(\gamma^5)^{-1} = -\gamma^\mu$ — so there are in fact two transpose-intertwiners, C (for $-\gamma^{\mu T}$) and $C' \propto C\gamma^5$ (for $+\gamma^{\mu T}$). In odd dimensions only one of the two signs occurs. The symmetry $C^T = \pm C$ derived next, and the existence of Majorana spinors, depend on d mod 8; we prove the $d = 4$ facts and flag the general pattern where it matters.

C.4.4 Symmetry of C and the bilinear symmetry table

Theorem C.37 (Antisymmetry of C in $d = 4$)**.** *The charge-conjugation matrix of Theorem C.36 satisfies $C^T = -C$.*

Proof. We proceed in two steps: first that $C^T = \lambda C$ with $\lambda = \pm 1$, then that $\lambda = -1$ by counting.

Step 1 — $C^T = \pm C$. Transpose the defining relation $C\gamma^\mu C^{-1} = -\gamma^{\mu T}$. The left side transposes to $(C^{-1})^T\gamma^{\mu T}C^T = (C^T)^{-1}\gamma^{\mu T}C^T$, the right side to $-\gamma^\mu$, so

$$(C^T)^{-1}\gamma^{\mu T}C^T = -\gamma^\mu. \tag{C.85}$$

Now substitute $\gamma^{\mu T} = -C\gamma^\mu C^{-1}$ (the defining relation rearranged):

$$(C^T)^{-1}(-C\gamma^\mu C^{-1})C^T = -\gamma^\mu \implies (C^{-1}C^T)^{-1}\gamma^\mu(C^{-1}C^T) = \gamma^\mu. \tag{C.86}$$

Thus $M \equiv C^{-1}C^T$ commutes with all γ^μ, and Schur's lemma gives $M = \lambda\mathbf{1}$, i.e. $C^T = \lambda C$. Transposing this once more, $C = \lambda C^T = \lambda^2 C$, so $\lambda^2 = 1$ and $\lambda = \pm 1$.

Step 2 — the counting argument fixes $\lambda = -1$. Recall the sixteen matrices

$$\{\Gamma_A\} = \{\mathbf{1},\ \gamma^\mu,\ \sigma^{\mu\nu}\ (\mu<\nu),\ \gamma^\mu\gamma^5,\ \gamma^5\}, \qquad \sigma^{\mu\nu} = \tfrac{i}{2}[\gamma^\mu,\gamma^\nu], \tag{C.87}$$

which are linearly independent and span the full algebra of 4×4 matrices (Section C.5). Since C is invertible, the sixteen products $\{C\Gamma_A\}$ are likewise a basis of the 16-dimensional matrix algebra. Every 4×4 matrix splits uniquely into a symmetric and an antisymmetric part, and the spaces of symmetric and antisymmetric 4×4 matrices have dimensions $\binom{4+1}{2} = 10$ and $\binom{4}{2} = 6$ respectively. A basis of the full algebra in which every element is purely symmetric or purely antisymmetric must therefore contain exactly 10 symmetric and 6 antisymmetric members. We now show that each $C\Gamma_A$ *is* purely symmetric or antisymmetric, compute which, and read off λ from the requirement that the totals come out $(10, 6)$.

For each basis element define the sign $t_A = \pm 1$ by $\Gamma_A^T = t_A\, C\Gamma_A C^{-1}$; this is well defined because conjugation by C maps each Γ_A to a multiple of its own transpose, as we verify. Then

$$(C\Gamma_A)^T = \Gamma_A^T C^T = (t_A\, C\Gamma_A C^{-1})(\lambda C) = \lambda t_A\, C\Gamma_A, \tag{C.88}$$

so $C\Gamma_A$ is symmetric when $\lambda t_A = +1$ and antisymmetric when $\lambda t_A = -1$. The signs t_A follow directly from (C.80):

- $\Gamma = \mathbf{1}$: trivially $\mathbf{1}^T = \mathbf{1} = C\mathbf{1}C^{-1}$, so $t = +1$.
- $\Gamma = \gamma^\mu$: $\gamma^{\mu T} = -C\gamma^\mu C^{-1}$, so $t = -1$.
- $\Gamma = \sigma^{\mu\nu}$: using $[\gamma^\mu,\gamma^\nu]^T = [\gamma^{\nu T},\gamma^{\mu T}] = -[\gamma^{\mu T},\gamma^{\nu T}]$ and $\gamma^{\mu T} = -C\gamma^\mu C^{-1}$,

  $$[\gamma^{\mu T},\gamma^{\nu T}] = [\,-C\gamma^\mu C^{-1}, -C\gamma^\nu C^{-1}] = C[\gamma^\mu,\gamma^\nu]C^{-1}, \tag{C.89}$$

  hence $\sigma^{\mu\nu T} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]^T = -C\,\frac{i}{2}[\gamma^\mu,\gamma^\nu]\,C^{-1} = -C\sigma^{\mu\nu}C^{-1}$, so $t = -1$.
- $\Gamma = \gamma^5$: with $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$,

  $$\gamma^{5T} = i\,\gamma^{3T}\gamma^{2T}\gamma^{1T}\gamma^{0T} = i(-1)^4 C\,\gamma^3\gamma^2\gamma^1\gamma^0\,C^{-1} = i\,C\,\gamma^0\gamma^1\gamma^2\gamma^3\,C^{-1} = C\gamma^5 C^{-1}, \tag{C.90}$$

  where reversing the order of four distinct anticommuting factors costs $(-1)^{4\cdot 3/2} = +1$, so $\gamma^3\gamma^2\gamma^1\gamma^0 = \gamma^0\gamma^1\gamma^2\gamma^3$; thus $t = +1$.
- $\Gamma = \gamma^\mu\gamma^5$: $(\gamma^\mu\gamma^5)^T = \gamma^{5T}\gamma^{\mu T} = (C\gamma^5C^{-1})(-C\gamma^\mu C^{-1}) = -C\gamma^5\gamma^\mu C^{-1} = C\gamma^\mu\gamma^5 C^{-1}$, using $\gamma^5\gamma^\mu = -\gamma^\mu\gamma^5$; so $t = +1$.

Collecting the multiplicities, $t_A = +1$ for the six matrices $\{\mathbf{1},\gamma^5,\gamma^\mu\gamma^5\}$ and $t_A = -1$ for the ten matrices $\{\gamma^\mu,\sigma^{\mu\nu}\}$. By (C.88) the number of *antisymmetric* $C\Gamma_A$ is the number with $\lambda t_A = -1$:

$$\lambda = +1: \quad \text{antisymmetric} \Leftrightarrow t_A = -1: \quad \# = 4 + 6 = 10, \tag{C.91}$$

$$\lambda = -1: \quad \text{antisymmetric} \Leftrightarrow t_A = +1: \quad \# = 1 + 1 + 4 = 6. \tag{C.92}$$

Only $\lambda = -1$ yields the required 6 antisymmetric (and correspondingly 10 symmetric) matrices; $\lambda = +1$ would demand 10 antisymmetric 4×4 matrices, which is impossible. Therefore $\lambda = -1$, i.e. $C^T = -C$. □

The same computation hands the next section its central tool. Combining $C^T = -C$ with the signs t_A gives the symmetry of each bilinear-form matrix $C\Gamma_A$, the data that controls Fierz rearrangement and the symmetry of Majorana bilinears.

Table C.1: Symmetry of the matrices $C\Gamma_A$ in $d = 4$, where $(C\Gamma_A)^T = \lambda t_A (C\Gamma_A)$ with $\lambda = -1$. "S" denotes symmetric, "A" antisymmetric. The counts (10 symmetric, 6 antisymmetric) saturate the dimensions of the symmetric and antisymmetric 4×4 matrices.

Γ_A	multiplicity	t_A	$(C\Gamma_A)^T/(C\Gamma_A)$	symmetry
$\mathbf{1}$	1	$+1$	-1	A
γ^μ	4	-1	$+1$	S
$\sigma^{\mu\nu}$	6	-1	$+1$	S
$\gamma^\mu\gamma^5$	4	$+1$	-1	A
γ^5	1	$+1$	-1	A
		totals:	S= 10	A= 6

C.4.5 The explicit form $C = i\gamma^2\gamma^0$

Theorems C.36–C.37 fix C abstractly; in the Dirac and chiral representations of Section C.3 it has a clean closed form. The relevant input is the transpose pattern of the standard matrices. In both the Dirac and chiral bases the spatial matrices are built from the Pauli matrices (Chapter B) as $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{smallmatrix}\right)$, with σ^1, σ^3 real symmetric and σ^2 imaginary antisymmetric, while γ^0 is symmetric. Consequently

$$\gamma^{0T} = \gamma^0, \quad \gamma^{1T} = -\gamma^1, \quad \gamma^{2T} = \gamma^2, \quad \gamma^{3T} = -\gamma^3, \tag{C.93}$$

so γ^0, γ^2 are symmetric and γ^1, γ^3 antisymmetric; equivalently, γ^2 is the unique imaginary matrix, $\gamma^{\mu *} = \gamma^\mu$ for $\mu \neq 2$ and $\gamma^{2*} = -\gamma^2$.

Lemma C.38 (Explicit charge-conjugation matrix)**.** *In any representation with the transpose pattern* (C.93) *(in particular the Dirac and chiral bases),*

$$C = i\gamma^2\gamma^0, \qquad C^{-1} = -C, \qquad C^2 = -\mathbf{1}, \qquad C^T = -C, \qquad C^\dagger = C^{-1}. \tag{C.94}$$

Proof. We verify the intertwining property and then each algebraic relation, carrying all the algebra. First, the auxiliary relations

$$(\gamma^0)^2 = \mathbf{1}, \qquad (\gamma^2)^2 = \eta^{22}\mathbf{1} = -\mathbf{1}, \qquad \gamma^0\gamma^2 = -\gamma^2\gamma^0. \tag{C.95}$$

From these, $C^{-1} = (i\gamma^2\gamma^0)^{-1} = -i\,(\gamma^0)^{-1}(\gamma^2)^{-1} = -i\,\gamma^0(-\gamma^2) = i\gamma^0\gamma^2 = -i\gamma^2\gamma^0 = -C$. Hence $C^{-1} = -C$, equivalently $C^2 = -\mathbf{1}$.

Next the intertwining. With $C^{-1} = -C = -i\gamma^2\gamma^0$,

$$C\gamma^\mu C^{-1} = (i\gamma^2\gamma^0)\gamma^\mu(-i\gamma^2\gamma^0) = \gamma^2\gamma^0\gamma^\mu\gamma^2\gamma^0. \tag{C.96}$$

Evaluate component by component using only the anticommutation rules and $\gamma^0\gamma^2\gamma^0 = -\gamma^2$:

$$\mu = 0: \quad \gamma^2\gamma^0\gamma^0\gamma^2\gamma^0 = \gamma^2\gamma^2\gamma^0 = (\gamma^2)^2\gamma^0 = -\gamma^0 = -\gamma^{0T}, \tag{C.97}$$

$$\mu = 1: \quad \gamma^2\gamma^0\gamma^1\gamma^2\gamma^0 = -\gamma^2\gamma^1(\gamma^0\gamma^2\gamma^0) = -\gamma^2\gamma^1(-\gamma^2) = \gamma^2\gamma^1\gamma^2 = -(\gamma^2)^2\gamma^1 = \gamma^1 = -\gamma^{1T}, \tag{C.98}$$

$$\mu = 2: \quad \gamma^2\gamma^0\gamma^2\gamma^2\gamma^0 = \gamma^2\gamma^0(-1)\gamma^0 = -\gamma^2 = -\gamma^{2T}, \tag{C.99}$$

$$\mu = 3: \quad \gamma^2\gamma^0\gamma^3\gamma^2\gamma^0 = -\gamma^2\gamma^3(\gamma^0\gamma^2\gamma^0) = -\gamma^2\gamma^3(-\gamma^2) = \gamma^2\gamma^3\gamma^2 = -(\gamma^2)^2\gamma^3 = \gamma^3 = -\gamma^{3T}, \tag{C.100}$$

where in the last column we used the transpose pattern (C.93). Thus $C\gamma^\mu C^{-1} = -\gamma^{\mu T}$ for all μ, confirming (C.80).

Finally the symmetry and unitarity. Using $\gamma^{0T} = \gamma^0$, $\gamma^{2T} = \gamma^2$,

$$C^T = (i\gamma^2\gamma^0)^T = i\,\gamma^{0T}\gamma^{2T} = i\gamma^0\gamma^2 = -i\gamma^2\gamma^0 = -C, \tag{C.101}$$

reproducing $C^T = -C$ from Theorem C.37 in this basis. For the adjoint, the Hermiticity convention $\gamma^{i\dagger} = -\gamma^i$, $\gamma^{0\dagger} = \gamma^0$ gives

$$C^\dagger = (i\gamma^2\gamma^0)^\dagger = -i\,\gamma^{0\dagger}\gamma^{2\dagger} = -i\,\gamma^0(-\gamma^2) = i\gamma^0\gamma^2 = -i\gamma^2\gamma^0 = -C = C^{-1}, \tag{C.102}$$

so C is unitary, $C^\dagger C = \mathbf{1}$. □

The relation $C^\dagger = C^{-1}$ is not forced by Theorems C.36 and C.37 alone — it is a normalization. The scalar freedom $C \to zC$ rescales $C^\dagger C \to |z|^2 C^\dagger C$, and one checks (by the same Schur argument as for uniqueness) that $C^\dagger C$ is a positive multiple of $\mathbf{1}$; choosing $|z|$ to make $C^\dagger C = \mathbf{1}$ is always possible and is the standard convention adopted here. With that and $C^T = -C$ fixed, the only remaining freedom is a phase, fixed by (C.94) to give $C^2 = -\mathbf{1}$.

C.4.6 Intertwining of γ^5 and $\sigma^{\mu\nu}$

The signs t_A computed in the proof of Theorem C.37 are exactly the statements of how C conjugates the higher basis elements, since $\Gamma_A^T = t_A C\Gamma_A C^{-1}$ means $C\Gamma_A C^{-1} = t_A\Gamma_A^T$. Reading them off:

Proposition C.39 (C on γ^5 and $\sigma^{\mu\nu}$)**.** *The charge-conjugation matrix satisfies*

$$C\gamma^5 C^{-1} = \gamma^{5T}, \qquad C\,\sigma^{\mu\nu}C^{-1} = -\sigma^{\mu\nu T}, \qquad C(\gamma^\mu\gamma^5)C^{-1} = (\gamma^\mu\gamma^5)^T = \gamma^{5T}\gamma^{\mu T}. \tag{C.103}$$

These three relations, together with $C\gamma^\mu C^{-1} = -\gamma^{\mu T}$ and $C\mathbf{1}C^{-1} = \mathbf{1}$, complete the action of C on the entire basis. In the chiral and Dirac representations γ^5 is symmetric, $\gamma^{5T} = \gamma^5$, so the first relation reads $C\gamma^5 C^{-1} = \gamma^5$: charge conjugation commutes with chirality up to the transpose, which is the algebraic root of the fact (used in Chapter 11 and in the anomaly analysis of Chapter 36) that charge conjugation maps a left-handed antiparticle to a right-handed particle. The second relation, $C\sigma^{\mu\nu}C^{-1} = -\sigma^{\mu\nu T}$, is what makes the tensor bilinear $\bar\psi\sigma^{\mu\nu}\psi$ odd under C.

C.4.7 Charge conjugation of spinors and the antiparticle solutions

We can now make precise the operation that turns a particle wavefunction into the corresponding antiparticle wavefunction at the level of the Dirac matrices — the algebraic backbone of every v-spinor manipulation in QED (Chapters 17 and 18). The charge-conjugate of a Dirac field is defined (Chapter 11) by

$$\psi^c \equiv C\bar\psi^{\,T}, \qquad \bar\psi = \psi^\dagger\gamma^0. \tag{C.104}$$

Worked Example C.40 (The v-spinors as charge conjugates of the u-spinors)**.** Let $u^s(p)$ be a positive-energy solution of the Dirac equation in momentum space, so that $(\not p - m)u^s(p) = 0$ and, taking the Dirac adjoint, $\bar u^s(p)(\not p - m) = 0$. We show that the charge conjugate of u solves the *negative*-energy equation $(\not p + m)v = 0$, which is the defining equation of the v-spinors, so that $v^s(p) \propto C\bar u^s(p)^T$.

Start from the adjoint equation and transpose it, using $(\not p)^T = p_\mu\gamma^{\mu T}$:

$$\bar u(\not p - m) = 0 \quad \xRightarrow{\text{transpose}} \quad (p_\mu\gamma^{\mu T} - m)\,\bar u^{\,T} = 0. \tag{C.105}$$

Now eliminate the transposed gammas with $\gamma^{\mu T} = -C\gamma^\mu C^{-1}$:

$$(-C\,p_\mu\gamma^\mu C^{-1} - m)\,\bar u^{\,T} = 0. \tag{C.106}$$

Left-multiply by $-C^{-1}$ and use $C^{-1}\bar u^T$ as the new object:

$$(p_\mu\gamma^\mu + m)\,C^{-1}\bar u^{\,T} = 0 \quad \Longleftrightarrow \quad (\not p + m)\,(C^{-1}\bar u^{\,T}) = 0. \tag{C.107}$$

Thus $C^{-1}\bar u^{\,T}$ satisfies exactly the v-spinor equation. Because $C^{-1} = -C$ (Theorem C.38) and (C.107) is homogeneous, $C\bar u^{\,T}$ solves it as well; the overall sign is the conventional phase, and the standard normalization fixes

$$v^s(p) = C\,\bar u^s(p)^{\,T}, \qquad u^s(p) = C\,\bar v^s(p)^{\,T}, \tag{C.108}$$

the second relation following by the same computation with $m \to -m$. This is the identity that lets every spin sum and trace over antiparticle states in Chapters 17 and 18 be rewritten in terms of u-spinors, and it is the reason the completeness relations $\sum_s u\bar{u} = \not{p} + m$ and $\sum_s v\bar{v} = \not{p} - m$ are charge conjugates of one another.

C.4.8 Time reversal and the reality matrix

Time reversal is antiunitary, so its matrix is naturally an intertwiner of $\{\gamma^\mu\}$ with the *complex-conjugate* set $\{\gamma^{\mu *}\}$, with one extra relative sign between the time and space components reflecting $t \mapsto -t$. Demanding that $\psi'(t,\mathbf{x}) = T\psi^*(-t,\mathbf{x})$ again solve the Dirac equation forces (the derivation is the operator content of Chapter 11; here we record the matrix condition)

$$T^{-1}\gamma^0 T = \gamma^{0*}, \qquad T^{-1}\gamma^i T = -\gamma^{i*}, \qquad \text{i.e.} \qquad T^{-1}\gamma^\mu T = \gamma^*_\mu. \tag{C.109}$$

In a representation with the transpose pattern (C.93), where only γ^2 is imaginary, (C.109) requires T to commute with γ^0, γ^2 and anticommute with γ^1, γ^3. The unique solution up to scale is

$$T = \gamma^1\gamma^3 \qquad \text{(up to a phase)}, \tag{C.110}$$

as one checks directly: $\gamma^1\gamma^3$ contains the two matrices it must anticommute with and none of those it must commute with, and using $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, $\gamma^1\gamma^3$ commutes with γ^0 and γ^2 (each crosses two anticommuting factors, $(-1)^2 = +1$) and anticommutes with γ^1 and γ^3 (each crosses one).

The time-reversal matrix is not independent of C. In these representations

$$C\gamma^5 = i\gamma^2\gamma^0 \cdot i\gamma^0\gamma^1\gamma^2\gamma^3 = -\gamma^2\gamma^1\gamma^2\gamma^3 = \gamma^1(\gamma^2)^2\gamma^3 = -\gamma^1\gamma^3 = -T, \tag{C.111}$$

so $T = -C\gamma^5$, displaying the discrete-symmetry matrices as a single algebraic family generated by γ^0, C, and γ^5. The reality matrix B of (C.81) belongs to the same family: from $\gamma^{\mu *} = (\gamma^{\mu\dagger})^T = (\gamma^0\gamma^\mu\gamma^0)^T = \gamma^{0T}\gamma^{\mu T}\gamma^{0T}$ and $\gamma^{\mu T} = -C\gamma^\mu C^{-1}$ one finds, after using $\gamma^{0T} = -C\gamma^0 C^{-1}$,

$$\gamma^{\mu *} = -(C\gamma^0)\,\gamma^\mu\,(C\gamma^0)^{-1}, \tag{C.112}$$

so $C\gamma^0$ intertwines $\{\gamma^\mu\}$ with $\{-\gamma^{\mu *}\}$; the matrix B of (C.81) for $\{+\gamma^{\mu *}\}$ is therefore $B \propto \gamma^5 C\gamma^0$ (prepending the γ^5 conjugation flips the overall sign). The product of the parity, charge-conjugation, and time-reversal matrices is proportional to γ^5, the matrix that accompanies the combined CPT transformation; we return to this in the discussion of CPT and of the chiral anomaly in Chapters 11 and 36.

C.4.9 The Majorana reality condition

A Dirac spinor carries four complex, i.e. eight real, components. The reality (Majorana) condition asks whether one can impose, consistently and Lorentz-covariantly, that a spinor equal its own charge conjugate, halving the count. With $\psi^c = C\bar{\psi}^T$ from (C.104), the condition is

Definition C.41 (Majorana spinor)**.** A Dirac spinor ψ is a *Majorana spinor* if it equals its charge conjugate,

$$\psi = \psi^c = C\bar{\psi}^T. \tag{C.113}$$

The right-hand side is antilinear in ψ, so (C.113) is a reality condition relating ψ to ψ^*, not a linear constraint within the complex vector space. Writing $\bar{\psi}^T = (\psi^\dagger\gamma^0)^T = \gamma^{0T}\psi^*$, it takes the form $\psi = M\psi^*$ with the antilinear-involution matrix

$$M \equiv C\gamma^{0T}. \tag{C.114}$$

For such a condition to be imposable, the map $\psi \mapsto M\psi^*$ must square to the identity, so that (C.113) has a nontrivial solution space rather than only $\psi = 0$.

Theorem C.42 (Consistency and component count of the Majorana condition)**.** *In $d = 4$ the antilinear map $K : \psi \mapsto M\psi^*$ with $M = C\gamma^{0T}$ satisfies $K^2 = \mathbf{1}$, so the Majorana condition* (C.113) *is consistent. Its solution space is a* real *vector space of dimension* 4*: a Majorana spinor has four independent real components, the same count as a single Weyl spinor.*

Proof. Consistency. Applying K twice, $K^2\psi = M\,(M\psi^*)^* = MM^*\psi$, so $K^2 = \mathbf{1}$ iff $MM^* = \mathbf{1}$. Compute MM^* using $\gamma^{0\dagger} = \gamma^0$ (hence $\gamma^{0T*} = \gamma^{0\dagger} = \gamma^0$), $C^* = -C^{-1}$ (from $C^\dagger = C^{-1}$ and $C^T = -C$, since $C^\dagger = (C^T)^* = -C^*$), $\gamma^{0T} = -C\gamma^0C^{-1}$, and $C^2 = -\mathbf{1}$:

$$\begin{aligned} MM^* &= C\gamma^{0T}\,(C\gamma^{0T})^* = C\gamma^{0T}\,C^*\gamma^{0\dagger} = C\big(-C\gamma^0C^{-1}\big)\big(-C^{-1}\big)\gamma^0 \\ &= C^2\,\gamma^0\,C^{-2}\,\gamma^0 = (-\mathbf{1})\gamma^0(-\mathbf{1})\gamma^0 = \gamma^0\gamma^0 = \mathbf{1}. \end{aligned} \tag{C.115}$$

Thus $K^2 = \mathbf{1}$; the condition is an honest involution. (Had $C^2 = +\mathbf{1}$ instead — the situation in dimensions $d \equiv 6 \bmod 8$ — one would find $MM^* = -\mathbf{1}$, $K^2 = -\mathbf{1}$, and no Majorana spinor could exist; the existence of the reality condition is thus tied to d mod 8, and $d = 4$ is one of the dimensions where it succeeds.)

Component count. The solution space is $V_M = \{\psi \in \mathbb{C}^4 : \psi = K\psi\}$, the $+1$ eigenspace of the antilinear involution K. It is closed under real but not complex scalar multiplication, so it is a real vector space; we compute its real dimension. Consider the real-linear map $P : \mathbb{C}^4 \to \mathbb{C}^4$, $P\varphi = \frac{1}{2}(\varphi + K\varphi)$. Because $K^2 = \mathbf{1}$, P is idempotent: $P^2\varphi = \frac{1}{4}(\varphi + K\varphi + K\varphi + K^2\varphi) = \frac{1}{2}(\varphi + K\varphi) = P\varphi$, and $KP\varphi = \frac{1}{2}(K\varphi + \varphi) = P\varphi$, so the image of P lies in V_M; conversely every $\psi \in V_M$

satisfies $P\psi = \psi$, so $\operatorname{im} P = V_M$. Likewise $Q = \frac{1}{2}(\mathbf{1} - K)$ is a complementary real idempotent with $\operatorname{im} Q = \{\psi : K\psi = -\psi\}$, and $P + Q = \mathbf{1}$. As a real vector space $\mathbb{C}^4$ has dimension 8 and splits as $\operatorname{im} P \oplus \operatorname{im} Q$. The two summands are isomorphic as real spaces via multiplication by i, which anticommutes with the antilinear K (i.e. $K(i\psi) = -iK\psi$) and so exchanges the ± 1 eigenspaces; hence $\dim_{\mathbb{R}} \operatorname{im} P = \dim_{\mathbb{R}} \operatorname{im} Q = 4$. Therefore $\dim_{\mathbb{R}} V_M = 4$. □

Worked Example C.43 (Why a Majorana spinor carries the data of one Weyl spinor)**.** We re-derive the count of Theorem C.42 constructively, exhibiting the four real degrees of freedom directly and showing how they organize into a single two-component object — the bridge to the Weyl description of Chapter 11 and the $(\frac{1}{2}, 0)$ representation of Chapter 3. Define, for an arbitrary $\varphi \in \mathbb{C}^4$,

$$\psi(\varphi) = \varphi + M\varphi^*. \tag{C.116}$$

Then $\psi(\varphi)$ is automatically Majorana: using $MM^* = \mathbf{1}$,

$$K\psi(\varphi) = M(\varphi + M\varphi^*)^* = M\varphi^* + MM^*\varphi = M\varphi^* + \varphi = \psi(\varphi). \tag{C.117}$$

The map $\varphi \mapsto \psi(\varphi)$ from $\mathbb{C}^4$ (real dimension 8) onto V_M is real-linear and surjective, since any $\psi \in V_M$ is its own image, $\psi(\psi) = \psi + M\psi^* = \psi + K\psi = 2\psi$, up to the harmless factor 2. Its kernel is $\{\varphi : \varphi = -M\varphi^*\} = \operatorname{im} Q$, of real dimension 4. By rank–nullity for real vector spaces, $\dim_{\mathbb{R}} V_M = 8 - 4 = 4$, confirming the count. In the chiral basis (Section C.3), where $\psi = (\psi_L, \psi_R)^T$ and chirality is block-diagonal, the Majorana condition (C.113) reduces to the single relation $\psi_R = i\sigma^2\psi_L^*$ (with σ^2 from Chapter B): the right-handed component is completely determined by the left-handed one, leaving the two complex — four real — components of ψ_L as the independent data. A Majorana spinor is thus precisely a Weyl spinor together with the reality condition that ties its conjugate to its other chirality, which is why a Majorana mass and a Weyl description carry the same information.

With the charge-conjugation matrix C constructed and its symmetry settled, we have in hand exactly the data the next section needs. The signs t_A and the symmetry table Table C.1 determine which of the sixteen bilinears $\bar{\psi}\Gamma_A\psi$ are symmetric or antisymmetric under interchange of the two spinors — the input for Majorana bilinears, for the relative signs in crossing, and for the Fierz rearrangement built next — while the matrices γ^0 (parity/Dirac), C (charge conjugation), and $\gamma^1\gamma^3$ (time reversal) are the discrete-symmetry intertwiners that Chapters 11 and 36 import as the matrix part of the C, P, T operators. We turn now to assembling the Γ_A into a complete, trace-orthogonal basis and reading off the Lorentz character of each bilinear.

C.5 The Sixteen Bilinear Covariants and Completeness

Sections C.3 and C.4 handed us everything we need to organize the Dirac algebra into a usable basis: the explicit matrices γ^μ, the spin generators $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]$, the chirality matrix $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, and the Dirac conjugation $\bar\psi = \psi^\dagger\gamma^0$ together with the discrete-symmetry intertwiners C and γ^0 (parity). What remains is to assemble these into a single algebraic statement that powers every spinor computation in the book: the space of 4×4 complex matrices is 16-dimensional, and a particular sixteen-element set built from the gammas is simultaneously (i) a basis, (ii) *trace-orthonormal*, and (iii) sorted by Lorentz character into a scalar, a vector, an antisymmetric tensor, an axial vector, and a pseudoscalar. The trace-orthonormality is the engine that lets us project any matrix onto the basis with a single trace; sorting by Lorentz character tells us, with no further work, how each fermion bilinear $\bar\psi\,\Gamma^A\psi$ behaves under the Lorentz group; and the resolution of the identity that follows — the spinor completeness, or Fierz, relation — is the master rearrangement identity that the amplitude chapters lean on constantly. Everything in this section is proved from the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}$ and the elementary trace facts it forces.

C.5.1 The sixteen matrices and the index set A

The four gamma matrices anticommute and square to $\pm\mathbf{1}$. Consequently any product of gammas can be reduced, using $\gamma^\mu\gamma^\nu = -\gamma^\nu\gamma^\mu$ for $\mu \neq \nu$ and $(\gamma^\mu)^2 = \eta^{\mu\mu}\mathbf{1}$ (no sum), to a product in which each of the four distinct matrices $\gamma^0, \gamma^1, \gamma^2, \gamma^3$ appears *at most once*, times a numerical factor ± 1 or $\pm i$. The distinct reduced products are labelled by the *subsets* of $\{0,1,2,3\}$: there are $\binom{4}{0} = 1$ empty product, $\binom{4}{1} = 4$ singletons, $\binom{4}{2} = 6$ pairs, $\binom{4}{3} = 4$ triples, and $\binom{4}{4} = 1$ quadruple, for a total of $1+4+6+4+1 = 16$. These sixteen reduced products, normalized as in the table below, are the *bilinear covariants*.

Definition C.44 (The sixteen Γ^A)**.** The basis matrices $\{\Gamma^A\}_{A=1}^{16}$ of the Dirac algebra, grouped by the number of distinct gammas they contain, are

$$\underbrace{\mathbf{1}}_{S}, \qquad \underbrace{\gamma^\mu}_{V}, \qquad \underbrace{\sigma^{\mu\nu}\ (\mu<\nu)}_{T}, \qquad \underbrace{\gamma^\mu\gamma^5}_{A}, \qquad \underbrace{\gamma^5}_{P}, \tag{C.118}$$

where $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu] = i\gamma^\mu\gamma^\nu$ for $\mu \neq \nu$ (cf. Section C.3), and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$. The labels S, V, T, A, P denote scalar, vector, tensor, axial-vector, and pseudoscalar; their multiplicities are $1, 4, 6, 4, 1$. We write the index A as a collective label ranging over these sixteen matrices, and we always sum a repeated A over all sixteen.

The triples are not listed separately because each product of three distinct gammas is, up to a factor, one of the $\gamma^\mu\gamma^5$: with all four indices distinct,

Type	Γ^A	# components	$(\Gamma^A)^2$	bilinear $\bar\psi\,\Gamma^A\psi$
Scalar S	$\mathbf{1}$	1	$+\mathbf{1}$	$\bar\psi\psi$
Vector V	γ^μ	4	$\eta^{\mu\mu}\mathbf{1}$	$\bar\psi\gamma^\mu\psi$
Tensor T	$\sigma^{\mu\nu}$	6	$\eta^{\mu\mu}\eta^{\nu\nu}\mathbf{1}$	$\bar\psi\sigma^{\mu\nu}\psi$
Axial vector A	$\gamma^\mu\gamma^5$	4	$-\eta^{\mu\mu}\mathbf{1}$	$\bar\psi\gamma^\mu\gamma^5\psi$
Pseudoscalar P	γ^5	1	$+\mathbf{1}$	$\bar\psi\gamma^5\psi$

Table C.2: The sixteen bilinear covariants. Repeated indices in the "$(\Gamma^A)^2$" column are *not* summed; each square is $\pm\mathbf{1}$. The Lorentz transformation laws of the bilinears are derived in Section C.5.5.

$\gamma^\nu\gamma^\rho\gamma^\sigma \propto \gamma^\mu\gamma^5$ where μ is the missing index. Thus the "four triples" are exactly the axial set A, and the single "quadruple" is γ^5 itself. The reorganization into Eq. (C.118) is the one adapted to the Lorentz group, as Section C.5.5 makes precise.

Squares and the dual basis. Every Γ^A squares to $\pm\mathbf{1}$, as recorded in Table C.2. The scalar and pseudoscalar are immediate: $\mathbf{1}^2 = \mathbf{1}$ and $(\gamma^5)^2 = \mathbf{1}$ (proved in Section C.3). For the vector, $(\gamma^\mu)^2 = \eta^{\mu\mu}\mathbf{1}$ (no sum) is the diagonal Clifford relation. For the tensor with $\mu \neq \nu$,

$$(\sigma^{\mu\nu})^2 = (i\gamma^\mu\gamma^\nu)^2 = -\gamma^\mu\gamma^\nu\gamma^\mu\gamma^\nu = +\gamma^\mu\gamma^\mu\gamma^\nu\gamma^\nu = \eta^{\mu\mu}\eta^{\nu\nu}\,\mathbf{1} \qquad \text{(no sum)}, \tag{C.119}$$

where we used $\gamma^\nu\gamma^\mu = -\gamma^\mu\gamma^\nu$ once. For the axial vector,

$$(\gamma^\mu\gamma^5)^2 = \gamma^\mu\gamma^5\gamma^\mu\gamma^5 = -\gamma^\mu\gamma^\mu(\gamma^5)^2 = -\eta^{\mu\mu}\,\mathbf{1} \qquad \text{(no sum)}, \tag{C.120}$$

using $\{\gamma^5, \gamma^\mu\} = 0$. Since each $(\Gamma^A)^2 = \eta_A\mathbf{1}$ with $\eta_A \in \{+1, -1\}$, every Γ^A is invertible with

$$\Gamma_A := (\Gamma^A)^{-1} = \eta_A\,\Gamma^A. \tag{C.121}$$

We call Γ_A the *dual* of Γ^A. Concretely, Γ_A is obtained by lowering all Lorentz indices: $\mathbf{1} \mapsto \mathbf{1}$, $\gamma^\mu \mapsto \gamma_\mu$, $\sigma^{\mu\nu} \mapsto \sigma_{\mu\nu}$, $\gamma^5 \mapsto \gamma^5$, while for the axial set the inverse carries an extra sign, $(\gamma^\mu\gamma^5)^{-1} = -\gamma_\mu\gamma^5$ (from Eq. (C.120), since $\eta_A = -\eta^{\mu\mu}$). This dual is exactly what makes the trace pairing in Section C.5.4 come out as a Kronecker delta.

C.5.2 Elementary trace facts

The trace pairing rests on a handful of facts, all consequences of the Clifford relation, of cyclicity of the trace, and of the properties of γ^5 established in

Section C.3. The systematic trace recursion — the full toolkit for traces of arbitrarily many gammas — is the business of Section C.6; here we need only the minimum.

Lemma C.45 (Basic traces)**.** *In the $d = 4$ representation,*

$$\operatorname{tr} \mathbf{1} = 4, \tag{C.122}$$

$$\operatorname{tr}(\gamma^{\mu_1} \cdots \gamma^{\mu_n}) = 0 \quad \text{for } n \text{ odd}, \tag{C.123}$$

$$\operatorname{tr}(\gamma^{\mu}\gamma^{\nu}) = 4\,\eta^{\mu\nu}, \tag{C.124}$$

$$\operatorname{tr} \gamma^5 = 0. \tag{C.125}$$

Proof. Equation (C.122) is the statement that the representation acts on $\mathbb{C}^4$. For Eq. (C.123), insert $\mathbf{1} = (\gamma^5)^2$ and slide one γ^5 through the string. Because $\{\gamma^5, \gamma^\mu\} = 0$, moving a γ^5 past n gammas costs $(-1)^n$:

$$\operatorname{tr}(\gamma^{\mu_1} \cdots \gamma^{\mu_n}) = \operatorname{tr}(\gamma^5 \gamma^5 \gamma^{\mu_1} \cdots \gamma^{\mu_n}) = (-1)^n \operatorname{tr}(\gamma^5 \gamma^{\mu_1} \cdots \gamma^{\mu_n} \gamma^5) = (-1)^n \operatorname{tr}(\gamma^{\mu_1} \cdots \gamma^{\mu_n}), \tag{C.126}$$

where the last equality used cyclicity to bring the trailing γ^5 to the front and $(\gamma^5)^2 = \mathbf{1}$. For n odd the trace equals minus itself, hence vanishes. Equation (C.124) follows by symmetrizing: by cyclicity $\operatorname{tr}(\gamma^\mu\gamma^\nu) = \operatorname{tr}(\gamma^\nu\gamma^\mu)$, so

$$\operatorname{tr}(\gamma^{\mu}\gamma^{\nu}) = \tfrac{1}{2} \operatorname{tr}\big(\{\gamma^\mu, \gamma^\nu\} \big) = \tfrac{1}{2} \operatorname{tr}(2\eta^{\mu\nu}\mathbf{1}) = \eta^{\mu\nu} \operatorname{tr} \mathbf{1} = 4\,\eta^{\mu\nu}. \tag{C.127}$$

Finally, for Eq. (C.125) choose any single index, say 0, and use $(\gamma^0)^2 = \mathbf{1}$:

$$\operatorname{tr} \gamma^5 = \operatorname{tr}(\gamma^5 \gamma^0 \gamma^0) = \operatorname{tr}(\gamma^0 \gamma^5 \gamma^0) = -\operatorname{tr}(\gamma^5 \gamma^0 \gamma^0) = -\operatorname{tr} \gamma^5, \tag{C.128}$$

where the middle step is cyclicity and the next uses $\{\gamma^5, \gamma^0\} = 0$. Hence $\operatorname{tr} \gamma^5 = 0$. □

C.5.3 Each nontrivial Γ^A is traceless, and the set closes under multiplication

The orthogonality theorem needs two structural facts about the sixteen matrices: every one except the identity is traceless, and the product of any two is again one of the sixteen up to a numerical factor. Both follow from a single observation — that for every $\Gamma^A \neq \mathbf{1}$ there is an *anticommuting partner* among the gammas.

Lemma C.46 (Tracelessness)**.** $\operatorname{tr} \Gamma^A = 0$ *for every A except* $\Gamma^A = \mathbf{1}$.

Proof. For each $\Gamma^A \neq \mathbf{1}$ we exhibit an invertible matrix B with $B\Gamma^A B^{-1} = -\Gamma^A$; then cyclicity gives $\operatorname{tr} \Gamma^A = \operatorname{tr}(B\Gamma^A B^{-1}) = -\operatorname{tr} \Gamma^A = 0$.

- $\Gamma^A = \gamma^\mu$: take $B = \gamma^5$. Then $\gamma^5\gamma^\mu(\gamma^5)^{-1} = -\gamma^\mu$ because $\{\gamma^5, \gamma^\mu\} = 0$.

- $\Gamma^A = \sigma^{\mu\nu} = i\gamma^\mu\gamma^\nu$ ($\mu \neq \nu$): take $B = \gamma^\mu$. Since $\gamma^\mu(\gamma^\mu\gamma^\nu) = (\gamma^\mu)^2\gamma^\nu$ while $(\gamma^\mu\gamma^\nu)\gamma^\mu = -(\gamma^\mu)^2\gamma^\nu$, the matrix γ^μ anticommutes with $\gamma^\mu\gamma^\nu$, hence with $\sigma^{\mu\nu}$.

- $\Gamma^A = \gamma^\mu\gamma^5$: take $B = \gamma^\mu$. As in Eq. (C.120), $\gamma^\mu(\gamma^\mu\gamma^5) = (\gamma^\mu)^2\gamma^5$ and $(\gamma^\mu\gamma^5)\gamma^\mu = -(\gamma^\mu)^2\gamma^5$, so they anticommute.

- $\Gamma^A = \gamma^5$: take any $B = \gamma^\rho$, since $\{\gamma^5, \gamma^\rho\} = 0$.

In every case B is one of γ^ρ or γ^5, all invertible by Eq. (C.121). □

Lemma C.47 (Closure)**.** *For any two basis matrices,* $\Gamma^A\Gamma^B = c_{AB}\,\Gamma^C$ *for some* Γ^C *in the set and some scalar* $c_{AB} \in \{\pm 1, \pm i\}$*. Moreover* $\Gamma^C = \mathbf{1}$ *if and only if* $A = B$*.*

Proof. Each Γ^A is, up to the normalizing factor $\pm i$ in Theorem C.44, the ordered product of the distinct gammas indexed by a subset $S_A \subseteq \{0, 1, 2, 3\}$. The product $\Gamma^A\Gamma^B$ is then the ordered product over S_A followed by that over S_B. Anticommuting the two strings into a single ordered product introduces signs only; whenever an index appears in both S_A and S_B the two copies meet and collapse via $(\gamma^\rho)^2 = \eta^{\rho\rho}\mathbf{1}$, again only a sign. What survives is the ordered product over the symmetric difference $S_A \triangle S_B$, times a factor in $\{\pm 1, \pm i\}$. That surviving product is, after restoring the normalization of Theorem C.44, exactly one of the sixteen Γ^C. Finally $\Gamma^A\Gamma^B \propto \mathbf{1}$ demands $S_A \triangle S_B = \varnothing$, i.e. $S_A = S_B$, i.e. $A = B$. □

C.5.4 Trace-orthogonality and the basis property

We can now state the central algebraic fact of the section.

Theorem C.48 (Trace-orthonormality)**.** *With the dual matrices* $\Gamma_A = (\Gamma^A)^{-1}$ *of Eq.* (C.121)*,*

$$\operatorname{tr}(\Gamma^A\,\Gamma_B) = 4\,\delta^A{}_B. \tag{C.129}$$

Proof. If $A = B$ then $\Gamma^A\Gamma_A = \Gamma^A(\Gamma^A)^{-1} = \mathbf{1}$ and the trace is $\operatorname{tr}\mathbf{1} = 4$ by Eq. (C.122). If $A \neq B$ then, writing $\Gamma_B = \eta_B\Gamma^B$ from Eq. (C.121) and applying Theorem C.47, $\Gamma^A\Gamma_B = \eta_B\,c_{AB}\,\Gamma^C$ with $\Gamma^C \neq \mathbf{1}$ (because $A \neq B$). By Theorem C.46 every nontrivial Γ^C is traceless, so the trace vanishes. Together these give Eq. (C.129). □

Trace-orthonormality forces linear independence, and with the dimension count this makes the sixteen matrices a basis.

Theorem C.49 (Basis and projection formula)**.** *The sixteen Γ^A are linearly independent and therefore form a basis of the 16-dimensional complex vector space of 4×4 matrices. Every 4×4 matrix M has the unique expansion*

$$M = \frac{1}{4}\sum_{A} \operatorname{tr}(M\,\Gamma_A)\,\Gamma^A. \tag{C.130}$$

Proof. Suppose $\sum_A c_A\,\Gamma^A = 0$ for scalars c_A. Multiply by Γ_B and take the trace; by Eq. (C.129),

$$0 = \operatorname{tr}\Big(\sum_A c_A\,\Gamma^A\,\Gamma_B\Big) = \sum_A c_A \cdot 4\,\delta^A{}_B = 4\,c_B, \tag{C.131}$$

so every $c_B = 0$ and the set is linearly independent. Sixteen linearly independent vectors in a sixteen-dimensional space form a basis, so we may write $M = \sum_A m_A\,\Gamma^A$ with unique coefficients m_A. Multiplying by Γ_B and tracing, $\operatorname{tr}(M\Gamma_B) = \sum_A m_A \cdot 4\,\delta^A{}_B = 4\,m_B$, which gives $m_A = \frac{1}{4}\operatorname{tr}(M\Gamma_A)$ and hence Eq. (C.130). □

Equation (C.130) is the workhorse: it reduces "what is this matrix?" to sixteen trace computations. The following worked example exercises it on a matrix that recurs in practice, the projector built from a unit timelike vector.

Worked Example C.50 (Expanding a chiral projector and a spin projector)**.** Consider first $M = P_R = \frac{1}{2}(\mathbf{1}+\gamma^5)$, the right-handed chirality projector. We expand it in the Γ^A basis using Eq. (C.130). The only nonzero traces are against the scalar and pseudoscalar duals: $\operatorname{tr}(P_R\,\mathbf{1}) = \frac{1}{2}(\operatorname{tr}\mathbf{1} + \operatorname{tr}\gamma^5) = 2$ and $\operatorname{tr}(P_R\,\gamma^5) = \frac{1}{2}(\operatorname{tr}\gamma^5 + \operatorname{tr}(\gamma^5)^2) = \frac{1}{2}\operatorname{tr}\mathbf{1} = 2$, while every trace against γ_μ, $\sigma_{\mu\nu}$ or $-\gamma_\mu\gamma^5$ vanishes by Theorem C.45 (odd numbers of gammas, or $\operatorname{tr}\gamma^5 = 0$, or $\operatorname{tr}(\gamma^5\gamma^\mu) = 0$ since $\gamma^5\gamma^\mu$ is a product of five gammas). Thus

$$P_R = \frac{1}{4}\big[\,2\cdot\mathbf{1} + 2\cdot\gamma^5\,\big] = \tfrac{1}{2}(\mathbf{1}+\gamma^5), \tag{C.132}$$

reproducing the input — a consistency check that the projection formula is faithful, and a reminder that P_R lives entirely in the $S\oplus P$ sector.

For a less trivial case take $M = \not{a}\not{b}$ with a, b four-vectors, where $\not{a} = a_\mu\gamma^\mu$. Using $\not{a}\not{b} = a_\mu b_\nu\gamma^\mu\gamma^\nu = a_\mu b_\nu(\frac{1}{2}\{\gamma^\mu,\gamma^\nu\} + \frac{1}{2}[\gamma^\mu,\gamma^\nu]) = (a\cdot b)\,\mathbf{1} - i\,a_\mu b_\nu\,\sigma^{\mu\nu}$, the expansion is read off directly: the scalar coefficient is $a\cdot b$ and the tensor coefficient is $-i\,a_\mu b_\nu$, with no vector, axial, or pseudoscalar part. Verifying through Eq. (C.130): $\frac{1}{4}\operatorname{tr}(\not{a}\not{b}) = \frac{1}{4}a_\mu b_\nu\operatorname{tr}(\gamma^\mu\gamma^\nu) = a\cdot b$ by Eq. (C.124), matching the scalar coefficient, and the antisymmetric remainder is carried by $\sigma^{\mu\nu}$. This decomposition of a product of two slashes into a symmetric (scalar) and antisymmetric (tensor) piece is used repeatedly when reducing amplitudes.

C.5.5 Lorentz classification of the bilinears

We now justify the names in Table C.2 by computing how each $\bar\psi\,\Gamma^A\psi$ transforms. Under a proper orthochronous Lorentz transformation Λ, the Dirac field transforms with the spinor representation $S(\Lambda)$ built in Section C.3,

$$\psi(x) \;\longmapsto\; S(\Lambda)\,\psi(\Lambda^{-1}x), \qquad S^{-1}(\Lambda)\,\gamma^\mu\,S(\Lambda) = \Lambda^\mu{}_\nu\,\gamma^\nu. \tag{C.133}$$

The Dirac conjugate transforms contragrediently: from $\bar\psi = \psi^\dagger\gamma^0$ and the property $S^\dagger\gamma^0 = \gamma^0 S^{-1}$ proved in Section C.4, one finds $\bar\psi(x) \mapsto \bar\psi(\Lambda^{-1}x)\,S^{-1}(\Lambda)$. Hence a bilinear transforms by *conjugating* the matrix sandwiched between the spinors,

$$\bar\psi\,\Gamma\,\psi \;\longmapsto\; \bar\psi\,(S^{-1}\Gamma\,S)\,\psi, \tag{C.134}$$

with the field arguments evaluated at $\Lambda^{-1}x$ throughout (we suppress them). Everything now reduces to computing $S^{-1}\Gamma^A S$ for each A.

Scalar. For $\Gamma = \mathbf{1}$, trivially $S^{-1}\mathbf{1}\,S = \mathbf{1}$, so $\bar\psi\psi$ is invariant: a Lorentz *scalar*.

Vector. For $\Gamma = \gamma^\mu$, Eq. (C.133) gives directly

$$\bar\psi\gamma^\mu\psi \;\longmapsto\; \Lambda^\mu{}_\nu\,\bar\psi\gamma^\nu\psi, \tag{C.135}$$

the transformation law of a contravariant *four-vector*.

Tensor. For $\Gamma = \sigma^{\mu\nu} = \frac{i}{2}\,[\gamma^\mu,\gamma^\nu]$, conjugation distributes over the commutator and we apply Eq. (C.133) to each factor:

$$S^{-1}\sigma^{\mu\nu}S = \tfrac{i}{2}\left[S^{-1}\gamma^\mu S, S^{-1}\gamma^\nu S\right] = \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\,\tfrac{i}{2}\left[\gamma^\alpha,\gamma^\beta\right] = \Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\,\sigma^{\alpha\beta}, \tag{C.136}$$

so $\bar\psi\sigma^{\mu\nu}\psi$ transforms as an antisymmetric rank-two *tensor*.

Axial vector and pseudoscalar. These two require the behaviour of γ^5 under conjugation. Write γ^5 in its manifestly covariant form (proved in Section C.3),

$$\gamma^5 = -\frac{i}{4!}\,\epsilon_{\mu\nu\rho\sigma}\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma, \qquad \epsilon^{0123} = +1. \tag{C.137}$$

Conjugating each gamma with Eq. (C.133) produces four factors of Λ contracted into ϵ, and the totally antisymmetric symbol satisfies the determinant identity $\epsilon_{\mu\nu\rho\sigma}\Lambda^\mu{}_\alpha\Lambda^\nu{}_\beta\Lambda^\rho{}_\gamma\Lambda^\sigma{}_\delta = (\det\Lambda)\,\epsilon_{\alpha\beta\gamma\delta}$. Hence

$$S^{-1}\gamma^5 S = -\frac{i}{4!}\,(\det\Lambda)\,\epsilon_{\alpha\beta\gamma\delta}\,\gamma^\alpha\gamma^\beta\gamma^\gamma\gamma^\delta = (\det\Lambda)\,\gamma^5. \tag{C.138}$$

For a proper transformation $\det\Lambda = +1$, so $S^{-1}\gamma^5 S = \gamma^5$; equivalently, γ^5 commutes with the spin generators $\sigma^{\mu\nu}$ (it anticommutes with each γ^μ, hence commutes with their products of two), which is the infinitesimal version of the same statement. Combining Eq. (C.138) with the vector and scalar results,

$$\bar\psi\gamma^\mu\gamma^5\psi \;\longmapsto\; (\det\Lambda)\,\Lambda^\mu{}_\nu\,\bar\psi\gamma^\nu\gamma^5\psi, \tag{C.139}$$

$$\bar\psi\gamma^5\psi \;\longmapsto\; (\det\Lambda)\,\bar\psi\gamma^5\psi. \tag{C.140}$$

Under the proper orthochronous subgroup these transform exactly like a vector and a scalar, but they carry the extra factor $\det\Lambda$ under improper transformations: they are an *axial vector* and a *pseudoscalar*.

The factor $\det\Lambda$ is invisible inside the connected Lorentz group and shows itself only under parity. From Section C.4, the parity intertwiner is γ^0, with $\psi(t,\mathbf{x}) \mapsto \gamma^0\psi(t,-\mathbf{x})$, so a bilinear transforms by sandwiching $\gamma^0\Gamma^A\gamma^0$. Using $\gamma^0\gamma^0 = \mathbf{1}$, $\gamma^0\gamma^i\gamma^0 = -\gamma^i$, and $\gamma^0\gamma^5\gamma^0 = -\gamma^5$, one reads off the parity eigenvalues: $\bar\psi\psi$ is even, $\bar\psi\gamma^\mu\psi$ transforms as $(+,-,-,-)$ on $(\mu = 0, i)$ like a polar vector, $\bar\psi\gamma^\mu\gamma^5\psi$ transforms as $(-,+,+,+)$ like an axial vector, and $\bar\psi\gamma^5\psi$ is odd. This is the precise content of $\det\Lambda_P = -1$ in Eqs. (C.139) and (C.140).

Worked Example C.51 (The vector and axial currents of the Dirac field)**.** The two bilinears that the book uses most are the vector current $j^\mu = \bar\psi\gamma^\mu\psi$ and the axial current $j^\mu_5 = \bar\psi\gamma^\mu\gamma^5\psi$. Equation (C.135) states that j^μ is a genuine four-vector: it can therefore be contracted with the photon field A_μ to form the Lorentz scalar $-e\,A_\mu\,\bar\psi\gamma^\mu\psi$ that is the QED interaction, and its conservation $\partial_\mu j^\mu = 0$ (the Noether current of the global $U(1)$ phase symmetry, Chapter 8) is a Lorentz-covariant statement. This is the structure that Chapter 17 feeds into the Feynman rules for $e^+e^- \to \mu^+\mu^-$: each vertex contributes one factor of γ^μ precisely because the conserved current is a vector bilinear.

Equation (C.139) states that j^μ_5 transforms identically under proper rotations and boosts but flips sign relative to j^μ under parity. Classically the massless Dirac Lagrangian has, alongside the vector symmetry, an axial $U(1)$ whose Noether current is j^μ_5; the chirality projectors of Theorem C.50 split a Dirac fermion into the two pieces this current weighs oppositely. That j^μ_5 is an axial vector — not a polar vector — is exactly why its naive conservation law cannot survive quantization in the presence of two vector currents: the would-be conserved $\partial_\mu j^\mu_5$ acquires the parity-odd, ϵ-tensor anomaly computed in Chapter 36. The Lorentz classification proved here is the kinematic input that makes the anomaly's $\epsilon^{\mu\nu\rho\sigma}F_{\mu\nu}F_{\rho\sigma}$ structure the only possibility.

C.5.6 The completeness (Fierz) relation

The projection formula Eq. (C.130) holds for *every* matrix M. Applying it to the elementary matrices — the matrix units E_{ij} with a single 1 in row i, column

j — packages the entire content of the basis into one tensor identity on the spinor indices. This is the spinor completeness relation, the seed of every Fierz rearrangement.

Theorem C.52 (Spinor completeness / Fierz identity)**.** *For all spinor indices* $i, j, k, l \in \{1, 2, 3, 4\}$,

$$\delta_{ij}\,\delta_{kl} = \frac{1}{4}\sum_{A} \left(\Gamma^{A}\right)_{il} \left(\Gamma_{A}\right)_{kj}, \tag{C.141}$$

the sum running over all sixteen basis matrices with their duals $\Gamma_A = (\Gamma^A)^{-1}$.

Proof. Equation (C.130) applied to an arbitrary M and read off in components gives, for every entry (i, l),

$$M_{il} = \frac{1}{4}\sum_{A} \left(\Gamma^{A}\right)_{il} \operatorname{tr}(M\,\Gamma_{A}) = \frac{1}{4}\sum_{A} \left(\Gamma^{A}\right)_{il} \sum_{a,b} M_{ba} \left(\Gamma_{A}\right)_{ab}, \tag{C.142}$$

where we wrote out $\operatorname{tr}(M\Gamma_A) = \sum_{a,b} M_{ba}(\Gamma_A)_{ab}$. Since this holds for an arbitrary matrix M, the coefficient of each independent entry M_{ba} on the two sides must agree. On the left, $M_{il} = \sum_{a,b} M_{ba}\,\delta_{ib}\delta_{la}$. Matching coefficients of M_{ba},

$$\delta_{ib}\,\delta_{la} = \frac{1}{4}\sum_{A} \left(\Gamma^{A}\right)_{il} \left(\Gamma_{A}\right)_{ab}. \tag{C.143}$$

Renaming the dummy indices $b \to j$, $a \to k$ (so that Γ_A carries indices k, j) yields $\delta_{ij}\,\delta_{kl} = \frac{1}{4}\sum_A (\Gamma^A)_{il}(\Gamma_A)_{kj}$, which is Eq. (C.141). □

Spelling out the dual matrices through Eq. (C.121), the relation takes its fully explicit form, with the multiplicities and index placements of Table C.2 made visible:

$$\delta_{ij}\,\delta_{kl} = \frac{1}{4}\Big[\delta_{il}\,\delta_{kj} + (\gamma^{\mu})_{il}(\gamma_{\mu})_{kj} + \tfrac{1}{2}(\sigma^{\mu\nu})_{il}(\sigma_{\mu\nu})_{kj} - (\gamma^{\mu}\gamma^{5})_{il}(\gamma_{\mu}\gamma^{5})_{kj} + (\gamma^{5})_{il}(\gamma^{5})_{kj}\Big]. \tag{C.144}$$

Three features deserve comment, each a direct consequence of the construction. The factor $\frac{1}{2}$ on the tensor term converts the constrained sum over $\mu < \nu$ (the six independent components) into the unconstrained sum over all μ, ν used here, since $\sigma^{\mu\nu}$ is antisymmetric and the unconstrained sum double-counts. The relative minus sign on the axial term is precisely the extra factor $\eta_A = -\eta^{\mu\mu}$ in the dual Eq. (C.121): $(\gamma^\mu\gamma^5)^{-1} = -\gamma_\mu\gamma^5$. And the contraction pattern (i, l) on Γ^A against (k, j) on Γ_A — the "crossed" pairing of Eq. (C.141) — is exactly what lets one trade $(\bar\psi_1\psi_2)(\bar\psi_3\psi_4)$ for sums of $(\bar\psi_1\psi_4)(\bar\psi_3\psi_2)$, which is the Fierz rearrangement.

Two contractions of Eq. (C.141) confirm the normalization. First set $j = i$ and $l = k$ and sum over i, k. The left side is $\sum_{i,k}\delta_{ii}\delta_{kk} = 4 \cdot 4 = 16$, while the right side becomes $\frac{1}{4}\sum_A \sum_{i,k}(\Gamma^A)_{ik}(\Gamma_A)_{ki} = \frac{1}{4}\sum_A \operatorname{tr}(\Gamma^A\Gamma_A) = \frac{1}{4}\cdot 16 \cdot 4 = 16$

by Eq. (C.129) (each of the sixteen terms contributes 4). Second, set $l = i$ and $j = k$ and sum: the left side is $\sum_{i,k} \delta_{ik}\delta_{ki} = 4$, while the right side is $\frac{1}{4}\sum_A \operatorname{tr}(\Gamma^A)\operatorname{tr}(\Gamma_A) = \frac{1}{4}(\operatorname{tr}\mathbf{1})^2 = 4$, since by Theorem C.46 only the scalar term survives. Both checks balance. The systematic exploitation of Eq. (C.141) for rearranging products of bilinears, together with the trace identities that Theorem C.48 makes available, is taken up next: the trace-orthonormality relation is the engine of Section C.6, and the completeness relation Eq. (C.141) is the engine of the Fierz rearrangements assembled there.

C.6 Trace and Contraction Technology in Four Dimensions

The previous section closed by establishing that the sixteen bilinear covariants $\{1, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5\}$ form a trace-orthogonal basis of the 4×4 matrix algebra, and in particular that every one of them except the identity is traceless: $\operatorname{tr}\gamma^\mu = 0$, $\operatorname{tr}\sigma^{\mu\nu} = 0$, $\operatorname{tr}(\gamma^\mu\gamma^5) = 0$, and $\operatorname{tr}\gamma^5 = 0$ (Section C.5). Those vanishing traces are the seeds of everything in this section. Trace orthogonality tells us that the trace of *any* product of gamma matrices is, after expansion in the basis, just 4 times the coefficient of the identity in that product; the entire technology below is the systematic bookkeeping of when and how the identity appears. The payoff is concrete and central to the rest of the book: when an amplitude is squared and summed over spins, the Dirac structure collapses—via the completeness relations of Chapter 11—into traces of strings of gamma matrices, and those traces must be reduced to ordinary Lorentz-covariant functions of the external momenta. This is precisely the machinery that turns the spin-summed $|\mathcal{M}|^2$ for $e^+e^- \to \mu^+\mu^-$ into a function of Mandelstam invariants in Chapter 17, and that evaluates the Dirac numerators of the one-loop QED diagrams in Chapter 19. We assemble the complete $d = 4$ toolkit here, proving each identity from the two structural facts that drive all of them—the cyclicity of the trace and the Clifford relation—and we finish with the trace that detects orientation, the one carrying γ^5 and the Levi-Civita tensor.

Throughout we use the conventions fixed in Chapter A and Section C.1: the mostly-minus metric $\eta_{\mu\nu} = \operatorname{diag}(+,-,-,-)$, the Clifford relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\,\eta^{\mu\nu}\,\mathbf{1}, \tag{C.145}$$

the irreducible representation acting on a four-dimensional spinor space, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ with $(\gamma^5)^2 = \mathbf{1}$ and $\{\gamma^5, \gamma^\mu\} = 0$.

C.6.1 The two pillars: cyclicity and the Clifford relation

Every identity in this section rests on exactly two facts. The first is purely algebraic and dimension-independent: the trace of a product is invariant under

cyclic permutation,

$$\operatorname{tr}(AB\cdots Z) = \operatorname{tr}(B\cdots ZA), \tag{C.146}$$

which follows at once from $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ applied to A and the block $B\cdots Z$. The second is the Clifford relation Eq. (C.145), which lets us trade the order of two adjacent gamma matrices for a c-number,

$$\gamma^\mu\gamma^\nu = 2\eta^{\mu\nu}\mathbf{1} - \gamma^\nu\gamma^\mu. \tag{C.147}$$

The strategy that recurs below is always the same: use Eq. (C.147) to march one gamma matrix through a string until it returns to its starting point, picking up metric tensors along the way, then use Eq. (C.146) to close the loop and solve the resulting linear relation for the trace. The only place the number 4—the dimension of the representation—enters is through

$$\operatorname{tr}\mathbf{1} = 4, \tag{C.148}$$

which holds because the irreducible representation of the $d = 4$ Clifford algebra acts on a space of dimension $2^{\lfloor 4/2\rfloor} = 4$ (Section C.1); the identity there is the 4×4 unit matrix. We carry $\operatorname{tr}\mathbf{1} = 4$ as a constant; in Section C.7 it will become a continuation parameter when we move off $d = 4$.

C.6.2 Vanishing of odd traces

Proposition C.53 (Odd traces vanish)**.** *The trace of a product of an odd number of gamma matrices is zero:*

$$\operatorname{tr}(\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}) = 0 \qquad (n \text{ odd}). \tag{C.149}$$

Proof. Write $M = \gamma^{\mu_1}\cdots\gamma^{\mu_n}$ and insert $\mathbf{1} = (\gamma^5)^2$ at the front:

$$\operatorname{tr} M = \operatorname{tr}(\gamma^5\,\gamma^5\,M). \tag{C.150}$$

By cyclicity Eq. (C.146) move the leftmost γ^5 to the far right, $\operatorname{tr}(\gamma^5\,\gamma^5 M) = \operatorname{tr}(\gamma^5 M\gamma^5)$. Now push the remaining left γ^5 rightward through the n gamma matrices of M; each pass costs a sign because $\{\gamma^5,\gamma^\mu\} = 0$, so $\gamma^5 M = (-1)^n M\gamma^5$. Hence

$$\operatorname{tr} M = \operatorname{tr}(\gamma^5 M\gamma^5) = (-1)^n\,\operatorname{tr}(M\,\gamma^5\,\gamma^5) = (-1)^n\operatorname{tr} M. \tag{C.151}$$

For odd n this reads $\operatorname{tr} M = -\operatorname{tr} M$, so $\operatorname{tr} M = 0$. □

This single argument disposes of $\operatorname{tr}\gamma^\mu = 0$, $\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho) = 0$, and every higher odd string at once, and it does so using only $(\gamma^5)^2 = \mathbf{1}$ and the anticommutation of γ^5—no explicit representation required. It is the same γ^5-insertion trick that will return below to show that short γ^5-traces vanish.

C.6.3 The even traces and the master recursion

The lowest even trace follows immediately from Eq. (C.147). Symmetrizing,

$$\operatorname{tr}(\gamma^\mu\gamma^\nu) = \tfrac{1}{2}\operatorname{tr}\left(\{\gamma^\mu,\gamma^\nu\}\right) = \tfrac{1}{2}\,(2\eta^{\mu\nu})\,\operatorname{tr}\mathbf{1} = 4\,\eta^{\mu\nu}, \tag{C.152}$$

where the antisymmetric part $\frac{1}{2}[\gamma^\mu,\gamma^\nu] = -i\sigma^{\mu\nu}$ dropped out because $\operatorname{tr}\sigma^{\mu\nu} = 0$ (Section C.5). For four gamma matrices we could symmetrize repeatedly, but it is cleaner—and sets up the general pattern—to establish the recursion that reduces any even trace to a sum of traces two matrices shorter.

Theorem C.54 (Trace recursion)**.** *For even n,*

$$\operatorname{tr}(\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}) = \sum_{k=2}^{n}(-1)^k\,\eta^{\mu_1\mu_k}\,\operatorname{tr}(\gamma^{\mu_2}\cdots\widehat{\gamma^{\mu_k}}\cdots\gamma^{\mu_n}), \tag{C.153}$$

where the hat denotes omission of the indicated factor.

Proof. Anticommute γ^{μ_1} rightward through the string using Eq. (C.147). Past the first neighbour,

$$\gamma^{\mu_1}\gamma^{\mu_2}\cdots = 2\eta^{\mu_1\mu_2}\gamma^{\mu_3}\cdots\gamma^{\mu_n} - \gamma^{\mu_2}\gamma^{\mu_1}\gamma^{\mu_3}\cdots\gamma^{\mu_n}. \tag{C.154}$$

Past the next, $\gamma^{\mu_1}\gamma^{\mu_3} = 2\eta^{\mu_1\mu_3} - \gamma^{\mu_3}\gamma^{\mu_1}$, generating $-2\eta^{\mu_1\mu_3}$ with the accumulated sign, and so on. After γ^{μ_1} has passed all of $\gamma^{\mu_2},\dots,\gamma^{\mu_n}$,

$$\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n} = 2\sum_{k=2}^{n}(-1)^k\,\eta^{\mu_1\mu_k}\,\gamma^{\mu_2}\cdots\widehat{\gamma^{\mu_k}}\cdots\gamma^{\mu_n} + (-1)^{n-1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}\gamma^{\mu_1}. \tag{C.155}$$

The alternating sign $(-1)^k$ records the parity of transpositions needed to bring γ^{μ_1} adjacent to γ^{μ_k}. Take the trace of Eq. (C.155). In the last term, cyclicity Eq. (C.146) returns $\operatorname{tr}(\gamma^{\mu_2}\cdots\gamma^{\mu_n}\gamma^{\mu_1}) = \operatorname{tr}(\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n})$, and for even n the prefactor is $(-1)^{n-1} = -1$. Thus

$$\operatorname{tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}) = 2\sum_{k=2}^{n}(-1)^k\eta^{\mu_1\mu_k}\,\operatorname{tr}(\gamma^{\mu_2}\cdots\widehat{\gamma^{\mu_k}}\cdots\gamma^{\mu_n}) - \operatorname{tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}), \tag{C.156}$$

and moving the last term to the left and dividing by two yields Eq. (C.153). □

The recursion is the whole story for γ^5-free even traces: applied repeatedly it terminates at $\operatorname{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ and expresses any $\operatorname{tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_{2m}})$ as 4 times a sum over all $(2m-1)!!$ ways of pairing the $2m$ indices into metric tensors, each pairing weighted by the sign of the permutation that brings the paired indices adjacent. For the four-index trace, Eq. (C.153) with $n = 4$ gives the workhorse identity

$$\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = \eta^{\mu\nu}\operatorname{tr}(\gamma^\rho\gamma^\sigma) - \eta^{\mu\rho}\operatorname{tr}(\gamma^\nu\gamma^\sigma) + \eta^{\mu\sigma}\operatorname{tr}(\gamma^\nu\gamma^\rho)$$

$$= 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}). \tag{C.157}$$

The three terms are the three pairings $\{(\mu\nu)(\rho\sigma), (\mu\rho)(\nu\sigma), (\mu\sigma)(\nu\rho)\}$ with the alternating signs; the middle one is negative because separating μ from ρ requires crossing ν.

C.6.4 The reversal identity

A symmetry of traces that saves real labour in amplitude calculations is that a string may be read backwards without changing its trace.

Proposition C.55 (Reversal)**.** *For any number of gamma matrices,*

$$\operatorname{tr}(\gamma^{\mu_1}\gamma^{\mu_2}\cdots\gamma^{\mu_n}) = \operatorname{tr}(\gamma^{\mu_n}\cdots\gamma^{\mu_2}\gamma^{\mu_1}). \tag{C.158}$$

Proof. For odd n both sides vanish by Theorem C.53, so take n even. The charge-conjugation matrix C constructed in Section C.4 satisfies $C\gamma^{\mu}C^{-1} = -(\gamma^{\mu})^{\mathsf{T}}$. Conjugating a product and using $C^{-1}C = \mathbf{1}$ between factors,

$$C\,\gamma^{\mu_1}\cdots\gamma^{\mu_n}\,C^{-1} = (-1)^n(\gamma^{\mu_1})^{\mathsf{T}}\cdots(\gamma^{\mu_n})^{\mathsf{T}} = (-1)^n(\gamma^{\mu_n}\cdots\gamma^{\mu_1})^{\mathsf{T}}, \tag{C.159}$$

where the last step is the reversal of order under transposition of a product. The trace is invariant under similarity transformations and under transposition, so

$$\operatorname{tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}) = \operatorname{tr}\Big(C\,\gamma^{\mu_1}\cdots\gamma^{\mu_n}\,C^{-1}\Big) = (-1)^n\,\operatorname{tr}(\gamma^{\mu_n}\cdots\gamma^{\mu_1}), \tag{C.160}$$

and $(-1)^n = 1$ for even n. □

One checks Eq. (C.158) directly on Eq. (C.157): reversing $\mu\nu\rho\sigma \to \sigma\rho\nu\mu$ maps the three pairings to themselves, leaving the result unchanged. In practice the reversal identity is used to recognize that, e.g., $\operatorname{tr}(\not{a}\not{b}\not{c}\not{d}) = \operatorname{tr}(\not{d}\not{c}\not{b}\not{a})$ without re-expanding.

C.6.5 Traces with γ^5 and the Levi-Civita tensor

The remaining traces are those that carry one factor of γ^5. They are what distinguish a theory from its mirror image: a nonzero γ^5-trace is the algebraic fingerprint of parity violation and of the chiral anomaly (Chapter 19 and beyond). We first dispose of the short ones, then compute the single nonvanishing case.

Proposition C.56 (Short γ^5-traces vanish)**.**

$$\operatorname{tr}\gamma^5 = 0, \qquad \operatorname{tr}(\gamma^5\gamma^{\mu}) = 0, \qquad \operatorname{tr}(\gamma^5\gamma^{\mu}\gamma^{\nu}) = 0, \qquad \operatorname{tr}(\gamma^5\gamma^{\mu}\gamma^{\nu}\gamma^{\rho}) = 0. \tag{C.161}$$

Proof. That $\operatorname{tr}\gamma^5 = 0$ was handed to us by Section C.5; it also follows directly from $(\gamma^0)^2 = \eta^{00}\mathbf{1} = \mathbf{1}$ together with $\gamma^0\gamma^5 = -\gamma^5\gamma^0$:

$$\operatorname{tr}\gamma^5 = \operatorname{tr}(\gamma^5(\gamma^0)^2) = \operatorname{tr}(\gamma^0\gamma^5\gamma^0) = -\operatorname{tr}(\gamma^5(\gamma^0)^2) = -\operatorname{tr}\gamma^5, \tag{C.162}$$

using cyclicity in the middle step. The traces $\operatorname{tr}(\gamma^5\gamma^\mu)$ and $\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho)$ contain an odd number of ordinary gamma matrices; since γ^5 is itself a product of four gammas, each is a trace of an odd total number of gamma matrices and vanishes by Theorem C.53. For $\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu)$ with $\mu = \nu$ the string collapses to $\eta^{\mu\mu}\operatorname{tr}\gamma^5 = 0$. For $\mu \neq \nu$, choose an index α distinct from both μ and ν—possible because spacetime is four-dimensional—so that $(\gamma^\alpha)^2 = \eta^{\alpha\alpha}\mathbf{1}$ with $\eta^{\alpha\alpha} = \pm 1 \neq 0$. Insert $\mathbf{1} = (\eta^{\alpha\alpha})^{-1}(\gamma^\alpha)^2$ at the right and cycle one γ^α to the front:

$$\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu) = \frac{1}{\eta^{\alpha\alpha}}\operatorname{tr}(\gamma^\alpha\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha). \tag{C.163}$$

Now move the leading γ^α rightward: it anticommutes with γ^5 (one sign) and with each of γ^μ, γ^ν (two signs, since $\alpha \neq \mu, \nu$), so $\gamma^\alpha\gamma^5\gamma^\mu\gamma^\nu = -\gamma^5\gamma^\mu\gamma^\nu\gamma^\alpha$. Hence the right side equals $(\eta^{\alpha\alpha})^{-1}(-1)\eta^{\alpha\alpha}\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu) = -\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu)$, forcing it to zero. □

The only γ^5-trace that survives in $d = 4$ is the one with four gamma matrices, and it produces the totally antisymmetric Levi-Civita tensor.

Theorem C.57 (The γ^5 four-trace)**.**

$$\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = -4i\,\epsilon^{\mu\nu\rho\sigma}, \tag{C.164}$$

with the convention $\epsilon^{0123} = +1$.

Proof. First, the left side is totally antisymmetric in $\mu\nu\rho\sigma$. If any two of the indices coincide, the corresponding pair of adjacent (after permutation) gamma matrices can be replaced via the Clifford relation by a metric factor times the identity, leaving a γ^5-trace with at most two gamma matrices, which vanishes by Theorem C.56; antisymmetry under interchange of unequal adjacent indices follows because $\gamma^\rho\gamma^\sigma = -\gamma^\sigma\gamma^\rho$ when $\rho \neq \sigma$. A totally antisymmetric four-index Lorentz tensor in four dimensions is unique up to scale, so

$$\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = K\,\epsilon^{\mu\nu\rho\sigma} \tag{C.165}$$

for some constant K. Fix K by evaluating at $(\mu\nu\rho\sigma) = (0123)$, where $\epsilon^{0123} = +1$. From the definition $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ we have $\gamma^0\gamma^1\gamma^2\gamma^3 = -i\gamma^5$, so

$$\operatorname{tr}(\gamma^5\gamma^0\gamma^1\gamma^2\gamma^3) = \operatorname{tr}(\gamma^5(-i\gamma^5)) = -i\,\operatorname{tr}((\gamma^5)^2) = -i\,\operatorname{tr}\mathbf{1} = -4i. \tag{C.166}$$

Therefore $K \cdot (+1) = -4i$, i.e. $K = -4i$, which is Eq. (C.164). □

Notation Watch

The sign and i in Eq. (C.164) are convention-laden and must be quoted together with the Levi-Civita normalization. We use $\epsilon^{0123} = +1$, which in the mostly-minus metric forces $\epsilon_{0123} = \eta_{0\alpha}\eta_{1\beta}\eta_{2\gamma}\eta_{3\delta}\,\epsilon^{\alpha\beta\gamma\delta} = (+1)(-1)(-1)(-1)\,\epsilon^{0123} = -1$. With $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ and $\operatorname{tr}\mathbf{1} = 4$ this fixes $\operatorname{tr}(\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = -4i\,\epsilon^{\mu\nu\rho\sigma}$. This matches the convention recorded in Chapter A; a text using $\epsilon_{0123} = +1$ or the opposite-sign γ^5 would report $+4i\,\epsilon^{\mu\nu\rho\sigma}$. Sign errors here propagate directly into parity-odd observables, so the convention is fixed once and used everywhere.

For amplitudes that square a parity-violating current one also needs the contraction of two Levi-Civita tensors. Because ϵ is totally antisymmetric, $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\alpha\beta\gamma\delta}$ equals the determinant of the 4×4 matrix of Kronecker deltas δ^μ_α, multiplied by the sign $\det\eta = -1$ that arises from the single index raising needed to compare $\epsilon^{0123} = +1$ with $\epsilon_{0123} = -1$. Contracting indices then gives the chain

$$\begin{aligned}
\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\sigma} &= -24, \qquad \epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\rho\delta} = -6\,\delta^\sigma_\delta,\\
\epsilon^{\mu\nu\rho\sigma}\epsilon_{\mu\nu\gamma\delta} &= -2\big(\delta^\rho_\gamma\delta^\sigma_\delta - \delta^\rho_\delta\delta^\sigma_\gamma\big),
\end{aligned} \tag{C.167}$$

each carrying the leading minus sign characteristic of Lorentzian signature; the full determinant form is recorded in Chapter A.

C.6.6 Index-contraction identities

When a gamma string is sandwiched between propagator numerators and one contracts a repeated Lorentz index—as happens whenever a photon or gluon line is summed over polarizations in Feynman gauge—the string shortens according to a second family of identities. Unlike the traces, these are matrix identities (no trace taken); they too follow directly from the Clifford relation, and unlike the traces they depend explicitly on the spacetime dimension through $\gamma^\mu\gamma_\mu = d$. In $d = 4$:

Theorem C.58 (Contraction identities, $d = 4$)**.**

$$\gamma^\mu\gamma_\mu = 4\,\mathbf{1}, \tag{C.168}$$
$$\gamma^\mu\gamma^\nu\gamma_\mu = -2\,\gamma^\nu, \tag{C.169}$$
$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 4\,\eta^{\nu\rho}\mathbf{1}, \tag{C.170}$$
$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\,\gamma^\sigma\gamma^\rho\gamma^\nu. \tag{C.171}$$

Proof. For Eq. (C.168), symmetrize the contracted product:

$$\gamma^\mu\gamma_\mu = \eta_{\mu\nu}\gamma^\mu\gamma^\nu = \tfrac{1}{2}\eta_{\mu\nu}\,\{\gamma^\mu,\gamma^\nu\} = \eta_{\mu\nu}\eta^{\mu\nu}\mathbf{1} = \delta^\mu_\mu\mathbf{1} = 4\,\mathbf{1}, \tag{C.172}$$

where $\delta^\mu_\mu = d = 4$. For Eq. (C.169), push γ_μ leftward past γ^ν using $\gamma^\nu\gamma_\mu = 2\delta^\nu_\mu \mathbf{1} - \gamma_\mu\gamma^\nu$:

$$\gamma^\mu\gamma^\nu\gamma_\mu = \gamma^\mu\big(2\delta^\nu_\mu - \gamma_\mu\gamma^\nu\big) = 2\gamma^\nu - (\gamma^\mu\gamma_\mu)\gamma^\nu = 2\gamma^\nu - 4\gamma^\nu = -2\gamma^\nu. \tag{C.173}$$

For Eq. (C.170), move γ_μ leftward one step at a time. Using $\gamma^\rho\gamma_\mu = 2\delta^\rho_\mu - \gamma_\mu\gamma^\rho$,

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 2\,\gamma^\rho\gamma^\nu - \gamma^\mu\gamma^\nu\gamma_\mu\gamma^\rho = 2\,\gamma^\rho\gamma^\nu - (-2\gamma^\nu)\gamma^\rho = 2\big(\gamma^\rho\gamma^\nu + \gamma^\nu\gamma^\rho\big) = 4\,\eta^{\nu\rho}\mathbf{1}, \tag{C.174}$$

where the first term used $\gamma^\mu\, 2\delta^\rho_\mu = 2\gamma^\rho$ and $\gamma^\mu\gamma^\nu\gamma_\mu = -2\gamma^\nu$ from Eq. (C.169). For Eq. (C.171), the same move on the outermost factor gives

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = 2\,\gamma^\sigma\gamma^\nu\gamma^\rho - \gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu\gamma^\sigma = 2\,\gamma^\sigma\gamma^\nu\gamma^\rho - 4\eta^{\nu\rho}\gamma^\sigma, \tag{C.175}$$

using Eq. (C.170). Finally rewrite the first term with $\gamma^\nu\gamma^\rho = 2\eta^{\nu\rho}\mathbf{1} - \gamma^\rho\gamma^\nu$:

$$2\gamma^\sigma\gamma^\nu\gamma^\rho = 2\gamma^\sigma\big(2\eta^{\nu\rho} - \gamma^\rho\gamma^\nu\big) = 4\eta^{\nu\rho}\gamma^\sigma - 2\gamma^\sigma\gamma^\rho\gamma^\nu, \tag{C.176}$$

so the two $4\eta^{\nu\rho}\gamma^\sigma$ pieces cancel and $\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu = -2\gamma^\sigma\gamma^\rho\gamma^\nu$. □

The pattern is transparent: contracting a pair of identical gamma matrices across a string of r matrices alternately contracts the enclosed indices (even r between them) or reverses the enclosed string (odd r between them), with the sign and the numerical coefficient set by $\gamma^\mu\gamma_\mu = d$. Keeping the dependence on d explicit—$\gamma^\mu\gamma_\mu = d$, $\gamma^\mu\gamma^\nu\gamma_\mu = (2-d)\gamma^\nu$, and so on—is exactly what Section C.7 will require for dimensional regularization; here we have set $d = 4$.

C.6.7 The toolkit in slash notation, and a worked trace

In amplitudes the gamma matrices are almost always contracted with momenta, $\not{p} \equiv \gamma^\mu p_\mu$. Re-expressing the identities above in slash form gives the shapes one actually meets. Contracting Eq. (C.152), Eq. (C.157), and Eq. (C.164) with four arbitrary four-vectors a, b, c, d,

$$\operatorname{tr}(\not{a}\not{b}) = 4\,(a\cdot b), \tag{C.177}$$

$$\operatorname{tr}(\not{a}\not{b}\not{c}\not{d}) = 4\big[(a\cdot b)(c\cdot d) - (a\cdot c)(b\cdot d) + (a\cdot d)(b\cdot c)\big], \tag{C.178}$$

$$\operatorname{tr}(\gamma^5\not{a}\not{b}\not{c}\not{d}) = -4i\,\epsilon_{\mu\nu\rho\sigma}\,a^\mu b^\nu c^\rho d^\sigma, \tag{C.179}$$

while the contraction identities become $\gamma^\mu\not{a}\gamma_\mu = -2\not{a}$ and $\gamma^\mu\not{a}\not{b}\gamma_\mu = 4\,(a\cdot b)\mathbf{1}$ and $\gamma^\mu\not{a}\not{b}\not{c}\gamma_\mu = -2\,\not{c}\not{b}\not{a}$. With Eq. (C.148), Eq. (C.177), Eq. (C.178), Eq. (C.179) and the contraction rules, every Dirac trace arising from a tree- or one-loop QED amplitude can be reduced to scalar invariants. The following worked example carries out the reduction that Chapter 17 needs.

Worked Example C.59 (The muon tensor in $e^+e^- \to \mu^+\mu^-$)**.** In $e^+e^- \to \mu^+\mu^-$ the squared amplitude factorizes, after summing over final muon spins and using the completeness relations $\sum_s u(p)\bar{u}(p) = \not{p} + m$ and $\sum_s v(p)\bar{v}(p) = \not{p} - m$ (Chapter 11), into a product of an electron tensor and a muon tensor. The muon tensor is

$$L^{\mu\nu} = \operatorname{tr}\Big[(\not{p}_3 + m)\,\gamma^\mu\,(\not{p}_4 - m)\,\gamma^\nu\Big], \tag{C.180}$$

where p_3 is the μ^- momentum, p_4 the μ^+ momentum, and $m = m_\mu$. Expand the product:

$$L^{\mu\nu} = \operatorname{tr}(\not{p}_3\gamma^\mu\not{p}_4\gamma^\nu) - m\,\operatorname{tr}(\not{p}_3\gamma^\mu\gamma^\nu) + m\,\operatorname{tr}(\gamma^\mu\not{p}_4\gamma^\nu) - m^2\,\operatorname{tr}(\gamma^\mu\gamma^\nu). \tag{C.181}$$

The two terms linear in m each contain three gamma matrices and vanish by Theorem C.53. The first term, written with explicit momentum indices, uses Eq. (C.157):

$$\begin{aligned}\operatorname{tr}(\not{p}_3\gamma^\mu\not{p}_4\gamma^\nu) &= p_{3\alpha}p_{4\beta}\,\operatorname{tr}(\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu)\\ &= 4\,p_{3\alpha}p_{4\beta}(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\mu\beta})\\ &= 4(p_3^\mu p_4^\nu - (p_3\cdot p_4)\,\eta^{\mu\nu} + p_3^\nu p_4^\mu).\end{aligned} \tag{C.182}$$

The last term is $-m^2\operatorname{tr}(\gamma^\mu\gamma^\nu) = -4m^2\eta^{\mu\nu}$ by Eq. (C.152). Adding,

$$\boxed{L^{\mu\nu} = 4\Big[p_3^\mu p_4^\nu + p_3^\nu p_4^\mu - (p_3\cdot p_4 + m^2)\eta^{\mu\nu}\Big].} \tag{C.183}$$

The structure is exactly what symmetry demands: $L^{\mu\nu}$ is symmetric in $\mu\nu$ and, contracting with the photon momentum $q = p_3 + p_4$, one verifies $q_\mu L^{\mu\nu} = 4[(q\cdot p_3)p_4^\nu + (q\cdot p_4)p_3^\nu - (p_3\cdot p_4 + m^2)q^\nu]$, which on shell ($p_3^2 = p_4^2 = m^2$, so $q\cdot p_3 = m^2 + p_3\cdot p_4 = q\cdot p_4$) reduces to $q_\mu L^{\mu\nu} = 4(q\cdot p_3)(p_3+p_4)^\nu - 4(p_3\cdot p_4 + m^2)q^\nu = 0$, the transversality forced by current conservation. The electron tensor has the same form with $p_3, p_4 \to p_1, p_2$ and $m \to m_e$; contracting $L^{\mu\nu}$ with its electron counterpart and dividing by the four initial spin states yields the unpolarized $|\mathcal{M}|^2$ assembled in Chapter 17. Every gamma matrix has disappeared; what remains is a polynomial in dot products of the external momenta, which is the entire purpose of the trace technology.

The toolkit of this section—Eq. (C.148), the recursion Eq. (C.153) with its descendants Eq. (C.152) and Eq. (C.157), the reversal identity Eq. (C.158), the γ^5-trace Eq. (C.164), and the contraction identities of Theorem C.58—is complete for any spin-summed calculation performed in exactly four dimensions. The moment a loop integral is regulated, however, the gamma algebra must be continued to $d = 4 - 2\varepsilon$ dimensions, and three of the results above develop a dimension dependence that cannot be ignored: $\gamma^\mu\gamma_\mu = d$ rather than 4, the contraction identities acquire d-dependent coefficients, and—most consequentially—the totally antisymmetric four-index object behind Eq. (C.164) ceases to make unambiguous sense, because

γ^5 has no natural anticommuting definition away from integer dimension. The next section takes up precisely this continuation and confronts the γ^5 problem head-on.

C.7 Continuation to $d = 4 - 2\varepsilon$ and the γ^5 Problem

The trace and contraction technology assembled in Section C.6 was built in exactly four spacetime dimensions: the contraction $\eta^\mu{}_\mu = 4$ appeared as a number, the Dirac matrices were 4×4, and $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$ was a concrete sixteenth element of the bilinear basis. Dimensional regularization, the workhorse of every loop chapter in this book (the master integrals live in Chapter D; their first use is the QED one-loop programme of Chapter 19), demands that we continue all of this to non-integer dimension $d = 4 - 2\varepsilon$. We must therefore ask, of each identity in turn, whether it survives the continuation, and if so in what d-dependent form. Most of the toolkit passes through cleanly, acquiring only d-dependent prefactors that the loop chapters quote and expand in ε. One element does *not* pass through: the matrix γ^5, which is intrinsically four-dimensional, cannot be continued while keeping all of its four-dimensional algebraic properties. This section gives the surviving d-dimensional rules in full, then confronts the γ^5 obstruction honestly — proving the inconsistency, exhibiting the standard 't Hooft–Veltman–Breitenlohner–Maison repair, and naming its price.

C.7.1 The formal Clifford algebra in d dimensions

There is no honest finite-dimensional representation of the Clifford algebra in non-integer dimension: γ^μ for "$d = 3.99$" is not a matrix. What dimensional regularization actually provides is an *algebra of formal manipulation rules*, fixed once and for all, together with the analytic continuation of the scalar loop integrals those manipulations produce. The rules are chosen so that (i) they reduce to the genuine four-dimensional algebra at $d = 4$, (ii) they are internally consistent enough to define every trace and contraction unambiguously, and (iii) they make the loop integrand a meromorphic function of d whose poles at $d = 4$ carry the ultraviolet divergences. A fully rigorous construction realizes the γ^μ as operators on an infinite-dimensional space with the index μ ranging over a formally d-dimensional metric space; for computation, only the contraction and trace rules below are ever needed.

Definition C.60 (Formal d-dimensional Clifford data)**.** The d-dimensional gamma matrices satisfy the Clifford relation

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{1}, \tag{C.184}$$

where the metric $\eta^{\mu\nu}$ now carries a formally d-dimensional trace,

$$\eta^\mu{}_\mu = \eta^{\mu\nu}\eta_{\mu\nu} = \delta^\mu{}_\mu = d. \tag{C.185}$$

The trace over the (suppressed) Dirac indices of the identity is fixed to

$$\operatorname{tr}\mathbf{1} = 4, \tag{C.186}$$

and the trace is linear and cyclic, $\operatorname{tr}(AB) = \operatorname{tr}(BA)$.

Two conventions inside Theorem C.60 deserve comment. First, Eq. (C.185) is the single structural change that distinguishes the d-dimensional algebra from its four-dimensional parent: every time an index pair is contracted around the loop, the bookkeeping value 4 is replaced by the symbol $d = 4 - 2\varepsilon$. This one substitution generates all the d-dependence in the contraction identities below. Second, the choice Eq. (C.186) retains $\operatorname{tr}\mathbf{1} = 4$ rather than the alternative $\operatorname{tr}\mathbf{1} = 2^{d/2}$ that one would obtain by literally continuing the dimension $2^{\lfloor d/2 \rfloor}$ of an irreducible Dirac representation. The two differ by $2^{d/2} = 4 \cdot 2^{-\varepsilon} = 4\,(1 - \varepsilon \ln 2 + \dots)$, i.e. only at $O(\varepsilon)$. Because an overall constant multiplying every trace is a global normalization that cancels in ratios and is absorbed into the (scheme-dependent) finite parts of counterterms, both choices give identical physics; we follow the universal convention $\operatorname{tr}\mathbf{1} = 4$ so that the four-dimensional trace formulae of Section C.6 continue to hold verbatim in form.

C.7.2 The d-dependent contraction identities

We now run the four-dimensional contraction recursion of Section C.6 with $\eta^{\mu}{}_{\mu} = d$ in place of 4. Every step uses only Eqs. (C.184) and (C.185); the proofs are short and we give all the algebra, since the loop chapters quote the results constantly.

Proposition C.61 (Contraction identities in d dimensions)**.** *With $d = 4 - 2\varepsilon$ understood,*

$$\gamma^{\mu}\gamma_{\mu} = d\,\mathbf{1}, \tag{C.187}$$

$$\gamma^{\mu}\gamma^{\nu}\gamma_{\mu} = (2 - d)\,\gamma^{\nu}, \tag{C.188}$$

$$\gamma^{\mu}\gamma^{\nu}\gamma^{\rho}\gamma_{\mu} = 4\,\eta^{\nu\rho}\mathbf{1} - (4 - d)\,\gamma^{\nu}\gamma^{\rho}, \tag{C.189}$$

$$\gamma^{\mu}\gamma^{\nu}\gamma^{\rho}\gamma^{\sigma}\gamma_{\mu} = -2\,\gamma^{\sigma}\gamma^{\rho}\gamma^{\nu} + (4 - d)\,\gamma^{\nu}\gamma^{\rho}\gamma^{\sigma}. \tag{C.190}$$

Proof. For Eq. (C.187), symmetrize the contracted product with Eq. (C.184): $\gamma^{\mu}\gamma_{\mu} = \frac{1}{2}\{\gamma^{\mu}, \gamma_{\mu}\} = \frac{1}{2} \cdot 2\,\eta^{\mu}{}_{\mu}\mathbf{1} = d\,\mathbf{1}$ by Eq. (C.185).

For Eq. (C.188), anticommute the left factor inward, $\gamma^{\mu}\gamma^{\nu} = 2\eta^{\mu\nu}\mathbf{1} - \gamma^{\nu}\gamma^{\mu}$, and then apply Eq. (C.187):

$$\gamma^{\mu}\gamma^{\nu}\gamma_{\mu} = (2\eta^{\mu\nu} - \gamma^{\nu}\gamma^{\mu})\gamma_{\mu} = 2\gamma^{\nu} - \gamma^{\nu}\,(\gamma^{\mu}\gamma_{\mu}) = 2\gamma^{\nu} - d\,\gamma^{\nu} = (2 - d)\gamma^{\nu}. \tag{C.191}$$

For Eq. (C.189), do the same with the leftmost pair and feed in Eq. (C.188):

$$\gamma^{\mu}\gamma^{\nu}\gamma^{\rho}\gamma_{\mu} = (2\eta^{\mu\nu} - \gamma^{\nu}\gamma^{\mu})\gamma^{\rho}\gamma_{\mu} = 2\gamma^{\rho}\gamma^{\nu} - \gamma^{\nu}\,(\gamma^{\mu}\gamma^{\rho}\gamma_{\mu})$$

$$= 2\gamma^\rho\gamma^\nu - (2-d)\,\gamma^\nu\gamma^\rho. \tag{C.192}$$

Now reduce the first term with $\gamma^\rho\gamma^\nu = 2\eta^{\nu\rho}\mathbf{1} - \gamma^\nu\gamma^\rho$, giving $2\gamma^\rho\gamma^\nu = 4\eta^{\nu\rho}\mathbf{1} - 2\gamma^\nu\gamma^\rho$, so that

$$\gamma^\mu\gamma^\nu\gamma^\rho\gamma_\mu = 4\eta^{\nu\rho}\mathbf{1} - 2\gamma^\nu\gamma^\rho - (2-d)\gamma^\nu\gamma^\rho = 4\eta^{\nu\rho}\mathbf{1} - (4-d)\gamma^\nu\gamma^\rho. \tag{C.193}$$

For Eq. (C.190), anticommute the left factor past γ^ν and apply Eq. (C.189):

$$\begin{aligned}\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\mu &= (2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu)\gamma^\rho\gamma^\sigma\gamma_\mu = 2\gamma^\rho\gamma^\sigma\gamma^\nu - \gamma^\nu\,(\gamma^\mu\gamma^\rho\gamma^\sigma\gamma_\mu)\\ &= 2\gamma^\rho\gamma^\sigma\gamma^\nu - \gamma^\nu\big[4\eta^{\rho\sigma}\mathbf{1} - (4-d)\gamma^\rho\gamma^\sigma\big]\\ &= 2\gamma^\rho\gamma^\sigma\gamma^\nu - 4\eta^{\rho\sigma}\gamma^\nu + (4-d)\gamma^\nu\gamma^\rho\gamma^\sigma.\end{aligned} \tag{C.194}$$

Writing $2\gamma^\rho\gamma^\sigma = 4\eta^{\rho\sigma}\mathbf{1} - 2\gamma^\sigma\gamma^\rho$ collapses the first two terms, $2\gamma^\rho\gamma^\sigma\gamma^\nu - 4\eta^{\rho\sigma}\gamma^\nu = -2\gamma^\sigma\gamma^\rho\gamma^\nu$, and Eq. (C.190) follows. □

The hallmark of these formulae is that the four-dimensional results are recovered by setting $d = 4$, whereupon Eqs. (C.187) to (C.190) read 4, $-2\gamma^\nu$, $4\eta^{\nu\rho}\mathbf{1}$, and $-2\gamma^\sigma\gamma^\rho\gamma^\nu$, exactly as in Section C.6. The corrections $(2-d) = 2(\varepsilon-1)$ and $(4-d) = 2\varepsilon$ are precisely the $O(\varepsilon)$ remnants that, multiplied against a $1/\varepsilon$ pole from a divergent integral, leave a finite contribution; this is why the loop chapters cannot drop them.

We shall need one identity at the next order, the contraction of a pair around a *four*-gamma string, for the γ^5 analysis to follow.

Lemma C.62 (Sandwich of four gammas)**.** *In d dimensions,*

$$\gamma^\alpha\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\alpha = 2\,\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu + 2\,\gamma^\mu\gamma^\sigma\gamma^\rho\gamma^\nu - (4-d)\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma. \tag{C.195}$$

Proof. Anticommute the left γ^α past γ^μ, $\gamma^\alpha\gamma^\mu = 2\eta^{\alpha\mu}\mathbf{1} - \gamma^\mu\gamma^\alpha$:

$$\gamma^\alpha\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\alpha = 2\,\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu - \gamma^\mu\,(\gamma^\alpha\gamma^\nu\gamma^\rho\gamma^\sigma\gamma_\alpha). \tag{C.196}$$

The bracketed factor is Eq. (C.190) with indices (ν,ρ,σ), equal to $-2\gamma^\sigma\gamma^\rho\gamma^\nu + (4-d)\gamma^\nu\gamma^\rho\gamma^\sigma$. Substituting and distributing the $-\gamma^\mu$ gives Eq. (C.195). □

C.7.3 Traces in d dimensions

The trace identities *without* γ^5 are unchanged in form. Their derivations in Section C.6 used only the Clifford relation, the cyclicity of the trace, and $\operatorname{tr}\mathbf{1} = 4$ — never the integer value of $\eta^\mu{}_\mu$ inside the trace itself — so they continue verbatim:

$$\operatorname{tr}\mathbf{1} = 4, \qquad \operatorname{tr}(\gamma^{\mu_1}\cdots\gamma^{\mu_n}) = 0 \quad (n \text{ odd}), \tag{C.197}$$

$$\operatorname{tr}(\gamma^\mu\gamma^\nu) = 4\,\eta^{\mu\nu}, \tag{C.198}$$

$$\operatorname{tr}(\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 4(\eta^{\mu\nu}\eta^{\rho\sigma} - \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}). \tag{C.199}$$

The dimension d enters traces only through Eq. (C.185), when an index is contracted *between* matrices inside the trace before the trace is taken. For instance $\mathrm{tr}(\gamma^\mu\gamma^\nu)\eta_{\mu\nu} = 4\,\eta^\mu{}_\mu = 4d$, and the contracted four-gamma trace becomes

$$\mathrm{tr}\big(\gamma^\mu\gamma^\nu\gamma_\mu\gamma_\nu\big) = \mathrm{tr}\big[(2-d)\gamma^\nu\gamma_\nu\big] = (2-d)\,d\;\mathrm{tr}\,\mathbf{1} = 4d(2-d), \tag{C.200}$$

using Eq. (C.188) then Eq. (C.187). Every such contracted factor produces a polynomial in $d = 4-2\varepsilon$ that, in a physical amplitude, multiplies the ε-expansion of a master integral. The loop chapters therefore carry these prefactors symbolically until the very end and only then expand: e.g. $d = 4-2\varepsilon$, $(2-d) = -2+2\varepsilon$, $(4-d) = 2\varepsilon$, $(d-2)/(d-1) = \frac{2}{3}(1+\frac{1}{2}\varepsilon+\dots)$, and so on. The discipline is simple but unforgiving: an $O(\varepsilon)$ slip in a prefactor times a $1/\varepsilon$ pole is a finite error in the answer.

Worked Example C.63 (The QED self-energy numerator)**.** The one-loop electron self-energy of Chapter 19 contains the Dirac structure

$$N(p) \;\equiv\; \gamma^\mu\,(\not{p}+m)\,\gamma_\mu, \tag{C.201}$$

the numerator left after the photon line contracts the two QED vertices. We reduce it in d dimensions. Split into the two pieces and apply Theorem C.61:

$$\gamma^\mu\not{p}\,\gamma_\mu = p_\alpha\,\gamma^\mu\gamma^\alpha\gamma_\mu = (2-d)\,p_\alpha\gamma^\alpha = (2-d)\,\not{p}, \tag{C.202}$$

$$\gamma^\mu\,m\,\gamma_\mu = m\,\gamma^\mu\gamma_\mu = m\,d. \tag{C.203}$$

Hence

$$N(p) = (2-d)\,\not{p} + d\,m. \tag{C.204}$$

Inserting $d = 4-2\varepsilon$,

$$N(p) = (-2+2\varepsilon)\,\not{p} + (4-2\varepsilon)\,m = -2\,(1-\varepsilon)\,\not{p} + 2\,(2-\varepsilon)\,m, \tag{C.205}$$

which at $d = 4$ collapses to the familiar $-2\not{p}+4m$. The retained ε-pieces are exactly what survive against the $1/\varepsilon$ pole of the accompanying scalar integral and feed the residue that fixes the field-strength and mass renormalization constants Z_2, Z_m in Chapter 19. This is the prototype for every numerator algebra in QED's loop computations (Chapters 17, 19 and 20): contract with Theorem C.61, keep d symbolic, expand last.

C.7.4 The γ^5 obstruction

Everything above continued without incident because γ^5 never appeared. The trouble is specific to $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, a product of *exactly four* fixed matrices, an object whose definition presupposes that spacetime has exactly four directions. Three properties of γ^5 are used pervasively in four-dimensional calculations, and one would like to keep all three under continuation:

(P1) total anticommutation, $\{\gamma^5, \gamma^\mu\} = 0$ for *every* $\mu = 0, 1, \ldots, d-1$;

(P2) the nonvanishing chiral trace, $\operatorname{tr}(\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = -4i\,\varepsilon^{\mu\nu\rho\sigma}$ (the convention and sign fixed in Section C.6, with $\varepsilon^{0123} = +1$ as in Chapter A);

(P3) cyclicity of the trace, together with $\operatorname{tr}\mathbf{1} = 4$.

These three demands are mutually inconsistent for $d \neq 4$. The proof is a single trace evaluated two ways.

Theorem C.64 (Incompatibility of the naive γ^5 rules)**.** *Assume (P1), (P3), and the contraction algebra of Theorem C.61, all taken to hold for general d. Then*

$$(d-4)\,\operatorname{tr}(\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = 0. \tag{C.206}$$

Consequently, for $d \neq 4$ the chiral trace must vanish, in direct contradiction with (P2). No continuation can keep all of (P1), (P2), (P3) at once.

Proof. Write $T^{\mu\nu\rho\sigma} \equiv \operatorname{tr}(\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma)$ and consider the single object

$$Y \;\equiv\; \operatorname{tr}(\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\,\gamma_\alpha\gamma^\alpha). \tag{C.207}$$

First evaluation. Contract the trailing pair immediately with Eq. (C.187), $\gamma_\alpha\gamma^\alpha = d\,\mathbf{1}$:

$$Y = d\,\operatorname{tr}(\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma) = d\,T^{\mu\nu\rho\sigma}. \tag{C.208}$$

Second evaluation. Leave the pair uncontracted and instead move the rightmost factor γ^α around the trace. By cyclicity (P3),

$$Y = \operatorname{tr}(\gamma^\alpha\,\gamma^5\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\,\gamma_\alpha). \tag{C.209}$$

By total anticommutation (P1), $\gamma^\alpha\gamma^5 = -\gamma^5\gamma^\alpha$, so

$$Y = -\operatorname{tr}(\gamma^5\,\gamma^\alpha\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\,\gamma_\alpha). \tag{C.210}$$

The remaining string is the four-gamma sandwich Eq. (C.195). Substituting it,

$$\begin{aligned} Y &= -\operatorname{tr}\Big(\gamma^5\big[\,2\gamma^\nu\gamma^\rho\gamma^\sigma\gamma^\mu + 2\gamma^\mu\gamma^\sigma\gamma^\rho\gamma^\nu - (4-d)\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma\big]\Big) \\ &= -2\,T^{\nu\rho\sigma\mu} - 2\,T^{\mu\sigma\rho\nu} + (4-d)\,T^{\mu\nu\rho\sigma}. \end{aligned} \tag{C.211}$$

Now use the total antisymmetry of $T^{\mu\nu\rho\sigma}$ in its four indices, which follows from (P2) but in fact only needs the anticommutator Eq. (C.184) inside the trace together with cyclicity: swapping any two adjacent gammas costs a sign (the symmetric remainder $2\eta\,\operatorname{tr}(\gamma^5\,\gamma\gamma) = 0$ vanishes by Eq. (C.197), as it is a γ^5-trace of two gammas). Thus T is alternating, and

$$T^{\nu\rho\sigma\mu} = -T^{\mu\nu\rho\sigma}, \qquad T^{\mu\sigma\rho\nu} = -T^{\mu\nu\rho\sigma}, \tag{C.212}$$

the first because $(\nu\rho\sigma\mu)$ is a single left-cyclic shift of $(\mu\nu\rho\sigma)$, an odd permutation of four objects, the second because $(\mu\sigma\rho\nu)$ differs from $(\mu\nu\rho\sigma)$ by the transposition $\nu \leftrightarrow \sigma$. Hence Eq. (C.211) becomes

$$Y = -2(-T^{\mu\nu\rho\sigma}) - 2(-T^{\mu\nu\rho\sigma}) + (4-d)T^{\mu\nu\rho\sigma} = (8-d)\,T^{\mu\nu\rho\sigma}. \tag{C.213}$$

Consistency. Equating the two evaluations Eq. (C.208) and Eq. (C.213),

$$d\,T^{\mu\nu\rho\sigma} = (8-d)\,T^{\mu\nu\rho\sigma} \quad\Longrightarrow\quad (2d-8)\,T^{\mu\nu\rho\sigma} = 0 \quad\Longrightarrow\quad 2(d-4)\,T^{\mu\nu\rho\sigma} = 0, \tag{C.214}$$

which is Eq. (C.206). For $d \neq 4$ this forces $T^{\mu\nu\rho\sigma} = 0$, contradicting (P2), where $T^{\mu\nu\rho\sigma} = -4i\,\varepsilon^{\mu\nu\rho\sigma} \neq 0$. □

Physical Insight

The contradiction is not a technical accident; it is the algebraic fingerprint of the chiral anomaly. The single piece of data that is genuinely four-dimensional — the Levi-Civita symbol $\varepsilon^{\mu\nu\rho\sigma}$ carried by the chiral trace — cannot be reconciled with a smooth, fully chirally symmetric continuation to general d. Whatever prescription one adopts, it must *break* something, and the breaking is exactly what surfaces as the nonconservation of the axial current computed in Chapter 36. A regulator that respected chiral symmetry exactly would predict no anomaly, in conflict with experiment ($\pi^0 \to \gamma\gamma$); the γ^5 problem of dimensional regularization is the price of having a regulator at all.

C.7.5 The 't Hooft–Veltman–Breitenlohner–Maison scheme

Theorem C.64 says we must sacrifice one of (P1), (P2), (P3). Cyclicity (P3) and the nonvanishing trace (P2) are too valuable — and too closely tied to the physical anomaly — to give up. The 't Hooft–Veltman prescription, made algebraically rigorous and proved consistent to all orders by Breitenlohner and Maison (the BMHV scheme), therefore sacrifices the *universality* of (P1): γ^5 is kept anticommuting with the four genuinely physical gamma matrices and made *commuting* with the remaining $(d-4)$ evanescent ones.

Definition C.65 (BMHV split)**.** Decompose the formally d-dimensional space into a four-dimensional "physical" subspace and a $(d-4)$-dimensional "evanescent" subspace. The metric and gamma matrices split as

$$\eta^{\mu\nu} = \bar{\eta}^{\mu\nu} + \hat{\eta}^{\mu\nu}, \qquad \bar{\eta}^{\mu}{}_{\mu} = 4, \quad \hat{\eta}^{\mu}{}_{\mu} = d-4 = -2\varepsilon, \tag{C.215}$$

$$\gamma^{\mu} = \bar{\gamma}^{\mu} + \hat{\gamma}^{\mu}, \qquad \bar{\gamma}^{\mu} \equiv \bar{\eta}^{\mu}{}_{\nu}\gamma^{\nu}, \quad \hat{\gamma}^{\mu} \equiv \hat{\eta}^{\mu}{}_{\nu}\gamma^{\nu}, \tag{C.216}$$

where the barred matrices $\bar{\gamma}^{\mu}$ obey the four-dimensional Clifford algebra $\{\bar{\gamma}^{\mu}, \bar{\gamma}^{\nu}\} = 2\bar{\eta}^{\mu\nu}$, the hatted matrices the evanescent one $\{\hat{\gamma}^{\mu}, \hat{\gamma}^{\nu}\} = 2\hat{\eta}^{\mu\nu}$, and the two sectors

anticommute, $\{\bar{\gamma}^\mu, \hat{\gamma}^\nu\} = 0$. The chiral matrix is defined purely in the physical sector,

$$\gamma^5 \equiv i\,\bar{\gamma}^0\bar{\gamma}^1\bar{\gamma}^2\bar{\gamma}^3. \tag{C.217}$$

The defining (an)ticommutators follow at once from Eq. (C.217) and the sector algebra: γ^5 is built from four barred matrices, so it anticommutes with each barred matrix and commutes with each hatted one,

$$\left\{\gamma^5, \bar{\gamma}^\mu\right\} = 0, \qquad \left[\gamma^5, \hat{\gamma}^\mu\right] = 0. \tag{C.218}$$

Recombining via Eq. (C.216),

$$\left\{\gamma^5, \gamma^\mu\right\} = \left\{\gamma^5, \bar{\gamma}^\mu\right\} + \left\{\gamma^5, \hat{\gamma}^\mu\right\} = 2\,\gamma^5\,\hat{\gamma}^\mu, \tag{C.219}$$

the right-hand side being purely evanescent (it is $O(\varepsilon)$ in the sense that it lives entirely in the $(d-4)$-dimensional sector). This is the precise statement that (P1) fails: γ^5 anticommutes with γ^μ only up to the evanescent remainder $2\gamma^5\hat{\gamma}^\mu$.

This single concession dissolves the contradiction of Theorem C.64. Retrace the second evaluation: the step $\gamma^\alpha\gamma^5 = -\gamma^5\gamma^\alpha$ is no longer exact, because $\{\gamma^5, \gamma^\alpha\} = 2\gamma^5\hat{\gamma}^\alpha \neq 0$. The extra evanescent term it generates is exactly what restores agreement with the first evaluation while leaving the genuinely four-dimensional chiral trace

$$\operatorname{tr}(\gamma^5\,\bar{\gamma}^\mu\bar{\gamma}^\nu\bar{\gamma}^\rho\bar{\gamma}^\sigma) = -4i\,\varepsilon^{\mu\nu\rho\sigma} \tag{C.220}$$

nonzero and consistent — the four matrices here are barred, so the obstruction, which required anticommuting an evanescent γ^α through γ^5, never triggers. Cyclicity and $\operatorname{tr}\mathbf{1} = 4$ are kept intact; the chiral trace is kept nonzero; and the algebra is provably consistent and unitary to all loop orders.

The price is stated honestly: because γ^5 no longer anticommutes with the full γ^μ, the scheme *breaks chiral symmetry by hand*. The chiral (and, in chiral gauge theories, the gauge) Ward identities acquire spurious violations of order ε from the evanescent remainder Eq. (C.219); multiplied by the $1/\varepsilon$ poles of divergent subgraphs, these leave *finite*, nonzero, symmetry-breaking residues. The Breitenlohner–Maison consistency theorem guarantees that all such breakings are local and can be cancelled, order by order in perturbation theory, by adding *finite, non-symmetric counterterms* tuned to restore the Ward identities. Where the violation is spurious — an artifact of the regulator — these finite counterterms remove it completely; where it is the genuine anomaly — as in the triangle graph of Chapter 36 — no symmetric counterterm exists, and the surviving finite piece is the physical anomaly coefficient, correctly normalized (the BMHV scheme automatically reproduces the Adler–Bardeen result).

Caution

The restoring counterterms in the BMHV scheme are an obligatory part of the calculation, not an optional polish. Omitting them does not merely shift a scheme convention; it leaves a wrong, gauge-variant answer. In any chiral calculation done with BMHV, one must check the relevant Ward–Takahashi identity (Chapter 19) at each order and add the finite counterterm needed to enforce it before quoting a physical result.

C.7.6 Naive dimensional regularization, Larin's scheme, and evanescent operators

The BMHV scheme is correct but cumbersome: the split into barred and hatted sectors multiplies the bookkeeping, and the restoring counterterms must be computed separately. Two pragmatic alternatives are in common use, each trading rigor for convenience in a controlled domain.

Naive dimensional regularization (NDR) keeps the four-dimensional rules wholesale: it imposes (P1) totally anticommuting γ^5 *and* retains cyclicity and $\operatorname{tr}\mathbf{1} = 4$, simply declining to apply both at the dangerous step. By Theorem C.64 this is inconsistent in general, and the inconsistency is concentrated entirely in traces containing a single γ^5 together with four or more other gamma matrices — precisely the "γ^5-odd" traces that probe the $\varepsilon^{\mu\nu\rho\sigma}$ structure. For amplitudes in which no such trace arises, or in which γ^5's appear in pairs that can be reduced away by $\gamma^5\gamma^5 = \mathbf{1}$ and $\{\gamma^5, \gamma^\mu\} = 0$ before any trace is closed, NDR is unambiguous and agrees with BMHV. It is the default in much of multiloop QCD, where chirality enters through vector and axial vertices in patterns that NDR handles cleanly. Its failure is exactly at the anomaly: an NDR calculation of the triangle graph must be supplemented by an external input (a "reading-point" prescription or the known Adler–Bardeen coefficient) to fix the ambiguity that Theorem C.64 guarantees is present.

Larin's scheme is a systematic prescription within dimensional regularization that sidesteps a free-floating γ^5 altogether by trading it, inside the d-dimensional algebra, for an explicit Levi-Civita tensor: one substitutes

$$\gamma^5\,\gamma^\mu \;\longrightarrow\; \frac{i}{3!}\,\varepsilon^{\mu\nu\rho\sigma}\,\gamma_\nu\gamma_\rho\gamma_\sigma, \qquad \gamma^5 \;\longrightarrow\; \frac{i}{4!}\,\varepsilon^{\mu\nu\rho\sigma}\,\gamma^\mu\gamma^\nu\gamma^\rho\gamma^\sigma, \tag{C.221}$$

treats the ε-tensor as a four-dimensional object contracted only at the end (with two ε's combining into a determinant of metrics carrying the full d-dependence), and then restores the correct Ward identities and the canonical normalization of the axial current by computing finite renormalization constants Z_5 and Z_A^{fin} once and for all. This is the scheme of choice for high-order axial-current and singlet-anomaly calculations in QCD, where its mechanical, manifestly d-dimensional rules make automation tractable.

Common to all three schemes is the notion of an *evanescent operator*: a Lorentz structure that vanishes identically at $d = 4$ but is nonzero at $d = 4 - 2\varepsilon$, such as the hatted bilinears built from $\hat{\gamma}^\mu$ in Eq. (C.216) or the difference between a d-dimensional Dirac string and its four-dimensional Fierz-reduced form. Evanescent operators are harmless at tree level but contribute through their $1/\varepsilon$ poles inside loops, mixing into physical operators with finite, scheme-dependent coefficients. They must be carried explicitly in the operator basis, renormalized like any other operator, and their finite mixing fixed by a stated convention; consistency between schemes is precisely the statement that physical quantities are independent of how the evanescent sector is renormalized once the same Ward identities are enforced. Their systematic treatment belongs to the operator-mixing machinery of Chapters 19 and 36; here we have established the algebra they live in and the reason they are unavoidable.

These rules — the d-dependent contractions Theorem C.61, the form-invariant traces Eqs. (C.198) and (C.199), and an explicitly chosen γ^5 scheme with its restoring counterterms — are the regularization-ready toolkit that QED's one-loop computations import in Chapters 19 and 20 and that the chiral-anomaly calculation of Chapter 36 cannot do without. With the algebra now continued and the γ^5 ambiguity pinned down rather than hidden, we return in the next section to four dimensions to turn the completeness relation into Fierz rearrangement and to assemble the external-state spinor sums that convert matrix elements into squared amplitudes.

C.8 Fierz Identities and External-State Spinor Technology

The completeness relation of Section C.5 is the statement that the sixteen matrices $\Gamma^A = \{\mathbf{1}, \gamma^\mu, \sigma^{\mu\nu}, \gamma^\mu\gamma^5, \gamma^5\}$ form a basis of the algebra of 4×4 matrices. Read forward, that result told us how to decompose any Dirac structure into bilinears. Read *sideways*, it tells us something more powerful: a product of two bilinears in which the four spinors are paired one way can be re-expanded as a sum of products in which they are paired the other way. This is the Fierz rearrangement, and it is the workhorse identity whenever a four-fermion operator must be regrouped — most famously when the charged-current weak interaction, written with the lepton paired to its neutrino, must be rewritten with the lepton paired to the other lepton. We derive the general rearrangement directly from completeness, paying close attention to the signs that the trace metric (Section C.6) and Fermi statistics (Section C.4, where the charge-conjugation data lives) introduce, and we carry the two canonical cases — scalar×scalar and the self-Fierzing $V-A$ current — in full.

We then assemble the second half of the section: the external-state spinor reference. Every cross section computed in this book contracts amplitudes against

the plane-wave solutions $u_s(p)$, $v_s(p)$ of the Dirac equation. We recall those solutions from Chapter 11 (their *existence* and physical meaning belong there; their *algebra* belongs here), establish their normalization and orthogonality, prove the spin-sum completeness relations and the energy projectors $\Lambda_\pm$, and finish with the Gordon decomposition — the identity that splits the electromagnetic current into a convection piece and a magnetic-moment piece, and so seeds the $g-2$ computation of Chapter 20.

C.8.1 The Fierz rearrangement problem

Fix four Dirac spinors and consider the product of two bilinears

$$(\bar\psi_1\,\Gamma^A\,\psi_2)(\bar\psi_3\,\Gamma^B\,\psi_4), \tag{C.222}$$

in which spinor 1 is contracted with 2 and spinor 3 with 4. We want to re-express this as a sum of products in which 1 is contracted with 4 and 3 with 2,

$$\sum_{C,D} F^{AB}{}_{CD}\,(\bar\psi_1\,\Gamma^C\,\psi_4)(\bar\psi_3\,\Gamma^D\,\psi_2). \tag{C.223}$$

The coefficients $F^{AB}{}_{CD}$ are pure numbers — they do not depend on the spinors — because (C.222) and (C.223) are both built from the same four spinors and the rearrangement is an identity at the level of the gamma matrices alone. Writing out the spinor indices (i,j,k,l running over the four Dirac components),

$$(\bar\psi_1)_i\,(\Gamma^A)_{ij}\,(\psi_2)_j\;(\bar\psi_3)_k\,(\Gamma^B)_{kl}\,(\psi_4)_l, \tag{C.224}$$

the entire content of the rearrangement is a statement about the tensor $(\Gamma^A)_{ij}(\Gamma^B)_{kl}$: we must re-expand it on the basis $(\Gamma^C)_{il}(\Gamma^D)_{kj}$, which carries the indices grouped so that i (spinor 1) sits with l (spinor 4) and k (spinor 3) with j (spinor 2).

C.8.2 The master formula and the 16×16 Fierz matrix

Recall from Section C.5 the normalized basis and its dual. With $\Gamma^A\in\{\mathbf{1},\gamma^\mu,\sigma^{\mu\nu},\gamma^\mu\gamma^5,\gamma^5\}$ we have the orthogonality

$$\operatorname{tr}\!\big(\Gamma^A\,\Gamma_B\big)=4\,\delta^A{}_B, \tag{C.225}$$

where the dual basis Γ_A is obtained by lowering with the diagonal Clifford metric g_{AB} defined by $\operatorname{tr}(\Gamma^A\Gamma^B)=4\,g^{AB}$. Explicitly, in the mostly-minus signature,

$$\Gamma_S=\mathbf{1},\qquad \Gamma_V=\gamma_\mu,\qquad \Gamma_T=\sigma_{\mu\nu},\qquad \Gamma_A=-\gamma_\mu\gamma^5,\qquad \Gamma_P=\gamma^5, \tag{C.226}$$

the crucial sign being the -1 on the axial sector, which follows from $\operatorname{tr}(\gamma^\mu\gamma^5\,\gamma^\nu\gamma^5)=-4\eta^{\mu\nu}$ (the two γ^5's and the anticommutation flip the sign relative to the vector sector). Completeness then states that any 4×4 matrix M obeys

$$M=\tfrac{1}{4}\sum_A \operatorname{tr}(\Gamma_A M)\,\Gamma^A. \tag{C.227}$$

Proposition C.66 (Fierz master formula)**.** *For the sixteen-element basis* $\{\Gamma^A\}$*,*

$$(\Gamma^A)_{ij}\,(\Gamma^B)_{kl} \;=\; \sum_{C,D} F^{AB}{}_{CD}\,(\Gamma^C)_{il}\,(\Gamma^D)_{kj}, \qquad F^{AB}{}_{CD} = \frac{1}{16}\,\mathrm{tr}(\Gamma^A\,\Gamma_D\,\Gamma^B\,\Gamma_C). \tag{C.228}$$

The 256 *numbers* $F^{AB}{}_{CD}$ *are the entries of the* 16×16 *Fierz matrix.*

Proof. Because the sixteen matrices Γ^C are a basis, and because for fixed i,l the object $(\Gamma^C)_{il}$ ranges over a basis of one index-pair while $(\Gamma^D)_{kj}$ ranges over a basis of the other, the products $(\Gamma^C)_{il}(\Gamma^D)_{kj}$ are a basis for the space of tensors with the index structure of the left-hand side. Hence an expansion of the form (C.228) exists with *some* coefficients $F^{AB}{}_{CD}$; it remains to extract them. Multiply both sides by $(\Gamma_{C'})_{li}\,(\Gamma_{D'})_{jk}$ and sum over all four indices. On the left,

$$\sum_{ijkl}(\Gamma^A)_{ij}(\Gamma^B)_{kl}(\Gamma_{C'})_{li}(\Gamma_{D'})_{jk} = \sum_{ijkl}(\Gamma^A)_{ij}(\Gamma_{D'})_{jk}(\Gamma^B)_{kl}(\Gamma_{C'})_{li} = \mathrm{tr}\big(\Gamma^A\,\Gamma_{D'}\,\Gamma^B\,\Gamma_{C'}\big), \tag{C.229}$$

where we have simply followed the index chain $i\to j\to k\to l\to i$. On the right,

$$\begin{aligned}\sum_{C,D} F^{AB}{}_{CD}\sum_{ijkl}(\Gamma^C)_{il}(\Gamma_{C'})_{li}\,(\Gamma^D)_{kj}(\Gamma_{D'})_{jk} &= \sum_{C,D} F^{AB}{}_{CD}\;\mathrm{tr}(\Gamma^C\Gamma_{C'})\;\mathrm{tr}(\Gamma^D\Gamma_{D'})\\ &= \sum_{C,D} F^{AB}{}_{CD}\,(4\delta^C{}_{C'})(4\delta^D{}_{D'}) = 16\,F^{AB}{}_{C'D'},\end{aligned} \tag{C.230}$$

using (C.225). Equating the two and renaming $C'D'\to CD$ gives the stated formula. □

Equation (C.228) is exact and completely general; everything below is its evaluation in physically important cases, together with the bookkeeping of Fermi statistics.

The sign from Fermi statistics. The identity (C.228) is a statement about gamma matrices, and it holds verbatim when the ψ's are commuting (c-number) external-state spinors — the situation in a scattering amplitude, where $u_s(p)$ and $v_s(p)$ are ordinary numbers. When instead the four bilinears are built from *anticommuting field operators*, as in an effective four-fermion operator $(\bar\psi_1\Gamma^A\psi_2)(\bar\psi_3\Gamma^B\psi_4)$, regrouping the fields to bring (C.222) into the form (C.223) requires moving ψ_2 past ψ_3 and ψ_4 (equivalently, reordering the four Grassmann factors from 1234 to 1432), which is an *odd* permutation and supplies an overall factor -1. We will state both versions explicitly; the -1 is exactly what makes the $V-A$ interaction self-Fierzing below.

C.8.3 Canonical case I: the scalar×scalar Fierz

Set $\Gamma^A = \Gamma^B = \mathbf{1}$. Then $F^{SS}{}_{CD} = \frac{1}{16}\,\mathrm{tr}(\Gamma_D\Gamma_C) = \frac{1}{16}\cdot 4\,g_{DC} = \frac{1}{4}g_{DC}$, using $\mathrm{tr}(\Gamma_D\Gamma_C) = 4g_{DC}$. Since $\sum_D g_{DC}\,\Gamma^D = \Gamma_C$, the sum over D collapses and

$$(\mathbf{1})_{ij}(\mathbf{1})_{kl} = \tfrac{1}{4}\sum_C (\Gamma^C)_{il}(\Gamma_C)_{kj}. \tag{C.231}$$

Spelling out the five sectors — and using the dual basis (C.226), so that the axial sector enters with a relative minus sign and the tensor sector carries the $\frac{1}{2}$ that compensates the double count of $\sigma_{\mu\nu} = -\sigma_{\nu\mu}$ when μ,ν are summed independently —

$$(\mathbf{1})_{ij}(\mathbf{1})_{kl} = \tfrac{1}{4}\Big[(\mathbf{1})_{il}(\mathbf{1})_{kj} + (\gamma^\mu)_{il}(\gamma_\mu)_{kj} + \tfrac{1}{2}(\sigma^{\mu\nu})_{il}(\sigma_{\mu\nu})_{kj} - (\gamma^\mu\gamma^5)_{il}(\gamma_\mu\gamma^5)_{kj} + (\gamma^5)_{il}(\gamma^5)_{kj}\Big]. \tag{C.232}$$

Sandwiching between the four spinors — i,j contracted with $\bar\psi_1,\psi_2$ and k,l with $\bar\psi_3,\psi_4$ — gives the scalar Fierz identity for c-number spinors,

$$(\bar1\,2)(\bar3\,4) = \tfrac{1}{4}\Big[(\bar1\,4)(\bar3\,2) + (\bar1\gamma^\mu 4)(\bar3\gamma_\mu 2) + \tfrac{1}{2}(\bar1\sigma^{\mu\nu}4)(\bar3\sigma_{\mu\nu}2) - (\bar1\gamma^\mu\gamma^5 4)(\bar3\gamma_\mu\gamma^5 2) + (\bar1\gamma^5 4)(\bar3\gamma^5 2)\Big], \tag{C.233}$$

where we abbreviate $(\bar1\,\Gamma\,2) \equiv \bar\psi_1\Gamma\psi_2$ and so on. For anticommuting fields the right-hand side acquires the overall -1 discussed above. The same evaluation applied to $\Gamma^A = \Gamma^B = \gamma^\rho$ (contracting the Lorentz index) uses the central contraction identity $\gamma^\rho\Gamma_D\gamma_\rho = \kappa_D\,\Gamma_D$ from Section C.6, with eigenvalues

$$\kappa_S = 4, \quad \kappa_V = -2, \quad \kappa_T = 0, \quad \kappa_A = 2, \quad \kappa_P = -4 \tag{C.234}$$

(these are $\gamma^\rho\gamma_\rho = 4$, $\gamma^\rho\gamma^\alpha\gamma_\rho = -2\gamma^\alpha$, $\gamma^\rho\sigma^{\alpha\beta}\gamma_\rho = 0$, and the γ^5-dressed versions). The result is the vector×vector Fierz,

$$(\bar1\gamma^\rho 2)(\bar3\gamma_\rho 4) = (\bar1\,4)(\bar3\,2) - \tfrac{1}{2}(\bar1\gamma^\mu 4)(\bar3\gamma_\mu 2) - \tfrac{1}{2}(\bar1\gamma^\mu\gamma^5 4)(\bar3\gamma_\mu\gamma^5 2) - (\bar1\gamma^5 4)(\bar3\gamma^5 2). \tag{C.235}$$

Iterating the same trace evaluation over all twenty-five sector pairs $\{S,V,T,A,P\}\times\{S,V,T,A,P\}$ generates the complete 5×5 Fierz table; the two rows (C.233), (C.235) exhibit the pattern, and every other row follows from (C.228) with the eigenvalue data (C.234) in exactly the same way. We do not tabulate the rest because the cases the book actually needs are the scalar identity above and the chiral identity we derive next.

C.8.4 Canonical case II: the $V-A$ self-Fierz

The charged-current weak interaction couples left-handed currents $\bar\psi\,\gamma^\mu(1-\gamma^5)\psi$. The remarkable fact — which is why the Fermi theory can be written equivalently with either pairing of the four fermions — is that the product of two such currents Fierzes *into itself*. We prove it cleanly using the chiral projectors $P_L = \frac{1}{2}(1-\gamma^5)$, $P_R = \frac{1}{2}(1+\gamma^5)$ (recall $P_L^2 = P_L$, $P_LP_R = 0$, $P_L\gamma^\mu = \gamma^\mu P_R$, from Section C.5).

Theorem C.67 ($V-A$ self-Fierz)**.** *As an identity of gamma matrices,*

$$(\gamma^\rho P_L)_{ij}\,(\gamma_\rho P_L)_{kl} = -\,(\gamma^\rho P_L)_{il}\,(\gamma_\rho P_L)_{kj}. \tag{C.236}$$

Consequently, for anticommuting fields the two left-handed currents obey

$$[\bar\psi_1\gamma^\rho(1-\gamma^5)\psi_2]\,[\bar\psi_3\gamma_\rho(1-\gamma^5)\psi_4] = [\bar\psi_1\gamma^\rho(1-\gamma^5)\psi_4]\,[\bar\psi_3\gamma_\rho(1-\gamma^5)\psi_2]. \tag{C.237}$$

Proof. First, a chirality argument fixes the *structure*. Because $\gamma^\rho P_L = P_R\gamma^\rho P_L$, the matrix $(\gamma^\rho P_L)_{ij}$ is nonzero only for i in the right-handed and j in the left-handed subspace; likewise k right, l left. Any Fierz expansion $\sum_{C,D} G_{CD}(\Gamma^C)_{il}(\Gamma^D)_{kj}$ therefore needs $(\Gamma^C)_{il}$ to connect a right index i to a left index l, i.e. Γ^C must anticommute with γ^5 ($P_R\Gamma^C P_L \neq 0$ forces $\Gamma^C \in \{V, A\}$), and similarly for Γ^D. But on this chirality block $(\gamma^\mu\gamma^5)_{il} = -(\gamma^\mu)_{il}$, so the vector and axial structures are not independent: the surviving content is precisely $(\gamma^\mu P_L)_{il}(\gamma_\mu P_L)_{kj}$. Hence

$$(\gamma^\rho P_L)_{ij}(\gamma_\rho P_L)_{kl} = c\,(\gamma^\mu P_L)_{il}(\gamma_\mu P_L)_{kj} \tag{C.238}$$

for a single constant c, which we now fix by contraction. Multiply both sides by $(\gamma_\nu)_{ji}(\gamma^\nu)_{lk}$ and sum all indices. The left-hand side factorizes into two traces,

$$\sum_{\rho,\nu} \operatorname{tr}(\gamma^\rho P_L\,\gamma_\nu)\ \operatorname{tr}(\gamma_\rho P_L\,\gamma^\nu). \tag{C.239}$$

Now $\operatorname{tr}(\gamma^\rho P_L\gamma_\nu) = \operatorname{tr}(\gamma^\rho\gamma_\nu P_R) = \frac{1}{2}\operatorname{tr}(\gamma^\rho\gamma_\nu) - \frac{1}{2}\operatorname{tr}(\gamma^\rho\gamma_\nu\gamma^5) = 2\delta^\rho{}_\nu$, the γ^5 trace vanishing (Section C.6). Therefore the left-hand side equals $\sum_{\rho,\nu}(2\delta^\rho{}_\nu)(2\delta_\rho{}^\nu) = 4\sum_\rho 1 = 16$.

For the right-hand side, the index contractions assemble a single trace,

$$c\sum_{\mu,\nu}\operatorname{tr}(\gamma^\mu P_L\,\gamma^\nu\,\gamma_\mu P_L\,\gamma_\nu). \tag{C.240}$$

Push the projectors through using $P_L\gamma = \gamma P_R$ and $P_R\gamma = \gamma P_L$:

$$\gamma^\mu P_L\gamma^\nu\gamma_\mu P_L\gamma_\nu = \gamma^\mu\gamma^\nu P_R\gamma_\mu P_L\gamma_\nu = \gamma^\mu\gamma^\nu\gamma_\mu P_L P_L\gamma_\nu = \gamma^\mu\gamma^\nu\gamma_\mu\gamma_\nu P_R, \tag{C.241}$$

where $P_L^2 = P_L$ and a final $P_L\gamma_\nu = \gamma_\nu P_R$ were used. The contraction $\sum_\mu \gamma^\mu\gamma^\nu\gamma_\mu = -2\gamma^\nu$ then gives $\sum_{\mu,\nu}\gamma^\mu\gamma^\nu\gamma_\mu\gamma_\nu = -2\sum_\nu\gamma^\nu\gamma_\nu = -2\cdot 4 = -8$, so the trace is $\operatorname{tr}(-8\,P_R) = -8\cdot 2 = -16$. The right-hand side is therefore $-16\,c$. Matching, $16 = -16\,c$, i.e. $c = -1$, which is (C.236).

Finally, sandwich (C.236) between c-number spinors and multiply through by $2\times 2 = 4$ to convert $P_L \to (1-\gamma^5)$ on each factor:

$$[\bar 1\gamma^\rho(1-\gamma^5)2]\,[\bar 3\gamma_\rho(1-\gamma^5)4] = -[\bar 1\gamma^\rho(1-\gamma^5)4]\,[\bar 3\gamma_\rho(1-\gamma^5)2] \quad \text{(c-number)}. \tag{C.242}$$

For anticommuting fields the regrouping supplies the extra -1 of Section C.8.2, and the two minus signs cancel, yielding (C.237). □

Physical Insight

The cancellation of the two minus signs — the $c = -1$ of the gamma algebra against the -1 of Fermi statistics — is the statement that the four-fermion left-handed interaction has no memory of how its fermions were originally paired. In the low-energy Fermi theory this is what lets one write the muon-decay interaction $(\bar{\nu}_\mu \Gamma \mu)(\bar{e} \Gamma \nu_e)$ and the rearranged $(\bar{\nu}_\mu \Gamma \nu_e)(\bar{e} \Gamma \mu)$ with identical $V-A$ structure and coefficient, a fact that simplifies every tree-level weak rate.

C.8.5 External-state spinors: solutions, normalization, orthogonality

We now turn to the plane-wave Dirac spinors. The Dirac field of Chapter 11 is expanded in modes $u_s(p)e^{-ip\cdot x}$ (positive frequency) and $v_s(p)e^{+ip\cdot x}$ (negative frequency), with $s \in \{1,2\}$ the spin label and $p^0 = E_p = \sqrt{\mathbf{p}^2 + m^2} > 0$. Substituting into the free Dirac equation $(i\partial\!\!\!/ - m)\psi = 0$ shows that the constant spinors satisfy the momentum-space Dirac equations

$$(\not{p} - m)\, u_s(p) = 0, \qquad (\not{p} + m)\, v_s(p) = 0, \qquad \bar{u}_s(p)(\not{p} - m) = 0, \qquad \bar{v}_s(p)(\not{p} + m) = 0, \tag{C.243}$$

the barred equations following from the unbarred by Dirac conjugation (Section C.4). The existence and explicit rest-frame form of these solutions are established in Chapter 11; we recall only that, in the Dirac representation of Section C.5, the rest-frame ($\mathbf{p} = 0$, $p = (m, \mathbf{0})$) solutions normalized to $\bar{u}u = 2m$ are

$$u_s(m,\mathbf{0}) = \sqrt{2m}\begin{pmatrix} \xi_s \\ 0 \end{pmatrix}, \qquad v_s(m,\mathbf{0}) = \sqrt{2m}\begin{pmatrix} 0 \\ \eta_s \end{pmatrix}, \tag{C.244}$$

with ξ_s, η_s two orthonormal two-component spinors, $\xi_s^\dagger \xi_{s'} = \delta_{ss'}$, $\sum_s \xi_s \xi_s^\dagger = \mathbf{1}_{2\times 2}$; the spinor for general p is the Lorentz boost $u_s(p) = S(\Lambda_p) u_s(m,\mathbf{0})$, with $S(\Lambda)$ the spinor representation of Chapter 11 obeying $S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu$.

From (C.243) the Lorentz-invariant normalizations and orthogonality relations follow algebraically. Contracting $\bar{u}_s(\not{p} - m) = 0$ on the right with $u_{s'}$ and $(\not{p} - m)u_{s'} = 0$ on the left with $\bar{u}_s$, then taking the sum and difference, isolates the scalar and vector bilinears; carried out in the rest frame with (C.244) and boosted, the results are

$$\bar{u}_s(p)u_{s'}(p) = 2m\,\delta_{ss'}, \quad \bar{v}_s(p)v_{s'}(p) = -2m\,\delta_{ss'}, \quad \bar{u}_s(p)v_{s'}(p) = \bar{v}_s(p)u_{s'}(p) = 0, \tag{C.245}$$

while the components of the conserved current give the Dirac-adjoint (i.e. $u^\dagger u$) normalizations

$$u_s^\dagger(p)u_{s'}(p) = 2E_p\,\delta_{ss'}, \qquad v_s^\dagger(p)v_{s'}(p) = 2E_p\,\delta_{ss'}, \qquad u_s^\dagger(p)v_{s'}(\tilde{p}) = 0, \tag{C.246}$$

where $\tilde{p} \equiv (E_p, -\mathbf{p})$. The factor $2E_p$ is the relativistic state normalization that pairs with the $1/(2E_p)$ in the Lorentz-invariant phase space; its appearance here is exactly what makes the one-particle states of Chapter 11 covariantly normalized. We verify (C.245) explicitly in the rest frame: $\bar{u}_s u_{s'} = 2m\,(\xi_s^\dagger, 0)\gamma^0(\xi_{s'}, 0)^{\mathsf{T}} = 2m\,\xi_s^\dagger \xi_{s'} = 2m\delta_{ss'}$ using $\gamma^0 = \mathrm{diag}(\mathbf{1}, -\mathbf{1})$ on the upper block; the boost preserves it because $\bar{u}(p)u(p) = \bar{u}(0)S^{-1}Su(0) = \bar{u}(0)u(0)$.

C.8.6 Spin sums and the energy projectors

The single most-used identity in cross-section computations is the spin sum, which closes the two-spinor outer product into a polynomial in $\not{p}$.

Theorem C.68 (Spin-sum completeness)**.** *The plane-wave spinors satisfy*

$$\sum_{s=1,2} u_s(p)\,\bar{u}_s(p) = \not{p} + m, \qquad \sum_{s=1,2} v_s(p)\,\bar{v}_s(p) = \not{p} - m. \tag{C.247}$$

Proof. Both sides are 4×4 matrices; we establish the first identity in the rest frame and then propagate it covariantly. Using (C.244) and $\sum_s \xi_s \xi_s^\dagger = \mathbf{1}_{2\times 2}$,

$$\sum_s u_s(m,\mathbf{0})\bar{u}_s(m,\mathbf{0}) = 2m \sum_s \begin{pmatrix} \xi_s \\ 0 \end{pmatrix} (\xi_s^\dagger, 0)\,\gamma^0 = 2m \begin{pmatrix} \mathbf{1} & 0 \\ 0 & 0 \end{pmatrix} \gamma^0 = 2m \begin{pmatrix} \mathbf{1} & 0 \\ 0 & 0 \end{pmatrix}, \tag{C.248}$$

which equals $\mathrm{diag}(2m, 2m, 0, 0)$. In the rest frame $\not{p} + m = m\gamma^0 + m = m\,\mathrm{diag}(\mathbf{1}, -\mathbf{1}) + m\,\mathbf{1} = \mathrm{diag}(2m, 2m, 0, 0)$, so the two agree. Under a boost, $u_s(p)\bar{u}_s(p) = S\,u_s(0)\bar{u}_s(0)\,S^{-1}$, while $S(\not{p}_{\text{rest}} + m)S^{-1} = \not{p} + m$ by $S^{-1}\gamma^\mu S = \Lambda^\mu{}_\nu \gamma^\nu$; since the rest-frame identity holds and both sides transform identically, the identity holds in every frame. The v sum is identical with η_s on the lower block: $\sum_s v_s(m,\mathbf{0})\bar{v}_s(m,\mathbf{0}) = 2m\,\mathrm{diag}(0,0,-1,-1) = \not{p} - m$ at rest, then boosted. □

The spin sums are precisely the residue numerators of the Dirac propagator, $i(\not{p}+m)/(p^2 - m^2 + i\epsilon)$, evaluated on shell — the bridge between the propagator pole and the external states it amputates to in LSZ. Normalizing them produces the *energy projectors*

$$\Lambda_+(p) \equiv \frac{\not{p} + m}{2m} = \frac{1}{2m}\sum_s u_s(p)\bar{u}_s(p), \qquad \Lambda_-(p) \equiv \frac{m - \not{p}}{2m} = -\frac{1}{2m}\sum_s v_s(p)\bar{v}_s(p). \tag{C.249}$$

That these deserve the name follows from the Clifford relation $\not{p}\not{p} = p^2 = m^2$ on shell. Idempotency:

$$\Lambda_+^2 = \frac{(\not{p}+m)^2}{4m^2} = \frac{p^2 + 2m\not{p} + m^2}{4m^2} = \frac{2m^2 + 2m\not{p}}{4m^2} = \frac{\not{p}+m}{2m} = \Lambda_+, \tag{C.250}$$

and identically $\Lambda_\pm^2 = \Lambda_\pm$. Completeness and orthogonality:

$$\Lambda_+ + \Lambda_- = \frac{(\not p + m) + (m - \not p)}{2m} = \mathbf{1}, \qquad \Lambda_+ \Lambda_- = \frac{(\not p + m)(m - \not p)}{4m^2} = \frac{m^2 - p^2}{4m^2} = 0. \tag{C.251}$$

Finally $\operatorname{tr} \Lambda_\pm = \frac{1}{2m} \operatorname{tr}(\pm \not p + m) = \frac{1}{2m}\, 4m = 2$, counting the two physical spin states each projector retains. These projectors let one compute polarized rates without ever constructing the spinors explicitly: a sum over a single spin orientation is $u_s \bar u_s = \Lambda_+ \, \Sigma(s)$ with $\Sigma(s)$ the spin projector of Chapter 11, but the *unpolarized* sums (C.247) are all that the cross sections of Chapters 17 to 19 require.

C.8.7 The Gordon decomposition

The last identity converts the vector current $\bar u(p') \gamma^\mu u(p)$ into a form that exhibits its physical content: a charge (convection) current proportional to the average momentum, plus a magnetic-moment current proportional to the momentum transfer through $\sigma^{\mu\nu}$.

Theorem C.69 (Gordon decomposition)**.** *For on-shell spinors of equal mass m, with $q \equiv p' - p$,*

$$\bar u(p')\, \gamma^\mu\, u(p) = \bar u(p') \left[\frac{(p' + p)^\mu}{2m} + \frac{i\, \sigma^{\mu\nu} q_\nu}{2m} \right] u(p), \qquad \sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu, \gamma^\nu]. \tag{C.252}$$

Proof. Use the Dirac equations $\not p\, u(p) = m\, u(p)$ and $\bar u(p')\, \not p' = m\, \bar u(p')$ from (C.243). Then

$$2m\, \bar u(p') \gamma^\mu u(p) = \bar u(p') (\gamma^\mu \not p + \not p' \gamma^\mu) u(p), \tag{C.253}$$

since $\gamma^\mu \not p\, u = m \gamma^\mu u$ and $\bar u' \not p' \gamma^\mu = m \bar u' \gamma^\mu$. Now decompose the two products of gamma matrices into symmetric and antisymmetric parts using the Clifford algebra. From $\gamma^\mu \gamma^\nu = \frac{1}{2}\{\gamma^\mu, \gamma^\nu\} + \frac{1}{2}[\gamma^\mu, \gamma^\nu] = \eta^{\mu\nu} - i\sigma^{\mu\nu}$ and, swapping the order, $\gamma^\nu \gamma^\mu = \eta^{\mu\nu} + i\sigma^{\mu\nu}$,

$$\gamma^\mu \not p = \gamma^\mu \gamma^\nu p_\nu = (\eta^{\mu\nu} - i\sigma^{\mu\nu}) p_\nu = p^\mu - i\sigma^{\mu\nu} p_\nu, \tag{C.254}$$

$$\not p' \gamma^\mu = \gamma^\nu \gamma^\mu p'_\nu = (\eta^{\mu\nu} + i\sigma^{\mu\nu}) p'_\nu = p'^\mu + i\sigma^{\mu\nu} p'_\nu. \tag{C.255}$$

Adding, the symmetric parts give $(p + p')^\mu$ and the antisymmetric parts combine into $i\sigma^{\mu\nu}(p'_\nu - p_\nu) = i\sigma^{\mu\nu} q_\nu$:

$$\gamma^\mu \not p + \not p' \gamma^\mu = (p + p')^\mu + i\sigma^{\mu\nu} q_\nu. \tag{C.256}$$

Therefore $2m\, \bar u' \gamma^\mu u = \bar u' \big[(p + p')^\mu + i\sigma^{\mu\nu} q_\nu\big] u$; dividing by $2m$ gives (C.252). $\square$

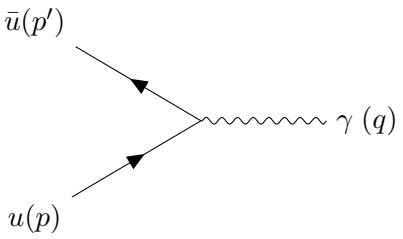


Figure C.3: The electromagnetic vertex $\bar{u}(p')\gamma^\mu u(p)$ with momentum transfer $q = p' - p$. The Gordon decomposition (C.252) splits this single matrix element into a convection current $\propto (p + p')^\mu$ (the Dirac, charge-coupling part) and a spin current $\propto i\sigma^{\mu\nu}q_\nu$ (the Pauli, magnetic-moment part). The coefficient of the latter is the seed of the anomalous magnetic moment computed in Chapter 20.

Physical Insight

At $q = 0$ only the first term survives and $\bar{u}(p)\gamma^\mu u(p) = (p^\mu/m)\,\bar{u}u = 2p^\mu$, the current of a structureless charge moving with momentum p (this also reproduces (C.245) for $\bar{u}\gamma^0 u = 2E_p$). The second term, $i\sigma^{\mu\nu}q_\nu/2m$, is the coupling of the spin magnetic moment to the field gradient; when a loop dresses the vertex it generates an additional form factor $F_2(q^2)$ multiplying exactly this structure, and $F_2(0) = a_e = (g-2)/2$ is the anomalous moment. The Gordon identity is therefore the algebraic skeleton onto which the $g - 2$ computation of Chapter 20 is hung.

C.8.8 Worked example: the spin-summed lepton tensor

The following computation assembles the spin sums and trace technology into the object that drives every QED cross section in the book.

Worked Example C.70. Compute the spin sum of the squared current matrix element,

$$L^{\mu\nu} \equiv \sum_{s,s'} \left[\bar{u}_{s'}(p')\gamma^\mu u_s(p)\right]\left[\bar{u}_{s'}(p')\gamma^\nu u_s(p)\right]^*, \tag{C.257}$$

which appears (with the muon-line analogue) in the unpolarized $e^+e^- \to \mu^+\mu^-$ cross section of Chapter 17.

Step 1 — complex-conjugate the second factor. A spinor bilinear is a 1×1 matrix, so its complex conjugate equals its Hermitian conjugate:

$$\left[\bar{u}_{s'}(p')\gamma^\nu u_s(p)\right]^* = \left[\bar{u}_{s'}(p')\gamma^\nu u_s(p)\right]^\dagger = u_s^\dagger(p)\,\gamma^{\nu\dagger}\,\gamma^{0\dagger}\,u_{s'}(p'). \tag{C.258}$$

With $\gamma^{0\dagger} = \gamma^0$ and the conjugation property $\gamma^0\gamma^{\nu\dagger}\gamma^0 = \gamma^\nu$ from Section C.4, insert $\gamma^0\gamma^0 = \mathbf{1}$:

$$= u_s^\dagger(p)\gamma^{\nu\dagger}\gamma^0 u_{s'}(p') = u_s^\dagger(p)\gamma^0\,(\gamma^0\gamma^{\nu\dagger}\gamma^0)\,u_{s'}(p') = \bar{u}_s(p)\,\gamma^\nu\,u_{s'}(p'). \tag{C.259}$$

Step 2 — close the spin sums into traces. Writing out the Dirac indices and using Theorem C.68,

$$\begin{aligned} L^{\mu\nu} &= \sum_{s,s'} \big[\bar{u}_{s'}(p')\gamma^{\mu}u_{s}(p)\big]\big[\bar{u}_{s}(p)\gamma^{\nu}u_{s'}(p')\big] \\ &= \sum_{s,s'} \bar{u}_{s'}(p')\gamma^{\mu}\,u_{s}(p)\bar{u}_{s}(p)\,\gamma^{\nu}\,u_{s'}(p') = \mathrm{tr}\Big[\Big(\textstyle\sum_{s'} u_{s'}(p')\bar{u}_{s'}(p')\Big)\gamma^{\mu}\Big(\textstyle\sum_{s} u_{s}(p)\bar{u}_{s}(p)\Big)\gamma^{\nu}\Big] \\ &= \mathrm{tr}\big[(\not{p}' + m)\,\gamma^{\mu}\,(\not{p} + m)\,\gamma^{\nu}\big], \end{aligned} \tag{C.260}$$

the reordering into a single trace being legitimate because the bilinears are c-numbers.

Step 3 — evaluate the trace. Expand the product; traces of an odd number of gamma matrices vanish (Section C.6), killing the two cross terms $m\ \mathrm{tr}(\not{p}'\gamma^{\mu}\gamma^{\nu})$ and $m\ \mathrm{tr}(\gamma^{\mu}\not{p}\gamma^{\nu})$. The two survivors are

$$L^{\mu\nu} = \mathrm{tr}(\not{p}'\gamma^{\mu}\not{p}\gamma^{\nu}) + m^{2}\ \mathrm{tr}(\gamma^{\mu}\gamma^{\nu}). \tag{C.261}$$

With $\mathrm{tr}(\gamma^{\alpha}\gamma^{\mu}\gamma^{\beta}\gamma^{\nu}) = 4(\eta^{\alpha\mu}\eta^{\beta\nu} - \eta^{\alpha\beta}\eta^{\mu\nu} + \eta^{\alpha\nu}\eta^{\mu\beta})$ and $\mathrm{tr}(\gamma^{\mu}\gamma^{\nu}) = 4\eta^{\mu\nu}$,

$$\mathrm{tr}(\not{p}'\gamma^{\mu}\not{p}\gamma^{\nu}) = p'_{\alpha}p_{\beta}\ \mathrm{tr}(\gamma^{\alpha}\gamma^{\mu}\gamma^{\beta}\gamma^{\nu}) = 4(p'^{\mu}p^{\nu} - (p'\cdot p)\,\eta^{\mu\nu} + p'^{\nu}p^{\mu}), \tag{C.262}$$

$$m^{2}\ \mathrm{tr}(\gamma^{\mu}\gamma^{\nu}) = 4m^{2}\eta^{\mu\nu}. \tag{C.263}$$

Result. Combining,

$$L^{\mu\nu} = 4\Big[p'^{\mu}p^{\nu} + p'^{\nu}p^{\mu} - (p'\cdot p - m^{2})\eta^{\mu\nu}\Big]. \tag{C.264}$$

This is the massive QED lepton tensor. It is manifestly symmetric in $\mu\nu$, and it is transverse: with $q = p' - p$ and $a \equiv p'\cdot p$ one has $q\cdot p' = m^{2} - a$, $q\cdot p = a - m^{2}$, so

$$q_{\mu}L^{\mu\nu} = 4\big[(q\cdot p')\,p^{\nu} + (q\cdot p)\,p'^{\nu} - (a - m^{2})\,q^{\nu}\big] = 4(a - m^{2})\big[-p^{\nu} + p'^{\nu} - q^{\nu}\big] = 0, \tag{C.265}$$

which is the algebraic shadow of current conservation $q_{\mu}\,\bar{u}(p')\gamma^{\mu}u(p) = \bar{u}(p')(\not{p}' - \not{p})u(p) = (m-m)\,\bar{u}(p')u(p) = 0$ enforced by the Dirac equation (C.243). Equation (C.264) is the exact object Chapters 17 and 19 contract to obtain the unpolarized $|\mathcal{M}|^{2}$.

The technology assembled here — spin sums, energy projectors, the Gordon split, and the trace reductions that close them — is the complete four-component external-state toolkit. For massless and helicity-resolved computations it is, however, more machinery than the physics requires: a Weyl fermion has a single two-component spinor and a single helicity, and the four-component spin sums collapse onto 2×2 structures. The next section recasts everything in the two-component (van der Waerden) language, where the spin sums become the $\sigma^{\mu}/\bar{\sigma}^{\mu}$ contraction identities and the Fierz rearrangement (C.236) reduces to a one-line ϵ-tensor identity — the foundation on which the spinor-helicity formalism of Chapter 58 is built.

C.9 Two-Component Spinor Calculus and the Bridge to Helicity

Every object built so far in this appendix lives on the four-dimensional spinor space: the Dirac matrices γ^μ, the sixteen bilinear covariants, the trace and Fierz technology, the external-state spinors u, v. Yet Chapter 3 taught us that the four-dimensional Dirac representation is *reducible* as a representation of the Lorentz group: under the universal cover $\mathrm{SL}(2,\mathbb{C})$ it splits into the two inequivalent two-dimensional irreducibles $(\tfrac{1}{2},0)$ and $(0,\tfrac{1}{2})$, the left- and right-handed Weyl spinors. The chiral basis of Section C.3 made this split block-diagonal in the γ^μ, and the Fierz machinery of Section C.8 repeatedly forced us to track left and right chirality separately. The natural conclusion is to stop carrying the reducible four-component object and work directly with its irreducible halves. This is the two-component, or van der Waerden, calculus: the minimal bookkeeping in which a Weyl spinor carries a single two-valued index, chirality is the type of the index rather than a projector, and the metric of the spinor space is the antisymmetric ϵ tensor. The payoff is twofold. Conceptually, the algebra becomes transparent: the Pauli matrices reappear as the only Lorentz-covariant objects with one undotted and one dotted index, and every four-component identity proved above is recovered as a statement about σ^μ and $\bar\sigma^\mu$. Practically, this is the language in which the spinor-helicity formalism of Chapter 58 is written; the angle and square brackets $\langle i\,j\rangle, [i\,j]$ that will compress an entire QED or Yang–Mills amplitude into a one-line rational function are nothing but ϵ-contractions of the two-component spinors introduced here. We therefore close the appendix by re-expressing the spin technology in two-component variables, proving the σ-matrix identities from the Pauli algebra of Chapter B, and verifying the dictionary back to the four-component results, so that Chapter 58 can introduce its bracket notation as shorthand on top of a fully established algebra.

Throughout we keep the global conventions of Chapter A: mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$, $\bar\psi = \psi^\dagger\gamma^0$, and the Levi-Civita normalization $\epsilon^{0123} = +1$. The Pauli matrices σ^i and their algebra are imported from Chapter B; we never reprove $\{\sigma^i,\sigma^j\} = 2\delta^{ij}\mathbf{1}$ or $\sigma^i\sigma^j = \delta^{ij}\mathbf{1} + i\epsilon^{ijk}\sigma^k$.

C.9.1 Undotted and dotted indices and the ϵ spinor metric

A left-handed Weyl spinor transforms in the $(\tfrac{1}{2},0)$ representation of $\mathrm{SL}(2,\mathbb{C})$. We write its components with a lowered *undotted* index,

$$\psi_a, \qquad a = 1,2, \qquad \psi_a \;\longmapsto\; M_a{}^b\,\psi_b, \quad M \in \mathrm{SL}(2,\mathbb{C}). \tag{C.266}$$

A right-handed Weyl spinor transforms in the complex-conjugate representation $(0,\tfrac{1}{2})$; its components carry a *dotted* index,

$$\bar\chi_{\dot a}, \qquad \dot a = \dot 1, \dot 2, \qquad \bar\chi_{\dot a} \;\longmapsto\; (M^*)_{\dot a}{}^{\dot b}\,\bar\chi_{\dot b}. \tag{C.267}$$

The two index types are genuinely distinct because M and M^* are inequivalent representations of $\mathrm{SL}(2,\mathbb{C})$ (unlike SU(2), where every representation is self-conjugate); the dot is the typographical record of which one acts. Complex conjugation interchanges the two: by definition $(\psi_a)^* \equiv \bar{\psi}_{\dot{a}}$, so conjugating an undotted spinor produces a dotted one and conjugating a dotted spinor an undotted one. This is the two-component face of the statement, proved in Chapter 3, that the $(\frac{1}{2},0)$ and $(0,\frac{1}{2})$ representations are exchanged by complex conjugation (equivalently, by parity).

Both representations are pseudo-real in the sharp sense that each carries an invariant antisymmetric bilinear form. Because $\det M = 1$, the totally antisymmetric symbol is invariant,

$$\epsilon^{ab} M_a{}^c M_b{}^d = (\det M)\,\epsilon^{cd} = \epsilon^{cd}, \tag{C.268}$$

and likewise for the dotted symbol under M^* (since $\det M^* = 1$ as well). This ϵ is the metric of spinor space: it raises and lowers indices exactly as $\eta_{\mu\nu}$ does for vectors. We fix the numerical conventions

$$\epsilon^{12} = -\epsilon^{21} = +1, \qquad \epsilon_{12} = -\epsilon_{21} = -1, \tag{C.269}$$

with the identical numerical values for the dotted symbols $\epsilon^{\dot{a}\dot{b}}, \epsilon_{\dot{a}\dot{b}}$. The signs in Eq. (C.269) are chosen so that raising followed by lowering is the identity:

$$\psi^a = \epsilon^{ab}\psi_b, \qquad \psi_a = \epsilon_{ab}\psi^b, \qquad \epsilon_{ab}\epsilon^{bc} = \delta_a^{\,c}. \tag{C.270}$$

We verify the contraction in Eq. (C.270) directly: $\epsilon_{ab}\epsilon^{bc}$ has its only contributions from $b \neq a$, and for $a = c = 1$ it is $\epsilon_{12}\epsilon^{21} = (-1)(-1) = 1 = \delta_1^1$, while for $a = 1, c = 2$ the sum $\epsilon_{12}\epsilon^{22} = 0 = \delta_1^2$; the remaining entries follow by the same count. The mnemonic forced by Eq. (C.269) is that undotted indices are contracted "northwest to southeast" (upper-left index summed against lower-right),

$$\psi\chi \equiv \psi^a\chi_a, \qquad \bar{\psi}\bar{\chi} \equiv \bar{\psi}_{\dot{a}}\bar{\chi}^{\dot{a}}, \tag{C.271}$$

the dotted contraction running the other way ("southwest to northeast"), so that the bars line up. These conventions are precisely those that Chapter 58 will inherit when it abbreviates $\psi^a\chi_a = \langle\psi\,\chi\rangle$.

The antisymmetry of ϵ has an immediate and important consequence for reordering. For two spinors,

$$\psi\chi = \psi^a\chi_a = \epsilon^{ab}\psi_b\chi_a = -\epsilon^{ba}\psi_b\chi_a = -\chi^b\psi_b \cdot (\pm 1), \tag{C.272}$$

where the final sign is $+1$ if ψ, χ are ordinary commuting spinors and -1 if they are anticommuting (Grassmann) field components, the extra minus arising from $\psi_b\chi_a = -\chi_a\psi_b$. Thus

$$\psi\chi = \begin{cases} -\chi\psi & \text{(commuting spinors, e.g. helicity spinors)}, \\ +\chi\psi & \text{(anticommuting spinors, e.g. Dirac field components)}. \end{cases} \tag{C.273}$$

The commuting case is the origin of the antisymmetry $\langle i\,j\rangle = -\langle j\,i\rangle$ of angle brackets in Chapter 58; the anticommuting case is what makes the Majorana mass term $\psi\psi = \psi^a\psi_a$ nonzero (it would vanish for commuting components).

C.9.2 The Pauli four-vectors and their algebra

To couple spinors to spacetime vectors we need an object with one undotted and one dotted index that transforms as a four-vector. The only Lorentz-covariant such objects are the *Pauli four-vectors*. Define

$$(\sigma^\mu)_{a\dot b} = (\mathbf{1},\ \sigma^i)_{a\dot b}, \qquad (\bar\sigma^\mu)^{\dot a b} = (\mathbf{1},\ -\sigma^i)^{\dot a b}, \tag{C.274}$$

so that $\sigma^0 = \bar\sigma^0 = \mathbf{1}$ and $\bar\sigma^i = -\sigma^i$, with the index heights fixed once and for all as shown: σ^μ carries a lower undotted and a lower dotted index, $\bar\sigma^\mu$ an upper dotted and an upper undotted index. These heights are not arbitrary; they are exactly what is needed for the bilinears $\bar\psi_{\dot a}(\bar\sigma^\mu)^{\dot a b}\psi_b$ and $\chi^a(\sigma^\mu)_{a\dot b}\bar\chi^{\dot b}$ to be index-free four-vectors. The two are related by raising both indices with ϵ:

$$(\bar\sigma^\mu)^{\dot a a} = \epsilon^{\dot a\dot b}\,\epsilon^{ab}\,(\sigma^\mu)_{b\dot b}. \tag{C.275}$$

One checks Eq. (C.275) componentwise: for $\mu = 0$ the right side is $\epsilon^{\dot a\dot b}\epsilon^{ab}\delta_{b\dot b} = \delta^{\dot a a}$ (the same count as in Eq. (C.270)), reproducing $\bar\sigma^0 = \mathbf{1}$; for $\mu = 2$ and $(\dot a, a) = (\dot 1, 2)$ the only surviving term is $\epsilon^{\dot 1\dot 2}\epsilon^{21}(\sigma^2)_{1\dot 2} = (+1)(-1)(-i) = +i$, which is exactly the $(\dot 1, 2)$ entry of $\bar\sigma^2 = -\sigma^2$. The remaining components follow identically. Thus σ^μ and $\bar\sigma^\mu$ are not independent: $\bar\sigma^\mu$ is the ϵ-transpose of σ^μ, the same data with both indices raised.

Everything in two-component calculus reduces to four identities for these matrices, all of which descend from the Pauli algebra.

Proposition C.71 (The $\sigma/\bar\sigma$ Clifford pair)**.**

$$(\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu)_a{}^b = 2\eta^{\mu\nu}\,\delta_a^{\ b}, \tag{C.276}$$

$$(\bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu)^{\dot a}{}_{\dot b} = 2\eta^{\mu\nu}\,\delta^{\dot a}_{\ \dot b}. \tag{C.277}$$

Proof. In $\sigma^\mu\bar\sigma^\nu$ the dotted indices are contracted, $(\sigma^\mu)_{a\dot c}(\bar\sigma^\nu)^{\dot c b}$, leaving one lower and one upper undotted index; this is an ordinary product of 2×2 matrices. Take $\mu = \nu = 0$: $\sigma^0\bar\sigma^0 + \sigma^0\bar\sigma^0 = 2\,\mathbf{1} = 2\eta^{00}\mathbf{1}$. Take $\mu = 0, \nu = j$: $\sigma^0\bar\sigma^j + \sigma^j\bar\sigma^0 = -\sigma^j + \sigma^j = 0 = 2\eta^{0j}\mathbf{1}$. Take $\mu = i, \nu = j$:

$$\sigma^i\bar\sigma^j + \sigma^j\bar\sigma^i = \sigma^i(-\sigma^j) + \sigma^j(-\sigma^i) = -\left\{\sigma^i, \sigma^j\right\} = -2\delta^{ij}\mathbf{1} = 2\eta^{ij}\mathbf{1}, \tag{C.278}$$

using $\left\{\sigma^i, \sigma^j\right\} = 2\delta^{ij}\mathbf{1}$ from Chapter B and $\eta^{ij} = -\delta^{ij}$. This proves Eq. (C.276). The computation for Eq. (C.277) is identical with $\sigma \leftrightarrow \bar\sigma$, the spatial case again giving $\bar\sigma^i\sigma^j + \bar\sigma^j\sigma^i = -\left\{\sigma^i, \sigma^j\right\} = 2\eta^{ij}\mathbf{1}$. □

Equations (C.276) and (C.277) are the two-component shadow of the Clifford relation $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$; we will see in Section C.9.4 that they are literally its block-diagonal entries. Next, the trace identity.

Proposition C.72 (σ-trace)**.**

$$\operatorname{tr}(\sigma^\mu \bar\sigma^\nu) \equiv (\sigma^\mu)_{a\dot b}(\bar\sigma^\nu)^{\dot b a} = 2\,\eta^{\mu\nu}. \tag{C.279}$$

Proof. Symmetrize using Eq. (C.276): $\operatorname{tr}(\sigma^\mu\bar\sigma^\nu) = \frac{1}{2}\operatorname{tr}(\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu)$ holds because the antisymmetric part $\frac{1}{2}(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu)$ is traceless—it is proportional to $\sigma^{\mu\nu}$ of Section C.9.5, a generator, and generators of $\mathrm{SL}(2,\mathbb{C})$ are traceless. Hence $\operatorname{tr}(\sigma^\mu\bar\sigma^\nu) = \frac{1}{2}\,2\eta^{\mu\nu}\operatorname{tr}\mathbf{1} = \eta^{\mu\nu}\cdot 2 = 2\eta^{\mu\nu}$, using $\operatorname{tr}\mathbf{1} = 2$ on the two-dimensional spinor space. □

The factor here is 2, not the 4 of the four-component trace $\operatorname{tr}(\gamma^\mu\gamma^\nu) = 4\eta^{\mu\nu}$ (Eq. (C.152)); the difference is exactly the dimension of the representation, and it is the cleanest single check that one is in the irreducible two-component world rather than the reducible four-component one.

C.9.3 The completeness (Fierz) relation

The trace identity says the four matrices σ^μ are orthonormal with respect to the trace form weighted by $\eta_{\mu\nu}$. Since the space of 2×2 complex matrices is four-dimensional and the σ^μ are manifestly linearly independent ($\mathbf{1}$ and the three traceless Pauli matrices), they form a basis. The statement that this basis is complete is the central two-component identity.

Theorem C.73 (Two-component completeness / Fierz relation)**.**

$$(\sigma^\mu)_{a\dot b}\,(\bar\sigma_\mu)^{\dot c d} = 2\,\delta_a^{\ d}\,\delta_{\dot b}^{\ \dot c}. \tag{C.280}$$

Proof. By the basis property, every 2×2 matrix M with indices $M^{\dot b a}$ (one upper dotted, one upper undotted) can be expanded on the σ^μ. We claim the expansion is

$$M^{\dot c d} = \tfrac{1}{2}\,(\bar\sigma_\mu)^{\dot c d}\,[(\sigma^\mu)_{a\dot b}M^{\dot b a}]. \tag{C.281}$$

To verify the coefficient, test Eq. (C.281) on the basis itself, $M = \bar\sigma^\nu$: the bracket becomes $(\sigma^\mu)_{a\dot b}(\bar\sigma^\nu)^{\dot b a} = \operatorname{tr}(\sigma^\mu\bar\sigma^\nu) = 2\eta^{\mu\nu}$ by Eq. (C.279), so the right side is $\frac{1}{2}(\bar\sigma_\mu)^{\dot c d}\,2\eta^{\mu\nu} = (\bar\sigma^\nu)^{\dot c d} = M^{\dot c d}$, confirming Eq. (C.281) for every M since the $\bar\sigma^\nu$ span. Now Eq. (C.281) must hold as an identity in the arbitrary matrix $M^{\dot b a}$, so the linear operators on both sides agree:

$$\tfrac{1}{2}\,(\bar\sigma_\mu)^{\dot c d}\,(\sigma^\mu)_{a\dot b} = \delta_{\dot b}^{\ \dot c}\,\delta_a^{\ d}, \tag{C.282}$$

which is Eq. (C.280) after multiplying by 2 and reordering the two scalar factors. □

As a consistency check, contract Eq. (C.280) on $a = d$ and $\dot{b} = \dot{c}$: the left side is $\mathrm{tr}(\sigma^\mu\bar\sigma_\mu) = \sum_\mu \eta_{\mu\mu}\,\mathrm{tr}(\sigma^\mu\bar\sigma^\mu)$, but it is cleaner to use Eq. (C.279), $\mathrm{tr}(\sigma^\mu\bar\sigma_\mu) = 2\eta^\mu{}_\mu = 2\delta^\mu_\mu = 8$; the right side gives $2\,\delta^a_a\,\delta^{\dot b}_{\dot b} = 2\cdot 2\cdot 2 = 8$. The identity Eq. (C.280) is the two-component master Fierz identity: it lets any pair of σ-matrix bilinears sharing a contracted vector index be rearranged into a product of two contractions of the spinors. It is the exact two-component image of the four-component completeness relation of Section C.8: there, the 16 bilinear covariants resolved the identity on the four-dimensional spinor space; projecting that resolution onto a definite chirality on each external pair of indices kills every covariant except $\mathbf{1}$ and γ^μ restricted to that chirality, and the surviving piece is precisely Eq. (C.280). We make the most useful rearrangement explicit.

Worked Example C.74 (The vector-current Fierz rearrangement)**.** Consider the product of two left-handed currents, $(\bar\psi_1\bar\sigma^\mu\psi_2)(\bar\psi_3\bar\sigma_\mu\psi_4)$, written with explicit indices,

$$(\bar\psi_{1\dot a}(\bar\sigma^\mu)^{\dot a b}\psi_{2b})\,(\bar\psi_{3\dot c}(\bar\sigma_\mu)^{\dot c d}\psi_{4d}). \tag{C.283}$$

The two $\bar\sigma^\mu$ share the contracted Lorentz index μ. To bring them together apply the completeness relation in its all-raised form, obtained from Eq. (C.280) by raising indices with ϵ:

$$(\bar\sigma^\mu)^{\dot a b}(\bar\sigma_\mu)^{\dot c d} = 2\,\epsilon^{bd}\,\epsilon^{\dot a\dot c}. \tag{C.284}$$

This follows by raising the two lower indices of Eq. (C.280): $(\bar\sigma^\mu)^{\dot a b}(\bar\sigma_\mu)^{\dot c d} = \epsilon^{\dot a\dot e}\epsilon^{bf}\epsilon^{\dot c\dot g}\epsilon^{dh}(\sigma_\mu)_{f\dot e}(\sigma^\mu)_{\dots}$; tracking the index pattern with Eq. (C.280) collapses it to $2\epsilon^{bd}\epsilon^{\dot a\dot c}$, and one checks the overall sign on the $(\dot a b\dot c d) = (\dot 1\,2\,\dot 1\,2)$ component, where the left side is $(\bar\sigma^\mu)^{\dot 1 2}(\bar\sigma_\mu)^{\dot 1 2}$ and the right side is $2\epsilon^{22}\epsilon^{\dot 1\dot 1} = 0$, consistent with $(\bar\sigma^\mu)^{\dot 1 2}$ summing to zero. Substituting Eq. (C.284),

$$(\bar\psi_1\bar\sigma^\mu\psi_2)(\bar\psi_3\bar\sigma_\mu\psi_4) = 2\,(\epsilon^{bd}\psi_{2b}\psi_{4d})\,(\epsilon^{\dot a\dot c}\bar\psi_{1\dot a}\bar\psi_{3\dot c}) = 2\,(\psi_4\psi_2)(\bar\psi_1\bar\psi_3), \tag{C.285}$$

the two currents having been Fierzed into a product of a purely undotted and a purely dotted scalar bilinear. This is the two-component analogue of the four-component Fierz rearrangement of Section C.8, accomplished here in a single step with no 16×16 rearrangement matrix to invert.

A second identity, equally central to Chapter 58, is purely a fact about the two-dimensionality of spinor space.

Proposition C.75 (Schouten identity)**.** *For any undotted spinors and any fixed index a,*

$$\epsilon_{ab}\,\chi_c + \epsilon_{bc}\,\chi_a + \epsilon_{ca}\,\chi_b = 0, \tag{C.286}$$

and equivalently $\epsilon_{ab}\epsilon_{cd} + \epsilon_{ac}\epsilon_{db} + \epsilon_{ad}\epsilon_{bc} = 0$.

Proof. Regard the left side of Eq. (C.286) as a function $T_{abc} = \epsilon_{ab}\chi_c + \epsilon_{bc}\chi_a + \epsilon_{ca}\chi_b$. It is totally antisymmetric in (a,b,c): interchanging $a \leftrightarrow b$ sends it

to $\epsilon_{ba}\chi_c + \epsilon_{ac}\chi_b + \epsilon_{cb}\chi_a = -T_{abc}$, and cyclic invariance is manifest. A totally antisymmetric tensor with three indices each running over only two values must vanish, because any assignment forces two of the indices to coincide, killing the antisymmetrized object. The four-index form follows by setting $\chi_c = \epsilon_{cd}\eta^d$ for arbitrary η^d and stripping it. □

Contracting Eq. (C.286) with three further spinors $\lambda_i^a, \lambda_j^b, \lambda_k^c$ (commuting, as the on-shell helicity spinors of Chapter 58 are) and writing $\lambda_i\lambda_j \equiv \lambda_i^a\lambda_{ja}$ turns it into

$$(\lambda_i\lambda_j)\,\lambda_k + (\lambda_j\lambda_k)\,\lambda_i + (\lambda_k\lambda_i)\,\lambda_j = 0, \tag{C.287}$$

which is exactly the Schouten identity for angle brackets, $\langle i\,j\rangle\lambda_k + \langle j\,k\rangle\lambda_i + \langle k\,i\rangle\lambda_j = 0$, used to reduce amplitudes in Chapters 58 and 59. We have proved it here, before any bracket has been defined, as a triviality of two-dimensional linear algebra.

C.9.4 The chiral-basis dictionary

We now make precise the claim that the four-component technology is the two-component technology assembled into 2×2 blocks. In the chiral basis of Section C.3 the Dirac matrices are

$$\gamma^\mu = \begin{pmatrix} 0 & (\sigma^\mu)_{a\dot b} \\ (\bar\sigma^\mu)^{\dot a b} & 0 \end{pmatrix}, \qquad \gamma^5 = \begin{pmatrix} -\delta_a^{\,b} & 0 \\ 0 & \delta^{\dot a}_{\,\dot b} \end{pmatrix}, \tag{C.288}$$

where the upper-left block of γ^5 acts on undotted indices and the lower-right on dotted. The Clifford relation is now a corollary of Theorem C.71: computing

$$\gamma^\mu\gamma^\nu = \begin{pmatrix} \sigma^\mu\bar\sigma^\nu & 0 \\ 0 & \bar\sigma^\mu\sigma^\nu \end{pmatrix}, \tag{C.289}$$

the anticommutator $\{\gamma^\mu, \gamma^\nu\}$ is block-diagonal with entries $\sigma^\mu\bar\sigma^\nu + \sigma^\nu\bar\sigma^\mu = 2\eta^{\mu\nu}\mathbf{1}$ (upper) and $\bar\sigma^\mu\sigma^\nu + \bar\sigma^\nu\sigma^\mu = 2\eta^{\mu\nu}\mathbf{1}$ (lower), i.e. $2\eta^{\mu\nu}\,\mathbf{1}_4$. The two relations of Theorem C.71 are thus precisely the two diagonal blocks of the four-dimensional Clifford algebra. That γ^5 takes the form in Eq. (C.288) is checked from its definition: with $\gamma^0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ and $\gamma^i = \left(\begin{smallmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{smallmatrix}\right)$ a direct multiplication gives $i\gamma^0\gamma^1\gamma^2\gamma^3 = \mathrm{diag}(-\mathbf{1}, +\mathbf{1})$, so the chiral projectors $P_{L,R} = \tfrac{1}{2}(\mathbf{1} \mp \gamma^5)$ project onto the undotted (upper) and dotted (lower) blocks respectively. Chirality is thus the index type, exactly as promised.

A Dirac spinor is the direct sum of a left-handed and a right-handed Weyl spinor; writing the left part as ψ_a and the right part as the conjugate $\bar\chi^{\dot a}$ of a second left-handed spinor χ,

$$\Psi = \begin{pmatrix} \psi_a \\ \bar\chi^{\dot a} \end{pmatrix}, \qquad \Psi^\dagger = (\bar\psi_{\dot a},\ \chi^a), \qquad \bar\psi = \Psi^\dagger\gamma^0 = (\chi^a,\ \bar\psi_{\dot a}), \tag{C.290}$$

where $\Psi^\dagger$ uses $(\psi_a)^* = \bar\psi_{\dot a}$ and $(\bar\chi^{\dot a})^* = \chi^a$ from Section C.9.1, and $\bar\psi$ follows from $\gamma^0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ swapping the two blocks. A Majorana spinor is the special case $\chi = \psi$. With Eq. (C.290) every Dirac bilinear reduces to two-component contractions by matrix multiplication. The scalar and pseudoscalar:

$$\bar\psi\Psi = \chi^a\psi_a + \bar\psi_{\dot a}\bar\chi^{\dot a} = \chi\psi + \bar\psi\bar\chi, \tag{C.291}$$

$$\bar\psi\gamma^5\Psi = -\chi\psi + \bar\psi\bar\chi. \tag{C.292}$$

The vector and axial currents, using the off-diagonal blocks of γ^μ:

$$\bar\psi\gamma^\mu\Psi = \bar\psi\,\bar\sigma^\mu\psi + \chi\,\sigma^\mu\bar\chi, \tag{C.293}$$

$$\bar\psi\gamma^\mu\gamma^5\Psi = -\bar\psi\,\bar\sigma^\mu\psi + \chi\,\sigma^\mu\bar\chi, \tag{C.294}$$

where $\bar\psi\bar\sigma^\mu\psi \equiv \bar\psi_{\dot a}(\bar\sigma^\mu)^{\dot a b}\psi_b$ and $\chi\sigma^\mu\bar\chi \equiv \chi^a(\sigma^\mu)_{a\dot b}\bar\chi^{\dot b}$. The combinations $\frac{1}{2}(\bar\psi\gamma^\mu\Psi \mp \bar\psi\gamma^\mu\gamma^5\Psi)$ project onto the purely left current $\psi\bar\sigma^\mu\psi$ and the purely right current $\chi\sigma^\mu\bar\chi$: a left-handed Weyl fermion's vector current is built from $\bar\sigma^\mu$, a right-handed one's from σ^μ. This is the structural fact behind the chiral couplings of the electroweak theory, and it is read off the dictionary with no computation. The tensor bilinear $\bar\psi\sigma^{\mu\nu}\Psi$ decomposes through the generators of Section C.9.5; we record it there.

Finally, charge conjugation. The four-component matrix C of Section C.4, defined by $C\gamma^\mu C^{-1} = -(\gamma^\mu)^{\mathsf T}$, is in the chiral basis nothing but the ϵ spinor metric.

Worked Example C.76 (Charge conjugation is the ϵ metric)**.** Take the representation $C = -i\gamma^2\gamma^0$ of Section C.4. With $\gamma^2 = \left(\begin{smallmatrix} 0 & \sigma^2 \\ -\sigma^2 & 0 \end{smallmatrix}\right)$ and $\gamma^0 = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$,

$$\gamma^2\gamma^0 = \begin{pmatrix} \sigma^2 & 0 \\ 0 & -\sigma^2 \end{pmatrix}, \qquad C = -i\gamma^2\gamma^0 = \begin{pmatrix} -i\sigma^2 & 0 \\ 0 & i\sigma^2 \end{pmatrix}. \tag{C.295}$$

Now $-i\sigma^2 = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right) = \epsilon_{ab}$ and $i\sigma^2 = \left(\begin{smallmatrix} 0 & 1 \\ -1 & 0 \end{smallmatrix}\right) = \epsilon^{\dot a\dot b}$ in the conventions Eq. (C.269), so

$$C = \begin{pmatrix} \epsilon_{ab} & 0 \\ 0 & \epsilon^{\dot a\dot b} \end{pmatrix}. \tag{C.296}$$

The charge-conjugation matrix is therefore block-diagonal and equal to the spinor metric on each chirality: C *is* the operation of raising and lowering two-component indices. Acting on Ψ of Eq. (C.290), charge conjugation $\Psi \mapsto \Psi^c = C\bar\psi^{\mathsf T}$ produces, block by block, $\epsilon_{ab}\chi^b = \chi_a$ in the upper slot and $\epsilon^{\dot a\dot b}\bar\psi_{\dot b} = \bar\psi^{\dot a}$ in the lower, i.e.

$$\Psi^c = \begin{pmatrix} \chi_a \\ \bar\psi^{\dot a} \end{pmatrix}, \tag{C.297}$$

which is Eq. (C.290) with $\psi \leftrightarrow \chi$. Charge conjugation simply exchanges the two Weyl spinors making up the Dirac field, and the Majorana condition $\chi = \psi$ is

exactly the self-conjugacy $\Psi^c = \Psi$. To confirm C does its defining job, check the relation $C\gamma^\mu C^{-1} = -(\gamma^\mu)^{\mathsf{T}}$ on the off-diagonal blocks: with C block-diagonal and γ^μ off-diagonal, $C\gamma^\mu C^{-1}$ is off-diagonal with upper-right block $\epsilon_{ab}(\sigma^\mu)^b{}_{\dot c}(\epsilon^{-1})^{\dot c\dot d}$. Using $\epsilon\sigma^\mu\epsilon^{-1} = -(\sigma^\mu)^{\mathsf{T}}$ for each Pauli four-vector—the statement $\epsilon_{ab}\epsilon_{\dot c\dot d}(\sigma^\mu)^{\cdots}$ that is the index-raising relation Eq. (C.275) read as a similarity transform—this block becomes $-(\sigma^\mu)^{\mathsf{T}}$, and likewise for the other block, reproducing $-(\gamma^\mu)^{\mathsf{T}}$. The intertwiner of Section C.4 and the spinor metric of Section C.9.1 are one object.

C.9.5 The generators $\sigma^{\mu\nu}$, $\bar\sigma^{\mu\nu}$ and (anti)self-duality

The Lorentz generators on the two Weyl spaces are the antisymmetrized products of σ and $\bar\sigma$. Define

$$(\sigma^{\mu\nu})_a{}^b = \tfrac{i}{4}\big(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu\big)_a{}^b, \tag{C.298}$$

$$(\bar\sigma^{\mu\nu})^{\dot a}{}_{\dot b} = \tfrac{i}{4}\big(\bar\sigma^\mu\sigma^\nu - \bar\sigma^\nu\sigma^\mu\big)^{\dot a}{}_{\dot b}. \tag{C.299}$$

These are exactly the diagonal blocks of the four-component Lorentz generator. Recalling from Section C.5 that $\sigma^{\mu\nu} = \frac{i}{2}[\gamma^\mu,\gamma^\nu]$ so that the generator implementing Lorentz transformations on Ψ is $S^{\mu\nu} = \frac12\sigma^{\mu\nu} = \frac{i}{4}[\gamma^\mu,\gamma^\nu]$, the block computation $[\gamma^\mu,\gamma^\nu] = \mathrm{diag}(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu,\ \bar\sigma^\mu\sigma^\nu - \bar\sigma^\nu\sigma^\mu)$ gives

$$S^{\mu\nu} = \tfrac{i}{4}[\gamma^\mu,\gamma^\nu] = \begin{pmatrix} (\sigma^{\mu\nu})_a{}^b & 0 \\ 0 & (\bar\sigma^{\mu\nu})^{\dot a}{}_{\dot b} \end{pmatrix}. \tag{C.300}$$

Thus $\sigma^{\mu\nu}$ generates the $(\frac12,0)$ representation on ψ_a and $\bar\sigma^{\mu\nu}$ the $(0,\frac12)$ representation on $\bar\chi^{\dot a}$; the reducibility of the Dirac representation is the block-diagonality of Eq. (C.300). Both generators are traceless—$\mathrm{tr}\,\sigma^{\mu\nu} = \frac{i}{4}\,\mathrm{tr}(\sigma^\mu\bar\sigma^\nu - \sigma^\nu\bar\sigma^\mu) = \frac{i}{4}(2\eta^{\mu\nu} - 2\eta^{\nu\mu}) = 0$ by Eq. (C.279)—the fact used in proving Theorem C.72.

The defining algebraic feature of these two-by-two generators, the one that makes them irreducible and that the spinor-helicity three-point amplitudes of Chapter 58 exploit, is their definite duality under the Levi-Civita tensor.

Theorem C.77 ((Anti)self-duality)**.** *With* $\epsilon^{0123} = +1$,

$$\tfrac12\,\epsilon^{\mu\nu\rho\sigma}\,\sigma_{\rho\sigma} = +\,i\,\sigma^{\mu\nu}, \qquad \tfrac12\,\epsilon^{\mu\nu\rho\sigma}\,\bar\sigma_{\rho\sigma} = -\,i\,\bar\sigma^{\mu\nu}. \tag{C.301}$$

The undotted generator $\sigma^{\mu\nu}$ is self-dual; the dotted generator $\bar\sigma^{\mu\nu}$ is anti-self-dual.

Proof. We descend from the four-component identity, proved as follows. For $\mu \neq \nu$ one has $\sigma^{\mu\nu} = i\gamma^\mu\gamma^\nu$, and we claim

$$\gamma^5\,\sigma^{\mu\nu} = \tfrac{i}{2}\,\epsilon^{\mu\nu\rho\sigma}\,\sigma^{\rho\sigma}. \tag{C.302}$$

To establish Eq. (C.302) it suffices to check one independent component, since both sides are antisymmetric tensors built from the same Clifford algebra. Take

$(\mu\nu) = (01)$. The right side is $\frac{i}{2}\epsilon^{01\rho\sigma}\sigma^{\rho\sigma} = \frac{i}{2}(\epsilon^{0123}\sigma^{23} + \epsilon^{0132}\sigma^{32}) = i\,\epsilon^{0123}\sigma^{23} = i\,\sigma^{23}$, and lowering indices, $\sigma^{23} = \eta_{2\alpha}\eta_{3\beta}\sigma^{\alpha\beta} = (-1)(-1)\sigma^{23} = \sigma^{23} = i\gamma^2\gamma^3$, so the right side is $i \cdot i\gamma^2\gamma^3 = -\gamma^2\gamma^3$. The left side is $\gamma^5\sigma^{01} = i\,\gamma^5\gamma^0\gamma^1 = i\,(i\gamma^0\gamma^1\gamma^2\gamma^3)\gamma^0\gamma^1 = -\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0\gamma^1$. Marching the trailing γ^0 left past $\gamma^3\gamma^2\gamma^1$ (three anticommutations, sign -1) and using $(\gamma^0)^2 = \mathbf{1}$, then the trailing γ^1 left past $\gamma^3\gamma^2$ (two anticommutations, sign $+1$) and using $(\gamma^1)^2 = -\mathbf{1}$, gives $\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0\gamma^1 = +\gamma^2\gamma^3$, so the left side is $-\gamma^2\gamma^3$, matching. Thus Eq. (C.302) holds.

Now pass to blocks. In the chiral basis $\gamma^5 = \mathrm{diag}(-\mathbf{1}, +\mathbf{1})$ and, by Eq. (C.300) together with $\sigma^{\mu\nu} = 2S^{\mu\nu}$, the four-component tensor is $\sigma^{\mu\nu} = \mathrm{diag}(2\sigma^{\mu\nu},\, 2\bar\sigma^{\mu\nu})$. The left side of Eq. (C.302) is $\mathrm{diag}(-2\sigma^{\mu\nu},\, +2\bar\sigma^{\mu\nu})$, while the right side is $\frac{i}{2}\epsilon^{\mu\nu\rho\sigma}\,\mathrm{diag}(2\sigma_{\rho\sigma},\, 2\bar\sigma_{\rho\sigma})$. Equating the upper blocks,

$$-2\sigma^{\mu\nu} = \tfrac{i}{2}\epsilon^{\mu\nu\rho\sigma}\,2\sigma_{\rho\sigma} \;\Longrightarrow\; \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}\sigma_{\rho\sigma} = \frac{-2\sigma^{\mu\nu}}{2i}\cdot\frac{2}{2} = i\,\sigma^{\mu\nu}, \tag{C.303}$$

the first relation in Eq. (C.301); equating the lower blocks gives $+2\bar\sigma^{\mu\nu} = \frac{i}{2}\epsilon^{\mu\nu\rho\sigma}2\bar\sigma_{\rho\sigma}$, i.e. $\frac{1}{2}\epsilon^{\mu\nu\rho\sigma}\bar\sigma_{\rho\sigma} = -i\,\bar\sigma^{\mu\nu}$, the second. □

The duality operation squares to -1: applying it twice and using $\epsilon^{\mu\nu\rho\sigma}\epsilon_{\rho\sigma\alpha\beta} = -2(\delta^\mu_\alpha\delta^\nu_\beta - \delta^\mu_\beta\delta^\nu_\alpha)$ from Eq. (C.167) returns $\frac{1}{4}\epsilon^{\mu\nu\rho\sigma}\epsilon_{\rho\sigma\alpha\beta}\sigma^{\alpha\beta} = -\sigma^{\mu\nu}$, consistent with $(+i)^2 = -1$; the eigenvalues $\pm i$ are the only possibility for a real operation squaring to -1, and the two signs label the two irreducibles. A quick component check confirms the normalization: from Eq. (C.298), $\sigma^{01} = \frac{i}{4}(\sigma^0\bar\sigma^1 - \sigma^1\bar\sigma^0) = \frac{i}{4}(-\sigma^1 - \sigma^1) = -\frac{i}{2}\sigma^1$, while $\sigma^{23} = \frac{i}{4}(\sigma^2\bar\sigma^3 - \sigma^3\bar\sigma^2) = -\frac{i}{4}\,[\sigma^2, \sigma^3] = -\frac{i}{4}(2i\sigma^1) = \frac{1}{2}\sigma^1$; then $\frac{1}{2}\epsilon^{01\rho\sigma}\sigma_{\rho\sigma} = \sigma_{23} = \sigma^{23} = \frac{1}{2}\sigma^1$, which indeed equals $i\,\sigma^{01} = i(-\frac{i}{2}\sigma^1) = \frac{1}{2}\sigma^1$. The tensor bilinear of a Dirac field thus splits, via Eq. (C.300), into a self-dual piece $\chi\sigma^{\mu\nu}\psi$ and an anti-self-dual piece $\bar\psi\bar\sigma^{\mu\nu}\bar\chi$; for the field strength this is the familiar decomposition of $F_{\mu\nu}$ into self-dual and anti-self-dual parts that organizes instanton physics (Chapter 41) and the helicity classification of gauge fields in Chapter 58.

The pieces are now in place. We have the ϵ metric and its raising conventions, the Pauli four-vectors $\sigma^\mu, \bar\sigma^\mu$ with their Clifford, trace, and completeness identities, the full dictionary translating every four-component bilinear, intertwiner, and generator into two-component contractions, and the self-duality that isolates the two helicity sectors. Chapter 58 can therefore open not by constructing spinor algebra from scratch but by christening the contractions already proved here—setting $\langle i\,j\rangle = \lambda_i^a\lambda_{ja}$ and $[i\,j] = \bar\lambda_{i\dot a}\bar\lambda_j^{\dot a}$ for the on-shell spinors $\lambda, \bar\lambda$ that solve $p_{a\dot a} = p_\mu(\sigma^\mu)_{a\dot a} = \lambda_a\bar\lambda_{\dot a}$—and immediately deploy the antisymmetry Eq. (C.273), the Schouten identity Eq. (C.287), the Fierz relation Eq. (C.280), and the self-duality Eq. (C.301) to compress the QED and Yang–Mills theory amplitudes into angle and square brackets. This closes the appendix: every spinorial manipulation the book performs, from the four-dimensional traces of Section C.6 to the helicity brackets of Chapter 59, now rests on identities proved in full within it.

Further reading

The standard textbook treatments of gamma-matrix technology are Peskin and Schroeder [396] (appendix and chapter 3), Schwartz [432] (chapters 10–11), Srednicki [454] (chapters 33–50), and Weinberg [500] (chapter 5), whose conventions differ in detail from one another and from ours; the systematic representation-theoretic standpoint, including Pauli's fundamental theorem and the intertwining matrices, is developed most carefully in Weinberg [500]. The bilinear covariants, charge conjugation, and the discrete symmetries are treated thoroughly in Bertlmann [73], which is also the cleanest reference for the anomaly machinery built on the γ^5 trace. For dimensional regularization and the γ^5 problem, the foundational papers are 't Hooft and Veltman [469] and the rigorous all-orders construction of Breitenlohner and Maison [90] (the "BMHV" scheme), with the practical multiloop alternative set out in Larin [317]. The two-component calculus and its four-component dictionary are given exhaustively, with Feynman rules, in the review of Dreiner et al. [162]; the amplitudes-oriented development that the spinor-helicity chapters build on is Elvang and Huang [176].

Appendix D

Loop Integrals and Dimensional Regularization

This appendix is the shared computational engine of every loop chapter in the book. The Feynman rules assembled in Chapter 15 for the scalar theory and in Chapter 17 for gauge theory are, at tree level, a finite bookkeeping device: amputate, contract, multiply. Beyond tree level they begin to produce integrals over loop momenta that the rules leave undetermined—closed momentum loops are not fixed by the external kinematics and must be integrated over all of $\mathbb{R}^{1,3}$—and those integrals are generically divergent. The power counting of Chapter 21 has already told the reader *which* of them diverge and *how badly*, through the superficial degree of divergence; what it does not do is assign a number to the finite remainder, or even give a controlled meaning to the divergent part. That is the task here. We collect, once and in full, the technology that turns any one-loop integral arising anywhere in the book into a Laurent series in a regulator: a pole part that exposes the divergence and a finite remainder that carries the physics. Each loop chapter may then quote a master integral by cross-reference instead of re-deriving the same Wick rotation and angular integral from scratch.

The strategy is uniform and is developed step by step across the eight sections. A loop integrand is a product of propagator denominators raised to various powers, possibly with loop momenta in the numerator. We first combine the denominators into a single quadratic form using Feynman and Schwinger parameters (Section D.2), proved in full for n denominators with arbitrary powers rather than quoted for the few cases we happen to need. We then complete the square, shift the loop momentum, and rotate the contour to Euclidean signature (Section D.3), where the integral becomes radial and the angular part is the volume of a sphere. The radial integral is a Beta function, and assembling the pieces gives the master scalar integral in d dimensions (Section D.4), from which every tensor and numerator variant follows by differentiation. Expanding the result around $d = 4$ produces the Laurent series in ε (Section D.5); tensor integrals with

open Lorentz indices are reduced to this scalar basis by the Passarino–Veltman algorithm (Section D.6); and the same dimensional continuation that regulated the ultraviolet is shown to regulate the infrared as well (Section D.7). The final section carries the program to the other side of the cut, evaluating two- and three-body phase space in d dimensions (Section D.8) so that the optical theorem can be checked against the loop result it predicts.

The conventions are those collected in Chapter A, and we import them by cross-reference rather than restating them: the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, the Feynman propagator $i/(p^2 - m^2 + i\epsilon)$ with its $i\epsilon$ prescription, the analytic-continuation definition of the spacetime dimension as $d = 4 - 2\varepsilon$ with ε a complex parameter, the renormalization scale μ entering through the bookkeeping factor $\mu^{2\varepsilon}$ that keeps couplings dimensionless away from four dimensions, and the modified minimal subtraction scheme $\overline{\mathrm{MS}}$ defined by which combination of constants is removed along with each pole. The contour deformation underlying the Wick rotation, and the residue calculus it rests on, are taken from Chapter 5; the d-dimensional gamma-matrix traces and contractions that dress the QED numerators are taken from Chapter C (in particular Section C.6); the color factors and Casimirs that dress the Yang–Mills numerators are taken from Chapter B.

The scope of this appendix should be stated honestly. It *owns* the one-loop technology completely: every one-loop integral the book needs is reducible, by the methods built here, to the master scalar function and its descendants, and that reduction is carried out in full. The multi-loop and large-scale tensor-reduction algorithms—integration by parts, the Laporta reduction to master integrals, the Mellin–Barnes and sector-decomposition methods for evaluating those masters—are named and characterized at the level the book actually uses, but they are not developed here; this is a one-loop book in its explicit computations, and we say so plainly rather than gesturing at machinery we will not build. The results assembled here are consumed downstream at well-defined places: the scalar bubble and tadpole feed the ϕ^4 renormalization and beta function of Chapters 22 and 24; the vacuum polarization, self-energy, and vertex integrals feed the QED one-loop program, the Ward–Takahashi identity, and $g - 2$ in Chapters 19 and 20; the color-dressed versions feed the Yang–Mills beta function and asymptotic freedom of Chapter 27; and the phase-space formulae feed the cross-section computations wherever observables live (Chapter 28 among them).

One conceptual thread ties the later sections together and deserves to be flagged at the outset: dimensional regularization is simultaneously an ultraviolet and an infrared regulator. The same parameter ε that tames the large-momentum divergence of a loop also tames the small-momentum and collinear divergences that appear when massless particles are exchanged, and a single integral can carry poles of both origins. Keeping the two roles distinct—knowing which $1/\varepsilon$ is ultraviolet and removed by a counterterm, and which is infrared and destined

to cancel against real-emission phase space—is the discipline that Section D.7 installs and that the observable-level chapters then rely on. Throughout, three carried examples make the machinery concrete: the ϕ^4 bubble of ϕ^4 theory, the photon vacuum polarization, electron self-energy, and vertex integrals of QED, and the color-dressed scalar pieces of Yang–Mills theory. Each tool is built on the page, not handed over; the appendix is a reference, but a proof-carrying one.

D.1 The anatomy of a loop integral and the case for analytic continuation in dimension

The chapter introduction set this appendix a single concrete task: take any momentum-space loop integral the Feynman rules can produce and turn it into a Laurent series in a regulator—a pole part that records the divergence plus a finite, scheme-dependent remainder—with every step done once and in full. Before any of the manipulations that achieve this (combining denominators, rotating to Euclidean space, doing the angular and radial integrals, expanding in ε), we must say precisely what object we are manipulating, diagnose exactly how and why it fails to converge, and decide what we are entitled to mean by "the value" of a divergent integral. That is the work of this opening section. We write down the generic integral, sharpen the power-counting that tells us which integrals diverge and how badly, lay out the four properties any acceptable regulator must have, show by a short honest accounting that the obvious candidates (a hard cutoff, Pauli–Villars) each forfeit one of them, and arrive at dimensional regularization as the one scheme that keeps all four. The section ends by handing the next section a single, well-posed product-of-denominators integral together with the instruction that is the conceptual core of the whole appendix: treat the spacetime dimension d as a continuous complex parameter.

D.1.1 The generic loop integral

The Feynman rules of Chapter 15 (for the scalar threads) and Chapter 17 (for QED) assign to every diagram a product of vertices, external wavefunctions, and propagators, integrated over the momenta that internal lines carry but momentum conservation at the vertices does not fix. A connected diagram with I internal lines and V vertices has, after imposing one momentum-conserving delta function at each vertex (one of which only enforces overall external conservation and is stripped off with the amputation), exactly

$$L \;=\; I - V + 1 \tag{D.1}$$

undetermined internal momenta. These are the *loop momenta*; L is the number of loops, equal to the number of independent cycles in the diagram regarded as a graph. After routing (any consistent choice of how the external and loop

momenta flow through the internal lines), each internal line j carries a definite momentum q_j that is a linear combination of the loop momenta $\ell_1, \dots, \ell_L$ and the external momenta $p_1, \dots, p_n$, and the diagram's amplitude takes a universal shape.

Definition D.1 (Generic L-loop, N-propagator integral)**.** A Feynman diagram contributing at L loops, with N internal lines of momenta q_j and masses m_j, is associated with the momentum-space integral

$$I \;=\; \int \prod_{i=1}^{L} \frac{\mathrm{d}^d \ell_i}{(2\pi)^d} \; \frac{\mathcal{N}(\ell_i; p_a)}{\displaystyle\prod_{j=1}^{N} (q_j^2 - m_j^2 + i\varepsilon)}, \qquad q_j \;=\; \sum_{i=1}^{L} c_{ji}\, \ell_i \;+\; \sum_{a} d_{ja}\, p_a, \tag{D.2}$$

where the routing coefficients $c_{ji}, d_{ja} \in \{0, \pm 1\}$ are fixed by momentum conservation at every vertex, the metric is mostly-minus and the $i\varepsilon$ is the Feynman prescription of Theorem A.23 (so each denominator is $q_j^2 - m_j^2 + i\varepsilon$ with $q_j^2 = q_j^{0\,2} - \mathbf{q}_j^2$), and the numerator $\mathcal{N}$ is a polynomial in the loop momenta whose coefficients are built from external momenta, masses, polarization data, and—for fermion lines and gauge propagators—Dirac matrices and Lorentz tensors. The measure carries one factor of $(2\pi)^{-d}$ per loop, in accord with the Fourier convention of Theorem A.12.

Three features of (D.2) deserve immediate comment, because the rest of the appendix is organized around them. First, the integral is over a *Minkowski* momentum space: each ℓ_i^0 runs along the real axis, the denominators are indefinite quadratic forms, and the $i\varepsilon$ is what makes the ℓ_i^0 contour well-defined by pushing the poles off the real axis. Wick rotation (Theorem A.23), carried out explicitly in a later section, will convert this into a Euclidean integral with a positive-definite denominator, but the object we start from is Lorentzian. Second, the numerator is a polynomial, so a tensor integral—one with uncontracted loop momenta upstairs—can always be reduced to scalar integrals once the denominator has been brought to a standard form; that reduction is the business of the tensor-reduction section. For the present diagnostic purposes only the *degree* of the numerator in the loop momenta matters. Third, the denominator is a product of N factors, and essentially every technique in this appendix is a device for collapsing that product into a single power of one quadratic form. The next section does exactly this with Feynman and Schwinger parameters.

It is worth recording the bookkeeping that relates the data in (D.2) to the graph. With V vertices and $I = N$ internal lines, (D.1) gives $L = N - V + 1$, so a one-loop diagram ($L = 1$) has $N = V$: as many propagators as vertices, the lines forming a single closed ring threaded by one undetermined momentum ℓ. This is the case the appendix solves completely and to which all three benchmark theories first reduce. The one-loop master integral with which they are all built

is

$$\int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{(\ell^2)^a}{\prod_{j=1}^{N} \big((\ell + r_j)^2 - m_j^2 + i\varepsilon\big)}, \tag{D.3}$$

where $a \geq 0$ counts the loop momenta in the numerator and the r_j are partial sums of external momenta fixed by the routing. Everything downstream—the ϕ^4 bubble that feeds β_λ in Chapter 24, the QED self-energy, vacuum polarization, and vertex of Chapter 19, the colour-dressed gluon and ghost loops of Chapter 27—is an instance of (D.3) with a particular N, particular masses, and a particular numerator.

D.1.2 Superficial degree of divergence: the UV diagnosis

The integral (D.2) is over an unbounded momentum space, and the question of whether it converges is, in the first instance, a question about its behaviour when the loop momenta are taken large together. The crude but decisive instrument is power counting. Chapter 21 owns the general combinatorial statement of the superficial degree of divergence for the scalar thread, and Theorem A.6 recast it as a statement about mass dimensions; here we re-derive it directly from (D.2) by the cleanest argument available—scaling the loop momenta—both because the scaling argument is the one that generalizes verbatim to the infrared analysis below, and because we will need the explicit one-loop count when staging the thread integrals.

Definition D.2 (Superficial degree of divergence)**.** For the integral (D.2), the *superficial degree of (ultraviolet) divergence* D is the net power of loop momentum in the integrand and measure when all loop momenta are scaled uniformly to infinity. With L loops in d dimensions, a numerator of total degree P in the loop momenta, and N propagators (each contributing two inverse powers),

$$D \;=\; d\,L \;+\; P \;-\; 2N. \tag{D.4}$$

The integral is said to be superficially divergent if $D \geq 0$ (logarithmically when $D = 0$, linearly when $D = 1$, quadratically when $D = 2$, and so on) and superficially convergent if $D < 0$.

The word "superficial" is a genuine caveat, not a flourish: $D < 0$ does not guarantee convergence, because a subset of the loop momenta may be scaled to infinity while the rest are held fixed, producing a *subdivergence* that the overall count (D.4) misses; and $D \geq 0$ does not guarantee that the leading divergence is actually present, because symmetries (gauge invariance, a Lorentz index structure that must come out transverse) can force the would-be leading term to vanish, softening the real divergence. Weinberg's theorem makes the precise statement: a one-particle-irreducible integral converges absolutely if $D < 0$ for the full integral and for every subintegral obtained by sending any subset of loop momenta to

infinity. At one loop there are no proper subintegrals, so $D < 0$ is necessary and sufficient for ultraviolet convergence, and (D.4) is exact.

Proposition D.3 (One-loop degree of divergence by scaling)**.** *For the one-loop integral* (D.3) *in d dimensions, with numerator degree $P = 2a$ and N propagators, the ultraviolet behaviour is governed by*

$$D \;=\; d \;+\; 2a \;-\; 2N, \tag{D.5}$$

and the integral converges in the ultraviolet if and only if $D < 0$.

Proof. Write the one-loop integrand as $f(\ell) = \mathcal{N}(\ell)/\prod_{j=1}^{N}\left((\ell + r_j)^2 - m_j^2 + i\varepsilon\right)$ and isolate the region of large loop momentum by rescaling $\ell = t\,\hat{\ell}$ with $\hat{\ell}$ on a fixed unit shell and $t \to \infty$. The measure scales homogeneously,

$$\mathrm{d}^d\ell \;=\; t^{\,d-1}\,\mathrm{d}t\,\mathrm{d}\Omega_{d-1}\,(\text{angular part}), \tag{D.6}$$

contributing $t^{d-1}\,\mathrm{d}t$ to the radial behaviour. In the limit $t \to \infty$ each propagator denominator is dominated by its quadratic term, $(\ell + r_j)^2 - m_j^2 = t^2\hat{\ell}^2 + 2t\,\hat{\ell}\cdot r_j + (r_j^2 - m_j^2) \sim t^2\hat{\ell}^2$, so the product of N denominators scales as t^{2N}. The numerator $(\ell^2)^a$ scales as t^{2a}. Collecting powers, the integrand times the radial measure behaves for large t as

$$t^{\,d-1}\,\frac{t^{2a}}{t^{2N}}\,\mathrm{d}t \;=\; t^{\,d-1+2a-2N}\,\mathrm{d}t \;=\; t^{\,D-1}\,\mathrm{d}t, \qquad D \;=\; d + 2a - 2N. \tag{D.7}$$

The radial integral $\int^{\infty} t^{D-1}\,\mathrm{d}t$ converges at its upper limit precisely when $D - 1 < -1$, i.e. $D < 0$, and diverges as Λ^D (for $D > 0$) or as $\log\Lambda$ (for $D = 0$) if cut off at $t = \Lambda$. Since one loop has no nontrivial subintegrals, this single scaling exhausts the large-momentum regions and (D.5) is the exact convergence criterion. Setting $L = 1$ and $P = 2a$ in (D.4) reproduces (D.5), confirming the general count against the explicit scaling. □

Two consequences of (D.5) are the engine of all later one-loop work. First, D depends on d *linearly and positively*: lowering the dimension lowers the degree of divergence. A scalar bubble ($N = 2, a = 0$) has $D = d - 4$, logarithmically divergent at $d = 4$ ($D = 0$) but convergent for any $d < 4$ ($D < 0$); a scalar tadpole/triangle pattern follows the same rule. This is the first hint, made rigorous below, that continuing d downward is exactly the right knob to turn to tame an ultraviolet divergence: the divergence is not removed, it is converted into the $d \to 4$ limit of a function that is finite away from four dimensions. Second, the degree of divergence of a tensor integral is read off by counting powers of ℓ in the numerator: the quadratically divergent ($D = 2$) piece of a $d = 4$ vacuum polarization comes entirely from the two loop momenta in its numerator, and the transversality that gauge invariance demands is precisely the statement that this leading piece must cancel.

D.1.3 Infrared, soft, and collinear divergences: the second diagnosis

Power counting at large momentum is only half the diagnosis. A loop integral can also diverge because the integrand blows up at *small* or *lightlike* loop momentum, where a denominator $q_j^2 - m_j^2$ vanishes inside the integration region. These are infrared divergences, and they fall into two physically distinct families, both of which dimensional regularization will also control.

A *soft* divergence occurs when a loop momentum $\ell \to 0$ on a line whose mass is zero or which is sandwiched between on-shell external lines, so that two adjacent denominators degenerate simultaneously. The exemplar is a massless gauge boson of momentum ℓ exchanged between two on-shell charged particles: as $\ell \to 0$ the photon propagator $1/\ell^2$ and the two adjacent fermion propagators $1/(2p\cdot\ell)$ all become singular. A *collinear* divergence occurs when a massless internal momentum becomes parallel to a massless external (or another massless internal) momentum, so that a denominator $(\ell + p)^2 = 2\,\ell\cdot p$ with $p^2 = 0$ vanishes along a whole ray $\ell \parallel p$ rather than at a point. Both are genuine, physical singularities of the perturbative amplitude—they signal that the asymptotic states (a charged particle is always dressed by soft photons; a massless particle is indistinguishable from a collinear pair) have been idealized—and both are tamed by the same scaling logic that gave (D.7), now applied to the dangerous small-momentum region instead of the large one.

Consider the soft region of a generic line, where ℓ is scaled *down*, $\ell = t\,\hat{\ell}$ with $t \to 0$. The measure contributes $t^{d-1}\,\mathrm{d}t$ as before, but now a soft propagator $1/\ell^2$ contributes t^{-2} and each eikonal propagator $1/(2p\cdot\ell + \dots)$ adjacent to an on-shell line contributes t^{-1}. The soft integral behaves as $\int_0 t^{\,d-1-2-2}\,\mathrm{d}t = \int_0 t^{\,d-5}\,\mathrm{d}t$ for the canonical one-soft-gauge-boson configuration with two eikonal lines, which converges at $t = 0$ iff $d - 5 > -1$, i.e. $d > 4$. The collinear region admits the same treatment with a light-cone decomposition: writing $\ell = \alpha p + \beta n + \ell_\perp$ along a reference null direction n and scaling $\beta, \ell_\perp$ toward the p direction, one finds the collinear integral also converges only for $d > 4$. The conclusion is structural and is the conceptual thread tying this appendix together: ultraviolet divergences require $d < 4$ to converge, infrared (soft and collinear) divergences require $d > 4$, and dimensional regularization sits in between, regulating both at once by holding d at a generic value off the integers and letting each kind of divergence reveal itself as a pole at $d = 4$. The same ε that carries the $1/\varepsilon$ ultraviolet pole carries the $1/\varepsilon$ (and, for overlapping soft-collinear regions, $1/\varepsilon^2$) infrared poles, with the sign of ε at which each is finite recorded in Theorem A.45.

Caution

The single symbol ε does double duty. A given integral may be ultraviolet finite but infrared divergent, or vice versa, or both at once; the Laurent

expansion in ε does not, by itself, tell you which kind of pole you are looking at. The convention of Theorem A.45—$\varepsilon > 0$ (so $d < 4$) for the ultraviolet, $\varepsilon < 0$ (so $d > 4$) for the infrared—means that a strictly correct treatment computes an ultraviolet-divergent integral by continuing from $d < 4$ and an infrared-divergent one by continuing from $d > 4$, then takes the common $\varepsilon \to 0$ limit of the resulting meromorphic function. In practice one computes in a dimension generic enough that the integral converges and trusts analytic continuation to connect the regions; we make this trust precise in Section D.1.6. When both kinds of pole are present and must be separated, the physical bookkeeping—which poles are removed by ultraviolet counterterms and which cancel against real-emission phase space, as in the QED programme of Chapter 19—is what disambiguates them, not the dimensional regulator on its own.

D.1.4 What a regulator must do, and why the obvious ones fail

A regulator is a deformation of the divergent integral that makes it finite, controlled by a parameter, such that the original integral is recovered (divergences and all) as the parameter is removed. The point of regularization is to give every intermediate quantity a definite, manipulable value so that the cancellations of renormalization can be carried out unambiguously; the physics resides only in quantities that survive the removal of the regulator. Not every deformation is acceptable. The renormalization program of Part VI, and the gauge theories of Part VII in particular, impose four requirements, and it is the simultaneous satisfaction of all four that singles out dimensional regularization.

Definition D.4 (Regulator desiderata)**.** A regularization scheme suitable for the renormalization of relativistic gauge theories should:

(R1) **Preserve Lorentz invariance.** The regulated integrals should remain Lorentz scalars/tensors, so that the divergence structure respects the same symmetry as the theory and counterterms are themselves Lorentz-invariant local operators.

(R2) **Preserve gauge invariance and the Ward–Takahashi/Slavnov–Taylor identities.** The regulator must not break the gauge symmetry, since those identities (proved for QED in Chapter 19) are what guarantee the cancellations—transversality of the vacuum polarization, equality of charge and wavefunction renormalizations—that make the theory renormalizable and unitary. A regulator that violates them generates spurious, symmetry-breaking counterterms.

(R3) **Regulate ultraviolet and infrared divergences uniformly.** A single parameter controlling both kinds of divergence avoids introducing a sec-

ond, independent regulator (with its own consistency conditions) and lets soft/collinear cancellations be tracked analytically.

(R4) **Be calculable and admit a clean analytic continuation.** The regulated integrals must be computable in closed form, and the dependence on the regulator parameter must be analytic, so that the divergence is isolated as an honest Laurent pole rather than buried in a transcendental cutoff dependence.

Measured against this list, the two regulators a reader meets first both fail, each for an instructive reason.

The momentum cutoff. The most physical-seeming regulator simply forbids loop momenta above a scale Λ: replace $\int \mathrm{d}^d\ell$ by $\int_{|\ell|<\Lambda} \mathrm{d}^d\ell$, or smooth it with a form factor. Divergences become explicit positive powers of Λ and logarithms $\log\Lambda$, and the picture is appealingly concrete—Λ is "new physics at short distance." But the cutoff is imposed on the *magnitude* $|\ell| = \sqrt{\ell_0^2 + \boldsymbol{\ell}^2}$ of the Euclidean momentum, a quantity that is not Lorentz invariant in Minkowski space; restoring Lorentz invariance requires the rotation to Euclidean signature, after which a sphere $|\ell_E| < \Lambda$ is invariant only under the Euclidean rotation group, not under the original Lorentz boosts in any frame-independent way once external momenta are present. Worse, the cutoff fails (R2): translating the loop momentum, $\ell \to \ell + k$, which is a freedom one constantly uses to route momenta and to complete squares, changes the cutoff surface $|\ell| < \Lambda$ into $|\ell + k| < \Lambda$, so the value of a linearly or quadratically divergent integral depends on the routing. In a gauge theory this routing dependence is exactly what spoils the Ward identity: the photon self-energy computed with a hard cutoff acquires a gauge-noninvariant, quadratically divergent mass term $\propto \Lambda^2 \eta^{\mu\nu}$ that must be removed by hand. The cutoff violates (R1) and (R2) at once, and these are the non-negotiable ones for Part VII.

Pauli–Villars. A more sophisticated scheme subtracts, from each propagator, a copy with a large regulator mass M: $\dfrac{1}{q^2 - m^2} \;\to\; \dfrac{1}{q^2 - m^2} - \dfrac{1}{q^2 - M^2} = \dfrac{m^2 - M^2}{(q^2 - m^2)(q^2 - M^2)}$, which falls off two powers faster at large q and so improves convergence while leaving the low-energy physics untouched as $M \to \infty$. Pauli–Villars is Lorentz invariant by construction (R1), and for QED, with enough auxiliary fields, it can be arranged to preserve the abelian Ward identity (R2). It is, historically, how the first correct one-loop QED results were obtained. But it fails on two counts for the book's purposes. It is *not* an infrared regulator (R3): subtracting a heavy copy does nothing to a soft or collinear singularity at small loop momentum, so a second, independent infrared regulator (a photon

mass, an off-shellness) must be bolted on, and keeping two regulators mutually consistent is exactly the bookkeeping one wants to avoid. More seriously, in a *non-abelian* gauge theory there is no way to give the gauge bosons a Pauli–Villars mass without breaking the gauge symmetry: a massive vector field is not gauge invariant, and the auxiliary heavy gauge fields wreck the Slavnov–Taylor identities, violating (R2) precisely where it matters most, in Yang–Mills theory of Chapter 27. Pauli–Villars survives (R1) but fails (R2) non-abelianly and (R3) always.

The lesson is that the requirement which kills the naive schemes is the conjunction of (R2) for *non-abelian* gauge theory with (R3): one needs a regulator that touches neither the gauge symmetry nor the requirement of a single uniform parameter. Dimensional regularization is the scheme that meets the entire list. It deforms not the integrand but the *dimension of the integration measure*, a deformation that is manifestly Lorentz (indeed, $\mathrm{O}(d)$) covariant (R1), that preserves gauge invariance because it never singles out a momentum scale or breaks any internal symmetry—the only structure that must be handled with care is the dimension-specific object γ^5, treated in Section C.7—(R2), that regulates ultraviolet and infrared divergences with the single parameter ε as Section D.1.3 showed (R3), and that yields integrals expressible in closed form through Euler Γ-functions whose poles are honest analytic poles in ε (R4). The price—an integral defined by analytic continuation rather than as a convergent sum, the absence of any direct "Λ^2" signal of quadratic divergence (power divergences are simply absent, set to zero by scaleless integration), and the subtlety of continuing γ^5 and the totally antisymmetric tensor—is small next to keeping the whole list, and it is the price the book pays everywhere.

D.1.5 Dimensional analysis in d dimensions and the scale μ

To define the integrals in $d = 4 - 2\varepsilon$ dimensions we must first confront a consequence of leaving $d = 4$ that has nothing to do with convergence: the couplings stop being dimensionless. Sections A.1.5 and A.7.1 fix this convention once for the whole book; we re-derive it here from the requirement $[S] = 0$ because it is load-bearing for every thread integral below and because the derivation is two lines.

In natural units the action $S = \int \mathrm{d}^d x\, \mathcal{L}$ is dimensionless, $[S] = 0$. With $[\mathrm{d}^d x] = -d$ in mass units, the Lagrangian density must carry $[\mathcal{L}] = d$. The kinetic terms then fix the field dimensions: for a scalar, $[\,(\partial\phi)^2\,] = 2[\partial] + 2[\phi] = 2 + 2[\phi] = d$ gives

$$[\phi] \;=\; \frac{d-2}{2}, \tag{D.8}$$

and for a Dirac field, $[\,\bar\psi i\partial\!\!\!/\psi\,] = 2[\psi] + 1 = d$ gives $[\psi] = (d-1)/2$, while the gauge kinetic term $[\,F^2\,] = 2([\partial] + [A]) = d$ gives $[A] = (d-2)/2$, the same as a

scalar. These reproduce Eqs. (A.10) to (A.12). The coupling dimensions follow by demanding $[\mathcal{L}] = d$ term by term.

Proposition D.5 (d-dependent coupling dimensions)**.** *In $d = 4 - 2\varepsilon$ dimensions, with the field dimensions above, the ϕ^4 and gauge couplings carry*

$$[\lambda] \;=\; 4 - d \;=\; 2\varepsilon, \qquad [e] \;=\; [g] \;=\; \frac{4-d}{2} \;=\; \varepsilon. \tag{D.9}$$

Proof. For the quartic scalar interaction $\dfrac{\lambda}{4!}\phi^4$, dimensional homogeneity requires $[\lambda] + 4[\phi] = [\mathcal{L}] = d$. Using (D.8), $[\lambda] = d - 4 \cdot \frac{d-2}{2} = d - 2(d-2) = 4 - d$, which is 2ε at $d = 4 - 2\varepsilon$. For the gauge coupling, read it off the covariant-derivative interaction. The QED vertex term in $\bar\psi i \not{D} \psi$ is $-e\,\bar\psi \not{A} \psi$ (from $D_\mu = \partial_\mu + ieA_\mu$ acting on the electron, the standing convention of QED); its dimension is $[e] + 2[\psi] + [A] = [e] + (d-1) + \frac{d-2}{2} = [e] + \frac{3d-4}{2}$, and setting this to d gives $[e] = d - \frac{3d-4}{2} = \frac{4-d}{2} = \varepsilon$. The non-abelian coupling enters identically through $D_\mu = \partial_\mu - igA^a_\mu T^a$ and through the cubic and quartic self-interactions in $F^a_{\mu\nu} F^{a\,\mu\nu}$; the cubic term $g f^{abc} (\partial A) A A$ has dimension $[g] + 3[A] + 1 = [g] + 3\frac{d-2}{2} + 1$, set equal to d gives $[g] = \frac{4-d}{2} = \varepsilon$, consistent with the fermion vertex. Hence (D.9). □

A coupling carrying a small dimension proportional to ε cannot be the expansion parameter of a perturbation series, since a Feynman amplitude is a pure number (times the kinematic factor carrying its external dimension) and one cannot expand in a dimensionful quantity. The remedy, fixed in Theorem A.46, is to extract the dimension into a fixed power of an arbitrary mass scale μ, the renormalization scale, leaving a dimensionless coupling to expand in:

$$\lambda \;\longrightarrow\; \lambda\,\mu^{2\varepsilon}, \qquad e \;\longrightarrow\; e\,\mu^{\varepsilon}, \qquad g \;\longrightarrow\; g\,\mu^{\varepsilon}, \tag{D.10}$$

where on the right λ, e, g are now dimensionless for all ε and μ alone carries the dimension dictated by (D.9). The power of μ is exactly [coupling]: $\mu^{2\varepsilon}$ for the quartic, μ^{ε} per gauge vertex. In a one-loop amplitude with two vertices the insertions multiply, so the ϕ^4 four-point function at one loop carries $(\mu^{2\varepsilon})^2 = \mu^{4\varepsilon}$, the QED self-energy and vacuum polarization (two vertices) carry $(\mu^{\varepsilon})^2 = \mu^{2\varepsilon}$, and so on; the universal effect is that every loop integral acquires the prefactor $\mu^{2\varepsilon}$ *per loop*, which combines with the $(\mu^2)^{-\varepsilon}$ that the integral itself produces (a d-dimensional integral of mass dimension D is μ^D times a number, and the leftover dimension is supplied by external invariants) to make every logarithm appear in the dimensionless, μ-restoring combination $\log(-p^2/\mu^2)$. This is why μ is the scale at which the renormalized coupling is defined and the variable in which the renormalization group of Chapters 24 and 27 runs. The scale is arbitrary: no physical quantity depends on it, and that independence is the content of the Callan–Symanzik equation derived downstream.

D.1.6 Dimensional regularization as analytic continuation

We can now state what (D.2) *means* when d is not an integer. There is no measure $\mathrm{d}^d\ell$ on a space of non-integer dimension in any literal sense; what exists is a family of integrals, one for each d in a strip where the integral converges, that happens to be an analytic function of d and can be continued.

Definition D.6 (Dimensional regularization)**.** Fix the external momenta and masses. For the integral (D.2), regarded as a function of the complex variable d, there is (after Wick rotation and, where needed, an explicit subtraction of infrared singular regions) a nonempty strip $d_- < \operatorname{Re} d < d_+$ in which the integral converges absolutely and defines a holomorphic function $I(d)$ of d. The *dimensionally regularized* value of the integral, for any d outside that strip—in particular for d in a punctured neighbourhood of 4—is the analytic continuation of $I(d)$. Ultraviolet divergences of the $d = 4$ integral appear as poles of this continuation at $d = 4$ (equivalently $\varepsilon = 0$): a logarithmically ultraviolet-divergent one-loop integral has a simple pole $1/\varepsilon$, an L-loop integral whose subdivergences have been arranged has poles up to $1/\varepsilon^L$; infrared divergences appear as poles of the continuation from the strip $\operatorname{Re} d > 4$, in accord with Theorem A.45.

The existence of the convergence strip is what makes the definition more than a wish. For the one-loop master (D.3) the ultraviolet count (D.5) gives convergence at large ℓ for $d < 2N - 2a$, and the infrared count of Section D.1.3 gives convergence at small ℓ for d above some threshold d_- set by the on-shell structure; whenever $d_- < d_+ = 2N - 2a$ a strip exists, the integral is genuinely convergent there, and the closed form one obtains (a ratio of Γ-functions times a power of the external invariant, derived in the next sections) is a meromorphic function of d that *is* the analytic continuation. For a scaleless integral the strip is empty—there is no d for which $\int \mathrm{d}^d\ell\,(\ell^2)^{-n}$ converges, since whatever power makes it converge at infinity makes it diverge at the origin and vice versa—and dimensional regularization assigns it the value zero, the unique continuation consistent with the scaling $\int \mathrm{d}^d(s\ell)\,((s\ell)^2)^{-n} = s^{d-2n}\int \mathrm{d}^d\ell\,(\ell^2)^{-n}$, which can hold for all s only if the integral vanishes. This vanishing of scaleless integrals is not a convention bolted on; it is forced by the same continuation that defines everything else, and it is the technical reason power divergences leave no Λ^2 trace in dimensional regularization.

Physical Insight

The conceptual move that makes the whole appendix work is to stop asking "what is the value of this divergent four-dimensional integral?"—a question with no answer—and ask instead "what is the meromorphic function of d that agrees with this integral wherever it converges?"—a question with a unique answer. Renormalization is then the statement that the $1/\varepsilon$ poles

of that function are absorbed, order by order, into a redefinition of finitely many parameters, leaving an $\varepsilon \to 0$ limit that is finite and physical. The regulator ε is the coordinate on the line of dimensions through $d = 4$; the divergences of physics live at the single point $\varepsilon = 0$ on that line, and dimensional regularization is the device for approaching that point from a direction where everything is finite and analytic.

D.1.7 One-loop integrals in the three benchmark theories

We close by writing down, in the canonical form (D.3), the specific one-loop integrals the benchmark theories will feed into this appendix. No evaluation is done here—that is the work of the sections that follow—and no new thread layer is introduced; the point is only to fix the integrands so the thread chapters can import a master integral by cross-reference instead of re-deriving the integral from the Feynman rules each time.

ϕ^4 theory: the four-point bubble. The one-loop correction to $\phi\phi \to \phi\phi$ in the $\lambda\phi^4$ theory comes from three diagrams (the s-, t-, and u-channel bubbles); the s-channel one, with total incoming momentum $p = p_1 + p_2$ and $s = p^2$, is shown in Fig. D.1. With the vertex factor $-i\lambda\mu^{2\varepsilon}$ from Chapter 15 and the symmetry factor $\frac{1}{2}$ for the two-propagator loop, its amplitude is

$$i\mathcal{M}_s \;=\; \frac{1}{2}\,(-i\lambda\,\mu^{2\varepsilon})^2 \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\; \frac{i}{\ell^2 - m^2 + i\varepsilon}\; \frac{i}{(\ell+p)^2 - m^2 + i\varepsilon}, \tag{D.11}$$

so the entire nontrivial content is the scalar bubble

$$B_0(p^2, m^2) \;\equiv\; \mu^{2\varepsilon} \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\; \frac{1}{(\ell^2 - m^2 + i\varepsilon)\big((\ell+p)^2 - m^2 + i\varepsilon\big)}, \tag{D.12}$$

an instance of (D.3) with $N = 2$, $a = 0$, hence $D = d - 4 = -2\varepsilon$: logarithmically divergent at $d = 4$, with a simple $1/\varepsilon$ pole. Its evaluation, completed in the following sections, supplies the one-loop coupling renormalization and the beta function $\beta_\lambda = 3\lambda^2/16\pi^2 + O(\lambda^3)$ quoted for ϕ^4 theory and computed in Chapter 24.

QED: vacuum polarization, self-energy, and vertex. The three one-loop one-particle-irreducible functions of QED, built with the vertex $-ie\gamma^\mu$ and propagators of Chapter 17, are the photon self-energy (vacuum polarization) $\Pi^{\mu\nu}$, the electron self-energy Σ, and the vertex correction Γ^μ. The vacuum polarization, with external photon momentum q and the fermion loop carrying

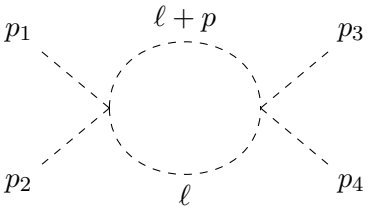


Figure D.1: The s-channel one-loop bubble correcting $\phi\phi \to \phi\phi$ in ϕ^4 theory. The two internal scalar lines carry ℓ and $\ell + p$ with $p = p_1 + p_2$; the loop momentum ℓ is undetermined and integrated over. This is the integral (D.12), the substrate for the ϕ^4 beta function of Chapter 24.

its mandatory overall minus sign, is

$$i\Pi^{\mu\nu}(q) \;=\; -(-ie\,\mu^{\varepsilon})^2 \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\;\mathrm{tr}\left[\gamma^\mu\,\frac{i(\not{\ell}+m)}{\ell^2-m^2+i\varepsilon}\,\gamma^\nu\,\frac{i(\not{\ell}+\not{q}+m)}{(\ell+q)^2-m^2+i\varepsilon}\right], \tag{D.13}$$

an $N = 2$, $P = 2$ integral with $D = d + 2 - 4 = 2 - 2\varepsilon$, superficially quadratically divergent; gauge invariance (the Ward identity proved in Chapter 19) forces the result into the transverse form $\Pi^{\mu\nu} = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\,\Pi(q^2)$, so the quadratic and linear divergences cancel and only a logarithmic $(1/\varepsilon)$ divergence survives in $\Pi(q^2)$. The diagram is drawn in Fig. D.2. The electron self-energy, in Feynman gauge ($\xi = 1$, the default for explicit computation in QED) with photon propagator $-i\eta_{\mu\nu}/(\ell^2 + i\varepsilon)$, is

$$-i\Sigma(p) \;=\; (-ie\,\mu^{\varepsilon})^2 \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\;\gamma^\mu\,\frac{i(\not{p}-\not{\ell}+m)}{(p-\ell)^2-m^2+i\varepsilon}\,\gamma^\nu\,\frac{-i\eta_{\mu\nu}}{\ell^2+i\varepsilon}, \tag{D.14}$$

an $N = 2$, $P = 1$ integral with $D = d + 1 - 4 = 1 - 2\varepsilon$, superficially linearly divergent but reducing (by Lorentz covariance, the result being a function of $\not{p}$ and m only) to a logarithmic divergence in the coefficients of $\not{p}$ and m; the massless photon makes (D.14) also infrared sensitive, regulated by the same ε as discussed in Section D.1.3. The vertex correction is the $N = 3$ integral

$$i\,\delta\Gamma^\mu(p',p) \;=\; (-ie\,\mu^{\varepsilon})^3 \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\;\gamma^\alpha\,\frac{i(\not{p}'-\not{\ell}+m)}{(p'-\ell)^2-m^2+i\varepsilon}\,\gamma^\mu\,\frac{i(\not{p}-\not{\ell}+m)}{(p-\ell)^2-m^2+i\varepsilon}\,\gamma^\beta\,\frac{-i\eta_{\alpha\beta}}{\ell^2+i\varepsilon}, \tag{D.15}$$

with $N = 3$, $P = 2$, hence $D = d + 2 - 6 = -2\varepsilon$, logarithmically (ultraviolet) divergent—the divergence that renormalizes the charge—and, like the self-energy, infrared divergent through the soft photon, the cancellation of which against bremsstrahlung is the subject of Chapter 19. The three integrals (D.13)–(D.15) are exactly what Chapter 19 consumes; their evaluation needs the Feynman-parameter reduction of the next section, the Wick rotation, the d-dimensional gamma algebra of Sections C.7.1 and C.7.3, and the Γ-function master formula.

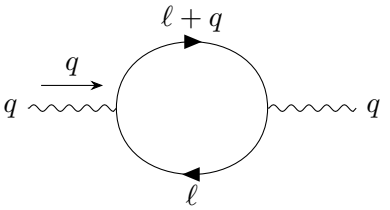


Figure D.2: The QED vacuum polarization (D.13): a photon of momentum q fluctuates into a virtual electron–positron pair carrying ℓ and $\ell + q$. The closed fermion loop supplies the trace and the overall minus sign; transversality $\Pi^{\mu\nu} \propto q^2\eta^{\mu\nu} - q^\mu q^\nu$ enforced by the Ward identity of Chapter 19 removes the would-be quadratic divergence.

Yang–Mills theory: colour-dressed scalar integrals. The one-loop diagrams of Yang–Mills theory—the gluon self-energy with its gluon loop, ghost loop, and (with matter) fermion loop, the ghost and quark self-energies, the triple-gluon and quark–gluon vertices—reduce, after the Lorentz and Dirac algebra is done, to exactly the same family of scalar integrals (D.3) as the abelian case, multiplied by a colour factor that factors out of the momentum integral entirely. A gluon loop contributes the adjoint Casimir through $f^{acd}f^{bcd} = C_A\,\delta^{ab}$, a quark loop the index through $\mathrm{tr}(T^aT^b) = T_F\,\delta^{ab}$, and a quark self-energy the fundamental Casimir through $(T^aT^a)_{ij} = C_F\,\delta_{ij}$; all of these constants, and the identities that produce them, are owned by Chapter B and quoted there. Schematically, every YM one-loop integral has the structure

$$(\text{colour factor: } C_A,\, T_F\,n_f,\, \text{or } C_F)\;\times\;\mu^{2\varepsilon}\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\mathcal{N}(\ell;p)}{\prod_j(q_j^2+i\varepsilon)}, \tag{D.16}$$

with the kinematic integral identical in form to (D.13)–(D.15). This is why the renormalization technology transfers from QED to QCD essentially unchanged, and why the appendix can supply a single set of master integrals that serves all three benchmark theories: the colour algebra is separated off by Chapter B before any momentum integration, and the surviving integrals are the ones this appendix evaluates once and for all. The combination of these integrals with their colour factors assembles, in Chapter 27, into the Yang–Mills beta-function coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$.

Worked Example D.7 (Reading off the degree of divergence of the QED one-loop functions)**.** We confirm the divergence diagnoses asserted for (D.13)–(D.15) directly from the one-loop count (D.5), $D = d + 2a - 2N$ with $2a = P$ the numerator degree, evaluated at $d = 4 - 2\varepsilon$.

Vacuum polarization (D.13). The loop has $N = 2$ fermion propagators; their numerators contribute $\not{\ell} + m$ and $\not{\ell} + \not{q} + m$, so the loop momentum appears to degree $P = 2$. Hence $D = d + 2 - 4 = d - 2 = 2 - 2\varepsilon$. The superficial divergence

is quadratic ($D = 2$ at $\varepsilon = 0$). The actual divergence is softer: writing the most general symmetric tensor built from q, the result is $A(q^2)\,\eta^{\mu\nu} + B(q^2)\,q^\mu q^\nu$, and the Ward identity $q_\mu \Pi^{\mu\nu} = 0$ forces $A = -q^2 B$, i.e. the transverse form $(q^2\eta^{\mu\nu} - q^\mu q^\nu)\Pi(q^2)$. The two powers of q pulled out front reduce the divergence of the scalar coefficient $\Pi(q^2)$ by two units, to $D_{\text{eff}} = 0$: logarithmic, a simple $1/\varepsilon$ pole. Dimensional regularization makes this automatic—it never generates the gauge-noninvariant $\propto \Lambda^2\eta^{\mu\nu}$ piece that a cutoff would, because, as Section D.1.6 showed, it assigns no power divergence at all.

Self-energy (D.14). Here $N = 2$ (one fermion, one photon propagator); the numerator carries $\not{p} - \not{\ell} + m$ from the fermion line and a constant from the photon, so $P = 1$ and $D = d + 1 - 4 = d - 3 = 1 - 2\varepsilon$, superficially linear. Lorentz covariance restricts the answer to $\Sigma(p) = \not{p}\,\Sigma_V(p^2) + m\,\Sigma_S(p^2)$; the coefficient functions are dimensionless, so each carries the residual logarithmic divergence, a single $1/\varepsilon$ pole. (The linear term that bare power counting allows is the one that would multiply a unit matrix with one power of external momentum; it is absent because there is no available vector to carry the loose index, leaving only the $\not{p}$ and m structures.)

Vertex (D.15). Now $N = 3$ (two fermion, one photon propagator) and the two fermion numerators give $P = 2$, so $D = d + 2 - 6 = d - 4 = -2\varepsilon$: superficially logarithmic, the lone $1/\varepsilon$ pole renormalizing the electric charge. Every count agrees with the assertions made above, and each is nothing but the scaling argument of Theorem D.3 applied to a specific numerator.

all three benchmark theories have thus been reduced to integrals of the single form (D.3): a numerator polynomial over a product of quadratic denominators, to be understood as a meromorphic function of the continued dimension d. What remains is to evaluate them, and the first obstacle is the product of denominators. The next section removes that obstacle, deriving in full the Feynman and Schwinger parametrizations that merge any product $\prod_j (q_j^2 - m_j^2 + i\varepsilon)$ into a single denominator raised to a power and, after completing the square in the loop momentum, expose the canonical form $(\ell^2 - \Delta)^{-n}$ with Δ a function of the external invariants, the masses, and the parameters—the form on which the Wick rotation and the Γ-function master integral act.

D.2 Combining denominators: Feynman and Schwinger parametrization

The previous section reduced every diagram of all three benchmark theories to a single template, the one-loop master integral (D.3): a numerator polynomial in the loop momentum over a *product* of quadratic denominators $\prod_j (q_j^2 - m_j^2 + i\varepsilon)$, to be read as a meromorphic function of the continued dimension d. That product is the one structural obstacle between us and a textbook radial integral. The

momenta q_j in the various factors differ by external shifts, so the denominator is not a function of a single quadratic form and no rotation or rescaling of ℓ can act on all factors at once. The remedy, due in its two complementary forms to Feynman and to Schwinger, is to trade the product of N denominators for a *single* denominator raised to a power, at the cost of introducing $N-1$ auxiliary integration variables. Once the denominator is a single quadratic form, the loop momentum enters only through that one form; a translation $\ell \to \ell + (\text{shift})$ completes the square and removes the linear term, leaving the canonical $(\ell^2 - \Delta + i\varepsilon)^{-n}$ with all dependence on external invariants, masses, and the new parameters sequestered into the single scalar Δ. This section derives both devices in full, proves the multi-denominator master formula by induction, establishes that the completing-the-square shift is a legitimate change of variables under the analytic continuation of Theorem D.6, and computes Δ explicitly for the ϕ^4 bubble and the QED self-energy, vacuum polarization, and vertex. Throughout, the $i\varepsilon$ is the Feynman prescription of Theorem A.23, and the Γ- and B-function facts we lean on (their integral representations, the recursion $\Gamma(z+1) = z\Gamma(z)$, analyticity away from the nonpositive integers) are the complex-analytic results reviewed in Chapter 5.

D.2.1 The two-denominator Feynman identity

We begin with the simplest and most frequently used case, two denominators each to the first power, and prove the formula that all of perturbation theory at one loop ultimately rests on.

Lemma D.8 (Feynman's two-denominator identity)**.** *Let A, B be complex numbers such that the straight-line segment joining them, $\{xA + (1-x)B : x \in [0,1]\}$, does not pass through the origin. Then*

$$\frac{1}{AB} \;=\; \int_0^1 \frac{\mathrm{d}x}{\big[xA + (1-x)B\big]^2}. \tag{D.17}$$

In the application to (D.3), A and B are the two Feynman denominators $q_1^2 - m_1^2 + i\varepsilon$ and $q_2^2 - m_2^2 + i\varepsilon$; because both carry the same *sign of the infinitesimal imaginary part, every point of the segment has imaginary part $+\varepsilon > 0$ and the hypothesis is automatically met, so no pole is crossed and (D.17) holds verbatim with the Feynman $i\varepsilon$ in place.*

Proof. Write the denominator of the integrand as $f(x) = xA + (1-x)B = B + x(A-B)$, an affine function of x. If $A \neq B$, then $f'(x) = A - B \neq 0$ and the integrand has the elementary antiderivative

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[-\frac{1}{A-B}\,\frac{1}{B + x(A-B)}\right] \;=\; \frac{1}{\big[B + x(A-B)\big]^2}, \tag{D.18}$$

valid as long as $f(x) \neq 0$ on $[0,1]$, which is exactly the hypothesis that the segment misses the origin. Evaluating between the limits,

$$\int_0^1 \frac{\mathrm{d}x}{f(x)^2} \;=\; -\frac{1}{A-B}\left[\frac{1}{A}-\frac{1}{B}\right] \;=\; -\frac{1}{A-B}\cdot\frac{B-A}{AB} \;=\; \frac{1}{AB}. \tag{D.19}$$

The case $A = B$ is recovered by continuity: the right side of (D.17) is $\int_0^1 \mathrm{d}x\, A^{-2} = A^{-2} = 1/(AB)$, and both sides are continuous in A, B wherever the segment avoids the origin. Finally, with $A = q_1^2 - m_1^2 + i\varepsilon$ and $B = q_2^2 - m_2^2 + i\varepsilon$, the segment is $\{x q_1^2 + (1-x) q_2^2 - x m_1^2 - (1-x) m_2^2 + i\varepsilon\}$, whose imaginary part is $\varepsilon > 0$ for every $x \in [0,1]$; it therefore never reaches the origin, the hypothesis holds, and the combined denominator inherits a single $+i\varepsilon$. □

The content of (D.17) is geometric: the reciprocal of a product is the average, over the unit interval, of the reciprocal of a *weighted* single denominator squared. The weight x is the first Feynman parameter. Two features generalize. First, the power of the combined denominator is the *sum* of the original powers (here $1 + 1 = 2$); for higher powers a Γ-function prefactor and a parameter weight $x^{\alpha-1}(1-x)^{\beta-1}$ appear. Second, for n denominators the single integral becomes an integral over the $(n-1)$-dimensional simplex of parameters $x_1, \dots, x_n$ constrained to sum to one. We now establish both generalizations, and the cleanest route to them passes through Schwinger's representation.

D.2.2 The Schwinger (proper-time) representation

Feynman's identity merges denominators by a clever interpolation; Schwinger's achieves the same by writing each denominator as an exponential integral, after which a product of denominators becomes a single Gaussian-type exponential of a *sum*. The two are not rivals: Schwinger's representation is the efficient engine from which the general Feynman formula follows in a few lines, and it is the form in which the loop integral is most transparent after Wick rotation. Its name ("proper time") comes from the worldline picture, in which the parameter s conjugate to $p^2 - m^2$ is the proper time elapsed along a particle's trajectory; we use it here purely as an integral identity.

Proposition D.9 (Schwinger representation)**.** *For any complex A with $\operatorname{Re} A > 0$ and any n with $\operatorname{Re} n > 0$,*

$$\frac{1}{A^n} \;=\; \frac{1}{\Gamma(n)} \int_0^\infty \mathrm{d}s\; s^{\,n-1}\, e^{-sA}. \tag{D.20}$$

For a Lorentzian Feynman denominator the relevant quantity has the form $A = -i(X + i\varepsilon)$ with $X = q^2 - m^2$ real; the convergent version reads

$$\frac{1}{(X+i\varepsilon)^n} \;=\; \frac{(-i)^n}{\Gamma(n)} \int_0^\infty \mathrm{d}s\; s^{\,n-1}\, e^{\,is(X+i\varepsilon)}, \tag{D.21}$$

in which the $i\varepsilon$ supplies the convergence factor $e^{-s\varepsilon}$ that damps the upper limit and, simultaneously, fixes the sense in which the s-contour may later be rotated.

Proof. Start from the Euler integral defining the Γ-function (reviewed in Chapter 5), $\Gamma(n) = \int_0^\infty \mathrm{d}u\; u^{n-1}e^{-u}$, valid for $\operatorname{Re} n > 0$. Substitute $u = sA$ with A a fixed complex number of positive real part; this is a rotation-plus-scaling of the integration ray by the fixed phase $\arg A \in (-\frac{\pi}{2}, \frac{\pi}{2})$, and since $\operatorname{Re} A > 0$ the integrand $u^{n-1}e^{-u}$ decays on the rotated ray and the contour may be rotated back to the positive real axis without crossing a singularity (the integrand is entire in u). Hence $\mathrm{d}u = A\,\mathrm{d}s$, $u^{n-1} = A^{n-1}s^{n-1}$, and

$$\Gamma(n) \;=\; \int_0^\infty A\,\mathrm{d}s\;(As)^{n-1}e^{-sA} \;=\; A^n\int_0^\infty \mathrm{d}s\; s^{n-1}e^{-sA}, \tag{D.22}$$

which rearranges to (D.20). For the Lorentzian form, the Feynman denominator $X + i\varepsilon$ has vanishing real part on shell, so (D.20) does not apply directly; instead write $X + i\varepsilon = i\big[-i(X + i\varepsilon)\big]$ and set $A = -i(X + i\varepsilon) = \varepsilon - iX$, which has $\operatorname{Re} A = \varepsilon > 0$. Then $A^{-n} = i^{-n}(X + i\varepsilon)^{-n} = (-i)^n(X + i\varepsilon)^{-n}$ and (D.20) gives

$$(X+i\varepsilon)^{-n} \;=\; i^{-n}A^{-n} \;=\; \frac{i^{-n}}{\Gamma(n)}\int_0^\infty \mathrm{d}s\; s^{n-1}e^{-s(\varepsilon-iX)} \;=\; \frac{(-i)^n}{\Gamma(n)}\int_0^\infty \mathrm{d}s\; s^{n-1}e^{\,is(X+i\varepsilon)}, \tag{D.23}$$

using $i^{-n} = (-i)^n$ and $e^{-s\varepsilon}e^{isX} = e^{is(X+i\varepsilon)}$. The factor $e^{-s\varepsilon}$ guarantees absolute convergence at $s \to \infty$; this is (D.21). □

Remark D.10 (The $i\varepsilon$ fixes the rotation sense). The sign of the exponent in (D.21) is not a matter of taste. Because the convergence factor is $e^{-s\varepsilon}$ with $\varepsilon > 0$, the s-integral converges on the positive real axis and on any ray rotated into the lower-half s-plane down to $\arg s = -\pi/2$, where $is \to s_E$ becomes real and the exponential becomes the manifestly convergent $e^{-s_E|X|}$ for spacelike $X < 0$. This is the Schwinger-parameter shadow of the Wick rotation carried out in Section D.3: the same $i\varepsilon$ that pushes the Feynman poles off the real energy axis dictates that one rotates the proper-time contour clockwise, equivalently the energy contour counterclockwise. Carrying the wrong sign of ε would demand the opposite rotation and produce the complex conjugate (anti-Feynman) propagator.

The Gaussian virtue of (D.20) is that a product of denominators exponentiates into a single exponential of a sum:

$$\frac{1}{A_1^{\alpha_1}\cdots A_N^{\alpha_N}} \;=\; \frac{1}{\prod_j \Gamma(\alpha_j)}\int_0^\infty\Big(\prod_{j=1}^N \mathrm{d}s_j\; s_j^{\alpha_j-1}\Big)\;\exp\Big(-\sum_{j=1}^N s_jA_j\Big). \tag{D.24}$$

This single formula is the source of everything that follows. Inserted directly into a loop integral it produces, after Wick rotation, a Gaussian momentum integral with the proper times as parameters — the route taken in the worldline and heat-kernel methods. Reorganized as below, it yields the Feynman simplex formula.

D.2.3 Feynman's formula for N denominators with arbitrary powers

We now state and prove the master parametric identity, the workhorse of the entire appendix.

Theorem D.11 (Feynman parametrization, general powers)**.** *Let $A_1, \dots, A_N$ be complex numbers whose convex hull excludes the origin, and let $\alpha_1, \dots, \alpha_N$ have positive real parts, with $S \equiv \sum_{j=1}^{N} \alpha_j$. Then*

$$\frac{1}{\prod_{j=1}^{N} A_j^{\alpha_j}} \;=\; \frac{\Gamma(S)}{\prod_{j=1}^{N} \Gamma(\alpha_j)} \int_0^1 \Big(\prod_{j=1}^{N} \mathrm{d}x_j \, x_j^{\alpha_j - 1} \Big) \, \frac{\delta\big(1 - \sum_{j=1}^{N} x_j\big)}{\big[\sum_{j=1}^{N} x_j A_j\big]^{S}}. \tag{D.25}$$

The integration runs over the standard $(N-1)$-simplex $\Delta_{N-1} = \{x_j \geq 0,\ \sum_j x_j = 1\}$ enforced by the δ-function. For $N = 2$, $\alpha_1 = \alpha_2 = 1$, (D.25) reduces to (D.17); setting all $\alpha_j = 1$ gives the unit-power form with prefactor $\Gamma(N) = (N-1)!$.

We give two complete proofs. The first derives (D.25) in a single stroke from the Schwinger representation (D.24); the second builds it up by induction on N from the two-denominator case, exhibiting the Γ-function recursion that organizes the simplex.

Proof via the Schwinger representation. Begin from (D.24). Insert into the right-hand side the identity $1 = \int_0^\infty \mathrm{d}\lambda \, \delta(\lambda - \sum_j s_j)$, which simply names the total proper time $\lambda = \sum_j s_j$. Then rescale each $s_j = \lambda x_j$ with $x_j \geq 0$; the delta becomes $\delta\big(\lambda(1 - \sum_j x_j)\big) = \lambda^{-1}\delta(1 - \sum_j x_j)$, and $\mathrm{d}s_j = \lambda \, \mathrm{d}x_j$ at fixed λ contributes one factor of λ for each of the N variables. Collecting powers of λ,

$$\begin{aligned}
\frac{1}{\prod_j A_j^{\alpha_j}} &= \frac{1}{\prod_j \Gamma(\alpha_j)} \int_0^\infty \mathrm{d}\lambda \int_0^\infty \Big(\prod_j \mathrm{d}s_j \, s_j^{\alpha_j - 1} \Big) \delta\Big(\lambda - \sum_j s_j\Big) e^{-\sum_j s_j A_j} \\
&= \frac{1}{\prod_j \Gamma(\alpha_j)} \int_0^\infty \mathrm{d}\lambda \int_0^1 \Big(\prod_j \mathrm{d}x_j \, (\lambda x_j)^{\alpha_j - 1} \Big) \, \lambda^{N} \, \lambda^{-1} \delta\Big(1 - \sum_j x_j\Big) \, e^{-\lambda \sum_j x_j A_j} \\
&= \frac{1}{\prod_j \Gamma(\alpha_j)} \int_0^1 \Big(\prod_j \mathrm{d}x_j \, x_j^{\alpha_j - 1} \Big) \delta\Big(1 - \sum_j x_j\Big) \int_0^\infty \mathrm{d}\lambda \, \lambda^{S-1} e^{-\lambda U},
\end{aligned} \tag{D.26}$$

where $U \equiv \sum_j x_j A_j$ and the total power of λ assembled is $\sum_j (\alpha_j - 1) + N - 1 = S - 1$. The remaining λ-integral is a single Schwinger integral, $\int_0^\infty \mathrm{d}\lambda \, \lambda^{S-1} e^{-\lambda U} = \Gamma(S)/U^S$ by (D.20), valid because $\mathrm{Re}\, U > 0$ on the simplex when the convex hull of the A_j excludes the origin. Substituting gives (D.25). The interchange of the λ- and x-integrations is justified by absolute convergence: with all $\mathrm{Re}\, A_j$ of one sign the double integral converges absolutely, and Fubini's theorem applies. In the Lorentzian application the same conclusion holds with $A_j \to -i(X_j + i\varepsilon)$, for which $\mathrm{Re}\, A_j = \varepsilon > 0$. $\square$

Proof by induction on N. It is convenient to work with the simplex in solved form, eliminating $x_N = 1 - \sum_{j<N} x_j$ and integrating the free parameters over $\Delta_{N-1} = \{x_1, \dots, x_{N-1} \geq 0,\ \sum_{j<N} x_j \leq 1\}$. Statement (D.25) is then

$$\frac{1}{\prod_{j=1}^{N} A_j^{\alpha_j}} = \frac{\Gamma(S)}{\prod_{j=1}^{N}\Gamma(\alpha_j)} \int_{\Delta_{N-1}} \mathrm{d}^{N-1}x \, \Big(\prod_{j=1}^{N} x_j^{\alpha_j - 1}\Big)\, U^{-S}, \qquad U = \sum_{j=1}^{N} x_j A_j. \tag{D.27}$$

The base case $N = 2$ is the general two-denominator identity, which we obtain at once from the $N = 2$ instance of the Schwinger derivation above (or, for $\alpha_1 = \alpha_2 = 1$, from Theorem D.8); explicitly, with $x_1 = x$, $x_2 = 1 - x$,

$$\frac{1}{A_1^{\alpha_1} A_2^{\alpha_2}} = \frac{\Gamma(\alpha_1 + \alpha_2)}{\Gamma(\alpha_1)\Gamma(\alpha_2)} \int_0^1 \mathrm{d}x \, \frac{x^{\alpha_1 - 1}(1-x)^{\alpha_2 - 1}}{\big[xA_1 + (1-x)A_2\big]^{\alpha_1 + \alpha_2}}. \tag{D.28}$$

Assume (D.27) holds for N denominators. For $N + 1$ denominators, apply the inductive hypothesis to the first N factors,

$$\frac{1}{\prod_{j=1}^{N} A_j^{\alpha_j}} = \frac{\Gamma(S)}{\prod_{j=1}^{N}\Gamma(\alpha_j)} \int_{\Delta_{N-1}} \mathrm{d}^{N-1}x \, \Big(\prod_{j=1}^{N} x_j^{\alpha_j - 1}\Big) U^{-S}, \qquad S = \sum_{j=1}^{N} \alpha_j, \tag{D.29}$$

multiply by $A_{N+1}^{-\alpha_{N+1}}$, and merge the single denominator U^{-S} with $A_{N+1}^{-\alpha_{N+1}}$ using the two-denominator formula (D.28):

$$U^{-S} A_{N+1}^{-\alpha_{N+1}} = \frac{\Gamma(S')}{\Gamma(S)\Gamma(\alpha_{N+1})} \int_0^1 \mathrm{d}t \, t^{\,S-1}(1-t)^{\alpha_{N+1} - 1} \big[tU + (1-t)A_{N+1}\big]^{-S'}, \tag{D.30}$$

where $S' = S + \alpha_{N+1} = \sum_{j=1}^{N+1} \alpha_j$. The two $\Gamma(S)$ factors cancel and $\prod_{j=1}^{N}\Gamma(\alpha_j) \cdot \Gamma(\alpha_{N+1}) = \prod_{j=1}^{N+1}\Gamma(\alpha_j)$, so

$$\frac{1}{\prod_{j=1}^{N+1} A_j^{\alpha_j}} = \frac{\Gamma(S')}{\prod_{j=1}^{N+1}\Gamma(\alpha_j)} \int_{\Delta_{N-1}} \mathrm{d}^{N-1}x \int_0^1 \mathrm{d}t \, \Big(\prod_{j=1}^{N} x_j^{\alpha_j - 1}\Big) t^{\,S-1}(1-t)^{\alpha_{N+1} - 1} \big[tU + (1-t)A_{N+1}\big]^{-S'}. \tag{D.31}$$

It remains to recognize the right-hand side as the $(N+1)$-denominator simplex integral. Introduce new parameters

$$u_j = t\, x_j \quad (j = 1, \dots, N), \qquad u_{N+1} = 1 - t, \tag{D.32}$$

a bijection from $\Delta_{N-1} \times (0,1)$ onto the N-simplex $\Delta_N = \{u_j \geq 0,\ \sum_{j=1}^{N+1} u_j = 1\}$. Three things must be checked: the combined denominator, the parameter weights, and the measure. *Denominator.* On the simplex $\sum_{j=1}^{N} x_j = 1$, so

$$tU + (1-t)A_{N+1} = \sum_{j=1}^{N} (t x_j) A_j + (1-t)A_{N+1} = \sum_{j=1}^{N+1} u_j A_j \equiv U', \tag{D.33}$$

and $\sum_{j=1}^{N+1} u_j = t\sum_{j=1}^{N} x_j + (1-t) = t + (1-t) = 1$, confirming $u \in \Delta_N$. *Measure.* Take $(x_1, \dots, x_{N-1}, t)$ as the N independent old variables (with $x_N = 1 - \sum_{j<N} x_j$) and $(u_1, \dots, u_N)$ as the N independent new variables (with $u_{N+1} = 1 - \sum_{j\le N} u_j$). From (D.32), $\partial u_j/\partial x_k = t\,\delta_{jk}$ and $\partial u_j/\partial t = x_j$ for $j,k \le N-1$, while $u_N = t\,x_N = t(1 - \sum_{k<N} x_k)$ gives $\partial u_N/\partial x_k = -t$ and $\partial u_N/\partial t = x_N$. Adding rows $1, \dots, N-1$ to row N clears its first $N-1$ entries and leaves $\sum_{j=1}^{N} x_j = 1$ in the last column, so the Jacobian determinant is the product of the diagonal t's,

$$\left|\frac{\partial(u_1,\dots,u_N)}{\partial(x_1,\dots,x_{N-1},t)}\right| = t^{\,N-1}, \qquad \Longrightarrow \qquad \mathrm{d}^{N-1}x\,\mathrm{d}t = t^{-(N-1)}\,\mathrm{d}^{N}u. \tag{D.34}$$

Weights. Using $x_j = u_j/t$ and $\sum_{j=1}^{N}(\alpha_j - 1) = S - N$,

$$\Big(\prod_{j=1}^{N} x_j^{\alpha_j-1}\Big) t^{\,S-1}(1-t)^{\alpha_{N+1}-1} = t^{-(S-N)}\Big(\prod_{j=1}^{N} u_j^{\alpha_j-1}\Big) t^{\,S-1} u_{N+1}^{\alpha_{N+1}-1} = t^{\,N-1}\prod_{j=1}^{N+1} u_j^{\alpha_j-1}. \tag{D.35}$$

The weight thus carries an overall t^{N-1} that cancels precisely against the $t^{-(N-1)}$ in the measure (D.34). Substituting all three into (D.31),

$$\frac{1}{\prod_{j=1}^{N+1} A_j^{\alpha_j}} = \frac{\Gamma(S')}{\prod_{j=1}^{N+1}\Gamma(\alpha_j)} \int_{\Delta_N} \mathrm{d}^{N}u\,\Big(\prod_{j=1}^{N+1} u_j^{\alpha_j-1}\Big) U'^{-S'}, \tag{D.36}$$

which is (D.27) with $N+1$ in place of N. By induction the formula holds for all $N \ge 2$. □

The two derivations illuminate the same object from opposite sides. The Schwinger route shows that the Feynman parameters are normalized proper times $x_j = s_j/\lambda$ and that the simplex constraint $\sum_j x_j = 1$ is nothing but the bookkeeping of total proper time; the inductive route shows that the Γ-function prefactor is assembled one denominator at a time, each merge contributing the recursion $\Gamma(S)\Gamma(\alpha_{N+1}) \to \Gamma(S')$ that telescopes into the single $\Gamma(S)$ on the front of (D.25).

D.2.4 The simplex and its volume

The domain of integration in (D.25) deserves a moment, because its measure recurs in every loop computation and its symmetry is what allows a numerator linear in the loop momentum to be discarded. The standard $(N-1)$-simplex $\Delta_{N-1} = \{x_j \ge 0,\ \sum_j x_j = 1\}$ is the convex hull of the N unit vectors; Fig. D.3 shows the $N = 2$ (a segment) and $N = 3$ (a triangle) cases.

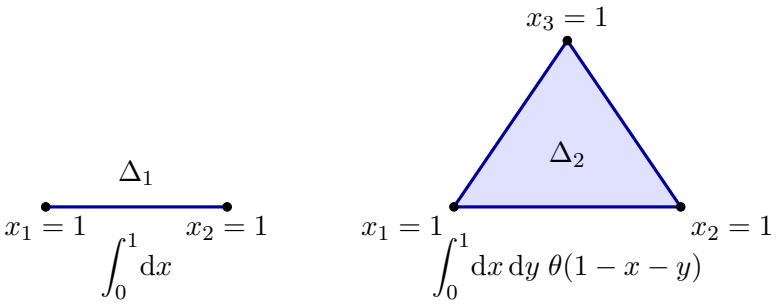


Figure D.3: The Feynman-parameter simplices for two and three denominators. For N denominators the parameters $(x_1, \dots, x_N)$, all nonnegative and summing to one, range over the $(N-1)$-simplex Δ_{N-1}, whose Euclidean volume in the solved coordinates is $1/(N-1)!$. The vertices are the configurations in which a single denominator dominates.

Proposition D.12 (Volume of the simplex)**.** *In the solved coordinates* $(x_1, \dots, x_{N-1})$ *with* $x_N = 1 - \sum_{j<N} x_j$,

$$\int_{\Delta_{N-1}} \mathrm{d}^{N-1}x \;=\; \int_0^1 \mathrm{d}x_1 \int_0^{1-x_1} \mathrm{d}x_2 \cdots \int_0^{1-x_1-\cdots-x_{N-2}} \mathrm{d}x_{N-1} \;=\; \frac{1}{(N-1)!}. \tag{D.37}$$

Proof. Set all $\alpha_j = 1$ and all $A_j = 1$ in the master formula (D.27). The left-hand side is 1, and $U = \sum_j x_j = 1$ on the simplex, so the right-hand side is $\Gamma(N) \int_{\Delta_{N-1}} \mathrm{d}^{N-1}x$. Hence $\int_{\Delta_{N-1}} \mathrm{d}^{N-1}x = 1/\Gamma(N) = 1/(N-1)!$. Equivalently, by direct iterated integration: let $V_k(a)$ denote the volume of the scaled simplex $\{x_j \geq 0,\ \sum_{j=1}^{k} x_j \leq a\}$. Then $V_1(a) = a$ and $V_k(a) = \int_0^a V_{k-1}(a - x_1)\,\mathrm{d}x_1 = \int_0^a (a-x_1)^{k-1}/(k-1)!\,\mathrm{d}x_1 = a^k/k!$, so by induction $V_k(a) = a^k/k!$. Taking $a = 1$, $k = N-1$ gives (D.37). □

A symmetry of the simplex measure will be used repeatedly. The domain Δ_{N-1} and the unit-power weight are invariant under permutations of the parameters, and after the loop momentum is shifted (next subsection) the integrand depends on ℓ only through the rotationally invariant ℓ^2. Consequently any factor of the shifted loop momentum that appears to an *odd* power in the numerator integrates to zero,

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, \frac{\ell^\mu}{(\ell^2 - \Delta + i\varepsilon)^n} = 0, \tag{D.38}$$

because the integrand is odd under $\ell \to -\ell$, a symmetry of the (dimensionally continued) measure and of the denominator; the integral, being absolutely convergent in the strip of Theorem D.6 and Lorentz-covariant, can only equal a vector built from the available data, and after the shift there is no external vector

left inside, so it must vanish. The same parity argument promotes a numerator $\ell^\mu \ell^\nu$ to its isotropic average $\ell^2 \eta^{\mu\nu}/d$, the rule that drives the tensor reduction of a later section. For the present section the relevant statement is the simplest one: (D.38) lets us drop the linear-in-ℓ remainder produced by completing the square.

D.2.5 Completing the square: the canonical form and Δ

We now apply the machinery to the one-loop master (D.3). After Feynman parametrization the N denominators have become a single factor U^{-N} (for unit powers) with $U = \sum_j x_j (q_j^2 - m_j^2 + i\varepsilon)$, and the loop momentum enters only through U. Route the momenta as $q_j = \ell + r_j$, where ℓ is the loop momentum and r_j is the partial sum of external momenta flowing through line j (so r_j is independent of ℓ). Then, using $\sum_j x_j = 1$,

$$\begin{aligned} U &= \sum_j x_j [(\ell + r_j)^2 - m_j^2] + i\varepsilon = \ell^2 + 2\ell \cdot \Big(\sum_j x_j r_j\Big) + \sum_j x_j r_j^2 - \sum_j x_j m_j^2 + i\varepsilon \\ &= \ell^2 + 2\ell \cdot R + \sum_j x_j r_j^2 - \sum_j x_j m_j^2 + i\varepsilon, \qquad R \equiv \sum_j x_j r_j. \end{aligned} \tag{D.39}$$

The first two terms are completed into a perfect square by the translation $\ell \to \ell - R$, i.e. defining the new loop variable $k = \ell + R$ so that $\ell^2 + 2\ell \cdot R = k^2 - R^2$. The combined denominator becomes

$$U = k^2 - \Delta + i\varepsilon, \qquad \boxed{\Delta \;=\; R^2 - \sum_j x_j r_j^2 + \sum_j x_j m_j^2 \;=\; \sum_j x_j m_j^2 - \tfrac{1}{2} \sum_{i,j} x_i x_j (r_i - r_j)^2} \tag{D.40}$$

where the second form follows from the algebraic identity $\sum_j x_j r_j^2 - \big(\sum_j x_j r_j\big)^2 = \frac{1}{2} \sum_{i,j} x_i x_j (r_i - r_j)^2$ valid when $\sum_j x_j = 1$ (expand the right side: $\frac{1}{2} \sum_{i,j} x_i x_j (r_i^2 + r_j^2 - 2 r_i \cdot r_j) = \sum_i x_i r_i^2 \sum_j x_j - \sum_{i,j} x_i x_j r_i \cdot r_j$, and the leftover factor $\sum_j x_j = 1$). The differences $r_i - r_j$ are sums of external momenta, so Δ depends only on the external invariants, the masses, and the Feynman parameters — exactly the segregation we sought. The $i\varepsilon$ rides along unchanged: Δ is real, and the prescription is equivalent to $\Delta \to \Delta - i\varepsilon$, so the denominator is $k^2 - \Delta + i\varepsilon$, which is what the Wick rotation of Section D.3 will act on.

Two points make this rigorous. First, the shift $\ell \to \ell - R$ must be a legitimate change of variables, which for a divergent integral is not automatic: shifting the variable of a linearly or quadratically divergent integral can generate a finite surface term. In dimensional regularization it is always legitimate, and the reason is the analytic continuation of Theorem D.6.

Proposition D.13 (Legitimacy of the loop-momentum shift)**.** *For the dimensionally regularized integral $I(d) = \int \mathrm{d}^d \ell \, f(\ell)$ defined by continuation from the strip $d_- < \operatorname{Re} d < d_+$ where it converges absolutely, and for any fixed R,*

$$\int \mathrm{d}^d \ell \; f(\ell + R) \;=\; \int \mathrm{d}^d \ell \; f(\ell). \tag{D.41}$$

Proof. For d in the convergence strip the integral is an ordinary absolutely convergent Lebesgue integral over $\mathbb{R}^d$ (after the Wick rotation that makes the measure Euclidean, carried out in Section D.3), and Lebesgue measure is translation invariant, so (D.41) holds there exactly, with no surface term — absolute convergence means the would-be boundary contributions at $|\ell| \to \infty$ vanish. Both sides of (D.41) are holomorphic functions of d on the strip (the integrand and its R-shift are holomorphic in d and dominated uniformly on compact d-subsets), and they agree there. By the identity theorem for analytic functions, their analytic continuations agree everywhere both are defined, in particular in a punctured neighbourhood of $d = 4$. Hence the shift is exact for the dimensionally regularized integral. This is the precise sense in which dimensional regularization "respects translations": it never produces the cutoff-dependent surface term $\propto \Lambda^{d-1}$ that a hard cutoff would, because such a term has no analytic continuation from the convergence strip — it is simply absent, consistent with the vanishing of scaleless integrals established in Section D.1.6. □

Second, with the shift legal, the numerator must be re-expressed in the new variable $k = \ell + R$, i.e. $\ell = k - R$. A numerator that was a polynomial in ℓ becomes a polynomial in k with R- and external-momentum-dependent coefficients; by the parity rule (D.38) every term odd in k integrates to zero, so only the even part survives. For the scalar bubble the numerator is unity and nothing further is needed; the tensor cases are handed to the reduction section, which uses exactly (D.38) and its $\ell^\mu \ell^\nu \to \ell^2 \eta^{\mu\nu}/d$ generalization.

The outcome of this subsection is the promised canonical form: every one-loop scalar integral, after Feynman parametrization and the shift, reads

$$\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \frac{1}{\prod_j (q_j^2 - m_j^2 + i\varepsilon)} = \Gamma(N)\, \mu^{2\varepsilon} \int_{\Delta_{N-1}} \mathrm{d}^{N-1} x \int \frac{\mathrm{d}^d k}{(2\pi)^d} \frac{1}{(k^2 - \Delta + i\varepsilon)^N}, \tag{D.42}$$

with Δ given by (D.40). The inner momentum integral is now identical for every diagram — a single power of one quadratic form — and depends on the kinematics only through the number Δ. We now compute Δ for the benchmark theories.

ϕ^4 theory: the bubble. The scalar bubble B_0 of (D.12) has $N = 2$ denominators with $q_1 = \ell$ ($r_1 = 0$), $q_2 = \ell + p$ ($r_2 = p$), and equal masses $m_1 = m_2 = m$. With Feynman parameter x on the second line and $1 - x$ on the first, $R = xp$, and (D.40) gives $\sum_j x_j m_j^2 = m^2$ and $\frac{1}{2} \sum_{i,j} x_i x_j (r_i - r_j)^2 = x(1-x)\, p^2$, so

$$\boxed{\Delta_{\phi^4} = m^2 - x(1-x)\, p^2 = m^2 - x(1-x)\, s} \qquad (s = p^2). \tag{D.43}$$

This is the Δ that Chapter 24 uses to extract the ϕ^4 beta function. Its physics is already visible in the formula: for spacelike $s < 0$, $\Delta > 0$ and the rotated

denominator $k_E^2 + \Delta$ is strictly positive; for timelike $s > 4m^2$ (above the two-particle threshold) Δ turns negative on part of the x-range, the $i\varepsilon$ becomes essential, and the integral develops the imaginary part demanded by unitarity — the branch cut whose discontinuity is the subject of the optical-theorem analysis downstream.

QED: vacuum polarization. The vacuum polarization (D.13) has, after the Dirac trace of Section C.7.3 is taken in the numerator, the same two-denominator structure as the bubble: $q_1 = \ell$ ($r_1 = 0$), $q_2 = \ell + q$ ($r_2 = q$), masses $m_1 = m_2 = m$. Hence

$$\boxed{\Delta_\Pi = m^2 - x(1-x)\,q^2} \tag{D.44}$$

identical in form to (D.43) with $s \to q^2$, as it must be since both are scalar bubbles of equal-mass lines; the difference between the two threads lives entirely in the numerators (a trivial 1 for ϕ^4, a Dirac trace for QED), not in Δ.

QED: the electron self-energy. The self-energy (D.14) combines a massless photon line, $A_\gamma = \ell^2 + i\varepsilon$ ($r_\gamma = 0$, $m_\gamma = 0$), with a massive fermion line, $A_e = (p-\ell)^2 - m^2 + i\varepsilon$. Writing the fermion momentum as $(\ell + r_e)^2$ with $r_e = -p$, and assigning Feynman parameter x to the photon and $1-x$ to the fermion, $R = (1-x)(-p) = -(1-x)p$. From (D.40), $\sum_j x_j m_j^2 = x\cdot 0 + (1-x)m^2 = (1-x)m^2$ and $\frac{1}{2}\sum_{i,j} x_i x_j (r_i - r_j)^2 = x(1-x)\,(0-(-p))^2 = x(1-x)\,p^2$, so

$$\boxed{\Delta_\Sigma = (1-x)\,m^2 - x(1-x)\,p^2} \tag{D.45}$$

This is the Δ consumed by the self-energy computation of Chapter 19. The masslessness of the photon shows itself in (D.45): at $x = 1$ (the photon carrying the full proper time) $\Delta_\Sigma = 0$, the vanishing that signals the infrared sensitivity flagged in Section D.1.3; a fictitious photon mass m_γ^2 would add $x\,m_\gamma^2$ and lift this zero, which is one standard way to regulate the infrared while the ultraviolet pole is handled by ε.

QED: the vertex. The vertex correction (D.15) is the first genuinely three-denominator case and exercises the $N = 3$ formula. The lines are the photon $A_\gamma = \ell^2 + i\varepsilon$ ($r_\gamma = 0$, $m_\gamma = 0$) and two fermion lines $(p'-\ell)^2 - m^2$ and $(p-\ell)^2 - m^2$, i.e. $r_1 = -p'$, $r_2 = -p$, both of mass m. Assign parameters x, y, z to the $(p'-\ell)$, $(p-\ell)$, and photon lines respectively, with $x + y + z = 1$. Then $\sum_j x_j m_j^2 = (x+y)m^2 = (1-z)m^2$, and the invariant piece is

$$\tfrac{1}{2}\sum_{i,j} x_i x_j (r_i - r_j)^2 = xy\,(p-p')^2 + xz\,p'^2 + yz\,p^2 = xy\,q^2 + z(1-z)\,m^2, \tag{D.46}$$

using $q = p' - p$, $q^2 = (p - p')^2$, and the on-shell conditions $p^2 = p'^2 = m^2$ for the external electron (so $xz\,p'^2 + yz\,p^2 = z(x+y)m^2 = z(1-z)m^2$). Therefore

$$\boxed{\Delta_\Gamma = (1-z)\,m^2 - \left[xy\,q^2 + z(1-z)m^2\right] = (1-z)^2\,m^2 - xy\,q^2} \tag{D.47}$$

a function on the triangle $\Delta_2 = \{x, y, z \geq 0,\ x+y+z = 1\}$, with the loop integral now of the form $\Gamma(3)\int \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\,\delta(1-x-y-z)\int \mathrm{d}^d k\,(k^2 - \Delta_\Gamma + i\varepsilon)^{-3}$. This is exactly the integrand the anomalous-magnetic-moment and charge-renormalization computations of Chapters 19 and 20 take as their starting point; the $g-2$ value $a_e = \alpha/2\pi$ falls out of integrating the numerator structure over this same triangle.

Yang–Mills theory. As recorded in (D.16), every Yang–Mills one-loop integrand is, after the colour algebra of Chapter B is factored off, one of the abelian scalar integrals above multiplied by a colour constant (C_A, $T_F n_f$, or C_F). The denominator combinations are identical to the QED ones — a gluon or ghost loop on the external gluon line produces the massless bubble with $\Delta = -x(1-x)q^2$, a quark loop the massive bubble (D.44), and the three-point functions the triangle of (D.47) — so no new Δ need be computed; the same parametrization serves all three benchmark theories, which is precisely why Chapter 27 can assemble the beta-function coefficient $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ from the master integral this appendix evaluates once.

We close with the worked computation the section was built around, carrying the ϕ^4 s-channel bubble explicitly from the parametric identity to the canonical form, so that the next section can act on a single, fully reduced integral.

Worked Example D.14 (The ϕ^4 s-channel bubble, parametrized)**.** We continue ϕ^4 theory from (D.11), reducing the scalar bubble $B_0(p^2, m^2)$ of (D.12) to canonical form. Starting point:

$$B_0(p^2, m^2) = \mu^{2\varepsilon}\int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - m^2 + i\varepsilon)\big((\ell+p)^2 - m^2 + i\varepsilon\big)}. \tag{D.48}$$

Step 1 — combine denominators. Apply Theorem D.8 with $A = (\ell+p)^2 - m^2 + i\varepsilon$ and $B = \ell^2 - m^2 + i\varepsilon$ (Feynman parameter x on A). Both denominators carry $+i\varepsilon$, so the hypothesis of Theorem D.8 holds and

$$\frac{1}{AB} = \int_0^1 \mathrm{d}x\,\frac{1}{\big[xA + (1-x)B\big]^2}, \qquad B_0 = \mu^{2\varepsilon}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\,\frac{1}{[xA + (1-x)B]^2}. \tag{D.49}$$

Step 2 — expand the combined denominator. With the parameters summing to one,

$$\begin{aligned} xA + (1-x)B &= x\big[(\ell+p)^2 - m^2\big] + (1-x)\big[\ell^2 - m^2\big] + i\varepsilon \\ &= \ell^2 + 2x\,\ell\cdot p + x\,p^2 - m^2 + i\varepsilon. \end{aligned} \tag{D.50}$$

Step 3 — complete the square. Set $k = \ell + xp$ (so $\ell = k - xp$ and $R = xp$ in the notation of (D.39)). Then $\ell^2 + 2x\,\ell\cdot p = k^2 - x^2p^2$, and

$$xA+(1-x)B = k^2-x^2p^2+xp^2-m^2+i\varepsilon = k^2-\Delta+i\varepsilon, \qquad \Delta = m^2-x(1-x)p^2, \tag{D.51}$$

recovering (D.43) with $p^2 = s$. *Step 4 — shift the measure.* By Theorem D.13 the translation $\ell \to k$ leaves the dimensionally regularized measure invariant, $\int \mathrm{d}^d\ell = \int \mathrm{d}^d k$, with no surface term. The integrand has no numerator (the bubble is scalar), so nothing odd in k arises and (D.38) is not even needed. Collecting,

$$\boxed{B_0(p^2,m^2) = \mu^{2\varepsilon}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{(k^2-\Delta+i\varepsilon)^2}, \qquad \Delta = m^2 - x(1-x)\,s.} \tag{D.52}$$

The product of denominators is gone; the loop momentum appears through the single quadratic form $k^2 - \Delta + i\varepsilon$, and the entire s- and m^2-dependence has been pushed into Δ and the trivial x-integral. The remaining momentum integral, $\int \mathrm{d}^d k\,(k^2 - \Delta + i\varepsilon)^{-2}$, is one number for each Δ; evaluating it is the task handed to the next section.

With (D.42), and explicitly (D.52), the product of propagator denominators has been collapsed to a single factor $(\ell^2 - \Delta + i\varepsilon)^{-N}$ and the Schwinger twin $\propto \int_0^\infty \mathrm{d}s\, s^{N-1} e^{is(\ell^2-\Delta+i\varepsilon)}$, with Δ a known function of external invariants, masses, and Feynman parameters for each thread. The remaining loop integral is now a single, manifestly Lorentz-invariant momentum integral over one quadratic form — but a *Minkowski* one, with the indefinite metric still in the measure and the denominator still able to change sign. The next section removes that last obstruction: it rotates the contour to Euclidean signature, turning $k^2 - \Delta + i\varepsilon$ into the strictly positive $k_E^2 + \Delta$, and constructs the d-dimensional measure, reducing (D.42) to a one-dimensional radial integral ready for the Γ-function master formula.

D.3 Wick rotation and the measure in d dimensions

The previous section did the algebraic work of collapsing a product of propagator denominators into a single factor, completing the square in the loop momentum to expose the canonical form $(\ell^2 - \Delta + i\epsilon)^{-n}$, with ℓ the shifted (integration) momentum and Δ a real function of the external invariants, masses, and Feynman parameters (Section D.2). What remains is a genuine integral over ℓ, and here we hit the obstruction that motivates everything in this appendix: the canonical integrand is Lorentzian. Its denominator changes sign as ℓ crosses the light cone, it vanishes on a hypersurface, and the $d^d\ell$ measure carries the indefinite Minkowski metric, so the integral is at best conditionally convergent and not in a form any real-variable technique can attack. The cure is to rotate the integration contour

out of Minkowski signature into the Euclidean domain, where the denominator becomes a strictly positive $(\ell_E^2 + \Delta)$ and the measure is positive-definite. This section carries out that rotation rigorously for the canonical integrand, tracks the factor of i and the sign flip it produces, and then builds the d-dimensional Euclidean measure by analytic continuation, reducing the whole integral to a one-dimensional radial integral with all angular information packaged into a single number, the solid angle Ω_d. The result is an object the next section can evaluate in closed form.

Throughout we use the standing conventions of Chapter A: mostly-minus metric $\eta = \mathrm{diag}(+,-,-,-)$, the Feynman prescription $i/(p^2 - m^2 + i\epsilon)$, and the Wick rotation as fixed in Theorem A.23. The contour technology—Cauchy's theorem, the residue theorem, Jordan-type arc estimates, and the legality of the energy-plane rotation itself—was built in Chapter 5; we import it here rather than reprove it, specializing the general statement of Theorem 5.31 to the loop integrand at hand.

D.3.1 Pole positions and the legality of the rotation

The rotation is, at bottom, a statement about a single complex variable: the energy component ℓ^0 of the loop momentum, with the spatial components $\boldsymbol{\ell}$ held fixed and real. We therefore begin by locating the singularities of the integrand in the complex ℓ^0 plane.

Write the canonical denominator in terms of its energy and spatial parts. With $\ell^2 = (\ell^0)^2 - \boldsymbol{\ell}^2$ in mostly-minus signature,

$$\ell^2 - \Delta + i\epsilon = (\ell^0)^2 - (\boldsymbol{\ell}^2 + \Delta) + i\epsilon. \tag{D.53}$$

The zeros of this quadratic in ℓ^0 are the poles of $(\ell^2 - \Delta + i\epsilon)^{-n}$, each of order n.

Proposition D.15 (Pole positions of the canonical integrand)**.** *Fix $\boldsymbol{\ell} \in \mathbb{R}^{d-1}$ and $\Delta \in \mathbb{R}$, and set $a \equiv \boldsymbol{\ell}^2 + \Delta$. For $\epsilon > 0$ the denominator (D.53) has exactly two zeros in the complex ℓ^0 plane,*

$$\ell^0_{\pm} = \pm\sqrt{a - i\epsilon}\,, \tag{D.54}$$

each a pole of order n of $(\ell^2 - \Delta + i\epsilon)^{-n}$. With the principal branch of the square root, ℓ^0_+ lies in the closed fourth quadrant ($\mathrm{Re}\,\ell^0_+ \geq 0$, $\mathrm{Im}\,\ell^0_+ \leq 0$) and ℓ^0_- in the closed second quadrant. In particular both poles avoid the open first and third quadrants for every value of $\boldsymbol{\ell}$ and every sign of Δ.

Proof. Setting (D.53) to zero gives $(\ell^0)^2 = a - i\epsilon$, whose two roots are (D.54). Write $a - i\epsilon = \rho\, e^{i\varphi}$ with $\rho > 0$. Because the imaginary part $-\epsilon$ is negative and the real part is a, the argument lies in $\varphi \in (-\pi, 0)$ for every real a: if $a > 0$ then $\varphi \in (-\frac{\pi}{2}, 0)$, if $a = 0$ then $\varphi = -\frac{\pi}{2}$, and if $a < 0$ then $\varphi \in (-\pi, -\frac{\pi}{2})$. The

principal square root $\sqrt{a-i\epsilon} = \sqrt{\rho}\, e^{i\varphi/2}$ then has argument $\varphi/2 \in (-\frac{\pi}{2}, 0)$, i.e. nonnegative real part and nonpositive imaginary part: ℓ^0_+ sits in the (closed) fourth quadrant. Its partner $\ell^0_- = -\ell^0_+$ is the reflection through the origin and sits in the second quadrant. Neither root can enter the open first or third quadrant, which is exactly the region the rotation will sweep. □

Remark D.16 (The $i\epsilon$ chooses the rotation sense). The placement in Theorem D.15 is the loop-integral image of the Feynman pole pattern fixed in Eq. (A.84): for the physical case $a > 0$ the poles sit just below the positive real axis and just above the negative real axis, precisely where a free propagator's poles $\pm(\omega_{\boldsymbol{p}} - i\delta)$ sit. It is the sign of the Feynman $i\epsilon$ that pushes the poles into quadrants II and IV and so leaves quadrants I and III clear; had we chosen the retarded or advanced prescription the clear quadrants would differ and the rotation below would run the other way. The $i\epsilon$ is thus not cosmetic: it selects the direction $\ell^0 \to i\ell^0_E$ (counterclockwise, through the empty first and third quadrants) as the legal one.

With the poles located we can state and prove the rotation for the canonical integrand. We treat the inner ℓ^0 integration as a contour integral in one complex variable; the outer spatial integrations are untouched because the rotation does not move $\boldsymbol{\ell}$, and we return to them when we build the measure in Section D.3.2.

Proposition D.17 (Wick rotation of the canonical integrand)**.** *Let $n \geq 1$ be an integer (or, by analytic continuation, $\operatorname{Re} n > \frac{1}{2}$), let $\Delta \in \mathbb{R}$, and fix $\boldsymbol{\ell} \in \mathbb{R}^{d-1}$. Then the energy integral may be rotated from the real to the imaginary ℓ^0 axis,*

$$\int_{-\infty}^{\infty} \frac{d\ell^0}{\left[(\ell^0)^2 - (\boldsymbol{\ell}^2 + \Delta) + i\epsilon\right]^n} = i \int_{-\infty}^{\infty} \frac{d\ell^0_E}{(-1)^n \left[(\ell^0_E)^2 + \boldsymbol{\ell}^2 + \Delta\right]^n}, \tag{D.55}$$

under the substitution $\ell^0 = i\,\ell^0_E$ with $\ell^0_E \in \mathbb{R}$. Carrying this through the full d-dimensional measure and writing $\ell_E = (\ell^0_E, \boldsymbol{\ell})$ with Euclidean square $\ell^2_E = (\ell^0_E)^2 + \boldsymbol{\ell}^2 \geq 0$ yields the dictionary

$$d^d\ell = i\, d^d\ell_E, \qquad \ell^2 = -\,\ell^2_E, \qquad \ell^2 - \Delta + i\epsilon \;\longrightarrow\; -(\ell^2_E + \Delta), \tag{D.56}$$

and therefore the master rotation formula

$$\int \frac{d^d\ell}{(2\pi)^d} \frac{1}{(\ell^2 - \Delta + i\epsilon)^n} = i\,(-1)^n \int \frac{d^d\ell_E}{(2\pi)^d} \frac{1}{(\ell^2_E + \Delta)^n}. \tag{D.57}$$

Proof. Abbreviate $g(\ell^0) = \left[(\ell^0)^2 - a + i\epsilon\right]^{-n}$ with $a = \boldsymbol{\ell}^2 + \Delta$ fixed real. By Theorem D.15 the only singularities of g lie in the open second and fourth quadrants. Consider the closed, positively oriented contour Γ_R in the first quadrant built from three pieces: the segment C_1 along the positive real axis from 0 to R; the quarter-arc C_2, $\ell^0 = Re^{i\theta}$ with $\theta : 0 \to \frac{\pi}{2}$; and the segment

C_3 down the positive imaginary axis from iR back to 0. The interior of Γ_R is contained in the open first quadrant, which holds no poles, so Cauchy's theorem (Theorem 5.23) gives

$$\int_{C_1} g\, d\ell^0 + \int_{C_2} g\, d\ell^0 + \int_{C_3} g\, d\ell^0 = 0. \tag{D.58}$$

On the arc we estimate g by its modulus. For $|\ell^0| = R$ large, $|(\ell^0)^2 - a + i\epsilon| \geq R^2 - |a| - \epsilon \geq \frac{1}{2}R^2$ once $R^2 \geq 2(|a| + \epsilon)$, so $|g(\ell^0)| \leq (\frac{1}{2}R^2)^{-n} = 2^n R^{-2n}$. The arc has length $\frac{\pi}{2}R$, and the ML estimate gives

$$\left| \int_{C_2} g\, d\ell^0 \right| \leq \frac{\pi}{2} R \cdot 2^n R^{-2n} = 2^{n-1}\pi\, R^{1-2n} \xrightarrow[R\to\infty]{} 0 \qquad (n \geq 1), \tag{D.59}$$

since $1 - 2n \leq -1 < 0$. This is the specialization of the decay hypothesis of Theorem 5.31 to the present integrand, with $\eta = 2n - 1 > 0$; for noninteger n the same bound holds whenever $\mathrm{Re}(2n) > 1$. Thus the arc contribution vanishes, and (D.58) collapses to $\int_{C_1} g = -\int_{C_3} g = \int_{C_3^{\mathrm{rev}}} g$, i.e. the integral up the positive real axis equals the integral up the positive imaginary axis. Repeating the argument verbatim in the third quadrant with the mirror contour (real axis from $-R$ to 0, arc $\theta : \pi \to \frac{3\pi}{2}$, imaginary axis from 0 to $-iR$) handles the negative half. Concatenating the two halves rotates the entire real-axis integration onto the entire imaginary axis:

$$\int_{-\infty}^{\infty} g(\ell^0)\, d\ell^0 = \int_{-i\infty}^{i\infty} g(\ell^0)\, d\ell^0. \tag{D.60}$$

Now parametrize the imaginary axis by $\ell^0 = i\ell_E^0$ with $\ell_E^0 \in \mathbb{R}$ running from $-\infty$ to $+\infty$ (the orientation matches: $\ell^0 = -i\infty$ at $\ell_E^0 = -\infty$ up to $\ell^0 = +i\infty$ at $\ell_E^0 = +\infty$). Then $d\ell^0 = i\, d\ell_E^0$ and $(\ell^0)^2 = -(\ell_E^0)^2$, so $(\ell^0)^2 - a + i\epsilon = -\big[(\ell_E^0)^2 + a\big]$ with the regulator now redundant because the bracket is strictly positive. Hence

$$\int_{-\infty}^{\infty} g(\ell^0)\, d\ell^0 = i \int_{-\infty}^{\infty} \frac{d\ell_E^0}{\big(-\big[(\ell_E^0)^2 + a\big]\big)^n} = i\,(-1)^{-n} \int_{-\infty}^{\infty} \frac{d\ell_E^0}{\big[(\ell_E^0)^2 + \boldsymbol{\ell}^2 + \Delta\big]^n}, \tag{D.61}$$

which is (D.55) once one notes $(-1)^{-n} = (-1)^n$ for integer n. The single factor of i comes from $d\ell^0 = i\, d\ell_E^0$ on the one rotated coordinate; the $(d-1)$ spatial integrations are identical on the two sides because $\boldsymbol{\ell}$ never moved. Collecting the spatial measure with the rotated energy measure gives $d^d\ell = i\, d^d\ell_E$ and the relations (D.56); restoring the $(2\pi)^{-d}$ and performing the unchanged spatial integral on both sides produces (D.57). □

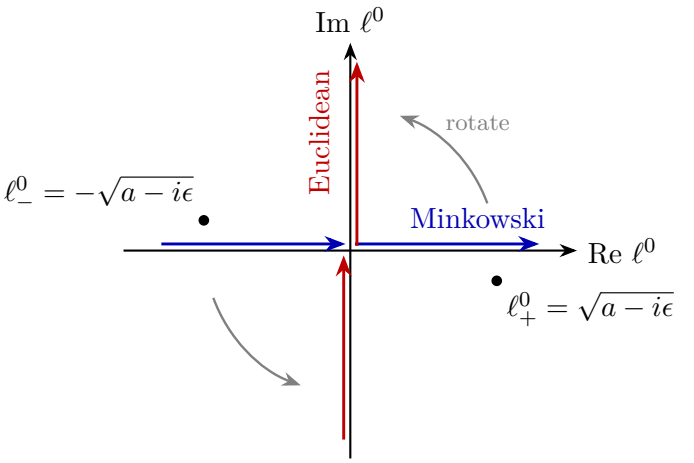


Figure D.4: Wick rotation of the energy contour for the canonical loop integrand. The Feynman $i\epsilon$ places the order-n poles $\ell^0_\pm$ in the second and fourth quadrants (Theorem D.15). The real-axis Minkowski contour (blue) is rotated counterclockwise, through the pole-free first and third quadrants, onto the imaginary-axis Euclidean contour (red). The quarter-arcs at infinity contribute nothing because the integrand falls as $|\ell^0|^{-2n}$ while the arc length grows only as R (Eq. (D.59)).

Caution

The factor of i and the sign flip are both physical. Two distinct relics of the rotation appear in (D.57), and both must be carried. The overall i comes from the single rotated coordinate $d\ell^0 = i\, d\ell^0_E$ and is *always* present; it is what turns the Minkowski amplitude's $i\mathcal{M}$ bookkeeping into real Euclidean integrals and, ultimately, what supplies the i in front of every one-loop self-energy and vertex. The sign $(-1)^n$ comes from $\ell^2 - \Delta \to -(\ell_E^2 + \Delta)$ raised to the power n and depends on the power of the denominator: it is $+1$ for the even powers $n = 2$ typical of bubbles and -1 for $n = 1, 3, \ldots$. Dropping either is a sign error of the most common kind (Section A.4); a reliable rule of thumb is to write (D.57) with both factors explicit before inserting numbers.

Figure D.4 summarizes the geometry: the blue real-axis contour swings through the empty quadrants onto the red imaginary axis, picking up the factor of i from the 90° turn and never crossing a pole.

D.3.2 The solid angle and the radial measure in d dimensions

Equation (D.57) has delivered a Euclidean integral over $\mathbb{R}^d$ (analytically continued in d) of a function that depends on ℓ_E only through the rotationally invariant

combination ℓ_E^2. Any such integral begs to be done in spherical coordinates, with the angular part performed once and for all. The angular part is the surface area of the unit $(d-1)$-sphere—the solid angle Ω_d—and the radial part is a single integral over $\ell_E \equiv |\ell_E| \in [0,\infty)$. The subtlety is that d is not an integer: the regulator has continued us to $d = 4 - 2\varepsilon$, where "the surface of a sphere" has no naive geometric meaning. We therefore *define* Ω_d by the one identity that makes sense for all d and reduces to the familiar areas at integer d, namely the Gaussian normalization.

Proposition D.18 (The d-dimensional solid angle)**.** *For* $\operatorname{Re} d > 0$ *define the solid angle* Ω_d *as the constant relating the Cartesian and radial Gaussian integrals,*

$$\int d^d x\, e^{-x^2} \;=\; \Omega_d \int_0^\infty dr\, r^{d-1} e^{-r^2}, \qquad x \in \mathbb{R}^d, \quad r = |x|. \tag{D.62}$$

Evaluating both sides yields the closed form

$$\boxed{\Omega_d = \frac{2\pi^{d/2}}{\Gamma(d/2)}}, \tag{D.63}$$

where Γ *is the Euler gamma function. The right-hand side is meromorphic in* d *and provides the* definition *of* Ω_d *at noninteger (in particular complex)* d*.*

Proof. Compute the left-hand side of (D.62) for positive integer d by factorizing the Cartesian Gaussian, then identify the answer with the radial form; the resulting algebraic identity (D.63) then *defines* Ω_d elsewhere by analytic continuation in d, the gamma function being the unique continuation respecting the functional equation. For integer d the measure factorizes, $d^d x = \prod_{j=1}^d dx_j$ and $x^2 = \sum_j x_j^2$, so the integral is a product of d identical one-dimensional Gaussians:

$$\int d^d x\, e^{-x^2} = \prod_{j=1}^{d} \int_{-\infty}^{\infty} dx_j\, e^{-x_j^2} = \left(\sqrt{\pi}\right)^d = \pi^{d/2}, \tag{D.64}$$

using the standard value $\int_{-\infty}^{\infty} e^{-u^2}\, du = \sqrt{\pi}$. For the radial side, substitute $t = r^2$, $dt = 2r\, dr$, so that $r^{d-1} dr = \tfrac{1}{2}\, t^{(d-1)/2}\, t^{-1/2}\, dt = \tfrac{1}{2}\, t^{d/2-1}\, dt$, and recognize the gamma function in its integral representation $\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\, dt$:

$$\int_0^\infty dr\, r^{d-1} e^{-r^2} = \frac{1}{2} \int_0^\infty dt\, t^{d/2-1} e^{-t} = \frac{1}{2}\, \Gamma\!\left(\frac{d}{2}\right). \tag{D.65}$$

Inserting (D.64) and (D.65) into the defining relation (D.62) gives $\pi^{d/2} = \Omega_d \cdot \frac{1}{2}\Gamma(d/2)$, hence (D.63). Both $\pi^{d/2}$ and $\Gamma(d/2)$ are entire/meromorphic in d, so the right-hand side of (D.63) continues the geometric quantity to all d where $\Gamma(d/2)$ is finite and nonzero; this is what we mean by Ω_d in dimensional regularization. □

Worked Example D.19 (The formula reproduces the elementary areas)**.** The continued Ω_d must agree with school geometry at small integer d, and it does. Using $\Gamma(1)=1$, $\Gamma(\tfrac12)=\sqrt{\pi}$, $\Gamma(\tfrac32)=\tfrac12\sqrt{\pi}$, and $\Gamma(2)=1$:

$$\begin{aligned}
\Omega_1 &= \frac{2\pi^{1/2}}{\Gamma(1/2)} = \frac{2\sqrt{\pi}}{\sqrt{\pi}} = 2, && \text{the two points } \{\pm1\} \text{ bounding } [-1,1];\\
\Omega_2 &= \frac{2\pi^{1}}{\Gamma(1)} = 2\pi, && \text{the circumference of the unit circle;}\\
\Omega_3 &= \frac{2\pi^{3/2}}{\Gamma(3/2)} = \frac{2\pi^{3/2}}{\frac12\sqrt{\pi}} = 4\pi, && \text{the area of the unit sphere;}\\
\Omega_4 &= \frac{2\pi^{2}}{\Gamma(2)} = 2\pi^2, && \text{the volume of the unit 3-sphere.}
\end{aligned} \tag{D.66}$$

At the physical value $d=4-2\varepsilon$ one has, expanding to the order needed in Section D.4, $\Omega_{4-2\varepsilon}=2\pi^{2-\varepsilon}/\Gamma(2-\varepsilon)$, which tends to $2\pi^2$ as $\varepsilon\to0$; the ε-dependence here is finite and will be combined with the genuinely divergent Γ from the radial integral in the next section. The point of (D.63) is precisely that it is an analytic function of d we can Taylor-expand, not a sequence of integer areas.

With Ω_d in hand the angular integration of any radially symmetric integrand is immediate.

Proposition D.20 (Reduction to a radial integral)**.** *Let $f\colon[0,\infty)\to\mathbb{C}$ be such that the integral below converges. Then*

$$\int d^d\ell_E\, f(\ell_E^2) = \Omega_d \int_0^\infty d\ell_E\, \ell_E^{d-1}\, f(\ell_E^2), \qquad \ell_E \equiv |\ell_E|. \tag{D.67}$$

Proof. For integer d, introduce spherical coordinates in $\mathbb{R}^d$: a radius $\ell_E\in[0,\infty)$ and angles $\hat n$ on the unit sphere S^{d-1}, with measure $d^d\ell_E=\ell_E^{d-1}\,d\ell_E\,d\Omega(\hat n)$, where $d\Omega$ is the surface element of S^{d-1}. Because the integrand depends only on $\ell_E^2=\ell_E^2$ and not on $\hat n$, the angular integral factors off and contributes the total surface area $\int_{S^{d-1}}d\Omega=\Omega_d$ established in Theorem D.18, leaving (D.67). For noninteger d this identity is the *definition* of the left-hand side: the d-dimensional integral of a radial function is declared to be the analytic continuation in d of the right-hand side, whose ingredients Ω_d (via (D.63)) and the one-dimensional radial integral are individually well defined for complex d. This is consistent with Theorem D.18—both rest on the same Gaussian identity—and it is the operational meaning of "integrating in d dimensions" used throughout this appendix. □

Combining the rotation (D.57) with the reduction (D.67) applied to $f(\ell_E^2)=(\ell_E^2+\Delta)^{-n}$ packages the entire angular content of the canonical loop integral

into Ω_d:

$$\int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^n} = \frac{i\,(-1)^n\,\Omega_d}{(2\pi)^d}\int_0^\infty d\ell_E\,\frac{\ell_E^{d-1}}{(\ell_E^2+\Delta)^n}. \tag{D.68}$$

Everything to the right of $\Omega_d/(2\pi)^d$ is a single, manifestly convergent (for suitable d) one-dimensional integral. We have turned a d-dimensional Lorentzian integral into a radial Euclidean one, with the price of the conversion—one factor of i, the sign $(-1)^n$, and the angular volume Ω_d—all displayed explicitly.

D.3.3 Carrying the ϕ^4 bubble across the rotation

To see the abstract machinery act on a real amplitude, we follow the one-loop bubble of ϕ^4 theory (with coupling λ and mass m) through the rotation. This is the integral that the previous section reduced to canonical form, and the object Chapter 24 will turn into the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$.

Worked Example D.21 (The ϕ^4 s-channel bubble, rotated)**.** The s-channel one-loop correction to $\phi\phi\to\phi\phi$ in $\lambda\phi^4$ theory carries the scalar integral

$$I(p^2) = \int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-m^2+i\epsilon)\big((\ell+p)^2-m^2+i\epsilon\big)}, \tag{D.69}$$

with p the total incoming momentum, $p^2 = s$. Feynman-combining the two denominators (Theorem D.11) and shifting $\ell\to\ell - xp$ to complete the square brings (D.69) to the canonical form with $n=2$,

$$I(p^2) = \int_0^1 dx \int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^2}, \qquad \Delta = m^2 - x(1-x)\,p^2, \tag{D.70}$$

where the shifted ℓ is again called ℓ and Δ depends on the Feynman parameter x and the external invariant. (The shift is legitimate because the d-dimensional measure is translation invariant; for the divergent piece this is part of what dimensional regularization buys, as the next sections make precise.)

Rotating the inner integral. The inner ℓ-integral is exactly the canonical object with $n=2$. We apply (D.57). Here $(-1)^n = (-1)^2 = +1$, so the sign flip is invisible and the *only* relic of the rotation is the overall factor of i:

$$\int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^2} = i\int \frac{d^d\ell_E}{(2\pi)^d}\,\frac{1}{(\ell_E^2+\Delta)^2}. \tag{D.71}$$

The Lorentzian denominator that vanished on a hypersurface has become the strictly positive $(\ell_E^2+\Delta)^2$, with $\Delta>0$ for $s<4m^2$ (below threshold) so that the Euclidean integral is genuinely convergent away from $d=4$; for $s>4m^2$ the Δ turns negative over part of the x-range and the $i\epsilon$ supplies the imaginary

part that becomes the bubble's discontinuity, but the rotation formula (D.71) is unchanged because it was derived for any real Δ.

Angular reduction. Applying (D.67) to the rotated integrand collapses the angles into Ω_d:

$$I(p^2) = i \int_0^1 dx \, \frac{\Omega_d}{(2\pi)^d} \int_0^\infty d\ell_E \, \frac{\ell_E^{d-1}}{(\ell_E^2 + \Delta)^2}. \tag{D.72}$$

This is the bubble reduced to a Feynman-parameter integral over a one-dimensional radial integral, with the factor of i explicit out front—the same i that will ride through to the renormalized vertex and, ultimately, into the sign and normalization of β_λ in Chapter 24. The radial integral $\int_0^\infty d\ell_E \, \ell_E^{d-1}(\ell_E^2 + \Delta)^{-2}$ is the object the next section evaluates in closed form.

Physical Insight

Why the Euclidean side is the honest one. The rotation is more than a computational trick. In Euclidean signature the integrand is a positive function of a positive-definite variable, so the integral is a genuine (if still UV-sensitive) Lebesgue integral rather than a conditionally convergent oscillatory one; convergence questions become questions about the behavior of $\ell_E^{d-1}(\ell_E^2 + \Delta)^{-n}$ at $\ell_E \to 0$ (the infrared) and $\ell_E \to \infty$ (the ultraviolet), which the power of ℓ_E makes transparent. This is the first place one sees, geometrically, the dual role of d as a regulator of *both* ends: lowering d tames the large-ℓ_E (UV) growth, while raising d tames the small-ℓ_E (IR) behavior. The single analytic continuation in d that the next section performs therefore regulates UV and IR simultaneously—the thread that ties the rest of this appendix together.

Equation (D.68), and its ϕ^4 instance (D.72), hand the next section a purely radial Euclidean integral $\int_0^\infty d\ell_E \, \ell_E^{d-1}(\ell_E^2+\Delta)^{-n}$ with every angular factor already absorbed into $\Omega_d = 2\pi^{d/2}/\Gamma(d/2)$. With the substitution $t = \ell_E^2/(\ell_E^2 + \Delta)$ this radial integral is a Beta function, and evaluating it produces the master scalar one-loop integral in closed form—the task of Section D.4.

D.4 The master one-loop integral and its tensor descendants

The previous section left us at the threshold of an honest calculation. The Wick rotation of Section D.3 turned the Minkowski loop integral into a Euclidean one, and the construction of the d-dimensional measure factored its angular part into the solid angle Ω_d, leaving a single one-dimensional integral over the Euclidean radius. That radial integral is the last formal obstacle between us

and a number: once it is done, every scalar one-loop integral in this book is a product of Γ-functions times a power of a kinematic invariant, and the entire ultraviolet content of the loop is concentrated in one of those Γ-functions. This section carries out the radial integral in closed form, assembles the result into *the* master scalar one-loop formula, and then extends it in the only direction the Feynman rules of Chapters 15 and 17 actually demand: to integrands carrying loop momenta in the numerator. Those numerator integrals are reduced to the scalar master by symmetric integration, a manoeuvre we prove rather than assert. With the master formula and its tensor descendants in hand we read off the standard scalar functions A_0 and B_0 that the loop chapters quote by name, and we identify exactly where the formula develops poles — the analytic signature of the ultraviolet divergences that the power counting of Section D.1 predicted.

D.4.1 The radial integral and the master formula

Fix the object of study. After combining denominators with Feynman or Schwinger parameters (Section D.2) and shifting the loop momentum to complete the square, every scalar one-loop integral in the book is reduced to a single canonical form, the integral of an inverse power of $\ell^2 - \Delta$ over all of d-dimensional Minkowski momentum space,

$$I_n(\Delta) \;\equiv\; \int \frac{\mathrm{d}^d \ell}{(2\pi)^d} \, \frac{1}{(\ell^2 - \Delta + i\epsilon)^{\,n}} \,, \tag{D.73}$$

where n is a positive integer (the total power of propagators after parametrization), Δ is the effective mass-squared produced by completing the square — a function of the external invariants, the internal masses, and the Feynman parameters, but *not* of the loop momentum — and the $i\epsilon$ is the Feynman prescription inherited from each propagator. The whole point of dimensional regularization is to treat the spacetime dimension d as a complex variable; Eq. (D.73) is then to be read as the analytic continuation, in d, of the integral evaluated wherever it converges.

The rotation to Euclidean signature was performed in Section D.3. Setting $\ell^0 = i\ell^0_E$ (so that $\mathrm{d}^d\ell = i\,\mathrm{d}^d\ell_E$) and $\ell^2 = (\ell^0)^2 - \boldsymbol{\ell}^2 = -\big[(\ell^0_E)^2 + \boldsymbol{\ell}^2\big] = -\ell_E^2$ rotates the denominator into $\ell^2 - \Delta \to -(\ell_E^2 + \Delta)$, with the $i\epsilon$ guaranteeing that the contour does not cross a pole during the rotation. Each inverse power therefore contributes a factor $(-1)^{-n} = (-1)^n$, and the Jacobian contributes the explicit i, so that

$$I_n(\Delta) \;=\; i\,(-1)^n \int \frac{\mathrm{d}^d \ell_E}{(2\pi)^d} \, \frac{1}{(\ell_E^2 + \Delta)^{\,n}} \,. \tag{D.74}$$

The Euclidean integrand depends on ℓ_E only through its length $\ell_E \equiv \sqrt{\ell_E^2} \geq 0$, so the angular integration is trivial: it produces the surface area of the unit

sphere in d dimensions, the solid angle

$$\Omega_d \;=\; \frac{2\pi^{d/2}}{\Gamma(d/2)}\,, \tag{D.75}$$

constructed by analytic continuation in Section D.3. What remains is purely radial,

$$I_n(\Delta) \;=\; i\,(-1)^n\,\frac{\Omega_d}{(2\pi)^d}\int_0^\infty \mathrm{d}\ell_E\;\frac{\ell_E^{\,d-1}}{(\ell_E^2+\Delta)^{\,n}}\,. \tag{D.76}$$

Everything now hinges on the one-dimensional integral in Eq. (D.76). We evaluate it exactly.

Theorem D.22 (The master scalar one-loop integral)**.** *For $\Delta>0$ and complex d with $0<\mathrm{Re}(d/2)<\mathrm{Re}(n)$ (the strip in which the radial integral converges absolutely, whence the result is continued analytically),*

$$\boxed{\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^{\,n}} \;=\; \frac{(-1)^n\,i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(n-\frac{d}{2}\right)}{\Gamma(n)}\,\Delta^{\,d/2-n}} \tag{D.77}$$

with the $i\epsilon$ understood throughout and the branch of $\Delta^{d/2-n}$ fixed by $\Delta>0$.

Proof. Start from the radial form Eq. (D.76) and change variables to expose a Beta-function integral. The substitution that does this is

$$t \;=\; \frac{\ell_E^2}{\Delta}\,,\qquad \ell_E=(\Delta t)^{1/2}\,,\qquad \mathrm{d}\ell_E=\tfrac{1}{2}\,\Delta^{1/2}\,t^{-1/2}\,\mathrm{d}t\,, \tag{D.78}$$

which maps $\ell_E\in(0,\infty)$ to $t\in(0,\infty)$. Each factor in the radial integrand becomes a power of Δ times a power of t:

$$\ell_E^{\,d-1}=(\Delta t)^{(d-1)/2}=\Delta^{(d-1)/2}\,t^{(d-1)/2}, \tag{D.79}$$

$$(\ell_E^2+\Delta)^{-n}=\big[\Delta(t+1)\big]^{-n}=\Delta^{-n}\,(1+t)^{-n}. \tag{D.80}$$

Collecting Eqs. (D.79) and (D.80) together with the Jacobian from Eq. (D.78), the powers of Δ combine as $\frac{d-1}{2}-n+\frac{1}{2}=\frac{d}{2}-n$, and the radial integral separates into a pure Δ-power times a t-integral,

$$\int_0^\infty \mathrm{d}\ell_E\;\frac{\ell_E^{\,d-1}}{(\ell_E^2+\Delta)^{\,n}} \;=\; \tfrac{1}{2}\,\Delta^{\,d/2-n}\int_0^\infty \mathrm{d}t\;\frac{t^{\,d/2-1}}{(1+t)^{\,n}}\,. \tag{D.81}$$

The remaining t-integral is exactly the standard integral representation of the Euler Beta function. Recall (Chapter 5) that for $\mathrm{Re}(\alpha)>0$ and $\mathrm{Re}(\beta)>0$ the Beta function admits the two representations

$$\mathrm{B}(\alpha,\beta) \;=\; \int_0^1 \mathrm{d}u\,u^{\alpha-1}(1-u)^{\beta-1} \;=\; \int_0^\infty \mathrm{d}t\;\frac{t^{\alpha-1}}{(1+t)^{\alpha+\beta}}\,, \tag{D.82}$$

the second following from the first under $u = t/(1+t)$, and that it is expressed through Gamma functions by

$$\mathrm{B}(\alpha,\beta) \;=\; \frac{\Gamma(\alpha)\,\Gamma(\beta)}{\Gamma(\alpha+\beta)}\,. \tag{D.83}$$

Matching the integrand in Eq. (D.81) to the second form of Eq. (D.82) requires $\alpha - 1 = \frac{d}{2} - 1$ and $\alpha + \beta = n$, that is $\alpha = \frac{d}{2}$ and $\beta = n - \frac{d}{2}$. Both arguments have positive real part precisely on the convergence strip $0 < \mathrm{Re}(d/2) < \mathrm{Re}(n)$ asserted in the statement; there

$$\int_0^\infty \mathrm{d}t\;\frac{t^{\,d/2-1}}{(1+t)^{\,n}} \;=\; \mathrm{B}\!\left(\tfrac{d}{2},\,n-\tfrac{d}{2}\right) \;=\; \frac{\Gamma\!\left(\frac{d}{2}\right)\Gamma\!\left(n-\frac{d}{2}\right)}{\Gamma(n)}\,. \tag{D.84}$$

Now reassemble. Insert Eqs. (D.81) and (D.84) into Eq. (D.76), using $\Omega_d = 2\pi^{d/2}/\Gamma(d/2)$ from Eq. (D.75):

$$I_n(\Delta) \;=\; i\,(-1)^n\,\frac{1}{(2\pi)^d}\,\frac{2\pi^{d/2}}{\Gamma(d/2)}\;\cdot\;\tfrac{1}{2}\,\Delta^{\,d/2-n}\,\frac{\Gamma(d/2)\,\Gamma(n-d/2)}{\Gamma(n)}\,. \tag{D.85}$$

Three simplifications finish the proof. The factor $\frac{1}{2}$ from the Jacobian cancels the 2 in Ω_d. The $\Gamma(d/2)$ from Ω_d cancels the $\Gamma(d/2)$ from the Beta function — this cancellation is the structural reason the answer is so clean. And the surviving powers of π and 2 combine into the canonical loop factor,

$$\frac{\pi^{d/2}}{(2\pi)^d} \;=\; \frac{\pi^{d/2}}{2^d\,\pi^d} \;=\; \frac{1}{2^d\,\pi^{d/2}} \;=\; \frac{1}{(4\pi)^{d/2}}\,, \tag{D.86}$$

where the last step uses $(4\pi)^{d/2} = 4^{d/2}\pi^{d/2} = 2^d\pi^{d/2}$. Substituting Eq. (D.86) into Eq. (D.85) leaves

$$I_n(\Delta) \;=\; \frac{(-1)^n\,i}{(4\pi)^{d/2}}\,\frac{\Gamma(n-d/2)}{\Gamma(n)}\,\Delta^{\,d/2-n}\,, \tag{D.87}$$

which is Eq. (D.77). Outside the convergence strip the right-hand side is a meromorphic function of d and n, and it *defines* the integral by analytic continuation; this is the operational content of dimensional regularization. □

A few features of Eq. (D.77) deserve to be named at once, because the rest of this appendix — and every loop chapter — leans on them. The dependence on the effective mass is the single power $\Delta^{d/2-n}$, fixed entirely by dimensional analysis: $[I_n] = [\text{mass}]^{d-2n}$, and indeed $d/2 - n$ is half of $d - 2n$ once the $\Delta^{d/2-n}$ is read as $(\Delta)^{(d-2n)/2}$. The dependence on the regulator is carried by the ratio of Gammas, and the divergence structure — the entire reason the integral needed regulating — sits in the single factor $\Gamma(n - d/2)$ in the numerator. We return to its poles in Section D.4.5.

Caution

The sign factor $(-1)^n$ and the explicit i in Eq. (D.77) are not optional decorations; they are the residue of the Wick rotation and the source of the overall i that makes one-loop amplitudes come out with the right reality properties. A common error is to quote the Euclidean version (without the i and with $\Delta^{d/2-n}$ already real and positive) and forget to rotate back. Throughout this book Eq. (D.77) is the Minkowski statement: the integral on the left is over Minkowski momenta with the Feynman $i\epsilon$, and the i on the right is genuinely there.

D.4.2 Numerator integrals: even powers of the loop momentum

Beyond the purely scalar integrand, the Feynman rules generate numerators built from the loop momentum ℓ^μ. The simplest are scalar numerators $(\ell^2)^k$, which arise after a tensor numerator has been reduced (next subsection) or directly in theories whose vertices carry derivatives. They are dispatched by the same radial integral, because $(\ell^2)^k$ merely shifts the effective dimension.

Proposition D.23 (Scalar numerator master)**.** *For integer $k \geq 0$ and n on the appropriate convergence strip,*

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d} \frac{(\ell^2)^k}{(\ell^2 - \Delta + i\epsilon)^{\,n}} = \frac{(-1)^{n+k}\, i}{(4\pi)^{d/2}} \frac{\Gamma\!\left(k + \frac{d}{2}\right)}{\Gamma\!\left(\frac{d}{2}\right)} \frac{\Gamma\!\left(n - k - \frac{d}{2}\right)}{\Gamma(n)} \Delta^{\,d/2+k-n}. \tag{D.88}$$

Proof. Under the Wick rotation of Eq. (D.74) the numerator rotates as $(\ell^2)^k = (-\ell_E^2)^k = (-1)^k(\ell_E^2)^k$, contributing an extra sign $(-1)^k$ relative to the scalar case and replacing the radial power ℓ_E^{d-1} by ℓ_E^{d-1+2k}. The Euclidean integral is therefore identical to Eq. (D.76) with $d-1 \mapsto d-1+2k$ in the radial measure — that is, with the replacement $d \mapsto d+2k$ *inside the radial integral only*, the prefactor $\Omega_d/(2\pi)^d$ being unchanged. Running the substitution Eq. (D.78) again, the powers of Δ now combine to $\frac{d+2k}{2} - n = \frac{d}{2} + k - n$, and the Beta integral becomes

$$\int_0^\infty \mathrm{d}t\, \frac{t^{\,d/2+k-1}}{(1+t)^{\,n}} = \mathrm{B}\!\left(\tfrac{d}{2}+k,\, n-k-\tfrac{d}{2}\right) = \frac{\Gamma\!\left(\frac{d}{2}+k\right)\Gamma\!\left(n-k-\frac{d}{2}\right)}{\Gamma(n)}, \tag{D.89}$$

matching $\alpha = \frac{d}{2}+k$, $\beta = n-k-\frac{d}{2}$ in Eqs. (D.82) and (D.83). Assembling exactly as in Theorem D.22, the cancellation that removed $\Gamma(d/2)$ before is now *incomplete*: the $\Gamma(d/2)$ from Ω_d cancels only one of the two, leaving the ratio $\Gamma(d/2+k)/\Gamma(d/2)$. This is the promised appearance of the dimension in the denominator. The extra sign $(-1)^k$ rides along, giving the overall $(-1)^{n+k}$, and

the loop factor $1/(4\pi)^{d/2}$ is produced by Eq. (D.86) as before. The result is Eq. (D.88). □

For $k = 0$ the ratio $\Gamma(d/2)/\Gamma(d/2) = 1$ and Eq. (D.88) collapses to Eq. (D.77), as it must. The factor $\Gamma(d/2+k)/\Gamma(d/2) = (d/2)(d/2+1)\cdots(d/2+k-1)$ is a polynomial of degree k in d; it is the algebraic trace of the angular average of $(\ell^2)^k$ over the sphere, and it is exactly the d-dependence that, expanded about $d = 4$ in the next section, can convert a $1/\varepsilon$ pole into a finite remainder.

D.4.3 Tensor numerators and symmetric integration

The numerators that actually occur in gauge theories and in any theory with derivative couplings are not scalars $(\ell^2)^k$ but tensors built from uncontracted loop momenta: ℓ^μ, $\ell^\mu\ell^\nu$, $\ell^\mu\ell^\nu\ell^\rho\ell^\sigma$, and so on, sitting over a scalar denominator $(\ell^2 - \Delta)^n$. The QED vacuum polarization and self-energy of Chapter 19 are precisely of this kind once the Dirac algebra of Section C.6 has been carried out on the numerator. The device that reduces every such tensor integral to the scalars already computed is *symmetric integration*: the observation that the integration domain and the denominator are invariant under $\ell \to -\ell$ and under d-dimensional Lorentz transformations, so the integral of a tensor numerator can only be the most general tensor with those symmetries, built from the metric alone.

Proposition D.24 (Symmetric integration, ranks one and two)**.** *Let $f(\ell^2)$ be any function for which the integrals below converge (in particular $f = (\ell^2 - \Delta)^{-n}$). Then under the loop integral,*

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, \ell^\mu\, f(\ell^2) = 0\,, \tag{D.90}$$

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, \ell^\mu\ell^\nu\, f(\ell^2) = \frac{\eta^{\mu\nu}}{d} \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, \ell^2\, f(\ell^2)\,. \tag{D.91}$$

Equivalently, inside such an integral one may make the replacements

$$\ell^\mu \;\longrightarrow\; 0\,, \qquad\qquad \ell^\mu\ell^\nu \;\longrightarrow\; \frac{\ell^2}{d}\,\eta^{\mu\nu}\,, \tag{D.92}$$

with the factor of d — the dimension of the spacetime over which we are integrating — in the denominator.

Proof. For Eq. (D.90): the measure $\mathrm{d}^d\ell$, the domain $\mathbb{R}^{1,d-1}$ (or its Euclidean rotation), and the weight $f(\ell^2)$ are all invariant under the reflection $\ell \to -\ell$, while the integrand factor ℓ^μ is odd. The integral of an odd function over a symmetric domain vanishes, provided it converges; in dimensional regularization convergence is arranged by the analytic continuation, and the statement holds

as an identity of meromorphic functions of d. More generally any numerator carrying an odd number of uncontracted ℓ's integrates to zero by the same parity argument.

For Eq. (D.91): write the integral as a Lorentz tensor,

$$T^{\mu\nu} \;\equiv\; \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\, \ell^\mu \ell^\nu\, f(\ell^2)\,. \tag{D.93}$$

The measure and the weight $f(\ell^2)$ are invariant under the (continued) Lorentz group $\mathrm{SO}(1, d-1)$, so $T^{\mu\nu}$ transforms as a rank-two symmetric Lorentz tensor that depends on *no* external vector — the only data left after integrating out ℓ is Δ (a scalar) and the dimension. The unique symmetric two-index tensor that can be built from no vectors is a multiple of the invariant metric. Hence

$$T^{\mu\nu} \;=\; C\,\eta^{\mu\nu} \tag{D.94}$$

for some scalar C depending on Δ, d, and f. To fix C, contract both sides of Eq. (D.94) with the metric $\eta_{\mu\nu}$. On the left, $\eta_{\mu\nu}\ell^\mu\ell^\nu = \ell^2$ under the integral, so $\eta_{\mu\nu}T^{\mu\nu} = \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\ell^2 f(\ell^2)$. On the right, the trace of the metric in d dimensions is

$$\eta_{\mu\nu}\eta^{\mu\nu} \;=\; \delta^\mu{}_\mu \;=\; d\,, \tag{D.95}$$

the defining bookkeeping rule of the d-dimensional formalism (Eq. (C.185)). Therefore $\eta_{\mu\nu}T^{\mu\nu} = C\,d$, and solving,

$$C \;=\; \frac{1}{d}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\ell^2 f(\ell^2)\,. \tag{D.96}$$

Inserting Eq. (D.96) into Eq. (D.94) gives Eq. (D.91), and reading the result back at the level of the integrand gives the replacement rule Eq. (D.92). It is the single appearance of $\eta_{\mu\nu}\eta^{\mu\nu} = d$, not 4, that makes this a genuinely d-dimensional manipulation: keeping the d exact here is essential, because it multiplies divergent scalar integrals and so contributes to finite parts after the ε-expansion. □

The same logic fixes higher even ranks. At rank four the unique symmetric four-index tensor built from the metric is the sum of the three pairings $\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}$, so

$$\ell^\mu\ell^\nu\ell^\rho\ell^\sigma \;\longrightarrow\; \frac{(\ell^2)^2}{d(d+2)}\left(\eta^{\mu\nu}\eta^{\rho\sigma} + \eta^{\mu\rho}\eta^{\nu\sigma} + \eta^{\mu\sigma}\eta^{\nu\rho}\right), \tag{D.97}$$

the coefficient determined by contracting both sides with $\eta_{\mu\nu}\eta_{\rho\sigma}$: the left side gives ℓ^4 while the right gives the coefficient times $(d\cdot d + d + d) = d(d+2)$, using $\eta_{\mu\nu}\eta^{\mu\nu} = d$ twice and one cross-contraction $\eta_{\mu\nu}\eta^{\mu\rho}\eta_{\rho\sigma}\eta^{\nu\sigma} = \eta_\nu{}^\rho\eta_\rho{}^\nu = d$. With Eqs. (D.92) and (D.97) every tensor numerator is converted into a sum of $(\ell^2)^k$ numerators, which Theorem D.23 then integrates in closed form. This is the complete reduction algorithm at one loop: parametrize, shift, symmetrize, and apply Eqs. (D.77) and (D.88).

D.4.4 The standard scalar functions A_0 and B_0

Two combinations of the master integral occur so often that they are given names and quoted as black boxes throughout the loop chapters. They are the scalar one-point function A_0 (the tadpole) and the scalar two-point function B_0 (the bubble). We define them as the bare loop integrals and read off their values from Theorem D.22; the renormalization-scale factor $\mu^{2\varepsilon}$ and the ε-expansion are restored in Section D.5.

The tadpole A_0. The scalar one-point integral has a single massive propagator and no external momentum:

$$A_0(m^2) \;\equiv\; \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{\ell^2 - m^2 + i\epsilon}\,. \tag{D.98}$$

This is Eq. (D.73) with $n = 1$ and $\Delta = m^2$, so Eq. (D.77) with $\Gamma(1) = 1$ gives immediately

$$A_0(m^2) \;=\; -\,\frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(1 - \tfrac{d}{2}\right)\,(m^2)^{\,d/2-1}\,. \tag{D.99}$$

Two properties are worth recording. First, A_0 vanishes identically for a massless line, $A_0(0) = 0$: with $m = 0$ the factor $(m^2)^{d/2-1} \to 0$ for $\mathrm{Re}(d) > 2$, and in dimensional regularization a scaleless integral is *defined* to be zero precisely because there is no scale to carry its dimension — a fact we use repeatedly to discard massless tadpoles. Second, the divergence lives in $\Gamma(1 - d/2)$, which at $d = 4$ is $\Gamma(-1 + \varepsilon)$, a double layer of pole structure (a simple pole with a specific residue) reflecting the quadratic ultraviolet divergence of the tadpole counted in Section D.1. A companion tensor tadpole that appears in self-energies is reduced by Eq. (D.92) together with the algebraic identity $\ell^2 = (\ell^2 - m^2) + m^2$:

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^\mu \ell^\nu}{\ell^2 - m^2 + i\epsilon} = \frac{\eta^{\mu\nu}}{d}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{\ell^2}{\ell^2 - m^2 + i\epsilon} = \frac{\eta^{\mu\nu}}{d}\,m^2\,A_0(m^2)\,, \tag{D.100}$$

where the term $\int (\ell^2 - m^2)^{-1}(\ell^2 - m^2) = \int 1$ is a scaleless integral and vanishes, leaving only $m^2 A_0$.

The bubble B_0. The scalar two-point integral has two propagators carrying internal masses m_1, m_2 and an external momentum p flowing through the loop:

$$B_0(p^2; m_1, m_2) \;\equiv\; \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - m_1^2 + i\epsilon)\big((\ell + p)^2 - m_2^2 + i\epsilon\big)}\,. \tag{D.101}$$

Combine the two denominators with a single Feynman parameter (Section D.2), $\frac{1}{AB} = \int_0^1 \mathrm{d}x\,[xB + (1-x)A]^{-2}$, choosing $A = \ell^2 - m_1^2$ and $B = (\ell + p)^2 - m_2^2$. The combined denominator is

$$xB + (1-x)A = \ell^2 + 2x\,\ell\cdot p + xp^2 - xm_2^2 - (1-x)m_1^2$$

$$= (\ell + xp)^2 - \Delta\,, \tag{D.102}$$

where completing the square introduced $-x^2p^2$ and the effective mass-squared is

$$\Delta(x) \;=\; (1-x)\,m_1^2 + x\,m_2^2 - x(1-x)\,p^2\,. \tag{D.103}$$

Shifting the loop momentum $\ell \to \ell - xp$ (legitimate in dimensional regularization, where the shift introduces no surface term) reduces the inner integral to Eq. (D.73) with $n = 2$, and Eq. (D.77) with $\Gamma(2) = 1$ gives

$$B_0(p^2; m_1, m_2) \;=\; \frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(2 - \tfrac{d}{2}\right) \int_0^1 \mathrm{d}x\,\left[\Delta(x)\right]^{d/2-2}, \tag{D.104}$$

with $\Delta(x)$ from Eq. (D.103). The entire ultraviolet divergence is the x-independent prefactor $\Gamma(2 - d/2) = \Gamma(\varepsilon)$ at $d = 4$; the parametric integral over x is finite and, once expanded in ε, produces the logarithm of Δ and its branch cut. In the equal-mass case $m_1 = m_2 = m$ relevant to most of our examples,

$$B_0(p^2; m, m) \;=\; \frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(2 - \tfrac{d}{2}\right) \int_0^1 \mathrm{d}x\,\left[m^2 - x(1-x)\,p^2\right]^{d/2-2}. \tag{D.105}$$

This single function is the integral backbone of the benchmark theories at one loop: it is the ϕ^4 bubble that feeds β_λ in Chapter 24, and — dressed with Dirac and (for Yang–Mills) color numerators — it is the scalar core of the QED vacuum polarization and self-energy of Chapter 19 and of the gluon self-energy of Chapter 27. We now work the ϕ^4 case explicitly.

Worked Example D.25 (The one-loop ϕ^4 bubble as B_0)**.** We continue ϕ^4 theory from the Feynman rules of Chapter 15. The one-loop correction to the four-point function in $\lambda\phi^4$ theory is the sum of three diagrams — one for each Mandelstam channel s, t, u — each the simple bubble shown in Fig. D.5. We evaluate the s-channel bubble to closed Γ-function form; the other two channels are obtained by $s \to t$ and $s \to u$.

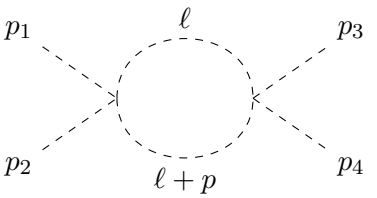


Figure D.5: The s-channel one-loop bubble in $\lambda\phi^4$ theory, with total incoming momentum $p = p_1 + p_2$ (so $p^2 = s$) circulating through the loop as ℓ and $\ell + p$. The t- and u-channel diagrams have the same topology with $p^2 = t$ and $p^2 = u$.

Each quartic vertex carries a factor $-i\lambda$, and the diagram has a symmetry factor $\frac{1}{2}$ from the interchange of the two internal lines (derived with the rest

of the ϕ^4 symmetry factors in Chapter 15). With total incoming momentum $p = p_1 + p_2$, $p^2 = s$, the amplitude contribution is

$$i\mathcal{M}_s \;=\; \frac{(-i\lambda)^2}{2}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2 - m^2 + i\epsilon}\,\frac{i}{(\ell+p)^2 - m^2 + i\epsilon}\,. \tag{D.106}$$

The two explicit propagator factors of i multiply to $i^2 = -1$, and $(-i\lambda)^2 = -\lambda^2$, so the kinematic prefactors combine to $(-\lambda^2)(-1)/2 = \lambda^2/2$, leaving exactly the scalar bubble:

$$i\mathcal{M}_s \;=\; \frac{\lambda^2}{2}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - m^2)((\ell+p)^2 - m^2)} \;=\; \frac{\lambda^2}{2}\,B_0(s; m, m)\,. \tag{D.107}$$

This is the promised identification of the carried ϕ^4 amplitude's loop with the master function B_0. Now apply the equal-mass result Eq. (D.105) with $p^2 = s$ and $\Delta(x) = m^2 - x(1-x)s$:

$$i\mathcal{M}_s \;=\; \frac{\lambda^2}{2}\,\frac{i}{(4\pi)^{d/2}}\,\Gamma\!\left(2 - \tfrac{d}{2}\right)\int_0^1 \mathrm{d}x\,\left[m^2 - x(1-x)\,s\right]^{d/2-2}\,. \tag{D.108}$$

Every step from the Feynman rules to a closed expression in Γ-functions and a one-dimensional parametric integral is now on the page. The divergence is the single $\Gamma(2 - d/2)$; at $d = 4$ this is $\Gamma(\varepsilon)$, a simple $1/\varepsilon$ pole, and its residue, summed over the three channels, is what Chapter 24 converts into the coefficient of $\beta_\lambda = 3\lambda^2/16\pi^2$. The parametric integral is finite and will, after the ε-expansion of Section D.5, deliver the physical logarithm $\ln\!\left[m^2 - x(1-x)s\right]$ whose branch cut for $s > 4m^2$ is the two-particle threshold demanded by unitarity.

Physical Insight

The reason a single function B_0 does so much work across three different theories is that the *denominator* of a one-loop diagram knows nothing about spin or color. After the Dirac traces of Section C.6 and the color contractions of Chapter B have been performed on the numerator, what is left to integrate is always a tensor built from ℓ over a scalar denominator $(\ell^2 - \Delta)^n$ — and that is precisely the object reduced by Theorems D.23 and D.24 to A_0 and B_0. The Yang–Mills scalar integrals are *identical* to the QED ones up to an overall color factor (a Casimir C_A, C_F, or $T_F n_f$ from Chapter B) multiplying the same B_0. Spin and gauge structure dress the building blocks; they never change the integrals.

D.4.5 Analytic structure: where the poles are, and what they mean

We close by reading the analytic structure of the master formula and tying it back to the power counting that opened this appendix. All the regulator dependence

in Eqs. (D.77) and (D.88) is carried by the Gamma function $\Gamma(n - d/2)$ (scalar numerator) or $\Gamma(n - k - d/2)$ (with a $(\ell^2)^k$ numerator). The Gamma function is meromorphic with simple poles at the non-positive integers, $\Gamma(z)$ blowing up at $z = 0, -1, -2, \ldots$. Hence the master integral develops a pole exactly when

$$n - k - \frac{d}{2} \;=\; -j\,, \qquad j = 0, 1, 2, \ldots\,, \qquad \text{i.e.} \qquad d \;=\; 2(n-k) + 2j\,. \tag{D.109}$$

Reading Eq. (D.109) the other way, at a *fixed* integer dimension d the integral $\int \mathrm{d}^d\ell\,(\ell^2)^k/(\ell^2 - \Delta)^n$ is ultraviolet divergent precisely when $d \geq 2(n-k)$ — which is the statement that its superficial degree of divergence

$$D \;=\; d + 2k - 2n \tag{D.110}$$

is non-negative, exactly the power counting of Section D.1 applied to a single loop. The integer j in Eq. (D.109) labels which subtraction the divergence requires: $j = 0$ ($D = 0$) is a logarithmic divergence, $j = 1$ ($D = 2$) a quadratic one, and so on. The poles in d are thus not artefacts of the continuation; they are the faithful image, in the analytic structure of the Γ-function, of the ultraviolet divergences that the integral genuinely has at integer dimension.

For the physically central case $d = 4$ this organizes the whole one-loop catalogue. Writing $d = 4 - 2\varepsilon$, the relevant Gamma function is $\Gamma(n - k - d/2) = \Gamma(n - k - 2 + \varepsilon)$, which has a pole as $\varepsilon \to 0$ whenever $n - k - 2 \leq 0$, i.e. $n - k \leq 2$. The three cases that exhaust the renormalizable one-loop divergences are tabulated by their effective propagator power $n - k$:

- $n - k = 1$: $\Gamma(-1 + \varepsilon)$, the quadratically divergent tadpole-type integral (A_0, Eq. (D.99); $D = 2$);
- $n - k = 2$: $\Gamma(\varepsilon)$, the logarithmically divergent bubble-type integral (B_0, Eq. (D.104); $D = 0$);
- $n - k \geq 3$: $\Gamma(1 + \varepsilon), \Gamma(2 + \varepsilon), \ldots$, all *finite* at $d = 4$ ($D < 0$, ultraviolet-convergent integrals such as the triangle and box).

This is the payoff of the whole construction. A divergent loop integral has been turned into a meromorphic function of d whose only singularity near the physical dimension is a single, explicit, isolated pole, with everything else — the powers of Δ, the parametric integral, the finite remainder — analytic and computable.

The master formula Eq. (D.77), the numerator extension Eq. (D.88), the symmetric-integration rules Eqs. (D.92) and (D.97), and the named functions A_0 and B_0 are the complete scalar toolkit at one loop. Each is now a product of Γ-functions times a power of Δ, with the entire ultraviolet divergence isolated in a single $\Gamma(n - k - d/2)$. What remains is to make that pole and its finite companion explicit: to Laurent-expand the master integral about $d = 4$, extract the universal $1/\varepsilon - \gamma_E + \ln 4\pi$ combination, and restore the dimensions with the scale μ. That expansion is the business of the next section.

D.5 Expansion in ε: poles, finite parts, and the meaning of the scale

The previous section closed the radial and angular integrations and handed us a single closed expression valid for general complex dimension: the master scalar one-loop integral

$$\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^n} \;=\; \frac{(-1)^n\,i}{(4\pi)^{d/2}}\,\frac{\Gamma\!\left(n-\frac{d}{2}\right)}{\Gamma(n)}\,\frac{1}{\Delta^{\,n-d/2}}, \tag{D.111}$$

established as Eq. (D.77), with $\Delta>0$ the Feynman-combined mass-squared and the Wick rotation $\ell^0=i\ell^0_E$ carried out there. This is exact, but it is also useless as it stands: it is a function of the continuous variable d, and physics lives at $d=4$, where the gamma function $\Gamma(n-d/2)$ has its poles for the very values of n that produced divergent integrals in the first place. To extract physics we must let $d\to 4$, that is, $\varepsilon\to 0$ in $d=4-2\varepsilon$, and watch the divergence resurface — now not as an ill-defined integral but as a pole in ε multiplying a perfectly finite remainder. The whole content of dimensional regularization is in this controlled limit. Performing it requires only two pieces of elementary analysis: the Laurent expansion of the gamma function about its poles, and the Taylor expansion of the two innocuous-looking powers $(4\pi)^{\varepsilon}$ and $\Delta^{-\varepsilon}$. Assembling them exposes a universal divergent combination that is the same in every theory and at every loop order, isolates it into a counterterm by the $\overline{\mathrm{MS}}$ prescription fixed in Chapter A, and leaves behind a finite logarithm of Δ whose branch cut — when Δ turns negative above a physical threshold — is the imaginary part demanded by unitarity. We carry all of this out in full, deriving the gamma-function expansions from first principles rather than quoting them, and finish by completing the ϕ^4 bubble to its renormalized $\overline{\mathrm{MS}}$ value.

D.5.1 The gamma function near its poles

Everything in the ε-expansion is downstream of one analytic fact: how $\Gamma(z)$ behaves near the nonpositive integers, where it has its only singularities. We need the expansion of $\Gamma(\varepsilon)$ about $\varepsilon=0$ (for the logarithmically divergent integrals, $n-d/2=\varepsilon$), of $\Gamma(-1+\varepsilon)$ (for the quadratically divergent ones, where $n-d/2=-1+\varepsilon$), and, once, the general $\Gamma(-m+\varepsilon)$ recursion that subsumes both. We derive them from the Weierstrass product, introducing the Euler–Mascheroni constant and the digamma function as we go, since both appear in the answers.

Definition D.26 (Euler–Mascheroni constant and the digamma function)**.** The *Euler–Mascheroni constant* is the limit

$$\gamma_{\mathrm{E}} \;\equiv\; \lim_{N\to\infty}\Big(\sum_{n=1}^{N}\frac{1}{n}-\ln N\Big) \;=\; 0.5772156649\ldots, \tag{D.112}$$

the finite excess of the harmonic sum over the logarithm it asymptotes to. The *digamma function* is the logarithmic derivative of the gamma function,

$$\psi(z) \;\equiv\; \frac{\mathrm{d}}{\mathrm{d}z}\ln\Gamma(z) \;=\; \frac{\Gamma'(z)}{\Gamma(z)}. \tag{D.113}$$

The link between the two is supplied by the Weierstrass product representation of $1/\Gamma$, which is the cleanest starting point because it displays all the poles of Γ at once as the zeros of its reciprocal.

Lemma D.27 (Log-derivative of the Weierstrass product)**.** *The reciprocal gamma function admits the absolutely convergent product*

$$\frac{1}{\Gamma(z)} \;=\; z\,e^{\gamma_{\mathrm{E}} z}\prod_{n=1}^{\infty}\Big(1+\frac{z}{n}\Big)\,e^{-z/n}, \tag{D.114}$$

entire in z, whose zeros at $z=0,-1,-2,\dots$ are the poles of Γ. Consequently the digamma function has the expansion

$$\psi(z) \;=\; -\gamma_{\mathrm{E}} - \frac{1}{z} + \sum_{n=1}^{\infty}\Big(\frac{1}{n}-\frac{1}{n+z}\Big), \tag{D.115}$$

so that in particular

$$\psi(1)=-\gamma_{\mathrm{E}}, \qquad \psi(m{+}1)=-\gamma_{\mathrm{E}}+\sum_{k=1}^{m}\frac{1}{k}=-\gamma_{\mathrm{E}}+H_m, \qquad \psi'(1)=\sum_{n=1}^{\infty}\frac{1}{n^2}=\frac{\pi^2}{6}, \tag{D.116}$$

where $H_m=\sum_{k=1}^{m}1/k$ is the mth harmonic number.

Proof. We take the product (D.114) as the definition of the entire function $1/\Gamma$; that it reproduces $\Gamma(z+1)=z\Gamma(z)$ and $\Gamma(1)=1$, and hence agrees with the gamma function, is the standard verification we recall here. Taking the logarithm of (D.114),

$$-\ln\Gamma(z)=\ln z+\gamma_{\mathrm{E}} z+\sum_{n=1}^{\infty}\Big[\ln\Big(1+\frac{z}{n}\Big)-\frac{z}{n}\Big], \tag{D.117}$$

and differentiating term by term, which is legitimate because the differentiated series $\sum_n\big(\frac{1}{n}-\frac{1}{n+z}\big)=\sum_n\frac{z}{n(n+z)}$ converges absolutely and uniformly on compacta away from the poles, gives

$$-\psi(z)=\frac{1}{z}+\gamma_{\mathrm{E}}+\sum_{n=1}^{\infty}\Big(\frac{1}{n+z}-\frac{1}{n}\Big), \tag{D.118}$$

which rearranges to (D.115). Setting $z=1$ collapses the telescoping sum $\sum_n(\frac{1}{n}-\frac{1}{n+1})=1$, leaving $\psi(1)=-\gamma_{\mathrm{E}}-1+1=-\gamma_{\mathrm{E}}$. For $\psi(m+1)$, the recursion $\psi(z+1)=\psi(z)+1/z$ (the log-derivative of $\Gamma(z+1)=z\Gamma(z)$) iterated m times from $z=1$ adds $\sum_{k=1}^{m}1/k=H_m$ to $\psi(1)$. Finally, differentiating (D.115) once more gives $\psi'(z)=\sum_{n=0}^{\infty}(n+z)^{-2}$, and at $z=1$ this is the Basel sum $\sum_{n\ge1}n^{-2}=\pi^2/6$. $\square$

With the digamma data in hand the Laurent expansion of Γ near $\varepsilon = 0$ follows by exponentiating its Taylor series at $z = 1$, which is regular there.

Proposition D.28 (Laurent expansion of $\Gamma(\varepsilon)$ and $\Gamma(1+\varepsilon)$)**.** *About $\varepsilon = 0$,*

$$\Gamma(1+\varepsilon) = 1 - \gamma_{\mathrm{E}}\,\varepsilon + \tfrac{1}{2}\Big(\gamma_{\mathrm{E}}^2 + \frac{\pi^2}{6}\Big)\varepsilon^2 + O(\varepsilon^3), \tag{D.119}$$

$$\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \tfrac{1}{2}\Big(\gamma_{\mathrm{E}}^2 + \frac{\pi^2}{6}\Big)\varepsilon + O(\varepsilon^2). \tag{D.120}$$

Proof. Because Γ is analytic and nonzero at $z = 1$ with $\Gamma(1) = 1$, the function $\ln\Gamma(1+\varepsilon)$ is analytic at $\varepsilon = 0$ and vanishes there, so it equals its Taylor series

$$\ln\Gamma(1+\varepsilon) = \psi(1)\,\varepsilon + \tfrac{1}{2}\,\psi'(1)\,\varepsilon^2 + O(\varepsilon^3) = -\gamma_{\mathrm{E}}\,\varepsilon + \frac{\pi^2}{12}\,\varepsilon^2 + O(\varepsilon^3), \tag{D.121}$$

using (D.116). (More generally the coefficients are the polygamma values $\psi^{(k-1)}(1)/k!$, and one shows $\ln\Gamma(1+\varepsilon) = -\gamma_{\mathrm{E}}\varepsilon + \sum_{k\geq 2}\frac{(-1)^k\zeta(k)}{k}\varepsilon^k$ with $\zeta(2) = \pi^2/6$, $\zeta(3)$ entering at the next order; we will not need beyond ε^2 until two-loop products, where the $O(\varepsilon)$ term of one factor multiplies the $1/\varepsilon$ pole of another.) Exponentiating (D.121),

$$\begin{aligned}\Gamma(1+\varepsilon) &= \exp\Big[-\gamma_{\mathrm{E}}\varepsilon + \frac{\pi^2}{12}\varepsilon^2 + O(\varepsilon^3)\Big] \\ &= 1 + \Big(-\gamma_{\mathrm{E}}\varepsilon + \frac{\pi^2}{12}\varepsilon^2\Big) + \tfrac{1}{2}\big(-\gamma_{\mathrm{E}}\varepsilon\big)^2 + O(\varepsilon^3) \\ &= 1 - \gamma_{\mathrm{E}}\varepsilon + \Big(\frac{\gamma_{\mathrm{E}}^2}{2} + \frac{\pi^2}{12}\Big)\varepsilon^2 + O(\varepsilon^3),\end{aligned} \tag{D.122}$$

which is (D.119) once $\pi^2/12 = \frac{1}{2}\cdot\pi^2/6$ is recognized inside the bracket. Dividing by ε through the functional equation $\Gamma(\varepsilon) = \Gamma(1+\varepsilon)/\varepsilon$ shifts every power down by one and produces the simple pole:

$$\Gamma(\varepsilon) = \frac{\Gamma(1+\varepsilon)}{\varepsilon} = \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \Big(\frac{\gamma_{\mathrm{E}}^2}{2} + \frac{\pi^2}{12}\Big)\varepsilon + O(\varepsilon^2), \tag{D.123}$$

which is (D.120). □

The leading two terms of (D.120) are the ones quoted as precise statements in Eq. (A.201); Theorem D.28 is the derivation that backs that quotation, and it carries one order further because the $O(\varepsilon)$ term of $\Gamma(\varepsilon)$ multiplies the $1/\varepsilon$ pole of a second factor in any two-loop product and so contributes to one-loop *finite parts* whenever a double pole is present. The same machinery handles the higher poles. Quadratic divergences, and more generally any integral with $n - d/2 = -m + \varepsilon$ for a nonnegative integer m, need Γ near a negative integer.

Proposition D.29 (The $\Gamma(-m+\varepsilon)$ recursion)**.** *For every integer $m \geq 0$,*

$$\Gamma(-m+\varepsilon) = \frac{(-1)^m}{m!}\Big[\frac{1}{\varepsilon} + \psi(m+1) + O(\varepsilon)\Big] = \frac{(-1)^m}{m!}\Big[\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + H_m + O(\varepsilon)\Big], \tag{D.124}$$

with $H_0 \equiv 0$. In particular $m = 0$ reproduces (D.120), and

$$\Gamma(-1+\varepsilon) = -\frac{1}{\varepsilon} - 1 + \gamma_{\mathrm{E}} + O(\varepsilon), \qquad \Gamma(-2+\varepsilon) = \frac{1}{2}\Big(\frac{1}{\varepsilon} + \frac{3}{2} - \gamma_{\mathrm{E}}\Big) + O(\varepsilon). \tag{D.125}$$

Proof. Iterating $\Gamma(z+1) = z\Gamma(z)$ downward from $z = \varepsilon$ a total of m times,

$$\Gamma(\varepsilon) = (\varepsilon-1)(\varepsilon-2)\cdots(\varepsilon-m)\,\Gamma(\varepsilon-m), \tag{D.126}$$

so that

$$\Gamma(-m+\varepsilon) = \frac{\Gamma(\varepsilon)}{\prod_{j=1}^{m}(\varepsilon-j)}. \tag{D.127}$$

Factor each denominator term as $\varepsilon - j = (-j)(1-\varepsilon/j)$, so the product is

$$\prod_{j=1}^{m}(\varepsilon-j) = (-1)^m\, m! \prod_{j=1}^{m}\Big(1-\frac{\varepsilon}{j}\Big) = (-1)^m\, m!\,\Big(1-\varepsilon\sum_{j=1}^{m}\frac{1}{j}+O(\varepsilon^2)\Big) = (-1)^m\, m!\,\big(1-\varepsilon H_m+O(\varepsilon^2)\big). \tag{D.128}$$

Its reciprocal is $\dfrac{(-1)^m}{m!}\big(1+\varepsilon H_m + O(\varepsilon^2)\big)$. Multiplying by $\Gamma(\varepsilon) = \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + O(\varepsilon)$ from (D.120),

$$\Gamma(-m+\varepsilon) = \frac{(-1)^m}{m!}\Big(\frac{1}{\varepsilon}-\gamma_{\mathrm{E}}+O(\varepsilon)\Big)\big(1+\varepsilon H_m+O(\varepsilon^2)\big) = \frac{(-1)^m}{m!}\Big(\frac{1}{\varepsilon}-\gamma_{\mathrm{E}}+H_m+O(\varepsilon)\Big), \tag{D.129}$$

where the cross term $\frac{1}{\varepsilon}\cdot\varepsilon H_m = H_m$ has joined the constant. Recognizing $-\gamma_{\mathrm{E}} + H_m = \psi(m+1)$ from (D.116) gives the first form in (D.124). Setting $m = 1$ ($H_1 = 1$) yields $\Gamma(-1+\varepsilon) = -(\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + 1)$, the first equation of (D.125); setting $m = 2$ ($H_2 = \frac{3}{2}$) yields the second. □

Remark D.30 (Why the pole structure is universal). Equation (D.124) contains the whole reason dimensional regularization is so economical. The single gamma function $\Gamma(n - d/2)$ in the master integral (D.111) carries *all* the ultraviolet information: its pole position records how badly the integral diverges (a simple pole at every even d for logarithmic divergences, the negative-integer poles for power divergences), and its residue is a pure rational number times the universal $-\gamma_{\mathrm{E}}$. No physics of the particular diagram — no masses, no external momenta — enters the pole; all of that sits in the accompanying $\Delta^{-(n-d/2)}$, which is finite at $\varepsilon = 0$ and contributes only to the finite part. This clean separation, divergence in Γ and physics in $\Delta^{-\varepsilon}$, is what makes the pole part of every one-loop integral computable by inspection.

D.5.2 The prefactor expansions

The gamma function supplies the pole; the two power-law prefactors in (D.111) — the $(4\pi)^{-d/2}$ from the angular measure and the $\Delta^{-(n-d/2)}$ from the radial integral — supply, on expansion, the constants and logarithms that ride on it. Both are handled by the elementary identity $a^{\varepsilon} = e^{\varepsilon \ln a} = 1 + \varepsilon \ln a + \tfrac{1}{2}\varepsilon^2 \ln^2 a + \cdots$, valid for any $a > 0$. Writing $d/2 = 2 - \varepsilon$, the measure factor is

$$\frac{1}{(4\pi)^{d/2}} = \frac{1}{(4\pi)^2}\,(4\pi)^{\varepsilon} = \frac{1}{(4\pi)^2}\Big(1 + \varepsilon \ln 4\pi + \tfrac{1}{2}\varepsilon^2 \ln^2 4\pi + O(\varepsilon^3)\Big), \tag{D.130}$$

exposing the constant $\ln 4\pi$ that will pair with the $1/\varepsilon$ pole. The radial factor, for the logarithmic case $n - d/2 = \varepsilon$, expands as

$$\Delta^{-\varepsilon} = \Big(\frac{\Delta}{\mu^2}\Big)^{-\varepsilon} \mu^{-2\varepsilon} = \mu^{-2\varepsilon}\Big(1 - \varepsilon \ln \frac{\Delta}{\mu^2} + \tfrac{1}{2}\varepsilon^2 \ln^2 \frac{\Delta}{\mu^2} + O(\varepsilon^3)\Big), \tag{D.131}$$

where we have deliberately inserted the renormalization scale μ of Theorem A.46 by writing $\Delta^{-\varepsilon} = (\Delta/\mu^2)^{-\varepsilon}\,\mu^{-2\varepsilon}$. This is not cosmetic. The quantity Δ has mass dimension two, so $\ln \Delta$ is the logarithm of a dimensionful quantity — a meaningless object until a scale is named. The factor $\mu^{2\varepsilon}$ that the coupling promotion of Theorem A.46 attaches to every loop is exactly what combines with $\mu^{-2\varepsilon}$ here to render the argument of the surviving logarithm the dimensionless ratio Δ/μ^2. We make that cancellation explicit in Section D.5.4; for now the point is bookkeeping discipline, that $\Delta^{-\varepsilon}$ never appears without its compensating $\mu^{2\varepsilon}$, and only their product, $(\Delta/\mu^2)^{-\varepsilon}$, has an ε-expansion whose logarithms are well defined. The generic positive power needed for arbitrary n is the same exercise,

$$\Big(\frac{\Delta}{\mu^2}\Big)^{-(n-d/2)} = \Big(\frac{\Delta}{\mu^2}\Big)^{2-n}\Big(\frac{\Delta}{\mu^2}\Big)^{-\varepsilon} = \Big(\frac{\Delta}{\mu^2}\Big)^{2-n}\Big(1 - \varepsilon \ln \frac{\Delta}{\mu^2} + O(\varepsilon^2)\Big), \tag{D.132}$$

the integer power $(\Delta/\mu^2)^{2-n}$ surviving to ε^0 and the ε-expansion riding on it as before.

D.5.3 The universal divergent combination and $\overline{\text{MS}}$

We can now assemble the three expansions. Specialize the master integral (D.111) to the logarithmically divergent case $n = 2$, where $\Gamma(n) = \Gamma(2) = 1$ and $n - d/2 = \varepsilon$, and reinstate the loop's compensating scale $\mu^{2\varepsilon}$:

$$\mu^{2\varepsilon} \int \frac{\mathrm{d}^d \ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^2} = \frac{i}{(4\pi)^2}\,\underbrace{\mu^{2\varepsilon}\,(4\pi)^{\varepsilon}\,\Gamma(\varepsilon)\,\Delta^{-\varepsilon}}_{=\ \Gamma(\varepsilon)\,(4\pi\mu^2/\Delta)^{\varepsilon}}. \tag{D.133}$$

The braced product is dimensionless — every factor of mass has cancelled — and its ε-expansion is the single most important computation of this appendix.

Multiplying (D.120) by the expansion of $(4\pi\mu^2/\Delta)^\varepsilon = 1 + \varepsilon \ln(4\pi\mu^2/\Delta) + O(\varepsilon^2)$ and keeping the finite part,

$$\begin{aligned}\Gamma(\varepsilon)\Big(\frac{4\pi\mu^2}{\Delta}\Big)^{\varepsilon} &= \Big(\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + O(\varepsilon)\Big)\Big(1 + \varepsilon \ln\frac{4\pi\mu^2}{\Delta} + O(\varepsilon^2)\Big) \\ &= \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln\frac{4\pi\mu^2}{\Delta} + O(\varepsilon) \\ &= \underbrace{\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi}_{\equiv\, 1/\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta} + O(\varepsilon). \end{aligned} \tag{D.134}$$

The structure of (D.134) is the lesson. The pole $1/\varepsilon$ never appears alone: it is rigidly accompanied by the two pure numbers $-\gamma_{\mathrm{E}}$ (from the gamma function) and $\ln 4\pi$ (from the angular measure), in a combination that is identical for every theory, every diagram, and every loop order, because it descends from the same two universal sources — the residue of Γ and the $(4\pi)^{d/2}$ of the d-dimensional sphere — and not from any physics. These constants are an artifact of the regulator. Carrying them through every calculation, only to have them clutter every finite answer, is a waste; the sensible thing is to bundle them into the counterterm together with the pole. That bundling is precisely the modified minimal-subtraction prescription fixed for the whole book in Chapter A: by Theorem A.49 the $\overline{\mathrm{MS}}$ counterterm removes not the bare pole but the combination

$$\frac{1}{\bar{\varepsilon}} \;\equiv\; \frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi, \tag{D.135}$$

which is Eq. (A.204), the universal bracket isolated in (D.134).

We keep $\bar{\varepsilon}$, not a new symbol

The combination $\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi$ recurs so relentlessly that it deserves a name, and the book's name for it is $1/\bar{\varepsilon}$, fixed in Eq. (A.204). We use that symbol here and nowhere introduce a second one for the same object; in particular the letter Δ in these sections always means the Feynman-combined mass-squared of (D.111), never the divergent combination, so that no Δ_ε shorthand is used.

The division of (D.134) into a universal piece and a physics-dependent piece is sharp and worth isolating as the operational statement of one-loop renormalization.

Proposition D.31 (Universal pole, scheme-dependent finite part)**.** *The ε-expansion* (D.134) *of the logarithmic master integral* (D.133) *splits as*

$$\mu^{2\varepsilon}\int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2 - \Delta + i\epsilon)^2} = \frac{i}{(4\pi)^2}\Big[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta}\Big] + O(\varepsilon), \tag{D.136}$$

in which the $1/\bar{\varepsilon}$ pole part is independent of the kinematics Δ and is the same in every renormalizable theory, while the finite part $\ln(\mu^2/\Delta)$ carries all the dependence on masses and external momenta and is what survives $\overline{MS}$ subtraction. Equivalently, in the $\bar{\mu}$ variable of Theorem A.50, $\bar{\mu}^2 = 4\pi\mu^2 e^{-\gamma_{\mathrm{E}}}$, the right-hand side reads $\frac{i}{(4\pi)^2}\left[\frac{1}{\varepsilon} + \ln(\bar{\mu}^2/\Delta)\right]$, the bare pole accompanied by a logarithm of the modified scale.

Proof. Insert (D.134) into the right-hand side of (D.133) and read off the two pieces by the definition (D.135) of $1/\bar{\varepsilon}$. The $\bar{\mu}$ form is the algebraic identity $\frac{1}{\varepsilon} - \gamma_{\mathrm{E}} + \ln 4\pi + \ln(\mu^2/\Delta) = \frac{1}{\varepsilon} + \ln(\bar{\mu}^2/\Delta)$ established in Theorem A.50, with $\ln \bar{\mu}^2 = \ln 4\pi - \gamma_{\mathrm{E}} + \ln \mu^2$. □

Why the pole must be universal and the finite part must not be

The split in Theorem D.31 is not an accident of the calculation but a structural feature with physical meaning, and it explains why $\overline{\mathrm{MS}}$ is a sensible thing to do at all. A counterterm is a local term added to the Lagrangian; it is a polynomial in fields and momenta with no logarithms, because the Lagrangian is local. The divergent part of a diagram must therefore be a pure pole times a polynomial — it cannot contain $\ln \Delta$, since Δ depends on external momenta in a way no local counterterm could reproduce. This is why the $\ln(\mu^2/\Delta)$ in (D.136) is automatically finite and the divergence is automatically a kinematics-free constant: locality forces it. The freedom in a renormalization scheme is exactly the freedom to move a *constant* (here $-\gamma_{\mathrm{E}} + \ln 4\pi$, or any other) from the finite part into the counterterm, since a constant is a legitimate local term; what no scheme can touch is the logarithm, whose μ-dependence is therefore physical and is precisely what the renormalization group of Chapter 24 tracks. The pole is universal because locality says it must be; the finite part is scheme-dependent because constants are ambiguous; the logarithm is sacred because it is the only piece that carries genuine physics.

D.5.4 The fully expanded scalar bubble

We now run the entire machine on the integral the loop chapters meet first: the scalar two-point bubble. It is the $n = 2$ master integral with a Feynman parameter not yet integrated, and exhibiting the $\mu^{2\varepsilon}$ cancellation in full is the cleanest demonstration of the scale's role.

Consider the scalar bubble with two propagators of mass m and external momentum p flowing through,

$$B(p^2) \;\equiv\; \mu^{2\varepsilon}\int \frac{\mathrm{d}^d k}{(2\pi)^d}\,\frac{1}{(k^2 - m^2 + i\epsilon)\big((k+p)^2 - m^2 + i\epsilon\big)}. \tag{D.137}$$

Combine the two denominators with the Feynman-parameter identity Eq. (D.17), $\frac{1}{AB} = \int_0^1 \mathrm{d}x\,[xA+(1-x)B]^{-2}$, and complete the square. With $A=(k+p)^2-m^2$ and $B=k^2-m^2$,

$$xA+(1-x)B = k^2+2x\,k\cdot p+xp^2-m^2 = (k+xp)^2-\big[m^2-x(1-x)p^2\big], \tag{D.138}$$

so shifting $\ell = k+xp$ (a translation of a dimensionally regularized integral, hence with unit Jacobian and no surface term — a property established in Section D.4) brings (D.137) to the standard form

$$B(p^2) = \mu^{2\varepsilon}\int_0^1 \mathrm{d}x \int \frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^2}, \qquad \Delta = m^2-x(1-x)\,p^2. \tag{D.139}$$

The inner integral is exactly (D.136). Inserting it and pulling the ε-expansion through the (finite, ε-independent) Feynman integral,

$$B(p^2) = \frac{i}{(4\pi)^2}\int_0^1 \mathrm{d}x\Big[\frac{1}{\bar\varepsilon}+\ln\frac{\mu^2}{\Delta}\Big]+O(\varepsilon) = \frac{i}{(4\pi)^2}\Big[\frac{1}{\bar\varepsilon}-\int_0^1 \mathrm{d}x\,\ln\frac{\Delta}{\mu^2}\Big]+O(\varepsilon), \tag{D.140}$$

where $\int_0^1 \mathrm{d}x = 1$ has turned the x-independent pole into a bare $1/\bar\varepsilon$ and $\ln(\mu^2/\Delta) = -\ln(\Delta/\mu^2)$ has been flipped to display the standard form. This is the canonical one-loop bubble: a universal $1/\bar\varepsilon$ pole plus the finite part $-\int_0^1 \mathrm{d}x\,\ln(\Delta/\mu^2)$ with $\Delta = m^2-x(1-x)p^2$. Every two-point function in the book is some dressing of (D.140).

Two features deserve emphasis. First, the role of $\mu^{2\varepsilon}$ is now completely explicit: had we omitted it, (D.140) would contain $\ln\Delta$ — the logarithm of a mass-squared, dimensionally illegal — and the ε-expansion would not even be definable, since one cannot Taylor-expand $\Delta^{-\varepsilon}$ about a dimensionful Δ. The single factor $\mu^{2\varepsilon}$, forced on us by coupling promotion (Theorem A.46) and nothing else, converts every loop logarithm into a logarithm of a ratio. The scale μ is not inserted by hand to fix dimensions; it is already present, carried in by the coupling, and the ε-expansion simply reveals it. Second, the finite part is a genuine, computable function of p^2. The Feynman integral $\int_0^1 \mathrm{d}x\,\ln[m^2-x(1-x)p^2]$ can be done in closed form, but it is the input to physics already as a one-dimensional integral, and we evaluate it explicitly for the model next.

Worked Example D.32 (The ϕ^4 bubble to its $\overline{\text{MS}}$ finite part)**.** We complete ϕ^4 theory at the one-loop level, taking up the s-channel bubble whose divergent part was extracted at convention-fixing level in Theorem A.52; here we carry it to the full finite $\overline{\text{MS}}$ result, including its evaluation in closed form and its behaviour above threshold. The model is the real scalar $\mathcal{L} = \frac{1}{2}(\partial\phi)^2 - \frac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4$.

The amputated one-loop correction to the four-point function in the s-channel ($s=p^2$ the squared total incoming momentum) is, with two vertices $-i\lambda\mu^{2\varepsilon}$ and symmetry factor $\frac{1}{2}$ and the two propagator numerators contributing $i\cdot i=-1$,

$$i\mathcal{M}_s = \frac{(-i\lambda\mu^{2\varepsilon})^2}{2}\int\frac{\mathrm{d}^dk}{(2\pi)^d}\,\frac{i}{k^2-m^2}\,\frac{i}{(k+p)^2-m^2} = \frac{\lambda^2\mu^{2\varepsilon}}{2}\,B(s), \tag{D.141}$$

the last equality recognizing the integral as $\mu^{-2\varepsilon}\,B(s)\cdot\mu^{2\varepsilon}$, i.e. exactly the bubble (D.137) after one factor $\mu^{2\varepsilon}$ has been kept outside to match the dimension of the tree vertex. Substituting (D.140),

$$i\mathcal{M}_s = \frac{i\lambda^2\mu^{2\varepsilon}}{2(4\pi)^2}\Big[\frac{1}{\bar\varepsilon} - \int_0^1 \mathrm{d}x\,\ln\frac{m^2 - x(1-x)s}{\mu^2}\Big] + O(\varepsilon). \tag{D.142}$$

The $\overline{\text{MS}}$ counterterm removes the $1/\bar\varepsilon$ by construction (Theorem A.49); the renormalized s-channel contribution is the finite remainder

$$i\mathcal{M}_s^{\overline{\text{MS}}} = -\frac{i\lambda^2}{32\pi^2}\int_0^1 \mathrm{d}x\,\ln\frac{m^2 - x(1-x)s}{\mu^2}, \tag{D.143}$$

using $(4\pi)^2 = 16\pi^2$ so that $1/2(4\pi)^2 = 1/32\pi^2$ and setting $\mu^{2\varepsilon}\to 1$ in the now-finite expression.

It remains to do the x-integral. Write $F(s)\equiv\int_0^1 \mathrm{d}x\,\ln[(m^2 - x(1-x)s)/\mu^2]$. Factor the quadratic $m^2 - x(1-x)s = s\,(x-x_+)(x-x_-)\cdot(-1) + \ldots$; more directly, complete the square in the argument, $m^2 - x(1-x)s = \frac{s}{4}\big[(2x-1)^2 + (4m^2/s - 1)\big]$, and substitute $u = 2x-1\in[-1,1]$, $\mathrm{d}x = \frac{1}{2}\mathrm{d}u$:

$$F(s) = \frac{1}{2}\int_{-1}^{1}\mathrm{d}u\,\ln\frac{s}{4\mu^2}\Big[u^2 + \Big(\frac{4m^2}{s} - 1\Big)\Big] = \ln\frac{s}{4\mu^2} + \frac{1}{2}\int_{-1}^{1}\mathrm{d}u\,\ln\left(u^2 + a^2\right), \tag{D.144}$$

with $a^2 \equiv 4m^2/s - 1$. The remaining integral is elementary,

$$\int \mathrm{d}u\,\ln(u^2 + a^2) = u\ln(u^2 + a^2) - 2u + 2a\arctan\frac{u}{a}, \tag{D.145}$$

verified by differentiation, so between ± 1 (the integrand is even, so the result is twice the value of the antiderivative's odd-in-u pieces at $u = 1$),

$$\frac{1}{2}\int_{-1}^{1}\mathrm{d}u\,\ln(u^2 + a^2) = \ln(1 + a^2) - 2 + 2a\arctan\frac{1}{a}. \tag{D.146}$$

Restoring $a^2 = 4m^2/s - 1$, hence $1 + a^2 = 4m^2/s$, and assembling,

$$F(s) = \ln\frac{s}{4\mu^2} + \ln\frac{4m^2}{s} - 2 + 2\sqrt{\frac{4m^2}{s} - 1}\,\arctan\frac{1}{\sqrt{4m^2/s - 1}} = \ln\frac{m^2}{\mu^2} - 2 + 2\,a\arctan\frac{1}{a}, \tag{D.147}$$

the two scale-logarithms combining to the clean $\ln(m^2/\mu^2)$. The renormalized one-loop s-channel amplitude is therefore

$$i\mathcal{M}_s^{\overline{\text{MS}}} = -\frac{i\lambda^2}{32\pi^2}\left[\ln\frac{m^2}{\mu^2} - 2 + 2\sqrt{\tfrac{4m^2}{s} - 1}\,\arctan\frac{1}{\sqrt{4m^2/s - 1}}\right], \tag{D.148}$$

valid for $s < 4m^2$, where $a^2 = 4m^2/s - 1 > 0$ and the arctan is real. The full one-loop vertex is the sum over the three channels s, t, u, each of this form with s replaced by the corresponding Mandelstam invariant, as recorded in Eq. (A.217).

Two carried numbers fall out. The pole of the *summed* amplitude is three times that of (D.142),

$$i\mathcal{M}^{\text{1-loop}}\big|_{\text{pole}} = \frac{3i\lambda^2}{32\pi^2}\,\frac{1}{\bar{\varepsilon}} = \frac{3i\lambda^2}{16\pi^2}\,\frac{1}{2\bar{\varepsilon}}, \tag{D.149}$$

whose residue is the input to the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ extracted in Chapter 24: demanding μ-independence of the physical amplitude converts the coefficient of $1/\bar{\varepsilon}$ here into the slope of λ in $\ln\mu$. And the finite part (D.148) furnishes the explicit one-loop logarithm $-\frac{\lambda^2}{32\pi^2}\ln(m^2/\mu^2)$ per channel that the running-coupling solution of Chapter 24 resums. We have thus advanced ϕ^4 theory from a formal Feynman integral to a numerically explicit renormalized amplitude.

The same scalar bubble, dressed with a numerator inherited from the electron loop, is the integral behind the QED vacuum polarization $\Pi(q^2)$ of QED, and every tool needed to expand it is now in hand. The fermion loop produces a rank-two numerator $\ell^\mu\ell^\nu$, but with a single external momentum it is reduced by the symmetric-integration rule Eq. (D.92) of Section D.4 — no full tensor reduction is required — while the Dirac trace that generates it is the owned computation of Chapter 19. We finish the carried QED loop integral here, completing the pole and logarithm of $\Pi(q^2)$ that Chapters 19 and 24 consume.

Worked Example D.33 (QED: the vacuum-polarization pole and logarithm)**.** We continue QED. The one-loop photon self-energy is the single fermion bubble; by the Ward identity proved in Chapter 19 it is transverse,

$$i\Pi^{\mu\nu}(q) = (q^2\eta^{\mu\nu} - q^\mu q^\nu)\, i\Pi(q^2), \tag{D.150}$$

so the would-be quadratic divergence that naive power counting assigns to the diagram cancels and all the content sits in the scalar $\Pi(q^2)$. Carrying out the Dirac trace over the numerator $\mathrm{tr}\big[\gamma^\mu(\not{\ell}+\not{q}+m)\gamma^\nu(\not{\ell}+m)\big]$ with the identities of Section C.6, combining the two denominators with a Feynman parameter exactly as in (D.138), shifting the loop momentum, and reducing the resulting $\ell^\mu\ell^\nu$ by the symmetric-integration replacement Eq. (D.92) — all carried out in full in Chapter 19 — collapse the transverse scalar onto the $\mu^{2\varepsilon}$-dressed bubble carrying the Feynman-parameter weight $x(1-x)$ that the trace supplies,

$$i\Pi(q^2) = -8e^2\,\mu^{2\varepsilon}\int_0^1 \mathrm{d}x\, x(1-x)\int\frac{\mathrm{d}^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2-\Delta+i\epsilon)^2}, \qquad \Delta = m^2 - x(1-x)\,q^2. \tag{D.151}$$

The only new feature relative to the ϕ^4 bubble is the polynomial weight $x(1-x)$; the divergence is untouched by it. The inner integral with its compensating $\mu^{2\varepsilon}$ is precisely the logarithmic master (D.136), so

$$i\Pi(q^2) = -8e^2\,\frac{i}{(4\pi)^2}\int_0^1 \mathrm{d}x\, x(1-x)\Big[\frac{1}{\bar{\varepsilon}} + \ln\frac{\mu^2}{\Delta}\Big] + O(\varepsilon), \tag{D.152}$$

and stripping the overall i and writing $8e^2/(4\pi)^2 = e^2/2\pi^2 = 2\alpha/\pi$ with $\alpha = e^2/4\pi$,

$$\Pi(q^2) = -\frac{2\alpha}{\pi}\int_0^1 \mathrm{d}x\, x(1-x)\Big[\frac{1}{\bar{\varepsilon}} - \ln\frac{\Delta}{\mu^2}\Big] + O(\varepsilon). \tag{D.153}$$

Both carried QED numbers fall out. With $\int_0^1 x(1-x)\,\mathrm{d}x = \frac{1}{6}$ the pole is

$$\Pi(q^2)\big|_{\text{pole}} = -\frac{2\alpha}{\pi}\cdot\frac{1}{6}\cdot\frac{1}{\varepsilon} = -\frac{\alpha}{3\pi}\frac{1}{\varepsilon}, \tag{D.154}$$

the celebrated $-\alpha/3\pi$ residue of QED vacuum polarization, removed by the photon field-strength counterterm δ_3 of Chapter 19 and converted by Chapter 24 into the QED beta function $\beta_e = e^3/12\pi^2$. The renormalized finite part in $\overline{\text{MS}}$,

$$\widehat{\Pi}(q^2) = \frac{2\alpha}{\pi}\int_0^1 \mathrm{d}x\, x(1-x)\,\ln\frac{m^2 - x(1-x)q^2}{\mu^2}, \tag{D.155}$$

is the scale-dependent screening that turns the Lagrangian charge e into the running coupling $\alpha(q^2)$ assembled in Chapters 19 and 24. The carried QED loop integral is thus delivered as a universal pole plus an explicit logarithm, with the same threshold branch cut at $q^2 = 4m^2$ studied below — there marking the onset of real e^+e^- pair production.

D.5.5 The imaginary part and the branch cut

The finite logarithm $\ln(\Delta/\mu^2)$ in (D.140) hides a discontinuity that is invisible at $\Delta > 0$ but is the seat of all the physics of particle production. The Feynman-combined mass-squared $\Delta = m^2 - x(1-x)p^2$ is positive when p^2 is small or spacelike, but for timelike p^2 large enough it can turn negative: since $x(1-x)$ ranges up to $\frac{1}{4}$ at $x = \frac{1}{2}$, Δ first reaches zero, for some x, when $p^2 = 4m^2$ — the two-particle production threshold. Above it, $\ln\Delta$ is the logarithm of a negative number and acquires an imaginary part. Which imaginary part is not a matter of choice: it is fixed by the $i\epsilon$ carried from the propagators all the way through the calculation.

Proposition D.34 (Branch of the logarithm and the discontinuity)**.** *Tracking the Feynman $i\epsilon$ of (D.111) through the ε-expansion, the mass-squared enters the finite part as $\Delta - i\epsilon$, and the logarithm is the boundary value of the principal branch approached from below the cut. Hence*

$$\ln(\Delta - i\epsilon) = \ln|\Delta| - i\pi\,\theta(-\Delta), \qquad \textit{i.e.} \qquad \operatorname{Im}\ln(\Delta - i\epsilon) = -\pi\,\theta(-\Delta), \tag{D.156}$$

and the discontinuity of the logarithm across the negative-Δ axis is

$$\operatorname{disc}_\Delta \ln\Delta \;\equiv\; \ln(\Delta - i\epsilon) - \ln(\Delta + i\epsilon) = -2\pi i\,\theta(-\Delta). \tag{D.157}$$

Proof. The master integral (D.111) carries the propagator prescription $\ell^2-\Delta+i\epsilon$, which after the Wick rotation and the radial integration of Section D.4 leaves the dependence $\Delta^{-(n-d/2)} \to (\Delta - i\epsilon)^{-(n-d/2)}$: the $+i\epsilon$ on the denominator is equivalent, for the purpose of the resulting power of Δ, to giving Δ a negative infinitesimal imaginary part, because $\ell^2 - \Delta + i\epsilon = \ell^2 - (\Delta - i\epsilon)$. Expanding $(\Delta - i\epsilon)^{-\varepsilon} = 1 - \varepsilon \ln(\Delta - i\epsilon) + O(\varepsilon^2)$ therefore produces $\ln(\Delta - i\epsilon)$ in the finite part. Now $\Delta - i\epsilon$ lies just below the real axis. For $\Delta > 0$ it is at argument 0^- and $\ln(\Delta - i\epsilon) = \ln \Delta$ is real. For $\Delta < 0$ it lies just below the negative real axis, at argument $-\pi$ (approached from below, the principal branch $\arg \in (-\pi, \pi]$ gives $-\pi$, not $+\pi$), so

$$\ln(\Delta - i\epsilon) = \ln|\Delta| + i \arg(\Delta - i\epsilon) = \ln|\Delta| - i\pi \qquad (\Delta < 0), \tag{D.158}$$

which combines with the $\Delta > 0$ case into (D.156). The discontinuity (D.157) follows by repeating the argument for $\Delta + i\epsilon$, which sits just above the negative axis at argument $+\pi$ and gives $\ln(\Delta + i\epsilon) = \ln|\Delta| + i\pi$ for $\Delta < 0$; subtracting, $\ln(\Delta - i\epsilon) - \ln(\Delta + i\epsilon) = -2\pi i$ there and zero for $\Delta > 0$. □

Applied to the bubble (D.140), Theorem D.34 says the finite part develops an imaginary piece exactly when $\Delta(x) = m^2 - x(1-x)p^2$ goes negative for some $x \in [0,1]$, i.e. for $p^2 > 4m^2$. There the imaginary part is the measure of the x-interval on which $\Delta < 0$, weighted by the π of (D.156):

$$\operatorname{Im}\Big[-\int_0^1 \mathrm{d}x\, \ln\frac{\Delta - i\epsilon}{\mu^2}\Big] = \pi \int_0^1 \mathrm{d}x\, \theta(x(1-x)p^2 - m^2) = \pi\sqrt{1 - \frac{4m^2}{p^2}}\;\theta(p^2 - 4m^2), \tag{D.159}$$

the last step solving $x(1-x)p^2 = m^2$ for the endpoints $x_\pm = \tfrac{1}{2}(1 \pm \sqrt{1 - 4m^2/p^2})$ and taking their separation $x_+ - x_- = \sqrt{1 - 4m^2/p^2}$. The square-root threshold factor is the unmistakable signature of two-body phase space opening up; it is plotted in Fig. D.6, where the imaginary part is seen to switch on precisely at $p^2 = 4m^2$ and grow as $\sqrt{1 - 4m^2/p^2}$.

The cut is the production of real particles

The appearance of an imaginary part above $p^2 = 4m^2$ is not a mathematical curiosity but the statement that the intermediate two-particle state in the loop can go on shell — the virtual pair becomes a real pair — once enough energy is available. Unitarity ties this directly to a physical rate. The optical theorem, stated in Chapter 5 from $\mathbf{S}^\dagger\mathbf{S} = \mathbf{1}$ and turned into a statement about a single loop integral in Section D.8.5, equates twice the imaginary part of a forward amplitude to a total cross section (or, for a self-energy, to a decay/production rate); equivalently the discontinuity (D.157) of a loop integral across its threshold cut equals the phase-space integral of the squared sub-amplitude for producing the on-shell intermediate state,

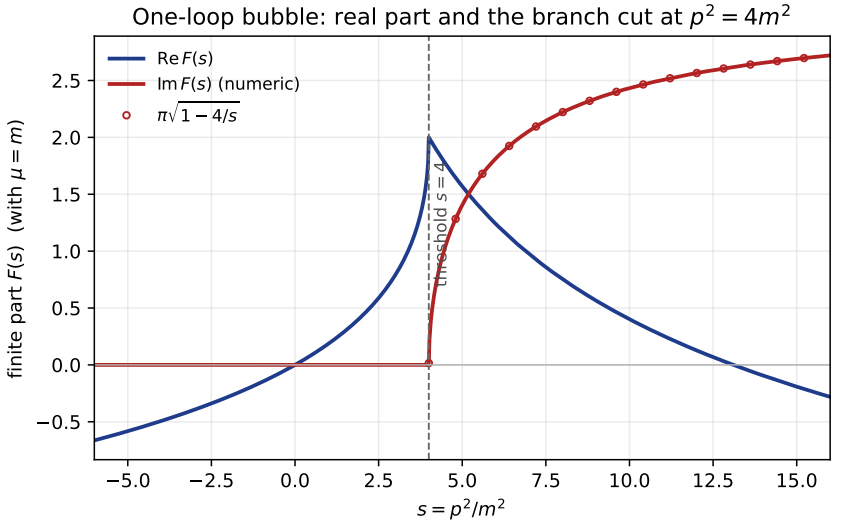


Figure D.6: The finite part $F(s)$ of the equal-mass one-loop bubble (D.140) as a function of $s = p^2/m^2$, with $\mu = m$. The real part (blue) is smooth across all s and grows logarithmically; the imaginary part (red) vanishes identically below the two-particle threshold $s = 4$ and, above it, follows the phase-space factor $\pi\sqrt{1-4/s}$ of (D.159) (open circles). The branch cut along $p^2 > 4m^2$ is the analytic shadow of the on-shell intermediate state, and its discontinuity is fixed by the optical theorem.

summed over that state. The factor $\sqrt{1-4m^2/p^2}$ in (D.159) is exactly the two-body phase-space volume the optical theorem predicts. The explicit check — that the discontinuity of the QED vacuum polarization computed by these methods reproduces the e^+e^- production cross section of Chapter 18 — is carried out in Chapter 19; here we have shown where the imaginary part comes from and that the $i\epsilon$ fixes its sign to be the one unitarity demands.

With the master integral now reduced to a numerically explicit Laurent series — a universal $1/\bar{\varepsilon}$ pole, a finite logarithm of Δ/μ^2 ready to be absorbed into counterterms or read as a physical running, and an imaginary part that switches on at threshold with the phase-space factor unitarity demands — the scalar one-loop technology is complete. The vacuum polarization just completed needed only the single-momentum symmetric integration of Section D.4; what this technology does not yet handle is a loop momentum in the numerator with free Lorentz indices contracted against *several* independent external momenta, as in the vertex correction, the box of $e^+e^- \to \mu^+\mu^-$, or any diagram with several

external legs: those tensor integrals must first be reduced to the scalar masters of this section. That reduction — decomposing a tensor integral onto a basis of external momenta and the metric, contracting to form a linear system, and inverting it at the price of a Gram determinant — is the subject of the next section.

D.6 Tensor reduction: the Passarino–Veltman algorithm

The master integrals assembled and ε-expanded in the previous sections are *scalar*: their numerators are unity. Real diagrams are not so kind. A photon–electron vertex, a triangle in ϕ^4 with derivative couplings, a box in $e^+e^- \to \mu^+\mu^-$ all produce loop integrands whose numerators carry the loop momentum ℓ with free Lorentz indices, contracted against gamma matrices and external momenta. A one-loop self-energy in QED already does this: the two fermion propagators put two powers of $\not{\ell}$ in the numerator. The question this section answers is how to convert such a tensor integral — an integral of $\ell^{\mu_1}\cdots\ell^{\mu_r}$ over a product of denominators — into the scalar masters we have already evaluated, with no new integration. The method is due to Passarino and Veltman: write the integral as the most general tensor its Lorentz structure permits, contract that ansatz with the available tensors (external momenta and the metric) to turn it into a linear system for the scalar coefficients, and solve the system. The solution divides by the determinant of the system — the *Gram determinant* of the external momenta — and we will flag honestly what that division costs when the kinematics degenerate.

D.6.1 The generic one-loop integral and the reduction philosophy

Fix the generic one-loop tensor integral with N denominators and rank r,

$$I_N^{\mu_1\cdots\mu_r} = \mu^{2\varepsilon}\int \frac{d^d\ell}{(2\pi)^d}\,\frac{\ell^{\mu_1}\cdots\ell^{\mu_r}}{D_0\,D_1\cdots D_{N-1}}, \qquad D_i = (\ell + k_i)^2 - m_i^2 + i\epsilon, \quad k_0 = 0, \tag{D.160}$$

where the k_i are partial sums of external momenta and $\mu^{2\varepsilon}$ is the dimensionful bookkeeping factor introduced in Section D.5. The whole reduction rests on a single elementary observation, which we call the *denominator identity*. Any scalar product of the loop momentum with an external momentum, and the square ℓ^2 itself, can be written as a linear combination of the denominators D_i plus constants, because the D_i are themselves quadratic in ℓ. Concretely, from $D_i - D_0 = (\ell + k_i)^2 - m_i^2 - \ell^2 + m_0^2$ we read off

$$2\,k_i\cdot\ell = D_i - D_0 - (k_i^2 + m_0^2 - m_i^2), \qquad \ell^2 = D_0 + m_0^2. \tag{D.161}$$

When such a combination appears in the numerator, each D_i it produces *cancels* the corresponding denominator, lowering both the rank and the number of propagators: the integral collapses onto a sub-integral with one fewer denominator. This is why the reduction terminates on scalars, and why those scalars have fewer propagators than the integral one started with. The Passarino–Veltman algorithm is the systematic bookkeeping of Eq. (D.161) organised so that the cancellation is automatic.

Notation Watch

We use the standard Passarino–Veltman names for the scalar coefficient functions: A for the one-point (tadpole) family, B for the two-point, C for the three-point, D for the four-point. Subscripts label the tensor coefficient: a single index i multiplies one power of external momentum, a double index a product, and the index 0 on a doubled slot (B_{00}, C_{00}) is reserved for the coefficient of the metric $\eta^{\mu\nu}$. All carry the measure and $\mu^{2\varepsilon}$ of Eq. (D.160), matching the scalar masters A_0, B_0, C_0, D_0 of Section D.5.

D.6.2 The two-point family: B^μ and $B^{\mu\nu}$

The cleanest place to see the machine work in full is the two-point family, which depends on a single external momentum p and two masses. Set $D_0 = \ell^2 - m_0^2$ and $D_1 = (\ell + p)^2 - m_1^2$, and define

$$\{B_0;\, B^\mu;\, B^{\mu\nu}\}(p^2; m_0, m_1) = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{\{1;\, \ell^\mu;\, \ell^\mu \ell^\nu\}}{D_0\, D_1}. \tag{D.162}$$

The only Lorentz-covariant vectors available to build B^μ are p^μ; the only symmetric tensors available for $B^{\mu\nu}$ are $p^\mu p^\nu$ and $\eta^{\mu\nu}$. Hence the integrals must take the form

$$B^\mu = p^\mu\, B_1, \tag{D.163}$$

$$B^{\mu\nu} = \eta^{\mu\nu}\, B_{00} + p^\mu p^\nu\, B_{11}. \tag{D.164}$$

These are exact statements about the integrals, not approximations: B^μ is a Lorentz vector built from the single vector p^μ on which the integrand depends, so it is proportional to p^μ; the scalar coefficients B_1, B_{00}, B_{11} are functions of the invariants p^2, m_0, m_1 alone. Our task is to express them through B_0 and the tadpole $A_0(m) = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d}\, (\ell^2 - m^2)^{-1}$ of Section D.5.

Rank one. Contract Eq. (D.163) with p_μ:

$$p^2\, B_1 = p_\mu B^\mu = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{p \cdot \ell}{D_0 D_1}. \tag{D.165}$$

Now apply the denominator identity Eq. (D.161) with $k_1 = p$. Writing $f \equiv p^2 + m_0^2 - m_1^2$,

$$p \cdot \ell = \tfrac{1}{2}(D_1 - D_0 - f), \tag{D.166}$$

so that

$$p^2 B_1 = \tfrac{1}{2}\,\mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d}\, \frac{D_1 - D_0 - f}{D_0 D_1} = \tfrac{1}{2}\left[\mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{D_0} - \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{D_1} - f\, B_0\right]. \tag{D.167}$$

In the first integral D_1 has cancelled, leaving $\int 1/D_0 = A_0(m_0)$. In the second D_0 has cancelled, leaving $\int 1/D_1$; shifting $\ell \to \ell - p$ turns D_1 into $\ell^2 - m_1^2$, so this is $A_0(m_1)$ (a tadpole is translation invariant). Therefore

$$\boxed{B_1(p^2; m_0, m_1) = \frac{1}{2p^2}\Big[\, A_0(m_0) - A_0(m_1) - (p^2 + m_0^2 - m_1^2) B_0 \,\Big].} \tag{D.168}$$

The factor $1/p^2$ in front is the two-point Gram determinant: p^2 is the determinant of the 1×1 Gram matrix $(p \cdot p)$ of the single external momentum. As a sanity check, when the masses are equal, $m_0 = m_1 = m$, the two tadpoles cancel and $f = p^2$, giving $B_1 = -\tfrac{1}{2} B_0$ with no leftover $1/p^2$ singularity, as it must be by the $\ell \to -\ell - p$ symmetry of the equal-mass integrand.

Rank two. Now contract the ansatz Eq. (D.164) in the two independent ways. Contracting with $\eta_{\mu\nu}$ uses $\eta_{\mu\nu}\eta^{\mu\nu} = d$ (the d-dimensional metric trace, Eq. (C.185)):

$$\eta_{\mu\nu} B^{\mu\nu} = d\, B_{00} + p^2\, B_{11} = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d}\, \frac{\ell^2}{D_0 D_1}. \tag{D.169}$$

The right-hand side is handled by the scalar part of Eq. (D.161), $\ell^2 = D_0 + m_0^2$:

$$\mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d}\, \frac{\ell^2}{D_0 D_1} = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{D_1} + m_0^2\, B_0 = A_0(m_1) + m_0^2\, B_0. \tag{D.170}$$

Contracting Eq. (D.164) instead with p_μ gives, on the left, $p_\mu B^{\mu\nu} = p^\nu B_{00} + p^2 p^\nu B_{11}$, and on the right

$$p_\mu B^{\mu\nu} = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d}\, \frac{(p \cdot \ell)\, \ell^\nu}{D_0 D_1} = \tfrac{1}{2}\left[\mu^{2\varepsilon} \int \frac{\ell^\nu}{D_0} - \mu^{2\varepsilon} \int \frac{\ell^\nu}{D_1} - f\, B^\nu\right], \tag{D.171}$$

using $p \cdot \ell = \tfrac{1}{2}(D_1 - D_0 - f)$ once more. The first integral, $\int \ell^\nu / D_0$, vanishes: it is odd in ℓ with a single mass scale and no external vector to align with. The second is $\int \ell^\nu / D_1 = -p^\nu A_0(m_1)$ by the shift used above. Hence $p_\mu B^{\mu\nu} = p^\nu\, \tfrac{1}{2}[A_0(m_1) - f\, B_1]$, and matching the p^ν coefficients,

$$B_{00} + p^2 B_{11} = \tfrac{1}{2}\big[A_0(m_1) - f\, B_1\big]. \tag{D.172}$$

We now have two linear equations, Eq. (D.169) and Eq. (D.172), for the two unknowns B_{00} and B_{11}. Subtracting Eq. (D.172) from Eq. (D.169) eliminates B_{11}:

$$(d-1)\,B_{00} = A_0(m_1) + m_0^2 B_0 - \tfrac{1}{2}\big[A_0(m_1) - f B_1\big] = \tfrac{1}{2}A_0(m_1) + m_0^2 B_0 + \tfrac{1}{2} f B_1, \tag{D.173}$$

so that

$$\boxed{B_{00} = \frac{1}{2(d-1)}\Big[A_0(m_1) + 2m_0^2 B_0 + (p^2 + m_0^2 - m_1^2) B_1\Big].} \tag{D.174}$$

Then Eq. (D.172) delivers the last coefficient:

$$\boxed{B_{11} = \frac{1}{p^2}\Big[\tfrac{1}{2}(A_0(m_1) - f B_1) - B_{00}\Big].} \tag{D.175}$$

Two features deserve comment, because they recur in every higher reduction. First, the metric coefficient B_{00} comes out with a factor $1/(d-1)$ but *no* factor $1/p^2$: it is finite as $p^2 \to 0$ and carries the genuine ultraviolet divergence of the rank-two integral (the A_0 and B_0 inside it both have $1/\varepsilon$ poles). Second, the momentum coefficient B_{11}, like B_1, divides by the Gram determinant p^2. The metric direction is always benign; the momentum directions always carry the Gram denominator. With B_1 from Eq. (D.168) substituted, Eqs. (D.174) and (D.175) express the complete rank-two two-point integral through the single scalar bubble B_0 and the tadpoles A_0, with no integration left to do.

D.6.3 The three-point family and the Gram determinant

The three-point family makes the Gram determinant visible as a genuine matrix determinant rather than a single invariant. Take denominators $D_0 = \ell^2 - m_0^2$, $D_1 = (\ell + p_1)^2 - m_1^2$, $D_2 = (\ell + p_2)^2 - m_2^2$ with two independent external momenta p_1, p_2, and define

$$\{C_0;\, C^\mu;\, C^{\mu\nu}\} = \mu^{2\varepsilon} \int \frac{d^d\ell}{(2\pi)^d} \frac{\{1;\, \ell^\mu;\, \ell^\mu \ell^\nu\}}{D_0\, D_1\, D_2}, \tag{D.176}$$

all functions of the invariants $p_1^2, p_2^2, (p_1 - p_2)^2$ and the masses. The covariant decompositions, built from the two vectors p_1^μ, p_2^μ and the metric, are

$$C^\mu = p_1^\mu\, C_1 + p_2^\mu\, C_2, \tag{D.177}$$

$$C^{\mu\nu} = \eta^{\mu\nu}\, C_{00} + p_1^\mu p_1^\nu\, C_{11} + p_2^\mu p_2^\nu\, C_{22} + (p_1^\mu p_2^\nu + p_2^\mu p_1^\nu)\, C_{12}. \tag{D.178}$$

Rank one: the Gram system. Contract Eq. (D.177) with $p_{1\mu}$ and with $p_{2\mu}$. The left-hand sides are

$$\begin{pmatrix} p_1^2 & p_1\cdot p_2 \\ p_1\cdot p_2 & p_2^2 \end{pmatrix}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} R_1 \\ R_2 \end{pmatrix}, \qquad R_i \equiv p_{i\mu}\, C^\mu = \mu^{2\varepsilon}\int \frac{d^d\ell}{(2\pi)^d}\,\frac{p_i\cdot\ell}{D_0 D_1 D_2}. \tag{D.179}$$

The matrix on the left is the *Gram matrix* $G_{ij} = p_i\cdot p_j$ of the external momenta. Its inverse is what the reduction needs, and its determinant

$$\det G = p_1^2\,p_2^2 - (p_1\cdot p_2)^2 \tag{D.180}$$

is the Gram determinant of the two-momentum kinematics. The right-hand sides R_i are evaluated by the denominator identity, $p_1\cdot\ell = \tfrac12(D_1 - D_0 - f_1)$ and $p_2\cdot\ell = \tfrac12(D_2 - D_0 - f_2)$ with

$$f_1 \equiv p_1^2 + m_0^2 - m_1^2, \qquad f_2 \equiv p_2^2 + m_0^2 - m_2^2. \tag{D.181}$$

Substituting and cancelling denominators,

$$R_1 = \tfrac12\left[\mu^{2\varepsilon}\int\frac{1}{D_0 D_2} - \mu^{2\varepsilon}\int\frac{1}{D_1 D_2} - f_1\, C_0\right] = \tfrac12\left[B_0^{(1)} - B_0^{(0)} - f_1\, C_0\right], \tag{D.182}$$

$$R_2 = \tfrac12\left[\mu^{2\varepsilon}\int\frac{1}{D_0 D_1} - \mu^{2\varepsilon}\int\frac{1}{D_1 D_2} - f_2\, C_0\right] = \tfrac12\left[B_0^{(2)} - B_0^{(0)} - f_2\, C_0\right], \tag{D.183}$$

where $B_0^{(i)}$ denotes the scalar bubble obtained by *pinching* (deleting) propagator i from the triangle: $B_0^{(0)} = \int 1/(D_1 D_2) = B_0((p_1-p_2)^2; m_1, m_2)$, $B_0^{(1)} = \int 1/(D_0 D_2) = B_0(p_2^2; m_0, m_2)$, and $B_0^{(2)} = \int 1/(D_0 D_1) = B_0(p_1^2; m_0, m_1)$, each reduced to standard form by the same shifts as before. The right-hand sides are thus made entirely of the scalar triangle C_0 and lower bubbles — objects we have already evaluated. Inverting the 2×2 Gram system Eq. (D.179) gives the coefficients in closed form,

$$\boxed{C_1 = \frac{p_2^2\, R_1 - (p_1\cdot p_2)\, R_2}{\det G}, \qquad C_2 = \frac{-(p_1\cdot p_2)\, R_1 + p_1^2\, R_2}{\det G},} \tag{D.184}$$

with $\det G$ from Eq. (D.180). This is the promised explicit appearance of the Gram determinant in the denominator of the reduced coefficients. The rank-two coefficients $C_{00}, C_{11}, C_{22}, C_{12}$ are obtained by exactly the same two-step procedure — contract Eq. (D.178) with $\eta_{\mu\nu}$ (which produces C_{00} free of any Gram denominator, again the carrier of the ultraviolet pole) and with $p_{1\mu}, p_{2\mu}$ (which produce 2×2 Gram systems for the pairs $\{C_{11}, C_{12}\}$ and $\{C_{12}, C_{22}\}$), each time using Eq. (D.161) to reduce the right-hand sides to the already-known C^μ, the pinched two-point tensors $B^{(i)}$, and C_0. We do not write the dozen resulting lines out here; the algorithm is mechanical once Eq. (D.184) is understood, and the worked example below exercises precisely the pieces a real diagram needs.

Caution: the price of the inversion

The factor $1/\det G$ in Eq. (D.184) is the honest cost of tensor reduction. When the external momenta become linearly dependent — $p_2 \to x\, p_1$ for some scalar x, i.e. a forward, collinear, or otherwise degenerate configuration — the Gram determinant Eq. (D.180) vanishes, $\det G = p_1^2 p_2^2 - (p_1 \cdot p_2)^2 \to 0$, while the numerators R_i do not generically vanish in the same way. The reduced coefficients C_1, C_2 then appear to blow up. This singularity is *spurious*: the tensor integral C^μ is perfectly finite at degenerate kinematics, and the apparent pole cancels among the scalar functions inside $p_2^2 R_1 - (p_1 \cdot p_2) R_2$ in the exact limit. But the cancellation is between large numbers, so a numerical evaluation near $\det G = 0$ suffers catastrophic loss of precision. This Gram-determinant problem is the standard practical obstruction to Passarino–Veltman reduction; the modern remedy is to expand around the degenerate point or to use integrand-level (unitarity-based) reduction instead, the subject of Chapter 60. For the analytic one-loop computations of this book the limits are taken in closed form and the cancellation is exact.

D.6.4 Completeness: everything reduces to A_0, B_0, C_0, D_0

The two- and three-point cases exhibit the entire pattern, and the same two moves — decompose into a covariant basis, then contract and solve — reduce the four-point (box) tensors $D^\mu, D^{\mu\nu}, D^{\mu\nu\rho}$ to the scalar box D_0 and lower functions, the Gram matrix now being the 3×3 matrix of three independent external momenta. We can state the general result.

Proposition D.35 (One-loop reduction)**.** *Every one-loop integral of the form Eq. (D.160), with any rank r and any number N of denominators, reduces in $d = 4 - 2\varepsilon$ dimensions to a finite linear combination of the scalar master integrals A_0, B_0, C_0, D_0, with coefficients that are rational functions of the external invariants, the masses, and d. In strictly four dimensions ($\varepsilon \to 0$) no scalar integral beyond the box D_0 is needed.*

Argument. Two independent reductions act. *Rank reduction*: each free index μ_j on the numerator ℓ^{μ_j} is removed by contracting the covariant ansatz with the basis tensors and applying the denominator identity Eq. (D.161) to the resulting scalar products $k_i \cdot \ell$ and squares ℓ^2. Every term so produced either cancels a denominator (lowering N) or is the metric coefficient (which closes on lower-rank functions). Iterating drives the rank to zero. *Propagator reduction*: a scalar integral with N denominators carries $N - 1$ independent external momenta. In $d = 4$ at most four vectors can be linearly independent, so for $N \geq 6$ the external momenta are necessarily dependent and the $(N-1)$-dimensional

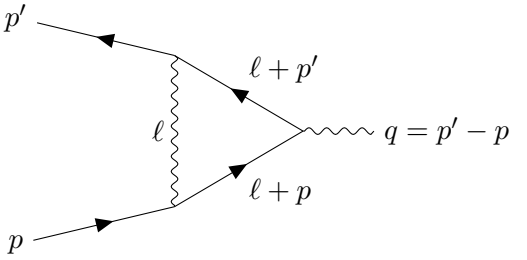


Figure D.7: The one-loop QED vertex. The electron enters with momentum p, exits with p', and the external photon injects $q = p' - p$ at the vertex on the fermion line. The internal photon carries loop momentum ℓ; the two fermion segments carry $\ell + p$ and $\ell + p'$. The three denominators are $D_0 = \ell^2$, $D_1 = (\ell + p)^2 - m^2$, $D_2 = (\ell + p')^2 - m^2$, the three-point family of Eq. (D.176) with $m_0 = 0$, $m_1 = m_2 = m$ and $p_1 = p$, $p_2 = p'$.

Gram determinant vanishes identically; the corresponding scalar integral is then expressible through integrals with fewer denominators. The pentagon, $N = 5$, is the boundary case: its Gram determinant is generically nonzero, but in exactly four dimensions the scalar pentagon reduces to a sum of five scalar boxes up to terms of order ε, which vanish at one loop as $\varepsilon \to 0$. Hence the four scalar masters span everything. The order-ε remainder of the pentagon reduction is genuine and must be retained when the box coefficients themselves carry $1/\varepsilon$ poles, as happens at higher loop order or in ε-dimensional unitarity; we flag this honestly and defer its systematic treatment to Chapter 60. □

This is the structural payoff of the section: the Feynman rules of any one-loop diagram, however many indices and momenta they carry, feed into a finite catalogue of four scalar functions whose ε-expansions are tabulated in Section D.5. The only remaining subtlety, taken up in the next section, is to tell which of the resulting $1/\varepsilon$ poles are ultraviolet and which are infrared.

D.6.5 QED: reducing the one-loop vertex

We now carry the photon–electron vertex to the scalar level, the computation Chapters 19 and 20 need for the Ward identity and for Schwinger's anomalous moment $a_e = \alpha/2\pi$. This advances QED from the tree amplitudes of Chapters 17 and 18 to its first form factors.

Worked Example D.36 (Passarino–Veltman reduction of the QED vertex)**.** With the routing of Fig. D.7 and in Feynman gauge ($\xi = 1$, the photon propagator $-i\eta_{\alpha\beta}/\ell^2$ of Chapter 13), the amputated one-loop vertex correction $-ie\,\Lambda^\mu$ is

the integral

$$\Lambda^{\mu} = -ie^{2}\,\mu^{2\varepsilon}\int \frac{d^{d}\ell}{(2\pi)^{d}}\,\frac{N^{\mu}}{\ell^{2}\left[(\ell+p)^{2}-m^{2}\right]\left[(\ell+p')^{2}-m^{2}\right]}, \tag{D.185}$$

where collecting the three vertex factors $(-ie\gamma)$, the two fermion numerators, and the photon propagator gives the overall constant $-ie^{2}$, and the Dirac numerator is the gamma sandwich

$$N^{\mu} = \gamma^{\alpha}\,(\not{\ell}+\not{p}'+m)\,\gamma^{\mu}\,(\not{\ell}+\not{p}+m)\,\gamma_{\alpha}. \tag{D.186}$$

We use the massless photon ($m_0 = 0$), letting the dimensional regulator ε also control the infrared singularity that the photon mass would otherwise screen; this is exactly the dual UV/IR role of ε taken up in the next section. The electron is on shell, $p^{2} = p'^{2} = m^{2}$, with $\not{p}\,u(p) = m\,u(p)$ and $\bar{u}(p')\not{p}' = m\,\bar{u}(p')$ understood once the structure is sandwiched between spinors.

Step 1: collapse the gamma sandwich. Abbreviate $a \equiv \ell + p'$, $b \equiv \ell + p$, and expand Eq. (D.186):

$$N^{\mu} = \gamma^{\alpha}\not{a}\,\gamma^{\mu}\not{b}\,\gamma_{\alpha} + m\,\gamma^{\alpha}\not{a}\,\gamma^{\mu}\gamma_{\alpha} + m\,\gamma^{\alpha}\gamma^{\mu}\not{b}\,\gamma_{\alpha} + m^{2}\,\gamma^{\alpha}\gamma^{\mu}\gamma_{\alpha}. \tag{D.187}$$

The contracted strings are reduced by the d-dimensional contraction identities of Section C.6, in the continued forms Eqs. (C.187) to (C.190): $\gamma^{\alpha}\gamma^{\mu}\gamma_{\alpha} = (2-d)\gamma^{\mu}$, $\gamma^{\alpha}\gamma^{\nu}\gamma^{\mu}\gamma_{\alpha} = 4\eta^{\nu\mu}\mathbf{1} - (4-d)\gamma^{\nu}\gamma^{\mu}$, and $\gamma^{\alpha}\gamma^{\nu}\gamma^{\mu}\gamma^{\rho}\gamma_{\alpha} = -2\gamma^{\rho}\gamma^{\mu}\gamma^{\nu} + (4-d)\gamma^{\nu}\gamma^{\mu}\gamma^{\rho}$. Applying these term by term,

$$m^{2}\,\gamma^{\alpha}\gamma^{\mu}\gamma_{\alpha} = -(d-2)\,m^{2}\,\gamma^{\mu}, \tag{D.188}$$

$$m\,\gamma^{\alpha}\not{a}\,\gamma^{\mu}\gamma_{\alpha} = m\big[4a^{\mu} - (4-d)\,\not{a}\,\gamma^{\mu}\big], \tag{D.189}$$

$$m\,\gamma^{\alpha}\gamma^{\mu}\not{b}\,\gamma_{\alpha} = m\big[4b^{\mu} - (4-d)\,\gamma^{\mu}\not{b}\big], \tag{D.190}$$

$$\gamma^{\alpha}\not{a}\,\gamma^{\mu}\not{b}\,\gamma_{\alpha} = -2\,\not{b}\,\gamma^{\mu}\not{a} + (4-d)\,\not{a}\,\gamma^{\mu}\not{b}. \tag{D.191}$$

Collecting, with $d = 4 - 2\varepsilon$ so that $4 - d = 2\varepsilon$ and $d - 2 = 2(1-\varepsilon)$,

$$N^{\mu} = -2\,\not{b}\,\gamma^{\mu}\not{a} + 4m\,(a^{\mu}+b^{\mu}) - 2(1-\varepsilon)\,m^{2}\,\gamma^{\mu} + 2\varepsilon\big[\not{a}\,\gamma^{\mu}\not{b} - m\,(\not{a}\,\gamma^{\mu} + \gamma^{\mu}\not{b})\big]. \tag{D.192}$$

The terms carrying an explicit 2ε cannot be dropped: multiplied against the $1/\varepsilon$ pole of the ultraviolet-divergent part of the integral they leave a finite remainder, exactly the mechanism flagged in Section C.7.3. Substituting $a = \ell + p'$, $b = \ell + p$ shows N^{μ} is a polynomial in ℓ of degree two: rank zero, one, and two pieces, which the integral Eq. (D.185) turns into C_0, C^{μ}, and $C^{\mu\nu}$ respectively, all in the family of Eq. (D.176) with $m_0 = 0$, $m_1 = m_2 = m$, $p_1 = p$, $p_2 = p'$.

Step 2: read off the rank-two piece. The highest-rank structure sits in $-2\,\not{b}\,\gamma^{\mu}\not{a}$. Writing $\not{b}\,\gamma^{\mu}\not{a} = (\not{\ell}+\not{p})\,\gamma^{\mu}(\not{\ell}+\not{p}')$ and isolating the part quadratic in ℓ,

$$\not{\ell}\,\gamma^{\mu}\not{\ell} = \ell_{\nu}\ell_{\rho}\,\gamma^{\nu}\gamma^{\mu}\gamma^{\rho} \;\longmapsto\; \gamma^{\nu}\gamma^{\mu}\gamma^{\rho}\,C_{\nu\rho}, \tag{D.193}$$

where under the integral $\ell_\nu \ell_\rho \to C_{\nu\rho}$ is the rank-two three-point tensor. Inserting its decomposition Eq. (D.178), the metric piece alone gives

$$\gamma^\nu \gamma^\mu \gamma^\rho \, \eta_{\nu\rho} \, C_{00} = \gamma^\nu \gamma^\mu \gamma_\nu \, C_{00} = (2-d)\, \gamma^\mu \, C_{00} = -2(1-\varepsilon)\, \gamma^\mu \, C_{00}, \tag{D.194}$$

using Eq. (C.188) again. This is the decisive line. The coefficient C_{00} is the metric coefficient of the three-point family — the analogue of B_{00} in Eq. (D.174) — and like B_{00} it is the *only* reduced coefficient that is ultraviolet divergent, carrying a $1/\varepsilon$ pole, with *no* Gram denominator. Equation (D.194) shows that this pole lands entirely on the Dirac structure γ^μ. The remaining rank-two pieces of $C_{\nu\rho}$ (the $p_i^\nu p_j^\rho$ terms with coefficients C_{11}, C_{22}, C_{12}, divided by $\det G$ per Eq. (D.184)) and the rank-one and rank-zero pieces are all ultraviolet finite.

Step 3: organise into form factors. Sandwiching the reduced numerator between $\bar{u}(p')$ and $u(p)$, using $\not{p}\, u = m\, u$, $\bar{u}'\not{p}' = m\, \bar{u}'$, the anticommutator $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$, and the Gordon identity

$$\bar{u}(p')\, \gamma^\mu \, u(p) = \bar{u}(p') \left[\frac{(p+p')^\mu}{2m} + \frac{i\sigma^{\mu\nu} q_\nu}{2m} \right] u(p), \qquad q = p' - p, \tag{D.195}$$

every term collapses onto exactly two independent on-shell structures, γ^μ and $i\sigma^{\mu\nu}q_\nu/2m$ (equivalently $(p+p')^\mu/2m$). The vertex therefore takes the form-factor form

$$\bar{u}(p')\, \Gamma^\mu \, u(p) = \bar{u}(p') \left[\gamma^\mu \, F_1(q^2) + \frac{i\sigma^{\mu\nu} q_\nu}{2m} \, F_2(q^2) \right] u(p), \tag{D.196}$$

where $\Gamma^\mu = \gamma^\mu + \Lambda^\mu$ at this order, and the form factors are the linear combinations of scalar functions produced by the reduction:

$$F_1(q^2) = 1 + \frac{e^2}{(4\pi)^{d/2}} \left[\#\, C_{00} + \#\, C_0 + \#\, C_i + \#\, B_0^{(i)} \right], \qquad F_2(q^2) = \frac{e^2}{(4\pi)^{d/2}} \left[\#\, C_0 + \#\, C_i + \#\, C_{ij} \right] \tag{D.197}$$

the symbols $\#$ standing for the definite rational coefficients (functions of m^2, q^2, d) generated by Steps 1–2 and the Gram inversion. The structural content, which is what this section is responsible for, is the *separation* visible in Eq. (D.197): F_1 contains C_{00} and hence the $1/\varepsilon$ ultraviolet pole, which is removed by the vertex (equivalently, by Ward, the field-strength) counterterm of Chapter 19; F_2 is built only from the ultraviolet-finite scalar functions C_0, C_i, C_{ij} and is therefore finite and unambiguous — it is a clean prediction with no counterterm freedom. Evaluating those scalar functions at $q^2 = 0$ and extracting the finite number is the task of Chapter 20, which obtains $F_2(0) = a_e = \alpha/2\pi$. We note the consistency of Theorem D.36 with the Gram-determinant caution above: at $q^2 = 0$ the three-point Gram determinant Eq. (D.180) reads $\det G = p^2 p'^2 - (p\cdot p')^2 = \frac{q^2}{4}(4m^2 - q^2) \to 0$, so the individual coefficients C_i, C_{ij} carry spurious $1/q^2$ singularities; that they cancel to leave the finite a_e is the explicit, physical instance of the spurious-pole cancellation the caution promised.

With every one-loop tensor integrand now reduced to the scalar masters A_0, B_0, C_0, D_0, the perturbative calculation is purely arithmetic: substitute the ε-expansions of Section D.5 and collect. The single remaining conceptual question is the provenance of the $1/\varepsilon$ poles that survive — whether a given pole signals an ultraviolet divergence to be absorbed by renormalization or an infrared divergence destined to cancel against real emission — and it is to that distinction, and to dimensional regularization in its second guise as an infrared regulator, that we turn next.

D.7 Infrared and collinear singularities; dimensional regularization as an IR regulator

The tensor reduction of Section D.6 delivered every one-loop integral of a realistic diagram as a linear combination of the scalar masters evaluated earlier in this appendix. Those masters carry $1/\varepsilon$ poles, and the natural reflex—trained by the power-counting of Chapter 21—is to read every pole as ultraviolet, a signal that some short-distance region of the loop integration diverges and must be absorbed into a counterterm. That reflex is wrong in general. Among the scalar integrals that survive reduction are some that are ultraviolet *convergent* and yet still produce $1/\varepsilon$: their poles come not from the region of large loop momentum but from regions where an internal line goes on shell over a finite stretch of the integration—a photon becoming soft, or a massless pair of momenta becoming collinear. The same analytic continuation in d that we introduced to tame the ultraviolet turns out to regulate these long-distance, infrared (IR) singularities as well. This is the dual role of dimensional regularization flagged in the chapter introduction, and this section is where we pay it off: we classify $1/\varepsilon$ poles by their physical origin, fix the sign of ε that regulates each, prove that the scaleless integral vanishes precisely because a UV pole and an IR pole cancel, evaluate a representative IR-divergent master in full, and then explain—at the level of precise statements, with the QED computation set up explicitly—how IR poles disappear from physical cross sections once virtual and real-emission contributions are summed. Throughout we keep the standing conventions of Chapter A: mostly-minus metric, Feynman $i\varepsilon$ in propagators, $d = 4 - 2\varepsilon$, the loop-measure bookkeeping factor $\mu^{2\varepsilon}$, and $\overline{\text{MS}}$.

D.7.1 The two origins of a $1/\varepsilon$ pole and the sign of ε

Every $1/\varepsilon$ pole in a dimensionally regulated integral is the failure of an ordinary integral to converge at one end of its range, exposed by analytic continuation in the dimension. The cleanest place to see both failure modes at once is the radial behaviour of a Euclidean loop integral. After Wick rotation (Section D.3) and the angular integration $\int d^d\ell_E = \frac{2\pi^{d/2}}{\Gamma(d/2)} \int_0^\infty d\ell_E\, \ell_E^{d-1}$, a scalar integrand that

behaves as a power ℓ_E^{-2n} contributes a radial integral

$$\int_0^\infty d\ell_E \, \ell_E^{d-1-2n}, \tag{D.198}$$

and this single integral encodes both diseases. At the upper end $\ell_E \to \infty$—the *ultraviolet* region of short distances—the integrand decays fast enough to converge if and only if $d - 1 - 2n < -1$, that is

$$\text{UV convergence:} \qquad d < 2n \quad \Longleftrightarrow \quad \varepsilon > \tfrac{4-2n}{2} \ \ (\text{e.g. } \varepsilon > 0 \text{ for } 2n = 4). \tag{D.199}$$

At the lower end $\ell_E \to 0$—the *infrared* region of long distances, reached physically when the integrand's power growth there is generated by a propagator sitting near its mass shell—the integral converges if and only if $d - 1 - 2n > -1$, that is

$$\text{IR convergence:} \qquad d > 2n \quad \Longleftrightarrow \quad \varepsilon < \tfrac{4-2n}{2} \ \ (\text{e.g. } \varepsilon < 0 \text{ for } 2n = 4). \tag{D.200}$$

The two requirements are mutually exclusive: no single real d makes Eq. (D.198) converge at both ends unless one of the two power behaviours is absent. Dimensional regularization defines the answer by analytic continuation from whichever region of d the relevant end converges in, and the resulting pole at $d = 2n$ is *ultraviolet* when reached from $d < 2n$ (small d) and *infrared* when reached from $d > 2n$ (large d). With the standard parametrization $d = 4 - 2\varepsilon$, a logarithmically UV-divergent integral ($2n = 4$) is regulated by $d < 4$, i.e. $\varepsilon > 0$, while a logarithmically IR-divergent one is regulated by $d > 4$, i.e. $\varepsilon < 0$. Ultraviolet poles want d *below* four; infrared poles want d *above* four. This is the single most important sentence of the section, and it forces a bookkeeping discipline.

Notation Watch

A bare symbol ε cannot simultaneously be positive (to regulate the UV) and negative (to regulate the IR). In dimensional regularization there is nonetheless only *one* dimension $d = 4 - 2\varepsilon$ and one ε; the resolution is that any given pole is reached by continuation from one side, and the Laurent coefficient is unambiguous once the integral is defined by continuation from its region of convergence. To keep the physics legible we tag a pole by its origin: we write $1/\varepsilon_{\mathrm{UV}}$ for a pole generated by the large-ℓ region (continued from $\varepsilon > 0$) and $1/\varepsilon_{\mathrm{IR}}$ for one generated by an on-shell region (continued from $\varepsilon < 0$). The tags are bookkeeping, not separate variables: numerically $\varepsilon_{\mathrm{UV}} = \varepsilon_{\mathrm{IR}} = \varepsilon$, and at the very end all poles are the same ε. The tags matter because UV poles are removed by counterterms (renormalization, Chapter 22) whereas IR poles must cancel against real emission (this section); confusing the two corrupts both the renormalization constants and the finite remainder. When a single integral carries both, as in Theorem D.37 below, we must keep the tags distinct

until the cancellation is exhibited.

D.7.2 The scaleless integral: a UV pole cancelling an IR pole

The most economical illustration of the two-origin structure is an integral with *no* scale: there is no mass, no external invariant, nothing for the answer to depend on. Such integrals occur constantly—massless on-shell self-energies, and any tadpole left after a propagator's mass is set to zero, reduce to them—and the rule that disposes of them is one line: a scaleless integral vanishes in dimensional regularization. The rule is true, but the reason is not that "nothing diverges." It is that a genuine ultraviolet divergence is being silently cancelled by a genuine infrared one, and we now prove exactly that.

Theorem D.37 (Vanishing of the scaleless integral)**.** *For any integer n and any d for which the expression is defined by analytic continuation,*

$$\int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2)^n} = 0, \tag{D.201}$$

and the vanishing is the cancellation of a $1/\varepsilon_{\mathrm{UV}}$ pole against a $1/\varepsilon_{\mathrm{IR}}$ pole of equal magnitude and opposite sign.

Proof. Wick-rotate to Euclidean signature, $\ell^0 = i\ell_E^0$, under which $\ell^2 \to -\ell_E^2$ and $d^d\ell \to i\,d^d\ell_E$, so the integral equals $i(-1)^n$ times the Euclidean integral $\int \frac{d^d\ell_E}{(2\pi)^d}(\ell_E^2)^{-n}$. Performing the angular integral leaves the radial integral Eq. (D.198) up to the positive constant $\frac{1}{(2\pi)^d}\frac{2\pi^{d/2}}{\Gamma(d/2)}$, which never vanishes and never introduces a pole at $d = 2n$; the entire content is in

$$R \equiv \int_0^\infty d\ell_E\, \ell_E^{d-1-2n}. \tag{D.202}$$

There is no value of d for which R converges, by the discussion of Eqs. (D.199) and (D.200). To define it we split the range at an arbitrary fiducial scale $\Lambda > 0$ and continue each piece from its own region of convergence:

$$R = \underbrace{\int_0^\Lambda d\ell_E\, \ell_E^{d-1-2n}}_{R_{\mathrm{IR}}} + \underbrace{\int_\Lambda^\infty d\ell_E\, \ell_E^{d-1-2n}}_{R_{\mathrm{UV}}}. \tag{D.203}$$

The lower piece R_{IR} converges at $\ell_E \to 0$ for $d > 2n$, where it evaluates to

$$R_{\mathrm{IR}} = \frac{\Lambda^{\,d-2n}}{d-2n} \qquad (\text{continued from } d > 2n), \tag{D.204}$$

with a pole at $d \to 2n^+$: this is the infrared pole, $R_{\mathrm{IR}} \sim 1/\varepsilon_{\mathrm{IR}}$ up to the constant relating $d - 2n$ to ε. The upper piece R_{UV} converges at $\ell_E \to \infty$ for $d < 2n$, where

$$R_{\mathrm{UV}} = \Big[\frac{\ell_E^{\,d-2n}}{d-2n}\Big]_{\Lambda}^{\infty} = \frac{-\Lambda^{\,d-2n}}{d-2n} = \frac{\Lambda^{\,d-2n}}{2n-d} \qquad \text{(continued from } d < 2n\text{)}, \tag{D.205}$$

with a pole at $d \to 2n^-$: the ultraviolet pole, $R_{\mathrm{UV}} \sim 1/\varepsilon_{\mathrm{UV}}$. Both expressions are now rational functions of d, analytic away from $d = 2n$, so their sum may be taken at common d:

$$R = R_{\mathrm{IR}} + R_{\mathrm{UV}} = \frac{\Lambda^{\,d-2n}}{d-2n} + \frac{\Lambda^{\,d-2n}}{2n-d} = \frac{\Lambda^{\,d-2n}}{d-2n} - \frac{\Lambda^{\,d-2n}}{d-2n} = 0. \tag{D.206}$$

The fiducial scale Λ has dropped out, as it must—the original integral knew of no scale—and the result is identically zero for all d, not merely at $d = 4$. Crucially, R vanishes because the residue of its UV pole is exactly minus the residue of its IR pole: writing $d - 2n = -2\delta$ near the singular dimension, $R_{\mathrm{UV}} = -\Lambda^{-2\delta}/2\delta \to -1/2\delta$ and $R_{\mathrm{IR}} = \Lambda^{-2\delta}/(2\delta) \to +1/2\delta$ as $\delta \to 0$. The zero in Eq. (D.201) is therefore the statement

$$\frac{1}{\varepsilon_{\mathrm{UV}}} - \frac{1}{\varepsilon_{\mathrm{IR}}} = 0 \qquad \text{(numerically, since } \varepsilon_{\mathrm{UV}} = \varepsilon_{\mathrm{IR}}\text{)}, \tag{D.207}$$

not the absence of divergence. □

The practical force of Theorem D.37 is twofold and opposite, and both directions are used later in the book. When a scaleless integral appears inside a finite physical quantity, one is free to drop it, knowing the discarded UV and IR poles were equal and opposite. But when one is computing a renormalization constant—an object that is, by construction, pure pole—the cancellation is a trap: the massless on-shell wavefunction renormalization Z_2 and the photon field renormalization receive contributions from integrals that vanish by Theorem D.37, and reading "= 0" as "no renormalization" would be a sign error. The honest statement keeps the tags,

$$\int \frac{d^d\ell}{(2\pi)^d}\,\frac{1}{(\ell^2)^n} \;\longrightarrow\; \frac{1}{\varepsilon_{\mathrm{UV}}} - \frac{1}{\varepsilon_{\mathrm{IR}}}, \tag{D.208}$$

so that the $1/\varepsilon_{\mathrm{UV}}$ is available to cancel a counterterm and the $1/\varepsilon_{\mathrm{IR}}$ is available to cancel a real-emission pole, even though their sum is zero. This is exactly the bookkeeping that makes the on-shell renormalization of massless QED and QCD consistent (Chapters 19 and 27).

Physical Insight

The cancellation in Eq. (D.206) is not a coincidence of dimensional regularization; it is the statement that, with no scale in the problem, "large momentum" and "small momentum" are not invariantly distinguishable. Any rescaling $\ell \to \kappa \ell$ is a symmetry of the integrand up to the overall power κ^{d-2n}, and a scale-invariant quantity that is also homogeneous of nonzero degree can only be zero. Dimensional regularization makes this visible by separating the two ends of the scaling and showing they are negatives of each other. The moment a scale is restored—a mass, an external virtuality—the two ends decouple, the UV pole stays (to be renormalized) and the IR pole either stays (to be cancelled by real emission) or is rendered finite, and the integral no longer vanishes.

D.7.3 A worked infrared master: the massless triangle

To see an unambiguous infrared pole—one that cannot be mistaken for ultraviolet—we evaluate an integral that is ultraviolet *convergent* by power counting and yet diverges, entirely from on-shell regions. The cleanest such master is the massless scalar triangle with two light-like external legs, which is precisely the scalar integral that the Passarino–Veltman reduction of the one-loop QED and Yang–Mills vertices deposits whenever the external fermions or gluons can be treated as massless. Its superficial degree of divergence is $d - 6 = -2 - 2\varepsilon < 0$, so every $1/\varepsilon$ it produces is infrared, and we will see it carry a *double* pole: the simultaneous soft and collinear singularity that is the hallmark of massless gauge theory.

Worked Example D.38 (The massless one-loop triangle and its double IR pole)**.** Evaluate, with $p_1^2 = p_2^2 = 0$ and the single invariant $s \equiv (p_1 + p_2)^2 = 2p_1 \cdot p_2$,

$$I_3 = \mu^{2\varepsilon} \int \frac{d^d \ell}{(2\pi)^d} \, \frac{1}{(\ell^2 + i\varepsilon)\,((\ell + p_1)^2 + i\varepsilon)\,((\ell - p_2)^2 + i\varepsilon)}. \tag{D.209}$$

Feynman parametrization. Using the three-denominator identity of Section D.2,

$$\frac{1}{ABC} = 2 \int_0^1 dx\, dy\, dz\; \delta(1 - x - y - z)\, \frac{1}{[\,xA + yB + zC\,]^3}, \tag{D.210}$$

with $A = \ell^2$, $B = (\ell + p_1)^2$, $C = (\ell - p_2)^2$. Because $p_1^2 = p_2^2 = 0$ the linear-in-p terms are all that survive in B and C:

$$xA + yB + zC = \ell^2 + 2\ell \cdot (yp_1 - zp_2) = \big(\ell + yp_1 - zp_2\big)^2 - \big(yp_1 - zp_2\big)^2. \tag{D.211}$$

The shift constant is $(yp_1 - zp_2)^2 = y^2 p_1^2 - 2yz\, p_1 \cdot p_2 + z^2 p_2^2 = -yz\, s$, so with $\ell' = \ell + yp_1 - zp_2$ the denominator becomes $\ell'^2 - \Delta$ with $\Delta = -yz\, s$.

Momentum integral. The shifted integral is the master of Theorem D.22 at $n = 3$,

$$\int \frac{d^d\ell'}{(2\pi)^d}\,\frac{1}{(\ell'^2 - \Delta + i\varepsilon)^3} = \frac{-\,i}{(4\pi)^{d/2}}\,\frac{\Gamma(3-\frac{d}{2})}{\Gamma(3)}\,\Delta^{\,d/2-3} = \frac{-\,i}{(4\pi)^{2-\varepsilon}}\,\frac{\Gamma(1+\varepsilon)}{2}\,\Delta^{-1-\varepsilon}, \tag{D.212}$$

where $\Delta^{-1-\varepsilon} = (-yz\,s)^{-1-\varepsilon} = (-s)^{-1-\varepsilon}(yz)^{-1-\varepsilon}$, the factor $(-s)^{-1-\varepsilon}$ carrying the branch cut that, continued to $s > 0$, supplies the physical imaginary part. Collecting the prefactors and the bookkeeping $\mu^{2\varepsilon}$,

$$I_3 = \frac{-\,i\,\Gamma(1+\varepsilon)}{(4\pi)^{2-\varepsilon}}\,\mu^{2\varepsilon}\,(-s)^{-1-\varepsilon}\,J, \qquad J \equiv \int_0^1 dx\,dy\,dz\;\delta(1-x-y-z)\,(yz)^{-1-\varepsilon}. \tag{D.213}$$

The Feynman-parameter integral is where the IR poles live. Doing the x integral against the delta function sets $x = 1-y-z$ over the simplex $y, z \geq 0$, $y + z \leq 1$:

$$J = \int_0^1 dy\; y^{-1-\varepsilon} \int_0^{1-y} dz\; z^{-1-\varepsilon}. \tag{D.214}$$

The inner integral $\int_0^{1-y} z^{-1-\varepsilon}dz = (1-y)^{-\varepsilon}/(-\varepsilon)$ converges at $z \to 0$ *only for* $\Re\varepsilon < 0$, i.e. $d > 4$: this is the infrared regulator at work, exactly the sign of Eq. (D.200). Hence

$$J = -\frac{1}{\varepsilon}\int_0^1 dy\; y^{-1-\varepsilon}(1-y)^{-\varepsilon} = -\frac{1}{\varepsilon}\,B(-\varepsilon,\,1-\varepsilon) = -\frac{1}{\varepsilon}\,\frac{\Gamma(-\varepsilon)\,\Gamma(1-\varepsilon)}{\Gamma(1-2\varepsilon)}, \tag{D.215}$$

the second factor of $y^{-1-\varepsilon}$ again convergent at $y \to 0$ only for $\Re\varepsilon < 0$, the second IR singularity. Using $\Gamma(-\varepsilon) = -\Gamma(1-\varepsilon)/\varepsilon$,

$$J = \frac{1}{\varepsilon^2}\,\frac{\Gamma(1-\varepsilon)^2}{\Gamma(1-2\varepsilon)}. \tag{D.216}$$

The two convergence conditions have produced the two factors of $1/\varepsilon$.

Result. Substituting Eq. (D.216) into Eq. (D.213) and grouping the Γ's into the ubiquitous one-loop constant $c_\Gamma \equiv \Gamma(1+\varepsilon)\Gamma(1-\varepsilon)^2/\Gamma(1-2\varepsilon)$,

$$\boxed{\,I_3 = \frac{-\,i\,c_\Gamma}{(4\pi)^2}\,\frac{1}{(-s)}\,\frac{1}{\varepsilon^2}\left(\frac{4\pi\mu^2}{-s}\right)^{\varepsilon}.\,} \tag{D.217}$$

The pole structure is laid bare by expanding $c_\Gamma = 1 - \gamma_E\,\varepsilon + (\frac{\gamma_E^2}{2} - \frac{\pi^2}{12})\varepsilon^2 + O(\varepsilon^3)$ and $(4\pi\mu^2/(-s))^{\varepsilon} = 1 + \varepsilon\ln\frac{4\pi\mu^2}{-s} + \ldots$, after which the standard $\overline{\text{MS}}$ rescaling $\mu^2 \to \mu^2 e^{\gamma_E}/4\pi$ (Chapter A) removes the 4π and γ_E—the product $c_\Gamma\,(4\pi\mu^2/(-s))^{\varepsilon}$ collapsing to $(1 - \frac{\pi^2}{12}\varepsilon^2)$ times powers of $\ln(-s/\mu^2)$—leaving

$$I_3 = \frac{-\,i}{(4\pi)^2}\,\frac{1}{(-s)}\left[\frac{1}{\varepsilon_{\text{IR}}^2} - \frac{1}{\varepsilon_{\text{IR}}}\,\ln\frac{-s}{\mu^2} + \frac{1}{2}\ln^2\frac{-s}{\mu^2} - \frac{\pi^2}{12} + \ldots\right]. \tag{D.218}$$

Both poles are infrared—the integral was UV-finite by power counting, and both factors of $1/\varepsilon$ traced to ℓ-regions where an internal line went on shell. The $1/\varepsilon^2$ is the simultaneous soft ($\ell \to 0$) and collinear ($\ell \parallel p_1$ or $\ell \parallel p_2$) singularity, the double logarithm $\frac{1}{2}\ln^2(-s/\mu^2)$ its finite shadow; these are the Sudakov double logarithms that govern the high-energy behaviour of every gauge-theory form factor.

Two features of Eq. (D.217) deserve emphasis. First, the scale dependence enters only through the ratio $(-s)/\mu^2$, exactly because the bookkeeping factor $\mu^{2\varepsilon}$ was attached: an IR pole multiplies $\ln(-s/\mu^2)$ just as a UV pole multiplies $\ln(\mu^2/\text{mass}^2)$, and the sign of the accompanying logarithm is the operational fingerprint of the pole's origin. Second, if either external leg is taken off shell, $p_1^2 \neq 0$, the collinear singularity it carried is screened: the corresponding factor in J becomes $\int_0^1 dy\, y^{-1-\varepsilon}(\cdots)$ with the $y \to 0$ endpoint cut off by the new scale, the double pole softens to a single pole, and a logarithm of the off-shellness appears in its place. The same screening happens for a massive internal line—the electron mass in QED cuts off the collinear region—which is why the QED vertex of Chapter 20 carries only a single $1/\varepsilon_{\text{IR}}$ (soft, no collinear), whereas the massless Yang–Mills vertices of Chapters 27 and 28 carry the full double pole. Mass is the physical IR cutoff that dimensional regularization replaces by ε.

D.7.4 Infrared cancellation: Bloch–Nordsieck and KLN

The infrared poles of Eq. (D.217) are not artefacts of regularization to be subtracted by a counterterm—there is no local operator whose insertion could cancel a $1/\varepsilon_{\text{IR}}$, and a renormalized but IR-divergent amplitude would still predict an infinite cross section. Their resolution is physical: the amplitudes themselves are not measurable. Any real detector has a finite energy resolution and a finite angular resolution, so it cannot distinguish a final state of n hard particles from one of n hard particles accompanied by photons (or gluons) too soft or too collinear to register. The measurable quantity is the *sum* over all such experimentally indistinguishable final states, and in that sum the virtual IR poles cancel against the IR poles of real emission. We state the two theorems that govern this cancellation, then exhibit the mechanism explicitly for the soft photon.

Theorem D.39 (Bloch–Nordsieck)**.** *In QED with massive charged particles, the infrared (soft) divergences arising from virtual photon loops cancel, order by order in α, against the divergences arising from the emission of real soft photons, provided one computes an inclusive transition probability that sums over the emission of any number of photons with total energy below the detector resolution ΔE. The resulting rate is finite as the IR regulator is removed; the $1/\varepsilon_{\text{IR}}$ poles are replaced by finite logarithms of ΔE over the hard scale.*

Theorem D.40 (Kinoshita–Lee–Nauenberg)**.** *In a theory with massless fields, the transition probability summed over all initial and final states that are degenerate in energy (in the limit of vanishing regulator) is free of infrared and mass (collinear) singularities, to all orders in perturbation theory. Bloch–Nordsieck is the special case in which only final-state soft degeneracies occur; with massless charged particles, or when collinear degeneracies are present, the sum over degenerate* initial *states is also required for finiteness.*

The hypotheses are not interchangeable, and the difference is exactly the difference between soft and collinear singularities. Bloch–Nordsieck suffices for an observable like the total cross section for $e^+e^- \to \mu^+\mu^-(+\text{soft } \gamma)$ with massive muons, where only soft singularities arise and summing over final-state photon emission is enough. It *fails* in the presence of collinear singularities from massless charged particles—the classic counterexample is an electron's transition probability in a theory with massless electrons, where final-state summation alone leaves a residual collinear pole and one must also average over the collinear-degenerate initial states, as KLN requires. The practical lesson, made precise in Chapter 28, is that a cross section is guaranteed finite only when the observable it computes is *infrared- and collinear-safe*.

Definition D.41 (Infrared-safe observable)**.** An observable built from final-state momenta is infrared- and collinear-safe if its value is unchanged, in the limit, by (i) the emission of an additional particle with vanishing energy and (ii) the splitting of any particle into two collinear particles carrying its total momentum. Formally, the measurement function $F_n(p_1, \dots, p_n)$ must satisfy $F_{n+1}(\dots, p_{n+1} \to 0) = F_n(\dots)$ and $F_{n+1}(\dots, p_i, p_j \parallel) = F_n(\dots, p_i + p_j)$. Only for such observables do the KLN cancellations occur diagram-class by diagram-class, leaving an IR-finite prediction.

The soft cancellation made explicit

We now demonstrate, for QED, that the virtual soft pole and the real soft pole have equal magnitude and opposite sign, setting up the real-emission phase-space integral whose completion belongs to Section D.8. Consider a charged particle line entering a hard process with momentum p and leaving with momentum p', both on shell, $p^2 = p'^2 = m^2$ (the kinematics of the QED form factor of Chapter 20). In the soft region the emission of a photon of momentum k factorizes: the amplitude with one extra soft photon is the hard amplitude $\mathcal{M}_0$ times the eikonal current,

$$\mathcal{M}_{1\gamma}(k \to 0) \;\longrightarrow\; \mathcal{M}_0 \, e \, \epsilon^*_\mu(k) \, j^\mu(k), \qquad j^\mu(k) = \frac{p'^\mu}{p' \cdot k} - \frac{p^\mu}{p \cdot k}, \tag{D.219}$$

the two terms coming from emission off the outgoing and incoming legs respectively, each the on-shell residue of the nearly-singular fermion propagator after

the spinor algebra collapses (the same eikonal limit that produced the soft part of the loop in Chapter 19). The eikonal current is conserved,

$$k_\mu j^\mu(k) = \frac{p'\cdot k}{p'\cdot k} - \frac{p\cdot k}{p\cdot k} = 1 - 1 = 0, \tag{D.220}$$

which is nothing but charge conservation $\sum_n \eta_n Q_n = 0$ written for the soft photon, and it is the Ward identity that guarantees the construction is gauge-invariant: the longitudinal part of the photon polarization sum, $\sum_\lambda \epsilon^*_\mu \epsilon_\nu = -\eta_{\mu\nu} + (\text{gauge})$, drops out against Eq. (D.220), so the soft factor is unambiguous and the cancellation we are about to display is scheme-independent. This is the precise sense in which the gauge/Ward structure of QED (Chapter 19) underwrites Bloch–Nordsieck.

Real emission. Squaring Eq. (D.219), summing photon polarizations with $-\eta_{\mu\nu}$, and integrating over the d-dimensional one-photon phase space gives the soft real-emission correction to the rate,

$$d\sigma^{\text{soft}}_{\text{real}} = d\sigma_0 \,(-e^2)\int \frac{d^{d-1}k}{(2\pi)^{d-1}\,2\omega_k}\; j(k) j(k), \qquad jj = \frac{m^2}{(p\cdot k)^2} + \frac{m^2}{(p'\cdot k)^2} - \frac{2\,p\cdot p'}{(p\cdot k)(p'\cdot k)}, \tag{D.221}$$

where $\omega_k = |\mathbf{k}|$ for the massless photon and the upper limit of the energy integration is set by the resolution ΔE. The Lorentz-invariant measure $d^{d-1}k/((2\pi)^{d-1}2\omega_k)$ is the d-dimensional one-particle phase space derived in Section D.8; here we need only its soft behaviour. Take the self-energy term $m^2/(p\cdot k)^2$ in the rest frame of p, where $p\cdot k = m\,\omega_k$, so the integrand is ω_k^{-2}, and write $d^{d-1}k = \omega_k^{d-2}\, d\omega_k\, d\Omega_{d-2}$:

$$\int \frac{d^{d-1}k}{(2\pi)^{d-1}2\omega_k}\,\frac{m^2}{(p\cdot k)^2} = \frac{\Omega_{d-2}}{2(2\pi)^{d-1}}\int_0^{\Delta E} d\omega_k\; \omega_k^{d-5} = \frac{\Omega_{d-2}}{2(2\pi)^{d-1}}\,\frac{(\Delta E)^{-2\varepsilon}}{-2\varepsilon}. \tag{D.222}$$

The radial integral $\int_0 \omega_k^{-1-2\varepsilon} d\omega_k$ converges at $\omega_k \to 0$ *only for* $\Re\varepsilon < 0$, i.e. $d > 4$—the infrared regulator, with precisely the sign Eq. (D.200)—and produces an explicit $1/\varepsilon_{\text{IR}}$. The upper cutoff supplies the finite $\ln \Delta E$ promised by Theorem D.39. Restoring $\Omega_{d-2} = 2\pi^{(d-1)/2}/\Gamma(\frac{d-1}{2}) \to 4\pi$ at $d = 4$, this representative term contributes $-\frac{1}{8\pi^2}\frac{1}{\varepsilon_{\text{IR}}} + \dots$ to $j\cdot j$ integrated, and the cross term $2p\cdot p'/((p\cdot k)(p'\cdot k))$ adds the velocity-dependent piece; assembled, Eq. (D.221) yields

$$d\sigma^{\text{soft}}_{\text{real}} = d\sigma_0\,\frac{\alpha}{\pi}\Big(+\frac{1}{\varepsilon_{\text{IR}}}\Big)\, f(v) + (\text{finite}), \tag{D.223}$$

with $f(v)$ the same kinematic function—built from the eikonal integrals of Eq. (D.221)—that appears below in the virtual contribution, v the relative velocity of p and p'.

Virtual correction. The one-loop virtual contribution to the same rate is $d\sigma_{\text{virt}} = 2\, d\sigma_0\, \Re F_1(q^2)$, with F_1 the Dirac form factor of Chapters 19 and 20. Its infrared pole comes entirely from the soft region $\ell \to 0$ of the loop, where the integrand reduces to the *same* eikonal current squared but integrated against the

off-shell photon propagator $i/(\ell^2+i\varepsilon)$ over all of d-dimensional loop momentum rather than over the on-shell phase space:

$$2\,\Re F_1\big|_{\text{soft}} = (-e^2)\,\Re\!\int\!\frac{d^d\ell}{(2\pi)^d}\,\frac{i}{\ell^2+i\varepsilon}\;j(\ell)\cdot j(\ell)+\dots \tag{D.224}$$

The decisive structural fact is that the propagator carries $\Im\big[i/(\ell^2+i\varepsilon)\big]=\pi\,\delta(\ell^2)$: cutting the photon line—putting it on shell, $\ell^2=0$—turns the loop integral Eq. (D.224) into precisely the real-emission phase-space integral Eq. (D.221), the optical-theorem relation that Section D.8.5 establishes between the imaginary part of a loop and a phase-space integral. The same eikonal integrand integrated over the off-shell loop ($d^d\ell$, which by the analysis of Section D.7.1 produces its $1/\varepsilon_{\text{IR}}$ from the soft endpoint with the *opposite* sign to the on-shell phase-space measure $d^{d-1}k/\omega_k$) gives

$$d\sigma_{\text{virt}} = d\sigma_0\,\frac{\alpha}{\pi}\Big(-\frac{1}{\varepsilon_{\text{IR}}}\Big)\,f(v)+(\text{finite}), \tag{D.225}$$

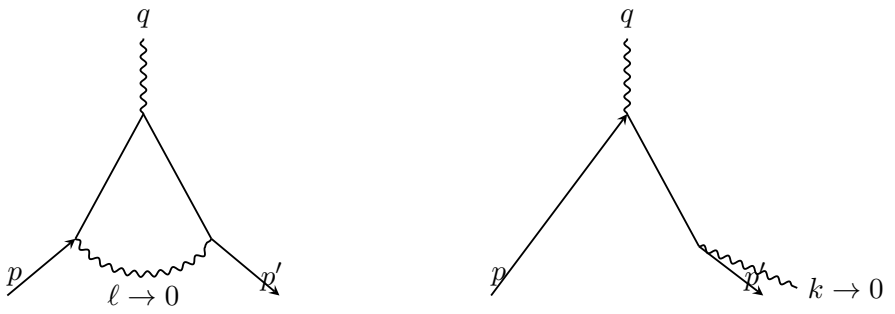
with the identical coefficient $f(v)$ because the soft virtual and soft real contributions are governed by one and the same eikonal current Eq. (D.219). The explicit one-loop evaluation of $f(v)$ is carried out in Chapter 19; here the load-bearing point is the relative sign, which follows from the IR-region analysis of Section D.7.1 and the cut relation above.

Cancellation. Adding Eqs. (D.223) and (D.225),

$$d\sigma_{\text{virt}}+d\sigma_{\text{real}}^{\text{soft}} = d\sigma_0\,\frac{\alpha}{\pi}\,f(v)\Big(-\frac{1}{\varepsilon_{\text{IR}}}+\frac{1}{\varepsilon_{\text{IR}}}\Big)+(\text{finite}) = d\sigma_0\,\frac{\alpha}{\pi}\,f(v)\,\ln\frac{\Delta E}{\dots}+(\text{finite}), \tag{D.226}$$

the $1/\varepsilon_{\text{IR}}$ poles cancel identically, exactly as Eq. (D.207) arranged for the scaleless integral, and the regulator-dependence is replaced by a finite logarithm of the detector resolution ΔE over the hard scale. This is Bloch–Nordsieck made arithmetic, and it is what renders the precision predictions of Chapter 20—the anomalous moment $a_e=\alpha/2\pi$ and the radiatively corrected cross sections—finite, unambiguous numbers. Figure D.8 shows the two diagram classes whose poles cancel.

For Yang–Mills theory the same mechanism operates, but with two complications stated here and developed in Chapters 27 and 28. First, the gluon is itself charged under the gauge group, so the eikonal current is colour-valued and its conservation Eq. (D.220) becomes the covariant statement that the colour charge is conserved at the soft vertex; the polarization sum requires the Faddeev–Popov ghosts of Chapter 26 to remove unphysical states. Second, with massless quarks and gluons the collinear singularities of Theorem D.38 are present in addition to the soft ones, so Bloch–Nordsieck alone is insufficient and the full KLN sum, together with the factorization of collinear poles into parton distributions, is what yields finite, infrared-safe QCD cross sections (Chapter 28). The conceptual



(a) virtual soft photon (b) real soft photon

Figure D.8: The two contributions whose infrared poles cancel in the inclusive rate. In (a) a virtual soft photon of momentum $\ell \to 0$ spans the incoming and outgoing legs of the hard vertex, giving $-\frac{1}{\varepsilon_{\rm IR}} f(v)$ via Eq. (D.224). In (b) a real soft photon of momentum $k \to 0$ is emitted into the unresolved final state, giving $+\frac{1}{\varepsilon_{\rm IR}} f(v)$ via Eq. (D.221). Both are controlled by the same conserved eikonal current Eq. (D.219); their sum Eq. (D.226) is finite. Cutting the internal photon in (a) on shell ($\ell^2 = 0$) turns it into (b)—the optical-theorem relation of Section D.8.5.

content—one ε regulating both UV and IR, virtual poles cancelling real-emission poles in inclusive observables—is identical.

The classification of $1/\varepsilon$ poles is now complete: a pole is ultraviolet (reached from $d < 4$, removed by a counterterm) or infrared (reached from $d > 4$, cancelled between virtual and real emission), and a scaleless integral is the limiting case in which one of each cancels the other. The arithmetic of the cancellation, however, has left one half of the calculation owed: Eqs. (D.221) and (D.223) required the d-dimensional final-state phase-space measure, used so far only in its soft limit. The remaining section builds that measure in closed form—the two-body phase space, the three-body Dalitz region, and the optical-theorem identity that ties the imaginary part of the bubble to two-body phase space and thereby closes the loop on Eq. (D.224)—so that the real-emission integrals set up here can be completed and the cross-check delivered.

D.8 Phase-space integrals in d dimensions

The previous section closed the virtual half of a one-loop calculation: it classified the poles of a loop integral, separated the ultraviolet from the infrared, disposed of the scaleless integral that vanishes because its soft and collinear divergences cancel one another, and explained, at the level of a statement, why the infrared poles of a virtual correction must be cancelled by the emission of soft and collinear

real radiation. That cancellation is a promise the loop chapters cannot keep on their own, because real emission is not a loop integral at all: it is an integral over the momenta of *extra final-state particles*, weighted by the squared matrix element. To discharge the promise we need the second half of the machine — the Lorentz-invariant final-state phase space, built in $d = 4 - 2\varepsilon$ dimensions so that its infrared poles are written in the same regulator as the virtual ones and can be added to them. The same construction does double duty: the two-body phase space in closed form is exactly the kinematic factor that turns the squared amplitude of $e^+e^- \to \mu^+\mu^-$ in Chapter 17 into a cross section, and its analytic structure is the right-hand side of the optical theorem. We therefore build the measure, evaluate the two-body case in full, set up the three-body (Dalitz) phase space and isolate its soft corner, complete the real-emission cancellation, and close the appendix by checking that the imaginary part of the ϕ^4 bubble computed in the ε-expansion section equals precisely the two-body phase-space volume — Cutkosky's rule verified on a single integral, with every factor of i and π tracked.

D.8.1 The Lorentz-invariant phase-space measure in d dimensions

Fix a total incoming four-momentum P (with $P^2 = s > 0$ in the physical region) and consider a final state of n particles with momenta $p_1, \dots, p_n$ and masses $m_1, \dots, m_n$. Each final particle is on shell, $p_i^2 = m_i^2$ with positive energy $p_i^0 = E_i > 0$, and the total four-momentum is conserved. The object that collects these constraints with the correct relativistic normalization is the *Lorentz-invariant phase-space measure*

$$d\Pi_n \;=\; (2\pi)^d\,\delta^d\Big(P - \sum_{i=1}^{n} p_i\Big)\;\prod_{i=1}^{n}\frac{d^{d-1}\mathbf{p}_i}{(2\pi)^{d-1}\,2E_i}\,, \qquad E_i = \sqrt{\mathbf{p}_i^{\,2} + m_i^2}\,. \tag{D.227}$$

This generalizes the four-dimensional measure derived in Chapter 16 by the single replacement $4 \to d$ in every counting of dimensions: each final particle carries a spatial measure $d^{d-1}\mathbf{p}_i$ divided by $(2\pi)^{d-1}$ and by the energy denominator $2E_i$, and overall energy–momentum conservation contributes one factor $(2\pi)^d\,\delta^d(\cdots)$. The bookkeeping is dictated by the relativistic state normalization $\langle \mathbf{p}|\mathbf{p}'\rangle = (2\pi)^{d-1}\,2E_{\mathbf{p}}\,\delta^{d-1}(\mathbf{p}-\mathbf{p}')$ inherited from the canonical quantization of Chapter 9; with this normalization the (2π) factors in Eq. (D.227) are exactly those required for $d\Pi_n$ to combine with a squared invariant amplitude $|\mathcal{M}|^2$ and a flux factor into a frame-independent rate.

The single most useful manipulation is to trade the spatial measure-plus-energy-denominator for a manifestly covariant on-shell delta function. For one particle,

$$\frac{d^{d-1}\mathbf{p}}{(2\pi)^{d-1}\,2E_{\mathbf{p}}} \;=\; \frac{d^d p}{(2\pi)^d}\,(2\pi)\,\delta^+(p^2 - m^2)\,, \qquad \delta^+(p^2-m^2) \equiv \theta(p^0)\,\delta(p^2-m^2)\,. \tag{D.228}$$

The identity is proved by performing the energy integral on the right. Writing $p^2-m^2=(p^0)^2-(\mathbf{p}^2+m^2)=(p^0)^2-E_{\mathbf{p}}^2$ and using $\delta(f(p^0))=\sum_{\text{roots}}|f'|^{-1}\delta(p^0-\text{root})$ with roots $p^0=\pm E_{\mathbf{p}}$,

$$\delta\big((p^0)^2-E_{\mathbf{p}}^2\big)=\frac{1}{2E_{\mathbf{p}}}\Big[\delta(p^0-E_{\mathbf{p}})+\delta(p^0+E_{\mathbf{p}})\Big]\,. \tag{D.229}$$

The factor $\theta(p^0)$ kills the negative-energy root, so $\int dp^0\,(2\pi)\,\delta^+(p^2-m^2)=2\pi/(2E_{\mathbf{p}})$, and dividing by the $(2\pi)^d$ that accompanies d^dp leaves $d^{d-1}\mathbf{p}/[(2\pi)^{d-1}2E_{\mathbf{p}}]$, which is the left-hand side. Each factor on the right of Eq. (D.228) is separately Lorentz invariant: d^dp is invariant under $SO(1,d-1)$, $\delta(p^2-m^2)$ is invariant because p^2 is, and $\theta(p^0)$ is invariant on the forward mass shell because no orthochronous Lorentz transformation can flip the sign of the energy of a timelike or null momentum. Hence $d^{d-1}\mathbf{p}/2E_{\mathbf{p}}$ is Lorentz invariant, the property that gives the measure its name. Substituting Eq. (D.228) into Eq. (D.227) gives the covariant form

$$d\Pi_n \;=\; (2\pi)^d\,\delta^d\Big(P-\sum_i p_i\Big)\,\prod_{i=1}^{n}\frac{d^dp_i}{(2\pi)^d}\,(2\pi)\,\delta^+(p_i^2-m_i^2)\,, \tag{D.230}$$

in which every constraint is an explicit delta function and the only frame-dependent step left to any computation is the choice of how to parametrize the remaining integrations. The total phase-space volume,

$$\Phi_n(s)\;\equiv\;\int d\Pi_n\,, \tag{D.231}$$

is a Lorentz scalar depending on $s=P^2$ and the masses; it carries mass dimension $(n-1)(d-2)-2$, as a quick scaling of Eq. (D.230) confirms, and it is the object the optical theorem and the inclusive rates of Chapter 28 are built from.

D.8.2 Two-body phase space in closed form

The case $n=2$ is the workhorse and the only one needed in closed form. Two final particles of masses m_1,m_2 are produced from P with $P^2=s$. Starting from the covariant measure Eq. (D.230),

$$\Phi_2(s)=\int\frac{d^dp_1}{(2\pi)^d}\frac{d^dp_2}{(2\pi)^d}\,(2\pi)^2\,\delta^+(p_1^2-m_1^2)\,\delta^+(p_2^2-m_2^2)\,(2\pi)^d\,\delta^d(P-p_1-p_2)\,. \tag{D.232}$$

The d-dimensional delta function does the entire p_2 integral, fixing $p_2=P-p_1$ and leaving

$$\Phi_2(s)=(2\pi)^{2-d}\int d^dp_1\;\delta^+(p_1^2-m_1^2)\,\delta^+((P-p_1)^2-m_2^2)\,. \tag{D.233}$$

Work in the centre-of-momentum frame, $P = (\sqrt{s}, \mathbf{0})$, and split $d^d p_1 = dp_1^0\, d^{d-1}\mathbf{p}$. The first delta is dispatched exactly as in Eq. (D.228): it sets $p_1^0 = E_1 = \sqrt{\mathbf{p}^2 + m_1^2}$ and supplies a Jacobian $1/(2E_1)$. With p_1^0 now fixed, the argument of the second delta is a function of the radial variable $|\mathbf{p}|$ alone,

$$(P - p_1)^2 - m_2^2 = s - 2\sqrt{s}\, E_1 + m_1^2 - m_2^2 \;\equiv\; g(|\mathbf{p}|)\,, \tag{D.234}$$

where we used $P \cdot p_1 = \sqrt{s}\, E_1$ and $p_1^2 = m_1^2$. Its derivative is $g'(|\mathbf{p}|) = -2\sqrt{s}\,(dE_1/d|\mathbf{p}|) = -2\sqrt{s}\,|\mathbf{p}|/E_1$, so

$$\delta(g(|\mathbf{p}|)) = \frac{E_1}{2\sqrt{s}\,|\mathbf{p}|}\,\delta(|\mathbf{p}| - |\mathbf{p}|_\star)\,, \tag{D.235}$$

with $|\mathbf{p}|_\star$ the unique positive root of g. Writing the angular part of the spatial measure as $d^{d-1}\mathbf{p} = |\mathbf{p}|^{d-2}\, d|\mathbf{p}|\, d\Omega_{d-1}$, the radial integral collapses against this delta and the two energy Jacobians combine, $1/(2E_1) \times E_1/(2\sqrt{s}\,|\mathbf{p}|_\star) = 1/(4\sqrt{s}\,|\mathbf{p}|_\star)$:

$$\Phi_2(s) = (2\pi)^{2-d}\,\Omega_{d-1}\,\frac{|\mathbf{p}|_\star^{d-3}}{4\sqrt{s}}\,, \qquad \Omega_{d-1} = \int d\Omega_{d-1} = \frac{2\pi^{(d-1)/2}}{\Gamma\!\left(\frac{d-1}{2}\right)}\,, \tag{D.236}$$

where Ω_{d-1} is the surface area of the unit sphere in the $(d-1)$ spatial dimensions, the same angular factor derived in the Wick-rotation section (Eq. (D.63)). The power $|\mathbf{p}|_\star^{d-3}$ is the d-dependent piece: in $d = 4$ it is linear in the centre-of-mass momentum, while away from four dimensions it carries the regulator into the threshold behaviour.

It remains to express the centre-of-mass momentum $|\mathbf{p}|_\star$ through the invariants. The root of g gives the energy of particle 1,

$$E_1 = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}}\,, \tag{D.237}$$

and then $|\mathbf{p}|_\star^2 = E_1^2 - m_1^2 = \dfrac{(s + m_1^2 - m_2^2)^2 - 4s\, m_1^2}{4s}$. The numerator is the Källén triangle function $\lambda(a,b,c) = a^2 + b^2 + c^2 - 2ab - 2bc - 2ca$ introduced with the two-body kinematics of Chapter 16: expanding $(s + m_1^2 - m_2^2)^2 - 4sm_1^2 = s^2 + m_1^4 + m_2^4 - 2sm_1^2 - 2sm_2^2 - 2m_1^2 m_2^2 = \lambda(s, m_1^2, m_2^2)$. Hence

$$|\mathbf{p}|_\star = \frac{\lambda^{1/2}(s, m_1^2, m_2^2)}{2\sqrt{s}}\,, \tag{D.238}$$

which is real precisely above threshold, $\sqrt{s} \geq m_1 + m_2$, where $\lambda \geq 0$. Collecting Eqs. (D.236) and (D.238) gives the two-body phase space in closed form for arbitrary d:

$$\boxed{\Phi_2(s) = \frac{(2\pi)^{2-d}}{4\sqrt{s}}\,\frac{2\pi^{(d-1)/2}}{\Gamma\!\left(\frac{d-1}{2}\right)}\,\left(\frac{\lambda^{1/2}(s, m_1^2, m_2^2)}{2\sqrt{s}}\right)^{d-3}} \tag{D.239}$$

For equal masses $m_1 = m_2 = m$ one has $\lambda(s, m^2, m^2) = s^2 - 4sm^2 = s^2\beta^2$ with the velocity $\beta = \sqrt{1 - 4m^2/s}$, so $|\mathbf{p}|_\star = \tfrac{1}{2}\sqrt{s}\,\beta$ and $\Phi_2 \propto \beta^{\,d-3}$; the threshold vanishing $\beta \to 0$ is softened or sharpened by the regulator, a fact that matters when a phase-space pole multiplies a virtual pole.

Reduction to $d = 4$. Setting $d = 4$ in Eq. (D.239) uses $\Omega_3 = 2\pi^{3/2}/\Gamma(3/2) = 2\pi^{3/2}/(\tfrac{1}{2}\sqrt{\pi}) = 4\pi$ and $(2\pi)^{2-d} = (2\pi)^{-2}$:

$$\Phi_2(s)\Big|_{d=4} = \frac{1}{(2\pi)^2}\,\frac{4\pi}{4\sqrt{s}}\,|\mathbf{p}|_\star = \frac{|\mathbf{p}|_\star}{4\pi\sqrt{s}} = \frac{1}{8\pi}\,\frac{2|\mathbf{p}|_\star}{\sqrt{s}} = \frac{1}{8\pi}\,\frac{\lambda^{1/2}(s, m_1^2, m_2^2)}{s}\,. \tag{D.240}$$

For equal masses this is $\Phi_2 = \beta/8\pi$, and in the massless limit $m \to 0$ it reduces to the familiar $1/8\pi$. The combination $2|\mathbf{p}|_\star/\sqrt{s}$ is the same kinematic factor that appears in the $e^+e^- \to \mu^+\mu^-$ cross section of Chapter 17: undoing the angular integration in Eq. (D.236) gives the differential measure

$$d\Phi_2 = \frac{(2\pi)^{2-d}}{4\sqrt{s}}\,|\mathbf{p}|_\star^{\,d-3}\,d\Omega_{d-1} \xrightarrow{\;d=4\;} \frac{1}{(2\pi)^2}\,\frac{|\mathbf{p}|_\star}{4\sqrt{s}}\,d\Omega = \frac{1}{32\pi^2}\,\frac{2|\mathbf{p}|_\star}{\sqrt{s}}\,d\Omega\,, \tag{D.241}$$

so that the differential cross section is $d\sigma/d\Omega = |\mathcal{M}|^2\,d\Phi_2/(d\Omega\cdot\mathcal{F})$ with the incident flux $\mathcal{F}$ supplied by Chapter 16. This is exactly the $1/64\pi^2 s$ prefactor of the $2 \to 2$ cross section that drives QED, now derived as a special case of the general d-dimensional measure.

Physical Insight

The whole content of two-body phase space is a single number, the centre-of-mass momentum $|\mathbf{p}|_\star$ of Eq. (D.238): it sets the size of the available sphere of directions, and Eq. (D.239) is just "solid angle times a one-dimensional Jacobian." Everything dimension-dependent sits in the angular factor Ω_{d-1} and the power $|\mathbf{p}|_\star^{\,d-3}$. When a $1/\varepsilon$ infrared pole multiplies this measure, it is the d-dependence of these two pieces that decides the finite remainder — which is why the cross section must be assembled in d dimensions and expanded only at the very end.

D.8.3 Three-body phase space and the Dalitz parametrization

Beyond two particles there is no single closed form, but there is an exact recursive structure that reduces every n-body phase space to nested two-body integrals. Inserting into Eq. (D.230) a complete set of states for an intermediate momentum $q = p_1 + \cdots + p_j$, written as $1 = \int \frac{d^dq}{(2\pi)^d}\,(2\pi)^d\delta^d(q - \sum_{i\le j} p_i)$ together with a unit factor $1 = \int \frac{dq^2}{2\pi}\,(2\pi)\,\delta(q^2 - (\text{its value}))$ implemented through the on-shell

decomposition, gives the factorization

$$d\Pi_n(P;p_1,\dots,p_n) = d\Pi_j(q;p_1,\dots,p_j)\,\frac{dq^2}{2\pi}\,d\Pi_{n-j+1}(P;q,p_{j+1},\dots,p_n)\,. \tag{D.242}$$

The invariant q^2 ranges over its kinematically allowed band, and each factor is an honest lower-multiplicity measure of the form already evaluated. For three particles ($n=3$, $j=2$) the recursion expresses $d\Pi_3$ as a two-body decay of P into q and p_3, an integral over the invariant mass $q^2 = s_{12} \equiv (p_1+p_2)^2$, and a two-body decay of q into p_1, p_2:

$$d\Pi_3 = \frac{ds_{12}}{2\pi}\,d\Pi_2(P;q,p_3)\,d\Pi_2(q;p_1,p_2)\,. \tag{D.243}$$

In $d=4$ this collapses to the Dalitz form. Carrying out both two-body angular integrals — using Eq. (D.240) for each and choosing the two independent invariants $s_{12}=(p_1+p_2)^2$ and $s_{23}=(p_2+p_3)^2$ as variables, with the third fixed by $s_{12}+s_{13}+s_{23} = s+m_1^2+m_2^2+m_3^2$ — one finds the flat Dalitz measure

$$d\Phi_3\Big|_{d=4} = \frac{1}{(2\pi)^3}\,\frac{1}{16\,s}\,ds_{12}\,ds_{23}\,. \tag{D.244}$$

The defining feature of the Dalitz plot is that the squared matrix element is the *only* non-constant weight: the phase space itself is uniform in the plane (s_{12},s_{23}), so the density of events directly maps the dynamics. The physical region is the interior of the kinematic boundary on which one of the relative angles reaches ± 1; for three massless final particles it degenerates to the triangle $s_{ij}\geq 0$, $s_{12}+s_{13}+s_{23}=s$ drawn in Fig. D.9.

For the infrared application we do not need the full d-dimensional three-body volume; we need only the behaviour near the boundary, where one emitted massless parton becomes soft or collinear. The recursion Eq. (D.242) isolates this region cleanly: write the emitted parton as particle 3 with momentum k and energy $\omega = |\mathbf{k}|$, and let $s_{12} = (P-k)^2$ be the invariant mass recoiling against it. The soft limit $\omega \to 0$ is the limit $s_{12}\to s$, in which the two-body factor $d\Pi_2(P;q,p_3)$ degenerates to the emission measure $d^{d-1}\mathbf{k}/[(2\pi)^{d-1}2\omega]$ of the single soft parton. This single-emission measure, kept in d dimensions, is the object that supplies the real-emission poles.

D.8.4 The soft region and completion of the infrared cancellation

We can now keep the promise left open by the previous section: that the $1/\varepsilon$ poles of a virtual correction are cancelled by real emission, and that dimensional regularization makes both sides simultaneously finite so that the cancellation is an algebraic identity rather than a hope. The arena is any process with a hard core dressed by the emission of one extra massless gauge boson (a soft

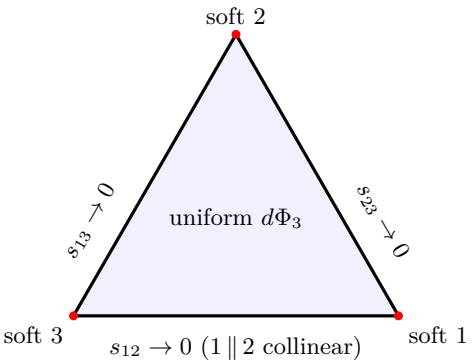


Figure D.9: Massless three-body Dalitz region. Phase space is uniform across the interior, Eq. (D.244). The three edges $s_{ij} \to 0$ are the collinear configurations (two partons parallel); the three corners, where two invariants vanish simultaneously, are the soft configurations (one parton's energy $\to 0$). The infrared poles of an inclusive rate live entirely on this boundary.

photon in QED, a soft gluon in the YM thread) of momentum k. As $k \to 0$ the $(n+1)$-particle amplitude factorizes into the n-particle amplitude times the classical eikonal current: for emission off external lines i carrying charges Q_i, signs $\eta_i = +1$ (outgoing) or -1 (incoming), and coupling e,

$$\mathcal{M}_{n+1} \xrightarrow{k\to 0} \mathcal{M}_n \times e \sum_i \eta_i Q_i \frac{p_i \cdot \varepsilon^*(k)}{p_i \cdot k}\,, \tag{D.245}$$

the leading $1/(p_i \cdot k)$ being the pole of the propagator of the hard line just before it emits. Squaring, summing over the boson polarizations with $\sum_\lambda \varepsilon^\mu \varepsilon^{*\nu} \to -\eta^{\mu\nu}$ (the gauge-dependent remainder cancels by the Ward identity proved in Chapter 19), and integrating over the soft boson's phase space, the real-emission correction to the rate is $|\mathcal{M}_n|^2$ times the eikonal phase-space integral

$$W \;=\; -\,e^2 \int \frac{d^{d-1}\mathbf{k}}{(2\pi)^{d-1}\,2\omega} \sum_{i,j} \eta_i \eta_j Q_i Q_j \frac{p_i \cdot p_j}{(p_i \cdot k)(p_j \cdot k)}\,. \tag{D.246}$$

The structure of the divergences is read off the integrand without choosing a frame. Parametrize $k = \omega(1, \mathbf{n})$ with $\mathbf{n}$ a unit vector in $d-1$ spatial dimensions, so $p_i \cdot k = \omega\,(E_i - \mathbf{p}_i \cdot \mathbf{n})$ and the measure is $d^{d-1}\mathbf{k}/2\omega = \tfrac{1}{2}\,\omega^{d-3}\,d\omega\,d\Omega_{d-1}$. Each term of Eq. (D.246) then has the radial profile

$$\frac{\omega^{d-3}}{(p_i \cdot k)(p_j \cdot k)} \;\propto\; \frac{\omega^{d-3}}{\omega^2} = \omega^{d-5} = \omega^{-1-2\varepsilon}\,, \tag{D.247}$$

so that, with an upper cutoff Δ on the soft energy that delimits the soft region (set, in a complete calculation, by the recoil kinematics; energies above it belong

to the hard region),

$$\int_0^{\Delta} d\omega\, \omega^{-1-2\varepsilon} = \frac{\Delta^{-2\varepsilon}}{-2\varepsilon} = -\frac{1}{2\varepsilon} - \ln\Delta + O(\varepsilon)\,. \tag{D.248}$$

This is the soft pole: the logarithmic divergence at $\omega \to 0$, which in four dimensions would be $\int d\omega/\omega$, is regulated by $d = 4 - 2\varepsilon$ into a single $1/\varepsilon$ with the residue carrying the soft logarithm $\ln\Delta$. The pole is of *infrared* origin — it comes from the lower limit $\omega \to 0$, the opposite end of the integration from the ultraviolet poles of the loop sections — yet it wears the same $1/\varepsilon$, which is the whole point of using one regulator for both.

The angular integral supplies the rest, and for massless emitters it generates the second, collinear pole. Consider the diagonal-pair term in Eq. (D.246) for two back-to-back massless hard legs p_1, p_2 in their centre-of-mass frame, $\mathbf{p}_1\cdot\mathbf{n} = E\cos\theta$, $\mathbf{p}_2\cdot\mathbf{n} = -E\cos\theta$, giving $(p_1\cdot k)(p_2\cdot k) \propto \omega^2 E^2(1-\cos\theta)(1+\cos\theta)$. The required angular integral is the d-dimensional master

$$\int d\Omega_{d-1}\,(1-\cos\theta)^{-a}(1+\cos\theta)^{-b} = \Omega_{d-2}\,2^{\,d-3-a-b}\,\frac{\Gamma\!\left(\frac{d-2}{2}-a\right)\Gamma\!\left(\frac{d-2}{2}-b\right)}{\Gamma(d-2-a-b)}\,, \tag{D.249}$$

with $\Omega_{d-2} = 2\pi^{(d-2)/2}/\Gamma(\frac{d-2}{2})$. The formula is proved by peeling off the polar angle, $d\Omega_{d-1} = \sin^{d-3}\theta\, d\theta\, d\Omega_{d-2}$, substituting $u = \cos\theta$ so that $\sin^{d-3}\theta\, d\theta = -(1-u^2)^{(d-4)/2}\, du$, and then $t = (1+u)/2$:

$$\begin{aligned}\int d\Omega_{d-1}(1-\cos\theta)^{-a}(1+\cos\theta)^{-b} &= \Omega_{d-2}\int_{-1}^{1} du\,(1-u)^{\frac{d-4}{2}-a}(1+u)^{\frac{d-4}{2}-b}\\ &= \Omega_{d-2}\,2^{\,d-3-a-b}\int_0^1 dt\,(1-t)^{\frac{d-2}{2}-a-1}t^{\frac{d-2}{2}-b-1}\\ &= \Omega_{d-2}\,2^{\,d-3-a-b}\,B\!\left(\tfrac{d-2}{2}-b,\ \tfrac{d-2}{2}-a\right),\end{aligned} \tag{D.250}$$

the last step being the Euler Beta integral $\int_0^1 t^{x-1}(1-t)^{y-1}dt = B(x,y) = \Gamma(x)\Gamma(y)/\Gamma(x+y)$, which reproduces Eq. (D.249). The eikonal back-to-back case is $a = b = 1$. With $d = 4 - 2\varepsilon$ one has $\frac{d-2}{2} - 1 = -\varepsilon$, so the angular factor carries $\Gamma(-\varepsilon)^2/\Gamma(-2\varepsilon)$, whose $\Gamma(-\varepsilon)$ each contribute a $1/\varepsilon$ collinear pole from the endpoints $\cos\theta = \pm 1$ where the soft boson is parallel to one of the hard legs. Multiplying by the radial $1/\varepsilon$ of Eq. (D.248) produces the characteristic

$$W \;\sim\; \frac{1}{\varepsilon^2} \quad (\text{soft} \times \text{collinear})\,, \tag{D.251}$$

the overlapping double pole of a massless gauge theory. When the emitter is massive, $p_i^2 = m_i^2 \neq 0$, the factor $1-\cos\theta$ never reaches zero ($E_i - \mathbf{p}_i\cdot\mathbf{n} \geq E_i - |\mathbf{p}_i| > 0$), the angular integral is finite, and only the single soft pole Eq. (D.248) survives — which is the QED case relevant to the electron, where

the collinear divergence is cut off by the electron mass and the residual infrared pole is purely soft.

The cancellation is now an identity. The virtual correction to the same observable — the vertex and self-energy graphs evaluated in Chapters 19 and 20 for QED and in Chapters 27 and 28 for QCD — produces, after renormalization of the ultraviolet subdivergences, exactly the infrared poles

$$2\,\mathrm{Re}\left(\mathcal{M}_n^{*}\,\mathcal{M}_n^{\mathrm{virtual}}\right) \;\supset\; |\mathcal{M}_n|^2\left(+\frac{c_2}{\varepsilon^2}+\frac{c_1}{\varepsilon}\right), \tag{D.252}$$

with coefficients c_2, c_1 fixed by the same eikonal/collinear data — the cusp and collinear anomalous dimensions — that govern W. Because the loop integral picks up its infrared pole from the region where the *virtual* boson is soft, and the phase-space integral picks up its pole from the region where the *real* boson is soft, the two regions are kinematically identical and the residues are equal and opposite:

$$\underbrace{\left(\frac{c_2}{\varepsilon^2}+\frac{c_1}{\varepsilon}\right)}_{\text{virtual}} + \underbrace{W = \left(-\frac{c_2}{\varepsilon^2}-\frac{c_1}{\varepsilon}+\text{finite}\right)}_{\text{real}} \;=\; \text{finite}\,, \tag{D.253}$$

the statement of the Bloch–Nordsieck theorem for soft photons and, with the collinear poles included, of the Kinoshita–Lee–Nauenberg theorem. The finite remainder contains the physical logarithms $\ln(\Delta/\sqrt{s})$ of the resolution Δ; their exponentiation is the Sudakov form factor. The mechanics matter for the present appendix only at the level just shown: the regulator ε that the loop sections introduced to tame the ultraviolet is the *same* parameter that, acting now at the lower end of the soft phase-space integral Eq. (D.248) and at the angular endpoints Eq. (D.249), regulates the infrared, so that virtual and real contributions are Laurent series in one ε and add term by term. This is the closure of the pipeline: parametrize, rotate, integrate, expand the virtual side; integrate the real phase space here; add and find the poles gone. The numerical payoff — Schwinger's $a_e = \alpha/2\pi$ in Chapter 20 and the inclusive R-ratio of Chapter 28 — rides on exactly this cancellation.

D.8.5 The optical theorem on a single integral

We close the appendix where the ε-expansion section pointed: the branch cut of the bubble. There we found that the one-loop two-point integral develops an imaginary part above threshold; unitarity, through the optical theorem of Chapter 18, demands that this imaginary part be a physical phase-space volume. The following worked example verifies the demand exactly for the ϕ^4 bubble — Cutkosky's rule Chapter 18 checked on this one integral, every factor of i and π kept.

Worked Example D.42 ($2\,\mathrm{Im}\,B$ equals the two-body phase space)**.** We continue ϕ^4 theory from the ε-expansion section. The s-channel one-loop correction to

$\phi\phi \to \phi\phi$ is the bubble of Fig. D.9's sibling diagram, built from two propagators $i/(p^2 - m^2 + i\epsilon)$ and two vertices $-i\lambda$, with the bubble symmetry factor $\frac{1}{2}$:

$$i\mathcal{M}_{\text{bub}}(s) = \frac{1}{2}(-i\lambda)^2 \int \frac{d^d\ell}{(2\pi)^d} \frac{i}{\ell^2 - m^2 + i\epsilon} \frac{i}{(P-\ell)^2 - m^2 + i\epsilon} = \frac{\lambda^2}{2} I(s)\,, \tag{D.254}$$

where $P^2 = s$, the two factors of i from the propagators and the $(-i\lambda)^2 = -\lambda^2$ combine to a net $+1/2$ times λ^2, and

$$I(s) \equiv \int \frac{d^d\ell}{(2\pi)^d} \frac{1}{(\ell^2 - m^2 + i\epsilon)\big((P-\ell)^2 - m^2 + i\epsilon\big)} \tag{D.255}$$

is the scalar bubble whose Feynman-parametrized value was established in the ε-expansion section,

$$I(s) = \frac{i}{16\pi^2}\left[\frac{1}{\bar{\varepsilon}} - \int_0^1 dx\, \ln\frac{m^2 - x(1-x)s - i\epsilon}{\mu^2}\right] + O(\varepsilon), \qquad \frac{1}{\bar{\varepsilon}} = \frac{1}{\varepsilon} - \gamma_E + \ln 4\pi\,, \tag{D.256}$$

with the $\mu^{2\varepsilon}$ bookkeeping of $\overline{\text{MS}}$ as in Chapter A. Solving Eq. (D.254) for the amplitude, $\mathcal{M}_{\text{bub}} = -i\,(\lambda^2/2)\, I(s)$, and inserting Eq. (D.256),

$$\mathcal{M}_{\text{bub}}(s) = \frac{\lambda^2}{32\pi^2}\left[\frac{1}{\bar{\varepsilon}} - \int_0^1 dx\, \ln\frac{m^2 - x(1-x)s - i\epsilon}{\mu^2}\right] + O(\varepsilon)\,, \tag{D.257}$$

where the explicit $-i$ has cancelled the i of Eq. (D.256) to leave a real prefactor: the divergence and the real part of the log are real, as a $2 \to 2$ amplitude's real part should be.

Imaginary part. The only source of an imaginary part is the logarithm, and it switches on exactly above threshold. The argument $m^2 - x(1-x)s$ is negative when $x(1-x) > m^2/s$, i.e. for $x \in (x_-, x_+)$ with $x_\pm = \frac{1}{2}(1 \pm \beta)$ the roots of $x(1-x) = m^2/s$, where $\beta = \sqrt{1 - 4m^2/s}$; such an interval exists only for $s > 4m^2$. For x in it, the $-i\epsilon$ prescription places the argument just below the negative real axis, so $\ln(\text{neg} - i\epsilon) = \ln|\cdots| - i\pi$ and $\operatorname{Im}\ln = -\pi$. Hence

$$\operatorname{Im}\mathcal{M}_{\text{bub}}(s) = \frac{\lambda^2}{32\pi^2}\Big(-\int_{x_-}^{x_+} dx\,(-\pi)\Big) = \frac{\lambda^2}{32\pi^2}\,\pi\,(x_+ - x_-) = \frac{\lambda^2\beta}{32\pi}\,, \tag{D.258}$$

using $x_+ - x_- = \beta$. Therefore

$$2\operatorname{Im}\mathcal{M}_{\text{bub}}(s) = \frac{\lambda^2\beta}{16\pi} \qquad (s > 4m^2)\,. \tag{D.259}$$

Phase-space side. The optical theorem of Chapter 18 states $2\operatorname{Im}\mathcal{M}(i \to i) = \sum_f \int d\Pi_f\, |\mathcal{M}(i \to f)|^2$. At one loop the only accessible intermediate state f is the two-particle state of two identical scalars; its tree amplitude is the

contact term $\mathcal{M}_{\text{tree}}(\phi\phi \to \phi\phi) = -\lambda$, and the sum over identical-particle final states carries the symmetry factor $1/2!$. With the equal-mass two-body volume Eq. (D.240), $\Phi_2 = \beta/8\pi$,

$$\sum_f \int d\Pi_f \, |\mathcal{M}(i \to f)|^2 = \frac{1}{2}\,\Phi_2(s)\,|-\lambda|^2 = \frac{1}{2}\cdot\frac{\beta}{8\pi}\cdot\lambda^2 = \frac{\lambda^2\beta}{16\pi}\,. \tag{D.260}$$

Equations (D.259) and (D.260) agree exactly:

$$\boxed{2\,\mathrm{Im}\,\mathcal{M}_{\text{bub}}(s) = \frac{1}{2}\,\Phi_2(s)\,\lambda^2 = \frac{\lambda^2\beta}{16\pi}} \tag{D.261}$$

the optical theorem satisfied identically by the bubble, with the threshold factor β appearing on both sides and the imaginary part proportional to the very phase space derived in Section D.8.2.

Cutkosky reading. The same number is the Cutkosky cut of $I(s)$: replacing each cut propagator by $1/(p^2 - m^2 + i\epsilon) \to -2\pi i\,\delta^+(p^2 - m^2)$ converts the loop into a phase-space integral,

$$\mathrm{Cut}\,I(s) = \int \frac{d^d\ell}{(2\pi)^d}\,(-2\pi i)\,\delta^+(\ell^2 - m^2)\,(-2\pi i)\,\delta^+((P-\ell)^2 - m^2) = -\,\Phi_2(s)\,, \tag{D.262}$$

because $(-2\pi i)^2 = -(2\pi)^2$ and the two on-shell deltas with their (2π) factors are precisely the integrand of Φ_2 in Eq. (D.233). The discontinuity $\mathrm{Disc}\,I = I(s+i0) - I(s-i0) = 2i\,\mathrm{Im}\,I$ computed directly from Eq. (D.256) equals $-\Phi_2(s)$ as well, so $\mathrm{Disc}\,\mathcal{M}_{\text{bub}} = -i\,(\lambda^2/2)\,\mathrm{Disc}\,I = i\,(\lambda^2/2)\Phi_2$ and $2\,\mathrm{Im}\,\mathcal{M}_{\text{bub}} = -i\,\mathrm{Disc}\,\mathcal{M}_{\text{bub}} = \frac{1}{2}\lambda^2\Phi_2$, reproducing Eq. (D.261) from the cut. Cutting the bubble is integrating the two-body phase space; this single integral is the optical theorem.

Theorem D.42 ties off the analytic-structure thread opened in the ε-expansion section: the branch cut of a loop integral above a physical threshold is not a calculational nuisance but the imaginary part demanded by unitarity, and its size is a phase-space volume computed here in closed form. With the two-body measure, the Dalitz setup, the soft real-emission integral, and this unitarity check in hand, the appendix has assembled both halves of a one-loop calculation under one set of conventions. Together, these eight sections provide a complete, self-contained reference for the one-loop calculations in the main text.

Further reading

The one-loop technology of this appendix is treated, with conventions close to ours, in Peskin and Schroeder [396] (chapters 6–7 and the dimensional-regularization

sections), Schwartz [432] (chapters 16–20), Srednicki [454] (chapters 14–16), and Weinberg [500] (chapter 11); these are the natural companions for the reader who wants a second pass through the same material. The foundational paper on dimensional regularization is 't Hooft and Veltman [469], and the systematic tensor reduction to the scalar $A_0/B_0/C_0/D_0$ basis originates with Passarino and Veltman [383]. The fully rigorous treatment of dimensional regularization as both an ultraviolet and an infrared regulator, including the operator-product and renormalization-group structure that the loop chapters build on, is Collins [134]. For the multi-loop machinery that lies beyond this book's one-loop scope—integration by parts, the reduction to master integrals, and the Mellin–Barnes and sector-decomposition methods for evaluating them, together with the modern function spaces (polylogarithms and beyond) in which the answers live—the standard reference is Smirnov [448].

Appendix E

Path Integral Technicalities

The path integral is the spine of this book's computational machinery, yet almost every time we wrote one down in the main text we deferred a debt. We declared a Gaussian functional integral equal to an inverse square root of a determinant; we wrote a one-loop correction as $\frac{1}{2}\ln\det$ of a fluctuation operator and pretended the determinant of a differential operator was a finite number; we changed variables in $\mathcal{D}\phi$ and assumed the measure transformed trivially; we expanded around a classical solution with a zero mode and quietly divided by infinity. Each of those moves is legitimate, but only when carried out with care, and the care is exactly what a running narrative has no room for. This appendix is where the debts are paid. It is the proof-carrying technical underpinning of the entire path-integral formalism: every formula that earlier chapters used on faith is here defined precisely, derived in full, and renormalized honestly, with all the intermediate algebra on the page.

The strategy is to build upward from a single piece of finite-dimensional linear algebra. The whole subject is, at bottom, the Gaussian integral $\int d^n x\, e^{-\frac{1}{2}x^T A x + J^T x} = (2\pi)^{n/2}(\det A)^{-1/2}\, e^{\frac{1}{2}J^T A^{-1} J}$ and its complex and Grassmann counterparts. Section E.1 establishes these master formulas with sources, in real-symmetric, complex-Hermitian, and Grassmann forms, and reads Wick's theorem off them as the statement that Gaussian moments factorize into pairings. Everything afterward is the disciplined promotion of $n \to \infty$. Section E.2 takes the continuum limit by time-slicing, turning the formal $\det A$ into the product over a spectrum and isolating the one number a path integral actually needs—a *ratio* of determinants—which it computes in closed form through the Gelfand–Yaglom theorem, proved here for one-dimensional operators. Section E.3 confronts the fact that the spectral product diverges and supplies the cure: the zeta-function definition $\ln\det K = -\zeta_K'(0)$, its Mellin-transform link to the heat kernel and the Seeley–DeWitt coefficients, the reading of $\zeta_K(0)$ as the encoder of the divergence and the scaling anomaly, and the dictionary translating this regulator into dimensional regularization and Pauli–Villars.

With the determinant under control, the appendix turns it into physics. Section E.4 identifies the one-loop effective action with $\frac{1}{2}\ln\det$ of the second functional derivative of the action and carries the Coleman–Weinberg effective potential of ϕ^4 theory from definition through regularization to a fully renormalized closed form. Section E.5 then scrutinizes the object we have been integrating against: it defines $\mathcal{D}\phi$ through the DeWitt field-space metric, computes the Jacobians of field redefinitions and shows why they vanish in dimensional regularization, constructs the fermionic Berezin measure with its characteristic inverse Jacobian, and lays out the Fujikawa setup that the anomaly chapter will detonate. Section E.6 closes the loop by handling the case the naive Gaussian formula cannot: when the fluctuation operator has a zero mode, we trade it for a collective coordinate, pay the per-mode Jacobian $\sqrt{S_{\mathrm{cl}}/2\pi}$, replace the determinant by its primed version with the zero eigenvalue struck out, and assemble the complete semiclassical amplitude—worked explicitly for the ϕ^4 kink.

What this appendix imports it imports cleanly. The formal generating functionals, the finite- and infinite-dimensional Gaussian heuristics, and the Legendre-transform effective action come from Chapter 4; the distribution theory, the $i\epsilon$ prescription, Wick rotation, asymptotic series, and the saddle-point method come from Chapter 5; the construction of $Z[J]$ from the quantum-mechanical path integral and the operational form of Wick's theorem come from Chapter 10; Grassmann variables and Berezin integration in their first form come from Chapter 12; the master one-loop integrals and Feynman parametrization that the Coleman–Weinberg computation leans on come from Chapter D; the γ^5 and chiral-projector data needed to pose the Fujikawa problem come from Chapter C, and the global conventions from Chapter A. The $\lambda\phi^4$ model and its parameters are exactly those fixed in Chapter 6; we continue that thread rather than restate it. What the appendix exports, in turn, is the toolkit several later chapters detonate: the Faddeev–Popov measure and ghost determinant (Chapters 13 and 26), the chiral anomaly as a measure Jacobian (Chapter 36), radiative symmetry breaking via the effective potential (Chapter 29), soliton and instanton fluctuation determinants (Chapters 41 and 42), and the curved-space heat kernel (Chapter 64). The appendix is a reference, but a proof-carrying one: nothing in it is tabulated without being established, because these are precisely the results other chapters use most and check least.

E.1 Gaussian Integrals: The Algebraic Bedrock

The saddle-point program announced in this appendix's introduction reduces every partition function we will ever write to a single arithmetic operation. One expands the action about a stationary configuration, discards the (vanishing) linear term, and is left with a sum over fluctuations weighted by the exponential of a quadratic form, corrected order by order by the higher variations treated

as a perturbation. The leading factor in that program—the one that fixes the normalization, the one-loop determinant, and the free propagator all at once—is a Gaussian integral. Everything technical in the rest of this appendix is the analysis of what that integral becomes when the number of fluctuation modes becomes infinite and the quadratic form becomes a differential operator. Before any of that can be made honest we must have the finite-dimensional statement in hand, in every guise the book will need it: a real form for neutral bosons, a complex form for charged bosons, and a Grassmann form for fermions. The first two were established in the main development of the path integral (Section 4.2); we recall them here so that this appendix stands on its own as a reference, and then build the fermionic case and the unifying principle that the main text only flagged.

The single fact the whole section orbits is already visible in the source term of Eq. (4.54): the value of a sourced Gaussian is the exponential of a quadratic in the source whose kernel is the *inverse* of the form in the exponent. That inverse is the propagator; the prefactor is a power of the determinant. The sign of that power—negative for bosons, positive for fermions—is the algebraic seed of the entire bose/fermi dichotomy, and isolating it is the second purpose of this section.

E.1.1 The real master formula

We work first in Euclidean signature, where the quadratic form is real, symmetric, and positive-definite and the integral converges absolutely as an ordinary Riemann integral. The oscillatory Lorentzian weight is recovered by analytic continuation in Section E.1.6.

Theorem E.1 (Real Gaussian master formula)**.** *Let A be a real symmetric positive-definite $n \times n$ matrix and $J \in \mathbb{R}^n$. Then*

$$\int_{\mathbb{R}^n} d^n x \, e^{-\frac{1}{2} x^{\mathsf{T}} A x + J^{\mathsf{T}} x} \;=\; \frac{(2\pi)^{n/2}}{\sqrt{\det A}} \, \exp\!\Big(\tfrac{1}{2}\, J^{\mathsf{T}} A^{-1} J\Big). \tag{E.1}$$

Proof. This is Theorem 4.15; we record the argument compactly because the rest of the appendix continues directly from it. Because $A = A^{\mathsf{T}} > 0$, the spectral theorem gives an orthogonal O ($\det O = \pm 1$, $|\det O| = 1$) with $A = O^{\mathsf{T}} D O$, $D = \mathrm{diag}(a_1, \dots, a_n)$, every $a_i > 0$. First set $J = 0$ and change variables $y = Ox$, whose Jacobian is $|\det O| = 1$, so $d^n x = d^n y$ and the domain is preserved. The form diagonalizes, $x^{\mathsf{T}} A x = \sum_i a_i y_i^2$, the integral factorizes into n copies of the one-dimensional Gaussian $\int_{-\infty}^{\infty} dy \, e^{-\frac{1}{2} a y^2} = \sqrt{2\pi/a}$ of Theorem 4.13, and the product is

$$\int_{\mathbb{R}^n} d^n x \, e^{-\frac{1}{2} x^{\mathsf{T}} A x} = \prod_{i=1}^{n} \sqrt{\frac{2\pi}{a_i}} = \frac{(2\pi)^{n/2}}{\sqrt{\prod_i a_i}} = \frac{(2\pi)^{n/2}}{\sqrt{\det A}}, \tag{E.2}$$

since $\det A = \det D = \prod_i a_i$. For general J shift $x = y + A^{-1}J$, a translation leaving $d^n x$ and the domain unchanged. Using $A^{\mathsf{T}} = A$ and $(A^{-1})^{\mathsf{T}} = A^{-1}$,

$$-\tfrac{1}{2}x^{\mathsf{T}}Ax + J^{\mathsf{T}}x = -\tfrac{1}{2}y^{\mathsf{T}}Ay + \tfrac{1}{2}J^{\mathsf{T}}A^{-1}J, \tag{E.3}$$

the cross terms cancelling exactly as in Eq. (4.55). The constant exits the integral and the remaining y-integral is Eq. (E.2); this gives Eq. (E.1). □

Two pieces of Eq. (E.1) are load-bearing for everything downstream. The prefactor $(2\pi)^{n/2}(\det A)^{-1/2}$ is the normalization that, in the continuum, becomes the one-loop functional determinant of Section E.1.5 and the next section. The kernel A^{-1} in the source exponent is the propagator: differentiating in J and setting $J = 0$ produces the moments, with $\langle x_i x_j \rangle = (A^{-1})_{ij}$ and all higher moments fixed by pairing—Wick's theorem, recalled in Section E.1.4.

E.1.2 The complex (Hermitian) form

Charged bosons are integrated against a Hermitian form in complex variables, and the determinant power changes: a complex field is two real fields, so its $(\det)^{-1/2}$ is squared to $(\det)^{-1}$.

Theorem E.2 (Complex Gaussian master formula)**.** *Let M be a Hermitian positive-definite $n \times n$ matrix, $M^\dagger = M > 0$, and $J \in \mathbb{C}^n$. With the measure*

$$\prod_{i=1}^{n} \frac{dz_i^*\, dz_i}{2\pi i} \;\equiv\; \prod_{i=1}^{n} \frac{d(\operatorname{Re} z_i)\, d(\operatorname{Im} z_i)}{\pi}, \tag{E.4}$$

one has

$$\int \Big(\prod_i \frac{dz_i^*\, dz_i}{2\pi i} \Big)\, e^{-z^\dagger M z + \bar{J} z + z^\dagger J} \;=\; \frac{1}{\det M}\, e^{\,\bar{J} M^{-1} J}, \tag{E.5}$$

where $\bar{J} \equiv J^\dagger$. In particular the source-free integral is $(\det M)^{-1}$.

Proof. This is Theorem 4.17 in the notation of the present section (there the matrix was called A and the source w); the measures agree because $dz^*\, dz/2\pi i = d(\operatorname{Re} z)\, d(\operatorname{Im} z)/\pi$ as a real two-form. For $J = 0$, diagonalize $M = U^\dagger D U$ by a unitary U and substitute $z = U^\dagger \zeta$; as a real-linear map $U^\dagger$ has Jacobian $|\det U^\dagger|^2 = 1$, so the measure is invariant. The form becomes $z^\dagger M z = \sum_i a_i |\zeta_i|^2$ and each factor is a planar Gaussian, $\int \frac{d(\operatorname{Re}\zeta)\, d(\operatorname{Im}\zeta)}{\pi} e^{-a|\zeta|^2} = 1/a$, whose product is $\prod_i a_i^{-1} = (\det M)^{-1}$. For general J shift $z = \zeta + M^{-1}J$ (so $z^\dagger = \zeta^\dagger + \bar{J}M^{-1}$, legitimate because M^{-1} is Hermitian); the cross terms cancel as in Eq. (4.62), leaving the constant $\bar{J}M^{-1}J$ and the source-free integral. □

Notation Watch

The contrast $(\det A)^{-1/2}$ for one real boson versus $(\det M)^{-1}$ for one complex boson is the statement that a complex degree of freedom carries twice the determinant power of a real one. The Grassmann integral below inverts the sign of that power outright, $(\det M)^{+1}$, and the source exponent keeps its *positive* sign $+\bar{\eta}M^{-1}\eta$, matching $+\bar{J}M^{-1}J$ here. The opposite determinant power, not the source sign, is what distinguishes a fermion loop from a boson loop.

E.1.3 The Grassmann form: $\det M$ from Berezin integration

We now build the fermionic case from scratch on top of the Berezin calculus. The algebra of anticommuting generators and the integration rules—that $\int d\theta = 0$, $\int d\theta\, \theta = 1$, and that the Berezin integral is translation-invariant—are the content of Theorem 12.17; we use them as given and emphasize only the two facts that drive the determinant. First, a Berezin integral over a generator θ_k annihilates everything except the part of the integrand *linear* in θ_k; consequently the full integral over $2n$ generators extracts precisely the coefficient of the top monomial in which every generator appears exactly once. Second, the integrators anticommute with each other and with all generators, so that signs must be tracked through every reordering.

Let $\theta_1, \dots, \theta_n$ and $\bar{\theta}_1, \dots, \bar{\theta}_n$ be $2n$ independent Grassmann generators (the bar is a label, not yet a conjugation), and let M be an arbitrary complex $n \times n$ matrix. The fermionic quadratic form is the Grassmann-even bilinear

$$\bar{\theta} M \theta \;=\; \sum_{i,j=1}^{n} \bar{\theta}_i \, M_{ij} \, \theta_j, \tag{E.6}$$

each summand $\bar{\theta}_i M_{ij} \theta_j$ being a product of two odd elements, hence even and mutually commuting. We integrate with the ordered measure

$$D[\bar{\theta}, \theta] \;\equiv\; d\bar{\theta}_1 \, d\theta_1 \, d\bar{\theta}_2 \, d\theta_2 \cdots d\bar{\theta}_n \, d\theta_n, \tag{E.7}$$

the integrators applied from the left in the written order.

Theorem E.3 (Grassmann Gaussian: the inverted determinant)**.** *For any complex $n \times n$ matrix M,*

$$\int D[\bar{\theta}, \theta] \; e^{-\bar{\theta} M \theta} \;=\; \det M. \tag{E.8}$$

Proof. Because each generator squares to zero, the exponential is a terminating sum,

$$e^{-\bar{\theta} M \theta} = \sum_{k=0}^{n} \frac{(-1)^k}{k!} \, (\bar{\theta} M \theta)^k, \tag{E.9}$$

and the Berezin integral over all $2n$ generators kills every term but the one containing each θ_j and each $\bar\theta_i$ exactly once—necessarily the $k=n$ term, since a power $(\bar\theta M\theta)^k$ contains exactly k of the θ's and k of the $\bar\theta$'s, and any repeated index vanishes. Expanding that term,

$$\frac{(-1)^n}{n!}(\bar\theta M\theta)^n = \frac{(-1)^n}{n!}\sum_{\substack{i_1,\dots,i_n\\ j_1,\dots,j_n}} \Big(\prod_{a=1}^{n} M_{i_a j_a}\Big)\,\bar\theta_{i_1}\theta_{j_1}\cdots\bar\theta_{i_n}\theta_{j_n}, \tag{E.10}$$

where the numbers $M_{i_a j_a}$ have been pulled to the front. A term survives the integration only if the $\bar\theta$ indices are all distinct and the θ indices are all distinct, i.e. $i_a=\pi(a)$ and $j_a=\sigma(a)$ for permutations $\pi,\sigma\in S_n$. It remains to integrate the surviving Grassmann monomials.

The monomial integral. Fix π,σ and consider $G(\pi,\sigma)=\bar\theta_{\pi(1)}\theta_{\sigma(1)}\cdots\bar\theta_{\pi(n)}\theta_{\sigma(n)}$. Each block $\bar\theta_{\pi(a)}\theta_{\sigma(a)}$ is even, so the blocks commute and may be reordered freely, at no sign cost, so that the bar indices increase: the block carrying $\bar\theta_m$ then sits in position m and its partner is $\theta_{\sigma(\pi^{-1}(m))}$. Writing $\nu\equiv\sigma\circ\pi^{-1}$,

$$G(\pi,\sigma)=\prod_{m=1}^{n}\bar\theta_m\,\theta_{\nu(m)}. \tag{E.11}$$

We claim

$$\int D[\bar\theta,\theta]\;\prod_{m=1}^{n}\bar\theta_m\theta_{\nu(m)} \;=\; (-1)^n\,\mathrm{sgn}(\nu). \tag{E.12}$$

The single-variable rule fixes the base value: from $\int d\theta\,\theta=1$ and the left-anticommuting integrators, $\int d\bar\theta\,d\theta\,\bar\theta\theta=-1$, while $\int d\bar\theta\,d\theta\,\theta\bar\theta=+1$. For $\nu=\mathrm{id}$ the blocks $\bar\theta_m\theta_m$ are even and the measure factorizes, giving $(\int d\bar\theta\,d\theta\,\bar\theta\theta)^n=(-1)^n$, which is Eq. (E.12) since $\mathrm{sgn}(\mathrm{id})=1$. A transposition is verified directly: for $n=2$, $\nu=(12)$, $\bar\theta_1\theta_2\bar\theta_2\theta_1=(\theta_2\bar\theta_2)(\bar\theta_1\theta_1)$ after commuting the even block $\theta_2\bar\theta_2$ leftward, whence the integral is $(+1)(-1)=-1=(-1)^2\,\mathrm{sgn}((12))$. In general, transposing two adjacent θ-partners exchanges $\theta_{\nu(m)}\leftrightarrow\theta_{\nu(m+1)}$ across an even number of intervening odd factors and so contributes one sign, exactly the sign of the corresponding transposition of ν; since every permutation is a product of transpositions and the value is multiplicative in them, Eq. (E.12) follows by induction on the number of transpositions reducing ν to the identity.

Assembly. Using $\mathrm{sgn}(\nu)=\mathrm{sgn}(\sigma)\,\mathrm{sgn}(\pi)$ (as $\mathrm{sgn}(\pi^{-1})=\mathrm{sgn}(\pi)$) and Eq. (E.12),

$$\int D[\bar\theta,\theta]\,e^{-\bar\theta M\theta}=\frac{(-1)^n}{n!}\sum_{\pi,\sigma}\Big(\prod_a M_{\pi(a)\sigma(a)}\Big)\,(-1)^n\,\mathrm{sgn}(\pi)\,\mathrm{sgn}(\sigma)=\frac{1}{n!}\sum_{\pi,\sigma}\mathrm{sgn}(\pi)\,\mathrm{sgn}(\sigma)\prod_a M_{\pi(} \tag{E.13}$$

Reindexing the product by $b=\pi(a)$ gives $\prod_a M_{\pi(a)\sigma(a)}=\prod_b M_{b,\,\nu(b)}$ with $\nu=\sigma\pi^{-1}$, and $\mathrm{sgn}(\pi)\,\mathrm{sgn}(\sigma)=\mathrm{sgn}(\nu)$. For each fixed ν there are exactly $n!$

pairs (π, σ) with $\sigma = \nu\pi$, so the double sum collapses,

$$\int D[\bar\theta, \theta]\, e^{-\bar\theta M\theta} = \frac{1}{n!}\cdot n! \sum_{\nu\in S_n} \operatorname{sgn}(\nu) \prod_{b=1}^{n} M_{b,\nu(b)} = \sum_{\nu\in S_n} \operatorname{sgn}(\nu) \prod_{b=1}^{n} M_{b,\nu(b)} = \det M, \tag{E.14}$$

the Leibniz formula for the determinant. This proves Eq. (E.8). □

The result deserves the same pause the bosonic one did. The very same Hermitian quadratic form that gave $(\det M)^{-1}$ in Theorem E.2 gives $(\det M)^{+1}$ once the integration variables anticommute. The inversion is not a convention; it is forced by the Berezin rule that integration extracts the top—rather than the constant—coefficient, which turns the geometric series $\prod a_i^{-1}$ into the polynomial $\prod a_i$. This single sign in the exponent of the determinant is, traced through the loop expansion, the origin of the minus sign attached to every closed fermion loop.

The source version follows by the same shift that worked for bosons, now with Grassmann-odd sources $\eta, \bar\eta$.

Theorem E.4 (Sourced Grassmann Gaussian)**.** *For invertible complex M and independent Grassmann sources $\eta_i, \bar\eta_i$ anticommuting with the θ's,*

$$\int D[\bar\theta, \theta]\; e^{-\bar\theta M\theta + \bar\eta\theta + \bar\theta\eta} \;=\; \det M \;\; e^{\bar\eta M^{-1}\eta}. \tag{E.15}$$

Proof. Shift $\theta = \theta' + M^{-1}\eta$ and $\bar\theta = \bar\theta' + \bar\eta M^{-1}$, which leaves the Berezin measure invariant (translation invariance, Theorem 12.17). Substituting,

$$\begin{aligned} -\bar\theta M\theta + \bar\eta\theta + \bar\theta\eta &= -(\bar\theta' + \bar\eta M^{-1})M(\theta' + M^{-1}\eta) + \bar\eta(\theta' + M^{-1}\eta) + (\bar\theta' + \bar\eta M^{-1})\eta \\ &= -\bar\theta' M\theta' - \bar\theta'\eta - \bar\eta\theta' - \bar\eta M^{-1}\eta + \bar\eta\theta' + \bar\eta M^{-1}\eta + \bar\theta'\eta + \bar\eta M^{-1}\eta \\ &= -\bar\theta' M\theta' + \bar\eta M^{-1}\eta, \end{aligned} \tag{E.16}$$

the four cross terms cancelling in pairs and the three $\bar\eta M^{-1}\eta$ pieces combining to one. The constant $\bar\eta M^{-1}\eta$ (even, hence an ordinary exponential) leaves the integral, and the remaining θ'-integral is Eq. (E.8). □

E.1.4 Wick's theorem: every Gaussian moment is a sum over pairings

A Gaussian distribution has no independent information beyond its two-point function: all higher moments are built from it by the combinatorics of pairing. This is Wick's theorem, and it is the same statement bosonically and fermionically up to the signs the anticommutation forces. The bosonic version was proved in full in the main text; we recall it and then prove the fermionic version that the appendix owes.

For bosons, write $\langle\,\cdot\,\rangle$ for the normalized real Gaussian average with respect to $A>0$, so that $\langle e^{J^{\mathsf T}x}\rangle = e^{\frac12 J^{\mathsf T}A^{-1}J}$ and $\langle x_i x_j\rangle = (A^{-1})_{ij}$.

Theorem E.5 (Bosonic Wick's theorem)**.** *With $G\equiv A^{-1}$, all odd moments vanish and*

$$\langle x_{i_1}x_{i_2}\cdots x_{i_{2k}}\rangle = \sum_{\text{pairings } P}\ \prod_{(a,b)\in P} G_{i_a i_b}, \tag{E.17}$$

the sum running over the $(2k-1)!!$ perfect matchings of $\{1,\dots,2k\}$ into unordered pairs.

This is Theorem 4.19, proved there both by differentiating the generating function $e^{\frac12 J^{\mathsf T}A^{-1}J}$ in the sources $J_{i_1},\dots,J_{i_m}$ and by the Schwinger–Dyson recursion that contracts the first index with each of the rest; we do not repeat the proof. The fermionic theorem differs in exactly two ways: the sources anticommute, so each pairing carries a sign equal to the signature of the permutation that brings the contracted partners together; and a θ pairs only with a $\bar\theta$, so the matchings are bijections of the θ-slots onto the $\bar\theta$-slots.

Theorem E.6 (Fermionic Wick's theorem)**.** *Let $\langle\,\cdot\,\rangle$ denote the normalized Grassmann average, $\langle F\rangle \equiv (\det M)^{-1}\int D[\bar\theta,\theta]\,F\,e^{-\bar\theta M\theta}$, with generating identity $\langle e^{\bar\eta\theta+\bar\theta\eta}\rangle = e^{\bar\eta M^{-1}\eta}$. Then the only nonvanishing elementary contraction is*

$$\langle\theta_i\,\bar\theta_j\rangle = (M^{-1})_{ij}, \qquad \langle\theta_i\theta_j\rangle = \langle\bar\theta_i\bar\theta_j\rangle = 0, \tag{E.18}$$

and the general moment is the signed sum over θ–$\bar\theta$ pairings,

$$\langle\theta_{i_1}\cdots\theta_{i_n}\,\bar\theta_{j_1}\cdots\bar\theta_{j_n}\rangle = \sum_{\sigma\in S_n}\operatorname{sgn}(\sigma)\,\prod_{a=1}^{n}(M^{-1})_{i_a\,j_{\sigma(a)}} = \det\big[(M^{-1})_{i_a j_b}\big]_{a,b=1}^{n}, \tag{E.19}$$

with any moment containing unequal numbers of θ's and $\bar\theta$'s vanishing.

Proof. Two-point function. Expand both sides of the generating identity to second order in the sources. The left side is $\langle 1+(\bar\eta\theta+\bar\theta\eta)+\frac12(\bar\eta\theta+\bar\theta\eta)^2+\cdots\rangle$. The linear terms vanish because $\langle\theta_i\rangle=\langle\bar\theta_i\rangle=0$ (odd integrand, no top monomial), and among the quadratic terms only the cross piece $(\bar\eta\theta)(\bar\theta\eta)$ survives for the same reason. Thus

$$\big\langle(\bar\eta\theta)(\bar\theta\eta)\big\rangle = \sum_{i,j}\bar\eta_i\,\langle\theta_i\bar\theta_j\rangle\,\eta_j, \tag{E.20}$$

where the $\bar\eta_i$ and η_j, being constants of the Grassmann average, have been moved outside leaving the c-number $\langle\theta_i\bar\theta_j\rangle$ between them. The right side expands to $1+\bar\eta M^{-1}\eta+\cdots = 1+\sum_{ij}\bar\eta_i(M^{-1})_{ij}\eta_j+\cdots$. Matching the coefficient of $\bar\eta_i\eta_j$ gives $\langle\theta_i\bar\theta_j\rangle=(M^{-1})_{ij}$. The averages $\langle\theta_i\theta_j\rangle$ and $\langle\bar\theta_i\bar\theta_j\rangle$ vanish because the generating function depends on η and $\bar\eta$ only through the bilinear $\bar\eta M^{-1}\eta$, which contains no $\eta\eta$ or $\bar\eta\bar\eta$ term.

Higher moments. Read Eq. (E.19) off the generating function by Grassmann source differentiation, or equivalently by induction on n; we give the induction, the fermionic mirror of the Schwinger–Dyson recursion. Insert one more factor and use anticommutation to expose how θ_{i_1} contracts. Concretely, inside the average θ_{i_1} can only be "used up" by pairing with one of the $\bar\theta_{j_b}$, since $\langle\theta\theta\rangle = 0$; moving θ_{i_1} to the right past the remaining $n-1$ factors $\theta_{i_2}\cdots\theta_{i_n}$ costs a sign $(-1)^{n-1}$, and bringing it adjacent to $\bar\theta_{j_b}$ costs a further $(-1)^{b-1}$ for the $b-1$ bar-factors it crosses. Replacing the contracted pair by $\langle\theta_{i_1}\bar\theta_{j_b}\rangle = (M^{-1})_{i_1 j_b}$ and the rest by the average of what remains yields the recursion

$$\langle\theta_{i_1}\cdots\theta_{i_n}\bar\theta_{j_1}\cdots\bar\theta_{j_n}\rangle = \sum_{b=1}^{n}(-1)^{n-1}(-1)^{b-1}(M^{-1})_{i_1 j_b}\,\langle\theta_{i_2}\cdots\theta_{i_n}\,\bar\theta_{j_1}\cdots\widehat{\bar\theta_{j_b}}\cdots\bar\theta_{j_n}\rangle, \tag{E.21}$$

the hat denoting omission. This is precisely the Laplace (cofactor) expansion of the determinant $\det[(M^{-1})_{i_a j_b}]$ along its first row: the sign $(-1)^{n-1}(-1)^{b-1} = (-1)^{n+b}$ is, up to the overall $(-1)^{n+1}$ common to the row, the checkerboard cofactor sign $(-1)^{1+b}$, and the reduced average is by the inductive hypothesis the corresponding minor. The base case $n=1$ is Eq. (E.18). Hence the moment equals $\det[(M^{-1})_{i_a j_b}]$, which expanded by Leibniz is the signed pairing sum in Eq. (E.19). □

Physical Insight

The bosonic moment is a *permanent* of two-point functions (all pairings with + sign), the fermionic moment a *determinant* (pairings weighted by sgn). The interchange permanent ↔ determinant is the statistics: symmetric wavefunctions for bosons, antisymmetric for fermions, read at the level of correlation functions. The Pauli principle is visible already here—if two fermionic indices coincide in a way that repeats a row, the determinant vanishes.

E.1.5 The unifying determinant rule and its threads

The three master formulas are one statement seen through three measures. Collecting the source-free normalizations,

$$\int_{\mathbb{R}^n} d^n x\, e^{-\frac{1}{2}x^{\mathsf T}Ax} = \frac{(2\pi)^{n/2}}{(\det A)^{1/2}}, \qquad \int \tfrac{d^{2n}z}{(2\pi i)^n}\, e^{-z^\dagger Mz} = \frac{1}{\det M}, \qquad \int D[\bar\theta,\theta]\, e^{-\bar\theta M\theta} = \det M, \tag{E.22}$$

exhibit the dichotomy in its cleanest form: a bosonic Gaussian returns the quadratic form's determinant to a *negative* power, a fermionic one to a *positive* power. Per real degree of freedom the exponent is $-\frac{1}{2}$ for bosons; per complex pair of Grassmann generators it is $+1$, i.e. $+\frac{1}{2}$ per real fermionic degree of freedom

counted with the doubling $\bar\theta, \theta$. The bosonic real case $(-\frac{1}{2})$ has a fermionic mirror of the same fractional power but opposite sign, realized by a real (Majorana) Grassmann integral against an antisymmetric form.

Proposition E.7 (The Pfaffian: real fermionic mirror)**.** *For $2m$ Grassmann generators $\theta_1, \dots, \theta_{2m}$ and a complex antisymmetric matrix $B = -B^{\mathsf T}$,*

$$\int d\theta_{2m} \cdots d\theta_1 \; e^{\frac{1}{2}\theta^{\mathsf T} B \theta} \;=\; \mathrm{Pf}(B), \qquad \big(\mathrm{Pf}(B)\big)^2 = \det B. \tag{E.23}$$

Proof. Only the top term $\frac{1}{m!}(\frac{1}{2}\theta^{\mathsf T} B\theta)^m$ of the exponential survives the integration, by the same top-monomial argument as in Theorem E.3. Expanding, $(\frac{1}{2}\sum_{i<j} B_{ij}(\theta_i\theta_j - \theta_j\theta_i))^m/m! = (\sum_{i<j} B_{ij}\theta_i\theta_j)^m/m!$, and collecting the coefficient of $\theta_1\theta_2\cdots\theta_{2m}$ gives, with the sign conventions of the ordered measure, the sum over perfect matchings $\{(i_1j_1), \dots, (i_mj_m)\}$ of $\{1, \dots, 2m\}$ weighted by the signature of the associated permutation, $\sum_{\text{matchings}} \mathrm{sgn} \prod_r B_{i_r j_r}$, which is by definition $\mathrm{Pf}(B)$. The identity $\mathrm{Pf}(B)^2 = \det B$ then follows from Theorem E.3: doubling the generators to a $\bar\theta, \theta$ pair and integrating $e^{-\bar\theta M\theta}$ with M built from B reproduces $\det B$ as the square of the real integral. The two-by-two case is the whole pattern in miniature: with $B_{12} = -B_{21} = b$, $\int d\theta_2 d\theta_1 \, e^{b\theta_1\theta_2} = \int d\theta_2 d\theta_1 \, (1 + b\theta_1\theta_2) = b = \mathrm{Pf}(B)$ and $\det B = b^2$. □

The one algebraic seed of all three benchmark theories

Every free propagator in this book is an A^{-1} and every loop normalization a power of $\det A$, with the power set by statistics through Eq. (E.22). In ϕ^4 theory the quadratic action is $\frac{1}{2}\phi(-\partial^2 + m^2)\phi$ in Euclidean signature, so the form is $A = -\partial^2 + m^2$ and its inverse A^{-1} is the scalar propagator built in Chapter 10; the free generating functional carries the bosonic $(\det A)^{-1/2}$. In QED the photon's quadratic operator (after gauge fixing) plays the role of A and its inverse is the photon propagator of Chapter 13, while the electron's bilinear $\bar\psi(i\not{D} - m)\psi$ is a Grassmann form with $M = i\not{D} - m$, whose inverse M^{-1} is the Dirac propagator and whose $\det M$—to the *positive* power, by Theorem E.3—is the fermion determinant integrated out in Chapter 12. In Yang–Mills theory the same two structures appear, now with the gauge field's quadratic operator and the Faddeev–Popov ghost determinant, the latter a fermionic det over commuting indices. The threads differ only in what A and M are; the integral that consumes them is the one proved in this section.

E.1.6 Analytic continuation to the oscillatory integral

The master formulas assumed a convergent, positive-definite Euclidean form. The physical path integral carries instead the oscillatory weight e^{iS}, whose quadratic

part is $e^{\frac{i}{2}x^{\mathsf{T}}Ax}$ with A real symmetric but not sign-definite. Such an integral does not converge absolutely; it is defined by the $i\epsilon$ prescription and the Wick rotation of Theorems 5.30 and 5.31, which we now apply to the Gaussian itself. The one-dimensional case carries the entire content.

Theorem E.8 (Fresnel/oscillatory Gaussian)**.** *For real $a \neq 0$, defined by the convergent regulator $a \to a - i\epsilon\ \mathrm{sgn}(a)$ with $\epsilon \to 0^+$,*

$$\int_{-\infty}^{\infty} dx\ e^{\frac{i}{2}ax^2} \;=\; \sqrt{\frac{2\pi}{|a|}}\ e^{\,i\frac{\pi}{4}\,\mathrm{sgn}(a)}, \tag{E.24}$$

and consequently, for real symmetric nonsingular A with n_+ positive and n_- negative eigenvalues (signature $\operatorname{sig} A = n_+ - n_-$),

$$\int_{\mathbb{R}^n} d^n x\ e^{\frac{i}{2}x^{\mathsf{T}}Ax} = \frac{(2\pi)^{n/2}}{|\det A|^{1/2}}\, e^{\,i\frac{\pi}{4}\operatorname{sig} A} = (2\pi)^{n/2}\big(\det(-iA)\big)^{-1/2}, \tag{E.25}$$

the branch of the last square root fixed by continuing each eigenvalue $a_k \to |a_k| e^{-i\frac{\pi}{2}\mathrm{sgn}(a_k)}$ from the convergent regulator.

Proof. Insert the convergence factor: for $\epsilon > 0$,

$$I_\epsilon = \int_{-\infty}^{\infty} dx\ e^{\frac{i}{2}ax^2 - \frac{\epsilon}{2}x^2} = \int_{-\infty}^{\infty} dx\ e^{-\frac{1}{2}(\epsilon - ia)x^2} = \sqrt{\frac{2\pi}{\epsilon - ia}}, \tag{E.26}$$

where the last step is the one-dimensional formula Theorem 4.13 applied with the complex coefficient $\epsilon - ia$, legitimate because $\mathrm{Re}(\epsilon - ia) = \epsilon > 0$ guarantees absolute convergence and the Gaussian is analytic in the coefficient throughout the right half-plane. Letting $\epsilon \to 0^+$ sends the coefficient to $-ia = |a|\, e^{-i\frac{\pi}{2}\mathrm{sgn}(a)}$, which lies on the boundary of the half-plane reached without crossing the branch cut along the negative real axis; hence

$$\int_{-\infty}^{\infty} dx\ e^{\frac{i}{2}ax^2} = \sqrt{\frac{2\pi}{|a|}}\left(e^{-i\frac{\pi}{2}\,\mathrm{sgn}(a)}\right)^{-1/2} = \sqrt{\frac{2\pi}{|a|}}\, e^{\,i\frac{\pi}{4}\,\mathrm{sgn}(a)}, \tag{E.27}$$

which is Eq. (E.24). For the multidimensional case, diagonalize the real symmetric A by an orthogonal O as in Theorem E.1; the measure is invariant and the form decouples into n one-dimensional Fresnel integrals with coefficients the eigenvalues a_k. Their product is

$$\prod_{k=1}^{n} \sqrt{\frac{2\pi}{|a_k|}}\, e^{\,i\frac{\pi}{4}\,\mathrm{sgn}(a_k)} = \frac{(2\pi)^{n/2}}{\big(\prod_k |a_k|\big)^{1/2}}\, e^{\,i\frac{\pi}{4}\sum_k \mathrm{sgn}(a_k)} = \frac{(2\pi)^{n/2}}{|\det A|^{1/2}}\, e^{\,i\frac{\pi}{4}\operatorname{sig} A}, \tag{E.28}$$

since $\prod_k |a_k| = |\det A|$ and $\sum_k \mathrm{sgn}(a_k) = \operatorname{sig} A$. Writing each phase as $e^{-i\frac{\pi}{2}\mathrm{sgn}(a_k)} = (-ia_k)/|a_k|$ recombines the modulus and phase into $\prod_k (-ia_k)^{-1/2} = (\det(-iA))^{-1/2}$ on the stated branch, giving the last equality of Eq. (E.25). □

The phase $e^{i\frac{\pi}{4}\operatorname{sig} A}$ is the finite-dimensional ancestor of the Maslov index that decorates semiclassical amplitudes; it counts, mode by mode, the negative directions of the quadratic form. The cleanest way to remove the oscillation altogether is the Wick rotation itself: continuing $x^0 \to -i\tau$ as in Theorem 5.30 turns e^{iS} into e^{-S_E} with S_E a positive Euclidean action, whereupon every formula of this section applies verbatim with a genuinely positive-definite A, and the physical answer is recovered by rotating back. We will adopt this Euclidean route as the default in the next section, where A becomes a differential operator and the only honest definition of its determinant is through a positive, real spectrum.

E.1.7 Worked computations

Worked Example E.9 (The two-by-two Grassmann determinant, by hand)**.** We verify Theorem E.3 for $n = 2$ with no appeal to the general permutation sum, exhibiting how the determinant assembles. Take generators $\bar\theta_1, \theta_1, \bar\theta_2, \theta_2$ and $\bar\theta M\theta = \sum_{i,j} \bar\theta_i M_{ij} \theta_j$. The exponential truncates at the quadratic term, and only the part with all four generators survives the integral:

$$e^{-\bar\theta M\theta}\Big|_{\text{top}} = \frac{1}{2}(\bar\theta M\theta)^2\Big|_{\text{top}}. \tag{E.29}$$

Expanding $(\bar\theta M\theta)^2$ and keeping only terms with each index once,

$$\begin{aligned}\tfrac{1}{2}(\bar\theta M\theta)^2\big|_{\text{top}} = \tfrac{1}{2}\Big[&M_{11}M_{22}\,\bar\theta_1\theta_1\bar\theta_2\theta_2 + M_{22}M_{11}\,\bar\theta_2\theta_2\bar\theta_1\theta_1 \\ &+ M_{12}M_{21}\,\bar\theta_1\theta_2\bar\theta_2\theta_1 + M_{21}M_{12}\,\bar\theta_2\theta_1\bar\theta_1\theta_2\Big].\end{aligned} \tag{E.30}$$

The two diagonal monomials are equal (even blocks commute), $\bar\theta_1\theta_1\bar\theta_2\theta_2 = \bar\theta_2\theta_2\bar\theta_1\theta_1$, and integrate against $D[\bar\theta,\theta]$ to $(-1)^2 = +1$. The two off-diagonal monomials are likewise equal and, by the transposition computation under Eq. (E.12), integrate to -1. Hence

$$\int D[\bar\theta,\theta]\, e^{-\bar\theta M\theta} = \tfrac{1}{2}\big[2M_{11}M_{22}(+1)+2M_{12}M_{21}(-1)\big] = M_{11}M_{22}-M_{12}M_{21} = \det M, \tag{E.31}$$

the 2×2 determinant, sign and all. The minus on the off-diagonal product is the sgn of the transposition (12)—the same sign that, promoted to fields, antisymmetrizes the fermion and supplies the closed-loop minus sign.

Worked Example E.10 (Fermionic four-point function as a 2×2 determinant)**.** We evaluate $\langle\theta_1\theta_2\,\bar\theta_1\bar\theta_2\rangle$ two ways and confirm Theorem E.6. By the determinant formula Eq. (E.19) with $n = 2$, $i_1 = 1, i_2 = 2, j_1 = 1, j_2 = 2$,

$$\langle\theta_1\theta_2\,\bar\theta_1\bar\theta_2\rangle = \det\begin{pmatrix}(M^{-1})_{11} & (M^{-1})_{12}\\ (M^{-1})_{21} & (M^{-1})_{22}\end{pmatrix} = (M^{-1})_{11}(M^{-1})_{22} - (M^{-1})_{12}(M^{-1})_{21}. \tag{E.32}$$

By the pairing form, θ_1 contracts with $\bar\theta_1$ or $\bar\theta_2$; the two matchings $\{(\theta_1\bar\theta_1),(\theta_2\bar\theta_2)\}$ and $\{(\theta_1\bar\theta_2),(\theta_2\bar\theta_1)\}$ carry signs $+1$ and -1 respectively (the second is an odd permutation of the contracted slots), giving

$$\langle\theta_1\theta_2\bar\theta_1\bar\theta_2\rangle = \langle\theta_1\bar\theta_1\rangle\langle\theta_2\bar\theta_2\rangle - \langle\theta_1\bar\theta_2\rangle\langle\theta_2\bar\theta_1\rangle, \tag{E.33}$$

identical to the determinant after Eq. (E.18). The contrast with the bosonic four-point function Eq. (4.74), where the analogous three (here two, the $\theta\theta$ pairing being absent) terms all enter with $+$ sign, is exactly the determinant-versus-permanent distinction of statistics.

Worked Example E.11 (The oscillatory integral and the i-epsilon propagator)**.** We connect Theorem E.8 to the Feynman propagator's $i\epsilon$. Consider the one-dimensional quantum-mechanical kernel whose action contributes $\frac{i}{2}(p^2-m^2)|\tilde x|^2$ per mode in momentum space; the relevant oscillatory integral is the $a = p^2 - m^2$ instance of Eq. (E.24). The regulator $a \to a - i\epsilon\,\mathrm{sgn}(a)$ that made Eq. (E.26) converge is the statement that the quadratic form is given an infinitesimal negative imaginary part, $p^2 - m^2 \to p^2 - m^2 + i\epsilon$ once the overall i in e^{iS} is accounted for. The propagator, being the inverse of the quadratic form, is then

$$(\text{form})^{-1} = \frac{i}{p^2 - m^2 + i\epsilon}, \tag{E.34}$$

which is exactly the Feynman propagator of the book's conventions. Thus the $i\epsilon$ is not an extra prescription bolted onto the propagator: it is the same convergence factor that defines the oscillatory Gaussian, viewed through the inverse rather than the determinant. The Euclidean alternative—rotate $p^0 \to ip^0_E$ per Theorem 5.30, so $p^2 - m^2 \to -(p_E^2 + m^2)$ with $p_E^2 + m^2 > 0$ strictly positive—turns the same form positive-definite and the oscillatory integral into the convergent Euclidean one, the route we follow in the remainder of the appendix.

We have now assembled the complete finite-dimensional kit: the real, complex, and Grassmann master formulas Eqs. (E.1), (E.5) and (E.8) with their sources, Wick's theorem in both statistics Eqs. (E.17) and (E.19), the unifying determinant rule Eq. (E.22) with its Pfaffian mirror, and the analytic continuation Eq. (E.25) that licenses the oscillatory physical integral. Each is a theorem about an ordinary (or Berezin) integral over finitely many variables. The two objects they hand forward are the determinant $\det A$, which fixes the normalization, and the inverse A^{-1}, which is the propagator. The next section takes the only remaining step that separates this algebra from quantum field theory: it lets the number of variables become infinite, so that A becomes a differential operator K and $\det A$ becomes an infinite product of eigenvalues that, taken naively, does not converge. Giving that product—and the functional determinant $(\det K)^{-1/2}$ it stands for—a finite, computable meaning is the business that begins immediately.

E.2 From Discrete to Continuum: Functional Determinants

The previous section (Section E.1) left us holding two finite-dimensional facts and one question. The facts: for a real symmetric positive matrix A on $\mathbb{R}^n$,

$$\int_{\mathbb{R}^n} \mathrm{d}^n x \, \exp\Big(-\tfrac{1}{2}\, x^{\mathsf{T}} A\, x + b^{\mathsf{T}} x \Big) = \frac{(2\pi)^{n/2}}{\sqrt{\det A}} \, \exp\Big(\tfrac{1}{2}\, b^{\mathsf{T}} A^{-1} b \Big), \tag{E.35}$$

so that every Gaussian moment is controlled by the inverse A^{-1} (the propagator) and the overall normalization by $(\det A)^{-1/2}$. The question is what becomes of Eq. (E.35) when $n \to \infty$ and the vector x is promoted to a field $\phi(\tau)$. The infinite-dimensional Gaussian heuristic of Section 4.2 already told us, formally, what to expect: the quadratic form $\tfrac{1}{2}\, x^{\mathsf{T}} A\, x$ becomes $\tfrac{1}{2}\int \phi\, K \phi$ for a differential operator K, the propagator becomes the Green's function K^{-1}, and the normalization becomes $(\det K)^{-1/2}$ with $\det K$ the product of the eigenvalues of K. The purpose of this section is to make the limit honest. We build the measure $\mathcal{D}\phi$ as a time-sliced limit, watch the discretized quadratic form converge to $\tfrac{1}{2}\int \phi K \phi$, discover that the resulting $\det K = \prod_n \lambda_n$ is literally infinite, isolate the ratio prescription that rescues finite physics from it, and then prove — in full — the one theorem that turns a one-dimensional functional determinant into a number you can compute by solving an ordinary differential equation: the Gelfand–Yaglom theorem.

E.2.1 Time slicing and the measure

The cleanest laboratory is single-particle quantum mechanics, which is field theory in $0+1$ dimensions: the "field" is the trajectory $x(\tau)$ of one degree of freedom. We work in Euclidean time from the outset, both because that is where the Gaussian integrals converge and because every functional determinant we shall meet is naturally Euclidean. The Wick rotation $t \to -i\tau$ (Section 5.4.1) sends the real-time weight e^{iS} to e^{-S_E}, and we restore real time at the end of any computation by continuing the answer back, $\tau \to it$. The Euclidean action for a particle of mass m in a potential V is

$$S_E[x] = \int_0^T \mathrm{d}\tau \, \Big[\tfrac{1}{2} m\, \dot{x}(\tau)^2 + V\big(x(\tau)\big) \Big], \qquad \dot{x} \equiv \frac{dx}{d\tau}, \tag{E.36}$$

and the object we wish to define is the Euclidean transition amplitude $\big\langle x_f \big| e^{-HT} \big| x_i \big\rangle = \int \mathcal{D}x \, e^{-S_E[x]}$, the imaginary-time continuation of the propagator built canonically in Section 10.1.

The measure has no a priori meaning; we *define* it by discretization. Partition the interval $[0, T]$ into N equal steps of width $\varepsilon = T/N$, with nodes $\tau_k = k\varepsilon$ and

values $x_k = x(\tau_k)$, the endpoints held fixed at $x_0 = x_i$ and $x_N = x_f$. The kinetic term becomes a sum of squared finite differences and the potential a Riemann sum,

$$S_E \;\longrightarrow\; S_N = \sum_{k=1}^{N} \Big[\frac{m}{2\varepsilon}(x_k - x_{k-1})^2 + \varepsilon\, V(x_k)\Big], \tag{E.37}$$

and the integral over all paths becomes an ordinary integral over the $N-1$ free interior heights, weighted by one factor of $\sqrt{m/2\pi\varepsilon}$ per slice:

$$\int \mathcal{D}x\, e^{-S_E[x]} \;:=\; \lim_{N\to\infty} \Big(\frac{m}{2\pi\varepsilon}\Big)^{N/2} \int_{-\infty}^{\infty} \prod_{k=1}^{N-1} \mathrm{d}x_k\; e^{-S_N}. \tag{E.38}$$

The per-slice prefactor is not decoration: it is exactly the normalization that makes the free short-time kernel a correctly normalized Gaussian, $\big\langle x_k \big| e^{-\varepsilon H_0} \big| x_{k-1} \big\rangle = \sqrt{m/2\pi\varepsilon}\, \exp[-\frac{m}{2\varepsilon}(x_k - x_{k-1})^2]$, so that composing N of them reproduces $e^{-H_0 T}$ in the limit. It is the source of the infinite normalization constant we will fight with below, and the reason functional determinants are only ever sensible as ratios.

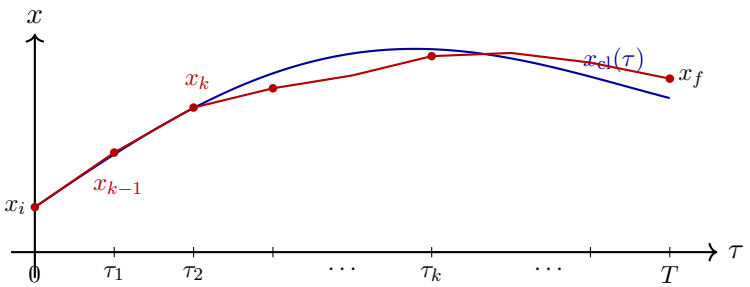


Figure E.1: The time-sliced path integral. The smooth classical trajectory x_{cl} (blue) extremizes S_E; the discretized measure Eq. (E.38) integrates over all broken-line configurations (red) pinned to the fixed endpoints x_i, x_f at the interior nodes $x_1, \ldots, x_{N-1}$. The continuum limit $N \to \infty$ ($\varepsilon = T/N \to 0$) defines $\int \mathcal{D}x$.

To expose the quadratic form, expand about the classical path. Let $x = x_{\mathrm{cl}} + \eta$, where x_{cl} solves the Euclidean equation of motion $m\ddot{x}_{\mathrm{cl}} = V'(x_{\mathrm{cl}})$ with $x_{\mathrm{cl}}(0) = x_i$, $x_{\mathrm{cl}}(T) = x_f$, and the fluctuation η vanishes at both ends, $\eta(0) = \eta(T) = 0$. Because the first variation vanishes at the classical path, Taylor expansion of Eq. (E.36) to second order gives, with no linear term,

$$S_E[x_{\mathrm{cl}}+\eta] = S_E[x_{\mathrm{cl}}] + \frac{1}{2}\int_0^T \mathrm{d}\tau\, \eta(\tau)\, K\, \eta(\tau) + O(\eta^3), \qquad K = -m\,\frac{d^2}{d\tau^2} + V''(x_{\mathrm{cl}}(\tau)), \tag{E.39}$$

the integration by parts in the cross term producing no boundary contribution precisely because η is pinned at the endpoints. This is the promised continuum

quadratic form $\frac{1}{2}\int \eta K \eta$, the infinite-dimensional avatar of $\frac{1}{2}\,x^{\mathsf{T}} A x$. The fluctuation operator K is a second-order differential operator on functions vanishing at 0 and T — a *Dirichlet* operator on $[0,T]$. For the harmonic oscillator $V = \frac{1}{2}m\omega^2 x^2$ the potential well is quadratic, $V'' = m\omega^2$ is constant, and $K = m(-\partial_\tau^2 + \omega^2)$; this is the model we carry through the section.

At finite N the quadratic form Eq. (E.39) is an honest $(N-1)$-dimensional quadratic form $\frac{1}{2}\,\eta^{\mathsf{T}} M \eta$ in the interior fluctuations $\eta_k = \eta(\tau_k)$. From Eq. (E.37) the matrix M is tridiagonal: the kinetic piece contributes $(m/\varepsilon)(2$ on the diagonal, -1 off-diagonal), and the potential adds $\varepsilon\, V_k''$ to the diagonal, where $V_k'' \equiv V''(x_{\mathrm{cl}}(\tau_k))$. Performing the Gaussian integral over the η_k with Eq. (E.35) and keeping the measure factors gives the one-loop fluctuation amplitude

$$\int \mathcal{D}\eta\, e^{-\frac{1}{2}\int \eta K \eta} = \lim_{N\to\infty} \Big(\frac{m}{2\pi\varepsilon}\Big)^{N/2} \frac{(2\pi)^{(N-1)/2}}{\sqrt{\det M}}, \tag{E.40}$$

and the bookkeeping of the powers of ε and 2π hidden in $\det M$ is exactly what we must now do carefully, because it is here that the would-be $(\det K)^{-1/2}$ appears together with a divergent constant.

The free determinant, done exactly. Take first the free particle, $V'' = 0$, $M = M^{(0)}$. The matrix $M^{(0)} = (m/\varepsilon)\,T_{N-1}$, where T_{N-1} is the $(N-1)\times(N-1)$ tridiagonal matrix with 2 on the diagonal and -1 on the first off-diagonals. Its determinant is computed by the elementary recursion for the leading principal minors $D_k = \det T_k$: expanding along the last row,

$$D_k = 2D_{k-1} - D_{k-2}, \qquad D_0 = 1,\ D_1 = 2 \implies D_k = k+1, \tag{E.41}$$

so $\det T_{N-1} = N$ and $\det M^{(0)} = (m/\varepsilon)^{N-1} N$. Substituting into Eq. (E.40) and using $\varepsilon N = T$, every power of 2π, m, and ε telescopes against the measure:

$$\Big(\frac{m}{2\pi\varepsilon}\Big)^{N/2} \frac{(2\pi)^{(N-1)/2}}{\sqrt{(m/\varepsilon)^{N-1}N}} = \sqrt{\frac{m}{2\pi\varepsilon}}\cdot\frac{1}{\sqrt{N}} = \sqrt{\frac{m}{2\pi\varepsilon N}} = \sqrt{\frac{m}{2\pi T}}. \tag{E.42}$$

This is the exact free Euclidean propagator $\big\langle x_f \big| e^{-H_0 T} \big| x_i \big\rangle$ prefactor; its real-time continuation $T \to iT$ reproduces the familiar $\sqrt{m/2\pi i T}$ of Section 10.1. Two lessons are worth pausing on. First, the recursion Eq. (E.41) is the discrete version of the homogeneous equation $-\ddot{\psi} = 0$: it is already the Gelfand–Yaglom construction in embryo. Second, the finite answer Eq. (E.42) emerged only because the divergent factor $(m/2\pi\varepsilon)^{N/2}$ from the measure cancelled the divergent factor $(m/\varepsilon)^{(N-1)/2}$ from $\sqrt{\det M^{(0)}}$. The determinant by itself is hopelessly divergent; only the measure-dressed combination is finite.

E.2.2 The formal determinant and its divergence

Strip away the measure and ask what the bare continuum object $\det K$ would be. By definition it is the product of the eigenvalues of the Dirichlet operator K on $[0, T]$. For the free operator $K_0 = -\partial_\tau^2$ (we set $m = 1$ in this subsection, restoring it trivially later) the eigenfunctions are $\sin(n\pi\tau/T)$ with $n = 1, 2, 3, \ldots$, and the eigenvalues are

$$\lambda_n^{(0)} = \left(\frac{n\pi}{T}\right)^2, \qquad n = 1, 2, 3, \ldots, \tag{E.43}$$

so the formal determinant is the infinite product

$$\det K_0 \overset{\text{formal}}{=} \prod_{n=1}^{\infty} \left(\frac{n\pi}{T}\right)^2. \tag{E.44}$$

This product diverges, and not subtly: its logarithm is $2\sum_n \ln(n\pi/T)$, whose terms grow without bound, so the partial products race to $+\infty$. Adding a potential only shifts each eigenvalue by a bounded amount (the eigenvalues of $-\partial_\tau^2 + W$ are $\lambda_n = (n\pi/T)^2 + O(1)$, as we prove below), so $\det K = \prod_n \lambda_n$ diverges in exactly the same way. The infinite-dimensional Gaussian of Eq. (E.35), taken at face value, returns $(\det K)^{-1/2} = 0$.

Caution

The divergence of Eq. (E.44) is *not* an artifact of bad bookkeeping that a cleverer choice of measure would remove. It reflects a genuine fact: a path integral assigns a weight to fluctuations of arbitrarily short wavelength, and there are infinitely many of them. The ultraviolet modes $n \to \infty$ are the source. Any sensible definition of $\det K$ must come with a prescription for taming them. There are two routes. The first, pursued in this section, is to never form $\det K$ alone but only the ratio $\det K / \det K_0$ against a reference operator, in which the ultraviolet modes cancel mode by mode. The second, pursued in Section E.3, is to give $\det K$ an intrinsic meaning through the spectral zeta function, which analytically continues the divergent product to a finite number. The two must agree on the ratios where both are defined, and they do.

E.2.3 The ratio prescription

Every physical quantity built from a Gaussian functional integral is a ratio of two such integrals in disguise. A normalized expectation value $\langle \cdot \rangle$ divides by the partition function; a transition amplitude is compared against the free one to fix its normalization; a one-loop effective action is the fluctuation determinant

relative to a reference. We therefore *define* the only object we will ever need, the determinant ratio

$$\frac{\det K}{\det K_0} \;:=\; \prod_{n=1}^{\infty} \frac{\lambda_n}{\lambda_n^{(0)}}, \tag{E.45}$$

where $\lambda_n, \lambda_n^{(0)}$ are the ordered Dirichlet eigenvalues of $K = -\partial_\tau^2 + W$ and of a reference operator $K_0 = -\partial_\tau^2 + W_0$ on $[0, T]$. This product *converges* whenever it should, for a reason the heuristic divergence of Eq. (E.44) hid: the ratio of the n-th eigenvalues tends to 1 fast. Writing $\lambda_n/\lambda_n^{(0)} = 1 + (\lambda_n - \lambda_n^{(0)})/\lambda_n^{(0)}$, the numerator is bounded (both eigenvalues equal $(n\pi/T)^2$ up to an $O(1)$ shift set by the potentials) while the denominator grows like n^2, so the correction is $O(1/n^2)$ and absolutely summable. An infinite product $\prod(1 + a_n)$ with $\sum |a_n| < \infty$ converges to a finite nonzero limit; this is the convergence fact imported from Section 5.4.1. The divergent measure factors that dressed each determinant in Eq. (E.40) are identical for K and K_0 and cancel in the ratio, so Eq. (E.45) is exactly the finite, measure-independent content of the one-loop integral. The remaining task is purely computational: evaluate the ratio without ever finding the individual eigenvalues. That is what Gelfand–Yaglom delivers.

E.2.4 The Gelfand–Yaglom theorem

Theorem E.12 (Gelfand–Yaglom)**.** *Let W, W_0 be continuous functions on $[0, T]$, and let $K = -\partial_\tau^2 + W$ and $K_0 = -\partial_\tau^2 + W_0$ act on functions obeying Dirichlet boundary conditions $\eta(0) = \eta(T) = 0$, with the determinant ratio defined by Eq. (E.45). Let ψ solve the homogeneous* initial-value *problem*

$$\big(-\partial_\tau^2 + W(\tau)\big)\psi(\tau) = 0, \qquad \psi(0) = 0, \quad \psi'(0) = 1, \tag{E.46}$$

and let ψ_0 solve the same problem with W replaced by W_0. Then, provided neither K nor K_0 has zero as an eigenvalue (equivalently $\psi(T) \neq 0$ and $\psi_0(T) \neq 0$),

$$\boxed{\frac{\det\big(-\partial_\tau^2 + W\big)}{\det\big(-\partial_\tau^2 + W_0\big)} = \frac{\psi(T)}{\psi_0(T)}\,.} \tag{E.47}$$

The statement is striking: an infinite product of eigenvalues equals the ratio of two numbers, each obtained by integrating a single ordinary differential equation forward from $\tau = 0$ to $\tau = T$. Note what Eq. (E.46) is and is not: it is an *initial*-value problem, not the eigenvalue (boundary-value) problem. We never impose $\psi(T) = 0$; we simply shoot from the left with a fixed slope and read off where we land at $\tau = T$. We give two proofs. The first is finite-dimensional and constructive — it is the continuum limit of the tridiagonal recursion we already met — and makes transparent why the theorem is true. The second is the spectral/analytic argument, which is intrinsically continuum and airtight; it also explains why only the ratio, never the determinant alone, has a chance of equalling a finite number.

Proof I: the discrete recursion and its limit

Discretize K exactly as in Section E.2.1. The Dirichlet operator $-\partial_\tau^2 + W$ on $[0, T]$ becomes the $(N-1) \times (N-1)$ symmetric tridiagonal matrix $\widehat{K}$ with diagonal entries $\widehat{K}_{kk} = \frac{2}{\varepsilon^2} + W_k$ (where $W_k = W(\tau_k)$) and off-diagonal entries $\widehat{K}_{k,k\pm1} = -\frac{1}{\varepsilon^2}$; the factors of $1/\varepsilon^2$ are the second-difference operator. Pull out the common scale and write $\widehat{K} = \varepsilon^{-2} B$, where B is tridiagonal with diagonal $2 + \varepsilon^2 W_k$ and off-diagonal -1. Let $d_k = \det B_k$ be the determinant of the leading $k \times k$ block. Cofactor expansion along the last row gives the three-term recursion

$$d_k = (2 + \varepsilon^2 W_k)\, d_{k-1} - d_{k-2}, \qquad d_0 = 1, \quad d_{-1} = 0. \tag{E.48}$$

Rearranging Eq. (E.48) as $\dfrac{d_k - 2d_{k-1} + d_{k-2}}{\varepsilon^2} = W_k\, d_{k-1}$ exhibits it as the discretization of the *continuum* equation $\psi'' = W\psi$, i.e. $(-\partial_\tau^2 + W)\psi = 0$, the very equation Eq. (E.46). To match initial data, set $\psi_k := \varepsilon\, d_{k-1}$. The seed values $d_{-1} = 0,\ d_0 = 1$ become $\psi_0 = \varepsilon d_{-1} = 0$ and $\psi_1 = \varepsilon d_0 = \varepsilon$, so $\psi_0 = 0$ and $(\psi_1 - \psi_0)/\varepsilon = 1$ — precisely $\psi(0) = 0,\ \psi'(0) = 1$. As $\varepsilon \to 0$ with $k\varepsilon = \tau$ held fixed, the solution of the difference equation converges to the solution of the differential equation (standard convergence of consistent, stable one-step schemes for a linear ODE with continuous coefficients), so $\psi_k \to \psi(\tau)$. In particular at the right endpoint $k = N$, $\tau_N = T$,

$$\det \widehat{K} = \varepsilon^{-2(N-1)}\, d_{N-1} = \varepsilon^{-2(N-1)}\, \frac{\psi_N}{\varepsilon} \xrightarrow{\varepsilon \to 0} \varepsilon^{-2(N-1)}\, \frac{\psi(T)}{\varepsilon}. \tag{E.49}$$

The prefactor $\varepsilon^{-(2N-1)}$ is divergent — it is the discrete shadow of the formal divergence of $\det K$ — but it depends only on ε and N, *not* on the potential W. Therefore in the ratio against the reference operator K_0 (same lattice, same N, same ε, diagonal $2 + \varepsilon^2 W_{0,k}$) it cancels identically:

$$\frac{\det \widehat{K}}{\det \widehat{K}_0} = \frac{d_{N-1}}{d_{N-1}^{(0)}} = \frac{\psi_N}{\psi_N^{(0)}} \xrightarrow{\varepsilon \to 0} \frac{\psi(T)}{\psi_0(T)}. \tag{E.50}$$

The left side converges to the determinant ratio Eq. (E.45), because the eigenvalues of the lattice operator $\widehat{K}$ converge to those of K mode by mode and the ratio is a convergent product. This proves Eq. (E.47). □

This proof is honest and complete, and its message is the one to remember: *the divergence of the functional determinant is entirely carried by a W-independent prefactor*, so it drops out of any ratio, leaving the finite number $\psi(T)/\psi_0(T)$. The second proof re-derives the same identity working directly in the continuum, and in doing so proves the eigenvalue asymptotics and the convergence facts we have so far only asserted.

Proof II: the spectral/analytic argument

The continuum proof studies how the determinant depends on a spectral parameter z, and uses one elementary fact from complex analysis. We first build the machinery, then assemble it.

The shooting solution as an entire function of z. For complex z, let $\psi_z(\tau)$ solve the initial-value problem

$$(-\partial_\tau^2 + W(\tau) - z)\psi_z = 0, \qquad \psi_z(0) = 0,\ \psi_z'(0) = 1. \tag{E.51}$$

Integrating twice and using the initial data converts Eq. (E.51) into the Volterra integral equation

$$\psi_z(\tau) = \frac{\sin(\sqrt{z}\,\tau)}{\sqrt{z}} + \int_0^\tau \frac{\sin(\sqrt{z}\,(\tau - s))}{\sqrt{z}}\, W(s)\, \psi_z(s)\, \mathrm{d}s, \tag{E.52}$$

in which $\sin(\sqrt{z}\,\tau)/\sqrt{z}$ is entire in z (only even powers of $\sqrt{z}$ appear, so the branch ambiguity cancels). Solving Eq. (E.52) by Picard iteration $\psi_z = \sum_{j\ge 0} \psi_z^{(j)}$, with $\psi_z^{(0)} = \sin(\sqrt{z}\,\tau)/\sqrt{z}$ and each $\psi_z^{(j)}$ obtained by inserting the previous term, gives terms bounded by

$$|\psi_z^{(j)}(\tau)| \le \frac{\left(\int_0^\tau |W|\right)^j}{j!}\, \frac{\sinh(|\sqrt{z}|\,\tau)}{|\sqrt{z}|}, \tag{E.53}$$

using $|\sin(\sqrt{z}\,u)/\sqrt{z}| \le \sinh(|\sqrt{z}|\,u)/|\sqrt{z}|$ for all complex z. The series therefore converges uniformly on compact sets to an *entire* function of z,

$$\psi_z(\tau) = \frac{\sin(\sqrt{z}\,\tau)}{\sqrt{z}}\Big(1 + O(1/\sqrt{z})\Big) \quad \text{as } |z| \to \infty, \tag{E.54}$$

the correction being uniform in $\tau \in [0, T]$ by Eq. (E.53). Since $\sin(\sqrt{z}\,T)/\sqrt{z}$ has order $\frac{1}{2}$ in z (it grows like $e^{|\sqrt{z}|\,T}$ along the negative real axis), so does $\psi_z(T)$: it is an entire function of order $\frac{1}{2}$.

Zeros of $\psi_z(T)$ are exactly the Dirichlet eigenvalues, and they are simple. By construction z is a Dirichlet eigenvalue of $K = -\partial_\tau^2 + W$ if and only if there is a nonzero solution vanishing at both endpoints, i.e. if and only if the shooting solution ψ_z — the unique solution with $\psi_z(0) = 0$ — also satisfies $\psi_z(T) = 0$. Hence the zeros of the entire function $z \mapsto \psi_z(T)$ are precisely the eigenvalues $\{\lambda_n\}$. They are simple as zeros: differentiating Eq. (E.51) in z and using a one-line Wronskian computation gives the Hadamard formula $\partial_z \psi_z(T)\big|_{z=\lambda_n} = -\psi_{\lambda_n}'(T) \int_0^T \psi_{\lambda_n}^2\, \mathrm{d}\tau / [\psi_{\lambda_n}'(T)]$, which is nonzero because the eigenfunction ψ_{λ_n} is nontrivial and, by uniqueness of the initial-value problem,

cannot have $\psi'_{\lambda_n}(T) = 0$ simultaneously with $\psi_{\lambda_n}(T) = 0$. (Concretely: a second-order linear ODE solution vanishing together with its derivative at a point is identically zero.) So each λ_n is a simple zero. From Eq. (E.54) the zeros sit asymptotically where $\sin(\sqrt{z}\,T) = 0$, i.e. $\sqrt{\lambda_n}\,T \to n\pi$, giving the promised asymptotics

$$\lambda_n = \Big(\frac{n\pi}{T}\Big)^2 + O(1), \tag{E.55}$$

which is what made the ratio product Eq. (E.45) converge and the bare product Eq. (E.44) diverge.

An elementary complex-analysis lemma.

Lemma E.13. *An entire function $g(z)$ of order $\rho < 1$ with no zeros is constant.*

Proof. If g has no zeros, $h := \ln g$ is entire. Order ρ means $|g(z)| \le \exp(C|z|^{\rho+\delta})$ for some C and any $\delta > 0$ with $\rho + \delta < 1$, hence $\operatorname{Re} h(z) \le C|z|^{\rho+\delta}$. The Borel–Carathéodory inequality bounds the maximum modulus of an analytic function on a circle by the maximum of its real part on a larger circle: for $0 < r < R$, $\max_{|z|=r} |h(z)| \le \frac{2r}{R-r} \max_{|z|=R} \operatorname{Re} h(z) + \frac{R+r}{R-r}|h(0)|$. Taking $R = 2r$ gives $\max_{|z|=r} |h| \le 4C(2r)^{\rho+\delta} + 3|h(0)|$, so h grows slower than $|z|$. Cauchy's estimate on the Taylor coefficients of h then forces every coefficient beyond the constant term to vanish (the k-th coefficient is bounded by $r^{-k} \max_{|z|=r} |h| \to 0$ for $k \ge 1$ as $r \to \infty$ since $\rho + \delta < 1$). Thus h is constant and so is g. □

Factorization and assembly. Let $\{\lambda_n\}$ and $\{\mu_n\}$ be the Dirichlet eigenvalues of K and K_0. Form the canonical product $P(z) := \prod_n (1 - z/\lambda_n)$, which converges and is entire of order $\frac{1}{2}$ because $\sum_n 1/\lambda_n < \infty$ by Eq. (E.55) (genus zero). Consider $g(z) := \psi_z(T)/P(z)$. It is entire (the simple zeros of the numerator are exactly cancelled by the simple zeros of P), it has no zeros, and it is a quotient of two entire functions of order $\frac{1}{2}$ that is itself entire, hence of order $\le \frac{1}{2} < 1$. By Theorem E.13, g is constant, equal to its value at $z = 0$: $g = \psi_0(T)/P(0) = \psi_0(T)$ since $P(0) = 1$. Therefore

$$\psi_z(T) = \psi_0(T) \prod_{n=1}^{\infty} \Big(1 - \frac{z}{\lambda_n}\Big), \qquad \psi^0_z(T) = \psi^0_0(T) \prod_{n=1}^{\infty} \Big(1 - \frac{z}{\mu_n}\Big), \tag{E.56}$$

the second identity being the same statement for K_0, with ψ^0_z its shooting solution and $\psi^0_0(T) \equiv \psi^0_{z=0}(T)$. (Here $\psi_0(T) = \psi_{z=0}(T)$ is the $z = 0$ shooting solution of Eq. (E.46), the very object on the right of Eq. (E.47).) Divide the two relations in Eq. (E.56) and use $(1 - z/\lambda_n)/(1 - z/\mu_n) = \big[(\lambda_n - z)/(\mu_n - z)\big] \cdot (\mu_n/\lambda_n)$ to separate the determinant-ratio product $R(z) := \prod_n (\lambda_n - z)/(\mu_n - z)$ from the z-independent constant $\prod_n (\mu_n/\lambda_n) = R(0)^{-1}$:

$$\frac{\psi_z(T)}{\psi^0_z(T)} = \frac{\psi_0(T)}{\psi^0_0(T)}\,\frac{R(z)}{R(0)}. \tag{E.57}$$

Now send $z \to -\infty$ along the real axis. On the left, Eq. (E.54) gives $\psi_z(T) \to \sinh(\sqrt{-z}\,T)/\sqrt{-z}$ for *both* operators with the same leading factor, so $\psi_z(T)/\psi_z^0(T) \to 1$. On the right, $R(z) = \prod_n (\lambda_n - z)/(\mu_n - z) \to 1$ as well, because each factor tends to 1 and $\ln R(z) = \sum_n \ln\!\left[1 + (\lambda_n - \mu_n)/(\mu_n - z)\right] \to 0$ by dominated convergence (the summand is bounded by $|\lambda_n - \mu_n|/|\mu_n - z|$, summable and vanishing as $z \to -\infty$). Taking the limit of Eq. (E.57) therefore yields $1 = \dfrac{\psi_0(T)}{\psi_0^0(T)} \cdot \dfrac{1}{R(0)}$, i.e.

$$\frac{\psi_0(T)}{\psi_0^0(T)} = R(0) = \prod_{n=1}^{\infty} \frac{\lambda_n}{\mu_n} = \frac{\det K}{\det K_0}, \tag{E.58}$$

which is exactly Eq. (E.47) (with $\psi_0 \equiv \psi$, $\psi_0^0 \equiv \psi_0$ in the notation of the theorem). □

The two proofs illuminate each other. Proof I shows the divergence is a potential-independent lattice prefactor; Proof II shows the same thing intrinsically, as the potential-independent asymptotic factor $\sinh(\sqrt{-z}\,T)/\sqrt{-z}$ that cancels in the ratio. Either way the content of a one-dimensional functional determinant is one number, $\psi(T)$, the endpoint value of a forward shoot.

Physical Insight

Gelfand–Yaglom replaces an infinite spectral problem with a single classical trajectory. The operator K is the second variation of the action — it governs how the action responds to fluctuations about the classical path — and $\psi(\tau)$, the solution of $K\psi = 0$ with $\psi(0) = 0$, is nothing but a *Jacobi field*: the linearized response of the classical trajectory to a kick in its initial velocity. The determinant of fluctuations is large exactly when neighbouring classical trajectories spread apart (large $\psi(T)$) and it develops a zero exactly at a *conjugate point*, where a nontrivial Jacobi field returns to zero ($\psi(T) = 0$) and the classical path ceases to be an isolated minimum. The vanishing of the determinant at conjugate points — the breakdown of the naive Gaussian approximation — is the same phenomenon as a caustic in optics and is the origin of the Maslov phase in semiclassical quantization.

E.2.5 Worked example: the harmonic oscillator

We now redeem the section's promise by recovering the harmonic-oscillator prefactor from Theorem E.12 and matching it against the canonical result of Section 10.1.

Worked Example E.14 (The oscillator fluctuation determinant and propagator prefactor)**.** The Euclidean fluctuation operator for the oscillator $V = \frac{1}{2}m\omega^2 x^2$ is, from Eq. (E.39), $K = m\,(-\partial_\tau^2 + \omega^2)$ with Dirichlet conditions on $[0, T]$; the

reference is the free operator $K_0 = m\,(-\partial_\tau^2)$. The overall constant m cancels in the ratio, so by Theorem E.12 we need only the shooting solutions of $(-\partial_\tau^2+\omega^2)\psi = 0$ and $(-\partial_\tau^2)\psi_0 = 0$.

Step 1 — solve the two initial-value problems. For $W = \omega^2$,

$$-\psi''+\omega^2\psi = 0,\ \ \psi(0) = 0,\ \psi'(0) = 1 \ \Longrightarrow\ \psi(\tau) = \frac{\sinh(\omega\tau)}{\omega},\ \ \psi(T) = \frac{\sinh(\omega T)}{\omega}. \tag{E.59}$$

For $W_0 = 0$, the equation is $-\psi_0'' = 0$, so $\psi_0(\tau) = \tau$ and $\psi_0(T) = T$.

Step 2 — form the determinant ratio. Equation (E.47) gives immediately

$$\frac{\det\,(-\partial_\tau^2+\omega^2)}{\det\,(-\partial_\tau^2)} = \frac{\psi(T)}{\psi_0(T)} = \frac{\sinh(\omega T)}{\omega T}. \tag{E.60}$$

It is worth checking this against the brute-force eigenvalue product, which we can do precisely because both spectra are known. The Dirichlet eigenvalues are $\lambda_n = (n\pi/T)^2+\omega^2$ and $\mu_n = (n\pi/T)^2$, so the ratio is the classical Euler product for the hyperbolic sine,

$$\prod_{n=1}^{\infty}\frac{(n\pi/T)^2+\omega^2}{(n\pi/T)^2} = \prod_{n=1}^{\infty}\Big(1+\frac{\omega^2T^2}{n^2\pi^2}\Big) = \frac{\sinh(\omega T)}{\omega T}, \tag{E.61}$$

in exact agreement with Eq. (E.60). The Gelfand–Yaglom theorem has summed this infinite product for us with a two-line ODE solution, and would have done so just as easily for a potential with no closed-form spectrum.

Step 3 — assemble the propagator prefactor. The full one-loop amplitude is the free normalization Eq. (E.42) times the inverse square root of the determinant ratio Eq. (E.60):

$$\sqrt{\frac{m}{2\pi T}}\,\Big(\frac{\sinh(\omega T)}{\omega T}\Big)^{-1/2} = \sqrt{\frac{m\,\omega}{2\pi\,\sinh(\omega T)}}. \tag{E.62}$$

The factors of T cancel cleanly, leaving the celebrated oscillator prefactor. With $m = 1$ this is $\sqrt{\omega/2\pi\sinh(\omega T)}$, exactly the form quoted in the section goal. Continuing back to real time by $T \to iT$ ($\sinh \to i\sin$) reproduces the Minkowski oscillator propagator prefactor $\sqrt{m\omega/2\pi i\sin(\omega T)}$ derived canonically (via the Mehler kernel and operator methods) in Section 10.1; the analytic continuation is licensed by Section 5.4.1. The path integral and the operator formalism agree, and the agreement rests entirely on the finite determinant ratio Eq. (E.60).

The same machine evaluates any one-loop prefactor in quantum mechanics, and — through the loop expansion of Section 4.2 — the leading semiclassical weight of any instanton or bounce, where $W = V''(x_{\mathrm{cl}})$ is read off the classical solution and $\psi(T)$ integrates the Jacobi equation along it. The boxed identity Eq. (E.47) is thus the first genuine computational handle on a functional determinant.

E.2.6 What survives, and what does not

We have extracted from the time-sliced measure a completely finite, regularization-independent number for any one-dimensional fluctuation operator: the determinant ratio $\det K/\det K_0 = \psi(T)/\psi_0(T)$, with the divergence quarantined in a universal prefactor that cancels. But the construction is irreducibly one-dimensional. Its engine, the reduction of a determinant to an initial-value problem, used that the eigenfunctions of $-\partial_\tau^2 + W$ on an interval are labelled by a single integer and that "shooting from one endpoint to the other" is a well-posed ODE. In two or more dimensions the spectrum of a Laplace-type operator is labelled by a lattice of momenta, there is no "initial" boundary to shoot from, and the eigenvalue density grows fast enough that even ratios $\det K/\det K_0$ can fail to converge as products. Gelfand–Yaglom gives us the answer in $d = 1$ and a sharp picture of why we cannot have it in general. To define $\det K$ for a differential operator in any dimension — and to confront, rather than cancel, the ultraviolet divergence of Eq. (E.44) — we need a definition that does not rely on a product at all. That is the spectral route of Section E.3: assemble the eigenvalues into a zeta function $\zeta_K(s) = \sum_n \lambda_n^{-s}$, analytically continue it past the divergent product, and *define* $\ln\det K = -\zeta_K'(0)$.

E.3 Regularizing the Determinant: Zeta Functions and the Heat Kernel

The previous section (Section E.2) left us in an uncomfortable position. Gaussian integration over the fluctuations of a field produces, as we saw in Section E.1, a factor $(\det K)^{-1/2}$ for a second-order differential operator K, and the logarithm of that determinant is the entire one-loop physics. In one dimension—the quantum mechanics of a single degree of freedom—the Gelfand–Yaglom theorem of Section E.2 gave a finite, computable answer for the *ratio* of two such determinants by solving an ordinary differential equation, sidestepping the eigenvalues altogether. That trick is precious but parochial: it is tied to the one-dimensional structure of an initial-value problem, and it computes only ratios. In a genuine field theory K is an elliptic operator on a d-dimensional space, its spectrum $\{\lambda_n\}$ grows without bound, and the object we are asked to make sense of is the divergent product $\prod_n \lambda_n$, or equivalently the divergent sum $\sum_n \ln\lambda_n$. We need a definition of $\det K$ that (i) reduces to the ordinary determinant whenever the spectrum is finite, (ii) is independent of the spatial dimension, (iii) does not presuppose a momentum cutoff or a lattice, and (iv) exposes, rather than hides, the ultraviolet divergences and the way they organize themselves into the renormalization of the theory. The spectral zeta function delivers exactly this, and the heat kernel is the engine that makes it computable. Throughout this section K is taken in Euclidean signature—the Wick rotation $t \to -i\tau$

of Section 5.4 turns the Minkowski fluctuation operator into a positive elliptic operator, which is the only regime in which e^{-tK} is a sensible (trace-class) object.

E.3.1 The spectral zeta function and the determinant

Let K be a non-negative, self-adjoint, elliptic operator on a compact d-dimensional Euclidean manifold (or a finite box with stated boundary conditions), with a discrete spectrum

$$0 \le \lambda_1 \le \lambda_2 \le \lambda_3 \le \cdots \to \infty, \tag{E.63}$$

each eigenvalue repeated according to its multiplicity. For the moment assume $\lambda_1 > 0$ (no zero modes); zero modes are reinstated in Section E.3.3. The growth rate of the eigenvalues is fixed by Weyl's law: for a second-order operator the n-th eigenvalue behaves as $\lambda_n \sim C\, n^{2/d}$ for large n, with C a constant built from the volume. This single fact controls every convergence question below.

Definition E.15 (Spectral zeta function)**.** The *spectral zeta function* of K is

$$\zeta_K(s) \;=\; \sum_n \lambda_n^{-s}, \tag{E.64}$$

defined initially for those complex s where the series converges absolutely.

By Weyl's law $\lambda_n^{-s} \sim C^{-s} n^{-2s/d}$, so Eq. (E.64) converges absolutely precisely when $\operatorname{Re} s > d/2$. In that half-plane $\zeta_K(s)$ is holomorphic, being a uniformly convergent sum of the entire functions $\lambda_n^{-s} = e^{-s \ln \lambda_n}$. The number $d/2$ is the *abscissa of convergence*; it is the first place the dimension of spacetime enters, and it will reappear as the location of the operator's leading ultraviolet divergence.

The reason to introduce Eq. (E.64) is a formal manipulation that becomes a definition. Differentiate term by term:

$$\zeta_K'(s) \;=\; \sum_n \frac{d}{ds}\, e^{-s \ln \lambda_n} \;=\; -\sum_n (\ln \lambda_n)\, \lambda_n^{-s}. \tag{E.65}$$

If we could set $s = 0$ in Eq. (E.65) we would obtain $\zeta_K'(0) = -\sum_n \ln \lambda_n = -\ln \prod_n \lambda_n = -\ln \det K$. The sum $\sum_n \ln \lambda_n$ diverges, and $s = 0$ lies far outside the domain of convergence $\operatorname{Re} s > d/2$; the manipulation is therefore purely formal as it stands. But it points to the right definition: provided $\zeta_K(s)$ extends to a function holomorphic at $s = 0$, the quantity $\zeta_K'(0)$ is perfectly well-defined, and we *adopt* it as the regularized value of the divergent sum.

Definition E.16 (Zeta-regularized determinant)**.** When $\zeta_K(s)$ admits an analytic continuation that is holomorphic in a neighbourhood of $s = 0$, the *zeta-regularized determinant* of K is

$$\boxed{\ln \det K \;=\; -\,\zeta_K'(0), \qquad \det K \;=\; \exp\!\big(-\,\zeta_K'(0)\big).} \tag{E.66}$$

This definition is forced on us in the sense that it is the *only* one consistent with the finite-dimensional case. If K is an $N\times N$ positive matrix with eigenvalues $\lambda_1,\dots,\lambda_N$, then $\zeta_K(s)=\sum_{n=1}^{N}\lambda_n^{-s}$ is an entire function (a finite sum of entire functions), so the continuation is trivial, and Eq. (E.65) at $s=0$ gives $\zeta_K'(0)=-\sum_{n=1}^{N}\ln\lambda_n$, whence $-\zeta_K'(0)=\ln\prod_n\lambda_n=\ln\det K$ exactly. Theorem E.16 reproduces the textbook determinant whenever the textbook determinant exists, and supplies a finite answer when it does not. It is also manifestly dimension-independent: nothing in Eq. (E.66) refers to a lattice spacing, a momentum cutoff, or even the value of d except through the analytic structure of ζ_K. The remaining work is to prove the continuation exists and to compute it; that is what the heat kernel is for.

E.3.2 From the heat trace: the Mellin transform

The bridge from the abstract sum Eq. (E.64) to something computable is the operator e^{-tK}, the *heat semigroup*, so named because for $K=-\nabla^2$ the kernel of e^{-tK} is the fundamental solution of the heat equation $\partial_t u=\nabla^2 u$. Its trace,

$$\Theta_K(t)\;=\;\operatorname{Tr}\,e^{-tK}\;=\;\sum_{n}e^{-t\lambda_n},\qquad t>0, \tag{E.67}$$

is called the *heat trace* (or partition function of the operator). For $\lambda_1>0$ the series converges for every $t>0$ and decays exponentially, $\Theta_K(t)\to e^{-t\lambda_1}\to 0$, as $t\to\infty$. Its $t\to 0^+$ behaviour is singular and encodes the ultraviolet physics; that is the subject of Section E.3.4.

The link to the zeta function is a single classical identity. Recall the integral representation of the Gamma function, $\Gamma(s)=\int_0^\infty dt\,t^{s-1}e^{-t}$, valid for $\operatorname{Re}s>0$. Rescaling $t\to\lambda t$ gives, for any $\lambda>0$,

$$\lambda^{-s}\;=\;\frac{1}{\Gamma(s)}\int_0^\infty dt\;t^{s-1}\,e^{-\lambda t},\qquad \operatorname{Re}s>0. \tag{E.68}$$

Summing Eq. (E.68) over the spectrum and exchanging sum and integral—legitimate for $\operatorname{Re}s>d/2$, where both the sum and the integral converge absolutely—we obtain the *Mellin transform* representation of the zeta function in terms of the heat trace.

Proposition E.17 (Mellin representation)**.** *For* $\operatorname{Re}s>d/2$,

$$\zeta_K(s)\;=\;\frac{1}{\Gamma(s)}\int_0^\infty dt\;t^{s-1}\,\Theta_K(t). \tag{E.69}$$

Proof. Insert Eq. (E.68) into Eq. (E.64):

$$\zeta_K(s)=\sum_n\lambda_n^{-s}=\sum_n\frac{1}{\Gamma(s)}\int_0^\infty dt\;t^{s-1}e^{-\lambda_n t}=\frac{1}{\Gamma(s)}\int_0^\infty dt\;t^{s-1}\sum_n e^{-\lambda_n t}. \tag{E.70}$$

The interchange of $\sum_n$ and $\int dt$ is justified by Tonelli's theorem: for $\operatorname{Re} s > d/2$ the integrand $t^{\operatorname{Re} s-1}e^{-\lambda_n t}$ is non-negative and the iterated integral $\sum_n \int_0^\infty t^{\operatorname{Re} s-1}e^{-\lambda_n t}\,dt = \Gamma(\operatorname{Re} s)\sum_n \lambda_n^{-\operatorname{Re} s} < \infty$ is finite. The inner sum is $\Theta_K(t)$ by Eq. (E.67). □

Equation (E.69) is the workhorse of the whole subject: it converts a question about the (inaccessible) individual eigenvalues into a question about the (computable) heat trace, a single function of one variable. Before continuing it is worth recording the heuristic that explains why $-\zeta_K'(0)$ "is" $\sum_n \ln\lambda_n$ at the level of the proper-time integral. Setting $s \to 0$ naively in Eq. (E.69) and using $1/\Gamma(s) \to s$, one has $\zeta_K(s) \approx s\int_0^\infty \frac{dt}{t}\,\Theta_K(t)$, so

$$\ln\det K = -\zeta_K'(0) \;\text{“}=\text{”}\; -\int_0^\infty \frac{dt}{t}\,\Theta_K(t) = -\int_0^\infty \frac{dt}{t}\sum_n e^{-\lambda_n t}. \tag{E.71}$$

The right-hand side is the *proper-time* (Schwinger) representation of the log-determinant. It diverges at the lower limit $t \to 0$, which is the ultraviolet end, and the divergence is purely additive: for two eigenvalues the combination is finite by Frullani's integral,

$$\int_0^\infty \frac{dt}{t}\left(e^{-\lambda t} - e^{-\lambda' t}\right) = \ln\frac{\lambda'}{\lambda}, \tag{E.72}$$

which is exactly $\ln(\lambda'/\lambda) = \ln\lambda' - \ln\lambda$, the difference of the "log-eigenvalues" the proper-time integral is meant to sum. (To verify Eq. (E.72), differentiate its left side with respect to λ, obtaining $-\int_0^\infty dt\, e^{-\lambda t} = -1/\lambda$, and integrate back with the boundary value 0 at $\lambda = \lambda'$.) Thus the proper-time integral computes log-eigenvalues correctly up to a common, eigenvalue-independent divergent constant, and the role of the zeta function is precisely to define that constant by analytic continuation rather than to throw it away by a subtraction. This is the same philosophy as dimensional regularization, a correspondence we make exact in Section E.3.7.

E.3.3 Analytic continuation and the value at zero

To turn Theorem E.16 into a theorem we must continue $\zeta_K(s)$ past the line $\operatorname{Re} s = d/2$ and establish that it is holomorphic at $s = 0$. The continuation is controlled entirely by the short-time behaviour of the heat trace, which (as proved in Section E.3.4) has an asymptotic expansion in powers of t,

$$\Theta_K(t) \overset{t\to 0^+}{\sim} \sum_{k=0}^{\infty} A_k\, t^{(k-d)/2}, \tag{E.73}$$

where the coefficients A_k are integrals of local geometric data over the manifold (only even k occur for the operators of interest, but nothing below needs that).

The term with $k = d$ is the t^0 (constant) term; we single out its coefficient A_d because it will turn out to govern $\zeta_K(0)$.

We must also restore zero modes. If K has $N_0 = \dim\ker K$ zero eigenvalues, they should be *omitted* from the determinant (a vanishing eigenvalue makes the literal product zero and signals a flat direction to be treated separately, typically by a collective coordinate). The zero modes contribute N_0 to the trace Eq. (E.67) at all times, spoiling the large-t decay. We therefore define the zeta function on the nonzero spectrum by subtracting them:

$$\zeta_K(s) \;=\; \frac{1}{\Gamma(s)}\int_0^\infty dt\, t^{s-1}\big[\Theta_K(t) - N_0\big], \qquad \det{}'K \equiv e^{-\zeta_K'(0)}, \tag{E.74}$$

the prime on $\det'$ recording that zero modes are excluded. We now prove the central analytic result.

Theorem E.18 (Meromorphic continuation and the value at zero)**.** *Let K be a positive elliptic operator with heat trace obeying Eq. (E.73), and let $N_0 = \dim\ker K$. Then $\zeta_K(s)$ defined by Eq. (E.74) extends to a meromorphic function on the whole s-plane whose only singularities are simple poles at*

$$s = \frac{d-k}{2}, \qquad k = 0, 1, 2, \dots, \quad \tfrac{d-k}{2} \notin \{0, -1, -2, \dots\}, \tag{E.75}$$

with residues $\operatorname{Res}_{s=(d-k)/2}\zeta_K(s) = A_k/\Gamma(\frac{d-k}{2})$. In particular ζ_K is holomorphic at $s = 0$, where

$$\boxed{\zeta_K(0) \;=\; A_d \;-\; \dim\ker K.} \tag{E.76}$$

Proof. Split the integral Eq. (E.74) at $t = 1$: $\zeta_K(s) = \frac{1}{\Gamma(s)}\big[I_<(s) + I_>(s)\big]$ with $I_< = \int_0^1$ and $I_> = \int_1^\infty$. The high-t piece $I_>(s) = \int_1^\infty t^{s-1}[\Theta_K(t) - N_0]\,dt$ converges for *every* complex s, because $\Theta_K(t) - N_0 = \sum_{\lambda_n>0} e^{-\lambda_n t} \le (\text{const})\, e^{-\lambda_1 t}$ decays exponentially while t^{s-1} grows only polynomially; differentiating under the integral shows $I_>(s)$ is entire.

For the low-t piece insert the asymptotic expansion Eq. (E.73). Fix an integer $N > d$ and write

$$\Theta_K(t) - N_0 = \sum_{k=0}^{N} A_k\, t^{(k-d)/2} - N_0 + R_N(t), \qquad R_N(t) = O\big(t^{(N+1-d)/2}\big)\ \ (t \to 0). \tag{E.77}$$

Integrate term by term against t^{s-1} over $(0, 1)$. For each power,

$$\int_0^1 t^{s-1}\, t^{(k-d)/2}\, dt = \frac{1}{s + \frac{k-d}{2}}, \qquad \int_0^1 t^{s-1}\,(-N_0)\, dt = -\frac{N_0}{s}, \tag{E.78}$$

both valid initially for $\operatorname{Re} s$ large enough and then by the explicit right-hand side for all s away from the indicated poles. The remainder contributes $\int_0^1 t^{s-1} R_N(t)\, dt$,

which converges and is holomorphic for $\operatorname{Re} s > (d-N-1)/2$. Collecting,

$$\zeta_K(s) = \frac{1}{\Gamma(s)}\left[\sum_{k=0}^{N} \frac{A_k}{s+\frac{k-d}{2}} \;-\; \frac{N_0}{s} \;+\; H_N(s)\right], \tag{E.79}$$

with $H_N(s) = I_>(s) + \int_0^1 t^{s-1} R_N(t)\,dt$ holomorphic for $\operatorname{Re} s > (d-N-1)/2$. Since N is arbitrary, Eq. (E.79) continues ζ_K to the entire plane. The bracket has simple poles at $s = (d-k)/2$ $(k = 0, \dots, N)$ and at $s = 0$. The prefactor $1/\Gamma(s)$ is *entire* (the reciprocal Gamma function has no poles) with simple zeros at $s = 0, -1, -2, \dots$. Wherever a pole of the bracket falls on such a zero it is cancelled, which removes exactly the values $(d-k)/2 \in \{0, -1, -2, \dots\}$ from the pole list; the surviving poles are Eq. (E.75), with residue $A_k \cdot \lim_{s\to(d-k)/2} 1/\Gamma(s) = A_k/\Gamma((d-k)/2)$.

It remains to evaluate $\zeta_K(0)$. Near $s = 0$ the reciprocal Gamma function behaves as

$$\frac{1}{\Gamma(s)} = s + \gamma_E\, s^2 + O(s^3), \tag{E.80}$$

which follows from $\Gamma(s) = \frac{1}{s} - \gamma_E + O(s)$ (the Laurent expansion of Γ at its pole, established with the other properties of the Gamma function in Section 5.4). Thus $1/\Gamma(s)$ vanishes linearly at $s = 0$, and only the residue of the bracket at $s = 0$ survives the product. The bracket's pole at $s = 0$ comes from two terms: the $k = d$ summand, $A_d/(s + \frac{d-d}{2}) = A_d/s$, and the zero-mode term $-N_0/s$. Hence

$$\zeta_K(0) = \lim_{s\to 0} s\left[\frac{A_d}{s} - \frac{N_0}{s} + (\text{regular at } 0)\right] = A_d - N_0, \tag{E.81}$$

which is Eq. (E.76). □

Two consequences deserve emphasis. First, Theorem E.18 guarantees that Theorem E.16 is well posed: ζ_K is holomorphic at $s = 0$, so $\zeta_K'(0)$ exists and $\det' K$ is a finite number. Second, Eq. (E.76) ties the value $\zeta_K(0)$—a pure number—to a single heat-kernel coefficient A_d, the t^0 term in the short-time expansion, minus the dimension of the kernel. When the spacetime dimension d is odd there is no t^0 term in Eq. (E.73) (the powers are half-odd-integers), so $A_d = 0$ and $\zeta_K(0) = -\dim\ker K$; odd-dimensional theories have no associated logarithmic divergence and no scaling anomaly, a fact we recover concretely in Theorem E.21. The location of the poles Eq. (E.75) is sketched in Fig. E.2.

E.3.4 The Seeley–DeWitt expansion

We owe a proof of the short-time expansion Eq. (E.73) and a computation of its leading coefficients, since everything above rests on it. The cleanest derivation

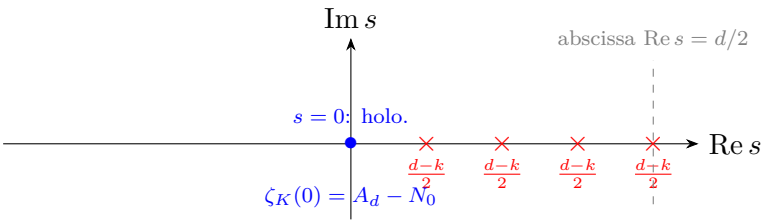


Figure E.2: Analytic structure of $\zeta_K(s)$ for a second-order operator in $d = 4$. The series Eq. (E.64) converges right of the abscissa $\mathrm{Re}\, s = d/2$ (here $s = 2$). Continuation produces simple poles (red crosses) at $s = (d-k)/2$ with residues set by the heat-kernel coefficients A_k; the would-be poles at the non-positive integers are killed by the zero of $1/\Gamma(s)$. The determinant is read off at the regular point $s = 0$ (blue), and $\zeta_K(0)$ measures the logarithmic divergence.

works directly with the heat kernel $\mathcal{K}(x, y; t)$, the integral kernel of e^{-tK}, defined by

$$(e^{-tK} f)(x) = \int d^d y\, \mathcal{K}(x, y; t)\, f(y), \tag{E.82}$$

so that the heat trace is the integral of the kernel over the diagonal, $\Theta_K(t) = \int d^d x\, \mathcal{K}(x, x; t)$. By construction $\mathcal{K}$ solves the heat equation in its first slot with a delta-function initial condition,

$$(\partial_t + K_x)\mathcal{K}(x, y; t) = 0, \qquad \mathcal{K}(x, y; 0^+) = \delta^{(d)}(x - y). \tag{E.83}$$

We carry out the analysis for the prototypical fluctuation operator

$$K = -\partial^2 + V(x) \tag{E.84}$$

on flat $\mathbb{R}^d$ (Euclidean), where $\partial^2 = \delta^{ij}\partial_i\partial_j$ is the Laplacian and V is a smooth potential; this is exactly the form of the one-loop operator $\delta^2 S/\delta\phi^2 = -\partial^2 + V''(\phi_{\mathrm{cl}})$ that the next section will need. The curved-space generalization is recorded at the end and proved in the curved-spacetime development (Chapter 64).

When $V \equiv 0$ the heat kernel is the Gaussian solution of the free heat equation,

$$\mathcal{K}_0(x, y; t) = \frac{1}{(4\pi t)^{d/2}} \exp\left(-\frac{|x - y|^2}{4t}\right), \tag{E.85}$$

as one checks by direct substitution into Eq. (E.83); its normalization is fixed by $\int d^d x\, \mathcal{K}_0 = 1$ and it reproduces $\delta^{(d)}(x - y)$ as $t \to 0^+$ because it is a nascent delta function of width $\sqrt{t}$. The presence of the potential is encoded by writing the full kernel as the free Gaussian times a regular prefactor,

$$\mathcal{K}(x, y; t) = \frac{1}{(4\pi t)^{d/2}}\, e^{-|x-y|^2/4t}\, \Phi(x, y; t), \qquad \Phi(x, y; t) = \sum_{k=0}^{\infty} b_k(x, y)\, t^k, \tag{E.86}$$

the *Seeley–DeWitt ansatz*. The matching to the free kernel at $t \to 0$ requires $\Phi \to 1$ on the diagonal, i.e. $b_0(x,x) = 1$, and the b_k are smooth. Substituting Eq. (E.86) into the heat equation determines the b_k recursively.

Proposition E.19 (Recursion for the heat-kernel coefficients)**.** *Write $u = x - y$. The coefficient functions in Eq. (E.86) satisfy $b_0 \equiv 1$ and, for $k \geq 1$,*

$$(k + u^i \partial_i)\, b_k(x,y) = (\partial^2 - V)\, b_{k-1}(x,y), \tag{E.87}$$

where ∂_i acts on the x variable. In particular the diagonal values are

$$b_0(x,x) = 1, \qquad b_1(x,x) = -V(x). \tag{E.88}$$

Proof. Write $\mathcal{K} = \mathcal{K}_0\,\Phi$ with $\mathcal{K}_0$ the free kernel Eq. (E.85), which itself satisfies the free heat equation $\partial_t \mathcal{K}_0 = \partial^2 \mathcal{K}_0$. We compute $(\partial_t + K)\mathcal{K} = (\partial_t - \partial^2 + V)(\mathcal{K}_0\Phi)$. Using the product rule and $\partial_t \mathcal{K}_0 = \partial^2 \mathcal{K}_0$,

$$\begin{aligned}(\partial_t - \partial^2)(\mathcal{K}_0\Phi) &= (\partial_t \mathcal{K}_0)\Phi + \mathcal{K}_0 \partial_t \Phi - (\partial^2 \mathcal{K}_0)\Phi - 2(\partial_i \mathcal{K}_0)(\partial_i \Phi) - \mathcal{K}_0 \partial^2 \Phi \\ &= \mathcal{K}_0 \Big[\partial_t \Phi - \tfrac{2}{\mathcal{K}_0}(\partial_i \mathcal{K}_0)(\partial_i \Phi) - \partial^2 \Phi \Big]. \end{aligned} \tag{E.89}$$

The gradient of the Gaussian is $\partial_i \mathcal{K}_0 = \mathcal{K}_0 \cdot (-u_i/2t)$, so $\frac{2}{\mathcal{K}_0}(\partial_i \mathcal{K}_0)(\partial_i \Phi) = -(u_i/t)\partial_i \Phi$. Demanding $(\partial_t + K)\mathcal{K} = 0$, i.e. $(\partial_t - \partial^2)(\mathcal{K}_0\Phi) = -V\mathcal{K}_0\Phi$, and dividing by $\mathcal{K}_0 \neq 0$ gives the transport equation

$$\partial_t \Phi + \frac{u^i}{t}\, \partial_i \Phi - \partial^2 \Phi = -V\, \Phi. \tag{E.90}$$

Insert $\Phi = \sum_k b_k t^k$ and read off powers of t. The operator ∂_t sends $b_k t^k \to k\, b_k t^{k-1}$; the term $\frac{u^i}{t}\partial_i$ sends $b_k t^k \to (u^i \partial_i b_k)\, t^{k-1}$; and $-\partial^2$ and $-V$ keep the power t^k. Matching the coefficient of t^{k-1} on both sides,

$$k\, b_k + u^i \partial_i b_k - \partial^2 b_{k-1} = -V\, b_{k-1}, \tag{E.91}$$

which rearranges to Eq. (E.87), with the convention $b_{-1} = 0$. At $k = 0$ this reads $u^i \partial_i b_0 = 0$, so b_0 is constant along rays emanating from y; smoothness at $u = 0$ together with $b_0(x,x) = 1$ forces $b_0 \equiv 1$. For $k = 1$, Eq. (E.87) gives $(1 + u^i\partial_i)b_1 = (\partial^2 - V)b_0 = -V$ (since $\partial^2 b_0 = 0$). Evaluating on the diagonal $u = 0$, the operator $u^i \partial_i$ annihilates, leaving $b_1(x,x) = -V(x)$. □

Integrating the diagonal kernel over the manifold and using Eq. (E.86),

$$\Theta_K(t) = \int d^d x\, \mathcal{K}(x,x;t) = \frac{1}{(4\pi t)^{d/2}} \int d^d x \Big[\, b_0(x,x) + b_1(x,x)\, t + \cdots \Big], \tag{E.92}$$

which is precisely the asymptotic series Eq. (E.73) once we identify the coefficients. It is conventional to factor out the $(4\pi)^{-d/2}$ and write

$$\Theta_K(t) \overset{t \to 0^+}{\sim} \frac{1}{(4\pi t)^{d/2}} \sum_{j=0}^{\infty} a_j\, t^j, \qquad a_j = \int d^d x\, b_j(x,x), \tag{E.93}$$

the a_j being the (integrated) *Seeley–DeWitt coefficients*; comparison with Eq. (E.73) gives $A_{2j} = (4\pi)^{-d/2} a_j$ and $A_{\text{odd}} = 0$. From Eq. (E.88),

$$a_0 = \int d^d x = \text{Vol}, \qquad a_1 = -\int d^d x \, V(x). \tag{E.94}$$

The leading coefficient is just the volume; the first correction is the integral of the potential. The constant (t^0) term in Θ_K is $A_d = (4\pi)^{-d/2} a_{d/2}$, so Eq. (E.76) reads, in this normalization,

$$\zeta_K(0) = \frac{a_{d/2}}{(4\pi)^{d/2}} - \dim\ker K. \tag{E.95}$$

This is the precise meaning of the slogan "$\zeta_K(0) = a_{d/2} - \dim\ker K$": $a_{d/2}$ is the half-dimension Seeley–DeWitt coefficient, the one whose power of t is exactly zero.

The curvature term. On a curved background the operator is $K = -\nabla^2 + V$ with ∇ the covariant Laplacian, the volume element carries $\sqrt{g}$, and the geodesic interval $\sigma(x,y) = \frac{1}{2}(\text{geodesic distance})^2$ replaces $\frac{1}{2}|x-y|^2$. The ansatz Eq. (E.86) acquires one extra factor, the square root of the van Vleck–Morette determinant $\Delta(x,y) = g^{-1/2} \det[-\partial^2\sigma/\partial x^i \partial y^j]$,

$$\mathcal{K}(x,y;t) = \frac{\Delta^{1/2}(x,y)}{(4\pi t)^{d/2}} \, e^{-\sigma(x,y)/2t} \sum_k b_k(x,y)\, t^k, \tag{E.96}$$

whose role is to keep $b_0 = 1$ in the curved transport equation. In Riemann normal coordinates centred at y one has the standard small-distance expansions $\sqrt{g} = 1 - \frac{1}{6} R_{ij} u^i u^j + \cdots$ and $\Delta^{1/2} = 1 + \frac{1}{12} R_{ij} u^i u^j + \cdots$, and repeating the $k = 1$ step of Theorem E.19 with the covariant Laplacian feeds the Ricci scalar R into the diagonal value. The result, derived in full in Chapter 64, is

$$b_1(x,x) = \frac{R(x)}{6} - V(x), \qquad a_1 = \int d^d x \, \sqrt{g} \left(\frac{R}{6} - V \right). \tag{E.97}$$

The coefficient $\frac{1}{6}$ is the conformal coupling in $d = 4$ ($\xi = (d-2)/4(d-1) = \frac{1}{6}$): the conformally coupled scalar is the one whose heat kernel has this universal curvature term, which is the first sign that $\zeta_K(0)$ will know about the conformal anomaly.

Physical Insight

The heat-kernel expansion is a geometry-by-time-resolution microscope. At short proper time t the kernel Eq. (E.85) probes the manifold only out to a distance $\sqrt{t}$, so the leading $\Theta_K \sim (4\pi t)^{-d/2}\text{Vol}$ just counts how many independent "cells" of size $\sqrt{t}$ fit in the volume—a mode-counting

that is the heat-kernel face of Weyl's law. Successive coefficients $a_1, a_2, \ldots$ are corrections that see ever-coarser local data: the potential, then the curvature, then its derivatives and squares. The deepest fact of this section is that the *same* coefficients that describe this geometric coarse graining are the coefficients of the ultraviolet divergences of the quantum theory. Short distance and short proper time are the same limit.

E.3.5 Ultraviolet divergences and the scaling anomaly

We can now read the renormalization structure of a one-loop determinant directly off the heat-kernel coefficients. The one-loop contribution to the effective action is $\Gamma^{(1)} = \frac{1}{2} \ln \det K$ (the $\frac{1}{2}$ from the $(\det K)^{-1/2}$ of bosonic Gaussian integration; the next section assembles this), and its proper-time form Eq. (E.71) is divergent at the lower limit. Cut the integral at $t = \epsilon^2$ (so ϵ is a short-distance length) and insert the short-time expansion Eq. (E.93):

$$\ln \det K\Big|_{\mathrm{UV}} = -\int_{\epsilon^2}^{1} \frac{dt}{t} \frac{1}{(4\pi t)^{d/2}} \sum_j a_j t^j = -\frac{1}{(4\pi)^{d/2}} \sum_j a_j \int_{\epsilon^2} dt\, t^{j-d/2-1}. \tag{E.98}$$

Each term diverges as $\epsilon \to 0$ whenever its exponent makes the integral blow up at the lower limit, i.e. for $j < d/2$ (power divergences) and for $j = d/2$ (a logarithm):

$$\int_{\epsilon^2} dt\, t^{j-d/2-1} = \begin{cases} \dfrac{1}{(d/2-j)\,\epsilon^{d-2j}} + (\text{finite}), & j < d/2, \\ -2\ln\epsilon + (\text{finite}), & j = d/2. \end{cases} \tag{E.99}$$

The dictionary is now explicit. The $j = 0$ term, $a_0 = \mathrm{Vol}$, produces a divergence $\propto \epsilon^{-d}\,\mathrm{Vol}$: this is the *cosmological-constant* (vacuum-energy) divergence, an additive shift of the free energy proportional to the volume. The $j = 1$ term, $a_1 = \int (R/6 - V)$, produces a divergence $\propto \epsilon^{-(d-2)}$ multiplying $\int V$ and $\int R$: this renormalizes the *mass* (the $\int V$ piece, since $V = m^2 + \cdots$) and, on a curved background, *Newton's constant* (the $\int R$ piece). The logarithmic term sits at $j = d/2$, with coefficient $a_{d/2}$—precisely the coefficient that Eq. (E.95) identified with $\zeta_K(0)$. Thus the single number $\zeta_K(0)$ is the coefficient of the logarithmic ultraviolet divergence of the determinant, and it is therefore the quantity that controls the *running* and the renormalization-group flow at one loop.

The zeta definition Theorem E.16 contains no cutoff ϵ at all: the power divergences are simply absent (the analytic continuation discards them, just as dimensional regularization sets scaleless power divergences to zero), while the logarithmic divergence reappears not as a divergence but as a finite, unavoidable dependence of the determinant on the unit in which eigenvalues are measured.

To see this, note that the determinant of a rescaled operator is governed by exactly $\zeta_K(0)$.

Theorem E.20 (Scaling law of the determinant)**.** *For any constant $\mu > 0$,*

$$\ln\det(\mu^2 K) = \ln\det K + 2\,\zeta_K(0)\,\ln\mu, \qquad \textit{equivalently} \qquad \det(\mu^2 K) = \mu^{2\zeta_K(0)}\det K. \tag{E.100}$$

Proof. The eigenvalues of $\mu^2 K$ are $\mu^2\lambda_n$, so its zeta function is $\zeta_{\mu^2 K}(s) = \sum_n (\mu^2\lambda_n)^{-s} = \mu^{-2s}\zeta_K(s)$. Differentiate in s:

$$\zeta'_{\mu^2 K}(s) = \frac{d}{ds}\left[\mu^{-2s}\zeta_K(s)\right] = -2\ln\mu\;\mu^{-2s}\zeta_K(s) + \mu^{-2s}\zeta'_K(s). \tag{E.101}$$

At $s = 0$, where $\mu^0 = 1$, $\zeta'_{\mu^2 K}(0) = -2\ln\mu\,\zeta_K(0) + \zeta'_K(0)$. Insert into $\ln\det(\mu^2 K) = -\zeta'_{\mu^2 K}(0)$ to obtain Eq. (E.100). □

Theorem E.20 is the heart of the matter. A determinant of eigenvalues is not a pure number: it carries a dimension, and rescaling the operator (say by absorbing a factor μ^2 to make the eigenvalues dimensionless, as one must to take a logarithm) changes it by $\mu^{2\zeta_K(0)}$. Classically the action of a scale-invariant theory is unchanged by $K \to \mu^2 K$; quantum-mechanically the determinant is multiplied by $\mu^{2\zeta_K(0)}$. When $\zeta_K(0) \neq 0$ this is an honest breaking of classical scale invariance by the regularized measure—the *scaling anomaly* of the determinant, and (in even d, where $a_{d/2}$ is generically nonzero) the microscopic origin of the trace/conformal anomaly. The anomaly coefficient is $\zeta_K(0) = a_{d/2}/(4\pi)^{d/2} - \dim\ker K$, a local geometric integral.

Physical Insight

The vanishing of $\zeta_K(0)$ in odd dimensions, proved below Theorem E.18, is the statement that odd-dimensional theories have no conformal anomaly. There is simply no local scalar of the right dimension—no t^0 heat-kernel coefficient—available to be the anomaly. The same counting underlies the absence of perturbative gauge anomalies in odd spacetime dimension and the special role of even d throughout this book.

This is also the precise point of contact with the chiral anomaly. In Chapter 36 the anomaly is computed as the Jacobian of the path-integral measure under a chiral rotation, and Fujikawa's regularized Jacobian is nothing but a heat-kernel trace, $\lim_{t\to 0}\mathrm{Tr}[\gamma^5 e^{-t\not{D}^2}]$; the surviving t^0 term is the coefficient $a_{d/2}$ of the Dirac operator $\not{D}^2$ evaluated against γ^5. The object we have just learned to extract is exactly the object that chapter evaluates to produce the $\int F\wedge F$ anomaly. We flag this forward: Chapter 36 *uses* $a_{d/2}$; here we have defined it and given the machinery to compute it.

E.3.6 A complete computation: the thermal harmonic oscillator

The cleanest non-trivial test of the entire apparatus is the determinant that computes the free energy of a single harmonic oscillator at temperature $T = 1/\beta$. The Euclidean partition function $Z = \operatorname{Tr} e^{-\beta H}$ is the imaginary-time path integral over loops of period β (the thermal circle S^1_β), and for the oscillator $H = \frac{1}{2}p^2 + \frac{1}{2}\omega^2 q^2$ the fluctuation operator is the one-dimensional ($d = 1$) operator $K = -\partial_\tau^2 + \omega^2$ acting on periodic functions on S^1_β. Gaussian integration gives $Z \propto (\det K)^{-1/2}$, hence the free energy $F = -T \ln Z = \frac{T}{2} \ln \det K$ (up to an ω-independent normalization that we will see is fixed unambiguously by the zeta procedure). The exact answer, known from elementary statistical mechanics, is

$$F = \frac{\omega}{2} + T \, \ln(1 - e^{-\beta\omega}), \tag{E.102}$$

the zero-point energy plus the thermal occupation. We now derive Eq. (E.102) from scratch by zeta-regularizing $\det K$, with the vacuum ($T \to 0$) piece carried in closed form.

Worked Example E.21 (Zeta-regularized free energy of the thermal oscillator)**.** The eigenfunctions of $K = -\partial_\tau^2 + \omega^2$ on S^1_β are the Matsubara modes $e^{i\omega_n \tau}$ with $\omega_n = 2\pi n/\beta$, $n \in \mathbb{Z}$, periodicity being the bosonic boundary condition. The eigenvalues are

$$\lambda_n = \omega_n^2 + \omega^2 = \Big(\frac{2\pi n}{\beta}\Big)^2 + \omega^2, \qquad n \in \mathbb{Z}, \tag{E.103}$$

all strictly positive ($\omega > 0$, so no zero modes, $N_0 = 0$). The heat trace is

$$\Theta_K(t) = \sum_{n\in\mathbb{Z}} e^{-t\lambda_n} = e^{-t\omega^2} \sum_{n\in\mathbb{Z}} \exp\Big[- t\Big(\tfrac{2\pi}{\beta}\Big)^2 n^2\Big]. \tag{E.104}$$

The sum is a theta function; its small-t behaviour is exposed by Poisson resummation. Applying the Jacobi identity $\sum_n e^{-\alpha n^2} = \sqrt{\pi/\alpha}\, \sum_m e^{-\pi^2 m^2/\alpha}$ (Poisson summation for the Gaussian, whose Fourier transform is again Gaussian) with $\alpha = t(2\pi/\beta)^2$,

$$\sum_n \exp\Big[-t\Big(\tfrac{2\pi}{\beta}\Big)^2 n^2\Big] = \frac{\beta}{\sqrt{4\pi t}} \sum_{m\in\mathbb{Z}} e^{-\beta^2 m^2/4t}, \qquad \Longrightarrow \qquad \Theta_K(t) = \frac{\beta}{\sqrt{4\pi t}}\, e^{-t\omega^2} \sum_{m\in\mathbb{Z}} e^{-\beta^2 m^2/4} \tag{E.105}$$

Consistency with Seeley–DeWitt. As $t \to 0^+$ only the $m = 0$ winding survives ($m \neq 0$ terms are suppressed by $e^{-\beta^2/4t}$), so $\Theta_K(t) \sim (4\pi t)^{-1/2} \beta\, e^{-t\omega^2} = (4\pi t)^{-1/2}(\beta - \beta\omega^2 t + \cdots)$. Comparing with Eq. (E.93) in $d = 1$ identifies $a_0 = \beta = \mathrm{Vol}(S^1_\beta)$ and $a_1 = -\beta\omega^2 = -\int_0^\beta V \, d\tau$ with $V = \omega^2$, exactly Eq. (E.94). Since $d = 1$ is odd there is no t^0 term, so $A_1 = 0$ and Eq. (E.76) gives $\zeta_K(0) = 0$: no scaling anomaly, as anticipated.

The Mellin integral. Feed Eq. (E.105) into the Mellin representation Eq. (E.69) and split off the $m = 0$ term:

$$\zeta_K(s) = \underbrace{\frac{\beta}{\sqrt{4\pi}}\,\frac{1}{\Gamma(s)}\int_0^\infty dt\; t^{s-3/2}e^{-t\omega^2}}_{\zeta_0(s)\quad (m=0,\ \text{vacuum})} + \underbrace{\frac{\beta}{\sqrt{4\pi}}\,\frac{1}{\Gamma(s)}\sum_{m\neq 0}\int_0^\infty dt\; t^{s-3/2}e^{-t\omega^2-\beta^2m^2/4t}}_{\zeta_{\rm th}(s)\quad (m\neq 0,\ \text{thermal})}. \tag{E.106}$$

Vacuum piece. The $m = 0$ integral is elementary: $\int_0^\infty t^{s-3/2}e^{-t\omega^2}dt = \Gamma(s-\frac{1}{2})\,\omega^{1-2s}$ by Eq. (E.68), so

$$\zeta_0(s) = \frac{\beta}{\sqrt{4\pi}}\,\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)}\,\omega^{1-2s} = \frac{\beta\omega}{\sqrt{4\pi}}\,\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)}\,\omega^{-2s}. \tag{E.107}$$

We need $\zeta_0'(0)$. Near $s = 0$ use $1/\Gamma(s) = s + O(s^2)$ from Eq. (E.80) and the finite value $\Gamma(-\frac{1}{2}) = -2\sqrt{\pi}$:

$$\frac{\Gamma(s-\frac{1}{2})}{\Gamma(s)} = s\,\Gamma(-\tfrac{1}{2})\,[1+O(s)] = -2\sqrt{\pi}\,s + O(s^2). \tag{E.108}$$

Hence, with $\omega^{-2s} = 1 - 2s\ln\omega + O(s^2)$ and $\beta/\sqrt{4\pi}\cdot(-2\sqrt{\pi}) = -\beta$,

$$\zeta_0(s) = -\beta\omega\, s\,(1 - 2s\ln\omega + \cdots)(1+\cdots) = -\beta\omega\, s + O(s^2), \tag{E.109}$$

so $\zeta_0(0) = 0$ and $\zeta_0'(0) = -\beta\omega$. The vacuum contribution to the log-determinant is $-\zeta_0'(0) = \beta\omega$, and to the free energy

$$F_{\rm vac} = \frac{T}{2}\,(-\zeta_0'(0)) = \frac{T}{2}\,\beta\omega = \frac{\omega}{2}. \tag{E.110}$$

The zeta procedure delivers the zero-point energy $\omega/2$ in closed form, with no leftover constant: every term that could have contributed an ambiguity (the $\psi(-\frac{1}{2})$, the $\ln\omega$, the γ_E) sits at order s^2 and never reaches $\zeta_0'(0)$.

Thermal piece. Because $1/\Gamma(s) = s + O(s^2)$ vanishes at $s = 0$ while the $m \neq 0$ integrals are finite there, the thermal zeta also satisfies $\zeta_{\rm th}(0) = 0$, and $\zeta_{\rm th}'(0)$ equals the bracket evaluated at $s = 0$ (the rule $\frac{d}{ds}[s\,f(s)]_{s=0} = f(0)$). We therefore need the $m \neq 0$ integrals only at $s = 0$, i.e. $\int_0^\infty t^{-3/2}e^{-\omega^2 t-\beta^2m^2/4t}dt$. This is the elementary half-line Gaussian integral of Theorem E.22 below,

$$\int_0^\infty dt\; t^{-3/2}\,e^{-at-b/t} = \sqrt{\frac{\pi}{b}}\;e^{-2\sqrt{ab}}, \qquad a = \omega^2,\quad b = \frac{\beta^2m^2}{4}, \tag{E.111}$$

giving $\sqrt{4\pi/\beta^2m^2}\,e^{-\beta\omega|m|} = \frac{2\sqrt{\pi}}{\beta|m|}e^{-\beta\omega|m|}$. Therefore

$$\zeta_{\rm th}'(0) = \frac{\beta}{\sqrt{4\pi}}\sum_{m\neq 0}\frac{2\sqrt{\pi}}{\beta|m|}\,e^{-\beta\omega|m|} = \sum_{m\neq 0}\frac{e^{-\beta\omega|m|}}{|m|} = 2\sum_{m=1}^{\infty}\frac{e^{-\beta\omega m}}{m} = -2\ln(1-e^{-\beta\omega}), \tag{E.112}$$

the last step being the Taylor series $-\ln(1-x) = \sum_{m\geq 1} x^m/m$ with $x = e^{-\beta\omega} < 1$. The thermal contribution to the free energy is then

$$F_{\rm th} = \frac{T}{2}\,\big(-\zeta'_{\rm th}(0)\big) = \frac{T}{2}\cdot 2\ln\!\big(1-e^{-\beta\omega}\big) = T\,\ln\!\big(1-e^{-\beta\omega}\big). \tag{E.113}$$

Total. Adding Eqs. (E.110) and (E.113),

$$F = F_{\rm vac} + F_{\rm th} = \frac{\omega}{2} + T\,\ln\!\big(1-e^{-\beta\omega}\big), \tag{E.114}$$

which is the exact free energy Eq. (E.102), with no undetermined additive constant whatsoever. The zeta-regularized determinant has reproduced the full thermodynamics of the oscillator—zero-point energy and Bose–Einstein occupation alike—from its eigenvalues, in a manifestly dimension-independent scheme.

The computation used one elementary integral, which we now prove for completeness.

Lemma E.22 (Half-line Gaussian integral)**.** *For $a, b > 0$,*

$$\int_0^\infty dt\; t^{-1/2}\, e^{-at-b/t} = \sqrt{\frac{\pi}{a}}\, e^{-2\sqrt{ab}}, \qquad \int_0^\infty dt\; t^{-3/2}\, e^{-at-b/t} = \sqrt{\frac{\pi}{b}}\, e^{-2\sqrt{ab}}. \tag{E.115}$$

Proof. For the first identity substitute $t = u^2$ ($dt = 2u\,du$, $t^{-1/2} = 1/u$):

$$\int_0^\infty t^{-1/2} e^{-at-b/t} dt = 2\int_0^\infty e^{-au^2-b/u^2}\, du. \tag{E.116}$$

Complete the square in the exponent, $au^2 + b/u^2 = (\sqrt{a}\,u - \sqrt{b}/u)^2 + 2\sqrt{ab}$, so the right side is $2e^{-2\sqrt{ab}}\int_0^\infty e^{-(\sqrt{a}\,u-\sqrt{b}/u)^2} du$. The remaining integral is evaluated by the Cauchy–Schlömilch substitution: setting $w = \sqrt{a}\,u - \sqrt{b}/u$, the map $u \mapsto \sqrt{b/a}/u$ sends the integrand to itself while $du \mapsto \sqrt{b/a}\,u^{-2}du$, and averaging the integral with its image gives

$$2\int_0^\infty e^{-(\sqrt{a}u-\sqrt{b}/u)^2} du = \int_0^\infty e^{-(\sqrt{a}u-\sqrt{b}/u)^2}\big(1+\sqrt{b/a}\,u^{-2}\big) du = \frac{1}{\sqrt{a}}\int_{-\infty}^\infty e^{-w^2} dw = \frac{\sqrt{\pi}}{\sqrt{a}}, \tag{E.117}$$

where we used $dw = (\sqrt{a} + \sqrt{b}\,u^{-2})du = \sqrt{a}\,(1+\sqrt{b/a}\,u^{-2})du$ and that w runs over all of $\mathbb{R}$ as $u : 0 \to \infty$. Hence $\int_0^\infty e^{-(\sqrt{a}u-\sqrt{b}/u)^2} du = \frac{\sqrt{\pi}}{2\sqrt{a}}$ and the first identity follows. The second is obtained by differentiating the first with respect to b: $\partial_b \int t^{-1/2} e^{-at-b/t} dt = -\int t^{-3/2} e^{-at-b/t} dt$, while $\partial_b[\sqrt{\pi/a}\, e^{-2\sqrt{ab}}] = -\sqrt{\pi/a}\,\sqrt{a/b}\, e^{-2\sqrt{ab}} = -\sqrt{\pi/b}\, e^{-2\sqrt{ab}}$. □

E.3.7 Dictionary: dimensional regularization and Pauli–Villars

Zeta-function regularization is one scheme among several, and it is essential to know that it sees the *same* ultraviolet divergences as the schemes used in the body of the book—it differs only in finite, scheme-dependent constants and in how it parametrizes the running. We make the translation precise on the simplest field-theory determinant, the constant-mass operator $K = -\partial^2 + m^2$ on flat $\mathbb{R}^d$ (Euclidean), whose one-loop vacuum energy density is the building block of the Coleman–Weinberg potential of the next section.

The common proper-time master integral. For $K = -\partial^2 + m^2$ the spectrum is continuous, labelled by Euclidean momenta k with eigenvalue k^2+m^2; the trace becomes a momentum integral times the volume. The log-determinant per unit volume is

$$\frac{1}{\mathrm{Vol}} \ln\det K = \int \frac{d^d k}{(2\pi)^d}\, \ln(k^2+m^2), \tag{E.118}$$

and all three schemes regulate the *same* object, the divergent integral $\int \frac{d^dk}{(2\pi)^d} \ln(k^2+m^2)$. The unifying representation is again proper time. Using $\ln\Lambda = -\frac{d}{ds}\Lambda^{-s}|_{s=0}$ with $\Lambda = k^2+m^2$ and the Gaussian momentum integral $\int \frac{d^dk}{(2\pi)^d} e^{-t(k^2+m^2)} = (4\pi t)^{-d/2} e^{-tm^2}$, one finds the per-volume heat trace $\Theta_K(t)/\mathrm{Vol} = (4\pi t)^{-d/2} e^{-tm^2}$ and hence the per-volume zeta function

$$\frac{\zeta_K(s)}{\mathrm{Vol}} = \frac{1}{\Gamma(s)} \int_0^\infty dt\, t^{s-1} (4\pi t)^{-d/2} e^{-tm^2} = \frac{1}{(4\pi)^{d/2}} \frac{\Gamma(s-\frac{d}{2})}{\Gamma(s)}\, m^{d-2s}. \tag{E.119}$$

Zeta scheme. Apply Theorem E.16 to Eq. (E.119). With $1/\Gamma(s) = s + O(s^2)$, write $\frac{\Gamma(s-d/2)}{\Gamma(s)} m^{d-2s} = s\, g(s)$ with $g(0) = \Gamma(-\frac{d}{2}) m^d$, so $-\zeta_K'(0)/\mathrm{Vol} = -\frac{1}{(4\pi)^{d/2}} g(0)$ and

$$\frac{1}{\mathrm{Vol}} \ln\det K = -\frac{\Gamma(-\frac{d}{2})}{(4\pi)^{d/2}}\, m^d. \tag{E.120}$$

This is finite for non-even d. Its information about divergences is encoded in the poles of $\Gamma(-d/2)$ at $d = 0, 2, 4, \ldots$, exactly the even dimensions at which the Seeley–DeWitt coefficient $a_{d/2}$ exists.

Dimensional regularization. Dimensional regularization (Chapter D) takes precisely Eq. (E.119) and continues in $d = 4 - 2\varepsilon$ rather than in s. The one-loop vacuum energy is Eq. (E.120) continued to $d = 4 - 2\varepsilon$, with the $\overline{\mathrm{MS}}$ scale μ restored on dimensional grounds ($m^d \to \mu^{2\varepsilon} m^d$):

$$\frac{1}{\mathrm{Vol}} \ln\det K = -\frac{m^4}{(4\pi)^2} (4\pi)^\varepsilon \left(\frac{\mu^2}{m^2}\right)^\varepsilon \Gamma(\varepsilon - 2). \tag{E.121}$$

Expanding $\Gamma(\varepsilon-2)=\frac{1}{2}\big(\frac{1}{\varepsilon}+\frac{3}{2}-\gamma_E+O(\varepsilon)\big)$ and the prefactors,

$$\frac{1}{\mathrm{Vol}}\ln\det K=-\frac{m^4}{2(4\pi)^2}\left[\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi+\frac{3}{2}-\ln\frac{m^2}{\mu^2}+O(\varepsilon)\right]. \tag{E.122}$$

The $1/\varepsilon$ pole is the logarithmic divergence; comparing the residue with Eq. (E.99) at $j=d/2=2$ shows its coefficient is fixed by exactly the same datum, the half-dimension coefficient a_2 ($\propto m^4$ here, since $V=m^2$ contributes $a_2\sim\int V^2$ at this order). The map between schemes is the standard one,

$$\boxed{\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi\;\longleftrightarrow\;-\ln\frac{m^2}{\mu^2}\;+\;(\text{scheme const}),} \tag{E.123}$$

i.e. the $\overline{\mathrm{MS}}$ pole is replaced, in the zeta scheme, by the finite $\ln(\mu^2/m^2)$ generated by the scaling law Theorem E.20: the arbitrary scale μ that dimensional regularization introduces to keep the coupling dimensionless is the *same* μ as the rescaling $K\to\mu^{-2}K$ that Theorem E.20 shows the determinant cannot avoid. The coefficient of $\ln\mu$ in both schemes is $\zeta_K(0)$. Both schemes therefore predict identical β-functions and anomalous dimensions; only the finite constant ($-\gamma_E+\ln 4\pi$ in $\overline{\mathrm{MS}}$, $+\frac{3}{2}$ and the like in zeta) differs, and that constant is reshuffled by a finite renormalization.

Pauli–Villars. Pauli–Villars regularization keeps $d=4$ fixed and instead subtracts heavy regulator determinants. In proper-time language it replaces the divergent small-t end of Eq. (E.71) by the regulated combination

$$\ln\det K\big|_{\mathrm{PV}}=-\int_0^\infty\frac{dt}{t}\,\mathrm{Vol}\,(4\pi t)^{-d/2}\Big(e^{-tm^2}-\sum_i c_i\,e^{-tM_i^2}\Big), \tag{E.124}$$

with regulator masses $M_i\to\infty$ and coefficients c_i chosen so that the bracket vanishes fast enough as $t\to 0$ to kill the divergences of Eq. (E.99): cancelling the power divergences and the logarithm requires $\sum_i c_i=1$ and $\sum_i c_i M_i^2=m^2$ in $d=4$ (one auxiliary mass with a single subtraction suffices for the logarithm alone). Carrying out the t integral with these conditions, the cutoff-dependence organizes into a single logarithm of the regulator scale,

$$\frac{1}{\mathrm{Vol}}\ln\det K\big|_{\mathrm{PV}}=-\frac{m^4}{2(4\pi)^2}\left[\ln\frac{M_{\mathrm{PV}}^2}{m^2}+(\text{scheme const})\right], \tag{E.125}$$

with M_{PV} a representative regulator mass. Comparison with Eq. (E.122) gives the final entry of the dictionary,

$$\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi\;\longleftrightarrow\;\ln\frac{M_{\mathrm{PV}}^2}{\mu^2}\;+\;(\text{scheme const}), \tag{E.126}$$

so that $\ln M_{\mathrm{PV}}^2$ plays the role of $1/\varepsilon$ and of the zeta scale-logarithm alike. In every scheme the divergence is the *same* number, fixed by the heat-kernel coefficient $a_{d/2}$; the three regulators differ only in a calculable finite constant and in whether the running is displayed as a pole, a $\ln \mu$, or a $\ln M_{\mathrm{PV}}$.

Caution

The scaling law Theorem E.20 makes manifest a fact that the zeta scheme can otherwise hide: $\det K$ is dimensionful when the eigenvalues are, so a bare number for $\ln \det K$ is meaningful only *relative* to a chosen scale. When quoting a zeta-regularized determinant one must state the unit in which K is measured; comparing two schemes, or two operators, means comparing their $\det(\mu^{-2}K)$ at a common μ. This is the regularization-independent core that survives, and it is the only piece that enters a physical observable.

With Theorems E.16 and E.18 we have a finite, dimension- and scheme-independent $\ln \det K = -\zeta_K'(0)$ for any positive elliptic operator, together with the heat-kernel dictionary Eq. (E.93) that exhibits its divergences and the scaling law Theorem E.20 that captures its anomaly. This is exactly the input the next section requires: it identifies the one-loop effective action with $\frac{1}{2}\ln \det K$ and feeds the present machinery—now including the fermionic sign from the Grassmann Gaussian integral—into the Coleman–Weinberg potential of ϕ^4 theory, where the abstract determinant finally becomes a curve one can plot.

E.4 The One-Loop Effective Action and the Coleman–Weinberg Potential

The previous section closed by handing us a precise, finite object: for any elliptic operator K the spectral zeta function ζ_K is analytic at the origin, and

$$\ln \det K \;=\; -\,\zeta_K'(0) \tag{E.127}$$

is a well-defined number, with its scheme dependence (the μ that the zeta and heat-kernel constructions both carry) made explicit there (Section E.3). We now spend that capital. The whole point of building the Gaussian-integral machinery of Section E.1 and the determinant-regularization machinery of Section E.3 was to compute one physical object: the *one-loop effective action*, the leading quantum correction to the classical action that governs the dynamics of field expectation values. This section assembles the pieces. We derive the effective action as a functional determinant directly from the fluctuation integral, fix the sign that distinguishes a bosonic loop from a fermionic one, and then carry ϕ^4 theory all the way to its one-loop effective potential — regularized, renormalized,

and cross-checked against the renormalization group of Chapter 24. The result, the Coleman–Weinberg potential, is the prototype for every radiative-correction-driven phenomenon in the book: vacuum energy, spontaneous symmetry breaking by quantum effects, and the running of couplings read off from a potential.

The effective action $\Gamma[\phi_c]$ itself was defined in Chapter 4 as the Legendre transform of the connected generating functional $W[J]$ (Theorem 4.33), and its one-loop form was already extracted there as a functional determinant, Eq. (4.211). We do not re-derive the Legendre theory. What we do here is the part that belongs to this appendix: turn the formal determinant into a finite, renormalized number through the regularization apparatus just built, and run the computation to its closed form for the carried scalar model. To make the determinant manifestly real and positive we work in the Euclidean theory throughout the derivation, rotating back to Lorentzian signature only at the end where physical statements are made.

E.4.1 The background-field split and the Gaussian fluctuation integral

Fix the conventions. The Lorentzian generating functional is $Z[J] = \int \mathcal{D}\phi\, \exp\big(iS[\phi] + i\int J\phi\big)$ with $\hbar$ restored as $\exp\big(\frac{i}{\hbar}(S + \int J\phi)\big)$; the Euclidean version, after the Wick rotation $t \to -i\tau$ defined in Chapter 10, is

$$Z_E[J] \;=\; \int \mathcal{D}\phi\, \exp\Big[-\tfrac{1}{\hbar}\Big(S_E[\phi] - \int J\phi\Big)\Big], \tag{E.128}$$

where S_E is the Euclidean action, real and (for the models of interest) bounded below. We compute $W_E = -\hbar \ln Z_E$ and its Legendre transform $\Gamma_E[\phi_c]$, the Euclidean effective action.

The *background-field method* evaluates these by saddle point. Choose the source J so that the field expectation value equals a prescribed background, $\langle\phi\rangle = \phi_c$, and split the integration variable into the background plus a fluctuation,

$$\phi(x) \;=\; \phi_c(x) + \sqrt{\hbar}\,\eta(x), \tag{E.129}$$

the factor $\sqrt{\hbar}$ inserted so that the quadratic fluctuation term comes out $\hbar$-independent. Taylor-expanding S_E about ϕ_c,

$$S_E[\phi_c + \sqrt{\hbar}\,\eta] = S_E[\phi_c] + \sqrt{\hbar}\int \frac{\delta S_E}{\delta\phi(x)}\Big|_{\phi_c}\eta(x)\,\mathrm{d}^d x + \frac{\hbar}{2}\iint \eta(x)\,\mathcal{O}(x,y)\,\eta(y)\,\mathrm{d}^d x\,\mathrm{d}^d y + O(\hbar^{3/2}), \tag{E.130}$$

with the second-variation operator

$$\mathcal{O}(x,y) \;\equiv\; \frac{\delta^2 S_E}{\delta\phi(x)\,\delta\phi(y)}\Big|_{\phi=\phi_c}. \tag{E.131}$$

The defining property of the Legendre transform is that the term linear in the fluctuation is cancelled against the source: writing $\Gamma_E[\phi_c] = W_E[J] + \int J\phi_c$ and using $\delta W_E/\delta J = \phi_c$, the $O(\sqrt{\hbar})$ piece drops out identically (this is the statement, proved in Section 4.5, that Γ_E collects one-particle-irreducible graphs in the background ϕ_c, with no tadpoles). What survives at the first nontrivial order is a pure Gaussian in η:

$$e^{-\Gamma_E[\phi_c]/\hbar} \;=\; e^{-S_E[\phi_c]/\hbar} \int \mathcal{D}\eta \, \exp\Big[-\tfrac{1}{2} \iint \eta \, \mathcal{O} \, \eta \Big] \, (1 + O(\hbar)). \tag{E.132}$$

The fluctuation integral is exactly the infinite-dimensional Gaussian evaluated in Section E.1: for a positive symmetric operator $\mathcal{O}$,

$$\int \mathcal{D}\eta \, \exp\Big[-\tfrac{1}{2} \iint \eta \, \mathcal{O} \, \eta \Big] \;=\; \mathcal{N} \, (\det \mathcal{O})^{-1/2}, \tag{E.133}$$

with $\mathcal{N}$ a field-independent normalization (an overall constant times powers of 2π per mode) that we absorb into the additive normalization of Γ_E. Taking $-\hbar \ln$ of Eq. (E.132) and using Eq. (E.133),

$$\boxed{\Gamma_E[\phi_c] \;=\; S_E[\phi_c] \;+\; \frac{\hbar}{2} \ln \det \mathcal{O} \;+\; O(\hbar^2) \,, \qquad \mathcal{O} = \frac{\delta^2 S_E}{\delta\phi \, \delta\phi}\Big|_{\phi_c} .} \tag{E.134}$$

This is the same content as the Lorentzian formula Eq. (4.211) of Chapter 4 — the factor $i/2$ there is replaced by $1/2$ here because Euclidean signature has already absorbed the i — but now $\mathcal{O}$ is a real elliptic operator and $\ln\det\mathcal{O}$ is exactly the finite object (E.127) that Section E.3 taught us to compute. The whole one-loop problem is reduced to evaluating one functional determinant. We set $\hbar = 1$ from here on.

Theorem E.23 (The one-loop effective action)**.** *Let S_E be a Euclidean action whose second variation $\mathcal{O} = \delta^2 S_E/\delta\phi\,\delta\phi|_{\phi_c}$ is, for every smooth background ϕ_c, a positive elliptic operator with the spectral zeta function and heat kernel of Section E.3. Then the effective action admits the loop expansion $\Gamma_E[\phi_c] = \sum_{L\geq 0} \hbar^L \Gamma_E^{(L)}[\phi_c]$ with $\Gamma_E^{(0)} = S_E$ and*

$$\Gamma_E^{(1)}[\phi_c] \;=\; \tfrac{1}{2} \ln \det \mathcal{O} \;=\; -\tfrac{1}{2}\, \zeta_{\mathcal{O}}'(0), \tag{E.135}$$

the zeta-regularized determinant being independent of any mode ordering and finite. For a free (Grassmann-even) field this is the complete one-loop contribution; the higher $\Gamma_E^{(L)}$, $L \geq 2$, are the sums of connected one-particle-irreducible vacuum graphs with L loops built on the propagator $\mathcal{O}^{-1}$ and the vertices $\delta^{n\geq 3} S_E/\delta\phi^n|_{\phi_c}$.

Proof. Everything but the loop counting is Eqs. (E.129) to (E.135) together with $\ln\det\mathcal{O} = -\zeta_{\mathcal{O}}'(0)$ from (E.127). The loop counting is the statement of Theorem 4.41: after the rescaling $\eta \to \sqrt{\hbar}\,\eta$ in (E.129), each propagator

$\mathcal{O}^{-1}$ carries one power of $\hbar$ and each interaction vertex $\delta^n S_E$ carries $\hbar^{-1}$, so a connected vacuum graph with V vertices, I internal lines, and $L = I - V + 1$ loops carries $\hbar^{I-V} = \hbar^{L-1}$; the prefactor $e^{-S_E/\hbar}$ supplies the missing $\hbar^{-1}$, leaving $\hbar^L$ in Γ_E. At $L = 1$ the only connected 1PI vacuum graph is the single closed η loop weighted by the quadratic form $\mathcal{O}$, whose value is precisely $\frac{1}{2}\ln\det\mathcal{O}$ (a closed loop of the free η propagator contracted against itself, with the $\frac{1}{2}$ the symmetry factor of the figure-eight-free single loop); the linear term is absent by the Legendre condition. Hence (E.135). □

Physical Insight

The one-loop term is the summed zero-point energy of the fluctuations in the background. Diagonalize $\mathcal{O}$ on the background ϕ_c with eigenvalues $\lambda_n(\phi_c)$; then $\frac{1}{2}\ln\det\mathcal{O} = \frac{1}{2}\sum_n \ln\lambda_n(\phi_c)$ is, mode by mode, the logarithm of a fluctuation frequency, and its ϕ_c-dependence is the back-reaction of the vacuum fluctuations on the background. A heavier background (larger $M^2(\phi_c)$) stiffens every mode; the integrated cost of that stiffening is the Coleman–Weinberg potential. This is why the effective potential is sometimes called the "vacuum energy as a function of the field."

E.4.2 The fermionic determinant and its sign

The construction above used a Grassmann-even (bosonic) field, for which the Gaussian integral produces $(\det\mathcal{O})^{-1/2}$. For a fermion the Berezin integral of Chapter 12 inverts this in two ways at once, and tracking both inversions is the whole content of "the fermion sign."

Let $\psi, \bar\psi$ be a Dirac field with Euclidean quadratic action $S_E^{\mathrm{f}} = \int \bar\psi\, D\, \psi$, where D is the (Euclidean) Dirac operator in whatever background is present — for the scalar models below, $D = \not\partial_E + M_{\mathrm{f}}(\phi_c)$ with a field-dependent mass $M_{\mathrm{f}}(\phi_c)$ from a Yukawa coupling; for gauge backgrounds, $D = \not\partial_E - ig\not{A} + m$. The Berezin Gaussian formula derived in Chapter 12 reads, for independent generators $\psi, \bar\psi$,

$$\int \mathcal{D}\bar\psi\,\mathcal{D}\psi\, \exp\Big(-\int \bar\psi\, D\,\psi\Big) \;=\; \det D, \tag{E.136}$$

with the determinant appearing to the *positive* first power, in contrast to the $(\det\mathcal{O})^{-1/2}$ of the bosonic case (E.133). Two differences are visible and both matter:

1. the *exponent of the determinant* flips from $-\frac{1}{2}$ to $+1$ — the $-\frac{1}{2}$ becomes $+1$, a relative factor of -2;

2. there is no field-independent $\mathcal{N}$ to discard differently; the determinant is the entire answer.

Repeating the saddle-point steps Eqs. (E.132) to (E.134) with (E.136) in place of (E.133) gives the fermionic one-loop effective action

$$\boxed{\Gamma_E^{(1),\,\mathrm{ferm}}[\phi_c] \;=\; -\,\ln\det D \;=\; -\,\mathrm{Tr}\ln D\,,} \tag{E.137}$$

to be compared with the bosonic $+\frac{1}{2}\ln\det\mathcal{O}$. The overall minus sign is the celebrated minus sign of a closed fermion loop; it is here seen to originate entirely in the Grassmann (Berezin) measure, not in any analytic continuation. The factor of 2 between $|-1|$ and $\frac{1}{2}$ is the statement that a Dirac fermion carries twice the determinantal weight of a real scalar before the Dirac trace is even taken.

It is convenient to rewrite (E.137) in the same "$\frac{1}{2}\ln\det$ of a second-order operator" form as the boson, so that bosons and fermions can be compared term by term. Using $\det D = \big(\det D\,\det D\big)^{1/2}$ and the identity $\det(\not{\partial}_E + M_\mathrm{f}) = \det(-\not{\partial}_E + M_\mathrm{f})$ (which holds because γ_5 anticommutes with $\not{\partial}_E$ and has unit determinant, $\gamma_5\not{\partial}_E\gamma_5 = -\not{\partial}_E$, so the two operators are similar),

$$\ln\det D = \tfrac{1}{2}\ln\det\big[(\not{\partial}_E+M_\mathrm{f})(-\not{\partial}_E+M_\mathrm{f})\big] = \tfrac{1}{2}\ln\det(-\not{\partial}_E^2+M_\mathrm{f}^2) = \tfrac{1}{2}\ln\det\big[(-\partial_E^2+M_\mathrm{f}^2)\,\mathbf{1}_4\big], \tag{E.138}$$

where in the last step $\not{\partial}_E^2 = \partial_E^2$ for a free Dirac operator (the gamma-matrix algebra $\{\gamma^\mu,\gamma^\nu\} = 2\delta^{\mu\nu}$ in Euclidean signature collapses $\not{\partial}_E^2$ to ∂_E^2 times the 4×4 identity; in a gauge background the same step produces $-D_E^2 + \frac{1}{2}\sigma_{\mu\nu}F^{\mu\nu}$, the Seeley–DeWitt input of Section E.3). The trace over the four Dirac indices supplies a factor of 4. Collecting the sign (E.137), the $\frac{1}{2}$ of (E.138), and the Dirac trace, a single Dirac fermion contributes to the one-loop vacuum energy exactly

$$\Gamma_E^{(1),\,\mathrm{ferm}} = -\,4\cdot\tfrac{1}{2}\,\ln\det(-\partial_E^2 + M_\mathrm{f}^2) = -\,4\,\Gamma_E^{(1),\,\text{real scalar}}\big|_{M\to M_\mathrm{f}}, \tag{E.139}$$

i.e. minus four times one real-scalar loop of the same mass: the sign from Grassmann statistics, the four from the on-shell degrees of freedom of a Dirac field. We will use this once, to write the general one-loop master formula; the rest of the section is the scalar case in full detail.

E.4.3 The effective potential as a single momentum integral

We now specialize to ϕ^4 theory, the real scalar with the standard Lagrangian Eq. (6.56),

$$\mathcal{L} = \tfrac{1}{2}(\partial_\mu\phi)(\partial^\mu\phi) - \tfrac{1}{2}m^2\phi^2 - \frac{\lambda}{4!}\phi^4, \tag{E.140}$$

in Euclidean signature $S_E = \int \mathrm{d}^d x\,[\frac{1}{2}(\partial_E\phi)^2 + \frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4]$. The *effective potential* $V_\mathrm{eff}(\phi_c)$ is the effective action evaluated on a constant background, per

unit spacetime volume. Writing $\Omega = \int \mathrm{d}^d x$ for the (regularized, large) volume, a constant ϕ_c has no kinetic energy and

$$\Gamma_E[\phi_c] = \Omega\, V_{\mathrm{eff}}(\phi_c), \qquad V_{\mathrm{eff}}(\phi_c) = U(\phi_c) + \frac{1}{2\Omega}\ln\det \mathcal{O}, \qquad U(\phi_c) = \tfrac{1}{2}m^2\phi_c^2 + \frac{\lambda}{4!}\phi_c^4, \tag{E.141}$$

U being the classical (tree) potential. The second variation of S_E at constant ϕ_c is the local operator

$$\mathcal{O}(x,y) = [-\partial_E^2 + U''(\phi_c)]\,\delta(x-y) = [-\partial_E^2 + M^2(\phi_c)]\,\delta(x-y), \qquad M^2(\phi_c) \equiv m^2 + \tfrac{\lambda}{2}\,\phi_c^2, \tag{E.142}$$

where $U'' = m^2 + \frac{\lambda}{2}\phi_c^2$ defines the background-dependent effective mass $M^2(\phi_c)$. This $M^2(\phi_c)$ is the single most important object in the section: it is the squared frequency of the scalar fluctuation in the presence of the background, and the entire Coleman–Weinberg potential is a function of it.

The operator (E.142) is diagonalized by plane waves $e^{ip\cdot x}$ with eigenvalue $p_E^2 + M^2(\phi_c)$. Its log-determinant is the trace of the logarithm,

$$\ln\det\mathcal{O} = \mathrm{Tr}\ln(-\partial_E^2 + M^2) = \int \mathrm{d}^d x \int \frac{\mathrm{d}^d p}{(2\pi)^d}\,\ln(p_E^2 + M^2) = \Omega \int \frac{\mathrm{d}^d p}{(2\pi)^d}\,\ln(p_E^2 + M^2), \tag{E.143}$$

the spatial integral giving the volume Ω (this is the continuum form of $\sum_n \ln\lambda_n$; the momentum density of states per unit volume is $\mathrm{d}^d p/(2\pi)^d$). The volume cancels against the $1/\Omega$ in (E.141), leaving the one-loop effective potential as a single momentum integral,

$$V_{\mathrm{eff}}(\phi_c) = U(\phi_c) + V_1(\phi_c), \qquad V_1(\phi_c) = \frac{1}{2}\int \frac{\mathrm{d}^d p}{(2\pi)^d}\,\ln(p_E^2 + M^2(\phi_c)). \tag{E.144}$$

This V_1 is the integral the section spec demands we evaluate, and the rest of the present subsection does so by dimensional regularization, anchored in the master integrals of Chapter D.

Wick rotation, for the record. Equation (E.144) is already Euclidean, so no rotation is needed to evaluate it. It is worth recording, however, how it follows from the Lorentzian determinant of Eq. (4.211), because the signs are a frequent source of error. In Lorentzian signature the second-variation operator is $-(\Box + M^2)$, with eigenvalue $p^2 - M^2$ under $\Box e^{-ip\cdot x} = -p^2 e^{-ip\cdot x}$, and the Lorentzian one-loop potential reads $V_1 = -\frac{i}{2}\int \frac{\mathrm{d}^d p}{(2\pi)^d}\ln(p^2 - M^2 + i\epsilon)$ (the $i\epsilon$ inherited from the Feynman prescription). Rotate $p^0 = ip_E^0$, so $\mathrm{d}^d p = i\,\mathrm{d}^d p_E$ and $p^2 = -p_E^2$; then $\ln(p^2 - M^2 + i\epsilon) \to \ln(-(p_E^2 + M^2)) = \ln(p_E^2 + M^2) +$ (field-independent imaginary constant), and

$$-\frac{i}{2}\int \frac{\mathrm{d}^d p}{(2\pi)^d}\ln(p^2 - M^2 + i\epsilon) \xrightarrow{\text{Wick}} -\frac{i}{2}\cdot i\int \frac{\mathrm{d}^d p_E}{(2\pi)^d}\ln(p_E^2 + M^2) = \frac{1}{2}\int \frac{\mathrm{d}^d p_E}{(2\pi)^d}\ln(p_E^2 + M^2), \tag{E.145}$$

reproducing (E.144) with the correct sign. The discarded constant shifts the (unobservable) zero of the vacuum energy.

Evaluating the log integral. The integrand $\ln(p_E^2+M^2)$ is not directly one of the algebraic master integrals of Chapter D; the standard device is to differentiate with respect to M^2, reducing the logarithm to a power, integrate that against the master formula, and integrate back in M^2. Differentiating (E.144),

$$\frac{\partial V_1}{\partial M^2} = \frac{1}{2}\int \frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{p_E^2+M^2}. \tag{E.146}$$

The Euclidean master integral (Section D.4, eq. (D.77), here the n-fold massive tadpole)

$$\int \frac{\mathrm{d}^d p}{(2\pi)^d}\,\frac{1}{(p_E^2+M^2)^n} = \frac{1}{(4\pi)^{d/2}}\,\frac{\Gamma(n-\frac{d}{2})}{\Gamma(n)}\,(M^2)^{\frac{d}{2}-n} \tag{E.147}$$

at $n=1$ gives

$$\frac{\partial V_1}{\partial M^2} = \frac{1}{2}\,\frac{1}{(4\pi)^{d/2}}\,\Gamma\!\left(1-\tfrac{d}{2}\right)(M^2)^{\frac{d}{2}-1}. \tag{E.148}$$

Integrating in M^2 (the constant of integration is field-independent and absorbed into the vacuum energy),

$$V_1 = \frac{1}{2}\,\frac{1}{(4\pi)^{d/2}}\,\Gamma\!\left(1-\tfrac{d}{2}\right)\frac{(M^2)^{d/2}}{d/2} = -\frac{1}{2}\,\frac{1}{(4\pi)^{d/2}}\,\Gamma\!\left(-\tfrac{d}{2}\right)(M^2)^{d/2}, \tag{E.149}$$

where the last equality used $\Gamma(1-\frac{d}{2})/(d/2) = -\Gamma(-\frac{d}{2})$, which follows from the recursion $\Gamma(z+1)=z\Gamma(z)$ at $z=-\frac{d}{2}$: $\Gamma(1-\frac{d}{2}) = (-\frac{d}{2})\Gamma(-\frac{d}{2})$, hence $\Gamma(1-\frac{d}{2})/(d/2) = -\Gamma(-\frac{d}{2})$. Equation (E.149) is the one-loop effective potential in d dimensions in closed form. All the physics is in expanding it about $d=4$.

The $d\to 4$ expansion. Set $d=4-2\varepsilon$ and reinstate the renormalization scale through the replacement $\int \frac{\mathrm{d}^d p}{(2\pi)^d} \to \mu^{2\varepsilon}\int \frac{\mathrm{d}^d p}{(2\pi)^d}$ that keeps V_1 at mass dimension four (the coupling λ carries the compensating $\mu^{2\varepsilon}$ in $\overline{\mathrm{MS}}$; see Chapter 22). Then $V_1 = -\frac{1}{2}\,\mu^{2\varepsilon}(4\pi)^{-d/2}\Gamma(-\frac{d}{2})(M^2)^{d/2}$, and we expand each factor to the order that survives the $1/\varepsilon$ pole. Write $L \equiv \ln(4\pi\mu^2/M^2)$. The mass and coupling powers give

$$(M^2)^{d/2} = (M^2)^{2-\varepsilon} = M^4\,e^{-\varepsilon \ln M^2},\quad (4\pi)^{-d/2} = (4\pi)^{-2}\,e^{\varepsilon \ln 4\pi},\quad \mu^{2\varepsilon} = e^{\varepsilon \ln \mu^2}, \tag{E.150}$$

so that the combined ε-dependent prefactor is $(16\pi^2)^{-1}M^4\,e^{\varepsilon L} = (16\pi^2)^{-1}M^4(1+\varepsilon L + O(\varepsilon^2))$. The Gamma function is expanded from the recursion $\Gamma(-2+\varepsilon) =$

$\dfrac{\Gamma(1+\varepsilon)}{\varepsilon(\varepsilon-1)(\varepsilon-2)}$, derived by applying $\Gamma(z)=\Gamma(z+1)/z$ three times:

$$\Gamma(-2+\varepsilon)=\frac{\Gamma(1+\varepsilon)}{\varepsilon\,(\varepsilon-1)(\varepsilon-2)}=\frac{1}{2\varepsilon}(1-\gamma_E\,\varepsilon)\Big(1+\tfrac{3}{2}\varepsilon\Big)+O(\varepsilon)=\frac{1}{2\varepsilon}+\frac{1}{2}\Big(\tfrac{3}{2}-\gamma_E\Big)+O(\varepsilon), \tag{E.151}$$

where we used $\Gamma(1+\varepsilon)=1-\gamma_E\varepsilon+O(\varepsilon^2)$ and $\dfrac{1}{(\varepsilon-1)(\varepsilon-2)}=\dfrac{1}{2-3\varepsilon+\varepsilon^2}=\dfrac{1}{2}(1+\tfrac{3}{2}\varepsilon+O(\varepsilon^2))$. Multiplying the prefactor and (E.151),

$$\begin{aligned}V_1&=-\frac{M^4}{2\,(16\pi^2)}(1+\varepsilon L)\Big[\frac{1}{2\varepsilon}+\frac{1}{2}\Big(\tfrac{3}{2}-\gamma_E\Big)\Big]+O(\varepsilon)\\&=-\frac{M^4}{32\pi^2}\Big[\frac{1}{2\varepsilon}+\frac{1}{2}\Big(L+\tfrac{3}{2}-\gamma_E\Big)\Big]=-\frac{M^4}{64\pi^2}\Big[\frac{1}{\varepsilon}+\ln\frac{4\pi\mu^2}{M^2}-\gamma_E+\frac{3}{2}\Big].\end{aligned} \tag{E.152}$$

Group the universal $\overline{\text{MS}}$ combination $\dfrac{1}{\bar\varepsilon}\equiv\dfrac{1}{\varepsilon}-\gamma_E+\ln 4\pi$ and rewrite $\ln(4\pi\mu^2/M^2)-\gamma_E=\dfrac{1}{\bar\varepsilon}-\dfrac{1}{\varepsilon}+\ln(\mu^2/M^2)=\ldots$; more directly, collecting the μ-dependent logarithm separately,

$$\boxed{V_1(\phi_c)=\frac{M^4(\phi_c)}{64\pi^2}\Big[\ln\frac{M^2(\phi_c)}{\mu^2}-\frac{3}{2}\Big]-\frac{M^4(\phi_c)}{64\pi^2}\,\frac{1}{\bar\varepsilon}\,,\qquad \frac{1}{\bar\varepsilon}=\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi.} \tag{E.153}$$

The first bracket is the finite, μ-dependent physics; the second term is the pure pole, local in ϕ_c because $M^4=(m^2+\frac{\lambda}{2}\phi_c^2)^2$ is a polynomial of degree four in the field, hence absorbable into the same operators ($\mathbf{1}$, ϕ_c^2, ϕ_c^4) that already appear in the action. That is the content of renormalizability, made concrete in the next subsection.

Caution

The closed form (E.153) carries $\ln M^2(\phi_c)$, and $M^2(\phi_c)=m^2+\frac{\lambda}{2}\phi_c^2$ can vanish or go negative. For $m^2>0$ and small ϕ_c it is positive and the logarithm is real. When $M^2(\phi_c)<0$ — as happens for a tachyonic tree mass, the case of interest for spontaneous symmetry breaking — the logarithm develops an imaginary part, $\ln M^2=\ln|M^2|+i\pi$. This imaginary part is not a mistake: it is the decay rate of an unstable (non-convex) field configuration, the statement that a homogeneous field sitting on a concave part of the potential is not a stationary state. The true effective potential, as the Legendre transform of a convex W, is convex (Section 4.5); the one-loop formula reconstructs the convex envelope only after the Maxwell construction. We use (E.153) in its region of validity, $M^2(\phi_c)>0$, and flag the imaginary part where it appears.

E.4.4 Renormalization: counterterms and conditions

The bare action carries bare parameters; we split them into renormalized parameters plus counterterms in the standard way (Chapter 22). At one loop the field strength is not renormalized in ϕ^4 (the self-energy is momentum-independent at this order, so $\delta_Z = O(\lambda^2)$), and we need only a mass and a coupling counterterm, together with a field-independent vacuum-energy counterterm δ_Λ:

$$V_{\rm ct}(\phi_c) = \delta_\Lambda + \tfrac{1}{2}\,\delta m^2\,\phi_c^2 + \frac{\delta\lambda}{4!}\,\phi_c^4. \tag{E.154}$$

The renormalized effective potential is $V_{\rm eff} = U + V_1 + V_{\rm ct}$. Expand the pole term of (E.153) in powers of the field using $M^4 = m^4 + m^2\lambda\,\phi_c^2 + \frac{\lambda^2}{4}\phi_c^4$:

$$-\frac{M^4}{64\pi^2\,\bar\varepsilon} = -\frac{1}{64\pi^2\,\bar\varepsilon}\Big(m^4 + m^2\lambda\,\phi_c^2 + \tfrac{\lambda^2}{4}\phi_c^4\Big). \tag{E.155}$$

The three pieces are exactly cancellable by the three counterterms in (E.154).

Minimal subtraction ($\overline{\rm MS}$). Choosing the counterterms to remove precisely the $1/\bar\varepsilon$ poles,

$$\delta_\Lambda = \frac{m^4}{64\pi^2\,\bar\varepsilon}, \qquad \delta m^2 = \frac{\lambda\,m^2}{32\pi^2\,\bar\varepsilon}, \qquad \delta\lambda = \frac{3\lambda^2}{32\pi^2\,\bar\varepsilon}, \tag{E.156}$$

(the factors follow by matching $\frac{1}{2}\delta m^2\phi_c^2$ to $+\frac{m^2\lambda}{64\pi^2\bar\varepsilon}\phi_c^2$ and $\frac{\delta\lambda}{4!}\phi_c^4$ to $+\frac{\lambda^2}{256\pi^2\bar\varepsilon}\phi_c^4$, i.e. $\delta\lambda = \frac{4!\,\lambda^2}{256\pi^2\bar\varepsilon} = \frac{3\lambda^2}{32\pi^2\bar\varepsilon}$), yields the renormalized Coleman–Weinberg potential

$$\boxed{V_{\rm eff}(\phi_c) = \frac{1}{2}m^2\phi_c^2 + \frac{\lambda}{4!}\phi_c^4 + \frac{M^4(\phi_c)}{64\pi^2}\Big[\ln\frac{M^2(\phi_c)}{\mu^2} - \frac{3}{2}\Big], \qquad M^2(\phi_c) = m^2 + \frac{\lambda}{2}\phi_c^2.} \tag{E.157}$$

This is the central result of the section: the one-loop effective potential of ϕ^4 theory, regularized by dimensional regularization, renormalized in $\overline{\rm MS}$, in the closed "$\ln - \frac{3}{2}$" form. The $-\frac{3}{2}$ is a scheme-dependent constant (a $-\frac{5}{6}$ would arise in a different but equally common convention for the master integral; ours is the standard $\overline{\rm MS}$ value), tied to the finite part of $\Gamma(-2+\varepsilon)$ in (E.151).

Caution

The pole coefficient of $\delta\lambda$ in (E.156) already contains the one-loop beta function. With $\lambda_0 = \mu^{2\varepsilon}(\lambda+\delta\lambda)$ and $\delta\lambda = a_1/\bar\varepsilon$, $a_1 = \frac{3\lambda^2}{32\pi^2}$, μ-independence of λ_0 forces $\beta_\lambda = 2a_1 = \dfrac{3\lambda^2}{16\pi^2}$, exactly the result of Chapter 24. The effective potential is not an independent computation of the beta function so much as a different bookkeeping of the same divergence; that they agree

is the first of two consistency checks below.

Physical (zero-field) renormalization conditions. Minimal subtraction leaves the finite parts of the counterterms fixed by fiat. An alternative, often more physical, scheme fixes them by conditions on measurable quantities. For $m^2 > 0$ the natural conditions are imposed at zero field: the curvature of V_{eff} at the origin is the physical mass, and its quartic derivative there is the physical coupling,

$$\left.\frac{\mathrm{d}^2 V_{\text{eff}}}{\mathrm{d}\phi_c^2}\right|_{\phi_c=0} = m^2, \qquad \left.\frac{\mathrm{d}^4 V_{\text{eff}}}{\mathrm{d}\phi_c^4}\right|_{\phi_c=0} = \lambda. \tag{E.158}$$

Theorem E.24 solves these for $\delta m^2, \delta\lambda$ in full, exhibiting both the pole and finite parts and confirming that the pole parts coincide with (E.156). The two schemes differ only by a finite, μ-dependent reparametrization of (m^2, λ) — the freedom that the renormalization group organizes.

Worked Example E.24 (Counterterms from the physical conditions)**.** We impose (E.158) on $V_{\text{eff}} = U + V_1 + V_{\text{ct}}$ with V_1 in the regularized form (E.152) and solve for the counterterms. Write $g(\phi_c) \equiv M^4[\ln(M^2/\mu^2) - \frac{3}{2}]$, so that the finite part of V_1 is $g/(64\pi^2)$, and keep the pole separately. Differentiate g using $M^2 = m^2 + \frac{\lambda}{2}\phi_c^2$, $\mathrm{d}M^2/\mathrm{d}\phi_c = \lambda\phi_c$, and $L \equiv \ln(M^2/\mu^2)$. A first derivative gives

$$g' = 2M^2\lambda\phi_c\,(L-1), \tag{E.159}$$

having used $\mathrm{d}M^4/\mathrm{d}\phi_c = 2M^2\lambda\phi_c$ and $\mathrm{d}L/\mathrm{d}\phi_c = \lambda\phi_c/M^2$, so the $+\frac{3}{2}$ combines with the $+1$ from differentiating the logarithm into -1 inside the bracket. In particular $g'(0) = 0$: there is no one-loop tadpole at the symmetric point, as the unbroken $\mathbb{Z}_2$ requires. Differentiating again,

$$g'' = 2\lambda\big[M^2(L-1) + \lambda\phi_c^2\,L\big], \qquad g''(0) = 2\lambda\,m^2\big[\ln(m^2/\mu^2) - 1\big]. \tag{E.160}$$

Two further derivatives, each time using $\mathrm{d}M^2/\mathrm{d}\phi_c = \lambda\phi_c$ and $\mathrm{d}L/\mathrm{d}\phi_c = \lambda\phi_c/M^2$, give $g''' = 2\lambda[3\lambda\phi_c L + \lambda^2\phi_c^3/M^2]$ (so $g'''(0) = 0$) and

$$g'''' = 2\lambda\Big[3\lambda L + \frac{6\lambda^2\phi_c^2}{M^2} - \frac{\lambda^3\phi_c^4}{M^4}\Big], \qquad g''''(0) = 6\lambda^2\ln(m^2/\mu^2). \tag{E.161}$$

Hence the one-loop potential contributes to the curvatures at the origin

$$V_1''(0) = \frac{g''(0)}{64\pi^2} - \frac{(M^4)''(0)}{64\pi^2\bar{\varepsilon}} = \frac{\lambda m^2}{32\pi^2}\big[\ln(m^2/\mu^2) - 1\big] - \frac{\lambda m^2}{32\pi^2\bar{\varepsilon}}, \tag{E.162}$$

where $(M^4)''(0) = 2m^2\lambda$, and

$$V_1''''(0) = \frac{g''''(0)}{64\pi^2} - \frac{(M^4)''''(0)}{64\pi^2\bar{\varepsilon}} = \frac{3\lambda^2}{32\pi^2}\ln(m^2/\mu^2) - \frac{3\lambda^2}{32\pi^2\bar{\varepsilon}}, \tag{E.163}$$

using $(M^4)''''(0) = 6\lambda^2$. Since $U''(0) = m^2$, $U''''(0) = \lambda$, the conditions (E.158) read $m^2 + \delta m^2 + V_1''(0) = m^2$ and $\lambda + \delta\lambda + V_1''''(0) = \lambda$, so

$$\delta m^2 = \frac{\lambda m^2}{32\pi^2\bar{\varepsilon}} - \frac{\lambda m^2}{32\pi^2}\big[\ln(m^2/\mu^2) - 1\big], \qquad \delta\lambda = \frac{3\lambda^2}{32\pi^2\bar{\varepsilon}} - \frac{3\lambda^2}{32\pi^2}\ln(m^2/\mu^2). \tag{E.164}$$

The pole parts reproduce (E.156) exactly; the finite parts implement the physical conditions and depend on μ in just the way that compensates the explicit μ in (E.157), the cancellation made systematic below. This is the promised explicit counterterm subtraction: every constant is fixed, nothing is left to the reader.

E.4.5 Scale dependence and the renormalization group

The renormalized potential (E.157) depends on the arbitrary scale μ, both explicitly through $\ln(M^2/\mu^2)$ and implicitly through the running parameters $\lambda = \lambda(\mu)$, $m^2 = m^2(\mu)$. A physical quantity — the value of V_{eff} at a given ϕ_c — cannot depend on the bookkeeping scale, so V_{eff} obeys a Callan–Symanzik equation,

$$\Big[\mu\frac{\partial}{\partial\mu} + \beta_\lambda\frac{\partial}{\partial\lambda} + \beta_{m^2}\frac{\partial}{\partial m^2} - \gamma_\phi\,\phi_c\frac{\partial}{\partial\phi_c}\Big]V_{\text{eff}} = 0, \tag{E.165}$$

with $\beta_\lambda = \mu\,\mathrm{d}\lambda/\mathrm{d}\mu$, $\beta_{m^2} = \mu\,\mathrm{d}m^2/\mathrm{d}\mu$, and $\gamma_\phi = -\frac{1}{2}\mu\,\mathrm{d}\ln Z_\phi/\mathrm{d}\mu$ the field anomalous dimension. At one loop $\gamma_\phi = O(\lambda^2)$ vanishes (the $\phi_c\partial_{\phi_c}$ term first acts at two loops), so (E.165) becomes a clean consistency check on β_λ and β_{m^2}.

Apply $\mu\partial_\mu$ to (E.157) at fixed parameters. Only the explicit logarithm responds, $\mu\partial_\mu\ln(M^2/\mu^2) = -2$, so

$$\mu\frac{\partial V_{\text{eff}}}{\partial\mu}\Big|_{\lambda,m^2} = \frac{M^4}{64\pi^2}\cdot(-2) = -\frac{M^4}{32\pi^2} = -\frac{1}{32\pi^2}\Big(m^4 + m^2\lambda\,\phi_c^2 + \tfrac{\lambda^2}{4}\phi_c^4\Big). \tag{E.166}$$

This must be cancelled by the parameter running acting on the tree potential $U = \frac{1}{2}m^2\phi_c^2 + \frac{\lambda}{4!}\phi_c^4$ (the running acting on V_1 is higher order). Matching powers of ϕ_c:

- ϕ_c^4: $\beta_\lambda\,\partial_\lambda U = \beta_\lambda\,\frac{\phi_c^4}{4!}$ must equal $+\frac{\lambda^2}{4}\frac{\phi_c^4}{32\pi^2} = \frac{\lambda^2\phi_c^4}{128\pi^2}$, giving $\beta_\lambda = \dfrac{4!\,\lambda^2}{4\cdot 32\pi^2} = \dfrac{3\lambda^2}{16\pi^2}$;
- ϕ_c^2: $\beta_{m^2}\,\partial_{m^2}U = \beta_{m^2}\,\frac{1}{2}\phi_c^2$ must equal $\frac{m^2\lambda}{32\pi^2}\phi_c^2$, giving $\beta_{m^2} = \dfrac{\lambda m^2}{16\pi^2}$;
- ϕ_c^0: the constant $m^4/(32\pi^2)$ is cancelled by the running of the vacuum-energy counterterm δ_Λ, which is field-independent and physically inert.

The ϕ_c^4 matching reproduces the one-loop beta function $\beta_\lambda = 3\lambda^2/16\pi^2$ of Chapter 24 — the carried number of ϕ^4 theory — now obtained a third way, from the μ-dependence of the effective potential. The ϕ_c^2 matching yields the one-loop mass anomalous running $\beta_{m^2} = \lambda m^2/16\pi^2$ (equivalently $\gamma_{m^2} \equiv \beta_{m^2}/m^2 = \lambda/16\pi^2$), consistent with the mass renormalization of Chapter 22. That the single function (E.157) encodes both is the strongest internal check available at this order: the same pole that fixed the counterterms (E.156) fixes the running that makes the potential μ-independent.

Physical Insight

The effective potential is the renormalization group made visible. Reading (E.157) at ϕ_c such that $M^2(\phi_c) = \mu^2$ kills the logarithm and leaves $V_{\text{eff}} = U - \frac{3}{128\pi^2}M^4$, the "tree potential evaluated at the running couplings of the scale set by the field." Choosing μ to track the field is the leading-log improvement: the large logarithms $\ln(M^2/\mu^2)$ that spoil naive perturbation theory when ϕ_c is far from μ are resummed by running $\lambda(\mu)$ down to $\mu \sim M(\phi_c)$. The Coleman–Weinberg potential thus carries the whole one-loop RG of the model in a single closed expression.

E.4.6 The general one-loop master formula and the massless limit

Assembling the bosonic result (E.157) with the fermionic sign and degeneracy (E.139), and allowing several fields of various spins with field-dependent masses $M_i(\phi_c)$, the one-loop effective potential of any renormalizable theory is the supertrace

$$\boxed{V_1(\phi_c) = \frac{1}{64\pi^2}\sum_i (-1)^{2s_i}\, n_i\, M_i^4(\phi_c)\Big[\ln\frac{M_i^2(\phi_c)}{\mu^2} - c_i\Big]}\,, \tag{E.167}$$

with n_i the number of (real, on-shell) degrees of freedom of species i, $(-1)^{2s_i} = +1$ for bosons and -1 for fermions, and $c_i = \frac{3}{2}$ for spin 0 and spin $\frac{1}{2}$, $c_i = \frac{5}{6}$ for a massive gauge boson in the standard convention (the constant tracks the spin content of the determinant and the gauge-fixing sector; we computed the scalar case and quoted the gauge value, derivable by the identical heat-kernel route on $-D^2 + \dots$ of Section E.3—the 5/6 rather than 3/2 reflecting the $\overline{\text{MS}}$ (Landau-gauge) disentanglement of the transverse and would-be-Goldstone/ghost sectors of a massive vector, as spelled out in the broken-phase application Eq. (29.142)). For a single Dirac fermion of mass $M_{\text{f}} = y\phi_c$ this gives $V_1 = -\frac{4}{64\pi^2}M_{\text{f}}^4[\ln(M_{\text{f}}^2/\mu^2) - \frac{3}{2}]$, the minus sign and the four being exactly (E.139). The relative sign is the engine of the supersymmetric cancellation: when bosons and fermions are degenerate ($M_{\text{b}} = M_{\text{f}}$) and balanced ($n_{\text{b}} = n_{\text{f}}$), the quartic-times-log pieces cancel in (E.167) and the vacuum energy is protected.

We close by carrying ϕ^4 theory to the phenomenon that made the Coleman–Weinberg potential famous: *dimensional transmutation*, the radiative generation of a mass scale in a classically scale-invariant theory.

Worked Example E.25 (Massless ϕ^4 and dimensional transmutation)**.** Set $m^2 = 0$ in (E.157), so the classical theory has a single dimensionless coupling λ and no scale. Then $M^2(\phi_c) = \frac{\lambda}{2}\phi_c^2$ and

$$V_{\text{eff}}(\phi_c) = \frac{\lambda}{4!}\phi_c^4 + \frac{(\lambda\phi_c^2/2)^2}{64\pi^2}\Big[\ln\frac{\lambda\phi_c^2/2}{\mu^2} - \frac{3}{2}\Big] = \frac{\lambda}{4!}\phi_c^4 + \frac{\lambda^2\phi_c^4}{256\pi^2}\Big[\ln\frac{\lambda\phi_c^2}{2\mu^2} - \frac{3}{2}\Big]. \tag{E.168}$$

Classically V is a pure quartic with its only stationary point at the origin. The one-loop term proportional to $\phi_c^4 \ln\phi_c^2$ changes this: the logarithm drives V_{eff} negative for a range of ϕ_c before the positive quartic recaptures it, producing a pair of minima at $\phi_c = \pm v$ with $v \neq 0$, set by $V'_{\text{eff}}(v) = 0$. Differentiating (E.168) and using $\mathrm{d}(\phi_c^4 \ln\phi_c^2)/\mathrm{d}\phi_c = \phi_c^3(4\ln\phi_c^2 + 2)$,

$$V'_{\text{eff}}(\phi_c) = \frac{\lambda}{6}\phi_c^3 + \frac{\lambda^2\phi_c^3}{256\pi^2}\Big[4\ln\frac{\lambda\phi_c^2}{2\mu^2} - 6 + 2\Big] = \phi_c^3\Big\{\frac{\lambda}{6} + \frac{\lambda^2}{64\pi^2}\Big[\ln\frac{\lambda\phi_c^2}{2\mu^2} - 1\Big]\Big\}, \tag{E.169}$$

whose nontrivial zero $\phi_c = v$ satisfies $\ln(\lambda v^2/2\mu^2) = 1 - \frac{32\pi^2}{3\lambda}$. This is dimensional transmutation: the dimensionless input λ has been traded for a dimensionful vacuum expectation value v,

$$v^2 = \frac{2\mu^2}{\lambda}\,\exp\Big(1 - \frac{32\pi^2}{3\lambda}\Big), \tag{E.170}$$

a scale generated entirely by quantum effects from a scale-free classical theory. The minimum lies where the one-loop term is comparable in magnitude to the tree term, i.e. where $\lambda\ln(\lambda v^2/2\mu^2) \sim O(1)$ relative to the loop counting; for pure scalar ϕ^4 this places v at the boundary of perturbative reliability, and the honest statement (Coleman and Weinberg's own) is that the pure-scalar example is illustrative rather than quantitative. The trustworthy realization adds a gauge field so that the analogous balance occurs at weak coupling — scalar electrodynamics, the original setting — but the mechanism is the one shown here. Figure E.3 plots both the massive potential (E.157) and the massless potential (E.168), the latter exhibiting the radiatively generated double well.

The effective potential (E.157) is now complete: derived from the Gaussian fluctuation integral, regularized by the determinant machinery of the previous section, renormalized by explicit counterterms, and shown consistent with the renormalization group of Chapter 24. It feeds directly into the spontaneous-symmetry-breaking analysis of Chapter 29, where the same formula — now with $m^2 < 0$ at tree level — selects the broken vacuum and dresses Goldstone's theorem with radiative corrections. Yet the entire derivation rested on a step we wrote down without scrutiny: that the measure $\mathcal{D}\phi$ is translation-invariant

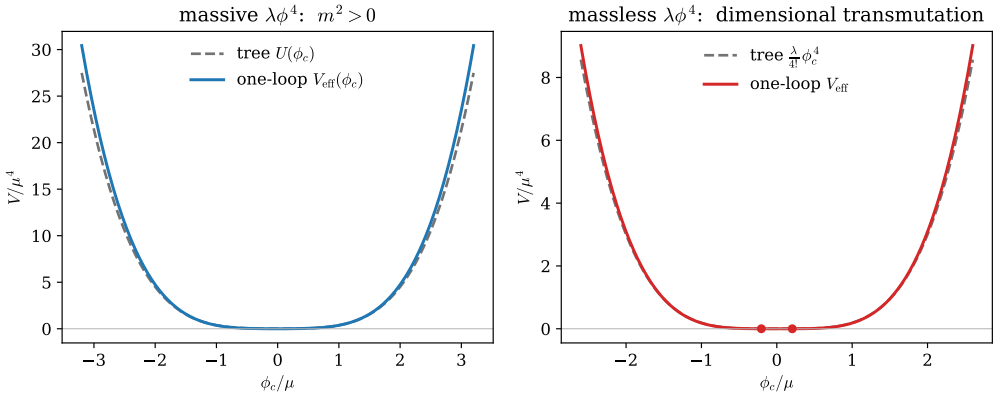


Figure E.3: The one-loop effective potential of ϕ^4 theory, Eq. (E.157). *Left:* the massive case $m^2 > 0$; the one-loop curve (solid) is a smooth deformation of the convex tree potential (dashed) near the origin. *Right:* the massless case (E.168); the classical pure quartic (dashed) has its minimum at the origin, while the one-loop term curves the potential into a double well with radiatively generated minima (dots) at $\phi_c = \pm v$ — dimensional transmutation, (E.170). A deliberately large coupling is used so the effect is visible; for pure scalar ϕ^4 the minimum sits at the edge of perturbative control (see Theorem E.25).

under the background split (E.129), $\mathcal{D}\phi = \mathcal{D}\eta$, and that the field-independent normalization $\mathcal{N}$ in (E.133) could be discarded with impunity. Whether $\mathcal{D}\phi$ deserves that trust — and what it costs when a field redefinition is nonlinear — is precisely what the next section confronts.

E.5 The Measure: Definition, Field Redefinitions, and Jacobians

Every computation so far has written $\int \mathcal{D}\phi$ as though the symbol were self-explanatory—the formal functional measure was introduced exactly that way in Section 4.2 and used unexamined ever since—and in the one-loop effective action of Section E.4 we leaned on a tacit and unexamined assumption: that the measure is *trivial*, an undistinguished product $\prod_x \mathrm{d}\phi(x)$ that is moreover *invariant* under the shift $\phi \to \phi_{\rm cl} + \eta$ separating the saddle from its fluctuation. Both properties were used—the shift invariance to write $\int \mathcal{D}\phi = \int \mathcal{D}\eta$, the triviality to drop a would-be Jacobian and read the one-loop correction off as nothing but $(\det K)^{-1/2}$. Neither property is automatic, and neither survives contact with a nonlinear change of field variables or with a chiral rotation of a fermion. This section pins down what $\mathcal{D}\phi$ *is* as a continuum object, computes the Jacobian that a field redefinition produces, and isolates the one place where

that Jacobian fails to be unity even in dimensional regularization: the fermionic measure under a γ^5 rotation. The payoff is not a single theorem but a piece of infrastructure—the exact groundwork that the gauge-fixing chapters (Chapters 13 and 26) and the anomaly chapter (Chapter 36) consume. We build the setup here; we deliberately do *not* compute their results.

E.5.1 Field space as a Riemannian manifold and the definition of $\mathcal{D}\phi$

The configurations of a theory with fields $\phi = (\phi^1, \dots, \phi^n)$ on a d-dimensional spacetime form an infinite-dimensional space $\mathcal{F}$, each point of which is one entire field configuration $x \mapsto \phi(x)$. A measure on $\mathcal{F}$ is not handed to us for free: there is no Lebesgue measure on an infinite-dimensional vector space. The construction that does work, and that makes every later manipulation honest, is to treat $\mathcal{F}$ as a Riemannian manifold and take $\mathcal{D}\phi$ to be its *Riemannian volume element*. For that we need a metric—an inner product on the tangent space at each configuration, i.e. on infinitesimal deformations $\delta\phi$.

The natural metric is the one inherited from the spacetime integral that already appears in every kinetic term. A deformation $\delta\phi(x)$ is assigned the squared length

$$\|\delta\phi\|^2 \;=\; \int \mathrm{d}^d x\, \big(\delta\phi(x)\big)^2 \;=\; \int \mathrm{d}^d x\, \delta_{ab}\, \delta\phi^a(x)\, \delta\phi^b(x). \tag{E.171}$$

This is the *DeWitt metric* in its simplest, flat-target form. Two features are decisive. First, it is *ultralocal*: the squared length is a single spacetime integral of a quantity built at one point x, with no derivatives of $\delta\phi$ and no coupling between $\delta\phi(x)$ and $\delta\phi(y)$ for $x \neq y$. In the language of a metric tensor on field space,

$$\|\delta\phi\|^2 = \int \mathrm{d}^d x\, \mathrm{d}^d y\; G_{ab}(x,y)\, \delta\phi^a(x)\, \delta\phi^b(y), \qquad G_{ab}(x,y) = \delta_{ab}\, \delta^d(x-y), \tag{E.172}$$

the field-space metric is diagonal in spacetime, proportional to $\delta^d(x-y)$. Second, for a theory whose fields are linear coordinates (a free scalar, or any polynomial-interacting scalar in Cartesian field variables) the metric (E.171) is *field-independent*: G_{ab} does not depend on ϕ. A field-independent metric has a constant volume element, and the Riemannian volume reduces to the flat product measure.

Definition E.26 (The continuum measure)**.** Let $\mathcal{F}$ be field space equipped with the ultralocal DeWitt metric (E.172), possibly with a configuration-dependent target metric $G_{ab}(\phi(x))$ in place of δ_{ab}. The *path-integral measure* is the associated Riemannian volume element,

$$\mathcal{D}\phi \;=\; \mathcal{N} \prod_x \sqrt{\det G(\phi(x))} \prod_a \mathrm{d}\phi^a(x), \tag{E.173}$$

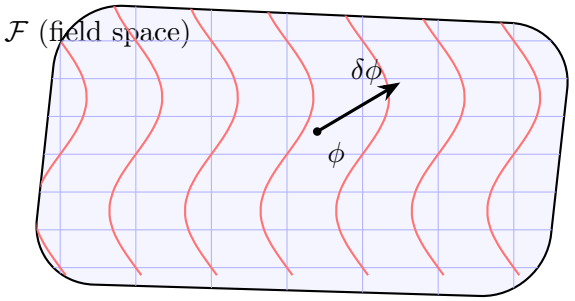


Figure E.4: Field space $\mathcal{F}$ as an (infinite-dimensional) Riemannian manifold. A field redefinition $\phi = F(\chi)$ is a change of *coordinates* on the same manifold (rectilinear ϕ-grid versus curvilinear χ-grid), not a change of the manifold. The DeWitt volume element $\mathcal{D}\phi$ of Theorem E.26 is coordinate-independent by construction; the apparent Jacobian of Theorem E.27 is the statement that the *naive flat* product measure, ignoring $\sqrt{\det G}$, is not.

where $\det G$ is the determinant of the finite $n \times n$ target metric and $\mathcal{N}$ is a field-independent (hence physically irrelevant) normalization that absorbs the ultralocal $\delta^d(x-y)$ factors. For a flat target, $G_{ab} = \delta_{ab}$, $\det G = 1$, and $\mathcal{D}\phi = \mathcal{N}\prod_x \mathrm{d}\phi(x)$ is the trivial product measure assumed in Section E.4.

Two things must now be made precise before (E.173) can be trusted. The continuum product $\prod_x$ and the determinant $\det G$ taken over an uncountable set of points are formal until they are given meaning by a regulator; and the claim that $\mathcal{N}$ is field-independent must be checked, because that is exactly the assumption that makes the measure split-invariant. Both are settled by discretization.

E.5.2 Discretization, the normalization, and the meaning of $\delta^d(0)$

Replace continuous spacetime by a hypercubic lattice of spacing a with sites x_i, $i = 1, \dots, M$, occupying a large volume $V = Ma^d$. A field configuration becomes the finite collection of numbers $\phi_i^a \equiv \phi^a(x_i)$, and field space becomes the ordinary finite-dimensional vector space $\mathbb{R}^{nM}$, on which Lebesgue measure unambiguously exists. The continuum integral $\int \mathrm{d}^d x$ becomes the Riemann sum $a^d \sum_i$, so the DeWitt length (E.171) discretizes to

$$\|\delta\phi\|^2 \;=\; a^d \sum_i \delta_{ab}\, \delta\phi_i^a\, \delta\phi_i^b \;=\; \sum_{i,a} G^{(\mathrm{lat})}_{(ia)(jb)}\, \delta\phi_i^a\, \delta\phi_j^b, \qquad G^{(\mathrm{lat})}_{(ia)(jb)} = a^d\, \delta_{ij}\, \delta_{ab}. \tag{E.174}$$

The lattice field-space metric is therefore a^d times the identity on the nM-dimensional space. Its determinant is $\det G^{(\mathrm{lat})} = (a^d)^{nM}$, and the Riemannian

volume element is

$$\mathcal{D}\phi \;=\; \sqrt{\det G^{(\mathrm{lat})}}\,\prod_{i,a}\mathrm{d}\phi_i^a \;=\; \prod_{i,a} a^{d/2}\,\mathrm{d}\phi_i^a. \tag{E.175}$$

The factors $a^{d/2}$ are field-independent constants; they are exactly the content of the normalization $\mathcal{N} = \prod_{i,a} a^{d/2}$ in Theorem E.26. Because $\mathcal{N}$ multiplies the partition function as an overall constant, it cancels in every normalized correlator $\langle\,\cdot\,\rangle = Z^{-1}\int\mathcal{D}\phi\,(\,\cdot\,)e^{iS}$ and is discarded. This is the precise sense in which "the measure is trivial": for a flat target the volume element is constant, the constant is irrelevant, and $\mathcal{D}\phi = \prod_x \mathrm{d}\phi(x)$ up to that constant. Crucially, the same field-independence makes the measure invariant under the shift $\phi \to \phi_{\mathrm{cl}} + \eta$, $\mathcal{D}\phi = \mathcal{D}\eta$, since a translation of $\mathbb{R}^{nM}$ has unit Jacobian—this is what licensed $\int\mathcal{D}\phi = \int\mathcal{D}\eta$ in Section E.4.

The discretization also fixes the meaning of the one object that controls every measure Jacobian below: the coincident-point delta function $\delta^d(0)$. On the lattice the Dirac delta is the Kronecker delta divided by the cell volume,

$$\delta^d(x_i - x_j) \;\longleftrightarrow\; \frac{1}{a^d}\,\delta_{ij}, \qquad \text{so} \qquad \delta^d(0) \;\longleftrightarrow\; \frac{1}{a^d} \;=\; \int\frac{\mathrm{d}^d k}{(2\pi)^d}\,1, \tag{E.176}$$

the number of field-space dimensions per unit spacetime volume. As $a \to 0$ this diverges as $a^{-d} \sim \Lambda^d$; it is the most ultraviolet-sensitive quantity in the formalism. The momentum-space form on the right makes the regulator dependence explicit: $\delta^d(0)$ is a scaleless integral. In a hard-cutoff scheme it equals a number of order Λ^d; in *dimensional regularization*, where every scaleless integral vanishes,

$$\delta^d(0) \;=\; \int\frac{\mathrm{d}^d k}{(2\pi)^d}\,1 \;=\; 0 \qquad \text{(dim. reg.)}, \tag{E.177}$$

because there is no scale from which to build a dimensionful answer. Holding onto (E.176) and (E.177) is the entire technical content of the next subsection.

E.5.3 The nonlinear sigma-model measure

When the fields are coordinates on a curved target manifold—a nonlinear sigma model, the prototypical example being the unit sphere S^{n-1} swept out by an $O(n)$ model after the radial mode is frozen—the kinetic term is $\frac{1}{2}\,G_{ab}(\phi)\,\partial_\mu\phi^a\partial^\mu\phi^b$ with a configuration-dependent target metric $G_{ab}(\phi)$. The DeWitt length then inherits that metric,

$$\|\delta\phi\|^2 = \int\mathrm{d}^d x\; G_{ab}(\phi(x))\,\delta\phi^a(x)\,\delta\phi^b(x), \tag{E.178}$$

and the volume element (E.173) acquires a genuinely field-dependent factor. Discretizing exactly as before, the lattice metric is now block-diagonal in sites but

each block is the local target metric: $G^{(\mathrm{lat})}_{(ia)(jb)} = a^d\,\delta_{ij}\,G_{ab}(\phi_i)$, with determinant $\prod_i (a^d)^n \det G(\phi_i)$. The Riemannian volume element is therefore

$$\mathcal{D}\phi = \mathcal{N}\prod_x \sqrt{\det G(\phi(x))}\prod_a \mathrm{d}\phi^a(x), \qquad \mathcal{N} = \prod_x a^{nd/2}, \tag{E.179}$$

which is the ultralocal DeWitt measure for a sigma model: one factor of $\sqrt{\det G(\phi(x))}$ per spacetime point. This factor is *not* an optional decoration. It is what makes the measure a coordinate-independent geometric object on the target, and—as we now show—it is exactly the factor that cancels the Jacobian of a nonlinear field redefinition, so that the combination $\mathcal{D}\phi$ is invariant while the naive $\prod_x \mathrm{d}\phi$ is not.

E.5.4 The Jacobian of a field redefinition

Consider an ultralocal (no-derivative) field redefinition $\phi \to \chi$ defined pointwise by an invertible map F,

$$\phi^a(x) = F^a\big(\chi(x)\big), \tag{E.180}$$

the same function F at every spacetime point. Such redefinitions are constantly performed: rescaling a field by its wavefunction renormalization, passing to the radial–angular variables of a sigma model, "integrating in" an auxiliary field, or eliminating a redundant operator by absorbing it into a shift of another field. The question is what the measure does.

Theorem E.27 (Jacobian of an ultralocal field redefinition)**.** *Under* (E.180)*, the naive flat product measure transforms as*

$$\prod_x \mathrm{d}\phi(x) \;=\; \Big(\prod_x \det F'(\chi(x))\Big)\prod_x \mathrm{d}\chi(x) \;=\; \exp\Big[\delta^d(0)\int \mathrm{d}^d x\ \operatorname{tr}\ln F'(\chi(x))\Big]\prod_x \mathrm{d}\chi(x), \tag{E.181}$$

where $F'^{\,a}{}_b = \partial F^a/\partial\chi^b$ *is the finite* $n\times n$ *Jacobian matrix of* F *and* $\operatorname{tr}\ln = \ln\det$ *acts on the internal indices. In dimensional regularization* $\delta^d(0) = 0$ *by* (E.177)*, so the Jacobian is unity and the redefinition is "free." The DeWitt measure* (E.179) *is invariant in any scheme: the factor* $\prod_x \det F'$ *is exactly cancelled by the transformation of* $\sqrt{\det G}$*.*

Proof. Work on the lattice, where everything is finite-dimensional and Lebesgue's change-of-variables theorem applies literally. The map (E.180) acts site by site: $\phi^a_i = F^a(\chi_i)$, so the full functional Jacobian matrix $\partial\phi^a_i/\partial\chi^b_j$ is *block-diagonal* across sites,

$$\frac{\partial\phi^a_i}{\partial\chi^b_j} \;=\; \delta_{ij}\,F'^{\,a}{}_b(\chi_i), \tag{E.182}$$

because the field at site i depends only on χ at the same site. The determinant of a block-diagonal matrix is the product of the block determinants, one per site:

$$\det\Big(\frac{\partial\phi_i}{\partial\chi_j}\Big) \;=\; \prod_i \det F'(\chi_i), \qquad \prod_{i,a} \mathrm{d}\phi_i^a = \Big(\prod_i \det F'(\chi_i)\Big)\prod_{i,a}\mathrm{d}\chi_i^a. \tag{E.183}$$

This is the lattice statement of the first equality in (E.181). To exhibit the $\delta^d(0)$ structure, exponentiate the product using $\det = \exp\operatorname{tr}\ln$:

$$\prod_i \det F'(\chi_i) = \exp\Big[\sum_i \operatorname{tr}\ln F'(\chi_i)\Big] = \exp\Big[\frac{1}{a^d}\sum_i a^d\,\operatorname{tr}\ln F'(\chi_i)\Big] \xrightarrow{a\to 0} \exp\Big[\frac{1}{a^d}\int \mathrm{d}^d x\,\operatorname{tr}\ln F'(\chi \tag{E.184}$$

The factor a^{-d} multiplying the integral is precisely the regularized $\delta^d(0)$ of (E.176); restoring continuum notation gives the middle expression in (E.181). In a scheme that sends $\delta^d(0)\to 0$—dimensional regularization—the exponent vanishes identically and the Jacobian is 1.

For the covariant statement, note that under (E.180) the target metric pulls back as $\widetilde{G}_{cd}(\chi) = G_{ab}(F(\chi))\,F'^{\,a}{}_c\,F'^{\,b}{}_d$, so $\det\widetilde{G} = (\det F')^2\det G$ and $\sqrt{\det\widetilde{G}} = |\det F'|\,\sqrt{\det G}$. Therefore

$$\prod_x \sqrt{\det\widetilde{G}(\chi(x))}\prod_a \mathrm{d}\chi^a(x) = \prod_x |\det F'|\,\sqrt{\det G}\prod_a \mathrm{d}\chi^a(x) = \prod_x \sqrt{\det G(\phi(x))}\prod_a \mathrm{d}\phi^a(x), \tag{E.185}$$

the last step undoing (E.183). The DeWitt volume element is the same whether expressed in ϕ or χ coordinates—it is the geometric volume of Fig. E.4, independent of the chart. □

The theorem disentangles three statements that are routinely conflated. (i) The naive flat measure $\prod_x \mathrm{d}\phi$ is *not* invariant under a nonlinear redefinition; it picks up $\exp[\delta^d(0)\int\operatorname{tr}\ln F']$. (ii) The geometric DeWitt measure *is* invariant, always, because the $\sqrt{\det G}$ factor was built to cancel the Jacobian. (iii) In dimensional regularization the two prescriptions coincide perturbatively, because $\delta^d(0) = 0$ kills the only difference between them. Statement (iii) is why, in practice, one may use the convenient flat measure and still perform nonlinear field redefinitions with impunity *provided dimensional regularization is the scheme*—the redefinition theorem of the S-matrix (its invariance under field reparametrizations) then needs no measure correction. In a cutoff scheme the $\delta^d(0)$ terms are real and must either be kept (the covariant measure) or cancelled against the explicit $\delta^d(0)$ tadpoles generated by the same redefinition in the action.

Worked Example E.28 (Wavefunction rescaling)**.** The simplest redefinition is the linear rescaling $\phi(x) = Z^{1/2}\chi(x)$ with Z a constant, which is exactly the field-strength renormalization performed in passing from bare to renormalized

fields. Here $F(\chi) = Z^{1/2}\chi$ for a single component, so $F' = Z^{1/2}$ is constant and $\operatorname{tr}\ln F' = \frac{1}{2}\ln Z$. Theorem E.27 gives

$$\prod_x \mathrm{d}\phi = \exp\Big[\tfrac{1}{2}(\ln Z)\,\delta^d(0)\int \mathrm{d}^d x\Big]\prod_x \mathrm{d}\chi = Z^{\frac{1}{2}\,\delta^d(0)\,V}\prod_x \mathrm{d}\chi, \tag{E.186}$$

with $V = \int \mathrm{d}^d x$ the spacetime volume. The exponent is a pure number times the volume: it is a field-independent constant, so even in a cutoff scheme it merely renormalizes the (irrelevant) normalization $\mathcal{N}$ and drops out of every correlator. In dimensional regularization it is exactly 1. This is the rigorous version of the everyday statement that constant field rescalings "do nothing" to the path integral—they multiply Z by a volume-divergent but physically inert constant.

Worked Example E.29 (A genuinely nonlinear redefinition)**.** Take the cubic redefinition $\phi(x) = \chi(x) + g\,\chi(x)^3$, used to remove a redundant operator. Now $F'(\chi) = 1 + 3g\chi^2$, and the Jacobian is

$$J = \exp\Big[\delta^d(0)\int \mathrm{d}^d x\,\ln(1+3g\chi^2(x))\Big] = \exp\Big[\delta^d(0)\int \mathrm{d}^d x\,(3g\chi^2 - \tfrac{9}{2}g^2\chi^4 + \cdots)\Big]. \tag{E.187}$$

Exponentiated, J contributes new *local* vertices to the action—a $\delta^d(0)$-weighted χ^2 mass term, a $\delta^d(0)$-weighted χ^4 vertex, and so on. In a cutoff scheme these are precisely the counterterms that cancel the $\delta^d(0)$ tadpole graphs (χ-loops closing on a single point) generated when $\phi = \chi + g\chi^3$ is substituted into the action; the cancellation is the field-redefinition invariance of the S-matrix made manifest. In dimensional regularization both sides vanish term by term: $J = 1$ because $\delta^d(0) = 0$, and the would-be tadpoles are themselves scaleless and vanish. Either way the on-shell physics is untouched, which is the entire point of allowing the redefinition.

E.5.5 The fermionic Berezin measure and its inverse Jacobian

For anticommuting fields the analysis runs in parallel but with one decisive sign flip: the Berezin measure transforms with the *inverse* of the Jacobian. Recall (Theorem 12.17) that for Grassmann generators $\{\theta_i\}$ the Berezin rules are $\int \mathrm{d}\theta_i = 0$ and $\int \mathrm{d}\theta_i\,\theta_i = 1$, and that for a complex Grassmann pair $(\psi, \bar\psi)$ these reproduce the fermionic Gaussian integral established in Section E.1,

$$\int \prod_i \mathrm{d}\bar\psi_i\,\mathrm{d}\psi_i\;\exp\!\big(-\bar\psi_i M_{ij}\psi_j\big) = \det M, \tag{E.188}$$

with the determinant appearing to the *positive* power, opposite to the $(\det)^{-1/2}$ of a bosonic Gaussian. The inverse-Jacobian rule is what forces that opposite sign, and we derive it directly from the Berezin definition.

Proposition E.30 (Inverse Jacobian of the Berezin measure)**.** *Let $\theta_i = A_{ij}\eta_j$ be an invertible linear change of n Grassmann variables (A an ordinary $n \times n$ matrix). Then the Berezin measures are related by*

$$\prod_i \mathrm{d}\theta_i = (\det A)^{-1} \prod_i \mathrm{d}\eta_i. \tag{E.189}$$

Proof. The Berezin measure is fixed by its action on the unique top-degree monomial. From $\int \mathrm{d}\theta_i\, \theta_i = 1$ one has the normalization

$$\int \Big(\prod_i \mathrm{d}\theta_i\Big)\, \theta_1\theta_2\cdots\theta_n = 1, \tag{E.190}$$

and likewise $\int (\prod_i \mathrm{d}\eta_i)\, \eta_1\cdots\eta_n = 1$. Now expand the product $\theta_1\cdots\theta_n$ in the η's. Because the η_j anticommute, only fully antisymmetrized terms survive, and the antisymmetrization of a product of matrix entries is exactly a determinant:

$$\theta_1\theta_2\cdots\theta_n = A_{1j_1}A_{2j_2}\cdots A_{nj_n}\,\eta_{j_1}\eta_{j_2}\cdots\eta_{j_n} = \Big(\sum_\sigma \mathrm{sgn}(\sigma)\, A_{1\sigma(1)}\cdots A_{n\sigma(n)}\Big)\,\eta_1\eta_2\cdots\eta_n = (\det A)\,\eta_1\eta_2\cdots\eta_n \tag{E.191}$$

where rearranging $\eta_{j_1}\cdots\eta_{j_n}$ into the standard order $\eta_1\cdots\eta_n$ produced the sign $\mathrm{sgn}(\sigma)$. Insert (E.191) into the normalization (E.190):

$$1 = \int \Big(\prod_i \mathrm{d}\theta_i\Big)\, \theta_1\cdots\theta_n = (\det A) \int \Big(\prod_i \mathrm{d}\theta_i\Big)\, \eta_1\cdots\eta_n. \tag{E.192}$$

Comparing with $\int (\prod_i \mathrm{d}\eta_i)\, \eta_1\cdots\eta_n = 1$ forces $\prod_i \mathrm{d}\theta_i = (\det A)^{-1} \prod_i \mathrm{d}\eta_i$, which is (E.189). The inverse power is the hallmark of Grassmann integration: where a bosonic integral $\int \mathrm{d}^n x$ over $x = Ay$ gains $|\det A|$, the Berezin integral over $\theta = A\eta$ gains $(\det A)^{-1}$. Applied to $\psi \to M^{1/2}\psi$ in (E.188), the two measure factors $\mathrm{d}\bar\psi\,\mathrm{d}\psi$ each contribute $(\det M^{1/2})^{-1}$, while the Gaussian core integrates to a constant; reassembling gives the positive power $\det M$, consistent with the Grassmann determinant of Section E.1. □

E.5.6 The chiral rotation and the Fujikawa setup

The one field redefinition whose Jacobian survives even in dimensional regularization—because it is not local-multiplicative but mixes the Dirac structure against a regulator that cannot be made chirally symmetric—is the chiral rotation of a Dirac fermion. We set it up exactly, identify the object that carries the anomaly, and stop precisely where the value of the anomaly would begin (that computation is owned by Chapter 36). Work in Euclidean signature, where the Dirac operator $\not{D} = \gamma^\mu(\partial_\mu - igA_\mu)$ is anti-Hermitian and $i\not{D}$ is Hermitian with a complete real spectrum, the setting in which the mode expansion below is rigorous; the Wick rotation is the one defined in Chapter A.

A chiral rotation with parameter $\alpha(x)$ acts as

$$\psi(x) \to \psi'(x) = e^{i\alpha(x)\gamma^5}\psi(x), \qquad \bar\psi(x) \to \bar\psi'(x) = \bar\psi(x)\, e^{i\alpha(x)\gamma^5}, \tag{E.193}$$

where the second relation follows from the first because γ^5 is Hermitian and anticommutes with γ^0 (Section C.3.5): $\bar\psi' = \psi'^\dagger\gamma^0 = \psi^\dagger e^{-i\alpha\gamma^5}\gamma^0 = \psi^\dagger\gamma^0 e^{i\alpha\gamma^5} = \bar\psi\, e^{i\alpha\gamma^5}$. The two fields rotate *the same way*, not oppositely, which is why the two measure factors will add rather than cancel.

Expand the fields in the orthonormal eigenmodes $\{\varphi_n\}$ of the Hermitian operator $i\not{D}$, with $i\not{D}\,\varphi_n = \lambda_n\varphi_n$ and $\int \mathrm{d}^d x\, \varphi_m^\dagger\varphi_n = \delta_{mn}$:

$$\psi(x) = \sum_n a_n\, \varphi_n(x), \qquad \bar\psi(x) = \sum_n \bar b_n\, \varphi_n^\dagger(x), \tag{E.194}$$

with $a_n, \bar b_n$ independent Grassmann coefficients. The fermionic measure is the Berezin measure in these coefficients, $\mathcal{D}\bar\psi\,\mathcal{D}\psi = \prod_n \mathrm{d}\bar b_n\, \mathrm{d}a_n$. Under (E.193) the coefficients transform linearly, $a_n \to a'_n = \sum_m C_{nm} a_m$, with

$$C_{nm} = \int \mathrm{d}^d x\, \varphi_n^\dagger(x)\, e^{i\alpha(x)\gamma^5}\, \varphi_m(x) = \delta_{nm} + i\int \mathrm{d}^d x\, \alpha(x)\, \varphi_n^\dagger\gamma^5\varphi_m + O(\alpha^2), \tag{E.195}$$

and the coefficients $\bar b_n$ transform by the same matrix C. By Theorem E.30 the Berezin measure picks up the *inverse* determinant once for the a's and once for the $\bar b$'s:

$$\prod_n \mathrm{d}\bar b'_n\, \mathrm{d}a'_n = (\det C)^{-2} \prod_n \mathrm{d}\bar b_n\, \mathrm{d}a_n. \tag{E.196}$$

Now $(\det C)^{-2} = \exp(-2\,\mathrm{tr}\ln C)$, and to first order $\mathrm{tr}\ln C = \mathrm{tr}(C - \mathbf{1}) = i\int \mathrm{d}^d x\, \alpha(x) \sum_n \varphi_n^\dagger(x)\gamma^5\varphi_n(x)$. Therefore

$$\mathcal{D}\bar\psi'\,\mathcal{D}\psi' = \exp\Big[-2i\int \mathrm{d}^d x\, \alpha(x) \sum_n \varphi_n^\dagger(x)\, \gamma^5\, \varphi_n(x)\Big]\, \mathcal{D}\bar\psi\,\mathcal{D}\psi. \tag{E.197}$$

This is the Fujikawa Jacobian. The exponent is controlled entirely by the mode sum

$$\mathcal{A}(x) \equiv \sum_n \varphi_n^\dagger(x)\, \gamma^5\, \varphi_n(x) = \langle x|\, \gamma^5\, |x\rangle = \mathrm{tr}\big[\gamma^5\, \delta^d(0)\big], \tag{E.198}$$

where the trace is over Dirac indices. Naively $\mathcal{A}(x) = 0$ twice over: $\mathrm{tr}\,\gamma^5 = 0$ in any dimension (Section C.3.5), and $\delta^d(0) = 0$ in dimensional regularization. The naive answer is wrong because the mode sum (E.198) is divergent and *must be regulated before the Dirac trace is taken*, and any regulator that respects the spectrum of $i\not{D}$ necessarily fails to commute with γ^5.

The regulator that does the job is the gauge-invariant, eigenvalue-damping factor $f(\lambda_n^2/M^2)$ with $f(0) = 1$, $f(\infty) = 0$, inserted into the sum and removed only after $M \to \infty$:

$$\mathcal{A}(x) = \lim_{M\to\infty} \sum_n f\Big(\frac{\lambda_n^2}{M^2}\Big)\, \varphi_n^\dagger(x)\, \gamma^5\, \varphi_n(x) = \lim_{M\to\infty} \mathrm{Tr}\Big[\gamma^5\, f\Big(\frac{(i\not{D})^2}{M^2}\Big)\Big]_{x,x}, \tag{E.199}$$

where $\mathrm{Tr}_{x,x}$ is the full trace (Dirac plus spacetime) evaluated at coincident points. The choice $f(s) = e^{-s}$ makes the operator inside the trace into the heat kernel of $(i\not{D})^2 = -\not{D}^2$,

$$\mathcal{A}(x) = \lim_{M\to\infty} \langle x|\, \gamma^5\, e^{\not{D}^2/M^2}\, |x\rangle\,, \tag{E.200}$$

which is *exactly* the heat-kernel object constructed in Section E.3. The coincidence limit of the heat kernel of a Laplace-type operator $\not{D}^2$ admits the Seeley–DeWitt expansion in inverse powers of M^2, and the only term that survives both the $M \to \infty$ limit and the γ^5 Dirac trace is the coefficient $a_{d/2}$ of the volume term: the lower coefficients carry positive powers of M and would diverge but are killed by the vanishing γ^5 trace of their Dirac structure, while the higher coefficients carry negative powers of M and vanish in the limit. The chiral Jacobian therefore reduces, with no further input, to

$$\mathcal{A}(x) = \mathrm{tr}_{\mathrm{Dirac}}\big[\gamma^5\, a_{d/2}(x)\big], \tag{E.201}$$

the Seeley–DeWitt coefficient $a_{d/2}$ of Section E.3 contracted against γ^5. This is the precise sense in which the anomaly is "set up" here: Eqs. (E.197), (E.199) and (E.201) reduce the entire computation to inserting one heat-kernel coefficient. We deliberately do not evaluate $a_{d/2}$ or take its γ^5 trace; that is the computation of Chapter 36, which imports this setup and finds, in $d = 4$, the ABJ result $\mathcal{A} = \frac{g^2}{32\pi^2}\epsilon^{\mu\nu\rho\sigma}F^a_{\mu\nu}F^a_{\rho\sigma}$ up to normalization. The structural lesson is what matters now: a chiral rotation is the one field redefinition whose measure Jacobian is nonzero even in dimensional regularization, because its $\delta^d(0)$ is multiplied by the γ^5 trace of a regulator and the product is a finite, geometric, and physical number.

Caution: why dimensional regularization does not kill this one

Theorem E.27 might suggest that, since $\delta^d(0) = 0$ in dimensional regularization, *every* measure Jacobian is trivial there, including the chiral one. The escape is that the chiral Jacobian is not of the form $\delta^d(0) \times$ (finite local density): it is $\delta^d(0)$ multiplied by $\mathrm{tr}\,\gamma^5$, an indeterminate 0×0 that the regulator resolves to a finite answer. Equivalently, in dimensional regularization the trouble migrates from $\delta^d(0)$ to the γ^5 problem itself—the impossibility of continuing γ^5 to $d \neq 4$ while keeping cyclicity and the four-dimensional chiral trace (Section C.7). The anomaly is scheme-independent precisely because it is forced by an obstruction no scheme removes.

E.5.7 The Faddeev–Popov insertion as a measure manipulation

A second place where the measure is actively manipulated, rather than merely defined, is gauge fixing. For a gauge theory the integration $\int \mathcal{D}A$ over all con-

nections overcounts by the volume of the gauge group, because gauge-equivalent configurations A and A^g (the gauge transform of A by g) contribute identically. The Faddeev–Popov procedure repairs this by inserting into the measure a cleverly chosen factor of unity,

$$1 = \Delta_{\mathrm{FP}}[A] \int \mathcal{D}g\, \delta(G(A^g)), \qquad \Delta_{\mathrm{FP}}[A] = \det \frac{\delta G(A^g)}{\delta g}\bigg|_{G=0}, \tag{E.202}$$

where $G(A) = 0$ is the gauge-fixing condition and Δ_{FP}, the Faddeev–Popov determinant, is exactly the Jacobian relating the delta function on the gauge condition to a delta function on the group. Reorganizing $\int \mathcal{D}A = \int \mathcal{D}g \int \mathcal{D}A\, \Delta_{\mathrm{FP}}\, \delta(G(A^g))$ and factoring out the (now finite) group volume leaves a gauge-fixed measure weighted by Δ_{FP}, which is subsequently re-exponentiated as a Grassmann integral over ghost fields using the inverse-Jacobian rule (E.189). This is once again a statement about how the measure transforms—here under the split of $\mathcal{D}A$ into gauge orbits and a transverse slice—and the apparatus we have built (the Riemannian volume element, the Jacobian of a change of variables, and the Berezin inverse Jacobian that turns a determinant into a ghost integral) is exactly what makes it precise. The full treatment, including the choice of gauge slice, the ghost Lagrangian, and the residual BRST symmetry, is the business of Chapter 13 (abelian) and Chapter 26 (non-abelian); they import the setup of this section rather than rebuild it.

We now have a complete and honest account of the measure and of the determinant it produces for a *non-degenerate* fluctuation operator K: the Gaussian integration of Section E.1 yields $(\det K)^{-1/2}$, the measure is split-invariant by Theorem E.26, and field redefinitions cost nothing in dimensional regularization save the single anomalous exception just isolated. One assumption remains unexamined, and it is the one that fails whenever a continuous symmetry of the saddle is present: that K has no zero eigenvalues. A zero mode makes $\det K$ vanish and the Gaussian formula meaningless. Repairing this—trading each zero mode for a collective coordinate, and recovering a finite, primed determinant—is the subject of Section E.6.

E.6 Zero Modes, Collective Coordinates, and the Primed Determinant

The previous section closed with a clean and complete account of the Gaussian integral about a *non-degenerate* saddle: with the measure of Theorem E.26 and the master formula Eq. (E.1), the one-loop weight of an isolated stationary point ϕ_{cl} is $e^{-S[\phi_{\mathrm{cl}}]}(\det K)^{-1/2}$, where $K = \delta^2 S/\delta\phi^2$ is the fluctuation operator. That account rested on one tacit hypothesis: that K is invertible, i.e. that it has no zero eigenvalue. The hypothesis fails—and fails for a structural, not accidental,

reason—precisely when the saddle is a *soliton* or an *instanton*: a finite-action solution that is not invariant under some continuous symmetry of the action. Translation invariance is the universal culprit. A kink centred at x_0 is a perfectly good solution for every x_0, so the saddle is not a point in field space but a whole orbit; moving along that orbit costs no action; and a costless direction of fluctuation is exactly a zero eigenvalue of K. The Gaussian formula then returns $(\det K)^{-1/2} = \infty$, which is not a divergence to be renormalized but a signal that the wrong integral was attempted. This section repairs it. We prove the existence and fix the normalization of the translational zero mode, trade the offending flat direction for a *collective coordinate* by a Faddeev–Popov manoeuvre, extract the resulting Jacobian $(S_{\mathrm{cl}}/2\pi)^{1/2}$ per zero mode, and assemble the complete semiclassical amplitude in terms of the *primed determinant* $\det' K$, the determinant with the zero modes deleted. The whole machine is then run, with every number shown, on the ϕ^4 kink and its avatar the quartic double-well instanton.

E.6.1 The saddle, its symmetry, and the translational zero mode

We work in the Euclidean theory throughout, where the weight is e^{-S_E} and saddles are genuine minima (or low-action critical points) of a real, bounded-below action. To keep the argument general and one-dimensional—the case that the Gelfand–Yaglom machinery of Section E.2 can actually evaluate—take a single field ϕ on a coordinate z (which will be a spatial coordinate for a static soliton, or Euclidean time for an instanton), with action of the standard form

$$S[\phi] = \int \mathrm{d}z \, \Big[\tfrac{1}{2}\, \phi'(z)^2 + V\big(\phi(z)\big) \Big], \qquad \phi' \equiv \frac{d\phi}{dz}, \tag{E.203}$$

which is Eq. (E.36) specialized to one dimension. The saddle equation $\delta S/\delta\phi = 0$ is the Euclidean equation of motion

$$-\phi'' + V'(\phi) = 0. \tag{E.204}$$

Let $\phi_{\mathrm{cl}}(z)$ be a finite-action solution of Eq. (E.204). Expanding $\phi = \phi_{\mathrm{cl}} + \eta$ and using Theorem 5.44 together with the n-dimensional Gaussian prefactor $(\det H)^{-1/2}$ of Theorem 4.40 produces, term by term,

$$S[\phi_{\mathrm{cl}} + \eta] = S[\phi_{\mathrm{cl}}] + \tfrac{1}{2} \int \mathrm{d}z \, \eta \, K \, \eta + O(\eta^3), \qquad K = -\partial_z^2 + V''(\phi_{\mathrm{cl}}(z)), \tag{E.205}$$

the linear term vanishing because ϕ_{cl} solves Eq. (E.204). The operator K is the second variation: a Schrödinger operator on the line with potential $V''(\phi_{\mathrm{cl}}(z))$, which tends to the constant $V''(\phi_{\mathrm{vac}}) \equiv m^2 > 0$ as $z \to \pm\infty$ (the curvature of V at the vacuum the soliton interpolates between or returns to). Its spectrum therefore consists of a finite number of discrete bound states below the threshold m^2 and a continuum above it.

The action Eq. (E.203) is invariant under the translation $z \mapsto z + a$: if $\phi_{\rm cl}(z)$ is a solution, so is $\phi_{\rm cl}(z-a)$, with identical action, for every real a. This is the continuous symmetry that forces the zero mode.

Proposition E.31 (The translational zero mode and its normalization)**.** *Let $\phi_{\rm cl}$ be a finite-action solution of Eq. (E.204) that is not constant. Then $\phi'_{\rm cl}(z)$ is a normalizable zero eigenfunction of the fluctuation operator K of Eq. (E.205),*

$$K\,\phi'_{\rm cl} = 0, \tag{E.206}$$

and its norm is the square root of the classical action,

$$\int \mathrm{d}z\,\left(\phi'_{\rm cl}\right)^2 = S_{\rm cl} \equiv S[\phi_{\rm cl}], \qquad \text{so the unit-normalized zero mode is} \quad \eta_0 = \frac{\phi'_{\rm cl}}{\sqrt{S_{\rm cl}}}. \tag{E.207}$$

Proof. Existence. The family $\phi_a(z) \equiv \phi_{\rm cl}(z-a)$ solves Eq. (E.204) for every a: $-\phi_a'' + V'(\phi_a) = 0$. Differentiate this identity with respect to a and set $a = 0$. Since $\partial_a \phi_a|_{a=0} = -\phi'_{\rm cl}(z)$, and using $\partial_a V'(\phi_a) = V''(\phi_a)\,\partial_a\phi_a$,

$$-(-\phi'_{\rm cl})'' + V''(\phi_{\rm cl})(-\phi'_{\rm cl}) = 0 \quad\Longrightarrow\quad (-\partial_z^2 + V''(\phi_{\rm cl}))\phi'_{\rm cl} = 0, \tag{E.208}$$

which is Eq. (E.206). Equivalently: differentiating the equation of motion is the same as acting with its linearization K, and the generator of the symmetry, $\phi'_{\rm cl}$, is annihilated. Because $\phi_{\rm cl} \to \phi_{\rm vac}$ exponentially at infinity (the approach to a nondegenerate vacuum of V is governed by $e^{-m|z|}$), $\phi'_{\rm cl}$ decays exponentially and is square-integrable, hence a genuine bound state. That it is the *ground* state of K—the eigenfunction of lowest eigenvalue, here 0—follows because $\phi'_{\rm cl}$ has no nodes for a monotonic kink: a nodeless eigenfunction is necessarily the ground state of a Schrödinger operator, so no negative eigenvalue can lie below it. (For a bounce solution that is not monotonic, $\phi'_{\rm cl}$ has a node, and exactly one negative mode sits below the zero mode; that single negative mode is the seat of the imaginary part responsible for a decay rate, the subject of Chapter 42.)

Normalization. Multiply Eq. (E.204) by $\phi'_{\rm cl}$ and integrate: $\phi'_{\rm cl}(-\phi''_{\rm cl} + V'(\phi_{\rm cl})) = 0$ is the total derivative $-\frac{\mathrm{d}}{\mathrm{d}z}\left[\frac{1}{2}(\phi'_{\rm cl})^2 - V(\phi_{\rm cl})\right] = 0$, so the "Euclidean energy"

$$\tfrac{1}{2}\left(\phi'_{\rm cl}\right)^2 - V(\phi_{\rm cl}) = \text{const} = 0, \tag{E.209}$$

the constant being fixed to zero by the boundary condition $\phi'_{\rm cl} \to 0$, $V \to 0$ at infinity. Hence $\frac{1}{2}(\phi'_{\rm cl})^2 = V(\phi_{\rm cl})$, the Bogomolny first-order condition, and the action Eq. (E.203) becomes

$$S_{\rm cl} = \int \mathrm{d}z\,\left[\tfrac{1}{2}(\phi'_{\rm cl})^2 + V(\phi_{\rm cl})\right] = \int \mathrm{d}z\,\left[\tfrac{1}{2}(\phi'_{\rm cl})^2 + \tfrac{1}{2}(\phi'_{\rm cl})^2\right] = \int \mathrm{d}z\,(\phi'_{\rm cl})^2, \tag{E.210}$$

which is exactly Eq. (E.207). □

The content of Theorem E.31 is that the would-be flat direction in field space is the direction $\eta_0 \propto \phi_{\rm cl}'$, and that the geometry of that direction is fixed entirely by the classical action: the "length" of the zero-mode direction, $\|\phi_{\rm cl}'\| = \sqrt{S_{\rm cl}}$, is what will set the Jacobian below.

E.6.2 Why the Gaussian integral diverges

Make the breakdown explicit in the eigenbasis. Let $\{\eta_n\}_{n\geq 0}$ be the orthonormal eigenfunctions of K, $K\eta_n = \epsilon_n \eta_n$, ordered so that $\epsilon_0 = 0$ is the translational mode of Theorem E.31 and $\epsilon_n > 0$ for $n \geq 1$. Expand the fluctuation in this basis,

$$\eta(z) = \sum_{n\geq 0} c_n\, \eta_n(z), \qquad \tfrac{1}{2}\int \mathrm{d}z\; \eta K \eta = \tfrac{1}{2}\sum_{n\geq 0} \epsilon_n c_n^2, \tag{E.211}$$

and recall that the continuum measure of Theorem E.26, in this basis, is the product $\mathcal{D}\eta = \prod_{n\geq 0} \mathrm{d}c_n/\sqrt{2\pi}$, normalized exactly so that each Gaussian factor integrates by Eq. (E.1) to $\int (\mathrm{d}c_n/\sqrt{2\pi})\, e^{-\frac{1}{2}\epsilon_n c_n^2} = \epsilon_n^{-1/2}$. The one-loop integral is then the formal product

$$\int \mathcal{D}\eta\; e^{-\frac{1}{2}\sum_n \epsilon_n c_n^2} = \prod_{n\geq 0} \epsilon_n^{-1/2} = (\det K)^{-1/2}. \tag{E.212}$$

The $n = 0$ factor is the disaster: $\epsilon_0 = 0$, so the integrand along c_0 is the constant $e^0 = 1$ and

$$\int_{-\infty}^{\infty} \frac{\mathrm{d}c_0}{\sqrt{2\pi}}\, 1 = \infty, \qquad \text{equivalently} \qquad \det K = \epsilon_0 \cdot \prod_{n\geq 1} \epsilon_n = 0. \tag{E.213}$$

The divergence is not ultraviolet—it is wholly carried by the single zero mode—and it has a transparent meaning. Moving along c_0 is, to linear order, the same as translating the soliton: from Eq. (E.205) and Eq. (E.207), $\delta\phi = \eta_0\, \mathrm{d}c_0$, while a translation by $\mathrm{d}a$ gives $\delta\phi = -\phi_{\rm cl}'\, \mathrm{d}a = -\sqrt{S_{\rm cl}}\, \eta_0\, \mathrm{d}a$. The c_0-integral is therefore an integral over the position of the soliton, and it diverges simply because the soliton can sit anywhere on an infinite line. The cure is to do that integral honestly, as an integral over the collective coordinate a, extracting the (finite, physical) volume of the symmetry orbit instead of integrating a redundant Gaussian over it.

E.6.3 Collective coordinates and the Faddeev–Popov Jacobian

The clean way to convert the dangerous c_0-integral into an a-integral is the Faddeev–Popov insertion of unity—the very device the gauge-fixing discussion of Section E.5.7 previewed, here applied not to a gauge orbit but to the translation orbit of the soliton. We insert into the path integral the identity

$$1 = \int_{-\infty}^{\infty} \mathrm{d}a \left|\frac{\partial F}{\partial a}\right| \delta(F(a)), \qquad F(a) \equiv \int \mathrm{d}z\; \big[\phi(z) - \phi_{\rm cl}(z-a)\big]\, \eta_0(z-a), \tag{E.214}$$

where $\eta_0(\cdot - a) = \phi_{\rm cl}'(\cdot - a)/\sqrt{S_{\rm cl}}$ is the normalized zero mode *of the soliton centred at* a. The function $F(a)$ measures the overlap of the deviation $\phi - \phi_a$ with the zero mode of ϕ_a; the gauge-fixing condition $F(a) = 0$ states that, having pulled out the centre a, what remains has no component along the costless direction. For a configuration ϕ close to some member of the orbit there is exactly one a with $F(a) = 0$, so Eq. (E.214) is a true resolution of unity. The Faddeev–Popov determinant in one collective coordinate is just the ordinary derivative $|\partial F/\partial a|$, evaluated on the constraint surface.

Compute that derivative at the solution. Differentiating F in Eq. (E.214) with respect to a,

$$\frac{\partial F}{\partial a} = \int \mathrm{d}z \, \Big[- \partial_a \phi_{\rm cl}(z-a)\, \eta_0(z-a) + \big(\phi(z) - \phi_{\rm cl}(z-a)\big)\, \partial_a \eta_0(z-a) \Big]. \tag{E.215}$$

The second term is proportional to the deviation $\phi - \phi_a$, which vanishes on the orbit ($\phi = \phi_a$) and is first order in the transverse fluctuation off it, hence does not contribute at the leading saddle order we are computing. In the first term, $\partial_a \phi_{\rm cl}(z-a) = -\phi_{\rm cl}'(z-a) = -\sqrt{S_{\rm cl}}\,\eta_0(z-a)$, so

$$\left.\frac{\partial F}{\partial a}\right|_{F=0} = \sqrt{S_{\rm cl}} \int \mathrm{d}z \; \eta_0(z-a)^2 = \sqrt{S_{\rm cl}}, \tag{E.216}$$

the last step using $\int \eta_0^2 = 1$. The Jacobian is the constant $\sqrt{S_{\rm cl}}$, independent of a. Now insert Eq. (E.214) into the fluctuation integral. Writing $\phi = \phi_a + \sum_{n \geq 1} c_n \eta_n(\cdot - a)$, the constraint $\delta(F(a))$ is $\delta(c_0)$ (it sets the zero-mode coefficient about the chosen centre to zero), and the measure factorizes as $\mathcal{D}\eta = (\mathrm{d}c_0/\sqrt{2\pi}) \prod_{n \geq 1} (\mathrm{d}c_n/\sqrt{2\pi})$. The c_0 integral is now performed against the delta function,

$$\int \frac{\mathrm{d}c_0}{\sqrt{2\pi}}\, \delta(c_0) = \frac{1}{\sqrt{2\pi}}, \tag{E.217}$$

and the leftover pieces are the collective-coordinate integral $\int \mathrm{d}a$, the Jacobian $\sqrt{S_{\rm cl}}$ of Eq. (E.216), and the *transverse* Gaussian over $n \geq 1$. Collecting the constants, each zero mode contributes the factor

$$\boxed{\frac{1}{\sqrt{2\pi}} \sqrt{S_{\rm cl}} \int \mathrm{d}a = \sqrt{\frac{S_{\rm cl}}{2\pi}} \int \mathrm{d}a \;,} \tag{E.218}$$

which is the advertised Jacobian $(S_{\rm cl}/2\pi)^{1/2}$ per zero mode, multiplying the integral $\int \mathrm{d}a$ over the modulus.

Theorem E.32 (Collective-coordinate change of variables)**.** *Let $\phi_{\rm cl}$ be a soliton/instanton saddle of Eq. (E.203) with d_0 exact zero modes $\eta_0^{(1)}, \ldots, \eta_0^{(d_0)}$ generated by a d_0-parameter continuous symmetry with collective coordinates*

$a_1, \dots, a_{d_0}$, each zero mode normalized to unity. Then the Gaussian fluctuation integral about ϕ_{cl} equals

$$\int \mathcal{D}\eta \, e^{-\frac{1}{2}\int \eta K \eta} = \left(\prod_{i=1}^{d_0} \sqrt{\frac{\mathcal{J}_i}{2\pi}} \right) \int \mathrm{d}^{d_0} a \, \left(\det{}' K \right)^{-1/2}, \qquad \mathcal{J}_i = \int \mathrm{d}z \, \left(\frac{\partial \phi_{\mathrm{cl}}}{\partial a_i} \right)^2, \tag{E.219}$$

where $\det' K \equiv \prod_{\epsilon_n > 0} \epsilon_n$ is the determinant with the zero modes deleted, $\int \mathrm{d}^{d_0} a$ runs over the moduli space (the symmetry orbit), and the per-mode Jacobian $\mathcal{J}_i$ is the norm-squared of the i-th unnormalized zero mode. For a single translational mode, $\partial \phi_{\mathrm{cl}} / \partial a = -\phi_{\mathrm{cl}}'$ and $\mathcal{J} = \int (\phi_{\mathrm{cl}}')^2 = S_{\mathrm{cl}}$ by Eq. (E.207), reproducing Eq. (E.218).

Proof. Repeat the construction of Eqs. (E.214) to (E.216) once per symmetry direction. The multidimensional Faddeev–Popov determinant is $\det(\partial F_i / \partial a_j)$, whose entries are $\partial F_i / \partial a_j = \int (\partial_{a_j} \phi_{\mathrm{cl}}) \, \eta_0^{(i)} = \sqrt{\mathcal{J}_j} \, \langle \hat{\eta}_0^{(j)}, \eta_0^{(i)} \rangle$ with $\hat{\eta}_0^{(j)} = \partial_{a_j} \phi_{\mathrm{cl}} / \sqrt{\mathcal{J}_j}$ the normalized j-th mode. When the zero modes are orthonormal ($\langle \hat{\eta}_0^{(j)}, \eta_0^{(i)} \rangle = \delta_{ij}$) the determinant is the product $\prod_i \sqrt{\mathcal{J}_i}$; each $\delta(F_i)$ does its $c_0^{(i)}$-integral against the measure factor $1/\sqrt{2\pi}$ as in Eq. (E.217); and the surviving $n \notin \{0\}$ Gaussians give $\prod_{\epsilon_n > 0} \epsilon_n^{-1/2} = (\det' K)^{-1/2}$. (Should the natural collective coordinates fail to be orthonormal—as for the dilatation and rotation moduli of higher-dimensional instantons—the determinant $\det(\partial F_i / \partial a_j)$ supplies precisely the correct metric volume on moduli space, $\sqrt{\det g}$ with $g_{ij} = \langle \partial_{a_i} \phi_{\mathrm{cl}}, \partial_{a_j} \phi_{\mathrm{cl}} \rangle$; this is the DeWitt metric of Eq. (E.171) restricted to the zero-mode subspace, and is the moduli-space measure quoted without derivation in Chapter 42.) □

Physical Insight

The Jacobian $(S_{\mathrm{cl}}/2\pi)^{1/2}$ is not a formal artefact; it is the statement that one quantum of motion along the soliton's translation mode has a finite "inertia" set by the classical action. In the soliton's rest frame S_{cl} is its mass M, and the collective coordinate a is its centre of mass: promoting a to a dynamical variable $a(t)$ and expanding the action recovers the nonrelativistic kinetic term $\frac{1}{2} M \dot{a}^2$. The collective-coordinate trick is thus the first step of soliton quantization—it extracts the soliton's translational degree of freedom from the field and hands it back as an ordinary particle coordinate—which is precisely how Chapter 41 builds the quantum kink's Hilbert space and momentum eigenstates.

E.6.4 The primed determinant and the complete semiclassical amplitude

Definition E.33 (Primed determinant)**.** The *primed determinant* of a fluctuation operator K with zero modes is the product of its nonzero eigenvalues,

$$\det{}' K \equiv \prod_{n:\,\epsilon_n \neq 0} \epsilon_n, \tag{E.220}$$

defined, like every functional determinant in this appendix, only through the finite ratio $\det' K / \det K_0$ against a reference operator K_0 with no zero modes, in the sense of Eq. (E.45).

We can now write the complete one-loop amplitude. Two saddles compete: the trivial vacuum ϕ_{vac} (a constant, with no zero mode and fluctuation operator $K_0 = -\partial_z^2 + m^2$), and the soliton/instanton ϕ_{cl}. Normalizing the soliton sector against the vacuum—as one always does, so that the divergent measure factors of Section E.2.2 cancel in the ratio—and combining Theorem E.32 with the vacuum weight $e^{-S[\phi_{\text{vac}}]}(\det K_0)^{-1/2}$, the ratio of the two functional integrals is

$$\boxed{\frac{Z_{\text{sol}}}{Z_{\text{vac}}} = \left(\prod_{i=1}^{d_0} \sqrt{\frac{\mathcal{J}_i}{2\pi}}\right) \int \mathrm{d}^{d_0} a \;\; e^{-(S_{\text{cl}} - S_{\text{vac}})} \left(\frac{\det{}' K}{\det K_0}\right)^{-1/2}.} \tag{E.221}$$

This is the master formula the appendix has been building toward for the degenerate case. Every factor is finite and computable: $S_{\text{cl}} - S_{\text{vac}}$ is the classical action of the soliton above the vacuum (its mass, or its tunneling exponent); the product of $(\mathcal{J}_i/2\pi)^{1/2}$ is the collective-coordinate Jacobian, equal to $(S_{\text{cl}}/2\pi)^{1/2}$ for a single translational mode; the integral $\int \mathrm{d}^{d_0} a$ is the volume of moduli space (for time translation it yields the total Euclidean time T, the source of a *rate*); and the ratio $\det' K / \det K_0$ is a finite ratio of determinants with the zero modes removed from the numerator, evaluated by Gelfand–Yaglom in $d = 1$ exactly as in the non-degenerate case, only with the zero eigenvalue divided out by hand. The remainder of the section computes that ratio, and assembles the prefactor, for the ϕ^4 kink.

E.6.5 Worked example: the ϕ^4 kink and the quartic double-well instanton

Worked Example E.34 (The kink fluctuation determinant and the tunneling prefactor)**.** We run the entire machine on ϕ^4 theory. The standard Lagrangian Eq. (6.56) carries the potential $\frac{1}{2}m^2\phi^2 + \frac{\lambda}{4!}\phi^4$; a kink requires the spontaneously broken regime in which the mass term has the opposite sign, so that V has two degenerate minima. We write the degenerate double well in the standard form

$$V(\phi) = \frac{\lambda}{4!}\left(\phi^2 - v^2\right)^2, \qquad V(\pm v) = 0, \qquad m^2 \equiv V''(\pm v) = \frac{\lambda v^2}{3}, \tag{E.222}$$

the two vacua at $\phi = \pm v$ and m the mass of the elementary meson (fluctuations about either vacuum). Expanding the square reproduces the frozen quartic $\frac{\lambda}{4!}\phi^4$ with a broken mass term $-\frac{\lambda v^2}{12}\phi^2 = -\frac{1}{2}(m^2/2)\phi^2$ and a constant chosen so that V vanishes at the minima; this is exactly the model of Eq. (6.56) in its broken phase. (See the notation watch below on the sign of the mass term.)

Step 1 — the kink solution. The Bogomolny condition $\frac{1}{2}(\phi_{\rm cl}')^2 = V(\phi_{\rm cl})$ of Eq. (E.209) gives $\phi_{\rm cl}' = \sqrt{2V} = \sqrt{\lambda/12}\,(v^2 - \phi_{\rm cl}^2)$ for the kink that climbs from $-v$ to $+v$. Separating variables, $\int \mathrm{d}\phi/(v^2-\phi^2) = \sqrt{\lambda/12}\int \mathrm{d}z$, i.e. $v^{-1}\,\mathrm{artanh}(\phi/v) = \sqrt{\lambda/12}\,(z - z_0)$, so with $v\sqrt{\lambda/12} = m/2$,

$$\phi_{\rm cl}(z) = v\,\tanh\Big[\frac{m}{2}\,(z - z_0)\Big], \tag{E.223}$$

a kink of width $2/m$ centred at the collective coordinate z_0.

Step 2 — the classical action. By Eq. (E.210), $S_{\rm cl} = \int \mathrm{d}z\,(\phi_{\rm cl}')^2$. With $\phi_{\rm cl}' = \frac{mv}{2}\,\mathrm{sech}^2(mz/2)$ and $\int_{-\infty}^{\infty}\mathrm{sech}^4 u\,\mathrm{d}u = 4/3$,

$$S_{\rm cl} = \frac{m^2v^2}{4}\int \mathrm{d}z\,\mathrm{sech}^4\frac{mz}{2} = \frac{m^2v^2}{4}\cdot\frac{2}{m}\cdot\frac{4}{3} = \frac{2}{3}\,mv^2 = \frac{2m^3}{\lambda}, \tag{E.224}$$

using $v^2 = 3m^2/\lambda$ at the last step. In the soliton reading $S_{\rm cl}$ is the classical kink mass $M_{\rm kink} = 2m^3/\lambda$; in the instanton reading it is the tunneling exponent S_0.

Step 3 — the fluctuation operator and its spectrum. From Eq. (E.205) with $V''(\phi) = \frac{\lambda}{6}(3\phi^2 - v^2)$ and $\phi_{\rm cl} = v\tanh(mz/2)$, using $3\tanh^2 - 1 = 2 - 3\,\mathrm{sech}^2$ and $\frac{\lambda v^2}{6} = \frac{m^2}{2}$,

$$K = -\partial_z^2 + m^2 - \frac{3m^2}{2}\,\mathrm{sech}^2\frac{mz}{2}. \tag{E.225}$$

This is the reflectionless Pöschl–Teller operator. Rescaling $s = mz/2$ turns it into $\frac{m^2}{4}\big[-\partial_s^2 + 4 - 6\,\mathrm{sech}^2 s\big]$; the bracket is $-\partial_s^2 - \ell(\ell+1)\mathrm{sech}^2 s$ shifted by 4, with $\ell(\ell+1) = 6$, i.e. $\ell = 2$. Its two bound states sit at $-\partial_s^2$ energies $-(\ell - j)^2 = -4, -1$ for $j = 0, 1$, and the continuum begins at 0; adding the shift 4 and multiplying by $m^2/4$ gives the spectrum of K,

$$\epsilon_0 = 0\ \text{(zero mode)}, \qquad \epsilon_1 = \frac{3m^2}{4}\ \text{(shape mode)}, \qquad \epsilon_k = k^2 + m^2\ \text{(continuum)}, \tag{E.226}$$

sketched in Fig. E.5. The zero mode is $\eta_0 \propto \mathrm{sech}^2(mz/2) \propto \phi_{\rm cl}'$, exactly as Theorem E.31 demands; the nonzero "shape mode" ϵ_1 is a genuine internal vibration of the kink; and the continuum is that of the free operator $K_0 = -\partial_z^2 + m^2$, shifted only by reflectionless phase shifts. There is no negative eigenvalue: the kink/instanton is a true minimum, and the prefactor will be real.

Step 4 — the primed determinant via Gelfand–Yaglom. We need the ratio $\det' K/\det K_0$ with $K_0 = -\partial_z^2 + m^2$. Place the system in a box $z \in [-T/2, T/2]$ with Dirichlet conditions, so $\det K$ and $\det K_0$ are well-defined and Theorem E.12

applies, and take $T \to \infty$ at the end. Two ingredients are needed: the *full* Gelfand–Yaglom ratio $\det K / \det K_0$ (which vanishes, because K has the would-be zero mode), and the small eigenvalue $\epsilon_0(T)$ into which the box lifts that mode; their quotient is the primed ratio, since $\det K = \epsilon_0(T)\, \det' K$.

For the full ratio, use the two independent solutions of $K\psi = 0$ in the rescaled variable $s = mz/2$, where the equation is $y'' = (4 - 6\,\mathrm{sech}^2 s)\, y$. The decaying solution is the zero mode $y_a(s) = \mathrm{sech}^2 s$ (even); reduction of order, $y_b = y_a \int \mathrm{d}s / y_a^2$ with $\int \cosh^4 s \,\mathrm{d}s = \frac{3}{8}s + \frac{1}{4}\sinh 2s + \frac{1}{32}\sinh 4s$, gives the growing solution

$$y_b(s) = \mathrm{sech}^2 s \left[\tfrac{3}{8}s + \tfrac{1}{4}\sinh 2s + \tfrac{1}{32}\sinh 4s\right] \xrightarrow{s\to+\infty} \tfrac{1}{16} e^{2s}, \tag{E.227}$$

which is odd and has unit Wronskian $W = y_a y_b' - y_a' y_b = 1$ (checked at $s = 0$, where $y_a = 1$, $y_a' = 0$, $y_b = 0$, $y_b' = 1$). The Gelfand–Yaglom shooting solution on the symmetric interval $s \in [-S, S]$, $S = mT/4$, with $u(-S) = 0$, $u'(-S) = 1$, is $u = \alpha y_a + \beta y_b$ with $\alpha = -y_b(-S)$, $\beta = y_a(-S)$ (Cramer's rule with denominator $W = 1$). Using the parities $y_a(-S) = y_a(S)$ and $y_b(-S) = -y_b(S)$,

$$u(S) = -y_b(-S)\, y_a(S) + y_a(-S)\, y_b(S) = 2\, y_a(S)\, y_b(S) \xrightarrow{S\to\infty} 2\cdot 4e^{-2S}\cdot \tfrac{1}{16} e^{2S} = \tfrac{1}{2}. \tag{E.228}$$

The reference shoot for K_0 (rescaled to $-\partial_s^2 + 4$) is $u_0(s) = \frac{1}{2}\sinh\big(2(s+S)\big)$, so $u_0(S) = \frac{1}{2}\sinh 4S = \frac{1}{2}\sinh(mT)$. By Eq. (E.47) the full ratio is therefore

$$\frac{\det K}{\det K_0} = \frac{u(S)}{u_0(S)} = \frac{1/2}{\frac{1}{2}\sinh(mT)} = \frac{1}{\sinh(mT)} \xrightarrow{T\to\infty} 2\, e^{-mT}, \tag{E.229}$$

which indeed $\to 0$: the zero mode has driven the determinant to zero, exactly as in Eq. (E.213).

For the lifted eigenvalue, the exact zero-energy solution $y_a = \mathrm{sech}^2(mz/2)$ does not vanish at the walls, so the true Dirichlet ground state φ has a small nonzero eigenvalue $\epsilon_0(T)$. Green's identity on $[-T/2, T/2]$ with $K\,\mathrm{sech}^2 = 0$ and $K\varphi = \epsilon_0 \varphi$, $\varphi(\pm T/2) = 0$, gives

$$\epsilon_0(T) \int \mathrm{d}z\; y_a \varphi = \big[\varphi\, y_a' - y_a \varphi'\big]_{-T/2}^{T/2} = -\big[y_a \varphi'\big]_{-T/2}^{T/2}, \tag{E.230}$$

the surface terms with φ dropping out by the Dirichlet condition. Matching $\varphi \simeq y_a \simeq 4e^{-m|z|}$ in the interior to a solution $A\sinh\big(m(T/2 - |z|)\big)$ vanishing at the wall fixes $A = 8e^{-mT/2}$ and $\varphi'(\pm T/2) = \mp 8 m e^{-mT/2}$; with $y_a(\pm T/2) = 4e^{-mT/2}$ and $\int y_a^2 = \int \mathrm{sech}^4(mz/2)\,\mathrm{d}z = 8/(3m)$, Eq. (E.230) gives

$$\epsilon_0(T) = \frac{64\, m\, e^{-mT}}{8/(3m)} = 24\, m^2\, e^{-mT}. \tag{E.231}$$

Dividing Eq. (E.229) by Eq. (E.231),

$$\boxed{\frac{\det' K}{\det K_0} = \frac{1}{\epsilon_0(T)} \frac{\det K}{\det K_0} = \frac{2\, e^{-mT}}{24\, m^2\, e^{-mT}} = \frac{1}{12\, m^2}\,,} \tag{E.232}$$

finite and T-independent, the celebrated determinant ratio for the ϕ^4/double-well fluctuation operator. (The exponentials cancel, confirming that the apparent collapse of $\det K$ was nothing but the zero mode, now correctly excised.)

Step 5 — assembling the prefactor. Substitute Eqs. (E.224) and (E.232) into the master amplitude Eq. (E.221) with a single translational mode ($d_0 = 1$, $\mathcal{J} = S_{\rm cl}$, $\int \mathrm{d}a = T$ in the instanton reading where z is Euclidean time), and write $m = \omega$ in QM language (the curvature of the well). The one-instanton amplitude relative to the no-instanton (vacuum) amplitude is $T\,K\,e^{-S_0}$ with

$$K = \sqrt{\frac{S_0}{2\pi}}\left(\frac{\det{}' K}{\det K_0}\right)^{-1/2} = \sqrt{\frac{S_0}{2\pi}}\,\sqrt{12\,\omega^2} = \omega\sqrt{\frac{6\,S_0}{\pi}}, \tag{E.233}$$

where the algebra is $2\sqrt{3}\cdot(2\pi)^{-1/2} = (12/2\pi)^{1/2} = (6/\pi)^{1/2}$. Summing the dilute gas of alternating instantons and anti-instantons exponentiates this into the two lowest levels $E_\pm = \frac{1}{2}\omega \mp K\,e^{-S_0}$, so the ground-state splitting of the symmetric double well is

$$\Delta E = E_- - E_+ = 2K\,e^{-S_0} = 2\,\omega\sqrt{\frac{6\,S_0}{\pi}}\;e^{-S_0}, \qquad S_0 = \frac{2}{3}\,\omega\,a^2, \tag{E.234}$$

the standard double-well tunneling result, here derived with every factor accounted for: the exponential e^{-S_0} from the classical action, the $\sqrt{S_0/2\pi}$ from the collective-coordinate Jacobian, and the $\sqrt{12\,\omega^2}$ from the primed determinant. In the soliton reading the same numbers feed Eq. (E.221) for the quantum kink: $\det' K/\det K_0 = 1/12m^2$ fixes the one-loop fluctuation correction to the kink mass and the normalization of the kink state, the computation completed in Chapter 41.

Notation watch: the sign of the mass term

The frozen ϕ^4 Lagrangian Eq. (6.56) is written with a positive mass term $+\frac{1}{2}m^2\phi^2$, describing a single vacuum at $\phi = 0$. A kink/instanton exists only in the spontaneously broken phase, where that term is negative and the potential is the double well Eq. (E.222). We therefore use m in this section for the *physical meson mass* $\sqrt{V''(\pm v)} = \sqrt{\lambda v^2/3}$ about a true vacuum, not for the (now tachyonic) coefficient in the symmetric Lagrangian. This is the same convention Chapters 41 and 42 adopt; no global constant is redefined.

Physical Insight

The number $\det' K/\det K_0 = 1/12m^2$ packages two physically distinct objects in one ratio. Read with z a spatial coordinate, it is the one-loop

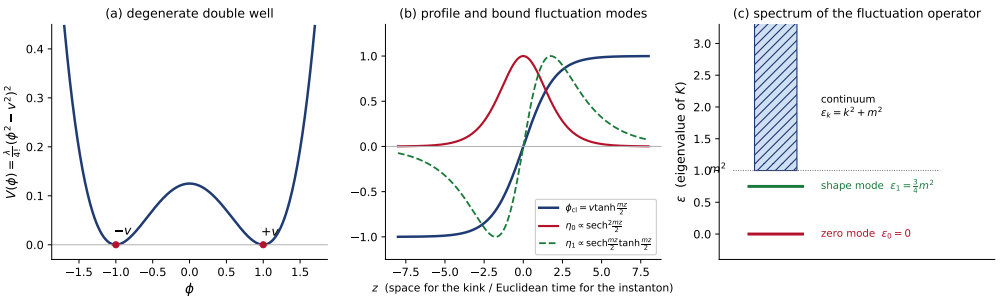


Figure E.5: The ϕ^4 kink / quartic double-well instanton. **(a)** the degenerate double well Eq. (E.222) with vacua at $\pm v$. **(b)** the kink profile $\phi_{\rm cl} = v\tanh(mz/2)$ together with the normalizable translational zero mode $\eta_0 \propto \operatorname{sech}^2(mz/2) \propto \phi_{\rm cl}'$ (Theorem E.31) and the discrete shape mode; here z is space for the static kink and Euclidean time for the instanton. **(c)** the spectrum Eq. (E.226) of the fluctuation operator K: a zero mode at 0 (excised into the collective coordinate), a bound shape mode at $\frac{3}{4}m^2$, and the reflectionless continuum above the meson threshold m^2.

correction to a soliton's mass and the normalization of its quantum state—a particle built by a classical lump dressed in quantum fluctuations. Read with z a Euclidean time, the very same operator and the very same ratio give the amplitude for a particle to tunnel through a barrier, and the splitting Eq. (E.234). That a static soliton and a tunneling event are governed by identical mathematics is not a coincidence: both are finite-action critical points of the same Euclidean functional, and the fluctuation determinant cannot tell space from imaginary time. This duality is the conceptual bridge from Chapter 41 to Chapter 42.

We have now closed the last gap left open by the non-degenerate treatment of Section E.5: a fluctuation operator with zero modes is handled by trading each costless direction for a collective coordinate, paying the Jacobian $(S_{\rm cl}/2\pi)^{1/2}$ per mode, and replacing $\det K$ by the finite primed determinant $\det' K$, with the complete semiclassical amplitude assembled in Eq. (E.221). Together with the Gaussian algebra of Section E.1, the Gelfand–Yaglom and zeta-function evaluations of Section E.2, the one-loop effective action of Section E.4, and the measure theory of Section E.5, this furnishes a complete toolkit for evaluating $\int \mathcal{D}\phi\, e^{-S}$ beyond tree level. Its principal consumers are the soliton quantization of Chapter 41, the instanton calculus and the θ-vacuum of Chapter 42, and the curved-space functional determinants—heat kernels, conformal anomalies, and the gravitational path integral—of Chapter 64, each of which inherits the primed-determinant prefactor derived here rather than rebuilding it.

Further reading

The path-integral and effective-action machinery of this appendix is developed at textbook length in Peskin and Schroeder [396] (chapters 9 and 11), Weinberg [500] (chapter 9 and volume II), Zee [532], and—most thoroughly for functional methods, the effective potential, and instanton determinants—Zinn-Justin [534]. The semiclassical lectures of Coleman [129] remain the clearest narrative source for the steepest-descent expansion, zero modes, and the dilute-instanton gas, with Rajaraman [412] the standard reference for soliton and instanton fluctuation determinants in full. The original radiative-symmetry-breaking computation is Coleman and Weinberg [132]; the Gelfand–Yaglom determinant formula is Gelfand and Yaglom [213]; zeta-function regularization of functional determinants in field theory traces to Hawking [257], and the heat-kernel expansion with its Seeley–DeWitt coefficients is reviewed exhaustively in Vassilevich [484]. For the geometric and topological language underlying the measure, zero modes, and the index theorem the anomaly chapter draws on, see Nakahara [355].

Appendix F

Standard Model Data

Parts VII through IX of this book built the Standard Model one mechanism at a time, and each mechanism was *earned* where it appears. Chapters 25 to 28 gave us non-abelian gauge theory: the connection and curvature, Faddeev–Popov quantization and BRST, the running coupling and asymptotic freedom that make QCD a predictive theory. Chapters 29 to 32 gave us the symmetry-breaking half of the story: Goldstone's theorem and the effective potential, the Higgs mechanism in the R_ξ and unitary gauges, the Glashow–Weinberg–Salam unification of the electroweak interactions, and the flavor sector with its Yukawa couplings, mass-versus-flavor bases, and the origin of CP violation. Chapter 33 then assembled the complete theory, identified its accidental baryon- and lepton-number symmetries, wrote down the θ term, and counted its parameters. The physics is therefore already proved. *Nothing physical is proved in this appendix.*

What this appendix does instead is collect, in one unambiguous place, the model's defining *data* and the conventions that pin every number down. It records the gauge group and the full representation content—every fermion field with its $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ quantum numbers, and the single scalar doublet; it writes the complete Lagrangian term by term, with every sign, normalization, and gauge-fixing choice fixed; it derives the free-parameter count in full; and it tabulates the measured numerical values *together with the renormalization scheme in which each one is quoted.* The organizing purpose is reproducibility: every mass, every coupling, and every cross section computed anywhere in this book should be reconstructible from the numbers and conventions on these pages, with no recourse to an external table.

The division of labor is worth stating plainly, because it governs what is and is not argued here. Load-bearing *physics*—the gauge principle and minimal coupling, the Higgs mechanism and mass generation, the GIM structure and the CKM origin of flavor mixing, gauge-anomaly cancellation—is imported by Chapter 33 from the chapters that prove it and is never reproved. But the *data-organizing derivations* are carried here in full, because they are themselves

load-bearing and belong nowhere else: the parameter count via the breaking of the global flavor symmetry, the standard parametrizations of the CKM and PMNS matrices, the Wolfenstein expansion, the rephasing-invariant Jarlskog measure of CP violation, and the scheme conversions—pole mass versus $\overline{\text{MS}}$ mass, and the $\{\alpha, G_F, m_Z\}$ versus $\{g, g', v\}$ input dictionary—that let the body chapters' numbers be compared at all. These are computations, not tabulations, and they are done step by step.

This appendix also freezes the standing conventions that the rest of the book quietly assumes. The gauge group is written in the fixed order $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. The hypercharge normalization is $Q = T^3 + Y$, so that the lepton doublet carries $Y_L = -\frac{1}{2}$ and the Gell-Mann–Nishijima relation holds with no stray factor of two; the alternative $Y' = 2Y$ convention is recorded only to translate other sources, never used. The covariant-derivative sign convention is the one fixed in Chapter A, $D_\mu = \partial_\mu - igA^a_\mu T^a$ (and $D_\mu = \partial_\mu - iqA_\mu$ in the abelian case), so that the gauge-boson couplings carry their canonical signs throughout. Unless stated otherwise, every measured number quoted here is a mid-2020s Particle Data Group value, dated as such and reported with its uncertainty; where a quantity is scheme-dependent its scheme is named on the spot.

F.1 The Defining Data: Gauge Group and Representation Content

A quantum field theory is fixed, before any Lagrangian is written, by a small amount of group-theoretic data: a gauge group, a list of matter fields together with the representations they carry, and whatever scalars are needed to break the symmetry. Everything else—the kinetic terms, the interactions allowed by gauge invariance, the pattern of masses after symmetry breaking—is then forced, up to a finite list of couplings. The Standard Model is no exception, and this appendix is organized around that fact. We begin, as the construction itself does, with the defining data. The physics that makes this data the *right* data was established in Parts VII–IX: the gauge principle and non-abelian Yang–Mills structure in Chapter 25, electroweak unification and the hypercharge assignment in Chapter 31, the requirement that gauge anomalies cancel in Chapter 37, and the assembly of the complete theory in Chapter 33. None of that is reproved here. What this section does is collect the result of all of it in one unambiguous table, fix once and for all the conventions—factor ordering, hypercharge normalization, the sign of the electric-charge relation—that the rest of the appendix will use without comment, and verify by direct arithmetic two facts on which the consistency of the whole edifice rests: that the tabulated hypercharges reproduce the observed electric charges, and that the four gauge and gravitational anomaly sums vanish generation by generation.

F.1.1 The gauge group

Definition F.1 (Standard Model gauge group)**.** The Standard Model is a gauge theory with compact gauge group

$$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y, \tag{F.1}$$

written always in this factor order: the strong (color) factor $\mathrm{SU}(3)_c$ first, the weak-isospin factor $\mathrm{SU}(2)_L$ second, and the weak-hypercharge factor $\mathrm{U}(1)_Y$ third. The subscripts name the physical role of each factor and are part of the data: c for color, L because SU(2) acts only on left-handed fields, Y for hypercharge. A representation of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ is specified by a triple

$$(\, d_3,\, d_2,\, Y\,), \tag{F.2}$$

where d_3 is the dimension (or name) of the $\mathrm{SU}(3)_c$ representation, d_2 the dimension of the $\mathrm{SU}(2)_L$ representation, and $Y \in \mathbb{Q}$ the hypercharge, i.e. the (one-dimensional) charge under $\mathrm{U}(1)_Y$.

Three remarks fix the meaning of Eq. (F.2) precisely.

First, the color and weak factors are non-abelian, and we label their irreducible representations by dimension in boldface: **1** for the singlet (trivial representation), **2** for the fundamental doublet of SU(2), **3** for the fundamental triplet of SU(3), and $\overline{\mathbf{3}}$ for its complex conjugate, the antifundamental. The conjugate matters because SU(3) has inequivalent fundamental and antifundamental representations ($\mathbf{3} \not\cong \overline{\mathbf{3}}$), whereas SU(2) has $\mathbf{2} \cong \overline{\mathbf{2}}$ (it is pseudoreal), so no bar is ever needed on a weak-isospin label. The generators, Dynkin indices, and Casimirs of these representations live in Chapter B and are quoted from there when the anomaly arithmetic needs them.

Second, hypercharge is a continuous parameter only in the abstract: the requirement that the theory be anomaly-free, imported from Chapter 37, quantizes the allowed values to the rationals tabulated below. We treat Y as a fixed rational number attached to each multiplet.

Third—and this is the convention the entire appendix freezes—the normalization of $\mathrm{U}(1)_Y$ is set by the Gell-Mann–Nishijima relation in the form

$$Q \;=\; T^3 + Y, \tag{F.3}$$

where Q is the electric charge in units of the proton charge $|e|$, T^3 is the diagonal (third) generator of weak isospin $\mathrm{SU}(2)_L$ acting on the multiplet, and Y is the hypercharge of Eq. (F.2). There is no factor of two and no extra coefficient in Eq. (F.3): hypercharge is normalized so that it appears with unit coefficient. With this choice the left-handed lepton doublet carries $Y_{L_L} = -\frac{1}{2}$, the value the chapter introduction singles out as the convention check. Other texts use $Y' = 2Y$ (so that $Q = T^3 + Y'/2$); we never do, and any number quoted in this appendix uses Eq. (F.3) as written.

Notation Watch

A persistent source of factor-of-two confusion is the normalization of $\mathrm{U}(1)_Y$. Two conventions are in wide use:

$Q = T^3 + Y$ (this book, "small" hypercharge), $\qquad Q = T^3 + \frac{Y'}{2}$ ("large" hypercharge, Y'

In the small-hypercharge convention, Y equals the average electric charge of the multiplet (the T^3 values average to zero over a complete SU(2) multiplet), a fact we use repeatedly below. Throughout this appendix Y always means the small-hypercharge value of Eq. (F.3); numerically $Y_{L_L} = -\frac{1}{2}$, $Y_{Q_L} = +\frac{1}{6}$. When comparing with a source that tabulates Y', multiply by two.

The covariant derivative that gauges Eq. (F.1) uses the sign convention fixed in Chapter A, $D_\mu = \partial_\mu - ig_s A^a_\mu T^a - ig\, W^i_\mu T^i - ig' Y B_\mu$, with one coupling per simple factor (g_s, g, g'); writing it out multiplet by multiplet is the business of Section F.2 and is not anticipated here. What we need first is the list of multiplets it acts on.

F.1.2 Chirality and the choice of field basis

The Standard Model is a chiral gauge theory: left- and right-handed components of a Dirac fermion transform under *different* representations of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$. This is the single structural fact that makes the field content nontrivial to write down, and it forces a convention. A four-component Dirac field Ψ decomposes under the chirality projectors $P_{L,R} = \frac{1}{2}(1 \mp \gamma^5)$ (Chapter A) into $\psi_L = P_L \Psi$ and $\psi_R = P_R \Psi$. We assign gauge representations to these chiral halves separately. The physical fields and their handedness are:

- the left-handed quark doublet $Q_L = (u_L,\, d_L)^{\mathsf{T}}$, a weak doublet;
- the right-handed up-type quark u_R and right-handed down-type quark d_R, weak singlets;
- the left-handed lepton doublet $L_L = (\nu_L,\, e_L)^{\mathsf{T}}$, a weak doublet;
- the right-handed charged lepton e_R, a weak singlet.

There is no right-handed neutrino in the minimal Standard Model; we note this explicitly because its absence is what makes the neutrino exactly massless at the renormalizable level (the matter is taken up in Section F.2 and the parameter-count section).

For the anomaly bookkeeping of Section F.1.5 it is cleaner to put every fermion on the same footing. We use the standard device: a right-handed Weyl

field ψ_R in representation R with hypercharge Y is equivalent to a left-handed Weyl field $\psi_R^c \equiv (\psi_R)^c$ in the conjugate representation $\bar{R}$ with hypercharge $-Y$, because charge conjugation exchanges a representation for its complex conjugate and flips the sign of every additive charge. Thus the complete fermion content of one generation can be presented as five left-handed Weyl multiplets,

$$Q_L, \quad u_R^c, \quad d_R^c, \quad L_L, \quad e_R^c, \tag{F.4}$$

and the gauge anomaly is the totally symmetric trace of three generators taken over exactly this list. We give both presentations below: the physical Dirac-handedness table (Table F.1), which is what one reads when writing the Lagrangian, and its all-left-handed recasting (Table F.2), which is what one sums when checking consistency.

F.1.3 The fermion content over three generations

Definition F.2 (Fermion representation content)**.** One *generation* of Standard Model fermions consists of the five chiral multiplets

$$\begin{aligned} Q_L = \begin{pmatrix} u_L \\ d_L \end{pmatrix} &\sim (\mathbf{3}, \mathbf{2}, +\tfrac{1}{6}), \qquad u_R \sim (\mathbf{3}, \mathbf{1}, +\tfrac{2}{3}), \qquad d_R \sim (\mathbf{3}, \mathbf{1}, -\tfrac{1}{3}), \\ L_L = \begin{pmatrix} \nu_L \\ e_L \end{pmatrix} &\sim (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \qquad e_R \sim (\mathbf{1}, \mathbf{1}, -1). \end{aligned} \tag{F.5}$$

This content is replicated identically across three generations, $g = 1, 2, 3$, the generation label distinguishing the otherwise gauge-identical copies

$$(u, d, \nu_e, e), \qquad (c, s, \nu_\mu, \mu), \qquad (t, b, \nu_\tau, \tau). \tag{F.6}$$

The gauge interactions are blind to g: the three generations differ only through the Yukawa couplings introduced in Section F.2.

Counting Weyl degrees of freedom, one generation contains $Q_L\,(3 \times 2 = 6) + u_R\,(3) + d_R\,(3) + L_L\,(2) + e_R\,(1) = 15$ left-handed Weyl fermions; three generations give 45. The single scalar completes the data.

Definition F.3 (The scalar sector)**.** The Standard Model contains exactly one scalar multiplet, the Higgs doublet

$$H = \begin{pmatrix} \phi^+ \\ \phi^0 \end{pmatrix} \sim (\mathbf{1}, \mathbf{2}, +\tfrac{1}{2}), \tag{F.7}$$

a color singlet, weak doublet, hypercharge $+\frac{1}{2}$ complex scalar with four real components. It is not replicated across generations: there is one Higgs doublet, period. Its potential and the spontaneous breaking $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_{\mathrm{em}}$ are imported from Chapter 31; here the doublet enters only as another row in the quantum-number table.

The complete per-field table assembles Eqs. (F.5) and (F.7) and appends the columns that Eq. (F.3) determines. We read off T^3 from the weak-isospin assignment—$T^3 = \pm\frac{1}{2}$ on the upper/lower component of a doublet, $T^3 = 0$ on a singlet—and compute $Q = T^3 + Y$ component by component.

Table F.1: Standard Model field content of one generation (identical for $g = 1, 2, 3$) and the single Higgs doublet. Representations are written as $(\mathrm{SU}(3)_c, \mathrm{SU}(2)_L)_Y$ with the small-hypercharge normalization Eq. (F.3). The chirality column gives the handedness of the field as written; T^3 and Q are listed per component for multiplets. Electric charge Q is in units of $|e|$.

Field	Components	$\mathrm{SU}(3)_c$	$\mathrm{SU}(2)_L$	Y	T^3	$Q = T^3 + Y$
Q_L	u_L	**3**	**2**	$+\frac{1}{6}$	$+\frac{1}{2}$	$+\frac{2}{3}$
	d_L	**3**			$-\frac{1}{2}$	$-\frac{1}{3}$
u_R	u_R	**3**	**1**	$+\frac{2}{3}$	0	$+\frac{2}{3}$
d_R	d_R	**3**	**1**	$-\frac{1}{3}$	0	$-\frac{1}{3}$
L_L	ν_L	**1**	**2**	$-\frac{1}{2}$	$+\frac{1}{2}$	0
	e_L	**1**			$-\frac{1}{2}$	-1
e_R	e_R	**1**	**1**	-1	0	-1
H	ϕ^+	**1**	**2**	$+\frac{1}{2}$	$+\frac{1}{2}$	$+1$
	ϕ^0	**1**			$-\frac{1}{2}$	0

The chirality assignments not displayed in the table are those of Section F.1.2: Q_L and L_L are left-handed, u_R, d_R, e_R are right-handed, and H is a complex scalar (no chirality). The table is the central object of this appendix; every covariant derivative, mass term, and coupling in Section F.2 is built from it.

F.1.4 Worked check: hypercharges reproduce the observed charges

The hypercharges in Table F.1 are not free inputs to be fitted; they are fixed by demanding that Eq. (F.3) return the electric charges measured in the laboratory. We verify this directly, because it is the one place where an arbitrary-looking choice of fractions $(+\frac{1}{6}, +\frac{2}{3}, -\frac{1}{3}, \dots)$ is seen to be forced.

Worked Example F.4 (The Gell-Mann–Nishijima relation on every field)**.** We evaluate $Q = T^3 + Y$ for each component of each multiplet in Eqs. (F.5) and (F.7) and compare with the known charges of the quarks, leptons, and Higgs components.

Quark doublet Q_L, $Y = +\frac{1}{6}$. The upper component is the weak-isospin-up state, $T^3 = +\frac{1}{2}$:

$$Q(u_L) = T^3 + Y = +\tfrac{1}{2} + \tfrac{1}{6} = \tfrac{3}{6} + \tfrac{1}{6} = +\tfrac{2}{3}. \tag{F.8}$$

The lower component has $T^3 = -\frac{1}{2}$:

$$Q(d_L) = -\tfrac{1}{2} + \tfrac{1}{6} = -\tfrac{3}{6} + \tfrac{1}{6} = -\tfrac{2}{6} = -\tfrac{1}{3}. \tag{F.9}$$

These are the measured charges of the up and down quarks, $+\frac{2}{3}$ and $-\frac{1}{3}$.

Right-handed quarks, weak singlets ($T^3 = 0$). For u_R with $Y = +\frac{2}{3}$ and d_R with $Y = -\frac{1}{3}$,

$$Q(u_R) = 0 + \tfrac{2}{3} = +\tfrac{2}{3}, \qquad Q(d_R) = 0 - \tfrac{1}{3} = -\tfrac{1}{3}. \tag{F.10}$$

The right-handed charges match the left-handed ones, as they must: electric charge is vectorial, so Q does not depend on chirality. This matching is precisely what *fixes* $Y_{u_R} = Q(u_R) = +\frac{2}{3}$ and $Y_{d_R} = Q(d_R) = -\frac{1}{3}$ for a weak singlet, where $T^3 = 0$ leaves $Y = Q$.

Lepton doublet L_L, $Y = -\frac{1}{2}$.

$$Q(\nu_L) = +\tfrac{1}{2} - \tfrac{1}{2} = 0, \qquad Q(e_L) = -\tfrac{1}{2} - \tfrac{1}{2} = -1. \tag{F.11}$$

The neutrino is neutral and the electron carries charge -1; this is the line that pins $Y_{L_L} = -\frac{1}{2}$, the convention-defining value.

Right-handed electron, singlet. With $Y = -1$ and $T^3 = 0$,

$$Q(e_R) = 0 - 1 = -1, \tag{F.12}$$

again matching the left-handed electron, which forces $Y_{e_R} = -1$.

Higgs doublet, $Y = +\frac{1}{2}$.

$$Q(\phi^+) = +\tfrac{1}{2} + \tfrac{1}{2} = +1, \qquad Q(\phi^0) = -\tfrac{1}{2} + \tfrac{1}{2} = 0. \tag{F.13}$$

The upper component is the charged Higgs (eaten by the W^+) and the lower is the neutral one whose vacuum expectation value $\langle\phi^0\rangle$ leaves Q unbroken—which is the whole point of choosing $Y_H = +\frac{1}{2}$: the component that condenses must be electrically neutral, $Q(\phi^0) = 0$, or it would break $\mathrm{U}(1)_{\mathrm{em}}$.

Every entry reproduces the observed charge, confirming the Q column of Table F.1.

Physical Insight

The pattern in Theorem F.4 is not a coincidence to be memorized but a consequence of one fact: in the small-hypercharge normalization, Y *is* the average electric charge of the multiplet. Averaging $Q = T^3 + Y$ over a complete SU(2) multiplet kills the T^3 term (the T^3 eigenvalues of any irreducible SU(2) representation sum to zero), leaving $\langle Q\rangle = Y$. Thus $Y_{Q_L} = \frac{1}{2}(\frac{2}{3} - \frac{1}{3}) = +\frac{1}{6}$, $Y_{L_L} = \frac{1}{2}(0 - 1) = -\frac{1}{2}$, $Y_H = \frac{1}{2}(1 + 0) = +\frac{1}{2}$, and for the singlets $\langle Q\rangle = Q = Y$ trivially. This is the quickest way to read

a hypercharge off a charge assignment, and it is the reason the fractions take the values they do.

F.1.5 Anomaly cancellation as a consistency check

A chiral gauge theory is quantum-mechanically consistent only if its gauge anomalies cancel; the proof that they *must*, and the construction of the triangle-diagram coefficients that measure them, is the content of Chapter 37 and is not reproduced here. What we do here is exhibit the arithmetic: with the hypercharges of Table F.1 fixed, the relevant anomaly sums vanish, generation by generation, and we show each one.

The gauge anomaly is the totally symmetric trace $\mathcal{A}^{abc} \propto \sum_{\text{L Weyl}} \operatorname{tr}(T^a\{T^b, T^c\})$ taken over all left-handed Weyl fermions, with right-handed fields entering as their left-handed conjugates per Eq. (F.4); a left-handed antiparticle contributes with the opposite sign of a left-handed particle, which is automatic once we conjugate the representation and flip Y. For the gauge group Eq. (F.1) the independent potentially-anomalous combinations of three currents, together with the mixed gauge–gravitational one, are

$$[\mathrm{SU}(3)_c]^3, \quad [\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{U}(1)_Y]^3, \quad [\mathrm{grav}]^2\,\mathrm{U}(1)_Y, \quad [\mathrm{SU}(2)_L]^3. \tag{F.14}$$

Two of these vanish for group-theoretic reasons before any hypercharge is used, and we dispose of them first.

The $[\mathrm{SU}(2)_L]^3$ *anomaly vanishes identically.* SU(2) has no totally symmetric cubic invariant: $d^{abc} \equiv \operatorname{tr}(T^a\{T^b, T^c\}) = 0$ for every representation, because all SU(2) representations are pseudoreal and a pseudoreal representation has vanishing cubic anomaly (Chapter B). No contribution from any multiplet.

The $[\mathrm{SU}(3)_c]^3$ *anomaly vanishes because color is vectorial.* The color-charged left-handed Weyl fields are Q_L in $\mathbf{3}$ (with weak multiplicity 2) and u_R^c, d_R^c in $\overline{\mathbf{3}}$ (one each). The cubic SU(3) anomaly of $\overline{\mathbf{3}}$ is minus that of $\mathbf{3}$, so the color content is anomaly-free provided the number of $\mathbf{3}$'s equals the number of $\overline{\mathbf{3}}$'s. Counting: Q_L supplies two color triplets (one per weak component), while u_R^c and d_R^c supply one antitriplet each, two in all. Two triplets, two antitriplets: the anomaly cancels. Equivalently, the QCD content is left–right symmetric (vectorlike), and a vectorlike theory has no anomaly.

The remaining four sums in Eq. (F.14) involve hypercharge and are the nontrivial checks. We organize the arithmetic in Table F.2, which lists, for each of the five left-handed multiplets, its total multiplicity (color × weak dimension), its hypercharge, the relevant group-theory index, and the contribution to each sum. We use the index normalization of Chapter B: the Dynkin index is $T(\mathbf{2}) = \frac{1}{2}$ for the SU(2) doublet and $T(\mathbf{3}) = T(\overline{\mathbf{3}}) = \frac{1}{2}$ for the SU(3) (anti)triplet, and $T(\mathbf{1}) = 0$ for any singlet.

Table F.2: Anomaly bookkeeping for one generation, with every fermion written as a left-handed Weyl multiplet (Eq. (F.4)). Right-handed physical fields appear conjugated: $u_R^c \sim (\overline{\mathbf{3}}, \mathbf{1}, -\frac{2}{3})$, $d_R^c \sim (\overline{\mathbf{3}}, \mathbf{1}, +\frac{1}{3})$, $e_R^c \sim (\mathbf{1}, \mathbf{1}, +1)$. Column "mult." is the total number of Weyl components (color dimension × weak dimension). The four hypercharge-dependent anomaly contributions are tabulated; each column sums to zero.

Multiplet	rep $(\mathrm{SU}(3), \mathrm{SU}(2))_Y$	mult.	Y	$[\mathrm{SU}(3)]^2\mathrm{U}(1)$ $\sum T_3\, Y$	$[\mathrm{SU}(2)]^2\mathrm{U}(1)$ $\sum T_2\, Y$	$[\mathrm{U}(1)]^3$ $\sum Y^3$	$\mathrm{grav}^2\,\mathrm{U}($ $\sum$
Q_L	$(\mathbf{3}, \mathbf{2})_{+1/6}$	6	$+\frac{1}{6}$	$+\frac{1}{6}$	$+\frac{1}{4}$	$+\frac{1}{36}$	+
u_R^c	$(\overline{\mathbf{3}}, \mathbf{1})_{-2/3}$	3	$-\frac{2}{3}$	$-\frac{1}{3}$	0	$-\frac{8}{9}$	−
d_R^c	$(\overline{\mathbf{3}}, \mathbf{1})_{+1/3}$	3	$+\frac{1}{3}$	$+\frac{1}{6}$	0	$+\frac{1}{9}$	+
L_L	$(\mathbf{1}, \mathbf{2})_{-1/2}$	2	$-\frac{1}{2}$	0	$-\frac{1}{4}$	$-\frac{1}{4}$	−
e_R^c	$(\mathbf{1}, \mathbf{1})_{+1}$	1	$+1$	0	0	$+1$	+
Sum				**0**	**0**	**0**	

Each column of Table F.2 is a separate consistency condition; we now derive each entry and confirm the vanishing sum.

$[\mathrm{SU}(3)_c]^2\,\mathrm{U}(1)_Y$. Only color-charged fields contribute. The contribution of a multiplet is its SU(3) Dynkin index times its hypercharge, summed over the weak components. For the SU(3) index, both $\mathbf{3}$ and $\overline{\mathbf{3}}$ give $\frac{1}{2}$, and the weak doublet Q_L counts twice:

$$\begin{aligned} Q_L: &\quad 2 \times \tfrac{1}{2} \times \left(+\tfrac{1}{6}\right) = +\tfrac{1}{6}, \\ u_R^c: &\quad 1 \times \tfrac{1}{2} \times \left(-\tfrac{2}{3}\right) = -\tfrac{1}{3}, \\ d_R^c: &\quad 1 \times \tfrac{1}{2} \times \left(+\tfrac{1}{3}\right) = +\tfrac{1}{6}, \\ \text{sum}: &\quad \tfrac{1}{6} - \tfrac{1}{3} + \tfrac{1}{6} = \tfrac{1}{6} - \tfrac{2}{6} + \tfrac{1}{6} = 0. \end{aligned} \tag{F.15}$$

$[\mathrm{SU}(2)_L]^2\,\mathrm{U}(1)_Y$. Only weak doublets contribute (singlets have SU(2) index zero). The contribution is the SU(2) Dynkin index $\frac{1}{2}$ times the hypercharge, with color multiplicity in front. Q_L is a color triplet (multiplicity 3); L_L a color singlet:

$$\begin{aligned} Q_L: &\quad 3 \times \tfrac{1}{2} \times \left(+\tfrac{1}{6}\right) = +\tfrac{1}{4}, \\ L_L: &\quad 1 \times \tfrac{1}{2} \times \left(-\tfrac{1}{2}\right) = -\tfrac{1}{4}, \\ \text{sum}: &\quad \tfrac{1}{4} - \tfrac{1}{4} = 0. \end{aligned} \tag{F.16}$$

$[\mathrm{U}(1)_Y]^3$**.** Every field contributes Y^3 weighted by its full multiplicity (color $\times$ weak):

$$\begin{aligned} Q_L: &\quad 6 \times \left(\tfrac{1}{6}\right)^3 = \tfrac{6}{216} = \tfrac{1}{36}, \\ u_R^c: &\quad 3 \times \left(-\tfrac{2}{3}\right)^3 = 3 \times \left(-\tfrac{8}{27}\right) = -\tfrac{8}{9}, \\ d_R^c: &\quad 3 \times \left(\tfrac{1}{3}\right)^3 = \tfrac{3}{27} = \tfrac{1}{9}, \\ L_L: &\quad 2 \times \left(-\tfrac{1}{2}\right)^3 = 2 \times \left(-\tfrac{1}{8}\right) = -\tfrac{1}{4}, \\ e_R^c: &\quad 1 \times (1)^3 = 1. \end{aligned} \tag{F.17}$$

Putting these over a common denominator of 36,

$$\tfrac{1}{36} - \tfrac{8}{9} + \tfrac{1}{9} - \tfrac{1}{4} + 1 = \frac{1 - 32 + 4 - 9 + 36}{36} = \frac{0}{36} = 0. \tag{F.18}$$

$[\mathrm{grav}]^2\,\mathrm{U}(1)_Y$**.** The mixed gauge–gravitational anomaly (two gravitons, one hypercharge current) is proportional to the sum of hypercharges over all left-handed Weyl fermions, weighted by multiplicity—the trace of Y:

$$\begin{aligned} &Q_L: 6 \times \left(\tfrac{1}{6}\right) = 1, \qquad && u_R^c: 3 \times \left(-\tfrac{2}{3}\right) = -2, \qquad && d_R^c: 3 \times \left(\tfrac{1}{3}\right) = 1, \\ &L_L: 2 \times \left(-\tfrac{1}{2}\right) = -1, && e_R^c: 1 \times (1) = 1, \\ &\text{sum}: 1 - 2 + 1 - 1 + 1 = 0. \end{aligned} \tag{F.19}$$

All four hypercharge-dependent anomaly sums vanish, and the two pure non-abelian sums vanished by representation theory. The single generation of Table F.1 is therefore gauge-consistent; since the content is replicated identically across the three generations of Eq. (F.6) and the anomaly is linear in the number of generations, the full Standard Model fermion content is anomaly-free. This is the arithmetic the chapter introduction promised the appendix would exhibit; the reason these particular combinations are the ones required to vanish, and the proof that their vanishing is necessary for a consistent quantum theory, are imported from Chapter 37.

Physical Insight

The four cancellations are not independent miracles. The mixed $[\mathrm{grav}]^2\,\mathrm{U}(1)$ and the $[\mathrm{SU}(2)]^2\mathrm{U}(1)$ conditions, together with the requirement that each multiplet have integer or familiar fractional charges, are enough to fix the relative hypercharges of one generation up to overall normalization; the $[\mathrm{U}(1)]^3$ and $[\mathrm{SU}(3)]^2\mathrm{U}(1)$ conditions are then automatically satisfied. In this precise sense the hypercharge assignments of Table F.1 are not chosen to fit the data—they are forced by demanding a consistent chiral gauge theory with this gauge group and this **3**/**2**/**1** representation

skeleton. That the charges so determined are exactly the ones measured (Theorem F.4) is the deep consistency the Standard Model passes.

With the gauge group fixed (Theorem F.1), the field content tabulated (Table F.1), the hypercharge normalization frozen by Eq. (F.3), and the data shown to be gauge-consistent, we have everything the construction of the Lagrangian requires. The next section turns each row of the representation table into an explicit gauge-covariant derivative—one coupling and one set of generators per multiplet—and from there writes the complete Standard Model Lagrangian term by term.

F.2 The Complete Lagrangian

The previous section fixed the model's defining data: the gauge group $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, the chiral fermion content replicated over three generations, the single complex scalar doublet, and the hypercharge normalization $Q = T^3 + Y$ (so that $Y_L = -\frac{1}{2}$ for the lepton doublet) recorded in the per-field table of Section F.1. That table is a list of representations; it is not yet a theory. A theory is a Lagrangian, and the Standard Model is precisely the most general renormalizable Lagrangian that one can write from those representations consistent with the gauge symmetry and with Lorentz invariance. This section writes it out in full, term by term, with every field strength, every covariant derivative, every coupling and every sign fixed unambiguously, so that any cross section or running coupling computed anywhere in this book can be traced back to a definite line below.

Nothing physical is proved here. The gauge principle that forces minimal coupling was established for the abelian case in Chapter 8 and for the non-abelian case in Chapter 25; the Higgs mechanism that will give these fields masses was worked in Chapter 30; the electroweak rotation to mass eigenstates was carried out in Chapter 31; the flavor structure of the Yukawa couplings was analyzed in Chapter 32; and the complete model was assembled in Chapter 33. What this section owes is organizational, not physical: an unambiguous transcription, in one place, with the conventions of this book, of every operator in

$$\mathcal{L}_{\mathrm{SM}} = \mathcal{L}_{\mathrm{gauge}} + \mathcal{L}_{\theta} + \mathcal{L}_{\mathrm{fermion}} + \mathcal{L}_{\mathrm{Higgs}} + \mathcal{L}_{\mathrm{Yukawa}} + \mathcal{L}_{\mathrm{gf}} + \mathcal{L}_{\mathrm{ghost}}, \tag{F.20}$$

together with two derivations that are genuinely load-bearing and belong nowhere else: the explicit expansion of the covariant derivative on a chiral multiplet, which exhibits how the photon, $W^{\pm}$ and Z couplings emerge from the data above (consolidating QED's $U(1)_{\mathrm{em}}$ and Yang–Mills theory's QCD term into one object), and the verification that the scalar potential written below is the unique renormalizable gauge invariant available. We treat the seven pieces of Eq. (F.20) in turn.

Throughout we use the mostly-minus metric $\eta_{\mu\nu} = \mathrm{diag}(+,-,-,-)$, natural units $\hbar = c = 1$, the gamma-matrix conventions of Chapter C, and the covariant-derivative sign convention of Chapter 25: a field Φ in a representation R of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ has

$$D_\mu \Phi = \Big(\partial_\mu - i g_s\, G_\mu^A\, T_{(R)}^A - i g\, W_\mu^I\, T_{(R)}^I - i g'\, Y_\Phi\, B_\mu\Big)\Phi, \tag{F.21}$$

where $T_{(R)}^A$ ($A = 1,\dots,8$) are the $\mathrm{SU}(3)_c$ generators in the representation R, $T_{(R)}^I$ ($I = 1,2,3$) the $\mathrm{SU}(2)_L$ generators, Y_Φ the hypercharge, and g_s, g, g' the three gauge couplings. On a color triplet $T_{(R)}^A = \lambda^A/2$ with λ^A the Gell-Mann matrices (Chapter B); on a color singlet $T_{(R)}^A = 0$. On a weak doublet $T_{(R)}^I = \sigma^I/2$ with σ^I the Pauli matrices; on a weak singlet $T_{(R)}^I = 0$. The single object Eq. (F.21) encodes every gauge coupling in the model once we specialize it to each multiplet; doing so explicitly is the content of Section F.2.2.

F.2.1 The gauge sector and the theta term

To each simple (or abelian) factor of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ belongs a gauge field and its field strength. For $\mathrm{SU}(3)_c$ we have the eight gluons G_μ^A with the non-abelian field strength

$$G_{\mu\nu}^A = \partial_\mu G_\nu^A - \partial_\nu G_\mu^A + g_s\, f^{ABC}\, G_\mu^B G_\nu^C, \qquad A,B,C = 1,\dots,8, \tag{F.22}$$

where f^{ABC} are the SU(3) structure constants of Chapter B. For $\mathrm{SU}(2)_L$ we have the three weak isospin bosons W_μ^I,

$$W_{\mu\nu}^I = \partial_\mu W_\nu^I - \partial_\nu W_\mu^I + g\, \epsilon^{IJK}\, W_\mu^J W_\nu^K, \qquad I,J,K = 1,2,3, \tag{F.23}$$

with ϵ^{IJK} the totally antisymmetric symbol that serves as the SU(2) structure constants. For $\mathrm{U}(1)_Y$ we have the single hypercharge boson B_μ with the abelian field strength

$$B_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu, \tag{F.24}$$

which carries no self-coupling because $\mathrm{U}(1)_Y$ is abelian. These are curvatures in the precise sense of Chapter 25: $G_{\mu\nu} = \frac{i}{g_s}[D_\mu, D_\nu]$ on the relevant representation, and likewise for the others. The gauge kinetic Lagrangian is the unique dimension-four, gauge-invariant, Lorentz-invariant, parity-even quadratic in these curvatures, normalized as fixed in Chapter 25:

$$\mathcal{L}_{\text{gauge}} = -\tfrac{1}{4}\, G_{\mu\nu}^A G^{A\,\mu\nu} - \tfrac{1}{4}\, W_{\mu\nu}^I W^{I\,\mu\nu} - \tfrac{1}{4}\, B_{\mu\nu} B^{\mu\nu}, \tag{F.25}$$

with the gauge indices summed (these are Cartan-Killing-orthonormal in the normalization $\mathrm{tr}(T^a T^b) = \frac{1}{2}\delta^{ab}$ of Chapter B). Expanding Eq. (F.25) with Eqs. (F.22) to (F.24) reproduces the cubic and quartic gluon and W self-interactions whose Feynman rules were derived in Chapter 26; we do not re-expand them here.

There is one further dimension-four gauge invariant that is Lorentz scalar but *parity- and time-reversal-odd*: the contraction of a field strength with its dual. For QCD this is the theta term,

$$\mathcal{L}_\theta = \frac{g_s^2\,\theta}{32\pi^2}\,G^A_{\mu\nu}\widetilde{G}^{A\,\mu\nu}, \qquad \widetilde{G}^{A\,\mu\nu} \equiv \tfrac{1}{2}\,\epsilon^{\mu\nu\rho\sigma}G^A_{\rho\sigma}, \tag{F.26}$$

with the orientation $\epsilon^{0123} = +1$. As shown in Chapter 42, $G\widetilde{G}$ is a total derivative, so it does not affect the classical equations of motion or any order of perturbation theory; it contributes only through non-perturbative (instanton) configurations, and the prefactor $g_s^2/32\pi^2$ is chosen so that θ shifts by 2π under a large gauge transformation, making the physics periodic in θ. The analogous $\mathrm{SU}(2)_L$ theta term can be rotated away using the chiral fermion content (baryon-plus-lepton-number rotation) and is conventionally set to zero; the $\mathrm{U}(1)_Y$ "theta term" $B\widetilde{B}$ is itself a total derivative with trivial topology and is dropped. Thus the only physical theta angle in the Standard Model is the QCD θ of Eq. (F.26), and the smallness of its measured value is the strong-CP problem flagged in Chapters 35 and 42. We retain it in Eq. (F.20) for completeness; the parameter count of Section F.4 includes it.

Notation Watch

The field-strength definitions Eqs. (F.22) to (F.24) carry the coupling *inside* the curvature, matching the covariant derivative Eq. (F.21) (the "g-inside" convention of Chapter 25). Some references instead rescale $A \to A/g$ so the coupling appears only as an overall $1/g^2$ in front of the kinetic term. The two are related by a field redefinition and give identical physics, but Feynman rules, the sign in D_μ, and the normalization of Eq. (F.26) all change. This book uses the g-inside convention everywhere; do not mix.

F.2.2 Gauge-covariant fermion kinetic terms

The fermion content of Section F.1, repeated for each generation $i = 1, 2, 3$, consists of five chiral multiplets: the left-handed quark doublet $Q_{Li} = (u_{Li}, d_{Li})^{\mathsf{T}}$ in $(\mathbf{3}, \mathbf{2}, +\frac{1}{6})$, the right-handed up- and down-type quark singlets u_{Ri} in $(\mathbf{3}, \mathbf{1}, +\frac{2}{3})$ and d_{Ri} in $(\mathbf{3}, \mathbf{1}, -\frac{1}{3})$, the left-handed lepton doublet $L_{Li} = (\nu_{Li}, e_{Li})^{\mathsf{T}}$ in $(\mathbf{1}, \mathbf{2}, -\frac{1}{2})$, and the right-handed charged-lepton singlet e_{Ri} in $(\mathbf{1}, \mathbf{1}, -1)$. There is no right-handed neutrino in the minimal model. Each multiplet enters through the gauge-covariant Dirac kinetic term $\bar{\psi}\, i\not{D}\,\psi$, with $\not{D} = \gamma^\mu D_\mu$ and D_μ the specialization of Eq. (F.21) to that multiplet's representation. Summing over the five multiplets and over the generation index i (summation implied),

$$\mathcal{L}_{\text{fermion}} = \bar{Q}_{Li}\, i\not{D}\, Q_{Li} + \bar{u}_{Ri}\, i\not{D}\, u_{Ri} + \bar{d}_{Ri}\, i\not{D}\, d_{Ri} + \bar{L}_{Li}\, i\not{D}\, L_{Li} + \bar{e}_{Ri}\, i\not{D}\, e_{Ri}. \tag{F.27}$$

The covariant derivatives differ from line to line only through which generators act and through the hypercharge; making each explicit:

$$D_\mu Q_{Li} = \Big(\partial_\mu - i g_s\, G_\mu^A \tfrac{\lambda^A}{2} - i g\, W_\mu^I \tfrac{\sigma^I}{2} - i g'\, \tfrac{1}{6}\, B_\mu\Big) Q_{Li}, \tag{F.28}$$

$$D_\mu u_{Ri} = \Big(\partial_\mu - i g_s\, G_\mu^A \tfrac{\lambda^A}{2} - i g'\, \tfrac{2}{3}\, B_\mu\Big) u_{Ri}, \tag{F.29}$$

$$D_\mu d_{Ri} = \Big(\partial_\mu - i g_s\, G_\mu^A \tfrac{\lambda^A}{2} - i g'\, (-\tfrac{1}{3})\, B_\mu\Big) d_{Ri}, \tag{F.30}$$

$$D_\mu L_{Li} = \Big(\partial_\mu - i g\, W_\mu^I \tfrac{\sigma^I}{2} - i g'\, (-\tfrac{1}{2})\, B_\mu\Big) L_{Li}, \tag{F.31}$$

$$D_\mu e_{Ri} = \Big(\partial_\mu - i g'\, (-1)\, B_\mu\Big) e_{Ri}. \tag{F.32}$$

The Higgs doublet, with the same isospin and a hypercharge $Y_H = +\tfrac{1}{2}$, carries

$$D_\mu H = \Big(\partial_\mu - i g\, W_\mu^I \tfrac{\sigma^I}{2} - i g'\, \tfrac{1}{2}\, B_\mu\Big) H, \tag{F.33}$$

which we will need in Section F.2.3. Color singlets (L_L, e_R, H) carry no gluon term; weak singlets (u_R, d_R, e_R) carry no W term; every multiplet couples to B_μ through its hypercharge, the right-handed singlets coupling to B_μ *only*.

Two structural facts about Eq. (F.27) deserve to be stated plainly, because they are the reason the Standard Model has the shape it does. First, the theory is *chiral*: left- and right-handed components of a given flavor sit in different representations of $\mathrm{SU}(2)_L$ and carry different hypercharges. Consequently a Dirac mass term $m\,\bar\psi\psi = m(\bar\psi_L\psi_R + \bar\psi_R\psi_L)$ is *not* gauge invariant for any Standard Model fermion: it would pair, say, the doublet Q_L with the singlet u_R into an object transforming as $(\mathbf{2})$ under $\mathrm{SU}(2)_L$ with net hypercharge $-\tfrac{1}{6} + \tfrac{2}{3} = +\tfrac{1}{2} \neq 0$, which is not a singlet. Fermion masses are therefore forbidden in Eq. (F.27) and can only arise after electroweak breaking, through the Yukawa couplings of Section F.2.4. Second, the kinetic term is, before the Yukawa couplings are switched on, invariant under independent unitary rotations of the three generations within each of the five multiplets—a $\mathrm{U}(3)^5$ global flavor symmetry. That symmetry is exactly what the Yukawa matrices break, and counting its breaking is how the flavor parameters are enumerated in Section F.4.

That the same covariant derivative Eq. (F.21) simultaneously produces the QCD coupling of Yang–Mills theory and, after electroweak rotation, the $U(1)_{\mathrm{em}}$ coupling of QED is best seen by expanding it on a single multiplet. We do this in full for the lepton doublet, where there is no color to clutter the algebra, and then read off the modifications for a quark doublet.

Worked Example F.5 (The neutral and charged currents from $D_\mu L_L$)**.** Take the left-handed lepton doublet $L_L = (\nu_L, e_L)^{\mathsf T}$ with $Y = -\tfrac{1}{2}$, drop the generation index, and read the gauge interactions off the kinetic term $\bar L_L\, i\not{D}\, L_L$. Writing $\not{D} = \not{\partial} - i g\, \not{W}^I \tfrac{\sigma^I}{2} - i g' Y \not{B}$, the interaction piece is

$$\mathcal{L}_L^{\mathrm{int}} = \bar L_L\, \gamma^\mu \Big(g\, W_\mu^I \tfrac{\sigma^I}{2} + g'\, Y\, B_\mu\Big) L_L, \qquad Y = -\tfrac{1}{2}, \tag{F.34}$$

the sign flipping relative to D_μ because $i\not{D}$ multiplies the $-i$ in the derivative to $+1$. Split the $\mathrm{SU}(2)_L$ part into its diagonal ($I=3$) and off-diagonal ($I=1,2$) pieces. The off-diagonal part is

$$\tfrac{g}{2}(W_\mu^1\sigma^1+W_\mu^2\sigma^2)=\frac{g}{2}\begin{pmatrix}0 & W_\mu^1-iW_\mu^2\\ W_\mu^1+iW_\mu^2 & 0\end{pmatrix}. \tag{F.35}$$

Introduce the charged eigenstates of definite electric charge, exactly as in Chapter 31,

$$W_\mu^\pm=\frac{1}{\sqrt{2}}(W_\mu^1\mp iW_\mu^2),\qquad\Longrightarrow\qquad W_\mu^1-iW_\mu^2=\sqrt{2}\,W_\mu^+,\quad W_\mu^1+iW_\mu^2=\sqrt{2}\,W_\mu^-, \tag{F.36}$$

so the off-diagonal block becomes $\frac{g}{\sqrt{2}}\left(\begin{smallmatrix}0 & W^+\\ W^- & 0\end{smallmatrix}\right)$. Sandwiched between $\bar L_L\gamma^\mu(\cdots)L_L$ this yields the *charged-current* interaction

$$\mathcal{L}_L^{\mathrm{CC}}=\frac{g}{\sqrt{2}}\Big(\bar\nu_L\gamma^\mu e_L\,W_\mu^+ +\bar e_L\gamma^\mu\nu_L\,W_\mu^-\Big), \tag{F.37}$$

the $V-A$ structure (left-handed fields only) being automatic because L_L is purely left-chiral. The coupling strength is $g/\sqrt{2}$, which is the origin of the Fermi constant relation recorded in Section F.3.

The diagonal, neutral part is $\bar L_L\gamma^\mu(g\,W_\mu^3\frac{\sigma^3}{2}+g'YB_\mu)L_L$. With $\sigma^3=\mathrm{diag}(+1,-1)$ acting as $T^3=\pm\frac12$ on (ν_L,e_L) and $Y=-\frac12$ common to both, the neutrino and electron pieces are

$$\nu_L:\quad \tfrac12(g\,W_\mu^3-g'\,B_\mu), \tag{F.38}$$

$$e_L:\quad -\tfrac12(g\,W_\mu^3+g'\,B_\mu). \tag{F.39}$$

Now rotate to the mass eigenstates A_μ (photon) and Z_μ, importing the EWSB rotation from Chapter 31 (we do *not* re-derive it):

$$\begin{pmatrix}W_\mu^3\\ B_\mu\end{pmatrix}=\begin{pmatrix}\cos\theta_{\mathrm W} & \sin\theta_{\mathrm W}\\ -\sin\theta_{\mathrm W} & \cos\theta_{\mathrm W}\end{pmatrix}\begin{pmatrix}Z_\mu\\ A_\mu\end{pmatrix},\qquad \tan\theta_{\mathrm W}=\frac{g'}{g},\quad e=g\sin\theta_{\mathrm W}=g'\cos\theta_{\mathrm W}. \tag{F.40}$$

Substitute into Eq. (F.38). The neutrino's photon coupling is proportional to $g\sin\theta_{\mathrm W}-g'\cos\theta_{\mathrm W}=e-e=0$: the neutrino is electrically neutral, as it must be since $Q_\nu=T^3+Y=\frac12-\frac12=0$. Its Z coupling is

$$\tfrac12(g\cos\theta_{\mathrm W}+g'\sin\theta_{\mathrm W})=\tfrac12\,\frac{g\cos^2\theta_{\mathrm W}+g'\sin\theta_{\mathrm W}\cos\theta_{\mathrm W}}{\cos\theta_{\mathrm W}}=\tfrac12\,\frac{g(\cos^2\theta_{\mathrm W}+\sin^2\theta_{\mathrm W})}{\cos\theta_{\mathrm W}}=\frac{g}{2\cos\theta_{\mathrm W}}, \tag{F.41}$$

using $g'\sin\theta_{\mathrm W}=g\,\sin^2\theta_{\mathrm W}/\cos\theta_{\mathrm W}$. For the electron, Eq. (F.39) gives a photon coupling $-\frac12(g\sin\theta_{\mathrm W}+g'\cos\theta_{\mathrm W})=-\frac12(e+e)=-e$, the correct electromagnetic charge $Q_{e_L}=T^3+Y=-\frac12-\frac12=-1$ times e; and a Z coupling

$$-\tfrac12(g\cos\theta_{\mathrm W}-g'\sin\theta_{\mathrm W})=-\tfrac{g}{2\cos\theta_{\mathrm W}}(\cos^2\theta_{\mathrm W}-\sin^2\theta_{\mathrm W})=\frac{g}{\cos\theta_{\mathrm W}}\Big(-\tfrac12+\sin^2\theta_{\mathrm W}\Big). \tag{F.42}$$

Both neutral results fit the universal pattern

$$\mathcal{L}^{\mathrm{NC}} = e\,Q\,\bar{\psi}\gamma^{\mu}\psi\,A_{\mu} + \frac{g}{\cos\theta_{\mathrm{W}}}(T^{3} - Q\,\sin^{2}\theta_{\mathrm{W}})\,\bar{\psi}\gamma^{\mu}\psi\,Z_{\mu}, \tag{F.43}$$

which one verifies term by term: for ν_L, $Q=0$, $T^3=\frac{1}{2}$ give photon coupling 0 and Z coupling $g/2\cos\theta_{\mathrm{W}}$; for e_L, $Q=-1$, $T^3=-\frac{1}{2}$ give photon coupling $-e$ and Z coupling $\frac{g}{\cos\theta_{\mathrm{W}}}(-\frac{1}{2}+\sin^2\theta_{\mathrm{W}})$, both matching. The photon thus couples strictly to the electric charge $Q = T^3 + Y$—this is QED's $U(1)_{\mathrm{em}}$, now seen to be the unbroken combination $A_\mu = \sin\theta_{\mathrm{W}}\,W^3_\mu + \cos\theta_{\mathrm{W}}\,B_\mu$ rather than B_μ itself—while the orthogonal combination Z_μ couples to $T^3 - Q\,\sin^2\theta_{\mathrm{W}}$. Every neutral-current coupling in the Standard Model follows from Eq. (F.43) by inserting the relevant (T^3, Q).

Worked Example F.6 (The quark doublet: color plus the same electroweak structure)**.** For the left-handed quark doublet $Q_L = (u_L, d_L)^{\mathsf{T}}$ with $Y = +\frac{1}{6}$, the covariant derivative Eq. (F.28) differs from the lepton case in two ways: it carries the gluon term $-ig_s G^A_\mu \lambda^A/2$, and the hypercharge is $+\frac{1}{6}$ rather than $-\frac{1}{2}$. The gluon term produces, in $\bar{Q}_L i\not{D} Q_L$, exactly the QCD quark–gluon vertex of Yang–Mills theory,

$$\mathcal{L}^{\mathrm{QCD}}_{Q} = g_s\,\bar{Q}_L\gamma^{\mu}\tfrac{\lambda^A}{2}Q_L\,G^A_\mu, \tag{F.44}$$

color-diagonal in the weak indices and identical for u_L and d_L; the same term appears for u_R, d_R from Eqs. (F.29) and (F.30), so QCD is vectorlike and flavor-diagonal, as established in Chapter 27. The charged-current piece is identical in form to Eq. (F.37) with $(\nu_L, e_L) \to (u_L, d_L)$,

$$\mathcal{L}^{\mathrm{CC}}_{Q} = \frac{g}{\sqrt{2}}(\bar{u}_L\gamma^{\mu}d_L\,W^{+}_{\mu} + \bar{d}_L\gamma^{\mu}u_L\,W^{-}_{\mu}), \tag{F.45}$$

the flavor rotation that turns these gauge eigenstates into mass eigenstates and produces the CKM matrix being deferred to Section F.3 and proved in Chapter 32. For the neutral currents, the diagonal couplings again take the universal form Eq. (F.43), now with the quark charges fixed by $Q = T^3 + Y$ with $Y = +\frac{1}{6}$:

$$Q_{u_L} = +\tfrac{1}{2} + \tfrac{1}{6} = +\tfrac{2}{3}, \qquad Q_{d_L} = -\tfrac{1}{2} + \tfrac{1}{6} = -\tfrac{1}{3}, \tag{F.46}$$

so the photon couples to u_L with strength $+\frac{2}{3}e$ and to d_L with $-\frac{1}{3}e$, the familiar quark charges. The right-handed quarks u_R, d_R are weak singlets ($T^3 = 0$), so Eq. (F.43) gives them photon couplings $+\frac{2}{3}e, -\frac{1}{3}e$ as required and Z couplings $-\frac{g}{\cos\theta_{\mathrm{W}}}Q\sin^2\theta_{\mathrm{W}}$; their hypercharges $+\frac{2}{3}, -\frac{1}{3}$ were chosen in Section F.1 precisely so that $Q = Y$ for these singlets, ensuring the electric charges of u_L and u_R (and of d_L, d_R) agree, as a consistent Dirac fermion demands.

F.2.3 The Higgs sector

The single scalar is the complex doublet H in $(\mathbf{1},\mathbf{2},+\frac{1}{2})$, with components $H=(H^+,H^0)^{\mathsf{T}}$. Its dynamics are the gauge-covariant kinetic term built from Eq. (F.33) together with the scalar potential:

$$\mathcal{L}_{\text{Higgs}}=(D_\mu H)^\dagger(D^\mu H)-V(H),\qquad V(H)=-\mu^2\,H^\dagger H+\lambda\,(H^\dagger H)^2,\quad\text{(F.47)}$$

with $\mu^2>0$ and $\lambda>0$. The sign of the quadratic term is the crucial one: with $-\mu^2 H^\dagger H$ and $\mu^2>0$ the origin $H=0$ is a maximum, the potential is minimized on the sphere $H^\dagger H=\mu^2/2\lambda\equiv v^2/2$, and electroweak symmetry breaks spontaneously, as worked in Chapters 30 and 31. We retain the symbols μ^2,λ,v from those chapters; the relation $v^2=\mu^2/\lambda$ and the physical Higgs mass $m_h^2=2\lambda v^2=2\mu^2$ are recorded in Section F.3. That $\lambda>0$ is required for the potential to be bounded below (so that a stable vacuum exists at all) and is what makes the quartic the highest term allowed.

It is worth verifying explicitly that Eq. (F.47) is not a choice among several renormalizable possibilities but the *only* one, given the field content.

Proposition F.7 (Uniqueness of the Higgs potential)**.** *The most general Lorentz-invariant,* $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$*-invariant scalar potential of operator dimension* ≤ 4 *built from the single doublet* $H\in(\mathbf{1},\mathbf{2},+\frac{1}{2})$ *is, up to an additive constant,*

$$V(H)=-\mu^2\,H^\dagger H+\lambda\,(H^\dagger H)^2.$$

Proof. Renormalizability in four dimensions restricts the potential to a polynomial in the fields of total operator dimension at most four; a scalar field has dimension one, so V is a polynomial of degree at most four in the components of H and $H^\dagger$. Gauge invariance requires every term to be a singlet of $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$. The doublet H is a color singlet, so $\mathrm{SU}(3)_c$ is automatic. Under $\mathrm{SU}(2)_L$, H is a $\mathbf{2}$ and $H^\dagger$ a $\bar{\mathbf{2}}$; under $\mathrm{U}(1)_Y$, H carries $Y=+\frac{1}{2}$ and $H^\dagger$ carries $-\frac{1}{2}$. We enumerate by degree.

Degree 0: a constant; physically irrelevant (it shifts the vacuum energy), dropped.

Degree 1: a term linear in H or $H^\dagger$ carries $Y=\pm\frac{1}{2}\neq 0$ and transforms as a doublet, never a singlet. None.

Degree 2: the available bilinears are $H^\dagger H$, $H^{\mathsf{T}}H$, $H^\dagger H^\dagger$ and their conjugates. The combination $H^\dagger H=|H^+|^2+|H^0|^2$ is the $\mathrm{SU}(2)_L$ singlet $\bar{\mathbf{2}}\otimes\mathbf{2}\supset\mathbf{1}$ with $Y=-\frac{1}{2}+\frac{1}{2}=0$: invariant. Could one instead contract two H's with the antisymmetric $\mathrm{SU}(2)$ invariant ϵ_{ab}, forming $\epsilon_{ab}H^aH^b$? This is the only $\mathrm{SU}(2)$-singlet bilinear in two doublets, but it is antisymmetric in the two factors, which are the *same* field, so $\epsilon_{ab}H^aH^b=0$ identically; and even were it nonzero it would carry $Y=+1\neq 0$. Hence the unique degree-two invariant is $H^\dagger H$, with real coefficient (hermiticity of V), which we call $-\mu^2$.

Degree 3: an odd polynomial in $H, H^\dagger$ cannot have net hypercharge zero with an odd number of fields unless the hypercharges cancel, but with three doublets the hypercharge is half-integer-odd ($\pm\frac{1}{2}$ summed an odd number of times gives a half-odd-integer, never zero). None.

Degree 4: the only invariant is built from two factors of the unique degree-two invariant, $(H^\dagger H)^2$, with real coefficient λ. Any other degree-four contraction either fails to be an $\mathrm{SU}(2)_L$ singlet (e.g. uses an uncontracted isospin index) or fails hypercharge neutrality (e.g. $(H^\dagger H)(H^{\mathsf{T}}\epsilon H)$ carries $Y=+1$, and $H^{\mathsf{T}}\epsilon H = 0$ anyway); the genuinely independent quartic Casimir of a single doublet collapses to $(H^\dagger H)^2$.

Collecting the surviving terms gives $V = -\mu^2\, H^\dagger H + \lambda\,(H^\dagger H)^2$ up to the dropped constant, with μ^2, λ real, as claimed. □

The conjugate doublet that the Yukawa sector will need is fixed here for definiteness. Because SU(2) is pseudoreal, the antisymmetric tensor $\epsilon_{ab} = i(\sigma^2)_{ab}$ maps the $\bar{\mathbf{2}}$ back to a $\mathbf{2}$; applied to H^* it produces a doublet of *opposite* hypercharge,

$$\widetilde{H} \equiv i\sigma^2 H^* = \begin{pmatrix} H^{0\,*} \\ -H^- \end{pmatrix}, \qquad \widetilde{H} \in (\mathbf{1}, \mathbf{2}, -\tfrac{1}{2}), \tag{F.48}$$

where $H^- \equiv (H^+)^*$. That $\widetilde{H}$ transforms as a genuine $\mathrm{SU}(2)_L$ doublet with $Y=-\frac{1}{2}$—and not as a $\bar{\mathbf{2}}$—is the standard fact about the pseudoreality of SU(2) established in Chapter B; it is what makes the up-type Yukawa coupling below gauge invariant.

F.2.4 The Yukawa sector

With fermion masses forbidden in $\mathcal{L}_{\text{fermion}}$, the only renormalizable couplings that can connect the left-handed doublets to the right-handed singlets are the dimension-four Yukawa terms, each pairing a fermion bilinear (carrying half-integer isospin and some hypercharge) with the scalar doublet (or its conjugate) so that the product is a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet. Writing the generation indices $i, j = 1, 2, 3$ explicitly and the three couplings as general complex 3×3 matrices Y_u, Y_d, Y_e,

$$\mathcal{L}_{\text{Yukawa}} = -\,\bar{Q}_{Li}\,(Y_u)_{ij}\,\widetilde{H}\,u_{Rj} - \bar{Q}_{Li}\,(Y_d)_{ij}\,H\,d_{Rj} - \bar{L}_{Li}\,(Y_e)_{ij}\,H\,e_{Rj} + \text{h.c.}, \tag{F.49}$$

the $\mathrm{SU}(2)_L$ doublet indices on $\bar{Q}_L$ (or $\bar{L}_L$) and on $H, \widetilde{H}$ being contracted, and the color indices on $\bar{Q}_L$ and the quark singlets being contracted. Each term is a $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ singlet, which one checks by adding hypercharges and contracting representations:

- Down-type quark, $\bar{Q}_L\, H\, d_R$: color $\bar{\mathbf{3}} \otimes \mathbf{3} = \mathbf{1} \oplus \mathbf{8} \supset \mathbf{1}$ (singlet kept); isospin $\bar{\mathbf{2}} \otimes \mathbf{2} = \mathbf{1} \oplus \mathbf{3} \supset \mathbf{1}$; hypercharge $-\frac{1}{6} + \frac{1}{2} - \frac{1}{3} = 0$. Invariant.

- Up-type quark, $\bar{Q}_L\,\widetilde{H}\,u_R$: color singlet as above; isospin singlet using $\widetilde{H} \in \mathbf{2}$; hypercharge $-\frac{1}{6} + (-\frac{1}{2}) + \frac{2}{3} = 0$. The conjugate doublet $\widetilde{H}$ is needed precisely because u_R has $Y = +\frac{2}{3}$, which H (with $Y = +\frac{1}{2}$) cannot neutralize but $\widetilde{H}$ (with $Y = -\frac{1}{2}$) can. Invariant.

- Charged lepton, $\bar{L}_L\,H\,e_R$: color trivially singlet; isospin singlet; hypercharge $+\frac{1}{2} + \frac{1}{2} - 1 = 0$. Invariant.

There is no $\bar{L}_L\,\widetilde{H}\,\nu_R$ term, because the minimal model contains no ν_R; this is why neutrinos are massless in the renormalizable Standard Model, and the leading gauge-invariant operator that gives them a (Majorana) mass is the dimension-five Weinberg operator discussed in Chapters 32 and 35, which we do not include in Eq. (F.20). The three matrices Y_u, Y_d, Y_e are entirely general: a priori arbitrary complex matrices in generation space, with no reality or hermiticity assumed. After electroweak breaking H acquires its vev and these matrices become the fermion mass matrices; their diagonalization, the mismatch between up- and down-type rotations, and the resulting CKM matrix were derived in Chapter 32 and are summarized in Section F.3. The counting of the physical parameters hidden in Y_u, Y_d, Y_e—which uses the $\mathrm{U}(3)^5$ flavor symmetry noted after Eq. (F.27)—is carried out in full in Section F.4.

F.2.5 Gauge fixing and ghosts (for reference)

The Lagrangian Eqs. (F.25) to (F.49) is the classical, gauge-invariant Standard Model; it is what the next section breaks. For quantization the gauge redundancy must be fixed, introducing the Faddeev–Popov ghosts derived in Chapter 26. We record the gauge-fixing and ghost terms here for completeness but neither re-derive nor use them in this appendix.

In a covariant R_ξ gauge one adds, for each gauge factor, a gauge-fixing term that softens the longitudinal propagator with a free parameter ξ. In the unbroken phase the simplest such choice is the linear covariant gauge,

$$\mathcal{L}_{\mathrm{gf}} = -\frac{1}{2\xi_3}\big(\partial^\mu G^A_\mu\big)^2 - \frac{1}{2\xi_2}\big(\partial^\mu W^I_\mu\big)^2 - \frac{1}{2\xi_1}\big(\partial^\mu B_\mu\big)^2, \tag{F.50}$$

with independent parameters ξ_3, ξ_2, ξ_1 for the three factors (Feynman gauge being $\xi = 1$, the default for explicit computation in this book). The associated ghost Lagrangian, one anticommuting complex scalar $c^a, \bar{c}^a$ per gauge generator, is

$$\mathcal{L}_{\mathrm{ghost}} = \bar{c}^A\big(-\partial^\mu D^{AB}_\mu\big)c^B + \bar{c}^I\big(-\partial^\mu D^{IJ}_\mu\big)c^J, \tag{F.51}$$

where $D^{AB}_\mu = \partial_\mu\delta^{AB} + g_s f^{ACB} G^C_\mu$ and $D^{IJ}_\mu = \partial_\mu\delta^{IJ} + g\,\epsilon^{IKJ} W^K_\mu$ are the adjoint covariant derivatives of Chapter 26; the abelian $\mathrm{U}(1)_Y$ ghost is free and decouples, so it is omitted. The full gauge-fixed action is invariant under the BRST symmetry

of Chapter 26, which is what guarantees the gauge independence of physical observables.

After electroweak breaking one uses instead the 't Hooft R_ξ gauge of Chapter 30, in which the gauge-fixing functionals are shifted by terms proportional to the would-be Goldstone bosons, $\mathcal{G}_a = \partial^\mu V_\mu^a - \xi\, M_a\, \varphi_a$ (schematically, with V_a a massive vector, M_a its mass and φ_a the eaten Goldstone), precisely so as to cancel the quadratic gauge–Goldstone mixing produced by the vev. The resulting massive-gauge and Goldstone propagators, and the ξ-dependent ghost masses, are recorded in Chapter 30; we point to them rather than reproduce them, since this appendix's task is the defining Lagrangian, not its quantization.

F.2.6 The assembled Lagrangian

Collecting Eq. (F.25), Eq. (F.26), Eq. (F.27), Eq. (F.47), Eq. (F.49) (and, for quantization, Eq. (F.50), Eq. (F.51)), the complete renormalizable Standard Model Lagrangian is

$$\begin{aligned}\mathcal{L}_{\rm SM} = \; & -\tfrac{1}{4} G^A_{\mu\nu} G^{A\,\mu\nu} - \tfrac{1}{4} W^I_{\mu\nu} W^{I\,\mu\nu} - \tfrac{1}{4} B_{\mu\nu} B^{\mu\nu} + \frac{g_s^2\,\theta}{32\pi^2} G^A_{\mu\nu} \widetilde{G}^{A\,\mu\nu} \\ & + \bar{Q}_{Li}\, i\not{D}\, Q_{Li} + \bar{u}_{Ri}\, i\not{D}\, u_{Ri} + \bar{d}_{Ri}\, i\not{D}\, d_{Ri} + \bar{L}_{Li}\, i\not{D}\, L_{Li} + \bar{e}_{Ri}\, i\not{D}\, e_{Ri} \\ & + (D_\mu H)^\dagger (D^\mu H) + \mu^2\, H^\dagger H - \lambda\, (H^\dagger H)^2 \\ & - \Big(\bar{Q}_{Li}\, (Y_u)_{ij}\, \widetilde{H}\, u_{Rj} + \bar{Q}_{Li}\, (Y_d)_{ij}\, H\, d_{Rj} + \bar{L}_{Li}\, (Y_e)_{ij}\, H\, e_{Rj} + \text{h.c.} \Big),\end{aligned} \tag{F.52}$$

each covariant derivative being the specialization Eqs. (F.28) to (F.33) appropriate to the multiplet it acts on, and the generation indices i, j summed. Every parameter that appears—the three gauge couplings g_s, g, g', the strong CP angle θ, the two scalar parameters μ^2, λ, and the three Yukawa matrices Y_u, Y_d, Y_e—is collected and counted in Section F.4. This is the entire content of the Standard Model: there is no operator of dimension four consistent with $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ gauge invariance and Lorentz invariance that has been omitted, and (Chapter 33) the accidental conservation of baryon and lepton number is a consequence of this exhaustiveness rather than an input.

Equation (F.52) is written in the unbroken phase, where the gauge symmetry is manifest and every fermion and gauge boson is massless. The next section takes the Higgs field to its vacuum expectation value, $\langle H \rangle = (0, v/\sqrt{2})^{\mathsf{T}}$, and reads off the consequences term by term: the gauge-boson mass eigenstates $W^\pm, Z, \gamma$ and the weak mixing angle from $(D_\mu H)^\dagger (D^\mu H)$, the physical Higgs mass from $V(H)$, and the fermion mass matrices from $\mathcal{L}_{\rm Yukawa}$—thereby tying every measured mass and mixing, and every numerical value tabulated later in this appendix, back to a definite parameter of Eq. (F.52).

F.3 Electroweak Breaking and the Physical Spectrum

The previous section wrote the complete Standard Model Lagrangian term by term (Section F.2), and among its pieces sat two that are inert in perturbation theory until the scalar field develops a vacuum expectation value: the Higgs potential $V(H)$ and the Yukawa couplings $y_f\,\bar{L}Hf_R+\text{h.c.}$ Written in terms of the symmetric fields W^a_μ, B_μ and the gauge-symmetric doublet H, that Lagrangian contains no mass term for any gauge boson and no mass term for any charged fermion—both would violate $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$ gauge invariance outright. The masses are not absent; they are stored in the vacuum. This section spends them. It records what the theory looks like once H settles into its minimum: the value v of the condensate, the rotation that turns (W^3,B) into the photon and the Z, the weak mixing angle, and the resulting tree-level dictionary that expresses every gauge-boson and fermion mass in terms of Lagrangian parameters. The *physics* of why a gauge symmetry may be hidden by a vacuum—the Higgs mechanism, the count of would-be Goldstone bosons eaten, the unitarity of the resulting massive theory—is proved in Chapter 30 and applied to the electroweak group in Chapter 31; we import those results by reference and never reprove them. What we carry here in full is the comparatively elementary but load-bearing linear algebra that converts the symmetric Lagrangian into the physical spectrum, because that algebra is the bridge between the parameters $\{g,g',v,\lambda,y_f\}$ and the numbers measured in the laboratory, and it belongs nowhere else in the book.

Throughout we use the standing conventions frozen for this appendix: the gauge group is $\mathrm{SU}(3)_c\times\mathrm{SU}(2)_L\times\mathrm{U}(1)_Y$ in that order; the electric charge is $Q=T^3+Y$ with the hypercharge of the lepton doublet $Y_L=-\frac{1}{2}$ and of the Higgs doublet $Y_H=+\frac{1}{2}$ (Section F.1); and the covariant derivative carries the sign convention of Chapter 30, namely

$$D_\mu H=\Big(\partial_\mu-ig\,W^a_\mu T^a-ig'\,Y_H\,B_\mu\Big)H,\qquad T^a=\tfrac{1}{2}\sigma^a,\quad Y_H=+\tfrac{1}{2},\tag{F.53}$$

where σ^a are the Pauli matrices (Chapter B) and g, g' are the $\mathrm{SU}(2)_L$ and $\mathrm{U}(1)_Y$ couplings.

F.3.1 The Higgs vacuum expectation value

The scalar sector is a single complex doublet H with $Y_H=+\frac{1}{2}$, governed by the potential

$$V(H)=-\mu^2\,H^\dagger H+\lambda\,(H^\dagger H)^2,\qquad \mu^2>0,\quad \lambda>0.\tag{F.54}$$

The sign of the quadratic term is the whole point: with $\mu^2>0$ the origin $H=0$ is a local maximum of the radial potential, and stability is supplied by the quartic with $\lambda>0$. That spontaneous symmetry breaking occurs—that the vacuum is

a point on the degenerate minimum circle rather than the symmetric point—is the content of Chapter 29; here we only locate the minimum and read off the spectrum about it.

Because V depends on H only through the gauge invariant $H^\dagger H \geq 0$, it suffices to minimize over that single nonnegative variable. Writing $u \equiv H^\dagger H$,

$$\frac{dV}{du} = -\mu^2 + 2\lambda u = 0 \quad \Longrightarrow \quad u_{\min} = H^\dagger H\big|_{\min} = \frac{\mu^2}{2\lambda}, \tag{F.55}$$

and $d^2V/du^2 = 2\lambda > 0$ confirms it is a minimum. The minimum is a three-sphere's worth of degenerate vacua related by $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$; Chapter 31 fixes the gauge that aligns the condensate with the lower, electrically neutral component, so that Q annihilates the vacuum and $\mathrm{U}(1)_{\mathrm{em}}$ survives unbroken. In that unitary gauge the doublet is parametrized by the single physical real scalar $h(x)$,

$$H(x) = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v + h(x) \end{pmatrix}, \qquad \langle H \rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix}, \tag{F.56}$$

the three would-be Goldstone bosons having been removed by the gauge choice and reappearing as the longitudinal polarizations of $W^\pm$ and Z (Chapter 30). With Eq. (F.56) one has $H^\dagger H = \frac{1}{2}(v+h)^2$, so the condition $H^\dagger H|_{\min} = v^2/2 = \mu^2/2\lambda$ fixes the *vacuum expectation value*

$$\boxed{v^2 = \frac{\mu^2}{\lambda}} \tag{F.57}$$

This is the number that sets the scale of all electroweak masses; numerically $v \simeq 246$ GeV, recorded with its provenance in Section F.6.

Expanding V about $H^\dagger H = \frac{1}{2}(v+h)^2$ gives the physical scalar mass. Substituting into Eq. (F.54),

$$\begin{aligned} V &= -\mu^2 \cdot \tfrac{1}{2}(v+h)^2 + \lambda \cdot \tfrac{1}{4}(v+h)^4 \\ &= -\tfrac{1}{2}\lambda v^2 (v+h)^2 + \tfrac{1}{4}\lambda (v+h)^4, \end{aligned} \tag{F.58}$$

where we used $\mu^2 = \lambda v^2$ from Eq. (F.57). The coefficient of h^2 is read off by differentiating twice at $h = 0$, or by direct expansion:

$$V \supset -\tfrac{1}{2}\lambda v^2 (2vh + h^2) + \tfrac{1}{4}\lambda (6v^2 h^2 + \dots) = \underbrace{\left(-\lambda v^3 + \lambda v^3\right)}_{=0} h + \tfrac{1}{2}(-\lambda v^2 + 3\lambda v^2) h^2 + \dots \tag{F.59}$$

The linear term cancels—confirming that $h = 0$ is the stationary point—and the quadratic term identifies the tree-level Higgs mass via $V \supset \frac{1}{2} m_h^2 h^2$:

$$\boxed{m_h^2 = 2\lambda v^2 = 2\mu^2} \tag{F.60}$$

Thus the two parameters of the scalar sector, (μ^2, λ), are traded for the two observables (v, m_h): $\lambda = m_h^2/2v^2$ and $\mu^2 = m_h^2/2$.

F.3.2 Gauge-boson masses and the weak mixing angle

Gauge-boson masses arise entirely from the scalar kinetic term $(D_\mu H)^\dagger (D^\mu H)$ once H is replaced by its vacuum value. Evaluating the covariant derivative Eq. (F.53) on the constant vacuum $\langle H \rangle$ kills the ∂_μ piece and leaves the gauge fields acting algebraically:

$$D_\mu \langle H \rangle = -i\Big(g\, W_\mu^a T^a + g'\, Y_H\, B_\mu\Big)\, \langle H \rangle = -\frac{i}{2}\Big(g\, W_\mu^a \sigma^a + g'\, B_\mu \mathbf{1}\Big)\, \langle H \rangle\,, \quad \text{(F.61)}$$

using $T^a = \sigma^a/2$ and $Y_H = \frac{1}{2}$. Write out the 2×2 matrix in the doublet space, with $\sigma^3 = \mathrm{diag}(1, -1)$,

$$g\, W_\mu^a \sigma^a + g'\, B_\mu \mathbf{1} = \begin{pmatrix} gW_\mu^3 + g'B_\mu & g(W_\mu^1 - iW_\mu^2) \\ g(W_\mu^1 + iW_\mu^2) & -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad \text{(F.62)}$$

Acting on $\langle H \rangle = \frac{1}{\sqrt{2}}(0, v)^{\mathsf{T}}$ picks out the second column:

$$D_\mu \langle H \rangle = -\frac{i}{2} \cdot \frac{v}{\sqrt{2}} \begin{pmatrix} g(W_\mu^1 - iW_\mu^2) \\ -gW_\mu^3 + g'B_\mu \end{pmatrix}. \quad \text{(F.63)}$$

The mass Lagrangian is the squared norm of this column vector,

$$\mathcal{L}_{\text{mass}} = (D_\mu \langle H \rangle)^\dagger (D^\mu \langle H \rangle) = \frac{1}{8} v^2 \Big[\, g^2 \big((W_\mu^1)^2 + (W_\mu^2)^2\big) + \big(gW_\mu^3 - g'B_\mu\big)^2 \,\Big], \quad \text{(F.64)}$$

where the cross term between the off-diagonal and diagonal entries vanishes because they live in orthogonal components of the doublet. This is the entire source of tree-level gauge-boson masses, and it splits cleanly into a charged sector (the first two terms) and a neutral sector (the last).

Charged sector. Define the charge eigenfields

$$W_\mu^\pm = \frac{1}{\sqrt{2}}(W_\mu^1 \mp i W_\mu^2), \quad \text{(F.65)}$$

so that W^+ carries $Q = +1$ and W^- carries $Q = -1$ (the assignment follows from $T^\pm = T^1 \pm i T^2$; see Chapter 31). Then $(W_\mu^1)^2 + (W_\mu^2)^2 = 2\, W_\mu^+ W^{-\mu}$, and the charged part of Eq. (F.64) becomes $\frac{1}{4} g^2 v^2\, W_\mu^+ W^{-\mu}$. A complex (charged) vector field of mass m_W has mass term $m_W^2\, W_\mu^+ W^{-\mu}$, with no factor of $\frac{1}{2}$, so

$$\boxed{m_W = \frac{gv}{2}} \quad \text{(F.66)}$$

Neutral sector. The last term of Eq. (F.64) is a quadratic form in the two real neutral fields (W^3_μ, B_μ). A real vector field of mass m has mass term $\frac{1}{2}m^2 V_\mu V^\mu$, so we read the symmetric mass-squared matrix $\mathcal{M}^2$ from $\mathcal{L}_{\text{mass}} \supset \frac{1}{2}\,(W^3, B)\,\mathcal{M}^2\,(W^3, B)^{\mathsf{T}}$. Expanding $(gW^3 - g'B)^2 = g^2(W^3)^2 - 2gg'\,W^3B + g'^2B^2$ and matching to $\frac{1}{8}v^2$ times this,

$$\mathcal{M}^2 = \frac{v^2}{4}\begin{pmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{pmatrix}. \tag{F.67}$$

The eigenstructure of this matrix is the worked diagonalization of Theorem F.8; its output is a massless photon A_μ, a massive Z_μ with the eigenvectors of Eq. (F.73), and the weak mixing angle of Eq. (F.72). We carry the diagonalization in full now.

Worked Example F.8 (Diagonalizing the neutral mass matrix)**.** We diagonalize $\mathcal{M}^2$ of Eq. (F.67) completely: eigenvalues, eigenvectors, mixing angle, and the masslessness of the photon.

Eigenvalues. The matrix $\mathcal{M}^2 = \frac{v^2}{4}\left(\begin{smallmatrix} g^2 & -gg' \\ -gg' & g'^2 \end{smallmatrix}\right)$ is, up to the prefactor, the outer product $(g, -g')^{\mathsf{T}}(g, -g')$, hence rank one. Its determinant is

$$\det \mathcal{M}^2 = \Big(\frac{v^2}{4}\Big)^2 (g^2 g'^2 - (gg')^2) = 0, \tag{F.68}$$

so one eigenvalue vanishes. The trace gives the other:

$$\operatorname{tr} \mathcal{M}^2 = \frac{v^2}{4}(g^2 + g'^2) = m_Z^2 + 0, \qquad \text{so} \qquad m_Z^2 = \frac{v^2}{4}(g^2 + g'^2), \quad m_\gamma^2 = 0. \tag{F.69}$$

The vanishing eigenvalue is the unbroken photon, guaranteed by the survival of $\mathrm{U}(1)_{\text{em}}$; the nonzero one is the Z.

Eigenvectors. The massless eigenvector $\hat{e}_A$ satisfies $\mathcal{M}^2 \hat{e}_A = 0$. Since each row of $\mathcal{M}^2$ is proportional to $(g, -g')$, the null vector is orthogonal to $(g, -g')$, i.e. proportional to (g', g). Normalizing,

$$A_\mu = \frac{g'\,W^3_\mu + g\,B_\mu}{\sqrt{g^2 + g'^2}}. \tag{F.70}$$

The Z eigenvector is the orthogonal combination,

$$Z_\mu = \frac{g\,W^3_\mu - g'\,B_\mu}{\sqrt{g^2 + g'^2}}, \tag{F.71}$$

and one checks $\mathcal{M}^2 \hat{e}_Z = \frac{v^2}{4}(g^2 + g'^2)\,\hat{e}_Z = m_Z^2\,\hat{e}_Z$ directly. Introducing the *weak mixing angle* θ_{W} through

$$\cos\theta_{\mathrm{W}} = \frac{g}{\sqrt{g^2 + g'^2}}, \qquad \sin\theta_{\mathrm{W}} = \frac{g'}{\sqrt{g^2 + g'^2}}, \qquad \tan\theta_{\mathrm{W}} = \frac{g'}{g}, \tag{F.72}$$

the eigenstates take their canonical form

$$\begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix} = \begin{pmatrix} \sin\theta_{\mathrm{W}} & \cos\theta_{\mathrm{W}} \\ \cos\theta_{\mathrm{W}} & -\sin\theta_{\mathrm{W}} \end{pmatrix} \begin{pmatrix} W^3_\mu \\ B_\mu \end{pmatrix}, \qquad \text{i.e.} \qquad \begin{aligned} A_\mu &= \sin\theta_{\mathrm{W}}\, W^3_\mu + \cos\theta_{\mathrm{W}}\, B_\mu, \\ Z_\mu &= \cos\theta_{\mathrm{W}}\, W^3_\mu - \sin\theta_{\mathrm{W}}\, B_\mu. \end{aligned} \tag{F.73}$$

The rotation is orthogonal (its matrix is symmetric with unit determinant up to sign), so it preserves the canonically normalized kinetic terms of W^3 and B.

The Z mass in terms of m_W. Using $m_Z^2 = \frac{v^2}{4}(g^2 + g'^2)$ and $m_W^2 = \frac{v^2}{4} g^2$ from Eq. (F.66),

$$\frac{m_W^2}{m_Z^2} = \frac{g^2}{g^2 + g'^2} = \cos^2\theta_{\mathrm{W}} \qquad \Longrightarrow \qquad m_Z = \frac{m_W}{\cos\theta_{\mathrm{W}}}. \tag{F.74}$$

This completes the diagonalization.

The electromagnetic coupling. The combination of couplings that multiplies the photon is the electric charge e. To see it, rewrite the neutral part of the covariant derivative $g W^3_\mu T^3 + g' B_\mu Y$ in the mass basis. Inverting Eq. (F.73),

$$W^3_\mu = \sin\theta_{\mathrm{W}}\, A_\mu + \cos\theta_{\mathrm{W}}\, Z_\mu, \qquad B_\mu = \cos\theta_{\mathrm{W}}\, A_\mu - \sin\theta_{\mathrm{W}}\, Z_\mu, \tag{F.75}$$

and substituting,

$$\begin{aligned} g\, W^3_\mu T^3 + g'\, B_\mu Y = {} & (g \sin\theta_{\mathrm{W}}\, T^3 + g' \cos\theta_{\mathrm{W}}\, Y)\, A_\mu \\ & + (g \cos\theta_{\mathrm{W}}\, T^3 - g' \sin\theta_{\mathrm{W}}\, Y)\, Z_\mu. \end{aligned} \tag{F.76}$$

For the photon to couple universally to the electric charge $Q = T^3 + Y$, the two coefficients in front of A_μ must be equal, $g \sin\theta_{\mathrm{W}} = g' \cos\theta_{\mathrm{W}}$, which is exactly Eq. (F.72); their common value *is* the electric charge:

$$\boxed{e = g \sin\theta_{\mathrm{W}} = g' \cos\theta_{\mathrm{W}} = \frac{g g'}{\sqrt{g^2 + g'^2}}} \tag{F.77}$$

so that the photon coupling reduces to $e\,(T^3 + Y) A_\mu = e\, Q\, A_\mu$, as required, and the unbroken interaction is precisely QED with charge e. The same substitution exhibits the neutral-current coupling to the Z as $\frac{g}{\cos\theta_{\mathrm{W}}}(T^3 - \sin^2\theta_{\mathrm{W}} Q) Z_\mu$, the structure analyzed in Chapter 31; it is the experimental handle on $\sin^2\theta_{\mathrm{W}}$.

It is worth recording the tree-level bridge from v to the Fermi constant, since it is what makes v measurable. Integrating out the W in muon decay (Chapter 31) matches the low-energy four-fermion interaction onto the Fermi theory with

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8 m_W^2} = \frac{1}{2v^2} \qquad \Longrightarrow \qquad v = (\sqrt{2}\, G_F)^{-1/2}, \tag{F.78}$$

where the middle equality used $m_W = g v/2$. Thus the precisely measured G_F fixes v directly; the scheme-conversion subtleties of trading $\{g, g', v\}$ for $\{\alpha, G_F, m_Z\}$ are carried in Section F.7.

F.3.3 Fermion masses from the Yukawa sector

Charged-fermion masses are forbidden as bare terms by $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ invariance—a Dirac mass $m\,\bar{\psi}_L \psi_R$ pairs a doublet with a singlet of different hypercharge—and are generated instead by the gauge-invariant Yukawa couplings of Section F.2. For a down-type fermion with right-handed singlet f_R and the left-handed doublet L_f,

$$\mathcal{L}_{\mathrm{Y}} \supset -y_f\, \bar{L}_f\, H\, f_R + \mathrm{h.c.}, \tag{F.79}$$

the doublet contraction $\bar{L}_f H$ is an $\mathrm{SU}(2)_L$ singlet and carries vanishing hypercharge, so the term is gauge invariant. Replacing H by its vacuum value Eq. (F.56) projects onto the lower component, $\bar{L}_f H \to \frac{1}{\sqrt{2}}(v+h)\,\bar{f}_L$, giving

$$\mathcal{L}_{\mathrm{Y}} \supset -\frac{y_f}{\sqrt{2}}(v+h)\,\bar{f}_L f_R + \mathrm{h.c.} = -\frac{y_f v}{\sqrt{2}}\,\bar{f}f - \frac{y_f}{\sqrt{2}}\,h\,\bar{f}f, \tag{F.80}$$

where $\bar{f}f = \bar{f}_L f_R + \bar{f}_R f_L$. The first term is a Dirac mass, the second the Higgs–fermion coupling. Hence

$$\boxed{m_f = \frac{y_f\, v}{\sqrt{2}}} \tag{F.81}$$

For up-type fermions the conjugate doublet $\widetilde{H} = i\sigma^2 H^*$ (hypercharge $-\frac{1}{2}$) is used in place of H so that the upper component acquires the vev; the mass formula Eq. (F.81) is unchanged. Equation Eq. (F.81) is written here for a single generation; in the three-generation theory y_f is a 3×3 matrix in flavor space, and diagonalizing it is what introduces the CKM matrix—that diagonalization, the flavor-basis rotations, and the parameter count it generates are the business of Chapter 32 and of Section F.4. The crucial structural fact recorded here is that *every* charged-fermion mass is $v/\sqrt{2}$ times a dimensionless Yukawa coupling, so the enormous hierarchy of fermion masses is encoded entirely in the hierarchy of the y_f. A coupling-derived corollary worth noting: the Higgs couples to each fermion with strength $y_f/\sqrt{2} = m_f/v$, proportional to the mass it generates—the prediction that makes the Higgs's decay pattern a sharp test (Chapter 34).

F.3.4 The ρ parameter and custodial symmetry

The mass relations Eqs. (F.66) and (F.74) are not independent: they imply a rigid prediction. Define the ρ parameter

$$\rho \equiv \frac{m_W^2}{m_Z^2\, \cos^2\theta_{\mathrm{W}}}. \tag{F.82}$$

This combination is constructed to be unity if and only if the gauge-boson masses obey the single-doublet pattern just derived.

Proposition F.9 (Tree-level $\rho = 1$)**.** *In the Standard Model with a single Higgs doublet, $\rho = 1$ at tree level.*

Proof. Substitute the diagonalization results. From Eq. (F.66), $m_W^2 = \frac{v^2}{4} g^2$; from Theorem F.8, $m_Z^2 = \frac{v^2}{4}(g^2 + g'^2)$; and from Eq. (F.72), $\cos^2\theta_{\mathrm{W}} = g^2/(g^2 + g'^2)$. Then

$$m_Z^2 \cos^2\theta_{\mathrm{W}} = \frac{v^2}{4}(g^2 + g'^2) \cdot \frac{g^2}{g^2 + g'^2} = \frac{v^2}{4} g^2 = m_W^2, \tag{F.83}$$

so $\rho = m_W^2/m_W^2 = 1$. □

The algebra is immediate, but its robustness is the interesting point: $\rho = 1$ is not an accident of the numbers, it is protected by a symmetry. The Higgs sector has, beyond the gauged $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$, an accidental *global* symmetry larger than what is gauged, and the surviving subgroup after breaking—*custodial* $\mathrm{SU}(2)_V$—forces the three weak gauge bosons to share a common mass before hypercharge mixing is switched on. The full custodial analysis, including its breaking by g' and by the Yukawa splittings, is carried in Chapter 31; the worked example below isolates exactly the part that controls ρ.

Worked Example F.10 (Custodial symmetry and the protection of ρ)**.** Assemble the Higgs doublet H (hypercharge $+\frac{1}{2}$) and its conjugate $\widetilde{H} = i\sigma^2 H^*$ (hypercharge $-\frac{1}{2}$) into a single 2×2 matrix

$$\Phi \equiv (\widetilde{H},\ H) = \begin{pmatrix} \phi_0^* & \phi^+ \\ -\phi^- & \phi_0 \end{pmatrix}. \tag{F.84}$$

A short computation gives $\mathrm{tr}(\Phi^\dagger \Phi) = 2\, H^\dagger H$, so the potential Eq. (F.54) depends on Φ only through $\mathrm{tr}(\Phi^\dagger \Phi)$. Under the transformation $\Phi \to L\, \Phi\, R^\dagger$ with $L \in \mathrm{SU}(2)_L$ and $R \in \mathrm{SU}(2)_R$, the trace is invariant because $\mathrm{tr}(R\Phi^\dagger L^\dagger L \Phi R^\dagger) = \mathrm{tr}(\Phi^\dagger \Phi)$. Thus the scalar potential and—when $g' = 0$—the scalar kinetic term enjoy a global $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$ symmetry, larger than the gauged $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$.

The vacuum Eq. (F.56) corresponds to $\langle \Phi \rangle = \frac{v}{\sqrt{2}}\, \mathbf{1}$, which is invariant under $\Phi \to L \Phi R^\dagger$ precisely when $L = R$: the diagonal subgroup $\mathrm{SU}(2)_{L+R} \equiv \mathrm{SU}(2)_V$ survives. This is the custodial symmetry.

Now examine the gauge mass term Eq. (F.64) with hypercharge turned off, $g' = 0$. The neutral matrix Eq. (F.67) becomes $\mathcal{M}^2 = \frac{v^2}{4}\, \mathrm{diag}(g^2, 0)$, so W^3 acquires mass $m_W = gv/2$ and B stays massless and decoupled. Together with the charged result Eq. (F.66), all three fields $W^{1,2,3}$ then have the identical mass $m_W = gv/2$: they form a degenerate triplet. This degeneracy is no coincidence—under custodial $\mathrm{SU}(2)_V$ the three W^a transform as a vector (adjoint) triplet, and a symmetry-invariant mass matrix on an irreducible triplet must be proportional to the identity. The common mass is the algebraic statement of custodial symmetry.

Switching g' back on gauges the T_R^3 generator inside $\mathrm{SU}(2)_R$ (with Y identified as $-T_R^3$ plus a singlet piece), explicitly breaking $\mathrm{SU}(2)_R \to \mathrm{U}(1)_Y$ and hence custodial $\mathrm{SU}(2)_V$. The effect is precisely the W^3–B mixing of Theorem F.8: the would-be degenerate W^3 mixes with B to form Z and γ. Crucially, this mixing rotates the triplet without changing the relation between the masses in the symmetric direction, so the prediction

$$\rho = \frac{m_W^2}{m_Z^2 \cos^2\theta_{\mathrm{W}}} = 1 \tag{F.85}$$

survives at tree level even though custodial symmetry is broken by g'. The reason it is called *custodial* is now visible: it is the symmetry that guards the tree-level relation $\rho = 1$. Were the scalar sector to contain, say, a hypercharge-± 1 triplet, this protection would fail and ρ would deviate from one already at tree level—a sensitivity that makes the measured $\rho \approx 1$ strong evidence that electroweak breaking is dominated by doublets. Custodial breaking by the large top–bottom Yukawa splitting, and by g' at loop level, generates the small calculable $\rho - 1$ corrections of the precision program (Chapter 34).

The tree-level dictionary is now complete. The Lagrangian parameters $\{g, g', v, \lambda, y_f\}$ have been mapped to the physical quantities $\{m_W, m_Z, \theta_{\mathrm{W}}, e, m_h, m_f\}$ through Eqs. (F.60), (F.66), (F.72), (F.74), (F.77) and (F.81), with the single doublet enforcing $\rho = 1$. These maps—together with the flavor rotations deferred to Chapter 32—are the relations whose independent inputs the next section counts, turning the informal slogan that the Standard Model has nineteen parameters into a derived number; and the values m_W, m_Z, θ_{W} established here are the electroweak anchors that the numerical section will feed into the running-coupling thread.

F.4 Free Parameters: the Counting

The Lagrangian-to-physical dictionary assembled in the preceding section made explicit which Lagrangian quantities are genuine independent inputs and which are derived: the gauge-boson and fermion masses, the weak mixing angle, the vacuum expectation value, all of them are *functions* of a smaller set of couplings written in the Lagrangian. The natural question is then exactly how small that set is. The folklore answer — "the Standard Model has nineteen free parameters" — is usually quoted without proof, as though it were a fact to be memorized. It is not folklore; it is a theorem about the orbit space of the flavor symmetry group acting on the Yukawa matrices, and in this section we prove it in full. We first isolate the handful of parameters in the gauge–Higgs–$\bar{\theta}$ sector, where the count is immediate, and then do the real work in the flavor sector, where the raw Lagrangian contains far more numbers than are physical and the redundancy

must be removed by a careful field-redefinition argument. The same machine, run again with the field content enlarged, delivers the neutrino-mass extensions and explains the otherwise mysterious extra phases of the Majorana case.

Nothing physical is established here that was not already proved in Part IX: the gauge principle, the Higgs mechanism, the origin of the CKM matrix and the structure of the Yukawa sector are imported by Chapters 31 to 33, and the anomalous non-conservation of the axial current — which we shall need to dispose cleanly of the strong-CP angle — is imported from Chapters 36 and 37. What is carried here in full is the *counting*, because it is a load-bearing organizing computation that belongs nowhere else: Chapter 33, which owns the one-line summary "nineteen parameters," cites back to this section for the derivation.

F.4.1 What counts as a physical parameter

A Lagrangian is a presentation of a theory in a particular choice of field variables. Two Lagrangians related by an invertible, local redefinition of the fields that leaves the path-integral measure and the action functionally identical describe *the same physics*: every correlation function, every S-matrix element, every cross section is the same. Consequently a number appearing in the Lagrangian is a genuine physical parameter only if it cannot be removed or altered by such a redefinition. The count of physical parameters is therefore not the count of symbols in the Lagrangian but the dimension of the space of Lagrangians *modulo* the group of allowed field redefinitions.

For the Standard Model the relevant redefinitions split into two kinds. The gauge and scalar couplings $(g_s, g, g', \mu^2, \lambda)$ and the topological angle multiply operators that are essentially rigid: there is no field redefinition that rotates one into another or sets one to a fixed value, so each is physical as written, and the only subtlety is the interplay between the chiral field rotations and the θ-term, treated below. The Yukawa couplings are entirely different. They are arbitrary complex matrices in the three-dimensional space of fermion generations, and the kinetic terms of the fermions are invariant under a large continuous group of unitary rotations among the generations. That group can be used to bring the Yukawa matrices to a canonical form, eliminating most of their entries. The physical Yukawa parameters are the coordinates that survive — the coordinates on the quotient of the raw Yukawa space by the flavor group. Making this precise and computing the dimension of that quotient is the content of Section F.4.3.

F.4.2 The gauge, Higgs and topological sector: six parameters

We begin with the part of the count that requires no quotient. The gauge sector of $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ carries one coupling constant per simple (or abelian) factor, since the normalization of each gauge field is fixed by its canonically

normalized kinetic term $-\frac{1}{4}F^a_{\mu\nu}F^{a\,\mu\nu}$ and the structure constants are fixed by the algebra. These are

$$g_s \text{ (for SU(3)}_c\text{)}, \qquad g \text{ (for SU(2)}_L\text{)}, \qquad g' \text{ (for U(1)}_Y\text{)}, \tag{F.86}$$

three real numbers. The hypercharge normalization is the one frozen for this appendix, $Q = T^3 + Y$ with $Y_L = -\frac{1}{2}$ for the lepton doublet; with that convention g' is unambiguous. The single Higgs doublet $H \sim (\mathbf{1}, \mathbf{2}, +\frac{1}{2})$ contributes the two real parameters of its renormalizable potential,

$$V(H) = -\mu^2\, H^\dagger H + \lambda\,(H^\dagger H)^2, \qquad \mu^2 > 0,\ \lambda > 0, \tag{F.87}$$

namely μ^2 (mass dimension two) and the quartic λ (dimensionless). These fix the vacuum expectation value $v^2 = \mu^2/\lambda$ and the physical Higgs mass $m_h^2 = 2\mu^2 = 2\lambda v^2$ recorded in Section F.3; they are independent inputs because no field redefinition relates μ^2 to λ. The coefficient of the Higgs kinetic term and the overall normalization of H are not parameters: they are fixed by canonical normalization, exactly as for the gauge fields.

Finally there is one parameter with no analogue in QED, the QCD vacuum angle. The Lagrangian admits the gauge-invariant, Lorentz-invariant, marginal operator

$$\mathcal{L}_\theta = \theta\,\frac{g_s^2}{32\pi^2}\,G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu}, \qquad \widetilde{G}^{a\,\mu\nu} = \tfrac{1}{2}\epsilon^{\mu\nu\rho\sigma}G^a_{\rho\sigma}, \tag{F.88}$$

a total derivative locally but with physical effects through the topologically nontrivial gauge configurations established in Chapter 37. Naively this adds a seventh number, θ. It does not add an *independent* one, because θ mixes with the phase of the quark mass determinant under the chiral field rotations we are about to use in the flavor sector. The axial rotation $q_i \to e^{i\alpha_i\gamma_5}q_i$ is not a symmetry of the quantum theory: by the Adler–Bell–Jackiw anomaly proved in Chapter 36 its non-invariance of the path-integral measure shifts the θ-angle. Performing the chiral rotations that make all quark masses real and positive therefore moves the entire CP-violating content of the strong sector into the single rephasing-invariant combination

$$\bar\theta = \theta + \arg\det(M_u M_d), \tag{F.89}$$

where $M_u = (v/\sqrt{2})Y_u$ and $M_d = (v/\sqrt{2})Y_d$ are the quark mass matrices. It is $\bar\theta$, not θ, that is observable (through, e.g., the neutron electric dipole moment), and it is one real parameter. We will keep careful track of this in Section F.4.4 to be sure it is counted exactly once: the chiral rotation that removes the phase $\arg\det(M_u M_d)$ from the Yukawa sector is precisely the rotation that, by the anomaly, generates the shift in θ, so the two bookkeepings agree on a single physical strong-CP angle.

The bosonic and topological tally is therefore

$$\underbrace{g_s,\ g,\ g'}_{3} + \underbrace{\mu^2,\ \lambda}_{2} + \underbrace{\bar{\theta}}_{1} = 6 \tag{F.90}$$

physical parameters.

F.4.3 The flavor sector: broken flavor symmetry

We now turn to the Yukawa couplings, where almost all of the apparent parameters are gauge for the global flavor group and must be subtracted. We work first with the minimal model: three generations and *no* right-handed neutrinos, so neutrinos are exactly massless. The Yukawa Lagrangian is

$$\mathcal{L}_Y = -\,\bar{Q}_L\,\widetilde{H}\,Y_u\,u_R \;-\; \bar{Q}_L\,H\,Y_d\,d_R \;-\; \bar{L}_L\,H\,Y_e\,e_R \;+\; \text{h.c.}, \tag{F.91}$$

with $\widetilde{H} = i\sigma^2 H^*$ the charge-conjugate doublet, and the generation indices suppressed: Y_u, Y_d, Y_e are each a general complex 3×3 matrix in flavor space. A complex 3×3 matrix has 9 complex entries, hence 18 real parameters; three such matrices give the raw count

$$\dim_{\mathbb{R}} \mathcal{Y} \;=\; 3\times 18 \;=\; 54 \tag{F.92}$$

real parameters before any redundancy is removed. We must now remove the redundancy.

The flavor group and how the Yukawas transform. In the limit $Y_u = Y_d = Y_e = 0$ the fermion sector is just five copies (three generations) of canonically normalized chiral kinetic terms, one for each gauge multiplet

$$Q_L \sim (\mathbf{3},\mathbf{2},\tfrac{1}{6}),\quad u_R \sim (\mathbf{3},\mathbf{1},\tfrac{2}{3}),\quad d_R \sim (\mathbf{3},\mathbf{1},-\tfrac{1}{3}),\quad L_L \sim (\mathbf{1},\mathbf{2},-\tfrac{1}{2}),\quad e_R \sim (\mathbf{1},\mathbf{1},-1). \tag{F.93}$$

Each kinetic term $\bar{\psi}\,i\not{D}\,\psi$ is invariant under an independent U(3) rotation acting on the three-dimensional generation index of that multiplet, because U(3) rotations commute with the gauge structure (they act only on flavor) and leave $\bar{\psi}\,i\not{D}\,\psi$ invariant by unitarity. The full global symmetry of the gauge-kinetic terms is therefore

$$G_{\mathrm{F}} = \mathrm{U}(3)_Q\times\mathrm{U}(3)_u\times\mathrm{U}(3)_d\times\mathrm{U}(3)_L\times\mathrm{U}(3)_e \;\equiv\; \mathrm{U}(3)^5, \qquad \dim_{\mathbb{R}} G_{\mathrm{F}} = 5\times 9 = 45, \tag{F.94}$$

using $\dim_{\mathbb{R}}\mathrm{U}(3) = 9$. Under $\psi \to V_\psi\psi$ with $V_\psi \in \mathrm{U}(3)$, the conjugate field transforms as $\bar{\psi}\to\bar{\psi}\,V_\psi^\dagger$, so that the Yukawa terms in (F.91) retain their form provided the coupling matrices are redefined by

$$Y_u \to V_Q^\dagger\,Y_u\,V_u, \qquad Y_d \to V_Q^\dagger\,Y_d\,V_d, \qquad Y_e \to V_L^\dagger\,Y_e\,V_e. \tag{F.95}$$

That is: G_{F} acts on the raw parameter space $\mathcal{Y} \cong \mathbb{C}^{3\times3}\oplus\mathbb{C}^{3\times3}\oplus\mathbb{C}^{3\times3}$, and two points of $\mathcal{Y}$ related by (F.95) are the *same* theory written in two flavor bases. The physical Yukawa parameters are the coordinates on the quotient $\mathcal{Y}/G_{\mathrm{F}}$.

The counting lemma. The dimension of the quotient near a generic point is governed by the stabilizer of that point, by the Lie-algebraic form of the orbit–stabilizer theorem.

Lemma F.11 (Parameter count by broken symmetry)**.** *Let a Lie group G act smoothly on a finite-dimensional real vector space $\mathcal{Y}$, and let $Y_0 \in \mathcal{Y}$ have stabilizer $H = \{h \in G : \ h \cdot Y_0 = Y_0\}$. Then the orbit $G \cdot Y_0$ is a manifold of dimension $\dim G - \dim H$, and the number of physical parameters — the local dimension of the orbit space $\mathcal{Y}/G$ at $[Y_0]$ — is*

$$N_{\text{phys}} = \dim_{\mathbb{R}} \mathcal{Y} - \big(\dim_{\mathbb{R}} G - \dim_{\mathbb{R}} H\big). \tag{F.96}$$

Proof. Consider the orbit map $\mu : G \to \mathcal{Y}$, $\mu(g) = g \cdot Y_0$, with image the orbit $\mathcal{O} = G \cdot Y_0$. Its differential at the identity is the linear map

$$d\mu_e : \ \mathfrak{g} \to T_{Y_0}\mathcal{Y}, \qquad X \mapsto \frac{d}{dt}\Big|_{t=0} e^{tX} \cdot Y_0, \tag{F.97}$$

the infinitesimal action of the Lie algebra $\mathfrak{g}$ on Y_0. The kernel of $d\mu_e$ consists of exactly those X that fix Y_0 to first order; these are the generators of the stabilizer, $\ker d\mu_e = \mathfrak{h} = \mathrm{Lie}(H)$. (An element X with $X \cdot Y_0 = 0$ generates a one-parameter subgroup e^{tX} fixing Y_0, hence lies in $\mathfrak{h}$; conversely every element of $\mathfrak{h}$ annihilates Y_0.) By rank–nullity the image $\operatorname{im} d\mu_e$, which is the tangent space $T_{Y_0}\mathcal{O}$ to the orbit, has dimension $\dim \mathfrak{g} - \dim \mathfrak{h} = \dim G - \dim H$. The orbit directions are precisely the unphysical directions, those along which Y_0 can be moved by a field redefinition without changing the physics; the physical directions are the complement inside $T_{Y_0}\mathcal{Y}$, of dimension $\dim \mathcal{Y} - \dim \mathcal{O}$. This is (F.96). For generic Y_0 the stabilizer dimension is locally constant, so this is the dimension of the quotient near $[Y_0]$. □

Equation (F.96) is the entire engine of the count: physical parameters = raw parameters − broken generators, where a "broken" generator is one that moves Y_0, i.e. one not in the stabilizer. We need only identify the stabilizer of a generic Yukawa configuration.

Worked Example F.12 (The thirteen flavor parameters of the minimal Standard Model)**.** We apply Theorem F.11 with $\mathcal{Y}$ the space of triples (Y_u, Y_d, Y_e), $\dim_{\mathbb{R}} \mathcal{Y} = 54$ from (F.92), and $G = G_{\text{F}} = \mathrm{U}(3)^5$, $\dim_{\mathbb{R}} G = 45$ from (F.94).

Step 1 — the stabilizer. We seek the subgroup of G_{F} that leaves a generic (Y_u, Y_d, Y_e) invariant under (F.95). Because no Yukawa matrix in (F.91) couples the lepton multiplets (L_L, e_R) to the quark multiplets (Q_L, u_R, d_R), the stabilizer factorizes into a quark part inside $\mathrm{U}(3)_Q \times \mathrm{U}(3)_u \times \mathrm{U}(3)_d$ and a lepton part inside $\mathrm{U}(3)_L \times \mathrm{U}(3)_e$, and we treat them separately.

Quark part. A generic pair (Y_u, Y_d) has trivial stabilizer except for an overall phase. Indeed, take V_Q, V_u, V_d with $V_Q^\dagger Y_u V_u = Y_u$ and $V_Q^\dagger Y_d V_d = Y_d$. For generic,

invertible Y_u, Y_d this forces $V_u = Y_u^{-1} V_Q Y_u$ and $V_d = Y_d^{-1} V_Q Y_d$; unitarity of V_u, V_d together with the requirement that the same V_Q work for the two *independent* matrices Y_u, Y_d leaves only $V_Q = e^{i\beta}\mathbf{1}$ (and then $V_u = V_d = e^{i\beta}\mathbf{1}$). This common phase is baryon number $\mathrm{U}(1)_B$: under it Q_L, u_R, d_R all rotate by the same phase, leaving every term of (F.91) invariant. Hence the quark stabilizer is $\mathrm{U}(1)_B$, $\dim = 1$.

Lepton part. With no neutrino Yukawa, only Y_e constrains (V_L, V_e), through $V_L^\dagger Y_e V_e = Y_e$. A single generic invertible matrix Y_e can be brought to diagonal, positive form by its singular-value decomposition, and in that basis the residual freedom is the diagonal phase rotations with $V_L = V_e = \mathrm{diag}(e^{i\gamma_1}, e^{i\gamma_2}, e^{i\gamma_3})$. These are the three separate lepton numbers $\mathrm{U}(1)_{L_e} \times \mathrm{U}(1)_{L_\mu} \times \mathrm{U}(1)_{L_\tau}$, which survive because each charged-lepton flavor is conserved when there is no mechanism to mix the generations. The lepton stabilizer is thus $\mathrm{U}(1)^3$, $\dim = 3$.

Total stabilizer. Therefore

$$H = \mathrm{U}(1)_B \times \mathrm{U}(1)_{L_e} \times \mathrm{U}(1)_{L_\mu} \times \mathrm{U}(1)_{L_\tau}, \qquad \dim_{\mathbb{R}} H = 1 + 3 = 4. \tag{F.98}$$

These four unbroken generators are exactly the accidental global symmetries of the minimal Standard Model identified in Chapter 33: baryon number and the three lepton numbers. They are precisely the directions in G_{F} that do *not* remove a Yukawa parameter.

Step 2 — the count. Inserting into (F.96),

$$N_{\text{flavor}} = \dim \mathcal{Y} - (\dim G_{\mathrm{F}} - \dim H) = 54 - (45 - 4) = 54 - 41 = 13. \tag{F.99}$$

There are thirteen physical flavor parameters.

Step 3 — naming the thirteen. It is worth seeing that these thirteen are exactly the masses and mixings one expects. Run the field redefinitions explicitly, sector by sector.

Quarks. By singular-value decomposition write $Y_u = U_u^L \hat{Y}_u U_u^{R\dagger}$ and $Y_d = U_d^L \hat{Y}_d U_d^{R\dagger}$ with $\hat{Y}_u, \hat{Y}_d$ diagonal, real and non-negative. Choosing $V_Q = U_u^L$, $V_u = U_u^R$ in (F.95) sets $Y_u \to \hat{Y}_u$, diagonal. The same $V_Q = U_u^L$ is now fixed, but we are still free in V_d; choosing $V_d = U_d^R$ removes the right-handed rotation of Y_d, leaving

$$Y_d \to U_u^{L\dagger} U_d^L \hat{Y}_d \;\equiv\; V_{\mathrm{CKM}} \hat{Y}_d, \qquad V_{\mathrm{CKM}} = U_u^{L\dagger} U_d^L \in \mathrm{U}(3). \tag{F.100}$$

The three eigenvalues of $\hat{Y}_u$ and the three of $\hat{Y}_d$ give the **six quark masses** (times $v/\sqrt{2}$). The leftover unitary matrix V_{CKM} is the Cabibbo–Kobayashi–Maskawa matrix; its physical content is counted next.

Leptons. With Y_e alone, the SVD $Y_e = U_e^L \hat{Y}_e U_e^{R\dagger}$ is undone completely by $V_L = U_e^L$, $V_e = U_e^R$, leaving $Y_e \to \hat{Y}_e$ diagonal with no leftover matrix. This yields the **three charged-lepton masses** and *no* lepton mixing: with massless neutrinos the analogue of V_{CKM} is unobservable because any unitary rotation

of the degenerate massless neutrinos can absorb it. That is the field-theoretic statement of why the PMNS matrix is trivial in the minimal model.

Physical content of $V_{\rm CKM}$. We must count how many of the nine real parameters of a 3×3 unitary matrix survive the residual rephasing freedom. After the diagonalizations above, the mass terms $\bar u_L \hat M_u u_R$ and $\bar d_L \hat M_d d_R$ are invariant under independent diagonal phase rotations $u_{L,R}\to P_u\, u_{L,R}$ and $d_{L,R}\to P_d\, d_{L,R}$ with $P_u, P_d \in \mathrm{U}(1)^3$ (the left and right rotations must coincide to keep the diagonal masses real). Under these, $V_{\rm CKM}$ appears in the charged current $\bar u_L \gamma^\mu V_{\rm CKM} d_L\, W^+_\mu$ and transforms as $V_{\rm CKM}\to P_u^\dagger\, V_{\rm CKM}\, P_d$. The $3+3=6$ phases of (P_u, P_d) can absorb six phases of $V_{\rm CKM}$, except that the overall common phase $P_u = P_d = e^{i\alpha}\mathbf{1}$ leaves $V_{\rm CKM}$ untouched — this is again $\mathrm{U}(1)_B$ — so only $6-1=5$ phases are removable. A general U(3) matrix has 9 parameters, decomposed (Theorem F.13 below) into 3 rotation angles and 6 phases; removing 5 phases leaves

$$\#(\mathrm{CKM}) = \underbrace{3}_{\text{angles}} + \underbrace{(6-5)}_{\text{phase}} = 3+1 = 4. \tag{F.101}$$

Three mixing angles and one CP-violating phase.

Assembly. The thirteen physical flavor parameters are

$$\underbrace{6}_{\text{quark masses}} + \underbrace{3}_{\text{lepton masses}} + \underbrace{4}_{\text{CKM: 3 angles+1 phase}} = 13, \tag{F.102}$$

in agreement with (F.99) obtained abstractly from the stabilizer. The two routes — subtract broken generators, or diagonalize and rephase — are the same computation seen from two sides.

The angle/phase decomposition used in (F.101) deserves its own statement, since it recurs for every mixing matrix in the book.

Proposition F.13 (Angles and phases of a unitary mixing matrix)**.** *A general $N\times N$ unitary matrix carries N^2 real parameters, of which*

$$\frac{N(N-1)}{2} \text{ are rotation angles} \qquad \text{and} \qquad \frac{N(N+1)}{2} \text{ are phases.} \tag{F.103}$$

Proof. The Lie group $\mathrm{U}(N)$ has real dimension N^2. Inside it sits the real orthogonal group $\mathrm{O}(N)$, of dimension $\tfrac12 N(N-1)$; its generators are the real antisymmetric matrices, which generate the genuine rotations (the "angles"), and these are precisely the parameters that remain in the special case of a real unitary — i.e. orthogonal — matrix. The remaining $N^2 - \tfrac12 N(N-1) = \tfrac12 N(N+1)$ generators of $\mathrm{U}(N)$ lie in the coset directions $\mathrm{U}(N)/\mathrm{O}(N)$; they are the imaginary symmetric generators and parametrize the phases. The split $N^2 = \tfrac12 N(N-1) + \tfrac12 N(N+1)$ is exact. □

For $N=3$, (F.103) gives 3 angles and 6 phases, the inputs to (F.101).

F.4.4 The total: nineteen

Adding the bosonic-plus-topological tally (F.90) to the flavor tally (F.102),

$$N_{\mathrm{SM}} = \underbrace{6}_{g_s,g,g',\mu^2,\lambda,\bar{\theta}} + \underbrace{13}_{\text{Yukawa sector}} = 19 \tag{F.104}$$

physical parameters in the minimal Standard Model with massless neutrinos. Table F.3 records the breakdown.

Sector	Parameters	Count
Gauge couplings	$g_s,\ g,\ g'$	3
Higgs potential	$\mu^2,\ \lambda$	2
Strong CP	$\bar{\theta}$	1
Quark masses	$m_u, m_c, m_t,\ m_d, m_s, m_b$	6
Charged-lepton masses	m_e, m_μ, m_τ	3
CKM mixing	3 angles + 1 phase	4
Total		**19**

Table F.3: The nineteen physical parameters of the minimal Standard Model (massless neutrinos). The flavor block (last three rows, thirteen parameters) is the orbit-space dimension computed in Theorem F.12.

A word is owed on the consistency of the $\bar{\theta}$ entry with the flavor count, because the same chiral rotations enter both.

Caution: no double counting of the strong-CP angle

In Theorem F.12 the full group $G_{\mathrm{F}} = \mathrm{U}(3)^5$ was treated as legitimate field-redefinition freedom, including the overall axial rotation $q_i \to e^{i\alpha\gamma_5} q_i$ of all quarks — the generator $\mathrm{U}(1)_A$ that rotates left and right quark fields by opposite phases. That rotation is *not* in the stabilizer (F.98); it changes $\arg\det(Y_u Y_d)$, and we duly used it (as part of the diagonalization and rephasing) to make the quark masses real. But $\mathrm{U}(1)_A$ is anomalous: by the result imported from Chapter 36, performing it shifts the topological angle, $\theta \to \theta - 2N_f\,\alpha = \theta - 4N_g\,\alpha$ in the normalization of (F.88) (with $N_f = 2N_g = 6$ quark flavors, each Dirac mass picking up a phase $e^{2i\alpha}$ so that $\arg\det(M_u M_d)$ shifts by $2N_f\alpha$). Hence the phase that one "removes" from the Yukawa sector by an axial rotation does not vanish from the theory: it reappears as a shift of θ. What is invariant under the entire chiral redefinition group — and therefore physical — is the combination $\bar{\theta} = \theta + \arg\det(M_u M_d)$ of (F.89). The bookkeeping is now unambiguous: the flavor count of 13 uses all of G_{F} to reach the mass basis with real masses, and the residual physical strong-CP content is the single number

$\bar{\theta}$, counted once in the bosonic block. Equivalently, one may count 12 flavor parameters using only the non-anomalous $\mathrm{SU}(3)^5 \times \mathrm{U}(1)^4$ field redefinitions and keep θ as a separate input; the total is unchanged. Either way $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$ has nineteen physical parameters.

F.4.5 The neutrino-mass extensions

The observation of neutrino oscillations forces nonzero neutrino masses, and the count grows. The two minimal mechanisms — a Dirac mass from new right-handed neutrinos, or a Majorana mass from the dimension-five Weinberg operator — give different counts, and the difference is instructive: it is entirely about which rephasings remain available. We record the counts here; the mechanisms themselves (the seesaw, the origin of the Weinberg operator) are deferred to Chapter 35.

Worked Example F.14 (Dirac neutrinos add seven parameters)**.** Introduce three right-handed neutrinos $\nu_R \sim (\mathbf{1}, \mathbf{1}, 0)$, gauge singlets, and forbid a Majorana mass for them (e.g. by imposing exact lepton number). The Yukawa Lagrangian (F.91) gains the term $-\bar{L}_L \widetilde{H}\, Y_\nu\, \nu_R + \text{h.c.}$, with Y_ν a fourth general complex 3×3 matrix.

Raw count. Four complex 3×3 matrices: $\dim \mathcal{Y} = 4 \times 18 = 72$.

Flavor group. The new singlet field carries its own $\mathrm{U}(3)_\nu$ flavor rotation, so $G_\mathrm{F} \to \mathrm{U}(3)^6$, $\dim = 6 \times 9 = 54$. The new transformation law is $Y_\nu \to V_L^\dagger Y_\nu V_\nu$.

Stabilizer. The quark part is unchanged, $\mathrm{U}(1)_B$, $\dim = 1$. In the lepton sector Y_e and Y_ν now play exactly the roles Y_u, Y_d played for quarks: two independent generic matrices sharing the common V_L. By the same argument as the quark case, the only surviving lepton rotation is the overall common phase, total lepton number $\mathrm{U}(1)_L$, $\dim = 1$. The three separate lepton numbers are now broken — Y_e and Y_ν cannot be simultaneously diagonalized, exactly as Y_u, Y_d could not — leaving only their sum. So

$$H = \mathrm{U}(1)_B \times \mathrm{U}(1)_L, \qquad \dim H = 2. \tag{F.105}$$

Count. By (F.96),

$$N_\mathrm{flavor}^\mathrm{Dirac} = 72 - (54 - 2) = 72 - 52 = 20, \tag{F.106}$$

that is $20 - 13 = 7$ more than the minimal model. Naming them: the lepton sector now mirrors the quark sector exactly, giving 3 charged-lepton masses, 3 neutrino masses, and a lepton mixing matrix U_PMNS with — by (F.101) run identically — 3 angles and 1 phase. The seven new parameters are 3 neutrino masses $+3$ PMNS angles $+1$ PMNS (Dirac) phase. The grand total becomes $19 + 7 = 26$.

Worked Example F.15 (Majorana neutrinos add nine parameters)**.** Now give neutrinos mass without new light fields, through the unique dimension-five operator built from Standard Model fields, the Weinberg operator

$$\mathcal{L}_5 = \frac{c_{ij}}{2\Lambda}\,(\overline{L^c_{L\,i}}\,\widetilde{H}^*)(\widetilde{H}^\dagger L_{L\,j}) + \text{h.c.}, \tag{F.107}$$

which after electroweak symmetry breaking produces a Majorana mass matrix $M_\nu = (v^2/\Lambda)\,c$ for the left-handed neutrinos. Because the operator is symmetric under $i \leftrightarrow j$ (the two lepton doublets are contracted symmetrically; the antisymmetric part vanishes by Fermi statistics together with the $\mathrm{SU}(2)_L$ contraction), c is a complex *symmetric* 3×3 matrix.

Raw count, lepton sector. A complex symmetric 3×3 matrix has 6 independent complex entries, i.e. 12 real parameters; together with Y_e (18 real), the lepton-sector raw count is $12 + 18 = 30$. (The quark sector is untouched: 36 raw, stabilizer $\mathrm{U}(1)_B$, giving its 10 as before.)

Flavor group, lepton sector. There is no ν_R to rotate, so the lepton flavor group is just $\mathrm{U}(3)_L \times \mathrm{U}(3)_e$, dim = 18. Under it $Y_e \to V_L^\dagger Y_e V_e$ as before, while the symmetric coupling transforms as $c \to V_L^T\, c\, V_L$ — note *two* factors of V_L and a transpose, because both doublets in (F.107) are L_L and one is charge-conjugated.

Stabilizer. Here is the crucial difference. The Majorana mass term is not invariant under any rephasing of the neutrinos: a phase rotation $L_L \to e^{i\alpha} L_L$ sends $c \to e^{2i\alpha} c \neq c$. Lepton number is violated by (F.107), so $\mathrm{U}(1)_L$ is gone. A generic symmetric c together with a generic Y_e has no continuous stabilizer at all in $\mathrm{U}(3)_L \times \mathrm{U}(3)_e$:

$$H_{\text{lepton}} = \{\,\mathbf{1}\,\}, \qquad \dim H_{\text{lepton}} = 0. \tag{F.108}$$

Count, lepton sector. By (F.96),

$$N^{\text{Maj}}_{\text{lepton}} = 30 - (18 - 0) = 30 - 18 = 12, \tag{F.109}$$

two more than the Dirac lepton count of 10 found in Theorem F.14. Adding the unchanged quark sector (10), the flavor total is 22, i.e. 9 more than the minimal model. The grand total is $19 + 9 = 28$.

Naming the twelve, and the extra phases. The twelve are 3 charged-lepton masses + 3 neutrino masses + 6 in the PMNS matrix. To see why the mixing matrix now holds 6 rather than 4, repeat the rephasing argument of (F.101) but with the Majorana constraint. In the charged-lepton mass basis the mixing matrix U_{PMNS} appears in the charged current $\bar{e}_L \gamma^\mu U_{\text{PMNS}}\, \nu_L W^-_\mu$. We may still rephase the charged leptons, $e_{L,R} \to P_e\, e_{L,R}$ with $P_e \in \mathrm{U}(1)^3$, sending $U_{\text{PMNS}} \to P_e^\dagger U_{\text{PMNS}}$. But we *cannot* rephase the neutrinos: a phase on ν_L would have to be undone in the Majorana mass term, where it does not cancel. So only $N = 3$ phases are removable, not the $2N - 1 = 5$ of the Dirac case. With

$U_{\mathrm{PMNS}} \in \mathrm{U}(3)$ carrying 3 angles and 6 phases by Theorem F.13,

$$\#(\text{PMNS, Majorana}) = \underbrace{3}_{\text{angles}} + \underbrace{(6-3)}_{\text{phases}} = 3+3, \tag{F.110}$$

namely 3 angles, 1 Dirac phase, and 2 further *Majorana* phases. The two extra phases relative to the Dirac case are precisely the $N-1=2$ neutrino rephasings that the Dirac theory could use to clean the matrix but the Majorana theory cannot. This is the promised origin of the Majorana phases: they are physical only because the Majorana mass removes the right-handed rephasing freedom.

Remark F.16 (The general pattern). For N generations the difference between the Majorana and Dirac mixing phases is always $N-1$: the Dirac count of removable phases is $2N-1$ (left and right rephasings minus one overall conserved $\mathrm{U}(1)$), the Majorana count is N (charged-lepton rephasings only, no conserved $\mathrm{U}(1)$ to subtract), and $(2N-1)-N=N-1$. Equivalently, a Majorana U_{PMNS} has $\frac{1}{2}N(N-1)$ physical phases against the Dirac $\frac{1}{2}(N-1)(N-2)$; their difference is $N-1$. For $N=3$ this is the two Majorana phases of (F.110).

We have now turned the slogan "nineteen parameters" into a theorem and tracked exactly how that number grows — by seven for Dirac neutrinos, by nine for Majorana — once the lepton sector is given mass. In both extensions the new physics is carried by a lepton mixing matrix with a definite number of angles and phases, just as the quark sector carries the CKM matrix with its three angles and one phase. Those mixing matrices — their standard angular parametrizations, the placement of the physical phases, and the rephasing-invariant measure of CP violation that distinguishes them — are the objects the next section makes explicit.

F.5 Mixing Matrices: CKM and PMNS

The parameter count of Section F.4 returned a sharp verdict about the flavor sector: once the three families of quarks are given their physical masses, the residual freedom in how the charged weak current connects up-type to down-type quarks is exactly *three mixing angles and one CP-violating phase*, and not one parameter more. The leptonic copy of the same argument returns, in the Majorana case, three angles and three phases. Those are *counts*: they tell us how many real numbers the data must supply, but they do not yet tell us in what shape to write the matrix that carries them, nor which combination of entries is the genuinely physical, convention-independent signature of CP violation. The job of this section is to convert the count into concrete, standard objects. We construct the explicit standard parametrization of the quark mixing matrix V_{CKM} by exhibiting how a general 3×3 unitary matrix collapses, under the maximal allowed rephasing of the quark fields, to precisely three angles and

one phase; we record the Wolfenstein expansion that organizes the measured hierarchy of V_{CKM} in powers of a single small number and the unitarity-triangle picture it makes vivid; we define the Jarlskog invariant and prove that it is rephasing invariant and equal to twice the area of any unitarity triangle; and we write the leptonic mixing matrix U_{PMNS} as the direct analog, flagging the extra Majorana phases that the count above already anticipated. The *origin* of these matrices—that V_{CKM} is the residue of diagonalizing the quark Yukawa couplings after electroweak symmetry breaking, and U_{PMNS} the residue of diagonalizing the charged-lepton and neutrino mass terms—is proved in Chapter 32 (quarks) and Chapter 35 (the neutrino-mass extensions); we import those results by reference and never reprove them here. What we owe in full, because it is load-bearing data-organizing machinery that lives nowhere else in the book, is the passage from "a unitary matrix" to "the standard matrix with its three angles, one phase, and a single invariant."

F.5.1 From a unitary matrix to three angles and one phase

We begin with the object handed to us by the diagonalization of the quark Yukawa matrices. Recall from Chapter 32 that after electroweak symmetry breaking the up- and down-type quark mass matrices M_u, M_d are brought to real, positive, diagonal form by independent unitary rotations of the left- and right-handed quark fields,

$$U_{uL}^{\dagger}\, M_u\, U_{uR} = \mathrm{diag}(m_u, m_c, m_t), \qquad U_{dL}^{\dagger}\, M_d\, U_{dR} = \mathrm{diag}(m_d, m_s, m_b), \tag{F.111}$$

and that the only place this biunitary freedom fails to cancel is in the charged weak current $J_{+}^{\mu} \propto \bar{u}_L \gamma^{\mu} d_L$, where the two *left-handed* rotations survive in the combination

$$V_{\mathrm{CKM}} \;\equiv\; U_{uL}^{\dagger}\, U_{dL}, \tag{F.112}$$

the Cabibbo–Kobayashi–Maskawa matrix [101, 298]. Being the product of two unitary matrices, V_{CKM} is itself unitary, $V_{\mathrm{CKM}}^{\dagger} V_{\mathrm{CKM}} = \mathbf{1}$. That is the entire input from Chapter 32. From here the analysis is pure linear algebra and rephasing bookkeeping, and we carry it out in full.

A unitary $N \times N$ matrix has, as a real manifold, dimension N^2: the defining condition $V^{\dagger}V = \mathbf{1}$ imposes N^2 real constraints on the $2N^2$ real parameters of a general complex matrix. We sort those N^2 real parameters into two kinds, because they behave differently under rephasing. Write $V = \exp(iH)$ with H Hermitian; split H into its real symmetric part and its imaginary antisymmetric part. The real, antisymmetric directions generate the real orthogonal subgroup $\mathrm{O}(N)$ and correspond to genuine *rotation angles*; there are

$$n_{\mathrm{angle}} = \dim \mathrm{O}(N) = \frac{N(N-1)}{2} \tag{F.113}$$

of them. The remaining

$$n_{\text{phase}} = N^2 - \frac{N(N-1)}{2} = \frac{N(N+1)}{2} \tag{F.114}$$

parameters are *phases*. The split is exhaustive and the two numbers sum to N^2, as they must. For $N = 3$ this is 3 angles and 6 phases.

Not all of those phases are physical, because the matrix V_{CKM} is not an observable in isolation: it appears sandwiched between quark fields whose phases we are free to redefine without changing any physics. A rephasing of the mass-eigenstate quark fields,

$$u_i \;\to\; e^{i\alpha_i}\, u_i, \qquad d_j \;\to\; e^{i\beta_j}\, d_j \qquad (i, j = 1, \ldots, N), \tag{F.115}$$

leaves the kinetic terms, the diagonal mass terms, and the neutral currents invariant—each of those involves a field paired with its own conjugate, so the phase cancels—while the charged current $\bar{u}_{Li}(V_{\text{CKM}})_{ij}\gamma^\mu d_{Lj}$ is invariant if and only if we simultaneously transform

$$(V_{\text{CKM}})_{ij} \;\to\; e^{-i\alpha_i}\,(V_{\text{CKM}})_{ij}\,e^{i\beta_j} \;=\; e^{\,i(\beta_j-\alpha_i)}\,(V_{\text{CKM}})_{ij}. \tag{F.116}$$

Two matrices related by Eq. (F.116) describe identical physics; the physical content of V_{CKM} is the equivalence class. The transformation Eq. (F.116) depends on $2N$ phases $\{\alpha_i, \beta_j\}$, but not all $2N$ act: the diagonal shift $\alpha_i \to \alpha_i + c$, $\beta_j \to \beta_j + c$ by a common constant c leaves every $(V_{\text{CKM}})_{ij}$ untouched, because only the difference $\beta_j - \alpha_i$ enters. Exactly one combination is therefore inert, so the number of independent rephasings that can be used to remove parameters from V_{CKM} is

$$n_{\text{rephase}} = 2N - 1. \tag{F.117}$$

Theorem F.17 (Physical content of an N-family quark mixing matrix)**.** *A unitary $N \times N$ quark mixing matrix, modulo the rephasing freedom Eq. (F.115) of the quark mass eigenstates, carries exactly*

$$n_{angle} = \frac{N(N-1)}{2} \quad \textit{rotation angles} \qquad \textit{and} \qquad n_\delta = \frac{(N-1)(N-2)}{2} \quad \textit{physical phases,} \tag{F.118}$$

for a total of $(N-1)^2$ physical parameters. For $N = 3$ this is 3 angles and 1 phase.

Proof. Rephasings Eq. (F.115) are pure phase transformations: they shift the n_{phase} phases of V but cannot touch a rotation angle, since multiplying matrix entries by pure phases never changes the moduli $|V_{ij}|$ on which the angles depend. The angle count Eq. (F.113) is therefore physical as it stands. Of the $n_{\text{phase}} = N(N+1)/2$ phases in V, we may remove $n_{\text{rephase}} = 2N-1$ by a suitable choice of $\{\alpha_i, \beta_j\}$—and we can remove that many and no more, because the

$2N-1$ independent differences $\beta_j-\alpha_i$ that act on V are linearly independent as functions on the phase directions, so they fix $2N-1$ of the phases to any desired values (e.g. zero). The number of phases that survive every rephasing is thus

$$n_\delta = \frac{N(N+1)}{2} - (2N-1) = \frac{N^2+N-4N+2}{2} = \frac{N^2-3N+2}{2} = \frac{(N-1)(N-2)}{2}. \tag{F.119}$$

Adding the angles, the total physical parameter count is

$$n_{\text{angle}} + n_\delta = \frac{N(N-1)}{2} + \frac{(N-1)(N-2)}{2} = \frac{(N-1)\big[N+(N-2)\big]}{2} = \frac{(N-1)(2N-2)}{2} = (\tag{F.120}$$

Setting $N=3$ gives $n_{\text{angle}}=3$, $n_\delta=1$, total 4. □

Theorem F.17 is the linear-algebra core behind the verdict of Section F.4, now exhibited entry by entry. Two features deserve emphasis. First, the phase count n_δ *vanishes for* $N=1$ *and* $N=2$: a single family has no mixing, and—this is the Kobayashi–Maskawa observation [298]—two families have only the single Cabibbo angle and *no* physical phase, so CP violation in the quark sector is impossible with fewer than three families. The existence of a CP-violating phase in the charged current is, by this counting, a prediction that there be at least a third quark family. Second, the count is silent about *how* to distribute the surviving 3 angles and 1 phase among the entries; that distribution is a convention, and we fix it now.

F.5.2 The standard parametrization

The canonical convention factors V_{CKM} into three complex rotations in the $(1,2)$, $(1,3)$, and $(2,3)$ planes, with the single physical phase δ attached to the $(1,3)$ rotation. Write $c_{ij}=\cos\theta_{ij}$ and $s_{ij}=\sin\theta_{ij}$ for the three angles $\theta_{12},\theta_{13},\theta_{23}\in[0,\pi/2]$, and $\delta\in[0,2\pi)$ for the phase.

Definition F.18 (Standard parametrization of the CKM matrix)**.**

$$V_{\text{CKM}} = \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & c_{23} & s_{23} \\ 0 & -s_{23} & c_{23} \end{pmatrix}}_{R_{23}} \underbrace{\begin{pmatrix} c_{13} & 0 & s_{13}e^{-i\delta} \\ 0 & 1 & 0 \\ -s_{13}e^{i\delta} & 0 & c_{13} \end{pmatrix}}_{U_{13}(\delta)} \underbrace{\begin{pmatrix} c_{12} & s_{12} & 0 \\ -s_{12} & c_{12} & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{R_{12}}. \tag{F.121}$$

Each factor is manifestly unitary, so the product is unitary; R_{23} and R_{12} are real rotations carrying the angles θ_{23},θ_{12}, while the central factor $U_{13}(\delta)$ carries both the angle θ_{13} and the phase δ. Multiplying the three factors out—first $U_{13}R_{12}$, then left-multiplying by R_{23}—gives the explicit matrix that the

numerical section will populate:

$$V_{\mathrm{CKM}} = \begin{pmatrix} V_{ud} & V_{us} & V_{ub} \\ V_{cd} & V_{cs} & V_{cb} \\ V_{td} & V_{ts} & V_{tb} \end{pmatrix} = \begin{pmatrix} c_{12}c_{13} & s_{12}c_{13} & s_{13}e^{-i\delta} \\ -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} & c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} & s_{23}c_{13} \\ s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} & -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} & c_{23}c_{13} \end{pmatrix}. \tag{F.122}$$

The labeling of rows by (u, c, t) and columns by (d, s, b) is fixed: V_{ij} is the amplitude for the charged-current transition between up-type quark i and down-type quark j. Three entries are especially transparent: the corners $V_{us} = s_{12}c_{13}$, $V_{cb} = s_{23}c_{13}$, and $V_{ub} = s_{13}e^{-i\delta}$ isolate the three angles directly (up to the small $c_{13} \approx 1$ corrections), and the phase δ sits, in this convention, on the smallest entry V_{ub} (and on V_{td}). One verifies unitarity of Eq. (F.122) by direct computation; we record the diagonal check for the first column, since it exhibits the cancellation pattern that the unitarity triangles below make geometric:

$$\begin{aligned} |V_{ud}|^2 + |V_{cd}|^2 + |V_{td}|^2 &= c_{12}^2 c_{13}^2 + \left|-s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta}\right|^2 + \left|s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta}\right|^2 \\ &= c_{12}^2 c_{13}^2 + s_{12}^2 c_{23}^2 + c_{12}^2 s_{23}^2 s_{13}^2 + 2 s_{12}c_{23}c_{12}s_{23}s_{13}\cos\delta \\ &\quad + s_{12}^2 s_{23}^2 + c_{12}^2 c_{23}^2 s_{13}^2 - 2 s_{12}s_{23}c_{12}c_{23}s_{13}\cos\delta \\ &= c_{12}^2 c_{13}^2 + s_{12}^2(c_{23}^2 + s_{23}^2) + c_{12}^2 s_{13}^2 (s_{23}^2 + c_{23}^2) \\ &= c_{12}^2 c_{13}^2 + s_{12}^2 + c_{12}^2 s_{13}^2 = c_{12}^2(c_{13}^2 + s_{13}^2) + s_{12}^2 = c_{12}^2 + s_{12}^2 = 1, \end{aligned} \tag{F.123}$$

where the two $\cos\delta$ cross terms cancelled identically—a foretaste of the fact that the phase δ drops out of the moduli-squared sums but survives in the imaginary cross terms that measure CP violation.

F.5.3 The Wolfenstein expansion

Measurement shows the CKM matrix is strongly hierarchical: it is nearly the identity, with the off-diagonal entries falling off in a regular pattern. The $(1,2)$ mixing $|V_{us}| \approx 0.22$ is the largest off-diagonal element; the $(2,3)$ mixing $|V_{cb}|$ is roughly its square, and the $(1,3)$ mixing $|V_{ub}|$ roughly its cube. Wolfenstein [526] packaged this hierarchy by introducing the single small parameter

$$\lambda \equiv s_{12} = \frac{|V_{us}|}{\sqrt{|V_{ud}|^2 + |V_{us}|^2}} \approx 0.225, \tag{F.124}$$

and expressing the remaining three parameters through numbers A, ρ, η of order unity defined by the *exact* relations

$$s_{12} = \lambda, \qquad s_{23} = A\lambda^2, \qquad s_{13}e^{-i\delta} = A\lambda^3(\rho - i\eta). \tag{F.125}$$

Equation (F.125) is a change of variables, not an approximation: given the four measured quantities $(\theta_{12}, \theta_{13}, \theta_{23}, \delta)$ it defines (λ, A, ρ, η) uniquely, and

conversely. The utility appears when we substitute Eq. (F.125) into the explicit matrix Eq. (F.122) and expand in λ. Using $c_{12} = \sqrt{1-\lambda^2} = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4)$, $c_{13} = 1 + O(\lambda^6)$, $c_{23} = 1 + O(\lambda^4)$, one finds entry by entry:

$$V_{ud} = c_{12}c_{13} = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4), \tag{F.126}$$

$$V_{us} = s_{12}c_{13} = \lambda + O(\lambda^7), \tag{F.127}$$

$$V_{ub} = s_{13}e^{-i\delta} = A\lambda^3(\rho - i\eta), \tag{F.128}$$

$$V_{cb} = s_{23}c_{13} = A\lambda^2 + O(\lambda^8), \tag{F.129}$$

$$V_{cd} = -s_{12}c_{23} - c_{12}s_{23}s_{13}e^{i\delta} = -\lambda + O(\lambda^5), \tag{F.130}$$

$$V_{cs} = c_{12}c_{23} - s_{12}s_{23}s_{13}e^{i\delta} = 1 - \tfrac{1}{2}\lambda^2 + O(\lambda^4), \tag{F.131}$$

$$V_{td} = s_{12}s_{23} - c_{12}c_{23}s_{13}e^{i\delta} = A\lambda^3(1 - \rho - i\eta) + O(\lambda^5), \tag{F.132}$$

$$V_{ts} = -c_{12}s_{23} - s_{12}c_{23}s_{13}e^{i\delta} = -A\lambda^2 + O(\lambda^4), \tag{F.133}$$

$$V_{tb} = c_{23}c_{13} = 1 + O(\lambda^4). \tag{F.134}$$

The single subtle entry is V_{td}. Its leading piece is $s_{12}s_{23} - c_{12}c_{23}\,s_{13}e^{i\delta}$; with $s_{13}e^{i\delta} = (s_{13}e^{-i\delta})^* = A\lambda^3(\rho + i\eta)$ this is $\lambda \cdot A\lambda^2 - 1 \cdot A\lambda^3(\rho + i\eta) = A\lambda^3(1 - \rho - i\eta)$, as quoted. Collecting the results gives the standard Wolfenstein matrix,

$$V_{\mathrm{CKM}} = \begin{pmatrix} 1 - \tfrac{1}{2}\lambda^2 & \lambda & A\lambda^3(\rho - i\eta) \\ -\lambda & 1 - \tfrac{1}{2}\lambda^2 & A\lambda^2 \\ A\lambda^3(1 - \rho - i\eta) & -A\lambda^2 & 1 \end{pmatrix} + O(\lambda^4). \tag{F.135}$$

The hierarchy is now visible at a glance: the diagonal is 1 up to $O(\lambda^2)$, the Cabibbo block is $O(\lambda)$, the $(2,3)$ mixing is $O(\lambda^2)$, and the $(1,3)$ corners are $O(\lambda^3)$ and complex. CP violation lives entirely in those small complex corners; the imaginary part is controlled by η, and $\eta \neq 0$ is the statement that the Standard Model violates CP.

It is worth recording that the variables (ρ, η) defined by Eq. (F.125) carry $O(\lambda^2)$ convention dependence, which the phenomenology removes by working instead with the *barred* variables

$$\bar{\rho} + i\bar{\eta} \;\equiv\; -\frac{V_{ud}V_{ub}^*}{V_{cd}V_{cb}^*}, \tag{F.136}$$

a ratio that is manifestly rephasing invariant (each quark phase cancels between numerator and denominator) and hence a genuine observable. Expanding the right-hand side with Eq. (F.135) gives $\bar{\rho} + i\bar{\eta} = (\rho + i\eta)(1 - \tfrac{1}{2}\lambda^2) + O(\lambda^4)$, so the barred and unbarred pairs agree at leading order and differ only at the percent level; the numerical section quotes $\bar{\rho}, \bar{\eta}$. The definition Eq. (F.136) is also exactly the apex of the unitarity triangle, to which we now turn.

F.5.4 Unitarity triangles

Unitarity of V_{CKM} means its rows are orthonormal and so are its columns. The six off-diagonal orthogonality relations each assert that three complex numbers sum to zero, and three complex numbers summing to zero form a closed triangle in the complex plane—a *unitarity triangle*. The phenomenologically central one comes from the orthogonality of the first (d) and third (b) columns:

$$V_{ud}V_{ub}^{*} + V_{cd}V_{cb}^{*} + V_{td}V_{tb}^{*} = 0. \tag{F.137}$$

Each of the three terms is, from Eq. (F.135), of order λ^3: this is the one triangle whose three sides are comparable in magnitude (the others are badly "squashed," with one side much shorter than the others), which is why it is the one drawn and measured. Dividing Eq. (F.137) by the (essentially real) middle term $V_{cd}V_{cb}^{*}$ normalizes the base of the triangle to unit length and places its two ends at 0 and 1 on the real axis:

$$\underbrace{\frac{V_{ud}V_{ub}^{*}}{V_{cd}V_{cb}^{*}}}_{=-(\bar{\rho}+i\bar{\eta})} + 1 + \underbrace{\frac{V_{td}V_{tb}^{*}}{V_{cd}V_{cb}^{*}}}_{=-(1-\bar{\rho}-i\bar{\eta})} = 0, \tag{F.138}$$

where the first ratio is $-(\bar{\rho}+i\bar{\eta})$ by Eq. (F.136), and the third follows from the same expansion. Reading Eq. (F.138) as "vector from 1 to apex" plus "base" plus "vector from apex to 0," the apex of the normalized triangle sits at the point $(\bar{\rho},\bar{\eta})$ in the complex plane. The three interior angles, named α,β,γ (equivalently ϕ_2,ϕ_1,ϕ_3), are themselves rephasing-invariant phase differences of CKM products and are measured independently in B-meson decays; their sum is π because the figure is a Euclidean triangle, and the experimental confirmation that $\alpha+\beta+\gamma=\pi$ within errors is a nontrivial test of the three-family unitarity of V_{CKM}. Figure F.1 draws the triangle.

F.5.5 The Jarlskog invariant

The phase δ in Eq. (F.122), and the parameter η in Eq. (F.135), are convention-dependent: a rephasing Eq. (F.116) shuffles where the phase "sits" and can even make a chosen entry real. What cannot be shuffled away—if CP is genuinely violated—is an overall, rephasing-invariant measure. Jarlskog [284] identified the unique (up to sign) such measure for three families.

Definition F.19 (Jarlskog invariant)**.**

$$J \;\equiv\; \mathrm{Im}\!\left(V_{us}\,V_{cb}\,V_{ub}^{*}\,V_{cs}^{*}\right). \tag{F.139}$$

The combination is built from a 2×2 minor: two entries and the complex conjugates of the two entries obtained by swapping both their row and their

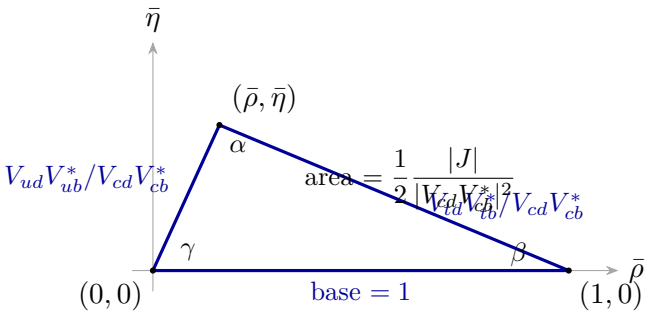


Figure F.1: The rescaled db unitarity triangle in the $(\bar\rho,\bar\eta)$ plane, from the column-orthogonality relation Eq. (F.137) divided by $V_{cd}V_{cb}^*$. The base runs from $(0,0)$ to $(1,0)$ and the apex sits at $(\bar\rho,\bar\eta)$; the interior angles α,β,γ satisfy $\alpha+\beta+\gamma=\pi$. The triangle has nonzero area precisely when $\bar\eta\neq 0$, i.e. when CP is violated; its area equals $|J|/2$ once the sides are restored to their unnormalized lengths, with J the Jarlskog invariant of Theorem F.19.

column indices $(us, cb$ paired with $ub, cs)$. This index structure is what makes it rephasing invariant, and the imaginary part is what makes it the CP-odd observable. We prove both facts, and then connect J to the triangle area.

Theorem F.20 (Properties of the Jarlskog invariant)**.** *The quantity J of Eq. (F.139) is invariant under the quark rephasing Eq. (F.116); it equals*

$$J = c_{12}c_{23}c_{13}^2\, s_{12}s_{23}s_{13}\,\sin\delta \tag{F.140}$$

in the standard parametrization, hence vanishes if and only if some angle is 0 or $\pi/2$ or $\sin\delta = 0$; and all six CKM unitarity triangles have the same area, equal to $|J|/2$.

Proof. Rephasing invariance. Under Eq. (F.116) each entry acquires a phase $V_{ij}\to e^{i(\beta_j-\alpha_i)}V_{ij}$, so the product in Eq. (F.139) acquires the phase

$$\begin{aligned}&\big[(\beta_s-\alpha_u)\big]_{V_{us}} + \big[(\beta_b-\alpha_c)\big]_{V_{cb}} - \big[(\beta_b-\alpha_u)\big]_{V_{ub}^*} - \big[(\beta_s-\alpha_c)\big]_{V_{cs}^*}\\ &\quad= (\beta_s-\alpha_u)+(\beta_b-\alpha_c)-(\beta_b-\alpha_u)-(\beta_s-\alpha_c) = 0,\end{aligned} \tag{F.141}$$

the rows (u,c) and columns (s,b) each appearing once with $+$ and once with $-$. The product $V_{us}V_{cb}V_{ub}^*V_{cs}^*$ is therefore invariant, and so is its imaginary part J. This is the algebraic reason Theorem F.19 is built from a closed "box" of indices: every rephasing phase cancels.

Standard-parametrization value. Insert the entries from Eq. (F.122): $V_{us} = s_{12}c_{13}$, $V_{cb} = s_{23}c_{13}$, $V_{ub}^* = s_{13}e^{i\delta}$, and $V_{cs}^* = c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta}$. The product is

$$V_{us}V_{cb}V_{ub}^*V_{cs}^* = (s_{12}c_{13})(s_{23}c_{13})(s_{13}e^{i\delta})(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta})$$

$$= s_{12}s_{23}s_{13}c_{13}^2\, e^{i\delta}\left(c_{12}c_{23} - s_{12}s_{23}s_{13}e^{-i\delta}\right)$$
$$= s_{12}s_{23}s_{13}c_{13}^2\left(c_{12}c_{23}\, e^{i\delta} - s_{12}s_{23}s_{13}\right). \tag{F.142}$$

The second term inside the brackets is real; taking the imaginary part keeps only $c_{12}c_{23}\sin\delta$ from the first, giving

$$J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*) = c_{12}c_{23}c_{13}^2\, s_{12}s_{23}s_{13}\,\sin\delta, \tag{F.143}$$

which is Eq. (F.140). Since every factor of a cosine or sine of an angle appears, $J = 0$ whenever any $\theta_{ij} \in \{0, \pi/2\}$ (a degenerate rotation) or $\sin\delta = 0$ (no genuine phase); CP violation requires all three angles to be nondegenerate *and* a nontrivial phase, exactly as the count of Theorem F.17 demanded.

Area of a unitarity triangle. Take any column-orthogonality relation, $\sum_i V_{i\alpha}V_{i\beta}^* = 0$ with $\alpha \neq \beta$ fixed, and write its three terms as the directed sides $z_i = V_{i\alpha}V_{i\beta}^*$ of a triangle in the complex plane, $z_1 + z_2 + z_3 = 0$. The (unsigned) area of a triangle whose two edge vectors emanating from one vertex are the complex numbers z_1 and $-z_2$ is $\frac{1}{2}|\,\mathrm{Im}(z_1\,\overline{(-z_2)})| = \frac{1}{2}|\,\mathrm{Im}(z_1 z_2^*)|$. Now

$$z_1 z_2^* = V_{1\alpha}V_{1\beta}^*\left(V_{2\alpha}V_{2\beta}^*\right)^* = V_{1\alpha}V_{2\beta}\,V_{1\beta}^*\,V_{2\alpha}^*, \tag{F.144}$$

which is exactly a Jarlskog box for rows $1, 2$ and columns α, β. By the standard reduction of such boxes for a 3×3 unitary matrix, $\mathrm{Im}(V_{i\alpha}V_{j\beta}V_{i\beta}^*V_{j\alpha}^*) = J\sum_{k,\gamma}\varepsilon_{ijk}\varepsilon_{\alpha\beta\gamma}$, the right-hand side equals $\pm J$ for any choice of two distinct rows and two distinct columns. Hence $|\,\mathrm{Im}(z_1 z_2^*)| = |J|$ and the area is $|J|/2$, *independent* of which orthogonality relation (which triangle) we chose. All six triangles share the common area $|J|/2$; they differ only in shape, the *db* triangle of Fig. F.1 being the one whose three sides are of comparable length. □

The reduction $\mathrm{Im}(V_{i\alpha}V_{j\beta}V_{i\beta}^*V_{j\alpha}^*) = J\sum_{k\gamma}\varepsilon_{ijk}\varepsilon_{\alpha\beta\gamma}$ used above is itself a consequence of unitarity; the following worked example both establishes it for the case at hand and extracts the numerical size of J in the Wolfenstein expansion, the number the data section will quote.

Worked Example F.21 (The size of CP violation: J in the Wolfenstein expansion)**.** We compute J in powers of λ and confirm directly that the imaginary parts of all the Jarlskog boxes coincide. From Eq. (F.140) and the Wolfenstein definitions Eq. (F.125), with $c_{12}, c_{23}, c_{13} \to 1$ at leading order,

$$J = c_{12}c_{23}c_{13}^2\, s_{12}\, s_{23}\left(s_{13}\sin\delta\right). \tag{F.145}$$

The combination $s_{13}\sin\delta$ is read off from $s_{13}e^{-i\delta} = A\lambda^3(\rho - i\eta)$, whose imaginary part is $-s_{13}\sin\delta = -A\lambda^3\eta$, i.e. $s_{13}\sin\delta = A\lambda^3\eta$. With $s_{12} = \lambda$ and $s_{23} = A\lambda^2$,

$$J = (1)\,(\lambda)\,(A\lambda^2)\,(A\lambda^3\eta) + O(\lambda^8) = A^2\lambda^6\,\eta + O(\lambda^8) \;\approx\; A^2\lambda^6\,\bar{\eta}. \tag{F.146}$$

The striking feature is the power λ^6: CP violation in the quark sector is suppressed by the *sixth* power of the Cabibbo parameter, because it requires the simultaneous participation of all three mixing angles, each contributing its own power of λ, with the $(1,3)$ angle entering twice (once as s_{13}, once because the relevant box necessarily reaches across all three families). Numerically, with the mid-2020s values $\lambda \approx 0.225$, $A \approx 0.83$, $\bar{\eta} \approx 0.355$ that the next section will pin down, $J \approx (0.83)^2(0.225)^6(0.355) \approx 3.1 \times 10^{-5}$—a tiny number, and the quantitative statement of why CP violation is a small effect in the Standard Model.

To verify the box reduction explicitly, compare two different boxes. The defining box is $J = \mathrm{Im}(V_{us}V_{cb}V_{ub}^*V_{cs}^*) \approx \mathrm{Im}[\lambda \cdot A\lambda^2 \cdot A\lambda^3(\rho+i\eta)\cdot 1] = A^2\lambda^6\eta$, using $V_{ub}^* = A\lambda^3(\rho+i\eta)$ and $V_{cs} \approx 1$. A different box, built from rows u,t and columns d,b, is $\mathrm{Im}(V_{ud}V_{tb}V_{ub}^*V_{td}^*)$. With $V_{ud} \approx 1$, $V_{tb} \approx 1$, $V_{ub}^* = A\lambda^3(\rho+i\eta)$, and $V_{td}^* = A\lambda^3(1-\rho+i\eta)$,

$$\begin{aligned}\mathrm{Im}(V_{ud}V_{tb}V_{ub}^*V_{td}^*) &\approx \mathrm{Im}\Big[A\lambda^3(\rho+i\eta)\cdot A\lambda^3(1-\rho+i\eta)\Big] \\ &= A^2\lambda^6\ \mathrm{Im}\big[(\rho+i\eta)(1-\rho+i\eta)\big] \\ &= A^2\lambda^6\big[\eta(1-\rho)+\eta\rho\big] = A^2\lambda^6\,\eta, \end{aligned} \tag{F.147}$$

the same value, confirming that the imaginary part is common to all boxes and equal to J, in agreement with Theorem F.20. The cancellation of the ρ-dependence in the last line is the algebraic shadow of the geometric fact that the area of the triangle does not depend on where its apex sits along a line of fixed height $\bar{\eta}$.

F.5.6 The leptonic analog: the PMNS matrix

The same construction runs in the lepton sector, with one decisive difference that the count of Section F.4 already flagged. Diagonalizing the charged-lepton mass matrix and the neutrino mass matrix by left-handed unitary rotations $U_{eL}, U_{\nu L}$ leaves, in the leptonic charged current, the residual combination

$$U_{\mathrm{PMNS}} \;\equiv\; U_{eL}^{\dagger}\,U_{\nu L}, \tag{F.148}$$

the Pontecorvo–Maki–Nakagawa–Sakata matrix [330, 408], whose origin is established in Chapter 35. If the neutrinos are *Dirac* particles—mass term $\bar{\nu}_L M_\nu \nu_R$ with an independent right-handed field—then the rephasing bookkeeping is identical to the quark case: both charged leptons and neutrinos can be freely rephased, the count of Theorem F.17 applies verbatim, and U_{PMNS} has three angles and one phase. Its standard parametrization is the literal transcription of Eq. (F.121),

$$U_{\mathrm{PMNS}} = R_{23}(\theta_{23})\,U_{13}(\theta_{13},\delta)\,R_{12}(\theta_{12}), \tag{F.149}$$

with the same 3×3 explicit form as Eq. (F.122) (now with rows labeled by the charged-lepton flavors e,μ,τ and columns by the mass eigenstates ν_1,ν_2,ν_3, and

the phase usually written $\delta = \delta_{\mathrm{CP}}$). One important contrast with the quark sector is quantitative rather than structural: the measured leptonic angles are *large*—θ_{12} and θ_{23} are tens of degrees, not the few degrees of the quark sector—so the PMNS matrix is far from the identity and admits no useful Wolfenstein-style hierarchy expansion. The Jarlskog construction nonetheless carries over unchanged: a leptonic invariant $J_{\mathrm{lep}} = \mathrm{Im}(U_{\mu 3} U_{e2} U_{e3}^{*} U_{\mu 2}^{*}) = c_{12} c_{23} c_{13}^{2} s_{12} s_{23} s_{13} \sin\delta_{\mathrm{CP}}$ measures CP violation in neutrino oscillations and can be of order 10^{-2}, far larger than its quark counterpart because the angles are large.

Caution: Majorana neutrinos carry extra phases

If instead the neutrinos are *Majorana* particles—each neutrino its own antiparticle, with a mass term $\frac{1}{2}\,\overline{\nu_L^{c}}\, M_\nu\, \nu_L$ built from the left-handed field alone (Chapter 35)—then the neutrino fields can no longer be rephased: a phase $\nu_i \to e^{i\phi_i}\nu_i$ multiplies the Majorana mass term by $e^{2i\phi_i} \neq 1$ and is not a symmetry. Only the three charged-lepton phases remain available, and there is now *no* overall common phase to subtract (the would-be common shift cannot be compensated on the neutrino side), so $n_{\mathrm{rephase}} = N$ rather than $2N-1$. Repeating the count of Theorem F.17 with this single change, the number of physical phases becomes

$$n_{\delta}^{\mathrm{Maj}} = \frac{N(N+1)}{2} - N = \frac{N(N-1)}{2}, \tag{F.150}$$

which for $N = 3$ is 3 phases: the one "Dirac" phase δ_{CP} already present in Eq. (F.149), plus *two* additional "Majorana" phases. They are conventionally collected in a diagonal matrix on the right,

$$U_{\mathrm{PMNS}}^{\mathrm{Maj}} = U_{\mathrm{PMNS}} \cdot \mathrm{diag}\big(1,\, e^{i\alpha_{21}/2},\, e^{i\alpha_{31}/2}\big), \tag{F.151}$$

with U_{PMNS} the standard matrix Eq. (F.149). The Majorana phases α_{21}, α_{31} drop out of all oscillation observables—oscillation amplitudes pair each $U_{\alpha i}$ with a conjugate $U_{\beta i}^{*}$ carrying the same neutrino index i, so the right-diagonal phase cancels—and therefore cannot be measured in oscillation experiments. They survive only in lepton-number violating processes, most notably neutrinoless double beta decay, whose rate depends on the effective mass $m_{\beta\beta} = |\sum_i U_{ei}^{2} m_i|$ in which the Majorana phases do *not* cancel. The angle count $n_{\mathrm{angle}} = N(N-1)/2 = 3$ is unaffected: Majorana neutrinos still mix through three angles, but through three phases rather than one.

The structures assembled here—the standard 3×3 matrix Eq. (F.122) with its three angles and one phase, the Wolfenstein hierarchy Eq. (F.135) with parameters $(\lambda, A, \bar\rho, \bar\eta)$, the rephasing-invariant Jarlskog measure J, and their leptonic counterparts including the Majorana phases—are exactly the slots that

the numerical section fills. Section F.6 now quotes the measured magnitudes $|V_{ij}|$, the fitted angles and phase, the apex $(\bar{\rho}, \bar{\eta})$ with the value of J, and the mid-2020s leptonic angles and δ_{CP}, each accompanied by the scheme and convention that make it a meaningful number.

F.6 Numerical Values and Their Defining Schemes

The previous sections built the empty slots: the standard CKM and PMNS parametrizations of Section F.5, the Jarlskog invariant Eq. (F.139), and the mass and coupling relations that the Higgs mechanism and electroweak unification of Chapters 31 and 33 fix in terms of the underlying Lagrangian parameters. It remains to fill those slots with numbers. This is less innocent than it sounds. A measured quantity in a renormalizable quantum field theory is almost never a bare invariant of nature; it is the value of a parameter *in a scheme*, and the same physical fact reported in two schemes is two different numbers. The fine-structure constant is 1/137.036 at zero momentum and about 1/128 at the Z mass; the weak mixing angle has one value if defined by a ratio of pole masses and a slightly different value if defined by $\overline{\mathrm{MS}}$ subtraction at $\mu = m_Z$; a quark "mass" of 4.18 GeV in $\overline{\mathrm{MS}}$ becomes a pole mass near 4.8 GeV, and the latter is not even unambiguously defined beyond a few loops. *A number without its scheme is not a measurement; it is an ambiguity.* The discipline of this section is therefore to quote every value together with the definition that makes it mean something, and to date everything honestly: all numbers below are the mid-2020s Particle Data Group values [382], with neutrino mixing taken from the contemporary global fits [180], and they will drift as measurements improve.

F.6.1 Three kinds of number, and the duty to label

It helps to keep three categories distinct, because the word "value" means something different in each.

The first kind is a genuine *physical observable*: a pole position, a decay rate, a cross section, a ratio of pole masses. These are scheme-independent by construction—they are properties of the S-matrix, not of a Lagrangian parametrization—and a number reported for one of them needs only a statement of *what* was measured, not of any subtraction scheme. The Z and W masses as Breit–Wigner pole parameters, the charged-lepton pole masses, and the Fermi constant extracted from the muon lifetime are of this type.

The second kind is a *short-distance Lagrangian parameter*: a running coupling $\alpha_s(\mu)$, a running mass $\overline{m}(\mu)$, an $\overline{\mathrm{MS}}$ mixing angle. Such a quantity is finite only after renormalization, and its value depends on the renormalization scheme and on the scale μ at which it is quoted. Reporting it requires naming the scheme ($\overline{\mathrm{MS}}$, on-shell, momentum subtraction) *and* the scale. The running of these

objects with μ is governed by the renormalization-group machinery of Chapter 24 for the electroweak couplings and Chapter 27 for the strong coupling; here we record values at conventional reference scales and rely on those chapters for the flow.

The third kind is a *derived quantity*: a number computed from the first two through a fixed relation, such as the Higgs vacuum expectation value v obtained from G_F, or the QCD scale Λ obtained from $\alpha_s(m_Z)$. These inherit their scheme from the inputs and from the order of the conversion formula, and the worked examples below show how much that matters.

Caution

The single most common error in using tabulated Standard Model data is to take a number quoted in one scheme and feed it into a formula derived in another. A pole mass inserted where an $\overline{\text{MS}}$ mass is expected, or $\alpha(0)$ used where $\alpha(m_Z)$ belongs, produces an answer wrong at the level of the very loop corrections one is trying to compute. Every entry below carries its scheme label for exactly this reason; the conversion dictionary that lets one move safely between schemes is assembled in Section F.7.

F.6.2 Electroweak and strong couplings

The gauge sector is fixed numerically by a small set of couplings, each of which exists in more than one definition. We list them with their schemes.

The *fine-structure constant* is best known at zero momentum transfer, from the electron anomalous magnetic moment and atom-interferometry determinations of α (QED, Chapter 24):

$$\alpha(0) = \frac{e^2}{4\pi}\bigg|_{q^2=0} = \frac{1}{137.035\,999\,18(2)} = 7.297\,352\,6 \times 10^{-3}. \tag{F.152}$$

This is the Thomson-limit coupling. For processes at the electroweak scale the relevant quantity is the effective coupling at $q^2 = m_Z^2$, into which the vacuum-polarization of all charged fermions has been run up per the QED RG of Chapter 24. In the $\overline{\text{MS}}$ scheme with five active quark flavors,

$$\alpha^{(5)}(m_Z)|^{-1}_{\overline{\text{MS}}} = 127.951(9), \qquad \alpha(m_Z)^{-1}\big|_{\text{eff., on-shell}} \approx 128.95, \tag{F.153}$$

where the two numbers differ because the on-shell effective coupling and the $\overline{\text{MS}}$ coupling resum the same logarithms with different finite parts and different treatments of the top and the W. The running from Eq. (F.152) to Eq. (F.153)—a change of nearly seven percent in α—is the numerical anchor for QED's running coupling Chapter 24; the dominant uncertainty in Eq. (F.153) is hadronic, from the light-quark contribution to the vacuum polarization that cannot be computed perturbatively.

The *strong coupling* is quoted at the Z mass in $\overline{\text{MS}}$ with five active flavors, the standard reference point for Yang–Mills theory:

$$\alpha_s^{(5)}(m_Z)|_{\overline{\text{MS}}} = 0.1180(9). \tag{F.154}$$

Its running, and the extraction of the QCD scale Λ, are governed by the one-loop beta function $\beta_g = -\frac{g^3}{16\pi^2}\, b_0$ with $b_0 = \frac{11}{3}C_A - \frac{4}{3}T_F n_f$ computed in Chapter 27; for SU(3) ($C_A = 3$, $T_F = \frac{1}{2}$) and n_f flavors this is $b_0 = 11 - \frac{2}{3}n_f$. We make this anchor concrete in Theorem F.24.

The *Fermi constant* is the cleanest electroweak input, extracted from the muon lifetime through the muon-decay rate with QED radiative corrections removed in the standard Fermi-theory convention:

$$G_F = 1.166\,378\,7(6) \times 10^{-5}\,\text{GeV}^{-2}. \tag{F.155}$$

It is measured to better than a part per million and is therefore taken as one of the three defining electroweak inputs $\{\alpha,\, G_F,\, m_Z\}$ of the input-scheme dictionary in Section F.7.

The *weak mixing angle* is the canonical illustration that a number is nothing without its scheme. Two definitions are in common use and give different numbers for the same physics. The *on-shell* definition fixes $\sin^2\theta_{\text{W}}$ by the ratio of the physical (pole) gauge-boson masses,

$$\sin^2\theta_{\text{W}}\big|_{\text{on-shell}} \equiv 1 - \frac{m_W^2}{m_Z^2} = 0.2233(6), \tag{F.156}$$

a manifestly scheme-independent combination of observables. The $\overline{\text{MS}}$ definition fixes it instead through the renormalized couplings at $\mu = m_Z$,

$$\sin^2\theta_{\text{W}}|_{\overline{\text{MS}}} \equiv \frac{\hat{g}'^2}{\hat{g}^2 + \hat{g}'^2}\bigg|_{\mu = m_Z} = \hat{s}_Z^2 = 0.231\,21(4), \tag{F.157}$$

where $\hat{g}, \hat{g}'$ are the $\overline{\text{MS}}$ gauge couplings. The two values differ by about 0.008, an eight-per-mille gap that is entirely real physics—the finite parts of the radiative corrections—and not experimental error: Eq. (F.156) is smaller because the on-shell definition absorbs the large top-quark contribution to the ρ parameter differently than $\overline{\text{MS}}$ does. A prediction quoted to per-mille accuracy that does not say which $\sin^2\theta_{\text{W}}$ it uses is simply not a prediction.

F.6.3 Gauge-boson and Higgs masses

The massive bosons are quoted as pole masses—physical observables, no scheme label needed beyond "pole":

$$m_Z = 91.1880(20)\,\text{GeV}, \tag{F.158}$$

Table F.4: Standard Model gauge couplings, each with its defining scheme, as of the mid-2020s [382]. Running quantities are quoted at their conventional reference scale. Schemes: "$q^2 = 0$" is the Thomson limit; "$\overline{\text{MS}}$" denotes modified minimal subtraction at the stated scale with five active quark flavors; "on-shell" denotes a definition through pole masses.

Quantity	Value	Scheme / definition
$\alpha(0)$	$1/137.035\,999\,18(2)$	$q^2 = 0$ (Thomson)
$\alpha^{(5)}(m_Z)$	$1/127.951(9)$	$\overline{\text{MS}}$, $\mu = m_Z$
$\alpha_s^{(5)}(m_Z)$	$0.1180(9)$	$\overline{\text{MS}}$, $\mu = m_Z$
G_F	$1.166\,378\,7(6) \times 10^{-5}\,\text{GeV}^{-2}$	muon lifetime
$\sin^2\theta_{\text{W}}$	$0.2233(6)$	on-shell, $1 - m_W^2/m_Z^2$
$\sin^2\theta_{\text{W}}$	$0.231\,21(4)$	$\overline{\text{MS}}$, $\mu = m_Z$

$$m_W = 80.3692(133)\,\text{GeV}, \tag{F.159}$$

$$m_h = 125.20(11)\,\text{GeV}. \tag{F.160}$$

The Z mass Eq. (F.158) is the single most precisely known electroweak mass and serves, with α and G_F, as a defining input. The W mass Eq. (F.159) is the world average; it carries a live experimental tension that must be flagged honestly.

Caution

The 2022 CDF II measurement reported $m_W = 80.4335(94)\,\text{GeV}$, some seven standard deviations above the average Eq. (F.159) and in tension with the Standard-Model electroweak fit. Subsequent measurements (ATLAS, LHCb) sit close to the fit value, and the discrepancy is unresolved as of the mid-2020s. The value Eq. (F.159) is the PDG average that does not include the discrepant CDF point; any electroweak fit using m_W should state which value it adopts.

The Higgs mass Eq. (F.160) is measured directly in the $\gamma\gamma$ and four-lepton channels and, like the gauge-boson masses, is a pole observable. With m_h, G_F, and the gauge couplings fixed, the quartic Higgs self-coupling λ and the vacuum expectation value v are determined; the extraction of v is Theorem F.23.

F.6.4 Fermion masses: the scheme is half the number

The charged leptons are quoted as pole masses, which for leptons are clean physical observables (QED corrections to the pole are tiny and well controlled):

$$m_e = 0.510\,998\,950\,0(2)\,\text{MeV}, \quad m_\mu = 105.658\,375\,5(2)\,\text{MeV}, \quad m_\tau = 1776.86(12)\,\text{MeV}. \tag{F.161}$$

The quarks are an entirely different story, because a confined particle has no asymptotic one-particle state and hence no unambiguous pole mass at the non-perturbative level. The mass that is short-distance well-defined is the *running* $\overline{\text{MS}}$ mass $\overline{m}(\mu)$, and the scale at which it is quoted is part of the datum. The convention is:

- the light quarks u, d, s, whose intrinsic mass scale lies below the regime where perturbation theory is trustworthy, are quoted as $\overline{\text{MS}}$ masses at the fixed reference scale $\mu = 2\,\text{GeV}$;
- the charm and bottom quarks are quoted as "scale-invariant" $\overline{\text{MS}}$ masses evaluated at their own value, $\overline{m}_c(\overline{m}_c)$ and $\overline{m}_b(\overline{m}_b)$, the natural self-consistent choice that makes the log in the pole relation vanish;
- the top quark, heavy enough to decay before it hadronizes, is quoted both as a (Monte-Carlo/direct) mass close to the pole mass and as an $\overline{\text{MS}}$ mass $\overline{m}_t(\overline{m}_t)$.

Table F.5: Quark masses with their schemes, mid-2020s [382]. Light quarks u, d, s are $\overline{\text{MS}}$ masses at $\mu = 2\,\text{GeV}$; charm and bottom are scale-invariant $\overline{\text{MS}}$ masses $\overline{m}(\overline{m})$; the top is given as the direct (pole-like) mass and as the $\overline{\text{MS}}$ mass. The asymmetric errors are reproduced as quoted.

Quark	Value	Scheme / scale
u	$2.16^{+0.49}_{-0.26}\,\text{MeV}$	$\overline{\text{MS}}$, $\mu = 2\,\text{GeV}$
d	$4.70^{+0.48}_{-0.17}\,\text{MeV}$	$\overline{\text{MS}}$, $\mu = 2\,\text{GeV}$
s	$93.5^{+8.6}_{-3.4}\,\text{MeV}$	$\overline{\text{MS}}$, $\mu = 2\,\text{GeV}$
c	$1.273(46)\,\text{GeV}$	$\overline{\text{MS}}$, $\overline{m}_c(\overline{m}_c)$
b	$4.183^{+0.037}_{-0.015}\,\text{GeV}$	$\overline{\text{MS}}$, $\overline{m}_b(\overline{m}_b)$
t	$172.57(29)\,\text{GeV}$	direct (pole-like)
t	$162.5^{+2.1}_{-1.5}\,\text{GeV}$	$\overline{\text{MS}}$, $\overline{m}_t(\overline{m}_t)$

The top entries make the point unmistakably: 172.6 GeV and 162.5 GeV are the *same quark*, the ten-GeV gap being nothing but the difference between the pole-type and $\overline{\text{MS}}$ definitions. Anyone who plugs the direct mass into a formula expecting $\overline{m}_t(\overline{m}_t)$ commits a six-percent error in the top mass—larger than every measurement uncertainty combined. We now show, at one loop, exactly where that gap comes from.

F.6.5 Worked example: $\overline{\text{MS}}$ mass to pole mass

Worked Example F.22 (From $\overline{m}_b(\overline{m}_b)$ to the bottom pole mass)**.** We convert the tabulated $\overline{\text{MS}}$ bottom mass $\overline{m}_b(\overline{m}_b) = 4.18\,\text{GeV}$ into the corresponding pole

mass at one loop in QCD, using the relation established in Chapter 22 and the master one-loop two-point integral of Chapter D, and we read off the lesson about scheme labels.

The relation. The pole mass M is defined as the location of the pole of the full quark propagator, fixed by the on-shell renormalization condition of Chapter 22; the $\overline{\text{MS}}$ mass $\overline{m}(\mu)$ is fixed by subtracting only the $1/\varepsilon$ pole of the self-energy. The difference between the two finite parts is the conversion factor. The one-loop quark self-energy $\Sigma(\not p)$ in dimensional regularization is built, after Feynman parametrization, on the standard scalar two-point master integral

$$B_0(p^2;m,0)=\frac{1}{i}\int\frac{d^dk}{(2\pi)^d}\,\frac{1}{(k^2-m^2)\,(k+p)^2}=\frac{1}{(4\pi)^2}\left[\frac{1}{\varepsilon}-\gamma_E+\ln 4\pi+2-\ln\frac{m^2}{\mu^2}+\dots\right], \tag{F.162}$$

whose value is taken from Chapter D. Carrying the Dirac algebra of Chapter 22 (in $d=4-2\varepsilon$, with $\gamma^\mu\not k\gamma_\mu=-(d-2)\not k$ and the color factor $C_F=\frac{N_c^2-1}{2N_c}=\frac{4}{3}$ for the fundamental of SU(3)) and imposing the two renormalization conditions yields the one-loop relation

$$M=\overline{m}(\mu)\left[1+\frac{\alpha_s(\mu)}{\pi}\,C_F\left(1+\frac{3}{4}\ln\frac{\mu^2}{\overline{m}^2(\mu)}\right)+\mathcal{O}(\alpha_s^2)\right]. \tag{F.163}$$

The structure is transparent: the color factor $C_F=4/3$ is the same one that appears in the QCD self-energy, the constant "1" is the finite part of Eq. (F.162), and the logarithm carries the running between μ and the mass scale—its coefficient $\frac{3}{4}C_F=1$ (the coefficient of $\ln(\mu^2/\overline{m}^2)$) is exactly *half* the one-loop mass anomalous dimension $\gamma_m=\frac{3}{2}C_F\,\alpha_s/\pi=2\alpha_s/\pi$ of Chapter 24, the factor of two arising because $\ln\mu^2=2\ln\mu$; this is what makes M (a physical pole) μ-independent order by order.

The self-consistent scale. Evaluating Eq. (F.163) at the scale-invariant point $\mu=\overline{m}(\overline{m})$ kills the logarithm and leaves the clean relation

$$M=\overline{m}(\overline{m})\left[1+\frac{4}{3}\,\frac{\alpha_s(\overline{m})}{\pi}+\mathcal{O}(\alpha_s^2)\right]. \tag{F.164}$$

This is why the bottom and charm masses are tabulated at their own scale: it makes the one-loop conversion a single multiplication.

The number. For the bottom quark, $\alpha_s(\overline{m}_b)\approx 0.225$ (running Eq. (F.154) down to 4.18 GeV with the QCD RG of Chapter 27). Then

$$\frac{4}{3}\,\frac{\alpha_s(\overline{m}_b)}{\pi}=\frac{4}{3}\cdot\frac{0.225}{3.1416}=\frac{4}{3}\times 0.0716=0.0955, \tag{F.165}$$

so

$$M_b^{\text{1-loop}}=4.18\,\text{GeV}\times(1+0.0955)=4.18\times 1.0955=4.58\,\text{GeV}. \tag{F.166}$$

A nearly half-GeV shift from a one-loop correction. Including the known two- and three-loop terms pushes the result up to roughly 4.8 GeV, and the series converges slowly: each order in α_s contributes a comparable fraction, a symptom of the *infrared renormalon* that makes the pole mass intrinsically ambiguous at the level of $\mathcal{O}(\Lambda_{\rm QCD}) \sim 200\,\text{MeV}$. The charm quark, with $\alpha_s(\overline{m}_c) \approx 0.39$, is worse: Eq. (F.164) gives $M_c^{\text{1-loop}} = 1.27\,(1 + \frac{4}{3}\cdot 0.124) = 1.27 \times 1.165 = 1.48\,\text{GeV}$, with a higher-order expansion that is barely under control.

The lesson. The $\overline{\rm MS}$ mass 4.18 GeV is short-distance, unambiguous, and scheme-labeled; the pole mass ≈ 4.8 GeV is a different number, definable only order by order, and ambiguous at a few hundred MeV. The two are not interchangeable, and a table entry that said merely "$m_b = 4.18\,\text{GeV}$" or "$m_b = 4.8\,\text{GeV}$" without its scheme would be useless for any computation accurate to better than ten percent. Scheme labels are not bureaucratic decoration; they are part of the measurement.

F.6.6 Worked example: the electroweak scale from G_F

Worked Example F.23 (Extracting $v = (\sqrt{2}\,G_F)^{-1/2}$)**.** The Higgs vacuum expectation value v sets the scale of electroweak symmetry breaking, yet it is not measured directly; it is derived from the precisely known Fermi constant Eq. (F.155). We carry out the extraction.

The relation. The Fermi constant was defined historically through the low-energy four-fermion muon-decay amplitude, which the full electroweak theory reproduces at tree level by exchanging a W boson. Matching the two (Chapter 31) gives

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\,m_W^2}. \tag{F.167}$$

Using the tree-level mass relation $m_W = \frac{1}{2} g\,v$ from the Higgs mechanism of Chapter 31 (the W acquires its mass from the doublet vev),

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\,(\frac{1}{2}gv)^2} = \frac{g^2}{8\cdot\frac{1}{4}g^2v^2} = \frac{1}{2v^2}, \tag{F.168}$$

so that the gauge coupling cancels and

$$v^2 = \frac{1}{\sqrt{2}\,G_F}, \qquad v = (\sqrt{2}\,G_F)^{-1/2}. \tag{F.169}$$

This is a tree-level definition; it fixes v as a derived quantity inheriting the sub-ppm precision of G_F, with loop corrections absorbed into the precise definition of G_F itself.

The number. Insert Eq. (F.155). First,

$$\sqrt{2}\,G_F = 1.414\,214 \times 1.166\,378\,7 \times 10^{-5}\,\text{GeV}^{-2} = 1.649\,38 \times 10^{-5}\,\text{GeV}^{-2}. \tag{F.170}$$

Then

$$v^2 = \frac{1}{1.649\,38 \times 10^{-5}\,\mathrm{GeV}^{-2}} = 6.0629 \times 10^4\,\mathrm{GeV}^2, \tag{F.171}$$

and taking the square root,

$$\boxed{v = \left(\sqrt{2}\,G_F\right)^{-1/2} = 246.22\,\mathrm{GeV}.} \tag{F.172}$$

This is the electroweak scale: every gauge-boson and fermion mass in the Standard Model is v times a dimensionless coupling. With v in hand one reads off, for example, the top Yukawa $y_t = \sqrt{2}\,\overline{m}_t/v \approx \sqrt{2} \times 162.5/246.2 \approx 0.93$, strikingly close to unity, and the Higgs self-coupling $\lambda = m_h^2/2v^2 \approx (125.2)^2/(2\times 246.22^2) \approx 0.129$. The number Eq. (F.172) is the anchor of the whole mass spectrum.

F.6.7 The CKM matrix in numbers

The quark mixing matrix V_{CKM}, whose standard parametrization in terms of three angles and one phase was given in Section F.5, is dominantly diagonal with a hierarchical off-diagonal structure captured by the Wolfenstein expansion in the small parameter $\lambda = \sin\theta_{12} \approx |V_{us}|$. The mid-2020s global-fit values are

$$\lambda = 0.22501(68), \quad A = 0.826^{+0.016}_{-0.015}, \quad \bar{\rho} = 0.1591(94), \quad \bar{\eta} = 0.3523(60), \tag{F.173}$$

which translate to the standard mixing angles $\theta_{12} \approx 13.0^\circ$, $\theta_{23} \approx 2.4^\circ$, $\theta_{13} \approx 0.20^\circ$, and a large CP phase $\delta \approx 66^\circ$. The magnitudes of the entries are

$$|V_{\mathrm{CKM}}| \approx \begin{pmatrix} 0.9743 & 0.2250 & 0.0037 \\ 0.2249 & 0.9735 & 0.0409 \\ 0.0086 & 0.0402 & 0.9991 \end{pmatrix}, \tag{F.174}$$

and the rephasing-invariant measure of CP violation, the Jarlskog invariant of Eq. (F.139), takes the value

$$J = (3.08^{+0.15}_{-0.13}) \times 10^{-5}. \tag{F.175}$$

That J is small not because the phase δ is small—it is of order one— but because J is suppressed by the product of all three small mixing angles, is the quantitative statement of why CP violation in the quark sector is a tiny effect despite an $\mathcal{O}(1)$ phase.

Caution

The third-row/third-column elements $|V_{cb}|$ and $|V_{ub}|$ carry a persistent *inclusive-versus-exclusive* tension: determinations from inclusive semileptonic B decays sit systematically above those from exclusive ($B \to D^{(*)}\ell\nu$, $B \to \pi\ell\nu$) channels, by roughly $|V_{cb}|_{\mathrm{incl}} \approx 42.2 \times 10^{-3}$ versus

$|V_{cb}|_{\text{excl}} \approx 39.8 \times 10^{-3}$ (a $\sim 3\sigma$ gap), and similarly for $|V_{ub}|$ (≈ 4.1 versus $\approx 3.7 \times 10^{-3}$). The averaged values in Eq. (F.174) paper over a discrepancy whose origin—experimental, or in the hadronic form factors—is unsettled as of the mid-2020s. A computation sensitive to $|V_{cb}|$ should state which determination it uses. This is also why the averaged entries $|V_{cb}| \approx 0.0409$ and $|V_{ts}| \approx 0.0402$ in Eq. (F.174) sit about 2% below the value $A\lambda^2 \approx 0.0418$ that the appendix's own Wolfenstein relation $|V_{cb}| = s_{23}c_{13} = A\lambda^2$ would give from the global-fit $A = 0.826$ of Eq. (F.173): the tabulated matrix elements are direct-measurement averages, whereas A is fixed by the global unitarity fit, so the gap is exactly this inclusive-versus-exclusive tension and not a failure of the Wolfenstein relation. The other entries ($|V_{ub}|$, $|V_{td}|$) are consistent with $A = 0.826$ at the quoted precision.

F.6.8 The PMNS matrix and neutrino masses

The lepton mixing matrix U_{PMNS}, parametrized exactly as V_{CKM} in Section F.5 (with possible additional Majorana phases that drop out of oscillations), is strikingly non-hierarchical: its mixing angles are all large. The global-fit values for normal ordering [180] are

$$\sin^2\theta_{12} = 0.307(12), \quad \sin^2\theta_{23} = 0.553^{+0.016}_{-0.024}, \quad \sin^2\theta_{13} = 0.02203(56), \tag{F.176}$$

corresponding to $\theta_{12} \approx 33.7^\circ$, $\theta_{23} \approx 48.0^\circ$, $\theta_{13} \approx 8.5^\circ$, with the CP-violating Dirac phase still poorly constrained, $\delta_{\text{CP}} \approx 197^\circ$ (consistent with a broad range and not yet established as nonzero). The two measured mass-squared differences are

$$\Delta m^2_{21} = 7.41^{+0.21}_{-0.20} \times 10^{-5}\,\text{eV}^2, \qquad |\Delta m^2_{31}| = 2.51^{+0.03}_{-0.03} \times 10^{-3}\,\text{eV}^2. \tag{F.177}$$

Table F.6: Lepton mixing parameters and neutrino mass-squared differences, normal ordering, from the mid-2020s global fit [180]. Contrast the large PMNS angles with the small CKM angles of Eq. (F.173).

Parameter	Value (normal ordering)
$\sin^2\theta_{12}$	$0.307(12)$
$\sin^2\theta_{23}$	$0.553^{+0.016}_{-0.024}$
$\sin^2\theta_{13}$	$0.022\,03(56)$
δ_{CP}	$\approx 197^\circ$ (weakly constrained)
Δm^2_{21}	$7.41 \times 10^{-5}\,\text{eV}^2$
$\lvert\Delta m^2_{31}\rvert$	$2.51 \times 10^{-3}\,\text{eV}^2$

Three honest caveats accompany these numbers, all reflecting genuine present ignorance rather than measurement imprecision. First, the *mass ordering* is undetermined: only Δm^2_{21} and $|\Delta m^2_{31}|$ are known, and whether $m_3 > m_2 > m_1$ (normal) or $m_3 < m_1 < m_2$ (inverted) is an open experimental question. Second, the *absolute mass scale* is unmeasured; oscillations fix only differences, and the lightest neutrino mass could be zero. Third, the θ_{23} *octant* (whether θ_{23} is above or below 45°) and the Dirac phase δ_{CP} are only weakly constrained. The Standard Model as defined in Chapter 33 has massless neutrinos; the very existence of Eq. (F.177) is therefore physics beyond it, recorded here as data and discussed as such in Section F.7.

F.6.9 Benchmark anchors: running α and the QCD scale

The numbers above are the numerical anchors for the two gauge threads carried through the book. QED is anchored by the running of α from Eq. (F.152) to Eq. (F.153), a flow computed from the QED beta function of Chapter 24; we do not recompute that beta function here. Yang–Mills theory is anchored by $\alpha_s(m_Z)$ of Eq. (F.154) together with the one-loop coefficient b_0 of Chapter 27, which together fix the QCD scale. We make the latter concrete, plugging measured numbers into the imported running formula.

Worked Example F.24 (The QCD scale Λ from $\alpha_s(m_Z)$)**.** The one-loop solution of the renormalization-group equation for the strong coupling, derived in Chapter 27, is

$$\alpha_s(\mu) = \frac{2\pi}{b_0 \ln(\mu/\Lambda)} = \frac{4\pi}{b_0 \ln(\mu^2/\Lambda^2)}, \qquad b_0 = 11 - \tfrac{2}{3}n_f, \tag{F.178}$$

which defines the QCD scale Λ as the integration constant of the flow. We do not rederive Eq. (F.178); we invert it. With five active flavors, $b_0 = 11 - \frac{10}{3} = \frac{23}{3} = 7.667$. Solving for Λ at $\mu = m_Z$ with $\alpha_s(m_Z) = 0.118$,

$$\ln \frac{m_Z^2}{\Lambda^2} = \frac{4\pi}{b_0\, \alpha_s(m_Z)} = \frac{12.566}{7.667 \times 0.118} = \frac{12.566}{0.9047} = 13.89, \tag{F.179}$$

so that $m_Z^2/\Lambda^2 = e^{13.89} = 1.08 \times 10^6$ and, with $m_Z = 91.19\,\mathrm{GeV}$,

$$\Lambda^{(5)}_{\text{1-loop}} = \frac{m_Z}{\sqrt{1.08 \times 10^6}} = \frac{91.19\,\mathrm{GeV}}{1039} = 0.088\,\mathrm{GeV} \approx 90\,\mathrm{MeV}. \tag{F.180}$$

The one-loop value is honestly small: the full two-loop $\overline{\mathrm{MS}}$ result is $\Lambda^{(5)}_{\overline{\mathrm{MS}}} \approx 0.21\,\mathrm{GeV}$, and the factor-of-two gap between one- and two-loop is itself a scheme-and-order statement of the kind this section exists to flag. The lesson rhymes with Theorem F.22: Λ is not a measured constant but a derived quantity whose value depends on the loop order and the flavor number n_f used to extract it. The $b_0 = 11 - \frac{2}{3}n_f$ input is supplied by Chapter 27 (Yang–Mills theory); the number it produces is anchored here.

Caution

A further open input, flagged for honesty, is the muon anomalous magnetic moment $a_\mu = (g-2)_\mu/2$. The Fermilab measurement [353] reaches sub-ppm precision, but the Standard-Model prediction it is compared against is itself unsettled: the data-driven (hadronic e^+e^-) and lattice evaluations of the hadronic vacuum polarization disagree, and a 2023 remeasurement of the $e^+e^- \to \pi^+\pi^-$ cross section shifted the data-driven value. As of the mid-2020s the size—and even the existence—of a discrepancy with the Standard Model is undetermined. This is a theory-input ambiguity, not a measurement error, and we record a_μ as an open number rather than a clean anchor.

These numbers, each now carrying the scheme that makes it meaningful, are the raw material of every reproducible computation in the book. The final section gathers the conversion relations—between $\overline{\text{MS}}$ and pole masses, between the $\{\alpha, G_F, m_Z\}$ and $\{g, g', v\}$ input sets, between on-shell and $\overline{\text{MS}}$ mixing angles—into a single input-scheme dictionary, so that any value used anywhere in the book can be traced back to, and reproduced from, this one place, and it records plainly which of these inputs the Standard Model explains and which it merely absorbs.

F.7 Conventions, Conversions, and the Input-Scheme Dictionary

The previous section made a point that this one must operationalize: a Standard Model number is meaningless without the scheme that defines it. We quoted the fine-structure constant in the Thomson limit, the weak mixing angle in three inequivalent definitions, the quark masses as $\overline{\text{MS}}$ running masses, and the strong coupling at the reference scale m_Z, and we insisted each be carried with its label. That discipline is only useful if one can also *move* between schemes and between the two natural sets of variables in which the electroweak theory is parametrized—because a cross section computed in Chapter 17 uses the Lagrangian couplings g, g', v, whereas the most precisely measured inputs are α, the Fermi constant G_F, and m_Z, and the two never coincide numerically until one writes the map between them. This closing section collects that map, and every other conversion the book relies on, in one place. Nothing physical is proved here: the Higgs mechanism and the tree-level mass relations come from Chapter 31, the radiative corrections from Chapters 31 and 34, the running of α_s and the definition of Λ_{QCD} from Chapter 27, and the anomalous dimensions behind the $\overline{\text{MS}}$ mass from Chapter 24. What is carried here in full is the *conversion algebra*—the inversion of the mass relations, the round-trip prediction of m_W, the factor-of-two hypercharge dictionary, and the pole/$\overline{\text{MS}}$ translation—because it is load-bearing

and belongs nowhere else. We end by recording, honestly and dated, the inputs the Standard Model takes from experiment without explaining: the smallness of $\bar{\theta}$ and the entire neutrino-mass sector.

F.7.1 The electroweak input dictionary

The electroweak sector of the Standard Model is fixed, at tree level, by three real parameters. In the Lagrangian they are the $\mathrm{SU}(2)_L$ gauge coupling g, the $\mathrm{U}(1)_Y$ gauge coupling g', and the Higgs vacuum expectation value $v \equiv \sqrt{2}\,\langle \phi^0 \rangle$ (the normalization fixed in Chapter 31, so that $v \approx 246$ GeV). Call this the *Lagrangian set* $\{g, g', v\}$. Every electroweak amplitude in the book is a function of these three numbers. They are, however, not the quantities measured most accurately. The *experimental set* $\{\alpha, G_F, m_Z\}$—the fine-structure constant in the Thomson limit, the Fermi constant from the muon lifetime, and the Z pole mass—is known to far higher relative precision, and it is in this set that the inputs of the previous section are quoted. The two sets carry the same information; the dictionary between them is what follows.

The physics that relates them is the electroweak symmetry breaking of Chapter 31. We import its tree-level output, which we will not reprove. After $\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y \to \mathrm{SU}(3)_c \times \mathrm{U}(1)_{\mathrm{em}}$, the mixing of the neutral gauge fields defines the weak mixing angle θ_{W} through

$$\cos\theta_{\mathrm{W}} = \frac{g}{\sqrt{g^2 + g'^2}}, \qquad \sin\theta_{\mathrm{W}} = \frac{g'}{\sqrt{g^2 + g'^2}}, \qquad \tan\theta_{\mathrm{W}} = \frac{g'}{g}, \tag{F.181}$$

the photon couples with strength

$$e = \frac{g\,g'}{\sqrt{g^2 + g'^2}} = g\sin\theta_{\mathrm{W}} = g'\cos\theta_{\mathrm{W}}, \qquad \alpha = \frac{e^2}{4\pi}, \tag{F.182}$$

the massive gauge bosons acquire

$$m_W = \tfrac{1}{2}\,g\,v, \qquad m_Z = \tfrac{1}{2}\,v\sqrt{g^2 + g'^2} = \frac{m_W}{\cos\theta_{\mathrm{W}}}, \tag{F.183}$$

and the strength of the charged-current four-fermion interaction that governs muon decay is, at tree level,

$$\frac{G_F}{\sqrt{2}} = \frac{g^2}{8\,m_W^2} = \frac{1}{2\,v^2}. \tag{F.184}$$

The last equality in Eq. (F.184) follows from Eq. (F.183): $g^2/(8m_W^2) = g^2/(8 \cdot g^2 v^2/4) = 1/(2v^2)$, so the Fermi constant measures the vacuum expectation value directly and nothing else. Equations (F.181) to (F.184) are the forward map: given the Lagrangian set $\{g, g', v\}$ they produce $\{\alpha, G_F, m_Z\}$ (and m_W, and θ_{W}) by arithmetic.

Inverting the map. The experimental set is the input; we need the inverse. The cleanest route uses $\theta_{\rm W}$ as an intermediate. From Eq. (F.182) and the first equality of Eq. (F.183),

$$m_W^2 = \tfrac{1}{4} g^2 v^2 = \frac{e^2}{4\sin^2\theta_{\rm W}}\, v^2 = \frac{\pi\alpha}{\sqrt{2}\, G_F}\, \frac{1}{\sin^2\theta_{\rm W}}, \tag{F.185}$$

where in the last step we used $e^2 = 4\pi\alpha$ from Eq. (F.182) and $v^2 = 1/(\sqrt{2}\, G_F)$ from Eq. (F.184). Combining Eq. (F.185) with $m_W = m_Z \cos\theta_{\rm W}$ from Eq. (F.183) eliminates m_W and isolates the mixing angle:

$$m_Z^2 \cos^2\theta_{\rm W} \sin^2\theta_{\rm W} = \frac{\pi\alpha}{\sqrt{2}\, G_F} \quad \Longrightarrow \quad \boxed{\sin^2\theta_{\rm W} \cos^2\theta_{\rm W} = \frac{\pi\alpha}{\sqrt{2}\, G_F\, m_Z^2}} \tag{F.186}$$

This is the master relation of the input dictionary: its right-hand side is built entirely from the experimental set, and it determines $\sin^2\theta_{\rm W}$ as the smaller root of a quadratic. Once $\theta_{\rm W}$ is known, the rest of the inverse map is explicit:

$$\begin{aligned} e &= \sqrt{4\pi\alpha}, & v &= (\sqrt{2}\, G_F)^{-1/2}, \\ g &= \frac{e}{\sin\theta_{\rm W}}, & g' &= \frac{e}{\cos\theta_{\rm W}}, \end{aligned} \tag{F.187}$$

and $m_W = m_Z \cos\theta_{\rm W}$ closes the system. Equations (F.186) and (F.187) are the inverse map: given $\{\alpha, G_F, m_Z\}$ they produce $\{g, g', v\}$, the mixing angle, and a *prediction* for m_W. That m_W emerges as a prediction rather than an input is the content of the next worked example, and the gap between the predicted and the measured m_W is the doorway through which loop corrections enter.

Table F.7: The electroweak input dictionary at tree level. The forward map takes the Lagrangian set $\{g, g', v\}$ to the experimental set $\{\alpha, G_F, m_Z\}$; the inverse map runs the other way through the master relation Eq. (F.186). All relations are tree-level and imported from Chapter 31; loop corrections are discussed in Section F.7.2.

Forward: $\{g, g', v\} \to \{\alpha, G_F, m_Z\}$	Inverse: $\{\alpha, G_F, m_Z\} \to \{g, g', v\}$
$e = gg'/\sqrt{g^2 + g'^2}$	$\sin^2\theta_{\rm W}\cos^2\theta_{\rm W} = \pi\alpha/(\sqrt{2}\, G_F m_Z^2)$
$\alpha = e^2/4\pi$	$e = \sqrt{4\pi\alpha}$
$\cos\theta_{\rm W} = g/\sqrt{g^2 + g'^2}$	$v = (\sqrt{2}\, G_F)^{-1/2}$
$m_Z = \tfrac{1}{2} v \sqrt{g^2 + g'^2}$	$g = e/\sin\theta_{\rm W}, \quad g' = e/\cos\theta_{\rm W}$
$G_F/\sqrt{2} = 1/(2v^2)$	$m_W = m_Z \cos\theta_{\rm W}$ (predicted)

F.7.2 Worked round-trip: predicting m_W at tree level

We now carry the inverse map through with the dated numbers of the previous section, compute the Lagrangian set, and predict m_W. The exercise is worth

doing in full because the answer is both a triumph and a warning: the tree-level prediction lands within a percent of the measurement, which is why the $\{\alpha, G_F, m_Z\}$ scheme is the standard one, but the residual percent is a genuine physical effect—the electroweak radiative corrections—and not numerical slop.

Worked Example F.25 (From $\{\alpha, G_F, m_Z\}$ to $\{g, g', v\}$ and a prediction for m_W)**.** We use the mid-2020s (PDG 2024) values quoted in the previous section,

$$\alpha^{-1} = 137.035\,999, \qquad G_F = 1.166\,379 \times 10^{-5}\ \text{GeV}^{-2}, \qquad m_Z = 91.1880\ \text{GeV}, \tag{F.188}$$

with $\alpha = \alpha(0)$ the Thomson-limit coupling, G_F the muon-decay constant, and m_Z the on-shell Z mass. All three carry the schemes fixed in the previous section; we propagate them faithfully.

Step 1: the master right-hand side. Evaluate the dimensionless combination on the right of Eq. (F.186). The numerator is

$$\pi\alpha = \frac{3.141\,593}{137.035\,999} = 2.292\,53 \times 10^{-2}. \tag{F.189}$$

The denominator factor $\sqrt{2}\,G_F = 1.414\,214 \times 1.166\,379 \times 10^{-5} = 1.649\,51 \times 10^{-5}\ \text{GeV}^{-2}$, and $m_Z^2 = (91.1880)^2 = 8315.25\ \text{GeV}^2$, so

$$\sqrt{2}\,G_F\,m_Z^2 = 1.649\,51 \times 10^{-5} \times 8315.25 = 0.137\,16, \tag{F.190}$$

which is dimensionless as it must be. Hence

$$A \equiv \frac{\pi\alpha}{\sqrt{2}\,G_F\,m_Z^2} = \frac{2.292\,53 \times 10^{-2}}{0.137\,16} = 0.167\,15. \tag{F.191}$$

Step 2: solve the quadratic for $\sin^2\theta_{\mathrm{W}}$. Writing $s \equiv \sin^2\theta_{\mathrm{W}}$, Eq. (F.186) reads $s(1-s) = A$, i.e. $s^2 - s + A = 0$, with roots

$$s = \frac{1 \pm \sqrt{1 - 4A}}{2} = \frac{1 \pm \sqrt{1 - 0.668\,60}}{2} = \frac{1 \pm 0.575\,67}{2}. \tag{F.192}$$

The two roots are $s = 0.2122$ and $s = 0.7879$; they are $\sin^2\theta_{\mathrm{W}}$ and $\cos^2\theta_{\mathrm{W}}$ respectively, and the physical mixing angle is the smaller one,

$$\sin^2\theta_{\mathrm{W}}\big|_{\text{tree}} = 0.2122, \tag{F.193}$$

so that $\cos^2\theta_{\mathrm{W}} = 0.7878$.

Step 3: the Lagrangian set. From Eq. (F.187),

$$e = \sqrt{4\pi\alpha} = \sqrt{4 \times 2.292\,53 \times 10^{-2} \times \pi/\pi} = \sqrt{0.091\,70} = 0.302\,82,$$
$$v = \left(\sqrt{2}\,G_F\right)^{-1/2} = \left(1.649\,51 \times 10^{-5}\right)^{-1/2} = 246.2\ \text{GeV},$$

$$g = \frac{e}{\sin\theta_{\rm W}} = \frac{0.302\,82}{\sqrt{0.2122}} = \frac{0.302\,82}{0.460\,6} = 0.6574,$$
$$g' = \frac{e}{\cos\theta_{\rm W}} = \frac{0.302\,82}{\sqrt{0.7878}} = \frac{0.302\,82}{0.887\,6} = 0.3412. \tag{F.194}$$

(The middle line in e is just $e = \sqrt{4\pi\alpha}$ with $4\pi\alpha = 0.091\,70$; we wrote the factor of π explicitly to make the cancellation visible.) These are the canonical values $\{g, g', v\} \approx \{0.66,\ 0.34,\ 246\ \text{GeV}\}$ used throughout the body of the book. A consistency check: $\tan\theta_{\rm W} = g'/g = 0.3412/0.6574 = 0.519$, so $\tan^2\theta_{\rm W} = 0.269$ and $\sin^2\theta_{\rm W} = \tan^2\theta_{\rm W}/(1+\tan^2\theta_{\rm W}) = 0.269/1.269 = 0.2122$, reproducing Eq. (F.193).

Step 4: predict m_W. The mass relation $m_W = m_Z\cos\theta_{\rm W}$ now gives a number that was *not* an input:

$$m_W\big|_{\rm tree} = m_Z\sqrt{\cos^2\theta_{\rm W}} = 91.1880 \times \sqrt{0.7878} = 91.1880 \times 0.887\,6 = 80.94\ \text{GeV}. \tag{F.195}$$

Equivalently, from the master form Eq. (F.185), $m_W^2 = \pi\alpha/(\sqrt{2}\,G_F\sin^2\theta_{\rm W}) = 2.292\,53\times10^{-2}/(1.649\,51\times10^{-5}\times0.2122) = 6551\ \text{GeV}^2$, giving the same $m_W = 80.94$ GeV.

The measured value, dated to the same epoch, is $m_W^{\rm exp} = 80.369 \pm 0.013$ GeV (PDG 2024 world average). The tree-level prediction Eq. (F.195) overshoots it by

$$\frac{m_W^{\rm tree} - m_W^{\rm exp}}{m_W^{\rm exp}} = \frac{80.94 - 80.369}{80.369} = 0.71\%. \tag{F.196}$$

A sub-percent discrepancy is precisely what makes the $\{\alpha, G_F, m_Z\}$ scheme the gold-standard input set: these three quantities are measured to parts in 10^5 or better, so almost the entire residual in Eq. (F.196) is theory, not experimental error. That residual is the electroweak radiative correction, conventionally packaged into a single quantity Δr that modifies the master relation Eq. (F.185) to

$$m_W^2\Big(1 - \frac{m_W^2}{m_Z^2}\Big) = \frac{\pi\alpha}{\sqrt{2}\,G_F}\,\frac{1}{1-\Delta r}, \tag{F.197}$$

whose derivation and numerical value $\Delta r \approx 0.036$ are imported from Chapter 31 and confronted with the global electroweak fit in Chapter 34. The single largest piece of Δr is the running of the electromagnetic coupling from the Thomson limit up to m_Z, $\alpha(0) \to \alpha(m_Z)$ with $\alpha(m_Z)^{-1} \approx 128$, an effect computed in Chapter 24; the remainder carries the genuine sensitivity to m_t and m_H that made the electroweak fit a prediction machine for both. The lesson for this appendix is structural: the *tree-level* dictionary of Table F.7 is exact as algebra and accurate to $\sim 1\%$ as physics, and the named scheme on each input is exactly what tells the reader at which loop order the residual percent must be supplied.

Caution

The mixing angle is the worst offender for scheme ambiguity, and the round-trip makes the size of the ambiguity concrete. The *tree value derived from the inputs*, Eq. (F.193), is $\sin^2\theta_{\rm W} = 0.2122$. The *on-shell* value, defined by $\sin^2\theta_{\rm W} \equiv 1 - m_W^2/m_Z^2$ with the *measured* m_W, is $1-(80.369/91.1880)^2 = 0.2232$. The $\overline{\rm MS}$ value $\hat{s}^2(m_Z) \approx 0.231$ and the effective leptonic value $\sin^2\theta_{\rm eff}^{\rm lept} \approx 0.2315$ differ again. These four numbers describe the same physics and differ by up to $\sim 9\%$; quoting "$\sin^2\theta_{\rm W}$" without its definition is therefore not a rounding question but a several-percent error. This is the previous section's thesis made arithmetic.

F.7.3 The two hypercharge normalizations

This appendix froze the hypercharge normalization in Eq. (F.3) as $Q = T^3 + Y$, the "small-Y" convention, in which the left-handed lepton doublet carries $Y_{L_L} = -\frac{1}{2}$. A large fraction of the literature instead writes the Gell-Mann–Nishijima relation as

$$Q = T^3 + \frac{Y'}{2}, \tag{F.198}$$

the "large-Y" or "$Y/2$" convention. The two are related by an overall factor of two,

$$Y' = 2Y, \tag{F.199}$$

and they describe identical physics: the rescaling of the abelian charge is compensated by a rescaling of the coupling, $g'_{\rm (large)} = \frac{1}{2}\, g'_{\rm (small)}$, so that the covariant-derivative term $g' Y B_\mu = g'_{\rm (large)}\,(Y'/2)\, B_\mu$ is invariant. Neither the electric charges nor any observable changes; only the bookkeeping label on each multiplet does. The conversion is therefore a pure dictionary entry, and we record it once and demonstrate it on a multiplet so that no reader is tripped by the factor of two when comparing this appendix to an outside source.

Worked Example F.26 (Hypercharge conversion on the lepton and quark doublets)**.** *Lepton doublet.* In the small-Y convention of Table F.1, $L_L \sim (\mathbf{1}, \mathbf{2})$ with $Y_{L_L} = -\frac{1}{2}$. The components have $T^3 = +\frac{1}{2}$ (the ν_L) and $T^3 = -\frac{1}{2}$ (the e_L), so Eq. (F.3) gives

$$Q(\nu_L) = +\tfrac{1}{2} - \tfrac{1}{2} = 0, \qquad Q(e_L) = -\tfrac{1}{2} - \tfrac{1}{2} = -1. \tag{F.200}$$

Convert with Eq. (F.199): $Y'_{L_L} = 2 \times (-\frac{1}{2}) = -1$. Now use the large-$Y$ relation Eq. (F.198):

$$Q(\nu_L) = +\tfrac{1}{2} + \frac{-1}{2} = 0, \qquad Q(e_L) = -\tfrac{1}{2} + \frac{-1}{2} = -1. \tag{F.201}$$

The charges are identical, as they must be—only the label changed from $Y = -\frac{1}{2}$ to $Y' = -1$.

Quark doublet. In the small-Y convention $Q_L \sim (\mathbf{3}, \mathbf{2})$ with $Y_{Q_L} = +\frac{1}{6}$, giving $Q(u_L) = \frac{1}{2} + \frac{1}{6} = +\frac{2}{3}$ and $Q(d_L) = -\frac{1}{2} + \frac{1}{6} = -\frac{1}{3}$. Converting, $Y'_{Q_L} = 2 \times \frac{1}{6} = +\frac{1}{3}$, and

$$Q(u_L) = +\tfrac{1}{2} + \frac{1/3}{2} = \tfrac{1}{2} + \tfrac{1}{6} = +\tfrac{2}{3}, \qquad Q(d_L) = -\tfrac{1}{2} + \tfrac{1}{6} = -\tfrac{1}{3}, \tag{F.202}$$

again unchanged. The full dictionary for one generation is then read off by doubling every entry of the Y column of Table F.1: $Y'_{Q_L} = \frac{1}{3}$, $Y'_{u_R} = \frac{4}{3}$, $Y'_{d_R} = -\frac{2}{3}$, $Y'_{L_L} = -1$, $Y'_{e_R} = -2$, $Y'_H = +1$.

Notation Watch

When importing a hypercharge from any external table, first identify the convention by the lepton doublet: if $Y_{L_L} = -\frac{1}{2}$ it is the small-Y convention of this book (Eq. (F.3)); if $Y_{L_L} = -1$ it is the large-Y convention (Eq. (F.198)). Then convert with $Y' = 2Y$ if and only if the two conventions disagree. The same factor of two propagates into the abelian coupling ($g'_{\text{large}} = \frac{1}{2} g'_{\text{small}}$) and into the $[\mathrm{U}(1)_Y]$ entries of any anomaly sum, so a mixed convention silently breaks the cancellations of Table F.2; never mix them within a single calculation.

F.7.4 Mass schemes and the strong coupling

The masses tabulated in the previous section are not all defined the same way, and the difference matters most for the quarks. Two conventions are in play, and the appendix must record both and the bridge between them.

Pole versus $\overline{\text{MS}}$ mass. The *pole mass* M is the position of the pole of the full propagator, a gauge-invariant and renormalization-scheme-independent quantity; it is the natural mass for a particle that exists as an asymptotic state. The *$\overline{MS}$ mass* $\overline{m}(\mu)$ is the renormalized mass parameter in the $\overline{\text{MS}}$ scheme at scale μ; it runs with μ through the mass anomalous dimension γ_m computed in Chapter 24. For a quark these differ at order α_s because of the gluonic self-energy. The one-loop relation, whose derivation from the on-shell self-energy and the on-shell-versus-$\overline{\text{MS}}$ conditions is imported from Chapter 22 (with the color factor C_F replacing the abelian charge), is

$$M = \overline{m}(\overline{m}) \left[1 + \frac{4}{3} \frac{\alpha_s(\overline{m})}{\pi} + \left(K - \tfrac{4}{3} \right) \left(\frac{\alpha_s}{\pi} \right)^2 + \cdots \right], \tag{F.203}$$

where the coefficient $\frac{4}{3} = C_F$ is the fundamental Casimir (Chapter B) and $K \approx 13.4 - 1.0\, n_\ell$ collects the two-loop constant for n_ℓ lighter flavors. The shift is

sizable: for the top quark $\alpha_s \approx 0.108$ at $\mu = m_t$ makes $M_t - \overline{m}_t(\overline{m}_t) \approx 10$ GeV, so quoting a top mass of "173 GeV" without saying pole or $\overline{\text{MS}}$ is a $\sim 6\%$ ambiguity. The running of the $\overline{\text{MS}}$ mass between scales is the solution of the γ_m equation of Chapter 24; at one loop, for n_f active flavors,

$$\overline{m}(\mu_2) = \overline{m}(\mu_1)\left[\frac{\alpha_s(\mu_2)}{\alpha_s(\mu_1)}\right]^{12/(33-2n_f)}, \tag{F.204}$$

which is the formula one uses to translate, say, the bottom mass $\overline{m}_b(\overline{m}_b)$ quoted at its own scale up to the reference scale m_Z.

The strong coupling and Λ_{QCD}. The strong coupling is quoted as $\alpha_s(m_Z) \approx 0.1180$, a number that is only meaningful with its scale and its scheme ($\overline{\text{MS}}$, $n_f = 5$) attached. The alternative parametrization replaces the dimensionless $\alpha_s(\mu)$ by the dimensionful Λ_{QCD}, the scale at which the perturbative coupling formally diverges. Solving the one-loop renormalization-group equation of Chapter 27, $\mu\,\mathrm{d}\alpha_s/\mathrm{d}\mu = -(b_0/2\pi)\alpha_s^2$ with $b_0 = 11 - \tfrac{2}{3}n_f$, gives

$$\alpha_s(\mu) = \frac{2\pi}{b_0 \ln(\mu/\Lambda_{\text{QCD}})}, \qquad \Lambda_{\text{QCD}} = \mu \exp\!\left[-\frac{2\pi}{b_0\,\alpha_s(\mu)}\right], \tag{F.205}$$

which is the dictionary between the two ways of stating the same coupling. Inserting $\mu = m_Z$, $n_f = 5$ (so $b_0 = 11 - \tfrac{10}{3} = \tfrac{23}{3}$), and $\alpha_s(m_Z) = 0.1180$,

$$\ln\frac{m_Z}{\Lambda^{(5)}} = \frac{2\pi}{b_0\,\alpha_s(m_Z)} = \frac{6.2832}{7.667 \times 0.1180} = 6.94, \quad \Longrightarrow \quad \Lambda^{(5)} \approx 91188\,\text{MeV} \times e^{-6.94} \approx 88\,\text{MeV}. \tag{F.206}$$

This one-loop value is deliberately crude: the physical $\Lambda^{(5)}_{\overline{\text{MS}}} \approx 210$ MeV requires the two-loop running and the flavor-threshold matching that Chapter 27 carries, and the very definition of Λ is scheme- and order-dependent. We record Eq. (F.205) as the structural dictionary and defer the precise number to Chapter 27; the point for this appendix is only that "$\alpha_s(m_Z) = 0.118$" and "$\Lambda^{(5)} \approx 0.2$ GeV" are two encodings of one input, related by Eq. (F.205).

F.7.5 What the Standard Model does not explain

The appendix has so far presented the Standard Model as a closed accounting system: a gauge group, a representation table, one scalar, a counted list of parameters, and the scheme-labeled numbers that fix them. It would be dishonest to end without recording, plainly and dated, that two of those inputs are not merely *measured* rather than predicted—all of the parameters are that—but are measured to take values the theory gives no reason to expect, or cannot accommodate at all without extension. These are the open inputs, and they are the natural exit from this appendix into the frontier of Chapter 35.

The smallness of $\bar{\theta}$ (the strong-CP problem). The QCD Lagrangian admits a CP-violating term $\theta\,(g_s^2/32\pi^2)\,G^a_{\mu\nu}\widetilde{G}^{a\,\mu\nu}$, and the physical, basis-independent angle is the combination

$$\bar{\theta} = \theta + \arg\det(M_u M_d), \tag{F.207}$$

already identified in the parameter count (Eq. (F.89)), where M_u, M_d are the up- and down-type quark mass matrices, so that a chiral rotation that removes a phase from the masses moves it into θ and only the sum is observable (this is the accidental-symmetry analysis of Chapter 33). A nonzero $\bar{\theta}$ generates a neutron electric dipole moment; the experimental bound, as of the mid-2020s, is $|\bar{\theta}| \lesssim 10^{-10}$. The Standard Model contains no symmetry forcing $\bar{\theta}$ to vanish—it is a free parameter that could have been of order unity—yet it is measured to be smaller than 10^{-10}. Why a CP-violating angle that the theory permits, and that the electroweak sector demonstrably uses elsewhere (the CKM phase), should be so finely tuned to zero in the strong sector is the *strong-CP problem*. It is unexplained within the Standard Model. The leading proposed resolution promotes $\bar{\theta}$ to a dynamical field (the axion); the status of both problem and proposal as of the mid-2020s, and the instanton physics that makes $\bar{\theta}$ physical in the first place, are taken up in Chapter 35.

The neutrino sector: scale, ordering, and nature. The minimal Standard Model of Table F.1 contains no right-handed neutrino and forbids a renormalizable neutrino mass; the neutrinos are exactly massless at the renormalizable level. The observation of neutrino oscillations establishes that they are not, and this is the one place where the data *require* physics beyond the field content tabulated here. Three facts are open as of the mid-2020s. *(i) Scale.* Oscillations measure only the mass-squared splittings, $\Delta m^2_{21} \approx 7.5 \times 10^{-5}$ eV^2 and $|\Delta m^2_{31}| \approx 2.5 \times 10^{-3}$ eV^2, not the absolute scale; the sum of masses is bounded from above by cosmology at $\sum m_\nu \lesssim 0.1$ eV and from below by the larger splitting at $\gtrsim 0.05$ eV, but no individual mass is known. *(ii) Ordering.* Whether the spectrum is normal ($m_1 < m_2 < m_3$) or inverted ($m_3 < m_1 < m_2$)—the sign of Δm^2_{31}—is not yet settled. *(iii) Nature.* Whether neutrinos are Dirac particles (requiring a new ν_R and lepton-number conservation) or Majorana particles (their own antiparticles, with lepton number violated) is unknown; the decisive experiment, neutrinoless double-beta decay, has not observed a signal. The mixing in the lepton sector is parametrized by the PMNS matrix $U_{\rm PMNS}$ introduced earlier in this appendix, in exact analogy to the CKM matrix $V_{\rm CKM}$, but the absolute scale, the ordering, and the Dirac-versus-Majorana question are inputs the Standard Model neither predicts nor, in its minimal form, even accommodates. The mass mechanisms that extend it—most economically the seesaw—are the subject of Chapter 35.

This completes the conversion machinery. Any number used anywhere in the book—a coupling in a cross section, a mixing angle in a width, a quark mass in

a threshold—can now be traced back through Table F.7, Eqs. (F.203) to (F.205), and the hypercharge dictionary to the scheme-labeled inputs of the previous section, and converted to whatever scheme a given computation demands. With the defining data, the complete Lagrangian, the parameter count, the measured values, and now the conversions all collected in one place, this appendix provides a consolidated Standard Model reference for the entire book.

Further reading

The authoritative source for every measured number in this appendix, with its scheme, its date, and its uncertainty, is the Particle Data Group's *Review of Particle Physics* [382]; it is the table one consults to update any value quoted here. The construction of the Standard Model Lagrangian and its electroweak and flavor sectors is developed at textbook length in Peskin and Schroeder [396], Schwartz [432], Srednicki [454] and, with particular care for renormalization and the symmetry structure, in Weinberg [501]. The historical foundations of the electroweak theory are the original papers of Glashow [224], Salam [426], Weinberg [496]; the Higgs mechanism is Higgs [267]; and the three-generation origin of CP violation through the quark mixing matrix is Kobayashi and Maskawa [298]. These are for going beyond the present book; every result this appendix relies upon is proved in the chapters cited throughout the text.

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